FORM

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Generated by Doxygen 1.9.8

Chapter 1

Data Structure Index

1.1 Data Structures

Here are the data structures with brief descriptions:

Chapter 2

File Index

2.1 File List

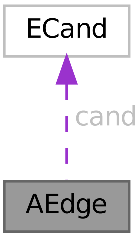
Here is a list of all documented files with brief descriptions:

Chapter 3

Data Structure Documentation

3.1 AEdge Class Reference

Collaboration diagram for AEdge:



Public Member Functions

- AEdge (int n0, int l0, int n1, int l1)

Data Fields

- ECand ∗ cand
- int nodes [2]
- int nlegs [2]
- int ptcl
- int DUMMYPADDING

3.1.1 Detailed Description

Definition at line 1102 of file grcc.h.

3.1.2 Constructor & Destructor Documentation

3.1.2.1 AEdge()

```
AEdge::AEdge (
            int n0,
            int l0,
            int n1,
            int l1 )
```

Definition at line 7413 of file grcc.cc.

3.1.2.2 ∼AEdge()

```
AEdge::∼AEdge (
            void )
```

Definition at line 7424 of file grcc.cc.

3.1.3 Field Documentation

3.1.3.1 cand

```
ECand* AEdge::cand
```

Definition at line 1109 of file grcc.h.

3.1.3.2 nodes

```
int AEdge::nodes[2]
```

Definition at line 1110 of file grcc.h.

3.1.3.3 nlegs

```
int AEdge::nlegs[2]
```

Definition at line 1111 of file grcc.h.

3.1.3.4 ptcl

```
int AEdge::ptcl
```

Definition at line 1112 of file grcc.h.

3.1.3.5 DUMMYPADDING

```
int AEdge::DUMMYPADDING
```

Definition at line 1114 of file grcc.h.

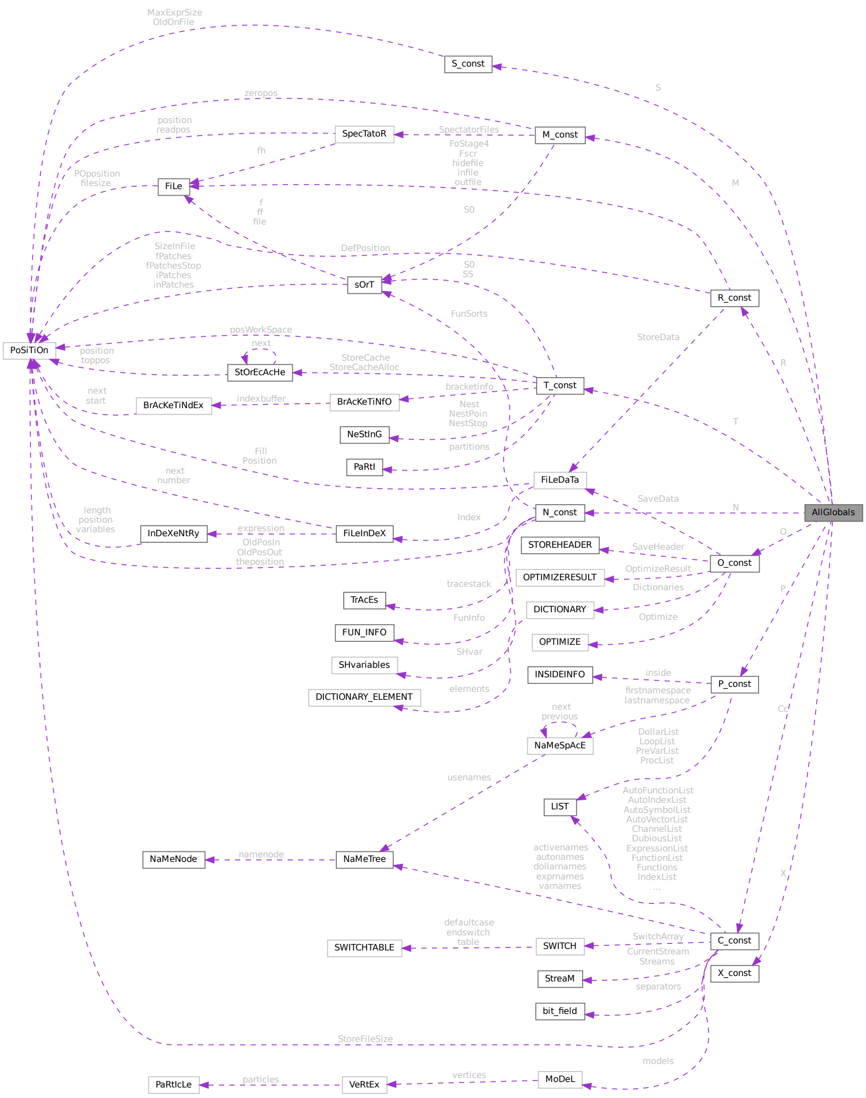
The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.2 AllGlobals Struct Reference

```
#include <structs.h>
```

Collaboration diagram for AllGlobals:



Public Member Functions

- **PADPOSITION** (0, 0, 0, 0, sizeof(struct P_const)+sizeof(struct T_const)+sizeof(struct X_const))

Data Fields

- struct M_const M
- struct C_const Cc
- struct S_const S
- struct R_const R
- struct N_const N
- struct O_const O
- struct P_const P
- struct T_const T
- struct X_const X

3.2.1 Detailed Description

Without pthreads (FORM) the ALLGLOBALS struct has all the global variables

Definition at line 2620 of file structs.h.

3.2.2 Field Documentation

3.2.2.1 M

`struct M_const AllGlobals::M`

Definition at line 2621 of file structs.h.

3.2.2.2 Cc

`struct C_const AllGlobals::Cc`

Definition at line 2622 of file structs.h.

3.2.2.3 S

`struct S_const AllGlobals::S`

Definition at line 2623 of file structs.h.

3.2.2.4 R

`struct R_const AllGlobals::R`

Definition at line 2624 of file structs.h.

3.2.2.5 N

`struct N_const AllGlobals::N`

Definition at line 2625 of file structs.h.

3.2.2.6 O

`struct O_const AllGlobals::O`

Definition at line 2626 of file structs.h.

3.2.2.7 P

`struct P_const AllGlobals::P`

Definition at line 2627 of file structs.h.

3.2.2.8 T

`struct T_const AllGlobals::T`

Definition at line 2628 of file structs.h.

3.2.2.9 X

`struct X_const AllGlobals::X`

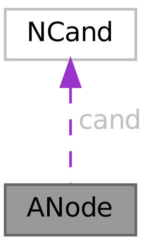
Definition at line 2629 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.3 ANode Class Reference

Collaboration diagram for ANode:



Public Member Functions

- ANode (int dg)
- int newleg (void)

Data Fields

- int deg
- int nlegs
- int * anodes
- int * aedges
- int * aelegs
- NCand * cand

3.3.1 Detailed Description

Definition at line 1078 of file grcc.h.

3.3.2 Constructor & Destructor Documentation

3.3.2.1 ANode()

```
ANode::ANode (
        int dg )
```

Definition at line 7358 of file grcc.cc.

3.3.2.2 ∼ANode()

```
ANode::∼ANode (
        void )
```

Definition at line 7376 of file grcc.cc.

3.3.3 Member Function Documentation

3.3.3.1 newleg()

```
int ANode::newleg (
        void )
```

Definition at line 7393 of file grcc.cc.

3.3.4 Field Documentation

3.3.4.1 deg

`int ANode::deg`

Definition at line 1088 of file grcc.h.

3.3.4.2 nlegs

`int ANode::nlegs`

Definition at line 1089 of file grcc.h.

3.3.4.3 anodes

`int* ANode::anodes`

Definition at line 1090 of file grcc.h.

3.3.4.4 aedges

`int* ANode::aedges`

Definition at line 1091 of file grcc.h.

3.3.4.5 aelegs

`int* ANode::aelegs`

Definition at line 1092 of file grcc.h.

3.3.4.6 cand

`NCand* ANode::cand`

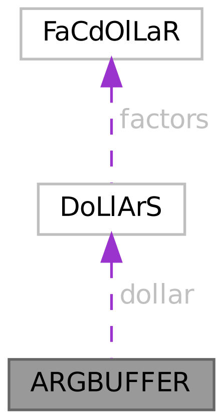
Definition at line 1093 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.4 ARGBUFFER Struct Reference

Collaboration diagram for ARGBUFFER:



Data Fields

- WORD ∗ buffer
- DOLLARS dollar
- LONG size
- int type
- int dummy

3.4.1 Detailed Description

Definition at line 601 of file ratio.c.

3.4.2 Field Documentation

3.4.2.1 buffer

```
WORD* ARGBUFFER::buffer
```

Definition at line 602 of file ratio.c.

3.4.2.2 dollar

```
DOLLARS ARGBUFFER::dollar
```

Definition at line 603 of file ratio.c.

3.4.2.3 size

```
LONG ARGBUFFER::size
```

Definition at line 604 of file ratio.c.

3.4.2.4 type

```
int ARGBUFFER::type
```

Definition at line 605 of file ratio.c.

3.4.2.5 dummy

```
int ARGBUFFER::dummy
```

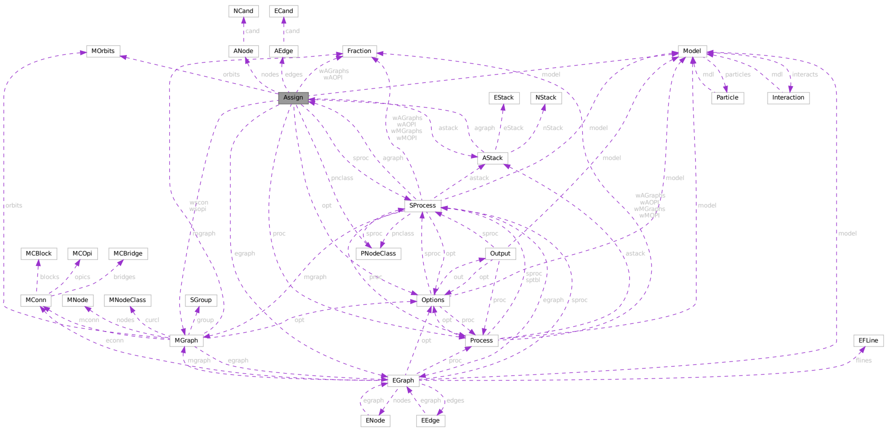
Definition at line 606 of file ratio.c.

The documentation for this struct was generated from the following file:

- ratio.c

3.5 Assign Class Reference

Collaboration diagram for Assign:



Public Member Functions

- Assign (SProcess *sprc, MGraph *mgr, PNodeClass *pnc)
- void prCand (const char *msg)
- void checkAG (const char *msg)
- Bool assignAllVertices (void)
- Bool selectVertex (void)
- Bool selectVertexSimp (int lastv)
- Bool selectLeg (int v, int lastlg)
- Bool assignVertex (int v)
- Bool allAssigned (void)
- Bool fromMGraph (void)
- void addEdge (int n0, int n1, int nplist, int *plist)
- void connect (int n0, int l0, int eg, int el, int n1, int l1)
- Bool fillEGraph (int aid, BigInt nsym, BigInt esym, BigInt nsym1)
- int * reordLeg (int n, int *reord, int *plist, int *used)
- int getLegParticle (int n, int ln)
- int legEdgeParticle (int n, int ln, int pt)
- int legPart (int v, int lg, int nplst, int *plst, int *rlist, const int size)
- int candPart (int v, int ln, int *plist, const int size)
- int selUnAssVertex (void)
- int selUnAssVertexSimp (int lastv)
- int selUnAssLeg (int v, int lastlg)
- NCandSt assignIVertex (int v, int ia)
- Bool assignPLeg (int n, int ln, int pt)
- Bool isOrdPLeg (int n, int ln, int pt)
- Bool detEdge (int e)
- Bool candPartClassify (int v, int *npdass, int *pdass, int *npuass, int *puass, const int size)
- Bool updateCandNode (int v)
- Bool checkOrderCpl (void)
- Bool isOrdLegs (void)
- Bool isIsomorphic (MNodeClass *cl, BigInt *nsym, BigInt *esym, BigInt *nsym1)
- int cmpPermGraph (int *p, MNodeClass *cl)
- int cmpNodes (int nd0, int nd1, MNodeClass *cn)
- BigInt edgeSym (void)
- void saveCouple (int *sav)
- void restoreCouple (int *sav)
- Bool subCouple (int *cpl)

Data Fields

- EGraph * egraph
- MGraph * mgraph
- SProcess * sproc
- Process * proc
- Model * model
- Options * opt
- AStack * astack
- PNodeClass * pnclass
- MOrbits * orbits
- int nNodes
- int nEdges
- int nExtern
- int nETotal

- BigInt nAGraphs
- Fraction wAGraphs
- BigInt nAOPI
- Fraction wAOPI
- ANode ** nodes
- AEdge ** edges
- CheckPt checkpoint0
- int cplleft [GRCC_MAXNCPLG]

3.5.1 Detailed Description

Definition at line 1121 of file grcc.h.

3.5.2 Constructor & Destructor Documentation

3.5.2.1 Assign()

```
Assign::Assign (
            SProcess * sprc,
            MGraph * mgr,
            PNodeClass * pnc )
```

Definition at line 7433 of file grcc.cc.

3.5.2.2 ∼Assign()

```
Assign::∼Assign (
            void )
```

Definition at line 7533 of file grcc.cc.

3.5.3 Member Function Documentation

3.5.3.1 prCand()

```
void Assign::prCand (
            const char * msg )
```

Definition at line 7553 of file grcc.cc.

3.5.3.2 checkAG()

```
void Assign::checkAG (
            const char * msg )
```

Definition at line 7589 of file grcc.cc.

3.5.3.3 assignAllVertices()

```
Bool Assign::assignAllVertices (
             void )
```

Definition at line 7613 of file grcc.cc.

3.5.3.4 selectVertex()

```
Bool Assign::selectVertex (
             void )
```

Definition at line 7686 of file grcc.cc.

3.5.3.5 selectVertexSimp()

```
Bool Assign::selectVertexSimp (
             int lastv )
```

Definition at line 7651 of file grcc.cc.

3.5.3.6 selectLeg()

```
Bool Assign::selectLeg (
             int v,
             int lastlg )
```

Definition at line 7721 of file grcc.cc.

3.5.3.7 assignVertex()

```
Bool Assign::assignVertex (
             int v )
```

Definition at line 7823 of file grcc.cc.

3.5.3.8 allAssigned()

```
Bool Assign::allAssigned (
             void )
```

Definition at line 7946 of file grcc.cc.

3.5.3.9 fromMGraph()

```
Bool Assign::fromMGraph (
             void )
```

Definition at line 8050 of file grcc.cc.

3.5.3.10 addEdge()

```
void Assign::addEdge (
             int n0,
             int n1,
             int nplist,
             int * plist )
```

Definition at line 8196 of file grcc.cc.

3.5.3.11 connect()

```
void Assign::connect (
             int n0,
             int l0,
             int eg,
             int el,
             int n1,
             int l1 )
```

Definition at line 8241 of file grcc.cc.

3.5.3.12 fillEGraph()

```
Bool Assign::fillEGraph (
             int aid,
             BigInt nsym,
             BigInt esym,
             BigInt nsym1 )
```

Definition at line 8270 of file grcc.cc.

3.5.3.13 reordLeg()

```
int * Assign::reordLeg (
             int n,
             int * reord,
             int * plist,
             int * used )
```

Definition at line 8417 of file grcc.cc.

3.5.3.14 getLegParticle()

```
int Assign::getLegParticle (
             int n,
             int ln )
```

Definition at line 8492 of file grcc.cc.

3.5.3.15 legEdgeParticle()

```
int Assign::legEdgeParticle (
             int n,
             int ln,
             int pt )
```

Definition at line 8519 of file grcc.cc.

3.5.3.16 legPart()

```
int Assign::legPart (
             int v,
             int lg,
             int nplst,
             int * plst,
             int * rlist,
             const int size )
```

Definition at line 8539 of file grcc.cc.

3.5.3.17 candPart()

```
int Assign::candPart (
             int v,
             int ln,
             int * plist,
             const int size )
```

Definition at line 8555 of file grcc.cc.

3.5.3.18 selUnAssVertex()

```
int Assign::selUnAssVertex (
             void )
```

Definition at line 8608 of file grcc.cc.

3.5.3.19 selUnAssVertexSimp()

```
int Assign::selUnAssVertexSimp (
             int lastv )
```

Definition at line 8588 of file grcc.cc.

3.5.3.20 selUnAssLeg()

```
int Assign::selUnAssLeg (
             int v,
             int lastlg )
```

Definition at line 8637 of file grcc.cc.

3.5.3.21 assignIVertex()

```
NCandSt Assign::assignIVertex (
             int v,
             int ia )
```

Definition at line 8676 of file grcc.cc.

3.5.3.22 assignPLeg()

```
Bool Assign::assignPLeg (
             int n,
             int ln,
             int pt )
```

Definition at line 8722 of file grcc.cc.

3.5.3.23 isOrdPLeg()

```
Bool Assign::isOrdPLeg (
             int n,
             int ln,
             int pt )
```

Definition at line 8798 of file grcc.cc.

3.5.3.24 detEdge()

```
Bool Assign::detEdge (
             int e )
```

Definition at line 8777 of file grcc.cc.

3.5.3.25 candPartClassify()

```
Bool Assign::candPartClassify (
             int v,
             int * npdass,
             int * pdass,
             int * npuass,
             int * puass,
             const int size )
```

Definition at line 8842 of file grcc.cc.

3.5.3.26 updateCandNode()

```
Bool Assign::updateCandNode (
             int v )
```

Definition at line 8879 of file grcc.cc.

3.5.3.27 checkOrderCpl()

```
Bool Assign::checkOrderCpl (
             void  )
```

Definition at line 9018 of file grcc.cc.

3.5.3.28 isOrdLegs()

```
Bool Assign::isOrdLegs (
             void  )
```

Definition at line 9056 of file grcc.cc.

3.5.3.29 isIsomorphic()

```
Bool Assign::isIsomorphic (
             MNodeClass * cl,
             BigInt * nsym,
             BigInt * esym,
             BigInt * nsym1 )
```

Definition at line 9094 of file grcc.cc.

3.5.3.30 cmpPermGraph()

```
int Assign::cmpPermGraph (
             int * p,
             MNodeClass * cl )
```

Definition at line 9176 of file grcc.cc.

3.5.3.31 cmpNodes()

```
int Assign::cmpNodes (
             int nd0,
             int nd1,
             MNodeClass * cn )
```

Definition at line 9274 of file grcc.cc.

3.5.3.32 edgeSym()

```
BigInt Assign::edgeSym (
             void  )
```

Definition at line 9300 of file grcc.cc.

3.5.3.33 saveCouple()

```
void Assign::saveCouple (
              int * sav )
```

Definition at line 7783 of file grcc.cc.

3.5.3.34 restoreCouple()

```
void Assign::restoreCouple (
              int * sav )
```

Definition at line 7795 of file grcc.cc.

3.5.3.35 subCouple()

```
Bool Assign::subCouple (
              int * cpl )
```

Definition at line 7807 of file grcc.cc.

3.5.4 Field Documentation

3.5.4.1 egraph

```
EGraph* Assign::egraph
```

Definition at line 1126 of file grcc.h.

3.5.4.2 mgraph

```
MGraph* Assign::mgraph
```

Definition at line 1127 of file grcc.h.

3.5.4.3 sproc

```
SProcess* Assign::sproc
```

Definition at line 1128 of file grcc.h.

3.5.4.4 proc

```
Process* Assign::proc
```

Definition at line 1129 of file grcc.h.

3.5.4.5 model

`Model* Assign::model`

Definition at line 1130 of file grcc.h.

3.5.4.6 opt

`Options* Assign::opt`

Definition at line 1131 of file grcc.h.

3.5.4.7 astack

`AStack* Assign::astack`

Definition at line 1132 of file grcc.h.

3.5.4.8 pnclass

`PNodeClass* Assign::pnclass`

Definition at line 1133 of file grcc.h.

3.5.4.9 orbits

`MOrbits* Assign::orbits`

Definition at line 1134 of file grcc.h.

3.5.4.10 nNodes

`int Assign::nNodes`

Definition at line 1136 of file grcc.h.

3.5.4.11 nEdges

`int Assign::nEdges`

Definition at line 1137 of file grcc.h.

3.5.4.12 nExtern

`int Assign::nExtern`

Definition at line 1138 of file grcc.h.

3.5.4.13 nETotal

`int Assign::nETotal`

Definition at line 1139 of file grcc.h.

3.5.4.14 nAGraphs

`BigInt Assign::nAGraphs`

Definition at line 1141 of file grcc.h.

3.5.4.15 wAGraphs

`Fraction Assign::wAGraphs`

Definition at line 1142 of file grcc.h.

3.5.4.16 nAOPI

`BigInt Assign::nAOPI`

Definition at line 1143 of file grcc.h.

3.5.4.17 wAOPI

`Fraction Assign::wAOPI`

Definition at line 1144 of file grcc.h.

3.5.4.18 nodes

`ANode** Assign::nodes`

Definition at line 1146 of file grcc.h.

3.5.4.19 edges

`AEdge** Assign::edges`

Definition at line 1147 of file grcc.h.

3.5.4.20 checkpoint0

`CheckPt Assign::checkpoint0`

Definition at line 1149 of file grcc.h.

3.5.4.21 cplleft

```
int Assign::cplleft[GRCC_MAXNCPLG]
```

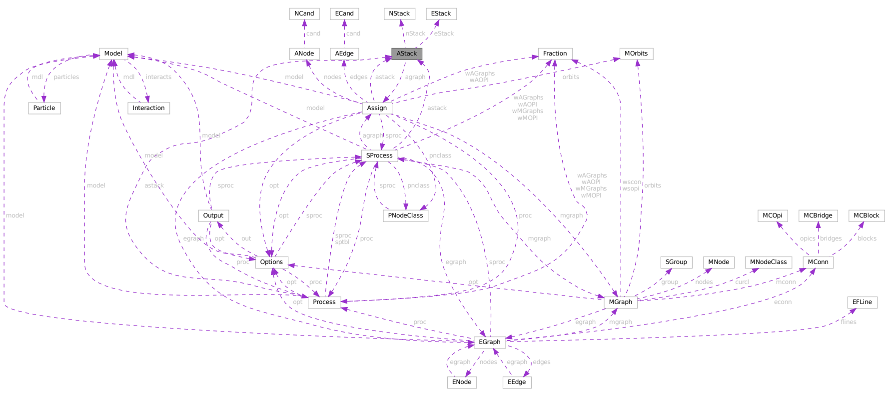
Definition at line 1151 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.6 AStack Class Reference

Collaboration diagram for AStack:



Public Member Functions

- AStack (int nSize, int eSize)
- void setAGraph (Assign ∗ag)
- void checkPoint (CheckPt sav)
- void restore (CheckPt sav)
- void **restoreMsg** (CheckPt sav, const char ∗msg)
- void prStack (void)
- void pushNode (int n)
- void pushEdge (int e)

Data Fields

- Assign ∗ agraph
- NStack ∗∗ nStack
- int nStackP
- int nSize
- EStack ∗∗ eStack
- int eStackP
- int eSize

3.6.1 Detailed Description

Definition at line 1305 of file grcc.h.

3.6.2 Constructor & Destructor Documentation

3.6.2.1 AStack()

```
AStack::AStack (
            int nSize,
            int eSize )
```

Definition at line 9470 of file grcc.cc.

3.6.2.2 ∼AStack()

```
AStack::∼AStack (
            void )
```

Definition at line 9502 of file grcc.cc.

3.6.3 Member Function Documentation

3.6.3.1 setAGraph()

```
void AStack::setAGraph (
            Assign * ag )
```

Definition at line 9530 of file grcc.cc.

3.6.3.2 checkPoint()

```
void AStack::checkPoint (
            CheckPt sav )
```

Definition at line 9595 of file grcc.cc.

3.6.3.3 restore()

```
void AStack::restore (
            CheckPt sav )
```

Definition at line 9639 of file grcc.cc.

3.6.3.4 prStack()

```
void AStack::prStack (
             void  )
```

Definition at line 9662 of file grcc.cc.

3.6.3.5 pushNode()

```
void AStack::pushNode (
             int n )
```

Definition at line 9536 of file grcc.cc.

3.6.3.6 pushEdge()

```
void AStack::pushEdge (
             int e )
```

Definition at line 9566 of file grcc.cc.

3.6.4 Field Documentation

3.6.4.1 agraph

```
Assign* AStack::agraph
```

Definition at line 1309 of file grcc.h.

3.6.4.2 nStack

```
NStack** AStack::nStack
```

Definition at line 1311 of file grcc.h.

3.6.4.3 nStackP

```
int AStack::nStackP
```

Definition at line 1312 of file grcc.h.

3.6.4.4 nSize

```
int AStack::nSize
```

Definition at line 1313 of file grcc.h.

3.6.4.5 eStack

```
EStack** AStack::eStack
```

Definition at line 1315 of file grcc.h.

3.6.4.6 eStackP

```
int AStack::eStackP
```

Definition at line 1316 of file grcc.h.

3.6.4.7 eSize

```
int AStack::eSize
```

Definition at line 1317 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.7 bit_field Struct Reference

```
#include <structs.h>
```

Data Fields

- UBYTE bit_0: 1
- UBYTE bit_1: 1
- UBYTE bit_2: 1
- UBYTE bit_3: 1
- UBYTE bit_4: 1
- UBYTE bit_5: 1
- UBYTE bit_6: 1
- UBYTE bit_7: 1

3.7.1 Detailed Description

The struct bit_field is used by set_in, set_set, set_del and set_sub. They in turn are used in pre.c to toggle bits that indicate whether a character can be used as a separator of function arguments. This facility is used in the communication with external channels.

Definition at line 927 of file structs.h.

3.7.2 Field Documentation

3.7.2.1 bit_0

`UBYTE bit_field::bit_0`

Definition at line 928 of file structs.h.

3.7.2.2 bit_1

`UBYTE bit_field::bit_1`

Definition at line 929 of file structs.h.

3.7.2.3 bit_2

`UBYTE bit_field::bit_2`

Definition at line 930 of file structs.h.

3.7.2.4 bit_3

`UBYTE bit_field::bit_3`

Definition at line 931 of file structs.h.

3.7.2.5 bit_4

`UBYTE bit_field::bit_4`

Definition at line 932 of file structs.h.

3.7.2.6 bit_5

`UBYTE bit_field::bit_5`

Definition at line 933 of file structs.h.

3.7.2.7 bit_6

`UBYTE bit_field::bit_6`

Definition at line 934 of file structs.h.

3.7.2.8 bit_7

```
UBYTE bit_field::bit_7
```

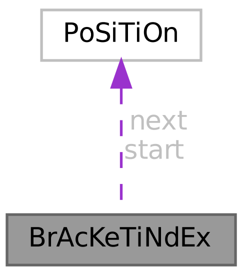
Definition at line 935 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.8 BrAcKeTiNdEx Struct Reference

Collaboration diagram for BrAcKeTiNdEx:



Public Member Functions

- **PADPOSITION** (0, 2, 0, 0, 0)

Data Fields

- POSITION start
- POSITION next
- LONG bracket
- LONG termsinbracket

3.8.1 Detailed Description

Definition at line 317 of file structs.h.

3.8.2 Field Documentation

3.8.2.1 start

`POSITION BrAcKeTiNdEx::start`

Definition at line 318 of file structs.h.

3.8.2.2 next

`POSITION BrAcKeTiNdEx::next`

Definition at line 319 of file structs.h.

3.8.2.3 bracket

`LONG BrAcKeTiNdEx::bracket`

Definition at line 320 of file structs.h.

3.8.2.4 termsinbracket

`LONG BrAcKeTiNdEx::termsinbracket`

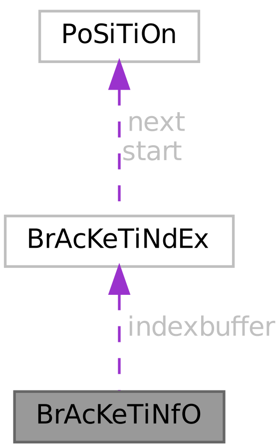
Definition at line 321 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.9 BrAcKeTiNfO Struct Reference

Collaboration diagram for BrAcKeTiNfO:



Data Fields

- BRACKETINDEX ∗ indexbuffer
- WORD ∗ bracketbuffer
- LONG bracketbuffersize
- LONG indexbuffersize
- LONG bracketfill
- LONG indexfill
- WORD SortType

3.9.1 Detailed Description

Definition at line 329 of file structs.h.

3.9.2 Field Documentation

3.9.2.1 indexbuffer

```
BRACKETINDEX* BrAcKeTiNfO::indexbuffer
```

[D]

Definition at line 330 of file structs.h.

Referenced by DoRecovery(), and FlushOut().

3.9.2.2 bracketbuffer

```
WORD* BrAcKeTiNfO::bracketbuffer
```

[D]

Definition at line 331 of file structs.h.

Referenced by DoRecovery().

3.9.2.3 bracketbuffersize

```
LONG BrAcKeTiNfO::bracketbuffersize
```

Definition at line 332 of file structs.h.

3.9.2.4 indexbuffersize

```
LONG BrAcKeTiNfO::indexbuffersize
```

Definition at line 333 of file structs.h.

3.9.2.5 bracketfill

`LONG BrAcKeTiNfO::bracketfill`

Definition at line 334 of file structs.h.

3.9.2.6 indexfill

`LONG BrAcKeTiNfO::indexfill`

Definition at line 335 of file structs.h.

3.9.2.7 SortType

`WORD BrAcKeTiNfO::SortType`

The sorting criterium used (like POWERFIRST etc)

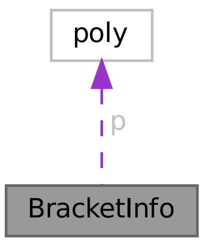
Definition at line 336 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.10 BracketInfo Struct Reference

Collaboration diagram for BracketInfo:



Public Member Functions

- BracketInfo (const std::vector< int > &pattern, int num_terms, const poly ∗p)
- bool operator< (const BracketInfo &rhs) const

Data Fields

- std::vector< int > pattern
- int num_terms
- int dummy = 0
- const poly * p

3.10.1 Detailed Description

Definition at line 1389 of file polygcd.cc.

3.10.2 Constructor & Destructor Documentation

3.10.2.1 BracketInfo()

```
BracketInfo::BracketInfo (
            const std::vector< int > & pattern,
            int num_terms,
            const poly * p )  [inline]
```

Definition at line 1394 of file polygcd.cc.

3.10.3 Member Function Documentation

3.10.3.1 operator<()

```
bool BracketInfo::operator< (
            const BracketInfo & rhs ) const  [inline]
```

Definition at line 1395 of file polygcd.cc.

3.10.4 Field Documentation

3.10.4.1 pattern

```
std::vector<int> BracketInfo::pattern
```

Definition at line 1390 of file polygcd.cc.

3.10.4.2 num_terms

```
int BracketInfo::num_terms
```

Definition at line 1391 of file polygcd.cc.

3.10.4.3 dummy

```
int BracketInfo::dummy = 0
```

Definition at line 1391 of file polygcd.cc.

3.10.4.4 p

```
const poly* BracketInfo::p
```

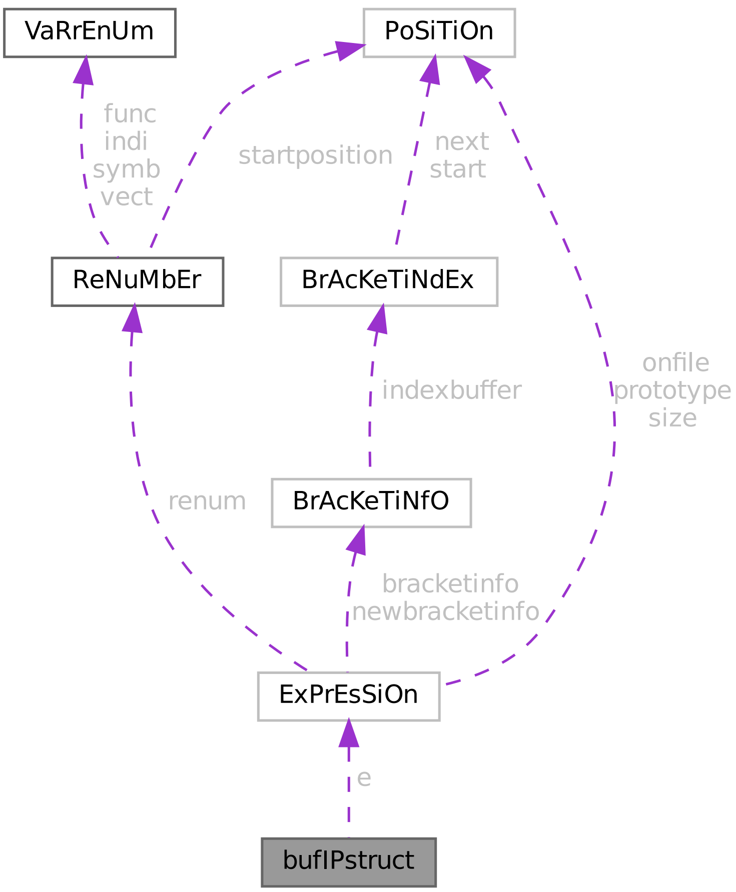
Definition at line 1392 of file polygcd.cc.

The documentation for this struct was generated from the following file:

- polygcd.cc

3.11 buflPstruct Struct Reference

Collaboration diagram for buflPstruct:



Data Fields

- LONG i
- struct ExPrEsSiOn e

3.11.1 Detailed Description

Definition at line 3921 of file parallel.c.

3.11.2 Field Documentation

3.11.2.1 i

```
LONG bufIPstruct::i
```

Definition at line 3922 of file parallel.c.

3.11.2.2 e

```
struct ExPrEsSiOn bufIPstruct::e
```

Definition at line 3923 of file parallel.c.

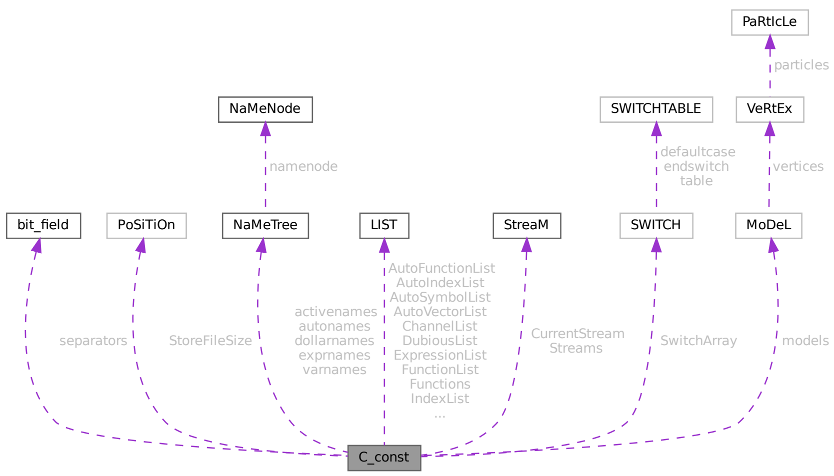
The documentation for this struct was generated from the following file:

- parallel.c

3.12 C_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for C_const:



Public Member Functions

- **PADPOSITION** (48, 8+3 ∗MAXNEST, 75, 48+3 ∗MAXNEST+MAXREPEAT, COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST) ∗17)

Data Fields

- set_of_char separators
- POSITION StoreFileSize
- NAMETREE ∗ dollarnames
- NAMETREE ∗ exprnames
- NAMETREE ∗ varnames
- LIST ChannelList
- LIST DubiousList
- LIST FunctionList
- LIST ExpressionList
- LIST IndexList
- LIST SetElementList
- LIST SetList
- LIST SymbolList
- LIST VectorList
- LIST PotModDolList
- LIST ModOptDolList
- LIST TableBaseList
- LIST cbufList
- LIST AutoSymbolList
- LIST AutoIndexList
- LIST AutoVectorList
- LIST AutoFunctionList
- NAMETREE ∗ autonames
- LIST ∗ Symbols
- LIST ∗ Indices
- LIST ∗ Vectors
- LIST ∗ Functions
- NAMETREE ∗∗ activenames
- STREAM ∗ Streams
- STREAM ∗ CurrentStream
- SWITCH ∗ SwitchArray
- MODEL ∗∗ models
- WORD ∗ SwitchHeap
- LONG ∗ termstack
- LONG ∗ termsortstack
- UWORD ∗ cmod
- UWORD ∗ powmod
- UWORD ∗ modpowers
- UWORD ∗ halfmod
- WORD ∗ ProtoType
- WORD ∗ WildC
- LONG ∗ IfHeap
- LONG ∗ IfCount
- LONG ∗ IfStack
- UBYTE ∗ iBuffer
- UBYTE ∗ iPointer
- UBYTE ∗ iStop

- UBYTE ∗∗ LabelNames
- WORD ∗ FixIndices
- WORD ∗ termsumcheck
- UBYTE ∗ WildcardNames
- int ∗ Labels
- SBYTE ∗ tokens
- SBYTE ∗ toptokens
- SBYTE ∗ endoftokens
- WORD ∗ tokenarglevel
- UWORD ∗ modinverses
- UBYTE ∗ Fortran90Kind
- WORD ∗∗ MultiBracketBuf
- UBYTE ∗ extrasym
- WORD ∗ doloopstack
- WORD ∗ doloopnest
- char ∗ CheckpointRunAfter
- char ∗ CheckpointRunBefore
- WORD ∗ IfSumCheck
- WORD ∗ CommuteInSet
- UBYTE ∗ TestValue
- LONG argstack [MAXNEST]
- LONG insidestack [MAXNEST]
- LONG inexprstack [MAXNEST]
- LONG iBufferSize
- LONG TransEname
- LONG ProcessBucketSize
- LONG mProcessBucketSize
- LONG CModule
- LONG ThreadBucketSize
- LONG CheckpointStamp
- LONG CheckpointInterval
- int cbufnum
- int AutoDeclareFlag
- int NoShowInput
- int ShortStats
- int compiletype
- int firstconstindex
- int insidefirst
- int minsidefirst
- int wildflag
- int NumLabels
- int MaxLabels
- int lDefDim
- int lDefDim4
- int NumWildcardNames
- int WildcardBufferSize
- int MaxIf
- int NumStreams
- int MaxNumStreams
- int firstctypemessage
- int tablecheck
- int idoption
- int BottomLevel
- int CompileLevel
- int TokensWriteFlag

- int UnsureDollarMode
- int outsidefun
- int funpowers
- int WarnFlag
- int StatsFlag
- int NamesFlag
- int CodesFlag
- int SetupFlag
- int SortType
- int lSortType
- int ThreadStats
- int FinalStats
- int OldParallelStats
- int ThreadsFlag
- int ThreadBalancing
- int ThreadSortFileSynch
- int ProcessStats
- int BracketNormalize
- int maxtermlevel
- int dumnumflag
- int bracketindexflag
- int parallelflag
- int mparallelflag
- int inparallelflag
- int partodoflag
- int properorderflag
- int vetofilling
- int tablefilling
- int vetotablebasefill
- int exprfillwarning
- int lhdollarflag
- int NoCompress
- int IsFortran90
- int MultiBracketLevels
- int topolynomialflag
- int ffbufnum
- int OldFactArgFlag
- int MemDebugFlag
- int OldGCDflag
- int WTimeStatsFlag
- int SortReallocateFlag
- int PrintBacktraceFlag
- int FlintPolyFlag
- int HumanStatsFlag
- int doloopstacksize
- int dolooplevel
- int CheckpointFlag
- int SizeCommuteInSet
- int origin
- int vectorlikeLHS
- int nummodels
- int modelspace
- int ModelLevel
- WORD argsumcheck [MAXNEST]
- WORD insidesumcheck [MAXNEST]

- WORD inexprsumcheck [MAXNEST]
- WORD RepSumCheck [MAXREPEAT]
- WORD lUniTrace [4]
- WORD RepLevel
- WORD arglevel
- WORD insidelevel
- WORD inexprlevel
- WORD termlevel
- WORD MustTestTable
- WORD DumNum
- WORD ncmod
- WORD npowmod
- WORD modmode
- WORD nhalfmod
- WORD DirtPow
- WORD lUnitTrace
- WORD NwildC
- WORD ComDefer
- WORD CollectFun
- WORD AltCollectFun
- WORD OutputMode
- WORD Cnumpows
- WORD OutputSpaces
- WORD OutNumberType
- WORD DidClean
- WORD IfLevel
- WORD WhileLevel
- WORD SwitchLevel
- WORD SwitchInArray
- WORD MaxSwitch
- WORD LogHandle
- WORD LineLength
- WORD StoreHandle
- WORD HideLevel
- WORD lPolyFun
- WORD lPolyFunInv
- WORD lPolyFunType
- WORD lPolyFunExp
- WORD lPolyFunVar
- WORD lPolyFunPow
- WORD SymChangeFlag
- WORD CollectPercentage
- WORD ShortStatsMax
- WORD extrasymbols
- WORD PolyRatFunChanged
- WORD ToBeInFactors
- WORD InnerTest
- UBYTE Commercial [COMMERCIALSIZE+2]
- UBYTE debugFlags [MAXFLAGS+2]

3.12.1 Detailed Description

The C_const struct is part of the global data and resides in the ALLGLOBALS struct A under the name #C. We see it used with the macro AC as in AC.exprnames. It contains variables that involve the compiler and objects set during compilation.

Definition at line 1697 of file structs.h.

3.12.2 Field Documentation

3.12.2.1 separators

`set_of_char C_const::separators`

Separators in #call and #do

Definition at line 1698 of file structs.h.

3.12.2.2 StoreFileSize

`POSITION C_const::StoreFileSize`

Definition at line 1699 of file structs.h.

3.12.2.3 dollarnames

`NAMETREE* C_const::dollarnames`

[D] Names of dollar variables

Definition at line 1700 of file structs.h.

3.12.2.4 exprnames

`NAMETREE* C_const::exprnames`

[D] Names of expressions

Definition at line 1701 of file structs.h.

3.12.2.5 varnames

`NAMETREE* C_const::varnames`

[D] Names of regular variables

Definition at line 1702 of file structs.h.

3.12.2.6 ChannelList

`LIST C_const::ChannelList`

Used for the #write statement. Contains CHANNEL

Definition at line 1703 of file structs.h.

3.12.2.7 DubiousList

`LIST C_const::DubiousList`

List of dubious variables. Contains DUBIOUSV. If not empty -> no execution

Definition at line 1705 of file structs.h.

3.12.2.8 FunctionList

`LIST C_const::FunctionList`

List of functions and properties. Contains FUNCTIONS

Definition at line 1707 of file structs.h.

3.12.2.9 ExpressionList

`LIST C_const::ExpressionList`

List of expressions, locations etc.

Definition at line 1708 of file structs.h.

3.12.2.10 IndexList

`LIST C_const::IndexList`

List of indices

Definition at line 1709 of file structs.h.

3.12.2.11 SetElementList

`LIST C_const::SetElementList`

List of all elements of all sets

Definition at line 1710 of file structs.h.

3.12.2.12 SetList

`LIST C_const::SetList`

List of the sets

Definition at line 1711 of file structs.h.

3.12.2.13 SymbolList

`LIST C_const::SymbolList`

List of the symbols and their properties

Definition at line 1712 of file structs.h.

3.12.2.14 VectorList

`LIST C_const::VectorList`

List of the vectors

Definition at line 1713 of file structs.h.

3.12.2.15 PotModDolList

`LIST C_const::PotModDolList`

Potentially changed dollars

Definition at line 1714 of file structs.h.

3.12.2.16 ModOptDolList

`LIST C_const::ModOptDolList`

Module Option Dollars list

Definition at line 1715 of file structs.h.

3.12.2.17 TableBaseList

`LIST C_const::TableBaseList`

TableBase list

Definition at line 1716 of file structs.h.

3.12.2.18 cbufList

```
LIST C_const::cbufList
```

List of compiler buffers

Definition at line 1720 of file structs.h.

3.12.2.19 AutoSymbolList

```
LIST C_const::AutoSymbolList
```

Definition at line 1724 of file structs.h.

3.12.2.20 AutoIndexList

```
LIST C_const::AutoIndexList
```

Definition at line 1725 of file structs.h.

3.12.2.21 AutoVectorList

```
LIST C_const::AutoVectorList
```

Definition at line 1726 of file structs.h.

3.12.2.22 AutoFunctionList

```
LIST C_const::AutoFunctionList
```

Definition at line 1727 of file structs.h.

3.12.2.23 autonames

```
NAMETREE* C_const::autonames
```

[D] Names in autodeclare

Definition at line 1728 of file structs.h.

3.12.2.24 Symbols

```
LIST* C_const::Symbols
```

Definition at line 1730 of file structs.h.

3.12.2.25 Indices

`LIST* C_const::Indices`

Definition at line 1732 of file structs.h.

3.12.2.26 Vectors

`LIST* C_const::Vectors`

Definition at line 1733 of file structs.h.

3.12.2.27 Functions

`LIST* C_const::Functions`

Definition at line 1734 of file structs.h.

3.12.2.28 activenames

`NAMETREE** C_const::activenames`

Definition at line 1735 of file structs.h.

3.12.2.29 Streams

`STREAM* C_const::Streams`

(C) Pointer for AutoDeclare statement. Points either to varnames or autonames. [D] The input streams.

Definition at line 1737 of file structs.h.

3.12.2.30 CurrentStream

`STREAM* C_const::CurrentStream`

(C) The current input stream. Streams are: do loop, file, prevariable. points into Streams memory.

Definition at line 1738 of file structs.h.

3.12.2.31 SwitchArray

`SWITCH* C_const::SwitchArray`

Definition at line 1740 of file structs.h.

3.12.2.32 models

```
MODEL** C_const::models
```

Definition at line 1741 of file structs.h.

3.12.2.33 SwitchHeap

```
WORD* C_const::SwitchHeap
```

Definition at line 1742 of file structs.h.

3.12.2.34 termstack

```
LONG* C_const::termstack
```

[D] Last term statement {offset}

Definition at line 1743 of file structs.h.

3.12.2.35 termsortstack

```
LONG* C_const::termsortstack
```

[D] Last sort statement {offset}

Definition at line 1744 of file structs.h.

3.12.2.36 cmod

```
UWORD* C_const::cmod
```

[D] Local setting of modulus. Pointer to value.

Definition at line 1745 of file structs.h.

3.12.2.37 powmod

```
UWORD* C_const::powmod
```

Local setting printing as powers. Points into cmod memory

Definition at line 1746 of file structs.h.

3.12.2.38 modpowers

`UWORD* C_const::modpowers`

[D] The conversion table for mod-> powers.

Definition at line 1747 of file structs.h.

3.12.2.39 halfmod

`UWORD* C_const::halfmod`

Definition at line 1748 of file structs.h.

3.12.2.40 ProtoType

`WORD* C_const::ProtoType`

Definition at line 1749 of file structs.h.

3.12.2.41 WildC

`WORD* C_const::WildC`

Definition at line 1750 of file structs.h.

3.12.2.42 IfHeap

`LONG* C_const::IfHeap`

[D] Keeps track of where to go in if

Definition at line 1751 of file structs.h.

3.12.2.43 IfCount

`LONG* C_const::IfCount`

[D] Keeps track of where to go in if

Definition at line 1752 of file structs.h.

3.12.2.44 IfStack

`LONG* C_const::IfStack`

Keeps track of where to go in if. Points into IfHeap-memory

Definition at line 1753 of file structs.h.

3.12.2.45 iBuffer

```
UBYTE* C_const::iBuffer
```

[D] Compiler input buffer

Definition at line 1754 of file structs.h.

3.12.2.46 iPointer

```
UBYTE* C_const::iPointer
```

Running pointer in the compiler input buffer

Definition at line 1755 of file structs.h.

3.12.2.47 iStop

```
UBYTE* C_const::iStop
```

Top of iBuffer

Definition at line 1756 of file structs.h.

3.12.2.48 LabelNames

```
UBYTE** C_const::LabelNames
```

[D] List of names in label statements

Definition at line 1757 of file structs.h.

3.12.2.49 FixIndices

```
WORD* C_const::FixIndices
```

[D] Buffer of fixed indices

Definition at line 1758 of file structs.h.

3.12.2.50 termsumcheck

```
WORD* C_const::termsumcheck
```

[D] Checking of nesting

Definition at line 1759 of file structs.h.

3.12.2.51 WildcardNames

```
UBYTE* C_const::WildcardNames
```

[D] Names of ?a variables

Definition at line 1760 of file structs.h.

3.12.2.52 Labels

```
int* C_const::Labels
```

Label information for during run. Pointer into LabelNames memory.

Definition at line 1761 of file structs.h.

3.12.2.53 tokens

```
SBYTE* C_const::tokens
```

[D] Array with tokens for tokenizer

Definition at line 1762 of file structs.h.

3.12.2.54 toptokens

```
SBYTE* C_const::toptokens
```

Top of tokens

Definition at line 1763 of file structs.h.

3.12.2.55 endoftokens

```
SBYTE* C_const::endoftokens
```

End of the actual tokens

Definition at line 1764 of file structs.h.

3.12.2.56 tokenarglevel

```
WORD* C_const::tokenarglevel
```

[D] Keeps track of function arguments

Definition at line 1765 of file structs.h.

3.12.2.57 modinverses

```
UWORD* C_const::modinverses
```

Definition at line 1766 of file structs.h.

3.12.2.58 Fortran90Kind

```
UBYTE* C_const::Fortran90Kind
```

Definition at line 1767 of file structs.h.

3.12.2.59 MultiBracketBuf

```
WORD** C_const::MultiBracketBuf
```

Definition at line 1768 of file structs.h.

3.12.2.60 extrasym

```
UBYTE* C_const::extrasym
```

Definition at line 1769 of file structs.h.

3.12.2.61 doloopstack

```
WORD* C_const::doloopstack
```

Definition at line 1770 of file structs.h.

3.12.2.62 doloopnest

```
WORD* C_const::doloopnest
```

Definition at line 1771 of file structs.h.

3.12.2.63 CheckpointRunAfter

```
char* C_const::CheckpointRunAfter
```

[D] Filename of script to be executed *before* creating the snapshot. =0 if no script shall be executed.

Definition at line 1772 of file structs.h.

3.12.2.64 CheckpointRunBefore

`char* C_const::CheckpointRunBefore`

[D] Filename of script to be executed *after* having created the snapshot. =0 if no script shall be executed.

Definition at line 1774 of file structs.h.

3.12.2.65 IfSumCheck

`WORD* C_const::IfSumCheck`

[D] Keeps track of if-nesting

Definition at line 1776 of file structs.h.

3.12.2.66 CommuteInSet

`WORD* C_const::CommuteInSet`

Definition at line 1777 of file structs.h.

3.12.2.67 TestValue

`UBYTE* C_const::TestValue`

Definition at line 1778 of file structs.h.

3.12.2.68 argstack

`LONG C_const::argstack[MAXNEST]`

Definition at line 1786 of file structs.h.

3.12.2.69 insidestack

`LONG C_const::insidestack[MAXNEST]`

Definition at line 1787 of file structs.h.

3.12.2.70 inexprstack

`LONG C_const::inexprstack[MAXNEST]`

Definition at line 1788 of file structs.h.

3.12.2.71 iBufferSize

```
LONG C_const::iBufferSize
```

Definition at line 1789 of file structs.h.

3.12.2.72 TransEname

```
LONG C_const::TransEname
```

Definition at line 1790 of file structs.h.

3.12.2.73 ProcessBucketSize

```
LONG C_const::ProcessBucketSize
```

Definition at line 1792 of file structs.h.

3.12.2.74 mProcessBucketSize

```
LONG C_const::mProcessBucketSize
```

Definition at line 1793 of file structs.h.

3.12.2.75 CModule

```
LONG C_const::CModule
```

Definition at line 1794 of file structs.h.

3.12.2.76 ThreadBucketSize

```
LONG C_const::ThreadBucketSize
```

Definition at line 1795 of file structs.h.

3.12.2.77 CheckpointStamp

```
LONG C_const::CheckpointStamp
```

Timestamp of the last created snapshot (set to Timer(0)).

Definition at line 1796 of file structs.h.

3.12.2.78 CheckpointInterval

```
LONG C_const::CheckpointInterval
```

Time interval in milliseconds for snapshots. =0 if snapshots shall be created at the end of *every* module.

Definition at line 1797 of file structs.h.

3.12.2.79 cbufnum

```
int C_const::cbufnum
```

Current compiler buffer

Definition at line 1805 of file structs.h.

3.12.2.80 AutoDeclareFlag

```
int C_const::AutoDeclareFlag
```

Definition at line 1806 of file structs.h.

3.12.2.81 NoShowInput

```
int C_const::NoShowInput
```

(C) Mode of looking for names. Set to NOAUTO (=0) or WITHAUTO (=2), cf. AutoDeclare statement

Definition at line 1808 of file structs.h.

3.12.2.82 ShortStats

```
int C_const::ShortStats
```

Definition at line 1809 of file structs.h.

3.12.2.83 compiletype

```
int C_const::compiletype
```

Definition at line 1810 of file structs.h.

3.12.2.84 firstconstindex

```
int C_const::firstconstindex
```

Definition at line 1811 of file structs.h.

3.12.2.85 insidefirst

`int C_const::insidefirst`

Definition at line 1812 of file structs.h.

3.12.2.86 minsidefirst

`int C_const::minsidefirst`

Definition at line 1813 of file structs.h.

3.12.2.87 wildflag

`int C_const::wildflag`

Definition at line 1814 of file structs.h.

3.12.2.88 NumLabels

`int C_const::NumLabels`

Definition at line 1815 of file structs.h.

3.12.2.89 MaxLabels

`int C_const::MaxLabels`

Definition at line 1816 of file structs.h.

3.12.2.90 lDefDim

`int C_const::lDefDim`

Definition at line 1817 of file structs.h.

3.12.2.91 lDefDim4

`int C_const::lDefDim4`

Definition at line 1818 of file structs.h.

3.12.2.92 NumWildcardNames

`int C_const::NumWildcardNames`

Definition at line 1819 of file structs.h.

3.12.2.93 WildcardBufferSize

`int C_const::WildcardBufferSize`

Definition at line 1820 of file structs.h.

3.12.2.94 MaxIf

`int C_const::MaxIf`

Definition at line 1821 of file structs.h.

3.12.2.95 NumStreams

`int C_const::NumStreams`

Definition at line 1822 of file structs.h.

3.12.2.96 MaxNumStreams

`int C_const::MaxNumStreams`

Definition at line 1823 of file structs.h.

3.12.2.97 firstctypemessage

`int C_const::firstctypemessage`

Definition at line 1824 of file structs.h.

3.12.2.98 tablecheck

`int C_const::tablecheck`

Definition at line 1825 of file structs.h.

3.12.2.99 idoption

`int C_const::idoption`

Definition at line 1826 of file structs.h.

3.12.2.100 BottomLevel

`int C_const::BottomLevel`

Definition at line 1827 of file structs.h.

3.12.2.101 CompileLevel

```
int C_const::CompileLevel
```

Definition at line 1828 of file structs.h.

3.12.2.102 TokensWriteFlag

```
int C_const::TokensWriteFlag
```

Definition at line 1829 of file structs.h.

3.12.2.103 UnsureDollarMode

```
int C_const::UnsureDollarMode
```

Definition at line 1830 of file structs.h.

3.12.2.104 outsidefun

```
int C_const::outsidefun
```

Definition at line 1831 of file structs.h.

3.12.2.105 funpowers

```
int C_const::funpowers
```

Definition at line 1832 of file structs.h.

3.12.2.106 WarnFlag

```
int C_const::WarnFlag
```

Definition at line 1833 of file structs.h.

3.12.2.107 StatsFlag

```
int C_const::StatsFlag
```

Definition at line 1834 of file structs.h.

3.12.2.108 NamesFlag

```
int C_const::NamesFlag
```

Definition at line 1835 of file structs.h.

3.12.2.109 CodesFlag

`int C_const::CodesFlag`

Definition at line 1836 of file structs.h.

3.12.2.110 SetupFlag

`int C_const::SetupFlag`

Definition at line 1837 of file structs.h.

3.12.2.111 SortType

`int C_const::SortType`

Definition at line 1838 of file structs.h.

3.12.2.112 lSortType

`int C_const::lSortType`

Definition at line 1839 of file structs.h.

3.12.2.113 ThreadStats

`int C_const::ThreadStats`

Definition at line 1840 of file structs.h.

3.12.2.114 FinalStats

`int C_const::FinalStats`

Definition at line 1841 of file structs.h.

3.12.2.115 OldParallelStats

`int C_const::OldParallelStats`

Definition at line 1842 of file structs.h.

3.12.2.116 ThreadsFlag

`int C_const::ThreadsFlag`

Definition at line 1843 of file structs.h.

3.12.2.117 ThreadBalancing

`int C_const::ThreadBalancing`

Definition at line 1844 of file structs.h.

3.12.2.118 ThreadSortFileSynch

`int C_const::ThreadSortFileSynch`

Definition at line 1845 of file structs.h.

3.12.2.119 ProcessStats

`int C_const::ProcessStats`

Definition at line 1846 of file structs.h.

3.12.2.120 BracketNormalize

`int C_const::BracketNormalize`

Definition at line 1847 of file structs.h.

3.12.2.121 maxtermlevel

`int C_const::maxtermlevel`

Definition at line 1848 of file structs.h.

3.12.2.122 dumnumflag

`int C_const::dumnumflag`

Definition at line 1849 of file structs.h.

3.12.2.123 bracketindexflag

`int C_const::bracketindexflag`

Definition at line 1850 of file structs.h.

3.12.2.124 parallelflag

`int C_const::parallelflag`

Definition at line 1851 of file structs.h.

3.12.2.125 mparallelflag

```
int C_const::mparallelflag
```

Definition at line 1852 of file structs.h.

3.12.2.126 inparallelflag

```
int C_const::inparallelflag
```

Definition at line 1853 of file structs.h.

3.12.2.127 partodoflag

```
int C_const::partodoflag
```

Definition at line 1854 of file structs.h.

3.12.2.128 properorderflag

```
int C_const::properorderflag
```

Definition at line 1855 of file structs.h.

3.12.2.129 vetofilling

```
int C_const::vetofilling
```

Definition at line 1856 of file structs.h.

3.12.2.130 tablefilling

```
int C_const::tablefilling
```

Definition at line 1857 of file structs.h.

3.12.2.131 vetotablebasefill

```
int C_const::vetotablebasefill
```

Definition at line 1858 of file structs.h.

3.12.2.132 exprfillwarning

```
int C_const::exprfillwarning
```

Definition at line 1859 of file structs.h.

3.12.2.133 lhdollarflag

```
int C_const::lhdollarflag
```

Definition at line 1860 of file structs.h.

3.12.2.134 NoCompress

```
int C_const::NoCompress
```

Definition at line 1861 of file structs.h.

3.12.2.135 IsFortran90

```
int C_const::IsFortran90
```

Definition at line 1862 of file structs.h.

3.12.2.136 MultiBracketLevels

```
int C_const::MultiBracketLevels
```

Definition at line 1863 of file structs.h.

3.12.2.137 topolynomialflag

```
int C_const::topolynomialflag
```

Definition at line 1864 of file structs.h.

3.12.2.138 ffbufnum

```
int C_const::ffbufnum
```

Definition at line 1865 of file structs.h.

3.12.2.139 OldFactArgFlag

```
int C_const::OldFactArgFlag
```

Definition at line 1866 of file structs.h.

3.12.2.140 MemDebugFlag

```
int C_const::MemDebugFlag
```

Definition at line 1867 of file structs.h.

3.12.2.141 OldGCDflag

```
int C_const::OldGCDflag
```

Definition at line 1868 of file structs.h.

3.12.2.142 WTimeStatsFlag

```
int C_const::WTimeStatsFlag
```

Definition at line 1869 of file structs.h.

3.12.2.143 SortReallocateFlag

```
int C_const::SortReallocateFlag
```

Definition at line 1870 of file structs.h.

3.12.2.144 PrintBacktraceFlag

```
int C_const::PrintBacktraceFlag
```

Definition at line 1874 of file structs.h.

3.12.2.145 FlintPolyFlag

```
int C_const::FlintPolyFlag
```

Definition at line 1875 of file structs.h.

3.12.2.146 HumanStatsFlag

```
int C_const::HumanStatsFlag
```

Definition at line 1876 of file structs.h.

3.12.2.147 doloopstacksize

```
int C_const::doloopstacksize
```

Definition at line 1877 of file structs.h.

3.12.2.148 dolooplevel

```
int C_const::dolooplevel
```

Definition at line 1878 of file structs.h.

3.12.2.149 CheckpointFlag

```
int C_const::CheckpointFlag
```

Tells preprocessor whether checkpoint code must executed. -1 : do recovery from snapshot, set by command line option; 0 : do nothing; 1 : create snapshots, set by On checkpoint statement

Definition at line 1879 of file structs.h.

3.12.2.150 SizeCommuteInSet

```
int C_const::SizeCommuteInSet
```

Definition at line 1883 of file structs.h.

3.12.2.151 origin

```
int C_const::origin
```

Definition at line 1888 of file structs.h.

3.12.2.152 vectorlikeLHS

```
int C_const::vectorlikeLHS
```

Definition at line 1889 of file structs.h.

3.12.2.153 nummodels

```
int C_const::nummodels
```

Definition at line 1890 of file structs.h.

3.12.2.154 modelspace

```
int C_const::modelspace
```

Definition at line 1891 of file structs.h.

3.12.2.155 ModelLevel

```
int C_const::ModelLevel
```

Definition at line 1892 of file structs.h.

3.12.2.156 argsumcheck

```
WORD C_const::argsumcheck[MAXNEST]
```

Definition at line 1893 of file structs.h.

3.12.2.157 insidesumcheck

```
WORD C_const::insidesumcheck[MAXNEST]
```

Definition at line 1894 of file structs.h.

3.12.2.158 inexprsumcheck

```
WORD C_const::inexprsumcheck[MAXNEST]
```

Definition at line 1895 of file structs.h.

3.12.2.159 RepSumCheck

```
WORD C_const::RepSumCheck[MAXREPEAT]
```

Definition at line 1896 of file structs.h.

3.12.2.160 lUniTrace

```
WORD C_const::lUniTrace[4]
```

Definition at line 1897 of file structs.h.

3.12.2.161 RepLevel

```
WORD C_const::RepLevel
```

Definition at line 1898 of file structs.h.

3.12.2.162 arglevel

```
WORD C_const::arglevel
```

Definition at line 1899 of file structs.h.

3.12.2.163 insidelevel

```
WORD C_const::insidelevel
```

Definition at line 1900 of file structs.h.

3.12.2.164 inexprlevel

```
WORD C_const::inexprlevel
```

Definition at line 1901 of file structs.h.

3.12.2.165 termlevel

```
WORD C_const::termlevel
```

Definition at line 1902 of file structs.h.

3.12.2.166 MustTestTable

```
WORD C_const::MustTestTable
```

Definition at line 1903 of file structs.h.

3.12.2.167 DumNum

```
WORD C_const::DumNum
```

Definition at line 1904 of file structs.h.

3.12.2.168 ncmod

```
WORD C_const::ncmod
```

Definition at line 1905 of file structs.h.

3.12.2.169 npowmod

```
WORD C_const::npowmod
```

Definition at line 1906 of file structs.h.

3.12.2.170 modmode

```
WORD C_const::modmode
```

Definition at line 1907 of file structs.h.

3.12.2.171 nhalfmod

```
WORD C_const::nhalfmod
```

Definition at line 1908 of file structs.h.

3.12.2.172 DirtPow

`WORD C_const::DirtPow`

Definition at line 1909 of file structs.h.

3.12.2.173 lUnitTrace

`WORD C_const::lUnitTrace`

Definition at line 1910 of file structs.h.

3.12.2.174 NwildC

`WORD C_const::NwildC`

Definition at line 1911 of file structs.h.

3.12.2.175 ComDefer

`WORD C_const::ComDefer`

Definition at line 1912 of file structs.h.

3.12.2.176 CollectFun

`WORD C_const::CollectFun`

Definition at line 1913 of file structs.h.

3.12.2.177 AltCollectFun

`WORD C_const::AltCollectFun`

Definition at line 1914 of file structs.h.

3.12.2.178 OutputMode

`WORD C_const::OutputMode`

Definition at line 1915 of file structs.h.

3.12.2.179 Cnumpows

`WORD C_const::Cnumpows`

Definition at line 1916 of file structs.h.

3.12.2.180 OutputSpaces

`WORD C_const::OutputSpaces`

Definition at line 1917 of file structs.h.

3.12.2.181 OutNumberType

`WORD C_const::OutNumberType`

Definition at line 1918 of file structs.h.

3.12.2.182 DidClean

`WORD C_const::DidClean`

Definition at line 1919 of file structs.h.

3.12.2.183 IfLevel

`WORD C_const::IfLevel`

Definition at line 1920 of file structs.h.

3.12.2.184 WhileLevel

`WORD C_const::WhileLevel`

Definition at line 1921 of file structs.h.

3.12.2.185 SwitchLevel

`WORD C_const::SwitchLevel`

Definition at line 1922 of file structs.h.

3.12.2.186 SwitchInArray

`WORD C_const::SwitchInArray`

Definition at line 1923 of file structs.h.

3.12.2.187 MaxSwitch

`WORD C_const::MaxSwitch`

Definition at line 1924 of file structs.h.

3.12.2.188 LogHandle

`WORD C_const::LogHandle`

Definition at line 1925 of file structs.h.

3.12.2.189 LineLength

`WORD C_const::LineLength`

Definition at line 1926 of file structs.h.

3.12.2.190 StoreHandle

`WORD C_const::StoreHandle`

Definition at line 1927 of file structs.h.

3.12.2.191 HideLevel

`WORD C_const::HideLevel`

Definition at line 1928 of file structs.h.

3.12.2.192 lPolyFun

`WORD C_const::lPolyFun`

Definition at line 1929 of file structs.h.

3.12.2.193 lPolyFunInv

`WORD C_const::lPolyFunInv`

Definition at line 1930 of file structs.h.

3.12.2.194 lPolyFunType

`WORD C_const::lPolyFunType`

Definition at line 1931 of file structs.h.

3.12.2.195 lPolyFunExp

`WORD C_const::lPolyFunExp`

Definition at line 1932 of file structs.h.

3.12.2.196 lPolyFunVar

`WORD C_const::lPolyFunVar`

Definition at line 1933 of file structs.h.

3.12.2.197 lPolyFunPow

`WORD C_const::lPolyFunPow`

Definition at line 1934 of file structs.h.

3.12.2.198 SymChangeFlag

`WORD C_const::SymChangeFlag`

Definition at line 1935 of file structs.h.

3.12.2.199 CollectPercentage

`WORD C_const::CollectPercentage`

Definition at line 1936 of file structs.h.

3.12.2.200 ShortStatsMax

`WORD C_const::ShortStatsMax`

Definition at line 1937 of file structs.h.

3.12.2.201 extrasymbols

`WORD C_const::extrasymbols`

Definition at line 1938 of file structs.h.

3.12.2.202 PolyRatFunChanged

`WORD C_const::PolyRatFunChanged`

Definition at line 1939 of file structs.h.

3.12.2.203 ToBeInFactors

`WORD C_const::ToBeInFactors`

Definition at line 1940 of file structs.h.

3.12.2.204 InnerTest

`WORD C_const::InnerTest`

Definition at line 1941 of file structs.h.

3.12.2.205 Commercial

`UBYTE C_const::Commercial[COMMERCIALSIZE+2]`

Definition at line 1945 of file structs.h.

3.12.2.206 debugFlags

`UBYTE C_const::debugFlags[MAXFLAGS+2]`

Definition at line 1946 of file structs.h.

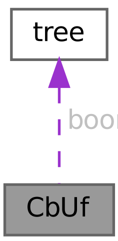
The documentation for this struct was generated from the following file:

- structs.h

3.13 CbUf Struct Reference

`#include <structs.h>`

Collaboration diagram for CbUf:



Data Fields

- WORD * Buffer
- WORD * Top
- WORD * Pointer
- WORD ** lhs
- WORD ** rhs
- LONG * CanCommu
- LONG * NumTerms
- WORD * numdum
- WORD * dimension
- COMPTREE * boomlijst
- LONG BufferSize
- int numlhs
- int numrhs
- int maxlhs
- int maxrhs
- int mnumlhs
- int mnumrhs
- int numtree
- int rootnum
- int MaxTreeSize

3.13.1 Detailed Description

The CBUF struct is used by the compiler. It is a compiler buffer of which since version 3.0 there can be many.

Definition at line 992 of file structs.h.

3.13.2 Field Documentation

3.13.2.1 Buffer

```
WORD* CbUf::Buffer
```

[D] Size in BufferSize

Definition at line 993 of file structs.h.

Referenced by CleanupArgCache(), clearcbuf(), DoubleCbuffer(), finishcbuf(), inicbufs(), PF_BroadcastCBuf(), and Processor().

3.13.2.2 Top

```
WORD* CbUf::Top
```

pointer to the end of the Buffer memory

Definition at line 994 of file structs.h.

Referenced by AddNtoC(), AddNtoL(), DoubleCbuffer(), finishcbuf(), inicbufs(), and PF_BroadcastCBuf().

3.13.2.3 Pointer

`WORD* CbUf::Pointer`

pointer into the Buffer memory

Definition at line 995 of file structs.h.

Referenced by AddLHS(), AddNtoC(), AddNtoL(), AddRHS(), CleanupArgCache(), clearcbuf(), DoubleCbuffer(), finishcbuf(), Generator(), inicbufs(), PF_BroadcastCBuf(), and Processor().

3.13.2.4 lhs

`WORD** CbUf::lhs`

[D] Size in maxlhs. list of pointers into Buffer.

Definition at line 996 of file structs.h.

Referenced by AddLHS(), DoubleCbuffer(), finishcbuf(), Generator(), inicbufs(), PF_BroadcastCBuf(), Processor(), and TestMatch().

3.13.2.5 rhs

`WORD** CbUf::rhs`

[D] Size in maxrhs. list of pointers into Buffer.

Definition at line 997 of file structs.h.

Referenced by AddRHS(), CleanupArgCache(), clearcbuf(), DoubleCbuffer(), finishcbuf(), Generator(), inicbufs(), IniFbuffer(), PF_BroadcastCBuf(), and Processor().

3.13.2.6 CanCommu

`LONG* CbUf::CanCommu`

points into rhs memory behind WORD* area.

Definition at line 998 of file structs.h.

Referenced by AddRHS(), finishcbuf(), inicbufs(), IniFbuffer(), and PF_BroadcastCBuf().

3.13.2.7 NumTerms

`LONG* CbUf::NumTerms`

points into rhs memory behind CanCommu area

Definition at line 999 of file structs.h.

Referenced by AddRHS(), finishcbuf(), inicbufs(), IniFbuffer(), and PF_BroadcastCBuf().

3.13.2.8 numdum

`WORD* CbUf::numdum`

points into rhs memory behind NumTerms

Definition at line 1000 of file structs.h.

Referenced by AddRHS(), inicbufs(), IniFbuffer(), and PF_BroadcastCBuf().

3.13.2.9 dimension

`WORD* CbUf::dimension`

points into rhs memory behind numdum

Definition at line 1001 of file structs.h.

Referenced by AddRHS(), inicbufs(), IniFbuffer(), and PF_BroadcastCBuf().

3.13.2.10 boomlijst

`COMPTREE* CbUf::boomlijst`

[D] Number elements in MaxTreeSize

Definition at line 1002 of file structs.h.

Referenced by CleanupArgCache(), clearcbuf(), finishcbuf(), inicbufs(), IniFbuffer(), and PF_BroadcastCBuf().

3.13.2.11 BufferSize

`LONG CbUf::BufferSize`

Number of allocated WORD's in Buffer

Definition at line 1003 of file structs.h.

Referenced by DoubleCbuffer(), finishcbuf(), inicbufs(), and PF_BroadcastCBuf().

3.13.2.12 numlhs

`int CbUf::numlhs`

Definition at line 1004 of file structs.h.

3.13.2.13 numrhs

`int CbUf::numrhs`

Definition at line 1005 of file structs.h.

3.13.2.14 maxlhs

```
int CbUf::maxlhs
```

Definition at line 1006 of file structs.h.

3.13.2.15 maxrhs

```
int CbUf::maxrhs
```

Definition at line 1007 of file structs.h.

3.13.2.16 mnumlhs

```
int CbUf::mnumlhs
```

Definition at line 1008 of file structs.h.

3.13.2.17 mnumrhs

```
int CbUf::mnumrhs
```

Definition at line 1009 of file structs.h.

3.13.2.18 numtree

```
int CbUf::numtree
```

Definition at line 1010 of file structs.h.

3.13.2.19 rootnum

```
int CbUf::rootnum
```

Definition at line 1011 of file structs.h.

3.13.2.20 MaxTreeSize

```
int CbUf::MaxTreeSize
```

Definition at line 1012 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.14 ChAnNeL Struct Reference

```
#include <structs.h>
```

Data Fields

- char * name
- int handle

3.14.1 Detailed Description

When we read input from text files we have to remember not only their handle but also their name. This is needed for error messages. Hence we call such a file a channel and reserve a struct of type CHANNEL to allow to lay this link.

Definition at line 1023 of file structs.h.

3.14.2 Field Documentation

3.14.2.1 name

```
char* ChAnNeL::name
```

[D] Name of the channel

Definition at line 1024 of file structs.h.

3.14.2.2 handle

```
int ChAnNeL::handle
```

File handle

Definition at line 1025 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.15 COST Struct Reference

Data Fields

- LONG add
- LONG mul
- LONG div
- LONG pow

3.15.1 Detailed Description

Definition at line 1294 of file structs.h.

3.15.2 Field Documentation

3.15.2.1 add

```
LONG COST::add
```

Definition at line 1295 of file structs.h.

3.15.2.2 mul

```
LONG COST::mul
```

Definition at line 1296 of file structs.h.

3.15.2.3 div

```
LONG COST::div
```

Definition at line 1297 of file structs.h.

3.15.2.4 pow

```
LONG COST::pow
```

Definition at line 1298 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.16 CSEEq Struct Reference

Public Member Functions

- bool operator() (const vector< WORD > &lhs, const vector< WORD > &rhs) const

3.16.1 Detailed Description

Definition at line 1060 of file optimize.cc.

3.16.2 Member Function Documentation

3.16.2.1 operator()()

```
bool CSEEq::operator() (
            const vector< WORD > & lhs,
            const vector< WORD > & rhs ) const  [inline]
```

Definition at line 1061 of file optimize.cc.

The documentation for this struct was generated from the following file:

- optimize.cc

3.17 CSEHash Struct Reference

Public Member Functions

- size_t operator() (const vector< WORD > &n) const

3.17.1 Detailed Description

Definition at line 1054 of file optimize.cc.

3.17.2 Member Function Documentation

3.17.2.1 operator()()

```
size_t CSEHash::operator() (
            const vector< WORD > & n ) const  [inline]
```

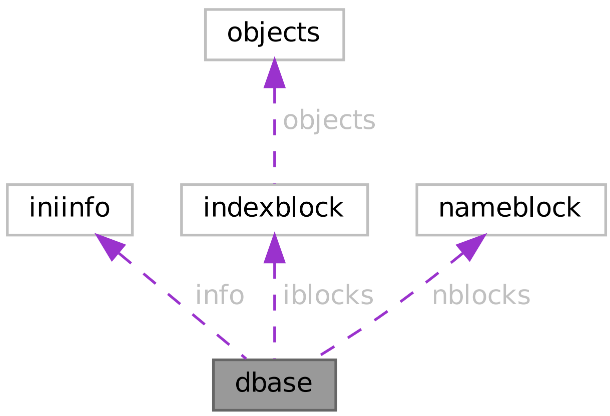
Definition at line 1055 of file optimize.cc.

The documentation for this struct was generated from the following file:

- optimize.cc

3.18 dbase Struct Reference

Collaboration diagram for dbase:



Data Fields

- INIINFO info
- MLONG mode
- MLONG tablenamessize
- MLONG topnumber
- MLONG tablenamefill
- MLONG rwmode
- INDEXBLOCK ** iblocks
- NAMESBLOCK ** nblocks
- FILE * handle
- char * name
- char * fullname
- char * tablenames

3.18.1 Detailed Description

Definition at line 121 of file minos.h.

3.18.2 Field Documentation

3.18.2.1 info

```
INIINFO dbase::info
```

Definition at line 122 of file minos.h.

3.18.2.2 mode

```
MLONG dbase::mode
```

Definition at line 123 of file minos.h.

3.18.2.3 tablenamessize

```
MLONG dbase::tablenamessize
```

Definition at line 124 of file minos.h.

3.18.2.4 topnumber

```
MLONG dbase::topnumber
```

Definition at line 125 of file minos.h.

3.18.2.5 tablenamefill

```
MLONG dbase::tablenamefill
```

Definition at line 126 of file minos.h.

3.18.2.6 rwmode

```
MLONG dbase::rwmode
```

Definition at line 127 of file minos.h.

3.18.2.7 iblocks

```
INDEXBLOCK** dbase::iblocks
```

Definition at line 128 of file minos.h.

3.18.2.8 nblocks

```
NAMESBLOCK** dbase::nblocks
```

Definition at line 129 of file minos.h.

3.18.2.9 handle

```
FILE* dbase::handle
```

Definition at line 130 of file minos.h.

3.18.2.10 name

```
char* dbase::name
```

Definition at line 131 of file minos.h.

3.18.2.11 fullname

```
char* dbase::fullname
```

Definition at line 132 of file minos.h.

3.18.2.12 tablenames

```
char* dbase::tablenames
```

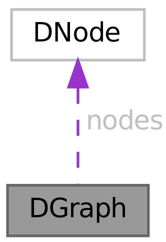
Definition at line 133 of file minos.h.

The documentation for this struct was generated from the following file:

- minos.h

3.19 DGraph Struct Reference

Collaboration diagram for DGraph:



Data Fields

- int nnodes
- int nedges
- DNode ∗ nodes
- DEdge ∗ edges

3.19.1 Detailed Description

Definition at line 199 of file grccparam.h.

3.19.2 Field Documentation

3.19.2.1 nnodes

`int DGraph::nnodes`

Definition at line 200 of file grccparam.h.

3.19.2.2 nedges

`int DGraph::nedges`

Definition at line 201 of file grccparam.h.

3.19.2.3 nodes

`DNode* DGraph::nodes`

Definition at line 202 of file grccparam.h.

3.19.2.4 edges

`DEdge* DGraph::edges`

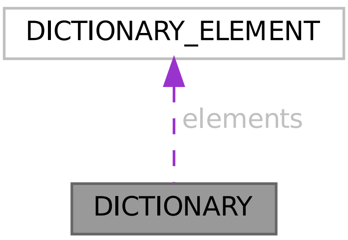
Definition at line 203 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.20 DICTIONARY Struct Reference

Collaboration diagram for DICTIONARY:



Data Fields

- DICTIONARY_ELEMENT ** elements
- UBYTE * name
- int sizeelements
- int numelements
- int numbers
- int variables
- int characters
- int funwith
- int gnumelements
- int ranges

3.20.1 Detailed Description

Definition at line 1361 of file structs.h.

3.20.2 Field Documentation

3.20.2.1 elements

```
DICTIONARY_ELEMENT** DICTIONARY::elements
```

Definition at line 1362 of file structs.h.

3.20.2.2 name

```
UBYTE* DICTIONARY::name
```

Definition at line 1363 of file structs.h.

3.20.2.3 sizeelements

```
int DICTIONARY::sizeelements
```

Definition at line 1364 of file structs.h.

3.20.2.4 numelements

```
int DICTIONARY::numelements
```

Definition at line 1365 of file structs.h.

3.20.2.5 numbers

```
int DICTIONARY::numbers
```

Definition at line 1366 of file structs.h.

3.20.2.6 variables

```
int DICTIONARY::variables
```

Definition at line 1367 of file structs.h.

3.20.2.7 characters

```
int DICTIONARY::characters
```

Definition at line 1368 of file structs.h.

3.20.2.8 funwith

```
int DICTIONARY::funwith
```

Definition at line 1369 of file structs.h.

3.20.2.9 gnumelements

```
int DICTIONARY::gnumelements
```

Definition at line 1370 of file structs.h.

3.20.2.10 ranges

```
int DICTIONARY::ranges
```

Definition at line 1371 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.21 DICTIONARY_ELEMENT Struct Reference

Data Fields

- WORD * lhs
- WORD * rhs
- int type
- int size

3.21.1 Detailed Description

Definition at line 1354 of file structs.h.

3.21.2 Field Documentation

3.21.2.1 lhs

`WORD* DICTIONARY_ELEMENT::lhs`

Definition at line 1355 of file structs.h.

3.21.2.2 rhs

`WORD* DICTIONARY_ELEMENT::rhs`

Definition at line 1356 of file structs.h.

3.21.2.3 type

`int DICTIONARY_ELEMENT::type`

Definition at line 1357 of file structs.h.

3.21.2.4 size

`int DICTIONARY_ELEMENT::size`

Definition at line 1358 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.22 DiStRiBuTe Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD * obj1
- WORD * obj2
- WORD * out
- WORD sign
- WORD n1
- WORD n2
- WORD n
- WORD cycle [MAXMATCH]

3.22.1 Detailed Description

The struct DISTRIBUTE is used to help the pattern matcher when matching antisymmetric tensors.

Definition at line 1103 of file structs.h.

3.22.2 Field Documentation

3.22.2.1 obj1

`WORD* DiStRiBuTe::obj1`

Definition at line 1104 of file structs.h.

3.22.2.2 obj2

`WORD* DiStRiBuTe::obj2`

Definition at line 1105 of file structs.h.

3.22.2.3 out

`WORD* DiStRiBuTe::out`

Definition at line 1106 of file structs.h.

3.22.2.4 sign

`WORD DiStRiBuTe::sign`

Definition at line 1107 of file structs.h.

3.22.2.5 n1

`WORD DiStRiBuTe::n1`

Definition at line 1108 of file structs.h.

3.22.2.6 n2

`WORD DiStRiBuTe::n2`

Definition at line 1109 of file structs.h.

3.22.2.7 n

`WORD DiStRiBuTe::n`

Definition at line 1110 of file structs.h.

3.22.2.8 cycle

`WORD DiStRiBuTe::cycle[MAXMATCH]`

Definition at line 1111 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.23 DNode Struct Reference

Data Fields

- int extloop
- int intrct

3.23.1 Detailed Description

Definition at line 192 of file grccparam.h.

3.23.2 Field Documentation

3.23.2.1 extloop

`int DNode::extloop`

Definition at line 193 of file grccparam.h.

3.23.2.2 intrct

`int DNode::intrct`

Definition at line 194 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.24 dollar_buf Struct Reference

3.24.1 Detailed Description

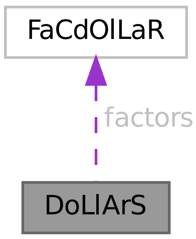
Definition at line 2479 of file parallel.c.

The documentation for this struct was generated from the following file:

- parallel.c

3.25 DoLIArS Struct Reference

Collaboration diagram for DoLIArS:



Data Fields

- WORD ∗ where
- FACDOLLAR ∗ factors
- LONG size
- LONG name
- WORD type
- WORD node
- WORD index
- WORD zero
- WORD numdummies
- WORD nfactors

3.25.1 Detailed Description

Definition at line 530 of file structs.h.

3.25.2 Field Documentation

3.25.2.1 where

`WORD* DoLlArS::where`

Definition at line 531 of file structs.h.

3.25.2.2 factors

`FACDOLLAR* DoLlArS::factors`

Definition at line 532 of file structs.h.

3.25.2.3 size

`LONG DoLlArS::size`

Definition at line 537 of file structs.h.

3.25.2.4 name

`LONG DoLlArS::name`

Definition at line 538 of file structs.h.

3.25.2.5 type

`WORD DoLlArS::type`

Definition at line 539 of file structs.h.

3.25.2.6 node

`WORD DoLlArS::node`

Definition at line 540 of file structs.h.

3.25.2.7 index

`WORD DoLlArS::index`

Definition at line 541 of file structs.h.

3.25.2.8 zero

```
WORD DoLlArS::zero
```

Definition at line 542 of file structs.h.

3.25.2.9 numdummies

```
WORD DoLlArS::numdummies
```

Definition at line 543 of file structs.h.

3.25.2.10 nfactors

```
WORD DoLlArS::nfactors
```

Definition at line 544 of file structs.h.

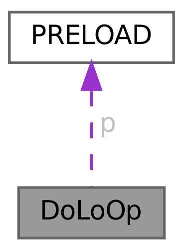
The documentation for this struct was generated from the following file:

- structs.h

3.26 DoLoOp Struct Reference

```
#include <structs.h>
```

Collaboration diagram for DoLoOp:



Data Fields

- PRELOAD p
- UBYTE * name
- UBYTE * vars
- UBYTE * contents
- UBYTE * dollarname
- LONG startlinenumber
- LONG firstnum
- LONG lastnum
- LONG incnum
- int type
- int NoShowInput
- int errorsinloop
- int firstloopcall
- WORD firstdollar
- WORD lastdollar
- WORD incdollar
- WORD NumPreTypes
- WORD PreIfLevel
- WORD PreSwitchLevel

3.26.1 Detailed Description

Each preprocessor do loop has a struct of type DOLOOP to keep track of all relevant parameters like where the beginning of the loop is, what the boundaries, increment and value of the loop parameter are, etc. Also we keep the whole loop inside a buffer of type PRELOAD

Definition at line 897 of file structs.h.

3.26.2 Field Documentation

3.26.2.1 p

```
PRELOAD DoLoOp::p
```

size, name and buffer

Definition at line 898 of file structs.h.

3.26.2.2 name

```
UBYTE* DoLoOp::name
```

pointer into PRELOAD buffer

Definition at line 899 of file structs.h.

3.26.2.3 vars

```
UBYTE* DoLoOp::vars
```

Definition at line 900 of file structs.h.

3.26.2.4 contents

```
UBYTE* DoLoOp::contents
```

Definition at line 901 of file structs.h.

3.26.2.5 dollarname

```
UBYTE* DoLoOp::dollarname
```

For loop over terms in expression. Allocated with Malloc1()

Definition at line 902 of file structs.h.

3.26.2.6 startlinenumber

```
LONG DoLoOp::startlinenumber
```

Definition at line 903 of file structs.h.

3.26.2.7 firstnum

```
LONG DoLoOp::firstnum
```

Definition at line 904 of file structs.h.

3.26.2.8 lastnum

```
LONG DoLoOp::lastnum
```

Definition at line 905 of file structs.h.

3.26.2.9 incnum

```
LONG DoLoOp::incnum
```

Definition at line 906 of file structs.h.

3.26.2.10 type

`int DoLoOp::type`

Definition at line 907 of file structs.h.

3.26.2.11 NoShowInput

`int DoLoOp::NoShowInput`

Definition at line 908 of file structs.h.

3.26.2.12 errorsinloop

`int DoLoOp::errorsinloop`

Definition at line 909 of file structs.h.

3.26.2.13 firstloopcall

`int DoLoOp::firstloopcall`

Definition at line 910 of file structs.h.

3.26.2.14 firstdollar

`WORD DoLoOp::firstdollar`

Definition at line 911 of file structs.h.

3.26.2.15 lastdollar

`WORD DoLoOp::lastdollar`

Definition at line 912 of file structs.h.

3.26.2.16 incdollar

`WORD DoLoOp::incdollar`

Definition at line 913 of file structs.h.

3.26.2.17 NumPreTypes

`WORD DoLoOp::NumPreTypes`

Definition at line 914 of file structs.h.

3.26.2.18 PreIfLevel

```
WORD DoLoOp::PreIfLevel
```

Definition at line 915 of file structs.h.

3.26.2.19 PreSwitchLevel

```
WORD DoLoOp::PreSwitchLevel
```

Definition at line 916 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.27 DuBiOuS Struct Reference

Data Fields

- LONG name
- WORD node
- WORD dummy

3.27.1 Detailed Description

Definition at line 515 of file structs.h.

3.27.2 Field Documentation

3.27.2.1 name

```
LONG DuBiOuS::name
```

Definition at line 516 of file structs.h.

3.27.2.2 node

```
WORD DuBiOuS::node
```

Definition at line 517 of file structs.h.

3.27.2.3 dummy

```
WORD DuBiOuS::dummy
```

Definition at line 518 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.28 ECand Class Reference

Public Member Functions

- ECand (int dt, int nplist, int ∗plst)
- void prECand (const char ∗msg)

Data Fields

- Bool det
- int nplist
- int plist [GRCC_MAXMPARTICLES2]

3.28.1 Detailed Description

Definition at line 1062 of file grcc.h.

3.28.2 Constructor & Destructor Documentation

3.28.2.1 ECand()

```
ECand::ECand (
            int dt,
            int nplist,
            int * plst )
```

Definition at line 7325 of file grcc.cc.

3.28.2.2 ∼ECand()

```
ECand::∼ECand (
            void )
```

Definition at line 7345 of file grcc.cc.

3.28.3 Member Function Documentation

3.28.3.1 prECand()

```
void ECand::prECand (
            const char * msg )
```

Definition at line 7350 of file grcc.cc.

3.28.4 Field Documentation

3.28.4.1 det

```
Bool ECand::det
```

Definition at line 1066 of file grcc.h.

3.28.4.2 nplist

```
int ECand::nplist
```

Definition at line 1067 of file grcc.h.

3.28.4.3 plist

```
int ECand::plist[GRCC_MAXMPARTICLES2]
```

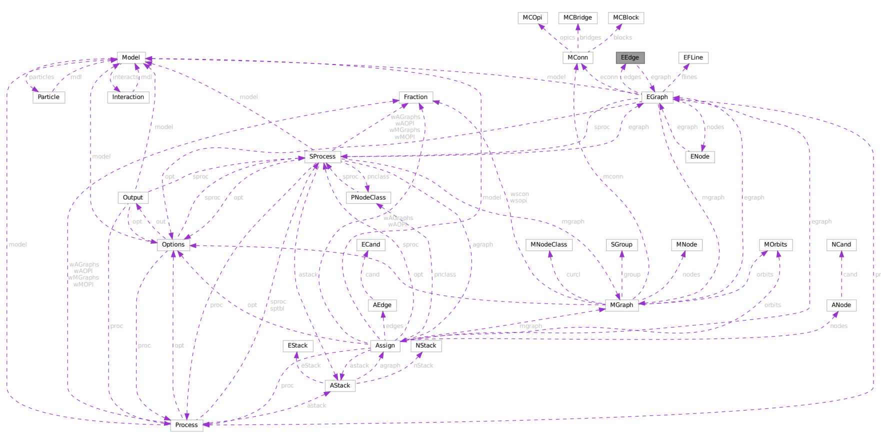
Definition at line 1068 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.29 EEdge Class Reference

Collaboration diagram for EEdge:



Public Member Functions

- EEdge (EGraph *egrph, int nedges, int nloops)
- void copy (EEdge *ee)
- void setId (EGraph *egrph, const int eid)
- void print (void)
- void setType (int typ)
- void setLMom (int k, int dir)
- void setEMom (int nedges, int *extn, int dir)

Data Fields

- EGraph * egraph
- int id
- int ext
- int ptcl
- int deleted
- int nodes [2]
- int nlegs [2]
- int * emom
- int * lmom
- int * extMom
- Bool cut
- int visited
- int conid
- int edtype
- int opicomp
- int dir

3.29.1 Detailed Description

Definition at line 538 of file grcc.h.

3.29.2 Constructor & Destructor Documentation

3.29.2.1 EEdge() [1/2]

```
EEdge::EEdge (
        void )
```

Definition at line 5640 of file grcc.cc.

3.29.2.2 EEdge() [2/2]

```
EEdge::EEdge (
        EGraph * egrph,
        int nedges,
        int nloops )
```

Definition at line 5659 of file grcc.cc.

3.29.2.3 ∼EEdge()

```
EEdge::∼EEdge (
            void )
```

Definition at line 5678 of file grcc.cc.

3.29.3 Member Function Documentation

3.29.3.1 copy()

```
void EEdge::copy (
            EEdge * ee )
```

Definition at line 5689 of file grcc.cc.

3.29.3.2 setId()

```
void EEdge::setId (
            EGraph * egrph,
            const int eid )
```

Definition at line 5701 of file grcc.cc.

3.29.3.3 print()

```
void EEdge::print (
            void )
```

Definition at line 5737 of file grcc.cc.

3.29.3.4 setType()

```
void EEdge::setType (
            int typ )
```

Definition at line 5708 of file grcc.cc.

3.29.3.5 setLMom()

```
void EEdge::setLMom (
            int k,
            int dir )
```

Definition at line 5731 of file grcc.cc.

3.29.3.6 setEMom()

```
void EEdge::setEMom (
            int nedges,
            int * extn,
            int dir )
```

Definition at line 5719 of file grcc.cc.

3.29.4 Field Documentation

3.29.4.1 egraph

```
EGraph* EEdge::egraph
```

Definition at line 541 of file grcc.h.

3.29.4.2 id

```
int EEdge::id
```

Definition at line 543 of file grcc.h.

3.29.4.3 ext

```
int EEdge::ext
```

Definition at line 544 of file grcc.h.

3.29.4.4 ptcl

```
int EEdge::ptcl
```

Definition at line 545 of file grcc.h.

3.29.4.5 deleted

```
int EEdge::deleted
```

Definition at line 546 of file grcc.h.

3.29.4.6 nodes

```
int EEdge::nodes[2]
```

Definition at line 547 of file grcc.h.

3.29.4.7 nlegs

```
int EEdge::nlegs[2]
```

Definition at line 548 of file grcc.h.

3.29.4.8 emom

```
int* EEdge::emom
```

Definition at line 551 of file grcc.h.

3.29.4.9 lmom

```
int* EEdge::lmom
```

Definition at line 552 of file grcc.h.

3.29.4.10 extMom

```
int* EEdge::extMom
```

Definition at line 553 of file grcc.h.

3.29.4.11 cut

```
Bool EEdge::cut
```

Definition at line 555 of file grcc.h.

3.29.4.12 visited

```
int EEdge::visited
```

Definition at line 556 of file grcc.h.

3.29.4.13 conid

```
int EEdge::conid
```

Definition at line 557 of file grcc.h.

3.29.4.14 edtype

```
int EEdge::edtype
```

Definition at line 558 of file grcc.h.

3.29.4.15 opicomp

```
int EEdge::opicomp
```

Definition at line 559 of file grcc.h.

3.29.4.16 dir

```
int EEdge::dir
```

Definition at line 560 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.30 EFLine Class Reference

Public Member Functions

- void print (const char *msg)

Data Fields

- int elist [GRCC_MAXNODES]
- FLType ftype
- int fkind
- int nlist

3.30.1 Detailed Description

Definition at line 583 of file grcc.h.

3.30.2 Constructor & Destructor Documentation

3.30.2.1 EFLine()

```
EFLine::EFLine (
            void )
```

Definition at line 5779 of file grcc.cc.

3.30.3 Member Function Documentation

3.30.3.1 print()

```
void EFLine::print (
            const char * msg )
```

Definition at line 5787 of file grcc.cc.

3.30.4 Field Documentation

3.30.4.1 elist

```
int EFLine::elist[GRCC_MAXNODES]
```

Definition at line 585 of file grcc.h.

3.30.4.2 ftype

```
FLType EFLine::ftype
```

Definition at line 586 of file grcc.h.

3.30.4.3 fkind

```
int EFLine::fkind
```

Definition at line 587 of file grcc.h.

3.30.4.4 nlist

```
int EFLine::nlist
```

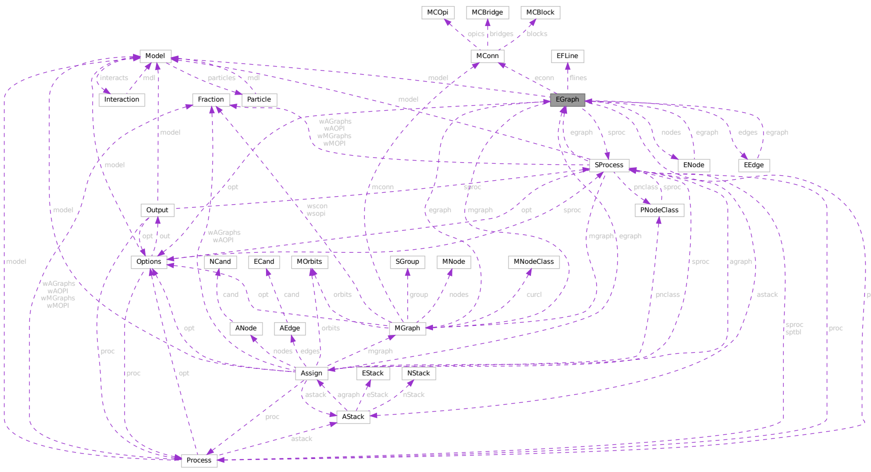
Definition at line 588 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.31 EGraph Class Reference

Collaboration diagram for EGraph:



Public Member Functions

- EGraph (int nnodes, int nedges, int mxdeg)
- void copy (EGraph ∗eg)
- void print (void)
- void fromDGraph (DGraph ∗dg)
- void fromMGraph (MGraph ∗mgraph)
- Bool isOptE (void)
- ENode ∗ setExtern (int n0, int pt, int ndtp)
- Bool isExternal (int nd)
- Bool isFermion (int nd)
- void setExtLoop (int nd, int val)
- void endSetExtLoop (void)
- int connComp (void)
- int connVisit (int nd, int ncc)
- void biconnE (void)
- void biinitE (void)
- void bisearchE (int nd, int ∗extlst, int ∗intlst, int ∗opiext, int ∗opiloop)
- int findRoot (void)
- int dirEdge (int n, int e)
- void extMomConsv (void)
- int cmpMom (int ∗lm0, int ∗em0, int ∗lm1, int ∗em1)
- int groupLMom (int ∗grp, int ∗ed2gr)
- void chkMomConsv (void)
- void prFLines (void)
- void getFLines (void)
- int fltrace (int fk, int nd0, int ∗fl)
- void addFLine (const FLType ft, int fk, int nfl, int ∗fl)
- int legParticle (int ed, int lg)

Data Fields

- Options ∗ opt
- Model ∗ model
- Process ∗ proc
- SProcess ∗ sproc
- MGraph ∗ mgraph
- MConn ∗ econn
- ENode ∗∗ nodes
- EEdge ∗∗ edges
- BigInt mId
- BigInt aId
- BigInt sId
- BigInt gSubId
- Bool assigned
- int fsign
- BigInt nsym
- BigInt esym
- BigInt nsym1
- BigInt extperm
- BigInt multp
- int pId
- int sNodes
- int sEdges
- int sMaxdeg
- int sLoops
- int nNodes
- int nEdges
- int nExtern
- int maxdeg
- int nLoops
- int totalc
- int nopicomp
- int opi2plp
- int nopi2p
- int nadj2ptv
- int DUMMYPADDING
- int ∗ bidef
- int ∗ bilow
- int ∗ extMom
- int bconn
- int bicount
- int loopm
- int opiCount
- EFLine ∗ flines [GRCC_MAXFLINES]
- int nflines
- int DUMMYPADDING1

3.31.1 Detailed Description

Definition at line 595 of file grcc.h.

3.31.2 Constructor & Destructor Documentation

3.31.2.1 EGraph()

```
EGraph::EGraph (
            int nnodes,
            int nedges,
            int mxdeg )
```

Definition at line 5812 of file grcc.cc.

3.31.2.2 ∼EGraph()

```
EGraph::~EGraph (
            void )
```

Definition at line 5873 of file grcc.cc.

3.31.3 Member Function Documentation

3.31.3.1 copy()

```
void EGraph::copy (
            EGraph * eg )
```

Definition at line 5904 of file grcc.cc.

3.31.3.2 print()

```
void EGraph::print (
            void )
```

Definition at line 6269 of file grcc.cc.

3.31.3.3 fromDGraph()

```
void EGraph::fromDGraph (
            DGraph * dg )
```

Definition at line 5969 of file grcc.cc.

3.31.3.4 fromMGraph()

```
void EGraph::fromMGraph (
            MGraph * mgraph )
```

Definition at line 6074 of file grcc.cc.

3.31.3.5 isOptE()

```
Bool EGraph::isOptE (
             void  )
```

Definition at line 6440 of file grcc.cc.

3.31.3.6 setExtern()

```
ENode * EGraph::setExtern (
             int n0,
             int pt,
             int ndtp )
```

Definition at line 6222 of file grcc.cc.

3.31.3.7 isExternal()

```
Bool EGraph::isExternal (
             int nd )  [inline]
```

Definition at line 667 of file grcc.h.

3.31.3.8 isFermion()

```
int EGraph::isFermion (
             int nd )
```

Definition at line 7056 of file grcc.cc.

3.31.3.9 setExtLoop()

```
void EGraph::setExtLoop (
             int nd,
             int val )
```

Definition at line 5946 of file grcc.cc.

3.31.3.10 endSetExtLoop()

```
void EGraph::endSetExtLoop (
             void  )
```

Definition at line 5954 of file grcc.cc.

3.31.3.11 connComp()

```
int EGraph::connComp (
            void )
```

Definition at line 6586 of file grcc.cc.

3.31.3.12 connVisit()

```
int EGraph::connVisit (
            int nd,
            int ncc )
```

Definition at line 6618 of file grcc.cc.

3.31.3.13 biconnE()

```
void EGraph::biconnE (
            void )
```

Definition at line 6639 of file grcc.cc.

3.31.3.14 biinitE()

```
void EGraph::biinitE (
            void )
```

Definition at line 6708 of file grcc.cc.

3.31.3.15 bisearchE()

```
void EGraph::bisearchE (
            int nd,
            int * extlst,
            int * intlst,
            int * opiext,
            int * opiloop )
```

Definition at line 6744 of file grcc.cc.

3.31.3.16 findRoot()

```
int EGraph::findRoot (
            void )
```

Definition at line 6537 of file grcc.cc.

3.31.3.17 dirEdge()

```
int EGraph::dirEdge (
             int n,
             int e )
```

Definition at line 6347 of file grcc.cc.

3.31.3.18 extMomConsv()

```
void EGraph::extMomConsv (
             void )
```

Definition at line 6949 of file grcc.cc.

3.31.3.19 cmpMom()

```
int EGraph::cmpMom (
             int * lm0,
             int * em0,
             int * lm1,
             int * em1 )
```

Definition at line 6357 of file grcc.cc.

3.31.3.20 groupLMom()

```
int EGraph::groupLMom (
             int * grp,
             int * ed2gr )
```

Definition at line 6407 of file grcc.cc.

3.31.3.21 chkMomConsv()

```
void EGraph::chkMomConsv (
             void )
```

Definition at line 6977 of file grcc.cc.

3.31.3.22 prFLines()

```
void EGraph::prFLines (
             void )
```

Definition at line 6334 of file grcc.cc.

3.31.3.23 getFLines()

```
void EGraph::getFLines (
             void )
```

Definition at line 7154 of file grcc.cc.

3.31.3.24 fltrace()

```
int EGraph::fltrace (
             int fk,
             int nd0,
             int * fl )
```

Definition at line 7069 of file grcc.cc.

3.31.3.25 addFLine()

```
void EGraph::addFLine (
             const FLType ft,
             int fk,
             int nfl,
             int * fl )
```

Definition at line 7240 of file grcc.cc.

3.31.3.26 legParticle()

```
int EGraph::legParticle (
             int ed,
             int lg )
```

Definition at line 7048 of file grcc.cc.

3.31.4 Field Documentation

3.31.4.1 opt

```
Options* EGraph::opt
```

Definition at line 597 of file grcc.h.

3.31.4.2 model

```
Model* EGraph::model
```

Definition at line 598 of file grcc.h.

3.31.4.3 proc

`Process* EGraph::proc`

Definition at line 599 of file grcc.h.

3.31.4.4 sproc

`SProcess* EGraph::sproc`

Definition at line 600 of file grcc.h.

3.31.4.5 mgraph

`MGraph* EGraph::mgraph`

Definition at line 601 of file grcc.h.

3.31.4.6 econn

`MConn* EGraph::econn`

Definition at line 602 of file grcc.h.

3.31.4.7 nodes

`ENode** EGraph::nodes`

Definition at line 604 of file grcc.h.

3.31.4.8 edges

`EEdge** EGraph::edges`

Definition at line 605 of file grcc.h.

3.31.4.9 mId

`BigInt EGraph::mId`

Definition at line 607 of file grcc.h.

3.31.4.10 aId

`BigInt EGraph::aId`

Definition at line 608 of file grcc.h.

3.31.4.11 sId

```
BigInt EGraph::sId
```

Definition at line 609 of file grcc.h.

3.31.4.12 gSubId

```
BigInt EGraph::gSubId
```

Definition at line 610 of file grcc.h.

3.31.4.13 assigned

```
Bool EGraph::assigned
```

Definition at line 611 of file grcc.h.

3.31.4.14 fsign

```
int EGraph::fsign
```

Definition at line 613 of file grcc.h.

3.31.4.15 nsym

```
BigInt EGraph::nsym
```

Definition at line 614 of file grcc.h.

3.31.4.16 esym

```
BigInt EGraph::esym
```

Definition at line 614 of file grcc.h.

3.31.4.17 nsym1

```
BigInt EGraph::nsym1
```

Definition at line 615 of file grcc.h.

3.31.4.18 extperm

```
BigInt EGraph::extperm
```

Definition at line 616 of file grcc.h.

3.31.4.19 multp

```
BigInt EGraph::multp
```

Definition at line 617 of file grcc.h.

3.31.4.20 pId

```
int EGraph::pId
```

Definition at line 619 of file grcc.h.

3.31.4.21 sNodes

```
int EGraph::sNodes
```

Definition at line 621 of file grcc.h.

3.31.4.22 sEdges

```
int EGraph::sEdges
```

Definition at line 622 of file grcc.h.

3.31.4.23 sMaxdeg

```
int EGraph::sMaxdeg
```

Definition at line 623 of file grcc.h.

3.31.4.24 sLoops

```
int EGraph::sLoops
```

Definition at line 624 of file grcc.h.

3.31.4.25 nNodes

```
int EGraph::nNodes
```

Definition at line 626 of file grcc.h.

3.31.4.26 nEdges

```
int EGraph::nEdges
```

Definition at line 627 of file grcc.h.

3.31.4.27 nExtern

`int EGraph::nExtern`

Definition at line 628 of file grcc.h.

3.31.4.28 maxdeg

`int EGraph::maxdeg`

Definition at line 629 of file grcc.h.

3.31.4.29 nLoops

`int EGraph::nLoops`

Definition at line 630 of file grcc.h.

3.31.4.30 totalc

`int EGraph::totalc`

Definition at line 631 of file grcc.h.

3.31.4.31 nopicomp

`int EGraph::nopicomp`

Definition at line 634 of file grcc.h.

3.31.4.32 opi2plp

`int EGraph::opi2plp`

Definition at line 635 of file grcc.h.

3.31.4.33 nopi2p

`int EGraph::nopi2p`

Definition at line 636 of file grcc.h.

3.31.4.34 nadj2ptv

`int EGraph::nadj2ptv`

Definition at line 637 of file grcc.h.

3.31.4.35 DUMMYPADDING

```
int EGraph::DUMMYPADDING
```

Definition at line 639 of file grcc.h.

3.31.4.36 bidef

```
int* EGraph::bidef
```

Definition at line 641 of file grcc.h.

3.31.4.37 bilow

```
int* EGraph::bilow
```

Definition at line 642 of file grcc.h.

3.31.4.38 extMom

```
int* EGraph::extMom
```

Definition at line 643 of file grcc.h.

3.31.4.39 bconn

```
int EGraph::bconn
```

Definition at line 644 of file grcc.h.

3.31.4.40 bicount

```
int EGraph::bicount
```

Definition at line 645 of file grcc.h.

3.31.4.41 loopm

```
int EGraph::loopm
```

Definition at line 646 of file grcc.h.

3.31.4.42 opiCount

```
int EGraph::opiCount
```

Definition at line 647 of file grcc.h.

3.31.4.43 flines

`EFLine* EGraph::flines[GRCC_MAXFLINES]`

Definition at line 650 of file grcc.h.

3.31.4.44 nflines

`int EGraph::nflines`

Definition at line 651 of file grcc.h.

3.31.4.45 DUMMYPADDING1

`int EGraph::DUMMYPADDING1`

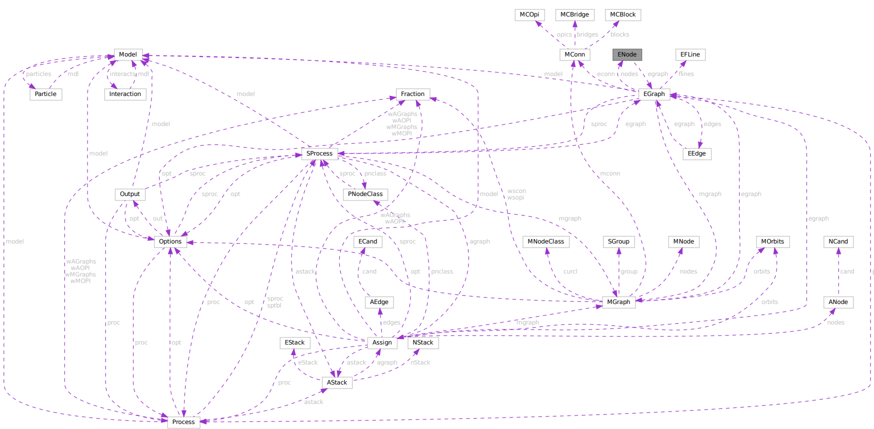
Definition at line 653 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.32 ENode Class Reference

Collaboration diagram for ENode:



Public Member Functions

- ENode (EGraph ∗egrph, int loops, int sdeg)
- void initAss (EGraph ∗egrph, int nid, int sdg)
- void setId (EGraph ∗egrph, const int nid)
- void copy (ENode ∗en)
- void setExtern (int typ, int pt)
- void setType (int typ)
- void print (void)

Data Fields

- EGraph ∗ egraph
- int ∗ edges
- int id
- int maxdeg
- int deg
- int extloop
- int ndtype
- int intrct
- int ∗ klow
- int visited
- int DUMMYPADDING

3.32.1 Detailed Description

Definition at line 504 of file grcc.h.

3.32.2 Constructor & Destructor Documentation

3.32.2.1 ENode() [1/2]

```
ENode::ENode (
        void )
```

Definition at line 5526 of file grcc.cc.

3.32.2.2 ENode() [2/2]

```
ENode::ENode (
        EGraph * egrph,
        int loops,
        int sdeg )
```

Definition at line 5542 of file grcc.cc.

3.32.2.3 ∼ENode()

```
ENode::∼ENode (
        void )
```

Definition at line 5562 of file grcc.cc.

3.32.3 Member Function Documentation

3.32.3.1 initAss()

```
void ENode::initAss (
            EGraph * egrph,
            int nid,
            int sdg )
```

Definition at line 5586 of file grcc.cc.

3.32.3.2 setId()

```
void ENode::setId (
            EGraph * egrph,
            const int nid )
```

Definition at line 5594 of file grcc.cc.

3.32.3.3 copy()

```
void ENode::copy (
            ENode * en )
```

Definition at line 5573 of file grcc.cc.

3.32.3.4 setExtern()

```
void ENode::setExtern (
            int typ,
            int pt )
```

Definition at line 5601 of file grcc.cc.

3.32.3.5 setType()

```
void ENode::setType (
            int typ )
```

Definition at line 5608 of file grcc.cc.

3.32.3.6 print()

```
void ENode::print (
            void )
```

Definition at line 5621 of file grcc.cc.

3.32.4 Field Documentation

3.32.4.1 egraph

`EGraph* ENode::egraph`

Definition at line 506 of file grcc.h.

3.32.4.2 edges

`int* ENode::edges`

Definition at line 507 of file grcc.h.

3.32.4.3 id

`int ENode::id`

Definition at line 510 of file grcc.h.

3.32.4.4 maxdeg

`int ENode::maxdeg`

Definition at line 511 of file grcc.h.

3.32.4.5 deg

`int ENode::deg`

Definition at line 512 of file grcc.h.

3.32.4.6 extloop

`int ENode::extloop`

Definition at line 513 of file grcc.h.

3.32.4.7 ndtype

`int ENode::ndtype`

Definition at line 514 of file grcc.h.

3.32.4.8 intrct

```
int ENode::intrct
```

Definition at line 515 of file grcc.h.

3.32.4.9 klow

```
int* ENode::klow
```

Definition at line 518 of file grcc.h.

3.32.4.10 visited

```
int ENode::visited
```

Definition at line 519 of file grcc.h.

3.32.4.11 DUMMYPADDING

```
int ENode::DUMMYPADDING
```

Definition at line 521 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.33 EStack Class Reference

Public Member Functions

- void print (const char *msg)

Data Fields

- int edgen
- int det
- int nplist
- int plist [GRCC_MAXMPARTICLES2]

3.33.1 Detailed Description

Definition at line 1288 of file grcc.h.

3.33.2 Constructor & Destructor Documentation

3.33.2.1 EStack()

```
EStack::EStack ( )  [inline]
```

Definition at line 1297 of file grcc.h.

3.33.2.2 ∼EStack()

```
EStack::∼EStack ( )  [inline]
```

Definition at line 1298 of file grcc.h.

3.33.3 Member Function Documentation

3.33.3.1 print()

```
void EStack::print (
             const char * msg )
```

Definition at line 9463 of file grcc.cc.

3.33.4 Field Documentation

3.33.4.1 edgen

```
int EStack::edgen
```

Definition at line 1292 of file grcc.h.

3.33.4.2 det

```
int EStack::det
```

Definition at line 1293 of file grcc.h.

3.33.4.3 nplist

```
int EStack::nplist
```

Definition at line 1294 of file grcc.h.

3.33.4.4 plist

```
int EStack::plist[GRCC_MAXMPARTICLES2]
```

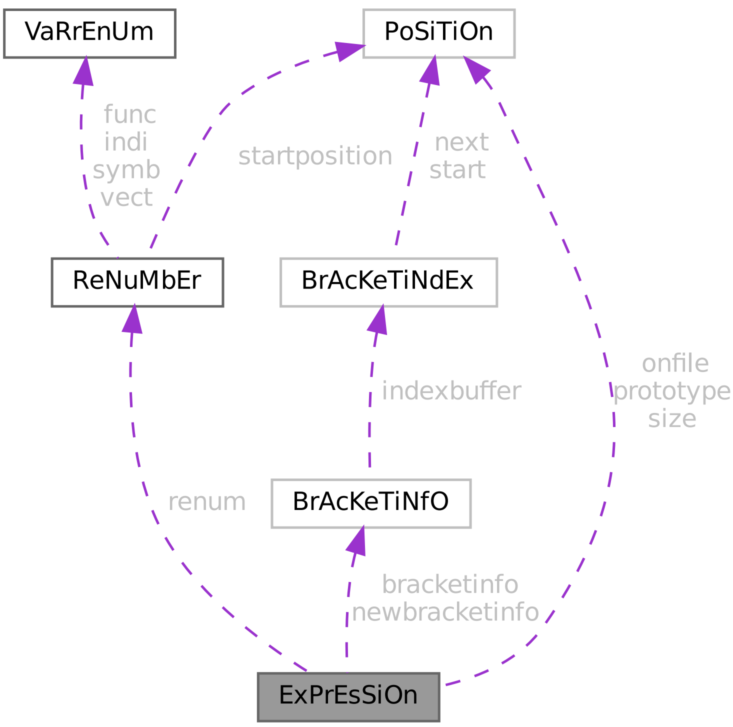
Definition at line 1295 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.34 ExPrEsSiOn Struct Reference

Collaboration diagram for ExPrEsSiOn:



Public Member Functions

- **PADPOSITION** (5, 2, 0, 13, 0)

Data Fields

- POSITION onfile
- POSITION prototype
- POSITION size
- RENUMBER renum
- BRACKETINFO * bracketinfo
- BRACKETINFO * newbracketinfo
- WORD * renumlists
- WORD * inmem
- LONG counter
- LONG name
- WORD hidelevel
- WORD vflags
- WORD printflag
- WORD status
- WORD replace
- WORD node
- WORD whichbuffer
- WORD namesize
- WORD compression
- WORD numdummies
- WORD numfactors
- WORD sizeprototype
- WORD uflags

3.34.1 Detailed Description

Definition at line 391 of file structs.h.

3.34.2 Field Documentation

3.34.2.1 onfile

```
POSITION ExPrEsSiOn::onfile
```

Definition at line 392 of file structs.h.

3.34.2.2 prototype

```
POSITION ExPrEsSiOn::prototype
```

Definition at line 393 of file structs.h.

3.34.2.3 size

```
POSITION ExPrEsSiOn::size
```

Definition at line 394 of file structs.h.

3.34.2.4 renum

`RENUMBER ExPrEsSiOn::renum`

Definition at line 395 of file structs.h.

3.34.2.5 bracketinfo

`BRACKETINFO* ExPrEsSiOn::bracketinfo`

Definition at line 396 of file structs.h.

3.34.2.6 newbracketinfo

`BRACKETINFO* ExPrEsSiOn::newbracketinfo`

Definition at line 397 of file structs.h.

3.34.2.7 renumlists

`WORD* ExPrEsSiOn::renumlists`

Allocated only for threaded version if variables exist, else points to AN.dummyrenumlist

Definition at line 398 of file structs.h.

Referenced by DoRecovery().

3.34.2.8 inmem

`WORD* ExPrEsSiOn::inmem`

Definition at line 400 of file structs.h.

3.34.2.9 counter

`LONG ExPrEsSiOn::counter`

Definition at line 401 of file structs.h.

3.34.2.10 name

`LONG ExPrEsSiOn::name`

Definition at line 402 of file structs.h.

3.34.2.11 hidelevel

`WORD ExPrEsSiOn::hidelevel`

Definition at line 403 of file structs.h.

3.34.2.12 vflags

`WORD ExPrEsSiOn::vflags`

Definition at line 404 of file structs.h.

3.34.2.13 printflag

`WORD ExPrEsSiOn::printflag`

Definition at line 405 of file structs.h.

3.34.2.14 status

`WORD ExPrEsSiOn::status`

Definition at line 406 of file structs.h.

3.34.2.15 replace

`WORD ExPrEsSiOn::replace`

Definition at line 407 of file structs.h.

3.34.2.16 node

`WORD ExPrEsSiOn::node`

Definition at line 408 of file structs.h.

3.34.2.17 whichbuffer

`WORD ExPrEsSiOn::whichbuffer`

Definition at line 409 of file structs.h.

3.34.2.18 namesize

`WORD ExPrEsSiOn::namesize`

Definition at line 410 of file structs.h.

3.34.2.19 compression

```
WORD ExPrEsSiOn::compression
```

Definition at line 411 of file structs.h.

3.34.2.20 numdummies

```
WORD ExPrEsSiOn::numdummies
```

Definition at line 412 of file structs.h.

3.34.2.21 numfactors

```
WORD ExPrEsSiOn::numfactors
```

Definition at line 413 of file structs.h.

3.34.2.22 sizeprototype

```
WORD ExPrEsSiOn::sizeprototype
```

Definition at line 414 of file structs.h.

3.34.2.23 uflags

```
WORD ExPrEsSiOn::uflags
```

Definition at line 415 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.35 FaCdOIlaR Struct Reference

Data Fields

- WORD * where
- LONG size
- WORD type
- WORD value

3.35.1 Detailed Description

Definition at line 522 of file structs.h.

3.35.2 Field Documentation

3.35.2.1 where

```
WORD* FaCdOlLaR::where
```

Definition at line 523 of file structs.h.

3.35.2.2 size

```
LONG FaCdOlLaR::size
```

Definition at line 524 of file structs.h.

3.35.2.3 type

```
WORD FaCdOlLaR::type
```

Definition at line 525 of file structs.h.

3.35.2.4 value

```
WORD FaCdOlLaR::value
```

Definition at line 526 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.36 factorized_poly Class Reference

Public Member Functions

- void add_factor (const poly &f, int p=1)
- const std::string tostring () const

Data Fields

- std::vector< poly > factor
- std::vector< int > power

Friends

- std::ostream & **operator<<** (std::ostream &out, const poly &p)

3.36.1 Detailed Description

Definition at line 54 of file polyfact.h.

3.36.2 Member Function Documentation

3.36.2.1 add_factor()

```
void factorized_poly::add_factor (
            const poly & f,
            int p = 1 )
```

Definition at line 125 of file polyfact.cc.

3.36.2.2 tostring()

```
const string factorized_poly::tostring ( ) const
```

Definition at line 59 of file polyfact.cc.

3.36.3 Field Documentation

3.36.3.1 factor

```
std::vector<poly> factorized_poly::factor
```

Definition at line 59 of file polyfact.h.

3.36.3.2 power

```
std::vector<int> factorized_poly::power
```

Definition at line 60 of file polyfact.h.

The documentation for this class was generated from the following files:

- polyfact.h
- polyfact.cc

3.37 FGInput Struct Reference

Data Fields

- long ninitl
- const char ∗ initln [GRCC_MAXNODES]
- int initlc [GRCC_MAXNODES]
- long nfinal
- const char ∗ finaln [GRCC_MAXNODES]
- int finalc [GRCC_MAXNODES]
- int coupl [GRCC_MAXNCPLG]

3.37.1 Detailed Description

Definition at line 165 of file grccparam.h.

3.37.2 Field Documentation

3.37.2.1 ninitl

```
long FGInput::ninitl
```

Definition at line 166 of file grccparam.h.

3.37.2.2 initln

```
const char* FGInput::initln[GRCC_MAXNODES]
```

Definition at line 167 of file grccparam.h.

3.37.2.3 initlc

```
int FGInput::initlc[GRCC_MAXNODES]
```

Definition at line 168 of file grccparam.h.

3.37.2.4 nfinal

```
long FGInput::nfinal
```

Definition at line 169 of file grccparam.h.

3.37.2.5 finaln

```
const char* FGInput::finaln[GRCC_MAXNODES]
```

Definition at line 170 of file grccparam.h.

3.37.2.6 finalc

```
int FGInput::finalc[GRCC_MAXNODES]
```

Definition at line 171 of file grccparam.h.

3.37.2.7 coupl

```
int FGInput::coupl[GRCC_MAXNCPLG]
```

Definition at line 172 of file grccparam.h.

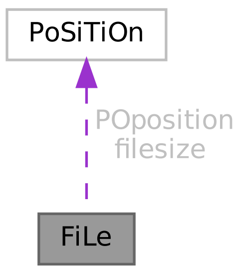
The documentation for this struct was generated from the following file:

- grccparam.h

3.38 FiLe Struct Reference

```
#include <structs.h>
```

Collaboration diagram for FiLe:



Public Member Functions

- **PADPOSITION** (5, 3, 2, 0, 0)

Data Fields

- POSITION POposition
- POSITION filesize
- WORD ∗ PObuffer
- WORD ∗ POstop
- WORD ∗ POfill
- WORD ∗ POfull
- char ∗ name
- ULONG numblocks
- ULONG inbuffer
- LONG POsize
- int handle
- int active

3.38.1 Detailed Description

The type FILEHANDLE is the struct that controls all relevant information of a file, whether it is open or not. The file may even not yet exist. There is a system of caches (PObuffer) and as long as the information to be written still fits inside the cache the file may never be created. There are variables that can store information about different types of files, like scratch files or sort files. Depending on what is available in the system we may also have information about gzip compression (currently sort file only) or locks (TFORM).

Definition at line 682 of file structs.h.

3.38.2 Field Documentation

3.38.2.1 POposition

```
POSITION FiLe::POposition
```

Definition at line 683 of file structs.h.

3.38.2.2 filesize

```
POSITION FiLe::filesize
```

Definition at line 684 of file structs.h.

3.38.2.3 PObuffer

```
WORD* FiLe::PObuffer
```

Definition at line 685 of file structs.h.

3.38.2.4 POstop

```
WORD* FiLe::POstop
```

Definition at line 686 of file structs.h.

3.38.2.5 POfill

```
WORD* FiLe::POfill
```

Definition at line 687 of file structs.h.

3.38.2.6 POfull

```
WORD* FiLe::POfull
```

Definition at line 688 of file structs.h.

3.38.2.7 name

```
char* FiLe::name
```

Definition at line 695 of file structs.h.

3.38.2.8 numblocks

```
ULONG FiLe::numblocks
```

Definition at line 700 of file structs.h.

3.38.2.9 inbuffer

```
ULONG FiLe::inbuffer
```

Definition at line 701 of file structs.h.

3.38.2.10 POsize

```
LONG FiLe::POsize
```

Definition at line 702 of file structs.h.

3.38.2.11 handle

```
int FiLe::handle
```

Our own handle. Equal -1 if no file exists.

Definition at line 710 of file structs.h.

Referenced by DoOnePow(), FlushOut(), GetFirstTerm(), MergePatches(), poly_factorize_expression(), poly_unfactorize_expression(), PutIn(), PutOut(), Sflush(), and StageSort().

3.38.2.12 active

```
int FiLe::active
```

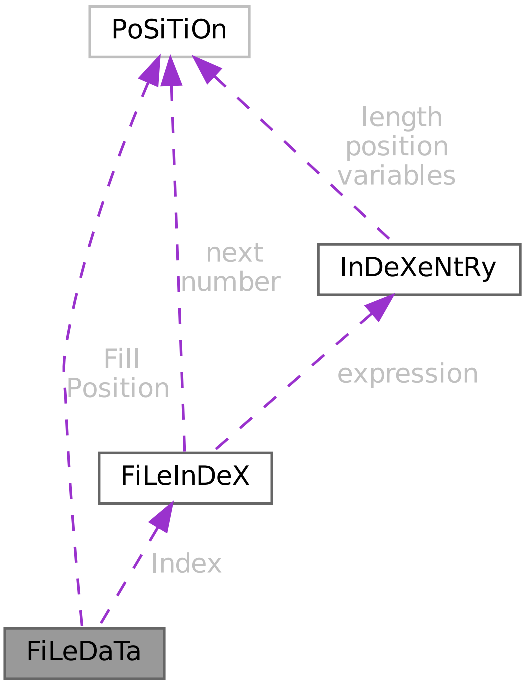
Definition at line 711 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.39 FiLeDaTa Struct Reference

Collaboration diagram for FiLeDaTa:



Public Member Functions

- **PADPOSITION** (0, 0, 0, 2, 0)

Data Fields

- FILEINDEX Index
- POSITION Fill
- POSITION Position
- WORD Handle
- WORD dirtyflag

3.39.1 Detailed Description

Definition at line 151 of file structs.h.

3.39.2 Field Documentation

3.39.2.1 Index

`FILEINDEX FiLeDaTa::Index`

Definition at line 152 of file structs.h.

3.39.2.2 Fill

`POSITION FiLeDaTa::Fill`

Definition at line 153 of file structs.h.

3.39.2.3 Position

`POSITION FiLeDaTa::Position`

Definition at line 154 of file structs.h.

3.39.2.4 Handle

`WORD FiLeDaTa::Handle`

Definition at line 155 of file structs.h.

3.39.2.5 dirtyflag

`WORD FiLeDaTa::dirtyflag`

Definition at line 156 of file structs.h.

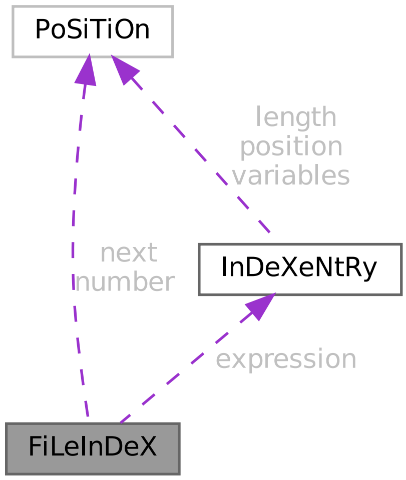
The documentation for this struct was generated from the following file:

- structs.h

3.40 FiLeInDeX Struct Reference

`#include <structs.h>`

Collaboration diagram for FiLeInDeX:



Data Fields

- POSITION next
- POSITION number
- INDEXENTRY expression [INFILEINDEX]
- SBYTE empty [EMPTYININDEX]

3.40.1 Detailed Description

Defines the structure of a file index in store-files and save-files.

It contains several entries (see struct InDeXeNtRy) up to a maximum of INFILEINDEX.

The variable number has been made of type POSITION to avoid padding problems with some types of computers/↩
OS and keep system independence of the .sav files.

This struct is always 512 bytes long.

Definition at line 138 of file structs.h.

3.40.2 Field Documentation

3.40.2.1 next

```
POSITION FiLeInDeX::next
```

Position of next FILEINDEX if any

Definition at line 139 of file structs.h.

Referenced by ReadSaveIndex().

3.40.2.2 number

```
POSITION FiLeInDeX::number
```

Number of used entries in this index

Definition at line 140 of file structs.h.

Referenced by ReadSaveIndex().

3.40.2.3 expression

```
INDEXENTRY FiLeInDeX::expression[INFILEINDEX]
```

File index entries

Definition at line 141 of file structs.h.

3.40.2.4 empty

```
SBYTE FiLeInDeX::empty[EMPTYININDEX]
```

Padding to 512 bytes

Definition at line 142 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.41 FixedGlobals Struct Reference

```
#include <structs.h>
```

Data Fields

- WCN Operation [8]
- WCN2 OperaFind [6]
- char ∗ VarType [10]
- char ∗ ExprStat [21]
- char ∗ FunNam [2]
- char ∗ swmes [3]
- char ∗ fname
- char ∗ fname2
- UBYTE ∗ s_one
- WORD fnamebase
- WORD fname2base
- WORD fnamesize
- WORD fname2size
- UINT cTable [256]

3.41.1 Detailed Description

The FIXEDGLOBALS struct is an anachronism. It started as the struct with global variables that needed initialization. It contains the elements Operation and OperaFind which define a very early way of automatically jumping to the proper operation. We find the results of it in parts of the file opera.c Later operations were treated differently in a more transparent way. We never changed the existing code. The most important part is currently the cTable which is used intensively in the compiler.

Definition at line 2674 of file structs.h.

3.41.2 Field Documentation

3.41.2.1 Operation

```
WCN FixedGlobals::Operation[8]
```

Definition at line 2675 of file structs.h.

3.41.2.2 OperaFind

```
WCN2 FixedGlobals::OperaFind[6]
```

Definition at line 2676 of file structs.h.

3.41.2.3 VarType

```
char* FixedGlobals::VarType[10]
```

Definition at line 2677 of file structs.h.

3.41.2.4 ExprStat

```
char* FixedGlobals::ExprStat[21]
```

Definition at line 2678 of file structs.h.

3.41.2.5 FunNam

```
char* FixedGlobals::FunNam[2]
```

Definition at line 2679 of file structs.h.

3.41.2.6 swmes

```
char* FixedGlobals::swmes[3]
```

Definition at line 2680 of file structs.h.

3.41.2.7 fname

```
char* FixedGlobals::fname
```

Definition at line 2681 of file structs.h.

3.41.2.8 fname2

```
char* FixedGlobals::fname2
```

Definition at line 2682 of file structs.h.

3.41.2.9 s_one

```
UBYTE* FixedGlobals::s_one
```

Definition at line 2683 of file structs.h.

3.41.2.10 fnamebase

```
WORD FixedGlobals::fnamebase
```

Definition at line 2684 of file structs.h.

3.41.2.11 fname2base

```
WORD FixedGlobals::fname2base
```

Definition at line 2685 of file structs.h.

3.41.2.12 fnamesize

```
WORD FixedGlobals::fnamesize
```

Definition at line 2686 of file structs.h.

3.41.2.13 fname2size

```
WORD FixedGlobals::fname2size
```

Definition at line 2687 of file structs.h.

3.41.2.14 cTable

```
UINT FixedGlobals::cTable[256]
```

Definition at line 2688 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.42 fixedset Struct Reference

Data Fields

- char $*$ name
- char $*$ description
- int type
- int dimension

3.42.1 Detailed Description

Definition at line 571 of file structs.h.

3.42.2 Field Documentation

3.42.2.1 name

```
char* fixedset::name
```

Definition at line 572 of file structs.h.

3.42.2.2 description

`char* fixedset::description`

Definition at line 573 of file structs.h.

3.42.2.3 type

`int fixedset::type`

Definition at line 574 of file structs.h.

3.42.2.4 dimension

`int fixedset::dimension`

Definition at line 575 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.43 flint::fmpz Class Reference

Public Member Functions

- void print (const string &text)

Data Fields

- fmpz_t d

3.43.1 Detailed Description

Definition at line 54 of file flintinterface.h.

3.43.2 Constructor & Destructor Documentation

3.43.2.1 fmpz()

`flint::fmpz::fmpz ( ) [inline]`

Definition at line 57 of file flintinterface.h.

3.43.2.2 ∼fmpz()

```
flint::fmpz::∼fmpz ( )  [inline]
```

Definition at line 58 of file flintinterface.h.

3.43.3 Member Function Documentation

3.43.3.1 print()

```
void flint::fmpz::print (
            const string & text )  [inline]
```

Definition at line 59 of file flintinterface.h.

3.43.4 Field Documentation

3.43.4.1 d

```
fmpz_t flint::fmpz::d
```

Definition at line 56 of file flintinterface.h.

The documentation for this class was generated from the following file:

- flintinterface.h

3.44 Fraction Class Reference

Public Member Functions

- Fraction (BigInt n, BigInt d)
- void print (const char *msg)
- void setValue (BigInt n, BigInt d)
- void setValue (Fraction &f)
- void add (BigInt n, BigInt d)
- void add (Fraction f)
- void sub (Fraction f)
- BigInt gcd (BigInt n0, BigInt n1)
- void normal (void)
- Bool isEq (Fraction f)

Data Fields

- BigInt num
- BigInt den
- double ratio

3.44.1 Detailed Description

Definition at line 182 of file grcc.h.

3.44.2 Constructor & Destructor Documentation

3.44.2.1 Fraction() [1/2]

```
Fraction::Fraction ( )  [inline]
```

Definition at line 187 of file grcc.h.

3.44.2.2 Fraction() [2/2]

```
Fraction::Fraction (
             BigInt n,
             BigInt d )
```

Definition at line 9680 of file grcc.cc.

3.44.3 Member Function Documentation

3.44.3.1 print()

```
void Fraction::print (
             const char * msg )
```

Definition at line 9691 of file grcc.cc.

3.44.3.2 setValue() [1/2]

```
void Fraction::setValue (
             BigInt n,
             BigInt d )
```

Definition at line 9703 of file grcc.cc.

3.44.3.3 setValue() [2/2]

```
void Fraction::setValue (
             Fraction & f )
```

Definition at line 9711 of file grcc.cc.

3.44.3.4 add() [1/2]

```
void Fraction::add (
            BigInt n,
            BigInt d )
```

Definition at line 9756 of file grcc.cc.

3.44.3.5 add() [2/2]

```
void Fraction::add (
            Fraction f )
```

Definition at line 9779 of file grcc.cc.

3.44.3.6 sub()

```
void Fraction::sub (
            Fraction f )
```

Definition at line 9788 of file grcc.cc.

3.44.3.7 gcd()

```
BigInt Fraction::gcd (
            BigInt n0,
            BigInt n1 )
```

Definition at line 9719 of file grcc.cc.

3.44.3.8 normal()

```
void Fraction::normal (
            void )
```

Definition at line 9739 of file grcc.cc.

3.44.3.9 isEq()

```
Bool Fraction::isEq (
            Fraction f )
```

Definition at line 9797 of file grcc.cc.

3.44.4 Field Documentation

3.44.4.1 num

```
BigInt Fraction::num
```

Definition at line 184 of file grcc.h.

3.44.4.2 den

```
BigInt Fraction::den
```

Definition at line 184 of file grcc.h.

3.44.4.3 ratio

```
double Fraction::ratio
```

Definition at line 185 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.45 FUN_INFO Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD * location
- int numargs
- int numfunnies
- int numwildcards
- int symmet
- int tensor
- int commute

3.45.1 Detailed Description

The struct FUN_INFO is used for information about functions in the file smart.c which is supposed to intelligently look for patterns in complicated wildcard situations involving symmetric functions.

Definition at line 608 of file structs.h.

3.45.2 Field Documentation

3.45.2.1 location

```
WORD* FUN_INFO::location
```

Definition at line 609 of file structs.h.

3.45.2.2 numargs

```
int FUN_INFO::numargs
```

Definition at line 610 of file structs.h.

3.45.2.3 numfunnies

```
int FUN_INFO::numfunnies
```

Definition at line 611 of file structs.h.

3.45.2.4 numwildcards

```
int FUN_INFO::numwildcards
```

Definition at line 612 of file structs.h.

3.45.2.5 symmet

```
int FUN_INFO::symmet
```

Definition at line 613 of file structs.h.

3.45.2.6 tensor

```
int FUN_INFO::tensor
```

Definition at line 614 of file structs.h.

3.45.2.7 commute

```
int FUN_INFO::commute
```

Definition at line 615 of file structs.h.

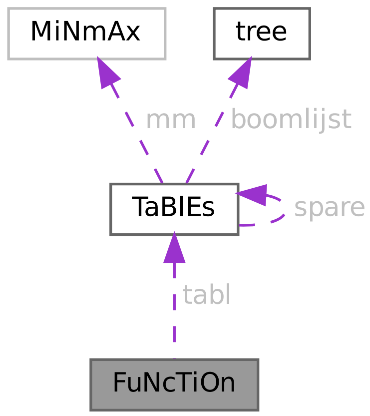
The documentation for this struct was generated from the following file:

- structs.h

3.46 FuNcTiOn Struct Reference

```
#include <structs.h>
```

Collaboration diagram for FuNcTiOn:



Data Fields

- TABLES tabl
- LONG symminfo
- LONG name
- WORD commute
- WORD complex
- WORD number
- WORD flags
- WORD spec
- WORD symmetric
- WORD node
- WORD namesize
- WORD dimension
- WORD maxnumargs
- WORD minnumargs

3.46.1 Detailed Description

Contains all information about a function. Also used for tables. It is used in the LIST elements of AC.

Definition at line 477 of file structs.h.

3.46.2 Field Documentation

3.46.2.1 tabl

`TABLES FuNcTiOn::tabl`

Used if redefined as table. != 0 if function is a table

Definition at line 478 of file structs.h.

3.46.2.2 symminfo

`LONG FuNcTiOn::symminfo`

Info regarding symm properties offset in buffer

Definition at line 479 of file structs.h.

3.46.2.3 name

`LONG FuNcTiOn::name`

Location in namebuffer of NAMETREE

Definition at line 480 of file structs.h.

Referenced by StartVariables().

3.46.2.4 commute

`WORD FuNcTiOn::commute`

Commutation properties

Definition at line 481 of file structs.h.

3.46.2.5 complex

`WORD FuNcTiOn::complex`

Properties under complex conjugation

Definition at line 482 of file structs.h.

3.46.2.6 number

```
WORD FuNcTiOn::number
```

Number when stored in file

Definition at line 483 of file structs.h.

Referenced by ReadSaveIndex().

3.46.2.7 flags

```
WORD FuNcTiOn::flags
```

Used to indicate usage when storing

Definition at line 484 of file structs.h.

3.46.2.8 spec

```
WORD FuNcTiOn::spec
```

Regular, Tensor, etc. See FunSpecs.

Definition at line 485 of file structs.h.

Referenced by ReadSaveVariables().

3.46.2.9 symmetric

```
WORD FuNcTiOn::symmetric
```

0 if symmetric properties

Definition at line 486 of file structs.h.

Referenced by StartVariables().

3.46.2.10 node

```
WORD FuNcTiOn::node
```

Location in namenode of NAMETREE

Definition at line 487 of file structs.h.

3.46.2.11 namesize

```
WORD FuNcTiOn::namesize
```

Length of the name

Definition at line 488 of file structs.h.

3.46.2.12 dimension

```
WORD FuNcTiOn::dimension
```

Definition at line 489 of file structs.h.

3.46.2.13 maxnumargs

```
WORD FuNcTiOn::maxnumargs
```

Definition at line 490 of file structs.h.

3.46.2.14 minnumargs

```
WORD FuNcTiOn::minnumargs
```

Definition at line 491 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.47 gcd_heuristic_failed Class Reference

3.47.1 Detailed Description

Definition at line 33 of file polygcd.h.

The documentation for this class was generated from the following file:

- polygcd.h

3.48 HANDLERS Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD newlogonly
- WORD newhandle
- WORD oldhandle
- WORD oldlogonly
- WORD oldprinttype
- WORD oldsilent

3.48.1 Detailed Description

The struct HANDLERS is used in the communication with external channels.

Definition at line 964 of file structs.h.

3.48.2 Field Documentation

3.48.2.1 newlogonly

`WORD HANDLERS::newlogonly`

Definition at line 965 of file structs.h.

3.48.2.2 newhandle

`WORD HANDLERS::newhandle`

Definition at line 966 of file structs.h.

3.48.2.3 oldhandle

`WORD HANDLERS::oldhandle`

Definition at line 967 of file structs.h.

3.48.2.4 oldlogonly

`WORD HANDLERS::oldlogonly`

Definition at line 968 of file structs.h.

3.48.2.5 oldprinttype

`WORD HANDLERS::oldprinttype`

Definition at line 969 of file structs.h.

3.48.2.6 oldsilent

```
WORD HANDLERS::oldsilent
```

Definition at line 970 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.49 IInput Struct Reference

Data Fields

- const char ∗ name
- int icode
- int nplistn
- const char ∗ plistn [GRCC_MAXLEGS]
- int plistc [GRCC_MAXLEGS]
- int cvallist [GRCC_MAXNCPLG]

3.49.1 Detailed Description

Definition at line 137 of file grccparam.h.

3.49.2 Field Documentation

3.49.2.1 name

```
const char* IInput::name
```

Definition at line 138 of file grccparam.h.

3.49.2.2 icode

```
int IInput::icode
```

Definition at line 139 of file grccparam.h.

3.49.2.3 nplistn

```
int IInput::nplistn
```

Definition at line 140 of file grccparam.h.

3.49.2.4 plistn

```
const char* IInput::plistn[GRCC_MAXLEGS]
```

Definition at line 141 of file grccparam.h.

3.49.2.5 plistc

```
int IInput::plistc[GRCC_MAXLEGS]
```

Definition at line 142 of file grccparam.h.

3.49.2.6 cvallist

```
int IInput::cvallist[GRCC_MAXNCPLG]
```

Definition at line 143 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.50 InDeX Struct Reference

Data Fields

- LONG name
- WORD type
- WORD dimension
- WORD number
- WORD flags
- WORD nmin4
- WORD node
- WORD namesize

3.50.1 Detailed Description

Definition at line 445 of file structs.h.

3.50.2 Field Documentation

3.50.2.1 name

```
LONG InDeX::name
```

Definition at line 446 of file structs.h.

3.50.2.2 type

```
WORD InDeX::type
```

Definition at line 447 of file structs.h.

3.50.2.3 dimension

```
WORD InDeX::dimension
```

Definition at line 448 of file structs.h.

3.50.2.4 number

```
WORD InDeX::number
```

Definition at line 449 of file structs.h.

3.50.2.5 flags

```
WORD InDeX::flags
```

Definition at line 450 of file structs.h.

3.50.2.6 nmin4

```
WORD InDeX::nmin4
```

Definition at line 451 of file structs.h.

3.50.2.7 node

```
WORD InDeX::node
```

Definition at line 452 of file structs.h.

3.50.2.8 namesize

```
WORD InDeX::namesize
```

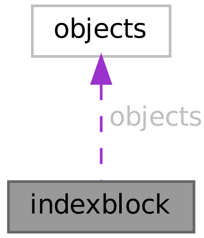
Definition at line 453 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.51 indexblock Struct Reference

Collaboration diagram for indexblock:



Data Fields

- MLONG flags
- MLONG previousblock
- MLONG position
- OBJECTS objects [NUMOBJECTS]

3.51.1 Detailed Description

Definition at line 108 of file minos.h.

3.51.2 Field Documentation

3.51.2.1 flags

```
MLONG indexblock::flags
```

Definition at line 109 of file minos.h.

3.51.2.2 previousblock

```
MLONG indexblock::previousblock
```

Definition at line 110 of file minos.h.

3.51.2.3 position

```
MLONG indexblock::position
```

Definition at line 111 of file minos.h.

3.51.2.4 objects

```
OBJECTS indexblock::objects[NUMOBJECTS]
```

Definition at line 112 of file minos.h.

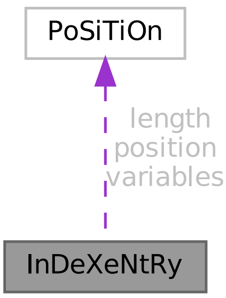
The documentation for this struct was generated from the following file:

- minos.h

3.52 InDeXeNtRy Struct Reference

```
#include <structs.h>
```

Collaboration diagram for InDeXeNtRy:



Public Member Functions

- **PADPOSITION** (0, 1, 0, 5, MAXENAME+1)

Data Fields

- POSITION position
- POSITION length
- POSITION variables
- LONG CompressSize
- WORD nsymbols
- WORD nindices
- WORD nvectors
- WORD nfunctions
- WORD size
- SBYTE name [MAXENAME+1]

3.52.1 Detailed Description

Defines the structure of an entry in a file index (see struct FiLeInDeX).

It represents one expression in the file.

Definition at line 100 of file structs.h.

3.52.2 Field Documentation

3.52.2.1 position

```
POSITION InDeXeNtRy::position
```

Position of the expression itself

Definition at line 101 of file structs.h.

3.52.2.2 length

```
POSITION InDeXeNtRy::length
```

Length of the expression itself

Definition at line 102 of file structs.h.

3.52.2.3 variables

```
POSITION InDeXeNtRy::variables
```

Position of the list with variables

Definition at line 103 of file structs.h.

3.52.2.4 CompressSize

```
LONG InDeXeNtRy::CompressSize
```

Size of buffer before compress

Definition at line 104 of file structs.h.

3.52.2.5 nsymbols

```
WORD InDeXeNtRy::nsymbols
```

Number of symbols in the list

Definition at line 105 of file structs.h.

Referenced by ReadSaveVariables().

3.52.2.6 nindices

`WORD InDeXeNtRy::nindices`

Number of indices in the list

Definition at line 106 of file structs.h.

Referenced by ReadSaveVariables().

3.52.2.7 nvectors

`WORD InDeXeNtRy::nvectors`

Number of vectors in the list

Definition at line 107 of file structs.h.

Referenced by ReadSaveVariables().

3.52.2.8 nfunctions

`WORD InDeXeNtRy::nfunctions`

Number of functions in the list

Definition at line 108 of file structs.h.

Referenced by ReadSaveVariables().

3.52.2.9 size

`WORD InDeXeNtRy::size`

Size of variables field

Definition at line 109 of file structs.h.

3.52.2.10 name

`SBYTE InDeXeNtRy::name[MAXENAME+1]`

Name of expression

Definition at line 110 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.53 iniinfo Struct Reference

Data Fields

- MLONG entriesinindex
- MLONG numberofindexblocks
- MLONG firstindexblock
- MLONG lastindexblock
- MLONG numberoftables
- MLONG numberofnamesblocks
- MLONG firstnameblock
- MLONG lastnameblock

3.53.1 Detailed Description

Definition at line 83 of file minos.h.

3.53.2 Field Documentation

3.53.2.1 entriesinindex

```
MLONG iniinfo::entriesinindex
```

Definition at line 85 of file minos.h.

3.53.2.2 numberofindexblocks

```
MLONG iniinfo::numberofindexblocks
```

Definition at line 86 of file minos.h.

3.53.2.3 firstindexblock

```
MLONG iniinfo::firstindexblock
```

Definition at line 87 of file minos.h.

3.53.2.4 lastindexblock

```
MLONG iniinfo::lastindexblock
```

Definition at line 88 of file minos.h.

3.53.2.5 numberoftables

```
MLONG iniinfo::numberoftables
```

Definition at line 89 of file minos.h.

3.53.2.6 numberofnamesblocks

```
MLONG iniinfo::numberofnamesblocks
```

Definition at line 90 of file minos.h.

3.53.2.7 firstnameblock

```
MLONG iniinfo::firstnameblock
```

Definition at line 91 of file minos.h.

3.53.2.8 lastnameblock

```
MLONG iniinfo::lastnameblock
```

Definition at line 92 of file minos.h.

The documentation for this struct was generated from the following file:

- minos.h

3.54 INSIDEINFO Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ∗ buffer
- int oldcompiletype
- int oldparallelflag
- int oldnumpotmoddollars
- WORD size
- WORD numdollars
- WORD oldcbuf
- WORD oldrbuf
- WORD inscbuf
- WORD oldcnumlhs

3.54.1 Detailed Description

Used for #inside

Definition at line 854 of file structs.h.

3.54.2 Field Documentation

3.54.2.1 buffer

```
WORD* INSIDEINFO::buffer
```

Definition at line 855 of file structs.h.

3.54.2.2 oldcompiletype

```
int INSIDEINFO::oldcompiletype
```

Definition at line 856 of file structs.h.

3.54.2.3 oldparallelflag

```
int INSIDEINFO::oldparallelflag
```

Definition at line 857 of file structs.h.

3.54.2.4 oldnumpotmoddollars

```
int INSIDEINFO::oldnumpotmoddollars
```

Definition at line 858 of file structs.h.

3.54.2.5 size

```
WORD INSIDEINFO::size
```

Definition at line 859 of file structs.h.

3.54.2.6 numdollars

```
WORD INSIDEINFO::numdollars
```

Definition at line 860 of file structs.h.

3.54.2.7 oldcbuf

`WORD INSIDEINFO::oldcbuf`

Definition at line 861 of file structs.h.

3.54.2.8 oldrbuf

`WORD INSIDEINFO::oldrbuf`

Definition at line 862 of file structs.h.

3.54.2.9 inscbuf

`WORD INSIDEINFO::inscbuf`

Definition at line 863 of file structs.h.

3.54.2.10 oldcnumlhs

`WORD INSIDEINFO::oldcnumlhs`

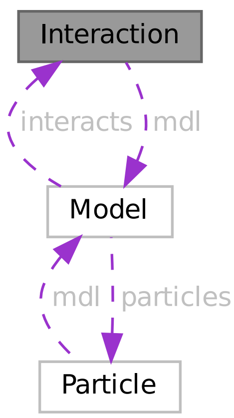
Definition at line 864 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.55 Interaction Class Reference

Collaboration diagram for Interaction:



Public Member Functions

- Interaction (Model ∗modl, int iid, const char ∗nam, int icode, int ∗cpl, int nlgs, int ∗plst, int csm, int lp)
- void prInteraction (void)

Data Fields

- Model ∗ mdl
- char ∗ name
- int ∗ plist
- int ∗ clist
- int ∗ slist
- int id
- int csum
- int nlegs
- int loop
- int icode
- int nplist
- int nclist
- int nslist

3.55.1 Detailed Description

Definition at line 236 of file grcc.h.

3.55.2 Constructor & Destructor Documentation

3.55.2.1 Interaction()

```
Interaction::Interaction (
            Model * modl,
            int iid,
            const char * nam,
            int icode,
            int * cpl,
            int nlgs,
            int * plst,
            int csm,
            int lp )
```

Definition at line 1570 of file grcc.cc.

3.55.2.2 ∼Interaction()

```
Interaction::∼Interaction (
            void )
```

Definition at line 1691 of file grcc.cc.

3.55.3 Member Function Documentation

3.55.3.1 prInteraction()

```
void Interaction::prInteraction (
            void )
```

Definition at line 1706 of file grcc.cc.

3.55.4 Field Documentation

3.55.4.1 mdl

```
Model* Interaction::mdl
```

Definition at line 238 of file grcc.h.

3.55.4.2 name

```
char* Interaction::name
```

Definition at line 239 of file grcc.h.

3.55.4.3 plist

```
int* Interaction::plist
```

Definition at line 240 of file grcc.h.

3.55.4.4 clist

```
int* Interaction::clist
```

Definition at line 241 of file grcc.h.

3.55.4.5 slist

```
int* Interaction::slist
```

Definition at line 242 of file grcc.h.

3.55.4.6 id

```
int Interaction::id
```

Definition at line 244 of file grcc.h.

3.55.4.7 csum

```
int Interaction::csum
```

Definition at line 245 of file grcc.h.

3.55.4.8 nlegs

```
int Interaction::nlegs
```

Definition at line 246 of file grcc.h.

3.55.4.9 loop

```
int Interaction::loop
```

Definition at line 247 of file grcc.h.

3.55.4.10 icode

```
int Interaction::icode
```

Definition at line 248 of file grcc.h.

3.55.4.11 nplist

```
int Interaction::nplist
```

Definition at line 250 of file grcc.h.

3.55.4.12 nclist

```
int Interaction::nclist
```

Definition at line 251 of file grcc.h.

3.55.4.13 nslist

```
int Interaction::nslist
```

Definition at line 252 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.56 KEYWORD Struct Reference

```
#include <structs.h>
```

Data Fields

- char * name
- TFUN func
- int type
- int flags

3.56.1 Detailed Description

The KEYWORD struct defines names of commands/statements and the routine to be called when they are encountered by the compiler or preprocessor.

Definition at line 222 of file structs.h.

3.56.2 Field Documentation

3.56.2.1 name

```
char* KEYWORD::name
```

Definition at line 223 of file structs.h.

3.56.2.2 func

```
TFUN KEYWORD::func
```

Definition at line 224 of file structs.h.

3.56.2.3 type

```
int KEYWORD::type
```

Definition at line 225 of file structs.h.

3.56.2.4 flags

```
int KEYWORD::flags
```

Definition at line 226 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.57 KEYWORDV Struct Reference

`#include <structs.h>`

Data Fields

- char * name
- int * var
- int type
- int flags

3.57.1 Detailed Description

The KEYWORDV struct defines names of commands/statements and the variable to be affected when they are encountered by the compiler or preprocessor.

Definition at line 234 of file structs.h.

3.57.2 Field Documentation

3.57.2.1 name

`char* KEYWORDV::name`

Definition at line 235 of file structs.h.

3.57.2.2 var

`int* KEYWORDV::var`

Definition at line 236 of file structs.h.

3.57.2.3 type

`int KEYWORDV::type`

Definition at line 237 of file structs.h.

3.57.2.4 flags

`int KEYWORDV::flags`

Definition at line 238 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.58 LIST Struct Reference

```
#include <structs.h>
```

Data Fields

- void * lijst
- char * message
- int num
- int maxnum
- int size
- int numglobal
- int numtemp
- int numclear

3.58.1 Detailed Description

Much information is stored in arrays of which we can double the size if the array proves to be too small. Such arrays are controlled by a variable of type LIST. The routines that expand the lists are in the file tools.c

Definition at line 205 of file structs.h.

3.58.2 Field Documentation

3.58.2.1 lijst

```
void* LIST::lijst
```

[D] Holds space for "maxnum" elements of size "size" each

Definition at line 206 of file structs.h.

3.58.2.2 message

```
char* LIST::message
```

Text for Malloc1 when allocating lijst. Set to constant string.

Definition at line 207 of file structs.h.

3.58.2.3 num

```
int LIST::num
```

Number of elements in lijst.

Definition at line 208 of file structs.h.

3.58.2.4 maxnum

```
int LIST::maxnum
```

Maximum number of elements in lijst.

Definition at line 209 of file structs.h.

3.58.2.5 size

```
int LIST::size
```

Size of one element in lijst.

Definition at line 210 of file structs.h.

3.58.2.6 numglobal

```
int LIST::numglobal
```

Marker for position when .global is executed.

Definition at line 211 of file structs.h.

3.58.2.7 numtemp

```
int LIST::numtemp
```

At the moment only needed for sets and setstore.

Definition at line 212 of file structs.h.

3.58.2.8 numclear

```
int LIST::numclear
```

Only for the clear instruction.

Definition at line 213 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.59 longMultiStruct Struct Reference

Collaboration diagram for longMultiStruct:

Data Fields

- UBYTE * buffer
- int bufpos
- int packpos
- int nPacks
- int lastLen
- struct longMultiStruct * next

3.59.1 Detailed Description

Definition at line 1013 of file mpi.c.

3.59.2 Field Documentation

3.59.2.1 buffer

`UBYTE* longMultiStruct::buffer`

Definition at line 1014 of file mpi.c.

3.59.2.2 bufpos

`int longMultiStruct::bufpos`

Definition at line 1015 of file mpi.c.

3.59.2.3 packpos

`int longMultiStruct::packpos`

Definition at line 1016 of file mpi.c.

3.59.2.4 nPacks

`int longMultiStruct::nPacks`

Definition at line 1017 of file mpi.c.

3.59.2.5 lastLen

`int longMultiStruct::lastLen`

Definition at line 1018 of file mpi.c.

3.59.2.6 next

`struct longMultiStruct* longMultiStruct::next`

Definition at line 1021 of file mpi.c.

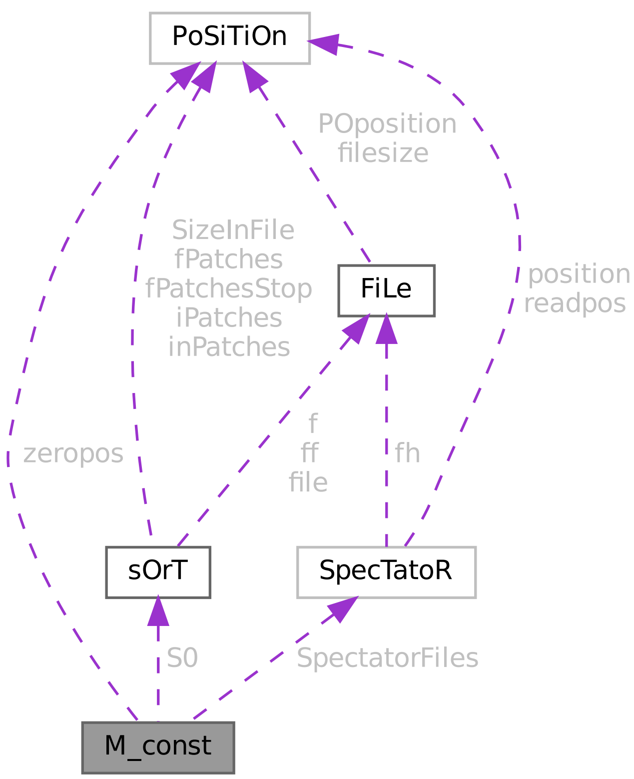
The documentation for this struct was generated from the following file:

- mpi.c

3.60 M_const Struct Reference

`#include <structs.h>`

Collaboration diagram for M_const:



Public Member Functions

- **PADPOSITION** (17, 28, 62, 84, 2)

Data Fields

- POSITION zeropos
- SORTING ∗ S0
- UWORD ∗ gcmod
- UWORD ∗ gpowmod
- UBYTE ∗ TempDir
- UBYTE ∗ TempSortDir
- UBYTE ∗ IncDir
- UBYTE ∗ InputFileName
- UBYTE ∗ LogFileName
- UBYTE ∗ OutBuffer
- UBYTE ∗ Path
- UBYTE ∗ SetupDir
- UBYTE ∗ SetupFile
- UBYTE ∗ gFortran90Kind
- UBYTE ∗ gextrasym
- UBYTE ∗ ggextrasym
- UBYTE ∗ oldnumextrasymbols
- SPECTATOR ∗ SpectatorFiles
- LONG MaxTer
- LONG CompressSize
- LONG ScratSize
- LONG HideSize
- LONG SizeStoreCache
- LONG MaxStreamSize
- LONG SIOsize
- LONG SLargeSize
- LONG SSmallEsize
- LONG SSmallSize
- LONG STermsInSmall
- LONG MaxBracketBufferSize
- LONG hProcessBucketSize
- LONG gProcessBucketSize
- LONG shmWinSize
- LONG OldChildTime
- LONG OldSecTime
- LONG OldMilliTime
- LONG WorkSize
- LONG gThreadBucketSize
- LONG ggThreadBucketSize
- LONG SumTime
- LONG SpectatorSize
- LONG TimeLimit
- int FileOnlyFlag
- int Interact
- int MaxParLevel
- int OutBufSize
- int SMaxFpatches
- int SMaxPatches

- int StdOut
- int ginsidefirst
- int gDefDim
- int gDefDim4
- int NumFixedSets
- int NumFixedFunctions
- int rbufnum
- int dbufnum
- int sbufnum
- int zbufnum
- int SkipClears
- int gTokensWriteFlag
- int gfunpowers
- int gStatsFlag
- int gNamesFlag
- int gCodesFlag
- int gSortType
- int gproperorderflag
- int hparallelflag
- int gparallelflag
- int totalnumberofthreads
- int gSizeCommuteInSet
- int gThreadStats
- int ggThreadStats
- int gFinalStats
- int ggFinalStats
- int gThreadsFlag
- int ggThreadsFlag
- int gThreadBalancing
- int ggThreadBalancing
- int gThreadSortFileSynch
- int ggThreadSortFileSynch
- int gProcessStats
- int ggProcessStats
- int gOldParallelStats
- int ggOldParallelStats
- int maxFlevels
- int resetTimeOnClear
- int gcNumDollars
- int MultiRun
- int gNoSpacesInNumbers
- int ggNoSpacesInNumbers
- int gIsFortran90
- int PrintTotalSize
- int fbuffersize
- int gOldFactArgFlag
- int ggOldFactArgFlag
- int gnumextrasym
- int ggnumextrasym
- int NumSpectatorFiles
- int SizeForSpectatorFiles
- int gOldGCDflag
- int ggOldGCDflag
- int gWTimeStatsFlag
- int ggWTimeStatsFlag

- int jumpratio
- WORD MaxTal
- WORD IndDum
- WORD DumInd
- WORD WilInd
- WORD gncmod
- WORD gnpowmod
- WORD gmodmode
- WORD gUnitTrace
- WORD gOutputMode
- WORD gOutputSpaces
- WORD gOutNumberType
- WORD gCnumpows
- WORD gUniTrace [4]
- WORD MaxWildcards
- WORD mTraceDum
- WORD OffsetIndex
- WORD OffsetVector
- WORD RepMax
- WORD LogType
- WORD ggStatsFlag
- WORD gLineLength
- WORD qError
- WORD FortranCont
- WORD HoldFlag
- WORD Ordering [15]
- WORD silent
- WORD tracebackflag
- WORD expnum
- WORD denomnum
- WORD facnum
- WORD invfacnum
- WORD sumnum
- WORD sumpnum
- WORD OldOrderFlag
- WORD termfunnum
- WORD matchfunnum
- WORD countfunnum
- WORD gPolyFun
- WORD gPolyFunInv
- WORD gPolyFunType
- WORD gPolyFunExp
- WORD gPolyFunVar
- WORD gPolyFunPow
- WORD dollarzero
- WORD atstartup
- WORD exitflag
- WORD NumStoreCaches
- WORD gIndentSpace
- WORD ggIndentSpace
- WORD gShortStatsMax
- WORD ggShortStatsMax
- WORD gextrasymbols
- WORD ggextrasymbols
- WORD zerorhs

- WORD onerhs
- WORD havesortdir
- WORD vectorzero
- WORD ClearStore
- WORD BracketFactors [8]
- BOOL FromStdin
- BOOL IgnoreDeprecation

3.60.1 Detailed Description

The M_const struct is part of the global data and resides in the ALLGLOBALS struct A under the name #M. We see it used with the macro AM as in AM.S0. It contains global settings at startup or .clear.

Definition at line 1424 of file structs.h.

3.60.2 Field Documentation

3.60.2.1 zeropos

```
POSITION M_const::zeropos
```

Definition at line 1425 of file structs.h.

3.60.2.2 S0

```
SORTING* M_const::S0
```

[D] The main sort buffer

Definition at line 1426 of file structs.h.

3.60.2.3 gcmod

```
UWORD* M_const::gcmod
```

Global setting of modulus. Uses AC.cmod's memory

Definition at line 1427 of file structs.h.

3.60.2.4 gpowmod

```
UWORD* M_const::gpowmod
```

Global setting printing as powers. Uses AC.cmod's memory

Definition at line 1428 of file structs.h.

3.60.2.5 TempDir

```
UBYTE* M_const::TempDir
```

Definition at line 1429 of file structs.h.

3.60.2.6 TempSortDir

```
UBYTE* M_const::TempSortDir
```

Definition at line 1430 of file structs.h.

3.60.2.7 IncDir

```
UBYTE* M_const::IncDir
```

Definition at line 1431 of file structs.h.

3.60.2.8 InputFileName

```
UBYTE* M_const::InputFileName
```

Definition at line 1432 of file structs.h.

3.60.2.9 LogFileName

```
UBYTE* M_const::LogFileName
```

Definition at line 1433 of file structs.h.

3.60.2.10 OutBuffer

```
UBYTE* M_const::OutBuffer
```

Definition at line 1434 of file structs.h.

3.60.2.11 Path

```
UBYTE* M_const::Path
```

Definition at line 1435 of file structs.h.

3.60.2.12 SetupDir

```
UBYTE* M_const::SetupDir
```

Definition at line 1436 of file structs.h.

3.60.2.13 SetupFile

`UBYTE* M_const::SetupFile`

Definition at line 1437 of file structs.h.

3.60.2.14 gFortran90Kind

`UBYTE* M_const::gFortran90Kind`

Definition at line 1438 of file structs.h.

3.60.2.15 gextrasym

`UBYTE* M_const::gextrasym`

Definition at line 1439 of file structs.h.

3.60.2.16 ggextrasym

`UBYTE* M_const::ggextrasym`

Definition at line 1440 of file structs.h.

3.60.2.17 oldnumextrasymbols

`UBYTE* M_const::oldnumextrasymbols`

Definition at line 1441 of file structs.h.

3.60.2.18 SpectatorFiles

`SPECTATOR* M_const::SpectatorFiles`

Definition at line 1442 of file structs.h.

3.60.2.19 MaxTer

`LONG M_const::MaxTer`

Definition at line 1450 of file structs.h.

3.60.2.20 CompressSize

`LONG M_const::CompressSize`

Definition at line 1451 of file structs.h.

3.60.2.21 ScratSize

```
LONG M_const::ScratSize
```

Definition at line 1452 of file structs.h.

3.60.2.22 HideSize

```
LONG M_const::HideSize
```

Definition at line 1453 of file structs.h.

3.60.2.23 SizeStoreCache

```
LONG M_const::SizeStoreCache
```

Definition at line 1454 of file structs.h.

3.60.2.24 MaxStreamSize

```
LONG M_const::MaxStreamSize
```

Definition at line 1455 of file structs.h.

3.60.2.25 SIOsize

```
LONG M_const::SIOsize
```

Definition at line 1456 of file structs.h.

3.60.2.26 SLargeSize

```
LONG M_const::SLargeSize
```

Definition at line 1457 of file structs.h.

3.60.2.27 SSmallEsize

```
LONG M_const::SSmallEsize
```

Definition at line 1458 of file structs.h.

3.60.2.28 SSmallSize

```
LONG M_const::SSmallSize
```

Definition at line 1459 of file structs.h.

3.60.2.29 STermsInSmall

`LONG M_const::STermsInSmall`

Definition at line 1460 of file structs.h.

3.60.2.30 MaxBracketBufferSize

`LONG M_const::MaxBracketBufferSize`

Definition at line 1461 of file structs.h.

3.60.2.31 hProcessBucketSize

`LONG M_const::hProcessBucketSize`

Definition at line 1462 of file structs.h.

3.60.2.32 gProcessBucketSize

`LONG M_const::gProcessBucketSize`

Definition at line 1463 of file structs.h.

3.60.2.33 shmWinSize

`LONG M_const::shmWinSize`

Definition at line 1464 of file structs.h.

3.60.2.34 OldChildTime

`LONG M_const::OldChildTime`

Definition at line 1465 of file structs.h.

3.60.2.35 OldSecTime

`LONG M_const::OldSecTime`

Definition at line 1466 of file structs.h.

3.60.2.36 OldMilliTime

`LONG M_const::OldMilliTime`

Definition at line 1467 of file structs.h.

3.60.2.37 WorkSize

```
LONG M_const::WorkSize
```

Definition at line 1468 of file structs.h.

3.60.2.38 gThreadBucketSize

```
LONG M_const::gThreadBucketSize
```

Definition at line 1469 of file structs.h.

3.60.2.39 ggThreadBucketSize

```
LONG M_const::ggThreadBucketSize
```

Definition at line 1470 of file structs.h.

3.60.2.40 SumTime

```
LONG M_const::SumTime
```

Definition at line 1471 of file structs.h.

3.60.2.41 SpectatorSize

```
LONG M_const::SpectatorSize
```

Definition at line 1472 of file structs.h.

3.60.2.42 TimeLimit

```
LONG M_const::TimeLimit
```

Definition at line 1473 of file structs.h.

3.60.2.43 FileOnlyFlag

```
int M_const::FileOnlyFlag
```

Definition at line 1480 of file structs.h.

3.60.2.44 Interact

```
int M_const::Interact
```

Definition at line 1481 of file structs.h.

3.60.2.45 MaxParLevel

`int M_const::MaxParLevel`

Definition at line 1482 of file structs.h.

3.60.2.46 OutBufSize

`int M_const::OutBufSize`

Definition at line 1483 of file structs.h.

3.60.2.47 SMaxFpatches

`int M_const::SMaxFpatches`

Definition at line 1484 of file structs.h.

3.60.2.48 SMaxPatches

`int M_const::SMaxPatches`

Definition at line 1485 of file structs.h.

3.60.2.49 StdOut

`int M_const::StdOut`

Definition at line 1486 of file structs.h.

3.60.2.50 ginsidefirst

`int M_const::ginsidefirst`

Definition at line 1487 of file structs.h.

3.60.2.51 gDefDim

`int M_const::gDefDim`

Definition at line 1488 of file structs.h.

3.60.2.52 gDefDim4

`int M_const::gDefDim4`

Definition at line 1489 of file structs.h.

3.60.2.53 NumFixedSets

```
int M_const::NumFixedSets
```

Definition at line 1490 of file structs.h.

3.60.2.54 NumFixedFunctions

```
int M_const::NumFixedFunctions
```

Definition at line 1491 of file structs.h.

3.60.2.55 rbufnum

```
int M_const::rbufnum
```

Definition at line 1492 of file structs.h.

3.60.2.56 dbufnum

```
int M_const::dbufnum
```

Definition at line 1493 of file structs.h.

3.60.2.57 sbufnum

```
int M_const::sbufnum
```

Definition at line 1494 of file structs.h.

3.60.2.58 zbufnum

```
int M_const::zbufnum
```

Definition at line 1495 of file structs.h.

3.60.2.59 SkipClears

```
int M_const::SkipClears
```

Definition at line 1496 of file structs.h.

3.60.2.60 gTokensWriteFlag

```
int M_const::gTokensWriteFlag
```

Definition at line 1497 of file structs.h.

3.60.2.61 gfunpowers

`int M_const::gfunpowers`

Definition at line 1498 of file structs.h.

3.60.2.62 gStatsFlag

`int M_const::gStatsFlag`

Definition at line 1499 of file structs.h.

3.60.2.63 gNamesFlag

`int M_const::gNamesFlag`

Definition at line 1500 of file structs.h.

3.60.2.64 gCodesFlag

`int M_const::gCodesFlag`

Definition at line 1501 of file structs.h.

3.60.2.65 gSortType

`int M_const::gSortType`

Definition at line 1502 of file structs.h.

3.60.2.66 gproperorderflag

`int M_const::gproperorderflag`

Definition at line 1503 of file structs.h.

3.60.2.67 hparallelflag

`int M_const::hparallelflag`

Definition at line 1504 of file structs.h.

3.60.2.68 gparallelflag

`int M_const::gparallelflag`

Definition at line 1505 of file structs.h.

3.60.2.69 totalnumberofthreads

`int M_const::totalnumberofthreads`

Definition at line 1506 of file structs.h.

3.60.2.70 gSizeCommuteInSet

`int M_const::gSizeCommuteInSet`

Definition at line 1507 of file structs.h.

3.60.2.71 gThreadStats

`int M_const::gThreadStats`

Definition at line 1508 of file structs.h.

3.60.2.72 ggThreadStats

`int M_const::ggThreadStats`

Definition at line 1509 of file structs.h.

3.60.2.73 gFinalStats

`int M_const::gFinalStats`

Definition at line 1510 of file structs.h.

3.60.2.74 ggFinalStats

`int M_const::ggFinalStats`

Definition at line 1511 of file structs.h.

3.60.2.75 gThreadsFlag

`int M_const::gThreadsFlag`

Definition at line 1512 of file structs.h.

3.60.2.76 ggThreadsFlag

`int M_const::ggThreadsFlag`

Definition at line 1513 of file structs.h.

3.60.2.77 gThreadBalancing

```
int M_const::gThreadBalancing
```

Definition at line 1514 of file structs.h.

3.60.2.78 ggThreadBalancing

```
int M_const::ggThreadBalancing
```

Definition at line 1515 of file structs.h.

3.60.2.79 gThreadSortFileSynch

```
int M_const::gThreadSortFileSynch
```

Definition at line 1516 of file structs.h.

3.60.2.80 ggThreadSortFileSynch

```
int M_const::ggThreadSortFileSynch
```

Definition at line 1517 of file structs.h.

3.60.2.81 gProcessStats

```
int M_const::gProcessStats
```

Definition at line 1518 of file structs.h.

3.60.2.82 ggProcessStats

```
int M_const::ggProcessStats
```

Definition at line 1519 of file structs.h.

3.60.2.83 gOldParallelStats

```
int M_const::gOldParallelStats
```

Definition at line 1520 of file structs.h.

3.60.2.84 ggOldParallelStats

```
int M_const::ggOldParallelStats
```

Definition at line 1521 of file structs.h.

3.60.2.85 maxFlevels

```
int M_const::maxFlevels
```

Definition at line 1522 of file structs.h.

3.60.2.86 resetTimeOnClear

```
int M_const::resetTimeOnClear
```

Definition at line 1523 of file structs.h.

3.60.2.87 gcNumDollars

```
int M_const::gcNumDollars
```

Definition at line 1524 of file structs.h.

3.60.2.88 MultiRun

```
int M_const::MultiRun
```

Definition at line 1525 of file structs.h.

3.60.2.89 gNoSpacesInNumbers

```
int M_const::gNoSpacesInNumbers
```

Definition at line 1526 of file structs.h.

3.60.2.90 ggNoSpacesInNumbers

```
int M_const::ggNoSpacesInNumbers
```

Definition at line 1527 of file structs.h.

3.60.2.91 gIsFortran90

```
int M_const::gIsFortran90
```

Definition at line 1528 of file structs.h.

3.60.2.92 PrintTotalSize

```
int M_const::PrintTotalSize
```

Definition at line 1529 of file structs.h.

3.60.2.93 fbuffersize

`int M_const::fbuffersize`

Definition at line 1530 of file structs.h.

3.60.2.94 gOldFactArgFlag

`int M_const::gOldFactArgFlag`

Definition at line 1531 of file structs.h.

3.60.2.95 ggOldFactArgFlag

`int M_const::ggOldFactArgFlag`

Definition at line 1532 of file structs.h.

3.60.2.96 gnumextrasym

`int M_const::gnumextrasym`

Definition at line 1533 of file structs.h.

3.60.2.97 ggnumextrasym

`int M_const::ggnumextrasym`

Definition at line 1534 of file structs.h.

3.60.2.98 NumSpectatorFiles

`int M_const::NumSpectatorFiles`

Definition at line 1535 of file structs.h.

3.60.2.99 SizeForSpectatorFiles

`int M_const::SizeForSpectatorFiles`

Definition at line 1536 of file structs.h.

3.60.2.100 gOldGCDflag

`int M_const::gOldGCDflag`

Definition at line 1537 of file structs.h.

3.60.2.101 ggOldGCDflag

`int M_const::ggOldGCDflag`

Definition at line 1538 of file structs.h.

3.60.2.102 gWTimeStatsFlag

`int M_const::gWTimeStatsFlag`

Definition at line 1539 of file structs.h.

3.60.2.103 ggWTimeStatsFlag

`int M_const::ggWTimeStatsFlag`

Definition at line 1540 of file structs.h.

3.60.2.104 jumpratio

`int M_const::jumpratio`

Definition at line 1541 of file structs.h.

3.60.2.105 MaxTal

`WORD M_const::MaxTal`

Definition at line 1542 of file structs.h.

3.60.2.106 IndDum

`WORD M_const::IndDum`

Definition at line 1543 of file structs.h.

3.60.2.107 DumInd

`WORD M_const::DumInd`

Definition at line 1544 of file structs.h.

3.60.2.108 WilInd

`WORD M_const::WilInd`

Definition at line 1545 of file structs.h.

3.60.2.109 gncmod

```
WORD M_const::gncmod
```

Definition at line 1546 of file structs.h.

3.60.2.110 gnpowmod

```
WORD M_const::gnpowmod
```

Definition at line 1547 of file structs.h.

3.60.2.111 gmodmode

```
WORD M_const::gmodmode
```

Definition at line 1548 of file structs.h.

3.60.2.112 gUnitTrace

```
WORD M_const::gUnitTrace
```

Definition at line 1549 of file structs.h.

3.60.2.113 gOutputMode

```
WORD M_const::gOutputMode
```

Definition at line 1550 of file structs.h.

3.60.2.114 gOutputSpaces

```
WORD M_const::gOutputSpaces
```

Definition at line 1551 of file structs.h.

3.60.2.115 gOutNumberType

```
WORD M_const::gOutNumberType
```

Definition at line 1552 of file structs.h.

3.60.2.116 gCnumpows

```
WORD M_const::gCnumpows
```

Definition at line 1553 of file structs.h.

3.60.2.117 gUniTrace

```
WORD M_const::gUniTrace[4]
```

Definition at line 1554 of file structs.h.

3.60.2.118 MaxWildcards

```
WORD M_const::MaxWildcards
```

Definition at line 1555 of file structs.h.

3.60.2.119 mTraceDum

```
WORD M_const::mTraceDum
```

Definition at line 1556 of file structs.h.

3.60.2.120 OffsetIndex

```
WORD M_const::OffsetIndex
```

Definition at line 1557 of file structs.h.

3.60.2.121 OffsetVector

```
WORD M_const::OffsetVector
```

Definition at line 1558 of file structs.h.

3.60.2.122 RepMax

```
WORD M_const::RepMax
```

Definition at line 1559 of file structs.h.

3.60.2.123 LogType

```
WORD M_const::LogType
```

Definition at line 1560 of file structs.h.

3.60.2.124 ggStatsFlag

```
WORD M_const::ggStatsFlag
```

Definition at line 1561 of file structs.h.

3.60.2.125 gLineLength

```
WORD M_const::gLineLength
```

Definition at line 1562 of file structs.h.

3.60.2.126 qError

```
WORD M_const::qError
```

Definition at line 1563 of file structs.h.

3.60.2.127 FortranCont

```
WORD M_const::FortranCont
```

Definition at line 1564 of file structs.h.

3.60.2.128 HoldFlag

```
WORD M_const::HoldFlag
```

Definition at line 1565 of file structs.h.

3.60.2.129 Ordering

```
WORD M_const::Ordering[15]
```

Definition at line 1566 of file structs.h.

3.60.2.130 silent

```
WORD M_const::silent
```

Definition at line 1567 of file structs.h.

3.60.2.131 tracebackflag

```
WORD M_const::tracebackflag
```

Definition at line 1568 of file structs.h.

3.60.2.132 expnum

```
WORD M_const::expnum
```

Definition at line 1569 of file structs.h.

3.60.2.133 denomnum

```
WORD M_const::denomnum
```

Definition at line 1570 of file structs.h.

3.60.2.134 facnum

```
WORD M_const::facnum
```

Definition at line 1571 of file structs.h.

3.60.2.135 invfacnum

```
WORD M_const::invfacnum
```

Definition at line 1572 of file structs.h.

3.60.2.136 sumnum

```
WORD M_const::sumnum
```

Definition at line 1573 of file structs.h.

3.60.2.137 sumpnum

```
WORD M_const::sumpnum
```

Definition at line 1574 of file structs.h.

3.60.2.138 OldOrderFlag

```
WORD M_const::OldOrderFlag
```

Definition at line 1575 of file structs.h.

3.60.2.139 termfunnum

```
WORD M_const::termfunnum
```

Definition at line 1576 of file structs.h.

3.60.2.140 matchfunnum

```
WORD M_const::matchfunnum
```

Definition at line 1577 of file structs.h.

3.60.2.141 countfunnum

```
WORD M_const::countfunnum
```

Definition at line 1578 of file structs.h.

3.60.2.142 gPolyFun

```
WORD M_const::gPolyFun
```

Definition at line 1579 of file structs.h.

3.60.2.143 gPolyFunInv

```
WORD M_const::gPolyFunInv
```

Definition at line 1580 of file structs.h.

3.60.2.144 gPolyFunType

```
WORD M_const::gPolyFunType
```

Definition at line 1581 of file structs.h.

3.60.2.145 gPolyFunExp

```
WORD M_const::gPolyFunExp
```

Definition at line 1582 of file structs.h.

3.60.2.146 gPolyFunVar

```
WORD M_const::gPolyFunVar
```

Definition at line 1583 of file structs.h.

3.60.2.147 gPolyFunPow

```
WORD M_const::gPolyFunPow
```

Definition at line 1584 of file structs.h.

3.60.2.148 dollarzero

```
WORD M_const::dollarzero
```

Definition at line 1585 of file structs.h.

3.60.2.149 atstartup

```
WORD M_const::atstartup
```

Definition at line 1586 of file structs.h.

3.60.2.150 exitflag

```
WORD M_const::exitflag
```

Definition at line 1587 of file structs.h.

3.60.2.151 NumStoreCaches

```
WORD M_const::NumStoreCaches
```

Definition at line 1588 of file structs.h.

3.60.2.152 gIndentSpace

```
WORD M_const::gIndentSpace
```

Definition at line 1589 of file structs.h.

3.60.2.153 ggIndentSpace

```
WORD M_const::ggIndentSpace
```

Definition at line 1590 of file structs.h.

3.60.2.154 gShortStatsMax

```
WORD M_const::gShortStatsMax
```

For On FewerStatistics 10;

Definition at line 1591 of file structs.h.

3.60.2.155 ggShortStatsMax

```
WORD M_const::ggShortStatsMax
```

For On FewerStatistics 10;

Definition at line 1592 of file structs.h.

3.60.2.156 gextrasymbols

```
WORD M_const::gextrasymbols
```

Definition at line 1593 of file structs.h.

3.60.2.157 ggextrasymbols

```
WORD M_const::ggextrasymbols
```

Definition at line 1594 of file structs.h.

3.60.2.158 zerorhs

```
WORD M_const::zerorhs
```

Definition at line 1595 of file structs.h.

3.60.2.159 onerhs

```
WORD M_const::onerhs
```

Definition at line 1596 of file structs.h.

3.60.2.160 havesortdir

```
WORD M_const::havesortdir
```

Definition at line 1597 of file structs.h.

3.60.2.161 vectorzero

```
WORD M_const::vectorzero
```

Definition at line 1598 of file structs.h.

3.60.2.162 ClearStore

```
WORD M_const::ClearStore
```

Definition at line 1599 of file structs.h.

3.60.2.163 BracketFactors

```
WORD M_const::BracketFactors[8]
```

Definition at line 1600 of file structs.h.

3.60.2.164 FromStdin

```
BOOL M_const::FromStdin
```

Definition at line 1601 of file structs.h.

3.60.2.165 IgnoreDeprecation

```
BOOL M_const::IgnoreDeprecation
```

Definition at line 1602 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.61 MCBlock Class Reference

Public Member Functions

- void init (void)

Data Fields

- Edge2n ∗ edges
- int nmedges
- int nartps
- int loop
- int DUMMYPADDING

3.61.1 Detailed Description

Definition at line 933 of file grcc.h.

3.61.2 Constructor & Destructor Documentation

3.61.2.1 MCBlock()

```
MCBlock::MCBlock (
             void  )
```

Definition at line 5106 of file grcc.cc.

3.61.2.2 ∼MCBlock()

```
MCBlock::∼MCBlock (
              void )
```

Definition at line 5112 of file grcc.cc.

3.61.3 Member Function Documentation

3.61.3.1 init()

```
void MCBlock::init (
              void )
```

Definition at line 5117 of file grcc.cc.

3.61.4 Field Documentation

3.61.4.1 edges

```
Edge2n* MCBlock::edges
```

Definition at line 935 of file grcc.h.

3.61.4.2 nmedges

```
int MCBlock::nmedges
```

Definition at line 936 of file grcc.h.

3.61.4.3 nartps

```
int MCBlock::nartps
```

Definition at line 937 of file grcc.h.

3.61.4.4 loop

```
int MCBlock::loop
```

Definition at line 938 of file grcc.h.

3.61.4.5 DUMMYPADDING

```
int MCBlock::DUMMYPADDING
```

Definition at line 940 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.62 MCBridge Class Reference

Data Fields

- Edge2n nodes
- int next

3.62.1 Detailed Description

Definition at line 921 of file grcc.h.

3.62.2 Constructor & Destructor Documentation

3.62.2.1 MCBridge()

```
MCBridge::MCBridge (
             void )
```

Definition at line 5094 of file grcc.cc.

3.62.2.2 ∼MCBridge()

```
MCBridge::∼MCBridge (
             void )
```

Definition at line 5099 of file grcc.cc.

3.62.3 Field Documentation

3.62.3.1 nodes

```
Edge2n MCBridge::nodes
```

Definition at line 923 of file grcc.h.

3.62.3.2 next

`int MCBridge::next`

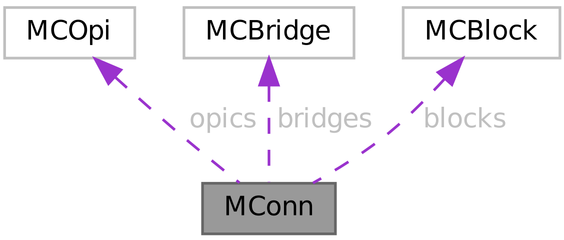
Definition at line 924 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.63 MConn Class Reference

Collaboration diagram for MConn:



Public Member Functions

- MConn (int nnod, int nedg)
- void init (void)
- void pushNode (int nd)
- void pushEdge (int n0, int n1)
- void addOPIc (MCOpi *mopi, int stp)
- void addBridge (int n0, int n1, int nex, int nextot)
- void addArtic (int nd, int mul)
- void addBlock (MCBlock *eblk, int stp)
- void addBlockSelf (int nd, int mul)
- void print (void)

Data Fields

- MCOpi ∗ opics
- MCBridge ∗ bridges
- MCBlock ∗ blocks
- int ∗ articuls
- int ∗ opisp
- int ∗ opistk
- Edge2n ∗ blksp
- Edge2n ∗ blkstk
- int snodes
- int sedges
- int nopic
- int nlpopic
- int nctopic
- int nbridges
- int ne0bridges
- int ne1bridges
- int nselfloops
- int nblocks
- int na1blocks
- int narticuls
- int neblocks
- int nopisp
- int opistkptr
- int nblksp
- int blkstkptr
- int DUMMYPADDING

3.63.1 Detailed Description

Definition at line 950 of file grcc.h.

3.63.2 Constructor & Destructor Documentation

3.63.2.1 MConn()

```
MConn::MConn (
            int nnod,
            int nedg )
```

Definition at line 5128 of file grcc.cc.

3.63.2.2 ∼MConn()

```
MConn::∼MConn (
            void )
```

Definition at line 5147 of file grcc.cc.

3.63.3 Member Function Documentation

3.63.3.1 init()

```
void MConn::init (
             void )
```

Definition at line 5161 of file grcc.cc.

3.63.3.2 pushNode()

```
void MConn::pushNode (
             int nd )
```

Definition at line 5189 of file grcc.cc.

3.63.3.3 pushEdge()

```
void MConn::pushEdge (
             int n0,
             int n1 )
```

Definition at line 5198 of file grcc.cc.

3.63.3.4 addOPIc()

```
void MConn::addOPIc (
             MCOpi * mopi,
             int stp )
```

Definition at line 5214 of file grcc.cc.

3.63.3.5 addBridge()

```
void MConn::addBridge (
             int n0,
             int n1,
             int nex,
             int nextot )
```

Definition at line 5240 of file grcc.cc.

3.63.3.6 addArtic()

```
void MConn::addArtic (
             int nd,
             int mul )
```

Definition at line 5257 of file grcc.cc.

3.63.3.7 addBlock()

```
void MConn::addBlock (
            MCBlock * eblk,
            int stp )
```

Definition at line 5268 of file grcc.cc.

3.63.3.8 addBlockSelf()

```
void MConn::addBlockSelf (
            int nd,
            int mul )
```

Definition at line 5292 of file grcc.cc.

3.63.3.9 print()

```
void MConn::print (
            void )
```

Definition at line 5310 of file grcc.cc.

3.63.4 Field Documentation

3.63.4.1 opics

```
MCOpi* MConn::opics
```

Definition at line 953 of file grcc.h.

3.63.4.2 bridges

```
MCBridge* MConn::bridges
```

Definition at line 954 of file grcc.h.

3.63.4.3 blocks

```
MCBlock* MConn::blocks
```

Definition at line 955 of file grcc.h.

3.63.4.4 articuls

```
int* MConn::articuls
```

Definition at line 956 of file grcc.h.

3.63.4.5 opisp

`int* MConn::opisp`

Definition at line 958 of file grcc.h.

3.63.4.6 opistk

`int* MConn::opistk`

Definition at line 959 of file grcc.h.

3.63.4.7 blksp

`Edge2n* MConn::blksp`

Definition at line 960 of file grcc.h.

3.63.4.8 blkstk

`Edge2n* MConn::blkstk`

Definition at line 961 of file grcc.h.

3.63.4.9 snodes

`int MConn::snodes`

Definition at line 962 of file grcc.h.

3.63.4.10 sedges

`int MConn::sedges`

Definition at line 963 of file grcc.h.

3.63.4.11 nopic

`int MConn::nopic`

Definition at line 966 of file grcc.h.

3.63.4.12 nlpopic

`int MConn::nlpopic`

Definition at line 967 of file grcc.h.

3.63.4.13 nctopic

```
int MConn::nctopic
```

Definition at line 968 of file grcc.h.

3.63.4.14 nbridges

```
int MConn::nbridges
```

Definition at line 969 of file grcc.h.

3.63.4.15 ne0bridges

```
int MConn::ne0bridges
```

Definition at line 970 of file grcc.h.

3.63.4.16 ne1bridges

```
int MConn::ne1bridges
```

Definition at line 971 of file grcc.h.

3.63.4.17 nselfloops

```
int MConn::nselfloops
```

Definition at line 972 of file grcc.h.

3.63.4.18 nblocks

```
int MConn::nblocks
```

Definition at line 975 of file grcc.h.

3.63.4.19 na1blocks

```
int MConn::na1blocks
```

Definition at line 976 of file grcc.h.

3.63.4.20 narticuls

```
int MConn::narticuls
```

Definition at line 977 of file grcc.h.

3.63.4.21 neblocks

`int MConn::neblocks`

Definition at line 978 of file grcc.h.

3.63.4.22 nopisp

`int MConn::nopisp`

Definition at line 981 of file grcc.h.

3.63.4.23 opistkptr

`int MConn::opistkptr`

Definition at line 982 of file grcc.h.

3.63.4.24 nblksp

`int MConn::nblksp`

Definition at line 983 of file grcc.h.

3.63.4.25 blkstkptr

`int MConn::blkstkptr`

Definition at line 984 of file grcc.h.

3.63.4.26 DUMMYPADDING

`int MConn::DUMMYPADDING`

Definition at line 986 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.64 MCOpi Class Reference

Public Member Functions

- void init (void)

Data Fields

- int * nodes
- int nnodes
- int nlegs
- int next
- int nedges
- int loop
- int ctloop

3.64.1 Detailed Description

Definition at line 902 of file grcc.h.

3.64.2 Constructor & Destructor Documentation

3.64.2.1 MCOpi()

```
MCOpi::MCOpi (
        void )
```

Definition at line 5067 of file grcc.cc.

3.64.2.2 ∼MCOpi()

```
MCOpi::∼MCOpi (
        void )
```

Definition at line 5073 of file grcc.cc.

3.64.3 Member Function Documentation

3.64.3.1 init()

```
void MCOpi::init (
        void )
```

Definition at line 5079 of file grcc.cc.

3.64.4 Field Documentation

3.64.4.1 nodes

```
int* MCOpi::nodes
```

Definition at line 904 of file grcc.h.

3.64.4.2 nnodes

```
int MCOpi::nnodes
```

Definition at line 905 of file grcc.h.

3.64.4.3 nlegs

```
int MCOpi::nlegs
```

Definition at line 906 of file grcc.h.

3.64.4.4 next

```
int MCOpi::next
```

Definition at line 907 of file grcc.h.

3.64.4.5 nedges

```
int MCOpi::nedges
```

Definition at line 908 of file grcc.h.

3.64.4.6 loop

```
int MCOpi::loop
```

Definition at line 909 of file grcc.h.

3.64.4.7 ctloop

```
int MCOpi::ctloop
```

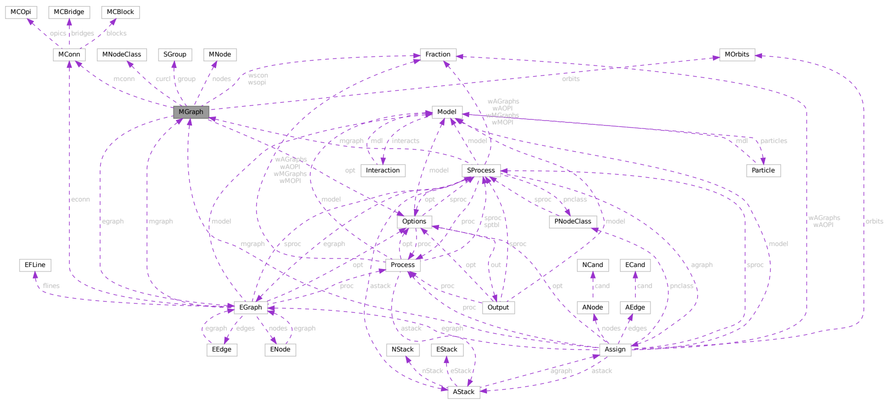
Definition at line 910 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.65 MGraph Class Reference

Collaboration diagram for MGraph:



Public Member Functions

- MGraph (int pid, int ncl, NCInput ∗mgi, Options ∗opt)
- MGraph (int pid, int ncl, int ∗cldeg, int ∗clnum, int ∗clexl, int ∗cmind, int ∗cmaxd, Options ∗opt)
- void init (void)
- BigInt generate (void)
- Bool isExternal (int nd)
- void printAdjMat (MNodeClass ∗cl)
- void print (void)
- Bool isConnected (void)
- Bool visit (int nd)
- Bool isIsomorphic (MNodeClass ∗cl)
- void permMat (int size, int ∗perm, int ∗∗mat0, int ∗∗mat1)
- int compMat (int size, int ∗∗mat0, int ∗∗mat1)
- MNodeClass ∗ refineClass (MNodeClass ∗cl)
- void bisearchME (int nd, int pd, int ned, MCOpi ∗mopi, MCBlock ∗mblk, int ∗next, int ∗nart)
- void biconnME (void)
- Bool isOptM (void)
- void connectClass (MNodeClass ∗cl)
- void connectNode (int sc, int ss, MNodeClass ∗cl)
- void connectLeg (int sc, int sn, int tc, int ts, MNodeClass ∗cl)
- void newGraph (MNodeClass ∗cl)

Data Fields

- Options ∗ opt
- SGroup ∗ group
- MOrbits ∗ orbits
- MNode ∗∗ nodes
- BigInt mId
- int ∗ clist
- int pId
- int nNodes

- int nEdges
- int nLoops
- int nExtern
- int nClasses
- int mindeg
- int maxdeg
- int ** adjMat
- MNodeClass * curcl
- EGraph * egraph
- BigInt nsym
- BigInt esym
- BigInt cDiag
- BigInt c1PI
- BigInt cNoTadpole
- BigInt cNoTadBlock
- BigInt c1PINoTadBlock
- Fraction wscon
- Fraction wsopi
- BigInt ngen
- BigInt ngconn
- BigInt nCallRefine
- BigInt discardRefine
- BigInt discardDisc
- BigInt discardIso
- Bool opi
- Bool opiloop
- Bool extself
- Bool selfloop
- Bool tadpole
- Bool tadblock
- Bool block
- int DUMMYPADDING1
- MConn * mconn
- int ** modmat
- int * bidef
- int * bilow
- int bicount
- int DUMMYPADDING2

3.65.1 Detailed Description

Definition at line 761 of file grcc.h.

3.65.2 Constructor & Destructor Documentation

3.65.2.1 MGraph() [1/2]

```
MGraph::MGraph (
            int pid,
            int ncl,
            NCInput * mgi,
            Options * opt )
```

Definition at line 3732 of file grcc.cc.

3.65.2.2 MGraph() [2/2]

```
MGraph::MGraph (
        int pid,
        int ncl,
        int * cldeg,
        int * clnum,
        int * clexl,
        int * cmind,
        int * cmaxd,
        Options * opt )
```

Definition at line 3649 of file grcc.cc.

3.65.2.3 ∼MGraph()

```
MGraph::∼MGraph (
        void )
```

Definition at line 3860 of file grcc.cc.

3.65.3 Member Function Documentation

3.65.3.1 init()

```
void MGraph::init (
        void )
```

Definition at line 3817 of file grcc.cc.

3.65.3.2 generate()

```
BigInt MGraph::generate (
        void )
```

Definition at line 4479 of file grcc.cc.

3.65.3.3 isExternal()

```
Bool MGraph::isExternal (
        int nd ) [inline]
```

Definition at line 841 of file grcc.h.

3.65.3.4 printAdjMat()

```
void MGraph::printAdjMat (
        MNodeClass * cl )
```

Definition at line 3889 of file grcc.cc.

3.65.3.5 print()

```
void MGraph::print (
            void )
```

Definition at line 3908 of file grcc.cc.

3.65.3.6 isConnected()

```
Bool MGraph::isConnected (
            void )
```

Definition at line 3932 of file grcc.cc.

3.65.3.7 visit()

```
Bool MGraph::visit (
            int nd )
```

Definition at line 3956 of file grcc.cc.

3.65.3.8 isIsomorphic()

```
Bool MGraph::isIsomorphic (
            MNodeClass * cl )
```

Definition at line 3981 of file grcc.cc.

3.65.3.9 permMat()

```
void MGraph::permMat (
            int size,
            int * perm,
            int ** mat0,
            int ** mat1 )
```

Definition at line 4039 of file grcc.cc.

3.65.3.10 compMat()

```
int MGraph::compMat (
            int size,
            int ** mat0,
            int ** mat1 )
```

Definition at line 4053 of file grcc.cc.

3.65.3.11 refineClass()

```
MNodeClass * MGraph::refineClass (
            MNodeClass * cl )
```

Definition at line 4071 of file grcc.cc.

3.65.3.12 bisearchME()

```
void MGraph::bisearchME (
            int nd,
            int pd,
            int ned,
            MCOpi * mopi,
            MCBlock * mblk,
            int * next,
            int * nart )
```

Definition at line 4239 of file grcc.cc.

3.65.3.13 biconnME()

```
void MGraph::biconnME (
            void )
```

Definition at line 4153 of file grcc.cc.

3.65.3.14 isOptM()

```
Bool MGraph::isOptM (
            void )
```

Definition at line 4731 of file grcc.cc.

3.65.3.15 connectClass()

```
void MGraph::connectClass (
            MNodeClass * cl )
```

Definition at line 4505 of file grcc.cc.

3.65.3.16 connectNode()

```
void MGraph::connectNode (
            int sc,
            int ss,
            MNodeClass * cl )
```

Definition at line 4537 of file grcc.cc.

3.65.3.17 connectLeg()

```
void MGraph::connectLeg (
            int sc,
            int sn,
            int tc,
            int ts,
            MNodeClass * cl )
```

Definition at line 4558 of file grcc.cc.

3.65.3.18 newGraph()

```
void MGraph::newGraph (
            MNodeClass * cl )
```

Definition at line 4815 of file grcc.cc.

3.65.4 Field Documentation

3.65.4.1 opt

```
Options* MGraph::opt
```

Definition at line 766 of file grcc.h.

3.65.4.2 group

```
SGroup* MGraph::group
```

Definition at line 769 of file grcc.h.

3.65.4.3 orbits

```
MOrbits* MGraph::orbits
```

Definition at line 770 of file grcc.h.

3.65.4.4 nodes

```
MNode** MGraph::nodes
```

Definition at line 772 of file grcc.h.

3.65.4.5 mId

```
BigInt MGraph::mId
```

Definition at line 773 of file grcc.h.

3.65.4.6 clist

```
int* MGraph::clist
```

Definition at line 774 of file grcc.h.

3.65.4.7 pId

```
int MGraph::pId
```

Definition at line 775 of file grcc.h.

3.65.4.8 nNodes

```
int MGraph::nNodes
```

Definition at line 776 of file grcc.h.

3.65.4.9 nEdges

```
int MGraph::nEdges
```

Definition at line 777 of file grcc.h.

3.65.4.10 nLoops

```
int MGraph::nLoops
```

Definition at line 778 of file grcc.h.

3.65.4.11 nExtern

```
int MGraph::nExtern
```

Definition at line 779 of file grcc.h.

3.65.4.12 nClasses

```
int MGraph::nClasses
```

Definition at line 780 of file grcc.h.

3.65.4.13 mindeg

```
int MGraph::mindeg
```

Definition at line 781 of file grcc.h.

3.65.4.14 maxdeg

```
int MGraph::maxdeg
```

Definition at line 782 of file grcc.h.

3.65.4.15 adjMat

```
int** MGraph::adjMat
```

Definition at line 785 of file grcc.h.

3.65.4.16 curcl

```
MNodeClass* MGraph::curcl
```

Definition at line 786 of file grcc.h.

3.65.4.17 egraph

```
EGraph* MGraph::egraph
```

Definition at line 787 of file grcc.h.

3.65.4.18 nsym

```
BigInt MGraph::nsym
```

Definition at line 788 of file grcc.h.

3.65.4.19 esym

```
BigInt MGraph::esym
```

Definition at line 789 of file grcc.h.

3.65.4.20 cDiag

```
BigInt MGraph::cDiag
```

Definition at line 792 of file grcc.h.

3.65.4.21 c1PI

```
BigInt MGraph::c1PI
```

Definition at line 793 of file grcc.h.

3.65.4.22 cNoTadpole

```
BigInt MGraph::cNoTadpole
```

Definition at line 794 of file grcc.h.

3.65.4.23 cNoTadBlock

```
BigInt MGraph::cNoTadBlock
```

Definition at line 795 of file grcc.h.

3.65.4.24 c1PINoTadBlock

```
BigInt MGraph::c1PINoTadBlock
```

Definition at line 796 of file grcc.h.

3.65.4.25 wscon

```
Fraction MGraph::wscon
```

Definition at line 797 of file grcc.h.

3.65.4.26 wsopi

```
Fraction MGraph::wsopi
```

Definition at line 798 of file grcc.h.

3.65.4.27 ngen

```
BigInt MGraph::ngen
```

Definition at line 801 of file grcc.h.

3.65.4.28 ngconn

```
BigInt MGraph::ngconn
```

Definition at line 802 of file grcc.h.

3.65.4.29 nCallRefine

`BigInt MGraph::nCallRefine`

Definition at line 804 of file grcc.h.

3.65.4.30 discardRefine

`BigInt MGraph::discardRefine`

Definition at line 805 of file grcc.h.

3.65.4.31 discardDisc

`BigInt MGraph::discardDisc`

Definition at line 806 of file grcc.h.

3.65.4.32 discardIso

`BigInt MGraph::discardIso`

Definition at line 807 of file grcc.h.

3.65.4.33 opi

`Bool MGraph::opi`

Definition at line 810 of file grcc.h.

3.65.4.34 opiloop

`Bool MGraph::opiloop`

Definition at line 811 of file grcc.h.

3.65.4.35 extself

`Bool MGraph::extself`

Definition at line 812 of file grcc.h.

3.65.4.36 selfloop

`Bool MGraph::selfloop`

Definition at line 813 of file grcc.h.

3.65.4.37 tadpole

`Bool MGraph::tadpole`

Definition at line 814 of file grcc.h.

3.65.4.38 tadblock

`Bool MGraph::tadblock`

Definition at line 815 of file grcc.h.

3.65.4.39 block

`Bool MGraph::block`

Definition at line 816 of file grcc.h.

3.65.4.40 DUMMYPADDING1

`int MGraph::DUMMYPADDING1`

Definition at line 818 of file grcc.h.

3.65.4.41 mconn

`MConn* MGraph::mconn`

Definition at line 821 of file grcc.h.

3.65.4.42 modmat

`int** MGraph::modmat`

Definition at line 824 of file grcc.h.

3.65.4.43 bidef

`int* MGraph::bidef`

Definition at line 827 of file grcc.h.

3.65.4.44 bilow

`int* MGraph::bilow`

Definition at line 828 of file grcc.h.

3.65.4.45 bicount

```
int MGraph::bicount
```

Definition at line 829 of file grcc.h.

3.65.4.46 DUMMYPADDING2

```
int MGraph::DUMMYPADDING2
```

Definition at line 831 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.66 MiNmAx Struct Reference

Data Fields

- WORD mini
- WORD maxi
- WORD size

3.66.1 Detailed Description

Definition at line 307 of file structs.h.

3.66.2 Field Documentation

3.66.2.1 mini

```
WORD MiNmAx::mini
```

Minimum value

Definition at line 308 of file structs.h.

3.66.2.2 maxi

```
WORD MiNmAx::maxi
```

Maximum value

Definition at line 309 of file structs.h.

3.66.2.3 size

```
WORD MiNmAx::size
```

Value of one unit in this position.

Definition at line 310 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.67 MInput Struct Reference

Data Fields

- const char * name
- int defpart
- int ncouple
- const char * cnamlist [GRCC_MAXNCPLG]

3.67.1 Detailed Description

Definition at line 119 of file grccparam.h.

3.67.2 Field Documentation

3.67.2.1 name

```
const char* MInput::name
```

Definition at line 120 of file grccparam.h.

3.67.2.2 defpart

```
int MInput::defpart
```

Definition at line 121 of file grccparam.h.

3.67.2.3 ncouple

```
int MInput::ncouple
```

Definition at line 123 of file grccparam.h.

3.67.2.4 cnamlist

```
const char* MInput::cnamlist[GRCC_MAXNCPLG]
```

Definition at line 124 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.68 MNode Class Reference

Public Member Functions

- MNode (int id, int clss, NCInput *mgi)
- MNode (int id, int deg, int extloop, int clss, int cmind, int cmaxd)

Data Fields

- int id
- int deg
- int clss
- int extloop
- int cmindeg
- int cmaxdeg
- int freelg
- int visited

3.68.1 Detailed Description

Definition at line 739 of file grcc.h.

3.68.2 Constructor & Destructor Documentation

3.68.2.1 MNode() [1/2]

```
MNode::MNode (
            int id,
            int clss,
            NCInput * mgi )
```

Definition at line 3605 of file grcc.cc.

3.68.2.2 MNode() [2/2]

```
MNode::MNode (
             int id,
             int deg,
             int extloop,
             int clss,
             int cmind,
             int cmaxd )
```

Definition at line 3626 of file grcc.cc.

3.68.3 Field Documentation

3.68.3.1 id

```
int MNode::id
```

Definition at line 741 of file grcc.h.

3.68.3.2 deg

```
int MNode::deg
```

Definition at line 742 of file grcc.h.

3.68.3.3 clss

```
int MNode::clss
```

Definition at line 743 of file grcc.h.

3.68.3.4 extloop

```
int MNode::extloop
```

Definition at line 744 of file grcc.h.

3.68.3.5 cmindeg

```
int MNode::cmindeg
```

Definition at line 745 of file grcc.h.

3.68.3.6 cmaxdeg

`int MNode::cmaxdeg`

Definition at line 746 of file grcc.h.

3.68.3.7 freelg

`int MNode::freelg`

Definition at line 748 of file grcc.h.

3.68.3.8 visited

`int MNode::visited`

Definition at line 749 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.69 MNodeClass Class Reference

Public Member Functions

- MNodeClass (int nnodes, int nclasses)
- void init (int ∗cl, int mxdeg, int ∗∗adjmat)
- void copy (MNodeClass ∗mnc)
- int clCmp (int nd0, int nd1, int cn)
- void printMat (void)
- void mkFlist (void)
- void mkNdCl (void)
- void mkClMat (int ∗∗adjmat)
- void incMat (int nd, int td, int val)
- int cmpMNCArray (int ∗a0, int ∗a1, int ma)
- void reorder (MGraph ∗mg)

Data Fields

- int clmat [GRCC_MAXNODES][GRCC_MAXNODES]
- int clist [GRCC_MAXNODES]
- int ndcl [GRCC_MAXNODES]
- int flist [GRCC_MAXNODES+1]
- int clord [GRCC_MAXNODES]
- int cmindeg [GRCC_MAXNODES]
- int cmaxdeg [GRCC_MAXNODES]
- int nNodes
- int nClasses
- int maxdeg
- int flg0
- int flg1
- int flg2

3.69.1 Detailed Description

Definition at line 865 of file grcc.h.

3.69.2 Constructor & Destructor Documentation

3.69.2.1 MNodeClass()

```
MNodeClass::MNodeClass (
             int nnodes,
             int nclasses )
```

Definition at line 3358 of file grcc.cc.

3.69.2.2 ∼MNodeClass()

```
MNodeClass::∼MNodeClass (
             void )
```

Definition at line 3387 of file grcc.cc.

3.69.3 Member Function Documentation

3.69.3.1 init()

```
void MNodeClass::init (
             int * cl,
             int mxdeg,
             int ** adjmat )
```

Definition at line 3394 of file grcc.cc.

3.69.3.2 copy()

```
void MNodeClass::copy (
             MNodeClass * mnc )
```

Definition at line 3418 of file grcc.cc.

3.69.3.3 clCmp()

```
int MNodeClass::clCmp (
             int nd0,
             int nd1,
             int cn )
```

Definition at line 3474 of file grcc.cc.

3.69.3.4 printMat()

```
void MNodeClass::printMat (
            void )
```

Definition at line 3525 of file grcc.cc.

3.69.3.5 mkFlist()

```
void MNodeClass::mkFlist (
            void )
```

Definition at line 3442 of file grcc.cc.

3.69.3.6 mkNdCl()

```
void MNodeClass::mkNdCl (
            void )
```

Definition at line 3458 of file grcc.cc.

3.69.3.7 mkClMat()

```
void MNodeClass::mkClMat (
            int ** adjmat )
```

Definition at line 3496 of file grcc.cc.

3.69.3.8 incMat()

```
void MNodeClass::incMat (
            int nd,
            int td,
            int val )
```

Definition at line 3516 of file grcc.cc.

3.69.3.9 cmpMNCArray()

```
int MNodeClass::cmpMNCArray (
            int * a0,
            int * a1,
            int ma )
```

Definition at line 3556 of file grcc.cc.

3.69.3.10 reorder()

```
void MNodeClass::reorder (
            MGraph * mg )
```

Definition at line 3572 of file grcc.cc.

3.69.4 Field Documentation

3.69.4.1 clmat

```
int MNodeClass::clmat[GRCC_MAXNODES][GRCC_MAXNODES]
```

Definition at line 867 of file grcc.h.

3.69.4.2 clist

```
int MNodeClass::clist[GRCC_MAXNODES]
```

Definition at line 868 of file grcc.h.

3.69.4.3 ndcl

```
int MNodeClass::ndcl[GRCC_MAXNODES]
```

Definition at line 869 of file grcc.h.

3.69.4.4 flist

```
int MNodeClass::flist[GRCC_MAXNODES+1]
```

Definition at line 870 of file grcc.h.

3.69.4.5 clord

```
int MNodeClass::clord[GRCC_MAXNODES]
```

Definition at line 871 of file grcc.h.

3.69.4.6 cmindeg

```
int MNodeClass::cmindeg[GRCC_MAXNODES]
```

Definition at line 872 of file grcc.h.

3.69.4.7 cmaxdeg

```
int MNodeClass::cmaxdeg[GRCC_MAXNODES]
```

Definition at line 873 of file grcc.h.

3.69.4.8 nNodes

```
int MNodeClass::nNodes
```

Definition at line 874 of file grcc.h.

3.69.4.9 nClasses

```
int MNodeClass::nClasses
```

Definition at line 875 of file grcc.h.

3.69.4.10 maxdeg

```
int MNodeClass::maxdeg
```

Definition at line 876 of file grcc.h.

3.69.4.11 flg0

```
int MNodeClass::flg0
```

Definition at line 878 of file grcc.h.

3.69.4.12 flg1

```
int MNodeClass::flg1
```

Definition at line 879 of file grcc.h.

3.69.4.13 flg2

```
int MNodeClass::flg2
```

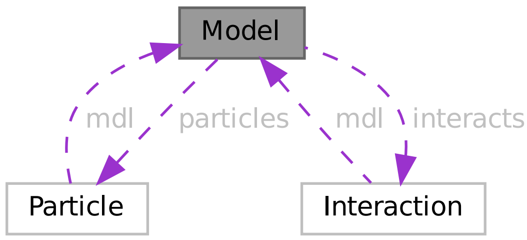
Definition at line 880 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.70 Model Class Reference

Collaboration diagram for Model:



Public Member Functions

- Model (MInput ∗minp)
- void prModel (void)
- void addParticle (PInput ∗pinp)
- void addParticleEnd (void)
- void addInteraction (IInput ∗iinp)
- void addInteractionEnd (void)
- int findParticleName (const char ∗name)
- int findParticleCode (int pcd)
- int findInteractionName (const char ∗name)
- int findInteractionCode (int icd)
- char ∗ particleName (int p)
- int particleCode (int p)
- int normalParticle (int pt)
- int antiParticle (int pt)
- int ∗ allParticles (int ∗len)
- int findMClass (const int cpl, const int dgr)
- void prParticleArray (int n, int ∗a, const char ∗msg)

Data Fields

- char ∗ name
- char ∗∗ cnlist
- int ncouple
- int nParticles
- Particle ∗∗ particles
- int pdef
- int nallPart
- int allPart [GRCC_MAXMPARTICLES2]
- int nintPart
- int intPart [GRCC_MAXMPARTICLES2]
- int nInteracts

- Interaction ** interacts
- int vdef
- int maxnlegs
- int maxcpl
- int maxloop
- int * cplgcp
- int * cplglg
- int * cplgnvl
- int ** cplgvl
- int ncplgcp
- int defpart
- int skipFLine
- int DUMMYPADDING

3.70.1 Detailed Description

Definition at line 262 of file grcc.h.

3.70.2 Constructor & Destructor Documentation

3.70.2.1 Model()

```
Model::Model (
            MInput * minp )
```

Definition at line 1726 of file grcc.cc.

3.70.2.2 ∼Model()

```
Model::∼Model (
            void )
```

Definition at line 1787 of file grcc.cc.

3.70.3 Member Function Documentation

3.70.3.1 prModel()

```
void Model::prModel (
            void )
```

Definition at line 1827 of file grcc.cc.

3.70.3.2 addParticle()

```
void Model::addParticle (
            PInput * pinp )
```

Definition at line 1876 of file grcc.cc.

3.70.3.3 addParticleEnd()

```
void Model::addParticleEnd (
             void )
```

Definition at line 1918 of file grcc.cc.

3.70.3.4 addInteraction()

```
void Model::addInteraction (
             IInput * iinp )
```

Definition at line 1956 of file grcc.cc.

3.70.3.5 addInteractionEnd()

```
void Model::addInteractionEnd (
             void )
```

Definition at line 2084 of file grcc.cc.

3.70.3.6 findParticleName()

```
int Model::findParticleName (
             const char * name )
```

Definition at line 2157 of file grcc.cc.

3.70.3.7 findParticleCode()

```
int Model::findParticleCode (
             int pcd )
```

Definition at line 2175 of file grcc.cc.

3.70.3.8 findInteractionName()

```
int Model::findInteractionName (
             const char * name )
```

Definition at line 2294 of file grcc.cc.

3.70.3.9 findInteractionCode()

```
int Model::findInteractionCode (
             int icd )
```

Definition at line 2307 of file grcc.cc.

3.70.3.10 particleName()

```
char * Model::particleName (
              int p )
```

Definition at line 2190 of file grcc.cc.

3.70.3.11 particleCode()

```
int Model::particleCode (
              int p )
```

Definition at line 2210 of file grcc.cc.

3.70.3.12 normalParticle()

```
int Model::normalParticle (
              int pt )
```

Definition at line 2256 of file grcc.cc.

3.70.3.13 antiParticle()

```
int Model::antiParticle (
              int pt )
```

Definition at line 2273 of file grcc.cc.

3.70.3.14 allParticles()

```
int * Model::allParticles (
              int * len )
```

Definition at line 2287 of file grcc.cc.

3.70.3.15 findMClass()

```
int Model::findMClass (
              const int cpl,
              const int dgr )
```

Definition at line 2243 of file grcc.cc.

3.70.3.16 prParticleArray()

```
void Model::prParticleArray (
              int n,
              int * a,
              const char * msg )
```

Definition at line 2228 of file grcc.cc.

3.70.4 Field Documentation

3.70.4.1 name

`char* Model::name`

Definition at line 264 of file grcc.h.

3.70.4.2 cnlist

`char** Model::cnlist`

Definition at line 265 of file grcc.h.

3.70.4.3 ncouple

`int Model::ncouple`

Definition at line 266 of file grcc.h.

3.70.4.4 nParticles

`int Model::nParticles`

Definition at line 268 of file grcc.h.

3.70.4.5 particles

`Particle** Model::particles`

Definition at line 269 of file grcc.h.

3.70.4.6 pdef

`int Model::pdef`

Definition at line 270 of file grcc.h.

3.70.4.7 nallPart

`int Model::nallPart`

Definition at line 273 of file grcc.h.

3.70.4.8 allPart

`int Model::allPart[GRCC_MAXMPARTICLES2]`

Definition at line 274 of file grcc.h.

3.70.4.9 nintPart

`int Model::nintPart`

Definition at line 277 of file grcc.h.

3.70.4.10 intPart

`int Model::intPart[GRCC_MAXMPARTICLES2]`

Definition at line 278 of file grcc.h.

3.70.4.11 nInteracts

`int Model::nInteracts`

Definition at line 280 of file grcc.h.

3.70.4.12 interacts

`Interaction** Model::interacts`

Definition at line 281 of file grcc.h.

3.70.4.13 vdef

`int Model::vdef`

Definition at line 282 of file grcc.h.

3.70.4.14 maxnlegs

`int Model::maxnlegs`

Definition at line 284 of file grcc.h.

3.70.4.15 maxcpl

`int Model::maxcpl`

Definition at line 285 of file grcc.h.

3.70.4.16 maxloop

```
int Model::maxloop
```

Definition at line 286 of file grcc.h.

3.70.4.17 cplgcp

```
int* Model::cplgcp
```

Definition at line 289 of file grcc.h.

3.70.4.18 cplglg

```
int* Model::cplglg
```

Definition at line 290 of file grcc.h.

3.70.4.19 cplgnvl

```
int* Model::cplgnvl
```

Definition at line 291 of file grcc.h.

3.70.4.20 cplgvl

```
int** Model::cplgvl
```

Definition at line 292 of file grcc.h.

3.70.4.21 ncplgcp

```
int Model::ncplgcp
```

Definition at line 293 of file grcc.h.

3.70.4.22 defpart

```
int Model::defpart
```

Definition at line 295 of file grcc.h.

3.70.4.23 skipFLine

```
int Model::skipFLine
```

Definition at line 296 of file grcc.h.

3.70.4.24 DUMMYPADDING

```
int Model::DUMMYPADDING
```

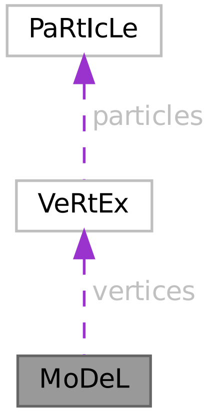
Definition at line 298 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.71 MoDeL Struct Reference

Collaboration diagram for MoDeL:



Data Fields

- VERTEX ** vertices
- WORD * couplings
- UBYTE * name
- void * grccmodel
- WORD legcouple [MAXLEGS+2]
- WORD nparticles
- WORD nvertices
- WORD invertices
- WORD sizevertices
- WORD sizecouplings
- WORD ncouplings
- WORD error
- WORD dummy

3.71.1 Detailed Description

Definition at line 637 of file structs.h.

3.71.2 Field Documentation

3.71.2.1 vertices

```
VERTEX** MoDeL::vertices
```

Definition at line 638 of file structs.h.

3.71.2.2 couplings

```
WORD* MoDeL::couplings
```

Definition at line 639 of file structs.h.

3.71.2.3 name

```
UBYTE* MoDeL::name
```

Definition at line 640 of file structs.h.

3.71.2.4 grccmodel

```
void* MoDeL::grccmodel
```

Definition at line 641 of file structs.h.

3.71.2.5 legcouple

```
WORD MoDeL::legcouple[MAXLEGS+2]
```

Definition at line 642 of file structs.h.

3.71.2.6 nparticles

```
WORD MoDeL::nparticles
```

Definition at line 643 of file structs.h.

3.71.2.7 nvertices

```
WORD MoDeL::nvertices
```

Definition at line 644 of file structs.h.

3.71.2.8 invertices

```
WORD MoDeL::invertices
```

Definition at line 645 of file structs.h.

3.71.2.9 sizevertices

```
WORD MoDeL::sizevertices
```

Definition at line 646 of file structs.h.

3.71.2.10 sizecouplings

```
WORD MoDeL::sizecouplings
```

Definition at line 647 of file structs.h.

3.71.2.11 ncouplings

```
WORD MoDeL::ncouplings
```

Definition at line 648 of file structs.h.

3.71.2.12 error

```
WORD MoDeL::error
```

Definition at line 649 of file structs.h.

3.71.2.13 dummy

```
WORD MoDeL::dummy
```

Definition at line 650 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.72 MODNUM Struct Reference

Data Fields

- UWORD * a
- UWORD * m
- WORD na
- WORD nm

3.72.1 Detailed Description

Definition at line 1301 of file structs.h.

3.72.2 Field Documentation

3.72.2.1 a

`UWORD* MODNUM::a`

Definition at line 1302 of file structs.h.

3.72.2.2 m

`UWORD* MODNUM::m`

Definition at line 1303 of file structs.h.

3.72.2.3 na

`WORD MODNUM::na`

Definition at line 1304 of file structs.h.

3.72.2.4 nm

`WORD MODNUM::nm`

Definition at line 1305 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.73 MoDoPtDoLlArS Struct Reference

Data Fields

- WORD number
- WORD type

3.73.1 Detailed Description

Definition at line 556 of file structs.h.

3.73.2 Field Documentation

3.73.2.1 number

```
WORD MoDoPtDoLlArS::number
```

Definition at line 560 of file structs.h.

3.73.2.2 type

```
WORD MoDoPtDoLlArS::type
```

Definition at line 561 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.74 monomial_larger Class Reference

Public Member Functions

- bool operator() (const WORD ∗a, const WORD ∗b)

3.74.1 Detailed Description

Definition at line 179 of file poly.h.

3.74.2 Constructor & Destructor Documentation

3.74.2.1 monomial_larger()

```
monomial_larger::monomial_larger ( )  [inline]
```

Definition at line 184 of file poly.h.

3.74.3 Member Function Documentation

3.74.3.1 operator()()

```
bool monomial_larger::operator() (
            const WORD * a,
            const WORD * b ) [inline]
```

Definition at line 190 of file poly.h.

The documentation for this class was generated from the following file:

- poly.h

3.75 MOrbits Class Reference

Public Member Functions

- void print (void)
- void initPerm (int nnodes)
- void fromPerm (int *perm)
- void toOrbits (void)

Data Fields

- int nOrbits
- int nNodes
- int nd2or [GRCC_MAXNODES]
- int or2nd [GRCC_MAXNODES]
- int flist [GRCC_MAXNODES+1]

3.75.1 Detailed Description

Definition at line 1005 of file grcc.h.

3.75.2 Constructor & Destructor Documentation

3.75.2.1 MOrbits()

```
MOrbits::MOrbits (
            void )
```

Definition at line 4964 of file grcc.cc.

3.75.2.2 ∼MOrbits()

```
MOrbits::∼MOrbits (
            void )
```

Definition at line 4971 of file grcc.cc.

3.75.3 Member Function Documentation

3.75.3.1 print()

```
void MOrbits::print (
            void )
```

Definition at line 4976 of file grcc.cc.

3.75.3.2 initPerm()

```
void MOrbits::initPerm (
            int nnodes )
```

Definition at line 4989 of file grcc.cc.

3.75.3.3 fromPerm()

```
void MOrbits::fromPerm (
            int * perm )
```

Definition at line 5000 of file grcc.cc.

3.75.3.4 toOrbits()

```
void MOrbits::toOrbits (
            void )
```

Definition at line 5039 of file grcc.cc.

3.75.4 Field Documentation

3.75.4.1 nOrbits

```
int MOrbits::nOrbits
```

Definition at line 1017 of file grcc.h.

3.75.4.2 nNodes

```
int MOrbits::nNodes
```

Definition at line 1018 of file grcc.h.

3.75.4.3 nd2or

```
int MOrbits::nd2or[GRCC_MAXNODES]
```

Definition at line 1019 of file grcc.h.

3.75.4.4 or2nd

```
int MOrbits::or2nd[GRCC_MAXNODES]
```

Definition at line 1020 of file grcc.h.

3.75.4.5 flist

```
int MOrbits::flist[GRCC_MAXNODES+1]
```

Definition at line 1021 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.76 flint::mpoly Class Reference

Public Member Functions

- mpoly (fmpz_mpoly_ctx_struct ∗ctx_in)
- void print (const string &text)

Data Fields

- fmpz_mpoly_t d

3.76.1 Detailed Description

Definition at line 76 of file flintinterface.h.

3.76.2 Constructor & Destructor Documentation

3.76.2.1 mpoly()

```
flint::mpoly::mpoly (
            fmpz_mpoly_ctx_struct * ctx_in )  [inline], [explicit]
```

Definition at line 81 of file flintinterface.h.

3.76.2.2 ∼mpoly()

```
flint::mpoly::~mpoly ( )  [inline]
```

Definition at line 82 of file flintinterface.h.

3.76.3 Member Function Documentation

3.76.3.1 print()

```
void flint::mpoly::print (
            const string & text )  [inline]
```

Definition at line 83 of file flintinterface.h.

3.76.4 Field Documentation

3.76.4.1 d

```
fmpz_mpoly_t flint::mpoly::d
```

Definition at line 80 of file flintinterface.h.

The documentation for this class was generated from the following file:

- flintinterface.h

3.77 flint::mpoly_ctx Class Reference

Public Member Functions

- mpoly_ctx (int64_t nvars)

Data Fields

- fmpz_mpoly_ctx_t d

3.77.1 Detailed Description

Definition at line 99 of file flintinterface.h.

3.77.2 Constructor & Destructor Documentation

3.77.2.1 mpoly_ctx()

```
flint::mpoly_ctx::mpoly_ctx (
            int64_t nvars )  [inline], [explicit]
```

Definition at line 102 of file flintinterface.h.

3.77.2.2 ∼mpoly_ctx()

```
flint::mpoly_ctx::~mpoly_ctx ( )  [inline]
```

Definition at line 103 of file flintinterface.h.

3.77.3 Field Documentation

3.77.3.1 d

```
fmpz_mpoly_ctx_t flint::mpoly_ctx::d
```

Definition at line 101 of file flintinterface.h.

The documentation for this class was generated from the following file:

- flintinterface.h

3.78 flint::mpoly_factor Class Reference

Public Member Functions

- mpoly_factor (fmpz_mpoly_ctx_struct ∗ctx_in)

Data Fields

- fmpz_mpoly_factor_t d

3.78.1 Detailed Description

Definition at line 89 of file flintinterface.h.

3.78.2 Constructor & Destructor Documentation

3.78.2.1 mpoly_factor()

```
flint::mpoly_factor::mpoly_factor (
            fmpz_mpoly_ctx_struct * ctx_in ) [inline], [explicit]
```

Definition at line 94 of file flintinterface.h.

3.78.2.2 ∼mpoly_factor()

```
flint::mpoly_factor::∼mpoly_factor ( ) [inline]
```

Definition at line 97 of file flintinterface.h.

3.78.3 Field Documentation

3.78.3.1 d

```
fmpz_mpoly_factor_t flint::mpoly_factor::d
```

Definition at line 93 of file flintinterface.h.

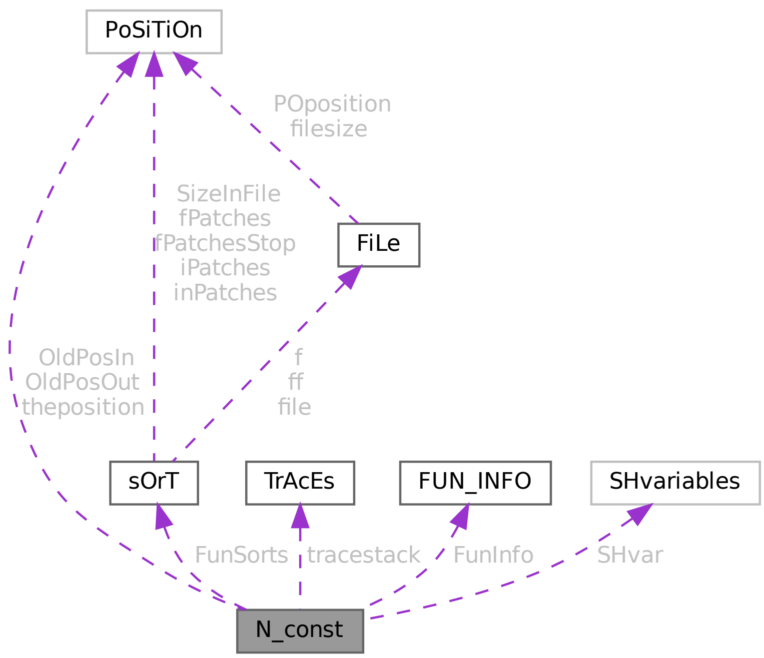
The documentation for this class was generated from the following file:

- flintinterface.h

3.79 N_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for N_const:



Public Member Functions

- **PADPOSITION** (50, 7, 23, 26, sizeof(SHvariables))

Data Fields

- POSITION OldPosIn
- POSITION OldPosOut
- POSITION theposition
- WORD ∗ EndNest
- WORD ∗ Frozen
- WORD ∗ FullProto
- WORD ∗ cTerm
- int ∗ RepPoint
- WORD ∗ WildValue
- WORD ∗ WildStop
- WORD ∗ argaddress
- WORD ∗ RepFunList
- WORD ∗ patstop
- WORD ∗ terstop
- WORD ∗ terstart
- WORD ∗ terfirstcomm
- WORD ∗ DumFound
- WORD ∗∗ DumPlace
- WORD ∗∗ DumFunPlace
- WORD ∗ UsedSymbol
- WORD ∗ UsedVector
- WORD ∗ UsedIndex
- WORD ∗ UsedFunction
- WORD ∗ MaskPointer
- WORD ∗ ForFindOnly
- WORD ∗ findTerm
- WORD ∗ findPattern
- WORD ∗ dummyrenumlist
- int ∗ funargs
- WORD ∗∗ funlocs
- int ∗ funinds
- UWORD ∗ NoScrat2
- WORD ∗ ReplaceScrat
- TRACES ∗ tracestack
- WORD ∗ selecttermundo
- WORD ∗ patternbuffer
- WORD ∗ termbuffer
- WORD ∗∗ PoinScratch
- WORD ∗∗ FunScratch
- WORD ∗ RenumScratch
- FUN_INFO ∗ FunInfo
- WORD ∗∗ SplitScratch
- WORD ∗∗ SplitScratch1
- SORTING ∗∗ FunSorts
- UWORD ∗ SoScratC
- WORD ∗ listinprint
- WORD ∗ currentTerm
- WORD ∗∗ arglist

- int * tlistbuf
- WORD * compressSpace
- UWORD * SHcombi
- WORD * poly_vars
- UWORD * cmod
- SHvariables SHvar
- LONG deferskipped
- LONG InScratch
- LONG SplitScratchSize
- LONG InScratch1
- LONG SplitScratchSize1
- LONG ninterms
- LONG SHcombisize
- int NumTotWildArgs
- int UseFindOnly
- int UsedOtherFind
- int ErrorInDollar
- int numfargs
- int numflocs
- int nargs
- int tohunt
- int numoffuns
- int funisize
- int RSsize
- int numtracesctack
- int intracestack
- int numfuninfo
- int NumFunSorts
- int MaxFunSorts
- int arglistsize
- int tlistsize
- int filenum
- int compressSize
- int polysortflag
- int nogroundlevel
- int subsubveto
- WORD MaxRenumScratch
- WORD oldtype
- WORD oldvalue
- WORD NumWild
- WORD RepFunNum
- WORD DisOrderFlag
- WORD WildDirt
- WORD NumFound
- WORD WildReserve
- WORD TeInFun
- WORD TeSuOut
- WORD WildArgs
- WORD WildEat
- WORD PolyNormFlag
- WORD PolyFunTodo
- WORD sizeselecttermundo
- WORD patternbuffersize
- WORD numlistinprint
- WORD ncmod

- WORD ExpectedSign
- WORD SignCheck
- WORD IndDum
- WORD poly_num_vars
- WORD idfunctionflag
- WORD poly_vars_type
- WORD tryterm

3.79.1 Detailed Description

The N_const struct is part of the global data and resides either in the ALLGLOBALS struct A, or the ALLPRIVATES struct B (TFORM) under the name N We see it used with the macro AN as in AN.RepFunNum It has variables that are private to each thread and are used as temporary storage during the expansion of the terms tree.

Definition at line 2276 of file structs.h.

3.79.2 Field Documentation

3.79.2.1 OldPosIn

`POSITION N_const::OldPosIn`

Definition at line 2277 of file structs.h.

3.79.2.2 OldPosOut

`POSITION N_const::OldPosOut`

Definition at line 2278 of file structs.h.

3.79.2.3 theposition

`POSITION N_const::theposition`

Definition at line 2279 of file structs.h.

3.79.2.4 EndNest

`WORD* N_const::EndNest`

Definition at line 2280 of file structs.h.

3.79.2.5 Frozen

`WORD* N_const::Frozen`

Definition at line 2281 of file structs.h.

3.79.2.6 FullProto

```
WORD* N_const::FullProto
```

Definition at line 2282 of file structs.h.

3.79.2.7 cTerm

```
WORD* N_const::cTerm
```

Definition at line 2283 of file structs.h.

3.79.2.8 RepPoint

```
int* N_const::RepPoint
```

Definition at line 2284 of file structs.h.

3.79.2.9 WildValue

```
WORD* N_const::WildValue
```

Definition at line 2285 of file structs.h.

3.79.2.10 WildStop

```
WORD* N_const::WildStop
```

Definition at line 2286 of file structs.h.

3.79.2.11 argaddress

```
WORD* N_const::argaddress
```

Definition at line 2287 of file structs.h.

3.79.2.12 RepFunList

```
WORD* N_const::RepFunList
```

Definition at line 2288 of file structs.h.

3.79.2.13 patstop

```
WORD* N_const::patstop
```

Definition at line 2289 of file structs.h.

3.79.2.14 terstop

```
WORD* N_const::terstop
```

Definition at line 2290 of file structs.h.

3.79.2.15 terstart

```
WORD* N_const::terstart
```

Definition at line 2291 of file structs.h.

3.79.2.16 terfirstcomm

```
WORD* N_const::terfirstcomm
```

Definition at line 2292 of file structs.h.

3.79.2.17 DumFound

```
WORD* N_const::DumFound
```

Definition at line 2293 of file structs.h.

3.79.2.18 DumPlace

```
WORD** N_const::DumPlace
```

Definition at line 2294 of file structs.h.

3.79.2.19 DumFunPlace

```
WORD** N_const::DumFunPlace
```

Definition at line 2295 of file structs.h.

3.79.2.20 UsedSymbol

```
WORD* N_const::UsedSymbol
```

Definition at line 2296 of file structs.h.

3.79.2.21 UsedVector

```
WORD* N_const::UsedVector
```

Definition at line 2297 of file structs.h.

3.79.2.22 UsedIndex

`WORD* N_const::UsedIndex`

Definition at line 2298 of file structs.h.

3.79.2.23 UsedFunction

`WORD* N_const::UsedFunction`

Definition at line 2299 of file structs.h.

3.79.2.24 MaskPointer

`WORD* N_const::MaskPointer`

Definition at line 2300 of file structs.h.

3.79.2.25 ForFindOnly

`WORD* N_const::ForFindOnly`

Definition at line 2301 of file structs.h.

3.79.2.26 findTerm

`WORD* N_const::findTerm`

Definition at line 2302 of file structs.h.

3.79.2.27 findPattern

`WORD* N_const::findPattern`

Definition at line 2303 of file structs.h.

3.79.2.28 dummyrenumlist

`WORD* N_const::dummyrenumlist`

Definition at line 2308 of file structs.h.

3.79.2.29 funargs

`int* N_const::funargs`

Definition at line 2309 of file structs.h.

3.79.2.30 funlocs

```
WORD** N_const::funlocs
```

Definition at line 2310 of file structs.h.

3.79.2.31 funinds

```
int* N_const::funinds
```

Definition at line 2311 of file structs.h.

3.79.2.32 NoScrat2

```
UWORD* N_const::NoScrat2
```

Definition at line 2312 of file structs.h.

3.79.2.33 ReplaceScrat

```
WORD* N_const::ReplaceScrat
```

Definition at line 2313 of file structs.h.

3.79.2.34 tracestack

```
TRACES* N_const::tracestack
```

Definition at line 2314 of file structs.h.

3.79.2.35 selecttermundo

```
WORD* N_const::selecttermundo
```

Definition at line 2315 of file structs.h.

3.79.2.36 patternbuffer

```
WORD* N_const::patternbuffer
```

Definition at line 2316 of file structs.h.

3.79.2.37 termbuffer

```
WORD* N_const::termbuffer
```

Definition at line 2317 of file structs.h.

3.79.2.38 PoinScratch

`WORD** N_const::PoinScratch`

Definition at line 2318 of file structs.h.

3.79.2.39 FunScratch

`WORD** N_const::FunScratch`

Definition at line 2319 of file structs.h.

3.79.2.40 RenumScratch

`WORD* N_const::RenumScratch`

Definition at line 2320 of file structs.h.

3.79.2.41 FunInfo

`FUN_INFO* N_const::FunInfo`

Definition at line 2321 of file structs.h.

3.79.2.42 SplitScratch

`WORD** N_const::SplitScratch`

Definition at line 2322 of file structs.h.

3.79.2.43 SplitScratch1

`WORD** N_const::SplitScratch1`

Definition at line 2323 of file structs.h.

3.79.2.44 FunSorts

`SORTING** N_const::FunSorts`

Definition at line 2324 of file structs.h.

3.79.2.45 SoScratC

`UWORD* N_const::SoScratC`

Definition at line 2325 of file structs.h.

3.79.2.46 listinprint

```
WORD* N_const::listinprint
```

Definition at line 2326 of file structs.h.

3.79.2.47 currentTerm

```
WORD* N_const::currentTerm
```

Definition at line 2327 of file structs.h.

3.79.2.48 arglist

```
WORD** N_const::arglist
```

Definition at line 2328 of file structs.h.

3.79.2.49 tlistbuf

```
int* N_const::tlistbuf
```

Definition at line 2329 of file structs.h.

3.79.2.50 compressSpace

```
WORD* N_const::compressSpace
```

Definition at line 2333 of file structs.h.

3.79.2.51 SHcombi

```
UWORD* N_const::SHcombi
```

Definition at line 2338 of file structs.h.

3.79.2.52 poly_vars

```
WORD* N_const::poly_vars
```

Definition at line 2339 of file structs.h.

3.79.2.53 cmod

```
UWORD* N_const::cmod
```

Definition at line 2340 of file structs.h.

3.79.2.54 SHvar

```
SHvariables N_const::SHvar
```

Definition at line 2341 of file structs.h.

3.79.2.55 deferskipped

```
LONG N_const::deferskipped
```

Definition at line 2342 of file structs.h.

3.79.2.56 InScratch

```
LONG N_const::InScratch
```

Definition at line 2343 of file structs.h.

3.79.2.57 SplitScratchSize

```
LONG N_const::SplitScratchSize
```

Definition at line 2344 of file structs.h.

3.79.2.58 InScratch1

```
LONG N_const::InScratch1
```

Definition at line 2345 of file structs.h.

3.79.2.59 SplitScratchSize1

```
LONG N_const::SplitScratchSize1
```

Definition at line 2346 of file structs.h.

3.79.2.60 ninterms

```
LONG N_const::ninterms
```

Definition at line 2347 of file structs.h.

3.79.2.61 SHcombisize

```
LONG N_const::SHcombisize
```

Definition at line 2356 of file structs.h.

3.79.2.62 NumTotWildArgs

```
int N_const::NumTotWildArgs
```

Definition at line 2357 of file structs.h.

3.79.2.63 UseFindOnly

```
int N_const::UseFindOnly
```

Definition at line 2358 of file structs.h.

3.79.2.64 UsedOtherFind

```
int N_const::UsedOtherFind
```

Definition at line 2359 of file structs.h.

3.79.2.65 ErrorInDollar

```
int N_const::ErrorInDollar
```

Definition at line 2360 of file structs.h.

3.79.2.66 numfargs

```
int N_const::numfargs
```

Definition at line 2361 of file structs.h.

3.79.2.67 numflocs

```
int N_const::numflocs
```

Definition at line 2362 of file structs.h.

3.79.2.68 nargs

```
int N_const::nargs
```

Definition at line 2363 of file structs.h.

3.79.2.69 tohunt

```
int N_const::tohunt
```

Definition at line 2364 of file structs.h.

3.79.2.70 numoffuns

```
int N_const::numoffuns
```

Definition at line 2365 of file structs.h.

3.79.2.71 funisize

```
int N_const::funisize
```

Definition at line 2366 of file structs.h.

3.79.2.72 RSsize

```
int N_const::RSsize
```

Definition at line 2367 of file structs.h.

3.79.2.73 numtracesctack

```
int N_const::numtracesctack
```

Definition at line 2368 of file structs.h.

3.79.2.74 intracestack

```
int N_const::intracestack
```

Definition at line 2369 of file structs.h.

3.79.2.75 numfuninfo

```
int N_const::numfuninfo
```

Definition at line 2370 of file structs.h.

3.79.2.76 NumFunSorts

```
int N_const::NumFunSorts
```

Definition at line 2371 of file structs.h.

3.79.2.77 MaxFunSorts

```
int N_const::MaxFunSorts
```

Definition at line 2372 of file structs.h.

3.79.2.78 arglistsize

```
int N_const::arglistsize
```

Definition at line 2373 of file structs.h.

3.79.2.79 tlistsize

```
int N_const::tlistsize
```

Definition at line 2374 of file structs.h.

3.79.2.80 filenum

```
int N_const::filenum
```

Definition at line 2375 of file structs.h.

3.79.2.81 compressSize

```
int N_const::compressSize
```

Definition at line 2376 of file structs.h.

3.79.2.82 polysortflag

```
int N_const::polysortflag
```

Definition at line 2377 of file structs.h.

3.79.2.83 nogroundlevel

```
int N_const::nogroundlevel
```

Definition at line 2378 of file structs.h.

3.79.2.84 subsubveto

```
int N_const::subsubveto
```

Definition at line 2379 of file structs.h.

3.79.2.85 MaxRenumScratch

```
WORD N_const::MaxRenumScratch
```

Definition at line 2380 of file structs.h.

3.79.2.86 oldtype

`WORD N_const::oldtype`

Definition at line 2381 of file structs.h.

3.79.2.87 oldvalue

`WORD N_const::oldvalue`

Definition at line 2382 of file structs.h.

3.79.2.88 NumWild

`WORD N_const::NumWild`

Definition at line 2383 of file structs.h.

3.79.2.89 RepFunNum

`WORD N_const::RepFunNum`

Definition at line 2384 of file structs.h.

3.79.2.90 DisOrderFlag

`WORD N_const::DisOrderFlag`

Definition at line 2385 of file structs.h.

3.79.2.91 WildDirt

`WORD N_const::WildDirt`

Definition at line 2386 of file structs.h.

3.79.2.92 NumFound

`WORD N_const::NumFound`

Definition at line 2387 of file structs.h.

3.79.2.93 WildReserve

`WORD N_const::WildReserve`

Definition at line 2388 of file structs.h.

3.79.2.94 TeInFun

`WORD N_const::TeInFun`

Definition at line 2389 of file structs.h.

3.79.2.95 TeSuOut

`WORD N_const::TeSuOut`

Definition at line 2390 of file structs.h.

3.79.2.96 WildArgs

`WORD N_const::WildArgs`

Definition at line 2391 of file structs.h.

3.79.2.97 WildEat

`WORD N_const::WildEat`

Definition at line 2392 of file structs.h.

3.79.2.98 PolyNormFlag

`WORD N_const::PolyNormFlag`

Definition at line 2393 of file structs.h.

3.79.2.99 PolyFunTodo

`WORD N_const::PolyFunTodo`

Definition at line 2394 of file structs.h.

3.79.2.100 sizeselecttermundo

`WORD N_const::sizeselecttermundo`

Definition at line 2395 of file structs.h.

3.79.2.101 patternbuffersize

`WORD N_const::patternbuffersize`

Definition at line 2396 of file structs.h.

3.79.2.102 numlistinprint

`WORD N_const::numlistinprint`

Definition at line 2397 of file structs.h.

3.79.2.103 ncmod

`WORD N_const::ncmod`

Definition at line 2398 of file structs.h.

3.79.2.104 ExpectedSign

`WORD N_const::ExpectedSign`

Definition at line 2399 of file structs.h.

3.79.2.105 SignCheck

`WORD N_const::SignCheck`

Used in pattern matching of antisymmetric functions

Definition at line 2400 of file structs.h.

3.79.2.106 IndDum

`WORD N_const::IndDum`

Used in pattern matching of antisymmetric functions

Definition at line 2401 of file structs.h.

3.79.2.107 poly_num_vars

`WORD N_const::poly_num_vars`

Definition at line 2402 of file structs.h.

3.79.2.108 idfunctionflag

`WORD N_const::idfunctionflag`

Definition at line 2403 of file structs.h.

3.79.2.109 poly_vars_type

```
WORD N_const::poly_vars_type
```

Definition at line 2404 of file structs.h.

3.79.2.110 tryterm

```
WORD N_const::tryterm
```

Definition at line 2405 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.80 nameblock Struct Reference

Data Fields

- MLONG previousblock
- MLONG position
- char names [NAMETABLESIZE]

3.80.1 Detailed Description

Definition at line 115 of file minos.h.

3.80.2 Field Documentation

3.80.2.1 previousblock

```
MLONG nameblock::previousblock
```

Definition at line 116 of file minos.h.

3.80.2.2 position

```
MLONG nameblock::position
```

Definition at line 117 of file minos.h.

3.80.2.3 names

```
char nameblock::names[NAMETABLESIZE]
```

Definition at line 118 of file minos.h.

The documentation for this struct was generated from the following file:

- minos.h

3.81 NaMeNode Struct Reference

```
#include <structs.h>
```

Data Fields

- LONG name
- WORD parent
- WORD left
- WORD right
- WORD balance
- WORD type
- WORD number

3.81.1 Detailed Description

The names of variables are kept in an array. Elements of type NAMENODE define a tree (that is kept balanced) that make it easy and fast to look for variables. See also NAMETREE.

Definition at line 247 of file structs.h.

3.81.2 Field Documentation

3.81.2.1 name

```
LONG NaMeNode::name
```

Offset into NAMETREE::namebuffer.

Definition at line 248 of file structs.h.

3.81.2.2 parent

```
WORD NaMeNode::parent
```

=-1 if no parent.

Definition at line 249 of file structs.h.

3.81.2.3 left

`WORD NaMeNode::left`

=-1 if no child.

Definition at line 250 of file structs.h.

3.81.2.4 right

`WORD NaMeNode::right`

=-1 if no child.

Definition at line 251 of file structs.h.

3.81.2.5 balance

`WORD NaMeNode::balance`

Used for the balancing of the tree.

Definition at line 252 of file structs.h.

3.81.2.6 type

`WORD NaMeNode::type`

Type associated with the name. See compiler types.

Definition at line 253 of file structs.h.

3.81.2.7 number

`WORD NaMeNode::number`

Number of variable in LIST's like for example C_const::SymbolList.

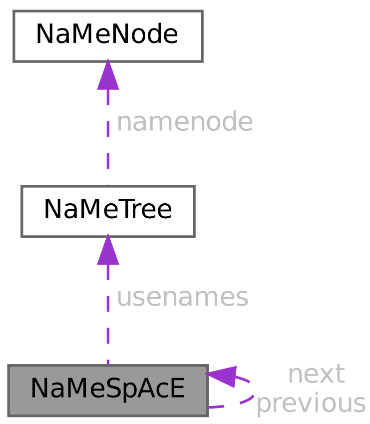
Definition at line 254 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.82 NaMeSpAcE Struct Reference

Collaboration diagram for NaMeSpAcE:



Data Fields

- struct NaMeSpAcE ∗ previous
- struct NaMeSpAcE ∗ next
- NAMETREE ∗ usenames
- UBYTE ∗ name

3.82.1 Detailed Description

Definition at line 658 of file structs.h.

3.82.2 Field Documentation

3.82.2.1 previous

```
struct NaMeSpAcE* NaMeSpAcE::previous
```

Definition at line 659 of file structs.h.

3.82.2.2 next

```
struct NaMeSpAcE* NaMeSpAcE::next
```

Definition at line 660 of file structs.h.

3.82.2.3 usenames

`NAMETREE* NaMeSpAcE::usenames`

Definition at line 661 of file structs.h.

3.82.2.4 name

`UBYTE* NaMeSpAcE::name`

Definition at line 662 of file structs.h.

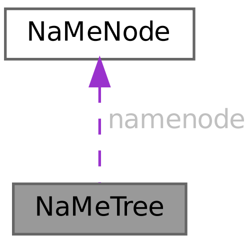
The documentation for this struct was generated from the following file:

- structs.h

3.83 NaMeTree Struct Reference

`#include <structs.h>`

Collaboration diagram for NaMeTree:



Data Fields

- NAMENODE * namenode
- UBYTE * namebuffer
- LONG nodesize
- LONG nodefill
- LONG namesize
- LONG namefill
- LONG oldnamefill
- LONG oldnodefill
- LONG globalnamefill
- LONG globalnodefill
- LONG clearnamefill
- LONG clearnodefill
- WORD headnode

3.83.1 Detailed Description

A struct of type NAMETREE controls a complete (balanced) tree of names for the compiler. The compiler maintains several of such trees and the system has been set up in such a way that one could define more of them if we ever want to work with local name spaces.

Definition at line 265 of file structs.h.

3.83.2 Field Documentation

3.83.2.1 namenode

`NAMENODE* NaMeTree::namenode`

[D] Vector of NAMENODE's. Number of elements is nodesize. =0 if no memory has been allocated.

Definition at line 266 of file structs.h.

3.83.2.2 namebuffer

`UBYTE* NaMeTree::namebuffer`

[D] Buffer that holds all the name strings referred to by the NAMENODE's. Allocation size is namesize. =0 if no memory has been allocated.

Definition at line 268 of file structs.h.

3.83.2.3 nodesize

`LONG NaMeTree::nodesize`

Maximum number of elements in namenode.

Definition at line 271 of file structs.h.

3.83.2.4 nodefill

`LONG NaMeTree::nodefill`

Number of currently used nodes in namenode.

Definition at line 272 of file structs.h.

3.83.2.5 namesize

`LONG NaMeTree::namesize`

Allocation size of namebuffer in bytes.

Definition at line 273 of file structs.h.

3.83.2.6 namefill

`LONG NaMeTree::namefill`

Number of bytes occupied.

Definition at line 274 of file structs.h.

3.83.2.7 oldnamefill

`LONG NaMeTree::oldnamefill`

UNUSED

Definition at line 275 of file structs.h.

3.83.2.8 oldnodefill

`LONG NaMeTree::oldnodefill`

UNUSED

Definition at line 276 of file structs.h.

3.83.2.9 globalnamefill

`LONG NaMeTree::globalnamefill`

Set by .global statement to the value of namefill. When a .store command is processed, this value will be used to reset namefill.

Definition at line 277 of file structs.h.

3.83.2.10 globalnodefill

`LONG NaMeTree::globalnodefill`

Same usage as globalnamefill, but for nodefill.

Definition at line 279 of file structs.h.

3.83.2.11 clearnamefill

`LONG NaMeTree::clearnamefill`

Marks the reset point used by the .clear statement.

Definition at line 280 of file structs.h.

3.83.2.12 clearnodefill

```
LONG NaMeTree::clearnodefill
```

Marks the reset point used by the .clear statement.

Definition at line 281 of file structs.h.

3.83.2.13 headnode

```
WORD NaMeTree::headnode
```

Offset in namenode of head node. =-1 if tree is empty.

Definition at line 282 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.84 NCand Class Reference

Public Member Functions

- NCand (const NCandSt sta, const int dega, const int nilst, int ∗ilst)
- void prNCand (const char ∗msg)

Data Fields

- int deg
- NCandSt st
- int nilist
- int ilist [GRCC_MAXMINTERACT]

3.84.1 Detailed Description

Definition at line 1044 of file grcc.h.

3.84.2 Constructor & Destructor Documentation

3.84.2.1 NCand()

```
NCand::NCand (
            const NCandSt sta,
            const int dega,
            const int nilst,
            int * ilst )
```

Definition at line 7285 of file grcc.cc.

3.84.2.2 ∼NCand()

```
NCand::~NCand (
            void )
```

Definition at line 7312 of file grcc.cc.

3.84.3 Member Function Documentation

3.84.3.1 prNCand()

```
void NCand::prNCand (
            const char * msg )
```

Definition at line 7317 of file grcc.cc.

3.84.4 Field Documentation

3.84.4.1 deg

```
int NCand::deg
```

Definition at line 1048 of file grcc.h.

3.84.4.2 st

```
NCandSt NCand::st
```

Definition at line 1049 of file grcc.h.

3.84.4.3 nilist

```
int NCand::nilist
```

Definition at line 1050 of file grcc.h.

3.84.4.4 ilist

```
int NCand::ilist[GRCC_MAXMINTERACT]
```

Definition at line 1051 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.85 NCInput Struct Reference

Data Fields

- int cldeg
- int clnum
- int cltyp
- int cmind
- int cmaxd
- int ptcl
- int cple

3.85.1 Detailed Description

Definition at line 149 of file grccparam.h.

3.85.2 Field Documentation

3.85.2.1 cldeg

```
int NCInput::cldeg
```

Definition at line 151 of file grccparam.h.

3.85.2.2 clnum

```
int NCInput::clnum
```

Definition at line 152 of file grccparam.h.

3.85.2.3 cltyp

```
int NCInput::cltyp
```

Definition at line 153 of file grccparam.h.

3.85.2.4 cmind

```
int NCInput::cmind
```

Definition at line 154 of file grccparam.h.

3.85.2.5 cmaxd

```
int NCInput::cmaxd
```

Definition at line 155 of file grccparam.h.

3.85.2.6 ptcl

```
int NCInput::ptcl
```

Definition at line 158 of file grccparam.h.

3.85.2.7 cple

```
int NCInput::cple
```

Definition at line 159 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.86 NeStInG Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ∗ termsize
- WORD ∗ funsize
- WORD ∗ argsize

3.86.1 Detailed Description

The NESTING struct is used when we enter the argument of functions and there is the possibility that we have to change something there. Because functions can be nested we have to keep track of all levels of functions in case we have to move the outer layers to make room for a larger function argument.

Definition at line 1053 of file structs.h.

3.86.2 Field Documentation

3.86.2.1 termsize

```
WORD* NeStInG::termsize
```

Definition at line 1054 of file structs.h.

3.86.2.2 funsize

```
WORD* NeStInG::funsize
```

Definition at line 1055 of file structs.h.

3.86.2.3 argsize

```
WORD* NeStInG::argsize
```

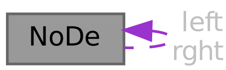
Definition at line 1056 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.87 NoDe Struct Reference

Collaboration diagram for NoDe:



Data Fields

- struct NoDe ∗ left
- struct NoDe ∗ rght
- int lloser
- int rloser
- int lsrc
- int rsrc

3.87.1 Detailed Description

A node for the tree of losers in the final sorting on the master.

Definition at line 265 of file parallel.c.

3.87.2 Field Documentation

3.87.2.1 left

```
struct NoDe* NoDe::left
```

Definition at line 266 of file parallel.c.

3.87.2.2 rght

```
struct NoDe* NoDe::rght
```

Definition at line 267 of file parallel.c.

3.87.2.3 lloser

```
int NoDe::lloser
```

Definition at line 268 of file parallel.c.

3.87.2.4 rloser

```
int NoDe::rloser
```

Definition at line 269 of file parallel.c.

3.87.2.5 lsrc

```
int NoDe::lsrc
```

Definition at line 270 of file parallel.c.

3.87.2.6 rsrc

```
int NoDe::rsrc
```

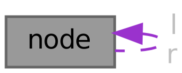
Definition at line 271 of file parallel.c.

The documentation for this struct was generated from the following file:

- parallel.c

3.88 node Struct Reference

Collaboration diagram for node:



Public Member Functions

- node (const WORD ∗data)
- int cmp (const struct node ∗rhs) const
- bool operator() (const struct node ∗lhs, const struct node ∗rhs) const
- void calcHash ()

Data Fields

- const WORD ∗ data
- struct node ∗ l
- struct node ∗ r
- WORD sign
- UWORD hash

3.88.1 Detailed Description

Definition at line 1496 of file optimize.cc.

3.88.2 Constructor & Destructor Documentation

3.88.2.1 node() [1/2]

```
node::node ( )  [inline]
```

Definition at line 1503 of file optimize.cc.

3.88.2.2 node() [2/2]

```
node::node (
            const WORD * data )  [inline]
```

Definition at line 1504 of file optimize.cc.

3.88.3 Member Function Documentation

3.88.3.1 cmp()

```
int node::cmp (
            const struct node * rhs ) const  [inline]
```

Definition at line 1508 of file optimize.cc.

3.88.3.2 operator()()

```
bool node::operator() (
            const struct node * lhs,
            const struct node * rhs ) const  [inline]
```

Definition at line 1532 of file optimize.cc.

3.88.3.3 calcHash()

```
void node::calcHash ( )  [inline]
```

Definition at line 1537 of file optimize.cc.

3.88.4 Field Documentation

3.88.4.1 data

```
const WORD* node::data
```

Definition at line 1497 of file optimize.cc.

3.88.4.2 l

```
struct node* node::l
```

Definition at line 1498 of file optimize.cc.

3.88.4.3 r

```
struct node* node::r
```

Definition at line 1499 of file optimize.cc.

3.88.4.4 sign

```
WORD node::sign
```

Definition at line 1500 of file optimize.cc.

3.88.4.5 hash

```
UWORD node::hash
```

Definition at line 1501 of file optimize.cc.

The documentation for this struct was generated from the following file:

- optimize.cc

3.89 NodeEq Struct Reference

Public Member Functions

- bool operator() (const NODE ∗lhs, const NODE ∗rhs) const

3.89.1 Detailed Description

Definition at line 1558 of file optimize.cc.

3.89.2 Member Function Documentation

3.89.2.1 operator()()

```
bool NodeEq::operator() (
            const NODE * lhs,
            const NODE * rhs ) const  [inline]
```

Definition at line 1559 of file optimize.cc.

The documentation for this struct was generated from the following file:

- optimize.cc

3.90 NodeHash Struct Reference

Public Member Functions

- size_t operator() (const NODE ∗n) const

3.90.1 Detailed Description

Definition at line 1552 of file optimize.cc.

3.90.2 Member Function Documentation

3.90.2.1 operator()()

```
size_t NodeHash::operator() (
            const NODE * n ) const  [inline]
```

Definition at line 1553 of file optimize.cc.

The documentation for this struct was generated from the following file:

- optimize.cc

3.91 NStack Class Reference

Public Member Functions

- void print (const char *msg)

Data Fields

- int noden
- int deg
- NCandSt st
- int nilist
- int ilist [GRCC_MAXMINTERACT]

3.91.1 Detailed Description

Definition at line 1270 of file grcc.h.

3.91.2 Constructor & Destructor Documentation

3.91.2.1 NStack()

```
NStack::NStack ( )  [inline]
```

Definition at line 1280 of file grcc.h.

3.91.2.2 ~NStack()

```
NStack::~NStack ( )  [inline]
```

Definition at line 1281 of file grcc.h.

3.91.3 Member Function Documentation

3.91.3.1 print()

```
void NStack::print (
             const char * msg )
```

Definition at line 9456 of file grcc.cc.

3.91.4 Field Documentation

3.91.4.1 noden

```
int NStack::noden
```

Definition at line 1274 of file grcc.h.

3.91.4.2 deg

```
int NStack::deg
```

Definition at line 1275 of file grcc.h.

3.91.4.3 st

```
NCandSt NStack::st
```

Definition at line 1276 of file grcc.h.

3.91.4.4 nilist

```
int NStack::nilist
```

Definition at line 1277 of file grcc.h.

3.91.4.5 ilist

```
int NStack::ilist[GRCC_MAXMINTERACT]
```

Definition at line 1278 of file grcc.h.

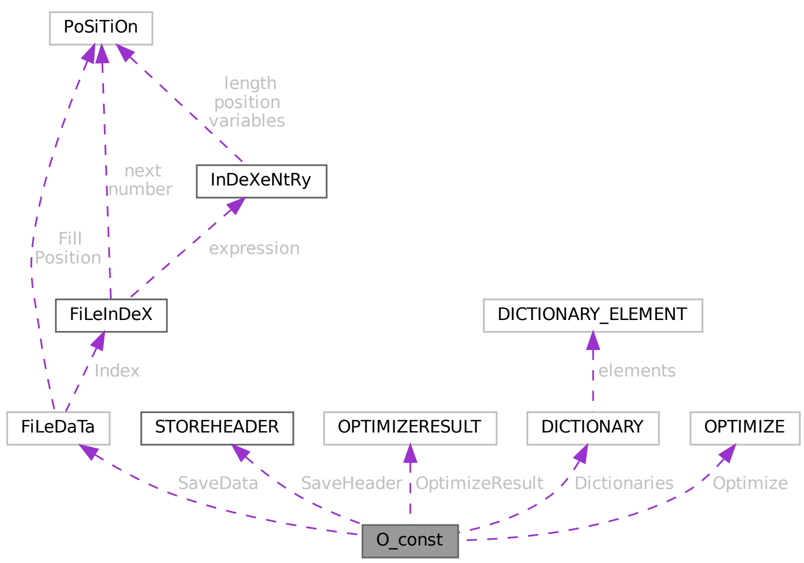
The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.92 O_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for O_const:



Public Member Functions

- **PADPOSITION** (25, 4, 36, 17, 1)

Data Fields

- FILEDATA SaveData
- STOREHEADER SaveHeader
- OPTIMIZERESULT OptimizeResult
- UBYTE ∗ OutputLine
- UBYTE ∗ OutStop
- UBYTE ∗ OutFill
- WORD ∗ bracket
- WORD ∗ termbuf
- WORD ∗ tabstring
- UBYTE ∗ wpos
- UBYTE ∗ wpoin
- UBYTE ∗ DollarOutBuffer
- UBYTE ∗ CurBufWrt
- void(∗ FlipWORD)(UBYTE ∗)
- void(∗ FlipLONG)(UBYTE ∗)
- void(∗ FlipPOS)(UBYTE ∗)
- void(∗ FlipPOINTER)(UBYTE ∗)
- void(∗ ResizeData)(UBYTE ∗, int, UBYTE ∗, int)
- void(∗ ResizeWORD)(UBYTE ∗, UBYTE ∗)
- void(∗ ResizeNCWORD)(UBYTE ∗, UBYTE ∗)
- void(∗ ResizeLONG)(UBYTE ∗, UBYTE ∗)
- void(∗ ResizePOS)(UBYTE ∗, UBYTE ∗)
- void(∗ ResizePOINTER)(UBYTE ∗, UBYTE ∗)
- void(∗ CheckPower)(UBYTE ∗)
- void(∗ RenumberVec)(UBYTE ∗)
- DICTIONARY ∗∗ Dictionaries
- UBYTE ∗ tensorList
- WORD ∗ inscheme
- LONG NumInBrack
- LONG wlen
- LONG DollarOutSizeBuffer
- LONG DollarInOutBuffer
- OPTIMIZE Optimize
- int OutInBuffer
- int NoSpacesInNumbers
- int BlockSpaces
- int CurrentDictionary
- int SizeDictionaries
- int NumDictionaries
- int CurDictNumbers
- int CurDictVariables
- int CurDictSpecials
- int CurDictFunWithArgs
- int CurDictNumberWarning
- int CurDictNotInFunctions
- int CurDictInDollars
- int gNumDictionaries
- WORD schemenum

- WORD transFlag
- WORD powerFlag
- WORD mpower
- WORD resizeFlag
- WORD bufferedInd
- WORD OutSkip
- WORD IsBracket
- WORD InFbrack
- WORD PrintType
- WORD FortFirst
- WORD DoubleFlag
- WORD IndentSpace
- WORD FactorMode
- WORD FactorNum
- WORD ErrorBlock
- WORD OptimizationLevel
- UBYTE FortDotChar

3.92.1 Detailed Description

The O_const struct is part of the global data and resides in the ALLGLOBALS struct A under the name O We see it used with the macro AO as in AO.OutputLine It contains variables that involve the writing of text output and save/store files.

Definition at line 2453 of file structs.h.

3.92.2 Field Documentation

3.92.2.1 SaveData

```
FILEDATA O_const::SaveData
```

Definition at line 2454 of file structs.h.

3.92.2.2 SaveHeader

```
STOREHEADER O_const::SaveHeader
```

Definition at line 2455 of file structs.h.

3.92.2.3 OptimizeResult

```
OPTIMIZERESULT O_const::OptimizeResult
```

Definition at line 2456 of file structs.h.

3.92.2.4 OutputLine

```
UBYTE* O_const::OutputLine
```

Definition at line 2457 of file structs.h.

3.92.2.5 OutStop

```
UBYTE* O_const::OutStop
```

Definition at line 2458 of file structs.h.

3.92.2.6 OutFill

```
UBYTE* O_const::OutFill
```

Definition at line 2459 of file structs.h.

3.92.2.7 bracket

```
WORD* O_const::bracket
```

Definition at line 2460 of file structs.h.

3.92.2.8 termbuf

```
WORD* O_const::termbuf
```

Definition at line 2461 of file structs.h.

3.92.2.9 tabstring

```
WORD* O_const::tabstring
```

Definition at line 2462 of file structs.h.

3.92.2.10 wpos

```
UBYTE* O_const::wpos
```

Definition at line 2463 of file structs.h.

3.92.2.11 wpoin

```
UBYTE* O_const::wpoin
```

Definition at line 2464 of file structs.h.

3.92.2.12 DollarOutBuffer

```
UBYTE* O_const::DollarOutBuffer
```

Definition at line 2465 of file structs.h.

3.92.2.13 CurBufWrt

```
UBYTE* O_const::CurBufWrt
```

Definition at line 2466 of file structs.h.

3.92.2.14 FlipWORD

```
void(* O_const::FlipWORD) (UBYTE *)
```

Definition at line 2467 of file structs.h.

3.92.2.15 FlipLONG

```
void(* O_const::FlipLONG) (UBYTE *)
```

Definition at line 2468 of file structs.h.

3.92.2.16 FlipPOS

```
void(* O_const::FlipPOS) (UBYTE *)
```

Definition at line 2469 of file structs.h.

3.92.2.17 FlipPOINTER

```
void(* O_const::FlipPOINTER) (UBYTE *)
```

Definition at line 2470 of file structs.h.

3.92.2.18 ResizeData

```
void(* O_const::ResizeData) (UBYTE *, int, UBYTE *, int)
```

Definition at line 2471 of file structs.h.

3.92.2.19 ResizeWORD

```
void(* O_const::ResizeWORD) (UBYTE *, UBYTE *)
```

Definition at line 2472 of file structs.h.

3.92.2.20 ResizeNCWORD

```
void(* O_const::ResizeNCWORD) (UBYTE *, UBYTE *)
```

Definition at line 2473 of file structs.h.

3.92.2.21 ResizeLONG

```
void(* O_const::ResizeLONG) (UBYTE *, UBYTE *)
```

Definition at line 2474 of file structs.h.

3.92.2.22 ResizePOS

```
void(* O_const::ResizePOS) (UBYTE *, UBYTE *)
```

Definition at line 2475 of file structs.h.

3.92.2.23 ResizePOINTER

```
void(* O_const::ResizePOINTER) (UBYTE *, UBYTE *)
```

Definition at line 2476 of file structs.h.

3.92.2.24 CheckPower

```
void(* O_const::CheckPower) (UBYTE *)
```

Definition at line 2477 of file structs.h.

3.92.2.25 RenumberVec

```
void(* O_const::RenumberVec) (UBYTE *)
```

Definition at line 2478 of file structs.h.

3.92.2.26 Dictionaries

```
DICTIONARY** O_const::Dictionaries
```

Definition at line 2479 of file structs.h.

3.92.2.27 tensorList

```
UBYTE* O_const::tensorList
```

Definition at line 2480 of file structs.h.

3.92.2.28 inscheme

`WORD* O_const::inscheme`

Definition at line 2481 of file structs.h.

3.92.2.29 NumInBrack

`LONG O_const::NumInBrack`

Definition at line 2487 of file structs.h.

3.92.2.30 wlen

`LONG O_const::wlen`

Definition at line 2488 of file structs.h.

3.92.2.31 DollarOutSizeBuffer

`LONG O_const::DollarOutSizeBuffer`

Definition at line 2489 of file structs.h.

3.92.2.32 DollarInOutBuffer

`LONG O_const::DollarInOutBuffer`

Definition at line 2490 of file structs.h.

3.92.2.33 Optimize

`OPTIMIZE O_const::Optimize`

Definition at line 2495 of file structs.h.

3.92.2.34 OutInBuffer

`int O_const::OutInBuffer`

Definition at line 2496 of file structs.h.

3.92.2.35 NoSpacesInNumbers

`int O_const::NoSpacesInNumbers`

Definition at line 2497 of file structs.h.

3.92.2.36 BlockSpaces

```
int O_const::BlockSpaces
```

Definition at line 2498 of file structs.h.

3.92.2.37 CurrentDictionary

```
int O_const::CurrentDictionary
```

Definition at line 2499 of file structs.h.

3.92.2.38 SizeDictionaries

```
int O_const::SizeDictionaries
```

Definition at line 2500 of file structs.h.

3.92.2.39 NumDictionaries

```
int O_const::NumDictionaries
```

Definition at line 2501 of file structs.h.

3.92.2.40 CurDictNumbers

```
int O_const::CurDictNumbers
```

Definition at line 2502 of file structs.h.

3.92.2.41 CurDictVariables

```
int O_const::CurDictVariables
```

Definition at line 2503 of file structs.h.

3.92.2.42 CurDictSpecials

```
int O_const::CurDictSpecials
```

Definition at line 2504 of file structs.h.

3.92.2.43 CurDictFunWithArgs

```
int O_const::CurDictFunWithArgs
```

Definition at line 2505 of file structs.h.

3.92.2.44 CurDictNumberWarning

`int O_const::CurDictNumberWarning`

Definition at line 2506 of file structs.h.

3.92.2.45 CurDictNotInFunctions

`int O_const::CurDictNotInFunctions`

Definition at line 2507 of file structs.h.

3.92.2.46 CurDictInDollars

`int O_const::CurDictInDollars`

Definition at line 2508 of file structs.h.

3.92.2.47 gNumDictionaries

`int O_const::gNumDictionaries`

Definition at line 2509 of file structs.h.

3.92.2.48 schemenum

`WORD O_const::schemenum`

Definition at line 2513 of file structs.h.

3.92.2.49 transFlag

`WORD O_const::transFlag`

Definition at line 2514 of file structs.h.

3.92.2.50 powerFlag

`WORD O_const::powerFlag`

Definition at line 2515 of file structs.h.

3.92.2.51 mpower

`WORD O_const::mpower`

Definition at line 2516 of file structs.h.

3.92.2.52 resizeFlag

```
WORD O_const::resizeFlag
```

Definition at line 2517 of file structs.h.

3.92.2.53 bufferedInd

```
WORD O_const::bufferedInd
```

Definition at line 2518 of file structs.h.

3.92.2.54 OutSkip

```
WORD O_const::OutSkip
```

Definition at line 2519 of file structs.h.

3.92.2.55 IsBracket

```
WORD O_const::IsBracket
```

Definition at line 2520 of file structs.h.

3.92.2.56 InFbrack

```
WORD O_const::InFbrack
```

Definition at line 2521 of file structs.h.

3.92.2.57 PrintType

```
WORD O_const::PrintType
```

Definition at line 2522 of file structs.h.

3.92.2.58 FortFirst

```
WORD O_const::FortFirst
```

Definition at line 2523 of file structs.h.

3.92.2.59 DoubleFlag

```
WORD O_const::DoubleFlag
```

Definition at line 2524 of file structs.h.

3.92.2.60 IndentSpace

```
WORD O_const::IndentSpace
```

Definition at line 2525 of file structs.h.

3.92.2.61 FactorMode

```
WORD O_const::FactorMode
```

Definition at line 2526 of file structs.h.

3.92.2.62 FactorNum

```
WORD O_const::FactorNum
```

Definition at line 2527 of file structs.h.

3.92.2.63 ErrorBlock

```
WORD O_const::ErrorBlock
```

Definition at line 2528 of file structs.h.

3.92.2.64 OptimizationLevel

```
WORD O_const::OptimizationLevel
```

Definition at line 2529 of file structs.h.

3.92.2.65 FortDotChar

```
UBYTE O_const::FortDotChar
```

Definition at line 2530 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.93 objects Struct Reference

Data Fields

- MLONG position
- MLONG size
- MLONG date
- MLONG tablenumber
- MLONG uncompressed
- MLONG spare1
- MLONG spare2
- MLONG spare3
- char element [ELEMENTSIZE]

3.93.1 Detailed Description

Definition at line 95 of file minos.h.

3.93.2 Field Documentation

3.93.2.1 position

```
MLONG objects::position
```

Definition at line 97 of file minos.h.

3.93.2.2 size

```
MLONG objects::size
```

Definition at line 98 of file minos.h.

3.93.2.3 date

```
MLONG objects::date
```

Definition at line 99 of file minos.h.

3.93.2.4 tablenumber

```
MLONG objects::tablenumber
```

Definition at line 100 of file minos.h.

3.93.2.5 uncompressed

```
MLONG objects::uncompressed
```

Definition at line 101 of file minos.h.

3.93.2.6 spare1

```
MLONG objects::spare1
```

Definition at line 102 of file minos.h.

3.93.2.7 spare2

```
MLONG objects::spare2
```

Definition at line 103 of file minos.h.

3.93.2.8 spare3

```
MLONG objects::spare3
```

Definition at line 104 of file minos.h.

3.93.2.9 element

```
char objects::element[ELEMENTSIZE]
```

Definition at line 105 of file minos.h.

The documentation for this struct was generated from the following file:

- minos.h

3.94 OptDef Struct Reference

Data Fields

- const char * name
- const char * mean
- int defaultv
- int DUMMYPADDING

3.94.1 Detailed Description

Definition at line 101 of file grccparam.h.

3.94.2 Field Documentation

3.94.2.1 name

```
const char* OptDef::name
```

Definition at line 102 of file grccparam.h.

3.94.2.2 mean

```
const char* OptDef::mean
```

Definition at line 103 of file grccparam.h.

3.94.2.3 defaultv

```
int OptDef::defaultv
```

Definition at line 104 of file grccparam.h.

3.94.2.4 DUMMYPADDING

```
int OptDef::DUMMYPADDING
```

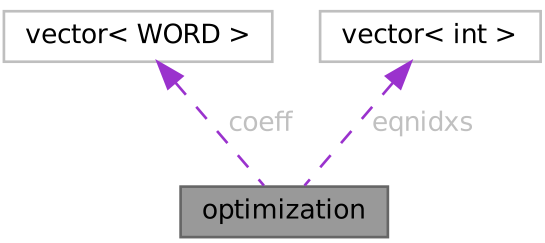
Definition at line 105 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.95 optimization Class Reference

Collaboration diagram for optimization:



Public Member Functions

- bool operator< (const optimization &a) const

Data Fields

- int type
- int arg1
- int arg2
- int improve
- vector< WORD > coeff
- vector< int > eqnidxs

3.95.1 Detailed Description

class Optimization

3.95.2 Description

This object represents an optimization. Its type is a number in the range 0 to 5. Depending on this type, the variables arg1, arg2 and coeff indicate:

type==0 : optimization of the form x[arg1] ^ arg2 (coeff=empty) type==1 : optimization of the form x[arg1] * x[arg2] (coeff=empty) type==2 : optimization of the form x[arg1] * coeff (arg2=0) type==3 : optimization of the form x[arg1] + coeff (arg2=0) type==4 : optimization of the form x[arg1] + x[arg2] (coeff=empty) type==5 : optimization of the form x[arg1] - x[arg2] (coeff=empty)

Here, "x[arg]" represents a symbol (if positive) or an extrasymbol (if negative). The represented symbol's id is ABS(x[arg])-1.

"eqns" is a list of equation, where this optimization can be performed.

"improve" is the total improvement of this optimization.

Definition at line 2670 of file optimize.cc.

3.95.3 Member Function Documentation

3.95.3.1 operator<()

```
bool optimization::operator< (
            const optimization & a ) const  [inline]
```

Definition at line 2676 of file optimize.cc.

3.95.4 Field Documentation

3.95.4.1 type

```
int optimization::type
```

Definition at line 2672 of file optimize.cc.

3.95.4.2 arg1

```
int optimization::arg1
```

Definition at line 2672 of file optimize.cc.

3.95.4.3 arg2

```
int optimization::arg2
```

Definition at line 2672 of file optimize.cc.

3.95.4.4 improve

```
int optimization::improve
```

Definition at line 2672 of file optimize.cc.

3.95.4.5 coeff

```
vector<WORD> optimization::coeff
```

Definition at line 2673 of file optimize.cc.

3.95.4.6 eqnidxs

```
vector<int> optimization::eqnidxs
```

Definition at line 2674 of file optimize.cc.

The documentation for this class was generated from the following file:

- optimize.cc

3.96 OPTIMIZE Struct Reference

Data Fields

- union {
    float fval
    int ival [2]
  } **mctsconstant**
- int horner
- int hornerdirection
- int method
- int mctstimelimit
- int mctsnumexpand
- int mctsnumkeep
- int mctsnumrepeat
- int greedytimelimit
- int greedyminnum
- int greedymaxperc
- int printstats
- int debugflags
- int schemeflags
- int mctsdecaymode
- int salter
- union {
    float fval
    int ival [2]
  } **saMaxT**
- union {
    float fval
    int ival [2]
  } **saMinT**
- int spare

3.96.1 Detailed Description

Definition at line 1312 of file structs.h.

3.96.2 Field Documentation

3.96.2.1 fval

```
float OPTIMIZE::fval
```

Definition at line 1314 of file structs.h.

3.96.2.2 ival

```
int OPTIMIZE::ival[2]
```

Definition at line 1315 of file structs.h.

3.96.2.3 horner

```
int OPTIMIZE::horner
```

Definition at line 1317 of file structs.h.

3.96.2.4 hornerdirection

```
int OPTIMIZE::hornerdirection
```

Definition at line 1318 of file structs.h.

3.96.2.5 method

```
int OPTIMIZE::method
```

Definition at line 1319 of file structs.h.

3.96.2.6 mctstimelimit

```
int OPTIMIZE::mctstimelimit
```

Definition at line 1320 of file structs.h.

3.96.2.7 mctsnumexpand

```
int OPTIMIZE::mctsnumexpand
```

Definition at line 1321 of file structs.h.

3.96.2.8 mctsnumkeep

```
int OPTIMIZE::mctsnumkeep
```

Definition at line 1322 of file structs.h.

3.96.2.9 mctsnumrepeat

```
int OPTIMIZE::mctsnumrepeat
```

Definition at line 1323 of file structs.h.

3.96.2.10 greedytimelimit

```
int OPTIMIZE::greedytimelimit
```

Definition at line 1324 of file structs.h.

3.96.2.11 greedyminnum

```
int OPTIMIZE::greedyminnum
```

Definition at line 1325 of file structs.h.

3.96.2.12 greedymaxperc

```
int OPTIMIZE::greedymaxperc
```

Definition at line 1326 of file structs.h.

3.96.2.13 printstats

```
int OPTIMIZE::printstats
```

Definition at line 1327 of file structs.h.

3.96.2.14 debugflags

```
int OPTIMIZE::debugflags
```

Definition at line 1328 of file structs.h.

3.96.2.15 schemeflags

```
int OPTIMIZE::schemeflags
```

Definition at line 1329 of file structs.h.

3.96.2.16 mctsdecaymode

```
int OPTIMIZE::mctsdecaymode
```

Definition at line 1330 of file structs.h.

3.96.2.17 salter

```
int OPTIMIZE::saIter
```

Definition at line 1331 of file structs.h.

3.96.2.18 spare

```
int OPTIMIZE::spare
```

Definition at line 1340 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.97 OPTIMIZERESULT Struct Reference

Public Member Functions

- **PADPOSITION** (2, 1, 0, 3, 0)

Data Fields

- WORD ∗ code
- UBYTE ∗ nameofexpr
- LONG codesize
- WORD exprnr
- WORD minvar
- WORD maxvar

3.97.1 Detailed Description

Definition at line 1343 of file structs.h.

3.97.2 Field Documentation

3.97.2.1 code

```
WORD* OPTIMIZERESULT::code
```

Definition at line 1344 of file structs.h.

3.97.2.2 nameofexpr

```
UBYTE* OPTIMIZERESULT::nameofexpr
```

Definition at line 1345 of file structs.h.

3.97.2.3 codesize

```
LONG OPTIMIZERESULT::codesize
```

Definition at line 1346 of file structs.h.

3.97.2.4 exprnr

```
WORD OPTIMIZERESULT::exprnr
```

Definition at line 1347 of file structs.h.

3.97.2.5 minvar

```
WORD OPTIMIZERESULT::minvar
```

Definition at line 1348 of file structs.h.

3.97.2.6 maxvar

```
WORD OPTIMIZERESULT::maxvar
```

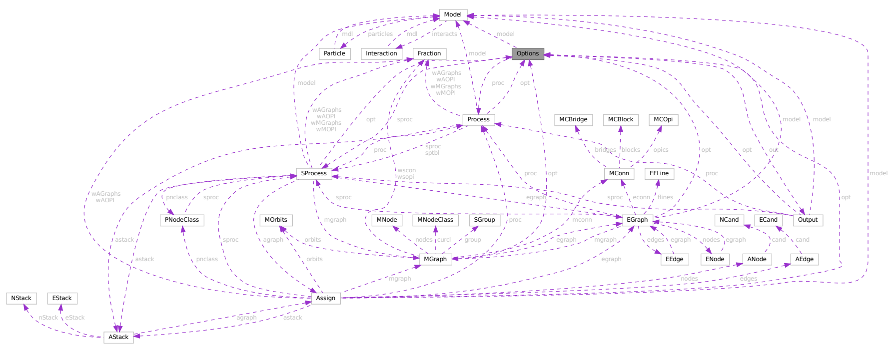
Definition at line 1349 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.98 Options Class Reference

Collaboration diagram for Options:



Public Member Functions

- void setDefaultValue (void)
- void setValue (int ind, int val)
- int getValue (int ind)
- void print (void)
- void setOutputF (Bool outgrf, const char *fname)
- void setOutputP (Bool outgrp, const char *fname)
- void printLevel (int l)
- void printModel (void)
- void setOutMG (OutEGB *omg, void *pt)
- void setOutAG (OutEG *oag, void *pt)
- void setEndMG (OutEGB *omg, void *pt)
- void setErExit (ErExit *ere, void *pt)
- const OptDef * getDef (void)
- void begin (Model *mdl)
- void end (void)
- void beginProc (Process *prc)
- void endProc (void)
- void beginSubProc (SProcess *sprc)
- void endSubProc (void)
- void newMGraph (MGraph *mgr)
- void newAGraph (EGraph *egr)
- void outModel (void)

Data Fields

- Model * model
- Process * proc
- SProcess * sproc
- Output * out
- OutEGB * outmg
- OutEGB * endmg
- OutEG * outag
- void * argmg
- void * argemg
- void * argag
- int values [GRCC_OPT_Size]
- int DUMMYPADDING
- double time0
- double time1

3.98.1 Detailed Description

Definition at line 88 of file grcc.h.

3.98.2 Constructor & Destructor Documentation

3.98.2.1 Options()

```
Options::Options (
            void )
```

Definition at line 117 of file grcc.cc.

3.98.2.2 ∼Options()

```
Options::∼Options (
            void )
```

Definition at line 149 of file grcc.cc.

3.98.3 Member Function Documentation

3.98.3.1 setDefaultValue()

```
void Options::setDefaultValue (
            void )
```

Definition at line 161 of file grcc.cc.

3.98.3.2 setValue()

```
void Options::setValue (
            int ind,
            int val )
```

Definition at line 212 of file grcc.cc.

3.98.3.3 getValue()

```
int Options::getValue (
            int ind )
```

Definition at line 227 of file grcc.cc.

3.98.3.4 print()

```
void Options::print (
            void )
```

Definition at line 241 of file grcc.cc.

3.98.3.5 setOutputF()

```
void Options::setOutputF (
            Bool outgrf,
            const char * fname )
```

Definition at line 264 of file grcc.cc.

3.98.3.6 setOutputP()

```
void Options::setOutputP (
            Bool outgrp,
            const char * fname )
```

Definition at line 271 of file grcc.cc.

3.98.3.7 printLevel()

```
void Options::printLevel (
            int l )
```

Definition at line 206 of file grcc.cc.

3.98.3.8 printModel()

```
void Options::printModel (
            void   )
```

Definition at line 278 of file grcc.cc.

3.98.3.9 setOutMG()

```
void Options::setOutMG (
            OutEGB * omg,
            void * pt )
```

Definition at line 178 of file grcc.cc.

3.98.3.10 setOutAG()

```
void Options::setOutAG (
            OutEG * oag,
            void * pt )
```

Definition at line 192 of file grcc.cc.

3.98.3.11 setEndMG()

```
void Options::setEndMG (
            OutEGB * omg,
            void * pt )
```

Definition at line 185 of file grcc.cc.

3.98.3.12 setErExit()

```
void Options::setErExit (
            ErExit * ere,
            void * pt )
```

Definition at line 199 of file grcc.cc.

3.98.3.13 getDef()

```
const OptDef * Options::getDef (
            void )
```

Definition at line 257 of file grcc.cc.

3.98.3.14 begin()

```
void Options::begin (
            Model * mdl )
```

Definition at line 299 of file grcc.cc.

3.98.3.15 end()

```
void Options::end (
            void )
```

Definition at line 319 of file grcc.cc.

3.98.3.16 beginProc()

```
void Options::beginProc (
            Process * prc )
```

Definition at line 351 of file grcc.cc.

3.98.3.17 endProc()

```
void Options::endProc (
            void )
```

Definition at line 367 of file grcc.cc.

3.98.3.18 beginSubProc()

```
void Options::beginSubProc (
            SProcess * sprc )
```

Definition at line 456 of file grcc.cc.

3.98.3.19 endSubProc()

```
void Options::endSubProc (
            void )
```

Definition at line 477 of file grcc.cc.

3.98.3.20 newMGraph()

```
void Options::newMGraph (
            MGraph * mgr )
```

Definition at line 546 of file grcc.cc.

3.98.3.21 newAGraph()

```
void Options::newAGraph (
            EGraph * egr )
```

Definition at line 583 of file grcc.cc.

3.98.3.22 outModel()

```
void Options::outModel (
            void )
```

Definition at line 290 of file grcc.cc.

3.98.4 Field Documentation

3.98.4.1 model

```
Model* Options::model
```

Definition at line 91 of file grcc.h.

3.98.4.2 proc

```
Process* Options::proc
```

Definition at line 92 of file grcc.h.

3.98.4.3 sproc

```
SProcess* Options::sproc
```

Definition at line 93 of file grcc.h.

3.98.4.4 out

```
Output* Options::out
```

Definition at line 94 of file grcc.h.

3.98.4.5 outmg

```
OutEGB* Options::outmg
```

Definition at line 95 of file grcc.h.

3.98.4.6 endmg

```
OutEGB* Options::endmg
```

Definition at line 96 of file grcc.h.

3.98.4.7 outag

```
OutEG* Options::outag
```

Definition at line 97 of file grcc.h.

3.98.4.8 argmg

```
void* Options::argmg
```

Definition at line 98 of file grcc.h.

3.98.4.9 argemg

```
void* Options::argemg
```

Definition at line 99 of file grcc.h.

3.98.4.10 argag

```
void* Options::argag
```

Definition at line 100 of file grcc.h.

3.98.4.11 values

```
int Options::values[GRCC_OPT_Size]
```

Definition at line 103 of file grcc.h.

3.98.4.12 DUMMYPADDING

```
int Options::DUMMYPADDING
```

Definition at line 105 of file grcc.h.

3.98.4.13 time0

```
double Options::time0
```

Definition at line 108 of file grcc.h.

3.98.4.14 time1

```
double Options::time1
```

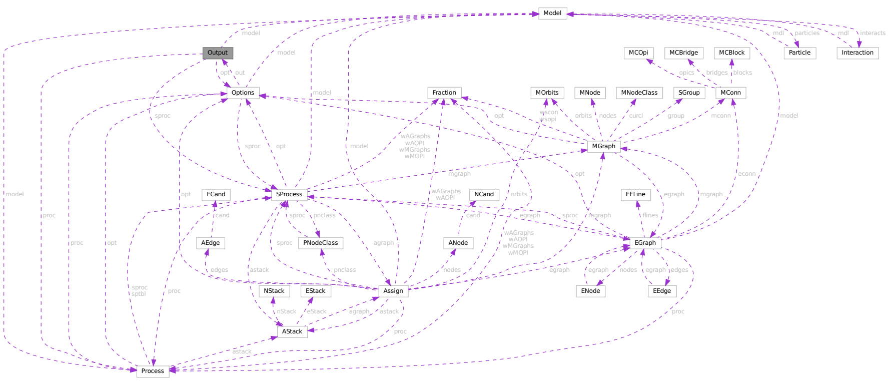
Definition at line 109 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.99 Output Class Reference

Collaboration diagram for Output:



Public Member Functions

- Output (Options *optn)
- void setOutgrf (const char *fname)
- void setOutgrp (const char *fname)
- Bool outBeginF (Model *mdl, Bool pr)
- Bool outBeginP (Model *mdl, Bool pr)
- void outEndF (void)
- void outEndP (void)
- void outProcBeginF (Process *prc)
- void outProcBeginP (Process *prc)
- void outProcBegin0 (int next, int couple, int loop)
- void outSProcBeginF (SProcess *sprc)
- void outSProcBeginP (SProcess *sprc)
- void outProcEndF (void)
- void outProcEndP (void)
- void outEGraphF (EGraph *egraph)
- void outEGraphP (EGraph *egraph)
- void outModelF (void)
- void outModelP (void)

Data Fields

- Options * opt
- Model * model
- Process * proc
- SProcess * sproc
- char * outgrf
- FILE * outgrfp
- char * outgrp
- FILE * outgrpp
- int procId
- Bool outproc

3.99.1 Detailed Description

Definition at line 144 of file grcc.h.

3.99.2 Constructor & Destructor Documentation

3.99.2.1 Output()

```
Output::Output (
            Options * optn )
```

Definition at line 637 of file grcc.cc.

3.99.2.2 ∼Output()

```
Output::~Output (
              void )
```

Definition at line 653 of file grcc.cc.

3.99.3 Member Function Documentation

3.99.3.1 setOutgrf()

```
void Output::setOutgrf (
              const char * fname )
```

Definition at line 684 of file grcc.cc.

3.99.3.2 setOutgrp()

```
void Output::setOutgrp (
              const char * fname )
```

Definition at line 701 of file grcc.cc.

3.99.3.3 outBeginF()

```
Bool Output::outBeginF (
              Model * mdl,
              Bool pr )
```

Definition at line 718 of file grcc.cc.

3.99.3.4 outBeginP()

```
Bool Output::outBeginP (
              Model * mdl,
              Bool pr )
```

Definition at line 756 of file grcc.cc.

3.99.3.5 outEndF()

```
void Output::outEndF (
              void )
```

Definition at line 787 of file grcc.cc.

3.99.3.6 outEndP()

```
void Output::outEndP (
            void )
```

Definition at line 803 of file grcc.cc.

3.99.3.7 outProcBeginF()

```
void Output::outProcBeginF (
            Process * prc )
```

Definition at line 819 of file grcc.cc.

3.99.3.8 outProcBeginP()

```
void Output::outProcBeginP (
            Process * prc )
```

Definition at line 864 of file grcc.cc.

3.99.3.9 outProcBegin0()

```
void Output::outProcBegin0 (
            int next,
            int couple,
            int loop )
```

Definition at line 886 of file grcc.cc.

3.99.3.10 outSProcBeginF()

```
void Output::outSProcBeginF (
            SProcess * sprc )
```

Definition at line 918 of file grcc.cc.

3.99.3.11 outSProcBeginP()

```
void Output::outSProcBeginP (
            SProcess * sprc )
```

Definition at line 973 of file grcc.cc.

3.99.3.12 outProcEndF()

```
void Output::outProcEndF (
            void )
```

Definition at line 993 of file grcc.cc.

3.99.3.13 outProcEndP()

```
void Output::outProcEndP (
            void )
```

Definition at line 1012 of file grcc.cc.

3.99.3.14 outEGraphF()

```
void Output::outEGraphF (
            EGraph * egraph )
```

Definition at line 1024 of file grcc.cc.

3.99.3.15 outEGraphP()

```
void Output::outEGraphP (
            EGraph * egraph )
```

Definition at line 1152 of file grcc.cc.

3.99.3.16 outModelF()

```
void Output::outModelF (
            void )
```

Definition at line 1254 of file grcc.cc.

3.99.3.17 outModelP()

```
void Output::outModelP (
            void )
```

Definition at line 1335 of file grcc.cc.

3.99.4 Field Documentation

3.99.4.1 opt

```
Options* Output::opt
```

Definition at line 147 of file grcc.h.

3.99.4.2 model

```
Model* Output::model
```

Definition at line 148 of file grcc.h.

3.99.4.3 proc

```
Process* Output::proc
```

Definition at line 149 of file grcc.h.

3.99.4.4 sproc

```
SProcess* Output::sproc
```

Definition at line 150 of file grcc.h.

3.99.4.5 outgrf

```
char* Output::outgrf
```

Definition at line 151 of file grcc.h.

3.99.4.6 outgrfp

```
FILE* Output::outgrfp
```

Definition at line 152 of file grcc.h.

3.99.4.7 outgrp

```
char* Output::outgrp
```

Definition at line 153 of file grcc.h.

3.99.4.8 outgrpp

```
FILE* Output::outgrpp
```

Definition at line 154 of file grcc.h.

3.99.4.9 procId

```
int Output::procId
```

Definition at line 155 of file grcc.h.

3.99.4.10 outproc

```
Bool Output::outproc
```

Definition at line 156 of file grcc.h.

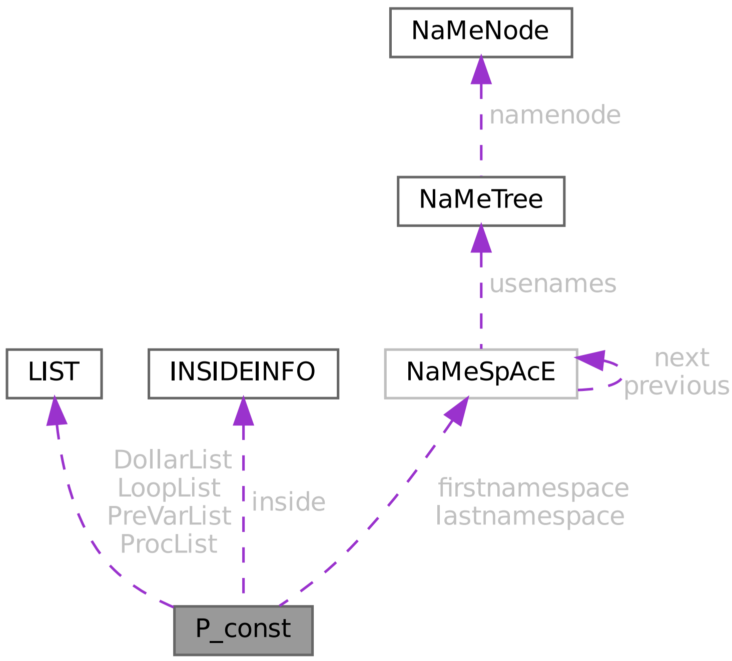
The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.100 P_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for P_const:



Data Fields

- LIST DollarList
- LIST PreVarList
- LIST LoopList
- LIST ProcList
- INSIDEINFO inside

- NAMESPACE * firstnamespace
- NAMESPACE * lastnamespace
- UBYTE * fullname
- UBYTE ** PreSwitchStrings
- UBYTE * preStart
- UBYTE * preStop
- UBYTE * preFill
- UBYTE * procedureExtension
- UBYTE * cprocedureExtension
- LONG * PreAssignStack
- int * PreIfStack
- int * PreSwitchModes
- int * PreTypes
- LONG StopWatchZero
- LONG InOutBuf
- LONG pSize
- int PreAssignFlag
- int PreContinuation
- int PreproFlag
- int iBufError
- int PreOut
- int PreSwitchLevel
- int NumPreSwitchStrings
- int MaxPreTypes
- int NumPreTypes
- int MaxPreIfLevel
- int PreIfLevel
- int PreInsideLevel
- int DelayPrevar
- int AllowDelay
- int lhdollarerror
- int eat
- int gNumPre
- int PreDebug
- int OpenDictionary
- int PreAssignLevel
- int MaxPreAssignLevel
- int fullnamesize
- WORD DebugFlag
- WORD preError
- UBYTE ComChar
- UBYTE cComChar

3.100.1 Detailed Description

The P_const struct is part of the global data and resides in the ALLGLOBALS struct A under the name P We see it used with the macro AP as in AP.InOutBuf It contains objects that have dealings with the preprocessor.

Definition at line 1628 of file structs.h.

3.100.2 Field Documentation

3.100.2.1 DollarList

```
LIST P_const::DollarList
```

Definition at line 1629 of file structs.h.

3.100.2.2 PreVarList

```
LIST P_const::PreVarList
```

Definition at line 1631 of file structs.h.

3.100.2.3 LoopList

```
LIST P_const::LoopList
```

Definition at line 1633 of file structs.h.

3.100.2.4 ProcList

```
LIST P_const::ProcList
```

Definition at line 1634 of file structs.h.

3.100.2.5 inside

```
INSIDEINFO P_const::inside
```

Definition at line 1635 of file structs.h.

3.100.2.6 firstnamespace

```
NAMESPACE* P_const::firstnamespace
```

Definition at line 1636 of file structs.h.

3.100.2.7 lastnamespace

```
NAMESPACE* P_const::lastnamespace
```

Definition at line 1637 of file structs.h.

3.100.2.8 fullname

```
UBYTE* P_const::fullname
```

Definition at line 1638 of file structs.h.

3.100.2.9 PreSwitchStrings

```
UBYTE** P_const::PreSwitchStrings
```

Definition at line 1639 of file structs.h.

3.100.2.10 preStart

```
UBYTE* P_const::preStart
```

Definition at line 1640 of file structs.h.

3.100.2.11 preStop

```
UBYTE* P_const::preStop
```

Definition at line 1641 of file structs.h.

3.100.2.12 preFill

```
UBYTE* P_const::preFill
```

Definition at line 1642 of file structs.h.

3.100.2.13 procedureExtension

```
UBYTE* P_const::procedureExtension
```

Definition at line 1643 of file structs.h.

3.100.2.14 cprocedureExtension

```
UBYTE* P_const::cprocedureExtension
```

Definition at line 1644 of file structs.h.

3.100.2.15 PreAssignStack

```
LONG* P_const::PreAssignStack
```

Definition at line 1645 of file structs.h.

3.100.2.16 PreIfStack

`int* P_const::PreIfStack`

Definition at line 1646 of file structs.h.

3.100.2.17 PreSwitchModes

`int* P_const::PreSwitchModes`

Definition at line 1647 of file structs.h.

3.100.2.18 PreTypes

`int* P_const::PreTypes`

Definition at line 1648 of file structs.h.

3.100.2.19 StopWatchZero

`LONG P_const::StopWatchZero`

Definition at line 1652 of file structs.h.

3.100.2.20 InOutBuf

`LONG P_const::InOutBuf`

Definition at line 1653 of file structs.h.

3.100.2.21 pSize

`LONG P_const::pSize`

Definition at line 1654 of file structs.h.

3.100.2.22 PreAssignFlag

`int P_const::PreAssignFlag`

Definition at line 1655 of file structs.h.

3.100.2.23 PreContinuation

`int P_const::PreContinuation`

Definition at line 1656 of file structs.h.

3.100.2.24 PreproFlag

```
int P_const::PreproFlag
```

Definition at line 1657 of file structs.h.

3.100.2.25 iBufError

```
int P_const::iBufError
```

Definition at line 1658 of file structs.h.

3.100.2.26 PreOut

```
int P_const::PreOut
```

Definition at line 1659 of file structs.h.

3.100.2.27 PreSwitchLevel

```
int P_const::PreSwitchLevel
```

Definition at line 1660 of file structs.h.

3.100.2.28 NumPreSwitchStrings

```
int P_const::NumPreSwitchStrings
```

Definition at line 1661 of file structs.h.

3.100.2.29 MaxPreTypes

```
int P_const::MaxPreTypes
```

Definition at line 1662 of file structs.h.

3.100.2.30 NumPreTypes

```
int P_const::NumPreTypes
```

Definition at line 1663 of file structs.h.

3.100.2.31 MaxPreIfLevel

```
int P_const::MaxPreIfLevel
```

Definition at line 1664 of file structs.h.

3.100.2.32 PreIfLevel

```
int P_const::PreIfLevel
```

Definition at line 1665 of file structs.h.

3.100.2.33 PreInsideLevel

```
int P_const::PreInsideLevel
```

Definition at line 1666 of file structs.h.

3.100.2.34 DelayPrevar

```
int P_const::DelayPrevar
```

Definition at line 1667 of file structs.h.

3.100.2.35 AllowDelay

```
int P_const::AllowDelay
```

Definition at line 1668 of file structs.h.

3.100.2.36 lhdollarerror

```
int P_const::lhdollarerror
```

Definition at line 1669 of file structs.h.

3.100.2.37 eat

```
int P_const::eat
```

Definition at line 1670 of file structs.h.

3.100.2.38 gNumPre

```
int P_const::gNumPre
```

Definition at line 1671 of file structs.h.

3.100.2.39 PreDebug

```
int P_const::PreDebug
```

Definition at line 1672 of file structs.h.

3.100.2.40 OpenDictionary

`int P_const::OpenDictionary`

Definition at line 1673 of file structs.h.

3.100.2.41 PreAssignLevel

`int P_const::PreAssignLevel`

Definition at line 1674 of file structs.h.

3.100.2.42 MaxPreAssignLevel

`int P_const::MaxPreAssignLevel`

Definition at line 1675 of file structs.h.

3.100.2.43 fullnamesize

`int P_const::fullnamesize`

Definition at line 1676 of file structs.h.

3.100.2.44 DebugFlag

`WORD P_const::DebugFlag`

Definition at line 1677 of file structs.h.

3.100.2.45 preError

`WORD P_const::preError`

Definition at line 1678 of file structs.h.

3.100.2.46 ComChar

`UBYTE P_const::ComChar`

Definition at line 1679 of file structs.h.

3.100.2.47 cComChar

```
UBYTE P_const::cComChar
```

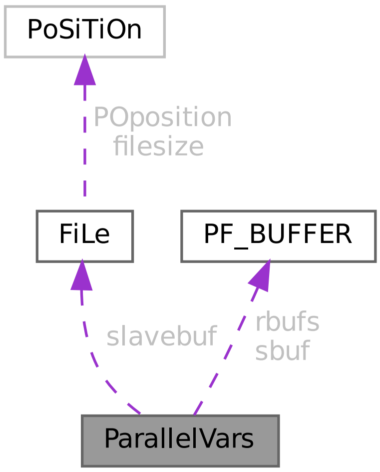
Definition at line 1680 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.101 ParallelVars Struct Reference

Collaboration diagram for ParallelVars:



Public Member Functions

- **PADPOSITION** (2, 0, 8, 2, 0)

Data Fields

- FILEHANDLE slavebuf
- PF_BUFFER ∗ sbuf
- PF_BUFFER ∗∗ rbufs
- int me
- int numtasks
- int parallel
- int rhsInParallel
- int mkSlaveInfile
- int exprbufsize
- int exprtodo
- int log
- WORD numsbufs
- WORD numrbufs

3.101.1 Detailed Description

Definition at line 170 of file parallel.h.

3.101.2 Field Documentation

3.101.2.1 slavebuf

`FILEHANDLE ParallelVars::slavebuf`

Definition at line 171 of file parallel.h.

3.101.2.2 sbuf

`PF_BUFFER* ParallelVars::sbuf`

Definition at line 173 of file parallel.h.

3.101.2.3 rbufs

`PF_BUFFER** ParallelVars::rbufs`

Definition at line 174 of file parallel.h.

3.101.2.4 me

`int ParallelVars::me`

Definition at line 175 of file parallel.h.

3.101.2.5 numtasks

`int ParallelVars::numtasks`

Definition at line 176 of file parallel.h.

3.101.2.6 parallel

`int ParallelVars::parallel`

Definition at line 177 of file parallel.h.

3.101.2.7 rhsInParallel

`int ParallelVars::rhsInParallel`

Definition at line 179 of file parallel.h.

3.101.2.8 mkSlaveInfile

`int ParallelVars::mkSlaveInfile`

Definition at line 180 of file parallel.h.

3.101.2.9 exprbufsize

`int ParallelVars::exprbufsize`

Definition at line 181 of file parallel.h.

3.101.2.10 exprtodo

`int ParallelVars::exprtodo`

Definition at line 182 of file parallel.h.

3.101.2.11 log

`int ParallelVars::log`

Definition at line 183 of file parallel.h.

3.101.2.12 numsbufs

`WORD ParallelVars::numsbufs`

Definition at line 184 of file parallel.h.

3.101.2.13 numrbufs

`WORD ParallelVars::numrbufs`

Definition at line 185 of file parallel.h.

The documentation for this struct was generated from the following file:

- parallel.h

3.102 PaRtI Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ∗ psize
- WORD ∗ args
- WORD ∗ nargs
- WORD ∗ nfun
- WORD numargs
- WORD numpart
- WORD where

3.102.1 Detailed Description

The struct PARTI is used to help determining whether a partition_ function can be replaced.

Definition at line 1120 of file structs.h.

3.102.2 Field Documentation

3.102.2.1 psize

```
WORD* PaRtI::psize
```

Definition at line 1121 of file structs.h.

3.102.2.2 args

```
WORD* PaRtI::args
```

Definition at line 1122 of file structs.h.

3.102.2.3 nargs

```
WORD* PaRtI::nargs
```

Definition at line 1123 of file structs.h.

3.102.2.4 nfun

```
WORD* PaRtI::nfun
```

Definition at line 1124 of file structs.h.

3.102.2.5 numargs

```
WORD PaRtI::numargs
```

Definition at line 1125 of file structs.h.

3.102.2.6 numpart

```
WORD PaRtI::numpart
```

Definition at line 1126 of file structs.h.

3.102.2.7 where

```
WORD PaRtI::where
```

Definition at line 1127 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.103 PaRtIcLe Struct Reference

Data Fields

- WORD number
- WORD spin
- WORD mass
- WORD type

3.103.1 Detailed Description

Definition at line 619 of file structs.h.

3.103.2 Field Documentation

3.103.2.1 number

```
WORD PaRtIcLe::number
```

Definition at line 620 of file structs.h.

3.103.2.2 spin

```
WORD PaRtIcLe::spin
```

Definition at line 621 of file structs.h.

3.103.2.3 mass

```
WORD PaRtIcLe::mass
```

Definition at line 622 of file structs.h.

3.103.2.4 type

```
WORD PaRtIcLe::type
```

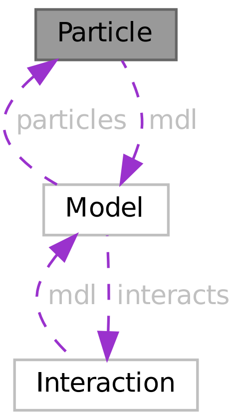
Definition at line 623 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.104 Particle Class Reference

Collaboration diagram for Particle:



Public Member Functions

- Particle (Model ∗modl, int pid, PInput ∗pinp)
- char ∗ particleName (int p)
- int particleCode (int p)
- char ∗ **interactionName** (int p)
- char ∗ aparticle (void)
- void prParticle (void)
- int isNeutral (void)
- const char ∗ typeName (void)
- const char ∗ typeGName (void)

Data Fields

- Model ∗ mdl
- char ∗ name
- char ∗ aname
- int id
- int ptype
- int neutral
- int pcode
- int acode
- int cmindeg
- int cmaxdeg
- int extonly

3.104.1 Detailed Description

Definition at line 205 of file grcc.h.

3.104.2 Constructor & Destructor Documentation

3.104.2.1 Particle()

```
Particle::Particle (
            Model * modl,
            int pid,
            PInput * pinp )
```

Definition at line 1422 of file grcc.cc.

3.104.2.2 ∼Particle()

```
Particle::~Particle (
            void )
```

Definition at line 1499 of file grcc.cc.

3.104.3 Member Function Documentation

3.104.3.1 particleName()

```
char * Particle::particleName (
             int p )
```

Definition at line 1506 of file grcc.cc.

3.104.3.2 particleCode()

```
int Particle::particleCode (
             int p )
```

Definition at line 1516 of file grcc.cc.

3.104.3.3 aparticle()

```
char * Particle::aparticle (
             void )
```

Definition at line 1544 of file grcc.cc.

3.104.3.4 prParticle()

```
void Particle::prParticle (
             void )
```

Definition at line 1556 of file grcc.cc.

3.104.3.5 isNeutral()

```
int Particle::isNeutral (
             void )
```

Definition at line 1526 of file grcc.cc.

3.104.3.6 typeName()

```
const char * Particle::typeName (
             void )
```

Definition at line 1532 of file grcc.cc.

3.104.3.7 typeGName()

```
const char * Particle::typeGName (
            void )
```

Definition at line 1538 of file grcc.cc.

3.104.4 Field Documentation

3.104.4.1 mdl

```
Model* Particle::mdl
```

Definition at line 207 of file grcc.h.

3.104.4.2 name

```
char* Particle::name
```

Definition at line 208 of file grcc.h.

3.104.4.3 aname

```
char* Particle::aname
```

Definition at line 209 of file grcc.h.

3.104.4.4 id

```
int Particle::id
```

Definition at line 210 of file grcc.h.

3.104.4.5 ptype

```
int Particle::ptype
```

Definition at line 211 of file grcc.h.

3.104.4.6 neutral

```
int Particle::neutral
```

Definition at line 212 of file grcc.h.

3.104.4.7 pcode

```
int Particle::pcode
```

Definition at line 213 of file grcc.h.

3.104.4.8 acode

```
int Particle::acode
```

Definition at line 214 of file grcc.h.

3.104.4.9 cmindeg

```
int Particle::cmindeg
```

Definition at line 215 of file grcc.h.

3.104.4.10 cmaxdeg

```
int Particle::cmaxdeg
```

Definition at line 216 of file grcc.h.

3.104.4.11 extonly

```
int Particle::extonly
```

Definition at line 218 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.105 PeRmUtE Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ∗ objects
- WORD sign
- WORD n
- WORD cycle [MAXMATCH]

3.105.1 Detailed Description

The struct PERM is used to generate all permutations when the pattern matcher has to try to match (anti)symmetric functions.

Definition at line 1078 of file structs.h.

3.105.2 Field Documentation

3.105.2.1 objects

```
WORD* PeRmUtE::objects
```

Definition at line 1079 of file structs.h.

3.105.2.2 sign

```
WORD PeRmUtE::sign
```

Definition at line 1080 of file structs.h.

3.105.2.3 n

```
WORD PeRmUtE::n
```

Definition at line 1081 of file structs.h.

3.105.2.4 cycle

```
WORD PeRmUtE::cycle[MAXMATCH]
```

Definition at line 1082 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.106 PeRmUtEp Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ** objects
- WORD sign
- WORD n
- WORD cycle [MAXMATCH]

3.106.1 Detailed Description

Like struct PERM but works with pointers.

Definition at line 1090 of file structs.h.

3.106.2 Field Documentation

3.106.2.1 objects

```
WORD** PeRmUtEp::objects
```

Definition at line 1091 of file structs.h.

3.106.2.2 sign

```
WORD PeRmUtEp::sign
```

Definition at line 1092 of file structs.h.

3.106.2.3 n

```
WORD PeRmUtEp::n
```

Definition at line 1093 of file structs.h.

3.106.2.4 cycle

```
WORD PeRmUtEp::cycle[MAXMATCH]
```

Definition at line 1094 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.107 PF_BUFFER Struct Reference

```
#include <parallel.h>
```

Data Fields

- WORD ∗∗ buff
- WORD ∗∗ fill
- WORD ∗∗ full
- WORD ∗∗ stop
- MPI_Status ∗ status
- MPI_Status ∗ retstat
- MPI_Request ∗ request
- MPI_Datatype ∗ type
- int ∗ index
- int ∗ tag
- int ∗ from
- int numbufs
- int active

3.107.1 Detailed Description

A struct for nonblocking, unbuffered send of the sorted terms in the PObuffers back to the master using several "rotating" PObuffers.

Definition at line 148 of file parallel.h.

3.107.2 Field Documentation

3.107.2.1 buff

```
WORD** PF_BUFFER::buff
```

Definition at line 149 of file parallel.h.

3.107.2.2 fill

```
WORD** PF_BUFFER::fill
```

Definition at line 150 of file parallel.h.

3.107.2.3 full

```
WORD** PF_BUFFER::full
```

Definition at line 151 of file parallel.h.

3.107.2.4 stop

```
WORD** PF_BUFFER::stop
```

Definition at line 152 of file parallel.h.

3.107.2.5 status

`MPI_Status* PF_BUFFER::status`

Definition at line 153 of file parallel.h.

3.107.2.6 retstat

`MPI_Status* PF_BUFFER::retstat`

Definition at line 154 of file parallel.h.

3.107.2.7 request

`MPI_Request* PF_BUFFER::request`

Definition at line 155 of file parallel.h.

3.107.2.8 type

`MPI_Datatype* PF_BUFFER::type`

Definition at line 156 of file parallel.h.

3.107.2.9 index

`int* PF_BUFFER::index`

Definition at line 157 of file parallel.h.

3.107.2.10 tag

`int* PF_BUFFER::tag`

Definition at line 158 of file parallel.h.

3.107.2.11 from

`int* PF_BUFFER::from`

Definition at line 159 of file parallel.h.

3.107.2.12 numbufs

`int PF_BUFFER::numbufs`

Definition at line 160 of file parallel.h.

3.107.2.13 active

```
int PF_BUFFER::active
```

Definition at line 161 of file parallel.h.

The documentation for this struct was generated from the following file:

- parallel.h

3.108 PGInput Struct Reference

Data Fields

- int cdeg
- int ctyp
- int cnum
- int cple
- const char * pname
- long pcode

3.108.1 Detailed Description

Definition at line 176 of file grccparam.h.

3.108.2 Field Documentation

3.108.2.1 cdeg

```
int PGInput::cdeg
```

Definition at line 177 of file grccparam.h.

3.108.2.2 ctyp

```
int PGInput::ctyp
```

Definition at line 178 of file grccparam.h.

3.108.2.3 cnum

```
int PGInput::cnum
```

Definition at line 179 of file grccparam.h.

3.108.2.4 cple

```
int PGInput::cple
```

Definition at line 180 of file grccparam.h.

3.108.2.5 pname

```
const char* PGInput::pname
```

Definition at line 182 of file grccparam.h.

3.108.2.6 pcode

```
long PGInput::pcode
```

Definition at line 184 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.109 PInput Struct Reference

Data Fields

- const char * name
- const char * aname
- int pcode
- int acode
- const char * ptypen
- int ptypec
- int extonly

3.109.1 Detailed Description

Definition at line 127 of file grccparam.h.

3.109.2 Field Documentation

3.109.2.1 name

```
const char* PInput::name
```

Definition at line 128 of file grccparam.h.

3.109.2.2 aname

```
const char* PInput::aname
```

Definition at line 129 of file grccparam.h.

3.109.2.3 pcode

```
int PInput::pcode
```

Definition at line 130 of file grccparam.h.

3.109.2.4 acode

```
int PInput::acode
```

Definition at line 131 of file grccparam.h.

3.109.2.5 ptypen

```
const char* PInput::ptypen
```

Definition at line 132 of file grccparam.h.

3.109.2.6 ptypec

```
int PInput::ptypec
```

Definition at line 133 of file grccparam.h.

3.109.2.7 extonly

```
int PInput::extonly
```

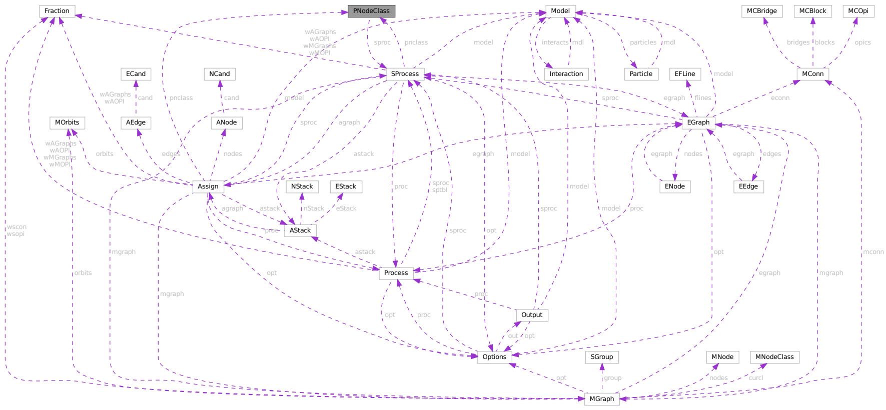
Definition at line 134 of file grccparam.h.

The documentation for this struct was generated from the following file:

- grccparam.h

3.110 PNodeClass Class Reference

Collaboration diagram for PNodeClass:



Public Member Functions

- PNodeClass (SProcess ∗sprc, int nnods, int nclss, NCInput ∗cls)
- PNodeClass (SProcess ∗sprc, int nnods, int nclss, int ∗dgs, int ∗typ, int ∗ptcl, int ∗cpl, int ∗cnt, int ∗cmind, int ∗cmaxd)
- void prPNodeClass (void)
- void prElem (int e)

Data Fields

- SProcess ∗ sproc
- int ∗ deg
- int ∗ type
- int ∗ particle
- int ∗ couple
- int ∗ cmindeg
- int ∗ cmaxdeg
- int ∗ count
- int ∗ cl2nd
- int ∗ nd2cl
- int ∗ cl2mcl
- int nnodes
- int nclass

3.110.1 Detailed Description

Definition at line 325 of file grcc.h.

3.110.2 Constructor & Destructor Documentation

3.110.2.1 PNodeClass() [1/2]

```
PNodeClass::PNodeClass (
            SProcess * sprc,
            int nnods,
            int nclss,
            NCInput * cls )
```

Definition at line 2325 of file grcc.cc.

3.110.2.2 PNodeClass() [2/2]

```
PNodeClass::PNodeClass (
            SProcess * sprc,
            int nnods,
            int nclss,
            int * dgs,
            int * typ,
            int * ptcl,
            int * cpl,
            int * cnt,
            int * cmind,
            int * cmaxd )
```

Definition at line 2399 of file grcc.cc.

3.110.2.3 ∼PNodeClass()

```
PNodeClass::~PNodeClass (
            void )
```

Definition at line 2477 of file grcc.cc.

3.110.3 Member Function Documentation

3.110.3.1 prPNodeClass()

```
void PNodeClass::prPNodeClass (
            void )
```

Definition at line 2492 of file grcc.cc.

3.110.3.2 prElem()

```
void PNodeClass::prElem (
            int e )
```

Definition at line 2504 of file grcc.cc.

3.110.4 Field Documentation

3.110.4.1 sproc

`SProcess* PNodeClass::sproc`

Definition at line 327 of file grcc.h.

3.110.4.2 deg

`int* PNodeClass::deg`

Definition at line 328 of file grcc.h.

3.110.4.3 type

`int* PNodeClass::type`

Definition at line 329 of file grcc.h.

3.110.4.4 particle

`int* PNodeClass::particle`

Definition at line 330 of file grcc.h.

3.110.4.5 couple

`int* PNodeClass::couple`

Definition at line 331 of file grcc.h.

3.110.4.6 cmindeg

`int* PNodeClass::cmindeg`

Definition at line 332 of file grcc.h.

3.110.4.7 cmaxdeg

`int* PNodeClass::cmaxdeg`

Definition at line 333 of file grcc.h.

3.110.4.8 count

`int* PNodeClass::count`

Definition at line 334 of file grcc.h.

3.110.4.9 cl2nd

`int* PNodeClass::cl2nd`

Definition at line 335 of file grcc.h.

3.110.4.10 nd2cl

`int* PNodeClass::nd2cl`

Definition at line 336 of file grcc.h.

3.110.4.11 cl2mcl

`int* PNodeClass::cl2mcl`

Definition at line 337 of file grcc.h.

3.110.4.12 nnodes

`int PNodeClass::nnodes`

Definition at line 338 of file grcc.h.

3.110.4.13 nclass

`int PNodeClass::nclass`

Definition at line 339 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.111 flint::poly Class Reference

Public Member Functions

- void print (const string &text)

Data Fields

- fmpz_poly_t d

3.111.1 Detailed Description

Definition at line 61 of file flintinterface.h.

3.111.2 Constructor & Destructor Documentation

3.111.2.1 poly()

```
flint::poly::poly ( ) [inline]
```

Definition at line 64 of file flintinterface.h.

3.111.2.2 ∼poly()

```
flint::poly::∼poly ( ) [inline]
```

Definition at line 65 of file flintinterface.h.

3.111.3 Member Function Documentation

3.111.3.1 print()

```
void flint::poly::print (
            const string & text ) [inline]
```

Definition at line 66 of file flintinterface.h.

3.111.4 Field Documentation

3.111.4.1 d

```
fmpz_poly_t flint::poly::d
```

Definition at line 63 of file flintinterface.h.

The documentation for this class was generated from the following file:

- flintinterface.h

3.112 poly Class Reference

Public Member Functions

- poly (PHEAD int, WORD=-1, WORD=1)
- poly (PHEAD const UWORD *, WORD, WORD=-1, WORD=1)
- poly (const poly &, WORD=-1, WORD=1)
- poly & operator+= (const poly &)
- poly & operator-= (const poly &)
- poly & operator*= (const poly &)
- poly & operator/= (const poly &)
- poly & operator%= (const poly &)
- const poly operator+ (const poly &) const
- const poly operator- (const poly &) const
- const poly operator* (const poly &) const
- const poly operator/ (const poly &) const
- const poly operator% (const poly &) const
- bool operator== (const poly &) const
- bool operator!= (const poly &) const
- poly & operator= (const poly &)
- WORD & operator[ ] (int)
- const WORD & operator[ ] (int) const
- void termscopy (const WORD *, int, int)
- void check_memory (int)
- void expand_memory (int)
- bool is_zero () const
- bool is_one () const
- bool is_integer () const
- bool is_monomial () const
- int is_dense_univariate () const
- int sign () const
- int degree (int) const
- int total_degree () const
- int first_variable () const
- int number_of_terms () const
- const std::vector< int > all_variables () const
- const poly integer_lcoeff () const
- const poly lcoeff_univar (int) const
- const poly lcoeff_multivar (int) const
- const poly coefficient (int, int) const
- const poly derivative (int) const
- void setmod (WORD, WORD=1)
- void coefficients_modulo (UWORD *, WORD, bool)
- int size_of_form_notation () const
- int size_of_form_notation_with_den (WORD) const
- const poly & normalize ()
- const std::string to_string () const
- WORD last_monomial_index () const
- WORD * last_monomial () const
- int compare_degree_vector (const poly &) const
- std::vector< int > degree_vector () const
- int **compare_degree_vector** (const std::vector< int > &) const

Static Public Member Functions

- static const poly simple_poly (PHEAD int, int=0, int=1, int=0, int=1)
- static const poly simple_poly (PHEAD int, const poly &, int=1, int=0, int=1)
- static void get_variables (PHEAD const std::vector< WORD * > &, bool, bool)
- static const poly argument_to_poly (PHEAD WORD *, bool, bool, poly *den=NULL)
- static void poly_to_argument (const poly &, WORD *, bool)
- static void poly_to_argument_with_den (const poly &, WORD, const UWORD *, WORD *, bool)
- static const poly from_coefficient_list (PHEAD const std::vector< WORD > &, int, WORD)
- static const std::vector< WORD > to_coefficient_list (const poly &)
- static const std::vector< WORD > coefficient_list_divmod (const std::vector< WORD > &, const std::vector< WORD > &, WORD, int)
- static int monomial_compare (PHEAD const WORD *, const WORD *)
- static void add (const poly &, const poly &, poly &)
- static void sub (const poly &, const poly &, poly &)
- static void mul (const poly &, const poly &, poly &)
- static void div (const poly &, const poly &, poly &)
- static void mod (const poly &, const poly &, poly &)
- static void divmod (const poly &, const poly &, poly &, poly &, bool only_divides)
- static bool divides (const poly &, const poly &)
- static void mul_one_term (const poly &, const poly &, poly &)
- static void mul_univar (const poly &, const poly &, poly &, int)
- static void mul_heap (const poly &, const poly &, poly &)
- static void divmod_one_term (const poly &, const poly &, poly &, poly &, bool)
- static void divmod_univar (const poly &, const poly &, poly &, poly &, int, bool)
- static void divmod_heap (const poly &, const poly &, poly &, poly &, bool, bool, bool &)
- static void push_heap (PHEAD WORD **, int)
- static void pop_heap (PHEAD WORD **, int)

Data Fields

- WORD * terms
- LONG size_of_terms
- WORD modp
- WORD modn

3.112.1 Detailed Description

Definition at line 53 of file poly.h.

3.112.2 Constructor & Destructor Documentation

3.112.2.1 poly() [1/3]

```
poly::poly (
            PHEAD int a,
            WORD modp = -1,
            WORD modn = 1 )
```

Definition at line 54 of file poly.cc.

3.112.2.2 poly() [2/3]

```
poly::poly (
            PHEAD const UWORD * a,
            WORD na,
            WORD modp = -1,
            WORD modn = 1 )
```

Definition at line 83 of file poly.cc.

3.112.2.3 poly() [3/3]

```
poly::poly (
            const poly & a,
            WORD modp = -1,
            WORD modn = 1 )
```

Definition at line 111 of file poly.cc.

3.112.2.4 ∼poly()

```
poly::∼poly ( )
```

Definition at line 133 of file poly.cc.

3.112.3 Member Function Documentation

3.112.3.1 operator+=()

```
poly & poly::operator+= (
            const poly & a )
```

Definition at line 1966 of file poly.cc.

3.112.3.2 operator-=()

```
poly & poly::operator-= (
            const poly & a )
```

Definition at line 1967 of file poly.cc.

3.112.3.3 operator∗=()

```
poly & poly::operator*= (
            const poly & a )
```

Definition at line 1968 of file poly.cc.

3.112.3.4 operator/=()

```
poly & poly::operator/= (
            const poly & a )
```

Definition at line 1969 of file poly.cc.

3.112.3.5 operator%=()

```
poly & poly::operator%= (
            const poly & a )
```

Definition at line 1970 of file poly.cc.

3.112.3.6 operator+()

```
const poly poly::operator+ (
            const poly & a ) const
```

Definition at line 1959 of file poly.cc.

3.112.3.7 operator-()

```
const poly poly::operator- (
            const poly & a ) const
```

Definition at line 1960 of file poly.cc.

3.112.3.8 operator∗()

```
const poly poly::operator* (
            const poly & a ) const
```

Definition at line 1961 of file poly.cc.

3.112.3.9 operator/()

```
const poly poly::operator/ (
            const poly & a ) const
```

Definition at line 1962 of file poly.cc.

3.112.3.10 operator%()

```
const poly poly::operator% (
            const poly & a ) const
```

Definition at line 1963 of file poly.cc.

3.112.3.11 operator==()

```
bool poly::operator== (
            const poly & a ) const
```

Definition at line 1973 of file poly.cc.

3.112.3.12 operator"!=()

```
bool poly::operator!= (
            const poly & a ) const
```

Definition at line 1979 of file poly.cc.

3.112.3.13 operator=()

```
poly & poly::operator= (
            const poly & a )
```

Definition at line 1939 of file poly.cc.

3.112.3.14 operator[]() [1/2]

```
WORD & poly::operator[] (
            int i ) [inline]
```

Definition at line 249 of file poly.h.

3.112.3.15 operator[]() [2/2]

```
const WORD & poly::operator[] (
            int i ) const [inline]
```

Definition at line 253 of file poly.h.

3.112.3.16 termscopy()

```
void poly::termscopy (
            const WORD * source,
            int dest,
            int num ) [inline]
```

Definition at line 260 of file poly.h.

3.112.3.17 check_memory()

```
void poly::check_memory (
            int i ) [inline]
```

Definition at line 242 of file poly.h.

3.112.3.18 expand_memory()

```
void poly::expand_memory (
            int i )
```

Definition at line 149 of file poly.cc.

3.112.3.19 is_zero()

```
bool poly::is_zero ( ) const
```

Definition at line 2209 of file poly.cc.

3.112.3.20 is_one()

```
bool poly::is_one ( ) const
```

Definition at line 2219 of file poly.cc.

3.112.3.21 is_integer()

```
bool poly::is_integer ( ) const
```

Definition at line 2239 of file poly.cc.

3.112.3.22 is_monomial()

```
bool poly::is_monomial ( ) const
```

Definition at line 2259 of file poly.cc.

3.112.3.23 is_dense_univariate()

```
int poly::is_dense_univariate ( ) const
```

Dense univariate detection

3.112.4 Description

This method returns whether the polynomial is dense and univariate. The possible return values are:

-2 is not dense univariate -1 is no variables n>=0 is univariate in n

3.112.5 Notes

A univariate polynomial is considered dense iff more than half of the coefficients a_0...a_deg are non-zero.

Definition at line 2284 of file poly.cc.

Referenced by divmod(), and mul().

3.112.5.1 sign()

```
int poly::sign ( ) const
```

Definition at line 2039 of file poly.cc.

3.112.5.2 degree()

```
int poly::degree (
            int x ) const
```

Definition at line 2050 of file poly.cc.

3.112.5.3 total_degree()

```
int poly::total_degree ( ) const
```

Definition at line 2063 of file poly.cc.

3.112.5.4 first_variable()

```
int poly::first_variable ( ) const
```

Definition at line 2000 of file poly.cc.

3.112.5.5 number_of_terms()

```
int poly::number_of_terms ( ) const
```

Definition at line 1986 of file poly.cc.

3.112.5.6 all_variables()

```
const vector< int > poly::all_variables ( ) const
```

Definition at line 2016 of file poly.cc.

3.112.5.7 integer_lcoeff()

```
const poly poly::integer_lcoeff ( ) const
```

Definition at line 2083 of file poly.cc.

3.112.5.8 lcoeff_univar()

```
const poly poly::lcoeff_univar (
            int x ) const
```

Definition at line 2133 of file poly.cc.

3.112.5.9 lcoeff_multivar()

```
const poly poly::lcoeff_multivar (
            int x ) const
```

Definition at line 2140 of file poly.cc.

3.112.5.10 coefficient()

```
const poly poly::coefficient (
            int x,
            int n ) const
```

Definition at line 2105 of file poly.cc.

3.112.5.11 derivative()

```
const poly poly::derivative (
            int x ) const
```

Definition at line 2172 of file poly.cc.

3.112.5.12 setmod()

```
void poly::setmod (
            WORD _modp,
            WORD _modn = 1 )
```

Definition at line 179 of file poly.cc.

3.112.5.13 coefficients_modulo()

```
void poly::coefficients_modulo (
            UWORD * a,
            WORD na,
            bool small )
```

Definition at line 215 of file poly.cc.

3.112.5.14 simple_poly() [1/2]

```
const poly poly::simple_poly (
            PHEAD int x,
            int a = 0,
            int b = 1,
            int p = 0,
            int n = 1 )  [static]
```

Definition at line 2317 of file poly.cc.

3.112.5.15 simple_poly() [2/2]

```
const poly poly::simple_poly (
            PHEAD int x,
            const poly & a,
            int b = 1,
            int p = 0,
            int n = 1 )  [static]
```

Definition at line 2356 of file poly.cc.

3.112.5.16 get_variables()

```
void poly::get_variables (
            PHEAD const std::vector< WORD * > & ,
            bool ,
            bool  )  [static]
```

Definition at line 2395 of file poly.cc.

3.112.5.17 argument_to_poly()

```
const poly poly::argument_to_poly (
            PHEAD WORD * e,
            bool with_arghead,
            bool sort_univar,
            poly * den = NULL )  [static]
```

Definition at line 2498 of file poly.cc.

3.112.5.18 poly_to_argument()

```
void poly::poly_to_argument (
            const poly & a,
            WORD * res,
            bool with_arghead )  [static]
```

Definition at line 2611 of file poly.cc.

3.112.5.19 poly_to_argument_with_den()

```
void poly::poly_to_argument_with_den (
            const poly & a,
            WORD nden,
            const UWORD * den,
            WORD * res,
            bool with_arghead ) [static]
```

Definition at line 2683 of file poly.cc.

3.112.5.20 size_of_form_notation()

```
int poly::size_of_form_notation ( ) const
```

Definition at line 2766 of file poly.cc.

3.112.5.21 size_of_form_notation_with_den()

```
int poly::size_of_form_notation_with_den (
            WORD nden ) const
```

Definition at line 2795 of file poly.cc.

3.112.5.22 normalize()

```
const poly & poly::normalize ( )
```

Definition at line 362 of file poly.cc.

3.112.5.23 from_coefficient_list()

```
const poly poly::from_coefficient_list (
            PHEAD const std::vector< WORD > & ,
            int ,
            WORD ) [static]
```

Definition at line 2885 of file poly.cc.

3.112.5.24 to_coefficient_list()

```
const vector< WORD > poly::to_coefficient_list (
            const poly & a ) [static]
```

Definition at line 2823 of file poly.cc.

3.112.5.25 coefficient_list_divmod()

```
const vector< WORD > poly::coefficient_list_divmod (
            const std::vector< WORD > & ,
            const std::vector< WORD > & ,
            WORD ,
            int ) [static]
```

Definition at line 2846 of file poly.cc.

3.112.5.26 to_string()

```
const string poly::to_string ( ) const
```

Definition at line 267 of file poly.cc.

3.112.5.27 monomial_compare()

```
int poly::monomial_compare (
            PHEAD const WORD * a,
            const WORD * b ) [static]
```

Definition at line 350 of file poly.cc.

3.112.5.28 last_monomial_index()

```
WORD poly::last_monomial_index ( ) const
```

Definition at line 465 of file poly.cc.

3.112.5.29 last_monomial()

```
WORD * poly::last_monomial ( ) const
```

Definition at line 472 of file poly.cc.

3.112.5.30 compare_degree_vector()

```
int poly::compare_degree_vector (
            const poly & b ) const
```

Definition at line 481 of file poly.cc.

3.112.5.31 degree_vector()

```
std::vector< int > poly::degree_vector ( ) const
```

Definition at line 503 of file poly.cc.

3.112.5.32 add()

```
void poly::add (
            const poly & a,
            const poly & b,
            poly & c ) [static]
```

Definition at line 538 of file poly.cc.

3.112.5.33 sub()

```
void poly::sub (
            const poly & a,
            const poly & b,
            poly & c ) [static]
```

Definition at line 622 of file poly.cc.

3.112.5.34 mul()

```
void poly::mul (
            const poly & a,
            const poly & b,
            poly & c ) [static]
```

Polynomial multiplication

3.112.6 Description

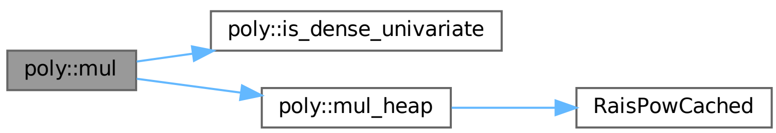
This routine determines which multiplication routine to use for multiplying two polynomials. The logic is as follows:

- If a or b consists of only one term, call mul_one_term;
- Otherwise, if both are univariate and dense, call mul_univar;
- Otherwise, call mul_heap.

Definition at line 1195 of file poly.cc.

References is_dense_univariate(), and mul_heap().

Here is the call graph for this function:



3.112.6.1 div()

```
void poly::div (
            const poly & a,
            const poly & b,
            poly & c )  [static]
```

Definition at line 1915 of file poly.cc.

3.112.6.2 mod()

```
void poly::mod (
            const poly & a,
            const poly & b,
            poly & c )  [static]
```

Definition at line 1927 of file poly.cc.

3.112.6.3 divmod()

```
void poly::divmod (
            const poly & a,
            const poly & b,
            poly & q,
            poly & r,
            bool only_divides )  [static]
```

Polynomial division

3.112.7 Description

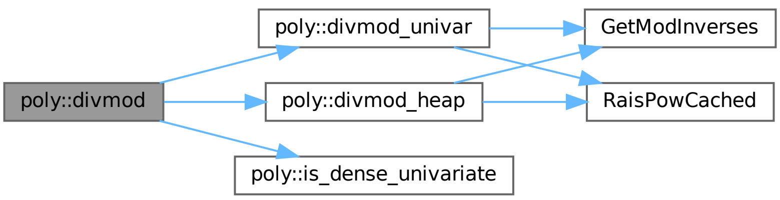
This routine determines which division routine to use for dividing two polynomials. The logic is as follows:

- If b consists of only one term, call divmod_one_term;
- Otherwise, if both are univariate and dense, call divmod_univar;
- Otherwise, call divmod_heap.

Definition at line 1856 of file poly.cc.

References divmod_heap(), divmod_univar(), and is_dense_univariate().

Here is the call graph for this function:



3.112.7.1 divides()

```
bool poly::divides (
            const poly & a,
            const poly & b )  [static]
```

Definition at line 1902 of file poly.cc.

3.112.7.2 mul_one_term()

```
void poly::mul_one_term (
            const poly & a,
            const poly & b,
            poly & c )  [static]
```

Definition at line 755 of file poly.cc.

3.112.7.3 mul_univar()

```
void poly::mul_univar (
            const poly & a,
            const poly & b,
            poly & c,
            int var )  [static]
```

Definition at line 829 of file poly.cc.

3.112.7.4 mul_heap()

```
void poly::mul_heap (
            const poly & a,
            const poly & b,
            poly & c )  [static]
```

Multiplication of polynomials with a heap

3.112.8 Description

Multiplies two multivariate polynomials. The next element of the product is efficiently determined by using a heap. If the product of the maximum power in all variables is small, a hash table is used to add equal terms for extra speed.

A heap element h is formatted as follows:

- h[0] = index in a
- h[1] = index in b
- h[2] = hash code (-1 if no hash is used)
- h[3] = length of coefficient with sign
- h[4...4+AN.poly_num_vars-1] = powers

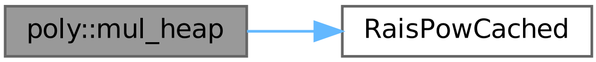
- h[4+AN.poly_num_vars...4+h[3]-1] = coefficient

Definition at line 961 of file poly.cc.

References RaisPowCached().

Referenced by mul().

Here is the call graph for this function:



3.112.8.1 divmod_one_term()

```
void poly::divmod_one_term (
            const poly & a,
            const poly & b,
            poly & q,
            poly & r,
            bool only_divides )  [static]
```

Definition at line 1245 of file poly.cc.

3.112.8.2 divmod_univar()

```
void poly::divmod_univar (
            const poly & a,
            const poly & b,
            poly & q,
            poly & r,
            int var,
            bool only_divides )  [static]
```

Division of dense univariate polynomials.

3.112.9 Description

Divides two dense univariate polynomials. For each power, the method collects all terms that result in that power.

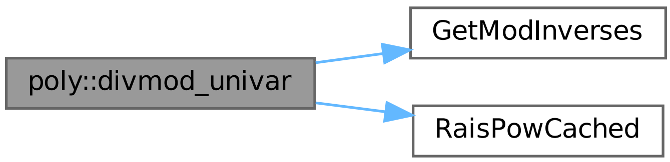
Relevant formula [Q=A/B, P=SUM(p_i∗x^i), n=deg(A), m=deg(B)]: q_k = [a_{m+k} - SUM(i=k+1...n-m, b_{m+k-i}∗q_i)] / b_m

Definition at line 1376 of file poly.cc.

References GetModInverses(), and RaisPowCached().

Referenced by divmod().

Here is the call graph for this function:



3.112.9.1 divmod_heap()

```
void poly::divmod_heap (
            const poly & a,
            const poly & b,
            poly & q,
            poly & r,
            bool only_divides,
            bool check_div,
            bool & div_fail )  [static]
```

Division of polynomials with a heap

3.112.10 Description

Divides two multivariate polynomials. The next element of the quotient/remainder is efficiently determined by using a heap. If the product of the maximum power in all variables is small, a hash table is used to add equal terms for extra speed.

If the input flag check_div is set then if the result of any coefficient division results in a non-zero remainder (indicating that division has failed over the integers) the output flag div_fail will be set and the division will be terminated early (q, r will be incorrect). If check_div is not set then non-zero remainders from coefficient division will be written into r.

A heap element h is formatted as follows:

- h[0] = index in a
- h[1] = index in b
- h[2] = -1 (no hash is used)
- h[3] = length of coefficient with sign
- h[4...4+AN.poly_num_vars-1] = powers
- h[4+AN.poly_num_vars...4+h[3]-1] = coefficient

Note: the hashing trick as in multiplication cannot be used easily, since there is no tight upperbound on the exponents in the answer.

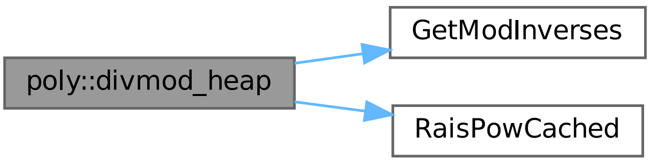
For details, see M. Monagan, "Polynomial Division using Dynamic Array, Heaps, and Packed Exponent Vectors"

Definition at line 1579 of file poly.cc.

References GetModInverses(), and RaisPowCached().

Referenced by divmod().

Here is the call graph for this function:



3.112.10.1 push_heap()

```
void poly::push_heap (
            PHEAD WORD ** heap,
            int n )  [static]
```

Definition at line 739 of file poly.cc.

3.112.10.2 pop_heap()

```
void poly::pop_heap (
            PHEAD WORD ** heap,
            int n )  [static]
```

Definition at line 707 of file poly.cc.

3.112.11 Field Documentation

3.112.11.1 terms

```
WORD* poly::terms
```

Definition at line 62 of file poly.h.

3.112.11.2 size_of_terms

```
LONG poly::size_of_terms
```

Definition at line 63 of file poly.h.

3.112.11.3 modp

```
WORD poly::modp
```

Definition at line 64 of file poly.h.

3.112.11.4 modn

```
WORD poly::modn
```

Definition at line 64 of file poly.h.

The documentation for this class was generated from the following files:

- poly.h
- poly.cc

3.113 flint::poly_factor Class Reference

Data Fields

- fmpz_poly_factor_t d

3.113.1 Detailed Description

Definition at line 70 of file flintinterface.h.

3.113.2 Constructor & Destructor Documentation

3.113.2.1 poly_factor()

```
flint::poly_factor::poly_factor ( )  [inline]
```

Definition at line 73 of file flintinterface.h.

3.113.2.2 ~poly_factor()

```
flint::poly_factor::~poly_factor ( )  [inline]
```

Definition at line 74 of file flintinterface.h.

3.113.3 Field Documentation

3.113.3.1 d

```
fmpz_poly_factor_t flint::poly_factor::d
```

Definition at line 72 of file flintinterface.h.

The documentation for this class was generated from the following file:

- flintinterface.h

3.114 POLYMOD Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ∗ coefs
- WORD numsym
- WORD arraysize
- WORD polysize
- WORD modnum

3.114.1 Detailed Description

The POLYMOD struct controls one univariate polynomial of which the coefficients have been taken modulus a (prime) number that fits inside a variable of type WORD. The polynomial is stored as an array of coefficients of size WORD.

Definition at line 1268 of file structs.h.

3.114.2 Field Documentation

3.114.2.1 coefs

```
WORD* POLYMOD::coefs
```

Definition at line 1269 of file structs.h.

3.114.2.2 numsym

```
WORD POLYMOD::numsym
```

Definition at line 1270 of file structs.h.

3.114.2.3 arraysize

```
WORD POLYMOD::arraysize
```

Definition at line 1271 of file structs.h.

3.114.2.4 polysize

```
WORD POLYMOD::polysize
```

Definition at line 1272 of file structs.h.

3.114.2.5 modnum

```
WORD POLYMOD::modnum
```

Definition at line 1273 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.115 PoSiTiOn Struct Reference

Data Fields

- off_t p1

3.115.1 Detailed Description

Definition at line 59 of file structs.h.

3.115.2 Field Documentation

3.115.2.1 p1

```
off_t PoSiTiOn::p1
```

Definition at line 60 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.116 PRELOAD Struct Reference

```
#include <structs.h>
```

Data Fields

- UBYTE ∗ buffer
- LONG size

3.116.1 Detailed Description

Used by the preprocessor to load the contents of a doloop or a procedure. The struct PRELOAD is used both in the DOLOOP and PROCEDURE structs.

Definition at line 873 of file structs.h.

3.116.2 Field Documentation

3.116.2.1 buffer

```
UBYTE* PRELOAD::buffer
```

Definition at line 874 of file structs.h.

3.116.2.2 size

```
LONG PRELOAD::size
```

Definition at line 875 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.117 pReVaR Struct Reference

```
#include <structs.h>
```

Data Fields

- UBYTE * name
- UBYTE * value
- UBYTE * argnames
- int nargs
- int wildarg

3.117.1 Detailed Description

An element of the type PREVAR is needed for each preprocessor variable.

Definition at line 841 of file structs.h.

3.117.2 Field Documentation

3.117.2.1 name

```
UBYTE* pReVaR::name
```

allocated

Definition at line 842 of file structs.h.

3.117.2.2 value

```
UBYTE* pReVaR::value
```

points into memory of name

Definition at line 843 of file structs.h.

3.117.2.3 argnames

```
UBYTE* pReVaR::argnames
```

names of arguments, zero separated. points into memory of name

Definition at line 844 of file structs.h.

3.117.2.4 nargs

`int pReVaR::nargs`

0 = regular, >= 1: number of macro arguments. total number

Definition at line 845 of file structs.h.

3.117.2.5 wildarg

`int pReVaR::wildarg`

The number of a potential ?var. If none: 0. wildarg<nargs

Definition at line 846 of file structs.h.

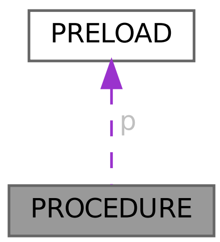
The documentation for this struct was generated from the following file:

- structs.h

3.118 PROCEDURE Struct Reference

`#include <structs.h>`

Collaboration diagram for PROCEDURE:



Data Fields

- PRELOAD p
- UBYTE ∗ name
- int loadmode

3.118.1 Detailed Description

An element of the type PROCEDURE is needed for each procedure in the system.

Definition at line 883 of file structs.h.

3.118.2 Field Documentation

3.118.2.1 p

`PRELOAD PROCEDURE::p`

Definition at line 884 of file structs.h.

3.118.2.2 name

`UBYTE* PROCEDURE::name`

Definition at line 885 of file structs.h.

3.118.2.3 loadmode

`int PROCEDURE::loadmode`

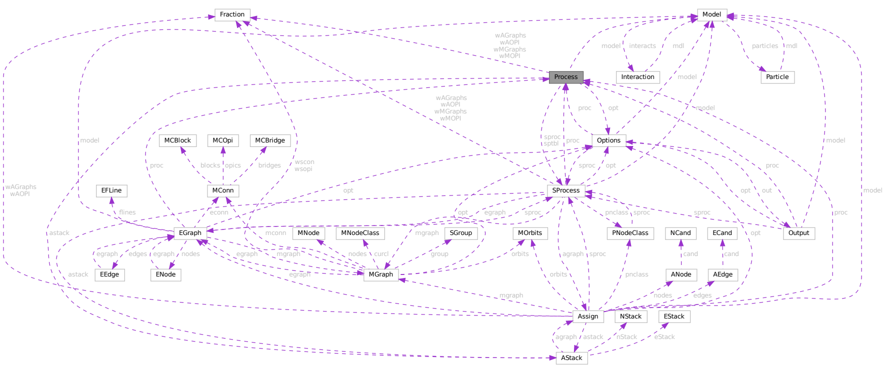
Definition at line 886 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.119 Process Class Reference

Collaboration diagram for Process:



Public Member Functions

- Process (int pid, Model ∗model, Options ∗opt, int nin, int ∗initlPart, int nfin, int ∗finalPart, int ∗coupling)
- **Process** (int pid, Model ∗model, Options ∗opt, FGInput ∗fgi)
- void prProcess (void)
- void mkSProcess (void)

Data Fields

- Model ∗ model
- Options ∗ opt
- BigInt mgrcount
- BigInt agrcount
- int ∗ initlPart
- int ∗ finalPart
- int id
- int ninitl
- int nfinal
- int ctotal
- int nExtern
- int loop
- int maxnlegs
- int clist [GRCC_MAXNCPLG]
- int nSubproc
- SProcess ∗ sptbl [GRCC_MAXSUBPROCS]
- SProcess ∗ sproc
- AStack ∗ astack
- BigInt ngraphs
- BigInt nopi
- BigInt wgraphs
- BigInt wopi
- BigInt nMGraphs
- BigInt nMOPI
- Fraction wMGraphs
- Fraction wMOPI
- BigInt nAGraphs
- BigInt nAOPI
- Fraction wAGraphs
- Fraction wAOPI
- double sec

3.119.1 Detailed Description

Definition at line 424 of file grcc.h.

3.119.2 Constructor & Destructor Documentation

3.119.2.1 Process()

```
Process::Process (
            int pid,
            Model * model,
            Options * opt,
            int nin,
            int * initlPart,
            int nfin,
            int * finalPart,
            int * coupling )
```

Definition at line 3036 of file grcc.cc.

3.119.2.2 ∼Process()

```
Process::∼Process (
            void )
```

Definition at line 3134 of file grcc.cc.

3.119.3 Member Function Documentation

3.119.3.1 prProcess()

```
void Process::prProcess (
            void )
```

Definition at line 3142 of file grcc.cc.

3.119.3.2 mkSProcess()

```
void Process::mkSProcess (
            void )
```

Definition at line 3150 of file grcc.cc.

3.119.4 Field Documentation

3.119.4.1 model

```
Model* Process::model
```

Definition at line 426 of file grcc.h.

3.119.4.2 opt

`Options* Process::opt`

Definition at line 427 of file grcc.h.

3.119.4.3 mgrcount

`BigInt Process::mgrcount`

Definition at line 428 of file grcc.h.

3.119.4.4 agrcount

`BigInt Process::agrcount`

Definition at line 429 of file grcc.h.

3.119.4.5 initlPart

`int* Process::initlPart`

Definition at line 430 of file grcc.h.

3.119.4.6 finalPart

`int* Process::finalPart`

Definition at line 431 of file grcc.h.

3.119.4.7 id

`int Process::id`

Definition at line 433 of file grcc.h.

3.119.4.8 ninitl

`int Process::ninitl`

Definition at line 434 of file grcc.h.

3.119.4.9 nfinal

`int Process::nfinal`

Definition at line 435 of file grcc.h.

3.119.4.10 ctotal

```
int Process::ctotal
```

Definition at line 436 of file grcc.h.

3.119.4.11 nExtern

```
int Process::nExtern
```

Definition at line 437 of file grcc.h.

3.119.4.12 loop

```
int Process::loop
```

Definition at line 438 of file grcc.h.

3.119.4.13 maxnlegs

```
int Process::maxnlegs
```

Definition at line 439 of file grcc.h.

3.119.4.14 clist

```
int Process::clist[GRCC_MAXNCPLG]
```

Definition at line 440 of file grcc.h.

3.119.4.15 nSubproc

```
int Process::nSubproc
```

Definition at line 444 of file grcc.h.

3.119.4.16 sptbl

```
SProcess* Process::sptbl[GRCC_MAXSUBPROCS]
```

Definition at line 445 of file grcc.h.

3.119.4.17 sproc

```
SProcess* Process::sproc
```

Definition at line 446 of file grcc.h.

3.119.4.18 astack

`AStack* Process::astack`

Definition at line 449 of file grcc.h.

3.119.4.19 ngraphs

`BigInt Process::ngraphs`

Definition at line 452 of file grcc.h.

3.119.4.20 nopi

`BigInt Process::nopi`

Definition at line 453 of file grcc.h.

3.119.4.21 wgraphs

`BigInt Process::wgraphs`

Definition at line 454 of file grcc.h.

3.119.4.22 wopi

`BigInt Process::wopi`

Definition at line 455 of file grcc.h.

3.119.4.23 nMGraphs

`BigInt Process::nMGraphs`

Definition at line 458 of file grcc.h.

3.119.4.24 nMOPI

`BigInt Process::nMOPI`

Definition at line 459 of file grcc.h.

3.119.4.25 wMGraphs

`Fraction Process::wMGraphs`

Definition at line 460 of file grcc.h.

3.119.4.26 wMOPI

```
Fraction Process::wMOPI
```

Definition at line 461 of file grcc.h.

3.119.4.27 nAGraphs

```
BigInt Process::nAGraphs
```

Definition at line 463 of file grcc.h.

3.119.4.28 nAOPI

```
BigInt Process::nAOPI
```

Definition at line 464 of file grcc.h.

3.119.4.29 wAGraphs

```
Fraction Process::wAGraphs
```

Definition at line 465 of file grcc.h.

3.119.4.30 wAOPI

```
Fraction Process::wAOPI
```

Definition at line 466 of file grcc.h.

3.119.4.31 sec

```
double Process::sec
```

Definition at line 468 of file grcc.h.

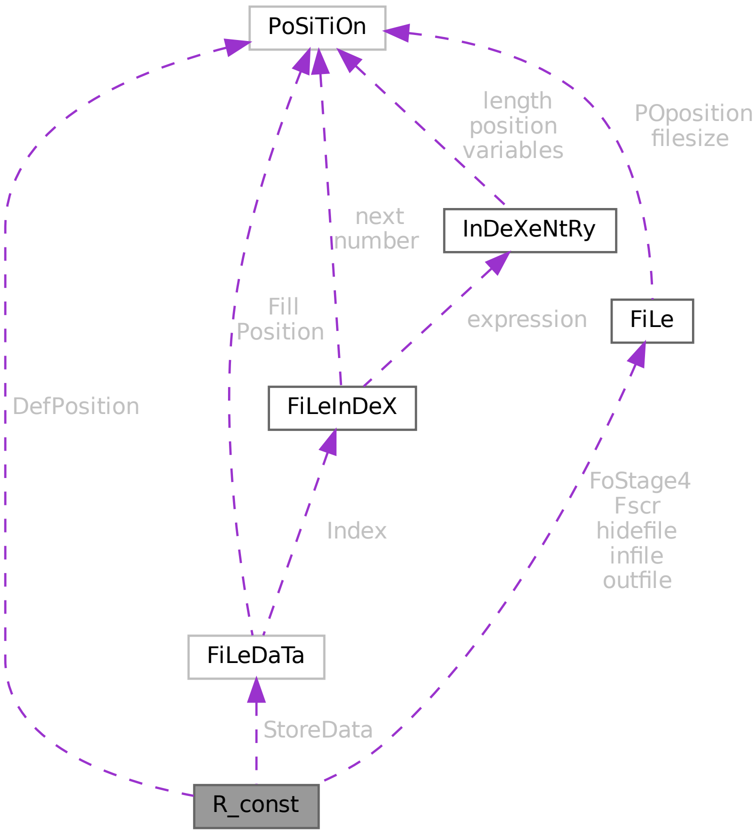
The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.120 R_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for R_const:



Public Member Functions

- **PADPOSITION** (8, 7, 7, 27, 0)

Data Fields

- FILEDATA StoreData
- FILEHANDLE Fscr [3]
- FILEHANDLE FoStage4 [2]
- POSITION DefPosition
- FILEHANDLE ∗ infile
- FILEHANDLE ∗ outfile
- FILEHANDLE ∗ hidefile
- WORD ∗ CompressBuffer
- WORD ∗ ComprTop

- WORD ∗ CompressPointer
- COMPAREDUMMY CompareRoutine
- ULONG ∗ wranfia
- SBYTE ∗ moebiustable
- LONG OldTime
- LONG InInBuf
- LONG InHiBuf
- LONG pWorkSize
- LONG lWorkSize
- LONG posWorkSize
- ULONG wranfseed
- int NoCompress
- int gzipCompress
- int Cnumlhs
- int outtohide
- int wranfcall
- int wranfnpair1
- int wranfnpair2
- WORD GetFile
- WORD KeptInHold
- WORD BracketOn
- WORD MaxBracket
- WORD CurDum
- WORD DeferFlag
- WORD TePos
- WORD sLevel
- WORD Stage4Name
- WORD GetOneFile
- WORD PolyFun
- WORD PolyFunInv
- WORD PolyFunType
- WORD PolyFunExp
- WORD PolyFunVar
- WORD PolyFunPow
- WORD Eside
- WORD MaxDum
- WORD level
- WORD expchanged
- WORD expflags
- WORD CurExpr
- WORD SortType
- WORD ShortSortCount
- WORD modeloptions
- WORD funoffset
- WORD moebiustablesize

3.120.1 Detailed Description

The R_const struct is part of the global data and resides either in the ALLGLOBALS struct A, or the ALLPRIVATES struct B (TFORM) under the name R We see it used with the macro AR as in AR.infile It has the variables that define the running environment and that should be transferred with a term in a multithreaded run.

Definition at line 2029 of file structs.h.

3.120.2 Field Documentation

3.120.2.1 StoreData

`FILEDATA R_const::StoreData`

Definition at line 2030 of file structs.h.

3.120.2.2 Fscr

`FILEHANDLE R_const::Fscr[3]`

Definition at line 2031 of file structs.h.

3.120.2.3 FoStage4

`FILEHANDLE R_const::FoStage4[2]`

Definition at line 2032 of file structs.h.

3.120.2.4 DefPosition

`POSITION R_const::DefPosition`

Definition at line 2033 of file structs.h.

3.120.2.5 infile

`FILEHANDLE* R_const::infile`

Definition at line 2034 of file structs.h.

3.120.2.6 outfile

`FILEHANDLE* R_const::outfile`

Definition at line 2035 of file structs.h.

3.120.2.7 hidefile

`FILEHANDLE* R_const::hidefile`

Definition at line 2036 of file structs.h.

3.120.2.8 CompressBuffer

`WORD* R_const::CompressBuffer`

Definition at line 2038 of file structs.h.

3.120.2.9 ComprTop

`WORD* R_const::ComprTop`

Definition at line 2039 of file structs.h.

3.120.2.10 CompressPointer

`WORD* R_const::CompressPointer`

Definition at line 2040 of file structs.h.

3.120.2.11 CompareRoutine

`COMPAREDUMMY R_const::CompareRoutine`

Definition at line 2041 of file structs.h.

3.120.2.12 wranfia

`ULONG* R_const::wranfia`

Definition at line 2042 of file structs.h.

3.120.2.13 moebiustable

`SBYTE* R_const::moebiustable`

Definition at line 2043 of file structs.h.

3.120.2.14 OldTime

`LONG R_const::OldTime`

Definition at line 2045 of file structs.h.

3.120.2.15 InInBuf

`LONG R_const::InInBuf`

Definition at line 2046 of file structs.h.

3.120.2.16 InHiBuf

```
LONG R_const::InHiBuf
```

Definition at line 2047 of file structs.h.

3.120.2.17 pWorkSize

```
LONG R_const::pWorkSize
```

Definition at line 2048 of file structs.h.

3.120.2.18 lWorkSize

```
LONG R_const::lWorkSize
```

Definition at line 2049 of file structs.h.

3.120.2.19 posWorkSize

```
LONG R_const::posWorkSize
```

Definition at line 2050 of file structs.h.

3.120.2.20 wranfseed

```
ULONG R_const::wranfseed
```

Definition at line 2051 of file structs.h.

3.120.2.21 NoCompress

```
int R_const::NoCompress
```

Definition at line 2052 of file structs.h.

3.120.2.22 gzipCompress

```
int R_const::gzipCompress
```

Definition at line 2053 of file structs.h.

3.120.2.23 Cnumlhs

```
int R_const::Cnumlhs
```

Definition at line 2054 of file structs.h.

3.120.2.24 outtohide

```
int R_const::outtohide
```

Definition at line 2055 of file structs.h.

3.120.2.25 wranfcall

```
int R_const::wranfcall
```

Definition at line 2059 of file structs.h.

3.120.2.26 wranfnpair1

```
int R_const::wranfnpair1
```

Definition at line 2060 of file structs.h.

3.120.2.27 wranfnpair2

```
int R_const::wranfnpair2
```

Definition at line 2061 of file structs.h.

3.120.2.28 GetFile

```
WORD R_const::GetFile
```

Definition at line 2067 of file structs.h.

3.120.2.29 KeptInHold

```
WORD R_const::KeptInHold
```

Definition at line 2068 of file structs.h.

3.120.2.30 BracketOn

```
WORD R_const::BracketOn
```

Definition at line 2069 of file structs.h.

3.120.2.31 MaxBracket

```
WORD R_const::MaxBracket
```

Definition at line 2070 of file structs.h.

3.120.2.32 CurDum

```
WORD R_const::CurDum
```

Definition at line 2071 of file structs.h.

3.120.2.33 DeferFlag

```
WORD R_const::DeferFlag
```

Definition at line 2072 of file structs.h.

3.120.2.34 TePos

```
WORD R_const::TePos
```

Definition at line 2073 of file structs.h.

3.120.2.35 sLevel

```
WORD R_const::sLevel
```

Definition at line 2074 of file structs.h.

3.120.2.36 Stage4Name

```
WORD R_const::Stage4Name
```

Definition at line 2075 of file structs.h.

3.120.2.37 GetOneFile

```
WORD R_const::GetOneFile
```

Definition at line 2076 of file structs.h.

3.120.2.38 PolyFun

```
WORD R_const::PolyFun
```

Definition at line 2077 of file structs.h.

3.120.2.39 PolyFunInv

```
WORD R_const::PolyFunInv
```

Definition at line 2078 of file structs.h.

3.120.2.40 PolyFunType

```
WORD R_const::PolyFunType
```

Definition at line 2079 of file structs.h.

3.120.2.41 PolyFunExp

```
WORD R_const::PolyFunExp
```

Definition at line 2080 of file structs.h.

3.120.2.42 PolyFunVar

```
WORD R_const::PolyFunVar
```

Definition at line 2081 of file structs.h.

3.120.2.43 PolyFunPow

```
WORD R_const::PolyFunPow
```

Definition at line 2082 of file structs.h.

3.120.2.44 Eside

```
WORD R_const::Eside
```

Definition at line 2083 of file structs.h.

3.120.2.45 MaxDum

```
WORD R_const::MaxDum
```

Definition at line 2084 of file structs.h.

3.120.2.46 level

```
WORD R_const::level
```

Definition at line 2085 of file structs.h.

3.120.2.47 expchanged

```
WORD R_const::expchanged
```

Definition at line 2086 of file structs.h.

3.120.2.48 expflags

```
WORD R_const::expflags
```

Definition at line 2087 of file structs.h.

3.120.2.49 CurExpr

```
WORD R_const::CurExpr
```

Definition at line 2088 of file structs.h.

3.120.2.50 SortType

```
WORD R_const::SortType
```

Definition at line 2089 of file structs.h.

3.120.2.51 ShortSortCount

```
WORD R_const::ShortSortCount
```

Definition at line 2090 of file structs.h.

3.120.2.52 modeloptions

```
WORD R_const::modeloptions
```

Definition at line 2091 of file structs.h.

3.120.2.53 funoffset

```
WORD R_const::funoffset
```

Definition at line 2092 of file structs.h.

3.120.2.54 moebiustablesize

```
WORD R_const::moebiustablesize
```

Definition at line 2093 of file structs.h.

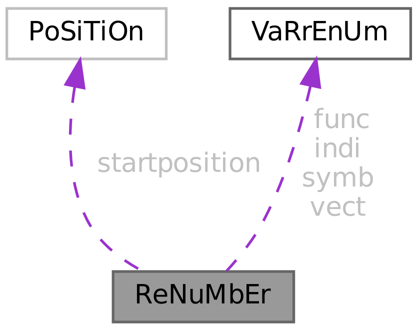
The documentation for this struct was generated from the following file:

- structs.h

3.121 ReNuMbEr Struct Reference

`#include <structs.h>`

Collaboration diagram for ReNuMbEr:



Public Member Functions

- **PADPOSITION** (4, 0, 0, 0, sizeof(VARRENUM) ∗4)

Data Fields

- POSITION startposition
- VARRENUM symb
- VARRENUM indi
- VARRENUM vect
- VARRENUM func
- WORD ∗ symnum
- WORD ∗ indnum
- WORD ∗ vecnum
- WORD ∗ funnum

3.121.1 Detailed Description

Only symb.lo gets dynamically allocated. All other pointers points into this memory.

Definition at line 178 of file structs.h.

3.121.2 Field Documentation

3.121.2.1 startposition

`POSITION ReNuMbEr::startposition`

Definition at line 179 of file structs.h.

3.121.2.2 symb

`VARRENUM ReNuMbEr::symb`

Symbols

Definition at line 181 of file structs.h.

Referenced by DoRecovery(), Generator(), GetFirstTerm(), and TermRenumber().

3.121.2.3 indi

`VARRENUM ReNuMbEr::indi`

Indices

Definition at line 182 of file structs.h.

Referenced by DoRecovery(), and TermRenumber().

3.121.2.4 vect

`VARRENUM ReNuMbEr::vect`

Vectors

Definition at line 183 of file structs.h.

Referenced by DoRecovery(), and TermRenumber().

3.121.2.5 func

`VARRENUM ReNuMbEr::func`

Functions

Definition at line 184 of file structs.h.

Referenced by DoRecovery(), and TermRenumber().

3.121.2.6 symnum

`WORD* ReNuMbEr::symnum`

Renumbered symbols

Definition at line 186 of file structs.h.

Referenced by DoRecovery(), and TermRenumber().

3.121.2.7 indnum

`WORD* ReNuMbEr::indnum`

Renumbered indices

Definition at line 187 of file structs.h.

Referenced by DoRecovery(), and TermRenumber().

3.121.2.8 vecnum

`WORD* ReNuMbEr::vecnum`

Renumbered vectors

Definition at line 188 of file structs.h.

Referenced by DoRecovery(), and TermRenumber().

3.121.2.9 funnum

`WORD* ReNuMbEr::funnum`

Renumbered functions

Definition at line 189 of file structs.h.

Referenced by DoRecovery(), and TermRenumber().

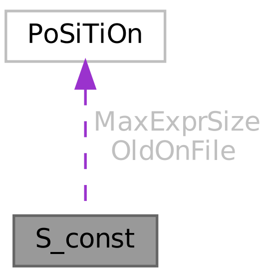
The documentation for this struct was generated from the following file:

- structs.h

3.122 S_const Struct Reference

`#include <structs.h>`

Collaboration diagram for S_const:



Public Member Functions

- **PADPOSITION** (4, 0, 4, 2, 0)

Data Fields

- POSITION MaxExprSize
- POSITION ∗ OldOnFile
- WORD ∗ OldNumFactors
- WORD ∗ Oldvflags
- WORD ∗ Olduflags
- int NumOldOnFile
- int NumOldNumFactors
- int MultiThreaded
- int Balancing
- WORD ExecMode
- WORD CollectOverFlag

3.122.1 Detailed Description

The S_const struct is part of the global data and resides in the ALLGLOBALS struct A under the name S We see it used with the macro AS as in AS.ExecMode It has some variables used by the master in multithreaded runs

Definition at line 1977 of file structs.h.

3.122.2 Field Documentation

3.122.2.1 MaxExprSize

```
POSITION S_const::MaxExprSize
```

Definition at line 1978 of file structs.h.

3.122.2.2 OldOnFile

```
POSITION* S_const::OldOnFile
```

Definition at line 1984 of file structs.h.

3.122.2.3 OldNumFactors

```
WORD* S_const::OldNumFactors
```

Definition at line 1985 of file structs.h.

3.122.2.4 Oldvflags

```
WORD* S_const::Oldvflags
```

Definition at line 1986 of file structs.h.

3.122.2.5 Olduflags

```
WORD* S_const::Olduflags
```

Definition at line 1987 of file structs.h.

3.122.2.6 NumOldOnFile

```
int S_const::NumOldOnFile
```

Definition at line 1991 of file structs.h.

3.122.2.7 NumOldNumFactors

```
int S_const::NumOldNumFactors
```

Definition at line 1992 of file structs.h.

3.122.2.8 MultiThreaded

```
int S_const::MultiThreaded
```

Definition at line 1993 of file structs.h.

3.122.2.9 Balancing

```
int S_const::Balancing
```

Definition at line 2000 of file structs.h.

3.122.2.10 ExecMode

```
WORD S_const::ExecMode
```

Definition at line 2001 of file structs.h.

3.122.2.11 CollectOverFlag

```
WORD S_const::CollectOverFlag
```

Definition at line 2003 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.123 SeTs Struct Reference

Data Fields

- LONG name
- WORD type
- WORD first
- WORD last
- WORD node
- WORD namesize
- WORD dimension
- WORD flags

3.123.1 Detailed Description

Definition at line 499 of file structs.h.

3.123.2 Field Documentation

3.123.2.1 name

```
LONG SeTs::name
```

Definition at line 500 of file structs.h.

3.123.2.2 type

```
WORD SeTs::type
```

Definition at line 501 of file structs.h.

3.123.2.3 first

```
WORD SeTs::first
```

Definition at line 502 of file structs.h.

3.123.2.4 last

`WORD SeTs::last`

Definition at line 503 of file structs.h.

3.123.2.5 node

`WORD SeTs::node`

Definition at line 504 of file structs.h.

3.123.2.6 namesize

`WORD SeTs::namesize`

Definition at line 505 of file structs.h.

3.123.2.7 dimension

`WORD SeTs::dimension`

Definition at line 506 of file structs.h.

3.123.2.8 flags

`WORD SeTs::flags`

Definition at line 507 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.124 SETUPPARAMETERS Struct Reference

```
#include <structs.h>
```

Data Fields

- UBYTE ∗ parameter
- int type
- int flags
- LONG value

3.124.1 Detailed Description

Each setup parameter has one element of the struct SETUPPARAMETERS assigned to it. By binary search in the array of them we can then locate the proper element by name. We have to assume that two ints make a long and either one or two longs make a pointer. The long before the ints and padding gives a problem in the initialization.

Definition at line 1038 of file structs.h.

3.124.2 Field Documentation

3.124.2.1 parameter

`UBYTE* SETUPPARAMETERS::parameter`

Definition at line 1039 of file structs.h.

3.124.2.2 type

`int SETUPPARAMETERS::type`

Definition at line 1040 of file structs.h.

3.124.2.3 flags

`int SETUPPARAMETERS::flags`

Definition at line 1041 of file structs.h.

3.124.2.4 value

`LONG SETUPPARAMETERS::value`

Definition at line 1042 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.125 SGroup Class Reference

Public Member Functions

- void print (void)
- void newGroup (int nelm, int nclss, int *clss)
- void clearGroup (void)
- void delGroup (void)
- int * genNext (void)
- void addGroup (int *p)
- BigInt nElem (void)
- int * nextElem (void)

Data Fields

- BigInt size
- BigInt nelem
- int ** elem
- int nnodes
- int neclass
- int eclass [GRCC_MAXNODES]
- int cgen
- int csav
- int permg [GRCC_MAXNODES]
- int perms [GRCC_MAXNODES]
- int pgr [GRCC_MAXNODES]
- int pgq [GRCC_MAXNODES]
- int psr [GRCC_MAXNODES]
- int psq [GRCC_MAXNODES]

3.125.1 Detailed Description

Definition at line 700 of file grcc.h.

3.125.2 Constructor & Destructor Documentation

3.125.2.1 SGroup()

```
SGroup::SGroup (
            void )
```

Definition at line 5376 of file grcc.cc.

3.125.2.2 ∼SGroup()

```
SGroup::∼SGroup (
            void )
```

Definition at line 5385 of file grcc.cc.

3.125.3 Member Function Documentation

3.125.3.1 print()

```
void SGroup::print (
            void )
```

Definition at line 5402 of file grcc.cc.

3.125.3.2 newGroup()

```
void SGroup::newGroup (
              int nelm,
              int nclss,
              int * clss )
```

Definition at line 5417 of file grcc.cc.

3.125.3.3 clearGroup()

```
void SGroup::clearGroup (
              void )
```

Definition at line 5442 of file grcc.cc.

3.125.3.4 delGroup()

```
void SGroup::delGroup (
              void )
```

Definition at line 5450 of file grcc.cc.

3.125.3.5 genNext()

```
int * SGroup::genNext (
              void )
```

Definition at line 5466 of file grcc.cc.

3.125.3.6 addGroup()

```
void SGroup::addGroup (
              int * p )
```

Definition at line 5476 of file grcc.cc.

3.125.3.7 nElem()

```
BigInt SGroup::nElem (
              void )
```

Definition at line 5493 of file grcc.cc.

3.125.3.8 nextElem()

```
int * SGroup::nextElem (
              void )
```

Definition at line 5499 of file grcc.cc.

3.125.4 Field Documentation

3.125.4.1 size

```
BigInt SGroup::size
```

Definition at line 702 of file grcc.h.

3.125.4.2 nelem

```
BigInt SGroup::nelem
```

Definition at line 703 of file grcc.h.

3.125.4.3 elem

```
int** SGroup::elem
```

Definition at line 704 of file grcc.h.

3.125.4.4 nnodes

```
int SGroup::nnodes
```

Definition at line 705 of file grcc.h.

3.125.4.5 neclass

```
int SGroup::neclass
```

Definition at line 706 of file grcc.h.

3.125.4.6 eclass

```
int SGroup::eclass[GRCC_MAXNODES]
```

Definition at line 707 of file grcc.h.

3.125.4.7 cgen

```
int SGroup::cgen
```

Definition at line 708 of file grcc.h.

3.125.4.8 csav

```
int SGroup::csav
```

Definition at line 709 of file grcc.h.

3.125.4.9 permg

```
int SGroup::permg[GRCC_MAXNODES]
```

Definition at line 710 of file grcc.h.

3.125.4.10 perms

```
int SGroup::perms[GRCC_MAXNODES]
```

Definition at line 711 of file grcc.h.

3.125.4.11 pgr

```
int SGroup::pgr[GRCC_MAXNODES]
```

Definition at line 713 of file grcc.h.

3.125.4.12 pgq

```
int SGroup::pgq[GRCC_MAXNODES]
```

Definition at line 714 of file grcc.h.

3.125.4.13 psr

```
int SGroup::psr[GRCC_MAXNODES]
```

Definition at line 715 of file grcc.h.

3.125.4.14 psq

```
int SGroup::psq[GRCC_MAXNODES]
```

Definition at line 716 of file grcc.h.

The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.126 SHvariables Struct Reference

Data Fields

- WORD ∗ outterm
- WORD ∗ outfun
- WORD ∗ incoef
- WORD ∗ stop1
- WORD ∗ stop2
- WORD ∗ ststop1
- WORD ∗ ststop2
- FINISHUFFLE finishuf
- DO_UFFLE do_uffle
- LONG combilast
- WORD nincoef
- WORD level
- WORD thefunction
- WORD option

3.126.1 Detailed Description

Definition at line 1276 of file structs.h.

3.126.2 Field Documentation

3.126.2.1 outterm

```
WORD* SHvariables::outterm
```

Definition at line 1277 of file structs.h.

3.126.2.2 outfun

```
WORD* SHvariables::outfun
```

Definition at line 1278 of file structs.h.

3.126.2.3 incoef

```
WORD* SHvariables::incoef
```

Definition at line 1279 of file structs.h.

3.126.2.4 stop1

```
WORD* SHvariables::stop1
```

Definition at line 1280 of file structs.h.

3.126.2.5 stop2

`WORD* SHvariables::stop2`

Definition at line 1281 of file structs.h.

3.126.2.6 ststop1

`WORD* SHvariables::ststop1`

Definition at line 1282 of file structs.h.

3.126.2.7 ststop2

`WORD* SHvariables::ststop2`

Definition at line 1283 of file structs.h.

3.126.2.8 finishuf

`FINISHUFFLE SHvariables::finishuf`

Definition at line 1284 of file structs.h.

3.126.2.9 do_uffle

`DO_UFFLE SHvariables::do_uffle`

Definition at line 1285 of file structs.h.

3.126.2.10 combilast

`LONG SHvariables::combilast`

Definition at line 1286 of file structs.h.

3.126.2.11 nincoef

`WORD SHvariables::nincoef`

Definition at line 1287 of file structs.h.

3.126.2.12 level

`WORD SHvariables::level`

Definition at line 1288 of file structs.h.

3.126.2.13 thefunction

```
WORD SHvariables::thefunction
```

Definition at line 1289 of file structs.h.

3.126.2.14 option

```
WORD SHvariables::option
```

Definition at line 1290 of file structs.h.

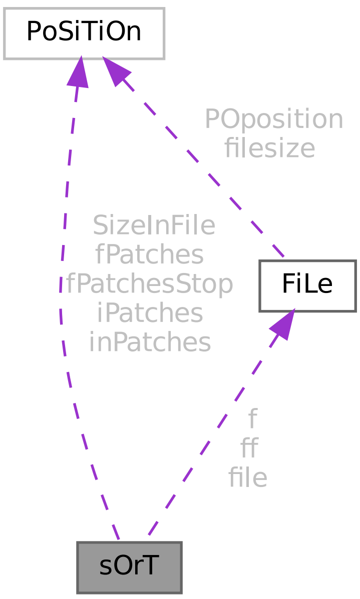
The documentation for this struct was generated from the following file:

- structs.h

3.127 sOrT Struct Reference

```
#include <structs.h>
```

Collaboration diagram for sOrT:



Public Member Functions

- **PADPOSITION** (24, 12, 12, 3, 0)

Data Fields

- FILEHANDLE file
- POSITION SizeInFile [3]
- WORD ∗ lBuffer
- WORD ∗ lTop
- WORD ∗ lFill
- WORD ∗ used
- WORD ∗ sBuffer
- WORD ∗ sTop
- WORD ∗ sTop2
- WORD ∗ sHalf
- WORD ∗ sFill
- WORD ∗∗ sPointer
- WORD ∗∗ PoinFill
- WORD ∗ cBuffer
- WORD ∗∗ Patches
- WORD ∗∗ pStop
- WORD ∗∗ poina
- WORD ∗∗ poin2a
- WORD ∗ ktoi
- WORD ∗ tree
- POSITION ∗ fPatches
- POSITION ∗ inPatches
- POSITION ∗ fPatchesStop
- POSITION ∗ iPatches
- FILEHANDLE ∗ f
- FILEHANDLE ∗∗ ff
- LONG sTerms
- LONG LargeSize
- LONG SmallSize
- LONG SmallEsize
- LONG TermsInSmall
- LONG Terms2InSmall
- LONG GenTerms
- LONG TermsLeft
- LONG GenSpace
- LONG SpaceLeft
- LONG putinsize
- LONG ninterms
- int MaxPatches
- int MaxFpatches
- int type
- int lPatch
- int fPatchN1
- int PolyWise
- int PolyFlag
- int cBufferSize
- int maxtermsize
- int newmaxtermsize
- int outputmode
- int stagelevel
- WORD fPatchN
- WORD inNum
- WORD stage4

3.127.1 Detailed Description

The struct SORTING is used to control a sort operation. It includes a small and a large buffer and arrays for keeping track of various stages of the (merge) sorts. Each sort level has its own struct and different levels can have different sizes for its arrays. Also different threads have their own set of SORTING structs.

Definition at line 1140 of file structs.h.

3.127.2 Field Documentation

3.127.2.1 file

```
FILEHANDLE sOrT::file
```

Definition at line 1141 of file structs.h.

3.127.2.2 SizeInFile

```
POSITION sOrT::SizeInFile[3]
```

Definition at line 1142 of file structs.h.

3.127.2.3 lBuffer

```
WORD* sOrT::lBuffer
```

Definition at line 1143 of file structs.h.

3.127.2.4 lTop

```
WORD* sOrT::lTop
```

Definition at line 1144 of file structs.h.

3.127.2.5 lFill

```
WORD* sOrT::lFill
```

Definition at line 1145 of file structs.h.

3.127.2.6 used

```
WORD* sOrT::used
```

Definition at line 1146 of file structs.h.

3.127.2.7 sBuffer

`WORD* sOrT::sBuffer`

Definition at line 1147 of file structs.h.

3.127.2.8 sTop

`WORD* sOrT::sTop`

Definition at line 1148 of file structs.h.

3.127.2.9 sTop2

`WORD* sOrT::sTop2`

Definition at line 1149 of file structs.h.

3.127.2.10 sHalf

`WORD* sOrT::sHalf`

Definition at line 1150 of file structs.h.

3.127.2.11 sFill

`WORD* sOrT::sFill`

Definition at line 1151 of file structs.h.

3.127.2.12 sPointer

`WORD** sOrT::sPointer`

Definition at line 1152 of file structs.h.

3.127.2.13 PoinFill

`WORD** sOrT::PoinFill`

Definition at line 1153 of file structs.h.

3.127.2.14 cBuffer

`WORD* sOrT::cBuffer`

Definition at line 1154 of file structs.h.

3.127.2.15 Patches

```
WORD** sOrT::Patches
```

Definition at line 1155 of file structs.h.

3.127.2.16 pStop

```
WORD** sOrT::pStop
```

Definition at line 1156 of file structs.h.

3.127.2.17 poina

```
WORD** sOrT::poina
```

Definition at line 1157 of file structs.h.

3.127.2.18 poin2a

```
WORD** sOrT::poin2a
```

Definition at line 1158 of file structs.h.

3.127.2.19 ktoi

```
WORD* sOrT::ktoi
```

Definition at line 1159 of file structs.h.

3.127.2.20 tree

```
WORD* sOrT::tree
```

Definition at line 1160 of file structs.h.

3.127.2.21 fPatches

```
POSITION* sOrT::fPatches
```

Definition at line 1166 of file structs.h.

3.127.2.22 inPatches

```
POSITION* sOrT::inPatches
```

Definition at line 1167 of file structs.h.

3.127.2.23 fPatchesStop

`POSITION* sOrT::fPatchesStop`

Definition at line 1168 of file structs.h.

3.127.2.24 iPatches

`POSITION* sOrT::iPatches`

Definition at line 1169 of file structs.h.

3.127.2.25 f

`FILEHANDLE* sOrT::f`

Definition at line 1170 of file structs.h.

3.127.2.26 ff

`FILEHANDLE** sOrT::ff`

Definition at line 1171 of file structs.h.

3.127.2.27 sTerms

`LONG sOrT::sTerms`

Definition at line 1172 of file structs.h.

3.127.2.28 LargeSize

`LONG sOrT::LargeSize`

Definition at line 1173 of file structs.h.

3.127.2.29 SmallSize

`LONG sOrT::SmallSize`

Definition at line 1174 of file structs.h.

3.127.2.30 SmallEsize

`LONG sOrT::SmallEsize`

Definition at line 1175 of file structs.h.

3.127.2.31 TermsInSmall

`LONG sOrT::TermsInSmall`

Definition at line 1176 of file structs.h.

3.127.2.32 Terms2InSmall

`LONG sOrT::Terms2InSmall`

Definition at line 1177 of file structs.h.

3.127.2.33 GenTerms

`LONG sOrT::GenTerms`

Definition at line 1178 of file structs.h.

3.127.2.34 TermsLeft

`LONG sOrT::TermsLeft`

Definition at line 1179 of file structs.h.

3.127.2.35 GenSpace

`LONG sOrT::GenSpace`

Definition at line 1180 of file structs.h.

3.127.2.36 SpaceLeft

`LONG sOrT::SpaceLeft`

Definition at line 1181 of file structs.h.

3.127.2.37 putinsize

`LONG sOrT::putinsize`

Definition at line 1182 of file structs.h.

3.127.2.38 ninterms

`LONG sOrT::ninterms`

Definition at line 1183 of file structs.h.

3.127.2.39 MaxPatches

```
int sOrT::MaxPatches
```

Definition at line 1184 of file structs.h.

3.127.2.40 MaxFpatches

```
int sOrT::MaxFpatches
```

Definition at line 1185 of file structs.h.

3.127.2.41 type

```
int sOrT::type
```

Definition at line 1186 of file structs.h.

3.127.2.42 lPatch

```
int sOrT::lPatch
```

Definition at line 1187 of file structs.h.

3.127.2.43 fPatchN1

```
int sOrT::fPatchN1
```

Definition at line 1188 of file structs.h.

3.127.2.44 PolyWise

```
int sOrT::PolyWise
```

Definition at line 1189 of file structs.h.

3.127.2.45 PolyFlag

```
int sOrT::PolyFlag
```

Definition at line 1190 of file structs.h.

3.127.2.46 cBufferSize

```
int sOrT::cBufferSize
```

Definition at line 1191 of file structs.h.

3.127.2.47 maxtermsize

```
int sOrT::maxtermsize
```

Definition at line 1192 of file structs.h.

3.127.2.48 newmaxtermsize

```
int sOrT::newmaxtermsize
```

Definition at line 1193 of file structs.h.

3.127.2.49 outputmode

```
int sOrT::outputmode
```

Definition at line 1194 of file structs.h.

3.127.2.50 stagelevel

```
int sOrT::stagelevel
```

Definition at line 1195 of file structs.h.

3.127.2.51 fPatchN

```
WORD sOrT::fPatchN
```

Definition at line 1196 of file structs.h.

3.127.2.52 inNum

```
WORD sOrT::inNum
```

Definition at line 1197 of file structs.h.

3.127.2.53 stage4

```
WORD sOrT::stage4
```

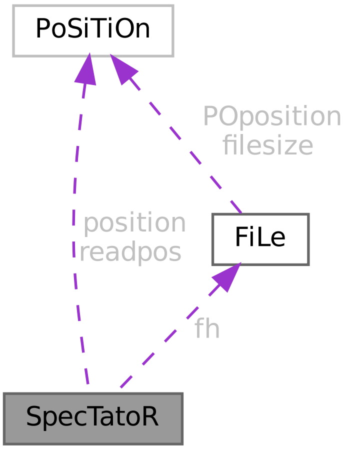
Definition at line 1198 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.128 SpecTatoR Struct Reference

Collaboration diagram for SpecTatoR:



Public Member Functions

- **PADPOSITION** (2, 0, 0, 2, 0)

Data Fields

- POSITION position
- POSITION readpos
- FILEHANDLE ∗ fh
- char ∗ name
- WORD exprnumber
- WORD flags

3.128.1 Detailed Description

Definition at line 765 of file structs.h.

3.128.2 Field Documentation

3.128.2.1 position

```
POSITION SpecTatoR::position
```

Definition at line 766 of file structs.h.

3.128.2.2 readpos

`POSITION SpecTatoR::readpos`

Definition at line 767 of file structs.h.

3.128.2.3 fh

`FILEHANDLE* SpecTatoR::fh`

Definition at line 768 of file structs.h.

3.128.2.4 name

`char* SpecTatoR::name`

Definition at line 769 of file structs.h.

3.128.2.5 exprnumber

`WORD SpecTatoR::exprnumber`

Definition at line 770 of file structs.h.

3.128.2.6 flags

`WORD SpecTatoR::flags`

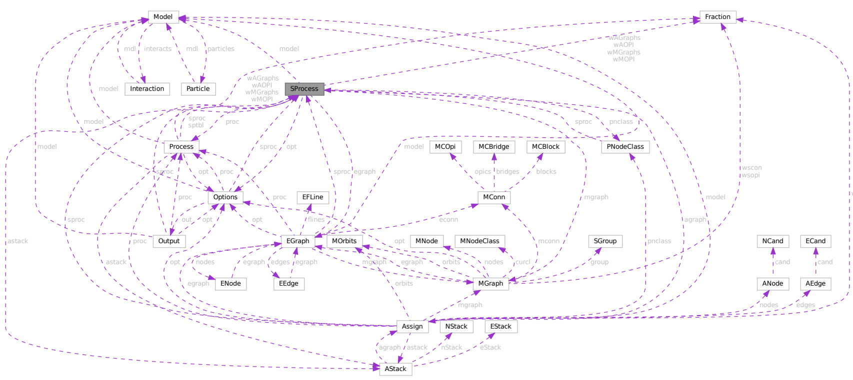
Definition at line 771 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.129 SProcess Class Reference

Collaboration diagram for SProcess:



Public Member Functions

- SProcess (Model ∗mdl, Process ∗prc, Options ∗opts, int sid, int ∗clst, int ncls, NCInput ∗cls)
- SProcess (Model ∗mdl, Process ∗prc, Options ∗opts, int sid, int ∗clst, int ncls, int ∗cdeg, int ∗ctyp, int ∗ptcl, int ∗cpl, int ∗cnum, int ∗cmind, int ∗cmaxd)
- void prSProcess (void)
- BigInt generate (void)
- void assign (MGraph ∗mgr)
- int toMNodeClass (int ∗ctyp, int ∗cldeg, int ∗clnum, int ∗cmind, int ∗cmaxd)
- PNodeClass ∗ match (MGraph ∗mgr)
- void endMGraph (MGraph ∗mgr)
- void endAGraph (EGraph ∗egr)
- void resultMGraph (BigInt nmgraphs, Fraction mwsum, BigInt nmopi, Fraction mwopi)
- void resultAGraph (BigInt nagraphs, Fraction awsum, BigInt naopi, Fraction awopi)

Data Fields

- Model ∗ model
- Process ∗ proc
- Options ∗ opt
- PNodeClass ∗ pnclass
- AStack ∗ astack
- MGraph ∗ mgraph
- EGraph ∗ egraph
- Assign ∗ agraph
- int ∗ cl2nd
- int ∗ nd2cl
- int nclass
- int ninitl
- int nfinal
- int nvert
- int clist [GRCC_MAXNCPLG]
- int id
- int loop
- int nNodes
- int nEdges
- int nExtern
- int ncouple
- int tCouple
- int DUMMYPADDING
- BigInt mgrcount
- BigInt agrcount
- BigInt extperm
- BigInt nMGraphs
- BigInt nMOPI
- Fraction wMGraphs
- Fraction wMOPI
- BigInt nAGraphs
- BigInt nAOPI
- Fraction wAGraphs
- Fraction wAOPI

3.129.1 Detailed Description

Definition at line 350 of file grcc.h.

3.129.2 Constructor & Destructor Documentation

3.129.2.1 SProcess() [1/2]

```
SProcess::SProcess (
            Model * mdl,
            Process * prc,
            Options * opts,
            int sid,
            int * clst,
            int ncls,
            NCInput * cls )
```

Definition at line 2695 of file grcc.cc.

3.129.2.2 SProcess() [2/2]

```
SProcess::SProcess (
            Model * mdl,
            Process * prc,
            Options * opts,
            int sid,
            int * clst,
            int ncls,
            int * cdeg,
            int * ctyp,
            int * ptcl,
            int * cpl,
            int * cnum,
            int * cmind,
            int * cmaxd )
```

Definition at line 2526 of file grcc.cc.

3.129.2.3 ∼SProcess()

```
SProcess::∼SProcess (
            void )
```

Definition at line 2857 of file grcc.cc.

3.129.3 Member Function Documentation

3.129.3.1 prSProcess()

```
void SProcess::prSProcess (
            void )
```

Definition at line 2865 of file grcc.cc.

3.129.3.2 generate()

```
BigInt SProcess::generate (
             void )
```

Definition at line 2877 of file grcc.cc.

3.129.3.3 assign()

```
void SProcess::assign (
             MGraph * mgr )
```

Definition at line 2921 of file grcc.cc.

3.129.3.4 toMNodeClass()

```
int SProcess::toMNodeClass (
             int * ctyp,
             int * cldeg,
             int * clnum,
             int * cmind,
             int * cmaxd )
```

Definition at line 2902 of file grcc.cc.

3.129.3.5 match()

```
PNodeClass * SProcess::match (
             MGraph * mgr )
```

Definition at line 2938 of file grcc.cc.

3.129.3.6 endMGraph()

```
void SProcess::endMGraph (
             MGraph * mgr )
```

Definition at line 3017 of file grcc.cc.

3.129.3.7 endAGraph()

```
void SProcess::endAGraph (
             EGraph * egr )
```

Definition at line 3027 of file grcc.cc.

3.129.3.8 resultMGraph()

```
void SProcess::resultMGraph (
            BigInt nmgraphs,
            Fraction mwsum,
            BigInt nmopi,
            Fraction mwopi )
```

Definition at line 2999 of file grcc.cc.

3.129.3.9 resultAGraph()

```
void SProcess::resultAGraph (
            BigInt nagraphs,
            Fraction awsum,
            BigInt naopi,
            Fraction awopi )
```

Definition at line 3008 of file grcc.cc.

3.129.4 Field Documentation

3.129.4.1 model

```
Model* SProcess::model
```

Definition at line 357 of file grcc.h.

3.129.4.2 proc

```
Process* SProcess::proc
```

Definition at line 358 of file grcc.h.

3.129.4.3 opt

```
Options* SProcess::opt
```

Definition at line 359 of file grcc.h.

3.129.4.4 pnclass

```
PNodeClass* SProcess::pnclass
```

Definition at line 360 of file grcc.h.

3.129.4.5 astack

`AStack* SProcess::astack`

Definition at line 361 of file grcc.h.

3.129.4.6 mgraph

`MGraph* SProcess::mgraph`

Definition at line 363 of file grcc.h.

3.129.4.7 egraph

`EGraph* SProcess::egraph`

Definition at line 364 of file grcc.h.

3.129.4.8 agraph

`Assign* SProcess::agraph`

Definition at line 365 of file grcc.h.

3.129.4.9 cl2nd

`int* SProcess::cl2nd`

Definition at line 367 of file grcc.h.

3.129.4.10 nd2cl

`int* SProcess::nd2cl`

Definition at line 369 of file grcc.h.

3.129.4.11 nclass

`int SProcess::nclass`

Definition at line 370 of file grcc.h.

3.129.4.12 ninitl

`int SProcess::ninitl`

Definition at line 372 of file grcc.h.

3.129.4.13 nfinal

```
int SProcess::nfinal
```

Definition at line 373 of file grcc.h.

3.129.4.14 nvert

```
int SProcess::nvert
```

Definition at line 374 of file grcc.h.

3.129.4.15 clist

```
int SProcess::clist[GRCC_MAXNCPLG]
```

Definition at line 376 of file grcc.h.

3.129.4.16 id

```
int SProcess::id
```

Definition at line 377 of file grcc.h.

3.129.4.17 loop

```
int SProcess::loop
```

Definition at line 378 of file grcc.h.

3.129.4.18 nNodes

```
int SProcess::nNodes
```

Definition at line 379 of file grcc.h.

3.129.4.19 nEdges

```
int SProcess::nEdges
```

Definition at line 380 of file grcc.h.

3.129.4.20 nExtern

```
int SProcess::nExtern
```

Definition at line 381 of file grcc.h.

3.129.4.21 ncouple

```
int SProcess::ncouple
```

Definition at line 382 of file grcc.h.

3.129.4.22 tCouple

```
int SProcess::tCouple
```

Definition at line 383 of file grcc.h.

3.129.4.23 DUMMYPADDING

```
int SProcess::DUMMYPADDING
```

Definition at line 385 of file grcc.h.

3.129.4.24 mgrcount

```
BigInt SProcess::mgrcount
```

Definition at line 387 of file grcc.h.

3.129.4.25 agrcount

```
BigInt SProcess::agrcount
```

Definition at line 388 of file grcc.h.

3.129.4.26 extperm

```
BigInt SProcess::extperm
```

Definition at line 389 of file grcc.h.

3.129.4.27 nMGraphs

```
BigInt SProcess::nMGraphs
```

Definition at line 392 of file grcc.h.

3.129.4.28 nMOPI

```
BigInt SProcess::nMOPI
```

Definition at line 393 of file grcc.h.

3.129.4.29 wMGraphs

`Fraction SProcess::wMGraphs`

Definition at line 394 of file grcc.h.

3.129.4.30 wMOPI

`Fraction SProcess::wMOPI`

Definition at line 395 of file grcc.h.

3.129.4.31 nAGraphs

`BigInt SProcess::nAGraphs`

Definition at line 397 of file grcc.h.

3.129.4.32 nAOPI

`BigInt SProcess::nAOPI`

Definition at line 398 of file grcc.h.

3.129.4.33 wAGraphs

`Fraction SProcess::wAGraphs`

Definition at line 399 of file grcc.h.

3.129.4.34 wAOPI

`Fraction SProcess::wAOPI`

Definition at line 400 of file grcc.h.

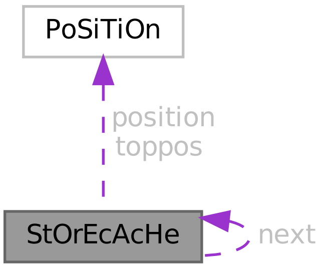
The documentation for this class was generated from the following files:

- grcc.h
- grcc.cc

3.130 StOrEcAcHe Struct Reference

```
#include <structs.h>
```

Collaboration diagram for StOrEcAcHe:



Public Member Functions

- **PADPOSITION** (1, 0, 0, 2, 0)

Data Fields

- POSITION position
- POSITION toppos
- struct StOrEcAcHe ∗ next
- WORD buffer [2]

3.130.1 Detailed Description

The struct of type STORECACHE is used by a caching system for reading terms from stored expressions. Each thread should have its own system of caches.

Definition at line 1065 of file structs.h.

3.130.2 Field Documentation

3.130.2.1 position

```
POSITION StOrEcAcHe::position
```

Definition at line 1066 of file structs.h.

3.130.2.2 toppos

`POSITION StOrEcAcHe::toppos`

Definition at line 1067 of file structs.h.

3.130.2.3 next

`struct StOrEcAcHe* StOrEcAcHe::next`

Definition at line 1068 of file structs.h.

3.130.2.4 buffer

`WORD StOrEcAcHe::buffer[2]`

Definition at line 1069 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.131 STOREHEADER Struct Reference

`#include <structs.h>`

Data Fields

- UBYTE headermark [8]
- UBYTE lenWORD
- UBYTE lenLONG
- UBYTE lenPOS
- UBYTE lenPOINTER
- UBYTE endianness [16]
- UBYTE sSym
- UBYTE sInd
- UBYTE sVec
- UBYTE sFun
- UBYTE maxpower [16]
- UBYTE wildoffset [16]
- UBYTE revision
- UBYTE reserved [512-8-4-16-4-16-16-1]

3.131.1 Detailed Description

Defines the structure of the file header for store-files and save-files.

The first 8 bytes serve as a unique mark to identity save-files that contain such a header. Older versions of FORM don't have this header and will write the POSITION of the next file index (struct FiLeInDeX) here, which is always different from this pattern.

It is always 512 bytes long.

Definition at line 75 of file structs.h.

3.131.2 Field Documentation

3.131.2.1 headermark

```
UBYTE STOREHEADER::headermark[8]
```

Pattern for header identification. Old versions of FORM have a maximum sizeof(POSITION) of 8

Definition at line 76 of file structs.h.

3.131.2.2 lenWORD

```
UBYTE STOREHEADER::lenWORD
```

Number of bytes for WORD

Definition at line 78 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.3 lenLONG

```
UBYTE STOREHEADER::lenLONG
```

Number of bytes for LONG

Definition at line 79 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.4 lenPOS

```
UBYTE STOREHEADER::lenPOS
```

Number of bytes for POSITION

Definition at line 80 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.5 lenPOINTER

`UBYTE STOREHEADER::lenPOINTER`

Number of bytes for void ∗

Definition at line 81 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.6 endianness

`UBYTE STOREHEADER::endianness[16]`

Used to determine endianness, sizeof(int) should be <= 16

Definition at line 82 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.7 sSym

`UBYTE STOREHEADER::sSym`

sizeof(struct SyMbOl)

Definition at line 83 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.8 sInd

`UBYTE STOREHEADER::sInd`

sizeof(struct InDeX)

Definition at line 84 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.9 sVec

`UBYTE STOREHEADER::sVec`

sizeof(struct VeCtOr)

Definition at line 85 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.10 sFun

```
UBYTE STOREHEADER::sFun
```

sizeof(struct FuNcTiOn)

Definition at line 86 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.11 maxpower

```
UBYTE STOREHEADER::maxpower[16]
```

Maximum power, see MAXPOWER

Definition at line 87 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.12 wildoffset

```
UBYTE STOREHEADER::wildoffset[16]
```

WILDOFFSET macro

Definition at line 88 of file structs.h.

Referenced by WriteStoreHeader().

3.131.2.13 revision

```
UBYTE STOREHEADER::revision
```

Revision number of save-file system

Definition at line 89 of file structs.h.

3.131.2.14 reserved

```
UBYTE STOREHEADER::reserved[512-8-4-16-4-16-16-1]
```

Padding to 512 bytes

Definition at line 90 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.132 StreaM Struct Reference

```
#include <structs.h>
```

Public Member Functions

- **PADPOSITION** (6, 3, 9, 0, 4)

Data Fields

- off_t fileposition
- off_t linenumber
- off_t prevline
- UBYTE * buffer
- UBYTE * pointer
- UBYTE * top
- UBYTE * FoldName
- UBYTE * name
- UBYTE * pname
- LONG buffersize
- LONG bufferposition
- LONG inbuffer
- int previous
- int handle
- int type
- int prevars
- int previousNoShowInput
- int eqnum
- int afterwards
- int olddelay
- int oldnoshowinput
- UBYTE isnextchar
- UBYTE nextchar [2]
- UBYTE reserved

3.132.1 Detailed Description

Input is read from 'streams' which are represented by objects of type STREAM. A stream can be a file, a do-loop, a procedure, the string value of a preprocessor variable ..... When a new stream is opened we have to keep information about where to fall back in the parent stream to allow this to happen even in the middle of reading names etc as would be the case with a`i'b

Definition at line 736 of file structs.h.

3.132.2 Field Documentation

3.132.2.1 fileposition

```
off_t StreaM::fileposition
```

Definition at line 737 of file structs.h.

3.132.2.2 linenumber

`off_t StreaM::linenumber`

Definition at line 738 of file structs.h.

3.132.2.3 prevline

`off_t StreaM::prevline`

Definition at line 739 of file structs.h.

3.132.2.4 buffer

`UBYTE* StreaM::buffer`

[D] Size in buffersize

Definition at line 740 of file structs.h.

3.132.2.5 pointer

`UBYTE* StreaM::pointer`

pointer into buffer memory

Definition at line 741 of file structs.h.

3.132.2.6 top

`UBYTE* StreaM::top`

pointer into buffer memory

Definition at line 742 of file structs.h.

3.132.2.7 FoldName

`UBYTE* StreaM::FoldName`

[D]

Definition at line 743 of file structs.h.

3.132.2.8 name

`UBYTE* StreaM::name`

[D]

Definition at line 744 of file structs.h.

3.132.2.9 pname

`UBYTE* StreaM::pname`

for DOLLARSTREAM and PREVARSTREAM it points always to name, else it is undefined

Definition at line 745 of file structs.h.

3.132.2.10 buffersize

`LONG StreaM::buffersize`

Definition at line 747 of file structs.h.

3.132.2.11 bufferposition

`LONG StreaM::bufferposition`

Definition at line 748 of file structs.h.

3.132.2.12 inbuffer

`LONG StreaM::inbuffer`

Definition at line 749 of file structs.h.

3.132.2.13 previous

`int StreaM::previous`

Definition at line 750 of file structs.h.

3.132.2.14 handle

`int StreaM::handle`

Definition at line 751 of file structs.h.

3.132.2.15 type

`int StreaM::type`

Definition at line 752 of file structs.h.

3.132.2.16 prevars

`int StreaM::prevars`

Definition at line 753 of file structs.h.

3.132.2.17 previousNoShowInput

`int StreaM::previousNoShowInput`

Definition at line 754 of file structs.h.

3.132.2.18 eqnum

`int StreaM::eqnum`

Definition at line 755 of file structs.h.

3.132.2.19 afterwards

`int StreaM::afterwards`

Definition at line 756 of file structs.h.

3.132.2.20 olddelay

`int StreaM::olddelay`

Definition at line 757 of file structs.h.

3.132.2.21 oldnoshowinput

`int StreaM::oldnoshowinput`

Definition at line 758 of file structs.h.

3.132.2.22 isnextchar

`UBYTE StreaM::isnextchar`

Definition at line 759 of file structs.h.

3.132.2.23 nextchar

```
UBYTE StreaM::nextchar[2]
```

Definition at line 760 of file structs.h.

3.132.2.24 reserved

```
UBYTE StreaM::reserved
```

Definition at line 761 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.133 SuBbUf Struct Reference

Data Fields

- WORD subexpnum
- WORD buffernum

3.133.1 Detailed Description

Definition at line 257 of file compiler.c.

3.133.2 Field Documentation

3.133.2.1 subexpnum

```
WORD SuBbUf::subexpnum
```

Definition at line 258 of file compiler.c.

3.133.2.2 buffernum

```
WORD SuBbUf::buffernum
```

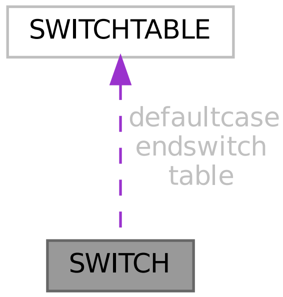
Definition at line 259 of file compiler.c.

The documentation for this struct was generated from the following file:

- compiler.c

3.134 SWITCH Struct Reference

Collaboration diagram for SWITCH:



Data Fields

- SWITCHTABLE * table
- SWITCHTABLE defaultcase
- SWITCHTABLE endswitch
- WORD typetable
- WORD maxcase
- WORD mincase
- WORD numcases
- WORD tablesize
- WORD caseoffset
- WORD iflevel
- WORD whilelevel
- WORD nestingsum
- WORD padding

3.134.1 Detailed Description

Definition at line 1380 of file structs.h.

3.134.2 Field Documentation

3.134.2.1 table

```
SWITCHTABLE* SWITCH::table
```

Definition at line 1381 of file structs.h.

3.134.2.2 defaultcase

`SWITCHTABLE SWITCH::defaultcase`

Definition at line 1382 of file structs.h.

3.134.2.3 endswitch

`SWITCHTABLE SWITCH::endswitch`

Definition at line 1383 of file structs.h.

3.134.2.4 typetable

`WORD SWITCH::typetable`

Definition at line 1384 of file structs.h.

3.134.2.5 maxcase

`WORD SWITCH::maxcase`

Definition at line 1385 of file structs.h.

3.134.2.6 mincase

`WORD SWITCH::mincase`

Definition at line 1386 of file structs.h.

3.134.2.7 numcases

`WORD SWITCH::numcases`

Definition at line 1387 of file structs.h.

3.134.2.8 tablesize

`WORD SWITCH::tablesize`

Definition at line 1388 of file structs.h.

3.134.2.9 caseoffset

`WORD SWITCH::caseoffset`

Definition at line 1389 of file structs.h.

3.134.2.10 iflevel

`WORD SWITCH::iflevel`

Definition at line 1390 of file structs.h.

3.134.2.11 whilelevel

`WORD SWITCH::whilelevel`

Definition at line 1391 of file structs.h.

3.134.2.12 nestingsum

`WORD SWITCH::nestingsum`

Definition at line 1392 of file structs.h.

3.134.2.13 padding

`WORD SWITCH::padding`

Definition at line 1393 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.135 SWITCHTABLE Struct Reference

Data Fields

- WORD ncase
- WORD value
- WORD compbuffer

3.135.1 Detailed Description

Definition at line 1374 of file structs.h.

3.135.2 Field Documentation

3.135.2.1 ncase

`WORD SWITCHTABLE::ncase`

Definition at line 1375 of file structs.h.

3.135.2.2 value

```
WORD SWITCHTABLE::value
```

Definition at line 1376 of file structs.h.

3.135.2.3 compbuffer

```
WORD SWITCHTABLE::compbuffer
```

Definition at line 1377 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.136 SyMbOl Struct Reference

Data Fields

- LONG name
- WORD minpower
- WORD maxpower
- WORD complex
- WORD number
- WORD flags
- WORD node
- WORD namesize
- WORD dimension

3.136.1 Detailed Description

Definition at line 428 of file structs.h.

3.136.2 Field Documentation

3.136.2.1 name

```
LONG SyMbOl::name
```

Definition at line 429 of file structs.h.

3.136.2.2 minpower

```
WORD SyMbOl::minpower
```

Definition at line 430 of file structs.h.

3.136.2.3 maxpower

`WORD SyMbOl::maxpower`

Definition at line 431 of file structs.h.

3.136.2.4 complex

`WORD SyMbOl::complex`

Definition at line 432 of file structs.h.

3.136.2.5 number

`WORD SyMbOl::number`

Definition at line 433 of file structs.h.

3.136.2.6 flags

`WORD SyMbOl::flags`

Definition at line 434 of file structs.h.

3.136.2.7 node

`WORD SyMbOl::node`

Definition at line 435 of file structs.h.

3.136.2.8 namesize

`WORD SyMbOl::namesize`

Definition at line 436 of file structs.h.

3.136.2.9 dimension

`WORD SyMbOl::dimension`

Definition at line 437 of file structs.h.

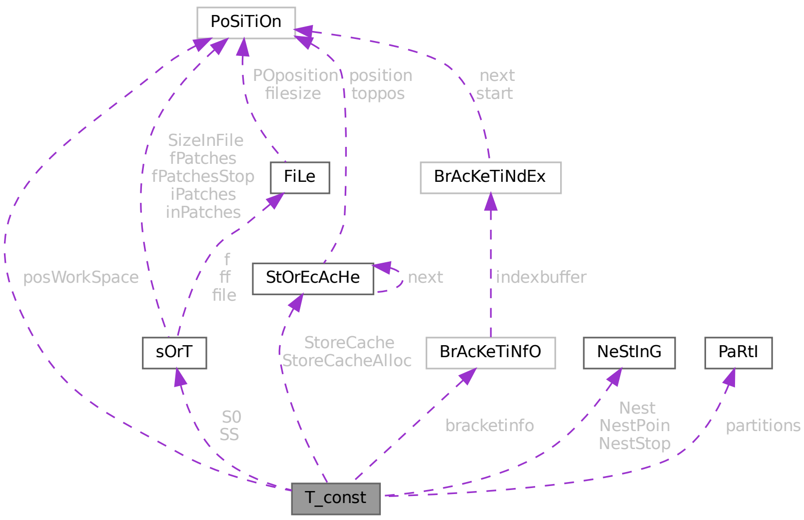
The documentation for this struct was generated from the following file:

- structs.h

3.137 T_const Struct Reference

```
#include <structs.h>
```

Collaboration diagram for T_const:



Data Fields

- SORTING ∗ S0
- SORTING ∗ SS
- NESTING Nest
- NESTING NestStop
- NESTING NestPoin
- WORD ∗ BrackBuf
- STORECACHE StoreCache
- STORECACHE StoreCacheAlloc
- WORD ∗∗ pWorkSpace
- LONG ∗ lWorkSpace
- POSITION ∗ posWorkSpace
- WORD ∗ WorkSpace
- WORD ∗ WorkTop
- WORD ∗ WorkPointer
- int ∗ RepCount
- int ∗ RepTop
- WORD ∗ WildArgTaken
- UWORD ∗ factorials
- WORD ∗ small_power_n
- UWORD ∗∗ small_power
- UWORD ∗ bernoullis
- WORD ∗ primelist
- LONG ∗ pfac
- LONG ∗ pBer
- WORD ∗ TMaddr

- WORD * WildMask
- WORD * previousEfactor
- WORD ** TermMemHeap
- UWORD ** NumberMemHeap
- UWORD ** CacheNumberMemHeap
- BRACKETINFO * bracketinfo
- WORD ** ListPoly
- WORD * ListSymbols
- UWORD * NumMem
- WORD * TopologiesTerm
- WORD * TopologiesStart
- PARTI partitions
- LONG sBer
- LONG pWorkPointer
- LONG lWorkPointer
- LONG posWorkPointer
- LONG InNumMem
- int sfact
- int mfac
- int ebufnum
- int fbufnum
- int allbufnum
- int aebufnum
- int idallflag
- int idallnum
- int idallmaxnum
- int WildcardBufferSize
- int TermMemMax
- int TermMemTop
- int NumberMemMax
- int NumberMemTop
- int CacheNumberMemMax
- int CacheNumberMemTop
- int bracketindexflag
- int optimtimes
- int ListSymbolsSize
- int NumListSymbols
- int numpoly
- int LeaveNegative
- int TrimPower
- WORD small_power_maxx
- WORD small_power_maxn
- WORD dummysubexp [SUBEXPSIZE+4]
- WORD comsym [8]
- WORD comnum [4]
- WORD comfun [FUNHEAD+4]
- WORD comind [7]
- WORD MinVecArg [7+ARGHEAD]
- WORD FunArg [4+ARGHEAD+FUNHEAD]
- WORD locwildvalue [SUBEXPSIZE]
- WORD mulpat [SUBEXPSIZE+5]
- WORD proexp [SUBEXPSIZE+5]
- WORD TMout [40]
- WORD TMbuff
- WORD TMdolfac

- WORD nfac
- WORD nBer
- WORD mBer
- WORD PolyAct
- WORD RecFlag
- WORD inprimelist
- WORD sizeprimelist
- WORD fromindex
- WORD setinterntopo
- WORD setexterntopo
- WORD TopologiesLevel
- WORD TopologiesOptions [2]

3.137.1 Detailed Description

The T_const struct is part of the global data and resides either in the ALLGLOBALS struct A, or the ALLPRIVATES struct B (TFORM) under the name T We see it used with the macro AT as in AT.WorkPointer It has variables that are private to each thread, most of which have to be defined at startup.

Definition at line 2121 of file structs.h.

3.137.2 Field Documentation

3.137.2.1 S0

```
SORTING* T_const::S0
```

Definition at line 2125 of file structs.h.

3.137.2.2 SS

```
SORTING* T_const::SS
```

Definition at line 2126 of file structs.h.

3.137.2.3 Nest

```
NESTING T_const::Nest
```

Definition at line 2127 of file structs.h.

3.137.2.4 NestStop

```
NESTING T_const::NestStop
```

Definition at line 2128 of file structs.h.

3.137.2.5 NestPoin

`NESTING T_const::NestPoin`

Definition at line 2129 of file structs.h.

3.137.2.6 BrackBuf

`WORD* T_const::BrackBuf`

Definition at line 2130 of file structs.h.

3.137.2.7 StoreCache

`STORECACHE T_const::StoreCache`

Definition at line 2131 of file structs.h.

3.137.2.8 StoreCacheAlloc

`STORECACHE T_const::StoreCacheAlloc`

Definition at line 2132 of file structs.h.

3.137.2.9 pWorkSpace

`WORD** T_const::pWorkSpace`

Definition at line 2133 of file structs.h.

3.137.2.10 lWorkSpace

`LONG* T_const::lWorkSpace`

Definition at line 2134 of file structs.h.

3.137.2.11 posWorkSpace

`POSITION* T_const::posWorkSpace`

Definition at line 2135 of file structs.h.

3.137.2.12 WorkSpace

`WORD* T_const::WorkSpace`

Definition at line 2136 of file structs.h.

3.137.2.13 WorkTop

`WORD* T_const::WorkTop`

Definition at line 2137 of file structs.h.

3.137.2.14 WorkPointer

`WORD* T_const::WorkPointer`

Definition at line 2138 of file structs.h.

3.137.2.15 RepCount

`int* T_const::RepCount`

Definition at line 2139 of file structs.h.

3.137.2.16 RepTop

`int* T_const::RepTop`

Definition at line 2140 of file structs.h.

3.137.2.17 WildArgTaken

`WORD* T_const::WildArgTaken`

Definition at line 2141 of file structs.h.

3.137.2.18 factorials

`UWORD* T_const::factorials`

Definition at line 2142 of file structs.h.

3.137.2.19 small_power_n

`WORD* T_const::small_power_n`

Definition at line 2143 of file structs.h.

3.137.2.20 small_power

`UWORD** T_const::small_power`

Definition at line 2144 of file structs.h.

3.137.2.21 bernoullis

UWORD* T_const::bernoullis

Definition at line 2145 of file structs.h.

3.137.2.22 primelist

WORD* T_const::primelist

Definition at line 2146 of file structs.h.

3.137.2.23 pfac

LONG* T_const::pfac

Definition at line 2147 of file structs.h.

3.137.2.24 pBer

LONG* T_const::pBer

Definition at line 2148 of file structs.h.

3.137.2.25 TMaddr

WORD* T_const::TMaddr

Definition at line 2149 of file structs.h.

3.137.2.26 WildMask

WORD* T_const::WildMask

Definition at line 2150 of file structs.h.

3.137.2.27 previousEfactor

WORD* T_const::previousEfactor

Definition at line 2151 of file structs.h.

3.137.2.28 TermMemHeap

WORD** T_const::TermMemHeap

Definition at line 2152 of file structs.h.

3.137.2.29 NumberMemHeap

`UWORD** T_const::NumberMemHeap`

Definition at line 2153 of file structs.h.

3.137.2.30 CacheNumberMemHeap

`UWORD** T_const::CacheNumberMemHeap`

Definition at line 2154 of file structs.h.

3.137.2.31 bracketinfo

`BRACKETINFO* T_const::bracketinfo`

Definition at line 2155 of file structs.h.

3.137.2.32 ListPoly

`WORD** T_const::ListPoly`

Definition at line 2156 of file structs.h.

3.137.2.33 ListSymbols

`WORD* T_const::ListSymbols`

Definition at line 2157 of file structs.h.

3.137.2.34 NumMem

`UWORD* T_const::NumMem`

Definition at line 2158 of file structs.h.

3.137.2.35 TopologiesTerm

`WORD* T_const::TopologiesTerm`

Definition at line 2159 of file structs.h.

3.137.2.36 TopologiesStart

`WORD* T_const::TopologiesStart`

Definition at line 2160 of file structs.h.

3.137.2.37 partitions

`PARTI T_const::partitions`

Definition at line 2169 of file structs.h.

3.137.2.38 sBer

`LONG T_const::sBer`

Definition at line 2170 of file structs.h.

3.137.2.39 pWorkPointer

`LONG T_const::pWorkPointer`

Definition at line 2171 of file structs.h.

3.137.2.40 lWorkPointer

`LONG T_const::lWorkPointer`

Definition at line 2172 of file structs.h.

3.137.2.41 posWorkPointer

`LONG T_const::posWorkPointer`

Definition at line 2173 of file structs.h.

3.137.2.42 InNumMem

`LONG T_const::InNumMem`

Definition at line 2174 of file structs.h.

3.137.2.43 sfact

`int T_const::sfact`

Definition at line 2175 of file structs.h.

3.137.2.44 mfac

`int T_const::mfac`

Definition at line 2176 of file structs.h.

3.137.2.45 ebufnum

```
int T_const::ebufnum
```

Definition at line 2177 of file structs.h.

3.137.2.46 fbufnum

```
int T_const::fbufnum
```

Definition at line 2178 of file structs.h.

3.137.2.47 allbufnum

```
int T_const::allbufnum
```

Definition at line 2179 of file structs.h.

3.137.2.48 aebufnum

```
int T_const::aebufnum
```

Definition at line 2180 of file structs.h.

3.137.2.49 idallflag

```
int T_const::idallflag
```

Definition at line 2181 of file structs.h.

3.137.2.50 idallnum

```
int T_const::idallnum
```

Definition at line 2182 of file structs.h.

3.137.2.51 idallmaxnum

```
int T_const::idallmaxnum
```

Definition at line 2183 of file structs.h.

3.137.2.52 WildcardBufferSize

```
int T_const::WildcardBufferSize
```

Definition at line 2184 of file structs.h.

3.137.2.53 TermMemMax

`int T_const::TermMemMax`

Definition at line 2193 of file structs.h.

3.137.2.54 TermMemTop

`int T_const::TermMemTop`

Definition at line 2194 of file structs.h.

3.137.2.55 NumberMemMax

`int T_const::NumberMemMax`

Definition at line 2195 of file structs.h.

3.137.2.56 NumberMemTop

`int T_const::NumberMemTop`

Definition at line 2196 of file structs.h.

3.137.2.57 CacheNumberMemMax

`int T_const::CacheNumberMemMax`

Definition at line 2197 of file structs.h.

3.137.2.58 CacheNumberMemTop

`int T_const::CacheNumberMemTop`

Definition at line 2198 of file structs.h.

3.137.2.59 bracketindexflag

`int T_const::bracketindexflag`

Definition at line 2199 of file structs.h.

3.137.2.60 optimtimes

`int T_const::optimtimes`

Definition at line 2200 of file structs.h.

3.137.2.61 ListSymbolsSize

```
int T_const::ListSymbolsSize
```

Definition at line 2201 of file structs.h.

3.137.2.62 NumListSymbols

```
int T_const::NumListSymbols
```

Definition at line 2202 of file structs.h.

3.137.2.63 numpoly

```
int T_const::numpoly
```

Definition at line 2203 of file structs.h.

3.137.2.64 LeaveNegative

```
int T_const::LeaveNegative
```

Definition at line 2204 of file structs.h.

3.137.2.65 TrimPower

```
int T_const::TrimPower
```

Definition at line 2205 of file structs.h.

3.137.2.66 small_power_maxx

```
WORD T_const::small_power_maxx
```

Definition at line 2206 of file structs.h.

3.137.2.67 small_power_maxn

```
WORD T_const::small_power_maxn
```

Definition at line 2207 of file structs.h.

3.137.2.68 dummysubexp

```
WORD T_const::dummysubexp[SUBEXPSIZE+4]
```

Definition at line 2208 of file structs.h.

3.137.2.69 comsym

```
WORD T_const::comsym[8]
```

Definition at line 2209 of file structs.h.

3.137.2.70 comnum

```
WORD T_const::comnum[4]
```

Definition at line 2210 of file structs.h.

3.137.2.71 comfun

```
WORD T_const::comfun[FUNHEAD+4]
```

Definition at line 2211 of file structs.h.

3.137.2.72 comind

```
WORD T_const::comind[7]
```

Definition at line 2213 of file structs.h.

3.137.2.73 MinVecArg

```
WORD T_const::MinVecArg[7+ARGHEAD]
```

Definition at line 2214 of file structs.h.

3.137.2.74 FunArg

```
WORD T_const::FunArg[4+ARGHEAD+FUNHEAD]
```

Definition at line 2215 of file structs.h.

3.137.2.75 locwildvalue

```
WORD T_const::locwildvalue[SUBEXPSIZE]
```

Definition at line 2216 of file structs.h.

3.137.2.76 mulpat

```
WORD T_const::mulpat[SUBEXPSIZE+5]
```

Definition at line 2217 of file structs.h.

3.137.2.77 proexp

```
WORD T_const::proexp[SUBEXPSIZE+5]
```

Definition at line 2219 of file structs.h.

3.137.2.78 TMout

```
WORD T_const::TMout[40]
```

Definition at line 2220 of file structs.h.

3.137.2.79 TMbuff

```
WORD T_const::TMbuff
```

Definition at line 2221 of file structs.h.

3.137.2.80 TMdolfac

```
WORD T_const::TMdolfac
```

Definition at line 2222 of file structs.h.

3.137.2.81 nfac

```
WORD T_const::nfac
```

Definition at line 2223 of file structs.h.

3.137.2.82 nBer

```
WORD T_const::nBer
```

Definition at line 2224 of file structs.h.

3.137.2.83 mBer

```
WORD T_const::mBer
```

Definition at line 2225 of file structs.h.

3.137.2.84 PolyAct

```
WORD T_const::PolyAct
```

Definition at line 2226 of file structs.h.

3.137.2.85 RecFlag

```
WORD T_const::RecFlag
```

Definition at line 2227 of file structs.h.

3.137.2.86 inprimelist

```
WORD T_const::inprimelist
```

Definition at line 2228 of file structs.h.

3.137.2.87 sizeprimelist

```
WORD T_const::sizeprimelist
```

Definition at line 2229 of file structs.h.

3.137.2.88 fromindex

```
WORD T_const::fromindex
```

Definition at line 2230 of file structs.h.

3.137.2.89 setinterntopo

```
WORD T_const::setinterntopo
```

Definition at line 2231 of file structs.h.

3.137.2.90 setexterntopo

```
WORD T_const::setexterntopo
```

Definition at line 2232 of file structs.h.

3.137.2.91 TopologiesLevel

```
WORD T_const::TopologiesLevel
```

Definition at line 2233 of file structs.h.

3.137.2.92 TopologiesOptions

```
WORD T_const::TopologiesOptions[2]
```

Definition at line 2234 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.138 T_EEdge Class Reference

Data Fields

- int nodes [2]
- int ext
- int momn
- char momc [2]
- char padding [6]

3.138.1 Detailed Description

Definition at line 24 of file gentopo.h.

3.138.2 Field Documentation

3.138.2.1 nodes

```
int T_EEdge::nodes[2]
```

Definition at line 26 of file gentopo.h.

3.138.2.2 ext

```
int T_EEdge::ext
```

Definition at line 27 of file gentopo.h.

3.138.2.3 momn

```
int T_EEdge::momn
```

Definition at line 28 of file gentopo.h.

3.138.2.4 momc

```
char T_EEdge::momc[2]
```

Definition at line 29 of file gentopo.h.

3.138.2.5 padding

```
char T_EEdge::padding[6]
```

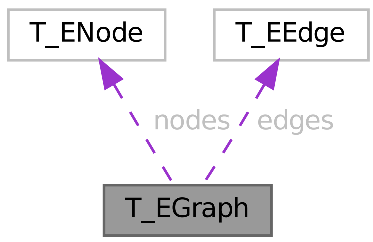
Definition at line 31 of file gentopo.h.

The documentation for this class was generated from the following file:

- gentopo.h

3.139 T_EGraph Class Reference

Collaboration diagram for T_EGraph:



Public Member Functions

- T_EGraph (int nnodes, int nedges, int mxdeg)
- void print (void)
- void init (int pid, long gid, int **adjmat, int sopi, BigInt nsym, BigInt esym)
- void setExtern (int nd, int val)
- void endSetExtern (void)

Data Fields

- long gId
- int pId
- int nNodes
- int nEdges
- int maxdeg
- int nExtern
- int opi
- BigInt nsym
- BigInt esym
- T_ENode ∗ nodes
- T_EEdge ∗ edges

3.139.1 Detailed Description

Definition at line 34 of file gentopo.h.

3.139.2 Constructor & Destructor Documentation

3.139.2.1 T_EGraph()

```
T_EGraph::T_EGraph (
              int nnodes,
              int nedges,
              int mxdeg )
```

Definition at line 88 of file gentopo.cc.

3.139.2.2 ∼T_EGraph()

```
T_EGraph::∼T_EGraph ( )
```

Definition at line 109 of file gentopo.cc.

3.139.3 Member Function Documentation

3.139.3.1 print()

```
void T_EGraph::print (
              void )
```

Definition at line 200 of file gentopo.cc.

3.139.3.2 init()

```
void T_EGraph::init (
             int pid,
             long gid,
             int ** adjmat,
             int sopi,
             BigInt nsym,
             BigInt esym )
```

Definition at line 124 of file gentopo.cc.

3.139.3.3 setExtern()

```
void T_EGraph::setExtern (
             int nd,
             int val )
```

Definition at line 181 of file gentopo.cc.

3.139.3.4 endSetExtern()

```
void T_EGraph::endSetExtern (
             void  )
```

Definition at line 187 of file gentopo.cc.

3.139.4 Field Documentation

3.139.4.1 gId

```
long T_EGraph::gId
```

Definition at line 36 of file gentopo.h.

3.139.4.2 pId

```
int T_EGraph::pId
```

Definition at line 37 of file gentopo.h.

3.139.4.3 nNodes

```
int T_EGraph::nNodes
```

Definition at line 38 of file gentopo.h.

3.139.4.4 nEdges

```
int T_EGraph::nEdges
```

Definition at line 39 of file gentopo.h.

3.139.4.5 maxdeg

```
int T_EGraph::maxdeg
```

Definition at line 40 of file gentopo.h.

3.139.4.6 nExtern

```
int T_EGraph::nExtern
```

Definition at line 41 of file gentopo.h.

3.139.4.7 opi

```
int T_EGraph::opi
```

Definition at line 43 of file gentopo.h.

3.139.4.8 nsym

```
BigInt T_EGraph::nsym
```

Definition at line 44 of file gentopo.h.

3.139.4.9 esym

```
BigInt T_EGraph::esym
```

Definition at line 44 of file gentopo.h.

3.139.4.10 nodes

```
T_ENode* T_EGraph::nodes
```

Definition at line 46 of file gentopo.h.

3.139.4.11 edges

```
T_EEdge* T_EGraph::edges
```

Definition at line 47 of file gentopo.h.

The documentation for this class was generated from the following files:

- gentopo.h
- gentopo.cc

3.140 T_ENode Class Reference

Data Fields

- int deg
- int ext
- int * edges

3.140.1 Detailed Description

Definition at line 17 of file gentopo.h.

3.140.2 Field Documentation

3.140.2.1 deg

```
int T_ENode::deg
```

Definition at line 19 of file gentopo.h.

3.140.2.2 ext

```
int T_ENode::ext
```

Definition at line 20 of file gentopo.h.

3.140.2.3 edges

```
int* T_ENode::edges
```

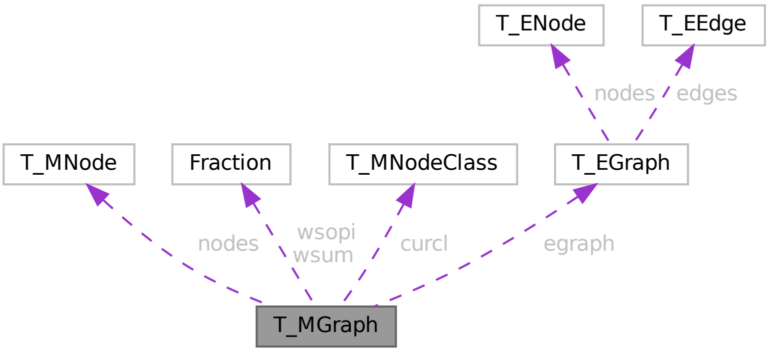
Definition at line 21 of file gentopo.h.

The documentation for this class was generated from the following file:

- gentopo.h

3.141 T_MGraph Class Reference

Collaboration diagram for T_MGraph:



Public Member Functions

- T_MGraph (int pid, int ncl, int ∗cldeg, int ∗clnum, int ∗clext, int sopi)
- long generate (void)

Data Fields

- int pId
- int nNodes
- int nEdges
- int nLoops
- T_MNode ∗∗ nodes
- int ∗ clist
- int nClasses
- int mindeg
- int maxdeg
- int selOPI
- long ndiag
- long n1PI
- int nBridges
- int c1PI
- int ∗∗ adjMat
- BigInt nsym
- BigInt esym
- Fraction wsum
- Fraction wsopi
- T_MNodeClass ∗ curcl
- T_EGraph ∗ egraph
- long ngen
- long ngconn
- long nCallRefine
- long discardOrd
- long discardRefine
- long discardDisc
- long discardIso

3.141.1 Detailed Description

Definition at line 84 of file gentopo.h.

3.141.2 Constructor & Destructor Documentation

3.141.2.1 T_MGraph()

```
T_MGraph::T_MGraph (
              int pid,
              int ncl,
              int * cldeg,
              int * clnum,
              int * clext,
              int sopi )
```

Definition at line 473 of file gentopo.cc.

3.141.2.2 ~T_MGraph()

```
T_MGraph::~T_MGraph ( )
```

Definition at line 572 of file gentopo.cc.

3.141.3 Member Function Documentation

3.141.3.1 generate()

```
long T_MGraph::generate (
              void  )
```

Definition at line 911 of file gentopo.cc.

3.141.4 Field Documentation

3.141.4.1 pId

```
int T_MGraph::pId
```

Definition at line 89 of file gentopo.h.

3.141.4.2 nNodes

```
int T_MGraph::nNodes
```

Definition at line 90 of file gentopo.h.

3.141.4.3 nEdges

`int T_MGraph::nEdges`

Definition at line 91 of file gentopo.h.

3.141.4.4 nLoops

`int T_MGraph::nLoops`

Definition at line 92 of file gentopo.h.

3.141.4.5 nodes

`T_MNode** T_MGraph::nodes`

Definition at line 93 of file gentopo.h.

3.141.4.6 clist

`int* T_MGraph::clist`

Definition at line 94 of file gentopo.h.

3.141.4.7 nClasses

`int T_MGraph::nClasses`

Definition at line 95 of file gentopo.h.

3.141.4.8 mindeg

`int T_MGraph::mindeg`

Definition at line 97 of file gentopo.h.

3.141.4.9 maxdeg

`int T_MGraph::maxdeg`

Definition at line 98 of file gentopo.h.

3.141.4.10 selOPI

`int T_MGraph::selOPI`

Definition at line 100 of file gentopo.h.

3.141.4.11 ndiag

```
long T_MGraph::ndiag
```

Definition at line 103 of file gentopo.h.

3.141.4.12 n1PI

```
long T_MGraph::n1PI
```

Definition at line 104 of file gentopo.h.

3.141.4.13 nBridges

```
int T_MGraph::nBridges
```

Definition at line 105 of file gentopo.h.

3.141.4.14 c1PI

```
int T_MGraph::c1PI
```

Definition at line 108 of file gentopo.h.

3.141.4.15 adjMat

```
int** T_MGraph::adjMat
```

Definition at line 109 of file gentopo.h.

3.141.4.16 nsym

```
BigInt T_MGraph::nsym
```

Definition at line 110 of file gentopo.h.

3.141.4.17 esym

```
BigInt T_MGraph::esym
```

Definition at line 111 of file gentopo.h.

3.141.4.18 wsum

```
Fraction T_MGraph::wsum
```

Definition at line 112 of file gentopo.h.

3.141.4.19 wsopi

`Fraction T_MGraph::wsopi`

Definition at line 113 of file gentopo.h.

3.141.4.20 curcl

`T_MNodeClass* T_MGraph::curcl`

Definition at line 114 of file gentopo.h.

3.141.4.21 egraph

`T_EGraph* T_MGraph::egraph`

Definition at line 115 of file gentopo.h.

3.141.4.22 ngen

`long T_MGraph::ngen`

Definition at line 118 of file gentopo.h.

3.141.4.23 ngconn

`long T_MGraph::ngconn`

Definition at line 119 of file gentopo.h.

3.141.4.24 nCallRefine

`long T_MGraph::nCallRefine`

Definition at line 121 of file gentopo.h.

3.141.4.25 discardOrd

`long T_MGraph::discardOrd`

Definition at line 122 of file gentopo.h.

3.141.4.26 discardRefine

`long T_MGraph::discardRefine`

Definition at line 123 of file gentopo.h.

3.141.4.27 discardDisc

```
long T_MGraph::discardDisc
```

Definition at line 124 of file gentopo.h.

3.141.4.28 discardIso

```
long T_MGraph::discardIso
```

Definition at line 125 of file gentopo.h.

The documentation for this class was generated from the following files:

- gentopo.h
- gentopo.cc

3.142 T_MNode Class Reference

Public Member Functions

- T_MNode (int id, int deg, int ext, int clss)

Data Fields

- int id
- int deg
- int clss
- int ext
- int freelg
- int visited

3.142.1 Detailed Description

Definition at line 61 of file gentopo.h.

3.142.2 Constructor & Destructor Documentation

3.142.2.1 T_MNode()

```
T_MNode::T_MNode (
             int id,
             int deg,
             int ext,
             int clss )
```

Definition at line 237 of file gentopo.cc.

3.142.3 Field Documentation

3.142.3.1 id

```
int T_MNode::id
```

Definition at line 63 of file gentopo.h.

3.142.3.2 deg

```
int T_MNode::deg
```

Definition at line 64 of file gentopo.h.

3.142.3.3 clss

```
int T_MNode::clss
```

Definition at line 65 of file gentopo.h.

3.142.3.4 ext

```
int T_MNode::ext
```

Definition at line 66 of file gentopo.h.

3.142.3.5 freelg

```
int T_MNode::freelg
```

Definition at line 68 of file gentopo.h.

3.142.3.6 visited

```
int T_MNode::visited
```

Definition at line 69 of file gentopo.h.

The documentation for this class was generated from the following files:

- gentopo.h
- gentopo.cc

3.143 T_MNodeClass Class Reference

Public Member Functions

- T_MNodeClass (int nnodes, int ncl)
- void init (int *cl, int mxdeg, int **adjmat)
- void copy (T_MNodeClass *mnc)
- int clCmp (int nd0, int nd1, int cn)
- void printMat (void)
- void mkFlist (void)
- void mkNdCl (void)
- void mkClMat (int **adjmat)
- void incMat (int nd, int td, int val)
- Bool chkOrd (int nd, int ndc, T_MNodeClass *cl, int *dtcl)
- int cmpArray (int *a0, int *a1, int ma)

Data Fields

- int nNodes
- int nClasses
- int * clist
- int * ndcl
- int ** clmat
- int * flist
- int maxdeg
- int forallignment

3.143.1 Detailed Description

Definition at line 249 of file gentopo.cc.

3.143.2 Constructor & Destructor Documentation

3.143.2.1 T_MNodeClass()

```
T_MNodeClass::T_MNodeClass (
              int nnodes,
              int ncl )
```

Definition at line 275 of file gentopo.cc.

3.143.2.2 ∼T_MNodeClass()

```
T_MNodeClass::∼T_MNodeClass ( )
```

Definition at line 290 of file gentopo.cc.

3.143.3 Member Function Documentation

3.143.3.1 init()

```
void T_MNodeClass::init (
            int * cl,
            int mxdeg,
            int ** adjmat )
```

Definition at line 302 of file gentopo.cc.

3.143.3.2 copy()

```
void T_MNodeClass::copy (
            T_MNodeClass * mnc )
```

Definition at line 315 of file gentopo.cc.

3.143.3.3 clCmp()

```
int T_MNodeClass::clCmp (
            int nd0,
            int nd1,
            int cn )
```

Definition at line 364 of file gentopo.cc.

3.143.3.4 printMat()

```
void T_MNodeClass::printMat (
            void )
```

Definition at line 435 of file gentopo.cc.

3.143.3.5 mkFlist()

```
void T_MNodeClass::mkFlist (
            void )
```

Definition at line 336 of file gentopo.cc.

3.143.3.6 mkNdCl()

```
void T_MNodeClass::mkNdCl (
            void )
```

Definition at line 350 of file gentopo.cc.

3.143.3.7 mkClMat()

```
void T_MNodeClass::mkClMat (
             int ** adjmat )
```

Definition at line 384 of file gentopo.cc.

3.143.3.8 incMat()

```
void T_MNodeClass::incMat (
             int nd,
             int td,
             int val )
```

Definition at line 398 of file gentopo.cc.

3.143.3.9 chkOrd()

```
Bool T_MNodeClass::chkOrd (
             int nd,
             int ndc,
             T_MNodeClass * cl,
             int * dtcl )
```

Definition at line 405 of file gentopo.cc.

3.143.3.10 cmpArray()

```
int T_MNodeClass::cmpArray (
             int * a0,
             int * a1,
             int ma )
```

Definition at line 458 of file gentopo.cc.

3.143.4 Field Documentation

3.143.4.1 nNodes

```
int T_MNodeClass::nNodes
```

Definition at line 251 of file gentopo.cc.

3.143.4.2 nClasses

```
int T_MNodeClass::nClasses
```

Definition at line 252 of file gentopo.cc.

3.143.4.3 clist

```
int* T_MNodeClass::clist
```

Definition at line 253 of file gentopo.cc.

3.143.4.4 ndcl

```
int* T_MNodeClass::ndcl
```

Definition at line 254 of file gentopo.cc.

3.143.4.5 clmat

```
int** T_MNodeClass::clmat
```

Definition at line 255 of file gentopo.cc.

3.143.4.6 flist

```
int* T_MNodeClass::flist
```

Definition at line 256 of file gentopo.cc.

3.143.4.7 maxdeg

```
int T_MNodeClass::maxdeg
```

Definition at line 257 of file gentopo.cc.

3.143.4.8 forallignment

```
int T_MNodeClass::forallignment
```

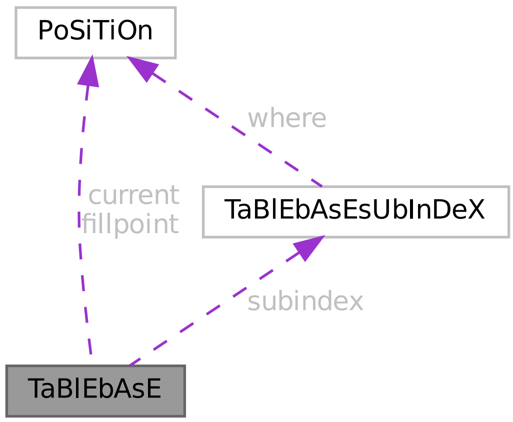
Definition at line 258 of file gentopo.cc.

The documentation for this class was generated from the following file:

- gentopo.cc

3.144 TaBlEbAsE Struct Reference

Collaboration diagram for TaBlEbAsE:



Public Member Functions

- **PADPOSITION** (3, 0, 1, 0, 0)

Data Fields

- POSITION fillpoint
- POSITION current
- UBYTE ∗ name
- int ∗ tablenumbers
- TABLEBASESUBINDEX ∗ subindex
- int numtables

3.144.1 Detailed Description

Definition at line 592 of file structs.h.

3.144.2 Field Documentation

3.144.2.1 fillpoint

```
POSITION TaBlEbAsE::fillpoint
```

Definition at line 593 of file structs.h.

3.144.2.2 current

`POSITION TaBlEbAsE::current`

Definition at line 594 of file structs.h.

3.144.2.3 name

`UBYTE* TaBlEbAsE::name`

Definition at line 595 of file structs.h.

3.144.2.4 tablenumbers

`int* TaBlEbAsE::tablenumbers`

Definition at line 596 of file structs.h.

3.144.2.5 subindex

`TABLEBASESUBINDEX* TaBlEbAsE::subindex`

Definition at line 597 of file structs.h.

3.144.2.6 numtables

`int TaBlEbAsE::numtables`

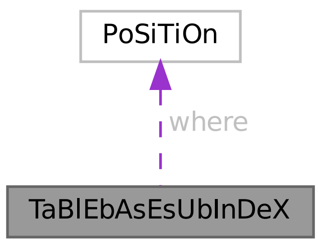
Definition at line 598 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.145 TaBlEbAsEsUbInDeX Struct Reference

Collaboration diagram for TaBlEbAsEsUbInDeX:



Public Member Functions

- **PADPOSITION** (0, 1, 0, 0, 0)

Data Fields

- POSITION where
- LONG size

3.145.1 Detailed Description

Definition at line 582 of file structs.h.

3.145.2 Field Documentation

3.145.2.1 where

```
POSITION TaBlEbAsEsUbInDeX::where
```

Definition at line 583 of file structs.h.

3.145.2.2 size

```
LONG TaBlEbAsEsUbInDeX::size
```

Definition at line 584 of file structs.h.

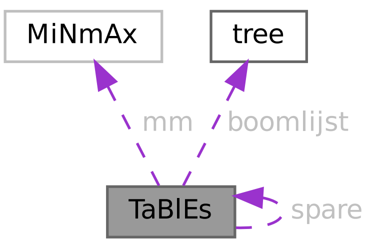
The documentation for this struct was generated from the following file:

- structs.h

3.146 TaBlEs Struct Reference

```
#include <structs.h>
```

Collaboration diagram for TaBlEs:



Data Fields

- WORD ∗ tablepointers
- WORD ∗ prototype
- WORD ∗ pattern
- MINMAX ∗ mm
- WORD ∗ flags
- COMPTREE ∗ boomlijst
- UBYTE ∗ argtail
- struct TaBlEs ∗ spare
- WORD ∗ buffers
- LONG totind
- LONG reserved
- LONG defined
- LONG mdefined
- int prototypeSize
- int numind
- int bounds
- int strict
- int sparse
- int numtree
- int rootnum
- int MaxTreeSize
- WORD bufnum
- WORD bufferssize
- WORD buffersfill
- WORD tablenum
- WORD mode
- WORD numdummies

3.146.1 Detailed Description

buffers, mm, flags, and prototype are always dynamically allocated, tablepointers only if needed (=0 if unallocated), boomlijst and argtail only for sparse tables.

Allocation is done for both the normal and the stub instance (spare), except for prototype and argtail which share memory.

Definition at line 350 of file structs.h.

3.146.2 Field Documentation

3.146.2.1 tablepointers

```
WORD* TaBlEs::tablepointers
```

[D] Start in tablepointers table.

Definition at line 351 of file structs.h.

Referenced by DoRecovery().

3.146.2.2 prototype

`WORD* TaBlEs::prototype`

[D] The wildcard prototyping for arguments

Definition at line 356 of file structs.h.

Referenced by DoRecovery().

3.146.2.3 pattern

`WORD* TaBlEs::pattern`

The pattern with which to match the arguments

Definition at line 357 of file structs.h.

Referenced by DoRecovery().

3.146.2.4 mm

`MINMAX* TaBlEs::mm`

[D] Array bounds, dimension by dimension. # elements = numind.

Definition at line 359 of file structs.h.

Referenced by DoRecovery().

3.146.2.5 flags

`WORD* TaBlEs::flags`

[D] Is element in use ? etc. # elements = numind.

Definition at line 360 of file structs.h.

Referenced by DoRecovery().

3.146.2.6 boomlijst

`COMPTREE* TaBlEs::boomlijst`

[D] Tree for searching in sparse tables

Definition at line 361 of file structs.h.

Referenced by DoRecovery().

3.146.2.7 argtail

```
UBYTE* TaBlEs::argtail
```

[D] The arguments in characters. Starts for tablebase with parenthesis to indicate tail

Definition at line 362 of file structs.h.

Referenced by DoRecovery().

3.146.2.8 spare

```
struct TaBlEs* TaBlEs::spare
```

[D] For tablebase. Alternatingly stubs and real

Definition at line 364 of file structs.h.

Referenced by DoRecovery().

3.146.2.9 buffers

```
WORD* TaBlEs::buffers
```

[D] When we use more than one compiler buffer.

Definition at line 365 of file structs.h.

Referenced by AddRHS(), and DoRecovery().

3.146.2.10 totind

```
LONG TaBlEs::totind
```

Total number requested

Definition at line 366 of file structs.h.

Referenced by DoRecovery().

3.146.2.11 reserved

```
LONG TaBlEs::reserved
```

Total reservation in tablepointers for sparse

Definition at line 367 of file structs.h.

Referenced by DoRecovery().

3.146.2.12 defined

`LONG TaBlEs::defined`

Number of table elements that are defined

Definition at line 368 of file structs.h.

3.146.2.13 mdefined

`LONG TaBlEs::mdefined`

Same as defined but after .global

Definition at line 369 of file structs.h.

3.146.2.14 prototypeSize

`int TaBlEs::prototypeSize`

Size of allocated memory for prototype in bytes.

Definition at line 370 of file structs.h.

Referenced by DoRecovery().

3.146.2.15 numind

`int TaBlEs::numind`

Number of array indices

Definition at line 371 of file structs.h.

Referenced by DoRecovery().

3.146.2.16 bounds

`int TaBlEs::bounds`

Array bounds check on/off.

Definition at line 372 of file structs.h.

3.146.2.17 strict

`int TaBlEs::strict`

$>$0: all must be defined. $<$0: undefined not substitute

Definition at line 373 of file structs.h.

3.146.2.18 sparse

```
int TaBlEs::sparse
```

0 --> sparse table

Definition at line 374 of file structs.h.

Referenced by DoRecovery().

3.146.2.19 numtree

```
int TaBlEs::numtree
```

For the tree for sparse tables

Definition at line 375 of file structs.h.

3.146.2.20 rootnum

```
int TaBlEs::rootnum
```

For the tree for sparse tables

Definition at line 376 of file structs.h.

3.146.2.21 MaxTreeSize

```
int TaBlEs::MaxTreeSize
```

For the tree for sparse tables

Definition at line 377 of file structs.h.

Referenced by DoRecovery().

3.146.2.22 bufnum

```
WORD TaBlEs::bufnum
```

Each table potentially its own buffer

Definition at line 378 of file structs.h.

Referenced by AddRHS().

3.146.2.23 bufferssize

`WORD TaBlEs::bufferssize`

When we use more than one compiler buffer

Definition at line 379 of file structs.h.

Referenced by AddRHS(), and DoRecovery().

3.146.2.24 buffersfill

`WORD TaBlEs::buffersfill`

When we use more than one compiler buffer

Definition at line 380 of file structs.h.

Referenced by AddRHS().

3.146.2.25 tablenum

`WORD TaBlEs::tablenum`

For testing of tableuse

Definition at line 381 of file structs.h.

3.146.2.26 mode

`WORD TaBlEs::mode`

0: normal, 1: stub

Definition at line 382 of file structs.h.

3.146.2.27 numdummies

`WORD TaBlEs::numdummies`

Definition at line 383 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.147 TERMINFO Struct Reference

Data Fields

- WORD ∗ term
- void ∗ currentModel
- void ∗ currentMODEL
- WORD ∗ legcouple [MAXLEGS+1]
- LONG numtopo
- LONG numdia
- WORD diaoffset
- WORD level
- WORD externalset
- WORD internalset
- WORD numextern
- WORD flags

3.147.1 Detailed Description

Definition at line 1396 of file structs.h.

3.147.2 Field Documentation

3.147.2.1 term

```
WORD* TERMINFO::term
```

Definition at line 1397 of file structs.h.

3.147.2.2 currentModel

```
void* TERMINFO::currentModel
```

Definition at line 1398 of file structs.h.

3.147.2.3 currentMODEL

```
void* TERMINFO::currentMODEL
```

Definition at line 1399 of file structs.h.

3.147.2.4 legcouple

```
WORD* TERMINFO::legcouple[MAXLEGS+1]
```

Definition at line 1400 of file structs.h.

3.147.2.5 numtopo

`LONG TERMINFO::numtopo`

Definition at line 1401 of file structs.h.

3.147.2.6 numdia

`LONG TERMINFO::numdia`

Definition at line 1402 of file structs.h.

3.147.2.7 diaoffset

`WORD TERMINFO::diaoffset`

Definition at line 1403 of file structs.h.

3.147.2.8 level

`WORD TERMINFO::level`

Definition at line 1404 of file structs.h.

3.147.2.9 externalset

`WORD TERMINFO::externalset`

Definition at line 1405 of file structs.h.

3.147.2.10 internalset

`WORD TERMINFO::internalset`

Definition at line 1406 of file structs.h.

3.147.2.11 numextern

`WORD TERMINFO::numextern`

Definition at line 1407 of file structs.h.

3.147.2.12 flags

```
WORD TERMINFO::flags
```

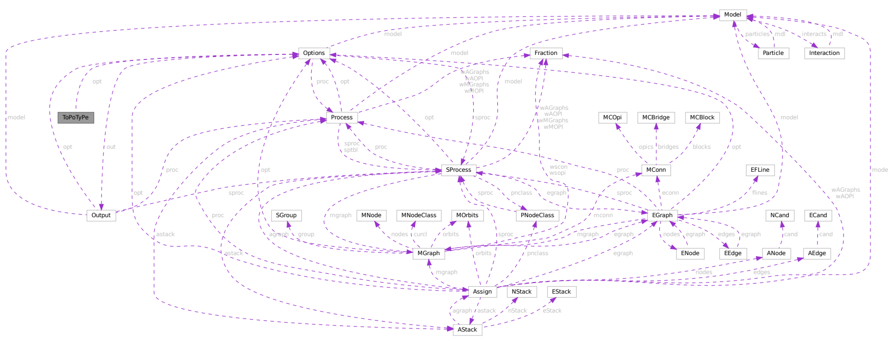
Definition at line 1408 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.148 ToPoTyPe Struct Reference

Collaboration diagram for ToPoTyPe:



Data Fields

- WORD ∗ vert
- WORD ∗ vertmax
- Options ∗ opt
- int cldeg [MAXPOINTS]
- int clnum [MAXPOINTS]
- int clext [MAXPOINTS]
- int cmind [MAXLEGS+1]
- int cmaxd [MAXLEGS+1]
- int ncl
- int nvert
- int nloops
- int nlegs
- int npadding
- WORD nvert
- WORD sopi

3.148.1 Detailed Description

Definition at line 12 of file diawrap.cc.

3.148.2 Field Documentation

3.148.2.1 vert

```
WORD * ToPoTyPe::vert
```

Definition at line 13 of file diawrap.cc.

3.148.2.2 vertmax

```
WORD * ToPoTyPe::vertmax
```

Definition at line 14 of file diawrap.cc.

3.148.2.3 opt

```
Options* ToPoTyPe::opt
```

Definition at line 15 of file diawrap.cc.

3.148.2.4 cldeg

```
int ToPoTyPe::cldeg
```

Definition at line 16 of file diawrap.cc.

3.148.2.5 clnum

```
int ToPoTyPe::clnum
```

Definition at line 16 of file diawrap.cc.

3.148.2.6 clext

```
int ToPoTyPe::clext
```

Definition at line 16 of file diawrap.cc.

3.148.2.7 cmind

```
int ToPoTyPe::cmind[MAXLEGS+1]
```

Definition at line 17 of file diawrap.cc.

3.148.2.8 cmaxd

```
int ToPoTyPe::cmaxd[MAXLEGS+1]
```

Definition at line 17 of file diawrap.cc.

3.148.2.9 ncl

```
int ToPoTyPe::ncl
```

Definition at line 18 of file diawrap.cc.

3.148.2.10 nvert [1/2]

```
int ToPoTyPe::nvert
```

Definition at line 18 of file diawrap.cc.

3.148.2.11 nloops

```
int ToPoTyPe::nloops
```

Definition at line 48 of file topowrap.cc.

3.148.2.12 nlegs

```
int ToPoTyPe::nlegs
```

Definition at line 48 of file topowrap.cc.

3.148.2.13 npadding

```
int ToPoTyPe::npadding
```

Definition at line 48 of file topowrap.cc.

3.148.2.14 nvert [2/2]

```
WORD ToPoTyPe::nvert
```

Definition at line 51 of file topowrap.cc.

3.148.2.15 sopi

```
WORD ToPoTyPe::sopi
```

Definition at line 52 of file topowrap.cc.

The documentation for this struct was generated from the following files:

- diawrap.cc
- topowrap.cc

3.149 TrAcEn Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD * accu
- WORD * accup
- WORD * termp
- WORD * perm
- WORD * inlist
- WORD sgn
- WORD num
- WORD level
- WORD factor
- WORD allsign

3.149.1 Detailed Description

The struct TRACEN keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue.

Definition at line 822 of file structs.h.

3.149.2 Field Documentation

3.149.2.1 accu

```
WORD* TrAcEn::accu
```

Definition at line 823 of file structs.h.

3.149.2.2 accup

```
WORD* TrAcEn::accup
```

Definition at line 824 of file structs.h.

3.149.2.3 termp

`WORD* TrAcEn::termp`

Definition at line 825 of file structs.h.

3.149.2.4 perm

`WORD* TrAcEn::perm`

Definition at line 826 of file structs.h.

3.149.2.5 inlist

`WORD* TrAcEn::inlist`

Definition at line 827 of file structs.h.

3.149.2.6 sgn

`WORD TrAcEn::sgn`

Definition at line 828 of file structs.h.

3.149.2.7 num

`WORD TrAcEn::num`

Definition at line 828 of file structs.h.

3.149.2.8 level

`WORD TrAcEn::level`

Definition at line 828 of file structs.h.

3.149.2.9 factor

`WORD TrAcEn::factor`

Definition at line 828 of file structs.h.

3.149.2.10 allsign

```
WORD TrAcEn::allsign
```

Definition at line 828 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.150 TrAcEs Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ∗ accu
- WORD ∗ accup
- WORD ∗ termp
- WORD ∗ perm
- WORD ∗ inlist
- WORD ∗ nt3
- WORD ∗ nt4
- WORD ∗ j3
- WORD ∗ j4
- WORD ∗ e3
- WORD ∗ e4
- WORD ∗ eers
- WORD ∗ mepf
- WORD ∗ mdel
- WORD ∗ pepf
- WORD ∗ pdel
- WORD sgn
- WORD stap
- WORD step1
- WORD kstep
- WORD mdum
- WORD gamm
- WORD ad
- WORD a3
- WORD a4
- WORD lc3
- WORD lc4
- WORD sign1
- WORD sign2
- WORD gamma5
- WORD num
- WORD level
- WORD factor
- WORD allsign
- WORD finalstep

3.150.1 Detailed Description

The struct TRACES keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue. The 4-dimensional traces are more complicated than the n-dimensional traces (see TRACEN) because of the extra tricks that can be used. They are responsible for the shorter final expressions.

Definition at line 789 of file structs.h.

3.150.2 Field Documentation

3.150.2.1 accu

`WORD* TrAcEs::accu`

Definition at line 790 of file structs.h.

3.150.2.2 accup

`WORD* TrAcEs::accup`

Definition at line 791 of file structs.h.

3.150.2.3 termp

`WORD* TrAcEs::termp`

Definition at line 792 of file structs.h.

3.150.2.4 perm

`WORD* TrAcEs::perm`

Definition at line 793 of file structs.h.

3.150.2.5 inlist

`WORD* TrAcEs::inlist`

Definition at line 794 of file structs.h.

3.150.2.6 nt3

`WORD* TrAcEs::nt3`

Definition at line 795 of file structs.h.

3.150.2.7 nt4

`WORD* TrAcEs::nt4`

Definition at line 796 of file structs.h.

3.150.2.8 j3

`WORD* TrAcEs::j3`

Definition at line 797 of file structs.h.

3.150.2.9 j4

`WORD* TrAcEs::j4`

Definition at line 798 of file structs.h.

3.150.2.10 e3

`WORD* TrAcEs::e3`

Definition at line 799 of file structs.h.

3.150.2.11 e4

`WORD* TrAcEs::e4`

Definition at line 800 of file structs.h.

3.150.2.12 eers

`WORD* TrAcEs::eers`

Definition at line 801 of file structs.h.

3.150.2.13 mepf

`WORD* TrAcEs::mepf`

Definition at line 802 of file structs.h.

3.150.2.14 mdel

`WORD* TrAcEs::mdel`

Definition at line 803 of file structs.h.

3.150.2.15 pepf

```
WORD* TrAcEs::pepf
```

Definition at line 804 of file structs.h.

3.150.2.16 pdel

```
WORD* TrAcEs::pdel
```

Definition at line 805 of file structs.h.

3.150.2.17 sgn

```
WORD TrAcEs::sgn
```

Definition at line 806 of file structs.h.

3.150.2.18 stap

```
WORD TrAcEs::stap
```

Definition at line 807 of file structs.h.

3.150.2.19 step1

```
WORD TrAcEs::step1
```

Definition at line 808 of file structs.h.

3.150.2.20 kstep

```
WORD TrAcEs::kstep
```

Definition at line 808 of file structs.h.

3.150.2.21 mdum

```
WORD TrAcEs::mdum
```

Definition at line 808 of file structs.h.

3.150.2.22 gamm

```
WORD TrAcEs::gamm
```

Definition at line 809 of file structs.h.

3.150.2.23 ad

```
WORD TrAcEs::ad
```

Definition at line 809 of file structs.h.

3.150.2.24 a3

```
WORD TrAcEs::a3
```

Definition at line 809 of file structs.h.

3.150.2.25 a4

```
WORD TrAcEs::a4
```

Definition at line 809 of file structs.h.

3.150.2.26 lc3

```
WORD TrAcEs::lc3
```

Definition at line 809 of file structs.h.

3.150.2.27 lc4

```
WORD TrAcEs::lc4
```

Definition at line 809 of file structs.h.

3.150.2.28 sign1

```
WORD TrAcEs::sign1
```

Definition at line 810 of file structs.h.

3.150.2.29 sign2

```
WORD TrAcEs::sign2
```

Definition at line 810 of file structs.h.

3.150.2.30 gamma5

```
WORD TrAcEs::gamma5
```

Definition at line 810 of file structs.h.

3.150.2.31 num

```
WORD TrAcEs::num
```

Definition at line 810 of file structs.h.

3.150.2.32 level

```
WORD TrAcEs::level
```

Definition at line 810 of file structs.h.

3.150.2.33 factor

```
WORD TrAcEs::factor
```

Definition at line 810 of file structs.h.

3.150.2.34 allsign

```
WORD TrAcEs::allsign
```

Definition at line 810 of file structs.h.

3.150.2.35 finalstep

```
WORD TrAcEs::finalstep
```

Definition at line 811 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.151 tree Struct Reference

```
#include <structs.h>
```

Data Fields

- int parent
- int left
- int right
- int value
- int blnce
- int usage

3.151.1 Detailed Description

The subexpressions in the compiler are kept track of in a (balanced) tree to reduce the need for subexpressions and hence save much space in large rhs expressions (like when we have xxxxxxx occurrences of objects like f(x+1,x+1) in which each x+1 becomes a subexpression. The struct that controls this tree is COMPTREE.

Definition at line 294 of file structs.h.

3.151.2 Field Documentation

3.151.2.1 parent

```
int tree::parent
```

Index of parent

Definition at line 295 of file structs.h.

Referenced by IniFbuffer().

3.151.2.2 left

```
int tree::left
```

Left child (if not -1)

Definition at line 296 of file structs.h.

Referenced by IniFbuffer().

3.151.2.3 right

```
int tree::right
```

Right child (if not -1)

Definition at line 297 of file structs.h.

Referenced by IniFbuffer().

3.151.2.4 value

```
int tree::value
```

The object to be sorted and searched

Definition at line 298 of file structs.h.

Referenced by IniFbuffer().

3.151.2.5 blnce

```
int tree::blnce
```

Balance factor

Definition at line 299 of file structs.h.

Referenced by IniFbuffer().

3.151.2.6 usage

```
int tree::usage
```

Number of uses in some types of trees

Definition at line 300 of file structs.h.

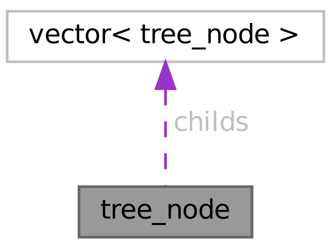
Referenced by CleanupArgCache(), and IniFbuffer().

The documentation for this struct was generated from the following file:

- structs.h

3.152 tree_node Class Reference

Collaboration diagram for tree_node:



Public Member Functions

- tree_node (int _var=0)
- tree_node (const tree_node &other)
- tree_node & operator= (const tree_node &other)

Data Fields

- vector< tree_node > childs
- double sum_results
- int num_visits
- WORD var
- bool finished

3.152.1 Detailed Description

Definition at line 125 of file optimize.cc.

3.152.2 Constructor & Destructor Documentation

3.152.2.1 tree_node() [1/2]

```
tree_node::tree_node (
            int _var = 0 )  [inline]
```

Definition at line 134 of file optimize.cc.

3.152.2.2 tree_node() [2/2]

```
tree_node::tree_node (
            const tree_node & other )  [inline]
```

Definition at line 137 of file optimize.cc.

3.152.3 Member Function Documentation

3.152.3.1 operator=()

```
tree_node & tree_node::operator= (
            const tree_node & other )  [inline]
```

Definition at line 144 of file optimize.cc.

3.152.4 Field Documentation

3.152.4.1 childs

```
vector<tree_node> tree_node::childs
```

Definition at line 127 of file optimize.cc.

3.152.4.2 sum_results

```
double tree_node::sum_results
```

Definition at line 128 of file optimize.cc.

3.152.4.3 num_visits

```
int tree_node::num_visits
```

Definition at line 129 of file optimize.cc.

3.152.4.4 var

```
WORD tree_node::var
```

Definition at line 130 of file optimize.cc.

3.152.4.5 finished

```
bool tree_node::finished
```

Definition at line 131 of file optimize.cc.

The documentation for this class was generated from the following file:

- optimize.cc

3.153 VaRrEnUm Struct Reference

```
#include <structs.h>
```

Data Fields

- WORD ∗ start
- WORD ∗ lo
- WORD ∗ hi

3.153.1 Detailed Description

Contains the pointers to an array in which a binary search will be performed.

Definition at line 166 of file structs.h.

3.153.2 Field Documentation

3.153.2.1 start

```
WORD* VaRrEnUm::start
```

Start point for search. Points inbetween lo and hi

Definition at line 167 of file structs.h.

Referenced by DoRecovery().

3.153.2.2 lo

```
WORD* VaRrEnUm::lo
```

Start of memory area

Definition at line 168 of file structs.h.

Referenced by DoRecovery(), Generator(), and GetFirstTerm().

3.153.2.3 hi

```
WORD* VaRrEnUm::hi
```

End of memory area

Definition at line 169 of file structs.h.

Referenced by DoRecovery().

The documentation for this struct was generated from the following file:

- structs.h

3.154 VeCtOr Struct Reference

Data Fields

- LONG name
- WORD complex
- WORD number
- WORD flags
- WORD node
- WORD namesize
- WORD dimension

3.154.1 Detailed Description

Definition at line 461 of file structs.h.

3.154.2 Field Documentation

3.154.2.1 name

```
LONG VeCtOr::name
```

Definition at line 462 of file structs.h.

3.154.2.2 complex

```
WORD VeCtOr::complex
```

Definition at line 463 of file structs.h.

3.154.2.3 number

```
WORD VeCtOr::number
```

Definition at line 464 of file structs.h.

3.154.2.4 flags

```
WORD VeCtOr::flags
```

Definition at line 465 of file structs.h.

3.154.2.5 node

```
WORD VeCtOr::node
```

Definition at line 466 of file structs.h.

3.154.2.6 namesize

```
WORD VeCtOr::namesize
```

Definition at line 467 of file structs.h.

3.154.2.7 dimension

```
WORD VeCtOr::dimension
```

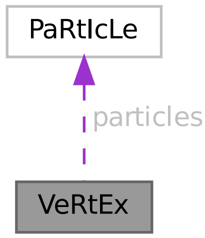
Definition at line 468 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.155 VeRtEx Struct Reference

Collaboration diagram for VeRtEx:



Data Fields

- PARTICLE particles [MAXPARTICLES]
- WORD couplings [2 ∗MAXCOUPLINGS]
- WORD nparticles
- WORD ncouplings
- WORD type
- WORD error
- WORD externonly
- WORD spare

3.155.1 Detailed Description

Definition at line 626 of file structs.h.

3.155.2 Field Documentation

3.155.2.1 particles

```
PARTICLE VeRtEx::particles[MAXPARTICLES]
```

Definition at line 627 of file structs.h.

3.155.2.2 couplings

```
WORD VeRtEx::couplings[2 *MAXCOUPLINGS]
```

Definition at line 628 of file structs.h.

3.155.2.3 nparticles

```
WORD VeRtEx::nparticles
```

Definition at line 629 of file structs.h.

3.155.2.4 ncouplings

```
WORD VeRtEx::ncouplings
```

Definition at line 630 of file structs.h.

3.155.2.5 type

```
WORD VeRtEx::type
```

Definition at line 631 of file structs.h.

3.155.2.6 error

```
WORD VeRtEx::error
```

Definition at line 632 of file structs.h.

3.155.2.7 externonly

```
WORD VeRtEx::externonly
```

Definition at line 633 of file structs.h.

3.155.2.8 spare

```
WORD VeRtEx::spare
```

Definition at line 634 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

3.156 X_const Struct Reference

```
#include <structs.h>
```

Data Fields

- UBYTE ∗ currentPrompt
- UBYTE ∗ shellname
- UBYTE ∗ stderrname
- int timeout
- int killSignal
- int killWholeGroup
- int daemonize
- int currentExternalChannel

3.156.1 Detailed Description

The X_const struct is part of the global data and resides in the ALLGLOBALS struct A under the name X We see it used with the macro AX as in AX.timeout It contains variables that involve communication with external programs

Definition at line 2559 of file structs.h.

3.156.2 Field Documentation

3.156.2.1 currentPrompt

```
UBYTE* X_const::currentPrompt
```

Definition at line 2560 of file structs.h.

3.156.2.2 shellname

```
UBYTE* X_const::shellname
```

Definition at line 2561 of file structs.h.

3.156.2.3 stderrname

```
UBYTE* X_const::stderrname
```

Definition at line 2563 of file structs.h.

3.156.2.4 timeout

```
int X_const::timeout
```

Definition at line 2565 of file structs.h.

3.156.2.5 killSignal

```
int X_const::killSignal
```

Definition at line 2568 of file structs.h.

3.156.2.6 killWholeGroup

```
int X_const::killWholeGroup
```

Definition at line 2569 of file structs.h.

3.156.2.7 daemonize

```
int X_const::daemonize
```

Definition at line 2571 of file structs.h.

3.156.2.8 currentExternalChannel

```
int X_const::currentExternalChannel
```

Definition at line 2572 of file structs.h.

The documentation for this struct was generated from the following file:

- structs.h

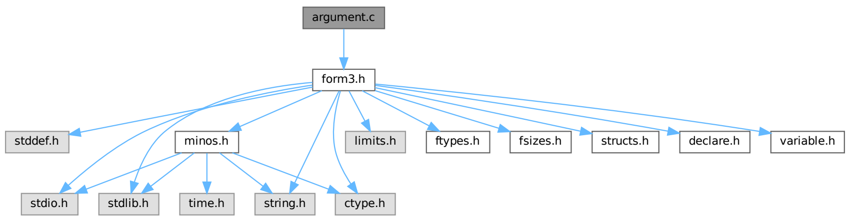
Chapter 4

File Documentation

4.1 argument.c File Reference

`#include "form3.h"`

Include dependency graph for argument.c:



Macros

- #define NEWORDER

Functions

- WORD execarg (PHEAD WORD ∗term, WORD level)
- WORD execterm (PHEAD WORD ∗term, WORD level)
- int ArgumentImplode (PHEAD WORD ∗term, WORD ∗thelist)
- int ArgumentExplode (PHEAD WORD ∗term, WORD ∗thelist)
- int ArgFactorize (PHEAD WORD ∗argin, WORD ∗argout)
- WORD FindArg (PHEAD WORD ∗a)
- WORD InsertArg (PHEAD WORD ∗argin, WORD ∗argout, int par)
- int CleanupArgCache (PHEAD WORD bufnum)
- int ArgSymbolMerge (WORD ∗t1, WORD ∗t2)
- int ArgDotproductMerge (WORD ∗t1, WORD ∗t2)
- WORD ∗ TakeArgContent (PHEAD WORD ∗argin, WORD ∗argout)
- WORD ∗ MakeInteger (PHEAD WORD ∗argin, WORD ∗argout, WORD ∗argfree)
- WORD ∗ MakeMod (PHEAD WORD ∗argin, WORD ∗argout, WORD ∗argfree)
- void SortWeights (LONG ∗weights, LONG ∗extraspace, WORD number)

4.1.1 Detailed Description

Contains the routines that deal with the execution phase of the argument and related statements (like term)

Definition in file argument.c.

4.1.2 Macro Definition Documentation

4.1.2.1 NEWORDER

```
#define NEWORDER
```

Factorizes an argument in general notation (meaning that the first word of the argument is a positive size indicator) Input (argin): pointer to the complete argument Output (argout): Pointer to where the output should be written. This is in the WorkSpace Return value should be negative if anything goes wrong.

The notation of the output should be a string of arguments terminated by the number zero.

Originally we sorted in a way that the constants came last. This gave conflicts with the dollar and expression factorizations (in the expressions we wanted the zero first and then followed by the constants).

Definition at line 2058 of file argument.c.

4.1.3 Function Documentation

4.1.3.1 execarg()

```
WORD execarg (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 56 of file argument.c.

4.1.3.2 execterm()

```
WORD execterm (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1769 of file argument.c.

4.1.3.3 ArgumentImplode()

```
int ArgumentImplode (
            PHEAD WORD * term,
            WORD * thelist )
```

Definition at line 1845 of file argument.c.

4.1.3.4 ArgumentExplode()

```
int ArgumentExplode (
            PHEAD WORD * term,
            WORD * thelist )
```

Definition at line 1949 of file argument.c.

4.1.3.5 ArgFactorize()

```
int ArgFactorize (
            PHEAD WORD * argin,
            WORD * argout )
```

Definition at line 2060 of file argument.c.

4.1.3.6 FindArg()

```
WORD FindArg (
            PHEAD WORD * a )
```

Looks the argument up in the (workers) table. If it is found the number in the table is returned (plus one to make it positive). If it is not found we look in the compiler provided table. If it is found - the number in the table is returned (minus one to make it negative). If in neither table we return zero.

Definition at line 2513 of file argument.c.

4.1.3.7 InsertArg()

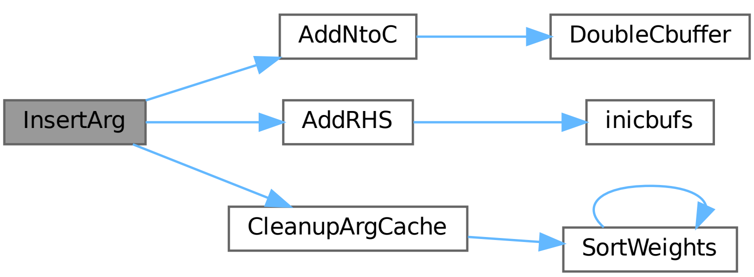
```
WORD InsertArg (
            PHEAD WORD * argin,
            WORD * argout,
            int par )
```

Inserts the argument into the (workers) table. If the table is too full we eliminate half of it. The eliminated elements are the ones that have not been used most recently, weighted by their total use and age(?). If par == 0 it inserts in the regular factorization cache If par == 1 it inserts in the cache defined with the FactorCache statement

Definition at line 2537 of file argument.c.

References AddNtoC(), AddRHS(), and CleanupArgCache().

Here is the call graph for this function:



4.1.3.8 CleanupArgCache()

```
int CleanupArgCache (
            PHEAD WORD bufnum )
```

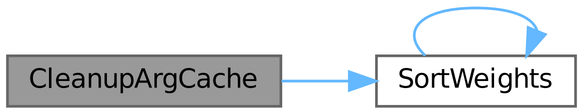
Cleans up the argument factorization cache. We throw half the elements. For a weight of what we want to keep we use the product of usage and the number in the buffer.

Definition at line 2572 of file argument.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::Pointer, CbUf::rhs, SortWeights(), and tree::usage.

Referenced by InsertArg().

Here is the call graph for this function:



4.1.3.9 ArgSymbolMerge()

```
int ArgSymbolMerge (
            WORD * t1,
            WORD * t2 )
```

Definition at line 2643 of file argument.c.

4.1.3.10 ArgDotproductMerge()

```
int ArgDotproductMerge (
            WORD * t1,
            WORD * t2 )
```

Definition at line 2698 of file argument.c.

4.1.3.11 TakeArgContent()

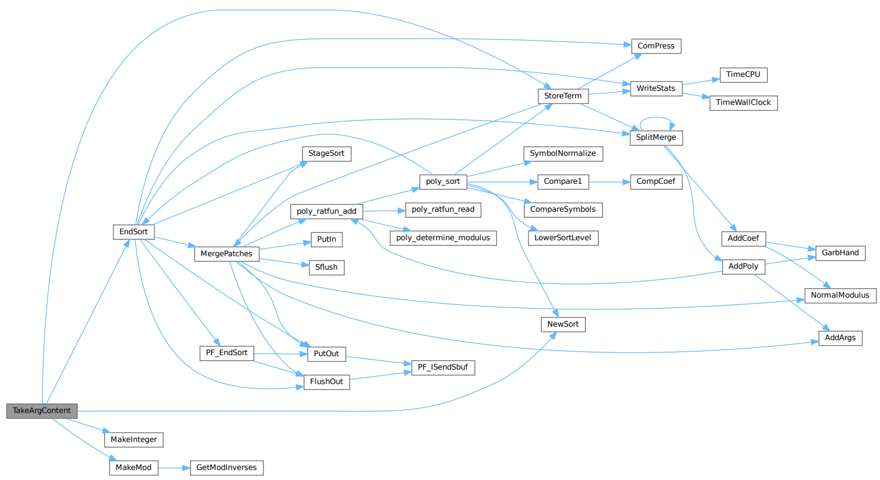
```
WORD * TakeArgContent (
            PHEAD WORD * argin,
            WORD * argout )
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. The common pieces are put in argout as a sequence of arguments. The part with the multiple terms that are now relative prime is put in argfree which is allocated via TermMalloc and is given as the return value. The difference with the old code is that negative powers are always removed. Hence it is as in MakeInteger in which only numerators will be left: now only zero or positive powers will be remaining.

Definition at line 2766 of file argument.c.

References EndSort(), MakeInteger(), MakeMod(), NewSort(), and StoreTerm().

Here is the call graph for this function:



4.1.3.12 MakeInteger()

```
WORD * MakeInteger (
            PHEAD WORD * argin,
            WORD * argout,
            WORD * argfree )
```

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in argin. The number that comes out should go to argout. The new pointer in the argout buffer is the return value. The normalized argument is in argfree.

Definition at line 3312 of file argument.c.

Referenced by TakeArgContent().

4.1.3.13 MakeMod()

```
WORD * MakeMod (
            PHEAD WORD * argin,
            WORD * argout,
            WORD * argfree )
```

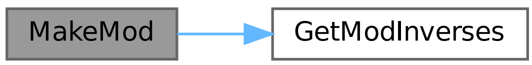
Similar to MakeInteger but now with modulus arithmetic using only a one WORD 'prime'. We make the coefficient of the first term in the argument equal to one. Already the coefficients are taken modulus AN.cmod and AN.ncmod == 1

Definition at line 3483 of file argument.c.

References GetModInverses().

Referenced by TakeArgContent().

Here is the call graph for this function:



4.1.3.14 SortWeights()

```
void SortWeights (
            LONG * weights,
            LONG * extraspace,
            WORD number )
```

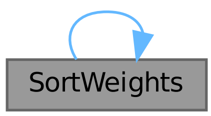
Sorts an array of LONGS in the same way SplitMerge (in sort.c) works We use gradual division in two.

Definition at line 3528 of file argument.c.

References SortWeights().

Referenced by CleanupArgCache(), and SortWeights().

Here is the call graph for this function:



4.2 argument.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 */
/* #] License : */

/*
    #[ include : argument.c
*/

#include "form3.h"

/*
    #] include :
    #[ execarg :

    Executes the subset of statements in an argument environment.
    The calling routine should be of the type
    if ( C->lhs[level][0] == TYPEARG ) {
        if ( execarg(term,level) ) goto GenCall;
        level = C->lhs[level][2];
        goto SkipCount;
    }
    Note that there will be cases in which extra space is needed.
    In addition the compare with C->numlhs isn't very fine, because we
    need to insert a different value (C->lhs[level][2]).
*/

WORD execarg(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD *t, *r, *m, *v;
    WORD *start, *stop, *rstop, *r1, *r2 = 0, *r3 = 0, *r4, *r5, *r6, *r7, *r8, *r9;
    WORD *mm, *mstop, *rnext, *rr, *factor, type, ngcd, nq;
    CBUF *C = cbuf+AM.rbufnum, *CC = cbuf+AT.ebufnum;
    WORD i, j, k, oldnumlhs = AR.Cnumlhs, count, action = 0, olddefer = AR.DeferFlag;
    WORD oldnumrhs = CC->numrhs, size, pow, jj;
    LONG oldcpointer = CC->Pointer - CC->Buffer, oldppointer = AT.pWorkPointer, lp;
    WORD *oldwork = AT.WorkPointer, *oldwork2, scale, renorm;
    WORD kLCM = 0, kGCD = 0, kGCD2, kkLCM = 0, jLCM = 0, jGCD, sign = 1;
    int ii, didpolyratfun;
    UWORD *EAscrat, *GCDbuffer = 0, *GCDbuffer2 = 0, *LCMbuffer = 0, *LCMb = 0, *LCMc = 0;
    AT.WorkPointer += *term;
    start = C->lhs[level];
    AR.Cnumlhs = start[2];
    stop = start + start[1];
    type = *start;
    scale = start[4];
    renorm = start[5];
    start += TYPEARGHEADSIZE;
/*
    #[ Dollars :
*/
    if ( renorm && start[1] != 0 ) {/* We have to evaluate $ symbols inside () */
        t = start+1; factor = oldwork2 = v = AT.WorkPointer;
        i = *t; t++;
        *v++ = i+3; i--; NCOPY(v,t,i);
        *v++ = 1; *v++ = 1; *v++ = 3;
        AT.WorkPointer = v;
        start = t; AR.Eside = LHSIDEX;
```

```
        NewSort(BHEAD0);
        if ( Generator(BHEAD factor,AR.Cnumlhs) ) {
            LowerSortLevel();
            AT.WorkPointer = oldwork;
            return(-1);
        }
        AT.WorkPointer = v;
        if ( EndSort(BHEAD factor,0) < 0 ) {}
        if ( *factor && *(factor+*factor) != 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("&$ in () does not evaluate into a single term");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        AR.Eside = RHSIDE;
        if ( *factor > 0 ) {
            v = factor+*factor;
            v -= ABS(v[-1]);
            *factor = v-factor;
        }
        AT.WorkPointer = v;
    }
    else {
        if ( *start < 0 ) {
            factor = start + 1;
            start += -*start;
        }
        else factor = 0;
    }
/*
    #] Dollars :
*/
    t = term;
    r = t + *t;
    rstop = r - ABS(r[-1]);
    t++;
/*
    #[ Argument detection : + argument statement
*/
/*
    Allocate Numbers for MakeInteger here, for re-use in case multiple
    functions are treated in the same term.
*/
    if ( type == TYPENORM4 ) {
        GCDbuffer = NumberMalloc("execarg");
        GCDbuffer2 = NumberMalloc("execarg");
        LCMbuffer = NumberMalloc("execarg");
        LCMb = NumberMalloc("execarg"); LCMc = NumberMalloc("execarg");
    }
    didpolyratfun = 0;
    while ( t < rstop ) {
        if ( *t >= FUNCTION && functions[*t-FUNCTION].spec <= 0 ) {
/*
            We have a function. First count the number of arguments.
            Tensors are excluded.
*/
            count = 0;
            v = t;
            m = t + FUNHEAD;
            r = t + t[1];
            while ( m < r ) {
                count++;
                NEXTARG(m)
            }
            if ( count <= 0 ) { t += t[1]; continue; }
/*
            Now we take the arguments one by one and test for a match
*/
            for ( i = 1; i <= count; i++ ) {
                m = start;
                while ( m < stop ) {
                    r = m + m[1];
                    j = *r++;
                    if ( j > 1 ) {
                        while ( --j > 0 ) {
                            if ( *r == i ) goto RightNum;
                            r++;
                        }
                        m = r;
                        continue;
                    }
RightNum:
                    if ( m[1] == 2 ) {
#ifdef WITHFLOAT
                        if ( *t != FLOATFUN || AT.aux_ == 0 || TestFloat(t) == 0 )
#endif
                        {
```

```
                            m += 2;
                            m += *m;
                            goto HaveTodo;
                        }
                    }
                    else {
                        r = m + m[1];
                        m += 2;
                        while ( m < r ) {
                            if ( *m == CSET ) {
                                r1 = SetElements + Sets[m[1]].first;
                                r2 = SetElements + Sets[m[1]].last;
                                while ( r1 < r2 ) {
                                    if ( *r1++ == *t ) goto HaveTodo;
                                }
                            }
                            else if ( m[1] == *t ) goto HaveTodo;
                            m += 2;
                        }
                    }
                    m += *m;
                }
                continue;
HaveTodo:
/*
                If we come here we have to do the argument i (first is 1).
*/
                sign = 1;
                action = 1;
                if ( *t == AR.PolyFun ) didpolyratfun = 1;
                v[2] |= DIRTYFLAG;
                r = t + FUNHEAD;
                j = i;
                while ( --j > 0 ) { NEXTARG(r) }
                if ( ( type == TYPESPLITARG ) || ( type == TYPESPLITFIRSTARG )
                 || ( type == TYPESPLITLASTARG ) ) {
                    if ( *t > FUNCTION && *r > 0 ) {
                        WantAddPointers(2);
                        AT.pWorkSpace[AT.pWorkPointer++] = t;
                        AT.pWorkSpace[AT.pWorkPointer++] = r;
                    }
                    continue;
                }
                else if ( type == TYPESPLITARG2 ) {
                    if ( *t > FUNCTION && *r > 0 ) {
                        WantAddPointers(2);
                        AT.pWorkSpace[AT.pWorkPointer++] = t;
                        AT.pWorkSpace[AT.pWorkPointer++] = r;
                    }
                    continue;
                }
                else if ( type == TYPEFACTARG || type == TYPEFACTARG2 ) {
                    if ( *t > FUNCTION || *t == DENOMINATOR ) {
                        if ( *r > 0 ) {
                        mm = r + ARGHEAD; mstop = r + *r;
                        if ( mm + *mm < mstop ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        if ( *mm == 1+ABS(mstop[-1]) ) continue;
                        if ( mstop[-3] != 1 || mstop[-2] != 1
                            || mstop[-1] != 3 ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        GETSTOP(mm,mstop); mm++;
                        if ( mm + mm[1] < mstop ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        if ( *mm == SYMBOL && ( mm[1] > 4 ||
                            ( mm[3] != 1 && mm[3] != -1 ) ) ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        else if ( *mm == DOTPRODUCT && ( mm[1] > 5 ||
                            ( mm[4] != 1 && mm[4] != -1 ) ) ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
```

```
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        else if ( ( *mm == DELTA || *mm == VECTOR )
                             && mm[1] > 4 ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        }
                        else if ( factor && *factor == 4 && factor[2] == 1 ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        else if ( factor && *factor == 0
                        && ( *r == -SNUMBER && r[1] != 1 ) ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                        else if ( *r == -MINVECTOR ) {
                            WantAddPointers(2);
                            AT.pWorkSpace[AT.pWorkPointer++] = t;
                            AT.pWorkSpace[AT.pWorkPointer++] = r;
                            continue;
                        }
                    }
                    continue;
                }
                else if ( type == TYPENORM || type == TYPENORM2 || type == TYPENORM3 || type == TYPENORM4 ) {
                    if ( *r < 0 ) {
                        WORD rone;
                        if ( *r == -MINVECTOR ) { rone = -1; *r = -INDEX; }
                        else if ( *r != -SNUMBER || r[1] == 1 || r[1] == 0 ) continue;
                        else { rone = r[1]; r[1] = 1; }
/*
                        Now we must multiply the general coefficient by r[1]
*/
                        if ( scale && ( factor == 0 || *factor ) ) {
                            action = 1;
                            v[2] |= DIRTYFLAG;
                            if ( rone < 0 ) {
                                if ( type == TYPENORM3 ) k = 1;
                                else k = -1;
                                rone = -rone;
                            }
                            else k = 1;
                            r1 = term + *term;
                            size = r1[-1];
                            size = REDLENG(size);
                            if ( scale > 0 ) {
                                for ( jj = 0; jj < scale; jj++ ) {
                                    if ( Mully(BHEAD (UWORD *)rstop,&size,(UWORD *)(&rone),k) )
                                        goto execargerr;
                                }
                            }
                            else {
                                for ( jj = 0; jj > scale; jj-- ) {
                                    if ( Divvy(BHEAD (UWORD *)rstop,&size,(UWORD *)(&rone),k) )
                                        goto execargerr;
                                }
                            }
                            size = INCLENG(size);
                            k = size < 0 ? -size: size;
                            rstop[k-1] = size;
                            *term = (WORD)(rstop - term) + k;
                        }
                        continue;
                    }
/*
                    Now we have to find a reference term.
                    If factor is defined and *factor != 0 we have to
                    look for the first term that matches the pattern exactly
                    Otherwise the first term plays this role
                    If its coefficient is not one,
                    we must set up a division of the whole argument by
                    this coefficient, and a multiplication of the term
                    when the type is not equal to TYPENORM2.
                    We first multiply the coefficient of the term.
                    Then we set up the division.

                    First find the magic term
```

```
*/
                    if ( type == TYPENORM4 ) {
/*
                        For normalizing everything to integers we have to
                        determine for all elements of this argument the LCM of
                        the denominators and the GCD of the numerators.
                        The buffers have been allocated already.
*/
                        r4 = r + *r;
                        r1 = r + ARGHEAD;
/*
                        First take the first term to load up the LCM and the GCD
*/
                        r2 = r1 + *r1;
                        j = r2[-1];
                        if ( j < 0 ) sign = -1;
                        r3 = r2 - ABS(j);
                        k = REDLENG(j);
                        if ( k < 0 ) k = -k;
                        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
                        for ( kGCD = 0; kGCD < k; kGCD++ ) GCDbuffer[kGCD] = r3[kGCD];
                        k = REDLENG(j);
                        if ( k < 0 ) k = -k;
                        r3 += k;
                        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
                        for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
                        r1 = r2;
/*
                        Now go through the rest of the terms in this argument.
*/
                        while ( r1 < r4 ) {
                            r2 = r1 + *r1;
                            j = r2[-1];
                            r3 = r2 - ABS(j);
                            k = REDLENG(j);
                            if ( k < 0 ) k = -k;
                            while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
                            if ( ( ( GCDbuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
/*
                                GCD is already 1
*/
                            }
                            else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
                                if ( GcdLong(BHEAD GCDbuffer,kGCD,(UWORD *)r3,k,GCDbuffer2,&kGCD2) ) {
                                    NumberFree(GCDbuffer,"execarg");
                                    NumberFree(GCDbuffer2,"execarg");
                                    NumberFree(LCMbuffer,"execarg");
                                    NumberFree(LCMb,"execarg"); NumberFree(LCMc,"execarg");
                                    goto execargerr;
                                }
                                kGCD = kGCD2;
                                for ( ii = 0; ii < kGCD; ii++ ) GCDbuffer[ii] = GCDbuffer2[ii];
                            }
                            else {
                                kGCD = 1; GCDbuffer[0] = 1;
                            }
                            k = REDLENG(j);
                            if ( k < 0 ) k = -k;
                            r3 += k;
                            while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
                            if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
                                for ( kLCM = 0; kLCM < k; kLCM++ )
                                    LCMbuffer[kLCM] = r3[kLCM];
                            }
                            else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
                                if ( GcdLong(BHEAD LCMbuffer,kLCM,(UWORD *)r3,k,LCMb,&kkLCM) ) {
                                    NumberFree(GCDbuffer,"execarg"); NumberFree(GCDbuffer2,"execarg");
                                    NumberFree(LCMbuffer,"execarg"); NumberFree(LCMb,"execarg");
NumberFree(LCMc,"execarg");
                                    goto execargerr;
                                }
                                DivLong((UWORD *)r3,k,LCMb,kkLCM,LCMb,&kkLCM,LCMc,&jLCM);
                                MulLong(LCMbuffer,kLCM,LCMb,kkLCM,LCMc,&jLCM);
                                for ( kLCM = 0; kLCM < jLCM; kLCM++ )
                                    LCMbuffer[kLCM] = LCMc[kLCM];
                            }
                            else {} /* LCM doesn't change */
                            r1 = r2;
                        }
/*
                        Now put the factor together: GCD/LCM
*/
                        r3 = (WORD *)(GCDbuffer);
                        if ( kGCD == kLCM ) {
                            for ( jGCD = 0; jGCD < kGCD; jGCD++ )
                                r3[jGCD+kGCD] = LCMbuffer[jGCD];
                            k = kGCD;
```

```
                        }
                        else if ( kGCD > kLCM ) {
                            for ( jGCD = 0; jGCD < kLCM; jGCD++ )
                                r3[jGCD+kGCD] = LCMbuffer[jGCD];
                            for ( jGCD = kLCM; jGCD < kGCD; jGCD++ )
                                r3[jGCD+kGCD] = 0;
                            k = kGCD;
                        }
                        else {
                            for ( jGCD = kGCD; jGCD < kLCM; jGCD++ )
                                r3[jGCD] = 0;
                            for ( jGCD = 0; jGCD < kLCM; jGCD++ )
                                r3[jGCD+kLCM] = LCMbuffer[jGCD];
                            k = kLCM;
                        }

                        j = 2*k+1;
/*
                        Now we have to correct the overall factor
*/
                        if ( scale && ( factor == 0 || *factor > 0 ) )
                            goto ScaledVariety;
/*
                        The if was added 28-nov-2012 to give MakeInteger also
                        the (0) option.
*/
                        if ( scale && ( factor == 0 || *factor ) ) {
                            size = term[*term-1];
                            size = REDLENG(size);
                            if ( MulRat(BHEAD (UWORD *)rstop,size,(UWORD *)r3,k,
                                    (UWORD *)rstop,&size) ) goto execargerr;
                            size = INCLENG(size);
                            k = size < 0 ? -size: size;
                            rstop[k-1] = size*sign;
                            *term = (WORD)(rstop - term) + k;
                        }
                    }
                    else {
                        if ( factor && *factor >= 1 ) {
                            r4 = r + *r;
                            r1 = r + ARGHEAD;
                            while ( r1 < r4 ) {
                                r2 = r1 + *r1;
                                r3 = r2 - ABS(r2[-1]);
                                j = r3 - r1;
                                r5 = factor;
                                if ( j != *r5 ) { r1 = r2; continue; }
                                r5++; r6 = r1+1;
                                while ( --j > 0 ) {
                                    if ( *r5 != *r6 ) break;
                                    r5++; r6++;
                                }
                                if ( j > 0 ) { r1 = r2; continue; }
                                break;
                            }
                            if ( r1 >= r4 ) continue;
                        }
                        else {
                            r1 = r + ARGHEAD;
                            r2 = r1 + *r1;
                            r3 = r2 - ABS(r2[-1]);
                        }
                        if ( *r3 == 1 && r3[1] == 1 ) {
                            if ( r2[-1] == 3 ) continue;
                            if ( r2[-1] == -3 && type == TYPENORM3 ) continue;
                        }
                        action = 1;
                        v[2] |= DIRTYFLAG;
                        j = r2[-1];
                        k = REDLENG(j);
                        if ( j < 0 ) j = -j;
                        if ( type == TYPENORM && scale && ( factor == 0 || *factor ) ) {
/*
                            Now we correct the overall factor
*/
ScaledVariety:;
                            size = term[*term-1];
                            size = REDLENG(size);
                            if ( scale > 0 ) {
                                for ( jj = 0; jj < scale; jj++ ) {
                                    if ( MulRat(BHEAD (UWORD *)rstop,size,(UWORD *)r3,k,
                                        (UWORD *)rstop,&size) ) goto execargerr;
                                }
                            }
                            else {
                                for ( jj = 0; jj > scale; jj-- ) {
                                    if ( DivRat(BHEAD (UWORD *)rstop,size,(UWORD *)r3,k,
```

```
                                        (UWORD *)rstop,&size) ) goto execargerr;
                                }
                            }
                            size = INCLENG(size);
                            k = size < 0 ? -size: size;
                            rstop[k-1] = size*sign;
                            *term = (WORD)(rstop - term) + k;
                        }
                    }
/*
                    We generate a statement for adapting all terms in the
                    argument successively
*/
                    r4 = AddRHS(AT.ebufnum,1);
                    while ( (r4+j+12) > CC->Top ) r4 = DoubleCbuffer(AT.ebufnum,r4,3);
                    *r4++ = j+1;
                    i = (j-1)/2;  /* was (j-1)*2  ????? 17-oct-2017 */
                    for ( k = 0; k < i; k++ ) *r4++ = r3[i+k];
                    for ( k = 0; k < i; k++ ) *r4++ = r3[k];
                    if ( ( type == TYPENORM3 ) || ( type == TYPENORM4 ) ) *r4++ = j*sign;
                    else *r4++ = r3[j-1];
                    *r4++ = 0;
                    CC->rhs[CC->numrhs+1] = r4;
                    CC->Pointer = r4;
                    AT.mulpat[5] = CC->numrhs;
                    AT.mulpat[7] = AT.ebufnum;
                }
                else if ( type == TYPEARGTOEXTRASYMBOL ) {
                    WORD n;
                    if ( r[0] < 0 ) {
                        /* The argument is in the fast notation. */
                        WORD tmp[MaX(9,FUNHEAD+5)];
                        switch ( r[0] ) {
                            case -SNUMBER:
                                if ( r[1] == 0 ) {
                                    tmp[0] = 0;
                                }
                                else {
                                    tmp[0] = 4;
                                    tmp[1] = ABS(r[1]);
                                    tmp[2] = 1;
                                    tmp[3] = r[1] > 0 ? 3 : -3;
                                    tmp[4] = 0;
                                }
                                break;
                            case -SYMBOL:
                                tmp[0] = 8;
                                tmp[1] = SYMBOL;
                                tmp[2] = 4;
                                tmp[3] = r[1];
                                tmp[4] = 1;
                                tmp[5] = 1;
                                tmp[6] = 1;
                                tmp[7] = 3;
                                tmp[8] = 0;
                                break;
                            case -INDEX:
                            case -VECTOR:
                            case -MINVECTOR:
                                tmp[0] = 7;
                                tmp[1] = INDEX;
                                tmp[2] = 3;
                                tmp[3] = r[1];
                                tmp[4] = 1;
                                tmp[5] = 1;
                                tmp[6] = r[0] != -MINVECTOR ? 3 : -3;
                                tmp[7] = 0;
                                break;
                            default:
                                if ( r[0] <= -FUNCTION ) {
                                    tmp[0] = FUNHEAD+4;
                                    tmp[1] = -r[0];
                                    tmp[2] = FUNHEAD;
                                    ZeroFillRange(tmp,3,1+FUNHEAD);
                                    tmp[FUNHEAD+1] = 1;
                                    tmp[FUNHEAD+2] = 1;
                                    tmp[FUNHEAD+3] = 3;
                                    tmp[FUNHEAD+4] = 0;
                                    break;
                                }
                                else {
                                    MLOCK(ErrorMessageLock);
                                    MesPrint("Unknown fast notation found (TYPEARGTOEXTRASYMBOL)");
                                    MUNLOCK(ErrorMessageLock);
                                    return(-1);
                                }
                        }
```

```
                        n = FindSubexpression(tmp);
                    }
                    else {
                        /*
                         * NOTE: writing to r[r[0]] is legal. As long as we work
                         * in a part of the term, at least the coefficient of
                         * the term must follow.
                         */
                        WORD old_rr0 = r[r[0]];
                        r[r[0]] = 0;  /* zero-terminated */
                        n = FindSubexpression(r+ARGHEAD);
                        r[r[0]] = old_rr0;
                    }
                    /* Put the new argument in the work space. */
                    if ( AT.WorkPointer+2 > AT.WorkTop ) {
                        MLOCK(ErrorMessageLock);
                        MesWork();
                        MUNLOCK(ErrorMessageLock);
                        return(-1);
                    }
                    r1 = AT.WorkPointer;
                    if ( scale ) {  /* means "tonumber" */
                        r1[0] = -SNUMBER;
                        r1[1] = n;
                    }
                    else {
                        r1[0] = -SYMBOL;
                        r1[1] = MAXVARIABLES-n;
                    }
                    /* We need r2, r3, m and k to shift the data. */
                    r2 = r + (r[0] > 0 ? r[0] : r[0] <= -FUNCTION ? 1 : 2);
                    r3 = r;
                    m = r1+ARGHEAD+2;
                    k = 2;
                    goto do_shift;
                }
                r3 = r;
                AR.DeferFlag = 0;
                if ( *r > 0 ) {
                    NewSort(BHEAD0);
                    action = 1;
                    r2 = r + *r;
                    r += ARGHEAD;
                    while ( r < r2 ) {  /* Sum over the terms */
                        m = AT.WorkPointer;
                        j = *r;
                        while ( --j >= 0 ) *m++ = *r++;
                        r1 = AT.WorkPointer;
                        AT.WorkPointer = m;
/*
                        What to do with dummy indices?
*/
                        if ( type == TYPENORM || type == TYPENORM2 || type == TYPENORM3 || type == TYPENORM4 ) {
                            if ( MultDo(BHEAD r1,AT.mulpat) ) goto execargerr;
                            AT.WorkPointer = r1 + *r1;
                        }
                        if ( Generator(BHEAD r1,level) ) goto execargerr;
                        AT.WorkPointer = r1;
                    }
                }
                else {
                    r2 = r + (( *r <= -FUNCTION ) ? 1:2);
                    r1 = AT.WorkPointer;
                    ToGeneral(r,r1,0);
                    m = r1 + ARGHEAD;
                    AT.WorkPointer = r1 + *r1;
                    NewSort(BHEAD0);
                    action = 1;
/*
                    What to do with dummy indices?
*/
                    if ( type == TYPENORM || type == TYPENORM2 || type == TYPENORM3 || type == TYPENORM4 ) {
                        if ( MultDo(BHEAD m,AT.mulpat) ) goto execargerr;
                        AT.WorkPointer = m + *m;
                    }
                    if ( (*m != 0 ) && Generator(BHEAD m,level) ) goto execargerr;
                    AT.WorkPointer = r1;
                }
                if ( EndSort(BHEAD AT.WorkPointer+ARGHEAD,1) < 0 ) goto execargerr;
                AR.DeferFlag = olddefer;
/*
                Now shift the sorted entity over the old argument.
*/
                m = AT.WorkPointer+ARGHEAD;
                while ( *m ) m += *m;
```

```
                k = WORDDIF(m,AT.WorkPointer);
                *AT.WorkPointer = k;
                AT.WorkPointer[1] = 0;
                if ( ToFast(AT.WorkPointer,AT.WorkPointer) ) {
                    if ( *AT.WorkPointer <= -FUNCTION ) k = 1;
                    else k = 2;
                }
do_shift:
                if ( *r3 > 0 ) j = k - *r3;
                else if ( *r3 <= -FUNCTION ) j = k - 1;
                else j = k - 2;

                t[1] += j;
                action = 1;
                v[2] |= DIRTYFLAG;
                if ( j > 0 ) {
                    r = m + j;
                    while ( m > AT.WorkPointer ) *--r = *--m;
                    AT.WorkPointer = r;
                    m = term + *term;
                    r = m + j;
                    while ( m > r2 ) *--r = *--m;
                }
                else if ( j < 0 ) {
                    r = r2 + j;
                    r1 = term + *term;
                    while ( r2 < r1 ) *r++ = *r2++;
                }
                r = r3;
                m = AT.WorkPointer;
                NCOPY(r,m,k);
                *term += j;
                rstop += j;
                CC->numrhs = oldnumrhs;
                CC->Pointer = CC->Buffer + oldcpointer;
            }
        }
        t += t[1];
    }
/*
        If TYPENORM4, we allocated Number buffers before the above while loop. Free them.
*/
    if ( type == TYPENORM4 ) {
        NumberFree(GCDbuffer,"execarg");
        NumberFree(GCDbuffer2,"execarg");
        NumberFree(LCMbuffer,"execarg");
        NumberFree(LCMb,"execarg"); NumberFree(LCMc,"execarg");
    }
    if ( didpolyratfun ) {
        PolyFunDirty(BHEAD term);
        didpolyratfun = 0;
    }
/*
    #] Argument detection :
    #[ SplitArg : + varieties
*/
    if ( ( type == TYPESPLITARG || type == TYPESPLITARG2
     || type == TYPESPLITFIRSTARG || type == TYPESPLITLASTARG ) &&
    AT.pWorkPointer > oldppointer ) {
        t = term+1;
        r1 = AT.WorkPointer + 1;
        lp = oldppointer;
        while ( t < rstop ) {
            if ( lp < AT.pWorkPointer && t == AT.pWorkSpace[lp] ) {
                v = t;
                m = t + FUNHEAD;
                r = t + t[1];
                r2 = r1; while ( t < m ) *r1++ = *t++;
                while ( m < r ) {
                    t = m;
                    NEXTARG(m)
                    if ( lp >= AT.pWorkPointer || t != AT.pWorkSpace[lp+1] ) {
                        if ( *t > 0 ) t[1] = 0;
                        while ( t < m ) *r1++ = *t++;
                        continue;
                    }
/*
                    Now we have a nontrivial argument that should be done.
*/
                    lp += 2;
                    action = 1;
                    v[2] |= DIRTYFLAG;
                    r3 = t + *t;
                    t += ARGHEAD;
                    if ( type == TYPESPLITFIRSTARG ) {
                        r4 = r1; r5 = t; r7 = oldwork;
                        *r1++ = *t + ARGHEAD;
```

```
                        for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
                        j = 0;
                        while ( t < r3 ) {
                            i = *t;
                            if ( j == 0 ) {
                                NCOPY(r7,t,i)
                                j++;
                            }
                            else {
                                NCOPY(r1,t,i)
                            }
                        }
                        *r4 = r1 - r4;
                        if ( j ) {
                            if ( ToFast(r4,r4) ) {
                                r1 = r4;
                                if ( *r1 > -FUNCTION ) r1++;
                                r1++;
                            }
                            r7 = oldwork;
                            while ( --j >= 0 ) {
                                r4 = r1; i = *r7;
                                *r1++ = i+ARGHEAD; *r1++ = 0;
                                FILLARG(r1);
                                NCOPY(r1,r7,i)
                                if ( ToFast(r4,r4) ) {
                                    r1 = r4;
                                    if ( *r1 > -FUNCTION ) r1++;
                                    r1++;
                                }
                            }
                        }
                        t = r3;
                    }
                    else if ( type == TYPESPLITLASTARG ) {
                        r4 = r1; r5 = t; r7 = oldwork;
                        *r1++ = *t + ARGHEAD;
                        for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
                        j = 0;
                        while ( t < r3 ) {
                            i = *t;
                            if ( t+i >= r3 ) {
                                NCOPY(r7,t,i)
                                j++;
                            }
                            else {
                                NCOPY(r1,t,i)
                            }
                        }
                        *r4 = r1 - r4;
                        if ( j ) {
                            if ( ToFast(r4,r4) ) {
                                r1 = r4;
                                if ( *r1 > -FUNCTION ) r1++;
                                r1++;
                            }
                            r7 = oldwork;
                            while ( --j >= 0 ) {
                                r4 = r1; i = *r7;
                                *r1++ = i+ARGHEAD; *r1++ = 0;
                                FILLARG(r1);
                                NCOPY(r1,r7,i)
                                if ( ToFast(r4,r4) ) {
                                    r1 = r4;
                                    if ( *r1 > -FUNCTION ) r1++;
                                    r1++;
                                }
                            }
                        }
                        t = r3;
                    }
                    else if ( factor == 0 || ( type == TYPESPLITARG2 && *factor == 0 ) ) {
                        while ( t < r3 ) {
                            r4 = r1;
                            *r1++ = *t + ARGHEAD;
                            for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
                            i = *t;
                            while ( --i >= 0 ) *r1++ = *t++;
                            if ( ToFast(r4,r4) ) {
                                r1 = r4;
                                if ( *r1 > -FUNCTION ) r1++;
                                r1++;
                            }
                        }
                    }
                    else if ( type == TYPESPLITARG2 ) {
/*
```

```
				Here we better put the pattern matcher at work?
				Remember: there are no wildcards.
*/
				WORD *oRepFunList = AN.RepFunList;
				WORD *oWildMask = AT.WildMask, *oWildValue = AN.WildValue;
				AN.WildValue = AT.locwildvalue; AT.WildMask = AT.locwildvalue+2;
				AN.NumWild = 0;
				r4 = r1; r5 = t; r7 = oldwork;
				*r1++ = *t + ARGHEAD;
				for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
				j = 0;
				while ( t < r3 ) {
					AN.UseFindOnly = 0; oldwork2 = AT.WorkPointer;
					AN.RepFunList = r1;
					AT.WorkPointer = r1+AN.RepFunNum+2;
					i = *t;
					if ( FindRest(BHEAD t,factor) &&
					 ( AN.UsedOtherFind || FindOnce(BHEAD t,factor) ) ) {
						NCOPY(r7,t,i)
						j++;
					}
					else if ( factor[0] == FUNHEAD+1 && factor[1] >= FUNCTION ) {
						WORD *rr1 = t+1, *rr2 = t+i;
						rr2 -= ABS(rr2[-1]);
						while ( rr1 < rr2 ) {
							if ( *rr1 == factor[1] ) break;
							rr1 += rr1[1];
						}
						if ( rr1 < rr2 ) {
							NCOPY(r7,t,i)
							j++;
						}
						else {
							NCOPY(r1,t,i)
						}
					}
					else {
						NCOPY(r1,t,i)
					}
					AT.WorkPointer = oldwork2;
				}
				AN.RepFunList = oRepFunList;
				*r4 = r1 - r4;
				if ( j ) {
					if ( ToFast(r4,r4) ) {
						r1 = r4;
						if ( *r1 > -FUNCTION ) r1++;
						r1++;
					}
					r7 = oldwork;
					while ( --j >= 0 ) {
						r4 = r1; i = *r7;
						*r1++ = i+ARGHEAD; *r1++ = 0;
						FILLARG(r1);
						NCOPY(r1,r7,i)
						if ( ToFast(r4,r4) ) {
							r1 = r4;
							if ( *r1 > -FUNCTION ) r1++;
							r1++;
						}
					}
				}
				t = r3;
				AT.WildMask = oWildMask; AN.WildValue = oWildValue;
			}
			else {
/*
				This code deals with splitting off a single term
*/
				r4 = r1; r5 = t;
				*r1++ = *t + ARGHEAD;
				for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
				j = 0;
				while ( t < r3 ) {
					r6 = t + *t; r6 -= ABS(r6[-1]);
					if ( (r6 - t) == *factor ) {
						k = *factor - 1;
						for ( ; k > 0; k-- ) {
							if ( t[k] != factor[k] ) break;
						}
						if ( k <= 0 ) {
							j = r3 - t; t += *t; continue;
						}
					}
					else if ( (r6 - t) == 1 && *factor == 0 ) {
						j = r3 - t; t += *t; continue;
					}
```

```
                            i = *t;
                            NCOPY(r1,t,i)
                        }
                        *r4 = r1 - r4;
                        if ( j ) {
                            if ( ToFast(r4,r4) ) {
                                r1 = r4;
                                if ( *r1 > -FUNCTION ) r1++;
                                r1++;
                            }
                            t = r3 - j;
                            r4 = r1;
                            *r1++ = *t + ARGHEAD;
                            for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
                            i = *t;
                            while ( --i >= 0 ) *r1++ = *t++;
                            if ( ToFast(r4,r4) ) {
                                r1 = r4;
                                if ( *r1 > -FUNCTION ) r1++;
                                r1++;
                            }
                        }
                        t = r3;
                    }
                }
                r2[1] = r1 - r2;
            }
            else {
                r = t + t[1];
                while ( t < r ) *r1++ = *t++;
            }
        }
        r = term + *term;
        while ( t < r ) *r1++ = *t++;
        m = AT.WorkPointer;
        i = m[0] = r1 - m;
        t = term;
        while ( --i >= 0 ) *t++ = *m++;
        if ( AT.WorkPointer < m ) AT.WorkPointer = m;
    }
/*
    #] SplitArg :
    #[ FACTARG :
*/
    if ( ( type == TYPEFACTARG || type == TYPEFACTARG2 ) &&
    AT.pWorkPointer > oldppointer ) {
        t = term+1;
        r1 = AT.WorkPointer + 1;
        lp = oldppointer;
        while ( t < rstop ) {
            if ( lp < AT.pWorkPointer && AT.pWorkSpace[lp] == t ) {
                v = t;
                m = t + FUNHEAD;
                r = t + t[1];
                r2 = r1; while ( t < m ) *r1++ = *t++;
                while ( m < r ) {
                    rr = t = m;
                    NEXTARG(m)
                    if ( lp >= AT.pWorkPointer || AT.pWorkSpace[lp+1] != t ) {
                        if ( *t > 0 ) t[1] = 0;
                        while ( t < m ) *r1++ = *t++;
                        continue;
                    }
/*
                    Now we have a nontrivial argument that should be studied.
                    Try to find common factors.
*/
                    lp += 2;
                    if ( *t < 0 ) {
                        if ( factor && ( *factor == 0 && *t == -SNUMBER ) ) {
                            *r1++ = *t++;
                            if ( *t == 0 ) *r1++ = *t++;
                            else { *r1++ = 1; t++; }
                            continue;
                        }
                        else if ( factor && *factor == 4 && factor[2] == 1 ) {
                            if ( *t == -SNUMBER ) {
                                if ( factor[3] < 0 || t[1] >= 0 ) {
                                    while ( t < m ) *r1++ = *t++;
                                }
                                else {
                                    *r1++ = -SNUMBER; *r1++ = -1;
                                    *r1++ = *t++; *r1++ = -*t++;
                                }
                            }
                            else {
                                while ( t < m ) *r1++ = *t++;
```

```
                    *r1++ = -SNUMBER; *r1++ = 1;
                }
                continue;
            }
            else if ( *t == -MINVECTOR ) {
                if ( AC.OldFactArgFlag == NEWFACTARG ) {
                    *r1++ = -SNUMBER; *r1++ = -1;
                    *r1++ = -VECTOR; t++; *r1++ = *t++;
                }
                else {
                    *r1++ = -VECTOR; t++; *r1++ = *t++;
                    *r1++ = -SNUMBER; *r1++ = -1;
                    *r1++ = -SNUMBER; *r1++ = 1;
                }
                continue;
            }
        }
/*
        Now we have a nontrivial argument
*/
        r3 = t + *t;
        t += ARGHEAD;  r5 = t; /* Store starting point */
        /* We have terms from r5 to r3 */
        if ( r5+*r5 == r3 && factor ) { /* One term only */
            if ( *factor == 0 ) {
                GETSTOP(t,r6);
                r9 = r1; *r1++ = 0; *r1++ = 1;
                FILLARG(r1);
                *r1++ = (r6-t)+3; t++;
                while ( t < r6 ) *r1++ = *t++;
                *r1++ = 1; *r1++ = 1; *r1++ = 3;
                *r9 = r1-r9;
                if ( ToFast(r9,r9) ) {
                    if ( *r9 <= -FUNCTION ) r1 = r9+1;
                    else r1 = r9+2;
                }
                t = r3; continue;
            }
            if ( factor[0] == 4 && factor[2] == 1 ) {
                GETSTOP(t,r6);
                r7 = r1; *r1++ = (r6-t)+3+ARGHEAD; *r1++ = 0;
                FILLARG(r1);
                *r1++ = (r6-t)+3; t++;
                while ( t < r6 ) *r1++ = *t++;
                *r1++ = 1; *r1++ = 1; *r1++ = 3;
                if ( ToFast(r7,r7) ) {
                    if ( *r7 <= -FUNCTION ) r1 = r7+1;
                    else r1 = r7+2;
                }
                if ( r3[-1] < 0 && factor[3] > 0 ) {
                    *r1++ = -SNUMBER; *r1++ = -1;
                    if ( r3[-1] == -3 && r3[-2] == 1
                    && ( r3[-3] & MAXPOSITIVE ) == r3[-3] ) {
                        *r1++ = -SNUMBER; *r1++ = r3[-3];
                    }
                    else {
                        *r1++ = (r3-r6)+1+ARGHEAD;
                        *r1++ = 0;
                        FILLARG(r1);
                        *r1++ = (r3-r6+1);
                        while ( t < r3 ) *r1++ = *t++;
                        r1[-1] = -r1[-1];
                    }
                }
                else {
                    if ( ( r3[-1] == -3 || r3[-1] == 3 )
                    && r3[-2] == 1
                    && ( r3[-3] & MAXPOSITIVE ) == r3[-3] ) {
                        *r1++ = -SNUMBER; *r1++ = r3[-3];
                        if ( r3[-1] < 0 ) r1[-1] = - r1[-1];
                    }
                    else {
                        *r1++ = (r3-r6)+1+ARGHEAD;
                        *r1++ = 0;
                        FILLARG(r1);
                        *r1++ = (r3-r6+1);
                        while ( t < r3 ) *r1++ = *t++;
                    }
                }
                t = r3; continue;
            }
        }
/*
        Now we take the first term and look for its pieces
        inside the other terms.

        It is at this point that a more general factorization
```

```
          routine could take over (allowing for writing the output
          properly of course).
*/
          if ( AC.OldFactArgFlag == NEWFACTARG ) {
          if ( factor == 0 ) {
            WORD *oldworkpointer2 = AT.WorkPointer;
            AT.WorkPointer = r1 + AM.MaxTer+FUNHEAD;
            if ( ArgFactorize(BHEAD t-ARGHEAD,r1) < 0 ) {
              MesCall("ExecArg");
              return(-1);
            }
            AT.WorkPointer = oldworkpointer2;
            t = r3;
            while ( *r1 ) { NEXTARG(r1) }
          }
          else {
            rnext = t + *t;
            GETSTOP(t,r6);
            t++;
            t = r5; pow = 1;
            while ( t < r3 ) {
              t += *t; if ( t[-1] > 0 ) { pow = 0; break; }
            }
/*
            We have to add here the code for computing the GCD
            and to divide it out.

      #[ Numerical factor :
*/
            t = r5;
            EAscrat = (UWORD *)(TermMalloc("execarg"));
            if ( t + *t == r3 ) {
              if ( factor == 0 || *factor > 2 ) {
               if ( pow > 0 ) {
                *r1++ = -SNUMBER; *r1++ = -1;
                t = r5;
                while ( t < r3 ) {
                  t += *t; t[-1] = -t[-1];
                }
               }
               t = rr; *r1++ = *t++; *r1++ = 1; t++;
               COPYARG(r1,t);
               while ( t < m ) *r1++ = *t++;
              }
            }
            else {
            GETSTOP(t,r6);
            ngcd = t[t[0]-1];
            i = abs(ngcd)-1;
            while ( --i >= 0 ) EAscrat[i] = r6[i];
            t += *t;
            while ( t < r3 ) {
              GETSTOP(t,r6);
              i = t[t[0]-1];
              if ( AccumGCD(BHEAD EAscrat,&ngcd,(UWORD *)r6,i) ) goto execargerr;
              if ( ngcd == 3 && EAscrat[0] == 1 && EAscrat[1] == 1 ) break;
              t += *t;
            }
/*
            if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 )
*/
            {
              if ( pow ) ngcd = -ngcd;
              t = r5; r9 = r1; *r1++ = t[-ARGHEAD]; *r1++ = 1;
              FILLARG(r1); ngcd = REDLENG(ngcd);
              while ( t < r3 ) {
                GETSTOP(t,r6);
                r7 = t; r8 = r1;
                while ( r7 < r6) *r1++ = *r7++;
                t += *t;
                i = REDLENG(t[-1]);
                if ( DivRat(BHEAD (UWORD *)r6,i,EAscrat,ngcd,(UWORD *)r1,&nq) ) goto execargerr;
                nq = INCLENG(nq);
                i = ABS(nq)-1;
                r1 += i; *r1++ = nq; *r8 = r1-r8;
              }
              *r9 = r1-r9;
              ngcd = INCLENG(ngcd);
              i = ABS(ngcd)-1;
              if ( factor && *factor == 0 ) {}
              else if ( ( factor && factor[0] == 4 && factor[2] == 1
              && factor[3] == -3 ) || pow == 0 ) {
                r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
                FILLARG(r1); *r1++ = i+2;
                for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
                *r1++ = ngcd;
```

```
                                if ( ToFast(r9,r9) ) r1 = r9+2;
                            }
                            else if ( factor && factor[0] == 4 && factor[2] == 1
                            && factor[3] > 0 && pow ) {
                                if ( ngcd < 0 ) ngcd = -ngcd;
                                *r1++ = -SNUMBER; *r1++ = -1;
                                r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
                                FILLARG(r1); *r1++ = i+2;
                                for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
                                *r1++ = ngcd;
                                if ( ToFast(r9,r9) ) r1 = r9+2;
                            }
                            else {
                                if ( ngcd < 0 ) ngcd = -ngcd;
                                if ( pow ) { *r1++ = -SNUMBER; *r1++ = -1; }
                                if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 ) {
                                    r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
                                    FILLARG(r1); *r1++ = i+2;
                                    for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
                                    *r1++ = ngcd;
                                    if ( ToFast(r9,r9) ) r1 = r9+2;
                                }
                            }
                        }
/*
            #] Numerical factor :
                        else {
onetermnew:;

                            if ( factor == 0 || *factor > 2 ) {
                              if ( pow > 0 ) {
                                *r1++ = -SNUMBER; *r1++ = -1;
                                t = r5;
                                while ( t < r3 ) {
                                    t += *t; t[-1] = -t[-1];
                                }
                              }
                              t = rr; *r1++ = *t++; *r1++ = 1; t++;
                              COPYARG(r1,t);
                              while ( t < m ) *r1++ = *t++;
                            }
                        }
onetermnew:;
*/
                        }
                        TermFree(EAscrat,"execarg");
                    }
                    }
                    else {  /* AC.OldFactArgFlag is ON */
                    {
                    WORD *mnext, ncom;
                    rnext = t + *t;
                    GETSTOP(t,r6);
                    t++;
                    if ( factor == 0 ) {
                      while ( t < r6 ) {
/*
            #[ SYMBOL :
*/
                        if ( *t == SYMBOL ) {
                            r7 = t; r8 = t + t[1]; t += 2;
                            while ( t < r8 ) {
                                pow = t[1];
                                mm = rnext;
                                while ( mm < r3 ) {
                                    mnext = mm + *mm;
                                    GETSTOP(mm,mstop); mm++;
                                    while ( mm < mstop ) {
                                        if ( *mm != SYMBOL ) mm += mm[1];
                                        else break;
                                    }
                                    if ( *mm == SYMBOL ) {
                                        mstop = mm + mm[1]; mm += 2;
                                        while ( *mm != *t && mm < mstop ) mm += 2;
                                        if ( mm >= mstop ) pow = 0;
                                        else if ( pow > 0 && mm[1] > 0 ) {
                                            if ( mm[1] < pow ) pow = mm[1];
                                        }
                                        else if ( pow < 0 && mm[1] < 0 ) {
                                            if ( mm[1] > pow ) pow = mm[1];
                                        }
                                        else pow = 0;
                                    }
                                    else pow = 0;
                                    if ( pow == 0 ) break;
                                    mm = mnext;
                                }
```

```
                                if ( pow == 0 ) { t += 2; continue; }
/*
                                We have a factor
*/
                                action = 1; i = pow;
                                if ( i > 0 ) {
                                    while ( --i >= 0 ) {
                                        *r1++ = -SYMBOL;
                                        *r1++ = *t;
                                    }
                                }
                                else {
                                    while ( i++ < 0 ) {
                                        *r1++ = 8 + ARGHEAD;
                                        for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
                                        *r1++ = 8; *r1++ = SYMBOL;
                                        *r1++ = 4; *r1++ = *t; *r1++ = -1;
                                        *r1++ = 1; *r1++ = 1; *r1++ = 3;
                                    }
                                }
/*
                                Now we have to remove the symbols
*/
                                t[1] -= pow;
                                mm = rnext;
                                while ( mm < r3 ) {
                                    mnext = mm + *mm;
                                    GETSTOP(mm,mstop); mm++;
                                    while ( mm < mstop ) {
                                        if ( *mm != SYMBOL ) mm += mm[1];
                                        else break;
                                    }
                                    mstop = mm + mm[1]; mm += 2;
                                    while ( mm < mstop && *mm != *t ) mm += 2;
                                    mm[1] -= pow;
                                    mm = mnext;
                                }
                                t += 2;
                            }
                        }
/*
            #] SYMBOL :
            #[ DOTPRODUCT :
*/
                        else if ( *t == DOTPRODUCT ) {
                            r7 = t; r8 = t + t[1]; t += 2;
                            while ( t < r8 ) {
                                pow = t[2];
                                mm = rnext;
                                while ( mm < r3 ) {
                                    mnext = mm + *mm;
                                    GETSTOP(mm,mstop); mm++;
                                    while ( mm < mstop ) {
                                        if ( *mm != DOTPRODUCT ) mm += mm[1];
                                        else break;
                                    }
                                    if ( *mm == DOTPRODUCT ) {
                                        mstop = mm + mm[1]; mm += 2;
                                        while ( ( *mm != *t || mm[1] != t[1] )
                                             && mm < mstop ) mm += 3;
                                        if ( mm >= mstop ) pow = 0;
                                        else if ( pow > 0 && mm[2] > 0 ) {
                                            if ( mm[2] < pow ) pow = mm[2];
                                        }
                                        else if ( pow < 0 && mm[2] < 0 ) {
                                            if ( mm[2] > pow ) pow = mm[2];
                                        }
                                        else pow = 0;
                                    }
                                    else pow = 0;
                                    if ( pow == 0 ) break;
                                    mm = mnext;
                                }
                                if ( pow == 0 ) { t += 3; continue; }
/*
                                We have a factor
*/
                                action = 1; i = pow;
                                if ( i > 0 ) {
                                    while ( --i >= 0 ) {
                                        *r1++ = 9 + ARGHEAD;
                                        for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
                                        *r1++ = 9; *r1++ = DOTPRODUCT;
                                        *r1++ = 5; *r1++ = *t; *r1++ = t[1]; *r1++ = 1;
                                        *r1++ = 1; *r1++ = 1; *r1++ = 3;
                                    }
                                }
```

```
                                else {
                                    while ( i++ < 0 ) {
                                        *r1++ = 9 + ARGHEAD;
                                        for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
                                        *r1++ = 9; *r1++ = DOTPRODUCT;
                                        *r1++ = 5; *r1++ = *t; *r1++ = t[1]; *r1++ = -1;
                                        *r1++ = 1; *r1++ = 1; *r1++ = 3;
                                    }
                                }
/*
                                Now we have to remove the dotproducts
*/
                                t[2] -= pow;
                                mm = rnext;
                                while ( mm < r3 ) {
                                    mnext = mm + *mm;
                                    GETSTOP(mm,mstop); mm++;
                                    while ( mm < mstop ) {
                                        if ( *mm != DOTPRODUCT ) mm += mm[1];
                                        else break;
                                    }
                                    mstop = mm + mm[1]; mm += 2;
                                    while ( mm < mstop && ( *mm != *t
                                        || mm[1] != t[1] ) ) mm += 3;
                                    mm[2] -= pow;
                                    mm = mnext;
                                }
                                t += 3;
                            }
                        }
/*
            #] DOTPRODUCT :
            #[ DELTA/VECTOR :
*/
                        else if ( *t == DELTA || *t == VECTOR ) {
                            r7 = t; r8 = t + t[1]; t += 2;
                            while ( t < r8 ) {
                                mm = rnext;
                                pow = 1;
                                while ( mm < r3 ) {
                                    mnext = mm + *mm;
                                    GETSTOP(mm,mstop); mm++;
                                    while ( mm < mstop ) {
                                        if ( *mm != *r7 ) mm += mm[1];
                                        else break;
                                    }
                                    if ( *mm == *r7 ) {
                                        mstop = mm + mm[1]; mm += 2;
                                        while ( ( *mm != *t || mm[1] != t[1] )
                                            && mm < mstop ) mm += 2;
                                        if ( mm >= mstop ) pow = 0;
                                    }
                                    else pow = 0;
                                    if ( pow == 0 ) break;
                                    mm = mnext;
                                }
                                if ( pow == 0 ) { t += 2; continue; }
/*
                                We have a factor
*/
                                action = 1;
                                *r1++ = 8 + ARGHEAD;
                                for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
                                *r1++ = 8; *r1++ = *r7;
                                *r1++ = 4; *r1++ = *t; *r1++ = t[1];
                                *r1++ = 1; *r1++ = 1; *r1++ = 3;
/*
                                Now we have to remove the delta's/vectors
*/
                                mm = rnext;
                                while ( mm < r3 ) {
                                    mnext = mm + *mm;
                                    GETSTOP(mm,mstop); mm++;
                                    while ( mm < mstop ) {
                                        if ( *mm != *r7 ) mm += mm[1];
                                        else break;
                                    }
                                    mstop = mm + mm[1]; mm += 2;
                                    while ( mm < mstop && (
                                     *mm != *t || mm[1] != t[1] ) ) mm += 2;
                                    *mm = mm[1] = NOINDEX;
                                    mm = mnext;
                                }
                                *t = t[1] = NOINDEX;
                                t += 2;
                            }
                        }
```

```
/*
          #] DELTA/VECTOR :
          #[ INDEX :
*/
                else if ( *t == INDEX ) {
                    r7 = t; r8 = t + t[1]; t += 2;
                    while ( t < r8 ) {
                        mm = rnext;
                        pow = 1;
                        while ( mm < r3 ) {
                            mnext = mm + *mm;
                            GETSTOP(mm,mstop); mm++;
                            while ( mm < mstop ) {
                                if ( *mm != *r7 ) mm += mm[1];
                                else break;
                            }
                            if ( *mm == *r7 ) {
                                mstop = mm + mm[1]; mm += 2;
                                while ( *mm != *t
                                    && mm < mstop ) mm++;
                                if ( mm >= mstop ) pow = 0;
                            }
                            else pow = 0;
                            if ( pow == 0 ) break;
                            mm = mnext;
                        }
                        if ( pow == 0 ) { t++; continue; }
/*
                        We have a factor
*/
                        action = 1;
/*
                        The next looks like an error.
                        We should have here a VECTOR or INDEX like object

                        *r1++ = 7 + ARGHEAD;
                        for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
                        *r1++ = 7; *r1++ = *r7;
                        *r1++ = 3; *r1++ = *t;
                        *r1++ = 1; *r1++ = 1; *r1++ = 3;

                        Replace this by:  (11-apr-2007)
*/
                        if ( *t < 0 ) { *r1++ = -VECTOR; }
                        else { *r1++ = -INDEX; }
                        *r1++ = *t;
/*
                        Now we have to remove the index
*/
                        *t = NOINDEX;
                        mm = rnext;
                        while ( mm < r3 ) {
                            mnext = mm + *mm;
                            GETSTOP(mm,mstop); mm++;
                            while ( mm < mstop ) {
                                if ( *mm != *r7 ) mm += mm[1];
                                else break;
                            }
                            mstop = mm + mm[1]; mm += 2;
                            while ( mm < mstop &&
                             *mm != *t ) mm += 1;
                            *mm = NOINDEX;
                            mm = mnext;
                        }
                        t += 1;
                    }
                }
/*
          #] INDEX :
          #[ FUNCTION :
*/
                else if ( *t >= FUNCTION ) {
/*
                    In the next code we should actually look inside
                    the DENOMINATOR or EXPONENT for noncommuting objects
*/
                    if ( *t >= FUNCTION &&
                        functions[*t-FUNCTION].commute == 0 ) ncom = 0;
                    else ncom = 1;
                    if ( ncom ) {
                        mm = r5 + 1;
                        while ( mm < t && ( *mm == DUMMYFUN
                        || *mm == DUMMYTEN ) ) mm += mm[1];
                        if ( mm < t ) { t += t[1]; continue; }
                    }
                    mm = rnext; pow = 1;
                    while ( mm < r3 ) {
```

```
                                mnext = mm + *mm;
                                GETSTOP(mm,mstop); mm++;
                                while ( mm < mstop ) {
                                    if ( *mm == *t && mm[1] == t[1] ) {
                                        for ( i = 2; i < t[1]; i++ ) {
                                            if ( mm[i] != t[i] ) break;
                                        }
                                        if ( i >= t[1] )
                                            { mm += mm[1]; goto nextmterm; }
                                    }
                                    if ( ncom && *mm != DUMMYFUN && *mm != DUMMYTEN )
                                        { pow = 0; break; }
                                    mm += mm[1];
                                }
                                if ( mm >= mstop ) pow = 0;
                                if ( pow == 0 ) break;
nextmterm:                      mm = mnext;
                            }
                            if ( pow == 0 ) { t += t[1]; continue; }
/*
                            Copy the function
*/
                            action = 1;
                            *r1++ = t[1] + 4 + ARGHEAD;
                            for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
                            *r1++ = t[1] + 4;
                            for ( i = 0; i < t[1]; i++ ) *r1++ = t[i];
                            *r1++ = 1; *r1++ = 1; *r1++ = 3;
/*
                            Now we have to take out the functions
*/
                            mm = rnext;
                            while ( mm < r3 ) {
                                mnext = mm + *mm;
                                GETSTOP(mm,mstop); mm++;
                                while ( mm < mstop ) {
                                    if ( *mm == *t && mm[1] == t[1] ) {
                                        for ( i = 2; i < t[1]; i++ ) {
                                            if ( mm[i] != t[i] ) break;
                                        }
                                        if ( i >= t[1] ) {
                                            if ( functions[*t-FUNCTION].spec > 0 )
                                                *mm = DUMMYTEN;
                                            else
                                                *mm = DUMMYFUN;
                                            mm += mm[1];
                                            goto nextterm;
                                        }
                                    }
                                    mm += mm[1];
                                }
nextterm:                       mm = mnext;
                            }
                            if ( functions[*t-FUNCTION].spec > 0 )
                                    *t = DUMMYTEN;
                            else
                                    *t = DUMMYFUN;
                            action = 1;
                            v[2] = DIRTYFLAG;
                            t += t[1];
                        }
/*
            #] FUNCTION :
*/
                        else {
                            t += t[1];
                        }
                      }
                    }
                    t = r5; pow = 1;
                    while ( t < r3 ) {
                        t += *t; if ( t[-1] > 0 ) { pow = 0; break; }
                    }
/*
                    We have to add here the code for computing the GCD
                    and to divide it out.
*/
/*
            #[ Numerical factor :
*/
                    t = r5;
                    EAscrat = (UWORD *)(TermMalloc("execarg"));
                    if ( t + *t == r3 ) goto oneterm;
                    GETSTOP(t,r6);
                    ngcd = t[t[0]-1];
                    i = abs(ngcd)-1;
                    while ( --i >= 0 ) EAscrat[i] = r6[i];
```

```
                    t += *t;
                    while ( t < r3 ) {
                        GETSTOP(t,r6);
                        i = t[t[0]-1];
                        if ( AccumGCD(BHEAD EAscrat,&ngcd,(UWORD *)r6,i) ) goto execargerr;
                        if ( ngcd == 3 && EAscrat[0] == 1 && EAscrat[1] == 1 ) break;
                        t += *t;
                    }
                    if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 ) {
                        if ( pow ) ngcd = -ngcd;
                        t = r5; r9 = r1; *r1++ = t[-ARGHEAD]; *r1++ = 1;
                        FILLARG(r1); ngcd = REDLENG(ngcd);
                        while ( t < r3 ) {
                            GETSTOP(t,r6);
                            r7 = t; r8 = r1;
                            while ( r7 < r6) *r1++ = *r7++;
                            t += *t;
                            i = REDLENG(t[-1]);
                            if ( DivRat(BHEAD (UWORD *)r6,i,EAscrat,ngcd,(UWORD *)r1,&nq) ) goto
execargerr;
                            nq = INCLENG(nq);
                            i = ABS(nq)-1;
                            r1 += i; *r1++ = nq; *r8 = r1-r8;
                        }
                        *r9 = r1-r9;
                        ngcd = INCLENG(ngcd);
                        i = ABS(ngcd)-1;
                        if ( factor && *factor == 0 ) {}
                        else if ( ( factor && factor[0] == 4 && factor[2] == 1
                        && factor[3] == -3 ) || pow == 0 ) {
                            r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
                            FILLARG(r1); *r1++ = i+2;
                            for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
                            *r1++ = ngcd;
                            if ( ToFast(r9,r9) ) r1 = r9+2;
                        }
                        else if ( factor && factor[0] == 4 && factor[2] == 1
                        && factor[3] > 0 && pow ) {
                            if ( ngcd < 0 ) ngcd = -ngcd;
                            *r1++ = -SNUMBER; *r1++ = -1;
                            r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
                            FILLARG(r1); *r1++ = i+2;
                            for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
                            *r1++ = ngcd;
                            if ( ToFast(r9,r9) ) r1 = r9+2;
                        }
                        else {
                            if ( ngcd < 0 ) ngcd = -ngcd;
                            if ( pow ) { *r1++ = -SNUMBER; *r1++ = -1; }
                            if ( ngcd != 3 || EAscrat[0] != 1 || EAscrat[1] != 1 ) {
                                r9 = r1; *r1++ = ARGHEAD+2+i; *r1++ = 0;
                                FILLARG(r1); *r1++ = i+2;
                                for ( j = 0; j < i; j++ ) *r1++ = EAscrat[j];
                                *r1++ = ngcd;
                                if ( ToFast(r9,r9) ) r1 = r9+2;
                            }
                        }
                    }
/*
            #] Numerical factor :
*/
                    else {
oneterm:;
                        if ( factor == 0 || *factor > 2 ) {
                        if ( pow > 0 ) {
                            *r1++ = -SNUMBER; *r1++ = -1;
                            t = r5;
                            while ( t < r3 ) {
                                t += *t; t[-1] = -t[-1];
                            }
                        }
                        t = rr; *r1++ = *t++; *r1++ = 1; t++;
                        COPYARG(r1,t);
                        while ( t < m ) *r1++ = *t++;
                        }
                    }
                    TermFree(EAscrat,"execarg");
                    }
                    } /* AC.OldFactArgFlag */
                }
/* r1 is fout in ons voorbeeld. */
                r2[1] = r1 - r2;
                action = 1;
                v[2] = DIRTYFLAG;
            }
            else {
                r = t + t[1];
```

```
                while ( t < r ) *r1++ = *t++;
            }
        }
        r = term + *term;
        while ( t < r ) *r1++ = *t++;
        m = AT.WorkPointer;
        i = m[0] = r1 - m;
        t = term;
        while ( --i >= 0 ) *t++ = *m++;
        if ( AT.WorkPointer < t ) AT.WorkPointer = t;
    }
/*
    #] FACTARG :
*/
    AR.Cnumlhs = oldnumlhs;
    if ( action && Normalize(BHEAD term) ) goto execargerr;
    AT.WorkPointer = oldwork;
    if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
    AT.pWorkPointer = oldppointer;
    return(action);
execargerr:
    AT.WorkPointer = oldwork;
    AT.pWorkPointer = oldppointer;
    MLOCK(ErrorMessageLock);
    MesCall("execarg");
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
    #] execarg :
    #[ execterm :
*/

WORD execterm(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    CBUF *C = cbuf+AM.rbufnum;
    WORD oldnumlhs = AR.Cnumlhs;
    WORD maxisat = C->lhs[level][2];
    WORD *buffer1 = 0;
    WORD *oldworkpointer = AT.WorkPointer;
    WORD *t1, i;
    WORD olddeferflag = AR.DeferFlag, tryterm = 0;
    AR.DeferFlag = 0;
    do {
        AR.Cnumlhs = C->lhs[level][3];
        NewSort(BHEAD0);
/*
        Normally for function arguments we do not use PolyFun/PolyRatFun.
        Hence NewSort sets the corresponding variables to zero.
        Here we overwrite that.
*/
        AN.FunSorts[AR.sLevel]->PolyFlag = ( AR.PolyFun != 0 ) ? AR.PolyFunType: 0;
        if ( AR.PolyFun == 0 ) { AN.FunSorts[AR.sLevel]->PolyFlag = 0; }
        else if ( AR.PolyFunType == 1 ) { AN.FunSorts[AR.sLevel]->PolyFlag = 1; }
        else if ( AR.PolyFunType == 2 ) {
            if ( AR.PolyFunExp == 2 ) AN.FunSorts[AR.sLevel]->PolyFlag = 1;
            else                      AN.FunSorts[AR.sLevel]->PolyFlag = 2;
        }
        if ( buffer1 ) {
            term = buffer1;
            while ( *term ) {
                t1 = oldworkpointer;
                i = *term; while ( --i >= 0 ) *t1++ = *term++;
                AT.WorkPointer = t1;
                if ( Generator(BHEAD oldworkpointer,level) ) goto exectermerr;
            }
        }
        else {
            if ( Generator(BHEAD term,level) ) goto exectermerr;
        }
        if ( buffer1 ) {
            if ( tryterm ) { TermFree(buffer1,"buffer in sort statement"); tryterm = 0; }
            else { M_free((void *)buffer1,"buffer in sort statement"); }
            buffer1 = 0;
        }
        AN.tryterm = 1;
        if ( EndSort(BHEAD (WORD *)((void *)(&buffer1)),2) < 0 ) goto exectermerr;
        tryterm = AN.tryterm; AN.tryterm = 0;
        level = AR.Cnumlhs;
    } while ( AR.Cnumlhs < maxisat );
    AR.Cnumlhs = oldnumlhs;
    AR.DeferFlag = olddeferflag;
    term = buffer1;
    while ( *term ) {
        t1 = oldworkpointer;
```

```
        i = *term; while ( --i >= 0 ) *t1++ = *term++;
        AT.WorkPointer = t1;
        if ( Generator(BHEAD oldworkpointer,level) ) goto exectermerr;
    }
    if ( tryterm ) { TermFree(buffer1,"buffer in term statement"); tryterm = 0; }
    else { M_free(buffer1,"buffer in term statement"); }
    buffer1 = 0;
    AT.WorkPointer = oldworkpointer;
    return(0);
exectermerr:
    AT.WorkPointer = oldworkpointer;
    AR.DeferFlag = olddeferflag;
    MLOCK(ErrorMessageLock);
    MesCall("execterm");
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
    #] execterm :
    #[ ArgumentImplode :
*/

int ArgumentImplode(PHEAD WORD *term, WORD *thelist)
{
    GETBIDENTITY
    WORD *liststart, *liststop, *inlist;
    WORD *w, *t, *tend, *tstop, *tt, *ttstop, *ttt, ncount, i;
    int action = 0;
    liststop = thelist + thelist[1];
    liststart = thelist + 2;
    t = term;
    tend = t + *t;
    tstop = tend - ABS(tend[-1]);
    t++;
    while ( t < tstop ) {
        if ( *t >= FUNCTION ) {
            inlist = liststart;
            while ( inlist < liststop && *inlist != *t ) inlist += inlist[1];
            if ( inlist < liststop ) {
                tt = t; ttstop = t + t[1]; w = AT.WorkPointer;
                for ( i = 0; i < FUNHEAD; i++ ) *w++ = *tt++;
                while ( tt < ttstop ) {
                    ncount = 0;
                    if ( *tt == -SNUMBER && tt[1] == 0 ) {
                        ncount = 1; ttt = tt; tt += 2;
                        while ( tt < ttstop && *tt == -SNUMBER && tt[1] == 0 ) {
                            ncount++; tt += 2;
                        }
                    }
                    if ( ncount > 0 ) {
                        if ( tt < ttstop && *tt == -SNUMBER && ( tt[1] == 1 || tt[1] == -1 ) ) {
                            *w++ = -SNUMBER;
                            *w++ = (ncount+1) * tt[1];
                            tt += 2;
                            action = 1;
                        }
                        else if ( ( tt[0] == tt[ARGHEAD] + ARGHEAD )
                        && ( ABS(tt[tt[0]-1]) == 3 )
                        && ( tt[tt[0]-2] == 1 )
                        && ( tt[tt[0]-3] == 1 ) ) { /* Single term with coef +/- 1 */
                            i = *tt; NCOPY(w,tt,i)
                            w[-3] = ncount+1;
                            action = 1;
                        }
                        else if ( *tt == -SYMBOL ) {
                            *w++ = ARGHEAD+8;
                            *w++ = 0;
                            FILLARG(w)
                            *w++ = 8;
                            *w++ = SYMBOL;
                            *w++ = tt[1];
                            *w++ = 1;
                            *w++ = ncount+1; *w++ = 1; *w++ = 3;
                            tt += 2;
                            action = 1;
                        }
                        else if ( *tt <= -FUNCTION ) {
                            *w++ = ARGHEAD+FUNHEAD+4;
                            *w++ = 0;
                            FILLARG(w)
                            *w++ = -*tt++;
                            *w++ = FUNHEAD+4;
                            FILLFUN(w)
                            *w++ = ncount+1; *w++ = 1; *w++ = 3;
                            action = 1;
                        }
```

```
                        else {
                            while ( ttt < tt ) *w++ = *ttt++;
                            if ( tt < ttstop && *tt == -SNUMBER ) {
                                *w++ = *tt++; *w++ = *tt++;
                            }
                        }
                    }
                    else if ( *tt <= -FUNCTION ) {
                        *w++ = *tt++;
                    }
                    else if ( *tt < 0 ) {
                        *w++ = *tt++;
                        *w++ = *tt++;
                    }
                    else {
                        i = *tt; NCOPY(w,tt,i)
                    }
                }
                AT.WorkPointer[1] = w - AT.WorkPointer;
                while ( tt < tend ) *w++ = *tt++;
                ttt = AT.WorkPointer; tt = t;
                while ( ttt < w ) *tt++ = *ttt++;
                term[0] = tt - term;
                AT.WorkPointer = tt;
                tend = tt; tstop = tt - ABS(tt[-1]);
            }
        }
        t += t[1];
    }
    if ( action ) {
        if ( Normalize(BHEAD term) ) return(-1);
    }
    return(0);
}

/*
    #] ArgumentImplode :
    #[ ArgumentExplode :
*/

int ArgumentExplode(PHEAD WORD *term, WORD *thelist)
{
    GETBIDENTITY
    WORD *liststart, *liststop, *inlist, *old;
    WORD *w, *t, *tend, *tstop, *tt, *ttstop, *ttt, ncount, i;
    int action = 0;
    LONG x;
    liststop = thelist + thelist[1];
    liststart = thelist + 2;
    t = term;
    tend = t + *t;
    tstop = tend - ABS(tend[-1]);
    t++;
    while ( t < tstop ) {
        if ( *t >= FUNCTION ) {
            inlist = liststart;
            while ( inlist < liststop && *inlist != *t ) inlist += inlist[1];
            if ( inlist < liststop ) {
                tt = t; ttstop = t + t[1]; w = AT.WorkPointer;
                for ( i = 0; i < FUNHEAD; i++ ) *w++ = *tt++;
                while ( tt < ttstop ) {
                    if ( *tt == -SNUMBER && tt[1] != 0 ) {
                        if ( tt[1] < AM.MaxTer/((WORD)sizeof(WORD)*4)
                            && tt[1] > -(AM.MaxTer/((WORD)sizeof(WORD)*4))
                            && ( tt[1] > 1 || tt[1] < -1 ) ) {
                            ncount = ABS(tt[1]);
                            while ( ncount > 1 ) {
                                *w++ = -SNUMBER; *w++ = 0; ncount--;
                            }
                            *w++ = -SNUMBER;
                            if ( tt[1] < 0 ) *w++ = -1;
                            else             *w++ =  1;
                            tt += 2;
                            action = 1;
                        }
                        else {
                            *w++ = *tt++; *w++ = *tt++;
                        }
                    }
                    else if ( *tt <= -FUNCTION ) {
                        *w++ = *tt++;
                    }
                    else if ( *tt < 0 ) {
                        *w++ = *tt++;
                        *w++ = *tt++;
                    }
                    else if ( tt[0] == tt[ARGHEAD]+ARGHEAD ) {
```

```
                        ttt = tt + tt[0] - 1;
                        i = (ABS(ttt[0])-1)/2;
                        if ( i > 1 ) {
TooMany:
                            old = AN.currentTerm;
                            AN.currentTerm = term;
                            MesPrint("Too many arguments in output of ArgExplode");
                            MesPrint("Term = %t");
                            AN.currentTerm = old;
                            return(-1);
                        }
                        if ( ttt[-1] != 1 ) goto NoExplode;
                        x = ttt[-2];
                        if ( 2*x > (AT.WorkTop-w)-*term ) goto TooMany;
                        ncount = x - 1;
                        while ( ncount > 0 ) {
                            *w++ = -SNUMBER; *w++ = 0; ncount--;
                        }
                        ttt[-2] = 1;
                        i = *tt; NCOPY(w,tt,i)
                        action = 1;
                    }
                    else {
NoExplode:
                        i = *tt; NCOPY(w,tt,i)
                    }
                }
                AT.WorkPointer[1] = w - AT.WorkPointer;
                while ( tt < tend ) *w++ = *tt++;
                ttt = AT.WorkPointer; tt = t;
                while ( ttt < w ) *tt++ = *ttt++;
                term[0] = tt - term;
                AT.WorkPointer = tt;
                tend = tt; tstop = tt - ABS(tt[-1]);
            }
        }
        t += t[1];
    }
    if ( action ) {
        if ( Normalize(BHEAD term) ) return(-1);
    }
    return(0);
}

/*
    #] ArgumentExplode :
    #[ ArgFactorize :
*/
#define NEWORDER

int ArgFactorize(PHEAD WORD *argin, WORD *argout)
{
/*
    #[ step 0 : Declarations and initializations
*/
    WORD *argfree, *argextra, *argcopy, *t, *tstop, *a, *a1, *a2;
#ifdef NEWORDER
    WORD *tt;
#endif
    WORD startebuf = cbuf[AT.ebufnum].numrhs,oldword;
    WORD oldsorttype = AR.SortType, numargs;
    int error = 0, action = 0, i, ii, number, sign = 1;

    *argout = 0;
/*
    #] step 0 :
    #[ step 1 : Take care of ordering
*/
    AR.SortType = SORTHIGHFIRST;
    if ( oldsorttype != AR.SortType ) {
        NewSort(BHEAD0);
        oldword = argin[*argin]; argin[*argin] = 0;
        t = argin+ARGHEAD;
        while ( *t ) {
            tstop = t + *t;
            if ( AN.ncmod != 0 ) {
                if ( AN.ncmod != 1 || ( (WORD)AN.cmod[0] < 0 ) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Factorization modulus a number, greater than a WORD not implemented.");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                if ( Modulus(t) ) {
                    MLOCK(ErrorMessageLock);
                    MesCall("ArgFactorize");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
```

```
                if ( !*t) { t = tstop; continue; }
            }
            StoreTerm(BHEAD t);
            t = tstop;
        }
        /* par = 1, in case the arg has more than SubTermsInSmall terms */
        EndSort(BHEAD argin+ARGHEAD,1);
        argin[*argin] = oldword;
    }
/*
    #] step 1 :
    #[ step 2 : take out the 'content'.
*/
    argfree = TakeArgContent(BHEAD argin,argout);
    {
        a1 = argout;
        while ( *a1 ) {
            if ( a1[0] == -SNUMBER && ( a1[1] == 1 || a1[1] == -1 ) ) {
                if ( a1[1] == -1 ) { sign = -sign; a1[1] = 1; }
                if ( a1[2] ) {
                    a = t = a1+2; while ( *t ) NEXTARG(t);
                    i = t - a1-2;
                    t = a1; NCOPY(t,a,i);
                    *t = 0;
                    continue;
                }
                else {
                    a1[0] = 0;
                }
                break;
            }
            else if ( a1[0] == FUNHEAD+ARGHEAD+4 && a1[ARGHEAD] == FUNHEAD+4
            && a1[*a1-1] == 3 && a1[*a1-2] == 1 && a1[*a1-3] == 1
            && a1[ARGHEAD+1] >= FUNCTION ) {
                a = t = a1+*a1; while ( *t ) NEXTARG(t);
                i = t - a;
                *a1 = -a1[ARGHEAD+1]; t = a1+1; NCOPY(t,a,i);
                *t = 0;
            }
            NEXTARG(a1);
        }
    }
    if ( argfree == 0 ) {
        argfree = argin;
    }
    else if ( argfree[0] == ( argfree[ARGHEAD]+ARGHEAD ) ) {
        Normalize(BHEAD argfree+ARGHEAD);
        argfree[0] = argfree[ARGHEAD]+ARGHEAD;
        argfree[1] = 0;
        if ( ( argfree[0] == ARGHEAD+4 ) && ( argfree[ARGHEAD+3] == 3 )
            && ( argfree[ARGHEAD+1] == 1 ) && ( argfree[ARGHEAD+2] == 1 ) ) {
            goto return0;
        }
    }
    else {
/*
        The way we took out objects is rather brutish. We have to
        normalize
*/
        NewSort(BHEAD0);
        t = argfree+ARGHEAD;
        while ( *t ) {
            tstop = t + *t;
            Normalize(BHEAD t);
            StoreTerm(BHEAD t);
            t = tstop;
        }
        /* par = 1, in case the arg has more than SubTermsInSmall terms */
        EndSort(BHEAD argfree+ARGHEAD,1);
        t = argfree+ARGHEAD;
        while ( *t ) t += *t;
        *argfree = t - argfree;
    }
/*
    #] step 2 :
    #[ step 3 : look whether we have done this one already.
*/
    if ( ( number = FindArg(BHEAD argfree) ) != 0 ) {
        if ( number > 0 ) t = cbuf[AT.fbufnum].rhs[number-1];
        else              t = cbuf[AC.ffbufnum].rhs[-number-1];
/*
        Now position on the result. Remember we have in the cache:
                    inputarg,0,outputargs,0
        t is currently at inputarg. *inputarg is always positive.
        in principle this holds also for the arguments in the output
        but we take no risks here (in case of future developments).
*/
```

```
        t += *t; t++;
        tstop = t;
        ii = 0;
        while ( *tstop ) {
            if ( *tstop == -SNUMBER && tstop[1] == -1 ) {
                sign = -sign; ii += 2;
            }
            NEXTARG(tstop);
        }
        a = argout; while ( *a ) NEXTARG(a);
#ifndef NEWORDER
        if ( sign == -1 ) { *a++ = -SNUMBER; *a++ = -1; *a = 0; sign = 1; }
#endif
        i = tstop - t - ii;
        ii = a - argout;
        a2 = a; a1 = a + i;
        *a1 = 0;
        while ( ii > 0 ) { *--a1 = *--a2; ii--; }
        a = argout;
        while ( *t ) {
            if ( *t == -SNUMBER && t[1] == -1 ) { t += 2; }
            else { COPY1ARG(a,t) }
        }
        goto return0;
    }
/*
    #] step 3 :
    #[ step 4 : invoke ConvertToPoly

                We make a copy first in case there are no factors
*/
    argcopy = TermMalloc("argcopy");
    for ( i = 0; i <= *argfree; i++ ) argcopy[i] = argfree[i];

    tstop = argfree + *argfree;
    {
        WORD sumcommu = 0;
        t = argfree + ARGHEAD;
        while ( t < tstop ) {
            sumcommu += DoesCommu(t);
            t += *t;
        }
        if ( sumcommu > 1 ) {
            MesPrint("ERROR: Cannot factorize an argument with more than one noncommuting object");
            Terminate(-1);
        }
    }
    t = argfree + ARGHEAD;

    while ( t < tstop ) {
        if ( ( t[1] != SYMBOL ) && ( *t != (ABS(t[*t-1])+1) ) ) {
            action = 1; break;
        }
        t += *t;
    }
    if ( action ) {
        t = argfree + ARGHEAD;
        argextra = AT.WorkPointer;
        NewSort(BHEAD0);
        while ( t < tstop ) {
            if ( LocalConvertToPoly(BHEAD t,argextra,startebuf,0) < 0 ) {
                error = -1;
getout:
                AR.SortType = oldsorttype;
                TermFree(argcopy,"argcopy");
                if ( argfree != argin ) TermFree(argfree,"argfree");
                MesCall("ArgFactorize");
                Terminate(-1);
                return(-1);
            }
            StoreTerm(BHEAD argextra);
            t += *t; argextra += *argextra;
        }
        /* par = 1, in case the arg has more than SubTermsInSmall terms */
        if ( EndSort(BHEAD argfree+ARGHEAD,1) < 0 ) { error = -2; goto getout; }
        t = argfree + ARGHEAD;
        while ( *t > 0 ) t += *t;
        *argfree = t - argfree;
    }
/*
    #] step 4 :
    #[ step 5 : If not in the tables, we have to do this by hard work.
*/

    a = argout;
    while ( *a ) NEXTARG(a);
    if ( poly_factorize_argument(BHEAD argfree,a) < 0 ) {
```

```
        MesCall("ArgFactorize");
        error = -1;
    }
/*
    #] step 5 :
    #[ step 6 : use now ConvertFromPoly

            Be careful: there should be more than one argument now.
*/
    if ( error == 0 && action ) {
        a1 = a; NEXTARG(a1);
        if ( *a1 != 0 ) {
            CBUF *C = cbuf+AC.cbufnum;
            CBUF *CC = cbuf+AT.ebufnum;
            WORD *oldworkpointer = AT.WorkPointer;
            WORD *argcopy2 = TermMalloc("argcopy2"), *a1, *a2;
            a1 = a; a2 = argcopy2;
            while ( *a1 ) {
                if ( *a1 < 0 ) {
                    if ( *a1 > -FUNCTION ) *a2++ = *a1++;
                    *a2++ = *a1++; *a2 = 0;
                    continue;
                }
                t = a1 + ARGHEAD;
                tstop = a1 + *a1;
                argextra = AT.WorkPointer;
                NewSort(BHEAD0);
                while ( t < tstop ) {
                    if ( ConvertFromPoly(BHEAD t,argextra,numxsymbol,CC->numrhs-startebuf+numxsymbol
                    ,startebuf-numxsymbol,1) <= 0 ) {
                        TermFree(argcopy2,"argcopy2");
                        LowerSortLevel();
                        error = -3;
                        goto getout;
                    }
                    t += *t;
                    AT.WorkPointer = argextra + *argextra;
/*
                    ConvertFromPoly leaves terms with subexpressions. Hence:
*/
                    if ( Generator(BHEAD argextra,C->numlhs) ) {
                        TermFree(argcopy2,"argcopy2");
                        LowerSortLevel();
                        error = -4;
                        goto getout;
                    }
                }
                AT.WorkPointer = oldworkpointer;
                /* par = 1, in case the factor has more than SubTermsInSmall terms */
                if ( EndSort(BHEAD a2+ARGHEAD,1) < 0 ) { error = -5; goto getout; }
                t = a2+ARGHEAD; while ( *t ) t += *t;
                *a2 = t - a2; a2[1] = 0; ZEROARG(a2);
                ToFast(a2,a2); NEXTARG(a2);
                a1 = tstop;
            }
            i = a2 - argcopy2;
            a2 = argcopy2; a1 = a;
            NCOPY(a1,a2,i);
            *a1 = 0;
            TermFree(argcopy2,"argcopy2");
/*
            Erase the entries we made temporarily in cbuf[AT.ebufnum]
*/
            CC->numrhs = startebuf;
        }
        else {  /* no factorization. recover the argument from before step 3. */
            for ( i = 0; i <= *argcopy; i++ ) a[i] = argcopy[i];
        }
    }
/*
    #] step 6 :
    #[ step 7 : Add this one to the tables.

            Possibly drop some elements in the tables
            when they become too full.
*/
    if ( error == 0 && AN.ncmod == 0 ) {
        if ( InsertArg(BHEAD argcopy,a,0) < 0 ) { error = -1; }
    }
/*
    #] step 7 :
    #[ step 8 : Clean up and return.

        Change the order of the arguments in argout and a.
        Use argcopy as spare space.
*/
    ii = a - argout;
```

```
    for ( i = 0; i < ii; i++ ) argcopy[i] = argout[i];
    a1 = a;
    while ( *a1 ) {
        if ( *a1 == -SNUMBER && a1[1] < 0 ) {
            sign = -sign; a1[1] = -a1[1];
            if ( a1[1] == 1 ) {
                a2 = a1+2; while ( *a2 ) NEXTARG(a2);
                i = a2-a1-2; a2 = a1+2;
                NCOPY(a1,a2,i);
                *a1 = 0;
            }
            while ( *a1 ) NEXTARG(a1);
            break;
        }
        else {
            if ( *a1 > 0 && *a1 == a1[ARGHEAD]+ARGHEAD && a1[*a1-1] < 0 ) {
                a1[*a1-1] = -a1[*a1-1]; sign = -sign;
            }
            if ( *a1 == ARGHEAD+4 && a1[ARGHEAD+1] == 1 && a1[ARGHEAD+2] == 1 ) {
                a2 = a1+ARGHEAD+4; while ( *a2 ) NEXTARG(a2);
                i = a2-a1-ARGHEAD-4; a2 = a1+ARGHEAD+4;
                NCOPY(a1,a2,i);
                *a1 = 0;
                break;
            }
            while ( *a1 ) NEXTARG(a1);
            break;
        }
        NEXTARG(a1);
    }
    i = a1 - a;
    a2 = argout;
    NCOPY(a2,a,i);
    for ( i = 0; i < ii; i++ ) *a2++ = argcopy[i];
#ifndef NEWORDER
    if ( sign == -1 ) { *a2++ = -SNUMBER; *a2++ = -1; sign = 1; }
#endif
    *a2 = 0;
    TermFree(argcopy,"argcopy");
return0:
    if ( argfree != argin ) TermFree(argfree,"argfree");
    if ( oldsorttype != AR.SortType ) {
        AR.SortType = oldsorttype;
        a = argout;
        while ( *a ) {
            if ( *a > 0 ) {
                NewSort(BHEAD0);
                oldword = a[*a]; a[*a] = 0;
                t = a+ARGHEAD;
                while ( *t ) {
                    tstop = t + *t;
                    StoreTerm(BHEAD t);
                    t = tstop;
                }
                /* par = 1, in case the factor has more than SubTermsInSmall terms */
                EndSort(BHEAD a+ARGHEAD,1);
                a[*a] = oldword;
                a += *a;
            }
            else { NEXTARG(a); }
        }
    }
#ifdef NEWORDER
    t = argout; numargs = 0;
    while ( *t ) {
        tt = t;
        NEXTARG(t);
        if ( *tt == ABS(t[-1])+1+ARGHEAD && sign == -1 ) { t[-1] = -t[-1]; sign = 1; }
        else if ( *tt == -SNUMBER && sign == -1 ) { tt[1] = -tt[1]; sign = 1; }
        numargs++;
    }
    if ( sign == -1 ) {
        *t++ = -SNUMBER; *t++ = -1; *t = 0; sign = 1; numargs++;
    }
#else
/*
    Now we have to sort the arguments
    First have the number of 'nontrivial/nonnumerical' arguments
    Then make a piece of code like in FullSymmetrize with that number
    of arguments to be symmetrized.
    Put a function in front
    Call the Symmetrize routine
*/
    t = argout; numargs = 0;
    while ( *t && *t != -SNUMBER && ( *t < 0 || ( ABS(t[*t-1]) != *t-1 ) ) ) {
        NEXTARG(t);
        numargs++;
```

```
    }
#endif
    if ( numargs > 1 ) {
        WORD *Lijst;
        WORD x[3];
        x[0] = argout[-FUNHEAD];
        x[1] = argout[-FUNHEAD+1];
        x[2] = argout[-FUNHEAD+2];
        while ( *t ) { NEXTARG(t); }
        argout[-FUNHEAD] = SQRTFUNCTION;
        argout[-FUNHEAD+1] = t-argout+FUNHEAD;
        argout[-FUNHEAD+2] = 0;
        AT.WorkPointer = t+1;
        Lijst = AT.WorkPointer;
        for ( i = 0; i < numargs; i++ ) Lijst[i] = i;
        AT.WorkPointer += numargs;
        error = Symmetrize(BHEAD argout-FUNHEAD,Lijst,numargs,1,SYMMETRIC);
        AT.WorkPointer = Lijst;
        argout[-FUNHEAD] = x[0];
        argout[-FUNHEAD+1] = x[1];
        argout[-FUNHEAD+2] = x[2];
#ifdef NEWORDER
/*
        Now we have to get a potential numerical argument to the first position
*/
        tstop = argout; while ( *tstop ) { NEXTARG(tstop); }
        t = argout; number = 0;
        while ( *t ) {
            tt = t; NEXTARG(t);
            if ( *tt == -SNUMBER ) {
                if ( number == 0 ) break;
                x[0] = tt[1];
                while ( tt > argout ) { *--t = *--tt; }
                argout[0] = -SNUMBER; argout[1] = x[0];
                break;
            }
            else if ( *tt == ABS(t[-1])+1+ARGHEAD ) {
                if ( number == 0 ) break;
                ii = t - tt;
                for ( i = 0; i < ii; i++ ) tstop[i] = tt[i];
                while ( tt > argout ) { *--t = *--tt; }
                for ( i = 0; i < ii; i++ ) argout[i] = tstop[i];
                *tstop = 0;
                break;
            }
            number++;
        }
#endif
    }
/*
    #] step 8 :
*/
    return(error);
}

/*
    #] ArgFactorize :
    #[ FindArg :
*/
WORD FindArg(PHEAD WORD *a)
{
    int number;
    if ( AN.ncmod != 0 ) return(0); /* no room for mod stuff */
    number = FindTree(AT.fbufnum,a);
    if ( number >= 0 ) return(number+1);
    number = FindTree(AC.ffbufnum,a);
    if ( number >= 0 ) return(-number-1);
    return(0);
}

/*
    #] FindArg :
    #[ InsertArg :
*/
WORD InsertArg(PHEAD WORD *argin, WORD *argout,int par)
{
    CBUF *C;
    WORD *a, i, bufnum;
    if ( par == 0 ) {
        bufnum = AT.fbufnum;
        C = cbuf+bufnum;
        if ( C->numrhs >= (C->maxrhs-2) ) CleanupArgCache(BHEAD AT.fbufnum);
    }
    else if ( par == 1 ) {
        bufnum = AC.ffbufnum;
        C = cbuf+bufnum;
    }
```

```
    else { return(-1); }
    AddRHS(bufnum,1);
    AddNtoC(bufnum,*argin,argin,1);
    AddToCB(C,0)
    a = argout; while ( *a ) NEXTARG(a);
    i = a - argout;
    AddNtoC(bufnum,i,argout,2);
    AddToCB(C,0)
    return(InsTree(bufnum,C->numrhs));
}

/*
    #] InsertArg :
    #[ CleanupArgCache :
*/
int CleanupArgCache(PHEAD WORD bufnum)
{
    CBUF *C = cbuf + bufnum;
    COMPTREE *boomlijst = C->boomlijst;
    LONG *weights = (LONG *)Malloc1(2*(C->numrhs+1)*sizeof(LONG),"CleanupArgCache");
    LONG w, whalf, *extraweights;
    WORD *a, *to, *from;
    int i,j,k;
    for ( i = 1; i <= C->numrhs; i++ ) {
        weights[i] = ((LONG)i) * (boomlijst[i].usage);
    }
/*
        Now sort the weights and determine the halfway weight
*/
    extraweights = weights+C->numrhs+1;
    SortWeights(weights+1,extraweights,C->numrhs);
    whalf = weights[C->numrhs/2+1];
/*
        We should drop everybody with a weight < whalf.
*/
    to = C->Buffer;
    k = 1;
    for ( i = 1; i <= C->numrhs; i++ ) {
        from = C->rhs[i]; w = ((LONG)i) * (boomlijst[i].usage);
        if ( w >= whalf ) {
            if ( i < C->numrhs-1 ) {
                if ( to == from ) {
                    to = C->rhs[i+1];
                }
                else {
                    j = C->rhs[i+1] - from;
                    C->rhs[k] = to;
                    NCOPY(to,from,j)
                }
            }
            else if ( to == from ) {
                to += *to + 1; while ( *to ) NEXTARG(to); to++;
            }
            else {
                a = from; a += *a+1; while ( *a ) NEXTARG(a); a++;
                j = a - from;
                C->rhs[k] = to;
                NCOPY(to,from,j)
            }
            weights[k++] = boomlijst[i].usage;
        }
    }
    C->numrhs = --k;
    C->Pointer = to;
/*
        Next we need to rebuild the tree.
        Note that this can probably be done much faster by using the
        remains of the old tree !!!!!!!!!!!!!!!!
*/
    ClearTree(AT.fbufnum);
    for ( i = 1; i <= k; i++ ) {
        InsTree(AT.fbufnum,i);
        boomlijst[i].usage = weights[i];
    }
/*
        And cleanup
*/
    M_free(weights,"CleanupArgCache");
    return(0);
}

/*
    #] CleanupArgCache :
    #[ ArgSymbolMerge :
*/

int ArgSymbolMerge(WORD *t1, WORD *t2)
```

```
{
    WORD *t1e = t1+t1[1];
    WORD *t2e = t2+t2[1];
    WORD *t1a = t1+2;
    WORD *t2a = t2+2;
    WORD *t3;
    while ( t1a < t1e && t2a < t2e ) {
        if ( *t1a < *t2a ) {
            if ( t1a[1] >= 0 ) {
                t3 = t1a+2;
                while ( t3 < t1e ) { t3[-2] = *t3; t3[-1] = t3[1]; t3 += 2; }
                t1e -= 2;
            }
            else t1a += 2;
        }
        else if ( *t1a > *t2a ) {
            if ( t2a[1] >= 0 ) t2a += 2;
            else {
                t3 = t1e;
                while ( t3 > t1a ) { *t3 = t3[-2]; t3[1] = t3[-1]; t3 -= 2; }
                *t1a++ = *t2a++;
                *t1a++ = *t2a++;
                t1e += 2;
            }
        }
        else {
            if ( t2a[1] < t1a[1] ) t1a[1] = t2a[1];
            t1a += 2; t2a += 2;
        }
    }
    while ( t2a < t2e ) {
        if ( t2a[1] < 0 ) {
            *t1a++ = *t2a++;
            *t1a++ = *t2a++;
        }
        else t2a += 2;
    }
    while ( t1a < t1e ) {
        if ( t1a[1] >= 0 ) {
            t3 = t1a+2;
            while ( t3 < t1e ) { t3[-2] = *t3; t3[-1] = t3[1]; t3 += 2; }
            t1e -= 2;
        }
        else t1a += 2;
    }
    t1[1] = t1a - t1;
    return(0);
}

/*
    #] ArgSymbolMerge :
    #[ ArgDotproductMerge :
*/

int ArgDotproductMerge(WORD *t1, WORD *t2)
{
    WORD *t1e = t1+t1[1];
    WORD *t2e = t2+t2[1];
    WORD *t1a = t1+2;
    WORD *t2a = t2+2;
    WORD *t3;
    while ( t1a < t1e && t2a < t2e ) {
        if ( *t1a < *t2a || ( *t1a == *t2a && t1a[1] < t2a[1] ) ) {
            if ( t1a[2] >= 0 ) {
                t3 = t1a+3;
                while ( t3 < t1e ) { t3[-3] = *t3; t3[-2] = t3[1]; t3[-1] = t3[2]; t3 += 3; }
                t1e -= 3;
            }
            else t1a += 3;
        }
        else if ( *t1a > *t2a || ( *t1a == *t2a && t1a[1] > t2a[1] ) ) {
            if ( t2a[2] >= 0 ) t2a += 3;
            else {
                t3 = t1e;
                while ( t3 > t1a ) { *t3 = t3[-3]; t3[1] = t3[-2]; t3[2] = t3[-1]; t3 -= 3; }
                *t1a++ = *t2a++;
                *t1a++ = *t2a++;
                *t1a++ = *t2a++;
                t1e += 3;
            }
        }
        else {
            if ( t2a[2] < t1a[2] ) t1a[2] = t2a[2];
            t1a += 3; t2a += 3;
        }
    }
    while ( t2a < t2e ) {
```

```
        if ( t2a[2] < 0 ) {
            *t1a++ = *t2a++;
            *t1a++ = *t2a++;
            *t1a++ = *t2a++;
        }
        else t2a += 3;
    }
    while ( t1a < t1e ) {
        if ( t1a[2] >= 0 ) {
            t3 = t1a+3;
            while ( t3 < t1e ) { t3[-3] = *t3; t3[-2] = t3[1]; t3[-1] = t3[2]; t3 += 3; }
            t1e -= 3;
        }
        else t1a += 2;
    }
    t1[1] = t1a - t1;
    return(0);
}

/*
    #] ArgDotproductMerge :
    #[ TakeArgContent :
*/
WORD *TakeArgContent(PHEAD WORD *argin, WORD *argout)
{
    GETBIDENTITY
    WORD *t, *rnext, *r1, *r2, *r3, *r5, *r6, *r7, *r8, *r9;
    WORD pow, *mm, *mnext, *mstop, *argin2 = argin, *argin3 = argin, *argfree;
    WORD ncom;
    int j, i, act;
    r5 = t = argin + ARGHEAD;
    r3 = argin + *argin;
    rnext = t + *t;
    GETSTOP(t,r6);
    r1 = argout;
    t++;
/*
        First pass: arrange everything but the symbols and dotproducts.
        They need separate treatment because we have to take out negative
        powers.
*/
    while ( t < r6 ) {
/*
            #[ DELTA/VECTOR :
*/
        if ( *t == DELTA || *t == VECTOR ) {
            r7 = t; r8 = t + t[1]; t += 2;
            while ( t < r8 ) {
                mm = rnext;
                pow = 1;
                while ( mm < r3 ) {
                    mnext = mm + *mm;
                    GETSTOP(mm,mstop); mm++;
                    while ( mm < mstop ) {
                        if ( *mm != *r7 ) mm += mm[1];
                        else break;
                    }
                    if ( *mm == *r7 ) {
                        mstop = mm + mm[1]; mm += 2;
                        while ( ( *mm != *t || mm[1] != t[1] )
                            && mm < mstop ) mm += 2;
                        if ( mm >= mstop ) pow = 0;
                    }
                    else pow = 0;
                    if ( pow == 0 ) break;
                    mm = mnext;
                }
                if ( pow == 0 ) { t += 2; continue; }
/*
                We have a factor
*/
                *r1++ = 8 + ARGHEAD;
                for ( j = 1; j < ARGHEAD; j++ ) *r1++ = 0;
                *r1++ = 8; *r1++ = *r7;
                *r1++ = 4; *r1++ = *t; *r1++ = t[1];
                *r1++ = 1; *r1++ = 1; *r1++ = 3;
                argout = r1;
/*
                Now we have to remove the delta's/vectors
*/
                mm = rnext;
                while ( mm < r3 ) {
                    mnext = mm + *mm;
                    GETSTOP(mm,mstop); mm++;
                    while ( mm < mstop ) {
                        if ( *mm != *r7 ) mm += mm[1];
                        else break;
```

```
                    }
                    mstop = mm + mm[1]; mm += 2;
                    while ( mm < mstop && (
                     *mm != *t || mm[1] != t[1] ) ) mm += 2;
                    *mm = mm[1] = NOINDEX;
                    mm = mnext;
                }
                *t = t[1] = NOINDEX;
                t += 2;
            }
        }
/*
            #] DELTA/VECTOR :
            #[ INDEX :
*/
        else if ( *t == INDEX ) {
            r7 = t; r8 = t + t[1]; t += 2;
            while ( t < r8 ) {
                mm = rnext;
                pow = 1;
                while ( mm < r3 ) {
                    mnext = mm + *mm;
                    GETSTOP(mm,mstop); mm++;
                    while ( mm < mstop ) {
                        if ( *mm != *r7 ) mm += mm[1];
                        else break;
                    }
                    if ( *mm == *r7 ) {
                        mstop = mm + mm[1]; mm += 2;
                        while ( *mm != *t
                            && mm < mstop ) mm++;
                        if ( mm >= mstop ) pow = 0;
                    }
                    else pow = 0;
                    if ( pow == 0 ) break;
                    mm = mnext;
                }
                if ( pow == 0 ) { t++; continue; }
/*
                We have a factor
*/
                if ( *t < 0 ) { *r1++ = -VECTOR; }
                else          { *r1++ = -INDEX; }
                *r1++ = *t;
                argout = r1;
/*
                Now we have to remove the index
*/
                *t = NOINDEX;
                mm = rnext;
                while ( mm < r3 ) {
                    mnext = mm + *mm;
                    GETSTOP(mm,mstop); mm++;
                    while ( mm < mstop ) {
                        if ( *mm != *r7 ) mm += mm[1];
                        else break;
                    }
                    mstop = mm + mm[1]; mm += 2;
                    while ( mm < mstop &&
                     *mm != *t ) mm += 1;
                    *mm = NOINDEX;
                    mm = mnext;
                }
                t += 1;
            }
        }
/*
            #] INDEX :
            #[ FUNCTION :
*/
        else if ( *t >= FUNCTION ) {
/*
            In the next code we should actually look inside
            the DENOMINATOR or EXPONENT for noncommuting objects
*/
            if ( *t >= FUNCTION &&
                functions[*t-FUNCTION].commute == 0 ) ncom = 0;
            else ncom = 1;
            if ( ncom ) {
                mm = r5 + 1;
                while ( mm < t && ( *mm == DUMMYFUN
                || *mm == DUMMYTEN ) ) mm += mm[1];
                if ( mm < t ) { t += t[1]; continue; }
            }
            mm = rnext; pow = 1;
            while ( mm < r3 ) {
                mnext = mm + *mm;
```

```
                GETSTOP(mm,mstop); mm++;
                while ( mm < mstop ) {
                    if ( *mm == *t && mm[1] == t[1] ) {
                        for ( i = 2; i < t[1]; i++ ) {
                            if ( mm[i] != t[i] ) break;
                        }
                        if ( i >= t[1] )
                            { mm += mm[1]; goto nextmterm; }
                    }
                    if ( ncom && *mm != DUMMYFUN && *mm != DUMMYTEN )
                        { pow = 0; break; }
                    mm += mm[1];
                }
                if ( mm >= mstop ) pow = 0;
                if ( pow == 0 ) break;
nextmterm:      mm = mnext;
            }
            if ( pow == 0 ) { t += t[1]; continue; }
/*
            Copy the function
*/
            *r1++ = t[1] + 4 + ARGHEAD;
            for ( i = 1; i < ARGHEAD; i++ ) *r1++ = 0;
            *r1++ = t[1] + 4;
            for ( i = 0; i < t[1]; i++ ) *r1++ = t[i];
            *r1++ = 1; *r1++ = 1; *r1++ = 3;
            argout = r1;
/*
            Now we have to take out the functions
*/
            mm = rnext;
            while ( mm < r3 ) {
                mnext = mm + *mm;
                GETSTOP(mm,mstop); mm++;
                while ( mm < mstop ) {
                    if ( *mm == *t && mm[1] == t[1] ) {
                        for ( i = 2; i < t[1]; i++ ) {
                            if ( mm[i] != t[i] ) break;
                        }
                        if ( i >= t[1] ) {
                            if ( functions[*t-FUNCTION].spec > 0 )
                                *mm = DUMMYTEN;
                            else
                                *mm = DUMMYFUN;
                            mm += mm[1];
                            goto nextterm;
                        }
                    }
                    mm += mm[1];
                }
nextterm:                       mm = mnext;
            }
            if ( functions[*t-FUNCTION].spec > 0 )
                    *t = DUMMYTEN;
            else
                    *t = DUMMYFUN;
            t += t[1];
        }
/*
            #] FUNCTION :
*/
        else {
            t += t[1];
        }
    }
/*
            #[ SYMBOL :

        Now collect all symbols. We can use the space after r1 as storage
*/
    t = argin+ARGHEAD;
    rnext = t + *t;
    r2 = r1;
    while ( t < r3 ) {
        GETSTOP(t,r6);
        t++;
        act = 0;
        while ( t < r6 ) {
            if ( *t == SYMBOL ) {
                act = 1;
                i = t[1];
                NCOPY(r2,t,i)
            }
            else { t += t[1]; }
        }
        if ( act == 0 ) {
            *r2++ = SYMBOL; *r2++ = 2;
```

```
        }
        t = rnext; rnext = rnext + *rnext;
    }
    *r2 = 0;
    argin2 = argin;
/*
        Now we have a list of all symbols as a sequence of SYMBOL subterms.
        Any symbol that is absent in a subterm has power zero.
        We now need a list of all minimum powers.
        This can be done by subsequent merges.
*/
    r7 = r1;            /* The first object into which we merge. */
    r8 = r7 + r7[1];    /* The object that gets merged into r7.  */
    while ( *r8 ) {
        r2 = r8 + r8[1]; /* Next object */
        ArgSymbolMerge(r7,r8);
        r8 = r2;
    }
/*
        Now we have to divide by the object in r7 and take it apart as factors.
        The division can be simple if there are no negative powers.
*/
    if ( r7[1] > 2 ) {
        r8 = r7+2;
        r2 = r7 + r7[1];
        act = 0;
        pow = 0;
        while ( r8 < r2 ) {
            if ( r8[1] < 0 ) { act = 1; pow += -r8[1]*(ARGHEAD+8); }
            else { pow += 2*r8[1]; }
            r8 += 2;
        }
/*
        The amount of space we need to move r7 is given in pow
*/
        if ( act == 0 ) {   /* this can be done 'in situ' */
            t = argin + ARGHEAD;
            while ( t < r3 ) {
                rnext = t + *t;
                GETSTOP(t,r6);
                t++;
                while ( t < r6 ) {
                    if ( *t != SYMBOL ) { t += t[1]; continue; }
                    r8 = r7+2; r9 = t + t[1]; t += 2;
                    while ( ( t < r9 ) && ( r8 < r2 ) ) {
                        if ( *t == *r8 ) {
                            t[1] -= r8[1]; t += 2; r8 += 2;
                        }
                        else { /* *t must be < than *r8 !!! */
                            t += 2;
                        }
                    }
                    t = r9;
                }
                t = rnext;
            }
/*
            And now the factors that go to argout.
            First we have to move r7 out of the way.
*/
            r8 = r7+pow; i = r7[1];
            while ( --i >= 0 ) r8[i] = r7[i];
            r2 += pow;
            r8 += 2;
            while ( r8 < r2 ) {
                for ( i = 0; i < r8[1]; i++ ) { *r1++ = -SYMBOL; *r1++ = *r8; }
                r8 += 2;
            }
        }
        else {  /* this needs a new location */
            argin2 = TermMalloc("TakeArgContent2");
/*
            We have to multiply the inverse of r7 into argin
            The answer should go to argin2.
*/
            r5 = argin2; *r5++ = 0; *r5++ = 0; FILLARG(r5);
            t = argin+ARGHEAD;
            while ( t < r3 ) {
                rnext = t + *t;
                GETSTOP(t,r6);
                r9 = r5;
                *r5++ = *t++ + r7[1];
                while ( t < r6 ) *r5++ = *t++;
                i = r7[1] - 2; r8 = r7+2;
                *r5++ = r7[0]; *r5++ = r7[1];
                while ( i > 0 ) { *r5++ = *r8++; *r5++ = -*r8++; i -= 2; }
                while ( t < rnext ) *r5++ = *t++;
```

```
                Normalize(BHEAD r9);
                r5 = r9 + *r9;
            }
            *r5 = 0;
            *argin2 = r5-argin2;
/*
            We may have to sort the terms in argin2.
*/
            NewSort(BHEAD0);
            t = argin2+ARGHEAD;
            while ( *t ) {
                StoreTerm(BHEAD t);
                t += *t;
            }
            t = argin2+ARGHEAD;
            if ( EndSort(BHEAD t,0) < 0 ) goto Irreg;
            while ( *t ) t += *t;
            *argin2 = t - argin2;
            r3 = t;
/*
            And now the factors that go to argout.
            First we have to move r7 out of the way.
*/
            r8 = r7+pow; i = r7[1];
            while ( --i >= 0 ) r8[i] = r7[i];
            r2 += pow;
            r8 += 2;
            while ( r8 < r2 ) {
                if ( r8[1] >= 0 ) {
                    for ( i = 0; i < r8[1]; i++ ) { *r1++ = -SYMBOL; *r1++ = *r8; }
                }
                else {
                    for ( i = 0; i < -r8[1]; i++ ) {
                        *r1++ = ARGHEAD+8; *r1++ = 0;
                        FILLARG(r1);
                        *r1++ = 8; *r1++ = SYMBOL; *r1++ = 4; *r1++ = *r8;
                        *r1++ = -1; *r1++ = 1; *r1++ = 1; *r1++ = 3;
                    }
                }
                r8 += 2;
            }
            argout = r1;
        }
    }
/*
            #] SYMBOL :
            #[ DOTPRODUCT :

        Now collect all dotproducts. We can use the space after r1 as storage
*/
      t = argin2+ARGHEAD;
      rnext = t + *t;
      r2 = r1;
      while ( t < r3 ) {
        GETSTOP(t,r6);
        t++;
        act = 0;
        while ( t < r6 ) {
            if ( *t == DOTPRODUCT ) {
                act = 1;
                i = t[1];
                NCOPY(r2,t,i)
            }
            else { t += t[1]; }
        }
        if ( act == 0 ) {
            *r2++ = DOTPRODUCT; *r2++ = 2;
        }
        t = rnext; rnext = rnext + *rnext;
      }
      *r2 = 0;
      argin3 = argin2;
/*
        Now we have a list of all dotproducts as a sequence of DOTPRODUCT
        subterms. Any dotproduct that is absent in a subterm has power zero.
        We now need a list of all minimum powers.
        This can be done by subsequent merges.
*/
      r7 = r1;          /* The first object into which we merge. */
      r8 = r7 + r7[1];  /* The object that gets merged into r7.  */
      while ( *r8 ) {
        r2 = r8 + r8[1]; /* Next object */
        ArgDotproductMerge(r7,r8);
        r8 = r2;
      }
/*
        Now we have to divide by the object in r7 and take it apart as factors.
```

```
        The division can be simple if there are no negative powers.
*/
      if ( r7[1] > 2 ) {
        r8 = r7+2;
        r2 = r7 + r7[1];
        act = 0;
        pow = 0;
        while ( r8 < r2 ) {
            if ( r8[2] < 0 ) { pow += -r8[2]*(ARGHEAD+9); }
            else             { pow +=  r8[2]*(ARGHEAD+9); }
            r8 += 3;
        }
/*
        The amount of space we need to move r7 is given in pow
        For dotproducts we always need a new location
*/
        {
            argin3 = TermMalloc("TakeArgContent3");
/*
            We have to multiply the inverse of r7 into argin
            The answer should go to argin2.
*/
            r5 = argin3; *r5++ = 0; *r5++ = 0; FILLARG(r5);
            t = argin2+ARGHEAD;
            while ( t < r3 ) {
                rnext = t + *t;
                GETSTOP(t,r6);
                r9 = r5;
                *r5++ = *t++ + r7[1];
                while ( t < r6 ) *r5++ = *t++;
                i = r7[1] - 2; r8 = r7+2;
                *r5++ = r7[0]; *r5++ = r7[1];
                while ( i > 0 ) { *r5++ = *r8++; *r5++ = *r8++; *r5++ = -*r8++; i -= 3; }
                while ( t < rnext ) *r5++ = *t++;
                Normalize(BHEAD r9);
                r5 = r9 + *r9;
            }
            *r5 = 0;
            *argin3 = r5-argin3;
/*
            We may have to sort the terms in argin3.
*/
            NewSort(BHEAD0);
            t = argin3+ARGHEAD;
            while ( *t ) {
                StoreTerm(BHEAD t);
                t += *t;
            }
            t = argin3+ARGHEAD;
            if ( EndSort(BHEAD t,0) < 0 ) goto Irreg;
            while ( *t ) t += *t;
            *argin3 = t - argin3;
            r3 = t;
/*
            And now the factors that go to argout.
            First we have to move r7 out of the way.
*/
            r8 = r7+pow; i = r7[1];
            while ( --i >= 0 ) r8[i] = r7[i];
            r2 += pow;
            r8 += 2;
            while ( r8 < r2 ) {
                for ( i = ABS(r8[2]); i > 0; i-- ) {
                    *r1++ = ARGHEAD+9; *r1++ = 0; FILLARG(r1);
                    *r1++ = 9; *r1++ = DOTPRODUCT; *r1++ = 5; *r1++ = *r8;
                    *r1++ = r8[1]; *r1++ = r8[2] < 0 ? -1: 1;
                    *r1++ = 1; *r1++ = 1; *r1++ = 3;
                }
                r8 += 3;
            }
            argout = r1;
        }
      }
/*
            #] DOTPRODUCT :

    We have now in argin3 the argument stripped of negative powers and
    common factors. The only thing left to deal with is to make the
    coefficients integer. For that we have to find the LCM of the denominators
    and the GCD of the numerators. And to start with, the sign.
    We force the sign of the first term to be positive.
*/
    t = argin3 + ARGHEAD; pow = 1;
    t += *t;
    if ( t[-1] < 0 ) {
        pow = 0;
        t[-1] = -t[-1];
```

```
        while ( t < r3 ) {
            t += *t; t[-1] = -t[-1];
        }
    }
/*
    Now the GCD of the numerators and the LCM of the denominators:
*/
    argfree = TermMalloc("TakeArgContent1");
    if ( AN.cmod != 0 ) {
        r1 = MakeMod(BHEAD argin3,r1,argfree);
    }
    else {
        r1 = MakeInteger(BHEAD argin3,r1,argfree);
    }
    if ( pow == 0 ) {
        *r1++ = -SNUMBER;
        *r1++ = -1;
    }
    *r1 = 0;
/*
    Cleanup
*/
    if ( argin3 != argin2 ) TermFree(argin3,"TakeArgContent3");
    if ( argin2 != argin  ) TermFree(argin2,"TakeArgContent2");
    return(argfree);
Irreg:
    MesPrint("Irregularity while sorting argument in TakeArgContent");
    if ( argin3 != argin2 ) TermFree(argin3,"TakeArgContent3");
    if ( argin2 != argin  ) TermFree(argin2,"TakeArgContent2");
    Terminate(-1);
    return(0);
}

/*
    #] TakeArgContent :
    #[ MakeInteger :
*/
WORD *MakeInteger(PHEAD WORD *argin,WORD *argout,WORD *argfree)
{
    UWORD *GCDbuffer, *GCDbuffer2, *LCMbuffer, *LCMb, *LCMc;
    WORD *r, *r1, *r2, *r3, *r4, *r5, *rnext, i, k, j;
    WORD kGCD, kLCM, kGCD2, kkLCM, jLCM, jGCD;
    GCDbuffer = NumberMalloc("MakeInteger");
    GCDbuffer2 = NumberMalloc("MakeInteger");
    LCMbuffer = NumberMalloc("MakeInteger");
    LCMb = NumberMalloc("MakeInteger");
    LCMc = NumberMalloc("MakeInteger");
    r4 = argin + *argin;
    r = argin + ARGHEAD;
/*
    First take the first term to load up the LCM and the GCD
*/
    r2 = r + *r;
    j = r2[-1];
    r3 = r2 - ABS(j);
    k = REDLENG(j);
    if ( k < 0 ) k = -k;
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kGCD = 0; kGCD < k; kGCD++ ) GCDbuffer[kGCD] = r3[kGCD];
    k = REDLENG(j);
    if ( k < 0 ) k = -k;
    r3 += k;
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
    r1 = r2;
/*
    Now go through the rest of the terms in this argument.
*/
    while ( r1 < r4 ) {
        r2 = r1 + *r1;
        j = r2[-1];
        r3 = r2 - ABS(j);
        k = REDLENG(j);
        if ( k < 0 ) k = -k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( GCDbuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
/*
            GCD is already 1
*/
        }
        else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
            if ( GcdLong(BHEAD GCDbuffer,kGCD,(UWORD *)r3,k,GCDbuffer2,&kGCD2) ) {
                NumberFree(GCDbuffer,"MakeInteger");
                NumberFree(GCDbuffer2,"MakeInteger");
                NumberFree(LCMbuffer,"MakeInteger");
                NumberFree(LCMb,"MakeInteger"); NumberFree(LCMc,"MakeInteger");
                goto MakeIntegerErr;
```

```
            }
            kGCD = kGCD2;
            for ( i = 0; i < kGCD; i++ ) GCDbuffer[i] = GCDbuffer2[i];
        }
        else {
            kGCD = 1; GCDbuffer[0] = 1;
        }
        k = REDLENG(j);
        if ( k < 0 ) k = -k;
        r3 += k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
            for ( kLCM = 0; kLCM < k; kLCM++ )
                LCMbuffer[kLCM] = r3[kLCM];
        }
        else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
            if ( GcdLong(BHEAD LCMbuffer,kLCM,(UWORD *)r3,k,LCMb,&kkLCM) ) {
                NumberFree(GCDbuffer,"MakeInteger"); NumberFree(GCDbuffer2,"MakeInteger");
                NumberFree(LCMbuffer,"MakeInteger"); NumberFree(LCMb,"MakeInteger");
NumberFree(LCMc,"MakeInteger");
                goto MakeIntegerErr;
            }
            DivLong((UWORD *)r3,k,LCMb,kkLCM,LCMb,&kkLCM,LCMc,&jLCM);
            MulLong(LCMbuffer,kLCM,LCMb,kkLCM,LCMc,&jLCM);
            for ( kLCM = 0; kLCM < jLCM; kLCM++ )
                LCMbuffer[kLCM] = LCMc[kLCM];
        }
        else {} /* LCM doesn't change */
        r1 = r2;
    }
/*
    Now put the factor together: GCD/LCM
*/
    r3 = (WORD *)(GCDbuffer);
    if ( kGCD == kLCM ) {
        for ( jGCD = 0; jGCD < kGCD; jGCD++ )
            r3[jGCD+kGCD] = LCMbuffer[jGCD];
        k = kGCD;
    }
    else if ( kGCD > kLCM ) {
        for ( jGCD = 0; jGCD < kLCM; jGCD++ )
            r3[jGCD+kGCD] = LCMbuffer[jGCD];
        for ( jGCD = kLCM; jGCD < kGCD; jGCD++ )
            r3[jGCD+kGCD] = 0;
        k = kGCD;
    }
    else {
        for ( jGCD = kGCD; jGCD < kLCM; jGCD++ )
            r3[jGCD] = 0;
        for ( jGCD = 0; jGCD < kLCM; jGCD++ )
            r3[jGCD+kLCM] = LCMbuffer[jGCD];
        k = kLCM;
    }
    j = 2*k+1;
/*
    Now we have to write this to argout
*/
    if ( ( j == 3 ) && ( r3[1] == 1 ) && ( (WORD)(r3[0]) > 0 ) ) {
        *argout = -SNUMBER;
        argout[1] = r3[0];
        r1 = argout+2;
    }
    else {
        r1 = argout;
        *r1++ = j+1+ARGHEAD; *r1++ = 0; FILLARG(r1);
        *r1++ = j+1; r2 = r3;
        for ( i = 0; i < k; i++ ) { *r1++ = *r2++; *r1++ = *r2++; }
        *r1++ = j;
    }
/*
    Next we have to take the factor out from the argument.
    This cannot be done in location, because the denominator stuff can make
    coefficients longer.
*/
    r2 = argfree + 2; FILLARG(r2)
    while ( r < r4 ) {
        rnext = r + *r;
        j = ABS(rnext[-1]);
        r5 = rnext - j;
        r3 = r2;
        while ( r < r5 ) *r2++ = *r++;
        j = (j-1)/2;    /* reduced length. Remember, k is the other red length */
        if ( DivRat(BHEAD (UWORD *)r5,j,GCDbuffer,k,(UWORD *)r2,&i) ) {
            goto MakeIntegerErr;
        }
        i = 2*i+1;
        r2 = r2 + i;
```

```
        if ( rnext[-1] < 0 ) r2[-1] = -i;
        else                 r2[-1] =  i;
        *r3 = r2-r3;
        r = rnext;
    }
    *r2 = 0;
    argfree[0] = r2-argfree;
    argfree[1] = 0;
/*
    Cleanup
*/
    NumberFree(LCMc,"MakeInteger");
    NumberFree(LCMb,"MakeInteger");
    NumberFree(LCMbuffer,"MakeInteger");
    NumberFree(GCDbuffer2,"MakeInteger");
    NumberFree(GCDbuffer,"MakeInteger");
    return(r1);

MakeIntegerErr:
    MesCall("MakeInteger");
    Terminate(-1);
    return(0);
}

/*
    #] MakeInteger :
    #[ MakeMod :
*/
WORD *MakeMod(PHEAD WORD *argin,WORD *argout,WORD *argfree)
{
    WORD *r, *instop, *r1, *m, x, xx, ix, ip;
    int i;
    r = argin; instop = r + *r; r += ARGHEAD;
    x = r[*r-3];
    if ( r[*r-1] < 0 ) x += AN.cmod[0];
    if ( GetModInverses(x,(WORD)(AN.cmod[0]),&ix,&ip) ) {
        Terminate(-1);
    }
    argout[0] = -SNUMBER;
    argout[1] = x;
    argout[2] = 0;
    r1 = argout+2;
/*
    Now we have to multiply all coefficients by ix.
    This does not make things longer, but we should keep to the conventions
    of MakeInteger.
*/
    m = argfree + ARGHEAD;
    while ( r < instop ) {
        xx = r[*r-3]; if ( r[*r-1] < 0 ) xx += AN.cmod[0];
        xx = (WORD)((((LONG)xx)*ix) % AN.cmod[0]);
        if ( xx != 0 ) {
            i = *r; NCOPY(m,r,i);
            m[-3] = xx; m[-1] = 3;
        }
        else { r += *r; }
    }
    *m = 0;
    *argfree = m - argfree;
    argfree[1] = 0;
    argfree += 2; FILLARG(argfree);
    return(r1);
}

/*
    #] MakeMod :
    #[ SortWeights :
*/
void SortWeights(LONG *weights,LONG *extraspace,WORD number)
{
    LONG w, *fill, *from1, *from2;
    int n1,n2,i;
    if ( number >= 4 ) {
        n1 = number/2; n2 = number - n1;
        SortWeights(weights,extraspace,n1);
        SortWeights(weights+n1,extraspace,n2);
/*
        We copy the first patch to the extra space. Then we merge
        Note that a potential remaining n2 objects are already in place.
*/
        for ( i = 0; i < n1; i++ ) extraspace[i] = weights[i];
        fill = weights; from1 = extraspace; from2 = weights+n1;
        while ( n1 > 0 && n2 > 0 ) {
            if ( *from1 <= *from2 ) { *fill++ = *from1++; n1--; }
            else                    { *fill++ = *from2++; n2--; }
        }
        while ( n1 > 0 ) { *fill++ = *from1++; n1--; }
```

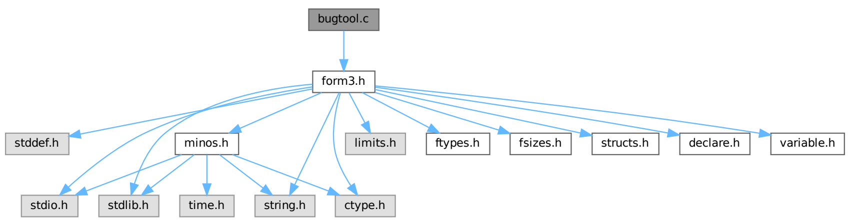
```
    }
/*
    Special cases
*/
    else if ( number == 3 ) { /* 6 permutations of which one is trivial */
        if ( weights[0] > weights[1] ) {
            if ( weights[1] > weights[2] ) {
                w = weights[0]; weights[0] = weights[2]; weights[2] = w;
            }
            else if ( weights[0] > weights[2] ) {
                w = weights[0]; weights[0] = weights[1];
                weights[1] = weights[2]; weights[2] = w;
            }
            else {
                w = weights[0]; weights[0] = weights[1]; weights[1] = w;
            }
        }
        else if ( weights[0] > weights[2] ) {
            w = weights[0]; weights[0] = weights[2];
            weights[2] = weights[1]; weights[1] = w;
        }
        else if ( weights[1] > weights[2] ) {
            w = weights[1]; weights[1] = weights[2]; weights[2] = w;
        }
    }
    else if ( number == 2 ) {
        if ( weights[0] > weights[1] ) {
            w = weights[0]; weights[0] = weights[1]; weights[1] = w;
        }
    }
    return;
}

/*
    #] SortWeights :
*/
```

4.3 bugtool.c File Reference

`#include "form3.h"`

Include dependency graph for bugtool.c:



Functions

- void ExprStatus (EXPRESSIONS e)

4.3.1 Detailed Description

Low level routines for debugging

Definition in file bugtool.c.

4.3.2 Function Documentation

4.3.2.1 ExprStatus()

```
void ExprStatus (
            EXPRESSIONS e )
```

Definition at line 66 of file bugtool.c.

4.4 bugtool.c

Go to the documentation of this file.

```
00001
00006 /* #[ License : */
00007 /*
00008  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009  *   When using this file you are requested to refer to the publication
00010  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011  *   This is considered a matter of courtesy as the development was paid
00012  *   for by FOM the Dutch physics granting agency and we would like to
00013  *   be able to track its scientific use to convince FOM of its value
00014  *   for the community.
00015  *
00016  *   This file is part of FORM.
00017  *
00018  *   FORM is free software: you can redistribute it and/or modify it under the
00019  *   terms of the GNU General Public License as published by the Free Software
00020  *   Foundation, either version 3 of the License, or (at your option) any later
00021  *   version.
00022  *
00023  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025  *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
00026  *   details.
00027  *
00028  *   You should have received a copy of the GNU General Public License along
00029  *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00030  */
00031 /* #] License : */
00032
00033 /*
00034     #[ Includes :
00035 */
00036
00037 #include "form3.h"
00038
00039 /*
00040     #] Includes :
00041     #[ ExprStatus :
00042 */
00043
00044 static UBYTE *statusexpr[] = {
00045      (UBYTE *)"LOCALEXPRESSION"
00046     ,(UBYTE *)"SKIPLEXPRESSION"
00047     ,(UBYTE *)"DROPLEXPRESSION"
00048     ,(UBYTE *)"DROPPEDEXPRESSION"
00049     ,(UBYTE *)"GLOBALEXPRESSION"
00050     ,(UBYTE *)"SKIPGEXPRESSION"
00051     ,(UBYTE *)"DROPGEXPRESSION"
00052     ,(UBYTE *)"UNKNOWN"
00053     ,(UBYTE *)"STOREDEXPRESSION"
00054     ,(UBYTE *)"HIDDENLEXPRESSION"
00055     ,(UBYTE *)"HIDELEXPRESSION"
00056     ,(UBYTE *)"DROPHLEXPRESSION"
00057     ,(UBYTE *)"UNHIDELEXPRESSION"
00058     ,(UBYTE *)"HIDDENGEXPRESSION"
00059     ,(UBYTE *)"HIDEGEXPRESSION"
00060     ,(UBYTE *)"DROPHGEXPRESSION"
00061     ,(UBYTE *)"UNHIDEGEXPRESSION"
00062     ,(UBYTE *)"INTOHIDELEXPRESSION"
00063     ,(UBYTE *)"INTOHIDEGEXPRESSION"
00064 };
00065
00066 void ExprStatus(EXPRESSIONS e)
00067 {
00068     MesPrint("Expression %s(%d) has status %s(%d,%d). Buffer: %d, Position: %15p",
```

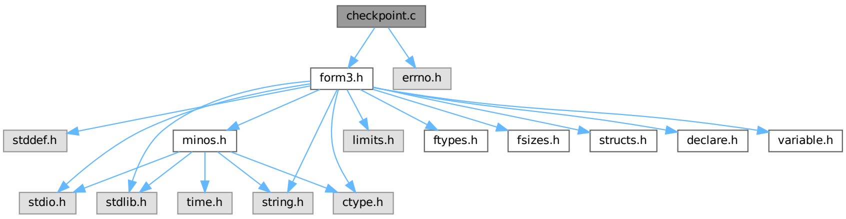
```
        AC.exprnames->namebuffer+e->name,(WORD)(e-Expressions),
        statusexpr[e->status],e->status,e->hidelevel,
        e->whichbuffer,&(e->onfile));
}

/*
    #] ExprStatus :
*/
```

4.5 checkpoint.c File Reference

```
#include "form3.h"
#include <errno.h>
```

Include dependency graph for checkpoint.c:



Macros

- #define CACHED_SNAPSHOT
- #define CACHE_SIZE 4096
- #define R_FREE(ARG) if (ARG) M_free(ARG, #ARG);
- #define R_FREE_NAMETREE(ARG)
- #define R_FREE_STREAM(ARG)
- #define R_SET(VAR, TYPE) VAR = ∗((TYPE∗)p); p = (unsigned char∗)p + sizeof(TYPE);
- #define R_COPY_B(VAR, SIZE, CAST)
- #define S_WRITE_B(BUF, LEN) if (fwrite_cached(BUF, 1, LEN, fd) != (size_t)(LEN)) return(__LINE__);
- #define S_FLUSH_B if (flush_cache(fd) != 1) return(__LINE__);
- #define R_COPY_S(VAR, CAST)
- #define S_WRITE_S(STR)
- #define R_COPY_LIST(ARG)
- #define S_WRITE_LIST(LST)
- #define R_COPY_NAMETREE(ARG)
- #define S_WRITE_NAMETREE(ARG)
- #define S_WRITE_DOLLAR(ARG)
- #define ANNOUNCE(str)

Functions

- int CheckRecoveryFile (void)
- void DeleteRecoveryFile (void)
- char ∗ RecoveryFilename (void)
- void InitRecovery (void)
- size_t fwrite_cached (const void ∗ptr, size_t size, size_t nmemb, FILE ∗fd)
- size_t flush_cache (FILE ∗fd)
- int DoRecovery (int ∗moduletype)
- void DoCheckpoint (int moduletype)

Variables

- unsigned char cache_buffer [CACHE_SIZE]
- size_t cache_fill = 0

4.5.1 Detailed Description

Contains all functions that deal with the recovery mechanism controlled and activated by the On Checkpoint switch.

The main function are DoCheckpoint, DoRecovery, and DoSnapshot. If the checkpoints are activated DoCheckpoint is called every time a module is finished executing. If the conditions for the creation of a recovery snapshot are met DoCheckpoint calls DoSnapshot. DoRecovery is called once when FORM starts up with the command line argument -R. Most of the other code contains debugging facilities that are only compiled if the macro PRINTDEBUG is defined.

The recovery mechanism is atomic, i.e. only if everything went well, the final recovery file is created (and the older one overwritten) in a single step (copying). If some errors occur, a warning is issued and the program continues without having created a new recovery file. The only situation in which the creation of the recovery data leads to a termination of the running program is if not enough disk or memory space is left.

For ParFORM each slave creates its own recovery file, sends it to the master and then it deletes the recovery file. The master stores all the recovery files and on recovery it feeds these files to the slaves. It is nearly impossible to recover after some MPI fault so ParFORM terminates on any recovery failure.

DoRecovery and DoSnapshot do the loading and saving of the recovery data, respectively. Every change in one functions needs to be accompanied by the appropriate change in the other function. The structure of both functions is quite similar. They handle the relevant global structs one after the other and then care about the copying of the hide and scratch files.

The names of the recovery, scratch and hide files are hard-coded in the variables in fold "filenames and system commands".

If the global structs AM,AP,AC,AR are changed, DoRecovery and DoSnapshot usually also have to be changed. Some structs are read/written as a whole (AP,AC), some are read/written only partly as a selection of their individual elements (AM,AR). If AM or AR have been changed by adding or removing an element that is important for the runtime status, then the reading/writing statements have to be added to or removed from DoRecovery and Do←
Snapshot. If AP or AC are changed, then for non-pointer variables (in the case of a struct it also means that none of its elements is a pointer) nothing has to be changed in the functions here. If pointers are involved, extra code has to be added (or removed). See the comments of DoRecovery and DoSnapshot.

Definition in file checkpoint.c.

4.5.2 Macro Definition Documentation

4.5.2.1 CACHED_SNAPSHOT

```
#define CACHED_SNAPSHOT
```

Definition at line 1214 of file checkpoint.c.

4.5.2.2 CACHE_SIZE

```
#define CACHE_SIZE 4096
```

Definition at line 1216 of file checkpoint.c.

4.5.2.3 R_FREE

```
#define R_FREE(
              ARG )   if ( ARG ) M_free(ARG, #ARG);
```

Definition at line 1281 of file checkpoint.c.

4.5.2.4 R_FREE_NAMETREE

```
#define R_FREE_NAMETREE(
              ARG )
```

Value:

```
R_FREE(ARG->namenode); \
R_FREE(ARG->namebuffer); \
R_FREE(ARG);
```

Definition at line 1284 of file checkpoint.c.

4.5.2.5 R_FREE_STREAM

```
#define R_FREE_STREAM(
              ARG )
```

Value:

```
R_FREE(ARG.buffer); \
R_FREE(ARG.FoldName); \
R_FREE(ARG.name);
```

Definition at line 1289 of file checkpoint.c.

4.5.2.6 R_SET

```
#define R_SET(
              VAR,
              TYPE )   VAR = *((TYPE*)p); p = (unsigned char*)p + sizeof(TYPE);
```

Definition at line 1296 of file checkpoint.c.

4.5.2.7 R_COPY_B

```
#define R_COPY_B(
              VAR,
              SIZE,
              CAST )
```

Value:

```
VAR = (CAST)Malloc1(SIZE,#VAR); \
memcpy(VAR, p, SIZE); p = (unsigned char*)p + SIZE;
```

Definition at line 1301 of file checkpoint.c.

4.5.2.8 S_WRITE_B

```
#define S_WRITE_B(
            BUF,
            LEN ) if ( fwrite_cached(BUF, 1, LEN, fd) != (size_t)(LEN) ) return(__LINE__);
```

Definition at line 1305 of file checkpoint.c.

4.5.2.9 S_FLUSH_B

```
#define S_FLUSH_B  if ( flush_cache(fd) != 1 ) return(__LINE__);
```

Definition at line 1308 of file checkpoint.c.

4.5.2.10 R_COPY_S

```
#define R_COPY_S(
            VAR,
            CAST )
```

Value:

```
if ( VAR ) { \
    VAR = (CAST)Malloc1(strlen(p)+1,"R_COPY_S"); \
    strcpy((char*)VAR, p); p = (unsigned char*)p + strlen(p) + 1; \
}
```

Definition at line 1313 of file checkpoint.c.

4.5.2.11 S_WRITE_S

```
#define S_WRITE_S(
            STR )
```

Value:

```
if ( STR ) { \
    l = strlen((char*)STR) + 1; \
    if ( fwrite_cached(STR, 1, l, fd) != (size_t)l ) return(__LINE__); \
}
```

Definition at line 1319 of file checkpoint.c.

4.5.2.12 R_COPY_LIST

```
#define R_COPY_LIST(
            ARG )
```

Value:

```
if ( ARG.maxnum ) { \
    R_COPY_B(ARG.lijst, ARG.size*ARG.maxnum, void*) \
}
```

Definition at line 1327 of file checkpoint.c.

4.5.2.13 S_WRITE_LIST

```
#define S_WRITE_LIST(
            LST )
```

Value:

```
if ( LST.maxnum ) { \
    S_WRITE_B((char*)LST.lijst, LST.maxnum*LST.size) \
}
```

Definition at line 1332 of file checkpoint.c.

4.5.2.14 R_COPY_NAMETREE

```
#define R_COPY_NAMETREE(
            ARG )
```

Value:

```
R_COPY_B(ARG, sizeof(NAMETREE), NAMETREE*); \
if ( ARG->namenode ) { \
    R_COPY_B(ARG->namenode, ARG->nodesize*sizeof(NAMENODE), NAMENODE*); \
} \
if ( ARG->namebuffer ) { \
    R_COPY_B(ARG->namebuffer, ARG->namesize, UBYTE*); \
}
```

Definition at line 1339 of file checkpoint.c.

4.5.2.15 S_WRITE_NAMETREE

```
#define S_WRITE_NAMETREE(
            ARG )
```

Value:

```
S_WRITE_B(ARG, sizeof(NAMETREE)); \
if ( ARG->namenode ) { \
    S_WRITE_B(ARG->namenode, ARG->nodesize*sizeof(struct NaMeNode)); \
} \
if ( ARG->namebuffer ) { \
    S_WRITE_B(ARG->namebuffer, ARG->namesize); \
}
```

Definition at line 1348 of file checkpoint.c.

4.5.2.16 S_WRITE_DOLLAR

```
#define S_WRITE_DOLLAR(
            ARG )
```

Value:

```
if ( ARG.size && ARG.where && ARG.where != &(AM.dollarzero) ) { \
    S_WRITE_B(ARG.where, ARG.size*sizeof(WORD)) \
}
```

Definition at line 1359 of file checkpoint.c.

4.5.2.17 ANNOUNCE

```
#define ANNOUNCE(
              str )
```

Definition at line 1370 of file checkpoint.c.

4.5.3 Function Documentation

4.5.3.1 CheckRecoveryFile()

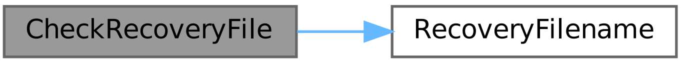
```
int CheckRecoveryFile (
             void   )
```

Checks whether a snapshot/recovery file exists. Returns 1 if it exists, 0 otherwise.

Definition at line 278 of file checkpoint.c.

References RecoveryFilename().

Here is the call graph for this function:



4.5.3.2 DeleteRecoveryFile()

```
void DeleteRecoveryFile (
             void   )
```

Deletes the recovery files. It is called by CleanUp() in the case of a successful completion.

Definition at line 333 of file checkpoint.c.

4.5.3.3 RecoveryFilename()

```
char * RecoveryFilename (
             void   )
```

Returns pointer to recovery filename.

Definition at line 364 of file checkpoint.c.

Referenced by CheckRecoveryFile(), and DoRecovery().

4.5.3.4 InitRecovery()

```
void InitRecovery (
            void )
```

Sets up the strings for the filenames of the recovery files. This functions should only be called once to avoid memory leaks and after AM.TempDir has been initialized.

Definition at line 399 of file checkpoint.c.

4.5.3.5 fwrite_cached()

```
size_t fwrite_cached (
            const void * ptr,
            size_t size,
            size_t nmemb,
            FILE * fd )
```

Definition at line 1222 of file checkpoint.c.

4.5.3.6 flush_cache()

```
size_t flush_cache (
            FILE * fd )
```

Definition at line 1247 of file checkpoint.c.

4.5.3.7 DoRecovery()

```
int DoRecovery (
            int * moduletype )
```

Reads from the recovery file and restores all necessary variables and states in FORM, so that the execution can recommence in preprocessor() as if no restart of FORM had occurred.

The recovery file is read into memory as a whole. The pointer p then points into this memory at the next non-processed data. The macros by which variables are restored, like R_SET, automatically increase p appropriately.

If something goes wrong, the function returns with a non-zero value.

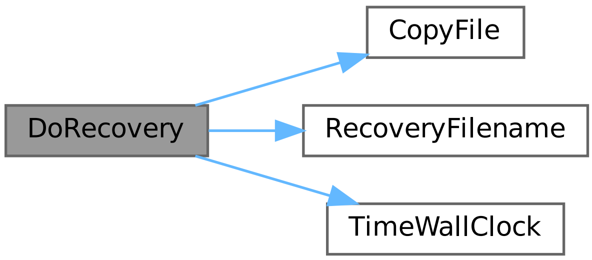
Allocated memory that would be lost when overwriting the global structs with data from the file is freed first. A major part of the code deals with the restoration of pointers. The idiom we use is to memorize the original pointer value (org), allocate new memory and copy the data from the file into this memory, calculate the offset between the old pointer value and the new allocated memory position (ofs), and then correct all affected pointers (+=ofs).

We rely on the fact that several variables (especially in AM) are already assigned the correct values by the startup functions. That means, in principle, that a change in the setup files between snapshot creation and recovery will be noticed.

Definition at line 1402 of file checkpoint.c.

References TaBlEs::argtail, TaBlEs::boomlijst, BrAcKeTiNfO::bracketbuffer, TaBlEs::buffers, TaBlEs::bufferssize, CopyFile(), TaBlEs::flags, ReNuMbEr::func, ReNuMbEr::funnum, VaRrEnUm::hi, BrAcKeTiNfO::indexbuffer, ReNuMbEr::indi, ReNuMbEr::indnum, VaRrEnUm::lo, TaBlEs::MaxTreeSize, TaBlEs::mm, TaBlEs::numind, TaBlEs::pattern, TaBlEs::prototype, TaBlEs::prototypeSize, RecoveryFilename(), ExPrEsSiOn::renumlists, TaBlEs::reserved, TaBlEs::spare, TaBlEs::sparse, VaRrEnUm::start, ReNuMbEr::symb, ReNuMbEr::symnum, TaBlEs::tablepointers, TimeWallClock(), TaBlEs::totind, ReNuMbEr::vecnum, and ReNuMbEr::vect.

Here is the call graph for this function:



4.5.3.8 DoCheckpoint()

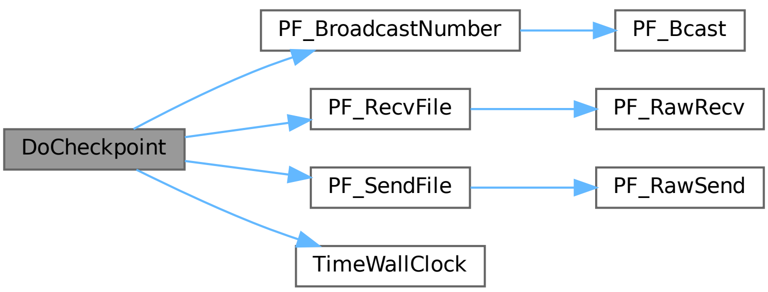
```
void DoCheckpoint (
            int moduletype )
```

Checks whether a snapshot should be done. Calls DoSnapshot() to create the snapshot.

Definition at line 3111 of file checkpoint.c.

References PF_BroadcastNumber(), PF_RecvFile(), PF_SendFile(), and TimeWallClock().

Here is the call graph for this function:



4.5.4 Variable Documentation

4.5.4.1 cache_buffer

```
unsigned char cache_buffer[CACHE_SIZE]
```

Definition at line 1219 of file checkpoint.c.

4.5.4.2 cache_fill

```
size_t cache_fill = 0
```

Definition at line 1220 of file checkpoint.c.

4.6 checkpoint.c

Go to the documentation of this file.

```
/*
    #[ Explanations :
*/
/*
    #] Explanations :
    #[ License :
 *
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   details.
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 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/*
    #] License :
    #[ Includes :
*/

#include "form3.h"

#include <errno.h>

/*
#define PRINTDEBUG
*/

/*
#define PRINTTIMEMARKS
*/

/*
    #] Includes :
    #[ filenames and system commands :
*/

#ifdef WITHMPI
#define BASENAME_FMT "%c%04dFORMrecv"
```

```
static char BaseName[] = BASENAME_FMT;
#else
static char *BaseName = "FORMrecv";
#endif
static char *recoveryfile = 0;
static char *intermedfile = 0;
static char *sortfile = 0;
static char *hidefile = 0;
static char *storefile = 0;

static int done_snapshot = 0;

#ifdef WITHMPI
static int PF_fmt_pos;

static const char *PF_recoveryfile(char prefix, int id, int intermed)
{
    /*
     * Assume that InitRecovery() has been already called, namely
     * recoveryfile, intermedfile and PF_fmt_pos are already initialized.
     */
    static char *tmp_recovery = NULL;
    static char *tmp_intermed  = NULL;
    char *tmp, c;
    if ( tmp_recovery == NULL ) {
        if ( PF.numtasks > 9999 ) {  /* see BASENAME_FMT */
            MesPrint("Checkpoint: too many number of processors.");
            Terminate(-1);
        }
        tmp_recovery = (char *)Malloc1(strlen(recoveryfile) + strlen(intermedfile) + 2, "PF_recoveryfile");
        tmp_intermed = tmp_recovery + strlen(recoveryfile) + 1;
        strcpy(tmp_recovery, recoveryfile);
        strcpy(tmp_intermed, intermedfile);
    }
    tmp = intermed ? tmp_intermed : tmp_recovery;
    c = tmp[PF_fmt_pos + 13];  /* The magic number 13 comes from BASENAME_FMT. */
    sprintf(tmp + PF_fmt_pos, BASENAME_FMT, prefix, id);
    tmp[PF_fmt_pos + 13] = c;
    return tmp;
}
#endif

/*
    #] filenames and system commands :
    #[ CheckRecoveryFile :
*/

#ifdef WITHMPI

static int PF_CheckRecoveryFile()
{
    int i,ret=0;
    FILE *fd;
    /* Check if the recovery file for the master exists. */
    if ( PF.me == MASTER ) {
        if ( (fd = fopen(recoveryfile, "r")) ) {
            fclose(fd);
            PF_BroadcastNumber(1);
        }
        else {
            PF_BroadcastNumber(0);
            return 0;
        }
    }
    else {
        if ( !PF_BroadcastNumber(0) )
            return 0;
    }
    /* Now the main part. */
    if (PF.me == MASTER){
        /*We have to have recovery files for the master and all the slaves:*/
        for(i=1; i<PF.numtasks;i++){
            const char *tmpnam = PF_recoveryfile('m', i, 0);
            if ( (fd = fopen(tmpnam, "r")) )
                fclose(fd);
            else
                break;
        }/*for(i=0; i<PF.numtasks;i++)*/
        if(i!=PF.numtasks){/*some files are absent*/
            int j;
            /*Send all slaves failure*/
            for(j=1; j<PF.numtasks;j++){
                ret=PF_SendFile(j, NULL);
                if(ret<0)
                    return(-1);
            }
```

```
            if(i==0)
                return(0);/*Ok, no recovery files at all.*/
            /*The master recovery file exists but some slave files are absent*/
            MesPrint("The file %s exists but some of the slave recovery files are absent.",
                            RecoveryFilename());
            return(-1);
        }/*if(i!=PF.numtasks)*/
        /*All the files are here.*/
        /*Send all slaves success and files:*/
        for(i=1; i<PF.numtasks;i++){
            const char *tmpnam = PF_recoveryfile('m', i, 0);
            fd = fopen(tmpnam, "r");
            ret=PF_SendFile(i, fd);/*if fd==NULL, PF_SendFile seds to a slave the failure tag*/
            if(fd == NULL)
                return(-1);
            else
                fclose(fd);
            if(ret<0)
                return(-1);
        }/*for(i=0; i<PF.numtasks;i++)*/
        return(1);
    }/*if (PF.me == MASTER)*/
    /*Slave:*/
    /*Get the answer from the master:*/
    fd=fopen(recoveryfile,"w");
    if(fd == NULL) {
        MesPrint("Failed to open %s in write mode in process %w", recoveryfile);
        return(-1);
    }
    ret=PF_RecvFile(MASTER,fd);
    if(ret<0)
        return(-1);
    fclose(fd);
    if(ret==0){
        /*Nothing is found by the master*/
        remove(recoveryfile);
        return(0);
    }
    /*Recovery file is successfully transferred*/
    return(1);
}
#endif

int CheckRecoveryFile(void)
{
    int ret = 0;
#ifdef WITHMPI
    ret = PF_CheckRecoveryFile();
#else
    FILE *fd;
    if ( (fd = fopen(recoveryfile, "r")) ) {
        fclose(fd);
        ret = 1;
    }
#endif
    if ( ret < 0 ){/*In ParFORM CheckRecoveryFile() may return a fatal error.*/
        MesPrint("Fail checking recovery file");
        Terminate(-1);
    }
    else if  ( ret > 0 ) {
        if ( AC.CheckpointFlag != -1 ) {
            /* recovery file exists but recovery option is not given */
#ifdef WITHMPI
            if ( PF.me == MASTER ) {
#endif
            MesPrint("The recovery file %s exists, but the recovery option -R has not been given!", RecoveryFilename());
            MesPrint("FORM will be terminated to avoid unintentional loss of data.");
            MesPrint("Delete the recovery file manually, if you want to start FORM without recovery.");
#ifdef WITHMPI
            }
            if(PF.me != MASTER)
                remove(RecoveryFilename());
#endif
            Terminate(-1);
        }
    }
    else {
        if ( AC.CheckpointFlag == -1 ) {
            /* recovery option given but recovery file does not exist */
#ifdef WITHMPI
            if ( PF.me == MASTER )
#endif
            MesPrint("Option -R for recovery has been given, but the recovery file %s does not exist!", RecoveryFilename());
            Terminate(-1);
```

```
        }
    }
    return(ret);
}

/*
    #] CheckRecoveryFile :
    #[ DeleteRecoveryFile :
*/

void DeleteRecoveryFile(void)
{
    if ( done_snapshot ) {
        remove(recoveryfile);
#ifdef WITHMPI
        if( PF.me == MASTER){
            int i;
            for(i=1; i<PF.numtasks;i++){
                const char *tmpnam = PF_recoveryfile('m', i, 0);
                remove(tmpnam);
            }/*for(i=1; i<PF.numtasks;i++)*/
            remove(storefile);
            remove(sortfile);
            remove(hidefile);
        }/*if( PF.me == MASTER)*/
#else
        remove(storefile);
        remove(sortfile);
        remove(hidefile);
#endif
    }
}

/*
    #] DeleteRecoveryFile :
    #[ RecoveryFilename :
*/

char *RecoveryFilename(void)
{
    return(recoveryfile);
}

/*
    #] RecoveryFilename :
    #[ InitRecovery :
*/

static char *InitName(char *str, char *ext)
{
    char *s, *d = str;
    if ( AM.TempDir ) {
        s = (char*)AM.TempDir;
        while ( *s ) { *d++ = *s++; }
        *d++ = SEPARATOR;
    }
    s = BaseName;
    while ( *s ) { *d++ = *s++; }
    *d++ = '.';
    s = ext;
    while ( *s ) { *d++ = *s++; }
    *d++ = 0;
    return d;
}

void InitRecovery(void)
{
    int lenpath = AM.TempDir ? strlen((char*)AM.TempDir)+1 : 0;
#ifdef WITHMPI
    sprintf(BaseName,BASENAME_FMT,(PF.me == MASTER)?'m':'s',PF.me);
    /*Now BaseName has a form ?XXXXFORMrecv where ? == 'm' for master and 's' for slave,
        XXXX is a zero - padded PF.me*/
    PF_fmt_pos = lenpath;
#endif
    recoveryfile = (char*)Malloc1(5*(lenpath+strlen(BaseName)+4+1),"InitRecovery");
    intermedfile = InitName(recoveryfile, "tmp");
    sortfile     = InitName(intermedfile, "XXX");
    hidefile     = InitName(sortfile, "out");
    storefile    = InitName(hidefile, "hid");
                   InitName(storefile, "str");
}

/*
    #] InitRecovery :
    #[ Debugging :
*/

```

```
#ifdef PRINTDEBUG

void print_BYTE(void *p)
{
    UBYTE h = (UBYTE)(*((UBYTE*)p) » 4);
    UBYTE l = (UBYTE)(*((UBYTE*)p) & 0x0F);
    if ( h > 9 ) h += 55; else h += 48;
    if ( l > 9 ) l += 55; else l += 48;
    printf("%c%c ", h, l);
}

static void print_STR(UBYTE *p)
{
    if ( p ) {
        MesPrint("%s", (char*)p);
    }
    else {
        MesPrint("NULL");
    }
}

static void print_WORDB(WORD *buf, WORD *top)
{
    LONG size = top-buf;
    int i;
    while ( size > 0 ) {
        if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
        else i = size;
        size -= i;
        MesPrint("%a",i,buf);
        buf += i;
    }
}

static void print_VOIDP(void *p, size_t size)
{
    int i;
    if ( p ) {
        while ( size > 0 ) {
            if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
            else i = size;
            size -= i;
            MesPrint("%b",i,(UBYTE *)p);
            p = ((UBYTE *)p)+i;
        }
    }
    else {
        MesPrint("NULL");
    }
}

static void print_CHARS(UBYTE *p, size_t size)
{
    int i;
    while ( size > 0 ) {
        if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
        else i = size;
        size -= i;
        MesPrint("%C",i,(char *)p);
        p += i;
    }
}

static void print_WORDV(WORD *p, size_t size)
{
    int i;
    if ( p ) {
        while ( size > 0 ) {
            if ( size > MAXPOSITIVE ) i = MAXPOSITIVE;
            else i = size;
            size -= i;
            MesPrint("%a",i,p);
            p += i;
        }
    }
    else {
        MesPrint("NULL");
    }
}

static void print_INTV(int *p, size_t size)
{
    int iarray[8];
    WORD i = 0;
    if ( p ) {
        while ( size > 0 ) {
            if ( i >= 8 ) {
```

```
                MesPrint("%I",i,iarray);
                i = 0;
            }
            iarray[i++] = *p++;
            size--;
        }
        if ( i > 0 ) MesPrint("%I",i,iarray);
    }
    else {
        MesPrint("NULL");
    }
}

static void print_LONGV(LONG *p, size_t size)
{
    LONG larray[8];
    WORD i = 0;
    if ( p ) {
        while ( size > 0 ) {
            if ( i >= 8 ) {
                MesPrint("%I",i,larray);
                i = 0;
            }
            larray[i++] = *p++;
            size--;
        }
        if ( i > 0 ) MesPrint("%I",i,larray);
    }
    else {
        MesPrint("NULL");
    }
}

static void print_PRELOAD(PRELOAD *l)
{
    if ( l->size ) {
        print_CHARS(l->buffer, l->size);
    }
    MesPrint("%ld", l->size);
}

static void print_PREVAR(PREVAR *l)
{
    MesPrint("%s", l->name);
    print_STR(l->value);
    if ( l->nargs ) print_STR(l->argnames);
    MesPrint("%d", l->nargs);
    MesPrint("%d", l->wildarg);
}

static void print_DOLLARS(DOLLARS l)
{
    print_VOIDP(l->where, l->size);
    MesPrint("%ld", l->size);
    MesPrint("%ld", l->name);
    MesPrint("%s", AC.dollarnames->namebuffer+l->name);
    MesPrint("%d", l->type);
    MesPrint("%d", l->node);
    MesPrint("%d", l->index);
    MesPrint("%d", l->zero);
    MesPrint("%d", l->numdummies);
    MesPrint("%d", l->nfactors);
}

static void print_LIST(LIST *l)
{
    print_VOIDP(l->lijst, l->size);
    MesPrint("%s", l->message);
    MesPrint("%d", l->num);
    MesPrint("%d", l->maxnum);
    MesPrint("%d", l->size);
    MesPrint("%d", l->numglobal);
    MesPrint("%d", l->numtemp);
    MesPrint("%d", l->numclear);
}

static void print_DOLOOP(DOLOOP *l)
{
    print_PRELOAD(&(l->p));
    print_STR(l->name);
    if ( l->type != NUMERICALLOOP ) {
        print_STR(l->vars);
    }
    print_STR(l->contents);
    if ( l->type != LISTEDLOOP && l->type != NUMERICALLOOP ) {
        print_STR(l->dollarname);
    }
```

```
    MesPrint("%l", l->startlinenumber);
    MesPrint("%l", l->firstnum);
    MesPrint("%l", l->lastnum);
    MesPrint("%l", l->incnum);
    MesPrint("%d", l->type);
    MesPrint("%d", l->NoShowInput);
    MesPrint("%d", l->errorsinloop);
    MesPrint("%d", l->firstloopcall);
}

static void print_PROCEDURE(PROCEDURE *l)
{
    if ( l->loadmode != 1 ) {
        print_PRELOAD(&(l->p));
    }
    print_STR(l->name);
    MesPrint("%d", l->loadmode);
}

static void print_NAMETREE(NAMETREE *t)
{
    int i;
    for ( i=0; i<t->nodefill; ++i ) {
        MesPrint("%l %d %d %d %d %d %d\n", t->namenode[i].name,
             t->namenode[i].parent, t->namenode[i].left, t->namenode[i].right,
             t->namenode[i].balance, t->namenode[i].type, t->namenode[i].number );
    }
    print_CHARS(t->namebuffer, t->namefill);
    MesPrint("%l", t->namesize);
    MesPrint("%l", t->namefill);
    MesPrint("%l", t->nodesize);
    MesPrint("%l", t->nodefill);
    MesPrint("%l", t->oldnamefill);
    MesPrint("%l", t->oldnodefill);
    MesPrint("%l", t->globalnamefill);
    MesPrint("%l", t->globalnodefill);
    MesPrint("%l", t->clearnamefill);
    MesPrint("%l", t->clearnodefill);
    MesPrint("%d", t->headnode);
}

void print_CBUF(CBUF *c)
{
    int i;
    print_WORDV(c->Buffer, c->BufferSize);
    /*
    MesPrint("%x", c->Buffer);
    MesPrint("%x", c->lhs);
    MesPrint("%x", c->rhs);
    */
    for ( i=0; i<c->numlhs; ++i ) {
        if ( c->lhs[i]) MesPrint("%d", *(c->lhs[i]));
    }
    for ( i=0; i<c->numrhs; ++i ) {
        if ( c->rhs[i]) MesPrint("%d", *(c->rhs[i]));
    }
    MesPrint("%l", *c->CanCommu);
    MesPrint("%l", *c->NumTerms);
    MesPrint("%d", *c->numdum);
    for ( i=0; i<c->MaxTreeSize; ++i ) {
        MesPrint("%d %d %d %d %d", c->boomlijst[i].parent, c->boomlijst[i].left, c->boomlijst[i].right,
                c->boomlijst[i].value, c->boomlijst[i].blnce);
    }
}

static void print_STREAM(STREAM *t)
{
    print_CHARS(t->buffer, t->inbuffer);
    MesPrint("%l", (LONG)(t->pointer-t->buffer));
    print_STR(t->FoldName);
    print_STR(t->name);
    if ( t->type == PREVARSTREAM || t->type == DOLLARSTREAM ) {
        print_STR(t->pname);
    }
    MesPrint("%l", (LONG)t->fileposition);
    MesPrint("%l", (LONG)t->linenumber);
    MesPrint("%l", (LONG)t->prevline);
    MesPrint("%l", t->buffersize);
    MesPrint("%l", t->bufferposition);
    MesPrint("%l", t->inbuffer);
    MesPrint("%d", t->previous);
    MesPrint("%d", t->handle);
    switch ( t->type ) {
        case FILESTREAM: MesPrint("%d == FILESTREAM", t->type); break;
        case PREVARSTREAM: MesPrint("%d == PREVARSTREAM", t->type); break;
        case PREREADSTREAM: MesPrint("%d == PREREADSTREAM", t->type); break;
```

```
        case PIPESTREAM: MesPrint("%d == PIPESTREAM", t->type); break;
        case PRECALCSTREAM: MesPrint("%d == PRECALCSTREAM", t->type); break;
        case DOLLARSTREAM: MesPrint("%d == DOLLARSTREAM", t->type); break;
        case PREREADSTREAM2: MesPrint("%d == PREREADSTREAM2", t->type); break;
        case EXTERNALCHANNELSTREAM: MesPrint("%d == EXTERNALCHANNELSTREAM", t->type); break;
        case PREREADSTREAM3: MesPrint("%d == PREREADSTREAM3", t->type); break;
        default: MesPrint("%d == UNKNOWN", t->type);
    }
}

static void print_M()
{
    MesPrint("%%%% M_const");
    MesPrint("%d", *AM.gcmod);
    MesPrint("%d", *AM.gpowmod);
    print_STR(AM.TempDir);
    print_STR(AM.TempSortDir);
    print_STR(AM.IncDir);
    print_STR(AM.InputFileName);
    print_STR(AM.LogFileName);
    print_STR(AM.OutBuffer);
    print_STR(AM.Path);
    print_STR(AM.SetupDir);
    print_STR(AM.SetupFile);
    MesPrint("--MARK  1");
    MesPrint("%l", (LONG)BASEPOSITION(AM.zeropos));
#ifdef WITHPTHREADS
    MesPrint("%l", AM.ThreadScratSize);
    MesPrint("%l", AM.ThreadScratOutSize);
#endif
    MesPrint("%l", AM.MaxTer);
    MesPrint("%l", AM.CompressSize);
    MesPrint("%l", AM.ScratSize);
    MesPrint("%l", AM.SizeStoreCache);
    MesPrint("%l", AM.MaxStreamSize);
    MesPrint("%l", AM.SIOsize);
    MesPrint("%l", AM.SLargeSize);
    MesPrint("%l", AM.SSmallEsize);
    MesPrint("%l", AM.SSmallSize);
    MesPrint("--MARK  2");
    MesPrint("%l", AM.STermsInSmall);
    MesPrint("%l", AM.MaxBracketBufferSize);
    MesPrint("%l", AM.hProcessBucketSize);
    MesPrint("%l", AM.gProcessBucketSize);
    MesPrint("%l", AM.shmWinSize);
    MesPrint("%l", AM.OldChildTime);
    MesPrint("%l", AM.OldSecTime);
    MesPrint("%l", AM.OldMilliTime);
    MesPrint("%l", AM.WorkSize);
    MesPrint("%l", AM.gThreadBucketSize);
    MesPrint("--MARK  3");
    MesPrint("%l", AM.ggThreadBucketSize);
    MesPrint("%d", AM.FileOnlyFlag);
    MesPrint("%d", AM.Interact);
    MesPrint("%d", AM.MaxParLevel);
    MesPrint("%d", AM.OutBufSize);
    MesPrint("%d", AM.SMaxFpatches);
    MesPrint("%d", AM.SMaxPatches);
    MesPrint("%d", AM.StdOut);
    MesPrint("%d", AM.ginsidefirst);
    MesPrint("%d", AM.gDefDim);
    MesPrint("%d", AM.gDefDim4);
    MesPrint("--MARK  4");
    MesPrint("%d", AM.NumFixedSets);
    MesPrint("%d", AM.NumFixedFunctions);
    MesPrint("%d", AM.rbufnum);
    MesPrint("%d", AM.dbufnum);
    MesPrint("%d", AM.SkipClears);
    MesPrint("%d", AM.gfunpowers);
    MesPrint("%d", AM.gStatsFlag);
    MesPrint("%d", AM.gNamesFlag);
    MesPrint("%d", AM.gCodesFlag);
    MesPrint("%d", AM.gTokensWriteFlag);
    MesPrint("%d", AM.gSortType);
    MesPrint("%d", AM.gproperorderflag);
    MesPrint("--MARK  5");
    MesPrint("%d", AM.hparallelflag);
    MesPrint("%d", AM.gparallelflag);
    MesPrint("%d", AM.totalnumberofthreads);
    MesPrint("%d", AM.gSizeCommuteInSet);
    MesPrint("%d", AM.gThreadStats);
    MesPrint("%d", AM.ggThreadStats);
    MesPrint("%d", AM.gFinalStats);
    MesPrint("%d", AM.ggFinalStats);
    MesPrint("%d", AM.gThreadsFlag);
    MesPrint("%d", AM.ggThreadsFlag);
    MesPrint("%d", AM.gThreadBalancing);
```

```
    MesPrint("%d", AM.ggThreadBalancing);
    MesPrint("%d", AM.gThreadSortFileSynch);
    MesPrint("%d", AM.ggThreadSortFileSynch);
    MesPrint("%d", AM.gProcessStats);
    MesPrint("%d", AM.ggProcessStats);
    MesPrint("%d", AM.gOldParallelStats);
    MesPrint("%d", AM.ggOldParallelStats);
    MesPrint("%d", AM.gWTimeStatsFlag);
    MesPrint("%d", AM.ggWTimeStatsFlag);
    MesPrint("%d", AM.maxFlevels);
    MesPrint("--MARK  6");
    MesPrint("%d", AM.resetTimeOnClear);
    MesPrint("%d", AM.gcNumDollars);
    MesPrint("%d", AM.MultiRun);
    MesPrint("%d", AM.gNoSpacesInNumbers);
    MesPrint("%d", AM.ggNoSpacesInNumbers);
    MesPrint("%d", AM.MaxTal);
    MesPrint("%d", AM.IndDum);
    MesPrint("%d", AM.DumInd);
    MesPrint("%d", AM.WilInd);
    MesPrint("%d", AM.gncmod);
    MesPrint("%d", AM.gnpowmod);
    MesPrint("%d", AM.gmodmode);
    MesPrint("--MARK  7");
    MesPrint("%d", AM.gUnitTrace);
    MesPrint("%d", AM.gOutputMode);
    MesPrint("%d", AM.gCnumpows);
    MesPrint("%d", AM.gOutputSpaces);
    MesPrint("%d", AM.gOutNumberType);
    MesPrint("%d %d %d %d", AM.gUniTrace[0], AM.gUniTrace[1], AM.gUniTrace[2], AM.gUniTrace[3]);
    MesPrint("%d", AM.MaxWildcards);
    MesPrint("%d", AM.mTraceDum);
    MesPrint("%d", AM.OffsetIndex);
    MesPrint("%d", AM.OffsetVector);
    MesPrint("%d", AM.RepMax);
    MesPrint("%d", AM.LogType);
    MesPrint("%d", AM.ggStatsFlag);
    MesPrint("%d", AM.gLineLength);
    MesPrint("%d", AM.qError);
    MesPrint("--MARK  8");
    MesPrint("%d", AM.FortranCont);
    MesPrint("%d", AM.HoldFlag);
    MesPrint("%d %d %d %d %d", AM.Ordering[0], AM.Ordering[1], AM.Ordering[2], AM.Ordering[3],
  AM.Ordering[4]);
    MesPrint("%d %d %d %d %d", AM.Ordering[5], AM.Ordering[6], AM.Ordering[7], AM.Ordering[8],
  AM.Ordering[9]);
    MesPrint("%d %d %d %d %d", AM.Ordering[10], AM.Ordering[11], AM.Ordering[12], AM.Ordering[13],
  AM.Ordering[14]);
    MesPrint("%d", AM.silent);
    MesPrint("%d", AM.tracebackflag);
    MesPrint("%d", AM.expnum);
    MesPrint("%d", AM.denomnum);
    MesPrint("%d", AM.facnum);
    MesPrint("%d", AM.invfacnum);
    MesPrint("%d", AM.sumnum);
    MesPrint("%d", AM.sumpnum);
    MesPrint("--MARK  9");
    MesPrint("%d", AM.OldOrderFlag);
    MesPrint("%d", AM.termfunnum);
    MesPrint("%d", AM.matchfunnum);
    MesPrint("%d", AM.countfunnum);
    MesPrint("%d", AM.gPolyFun);
    MesPrint("%d", AM.gPolyFunInv);
    MesPrint("%d", AM.gPolyFunType);
    MesPrint("%d", AM.gPolyFunExp);
    MesPrint("%d", AM.gPolyFunVar);
    MesPrint("%d", AM.gPolyFunPow);
    MesPrint("--MARK 10");
    MesPrint("%d", AM.dollarzero);
    MesPrint("%d", AM.atstartup);
    MesPrint("%d", AM.exitflag);
    MesPrint("%d", AM.NumStoreCaches);
    MesPrint("%d", AM.gIndentSpace);
    MesPrint("%d", AM.ggIndentSpace);
    MesPrint("%d", AM.gShortStatsMax);
    MesPrint("%d", AM.ggShortStatsMax);
    MesPrint("--MARK 11");
    MesPrint("%d", AM.FromStdin);
    MesPrint("%d", AM.IgnoreDeprecation);
    MesPrint("%%%% END M_const");
/*  fflush(0); */
}

static void print_P()
{
    int i;
    MesPrint("%%%% P_const");
```

```
    print_LIST(&AP.DollarList);
    for ( i=0; i<AP.DollarList.num; ++i ) {
        print_DOLLARS(&(Dollars[i]));
    }
    MesPrint("--MARK  1");
    print_LIST(&AP.PreVarList);
    for ( i=0; i<AP.PreVarList.num; ++i ) {
        print_PREVAR(&(PreVar[i]));
    }
    MesPrint("--MARK  2");
    print_LIST(&AP.LoopList);
    for ( i=0; i<AP.LoopList.num; ++i ) {
        print_DOLOOP(&(DoLoops[i]));
    }
    MesPrint("--MARK  3");
    print_LIST(&AP.ProcList);
    for ( i=0; i<AP.ProcList.num; ++i ) {
        print_PROCEDURE(&(Procedures[i]));
    }
    MesPrint("--MARK  4");
    for ( i=0; i<=AP.PreSwitchLevel; ++i ) {
        print_STR(AP.PreSwitchStrings[i]);
    }
    MesPrint("%l", AP.preStop-AP.preStart);
    if ( AP.preFill ) MesPrint("%l", AP.preFill-AP.preStart);
    print_CHARS(AP.preStart, AP.pSize);
    MesPrint("%s", AP.procedureExtension);
    MesPrint("%s", AP.cprocedureExtension);
    print_INTV(AP.PreIfStack, AP.MaxPreIfLevel);
    print_INTV(AP.PreSwitchModes, AP.NumPreSwitchStrings+1);
    print_INTV(AP.PreTypes, AP.NumPreTypes+1);
    MesPrint("%d", AP.PreAssignFlag);
    MesPrint("--MARK  5");
    MesPrint("%d", AP.PreContinuation);
    MesPrint("%l", AP.InOutBuf);
    MesPrint("%l", AP.pSize);
    MesPrint("%d", AP.PreproFlag);
    MesPrint("%d", AP.iBufError);
    MesPrint("%d", AP.PreOut);
    MesPrint("%d", AP.PreSwitchLevel);
    MesPrint("%d", AP.NumPreSwitchStrings);
    MesPrint("%d", AP.MaxPreTypes);
    MesPrint("--MARK  6");
    MesPrint("%d", AP.NumPreTypes);
    MesPrint("%d", AP.DelayPrevar);
    MesPrint("%d", AP.AllowDelay);
    MesPrint("%d", AP.lhdollarerror);
    MesPrint("%d", AP.eat);
    MesPrint("%d", AP.gNumPre);
    MesPrint("%d", AP.PreDebug);
    MesPrint("--MARK  7");
    MesPrint("%d", AP.DebugFlag);
    MesPrint("%d", AP.preError);
    MesPrint("%C", 1, &(AP.ComChar));
    MesPrint("%C", 1, &(AP.cComChar));
    MesPrint("%%%% END P_const");
/*  fflush(0); */
}

static void print_C()
{
    int i;
    UBYTE buf[40], *t;
    MesPrint("%%%% C_const");
    for ( i=0; i<32; ++i ) {
        t = buf;
        t = NumCopy((WORD)(AC.separators[i].bit_7),t);
        t = NumCopy((WORD)(AC.separators[i].bit_6),t);
        t = NumCopy((WORD)(AC.separators[i].bit_5),t);
        t = NumCopy((WORD)(AC.separators[i].bit_4),t);
        t = NumCopy((WORD)(AC.separators[i].bit_3),t);
        t = NumCopy((WORD)(AC.separators[i].bit_2),t);
        t = NumCopy((WORD)(AC.separators[i].bit_1),t);
        t = NumCopy((WORD)(AC.separators[i].bit_0),t);
        MesPrint("%s ",buf);
    }
    print_NAMETREE(AC.dollarnames);
    print_NAMETREE(AC.exprnames);
    print_NAMETREE(AC.varnames);
    MesPrint("--MARK  1");
    print_LIST(&AC.ChannelList);
    for ( i=0; i<AC.ChannelList.num; ++i ) {
        MesPrint("%s %d", channels[i].name, channels[i].handle);
    }
    MesPrint("--MARK  2");
    print_LIST(&AC.DubiousList);
    MesPrint("--MARK  3");
```

```
    print_LIST(&AC.FunctionList);
    for ( i=0; i<AC.FunctionList.num; ++i ) {
        if ( functions[i].tabl ) {

        }
        MesPrint("%l", functions[i].symminfo);
        MesPrint("%l", functions[i].name);
        MesPrint("%d", functions[i].namesize);
    }
    MesPrint("--MARK  4");
    print_LIST(&AC.ExpressionList);
    print_LIST(&AC.IndexList);
    print_LIST(&AC.SetElementList);
    print_LIST(&AC.SetList);
    MesPrint("--MARK  5");
    print_LIST(&AC.SymbolList);
    print_LIST(&AC.VectorList);
    print_LIST(&AC.PotModDolList);
    print_LIST(&AC.ModOptDolList);
    print_LIST(&AC.TableBaseList);

/*
    print_LIST(&AC.cbufList);
    for ( i=0; i<AC.cbufList.num; ++i ) {
        MesPrint("cbufList.num == %d", i);
        print_CBUF(cbuf+i);
    }
    MesPrint("%d", AC.cbufnum);
*/
    MesPrint("--MARK  6");

    print_LIST(&AC.AutoSymbolList);
    print_LIST(&AC.AutoIndexList);
    print_LIST(&AC.AutoVectorList);
    print_LIST(&AC.AutoFunctionList);

    print_NAMETREE(AC.autonames);
    MesPrint("--MARK  7");

    print_LIST(AC.Symbols);
    print_LIST(AC.Indices);
    print_LIST(AC.Vectors);
    print_LIST(AC.Functions);
    MesPrint("--MARK  8");

    print_NAMETREE(*AC.activenames);

    MesPrint("--MARK  9");

    MesPrint("%d", AC.AutoDeclareFlag);

    for ( i=0; i<AC.NumStreams; ++i ) {
        MesPrint("Stream %d\n", i);
        print_STREAM(AC.Streams+i);
    }
    print_STREAM(AC.CurrentStream);
    MesPrint("--MARK 10");

    print_LONGV(AC.termstack, AC.maxtermlevel);
    print_LONGV(AC.termsortstack, AC.maxtermlevel);
    print_VOIDP(AC.cmod, AM.MaxTal*4*sizeof(UWORD));
    print_WORDV((WORD *)(AC.cmod), 1);
    print_WORDV((WORD *)(AC.powmod), 1);
    print_WORDV((WORD*)AC.modpowers, 1);
    print_WORDV((WORD*)AC.halfmod, 1);
    MesPrint("--MARK 10-2");
    /*
    print_WORDV(AC.ProtoType, AC.ProtoType[1]);
    print_WORDV(AC.WildC, 1);
    */

    MesPrint("--MARK 11");
    /* IfHeap ... Labels */

    print_CHARS((UBYTE*)AC.tokens, AC.toptokens-AC.tokens);
    MesPrint("%l", AC.endoftokens-AC.tokens);
    print_WORDV(AC.tokenarglevel, AM.MaxParLevel);
    print_WORDV((WORD*)AC.modinverses, ABS(AC.ncmod));
#ifdef WITHPTHREADS
    print_LONGV(AC.inputnumbers, AC.sizepfirstnum+AC.sizepfirstnum*sizeof(WORD)/sizeof(LONG));
    print_WORDV(AC.pfirstnum, 1);
#endif
    MesPrint("--MARK 12");
    print_LONGV(AC.argstack, MAXNEST);
    print_LONGV(AC.insidestack, MAXNEST);
    print_LONGV(AC.inexprstack, MAXNEST);
    MesPrint("%l", AC.iBufferSize);
```

```
    MesPrint("%l", AC.TransEname);
    MesPrint("%l", AC.ProcessBucketSize);
    MesPrint("%l", AC.mProcessBucketSize);
    MesPrint("%l", AC.CModule);
    MesPrint("%l", AC.ThreadBucketSize);
    MesPrint("%d", AC.NoShowInput);
    MesPrint("%d", AC.ShortStats);
    MesPrint("%d", AC.compiletype);
    MesPrint("%d", AC.firstconstindex);
    MesPrint("%d", AC.insidefirst);
    MesPrint("%d", AC.minsidefirst);
    MesPrint("%d", AC.wildflag);
    MesPrint("%d", AC.NumLabels);
    MesPrint("%d", AC.MaxLabels);
    MesPrint("--MARK 13");
    MesPrint("%d", AC.lDefDim);
    MesPrint("%d", AC.lDefDim4);
    MesPrint("%d", AC.NumWildcardNames);
    MesPrint("%d", AC.WildcardBufferSize);
    MesPrint("%d", AC.MaxIf);
    MesPrint("%d", AC.NumStreams);
    MesPrint("%d", AC.MaxNumStreams);
    MesPrint("%d", AC.firstctypemessage);
    MesPrint("%d", AC.tablecheck);
    MesPrint("%d", AC.idoption);
    MesPrint("%d", AC.BottomLevel);
    MesPrint("%d", AC.CompileLevel);
    MesPrint("%d", AC.TokensWriteFlag);
    MesPrint("%d", AC.UnsureDollarMode);
    MesPrint("%d", AC.outsidefun);
    MesPrint("%d", AC.funpowers);
    MesPrint("--MARK 14");
    MesPrint("%d", AC.WarnFlag);
    MesPrint("%d", AC.StatsFlag);
    MesPrint("%d", AC.NamesFlag);
    MesPrint("%d", AC.CodesFlag);
    MesPrint("%d", AC.TokensWriteFlag);
    MesPrint("%d", AC.SetupFlag);
    MesPrint("%d", AC.SortType);
    MesPrint("%d", AC.lSortType);
    MesPrint("%d", AC.ThreadStats);
    MesPrint("%d", AC.FinalStats);
    MesPrint("%d", AC.ThreadsFlag);
    MesPrint("%d", AC.ThreadBalancing);
    MesPrint("%d", AC.ThreadSortFileSynch);
    MesPrint("%d", AC.ProcessStats);
    MesPrint("%d", AC.OldParallelStats);
    MesPrint("%d", AC.WTimeStatsFlag);
    MesPrint("%d", AC.BracketNormalize);
    MesPrint("%d", AC.maxtermlevel);
    MesPrint("%d", AC.dumnumflag);
    MesPrint("--MARK 15");
    MesPrint("%d", AC.bracketindexflag);
    MesPrint("%d", AC.parallelflag);
    MesPrint("%d", AC.mparallelflag);
    MesPrint("%d", AC.properorderflag);
    MesPrint("%d", AC.vetofilling);
    MesPrint("%d", AC.tablefilling);
    MesPrint("%d", AC.vetotablebasefill);
    MesPrint("%d", AC.exprfillwarning);
    MesPrint("%d", AC.lhdollarflag);
    MesPrint("%d", AC.NoCompress);
#ifdef WITHPTHREADS
    MesPrint("%d", AC.numpfirstnum);
    MesPrint("%d", AC.sizepfirstnum);
#endif
    MesPrint("%d", AC.RepLevel);
    MesPrint("%d", AC.arglevel);
    MesPrint("%d", AC.insidelevel);
    MesPrint("%d", AC.inexprlevel);
    MesPrint("%d", AC.termlevel);
    MesPrint("--MARK 16");
    print_WORDV(AC.argsumcheck, MAXNEST);
    print_WORDV(AC.insidesumcheck, MAXNEST);
    print_WORDV(AC.inexprsumcheck, MAXNEST);
    MesPrint("%d", AC.MustTestTable);
    MesPrint("%d", AC.DumNum);
    MesPrint("%d", AC.ncmod);
    MesPrint("%d", AC.npowmod);
    MesPrint("%d", AC.modmode);
    MesPrint("%d", AC.nhalfmod);
    MesPrint("%d", AC.DirtPow);
    MesPrint("%d", AC.lUnitTrace);
    MesPrint("%d", AC.NwildC);
    MesPrint("%d", AC.ComDefer);
    MesPrint("%d", AC.CollectFun);
    MesPrint("%d", AC.AltCollectFun);
```

```
    MesPrint("--MARK 17");
    MesPrint("%d", AC.OutputMode);
    MesPrint("%d", AC.Cnumpows);
    MesPrint("%d", AC.OutputSpaces);
    MesPrint("%d", AC.OutNumberType);
    print_WORDV(AC.lUniTrace, 4);
    print_WORDV(AC.RepSumCheck, MAXREPEAT);
    MesPrint("%d", AC.DidClean);
    MesPrint("%d", AC.IfLevel);
    MesPrint("%d", AC.WhileLevel);
    print_WORDV(AC.IfSumCheck, (AC.MaxIf+1));
    MesPrint("%d", AC.LogHandle);
    MesPrint("%d", AC.LineLength);
    MesPrint("%d", AC.StoreHandle);
    MesPrint("%d", AC.HideLevel);
    MesPrint("%d", AC.lPolyFun);
    MesPrint("%d", AC.lPolyFunInv);
    MesPrint("%d", AC.lPolyFunType);
    MesPrint("%d", AC.lPolyFunExp);
    MesPrint("%d", AC.lPolyFunVar);
    MesPrint("%d", AC.lPolyFunPow);
    MesPrint("%d", AC.SymChangeFlag);
    MesPrint("%d", AC.CollectPercentage);
    MesPrint("%d", AC.ShortStatsMax);
    MesPrint("--MARK 18");

    print_CHARS(AC.Commercial, COMMERCIALSIZE+2);

    MesPrint("%", AC.CheckpointFlag);
    MesPrint("%l", AC.CheckpointStamp);
    print_STR((unsigned char*)AC.CheckpointRunAfter);
    print_STR((unsigned char*)AC.CheckpointRunBefore);
    MesPrint("%l", AC.CheckpointInterval);

    MesPrint("%%%% END C_const");
/*  fflush(0); */
}

static void print_R()
{
    GETIDENTITY
    int i;
    MesPrint("%%%% R_const");
    MesPrint("%l", (LONG)(AR.infile-AR.Fscr));
    MesPrint("%s", AR.infile->name);
    MesPrint("%l", (LONG)(AR.outfile-AR.Fscr));
    MesPrint("%s", AR.outfile->name);
    MesPrint("%l", AR.hidefile-AR.Fscr);
    MesPrint("%s", AR.hidefile->name);
    for ( i=0; i<3; ++i ) {
        MesPrint("FSCR %d", i);
        print_WORDB(AR.Fscr[i].PObuffer, AR.Fscr[i].POfull);
    }
    /* ... */
    MesPrint("%l", AR.OldTime);
    MesPrint("%l", AR.InInBuf);
    MesPrint("%l", AR.InHiBuf);
    MesPrint("%l", AR.pWorkSize);
    MesPrint("%l", AR.lWorkSize);
    MesPrint("%l", AR.posWorkSize);
    MesPrint("%d", AR.NoCompress);
    MesPrint("%d", AR.gzipCompress);
    MesPrint("%d", AR.Cnumlhs);
#ifdef WITHPTHREADS
    MesPrint("%d", AR.exprtodo);
#endif
    MesPrint("%d", AR.GetFile);
    MesPrint("%d", AR.KeptInHold);
    MesPrint("%d", AR.BracketOn);
    MesPrint("%d", AR.MaxBracket);
    MesPrint("%d", AR.CurDum);
    MesPrint("%d", AR.DeferFlag);
    MesPrint("%d", AR.TePos);
    MesPrint("%d", AR.sLevel);
    MesPrint("%d", AR.Stage4Name);
    MesPrint("%d", AR.GetOneFile);
    MesPrint("%d", AR.PolyFun);
    MesPrint("%d", AR.PolyFunInv);
    MesPrint("%d", AR.PolyFunType);
    MesPrint("%d", AR.PolyFunExp);
    MesPrint("%d", AR.PolyFunVar);
    MesPrint("%d", AR.PolyFunPow);
    MesPrint("%d", AR.Eside);
    MesPrint("%d", AR.MaxDum);
    MesPrint("%d", AR.level);
    MesPrint("%d", AR.expchanged);
    MesPrint("%d", AR.expflags);
```

```
    MesPrint("%d", AR.CurExpr);
    MesPrint("%d", AR.SortType);
    MesPrint("%d", AR.ShortSortCount);
    MesPrint("%%%% END R_const");
/*  fflush(0); */
}

#endif /* ifdef PRINTDEBUG */

/*
    #] Debugging :
    #[ Cached file operation functions :
*/

#define CACHED_SNAPSHOT

#define CACHE_SIZE 4096

#ifdef CACHED_SNAPSHOT
unsigned char cache_buffer[CACHE_SIZE];
size_t cache_fill = 0;

size_t fwrite_cached(const void *ptr, size_t size, size_t nmemb, FILE *fd)
{
    size_t fullsize = size*nmemb;
    if ( fullsize+cache_fill >= CACHE_SIZE ) {
        size_t overlap = CACHE_SIZE-cache_fill;
        memcpy(cache_buffer+cache_fill, (unsigned char*)ptr, overlap);
        if ( fwrite(cache_buffer, 1, CACHE_SIZE, fd) != CACHE_SIZE ) return 0;
        fullsize -= overlap;
        if ( fullsize >= CACHE_SIZE ) {
            cache_fill = fullsize % CACHE_SIZE;
            if ( cache_fill ) memcpy(cache_buffer, (unsigned char*)ptr+overlap+fullsize-cache_fill, cache_fill);
            if ( fwrite((unsigned char*)ptr+overlap, 1, fullsize-cache_fill, fd) != fullsize-cache_fill ) return 0;
        }
        else {
            memcpy(cache_buffer, (unsigned char*)ptr+overlap, fullsize);
            cache_fill = fullsize;
        }
    }
    else {
        memcpy(cache_buffer+cache_fill, (unsigned char*)ptr, fullsize);
        cache_fill += fullsize;
    }
    return nmemb;
}

size_t flush_cache(FILE *fd)
{
    if ( cache_fill ) {
        size_t retval = fwrite(cache_buffer, 1, cache_fill, fd);
        if ( retval != cache_fill ) {
            cache_fill = 0;
            return 0;
        }
        cache_fill = 0;
    }
    return 1;
}
#else
size_t fwrite_cached(const void *ptr, size_t size, size_t nmemb, FILE *fd)
{
    return fwrite(ptr, size, nmemb, fd);
}

size_t flush_cache(FILE *fd)
{
    DUMMYUSE(fd)
    return 1;
}
#endif

/*
    #] Cached file operation functions :
    #[ Helper Macros :
*/

/* some helper macros to streamline the code in DoSnapshot() and DoRecovery() */

/* freeing memory */

#define R_FREE(ARG) \
    if ( ARG ) M_free(ARG, #ARG);

#define R_FREE_NAMETREE(ARG) \
```

```
    R_FREE(ARG->namenode); \
    R_FREE(ARG->namebuffer); \
    R_FREE(ARG);

#define R_FREE_STREAM(ARG) \
    R_FREE(ARG.buffer); \
    R_FREE(ARG.FoldName); \
    R_FREE(ARG.name);

/* reading a single variable */

#define R_SET(VAR,TYPE) \
    VAR = *((TYPE*)p); p = (unsigned char*)p + sizeof(TYPE);

/* general buffer */

#define R_COPY_B(VAR,SIZE,CAST) \
    VAR = (CAST)Malloc1(SIZE,#VAR); \
    memcpy(VAR, p, SIZE); p = (unsigned char*)p + SIZE;

#define S_WRITE_B(BUF,LEN) \
    if ( fwrite_cached(BUF, 1, LEN, fd) != (size_t)(LEN) ) return(__LINE__);

#define S_FLUSH_B \
    if ( flush_cache(fd) != 1 ) return(__LINE__);

/* character strings */

#define R_COPY_S(VAR,CAST) \
    if ( VAR ) { \
        VAR = (CAST)Malloc1(strlen(p)+1,"R_COPY_S"); \
        strcpy((char*)VAR, p); p = (unsigned char*)p + strlen(p) + 1; \
    }

#define S_WRITE_S(STR) \
    if ( STR ) { \
        l = strlen((char*)STR) + 1; \
        if ( fwrite_cached(STR, 1, l, fd) != (size_t)l ) return(__LINE__); \
    }

/* LIST */

#define R_COPY_LIST(ARG) \
    if ( ARG.maxnum ) { \
        R_COPY_B(ARG.lijst, ARG.size*ARG.maxnum, void*) \
    }

#define S_WRITE_LIST(LST) \
    if ( LST.maxnum ) { \
        S_WRITE_B((char*)LST.lijst, LST.maxnum*LST.size) \
    }

/* NAMETREE */

#define R_COPY_NAMETREE(ARG) \
    R_COPY_B(ARG, sizeof(NAMETREE), NAMETREE*); \
    if ( ARG->namenode ) { \
        R_COPY_B(ARG->namenode, ARG->nodesize*sizeof(NAMENODE), NAMENODE*); \
    } \
    if ( ARG->namebuffer ) { \
        R_COPY_B(ARG->namebuffer, ARG->namesize, UBYTE*); \
    }

#define S_WRITE_NAMETREE(ARG) \
    S_WRITE_B(ARG, sizeof(NAMETREE)); \
    if ( ARG->namenode ) { \
        S_WRITE_B(ARG->namenode, ARG->nodesize*sizeof(struct NaMeNode)); \
    } \
    if ( ARG->namebuffer ) { \
        S_WRITE_B(ARG->namebuffer, ARG->namesize); \
    }

/* DOLLAR */

#define S_WRITE_DOLLAR(ARG) \
    if ( ARG.size && ARG.where && ARG.where != &(AM.dollarzero) ) { \
        S_WRITE_B(ARG.where, ARG.size*sizeof(WORD)) \
    }

/* Printing time marks with ANNOUNCE macro */

#ifdef PRINTTIMEMARKS
time_t announce_time;
#define ANNOUNCE(str) time(&announce_time); MesPrint("TIMEMARK %s  %s", ctime(&announce_time), #str);
#else
#define ANNOUNCE(str)
#endif
```

```
/*
    #] Helper Macros :
    #[ DoRecovery :
*/

int DoRecovery(int *moduletype)
{
    GETIDENTITY
    FILE *fd;
    POSITION pos;
    void *buf, *p;
    LONG size, l;
    int i, j;
    UBYTE *org;
    char *namebufout, *namebufhide;
    LONG ofs;
    void *oldAMdollarzero;
    LIST PotModDolListBackup;
    LIST ModOptDolListBackup;
    WORD oldLogHandle;

    MesPrint("Recovering ... %"); fflush(0);

    if ( !(fd = fopen(recoveryfile, "r")) ) return(__LINE__);

    /* load the complete recovery file into a buffer */
    if ( fread(&pos, sizeof(POSITION), 1, fd) != 1 ) return(__LINE__);
    size = BASEPOSITION(pos) - sizeof(POSITION);
    buf = Malloc1(size, "recovery buffer");
    if ( fread(buf, size, 1, fd) != 1 ) return(__LINE__);

    /* pointer p will go through the buffer in the following */
    p = buf;

    /* read moduletype */
    R_SET(*moduletype, int);

    /*#[ AM : */

    /* only certain elements will be restored. the rest of AM should have gotten
     * the correct values at startup. */

    R_SET(AM.hparallelflag, int);
    R_SET(AM.gparallelflag, int);
    R_SET(AM.gCodesFlag, int);
    R_SET(AM.gNamesFlag, int);
    R_SET(AM.gStatsFlag, int);
    R_SET(AM.gTokensWriteFlag, int);
    R_SET(AM.gNoSpacesInNumbers, int);
    R_SET(AM.gIndentSpace, WORD);
    R_SET(AM.gUnitTrace, WORD);
    R_SET(AM.gDefDim, int);
    R_SET(AM.gDefDim4, int);
    R_SET(AM.gncmod, WORD);
    R_SET(AM.gnpowmod, WORD);
    R_SET(AM.gmodmode, WORD);
    R_SET(AM.gOutputMode, WORD);
    R_SET(AM.gCnumpows, WORD);
    R_SET(AM.gOutputSpaces, WORD);
    R_SET(AM.gOutNumberType, WORD);
    R_SET(AM.gfunpowers, int);
    R_SET(AM.gPolyFun, WORD);
    R_SET(AM.gPolyFunInv, WORD);
    R_SET(AM.gPolyFunType, WORD);
    R_SET(AM.gPolyFunExp, WORD);
    R_SET(AM.gPolyFunVar, WORD);
    R_SET(AM.gPolyFunPow, WORD);
    R_SET(AM.gProcessBucketSize, LONG);
    R_SET(AM.OldChildTime, LONG);
    R_SET(AM.OldSecTime, LONG);
    R_SET(AM.OldMilliTime, LONG);
    R_SET(AM.gproperorderflag, int);
    R_SET(AM.gThreadBucketSize, LONG);
    R_SET(AM.gSizeCommuteInSet, int);
    R_SET(AM.gThreadStats, int);
    R_SET(AM.gFinalStats, int);
    R_SET(AM.gThreadsFlag, int);
    R_SET(AM.gThreadBalancing, int);
    R_SET(AM.gThreadSortFileSynch, int);
    R_SET(AM.gProcessStats, int);
    R_SET(AM.gOldParallelStats, int);
    R_SET(AM.gSortType, int);
    R_SET(AM.gShortStatsMax, WORD);
    R_SET(AM.gIsFortran90, int);
    R_SET(oldAMdollarzero, void*);
    R_FREE(AM.gFortran90Kind);
```

```
    R_SET(AM.gFortran90Kind,UBYTE *);
    R_COPY_S(AM.gFortran90Kind,UBYTE *);

    R_COPY_S(AM.gextrasym,UBYTE *);
    R_COPY_S(AM.ggextrasym,UBYTE *);

    R_SET(AM.PrintTotalSize,int);
    R_SET(AM.fbuffersize,int);
    R_SET(AM.gOldFactArgFlag,int);
    R_SET(AM.ggOldFactArgFlag,int);

    R_SET(AM.gnumextrasym,int);
    R_SET(AM.ggnumextrasym,int);
    R_SET(AM.NumSpectatorFiles,int);
    R_SET(AM.SizeForSpectatorFiles,int);
    R_SET(AM.gOldGCDflag,int);
    R_SET(AM.ggOldGCDflag,int);
    R_SET(AM.gWTimeStatsFlag, int);

    R_FREE(AM.Path);
    R_SET(AM.Path,UBYTE *);
    R_COPY_S(AM.Path,UBYTE *);

    R_SET(AM.FromStdin, BOOL);
    R_SET(AM.IgnoreDeprecation, BOOL);

#ifdef PRINTDEBUG
    print_M();
#endif

    /*#] AM : */
    /*#[ AC : */

    /* #[ AC free pointers */

    /* AC will be overwritten by data from the recovery file, therefore
     * dynamically allocated memory must be freed first. */

    R_FREE_NAMETREE(AC.dollarnames);
    R_FREE_NAMETREE(AC.exprnames);
    R_FREE_NAMETREE(AC.varnames);
    for ( i=0; i<AC.ChannelList.num; ++i ) {
        R_FREE(channels[i].name);
    }
    R_FREE(AC.ChannelList.lijst);
    R_FREE(AC.DubiousList.lijst);
    for ( i=0; i<AC.FunctionList.num; ++i ) {
        TABLES T = functions[i].tabl;
        if ( T ) {
            R_FREE(T->buffers);
            R_FREE(T->mm);
            R_FREE(T->flags);
            R_FREE(T->prototype);
            R_FREE(T->tablepointers);
            if ( T->sparse ) {
                R_FREE(T->boomlijst);
                R_FREE(T->argtail);
            }
            if ( T->spare ) {
                R_FREE(T->spare->buffers);
                R_FREE(T->spare->mm);
                R_FREE(T->spare->flags);
                R_FREE(T->spare->tablepointers);
                if ( T->spare->sparse ) {
                    R_FREE(T->spare->boomlijst);
                }
                R_FREE(T->spare);
            }
            R_FREE(T);
        }
    }
    R_FREE(AC.FunctionList.lijst);
    for ( i=0; i<AC.ExpressionList.num; ++i ) {
        if ( Expressions[i].renum ) {
            R_FREE(Expressions[i].renum->symb.lo);
            R_FREE(Expressions[i].renum);
        }
        if ( Expressions[i].bracketinfo ) {
            R_FREE(Expressions[i].bracketinfo->indexbuffer);
            R_FREE(Expressions[i].bracketinfo->bracketbuffer);
            R_FREE(Expressions[i].bracketinfo);
        }
        if ( Expressions[i].newbracketinfo ) {
            R_FREE(Expressions[i].newbracketinfo->indexbuffer);
            R_FREE(Expressions[i].newbracketinfo->bracketbuffer);
            R_FREE(Expressions[i].newbracketinfo);
        }
```

```
        if ( Expressions[i].renumlists != AN.dummyrenumlist ) {
            R_FREE(Expressions[i].renumlists);
        }
        R_FREE(Expressions[i].inmem);
    }
    R_FREE(AC.ExpressionList.lijst);
    R_FREE(AC.IndexList.lijst);
    R_FREE(AC.SetElementList.lijst);
    R_FREE(AC.SetList.lijst);
    R_FREE(AC.SymbolList.lijst);
    R_FREE(AC.VectorList.lijst);
    for ( i=0; i<AC.TableBaseList.num; ++i ) {
        R_FREE(tablebases[i].iblocks);
        R_FREE(tablebases[i].nblocks);
        R_FREE(tablebases[i].name);
        R_FREE(tablebases[i].fullname);
        R_FREE(tablebases[i].tablenames);
    }
    R_FREE(AC.TableBaseList.lijst);
    for ( i=0; i<AC.cbufList.num; ++i ) {
        R_FREE(cbuf[i].Buffer);
        R_FREE(cbuf[i].lhs);
        R_FREE(cbuf[i].rhs);
        R_FREE(cbuf[i].boomlijst);
    }
    R_FREE(AC.cbufList.lijst);
    R_FREE(AC.AutoSymbolList.lijst);
    R_FREE(AC.AutoIndexList.lijst);
    R_FREE(AC.AutoVectorList.lijst);
    /* Tables cannot be auto-declared, therefore no extra code here */
    R_FREE(AC.AutoFunctionList.lijst);
    R_FREE_NAMETREE(AC.autonames);
    for ( i=0; i<AC.NumStreams; ++i ) {
        R_FREE_STREAM(AC.Streams[i]);
    }
    R_FREE(AC.Streams);
    R_FREE(AC.termstack);
    R_FREE(AC.termsortstack);
    R_FREE(AC.cmod);
    R_FREE(AC.modpowers);
    R_FREE(AC.halfmod);
    R_FREE(AC.IfHeap);
    R_FREE(AC.IfCount);
    R_FREE(AC.iBuffer);
    for ( i=0; i<AC.NumLabels; ++i ) {
        R_FREE(AC.LabelNames[i]);
    }
    R_FREE(AC.LabelNames);
    R_FREE(AC.FixIndices);
    R_FREE(AC.termsumcheck);
    R_FREE(AC.WildcardNames);
    R_FREE(AC.tokens);
    R_FREE(AC.tokenarglevel);
    R_FREE(AC.modinverses);
    R_FREE(AC.Fortran90Kind);
#ifdef WITHPTHREADS
    R_FREE(AC.inputnumbers);
#endif
    R_FREE(AC.IfSumCheck);
    R_FREE(AC.CommuteInSet);
    R_FREE(AC.CheckpointRunAfter);
    R_FREE(AC.CheckpointRunBefore);

    /* #] AC free pointers */

    /* backup some lists in order to restore it to the initial setup */
    PotModDolListBackup = AC.PotModDolList;
    ModOptDolListBackup = AC.ModOptDolList;
    oldLogHandle = AC.LogHandle;

    /* first we copy AC as a whole and then restore the pointer structures step
       by step. */

    AC = *((struct C_const*)p); p = (unsigned char*)p + sizeof(struct C_const);

    R_COPY_NAMETREE(AC.dollarnames);
    R_COPY_NAMETREE(AC.exprnames);
    R_COPY_NAMETREE(AC.varnames);

    R_COPY_LIST(AC.ChannelList);
    for ( i=0; i<AC.ChannelList.num; ++i ) {
        R_COPY_S(channels[i].name,char*);
        channels[i].handle = ReOpenFile(channels[i].name);
    }
    AC.ChannelList.message = "channel buffer";

    AC.DubiousList.lijst = 0;
```

```
    AC.DubiousList.message = "ambiguous variable";
    AC.DubiousList.num =
    AC.DubiousList.maxnum =
    AC.DubiousList.numglobal =
    AC.DubiousList.numtemp =
    AC.DubiousList.numclear = 0;

    R_COPY_LIST(AC.FunctionList);
    for ( i=0; i<AC.FunctionList.num; ++i ) {
        if ( functions[i].tabl ) {
            TABLES tabl;
            R_COPY_B(tabl, sizeof(struct TaBlEs), TABLES);
            functions[i].tabl = tabl;
            if ( tabl->tablepointers ) {
                if ( tabl->sparse ) {
                    R_COPY_B(tabl->tablepointers,
                        tabl->reserved*sizeof(WORD)*(tabl->numind+TABLEEXTENSION),
                        WORD*);
                }
                else {
                    R_COPY_B(tabl->tablepointers,
                        TABLEEXTENSION*sizeof(WORD)*(tabl->totind), WORD*);
                }
            }
            org = (UBYTE*)tabl->prototype;
#ifdef WITHPTHREADS
            R_COPY_B(tabl->prototype, tabl->prototypeSize, WORD**);
            ofs = (UBYTE*)tabl->prototype - org;
            for ( j=0; j<AM.totalnumberofthreads; ++j ) {
                if ( tabl->prototype[j] ) {
                    tabl->prototype[j] = (WORD*)((UBYTE*)tabl->prototype[j] + ofs);
                }
            }
            if ( tabl->pattern ) {
                tabl->pattern = (WORD**)((UBYTE*)tabl->pattern + ofs);
                for ( j=0; j<AM.totalnumberofthreads; ++j ) {
                    if ( tabl->pattern[j] ) {
                        tabl->pattern[j] = (WORD*)((UBYTE*)tabl->pattern[j] + ofs);
                    }
                }
            }
#else
            ofs = tabl->pattern - tabl->prototype;
            R_COPY_B(tabl->prototype, tabl->prototypeSize, WORD*);
            if ( tabl->pattern ) {
                tabl->pattern = tabl->prototype + ofs;
            }
#endif
            R_COPY_B(tabl->mm, tabl->numind*(LONG)sizeof(MINMAX), MINMAX*);
            R_COPY_B(tabl->flags, tabl->numind*(LONG)sizeof(WORD), WORD*);
            if ( tabl->sparse ) {
                R_COPY_B(tabl->boomlijst, tabl->MaxTreeSize*(LONG)sizeof(COMPTREE), COMPTREE*);
                R_COPY_S(tabl->argtail,UBYTE*);
            }
            R_COPY_B(tabl->buffers, tabl->bufferssize*(LONG)sizeof(WORD), WORD*);
            if ( tabl->spare ) {
                TABLES spare;
                R_COPY_B(spare, sizeof(struct TaBlEs), TABLES);
                tabl->spare = spare;
                if ( spare->tablepointers ) {
                    if ( spare->sparse ) {
                        R_COPY_B(spare->tablepointers,
                            spare->reserved*sizeof(WORD)*(spare->numind+TABLEEXTENSION),
                            WORD*);
                    }
                    else {
                        R_COPY_B(spare->tablepointers,
                            TABLEEXTENSION*sizeof(WORD)*(spare->totind), WORD*);
                    }
                }
                spare->prototype = tabl->prototype;
                spare->pattern = tabl->pattern;
                R_COPY_B(spare->mm, spare->numind*(LONG)sizeof(MINMAX), MINMAX*);
                R_COPY_B(spare->flags, spare->numind*(LONG)sizeof(WORD), WORD*);
                if ( tabl->sparse ) {
                    R_COPY_B(spare->boomlijst, spare->MaxTreeSize*(LONG)sizeof(COMPTREE), COMPTREE*);
                    spare->argtail = tabl->argtail;
                }
                spare->spare = tabl;
                R_COPY_B(spare->buffers, spare->bufferssize*(LONG)sizeof(WORD), WORD*);
            }
        }
    }
    AC.FunctionList.message = "function";

    R_COPY_LIST(AC.ExpressionList);
    for ( i=0; i<AC.ExpressionList.num; ++i ) {
```

```
        EXPRESSIONS ex = Expressions + i;
        if ( ex->renum ) {
            R_COPY_B(ex->renum, sizeof(struct ReNuMbEr), RENUMBER);
            org = (UBYTE*)ex->renum->symb.lo;
            R_SET(size, size_t);
            R_COPY_B(ex->renum->symb.lo, size, WORD*);
            ofs = (UBYTE*)ex->renum->symb.lo - org;
            ex->renum->symb.start = (WORD*)((UBYTE*)ex->renum->symb.start + ofs);
            ex->renum->symb.hi = (WORD*)((UBYTE*)ex->renum->symb.hi + ofs);
            ex->renum->indi.lo = (WORD*)((UBYTE*)ex->renum->indi.lo + ofs);
            ex->renum->indi.start = (WORD*)((UBYTE*)ex->renum->indi.start + ofs);
            ex->renum->indi.hi = (WORD*)((UBYTE*)ex->renum->indi.hi + ofs);
            ex->renum->vect.lo = (WORD*)((UBYTE*)ex->renum->vect.lo + ofs);
            ex->renum->vect.start = (WORD*)((UBYTE*)ex->renum->vect.start + ofs);
            ex->renum->vect.hi = (WORD*)((UBYTE*)ex->renum->vect.hi + ofs);
            ex->renum->func.lo = (WORD*)((UBYTE*)ex->renum->func.lo + ofs);
            ex->renum->func.start = (WORD*)((UBYTE*)ex->renum->func.start + ofs);
            ex->renum->func.hi = (WORD*)((UBYTE*)ex->renum->func.hi + ofs);
            ex->renum->symnum = (WORD*)((UBYTE*)ex->renum->symnum + ofs);
            ex->renum->indnum = (WORD*)((UBYTE*)ex->renum->indnum + ofs);
            ex->renum->vecnum = (WORD*)((UBYTE*)ex->renum->vecnum + ofs);
            ex->renum->funnum = (WORD*)((UBYTE*)ex->renum->funnum + ofs);
        }
        if ( ex->bracketinfo ) {
            R_COPY_B(ex->bracketinfo, sizeof(BRACKETINFO), BRACKETINFO*);
            R_COPY_B(ex->bracketinfo->indexbuffer,
      ex->bracketinfo->indexbuffersize*sizeof(BRACKETINDEX), BRACKETINDEX*);
            R_COPY_B(ex->bracketinfo->bracketbuffer, ex->bracketinfo->bracketbuffersize*sizeof(WORD),
      WORD*);
        }
        if ( ex->newbracketinfo ) {
            R_COPY_B(ex->newbracketinfo, sizeof(BRACKETINFO), BRACKETINFO*);
            R_COPY_B(ex->newbracketinfo->indexbuffer,
      ex->newbracketinfo->indexbuffersize*sizeof(BRACKETINDEX), BRACKETINDEX*);
            R_COPY_B(ex->newbracketinfo->bracketbuffer,
      ex->newbracketinfo->bracketbuffersize*sizeof(WORD), WORD*);
        }
#ifdef WITHPTHREADS
        ex->renumlists = 0;
#else
        ex->renumlists = AN.dummyrenumlist;
#endif
        if ( ex->inmem ) {
            R_SET(size, size_t);
            R_COPY_B(ex->inmem, size, WORD*);
        }
    }
    AC.ExpressionList.message = "expression";

    R_COPY_LIST(AC.IndexList);
    AC.IndexList.message = "index";
    R_COPY_LIST(AC.SetElementList);
    AC.SetElementList.message = "set element";
    R_COPY_LIST(AC.SetList);
    AC.SetList.message = "set";
    R_COPY_LIST(AC.SymbolList);
    AC.SymbolList.message = "symbol";
    R_COPY_LIST(AC.VectorList);
    AC.VectorList.message = "vector";

    AC.PotModDolList = PotModDolListBackup;
    AC.ModOptDolList = ModOptDolListBackup;

    R_COPY_LIST(AC.TableBaseList);
    for ( i=0; i<AC.TableBaseList.num; ++i ) {
        if ( tablebases[i].iblocks ) {
            R_COPY_B(tablebases[i].iblocks,
      tablebases[i].info.numberofindexblocks*sizeof(INDEXBLOCK*),INDEXBLOCK**);
            for ( j=0; j<tablebases[i].info.numberofindexblocks; ++j ) {
                if ( tablebases[i].iblocks[j] ) {
                    R_COPY_B(tablebases[i].iblocks[j], sizeof(INDEXBLOCK), INDEXBLOCK*);
                }
            }
        }
        if ( tablebases[i].nblocks ) {
            R_COPY_B(tablebases[i].nblocks,
      tablebases[i].info.numberofnamesblocks*sizeof(NAMESBLOCK*),NAMESBLOCK**);
            for ( j=0; j<tablebases[i].info.numberofindexblocks; ++j ) {
                if ( tablebases[i].nblocks[j] ) {
                    R_COPY_B(tablebases[i].nblocks[j], sizeof(NAMESBLOCK), NAMESBLOCK*);
                }
            }
        }
        /* reopen file */
        if ( ( tablebases[i].handle = fopen(tablebases[i].fullname, "r+b") ) == NULL ) {
            MesPrint("ERROR: Could not reopen tablebase %s!",tablebases[i].name);
            Terminate(-1);
```

```
        }
        R_COPY_S(tablebases[i].name,char*);
        R_COPY_S(tablebases[i].fullname,char*);
        R_COPY_S(tablebases[i].tablenames,char*);
    }
    AC.TableBaseList.message = "list of tablebases";

    R_COPY_LIST(AC.cbufList);
    for ( i=0; i<AC.cbufList.num; ++i ) {
        org = (UBYTE*)cbuf[i].Buffer;
        R_COPY_B(cbuf[i].Buffer, cbuf[i].BufferSize*sizeof(WORD), WORD*);
        ofs = (UBYTE*)cbuf[i].Buffer - org;
        cbuf[i].Top = (WORD*)((UBYTE*)cbuf[i].Top + ofs);
        cbuf[i].Pointer = (WORD*)((UBYTE*)cbuf[i].Pointer + ofs);
        R_COPY_B(cbuf[i].lhs, cbuf[i].maxlhs*(LONG)sizeof(WORD*), WORD**);
        for ( j=1; j<=cbuf[i].numlhs; ++j ) {
            if ( cbuf[i].lhs[j] ) cbuf[i].lhs[j] = (WORD*)((UBYTE*)cbuf[i].lhs[j] + ofs);
        }
        org = (UBYTE*)cbuf[i].rhs;
        R_COPY_B(cbuf[i].rhs, cbuf[i].maxrhs*(LONG)(sizeof(WORD*)+2*sizeof(LONG)+2*sizeof(WORD)),
WORD**);
        for ( j=1; j<=cbuf[i].numrhs; ++j ) {
            if ( cbuf[i].rhs[j] ) cbuf[i].rhs[j] = (WORD*)((UBYTE*)cbuf[i].rhs[j] + ofs);
        }
        ofs = (UBYTE*)cbuf[i].rhs - org;
        cbuf[i].CanCommu = (LONG*)((UBYTE*)cbuf[i].CanCommu + ofs);
        cbuf[i].NumTerms = (LONG*)((UBYTE*)cbuf[i].NumTerms + ofs);
        cbuf[i].numdum = (WORD*)((UBYTE*)cbuf[i].numdum + ofs);
        cbuf[i].dimension = (WORD*)((UBYTE*)cbuf[i].dimension + ofs);
        if ( cbuf[i].boomlijst ) {
            R_COPY_B(cbuf[i].boomlijst, cbuf[i].MaxTreeSize*sizeof(COMPTREE), COMPTREE*);
        }
    }
    AC.cbufList.message = "compiler buffer";

    R_COPY_LIST(AC.AutoSymbolList);
    AC.AutoSymbolList.message = "autosymbol";
    R_COPY_LIST(AC.AutoIndexList);
    AC.AutoIndexList.message = "autoindex";
    R_COPY_LIST(AC.AutoVectorList);
    AC.AutoVectorList.message = "autovector";
    R_COPY_LIST(AC.AutoFunctionList);
    AC.AutoFunctionList.message = "autofunction";

    R_COPY_NAMETREE(AC.autonames);

    AC.Symbols = &(AC.SymbolList);
    AC.Indices = &(AC.IndexList);
    AC.Vectors = &(AC.VectorList);
    AC.Functions = &(AC.FunctionList);
    AC.activenames = &(AC.varnames);

    org = (UBYTE*)AC.Streams;
    R_COPY_B(AC.Streams, AC.MaxNumStreams*(LONG)sizeof(STREAM), STREAM*);
    for ( i=0; i<AC.NumStreams; ++i ) {
        if ( AC.Streams[i].type != FILESTREAM ) {
            UBYTE *org2;
            org2 = AC.Streams[i].buffer;
            if ( AC.Streams[i].inbuffer ) {
                R_COPY_B(AC.Streams[i].buffer, AC.Streams[i].inbuffer, UBYTE*);
            }
            ofs = AC.Streams[i].buffer - org2;
            AC.Streams[i].pointer += ofs;
            AC.Streams[i].top += ofs;
        }
        else {
            p = (unsigned char*)p + AC.Streams[i].inbuffer;
        }
        AC.Streams[i].buffersize = AC.Streams[i].inbuffer;
        R_COPY_S(AC.Streams[i].FoldName,UBYTE*);
        R_COPY_S(AC.Streams[i].name,UBYTE*);
        if ( AC.Streams[i].type == PREVARSTREAM || AC.Streams[i].type == DOLLARSTREAM ) {
            AC.Streams[i].pname = AC.Streams[i].name;
        }
        else if ( AC.Streams[i].type == FILESTREAM ) {
            UBYTE *org2;
            org2 = AC.Streams[i].buffer;
            AC.Streams[i].buffer = (UBYTE*)Malloc1(AC.Streams[i].buffersize, "buffer");
            ofs = AC.Streams[i].buffer - org2;
            AC.Streams[i].pointer += ofs;
            AC.Streams[i].top += ofs;

            /* open file except for already opened main input file */
            if ( i ) {
                AC.Streams[i].handle = OpenFile((char *)(AC.Streams[i].name));
                if ( AC.Streams[i].handle == -1 ) {
                    MesPrint("ERROR: Could not reopen stream %s!",AC.Streams[i].name);
```

```
                    Terminate(-1);
                }
            }

            PUTZERO(pos);
            ADDPOS(pos, AC.Streams[i].bufferposition);
            SeekFile(AC.Streams[i].handle, &pos, SEEK_SET);

            AC.Streams[i].inbuffer = ReadFile(AC.Streams[i].handle, AC.Streams[i].buffer,
      AC.Streams[i].inbuffer);

            SETBASEPOSITION(pos, AC.Streams[i].fileposition);
            SeekFile(AC.Streams[i].handle, &pos, SEEK_SET);
        }
        /*
         * Ideally, we should check if we have a type PREREADSTREAM, PREREADSTREAM2, and
         * PRECALCSTREAM here. If so, we should free element name and point it
         * to the name element of the embracing stream's struct. In practice,
         * this is undoable without adding new data to STREAM. Since we create
         * only a small memory leak here (some few byte for each existing
         * stream of these types) and only once when we do a recovery, we
         * tolerate this leak and keep it STREAM as it is.
         */
    }
    ofs = (UBYTE*)AC.Streams - org;
    AC.CurrentStream = (STREAM*)((UBYTE*)AC.CurrentStream + ofs);

    if ( AC.termstack ) {
        R_COPY_B(AC.termstack, AC.maxtermlevel*(LONG)sizeof(LONG), LONG*);
    }

    if ( AC.termsortstack ) {
        R_COPY_B(AC.termsortstack, AC.maxtermlevel*(LONG)sizeof(LONG), LONG*);
    }

    /* exception: here we also change values from struct AM */
    R_COPY_B(AC.cmod, AM.MaxTal*4*(LONG)sizeof(UWORD), UWORD*);
    AM.gcmod = AC.cmod + AM.MaxTal;
    AC.powmod = AM.gcmod + AM.MaxTal;
    AM.gpowmod = AC.powmod + AM.MaxTal;

    AC.modpowers = 0;
    AC.halfmod = 0;

    /* we don't care about AC.ProtoType/WildC */

    if ( AC.IfHeap ) {
        ofs = AC.IfStack - AC.IfHeap;
        R_COPY_B(AC.IfHeap, (LONG)sizeof(LONG)*(AC.MaxIf+1), LONG*);
        AC.IfStack = AC.IfHeap + ofs;
        R_COPY_B(AC.IfCount, (LONG)sizeof(LONG)*(AC.MaxIf+1), LONG*);
    }

    org = AC.iBuffer;
    size = AC.iStop - AC.iBuffer + 2;
    R_COPY_B(AC.iBuffer, size, UBYTE*);
    ofs = AC.iBuffer - org;
    AC.iPointer += ofs;
    AC.iStop += ofs;

    if ( AC.LabelNames ) {
        org = (UBYTE*)AC.LabelNames;
        R_COPY_B(AC.LabelNames, AC.MaxLabels*(LONG)(sizeof(UBYTE*)+sizeof(WORD)), UBYTE**);
        for ( i=0; i<AC.NumLabels; ++i ) {
            R_COPY_S(AC.LabelNames[i],UBYTE*);
        }
        ofs = (UBYTE*)AC.LabelNames - org;
        AC.Labels = (int*)((UBYTE*)AC.Labels + ofs);
    }

    R_COPY_B(AC.FixIndices, AM.OffsetIndex*(LONG)sizeof(WORD), WORD*);

    if ( AC.termsumcheck ) {
        R_COPY_B(AC.termsumcheck, AC.maxtermlevel*(LONG)sizeof(WORD), WORD*);
    }

    R_COPY_B(AC.WildcardNames, AC.WildcardBufferSize, UBYTE*);

    if ( AC.tokens ) {
        size = AC.toptokens - AC.tokens;
        if ( size ) {
            org = (UBYTE*)AC.tokens;
            R_COPY_B(AC.tokens, size, SBYTE*);
            ofs = (UBYTE*)AC.tokens - org;
            AC.endoftokens += ofs;
            AC.toptokens += ofs;
        }
```

```
        else {
            AC.tokens = 0;
            AC.endoftokens = AC.tokens;
            AC.toptokens = AC.tokens;
        }
    }

    R_COPY_B(AC.tokenarglevel, AM.MaxParLevel*(LONG)sizeof(WORD), WORD*);

    AC.modinverses = 0;

    R_COPY_S(AC.Fortran90Kind,UBYTE *);

#ifdef WITHPTHREADS
    if ( AC.inputnumbers ) {
        org = (UBYTE*)AC.inputnumbers;
        R_COPY_B(AC.inputnumbers, AC.sizepfirstnum*(LONG)(sizeof(WORD)+sizeof(LONG)), LONG*);
        ofs = (UBYTE*)AC.inputnumbers - org;
        AC.pfirstnum = (WORD*)((UBYTE*)AC.pfirstnum + ofs);
    }
    AC.halfmodlock = dummylock;
#endif /* ifdef WITHPTHREADS */

    if ( AC.IfSumCheck ) {
        R_COPY_B(AC.IfSumCheck, (LONG)sizeof(WORD)*(AC.MaxIf+1), WORD*);
    }
    if ( AC.CommuteInSet ) {
        R_COPY_B(AC.CommuteInSet, (LONG)sizeof(WORD)*(AC.SizeCommuteInSet+1), WORD*);
    }

    AC.LogHandle = oldLogHandle;

    R_COPY_S(AC.CheckpointRunAfter,char*);
    R_COPY_S(AC.CheckpointRunBefore,char*);

    R_COPY_S(AC.extrasym,UBYTE *);

#ifdef PRINTDEBUG
    print_C();
#endif

    /*#] AC : */
    /*#[ AP : */

    /* #[ AP free pointers */

    /* AP will be overwritten by data from the recovery file, therefore
     * dynamically allocated memory must be freed first. */

    for ( i=0; i<AP.DollarList.num; ++i ) {
        if ( Dollars[i].size ) {
            R_FREE(Dollars[i].where);
        }
        CleanDollarFactors(Dollars+i);
    }
    R_FREE(AP.DollarList.lijst);

    for ( i=0; i<AP.PreVarList.num; ++i ) {
        R_FREE(PreVar[i].name);
    }
    R_FREE(AP.PreVarList.lijst);

    for ( i=0; i<AP.LoopList.num; ++i ) {
        R_FREE(DoLoops[i].p.buffer);
        if ( DoLoops[i].dollarname ) {
            R_FREE(DoLoops[i].dollarname);
        }
    }
    R_FREE(AP.LoopList.lijst);

    for ( i=0; i<AP.ProcList.num; ++i ) {
        R_FREE(Procedures[i].p.buffer);
        R_FREE(Procedures[i].name);
    }
    R_FREE(AP.ProcList.lijst);

    for ( i=1; i<=AP.PreSwitchLevel; ++i ) {
        R_FREE(AP.PreSwitchStrings[i]);
    }
    R_FREE(AP.PreSwitchStrings);
    R_FREE(AP.preStart);
    R_FREE(AP.procedureExtension);
    R_FREE(AP.cprocedureExtension);
    R_FREE(AP.PreIfStack);
    R_FREE(AP.PreSwitchModes);
    R_FREE(AP.PreTypes);

```

```
    /* #] AP free pointers */

    /* first we copy AP as a whole and then restore the pointer structures step
       by step. */

    AP = *((struct P_const*)p); p = (unsigned char*)p + sizeof(struct P_const);
#ifdef WITHPTHREADS
    AP.PreVarLock = dummylock;
#endif

    R_COPY_LIST(AP.DollarList);
    for ( i=0; i<AP.DollarList.num; ++i ) {
        DOLLARS d = Dollars + i;
        size = d->size * sizeof(WORD);
        if ( size && d->where && d->where != oldAMdollarzero ) {
            R_COPY_B(d->where, size, void*);
        }
#ifdef WITHPTHREADS
        d->pthreadslockread = dummylock;
        d->pthreadslockwrite = dummylock;
#endif
        if ( d->nfactors > 1 ) {
            R_COPY_B(d->factors,sizeof(FACDOLLAR)*d->nfactors,FACDOLLAR*);
            for ( j = 0; j < d->nfactors; j++ ) {
                if ( d->factors[j].size > 0 ) {
                    R_COPY_B(d->factors[i].where,sizeof(WORD)*(d->factors[j].size+1),WORD*);
                }
            }
        }
    }
    AP.DollarList.message = "$-variable";

    R_COPY_LIST(AP.PreVarList);
    for ( i=0; i<AP.PreVarList.num; ++i ) {
        R_SET(size, size_t);
        org = PreVar[i].name;
        R_COPY_B(PreVar[i].name, size, UBYTE*);
        ofs = PreVar[i].name - org;
        if ( PreVar[i].value ) PreVar[i].value += ofs;
        if ( PreVar[i].argnames ) PreVar[i].argnames += ofs;
    }
    AP.PreVarList.message = "PreVariable";

    R_COPY_LIST(AP.LoopList);
    for ( i=0; i<AP.LoopList.num; ++i ) {
        org = DoLoops[i].p.buffer;
        R_COPY_B(DoLoops[i].p.buffer, DoLoops[i].p.size, UBYTE*);
        ofs = DoLoops[i].p.buffer - org;
        if ( DoLoops[i].name ) DoLoops[i].name += ofs;
        if ( DoLoops[i].vars ) DoLoops[i].vars += ofs;
        if ( DoLoops[i].contents ) DoLoops[i].contents += ofs;
        if ( DoLoops[i].type == ONEEXPRESSION ) {
            R_COPY_S(DoLoops[i].dollarname,UBYTE*);
        }
    }
    AP.LoopList.message = "doloop";

    R_COPY_LIST(AP.ProcList);
    for ( i=0; i<AP.ProcList.num; ++i ) {
        if ( Procedures[i].p.size ) {
            if ( Procedures[i].loadmode != 1 ) {
                R_COPY_B(Procedures[i].p.buffer, Procedures[i].p.size, UBYTE*);
            }
            else {
                R_SET(j, int);
                Procedures[i].p.buffer = Procedures[j].p.buffer;
            }
        }
        R_COPY_S(Procedures[i].name,UBYTE*);
    }
    AP.ProcList.message = "procedure";

    size = (AP.NumPreSwitchStrings+1)*(LONG)sizeof(UBYTE*);
    R_COPY_B(AP.PreSwitchStrings, size, UBYTE**);
    for ( i=1; i<=AP.PreSwitchLevel; ++i ) {
        R_COPY_S(AP.PreSwitchStrings[i],UBYTE*);
    }

    org = AP.preStart;
    R_COPY_B(AP.preStart, AP.pSize, UBYTE*);
    ofs = AP.preStart - org;
    if ( AP.preFill ) AP.preFill += ofs;
    if ( AP.preStop ) AP.preStop += ofs;

    R_COPY_S(AP.procedureExtension,UBYTE*);
    R_COPY_S(AP.cprocedureExtension,UBYTE*);

```

```
    R_COPY_B(AP.PreAssignStack,AP.MaxPreAssignLevel*(LONG)sizeof(LONG),LONG*);
    R_COPY_B(AP.PreIfStack, AP.MaxPreIfLevel*(LONG)sizeof(int), int*);
    R_COPY_B(AP.PreSwitchModes, (AP.NumPreSwitchStrings+1)*(LONG)sizeof(int), int*);
    R_COPY_B(AP.PreTypes, (AP.MaxPreTypes+1)*(LONG)sizeof(int), int*);

#ifdef PRINTDEBUG
    print_P();
#endif

    /*#] AP : */
    /*#[ AR : */

    R_SET(ofs,LONG);
    if ( ofs ) {
        AR.infile = AR.Fscr+1;
        AR.outfile = AR.Fscr;
        AR.hidefile = AR.Fscr+2;
    }
    else {
        AR.infile = AR.Fscr;
        AR.outfile = AR.Fscr+1;
        AR.hidefile = AR.Fscr+2;
    }

    /* #[ AR free pointers */

    /* Parts of AR will be overwritten by data from the recovery file, therefore
     * dynamically allocated memory must be freed first. */

    namebufout = AR.outfile->name;
    namebufhide = AR.hidefile->name;
    R_FREE(AR.outfile->PObuffer);
#ifdef WITHZLIB
    R_FREE(AR.outfile->zsp);
    R_FREE(AR.outfile->ziobuffer);
#endif
    namebufhide = AR.hidefile->name;
    R_FREE(AR.hidefile->PObuffer);
#ifdef WITHZLIB
    R_FREE(AR.hidefile->zsp);
    R_FREE(AR.hidefile->ziobuffer);
#endif
    /* no files should be opened -> nothing to do with handle */

    /* #] AR free pointers */

    /* outfile */
    R_SET(*AR.outfile, FILEHANDLE);
    org = (UBYTE*)AR.outfile->PObuffer;
    size = AR.outfile->POfull - AR.outfile->PObuffer;
    AR.outfile->PObuffer = (WORD*)Malloc1(AR.outfile->POsize, "PObuffer");
    if ( size ) {
        memcpy(AR.outfile->PObuffer, p, size*sizeof(WORD));
        p = (unsigned char*)p + size*sizeof(WORD);
    }
    ofs = (UBYTE*)AR.outfile->PObuffer - org;
    AR.outfile->POstop = (WORD*)((UBYTE*)AR.outfile->POstop + ofs);
    AR.outfile->POfill = (WORD*)((UBYTE*)AR.outfile->POfill + ofs);
    AR.outfile->POfull = (WORD*)((UBYTE*)AR.outfile->POfull + ofs);
    AR.outfile->name = namebufout;
#ifdef WITHPTHREADS
    AR.outfile->wPObuffer = AR.outfile->PObuffer;
    AR.outfile->wPOstop = AR.outfile->POstop;
    AR.outfile->wPOfill = AR.outfile->POfill;
    AR.outfile->wPOfull = AR.outfile->POfull;
#endif
#ifdef WITHZLIB
    /* zsp and ziobuffer will be allocated when used */
    AR.outfile->zsp = 0;
    AR.outfile->ziobuffer = 0;
#endif
    /* reopen old outfile */
#ifdef WITHMPI
    if(PF.me==MASTER)
#endif
    if ( AR.outfile->handle >= 0 ) {
        if ( CopyFile(sortfile, AR.outfile->name) ) {
            MesPrint("ERROR: Could not copy old output sort file %s!",sortfile);
            Terminate(-1);
        }
        AR.outfile->handle = ReOpenFile(AR.outfile->name);
        if ( AR.outfile->handle == -1 ) {
            MesPrint("ERROR: Could not reopen output sort file %s!",AR.outfile->name);
            Terminate(-1);
        }
        SeekFile(AR.outfile->handle, &AR.outfile->POposition, SEEK_SET);
    }
```

```
    /* hidefile */
    R_SET(*AR.hidefile, FILEHANDLE);
    AR.hidefile->name = namebufhide;
    if ( AR.hidefile->PObuffer ) {
        org = (UBYTE*)AR.hidefile->PObuffer;
        size = AR.hidefile->POfull - AR.hidefile->PObuffer;
        AR.hidefile->PObuffer = (WORD*)Malloc1(AR.hidefile->POsize, "PObuffer");
        if ( size ) {
            memcpy(AR.hidefile->PObuffer, p, size*sizeof(WORD));
            p = (unsigned char*)p + size*sizeof(WORD);
        }
        ofs = (UBYTE*)AR.hidefile->PObuffer - org;
        AR.hidefile->POstop = (WORD*)((UBYTE*)AR.hidefile->POstop + ofs);
        AR.hidefile->POfill = (WORD*)((UBYTE*)AR.hidefile->POfill + ofs);
        AR.hidefile->POfull = (WORD*)((UBYTE*)AR.hidefile->POfull + ofs);
#ifdef WITHPTHREADS
        AR.hidefile->wPObuffer = AR.hidefile->PObuffer;
        AR.hidefile->wPOstop = AR.hidefile->POstop;
        AR.hidefile->wPOfill = AR.hidefile->POfill;
        AR.hidefile->wPOfull = AR.hidefile->POfull;
#endif
    }
#ifdef WITHZLIB
        /* zsp and ziobuffer will be allocated when used */
        AR.hidefile->zsp = 0;
        AR.hidefile->ziobuffer = 0;
#endif
    /* reopen old hidefile */
    if ( AR.hidefile->handle >= 0 ) {
        if ( CopyFile(hidefile, AR.hidefile->name) ) {
            MesPrint("ERROR: Could not copy old hide file %s!",hidefile);
            Terminate(-1);
        }
        AR.hidefile->handle = ReOpenFile(AR.hidefile->name);
        if ( AR.hidefile->handle == -1 ) {
            MesPrint("ERROR: Could not reopen hide file %s!",AR.hidefile->name);
            Terminate(-1);
        }
        SeekFile(AR.hidefile->handle, &AR.hidefile->POposition, SEEK_SET);
    }

    /* store file */
    R_SET(pos, POSITION);
    if ( ISNOTZEROPOS(pos) ) {
        CloseFile(AR.StoreData.Handle);
        R_SET(AR.StoreData, FILEDATA);
        if ( CopyFile(storefile, FG.fname) ) {
            MesPrint("ERROR: Could not copy old store file %s!",storefile);
            Terminate(-1);
        }
        AR.StoreData.Handle = (WORD)ReOpenFile(FG.fname);
        SeekFile(AR.StoreData.Handle, &AR.StoreData.Position, SEEK_SET);
    }

    R_SET(AR.DefPosition, POSITION);
    R_SET(AR.OldTime, LONG);
    R_SET(AR.InInBuf, LONG);
    R_SET(AR.InHiBuf, LONG);

    R_SET(AR.NoCompress, int);
    R_SET(AR.gzipCompress, int);

    R_SET(AR.outtohide, int);

    R_SET(AR.GetFile, WORD);
    R_SET(AR.KeptInHold, WORD);
    R_SET(AR.BracketOn, WORD);
    R_SET(AR.MaxBracket, WORD);
    R_SET(AR.CurDum, WORD);
    R_SET(AR.DeferFlag, WORD);
    R_SET(AR.TePos, WORD);
    R_SET(AR.sLevel, WORD);
    R_SET(AR.Stage4Name, WORD);
    R_SET(AR.GetOneFile, WORD);
    R_SET(AR.PolyFun, WORD);
    R_SET(AR.PolyFunInv, WORD);
    R_SET(AR.PolyFunType, WORD);
    R_SET(AR.PolyFunExp, WORD);
    R_SET(AR.PolyFunVar, WORD);
    R_SET(AR.PolyFunPow, WORD);
    R_SET(AR.Eside, WORD);
    R_SET(AR.MaxDum, WORD);
    R_SET(AR.level, WORD);
    R_SET(AR.expchanged, WORD);
    R_SET(AR.expflags, WORD);
    R_SET(AR.CurExpr, WORD);
```

```
    R_SET(AR.SortType, WORD);
    R_SET(AR.ShortSortCount, WORD);

    /* this is usually done in Process(), but sometimes FORM does not
       end up executing Process() before it uses the AR.CompressPointer,
       so we need to explicitly set it here. */
    AR.CompressPointer = AR.CompressBuffer;

#ifdef WITHPTHREADS
    for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
        R_SET(AB[j]->R.wranfnpair1, int);
        R_SET(AB[j]->R.wranfnpair2, int);
        R_SET(AB[j]->R.wranfcall, int);
        R_SET(AB[j]->R.wranfseed, ULONG);
        R_SET(AB[j]->R.wranfia,ULONG*);
        if ( AB[j]->R.wranfia ) {
            R_COPY_B(AB[j]->R.wranfia, sizeof(ULONG)*AB[j]->R.wranfnpair2, ULONG*);
        }
    }
#else
    R_SET(AR.wranfnpair1, int);
    R_SET(AR.wranfnpair2, int);
    R_SET(AR.wranfcall, int);
    R_SET(AR.wranfseed, ULONG);
    R_SET(AR.wranfia,ULONG*);
    if ( AR.wranfia ) {
        R_COPY_B(AR.wranfia, sizeof(ULONG)*AR.wranfnpair2, ULONG*);
    }
#endif

#ifdef PRINTDEBUG
    print_R();
#endif

    /*#] AR : */
    /*#[ AO :*/
/*
    We copy all non-pointer variables.
*/
    l = sizeof(A.O) - ((UBYTE *)(&(A.O.NumInBrack))-(UBYTE *)(&A.O));
    memcpy(&(A.O.NumInBrack), p, l); p = (unsigned char*)p + l;
/*
    Now the variables in OptimizeResult
*/
    memcpy(&(A.O.OptimizeResult),p,sizeof(OPTIMIZERESULT));
        p = (unsigned char*)p + sizeof(OPTIMIZERESULT);

    if ( A.O.OptimizeResult.codesize > 0 ) {
        R_COPY_B(A.O.OptimizeResult.code,A.O.OptimizeResult.codesize*sizeof(WORD),WORD *);
    }
    R_COPY_S(A.O.OptimizeResult.nameofexpr,UBYTE *);
/*
    And now the dictionaries. We know how many there are. We also know
    how many elements the array AO.Dictionaries should have.
*/
    if ( AO.SizeDictionaries > 0 ) {
        AO.Dictionaries = (DICTIONARY **)Malloc1(AO.SizeDictionaries*sizeof(DICTIONARY *),
                    "Dictionaries");
        for ( i = 0; i < AO.NumDictionaries; i++ ) {
            R_SET(l,LONG)
            AO.Dictionaries[i] = DictFromBytes(p);
            p = (char *)p + l;
        }
    }
    /*#] AO :*/
#ifdef WITHMPI
    /*#[ PF : */
    {/*Block*/
        int numtasks;
        R_SET(numtasks, int);
        if(numtasks!=PF.numtasks){
            MesPrint("%d number of tasks expected instead of %d; use mpirun -np %d",
                        numtasks,PF.numtasks,numtasks);
            if(PF.me!=MASTER)
                remove(RecoveryFilename());
            Terminate(-1);
        }
    }/*Block*/
    R_SET(PF.rhsInParallel, int);
    R_SET(PF.exprbufsize, int);
    R_SET(PF.log, int);
    /*#] PF : */
#endif

#ifdef WITHPTHREADS
    /* read timing information of individual threads */
    R_SET(i, int);
```

```
    for ( j=1; j<AM.totalnumberofthreads; ++j ) {
        /* ... and correcting OldTime */
        AB[j]->R.OldTime = -(*((LONG*)p+j));
    }
    WriteTimerInfo((LONG*)p,(LONG *)((unsigned char*)p + i*(LONG)sizeof(LONG)));
    p = (unsigned char*)p + 2*i*(LONG)sizeof(LONG);
#endif /* ifdef WITHPTHREADS */

    if ( fclose(fd) ) return(__LINE__);

    M_free(buf,"recovery buffer");

    /* cares about data in S_const */
    UpdatePositions();
    AT.SS = AT.S0;
/*
    Set the checkpoint parameter right for the next checkpoint.
*/
    AC.CheckpointStamp = TimeWallClock(1);

    done_snapshot = 1;
    MesPrint("done."); fflush(0);

    return(0);
}

/*
    #] DoRecovery :
    #[ DoSnapshot :
*/

static int DoSnapshot(int moduletype)
{
    GETIDENTITY
    FILE *fd;
    POSITION pos;
    int i, j;
    LONG l;
    WORD *w;
    void *adr;
#ifdef WITHPTHREADS
    LONG *longp,*longpp;
#endif /* ifdef WITHPTHREADS */

    MesPrint("Saving recovery point ... %"); fflush(0);
#ifdef PRINTTIMEMARKS
    MesPrint("\n");
#endif

    if ( !(fd = fopen(intermedfile, "wb")) ) return(__LINE__);

    /* reserve space in the file for a length field */
    if ( fwrite(&pos, 1, sizeof(POSITION), fd) != sizeof(POSITION) ) return(__LINE__);

    /* write moduletype */
    if ( fwrite(&moduletype, 1, sizeof(int), fd) != sizeof(int) ) return(__LINE__);

    /*#[ AM :*/

    /* since most values don't change during execution, AM doesn't need to be
     * written as a whole. all values will be correctly set when starting up
     * anyway. only the exceptions need to be taken care of. see MakeGlobal()
     * and PopVariables() in execute.c. */

    ANNOUNCE(AM)
    S_WRITE_B(&AM.hparallelflag, sizeof(int));
    S_WRITE_B(&AM.gparallelflag, sizeof(int));
    S_WRITE_B(&AM.gCodesFlag, sizeof(int));
    S_WRITE_B(&AM.gNamesFlag, sizeof(int));
    S_WRITE_B(&AM.gStatsFlag, sizeof(int));
    S_WRITE_B(&AM.gTokensWriteFlag, sizeof(int));
    S_WRITE_B(&AM.gNoSpacesInNumbers, sizeof(int));
    S_WRITE_B(&AM.gIndentSpace, sizeof(WORD));
    S_WRITE_B(&AM.gUnitTrace, sizeof(WORD));
    S_WRITE_B(&AM.gDefDim, sizeof(int));
    S_WRITE_B(&AM.gDefDim4, sizeof(int));
    S_WRITE_B(&AM.gncmod, sizeof(WORD));
    S_WRITE_B(&AM.gnpowmod, sizeof(WORD));
    S_WRITE_B(&AM.gmodmode, sizeof(WORD));
    S_WRITE_B(&AM.gOutputMode, sizeof(WORD));
    S_WRITE_B(&AM.gCnumpows, sizeof(WORD));
    S_WRITE_B(&AM.gOutputSpaces, sizeof(WORD));
    S_WRITE_B(&AM.gOutNumberType, sizeof(WORD));
    S_WRITE_B(&AM.gfunpowers, sizeof(int));
    S_WRITE_B(&AM.gPolyFun, sizeof(WORD));
    S_WRITE_B(&AM.gPolyFunInv, sizeof(WORD));
    S_WRITE_B(&AM.gPolyFunType, sizeof(WORD));
```

```
    S_WRITE_B(&AM.gPolyFunExp, sizeof(WORD));
    S_WRITE_B(&AM.gPolyFunVar, sizeof(WORD));
    S_WRITE_B(&AM.gPolyFunPow, sizeof(WORD));
    S_WRITE_B(&AM.gProcessBucketSize, sizeof(LONG));
    S_WRITE_B(&AM.OldChildTime, sizeof(LONG));
    S_WRITE_B(&AM.OldSecTime, sizeof(LONG));
    S_WRITE_B(&AM.OldMilliTime, sizeof(LONG));
    S_WRITE_B(&AM.gproperorderflag, sizeof(int));
    S_WRITE_B(&AM.gThreadBucketSize, sizeof(LONG));
    S_WRITE_B(&AM.gSizeCommuteInSet, sizeof(int));
    S_WRITE_B(&AM.gThreadStats, sizeof(int));
    S_WRITE_B(&AM.gFinalStats, sizeof(int));
    S_WRITE_B(&AM.gThreadsFlag, sizeof(int));
    S_WRITE_B(&AM.gThreadBalancing, sizeof(int));
    S_WRITE_B(&AM.gThreadSortFileSynch, sizeof(int));
    S_WRITE_B(&AM.gProcessStats, sizeof(int));
    S_WRITE_B(&AM.gOldParallelStats, sizeof(int));
    S_WRITE_B(&AM.gSortType, sizeof(int));
    S_WRITE_B(&AM.gShortStatsMax, sizeof(WORD));
    S_WRITE_B(&AM.gIsFortran90, sizeof(int));
    adr = &AM.dollarzero;
    S_WRITE_B(&adr, sizeof(void*));
    S_WRITE_B(&AM.gFortran90Kind,sizeof(UBYTE *));
    S_WRITE_S(AM.gFortran90Kind);

    S_WRITE_S(AM.gextrasym);
    S_WRITE_S(AM.ggextrasym);

    S_WRITE_B(&AM.PrintTotalSize,sizeof(int));
    S_WRITE_B(&AM.fbuffersize,sizeof(int));
    S_WRITE_B(&AM.gOldFactArgFlag,sizeof(int));
    S_WRITE_B(&AM.ggOldFactArgFlag,sizeof(int));

    S_WRITE_B(&AM.gnumextrasym,sizeof(int));
    S_WRITE_B(&AM.ggnumextrasym,sizeof(int));
    S_WRITE_B(&AM.NumSpectatorFiles,sizeof(int));
    S_WRITE_B(&AM.SizeForSpectatorFiles,sizeof(int));
    S_WRITE_B(&AM.gOldGCDflag,sizeof(int));
    S_WRITE_B(&AM.ggOldGCDflag,sizeof(int));
    S_WRITE_B(&AM.gWTimeStatsFlag, sizeof(int));

    S_WRITE_B(&AM.Path,sizeof(UBYTE *));
    S_WRITE_S(AM.Path);

    S_WRITE_B(&AM.FromStdin,sizeof(BOOL));
    S_WRITE_B(&AM.IgnoreDeprecation,sizeof(BOOL));

    /*#] AM :*/
    /*#[ AC :*/

    /* we write AC as a whole and then write all additional data step by step.
     * AC.DubiousList doesn't need to be treated, because it should be empty. */

    ANNOUNCE(AC)
    S_WRITE_B(&AC, sizeof(struct C_const));

    S_WRITE_NAMETREE(AC.dollarnames);
    S_WRITE_NAMETREE(AC.exprnames);
    S_WRITE_NAMETREE(AC.varnames);

    S_WRITE_LIST(AC.ChannelList);
    for ( i=0; i<AC.ChannelList.num; ++i ) {
        S_WRITE_S(channels[i].name);
    }

    ANNOUNCE(AC.FunctionList)
    S_WRITE_LIST(AC.FunctionList);
    for ( i=0; i<AC.FunctionList.num; ++i ) {
        /* if the function is a table */
        if ( functions[i].tabl ) {
            TABLES tabl = functions[i].tabl;
            S_WRITE_B(tabl, sizeof(struct TaBlEs));
            if ( tabl->tablepointers ) {
                if ( tabl->sparse ) {
                    /* sparse tables. reserved holds number of allocated
                     * elements. the size of an element is numind plus
                     * TABLEEXTENSION times the size of WORD. */
                    S_WRITE_B(tabl->tablepointers,
                        tabl->reserved*sizeof(WORD)*(tabl->numind+TABLEEXTENSION));
                }
                else {
                    /* matrix like tables. */
                    S_WRITE_B(tabl->tablepointers,
                        TABLEEXTENSION*sizeof(WORD)*(tabl->totind));
                }
            }
            S_WRITE_B(tabl->prototype, tabl->prototypeSize);
```

```
            S_WRITE_B(tabl->mm, tabl->numind*(LONG)sizeof(MINMAX));
            S_WRITE_B(tabl->flags, tabl->numind*(LONG)sizeof(WORD));
            if ( tabl->sparse ) {
                S_WRITE_B(tabl->boomlijst, tabl->MaxTreeSize*(LONG)sizeof(COMPTREE));
                S_WRITE_S(tabl->argtail);
            }
            S_WRITE_B(tabl->buffers, tabl->bufferssize*(LONG)sizeof(WORD));
            if ( tabl->spare ) {
                TABLES spare = tabl->spare;
                S_WRITE_B(spare, sizeof(struct TaBlEs));
                if ( spare->tablepointers ) {
                    if ( spare->sparse ) {
                        /* sparse tables */
                        S_WRITE_B(spare->tablepointers,
                            spare->reserved*sizeof(WORD)*(spare->numind+TABLEEXTENSION));
                    }
                    else {
                        /* matrix like tables */
                        S_WRITE_B(spare->tablepointers,
                            TABLEEXTENSION*sizeof(WORD)*(spare->totind));
                    }
                }
                S_WRITE_B(spare->mm, spare->numind*(LONG)sizeof(MINMAX));
                S_WRITE_B(spare->flags, spare->numind*(LONG)sizeof(WORD));
                if ( spare->sparse ) {
                    S_WRITE_B(spare->boomlijst, spare->MaxTreeSize*(LONG)sizeof(COMPTREE));
                }
                S_WRITE_B(spare->buffers, spare->bufferssize*(LONG)sizeof(WORD));
            }
        }
    }

    ANNOUNCE(AC.ExpressionList)
    S_WRITE_LIST(AC.ExpressionList);
    for ( i=0; i<AC.ExpressionList.num; ++i ) {
        EXPRESSIONS ex = Expressions + i;
        if ( ex->renum ) {
            S_WRITE_B(ex->renum, sizeof(struct ReNuMbEr));
            /* there is one dynamically allocated buffer for struct ReNuMbEr and
             * symb.lo points to its beginning. the size of the buffer is not
             * stored anywhere but we know it is 2*sizeof(WORD)*N, where N is
             * the number of all vectors, indices, functions and symbols. since
             * funum points into the buffer at a distance 2N-[Number of
             * functions] from symb.lo (see GetTable() in store.c), we can
             * calculate the buffer size by some pointer arithmetic. the size is
             * then written to the file. */
            l = ex->renum->funnum - ex->renum->symb.lo;
            l += ex->renum->funnum - ex->renum->func.lo;
            S_WRITE_B(&l, sizeof(size_t));
            S_WRITE_B(ex->renum->symb.lo, l);
        }
        if ( ex->bracketinfo ) {
            S_WRITE_B(ex->bracketinfo, sizeof(BRACKETINFO));
            S_WRITE_B(ex->bracketinfo->indexbuffer,
ex->bracketinfo->indexbuffersize*sizeof(BRACKETINDEX));
            S_WRITE_B(ex->bracketinfo->bracketbuffer,
ex->bracketinfo->bracketbuffersize*sizeof(WORD));
        }
        if ( ex->newbracketinfo ) {
            S_WRITE_B(ex->newbracketinfo, sizeof(BRACKETINFO));
            S_WRITE_B(ex->newbracketinfo->indexbuffer,
ex->newbracketinfo->indexbuffersize*sizeof(BRACKETINDEX));
            S_WRITE_B(ex->newbracketinfo->bracketbuffer,
ex->newbracketinfo->bracketbuffersize*sizeof(WORD));
        }
        /* don't need to write ex->renumlists */
        if ( ex->inmem ) {
            /* size of the inmem buffer has to be determined. we use the fact
             * that the end of an expression is marked by a zero. */
            w = ex->inmem;
            while ( *w++ ) ;
            l = w - ex->inmem;
            S_WRITE_B(&l, sizeof(size_t));
            S_WRITE_B(ex->inmem, l);
        }
    }

    ANNOUNCE(AC.IndexList)
    S_WRITE_LIST(AC.IndexList);
    S_WRITE_LIST(AC.SetElementList);
    S_WRITE_LIST(AC.SetList);
    S_WRITE_LIST(AC.SymbolList);
    S_WRITE_LIST(AC.VectorList);

    ANNOUNCE(AC.TableBaseList)
    S_WRITE_LIST(AC.TableBaseList);
    for ( i=0; i<AC.TableBaseList.num; ++i ) {
```

```
        /* see struct dbase in minos.h */
        if ( tablebases[i].iblocks ) {
            S_WRITE_B(tablebases[i].iblocks, tablebases[i].info.numberofindexblocks * sizeof(INDEXBLOCK*));
            for ( j=0; j<tablebases[i].info.numberofindexblocks; ++j ) {
                if ( tablebases[i].iblocks[j] ) {
                    S_WRITE_B(tablebases[i].iblocks[j], sizeof(INDEXBLOCK));
                }
            }
        }
        if ( tablebases[i].nblocks ) {
            S_WRITE_B(tablebases[i].nblocks, tablebases[i].info.numberofnamesblocks * sizeof(NAMESBLOCK*));
            for ( j=0; j<tablebases[i].info.numberofnamesblocks; ++j ) {
                if ( tablebases[i].nblocks[j] ) {
                    S_WRITE_B(tablebases[i].nblocks[j], sizeof(NAMESBLOCK));
                }
            }
        }
        S_WRITE_S(tablebases[i].name);
        S_WRITE_S(tablebases[i].fullname);
        S_WRITE_S(tablebases[i].tablenames);
    }

    ANNOUNCE(AC.cbufList)
    S_WRITE_LIST(AC.cbufList);
    for ( i=0; i<AC.cbufList.num; ++i ) {
        S_WRITE_B(cbuf[i].Buffer, cbuf[i].BufferSize*sizeof(WORD));
        /* see inicbufs in comtool.c */
        S_WRITE_B(cbuf[i].lhs, cbuf[i].maxlhs*(LONG)sizeof(WORD*));
        S_WRITE_B(cbuf[i].rhs, cbuf[i].maxrhs*(LONG)(sizeof(WORD*)+2*sizeof(LONG)+2*sizeof(WORD)));
        if ( cbuf[i].boomlijst ) {
            S_WRITE_B(cbuf[i].boomlijst, cbuf[i].MaxTreeSize*(LONG)sizeof(COMPTREE));
        }
    }

    S_WRITE_LIST(AC.AutoSymbolList);
    S_WRITE_LIST(AC.AutoIndexList);
    S_WRITE_LIST(AC.AutoVectorList);
    S_WRITE_LIST(AC.AutoFunctionList);

    S_WRITE_NAMETREE(AC.autonames);

    ANNOUNCE(AC.Streams)
    S_WRITE_B(AC.Streams, AC.MaxNumStreams*(LONG)sizeof(STREAM));
    for ( i=0; i<AC.NumStreams; ++i ) {
        if ( AC.Streams[i].inbuffer ) {
            S_WRITE_B(AC.Streams[i].buffer, AC.Streams[i].inbuffer);
        }
        S_WRITE_S(AC.Streams[i].FoldName);
        S_WRITE_S(AC.Streams[i].name);
    }

    if ( AC.termstack ) {
        S_WRITE_B(AC.termstack, AC.maxtermlevel*(LONG)sizeof(LONG));
    }

    if ( AC.termsortstack ) {
        S_WRITE_B(AC.termsortstack, AC.maxtermlevel*(LONG)sizeof(LONG));
    }

    S_WRITE_B(AC.cmod, AM.MaxTal*4*(LONG)sizeof(UWORD));

    if ( AC.IfHeap ) {
        S_WRITE_B(AC.IfHeap, (LONG)sizeof(LONG)*(AC.MaxIf+1));
        S_WRITE_B(AC.IfCount, (LONG)sizeof(LONG)*(AC.MaxIf+1));
    }

    l = AC.iStop - AC.iBuffer + 2;
    S_WRITE_B(AC.iBuffer, l);

    if ( AC.LabelNames ) {
        S_WRITE_B(AC.LabelNames, AC.MaxLabels*(LONG)(sizeof(UBYTE*)+sizeof(WORD)));
        for ( i=0; i<AC.NumLabels; ++i ) {
            S_WRITE_S(AC.LabelNames[i]);
        }
    }

    S_WRITE_B(AC.FixIndices, AM.OffsetIndex*(LONG)sizeof(WORD));

    if ( AC.termsumcheck ) {
        S_WRITE_B(AC.termsumcheck, AC.maxtermlevel*(LONG)sizeof(WORD));
    }

    S_WRITE_B(AC.WildcardNames, AC.WildcardBufferSize);

    if ( AC.tokens ) {
```

```
        l = AC.toptokens - AC.tokens;
        if ( l ) {
            S_WRITE_B(AC.tokens, l);
        }
    }

    S_WRITE_B(AC.tokenarglevel, AM.MaxParLevel*(LONG)sizeof(WORD));

    S_WRITE_S(AC.Fortran90Kind);

#ifdef WITHPTHREADS
    if ( AC.inputnumbers ) {
        S_WRITE_B(AC.inputnumbers, AC.sizepfirstnum*(LONG)(sizeof(WORD)+sizeof(LONG)));
    }
#endif /* ifdef WITHPTHREADS */

    if ( AC.IfSumCheck ) {
        S_WRITE_B(AC.IfSumCheck, (LONG)sizeof(WORD)*(AC.MaxIf+1));
    }
    if ( AC.CommuteInSet ) {
        S_WRITE_B(AC.CommuteInSet, (LONG)sizeof(WORD)*(AC.SizeCommuteInSet+1));
    }

    S_WRITE_S(AC.CheckpointRunAfter);
    S_WRITE_S(AC.CheckpointRunBefore);

    S_WRITE_S(AC.extrasym);

    /*#] AC :*/
    /*#[ AP :*/

    /* we write AP as a whole and then write all additional data step by step. */

    ANNOUNCE(AP)
    S_WRITE_B(&AP, sizeof(struct P_const));

    S_WRITE_LIST(AP.DollarList);
    for ( i=0; i<AP.DollarList.num; ++i ) {
        DOLLARS d = Dollars + i;
        S_WRITE_DOLLAR(Dollars[i]);
        if ( d->nfactors > 1 ) {
            S_WRITE_B(&(d->factors),sizeof(FACDOLLAR)*d->nfactors);
            for ( j = 0; j < d->nfactors; j++ ) {
                if ( d->factors[j].size > 0 ) {
                    S_WRITE_B(&(d->factors[i].where),sizeof(WORD)*(d->factors[j].size+1));
                }
            }
        }
    }

    S_WRITE_LIST(AP.PreVarList);
    for ( i=0; i<AP.PreVarList.num; ++i ) {
        /* there is one dynamically allocated buffer in struct pReVaR holding
         * the strings name, value and several argnames. the size of the buffer
         * can be calculated by adding up their string lengths. */
        l = strlen((char*)PreVar[i].name) + 1;
        if ( PreVar[i].value ) {
            l += strlen((char*)(PreVar[i].name+l)) + 1;
        }
        for ( j=0; j<PreVar[i].nargs; ++j ) {
            l += strlen((char*)(PreVar[i].name+l)) + 1;
        }
        S_WRITE_B(&l, sizeof(size_t));
        S_WRITE_B(PreVar[i].name, l);
    }

    ANNOUNCE(AP.LoopList)
    S_WRITE_LIST(AP.LoopList);
    for ( i=0; i<AP.LoopList.num; ++i ) {
        S_WRITE_B(DoLoops[i].p.buffer, DoLoops[i].p.size);
        if ( DoLoops[i].type == ONEEXPRESSION ) {
            /* do loops with an expression keep this expression in a dynamically
             * allocated buffer in dollarname */
            S_WRITE_S(DoLoops[i].dollarname);
        }
    }

    S_WRITE_LIST(AP.ProcList);
    for ( i=0; i<AP.ProcList.num; ++i ) {
        if ( Procedures[i].p.size ) {
            if ( Procedures[i].loadmode != 1 ) {
                S_WRITE_B(Procedures[i].p.buffer, Procedures[i].p.size);
            }
            else {
                for ( j=0; j<AP.ProcList.num; ++j ) {
                    if ( Procedures[i].p.buffer == Procedures[j].p.buffer ) {
                        break;
```

```
                    }
                }
                if ( j == AP.ProcList.num ) {
                    MesPrint("Error writing procedures to recovery file!");
                }
                S_WRITE_B(&j, sizeof(int));
            }
        }
        S_WRITE_S(Procedures[i].name);
    }

    S_WRITE_B(AP.PreSwitchStrings, (AP.NumPreSwitchStrings+1)*(LONG)sizeof(UBYTE*));
    for ( i=1; i<=AP.PreSwitchLevel; ++i ) {
        S_WRITE_S(AP.PreSwitchStrings[i]);
    }

    S_WRITE_B(AP.preStart, AP.pSize);

    S_WRITE_S(AP.procedureExtension);
    S_WRITE_S(AP.cprocedureExtension);

    S_WRITE_B(AP.PreAssignStack, AP.MaxPreAssignLevel*(LONG)sizeof(LONG));
    S_WRITE_B(AP.PreIfStack, AP.MaxPreIfLevel*(LONG)sizeof(int));
    S_WRITE_B(AP.PreSwitchModes, (AP.NumPreSwitchStrings+1)*(LONG)sizeof(int));
    S_WRITE_B(AP.PreTypes, (AP.MaxPreTypes+1)*(LONG)sizeof(int));

    /*#] AP :*/
    /*#[ AR :*/

    ANNOUNCE(AR)
    /* to remember which entry in AR.Fscr corresponds to the infile */
    l = AR.infile - AR.Fscr;
    S_WRITE_B(&l, sizeof(LONG));

    /* write the FILEHANDLEs */
    S_WRITE_B(AR.outfile, sizeof(FILEHANDLE));
    l = AR.outfile->POfull - AR.outfile->PObuffer;
    if ( l ) {
        S_WRITE_B(AR.outfile->PObuffer, l*sizeof(WORD));
    }
    S_WRITE_B(AR.hidefile, sizeof(FILEHANDLE));
    l = AR.hidefile->POfull - AR.hidefile->PObuffer;
    if ( l ) {
        S_WRITE_B(AR.hidefile->PObuffer, l*sizeof(WORD));
    }

    S_WRITE_B(&AR.StoreData.Fill, sizeof(POSITION));
    if ( ISNOTZEROPOS(AR.StoreData.Fill) ) {
        S_WRITE_B(&AR.StoreData, sizeof(FILEDATA));
    }

    S_WRITE_B(&AR.DefPosition, sizeof(POSITION));

    l = TimeCPU(1); l = -l;
    S_WRITE_B(&l, sizeof(LONG));

    ANNOUNCE(AR.InInBuf)
    S_WRITE_B(&AR.InInBuf, sizeof(LONG));
    S_WRITE_B(&AR.InHiBuf, sizeof(LONG));

    S_WRITE_B(&AR.NoCompress, sizeof(int));
    S_WRITE_B(&AR.gzipCompress, sizeof(int));

    S_WRITE_B(&AR.outtohide, sizeof(int));

    S_WRITE_B(&AR.GetFile, sizeof(WORD));
    S_WRITE_B(&AR.KeptInHold, sizeof(WORD));
    S_WRITE_B(&AR.BracketOn, sizeof(WORD));
    S_WRITE_B(&AR.MaxBracket, sizeof(WORD));
    S_WRITE_B(&AR.CurDum, sizeof(WORD));
    S_WRITE_B(&AR.DeferFlag, sizeof(WORD));
    S_WRITE_B(&AR.TePos, sizeof(WORD));
    S_WRITE_B(&AR.sLevel, sizeof(WORD));
    S_WRITE_B(&AR.Stage4Name, sizeof(WORD));
    S_WRITE_B(&AR.GetOneFile, sizeof(WORD));
    S_WRITE_B(&AR.PolyFun, sizeof(WORD));
    S_WRITE_B(&AR.PolyFunInv, sizeof(WORD));
    S_WRITE_B(&AR.PolyFunType, sizeof(WORD));
    S_WRITE_B(&AR.PolyFunExp, sizeof(WORD));
    S_WRITE_B(&AR.PolyFunVar, sizeof(WORD));
    S_WRITE_B(&AR.PolyFunPow, sizeof(WORD));
    S_WRITE_B(&AR.Eside, sizeof(WORD));
    S_WRITE_B(&AR.MaxDum, sizeof(WORD));
    S_WRITE_B(&AR.level, sizeof(WORD));
    S_WRITE_B(&AR.expchanged, sizeof(WORD));
    S_WRITE_B(&AR.expflags, sizeof(WORD));
    S_WRITE_B(&AR.CurExpr, sizeof(WORD));
```

```
    S_WRITE_B(&AR.SortType, sizeof(WORD));
    S_WRITE_B(&AR.ShortSortCount, sizeof(WORD));

#ifdef WITHPTHREADS
    for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
        S_WRITE_B(&(AB[j]->R.wranfnpair1), sizeof(int));
        S_WRITE_B(&(AB[j]->R.wranfnpair2), sizeof(int));
        S_WRITE_B(&(AB[j]->R.wranfcall), sizeof(int));
        S_WRITE_B(&(AB[j]->R.wranfseed), sizeof(ULONG));
        S_WRITE_B(&(AB[j]->R.wranfia),sizeof(ULONG *));
        if ( AB[j]->R.wranfia ) {
            S_WRITE_B(AB[j]->R.wranfia, sizeof(ULONG)*AB[j]->R.wranfnpair2);
        }
    }
#else
    S_WRITE_B(&(AR.wranfnpair1), sizeof(int));
    S_WRITE_B(&(AR.wranfnpair2), sizeof(int));
    S_WRITE_B(&(AR.wranfcall), sizeof(int));
    S_WRITE_B(&(AR.wranfseed), sizeof(ULONG));
    S_WRITE_B(&(AR.wranfia),sizeof(ULONG *));
    if ( AR.wranfia ) {
        S_WRITE_B(AR.wranfia, sizeof(ULONG)*AR.wranfnpair2);
    }
#endif

    /*#] AR :*/
    /*#[ AO :*/
/*
    We copy all non-pointer variables.
*/
    ANNOUNCE(AO)
    l = sizeof(A.O) - ((UBYTE *)(&(A.O.NumInBrack))-(UBYTE *)(&A.O));
    S_WRITE_B(&(A.O.NumInBrack),l);
/*
    Now the variables in OptimizeResult
*/
    S_WRITE_B(&(A.O.OptimizeResult),sizeof(OPTIMIZERESULT));
    if ( A.O.OptimizeResult.codesize > 0 ) {
        S_WRITE_B(A.O.OptimizeResult.code,A.O.OptimizeResult.codesize*sizeof(WORD));
    }
    S_WRITE_S(A.O.OptimizeResult.nameofexpr);
/*
    And now the dictionaries.
    We write each dictionary to a buffer and get the size of that buffer.
    Then we write the size and the buffer.
*/
    for ( i = 0; i < AO.NumDictionaries; i++ ) {
        l = DictToBytes(AO.Dictionaries[i],(UBYTE *)(AT.WorkPointer));
        S_WRITE_B(&l,sizeof(LONG));
        S_WRITE_B(AT.WorkPointer,l);
    }

    /*#] AO :*/
    /*#[ PF :*/
#ifdef WITHMPI
    S_WRITE_B(&PF.numtasks, sizeof(int));
    S_WRITE_B(&PF.rhsInParallel, sizeof(int));
    S_WRITE_B(&PF.exprbufsize, sizeof(int));
    S_WRITE_B(&PF.log, sizeof(int));
#endif
    /*#] PF :*/

#ifdef WITHPTHREADS

    ANNOUNCE(GetTimerInfo)
/*
    write timing information of individual threads
*/
    i = GetTimerInfo(&longp,&longpp);
    S_WRITE_B(&i, sizeof(int));
    S_WRITE_B(longp, i*(LONG)sizeof(LONG));
    S_WRITE_B(&i, sizeof(int));
    S_WRITE_B(longpp, i*(LONG)sizeof(LONG));

#endif

    S_FLUSH_B /* because we will call fwrite() directly in the following code */

    /* save length of data at the beginning of the file */
    ANNOUNCE(file length)
    SETBASEPOSITION(pos, (ftell(fd)));
    fseek(fd, 0, SEEK_SET);
    if ( fwrite(&pos, 1, sizeof(POSITION), fd) != sizeof(POSITION) ) return(__LINE__);
    fseek(fd, BASEPOSITION(pos), SEEK_SET);

    ANNOUNCE(file close)
    if ( fclose(fd) ) return(__LINE__);
```

```
#ifdef WITHMPI
    if ( PF.me == MASTER ) {
#endif
/*
        copy store file if necessary
*/
        ANNOUNCE(copy store file)
        if ( ISNOTZEROPOS(AR.StoreData.Fill) ) {
            if ( CopyFile(FG.fname, storefile) ) return(__LINE__);
        }
/*
        copy sort file if necessary
*/
        ANNOUNCE(copy sort file)
        if ( AR.outfile->handle >= 0 ) {
            if ( CopyFile(AR.outfile->name, sortfile) ) return(__LINE__);
        }
/*
        copy hide file if necessary
*/
        ANNOUNCE(copy hide file)
        if ( AR.hidefile->handle >= 0 ) {
            if ( CopyFile(AR.hidefile->name, hidefile) ) return(__LINE__);
        }
#ifdef WITHMPI
    }
    /*
     * For ParFORM, the renaming will be performed after the master got
     * all recovery files from the slaves.
     */
#else
/*
    make the intermediate file the recovery file
*/
    ANNOUNCE(rename intermediate file)
    if ( rename(intermedfile, recoveryfile) ) return(__LINE__);

    done_snapshot = 1;

    MesPrint("done."); fflush(0);
#endif

#ifdef PRINTDEBUG
    print_M();
    print_C();
    print_P();
    print_R();
#endif

    return(0);
}

/*
    #] DoSnapshot :
    #[ DoCheckpoint :
*/

void DoCheckpoint(int moduletype)
{
    int error;
    LONG timestamp = TimeWallClock(1);
#ifdef WITHMPI
    if(PF.me == MASTER){
#endif
    if ( timestamp - AC.CheckpointStamp >= AC.CheckpointInterval ) {
        char argbuf[20];
        int retvalue = 0;
        if ( AC.CheckpointRunBefore ) {
            size_t l, l2;
            char *str;
            l = strlen(AC.CheckpointRunBefore);
            NumToStr((UBYTE*)argbuf, AC.CModule);
            l2 = strlen(argbuf);
            str = (char*)Malloc1(l+l2+2, "callbefore");
            strcpy(str, AC.CheckpointRunBefore);
            *(str+l) = ' ';
            strcpy(str+l+1, argbuf);
            retvalue = system(str);
            M_free(str, "callbefore");
            if ( retvalue ) {
                MesPrint("Script returned error -> no recovery file will be created.");
            }
        }
#ifdef WITHMPI
        /* Confirm slaves to make snapshots. */
        PF_BroadcastNumber(retvalue == 0);
#endif
```

```
        if ( retvalue == 0 ) {
            if ( (error = DoSnapshot(moduletype)) ) {
                MesPrint("Error creating recovery files: %d", error);
            }
#ifdef WITHMPI
            {
            int i;
            /*get recovery files from slaves:*/
            for(i=1; i<PF.numtasks;i++){
                FILE *fd;
                const char *tmpnam = PF_recoveryfile('m', i, 1);
                fd = fopen(tmpnam, "w");
                if(fd == NULL){
                    MesPrint("Error opening recovery file for slave %d",i);
                    Terminate(-1);
                }/*if(fd == NULL)*/
                retvalue=PF_RecvFile(i,fd);
                if(retvalue<=0){
                    MesPrint("Error receiving recovery file from slave %d",i);
                    Terminate(-1);
                }/*if(retvalue<=0)*/
                fclose(fd);
            }/*for(i=0; i<PF.numtasks;i++)*/
            /*
             * Make the intermediate files the recovery files.
             */
            ANNOUNCE(rename intermediate file)
            for ( i = 0; i < PF.numtasks; i++ ) {
                const char *src = PF_recoveryfile('m', i, 1);
                const char *dst = PF_recoveryfile('m', i, 0);
                if ( rename(src, dst) ) {
                    MesPrint("Error renaming recovery file %s -> %s", src, dst);
                }
            }
            done_snapshot = 1;
            MesPrint("done."); fflush(0);
            }
#endif
        }
        if ( AC.CheckpointRunAfter ) {
            size_t l, l2;
            char *str;
            l = strlen(AC.CheckpointRunAfter);
            NumToStr((UBYTE*)argbuf, AC.CModule);
            l2 = strlen(argbuf);
            str = (char*)Malloc1(l+l2+2, "callafter");
            strcpy(str, AC.CheckpointRunAfter);
            *(str+l) = ' ';
            strcpy(str+l+1, argbuf);
            retvalue = system(str);
            M_free(str, "callafter");
            if ( retvalue ) {
                MesPrint("Error calling script after recovery.");
            }
        }
        AC.CheckpointStamp = TimeWallClock(1);
    }
#ifdef WITHMPI
    else{/* timestamp - AC.CheckpointStamp < AC.CheckpointInterval*/
        /* The slaves don't need to make snapshots. */
        PF_BroadcastNumber(0);
    }
    }/*if(PF.me == MASTER)*/
    else{/*Slave*/
        int i;
        /* Check if the slave needs to make a snapshot. */
        if ( PF_BroadcastNumber(0) ) {
            error = DoSnapshot(moduletype);
            if(error == 0){
                FILE *fd;
                /*
                 * Send the recovery file to the master. Note that no renaming
                 * has been performed and what we have to send is actually sitting
                 * in the intermediate file.
                 */
                fd = fopen(intermedfile, "r");
                i=PF_SendFile(MASTER, fd);/*if fd==NULL, PF_SendFile seds to a slave the failure tag*/
                if(fd == NULL)
                    Terminate(-1);
                fclose(fd);
                if(i<=0)
                    Terminate(-1);
                /*Now the slave need not the recovery file so remove it:*/
                remove(intermedfile);
            }
            else{
                /*send the error tag to the master:*/
```

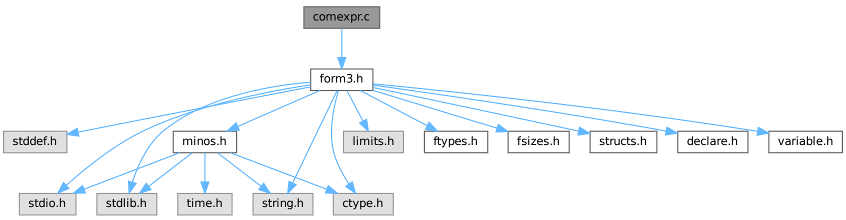
```
                PF_SendFile(MASTER,NULL);/*if fd==NULL, PF_SendFile seds to a slave the failure tag*/
                Terminate(-1);
            }
            done_snapshot = 1;
        }/*if(tag=PF_DATA_MSGTAG)*/
    }/*if(PF.me != MASTER)*/
#endif
}

/*
    #] DoCheckpoint :
*/
```

4.7 comexpr.c File Reference

`#include "form3.h"`

Include dependency graph for comexpr.c:



Data Structures

- struct **id_options**

Functions

- int CoLocal (UBYTE ∗inp)
- int CoGlobal (UBYTE ∗inp)
- int CoLocalFactorized (UBYTE ∗inp)
- int CoGlobalFactorized (UBYTE ∗inp)
- int DoExpr (UBYTE ∗inp, int type, int par)
- int CoIdOld (UBYTE ∗inp)
- int CoId (UBYTE ∗inp)
- int CoIdNew (UBYTE ∗inp)
- int CoDisorder (UBYTE ∗inp)
- int CoMany (UBYTE ∗inp)
- int CoMulti (UBYTE ∗inp)
- int CoIfMatch (UBYTE ∗inp)
- int CoIfNoMatch (UBYTE ∗inp)
- int CoOnce (UBYTE ∗inp)
- int CoOnly (UBYTE ∗inp)
- int CoSelect (UBYTE ∗inp)
- int CoIdExpression (UBYTE ∗inp, int type)
- int CoMultiply (UBYTE ∗inp)
- int CoFill (UBYTE ∗inp)
- int CoFillExpression (UBYTE ∗inp)
- int CoPrintTable (UBYTE ∗inp)
- int CoAssign (UBYTE ∗inp)
- int CoDeallocateTable (UBYTE ∗inp)

4.7.1 Detailed Description

Compiler routines for statements that involve algebraic expressions. These involve definitions, id-statements, the multiply statement and the fill statement.

Definition in file comexpr.c.

4.7.2 Function Documentation

4.7.2.1 CoLocal()

```
int CoLocal (
            UBYTE * inp )
```

Definition at line 66 of file comexpr.c.

4.7.2.2 CoGlobal()

```
int CoGlobal (
            UBYTE * inp )
```

Definition at line 73 of file comexpr.c.

4.7.2.3 CoLocalFactorized()

```
int CoLocalFactorized (
            UBYTE * inp )
```

Definition at line 80 of file comexpr.c.

4.7.2.4 CoGlobalFactorized()

```
int CoGlobalFactorized (
            UBYTE * inp )
```

Definition at line 87 of file comexpr.c.

4.7.2.5 DoExpr()

```
int DoExpr (
            UBYTE * inp,
            int type,
            int par )
```

Definition at line 96 of file comexpr.c.

4.7.2.6 CoIdOld()

```
int CoIdOld (
            UBYTE * inp )
```

Definition at line 384 of file comexpr.c.

4.7.2.7 CoId()

```
int CoId (
            UBYTE * inp )
```

Definition at line 395 of file comexpr.c.

4.7.2.8 CoIdNew()

```
int CoIdNew (
            UBYTE * inp )
```

Definition at line 406 of file comexpr.c.

4.7.2.9 CoDisorder()

```
int CoDisorder (
            UBYTE * inp )
```

Definition at line 417 of file comexpr.c.

4.7.2.10 CoMany()

```
int CoMany (
            UBYTE * inp )
```

Definition at line 428 of file comexpr.c.

4.7.2.11 CoMulti()

```
int CoMulti (
            UBYTE * inp )
```

Definition at line 439 of file comexpr.c.

4.7.2.12 CoIfMatch()

```
int CoIfMatch (
            UBYTE * inp )
```

Definition at line 450 of file comexpr.c.

4.7.2.13 CoIfNoMatch()

```
int CoIfNoMatch (
        UBYTE * inp )
```

Definition at line 461 of file comexpr.c.

4.7.2.14 CoOnce()

```
int CoOnce (
        UBYTE * inp )
```

Definition at line 472 of file comexpr.c.

4.7.2.15 CoOnly()

```
int CoOnly (
        UBYTE * inp )
```

Definition at line 483 of file comexpr.c.

4.7.2.16 CoSelect()

```
int CoSelect (
        UBYTE * inp )
```

Definition at line 494 of file comexpr.c.

4.7.2.17 CoIdExpression()

```
int CoIdExpression (
        UBYTE * inp,
        int type )
```

Definition at line 507 of file comexpr.c.

4.7.2.18 CoMultiply()

```
int CoMultiply (
        UBYTE * inp )
```

Definition at line 1031 of file comexpr.c.

4.7.2.19 CoFill()

```
int CoFill (
        UBYTE * inp )
```

Definition at line 1070 of file comexpr.c.

4.7.2.20 CoFillExpression()

```
int CoFillExpression (
            UBYTE * inp )
```

Definition at line 1356 of file comexpr.c.

4.7.2.21 CoPrintTable()

```
int CoPrintTable (
            UBYTE * inp )
```

Definition at line 1748 of file comexpr.c.

4.7.2.22 CoAssign()

```
int CoAssign (
            UBYTE * inp )
```

Definition at line 1924 of file comexpr.c.

4.7.2.23 CoDeallocateTable()

```
int CoDeallocateTable (
            UBYTE * inp )
```

Definition at line 1982 of file comexpr.c.

4.8 comexpr.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

/*
```

```
    #[ Includes : compi2.c

    File contains most of what has to do with compiling expressions.
    Main supporting file: token.c
*/

#include "form3.h"

static struct id_options {
    UBYTE *name;
    int code;
    int dummy;
} IdOptions[] = {
     {(UBYTE *)"multi",      SUBMULTI     ,0}
    ,{(UBYTE *)"many",       SUBMANY      ,0}
    ,{(UBYTE *)"only",       SUBONLY      ,0}
    ,{(UBYTE *)"once",       SUBONCE      ,0}
    ,{(UBYTE *)"ifmatch",    SUBAFTER     ,0}
    ,{(UBYTE *)"ifnomatch",  SUBAFTERNOT  ,0}
    ,{(UBYTE *)"ifnotmatch", SUBAFTERNOT  ,0}
    ,{(UBYTE *)"disorder",   SUBDISORDER  ,0}
    ,{(UBYTE *)"select",     SUBSELECT    ,0}
    ,{(UBYTE *)"all",        SUBALL       ,0}
};

/*
    #] Includes :
    #[ CoLocal :
*/

int CoLocal(UBYTE *inp) { return(DoExpr(inp,LOCALEXPRESSION,0)); }

/*
    #] CoLocal :
    #[ CoGlobal :
*/

int CoGlobal(UBYTE *inp) { return(DoExpr(inp,GLOBALEXPRESSION,0)); }

/*
    #] CoGlobal :
    #[ CoLocalFactorized :
*/

int CoLocalFactorized(UBYTE *inp) { return(DoExpr(inp,LOCALEXPRESSION,1)); }

/*
    #] CoLocalFactorized :
    #[ CoGlobalFactorized :
*/

int CoGlobalFactorized(UBYTE *inp) { return(DoExpr(inp,GLOBALEXPRESSION,1)); }

/*
    #] CoGlobalFactorized :
    #[ DoExpr:


*/

int DoExpr(UBYTE *inp, int type, int par)
{
    GETIDENTITY
    int error = 0;
    UBYTE *p, *q, c;
    WORD *w, i, j = 0, c1, c2, *OldWork = AT.WorkPointer, osize;
    WORD jold = 0;
    POSITION pos;
    while ( *inp == ',' ) inp++;
    if ( par ) AC.ToBeInFactors = 1;
    else       AC.ToBeInFactors = 0;
    p = inp;
    while ( *p && *p != '=' ) {
        if ( *p == '(' ) SKIPBRA4(p)
        else if ( *p == '{' ) SKIPBRA5(p)
        else if ( *p == '[' ) SKIPBRA1(p)
        else p++;
    }
    if ( *p ) {     /* Variety with the = sign */
        if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
            MesPrint("&Illegal name for expression");
            error = 1;
            if ( q[-1] == '_' ) {
                while ( FG.cTable[*q] < 2 || *q == '_' ) q++;
            }
        }
        else {
```

```
            c = *q; *q = 0;
            if ( GetVar(inp,&c1,&c2,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
                if ( c1 == CEXPRESSION ) {
                    if ( Expressions[c2].status == STOREDEXPRESSION ) {
                        MesPrint("&Illegal attempt to overwrite a stored expression");
                        error = 1;
                    }
                    else {
                        HighWarning("Expression is replaced by new definition");
                        if ( AO.OptimizeResult.nameofexpr != NULL &&
                            StrCmp(inp,AO.OptimizeResult.nameofexpr) == 0 ) {
                            ClearOptimize();
                        }
                        if ( Expressions[c2].status != DROPPEDEXPRESSION ) {
                            w = &(Expressions[c2].status);
                            if ( *w == LOCALEXPRESSION || *w == SKIPLEXPRESSION )
                                *w = DROPLEXPRESSION;
                            else if ( *w == GLOBALEXPRESSION || *w == SKIPGEXPRESSION )
                                *w = DROPGEXPRESSION;
                            else if ( *w == HIDDENLEXPRESSION )
                                *w = DROPHLEXPRESSION;
                            else if ( *w == HIDDENGEXPRESSION )
                                *w = DROPHGEXPRESSION;
                        }
                        AC.TransEname = Expressions[c2].name;
                        j = EntVar(CEXPRESSION,0,type,0,0,0);
                        Expressions[j].node = Expressions[c2].node;
                        Expressions[c2].replace = j;
                    }
                }
                else {
                    MesPrint("&name of expression is also name of a variable");
                    error = 1;
                    j = EntVar(CEXPRESSION,inp,type,0,0,0);
                }
                jold = c2;
            }
            else {
/*
                Here we have to worry about reuse of the expression in the
                same module. That will need AS.Oldvflags but that may not
                be defined or have the proper value.
*/
                j = EntVar(CEXPRESSION,inp,type,0,0,0);
                jold = j;
            }
            *q = c;
            OldWork = w = AT.WorkPointer;
            *w++ = TYPEEXPRESSION;
            *w++ = 3+SUBEXPSIZE;
            *w++ = j;
            AC.ProtoType = w;
            AR.CurExpr = j;             /* Block expression j */
            *w++ = SUBEXPRESSION;
            *w++ = SUBEXPSIZE;
            *w++ = j;
            *w++ = 1;
            *w++ = AC.cbufnum;
            FILLSUB(w)

            if ( c == '(' ) {
                while ( *q == ',' || *q == '(' ) {
                    inp = q+1;
                    if ( ( q = SkipAName(inp) ) == 0 ) {
                        MesPrint("&Illegal name for expression argument");
                        error = 1;
                        q = p - 1;
                        break;
                    }
                    c = *q; *q = 0;
                    if ( GetVar(inp,&c1,&c2,ALLVARIABLES,WITHAUTO) < 0 ) c1 = -1;
                    switch ( c1 ) {
                        case CSYMBOL :
                            *w++ = SYMTOSYM; *w++ = 4; *w++ = c2; *w++ = 0;
                            break;
                        case CINDEX :
                            *w++ = INDTOIND; *w++ = 4;
                            *w++ = c2 + AM.OffsetIndex; *w++ = 0;
                            break;
                        case CVECTOR :
                            *w++ = VECTOVEC; *w++ = 4;
                            *w++ = c2 + AM.OffsetVector; *w++ = 0;
                            break;
                        case CFUNCTION :
                            *w++ = FUNTOFUN; *w++ = 4; *w++ = c2 + FUNCTION; *w++ = 0;
                            break;
                        default :
```

```
                            MesPrint("&Illegal expression parameter: %s",inp);
                            error = 1;
                            break;
                        }
                        *q = c;
                    }
                    if ( *q != ')' || q+1 != p ) {
                        MesPrint("&Illegal use of arguments for expression");
                        error = 1;
                    }
                    AC.ProtoType[1] = w - AC.ProtoType;
                }
                else if ( c != '=' ) {
/*
                    The dummy accepted L F := RHS;
*/
                    MesPrint("&Illegal LHS for expression definition");
                    error = 1;
                }
                *w++ = 1;
                *w++ = 1;
                *w++ = 3;
                *w++ = 0;
                SeekScratch(AR.outfile,&pos);
                Expressions[j].counter = 1;
                Expressions[j].onfile = pos;
                Expressions[j].whichbuffer = 0;
#ifdef PARALLELCODE
                Expressions[j].partodo = AC.inparallelflag;
#endif
                OldWork[2] = w - OldWork - 3;
                AT.WorkPointer = w;
/*
                Writing the expression prototype to disk and to the compiler
                buffer is done only after the RHS has been compiled because
                we don't know the number of the main level RHS yet.
*/
            }
            inp = p+1;
            ClearWildcardNames();
            osize = AC.ProtoType[1]; AC.ProtoType[1] = SUBEXPSIZE;
            PutInVflags(jold);
            if ( ( i = CompileAlgebra(inp,RHSIDE,AC.ProtoType) ) < 0 ) {
                AC.ProtoType[1] = osize;
                error = 1;
            }
            else if ( error == 0 ) {
                AC.ProtoType[1] = osize;
                AC.ProtoType[2] = i;
                if ( PutOut(BHEAD OldWork+2,&pos,AR.outfile,0) < 0 ) {
                    MesPrint("&Cannot create expression");
                    error = -1;
                }
                else {
                    Expressions[j].sizeprototype = OldWork[2];
                    OldWork[2] = 4+SUBEXPSIZE;
                    OldWork[4] = SUBEXPSIZE;
                    OldWork[5] = i;
                    OldWork[SUBEXPSIZE+3] = 1;
                    OldWork[SUBEXPSIZE+4] = 1;
                    OldWork[SUBEXPSIZE+5] = 3;
                    OldWork[SUBEXPSIZE+6] = 0;
                    if ( PutOut(BHEAD OldWork+2,&pos,AR.outfile,0) < 0
                    || FlushOut(&pos,AR.outfile,0) ) {
                        MesPrint("&Cannot create expression");
                        error = -1;
                    }
                    AR.outfile->POfull = AR.outfile->POfill;
                }
                OldWork[2] = j;
/*
                Seems unnecessary (13-feb-2018)

                AddNtoL(OldWork[1],OldWork);
*/
                AT.WorkPointer = OldWork;
                if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
            }
            AC.ToBeInFactors = 0;
        }
        else {  /* Variety in which expressions change property */
/*
                This code got a major revision because it didn't
                take hidden expressions into account. (1-jun-2010 JV)
*/
            do {
                if ( ( q = SkipAName(inp) ) == 0 ) {
```

```
                MesPrint("&Illegal name(s) for expression(s)");
                return(1);
            }
            c = *q; *q = 0;
            if ( GetName(AC.exprnames,inp,&c2,NOAUTO) == NAMENOTFOUND ) {
                MesPrint("&%s is not a valid expression",inp);
                error = 1;
            }
            else {
                w = &(Expressions[c2].status);
                if ( type == LOCALEXPRESSION ) {
                    switch ( *w ) {
                        case GLOBALEXPRESSION:
                            *w = LOCALEXPRESSION;
                            break;
                        case SKIPGEXPRESSION:
                            *w = SKIPLEXPRESSION;
                            break;
                        case DROPGEXPRESSION:
                            *w = DROPLEXPRESSION;
                            break;
                        case HIDDENGEXPRESSION:
                            *w = HIDDENLEXPRESSION;
                            break;
                        case HIDEGEXPRESSION:
                            *w = HIDELEXPRESSION;
                            break;
                        case UNHIDEGEXPRESSION:
                            *w = UNHIDELEXPRESSION;
                            break;
                        case INTOHIDEGEXPRESSION:
                            *w = INTOHIDELEXPRESSION;
                            break;
                        case DROPHGEXPRESSION:
                            *w = DROPHLEXPRESSION;
                            break;
                    }
                }
                else if ( type == GLOBALEXPRESSION ) {
                    switch ( *w ) {
                        case LOCALEXPRESSION:
                            *w = GLOBALEXPRESSION;
                            break;
                        case SKIPLEXPRESSION:
                            *w = SKIPGEXPRESSION;
                            break;
                        case DROPLEXPRESSION:
                            *w = DROPGEXPRESSION;
                            break;
                        case HIDDENLEXPRESSION:
                            *w = HIDDENGEXPRESSION;
                            break;
                        case HIDELEXPRESSION:
                            *w = HIDEGEXPRESSION;
                            break;
                        case UNHIDELEXPRESSION:
                            *w = UNHIDEGEXPRESSION;
                            break;
                        case INTOHIDELEXPRESSION:
                            *w = INTOHIDEGEXPRESSION;
                            break;
                        case DROPHLEXPRESSION:
                            *w = DROPHGEXPRESSION;
                            break;
                    }
                }
/*
                old code
                if ( type != LOCALEXPRESSION || *w != STOREDEXPRESSION )
                        *w = type;
*/
            }
            *q = c; inp = q+1;
        } while ( c == ',' );
        if ( c ) {
            MesPrint("&Illegal object in local or global redefinition");
            error = 1;
        }
    }
    return(error);
}

/*
    #] DoExpr:
    #[ CoIdOld :
*/

```

```
int CoIdOld(UBYTE *inp)
{
    AC.idoption = 0;
    return(CoIdExpression(inp,TYPEIDOLD));
}

/*
    #] CoIdOld :
    #[ CoId :
*/

int CoId(UBYTE *inp)
{
    AC.idoption = 0;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoId :
    #[ CoIdNew :
*/

int CoIdNew(UBYTE *inp)
{
    AC.idoption = 0;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoIdNew :
    #[ CoDisorder :
*/

int CoDisorder(UBYTE *inp)
{
    AC.idoption = SUBDISORDER;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoDisorder :
    #[ CoMany :
*/

int CoMany(UBYTE *inp)
{
    AC.idoption = SUBMANY;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoMany :
    #[ CoMulti :
*/

int CoMulti(UBYTE *inp)
{
    AC.idoption = SUBMULTI;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoMulti :
    #[ CoIfMatch :
*/

int CoIfMatch(UBYTE *inp)
{
    AC.idoption = SUBAFTER;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoIfMatch :
    #[ CoIfNoMatch :
*/

int CoIfNoMatch(UBYTE *inp)
{
    AC.idoption = SUBAFTERNOT;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoIfNoMatch :
    #[ CoOnce :
*/
```

```

int CoOnce(UBYTE *inp)
{
    AC.idoption = SUBONCE;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoOnce :
    #[ CoOnly :
*/

int CoOnly(UBYTE *inp)
{
    AC.idoption = SUBONLY;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoOnly :
    #[ CoSelect :
*/

int CoSelect(UBYTE *inp)
{
    AC.idoption = SUBSELECT;
    return(CoIdExpression(inp,TYPEIDNEW));
}

/*
    #] CoSelect :
    #[ CoIdExpression :

    First finish dealing with secondary keywords
*/

int CoIdExpression(UBYTE *inp, int type)
{
    GETIDENTITY
    int i, j, idhead, error = 0, MinusSign = 0, opt, retcode;
    WORD *w, *s, *m, *mm, *ww, *FirstWork, *OldWork, c1, numsets = 0,
        oldnumrhs, *ow, oldEside;
    UBYTE *p, *pp, c;
    CBUF *C = cbuf+AC.cbufnum;
    LONG oldcpointer, x;
    FirstWork = OldWork = AT.WorkPointer;
/*
    Don't forget to change in StudyPattern if we change/add_to the
    following setup.
    if ( type == TYPEIF ) idhead = IDHEAD-1;
    else
*/
    idhead = IDHEAD;
    AR.CurExpr = -1;
    w = AT.WorkPointer;
    *w++ = type;
    *w++ = idhead + SUBEXPSIZE;
    w++;
    if ( idhead >= IDHEAD ) *w++ = -1;
#if IDHEAD > 4
    for ( i = 4; i < idhead; i++ ) *w++ = 0;
#endif
    while ( *inp == ',' ) inp++;
    p = inp;
    if ( AC.idoption == SUBSELECT ) {
        p--;
        goto findsets;
    }
    else if ( ( AC.idoption == SUBAFTER ) || ( AC.idoption == SUBAFTERNOT ) ) {
        while ( *p && *p != '=' && *p != ',' ) {
            if ( *p == '(' ) SKIPBRA4(p)
            else if ( *p == '{' ) SKIPBRA5(p)
            else if ( *p == '[' ) SKIPBRA1(p)
            else p++;
        }
        if ( *p == '=' || *inp != '-' || inp[1] != '>' ) {
            MesPrint("&Illegal use if if[no]match in id statement");
            error = 1; goto AllDone;
        }
        if ( *p == 0 ) {
            MesPrint("&id-statement without = sign");
            error = 1; goto AllDone;
        }
        inp += 2; pp = inp;
        goto readlabel;
    }
    for(;;) {
```

```
        while ( *p && *p != '=' && *p != ',' ) {
            if ( *p == '(' ) SKIPBRA4(p)
            else if ( *p == '{' ) SKIPBRA5(p)
            else if ( *p == '[' ) SKIPBRA1(p)
            else p++;
        }
        if ( *p == '=' ) break;
        if ( *p == 0 ) {
            MesPrint("&id-statement without = sign");
            error = 1; goto AllDone;
        }
/*
        We have either a secondary option or a syntax error
*/
        pp = inp;
        while ( FG.cTable[*pp] == 0 ) pp++;
        c = *pp; *pp = 0;
        i = sizeof(IdOptions)/sizeof(struct id_options);
        while ( --i >= 0 ) {
            if ( StrICmp(inp,IdOptions[i].name) == 0 ) break;
        }
        if ( i < 0 ) {
            MesPrint("&Illegal option %s in id-statement",inp);
            *pp = c; error = 1; p++; inp = p; continue;
        }
        opt = IdOptions[i].code;
        *pp = c;
        inp = pp+1;
        switch ( opt ) {
            case SUBDISORDER:
                if ( pp != p ) goto IllField;
                AC.idoption |= SUBDISORDER;
                p++; inp = p;
                break;
            case SUBSELECT:
                if ( p != pp ) goto IllField;
                if ( ( AC.idoption & SUBMASK ) != 0 ) {
                    if ( AC.idoption == SUBMULTI && type == TYPEIF ) {}
                    else {
                        MesPrint("&Conflicting options in id-statement");
                        error = 1;
                    }
                }
findsets:;
/*
                Now we read the sets
*/
                numsets = 0;
                for(;;) {
                    inp = ++p;
                    while ( *p && *p != '=' && *p != ',' ) {
                        if ( *p == '(' ) SKIPBRA4(p)
                        else if ( *p == '{' ) SKIPBRA5(p)
                        else if ( *p == '[' ) SKIPBRA1(p)
                        else p++;
                    }
                    if ( *p == '=' ) break;
                    if ( *p == 0 ) {
                        MesPrint("&id-statement without = sign");
                        error = 1; goto AllDone;
                    }
/*
                    We have a set at inp.
*/
                    if ( *inp == '{' ) {
                        if ( p[-1] != '}' ) {
                            c = *p; *p = 0;
                            MesPrint("&Illegal temporary set: %s",inp);
                            error = 1; *p = c;
                        }
                        else {
                            inp++;
                            c = p[-1]; p[-1] = 0;
                            c1 = DoTempSet(inp,p-1);
                            *w++ = c1;
                            p[-1] = c;
                            numsets++;
                            if ( w[-1] < 0 ) error = 1;
                        }
                    }
                    else {
                        c = *p; *p = 0;
                        if ( GetName(AC.varnames,inp,&c1,NOAUTO) != CSET ) {
                            MesPrint("&%s is not a set",inp);
                            error = 1;
                        }
                        else {
```

```
                        if ( c1 < AM.NumFixedSets ) {
                            MesPrint("&Built in sets are not allowed in the select option");
                            error = 1;
                        }
                        else if ( Sets[c1].type == CRANGE ) {
                            MesPrint("&Ranged sets are not allowed in the select option");
                            error = 1;
                        }
                        numsets++;
                        *w++ = c1;
                    }
                    *p = c;
                }
            }
/*
            Now exchange the positions a bit.
            Regular stuff at OldWork, numsets sets at FirstWork[idhead]
*/
            OldWork = w;
            for ( i = 0; i < idhead; i++ ) *w++ = FirstWork[i];
            AC.idoption = SUBSELECT;
            break;
        case SUBAFTER:
        case SUBAFTERNOT:
            if ( type == TYPEIF ) {
                MesPrint("&The if[no]match->label option is not allowed in an if statement");
                error = 1; goto AllDone;
            }
            if ( pp[0] != '-' || pp[1] != '>' ) goto IllField;
            pp += 2;    /* points now at the label */
            inp = pp;
            AC.idoption |= opt;
readlabel:
            while ( FG.cTable[*pp] <= 1 ) pp++;
            if ( pp != p ) {
                c = *p; *p = 0;
                MesPrint("&Illegal label %s in if[no]match option of id-statement",inp);
                *p = c; error = 1; inp = p+1; continue;
            }
            c = *p; *p = 0;
            OldWork[3] = GetLabel(inp);
            *p++ = c; inp = p;
            break;
        case SUBALL:
            x = 0;
            if ( *pp == '(' ) {
                if ( FG.cTable[*inp] == 1 ) {
                    while ( *inp >= '0' && *inp <= '9' ) x = 10*x+*inp++ - '0';
                }
                else {
                    pp++;
                    while ( FG.cTable[*inp] == 0 ) inp++;
                    c = *inp; *inp = 0;
                    if ( StrICont(pp,(UBYTE *)"normalize") != 0 ) goto IllOpt;
                    *inp = c;
                    OldWork[4] |= NORMALIZEFLAG;
                }
                if ( *inp != ')' || inp+1 != p ) {
                    c = *inp; *inp = 0;
IllOpt:
                    MesPrint("&Illegal ALL option in id-statement: ",pp);
                    *inp++ = c;
                    error = 1;
                    continue;
                }
                pp = inp;
                inp = pp+1;
            }
/*
            Note that the following statement limits x to
*/
            if ( x > MAXPOSITIVE ) {
                MesPrint("&Requested maximum number of matches %l in ALL option in id-statement is
greater than %l ",x,MAXPOSITIVE);
                error = 1;
            }
            OldWork[5] = x;
            if ( type != TYPEIDNEW ) {
              if ( type == TYPEIDOLD ) {
                MesPrint("&Requested ALL option not allowed in idold/also statement.");
                error = 1;
              }
              else if ( type == TYPEIF ) {
                MesPrint("&Requested ALL option not allowed in if(match())");
                error = 1;
              }
              else {
```

```
                    MesPrint("&ALL option only allowed in regular id-statement.");
                    error = 1;
                  }
                }
                p++; inp = p;
                AC.idoption = opt;
                break;
            default:
                if ( pp != p ) {
IllField:           c = *p; *p = 0;
                    MesPrint("&Illegal optionfield %s in id-statement",inp);
                    *p = c; error = 1; inp = p+1; continue;
                }
                i = AC.idoption & SUBMASK;
                if ( i && i != opt ) {
                    MesPrint("&Conflicting options in id-statement");
                    error = 1;  continue;
                }
                else AC.idoption |= opt;
                while ( *p == ',' ) p++;
                inp = p;
                break;
        }
    }
    if ( ( AC.idoption & SUBMASK ) == 0 ) AC.idoption |= SUBMULTI;
    OldWork[2] = AC.idoption;
/*
    Now we have a field till the = sign
    Now the subexpression prototype
*/
    AC.ProtoType = w;
    *w++ = SUBEXPRESSION;
    *w++ = SUBEXPSIZE;
    *w++ = C->numrhs+1;
    *w++ = 1;
    *w++ = AC.cbufnum;
    FILLSUB(w)
    AC.WildC = w;
    AC.NwildC = 0;
    AT.WorkPointer = s = w + 4*AM.MaxWildcards + 8;
/*
    Now read the LHS
*/
    ClearWildcardNames();
    oldcpointer = AddLHS(AC.cbufnum) - C->Buffer;

    *p = 0;
    oldnumrhs = C->numrhs;
    if ( ( retcode = CompileAlgebra(inp,LHSIDE,AC.ProtoType) ) < 0 ) { error = 1; }
    else AC.ProtoType[2] = retcode;
    *p = '='; inp = p+1;
    AT.WorkPointer = s;
    if ( AC.NwildC && SortWild(w,AC.NwildC) ) error = 1;

        /* Make the LHS pointers ready */

    OldWork[1] = AC.WildC-OldWork;
    OldWork[idhead+1] = OldWork[1] - idhead;
    w = AC.WildC;
    AT.WorkPointer = w;
    s = C->rhs[C->numrhs];
/*
    Now check whether wildcards get converted to dollars (for PARALLEL)
*/
    {
        WORD *tw, *twstop;
        tw = AC.ProtoType; twstop = tw + tw[1]; tw += SUBEXPSIZE;
        while ( tw < twstop ) {
            if ( *tw == LOADDOLLAR ) {
                AddPotModdollar(tw[2]);
            }
            tw += tw[1];
        }
    }
/*
    We have the expression in the compiler buffers.
    The main level is at lhs[numlhs]
    The partial lhs (including ProtoType) is in OldWork (in WorkSpace)
    We need to load the result at w after the prototype
    Because these sort routines don't use the WorkSpace
    there should not be a conflict
*/
    if ( !error && *s == 0 ) {
IllLeft:MesPrint("&Illegal LHS");
        AC.lhdollarflag = 0;
        return(1);
    }
```

```
    if ( !error && *(s+*s) != 0 ) {
        MesPrint("&LHS should be one term only");
        return(1);
    }
    if ( error == 0 ) {
        WORD oldpolyfun = AR.PolyFun;
        if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) {
            if ( !error ) error = 1;
            return(error);
        }
        AN.RepPoint = AT.RepCount + 1;
        ow = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
        mm = s; ww = ow; i = *mm;
        while ( --i >= 0 ) {*ww++ = *mm++;} AT.WorkPointer = ww;
        AC.lhdollarflag = 0; oldEside = AR.Eside; AR.Eside = LHSIDE;
        AR.Cnumlhs = C->numlhs;
        AR.PolyFun = 0;
        if ( Generator(BHEAD ow,C->numlhs) ) {
            AR.Eside = oldEside;
            LowerSortLevel(); LowerSortLevel(); AR.PolyFun = oldpolyfun; goto IllLeft;
        }
        AR.Eside = oldEside;
        AT.WorkPointer = w;
        if ( EndSort(BHEAD w,0) < 0 ) { LowerSortLevel(); AR.PolyFun = oldpolyfun; goto IllLeft; }
        AR.PolyFun = oldpolyfun;
        if ( *w == 0 || *(w+*w) != 0 ) {
            MesPrint("&LHS must be one term");
            AC.lhdollarflag = 0;
            return(1);
        }
        LowerSortLevel();
        if ( AC.lhdollarflag ) MarkDirty(w,DIRTYFLAG);
    }
    AT.WorkPointer = w + *w;
    AC.DumNum = 0;
/*
    Everything is now after OldWork. We can pop the compilerbuffer.
    Next test for illegal things like a coefficient
    At this point we have:
    w = the term of the LHS
*/
    C->Pointer = C->Buffer + oldcpointer;
    C->numrhs = oldnumrhs;
    C->numlhs--;

    m = w + *w - 3;
    AC.vectorlikeLHS = 0;
    if ( !error ) {
      if ( m[2] != 3 || m[1] != 1 || *m != 1 ) {
        if ( *m == 1 && m[1] == 1 && m[2] == -3 ) {
            MinusSign = 1;
        }
        else {
            MesPrint("&Coefficient in LHS");
            error = 1;
            AC.DumNum = 0;
            *w -= ABS(m[2])-3;
        }
      }
      if ( *w == 7 && w[1] == INDEX && w[3] < 0 ) {
        if ( ( AC.idoption & SUBMASK ) != 0 &&  ( AC.idoption & SUBMASK ) !=
        SUBMULTI ) {
            MesPrint("&Illegal option for substitution of a vector");
            error = 1;
        }
        AC.DumNum = AM.IndDum;
        OldWork[2] = ( OldWork[2] - ( OldWork[2] & SUBMASK ) ) | SUBVECTOR;
        c1 = w[3];
            /* We overwrite the LHS */
        *w++ = INDTOIND;
        *w++ = 4;
        *w++ = AC.DumNum + WILDOFFSET;
        *w++ = 0;
        w[0] = 5;
        w[1] = VECTOR;
        w[2] = 4;
        w[3] = c1;
        w[4] = AC.DumNum + WILDOFFSET;
        OldWork[idhead+1] = w - OldWork - idhead;
        AC.vectorlikeLHS = 1;
      }
      else {
        AC.DumNum = 0;
        *w -= 3;
        i = OldWork[2] & SUBMASK;
        m = w + *w;
        if ( i == 0 || i == SUBMULTI ) {
```

```
            s = w+1;
            while ( s < m ) {
                if ( *s == SYMBOL ) {
                    j = s[1]/2; s += 2;
                    while ( --j >= 0 ) {
                        if ( ABS(s[1]) > 2*MAXPOWER ) {
                            OldWork[2] = ( OldWork[2] - i ) | SUBONCE;
                            break;
                        }
                        s += 2;
                    }
                    if ( j >= 0 ) break;
                }
                else if ( *s == DOTPRODUCT ) {
                    j = s[1]/3; s += 2;
                    while ( --j >= 0 ) {
                        if ( ABS(s[2]) > 2*MAXPOWER ) {
                            OldWork[2] = ( OldWork[2] - i ) | SUBONCE;
                            break;
                        }
                        else if ( s[1] >= -(2*WILDOFFSET) || s[0] >= -(2*WILDOFFSET) ) {
                            OldWork[2] = ( OldWork[2] - i ) | SUBMANY;
                            i = SUBMANY;
                        }
                        s += 3;
                    }
                    if ( j >= 0 ) break;
                }
                else {
                    OldWork[2] = ( OldWork[2] - i ) | SUBMANY;
                    break;
                }
            }
        }
        if ( ( OldWork[2] & SUBMASK ) == 0 ) OldWork[2] |= SUBMULTI;
      }
      if ( ( OldWork[2] & SUBMASK ) == SUBSELECT ) {
/*
        Paste the SETSET information after the pattern.
        Important note: We will still get function information for the
        smart patternmatching after it. To distinguish them we need to have
        that SETSET != m*n+1 in which m is the number of words per function
        and n the number of functions. Currently (29-may-1997) m = 4.
*/
        *m++ = SETSET;
        *m++ = numsets+2;
        s = FirstWork + idhead;
        while ( --numsets >= 0 ) *m++ = *s++;
      }
      else {
        m = w + *w;
      }
    }
/*
    We keep the whole thing in OldWork for the moment.
    We still have to add the number of the RHS expression.
    There is also some opportunity now to be smart about the pattern.
    This is needed for complicated wildcarding with symmetric functions.
    We do this in a special routine during compile time to make sure
    that we loose as little time as possible (during running) if there
    is no need to be smart.
*/
    *m++ = 0;
    OldWork[1] = m - OldWork;
    AC.ProtoType = OldWork+idhead;
    if ( !error ) {
        if ( StudyPattern(OldWork) ) error = 1;
    }
    AT.WorkPointer = OldWork + OldWork[1];
    if ( AC.lhdollarflag ) OldWork[4] |= DOLLARFLAG;
    AC.lhdollarflag = 0;
/*
    Test whether the id/idold configuration is fine.
*/
    if ( type == TYPEIDOLD ) {
        WORD ci = C->numlhs;
        while ( ci >= 1 ) {
            if ( C->lhs[ci][0] == TYPEIDNEW ) {
                if ( (C->lhs[ci][2] & SUBMASK) == SUBALL ) {
                    MesPrint("&Idold/also cannot follow an id,all statement.");
                    error = 1;
                }
                break;
            }
            else if ( C->lhs[ci][0] == TYPEDETCURDUM ) { ci--; continue; }
            else if ( C->lhs[ci][0] == TYPEIDOLD ) { ci--; continue; }
            else ci = 0;
```

```
        }
        if ( ci < 1 ) {
            MesPrint("&Idold/also should follow an id/idnew statement.");
            error = 1;
        }
    }
/*
    Now the right hand side.
*/
    if ( type != TYPEIF ) {
        if ( ( retcode = CompileAlgebra(inp,RHSIDE,AC.ProtoType) ) < 0 ) error = 1;
        else {
            AC.ProtoType[2] = retcode;
            AC.DumNum = 0;
            if ( MinusSign ) {  /* Flip the sign of the RHS */
                w = C->rhs[retcode];
                while ( *w ) { w += *w; w[-1] = -w[-1]; }
            }
            if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
        }
    }
/*
    Actual adding happens only now after numrhs insertion
*/
    if ( !error ) { AddNtoL(OldWork[1],OldWork); }
AllDone:
    AC.lhdollarflag = 0;
    AT.WorkPointer = FirstWork;
    return(error);
}

/*
    #] CoIdExpression :
    #[ CoMultiply :
*/

static WORD mularray[13] = { TYPEMULT, SUBEXPSIZE+3, 0, SUBEXPRESSION,
        SUBEXPSIZE, 0, 1, 0, 0, 0, 0, 0, 0 };

int CoMultiply(UBYTE *inp)
{
    UBYTE *p;
    int error = 0, RetCode;
    mularray[2] = 0;         /* right multiply is default */
    while ( *inp == ',' ) inp++;
/*  if ( inp[-1] == '-' || inp[-1] == '+' ) inp--; */
    p = SkipField(inp,0);
    if ( *p ) {
        *p = 0;
        if ( StrICont(inp,(UBYTE *)"left") == 0 )       mularray[2] = 1;
        else if ( StrICont(inp,(UBYTE *)"right") == 0 ) mularray[2] = 0;
        else {
            MesPrint("&Illegal option in multiply statement or ; forgotten.");
            return(1);
        }
        *p = ',';
        inp = p + 1;
    }
    ClearWildcardNames();
    while ( *inp == ',' ) inp++;
    AC.ProtoType = mularray+3;
    mularray[7] = AC.cbufnum;
    if ( ( RetCode = CompileAlgebra(inp,RHSIDE,AC.ProtoType) ) < 0 ) error = 1;
    else {
        mularray[5] = RetCode;
        AddNtoL(SUBEXPSIZE+3,mularray);
        if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
    }
    return(error);
}

/*
    #] CoMultiply :
    #[ CoFill :

    Special additions for tablebase-like tables added 12-aug-2002
*/

int CoFill(UBYTE *inp)
{
    GETIDENTITY
    WORD error = 0, x, xx, funnum, type, *oldwp = AT.WorkPointer;
    int i, oldcbufnum = AC.cbufnum, nofill = 0, numover, redef = 0;
    WORD *w, *wold, *Tprototype;
    UBYTE *p = inp, c, *inp1;
    TABLES T = 0, oldT;
    LONG newreservation, sum = 0;
```

```
    UBYTE *p1, *p2, *p3, *p4, *fake = 0;
    int tablestub = 0;
    if ( AC.exprfillwarning == 1 ) AC.exprfillwarning = 0;
/*
    Read the name of the function and test that it is in the table.
*/
    p1 = inp;
    if ( ( p = SkipAName(inp) ) == 0 ) return(1);
    p2 = p;
    c = *p; *p = 0;
    if ( ( GetVar(inp,&type,&funnum,CFUNCTION,WITHAUTO) == NAMENOTFOUND )
    || ( T = functions[funnum].tabl ) == 0 || ( T->numind > 0 && c != '(' ) ) {
        MesPrint("&%s should be a table with argument(s)",inp);
        *p = c; return(1);
    }
    oldT = T;
    *p++ = c;
    if ( T->numind == 0 ) {
        if ( c == '(' ) {
            if ( *p != ')' ) {
                c = *p; *p = 0;
                MesPrint("&%s should be a table without arguments",inp);
                *p = c; return(1);
            }
            else { p++; }
        }
        else { p--; }
        sum = 0;
        p3 = p;
        goto andagain;
    }
    w = oldwp;
    if ( T->numind < 0 ) { /* Pick up the first index */
        ParseSignedNumber(xx,p);
        if ( FG.cTable[p[-1]] != 1 || *p != ',' || xx < 1 || ( xx > ( -T->numind - 1 ) ) ) {
            MesPrint("&No valid number of table indices in *-table fill statement.");
            return(1);
        }
        *w++ = xx;
        p++;
    }
    else { xx = T->numind; }
    for ( sum = 0, i = 0; i < xx; i++ ) {
        ParseSignedNumber(x,p);
        if ( FG.cTable[p[-1]] != 1 || ( *p != ',' && *p != ')' ) ) {
            MesPrint("&Table arguments in fill statement should be numbers");
            return(1);
        }
        if ( T->sparse ) *w++ = x;
        else if ( x < T->mm[i].mini || x > T->mm[i].maxi ) {
            MesPrint("&Value %d for argument %d of table out of bounds",x,i+1);
            error = 1; nofill = 1;
        }
        else sum += ( x - T->mm[i].mini ) * T->mm[i].size;
        if ( *p == ')' ) break;
        p++;
    }
    p3 = p;
    if ( T->numind < 0 ) {
        for ( ; i < ABS(T->numind)-1; i++ ) *w++ = 0;
        xx = -T->numind;
    }
    if ( *p != ')' || i < ( xx - 1 ) ) {
        MesPrint("&Incorrect number of table arguments in fill statement. Should be %d"
        ,T->numind);
        error = 1; nofill = 1;
    }
    AT.WorkPointer = w;
    if ( T->sparse == 0 ) sum *= TABLEEXTENSION;
andagain:;
    AC.cbufnum = T->bufnum;
    if ( T->sparse ) {
        i = FindTableTree(T,oldwp,1);
        if ( i >= 0 ) {
            sum = i + ABS(T->numind);
            if ( tablestub == 0 && ( ( T->sparse & 2 ) == 2 ) && ( T->mode != 0 )
            && ( AC.vetotablebasefill == 0 ) ) {
/*
                This redefinition does not need a new stub
*/
                functions[funnum].tabl = T = T->spare;
                tablestub = 1;
                goto andagain;
            }
            redef = 1;
            goto redef;
        }
```

```
        if ( T->totind >= T->reserved ) {
            if ( T->reserved == 0 ) newreservation = 20;
            else newreservation = T->reserved;
            while ( T->totind >= newreservation && newreservation < MAXTABLECOMBUF )
                    newreservation = 2*newreservation;
            if ( newreservation > MAXTABLECOMBUF ) newreservation = MAXTABLECOMBUF;
            if ( T->totind >= newreservation ) {
                MesPrint("@More than %ld elements in sparse table",MAXTABLECOMBUF);
                AC.cbufnum = oldcbufnum;
                Terminate(-1);
            }
            wold = (WORD *)Malloc1(newreservation*sizeof(WORD)*
                                (ABS(T->numind)+TABLEEXTENSION),"tablepointers");
            for ( i = T->reserved*(ABS(T->numind)+TABLEEXTENSION)-1; i >= 0; i-- )
                wold[i] = T->tablepointers[i];
            if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
            T->tablepointers = wold;
            T->reserved = newreservation;
        }
        w = oldwp;
        for ( sum = T->totind*(ABS(T->numind)+TABLEEXTENSION), i = 0; i < ABS(T->numind); i++ ) {
            T->tablepointers[sum++] = *w++;
        }
        InsTableTree(T,T->tablepointers+sum-ABS(T->numind));
#if TABLEEXTENSION == 2
        T->tablepointers[sum+TABLEEXTENSION-1] = -1;  /* New element! */
#else
        T->tablepointers[sum+1] = T->bufnum;
        T->tablepointers[sum+2] = -1;
        T->tablepointers[sum+3] = -1;
        T->tablepointers[sum+4] = 0;
        T->tablepointers[sum+5] = 0;
#endif
    }
    else {
        if ( !nofill && T->tablepointers[sum] >= 0 ) {
redef:;
            if ( AC.vetofilling ) nofill = 1;
            else {
                Warning("Table element was already defined. New definition will be used");
            }
        }
#if TABLEEXTENSION == 2
        T->tablepointers[sum+TABLEEXTENSION-1] = -1;  /* New element! */
#else
        T->tablepointers[sum+1] = T->bufnum;
        T->tablepointers[sum+2] = -1;
        T->tablepointers[sum+3] = -1;
        T->tablepointers[sum+4] = 0;
        T->tablepointers[sum+5] = 0;
#endif
    }
    if ( T->numind ) { p++; }
    if ( *p != '=' ) {
        MesPrint("&Fill statement misses = sign after the table element");
        AC.cbufnum = oldcbufnum;
        AT.WorkPointer = oldwp;
        functions[funnum].tabl = oldT;
        return(1);
    }
    if ( tablestub == 0 && T->mode == 1 && AC.vetotablebasefill == 0 ) {
/*
        Here we construct a righthandside from the indices and the wildcards
*/
        int numfake;
        tablestub = 1;
        p4 = T->argtail;
        while ( *p4 ) p4++;
        numfake = (p4-T->argtail)+(p3-p1)+10;

        fake = (UBYTE *)Malloc1(numfake*sizeof(UBYTE),"Fill fake rhs");
        p = fake;
        *p++ = 't'; *p++ = 'b'; *p++ = 'l'; *p++ = '_'; *p++ = '(';
        p4 = p1; while ( p4 < p2 ) *p++ = *p4++; *p++ = ',';
        p4 = p2+1; while ( p4 < p3 ) *p++ = *p4++;
        if ( T->argtail ) {
            p4 = T->argtail + 1;
            while ( FG.cTable[*p4] == 1 ) p4++;
            while ( *p4 ) {
                if ( *p4 == '?' && p[-1] != ',' ) {
                    p4++;
                    if ( FG.cTable[*p4] == 0 || *p4 == '$' || *p4 == '[' ) {
                        p4 = SkipAName(p4);
                        if ( *p4 == '[' ) {
                            SKIPBRA1(p4);
                        }
                    }
```

```
                    else if ( *p4 == '{' ) {
                        SKIPBRA2(p4);
                    }
                    else if ( *p4 ) { *p++ = *p4++; continue; }
                }
                else *p++ = *p4++;
            }
        }
        *p++ = ')';
        *p = 0;
        inp1 = fake;
    }
    else {
        inp1 = ++p;
    }
    c = 0;
/*
    Now we have the indices and p points to the rhs.
*/
    numover = 0;
    AC.tablefilling = funnum;
    while ( *inp1 ) {
        p = SkipField(inp1,0);
        c = *p; *p = 0;
#ifdef WITHPTHREADS
        Tprototype = T->prototype[0];
#else
        Tprototype = T->prototype;
#endif
        if ( ( i = CompileAlgebra(inp1,RHSIDE,Tprototype) ) < 0 ) { error = 1; i = 0; }
        if ( !nofill ) {
            T->tablepointers[sum] = i;
            T->tablepointers[sum+1] = T->bufnum;
        }
        AC.DumNum = 0;
        *p = c;
        if ( T->sparse || c == 0 ) break;
        inp1 = ++p;
#if ( TABLEEXTENSION == 2 )
        sum++;
#else
        sum += 2;
#endif
        if ( !nofill && T->tablepointers[sum] >= 0 ) numover++;
#if ( TABLEEXTENSION == 2 )
        sum++;
#else
        sum += TABLEEXTENSION-2;
#endif
    }
    if ( AC.exprfillwarning == 1 ) {
        AC.exprfillwarning = 2;
        Warning("Use of expressions and/or $variables in Fill statements is potentially very
dangerous.");
    }
    AC.tablefilling = 0;
    if ( T->sparse && c != 0 ) {
        MesPrint("&In sparse tables one can fill only one element at a time");
        error = 1;
    }
    else if ( numover ) {
        if ( numover == 1 )
            Warning("one element was overwritten. New definition will be used");
        else if ( AC.WarnFlag )
            MesPrint("&Warning: %d elements were overwritten. New definitions will be used",numover);
    }
    if ( T->sparse ) {
        if ( redef == 0 ) T->totind++;
    }
    else T->defined++;
/*
    NumSets = AC.SetList.numtemp;
    NumSetElements = AC.SetElementList.numtemp;
*/
    if ( fake ) {
        M_free(fake,"Fill fake rhs");
        fake = 0;
        functions[funnum].tabl = T = T->spare;
        p = p3;
        goto andagain;
    }
    AC.cbufnum = oldcbufnum;
    AC.SymChangeFlag = 1;
    AT.WorkPointer = oldwp;
    functions[funnum].tabl = oldT;
    return(error);
}
```

```
/*
    #] CoFill :
    #[ CoFillExpression :

    Syntax: FillExpression table = expression(x1,...,xn);
    The arguments should have been bracketed. Each corresponds to one
    of the dimensions of the table. Then the bracket with x1^2*x3^4
    will fill the (2,0,4) element of the table (if n=3 of course).
    Brackets that don't fit will be skipped. It just gives a warning.

    New option (13-jul-2005)
    Syntax: FillExpression table = expression(f);
    The table indices are arguments of the function f which should
    have been bracketed before.
*/

int CoFillExpression(UBYTE *inp)
{
    GETIDENTITY
    UBYTE *p, c;
    WORD type, funnum, expnum, symnum, numsym = 0, *oldwork = AT.WorkPointer;
    WORD *brackets, *term, brasize, *b, *m, *w, *pw, *tstop, zero = 0;
    WORD oldcbuf = AC.cbufnum, curelement = 0;
    int weneedit, i, j, numzero, pow, numfirst;
    TABLES T = 0;
    LONG newreservation, numcommu, sum;
    POSITION oldposition;
    FILEHANDLE *fi;
    CBUF *C;
    WORD numdummies;

    AN.IndDum = AM.IndDum;
    if ( ( p = SkipAName(inp) ) == 0 ) return(1);
    c = *p; *p = 0;
    if ( ( GetVar(inp,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
    || ( T = functions[funnum].tabl ) == 0 ) {
        MesPrint("&%s should be a previously declared table",inp);
        *p = c; return(1);
    }
    *p++ = c;
    if ( T->spare ) T = T->spare;
    C = cbuf + T->bufnum;
    if ( c != '=' ) {
        MesPrint("&No = sign in FillExpression statement");
        return(1);
    }
    inp = p;
    if ( ( p = SkipAName(inp) ) == 0 ) return(1);
    c = *p; *p = 0;
    if ( ( type = GetName(AC.exprnames,inp,&expnum,NOAUTO) ) == NAMENOTFOUND
    || c != '(' || (
        Expressions[expnum].status != LOCALEXPRESSION &&
        Expressions[expnum].status != SKIPLEXPRESSION &&
        Expressions[expnum].status != DROPLEXPRESSION &&
        Expressions[expnum].status != GLOBALEXPRESSION &&
        Expressions[expnum].status != SKIPGEXPRESSION &&
        Expressions[expnum].status != DROPGEXPRESSION ) ) {
        MesPrint("&%s should be an active expression with arguments",inp);
        *p = c; return(1);
    }
    if ( Expressions[expnum].inmem ) {
        MesPrint("&%s cannot be used in a FillExpression statement in the same %n\
        module that it has been redefined",inp);
        *p = c; return(1);
    }
    *p++ = c;
    while ( *p ) {
        inp = p;
        if ( ( p = SkipAName(inp) ) == 0 ) return(1);
        c = *p; *p = 0;

        if ( GetVar(inp,&type,&symnum,-1,NOAUTO) == NAMENOTFOUND ) {
            MesPrint("&%s should be a previously declared symbol or function",inp);
            *p = c; return(1);
        }
        else if ( type == CSYMBOL ) {
            *p++ = c;
            *AT.WorkPointer++ = symnum;
            numsym++;
        }
        else if ( type == CFUNCTION ) {
            numsym = -1;
            *p++ = c;
            if ( c != ')' ) {
                MesPrint("&Argument should be a single function or a list of symbols");
                return(1);
```

```
            }
            symnum += FUNCTION;
            *AT.WorkPointer++ = symnum;
        }
        else {
            MesPrint("&%s should be a previously declared symbol or function",inp);
            *p = c; return(1);
        }
        if ( c == ')' ) break;
        if ( c != ',' ) {
            MesPrint("&Illegal separator in FillExpression statement");
            goto noway;
        }
    }
    if ( *p ) {
        MesPrint("&Illegal end of FillExpression statement");
        goto noway;
    }
/*
    We have the number of the table in funnum.
    The number of the expression in expnum, the table struct in T
    and either the numbers of the symbols in oldwork (there are numsym of them)
    or the number of the function in oldwork (just one and numsym = -1).
    We don't sort them!!!!
*/
    if ( ( numsym > 0 ) && ( ABS(T->numind) != numsym ) ) {
        MesPrint("&This table needs %d symbols for its array indices");
        goto noway;
    }
    EXCHINOUT
#ifdef WITHMPI
    /*
     * The workers can't access to the data of the input expression. We need to
     * broadcast it to all the workers.
     */
    PF_BroadcastExpr(&Expressions[expnum], AR.infile);
    if ( PF.me == MASTER ) {
        /*
         * Restore the file position on the master.
         */
        POSITION pos;
        SetEndScratch(AR.infile, &pos);
    }
#endif
    fi = AR.infile;
    if ( fi->handle >= 0 ) {
        PUTZERO(oldposition);
        SeekFile(fi->handle,&oldposition,SEEK_CUR);
        SetScratch(fi,&(Expressions[expnum].onfile));
        if ( ISNEGPOS(Expressions[expnum].onfile) ) {
            MesPrint("&File error in FillExpression");
            BACKINOUT
            goto noway;
        }
    }
    else {
/*
        Note: Because everything fits inside memory we never get problems
        with excessive file sizes.
*/
        SETBASEPOSITION(oldposition,(UBYTE *)(fi->POfill)-(UBYTE *)(fi->PObuffer));
        fi->POfill = (WORD *)((UBYTE *)(fi->PObuffer) + BASEPOSITION(Expressions[expnum].onfile));
    }
    pw = AT.WorkPointer;
    if ( numsym < 0 ) { brackets = pw + 1; }
    else { brackets = pw + numsym; }
    brasize = -1; weneedit = 0; /* stands for we need it */
    term = (WORD *)(((UBYTE *)(brackets)) + AM.MaxTer);
    AT.WorkPointer = (WORD *)(((UBYTE *)(term)) + AM.MaxTer);
    AC.cbufnum = T->bufnum;
    AC.tablefilling = funnum;
    if ( GetTerm(BHEAD term) > 0 ) {            /* Skip prototype */
        while ( GetTerm(BHEAD term) > 0 ) {
            GETSTOP(term,tstop);
            w = m = term + 1;
            while ( m < tstop && *m != HAAKJE ) m += m[1];
            if ( *m != HAAKJE ) {
                MesPrint("&Illegal attempt to put an expression without brackets in a table");
                BACKINOUT
                goto noway;
            }
            if ( brasize == m - w ) {
                b = brackets;
                while ( *b == *w && w < m ) { b++; w++; }
                if ( w == m ) { /* Same as current bracket. Copy. */
                    if ( weneedit ) {
                        m += m[1] - 1;
```

```
                    *m = *term - (m-term);
                    AddNtoC(AC.cbufnum,*m,m,3);
                    numdummies = DetCurDum(BHEAD term) - AM.IndDum;
                    if ( numdummies > T->numdummies ) T->numdummies = numdummies;
                }
                continue; /* Next term */
            }
        }
        if ( weneedit ) {
            AddNtoC(AC.cbufnum,1,&zero,4);   /* Terminate old bracket */
            numcommu = numcommute(C->rhs[curelement],&(C->NumTerms[curelement]));
            C->CanCommu[curelement] = numcommu;
        }
        b = brackets; w = term + 1;
        if ( numsym < 0 ) pw = oldwork + 1;
        else              pw = oldwork + numsym;
        while ( w < m ) *b++ = *w++;
        brasize = b - brackets;
/*
        Now compute the element. See whether we need it
*/
        if ( numsym < 0 ) {
            WORD *bb, bnum;
            if ( *brackets != symnum || brasize != brackets[1] ) {
                weneedit = 0; continue; /* Cannot work! */
            }
/*
            Now count the number of arguments and whether they are numbers
*/
            b = brackets + FUNHEAD;
            bb = brackets+brackets[1];
            i = 0;
            if ( T->numind < 0 ) {
                bnum = b[1]+1;
                if ( bnum > -T->numind ) {
                    weneedit = 0; continue; /* Cannot work! */
                }
            }
            else bnum = T->numind;
            while ( b < bb ) {
                if ( *b != -SNUMBER ) break;
                i++;
                b += 2;
            }
            if ( b < bb || i != bnum ) {
                weneedit = 0; continue; /* Cannot work! */
            }
        }
        else if ( brasize > 0 && ( *brackets != SYMBOL
        || brackets[1] < brasize || (brackets[1]-2) > numsym*2 ) ) {
            weneedit = 0; continue; /* Cannot work! */
        }
        numzero = 0; sum = 0;
        numfirst = 0;
        if ( numsym > 0 ) {
          for ( i = 0; i < numsym; i++ ) {
            if ( brasize > 0 ) {
                b = brackets + 2; j = brackets[1]-2;
                while ( j > 0 ) {
                    if ( *b == oldwork[i] ) break;
                    j -= 2; b += 2;
                }
                if ( j <= 0 ) {  /* it was not there */
                    numzero++; pow = 0;
                    if ( 2*numzero+brackets[1]-2 > numsym*2 ) {
                        weneedit = 0; goto nextterm;
                    }
                }
                else pow = b[1];
            }
            else pow = 0;
            if ( T->sparse ) {
                if ( T->numind < 0 ) {
                    if ( i == 0 ) {
                        numfirst = pow;
                        if ( pow > -T->numind ) {
                            weneedit = 0; goto nextterm;
                        }
                    }
                    else if ( i > pow ) {
                        weneedit = 0; goto nextterm;
                    }
                }
                *pw++ = pow;
            }
            else if ( pow < T->mm[i].mini || pow > T->mm[i].maxi ) {
                weneedit = 0; goto nextterm;
```

```
                }
                else sum += ( pow - T->mm[i].mini ) * T->mm[i].size;
              }
            }
            else {
              WORD xx;
              b = brackets + FUNHEAD;
              sum = 0;
/*
              Now scan the arguments of the function.
              We did check already the number and type of the arguments.
*/
              xx = (brackets[1]-FUNHEAD)/2;
              for ( i = 0; i < xx; i++ ) {
                pow = b[1];
                b += 2;
                if ( T->sparse ) {
                    if ( T->numind < 0 ) {
                        if ( i == 0 ) {
                            numfirst = pow;
                            if ( pow >= -T->numind ) {
                                weneedit = 0; goto nextterm;
                            }
                        }
                    }
                    *pw++ = pow;
                }
                else if ( pow < T->mm[i].mini || pow > T->mm[i].maxi ) {
                    weneedit = 0; goto nextterm;
                }
                else sum += ( pow - T->mm[i].mini ) * T->mm[i].size;
              }
            }
            if ( T->numind < 0 ) {
                for ( i = numfirst+1; i < -T->numind; i++ ) *pw++ = 0;
            }
            weneedit = 1;
            if ( T->sparse ) {
                if ( numsym < 0 ) pw = oldwork + 1;
                else              pw = oldwork + ABS(T->numind);
                i = FindTableTree(T,pw,1);
                if ( i >= 0 ) {
                    sum = i+ABS(T->numind);
/*
Wrong!!!!           C->rhs[T->tablepointers[sum]] = C->Pointer;
*/
                    C->Pointer--; /* Back up over the zero */
                    goto newentry;
                }
                if ( T->totind >= T->reserved ) {
                    if ( T->reserved == 0 ) newreservation = 20;
                    else newreservation = T->reserved;
/*---Copied from Fill---------------------------*/
                    while ( T->totind >= newreservation && newreservation < MAXTABLECOMBUF )
                            newreservation = 2*newreservation;
                    if ( newreservation > MAXTABLECOMBUF ) newreservation = MAXTABLECOMBUF;
                    if ( T->totind >= newreservation ) {
                        MesPrint("@More than %ld elements in sparse table",MAXTABLECOMBUF);
                        AC.cbufnum = oldcbuf;
                        AT.WorkPointer = oldwork;
                        Terminate(-1);
                    }
/*---Copied from Fill---------------------------*/
                    if ( T->totind >= newreservation ) {
                        MesPrint("@More than %ld elements in sparse table",MAXTABLECOMBUF);
                        AC.cbufnum = oldcbuf;
                        AT.WorkPointer = oldwork;
                        Terminate(-1);
                    }
                    w = (WORD *)Malloc1(newreservation*sizeof(WORD)*
                            (ABS(T->numind)+TABLEEXTENSION),"tablepointers");
                    for ( i = T->reserved*(ABS(T->numind)+TABLEEXTENSION)-1; i >= 0; i-- )
                        w[i] = T->tablepointers[i];
                    if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
                    T->tablepointers = w;
                    T->reserved = newreservation;
                }
                if ( numsym < 0 ) pw = oldwork + 1;
                else              pw = oldwork + numsym;
                for ( sum = T->totind*(ABS(T->numind)+TABLEEXTENSION), i = 0; i < ABS(T->numind); i++ ) {
                    T->tablepointers[sum++] = *pw++;
                }
                InsTableTree(T,T->tablepointers+sum-ABS(T->numind));
                (T->totind)++;
            }
#if ( TABLEEXTENSION != 2 )
```

```
            else {
                sum *= TABLEEXTENSION;
            }
#endif
/*
            Start a new entry. Copy the element.
*/
            AddRHS(T->bufnum,0);
            T->tablepointers[sum] = C->numrhs;
#if ( TABLEEXTENSION == 2 )
            T->tablepointers[sum+TABLEEXTENSION-1] = -1;
#else
            T->tablepointers[sum+1] = T->bufnum;
            T->tablepointers[sum+2] = -1;
            T->tablepointers[sum+3] = -1;
            T->tablepointers[sum+4] = 0;
            T->tablepointers[sum+5] = 0;
#endif
newentry:   if ( *m == HAAKJE ) { m += m[1] - 1; }
            else m--;
            *m = *term - (m-term);
            AddNtoC(AC.cbufnum,*m,m,5);
            curelement = T->tablepointers[sum];
nextterm:;
        }
        if ( weneedit ) {
            AddNtoC(AC.cbufnum,1,&zero,6);   /* Terminate old bracket */
            numcommu = numcommute(C->rhs[curelement],&(C->NumTerms[curelement]));
            C->CanCommu[curelement] = numcommu;
        }
    }
    if ( fi->handle >= 0 ) {
        SetScratch(fi,&(oldposition));
    }
    else {
        fi->POfill = (WORD *)((UBYTE *)(fi->PObuffer) + BASEPOSITION(oldposition));
    }
    BACKINOUT
    AC.cbufnum = oldcbuf;
    AC.tablefilling = 0;
    AT.WorkPointer = oldwork;
    return(0);
noway:
    BACKINOUT
    AC.cbufnum = oldcbuf;
    AC.tablefilling = 0;
    AT.WorkPointer = oldwork;
    return(1);
}

/*
    #] CoFillExpression :
    #[ CoPrintTable :

    Syntax
        PrintTable [+f] [+s] tablename [>[>] file];
    All defined elements are written with individual Fill statements.
    If a file is specified, the result is written to file only.
    The flags of the print statement apply as much as possible.
    We make use of the regular write routines.
*/

int CoPrintTable(UBYTE *inp)
{
    GETIDENTITY
    int fflag = 0, sflag = 0, addflag = 0, error = 0, sum, i, j;
    UBYTE *filename, *p, c, buffer[100], *s, *oldoutputline = AO.OutputLine;
    WORD type, funnum, *expr, *m, num;
    TABLES T = 0;
    WORD oldSkip = AO.OutSkip, oldMode = AC.OutputMode, oldHandle = AC.LogHandle;
    WORD oldType = AO.PrintType, *oldwork = AT.WorkPointer;
    UBYTE *oldFill = AO.OutFill, *oldLine = AO.OutputLine;
#ifdef WITHMPI
    if ( PF.me != MASTER ) return 0;
#endif
/*
    First the flags
*/
    while ( *inp == '+' ) {
        inp++;
        if ( *inp == 'f' || *inp == 'F' ) { fflag = 1; inp++; }
        else if ( *inp == 's' || *inp == 'S' ) { sflag = PRINTONETERM; inp++; }
        else {
            MesPrint("&Illegal + option in PrintTable statement");
            error = 1; inp++;
        }
        while ( *inp != ',' && *inp && *inp != '+' ) {
```

```
                if ( !error ) {
                    if ( *inp ) {
                        MesPrint("&Illegal + option in PrintTable statement");
                        inp++;
                    }
                    else {
                        MesPrint("&Unfinished PrintTable statement");
                        return(1);
                    }
                    error = 1;
                }
                inp++;
            }
            if ( *inp == ',' ) inp++;
        }
/*
        Now the name of the table
*/
        if ( ( p = SkipAName(inp) ) == 0 ) return(1);
        c = *p; *p = 0;
        if ( ( GetVar(inp,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
        || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",inp);
            *p = c; return(1);
        }
        if ( T->spare && T->mode == 1 ) T = T->spare;
        *p++ = c;
/*
        Check for a filename. Runs to the end of the statement.
*/
        filename = 0;
        if ( c == '>' ) {
            if ( *p == '>' ) { addflag = 1; p++; }
            filename = p;
        }
        else filename = 0;

        if ( filename ) {
            if ( addflag ) AC.LogHandle = OpenAddFile((char *)filename);
            else           AC.LogHandle = CreateFile((char *)filename);
            if ( AC.LogHandle < 0 ) {
                MesPrint("&Cannot open file '%s' properly",filename);
                error = 1; goto finally;
            }
            AO.PrintType = PRINTLFILE;
        }
        else if ( fflag && AC.LogHandle >= 0 ) {
            AO.PrintType = PRINTLFILE;
        }
        AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer;
        AT.WorkPointer += 2*AC.LineLength;

        AO.PrintType |= sflag;
        AC.OutputMode = 0;
        AO.IsBracket = 0;
        AO.OutSkip = 0;
        AR.DeferFlag = 0;
        AC.outsidefun = 1;
        if ( AC.LogHandle == oldHandle ) FiniLine();
        AO.OutputLine = AO.OutFill = (UBYTE *)Malloc1(AC.LineLength+20,"PrintTable");
        AO.OutStop = AO.OutFill + AC.LineLength;
        for ( i = 0; i < T->totind; i++ ) {
            if ( !T->sparse && T->tablepointers[i*TABLEEXTENSION] < 0 ) continue;
            TokenToLine((UBYTE *)"Fill ");
            TokenToLine((UBYTE *)(VARNAME(functions,funnum)));
            TokenToLine((UBYTE *)"(");
            AO.OutSkip = 3;
            if ( T->sparse ) {
                sum = i * ( T->numind + TABLEEXTENSION );
                for ( j = 0; j < T->numind; j++, sum++ ) {
                    if ( j > 0 ) TokenToLine((UBYTE *)",");
                    num = T->tablepointers[sum];
                    s = buffer; s = NumCopy(num,s);
                    TokenToLine(buffer);
                }
                expr = cbuf[T->bufnum].rhs[T->tablepointers[sum]];
            }
            else {
                for ( j = 0; j < T->numind; j++ ) {
                    if ( j > 0 ) {
                        TokenToLine((UBYTE *)",");
                        num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;
                    }
                    else {
                        num = T->mm[j].mini + i / T->mm[j].size;
                    }
                    s = buffer; s = NumCopy(num,s);
```

```
                TokenToLine(buffer);
            }
            expr = cbuf[T->bufnum].rhs[T->tablepointers[TABLEEXTENSION*i]];
        }
        TOKENTOLINE(") =",")=");
        if ( sflag ) {
            FiniLine();
            if ( AC.OutputSpaces != NOSPACEFORMAT ) TokenToLine((UBYTE *)"    ");
        }
        m = expr;
/*
        WORD lbrac, first;
        lbrac = 0; first = 1;
        while ( *m ) {
            if ( WriteTerm(m,&lbrac,first,1,0) ) {
                MesPrint("Error while writing table");
                error = 1;
                goto finally;
            }
            first = 0;
            m += *m;
        }
        if ( first ) { TOKENTOLINE(" 0","0") }
        else if ( lbrac ) { TOKENTOLINE(" )",")") }
*/
        while ( *m ) m += *m;
        if ( m > expr ) {
            if ( WriteExpression(expr,(LONG)(m-expr)) ) { error = 1; goto finally; }
            AO.OutSkip = 0;
        }
        else {
            TokenToLine((UBYTE *)"0");
        }
        TokenToLine((UBYTE *)";");
        FiniLine();
    }
    M_free(AO.OutputLine,"PrintTable");
    AO.OutputLine = AO.OutFill = oldoutputline;
/*
    Reset the file pointers and parameters if any. Close file if needed.
*/
finally:
    AO.OutSkip     = oldSkip;
    AC.OutputMode  = oldMode;
    AC.LogHandle   = oldHandle;
    AO.PrintType   = oldType;
    AO.OutFill     = oldFill;
    AO.OutputLine  = oldLine;
    AT.WorkPointer = oldwork;
    AC.outsidefun  = 0;
    return(error);
}

/*
    #] CoPrintTable :
    #[ CoAssign :

    This statement has an easy syntax:
        $name = expression
*/

static WORD AssignLHS[14] = { TYPEASSIGN, 3+SUBEXPSIZE, 0,
                            SUBEXPRESSION, SUBEXPSIZE, 0, 1, 0, 0,0,0,0,0 };

int CoAssign(UBYTE *inp)
{
    int error = 0, retcode;
    UBYTE *name, c;
    WORD number;
    if ( *inp != '$' ) {
nolhs:  MesPrint("&assign statement should have a dollar variable in the LHS");
        return(1);
    }
    inp++; name = inp;
    if ( FG.cTable[*inp] != 0 ) goto nolhs;
    while ( FG.cTable[*inp] < 2 ) inp++;
    if ( AP.PreAssignFlag == 2 ) {
        if ( *inp == '_' ) inp++;
    }
    if ( ( *inp == ',' && inp[1] != '=' ) && ( *inp != '=' ) ) {
        MesPrint("&assign statement should have only a dollar variable in the LHS");
        return(1);
    }
    c = *inp;
    *inp = 0;
    if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
        number = AddDollar(name,DOLUNDEFINED,0,0);
```

```
    }
    *inp = c;
    if ( c == ',' ) inp++;
    *inp++ = '=';
    if ( *inp == ',' ) inp++;
/*
    Fake a Prototype and read the RHS
*/
    AssignLHS[7] = AC.cbufnum;
    retcode = CompileAlgebra(inp,RHSIDE,(AssignLHS+3));
    if ( retcode < 0 ) error = 1;
    AC.DumNum = 0;
/*
    Now add the LHS
*/
    AssignLHS[2] = number;
    AssignLHS[5] = retcode;
    AddNtoL(AssignLHS[1],AssignLHS);
/*
    Add to the list of potentially modified dollars (for PARALLEL)
*/
    AddPotModdollar(number);
    return(error);
}

/*
    #] CoAssign :
    #[ CoDeallocateTable :

    Syntax: DeallocateTable tablename(s);
    Should work only for sparse tables.
    Action: Cleans all definitions of elements of a table as if there have
            never been any fill statements.
*/

int CoDeallocateTable(UBYTE *inp)
{
    UBYTE *p, c;
    TABLES T = 0;
    WORD type, funnum, i;
    c = *inp;
    while ( c ) {
        while ( *inp == ',' ) inp++;
        if ( *inp == 0 ) break;
        if ( ( p = SkipAName(inp) ) == 0 ) return(1);
        c = *p; *p = 0;
        if ( ( GetVar(inp,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
        || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",inp);
            *p = c; return(1);
        }
        if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",inp);
            *p = c; return(1);
        }
        if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
        ClearTableTree(T);
        for (i = 0; i < T->buffersfill; i++ ) { /* was <= */
            finishcbuf(T->buffers[i]);
        }
        T->bufnum = inicbufs();
        T->buffersfill = 0;
        T->buffers[T->buffersfill++] = T->bufnum;
        T->tablepointers = 0;
        T->boomlijst = 0;
        T->totind = 0;
        T->reserved = 0;

        if ( T->spare ) {
            TABLES TT = T->spare;
            if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
            ClearTableTree(TT);
            for (i = 0; i < TT->buffersfill; i++ ) { /* was <= */
                finishcbuf(TT->buffers[i]);
            }
            TT->bufnum = inicbufs();
            TT->buffersfill = 0;
            TT->buffers[T->buffersfill++] = T->bufnum;
            TT->tablepointers = 0;
            TT->boomlijst = 0;
            TT->totind = 0;
            TT->reserved = 0;
        }
        *p++ = c;
        inp = p;
    }
    return(0);
```

```
}

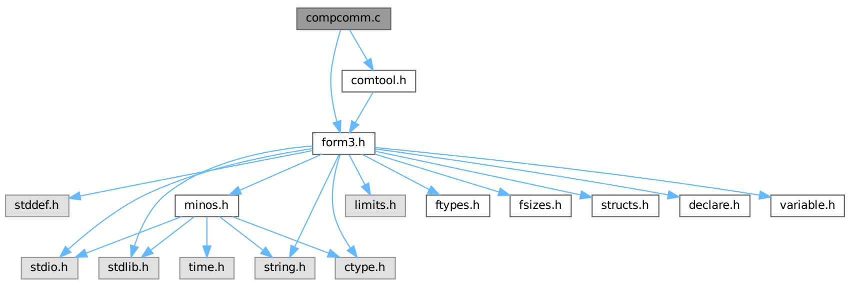
/*
    #] CoDeallocateTable :
    #[ CoFactorCache :
*/
/*
int CoFactorCache(UBYTE *inp)
{
    Code to be added in due time
    We need to read 'expression', get its terms through Generator and sort them.
    We store the result in the WorkSpace in argument notation.
    This will be argin.
    Then we do the same with the sequence of factors. They form argout.
    The whole is put in the buffer with the call
        InsertArg(BHEAD argin,argout,1)
    return(0);
}
*/

/*
    #] CoFactorCache :
*/
```

4.9 compcomm.c File Reference

```
#include "form3.h"
#include "comtool.h"
```

Include dependency graph for compcomm.c:



Functions

- int CoFormat (UBYTE ∗s)
- int CoCollect (UBYTE ∗s)
- int setonoff (UBYTE ∗s, int ∗flag, int onvalue, int offvalue)
- int CoCompress (UBYTE ∗s)
- int CoFlags (UBYTE ∗s, int value)
- int CoOff (UBYTE ∗s)
- int CoOn (UBYTE ∗s)
- int CoInsideFirst (UBYTE ∗s)
- int CoProperCount (UBYTE ∗s)
- int CoDelete (UBYTE ∗s)
- int CoKeep (UBYTE ∗s)
- int CoFixIndex (UBYTE ∗s)
- int CoMetric (UBYTE ∗s)
- int DoPrint (UBYTE ∗s, int par)
- int CoPrint (UBYTE ∗s)

- int CoPrintB (UBYTE *s)
- int CoNPrint (UBYTE *s)
- int CoPushHide (UBYTE *s)
- int CoPopHide (UBYTE *s)
- int SetExprCases (int par, int setunset, int val)
- int SetExpr (UBYTE *s, int setunset, int par)
- int CoDrop (UBYTE *s)
- int CoNoDrop (UBYTE *s)
- int CoSkip (UBYTE *s)
- int CoNoSkip (UBYTE *s)
- int CoHide (UBYTE *inp)
- int CoIntoHide (UBYTE *inp)
- int CoNoIntoHide (UBYTE *inp)
- int CoNoHide (UBYTE *inp)
- int CoUnHide (UBYTE *inp)
- int CoNoUnHide (UBYTE *inp)
- void AddToCom (int n, WORD *array)
- int AddComString (int n, WORD *array, UBYTE *thestring, int par)
- int Add2ComStrings (int n, WORD *array, UBYTE *string1, UBYTE *string2)
- int CoDiscard (UBYTE *s)
- int CoContract (UBYTE *s)
- int CoGoTo (UBYTE *inp)
- int CoLabel (UBYTE *inp)
- int DoArgument (UBYTE *s, int par)
- int CoArgument (UBYTE *s)
- int CoEndArgument (UBYTE *s)
- int CoInside (UBYTE *s)
- int CoEndInside (UBYTE *s)
- int CoNormalize (UBYTE *s)
- int CoMakeInteger (UBYTE *s)
- int CoSplitArg (UBYTE *s)
- int CoSplitFirstArg (UBYTE *s)
- int CoSplitLastArg (UBYTE *s)
- int CoFactArg (UBYTE *s)
- int DoSymmetrize (UBYTE *s, int par)
- int CoSymmetrize (UBYTE *s)
- int CoAntiSymmetrize (UBYTE *s)
- int CoCycleSymmetrize (UBYTE *s)
- int CoRCycleSymmetrize (UBYTE *s)
- int CoWrite (UBYTE *s)
- int CoNWrite (UBYTE *s)
- int CoRatio (UBYTE *s)
- int CoRedefine (UBYTE *s)
- int CoRenumber (UBYTE *s)
- int CoSum (UBYTE *s)
- int CoToTensor (UBYTE *s)
- int CoToVector (UBYTE *s)
- int CoTrace4 (UBYTE *s)
- int CoTraceN (UBYTE *s)
- int CoChisholm (UBYTE *s)
- int DoChain (UBYTE *s, int option)
- int CoChainin (UBYTE *s)
- int CoChainout (UBYTE *s)
- int CoExit (UBYTE *s)
- int CoInParallel (UBYTE *s)

- int CoNotInParallel (UBYTE ∗s)
- int DoInParallel (UBYTE ∗s, int par)
- int CoInExpression (UBYTE ∗s)
- int CoEndInExpression (UBYTE ∗s)
- int CoSetExitFlag (UBYTE ∗s)
- int CoTryReplace (UBYTE ∗p)
- int CoModulus (UBYTE ∗inp)
- int CoRepeat (UBYTE ∗inp)
- int CoEndRepeat (UBYTE ∗inp)
- int DoBrackets (UBYTE ∗inp, int par)
- int CoBracket (UBYTE ∗inp)
- int CoAntiBracket (UBYTE ∗inp)
- int CoMultiBracket (UBYTE ∗inp)
- WORD ∗ CountComp (UBYTE ∗inp, WORD ∗to)
- int CoIf (UBYTE ∗inp)
- int CoElse (UBYTE ∗p)
- int CoElseIf (UBYTE ∗inp)
- int CoEndIf (UBYTE ∗inp)
- int CoWhile (UBYTE ∗inp)
- int CoEndWhile (UBYTE ∗inp)
- int DoFindLoop (UBYTE ∗inp, int mode)
- int CoFindLoop (UBYTE ∗inp)
- int CoReplaceLoop (UBYTE ∗inp)
- int CoFunPowers (UBYTE ∗inp)
- int CoUnitTrace (UBYTE ∗s)
- int CoTerm (UBYTE ∗s)
- int CoEndTerm (UBYTE ∗s)
- int CoSort (UBYTE ∗s)
- int CoPolyFun (UBYTE ∗s)
- int CoPolyRatFun (UBYTE ∗s)
- int CoMerge (UBYTE ∗inp)
- int CoStuffle (UBYTE ∗inp)
- int CoProcessBucket (UBYTE ∗s)
- int CoThreadBucket (UBYTE ∗s)
- int DoArgPlode (UBYTE ∗s, int par)
- int CoArgExplode (UBYTE ∗s)
- int CoArgImplode (UBYTE ∗s)
- int CoClearTable (UBYTE ∗s)
- int CoDenominators (UBYTE ∗s)
- int CoDropCoefficient (UBYTE ∗s)
- int CoDropSymbols (UBYTE ∗s)
- int CoToPolynomial (UBYTE ∗inp)
- int CoFromPolynomial (UBYTE ∗inp)
- int CoArgToExtraSymbol (UBYTE ∗s)
- int CoExtraSymbols (UBYTE ∗inp)
- WORD ∗ GetIfDollarFactor (UBYTE ∗∗inp, WORD ∗w)
- UBYTE ∗ GetDoParam (UBYTE ∗inp, WORD ∗∗wp, int par)
- int CoDo (UBYTE ∗inp)
- int CoEndDo (UBYTE ∗inp)
- int CoFactDollar (UBYTE ∗inp)
- int CoFactorize (UBYTE ∗s)
- int CoNFactorize (UBYTE ∗s)
- int CoUnFactorize (UBYTE ∗s)
- int CoNUnFactorize (UBYTE ∗s)
- int DoFactorize (UBYTE ∗s, int par)

- int CoOptimizeOption (UBYTE *s)
- int CoPutInside (UBYTE *inp)
- int CoAntiPutInside (UBYTE *inp)
- int DoPutInside (UBYTE *inp, int par)
- int CoSwitch (UBYTE *s)
- int CoCase (UBYTE *s)
- int CoBreak (UBYTE *s)
- int CoDefault (UBYTE *s)
- int CoEndSwitch (UBYTE *s)
- int CoSetUserFlag (UBYTE *s)
- int CoClearUserFlag (UBYTE *s)
- int CoCreateAllLoops (UBYTE *s)
- int CoCreateAllPaths (UBYTE *s)
- int CoCreateAll (UBYTE *s)

4.9.1 Detailed Description

Compiler routines for most statements that don't involve algebraic expressions. Exceptions: all routines involving declarations are in the file names.c When making new statements one can add the compiler routines here and have a look whether there is already a routine that is similar. In that case one can make a copy and modify it.

Definition in file compcomm.c.

4.9.2 Function Documentation

4.9.2.1 CoFormat()

```
int CoFormat (
            UBYTE * s )
```

Definition at line 153 of file compcomm.c.

4.9.2.2 CoCollect()

```
int CoCollect (
            UBYTE * s )
```

Definition at line 428 of file compcomm.c.

4.9.2.3 setonoff()

```
int setonoff (
            UBYTE * s,
            int * flag,
            int onvalue,
            int offvalue )
```

Definition at line 490 of file compcomm.c.

4.9.2.4 CoCompress()

```
int CoCompress (
              UBYTE * s )
```

Definition at line 506 of file compcomm.c.

4.9.2.5 CoFlags()

```
int CoFlags (
              UBYTE * s,
              int value )
```

Definition at line 555 of file compcomm.c.

4.9.2.6 CoOff()

```
int CoOff (
              UBYTE * s )
```

Definition at line 590 of file compcomm.c.

4.9.2.7 CoOn()

```
int CoOn (
              UBYTE * s )
```

Definition at line 656 of file compcomm.c.

4.9.2.8 CoInsideFirst()

```
int CoInsideFirst (
              UBYTE * s )
```

Definition at line 928 of file compcomm.c.

4.9.2.9 CoProperCount()

```
int CoProperCount (
              UBYTE * s )
```

Definition at line 935 of file compcomm.c.

4.9.2.10 CoDelete()

```
int CoDelete (
              UBYTE * s )
```

Definition at line 942 of file compcomm.c.

4.9.2.11 CoKeep()

```
int CoKeep (
            UBYTE * s )
```

Definition at line 987 of file compcomm.c.

4.9.2.12 CoFixIndex()

```
int CoFixIndex (
            UBYTE * s )
```

Definition at line 999 of file compcomm.c.

4.9.2.13 CoMetric()

```
int CoMetric (
            UBYTE * s )
```

Definition at line 1033 of file compcomm.c.

4.9.2.14 DoPrint()

```
int DoPrint (
            UBYTE * s,
            int par )
```

Definition at line 1041 of file compcomm.c.

4.9.2.15 CoPrint()

```
int CoPrint (
            UBYTE * s )
```

Definition at line 1257 of file compcomm.c.

4.9.2.16 CoPrintB()

```
int CoPrintB (
            UBYTE * s )
```

Definition at line 1264 of file compcomm.c.

4.9.2.17 CoNPrint()

```
int CoNPrint (
            UBYTE * s )
```

Definition at line 1271 of file compcomm.c.

4.9.2.18 CoPushHide()

```
int CoPushHide (
            UBYTE * s )
```

Definition at line 1278 of file compcomm.c.

4.9.2.19 CoPopHide()

```
int CoPopHide (
            UBYTE * s )
```

Definition at line 1322 of file compcomm.c.

4.9.2.20 SetExprCases()

```
int SetExprCases (
            int par,
            int setunset,
            int val )
```

Definition at line 1357 of file compcomm.c.

4.9.2.21 SetExpr()

```
int SetExpr (
            UBYTE * s,
            int setunset,
            int par )
```

Definition at line 1512 of file compcomm.c.

4.9.2.22 CoDrop()

```
int CoDrop (
            UBYTE * s )
```

Definition at line 1557 of file compcomm.c.

4.9.2.23 CoNoDrop()

```
int CoNoDrop (
            UBYTE * s )
```

Definition at line 1564 of file compcomm.c.

4.9.2.24 CoSkip()

```
int CoSkip (
            UBYTE * s )
```

Definition at line 1571 of file compcomm.c.

4.9.2.25 CoNoSkip()

```
int CoNoSkip (
            UBYTE * s )
```

Definition at line 1578 of file compcomm.c.

4.9.2.26 CoHide()

```
int CoHide (
            UBYTE * inp )
```

Definition at line 1585 of file compcomm.c.

4.9.2.27 CoIntoHide()

```
int CoIntoHide (
            UBYTE * inp )
```

Definition at line 1603 of file compcomm.c.

4.9.2.28 CoNoIntoHide()

```
int CoNoIntoHide (
            UBYTE * inp )
```

Definition at line 1621 of file compcomm.c.

4.9.2.29 CoNoHide()

```
int CoNoHide (
            UBYTE * inp )
```

Definition at line 1628 of file compcomm.c.

4.9.2.30 CoUnHide()

```
int CoUnHide (
            UBYTE * inp )
```

Definition at line 1635 of file compcomm.c.

4.9.2.31 CoNoUnHide()

```
int CoNoUnHide (
             UBYTE * inp )
```

Definition at line 1642 of file compcomm.c.

4.9.2.32 AddToCom()

```
void AddToCom (
             int n,
             WORD * array )
```

Definition at line 1649 of file compcomm.c.

4.9.2.33 AddComString()

```
int AddComString (
             int n,
             WORD * array,
             UBYTE * thestring,
             int par )
```

Definition at line 1664 of file compcomm.c.

4.9.2.34 Add2ComStrings()

```
int Add2ComStrings (
             int n,
             WORD * array,
             UBYTE * string1,
             UBYTE * string2 )
```

Definition at line 1723 of file compcomm.c.

4.9.2.35 CoDiscard()

```
int CoDiscard (
             UBYTE * s )
```

Definition at line 1764 of file compcomm.c.

4.9.2.36 CoContract()

```
int CoContract (
             UBYTE * s )
```

Definition at line 1787 of file compcomm.c.

4.9.2.37 CoGoTo()

```
int CoGoTo (
            UBYTE * inp )
```

Definition at line 1816 of file compcomm.c.

4.9.2.38 CoLabel()

```
int CoLabel (
            UBYTE * inp )
```

Definition at line 1835 of file compcomm.c.

4.9.2.39 DoArgument()

```
int DoArgument (
            UBYTE * s,
            int par )
```

Definition at line 1862 of file compcomm.c.

4.9.2.40 CoArgument()

```
int CoArgument (
            UBYTE * s )
```

Definition at line 2096 of file compcomm.c.

4.9.2.41 CoEndArgument()

```
int CoEndArgument (
            UBYTE * s )
```

Definition at line 2103 of file compcomm.c.

4.9.2.42 CoInside()

```
int CoInside (
            UBYTE * s )
```

Definition at line 2129 of file compcomm.c.

4.9.2.43 CoEndInside()

```
int CoEndInside (
            UBYTE * s )
```

Definition at line 2136 of file compcomm.c.

4.9.2.44 CoNormalize()

```
int CoNormalize (
            UBYTE * s )
```

Definition at line 2162 of file compcomm.c.

4.9.2.45 CoMakeInteger()

```
int CoMakeInteger (
            UBYTE * s )
```

Definition at line 2169 of file compcomm.c.

4.9.2.46 CoSplitArg()

```
int CoSplitArg (
            UBYTE * s )
```

Definition at line 2176 of file compcomm.c.

4.9.2.47 CoSplitFirstArg()

```
int CoSplitFirstArg (
            UBYTE * s )
```

Definition at line 2183 of file compcomm.c.

4.9.2.48 CoSplitLastArg()

```
int CoSplitLastArg (
            UBYTE * s )
```

Definition at line 2190 of file compcomm.c.

4.9.2.49 CoFactArg()

```
int CoFactArg (
            UBYTE * s )
```

Definition at line 2197 of file compcomm.c.

4.9.2.50 DoSymmetrize()

```
int DoSymmetrize (
            UBYTE * s,
            int par )
```

Definition at line 2217 of file compcomm.c.

4.9.2.51 CoSymmetrize()

```
int CoSymmetrize (
            UBYTE * s )
```

Definition at line 2337 of file compcomm.c.

4.9.2.52 CoAntiSymmetrize()

```
int CoAntiSymmetrize (
            UBYTE * s )
```

Definition at line 2344 of file compcomm.c.

4.9.2.53 CoCycleSymmetrize()

```
int CoCycleSymmetrize (
            UBYTE * s )
```

Definition at line 2351 of file compcomm.c.

4.9.2.54 CoRCycleSymmetrize()

```
int CoRCycleSymmetrize (
            UBYTE * s )
```

Definition at line 2358 of file compcomm.c.

4.9.2.55 CoWrite()

```
int CoWrite (
            UBYTE * s )
```

Definition at line 2365 of file compcomm.c.

4.9.2.56 CoNWrite()

```
int CoNWrite (
            UBYTE * s )
```

Definition at line 2390 of file compcomm.c.

4.9.2.57 CoRatio()

```
int CoRatio (
            UBYTE * s )
```

Definition at line 2417 of file compcomm.c.

4.9.2.58 CoRedefine()

```
int CoRedefine (
            UBYTE * s )
```

Definition at line 2457 of file compcomm.c.

4.9.2.59 CoRenumber()

```
int CoRenumber (
            UBYTE * s )
```

Definition at line 2562 of file compcomm.c.

4.9.2.60 CoSum()

```
int CoSum (
            UBYTE * s )
```

Definition at line 2583 of file compcomm.c.

4.9.2.61 CoToTensor()

```
int CoToTensor (
            UBYTE * s )
```

Definition at line 2703 of file compcomm.c.

4.9.2.62 CoToVector()

```
int CoToVector (
            UBYTE * s )
```

Definition at line 2871 of file compcomm.c.

4.9.2.63 CoTrace4()

```
int CoTrace4 (
            UBYTE * s )
```

Definition at line 2941 of file compcomm.c.

4.9.2.64 CoTraceN()

```
int CoTraceN (
            UBYTE * s )
```

Definition at line 3028 of file compcomm.c.

4.9.2.65 CoChisholm()

```
int CoChisholm (
              UBYTE * s )
```

Definition at line 3088 of file compcomm.c.

4.9.2.66 DoChain()

```
int DoChain (
              UBYTE * s,
              int option )
```

Definition at line 3170 of file compcomm.c.

4.9.2.67 CoChainin()

```
int CoChainin (
              UBYTE * s )
```

Definition at line 3214 of file compcomm.c.

4.9.2.68 CoChainout()

```
int CoChainout (
              UBYTE * s )
```

Definition at line 3226 of file compcomm.c.

4.9.2.69 CoExit()

```
int CoExit (
              UBYTE * s )
```

Definition at line 3236 of file compcomm.c.

4.9.2.70 CoInParallel()

```
int CoInParallel (
              UBYTE * s )
```

Definition at line 3263 of file compcomm.c.

4.9.2.71 CoNotInParallel()

```
int CoNotInParallel (
              UBYTE * s )
```

Definition at line 3273 of file compcomm.c.

4.9.2.72 DoInParallel()

```
int DoInParallel (
            UBYTE * s,
            int par )
```

Definition at line 3288 of file compcomm.c.

4.9.2.73 CoInExpression()

```
int CoInExpression (
            UBYTE * s )
```

Definition at line 3357 of file compcomm.c.

4.9.2.74 CoEndInExpression()

```
int CoEndInExpression (
            UBYTE * s )
```

Definition at line 3410 of file compcomm.c.

4.9.2.75 CoSetExitFlag()

```
int CoSetExitFlag (
            UBYTE * s )
```

Definition at line 3436 of file compcomm.c.

4.9.2.76 CoTryReplace()

```
int CoTryReplace (
            UBYTE * p )
```

Definition at line 3450 of file compcomm.c.

4.9.2.77 CoModulus()

```
int CoModulus (
            UBYTE * inp )
```

Definition at line 3553 of file compcomm.c.

4.9.2.78 CoRepeat()

```
int CoRepeat (
            UBYTE * inp )
```

Definition at line 3691 of file compcomm.c.

4.9.2.79 CoEndRepeat()

```
int CoEndRepeat (
            UBYTE * inp )
```

Definition at line 3714 of file compcomm.c.

4.9.2.80 DoBrackets()

```
int DoBrackets (
            UBYTE * inp,
            int par )
```

Definition at line 3752 of file compcomm.c.

4.9.2.81 CoBracket()

```
int CoBracket (
            UBYTE * inp )
```

Definition at line 3879 of file compcomm.c.

4.9.2.82 CoAntiBracket()

```
int CoAntiBracket (
            UBYTE * inp )
```

Definition at line 3887 of file compcomm.c.

4.9.2.83 CoMultiBracket()

```
int CoMultiBracket (
            UBYTE * inp )
```

Definition at line 3898 of file compcomm.c.

4.9.2.84 CountComp()

```
WORD * CountComp (
            UBYTE * inp,
            WORD * to )
```

Definition at line 4030 of file compcomm.c.

4.9.2.85 CoIf()

```
int CoIf (
            UBYTE * inp )
```

Definition at line 4220 of file compcomm.c.

4.9.2.86 CoElse()

```
int CoElse (
            UBYTE * p )
```

Definition at line 4862 of file compcomm.c.

4.9.2.87 CoElseIf()

```
int CoElseIf (
            UBYTE * inp )
```

Definition at line 4889 of file compcomm.c.

4.9.2.88 CoEndIf()

```
int CoEndIf (
            UBYTE * inp )
```

Definition at line 4916 of file compcomm.c.

4.9.2.89 CoWhile()

```
int CoWhile (
            UBYTE * inp )
```

Definition at line 4961 of file compcomm.c.

4.9.2.90 CoEndWhile()

```
int CoEndWhile (
            UBYTE * inp )
```

Definition at line 4982 of file compcomm.c.

4.9.2.91 DoFindLoop()

```
int DoFindLoop (
            UBYTE * inp,
            int mode )
```

Definition at line 5010 of file compcomm.c.

4.9.2.92 CoFindLoop()

```
int CoFindLoop (
            UBYTE * inp )
```

Definition at line 5127 of file compcomm.c.

4.9.2.93 CoReplaceLoop()

```
int CoReplaceLoop (
            UBYTE * inp )
```

Definition at line 5135 of file compcomm.c.

4.9.2.94 CoFunPowers()

```
int CoFunPowers (
            UBYTE * inp )
```

Definition at line 5155 of file compcomm.c.

4.9.2.95 CoUnitTrace()

```
int CoUnitTrace (
            UBYTE * s )
```

Definition at line 5182 of file compcomm.c.

4.9.2.96 CoTerm()

```
int CoTerm (
            UBYTE * s )
```

Definition at line 5217 of file compcomm.c.

4.9.2.97 CoEndTerm()

```
int CoEndTerm (
            UBYTE * s )
```

Definition at line 5265 of file compcomm.c.

4.9.2.98 CoSort()

```
int CoSort (
            UBYTE * s )
```

Definition at line 5292 of file compcomm.c.

4.9.2.99 CoPolyFun()

```
int CoPolyFun (
            UBYTE * s )
```

Definition at line 5331 of file compcomm.c.

4.9.2.100 CoPolyRatFun()

```
int CoPolyRatFun (
            UBYTE * s )
```

Definition at line 5378 of file compcomm.c.

4.9.2.101 CoMerge()

```
int CoMerge (
            UBYTE * inp )
```

Definition at line 5548 of file compcomm.c.

4.9.2.102 CoStuffle()

```
int CoStuffle (
            UBYTE * inp )
```

Definition at line 5604 of file compcomm.c.

4.9.2.103 CoProcessBucket()

```
int CoProcessBucket (
            UBYTE * s )
```

Definition at line 5659 of file compcomm.c.

4.9.2.104 CoThreadBucket()

```
int CoThreadBucket (
            UBYTE * s )
```

Definition at line 5677 of file compcomm.c.

4.9.2.105 DoArgPlode()

```
int DoArgPlode (
            UBYTE * s,
            int par )
```

Definition at line 5707 of file compcomm.c.

4.9.2.106 CoArgExplode()

```
int CoArgExplode (
            UBYTE * s )
```

Definition at line 5763 of file compcomm.c.

4.9.2.107 CoArgImplode()

```
int CoArgImplode (
            UBYTE * s )
```

Definition at line 5770 of file compcomm.c.

4.9.2.108 CoClearTable()

```
int CoClearTable (
            UBYTE * s )
```

Definition at line 5777 of file compcomm.c.

4.9.2.109 CoDenominators()

```
int CoDenominators (
            UBYTE * s )
```

Definition at line 5859 of file compcomm.c.

4.9.2.110 CoDropCoefficient()

```
int CoDropCoefficient (
            UBYTE * s )
```

Definition at line 5888 of file compcomm.c.

4.9.2.111 CoDropSymbols()

```
int CoDropSymbols (
            UBYTE * s )
```

Definition at line 5902 of file compcomm.c.

4.9.2.112 CoToPolynomial()

```
int CoToPolynomial (
            UBYTE * inp )
```

Definition at line 5925 of file compcomm.c.

4.9.2.113 CoFromPolynomial()

```
int CoFromPolynomial (
            UBYTE * inp )
```

Definition at line 6000 of file compcomm.c.

4.9.2.114 CoArgToExtraSymbol()

```
int CoArgToExtraSymbol (
            UBYTE * s )
```

Definition at line 6026 of file compcomm.c.

4.9.2.115 CoExtraSymbols()

```
int CoExtraSymbols (
            UBYTE * inp )
```

Definition at line 6076 of file compcomm.c.

4.9.2.116 GetIfDollarFactor()

```
WORD * GetIfDollarFactor (
            UBYTE ** inp,
            WORD * w )
```

Definition at line 6149 of file compcomm.c.

4.9.2.117 GetDoParam()

```
UBYTE * GetDoParam (
            UBYTE * inp,
            WORD ** wp,
            int par )
```

Definition at line 6207 of file compcomm.c.

4.9.2.118 CoDo()

```
int CoDo (
            UBYTE * inp )
```

Definition at line 6284 of file compcomm.c.

4.9.2.119 CoEndDo()

```
int CoEndDo (
            UBYTE * inp )
```

Definition at line 6387 of file compcomm.c.

4.9.2.120 CoFactDollar()

```
int CoFactDollar (
              UBYTE * inp )
```

Definition at line 6418 of file compcomm.c.

4.9.2.121 CoFactorize()

```
int CoFactorize (
              UBYTE * s )
```

Definition at line 6447 of file compcomm.c.

4.9.2.122 CoNFactorize()

```
int CoNFactorize (
              UBYTE * s )
```

Definition at line 6454 of file compcomm.c.

4.9.2.123 CoUnFactorize()

```
int CoUnFactorize (
              UBYTE * s )
```

Definition at line 6461 of file compcomm.c.

4.9.2.124 CoNUnFactorize()

```
int CoNUnFactorize (
              UBYTE * s )
```

Definition at line 6468 of file compcomm.c.

4.9.2.125 DoFactorize()

```
int DoFactorize (
              UBYTE * s,
              int par )
```

Definition at line 6475 of file compcomm.c.

4.9.2.126 CoOptimizeOption()

```
int CoOptimizeOption (
              UBYTE * s )
```

Definition at line 6608 of file compcomm.c.

4.9.2.127 CoPutInside()

```
int CoPutInside (
            UBYTE * inp )
```

Definition at line 7044 of file compcomm.c.

4.9.2.128 CoAntiPutInside()

```
int CoAntiPutInside (
            UBYTE * inp )
```

Definition at line 7045 of file compcomm.c.

4.9.2.129 DoPutInside()

```
int DoPutInside (
            UBYTE * inp,
            int par )
```

Definition at line 7047 of file compcomm.c.

4.9.2.130 CoSwitch()

```
int CoSwitch (
            UBYTE * s )
```

Definition at line 7167 of file compcomm.c.

4.9.2.131 CoCase()

```
int CoCase (
            UBYTE * s )
```

Definition at line 7213 of file compcomm.c.

4.9.2.132 CoBreak()

```
int CoBreak (
            UBYTE * s )
```

Definition at line 7263 of file compcomm.c.

4.9.2.133 CoDefault()

```
int CoDefault (
            UBYTE * s )
```

Definition at line 7289 of file compcomm.c.

4.9.2.134 CoEndSwitch()

```
int CoEndSwitch (
              UBYTE * s )
```

Definition at line 7316 of file compcomm.c.

4.9.2.135 CoSetUserFlag()

```
int CoSetUserFlag (
              UBYTE * s )
```

Definition at line 7384 of file compcomm.c.

4.9.2.136 CoClearUserFlag()

```
int CoClearUserFlag (
              UBYTE * s )
```

Definition at line 7412 of file compcomm.c.

4.9.2.137 CoCreateAllLoops()

```
int CoCreateAllLoops (
              UBYTE * s )
```

Definition at line 7449 of file compcomm.c.

4.9.2.138 CoCreateAllPaths()

```
int CoCreateAllPaths (
              UBYTE * s )
```

Definition at line 7569 of file compcomm.c.

4.9.2.139 CoCreateAll()

```
int CoCreateAll (
              UBYTE * s )
```

Definition at line 7697 of file compcomm.c.

4.10 compcomm.c

Go to the documentation of this file.

```
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/*
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 */
/* #] License : */
/*
    #[ includes :
*/

#include "form3.h"
#include "comtool.h"
#ifdef WITHFLOAT
#include <gmp.h>
#endif

static KEYWORD formatoptions[] = {
     {"allfloat",        (TFUN)0,    ALLINTEGERDOUBLE,   0}
    ,{"c",               (TFUN)0,    CMODE,              0}
    ,{"doublefortran",   (TFUN)0,    DOUBLEFORTRANMODE,  0}
    ,{"float",           (TFUN)0,    0,                  2}
#ifdef WITHFLOAT
    ,{"floatprecision",  (TFUN)0,    0,                  5}
#endif
    ,{"fortran",         (TFUN)0,    FORTRANMODE,        0}
    ,{"fortran90",       (TFUN)0,    FORTRANMODE,        4}
    ,{"maple",           (TFUN)0,    MAPLEMODE,          0}
    ,{"mathematica",     (TFUN)0,    MATHEMATICAMODE,    0}
    ,{"normal",          (TFUN)0,    NORMALFORMAT,       1}
    ,{"nospaces",        (TFUN)0,    NOSPACEFORMAT,      3}
    ,{"pfortran",        (TFUN)0,    PFORTRANMODE,       0}
    ,{"quadfortran",     (TFUN)0,    QUADRUPLEFORTRANMODE,   0}
    ,{"quadruplefortran",    (TFUN)0,    QUADRUPLEFORTRANMODE,   0}
    ,{"rational",        (TFUN)0,    RATIONALMODE,       1}
    ,{"reduce",          (TFUN)0,    REDUCEMODE,         0}
    ,{"spaces",          (TFUN)0,    NORMALFORMAT,       3}
    ,{"vortran",         (TFUN)0,    VORTRANMODE,        0}
};

static KEYWORD trace4options[] = {
     {"contract",     (TFUN)0,    CHISHOLM,       0          }
    ,{"nocontract",   (TFUN)0,    0,              CHISHOLM   }
    ,{"nosymmetrize",(TFUN)0,    0,              ALSOREVERSE}
    ,{"notrick",      (TFUN)0,    NOTRICK,        0          }
    ,{"symmetrize",   (TFUN)0,    ALSOREVERSE,    0          }
    ,{"trick",        (TFUN)0,    0,              NOTRICK    }
};

static KEYWORD chisoptions[] = {
     {"nosymmetrize",(TFUN)0,    0,              ALSOREVERSE}
    ,{"symmetrize",   (TFUN)0,    ALSOREVERSE,    0          }
};

static KEYWORDV writeoptions[] = {
     {"stats",           &(AC.StatsFlag),    1,      0}
    ,{"statistics",      &(AC.StatsFlag),    1,      0}
    ,{"shortstats",      &(AC.ShortStats),   1,      0}
    ,{"shortstatistics",&(AC.ShortStats),   1,      0}
    ,{"warnings",        &(AC.WarnFlag), 1,      0}
    ,{"allwarnings",     &(AC.WarnFlag), 2,      0}
    ,{"setup",           &(AC.SetupFlag),    1,      0}
```

```
	,{"names",          &(AC.NamesFlag),     1,       0}
	,{"allnames",       &(AC.NamesFlag),     2,       0}
	,{"codes",          &(AC.CodesFlag),     1,       0}
	,{"highfirst",      &(AC.SortType), SORTHIGHFIRST,      SORTLOWFIRST}
	,{"lowfirst",       &(AC.SortType), SORTLOWFIRST,       SORTHIGHFIRST}
	,{"powerfirst",     &(AC.SortType), SORTPOWERFIRST,     SORTHIGHFIRST}
	,{"tokens",         &(AC.TokensWriteFlag),1,    0}
};

static KEYWORDV onoffoptions[] = {
	 {"compress",       &(AC.NoCompress),  0,  1}
	,{"checkpoint",     &(AC.CheckpointFlag),  1,  0}
	,{"insidefirst",    &(AC.insidefirst), 1,  0}
	,{"propercount",    &(AC.BottomLevel), 1,  0}
	,{"stats",          &(AC.StatsFlag),    1,  0}
	,{"statistics",     &(AC.StatsFlag),    1,  0}
	,{"shortstats",     &(AC.ShortStats),   1,  0}
	,{"shortstatistics",&(AC.ShortStats),   1,  0}
	,{"names",          &(AC.NamesFlag),    1,  0}
	,{"allnames",       &(AC.NamesFlag),    2,  0}
	,{"warnings",       &(AC.WarnFlag), 1,  0}
	,{"allwarnings",    &(AC.WarnFlag), 2,  0}
	,{"highfirst",      &(AC.SortType), SORTHIGHFIRST,  SORTLOWFIRST}
	,{"lowfirst",       &(AC.SortType), SORTLOWFIRST,   SORTHIGHFIRST}
	,{"powerfirst",     &(AC.SortType), SORTPOWERFIRST, SORTHIGHFIRST}
	,{"setup",          &(AC.SetupFlag),    1,  0}
	,{"codes",          &(AC.CodesFlag),    1,  0}
	,{"tokens",         &(AC.TokensWriteFlag),1,0}
	,{"properorder",    &(AC.properorderflag),1,0}
	,{"threadloadbalancing",&(AC.ThreadBalancing),1,    0}
	,{"threads",        &(AC.ThreadsFlag),1,    0}
	,{"threadsortfilesynch",&(AC.ThreadSortFileSynch),1,  0}
	,{"threadstats",    &(AC.ThreadStats),1,    0}
	,{"finalstats",     &(AC.FinalStats),1, 0}
	,{"fewerstats",     &(AC.ShortStatsMax),    10,     0}
	,{"fewerstatistics",&(AC.ShortStatsMax),    10,     0}
	,{"processstats",   &(AC.ProcessStats),1,   0}
	,{"oldparallelstats",&(AC.OldParallelStats),1,0}
	,{"parallel",       &(AC.parallelflag),PARALLELFLAG,NOPARALLEL_USER}
	,{"nospacesinnumbers",&(AO.NoSpacesInNumbers),1,0}
	,{"indentspace",    &(AO.IndentSpace),INDENTSPACE,0}
	,{"totalsize",      &(AM.PrintTotalSize),   1,  0}
	,{"flag",           (int *)&(AC.debugFlags),    1,  0}
	,{"oldfactarg",     &(AC.OldFactArgFlag),   1,  0}
	,{"memdebugflag",   &(AC.MemDebugFlag), 1,  0}
	,{"oldgcd",         &(AC.OldGCDflag),   1,  0}
	,{"innertest",      &(AC.InnerTest),  1,  0}
	,{"wtimestats",     &(AC.WTimeStatsFlag),  1,  0}
	,{"sortreallocate", &(AC.SortReallocateFlag), 1, 0}
	,{"backtrace",      &(AC.PrintBacktraceFlag), 1, 0}
	,{"flint",          &(AC.FlintPolyFlag), 1, 0}
	,{"humanstats", &(AC.HumanStatsFlag), 1, 0}
	,{"humanstatistics",    &(AC.HumanStatsFlag), 1, 0}
};

static WORD one = 1;

/*
	#] includes :
	#[ CoFormat :
*/

int CoFormat(UBYTE *s)
{
	int error = 0, x;
	KEYWORD *key;
	UBYTE *ss;
	while ( *s == ' ' || *s == ',' ) s++;
	if ( *s == 0 ) {
		AC.OutputMode = 72;
		AC.OutputSpaces = NORMALFORMAT;
		return(error);
	}
/*
	First the optimization level
*/
	if ( *s == 'O' || *s == 'o' ) {
		if ( ( FG.cTable[s[1]] == 1 ) ||
			 ( s[1] == '=' && FG.cTable[s[2]] == 1 ) ) {
			s++; if ( *s == '=' ) s++;
			x = 0;
			while ( *s >= '0' && *s <= '9' ) x = 10*x + *s++ - '0';
			while ( *s == ',' ) s++;
			AO.OptimizationLevel = x;
			AO.Optimize.greedytimelimit = 0;
			AO.Optimize.mctstimelimit = 0;
			AO.Optimize.printstats = 0;
```

```
            AO.Optimize.debugflags = 0;
            AO.Optimize.schemeflags = 0;
            AO.Optimize.mctsdecaymode = 1; /* default is decreasing C_p with iteration number */
            if ( AO.inscheme ) {
                M_free(AO.inscheme,"Horner input scheme");
                AO.inscheme = 0; AO.schemenum = 0;
            }
            switch ( x ) {
                case 0:
                    break;
                case 1:
                    AO.Optimize.mctsconstant.fval = -1.0;
                    AO.Optimize.horner = O_OCCURRENCE;
                    AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
                    AO.Optimize.method = O_CSE;
                    break;
                case 2:
                    AO.Optimize.horner = O_OCCURRENCE;
                    AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
                    AO.Optimize.method = O_GREEDY;
                    AO.Optimize.greedyminnum = 10;
                    AO.Optimize.greedymaxperc = 5;
                    break;
                case 3:
                    AO.Optimize.mctsconstant.fval = 1.0;
                    AO.Optimize.horner = O_MCTS;
                    AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
                    AO.Optimize.method = O_GREEDY;
                    AO.Optimize.mctsnumexpand = 1000;
                    AO.Optimize.mctsnumkeep = 10;
                    AO.Optimize.mctsnumrepeat = 1;
                    AO.Optimize.greedyminnum = 10;
                    AO.Optimize.greedymaxperc = 5;
                    break;
                case 4:
                    AO.Optimize.horner = O_SIMULATED_ANNEALING;
                    AO.Optimize.saIter = 1000;
                    AO.Optimize.saMaxT.fval = 2000;
                    AO.Optimize.saMinT.fval = 1;
                    break;
                default:
                    error = 1;
                    MesPrint("&Illegal optimization specification in format statement");
                    break;
            }
            if ( error == 0 && *s != 0 && x > 0 ) return(CoOptimizeOption(s));
            return(error);
        }
#ifdef EXPOPT
        { UBYTE c;
        ss = s;
        while ( FG.cTable[*s] == 0 ) s++;
        c = *s; *s = 0;
        if ( StrICont(ss,(UBYTE *)"optimize") == 0 ) {
            *s = c;
            while ( *s == ',' ) s++;
            if ( *s == '=' ) s++;
            AO.OptimizationLevel = 3;
            AO.Optimize.mctsconstant.fval = 1.0;
            AO.Optimize.horner = O_MCTS;
            AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
            AO.Optimize.method = O_GREEDY;
            AO.Optimize.mctstimelimit = 0;
            AO.Optimize.mctsnumexpand = 1000;
            AO.Optimize.mctsnumkeep = 10;
            AO.Optimize.mctsnumrepeat = 1;
            AO.Optimize.greedytimelimit = 0;
            AO.Optimize.greedyminnum = 10;
            AO.Optimize.greedymaxperc = 5;
            AO.Optimize.printstats = 0;
            AO.Optimize.debugflags = 0;
            AO.Optimize.schemeflags = 0;
            AO.Optimize.mctsdecaymode = 1;
            if ( AO.inscheme ) {
                M_free(AO.inscheme,"Horner input scheme");
                AO.inscheme = 0; AO.schemenum = 0;
            }
            return(CoOptimizeOption(s));
        }
        else {
            error = 1;
            MesPrint("&Illegal optimization specification in format statement");
            return(error);
        }
        }
#endif
    }
```

```
    else if ( FG.cTable[*s] == 1 ) {
        x = 0;
        while ( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
        if ( x <= 0 || x >= MAXLINELENGTH ) {
            error = 1;
            MesPrint("&Illegal value for linesize: %d",x);
            x = 72;
        }
        if ( x < 39 ) {
            MesPrint(" ... Too small value for linesize corrected to 39");
            x = 39;
        }
        AO.DoubleFlag = 0;
/*
        The next line resets the mode to normal. Because the special modes
        reset the line length we have a little problem with the special modes
        and customized line length. We try to improve by removing the next line
*/
/*      AC.OutputMode = 0;  */
        AC.LineLength = x;
        if ( *s != 0 ) {
            error = 1;
            MesPrint("&Illegal linesize field in format statement");
        }
    }
    else {
        key = FindKeyWord(s,formatoptions,
            sizeof(formatoptions)/sizeof(KEYWORD));
        if ( key ) {
            if ( key->type == FORTRANMODE || key->type == PFORTRANMODE || key->type == DOUBLEFORTRANMODE
                || key->type == QUADRUPLEFORTRANMODE || key->type == VORTRANMODE ) {
                    if (AC.LineLength > 72) AC.LineLength = 72;
            }
            if ( key->flags == 0 ) {
                if ( key->type == FORTRANMODE || key->type == PFORTRANMODE
                || key->type == DOUBLEFORTRANMODE || key->type == ALLINTEGERDOUBLE
                || key->type == QUADRUPLEFORTRANMODE || key->type == VORTRANMODE ) {
                    AC.IsFortran90 = ISNOTFORTRAN90;
                    if ( AC.Fortran90Kind ) {
                        M_free(AC.Fortran90Kind,"Fortran90 Kind");
                        AC.Fortran90Kind = 0;
                    }
                }
                if ( ( key->type == ALLINTEGERDOUBLE ) && AO.DoubleFlag != 0 ) {
                    AO.DoubleFlag |= 4;
                }
                else {
                    AO.DoubleFlag = 0;
                    AC.OutputMode = key->type & NODOUBLEMASK;
                    if ( ( key->type & DOUBLEPRECISIONFLAG ) != 0 ) {
                        AO.DoubleFlag = 1;
                    }
                    else if ( ( key->type & QUADRUPLEPRECISIONFLAG ) != 0 ) {
                        AO.DoubleFlag = 2;
                    }
                }
            }
            else if ( key->flags == 1 ) {
                AC.OutputMode = AC.OutNumberType = key->type;
            }
            else if ( key->flags == 2 ) {
                while ( FG.cTable[*s] == 0 ) s++;
                if ( *s == 0 ) AC.OutNumberType = 10;
                else if ( *s == ',' ) {
                    s++;
                    x = 0;
                    while ( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
                    if ( *s != 0 ) {
                        error = 1;
                        MesPrint("&Illegal float format specifier");
                    }
                    else {
                        if ( x < 3 ) {
                            x = 3;
                            MesPrint("& ... float format value corrected to 3");
                        }
                        if ( x > 100 ) {
                            x = 100;
                            MesPrint("& ... float format value corrected to 100");
                        }
                        AC.OutNumberType = x;
                    }
                }
            }
            else if ( key->flags == 3 ) {
                AC.OutputSpaces = key->type;
```

```
            }
            else if ( key->flags == 4 ) {
                AC.IsFortran90 = ISFORTRAN90;
                if ( AC.Fortran90Kind ) {
                    M_free(AC.Fortran90Kind,"Fortran90 Kind");
                    AC.Fortran90Kind = 0;
                }
                while ( FG.cTable[*s] <= 1 ) s++;
                if ( *s == ',' ) {
                    s++; ss = s;
                    while ( *ss && *ss != ',' ) ss++;
                    if ( *ss == ',' ) {
                        MesPrint("&No white space or comma's allowed in Fortran90 option: %s",s);
error = 1;
                    }
                    else {
                        AC.Fortran90Kind = strDup1(s,"Fortran90 Kind");
                    }
                }
                AO.DoubleFlag = 0;
                AC.OutputMode = key->type & NODOUBLEMASK;
            }
#ifdef WITHFLOAT
            else if ( key->flags == 5 ) {
/*
                Syntax: Format FloatPrecision number;
                        Format FloatPrecision off;
*/
                while ( FG.cTable[*s] == 0 ) s++;
                while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
                if ( *s == 0 ) {
                    AO.FloatPrec = 0;
                }
                else if ( tolower(*s) == 'o' && tolower(s[1]) == 'f'
                && tolower(s[2]) == 'f' ) {
                    ss = s;
                    s += 3;
                    while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
                    if ( *s ) { s = ss; goto WrongOption; }
                    AO.FloatPrec = -1;
                }
                else if ( FG.cTable[*s] == 1 ) {
                    ss = s;
                    AO.FloatPrec = 0;
                    while ( *s <= '9' && *s >= '0' )
                        AO.FloatPrec = 10*AO.FloatPrec + (*s++ - '0');
                    while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
                    if ( *s ) { s = ss; goto WrongOption; }
                }
                else {
WrongOption:        MesPrint("&Illegal option in Format FloatPrecision: %s",s);
                    error = 1;
                }
            }
#endif
        }
        else if ( ( *s == 'c' || *s == 'C' ) && ( FG.cTable[s[1]] == 1 ) ) {
            UBYTE *ss = s+1;
            WORD x = 0;
            while ( *ss >= '0' && *ss <= '9' ) x = 10*x + *ss++ - '0';
            if ( *ss != 0 ) goto Unknown;
            AC.OutputMode = CMODE;
            AC.Cnumpows = x;
        }
        else {
Unknown:    MesPrint("&Unknown option: %s",s); error = 1;
        }
    }
    return(error);
}

/*
    #] CoFormat :
    #[ CoCollect :

    Collect,functionname
*/

int CoCollect(UBYTE *s)
{
/*  --------------change 17-feb-2003 Added percentage */
    WORD numfun;
    int type,x = 0;
    UBYTE *t = SkipAName(s), *t1, *t2;
    AC.AltCollectFun = 0;
    if ( t == 0 ) goto syntaxerror;
    t1 = t; while ( *t1 == ',' || *t1 == ' ' || *t1 == '\t' ) t1++;
```

```
    *t = 0; t = t1;
    if ( *t1 && ( FG.cTable[*t1] == 0 || *t1 == '[' ) ) {
        t2 = SkipAName(t1);
        if ( t2 == 0 ) goto syntaxerror;
        t = t2;
        while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
        *t2 = 0;
    }
    else t1 = 0;
    if ( *t && FG.cTable[*t] == 1 ) {
        while ( *t >= '0' && *t <= '9' ) x = 10*x + *t++ - '0';
        if ( x > 100 ) x = 100;
        while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
        if ( *t ) goto syntaxerror;
    }
    else {
        if ( *t ) goto syntaxerror;
        x = 100;
    }
    if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
    || ( functions[numfun].spec != 0 ) ) {
        MesPrint("&%s should be a regular function",s);
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        return(1);
    }
    AC.CollectFun = numfun+FUNCTION;
    AC.CollectPercentage = (WORD)x;
    if ( t1 ) {
        if ( ( ( type = GetName(AC.varnames,t1,&numfun,WITHAUTO) ) != CFUNCTION )
        || ( functions[numfun].spec != 0 ) ) {
            MesPrint("&%s should be a regular function",t1);
            if ( type < 0 ) {
                if ( GetName(AC.exprnames,t1,&numfun,NOAUTO) == NAMENOTFOUND )
                    AddFunction(t1,0,0,0,0,0,-1,-1);
            }
            return(1);
        }
        AC.AltCollectFun = numfun+FUNCTION;
    }
    return(0);
syntaxerror:
    MesPrint("&Collect statement needs one or two functions (and a percentage) for its argument(s)");
    return(1);
}

/*
    #] CoCollect :
    #[ setonoff :
*/

int setonoff(UBYTE *s, int *flag, int onvalue, int offvalue)
{
    if ( StrICmp(s,(UBYTE *)"on") == 0 ) *flag = onvalue;
    else if ( StrICmp(s,(UBYTE *)"off") == 0 ) *flag = offvalue;
    else {
        MesPrint("&Unknown option: %s, on or off expected",s);
        return(1);
    }
    return(0);
}

/*
    #] setonoff :
    #[ CoCompress :
*/

int CoCompress(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c;
    if ( StrICmp(s,(UBYTE *)"on") == 0 ) {
        AC.NoCompress = 0;
        AR.gzipCompress = 0;
    }
    else if ( StrICmp(s,(UBYTE *)"off") == 0 ) {
        AC.NoCompress = 1;
        AR.gzipCompress = 0;
    }
    else {
        t = s; while ( FG.cTable[*t] <= 1 ) t++;
        c = *t; *t = 0;
        if ( StrICmp(s,(UBYTE *)"gzip") == 0 ) {
#ifndef WITHZLIB
            Warning("gzip compression not supported on this platform");
```

```
#endif
            s = t; *s = c;
            if ( *s == 0 ) {
                AR.gzipCompress = GZIPDEFAULT;  /* Normally should be 6 */
                return(0);
            }
            while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
            t = s;
            if ( FG.cTable[*s] == 1 ) {
                AR.gzipCompress = *s - '0';
                s++;
                while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
                if ( *s == 0 ) return(0);
            }
            MesPrint("&Unknown gzip option: %s, a digit was expected",t);
            return(1);

        }
        else {
            MesPrint("&Unknown option: %s, on, off or gzip expected",s);
            return(1);
        }
    }
    return(0);
}

/*
    #] CoCompress :
    #[ CoFlags :
*/

int CoFlags(UBYTE *s,int value)
{
    int i, error = 0;
    if ( *s != ',' ) {
        MesPrint("&Proper syntax is: On/Off Flag,number[s];");
        error = 1;
    }
    while ( *s == ',' ) {
        do { s++; } while ( *s == ',' );
        i = 0;
        if ( FG.cTable[*s] != 1 ) {
            MesPrint("&Proper syntax is: On/Off Flag,number[s];");
            error = 1;
            break;
        }
        while ( FG.cTable[*s] == 1 ) { i = 10*i + *s++ - '0'; }
        if ( i <= 0 || i > MAXFLAGS ) {
            MesPrint("&The number of a flag in On/Off Flag should be in the range 0-%d",(int)MAXFLAGS);
            error = 1;
            break;
        }
        AC.debugFlags[i] = value;
    }
    if ( *s ) {
        MesPrint("&Proper syntax is: On/Off Flag,number[s];");
        error = 1;
    }
    return(error);
}

/*
    #] CoFlags :
    #[ CoOff :
*/

int CoOff(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c;
    int i, num = sizeof(onoffoptions)/sizeof(KEYWORD);
    for (;;) {
        while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
        if ( *s == 0 ) return(0);
        if ( chartype[*s] != 0 ) {
            MesPrint("&Illegal character or option encountered in OFF statement");
            return(-1);
        }
        t = s;  while ( chartype[*s] == 0 ) s++;
        c = *s; *s = 0;
        for ( i = 0; i < num; i++ ) {
            if ( StrICont(t,(UBYTE *)(onoffoptions[i].name)) == 0 ) break;
        }
        if ( i >= num ) {
            MesPrint("&Unrecognized option in OFF statement: %s",t);
            *s = c; return(-1);
```

```
        }
        else if ( StrICont(t,(UBYTE *)"compress") == 0 ) {
            AR.gzipCompress = 0;
        }
        else if ( StrICont(t,(UBYTE *)"checkpoint") == 0 ) {
            PrintDeprecation("the checkpoint mechanism", "issues/626");
            AC.CheckpointInterval = 0;
            if ( AC.CheckpointRunBefore ) { free(AC.CheckpointRunBefore); AC.CheckpointRunBefore = NULL; }
            if ( AC.CheckpointRunAfter ) { free(AC.CheckpointRunAfter); AC.CheckpointRunAfter = NULL; }
            if ( AC.NoShowInput == 0 ) MesPrint("Checkpoints deactivated.");
        }
        else if ( StrICont(t,(UBYTE *)"threads") == 0 ) {
            AS.MultiThreaded = 0;
        }
        else if ( StrICont(t,(UBYTE *)"flag") == 0 ) {
            *s = c;
            return(CoFlags(s,0));
        }
        else if ( StrICont(t,(UBYTE *)"innertest") == 0 ) {
            *s = c;
            AC.InnerTest = 0;
            if ( AC.TestValue ) {
                M_free(AC.TestValue,"InnerTest");
                AC.TestValue = 0;
            }
        }
        else if ( StrICont(t,(UBYTE *)"sortreallocate") == 0 ) {
            if ( AC.SortReallocateFlag == 2 ) {
                /* The flag has been set by #sortreallocate, and it was off before. Leave it as 2,
                   so that the reallocation still happens in the current module. It will be turned
                   off after the reallocation is done. */
                return(0);
            }
        }
        *s = c;
        *onoffoptions[i].var = onoffoptions[i].flags;
        AR.SortType = AC.SortType;
        AC.mparallelflag = AC.parallelflag | AM.hparallelflag;
    }
}

/*
    #] CoOff :
    #[ CoOn :
*/

int CoOn(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c;
    int i, num = sizeof(onoffoptions)/sizeof(KEYWORD);
    LONG interval;
    for (;;) {
        while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
        if ( *s == 0 ) return(0);
        if ( chartype[*s] != 0 ) {
            MesPrint("&Illegal character or option encountered in ON statement");
            return(-1);
        }
        t = s;  while ( chartype[*s] == 0 ) s++;
        c = *s; *s = 0;
        for ( i = 0; i < num; i++ ) {
            if ( StrICont(t,(UBYTE *)(onoffoptions[i].name)) == 0 ) break;
        }
        if ( i >= num ) {
            MesPrint("&Unrecognized option in ON statement: %s",t);
            *s = c; return(-1);
        }
        if ( StrICont(t,(UBYTE *)"backtrace") == 0 ) {
#ifndef ENABLE_BACKTRACE
            Warning("backtrace not supported on this platform");
#endif
        }
        else if ( StrICont(t,(UBYTE *)"compress") == 0 ) {
            AR.gzipCompress = 0;
            *s = c;
            while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
            if ( *s ) {
                t = s;
                while ( FG.cTable[*s] <= 1 ) s++;
                c = *s; *s = 0;
                if ( StrICmp(t,(UBYTE *)"gzip") == 0 ) {
#ifndef WITHZLIB
                    Warning("gzip compression not supported on this platform");
#endif
```

```
#ifdef WITHZSTD
                    /* If gzip is specified, turn off zstd compression. zlib still goes via the wrapper. */
                    ZWRAP_useZSTDcompression(0);
#endif
                }
                else if ( StrICmp(t,(UBYTE *)"zstd") == 0 ) {
#ifdef WITHZSTD
                    ZWRAP_useZSTDcompression(1);
#else
                    Warning("zstd compression not supported on this platform");
#endif
                }
                else {
                    MesPrint("&Unrecognized option in ON compress statement: %s",t);
                    return(-1);
                }
                /* Whether we are using zlib or zstd, accept and use a compression level. */
                *s = c;
                while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
                if ( FG.cTable[*s] == 1 ) {
                    AR.gzipCompress = *s++ - '0';
                    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
                    if ( *s ) {
                        MesPrint("&Unrecognized option in ON compress gzip/zstd statement: %s",t);
                        return(-1);
                    }
                }
                else if ( *s == 0 ) {
                    AR.gzipCompress = GZIPDEFAULT;
                }
                else {
                    MesPrint("&Unrecognized option in ON compress gzip/zstd statement: %s, single digit expected",t);
                    return(-1);
                }
            }
        }
        else if ( StrICont(t,(UBYTE *)"checkpoint") == 0 ) {
            PrintDeprecation("the checkpoint mechanism", "issues/626");
            AC.CheckpointInterval = 0;
            if ( AC.CheckpointRunBefore ) { free(AC.CheckpointRunBefore); AC.CheckpointRunBefore = NULL; }
            if ( AC.CheckpointRunAfter ) { free(AC.CheckpointRunAfter); AC.CheckpointRunAfter = NULL; }
            *s = c;
            while ( *s ) {
                while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
                if ( FG.cTable[*s] == 1 ) {
                    interval = 0;
                    t = s;
                    do { interval = 10*interval + *s++ - '0'; } while ( FG.cTable[*s] == 1 );
                    if ( *s == 's' || *s == 'S' ) {
                        s++;
                    }
                    else if ( *s == 'm' || *s == 'M' ) {
                        interval *= 60; s++;
                    }
                    else if ( *s == 'h' || *s == 'H' ) {
                        interval *= 3600; s++;
                    }
                    else if ( *s == 'd' || *s == 'D' ) {
                        interval *= 86400; s++;
                    }
                    if ( *s != ',' && FG.cTable[*s] != 6 && FG.cTable[*s] != 10 ) {
                        MesPrint("&Unrecognized time interval in ON Checkpoint statement: %s", t);
                        return(-1);
                    }
                    AC.CheckpointInterval = interval * 100; /* in 1/100 of seconds */
                }
                else if ( FG.cTable[*s] == 0 ) {
                    int type;
                    t = s;
                    while ( FG.cTable[*s] == 0 ) s++;
                    c = *s; *s = 0;
                    if ( StrICmp(t,(UBYTE *)"run") == 0 ) {
                        type = 3;
                    }
                    else if ( StrICmp(t,(UBYTE *)"runafter") == 0 ) {
                        type = 2;
                    }
                    else if ( StrICmp(t,(UBYTE *)"runbefore") == 0 ) {
                        type = 1;
                    }
                    else {
                        MesPrint("&Unrecognized option in ON Checkpoint statement: %s", t);
                        *s = c; return(-1);
```

```
                    }
                    *s = c;
                    if ( *s != '=' && FG.cTable[*(s+1)] != 9 ) {
                        MesPrint("&Unrecognized option in ON Checkpoint statement: %s", t);
                        return(-1);
                    }
                    ++s;
                    t = ++s;
                    while ( *s ) {
                        if ( FG.cTable[*s] == 9 ) {
                            c = *s; *s = 0;
                            if ( type & 1 ) {
                                if ( AC.CheckpointRunBefore ) {
                                    free(AC.CheckpointRunBefore); AC.CheckpointRunBefore = NULL;
                                }
                                if ( s-t > 0 ) {
                                    AC.CheckpointRunBefore = Malloc1(s-t+1, "AC.CheckpointRunBefore");
                                    StrCopy(t, (UBYTE*)AC.CheckpointRunBefore);
                                }
                            }
                            if ( type & 2 ) {
                                if ( AC.CheckpointRunAfter ) {
                                    free(AC.CheckpointRunAfter); AC.CheckpointRunAfter = NULL;
                                }
                                if ( s-t > 0 ) {
                                    AC.CheckpointRunAfter = Malloc1(s-t+1, "AC.CheckpointRunAfter");
                                    StrCopy(t, (UBYTE*)AC.CheckpointRunAfter);
                                }
                            }
                            *s = c;
                            break;
                        }
                        ++s;
                    }
                    if ( FG.cTable[*s] != 9 ) {
                        MesPrint("&Unrecognized option in ON Checkpoint statement: %s", t);
                        return(-1);
                    }
                    ++s;
                }
            }
/*
            if ( AC.NoShowInput == 0 ) {
                MesPrint("Checkpoints activated.");
                if ( AC.CheckpointInterval ) {
                    MesPrint("-> Minimum saving interval: %l seconds.", AC.CheckpointInterval/100);
                }
                else {
                    MesPrint("-> No minimum saving interval given. Saving after EVERY module.");
                }
                if ( AC.CheckpointRunBefore ) {
                    MesPrint("-> Calling script \"%s\" before saving.", AC.CheckpointRunBefore);
                }
                if ( AC.CheckpointRunAfter ) {
                    MesPrint("-> Calling script \"%s\" after saving.", AC.CheckpointRunAfter);
                }
            }
*/
        }
        else if ( StrICont(t,(UBYTE *)"indentspace") == 0 ) {
            *s = c;
            while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
            if ( *s ) {
                i = 0;
                while ( FG.cTable[*s] == 1 ) { i = 10*i + *s++ - '0'; }
                if ( *s ) {
                    MesPrint("&Unrecognized option in ON IndentSpace statement: %s",t);
                    return(-1);
                }
                if ( i > 40 ) {
                    Warning("IndentSpace parameter adjusted to 40");
                    i = 40;
                }
                AO.IndentSpace = i;
            }
            else {
                AO.IndentSpace = AM.ggIndentSpace;
            }
            return(0);
        }
        else if ( ( StrICont(t,(UBYTE *)"fewerstats") == 0 ) ||
                  ( StrICont(t,(UBYTE *)"fewerstatistics") == 0 ) ) {
            *s = c;
            while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
            if ( *s ) {
                i = 0;
                while ( FG.cTable[*s] == 1 ) { i = 10*i + *s++ - '0'; }
```

```
                if ( *s ) {
                    MesPrint("&Unrecognized option in ON FewerStatistics statement: %s",t);
                    return(-1);
                }
                if ( i > AM.S0->MaxPatches ) {
                    if ( AC.WarnFlag )
                    MesPrint("&Warning: FewerStatistics parameter greater than MaxPatches(=%d). Adjusted to %d"
                    ,AM.S0->MaxPatches,(AM.S0->MaxPatches+1)/2);
                    i = (AM.S0->MaxPatches+1)/2;
                }
                AC.ShortStatsMax = i;
            }
            else {
                AC.ShortStatsMax = 10; /* default value */
            }
            return(0);
        }
        else if ( StrICont(t,(UBYTE *)"threads") == 0 ) {
            if ( AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
        }
        else if ( StrICont(t,(UBYTE *)"flag") == 0 ) {
            *s = c;
            return(CoFlags(s,1));
        }
        else if ( StrICont(t,(UBYTE *)"innertest") == 0 ) {
            UBYTE *t;
            *s = c;
            while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
            if ( *s ) {
                t = s; while ( *t ) t++;
                while ( t[-1] == ' ' || t[-1] == '\t' ) t--;
                c = *t; *t = 0;
                if ( AC.TestValue ) M_free(AC.TestValue,"InnerTest");
                AC.TestValue = strDup1(s,"InnerTest");
                *t = c;
                s = t;
                while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
            }
            else {
                if ( AC.TestValue ) {
                    M_free(AC.TestValue,"InnerTest");
                    AC.TestValue = 0;
                }
            }
        }
        else if ( StrICont(t,(UBYTE *)"flint") == 0 ) {
#ifndef WITHFLINT
            MesPrint("&Warning: FORM was not built with FLINT support.");
            MesPrint("Statement has no effect.");
#endif
        }
        else { *s = c; }
        *onoffoptions[i].var = onoffoptions[i].type;
        AR.SortType = AC.SortType;
        AC.mparallelflag = AC.parallelflag | AM.hparallelflag;
    }
}

/*
    #] CoOn :
    #[ CoInsideFirst :
*/

int CoInsideFirst(UBYTE *s) { return(setonoff(s,&AC.insidefirst,1,0)); }

/*
    #] CoInsideFirst :
    #[ CoProperCount :
*/

int CoProperCount(UBYTE *s) { return(setonoff(s,&AC.BottomLevel,1,0)); }

/*
    #] CoProperCount :
    #[ CoDelete :
*/

int CoDelete(UBYTE *s)
{
    int error = 0;
    if ( StrICmp(s,(UBYTE *)"storage") == 0 ) {
        if ( DeleteStore(1) < 0 ) {
            MesPrint("&Cannot restart storage file");
            error = 1;
        }
    }
```

```
    else {
        UBYTE *t = s, c;
        while ( *t && *t != ',' && *t != '>' ) t++;
        c = *t; *t = 0;
        if ( ( StrICmp(s,(UBYTE *)"extrasymbols") == 0 )
        || ( StrICmp(s,(UBYTE *)"extrasymbol") == 0 ) ) {
            WORD x = 0;
/*
            Either deletes all extra symbols or deletes above a given number
*/
            *t = c; s = t;
            if ( *s == '>' ) {
                s++;
                if ( FG.cTable[*s] != 1 ) goto unknown;
                while ( *s <= '9' && *s >= '0' ) x = 10*x + *s++ - '0';
                if ( *s ) goto unknown;
            }
            else if ( *s ) goto unknown;
            if ( x < AM.gnumextrasym ) x = AM.gnumextrasym;
            PruneExtraSymbols(x);
        }
        else {
            *t = c;
unknown:
            MesPrint("&Unknown option: %s",s);
            error = 1;
        }
    }
    return(error);
}

/*
    #] CoDelete :
    #[ CoKeep :
*/

int CoKeep(UBYTE *s)
{
    if ( StrICmp(s,(UBYTE *)"brackets") == 0 ) AC.ComDefer = 1;
    else { MesPrint("&Unknown option: '%s'",s); return(1); }
    return(0);
}

/*
    #] CoKeep :
    #[ CoFixIndex :
*/

int CoFixIndex(UBYTE *s)
{
    int x, y, error = 0;
    while ( *s ) {
        if ( FG.cTable[*s] != 1 ) {
proper:     MesPrint("&Proper syntax is: FixIndex,number:value[,number,value];");
            return(1);
        }
        ParseNumber(x,s)
        if ( *s != ':' ) goto proper;
        s++;
        if ( *s != '-' && *s != '+' && FG.cTable[*s] != 1 ) goto proper;
        ParseSignedNumber(y,s)
        if ( *s && *s != ',' ) goto proper;
        while ( *s == ',' ) s++;
        if ( x >= AM.OffsetIndex ) {
            MesPrint("&Fixed index out of allowed range. Change ConstIndex in setup file?");
            MesPrint("&Current value of ConstIndex = %d",AM.OffsetIndex-1);
            error = 1;
        }
        if ( y != (int)((WORD)y) ) {
            MesPrint("&Value of d_(%d,%d) outside range for this computer",x,x);
            error = 1;
        }
        if ( error == 0 ) AC.FixIndices[x] = y;
    }
    return(error);
}

/*
    #] CoFixIndex :
    #[ CoMetric :
*/

int CoMetric(UBYTE *s)
{ DUMMYUSE(s); MesPrint("&The metric statement does not do anything yet"); return(1); }

/*
    #] CoMetric :
```

```
    #[ DoPrint :
*/

int DoPrint(UBYTE *s, int par)
{
    int i, error = 0, numdol = 0, type;
    WORD handle = -1;
    UBYTE *name, c, *t;
    EXPRESSIONS e;
    WORD numexpr, tofile = 0, *w, par2 = 0;
    CBUF *C = cbuf + AC.cbufnum;
    while ( *s == ',' ) s++;
    if ( ( *s == '+' || *s == '-' ) && ( s[1] == 'f' || s[1] == 'F' ) ) {
        t = s + 2; while ( *t == ' ' || *t == ',' ) t++;
        if ( *t == '"' ) {
            if ( *s == '+' ) { tofile = 1; handle = AC.LogHandle; }
            s = t;
        }
    }
    else if ( *s == '<' ) {
        UBYTE *filename;
        s++; filename = s;
        while ( *s && *s != '>' ) s++;
        if ( *s == 0 ) {
            MesPrint("&Improper filename in print statement");
            return(1);
        }
        *s++ = 0;
        tofile = 1;
        if ( ( handle = GetChannel((char *)filename,1) ) < 0 ) return(1);
        SKIPBLANKS(s) if ( *s == ',' ) s++; SKIPBLANKS(s)
        if ( *s == '+' && ( s[1] == 's' || s[1] == 'S' ) ) {
            s += 2;
            par2 |= PRINTONETERM;
            if ( *s == 's' || *s == 'S' ) {
                s++;
                par2 |= PRINTONEFUNCTION;
                if ( *s == 's' || *s == 'S' ) {
                    s++;
                    par2 |= PRINTALL;
                }
            }
            SKIPBLANKS(s) if ( *s == ',' ) s++; SKIPBLANKS(s)
        }
    }
    if ( par == PRINTON && *s == '"' ) {
        WORD code[3];
        if ( tofile == 1 ) code[0] = TYPEFPRINT;
        else code[0] = TYPEPRINT;
        code[1] = handle;
        code[2] = par2;
        s++; name = s;
        while ( *s && *s != '"' ) {
            if ( *s == '\\' ) s++;
            if ( *s == '%' && s[1] == '$' ) numdol++;
            s++;
        }
        if ( *s != '"' ) {
            MesPrint("&String in print statement should be enclosed in \"");
            return(1);
        }
        *s = 0;
        AddComString(3,code,name,1);
        *s++ = '"';
        while ( *s == ',' ) {
            s++;
            if ( *s == '$' ) {
                s++; name = s; while ( FG.cTable[*s] <= 1 ) s++;
                c = *s; *s = 0;
                type = GetName(AC.dollarnames,name,&numexpr,NOAUTO);
                if ( type == NAMENOTFOUND ) {
                    MesPrint("&$ variable %s not (yet) defined",name);
                    error = 1;
                }
                else {
                    C->lhs[C->numlhs][1] += 2;
                    *(C->Pointer)++ = DOLLAREXPRESSION;
                    *(C->Pointer)++ = numexpr;
                    numdol--;
                }
            }
            else {
                MesPrint("&Illegal object in print statement");
                error = 1;
                return(error);
            }
            *s = c;
```

```
                if ( c == '[' ) {
                    w = C->Pointer;
                    s++;
                    s = GetDoParam(s,&(C->Pointer),-1);
                    if ( s == 0 ) return(1);
                    if ( *s != ']' ) {
                        MesPrint("&unmatched [] in $ factor");
                        return(1);
                    }
                    C->lhs[C->numlhs][1] += C->Pointer - w;
                    s++;
                }
            }
            if ( *s != 0 ) {
                MesPrint("&Illegal object in print statement");
                error = 1;
            }
            if ( numdol > 0 ) {
                MesPrint("&More $ variables asked for than provided");
                error = 1;
            }
            *(C->Pointer)++ = 0;
            return(error);
        }
        if ( *s == 0 ) {     /* All active expressions */
AllExpr:
            for ( e = Expressions, i = NumExpressions; i > 0; i--, e++ ) {
                if ( e->status == LOCALEXPRESSION || e->status ==
                GLOBALEXPRESSION || e->status == UNHIDELEXPRESSION
                || e->status == UNHIDEGEXPRESSION ) e->printflag = par;
            }
            return(error);
        }
        while ( *s ) {
            if ( *s == '+' ) {
                s++;
                if ( tolower(*s) == 'f' ) par |= PRINTLFILE;
                else if ( tolower(*s) == 's' ) {
                    if ( tolower(s[1]) == 's' ) {
                        if ( tolower(s[2]) == 's' ) {
                            par |= PRINTONEFUNCTION | PRINTONETERM | PRINTALL;
                            s++;
                        }
                        else if ( ( par & 3 ) < 2 ) par |= PRINTONEFUNCTION | PRINTONETERM;
                        s++;
                    }
                    else {
                        if ( ( par & 3 ) < 2 ) par |= PRINTONETERM;
                    }
                }
                else {
illeg:                  MesPrint("&Illegal option in (n)print statement");
                    error = 1;
                }
                s++;
                if ( *s == 0 ) goto AllExpr;
            }
            else if ( *s == '-' ) {
                s++;
                if ( tolower(*s) == 'f' ) par &= ~PRINTLFILE;
                else if ( tolower(*s) == 's' ) {
                    if ( tolower(s[1]) == 's' ) {
                        if ( tolower(s[2]) == 's' ) {
                            par &= ~PRINTALL;
                            s++;
                        }
                        else if ( ( par & 3 ) < 2 ) {
                            par &= ~PRINTONEFUNCTION;
                            par &= ~PRINTALL;
                        }
                        s++;
                    }
                    else {
                        if ( ( par & 3 ) < 2 ) {
                            par &= ~PRINTONETERM;
                            par &= ~PRINTONEFUNCTION;
                            par &= ~PRINTALL;
                        }
                    }
                }
                else goto illeg;
                s++;
                if ( *s == 0 ) goto AllExpr;
            }
            else if ( FG.cTable[*s] == 0 || *s == '[' ) {
                name = s;
                if ( ( s = SkipAName(s) ) == 0 ) {
```

```
                MesPrint("&Improper name in (n)print statement");
                return(1);
            }
            c = *s; *s = 0;
            if ( ( GetName(AC.exprnames,name,&numexpr,NOAUTO) == CEXPRESSION )
            && ( Expressions[numexpr].status == LOCALEXPRESSION
            || Expressions[numexpr].status == GLOBALEXPRESSION ) ) {
FoundExpr:;
                if ( c == '[' && s[1] == ']' ) {
                    Expressions[numexpr].printflag = par | PRINTCONTENTS;
                    *s++ = c; c = *++s;
                }
                else
                    Expressions[numexpr].printflag = par;
            }
            else if ( GetLastExprName(name,&numexpr)
            && ( Expressions[numexpr].status == LOCALEXPRESSION
            || Expressions[numexpr].status == GLOBALEXPRESSION
            || Expressions[numexpr].status == UNHIDELEXPRESSION
            || Expressions[numexpr].status == UNHIDEGEXPRESSION
            ) ) {
                goto FoundExpr;
            }
            else {
                MesPrint("&%s is not the name of an active expression",name);
                error = 1;
            }
            *s++ = c;
            if ( c == 0 ) return(0);
            if ( c == '-' || c == '+' ) s--;
        }
        else if ( *s == ',' ) s++;
        else {
            MesPrint("&Illegal object in (n)print statement");
            return(1);
        }
    }
    return(0);
}

/*
    #] DoPrint :
    #[ CoPrint :
*/

int CoPrint(UBYTE *s) { return(DoPrint(s,PRINTON)); }

/*
    #] CoPrint :
    #[ CoPrintB :
*/

int CoPrintB(UBYTE *s) { return(DoPrint(s,PRINTCONTENT)); }

/*
    #] CoPrintB :
    #[ CoNPrint :
*/

int CoNPrint(UBYTE *s) { return(DoPrint(s,PRINTOFF)); }

/*
    #] CoNPrint :
    #[ CoPushHide :
*/

int CoPushHide(UBYTE *s)
{
    GETIDENTITY
    WORD *ScratchBuf;
    int i;
    if ( AR.Fscr[2].PObuffer == 0 ) {
        ScratchBuf = (WORD *)Malloc1(AM.HideSize*sizeof(WORD),"hidesize");
        AR.Fscr[2].POsize = AM.HideSize * sizeof(WORD);
        AR.Fscr[2].POfull = AR.Fscr[2].POfill = AR.Fscr[2].PObuffer = ScratchBuf;
        AR.Fscr[2].POstop = AR.Fscr[2].PObuffer + AM.HideSize;
        PUTZERO(AR.Fscr[2].POposition);
    }
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    AC.HideLevel += 2;
    if ( *s ) {
        MesPrint("&PushHide statement should have no arguments");
        return(-1);
    }
    for ( i = 0; i < NumExpressions; i++ ) {
        switch ( Expressions[i].status ) {
            case DROPLEXPRESSION:
```

```
            case SKIPLEXPRESSION:
            case LOCALEXPRESSION:
                Expressions[i].status = HIDELEXPRESSION;
                Expressions[i].hidelevel = AC.HideLevel-1;
                break;
            case DROPGEXPRESSION:
            case SKIPGEXPRESSION:
            case GLOBALEXPRESSION:
                Expressions[i].status = HIDEGEXPRESSION;
                Expressions[i].hidelevel = AC.HideLevel-1;
                break;
            default:
                break;
        }
    }
    return(0);
}

/*
    #] CoPushHide :
    #[ CoPopHide :
*/

int CoPopHide(UBYTE *s)
{
    int i;
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( AC.HideLevel <= 0 ) {
        MesPrint("&PopHide statement without corresponding PushHide statement");
        return(-1);
    }
    AC.HideLevel -= 2;
    if ( *s ) {
        MesPrint("&PopHide statement should have no arguments");
        return(-1);
    }
    for ( i = 0; i < NumExpressions; i++ ) {
        switch ( Expressions[i].status ) {
            case HIDDENLEXPRESSION:
                if ( Expressions[i].hidelevel > AC.HideLevel )
                    Expressions[i].status = UNHIDELEXPRESSION;
                break;
            case HIDDENGEXPRESSION:
                if ( Expressions[i].hidelevel > AC.HideLevel )
                    Expressions[i].status = UNHIDEGEXPRESSION;
                break;
            default:
                break;
        }
    }
    return(0);
}

/*
    #] CoPopHide :
    #[ SetExprCases :
*/

int SetExprCases(int par, int setunset, int val)
{
    switch ( par ) {
        case SKIP:
            switch ( val ) {
                case SKIPLEXPRESSION:
                    if ( !setunset ) val = LOCALEXPRESSION;
                    break;
                case SKIPGEXPRESSION:
                    if ( !setunset ) val = GLOBALEXPRESSION;
                    break;
                case LOCALEXPRESSION:
                    if ( setunset ) val = SKIPLEXPRESSION;
                    break;
                case GLOBALEXPRESSION:
                    if ( setunset ) val = SKIPGEXPRESSION;
                    break;
                case INTOHIDEGEXPRESSION:
                case INTOHIDELEXPRESSION:
                default:
                    break;
            }
            break;
        case DROP:
            switch ( val ) {
                case SKIPLEXPRESSION:
                case LOCALEXPRESSION:
                case HIDELEXPRESSION:
                    if ( setunset ) val = DROPLEXPRESSION;
```

```
                    break;
                case DROPLEXPRESSION:
                    if ( !setunset ) val = LOCALEXPRESSION;
                    break;
                case SKIPGEXPRESSION:
                case GLOBALEXPRESSION:
                case HIDEGEXPRESSION:
                    if ( setunset ) val = DROPGEXPRESSION;
                    break;
                case DROPGEXPRESSION:
                    if ( !setunset ) val = GLOBALEXPRESSION;
                    break;
                case HIDDENLEXPRESSION:
                case UNHIDELEXPRESSION:
                    if ( setunset ) val = DROPHLEXPRESSION;
                    break;
                case HIDDENGEXPRESSION:
                case UNHIDEGEXPRESSION:
                    if ( setunset ) val = DROPHGEXPRESSION;
                    break;
                case DROPHLEXPRESSION:
                    if ( !setunset ) val = HIDDENLEXPRESSION;
                    break;
                case DROPHGEXPRESSION:
                    if ( !setunset ) val = HIDDENGEXPRESSION;
                    break;
                case INTOHIDEGEXPRESSION:
                case INTOHIDELEXPRESSION:
                default:
                    break;
            }
            break;
        case HIDE:
            switch ( val ) {
                case DROPLEXPRESSION:
                case SKIPLEXPRESSION:
                case LOCALEXPRESSION:
                    if ( setunset ) val = HIDELEXPRESSION;
                    break;
                case HIDELEXPRESSION:
                    if ( !setunset ) val = LOCALEXPRESSION;
                    break;
                case DROPGEXPRESSION:
                case SKIPGEXPRESSION:
                case GLOBALEXPRESSION:
                    if ( setunset ) val = HIDEGEXPRESSION;
                    break;
                case HIDEGEXPRESSION:
                    if ( !setunset ) val = GLOBALEXPRESSION;
                    break;
                case INTOHIDEGEXPRESSION:
                case INTOHIDELEXPRESSION:
                default:
                    break;
            }
            break;
        case UNHIDE:
            switch ( val ) {
                case HIDDENLEXPRESSION:
                case DROPHLEXPRESSION:
                    if ( setunset ) val = UNHIDELEXPRESSION;
                    break;
                case UNHIDELEXPRESSION:
                    if ( !setunset ) val = HIDDENLEXPRESSION;
                    break;
                case HIDDENGEXPRESSION:
                case DROPHGEXPRESSION:
                    if ( setunset ) val = UNHIDEGEXPRESSION;
                    break;
                case UNHIDEGEXPRESSION:
                    if ( !setunset ) val = HIDDENGEXPRESSION;
                    break;
                case INTOHIDEGEXPRESSION:
                case INTOHIDELEXPRESSION:
                default:
                    break;
            }
            break;
        case INTOHIDE:
            switch ( val ) {
                case HIDDENLEXPRESSION:
                case HIDDENGEXPRESSION:
                    MesPrint("&Expression is already hidden");
                    return(-1);
                case DROPHLEXPRESSION:
                case DROPHGEXPRESSION:
                case UNHIDELEXPRESSION:
```

```
                case UNHIDEGEXPRESSION:
                    if ( setunset ) {
                        MesPrint("&Cannot unhide/drop and put intohide expression in the same
module");
                        return(-1);
                    }
                    break;
                case LOCALEXPRESSION:
                case DROPLEXPRESSION:
                case SKIPLEXPRESSION:
                case HIDELEXPRESSION:
                    if ( setunset ) val = INTOHIDELEXPRESSION;
                    break;
                case GLOBALEXPRESSION:
                case DROPGEXPRESSION:
                case SKIPGEXPRESSION:
                case HIDEGEXPRESSION:
                    if ( setunset ) val = INTOHIDEGEXPRESSION;
                    break;
                case INTOHIDELEXPRESSION:
                    if ( !setunset ) val = LOCALEXPRESSION;
                    break;
                case INTOHIDEGEXPRESSION:
                    if ( !setunset ) val = GLOBALEXPRESSION;
                    break;
                default:
                    break;
            }
            break;
        default:
            break;
    }
    return(val);
}

/*
    #] SetExprCases :
    #[ SetExpr :
*/

int SetExpr(UBYTE *s, int setunset, int par)
{
    WORD *w, numexpr;
    int error = 0, i;
    UBYTE *name, c;
    if ( *s == 0 ) {
        for ( i = 0; i < NumExpressions; i++ ) {
            w = &(Expressions[i].status);
            *w = SetExprCases(par,setunset,*w);
            if ( *w < 0 ) error = 1;
            if ( ( par == HIDE || par == INTOHIDE ) && setunset == 1 )
                Expressions[i].hidelevel = AC.HideLevel;
        }
        return(0);
    }
    while ( *s ) {
        if ( *s == ',' ) { s++; continue; }
        if ( *s == '0' ) { s++; continue; }
        name = s;
        if ( ( s = SkipAName(s) ) == 0 ) {
            MesPrint("&Improper name for an expression: '%s'",name);
            return(1);
        }
        c = *s; *s = 0;
        if ( GetName(AC.exprnames,name,&numexpr,NOAUTO) == CEXPRESSION ) {
            w = &(Expressions[numexpr].status);
            *w = SetExprCases(par,setunset,*w);
            if ( *w < 0 ) error = 1;
            if ( ( par == HIDE || par == INTOHIDE ) && setunset == 1 )
                Expressions[numexpr].hidelevel = AC.HideLevel;
        }
        else if ( GetName(AC.varnames,name,&numexpr,NOAUTO) != NAMENOTFOUND ) {
            MesPrint("&%s is not an expression",name);
            error = 1;
        }
        *s = c;
    }
    return(error);
}

/*
    #] SetExpr :
    #[ CoDrop :
*/

int CoDrop(UBYTE *s) { return(SetExpr(s,1,DROP)); }

```

```
/*
    #] CoDrop :
    #[ CoNoDrop :
*/

int CoNoDrop(UBYTE *s) { return(SetExpr(s,0,DROP)); }

/*
    #] CoNoDrop :
    #[ CoSkip :
*/

int CoSkip(UBYTE *s) { return(SetExpr(s,1,SKIP)); }

/*
    #] CoSkip :
    #[ CoNoSkip :
*/

int CoNoSkip(UBYTE *s) { return(SetExpr(s,0,SKIP)); }

/*
    #] CoNoSkip :
    #[ CoHide :
*/

int CoHide(UBYTE *inp) {
    GETIDENTITY
    WORD *ScratchBuf;
    if ( AR.Fscr[2].PObuffer == 0 ) {
        ScratchBuf = (WORD *)Malloc1(AM.HideSize*sizeof(WORD),"hidesize");
        AR.Fscr[2].POsize = AM.HideSize * sizeof(WORD);
        AR.Fscr[2].POfull = AR.Fscr[2].POfill = AR.Fscr[2].PObuffer = ScratchBuf;
        AR.Fscr[2].POstop = AR.Fscr[2].PObuffer + AM.HideSize;
        PUTZERO(AR.Fscr[2].POposition);
    }
    return(SetExpr(inp,1,HIDE));
}

/*
    #] CoHide :
    #[ CoIntoHide :
*/

int CoIntoHide(UBYTE *inp) {
    GETIDENTITY
    WORD *ScratchBuf;
    if ( AR.Fscr[2].PObuffer == 0 ) {
        ScratchBuf = (WORD *)Malloc1(AM.HideSize*sizeof(WORD),"hidesize");
        AR.Fscr[2].POsize = AM.HideSize * sizeof(WORD);
        AR.Fscr[2].POfull = AR.Fscr[2].POfill = AR.Fscr[2].PObuffer = ScratchBuf;
        AR.Fscr[2].POstop = AR.Fscr[2].PObuffer + AM.HideSize;
        PUTZERO(AR.Fscr[2].POposition);
    }
    return(SetExpr(inp,1,INTOHIDE));
}

/*
    #] CoIntoHide :
    #[ CoNoIntoHide :
*/

int CoNoIntoHide(UBYTE *inp) { return(SetExpr(inp,0,INTOHIDE)); }

/*
    #] CoNoIntoHide :
    #[ CoNoHide :
*/

int CoNoHide(UBYTE *inp) { return(SetExpr(inp,0,HIDE)); }

/*
    #] CoNoHide :
    #[ CoUnHide :
*/

int CoUnHide(UBYTE *inp) { return(SetExpr(inp,1,UNHIDE)); }

/*
    #] CoUnHide :
    #[ CoNoUnHide :
*/

int CoNoUnHide(UBYTE *inp) { return(SetExpr(inp,0,UNHIDE)); }

/*
    #] CoNoUnHide :
```

```
    #[ AddToCom :
*/

void AddToCom(int n, WORD *array)
{
    CBUF *C = cbuf+AC.cbufnum;
#ifdef COMPBUFDEBUG
    MesPrint("  %a",n,array);
#endif
    while ( C->Pointer+n >= C->Top ) DoubleCbuffer(AC.cbufnum,C->Pointer,18);
    while ( --n >= 0 ) *(C->Pointer)++ = *array++;
}

/*
    #] AddToCom :
    #[ AddComString :
*/

int AddComString(int n, WORD *array, UBYTE *thestring, int par)
{
    CBUF *C = cbuf+AC.cbufnum;
    UBYTE *s = thestring, *w;
#ifdef COMPBUFDEBUG
    WORD *cc;
    UBYTE *ww;
#endif
    int i, numchars = 0, size, zeroes;
    while ( *s ) {
        if ( *s == '\\' ) s++;
        else if ( par == 1 &&
        ( ( *s == '%' && s[1] != 't' && s[1] != 'T' && s[1] != '$' &&
         s[1] != 'w' && s[1] != 'W' && s[1] != 'r' && s[1] != 0 ) || *s == '#'
        || *s == '@' || *s == '&' ) ) {
            numchars++;
        }
        s++; numchars++;
    }
    AddLHS(AC.cbufnum);
    size = numchars/sizeof(WORD)+1;
    while ( C->Pointer+size+n+2 >= C->Top ) DoubleCbuffer(AC.cbufnum,C->Pointer,19);
#ifdef COMPBUFDEBUG
    cc = C->Pointer;
#endif
    *(C->Pointer)++ = array[0];
    *(C->Pointer)++ = size+n+2;
    for ( i = 1; i < n; i++ ) *(C->Pointer)++ = array[i];
    *(C->Pointer)++ = size;
#ifdef COMPBUFDEBUG
    ww =
#endif
    w = (UBYTE *)(C->Pointer);
    zeroes = size*sizeof(WORD)-numchars;
    s = thestring;
    while ( *s ) {
        if ( *s == '\\' ) s++;
        else if ( par == 1 && ( ( *s == '%' &&
        s[1] != 't' && s[1] != 'T' && s[1] != '$' &&
        s[1] != 'w' && s[1] != 'W' && s[1] != 'r' && s[1] != 0 ) || *s == '#'
        || *s == '@' || *s == '&' ) ) {
            *w++ = '%';
        }
        *w++ = *s++;
    }
    while ( --zeroes >= 0 ) *w++ = 0;
    C->Pointer += size;
#ifdef COMPBUFDEBUG
    MesPrint("LH: %a",size+1+n,cc);
    MesPrint("        %s",thestring);
#endif
    return(0);
}

/*
    #] AddComString :
    #[ Add2ComStrings :
*/

int Add2ComStrings(int n, WORD *array, UBYTE *string1, UBYTE *string2)
{
    CBUF *C = cbuf+AC.cbufnum;
    UBYTE *s1 = string1, *s2 = string2, *w;
    int i, num1chars = 0, num2chars = 0, size1, size2, zeroes1, zeroes2;
    AddLHS(AC.cbufnum);
    while ( *s1 ) { s1++; num1chars++; }
    size1 = num1chars/sizeof(WORD)+1;
    if ( s2 ) {
        while ( *s2 ) { s2++; num2chars++; }
```

```
        size2 = num2chars/sizeof(WORD)+1;
    }
    else size2 = 0;
    while ( C->Pointer+size1+size2+n+3 >= C->Top ) DoubleCbuffer(AC.cbufnum,C->Pointer,20);
    *(C->Pointer)++ = array[0];
    *(C->Pointer)++ = size1+size2+n+3;
    for ( i = 1; i < n; i++ ) *(C->Pointer)++ = array[i];
    *(C->Pointer)++ = size1;
    w = (UBYTE *)(C->Pointer);
    zeroes1 = size1*sizeof(WORD)-num1chars;
    s1 = string1;
    while ( *s1 ) { *w++ = *s1++; }
    while ( --zeroes1 >= 0 ) *w++ = 0;
    C->Pointer += size1;
    *(C->Pointer)++ = size2;
    if ( size2 ) {
        w = (UBYTE *)(C->Pointer);
        zeroes2 = size2*sizeof(WORD)-num2chars;
        s2 = string2;
        while ( *s2 ) { *w++ = *s2++; }
        while ( --zeroes2 >= 0 ) *w++ = 0;
        C->Pointer += size2;
    }
    return(0);
}

/*
    #] Add2ComStrings :
    #[ CoDiscard :
*/

int CoDiscard(UBYTE *s)
{
    if ( *s == 0 ) {
        Add2Com(TYPEDISCARD)
        return(0);
    }
    MesPrint("&Illegal argument in discard statement: '%s'",s);
    return(1);
}

/*
    #] CoDiscard :
    #[ CoContract :

    Syntax:
        Contract
        Contract:#
        Contract #
        Contract:#,#
*/

static WORD ccarray[5] = { TYPEOPERATION,5,CONTRACT,0,0 };

int CoContract(UBYTE *s)
{
    int x;
    if ( *s == ':' ) {
        s++;
        ParseNumber(x,s)
        if ( *s != ',' && *s ) {
proper:     MesPrint("&Illegal number in contract statement");
            return(1);
        }
        if ( *s ) s++;
        ccarray[4] = x;
    }
    else ccarray[4] = 0;
    if ( FG.cTable[*s] == 1 ) {
        ParseNumber(x,s)
        if ( *s ) goto proper;
        ccarray[3] = x;
    }
    else if ( *s ) goto proper;
    else ccarray[3] = -1;
    return(AddNtoL(5,ccarray));
}

/*
    #] CoContract :
    #[ CoGoTo :
*/

int CoGoTo(UBYTE *inp)
{
    UBYTE *s = inp;
    int x;
```

```
    while ( FG.cTable[*s] <= 1 ) s++;
    if ( *s ) {
        MesPrint("&Label should be an alpha-numeric string");
        return(1);
    }
    x = GetLabel(inp);
    Add3Com(TYPEGOTO,x);
    return(0);
}

/*
    #] CoGoTo :
    #[ CoLabel :
*/

int CoLabel(UBYTE *inp)
{
    UBYTE *s = inp;
    int x;
    while ( FG.cTable[*s] <= 1 ) s++;
    if ( *s ) {
        MesPrint("&Label should be an alpha-numeric string");
        return(1);
    }
    x = GetLabel(inp);
    if ( AC.Labels[x] >= 0 ) {
        MesPrint("&Label %s defined more than once",AC.LabelNames[x]);
        return(1);
    }
    AC.Labels[x] = cbuf[AC.cbufnum].numlhs;
    return(0);
}

/*
    #] CoLabel :
    #[ DoArgument :

    Layout:
        par,full size,numlhs(+1),par,scale
        scale is for normalize
*/

int DoArgument(UBYTE *s, int par)
{
    GETIDENTITY
    UBYTE *name, *t, *v, c;
    WORD *oldworkpointer = AT.WorkPointer, *w, *ww, number, *scale;
    int error = 0, zeroflag, type, x;
    AC.lhdollarflag = 0;
    while ( *s == ',' ) s++;
    w = AT.WorkPointer;
    *w++ = par;
    w++;
    switch ( par ) {
        case TYPEARG:
            if ( AC.arglevel >= MAXNEST ) {
                MesPrint("@Nesting of argument statements more than %d levels"
                ,(WORD)MAXNEST);
                return(-1);
            }
            AC.argsumcheck[AC.arglevel] = NestingChecksum();
            AC.argstack[AC.arglevel] = cbuf[AC.cbufnum].Pointer
                                     - cbuf[AC.cbufnum].Buffer + 2;
            AC.arglevel++;
            *w++ = cbuf[AC.cbufnum].numlhs;
            break;
        case TYPENORM:
        case TYPENORM4:
        case TYPESPLITARG:
        case TYPESPLITFIRSTARG:
        case TYPESPLITLASTARG:
        case TYPEFACTARG:
        case TYPEARGTOEXTRASYMBOL:
            *w++ = cbuf[AC.cbufnum].numlhs+1;
            break;
    }
    *w++ = par;
    scale = w;
    *w++ = 1;
    *w++ = 0;
    if ( *s == '^' ) {
        s++; ParseSignedNumber(x,s)
        while ( *s == ',' ) s++;
        *scale = x;
    }
    if ( *s == '(' ) {
        t = s+1; SKIPBRA3(s)    /* We did check the brackets already */
```

```
        if ( par == TYPEARG ) {
            MesPrint("&Illegal () entry in argument statement");
            error = 1; s++; goto skipbracks;
        }
        else if ( par == TYPESPLITFIRSTARG ) {
            MesPrint("&Illegal () entry in splitfirstarg statement");
            error = 1; s++; goto skipbracks;
        }
        else if ( par == TYPESPLITLASTARG ) {
            MesPrint("&Illegal () entry in splitlastarg statement");
            error = 1; s++; goto skipbracks;
        }
        v = t;
        while ( v < s ) {
            if ( *v == '?' ) {
                MesPrint("&Wildcarding not allowed in this type of statement");
                error = 1; break;
            }
            v++;
        }
        v = s++;
        if ( *t == '(' && v[-1] == ')' ) {
            t++; v--;
            if ( par == TYPESPLITARG ) oldworkpointer[0] = TYPESPLITARG2;
            else if ( par == TYPEFACTARG ) oldworkpointer[0] = TYPEFACTARG2;
            else if ( par == TYPENORM4 ) oldworkpointer[0] = TYPENORM4;
            else if ( par == TYPENORM ) {
                if ( *t == '-' ) { oldworkpointer[0] = TYPENORM3; t++; }
                else             { oldworkpointer[0] = TYPENORM2; *scale = 0; }
            }
        }
        if ( error == 0 ) {
            CBUF *C = cbuf+AC.cbufnum;
            WORD oldnumrhs = C->numrhs, oldnumlhs = C->numlhs;
            WORD prototype[SUBEXPSIZE+40]; /* Up to 10 nested sums! */
            WORD *m, *mm;
            int i, retcode;
            LONG oldpointer = C->Pointer - C->Buffer;
            *v = 0;
            prototype[0] = SUBEXPRESSION;
            prototype[1] = SUBEXPSIZE;
            prototype[2] = C->numrhs+1;
            prototype[3] = 1;
            prototype[4] = AC.cbufnum;
            AT.WorkPointer += TYPEARGHEADSIZE+1;
            AddLHS(AC.cbufnum);
            if ( ( retcode = CompileAlgebra(t,LHSIDE,prototype) ) < 0 )
                error = 1;
            else {
                prototype[2] = retcode;
                ww = C->lhs[retcode];
                AC.lhdollarflag = 0;
                if ( *ww == 0 ) {
                    *w++ = -2; *w++ = 0;
                }
                else if ( ww[ww[0]] != 0 ) {
                    MesPrint("&There should be only one term between ()");
                    error = 1;
                }
                else if ( NewSort(BHEAD0) ) { if ( !error ) error = 1; }
                else if ( NewSort(BHEAD0) ) {
                    LowerSortLevel();
                    if ( !error ) error = 1;
                }
                else {
                    AN.RepPoint = AT.RepCount + 1;
                    m = AT.WorkPointer;
                    mm = ww; i = *mm;
                    while ( --i >= 0 ) *m++ = *mm++;
                    mm = AT.WorkPointer; AT.WorkPointer = m;
                    AR.Cnumlhs = C->numlhs;
                    if ( Generator(BHEAD mm,C->numlhs) ) {
                        LowerSortLevel(); error = 1;
                    }
                    else if ( EndSort(BHEAD mm,0) < 0 ) {
                        error = 1;
                        AT.WorkPointer = mm;
                    }
                    else if ( *mm == 0 ) {
                        *w++ = -2; *w++ = 0;
                        AT.WorkPointer = mm;
                    }
                    else if ( mm[mm[0]] != 0 ) {
                        error = 1;
                        AT.WorkPointer = mm;
                    }
                    else {
```

```
                        AT.WorkPointer = mm;
                        m = mm+*mm;
                        if ( par == TYPEFACTARG ) {
                            if ( *mm != ABS(m[-1])+1 ) {
                                *mm -= ABS(m[-1]);  /* Strip coefficient */
                            }
                            mm[-1] = -*mm-1; w += *mm+1;
                        }
                        else {
                            *mm -= ABS(m[-1]);  /* Strip coefficient */
/*
                            if ( *mm == 1 ) { *w++ = -2; *w++ = 0; }
                            else
*/
                            { mm[-1] = -*mm-1; w += *mm+1; }
                        }
                        oldworkpointer[1] = w - oldworkpointer;
                    }
                    LowerSortLevel();
                }
                oldworkpointer[5] = AC.lhdollarflag;
            }
            *v = ')';
            C->numrhs = oldnumrhs;
            C->numlhs = oldnumlhs;
            C->Pointer = C->Buffer + oldpointer;
        }
    }
skipbracks:
    if ( *s == 0 ) { *w++ = 0; *w++ = 2; *w++ = 1; }
    else {
        do {
            if ( *s == ',' ) { s++; continue; }
            ww = w; *w++ = 0; w++;
            if ( FG.cTable[*s] > 1 && *s != '[' && *s != '{' ) {
                MesPrint("&Illegal parameters in statement");
                error = 1;
                break;
            }
            while ( FG.cTable[*s] == 0 || *s == '[' || *s == '{' ) {
                if ( *s == '{' ) {
                    name = s+1;
                    SKIPBRA2(s)
                    c = *s; *s = 0;
                    number = DoTempSet(name,s);
                    name--; *s++ = c; c = *s; *s = 0;
                    goto doset;
                }
                else {
                    name = s;
                    if ( ( s = SkipAName(s) ) == 0 ) {
                        MesPrint("&Illegal name '%s'",name);
                        return(1);
                    }
                    c = *s; *s = 0;
                    if ( ( type = GetName(AC.varnames,name,&number,WITHAUTO) ) == CSET ) {
doset:                  if ( Sets[number].type != CFUNCTION ) goto nofun;
                        *w++ = CSET; *w++ = number;
                    }
                    else if ( type == CFUNCTION ) {
                        *w++ = CFUNCTION; *w++ = number + FUNCTION;
                    }
                    else {
nofun:                  MesPrint("&%s is not a function or a set of functions"
                        ,name);
                        error = 1;
                    }
                }
                *s = c;
                while ( *s == ',' ) s++;
            }
            ww[1] = w - ww;
            ww = w; w++; zeroflag = 0;
            while ( FG.cTable[*s] == 1 ) {
                ParseNumber(x,s)
                if ( *s && *s != ',' ) {
                    MesPrint("&Illegal separator after number");
                    error = 1;
                    while ( *s && *s != ',' ) s++;
                }
                while ( *s == ',' ) s++;
                if ( x == 0 ) zeroflag = 1;
                if ( !zeroflag ) *w++ = (WORD)x;
            }
            *ww = w - ww;
        } while ( *s );
    }
```

```
    oldworkpointer[1] = w - oldworkpointer;
    if ( par == TYPEARG ) {  /* To make sure. The Pointer might move in the future */
        AC.argstack[AC.arglevel-1] = cbuf[AC.cbufnum].Pointer
                                    - cbuf[AC.cbufnum].Buffer + 2;
    }
    AddNtoL(oldworkpointer[1],oldworkpointer);
    AT.WorkPointer = oldworkpointer;
    return(error);
}

/*
    #] DoArgument :
    #[ CoArgument :
*/

int CoArgument(UBYTE *s) { return(DoArgument(s,TYPEARG)); }

/*
    #] CoArgument :
    #[ CoEndArgument :
*/

int CoEndArgument(UBYTE *s)
{
    CBUF *C = cbuf+AC.cbufnum;
    while ( *s == ',' ) s++;
    if ( *s ) {
        MesPrint("&Illegal syntax for EndArgument statement");
        return(1);
    }
    if ( AC.arglevel <= 0 ) {
        MesPrint("&EndArgument without corresponding Argument statement");
        return(1);
    }
    AC.arglevel--;
    cbuf[AC.cbufnum].Buffer[AC.argstack[AC.arglevel]] = C->numlhs;
    if ( AC.argsumcheck[AC.arglevel] != NestingChecksum() ) {
        MesNesting();
        return(1);
    }
    return(0);
}

/*
    #] CoEndArgument :
    #[ CoInside :
*/

int CoInside(UBYTE *s) { return(ExecInside(s)); }

/*
    #] CoInside :
    #[ CoEndInside :
*/

int CoEndInside(UBYTE *s)
{
    CBUF *C = cbuf+AC.cbufnum;
    while ( *s == ',' ) s++;
    if ( *s ) {
        MesPrint("&Illegal syntax for EndInside statement");
        return(1);
    }
    if ( AC.insidelevel <= 0 ) {
        MesPrint("&EndInside without corresponding Inside statement");
        return(1);
    }
    AC.insidelevel--;
    cbuf[AC.cbufnum].Buffer[AC.insidestack[AC.insidelevel]] = C->numlhs;
    if ( AC.insidesumcheck[AC.insidelevel] != NestingChecksum() ) {
        MesNesting();
        return(1);
    }
    return(0);
}

/*
    #] CoEndInside :
    #[ CoNormalize :
*/

int CoNormalize(UBYTE *s) { return(DoArgument(s,TYPENORM)); }

/*
    #] CoNormalize :
    #[ CoMakeInteger :
*/
```

```
int CoMakeInteger(UBYTE *s) { return(DoArgument(s,TYPENORM4)); }

/*
    #] CoMakeInteger :
    #[ CoSplitArg :
*/

int CoSplitArg(UBYTE *s) { return(DoArgument(s,TYPESPLITARG)); }

/*
    #] CoSplitArg :
    #[ CoSplitFirstArg :
*/

int CoSplitFirstArg(UBYTE *s) { return(DoArgument(s,TYPESPLITFIRSTARG)); }

/*
    #] CoSplitFirstArg :
    #[ CoSplitLastArg :
*/

int CoSplitLastArg(UBYTE *s) { return(DoArgument(s,TYPESPLITLASTARG)); }

/*
    #] CoSplitLastArg :
    #[ CoFactArg :
*/

int CoFactArg(UBYTE *s) {
    if ( ( AC.topolynomialflag & TOPOLYNOMIALFLAG ) != 0 ) {
        MesPrint("&ToPolynomial statement and FactArg statement are not allowed in the same module");
        return(1);
    }
    AC.topolynomialflag |= FACTARGFLAG;
    return(DoArgument(s,TYPEFACTARG));
}

/*
    #] CoFactArg :
    #[ DoSymmetrize :

        Syntax:
        Symmetrize Fun[:[number]] [Fields]      -> par = 0;
        AntiSymmetrize Fun[:[number]] [Fields]  -> par = 1;
        CycleSymmetrize Fun[:[number]] [Fields] -> par = 2;
        RCycleSymmetrize Fun[:[number]] [Fields]-> par = 3;
*/

int DoSymmetrize(UBYTE *s, int par)
{
    GETIDENTITY
    int extra = 0, error = 0, err, fix, x, groupsize, num, i;
    UBYTE *name, c;
    WORD funnum, *w, *ww, type;
    for(;;) {
        name = s;
        if ( ( s = SkipAName(s) ) == 0 ) {
            MesPrint("&Improper function name");
            return(1);
        }
        c = *s; *s = 0;
        if ( c != ',' || ( FG.cTable[s[1]] != 0 && s[1] != '[' ) ) break;
        if ( par <= 0 && StrICmp(name,(UBYTE *)"cyclic") == 0 ) extra = 2;
        else if ( par <= 0 && StrICmp(name,(UBYTE *)"rcyclic") == 0 ) extra = 6;
        else {
            MesPrint("&Illegal option: '%s'",name);
            error = 1;
        }
        *s++ = c;
    }
    if ( ( err = GetVar(name,&type,&funnum,CFUNCTION,WITHAUTO) ) == NAMENOTFOUND ) {
        MesPrint("&Undefined function: %s",name);
        AddFunction(name,0,0,0,0,0,-1,-1);
        *s++ = c;
        return(1);
    }
    funnum += FUNCTION;
    if ( err == -1 ) error = 1;
    *s = c;
    if ( *s == ':' ) {
        s++;
        if ( *s == ',' || *s == '(' || *s == 0 ) fix = -1;
        else if ( FG.cTable[*s] == 1 ) {
            ParseNumber(fix,s)
            if ( fix == 0 )
                Warning("Restriction to zero arguments removed");
```

```
        }
        else {
            MesPrint("&Illegal character after :");
            return(1);
        }
    }
    else fix = 0;
    w = AT.WorkPointer;
    *w++ = TYPEOPERATION;
    w++;
    *w++ = SYMMETRIZE;
    *w++ = par | extra;
    *w++ = funnum;
    *w++ = fix;
/*
    And now the argument lists. We have either ,#,#,... or (#,#,..,#),(#,...
*/
    w += 2; ww = w; groupsize = -1;
    while ( *s == ',' ) s++;
    while ( *s ) {
        if ( *s == '(' ) {
            s++; num = 0;
            while ( *s && *s != ')' ) {
                if ( *s == ',' ) { s++; continue; }
                if ( FG.cTable[*s] != 1 ) goto illarg;
                ParseNumber(x,s)
                if ( x <= 0 || ( fix > 0 && x > fix ) ) goto illnum;
                num++;
                *w++ = x-1;
            }
            if ( *s == 0 ) {
                MesPrint("&Improper termination of statement");
                return(1);
            }
            if ( groupsize < 0 ) groupsize = num;
            else if ( groupsize != num ) goto group;
            s++;
        }
        else if ( FG.cTable[*s] == 1 ) {
            if ( groupsize < 0 ) groupsize = 1;
            else if ( groupsize != 1 ) {
group:          MesPrint("&All groups should have the same number of arguments");
                return(1);
            }
            ParseNumber(x,s)
            if ( x <= 0 || ( fix > 0 && x > fix ) ) {
illnum:         MesPrint("&Illegal argument number: %d",x);
                return(1);
            }
            *w++ = x-1;
        }
        else {
illarg:     MesPrint("&Illegal argument");
            return(1);
        }
        while ( *s == ',' ) s++;
    }
/*
    Now the completion
*/
    if ( w == ww ) {
        ww[-1] = 1;
        ww[-2] = 0;
        if ( fix > 0 ) {
            for ( i = 0; i < fix; i++ ) *w++ = i;
            ww[-2] = fix; /* Bugfix 31-oct-2001. Reported by York Schroeder */
        }
    }
    else {
        ww[-1] = groupsize;
        ww[-2] = (w-ww)/groupsize;
    }
    AT.WorkPointer[1] = w - AT.WorkPointer;
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    return(error);
}

/*
    #] DoSymmetrize :
    #[ CoSymmetrize :
*/

int CoSymmetrize(UBYTE *s) { return(DoSymmetrize(s,SYMMETRIC)); }

/*
    #] CoSymmetrize :
    #[ CoAntiSymmetrize :
```

```
*/

int CoAntiSymmetrize(UBYTE *s) { return(DoSymmetrize(s,ANTISYMMETRIC)); }

/*
    #] CoAntiSymmetrize :
    #[ CoCycleSymmetrize :
*/

int CoCycleSymmetrize(UBYTE *s) { return(DoSymmetrize(s,CYCLESYMMETRIC)); }

/*
    #] CoCycleSymmetrize :
    #[ CoRCycleSymmetrize :
*/

int CoRCycleSymmetrize(UBYTE *s) { return(DoSymmetrize(s,RCYCLESYMMETRIC)); }

/*
    #] CoRCycleSymmetrize :
    #[ CoWrite :
*/

int CoWrite(UBYTE *s)
{
    GETIDENTITY
    UBYTE *option;
    KEYWORDV *key;
    option = s;
    if ( ( ( s = SkipAName(s) ) == 0 ) || *s != 0 ) {
        MesPrint("&Proper use of write statement is: write option");
        return(1);
    }
    key = (KEYWORDV *)FindInKeyWord(option,(KEYWORD *)writeoptions,sizeof(writeoptions)/sizeof(KEYWORD));
    if ( key == 0 ) {
        MesPrint("&Unrecognized option in write statement");
        return(1);
    }
    *key->var = key->type;
    AR.SortType = AC.SortType;
    return(0);
}

/*
    #] CoWrite :
    #[ CoNWrite :
*/

int CoNWrite(UBYTE *s)
{
    GETIDENTITY
    UBYTE *option;
    KEYWORDV *key;
    option = s;
    if ( ( ( s = SkipAName(s) ) == 0 ) || *s != 0 ) {
        MesPrint("&Proper use of nwrite statement is: nwrite option");
        return(1);
    }
    key = (KEYWORDV *)FindInKeyWord(option,(KEYWORD *)writeoptions,sizeof(writeoptions)/sizeof(KEYWORD));
    if ( key == 0 ) {
        MesPrint("&Unrecognized option in nwrite statement");
        return(1);
    }
    *key->var = key->flags;
    AR.SortType = AC.SortType;
    return(0);
}

/*
    #] CoNWrite :
    #[ CoRatio :
*/

static WORD ratstring[6] = { TYPEOPERATION, 6, RATIO, 0, 0, 0 };

int CoRatio(UBYTE *s)
{
    UBYTE c, *t;
    int i, type, error = 0;
    WORD numsym, *rs;
    rs = ratstring+3;
    for ( i = 0; i < 3; i++ ) {
        if ( *s ) {
            t = s;
            s = SkipAName(s);
```

```
            c = *s; *s = 0;
            if ( ( ( type = GetName(AC.varnames,t,&numsym,WITHAUTO) ) != CSYMBOL )
            && type != CDUBIOUS ) {
                MesPrint("&%s is not a symbol",t);
                error = 4;
                if ( type < 0 ) numsym = AddSymbol(t,-MAXPOWER,MAXPOWER,0,0);
            }
            *s = c;
            if ( *s == ',' ) s++;
        }
        else {
            if ( error == 0 )
                MesPrint("&The ratio statement needs three symbols for its arguments");
            error++;
            numsym = 0;
        }
        *rs++ = numsym;
    }
    AddNtoL(6,ratstring);
    return(error);
}

/*
    #] CoRatio :
    #[ CoRedefine :

    We have a preprocessor variable and a (new) value for it.
    This value is inside a string that must be stored.
*/

int CoRedefine(UBYTE *s)
{
    UBYTE *name, c, *args = 0;
    int numprevar;
    WORD code[2];
    name = s;
    if ( FG.cTable[*s] || ( s = SkipAName(s) ) == 0 || s[-1] == '_' ) {
        MesPrint("&Illegal name for preprocessor variable in redefine statement");
        return(1);
    }
    c = *s; *s = 0;
    for ( numprevar = NumPre-1; numprevar >= 0; numprevar-- ) {
        if ( StrCmp(name,PreVar[numprevar].name) == 0 ) break;
    }
    if ( numprevar < 0 ) {
        MesPrint("&There is no preprocessor variable with the name `%s'",name);
        *s = c;
        return(1);
    }
    *s = c;
/*
    The next code worries about arguments.
    It is a direct copy of the code in TheDefine in the preprocessor.
*/
    if ( *s == '(' ) { /* arguments. scan for correctness */
        s++; args = s;
        for (;;) {
            if ( chartype[*s] != 0 ) goto illarg;
            s++;
            while ( chartype[*s] <= 1 ) s++;
            while ( *s == ' ' || *s == '\t' ) s++;
            if ( *s == ')' ) break;
            if ( *s != ',' ) goto illargs;
            s++;
            while ( *s == ' ' || *s == '\t' ) s++;
        }
        *s++ = 0;
        while ( *s == ' ' || *s == '\t' ) s++;
    }
    while ( *s == ',' ) s++;
    if ( *s != '"' ) {
encl:   MesPrint("&Value for %s should be enclosed in double quotes"
        ,PreVar[numprevar].name);
        return(1);
    }
    s++; name = s; /* actually name points to the new string */
    while ( *s && *s != '"' ) { if ( *s == '\\' ) s++; s++; }
    if ( *s != '"' ) goto encl;
    *s = 0;
    code[0] = TYPEREDEFPRE; code[1] = numprevar;
/*
    AddComString(2,code,name,0);
*/
    Add2ComStrings(2,code,name,args);
    *s = '"';
#ifdef PARALLELCODE
/*
```

```
    Now we prepare the input numbering system for pthreads.
    We need a list of preprocessor variables that are redefined in this
    module.
*/
    {
      int j;
      WORD *newpf;
      LONG *newin;
      for ( j = 0; j < AC.numpfirstnum; j++ ) {
        if ( numprevar == AC.pfirstnum[j] ) break;
      }
      if ( j >= AC.numpfirstnum ) {  /* add to list */
        if ( j >= AC.sizepfirstnum ) {
            if ( AC.sizepfirstnum <= 0 ) { AC.sizepfirstnum = 10; }
            else { AC.sizepfirstnum = 2 * AC.sizepfirstnum; }
            newin = (LONG *)Malloc1(AC.sizepfirstnum*(sizeof(WORD)+sizeof(LONG)),"AC.pfirstnum");
            newpf = (WORD *)(newin+AC.sizepfirstnum);
            for ( j = 0; j < AC.numpfirstnum; j++ ) {
                newpf[j] = AC.pfirstnum[j];
                newin[j] = AC.inputnumbers[j];
            }
            if ( AC.inputnumbers ) M_free(AC.inputnumbers,"AC.pfirstnum");
            AC.inputnumbers = newin;
            AC.pfirstnum = newpf;
        }
        AC.pfirstnum[AC.numpfirstnum] = numprevar;
        AC.inputnumbers[AC.numpfirstnum] = -1;
        AC.numpfirstnum++;
      }
    }
#endif
    return(0);
illarg:;
    MesPrint("&Illegally formed name in argument of redefine statement");
    return(1);
illargs:;
    MesPrint("&Illegally formed arguments in redefine statement");
    return(1);
}

/*
    #] CoRedefine :
    #[ CoRenumber :

    renumber    or renumber,0     Only exchanges (n^2 until no improvement)
    renumber,1                    All permutations (could be slow)
*/

int CoRenumber(UBYTE *s)
{
    int x;
    UBYTE *inp;
    while ( *s == ',' ) s++;
    inp = s;
    if ( *s == 0 ) { x = 0; }
    else ParseNumber(x,s)
    if ( *s == 0 && x >= 0 && x <= 1 ) {
        Add3Com(TYPERENUMBER,x);
        return(0);
    }
    MesPrint("&Illegal argument in Renumber statement: '%s'",inp);
    return(1);
}

/*
    #] CoRenumber :
    #[ CoSum :
*/

int CoSum(UBYTE *s)
{
    CBUF *C = cbuf+AC.cbufnum;
    UBYTE *ss = 0, c, *t;
    int error = 0, i = 0, type, x;
    WORD numindex,number;
    while ( *s ) {
        t = s;
        if ( *s == '$' ) {
            t++; s++; while ( FG.cTable[*s] < 2 ) s++;
            c = *s; *s = 0;
            if ( ( number = GetDollar(t) ) < 0 ) {
                MesPrint("&Undefined variable $%s",t);
                if ( !error ) error = 1;
                number = AddDollar(t,0,0,0);
            }
            numindex = -number;
        }
```

```
        else {
            if ( ( s = SkipAName(s) ) == 0 ) return(1);
            c = *s; *s = 0;
            if ( ( ( type = GetOName(AC.exprnames,t,&numindex,NOAUTO) ) != NAMENOTFOUND )
            || ( ( type = GetOName(AC.varnames,t,&numindex,WITHAUTO) ) != CINDEX ) ) {
                if ( type != NAMENOTFOUND ) error = NameConflict(type,t);
                else {
                    MesPrint("&%s should have been declared as an index",t);
                    error = 1;
                    numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
                }
            }
        }
        Add3Com(TYPESUM,numindex);
        i = 3; *s = c;
        if ( *s == 0 ) break;
        if ( *s != ',' ) {
            MesPrint("&Illegal separator between objects in sum statement.");
            return(1);
        }
        s++;
        if ( FG.cTable[*s] == 0 || *s == '[' || *s == '$' ) {
            while ( FG.cTable[*s] == 0 || *s == '[' || *s == '$' ) {
                if ( *s == '$' ) {
                    s++;
                    ss = t = s;
                    while ( FG.cTable[*s] < 2 ) s++;
                    c = *s; *s = 0;
                    if ( ( number = GetDollar(t) ) < 0 ) {
                        MesPrint("&Undefined variable $%s",t);
                        if ( !error ) error = 1;
                        number = AddDollar(t,0,0,0);
                    }
                    numindex = -number;
                }
                else {
                    ss = t = s;
                    if ( ( s = SkipAName(s) ) == 0 ) return(1);
                    c = *s; *s = 0;
                    if ( ( ( type = GetOName(AC.exprnames,t,&numindex,NOAUTO) ) != NAMENOTFOUND )
                    || ( ( type = GetOName(AC.varnames,t,&numindex,WITHAUTO) ) != CINDEX ) ) {
                        if ( type != NAMENOTFOUND ) error = NameConflict(type,t);
                        else {
                            MesPrint("&%s should have been declared as an index",t);
                            error = 1;
                            numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
                        }
                    }
                }
                AddToCB(C,numindex)
                i++;
                C->Pointer[-i+1] = i;
                *s = c;
                if ( *s == 0 ) return(error);
                if ( *s != ',' ) {
                    MesPrint("&Illegal separator between objects in sum statement.");
                    return(1);
                }
                s++;
            }
            if ( FG.cTable[*s] == 1 ) {
                C->Pointer[-i+1]--; C->Pointer--; s = ss;
            }
        }
        else if ( FG.cTable[*s] == 1 ) {
            while ( FG.cTable[*s] == 1 ) {
                t = s;
                x = *s++ - '0';
                while( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
                if ( *s && *s != ',' ) {
                    MesPrint("&%s is not a legal fixed index",t);
                    return(1);
                }
                else if ( x >= AM.OffsetIndex ) {
                    MesPrint("&%d is too large to be a fixed index",x);
                    error = 1;
                }
                else {
                    AddToCB(C,x)
                    i++;
                    C->Pointer[-i] = TYPESUMFIX;
                    C->Pointer[-i+1] = i;
                }
                if ( *s == 0 ) break;
                s++;
            }
        }
```

```
        else {
            MesPrint("&Illegal object in sum statement");
            error = 1;
        }
    }
    return(error);
}

/*
    #] CoSum :
    #[ CoToTensor :
*/

static WORD cttarray[7] = { TYPEOPERATION,7,TENVEC,0,0,1,0 };

int CoToTensor(UBYTE *s)
{
    UBYTE c, *t;
    int type, j, nargs, error = 0;
    WORD number, dol[2] = { 0, 0 };
    cttarray[1] = 6;  /* length */
    cttarray[3] = 0;  /* tensor */
    cttarray[4] = 0;  /* vector */
    cttarray[5] = 1;  /* option flags */
/*  cttarray[6] = 0;     set veto */
/*
    Count the number of the arguments. The validity of them is not checked here.
*/
    nargs = 0;
    t = s;
    for (;;) {
        while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
        if ( *s == 0 ) break;
        if ( *s == '!' ) {
            s++;
            if ( *s == '{' ) {
                SKIPBRA2(s)
                s++;
            } else {
                if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
            }
        } else {
            if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
        }
        nargs++;
    }
    if ( nargs < 2 ) goto not_enough_arguments;
    s = t;
/*
    Parse options, which are given as the arguments except the last two.
*/
    for ( j = 2; j < nargs; j++ ) {
        while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
        if ( *s == '!' ) {
/*
            Handle !set or !{vector,...}. Note: If two or more sets are
            specified, then only the last one is used.
*/
            s++;
            cttarray[1] = 7;
            cttarray[5] |= 8;
            if ( FG.cTable[*s] == 0 || *s == '[' || *s == '_' ) {
                t = s;
                if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
                c = *s; *s = 0;
                type = GetName(AC.varnames,t,&number,WITHAUTO);
                if ( type == CVECTOR ) {
/*
                    As written in the manual, "!p" (without "{}") should work.
*/
                    cttarray[6] = DoTempSet(t,s);
                    *s = c;
                    goto check_tempset;
                }
                else if ( type != CSET ) {
                    MesPrint("&%s is not the name of a set or a vector",t);
                    error = 1;
                }
                *s = c;
                cttarray[6] = number;
            }
            else if ( *s == '{' ) {
                t = ++s; SKIPBRA2(s) *s = 0;
                cttarray[6] = DoTempSet(t,s);
                *s++ = '}';
check_tempset:
                if ( cttarray[6] < 0 ) {
```

```
                        error = 1;
                    }
                    if ( AC.wildflag ) {
                        MesPrint("&Improper use of wildcard(s) in set specification");
                        error = 1;
                    }
                }
            } else {
/*
                Other options.
*/
                t = s;
                if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
                c = *s; *s = 0;
                if ( StrICmp(t,(UBYTE *)"nosquare") == 0 ) cttarray[5] |= 2;
                else if ( StrICmp(t,(UBYTE *)"functions") == 0 ) cttarray[5] |= 4;
                else {
                    MesPrint("&Unrecognized option in ToTensor statement: '%s'",t);
                    *s = c;
                    return(1);
                }
                *s = c;
            }
        }
/*
        Now parse a vector and a tensor. The ordering doesn't matter.
*/
        for ( j = 0; j < 2; j++ ) {
            while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
            t = s;
            if ( ( s = SkipAName(s) ) == 0 ) goto syntax_error;
            c = *s; *s = 0;
            if ( t[0] == '$' ) {
                dol[j] = GetDollar(t+1);
                if ( dol[j] < 0 ) dol[j] = AddDollar(t+1,DOLUNDEFINED,0,0);
            } else {
                type = GetName(AC.varnames,t,&number,WITHAUTO);
                if ( type == CVECTOR ) {
                    cttarray[4] = number + AM.OffsetVector;
                }
                else if ( type == CFUNCTION && ( functions[number].spec > 0 ) ) {
                    cttarray[3] = number + FUNCTION;
                }
                else {
                    MesPrint("&%s is not a vector or a tensor",t);
                    error = 1;
                }
            }
            *s = c;
        }
        if ( cttarray[3] == 0 || cttarray[4] == 0 ) {
            if ( dol[0] == 0 && dol[1] == 0 ) {
                goto not_enough_arguments;
            }
            else if ( cttarray[3] ) {
                if ( dol[1] )          cttarray[4] = dol[1];
                else if ( dol[0] ) { cttarray[4] = dol[0]; }
                else {
                    goto not_enough_arguments;
                }
            }
            else if ( cttarray[4] ) {
                if ( dol[1] )     { cttarray[3] = -dol[1]; }
                else if ( dol[0] ) cttarray[3] = -dol[0];
                else {
                    goto not_enough_arguments;
                }
            }
            else {
                if ( dol[0] == 0 || dol[1] == 0 ) {
                    goto not_enough_arguments;
                }
                else {
                    cttarray[3] = -dol[0]; cttarray[4] = dol[1];
                }
            }
        }
        AddNtoL(cttarray[1],cttarray);
        return(error);

syntax_error:
        MesPrint("&Syntax error in ToTensor statement");
        return(1);

not_enough_arguments:
        MesPrint("&ToTensor statement needs a vector and a tensor");
        return(1);
```

```
}

/*
    #] CoToTensor :
    #[ CoToVector :
*/

static WORD ctvarray[6] = { TYPEOPERATION,6,TENVEC,0,0,0 };

int CoToVector(UBYTE *s)
{
    UBYTE *t, c;
    int j, type, error = 0;
    WORD number, dol[2];
    dol[0] = dol[1] = 0;
    ctvarray[3] = ctvarray[4] = ctvarray[5] = 0;
    for ( j = 0; j < 2; j++ ) {
        t = s;
        if ( ( s = SkipAName(s) ) == 0 ) {
proper:     MesPrint("&Arguments of ToVector statement should be a vector and a tensor");
            return(1);
        }
        c = *s; *s = 0;
        if ( *t == '$' ) {
            dol[j] = GetDollar(t+1);
            if ( dol[j] < 0 ) dol[j] = AddDollar(t+1,DOLUNDEFINED,0,0);
        }
        else if ( ( type = GetName(AC.varnames,t,&number,WITHAUTO) ) == CVECTOR )
            ctvarray[4] = number + AM.OffsetVector;
        else if ( type == CFUNCTION && ( functions[number].spec > 0 ) )
            ctvarray[3] = number+FUNCTION;
        else {
            MesPrint("&%s is not a vector or a tensor",t);
            error = 1;
        }
        *s = c; if ( *s && *s != ',' ) goto proper;
        if ( *s ) s++;
    }
    if ( *s != 0 ) goto proper;
    if ( ctvarray[3] == 0 || ctvarray[4] == 0 ) {
        if ( dol[0] == 0 && dol[1] == 0 ) {
            MesPrint("&ToVector statement needs a vector and a tensor");
            error = 1;
        }
        else if ( ctvarray[3] ) {
            if ( dol[1] )      ctvarray[4] = dol[1];
            else if ( dol[0] ) ctvarray[4] = dol[0];
            else {
                MesPrint("&ToVector statement needs a vector and a tensor");
                error = 1;
            }
        }
        else if ( ctvarray[4] ) {
            if ( dol[1] )      ctvarray[3] = -dol[1];
            else if ( dol[0] ) ctvarray[3] = -dol[0];
            else {
                MesPrint("&ToVector statement needs a vector and a tensor");
                error = 1;
            }
        }
        else {
            if ( dol[0] == 0 || dol[1] == 0 ) {
                MesPrint("&ToVector statement needs a vector and a tensor");
                error = 1;
            }
            else {
                ctvarray[3] = -dol[0]; ctvarray[4] = dol[1];
            }
        }
    }
    AddNtoL(6,ctvarray);
    return(error);
}

/*
    #] CoToVector :
    #[ CoTrace4 :
*/

int CoTrace4(UBYTE *s)
{
    int error = 0, type, option = CHISHOLM;
    UBYTE *t, c;
    WORD numindex, one = 1;
    KEYWORD *key;
    for (;;) {
        t = s;
```

```
        if ( FG.cTable[*s] == 1 ) break;
        if ( ( s = SkipAName(s) ) == 0 ) {
proper:     MesPrint("&Proper syntax for Trace4 is 'Trace4[,options],index;'");
            return(1);
        }
        if ( *s == 0 ) break;
        c = *s; *s = 0;
        if ( ( key = FindKeyWord(t,trace4options,
            sizeof(trace4options)/sizeof(KEYWORD)) ) == 0 ) break;
        else {
            option |=  key->type;
            option &= ~key->flags;
        }
        if ( ( *s++ = c ) != ',' ) {
            MesPrint("&Illegal separator in Trace4 statement");
            return(1);
        }
        if ( *s == 0 ) goto proper;
    }
    s = t;
    if ( FG.cTable[*s] == 1 ) {
retry:
        ParseNumber(numindex,s)
        if ( *s != 0 ) {
            MesPrint("&Last argument of Trace4 should be an index");
            return(1);
        }
        if ( numindex >= AM.OffsetIndex ) {
            MesPrint("&fixed index >= %d. Change value of OffsetIndex in setup file"
            ,AM.OffsetIndex);
            return(1);
        }
    }
    else if ( *s == '$' ) {
        if ( ( type = GetName(AC.dollarnames,s+1,&numindex,NOAUTO) ) == CDOLLAR )
            numindex = -numindex;
        else {
            MesPrint("&%s is undefined",s);
            numindex = AddDollar(s+1,DOLINDEX,&one,1);
            return(1);
        }
tests:  s = SkipAName(s);
        if ( *s != 0 ) {
            MesPrint("&Trace4 should have a single index or $variable for its argument");
            return(1);
        }
    }
    else if ( ( type = GetName(AC.varnames,s,&numindex,WITHAUTO) ) == CINDEX ) {
        numindex += AM.OffsetIndex;
        goto tests;
    }
    else if ( type != -1 ) {
        if ( type != CDUBIOUS ) {
            if ( ( FG.cTable[*s] != 0 ) && ( *s != '[' ) ) {
                if ( *s == '+' && FG.cTable[s[1]] == 1 ) { s++; goto retry; }
                goto proper;
            }
            NameConflict(type,s);
            type = MakeDubious(AC.varnames,s,&numindex);
        }
        return(1);
    }
    else {
        MesPrint("&%s is not an index",s);
        numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
        return(1);
    }
    if ( error ) return(error);
    if ( ( option & CHISHOLM ) != 0 )
        Add4Com(TYPECHISHOLM,numindex,(option & ALSOREVERSE));
    Add5Com(TYPEOPERATION,TAKETRACE,4 + (option & NOTRICK),numindex);
    return(0);
}

/*
    #] CoTrace4 :
    #[ CoTraceN :
*/

int CoTraceN(UBYTE *s)
{
    WORD numindex, one = 1;
    int type;
    if ( FG.cTable[*s] == 1 ) {
retry:
        ParseNumber(numindex,s)
        if ( *s != 0 ) {
```

```
proper:     MesPrint("&TraceN should have a single index for its argument");
            return(1);
        }
        if ( numindex >= AM.OffsetIndex ) {
            MesPrint("&fixed index >= %d. Change value of OffsetIndex in setup file"
            ,AM.OffsetIndex);
            return(1);
        }
    }
    else if ( *s == '$' ) {
        if ( ( type = GetName(AC.dollarnames,s+1,&numindex,NOAUTO) ) == CDOLLAR )
            numindex = -numindex;
        else {
            MesPrint("&%s is undefined",s);
            numindex = AddDollar(s+1,DOLINDEX,&one,1);
            return(1);
        }
tests:  s = SkipAName(s);
        if ( *s != 0 ) {
            MesPrint("&TraceN should have a single index or $variable for its argument");
            return(1);
        }
    }
    else if ( ( type = GetName(AC.varnames,s,&numindex,WITHAUTO) ) == CINDEX ) {
        numindex += AM.OffsetIndex;
        goto tests;
    }
    else if ( type != -1 ) {
        if ( type != CDUBIOUS ) {
            if ( ( FG.cTable[*s] != 0 ) && ( *s != '[' ) ) {
                if ( *s == '+' && FG.cTable[s[1]] == 1 ) { s++; goto retry; }
                goto proper;
            }
            NameConflict(type,s);
            type = MakeDubious(AC.varnames,s,&numindex);
        }
        return(1);
    }
    else {
        MesPrint("&%s is not an index",s);
        numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
        return(1);
    }
    Add5Com(TYPEOPERATION,TAKETRACE,0,numindex);
    return(0);
}

/*
    #] CoTraceN :
    #[ CoChisholm :
*/

int CoChisholm(UBYTE *s)
{
    int error = 0, type, option = CHISHOLM;
    UBYTE *t, c;
    WORD numindex, one = 1;
    KEYWORD *key;
    for (;;) {
        t = s;
        if ( FG.cTable[*s] == 1 ) break;
        if ( ( s = SkipAName(s) ) == 0 ) {
proper:     MesPrint("&Proper syntax for Chisholm is 'Chisholm[,options],index;'");
            return(1);
        }
        if ( *s == 0 ) break;
        c = *s; *s = 0;
        if ( ( key = FindKeyWord(t,chisoptions,
            sizeof(chisoptions)/sizeof(KEYWORD)) ) == 0 ) break;
        else {
            option |=  key->type;
            option &= ~key->flags;
        }
        if ( ( *s++ = c ) != ',' ) {
            MesPrint("&Illegal separator in Chisholm statement");
            return(1);
        }
        if ( *s == 0 ) goto proper;
    }
    s = t;
    if ( FG.cTable[*s] == 1 ) {
        ParseNumber(numindex,s)
        if ( *s != 0 ) {
            MesPrint("&Last argument of Chisholm should be an index");
            return(1);
        }
        if ( numindex >= AM.OffsetIndex ) {
```

```
                MesPrint("&fixed index >= %d. Change value of OffsetIndex in setup file"
                ,AM.OffsetIndex);
                return(1);
            }
        }
        else if ( *s == '$' ) {
            if ( ( type = GetName(AC.dollarnames,s+1,&numindex,NOAUTO) ) == CDOLLAR )
                numindex = -numindex;
            else {
                MesPrint("&%s is undefined",s);
                numindex = AddDollar(s+1,DOLINDEX,&one,1);
                return(1);
            }
tests:  s = SkipAName(s);
            if ( *s != 0 ) {
                MesPrint("&Chisholm should have a single index or $variable for its argument");
                return(1);
            }
        }
        else if ( ( type = GetName(AC.varnames,s,&numindex,WITHAUTO) ) == CINDEX ) {
            numindex += AM.OffsetIndex;
            goto tests;
        }
        else if ( type != -1 ) {
            if ( type != CDUBIOUS ) {
                NameConflict(type,s);
                type = MakeDubious(AC.varnames,s,&numindex);
            }
            return(1);
        }
        else {
            MesPrint("&%s is not an index",s);
            numindex = AddIndex(s,AC.lDefDim,AC.lDefDim4) + AM.OffsetIndex;
            return(1);
        }
        if ( error ) return(error);
        Add4Com(TYPECHISHOLM,numindex,(option & ALSOREVERSE));
        return(0);
}

/*
        #] CoChisholm :
        #[ DoChain :

        Syntax: Chainxx functionname;
*/

int DoChain(UBYTE *s, int option)
{
        WORD numfunc,type;
        if ( *s == '$' ) {
            if ( ( type = GetName(AC.dollarnames,s+1,&numfunc,NOAUTO) ) == CDOLLAR )
                numfunc = -numfunc;
            else {
                MesPrint("&%s is undefined",s);
                numfunc = AddDollar(s+1,DOLINDEX,&one,1);
                return(1);
            }
tests:  s = SkipAName(s);
            if ( *s != 0 ) {
                MesPrint("&ChainIn/ChainOut should have a single function or $variable for its argument");
                return(1);
            }
        }
        else if ( ( type = GetName(AC.varnames,s,&numfunc,WITHAUTO) ) == CFUNCTION ) {
            numfunc += FUNCTION;
            goto tests;
        }
        else if ( type != -1 ) {
            if ( type != CDUBIOUS ) {
                NameConflict(type,s);
                type = MakeDubious(AC.varnames,s,&numfunc);
            }
            return(1);
        }
        else {
            MesPrint("&%s is not a function",s);
            numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
            return(1);
        }
        Add3Com(option,numfunc);
        return(0);
}

/*
        #] DoChain :
        #[ CoChainin :
```

```
    Syntax: Chainin functionname;
*/

int CoChainin(UBYTE *s)
{
    return(DoChain(s,TYPECHAININ));
}

/*
    #] CoChainin :
    #[ CoChainout :

    Syntax: Chainout functionname;
*/

int CoChainout(UBYTE *s)
{
    return(DoChain(s,TYPECHAINOUT));
}

/*
    #] CoChainout :
    #[ CoExit :
*/

int CoExit(UBYTE *s)
{
    UBYTE *name;
    WORD code = TYPEEXIT;
    while ( *s == ',' ) s++;
    if ( *s == 0 ) {
        Add3Com(TYPEEXIT,0);
        return(0);
    }
    name = s+1;
    s++;
    while ( *s ) { if ( *s == '\\' ) s++; s++; }
    if ( name[-1] != '"' || s[-1] != '"' ) {
        MesPrint("&Illegal syntax for exit statement");
        return(1);
    }
    s[-1] = 0;
    AddComString(1,&code,name,0);
    s[-1] = '"';
    return(0);
}

/*
    #] CoExit :
    #[ CoInParallel :
*/

int CoInParallel(UBYTE *s)
{
    return(DoInParallel(s,1));
}

/*
    #] CoInParallel :
    #[ CoNotInParallel :
*/

int CoNotInParallel(UBYTE *s)
{
    return(DoInParallel(s,0));
}

/*
    #] CoNotInParallel :
    #[ DoInParallel :

    InParallel;
    InParallel,names;
    NotInParallel;
    NotInParallel,names;
*/

int DoInParallel(UBYTE *s, int par)
{
#ifdef PARALLELCODE
    EXPRESSIONS e;
    WORD i;
#endif
    WORD number;
    UBYTE *t, c;
    int error = 0;
```

```
#ifndef WITHPTHREADS
    DUMMYUSE(par);
#endif
    if ( *s == 0 ) {
        AC.inparallelflag = par;
#ifdef PARALLELCODE
        for ( i = NumExpressions-1; i >= 0; i-- ) {
            e = Expressions+i;
            if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
            || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
            ) {
                e->partodo = par;
            }
        }
#endif
    }
    else {
        for(;;) {   /* Look for a (comma separated) list of variables */
            while ( *s == ',' ) s++;
            if ( *s == 0 ) break;
            if ( *s == '[' || FG.cTable[*s] == 0 ) {
                t = s;
                if ( ( s = SkipAName(s) ) == 0 ) {
                    MesPrint("&Improper name for an expression: '%s'",t);
                    return(1);
                }
                c = *s; *s = 0;
                if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
#ifdef PARALLELCODE
                    e = Expressions+number;
                    if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
                    || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
                    ) {
                        e->partodo = par;
                    }
#endif
                }
                else if ( GetName(AC.varnames,t,&number,NOAUTO) != NAMENOTFOUND ) {
                    MesPrint("&%s is not an expression",t);
                    error = 1;
                }
                *s = c;
            }
            else {
                MesPrint("&Illegal object in InExpression statement");
                error = 1;
                while ( *s && *s != ',' ) s++;
                if ( *s == 0 ) break;
            }
        }

    }
    return(error);
}

/*
    #] DoInParallel :
    #[ CoInExpression :
*/

int CoInExpression(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c;
    WORD *w, number;
    int error = 0;
    w = AT.WorkPointer;
    if ( AC.inexprlevel >= MAXNEST ) {
        MesPrint("@Nesting of inexpression statements more than %d levels",(WORD)MAXNEST);
        return(-1);
    }
    AC.inexprsumcheck[AC.inexprlevel] = NestingChecksum();
    AC.inexprstack[AC.inexprlevel] = cbuf[AC.cbufnum].Pointer
                                 - cbuf[AC.cbufnum].Buffer + 2;
    AC.inexprlevel++;
    *w++ = TYPEINEXPRESSION;
    w++; w++;
    for(;;) {   /* Look for a (comma separated) list of variables */
        while ( *s == ',' ) s++;
        if ( *s == 0 ) break;
        if ( *s == '[' || FG.cTable[*s] == 0 ) {
            t = s;
            if ( ( s = SkipAName(s) ) == 0 ) {
                MesPrint("&Improper name for an expression: '%s'",t);
                return(1);
            }
            c = *s; *s = 0;
```

```
            if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
                *w++ = number;
            }
            else if ( GetName(AC.varnames,t,&number,NOAUTO) != NAMENOTFOUND ) {
                MesPrint("&%s is not an expression",t);
                error = 1;
            }
            *s = c;
        }
        else {
            MesPrint("&Illegal object in InExpression statement");
            error = 1;
            while ( *s && *s != ',' ) s++;
            if ( *s == 0 ) break;
        }
    }
    AT.WorkPointer[1] = w - AT.WorkPointer;
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    return(error);
}

/*
    #] CoInExpression :
    #[ CoEndInExpression :
*/

int CoEndInExpression(UBYTE *s)
{
    CBUF *C = cbuf+AC.cbufnum;
    while ( *s == ',' ) s++;
    if ( *s ) {
        MesPrint("&Illegal syntax for EndInExpression statement");
        return(1);
    }
    if ( AC.inexprlevel <= 0 ) {
        MesPrint("&EndInExpression without corresponding InExpression statement");
        return(1);
    }
    AC.inexprlevel--;
    cbuf[AC.cbufnum].Buffer[AC.inexprstack[AC.inexprlevel]] = C->numlhs;
    if ( AC.inexprsumcheck[AC.inexprlevel] != NestingChecksum() ) {
        MesNesting();
        return(1);
    }
    return(0);
}

/*
    #] CoEndInExpression :
    #[ CoSetExitFlag :
*/

int CoSetExitFlag(UBYTE *s)
{
    if ( *s ) {
        MesPrint("&Illegal syntax for the SetExitFlag statement");
        return(1);
    }
    Add2Com(TYPESETEXIT);
    return(0);
}

/*
    #] CoSetExitFlag :
    #[ CoTryReplace :
*/
int CoTryReplace(UBYTE *p)
{
    GETIDENTITY
    UBYTE *name, c;
    WORD *w, error = 0, i, which = -1, c1, minvec = 0;
    w = AT.WorkPointer;
    *w++ = TYPETRY;
    *w++ = 3;
    *w++ = 0;
    *w++ = REPLACEMENT;
    *w++ = FUNHEAD;
    FILLFUN(w)
/*
    Now we have to read a function argument for the replace_ function.
    Current arguments that we allow involve only single arguments
    that do not expand further. No brackets!
*/
    while ( *p ) {
/*
        No numbers yet
*/
```

```
        if ( *p == '-' && minvec == 0 && which == (CVECTOR+1) ) {
            minvec = 1; p++;
        }
        if ( *p == '[' || FG.cTable[*p] == 0 ) {
            name = p;
            if ( ( p = SkipAName(p) )  == 0 ) return(1);
            c = *p; *p = 0;
            i = GetName(AC.varnames,name,&c1,WITHAUTO);
            if ( which >= 0 && i >= 0 && i != CDUBIOUS && which != (i+1) ) {
                MesPrint("&Illegal combination of objects in TryReplace");
                error = 1;
            }
            else if ( minvec && i != CVECTOR && i != CDUBIOUS ) {
                MesPrint("&Currently a - sign can be used only with a vector in TryReplace");
                error = 1;
            }
            else switch ( i ) {
                case CSYMBOL: *w++ = -SYMBOL; *w++ = c1; break;
                case CVECTOR:
                    if ( minvec ) *w++ = -MINVECTOR;
                    else          *w++ = -VECTOR;
                    *w++ = c1 + AM.OffsetVector;
                    minvec = 0;
                    break;
                case CINDEX: *w++ = -INDEX; *w++ = c1 + AM.OffsetIndex;
                    if ( c1 >= AM.WilInd && c == '?' ) { *p++ = c; c = *p; }
                    break;
                case CFUNCTION: *w++ = -c1-FUNCTION; break;
                case CDUBIOUS: minvec = 0; error = 1; break;
                default:
                    MesPrint("&Illegal object type in TryReplace: %s",name);
                    error = 1;
                    i = 0;
                    break;
            }
            if ( which < 0 ) which = i+1;
            else which = -1;
            *p = c;
            if ( *p == ',' ) p++;
            continue;
        }
        else {
            MesPrint("&Illegal object in TryReplace");
            error = 1;
            while ( *p && *p != ',' ) {
                if ( *p == '(' ) SKIPBRA3(p)
                else if ( *p == '{' ) SKIPBRA2(p)
                else if ( *p == '[' ) SKIPBRA1(p)
                else p++;
            }
        }
        if ( *p == ',' ) p++;
        if ( which < 0 ) which = 0;
        else which = -1;
    }
    if ( which >= 0 ) {
        MesPrint("&Odd number of arguments in TryReplace");
        error = 1;
    }
    i = w - AT.WorkPointer;
    AT.WorkPointer[1] = i;
    AT.WorkPointer[2] = i - 3;
    AT.WorkPointer[4] = i - 3;
    AddNtoL((int)i,AT.WorkPointer);
    return(error);
}

/*
    #] CoTryReplace :
    #[ CoModulus :

    Old syntax:  Modulus [-] number [:number]
    New syntax:  Modulus [option(s)] number
        Options are: NoFunctions/CoefficientsOnly/AlsoFunctions
                     PlusMin/Positive
                     InverseTable
                     PrintPowersOf(number)
                     AlsoPowers/NoPowers
                     AlsoDollars/NoDollars
    Notice: We change the defaults. This may cause problems to some.
*/

int CoModulus(UBYTE *inp)
{
    GETIDENTITY
    int Retval = 0, sign = 1;
    UBYTE *p, c;
```

```
    while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
    if ( *inp == 0 ) {
SwitchOff:
        if ( AC.modpowers ) M_free(AC.modpowers,"AC.modpowers");
        AC.modpowers = 0;
        AN.ncmod = AC.ncmod = 0;
        if ( AC.halfmod ) M_free(AC.halfmod,"halfmod");
        AC.halfmod = 0; AC.nhalfmod = 0;
        if ( AC.modinverses ) M_free(AC.modinverses,"modinverses");
        AC.modinverses = 0;
        AC.modmode = 0;
        return(0);
    }
#ifdef WITHFLOAT
    if ( AT.aux_ != 0 ) {
        MesPrint("&Simultaneous use of floating point and modulus arithmetic makes no sense.");
        Retval = 1;
    }
#endif
    AC.modmode = 0;
    if ( *inp == '-' ) {
        sign = -1;
        inp++;
    }
    else {
      while ( FG.cTable[*inp] == 0 ) {
        p = inp;
        while ( FG.cTable[*inp] == 0 ) inp++;
        c = *inp; *inp = 0;
        if ( StrICmp(p,(UBYTE *)"nofunctions") == 0 ) {
            AC.modmode &= ~ALSOFUNARGS;
        }
        else if ( StrICmp(p,(UBYTE *)"alsofunctions") == 0 ) {
            AC.modmode |= ALSOFUNARGS;
        }
        else if ( StrICmp(p,(UBYTE *)"coefficientsonly") == 0 ) {
            AC.modmode &= ~ALSOFUNARGS;
            AC.modmode &= ~ALSOPOWERS;
            sign = -1;
        }
        else if ( StrICmp(p,(UBYTE *)"plusmin") == 0 ) {
            AC.modmode |= POSNEG;
        }
        else if ( StrICmp(p,(UBYTE *)"positive") == 0 ) {
            AC.modmode &= ~POSNEG;
        }
        else if ( StrICmp(p,(UBYTE *)"inversetable") == 0 ) {
            AC.modmode |= INVERSETABLE;
        }
        else if ( StrICmp(p,(UBYTE *)"noinversetable") == 0 ) {
            AC.modmode &= ~INVERSETABLE;
        }
        else if ( StrICmp(p,(UBYTE *)"nodollars") == 0 ) {
            AC.modmode &= ~ALSODOLLARS;
        }
        else if ( StrICmp(p,(UBYTE *)"alsodollars") == 0 ) {
            AC.modmode |= ALSODOLLARS;
        }
        else if ( StrICmp(p,(UBYTE *)"printpowersof") == 0 ) {
            *inp = c;
            if ( *inp != '(' ) {
badsyntax:
                MesPrint("&Bad syntax in argument of PrintPowersOf(number) in Modulus statement");
                return(1);
            }
            while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
            inp++; p = inp;
            if ( FG.cTable[*inp] != 1 ) goto badsyntax;
            do { inp++; } while ( FG.cTable[*inp] == 1 );
            c = *inp; *inp = 0;
            if ( GetLong(p,(UWORD *)AC.powmod,&AC.npowmod) ) Retval = -1;
            if ( TakeModulus((UWORD *)AC.powmod,&AC.npowmod,AC.cmod,AC.ncmod,NOUNPACK) ) Retval = -1;
            if ( AC.npowmod == 0 ) {
                MesPrint("&Improper value for generator");
                Retval = -1;
            }
            if ( MakeModTable() ) Retval = -1;
            AC.DirtPow = 1;
            *inp = c;
            while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
            if ( *inp != ')' ) goto badsyntax;
            inp++;
            c = *inp;
        }
        else if ( StrICmp(p,(UBYTE *)"alsopowers") == 0 ) {
            AC.modmode |= ALSOPOWERS;
            sign =  1;
```

```
        }
        else if ( StrICmp(p,(UBYTE *)"nopowers") == 0 ) {
            AC.modmode &= ~ALSOPOWERS;
            sign = -1;
        }
        else {
            MesPrint("&Unrecognized option %s in Modulus statement",inp);
            return(1);
        }
        *inp = c;
        while ( *inp == ',' || *inp == ' ' || *inp == '\t' ) inp++;
        if ( *inp == 0 ) {
            MesPrint("&Modulus statement with no value!!!");
            return(1);
        }
      }
    }
    p = inp;
    if ( FG.cTable[*inp] != 1 ) {
        MesPrint("&Invalid value for modulus:%s",inp);
        if ( AC.modpowers ) M_free(AC.modpowers,"AC.modpowers");
        AC.modpowers = 0;
        AN.ncmod = AC.ncmod = 0;
        if ( AC.halfmod ) M_free(AC.halfmod,"halfmod");
        AC.halfmod = 0; AC.nhalfmod = 0;
        if ( AC.modinverses ) M_free(AC.modinverses,"modinverses");
        AC.modinverses = 0;
        return(1);
    }
    do { inp++; } while ( FG.cTable[*inp] == 1 );
    c = *inp; *inp = 0;
    Retval = GetLong(p,(UWORD *)AC.cmod,&AC.ncmod);
    if ( Retval == 0 && AC.ncmod == 0 ) goto SwitchOff;
    if ( sign < 0 ) AC.ncmod = -AC.ncmod;
    AN.ncmod = AC.ncmod;
    if ( ( AC.modmode & INVERSETABLE ) != 0 ) MakeInverses();
    if ( AC.halfmod ) M_free(AC.halfmod,"halfmod");
    AC.halfmod = 0; AC.nhalfmod = 0;
    return(Retval);
}

/*
    #] CoModulus :
    #[ CoRepeat :
*/

int CoRepeat(UBYTE *inp)
{
    int error = 0;
    AC.RepSumCheck[AC.RepLevel] = NestingChecksum();
    AC.RepLevel++;
    if ( AC.RepLevel > AM.RepMax ) {
        MesPrint("&Too many repeat levels. Maximum is %d",AM.RepMax);
        return(1);
    }
    Add3Com(TYPEREPEAT,-1)    /* Means indefinite */
    while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
    if ( *inp ) {
        error = CompileStatement(inp);
        if ( CoEndRepeat(inp) ) error = 1;
    }
    return(error);
}

/*
    #] CoRepeat :
    #[ CoEndRepeat :
*/

int CoEndRepeat(UBYTE *inp)
{
    CBUF *C = cbuf+AC.cbufnum;
    int level, error = 0, repeatlevel = 0;
    DUMMYUSE(inp);
    AC.RepLevel--;
    if ( AC.RepLevel < 0 ) {
        MesPrint("&EndRepeat without Repeat");
        AC.RepLevel = 0;
        return(1);
    }
    else if ( AC.RepSumCheck[AC.RepLevel] != NestingChecksum() ) {
        MesNesting();
        error = 1;
    }
    level = C->numlhs+1;
    while ( level > 0 ) {
        if ( C->lhs[--level][0] == TYPEREPEAT ) {
```

```
            if ( repeatlevel == 0 ) {
                Add3Com(TYPEENDREPEAT,level)
                return(error);
            }
            repeatlevel--;
        }
        else if ( C->lhs[level][0] == TYPEENDREPEAT ) repeatlevel++;
    }
    return(1);
}

/*
    #] CoEndRepeat :
    #[ DoBrackets :

        Reads in the bracket information.
        Storage is in the form of a regular term.
        No subterms and arguments are allowed.
*/

int DoBrackets(UBYTE *inp, int par)
{
    GETIDENTITY
    UBYTE *p, *pp, c;
    WORD *to, i, type, *w, error = 0;
    WORD c1,c2, *WorkSave;
    int biflag;
    p = inp;
    WorkSave = to = AT.WorkPointer;
    to++;
    if ( AT.BrackBuf == 0 ) {
        AR.MaxBracket = 100;
        AT.BrackBuf = (WORD *)Malloc1(sizeof(WORD)*(AR.MaxBracket+1),"bracket buffer");
    }
    *AT.BrackBuf = 0;
    AR.BracketOn = 0;
    AC.bracketindexflag = 0;
    AT.bracketindexflag = 0;
    if ( *p == '+' || *p == '-' ) p++;
    if ( p[-1] == ',' && *p ) p--;
    if ( p[-1] == '+' && *p ) { biflag = 1;  if ( *p != ',' ) { *--p = ','; } }
    else if ( p[-1] == '-' && *p ) { biflag = -1; if ( *p != ',' ) { *--p = ','; } }
    else biflag = 0;
    while ( *p == ',' ) {
redo:   AR.BracketOn++;
        while ( *p == ',' ) p++;
        if ( *p == 0 ) break;
        if ( *p == '0' ) {
            p++; while ( *p == '0' ) p++;
            continue;
        }
        inp = pp = p;
        p = SkipAName(p);
        if ( p == 0 ) return(1);
        c = *p;
        *p = 0;
        type = GetName(AC.varnames,inp,&c1,WITHAUTO);
        if ( c == '.' ) {
            if ( type == CVECTOR || type == CDUBIOUS ) {
                *p++ = c;
                inp = p;
                p = SkipAName(p);
                if ( p == 0 ) return(1);
                c = *p;
                *p = 0;
                type = GetName(AC.varnames,inp,&c2,WITHAUTO);
                if ( type != CVECTOR && type != CDUBIOUS ) {
                    MesPrint("&Not a vector in dotproduct in bracket statement: %s",inp);
                    error = 1;
                }
                else type = CDOTPRODUCT;
            }
            else {
                MesPrint("&Illegal use of . after %s in bracket statement",inp);
                error = 1;
                *p++ = c;
                goto redo;
            }
        }
        switch ( type ) {
            case CSYMBOL :
                *to++ = SYMBOL; *to++ = 4; *to++ = c1; *to++ = 1; break;
            case CVECTOR :
                *to++ = INDEX; *to++ = 3; *to++ = AM.OffsetVector + c1; break;
            case CFUNCTION :
                *to++ = c1+FUNCTION; *to++ = FUNHEAD; *to++ = 0;
                FILLFUN3(to)
```

```
				break;
			case CDOTPRODUCT :
				*to++ = DOTPRODUCT; *to++ = 5; *to++ = c1 + AM.OffsetVector;
				*to++ = c2 + AM.OffsetVector; *to++ = 1; break;
			case CDELTA :
				*to++ = DELTA; *to++ = 4; *to++ = EMPTYINDEX; *to++ = EMPTYINDEX; break;
			case CSET :
				*to++ = SETSET; *to++ = 4; *to++ = c1; *to++ = Sets[c1].type; break;
			default :
				MesPrint("&Illegal bracket request for %s",pp);
				error = 1; break;
		}
		*p = c;
	}
	if ( *p ) {
		MesCerr("separator",p);
		AC.BracketNormalize = 0;
		AT.WorkPointer = WorkSave;
		error = 1;
		return(error);
	}
	*to++ = 1; *to++ = 1; *to++ = 3;
	*AT.WorkPointer = to - AT.WorkPointer;
	AT.WorkPointer = to;
	AC.BracketNormalize = 1;
	if ( BracketNormalize(BHEAD WorkSave) ) { error = 1; AR.BracketOn = 0; }
	else {
		w = WorkSave;
		if ( *w == 4 || !*w ) { AR.BracketOn = 0; }
		else {
			i = *(w+*w-1);
			if ( i < 0 ) i = -i;
			*w -= i;
			i = *w;
			if ( i > AR.MaxBracket ) {
				WORD *newbuf;
				newbuf = (WORD *)Malloc1(sizeof(WORD)*(i+1),"bracket buffer");
				AR.MaxBracket = i;
				if ( AT.BrackBuf != 0 ) M_free(AT.BrackBuf,"bracket buffer");
				AT.BrackBuf = newbuf;
			}
			to = AT.BrackBuf;
			NCOPY(to,w,i);
		}
	}
	AC.BracketNormalize = 0;
	if ( par == 1 ) AR.BracketOn = -AR.BracketOn;
	if ( error == 0 ) {
		AC.bracketindexflag = biflag;
		AT.bracketindexflag = biflag;
	}
	AT.WorkPointer = WorkSave;
	return(error);
}

/*
	#] DoBrackets :
	#[ CoBracket :
*/

int CoBracket(UBYTE *inp)
{ return(DoBrackets(inp,0)); }

/*
	#] CoBracket :
	#[ CoAntiBracket :
*/

int CoAntiBracket(UBYTE *inp)
{ return(DoBrackets(inp,1)); }

/*
	#] CoAntiBracket :
	#[ CoMultiBracket :

	Syntax:
		MultiBracket:{A|B} bracketinfo:...:{A|B} bracketinfo;
*/

int CoMultiBracket(UBYTE *inp)
{
	GETIDENTITY
	int i, error = 0, error1, type, num;
	UBYTE *s, c;
	WORD *to, *from;

	if ( *inp != ':' ) {
```

```
        MesPrint("&Illegal Multiple Bracket separator: %s",inp);
        return(1);
    }
    inp++;
    if ( AC.MultiBracketBuf == 0 ) {
        AC.MultiBracketBuf = (WORD **)Malloc1(sizeof(WORD *)*MAXMULTIBRACKETLEVELS,"multi bracket buffer");
        for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
            AC.MultiBracketBuf[i] = 0;
        }
    }
    else {
      for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
        if ( AC.MultiBracketBuf[i] ) {
            M_free(AC.MultiBracketBuf[i],"bracket buffer i");
            AC.MultiBracketBuf[i] = 0;
        }
      }
      AC.MultiBracketLevels = 0;
    }
    AC.MultiBracketLevels = 0;
/*
        Start with disabling the regular brackets.
*/
    if ( AT.BrackBuf == 0 ) {
        AR.MaxBracket = 100;
        AT.BrackBuf = (WORD *)Malloc1(sizeof(WORD)*(AR.MaxBracket+1),"bracket buffer");
    }
    *AT.BrackBuf = 0;
    AR.BracketOn = 0;
    AC.bracketindexflag = 0;
    AT.bracketindexflag = 0;
/*
    Now loop through the various levels, separated by the colons.
*/
    for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
        if ( *inp == 0 ) goto RegEnd;
/*
        1: skip to ':', determine bracket or antibracket
*/
        s = inp;
        while ( *s && *s != ':' ) {
            if ( *s == '[' ) { SKIPBRA1(s) s++; }
            else if ( *s == '{' ) { SKIPBRA2(s) s++; }
            else s++;
        }
        c = *s; *s = 0;
        if ( StrICont(inp,(UBYTE *)"antibrackets") == 0 ) { type = 1; }
        else if ( StrICont(inp,(UBYTE *)"brackets") == 0 ) { type = 0; }
        else {
            MesPrint("&Illegal (anti)bracket specification in MultiBracket statement");
            if ( error == 0 ) error = 1;
            goto NextLevel;
        }
        while ( FG.cTable[*inp] == 0 ) inp++;
        if ( *inp != ',' ) {
            MesPrint("&Illegal separator after (anti)bracket specification in MultiBracket statement");
            if ( error == 0 ) error = 1;
            goto NextLevel;
        }
        inp++;
/*
        2: call DoBrackets.
*/
        error1 = DoBrackets(inp, type);
        if ( error < 0 ) return(error1);
        if ( error1 > error ) error = error1;
/*
        3: copy bracket information to the multi bracket arrays
*/
        if ( AR.BracketOn ) {
            num = AT.BrackBuf[0];
            to = AC.MultiBracketBuf[i] = (WORD *)Malloc1((num+2)*sizeof(WORD),"bracket buffer i");
            from = AT.BrackBuf;
            *to++ = AR.BracketOn;
            NCOPY(to,from,num);
            *to = 0;
        }
/*
        4: set ready for the next level
*/
NextLevel:
        *s = c; if ( c == ':' ) s++;
        inp = s;
        *AT.BrackBuf = 0;
        AR.BracketOn = 0;
```

```
    }
    if ( *inp != 0 ) {
        MesPrint("&More than %d levels in MultiBracket statement",(WORD)MAXMULTIBRACKETLEVELS);
        if ( error == 0 ) error = 1;
    }
RegEnd:
    AC.MultiBracketLevels = i;
    *AT.BrackBuf = 0;
    AR.BracketOn = 0;
    AC.bracketindexflag = 0;
    AT.bracketindexflag = 0;
    return(error);
}

/*
    #] CoMultiBracket :
    #[ CountComp :

        This routine reads the count statement. The syntax is:
        count minimum,object,size[,object,size]
        Objects can be:
            symbol
            dotproduct
            vector
            function
        Vectors can have the auxiliary flags:
            +v +f +d +?setname

        Output for the compiler:
        TYPECOUNT,size,minimum,objects
        with the objects:
        SYMBOL,4,number,size
        DOTPRODUCT,5,v1,v2,size
        FUNCTION,4,number,size
        VECTOR,5,number,bits,size or VECTOR,6,number,bits,setnumber,size

        Currently only used in the if statement
*/

WORD *CountComp(UBYTE *inp, WORD *to)
{
    GETIDENTITY
    UBYTE *p, c;
    WORD *w, mini = 0, type, c1, c2;
    int error = 0;
    p = inp;
    w = to;
    AR.Eside = 2;
    *w++ = TYPECOUNT;
    *w++ = 0;
    *w++ = 0;
    while ( *p == ',' ) {
        p++; inp = p;
        if ( *p == '[' || FG.cTable[*p] == 0 ) {
            if ( ( p = SkipAName(inp) ) == 0 ) return(0);
            c = *p; *p = 0;
            type = GetName(AC.varnames,inp,&c1,WITHAUTO);
            if ( c == '.' ) {
                if ( type == CVECTOR || type == CDUBIOUS ) {
                    *p++ = c;
                    inp = p;
                    p = SkipAName(p);
                    if ( p == 0 ) return(0);
                    c = *p;
                    *p = 0;
                    type = GetName(AC.varnames,inp,&c2,WITHAUTO);
                    if ( type != CVECTOR && type != CDUBIOUS ) {
                        MesPrint("&Not a vector in dotproduct in if statement: %s",inp);
                        error = 1;
                    }
                    else type = CDOTPRODUCT;
                }
                else {
                    MesPrint("&Illegal use of . after %s in if statement",inp);
                    if ( type == NAMENOTFOUND )
                        MesPrint("&%s is not a properly declared variable",inp);
                    error = 1;
                    *p++ = c;
                    while ( *p && *p != ')' && *p != ',' ) p++;
                    if ( *p == ',' && FG.cTable[p[1]] == 1 ) {
                        p++;
                        while ( *p && *p != ')' && *p != ',' ) p++;
                    }
                    continue;
                }
            }
            *p = c;
```

```
            switch ( type ) {
                case CSYMBOL:
                    *w++ = SYMBOL; *w++ = 4; *w++ = c1;
Sgetnum:            if ( *p != ',' ) {
                        MesCerr("sequence",p);
                        while ( *p && *p != ')' && *p != ',' ) p++;
                        error = 1;
                    }
                    p++; inp = p;
                    ParseSignedNumber(mini,p)
                    if ( FG.cTable[p[-1]] != 1 || ( *p && *p != ')' && *p != ',' ) ) {
                        while ( *p && *p != ')' && *p != ',' ) p++;
                        error = 1;
                        c = *p; *p = 0;
                        MesPrint("&Improper value in count: %s",inp);
                        *p = c;
                        while ( *p && *p != ')' && *p != ',' ) p++;
                    }
                    *w++ = mini;
                    break;
                case CFUNCTION:
                    *w++ = FUNCTION; *w++ = 4; *w++ = c1+FUNCTION; goto Sgetnum;
                case CDOTPRODUCT:
                    *w++ = DOTPRODUCT; *w++ = 5;
                    *w++ = c2 + AM.OffsetVector;
                    *w++ = c1 + AM.OffsetVector;
                    goto Sgetnum;
                case CVECTOR:
                    *w++ = VECTOR; *w++ = 5;
                    *w++ = c1 + AM.OffsetVector;
                    if ( *p == ',' ) {
                        *w++ = VECTBIT | DOTPBIT | FUNBIT;
                        goto Sgetnum;
                    }
                    else if ( *p == '+' ) {
                        p++;
                        *w = 0;
                        while ( *p && *p != ',' ) {
                            if ( *p == 'v' || *p == 'V' ) {
                                *w |= VECTBIT; p++;
                            }
                            else if ( *p == 'd' || *p == 'D' ) {
                                *w |= DOTPBIT; p++;
                            }
                            else if ( *p == 'f' || *p == 'F'
                            || *p == 't' || *p == 'T' ) {
                                *w |= FUNBIT; p++;
                            }
                            else if ( *p == '?' ) {
                                p++; inp = p;
                                if ( *p == '{' ) { /* } */
                                    SKIPBRA2(p)
                                    if ( p == 0 ) return(0);
                                    if ( ( c1 = DoTempSet(inp+1,p) ) < 0 ) return(0);
                                    if ( Sets[c1].type != CFUNCTION ) {
                                        MesPrint("&set type conflict: Function expected");
                                        return(0);
                                    }
                                    type = CSET;
                                    c = *++p;
                                }
                                else {
                                    p = SkipAName(p);
                                    if ( p == 0 ) return(0);
                                    c = *p; *p = 0;
                                    type = GetName(AC.varnames,inp,&c1,WITHAUTO);
                                }
                                if ( type != CSET && type != CDUBIOUS ) {
                                    MesPrint("&%s is not a set",inp);
                                    error = 1;
                                }
                                w[-2] = 6;
                                *w++ |= SETBIT;
                                *w++ = c1;
                                *p = c;
                                goto Sgetnum;
                            }
                            else {
                                MesCerr("specifier for vector",p);
                                error = 1;
                            }
                        }
                        w++;
                        goto Sgetnum;
                    }
                    else {
                        MesCerr("specifier for vector",p);
```

```
                        while ( *p && *p != ')' && *p != ',' ) p++;
                        error = 1;
                        *w++ = VECTBIT | DOTPBIT | FUNBIT;
                        goto Sgetnum;
                    }
                case CDUBIOUS:
                    goto skipfield;
                default:
                    *p = 0;
                    MesPrint("&%s is not a symbol, function, vector or dotproduct",inp);
                    error = 1;
skipfield:          while ( *p && *p != ')' && *p != ',' ) p++;
                    if ( *p && FG.cTable[p[1]] == 1 ) {
                        p++;
                        while ( *p && *p != ')' && *p != ',' ) p++;
                    }
                    break;
            }
        }
        else {
            MesCerr("name",p);
            while ( *p && *p != ',' ) p++;
            error = 1;
        }
    }
    to[1] = w-to;
    if ( *p == ')' ) p++;
    if ( *p ) { MesCerr("end of statement",p); return(0); }
    if ( error ) return(0);
    return(w);
}

/*
    #] CountComp :
    #[ CoIf :

        Reads the if statement: There must be a pair of parentheses.
        Much work is delegated to the routines in compi2 and CountComp.
        The goto is kept hanging as it is forward.
        The address in which the label must be written is pushed on
        the AC.IfStack.

        Here we allow statements of the type
        if ( condition ) single statement;
        compile the if statement.
        test character at end
        if not ; or )
        copy the statement after the proper parenthesis to the
        beginning of the AC.iBuffer.
        Have it compiled.
        generate an endif statement.
*/

static UWORD *CIscratC = 0;

int CoIf(UBYTE *inp)
{
    GETIDENTITY
    int error = 0, level;
    WORD *w, *ww, *u, *s, *OldWork, *OldSpace = AT.WorkSpace;
    WORD gotexp = 0;        /* Indicates whether there can be a condition */
    WORD lenpp, lenlev, ncoef, i, number;
    UBYTE *p, *pp, *ppp, c;
    CBUF *C = cbuf+AC.cbufnum;
    LONG x;
#ifdef WITHFLOAT
    int spec;
#endif
    if ( *inp == '(' && inp[1] == ',' ) inp += 2;
    else if ( *inp == '(' ) inp++;   /* Usually we enter at the bracket */

    if ( CIscratC == 0 )
        CIscratC = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD),"CoIf");
    lenpp = 0;
    lenlev = 1;
    if ( AC.IfLevel >= AC.MaxIf ) DoubleIfBuffers();
    AC.IfCount[lenpp++] = 0;
/*
    IfStack is used for organizing the 'go to' for the various if levels
*/
    *AC.IfStack++ = C->Pointer-C->Buffer+2;
/*
    IfSumCheck is used to test for illegal nesting of if, argument or repeat.
*/
    AC.IfSumCheck[AC.IfLevel] = NestingChecksum();
    AC.IfLevel++;
    w = OldWork = AT.WorkPointer;
```

```
    *w++ = TYPEIF;
    w += 2;
    p = inp;
    for(;;) {
        inp = p;
        level = 0;
ReDo:
        if ( FG.cTable[*p] == 1 ) {     /* Number */
            if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
#ifdef WITHFLOAT
            pp = CheckFloat(p,&spec);
            if ( pp > p ) { /* Got one */
HaveFloat:
                if ( spec == 1 ) { /* is zero */
                    *w++ = LONGNUMBER; *w++ = 3; *w++ = 0;
                }
                else if ( spec == -1 ) {
                    MesPrint("&The floating point system has not been started: %s",p);
                    if ( !error ) error = 1;
                }
                else {
                    WORD *ow = AT.WorkPointer;
                    AT.WorkPointer = w;
                    c = *pp; c = 0;
                    ReadFloat((SBYTE *)p);  /* Is now at AT.WorkPointer */
                    *pp = c;
                    AT.WorkPointer[0] = IFFLOATNUMBER;
                    w = AT.WorkPointer + AT.WorkPointer[1];
                    AT.WorkPointer = ow;
                    if ( level ) w[FUNHEAD+3] = -w[FUNHEAD+3];
                }
                goto DoneWithNumber;
            }
/*
            Notation: Same as FLOATFUN but FLOATFUN replaced by IFFLOATNUMBER.
*/

#endif
            u = w;
            *w++ = LONGNUMBER;
/*
            Notation:
            LONGNUMBER,size,reducedsize*sign,numerator,denominator
            with the length of denominator and numerator equal to reducedsize
*/
            w += 2;
            if ( GetLong(p,(UWORD *)w,&ncoef) ) { ncoef = 1; error = 1; }
            w[-1] = ncoef;
            while ( FG.cTable[*++p] == 1 );
            if ( *p == '/' ) {
                p++;
                if ( FG.cTable[*p] != 1 ) {
                    MesCerr("sequence",p); error = 1; goto OnlyNum;
                }
                if ( GetLong(p,CIscratC,&ncoef) ) {
                    ncoef = 1; error = 1;
                }
                while ( FG.cTable[*++p] == 1 );
                if ( ncoef == 0 ) {
                    MesPrint("&Division by zero!");
                    error = 1;
                }
                else {
                    if ( w[-1] != 0 ) {
                        if ( Simplify(BHEAD (UWORD *)w,(WORD *)(w-1),
                        CIscratC,&ncoef) ) error = 1;
                        else {
                            i = w[-1];
                            if ( i >= ncoef ) {
                                i = w[-1];
                                w += i;
                                i -= ncoef;
                                s = (WORD *)CIscratC;
                                NCOPY(w,s,ncoef);
                                while ( --i >= 0 ) *w++ = 0;
                            }
                            else {
                                w += i;
                                i = ncoef - i;
                                while ( --i >= 0 ) *w++ = 0;
                                s = (WORD *)CIscratC;
                                NCOPY(w,s,ncoef);
                            }
                        }
                    }
                }
            }
```

```
            else {
OnlyNum:
                w += ncoef;
                if ( ncoef > 0 ) {
                    ncoef--; *w++ = 1;
                    while ( --ncoef >= 0 ) *w++ = 0;
                }
            }
            u[1] = WORDDIF(w,u);
            u[2] = (u[1] - 3)/2;
            if ( level ) u[2] = -u[2];
#ifdef WITHFLOAT
DoneWithNumber:
#endif
            gotexp = 1;
        }
        else if ( *p == '+' ) { p++; goto ReDo; }
        else if ( *p == '-' ) { level ^= 1; p++; goto ReDo; }
        else if ( *p == 'c' || *p == 'C' ) {    /* Count or Coefficient */
            if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
            while ( FG.cTable[*++p] == 0 );
            c = *p; *p = 0;
            if ( !StrICmp(inp,(UBYTE *)"count") ) {
                *p = c;
                if ( c != '(' ) {
                    MesPrint("&no ( after count");
                    error = 1;
                    goto endofif;
                }
                inp = p;
                SKIPBRA4(p);
                c = *++p; *p = 0; *inp = ',';
                w = CountComp(inp,w);
                *p = c; *inp = '(';
                if ( w == 0 ) { error = 1; goto endofif; }
                gotexp = 1;
            }
            else if ( ConWord(inp,(UBYTE *)"coefficient") && ( p - inp ) > 3 ) {
                *w++ = COEFFI;
                *w++ = 2;
                *p = c;
                gotexp = 1;
            }
            else goto NoGood;
            inp = p;
        }
        else if ( *p == 'm' || *p == 'M' ) {    /* match */
            if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
            while ( !FG.cTable[*++p] );
            c = *p; *p = 0;
            if ( !StrICmp(inp,(UBYTE *)"match") ) {
                *p = c;
                if ( c != '(' ) {
                    MesPrint("&no ( after match");
                    error = 1;
                    goto endofif;
                }
                p++; inp = p;
                SKIPBRA4(p);
                *p = '=';
/*
                Now we can call the reading of the lhs of an id statement.
                This has to be modified in the future.
*/
                AT.WorkSpace = AT.WorkPointer = w;
                ppp = inp;
                while ( FG.cTable[*ppp] == 0 && ppp < p ) ppp++;
                if ( *ppp == ',' ) AC.idoption = 0;
                else AC.idoption = SUBMULTI;
                level = CoIdExpression(inp,TYPEIF);
                AT.WorkSpace = OldSpace;
                AT.WorkPointer = OldWork;
                if ( level != 0 ) {
                    if ( level < 0 ) { error = -1; goto endofif; }
                    error = 1;
                }
/*
                If we pop numlhs we are in good shape
*/
                s = u = C->lhs[C->numlhs];
                while ( u < C->Pointer ) *w++ = *u++;
                C->numlhs--; C->Pointer = s;
                *p++ = ')';
                inp = p;
                gotexp = 1;
            }
            else if ( !StrICmp(inp,(UBYTE *)"multipleof") ) {
```

```
                if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
                *p = c;
                if ( c != '(' ) {
                    MesPrint("&no ( after multipleof");
                    error = 1; goto endofif;
                }
                p++;
                if ( FG.cTable[*p] != 1 ) {
Nomulof:            MesPrint("&multipleof needs a short positive integer argument");
                    error = 1; goto endofif;
                }
                ParseNumber(x,p)
                if ( *p != ')' || x <= 0 || x > MAXPOSITIVE ) goto Nomulof;
                p++;
                *w++ = MULTIPLEOF; *w++ = 3; *w++ = (WORD)x;
                inp = p;
                gotexp = 1;
            }
            else {
NoGood:         MesPrint("&Unrecognized word: %s",inp);
                *p = c;
                error = 1;
                level = 0;
                if ( c == '(' ) SKIPBRA4(p)
                inp = ++p;
                gotexp = 1;
            }
        }
        else if ( *p == 'f' || *p == 'F' ) {    /* FindLoop */
            if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
            while ( FG.cTable[*++p] == 0 );
            c = *p; *p = 0;
            if ( !StrICmp(inp,(UBYTE *)"findloop") ) {
                *p = c;
                if ( c != '(' ) {
                    MesPrint("&no ( after findloop");
                    error = 1;
                    goto endofif;
                }
                inp = p;
                SKIPBRA4(p);
                c = *++p; *p = 0; *inp = ',';
                if ( CoFindLoop(inp) ) { error = 1; goto endofif; }
                s = u = C->lhs[C->numlhs];
                while ( u < C->Pointer ) *w++ = *u++;
                C->numlhs--; C->Pointer = s;
                *p = c; *inp = '(';
                if ( w == 0 ) { error = 1; goto endofif; }
                gotexp = 1;
            }
            else if ( !StrICmp(inp,(UBYTE *)"flag") ) {
                UBYTE cc = c, *pppp;
                *p = cc;
                if ( cc != '(' ) {
                    MesPrint("&no ( after flag");
                    error = 1;
                    goto endofif;
                }
                inp = p;
                SKIPBRA4(p);
                cc = *++p; *p = 0; *inp = ','; pppp = p;
                ww = w;
                *w++ = IFUSERFLAG; *w++ = 0;
                while ( *inp ) {
                    int x = 0;
                    while ( *inp == ',' ) inp++;
                    if ( *inp == 0 || *inp == ')' ) break;
                    while ( *inp >= '0' && *inp <= '9' ) x = 10*x+(*inp++-'0');
                    if ( x < 1 || x > BITSINWORD ) {
                        MesPrint("&Flag number %d outside the permitted range 1-%d.",BITSINWORD);
                        error = 1;
                    }
                    *w++ = x-1;
                }
                ww[1] = w-ww;
                p = pppp; *p = cc; *inp = '(';
                gotexp = 1;
                if ( ww[1] <= 2 ) {
                    MesPrint("&The userflag condition in the if statement needs arguments.");
                    error = 1;
                }
                inp = p;
                gotexp = 1;
            }
            else goto NoGood;
        }
        else if ( *p == 'e' || *p == 'E' ) { /* Expression */
```

```
            if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
            while ( FG.cTable[*++p] == 0 );
            c = *p; *p = 0;
            if ( !StrICmp(inp,(UBYTE *)"expression") ) {
                *p = c;
                if ( c != '(' ) {
                    MesPrint("&no ( after expression");
                    error = 1;
                    goto endofif;
                }
                p++; ww = w; *w++ = IFEXPRESSION; w++;
                while ( *p != ')' ) {
                    if ( *p == ',' ) { p++; continue; }
                    if ( *p == '[' || FG.cTable[*p] == 0 ) {
                        pp = p;
                        if ( ( p = SkipAName(p) ) == 0 ) {
                            MesPrint("&Improper name for an expression: '%s'",pp);
                            error = 1;
                            goto endofif;
                        }
                        c = *p; *p = 0;
                        if ( GetName(AC.exprnames,pp,&number,NOAUTO) == CEXPRESSION ) {
                            *w++ = number;
                        }
                        else if ( GetName(AC.varnames,pp,&number,NOAUTO) != NAMENOTFOUND ) {
                            MesPrint("&%s is not an expression",pp);
                            error = 1;
                            *w++ = number;
                        }
                        *p = c;
                    }
                    else {
                        MesPrint("&Illegal object in Expression in if-statement");
                        error = 1;
                        while ( *p && *p != ',' && *p != ')' ) p++;
                        if ( *p == 0 || *p == ')' ) break;
                    }
                }
                ww[1] = w - ww;
                p++;
                gotexp = 1;
            }
            else goto NoGood;
            inp = p;
        }
        else if ( *p == 'i' || *p == 'I' ) { /* IsFactorized */
            if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
            while ( FG.cTable[*++p] == 0 );
            c = *p; *p = 0;
            if ( !StrICmp(inp,(UBYTE *)"isfactorized") ) {
                *p = c;
                if ( c != '(' ) { /* No expression means current expression */
                    ww = w; *w++ = IFISFACTORIZED; w++;
                }
                else {
                  p++; ww = w; *w++ = IFISFACTORIZED; w++;
                  while ( *p != ')' ) {
                    if ( *p == ',' ) { p++; continue; }
                    if ( *p == '[' || FG.cTable[*p] == 0 ) {
                        pp = p;
                        if ( ( p = SkipAName(p) ) == 0 ) {
                            MesPrint("&Improper name for an expression: '%s'",pp);
                            error = 1;
                            goto endofif;
                        }
                        c = *p; *p = 0;
                        if ( GetName(AC.exprnames,pp,&number,NOAUTO) == CEXPRESSION ) {
                            *w++ = number;
                        }
                        else if ( GetName(AC.varnames,pp,&number,NOAUTO) != NAMENOTFOUND ) {
                            MesPrint("&%s is not an expression",pp);
                            error = 1;
                            *w++ = number;
                        }
                        *p = c;
                    }
                    else {
                        MesPrint("&Illegal object in IsFactorized in if-statement");
                        error = 1;
                        while ( *p && *p != ',' && *p != ')' ) p++;
                        if ( *p == 0 || *p == ')' ) break;
                    }
                  }
                  p++;
                }
                ww[1] = w - ww;
                gotexp = 1;
```

```
            }
            else goto NoGood;
            inp = p;
        }
        else if ( *p == 'o' || *p == 'O' ) { /* Occurs */
/*
            Tests whether variables occur inside a term.
            At the moment this is done one by one.
            If we want to do them in groups we should do the reading
            a bit different: each as a variable in a term, and then
            use Normalize to get the variables grouped and in order.
            That way FindVar (in if.c) can work more efficiently.
            Still to be done!!!
            TASK: Nice little task for someone to learn.
*/
            UBYTE cc;
            if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
            while ( FG.cTable[*++p] == 0 );
            c = cc = *p; *p = 0;
            if ( !StrICmp(inp,(UBYTE *)"occurs") ) {
                WORD c1, c2, type;
                *p = cc;
                if ( cc != '(' ) {
                    MesPrint("&no ( after occurs");
                    error = 1;
                    goto endofif;
                }
                inp = p;
                SKIPBRA4(p);
                cc = *++p; *p = 0; *inp = ','; pp = p;
                ww = w;
                *w++ = IFOCCURS; *w++ = 0;
                while ( *inp ) {
                    while ( *inp == ',' ) inp++;
                    if ( *inp == 0 || *inp == ')' ) break;
/*
                    Now read a list of names
                    We can have symbols, vectors, dotproducts, indices, functions.
                    There could also be dummy indices and/or extra symbols.
*/
                    if ( *inp == '[' || FG.cTable[*inp] == 0 ) {
                        if ( ( p = SkipAName(inp) ) == 0 ) return(0);
                        c = *p; *p = 0;
                        type = GetName(AC.varnames,inp,&c1,WITHAUTO);
                        if ( c == '.' ) {
                            if ( type == CVECTOR || type == CDUBIOUS ) {
                                *p++ = c;
                                inp = p;
                                p = SkipAName(p);
                                if ( p == 0 ) return(0);
                                c = *p;
                                *p = 0;
                                type = GetName(AC.varnames,inp,&c2,WITHAUTO);
                                if ( type != CVECTOR && type != CDUBIOUS ) {
                                    MesPrint("&Not a vector in dotproduct in if statement: %s",inp);
                                    error = 1;
                                }
                                else type = CDOTPRODUCT;
                            }
                            else {
                                MesPrint("&Illegal use of . after %s in if statement",inp);
                                if ( type == NAMENOTFOUND )
                                    MesPrint("&%s is not a properly declared variable",inp);
                                error = 1;
                                *p++ = c;
                                while ( *p && *p != ')' && *p != ',' ) p++;
                                if ( *p == ',' && FG.cTable[p[1]] == 1 ) {
                                    p++;
                                    while ( *p && *p != ')' && *p != ',' ) p++;
                                }
                                continue;
                            }
                        }
                        *p = c;
                        switch ( type ) {
                            case CSYMBOL: /* To worry about extra symbols */
                                *w++ = SYMBOL;
                                *w++ = c1;
                            break;
                            case CINDEX:
                                *w++ = INDEX;
                                *w++ = c1 + AM.OffsetIndex;
                            break;
                            case CVECTOR:
                                *w++ = VECTOR;
                                *w++ = c1 + AM.OffsetVector;
                            break;
```

```
                        case CDOTPRODUCT:
                            *w++ = DOTPRODUCT;
                            *w++ = c1 + AM.OffsetVector;
                            *w++ = c2 + AM.OffsetVector;
                        break;
                        case CFUNCTION:
                            *w++ = FUNCTION;
                            *w++ = c1+FUNCTION;
                        break;
                        default:
                            MesPrint("&Illegal variable %s in occurs condition in if
statement",inp);
                            error = 1;
                        break;
                    }
                    inp = p;
                }
                else {
                    MesPrint("&Illegal object %s in occurs condition in if statement",inp);
                    error = 1;
                    break;
                }
            }
            ww[1] = w-ww;
            p = pp; *p = cc; *inp = '(';
            gotexp = 1;
            if ( ww[1] <= 2 ) {
                MesPrint("&The occurs condition in the if statement needs arguments.");
                error = 1;
            }
        }
        else goto NoGood;
        inp = p;
    }
    else if ( *p == '$' ) {
        if ( gotexp == 1 ) { MesCerr("position for )",p); error = 1; }
        p++; inp = p;
        while ( FG.cTable[*p] == 0 || FG.cTable[*p] == 1 ) p++;
        c = *p; *p = 0;
        if ( ( i = GetDollar(inp) ) < 0 ) {
            MesPrint("&undefined dollar expression %s",inp);
            error = 1;
            i = AddDollar(inp,DOLUNDEFINED,0,0);
        }
        *p = c;
        *w++ = IFDOLLAR; *w++ = 3; *w++ = i;
/*
        And then the IFDOLLAREXTRA pieces for [1] [$y] etc
*/
        if ( *p == '[' ) {
            p++;
            if ( ( w = GetIfDollarFactor(&p,w) ) == 0 ) {
                error = 1;
                goto endofif;
            }
            else if ( *p != ']' ) {
                error = 1;
                goto endofif;
            }
            p++;
        }
        inp = p;
        gotexp = 1;
    }
#ifdef WITHFLOAT
    else if ( *p == '.' ) {
        pp = CheckFloat(p,&spec);
        if ( pp > p ) goto HaveFloat;
    }
#endif
    else if ( *p == '(' ) {
        if ( gotexp ) {
            MesCerr("parenthesis",p);
            error = 1;
            goto endofif;
        }
        gotexp = 0;
        if ( ++lenlev >= AC.MaxIf ) DoubleIfBuffers();
        AC.IfCount[lenpp++] = w-OldWork;
        *w++ = SUBEXPR;
        w += 2;
        p++;
    }
    else if ( *p == ')' ) {
        if ( gotexp == 0 ) { MesCerr("position for )",p); error = 1; }
        gotexp = 1;
        u = AC.IfCount[--lenpp]+OldWork;
```

```
            lenlev--;
            u[1] = w - u;
            if ( lenlev <= 0 ) {     /* End if condition */
                AT.WorkSpace = OldSpace;
                AT.WorkPointer = OldWork;
                AddNtoL(OldWork[1],OldWork);
                p++;
                if ( *p == ')' ) {
                    MesPrint("&unmatched parenthesis in if/while ()");
                    error = 1;
                    while ( *++p == ')' );
                }
                if ( *p ) {
                    level = CompileStatement(p);
                    if ( level ) error = level;
                    while ( *p ) p++;
                    if ( CoEndIf(p) && error == 0 ) error = 1;
                }
                return(error);
            }
            p++;
        }
        else if ( *p == '>' ) {
            if ( gotexp == 0 ) goto NoExp;
            if ( p[1] == '=' ) { *w++ = GREATEREQUAL; *w++ = 2; p += 2; }
            else               { *w++ = GREATER;      *w++ = 2; p++; }
            gotexp = 0;
        }
        else if ( *p == '<' ) {
            if ( gotexp == 0 ) goto NoExp;
            if ( p[1] == '=' ) { *w++ = LESSEQUAL; *w++ = 2; p += 2; }
            else               { *w++ = LESS;      *w++ = 2; p++; }
            gotexp = 0;
        }
        else if ( *p == '=' ) {
            if ( gotexp == 0 ) goto NoExp;
            if ( p[1] == '=' ) p++;
            *w++ = EQUAL; *w++ = 2; p++;
            gotexp = 0;
        }
        else if ( *p == '!' && p[1] == '=' ) {
            if ( gotexp == 0 ) { p++; goto NoExp; }
            *w++ = NOTEQUAL; *w++ = 2; p += 2;
            gotexp = 0;
        }
        else if ( *p == '|' && p[1] == '|' ) {
            if ( gotexp == 0 ) { p++; goto NoExp; }
            *w++ = ORCOND; *w++ = 2; p += 2;
            gotexp = 0;
        }
        else if ( *p == '&' && p[1] == '&' ) {
            if ( gotexp == 0 ) {
                p++;
NoExp:          p++;
                MesCerr("sequence",p);
                error = 1;
            }
            else {
                *w++ = ANDCOND; *w++ = 2; p += 2;
                gotexp = 0;
            }
        }
        else if ( *p == 0 ) {
            MesPrint("&Unmatched parentheses");
            error = 1;
            goto endofif;
        }
        else {
            if ( FG.cTable[*p] == 0 ) {
                WORD ij;
                inp = p;
                while ( ( ij = FG.cTable[*++p] ) == 0 || ij == 1 );
                c = *p; *p = 0;
                goto NoGood;
            }
            MesCerr("sequence",p);
            error = 1;
            p++;
        }
    }
endofif:;
    return(error);
}

/*
    #] CoIf :
    #[ CoElse :
```

```
*/

int CoElse(UBYTE *p)
{
    int error = 0;
    CBUF *C = cbuf+AC.cbufnum;
    if ( *p != 0 ) {
        while ( *p == ',' ) p++;
        if ( tolower(*p) == 'i' && tolower(p[1]) == 'f' && p[2] == '(' )
                                                    return(CoElseIf(p+2));
        MesPrint("&No extra text allowed as part of an else statement");
        error = 1;
    }
    if ( AC.IfLevel <= 0 ) { MesPrint("&else statement without if"); return(1); }
    if ( AC.IfSumCheck[AC.IfLevel-1] != NestingChecksum() - 1 ) {
        MesNesting();
        error = 1;
    }
    Add3Com(TYPEELSE,AC.IfLevel)
    C->Buffer[AC.IfStack[-1]] = C->numlhs;
    AC.IfStack[-1] = C->Pointer - C->Buffer - 1;
    return(error);
}

/*
    #] CoElse :
    #[ CoElseIf :
*/

int CoElseIf(UBYTE *inp)
{
    CBUF *C = cbuf+AC.cbufnum;
    if ( AC.IfLevel <= 0 ) { MesPrint("&elseif statement without if"); return(1); }
    Add3Com(TYPEELSE,-AC.IfLevel)
    AC.IfLevel--;
    C->Buffer[*--AC.IfStack] = C->numlhs;
    return(CoIf(inp));
}

/*
    #] CoElseIf :
    #[ CoEndIf :

        It puts a RHS-level at the position indicated in the AC.IfStack.
        This corresponds to the label belonging to a forward goto.
        It is the goto that belongs either to the failing condition
        of the if (no else statement), or the completion of the
        success path (with else statement)
        The code is a jump to the next statement. It is there to prevent
        problems with
        if ( .. )
            if ( .. )
            endif;
        elseif ( .. )
*/

int CoEndIf(UBYTE *inp)
{
    CBUF *C = cbuf+AC.cbufnum;
    WORD i = C->numlhs, to, k = -AC.IfLevel;
    int error = 0;
    while ( *inp == ',' ) inp++;
    if ( *inp != 0 ) {
        error = 1;
        MesPrint("&No extra text allowed as part of an endif/elseif statement");
    }
    if ( AC.IfLevel <= 0 ) {
        MesPrint("&Endif statement without corresponding if"); return(1);
    }
    AC.IfLevel--;
    C->Buffer[*--AC.IfStack] = i+1;
    if ( AC.IfSumCheck[AC.IfLevel] != NestingChecksum() ) {
        MesNesting();
        error = 1;
    }
    Add3Com(TYPEENDIF,i+1)
/*
    Now the search for the TYPEELSE in front of the elseif statements
*/
    to = C->numlhs;
    while ( i > 0 ) {
        if ( C->lhs[i][0] == TYPEELSE && C->lhs[i][2] == to ) to = i;
        if ( C->lhs[i][0] == TYPEIF ) {
            if ( C->lhs[i][2] == to ) {
                i--;
                if ( i <= 0 || C->lhs[i][0] != TYPEELSE
                || C->lhs[i][2] != k ) break;
```

```
                C->lhs[i][2] = C->numlhs;
                to = i;
            }
        }
        i--;
    }
    return(error);
}

/*
    #] CoEndIf :
    #[ CoWhile :
*/

int CoWhile(UBYTE *inp)
{
    CBUF *C = cbuf+AC.cbufnum;
    WORD startnum = C->numlhs + 1;
    int error;
    AC.WhileLevel++;
    error = CoIf(inp);
    if ( C->numlhs > startnum && C->lhs[startnum][2] == C->numlhs
                            && C->lhs[C->numlhs][0] == TYPEENDIF ) {
        C->lhs[C->numlhs][2] = startnum-1;
        AC.WhileLevel--;
    }
    else C->lhs[startnum][2] = startnum;
    return(error);
}

/*
    #] CoWhile :
    #[ CoEndWhile :
*/

int CoEndWhile(UBYTE *inp)
{
    int error = 0;
    WORD i;
    CBUF *C = cbuf+AC.cbufnum;
    if ( AC.WhileLevel <= 0 ) {
        MesPrint("&EndWhile statement without corresponding While"); return(1);
    }
    AC.WhileLevel--;
    i = C->Buffer[AC.IfStack[-1]];
    error = CoEndIf(inp);
    C->lhs[C->numlhs][2] = i - 1;
    return(error);
}

/*
    #] CoEndWhile :
    #[ DoFindLoop :

    Function,arguments=number,loopsize=number,outfun=function,include=index;
*/

static char *messfind[] = {
    "Findloop(function,arguments=#,loopsize(=#|<#)[,include=index])"
   ,"Replaceloop,function,arguments=#,loopsize(=#|<#),outfun=function[,include=index]"
    };
static WORD comfindloop[7] = { TYPEFINDLOOP,7,0,0,0,0,0 };

int DoFindLoop(UBYTE *inp, int mode)
{
    UBYTE *s, c;
    WORD funnum, nargs = 0, nloop = 0, indexnum = 0, outfun = 0;
    int type, aflag, lflag, indflag, outflag, error = 0, sym;
    while ( *inp == ',' ) inp++;
    if ( ( s = SkipAName(inp) ) == 0 ) {
syntax:
        MesPrint("&Proper syntax is:");
        MesPrint("%s",messfind[mode]);
        return(1);
    }
    c = *s; *s = 0;
    if ( ( ( type = GetName(AC.varnames,inp,&funnum,WITHAUTO) ) == NAMENOTFOUND )
        || type != CFUNCTION || ( ( sym = (functions[funnum].symmetric) & ~REVERSEORDER )
        != SYMMETRIC && sym != ANTISYMMETRIC ) ) {
        MesPrint("&%s should be a (anti)symmetric function or tensor",inp);
        error = 1;
    }
    funnum += FUNCTION;
    *s = c; inp = s;
    aflag = lflag = indflag = outflag = 0;
    while ( *inp == ',' ) {
        while ( *inp == ',' ) inp++;
```

```
        s = inp;
        if ( ( s = SkipAName(inp) ) == 0 ) goto syntax;
        c = *s; *s = 0;
        if ( StrICont(inp,(UBYTE *)"arguments") == 0 ) {
            if ( c != '=' ) goto syntax;
            *s++ = c;
            NeedNumber(nargs,s,syntax)
            aflag++;
            inp = s;
        }
        else if ( StrICont(inp,(UBYTE *)"loopsize") == 0 ) {
            if ( c != '=' && c != '<' ) goto syntax;
            *s++ = c;
            if ( FG.cTable[*s] == 1 ) {
                NeedNumber(nloop,s,syntax)
                if ( nloop < 2 ) {
                    MesPrint("&loopsize should be at least 2");
                    error = 1;
                }
                if ( c == '<' ) nloop = -nloop;
            }
            else if ( tolower(*s) == 'a' && tolower(s[1]) == 'l'
            && tolower(s[2]) == 'l' && FG.cTable[s[3]] > 1 ) {
                nloop = -1; s += 3;
                if ( c != '=' ) goto syntax;
            }
            inp = s;
            lflag++;
        }
        else if ( StrICont(inp,(UBYTE *)"include") == 0 ) {
            if ( c != '=' ) goto syntax;
            *s++ = c;
            if ( ( inp = SkipAName(s) ) == 0 ) goto syntax;
            c = *inp; *inp = 0;
            if ( ( type = GetName(AC.varnames,s,&indexnum,WITHAUTO) ) != CINDEX ) {
                MesPrint("&%s is not a proper index",s);
                error = 1;
            }
            else if ( indexnum < WILDOFFSET
            && indices[indexnum].dimension == 0 ) {
                MesPrint("&%s should be a summable index",s);
                error = 1;
            }
            indexnum += AM.OffsetIndex;
            *inp = c;
            indflag++;
        }
        else if ( StrICont(inp,(UBYTE *)"outfun") == 0 ) {
            if ( c != '=' ) goto syntax;
            *s++ = c;
            if ( ( inp = SkipAName(s) ) == 0 ) goto syntax;
            c = *inp; *inp = 0;
            if ( ( type = GetName(AC.varnames,s,&outfun,WITHAUTO) ) != CFUNCTION ) {
                MesPrint("&%s is not a proper function or tensor",s);
                error = 1;
            }
            outfun += FUNCTION;
            outflag++;
            *inp = c;
        }
        else {
            MesPrint("&Unrecognized option in FindLoop or ReplaceLoop: %s",inp);
            error = 1;
            *s = c; inp = s;
            while ( *inp && *inp != ',' ) inp++;
        }
    }
    if ( *inp != 0 && mode == REPLACELOOP ) goto syntax;
    if ( mode == FINDLOOP && outflag > 0 ) {
        MesPrint("&outflag option is illegal in FindLoop");
        error = 1;
    }
    if ( mode == REPLACELOOP && outflag == 0 ) goto syntax;
    if ( aflag == 0 || lflag == 0 ) goto syntax;
    comfindloop[3] = funnum;
    comfindloop[4] = nloop;
    comfindloop[5] = nargs;
    comfindloop[6] = outfun;
    comfindloop[1] = 7;
    if ( indflag ) {
        if ( mode == 0 ) comfindloop[2] =  indexnum + 5;
        else             comfindloop[2] = -indexnum - 5;
    }
    else comfindloop[2] = mode;
    AddNtoL(comfindloop[1],comfindloop);
    return(error);
}
```

```
/*
    #] DoFindLoop :
    #[ CoFindLoop :
*/

int CoFindLoop(UBYTE *inp)
{ return(DoFindLoop(inp,FINDLOOP)); }

/*
    #] CoFindLoop :
    #[ CoReplaceLoop :
*/

int CoReplaceLoop(UBYTE *inp)
{
    int error = DoFindLoop(inp,REPLACELOOP);
    if ( error ) {
        Terminate(-1);
    }
    return(error);
}

/*
    #] CoReplaceLoop :
    #[ CoFunPowers :
*/

static UBYTE *FunPowOptions[] = {
     (UBYTE *)"nofunpowers"
    ,(UBYTE *)"commutingonly"
    ,(UBYTE *)"allfunpowers"
    };

int CoFunPowers(UBYTE *inp)
{
    UBYTE *option, c;
    int i, maxoptions = sizeof(FunPowOptions)/sizeof(UBYTE *);
    while ( *inp == ',' ) inp++;
    option = inp;
    inp = SkipAName(inp); c = *inp; *inp = 0;
    for ( i = 0; i < maxoptions; i++ ) {
        if ( StrICont(option,FunPowOptions[i]) == 0 ) {
            if ( c ) {
                *inp = c;
                MesPrint("&Illegal FunPowers statement");
                return(1);
            }
            AC.funpowers = i;
            return(0);
        }
    }
    MesPrint("&Illegal option in FunPowers statement: %s",option);
    return(1);
}

/*
    #] CoFunPowers :
    #[ CoUnitTrace :
*/

int CoUnitTrace(UBYTE *s)
{
    WORD num;
    if ( FG.cTable[*s] == 1 ) {
        ParseNumber(num,s)
        if ( *s != 0 ) {
nogood:     MesPrint("&Value of UnitTrace should be a (positive) number or a symbol");
            return(1);
        }
        AC.lUniTrace[0] = SNUMBER;
        AC.lUniTrace[2] = num;
    }
    else {
        if ( GetName(AC.varnames,s,&num,WITHAUTO) == CSYMBOL ) {
            AC.lUniTrace[0] = SYMBOL;
            AC.lUniTrace[2] = num;
            num = -num;
        }
        else goto nogood;
        s = SkipAName(s);
        if ( *s ) goto nogood;
    }
    AC.lUnitTrace = num;
    return(0);
}

```

```
/*
    #] CoUnitTrace :
    #[ CoTerm :

    Note: termstack holds the offset of the term statement in the compiler
    buffer. termsortstack holds the offset of the last sort statement
        (or the corresponding term statement)
*/

int CoTerm(UBYTE *s)
{
    GETIDENTITY
    WORD *w = AT.WorkPointer;
    int error = 0;
    while ( *s == ',' ) s++;
    if ( *s ) {
        MesPrint("&Illegal syntax for Term statement");
        return(1);
    }
    if ( AC.termlevel+1 >= AC.maxtermlevel ) {
        if ( AC.maxtermlevel <= 0 ) {
            AC.maxtermlevel = 20;
            AC.termstack = (LONG *)Malloc1(AC.maxtermlevel*sizeof(LONG),"termstack");
            AC.termsortstack = (LONG *)Malloc1(AC.maxtermlevel*sizeof(LONG),"termsortstack");
            AC.termsumcheck = (WORD *)Malloc1(AC.maxtermlevel*sizeof(WORD),"termsumcheck");
        }
        else {
            DoubleBuffer((void **)AC.termstack,(void **)AC.termstack+AC.maxtermlevel,
                sizeof(LONG),"doubling termstack");
            DoubleBuffer((void **)AC.termsortstack,
                (void **)AC.termsortstack+AC.maxtermlevel,
                sizeof(LONG),"doubling termsortstack");
            DoubleBuffer((void **)AC.termsumcheck,
                (void **)AC.termsumcheck+AC.maxtermlevel,
                sizeof(LONG),"doubling termsumcheck");
            AC.maxtermlevel *= 2;
        }
    }
    AC.termsumcheck[AC.termlevel] = NestingChecksum();
    AC.termstack[AC.termlevel] = cbuf[AC.cbufnum].Pointer
                                 - cbuf[AC.cbufnum].Buffer + 2;
    AC.termsortstack[AC.termlevel] = AC.termstack[AC.termlevel] + 1;
    AC.termlevel++;
    *w++ = TYPETERM;
    w++;
    *w++ = cbuf[AC.cbufnum].numlhs;
    *w++ = cbuf[AC.cbufnum].numlhs;
    AT.WorkPointer[1] = w - AT.WorkPointer;
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    return(error);
}

/*
    #] CoTerm :
    #[ CoEndTerm :
*/

int CoEndTerm(UBYTE *s)
{
    CBUF *C = cbuf+AC.cbufnum;
    while ( *s == ',' ) s++;
    if ( *s ) {
        MesPrint("&Illegal syntax for EndTerm statement");
        return(1);
    }
    if ( AC.termlevel <= 0 ) {
        MesPrint("&EndTerm without corresponding Argument statement");
        return(1);
    }
    AC.termlevel--;
    cbuf[AC.cbufnum].Buffer[AC.termstack[AC.termlevel]] = C->numlhs;
    cbuf[AC.cbufnum].Buffer[AC.termsortstack[AC.termlevel]] = C->numlhs;
    if ( AC.termsumcheck[AC.termlevel] != NestingChecksum() ) {
        MesNesting();
        return(1);
    }
    return(0);
}

/*
    #] CoEndTerm :
    #[ CoSort :
*/

int CoSort(UBYTE *s)
{
    GETIDENTITY
```

```
    WORD *w = AT.WorkPointer;
    int error = 0;
    while ( *s == ',' ) s++;
    if ( *s ) {
        MesPrint("&Illegal syntax for Sort statement");
        error = 1;
    }
    if ( AC.termlevel <= 0 ) {
        MesPrint("&The Sort statement can only be used inside a term environment");
        error = 1;
    }
    if ( error ) return(error);
    *w++ = TYPESORT;
    w++;
    w++;
    cbuf[AC.cbufnum].Buffer[AC.termsortstack[AC.termlevel-1]] =
                                    *w = cbuf[AC.cbufnum].numlhs+1;
    w++;
    AC.termsortstack[AC.termlevel-1] = cbuf[AC.cbufnum].Pointer
                         - cbuf[AC.cbufnum].Buffer + 3;
    if ( AC.termsumcheck[AC.termlevel-1] != NestingChecksum() - 1 ) {
        MesNesting();
        return(1);
    }
    AT.WorkPointer[1] = w - AT.WorkPointer;
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    return(error);
}

/*
    #] CoSort :
    #[ CoPolyFun :

    Collect,functionname
*/

int CoPolyFun(UBYTE *s)
{
    GETIDENTITY
    WORD numfun;
    int type, error = 0;
    UBYTE *t;
    AR.PolyFun = AC.lPolyFun = 0;
    AR.PolyFunInv = AC.lPolyFunInv = 0;
    AR.PolyFunType = AC.lPolyFunType = 0;
    AR.PolyFunExp = AC.lPolyFunExp = 0;
    AR.PolyFunVar = AC.lPolyFunVar = 0;
    AR.PolyFunPow = AC.lPolyFunPow = 0;
    if ( *s == 0 ) { return(0); }
    t = SkipAName(s);
    if ( t == 0 || *t != 0 ) {
        MesPrint("&PolyFun statement needs a single commuting function for its argument");
        return(1);
    }
    if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
    || ( functions[numfun].spec != 0 ) || ( functions[numfun].commute != 0 ) ) {
        MesPrint("&%s should be a regular commuting function",s);
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        error = 1;
    }
    else {
        AR.PolyFun = AC.lPolyFun = numfun+FUNCTION;
        AR.PolyFunType = AC.lPolyFunType = 1;
    }
#ifdef WITHFLOAT
    if ( mpfaux_ != 0 ) {
        MesPrint("&Simultaneous use of PolyFun and float_ is not allowed.");
        error = 1;
    }
#endif
    return(error);
}

/*
    #] CoPolyFun :
    #[ CoPolyRatFun :

    PolyRatFun [,functionname[,functionname](option)]
*/

int CoPolyRatFun(UBYTE *s)
{
    GETIDENTITY
    WORD numfun;
```

```
	int type, error = 0;
	UBYTE *t, c;
	AR.PolyFun = AC.lPolyFun = 0;
	AR.PolyFunInv = AC.lPolyFunInv = 0;
	AR.PolyFunType = AC.lPolyFunType = 0;
	AR.PolyFunExp = AC.lPolyFunExp = 0;
	AR.PolyFunVar = AC.lPolyFunVar = 0;
	AR.PolyFunPow = AC.lPolyFunPow = 0;
	if ( *s == 0 ) return(error);
	t = SkipAName(s);
	if ( t == 0 ) goto NumErr;
	c = *t; *t = 0;
#ifdef WITHFLOAT
	if ( mpfaux_ != 0 ) {
		MesPrint("&Simultaneous use of PolyFun and float_ is not allowed.");
		error = 1;
	}
#endif
	if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
	|| ( functions[numfun].spec != 0 ) || ( functions[numfun].commute != 0 ) ) {
		MesPrint("&%s should be a regular commuting function",s);
		if ( type < 0 ) {
			if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
				AddFunction(s,0,0,0,0,0,-1,-1);
		}
		return(1);
	}
	AR.PolyFun = AC.lPolyFun = numfun+FUNCTION;
	AR.PolyFunInv = AC.lPolyFunInv = 0;
	AR.PolyFunType = AC.lPolyFunType = 2;
	AC.PolyRatFunChanged = 1;
	if ( c == 0 ) return(error);
	*t = c;
	if ( *t == '-' ) { AC.PolyRatFunChanged = 0; t++; }
	while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
	if ( *t == 0 ) return(error);
	if ( *t != '(' ) {
		s = t;
		t = SkipAName(s);
		if ( t == 0 ) goto NumErr;
		c = *t; *t = 0;
		if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
		|| ( functions[numfun].spec != 0 ) || ( functions[numfun].commute != 0 ) ) {
			MesPrint("&%s should be a regular commuting function",s);
			if ( type < 0 ) {
				if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
					AddFunction(s,0,0,0,0,0,-1,-1);
			}
			return(1);
		}
		AR.PolyFunInv = AC.lPolyFunInv = numfun+FUNCTION;
		if ( c == 0 ) return(error);
		*t = c;
		if ( *t == '-' ) { AC.PolyRatFunChanged = 0; t++; }
		while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
		if ( *t == 0 ) return(error);
	}
	if ( *t == '(' ) {
		t++;
		while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
/*
		Next we need a keyword like
			(divergence,ep)
			(expand,ep,maxpow)
*/
		s = t;
		t = SkipAName(s);
		if ( t == 0 ) goto NumErr;
		c = *t; *t = 0;
		if ( ( StrICmp(s,(UBYTE *)"divergence") == 0 )
		|| ( StrICmp(s,(UBYTE *)"finddivergence") == 0 ) ) {
			if ( c != ',' ) {
				MesPrint("&Illegal option field in PolyRatFun statement.");
				return(1);
			}
			*t = c;
			while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
			s = t;
			t = SkipAName(s);
			if ( t == 0 ) goto NumErr;
			c = *t; *t = 0;
			if ( ( type = GetName(AC.varnames,s,&AC.lPolyFunVar,WITHAUTO) ) != CSYMBOL ) {
				MesPrint("&Illegal symbol %s in option field in PolyRatFun statement.",s);
				return(1);
			}
			*t = c;
			while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
```

```
            if ( *t != ')' ) {
                MesPrint("&Illegal termination of option in PolyRatFun statement.");
                return(1);
            }
            AR.PolyFunExp = AC.lPolyFunExp = 1;
            AR.PolyFunVar = AC.lPolyFunVar;
            symbols[AC.lPolyFunVar].minpower = -MAXPOWER;
            symbols[AC.lPolyFunVar].maxpower =  MAXPOWER;
        }
        else if ( StrICmp(s,(UBYTE *)"expand") == 0 ) {
            WORD x = 0, etype = 2;
            if ( c != ',' ) {
                MesPrint("&Illegal option field in PolyRatFun statement.");
                return(1);
            }
            *t = c;
            while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
            s = t;
            t = SkipAName(s);
            if ( t == 0 ) goto NumErr;
            c = *t; *t = 0;
            if ( ( type = GetName(AC.varnames,s,&AC.lPolyFunVar,WITHAUTO) ) != CSYMBOL ) {
                MesPrint("&Illegal symbol %s in option field in PolyRatFun statement.",s);
                return(1);
            }
            *t = c;
            while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
            if ( *t > '9' || *t < '0' ) {
                MesPrint("&Illegal option field in PolyRatFun statement.");
                return(1);
            }
            while ( *t <= '9' && *t >= '0' ) x = 10*x + *t++ - '0';
            while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
            if ( *t != ')' ) {
                s = t;
                t = SkipAName(s);
                if ( t == 0 ) goto ParErr;
                c = *t; *t = 0;
                if ( StrICmp(s,(UBYTE *)"fixed") == 0 ) {
                    etype = 3;
                }
                else if ( StrICmp(s,(UBYTE *)"relative") == 0 ) {
                    etype = 2;
                }
                else {
                    MesPrint("&Illegal termination of option in PolyRatFun statement.");
                    return(1);
                }
                *t = c;
                while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
                if ( *t != ')' ) {
                    MesPrint("&Illegal termination of option in PolyRatFun statement.");
                    return(1);
                }
            }
            AR.PolyFunExp = AC.lPolyFunExp = etype;
            AR.PolyFunVar = AC.lPolyFunVar;
            AR.PolyFunPow = AC.lPolyFunPow = x;
            symbols[AC.lPolyFunVar].minpower = -MAXPOWER;
            symbols[AC.lPolyFunVar].maxpower =  MAXPOWER;
        }
        else {
ParErr:     MesPrint("&Illegal option %s in PolyRatFun statement.",s);
            return(1);
        }
        t++;
        while ( *t == ',' || *t == ' ' || *t == '\t' ) t++;
        if ( *t == 0 ) return(error);
    }
NumErr:;
    MesPrint("&PolyRatFun statement needs one or two commuting function(s) for its argument(s)");
    return(1);
}

/*
    #] CoPolyRatFun :
    #[ CoMerge :
*/

int CoMerge(UBYTE *inp)
{
    UBYTE *s = inp;
    int type;
    WORD numfunc, option = 0;
    if ( tolower(s[0]) == 'o' && tolower(s[1]) == 'n' && tolower(s[2]) == 'c' &&
         tolower(s[3]) == 'e' && tolower(s[4]) == ',' ) {
        option = 1; s += 5;
```

```
    }
    else if ( tolower(s[0]) == 'a' && tolower(s[1]) == 'l' && tolower(s[2]) == 'l' &&
         tolower(s[3]) == ',' ) {
        option = 0; s += 4;
    }
    if ( *s == '$' ) {
        if ( ( type = GetName(AC.dollarnames,s+1,&numfunc,NOAUTO) ) == CDOLLAR )
            numfunc = -numfunc;
        else {
            MesPrint("&%s is undefined",s);
            numfunc = AddDollar(s+1,DOLINDEX,&one,1);
            return(1);
        }
tests:  s = SkipAName(s);
        if ( *s != 0 ) {
            MesPrint("&Merge/shuffle should have a single function or $variable for its argument");
            return(1);
        }
    }
    else if ( ( type = GetName(AC.varnames,s,&numfunc,WITHAUTO) ) == CFUNCTION ) {
        numfunc += FUNCTION;
        goto tests;
    }
    else if ( type != -1 ) {
        if ( type != CDUBIOUS ) {
            NameConflict(type,s);
            type = MakeDubious(AC.varnames,s,&numfunc);
        }
        return(1);
    }
    else {
        MesPrint("&%s is not a function",s);
        numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
        return(1);
    }
    Add4Com(TYPEMERGE,numfunc,option);
    return(0);
}

/*
    #] CoMerge :
    #[ CoStuffle :

    Important for future options: The bit, given by 256 (bit 8) is reserved
    internally for keeping track of the sign in the number of Stuffle
    additions.
*/

int CoStuffle(UBYTE *inp)
{
    UBYTE *s = inp, *ss, c;
    int type;
    WORD numfunc, option = 0;
    if ( tolower(s[0]) == 'o' && tolower(s[1]) == 'n' && tolower(s[2]) == 'c' &&
         tolower(s[3]) == 'e' && tolower(s[4]) == ',' ) {
        option = 1; s += 5;
    }
    else if ( tolower(s[0]) == 'a' && tolower(s[1]) == 'l' && tolower(s[2]) == 'l' &&
         tolower(s[3]) == ',' ) {
        option = 0; s += 4;
    }
    ss = SkipAName(s);
    c = *ss; *ss = 0;
    if ( *s == '$' ) {
        if ( ( type = GetName(AC.dollarnames,s+1,&numfunc,NOAUTO) ) == CDOLLAR )
            numfunc = -numfunc;
        else {
            MesPrint("&%s is undefined",s);
            numfunc = AddDollar(s+1,DOLINDEX,&one,1);
            return(1);
        }
tests:  *ss = c;
        if ( *ss != '+' && *ss != '-' && ss[1] != 0 ) {
            MesPrint("&Stuffle should have a single function or $variable for its argument, followed by either + or -");
            return(1);
        }
        if ( *ss == '-' ) option += 2;
    }
    else if ( ( type = GetName(AC.varnames,s,&numfunc,WITHAUTO) ) == CFUNCTION ) {
        numfunc += FUNCTION;
        goto tests;
    }
    else if ( type != -1 ) {
        if ( type != CDUBIOUS ) {
            NameConflict(type,s);
            type = MakeDubious(AC.varnames,s,&numfunc);
```

```
        }
        return(1);
    }
    else {
        MesPrint("&%s is not a function",s);
        numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
        return(1);
    }
    Add4Com(TYPESTUFFLE,numfunc,option);
    return(0);
}

/*
    #] CoStuffle :
    #[ CoProcessBucket :
*/

int CoProcessBucket(UBYTE *s)
{
    LONG x;
    while ( *s == ',' || *s == '=' ) s++;
    ParseNumber(x,s)
    if ( *s && *s != ' ' && *s != '\t' ) {
        MesPrint("&Numerical value expected for ProcessBucketSize");
        return(1);
    }
    AC.ProcessBucketSize = x;
    return(0);
}

/*
    #] CoProcessBucket :
    #[ CoThreadBucket :
*/

int CoThreadBucket(UBYTE *s)
{
    LONG x;
    while ( *s == ',' || *s == '=' ) s++;
    ParseNumber(x,s)
    if ( *s && *s != ' ' && *s != '\t' ) {
        MesPrint("&Numerical value expected for ThreadBucketSize");
        return(1);
    }
    if ( x <= 0 ) {
        Warning("Negative of zero value not allowed for ThreadBucketSize. Adjusted to 1.");
        x = 1;
    }
    AC.ThreadBucketSize = x;
#ifdef WITHPTHREADS
    if ( AS.MultiThreaded ) MakeThreadBuckets(-1,1);
#endif
    return(0);
}

/*
    #] CoThreadBucket :
    #[ DoArgPlode :

    Syntax: a list of functions.
    If the functions have an argument it must be a function.
    In the case f(g) we treat f(g(...)) with g any argument.
      (not yet implemented)
*/

int DoArgPlode(UBYTE *s, int par)
{
    GETIDENTITY
    WORD numfunc, type, error = 0, *w, n;
    UBYTE *t,c;
    int i;
    w = AT.WorkPointer;
    *w++ = par;
    w++;
    while ( *s == ',' ) s++;
    while ( *s ) {
        if ( *s == '$' ) {
            MesPrint("&We don't do dollar variables yet in ArgImplode/ArgExplode");
            return(1);
        }
        t = s;
        if ( ( s = SkipAName(s) ) == 0 ) return(1);
        c = *s; *s = 0;
        if ( ( type = GetName(AC.varnames,t,&numfunc,WITHAUTO) ) == CFUNCTION ) {
            numfunc += FUNCTION;
        }
        else if ( type != -1 ) {
```

```
            if ( type != CDUBIOUS ) {
                NameConflict(type,t);
                type = MakeDubious(AC.varnames,t,&numfunc);
            }
            error = 1;
        }
        else {
            MesPrint("&%s is not a function",t);
            numfunc = AddFunction(s,0,0,0,0,0,-1,-1) + FUNCTION;
            return(1);
        }
        *s = c;
        *w++ = numfunc;
        *w++ = FUNHEAD;
#if FUNHEAD > 2
        for ( i = 2; i < FUNHEAD; i++ ) *w++ = 0;
#endif
        if ( *s && *s != ',' ) {
            MesPrint("&Illegal character in ArgImplode/ArgExplode statement: %s",s);
            return(1);
        }
        while ( *s == ',' ) s++;
    }
    n = w - AT.WorkPointer;
    AT.WorkPointer[1] = n;
    AddNtoL(n,AT.WorkPointer);
    return(error);
}

/*
    #] DoArgPlode :
    #[ CoArgExplode :
*/

int CoArgExplode(UBYTE *s) { return(DoArgPlode(s,TYPEARGEXPLODE)); }

/*
    #] CoArgExplode :
    #[ CoArgImplode :
*/

int CoArgImplode(UBYTE *s) { return(DoArgPlode(s,TYPEARGIMPLODE)); }

/*
    #] CoArgImplode :
    #[ CoClearTable :
*/

int CoClearTable(UBYTE *s)
{
    UBYTE c, *t;
    int j, type, error = 0;
    WORD numfun;
    TABLES T, TT;
    if ( *s == 0 ) {
        MesPrint("&The ClearTable statement needs at least one (table) argument.");
        return(1);
    }
    while ( *s ) {
        t = s;
        s = SkipAName(s);
        c = *s; *s = 0;
        if ( ( ( type = GetName(AC.varnames,t,&numfun,WITHAUTO) ) != CFUNCTION )
        && type != CDUBIOUS ) {
nofunc:     MesPrint("&%s is not a table",t);
            error = 4;
            if ( type < 0 ) numfun = AddFunction(t,0,0,0,0,0,-1,-1);
            *s = c;
            if ( *s == ',' ) s++;
            continue;
        }
/*
        else if ( ( ( T = functions[numfun].tabl ) == 0 )
         || ( T->sparse == 0 ) ) goto nofunc;
*/
        else if ( ( T = functions[numfun].tabl ) == 0 ) goto nofunc;
        numfun += FUNCTION;
        *s = c;
        if ( *s == ',' ) s++;
/*
        Now we clear the table.
*/
        if ( T->sparse ) {
        if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
        for (j = 0; j < T->buffersfill; j++ ) { /* was <= */
            finishcbuf(T->buffers[j]);
        }
```

```
        if ( T->buffers ) M_free(T->buffers,"Table buffers");
        finishcbuf(T->bufnum);

        T->boomlijst = 0;
        T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
        T->boomlijst = 0;
        T->bufnum = inicbufs();
        T->buffersize = 8;
        T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->bufferssize,"Table buffers");
        T->buffersfill = 0;
        T->buffers[T->buffersfill++] = T->bufnum;

        T->totind = 0;            /* At the moment there are this many */
        T->reserved = 0;

        ClearTableTree(T);

        if ( T->spare ) {
            if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
            T->tablepointers = 0;
            TT = T->spare;
            if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
            for (j = 0; j < TT->buffersfill; j++ ) {
                finishcbuf(TT->buffers[j]);
            }
            if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
            if ( TT->buffers )M_free(TT->buffers,"Table buffers");
            if ( TT->mm ) M_free(TT->mm,"tableminmax");
            if ( TT->flags ) M_free(TT->flags,"tableflags");
            M_free(TT,"table");
            SpareTable(T);
        }
        }
        else EmptyTable(T);
    }
    return(error);
}

/*
    #] CoClearTable :
    #[ CoDenominators :
*/

int CoDenominators(UBYTE *s)
{
    WORD numfun;
    int type;
    UBYTE *t = SkipAName(s), *t1;
    if ( t == 0 ) goto syntaxerror;
    t1 = t; while ( *t1 == ',' || *t1 == ' ' || *t1 == '\t' ) t1++;
    if ( *t1 ) goto syntaxerror;
    *t = 0;
    if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
    || ( functions[numfun].spec != 0 ) ) {
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        goto syntaxerror;
    }
    Add3Com(TYPEDENOMINATORS,numfun+FUNCTION);
    return(0);
syntaxerror:
    MesPrint("&Denominators statement needs one regular function for its argument");
    return(1);
}

/*
    #] CoDenominators :
    #[ CoDropCoefficient :
*/

int CoDropCoefficient(UBYTE *s)
{
    if ( *s == 0 ) {
        Add2Com(TYPEDROPCOEFFICIENT)
        return(0);
    }
    MesPrint("&Illegal argument in DropCoefficient statement: '%s'",s);
    return(1);
}
/*
    #] CoDropCoefficient :
    #[ CoDropSymbols :
*/

int CoDropSymbols(UBYTE *s)
```

```
{
    if ( *s == 0 ) {
        Add2Com(TYPEDROPSYMBOLS)
        return(0);
    }
    MesPrint("&Illegal argument in DropSymbols statement: '%s'",s);
    return(1);
}
/*
    #] CoDropSymbols :
    #[ CoToPolynomial :

    Converts the current term as much as possible to symbols.
    Keeps a list of all objects converted to symbols in AM.sbufnum.
    Note that this cannot be executed in parallel because we have only
    a single compiler buffer for this. Hence we switch on the noparallel
    module option.

    Option(s):
        OnlyFunctions [,name1][,name2][,...,namem];
*/

int CoToPolynomial(UBYTE *inp)
{
    int error = 0;
    while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
    if ( ( AC.topolynomialflag & ~TOPOLYNOMIALFLAG ) != 0 ) {
        MesPrint("&ToPolynomial statement and FactArg statement are not allowed in the same module");
        return(1);
    }
    if ( AO.OptimizeResult.code != NULL ) {
        MesPrint("&Using ToPolynomial statement when there are still optimization results active.");
        MesPrint("&Please use #ClearOptimize instruction first.");
        MesPrint("&This will loose the optimized expression.");
        return(1);
    }
    if ( *inp == 0 ) {
        Add3Com(TYPETOPOLYNOMIAL,DOALL)
    }
    else {
        int numargs = 0;
        WORD *funnums = 0, type, num;
        UBYTE *s, c;
        s = SkipAName(inp);
        if ( s == 0 ) return(1);
        c = *s; *s = 0;
        if ( StrICmp(inp,(UBYTE *)"onlyfunctions") ) {
            MesPrint("&Illegal option %s in ToPolynomial statement",inp);
            *s = c;
            return(1);
        }
        *s = c;
        inp = s;
        while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
        s = inp;
        while ( *s ) s++;
/*
        Get definitely enough space for the numbers of the functions
*/
        funnums = (WORD *)Malloc1(((LONG)(s-inp)+3)*sizeof(WORD),"ToPlynomial");
        while ( *inp ) {
            s = SkipAName(inp);
            if ( s == 0 ) return(1);
            c = *s; *s = 0;
            type = GetName(AC.varnames,inp,&num,WITHAUTO);
            if ( type != CFUNCTION ) {
                MesPrint("&%s is not a function in ToPolynomial statement",inp);
                error = 1;
            }
            funnums[3+numargs++] = num+FUNCTION;
            *s = c;
            inp = s;
            while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
        }
        funnums[0] = TYPETOPOLYNOMIAL;
        funnums[1] = numargs+3;
        funnums[2] = ONLYFUNCTIONS;

        AddNtoL(numargs+3,funnums);
        if ( funnums ) M_free(funnums,"ToPolynomial");
    }
    AC.topolynomialflag |= TOPOLYNOMIALFLAG;
#ifdef WITHMPI
    /* In ParFORM, ToPolynomial has to be executed on the master. */
    AC.mparallelflag |= NOPARALLEL_CONVPOLY;
#endif
    return(error);
```

```
}

/*
    #] CoToPolynomial :
    #[ CoFromPolynomial :

    Converts the current term as much as possible back from extra symbols
    to their original values. Does not look inside functions.
*/

int CoFromPolynomial(UBYTE *inp)
{
    while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
    if ( *inp == 0 ) {
        if ( AO.OptimizeResult.code != NULL ) {
            MesPrint("&Using FromPolynomial statement when there are still optimization results active.");
            MesPrint("&Please use #ClearOptimize instruction first.");
            MesPrint("&This will loose the optimized expression.");
            return(1);
        }
        Add2Com(TYPEFROMPOLYNOMIAL)
        return(0);
    }
    MesPrint("&Illegal argument in FromPolynomial statement: '%s'",inp);
    return(1);
}

/*
    #] CoFromPolynomial :
    #[ CoArgToExtraSymbol :

    Converts the specified function arguments into extra symbols.

    Syntax: ArgToExtraSymbol [ToNumber] [<argument specifications>]
*/

int CoArgToExtraSymbol(UBYTE *s)
{
    CBUF *C = cbuf + AC.cbufnum;
    WORD *lhs;

    /* TODO: resolve interference with rational arithmetic. (#138) */
    if ( ( AC.topolynomialflag & ~TOPOLYNOMIALFLAG ) != 0 ) {
        MesPrint("&ArgToExtraSymbol statement and FactArg statement are not allowed in the same module");
        return(1);
    }
    if ( AO.OptimizeResult.code != NULL ) {
        MesPrint("&Using ArgToExtraSymbol statement when there are still optimization results active.");
        MesPrint("&Please use #ClearOptimize instruction first.");
        MesPrint("&This will loose the optimized expression.");
        return(1);
    }

    SkipSpaces(&s);
    int tonumber = ConsumeOption(&s, "tonumber");

    int ret = DoArgument(s,TYPEARGTOEXTRASYMBOL);
    if ( ret ) return(ret);

    /*
     * The "scale" parameter is unused. Instead, we put the "tonumber"
     * parameter.
     */
    lhs = C->lhs[C->numlhs];
    if ( lhs[4] != 1 ) {
        Warning("scale parameter (^n) is ignored in ArgToExtraSymbol");
    }
    lhs[4] = tonumber;

    AC.topolynomialflag |= TOPOLYNOMIALFLAG;  /* This flag is also used in ParFORM. */
#ifdef WITHMPI
    /*
     * In ParFORM, the conversion to extra symbols has to be performed on
     * the master.
     */
    AC.mparallelflag |= NOPARALLEL_CONVPOLY;
#endif

    return(0);
}

/*
    #] CoArgToExtraSymbol :
    #[ CoExtraSymbols :
```

```
*/

int CoExtraSymbols(UBYTE *inp)
{
    UBYTE *arg1, *arg2, c, *s;
    WORD i, j, type, number;
    while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
    if ( FG.cTable[*inp] != 0 ) {
        MesPrint("&Illegal argument in ExtraSymbols statement: '%s'",inp);
        return(1);
    }
    arg1 = inp;
    while ( FG.cTable[*inp] == 0 ) inp++;
    c = *inp; *inp = 0;
    if ( ( StrICmp(arg1,(UBYTE *)"array") == 0 )
          || ( StrICmp(arg1,(UBYTE *)"vector") == 0 ) ) {
        AC.extrasymbols = 1;
    }
    else if ( StrICmp(arg1,(UBYTE *)"underscore") == 0 ) {
        AC.extrasymbols = 0;
    }
/*
    else if ( StrICmp(arg1,(UBYTE *)"nothing") == 0 ) {
        AC.extrasymbols = 2;
    }
*/
    else {
        MesPrint("&Illegal keyword in ExtraSymbols statement: '%s'",arg1);
        return(1);
    }
    *inp = c;
    while ( *inp == ' ' || *inp == ',' || *inp == '\t' ) inp++;
    if ( FG.cTable[*inp] != 0 ) {
        MesPrint("&Illegal argument in ExtraSymbols statement: '%s'",inp);
        return(1);
    }
    arg2 = inp;
    while ( FG.cTable[*inp] <= 1 ) inp++;
    if ( *inp != 0 ) {
        MesPrint("&Illegal end of ExtraSymbols statement: '%s'",inp);
        return(1);
    }
/*
        Now check whether this object has been declared already.
        That would not be allowed.
*/
    if ( AC.extrasymbols == 1 ) {
        type = GetName(AC.varnames,arg2,&number,NOAUTO);
        if ( type != NAMENOTFOUND ) {
            MesPrint("&ExtraSymbols statement: '%s' has already been declared before",arg2);
            return(1);
        }
    }
    else if ( AC.extrasymbols == 0 ) {
        if ( *arg2 == 'N' ) {
            s = arg2+1;
            while ( FG.cTable[*s] == 1 ) s++;
            if ( *s == 0 ) {
                MesPrint("&ExtraSymbols statement: '%s' creates conflicts with summed indices",arg2);
                return(1);
            }
        }
    }
    if ( AC.extrasym ) { M_free(AC.extrasym,"extrasym"); AC.extrasym = 0; }
    i = inp - arg2 + 1;
    AC.extrasym = (UBYTE *)Malloc1(i*sizeof(UBYTE),"extrasym");
    for ( j = 0; j < i; j++ ) AC.extrasym[j] = arg2[j];
    return(0);
}

/*
    #] CoExtraSymbols :
    #[ GetIfDollarFactor :
*/

WORD *GetIfDollarFactor(UBYTE **inp, WORD *w)
{
    LONG x;
    WORD number;
    UBYTE *name, c, *s;
    s = *inp;
    if ( FG.cTable[*s] == 1 ) {
        x = 0;
        while ( FG.cTable[*s] == 1 ) {
            x = 10*x + *s++ - '0';
            if ( x >= MAXPOSITIVE ) {
                MesPrint("&Value in dollar factor too large");
```

```
                while ( FG.cTable[*s] == 1 ) s++;
                *inp = s;
                return(0);
            }
        }
        *w++ = IFDOLLAREXTRA;
        *w++ = 3;
        *w++ = -x-1;
        *inp = s;
        return(w);
    }
    if ( *s != '$' ) {
        MesPrint("&Factor indicator for $-variable should be a number or a $-variable.");
        return(0);
    }
    s++; name = s;
    while ( FG.cTable[*s] < 2 ) s++;
    c = *s; *s = 0;
    if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
        MesPrint("&dollar in if statement should have been defined previously");
        return(0);
    }
    *s = c;
    *w++ = IFDOLLAREXTRA;
    *w++ = 3;
    *w++ = number;
    if ( c == '[' ) {
        s++;
        *inp = s;
        if ( ( w = GetIfDollarFactor(inp,w) ) == 0 ) return(0);
        s = *inp;
        if ( *s != ']' ) {
            MesPrint("&unmatched [] in $ in if statement");
            return(0);
        }
        s++;
        *inp = s;
    }
    return(w);
}

/*
    #] GetIfDollarFactor :
    #[ GetDoParam :
*/

UBYTE *GetDoParam(UBYTE *inp, WORD **wp, int par)
{
    LONG x;
    WORD number;
    UBYTE *name, c;
    if ( FG.cTable[*inp] == 1 ) {
        x = 0;
        while ( *inp >= '0' && *inp <= '9' ) {
            x = 10*x + *inp++ - '0';
            if ( x > MAXPOSITIVE ) {
                if ( par == -1 ) {
                    MesPrint("&Value in dollar factor too large");
                }
                else {
                    MesPrint("&Value in do loop boundaries too large");
                }
                while ( FG.cTable[*inp] == 1 ) inp++;
                return(0);
            }
        }
        if ( par > 0 ) {
            *(*wp)++ = SNUMBER;
            *(*wp)++ = (WORD)x;
        }
        else {
            *(*wp)++ = DOLLAREXPR2;
            *(*wp)++ = -((WORD)x)-1;
        }
        return(inp);
    }
    if ( *inp != '$' ) {
        return(0);
    }
    inp++; name = inp;
    while ( FG.cTable[*inp] < 2 ) inp++;
    c = *inp; *inp = 0;
    if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
        if ( par == -1 ) {
            MesPrint("&dollar in print statement should have been defined previously");
        }
        else {
```

```
            MesPrint("&dollar in do loop boundaries should have been defined previously");
        }
        return(0);
    }
    *inp = c;
    if ( par > 0 ) {
        *(*wp)++ = DOLLAREXPRESSION;
        *(*wp)++ = number;
    }
    else {
        *(*wp)++ = DOLLAREXPR2;
        *(*wp)++ = number;
    }
    if ( c == '[' ) {
        inp++;
        inp = GetDoParam(inp,wp,0);
        if ( inp == 0 ) return(0);
        if ( *inp != ']' ) {
            if ( par == -1 ) {
                MesPrint("&unmatched [] in $ in print statement");
            }
            else {
                MesPrint("&unmatched [] in do loop boundaries");
            }
            return(0);
        }
        inp++;
    }
    return(inp);
}

/*
    #] GetDoParam :
    #[ CoDo :
*/

int CoDo(UBYTE *inp)
{
    GETIDENTITY
    CBUF *C = cbuf+AC.cbufnum;
    WORD *w, numparam;
    int error = 0, i;
    UBYTE *name, c;
    if ( AC.doloopstack == 0 ) {
        AC.doloopstacksize = 20;
        AC.doloopstack = (WORD *)Malloc1(AC.doloopstacksize*2*sizeof(WORD),"doloop stack");
        AC.doloopnest = AC.doloopstack + AC.doloopstacksize;
    }
    if ( AC.dolooplevel >= AC.doloopstacksize ) {
        WORD *newstack, *newnest, newsize;
        newsize = AC.doloopstacksize * 2;
        newstack = (WORD *)Malloc1(newsize*2*sizeof(WORD),"doloop stack");
        newnest = newstack + newsize;
        for ( i = 0; i < newsize; i++ ) {
            newstack[i] = AC.doloopstack[i];
            newnest[i] = AC.doloopnest[i];
        }
        M_free(AC.doloopstack,"doloop stack");
        AC.doloopstack = newstack;
        AC.doloopnest = newnest;
        AC.doloopstacksize = newsize;
    }
    AC.doloopnest[AC.dolooplevel] = NestingChecksum();

    w = AT.WorkPointer;
    *w++ = TYPEDOLOOP;
    w++; /* Space for the length of the statement */
/*
    Now the $loopvariable
*/
    while ( *inp == ',' ) inp++;
    if ( *inp != '$' ) {
        error = 1;
        MesPrint("&do loop parameter should be a dollar variable");
    }
    else {
        inp++;
        name = inp;
        if ( FG.cTable[*inp] != 0 ) {
            error = 1;
            MesPrint("&illegal name for do loop parameter");
        }
        while ( FG.cTable[*inp] < 2 ) inp++;
        c = *inp; *inp = 0;
        if ( GetName(AC.dollarnames,name,&numparam,NOAUTO) == NAMENOTFOUND ) {
            numparam = AddDollar(name,DOLUNDEFINED,0,0);
        }
```

```
            *w++ = numparam;
            *inp = c;
            AddPotModdollar(numparam);
        }
        w++;  /* space for the level of the enddo statement */
        while ( *inp == ',' ) inp++;
        if ( *inp != '=' ) goto IllSyntax;
        inp++;
        while ( *inp == ',' ) inp++;
/*
        The start value
*/
        inp = GetDoParam(inp,&w,1);
        if ( inp == 0 || *inp != ',' ) goto IllSyntax;
        while ( *inp == ',' ) inp++;
/*
        The end value
*/
        inp = GetDoParam(inp,&w,1);
        if ( inp == 0 || ( *inp != 0 && *inp != ',' ) ) goto IllSyntax;
/*
        The increment value
*/
        if ( *inp != ',' ) {
            if ( *inp == 0 ) { *w++ = SNUMBER; *w++ = 1; }
            else goto IllSyntax;
        }
        else {
            while ( *inp == ',' ) inp++;
            inp = GetDoParam(inp,&w,1);
        }
        if ( inp == 0 || *inp != 0 ) goto IllSyntax;
        *w = 0;
        AT.WorkPointer[1] = w - AT.WorkPointer;
/*
        Put away and set information for placing enddo information.
*/
        AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
        AC.doloopstack[AC.dolooplevel++] = C->numlhs;

        return(error);

IllSyntax:
        MesPrint("&Illegal syntax for do statement");
        return(1);
}

/*
        #] CoDo :
        #[ CoEndDo :
*/

int CoEndDo(UBYTE *inp)
{
        CBUF *C = cbuf+AC.cbufnum;
        WORD scratch[3];
        while ( *inp == ',' ) inp++;
        if ( *inp ) {
            MesPrint("&Illegal syntax for EndDo statement");
            return(1);
        }
        if ( AC.dolooplevel <= 0 ) {
            MesPrint("&EndDo without corresponding Do statement");
            return(1);
        }
        AC.dolooplevel--;
        scratch[0] = TYPEENDDOLOOP;
        scratch[1] = 3;
        scratch[2] = AC.doloopstack[AC.dolooplevel];
        AddNtoL(3,scratch);
        cbuf[AC.cbufnum].lhs[AC.doloopstack[AC.dolooplevel]][3] = C->numlhs;
        if ( AC.doloopnest[AC.dolooplevel] != NestingChecksum() ) {
            MesNesting();
            return(1);
        }
        return(0);
}

/*
        #] CoEndDo :
        #[ CoFactDollar :
*/

int CoFactDollar(UBYTE *inp)
{
        WORD numdollar;
        if ( *inp == '$' ) {
```

```
        if ( GetName(AC.dollarnames,inp+1,&numdollar,NOAUTO) != CDOLLAR ) {
            MesPrint("&%s is undefined",inp);
            numdollar = AddDollar(inp+1,DOLINDEX,&one,1);
            return(1);
        }
        inp = SkipAName(inp+1);
        if ( *inp != 0 ) {
            MesPrint("&FactDollar should have a single $variable for its argument");
            return(1);
        }
        AddPotModdollar(numdollar);
    }
    else {
        MesPrint("&%s is not a $-variable",inp);
        return(1);
    }
    Add3Com(TYPEFACTOR,numdollar);
    return(0);
}

/*
    #] CoFactDollar :
    #[ CoFactorize :
*/

int CoFactorize(UBYTE *s) { return(DoFactorize(s,1)); }

/*
    #] CoFactorize :
    #[ CoNFactorize :
*/

int CoNFactorize(UBYTE *s) { return(DoFactorize(s,0)); }

/*
    #] CoNFactorize :
    #[ CoUnFactorize :
*/

int CoUnFactorize(UBYTE *s) { return(DoFactorize(s,3)); }

/*
    #] CoUnFactorize :
    #[ CoNUnFactorize :
*/

int CoNUnFactorize(UBYTE *s) { return(DoFactorize(s,2)); }

/*
    #] CoNUnFactorize :
    #[ DoFactorize :
*/

int DoFactorize(UBYTE *s,int par)
{
    EXPRESSIONS e;
    WORD i;
    WORD number;
    UBYTE *t, c;
    int error = 0, keepzeroflag = 0;
    if ( *s == '(' ) {
        s++;
        while ( *s != ')' && *s ) {
            if ( FG.cTable[*s] == 0 ) {
                t = s; while ( FG.cTable[*s] == 0 ) s++;
                c = *s; *s = 0;
                if ( StrICmp((UBYTE *)"keepzero",t) == 0 ) {
                    keepzeroflag = 1;
                }
                else {
                    MesPrint("&Illegal option in [N][Un]Factorize statement: %s",t);
                    error = 1;
                }
                *s = c;
            }
            while ( *s == ',' ) s++;
            if ( *s && *s != ')' && FG.cTable[*s] != 0 ) {
                MesPrint("&Illegal character in option field of [N][Un]Factorize statement");
                error = 1;
                return(error);
            }
        }
        if ( *s ) s++;
        while ( *s == ',' || *s == ' ' ) s++;
    }
    if ( *s == 0 ) {
        for ( i = NumExpressions-1; i >= 0; i-- ) {
```

```
            e = Expressions+i;
            if ( e->replace >= 0 ) {
                e = Expressions + e->replace;
            }
            if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
            || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
          || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION
            ) {
                switch ( par ) {
                    case 0:
                        e->vflags &= ~TOBEFACTORED;
                        break;
                    case 1:
                        e->vflags |=  TOBEFACTORED;
                        e->vflags &= ~TOBEUNFACTORED;
                        break;
                    case 2:
                        e->vflags &= ~TOBEUNFACTORED;
                        break;
                    case 3:
                        e->vflags |=  TOBEUNFACTORED;
                        e->vflags &= ~TOBEFACTORED;
                        break;
                }
            }
            if ( ( e->vflags & TOBEFACTORED ) != 0 ) {
                if ( keepzeroflag ) e->vflags |=  KEEPZERO;
                else                e->vflags &= ~KEEPZERO;
            }
            else                    e->vflags &= ~KEEPZERO;
        }
    }
    else {
        for(;;) {   /* Look for a (comma separated) list of variables */
            while ( *s == ',' ) s++;
            if ( *s == 0 ) break;
            if ( *s == '[' || FG.cTable[*s] == 0 ) {
                t = s;
                if ( ( s = SkipAName(s) ) == 0 ) {
                    MesPrint("&Improper name for an expression: '%s'",t);
                    return(1);
                }
                c = *s; *s = 0;
                if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
                    e = Expressions+number;
                    if ( e->replace >= 0 ) {
                        e = Expressions + e->replace;
                    }
                    if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
                    || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
                    || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION
                    ) {
                        switch ( par ) {
                            case 0:
                                e->vflags &= ~TOBEFACTORED;
                                break;
                            case 1:
                                e->vflags |=  TOBEFACTORED;
                                e->vflags &= ~TOBEUNFACTORED;
                                break;
                            case 2:
                                e->vflags &= ~TOBEUNFACTORED;
                                break;
                            case 3:
                                e->vflags |=  TOBEUNFACTORED;
                                e->vflags &= ~TOBEFACTORED;
                                break;
                        }
                    }
                    if ( ( e->vflags & TOBEFACTORED ) != 0 ) {
                        if ( keepzeroflag ) e->vflags |=  KEEPZERO;
                        else                e->vflags &= ~KEEPZERO;
                    }
                    else                    e->vflags &= ~KEEPZERO;
                }
                else if ( GetName(AC.varnames,t,&number,NOAUTO) != NAMENOTFOUND ) {
                    MesPrint("&%s is not an expression",t);
                    error = 1;
                }
                *s = c;
            }
            else {
                MesPrint("&Illegal object in (N)Factorize statement");
                error = 1;
                while ( *s && *s != ',' ) s++;
                if ( *s == 0 ) break;
            }
```

```
        }

    }
    return(error);
}

/*
    #] DoFactorize : 
    #[ CoOptimizeOption :

*/

int CoOptimizeOption(UBYTE *s)
{
    UBYTE *name, *t1, *t2, c1, c2, *value, *u;
    int error = 0, x;
    double d;
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    while ( *s ) {
        name = s; while ( FG.cTable[*s] == 0 ) s++;
        t1 = s; c1 = *t1;
        while ( *s == ' ' || *s == '\t' ) s++;
        if ( *s != '=' ) {
correctuse:
            MesPrint("&Correct use in Format,Optimize statement is Optionname=value");
            error = 1;
            while ( *s == ' ' || *s == ',' || *s == '\t' || *s == '=' ) s++;
            *t1 = c1;
            continue;
        }
        *t1 = 0;
        s++;
        while ( *s == ' ' || *s == '\t' ) s++;
        if ( *s == 0 ) goto correctuse;
        value = s;
        while ( FG.cTable[*s] <= 1 || *s=='.' || *s=='*' || *s == '(' || *s == ')' ) {
            if ( *s == '(' ) { SKIPBRA4(s) }
            s++;
        }
        t2 = s; c2 = *t2;
        while ( *s == ' ' || *s == '\t' ) s++;
        if ( *s && *s != ',' ) goto correctuse;
        if ( *s ) {
            s++;
            while ( *s == ' ' || *s == '\t' ) s++;
        }
        *t2 = 0;
/*
        Now we have name=value with name and value zero terminated strings.
*/
        if ( StrICmp(name,(UBYTE *)"horner") == 0 ) {
            if ( StrICmp(value,(UBYTE *)"occurrence") == 0 ) {
                AO.Optimize.horner = O_OCCURRENCE;
            }
            else if ( StrICmp(value,(UBYTE *)"mcts") == 0 ) {
                AO.Optimize.horner = O_MCTS;
            }
            else if ( StrICmp(value,(UBYTE *)"sa") == 0 ) {
                AO.Optimize.horner = O_SIMULATED_ANNEALING;
            }
            else {
                AO.Optimize.horner = -1;
                MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
                error = 1;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"hornerdirection") == 0 ) {
            if ( StrICmp(value,(UBYTE *)"forward") == 0 ) {
                AO.Optimize.hornerdirection = O_FORWARD;
            }
            else if ( StrICmp(value,(UBYTE *)"backward") == 0 ) {
                AO.Optimize.hornerdirection = O_BACKWARD;
            }
            else if ( StrICmp(value,(UBYTE *)"forwardorbackward") == 0 ) {
                AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
            }
            else if ( StrICmp(value,(UBYTE *)"forwardandbackward") == 0 ) {
                AO.Optimize.hornerdirection = O_FORWARDANDBACKWARD;
            }
            else {
                AO.Optimize.method = -1;
                MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
                error = 1;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"method") == 0 ) {
            if ( StrICmp(value,(UBYTE *)"none") == 0 ) {
```

```
                AO.Optimize.method = O_NONE;
            }
            else if ( StrICmp(value,(UBYTE *)"cse") == 0 ) {
                AO.Optimize.method = O_CSE;
            }
            else if ( StrICmp(value,(UBYTE *)"csegreedy") == 0 ) {
                AO.Optimize.method = O_CSEGREEDY;
            }
            else if ( StrICmp(value,(UBYTE *)"greedy") == 0 ) {
                AO.Optimize.method = O_GREEDY;
            }
            else {
                AO.Optimize.method = -1;
                MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
                error = 1;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"timelimit") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option TimeLimit in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.mctstimelimit = 0;
                AO.Optimize.greedytimelimit = 0;
                error = 1;
            }
            else {
                AO.Optimize.mctstimelimit = x/2;
                AO.Optimize.greedytimelimit = x/2;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"mctstimelimit") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option MCTSTimeLimit in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.mctstimelimit = 0;
                error = 1;
            }
            else {
                AO.Optimize.mctstimelimit = x;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"mctsnumexpand") == 0 ) {
            int y;
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u == '*' || *u == 'x' || *u == 'X' ) {
                u++; y = x;
                x = 0;
                while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            }
            else { y = 1; }
            if ( *u != 0 ) {
                MesPrint("&Option MCTSNumExpand in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.mctsnumexpand= 0;
                AO.Optimize.mctsnumrepeat= 1;
                error = 1;
            }
            else {
                AO.Optimize.mctsnumexpand= x;
                AO.Optimize.mctsnumrepeat= y;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"mctsnumrepeat") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option MCTSNumExpand in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.mctsnumrepeat= 1;
                error = 1;
            }
            else {
                AO.Optimize.mctsnumrepeat= x;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"mctsnumkeep") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option MCTSNumKeep in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.mctsnumkeep= 0;
```

```
                error = 1;
            }
            else {
                AO.Optimize.mctsnumkeep= x;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"mctsconstant") == 0 ) {
            d = 0;
            if ( sscanf ((char*)value, "%lf", &d) != 1 ) {
                MesPrint("&Option MCTSConstant in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.mctsconstant.fval = 0;
                error = 1;
            }
            else {
                AO.Optimize.mctsconstant.fval = d;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"greedytimelimit") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option GreedyTimeLimit in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.greedytimelimit = 0;
                error = 1;
            }
            else {
                AO.Optimize.greedytimelimit = x;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"greedyminnum") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option GreedyMinNum in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.greedyminnum= 0;
                error = 1;
            }
            else {
                AO.Optimize.greedyminnum= x;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"greedymaxperc") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option GreedyMaxPerc in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.greedymaxperc= 0;
                error = 1;
            }
            else {
                AO.Optimize.greedymaxperc= x;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"stats") == 0 ) {
            if ( StrICmp(value,(UBYTE *)"on") == 0 ) {
                AO.Optimize.printstats = 1;
            }
            else if ( StrICmp(value,(UBYTE *)"off") == 0 ) {
                AO.Optimize.printstats = 0;
            }
            else {
                AO.Optimize.printstats = 0;
                MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
                error = 1;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"printscheme") == 0 ) {
            if ( StrICmp(value,(UBYTE *)"on") == 0 ) {
                AO.Optimize.schemeflags |= 1;
            }
            else if ( StrICmp(value,(UBYTE *)"off") == 0 ) {
                AO.Optimize.schemeflags &= ~1;
            }
            else {
                AO.Optimize.schemeflags &= ~1;
                MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
                error = 1;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"debugflag") == 0 ) {
/*
            This option is for debugging purposes only. Not in the manual!
            0x1: Print statements in reverse order.
```

```
            0x2: Print the scheme of the variables.
*/
            x = 0;
            u = value;
            if ( FG.cTable[*u] == 1 ) {
                while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
                if ( *u != 0 ) {
                    MesPrint("&Numerical value for DebugFlag in Format,Optimize statement should be a
nonnegative number: %s",value);
                    AO.Optimize.debugflags = 0;
                    error = 1;
                }
                else {
                    AO.Optimize.debugflags = x;
                }
            }
            else if ( StrICmp(value,(UBYTE *)"on") == 0 ) {
                AO.Optimize.debugflags = 1;
            }
            else if ( StrICmp(value,(UBYTE *)"off") == 0 ) {
                AO.Optimize.debugflags = 0;
            }
            else {
                AO.Optimize.debugflags = 0;
                MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
                error = 1;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"scheme") == 0 ) {
            UBYTE *ss, *s1, c;
            WORD type, numsym;
            AO.schemenum = 0;
            u = value;
            if ( *u != '(' ) {
noscheme:
                MesPrint("&Option Scheme in Format,Optimize statement should be an array of names or
integers between (): %s",value);
                error = 1;
                break;
            }
            u++; ss = u;
            while ( *ss == ' ' || *ss == '\t' || *ss == ',' ) ss++;
            if ( FG.cTable[*ss] == 0 || *ss == '$' || *ss == '[' ) { /* Name */
                s1 = u; SKIPBRA3(s1)
                if ( *s1 != ')' ) goto noscheme;
                while ( ss < s1 ) { if ( *ss++ == ',' ) AO.schemenum++; }
                *ss++ = 0; while ( *ss == ' ' ) ss++;
                if ( *ss != 0 ) goto noscheme;
                ss = u;
                if ( AO.schemenum < 1 ) {
                    MesPrint("&Option Scheme in Format,Optimize statement should have at least one
name or number between ()");
                    error = 1;
                    break;
                }
                if ( AO.inscheme ) M_free(AO.inscheme,"Horner input scheme");
                AO.inscheme = (WORD *)Malloc1((AO.schemenum+1)*sizeof(WORD),"Horner input scheme");
                while ( *ss == ' ' || *ss == '\t' || *ss == ',' ) ss++;
                AO.schemenum = 0;
                for(;;) {
                    if ( *ss == 0 ) break;
                    s1 = ss; ss = SkipAName(s1); c = *ss; *ss = 0;

                    if ( ss[-1] == '_' ) {
/*
                        Now AC.extrasym followed by a number and _
*/
                        UBYTE *u1, *u2;
                        u1 = s1; u2 = AC.extrasym;
                        while ( *u1 == *u2 ) { u1++; u2++; }
                        if ( *u2 == 0 ) { /* Good start */
                            numsym = 0;
                            while ( *u1 >= '0' && *u1 <= '9' ) numsym = 10*numsym + *u1++ - '0';
                            if ( u1 != ss-1 || numsym == 0 || AC.extrasymbols != 0 ) {
                                MesPrint("&Improper use of extra symbol in scheme format option");
                                goto noscheme;
                            }
                            numsym = MAXVARIABLES-numsym;
                            ss++;
                            goto GotTheNumber;
                        }
                    }
                    else if ( *s1 == '$' ) {
                        GETIDENTITY
                        int numdollar;
                        if ( ( numdollar = GetDollar(s1+1) ) < 0 ) {
                            MesPrint("&Undefined variable %s",s1);
```

```
                            error = 5;
                        }
                        else if ( ( numsym = DolToSymbol(BHEAD numdollar) ) < 0 ) {
                            MesPrint("&$%s does not evaluate to a symbol",s1);
                            error = 5;
                        }
                        *ss = c;
                        goto GotTheNumber;
                    }
                    else if ( c == '(' ) {
                        if ( StrCmp(s1,AC.extrasym) == 0 ) {
                            if ( (AC.extrasymbols&1) != 1 ) {
                                MesPrint("&Improper use of extra symbol in scheme format option");
                                goto noscheme;
                            }
                            *ss++ = c;
                            numsym = 0;
                            while ( *ss >= '0' && *ss <= '9' ) numsym = 10*numsym + *ss++ - '0';
                            if ( *ss != ')' ) {
                                MesPrint("&Extra symbol should have a number for its argument.");
                                goto noscheme;
                            }
                            numsym = MAXVARIABLES-numsym;
                            ss++;
                            goto GotTheNumber;
                        }
                    }
                    type = GetName(AC.varnames,s1,&numsym,WITHAUTO);
                    if ( ( type != CSYMBOL ) && type != CDUBIOUS ) {
                        MesPrint("&%s is not a symbol",s1);
                        error = 4;
                        if ( type < 0 ) numsym = AddSymbol(s1,-MAXPOWER,MAXPOWER,0,0);
                    }
                    *ss = c;
GotTheNumber:
                    AO.inscheme[AO.schemenum++] = numsym;
                    while ( *ss == ' ' || *ss == '\t' || *ss == ',' ) ss++;
                }
            }
        }
        else if ( StrICmp(name,(UBYTE *)"mctsdecaymode") == 0 ) {
            x = 0;
            u = value;
            if ( FG.cTable[*u] == 1 ) {
                while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
                if ( *u != 0 ) {
                    MesPrint("&Option MCTSDecayMode in Format,Optimize statement should be a
nonnegative integer: %s",value);
                    AO.Optimize.mctsdecaymode = 0;
                    error = 1;
                }
                else {
                    AO.Optimize.mctsdecaymode = x;
                }
            }
            else {
                AO.Optimize.mctsdecaymode = 0;
                MesPrint("&Unrecognized option value in Format,Optimize statement: %s=%s",name,value);
                error = 1;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"saiter") == 0 ) {
            x = 0;
            u = value; while ( *u >= '0' && *u <= '9' ) x = 10*x + *u++ - '0';
            if ( *u != 0 ) {
                MesPrint("&Option SAIter in Format,Optimize statement should be a positive integer:
%s",value);
                AO.Optimize.saIter = 0;
                error = 1;
            }
            else {
                AO.Optimize.saIter= x;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"samaxt") == 0 ) {
            d = 0;
            if ( sscanf ((char*)value, "%lf", &d) != 1 ) {
                MesPrint("&Option SAMaxT in Format,Optimize statement should be a positive number:
%s",value);
                AO.Optimize.saMaxT.fval = 0;
                error = 1;
            }
            else {
                AO.Optimize.saMaxT.fval = d;
            }
        }
        else if ( StrICmp(name,(UBYTE *)"samint") == 0 ) {
```

```
            d = 0;
            if ( sscanf ((char*)value, "%lf", &d) != 1 ) {
                MesPrint("&Option SAMinT in Format,Optimize statement should be a positive number: %s",value);
                AO.Optimize.saMinT.fval = 0;
                error = 1;
            }
            else {
                AO.Optimize.saMinT.fval = d;
            }
        }
        else {
            MesPrint("&Unrecognized option name in Format,Optimize statement: %s",name);
            error = 1;
        }
        *t1 = c1; *t2 = c2;
    }
    return(error);
}

/*
    #] CoOptimizeOption :
    #[ DoPutInside :

    Syntax:
        PutIn[side],functionname[,brackets]  -> par = 1
        AntiPutIn[side],functionname,antibrackets  -> par = -1
*/

int CoPutInside(UBYTE *inp) { return(DoPutInside(inp,1)); }
int CoAntiPutInside(UBYTE *inp) { return(DoPutInside(inp,-1)); }

int DoPutInside(UBYTE *inp, int par)
{
    GETIDENTITY
    UBYTE *p, c;
    WORD *to, type, c1,c2,funnum, *WorkSave;
    int error = 0;
    while ( *inp == ' ' || *inp == '\t' || *inp == ',' ) inp++;
/*
    First we need the name of a function. (Not a tensor or table!)
*/
    p = SkipAName(inp);
    if ( p == 0 ) return(1);
    c = *p; *p = 0;
    type = GetName(AC.varnames,inp,&funnum,WITHAUTO);
    if ( type != CFUNCTION || functions[funnum].tabl != 0 || functions[funnum].spec ) {
        MesPrint("&PutInside/AntiPutInside expects a regular function for its first argument");
        MesPrint("&Argument is %s",inp);
        error = 1;
    }
    funnum += FUNCTION;
    *p = c;
    inp = p;
    while ( *inp == ' ' || *inp == '\t' || *inp == ',' ) inp++;
    if ( *inp == 0 ) {
        if ( par == 1 ) {
            WORD tocompiler[4];
            tocompiler[0] = TYPEPUTINSIDE;
            tocompiler[1] = 4;
            tocompiler[2] = 0;
            tocompiler[3] = funnum;
            AddNtoL(4,tocompiler);
        }
        else {
            MesPrint("&AntiPutInside needs inside information.");
            error = 1;
        }
        return(error);
    }
    WorkSave = to = AT.WorkPointer;
    *to++ = TYPEPUTINSIDE;
    *to++ = 4;
    *to++ = par;
    *to++ = funnum;
    to++;
    while ( *inp ) {
        while ( *inp == ' ' || *inp == '\t' || *inp == ',' ) inp++;
        if ( *inp == 0 ) break;
        p = SkipAName(inp);
        if ( p == 0 ) { error = 1; break; }
        c = *p; *p = 0;
        type = GetName(AC.varnames,inp,&c1,WITHAUTO);
        if ( c == '.' ) {
            if ( type == CVECTOR || type == CDUBIOUS ) {
                *p++ = c;
                inp = p;
```

```
                p = SkipAName(inp);
                if ( p == 0 ) return(1);
                c = *p; *p = 0;
                type = GetName(AC.varnames,inp,&c2,WITHAUTO);
                if ( type != CVECTOR && type != CDUBIOUS ) {
                    MesPrint("&Not a vector in dotproduct in PutInside/AntiPutInside statement:
    %s",inp);
                    error = 1;
                }
                else type = CDOTPRODUCT;
            }
            else {
                MesPrint("&Illegal use of . after %s in PutInside/AntiPutInside statement",inp);
                error = 1;
                *p = c; inp = p;
                continue;
            }
        }
        switch ( type ) {
            case CSYMBOL :
                *to++ = SYMBOL; *to++ = 4; *to++ = c1; *to++ = 1; break;
            case CVECTOR :
                *to++ = INDEX; *to++ = 3; *to++ = AM.OffsetVector + c1; break;
            case CFUNCTION :
                *to++ = c1+FUNCTION; *to++ = FUNHEAD; *to++ = 0;
                FILLFUN3(to)
                break;
            case CDOTPRODUCT :
                *to++ = DOTPRODUCT; *to++ = 5; *to++ = c1 + AM.OffsetVector;
                *to++ = c2 + AM.OffsetVector; *to++ = 1; break;
            case CDELTA :
                *to++ = DELTA; *to++ = 4; *to++ = EMPTYINDEX; *to++ = EMPTYINDEX; break;
            default :
                MesPrint("&Illegal variable request for %s in PutInside/AntiPutInside statement",inp);
                error = 1; break;
        }
        *p = c;
        inp = p;
    }
    *to++ = 1; *to++ = 1; *to++ = 3;
    AT.WorkPointer[1] = to - AT.WorkPointer;
    AT.WorkPointer[4] = AT.WorkPointer[1]-4;
    AT.WorkPointer = to;
    AC.BracketNormalize = 1;
    if ( Normalize(BHEAD WorkSave+4) ) { error = 1; }
    else {
        WorkSave[1] = WorkSave[4]+4;
        to = WorkSave + WorkSave[1] - 1;
        c1 = ABS(*to);
        WorkSave[1] -= c1;
        WorkSave[4] -= c1;
        AddNtoL(WorkSave[1],WorkSave);
    }
    AC.BracketNormalize = 0;
    AT.WorkPointer = WorkSave;
    return(error);
}

/*
    #] DoPutInside :
    #[ CoSwitch :

    Syntax: Switch $var;
    Be careful with illegal nestings with repeat, if, while.
*/

int CoSwitch(UBYTE *s)
{
    WORD numdollar;
    SWITCH *sw;
    if ( *s == '$' ) {
        if ( GetName(AC.dollarnames,s+1,&numdollar,NOAUTO) != CDOLLAR ) {
            MesPrint("&%s is undefined in switch statement",s);
            numdollar = AddDollar(s+1,DOLINDEX,&one,1);
            return(1);
        }
        s = SkipAName(s+1);
        if ( *s != 0 ) {
            MesPrint("&Switch should have a single $variable for its argument");
            return(1);
        }
/*      AddPotModdollar(numdollar);  */
    }
    else {
        MesPrint("&%s is not a $-variable in switch statement",s);
        return(1);
    }
```

```
/*
    Now create the switch table. We will add to it each time we run
    into a new case. It will all be sorted out the moment we run into
    the endswitch statement.
*/
    AC.SwitchLevel++;
    if ( AC.SwitchInArray >= AC.MaxSwitch ) DoubleSwitchBuffers();
    AC.SwitchHeap[AC.SwitchLevel] = AC.SwitchInArray;
    sw = AC.SwitchArray + AC.SwitchInArray;

    sw->iflevel = AC.IfLevel;
    sw->whilelevel = AC.WhileLevel;
    sw->nestingsum = NestingChecksum();

    Add4Com(TYPESWITCH,numdollar,AC.SwitchInArray);

    AC.SwitchInArray++;
    return(0);
}

/*
    #] CoSwitch :
    #[ CoCase :
*/

int CoCase(UBYTE *s)
{
    SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
    WORD x = 0, sign = 1;
    while ( *s == ',' ) s++;
    SKIPBLANKS(s);
    while ( *s == '-' || *s == '+' ) {
        if ( *s == '-' ) sign = -sign;
        s++;
    }
    while ( FG.cTable[*s] == 1 ) { x = 10*x + *s++ - '0'; }
    x = sign*x;

    if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
        || sw->nestingsum != NestingChecksum() ) {
        MesPrint("&Illegal nesting of switch/case/default with
 if/while/repeat/loop/argument/term/...");
        return(-1);
    }
/*
    Now add a case to the table with the current 'address'.
*/
    if ( sw->numcases >= sw->tablesize ) {
        int i;
        SWITCHTABLE *newtable;
        WORD newsize;
        if ( sw->tablesize == 0 ) newsize = 10;
        else                      newsize = 2*sw->tablesize;
        newtable = (SWITCHTABLE *)Malloc1(newsize*sizeof(SWITCHTABLE),"Switch table");
        if ( sw->table ) {
            for ( i = 0; i < sw->tablesize; i++ ) newtable[i] = sw->table[i];
            M_free(sw->table,"Switch table");
        }
        sw->table = newtable;
        sw->tablesize = newsize;
    }
    if ( sw->numcases == 0 ) { sw->mincase = sw->maxcase = x; }
    else if ( x > sw->maxcase ) sw->maxcase = x;
    else if ( x < sw->mincase ) sw->mincase = x;
    sw->table[sw->numcases].ncase = x;
    sw->table[sw->numcases].value = cbuf[AC.cbufnum].numlhs;
    sw->table[sw->numcases].compbuffer = AC.cbufnum;
    sw->numcases++;
    return(0);
}

/*
    #] CoCase :
    #[ CoBreak :
*/

int CoBreak(UBYTE *s)
{
/*
    This involves a 'postponed' jump to the end. This can be done
    in a special routine during execution.
    That routine should also pop the switch level.
*/
    SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
    if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
        || sw->nestingsum != NestingChecksum() ) {
        MesPrint("&Illegal nesting of switch/case/default with
```

```
if/while/repeat/loop/argument/term/...");
        return(-1);
    }
    if ( *s ) {
        MesPrint("&No parameters allowed in Break statement");
        return(-1);
    }
    Add3Com(TYPEENDSWITCH,AC.SwitchHeap[AC.SwitchLevel]);
    return(0);
}

/*
    #] CoBreak :
    #[ CoDefault :
*/

int CoDefault(UBYTE *s)
{
/*
    A bit like case, except that the address gets stored directly in the
    SWITCH struct.
*/
    SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
    if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
        || sw->nestingsum != NestingChecksum() ) {
        MesPrint("&Illegal nesting of switch/case/default with
if/while/repeat/loop/argument/term/...");
        return(-1);
    }
    if ( *s ) {
        MesPrint("&No parameters allowed in Default statement");
        return(-1);
    }
    sw->defaultcase.ncase = 0;
    sw->defaultcase.value = cbuf[AC.cbufnum].numlhs;
    sw->defaultcase.compbuffer = AC.cbufnum;
    return(0);
}

/*
    #] CoDefault :
    #[ CoEndSwitch :
*/

int CoEndSwitch(UBYTE *s)
{
/*
    We store this address in the SWITCH struct.
    Next we sort the table by ncase.
    Then we decide whether the table is DENSE or SPARSE.
    If it is dense we change the allocation and spread the cases is necessary.
    Finally we pop levels.
*/
    SWITCH *sw = AC.SwitchArray + AC.SwitchHeap[AC.SwitchLevel];
    WORD i;
    WORD totcases = sw->maxcase-sw->mincase+1;
    while ( *s == ',' ) s++;
    SKIPBLANKS(s)
    if ( *s ) {
        MesPrint("&No parameters allowed in EndSwitch statement");
        return(-1);
    }
    if ( sw->iflevel != AC.IfLevel || sw->whilelevel != AC.WhileLevel
        || sw->nestingsum != NestingChecksum() ) {
        MesPrint("&Illegal nesting of switch/case/default with
if/while/repeat/loop/argument/term/...");
        return(-1);
    }
    if ( sw->defaultcase.value == 0 ) CoDefault(s);
    if ( totcases > sw->numcases*AM.jumpratio ) { /* The factor is experimental */
        sw->caseoffset = 0;
        sw->typetable = SPARSETABLE;
/*
        Now we need to sort sw->table
*/
        SwitchSplitMerge(sw->table,sw->numcases);
    }
    else {  /* DENSE */
        SWITCHTABLE *ntable;
        sw->caseoffset = sw->mincase;
        sw->typetable = DENSETABLE;
        ntable = (SWITCHTABLE *)Malloc1(totcases*sizeof(SWITCHTABLE),"Switch table");
        for ( i = 0; i < totcases; i++ ) {
            ntable[i].ncase = i+sw->caseoffset;
            ntable[i].value = sw->defaultcase.value;
            ntable[i].compbuffer = sw->defaultcase.compbuffer;
        }
```

```
        for ( i = 0; i < sw->numcases; i++ ) {
            ntable[sw->table[i].ncase-sw->caseoffset] = sw->table[i];
        }
        M_free(sw->table,"Switch table");
        sw->table = ntable;
        sw->numcases = totcases;
    }
    sw->endswitch.ncase = 0;
    sw->endswitch.value = cbuf[AC.cbufnum].numlhs;
    sw->endswitch.compbuffer = AC.cbufnum;
    if ( sw->defaultcase.value == 0 ) {
        sw->defaultcase = sw->endswitch;
    }
    Add3Com(TYPEENDSWITCH,AC.SwitchHeap[AC.SwitchLevel]);
/*
    Now we need to pop.
*/
    AC.SwitchLevel--;
    return(0);
}

/*
    #] CoEndSwitch :
    #[ CoSetUserFlag :
*/

int CoSetUserFlag(UBYTE *s)
{
    int error = 0;
    while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
    while ( *s && ( FG.cTable[*s] == 1 ) ) {
        int x = 0;
        while ( *s && ( FG.cTable[*s] == 1 ) ) x = 10*x+(*s++ - '0');
        if ( x < 1 || x > BITSINWORD ) {
            MesPrint("&Flag number %d outside the permitted range 1-%d.",BITSINWORD);
            error = 1;
        }
        else {
            Add3Com(TYPESETUSERFLAG,x-1);
        }
        while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
    }
    if ( *s ) {
        MesPrint("&Illegal character in SetUserFlag statement: %s",s);
        error = 1;
    }
    return(error);
}

/*
    #] CoSetUserFlag :
    #[ CoClearUserFlag :
*/

int CoClearUserFlag(UBYTE *s)
{
    int error = 0;
    while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
    while ( *s && ( FG.cTable[*s] == 1 ) ) {
        int x = 0;
        while ( *s && ( FG.cTable[*s] == 1 ) ) x = 10*x+(*s++ - '0');
        if ( x < 1 || x > BITSINWORD ) {
            MesPrint("&Flag number %d outside the permitted range 1-%d.",BITSINWORD);
            error = 1;
        }
        else {
            Add3Com(TYPECLEARUSERFLAG,x);
        }
        while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
    }
    if ( *s ) {
        MesPrint("&Illegal character in SetUserFlag statement: %s",s);
        error = 1;
    }
    return(error);
}

/*
    #] CoClearUserFlag :
    #[ CoCreateAllLoops :

    Syntax:
        CoCreateAllLoops,in-function,out-function,type-argument,ifnoloop;
    Types allowed:
        vector, index, symbol, snumber
    ifnoloop:
        ifnoloop=0  or ifnoloop=1
```

```
    in-function can be a tensor. In that case type can be only vector or index.
    out-function can be a tensor. In that case type can be only vector or index.
*/

int CoCreateAllLoops(UBYTE *s)
{
    GETIDENTITY
    UBYTE *inname, *outname, *stype, c;
    WORD infun, outfun, x, type, tensorflag, typenum;
    WORD *WorkSave, *to;
    while ( *s == ',' || *s == ' ' ) s++;
    inname = s; s = SkipAName(s);
    c = *s; *s = 0;
    if ( ( ( type = GetName(AC.varnames,inname,&infun,WITHAUTO) ) != CFUNCTION )
    || ( ( functions[infun].spec != 0 ) && ( functions[infun].spec != TENSORFUNCTION ) ) ) {
        MesPrint("&%s should be a regular function or a tensor.",inname);
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&infun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        return(1);
    }
    infun += FUNCTION;
    *s++ = c;
    while ( *s == ',' || *s == ' ' ) s++;
    outname = s; s = SkipAName(s);
    c = *s; *s = 0;
    if ( ( ( type = GetName(AC.varnames,outname,&outfun,WITHAUTO) ) != CFUNCTION )
    || ( ( functions[outfun].spec != 0 ) && ( functions[outfun].spec != TENSORFUNCTION ) ) ) {
        MesPrint("&%s should be a regular function or a tensor.",outname);
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&outfun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        return(1);
    }
    outfun += FUNCTION;
    *s++ = c;
    if ( functions[infun].spec == TENSORFUNCTION ||
         functions[outfun].spec == TENSORFUNCTION ) tensorflag = 1;
    else tensorflag = 0;
/*
    Now the type: type=....
*/
    while ( *s == ',' || *s == ' ' ) s++;
    stype = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( StrICmp(stype,(UBYTE *)"type") != 0 || c != '=' ) {
        MesPrint("&In CreateAllLoops statement: expected type=vartype.");
        return(1);
    }
    *s++ = c;
    stype = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( StrICmp(stype,(UBYTE *)"vector") == 0 ) {
        typenum = -VECTOR;
    }
    else if ( StrICmp(stype,(UBYTE *)"index") == 0 ) {
        typenum = -INDEX;
    }
    else if ( StrICmp(stype,(UBYTE *)"symbol") == 0 ) {
        if ( tensorflag ) goto notintensor;
        typenum = -SYMBOL;
    }
    else if ( StrICmp(stype,(UBYTE *)"snumber") == 0 ) {
        if ( tensorflag ) goto notintensor;
        typenum = -SNUMBER;
    }
    else {
        MesPrint("&Unknown/not allowed variable type in CreateAllLoops: %s",stype);
        return(1);
    }
    *s = c;
    while ( *s == ',' || *s == ' ' ) s++;

    stype = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( StrICmp(stype,(UBYTE *)"ifnoloop") != 0 || c != '=' ) {
        MesPrint("&Unrecognised option in CreateAllLoops statement: %s",stype);
        return(1);
    }
    *s++ = c;
    x = -1;
    if ( FG.cTable[*s] == 1 ) {
```

```
        x = 0;
        do { x = 10*x + (*s++-'0'); } while (FG.cTable[*s] == 1);
    }
    if ( x != 0 && x != 1 ) {
        MesPrint("&Only options allowed for ifnoloop are 0 or 1.");
        return(1);
    }
    WorkSave = to = AT.WorkPointer;
    *to++ = TYPEALLLOOPS;
    *to++ = 6;
    *to++ = infun;
    *to++ = outfun;
    *to++ = typenum;
    *to++ = x;

    AddNtoL(WorkSave[1],WorkSave);

    return(0);
notintensor:
    MesPrint("&Variable type not allowed in tensors: %s",stype);
    return(1);
}

/*
    #] CoCreateAllLoops :
    #[ CoCreateAllPaths :

    Syntax:
        CreateAllPaths,end-function,intermediate-function,out-function,type-argument,ifnopath;
    Types allowed:
        vector, index, symbol, snumber
    ifnoloop:
        ifnoloop=0  or ifnoloop=1
    in-function can be a tensor. In that case type can be only vector or index.
    out-function can be a tensor. In that case type can be only vector or index.
*/

int CoCreateAllPaths(UBYTE *s)
{
    GETIDENTITY
    UBYTE *endname,*inname, *outname, *stype, c;
    WORD endfun, infun, outfun, x, type, tensorflag, typenum;
    WORD *WorkSave, *to;
    while ( *s == ',' || *s == ' ' ) s++;
    endname = s; s = SkipAName(s);
    c = *s; *s = 0;
    if ( ( ( type = GetName(AC.varnames,endname,&endfun,WITHAUTO) ) != CFUNCTION )
    || ( ( functions[endfun].spec != 0 ) && ( functions[endfun].spec != TENSORFUNCTION ) ) ) {
        MesPrint("&%s should be a regular function or a tensor.",endname);
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&endfun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        return(1);
    }
    endfun += FUNCTION;
    *s++ = c;
    while ( *s == ',' || *s == ' ' ) s++;
    inname = s; s = SkipAName(s);
    c = *s; *s = 0;
    if ( ( ( type = GetName(AC.varnames,inname,&infun,WITHAUTO) ) != CFUNCTION )
    || ( ( functions[infun].spec != 0 ) && ( functions[infun].spec != TENSORFUNCTION ) ) ) {
        MesPrint("&%s should be a regular function or a tensor.",inname);
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&infun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        return(1);
    }
    infun += FUNCTION;
    *s++ = c;
    while ( *s == ',' || *s == ' ' ) s++;
    outname = s; s = SkipAName(s);
    c = *s; *s = 0;
    if ( ( ( type = GetName(AC.varnames,outname,&outfun,WITHAUTO) ) != CFUNCTION )
    || ( ( functions[outfun].spec != 0 ) && ( functions[outfun].spec != TENSORFUNCTION ) ) ) {
        MesPrint("&%s should be a regular function or a tensor.",outname);
        if ( type < 0 ) {
            if ( GetName(AC.exprnames,s,&outfun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
        }
        return(1);
    }
    outfun += FUNCTION;
    *s++ = c;
    if ( functions[infun].spec == TENSORFUNCTION ||
         functions[outfun].spec == TENSORFUNCTION ) tensorflag = 1;
```

```
    else tensorflag = 0;
/*
    Now the type: type=....
*/
    while ( *s == ',' || *s == ' ' ) s++;
    stype = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( StrICmp(stype,(UBYTE *)"type") != 0 || c != '=' ) {
        MesPrint("&In CreateAllPaths statement: expected type=vartype.");
        return(1);
    }
    *s++ = c;
    stype = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( StrICmp(stype,(UBYTE *)"vector") == 0 ) {
        typenum = -VECTOR;
    }
    else if ( StrICmp(stype,(UBYTE *)"index") == 0 ) {
        typenum = -INDEX;
    }
    else if ( StrICmp(stype,(UBYTE *)"symbol") == 0 ) {
        if ( tensorflag ) goto notintensor;
        typenum = -SYMBOL;
    }
    else if ( StrICmp(stype,(UBYTE *)"snumber") == 0 ) {
        if ( tensorflag ) goto notintensor;
        typenum = -SNUMBER;
    }
    else {
        MesPrint("&Unknown/not allowed variable type in CreateAllPaths: %s",stype);
        return(1);
    }
    *s = c;
    while ( *s == ',' || *s == ' ' ) s++;

    stype = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( StrICmp(stype,(UBYTE *)"ifnopath") != 0 || c != '=' ) {
        MesPrint("&Unrecognised option in CreateAllPaths statement: %s",stype);
        return(1);
    }
    *s++ = c;
    x = -1;
    if ( FG.cTable[*s] == 1 ) {
        x = 0;
        do { x = 10*x + (*s++-'0'); } while (FG.cTable[*s] == 1);
    }
    if ( x != 0 && x != 1 ) {
        MesPrint("&Only options allowed for ifnopath are 0 or 1.");
        return(1);
    }
    WorkSave = to = AT.WorkPointer;
    *to++ = TYPEALLPATHS;
    *to++ = 7;
    *to++ = endfun;
    *to++ = infun;
    *to++ = outfun;
    *to++ = typenum;
    *to++ = x;

    AddNtoL(WorkSave[1],WorkSave);

    return(0);
notintensor:
    MesPrint("&CreateAllPaths: Variable type not allowed in tensors: %s",stype);
    return(1);
}

/*
    #] CoCreateAllPaths :
    #[ CoCreateAll :

    Syntax: subkey, two or three functions, type, ifnone
*/

int CoCreateAll(UBYTE *s)
{
    UBYTE *subkey;
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    subkey = s;
    while ( FG.cTable[*s] == 0 ) s++;
    if ( *s != ' ' && *s != ',' && *s != '\t' ) {
        MesPrint("&Illegal subkey in CoCreate statement.");
        return(1);
```

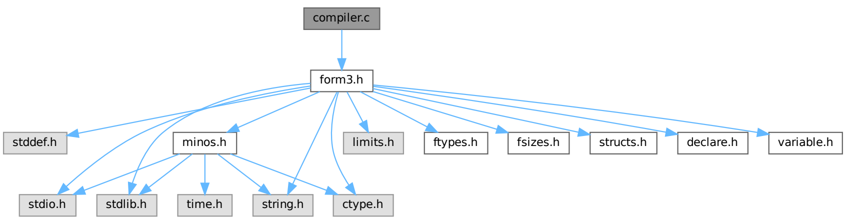
```
    }
    /* c = *s; */ *s++ = 0;
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( StrICmp(subkey,(UBYTE *)"loops") == 0 ) {
        return(CoCreateAllLoops(s));
    }
    else if ( StrICmp(subkey,(UBYTE *)"paths") == 0 ) {
        return(CoCreateAllPaths(s));
    }
/*
    else if ( StrICmp(subkey,(UBYTE *)"motics") == 0 ) {
    }
    else if ( StrICmp(subkey,(UBYTE *)"onepi") == 0 ) {
    }
    else if ( StrICmp(subkey,(UBYTE *)"cuts") == 0 ) {
    }
*/
    else {
        MesPrint("&Illegal subkey in CoCreate statement: %s.",subkey);
        return(1);
    }
}

/*
    #] CoCreateAll :
*/
```

4.11 compiler.c File Reference

`#include "form3.h"`

Include dependency graph for compiler.c:



Data Structures

- struct SuBbUf

Macros

- #define OPTION0 1
- #define OPTION1 2
- #define OPTION2 3
- #define REDUCESUBEXPBUFFERS

Typedefs

- typedef struct SuBbUf **SUBBUF**

Functions

- void inictable (void)
- KEYWORD * findcommand (UBYTE *in)
- int ParenthesesTest (UBYTE *sin)
- UBYTE * SkipAName (UBYTE *s)
- UBYTE * IsRHS (UBYTE *s, UBYTE c)
- int IsIdStatement (UBYTE *s)
- int CompileAlgebra (UBYTE *s, int leftright, WORD *prototype)
- int CompileStatement (UBYTE *in)
- int TestTables (void)
- int CompileSubExpressions (SBYTE *tokens)
- int CodeGenerator (SBYTE *tokens)
- int CompleteTerm (WORD *term, UWORD *numer, UWORD *denom, WORD nnum, WORD nden, int sign)
- int CodeFactors (SBYTE *tokens)
- WORD GenerateFactors (WORD n, WORD inc)

Variables

- int alfatable1 [27]
- SUBBUF * subexpbuffers = 0
- SUBBUF * topsubexpbuffers = 0
- LONG insubexpbuffers = 0

4.11.1 Detailed Description

The heart of the compiler. It contains the tables of statements. It finds the statements in the tables and calls the proper routines. For algebraic expressions it runs the compilation by first calling the tokenizer, splitting things into subexpressions and generating the code. There is a system for recognizing already existing subexpressions. This economizes on the length of the output.

Note: the compiler of FORM doesn't attempt to normalize the input. Hence x+1 and 1+x are different objects during compilation. Similarly (a+b-b) will not be simplified to (a).

Definition in file compiler.c.

4.11.2 Macro Definition Documentation

4.11.2.1 OPTION0

```
#define OPTION0 1
```

Definition at line 253 of file compiler.c.

4.11.2.2 OPTION1

```
#define OPTION1 2
```

Definition at line 254 of file compiler.c.

4.11.2.3 OPTION2

```
#define OPTION2 3
```

Definition at line 255 of file compiler.c.

4.11.2.4 REDUCESUBEXPBUFFERS

```
#define REDUCESUBEXPBUFFERS
```

Value:

```
{ if ( (topsubexpbuffers-subexpbuffers) > 256 ) {\
M_free(subexpbuffers,"subexpbuffers");\
subexpbuffers = (SUBBUF *)Malloc1(256*sizeof(SUBBUF),"subexpbuffers");\
topsubexpbuffers = subexpbuffers+256; } insubexpbuffers = 0; }
```

Definition at line 266 of file compiler.c.

4.11.3 Function Documentation

4.11.3.1 inictable()

```
void inictable (
        void )
```

Definition at line 310 of file compiler.c.

4.11.3.2 findcommand()

```
KEYWORD * findcommand (
        UBYTE * in )
```

Definition at line 336 of file compiler.c.

4.11.3.3 ParenthesesTest()

```
int ParenthesesTest (
        UBYTE * sin )
```

Definition at line 380 of file compiler.c.

4.11.3.4 SkipAName()

```
UBYTE * SkipAName (
        UBYTE * s )
```

Definition at line 430 of file compiler.c.

4.11.3.5 IsRHS()

```
UBYTE * IsRHS (
            UBYTE * s,
            UBYTE c )
```

Definition at line 458 of file compiler.c.

4.11.3.6 IsIdStatement()

```
int IsIdStatement (
            UBYTE * s )
```

Definition at line 504 of file compiler.c.

4.11.3.7 CompileAlgebra()

```
int CompileAlgebra (
            UBYTE * s,
            int leftright,
            WORD * prototype )
```

Definition at line 518 of file compiler.c.

4.11.3.8 CompileStatement()

```
int CompileStatement (
            UBYTE * in )
```

Definition at line 554 of file compiler.c.

4.11.3.9 TestTables()

```
int TestTables (
            void )
```

Definition at line 669 of file compiler.c.

4.11.3.10 CompileSubExpressions()

```
int CompileSubExpressions (
            SBYTE * tokens )
```

Definition at line 713 of file compiler.c.

4.11.3.11 CodeGenerator()

```
int CodeGenerator (
            SBYTE * tokens )
```

Definition at line 848 of file compiler.c.

4.11.3.12 CompleteTerm()

```
int CompleteTerm (
            WORD * term,
            UWORD * numer,
            UWORD * denom,
            WORD nnum,
            WORD nden,
            int sign )
```

Definition at line 1911 of file compiler.c.

4.11.3.13 CodeFactors()

```
int CodeFactors (
            SBYTE * tokens )
```

Definition at line 1948 of file compiler.c.

4.11.3.14 GenerateFactors()

```
WORD GenerateFactors (
            WORD n,
            WORD inc )
```

Definition at line 2245 of file compiler.c.

4.11.4 Variable Documentation

4.11.4.1 alfatable1

```
int alfatable1[27]
```

Definition at line 251 of file compiler.c.

4.11.4.2 subexpbuffers

```
SUBBUF* subexpbuffers = 0
```

Definition at line 262 of file compiler.c.

4.11.4.3 topsubexpbuffers

```
SUBBUF* topsubexpbuffers = 0
```

Definition at line 263 of file compiler.c.

4.11.4.4 insubexpbuffers

```
LONG insubexpbuffers = 0
```

Definition at line 264 of file compiler.c.

4.12 compiler.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ includes :
*/

#include "form3.h"

/*
    com1commands are the commands of which only part of the word has to
    be present. The order is rather important here.
    com2commands are the commands that must have their whole word match.
    here we can do a binary search.
    {[(
*/

static KEYWORD com1commands[] = {
     {"also",           (TFUN)CoIdOld,           STATEMENT,    PARTEST}
    ,{"abrackets",      (TFUN)CoAntiBracket,     TOOUTPUT,     PARTEST}
    ,{"antisymmetrize", (TFUN)CoAntiSymmetrize,  STATEMENT,    PARTEST}
    ,{"antibrackets",   (TFUN)CoAntiBracket,     TOOUTPUT,     PARTEST}
    ,{"brackets",       (TFUN)CoBracket,         TOOUTPUT,     PARTEST}
    ,{"cfunctions",     (TFUN)CoCFunction,       DECLARATION,  PARTEST|WITHAUTO}
    ,{"commuting",      (TFUN)CoCFunction,       DECLARATION,  PARTEST|WITHAUTO}
    ,{"compress",       (TFUN)CoCompress,        DECLARATION,  PARTEST}
    ,{"ctensors",       (TFUN)CoCTensor,         DECLARATION,  PARTEST|WITHAUTO}
    ,{"cyclesymmetrize",(TFUN)CoCycleSymmetrize, STATEMENT,    PARTEST}
    ,{"dimension",      (TFUN)CoDimension,       DECLARATION,  PARTEST}
    ,{"discard",        (TFUN)CoDiscard,         STATEMENT,    PARTEST}
    ,{"functions",      (TFUN)CoNFunction,       DECLARATION,  PARTEST|WITHAUTO}
    ,{"format",         (TFUN)CoFormat,          TOOUTPUT,     PARTEST}
    ,{"fixindex",       (TFUN)CoFixIndex,        DECLARATION,  PARTEST}
    ,{"global",         (TFUN)CoGlobal,          DEFINITION,   PARTEST}
```

```
    ,{"gfactorized",      (TFUN)CoGlobalFactorized, DEFINITION,    PARTEST}
    ,{"globalfactorized",(TFUN)CoGlobalFactorized,DEFINITION,    PARTEST}
    ,{"goto",             (TFUN)CoGoTo,             STATEMENT,     PARTEST}
    ,{"indexes",          (TFUN)CoIndex,            DECLARATION,   PARTEST|WITHAUTO}
    ,{"indices",          (TFUN)CoIndex,            DECLARATION,   PARTEST|WITHAUTO}
    ,{"identify",         (TFUN)CoId,               STATEMENT,     PARTEST}
    ,{"idnew",            (TFUN)CoIdNew,            STATEMENT,     PARTEST}
    ,{"idold",            (TFUN)CoIdOld,            STATEMENT,     PARTEST}
    ,{"local",            (TFUN)CoLocal,            DEFINITION,    PARTEST}
    ,{"lfactorized",      (TFUN)CoLocalFactorized,  DEFINITION,    PARTEST}
    ,{"localfactorized", (TFUN)CoLocalFactorized,  DEFINITION,    PARTEST}
    ,{"load",             (TFUN)CoLoad,             DECLARATION,   PARTEST}
    ,{"label",            (TFUN)CoLabel,            STATEMENT,     PARTEST}
    ,{"modulus",          (TFUN)CoModulus,          DECLARATION,   PARTEST}
    ,{"multiply",         (TFUN)CoMultiply,         STATEMENT,     PARTEST}
    ,{"nfunctions",       (TFUN)CoNFunction,        DECLARATION,   PARTEST|WITHAUTO}
    ,{"nprint",           (TFUN)CoNPrint,           TOOUTPUT,      PARTEST}
    ,{"ntensors",         (TFUN)CoNTensor,          DECLARATION,   PARTEST|WITHAUTO}
    ,{"nwrite",           (TFUN)CoNWrite,           DECLARATION,   PARTEST}
    ,{"print",            (TFUN)CoPrint,            MIXED,         0}
    ,{"redefine",         (TFUN)CoRedefine,         STATEMENT,     0}
    ,{"rcyclesymmetrize",(TFUN)CoRCycleSymmetrize,STATEMENT,     PARTEST}
    ,{"symbols",          (TFUN)CoSymbol,           DECLARATION,   PARTEST|WITHAUTO}
    ,{"save",             (TFUN)CoSave,             DECLARATION,   PARTEST}
    ,{"symmetrize",       (TFUN)CoSymmetrize,       STATEMENT,     PARTEST}
    ,{"tensors",          (TFUN)CoCTensor,          DECLARATION,   PARTEST|WITHAUTO}
    ,{"unittrace",        (TFUN)CoUnitTrace,        DECLARATION,   PARTEST}
    ,{"vectors",          (TFUN)CoVector,           DECLARATION,   PARTEST|WITHAUTO}
    ,{"write",            (TFUN)CoWrite,            DECLARATION,   PARTEST}
};

static KEYWORD com2commands[] = {
     {"antiputinside",    (TFUN)CoAntiPutInside,    STATEMENT,     PARTEST}
    ,{"apply",            (TFUN)CoApply,            STATEMENT,     PARTEST}
    ,{"aputinside",       (TFUN)CoAntiPutInside,    STATEMENT,     PARTEST}
    ,{"argexplode",       (TFUN)CoArgExplode,       STATEMENT,     PARTEST}
    ,{"argimplode",       (TFUN)CoArgImplode,       STATEMENT,     PARTEST}
    ,{"argtoextrasymbol",(TFUN)CoArgToExtraSymbol,STATEMENT,     PARTEST}
    ,{"argument",         (TFUN)CoArgument,         STATEMENT,     PARTEST}
    ,{"assign",           (TFUN)CoAssign,           STATEMENT,     PARTEST}
    ,{"auto",             (TFUN)CoAuto,             DECLARATION,   PARTEST}
    ,{"autodeclare",      (TFUN)CoAuto,             DECLARATION,   PARTEST}
    ,{"break",            (TFUN)CoBreak,            STATEMENT,     PARTEST}
    ,{"canonicalize",     (TFUN)CoCanonicalize,     STATEMENT,     PARTEST}
    ,{"case",             (TFUN)CoCase,             STATEMENT,     PARTEST}
    ,{"chainin",          (TFUN)CoChainin,          STATEMENT,     PARTEST}
    ,{"chainout",         (TFUN)CoChainout,         STATEMENT,     PARTEST}
    ,{"chisholm",         (TFUN)CoChisholm,         STATEMENT,     PARTEST}
    ,{"cleartable",       (TFUN)CoClearTable,       DECLARATION,   PARTEST}
    ,{"clearflag",        (TFUN)CoClearUserFlag,    STATEMENT,     PARTEST}
    ,{"collect",          (TFUN)CoCollect,          SPECIFICATION,PARTEST}
    ,{"commuteinset",     (TFUN)CoCommuteInSet,     DECLARATION,   PARTEST}
    ,{"contract",         (TFUN)CoContract,         STATEMENT,     PARTEST}
    ,{"copyspectator",    (TFUN)CoCopySpectator,    DEFINITION,    PARTEST}
    ,{"createall",        (TFUN)CoCreateAll,        STATEMENT,     PARTEST}
    ,{"createspectator", (TFUN)CoCreateSpectator,  DECLARATION,   PARTEST}
    ,{"ctable",           (TFUN)CoCTable,           DECLARATION,   PARTEST}
    ,{"deallocatetable", (TFUN)CoDeallocateTable,  DECLARATION,   PARTEST}
    ,{"default",          (TFUN)CoDefault,          STATEMENT,     PARTEST}
    ,{"delete",           (TFUN)CoDelete,           SPECIFICATION,PARTEST}
    ,{"denominators",     (TFUN)CoDenominators,     STATEMENT,     PARTEST}
    ,{"disorder",         (TFUN)CoDisorder,         STATEMENT,     PARTEST}
    ,{"do",               (TFUN)CoDo,               STATEMENT,     PARTEST}
    ,{"drop",             (TFUN)CoDrop,             SPECIFICATION,PARTEST}
    ,{"dropcoefficient", (TFUN)CoDropCoefficient,  STATEMENT,     PARTEST}
    ,{"dropsymbols",      (TFUN)CoDropSymbols,      STATEMENT,     PARTEST}
    ,{"else",             (TFUN)CoElse,             STATEMENT,     PARTEST}
    ,{"elseif",           (TFUN)CoElseIf,           STATEMENT,     PARTEST}
    ,{"emptyspectator",  (TFUN)CoEmptySpectator,   SPECIFICATION,PARTEST}
    ,{"endargument",      (TFUN)CoEndArgument,      STATEMENT,     PARTEST}
    ,{"enddo",            (TFUN)CoEndDo,            STATEMENT,     PARTEST}
    ,{"endif",            (TFUN)CoEndIf,            STATEMENT,     PARTEST}
    ,{"endinexpression", (TFUN)CoEndInExpression,  STATEMENT,     PARTEST}
    ,{"endinside",        (TFUN)CoEndInside,        STATEMENT,     PARTEST}
    ,{"endmodel",         (TFUN)CoEndModel,         DECLARATION,   PARTEST}
    ,{"endrepeat",        (TFUN)CoEndRepeat,        STATEMENT,     PARTEST}
    ,{"endswitch",        (TFUN)CoEndSwitch,        STATEMENT,     PARTEST}
    ,{"endterm",          (TFUN)CoEndTerm,          STATEMENT,     PARTEST}
    ,{"endwhile",         (TFUN)CoEndWhile,         STATEMENT,     PARTEST}
#ifdef WITHFLOAT
    ,{"evaluate",         (TFUN)CoEvaluate,         STATEMENT,     PARTEST}
#endif
    ,{"exit",             (TFUN)CoExit,             STATEMENT,     PARTEST}
    ,{"extrasymbols",     (TFUN)CoExtraSymbols,     DECLARATION,   PARTEST}
    ,{"factarg",          (TFUN)CoFactArg,          STATEMENT,     PARTEST}
    ,{"factdollar",       (TFUN)CoFactDollar,       STATEMENT,     PARTEST}
    ,{"factorize",        (TFUN)CoFactorize,        TOOUTPUT,      PARTEST}
```

```
    ,{"fill",            (TFUN)CoFill,            DECLARATION,  PARTEST}
    ,{"fillexpression",  (TFUN)CoFillExpression,  DECLARATION,  PARTEST}
    ,{"frompolynomial",  (TFUN)CoFromPolynomial,  STATEMENT,    PARTEST}
    ,{"funpowers",       (TFUN)CoFunPowers,       DECLARATION,  PARTEST}
    ,{"hide",            (TFUN)CoHide,            SPECIFICATION,PARTEST}
    ,{"if",              (TFUN)CoIf,              STATEMENT,    PARTEST}
    ,{"ifmatch",         (TFUN)CoIfMatch,         STATEMENT,    PARTEST}
    ,{"ifnomatch",       (TFUN)CoIfNoMatch,       STATEMENT,    PARTEST}
    ,{"ifnotmatch",      (TFUN)CoIfNoMatch,       STATEMENT,    PARTEST}
    ,{"inexpression",    (TFUN)CoInExpression,    STATEMENT,    PARTEST}
    ,{"inparallel",      (TFUN)CoInParallel,      SPECIFICATION,PARTEST}
    ,{"inside",          (TFUN)CoInside,          STATEMENT,    PARTEST}
    ,{"insidefirst",     (TFUN)CoInsideFirst,     DECLARATION,  PARTEST}
    ,{"intohide",        (TFUN)CoIntoHide,        SPECIFICATION,PARTEST}
    ,{"keep",            (TFUN)CoKeep,            SPECIFICATION,PARTEST}
    ,{"makeinteger",     (TFUN)CoMakeInteger,     STATEMENT,    PARTEST}
    ,{"many",            (TFUN)CoMany,            STATEMENT,    PARTEST}
    ,{"merge",           (TFUN)CoMerge,           STATEMENT,    PARTEST}
    ,{"metric",          (TFUN)CoMetric,          DECLARATION,  PARTEST}
    ,{"model",           (TFUN)CoModel,           DECLARATION,  PARTEST}
    ,{"moduleoption",    (TFUN)CoModuleOption,    ATENDOFMODULE,PARTEST}
    ,{"multi",           (TFUN)CoMulti,           STATEMENT,    PARTEST}
    ,{"multibracket",    (TFUN)CoMultiBracket,    STATEMENT,    PARTEST}
    ,{"ndrop",           (TFUN)CoNoDrop,          SPECIFICATION,PARTEST}
    ,{"nfactorize",      (TFUN)CoNFactorize,      TOOUTPUT,     PARTEST}
    ,{"nhide",           (TFUN)CoNoHide,          SPECIFICATION,PARTEST}
    ,{"nintohide",       (TFUN)CoNoIntoHide,      SPECIFICATION,PARTEST}
    ,{"normalize",       (TFUN)CoNormalize,       STATEMENT,    PARTEST}
    ,{"notinparallel",   (TFUN)CoNotInParallel,   SPECIFICATION,PARTEST}
    ,{"nskip",           (TFUN)CoNoSkip,          SPECIFICATION,PARTEST}
    ,{"ntable",          (TFUN)CoNTable,          DECLARATION,  PARTEST}
    ,{"nunfactorize",    (TFUN)CoNUnFactorize,    TOOUTPUT,     PARTEST}
    ,{"nunhide",         (TFUN)CoNoUnHide,        SPECIFICATION,PARTEST}
    ,{"off",             (TFUN)CoOff,             DECLARATION,  PARTEST}
    ,{"on",              (TFUN)CoOn,              DECLARATION,  PARTEST}
    ,{"once",            (TFUN)CoOnce,            STATEMENT,    PARTEST}
    ,{"only",            (TFUN)CoOnly,            STATEMENT,    PARTEST}
    ,{"particle",        (TFUN)CoParticle,        DECLARATION,  PARTEST}
    ,{"polyfun",         (TFUN)CoPolyFun,         DECLARATION,  PARTEST}
    ,{"polyratfun",      (TFUN)CoPolyRatFun,      DECLARATION,  PARTEST}
    ,{"pophide",         (TFUN)CoPopHide,         SPECIFICATION,PARTEST}
    ,{"print[]",         (TFUN)CoPrintB,          TOOUTPUT,     PARTEST}
    ,{"printtable",      (TFUN)CoPrintTable,      MIXED,        PARTEST}
    ,{"processbucketsize",(TFUN)CoProcessBucket,  DECLARATION,  PARTEST}
    ,{"propercount",     (TFUN)CoProperCount,     DECLARATION,  PARTEST}
    ,{"pushhide",        (TFUN)CoPushHide,        SPECIFICATION,PARTEST}
    ,{"putinside",       (TFUN)CoPutInside,       STATEMENT,    PARTEST}
    ,{"ratio",           (TFUN)CoRatio,           STATEMENT,    PARTEST}
    ,{"removespectator",(TFUN)CoRemoveSpectator,  SPECIFICATION,PARTEST}
    ,{"renumber",        (TFUN)CoRenumber,        STATEMENT,    PARTEST}
    ,{"repeat",          (TFUN)CoRepeat,          STATEMENT,    PARTEST}
    ,{"replaceloop",     (TFUN)CoReplaceLoop,     STATEMENT,    PARTEST}
    ,{"select",          (TFUN)CoSelect,          STATEMENT,    PARTEST}
    ,{"set",             (TFUN)CoSet,             DECLARATION,  PARTEST}
    ,{"setexitflag",     (TFUN)CoSetExitFlag,     STATEMENT,    PARTEST}
    ,{"setflag",         (TFUN)CoSetUserFlag,     STATEMENT,    PARTEST}
    ,{"shuffle",         (TFUN)CoMerge,           STATEMENT,    PARTEST}
    ,{"skip",            (TFUN)CoSkip,            SPECIFICATION,PARTEST}
    ,{"sort",            (TFUN)CoSort,            STATEMENT,    PARTEST}
    ,{"splitarg",        (TFUN)CoSplitArg,        STATEMENT,    PARTEST}
    ,{"splitfirstarg",   (TFUN)CoSplitFirstArg,   STATEMENT,    PARTEST}
    ,{"splitlastarg",    (TFUN)CoSplitLastArg,    STATEMENT,    PARTEST}
    ,{"stuffle",         (TFUN)CoStuffle,         STATEMENT,    PARTEST}
    ,{"sum",             (TFUN)CoSum,             STATEMENT,    PARTEST}
    ,{"switch",          (TFUN)CoSwitch,          STATEMENT,    PARTEST}
    ,{"table",           (TFUN)CoTable,           DECLARATION,  PARTEST}
    ,{"tablebase",       (TFUN)CoTableBase,       DECLARATION,  PARTEST}
    ,{"tb",              (TFUN)CoTableBase,       DECLARATION,  PARTEST}
    ,{"term",            (TFUN)CoTerm,            STATEMENT,    PARTEST}
    ,{"testuse",         (TFUN)CoTestUse,         STATEMENT,    PARTEST}
    ,{"threadbucketsize",(TFUN)CoThreadBucket,    DECLARATION,  PARTEST}
#ifdef WITHFLOAT
    ,{"tofloat",         (TFUN)CoToFloat,         STATEMENT,    PARTEST}
#endif
    ,{"topolynomial",    (TFUN)CoToPolynomial,    STATEMENT,    PARTEST}
#ifdef WITHFLOAT
    ,{"torat",           (TFUN)CoToRat,           STATEMENT,    PARTEST}
    ,{"torational",      (TFUN)CoToRat,           STATEMENT,    PARTEST}
#endif
    ,{"tospectator",     (TFUN)CoToSpectator,     STATEMENT,    PARTEST}
    ,{"totensor",        (TFUN)CoToTensor,        STATEMENT,    PARTEST}
    ,{"tovector",        (TFUN)CoToVector,        STATEMENT,    PARTEST}
    ,{"trace4",          (TFUN)CoTrace4,          STATEMENT,    PARTEST}
    ,{"tracen",          (TFUN)CoTraceN,          STATEMENT,    PARTEST}
    ,{"transform",       (TFUN)CoTransform,       STATEMENT,    PARTEST}
    ,{"tryreplace",      (TFUN)CoTryReplace,      STATEMENT,    PARTEST}
    ,{"unfactorize",     (TFUN)CoUnFactorize,     TOOUTPUT,     PARTEST}
```

```
    ,{"unhide",         (TFUN)CoUnHide,          SPECIFICATION,PARTEST}
    ,{"vertex",         (TFUN)CoVertex,          DECLARATION,  PARTEST}
    ,{"while",          (TFUN)CoWhile,           STATEMENT,    PARTEST}
};

int alfatable1[27];

#define OPTION0 1
#define OPTION1 2
#define OPTION2 3

typedef struct SuBbUf {
    WORD    subexpnum;
    WORD    buffernum;
} SUBBUF;

SUBBUF *subexpbuffers = 0;
SUBBUF *topsubexpbuffers = 0;
LONG insubexpbuffers = 0;

#define REDUCESUBEXPBUFFERS { if ( (topsubexpbuffers-subexpbuffers) > 256 ) {\
    M_free(subexpbuffers,"subexpbuffers");\
    subexpbuffers = (SUBBUF *)Malloc1(256*sizeof(SUBBUF),"subexpbuffers");\
    topsubexpbuffers = subexpbuffers+256; } insubexpbuffers = 0; }

#if defined(ILP32)

#define PUTNUMBER128(t,n) { if ( n >= 16384 ) { \
                *t++ = n/(128*128); *t++ = (n/128)%128; *t++ = n%128; } \
            else if ( n >= 128 ) { *t++ = n/128; *t++ = n%128; }      \
            else *t++ = n; }
#define PUTNUMBER100(t,n) { if ( n >= 10000 ) { \
                *t++ = n/10000; *t++ = (n/100)%100; *t++ = n%100; } \
            else if ( n >= 100 ) { *t++ = n/100; *t++ = n%100; }    \
            else *t++ = n; }

#elif ( defined(LLP64) || defined(LP64) )

#define PUTNUMBER128(t,n) { if ( n >= 2097152 ) { \
                *t++ = ((n/128)/128)/128; *t++ = ((n/128)/128)%128; *t++ = (n/128)%128; *t++ = n%128; } \
            else if ( n >= 16384 ) { \
                *t++ = n/(128*128); *t++ = (n/128)%128; *t++ = n%128; } \
            else if ( n >= 128 ) { *t++ = n/128; *t++ = n%128; }      \
            else *t++ = n; }
#define PUTNUMBER100(t,n) { if ( n >= 1000000 ) { \
                *t++ = ((n/100)/100)/100; *t++ = ((n/100)/100)%100; *t++ = (n/100)%100; *t++ = n%100; } \
            else if ( n >= 10000 ) { \
                *t++ = n/10000; *t++ = (n/100)%100; *t++ = n%100; } \
            else if ( n >= 100 ) { *t++ = n/100; *t++ = n%100; }    \
            else *t++ = n; }

#endif

/*
    )]}
    #] includes :
    #[ Compiler :
        #[ inictable :

        Routine sets the table for 1-st characters that allow a faster
        start in the search in table 1 which should be sequential.
        Search in table 2 can be binary.
*/

void inictable(void)
{
    KEYWORD *k = com1commands;
    int i, j, ksize;
    ksize = sizeof(com1commands)/sizeof(KEYWORD);
    j = 0;
    alfatable1[0] = 0;
    for ( i = 0; i < 26; i++ ) {
        while ( j < ksize && k[j].name[0] == 'a'+i ) j++;
        alfatable1[i+1] = j;
    }
}

/*
        #] inictable :
        #[ findcommand :

        Checks whether a command is in the command table.
        If so a pointer to the table element is returned.
        If not we return 0.
        Note that when a command is not in the table, we have
```

```
        to test whether it is an id command without id. It should
        then have the structure pattern = rhs. This should be done
        in the calling routine.
*/

KEYWORD *findcommand(UBYTE *in)
{
    int hi, med, lo, i;
    UBYTE *s, c;
    s = in;
    while ( FG.cTable[*s] <= 1 ) s++;
    if ( s > in && *s == '[' && s[1] == ']' ) s += 2;
    if ( *s ) { c = *s; *s = 0; }
    else c = 0;
/*
    First do a binary search in the second table
*/
    lo = 0;
    hi = sizeof(com2commands)/sizeof(KEYWORD)-1;
    do {
        med = ( hi + lo ) / 2;
        i = StrICmp(in,(UBYTE *)com2commands[med].name);
        if ( i == 0 ) { if ( c ) *s = c; return(com2commands+med); }
        if ( i < 0 ) hi = med-1;
        else         lo = med+1;
    } while ( hi >= lo );
/*
    Now do a 'hashed' search in the first table. It is sequential.
*/
    i = tolower(*in) - 'a';
    med = alfatable1[i];
    hi  = alfatable1[i+1];
    while ( med < hi ) {
        if ( StrICont(in,(UBYTE *)com1commands[med].name) == 0 )
            { if ( c ) *s = c; return(com1commands+med); }
        med++;
    }
    if ( c ) *s = c;
/*
    Unrecognized. Too bad!
*/
    return(0);
}

/*
        #] findcommand :
        #[ ParenthesesTest :
*/

int ParenthesesTest(UBYTE *sin)
{
    WORD L1 = 0, L2 = 0, L3 = 0;
    UBYTE *s = sin;
    while ( *s ) {
        if ( *s == '[' ) L1++;
        else if ( *s == ']' ) {
            L1--;
            if ( L1 < 0 ) { MesPrint("&Unmatched []"); return(1); }
        }
        s++;
    }
    if ( L1 > 0 ) { MesPrint("&Unmatched []"); return(1); }
    s = sin;
    while ( *s ) {
        if ( *s == '[' ) SKIPBRA1(s)
        else if ( *s == '(' ) { L2++; s++; }
        else if ( *s == ')' ) {
            L2--; s++;
            if ( L2 < 0 ) { MesPrint("&Unmatched ()"); return(1); }
        }
        else s++;
    }
    if ( L2 > 0 ) { MesPrint("&Unmatched ()"); return(1); }
    s = sin;
    while ( *s ) {
        if ( *s == '[' ) SKIPBRA1(s)
        else if ( *s == '[' ) SKIPBRA4(s)
        else if ( *s == '{' ) { L3++; s++; }
        else if ( *s == '}' ) {
            L3--; s++;
            if ( L3 < 0 ) { MesPrint("&Unmatched {}"); return(1); }
        }
        else s++;
    }
    if ( L3 > 0 ) { MesPrint("&Unmatched {}"); return(1); }
    return(0);
}
```

```
/*
  	#] ParenthesesTest :
  	#[ SkipAName :

  	Skips a name and gives a pointer to the object after the name.
  	If there is not a proper name, it returns a zero pointer.
  	In principle the brackets match already, so the `if ( *s == 0 )'
  	code is not really needed, but you never know how the program
  	is extended later.
*/

UBYTE *SkipAName(UBYTE *s)
{
	UBYTE *t = s;
	if ( *s == '[' ) {
		SKIPBRA1(s)
		if ( *s == 0 ) {
			MesPrint("&Illegal name: '%s'",t);
			return(0);
		}
		s++;
	}
	else if ( FG.cTable[*s] == 0 || *s == '_' || *s == '$' ) {
		if ( *s == '$' ) s++;
		while ( FG.cTable[*s] <= 1 ) s++;
		if ( *s == '_' ) s++;
	}
	else {
		MesPrint("&Illegal name: '%s'",t);
		return(0);
	}
	return(s);
}

/*
  	#] SkipAName :
  	#[ IsRHS :
*/

UBYTE *IsRHS(UBYTE *s, UBYTE c)
{
	while ( *s && *s != c ) {
		if ( *s == '[' ) {
			SKIPBRA1(s);
			if ( *s != ']' ) {
				MesPrint("&Unmatched []");
				return(0);
			}
		}
		else if ( *s == '{' ) {
			SKIPBRA2(s);
			if ( *s != '}' ) {
				MesPrint("&Unmatched {}");
				return(0);
			}
		}
		else if ( *s == '(' ) {
			SKIPBRA3(s);
			if ( *s != ')' ) {
				MesPrint("&Unmatched ()");
				return(0);
			}
		}
		else if ( *s == ')' ) {
			MesPrint("&Unmatched ()");
			return(0);
		}
		else if ( *s == '}' ) {
			MesPrint("&Unmatched {}");
			return(0);
		}
		else if ( *s == ']' ) {
			MesPrint("&Unmatched []");
			return(0);
		}
		s++;
	}
	return(s);
}

/*
  	#] IsRHS :
  	#[ IsIdStatement :
*/

int IsIdStatement(UBYTE *s)
```

```
{
    DUMMYUSE(s);
    return(0);
}

/*
        #] IsIdStatement :
        #[ CompileAlgebra :

        Returns either the number of the main level RHS (>= 0)
        or an error code (< 0)
*/

int CompileAlgebra(UBYTE *s, int leftright, WORD *prototype)
{
    GETIDENTITY
    int error;
    WORD *oldproto = AC.ProtoType;
    AC.ProtoType = prototype;
    if ( AC.TokensWriteFlag ) {
        MesPrint("To tokenize: %s",s);
        error = tokenize(s,leftright);
        MesPrint("  The contents of the token buffer are:");
        WriteTokens(AC.tokens);
    }
    else error = tokenize(s,leftright);
    if ( error == 0 ) {
        AR.Eside = leftright;
        AC.CompileLevel = 0;
        if ( leftright == LHSIDE ) { AC.DumNum = AR.CurDum = 0; }
        error = CompileSubExpressions(AC.tokens);
        REDUCESUBEXPBUFFERS
    }
    else {
        AC.ProtoType = oldproto;
        return(-1);
    }
    AC.ProtoType = oldproto;
    if ( error < 0 ) return(-1);
    else if ( leftright == LHSIDE ) return(cbuf[AC.cbufnum].numlhs);
    else                            return(cbuf[AC.cbufnum].numrhs);
}

/*
        #] CompileAlgebra :
        #[ CompileStatement :

*/

int CompileStatement(UBYTE *in)
{
    KEYWORD *k;
    UBYTE *s;
    int error1 = 0, error2;
    /* A.iStatement = */ s = in;
    if ( *s == 0 ) return(0);
    if ( *s == '$' ) {
        k = findcommand((UBYTE *)"assign");
    }
    else {
        if ( ( k = findcommand(s) ) == 0 && IsIdStatement(s) == 0 ) {
            MesPrint("&Unrecognized statement %s",s);
            return(1);
        }
        if ( k == 0 ) { /* Id statement without id. Note: id must be in table */
            k = com1commands + alfatable1['i'-'a'];
            while ( k->name[1] != 'd' || k->name[2] ) k++;
        }
        else {
            while ( FG.cTable[*s] <= 1 ) s++;
            if ( s > in && *s == '[' && s[1] == ']' ) s += 2;
/*
            The next statement is rather mysterious
            It is undone in DoPrint and CoMultiply, but it also causes effects
            in other (wrong) statements like dimension -4; or Trace4 -1;
            The code in pre.c (LoadStatement) has been changed 8-sep-2009
            to force a comma after the keyword. This means that the
            'mysterious' line is automatically inactive. Hence it is taken out.

            if ( *s == '+' || *s == '-' ) s++;
*/
            if ( *s == ',' ) s++;
        }
    }
/*
    First the test on the order of the statements.
    This is relatively new (2.2c) and may cause some problems with old
```

```
    programs. Hence the first error message should explain!
*/
    if ( AP.PreAssignFlag == 0 && AM.OldOrderFlag == 0 ) {
     if ( AP.PreInsideLevel ) {
      if ( k->type != STATEMENT && k->type != MIXED ) {
        MesPrint("&Only executable and print statements are allowed in an %#inside/%#endinside construction");
        return(-1);
      }
     }
     else {
      if ( ( AC.compiletype == DECLARATION || AC.compiletype == SPECIFICATION )
      && ( k->type == STATEMENT || k->type == DEFINITION || k->type == TOOUTPUT ) ) {
        if ( AC.tablecheck == 0 ) {
            AC.tablecheck = 1;
            if ( TestTables() ) error1 = 1;
        }
      }
      if ( k->type == MIXED ) {
        if ( AC.compiletype <= DEFINITION ) {
            AC.compiletype = STATEMENT;
        }
      }
      else if ( k->type == MIXED2 ) {}
      else if ( k->type > AC.compiletype ) {
        if ( StrCmp((UBYTE *)(k->name),(UBYTE *)"format") != 0 )
            AC.compiletype = k->type;
      }
      else if ( k->type < AC.compiletype ) {
        switch ( k->type ) {
            case DECLARATION:
                MesPrint("&Declaration out of order");
                MesPrint("&  %s",in);
                break;
            case DEFINITION:
                MesPrint("&Definition out of order");
                MesPrint("&  %s",in);
                break;
            case SPECIFICATION:
                MesPrint("&Specification out of order");
                MesPrint("&  %s",in);
                break;
            case STATEMENT:
                MesPrint("&Statement out of order");
                break;
            case TOOUTPUT:
                MesPrint("&Output control statement out of order");
                MesPrint("&  %s",in);
                break;
        }
        AC.compiletype = k->type;
        if ( AC.firstctypemessage == 0 ) {
            MesPrint("&Proper order inside a module is:");
            MesPrint("Declarations, specifications, definitions, statements, output control statements");
            AC.firstctypemessage = 1;
        }
        error1 = 1;
      }
     }
    }
/*
    Now we execute the tests that are prescribed by the flags.
*/
    if ( AC.AutoDeclareFlag && ( ( k->flags & WITHAUTO ) == 0 ) ) {
        MesPrint("&Illegal type of auto-declaration");
        return(1);
    }
    if ( ( ( k->flags & PARTEST ) != 0 ) && ParenthesesTest(s) ) return(1);
    error2 = (*k->func)(s);
    if ( error2 == 0 ) return(error1);
    return(error2);
}

/*
        #] CompileStatement :
        #[ TestTables :
*/

int TestTables(void)
{
    FUNCTIONS f = functions;
    TABLES t;
    WORD j;
    int error = 0, i;
    LONG x;
    i = NumFunctions + FUNCTION - MAXBUILTINFUNCTION - 1;
```

```
    f = f + MAXBUILTINFUNCTION - FUNCTION + 1;
    if ( AC.MustTestTable > 0 ) {
      while ( i > 0 ) {
        if ( ( t = f->tabl ) != 0 && t->strict > 0 && !t->sparse ) {
            for ( x = 0, j = 0; x < t->totind; x++ ) {
                if ( t->tablepointers[TABLEEXTENSION*x] < 0 ) j++;
            }
            if ( j > 0 ) {
                if ( j > 1 ) {
                    MesPrint("&In table %s there are %d unfilled elements",
                    AC.varnames->namebuffer+f->name,j);
                }
                else {
                    MesPrint("&In table %s there is one unfilled element",
                    AC.varnames->namebuffer+f->name);
                }
                error = 1;
            }
        }
        i--; f++;
      }
      AC.MustTestTable--;
    }
    return(error);
}

/*
        #] TestTables :
        #[ CompileSubExpressions :

        Now we attack the subexpressions from inside out.
        We try to see whether we had any of them already.
        We have to worry about adding the wildcard sum parameter
        to the prototype.
*/

int CompileSubExpressions(SBYTE *tokens)
{
    GETIDENTITY
    SBYTE *fill = tokens, *s = tokens, *t;
    WORD number[MAXNUMSIZE], *oldwork, *w1, *w2;
    int level, num, i, sumlevel = 0, sumtype = SYMTOSYM;
    int retval, error = 0;
/*
    Eliminate all subexpressions. They are marked by LPARENTHESIS,RPARENTHESIS
*/
    AC.CompileLevel++;
    while ( *s != TENDOFIT ) {
        if ( *s == TFUNOPEN ) {
            if ( fill < s ) *fill = TENDOFIT;
            t = fill - 1;
            while ( t >= tokens && t[0] >= 0 ) t--;
            if ( t >= tokens && *t == TFUNCTION ) {
                t++; i = 0; while ( *t >= 0 ) i = 128*i + *t++;
                if ( i == AM.sumnum || i == AM.sumpnum ) {
                    t = s + 1;
                    if ( *t == TSYMBOL || *t == TINDEX ) {
                        t++; i = 0; while ( *t >= 0 ) i = 128*i + *t++;
                        if ( s[1] == TINDEX ) {
                            i += AM.OffsetIndex;
                            sumtype = INDTOIND;
                        }
                        else sumtype = SYMTOSYM;
                        sumlevel = i;
                    }
                }
            }
            *fill++ = *s++;
        }
        else if ( *s == TFUNCLOSE ) { sumlevel = 0; *fill++ = *s++; }
        else if ( *s == LPARENTHESIS ) {
/*
            We must make an exception here.
            If the subexpression is just an integer, whatever its length,
            we should try to keep it.
            This is important when we have a function with an integer
            argument. In particular this is relevant for the MZV program.
*/
            t = s; level = 0;
            while ( level >= 0 ) {
                s++;
                if ( *s == LPARENTHESIS ) level++;
                else if ( *s == RPARENTHESIS ) level--;
                else if ( *s == TENDOFIT ) {
                    MesPrint("&Unbalanced subexpression parentheses");
                    return(-1);
                }
```

```
            }
            t++; *s = TENDOFIT;
            if ( sumlevel > 0 ) { /* Inside sum. Add wildcard to prototype */
                oldwork = w1 = AT.WorkPointer;
                w2 = AC.ProtoType;
                i = w2[1];
                while ( --i >= 0 ) *w1++ = *w2++;
                oldwork[1] += 4;
                *w1++ = sumtype; *w1++ = 4; *w1++ = sumlevel; *w1++ = sumlevel;
                w2 = AC.ProtoType; AT.WorkPointer = w1;
                AC.ProtoType = oldwork;
                num = CompileSubExpressions(t);
                AC.ProtoType = w2; AT.WorkPointer = oldwork;
            }
            else num = CompileSubExpressions(t);
            if ( num < 0 ) return(-1);
/*
            Note that the subexpression code should always fit.
            We had two parentheses and at least two bytes contents.
            There cannot be more than 2^21 subexpressions or we get outside
            this minimum. Ignoring this might lead to really rare and
            hard to find errors, years from now.
*/
            if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
                MesPrint("&More than %d subexpressions inside one expression",(WORD)MAXSUBEXPRESSIONS);
                Terminate(-1);
            }
            if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
                DoubleBuffer((void **)((void *)(&subexpbuffers))
                ,(void **)((void *)(&topsubexpbuffers)),sizeof(SUBBUF),"subexpbuffers");
            }
            subexpbuffers[insubexpbuffers].subexpnum = num;
            subexpbuffers[insubexpbuffers].buffernum = AC.cbufnum;
            num = insubexpbuffers++;
            *fill++ = TSUBEXP;
            i = 0;
            do { number[i++] = num & 0x7F; num >>= 7; } while ( num );
            while ( --i >= 0 ) *fill++ = (SBYTE)(number[i]);
            s++;
        }
        else if ( *s == TEMPTY ) s++;
        else *fill++ = *s++;
    }
    *fill = TENDOFIT;
/*
    At this stage there are no more subexpressions.
    Hence we can do the basic compilation.
*/
    if ( AC.CompileLevel == 1 && AC.ToBeInFactors ) {
        error = CodeFactors(tokens);
    }
    AC.CompileLevel--;
    retval = CodeGenerator(tokens);
    if ( error < 0 ) return(error);
    return(retval);
}

/*
        #] CompileSubExpressions :
        #[ CodeGenerator :

        This routine does the real code generation.
        It returns the number of the rhs subexpression.
        At this point we do not have to worry about subexpressions,
        sets, setelements, simple vs complicated function arguments
        simple vs complicated powers etc.

        The variable 'first' indicates whether we are starting a new term

        The major complication are the set elements of type set[n].
        We have marked them as TSETNUM,n,Ttype,setnum
        They go into
        SETSET,size,subterm,relocation list
        in which the subterm should be ready to become a regular
        subterm in which the sets have been replaced by their element
        The relocation list consists of pairs of numbers:
        1: offset in the subterm, 2: the symbol n.
        Note that such a subterm can be a whole function with its arguments.
        We use the variable inset to indicate that we have something going.
        The relocation list is collected in the top of the WorkSpace.
*/

static UWORD *CGscrat7 = 0;

int CodeGenerator(SBYTE *tokens)
{
```

```
    GETIDENTITY
    SBYTE *s = tokens, c;
    int i, sign = 1, first = 1, deno = 1, error = 0, minus, n, needarg, numexp, cc;
    int base, sumlevel = 0, sumtype = SYMTOSYM, firstsumarg, inset = 0;
    int funflag = 0, settype, x1, x2, mulflag = 0;
    WORD *t, *v, *r, *term, nnumerator, ndenominator, *oldwork, x3, y, nin;
    WORD *w1, *w2, *tsize = 0, *relo = 0;
    UWORD *numerator, *denominator, *innum;
    CBUF *C;
    POSITION position;
    WORD TMproto[SUBEXPSIZE];
/*
#ifdef WITHPTHREADS
    RENUMBER renumber;
#endif
*/
    RENUMBER renumber;
    if ( AC.TokensWriteFlag ) WriteTokens(tokens);
    if ( CGscrat7 == 0 )
        CGscrat7 = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(WORD),"CodeGenerator");
    AddRHS(AC.cbufnum,0);
    C = cbuf + AC.cbufnum;
    numexp = C->numrhs;
    C->NumTerms[numexp] = 0;
    C->numdum[numexp] = 0;
    oldwork = AT.WorkPointer;
    numerator = (UWORD *)(AT.WorkPointer);
    denominator = numerator + 2*AM.MaxTal;
    innum = denominator + 2*AM.MaxTal;
    term = (WORD *)(innum + 2*AM.MaxTal);
    AT.WorkPointer = term + AM.MaxTer/sizeof(WORD);
    if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
    cc = 0;
    t = term+1;
    numerator[0] = denominator[0] = 1;
    nnumerator = ndenominator = 1;
    while ( *s != TENDOFIT ) {
        if ( *s == TPLUS || *s == TMINUS ) {
            if ( first || mulflag ) { if ( *s == TMINUS ) sign = -sign; }
            else {
                *term = t-term;
                C->NumTerms[numexp]++;
                if ( cc && sign ) C->CanCommu[numexp]++;
                CompleteTerm(term,numerator,denominator,nnumerator,ndenominator,sign);
                first = 1; cc = 0; t = term + 1; deno = 1;
                numerator[0] = denominator[0] = 1;
                nnumerator = ndenominator = 1;
                if ( *s == TMINUS ) sign = -1;
                else sign = 1;
            }
            s++;
        }
        else {
            mulflag = first = 0; c = *s++;
            switch ( c ) {
            case TSYMBOL:
                x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
                if ( *s == TWILDCARD ) { s++; x1 += 2*MAXPOWER; }
                *t++ = SYMBOL; *t++ = 4; *t++ = x1;
                if ( inset ) *relo = 2;
TryPower:       if ( *s == TPOWER ) {
                    s++;
                    if ( *s == TMINUS ) { s++; deno = -deno; }
                    c = *s++;
                    base = ( c == TNUMBER ) ? 100: 128;
                    x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
                    if ( c == TSYMBOL ) {
                        if ( *s == TWILDCARD ) s++;
                        x2 += 2*MAXPOWER;
                    }
                    *t++ = deno*x2;
                }
                else *t++ = deno;
fin:            deno = 1;
                if ( inset ) {
                    while ( relo < AT.WorkTop ) *t++ = *relo++;
                    inset = 0; tsize[1] = t - tsize;
                }
                break;
            case TINDEX:
                x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
                *t++ = INDEX; *t++ = 3;
                if ( *s == TWILDCARD ) { s++; x1 += WILDOFFSET; }
                if ( inset ) { *t++ = x1; *relo = 2; }
                else           *t++ = x1 + AM.OffsetIndex;
                if ( t[-1] > AM.IndDum ) {
                    x1 = t[-1] - AM.IndDum;
```

```
                    if ( x1 > C->numdum[numexp] ) C->numdum[numexp] = x1;
                }
                goto fin;
            case TGENINDEX:
                *t++ = INDEX; *t++ = 3; *t++ = AC.DumNum+WILDOFFSET;
                deno = 1;
                break;
            case TVECTOR:
                x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
dovector:       if ( inset == 0 ) x1 += AM.OffsetVector;
                if ( *s == TWILDCARD ) { s++; x1 += WILDOFFSET; }
                if ( inset ) *relo = 2;
                if ( *s == TDOT ) {      /* DotProduct ? */
                    s++;
                    if ( *s == TSETNUM || *s == TSETDOL ) {
                        settype = ( *s == TSETDOL );
                        s++; x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
                        if ( settype ) x2 = -x2;
                        if ( inset == 0 ) {
                            tsize = t; *t++ = SETSET; *t++ = 0;
                            relo = AT.WorkTop;
                        }
                        inset += 2;
                        *--relo = x2; *--relo = 3;
                    }
                    if ( *s != TVECTOR && *s != TDUBIOUS ) {
                        MesPrint("&Illegally formed dotproduct");
                        error = 1;
                    }
                    s++; x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
                    if ( inset < 2 ) x2 += AM.OffsetVector;
                    if ( *s == TWILDCARD ) { s++; x2 += WILDOFFSET; }
                    *t++ = DOTPRODUCT; *t++ = 5; *t++ = x1; *t++ = x2;
                    goto TryPower;
                }
                else if ( *s == TFUNOPEN ) {
                    s++;
                    if ( *s == TSETNUM || *s == TSETDOL ) {
                        settype = ( *s == TSETDOL );
                        s++; x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
                        if ( settype ) x2 = -x2;
                        if ( inset == 0 ) {
                            tsize = t; *t++ = SETSET; *t++ = 0;
                            relo = AT.WorkTop;
                        }
                        inset += 2;
                        *--relo = x2; *--relo = 3;
                    }
                    if ( *s == TINDEX || *s == TDUBIOUS ) {
                        s++;
                        x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
                        if ( inset < 2 ) x2 += AM.OffsetIndex;
                        if ( *s == TWILDCARD ) { s++; x2 += WILDOFFSET; }
                        *t++ = VECTOR; *t++ = 4; *t++ = x1; *t++ = x2;
                        if ( t[-1] > AM.IndDum ) {
                            x2 = t[-1] - AM.IndDum;
                            if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                        }
                    }
                    else if ( *s == TGENINDEX ) {
                        *t++ = VECTOR; *t++ = 4; *t++ = x1;
                        *t++ = AC.DumNum + WILDOFFSET;
                    }
                    else if ( *s == TNUMBER || *s == TNUMBER1 ) {
                        base = ( *s == TNUMBER ) ? 100: 128;
                        s++;
                        x2 = 0; while ( *s >= 0 ) { x2 = x2*base + *s++; }
                        if ( x2 >= AM.OffsetIndex && inset < 2 ) {
                            MesPrint("&Fixed index in vector greater than %d",
                            AM.OffsetIndex);
                            return(-1);
                        }
                        *t++ = VECTOR; *t++ = 4; *t++ = x1; *t++ = x2;
                    }
                    else if ( *s == TVECTOR || ( *s == TMINUS && s[1] == TVECTOR ) ) {
                        if ( *s == TMINUS ) { s++; sign = -sign; }
                        s++;
                        x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
                        if ( inset < 2 ) x2 += AM.OffsetVector;
                        if ( *s == TWILDCARD ) { s++; x2 += WILDOFFSET; }
                        *t++ = DOTPRODUCT; *t++ = 5; *t++ = x1; *t++ = x2; *t++ = deno;
                    }
                    else {
                        MesPrint("&Illegal argument for vector");
                        return(-1);
                    }
                    if ( *s != TFUNCLOSE ) {
```

```
                        MesPrint("&Illegal argument for vector");
                        return(-1);
                    }
                    s++;
                }
                else {
                    if ( AC.DumNum ) {
                        *t++ = VECTOR; *t++ = 4; *t++ = x1;
                        *t++ = AC.DumNum + WILDOFFSET;
                    }
                    else {
                        *t++ = INDEX; *t++ = 3; *t++ = x1;
                    }
                }
                goto fin;
            case TDELTA:
                if ( *s != TFUNOPEN ) {
                    MesPrint("&d_ needs two arguments");
                    error = -1;
                }
                v = t; *t++ = DELTA; *t++ = 4;
                needarg = 2; x3 = x1 = -1;
                goto dotensor;
            case TFUNCTION:
                x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
                if ( x1 == AM.sumnum || x1 == AM.sumpnum ) sumlevel = x1;
                x1 += FUNCTION;
                if ( x1 == FIRSTBRACKET ) {
                    if ( s[0] == TFUNOPEN && s[1] == TEXPRESSION ) {
doexpr:                     s += 2;
                        *t++ = x1; *t++ = FUNHEAD+2; *t++ = 0;
                        if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
                                t[-1] |= MUSTCLEANPRF;
                        FILLFUN3(t)
                        x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
                        *t++ = -EXPRESSION; *t++ = x2;
/*
                        The next code is added to facilitate parallel processing
                        We need to call GetTable here to make sure all processors
                        have the same numbering of all variables.
*/
                        if ( Expressions[x2].status == STOREDEXPRESSION ) {
                            TMproto[0] = EXPRESSION;
                            TMproto[1] = SUBEXPSIZE;
                            TMproto[2] = x2;
                            TMproto[3] = 1;
                            { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
                            AT.TMaddr = TMproto;
                            PUTZERO(position);
/*
                            if ( (
#ifdef WITHPTHREADS
                                    renumber =
#endif
                                    GetTable(x2,&position,0) ) == 0 ) {
                                error = 1;
                                MesPrint("&Problems getting information about stored expression %s(1)"
                                ,EXPRNAME(x2));
                            }
#ifdef WITHPTHREADS
                            M_free(renumber->symb.lo,"VarSpace");
                            M_free(renumber,"Renumber");
#endif
*/
                            if ( ( renumber = GetTable(x2,&position,0) ) == 0 ) {
                                error = 1;
                                MesPrint("&Problems getting information about stored expression %s(1)"
                                ,EXPRNAME(x2));
                            }
                            if ( renumber->symb.lo != AN.dummyrenumlist )
                                M_free(renumber->symb.lo,"VarSpace");
                            M_free(renumber,"Renumber");
                            AR.StoreData.dirtyflag = 1;
                        }
                        if ( *s != TFUNCLOSE ) {
                            if ( x1 == FIRSTBRACKET )
                                MesPrint("&Problems with argument of FirstBracket_");
                            else if ( x1 == FIRSTTERM )
                                MesPrint("&Problems with argument of FirstTerm_");
                            else if ( x1 == CONTENTTERM )
                                MesPrint("&Problems with argument of FirstTerm_");
                            else if ( x1 == TERMSINEXPR )
                                MesPrint("&Problems with argument of TermsIn_");
                            else if ( x1 == SIZEOFFUNCTION )
                                MesPrint("&Problems with argument of SizeOf_");
                            else if ( x1 == NUMFACTORS )
                                MesPrint("&Problems with argument of NumFactors_");
```

```
                            else
                                MesPrint("&Problems with argument of FactorIn_");
                            error = 1;
                            while ( *s != TENDOFIT && *s != TFUNCLOSE ) s++;
                        }
                        if ( *s == TFUNCLOSE ) s++;
                        goto fin;
                    }
                }
                else if ( x1 == TERMSINEXPR || x1 == SIZEOFFUNCTION || x1 == FACTORIN
                 || x1 == NUMFACTORS || x1 == FIRSTTERM || x1 == CONTENTTERM ) {
                    if ( s[0] == TFUNOPEN && s[1] == TEXPRESSION ) goto doexpr;
                    if ( s[0] == TFUNOPEN && s[1] == TDOLLAR ) {
                        s += 2;
                        *t++ = x1; *t++ = FUNHEAD+2; *t++ = 0;
                        FILLFUN3(t)
                        x2 = 0; while ( *s >= 0 ) { x2 = x2*128 + *s++; }
                        *t++ = -DOLLAREXPRESSION; *t++ = x2;
                        if ( *s != TFUNCLOSE ) {
                            if ( x1 == TERMSINEXPR )
                                MesPrint("&Problems with argument of TermsIn_");
                            else if ( x1 == SIZEOFFUNCTION )
                                MesPrint("&Problems with argument of SizeOf_");
                            else if ( x1 == NUMFACTORS )
                                MesPrint("&Problems with argument of NumFactors_");
                            else
                                MesPrint("&Problems with argument of FactorIn_");
                            error = 1;
                            while ( *s != TENDOFIT && *s != TFUNCLOSE ) s++;
                        }
                        if ( *s == TFUNCLOSE ) s++;
                        goto fin;
                    }
                }
                x3 = x1;
                if ( inset && ( t-tsize == 2 ) ) x1 -= FUNCTION;
                if ( *s == TWILDCARD ) { x1 += WILDOFFSET; s++; }
                if ( functions[x3-FUNCTION].commute ) cc = 1;
                if ( *s != TFUNOPEN ) {
                    *t++ = x1; *t++ = FUNHEAD; *t++ = 0;
                    if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
                            t[-1] |= MUSTCLEANPRF;
                    FILLFUN3(t) sumlevel = 0; goto fin;
                }
                v = t; *t++ = x1; *t++ = FUNHEAD; *t++ = DIRTYFLAG;
                if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
                        t[-1] |= MUSTCLEANPRF;
                FILLFUN3(t)
                needarg = -1;
                if ( !inset && functions[x3-FUNCTION].spec >= TENSORFUNCTION ) {
dotensor:
                    do {
                        if ( needarg == 0 ) {
                            if ( x1 >= 0 ) {
                                x3 = x1;
                                if ( x3 >= FUNCTION+WILDOFFSET ) x3 -= WILDOFFSET;
                                MesPrint("&Too many arguments in function %s",
                                    VARNAME(functions,(x3-FUNCTION)) );
                            }
                            else
                                MesPrint("&d_ needs exactly two arguments");
                            error = -1;
                            needarg--;
                        }
                        else if ( needarg > 0 ) needarg--;
                        s++;
                        c = *s++;
                        if ( c == TMINUS && *s == TVECTOR ) { sign = -sign; c = *s++; }
                        base = ( c == TNUMBER ) ? 100: 128;
                        x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
                        if ( *s == TWILDCARD && c != TNUMBER ) { x2 += WILDOFFSET; s++; }
                        if ( c == TSETNUM || c == TSETDOL ) {
                            if ( c == TSETDOL ) x2 = -x2;
                            if ( inset == 0 ) {
                                w1 = t; t += 2; w2 = t;
                                while ( w1 > v ) *--w2 = *--w1;
                                tsize = v; relo = AT.WorkTop;
                                *v++ = SETSET; *v++ = 0;
                            }
                            inset = 2; *--relo = x2; *--relo = t - v;
                            c = *s++;
                            x2 = 0; while ( *s >= 0 ) x2 = 128*x2 + *s++;
                            switch ( c ) {
                                case TINDEX:
                                    *t++ = x2;
                                    if ( t[-1]+AM.OffsetIndex > AM.IndDum ) {
                                        x2 = t[-1]+AM.OffsetIndex - AM.IndDum;
```

```
                                        if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                                    }
                                    break;
                                case TVECTOR:
                                    *t++ = x2; break;
                                case TNUMBER1:
                                    if ( x2 >= 0 && x2 < AM.OffsetIndex ) {
                                        *t++ = x2; break;
                                    }
                                    /* fall through */
                                default:
                                    MesPrint("&Illegal type of set inside tensor");
                                    error = 1;
                                    *t++ = x2;
                                    break;
                            }
                        }
                        else { switch ( c ) {
                            case TINDEX:
                                if ( inset < 2 ) *t++ = x2 + AM.OffsetIndex;
                                else *t++ = x2;
                                if ( x2+AM.OffsetIndex > AM.IndDum ) {
                                    x2 = x2+AM.OffsetIndex - AM.IndDum;
                                    if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                                }
                                break;
                            case TGENINDEX:
                                *t++ = AC.DumNum + WILDOFFSET;
                                break;
                            case TVECTOR:
                                if ( inset < 2 ) *t++ = x2 + AM.OffsetVector;
                                else *t++ = x2;
                                break;
                            case TWILDARG:
                                *t++ = FUNNYWILD; *t++ = x2;
/*                              v[2] = 0; */
                                break;
                            case TDOLLAR:
                                *t++ = FUNNYDOLLAR; *t++ = x2;
                                break;
                            case TDUBIOUS:
                                if ( inset < 2 ) *t++ = x2 + AM.OffsetVector;
                                else *t++ = x2;
                                break;
                            case TSGAMMA:   /* Special gamma's */
                                if ( x3 != GAMMA ) {
                                    MesPrint("&5_,6_,7_ can only be used inside g_");
                                    error = -1;
                                }
                                *t++ = -x2;
                                break;
                            case TNUMBER:
                            case TNUMBER1:
                                if ( x2 >= AM.OffsetIndex && inset < 2 ) {
                                    MesPrint("&Value of constant index in tensor too large");
                                    error = -1;
                                }
                                *t++ = x2;
                                break;
                            default:
                                MesPrint("&Illegal object in tensor");
                                error = -1;
                                break;
                        }}
                        if ( inset >= 2 ) inset = 1;
                    } while ( *s == TCOMMA );
                }
                else {
dofunction:         firstsumarg = 1;
                    do {
                        unsigned int ux2;
                        s++;
                        c = *s++;
                        if ( c == TMINUS && ( *s == TVECTOR || *s == TNUMBER
                        || *s == TNUMBER1 || *s == TSUBEXP ) ) {
                            minus = 1; c = *s++;
                        }
                        else minus = 0;
                        base = ( c == TNUMBER ) ? 100: 128;
                        ux2 = 0; while ( *s >= 0 ) { ux2 = base*ux2 + *s++; }
                        x2 = ux2;  /* may cause an implementation-defined behaviour */
/*
        !!!!!!!!  What if it does not fit?
*/
                        if ( firstsumarg ) {
                            firstsumarg = 0;
                            if ( sumlevel > 0 ) {
```

```
                                if ( c == TSYMBOL ) {
                                    sumlevel = x2; sumtype = SYMTOSYM;
                                }
                                else if ( c == TINDEX ) {
                                    sumlevel = x2+AM.OffsetIndex; sumtype = INDTOIND;
                                    if ( sumlevel > AM.IndDum ) {
                                        x2 = sumlevel - AM.IndDum;
                                        if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                                    }
                                }
                            }
                        }
                        if ( *s == TWILDCARD ) {
                            if ( c == TSYMBOL ) x2 += 2*MAXPOWER;
                            else if ( c != TNUMBER ) x2 += WILDOFFSET;
                            s++;
                        }
                        switch ( c ) {
                            case TSYMBOL:
                                *t++ = -SYMBOL; *t++ = x2; break;
                            case TDOLLAR:
                                *t++ = -DOLLAREXPRESSION; *t++ = x2; break;
                            case TEXPRESSION:
                                *t++ = -EXPRESSION; *t++ = x2;
/*
                                The next code is added to facilitate parallel processing
                                We need to call GetTable here to make sure all processors
                                have the same numbering of all variables.
*/
                                if ( Expressions[x2].status == STOREDEXPRESSION ) {
                                    TMproto[0] = EXPRESSION;
                                    TMproto[1] = SUBEXPSIZE;
                                    TMproto[2] = x2;
                                    TMproto[3] = 1;
                                    { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
                                    AT.TMaddr = TMproto;
                                    PUTZERO(position);
/*
                                    if ( (
#ifdef WITHPTHREADS
                                        renumber =
#endif
                                            GetTable(x2,&position,0) ) == 0 ) {
                                        error = 1;
                                        MesPrint("&Problems getting information about stored
expression %s(2)"
                                        ,EXPRNAME(x2));
                                    }
#ifdef WITHPTHREADS
                                    M_free(renumber->symb.lo,"VarSpace");
                                    M_free(renumber,"Renumber");
#endif
*/
                                    if ( ( renumber = GetTable(x2,&position,0) ) == 0 ) {
                                        error = 1;
                                        MesPrint("&Problems getting information about stored
expression %s(2)"
                                        ,EXPRNAME(x2));
                                    }
                                    if ( renumber->symb.lo != AN.dummyrenumlist )
                                        M_free(renumber->symb.lo,"VarSpace");
                                    M_free(renumber,"Renumber");
                                    AR.StoreData.dirtyflag = 1;
                                }
                                break;
                            case TINDEX:
                                *t++ = -INDEX; *t++ = x2 + AM.OffsetIndex;
                                if ( t[-1] > AM.IndDum ) {
                                    x2 = t[-1] - AM.IndDum;
                                    if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                                }
                                break;
                            case TGENINDEX:
                                *t++ = -INDEX; *t++ = AC.DumNum + WILDOFFSET;
                                break;
                            case TVECTOR:
                                if ( minus ) *t++ = -MINVECTOR;
                                else *t++ = -VECTOR;
                                *t++ = x2 + AM.OffsetVector;
                                break;
                            case TSGAMMA:   /* Special gamma's */
                                MesPrint("&5_,6_,7_ can only be used inside g_");
                                error = -1;
                                *t++ = -INDEX;
                                *t++ = -x2;
                                break;
                            case TDUBIOUS:
```

```
                            *t++ = -SYMBOL; *t++ = x2; break;
                        case TFUNCTION:
                            *t++ = -x2-FUNCTION;
                            break;
                        case TSET:
                            *t++ = -SETSET;
                            *t++ = x2;
                            break;
                        case TWILDARG:
                            *t++ = -ARGWILD; *t++ = x2; break;
                        case TSETDOL:
                            x2 = -x2;
                            /* fall through */
                        case TSETNUM:
                            if ( inset == 0 ) {
                                w1 = t; t += 2; w2 = t;
                                while ( w1 > v ) *--w2 = *--w1;
                                tsize = v; relo = AT.WorkTop;
                                *v++ = SETSET; *v++ = 0;
                                inset = 1;
                            }
                            *--relo = x2; *--relo = t-v+1;
                            c = *s++;
                            x2 = 0; while ( *s >= 0 ) x2 = 128*x2 + *s++;
                            switch ( c ) {
                                case TFUNCTION:
                                    (*relo)--; *t++ = -x2-1; break;
                                case TSYMBOL:
                                    *t++ = -SYMBOL; *t++ = x2; break;
                                case TINDEX:
                                    *t++ = -INDEX; *t++ = x2;
                                    if ( x2+AM.OffsetIndex > AM.IndDum ) {
                                        x2 = x2+AM.OffsetIndex - AM.IndDum;
                                        if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                                    }
                                    break;
                                case TVECTOR:
                                    *t++ = -VECTOR; *t++ = x2; break;
                                case TNUMBER1:
                                    *t++ = -SNUMBER; *t++ = x2; break;
                                default:
                                    MesPrint("&Internal error 435");
                                    error = 1;
                                    *t++ = -SYMBOL; *t++ = x2; break;
                            }
                            break;
                        case TSUBEXP:
                            w2 = AC.ProtoType; i = w2[1];
                            w1 = t;
                            *t++ = i+ARGHEAD+4;
                            *t++ = 1;
                            FILLARG(t);
                            *t++ = i + 4;
                            while ( --i >= 0 ) *t++ = *w2++;
                            w1[ARGHEAD+3] = subexpbuffers[x2].subexpnum;
                            w1[ARGHEAD+5] = subexpbuffers[x2].buffernum;
                            if ( sumlevel > 0 ) {
                                w1[0] += 4;
                                w1[ARGHEAD] += 4;
                                w1[ARGHEAD+2] += 4;
                                *t++ = sumtype; *t++ = 4;
                                *t++ = sumlevel; *t++ = sumlevel;
                            }
                            *t++ = 1; *t++ = 1;
                            if ( minus ) *t++ = -3;
                            else         *t++ =  3;
                            break;
                        case TNUMBER:
                        case TNUMBER1:
                            if ( minus ) x2 = UnsignedToInt(-IntAbs(x2));
                            *t++ = -SNUMBER;
                            *t++ = x2;
                            break;
                        default:
                            MesPrint("&Illegal object in function");
                            error = -1;
                            break;
                    }
                } while ( *s == TCOMMA );
            }
            if ( *s != TFUNCLOSE ) {
                MesPrint("&Illegal argument field for function. Expected )");
                return(-1);
            }
            s++; sumlevel = 0;
            v[1] = t-v;
/*
```

```
                if ( *v == AM.termfunnum && ( v[1] != FUNHEAD+2 ||
                v[FUNHEAD] != -DOLLAREXPRESSION ) ) {
                    MesPrint("&The function term_ can only have one argument with a single
$-expression");
                    error = 1;
                }
*/
                goto fin;
            case TDUBIOUS:
                x1 = 0; while ( *s >= 0 ) x1 = 128*x1 + *s++;
                if ( *s == TWILDCARD ) s++;
                if ( *s == TDOT ) goto dovector;
                if ( *s == TFUNOPEN ) {
                    x1 += FUNCTION;
                    cc = 1;
                    v = t; *t++ = x1; *t++ = FUNHEAD; *t++ = DIRTYFLAG;
                    if ( x1 == AR.PolyFun && AR.PolyFunType == 2 && AR.Eside != LHSIDE )
                            t[-1] |= MUSTCLEANPRF;
                    FILLFUN3(t)
                    needarg = -1; goto dofunction;
                }
                *t++ = SYMBOL; *t++ = 4; *t++ = 0;
                if ( inset ) *relo = 2;
                goto TryPower;
            case TSUBEXP:
                x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
                if ( *s == TPOWER ) {
                    s++; c = *s++;
                    base = ( c == TNUMBER ) ? 100: 128;
                    x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
                    if ( *s == TWILDCARD ) { x2 += 2*MAXPOWER; s++; }
                    else if ( c == TSYMBOL ) x2 += 2*MAXPOWER;
                }
                else x2 = 1;
                r = AC.ProtoType; n = r[1] - 5; r += 5;
                *t++ = SUBEXPRESSION; *t++ = r[-4];
                *t++ = subexpbuffers[x1].subexpnum;
                *t++ = x2*deno;
                *t++ = subexpbuffers[x1].buffernum;
                NCOPY(t,r,n);
                if ( cbuf[subexpbuffers[x1].buffernum].CanCommu[subexpbuffers[x1].subexpnum] ) cc = 1;
                deno = 1;
                break;
            case TMULTIPLY:
                mulflag = 1;
                break;
            case TDIVIDE:
                mulflag = 1;
                deno = -deno;
                break;
            case TEXPRESSION:
                cc = 1;
                x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
                v = t;
                *t++ = EXPRESSION; *t++ = SUBEXPSIZE; *t++ = x1; *t++ = deno;
                *t++ = 0; FILLSUB(t)
/*
                Here we had some erroneous code before. It should be after
                the reading of the parameters as it is now (after 15-jan-2007).
                Thomas Hahn noticed this and reported it.
*/
                if ( *s == TFUNOPEN ) {
                    do {
                        s++; c = *s++;
                        base = ( c == TNUMBER ) ? 100: 128;
                        x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
                        switch ( c ) {
                            case TSYMBOL:
                                *t++ = SYMBOL; *t++ = 4; *t++ = x2; *t++ = 1;
                                break;
                            case TINDEX:
                                *t++ = INDEX; *t++ = 3; *t++ = x2+AM.OffsetIndex;
                                if ( t[-1] > AM.IndDum ) {
                                    x2 = t[-1] - AM.IndDum;
                                    if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                                }
                                break;
                            case TVECTOR:
                                *t++ = INDEX; *t++ = 3; *t++ = x2+AM.OffsetVector;
                                break;
                            case TFUNCTION:
                                *t++ = x2+FUNCTION; *t++ = 2; break;
                            case TNUMBER:
                            case TNUMBER1:
                                if ( x2 >= AM.OffsetIndex || x2 < 0 ) {
                                    MesPrint("&Index as argument of expression has illegal value");
                                    error = -1;
```

```
                                }
                                *t++ = INDEX; *t++ = 3; *t++ = x2; break;
                            case TSETDOL:
                                x2 = -x2;
                                /* fall through */
                            case TSETNUM:
                                if ( inset == 0 ) {
                                    w1 = t; t += 2; w2 = t;
                                    while ( w1 > v ) *--w2 = *--w1;
                                    tsize = v; relo = AT.WorkTop;
                                    *v++ = SETSET; *v++ = 0;
                                    inset = 1;
                                }
                                *--relo = x2; *--relo = t-v+2;
                                c = *s++;
                                x2 = 0; while ( *s >= 0 ) x2 = 128*x2 + *s++;
                                switch ( c ) {
                                    case TFUNCTION:
                                        *relo -= 2; *t++ = -x2-1; break;
                                    case TSYMBOL:
                                        *t++ = SYMBOL; *t++ = 4; *t++ = x2; *t++ = 1; break;
                                    case TINDEX:
                                        *t++ = INDEX; *t++ = 3; *t++ = x2;
                                        if ( x2+AM.OffsetIndex > AM.IndDum ) {
                                            x2 = x2+AM.OffsetIndex - AM.IndDum;
                                            if ( x2 > C->numdum[numexp] ) C->numdum[numexp] = x2;
                                        }
                                        break;
                                    case TVECTOR:
                                        *t++ = VECTOR; *t++ = 3; *t++ = x2; break;
                                    case TNUMBER1:
                                        *t++ = SNUMBER; *t++ = 4; *t++ = x2; *t++ = 1; break;
                                    default:
                                        MesPrint("&Internal error 435");
                                        error = 1;
                                        *t++ = SYMBOL; *t++ = 4; *t++ = x2; *t++ = 1; break;
                                }
                                break;
                            default:
                                MesPrint("&Argument of expression can only be symbol, index, vector or function");
                                error = -1;
                                break;
                        }
                    } while ( *s == TCOMMA );
                    if ( *s != TFUNCLOSE ) {
                        MesPrint("&Illegal object in argument field for expression");
                        error = -1;
                        while ( *s != TFUNCLOSE ) s++;
                    }
                    s++;
                }
                r = AC.ProtoType; n = r[1];
                if ( n > SUBEXPSIZE ) {
                    *t++ = WILDCARDS; *t++ = n+2;
                    NCOPY(t,r,n);
                }
/*
                Code added for parallel processing.
                This is different from the other occurrences to test immediately
                for renumbering. Here we have to read the parameters first.
*/
                if ( Expressions[x1].status == STOREDEXPRESSION ) {
                    v[1] = t-v;
                    AT.TMaddr = v;
                    PUTZERO(position);
/*
                    if ( (
#ifdef WITHPTHREADS
                        renumber =
#endif
                            GetTable(x1,&position,0) ) == 0 ) {
                        error = 1;
                        MesPrint("&Problems getting information about stored expression %s(3)"
                        ,EXPRNAME(x1));
                    }
#ifdef WITHPTHREADS
                    M_free(renumber->symb.lo,"VarSpace");
                    M_free(renumber,"Renumber");
#endif
*/
                    if ( ( renumber = GetTable(x1,&position,0) ) == 0 ) {
                        error = 1;
                        MesPrint("&Problems getting information about stored expression %s(3)"
                        ,EXPRNAME(x1));
                    }
                    if ( renumber->symb.lo != AN.dummyrenumlist )
```

```
                        M_free(renumber->symb.lo,"VarSpace");
                    M_free(renumber,"Renumber");
                    AR.StoreData.dirtyflag = 1;
                }
                if ( *s == LBRACE ) {
/*
                    This should be one term that should be inserted
                    FROMBRAC size+2 ( term )
                    Because this term should have been translated
                    already we can copy it from the 'subexpression'
*/
                    s++;
                    if ( *s != TSUBEXP ) {
                        MesPrint("&Internal error 23");
                        Terminate(-1);
                    }
                    s++; x2 = 0; while ( *s >= 0 ) { x2 = 128*x2 + *s++; }
                    r = cbuf[subexpbuffers[x2].buffernum].rhs[subexpbuffers[x2].subexpnum];
                    *t++ = FROMBRAC; *t++ = *r+2;
                    n = *r;
                    NCOPY(t,r,n);
                    if ( *r != 0 ) {
                        MesPrint("&Object between [] in expression should be a single term");
                        error = -1;
                    }
                    if ( *s != RBRACE ) {
                        MesPrint("&Internal error 23b");
                        Terminate(-1);
                    }
                    s++;
                }
                if ( *s == TPOWER ) {
                    s++; c = *s++;
                    base = ( c == TNUMBER ) ? 100: 128;
                    x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
                    if ( *s == TWILDCARD || c == TSYMBOL ) { x2 += 2*MAXPOWER; s++; }
                    v[3] = x2;
                }
                v[1] = t - v;
                deno = 1;
                break;
            case TNUMBER:
                if ( *s == 0 ) {
                    s++;
                    if ( *s == TPOWER ) {
                        s++; if ( *s == TMINUS ) { s++; deno = -deno; }
                        c = *s++; base = ( c == TNUMBER ) ? 100: 128;
                        x2 = 0; while ( *s >= 0 ) { x2 = x2*base + *s++; }
                        if ( x2 == 0 ) {
                            error = -1;
                            MesPrint("&Encountered 0^0 during compilation");
                        }
                        if ( deno < 0 ) {
                            error = -1;
                            MesPrint("&Division by zero during compilation (0 to the power negative
number)");
                        }
                    }
                    else if ( deno < 0 ) {
                        error = -1;
                        MesPrint("&Division by zero during compilation");
                    }
                    sign = 0; break; /* term is zero */
                }
                y = *s++;
                if ( *s >= 0 ) { y = 100*y + *s++; }
                innum[0] = y; nin = 1;
                while ( *s >= 0 ) {
                    y = *s++; x2 = 100;
                    if ( *s >= 0 ) { y = 100*y + *s++; x2 = 10000; }
                    Product(innum,&nin,(WORD)x2);
                    if ( y ) AddLong(innum,nin,(UWORD *)(&y),(WORD)1,innum,&nin);
                }
docoef:
                if ( *s == TPOWER ) {
                    s++; if ( *s == TMINUS ) { s++; deno = -deno; }
                    c = *s++; base = ( c == TNUMBER ) ? 100: 128;
                    x2 = 0; while ( *s >= 0 ) { x2 = x2*base + *s++; }
                    if ( x2 == 0 ) {
                        innum[0] = 1; nin = 1;
                    }
                    else if ( RaisPow(BHEAD innum,&nin,x2) ) {
                        error = -1; innum[0] = 1; nin = 1;
                    }
                }
                if ( deno > 0 ) {
                    Simplify(BHEAD innum,&nin,denominator,&ndenominator);
```

```
                    for ( i = 0; i < nnumerator; i++ ) CGscrat7[i] = numerator[i];
                    MulLong(innum,nin,CGscrat7,nnumerator,numerator,&nnumerator);
                }
                else if ( deno < 0 ) {
                    Simplify(BHEAD innum,&nin,numerator,&nnumerator);
                    for ( i = 0; i < ndenominator; i++ ) CGscrat7[i] = denominator[i];
                    MulLong(innum,nin,CGscrat7,ndenominator,denominator,&ndenominator);
                }
                deno = 1;
                break;
            case TNUMBER1:
                if ( *s == 0 ) { s++; sign = 0; break; /* term is zero */ }
                y = *s++;
                if ( *s >= 0 ) { y = 128*y + *s++; }
                if ( inset == 0 ) {
                    innum[0] = y; nin = 1;
                    while ( *s >= 0 ) {
                        y = *s++; x2 = 128;
                        if ( *s >= 0 ) { y = 128*y + *s++; x2 = 16384; }
                        Product(innum,&nin,(WORD)x2);
                        if ( y ) AddLong(innum,nin,(UWORD *)&y,(WORD)1,innum,&nin);
                    }
                    goto docoef;
                }
                *relo = 2; *t++ = SNUMBER; *t++ = 4; *t++ = y;
                goto TryPower;
#ifdef WITHFLOAT
            case TFLOAT:
                { WORD *w;
                s = ReadFloat(s);
                i = AT.WorkPointer[1];
                w = AT.WorkPointer;
                NCOPY(t,w,i);
/*
Power?
*/
                }
                break;
#endif
            case TDOLLAR:
            {
                WORD *powplace;
                x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
                if ( AR.Eside != LHSIDE ) {
                    *t++ = SUBEXPRESSION; *t++ = SUBEXPSIZE; *t++ = x1;
                }
                else {
                    *t++ = DOLLAREXPRESSION; *t++ = SUBEXPSIZE; *t++ = x1;
                }
                powplace = t; t++;
                *t++ = AM.dbufnum; FILLSUB(t)
/*
                Now we have to test for factors of dollars with [ ] and [ [ ]]
*/
                if ( *s == LBRACE ) {
                    int bracelevel = 1;
                    s++;
                    while ( bracelevel > 0 ) {
                        if ( *s == RBRACE ) {
                            bracelevel--; s++;
                        }
                        else if ( *s == TNUMBER ) {
                            s++;
                            x2 = 0; while ( *s >= 0 ) { x2 = 100*x2 + *s++; }
                            *t++ = DOLLAREXPR2; *t++ = 3; *t++ = -x2-1;
CloseBraces:
                            while ( bracelevel > 0 ) {
                                if ( *s != RBRACE ) {
ErrorBraces:
                                    error = -1;
                                    MesPrint("&Improper use of [] in $-variable.");
                                    return(error);
                                }
                                else {
                                    s++; bracelevel--;
                                }
                            }
                        }
                        else if ( *s == TDOLLAR ) {
                            s++;
                            x1 = 0; while ( *s >= 0 ) { x1 = x1*128 + *s++; }
                            *t++ = DOLLAREXPR2; *t++ = 3; *t++ = x1;
                            if ( *s == RBRACE ) goto CloseBraces;
                            else if ( *s == LBRACE ) {
                                s++; bracelevel++;
                            }
                        }
```

```
                        else goto ErrorBraces;
                    }
                }
/*
                Finally we can continue with the power
*/
                if ( *s == TPOWER ) {
                    s++;
                    if ( *s == TMINUS ) { s++; deno = -deno; }
                    c = *s++;
                    base = ( c == TNUMBER ) ? 100: 128;
                    x2 = 0; while ( *s >= 0 ) { x2 = base*x2 + *s++; }
                    if ( c == TSYMBOL ) {
                        if ( *s == TWILDCARD ) s++;
                        x2 += 2*MAXPOWER;
                    }
                    *powplace = deno*x2;
                }
                else *powplace = deno;
                deno = 1;
/*
                if ( inset ) {
                    while ( relo < AT.WorkTop ) *t++ = *relo++;
                    inset = 0; tsize[1] = t - tsize;
                }
*/
            }
                break;
            case TSETNUM:
                inset = 1; tsize = t; relo = AT.WorkTop;
                *t++ = SETSET; *t++ = 0;
                x1 = 0; while ( *s >= 0 ) x1 = x1*128 + *s++;
                *--relo = x1; *--relo = 0;
                break;
            case TSETDOL:
                inset = 1; tsize = t; relo = AT.WorkTop;
                *t++ = SETSET; *t++ = 0;
                x1 = 0; while ( *s >= 0 ) x1 = x1*128 + *s++;
                *--relo = -x1; *--relo = 0;
                break;
            case TFUNOPEN:
                MesPrint("&Illegal use of function arguments");
                error = -1;
                funflag = 1;
                deno = 1;
                break;
            case TFUNCLOSE:
                if ( funflag == 0 )
                    MesPrint("&Illegal use of function arguments");
                error = -1;
                funflag = 0;
                deno = 1;
                break;
            case TSGAMMA:
                MesPrint("&Illegal use special gamma symbols 5_, 6_, 7_");
                error = -1;
                funflag = 0;
                deno = 1;
                break;
            case TCONJUGATE:
                MesPrint("&Complex conjugate operator (%#) is not implemented");
                error = -1;
                deno = 1;
                break;
            default:
                MesPrint("&Internal error in code generator. Unknown object: %d",c);
                error = -1;
                deno = 1;
                break;
            }
        }
    }
    if ( mulflag ) {
        MesPrint("&Irregular end of statement.");
        error = 1;
    }
    if ( !first && error == 0 ) {
        *term = t-term;
        C->NumTerms[numexp]++;
        if ( cc && sign ) C->CanCommu[numexp]++;
        error = CompleteTerm(term,numerator,denominator,nnumerator,ndenominator,sign);
    }
    AT.WorkPointer = oldwork;
    if ( error ) return(-1);
    AddToCB(C,0)
    if ( AC.CompileLevel > 0 && AR.Eside != LHSIDE ) {
        /* See whether we have this one already */
```

```
        error = InsTree(AC.cbufnum,C->numrhs);
        if ( error < (C->numrhs) ) {
            C->Pointer = C->rhs[C->numrhs--];
            return(error);
        }
    }
    return(C->numrhs);
OverWork:
    MLOCK(ErrorMessageLock);
    MesWork();
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] CodeGenerator :
        #[ CompleteTerm :

        Completes the term
        Puts it in the buffer
*/

int CompleteTerm(WORD *term, UWORD *numer, UWORD *denom, WORD nnum, WORD nden, int sign)
{
    int nsize, i;
    WORD *t;
    if ( sign == 0 ) return(0);     /* Term is zero */
    if ( nnum >= nden ) nsize = nnum;
    else                nsize = nden;
    t = term + *term;
    for ( i = 0; i < nnum; i++ ) *t++ = numer[i];
    for ( ; i < nsize; i++ ) *t++ = 0;
    for ( i = 0; i < nden; i++ ) *t++ = denom[i];
    for ( ; i < nsize; i++ ) *t++ = 0;
    *t++ = (2*nsize+1)*sign;
    *term = t - term;
    AddNtoC(AC.cbufnum,*term,term,7);
    return(0);
}

/*
        #] CompleteTerm :
        #[ CodeFactors :

        This routine does the part of reading in in terms of factors.
        If there is more than one term at this level we have only one
        factor. In that case any expression should first be unfactorized.
        Then the whole expression gets read as a new subexpression and finally
        we generate factor_*subexpression.
        If the whole has only multiplications we have factors. Then the
        nasty thing is powers of objects and in particular powers of
        factorized expressions or dollars.
        For a power we generate a new subexpression of the type
          1+factor_+...+factor_^(power-1)
        with which we multiply.

        WE HAVE NOT YET WORRIED ABOUT SETS
*/

int CodeFactors(SBYTE *tokens)
{
    GETIDENTITY
    EXPRESSIONS e = Expressions + AR.CurExpr;
    int nfactor = 1, nparenthesis, i, last = 0, error = 0;
    SBYTE *t, *startobject, *tt, *s1, *out, *outtokens;
    WORD nexp, subexp = 0, power, pow, x2, powfactor, first;
/*
    First scan the number of factors
*/
    t = tokens;
    while ( *t != TENDOFIT ) {
        if ( *t >= 0 ) { while ( *t >= 0 ) t++; continue; }
        if ( *t == LPARENTHESIS || *t == LBRACE || *t == TSETOPEN || *t == TFUNOPEN ) {
            nparenthesis = 0; t++;
            while ( nparenthesis >= 0 ) {
                if ( *t == LPARENTHESIS || *t == LBRACE || *t == TSETOPEN || *t == TFUNOPEN )
nparenthesis++;
                else if ( *t == RPARENTHESIS || *t == RBRACE || *t == TSETCLOSE || *t == TFUNCLOSE )
nparenthesis--;
                t++;
            }
            continue;
        }
        else if ( ( *t == TPLUS || *t == TMINUS ) && ( t > tokens )
        && ( t[-1] != TPLUS && t[-1] != TMINUS ) ) {
            if ( t[-1] >= 0 || t[-1] == RPARENTHESIS || t[-1] == RBRACE
            || t[-1] == TSETCLOSE || t[-1] == TFUNCLOSE ) {
```

```
                subexp = CodeGenerator(tokens);
                if ( subexp < 0 ) error = -1;
                if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
                    MesPrint("&More than %d subexpressions inside one expression",(WORD)MAXSUBEXPRESSIONS);
                    Terminate(-1);
                }
                if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
                    DoubleBuffer((void **)((void *)(&subexpbuffers))
                    ,(void **)((void *)(&topsubexpbuffers)),sizeof(SUBBUF),"subexpbuffers");
                }
                subexpbuffers[insubexpbuffers].subexpnum = subexp;
                subexpbuffers[insubexpbuffers].buffernum = AC.cbufnum;
                subexp = insubexpbuffers++;
                t = tokens;
                *t++ = TSYMBOL; *t++ = FACTORSYMBOL;
                *t++ = TMULTIPLY; *t++ = TSUBEXP;
                PUTNUMBER128(t,subexp)
                *t++ = TENDOFIT;
                e->numfactors = 1;
                e->vflags |= ISFACTORIZED;
                return(subexp);
            }
        }
        else if ( ( *t == TMULTIPLY || *t == TDIVIDE ) && t > tokens ) {
            nfactor++;
        }
        else if ( *t == TEXPRESSION ) {
            t++;
            nexp = 0; while ( *t >= 0 ) { nexp = nexp*128 + *t++; }
            if ( *t == LBRACE ) continue;
            if ( ( AS.Oldvflags[nexp] & ISFACTORIZED ) != 0 ) {
                nfactor += AS.OldNumFactors[nexp];
            }
            else { nfactor++; }
            continue;
        }
        else if ( *t == TDOLLAR ) {
            t++;
            nexp = 0; while ( *t >= 0 ) { nexp = nexp*128 + *t++; }
            if ( *t == LBRACE ) continue;
            if ( Dollars[nexp].nfactors > 0 ) {
                nfactor += Dollars[nexp].nfactors;
            }
            else { nfactor++; }
            continue;
        }
        t++;
    }
/*
    Now the real pass.
    nfactor is a not so reliable measure for the space we need.
*/
    outtokens = (SBYTE *)Malloc1(((t-tokens)+(nfactor+2)*25)*sizeof(SBYTE),"CodeFactors");
    out = outtokens;
    t = tokens; first = 1; powfactor = 1;
    while ( *t == TPLUS || *t == TMINUS ) { if ( *t == TMINUS ) first = -first; t++; }
    if ( first < 0 ) {
        *out++ = TMINUS; *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
        *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,powfactor)
        powfactor++;
    }
    startobject = t; power = 1;
    while ( *t != TENDOFIT ) {
        if ( *t >= 0 ) { while ( *t >= 0 ) t++; continue; }
        if ( *t == LPARENTHESIS || *t == LBRACE || *t == TSETOPEN || *t == TFUNOPEN ) {
            nparenthesis = 0; t++;
            while ( nparenthesis >= 0 ) {
                if ( *t == LPARENTHESIS || *t == LBRACE || *t == TSETOPEN || *t == TFUNOPEN ) nparenthesis++;
                else if ( *t == RPARENTHESIS || *t == RBRACE || *t == TSETCLOSE || *t == TFUNCLOSE ) nparenthesis--;
                t++;
            }
            continue;
        }
        else if ( ( *t == TMULTIPLY || *t == TDIVIDE ) && ( t > tokens ) ) {
            if ( t[-1] >= 0 || t[-1] == RPARENTHESIS || t[-1] == RBRACE
            || t[-1] == TSETCLOSE || t[-1] == TFUNCLOSE ) {
dolast:
                if ( startobject ) {    /* apparently power is 1 or -1 */
                    *out++ = TPLUS;
                    if ( power < 0 ) { *out++ = TNUMBER; *out++ = 1; *out++ = TDIVIDE; }
                    s1 = startobject;
                    while ( s1 < t ) *out++ = *s1++;
                    *out++ = TMULTIPLY; *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
                    *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,powfactor)
```

```
                    powfactor++;
                }
                if ( last ) { startobject = 0; break; }
                startobject = t+1;
                if ( *t == TDIVIDE ) power = -1;
                if ( *t == TMULTIPLY ) power = 1;
            }
        }
        else if ( *t == TPOWER ) {
            pow = 1;
            tt = t+1;
            while ( ( *tt == TMINUS ) || ( *tt == TPLUS ) ) {
                if ( *tt == TMINUS ) pow = -pow;
                tt++;
            }
            if ( *tt == TSYMBOL ) {
                tt++; while ( *tt >= 0 ) tt++;
                t = tt; continue;
            }
            tt++; x2 = 0; while ( *tt >= 0 ) { x2 = 100*x2 + *tt++; }
/*
            We have an object in startobject till t. The power is
            power*pow*x2
*/
            power = power*pow*x2;
            if ( power < 0 ) { pow = -power; power = -1; }
            else if ( power == 0 ) { t = tt; startobject = tt; continue; }
            else { pow = power; power = 1; }
            *out++ = TPLUS;
            if ( pow > 1 ) {
                subexp = GenerateFactors(pow,1);
                if ( subexp < 0 ) { error = -1; subexp = 0; }
                *out++ = TSUBEXP; PUTNUMBER128(out,subexp);
            }
            *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
            *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,powfactor)
            powfactor += pow;
            if ( power > 0 ) *out++ = TMULTIPLY;
            else *out++ = TDIVIDE;
            s1 = startobject; while ( s1 < t ) *out++ = *s1++;
            startobject = 0; t = tt; continue;
        }
        else if ( *t == TEXPRESSION ) {
            startobject = t;
            t++;
            nexp = 0; while ( *t >= 0 ) { nexp = nexp*128 + *t++; }
            if ( *t == LBRACE ) continue;
            if ( *t == LPARENTHESIS ) {
                nparenthesis = 0; t++;
                while ( nparenthesis >= 0 ) {
                    if ( *t == LPARENTHESIS ) nparenthesis++;
                    else if ( *t == RPARENTHESIS ) nparenthesis--;
                    t++;
                }
            }
            if ( ( AS.Oldvflags[nexp] & ISFACTORIZED ) == 0 ) continue;
            if ( *t == TPOWER ) {
                pow = 1;
                tt = t+1;
                while ( ( *tt == TMINUS ) || ( *tt == TPLUS ) ) {
                    if ( *tt == TMINUS ) pow = -pow;
                    tt++;
                }
                if ( *tt != TNUMBER ) {
                    MesPrint("Internal problems(1) in CodeFactors");
                    return(-1);
                }
                tt++; x2 = 0; while ( *tt >= 0 ) { x2 = 100*x2 + *tt++; }
/*
                We have an object in startobject till t. The power is
                power*pow*x2
*/
dopower:
                power = power*pow*x2;
                if ( power < 0 ) { pow = -power; power = -1; }
                else if ( power == 0 ) { t = tt; startobject = tt; continue; }
                else { pow = power; power = 1; }
                *out++ = TPLUS;
                if ( pow > 1 ) {
                    subexp = GenerateFactors(pow,AS.OldNumFactors[nexp]);
                    if ( subexp < 0 ) { error = -1; subexp = 0; }
                    *out++ = TSUBEXP; PUTNUMBER128(out,subexp)
                    *out++ = TMULTIPLY;
                }
                i = powfactor-1;
                if ( i > 0 ) {
                    *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
```

```
                    if ( i > 1 ) {
                        *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,i)
                    }
                    *out++ = TMULTIPLY;
                }
                powfactor += AS.OldNumFactors[nexp]*pow;
                s1 = startobject;
                while ( s1 < t ) *out++ = *s1++;
                startobject = 0; t = tt; continue;
            }
            else {
                tt = t; pow = 1; x2 = 1; goto dopower;
            }
        }
        else if ( *t == TDOLLAR ) {
            startobject = t;
            t++;
            nexp = 0; while ( *t >= 0 ) { nexp = nexp*128 + *t++; }
            if ( *t == LBRACE ) continue;
            if ( Dollars[nexp].nfactors == 0 ) continue;
            if ( *t == TPOWER ) {
                pow = 1;
                tt = t+1;
                while ( ( *tt == TMINUS ) || ( *tt == TPLUS ) ) {
                    if ( *tt == TMINUS ) pow = -pow;
                    tt++;
                }
                if ( *tt != TNUMBER ) {
                    MesPrint("Internal problems(2) in CodeFactors");
                    return(-1);
                }
                tt++; x2 = 0; while ( *tt >= 0 ) { x2 = 100*x2 + *tt++; }
/*
                We have an object in startobject till t. The power is
                power*pow*x2
*/
dopowerd:
                power = power*pow*x2;
                if ( power < 0 ) { pow = -power; power = -1; }
                else if ( power == 0 ) { t = tt; startobject = tt; continue; }
                else { pow = power; power = 1; }
                if ( pow > 1 ) {
                    subexp = GenerateFactors(pow,1);
                    if ( subexp < 0 ) { error = -1; subexp = 0; }
                }
                for ( i = 1; i <= Dollars[nexp].nfactors; i++ ) {
                    s1 = startobject; *out++ = TPLUS;
                    while ( s1 < t ) *out++ = *s1++;
                    *out++ = LBRACE; *out++ = TNUMBER; PUTNUMBER128(out,i)
                    *out++ = RBRACE;
                    *out++ = TMULTIPLY;
                    *out++ = TSYMBOL; *out++ = FACTORSYMBOL;
                    *out++ = TPOWER; *out++ = TNUMBER; PUTNUMBER100(out,powfactor)
                    powfactor += pow;
                    if ( pow > 1 ) {
                        *out++ = TSUBEXP; PUTNUMBER128(out,subexp)
                    }
                }
                startobject = 0; t = tt; continue;
            }
            else {
                tt = t; pow = 1; x2 = 1; goto dopowerd;
            }
        }
        t++;
    }
    if ( last == 0 ) { last = 1; goto dolast; }
    *out = TENDOFIT;
    e->numfactors = powfactor-1;
    e->vflags |= ISFACTORIZED;
    subexp = CodeGenerator(outtokens);
    if ( subexp < 0 ) error = -1;
    if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
        MesPrint("&More than %d subexpressions inside one expression",(WORD)MAXSUBEXPRESSIONS);
        Terminate(-1);
    }
    if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
        DoubleBuffer((void **)((void *)(&subexpbuffers))
        ,(void **)((void *)(&topsubexpbuffers)),sizeof(SUBBUF),"subexpbuffers");
    }
    subexpbuffers[insubexpbuffers].subexpnum = subexp;
    subexpbuffers[insubexpbuffers].buffernum = AC.cbufnum;
    subexp = insubexpbuffers++;
    M_free(outtokens,"CodeFactors");
    s1 = tokens;
    *s1++ = TSUBEXP; PUTNUMBER128(s1,subexp); *s1++ = TENDOFIT;
    if ( error < 0 ) return(-1);
```

```
    else return(subexp);
}

/*
        #] CodeFactors :
        #[ GenerateFactors :

    Generates an expression of the type
      1+factor_+factor_^2+...+factor_^(n-1)
    (this is if inc=1)
    Returns the subexpression pointer of it.
*/

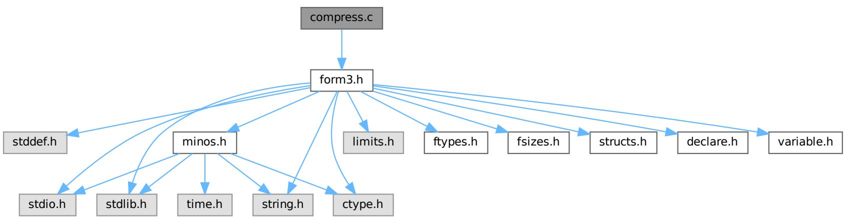
WORD GenerateFactors(WORD n,WORD inc)
{
    WORD subexp;
    int i, error = 0;
    SBYTE *s;
    SBYTE *tokenbuffer = (SBYTE *)Malloc1(8*n*sizeof(SBYTE),"GenerateFactors");
    s = tokenbuffer;
    *s++ = TNUMBER; *s++ = 1;
    for ( i = inc; i < n*inc; i += inc ) {
        *s++ = TPLUS; *s++ = TSYMBOL; *s++ = FACTORSYMBOL;
        if ( i > 1 ) {
            *s++ = TPOWER; *s++ = TNUMBER;
            PUTNUMBER100(s,i)
        }
    }
    *s++ = TENDOFIT;
    subexp = CodeGenerator(tokenbuffer);
    if ( subexp < 0 ) error = -1;
    M_free(tokenbuffer,"GenerateFactors");
    if ( insubexpbuffers >= MAXSUBEXPRESSIONS ) {
        MesPrint("&More than %d subexpressions inside one expression",(WORD)MAXSUBEXPRESSIONS);
        Terminate(-1);
    }
    if ( subexpbuffers+insubexpbuffers >= topsubexpbuffers ) {
        DoubleBuffer((void **)((void *)(&subexpbuffers))
        ,(void **)((void *)(&topsubexpbuffers)),sizeof(SUBBUF),"subexpbuffers");
    }
    subexpbuffers[insubexpbuffers].subexpnum = subexp;
    subexpbuffers[insubexpbuffers].buffernum = AC.cbufnum;
    subexp = insubexpbuffers++;
    if ( error < 0 ) return(error);
    return(subexp);
}

/*
        #] GenerateFactors :
    #] Compiler :
*/
```

4.13 compress.c File Reference

`#include "form3.h"`

Include dependency graph for compress.c:



4.13.1 Detailed Description

The routines for the use of gzip (de)compression of the information in the sort file.

Definition in file compress.c.

4.14 compress.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 */
/* #] License : */

#include "form3.h"

#ifdef WITHZLIB
/*
#define GZIPDEBUG

    Low level routines for dealing with zlib during sorting and handling
    the scratch files. Work started 5-sep-2005.
    The .sor file handling was more or less completed on 8-sep-2005
    The handling of the scratch files still needs some thinking.
    Complications are:
        gzip compression should be per expression, not per buffer.
        No gzip compression for expressions with a bracket index.
        Separate decompression buffers for expressions in the rhs.
        This last one will involve more buffer work and organization.
        Information about compression should be stored for each expr.
            (including what method/program was used)
    Note: Be careful with compression. By far the most compact method
    is the original problem....

    #[ Variables :

    The following variables are to contain the intermediate buffers
    for the inflation of the various patches in the sort file.
    There can be up to MaxFpatches (FilePatches in the setup) and hence
    we can have that many streams simultaneously. We set this up once
    and only when needed.
        (in struct A.N or AB[threadnum].N)
    Bytef **AN.ziobufnum;
    Bytef *AN.ziobuffers;
*/

/*
    #] Variables :
    #[ SetupOutputGZIP :

    Routine prepares a gzip output stream for the given file.
*/

int SetupOutputGZIP(FILEHANDLE *f)
{
    GETIDENTITY

    if ( AT.SS != AT.S0 ) return(0);
    if ( AR.NoCompress == 1 ) return(0);
    if ( AR.gzipCompress <= 0 ) return(0);

    if ( f->ziobuffer == 0 ) {
/*
        1: Allocate a struct for the gzip stream:
*/
        f->zsp = Malloc1(sizeof(z_stream),"output zstream");
/*
        2: Allocate the output buffer.
```

```
*/
        f->ziobuffer =
            (Bytef *)Malloc1(f->ziosize*sizeof(char),"output zbuffer");
        if ( f->zsp == 0 || f->ziobuffer == 0 ) {
            MLOCK(ErrorMessageLock);
            MesCall("SetupOutputGZIP");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    }
/*
    3: Set the default fields:
*/
    f->zsp->zalloc = Z_NULL;
    f->zsp->zfree  = Z_NULL;
    f->zsp->opaque = Z_NULL;
/*
    4: Set the output space:
*/
    f->zsp->next_out  = f->ziobuffer;
    f->zsp->avail_out = f->ziosize;
    f->zsp->total_out = 0;
/*
    5: Set the input space:
*/
    f->zsp->next_in  = (Bytef *)(f->PObuffer);
    f->zsp->avail_in = (Bytef *)(f->POfill) - (Bytef *)(f->PObuffer);
    f->zsp->total_in = 0;
/*
    6: Initiate the deflation
*/
    if ( deflateInit(f->zsp,AR.gzipCompress) != Z_OK ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Error from zlib: %s",f->zsp->msg);
        MesCall("SetupOutputGZIP");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    return(0);
}

/*
    #] SetupOutputGZIP :
    #[ PutOutputGZIP :

    Routine is called when the PObuffer of f is full.
    The contents of it will be compressed and whenever the output buffer
    f->ziobuffer is full it will be written and the output buffer
    will be reset.
    Upon exit the input buffer will be cleared.
*/

int PutOutputGZIP(FILEHANDLE *f)
{
    GETIDENTITY
    int zerror;
/*
    First set the number of bytes in the input
*/
    f->zsp->next_in  = (Bytef *)(f->PObuffer);
    f->zsp->avail_in = (Bytef *)(f->POfill) - (Bytef *)(f->PObuffer);
    f->zsp->total_in = 0;

    while ( ( zerror = deflate(f->zsp,Z_NO_FLUSH) ) == Z_OK ) {
        if ( f->zsp->avail_out == 0 ) {
/*
            ziobuffer is full. Write the output.
*/
#ifdef GZIPDEBUG
            {
                char *s = (char *)((UBYTE *)(f->ziobuffer)+f->ziosize);
                MLOCK(ErrorMessageLock);
                MesPrint("%wWriting %l bytes at %10p: %d %d %d %d %d"
                ,f->ziosize,&(f->POposition),s[-5],s[-4],s[-3],s[-2],s[-1]);
                MUNLOCK(ErrorMessageLock);
            }
#endif
#ifdef ALLLOCK
            LOCK(f->pthreadslock);
#endif
            if ( f == AR.hidefile ) {
                LOCK(AS.inputslock);
            }
            SeekFile(f->handle,&(f->POposition),SEEK_SET);
            if ( WriteFile(f->handle,(UBYTE *)(f->ziobuffer),f->ziosize)
                        != f->ziosize ) {
```

```
                if ( f == AR.hidefile ) {
                    UNLOCK(AS.inputslock);
                }
#ifdef ALLLOCK
                UNLOCK(f->pthreadslock);
#endif
                MLOCK(ErrorMessageLock);
                MesPrint("%wWrite error during compressed sort. Disk full?");
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            if ( f == AR.hidefile ) {
                UNLOCK(AS.inputslock);
            }
#ifdef ALLLOCK
            UNLOCK(f->pthreadslock);
#endif
            ADDPOS(f->filesize,f->ziosize);
            ADDPOS(f->POposition,f->ziosize);
#ifdef WITHPTHREADS
            if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
                if ( f->handle >= 0 ) SynchFile(f->handle);
            }
#endif
/*
            Reset the output
*/
            f->zsp->next_out  = f->ziobuffer;
            f->zsp->avail_out = f->ziosize;
            f->zsp->total_out = 0;
        }
        else if ( f->zsp->avail_in == 0 ) {
/*
            We compressed everything and it sits in ziobuffer. Finish
*/
            return(0);
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("%w avail_in = %d, avail_out = %d.",f->zsp->avail_in,f->zsp->avail_out);
            MUNLOCK(ErrorMessageLock);
            break;
        }
    }
    MLOCK(ErrorMessageLock);
    MesPrint("%wError in gzip handling of output. zerror = %d",zerror);
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
    #] PutOutputGZIP :
    #[ FlushOutputGZIP :

    Routine is called to flush a stream. The compression of the input buffer
    will be completed and the contents of f->ziobuffer will be written.
    Both buffers will be cleared.
*/

int FlushOutputGZIP(FILEHANDLE *f)
{
    GETIDENTITY
    int zerror;
/*
    Set the proper parameters
*/
    f->zsp->next_in  = (Bytef *)(f->PObuffer);
    f->zsp->avail_in = (Bytef *)(f->POfill) - (Bytef *)(f->PObuffer);
    f->zsp->total_in = 0;

    while ( ( zerror = deflate(f->zsp,Z_FINISH) ) == Z_OK ) {
        if ( f->zsp->avail_out == 0 ) {
/*
            Write the output
*/
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%wWriting %l bytes at %10p",f->ziosize,&(f->POposition));
            MUNLOCK(ErrorMessageLock);
#endif
#ifdef ALLLOCK
            LOCK(f->pthreadslock);
#endif
            if ( f == AR.hidefile ) {
                UNLOCK(AS.inputslock);
            }
            SeekFile(f->handle,&(f->POposition),SEEK_SET);
```

```
            if ( WriteFile(f->handle,(UBYTE *)(f->ziobuffer),f->ziosize)
                        != f->ziosize ) {
                if ( f == AR.hidefile ) {
                    UNLOCK(AS.inputslock);
                }
#ifdef ALLLOCK
                UNLOCK(f->pthreadslock);
#endif
                MLOCK(ErrorMessageLock);
                MesPrint("%wWrite error during compressed sort. Disk full?");
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            if ( f == AR.hidefile ) {
                UNLOCK(AS.inputslock);
            }
#ifdef ALLLOCK
            UNLOCK(f->pthreadslock);
#endif
            ADDPOS(f->filesize,f->ziosize);
            ADDPOS(f->POposition,f->ziosize);
#ifdef WITHPTHREADS
            if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
                if ( f->handle >= 0 ) SynchFile(f->handle);
            }
#endif
/*
            Reset the output
*/
            f->zsp->next_out  = f->ziobuffer;
            f->zsp->avail_out = f->ziosize;
            f->zsp->total_out = 0;
        }
    }
    if ( zerror == Z_STREAM_END ) {
/*
        Write the output
*/
#ifdef GZIPDEBUG
        MLOCK(ErrorMessageLock);
        MesPrint("%wWriting %l bytes at %10p",(LONG)(f->zsp->avail_out),&(f->POposition));
        MUNLOCK(ErrorMessageLock);
#endif
#ifdef ALLLOCK
        LOCK(f->pthreadslock);
#endif
        if ( f == AR.hidefile ) {
            LOCK(AS.inputslock);
        }
        SeekFile(f->handle,&(f->POposition),SEEK_SET);
        if ( WriteFile(f->handle,(UBYTE *)(f->ziobuffer),f->zsp->total_out)
                        != (LONG)(f->zsp->total_out) ) {
            if ( f == AR.hidefile ) {
                UNLOCK(AS.inputslock);
            }
#ifdef ALLLOCK
            UNLOCK(f->pthreadslock);
#endif
            MLOCK(ErrorMessageLock);
            MesPrint("%wWrite error during compressed sort. Disk full?");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        if ( f == AR.hidefile ) {
            LOCK(AS.inputslock);
        }
#ifdef ALLLOCK
        UNLOCK(f->pthreadslock);
#endif
        ADDPOS(f->filesize,f->zsp->total_out);
        ADDPOS(f->POposition,f->zsp->total_out);
#ifdef WITHPTHREADS
        if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
            if ( f->handle >= 0 ) SynchFile(f->handle);
        }
#endif
#ifdef GZIPDEBUG
        MLOCK(ErrorMessageLock);
        {   char *s = f->ziobuffer+f->zsp->total_out;
            MesPrint("%w   Last bytes written: %d %d %d %d %d",s[-5],s[-4],s[-3],s[-2],s[-1]);
        }
        MesPrint("%w     Perceived position in FlushOutputGZIP is %10p",&(f->POposition));
        MUNLOCK(ErrorMessageLock);
#endif
/*
            Reset the output
*/
```

```
        f->zsp->next_out  = f->ziobuffer;
        f->zsp->avail_out = f->ziosize;
        f->zsp->total_out = 0;
        if ( ( zerror = deflateEnd(f->zsp) ) == Z_OK ) return(0);
        MLOCK(ErrorMessageLock);
        if ( f->zsp->msg ) {
            MesPrint("%wError in finishing gzip handling of output: %s",f->zsp->msg);
        }
        else {
            MesPrint("%wError in finishing gzip handling of output.");
        }
        MUNLOCK(ErrorMessageLock);
    }
    else {
        MLOCK(ErrorMessageLock);
        MesPrint("%wError in gzip handling of output.");
        MUNLOCK(ErrorMessageLock);
    }
    return(-1);
}

/*
    #] FlushOutputGZIP :
    #[ SetupAllInputGZIP :

    Routine prepares all gzip input streams for a merge.

    Problem (29-may-2008): If we never use GZIP compression, this routine
    will still allocate the array space. This is an enormous amount!
    It places an effective restriction on the value of SortIOsize
*/

int SetupAllInputGZIP(SORTING *S)
{
    GETIDENTITY
    int i, NumberOpened = 0;
    z_streamp zsp;
/*
    This code was added 29-may-2008 by JV to prevent further processing if
    there is no compression at all (usually).
*/
    for ( i = 0; i < S->inNum; i++ ) {
        if ( S->fpincompressed[i] ) break;
    }
    if ( i >= S->inNum ) return(0);

    if ( S->zsparray == 0 ) {
        S->zsparray = (z_streamp)Malloc1(sizeof(z_stream)*S->MaxFpatches,"input zstreams");
        if ( S->zsparray == 0 ) {
            MLOCK(ErrorMessageLock);
            MesCall("SetupAllInputGZIP");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
/*
        We add 128 bytes in the hope that if it can happen that it goes
        outside the buffer during decompression, it does not do damage.
*/
        AN.ziobuffers = (Bytef *)Malloc1(S->MaxFpatches*(S->file.ziosize+128)*sizeof(Bytef),"input raw buffers");
/*
        This seems to be one of the really stupid errors:
        We allocate way too much space. Way way way too much.
        AN.ziobufnum  = (Bytef **)Malloc1(S->MaxFpatches*S->file.ziosize*sizeof(Bytef *),"input raw pointers");
*/
        AN.ziobufnum  = (Bytef **)Malloc1(S->MaxFpatches*sizeof(Bytef *),"input raw pointers");
        if ( AN.ziobuffers == 0 || AN.ziobufnum == 0 ) {
            MLOCK(ErrorMessageLock);
            MesCall("SetupAllInputGZIP");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        for ( i  = 0 ; i < S->MaxFpatches; i++ ) {
            AN.ziobufnum[i] = AN.ziobuffers + i * (S->file.ziosize+128);
        }
    }
    for ( i = 0; i < S->inNum; i++ ) {
#ifdef GZIPDEBUG
        MLOCK(ErrorMessageLock);
        MesPrint("%wPreparing z-stream %d with compression %d",i,S->fpincompressed[i]);
        MUNLOCK(ErrorMessageLock);
#endif
        if ( S->fpincompressed[i] ) {
            zsp = &(S->zsparray[i]);
/*
            1: Set the default fields:
```

```
*/
            zsp->zalloc = Z_NULL;
            zsp->zfree  = Z_NULL;
            zsp->opaque = Z_NULL;
/*
            2: Set the output space:
*/
            zsp->next_out  = Z_NULL;
            zsp->avail_out = 0;
            zsp->total_out = 0;
/*
            3: Set the input space temporarily:
*/
            zsp->next_in  = Z_NULL;
            zsp->avail_in = 0;
            zsp->total_in = 0;
/*
            4: Initiate the inflation
*/
            if ( inflateInit(zsp) != Z_OK ) {
                MLOCK(ErrorMessageLock);
                if ( zsp->msg ) MesPrint("%wError from inflateInit: %s",zsp->msg);
                else            MesPrint("%wError from inflateInit");
                MesCall("SetupAllInputGZIP");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            NumberOpened++;
        }
    }
    return(NumberOpened);
}

/*
    #] SetupAllInputGZIP :
    #[ FillInputGZIP :

    Routine is called when we need new input in the specified buffer.
    This buffer is used for the output and we keep reading and uncompressing
    input till either this buffer is full or the input stream is finished.
    The return value is the number of bytes in the buffer.
*/

LONG FillInputGZIP(FILEHANDLE *f, POSITION *position, UBYTE *buffer, LONG buffersize, int numstream)
{
    GETIDENTITY
    int zerror;
    LONG readsize, toread = 0;
    SORTING *S = AT.SS;
    z_streamp zsp;
    POSITION pos;
    if ( S->fpincompressed[numstream] ) {
        zsp = &(S->zsparray[numstream]);
        zsp->next_out = (Bytef *)buffer;
        zsp->avail_out = buffersize;
        zsp->total_out = 0;
        if ( zsp->avail_in == 0 ) {
/*
            First loading of the input
*/
            if ( ISGEPOSINC(S->fPatchesStop[numstream],*position,f->ziosize) ) {
                toread = f->ziosize;
            }
            else {
                DIFPOS(pos,S->fPatchesStop[numstream],*position);
                toread = (LONG)(BASEPOSITION(pos));
            }
            if ( toread > 0 ) {
#ifdef GZIPDEBUG
                MLOCK(ErrorMessageLock);
                MesPrint("%w-+Reading %l bytes in stream %d at position %10p; stop at %10p",toread,numstream,position,&(S->fPatchesStop[numstream]));
                MUNLOCK(ErrorMessageLock);
#endif
#ifdef ALLLOCK
                LOCK(f->pthreadslock);
#endif
                SeekFile(f->handle,position,SEEK_SET);
                readsize = ReadFile(f->handle,(UBYTE *)(AN.ziobufnum[numstream]),toread);
                SeekFile(f->handle,position,SEEK_CUR);
#ifdef ALLLOCK
                UNLOCK(f->pthreadslock);
#endif
#ifdef GZIPDEBUG
                MLOCK(ErrorMessageLock);
                {  char *s = AN.ziobufnum[numstream]+readsize;
                    MesPrint("%w read: %l +Last bytes read: %d %d %d %d %d in %s, newpos =
```

```
%10p",readsize,s[-5],s[-4],s[-3],s[-2],s[-1],f->name,position);
                }
                MUNLOCK(ErrorMessageLock);
#endif
                if ( readsize == 0 ) {
                    zsp->next_in  = AN.ziobufnum[numstream];
                    zsp->avail_in = f->ziosize;
                    zsp->total_in = 0;
                    return(zsp->total_out);
                }
                if ( readsize < 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("%wFillInputGZIP: Read error during compressed sort.");
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                ADDPOS(f->filesize,readsize);
                ADDPOS(f->POposition,readsize);
/*
                Set the input
*/
                zsp->next_in  = AN.ziobufnum[numstream];
                zsp->avail_in = readsize;
                zsp->total_in = 0;
            }
        }
        if ( toread > 0 || zsp->avail_in ) {
          while ( ( zerror = inflate(zsp,Z_NO_FLUSH) ) == Z_OK ) {
            if ( zsp->avail_out == 0 ) {
/*
                Finish
*/
                return((LONG)(zsp->total_out));
            }
            if ( zsp->avail_in == 0 ) {

                if ( ISEQUALPOS(S->fPatchesStop[numstream],*position) ) {
/*
                    We finished this stream. Try to terminate.
*/
                    zerror = Z_STREAM_END;
                    break;
/*
                    if ( ( zerror = inflate(zsp,Z_SYNC_FLUSH) ) == Z_OK ) {
                        return((LONG)(zsp->total_out));
                    }
                    else
                        break;
*/
/*
#ifdef GZIPDEBUG
                    MLOCK(ErrorMessageLock);
                    MesPrint("%wClosing stream %d",numstream);
#endif
                    readsize = zsp->total_out;
#ifdef GZIPDEBUG
                    if ( readsize > 0 ) {
                        WORD *s = (WORD *)(buffer+zsp->total_out);
                        MesPrint("%w   Last words: %d %d %d %d %d",s[-5],s[-4],s[-3],s[-2],s[-1]);
                    }
                    else {
                        MesPrint("%w No words");
                    }
                    MUNLOCK(ErrorMessageLock);
#endif
                    if ( ( zerror = inflateEnd(zsp) ) == Z_OK ) return(readsize);
                    break;
*/
                }
/*
                Read more input
*/
#ifdef GZIPDEBUG
                if ( numstream == 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("%wWant to read in stream 0 at position %10p",position);
                    MUNLOCK(ErrorMessageLock);
                }
#endif
                if ( ISGEPOSINC(S->fPatchesStop[numstream],*position,f->ziosize) ) {
                    toread = f->ziosize;
                }
                else {
                    DIFPOS(pos,S->fPatchesStop[numstream],*position);
                    toread = (LONG)(BASEPOSITION(pos));
                }
#ifdef GZIPDEBUG
```

```
                MLOCK(ErrorMessageLock);
                MesPrint("%w--Reading %l bytes in stream %d at position %10p",toread,numstream,position);
                MUNLOCK(ErrorMessageLock);
#endif
#ifdef ALLLOCK
                LOCK(f->pthreadslock);
#endif
                SeekFile(f->handle,position,SEEK_SET);
                readsize = ReadFile(f->handle,(UBYTE *)(AN.ziobufnum[numstream]),toread);
                SeekFile(f->handle,position,SEEK_CUR);
#ifdef ALLLOCK
                UNLOCK(f->pthreadslock);
#endif
#ifdef GZIPDEBUG
                MLOCK(ErrorMessageLock);
                {  char *s = AN.ziobufnum[numstream]+readsize;
                    MesPrint("%w   Last bytes read: %d %d %d %d %d",s[-5],s[-4],s[-3],s[-2],s[-1]);
                }
                MUNLOCK(ErrorMessageLock);
#endif
                if ( readsize == 0 ) {
                    zsp->next_in  = AN.ziobufnum[numstream];
                    zsp->avail_in = f->ziosize;
                    zsp->total_in = 0;
                    return(zsp->total_out);
                }
                if ( readsize < 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("%wFillInputGZIP: Read error during compressed sort.");
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                ADDPOS(f->filesize,readsize);
                ADDPOS(f->POposition,readsize);
/*
                Reset the input
*/
                zsp->next_in  = AN.ziobufnum[numstream];
                zsp->avail_in = readsize;
                zsp->total_in = 0;
            }
            else {
                break;
            }
          }
        }
        else {
            zerror = Z_STREAM_END;
            zsp->total_out = 0;
        }
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%w zerror = %d in stream %d. At position %10p",zerror,numstream,position);
            MUNLOCK(ErrorMessageLock);
#endif
        if ( zerror == Z_STREAM_END ) {
/*
            Reset the input
*/
            zsp->next_in  = Z_NULL;
            zsp->avail_in = 0;
            zsp->total_in = 0;
/*
            Make the final call and finish
*/
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%wClosing stream %d",numstream);
#endif
            readsize = zsp->total_out;
#ifdef GZIPDEBUG
            if ( readsize > 0 ) {
                WORD *s = (WORD *)(buffer+zsp->total_out);
                MesPrint("%w  -Last words: %d %d %d %d %d",s[-5],s[-4],s[-3],s[-2],s[-1]);
            }
            else {
                MesPrint("%w No words");
            }
            MUNLOCK(ErrorMessageLock);
#endif
            if ( zsp->zalloc != Z_NULL ) {
                zerror = inflateEnd(zsp);
                zsp->zalloc = Z_NULL;
            }
            if ( zerror == Z_OK || zerror == Z_STREAM_END ) return(readsize);
        }
```

```
        MLOCK(ErrorMessageLock);
        MesPrint("%wFillInputGZIP: Error in gzip handling of input. zerror = %d",zerror);
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    else {
#ifdef GZIPDEBUG
        MLOCK(ErrorMessageLock);
        MesPrint("%w++Reading %l bytes at position %10p",buffersize,position);
        MUNLOCK(ErrorMessageLock);
#endif
#ifdef ALLLOCK
        LOCK(f->pthreadslock);
#endif
        SeekFile(f->handle,position,SEEK_SET);
        readsize = ReadFile(f->handle,buffer,buffersize);
        SeekFile(f->handle,position,SEEK_CUR);
#ifdef ALLLOCK
        UNLOCK(f->pthreadslock);
#endif
        if ( readsize < 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("%wFillInputGZIP: Read error during uncompressed sort.");
            MesPrint("%w++Reading %l bytes at position %10p",buffersize,position);
            MUNLOCK(ErrorMessageLock);
        }
        return(readsize);
    }
}

/*
    #] FillInputGZIP :
    #[ ClearSortGZIP :
*/

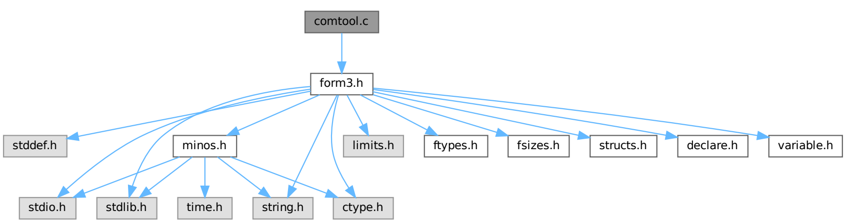
void ClearSortGZIP(FILEHANDLE *f)
{
    if ( f->ziobuffer ) {
        M_free(f->ziobuffer,"output zbuffer");
        M_free(f->zsp,"output zstream");
        f->ziobuffer = 0;
    }
}

/*
    #] ClearSortGZIP :
*/
#endif
```

4.15 comtool.c File Reference

`#include "form3.h"`

Include dependency graph for comtool.c:



Functions

- int inicbufs (void)
- void finishcbuf (WORD num)

- void clearcbuf (WORD num)
- WORD ∗ DoubleCbuffer (int num, WORD ∗w, int par)
- WORD ∗ AddLHS (int num)
- WORD ∗ AddRHS (int num, int type)
- int AddNtoL (int n, WORD ∗array)
- int AddNtoC (int bufnum, int n, WORD ∗array, int par)
- int InsTree (int bufnum, int h)
- int FindTree (int bufnum, WORD ∗subexpr)
- void RedoTree (CBUF ∗C, int size)
- void ClearTree (int i)
- int IniFbuffer (WORD bufnum)
- LONG numcommute (WORD ∗terms, LONG ∗numterms)

4.15.1 Detailed Description

Utility routines for the compiler.

Definition in file comtool.c.

4.15.2 Function Documentation

4.15.2.1 inicbufs()

```
int inicbufs (
            void  )
```

Creates a new compiler buffer and returns its ID number.

Returns

The ID number for the new compiler buffer.

Definition at line 47 of file comtool.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::BufferSize, CbUf::CanCommu, CbUf::dimension, CbUf::lhs, CbUf::numdum, CbUf::NumTerms, CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Referenced by AddRHS(), and StartVariables().

4.15.2.2 finishcbuf()

```
void finishcbuf (
            WORD num )
```

Frees a compiler buffer.

Parameters

num	The ID number for the buffer to be freed.

Definition at line 89 of file comtool.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::BufferSize, CbUf::CanCommu, CbUf::lhs, CbUf::NumTerms, CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Referenced by PF_BroadcastCBuf().

4.15.2.3 clearcbuf()

```
void clearcbuf (
            WORD num )
```

Clears contents in a compiler buffer.

Parameters

num	The ID number for the buffer to be cleared.

Definition at line 116 of file comtool.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::Pointer, and CbUf::rhs.

4.15.2.4 DoubleCbuffer()

```
WORD * DoubleCbuffer (
            int num,
            WORD * w,
            int par )
```

Doubles a compiler buffer.

Parameters

num	The ID number for the buffer to be doubled.
w	The pointer to the end (exclusive) of the current buffer. The contents in the range of [cbuf[num].Buffer,w) will be kept.

Definition at line 143 of file comtool.c.

References CbUf::Buffer, CbUf::BufferSize, CbUf::lhs, CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Referenced by AddNtoC(), and AddNtoL().

4.15.2.5 AddLHS()

```
WORD * AddLHS (
            int num )
```

Adds an LHS to a compiler buffer and returns the pointer to a buffer for the new LHS.

Parameters

num	The ID number for the buffer to get another LHS.

Definition at line 188 of file comtool.c.

References CbUf::lhs, and CbUf::Pointer.

Referenced by AddNtoL().

4.15.2.6 AddRHS()

```
WORD * AddRHS (
            int num,
            int type )
```

Adds an RHS to a compiler buffer and returns the pointer to a buffer for the new RHS.

Parameters

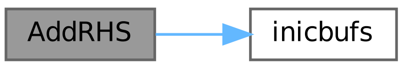
num	The ID number for the buffer to get another RHS.
type	If 0, the subexpression tree will be reallocated.

Definition at line 214 of file comtool.c.

References TaBlEs::buffers, TaBlEs::buffersfill, TaBlEs::bufferssize, TaBlEs::bufnum, CbUf::CanCommu, CbUf::dimension, inicbufs(), CbUf::numdum, CbUf::NumTerms, CbUf::Pointer, and CbUf::rhs.

Referenced by InsertArg(), StartVariables(), and TestMatch().

Here is the call graph for this function:



4.15.2.7 AddNtoL()

```
int AddNtoL (
            int n,
            WORD * array )
```

Adds an LHS with the given data to the current compiler buffer.

Parameters

n	The length of the data.
array	The data to be added.

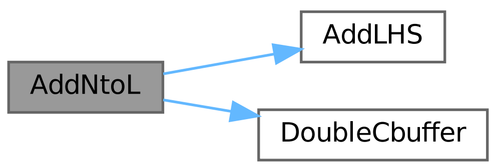
Returns

0 if succeeds.

Definition at line 288 of file comtool.c.

References AddLHS(), DoubleCbuffer(), CbUf::Pointer, and CbUf::Top.

Here is the call graph for this function:



4.15.2.8 AddNtoC()

```
int AddNtoC (
            int bufnum,
            int n,
            WORD * array,
            int par )
```

Adds the given data to the last LHS/RHS in a compiler buffer.

Parameters

bufnum	The ID number for the buffer where the data will be added.
n	The length of the data.
array	The data to be added.

Returns

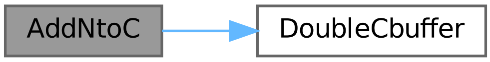
0 if succeeds.

Definition at line 317 of file comtool.c.

References DoubleCbuffer(), CbUf::Pointer, and CbUf::Top.

Referenced by InsertArg(), and StartVariables().

Here is the call graph for this function:



4.15.2.9 InsTree()

```
int InsTree (
            int bufnum,
            int h )
```

Definition at line 355 of file comtool.c.

4.15.2.10 FindTree()

```
int FindTree (
            int bufnum,
            WORD * subexpr )
```

Definition at line 533 of file comtool.c.

4.15.2.11 RedoTree()

```
void RedoTree (
            CBUF * C,
            int size )
```

Definition at line 569 of file comtool.c.

4.15.2.12 ClearTree()

```
void ClearTree (
            int i )
```

Definition at line 588 of file comtool.c.

4.15.2.13 IniFbuffer()

```
int IniFbuffer (
            WORD bufnum )
```

Initialize a factorization cache buffer. We set the size of the rhs and boomlijst buffers immediately to their final values.

Definition at line 614 of file comtool.c.

References tree::blnce, CbUf::boomlijst, CbUf::CanCommu, CbUf::dimension, tree::left, CbUf::numdum, CbUf::NumTerms, tree::parent, CbUf::rhs, tree::right, tree::usage, and tree::value.

4.15.2.14 numcommute()

```
LONG numcommute (
            WORD * terms,
            LONG * numterms )
```

Definition at line 659 of file comtool.c.

4.16 comtool.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
*/

#include "form3.h"

/*
    #] Includes :
    #[ inicbufs :
*/

int inicbufs(void)
{
    int i, num = AC.cbufList.num;
    CBUF *C = cbuf;
    for ( i = 0; i < num; i++, C++ ) {
        if ( C->Buffer == 0 ) break;
```

```
    }
    if ( i >= num ) C = (CBUF *)FromList(&AC.cbufList);
    else num = i;
    C->BufferSize = 2000;
    C->Buffer = (WORD *)Malloc1(C->BufferSize*sizeof(WORD),"compiler buffer-1");
    C->Pointer = C->Buffer;
    C->Top = C->Buffer + C->BufferSize;
    C->maxlhs = 10;
    C->lhs = (WORD **)Malloc1(C->maxlhs*sizeof(WORD *),"compiler buffer-2");
    C->numlhs = 0;
    C->mnumlhs = 0;
    C->maxrhs = 25;
    C->rhs = (WORD **)Malloc1(C->maxrhs*(sizeof(WORD *)+2*sizeof(LONG)+2*sizeof(WORD)),"compiler buffer-3");
    C->CanCommu = (LONG *)(C->rhs+C->maxrhs);
    C->NumTerms = C->CanCommu+C->maxrhs;
    C->numdum = (WORD *)(C->NumTerms+C->maxrhs);
    C->dimension = C->numdum + C->maxrhs;
    C->numrhs = 0;
    C->mnumrhs = 0;
    C->rhs[0] = C->rhs[1] = C->Pointer;
    C->boomlijst = 0;
    RedoTree(C,C->maxrhs);
    ClearTree(num);
    return(num);
}

/*
    #] inicbufs :
    #[ finishcbuf :
*/

void finishcbuf(WORD num)
{
    CBUF *C = cbuf+num;
    if ( C->Buffer ) M_free(C->Buffer,"compiler buffer-1");
    if ( C->rhs ) M_free(C->rhs,"compiler buffer-3");
    if ( C->lhs ) M_free(C->lhs,"compiler buffer-2");
    if ( C->boomlijst ) M_free(C->boomlijst,"boomlijst");
    C->Top = C->Pointer = C->Buffer = 0;
    C->rhs = C->lhs = 0;
    C->CanCommu = 0;
    C->NumTerms = 0;
    C->BufferSize = 0;
    C->boomlijst = 0;
    C->numlhs = C->numrhs = C->maxlhs = C->maxrhs = C->mnumlhs =
    C->mnumrhs = C->numtree = C->rootnum = C->MaxTreeSize = 0;
}

/*
    #] finishcbuf :
    #[ clearcbuf :
*/

void clearcbuf(WORD num)
{
    CBUF *C = cbuf+num;
    if ( C->boomlijst ) M_free(C->boomlijst,"boomlijst");
    C->Pointer = C->Buffer;
    C->numrhs = C->numlhs = 0;
    C->mnumlhs = 0;
    C->boomlijst = 0;
    C->mnumrhs = 0;
    C->rhs[0] = C->rhs[1] = C->Pointer;
    C->numtree = C->rootnum = C->MaxTreeSize = 0;
    RedoTree(C,C->maxrhs);
    ClearTree(num);
}

/*
    #] clearcbuf :
    #[ DoubleCbuffer :
*/

WORD *DoubleCbuffer(int num, WORD *w,int par)
{
    CBUF *C = cbuf + num;
    LONG newsize = C->BufferSize*2;
    WORD *newbuffer = (WORD *)Malloc1(newsize*sizeof(WORD),"compiler buffer-4");
    WORD *w1, *w2;
    LONG offset, j, i;
    DUMMYUSE(par)
/*
    MLOCK(ErrorMessageLock);
        MesPrint(" doubleCbuffer: par = %d",par);
    MUNLOCK(ErrorMessageLock);
*/
```

```
    w1 = C->Buffer; w2 = newbuffer;
    i = w - w1;
    j = i & 7;
    while ( --j >= 0 ) *w2++ = *w1++;
    i >>= 3;
    while ( --i >= 0 ) {
        *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++;
        *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++; *w2++ = *w1++;
    }
    offset = newbuffer - C->Buffer;
    for ( i = 0; i <= C->numlhs; i++ ) C->lhs[i] += offset;
    for ( i = 1; i <= C->numrhs; i++ ) C->rhs[i] += offset;
    w1 = C->Buffer;
    C->Pointer += offset;
    C->Top = newbuffer + newsize;
    C->BufferSize = newsize;
    C->Buffer = newbuffer;
    M_free(w1,"DoubleCbuffer");
    return(w2);
}

/*
    #] DoubleCbuffer :
    #[ AddLHS :
*/

WORD *AddLHS(int num)
{
    CBUF *C = cbuf + num;
    C->numlhs++;
    if ( C->numlhs >= (C->maxlhs-2) ) {
        WORD ***ppp = &(C->lhs); /* to avoid compiler warning */
        if ( DoubleList((void ***)ppp,&(C->maxlhs),sizeof(WORD *),
        "statement lists") ) Terminate(-1);
    }
    C->lhs[C->numlhs] = C->Pointer;
    C->lhs[C->numlhs+1] = 0;
    return(C->Pointer);
}

/*
    #] AddLHS :
    #[ AddRHS :
*/

WORD *AddRHS(int num, int type)
{
    LONG fullsize, *lold, newsize;
    int i;
    WORD **old, *wold;
    CBUF *C;
restart:;
    C = cbuf + num;
    if ( C->numrhs >= (C->maxrhs-2) ) {
        if ( C->maxrhs == 0 ) newsize = 100;
        else                  newsize = C->maxrhs * 2;
        if ( newsize > MAXCOMBUFRHS ) newsize = MAXCOMBUFRHS;
        if ( newsize == C->maxrhs ) {
            if ( AC.tablefilling ) {
                TABLES T = functions[AC.tablefilling].tabl;
/*
                We add a compiler buffer, change a few settings and continue.
*/
                if ( T->buffersfill >= T->bufferssize ) {
                    int new1 = 2*T->bufferssize;
                    WORD *nbufs = (WORD *)Malloc1(new1*sizeof(WORD),"Table compile buffers");
                    for ( i = 0; i < T->buffersfill; i++ )
                            nbufs[i] = T->buffers[i];
                    for ( ; i < new1; i++ ) nbufs[i] = 0;
                    M_free(T->buffers,"Table compile buffers");
                    T->buffers = nbufs;
                    T->bufferssize = new1;
                }
                T->buffers[T->buffersfill++] = T->bufnum = inicbufs();
                AC.cbufnum = num = T->bufnum;
                goto restart;
            }
            else {
                MesPrint("@Compiler buffer overflow. Try to make modules smaller");
                Terminate(-1);
            }
        }
        old = C->rhs;
        fullsize = newsize * (sizeof(WORD *) + 2*sizeof(LONG) + 2*sizeof(WORD));
        C->rhs = (WORD **)Malloc1(fullsize,"subexpression lists");
        for ( i = 0; i < C->maxrhs; i++ ) C->rhs[i] = old[i];
        lold = C->CanCommu; C->CanCommu = (LONG *)(C->rhs+newsize);
```

```
        for ( i = 0; i < C->maxrhs; i++ ) C->CanCommu[i] = lold[i];
        lold = C->NumTerms; C->NumTerms = (LONG *)(C->rhs+2*newsize);
        for ( i = 0; i < C->maxrhs; i++ ) C->NumTerms[i] = lold[i];
        wold = C->numdum; C->numdum = (WORD *)(C->NumTerms+newsize);
        for ( i = 0; i < C->maxrhs; i++ ) C->numdum[i] = wold[i];
        wold = C->dimension; C->dimension = (WORD *)(C->numdum+newsize);
        for ( i = 0; i < C->maxrhs; i++ ) C->dimension[i] = wold[i];
        if ( old ) M_free(old,"subexpression lists");
        C->maxrhs = newsize;
        if ( type == 0 ) RedoTree(C,C->maxrhs);
    }
    C->numrhs++;
    C->CanCommu[C->numrhs] = 0;
    C->NumTerms[C->numrhs] = 0;
    C->numdum[C->numrhs] = 0;
    C->dimension[C->numrhs] = 0;
    C->rhs[C->numrhs] = C->Pointer;
    return(C->Pointer);
}

/*
    #] AddRHS :
    #[ AddNtoL :
*/

int AddNtoL(int n, WORD *array)
{
    int i;
    CBUF *C = cbuf+AC.cbufnum;
#ifdef COMPBUFDEBUG
    MesPrint("LH: %a",n,array);
#endif
    AddLHS(AC.cbufnum);
    while ( C->Pointer+n >= C->Top ) DoubleCbuffer(AC.cbufnum,C->Pointer,1);
    for ( i = 0; i < n; i++ ) *(C->Pointer)++ = *array++;
    return(0);
}

/*
    #] AddNtoL :
    #[ AddNtoC :

    Commentary: added the bufnum on 14-sep-2010 to make the whole a bit
    more flexible (JV). Still to do with AddNtoL.
*/

int AddNtoC(int bufnum, int n, WORD *array,int par)
{
    int i;
    WORD *w;
    CBUF *C = cbuf+bufnum;
#ifdef COMPBUFDEBUG
    MesPrint("RH: %a",n,array);
#endif
    while ( C->Pointer+n+1 >= C->Top ) DoubleCbuffer(bufnum,C->Pointer,50+par);
    w = C->Pointer;
    for ( i = 0; i < n; i++ ) *w++ = *array++;
    C->Pointer = w;
    return(0);
}

/*
    #] AddNtoC :
    #[ InsTree :

    Routines for balanced tree searching and insertion.
    Compared to Knuth we have a parent link. This minimizes the
    number of compares. That is better for anything that is more
    complicated than just single numbers.
    There are no provisions for removing elements from the tree.
    The routines are:
    void RedoTree(size) Re-allocates the tree space. There will
                        be MaxTreeSize = size elements.
    void ClearTree()    Prunes the tree down to the root element.
    int InsTree(int,int)Searches for the requested element. If not found it
                        will allocate a new element, balance the tree if
                        necessary and return the called number.
                        If it was in the tree, it returns the tree 'value'.

    Commentary: added the bufnum on 14-sep-2010 to make the whole a bit
    more flexible (JV).
*/
static COMPTREE comptreezero = {0,0,0,0,0,0};

int InsTree(int bufnum, int h)
{
    CBUF *C = cbuf + bufnum;
```

```
    COMPTREE *boomlijst = C->boomlijst, *q = boomlijst + C->rootnum, *p, *s;
    WORD *v1, *v2, *v3;
    int ip, iq, is;

    if ( C->numtree + 1 >= C->MaxTreeSize ) {
        if ( C->MaxTreeSize == 0 ) {
            COMPTREE *root;
            C->MaxTreeSize = 125;
            C->boomlijst = (COMPTREE *)Malloc1((C->MaxTreeSize+1)*sizeof(COMPTREE),
                "ClearInsTree");
            root = C->boomlijst;
            C->numtree = 0;
            C->rootnum = 0;
            root->left = -1;
            root->right = -1;
            root->parent = -1;
            root->blnce = 0;
            root->value = -1;
            root->usage = 0;
            for ( ip = 1; ip < C->MaxTreeSize; ip++ ) { C->boomlijst[ip] = comptreezero; }
        }
        else {
            is = C->MaxTreeSize * 2;
            s  = (COMPTREE *)Malloc1((is+1)*sizeof(COMPTREE),"InsTree");
            for ( ip = 0; ip < C->MaxTreeSize; ip++ ) { s[ip] = C->boomlijst[ip]; }
            for ( ip = C->MaxTreeSize; ip <= is; ip++ ) { s[ip] = comptreezero; }
            if ( C->boomlijst ) M_free(C->boomlijst,"InsTree");
            C->boomlijst = s;
            C->MaxTreeSize = is;
        }
        boomlijst = C->boomlijst;
        q = boomlijst + C->rootnum;
    }

    if ( q->right == -1 ) { /* First element */
        C->numtree++;
        s = boomlijst+C->numtree;
        q->right = C->numtree;
        s->parent = C->rootnum;
        s->left = s->right = -1;
        s->blnce = 0;
        s->value = h;
        s->usage = 1;
        return(h);
    }
    ip = q->right;
    while ( ip >= 0 ) {
        p = boomlijst + ip;
        v1 = C->rhs[p->value]; v2 = v3 = C->rhs[h];
        while ( *v3 ) v3 += *v3;  /* find the 0 that indicates end-of-expr */
        while ( *v1 == *v2 && v2 < v3 ) { v1++; v2++; }
        if ( *v1 > *v2 ) {
            iq = p->right;
            if ( iq >= 0 ) { ip = iq; }
            else {
                C->numtree++;
                is = C->numtree;
                p->right = is;
                s = boomlijst + is;
                s->parent = ip; s->left = s->right = -1;
                s->blnce = 0;   s->value = h; s->usage = 1;
                p->blnce++;
                if ( p->blnce == 0 ) return(h);
                goto balance;
            }
        }
        else if ( *v1 < *v2 ) {
            iq = p->left;
            if ( iq >= 0 ) { ip = iq; }
            else {
                C->numtree++;
                is = C->numtree;
                s = boomlijst+is;
                p->left = is;
                s->parent = ip; s->left = s->right = -1;
                s->blnce = 0;   s->value = h; s->usage = 1;
                p->blnce--;
                if ( p->blnce == 0 ) return(h);
                goto balance;
            }
        }
        else {
            p->usage++;
            return(p->value);
        }
    }
    MesPrint("We vallen uit de boom!");
```

```
    Terminate(-1);
    return(h);
balance:;
    for (;;) {
        p = boomlijst + ip;
        iq = p->parent;
        if ( iq == C->rootnum ) break;
        q = boomlijst + iq;
        if ( ip == q->left ) q->blnce--;
        else                 q->blnce++;
        if ( q->blnce == 0 ) break;
        if ( q->blnce == -2 ) {
            if ( p->blnce == -1 ) { /* single rotation */
                q->left = p->right;
                p->right = iq;
                p->parent = q->parent;
                q->parent = ip;
                if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
                else                                   boomlijst[p->parent].right = ip;
                if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
                q->blnce = p->blnce = 0;
            }
            else {  /* double rotation */
                s = boomlijst + is;
                q->left = s->right;
                p->right = s->left;
                s->right = iq;
                s->left = ip;
                if ( p->right >= 0 ) boomlijst[p->right].parent = ip;
                if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
                s->parent = q->parent;
                q->parent = is;
                p->parent = is;
                if ( boomlijst[s->parent].left == iq )
                       boomlijst[s->parent].left = is;
                else   boomlijst[s->parent].right = is;
                if ( s->blnce > 0 ) { q->blnce = s->blnce = 0; p->blnce = -1; }
                else if ( s->blnce < 0 ) { p->blnce = s->blnce = 0; q->blnce = 1; }
                else { p->blnce = s->blnce = q->blnce = 0; }
            }
            break;
        }
        else if ( q->blnce == 2 ) {
            if ( p->blnce == 1 ) {  /* single rotation */
                q->right = p->left;
                p->left = iq;
                p->parent = q->parent;
                q->parent = ip;
                if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
                else                                   boomlijst[p->parent].right = ip;
                if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
                q->blnce = p->blnce = 0;
            }
            else {  /* double rotation */
                s = boomlijst + is;
                q->right = s->left;
                p->left = s->right;
                s->left = iq;
                s->right = ip;
                if ( p->left >= 0 ) boomlijst[p->left].parent = ip;
                if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
                s->parent = q->parent;
                q->parent = is;
                p->parent = is;
                if ( boomlijst[s->parent].left == iq ) boomlijst[s->parent].left = is;
                else                                   boomlijst[s->parent].right = is;
                if ( s->blnce < 0 ) { q->blnce = s->blnce = 0; p->blnce = 1; }
                else if ( s->blnce > 0 ) { p->blnce = s->blnce = 0; q->blnce = -1; }
                else { p->blnce = s->blnce = q->blnce = 0; }
            }
            break;
        }
        is = ip; ip = iq;
    }
    return(h);
}

/*
    #] InsTree :
    #[ FindTree :

    Routines for balanced tree searching.
    Is like InsTree but without the insertions.
    Returns -1 if the element is not in the tree.
    The advantage of this routine over InsTree is that this routine
    can be run in parallel.
*/
```

```
int FindTree(int bufnum, WORD *subexpr)
{
    CBUF *C = cbuf + bufnum;
    COMPTREE *boomlijst = C->boomlijst, *q = boomlijst + C->rootnum, *p;
    WORD *v1, *v2, *v3;
    int ip, iq;

    ip = q->right;
    while ( ip >= 0 ) {
        p = boomlijst + ip;
        v1 = C->rhs[p->value]; v2 = v3 = subexpr;
        while ( *v3 ) v3 += *v3;  /* find the 0 that indicates end-of-expr */
        while ( *v1 == *v2 && v2 < v3 ) { v1++; v2++; }
        if ( *v1 > *v2 ) {
            iq = p->right;
            if ( iq >= 0 ) { ip = iq; }
            else { return(-1); }
        }
        else if ( *v1 < *v2 ) {
            iq = p->left;
            if ( iq >= 0 ) { ip = iq; }
            else { return(-1); }
        }
        else {
            p->usage++;
            return(p->value);
        }
    }
    return(-1);
}

/*
    #] FindTree :
    #[ RedoTree :
*/

void RedoTree(CBUF *C, int size)
{
    COMPTREE *newboomlijst;
    int i;
    newboomlijst = (COMPTREE *)Malloc1((size+1)*sizeof(COMPTREE),"newboomlijst");
    if ( C->boomlijst ) {
        if ( C->MaxTreeSize > size ) C->MaxTreeSize = size;
        for ( i = 0; i < C->MaxTreeSize; i++ ) newboomlijst[i] = C->boomlijst[i];
        M_free(C->boomlijst,"boomlijst");
    }
    C->boomlijst = newboomlijst;
    C->MaxTreeSize = size;
}

/*
    #] RedoTree :
    #[ ClearTree :
*/

void ClearTree(int i)
{
    CBUF *C = cbuf + i;
    COMPTREE *root = C->boomlijst;
    if ( root ) {
        C->numtree = 0;
        C->rootnum = 0;
        root->left = -1;
        root->right = -1;
        root->parent = -1;
        root->blnce = 0;
        root->value = -1;
        root->usage = 0;
    }
}

/*
    #] ClearTree :
    #[ IniFbuffer :
*/
int IniFbuffer(WORD bufnum)
{
    CBUF *C = cbuf + bufnum;
    COMPTREE *root;
    int i;
    LONG fullsize;
    C->maxrhs = AM.fbuffersize;
    C->MaxTreeSize = AM.fbuffersize;

    /*
     * Note that bufnum is a return value of inicbufs(). So C has been already
```

```
     * initialized. (TU 20 Dec 2011)
     */
    if ( C->boomlijst ) M_free(C->boomlijst, "IniFbuffer-tree");
    if ( C->rhs ) M_free(C->rhs, "IniFbuffer-rhs");

    C->boomlijst = (COMPTREE *)Malloc1((C->MaxTreeSize+1)*sizeof(COMPTREE),"IniFbuffer-tree");
    root = C->boomlijst;
    C->numtree = 0;
    C->rootnum = 0;
    root->left = -1;
    root->right = -1;
    root->parent = -1;
    root->blnce = 0;
    root->value = -1;
    root->usage = 0;
    for ( i = 1; i < C->MaxTreeSize; i++ ) { C->boomlijst[i] = comptreezero; }

    fullsize = (C->maxrhs+1) * (sizeof(WORD *) + 2*sizeof(LONG) + 2*sizeof(WORD));
    C->rhs = (WORD **)Malloc1(fullsize,"IniFbuffer-rhs");
    C->CanCommu = (LONG *)(C->rhs+C->maxrhs);
    C->NumTerms = (LONG *)(C->rhs+2*C->maxrhs);
    C->numdum = (WORD *)(C->NumTerms+C->maxrhs);
    C->dimension = (WORD *)(C->numdum+C->maxrhs);

    return(0);
}

/*
    #] IniFbuffer :
    #[ numcommute :

    Returns the number of non-commuting terms in the expression
*/

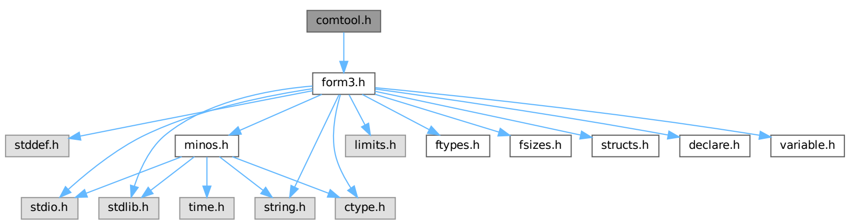
LONG numcommute(WORD *terms, LONG *numterms)
{
    LONG num = 0;
    WORD *t, *m;
    *numterms = 0;
    while ( *terms ) {
        *numterms += 1;
        t = terms + 1;
        GETSTOP(terms,m);
        while ( t < m ) {
            if ( *t >= FUNCTION ) {
                if ( functions[*t-FUNCTION].commute ) { num++; break; }
            }
            t += t[1];
        }
        terms = terms + *terms;
    }
    return(num);
}

/*
    #] numcommute :
*/
```

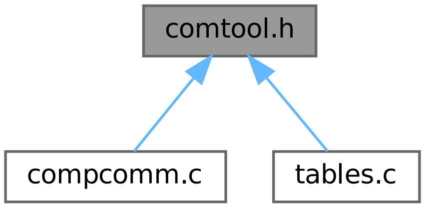
4.17 comtool.h File Reference

```
#include "form3.h"
```

Include dependency graph for comtool.h:



This graph shows which files directly or indirectly include this file:



4.17.1 Detailed Description

Utility routines for the compiler.

Definition in file comtool.h.

4.18 comtool.h

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
#ifndef FORM_COMTOOL_H_
#define FORM_COMTOOL_H_
/*
    #[ Includes :
*/

#include "form3.h"

/*
    #] Includes :
    #[ Inline functions :
*/

static inline void SkipSpaces(UBYTE **s)
{
```

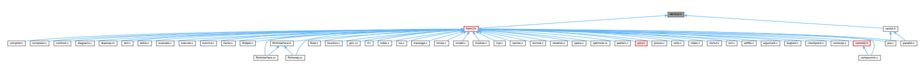
```
    const char *p = (const char *)*s;
    while ( *p == ' ' || *p == ',' || *p == '\t' ) p++;
    *s = (UBYTE *)p;
}

static inline int ConsumeOption(UBYTE **s, const char *opt)
{
    const char *p = (const char *)*s;
    while ( *p && *opt && tolower(*p) == tolower(*opt) ) {
        p++;
        opt++;
    }
    /* Check if `opt' ended. */
    if ( !*opt ) {
        /* Check if `*p' is a word boundary. */
        if ( !*p || !(FG.cTable[(unsigned char)*p] == 0 ||
                      FG.cTable[(unsigned char)*p] == 1 || *p == '_' ||
                      *p == '$') ) {
            /* Consume the option. Skip the trailing spaces. */
            *s = (UBYTE *)p;
            SkipSpaces(s);
            return(1);
        }
    }
    return(0);
}

/*
    #] Inline functions :
*/
#endif  /* FORM_COMTOOL_H_ */
```

4.19 declare.h File Reference

This graph shows which files directly or indirectly include this file:



Macros

- #define MaX(x, y) ((x) > (y) ? (x): (y))
- #define MiN(x, y) ((x) < (y) ? (x): (y))
- #define ABS(x) ((x) < 0 ? -(x): (x))
- #define SGN(x) ((x) > 0 ? 1 : (x) < 0 ? -1 : 0)
- #define REDLENG(x) ((((x)<0)?((x)+1):((x)-1))/2)
- #define INCLENG(x) (((x)<0)?(((x)*2)-1):(((x)*2)+1))
- #define GETCOEF(x, y) x += *x;y = x[-1];x -= ABS(y);y=REDLENG(y)
- #define GETSTOP(x, y) y=x+(*x)-1;y -= ABS(*y)-1
- #define StuffAdd(x, y) (((x)<0?-1:1)*(y)+((y)<0?-1:1)*(x))
- #define EXCHN(t1, t2, n) { WORD a,i; for(i=0;i<n;i++){a=t1[i];t1[i]=t2[i];t2[i]=a;} }
- #define EXCH(x, y) { WORD a = (x); (x) = (y); (y) = a; }
- #define TOKENTOLINE(x, y)
- #define UngetFromStream(stream, c) ((stream)->nextchar[(stream)->isnextchar++]=c)
- #define AddLineFeed(s, n) { (s)[(n)++] = LINEFEED; }
- #define TryRecover(x) Terminate(-1)
- #define UngetChar(c) { pushbackchar = c; }
- #define ParseNumber(x, s) {(x)=0;while(*(s)>='0'&&*(s)<='9')(x)=10*(x)+*(s)++ -'0';}
- #define ParseSign(sgn, s)
- #define ParseSignedNumber(x, s)
- #define NCOPY(s, t, n) { while ((n)-- > 0) { *s++ = *t++; } }
- #define NCOPYI(s, t, n) { while ((n)-- > 0) { *s++ = *t++; } }

- #define NCOPYB(s, t, n) { while ((n)-- > 0) { ∗s++ = ∗t++; } }
- #define NCOPYI32(s, t, n) { while ((n)-- > 0) { ∗s++ = ∗t++; } }
- #define WCOPY(s, t, n) { int nn=n; WORD ∗ss=(WORD ∗)s, ∗tt=(WORD ∗)t; while ((nn)-- > 0) { ∗ss++ = ∗tt++; } }
- #define NeedNumber(x, s, err)
- #define SKIPBLANKS(s) { while (∗(s) == ' ' || ∗(s) == '\t') (s)++; }
- #define FLUSHCONSOLE if (AP.InOutBuf > 0) CharOut(LINEFEED)
- #define SKIPBRA1(s)
- #define SKIPBRA2(s)
- #define SKIPBRA3(s)
- #define SKIPBRA4(s)
- #define SKIPBRA5(s)
- #define CYCLE1(t, a, i) {t iX=∗a; WORD jX; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX;}
- #define AddToCB(c, wx)
- #define EXCHINOUT
- #define BACKINOUT
- #define CopyArg(to, from)
- #define FILLARG(w)
- #define COPYARG(w, t)
- #define ZEROARG(w)
- #define FILLFUN(w)
- #define COPYFUN(w, t)
- #define COPYFUN3(w, t)
- #define FILLFUN3(w)
- #define FILLSUB(w)
- #define COPYSUB(w, ww)
- #define FILLEXPR(w)
- #define NEXTARG(x) if(∗x>0) x += ∗x; else if(∗x <= -FUNCTION)x++; else x += 2;
- #define COPY1ARG(s1, t1)
- #define ZeroFillRange(w, begin, end)
- #define TABLESIZE(a, b) (((WORD)sizeof(a))/((WORD)sizeof(b)))
- #define WORDDIF(x, y) (WORD)(x-y)
- #define wsizeof(a) ((WORD)sizeof(a))
- #define VARNAME(type, num) (AC.varnames->namebuffer+type[num].name)
- #define DOLLARNAME(type, num) (AC.dollarnames->namebuffer+type[num].name)
- #define EXPRNAME(num) (AC.exprnames->namebuffer+Expressions[num].name)
- #define PREV(x) prevorder?prevorder:x
- #define Terminate(x) do { TerminateImpl(x, __FILE__, __LINE__, __FUNCTION__); } while(0)
- #define SETERROR(x) { Terminate(-1); return(-1); }
- #define DUMMYUSE(x) (void)(x);
- #define ADDPOS(pp, x) (pp).p1 = ((pp).p1+(LONG)(x))
- #define SETBASELENGTH(ss, x) (ss).p1 = (LONG)(x)
- #define SETBASEPOSITION(pp, x) (pp).p1 = (LONG)(x)
- #define ISEQUALPOSINC(pp1, pp2, x) ((pp1).p1 == ((pp2).p1+(LONG)(x)))
- #define ISGEPOSINC(pp1, pp2, x) ((pp1).p1 >= ((pp2).p1+(LONG)(x)))
- #define DIVPOS(pp, n) ((pp).p1/(LONG)(n))
- #define MULPOS(pp, n) (pp).p1 ∗= (LONG)(n)
- #define DIFPOS(ss, pp1, pp2) (ss).p1 = ((pp1).p1-(pp2).p1)
- #define DIFBASE(pp1, pp2) ((pp1).p1-(pp2).p1)
- #define ADD2POS(pp1, pp2) (pp1).p1 += (pp2).p1
- #define PUTZERO(pp) (pp).p1 = 0
- #define BASEPOSITION(pp) ((pp).p1)
- #define SETSTARTPOS(pp) (pp).p1 = -2
- #define NOTSTARTPOS(pp) ((pp).p1 > -2)
- #define ISMINPOS(pp) ((pp).p1 == -1)

- #define ISEQUALPOS(pp1, pp2) ((pp1).p1 == (pp2).p1)
- #define ISNOTEQUALPOS(pp1, pp2) ((pp1).p1 != (pp2).p1)
- #define ISLESSPOS(pp1, pp2) ((pp1).p1 < (pp2).p1)
- #define ISGEPOS(pp1, pp2) ((pp1).p1 >= (pp2).p1)
- #define ISNOTZEROPOS(pp) ((pp).p1 != 0)
- #define ISZEROPOS(pp) ((pp).p1 == 0)
- #define ISPOSPOS(pp) ((pp).p1 > 0)
- #define ISNEGPOS(pp) ((pp).p1 < 0)
- #define TOLONG(x) ((LONG)(x))
- #define Add2Com(x) { WORD cod[2]; cod[0] = x; cod[1] = 2; AddNtoL(2,cod); }
- #define Add3Com(x1, x2) { WORD cod[3]; cod[0] = x1; cod[1] = 3; cod[2] = x2; AddNtoL(3,cod); }
- #define Add4Com(x1, x2, x3)
- #define Add5Com(x1, x2, x3, x4)
- #define WantAddPointers(x)
- #define WantAddLongs(x)
- #define WantAddPositions(x)
- #define FORM_INLINE inline
- #define MEMORYMACROS
- #define TermMalloc(x) ((AT.TermMemTop <= 0) ? TermMallocAddMemory(BHEAD0), AT.TermMemHeap[--AT.TermMemTop]: AT.TermMemHeap[--AT.TermMemTop])
- #define NumberMalloc(x) ((AT.NumberMemTop <= 0) ? NumberMallocAddMemory(BHEAD0), AT.↩ NumberMemHeap[--AT.NumberMemTop]: AT.NumberMemHeap[--AT.NumberMemTop])
- #define CacheNumberMalloc(x) ((AT.CacheNumberMemTop <= 0) ? CacheNumberMallocAdd↩ Memory(BHEAD0), AT.CacheNumberMemHeap[--AT.CacheNumberMemTop]: AT.CacheNumberMemHeap[--AT.CacheNumberMemTop])
- #define TermFree(TermMem, x) AT.TermMemHeap[AT.TermMemTop++] = (WORD ∗)(TermMem)
- #define NumberFree(NumberMem, x) AT.NumberMemHeap[AT.NumberMemTop++] = (UWORD ∗)(Number↩ Mem)
- #define CacheNumberFree(NumberMem, x) AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = (UWORD ∗)(NumberMem)
- #define NestingChecksum() (AC.IfLevel + AC.RepLevel + AC.arglevel + AC.insidelevel + AC.termlevel + AC.↩ inexprlevel + AC.dolooplevel +AC.SwitchLevel)
- #define MesNesting() MesPrint("&Illegal nesting of if, repeat, argument, inside, term, inexpression and do")
- #define MarkPolyRatFunDirty(T)
- #define PUSHPREASSIGNLEVEL
- #define POPPREASSIGNLEVEL
- #define EXTERNLOCK(x)
- #define INILOCK(x)
- #define LOCK(x)
- #define UNLOCK(x)
- #define EXTERNRWLOCK(x)
- #define INIRWLOCK(x)
- #define RWLOCKR(x)
- #define RWLOCKW(x)
- #define UNRWLOCK(x)
- #define MLOCK(x)
- #define MUNLOCK(x)
- #define GETIDENTITY
- #define GETBIDENTITY
- #define M_alloc(x) malloc((size_t)(x))
- #define CompareTerms ((COMPARE)AR.CompareRoutine)
- #define FiniShuffle AN.SHvar.finishuf
- #define DoShtuffle ((DO_UFFLE)AN.SHvar.do_uffle)

Typedefs

- typedef int(∗ WRITEBUFTOEXTCHANNEL) (char ∗, size_t)
- typedef int(∗ GETCFROMEXTCHANNEL) (void)
- typedef int(∗ SETTERMINATORFOREXTERNALCHANNEL) (char ∗)
- typedef int(∗ SETKILLMODEFOREXTERNALCHANNEL) (int, int)
- typedef LONG(∗ WRITEFILE) (int, UBYTE ∗, LONG)
- typedef WORD(∗ GETTERM) (PHEAD WORD ∗)

Functions

- void TELLFILE (int, POSITION ∗)
- void StartVariables (void)
- void **setSignalHandlers** (void)
- UBYTE ∗ CodeToLine (WORD, UBYTE ∗)
- UBYTE ∗ AddArrayIndex (WORD, UBYTE ∗)
- INDEXENTRY ∗ FindInIndex (WORD, FILEDATA ∗, WORD, WORD)
- INDEXENTRY ∗ NextFileIndex (POSITION ∗)
- WORD ∗ PasteTerm (PHEAD WORD, WORD ∗, WORD ∗, WORD, WORD)
- UBYTE ∗ StrCopy (UBYTE ∗, UBYTE ∗)
- UBYTE ∗ WrtPower (UBYTE ∗, WORD)
- WORD AccumGCD (PHEAD UWORD ∗, WORD ∗, UWORD ∗, WORD)
- void AddArgs (PHEAD WORD ∗, WORD ∗, WORD ∗)
- WORD AddCoef (PHEAD WORD ∗∗, WORD ∗∗)
- WORD AddLong (UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- WORD AddPLon (UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- WORD AddPoly (PHEAD WORD ∗∗, WORD ∗∗)
- WORD AddRat (PHEAD UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- void AddToLine (UBYTE ∗)
- WORD AddWild (PHEAD WORD, WORD, WORD)
- WORD BigLong (UWORD ∗, WORD, UWORD ∗, WORD)
- WORD BinomGen (PHEAD WORD ∗, WORD, WORD ∗∗, WORD, WORD, WORD, WORD, WORD, UWORD ∗, WORD)
- WORD CheckWild (PHEAD WORD, WORD, WORD, WORD ∗)
- WORD Chisholm (PHEAD WORD ∗, WORD)
- WORD CleanExpr (WORD)
- void CleanUp (WORD)
- void ClearWild (PHEAD0)
- WORD CompareFunctions (WORD ∗, WORD ∗)
- WORD Commute (WORD ∗, WORD ∗)
- WORD DetCommu (WORD ∗)
- WORD DoesCommu (WORD ∗)
- int CompArg (WORD ∗, WORD ∗)
- WORD CompCoef (WORD ∗, WORD ∗)
- WORD CompGroup (PHEAD WORD, WORD ∗∗, WORD ∗, WORD ∗, WORD)
- WORD Compare1 (PHEAD WORD ∗, WORD ∗, WORD)
- WORD CountDo (WORD ∗, WORD ∗)
- WORD CountFun (WORD ∗, WORD ∗)
- WORD DimensionSubterm (WORD ∗)
- WORD DimensionTerm (WORD ∗)
- WORD DimensionExpression (PHEAD WORD ∗)
- WORD Deferred (PHEAD WORD ∗, WORD)
- WORD DeleteStore (WORD)
- WORD DetCurDum (PHEAD WORD ∗)

- void DetVars (WORD ∗, WORD)
- WORD Distribute (DISTRIBUTE ∗, WORD)
- WORD DivLong (UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗, UWORD ∗, WORD ∗)
- WORD DivRat (PHEAD UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- WORD Divvy (PHEAD UWORD ∗, WORD ∗, UWORD ∗, WORD)
- WORD DoDelta (WORD ∗)
- WORD DoDelta3 (PHEAD WORD ∗, WORD)
- WORD TestPartitions (WORD ∗, PARTI ∗)
- WORD DoPartitions (PHEAD WORD ∗, WORD)
- int CoCanonicalize (UBYTE ∗)
- int DoCanonicalize (PHEAD WORD ∗, WORD ∗)
- WORD GenTopologies (PHEAD WORD ∗, WORD)
- WORD GenDiagrams (PHEAD WORD ∗, WORD)
- int DoTopologyCanonicalize (PHEAD WORD ∗, WORD, WORD, WORD ∗)
- int DoShattering (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- WORD GenerateTopologies (PHEAD WORD, WORD, WORD, WORD)
- WORD DoTableExpansion (WORD ∗, WORD)
- WORD DoDistrib (PHEAD WORD ∗, WORD)
- WORD DoShuffle (WORD ∗, WORD, WORD, WORD)
- WORD DoPermutations (PHEAD WORD ∗, WORD)
- int Shuffle (WORD ∗, WORD ∗, WORD ∗)
- int FinishShuffle (WORD ∗)
- WORD DoStuffle (WORD ∗, WORD, WORD, WORD)
- int Stuffle (WORD ∗, WORD ∗, WORD ∗)
- int FinishStuffle (WORD ∗)
- WORD ∗ StuffRootAdd (WORD ∗, WORD ∗, WORD ∗)
- WORD TestUse (WORD ∗, WORD)
- DBASE ∗ FindTB (UBYTE ∗)
- int CheckTableDeclarations (DBASE ∗)
- WORD Apply (WORD ∗, WORD)
- int ApplyExec (WORD ∗, int, WORD)
- WORD ApplyReset (WORD)
- WORD TableReset (void)
- void ReWorkT (WORD ∗, WORD ∗, WORD)
- WORD GetIfDollarNum (WORD ∗, WORD ∗)
- int FindVar (WORD ∗, WORD ∗)
- WORD DoIfStatement (PHEAD WORD ∗, WORD ∗)
- WORD DoOnePow (PHEAD WORD ∗, WORD, WORD, WORD ∗, WORD ∗, WORD, WORD ∗)
- void DoRevert (WORD ∗, WORD ∗)
- WORD DoSumF1 (PHEAD WORD ∗, WORD ∗, WORD, WORD)
- WORD DoSumF2 (PHEAD WORD ∗, WORD ∗, WORD, WORD)
- WORD DoTheta (PHEAD WORD ∗)
- LONG EndSort (PHEAD WORD ∗, int)
- WORD EntVar (WORD, UBYTE ∗, WORD, WORD, WORD, WORD)
- WORD EpfCon (PHEAD WORD ∗, WORD ∗, WORD, WORD)
- WORD EpfFind (PHEAD WORD ∗, WORD ∗)
- WORD EpfGen (WORD, WORD ∗, WORD ∗, WORD ∗, WORD)
- WORD EqualArg (WORD ∗, WORD, WORD)
- WORD **Evaluate** (UBYTE ∗∗)
- int Factorial (PHEAD WORD, UWORD ∗, WORD ∗)
- int Bernoulli (WORD, UWORD ∗, WORD ∗)
- int FactorIn (PHEAD WORD ∗, WORD)
- int FactorInExpr (PHEAD WORD ∗, WORD)
- WORD FindAll (PHEAD WORD ∗, WORD ∗, WORD, WORD ∗)
- WORD FindMulti (PHEAD WORD ∗, WORD ∗)

- WORD FindOnce (PHEAD WORD ∗, WORD ∗)
- WORD FindOnly (PHEAD WORD ∗, WORD ∗)
- WORD FindRest (PHEAD WORD ∗, WORD ∗)
- WORD **FindSpecial** (WORD ∗)
- WORD FindrNumber (WORD, VARRENUM ∗)
- void FiniLine (void)
- WORD FiniTerm (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD, WORD)
- WORD FlushOut (POSITION ∗, FILEHANDLE ∗, int)
- void FunLevel (PHEAD WORD ∗)
- void AdjustRenumScratch (PHEAD0)
- void GarbHand (void)
- WORD GcdLong (PHEAD UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- WORD LcmLong (PHEAD UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- void **GCD** (UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- ULONG **GCD2** (ULONG, ULONG)
- WORD Generator (PHEAD WORD ∗, WORD)
- WORD GetBinom (UWORD ∗, WORD ∗, WORD, WORD)
- WORD GetFromStore (WORD ∗, POSITION ∗, RENUMBER, WORD ∗, WORD)
- WORD GetLong (UBYTE ∗, UWORD ∗, WORD ∗)
- WORD GetMoreTerms (WORD ∗)
- WORD GetMoreFromMem (WORD ∗, WORD ∗∗)
- WORD GetOneTerm (PHEAD WORD ∗, FILEHANDLE ∗, POSITION ∗, int)
- RENUMBER GetTable (WORD, POSITION ∗, WORD)
- WORD GetTerm (PHEAD WORD ∗)
- WORD Glue (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- WORD InFunction (PHEAD WORD ∗, WORD ∗)
- void IniLine (WORD)
- WORD IniVars (void)
- void **Initialize** (void)
- WORD InsertTerm (PHEAD WORD ∗, WORD, WORD, WORD ∗, WORD ∗, WORD)
- void LongToLine (UWORD ∗, WORD)
- WORD MakeDirty (WORD ∗, WORD ∗, WORD)
- void MarkDirty (WORD ∗, WORD)
- void PolyFunDirty (PHEAD WORD ∗)
- void PolyFunClean (PHEAD WORD ∗)
- WORD MakeModTable (void)
- WORD MatchE (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- int MatchCy (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- int FunMatchCy (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- int FunMatchSy (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- int MatchArgument (PHEAD WORD ∗, WORD ∗)
- WORD MatchFunction (PHEAD WORD ∗, WORD ∗, WORD ∗)
- WORD MergePatches (WORD)
- WORD MesCerr (char ∗, UBYTE ∗)
- WORD MesComp (char ∗, UBYTE ∗, UBYTE ∗)
- WORD Modulus (WORD ∗)
- void MoveDummies (PHEAD WORD ∗, WORD)
- WORD MulLong (UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- WORD MulRat (PHEAD UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- WORD Mully (PHEAD UWORD ∗, WORD ∗, UWORD ∗, WORD)
- WORD MultDo (PHEAD WORD ∗, WORD ∗)
- WORD NewSort (PHEAD0)
- WORD ExtraSymbol (WORD, WORD, WORD, WORD ∗, WORD ∗)
- WORD Normalize (PHEAD WORD ∗)
- WORD BracketNormalize (PHEAD WORD ∗)

- void DropCoefficient (PHEAD WORD ∗)
- void DropSymbols (PHEAD WORD ∗)
- int PutInside (PHEAD WORD ∗, WORD ∗)
- WORD OpenTemp (void)
- void Pack (UWORD ∗, WORD ∗, UWORD ∗, WORD)
- LONG PasteFile (PHEAD WORD, WORD ∗, POSITION ∗, WORD ∗∗, RENUMBER, WORD ∗, WORD)
- WORD Permute (PERM ∗, WORD)
- WORD PermuteP (PERMP ∗, WORD)
- WORD PolyFunMul (PHEAD WORD ∗)
- WORD PopVariables (void)
- WORD PrepPoly (PHEAD WORD ∗, WORD)
- WORD Processor (void)
- WORD Product (UWORD ∗, WORD ∗, WORD)
- void PrtLong (UWORD ∗, WORD, UBYTE ∗)
- void PrtTerms (void)
- void PrintDeprecation (const char ∗, const char ∗)
- void PrintRunningTime (void)
- LONG GetRunningTime (void)
- WORD PutBracket (PHEAD WORD ∗)
- LONG PutIn (FILEHANDLE ∗, POSITION ∗, WORD ∗, WORD ∗∗, int)
- WORD PutInStore (INDEXENTRY ∗, WORD)
- WORD PutOut (PHEAD WORD ∗, POSITION ∗, FILEHANDLE ∗, WORD)
- UWORD Quotient (UWORD ∗, WORD ∗, WORD)
- WORD RaisPow (PHEAD UWORD ∗, WORD ∗, UWORD)
- void RaisPowCached (PHEAD WORD, WORD, UWORD ∗∗, WORD ∗)
- WORD RaisPowMod (WORD, WORD, WORD)
- int NormalModulus (UWORD ∗, WORD ∗)
- int MakeInverses (void)
- int GetModInverses (WORD, WORD, WORD ∗, WORD ∗)
- int GetLongModInverses (PHEAD UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗, UWORD ∗, WORD ∗)
- void RatToLine (UWORD ∗, WORD)
- WORD RatioFind (PHEAD WORD ∗, WORD ∗)
- WORD RatioGen (PHEAD WORD ∗, WORD ∗, WORD, WORD)
- WORD ReNumber (PHEAD WORD ∗)
- WORD ReadSnum (UBYTE ∗∗)
- WORD Remain10 (UWORD ∗, WORD ∗)
- WORD Remain4 (UWORD ∗, WORD ∗)
- WORD ResetScratch (void)
- WORD ResolveSet (PHEAD WORD ∗, WORD ∗, WORD ∗)
- WORD RevertScratch (void)
- WORD ScanFunctions (PHEAD WORD ∗, WORD ∗, WORD)
- void SeekScratch (FILEHANDLE ∗, POSITION ∗)
- void SetEndScratch (FILEHANDLE ∗, POSITION ∗)
- void SetEndHScratch (FILEHANDLE ∗, POSITION ∗)
- WORD SetFileIndex (void)
- WORD Sflush (FILEHANDLE ∗)
- WORD Simplify (PHEAD UWORD ∗, WORD ∗, UWORD ∗, WORD ∗)
- WORD SortWild (WORD ∗, WORD)
- FILE ∗ LocateBase (char ∗∗, char ∗∗, char ∗)
- LONG SplitMerge (PHEAD WORD ∗∗, LONG)
- WORD StoreTerm (PHEAD WORD ∗)
- void SubPLon (UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- void Substitute (PHEAD WORD ∗, WORD ∗, WORD)
- WORD SymFind (PHEAD WORD ∗, WORD ∗)

- WORD SymGen (PHEAD WORD *, WORD *, WORD, WORD)
- WORD Symmetrize (PHEAD WORD *, WORD *, WORD, WORD, WORD)
- int FullSymmetrize (PHEAD WORD *, int)
- WORD TakeModulus (UWORD *, WORD *, UWORD *, WORD, WORD)
- WORD TakeNormalModulus (UWORD *, WORD *, UWORD *, WORD, WORD)
- void TalToLine (UWORD)
- WORD TenVec (PHEAD WORD *, WORD *, WORD, WORD)
- WORD TenVecFind (PHEAD WORD *, WORD *)
- WORD TermRenumber (WORD *, RENUMBER, WORD)
- void TestDrop (void)
- void PutInVflags (WORD)
- WORD TestMatch (PHEAD WORD *, WORD *)
- WORD TestSub (PHEAD WORD *, WORD)
- LONG TimeCPU (WORD)
- LONG TimeChildren (WORD)
- LONG TimeWallClock (WORD)
- LONG **Timer** (int)
- int **GetTimerInfo** (LONG **, LONG **)
- void **WriteTimerInfo** (LONG *, LONG *)
- LONG **GetWorkerTimes** (void)
- WORD ToStorage (EXPRESSIONS, POSITION *)
- void TokenToLine (UBYTE *)
- WORD Trace4 (PHEAD WORD *, WORD *, WORD, WORD)
- WORD Trace4Gen (PHEAD TRACES *, WORD)
- WORD Trace4no (WORD, WORD *, TRACES *)
- WORD TraceFind (PHEAD WORD *, WORD *)
- WORD TraceN (PHEAD WORD *, WORD *, WORD, WORD)
- WORD TraceNgen (PHEAD TRACES *, WORD)
- WORD TraceNno (WORD, WORD *, TRACES *)
- WORD Traces (PHEAD WORD *, WORD *, WORD, WORD)
- WORD Trick (WORD *, TRACES *)
- WORD TryDo (PHEAD WORD *, WORD *, WORD)
- void UnPack (UWORD *, WORD, WORD *, WORD *)
- WORD VarStore (UBYTE *, WORD, WORD, WORD)
- WORD WildFill (PHEAD WORD *, WORD *, WORD *)
- WORD WriteAll (void)
- WORD WriteOne (UBYTE *, int, int, WORD)
- void WriteArgument (WORD *)
- WORD WriteExpression (WORD *, LONG)
- WORD WriteInnerTerm (WORD *, WORD)
- void WriteLists (void)
- void WriteSetup (void)
- void WriteStats (POSITION *, WORD, WORD)
- WORD WriteSubTerm (WORD *, WORD)
- WORD WriteTerm (WORD *, WORD *, WORD, WORD, WORD)
- WORD execarg (PHEAD WORD *, WORD)
- WORD execterm (PHEAD WORD *, WORD)
- void SpecialCleanup (PHEAD0)
- void SetMods (void)
- void UnSetMods (void)
- WORD DoExecute (WORD, WORD)
- void SetScratch (FILEHANDLE *, POSITION *)
- void Warning (char *)
- void HighWarning (char *)
- int SpareTable (TABLES)

- UBYTE * strDup1 (UBYTE *, char *)
- void * Malloc (LONG)
- void * Malloc1 (LONG, const char *)
- int DoTail (int, UBYTE **)
- int OpenInput (void)
- int PutPreVar (UBYTE *, UBYTE *, UBYTE *, int)
- void Error0 (char *)
- void Error1 (char *, UBYTE *)
- void Error2 (char *, char *, UBYTE *)
- UBYTE ReadFromStream (STREAM *)
- UBYTE GetFromStream (STREAM *)
- UBYTE LookInStream (STREAM *)
- STREAM * OpenStream (UBYTE *, int, int, int)
- int LocateFile (UBYTE **, int)
- STREAM * CloseStream (STREAM *)
- void PositionStream (STREAM *, LONG)
- int ReverseStatements (STREAM *)
- int ProcessOption (UBYTE *, UBYTE *, int)
- int DoSetups (void)
- void TerminateImpl (int, const char *, int, const char *)
- NAMENODE * GetNode (NAMETREE *, UBYTE *)
- int AddName (NAMETREE *, UBYTE *, WORD, WORD, int *)
- int GetName (NAMETREE *, UBYTE *, WORD *, int)
- UBYTE * GetFunction (UBYTE *, WORD *)
- UBYTE * GetNumber (UBYTE *, WORD *)
- int GetLastExprName (UBYTE *, WORD *)
- int GetAutoName (UBYTE *, WORD *)
- int GetVar (UBYTE *, WORD *, WORD *, int, int)
- int MakeDubious (NAMETREE *, UBYTE *, WORD *)
- int GetOName (NAMETREE *, UBYTE *, WORD *, int)
- void DumpTree (NAMETREE *)
- void DumpNode (NAMETREE *, WORD, WORD)
- void LinkTree (NAMETREE *, WORD, WORD)
- void CopyTree (NAMETREE *, NAMETREE *, WORD, WORD)
- int CompactifyTree (NAMETREE *, WORD)
- NAMETREE * MakeNameTree (void)
- void FreeNameTree (NAMETREE *)
- int AddExpression (UBYTE *, int, int)
- int AddSymbol (UBYTE *, int, int, int, int)
- int AddDollar (UBYTE *, WORD, WORD *, LONG)
- int ReplaceDollar (WORD, WORD, WORD *, LONG)
- int DollarRaiseLow (UBYTE *, LONG)
- int AddVector (UBYTE *, int, int)
- int AddDubious (UBYTE *)
- int AddIndex (UBYTE *, int, int)
- UBYTE * DoDimension (UBYTE *, int *, int *)
- int AddFunction (UBYTE *, int, int, int, int, int, int, int)
- int CoCommuteInSet (UBYTE *)
- int CoFunction (UBYTE *, int, int)
- int TestName (UBYTE *)
- int AddSet (UBYTE *, WORD)
- int DoElements (UBYTE *, SETS, UBYTE *)
- int DoTempSet (UBYTE *, UBYTE *)
- int NameConflict (int, UBYTE *)
- int OpenFile (char *)

- int OpenAddFile (char ∗)
- int ReOpenFile (char ∗)
- int CreateFile (char ∗)
- int CreateLogFile (char ∗)
- void CloseFile (int)
- int CopyFile (char ∗, char ∗)
- int CreateHandle (void)
- LONG ReadFile (int, UBYTE ∗, LONG)
- LONG ReadPosFile (PHEAD FILEHANDLE ∗, UBYTE ∗, LONG, POSITION ∗)
- LONG WriteFileToFile (int, UBYTE ∗, LONG)
- void SeekFile (int, POSITION ∗, int)
- LONG TellFile (int)
- void FlushFile (int)
- int GetPosFile (int, fpos_t ∗)
- int SetPosFile (int, fpos_t ∗)
- void SynchFile (int)
- void TruncateFile (int)
- int GetChannel (char ∗, int)
- int GetAppendChannel (char ∗)
- int CloseChannel (char ∗)
- void inictable (void)
- KEYWORD ∗ findcommand (UBYTE ∗)
- int inicbufs (void)
- void StartFiles (void)
- UBYTE ∗ MakeDate (void)
- void PreProcessor (void)
- void ∗ FromList (LIST ∗)
- void ∗ From0List (LIST ∗)
- void ∗ FromVarList (LIST ∗)
- int DoubleList (void ∗∗∗, int ∗, int, char ∗)
- int DoubleLList (void ∗∗∗, LONG ∗, int, char ∗)
- void DoubleBuffer (void ∗∗, void ∗∗, int, char ∗)
- void ExpandBuffer (void ∗∗, LONG ∗, int)
- LONG iexp (LONG, int)
- int IsLikeVector (WORD ∗)
- int AreArgsEqual (WORD ∗, WORD ∗)
- int CompareArgs (WORD ∗, WORD ∗)
- UBYTE ∗ SkipField (UBYTE ∗, int)
- int StrCmp (UBYTE ∗, UBYTE ∗)
- int StrICmp (UBYTE ∗, UBYTE ∗)
- int StrHICmp (UBYTE ∗, UBYTE ∗)
- int StrICont (UBYTE ∗, UBYTE ∗)
- int CmpArray (WORD ∗, WORD ∗, WORD)
- int ConWord (UBYTE ∗, UBYTE ∗)
- int StrLen (UBYTE ∗)
- UBYTE ∗ GetPreVar (UBYTE ∗, int)
- void ToGeneral (WORD ∗, WORD ∗, WORD)
- WORD ToPolyFunGeneral (PHEAD WORD ∗)
- int ToFast (WORD ∗, WORD ∗)
- SETUPPARAMETERS ∗ GetSetupPar (UBYTE ∗)
- int RecalcSetups (void)
- int AllocSetups (void)
- SORTING ∗ AllocSort (LONG, LONG, LONG, LONG, int, int, LONG, int)
- void AllocSortFileName (SORTING ∗)
- UBYTE ∗ LoadInputFile (UBYTE ∗, int)

- UBYTE GetInput (void)
- void ClearPushback (void)
- UBYTE GetChar (int)
- void CharOut (UBYTE)
- void UnsetAllowDelay (void)
- void PopPreVars (int)
- void IniModule (int)
- void IniSpecialModule (int)
- int ModuleInstruction (int ∗, int ∗)
- int PreProInstruction (void)
- int LoadInstruction (int)
- int LoadStatement (int)
- KEYWORD ∗ FindKeyWord (UBYTE ∗, KEYWORD ∗, int)
- KEYWORD ∗ FindInKeyWord (UBYTE ∗, KEYWORD ∗, int)
- int DoDefine (UBYTE ∗)
- int DoRedefine (UBYTE ∗)
- int TheDefine (UBYTE ∗, int)
- int TheUndefine (UBYTE ∗)
- int ClearMacro (UBYTE ∗)
- int DoUndefine (UBYTE ∗)
- int DoInclude (UBYTE ∗)
- int DoReverseInclude (UBYTE ∗)
- int Include (UBYTE ∗, int)
- int DoExternal (UBYTE ∗)
- int DoToExternal (UBYTE ∗)
- int DoFromExternal (UBYTE ∗)
- int DoPrompt (UBYTE ∗)
- int DoSetExternal (UBYTE ∗)
- int DoSetExternalAttr (UBYTE ∗)
- int DoRmExternal (UBYTE ∗)
- int DoFactDollar (UBYTE ∗)
- WORD GetDollarNumber (UBYTE ∗∗, DOLLARS)
- int DoSetRandom (UBYTE ∗)
- int DoOptimize (UBYTE ∗)
- int DoClearOptimize (UBYTE ∗)
- int DoSkipExtraSymbols (UBYTE ∗)
- int DoTimeOutAfter (UBYTE ∗)
- int DoMessage (UBYTE ∗)
- int DoPreOut (UBYTE ∗)
- int DoPreAppend (UBYTE ∗)
- int DoPreCreate (UBYTE ∗)
- int DoPreAssign (UBYTE ∗)
- int DoPreBreak (UBYTE ∗)
- int DoPreDefault (UBYTE ∗)
- int DoPreSwitch (UBYTE ∗)
- int DoPreEndSwitch (UBYTE ∗)
- int DoPreCase (UBYTE ∗)
- int DoPreShow (UBYTE ∗)
- int DoPreExchange (UBYTE ∗)
- int DoSystem (UBYTE ∗)
- int DoPipe (UBYTE ∗)
- void StartPrepro (void)
- int DoIfdef (UBYTE ∗, int)
- int DoIfydef (UBYTE ∗)
- int DoIfndef (UBYTE ∗)

- int DoElse (UBYTE ∗)
- int DoElseif (UBYTE ∗)
- int DoEndif (UBYTE ∗)
- int DoTerminate (UBYTE ∗)
- int DoIf (UBYTE ∗)
- int DoCall (UBYTE ∗)
- int DoDebug (UBYTE ∗)
- int DoContinueDo (UBYTE ∗)
- int DoDo (UBYTE ∗)
- int DoBreakDo (UBYTE ∗)
- int DoEnddo (UBYTE ∗)
- int DoEndprocedure (UBYTE ∗)
- int DoInside (UBYTE ∗)
- int DoEndInside (UBYTE ∗)
- int DoProcedure (UBYTE ∗)
- int DoPrePrintTimes (UBYTE ∗)
- int DoPreWrite (UBYTE ∗)
- int DoPreClose (UBYTE ∗)
- int DoPreRemove (UBYTE ∗)
- int DoPreSortReallocate (UBYTE ∗)
- int DoCommentChar (UBYTE ∗)
- int DoPrcExtension (UBYTE ∗)
- int DoPreReset (UBYTE ∗)
- void WriteString (int, UBYTE ∗, int)
- void WriteUnfinString (int, UBYTE ∗, int)
- UBYTE ∗ AddToString (UBYTE ∗, UBYTE ∗, int)
- UBYTE ∗ PreCalc (void)
- UBYTE ∗ PreEval (UBYTE ∗, LONG ∗)
- void NumToStr (UBYTE ∗, LONG)
- int PreCmp (int, int, UBYTE ∗, int, int, UBYTE ∗, int)
- int PreEq (int, int, UBYTE ∗, int, int, UBYTE ∗, int)
- UBYTE ∗ pParseObject (UBYTE ∗, int ∗, LONG ∗)
- UBYTE ∗ PreIfEval (UBYTE ∗, int ∗)
- int EvalPreIf (UBYTE ∗)
- int PreLoad (PRELOAD ∗, UBYTE ∗, UBYTE ∗, int, char ∗)
- int PreSkip (UBYTE ∗, UBYTE ∗, int)
- UBYTE ∗ EndOfToken (UBYTE ∗)
- void SetSpecialMode (int, int)
- void MakeGlobal (void)
- int ExecModule (int)
- int ExecStore (void)
- void FullCleanUp (void)
- int DoExecStatement (void)
- int DoPipeStatement (void)
- int DoPolyfun (UBYTE ∗)
- int DoPolyratfun (UBYTE ∗)
- int CompileStatement (UBYTE ∗)
- UBYTE ∗ ToToken (UBYTE ∗)
- int GetDollar (UBYTE ∗)
- int MesWork (void)
- int **MesPrint** (const char ∗,...)
- int MesCall (char ∗)
- UBYTE ∗ NumCopy (WORD, UBYTE ∗)
- char ∗ LongCopy (LONG, char ∗)
- char ∗ LongLongCopy (off_t ∗, char ∗)

- void ReserveTempFiles (int)
- void PrintTerm (WORD ∗, char ∗)
- void PrintTermC (WORD ∗, char ∗)
- void PrintSubTerm (WORD ∗, char ∗)
- void PrintWords (WORD ∗, LONG)
- void PrintSeq (WORD ∗, char ∗)
- int ExpandTripleDots (int)
- LONG ComPress (WORD ∗∗, LONG ∗)
- void StageSort (FILEHANDLE ∗)
- void M_free (void ∗, const char ∗)
- void ClearWildcardNames (void)
- int AddWildcardName (UBYTE ∗)
- int GetWildcardName (UBYTE ∗)
- void Globalize (int)
- void ResetVariables (int)
- void AddToPreTypes (int)
- void MessPreNesting (int)
- LONG GetStreamPosition (STREAM ∗)
- WORD ∗ DoubleCbuffer (int, WORD ∗, int)
- WORD ∗ AddLHS (int)
- WORD ∗ AddRHS (int, int)
- int AddNtoL (int, WORD ∗)
- int AddNtoC (int, int, WORD ∗, int)
- void DoubleIfBuffers (void)
- STREAM ∗ CreateStream (UBYTE ∗)
- int setonoff (UBYTE ∗, int ∗, int, int)
- int DoPrint (UBYTE ∗, int)
- int SetExpr (UBYTE ∗, int, int)
- void AddToCom (int, WORD ∗)
- int Add2ComStrings (int, WORD ∗, UBYTE ∗, UBYTE ∗)
- int DoSymmetrize (UBYTE ∗, int)
- int DoArgument (UBYTE ∗, int)
- int ArgFactorize (PHEAD WORD ∗, WORD ∗)
- WORD ∗ TakeArgContent (PHEAD WORD ∗, WORD ∗)
- WORD ∗ MakeInteger (PHEAD WORD ∗, WORD ∗, WORD ∗)
- WORD ∗ MakeMod (PHEAD WORD ∗, WORD ∗, WORD ∗)
- WORD FindArg (PHEAD WORD ∗)
- WORD InsertArg (PHEAD WORD ∗, WORD ∗, int)
- int CleanupArgCache (PHEAD WORD)
- int ArgSymbolMerge (WORD ∗, WORD ∗)
- int ArgDotproductMerge (WORD ∗, WORD ∗)
- void SortWeights (LONG ∗, LONG ∗, WORD)
- int DoBrackets (UBYTE ∗, int)
- int DoPutInside (UBYTE ∗, int)
- WORD ∗ CountComp (UBYTE ∗, WORD ∗)
- int CoAntiBracket (UBYTE ∗)
- int CoAntiSymmetrize (UBYTE ∗)
- int DoArgPlode (UBYTE ∗, int)
- int CoArgExplode (UBYTE ∗)
- int CoArgImplode (UBYTE ∗)
- int CoArgument (UBYTE ∗)
- int CoInside (UBYTE ∗)
- int ExecInside (UBYTE ∗)
- int CoInExpression (UBYTE ∗)
- int CoInParallel (UBYTE ∗)

- int CoNotInParallel (UBYTE ∗)
- int DoInParallel (UBYTE ∗, int)
- int CoEndInExpression (UBYTE ∗)
- int CoBracket (UBYTE ∗)
- int CoPutInside (UBYTE ∗)
- int CoAntiPutInside (UBYTE ∗)
- int CoMultiBracket (UBYTE ∗)
- int CoCFunction (UBYTE ∗)
- int CoCTensor (UBYTE ∗)
- int CoCollect (UBYTE ∗)
- int CoCompress (UBYTE ∗)
- int CoContract (UBYTE ∗)
- int CoCycleSymmetrize (UBYTE ∗)
- int CoDelete (UBYTE ∗)
- int CoTableBase (UBYTE ∗)
- int CoApply (UBYTE ∗)
- int CoDenominators (UBYTE ∗)
- int CoDimension (UBYTE ∗)
- int CoDiscard (UBYTE ∗)
- int CoDisorder (UBYTE ∗)
- int CoDrop (UBYTE ∗)
- int CoDropCoefficient (UBYTE ∗)
- int CoDropSymbols (UBYTE ∗)
- int CoElse (UBYTE ∗)
- int CoElseIf (UBYTE ∗)
- int CoEndArgument (UBYTE ∗)
- int CoEndInside (UBYTE ∗)
- int CoEndIf (UBYTE ∗)
- int CoEndRepeat (UBYTE ∗)
- int CoEndTerm (UBYTE ∗)
- int CoEndWhile (UBYTE ∗)
- int CoExit (UBYTE ∗)
- int CoFactArg (UBYTE ∗)
- int CoFactDollar (UBYTE ∗)
- int CoFactorize (UBYTE ∗)
- int CoNFactorize (UBYTE ∗)
- int CoUnFactorize (UBYTE ∗)
- int CoNUnFactorize (UBYTE ∗)
- int DoFactorize (UBYTE ∗, int)
- int CoFill (UBYTE ∗)
- int CoFillExpression (UBYTE ∗)
- int CoFixIndex (UBYTE ∗)
- int CoFormat (UBYTE ∗)
- int CoGlobal (UBYTE ∗)
- int CoGlobalFactorized (UBYTE ∗)
- int CoGoTo (UBYTE ∗)
- int CoId (UBYTE ∗)
- int CoIdNew (UBYTE ∗)
- int CoIdOld (UBYTE ∗)
- int CoIf (UBYTE ∗)
- int CoIfMatch (UBYTE ∗)
- int CoIfNoMatch (UBYTE ∗)
- int CoIndex (UBYTE ∗)
- int CoInsideFirst (UBYTE ∗)
- int CoKeep (UBYTE ∗)

- int CoLabel (UBYTE ∗)
- int CoLoad (UBYTE ∗)
- int CoLocal (UBYTE ∗)
- int CoLocalFactorized (UBYTE ∗)
- int CoMany (UBYTE ∗)
- int CoMerge (UBYTE ∗)
- int CoStuffle (UBYTE ∗)
- int CoMetric (UBYTE ∗)
- int CoModOption (UBYTE ∗)
- int CoModuleOption (UBYTE ∗)
- int CoModulus (UBYTE ∗)
- int CoMulti (UBYTE ∗)
- int CoMultiply (UBYTE ∗)
- int CoNFunction (UBYTE ∗)
- int CoNPrint (UBYTE ∗)
- int CoNTensor (UBYTE ∗)
- int CoNWrite (UBYTE ∗)
- int CoNoDrop (UBYTE ∗)
- int CoNoSkip (UBYTE ∗)
- int CoNormalize (UBYTE ∗)
- int CoMakeInteger (UBYTE ∗)
- int CoFlags (UBYTE ∗, int)
- int CoOff (UBYTE ∗)
- int CoOn (UBYTE ∗)
- int CoOnce (UBYTE ∗)
- int CoOnly (UBYTE ∗)
- int CoOptimizeOption (UBYTE ∗)
- int **CoOptimize** (UBYTE ∗)
- int CoPolyFun (UBYTE ∗)
- int CoPolyRatFun (UBYTE ∗)
- int CoPrint (UBYTE ∗)
- int CoPrintB (UBYTE ∗)
- int CoProperCount (UBYTE ∗)
- int CoUnitTrace (UBYTE ∗)
- int CoRCycleSymmetrize (UBYTE ∗)
- int CoRatio (UBYTE ∗)
- int CoRedefine (UBYTE ∗)
- int CoRenumber (UBYTE ∗)
- int CoRepeat (UBYTE ∗)
- int CoSave (UBYTE ∗)
- int CoSelect (UBYTE ∗)
- int CoSet (UBYTE ∗)
- int CoSetExitFlag (UBYTE ∗)
- int CoSkip (UBYTE ∗)
- int CoProcessBucket (UBYTE ∗)
- int CoPushHide (UBYTE ∗)
- int CoPopHide (UBYTE ∗)
- int CoHide (UBYTE ∗)
- int CoIntoHide (UBYTE ∗)
- int CoNoIntoHide (UBYTE ∗)
- int CoNoHide (UBYTE ∗)
- int CoUnHide (UBYTE ∗)
- int CoNoUnHide (UBYTE ∗)
- int CoSort (UBYTE ∗)
- int CoSplitArg (UBYTE ∗)

- int CoSplitFirstArg (UBYTE ∗)
- int CoSplitLastArg (UBYTE ∗)
- int CoSum (UBYTE ∗)
- int CoSymbol (UBYTE ∗)
- int CoSymmetrize (UBYTE ∗)
- int DoTable (UBYTE ∗, int)
- int CoTable (UBYTE ∗)
- int CoTerm (UBYTE ∗)
- int CoNTable (UBYTE ∗)
- int CoCTable (UBYTE ∗)
- void EmptyTable (TABLES)
- int CoToTensor (UBYTE ∗)
- int CoToVector (UBYTE ∗)
- int CoTrace4 (UBYTE ∗)
- int CoTraceN (UBYTE ∗)
- int CoChisholm (UBYTE ∗)
- int CoTransform (UBYTE ∗)
- int CoClearTable (UBYTE ∗)
- int DoChain (UBYTE ∗, int)
- int CoChainin (UBYTE ∗)
- int CoChainout (UBYTE ∗)
- int CoTryReplace (UBYTE ∗)
- int CoVector (UBYTE ∗)
- int CoWhile (UBYTE ∗)
- int CoWrite (UBYTE ∗)
- int CoAuto (UBYTE ∗)
- int CoSwitch (UBYTE ∗)
- int CoCase (UBYTE ∗)
- int CoBreak (UBYTE ∗)
- int CoDefault (UBYTE ∗)
- int CoEndSwitch (UBYTE ∗)
- int CoTBaddto (UBYTE ∗)
- int CoTBaudit (UBYTE ∗)
- int CoTBcleanup (UBYTE ∗)
- int CoTBcreate (UBYTE ∗)
- int CoTBenter (UBYTE ∗)
- int CoTBhelp (UBYTE ∗)
- int CoTBload (UBYTE ∗)
- int CoTBoff (UBYTE ∗)
- int CoTBon (UBYTE ∗)
- int CoTBopen (UBYTE ∗)
- int CoTBreplace (UBYTE ∗)
- int CoTBuse (UBYTE ∗)
- int CoTestUse (UBYTE ∗)
- int CoThreadBucket (UBYTE ∗)
- int AddComString (int, WORD ∗, UBYTE ∗, int)
- int CompileAlgebra (UBYTE ∗, int, WORD ∗)
- int IsIdStatement (UBYTE ∗)
- UBYTE ∗ IsRHS (UBYTE ∗, UBYTE)
- int ParenthesesTest (UBYTE ∗)
- int tokenize (UBYTE ∗, WORD)
- void WriteTokens (SBYTE ∗)
- int simp1token (SBYTE ∗)
- int simpwtoken (SBYTE ∗)
- int simp2token (SBYTE ∗)

- int simp3atoken (SBYTE ∗, int)
- int simp3btoken (SBYTE ∗, int)
- int simp4token (SBYTE ∗)
- int simp5token (SBYTE ∗, int)
- int simp6token (SBYTE ∗, int)
- UBYTE ∗ SkipAName (UBYTE ∗)
- int TestTables (void)
- int GetLabel (UBYTE ∗)
- int CoIdExpression (UBYTE ∗, int)
- int CoAssign (UBYTE ∗)
- int DoExpr (UBYTE ∗, int, int)
- int CompileSubExpressions (SBYTE ∗)
- int CodeGenerator (SBYTE ∗)
- int CompleteTerm (WORD ∗, UWORD ∗, UWORD ∗, WORD, WORD, int)
- int CodeFactors (SBYTE ∗s)
- WORD GenerateFactors (WORD, WORD)
- int InsTree (int, int)
- int FindTree (int, WORD ∗)
- void RedoTree (CBUF ∗, int)
- void ClearTree (int)
- int CatchDollar (int)
- int AssignDollar (PHEAD WORD ∗, WORD)
- UBYTE ∗ WriteDollarToBuffer (WORD, WORD)
- UBYTE ∗ WriteDollarFactorToBuffer (WORD, WORD, WORD)
- void AddToDollarBuffer (UBYTE ∗)
- int PutTermInDollar (WORD ∗, WORD)
- void TermAssign (WORD ∗)
- void WildDollars (PHEAD WORD ∗)
- LONG numcommute (WORD ∗, LONG ∗)
- int FullRenumber (PHEAD WORD ∗, WORD)
- int Lus (WORD ∗, WORD, WORD, WORD, WORD, WORD)
- int FindLus (int, int, int)
- int CoReplaceLoop (UBYTE ∗)
- int CoFindLoop (UBYTE ∗)
- int DoFindLoop (UBYTE ∗, int)
- int CoFunPowers (UBYTE ∗)
- int SortTheList (int ∗, int)
- int MatchIsPossible (WORD ∗, WORD ∗)
- int StudyPattern (WORD ∗)
- WORD DolToTensor (PHEAD WORD)
- WORD DolToFunction (PHEAD WORD)
- WORD DolToVector (PHEAD WORD)
- WORD DolToNumber (PHEAD WORD)
- WORD DolToSymbol (PHEAD WORD)
- WORD DolToIndex (PHEAD WORD)
- LONG DolToLong (PHEAD WORD)
- int DollarFactorize (PHEAD WORD)
- int CoPrintTable (UBYTE ∗)
- int CoDeallocateTable (UBYTE ∗)
- void CleanDollarFactors (DOLLARS)
- WORD ∗ TakeDollarContent (PHEAD WORD ∗, WORD ∗∗)
- WORD ∗ MakeDollarInteger (PHEAD WORD ∗, WORD ∗∗)
- WORD ∗ MakeDollarMod (PHEAD WORD ∗, WORD ∗∗)
- int GetDolNum (PHEAD WORD ∗, WORD ∗)
- void AddPotModdollar (WORD)

- int Optimize (WORD, int)
- int ClearOptimize (void)
- int **LoadOpti** (WORD)
- int **PutObject** (WORD ∗, int)
- void **CleanOptiBuffer** (void)
- int **PrintOptima** (WORD)
- int **FindScratchName** (void)
- WORD **MaxPowerOpti** (LONG)
- WORD **HuntNumFactor** (LONG, WORD ∗, int)
- WORD **HuntFactor** (LONG, WORD ∗, int)
- void **HuntPairs** (LONG, WORD)
- void **HuntBrackets** (LONG)
- int **AddToOpti** (WORD ∗, int)
- LONG **TestNewSca** (LONG, WORD ∗, WORD ∗)
- void **NormOpti** (WORD ∗)
- void **SortOpti** (LONG)
- void **SplitOpti** (WORD ∗∗, LONG)
- void **CombiOpti** (void)
- int TakeLongRoot (UWORD ∗, WORD ∗, WORD)
- int TakeRatRoot (UWORD ∗, WORD ∗, WORD)
- int MakeRational (WORD, WORD, WORD ∗, WORD ∗)
- int MakeLongRational (PHEAD UWORD ∗, WORD, UWORD ∗, WORD, UWORD ∗, WORD ∗)
- void **HuntPowers** (LONG, WORD)
- void **HuntNumBrackets** (LONG)
- void ClearTableTree (TABLES)
- int InsTableTree (TABLES, WORD ∗)
- void RedoTableTree (TABLES, int)
- int FindTableTree (TABLES, WORD ∗, int)
- void finishcbuf (WORD)
- void clearcbuf (WORD)
- void CleanUpSort (int)
- FILEHANDLE ∗ AllocFileHandle (WORD, char ∗)
- void DeAllocFileHandle (FILEHANDLE ∗)
- void LowerSortLevel (void)
- WORD ∗ PolyRatFunSpecial (PHEAD WORD ∗, WORD ∗)
- void SimpleSplitMergeRec (WORD ∗, WORD, WORD ∗)
- void SimpleSplitMerge (WORD ∗, WORD)
- WORD BinarySearch (WORD ∗, WORD, WORD)
- int InsideDollar (PHEAD WORD ∗, WORD)
- DOLLARS DolToTerms (PHEAD WORD)
- WORD EvalDoLoopArg (PHEAD WORD ∗, WORD)
- int SetExprCases (int, int, int)
- int TestSelect (WORD ∗, WORD ∗)
- void SubsInAll (PHEAD0)
- void TransferBuffer (int, int, int)
- int TakeIDfunction (PHEAD WORD ∗)
- int MakeSetupAllocs (void)
- int TryFileSetups (void)
- void ExchangeExpressions (int, int)
- void ExchangeDollars (int, int)
- int GetFirstBracket (WORD ∗, int)
- int GetFirstTerm (WORD ∗, int, int)
- int GetContent (WORD ∗, int)
- int CleanupTerm (WORD ∗)
- WORD ContentMerge (PHEAD WORD ∗, WORD ∗)

- UBYTE ∗ PreIfDollarEval (UBYTE ∗, int ∗)
- LONG TermsInDollar (WORD)
- LONG SizeOfDollar (WORD)
- LONG TermsInExpression (WORD)
- LONG SizeOfExpression (WORD)
- WORD ∗ TranslateExpression (UBYTE ∗)
- int IsSetMember (WORD ∗, WORD)
- int IsMultipleOf (WORD ∗, WORD ∗)
- int TwoExprCompare (WORD ∗, WORD ∗, int)
- void UpdatePositions (void)
- void **M_check** (void)
- void M_print (void)
- void M_check1 (void)
- void PrintTime (UBYTE ∗)
- POSITION ∗ FindBracket (WORD, WORD ∗)
- void PutBracketInIndex (PHEAD WORD ∗, POSITION ∗)
- void ClearBracketIndex (WORD)
- void OpenBracketIndex (WORD)
- int DoNoParallel (UBYTE ∗)
- int DoParallel (UBYTE ∗)
- int DoModSum (UBYTE ∗)
- int DoModMax (UBYTE ∗)
- int DoModMin (UBYTE ∗)
- int DoModLocal (UBYTE ∗)
- UBYTE ∗ DoModDollar (UBYTE ∗, int)
- int DoProcessBucket (UBYTE ∗)
- int DoinParallel (UBYTE ∗)
- int DonotinParallel (UBYTE ∗)
- int FlipTable (FUNCTIONS, int)
- int ChainIn (PHEAD WORD ∗, WORD)
- int ChainOut (PHEAD WORD ∗, WORD)
- int ArgumentImplode (PHEAD WORD ∗, WORD ∗)
- int ArgumentExplode (PHEAD WORD ∗, WORD ∗)
- int DenToFunction (WORD ∗, WORD)
- WORD HowMany (PHEAD WORD ∗, WORD ∗)
- void RemoveDollars (void)
- LONG CountTerms1 (PHEAD0)
- LONG TermsInBracket (PHEAD WORD ∗, WORD)
- int Crash (void)
- char ∗ str_dup (char ∗)
- void convertblock (INDEXBLOCK ∗, INDEXBLOCK ∗, int)
- void convertnamesblock (NAMESBLOCK ∗, NAMESBLOCK ∗, int)
- void convertiniinfo (INIINFO ∗, INIINFO ∗, int)
- int ReadIndex (DBASE ∗)
- int WriteIndexBlock (DBASE ∗, MLONG)
- int WriteNamesBlock (DBASE ∗, MLONG)
- int WriteIndex (DBASE ∗)
- int WriteIniInfo (DBASE ∗)
- int ReadIniInfo (DBASE ∗)
- int AddToIndex (DBASE ∗, MLONG)
- DBASE ∗ GetDbase (char ∗, MLONG)
- DBASE ∗ OpenDbase (char ∗)
- char ∗ ReadObject (DBASE ∗, MLONG, char ∗)
- char ∗ ReadijObject (DBASE ∗, MLONG, MLONG, char ∗)
- int ExistsObject (DBASE ∗, MLONG, char ∗)

- int DeleteObject (DBASE ∗, MLONG, char ∗)
- int WriteObject (DBASE ∗, MLONG, char ∗, char ∗, MLONG)
- MLONG AddObject (DBASE ∗, MLONG, char ∗, char ∗)
- int **Cleanup** (DBASE ∗)
- DBASE ∗ NewDbase (char ∗, MLONG)
- void FreeTableBase (DBASE ∗)
- int ComposeTableNames (DBASE ∗)
- int PutTableNames (DBASE ∗)
- MLONG AddTableName (DBASE ∗, char ∗, TABLES)
- MLONG GetTableName (DBASE ∗, char ∗)
- MLONG FindTableNumber (DBASE ∗, char ∗)
- int TryEnvironment (void)
- int CopyExpression (FILEHANDLE ∗, FILEHANDLE ∗)
- int set_in (UBYTE, set_of_char)
- one_byte set_set (UBYTE, set_of_char)
- one_byte set_del (UBYTE, set_of_char)
- one_byte set_sub (set_of_char, set_of_char, set_of_char)
- int DoPreAddSeparator (UBYTE ∗)
- int DoPreRmSeparator (UBYTE ∗)
- int openExternalChannel (UBYTE ∗, int, UBYTE ∗, UBYTE ∗)
- int initPresetExternalChannels (UBYTE ∗, int)
- int closeExternalChannel (int)
- int selectExternalChannel (int)
- int getCurrentExternalChannel (void)
- void closeAllExternalChannels (void)
- UBYTE ∗ defineChannel (UBYTE ∗, HANDLERS ∗)
- int writeToChannel (int, UBYTE ∗, HANDLERS ∗)
- int **writeBufToExtChannelOk** (char ∗, size_t)
- int **getcFromExtChannelOk** (void)
- int **setKillModeForExternalChannelOk** (int, int)
- int **setTerminatorForExternalChannelOk** (char ∗)
- int **getcFromExtChannelFailure** (void)
- int **setKillModeForExternalChannelFailure** (int, int)
- int **setTerminatorForExternalChannelFailure** (char ∗)
- int **writeBufToExtChannelFailure** (char ∗, size_t)
- int ReleaseTB (void)
- int SymbolNormalize (WORD ∗)
- int TestFunFlag (PHEAD WORD ∗)
- WORD CompareSymbols (PHEAD WORD ∗, WORD ∗, WORD)
- WORD CompareHSymbols (PHEAD WORD ∗, WORD ∗, WORD)
- WORD NextPrime (PHEAD WORD)
- WORD Moebius (PHEAD WORD)
- UWORD wranf (PHEAD0)
- UWORD iranf (PHEAD UWORD)
- void iniwranf (PHEAD0)
- UBYTE ∗ PreRandom (UBYTE ∗)
- WORD ∗ **PolyNormPoly** (PHEAD WORD)
- WORD ∗ **EvaluateGcd** (PHEAD WORD ∗)
- int TreatPolyRatFun (PHEAD WORD ∗)
- WORD ReadSaveHeader (void)
- WORD ReadSaveIndex (FILEINDEX ∗)
- WORD ReadSaveExpression (UBYTE ∗, UBYTE ∗, LONG ∗, LONG ∗)
- UBYTE ∗ ReadSaveTerm32 (UBYTE ∗, UBYTE ∗, UBYTE ∗∗, UBYTE ∗, UBYTE ∗, int)
- WORD ReadSaveVariables (UBYTE ∗, UBYTE ∗, LONG ∗, LONG ∗, INDEXENTRY ∗, LONG ∗)
- WORD WriteStoreHeader (WORD)

- void InitRecovery (void)
- int CheckRecoveryFile (void)
- void DeleteRecoveryFile (void)
- char ∗ RecoveryFilename (void)
- int DoRecovery (int ∗)
- void DoCheckpoint (int)
- void NumberMallocAddMemory (PHEAD0)
- void CacheNumberMallocAddMemory (PHEAD0)
- void TermMallocAddMemory (PHEAD0)
- void ExprStatus (EXPRESSIONS)
- void iniTools (void)
- int TestTerm (WORD ∗)
- WORD RunTransform (PHEAD WORD ∗term, WORD ∗params)
- WORD RunEncode (PHEAD WORD ∗fun, WORD ∗args, WORD ∗info)
- WORD RunDecode (PHEAD WORD ∗fun, WORD ∗args, WORD ∗info)
- WORD RunReplace (PHEAD WORD ∗fun, WORD ∗args, WORD ∗info)
- WORD RunImplode (WORD ∗fun, WORD ∗args)
- WORD RunExplode (PHEAD WORD ∗fun, WORD ∗args)
- int TestArgNum (int n, int totarg, WORD ∗args)
- WORD PutArgInScratch (WORD ∗arg, UWORD ∗scrat)
- UBYTE ∗ ReadRange (UBYTE ∗s, WORD ∗out, int par)
- int FindRange (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- WORD RunPermute (PHEAD WORD ∗fun, WORD ∗args, WORD ∗info)
- WORD RunReverse (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunCycle (PHEAD WORD ∗fun, WORD ∗args, WORD ∗info)
- WORD RunAddArg (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunMulArg (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunIsLyndon (PHEAD WORD ∗fun, WORD ∗args, int par)
- WORD RunToLyndon (PHEAD WORD ∗fun, WORD ∗args, int par)
- WORD RunDropArg (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunSelectArg (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunDedup (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunZtoHArg (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunHtoZArg (PHEAD WORD ∗fun, WORD ∗args)
- int NormPolyTerm (PHEAD WORD ∗)
- WORD **ComparePoly** (WORD ∗, WORD ∗, WORD)
- int ConvertToPoly (PHEAD WORD ∗, WORD ∗, WORD ∗, WORD)
- int LocalConvertToPoly (PHEAD WORD ∗, WORD ∗, WORD, WORD)
- int ConvertFromPoly (PHEAD WORD ∗, WORD ∗, WORD, WORD, WORD, WORD)
- WORD FindSubterm (WORD ∗)
- WORD FindLocalSubterm (PHEAD WORD ∗, WORD)
- void PrintSubtermList (int, int)
- void PrintExtraSymbol (int, WORD ∗, int)
- WORD FindSubexpression (WORD ∗)
- void UpdateMaxSize (void)
- int CoToPolynomial (UBYTE ∗)
- int CoFromPolynomial (UBYTE ∗)
- int CoArgToExtraSymbol (UBYTE ∗)
- int CoExtraSymbols (UBYTE ∗)
- UBYTE ∗ GetDoParam (UBYTE ∗, WORD ∗∗, int)
- WORD ∗ GetIfDollarFactor (UBYTE ∗∗, WORD ∗)
- int CoDo (UBYTE ∗)
- int CoEndDo (UBYTE ∗)
- int ExtraSymFun (PHEAD WORD ∗, WORD)
- int PruneExtraSymbols (WORD)

- int IniFbuffer (WORD)
- void **IniFbufs** (void)
- int GCDfunction (PHEAD WORD ∗, WORD)
- WORD ∗ GCDfunction3 (PHEAD WORD ∗, WORD ∗)
- WORD ∗ **GCDfunction4** (PHEAD WORD ∗, WORD ∗)
- int ReadPolyRatFun (PHEAD WORD ∗)
- int FromPolyRatFun (PHEAD WORD ∗, WORD ∗∗, WORD ∗∗)
- void **PRFnormalize** (PHEAD WORD ∗)
- WORD ∗ **PRFadd** (PHEAD WORD ∗, WORD ∗)
- WORD ∗ PolyDiv (PHEAD WORD ∗, WORD ∗, char ∗)
- WORD ∗ **PolyGCD** (PHEAD WORD ∗, WORD ∗)
- WORD ∗ **PolyAdd** (PHEAD WORD ∗, WORD ∗)
- void GCDclean (PHEAD WORD ∗, WORD ∗)
- int **RatFunNormalize** (PHEAD WORD ∗)
- WORD ∗ TakeSymbolContent (PHEAD WORD ∗, WORD ∗)
- int GCDterms (PHEAD WORD ∗, WORD ∗, WORD ∗)
- WORD ∗ PutExtraSymbols (PHEAD WORD ∗, WORD, int ∗)
- WORD ∗ TakeExtraSymbols (PHEAD WORD ∗, WORD)
- WORD ∗ MultiplyWithTerm (PHEAD WORD ∗, WORD ∗, WORD)
- WORD ∗ TakeContent (PHEAD WORD ∗, WORD ∗)
- int MergeSymbolLists (PHEAD WORD ∗, WORD ∗, int)
- int MergeDotproductLists (PHEAD WORD ∗, WORD ∗, int)
- WORD ∗ CreateExpression (PHEAD WORD)
- int DIVfunction (PHEAD WORD ∗, WORD, int)
- WORD ∗ MULfunc (PHEAD WORD ∗, WORD ∗)
- WORD ∗ ConvertArgument (PHEAD WORD ∗, int ∗)
- int ExpandRat (PHEAD WORD ∗)
- int InvPoly (PHEAD WORD ∗, WORD, WORD)
- WORD TestDoLoop (PHEAD WORD ∗, WORD)
- WORD TestEndDoLoop (PHEAD WORD ∗, WORD)
- WORD ∗ poly_gcd (PHEAD WORD ∗, WORD ∗, WORD)
- WORD ∗ poly_div (PHEAD WORD ∗, WORD ∗, WORD)
- WORD ∗ poly_rem (PHEAD WORD ∗, WORD ∗, WORD)
- WORD ∗ poly_inverse (PHEAD WORD ∗, WORD ∗)
- WORD ∗ poly_mul (PHEAD WORD ∗, WORD ∗)
- WORD ∗ poly_ratfun_add (PHEAD WORD ∗, WORD ∗)
- int poly_ratfun_normalize (PHEAD WORD ∗)
- int poly_factorize_argument (PHEAD WORD ∗, WORD ∗)
- WORD ∗ poly_factorize_dollar (PHEAD WORD ∗)
- int poly_factorize_expression (EXPRESSIONS)
- int poly_unfactorize_expression (EXPRESSIONS)
- void poly_free_poly_vars (PHEAD const char ∗)
- void optimize_print_code (int)
- int DoPreAdd (UBYTE ∗s)
- int DoPreUseDictionary (UBYTE ∗s)
- int DoPreCloseDictionary (UBYTE ∗s)
- int DoPreOpenDictionary (UBYTE ∗s)
- void RemoveDictionary (DICTIONARY ∗dict)
- void UnSetDictionary (void)
- int SetDictionaryOptions (UBYTE ∗options)
- int **SelectDictionary** (UBYTE ∗name, UBYTE ∗options)
- int AddToDictionary (DICTIONARY ∗dict, UBYTE ∗left, UBYTE ∗right)
- int AddDictionary (UBYTE ∗name)
- int FindDictionary (UBYTE ∗name)
- UBYTE ∗ IsExponentSign (void)

- UBYTE ∗ IsMultiplySign (void)
- void TransformRational (UWORD ∗a, WORD na)
- void WriteDictionary (DICTIONARY ∗)
- void ShrinkDictionary (DICTIONARY ∗)
- void MultiplyToLine (void)
- UBYTE ∗ FindSymbol (WORD num)
- UBYTE ∗ FindVector (WORD num)
- UBYTE ∗ FindIndex (WORD num)
- UBYTE ∗ FindFunction (WORD num)
- UBYTE ∗ FindFunWithArgs (WORD ∗t)
- UBYTE ∗ FindExtraSymbol (WORD num)
- LONG DictToBytes (DICTIONARY ∗dict, UBYTE ∗buf)
- DICTIONARY ∗ DictFromBytes (UBYTE ∗buf)
- int CoCreateSpectator (UBYTE ∗inp)
- int CoToSpectator (UBYTE ∗inp)
- int CoRemoveSpectator (UBYTE ∗inp)
- int CoEmptySpectator (UBYTE ∗inp)
- int CoCopySpectator (UBYTE ∗inp)
- int PutInSpectator (WORD ∗, WORD)
- void ClearSpectators (WORD)
- WORD GetFromSpectator (WORD ∗, WORD)
- void FlushSpectators (void)
- WORD ∗ **PreGCD** (PHEAD WORD ∗, WORD ∗, int)
- WORD ∗ **FindCommonVariables** (PHEAD int, int)
- void **AddToSymbolList** (PHEAD WORD)
- int **AddToListPoly** (PHEAD0)
- int ReadFromScratch (FILEHANDLE ∗, POSITION ∗, UBYTE ∗, POSITION ∗)
- int AddToScratch (FILEHANDLE ∗, POSITION ∗, UBYTE ∗, POSITION ∗, int)
- int DoPreAppendPath (UBYTE ∗)
- int DoPrePrependPath (UBYTE ∗)
- int DoSwitch (PHEAD WORD ∗, WORD ∗)
- int DoEndSwitch (PHEAD WORD ∗, WORD ∗)
- SWITCHTABLE ∗ FindCase (WORD, WORD)
- void SwitchSplitMergeRec (SWITCHTABLE ∗, WORD, SWITCHTABLE ∗)
- void SwitchSplitMerge (SWITCHTABLE ∗, WORD)
- int DoubleSwitchBuffers (void)
- int DistrN (int, int ∗, int, int ∗)
- int DoNamespace (UBYTE ∗)
- int DoEndNamespace (UBYTE ∗)
- int DoUse (UBYTE ∗)
- UBYTE ∗ SkipName (UBYTE ∗)
- UBYTE ∗ ConstructName (UBYTE ∗, UBYTE)
- int DoSetUserFlag (UBYTE ∗)
- int DoClearUserFlag (UBYTE ∗)
- int **DoUserFlag** (UBYTE ∗, int)
- VERTEX ∗ CreateVertex (MODEL ∗)
- UBYTE ∗ ReadParticle (UBYTE ∗, VERTEX ∗, MODEL ∗, int)
- int CoModel (UBYTE ∗)
- int CoParticle (UBYTE ∗)
- int CoVertex (UBYTE ∗)
- int CoEndModel (UBYTE ∗)
- int LoadModel (MODEL ∗)
- int CoSetUserFlag (UBYTE ∗)
- int CoClearUserFlag (UBYTE ∗)
- int CoCreateAllLoops (UBYTE ∗)

- int CoCreateAllPaths (UBYTE ∗)
- int CoCreateAll (UBYTE ∗)
- WORD AllLoops (PHEAD WORD ∗, WORD)
- LONG StartLoops (PHEAD WORD ∗, WORD, LONG, WORD, WORD ∗, WORD, WORD ∗, WORD)
- LONG GenLoops (PHEAD WORD ∗, WORD, LONG, WORD, WORD ∗, WORD, WORD ∗, WORD)
- void LoopOutput (PHEAD WORD ∗, WORD, WORD ∗, WORD)
- WORD AllPaths (PHEAD WORD ∗, WORD)
- LONG GenPaths (PHEAD WORD ∗, WORD, LONG, WORD, WORD ∗, WORD, WORD ∗, WORD)
- void PathOutput (PHEAD WORD ∗, WORD, WORD ∗, WORD)

4.19.1 Detailed Description

Contains macros and function declarations.

Definition in file declare.h.

4.19.2 Macro Definition Documentation

4.19.2.1 MaX

```
#define MaX(
              x,
              y ) ((x) > (y) ? (x): (y))
```

Definition at line 40 of file declare.h.

4.19.2.2 MiN

```
#define MiN(
              x,
              y ) ((x) < (y) ? (x): (y))
```

Definition at line 41 of file declare.h.

4.19.2.3 ABS

```
#define ABS(
              x ) ( (x) < 0 ? -(x): (x) )
```

Definition at line 42 of file declare.h.

4.19.2.4 SGN

```
#define SGN(
              x ) ( (x) > 0 ? 1 : (x) < 0 ? -1 : 0 )
```

Definition at line 43 of file declare.h.

4.19.2.5 REDLENG

```
#define REDLENG(
                x ) ((((x)<0)?((x)+1):((x)-1))/2)
```

Definition at line 44 of file declare.h.

4.19.2.6 INCLENG

```
#define INCLENG(
                x ) (((x)<0)?(((x)*2)-1):(((x)*2)+1))
```

Definition at line 45 of file declare.h.

4.19.2.7 GETCOEF

```
#define GETCOEF(
                x,
                y ) x += *x;y = x[-1];x -= ABS(y);y=REDLENG(y)
```

Definition at line 46 of file declare.h.

4.19.2.8 GETSTOP

```
#define GETSTOP(
                x,
                y ) y=x+(*x)-1;y -= ABS(*y)-1
```

Definition at line 47 of file declare.h.

4.19.2.9 StuffAdd

```
#define StuffAdd(
                x,
                y ) (((x)<0?-1:1)*(y)+((y)<0?-1:1)*(x))
```

Definition at line 48 of file declare.h.

4.19.2.10 EXCHN

```
#define EXCHN(
                t1,
                t2,
                n ) { WORD a,i; for(i=0;i<n;i++){a=t1[i];t1[i]=t2[i];t2[i]=a;} }
```

Definition at line 50 of file declare.h.

4.19.2.11 EXCH

```
#define EXCH(
            x,
            y ) { WORD a = (x); (x) = (y); (y) = a; }
```

Definition at line 51 of file declare.h.

4.19.2.12 TOKENTOLINE

```
#define TOKENTOLINE(
            x,
            y )
```

Value:

```
if ( AC.OutputSpaces == NOSPACEFORMAT ) { \
TokenToLine((UBYTE *)(y)); } else { TokenToLine((UBYTE *)(x)); }
```

Definition at line 53 of file declare.h.

4.19.2.13 UngetFromStream

```
#define UngetFromStream(
            stream,
            c ) ((stream)->nextchar[(stream)->isnextchar++]=c)
```

Definition at line 56 of file declare.h.

4.19.2.14 AddLineFeed

```
#define AddLineFeed(
            s,
            n ) { (s)[(n)++] = LINEFEED; }
```

Definition at line 60 of file declare.h.

4.19.2.15 TryRecover

```
#define TryRecover(
            x ) Terminate(-1)
```

Definition at line 62 of file declare.h.

4.19.2.16 UngetChar

```
#define UngetChar(
            c ) { pushbackchar = c; }
```

Definition at line 63 of file declare.h.

4.19.2.17 ParseNumber

```
#define ParseNumber(
              x,
              s ) {(x)=0;while(*(s)>='0'&&*(s)<='9')(x)=10*(x)+*(s)++ -'0';}
```

Definition at line 64 of file declare.h.

4.19.2.18 ParseSign

```
#define ParseSign(
              sgn,
              s )
```

Value:

```
{(sgn)=0;while(*(s)=='-'||*(s)=='+'){\
if ( *(s)++ == '-' ) sgn ^= 1;}}
```

Definition at line 65 of file declare.h.

4.19.2.19 ParseSignedNumber

```
#define ParseSignedNumber(
              x,
              s )
```

Value:

```
{ int sgn; ParseSign(sgn,s)\
ParseNumber(x,s) if ( sgn ) x = -x; }
```

Definition at line 67 of file declare.h.

4.19.2.20 NCOPY

```
#define NCOPY(
              s,
              t,
              n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 71 of file declare.h.

4.19.2.21 NCOPYI

```
#define NCOPYI(
              s,
              t,
              n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 73 of file declare.h.

4.19.2.22 NCOPYB

```
#define NCOPYB(
              s,
              t,
              n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 74 of file declare.h.

4.19.2.23 NCOPYI32

```
#define NCOPYI32(
              s,
              t,
              n ) { while ( (n)-- > 0 ) { *s++ = *t++; } }
```

Definition at line 75 of file declare.h.

4.19.2.24 WCOPY

```
#define WCOPY(
              s,
              t,
              n ) { int nn=n; WORD *ss=(WORD *)s, *tt=(WORD *)t; while ( (nn)-- > 0 ) { *ss++ =
*tt++; } }
```

Definition at line 76 of file declare.h.

4.19.2.25 NeedNumber

```
#define NeedNumber(
              x,
              s,
              err )
```

Value:

```
    { int sgn = 1;                                  \
    while ( *s == ' ' || *s == '\t' || *s == '-' || *s == '+' ) {       \
        if ( *s == '-' ) {sgn = -sgn;} s++; }                              \
    if ( chartype[*s] != 1 ) goto err;                                  \
    ParseNumber(x,s)                                                    \
    if ( sgn < 0 ) {(x) = -(x);} while ( *s == ' ' || *s == '\t' ) s++;\
}
```

Definition at line 78 of file declare.h.

4.19.2.26 SKIPBLANKS

```
#define SKIPBLANKS(
              s ) { while ( *(s) == ' ' || *(s) == '\t' ) (s)++; }
```

Definition at line 85 of file declare.h.

4.19.2.27 FLUSHCONSOLE

```
#define FLUSHCONSOLE if ( AP.InOutBuf > 0 ) CharOut(LINEFEED)
```

Definition at line 86 of file declare.h.

4.19.2.28 SKIPBRA1

```
#define SKIPBRA1(
              s )
```

Value:

```
{ int lev1=0; s++; while(*s) { if(*s=='['){lev1++;} \
else {if(*s==']'&&--lev1<0){break;}} s++;} }
```

Definition at line 88 of file declare.h.

4.19.2.29 SKIPBRA2

```
#define SKIPBRA2(
              s )
```

Value:

```
{ int lev2=0; s++; while(*s) { if(*s=='{'){lev2++;} \
else {if(*s=='}'&&--lev2<0){break;} \
else {if(*s=='['){SKIPBRA1(s)}}} s++;} }
```

Definition at line 90 of file declare.h.

4.19.2.30 SKIPBRA3

```
#define SKIPBRA3(
              s )
```

Value:

```
{ int lev3=0; s++; while(*s) { if(*s=='('){lev3++;} \
else {if(*s==')'&&--lev3<0){break;} \
else {if(*s=='{'){SKIPBRA2(s)} \
else {if(*s=='['){SKIPBRA1(s)}}}} s++;} }
```

Definition at line 93 of file declare.h.

4.19.2.31 SKIPBRA4

```
#define SKIPBRA4(
              s )
```

Value:

```
{ int lev4=0; s++; while(*s) { if(*s=='('){lev4++;} \
else {if(*s==')'&&--lev4<0){break;} \
else {if(*s=='['){SKIPBRA1(s)}}} s++;} }
```

Definition at line 97 of file declare.h.

4.19.2.32 SKIPBRA5

```
#define SKIPBRA5(
            s )
```

Value:

```
{ int lev5=0; s++; while(*s) { if(*s=='{'){lev5++;} \
else {if(*s=='}'&&--lev5<0){break;} \
else {if(*s=='('){SKIPBRA4(s)} \
else {if(*s=='['){SKIPBRA1(s)}}}} s++;} }
```

Definition at line 100 of file declare.h.

4.19.2.33 CYCLE1

```
#define CYCLE1(
            t,
            a,
            i ) {t iX=*a; WORD jX; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX;}
```

Definition at line 108 of file declare.h.

4.19.2.34 AddToCB

```
#define AddToCB(
            c,
            wx )
```

Value:

```
if(c->Pointer>=c->Top) \
{DoubleCbuffer(c-cbuf,c->Pointer,21);} \
*(c->Pointer)++ = wx;
```

Definition at line 110 of file declare.h.

4.19.2.35 EXCHINOUT

```
#define EXCHINOUT
```

Value:

```
{ FILEHANDLE *ffFi = AR.outfile; \
AR.outfile = AR.infile; AR.infile = ffFi; }
```

Definition at line 114 of file declare.h.

4.19.2.36 BACKINOUT

```
#define BACKINOUT
```

Value:

```
{ FILEHANDLE *ffFi = AR.outfile; POSITION posi; \
AR.outfile = AR.infile; AR.infile = ffFi; \
SetEndScratch(AR.infile,&posi); }
```

Definition at line 116 of file declare.h.

4.19.2.37 CopyArg

```
#define CopyArg(
            to,
            from )
```

Value:

```
{ if ( *from > 0 ) { int ica = *from; NCOPY(to,from,ica) } \
else if ( *from <= -FUNCTION ) *to++ = *from++;  \
else { *to++ = *from++; *to++ = *from++; } }
```

Definition at line 120 of file declare.h.

4.19.2.38 FILLARG

```
#define FILLARG(
            w )
```

Definition at line 129 of file declare.h.

4.19.2.39 COPYARG

```
#define COPYARG(
            w,
            t )
```

Definition at line 130 of file declare.h.

4.19.2.40 ZEROARG

```
#define ZEROARG(
            w )
```

Definition at line 131 of file declare.h.

4.19.2.41 FILLFUN

```
#define FILLFUN(
            w )
```

Definition at line 138 of file declare.h.

4.19.2.42 COPYFUN

```
#define COPYFUN(
            w,
            t )
```

Definition at line 139 of file declare.h.

4.19.2.43 COPYFUN3

```
#define COPYFUN3(
              w,
              t )
```

Definition at line 146 of file declare.h.

4.19.2.44 FILLFUN3

```
#define FILLFUN3(
              w )
```

Definition at line 147 of file declare.h.

4.19.2.45 FILLSUB

```
#define FILLSUB(
              w )
```

Definition at line 154 of file declare.h.

4.19.2.46 COPYSUB

```
#define COPYSUB(
              w,
              ww )
```

Definition at line 155 of file declare.h.

4.19.2.47 FILLEXPR

```
#define FILLEXPR(
              w )
```

Definition at line 161 of file declare.h.

4.19.2.48 NEXTARG

```
#define NEXTARG(
              x ) if(*x>0) x += *x; else if(*x <= -FUNCTION)x++; else x += 2;
```

Definition at line 164 of file declare.h.

4.19.2.49 COPY1ARG

```
#define COPY1ARG(
            s1,
            t1 )
```

Value:

```
{ int ica; if ( (ica=*t1) > 0 ) { NCOPY(s1,t1,ica) } \
else if(*t1<=-FUNCTION){*s1++=*t1++;} else{*s1++=*t1++;*s1++=*t1++;} }
```

Definition at line 165 of file declare.h.

4.19.2.50 ZeroFillRange

```
#define ZeroFillRange(
            w,
            begin,
            end )
```

Value:

```
    do { \
    int tmp_i; \
    for ( tmp_i = begin; tmp_i < end; tmp_i++ ) { (w)[tmp_i] = 0; } \
} while (0)
```

Fills a buffer by zero in the range [begin,end).

Parameters

w	The buffer.
begin	The index for the beginning of the range.
end	The index for the end of the range (exclusive).

Definition at line 175 of file declare.h.

4.19.2.51 TABLESIZE

```
#define TABLESIZE(
            a,
            b ) (((WORD)sizeof(a))/((WORD)sizeof(b)))
```

Definition at line 180 of file declare.h.

4.19.2.52 WORDDIF

```
#define WORDDIF(
            x,
            y ) (WORD)(x-y)
```

Definition at line 181 of file declare.h.

4.19.2.53 wsizeof

```
#define wsizeof(
                a ) ((WORD)sizeof(a))
```

Definition at line 182 of file declare.h.

4.19.2.54 VARNAME

```
#define VARNAME(
                type,
                num ) (AC.varnames->namebuffer+type[num].name)
```

Definition at line 183 of file declare.h.

4.19.2.55 DOLLARNAME

```
#define DOLLARNAME(
                type,
                num ) (AC.dollarnames->namebuffer+type[num].name)
```

Definition at line 184 of file declare.h.

4.19.2.56 EXPRNAME

```
#define EXPRNAME(
                num ) (AC.exprnames->namebuffer+Expressions[num].name)
```

Definition at line 185 of file declare.h.

4.19.2.57 PREV

```
#define PREV(
                x ) prevorder?prevorder:x
```

Definition at line 187 of file declare.h.

4.19.2.58 Terminate

```
#define Terminate(
                x ) do { TerminateImpl(x, __FILE__, __LINE__, __FUNCTION__); } while(0)
```

Definition at line 189 of file declare.h.

4.19.2.59 SETERROR

```
#define SETERROR(
                x ) { Terminate(-1); return(-1); }
```

Definition at line 190 of file declare.h.

4.19.2.60 DUMMYUSE

```
#define DUMMYUSE(
                x ) (void)(x);
```

Definition at line 193 of file declare.h.

4.19.2.61 ADDPOS

```
#define ADDPOS(
                pp,
                x ) (pp).p1 = ((pp).p1+(LONG)(x))
```

Definition at line 219 of file declare.h.

4.19.2.62 SETBASELENGTH

```
#define SETBASELENGTH(
                ss,
                x ) (ss).p1 = (LONG)(x)
```

Definition at line 220 of file declare.h.

4.19.2.63 SETBASEPOSITION

```
#define SETBASEPOSITION(
                pp,
                x ) (pp).p1 = (LONG)(x)
```

Definition at line 221 of file declare.h.

4.19.2.64 ISEQUALPOSINC

```
#define ISEQUALPOSINC(
                pp1,
                pp2,
                x ) ( (pp1).p1 == ((pp2).p1+(LONG)(x)) )
```

Definition at line 222 of file declare.h.

4.19.2.65 ISGEPOSINC

```
#define ISGEPOSINC(
                pp1,
                pp2,
                x ) ( (pp1).p1 >= ((pp2).p1+(LONG)(x)) )
```

Definition at line 223 of file declare.h.

4.19.2.66 DIVPOS

```
#define DIVPOS(
            pp,
            n ) ( (pp).p1/(LONG)(n) )
```

Definition at line 224 of file declare.h.

4.19.2.67 MULPOS

```
#define MULPOS(
            pp,
            n ) (pp).p1 *= (LONG)(n)
```

Definition at line 225 of file declare.h.

4.19.2.68 DIFPOS

```
#define DIFPOS(
            ss,
            pp1,
            pp2 ) (ss).p1 = ((pp1).p1-(pp2).p1)
```

Definition at line 228 of file declare.h.

4.19.2.69 DIFBASE

```
#define DIFBASE(
            pp1,
            pp2 ) ((pp1).p1-(pp2).p1)
```

Definition at line 229 of file declare.h.

4.19.2.70 ADD2POS

```
#define ADD2POS(
            pp1,
            pp2 ) (pp1).p1 += (pp2).p1
```

Definition at line 230 of file declare.h.

4.19.2.71 PUTZERO

```
#define PUTZERO(
            pp ) (pp).p1 = 0
```

Definition at line 231 of file declare.h.

4.19.2.72 BASEPOSITION

```
#define BASEPOSITION(
              pp ) ((pp).p1)
```

Definition at line 232 of file declare.h.

4.19.2.73 SETSTARTPOS

```
#define SETSTARTPOS(
              pp ) (pp).p1 = -2
```

Definition at line 233 of file declare.h.

4.19.2.74 NOTSTARTPOS

```
#define NOTSTARTPOS(
              pp ) ( (pp).p1 > -2 )
```

Definition at line 234 of file declare.h.

4.19.2.75 ISMINPOS

```
#define ISMINPOS(
              pp ) ( (pp).p1 == -1 )
```

Definition at line 235 of file declare.h.

4.19.2.76 ISEQUALPOS

```
#define ISEQUALPOS(
              pp1,
              pp2 ) ( (pp1).p1 == (pp2).p1 )
```

Definition at line 236 of file declare.h.

4.19.2.77 ISNOTEQUALPOS

```
#define ISNOTEQUALPOS(
              pp1,
              pp2 ) ( (pp1).p1 != (pp2).p1 )
```

Definition at line 237 of file declare.h.

4.19.2.78 ISLESSPOS

```
#define ISLESSPOS(
              pp1,
              pp2 ) ( (pp1).p1 < (pp2).p1 )
```

Definition at line 238 of file declare.h.

4.19.2.79 ISGEPOS

```
#define ISGEPOS(
              pp1,
              pp2 ) ( (pp1).p1 >= (pp2).p1 )
```

Definition at line 239 of file declare.h.

4.19.2.80 ISNOTZEROPOS

```
#define ISNOTZEROPOS(
              pp ) ( (pp).p1 != 0 )
```

Definition at line 240 of file declare.h.

4.19.2.81 ISZEROPOS

```
#define ISZEROPOS(
              pp ) ( (pp).p1 == 0 )
```

Definition at line 241 of file declare.h.

4.19.2.82 ISPOSPOS

```
#define ISPOSPOS(
              pp ) ( (pp).p1 > 0 )
```

Definition at line 242 of file declare.h.

4.19.2.83 ISNEGPOS

```
#define ISNEGPOS(
              pp ) ( (pp).p1 < 0 )
```

Definition at line 243 of file declare.h.

4.19.2.84 TOLONG

```
#define TOLONG(
              x ) ((LONG)(x))
```

Definition at line 246 of file declare.h.

4.19.2.85 Add2Com

```
#define Add2Com(
                x ) { WORD cod[2]; cod[0] = x; cod[1] = 2; AddNtoL(2,cod); }
```

Definition at line 248 of file declare.h.

4.19.2.86 Add3Com

```
#define Add3Com(
                x1,
                x2 ) { WORD cod[3]; cod[0] = x1; cod[1] = 3; cod[2] = x2; AddNtoL(3,cod); }
```

Definition at line 249 of file declare.h.

4.19.2.87 Add4Com

```
#define Add4Com(
                x1,
                x2,
                x3 )
```

Value:

```
{ WORD cod[4]; cod[0] = x1; cod[1] = 4; \
cod[2] = x2; cod[3] = x3; AddNtoL(4,cod); }
```

Definition at line 250 of file declare.h.

4.19.2.88 Add5Com

```
#define Add5Com(
                x1,
                x2,
                x3,
                x4 )
```

Value:

```
{ WORD cod[5]; cod[0] = x1; cod[1] = 5; \
cod[2] = x2; cod[3] = x3; cod[4] = x4; AddNtoL(5,cod); }
```

Definition at line 252 of file declare.h.

4.19.2.89 WantAddPointers

```
#define WantAddPointers(
                x )
```

Value:

```
while((AT.pWorkPointer+(x))>AR.pWorkSize){WORD ***ppp=&AT.pWorkSpace;\
ExpandBuffer((void **)ppp,&AR.pWorkSize,(int)(sizeof(WORD *)));}
```

Definition at line 258 of file declare.h.

4.19.2.90 WantAddLongs

```
#define WantAddLongs(
              x )
```

Value:
```
while((AT.lWorkPointer+(x))>AR.lWorkSize){LONG **ppp=&AT.lWorkSpace;\
ExpandBuffer((void **)ppp,&AR.lWorkSize,sizeof(LONG));}
```

Definition at line 260 of file declare.h.

4.19.2.91 WantAddPositions

```
#define WantAddPositions(
              x )
```

Value:
```
while((AT.posWorkPointer+(x))>AR.posWorkSize){POSITION **ppp=&AT.posWorkSpace;\
ExpandBuffer((void **)ppp,&AR.posWorkSize,sizeof(POSITION));}
```

Definition at line 262 of file declare.h.

4.19.2.92 FORM_INLINE

```
#define FORM_INLINE inline
```

Definition at line 266 of file declare.h.

4.19.2.93 MEMORYMACROS

```
#define MEMORYMACROS
```

Definition at line 274 of file declare.h.

4.19.2.94 TermMalloc

```
#define TermMalloc(
              x ) ( (AT.TermMemTop <= 0 ) ? TermMallocAddMemory(BHEAD0), AT.TermMemHeap[--AT.↩
TermMemTop]: AT.TermMemHeap[--AT.TermMemTop] )
```

Definition at line 278 of file declare.h.

4.19.2.95 NumberMalloc

```
#define NumberMalloc(
              x ) ( (AT.NumberMemTop <= 0 ) ? NumberMallocAddMemory(BHEAD0), AT.NumberMem↩
Heap[--AT.NumberMemTop]: AT.NumberMemHeap[--AT.NumberMemTop] )
```

Definition at line 279 of file declare.h.

4.19.2.96 CacheNumberMalloc

```
#define CacheNumberMalloc(
            x ) ( (AT.CacheNumberMemTop <= 0 ) ?  CacheNumberMallocAddMemory(BHEAD0), AT.↩
CacheNumberMemHeap[--AT.CacheNumberMemTop]:  AT.CacheNumberMemHeap[--AT.CacheNumberMemTop] )
```

Definition at line 280 of file declare.h.

4.19.2.97 TermFree

```
#define TermFree(
            TermMem,
            x ) AT.TermMemHeap[AT.TermMemTop++] = (WORD *)(TermMem)
```

Definition at line 281 of file declare.h.

4.19.2.98 NumberFree

```
#define NumberFree(
            NumberMem,
            x ) AT.NumberMemHeap[AT.NumberMemTop++] = (UWORD *)(NumberMem)
```

Definition at line 282 of file declare.h.

4.19.2.99 CacheNumberFree

```
#define CacheNumberFree(
            NumberMem,
            x ) AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = (UWORD *)(NumberMem)
```

Definition at line 283 of file declare.h.

4.19.2.100 NestingChecksum

```
#define NestingChecksum( ) (AC.IfLevel + AC.RepLevel + AC.arglevel + AC.insidelevel + AC.↩
termlevel + AC.inexprlevel + AC.dolooplevel +AC.SwitchLevel)
```

Definition at line 311 of file declare.h.

4.19.2.101 MesNesting

```
#define MesNesting( ) MesPrint("&Illegal nesting of if, repeat, argument, inside, term, inexpression
and do")
```

Definition at line 312 of file declare.h.

4.19.2.102 MarkPolyRatFunDirty

```
#define MarkPolyRatFunDirty(
            T )
```

Value:

```
{if(*T&&AR.PolyFunType==2){WORD *TP,*TT;TT=T+*T;TT-=ABS(TT[-1]);\
TP=T+1;while(TP<TT){if(*TP==AR.PolyFun){TP[2]|=(DIRTYFLAG|MUSTCLEANPRF);}TP+=TP[1];}}}
```

Definition at line 314 of file declare.h.

4.19.2.103 PUSHPREASSIGNLEVEL

```
#define PUSHPREASSIGNLEVEL
```

Value:

```
    AP.PreAssignLevel++; { GETIDENTITY \
    if ( AP.PreAssignLevel >= AP.MaxPreAssignLevel ) { int i; \
        LONG *ap = (LONG *)Malloc1(2*AP.MaxPreAssignLevel*sizeof(LONG *),"PreAssignStack"); \
        for ( i = 0; i < AP.MaxPreAssignLevel; i++ ) ap[i] = AP.PreAssignStack[i]; \
        M_free(AP.PreAssignStack,"PreAssignStack"); \
        AP.MaxPreAssignLevel *= 2; AP.PreAssignStack = ap; \
    } \
    *AT.WorkPointer++ = AP.PreContinuation; AP.PreContinuation = 0; \
    AP.PreAssignStack[AP.PreAssignLevel] = AC.iPointer - AC.iBuffer; }
```

Definition at line 321 of file declare.h.

4.19.2.104 POPPREASSIGNLEVEL

```
#define POPPREASSIGNLEVEL
```

Value:

```
    if ( AP.PreAssignLevel > 0 ) { GETIDENTITY \
    AC.iPointer = AC.iBuffer + AP.PreAssignStack[AP.PreAssignLevel--]; \
    AP.PreContinuation = *--AT.WorkPointer; \
    *AC.iPointer = 0; }
```

Definition at line 331 of file declare.h.

4.19.2.105 EXTERNLOCK

```
#define EXTERNLOCK(
            x )
```

NOTE: We have replaced LOCK(ErrorMessageLock) and UNLOCK(ErrorMessageLock) by MLOCK(Error↩ MessageLock) and MUNLOCK(ErrorMessageLock). They are used for the synchronised output in ParFORM. (TU 28 May 2011)

Definition at line 477 of file declare.h.

4.19.2.106 INILOCK

```
#define INILOCK(
            x )
```

Definition at line 478 of file declare.h.

4.19.2.107 LOCK

```
#define LOCK(
                x )
```

Definition at line 479 of file declare.h.

4.19.2.108 UNLOCK

```
#define UNLOCK(
                x )
```

Definition at line 480 of file declare.h.

4.19.2.109 EXTERNRWLOCK

```
#define EXTERNRWLOCK(
                x )
```

Definition at line 481 of file declare.h.

4.19.2.110 INIRWLOCK

```
#define INIRWLOCK(
                x )
```

Definition at line 482 of file declare.h.

4.19.2.111 RWLOCKR

```
#define RWLOCKR(
                x )
```

Definition at line 483 of file declare.h.

4.19.2.112 RWLOCKW

```
#define RWLOCKW(
                x )
```

Definition at line 484 of file declare.h.

4.19.2.113 UNRWLOCK

```
#define UNRWLOCK(
                x )
```

Definition at line 485 of file declare.h.

4.19.2.114 MLOCK

```
#define MLOCK(
              x )
```

Definition at line 490 of file declare.h.

4.19.2.115 MUNLOCK

```
#define MUNLOCK(
              x )
```

Definition at line 491 of file declare.h.

4.19.2.116 GETIDENTITY

```
#define GETIDENTITY
```

Definition at line 493 of file declare.h.

4.19.2.117 GETBIDENTITY

```
#define GETBIDENTITY
```

Definition at line 494 of file declare.h.

4.19.2.118 M_alloc

```
#define M_alloc(
              x ) malloc((size_t)(x))
```

Definition at line 1009 of file declare.h.

4.19.2.119 CompareTerms

```
#define CompareTerms ((COMPARE)AR.CompareRoutine)
```

Definition at line 1490 of file declare.h.

4.19.2.120 FiniShuffle

```
#define FiniShuffle AN.SHvar.finishuf
```

Definition at line 1491 of file declare.h.

4.19.2.121 DoShtuffle

```
#define DoShtuffle ((DO_UFFLE)AN.SHvar.do_uffle)
```

Definition at line 1492 of file declare.h.

4.19.3 Typedef Documentation

4.19.3.1 WRITEBUFTOEXTCHANNEL

```
typedef int(* WRITEBUFTOEXTCHANNEL) (char *, size_t)
```

Definition at line 1483 of file declare.h.

4.19.3.2 GETCFROMEXTCHANNEL

```
typedef int(* GETCFROMEXTCHANNEL) (void)
```

Definition at line 1484 of file declare.h.

4.19.3.3 SETTERMINATORFOREXTERNALCHANNEL

```
typedef int(* SETTERMINATORFOREXTERNALCHANNEL) (char *)
```

Definition at line 1485 of file declare.h.

4.19.3.4 SETKILLMODEFOREXTERNALCHANNEL

```
typedef int(* SETKILLMODEFOREXTERNALCHANNEL) (int, int)
```

Definition at line 1486 of file declare.h.

4.19.3.5 WRITEFILE

```
typedef LONG(* WRITEFILE) (int, UBYTE *, LONG)
```

Definition at line 1487 of file declare.h.

4.19.3.6 GETTERM

```
typedef WORD(* GETTERM) (PHEAD WORD *)
```

Definition at line 1488 of file declare.h.

4.19.4 Function Documentation

4.19.4.1 TELLFILE()

```
void TELLFILE (
            int handle,
            POSITION * position )  [extern]
```

Definition at line 1406 of file tools.c.

4.19.4.2 StartVariables()

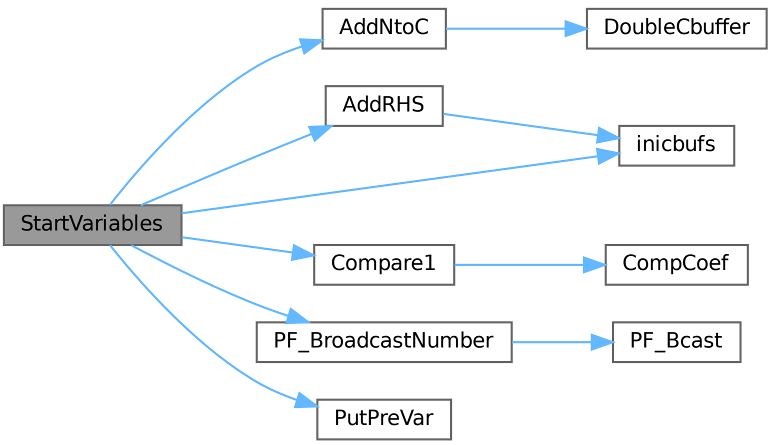
```
void StartVariables (
            void  )  [extern]
```

All functions (well, nearly all) are declared here.

Definition at line 905 of file startup.c.

References AddNtoC(), AddRHS(), Compare1(), inicbufs(), FuNcTiOn::name, PF_BroadcastNumber(), PutPreVar(), and FuNcTiOn::symmetric.

Here is the call graph for this function:



4.19.4.3 CodeToLine()

```
UBYTE * CodeToLine (
            WORD number,
            UBYTE * Out )  [extern]
```

Definition at line 658 of file sch.c.

4.19.4.4 AddArrayIndex()

```
UBYTE * AddArrayIndex (
             WORD num,
             UBYTE * out )  [extern]
```

Definition at line 696 of file sch.c.

4.19.4.5 FindInIndex()

```
INDEXENTRY * FindInIndex (
             WORD expr,
             FILEDATA * f,
             WORD par,
             WORD mode )  [extern]
```

Definition at line 2665 of file store.c.

4.19.4.6 NextFileIndex()

```
INDEXENTRY * NextFileIndex (
             POSITION * indexpos )  [extern]
```

Definition at line 2258 of file store.c.

4.19.4.7 PasteTerm()

```
WORD * PasteTerm (
             PHEAD WORD number,
             WORD * accum,
             WORD * position,
             WORD times,
             WORD divby )  [extern]
```

Puts the term at position in the accumulator accum at position 'number+1'. if times $>$ 0 the coefficient of this term is multiplied by times/divby.

Parameters

number	The number of term fragments in accum that should be skipped
accum	The accumulator of term fragments
position	A position in (typically) a compiler buffer from where a (piece of a) term comes.
times	Multiply the result by this
divby	Divide the result by this.

This routine is typically used when we have to replace a (sub)expression pointer by a power of a (sub)expression. This uses mostly a binomial expansion and the new term is the old term multiplied one by one by terms of the new expression. The factors times and divby keep track of the binomial coefficient. Once this is complete, the routine FiniTerm will make the contents of the accumulator into a proper term that still needs to be normalized.

Definition at line 3002 of file proces.c.

Referenced by Generator().

4.19.4.8 StrCopy()

```
UBYTE * StrCopy (
            UBYTE * from,
            UBYTE * to )  [extern]
```

Definition at line 67 of file sch.c.

4.19.4.9 WrtPower()

```
UBYTE * WrtPower (
            UBYTE * Out,
            WORD Power )  [extern]
```

Definition at line 738 of file sch.c.

4.19.4.10 AccumGCD()

```
WORD AccumGCD (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb )  [extern]
```

Definition at line 647 of file reken.c.

4.19.4.11 AddArgs()

```
void AddArgs (
            PHEAD WORD * s1,
            WORD * s2,
            WORD * m )  [extern]
```

Adds the arguments of two occurrences of the PolyFun.

Parameters

s1	Pointer to the first occurrence.
s2	Pointer to the second occurrence.
m	Pointer to where the answer should be.

Definition at line 2343 of file sort.c.

Referenced by AddPoly(), and MergePatches().

4.19.4.12 AddCoef()

```
WORD AddCoef (
            PHEAD WORD ** ps1,
            WORD ** ps2 )  [extern]
```

Adds the coefficients of the terms *ps1 and *ps2. The problem comes when there is not enough space for a new longer coefficient. First a local solution is tried. If this is not successful we need to move terms around. The possibility of a garbage collection should not be ignored, as avoiding this costs very much extra space which is nearly wasted otherwise.

If the return value is zero the terms cancelled.

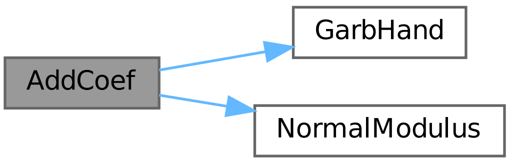
The resulting term is left in *ps1.

Definition at line 2042 of file sort.c.

References GarbHand(), and NormalModulus().

Referenced by SplitMerge().

Here is the call graph for this function:



4.19.4.13 AddLong()

```
WORD AddLong (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )  [extern]
```

Definition at line 705 of file reken.c.

4.19.4.14 AddPLon()

```
WORD AddPLon (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )  [extern]
```

Definition at line 750 of file reken.c.

4.19.4.15 AddPoly()

```
WORD AddPoly (
            PHEAD WORD ** ps1,
            WORD ** ps2 )  [extern]
```

Routine should be called when S->PolyWise != 0. It points then to the position of AR.PolyFun in both terms.

We add the contents of the arguments of the two polynomials. Special attention has to be given to special arguments. We have to reserve a space equal to the size of one term + the size of the argument of the other. The addition has to be done in this routine because not all objects are reentrant.

Newer addition (12-nov-2007). The PolyFun can have two arguments. In that case S->PolyFlag is 2 and we have to call the routine for adding rational polynomials. We have to be rather careful what happens with: The location of the output The order of the terms in the arguments At first we allow only univariate polynomials in the PolyFun. This restriction will be lifted a.s.a.p.

Parameters

ps1	A pointer to the position of the first term
ps2	A pointer to the position of the second term

Returns

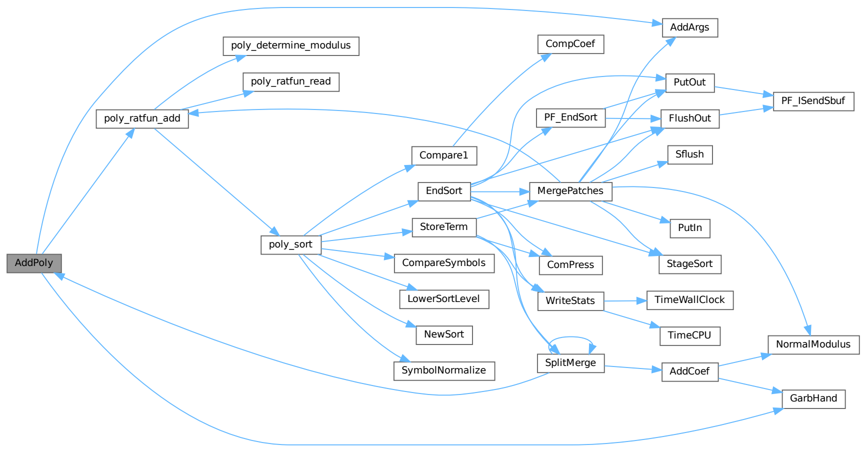
If zero the terms cancel. Otherwise the new term is in ∗ps1.

Definition at line 2171 of file sort.c.

References AddArgs(), GarbHand(), and poly_ratfun_add().

Referenced by SplitMerge().

Here is the call graph for this function:



4.19.4.16 AddRat()

```
WORD AddRat (
             PHEAD UWORD * a,
             WORD na,
             UWORD * b,
             WORD nb,
             UWORD * c,
             WORD * nc ) [extern]
```

Definition at line 210 of file reken.c.

4.19.4.17 AddToLine()

```
void AddToLine (
             UBYTE * s ) [extern]
```

Definition at line 82 of file sch.c.

4.19.4.18 AddWild()

```
WORD AddWild (
             PHEAD WORD,
             WORD type,
             WORD newnumber ) [extern]
```

Definition at line 1544 of file wildcard.c.

4.19.4.19 BigLong()

```
WORD BigLong (
             UWORD * a,
             WORD na,
             UWORD * b,
             WORD nb ) [extern]
```

Definition at line 944 of file reken.c.

4.19.4.20 BinomGen()

```
WORD BinomGen (
             PHEAD WORD * term,
             WORD level,
             WORD ** tstops,
             WORD x1,
             WORD x2,
             WORD pow1,
             WORD pow2,
             WORD sign,
             UWORD * coef,
             WORD ncoef ) [extern]
```

Definition at line 320 of file ratio.c.

4.19.4.21 CheckWild()

```
WORD CheckWild (
             PHEAD WORD,
             WORD type,
             WORD newnumber,
             WORD * newval )  [extern]
```

Definition at line 1791 of file wildcard.c.

4.19.4.22 Chisholm()

```
WORD Chisholm (
             PHEAD WORD * term,
             WORD level )  [extern]
```

Definition at line 1963 of file opera.c.

4.19.4.23 CleanExpr()

```
WORD CleanExpr (
             WORD par )  [extern]
```

Definition at line 47 of file execute.c.

4.19.4.24 CleanUp()

```
void CleanUp (
             WORD par )  [extern]
```

Definition at line 1761 of file startup.c.

4.19.4.25 ClearWild()

```
void ClearWild (
             PHEAD0  )  [extern]
```

Definition at line 1518 of file wildcard.c.

4.19.4.26 CompareFunctions()

```
WORD CompareFunctions (
             WORD * fleft,
             WORD * fright )  [extern]
```

Definition at line 54 of file normal.c.

4.19.4.27 Commute()

```
WORD Commute (
            WORD * fleft,
            WORD * fright ) [extern]
```

Definition at line 118 of file normal.c.

4.19.4.28 DetCommu()

```
WORD DetCommu (
            WORD * terms ) [extern]
```

Definition at line 4510 of file normal.c.

4.19.4.29 DoesCommu()

```
WORD DoesCommu (
            WORD * term ) [extern]
```

Definition at line 4569 of file normal.c.

4.19.4.30 CompArg()

```
int CompArg (
            WORD * s1,
            WORD * s2 ) [extern]
```

Definition at line 3307 of file tools.c.

4.19.4.31 CompCoef()

```
WORD CompCoef (
            WORD * term1,
            WORD * term2 ) [extern]
```

```
Routine takes a1 mod m1 and a2 mod m2 and returns a mod m1*m2 with
a mod m1 = a1 and a mod m2 = a2
```

Chinese remainder: a%(m1∗m2) = q1∗m1+a1 a%(m1∗m2) = q2∗m2+a2 Compute n1 such that (n1∗m1)m2 is one Compute n2 such that (n2∗m2)m1 is one Then (a1∗n2∗m2+a2∗n1∗m1)%(m1∗m2) is a%(m1∗m2)

Definition at line 3047 of file reken.c.

Referenced by Compare1().

4.19.4.32 CompGroup()

```
WORD CompGroup (
            PHEAD WORD,
            WORD ** args,
            WORD * a1,
            WORD * a2,
            WORD num )  [extern]
```

Definition at line 373 of file function.c.

4.19.4.33 Compare1()

```
WORD Compare1 (
            PHEAD WORD * term1,
            WORD * term2,
            WORD level )  [extern]
```

Compares two terms. The answer is: 0 equal (with exception of the coefficient if level == 0.) >0 term1 comes first. <0 term2 comes first. Some special precautions may be needed to keep the CompCoef routine from generating overflows, although this is very unlikely in subterms. This routine should not return an error condition.

Originally this routine was called Compare. With the treatment of special polynomials with terms that contain only symbols and the need for extreme speed for the polynomial routines we made a special compare routine and now we store the address of the current compare routine in AR.CompareRoutine and have a macro Compare which makes all existing code work properly and we can just replace the routine on a thread by thread basis (each thread has its own AR struct).

Parameters

term1	First input term
term2	Second input term
level	The sorting level (may influence on the result)

Returns

0 equal (with exception of the coefficient if level == 0.) >0 term1 comes first. <0 term2 comes first.

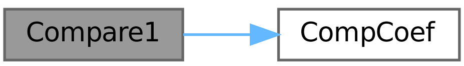
When there are floating point numbers active (float_ = FLOATFUN) the presence of one or more float_ functions is returned in AT.SortFloatMode: 0: no float_ 1: float_ in term1 only 2: float_ in term2 only 3: float_ in both terms

Definition at line 2636 of file sort.c.

References CompCoef().

Referenced by InFunction(), poly_sort(), and StartVariables().

Here is the call graph for this function:



4.19.4.34 CountDo()

```
WORD CountDo (
            WORD * term,
            WORD * instruct )  [extern]
```

Definition at line 552 of file reshuf.c.

4.19.4.35 CountFun()

```
WORD CountFun (
            WORD * term,
            WORD * countfun )  [extern]
```

Definition at line 706 of file reshuf.c.

4.19.4.36 DimensionSubterm()

```
WORD DimensionSubterm (
            WORD * subterm )  [extern]
```

Definition at line 867 of file reshuf.c.

4.19.4.37 DimensionTerm()

```
WORD DimensionTerm (
            WORD * term )  [extern]
```

Definition at line 968 of file reshuf.c.

4.19.4.38 DimensionExpression()

```
WORD DimensionExpression (
            PHEAD WORD * expr )  [extern]
```

Definition at line 1000 of file reshuf.c.

4.19.4.39 Deferred()

```
WORD Deferred (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Picks up the deferred brackets. These are the bracket contents of which we postpone the reading when we use the 'Keep Brackets' statement. These contents are multiplying the terms just before they are sent to the sorting system. Special attention goes to having it thread-safe We have to lock positioning the file and reading it in a thread specific buffer.

Parameters

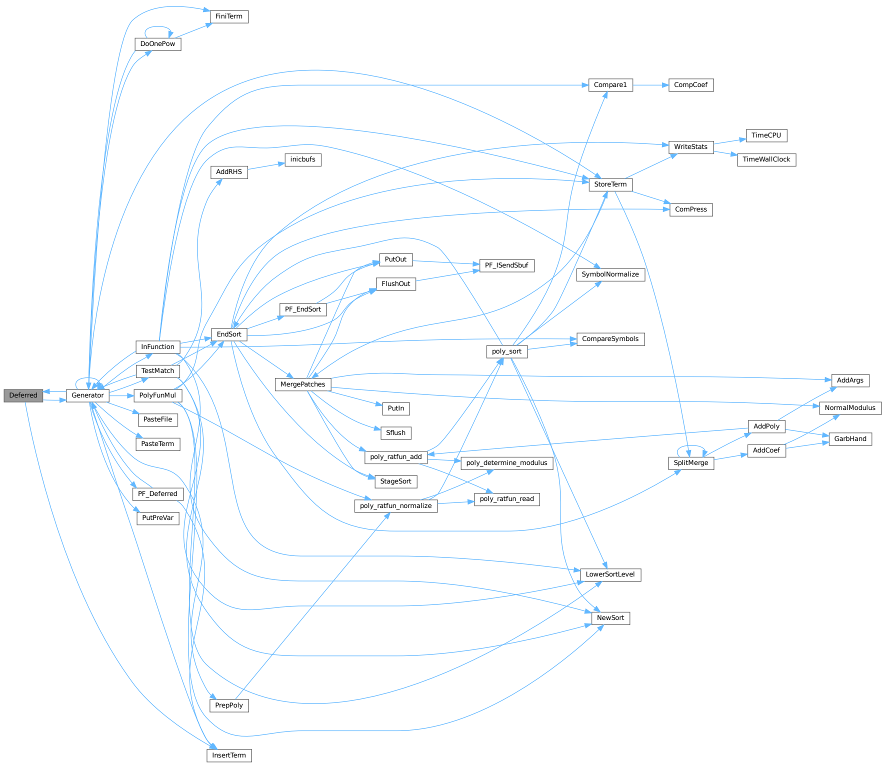
term	The term that must be multiplied by the contents of the current bracket
level	The compiler level. This is needed because after multiplying term by term we call Generator again.

Definition at line 4844 of file proces.c.

References Generator(), and InsertTerm().

Referenced by Generator().

Here is the call graph for this function:



4.19.4.40 DeleteStore()

```
WORD DeleteStore (
            WORD par )  [extern]
```

Definition at line 773 of file store.c.

4.19.4.41 DetCurDum()

```
WORD DetCurDum (
            PHEAD WORD * t )  [extern]
```

Definition at line 252 of file reshuf.c.

4.19.4.42 DetVars()

```
void DetVars (
            WORD * term,
            WORD par )  [extern]
```

Definition at line 1830 of file store.c.

4.19.4.43 Distribute()

```
WORD Distribute (
            DISTRIBUTE * d,
            WORD first )  [extern]
```

Definition at line 509 of file symmetr.c.

4.19.4.44 DivLong()

```
WORD DivLong (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc,
            UWORD * d,
            WORD * nd )  [extern]
```

Definition at line 972 of file reken.c.

4.19.4.45 DivRat()

```
WORD DivRat (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )  [extern]
```

Definition at line 500 of file reken.c.

4.19.4.46 Divvy()

```
WORD Divvy (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb )  [extern]
```

Definition at line 172 of file reken.c.

4.19.4.47 DoDelta()

```
WORD DoDelta (
            WORD * t ) [extern]
```

Definition at line 4355 of file normal.c.

4.19.4.48 DoDelta3()

```
WORD DoDelta3 (
            PHEAD WORD * term,
            WORD level ) [extern]
```

Definition at line 1405 of file reshuf.c.

4.19.4.49 TestPartitions()

```
WORD TestPartitions (
            WORD * tfun,
            PARTI * parti ) [extern]
```

Definition at line 1607 of file reshuf.c.

4.19.4.50 DoPartitions()

```
WORD DoPartitions (
            PHEAD WORD * term,
            WORD level ) [extern]
```

Definition at line 1722 of file reshuf.c.

4.19.4.51 CoCanonicalize()

```
int CoCanonicalize (
            UBYTE * s ) [extern]
```

Definition at line 64 of file diagrams.c.

4.19.4.52 DoCanonicalize()

```
int DoCanonicalize (
            PHEAD WORD * term,
            WORD * params ) [extern]
```

Definition at line 185 of file diagrams.c.

4.19.4.53 GenTopologies()

```
WORD GenTopologies (
             PHEAD WORD * term,
             WORD level ) [extern]
```

Definition at line 962 of file diawrap.cc.

4.19.4.54 GenDiagrams()

```
WORD GenDiagrams (
             PHEAD WORD * term,
             WORD level ) [extern]
```

Definition at line 732 of file diawrap.cc.

4.19.4.55 DoTopologyCanonicalize()

```
int DoTopologyCanonicalize (
             PHEAD WORD * term,
             WORD vert,
             WORD edge,
             WORD * args ) [extern]
```

Definition at line 417 of file diagrams.c.

4.19.4.56 DoShattering()

```
int DoShattering (
             PHEAD WORD * connect,
             WORD * environ,
             WORD * partitions,
             WORD nvert ) [extern]
```

Definition at line 626 of file diagrams.c.

4.19.4.57 GenerateTopologies()

```
WORD GenerateTopologies (
             PHEAD WORD,
             WORD nlegs,
             WORD setvert,
             WORD setmax ) [extern]
```

Definition at line 123 of file topowrap.cc.

4.19.4.58 DoTableExpansion()

```
WORD DoTableExpansion (
            WORD * term,
            WORD level )  [extern]
```

Definition at line 334 of file tables.c.

4.19.4.59 DoDistrib()

```
WORD DoDistrib (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 1119 of file reshuf.c.

4.19.4.60 DoShuffle()

```
WORD DoShuffle (
            WORD * term,
            WORD level,
            WORD fun,
            WORD option )  [extern]
```

Definition at line 2095 of file reshuf.c.

4.19.4.61 DoPermutations()

```
WORD DoPermutations (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 2007 of file reshuf.c.

4.19.4.62 Shuffle()

```
int Shuffle (
            WORD * from1,
            WORD * from2,
            WORD * to )  [extern]
```

Definition at line 2229 of file reshuf.c.

4.19.4.63 FinishShuffle()

```
int FinishShuffle (
            WORD * fini )  [extern]
```

Definition at line 2421 of file reshuf.c.

4.19.4.64 DoStuffle()

```
WORD DoStuffle (
            WORD * term,
            WORD level,
            WORD fun,
            WORD option )  [extern]
```

Definition at line 2487 of file reshuf.c.

4.19.4.65 Stuffle()

```
int Stuffle (
            WORD * from1,
            WORD * from2,
            WORD * to )  [extern]
```

Definition at line 2702 of file reshuf.c.

4.19.4.66 FinishStuffle()

```
int FinishStuffle (
            WORD * fini )  [extern]
```

Definition at line 2829 of file reshuf.c.

4.19.4.67 StuffRootAdd()

```
WORD * StuffRootAdd (
            WORD * t1,
            WORD * t2,
            WORD * to )  [extern]
```

Definition at line 2876 of file reshuf.c.

4.19.4.68 TestUse()

```
WORD TestUse (
            WORD * term,
            WORD level )  [extern]
```

Definition at line 1414 of file tables.c.

4.19.4.69 FindTB()

```
DBASE * FindTB (
            UBYTE * name )  [extern]
```

Definition at line 734 of file tables.c.

4.19.4.70 CheckTableDeclarations()

```
int CheckTableDeclarations (
            DBASE * d )  [extern]
```

Definition at line 1178 of file tables.c.

4.19.4.71 Apply()

```
WORD Apply (
            WORD * term,
            WORD level )  [extern]
```

Definition at line 1980 of file tables.c.

4.19.4.72 ApplyExec()

```
int ApplyExec (
            WORD * term,
            int maxtogo,
            WORD level )  [extern]
```

Definition at line 2039 of file tables.c.

4.19.4.73 ApplyReset()

```
WORD ApplyReset (
            WORD level )  [extern]
```

Definition at line 2256 of file tables.c.

4.19.4.74 TableReset()

```
WORD TableReset (
            void )  [extern]
```

Definition at line 2292 of file tables.c.

4.19.4.75 ReWorkT()

```
void ReWorkT (
            WORD * term,
            WORD * funs,
            WORD numfuns )  [extern]
```

Definition at line 1899 of file tables.c.

4.19.4.76 GetIfDollarNum()

```
WORD GetIfDollarNum (
            WORD * ifp,
            WORD * ifstop )  [extern]
```

Definition at line 106 of file if.c.

4.19.4.77 FindVar()

```
int FindVar (
            WORD * v,
            WORD * term )  [extern]
```

Definition at line 173 of file if.c.

4.19.4.78 DoIfStatement()

```
WORD DoIfStatement (
            PHEAD WORD * ifcode,
            WORD * term )  [extern]
```

Definition at line 280 of file if.c.

4.19.4.79 DoOnePow()

```
WORD DoOnePow (
            PHEAD WORD * term,
            WORD power,
            WORD nexp,
            WORD * accum,
            WORD * aa,
            WORD level,
            WORD * freeze )  [extern]
```

Routine gets one power of an expression in the scratch system. If there are more powers needed there will be a recursion.

No attempt is made to use binomials because we have no information about commuting properties.

There is a searching for the contents of brackets if needed. This searching may be rather slow because of the single links.

Parameters

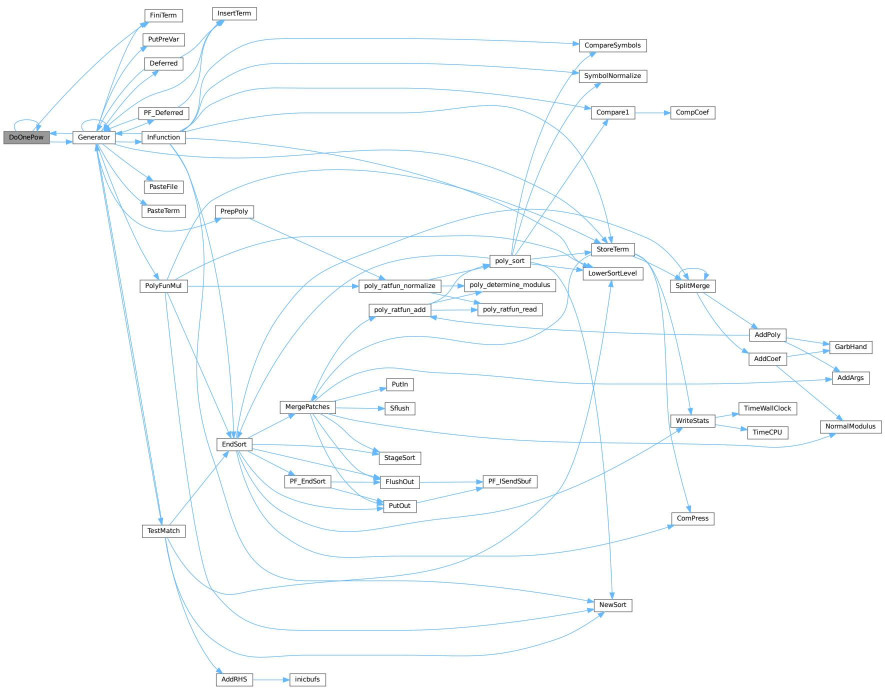
term	is the term we are adding to.
power	is the power of the expression that we need.
nexp	is the number of the expression.
accum	is the accumulator of terms. It accepts the termfragments that are made into a proper term in FiniTerm
aa	points to the start of the entire accumulator. In *aa we store the number of term fragments that are in the accumulator.
level	is the current depth in the tree of statements. It is needed to continue to the next operation/substitution with each generated term
freeze	is the pointer to the bracket information that should be matched.

Definition at line 4623 of file proces.c.

References DoOnePow(), FiniTerm(), Generator(), and FiLe::handle.

Referenced by DoOnePow(), and Generator().

Here is the call graph for this function:



4.19.4.80 DoRevert()

```
void DoRevert (
            WORD * fun,
            WORD * tmp )  [extern]
```

Definition at line 4422 of file normal.c.

4.19.4.81 DoSumF1()

```
WORD DoSumF1 (
            PHEAD WORD * term,
            WORD * params,
            WORD replac,
            WORD level )  [extern]
```

Definition at line 395 of file ratio.c.

4.19.4.82 DoSumF2()

```
WORD DoSumF2 (
            PHEAD WORD * term,
            WORD * params,
            WORD replac,
            WORD level )  [extern]
```

Definition at line 520 of file ratio.c.

4.19.4.83 DoTheta()

```
WORD DoTheta (
            PHEAD WORD * t )  [extern]
```

Definition at line 4258 of file normal.c.

4.19.4.84 EndSort()

```
LONG EndSort (
            PHEAD WORD * buffer,
            int par )  [extern]
```

Finishes a sort. At AR.sLevel == 0 the output is to the regular output stream. When AR.sLevel > 0, the parameter par determines the actual output. The AR.sLevel will be popped. All ongoing stages are finished and if the sortfile is open it is closed. The statistics are printed when AR.sLevel == 0 par == 0 Output to the buffer. par == 1 Sort for function arguments. The output will be copied into the buffer. It is assumed that this is in the WorkSpace. par == 2 Sort for $-variable. We return the address of the buffer that contains the output in buffer (treated like WORD **). We first catch the output in a file (unless we can intercept things after the small buffer has been sorted) Then we read from the file into a buffer. Only when par == 0 data compression can be attempted at AT.SS==AT.S0.

Parameters

buffer	buffer for output when needed
par	See above

Returns

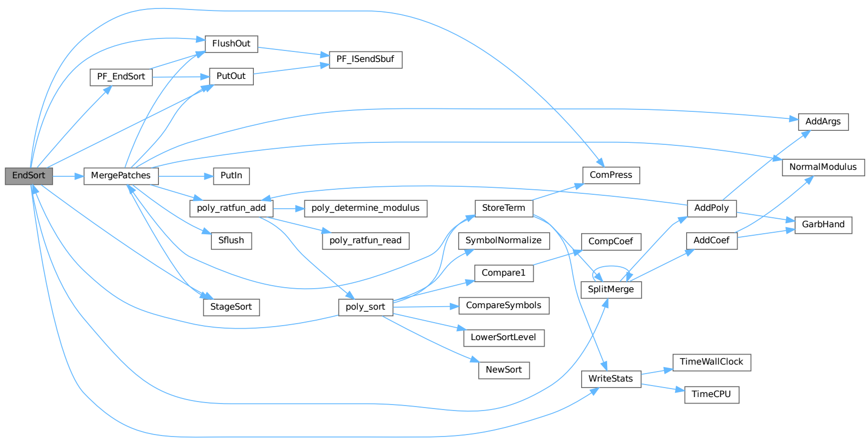
If negative: error. If positive: number of words in output.

Definition at line 733 of file sort.c.

References ComPress(), FlushOut(), MergePatches(), PF_EndSort(), PutOut(), SplitMerge(), StageSort(), and WriteStats().

Referenced by generate_expression(), get_expression(), InFunction(), MakeDollarInteger(), MakeDollarMod(), PF_CollectModifiedDollars(), PF_Processor(), poly_factorize_expression(), poly_sort(), poly_unfactorize_expression(), PolyFunMul(), Processor(), TakeArgContent(), and TestMatch().

Here is the call graph for this function:



4.19.4.85 EntVar()

```
WORD EntVar (
            WORD type,
            UBYTE * name,
            WORD x,
            WORD y,
            WORD z,
            WORD d ) [extern]
```

Definition at line 557 of file names.c.

4.19.4.86 EpfCon()

```
WORD EpfCon (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level ) [extern]
```

Definition at line 212 of file opera.c.

4.19.4.87 EpfFind()

```
WORD EpfFind (
            PHEAD WORD * term,
            WORD * params ) [extern]
```

Definition at line 58 of file opera.c.

4.19.4.88 EpfGen()

```
WORD EpfGen (
            WORD number,
            WORD * inlist,
            WORD * kron,
            WORD * perm,
            WORD sgn ) [extern]
```

Definition at line 282 of file opera.c.

4.19.4.89 EqualArg()

```
WORD EqualArg (
            WORD * parms,
            WORD num1,
            WORD num2 ) [extern]
```

Definition at line 1379 of file reshuf.c.

4.19.4.90 Factorial()

```
int Factorial (
            PHEAD WORD,
            UWORD * a,
            WORD * na ) [extern]
```

Definition at line 3449 of file reken.c.

4.19.4.91 Bernoulli()

```
int Bernoulli (
            WORD n,
            UWORD * a,
            WORD * na ) [extern]
```

Definition at line 3529 of file reken.c.

4.19.4.92 FactorIn()

```
int FactorIn (
            PHEAD WORD * term,
            WORD level ) [extern]
```

Definition at line 46 of file factor.c.

4.19.4.93 FactorInExpr()

```
int FactorInExpr (
            PHEAD WORD * term,
            WORD level ) [extern]
```

Definition at line 421 of file factor.c.

4.19.4.94 FindAll()

```
WORD FindAll (
            PHEAD WORD * term,
            WORD * pattern,
            WORD level,
            WORD * par ) [extern]
```

Definition at line 1197 of file pattern.c.

4.19.4.95 FindMulti()

```
WORD FindMulti (
            PHEAD WORD * term,
            WORD * pattern ) [extern]
```

Definition at line 954 of file findpat.c.

4.19.4.96 FindOnce()

```
WORD FindOnce (
            PHEAD WORD * term,
            WORD * pattern ) [extern]
```

Definition at line 419 of file findpat.c.

4.19.4.97 FindOnly()

```
WORD FindOnly (
            PHEAD WORD * term,
            WORD * pattern ) [extern]
```

Definition at line 61 of file findpat.c.

4.19.4.98 FindRest()

```
WORD FindRest (
            PHEAD WORD * term,
            WORD * pattern ) [extern]
```

Definition at line 1070 of file findpat.c.

4.19.4.99 FindrNumber()

```
WORD FindrNumber (
            WORD n,
            VARRENUM * v )  [extern]
```

Definition at line 2588 of file store.c.

4.19.4.100 FiniLine()

```
void FiniLine (
            void )  [extern]
```

Definition at line 182 of file sch.c.

4.19.4.101 FiniTerm()

```
WORD FiniTerm (
            PHEAD WORD * term,
            WORD * accum,
            WORD * termout,
            WORD number,
            WORD tepos )  [extern]
```

Concatenates the contents of the accumulator into a single legal term, which replaces the subexpression pointer

Parameters

term	the input term with the (sub)expression subterm
accum	the accumulator with the term fragments
termout	the location where the output should be written
number	the number of term fragments in the accumulator
tepos	the position of the subterm in term to be replaced

Definition at line 3067 of file proces.c.

Referenced by DoOnePow(), and Generator().

4.19.4.102 FlushOut()

```
WORD FlushOut (
            POSITION * position,
            FILEHANDLE * fi,
            int compr )  [extern]
```

Completes output to an output file and writes the trailing zero.

Parameters

position	The position in the file after writing
fi	The file (or its cache)
compr	Indicates whether there should be compression with gzip.

Returns

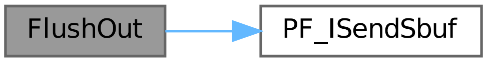
Regular conventions (OK -> 0).

Definition at line 1828 of file sort.c.

References FiLe::handle, BrAcKeTiNfO::indexbuffer, and PF_ISendSbuf().

Referenced by EndSort(), MergePatches(), PF_EndSort(), and Processor().

Here is the call graph for this function:



4.19.4.103 FunLevel()

```
void FunLevel (
            PHEAD WORD * term )  [extern]
```

Definition at line 130 of file reshuf.c.

4.19.4.104 AdjustRenumScratch()

```
void AdjustRenumScratch (
            PHEAD0  )  [extern]
```

Definition at line 506 of file reshuf.c.

4.19.4.105 GarbHand()

```
void GarbHand (
            void  )  [extern]
```

Garbage collection that takes place when the small extension is full and we need to place more terms there. When this is the case there are many holes in the small buffer and the whole can be compactified. The major complication is the buffer for SplitMerge. There are to options for temporary memory: 1: find some buffer that has enough space (maybe in the large buffer). 2: allocate a buffer. Give it back afterwards of course. If the small extension is properly dimensioned this routine should be called very rarely. Most of the time it will be called when the polyfun or polyratfun is active.

Definition at line 3642 of file sort.c.

Referenced by AddCoef(), and AddPoly().

4.19.4.106 GcdLong()

```
WORD GcdLong (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc ) [extern]
```

Definition at line 2276 of file reken.c.

4.19.4.107 LcmLong()

```
WORD LcmLong (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc ) [extern]
```

Definition at line 2701 of file reken.c.

4.19.4.108 Generator()

```
WORD Generator (
            PHEAD WORD * term,
            WORD level ) [extern]
```

The heart of the program. Here the expansion tree is set up in one giant recursion

Parameters

term	the input term. may be overwritten
level	the level in the compiler buffer (number of statement)

Returns

Normal conventions (OK = 0).

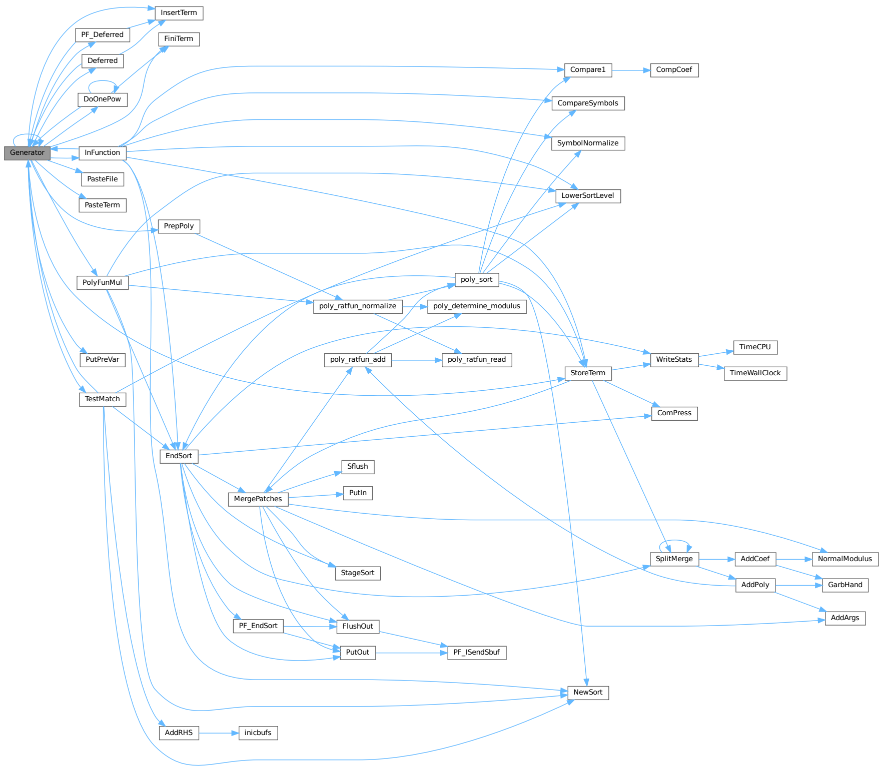
The routine looks first whether there are unsubstituted (sub)expressions. If so, one of them gets inserted term by term and the new term is used in a renewed call to Generator. If there are no (sub)expressions, the term is normalized, the compiler level is raised (next statement) and the program looks what type of statement this is. If this is a special statement it is either treated on the spot or the appropriate routine is called. If it is a substitution, the pattern matcher is called (TestMatch) which tells whether there was a match. If so we need to call TestSub again to test for (sub)expressions. If we run out of levels, the term receives a final treatment for modulus calculus and/or brackets and is then sent off to the sorting routines.

Definition at line 3266 of file proces.c.

References Deferred(), DoOnePow(), FiniTerm(), Generator(), InFunction(), InsertTerm(), CbUf::lhs, VaRrEnUm::lo, PasteFile(), PasteTerm(), PF_Deferred(), CbUf::Pointer, PolyFunMul(), PrepPoly(), PutPreVar(), CbUf::rhs, StoreTerm(), ReNuMbEr::symb, and TestMatch().

Referenced by Deferred(), DoOnePow(), generate_expression(), Generator(), InFunction(), MakeDollarInteger(), MakeDollarMod(), PF_CollectModifiedDollars(), PF_Deferred(), PF_Processor(), poly_factorize_expression(), poly_unfactorize_expression(), Processor(), and TestMatch().

Here is the call graph for this function:



4.19.4.109 GetBinom()

```
WORD GetBinom (
            UWORD * a,
            WORD * na,
            WORD i1,
            WORD i2 )  [extern]
```

Definition at line 2667 of file reken.c.

4.19.4.110 GetFromStore()

```
WORD GetFromStore (
            WORD * to,
            POSITION * position,
            RENUMBER renumber,
            WORD * InCompState,
            WORD nexpr )  [extern]
```

Definition at line 1629 of file store.c.

4.19.4.111 GetLong()

```
WORD GetLong (
            UBYTE * s,
            UWORD * a,
            WORD * na ) [extern]
```

Definition at line 1774 of file reken.c.

4.19.4.112 GetMoreTerms()

```
WORD GetMoreTerms (
            WORD * term ) [extern]
```

Definition at line 1426 of file store.c.

4.19.4.113 GetMoreFromMem()

```
WORD GetMoreFromMem (
            WORD * term,
            WORD ** tpoin ) [extern]
```

Definition at line 1522 of file store.c.

4.19.4.114 GetOneTerm()

```
WORD GetOneTerm (
            PHEAD WORD * term,
            FILEHANDLE * fi,
            POSITION * pos,
            int par ) [extern]
```

Definition at line 1268 of file store.c.

4.19.4.115 GetTable()

```
RENUMBER GetTable (
            WORD expr,
            POSITION * position,
            WORD mode ) [extern]
```

Definition at line 2826 of file store.c.

4.19.4.116 GetTerm()

```
WORD GetTerm (
            PHEAD WORD * term ) [extern]
```

Definition at line 993 of file store.c.

4.19.4.117 Glue()

```
WORD Glue (
            PHEAD WORD * term1,
            WORD * term2,
            WORD * sub,
            WORD insert )  [extern]
```

Definition at line 461 of file ratio.c.

4.19.4.118 InFunction()

```
WORD InFunction (
            PHEAD WORD * term,
            WORD * termout )  [extern]
```

Makes the replacement of the subexpression with the number 'replac' in a function argument. Additional information is passed in some of the AR, AN, AT variables.

Parameters

term	The input term
termout	The output term

Returns

0: everything is fine, Negative: fatal, Positive: error.

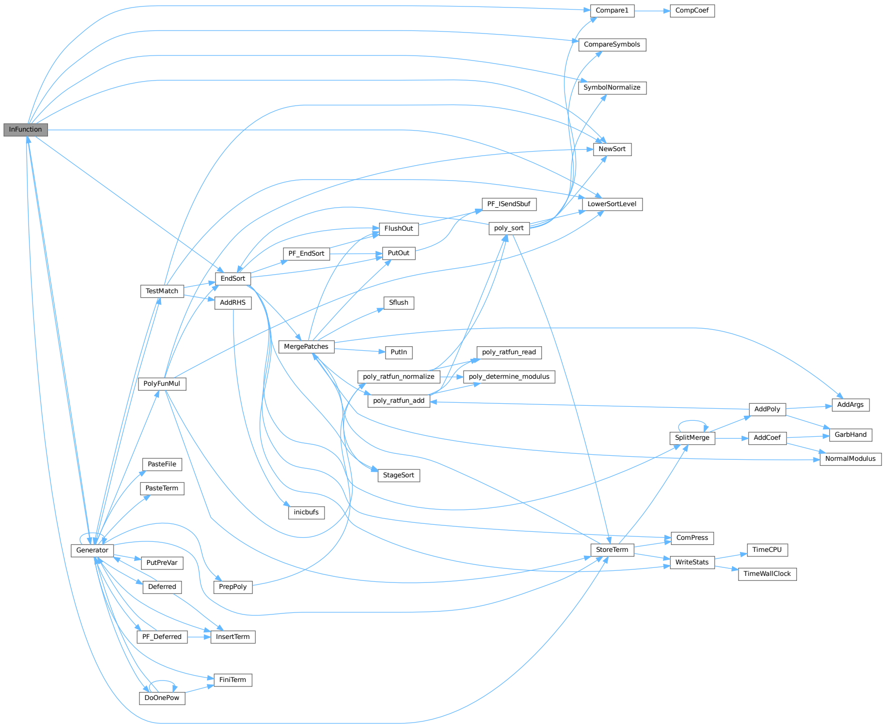
Special attention should be given to nested functions!

Definition at line 2174 of file proces.c.

References Compare1(), CompareSymbols(), EndSort(), Generator(), LowerSortLevel(), NewSort(), StoreTerm(), and SymbolNormalize().

Referenced by Generator().

Here is the call graph for this function:



4.19.4.119 IniLine()

```
void IniLine (
            WORD extrablank )  [extern]
```

Definition at line 267 of file sch.c.

4.19.4.120 IniVars()

```
WORD IniVars (
            void  )  [extern]
```

Definition at line 1354 of file startup.c.

4.19.4.121 InsertTerm()

```
WORD InsertTerm (
            PHEAD WORD * term,
            WORD replac,
            WORD extractbuff,
            WORD * position,
            WORD * termout,
            WORD tepos )  [extern]
```

Puts the terms 'term' and 'position' together into a single legal term in termout. replac is the number of the subexpression that should be replaced. It must be a positive term. When action is needed in the argument of a function all terms in that argument are dealt with recursively. The subexpression is sorted. Only one subexpression is done at a time this way.

Parameters

term	the input term
replac	number of the subexpression pointer to replace
extractbuff	number of the compiler buffer replac refers to
position	position from where to take the term in the compiler buffer
termout	the output term
tepos	offset in term where the subexpression is.

Returns

Normal conventions (OK = 0).

Definition at line 2744 of file proces.c.

Referenced by Deferred(), Generator(), and PF_Deferred().

4.19.4.122 LongToLine()

```
void LongToLine (
            UWORD * a,
            WORD na )  [extern]
```

Definition at line 303 of file sch.c.

4.19.4.123 MakeDirty()

```
WORD MakeDirty (
            WORD * term,
            WORD * x,
            WORD par )  [extern]
```

Definition at line 51 of file function.c.

4.19.4.124 MarkDirty()

```
void MarkDirty (
            WORD * term,
            WORD flags )  [extern]
```

Definition at line 97 of file function.c.

4.19.4.125 PolyFunDirty()

```
void PolyFunDirty (
            PHEAD WORD * term )  [extern]
```

Definition at line 139 of file function.c.

4.19.4.126 PolyFunClean()

```
void PolyFunClean (
            PHEAD WORD * term )  [extern]
```

Definition at line 174 of file function.c.

4.19.4.127 MakeModTable()

```
WORD MakeModTable (
            void  )  [extern]
```

Definition at line 3363 of file reken.c.

4.19.4.128 MatchE()

```
WORD MatchE (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )  [extern]
```

Definition at line 56 of file symmetr.c.

4.19.4.129 MatchCy()

```
int MatchCy (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )  [extern]
```

Definition at line 570 of file symmetr.c.

4.19.4.130 FunMatchCy()

```
int FunMatchCy (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )  [extern]
```

Definition at line 1066 of file symmetr.c.

4.19.4.131 FunMatchSy()

```
int FunMatchSy (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )  [extern]
```

Definition at line 1524 of file symmetr.c.

4.19.4.132 MatchArgument()

```
int MatchArgument (
            PHEAD WORD * arg,
            WORD * pat )  [extern]
```

Definition at line 2051 of file symmetr.c.

4.19.4.133 MatchFunction()

```
WORD MatchFunction (
            PHEAD WORD * pattern,
            WORD * interm,
            WORD * wilds )  [extern]
```

Definition at line 826 of file function.c.

4.19.4.134 MergePatches()

```
WORD MergePatches (
            WORD par )  [extern]
```

The general merge routine. Can be used for the large buffer and the file merging. The array S->Patches tells where the patches start S->pStop tells where they end (has to be computed first). The end of a 'line to be merged' is indicated by a zero. If the end is reached without running into a zero or a term runs over the boundary of a patch it is a file merging operation and a new piece from the file is read in.

Parameters

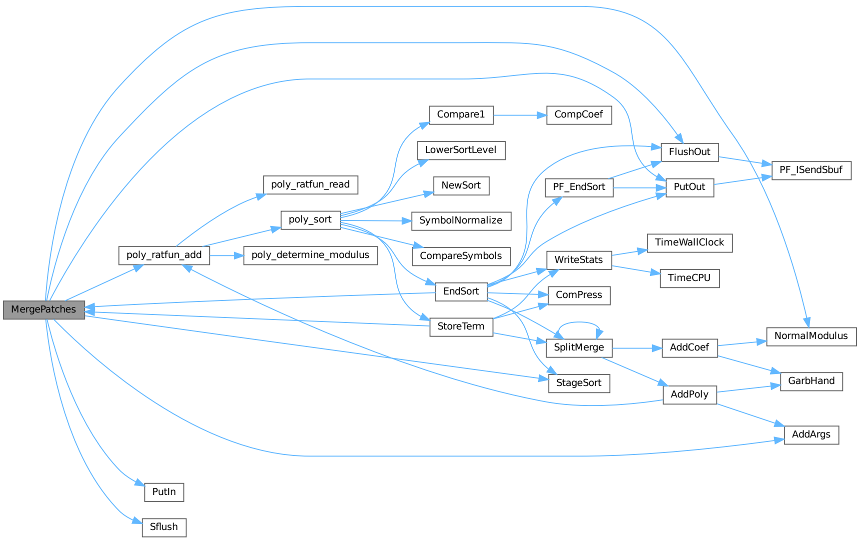
par	If par == 0 the sort is for file -> outputfile. If par == 1 the sort is for large buffer -> sortfile. If par == 2 the sort is for large buffer -> outputfile.

Definition at line 3757 of file sort.c.

References AddArgs(), FlushOut(), FiLe::handle, NormalModulus(), poly_ratfun_add(), PutIn(), PutOut(), Sflush(), and StageSort().

Referenced by EndSort(), and StoreTerm().

Here is the call graph for this function:



4.19.4.135 MesCerr()

```
WORD MesCerr (
            char * s,
            UBYTE * t )  [extern]
```

Definition at line 824 of file message.c.

4.19.4.136 MesComp()

```
WORD MesComp (
            char * s,
            UBYTE * p,
            UBYTE * q )  [extern]
```

Definition at line 843 of file message.c.

4.19.4.137 Modulus()

```
WORD Modulus (
            WORD * term )  [extern]
```

Definition at line 3102 of file reken.c.

4.19.4.138 MoveDummies()

```
void MoveDummies (
            PHEAD WORD * term,
            WORD shift )  [extern]
```

Definition at line 436 of file reshuf.c.

4.19.4.139 MulLong()

```
WORD MulLong (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )  [extern]
```

Definition at line 841 of file reken.c.

4.19.4.140 MulRat()

```
WORD MulRat (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )  [extern]
```

Definition at line 378 of file reken.c.

4.19.4.141 Mully()

```
WORD Mully (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb )  [extern]
```

Definition at line 126 of file reken.c.

4.19.4.142 MultDo()

```
WORD MultDo (
            PHEAD WORD * term,
            WORD * pattern )  [extern]
```

Definition at line 1044 of file reshuf.c.

4.19.4.143 NewSort()

```
WORD NewSort (
            PHEAD0  )  [extern]
```

Starts a new sort. At the lowest level this is a 'main sort' with the struct according to the parameters in S0. At higher levels this is a sort for functions, subroutines or dollars. We prepare the arrays and structs.

Returns

Regular convention (OK -> 0)

Definition at line 643 of file sort.c.

Referenced by generate_expression(), get_expression(), InFunction(), MakeDollarInteger(), MakeDollarMod(), PF_CollectModifiedDollars(), PF_Processor(), poly_factorize_expression(), poly_sort(), poly_unfactorize_expression(), PolyFunMul(), Processor(), TakeArgContent(), and TestMatch().

4.19.4.144 ExtraSymbol()

```
WORD ExtraSymbol (
            WORD sym,
            WORD pow,
            WORD nsym,
            WORD * ppsym,
            WORD * ncoef ) [extern]
```

Definition at line 4196 of file normal.c.

4.19.4.145 Normalize()

```
WORD Normalize (
            PHEAD WORD * term ) [extern]
```

Definition at line 193 of file normal.c.

4.19.4.146 BracketNormalize()

```
WORD BracketNormalize (
            PHEAD WORD * term ) [extern]
```

Definition at line 5271 of file normal.c.

4.19.4.147 DropCoefficient()

```
void DropCoefficient (
            PHEAD WORD * term ) [extern]
```

Definition at line 5096 of file normal.c.

4.19.4.148 DropSymbols()

```
void DropSymbols (
            PHEAD WORD * term ) [extern]
```

Definition at line 5114 of file normal.c.

4.19.4.149 PutInside()

```
int PutInside (
            PHEAD WORD * term,
            WORD * code ) [extern]
```

Definition at line 609 of file index.c.

4.19.4.150 OpenTemp()

```
WORD OpenTemp (
            void ) [extern]
```

Definition at line 48 of file store.c.

4.19.4.151 Pack()

```
void Pack (
            UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb ) [extern]
```

Definition at line 62 of file reken.c.

4.19.4.152 PasteFile()

```
LONG PasteFile (
            PHEAD WORD number,
            WORD * accum,
            POSITION * position,
            WORD ** accfill,
            RENUMBER renumber,
            WORD * freeze,
            WORD nexpr ) [extern]
```

Gets a term from stored expression expr and puts it in the accumulator at position number. It returns the length of the term that came from file.

Parameters

number	number of partial terms to skip in accum
accum	the accumulator
position	file position from where to get the stored term
accfill	returns tail position in accum
renumber	the renumber struct for the variables in the stored expression
freeze	information about if we need only the contents of a bracket
nexpr	the number of the stored expression

Returns

Normal conventions (OK = 0).

Definition at line 2880 of file proces.c.

Referenced by Generator().

4.19.4.153 Permute()

```
WORD Permute (
            PERM * perm,
            WORD first ) [extern]
```

Definition at line 443 of file symmetr.c.

4.19.4.154 PermuteP()

```
WORD PermuteP (
            PERMP * perm,
            WORD first ) [extern]
```

Definition at line 477 of file symmetr.c.

4.19.4.155 PolyFunMul()

```
WORD PolyFunMul (
            PHEAD WORD * term ) [extern]
```

Multiplies the arguments of multiple occurrences of the polyfun. In this routine we do the original PolyFun with one argument only. The PolyRatFun (PolyFunType = 2) is done in a dedicated routine in the file polywrap.cc The new result is written over the old result.

Parameters

term	It contains the input term and later the output.

Returns

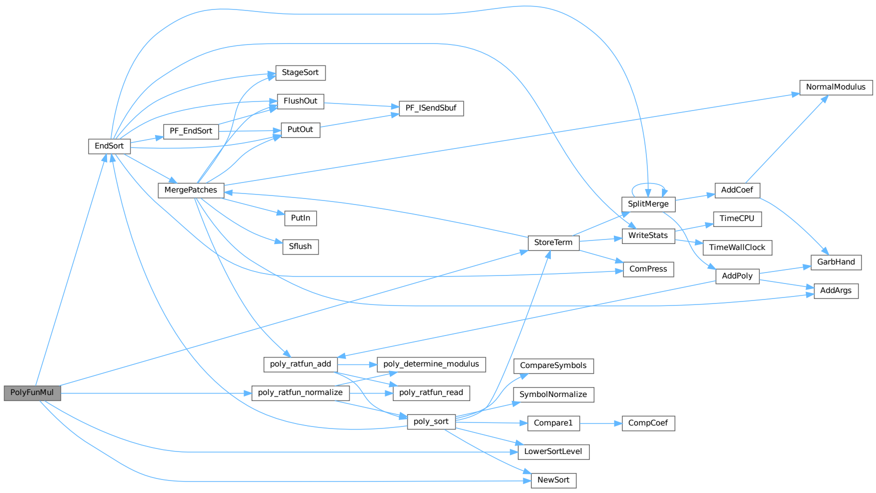
Normal conventions (OK = 0).

Definition at line 5360 of file proces.c.

References EndSort(), LowerSortLevel(), NewSort(), poly_ratfun_normalize(), and StoreTerm().

Referenced by Generator().

Here is the call graph for this function:



4.19.4.156 PopVariables()

```
WORD PopVariables (
            void ) [extern]
```

Definition at line 211 of file execute.c.

4.19.4.157 PrepPoly()

```
WORD PrepPoly (
            PHEAD WORD * term,
            WORD par ) [extern]
```

Routine checks whether the count of function AR.PolyFun is zero or one. If it is one and it has one scalarlike argument the coefficient of the term is pulled inside the argument. If the count is zero a new function is made with the coefficient as its only argument. The function should be placed at its proper position.

When this function is active it places the PolyFun as last object before the coefficient. This is needed because otherwise the compress algorithm has problems in MergePatches.

The bracket routine should also place the PolyFun at a comparable spot. The compression should then stop at the PolyFun. It doesn't really have to stop when writing the final result but this may be too complicated.

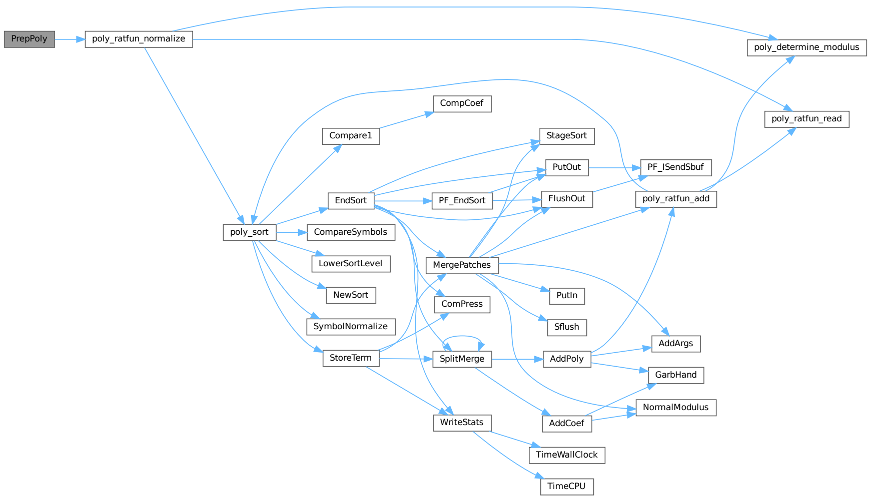
The parameter par tells whether we are at groundlevel or inside a function or dollar variable.

Definition at line 4972 of file proces.c.

References poly_ratfun_normalize().

Referenced by Generator().

Here is the call graph for this function:



4.19.4.158 Processor()

```
WORD Processor (
            void ) [extern]
```

This is the central processor. It accepts a stream of Expressions which is accessed by calls to GetTerm. The expressions reside either in AR.infile or AR.hidefile The definitions of an expression are seen as an id-statement, so the primary Expressions should be written to the system of scratch files as single terms with an expression pointer. Each expression is terminated with a zero and the whole is terminated by two zeroes.

The routine DoExecute should determine whether results are to be printed, should revert the scratch I/O directions etc. In principle it is DoExecute that calls Processor.

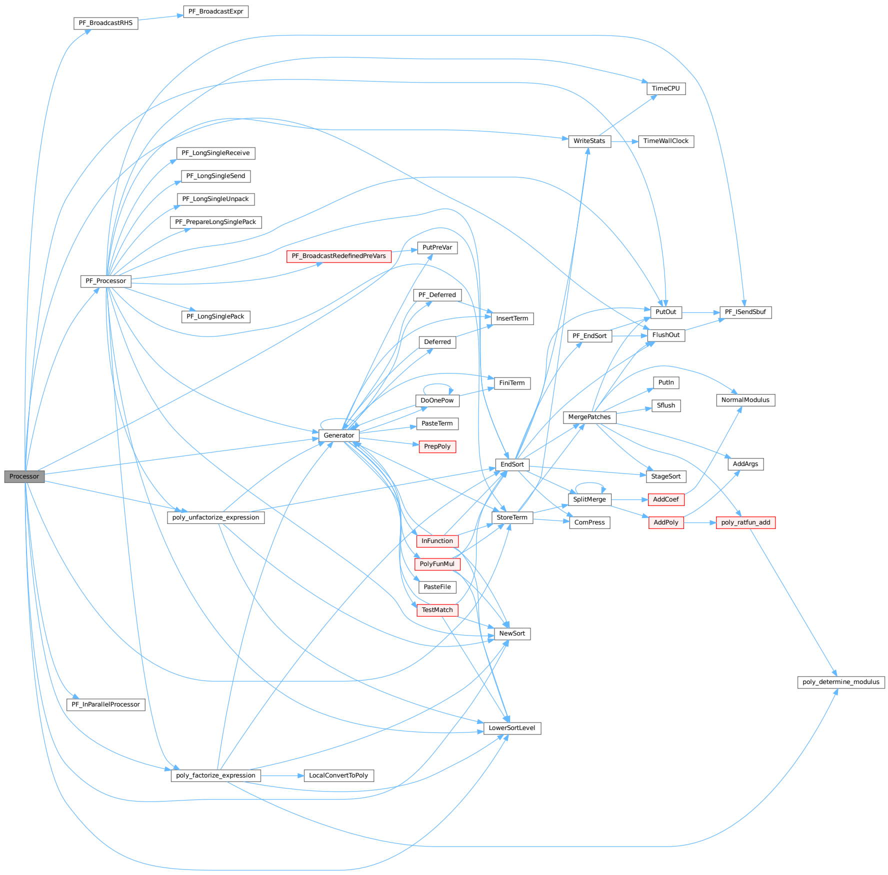
Returns

if everything OK: 0. Otherwise error. The preprocessor may continue with compilation though. Really fatal errors should return on the spot by calling Terminate.

Definition at line 64 of file proces.c.

References CbUf::Buffer, EndSort(), FlushOut(), Generator(), CbUf::lhs, LowerSortLevel(), NewSort(), PF_BroadcastRHS(), PF_InParallelProcessor(), PF_Processor(), CbUf::Pointer, poly_factorize_expression(), poly_unfactorize_expression(), PutOut(), CbUf::rhs, and StoreTerm().

Here is the call graph for this function:



4.19.4.159 Product()

```
WORD Product (
            UWORD * a,
            WORD * na,
            WORD b )  [extern]
```

Definition at line 1585 of file reken.c.

4.19.4.160 PrtLong()

```
void PrtLong (
            UWORD * a,
            WORD na,
            UBYTE * s )  [extern]
```

Definition at line 1712 of file reken.c.

4.19.4.161 PrtTerms()

```
void PrtTerms (
            void )  [extern]
```

Definition at line 716 of file sch.c.

4.19.4.162 PrintDeprecation()

```
void PrintDeprecation (
            const char * feature,
            const char * issue )  [extern]
```

Prints a deprecation warning for a given feature.

Parameters

feature	The name of the deprecated feature.
issue	The associated issue, e.g., "issues/700".

Definition at line 2063 of file startup.c.

4.19.4.163 PrintRunningTime()

```
void PrintRunningTime (
            void )  [extern]
```

Definition at line 2087 of file startup.c.

4.19.4.164 GetRunningTime()

```
LONG GetRunningTime (
            void )  [extern]
```

Definition at line 2128 of file startup.c.

4.19.4.165 PutBracket()

```
WORD PutBracket (
            PHEAD WORD * termin )  [extern]
```

Definition at line 1111 of file execute.c.

4.19.4.166 PutIn()

```
LONG PutIn (
            FILEHANDLE * file,
            POSITION * position,
            WORD * buffer,
            WORD ** take,
            int npat )  [extern]
```

Reads a new patch from position in file handle. It is put at buffer, anything after take is moved forward. This would be part of a term that hasn't been used yet. Because of this there should be some space before the start of the buffer

Parameters

file	The file system from which to read
position	The position from which to read
buffer	The buffer into which to read
take	The unused tail should be moved before the buffer
npat	The number of the patch. Is needed if the information was compressed with gzip, because each patch has its own independent gzip encoding.

Definition at line 1320 of file sort.c.

References FiLe::handle.

Referenced by MergePatches().

4.19.4.167 PutInStore()

```
WORD PutInStore (
            INDEXENTRY * ind,
            WORD num ) [extern]
```

Definition at line 874 of file store.c.

4.19.4.168 PutOut()

```
WORD PutOut (
            PHEAD WORD * term,
            POSITION * position,
            FILEHANDLE * fi,
            WORD ncomp ) [extern]
```

Routine writes one term to file handle at position. It returns the new value of the position.

NOTE: For 'final output' we may have to index the brackets. See the struct BRACKETINDEX. We should maintain: 1: a list with brackets array with the brackets 2: a list of objects of type BRACKETINDEX. It contains array with either pointers or offsets to the list of brackets. starting positions in the file. The index may be tied to a maximum size. In that case we may have to prune the list occasionally.

Parameters

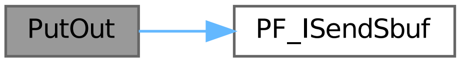
term	The term to be written
position	The position in the file. Afterwards it is updated
fi	The file (or its cache) to which should be written
ncomp	Information about what type of compression should be used

Definition at line 1466 of file sort.c.

References FiLe::handle, and PF_ISendSbuf().

Referenced by EndSort(), generate_expression(), MergePatches(), PF_EndSort(), PF_Processor(), and Processor().

Here is the call graph for this function:



4.19.4.169 Quotient()

```
UWORD Quotient (
            UWORD * a,
            WORD * na,
            WORD b )  [extern]
```

Definition at line 1620 of file reken.c.

4.19.4.170 RaisPow()

```
WORD RaisPow (
            PHEAD UWORD * a,
            WORD * na,
            UWORD b )  [extern]
```

Definition at line 1228 of file reken.c.

4.19.4.171 RaisPowCached()

```
void RaisPowCached (
            PHEAD WORD x,
            WORD n,
            UWORD ** c,
            WORD * nc )  [extern]
```

Computes power x^n and caches the value

4.19.5 Description

Calculates the power x^n and stores the results for caching purposes. The pointer c (i.e., the pointer, and not what it points to) is overwritten. What it points to should not be overwritten in the calling function.

4.19.6 Notes

- Caching is done in AT.small_power[]. This array is extended if necessary.

Definition at line 1296 of file reken.c.

Referenced by poly::divmod_heap(), poly::divmod_univar(), and poly::mul_heap().

4.19.6.1 RaisPowMod()

```
WORD RaisPowMod (
            WORD x,
            WORD n,
            WORD m )  [extern]
```

Definition at line 1383 of file reken.c.

4.19.6.2 NormalModulus()

```
int NormalModulus (
            UWORD * a,
            WORD * na )  [extern]
```

Brings a modular representation in the range -p/2 to +p/2 The return value tells whether anything was done. Routine made in the general modulus revamp of July 2008 (JV).

Definition at line 1403 of file reken.c.

Referenced by AddCoef(), and MergePatches().

4.19.6.3 MakeInverses()

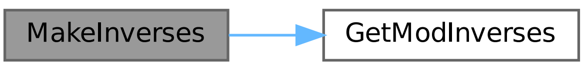
```
int MakeInverses (
            void )  [extern]
```

Makes a table of inverses in modular calculus The modulus is in AC.cmod and AC.ncmod One should notice that the table of inverses can only be made if the modulus fits inside a single FORM word. Otherwise the table lookup becomes too difficult and the table too long.

Definition at line 1440 of file reken.c.

References GetModInverses().

Here is the call graph for this function:



4.19.6.4 GetModInverses()

```
int GetModInverses (
            WORD m1,
            WORD m2,
            WORD * im1,
            WORD * im2 )  [extern]
```

Input m1 and m2, which are relative prime. determines a∗m1+b∗m2 = 1 (and 1 is the gcd of m1 and m2) then a∗m1 = 1 mod m2 and hence im1 = a. and b∗m2 = 1 mod m1 and hence im2 = b. Set m1 = 0∗m1+1∗m2 = a1∗m1+b1∗m2 m2 = 1∗m1+0∗m2 = a2∗m1+b2∗m2 If everything is OK, the return value is zero

Definition at line 1476 of file reken.c.

Referenced by poly::divmod_heap(), poly::divmod_univar(), MakeDollarMod(), MakeInverses(), MakeMod(), TakeContent(), and TakeSymbolContent().

4.19.6.5 GetLongModInverses()

```
int GetLongModInverses (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * ia,
            WORD * nia,
            UWORD * ib,
            WORD * nib )  [extern]
```

Definition at line 1508 of file reken.c.

4.19.6.6 RatToLine()

```
void RatToLine (
            UWORD * a,
            WORD na )  [extern]
```

Definition at line 337 of file sch.c.

4.19.6.7 RatioFind()

```
WORD RatioFind (
            PHEAD WORD * term,
            WORD * params )  [extern]
```

Definition at line 62 of file ratio.c.

4.19.6.8 RatioGen()

```
WORD RatioGen (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )  [extern]
```

Definition at line 170 of file ratio.c.

4.19.6.9 ReNumber()

```
WORD ReNumber (
            PHEAD WORD * term )  [extern]
```

Definition at line 80 of file reshuf.c.

4.19.6.10 ReadSnum()

```
WORD ReadSnum (
            UBYTE ** p )  [extern]
```

Definition at line 1973 of file tools.c.

4.19.6.11 Remain10()

```
WORD Remain10 (
            UWORD * a,
            WORD * na )  [extern]
```

Definition at line 1659 of file reken.c.

4.19.6.12 Remain4()

```
WORD Remain4 (
            UWORD * a,
            WORD * na )  [extern]
```

Definition at line 1686 of file reken.c.

4.19.6.13 ResetScratch()

```
WORD ResetScratch (
            void  )  [extern]
```

Definition at line 235 of file store.c.

4.19.6.14 ResolveSet()

```
WORD ResolveSet (
            PHEAD WORD * from,
            WORD * to,
            WORD * subs )  [extern]
```

Definition at line 1361 of file wildcard.c.

4.19.6.15 RevertScratch()

```
WORD RevertScratch (
            void )  [extern]
```

Definition at line 190 of file store.c.

4.19.6.16 ScanFunctions()

```
WORD ScanFunctions (
            PHEAD WORD * inpat,
            WORD * inter,
            WORD par )  [extern]
```

Definition at line 1616 of file function.c.

4.19.6.17 SeekScratch()

```
void SeekScratch (
            FILEHANDLE * fi,
            POSITION * pos )  [extern]
```

Definition at line 64 of file store.c.

4.19.6.18 SetEndScratch()

```
void SetEndScratch (
            FILEHANDLE * f,
            POSITION * position )  [extern]
```

Definition at line 75 of file store.c.

4.19.6.19 SetEndHScratch()

```
void SetEndHScratch (
            FILEHANDLE * f,
            POSITION * position )  [extern]
```

Definition at line 89 of file store.c.

4.19.6.20 SetFileIndex()

```
WORD SetFileIndex (
            void ) [extern]
```

Reads the next file index and puts it into AR.StoreData.Index. TODO

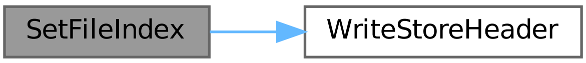
Returns

= 0 everything okay, != 0 an error occurred

Definition at line 2321 of file store.c.

References WriteStoreHeader().

Here is the call graph for this function:



4.19.6.21 Sflush()

```
WORD Sflush (
            FILEHANDLE * fi ) [extern]
```

Puts the contents of a buffer to output Only to be used when there is a single patch in the large buffer.

Parameters

fi	The filesystem (or its cache) to which the patch should be written

Definition at line 1380 of file sort.c.

References FiLe::handle.

Referenced by MergePatches().

4.19.6.22 Simplify()

```
WORD Simplify (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD * nb ) [extern]
```

Definition at line 530 of file reken.c.

4.19.6.23 SortWild()

```
WORD SortWild (
            WORD * w,
            WORD nw )  [extern]
```

Sorts the wildcard entries in the parameter w. Double entries are removed. Full space taken is nw words. Routine serves for the reading of wildcards in the compiler. The entries come in the format: (type,4,number,0) in which the zero is reserved for the future replacement of 'number'.

Parameters

w	buffer with wildcard entries.
nw	number of wildcard entries.

Returns

Normal conventions (OK -> 0)

Definition at line 4753 of file sort.c.

4.19.6.24 LocateBase()

```
FILE * LocateBase (
            char ** name,
            char ** newname,
            char * iomode )  [extern]
```

Definition at line 225 of file minos.c.

4.19.6.25 SplitMerge()

```
LONG SplitMerge (
            PHEAD WORD ** Pointer,
            LONG number )  [extern]
```

Algorithm by J.A.M.Vermaseren (31-7-1988)

Note that AN.SplitScratch and AN.InScratch are used also in GarbHand

Merge sort in memory. The input is an array of pointers. Sorting is done recursively by dividing the array in two equal parts and calling SplitMerge for each. When the parts are small enough we can do the compare and take the appropriate action. An addition is that we look for 'runs'. Sequences that are already ordered. This happens a lot when there is very little action in a module. This made FORM faster by a few percent.

Parameters

Pointer	The array of pointers to the terms to be sorted.
number	The number of pointers in Pointer.

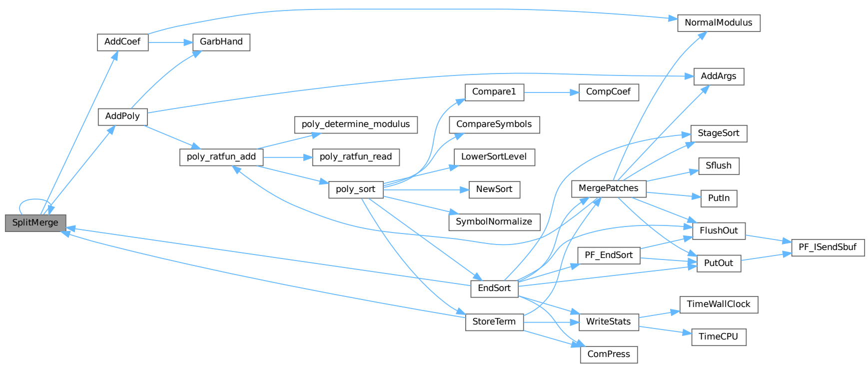
The terms are supposed to be sitting in the small buffer and there is supposed to be an extension to this buffer for when there are two terms that should be added and the result takes more space than each of the original terms. The notation guarantees that the result never needs more space than the sum of the spaces of the original terms.

Definition at line 3362 of file sort.c.

References AddCoef(), AddPoly(), and SplitMerge().

Referenced by EndSort(), SplitMerge(), and StoreTerm().

Here is the call graph for this function:



4.19.6.26 StoreTerm()

```
WORD StoreTerm (
            PHEAD WORD * term )  [extern]
```

The central routine to accept terms, store them and keep things at least partially sorted. A call to EndSort will then complete storing and sorting.

Parameters

term	The term to be stored

Returns

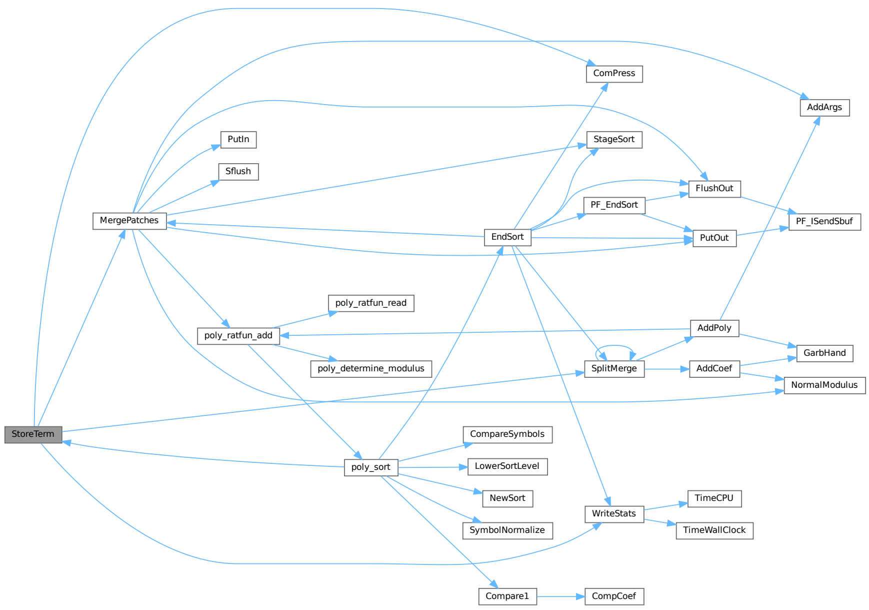
Regular return conventions (OK -> 0)

Definition at line 4540 of file sort.c.

References ComPress(), MergePatches(), SplitMerge(), and WriteStats().

Referenced by Generator(), get_expression(), InFunction(), PF_Processor(), poly_sort(), PolyFunMul(), Processor(), and TakeArgContent().

Here is the call graph for this function:



4.19.6.27 SubPLon()

```
void SubPLon (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )   [extern]
```

Definition at line 803 of file reken.c.

4.19.6.28 Substitute()

```
void Substitute (
            PHEAD WORD * term,
            WORD * pattern,
            WORD power )   [extern]
```

Definition at line 640 of file pattern.c.

4.19.6.29 SymFind()

```
WORD SymFind (
            PHEAD WORD * term,
            WORD * params )   [extern]
```

Definition at line 633 of file function.c.

4.19.6.30 SymGen()

```
WORD SymGen (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level ) [extern]
```

Definition at line 518 of file function.c.

4.19.6.31 Symmetrize()

```
WORD Symmetrize (
            PHEAD WORD * func,
            WORD * Lijst,
            WORD ngroups,
            WORD gsize,
            WORD type ) [extern]
```

Definition at line 213 of file function.c.

4.19.6.32 FullSymmetrize()

```
int FullSymmetrize (
            PHEAD WORD * fun,
            int type ) [extern]
```

Definition at line 473 of file function.c.

4.19.6.33 TakeModulus()

```
WORD TakeModulus (
            UWORD * a,
            WORD * na,
            UWORD * cmodvec,
            WORD ncmod,
            WORD par ) [extern]
```

Definition at line 3149 of file reken.c.

4.19.6.34 TakeNormalModulus()

```
WORD TakeNormalModulus (
            UWORD * a,
            WORD * na,
            UWORD * c,
            WORD nc,
            WORD par ) [extern]
```

Definition at line 3321 of file reken.c.

4.19.6.35 TalToLine()

```
void TalToLine (
            UWORD x )  [extern]
```

Definition at line 522 of file sch.c.

4.19.6.36 TenVec()

```
WORD TenVec (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )  [extern]
```

Definition at line 2312 of file opera.c.

4.19.6.37 TenVecFind()

```
WORD TenVecFind (
            PHEAD WORD * term,
            WORD * params )  [extern]
```

Definition at line 2178 of file opera.c.

4.19.6.38 TermRenumber()

```
WORD TermRenumber (
            WORD * term,
            RENUMBER renumber,
            WORD nexpr )  [extern]
```

!! WORD ∗memterm=term; static LONG ctrap=0; !!!

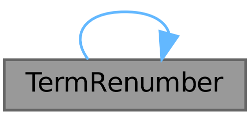
!! ctrap++; !!!

Definition at line 2428 of file store.c.

References ReNuMbEr::func, ReNuMbEr::funnum, ReNuMbEr::indi, ReNuMbEr::indnum, ReNuMbEr::symb, ReNuMbEr::symnum, TermRenumber(), ReNuMbEr::vecnum, and ReNuMbEr::vect.

Referenced by TermRenumber().

Here is the call graph for this function:



4.19.6.39 TestDrop()

```
void TestDrop (
            void ) [extern]
```

Definition at line 483 of file execute.c.

4.19.6.40 PutInVflags()

```
void PutInVflags (
            WORD nexpr ) [extern]
```

Definition at line 561 of file execute.c.

4.19.6.41 TestMatch()

```
WORD TestMatch (
            PHEAD WORD * term,
            WORD * level ) [extern]
```

This routine governs the pattern matching. If it decides that a substitution should be made, this can be either the insertion of a right hand side (C->rhs) or the automatic generation of terms as a result of an operation (like trace). The object to be replaced is removed from term and a subexpression pointer is inserted. If the substitution is made more than once there can be more subexpression pointers. Its number is positive as it corresponds to the level at which the C->rhs can be found in the compiler output. The subexpression pointer contains the wildcard substitution information. The power is found in ∗AT.TMout. For operations the subexpression pointer is negative and corresponds to an address in the array AT.TMout. In this array are then the instructions for the routine to be called and its number in the array 'Operations' The format is here: length,functionnumber,length-2 parameters

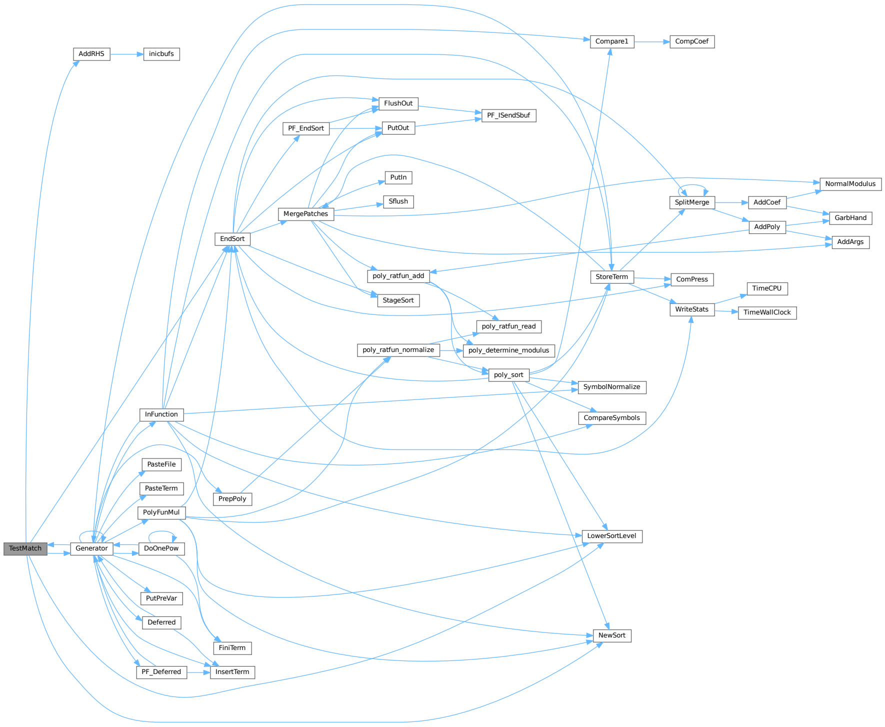
There is a certain complexity wrt repeat levels. Another complication is the poking of the wildcard values in the subexpression prototype in the compiler buffer. This was how things were done in the past with sequential FORM, but with the advent of TFORM this cannot be maintained. Now, for TFORM we make a copy of it. 7-may-2008 (JV): We cannot yet guarantee that this has been done 100% correctly. There are errors that occur in TFORM only and that may indicate problems.

Definition at line 97 of file pattern.c.

References AddRHS(), EndSort(), Generator(), CbUf::lhs, LowerSortLevel(), and NewSort().

Referenced by Generator().

Here is the call graph for this function:



4.19.6.42 TestSub()

```
WORD TestSub (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 684 of file proces.c.

4.19.6.43 TimeCPU()

```
LONG TimeCPU (
            WORD par )  [extern]
```

Returns the CPU time.

Parameters

par	If zero, the CPU time will be reset to 0.

Returns

The CPU time in milliseconds.

Definition at line 3551 of file tools.c.

Referenced by PF_GetSlaveTimes(), PF_Processor(), and WriteStats().

4.19.6.44 TimeChildren()

```
LONG TimeChildren (
            WORD par )  [extern]
```

Definition at line 3533 of file tools.c.

4.19.6.45 TimeWallClock()

```
LONG TimeWallClock (
            WORD par )  [extern]
```

Returns the wall-clock time.

Parameters

par	If zero, the wall-clock time will be reset to 0.

Returns

The wall-clock time in centiseconds.

Definition at line 3477 of file tools.c.

Referenced by DoCheckpoint(), DoRecovery(), find_Horner_MCTS(), optimize_expression_given_Horner(), optimize_greedy(), and WriteStats().

4.19.6.46 ToStorage()

```
WORD ToStorage (
            EXPRESSIONS e,
            POSITION * length )  [extern]
```

Definition at line 2017 of file store.c.

4.19.6.47 TokenToLine()

```
void TokenToLine (
            UBYTE * s )  [extern]
```

Definition at line 551 of file sch.c.

4.19.6.48 Trace4()

```
WORD Trace4 (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )  [extern]
```

Definition at line 694 of file opera.c.

4.19.6.49 Trace4Gen()

```
WORD Trace4Gen (
            PHEAD TRACES * t,
            WORD number ) [extern]
```

Definition at line 856 of file opera.c.

4.19.6.50 Trace4no()

```
WORD Trace4no (
            WORD number,
            WORD * kron,
            TRACES * t ) [extern]
```

Definition at line 418 of file opera.c.

4.19.6.51 TraceFind()

```
WORD TraceFind (
            PHEAD WORD * term,
            WORD * params ) [extern]
```

Definition at line 1853 of file opera.c.

4.19.6.52 TraceN()

```
WORD TraceN (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level ) [extern]
```

Definition at line 1382 of file opera.c.

4.19.6.53 TraceNgen()

```
WORD TraceNgen (
            PHEAD TRACES * t,
            WORD number ) [extern]
```

Definition at line 1446 of file opera.c.

4.19.6.54 TraceNno()

```
WORD TraceNno (
            WORD number,
            WORD * kron,
            TRACES * t ) [extern]
```

Definition at line 1341 of file opera.c.

4.19.6.55 Traces()

```
WORD Traces (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )  [extern]
```

Definition at line 1831 of file opera.c.

4.19.6.56 Trick()

```
WORD Trick (
            WORD * in,
            TRACES * t )  [extern]
```

Definition at line 331 of file opera.c.

4.19.6.57 TryDo()

```
WORD TryDo (
            PHEAD WORD * term,
            WORD * pattern,
            WORD level )  [extern]
```

Definition at line 1072 of file reshuf.c.

4.19.6.58 UnPack()

```
void UnPack (
            UWORD * a,
            WORD na,
            WORD * denom,
            WORD * numer )  [extern]
```

Definition at line 100 of file reken.c.

4.19.6.59 VarStore()

```
WORD VarStore (
            UBYTE * s,
            WORD n,
            WORD name,
            WORD namesize )  [extern]
```

Definition at line 2370 of file store.c.

4.19.6.60 WildFill()

```
WORD WildFill (
             PHEAD WORD * to,
             WORD * from,
             WORD * sub )  [extern]
```

Definition at line 65 of file wildcard.c.

4.19.6.61 WriteAll()

```
WORD WriteAll (
             void  )  [extern]
```

Definition at line 2501 of file sch.c.

4.19.6.62 WriteOne()

```
WORD WriteOne (
             UBYTE * name,
             int alreadyinline,
             int nosemi,
             WORD plus )  [extern]
```

Definition at line 2727 of file sch.c.

4.19.6.63 WriteArgument()

```
void WriteArgument (
             WORD * t )  [extern]
```

Definition at line 1424 of file sch.c.

4.19.6.64 WriteExpression()

```
WORD WriteExpression (
             WORD * terms,
             LONG ltot )  [extern]
```

Definition at line 2470 of file sch.c.

4.19.6.65 WriteInnerTerm()

```
WORD WriteInnerTerm (
             WORD * term,
             WORD first )  [extern]
```

Definition at line 1951 of file sch.c.

4.19.6.66 WriteLists()

```
void WriteLists (
            void ) [extern]
```

Definition at line 810 of file sch.c.

4.19.6.67 WriteSetup()

```
void WriteSetup (
            void ) [extern]
```

Definition at line 811 of file setfile.c.

4.19.6.68 WriteStats()

```
void WriteStats (
            POSITION * plspace,
            WORD par,
            WORD checkLogType ) [extern]
```

Writes the statistics.

Parameters

plspace	The size in bytes currently occupied
par	par = 0 = STATSSPLITMERGE after a splitmerge. par = 1 = STATSMERGETOFILE after merge to sortfile. par = 2 = STATSPOSTSORT after the sort
checkLogType	== CHECKLOGTYPE: The output should not go to the log file if AM.LogType is 1.

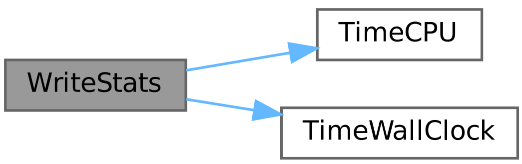
current expression is to be found in AR.CurExpr. terms are in S->TermsLeft. S->GenTerms.

Definition at line 121 of file sort.c.

References TimeCPU(), and TimeWallClock().

Referenced by EndSort(), PF_Processor(), and StoreTerm().

Here is the call graph for this function:



4.19.6.69 WriteSubTerm()

```
WORD WriteSubTerm (
            WORD * sterm,
            WORD first )  [extern]
```

Definition at line 1542 of file sch.c.

4.19.6.70 WriteTerm()

```
WORD WriteTerm (
            WORD * term,
            WORD * lbrac,
            WORD first,
            WORD prtf,
            WORD br )  [extern]
```

Definition at line 2148 of file sch.c.

4.19.6.71 execarg()

```
WORD execarg (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 56 of file argument.c.

4.19.6.72 execterm()

```
WORD execterm (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 1769 of file argument.c.

4.19.6.73 SpecialCleanup()

```
void SpecialCleanup (
            PHEAD0  )  [extern]
```

Definition at line 1623 of file execute.c.

4.19.6.74 SetMods()

```
void SetMods (
            void  )  [extern]
```

Definition at line 1637 of file execute.c.

4.19.6.75 UnSetMods()

```
void UnSetMods (
            void ) [extern]
```

Definition at line 1655 of file execute.c.

4.19.6.76 DoExecute()

```
WORD DoExecute (
            WORD par,
            WORD skip ) [extern]
```

Definition at line 615 of file execute.c.

4.19.6.77 SetScratch()

```
void SetScratch (
            FILEHANDLE * f,
            POSITION * position ) [extern]
```

Definition at line 115 of file store.c.

4.19.6.78 Warning()

```
void Warning (
            char * s ) [extern]
```

Definition at line 790 of file message.c.

4.19.6.79 HighWarning()

```
void HighWarning (
            char * s ) [extern]
```

Definition at line 802 of file message.c.

4.19.6.80 SpareTable()

```
int SpareTable (
            TABLES TT ) [extern]
```

Definition at line 694 of file tables.c.

4.19.6.81 strDup1()

```
UBYTE * strDup1 (
            UBYTE * instring,
            char * ifwrong )  [extern]
```

Definition at line 1904 of file tools.c.

4.19.6.82 Malloc()

```
void * Malloc (
            LONG size )  [extern]
```

Definition at line 2221 of file tools.c.

4.19.6.83 Malloc1()

```
void * Malloc1 (
            LONG size,
            const char * messageifwrong )  [extern]
```

Definition at line 2303 of file tools.c.

4.19.6.84 DoTail()

```
int DoTail (
            int argc,
            UBYTE ** argv )  [extern]
```

Definition at line 237 of file startup.c.

4.19.6.85 OpenInput()

```
int OpenInput (
            void  )  [extern]
```

Definition at line 542 of file startup.c.

4.19.6.86 PutPreVar()

```
int PutPreVar (
            UBYTE * name,
            UBYTE * value,
            UBYTE * args,
            int mode )  [extern]
```

Inserts/Updates a preprocessor variable in the name administration.

Parameters

name	Character string with the variable name.
value	Character string with a possible value. Special case: if this argument is zero, then we have no value. Note: This is different from having an empty argument! This should only occur when the name starts with a ?
args	Character string with possible arguments.
mode	=0: always create a new name entry, =1: try to do a redefinition if possible.

Returns

Index of used entry in name list.

Definition at line 718 of file pre.c.

Referenced by ClearOptimize(), Generator(), Optimize(), PF_BroadcastRedefinedPreVars(), StartVariables(), and TheDefine().

4.19.6.87 Error0()

```
void Error0 (
            char * s ) [extern]
```

Definition at line 56 of file message.c.

4.19.6.88 Error1()

```
void Error1 (
            char * s,
            UBYTE * t ) [extern]
```

Definition at line 67 of file message.c.

4.19.6.89 Error2()

```
void Error2 (
            char * s1,
            char * s2,
            UBYTE * t ) [extern]
```

Definition at line 78 of file message.c.

4.19.6.90 ReadFromStream()

```
UBYTE ReadFromStream (
            STREAM * stream ) [extern]
```

Definition at line 151 of file tools.c.

4.19.6.91 GetFromStream()

```
UBYTE GetFromStream (
            STREAM * stream )  [extern]
```

Definition at line 286 of file tools.c.

4.19.6.92 LookInStream()

```
UBYTE LookInStream (
            STREAM * stream )  [extern]
```

Definition at line 309 of file tools.c.

4.19.6.93 OpenStream()

```
STREAM * OpenStream (
            UBYTE * name,
            int type,
            int prevarmode,
            int raiselow )  [extern]
```

Definition at line 321 of file tools.c.

4.19.6.94 LocateFile()

```
int LocateFile (
            UBYTE ** name,
            int type )  [extern]
```

Definition at line 576 of file tools.c.

4.19.6.95 CloseStream()

```
STREAM * CloseStream (
            STREAM * stream )  [extern]
```

Definition at line 663 of file tools.c.

4.19.6.96 PositionStream()

```
void PositionStream (
            STREAM * stream,
            LONG position )  [extern]
```

Definition at line 823 of file tools.c.

4.19.6.97 ReverseStatements()

```
int ReverseStatements (
            STREAM * stream )  [extern]
```

Definition at line 855 of file tools.c.

4.19.6.98 ProcessOption()

```
int ProcessOption (
            UBYTE * s1,
            UBYTE * s2,
            int filetype )  [extern]
```

Definition at line 182 of file setfile.c.

4.19.6.99 DoSetups()

```
int DoSetups (
            void  )  [extern]
```

Definition at line 132 of file setfile.c.

4.19.6.100 TerminateImpl()

```
void TerminateImpl (
            int errorcode,
            const char * file,
            int line,
            const char * function )  [extern]
```

Definition at line 1850 of file startup.c.

4.19.6.101 GetNode()

```
NAMENODE * GetNode (
            NAMETREE * nametree,
            UBYTE * name )  [extern]
```

Definition at line 49 of file names.c.

4.19.6.102 AddName()

```
int AddName (
            NAMETREE * nametree,
            UBYTE * name,
            WORD type,
            WORD number,
            int * nodenum )  [extern]
```

Definition at line 71 of file names.c.

4.19.6.103 GetName()

```
int GetName (
            NAMETREE * nametree,
            UBYTE * namein,
            WORD * number,
            int par ) [extern]
```

Definition at line 252 of file names.c.

4.19.6.104 GetFunction()

```
UBYTE * GetFunction (
            UBYTE * s,
            WORD * funnum ) [extern]
```

Definition at line 342 of file names.c.

4.19.6.105 GetNumber()

```
UBYTE * GetNumber (
            UBYTE * s,
            WORD * num ) [extern]
```

Definition at line 388 of file names.c.

4.19.6.106 GetLastExprName()

```
int GetLastExprName (
            UBYTE * name,
            WORD * number ) [extern]
```

Definition at line 438 of file names.c.

4.19.6.107 GetAutoName()

```
int GetAutoName (
            UBYTE * name,
            WORD * number ) [extern]
```

Definition at line 479 of file names.c.

4.19.6.108 GetVar()

```
int GetVar (
            UBYTE * name,
            WORD * type,
            WORD * number,
            int wantedtype,
            int par ) [extern]
```

Definition at line 524 of file names.c.

4.19.6.109 MakeDubious()

```
int MakeDubious (
            NAMETREE * nametree,
            UBYTE * name,
            WORD * number ) [extern]
```

Definition at line 2693 of file names.c.

4.19.6.110 GetOName()

```
int GetOName (
            NAMETREE * nametree,
            UBYTE * name,
            WORD * number,
            int par ) [extern]
```

Definition at line 460 of file names.c.

4.19.6.111 DumpTree()

```
void DumpTree (
            NAMETREE * nametree ) [extern]
```

Definition at line 602 of file names.c.

4.19.6.112 DumpNode()

```
void DumpNode (
            NAMETREE * nametree,
            WORD node,
            WORD depth ) [extern]
```

Definition at line 615 of file names.c.

4.19.6.113 LinkTree()

```
void LinkTree (
            NAMETREE * tree,
            WORD offset,
            WORD numnodes ) [extern]
```

Definition at line 784 of file names.c.

4.19.6.114 CopyTree()

```
void CopyTree (
            NAMETREE * newtree,
            NAMETREE * oldtree,
            WORD node,
            WORD par ) [extern]
```

Definition at line 696 of file names.c.

4.19.6.115 CompactifyTree()

```
int CompactifyTree (
             NAMETREE * nametree,
             WORD par )  [extern]
```

Definition at line 634 of file names.c.

4.19.6.116 MakeNameTree()

```
NAMETREE * MakeNameTree (
             void  )  [extern]
```

Definition at line 815 of file names.c.

4.19.6.117 FreeNameTree()

```
void FreeNameTree (
             NAMETREE * n )  [extern]
```

Definition at line 834 of file names.c.

4.19.6.118 AddExpression()

```
int AddExpression (
             UBYTE * name,
             int x,
             int y )  [extern]
```

Definition at line 2744 of file names.c.

4.19.6.119 AddSymbol()

```
int AddSymbol (
             UBYTE * name,
             int minpow,
             int maxpow,
             int cplx,
             int dim )  [extern]
```

Definition at line 920 of file names.c.

4.19.6.120 AddDollar()

```
int AddDollar (
             UBYTE * name,
             WORD type,
             WORD * start,
             LONG size )  [extern]
```

Definition at line 2602 of file names.c.

4.19.6.121 ReplaceDollar()

```
int ReplaceDollar (
            WORD number,
            WORD newtype,
            WORD * newstart,
            LONG newsize ) [extern]
```

Definition at line 2646 of file names.c.

4.19.6.122 DollarRaiseLow()

```
int DollarRaiseLow (
            UBYTE * name,
            LONG value ) [extern]
```

Definition at line 2552 of file dollar.c.

4.19.6.123 AddVector()

```
int AddVector (
            UBYTE * name,
            int cplx,
            int dim ) [extern]
```

Definition at line 1244 of file names.c.

4.19.6.124 AddDubious()

```
int AddDubious (
            UBYTE * name ) [extern]
```

Definition at line 2679 of file names.c.

4.19.6.125 AddIndex()

```
int AddIndex (
            UBYTE * name,
            int dim,
            int dim4 ) [extern]
```

Definition at line 1088 of file names.c.

4.19.6.126 DoDimension()

```
UBYTE * DoDimension (
            UBYTE * s,
            int * dim,
            int * dim4 ) [extern]
```

Definition at line 1160 of file names.c.

4.19.6.127 AddFunction()

```
int AddFunction (
            UBYTE * name,
            int comm,
            int istensor,
            int cplx,
            int symprop,
            int dim,
            int argmax,
            int argmin ) [extern]
```

Definition at line 1326 of file names.c.

4.19.6.128 CoCommuteInSet()

```
int CoCommuteInSet (
            UBYTE * s ) [extern]
```

Definition at line 1356 of file names.c.

4.19.6.129 CoFunction()

```
int CoFunction (
            UBYTE * s,
            int comm,
            int istensor ) [extern]
```

Definition at line 1446 of file names.c.

4.19.6.130 TestName()

```
int TestName (
            UBYTE * name ) [extern]
```

Definition at line 3254 of file names.c.

4.19.6.131 AddSet()

```
int AddSet (
            UBYTE * name,
            WORD dim ) [extern]
```

Definition at line 2122 of file names.c.

4.19.6.132 DoElements()

```
int DoElements (
            UBYTE * s,
            SETS set,
            UBYTE * name ) [extern]
```

Definition at line 2155 of file names.c.

4.19.6.133 DoTempSet()

```
int DoTempSet (
            UBYTE * from,
            UBYTE * to )  [extern]
```

Definition at line 2480 of file names.c.

4.19.6.134 NameConflict()

```
int NameConflict (
            int type,
            UBYTE * name )  [extern]
```

Definition at line 2728 of file names.c.

4.19.6.135 OpenFile()

```
int OpenFile (
            char * name )  [extern]
```

Definition at line 983 of file tools.c.

4.19.6.136 OpenAddFile()

```
int OpenAddFile (
            char * name )  [extern]
```

Definition at line 1002 of file tools.c.

4.19.6.137 ReOpenFile()

```
int ReOpenFile (
            char * name )  [extern]
```

Definition at line 1023 of file tools.c.

4.19.6.138 CreateFile()

```
int CreateFile (
            char * name )  [extern]
```

Definition at line 1043 of file tools.c.

4.19.6.139 CreateLogFile()

```
int CreateLogFile (
            char * name )  [extern]
```

Definition at line 1060 of file tools.c.

4.19.6.140 CloseFile()

```
void CloseFile (
            int handle )  [extern]
```

Definition at line 1078 of file tools.c.

4.19.6.141 CopyFile()

```
int CopyFile (
            char * source,
            char * dest )  [extern]
```

Copies a file with name *source to a file named *dest. The involved files must not be open. Returns non-zero if an error occurred. Uses if possible the combined large and small sorting buffers as cache.

Definition at line 1101 of file tools.c.

Referenced by DoRecovery().

4.19.6.142 CreateHandle()

```
int CreateHandle (
            void )  [extern]
```

Definition at line 1162 of file tools.c.

4.19.6.143 ReadFile()

```
LONG ReadFile (
            int handle,
            UBYTE * buffer,
            LONG size )  [extern]
```

Definition at line 1217 of file tools.c.

4.19.6.144 ReadPosFile()

```
LONG ReadPosFile (
            PHEAD FILEHANDLE * fi,
            UBYTE * buffer,
            LONG size,
            POSITION * pos )  [extern]
```

Definition at line 1267 of file tools.c.

4.19.6.145 WriteFileToFile()

```
LONG WriteFileToFile (
             int handle,
             UBYTE * buffer,
             LONG size )  [extern]
```

Definition at line 1333 of file tools.c.

4.19.6.146 SeekFile()

```
void SeekFile (
             int handle,
             POSITION * offset,
             int origin )  [extern]
```

Definition at line 1371 of file tools.c.

4.19.6.147 TellFile()

```
LONG TellFile (
             int handle )  [extern]
```

Definition at line 1396 of file tools.c.

4.19.6.148 FlushFile()

```
void FlushFile (
             int handle )  [extern]
```

Definition at line 1420 of file tools.c.

4.19.6.149 GetPosFile()

```
int GetPosFile (
             int handle,
             fpos_t * pospointer )  [extern]
```

Definition at line 1434 of file tools.c.

4.19.6.150 SetPosFile()

```
int SetPosFile (
             int handle,
             fpos_t * pospointer )  [extern]
```

Definition at line 1448 of file tools.c.

4.19.6.151 SynchFile()

```
void SynchFile (
            int handle )  [extern]
```

Definition at line 1468 of file tools.c.

4.19.6.152 TruncateFile()

```
void TruncateFile (
            int handle )  [extern]
```

Definition at line 1490 of file tools.c.

4.19.6.153 GetChannel()

```
int GetChannel (
            char * name,
            int mode )  [extern]
```

Definition at line 1510 of file tools.c.

4.19.6.154 GetAppendChannel()

```
int GetAppendChannel (
            char * name )  [extern]
```

Definition at line 1546 of file tools.c.

4.19.6.155 CloseChannel()

```
int CloseChannel (
            char * name )  [extern]
```

Definition at line 1576 of file tools.c.

4.19.6.156 inictable()

```
void inictable (
            void )  [extern]
```

Definition at line 310 of file compiler.c.

4.19.6.157 findcommand()

```
KEYWORD * findcommand (
            UBYTE * in )  [extern]
```

Definition at line 336 of file compiler.c.

4.19.6.158 inicbufs()

```
int inicbufs (
            void ) [extern]
```

Creates a new compiler buffer and returns its ID number.

Returns

The ID number for the new compiler buffer.

Definition at line 47 of file comtool.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::BufferSize, CbUf::CanCommu, CbUf::dimension, CbUf::lhs, CbUf::numdum, CbUf::NumTerms, CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Referenced by AddRHS(), and StartVariables().

4.19.6.159 StartFiles()

```
void StartFiles (
            void ) [extern]
```

Definition at line 956 of file tools.c.

4.19.6.160 MakeDate()

```
UBYTE * MakeDate (
            void ) [extern]
```

Definition at line 2090 of file tools.c.

4.19.6.161 PreProcessor()

```
void PreProcessor (
            void ) [extern]
```

Definition at line 947 of file pre.c.

4.19.6.162 FromList()

```
void * FromList (
            LIST * L ) [extern]
```

Definition at line 2780 of file tools.c.

4.19.6.163 From0List()

```
void * From0List (
            LIST * L )  [extern]
```

Definition at line 2806 of file tools.c.

4.19.6.164 FromVarList()

```
void * FromVarList (
            LIST * L )  [extern]
```

Definition at line 2835 of file tools.c.

4.19.6.165 DoubleList()

```
int DoubleList (
            void *** lijst,
            int * oldsize,
            int objectsize,
            char * nameoftype )  [extern]
```

Definition at line 2877 of file tools.c.

4.19.6.166 DoubleLList()

```
int DoubleLList (
            void *** lijst,
            LONG * oldsize,
            int objectsize,
            char * nameoftype )  [extern]
```

Definition at line 2929 of file tools.c.

4.19.6.167 DoubleBuffer()

```
void DoubleBuffer (
            void ** start,
            void ** stop,
            int size,
            char * text )  [extern]
```

Definition at line 2977 of file tools.c.

4.19.6.168 ExpandBuffer()

```
void ExpandBuffer (
            void ** buffer,
            LONG * oldsize,
            int type )  [extern]
```

Definition at line 3002 of file tools.c.

4.19.6.169 iexp()

```
LONG iexp (
            LONG x,
            int p )  [extern]
```

Definition at line 3026 of file tools.c.

4.19.6.170 IsLikeVector()

```
int IsLikeVector (
            WORD * arg )  [extern]
```

Definition at line 3229 of file tools.c.

4.19.6.171 AreArgsEqual()

```
int AreArgsEqual (
            WORD * arg1,
            WORD * arg2 )  [extern]
```

Definition at line 3257 of file tools.c.

4.19.6.172 CompareArgs()

```
int CompareArgs (
            WORD * arg1,
            WORD * arg2 )  [extern]
```

Definition at line 3276 of file tools.c.

4.19.6.173 SkipField()

```
UBYTE * SkipField (
            UBYTE * s,
            int level )  [extern]
```

Definition at line 1946 of file tools.c.

4.19.6.174 StrCmp()

```
int StrCmp (
            UBYTE * s1,
            UBYTE * s2 )  [extern]
```

Definition at line 1700 of file tools.c.

4.19.6.175 StrICmp()

```
int StrICmp (
            UBYTE * s1,
            UBYTE * s2 )  [extern]
```

Definition at line 1711 of file tools.c.

4.19.6.176 StrHICmp()

```
int StrHICmp (
            UBYTE * s1,
            UBYTE * s2 )  [extern]
```

Definition at line 1722 of file tools.c.

4.19.6.177 StrICont()

```
int StrICont (
            UBYTE * s1,
            UBYTE * s2 )  [extern]
```

Definition at line 1733 of file tools.c.

4.19.6.178 CmpArray()

```
int CmpArray (
            WORD * t1,
            WORD * t2,
            WORD n )  [extern]
```

Definition at line 1745 of file tools.c.

4.19.6.179 ConWord()

```
int ConWord (
            UBYTE * s1,
            UBYTE * s2 )  [extern]
```

Definition at line 1759 of file tools.c.

4.19.6.180 StrLen()

```
int StrLen (
            UBYTE * s )  [extern]
```

Definition at line 1771 of file tools.c.

4.19.6.181 GetPreVar()

```
UBYTE * GetPreVar (
            UBYTE * name,
            int flag )  [extern]
```

Definition at line 536 of file pre.c.

4.19.6.182 ToGeneral()

```
void ToGeneral (
            WORD * r,
            WORD * m,
            WORD par )  [extern]
```

Definition at line 3057 of file tools.c.

4.19.6.183 ToPolyFunGeneral()

```
WORD ToPolyFunGeneral (
            PHEAD WORD * term )  [extern]
```

Definition at line 3154 of file tools.c.

4.19.6.184 ToFast()

```
int ToFast (
            WORD * r,
            WORD * m )  [extern]
```

Definition at line 3101 of file tools.c.

4.19.6.185 GetSetupPar()

```
SETUPPARAMETERS * GetSetupPar (
            UBYTE * s )  [extern]
```

Definition at line 337 of file setfile.c.

4.19.6.186 RecalcSetups()

```
int RecalcSetups (
            void  )  [extern]
```

Definition at line 357 of file setfile.c.

4.19.6.187 AllocSetups()

```
int AllocSetups (
            void ) [extern]
```

Definition at line 404 of file setfile.c.

4.19.6.188 AllocSort()

```
SORTING * AllocSort (
            LONG inLargeSize,
            LONG inSmallSize,
            LONG inSmallEsize,
            LONG inTermsInSmall,
            int inMaxPatches,
            int inMaxFpatches,
            LONG inIOsize,
            int level ) [extern]
```

Definition at line 869 of file setfile.c.

4.19.6.189 AllocSortFileName()

```
void AllocSortFileName (
            SORTING * sort ) [extern]
```

Definition at line 1019 of file setfile.c.

4.19.6.190 LoadInputFile()

```
UBYTE * LoadInputFile (
            UBYTE * filename,
            int type ) [extern]
```

Definition at line 112 of file tools.c.

4.19.6.191 GetInput()

```
UBYTE GetInput (
            void ) [extern]
```

Definition at line 132 of file pre.c.

4.19.6.192 ClearPushback()

```
void ClearPushback (
            void ) [extern]
```

Definition at line 161 of file pre.c.

4.19.6.193 GetChar()

```
UBYTE GetChar (
            int level ) [extern]
```

Definition at line 179 of file pre.c.

4.19.6.194 CharOut()

```
void CharOut (
            UBYTE c ) [extern]
```

Definition at line 489 of file pre.c.

4.19.6.195 UnsetAllowDelay()

```
void UnsetAllowDelay (
            void ) [extern]
```

Definition at line 510 of file pre.c.

4.19.6.196 PopPreVars()

```
void PopPreVars (
            int tonumber ) [extern]
```

Definition at line 820 of file pre.c.

4.19.6.197 IniModule()

```
void IniModule (
            int type ) [extern]
```

Definition at line 835 of file pre.c.

4.19.6.198 IniSpecialModule()

```
void IniSpecialModule (
            int type ) [extern]
```

Definition at line 937 of file pre.c.

4.19.6.199 ModuleInstruction()

```
int ModuleInstruction (
            int * moduletype,
            int * specialtype ) [extern]
```

Definition at line 76 of file module.c.

4.19.6.200 PreProInstruction()

```
int PreProInstruction (
            void )  [extern]
```

Definition at line 1166 of file pre.c.

4.19.6.201 LoadInstruction()

```
int LoadInstruction (
            int mode )  [extern]
```

Definition at line 1254 of file pre.c.

4.19.6.202 LoadStatement()

```
int LoadStatement (
            int type )  [extern]
```

Definition at line 1457 of file pre.c.

4.19.6.203 FindKeyWord()

```
KEYWORD * FindKeyWord (
            UBYTE * theword,
            KEYWORD * table,
            int size )  [extern]
```

Definition at line 1963 of file pre.c.

4.19.6.204 FindInKeyWord()

```
KEYWORD * FindInKeyWord (
            UBYTE * theword,
            KEYWORD * table,
            int size )  [extern]
```

Definition at line 1994 of file pre.c.

4.19.6.205 DoDefine()

```
int DoDefine (
            UBYTE * s )  [extern]
```

Definition at line 2159 of file pre.c.

4.19.6.206 DoRedefine()

```
int DoRedefine (
            UBYTE * s )  [extern]
```

Definition at line 2169 of file pre.c.

4.19.6.207 TheDefine()

```
int TheDefine (
            UBYTE * s,
            int mode )  [extern]
```

Preprocessor assignment. Possible arguments and values are treated and the new preprocessor variable is put into the name administration.

Parameters

s	Pointer to the character string following the preprocessor command.
mode	Bitmask. 0-bit clear: always create a new name entry, 0-bit set: try to redefine an existing name, 1-bit set: ignore preprocessor if/switch status.

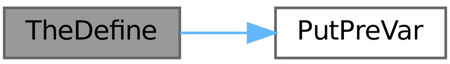
Returns

zero: no errors, negative number: errors.

Definition at line 2024 of file pre.c.

References PutPreVar().

Here is the call graph for this function:



4.19.6.208 TheUndefine()

```
int TheUndefine (
            UBYTE * name )  [extern]
```

Definition at line 2209 of file pre.c.

4.19.6.209 ClearMacro()

```
int ClearMacro (
              UBYTE * name )  [extern]
```

Definition at line 2181 of file pre.c.

4.19.6.210 DoUndefine()

```
int DoUndefine (
              UBYTE * s )  [extern]
```

Definition at line 2263 of file pre.c.

4.19.6.211 DoInclude()

```
int DoInclude (
              UBYTE * s )  [extern]
```

Definition at line 2315 of file pre.c.

4.19.6.212 DoReverseInclude()

```
int DoReverseInclude (
              UBYTE * s )  [extern]
```

Definition at line 2322 of file pre.c.

4.19.6.213 Include()

```
int Include (
              UBYTE * s,
              int type )  [extern]
```

Definition at line 2329 of file pre.c.

4.19.6.214 DoExternal()

```
int DoExternal (
              UBYTE * s )  [extern]
```

Definition at line 5424 of file pre.c.

4.19.6.215 DoToExternal()

```
int DoToExternal (
              UBYTE * s )  [extern]
```

Definition at line 5986 of file pre.c.

4.19.6.216 DoFromExternal()

```
int DoFromExternal (
            UBYTE * s )  [extern]
```

Definition at line 5791 of file pre.c.

4.19.6.217 DoPrompt()

```
int DoPrompt (
            UBYTE * s )  [extern]
```

Definition at line 5506 of file pre.c.

4.19.6.218 DoSetExternal()

```
int DoSetExternal (
            UBYTE * s )  [extern]
```

Definition at line 5539 of file pre.c.

4.19.6.219 DoSetExternalAttr()

```
int DoSetExternalAttr (
            UBYTE * s )  [extern]
```

Definition at line 5609 of file pre.c.

4.19.6.220 DoRmExternal()

```
int DoRmExternal (
            UBYTE * s )  [extern]
```

Definition at line 5726 of file pre.c.

4.19.6.221 DoFactDollar()

```
int DoFactDollar (
            UBYTE * s )  [extern]
```

Definition at line 6633 of file pre.c.

4.19.6.222 GetDollarNumber()

```
WORD GetDollarNumber (
            UBYTE ** inp,
            DOLLARS d )  [extern]
```

Definition at line 6670 of file pre.c.

4.19.6.223 DoSetRandom()

```
int DoSetRandom (
            UBYTE * s )  [extern]
```

Definition at line 6760 of file pre.c.

4.19.6.224 DoOptimize()

```
int DoOptimize (
            UBYTE * s )  [extern]
```

Definition at line 6803 of file pre.c.

4.19.6.225 DoClearOptimize()

```
int DoClearOptimize (
            UBYTE * s )  [extern]
```

Definition at line 6936 of file pre.c.

4.19.6.226 DoSkipExtraSymbols()

```
int DoSkipExtraSymbols (
            UBYTE * s )  [extern]
```

Definition at line 6956 of file pre.c.

4.19.6.227 DoTimeOutAfter()

```
int DoTimeOutAfter (
            UBYTE * s )  [extern]
```

Definition at line 7182 of file pre.c.

4.19.6.228 DoMessage()

```
int DoMessage (
            UBYTE * s )  [extern]
```

Definition at line 3676 of file pre.c.

4.19.6.229 DoPreOut()

```
int DoPreOut (
            UBYTE * s )  [extern]
```

Definition at line 3747 of file pre.c.

4.19.6.230 DoPreAppend()

```
int DoPreAppend (
            UBYTE * s )  [extern]
```

Definition at line 3804 of file pre.c.

4.19.6.231 DoPreCreate()

```
int DoPreCreate (
            UBYTE * s )  [extern]
```

Definition at line 3847 of file pre.c.

4.19.6.232 DoPreAssign()

```
int DoPreAssign (
            UBYTE * s )  [extern]
```

Definition at line 2128 of file pre.c.

4.19.6.233 DoPreBreak()

```
int DoPreBreak (
            UBYTE * s )  [extern]
```

Definition at line 4047 of file pre.c.

4.19.6.234 DoPreDefault()

```
int DoPreDefault (
            UBYTE * s )  [extern]
```

Definition at line 4110 of file pre.c.

4.19.6.235 DoPreSwitch()

```
int DoPreSwitch (
            UBYTE * s )  [extern]
```

Definition at line 4161 of file pre.c.

4.19.6.236 DoPreEndSwitch()

```
int DoPreEndSwitch (
            UBYTE * s )  [extern]
```

Definition at line 4134 of file pre.c.

4.19.6.237 DoPreCase()

```
int DoPreCase (
            UBYTE * s )  [extern]
```

Definition at line 4071 of file pre.c.

4.19.6.238 DoPreShow()

```
int DoPreShow (
            UBYTE * s )  [extern]
```

Definition at line 4215 of file pre.c.

4.19.6.239 DoPreExchange()

```
int DoPreExchange (
            UBYTE * s )  [extern]
```

Definition at line 2490 of file pre.c.

4.19.6.240 DoSystem()

```
int DoSystem (
            UBYTE * s )  [extern]
```

Definition at line 4254 of file pre.c.

4.19.6.241 DoPipe()

```
int DoPipe (
            UBYTE * s )  [extern]
```

Definition at line 3690 of file pre.c.

4.19.6.242 StartPrepro()

```
void StartPrepro (
            void )  [extern]
```

Definition at line 4432 of file pre.c.

4.19.6.243 DoIfdef()

```
int DoIfdef (
            UBYTE * s,
            int par )  [extern]
```

Definition at line 3391 of file pre.c.

4.19.6.244 DoIfydef()

```
int DoIfydef (
              UBYTE * s )  [extern]
```

Definition at line 3416 of file pre.c.

4.19.6.245 DoIfndef()

```
int DoIfndef (
              UBYTE * s )  [extern]
```

Definition at line 3426 of file pre.c.

4.19.6.246 DoElse()

```
int DoElse (
              UBYTE * s )  [extern]
```

Definition at line 3095 of file pre.c.

4.19.6.247 DoElseif()

```
int DoElseif (
              UBYTE * s )  [extern]
```

Definition at line 3132 of file pre.c.

4.19.6.248 DoEndif()

```
int DoEndif (
              UBYTE * s )  [extern]
```

Definition at line 3313 of file pre.c.

4.19.6.249 DoTerminate()

```
int DoTerminate (
              UBYTE * s )  [extern]
```

Definition at line 2757 of file pre.c.

4.19.6.250 DoIf()

```
int DoIf (
              UBYTE * s )  [extern]
```

Definition at line 3367 of file pre.c.

4.19.6.251 DoCall()

```
int DoCall (
            UBYTE * s )  [extern]
```

Definition at line 2551 of file pre.c.

4.19.6.252 DoDebug()

```
int DoDebug (
            UBYTE * s )  [extern]
```

Definition at line 2720 of file pre.c.

4.19.6.253 DoContinueDo()

```
int DoContinueDo (
            UBYTE * s )  [extern]
```

Definition at line 2780 of file pre.c.

4.19.6.254 DoDo()

```
int DoDo (
            UBYTE * s )  [extern]
```

Definition at line 2811 of file pre.c.

4.19.6.255 DoBreakDo()

```
int DoBreakDo (
            UBYTE * s )  [extern]
```

Definition at line 3026 of file pre.c.

4.19.6.256 DoEnddo()

```
int DoEnddo (
            UBYTE * s )  [extern]
```

Definition at line 3167 of file pre.c.

4.19.6.257 DoEndprocedure()

```
int DoEndprocedure (
            UBYTE * s )  [extern]
```

Definition at line 3341 of file pre.c.

4.19.6.258 DoInside()

```
int DoInside (
              UBYTE * s )  [extern]
```

Definition at line 3451 of file pre.c.

4.19.6.259 DoEndInside()

```
int DoEndInside (
              UBYTE * s )  [extern]
```

Definition at line 3544 of file pre.c.

4.19.6.260 DoProcedure()

```
int DoProcedure (
              UBYTE * s )  [extern]
```

Definition at line 4005 of file pre.c.

4.19.6.261 DoPrePrintTimes()

```
int DoPrePrintTimes (
              UBYTE * s )  [extern]
```

Definition at line 3770 of file pre.c.

4.19.6.262 DoPreWrite()

```
int DoPreWrite (
              UBYTE * s )  [extern]
```

Definition at line 3964 of file pre.c.

4.19.6.263 DoPreClose()

```
int DoPreClose (
              UBYTE * s )  [extern]
```

Definition at line 3920 of file pre.c.

4.19.6.264 DoPreRemove()

```
int DoPreRemove (
              UBYTE * s )  [extern]
```

Definition at line 3887 of file pre.c.

4.19.6.265 DoPreSortReallocate()

```
int DoPreSortReallocate (
            UBYTE * s )  [extern]
```

Definition at line 3784 of file pre.c.

4.19.6.266 DoCommentChar()

```
int DoCommentChar (
            UBYTE * s )  [extern]
```

Definition at line 2097 of file pre.c.

4.19.6.267 DoPrcExtension()

```
int DoPrcExtension (
            UBYTE * s )  [extern]
```

Definition at line 3713 of file pre.c.

4.19.6.268 DoPreReset()

```
int DoPreReset (
            UBYTE * s )  [extern]
```

Definition at line 6995 of file pre.c.

4.19.6.269 WriteString()

```
void WriteString (
            int type,
            UBYTE * str,
            int num )  [extern]
```

Definition at line 1807 of file tools.c.

4.19.6.270 WriteUnfinString()

```
void WriteUnfinString (
            int type,
            UBYTE * str,
            int num )  [extern]
```

Definition at line 1841 of file tools.c.

4.19.6.271 AddToString()

```
UBYTE * AddToString (
            UBYTE * outstring,
            UBYTE * extrastring,
            int par ) [extern]
```

Definition at line 1866 of file tools.c.

4.19.6.272 PreCalc()

```
UBYTE * PreCalc (
            void ) [extern]
```

Definition at line 5113 of file pre.c.

4.19.6.273 PreEval()

```
UBYTE * PreEval (
            UBYTE * s,
            LONG * x ) [extern]
```

Definition at line 5191 of file pre.c.

4.19.6.274 NumToStr()

```
void NumToStr (
            UBYTE * s,
            LONG x ) [extern]
```

Definition at line 1783 of file tools.c.

4.19.6.275 PreCmp()

```
int PreCmp (
            int type,
            int val,
            UBYTE * t,
            int type2,
            int val2,
            UBYTE * t2,
            int cmpop ) [extern]
```

Definition at line 4624 of file pre.c.

4.19.6.276 PreEq()

```
int PreEq (
            int type,
            int val,
            UBYTE * t,
            int type2,
            int val2,
            UBYTE * t2,
            int eqop ) [extern]
```

Definition at line 4648 of file pre.c.

4.19.6.277 pParseObject()

```
UBYTE * pParseObject (
            UBYTE * s,
            int * type,
            LONG * val2 ) [extern]
```

Definition at line 4677 of file pre.c.

4.19.6.278 PreIfEval()

```
UBYTE * PreIfEval (
            UBYTE * s,
            int * value ) [extern]
```

Definition at line 4495 of file pre.c.

4.19.6.279 EvalPreIf()

```
int EvalPreIf (
            UBYTE * s ) [extern]
```

Definition at line 4460 of file pre.c.

4.19.6.280 PreLoad()

```
int PreLoad (
            PRELOAD * p,
            UBYTE * start,
            UBYTE * stop,
            int mode,
            char * message ) [extern]
```

Definition at line 4294 of file pre.c.

4.19.6.281 PreSkip()

```
int PreSkip (
            UBYTE * start,
            UBYTE * stop,
            int mode )  [extern]
```

Definition at line 4377 of file pre.c.

4.19.6.282 EndOfToken()

```
UBYTE * EndOfToken (
            UBYTE * s )  [extern]
```

Definition at line 1919 of file tools.c.

4.19.6.283 SetSpecialMode()

```
void SetSpecialMode (
            int moduletype,
            int specialtype )  [extern]
```

Definition at line 257 of file module.c.

4.19.6.284 MakeGlobal()

```
void MakeGlobal (
            void  )  [extern]
```

Definition at line 382 of file execute.c.

4.19.6.285 ExecModule()

```
int ExecModule (
            int moduletype )  [extern]
```

Definition at line 274 of file module.c.

4.19.6.286 ExecStore()

```
int ExecStore (
            void  )  [extern]
```

Definition at line 299 of file module.c.

4.19.6.287 FullCleanUp()

```
void FullCleanUp (
             void ) [extern]
```

Definition at line 314 of file module.c.

4.19.6.288 DoExecStatement()

```
int DoExecStatement (
             void ) [extern]
```

Definition at line 780 of file module.c.

4.19.6.289 DoPipeStatement()

```
int DoPipeStatement (
             void ) [extern]
```

Definition at line 797 of file module.c.

4.19.6.290 DoPolyfun()

```
int DoPolyfun (
             UBYTE * s ) [extern]
```

Definition at line 395 of file module.c.

4.19.6.291 DoPolyratfun()

```
int DoPolyratfun (
             UBYTE * s ) [extern]
```

Definition at line 458 of file module.c.

4.19.6.292 CompileStatement()

```
int CompileStatement (
             UBYTE * in ) [extern]
```

Definition at line 554 of file compiler.c.

4.19.6.293 ToToken()

```
UBYTE * ToToken (
             UBYTE * s ) [extern]
```

Definition at line 1931 of file tools.c.

4.19.6.294 GetDollar()

```
int GetDollar (
            UBYTE * name )  [extern]
```

Definition at line 590 of file names.c.

4.19.6.295 MesWork()

```
int MesWork (
            void  )  [extern]
```

Definition at line 89 of file message.c.

4.19.6.296 MesCall()

```
int MesCall (
            char * s )  [extern]
```

Definition at line 814 of file message.c.

4.19.6.297 NumCopy()

```
UBYTE * NumCopy (
            WORD y,
            UBYTE * to )  [extern]
```

Definition at line 1997 of file tools.c.

4.19.6.298 LongCopy()

```
char * LongCopy (
            LONG y,
            char * to )  [extern]
```

Definition at line 2024 of file tools.c.

4.19.6.299 LongLongCopy()

```
char * LongLongCopy (
            off_t * y,
            char * to )  [extern]
```

Definition at line 2051 of file tools.c.

4.19.6.300 ReserveTempFiles()

```
void ReserveTempFiles (
             int par )  [extern]
```

Definition at line 666 of file startup.c.

4.19.6.301 PrintTerm()

```
void PrintTerm (
             WORD * term,
             char * where )  [extern]
```

Definition at line 857 of file message.c.

4.19.6.302 PrintTermC()

```
void PrintTermC (
             WORD * term,
             char * where )  [extern]
```

Definition at line 887 of file message.c.

4.19.6.303 PrintSubTerm()

```
void PrintSubTerm (
             WORD * term,
             char * where )  [extern]
```

Definition at line 921 of file message.c.

4.19.6.304 PrintWords()

```
void PrintWords (
             WORD * buffer,
             LONG number )  [extern]
```

Definition at line 943 of file message.c.

4.19.6.305 PrintSeq()

```
void PrintSeq (
             WORD * a,
             char * text )  [extern]
```

Definition at line 961 of file message.c.

4.19.6.306 ExpandTripleDots()

```
int ExpandTripleDots (
            int par ) [extern]
```

Definition at line 1595 of file pre.c.

4.19.6.307 ComPress()

```
LONG ComPress (
            WORD ** ss,
            LONG * n ) [extern]
```

Gets a list of pointers to terms and compresses the terms. In n it collects the number of terms and the return value of the function is the space that is occupied.

We have to pay some special attention to the compression of terms with a PolyFun. This PolyFun should occur only straight before the coefficient, so we can use the same trick as for the coefficient to sabotage compression of this object (Replace in the history the function pointer by zero. This is safe, because terms that would be identical otherwise would have been added).

Parameters

ss	Array of pointers to terms to be compressed.
n	Number of pointers in ss.

Returns

Total number of words needed for the compressed result.

Definition at line 3196 of file sort.c.

Referenced by EndSort(), and StoreTerm().

4.19.6.308 StageSort()

```
void StageSort (
            FILEHANDLE * fout ) [extern]
```

Prepares a stage 4 or higher sort. Stage 4 sorts occur when the sort file contains more patches than can be merged in one pass.

Definition at line 4654 of file sort.c.

References FiLe::handle.

Referenced by EndSort(), and MergePatches().

4.19.6.309 M_free()

```
void M_free (
            void * x,
            const char * where ) [extern]
```

Definition at line 2383 of file tools.c.

4.19.6.310 ClearWildcardNames()

```
void ClearWildcardNames (
            void ) [extern]
```

Definition at line 849 of file names.c.

4.19.6.311 AddWildcardName()

```
int AddWildcardName (
            UBYTE * name ) [extern]
```

Definition at line 854 of file names.c.

4.19.6.312 GetWildcardName()

```
int GetWildcardName (
            UBYTE * name ) [extern]
```

Definition at line 898 of file names.c.

4.19.6.313 Globalize()

```
void Globalize (
            int par ) [extern]
```

Definition at line 3155 of file names.c.

4.19.6.314 ResetVariables()

```
void ResetVariables (
            int par ) [extern]
```

Definition at line 2832 of file names.c.

4.19.6.315 AddToPreTypes()

```
void AddToPreTypes (
            int type ) [extern]
```

Definition at line 5341 of file pre.c.

4.19.6.316 MessPreNesting()

```
void MessPreNesting (
            int par )   [extern]
```

Definition at line 5359 of file pre.c.

4.19.6.317 GetStreamPosition()

```
LONG GetStreamPosition (
            STREAM * stream )   [extern]
```

Definition at line 813 of file tools.c.

4.19.6.318 DoubleCbuffer()

```
WORD * DoubleCbuffer (
            int num,
            WORD * w,
            int par )   [extern]
```

Doubles a compiler buffer.

Parameters

num	The ID number for the buffer to be doubled.
w	The pointer to the end (exclusive) of the current buffer. The contents in the range of [cbuf[num].Buffer,w) will be kept.

Definition at line 143 of file comtool.c.

References CbUf::Buffer, CbUf::BufferSize, CbUf::lhs, CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Referenced by AddNtoC(), and AddNtoL().

4.19.6.319 AddLHS()

```
WORD * AddLHS (
            int num )   [extern]
```

Adds an LHS to a compiler buffer and returns the pointer to a buffer for the new LHS.

Parameters

num	The ID number for the buffer to get another LHS.

Definition at line 188 of file comtool.c.

References CbUf::lhs, and CbUf::Pointer.

Referenced by AddNtoL().

4.19.6.320 AddRHS()

```
WORD * AddRHS (
            int num,
            int type )  [extern]
```

Adds an RHS to a compiler buffer and returns the pointer to a buffer for the new RHS.

Parameters

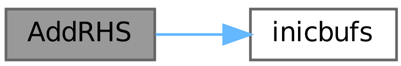
num	The ID number for the buffer to get another RHS.
type	If 0, the subexpression tree will be reallocated.

Definition at line 214 of file comtool.c.

References TaBlEs::buffers, TaBlEs::buffersfill, TaBlEs::bufferssize, TaBlEs::bufnum, CbUf::CanCommu, CbUf::dimension, inicbufs(), CbUf::numdum, CbUf::NumTerms, CbUf::Pointer, and CbUf::rhs.

Referenced by InsertArg(), StartVariables(), and TestMatch().

Here is the call graph for this function:



4.19.6.321 AddNtoL()

```
int AddNtoL (
            int n,
            WORD * array )  [extern]
```

Adds an LHS with the given data to the current compiler buffer.

Parameters

n	The length of the data.
array	The data to be added.

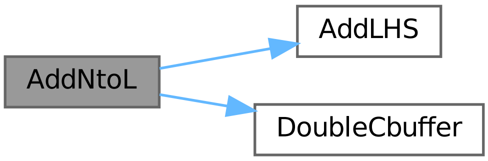
Returns

0 if succeeds.

Definition at line 288 of file comtool.c.

References AddLHS(), DoubleCbuffer(), CbUf::Pointer, and CbUf::Top.

Here is the call graph for this function:



4.19.6.322 AddNtoC()

```
int AddNtoC (
            int bufnum,
            int n,
            WORD * array,
            int par )  [extern]
```

Adds the given data to the last LHS/RHS in a compiler buffer.

Parameters

bufnum	The ID number for the buffer where the data will be added.
n	The length of the data.
array	The data to be added.

Returns

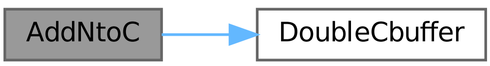
0 if succeeds.

Definition at line 317 of file comtool.c.

References DoubleCbuffer(), CbUf::Pointer, and CbUf::Top.

Referenced by InsertArg(), and StartVariables().

Here is the call graph for this function:



4.19.6.323 DoubleIfBuffers()

```
void DoubleIfBuffers (
            void ) [extern]
```

Definition at line 1033 of file if.c.

4.19.6.324 CreateStream()

```
STREAM * CreateStream (
            UBYTE * where ) [extern]
```

Definition at line 782 of file tools.c.

4.19.6.325 setonoff()

```
int setonoff (
            UBYTE * s,
            int * flag,
            int onvalue,
            int offvalue ) [extern]
```

Definition at line 490 of file compcomm.c.

4.19.6.326 DoPrint()

```
int DoPrint (
            UBYTE * s,
            int par ) [extern]
```

Definition at line 1041 of file compcomm.c.

4.19.6.327 SetExpr()

```
int SetExpr (
            UBYTE * s,
            int setunset,
            int par ) [extern]
```

Definition at line 1512 of file compcomm.c.

4.19.6.328 AddToCom()

```
void AddToCom (
            int n,
            WORD * array ) [extern]
```

Definition at line 1649 of file compcomm.c.

4.19.6.329 Add2ComStrings()

```
int Add2ComStrings (
            int n,
            WORD * array,
            UBYTE * string1,
            UBYTE * string2 )  [extern]
```

Definition at line 1723 of file compcomm.c.

4.19.6.330 DoSymmetrize()

```
int DoSymmetrize (
            UBYTE * s,
            int par )  [extern]
```

Definition at line 2217 of file compcomm.c.

4.19.6.331 DoArgument()

```
int DoArgument (
            UBYTE * s,
            int par )  [extern]
```

Definition at line 1862 of file compcomm.c.

4.19.6.332 ArgFactorize()

```
int ArgFactorize (
            PHEAD WORD * argin,
            WORD * argout )  [extern]
```

Definition at line 2060 of file argument.c.

4.19.6.333 TakeArgContent()

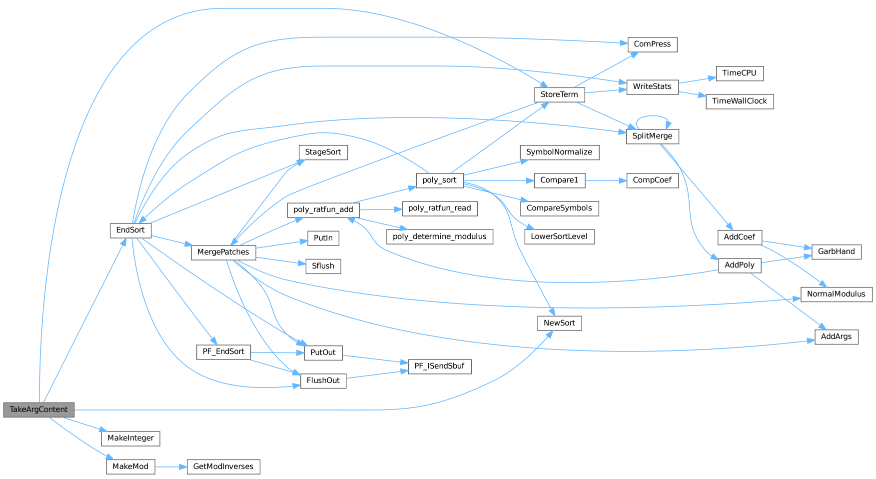
```
WORD * TakeArgContent (
            PHEAD WORD * argin,
            WORD * argout )  [extern]
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. The common pieces are put in argout as a sequence of arguments. The part with the multiple terms that are now relative prime is put in argfree which is allocated via TermMalloc and is given as the return value. The difference with the old code is that negative powers are always removed. Hence it is as in MakeInteger in which only numerators will be left: now only zero or positive powers will be remaining.

Definition at line 2766 of file argument.c.

References EndSort(), MakeInteger(), MakeMod(), NewSort(), and StoreTerm().

Here is the call graph for this function:



4.19.6.334 MakeInteger()

```
WORD * MakeInteger (
            PHEAD WORD * argin,
            WORD * argout,
            WORD * argfree )  [extern]
```

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in argin. The number that comes out should go to argout. The new pointer in the argout buffer is the return value. The normalized argument is in argfree.

Definition at line 3312 of file argument.c.

Referenced by TakeArgContent().

4.19.6.335 MakeMod()

```
WORD * MakeMod (
            PHEAD WORD * argin,
            WORD * argout,
            WORD * argfree )  [extern]
```

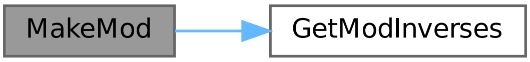
Similar to MakeInteger but now with modulus arithmetic using only a one WORD 'prime'. We make the coefficient of the first term in the argument equal to one. Already the coefficients are taken modulus AN.cmod and AN.ncmod == 1

Definition at line 3483 of file argument.c.

References GetModInverses().

Referenced by TakeArgContent().

Here is the call graph for this function:



4.19.6.336 FindArg()

```
WORD FindArg (
            PHEAD WORD * a )  [extern]
```

Looks the argument up in the (workers) table. If it is found the number in the table is returned (plus one to make it positive). If it is not found we look in the compiler provided table. If it is found - the number in the table is returned (minus one to make it negative). If in neither table we return zero.

Definition at line 2513 of file argument.c.

4.19.6.337 InsertArg()

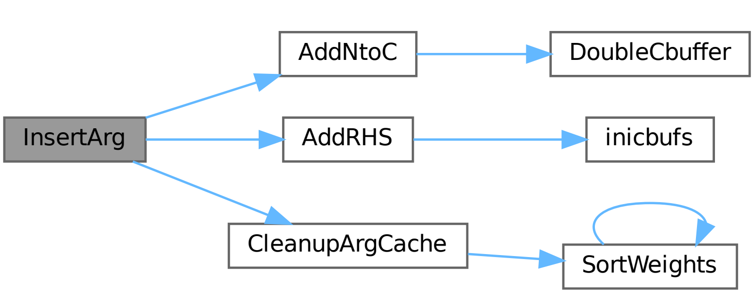
```
WORD InsertArg (
            PHEAD WORD * argin,
            WORD * argout,
            int par )  [extern]
```

Inserts the argument into the (workers) table. If the table is too full we eliminate half of it. The eliminated elements are the ones that have not been used most recently, weighted by their total use and age(?). If par == 0 it inserts in the regular factorization cache If par == 1 it inserts in the cache defined with the FactorCache statement

Definition at line 2537 of file argument.c.

References AddNtoC(), AddRHS(), and CleanupArgCache().

Here is the call graph for this function:



4.19.6.338 CleanupArgCache()

```
int CleanupArgCache (
            PHEAD WORD bufnum )  [extern]
```

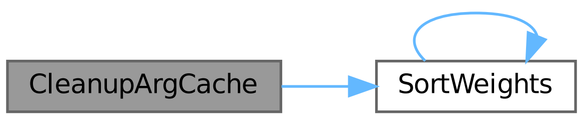
Cleans up the argument factorization cache. We throw half the elements. For a weight of what we want to keep we use the product of usage and the number in the buffer.

Definition at line 2572 of file argument.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::Pointer, CbUf::rhs, SortWeights(), and tree::usage.

Referenced by InsertArg().

Here is the call graph for this function:



4.19.6.339 ArgSymbolMerge()

```
int ArgSymbolMerge (
            WORD * t1,
            WORD * t2 )  [extern]
```

Definition at line 2643 of file argument.c.

4.19.6.340 ArgDotproductMerge()

```
int ArgDotproductMerge (
            WORD * t1,
            WORD * t2 )  [extern]
```

Definition at line 2698 of file argument.c.

4.19.6.341 SortWeights()

```
void SortWeights (
            LONG * weights,
            LONG * extraspace,
            WORD number ) [extern]
```

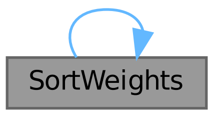
Sorts an array of LONGS in the same way SplitMerge (in sort.c) works We use gradual division in two.

Definition at line 3528 of file argument.c.

References SortWeights().

Referenced by CleanupArgCache(), and SortWeights().

Here is the call graph for this function:



4.19.6.342 DoBrackets()

```
int DoBrackets (
            UBYTE * inp,
            int par ) [extern]
```

Definition at line 3752 of file compcomm.c.

4.19.6.343 DoPutInside()

```
int DoPutInside (
            UBYTE * inp,
            int par ) [extern]
```

Definition at line 7047 of file compcomm.c.

4.19.6.344 CountComp()

```
WORD * CountComp (
            UBYTE * inp,
            WORD * to ) [extern]
```

Definition at line 4030 of file compcomm.c.

4.19.6.345 CoAntiBracket()

```
int CoAntiBracket (
            UBYTE * inp ) [extern]
```

Definition at line 3887 of file compcomm.c.

4.19.6.346 CoAntiSymmetrize()

```
int CoAntiSymmetrize (
            UBYTE * s ) [extern]
```

Definition at line 2344 of file compcomm.c.

4.19.6.347 DoArgPlode()

```
int DoArgPlode (
            UBYTE * s,
            int par ) [extern]
```

Definition at line 5707 of file compcomm.c.

4.19.6.348 CoArgExplode()

```
int CoArgExplode (
            UBYTE * s ) [extern]
```

Definition at line 5763 of file compcomm.c.

4.19.6.349 CoArgImplode()

```
int CoArgImplode (
            UBYTE * s ) [extern]
```

Definition at line 5770 of file compcomm.c.

4.19.6.350 CoArgument()

```
int CoArgument (
            UBYTE * s ) [extern]
```

Definition at line 2096 of file compcomm.c.

4.19.6.351 CoInside()

```
int CoInside (
            UBYTE * s ) [extern]
```

Definition at line 2129 of file compcomm.c.

4.19.6.352 ExecInside()

```
int ExecInside (
            UBYTE * s )  [extern]
```

Definition at line 1681 of file dollar.c.

4.19.6.353 CoInExpression()

```
int CoInExpression (
            UBYTE * s )  [extern]
```

Definition at line 3357 of file compcomm.c.

4.19.6.354 CoInParallel()

```
int CoInParallel (
            UBYTE * s )  [extern]
```

Definition at line 3263 of file compcomm.c.

4.19.6.355 CoNotInParallel()

```
int CoNotInParallel (
            UBYTE * s )  [extern]
```

Definition at line 3273 of file compcomm.c.

4.19.6.356 DoInParallel()

```
int DoInParallel (
            UBYTE * s,
            int par )  [extern]
```

Definition at line 3288 of file compcomm.c.

4.19.6.357 CoEndInExpression()

```
int CoEndInExpression (
            UBYTE * s )  [extern]
```

Definition at line 3410 of file compcomm.c.

4.19.6.358 CoBracket()

```
int CoBracket (
            UBYTE * inp )  [extern]
```

Definition at line 3879 of file compcomm.c.

4.19.6.359 CoPutInside()

```
int CoPutInside (
              UBYTE * inp )  [extern]
```

Definition at line 7044 of file compcomm.c.

4.19.6.360 CoAntiPutInside()

```
int CoAntiPutInside (
              UBYTE * inp )  [extern]
```

Definition at line 7045 of file compcomm.c.

4.19.6.361 CoMultiBracket()

```
int CoMultiBracket (
              UBYTE * inp )  [extern]
```

Definition at line 3898 of file compcomm.c.

4.19.6.362 CoCFunction()

```
int CoCFunction (
              UBYTE * s )  [extern]
```

Definition at line 1625 of file names.c.

4.19.6.363 CoCTensor()

```
int CoCTensor (
              UBYTE * s )  [extern]
```

Definition at line 1627 of file names.c.

4.19.6.364 CoCollect()

```
int CoCollect (
              UBYTE * s )  [extern]
```

Definition at line 428 of file compcomm.c.

4.19.6.365 CoCompress()

```
int CoCompress (
              UBYTE * s )  [extern]
```

Definition at line 506 of file compcomm.c.

4.19.6.366 CoContract()

```
int CoContract (
            UBYTE * s )  [extern]
```

Definition at line 1787 of file compcomm.c.

4.19.6.367 CoCycleSymmetrize()

```
int CoCycleSymmetrize (
            UBYTE * s )  [extern]
```

Definition at line 2351 of file compcomm.c.

4.19.6.368 CoDelete()

```
int CoDelete (
            UBYTE * s )  [extern]
```

Definition at line 942 of file compcomm.c.

4.19.6.369 CoTableBase()

```
int CoTableBase (
            UBYTE * s )  [extern]
```

Definition at line 627 of file tables.c.

4.19.6.370 CoApply()

```
int CoApply (
            UBYTE * s )  [extern]
```

Definition at line 1796 of file tables.c.

4.19.6.371 CoDenominators()

```
int CoDenominators (
            UBYTE * s )  [extern]
```

Definition at line 5859 of file compcomm.c.

4.19.6.372 CoDimension()

```
int CoDimension (
            UBYTE * s )  [extern]
```

Definition at line 1226 of file names.c.

4.19.6.373 CoDiscard()

```
int CoDiscard (
            UBYTE * s )  [extern]
```

Definition at line 1764 of file compcomm.c.

4.19.6.374 CoDisorder()

```
int CoDisorder (
            UBYTE * inp )  [extern]
```

Definition at line 417 of file comexpr.c.

4.19.6.375 CoDrop()

```
int CoDrop (
            UBYTE * s )  [extern]
```

Definition at line 1557 of file compcomm.c.

4.19.6.376 CoDropCoefficient()

```
int CoDropCoefficient (
            UBYTE * s )  [extern]
```

Definition at line 5888 of file compcomm.c.

4.19.6.377 CoDropSymbols()

```
int CoDropSymbols (
            UBYTE * s )  [extern]
```

Definition at line 5902 of file compcomm.c.

4.19.6.378 CoElse()

```
int CoElse (
            UBYTE * p )  [extern]
```

Definition at line 4862 of file compcomm.c.

4.19.6.379 CoElseIf()

```
int CoElseIf (
            UBYTE * inp )  [extern]
```

Definition at line 4889 of file compcomm.c.

4.19.6.380 CoEndArgument()

```
int CoEndArgument (
            UBYTE * s ) [extern]
```

Definition at line 2103 of file compcomm.c.

4.19.6.381 CoEndInside()

```
int CoEndInside (
            UBYTE * s ) [extern]
```

Definition at line 2136 of file compcomm.c.

4.19.6.382 CoEndIf()

```
int CoEndIf (
            UBYTE * inp ) [extern]
```

Definition at line 4916 of file compcomm.c.

4.19.6.383 CoEndRepeat()

```
int CoEndRepeat (
            UBYTE * inp ) [extern]
```

Definition at line 3714 of file compcomm.c.

4.19.6.384 CoEndTerm()

```
int CoEndTerm (
            UBYTE * s ) [extern]
```

Definition at line 5265 of file compcomm.c.

4.19.6.385 CoEndWhile()

```
int CoEndWhile (
            UBYTE * inp ) [extern]
```

Definition at line 4982 of file compcomm.c.

4.19.6.386 CoExit()

```
int CoExit (
            UBYTE * s ) [extern]
```

Definition at line 3236 of file compcomm.c.

4.19.6.387 CoFactArg()

```
int CoFactArg (
            UBYTE * s )   [extern]
```

Definition at line 2197 of file compcomm.c.

4.19.6.388 CoFactDollar()

```
int CoFactDollar (
            UBYTE * inp )   [extern]
```

Definition at line 6418 of file compcomm.c.

4.19.6.389 CoFactorize()

```
int CoFactorize (
            UBYTE * s )   [extern]
```

Definition at line 6447 of file compcomm.c.

4.19.6.390 CoNFactorize()

```
int CoNFactorize (
            UBYTE * s )   [extern]
```

Definition at line 6454 of file compcomm.c.

4.19.6.391 CoUnFactorize()

```
int CoUnFactorize (
            UBYTE * s )   [extern]
```

Definition at line 6461 of file compcomm.c.

4.19.6.392 CoNUnFactorize()

```
int CoNUnFactorize (
            UBYTE * s )   [extern]
```

Definition at line 6468 of file compcomm.c.

4.19.6.393 DoFactorize()

```
int DoFactorize (
            UBYTE * s,
            int par )   [extern]
```

Definition at line 6475 of file compcomm.c.

4.19.6.394 CoFill()

```
int CoFill (
            UBYTE * inp )  [extern]
```

Definition at line 1070 of file comexpr.c.

4.19.6.395 CoFillExpression()

```
int CoFillExpression (
            UBYTE * inp )  [extern]
```

Definition at line 1356 of file comexpr.c.

4.19.6.396 CoFixIndex()

```
int CoFixIndex (
            UBYTE * s )  [extern]
```

Definition at line 999 of file compcomm.c.

4.19.6.397 CoFormat()

```
int CoFormat (
            UBYTE * s )  [extern]
```

Definition at line 153 of file compcomm.c.

4.19.6.398 CoGlobal()

```
int CoGlobal (
            UBYTE * inp )  [extern]
```

Definition at line 73 of file comexpr.c.

4.19.6.399 CoGlobalFactorized()

```
int CoGlobalFactorized (
            UBYTE * inp )  [extern]
```

Definition at line 87 of file comexpr.c.

4.19.6.400 CoGoTo()

```
int CoGoTo (
            UBYTE * inp )  [extern]
```

Definition at line 1816 of file compcomm.c.

4.19.6.401 CoId()

```
int CoId (
            UBYTE * inp )  [extern]
```

Definition at line 395 of file comexpr.c.

4.19.6.402 CoIdNew()

```
int CoIdNew (
            UBYTE * inp )  [extern]
```

Definition at line 406 of file comexpr.c.

4.19.6.403 CoIdOld()

```
int CoIdOld (
            UBYTE * inp )  [extern]
```

Definition at line 384 of file comexpr.c.

4.19.6.404 CoIf()

```
int CoIf (
            UBYTE * inp )  [extern]
```

Definition at line 4220 of file compcomm.c.

4.19.6.405 CoIfMatch()

```
int CoIfMatch (
            UBYTE * inp )  [extern]
```

Definition at line 450 of file comexpr.c.

4.19.6.406 CoIfNoMatch()

```
int CoIfNoMatch (
            UBYTE * inp )  [extern]
```

Definition at line 461 of file comexpr.c.

4.19.6.407 CoIndex()

```
int CoIndex (
            UBYTE * s )  [extern]
```

Definition at line 1112 of file names.c.

4.19.6.408 CoInsideFirst()

```
int CoInsideFirst (
        UBYTE * s )  [extern]
```

Definition at line 928 of file compcomm.c.

4.19.6.409 CoKeep()

```
int CoKeep (
        UBYTE * s )  [extern]
```

Definition at line 987 of file compcomm.c.

4.19.6.410 CoLabel()

```
int CoLabel (
        UBYTE * inp )  [extern]
```

Definition at line 1835 of file compcomm.c.

4.19.6.411 CoLoad()

```
int CoLoad (
        UBYTE * inp )  [extern]
```

Definition at line 573 of file store.c.

4.19.6.412 CoLocal()

```
int CoLocal (
        UBYTE * inp )  [extern]
```

Definition at line 66 of file comexpr.c.

4.19.6.413 CoLocalFactorized()

```
int CoLocalFactorized (
        UBYTE * inp )  [extern]
```

Definition at line 80 of file comexpr.c.

4.19.6.414 CoMany()

```
int CoMany (
        UBYTE * inp )  [extern]
```

Definition at line 428 of file comexpr.c.

4.19.6.415 CoMerge()

```
int CoMerge (
            UBYTE * inp )  [extern]
```

Definition at line 5548 of file compcomm.c.

4.19.6.416 CoStuffle()

```
int CoStuffle (
            UBYTE * inp )  [extern]
```

Definition at line 5604 of file compcomm.c.

4.19.6.417 CoMetric()

```
int CoMetric (
            UBYTE * s )  [extern]
```

Definition at line 1033 of file compcomm.c.

4.19.6.418 CoModOption()

```
int CoModOption (
            UBYTE * s )  [extern]
```

Definition at line 212 of file module.c.

4.19.6.419 CoModuleOption()

```
int CoModuleOption (
            UBYTE * s )  [extern]
```

Definition at line 143 of file module.c.

4.19.6.420 CoModulus()

```
int CoModulus (
            UBYTE * inp )  [extern]
```

Definition at line 3553 of file compcomm.c.

4.19.6.421 CoMulti()

```
int CoMulti (
            UBYTE * inp )  [extern]
```

Definition at line 439 of file comexpr.c.

4.19.6.422 CoMultiply()

```
int CoMultiply (
            UBYTE * inp )  [extern]
```

Definition at line 1031 of file comexpr.c.

4.19.6.423 CoNFunction()

```
int CoNFunction (
            UBYTE * s )  [extern]
```

Definition at line 1624 of file names.c.

4.19.6.424 CoNPrint()

```
int CoNPrint (
            UBYTE * s )  [extern]
```

Definition at line 1271 of file compcomm.c.

4.19.6.425 CoNTensor()

```
int CoNTensor (
            UBYTE * s )  [extern]
```

Definition at line 1626 of file names.c.

4.19.6.426 CoNWrite()

```
int CoNWrite (
            UBYTE * s )  [extern]
```

Definition at line 2390 of file compcomm.c.

4.19.6.427 CoNoDrop()

```
int CoNoDrop (
            UBYTE * s )  [extern]
```

Definition at line 1564 of file compcomm.c.

4.19.6.428 CoNoSkip()

```
int CoNoSkip (
            UBYTE * s )  [extern]
```

Definition at line 1578 of file compcomm.c.

4.19.6.429 CoNormalize()

```
int CoNormalize (
            UBYTE * s )  [extern]
```

Definition at line 2162 of file compcomm.c.

4.19.6.430 CoMakeInteger()

```
int CoMakeInteger (
            UBYTE * s )  [extern]
```

Definition at line 2169 of file compcomm.c.

4.19.6.431 CoFlags()

```
int CoFlags (
            UBYTE * s,
            int value )  [extern]
```

Definition at line 555 of file compcomm.c.

4.19.6.432 CoOff()

```
int CoOff (
            UBYTE * s )  [extern]
```

Definition at line 590 of file compcomm.c.

4.19.6.433 CoOn()

```
int CoOn (
            UBYTE * s )  [extern]
```

Definition at line 656 of file compcomm.c.

4.19.6.434 CoOnce()

```
int CoOnce (
            UBYTE * inp )  [extern]
```

Definition at line 472 of file comexpr.c.

4.19.6.435 CoOnly()

```
int CoOnly (
            UBYTE * inp )  [extern]
```

Definition at line 483 of file comexpr.c.

4.19.6.436 CoOptimizeOption()

```
int CoOptimizeOption (
            UBYTE * s ) [extern]
```

Definition at line 6608 of file compcomm.c.

4.19.6.437 CoPolyFun()

```
int CoPolyFun (
            UBYTE * s ) [extern]
```

Definition at line 5331 of file compcomm.c.

4.19.6.438 CoPolyRatFun()

```
int CoPolyRatFun (
            UBYTE * s ) [extern]
```

Definition at line 5378 of file compcomm.c.

4.19.6.439 CoPrint()

```
int CoPrint (
            UBYTE * s ) [extern]
```

Definition at line 1257 of file compcomm.c.

4.19.6.440 CoPrintB()

```
int CoPrintB (
            UBYTE * s ) [extern]
```

Definition at line 1264 of file compcomm.c.

4.19.6.441 CoProperCount()

```
int CoProperCount (
            UBYTE * s ) [extern]
```

Definition at line 935 of file compcomm.c.

4.19.6.442 CoUnitTrace()

```
int CoUnitTrace (
            UBYTE * s ) [extern]
```

Definition at line 5182 of file compcomm.c.

4.19.6.443 CoRCycleSymmetrize()

```
int CoRCycleSymmetrize (
            UBYTE * s ) [extern]
```

Definition at line 2358 of file compcomm.c.

4.19.6.444 CoRatio()

```
int CoRatio (
            UBYTE * s ) [extern]
```

Definition at line 2417 of file compcomm.c.

4.19.6.445 CoRedefine()

```
int CoRedefine (
            UBYTE * s ) [extern]
```

Definition at line 2457 of file compcomm.c.

4.19.6.446 CoRenumber()

```
int CoRenumber (
            UBYTE * s ) [extern]
```

Definition at line 2562 of file compcomm.c.

4.19.6.447 CoRepeat()

```
int CoRepeat (
            UBYTE * inp ) [extern]
```

Definition at line 3691 of file compcomm.c.

4.19.6.448 CoSave()

```
int CoSave (
            UBYTE * inp ) [extern]
```

Definition at line 363 of file store.c.

4.19.6.449 CoSelect()

```
int CoSelect (
            UBYTE * inp ) [extern]
```

Definition at line 494 of file comexpr.c.

4.19.6.450 CoSet()

```
int CoSet (
            UBYTE * s )  [extern]
```

Definition at line 2355 of file names.c.

4.19.6.451 CoSetExitFlag()

```
int CoSetExitFlag (
            UBYTE * s )  [extern]
```

Definition at line 3436 of file compcomm.c.

4.19.6.452 CoSkip()

```
int CoSkip (
            UBYTE * s )  [extern]
```

Definition at line 1571 of file compcomm.c.

4.19.6.453 CoProcessBucket()

```
int CoProcessBucket (
            UBYTE * s )  [extern]
```

Definition at line 5659 of file compcomm.c.

4.19.6.454 CoPushHide()

```
int CoPushHide (
            UBYTE * s )  [extern]
```

Definition at line 1278 of file compcomm.c.

4.19.6.455 CoPopHide()

```
int CoPopHide (
            UBYTE * s )  [extern]
```

Definition at line 1322 of file compcomm.c.

4.19.6.456 CoHide()

```
int CoHide (
            UBYTE * inp )  [extern]
```

Definition at line 1585 of file compcomm.c.

4.19.6.457 CoIntoHide()

```
int CoIntoHide (
            UBYTE * inp )  [extern]
```

Definition at line 1603 of file compcomm.c.

4.19.6.458 CoNoIntoHide()

```
int CoNoIntoHide (
            UBYTE * inp )  [extern]
```

Definition at line 1621 of file compcomm.c.

4.19.6.459 CoNoHide()

```
int CoNoHide (
            UBYTE * inp )  [extern]
```

Definition at line 1628 of file compcomm.c.

4.19.6.460 CoUnHide()

```
int CoUnHide (
            UBYTE * inp )  [extern]
```

Definition at line 1635 of file compcomm.c.

4.19.6.461 CoNoUnHide()

```
int CoNoUnHide (
            UBYTE * inp )  [extern]
```

Definition at line 1642 of file compcomm.c.

4.19.6.462 CoSort()

```
int CoSort (
            UBYTE * s )  [extern]
```

Definition at line 5292 of file compcomm.c.

4.19.6.463 CoSplitArg()

```
int CoSplitArg (
            UBYTE * s )  [extern]
```

Definition at line 2176 of file compcomm.c.

4.19.6.464 CoSplitFirstArg()

```
int CoSplitFirstArg (
            UBYTE * s ) [extern]
```

Definition at line 2183 of file compcomm.c.

4.19.6.465 CoSplitLastArg()

```
int CoSplitLastArg (
            UBYTE * s ) [extern]
```

Definition at line 2190 of file compcomm.c.

4.19.6.466 CoSum()

```
int CoSum (
            UBYTE * s ) [extern]
```

Definition at line 2583 of file compcomm.c.

4.19.6.467 CoSymbol()

```
int CoSymbol (
            UBYTE * s ) [extern]
```

Definition at line 946 of file names.c.

4.19.6.468 CoSymmetrize()

```
int CoSymmetrize (
            UBYTE * s ) [extern]
```

Definition at line 2337 of file compcomm.c.

4.19.6.469 DoTable()

```
int DoTable (
            UBYTE * s,
            int par ) [extern]
```

Definition at line 1666 of file names.c.

4.19.6.470 CoTable()

```
int CoTable (
            UBYTE * s ) [extern]
```

Definition at line 2048 of file names.c.

4.19.6.471 CoTerm()

```
int CoTerm (
            UBYTE * s )  [extern]
```

Definition at line 5217 of file compcomm.c.

4.19.6.472 CoNTable()

```
int CoNTable (
            UBYTE * s )  [extern]
```

Definition at line 2058 of file names.c.

4.19.6.473 CoCTable()

```
int CoCTable (
            UBYTE * s )  [extern]
```

Definition at line 2068 of file names.c.

4.19.6.474 EmptyTable()

```
void EmptyTable (
            TABLES T )  [extern]
```

Definition at line 2078 of file names.c.

4.19.6.475 CoToTensor()

```
int CoToTensor (
            UBYTE * s )  [extern]
```

Definition at line 2703 of file compcomm.c.

4.19.6.476 CoToVector()

```
int CoToVector (
            UBYTE * s )  [extern]
```

Definition at line 2871 of file compcomm.c.

4.19.6.477 CoTrace4()

```
int CoTrace4 (
            UBYTE * s )  [extern]
```

Definition at line 2941 of file compcomm.c.

4.19.6.478 CoTraceN()

```
int CoTraceN (
            UBYTE * s )  [extern]
```

Definition at line 3028 of file compcomm.c.

4.19.6.479 CoChisholm()

```
int CoChisholm (
            UBYTE * s )  [extern]
```

Definition at line 3088 of file compcomm.c.

4.19.6.480 CoTransform()

```
int CoTransform (
            UBYTE * in )  [extern]
```

Definition at line 87 of file transform.c.

4.19.6.481 CoClearTable()

```
int CoClearTable (
            UBYTE * s )  [extern]
```

Definition at line 5777 of file compcomm.c.

4.19.6.482 DoChain()

```
int DoChain (
            UBYTE * s,
            int option )  [extern]
```

Definition at line 3170 of file compcomm.c.

4.19.6.483 CoChainin()

```
int CoChainin (
            UBYTE * s )  [extern]
```

Definition at line 3214 of file compcomm.c.

4.19.6.484 CoChainout()

```
int CoChainout (
            UBYTE * s )  [extern]
```

Definition at line 3226 of file compcomm.c.

4.19.6.485 CoTryReplace()

```
int CoTryReplace (
            UBYTE * p )  [extern]
```

Definition at line 3450 of file compcomm.c.

4.19.6.486 CoVector()

```
int CoVector (
            UBYTE * s )  [extern]
```

Definition at line 1267 of file names.c.

4.19.6.487 CoWhile()

```
int CoWhile (
            UBYTE * inp )  [extern]
```

Definition at line 4961 of file compcomm.c.

4.19.6.488 CoWrite()

```
int CoWrite (
            UBYTE * s )  [extern]
```

Definition at line 2365 of file compcomm.c.

4.19.6.489 CoAuto()

```
int CoAuto (
            UBYTE * inp )  [extern]
```

Definition at line 2572 of file names.c.

4.19.6.490 CoSwitch()

```
int CoSwitch (
            UBYTE * s )  [extern]
```

Definition at line 7167 of file compcomm.c.

4.19.6.491 CoCase()

```
int CoCase (
            UBYTE * s )  [extern]
```

Definition at line 7213 of file compcomm.c.

4.19.6.492 CoBreak()

```
int CoBreak (
            UBYTE * s )  [extern]
```

Definition at line 7263 of file compcomm.c.

4.19.6.493 CoDefault()

```
int CoDefault (
            UBYTE * s )  [extern]
```

Definition at line 7289 of file compcomm.c.

4.19.6.494 CoEndSwitch()

```
int CoEndSwitch (
            UBYTE * s )  [extern]
```

Definition at line 7316 of file compcomm.c.

4.19.6.495 CoTBaddto()

```
int CoTBaddto (
            UBYTE * s )  [extern]
```

Definition at line 801 of file tables.c.

4.19.6.496 CoTBaudit()

```
int CoTBaudit (
            UBYTE * s )  [extern]
```

Definition at line 1473 of file tables.c.

4.19.6.497 CoTBcleanup()

```
int CoTBcleanup (
            UBYTE * s )  [extern]
```

Definition at line 1575 of file tables.c.

4.19.6.498 CoTBcreate()

```
int CoTBcreate (
            UBYTE * s )  [extern]
```

Definition at line 756 of file tables.c.

4.19.6.499 CoTBenter()

```
int CoTBenter (
              UBYTE * s )  [extern]
```

Definition at line 974 of file tables.c.

4.19.6.500 CoTBhelp()

```
int CoTBhelp (
              UBYTE * s )  [extern]
```

Definition at line 1883 of file tables.c.

4.19.6.501 CoTBload()

```
int CoTBload (
              UBYTE * ss )  [extern]
```

Definition at line 1252 of file tables.c.

4.19.6.502 CoTBoff()

```
int CoTBoff (
              UBYTE * s )  [extern]
```

Definition at line 1544 of file tables.c.

4.19.6.503 CoTBon()

```
int CoTBon (
              UBYTE * s )  [extern]
```

Definition at line 1513 of file tables.c.

4.19.6.504 CoTBopen()

```
int CoTBopen (
              UBYTE * s )  [extern]
```

Definition at line 772 of file tables.c.

4.19.6.505 CoTBreplace()

```
int CoTBreplace (
              UBYTE * s )  [extern]
```

Definition at line 1587 of file tables.c.

4.19.6.506 CoTBuse()

```
int CoTBuse (
            UBYTE * s ) [extern]
```

Definition at line 1602 of file tables.c.

4.19.6.507 CoTestUse()

```
int CoTestUse (
            UBYTE * s ) [extern]
```

Definition at line 1133 of file tables.c.

4.19.6.508 CoThreadBucket()

```
int CoThreadBucket (
            UBYTE * s ) [extern]
```

Definition at line 5677 of file compcomm.c.

4.19.6.509 AddComString()

```
int AddComString (
            int n,
            WORD * array,
            UBYTE * thestring,
            int par ) [extern]
```

Definition at line 1664 of file compcomm.c.

4.19.6.510 CompileAlgebra()

```
int CompileAlgebra (
            UBYTE * s,
            int leftright,
            WORD * prototype ) [extern]
```

Definition at line 518 of file compiler.c.

4.19.6.511 IsIdStatement()

```
int IsIdStatement (
            UBYTE * s ) [extern]
```

Definition at line 504 of file compiler.c.

4.19.6.512 IsRHS()

```
UBYTE * IsRHS (
            UBYTE * s,
            UBYTE c ) [extern]
```

Definition at line 458 of file compiler.c.

4.19.6.513 ParenthesesTest()

```
int ParenthesesTest (
            UBYTE * sin ) [extern]
```

Definition at line 380 of file compiler.c.

4.19.6.514 tokenize()

```
int tokenize (
            UBYTE * in,
            WORD leftright ) [extern]
```

Definition at line 58 of file token.c.

4.19.6.515 WriteTokens()

```
void WriteTokens (
            SBYTE * in ) [extern]
```

Definition at line 657 of file token.c.

4.19.6.516 simp1token()

```
int simp1token (
            SBYTE * s ) [extern]
```

Definition at line 705 of file token.c.

4.19.6.517 simpwtoken()

```
int simpwtoken (
            SBYTE * s ) [extern]
```

Definition at line 794 of file token.c.

4.19.6.518 simp2token()

```
int simp2token (
            SBYTE * s ) [extern]
```

Definition at line 948 of file token.c.

4.19.6.519 simp3atoken()

```
int simp3atoken (
            SBYTE * s,
            int mode ) [extern]
```

Definition at line 1196 of file token.c.

4.19.6.520 simp3btoken()

```
int simp3btoken (
            SBYTE * s,
            int mode ) [extern]
```

Definition at line 1437 of file token.c.

4.19.6.521 simp4token()

```
int simp4token (
            SBYTE * s ) [extern]
```

Definition at line 1822 of file token.c.

4.19.6.522 simp5token()

```
int simp5token (
            SBYTE * s,
            int mode ) [extern]
```

Definition at line 1982 of file token.c.

4.19.6.523 simp6token()

```
int simp6token (
            SBYTE * tokens,
            int mode ) [extern]
```

Definition at line 2028 of file token.c.

4.19.6.524 SkipAName()

```
UBYTE * SkipAName (
            UBYTE * s ) [extern]
```

Definition at line 430 of file compiler.c.

4.19.6.525 TestTables()

```
int TestTables (
            void ) [extern]
```

Definition at line 669 of file compiler.c.

4.19.6.526 GetLabel()

```
int GetLabel (
            UBYTE * name ) [extern]
```

Definition at line 2787 of file names.c.

4.19.6.527 CoIdExpression()

```
int CoIdExpression (
            UBYTE * inp,
            int type ) [extern]
```

Definition at line 507 of file comexpr.c.

4.19.6.528 CoAssign()

```
int CoAssign (
            UBYTE * inp ) [extern]
```

Definition at line 1924 of file comexpr.c.

4.19.6.529 DoExpr()

```
int DoExpr (
            UBYTE * inp,
            int type,
            int par ) [extern]
```

Definition at line 96 of file comexpr.c.

4.19.6.530 CompileSubExpressions()

```
int CompileSubExpressions (
            SBYTE * tokens ) [extern]
```

Definition at line 713 of file compiler.c.

4.19.6.531 CodeGenerator()

```
int CodeGenerator (
            SBYTE * tokens )  [extern]
```

Definition at line 848 of file compiler.c.

4.19.6.532 CompleteTerm()

```
int CompleteTerm (
            WORD * term,
            UWORD * numer,
            UWORD * denom,
            WORD nnum,
            WORD nden,
            int sign )  [extern]
```

Definition at line 1911 of file compiler.c.

4.19.6.533 CodeFactors()

```
int CodeFactors (
            SBYTE * s )  [extern]
```

Definition at line 1948 of file compiler.c.

4.19.6.534 GenerateFactors()

```
WORD GenerateFactors (
            WORD n,
            WORD inc )  [extern]
```

Definition at line 2245 of file compiler.c.

4.19.6.535 InsTree()

```
int InsTree (
            int bufnum,
            int h )  [extern]
```

Definition at line 355 of file comtool.c.

4.19.6.536 FindTree()

```
int FindTree (
            int bufnum,
            WORD * subexpr )  [extern]
```

Definition at line 533 of file comtool.c.

4.19.6.537 RedoTree()

```
void RedoTree (
            CBUF * C,
            int size )  [extern]
```

Definition at line 569 of file comtool.c.

4.19.6.538 ClearTree()

```
void ClearTree (
            int i )  [extern]
```

Definition at line 588 of file comtool.c.

4.19.6.539 CatchDollar()

```
int CatchDollar (
            int par )  [extern]
```

Definition at line 54 of file dollar.c.

4.19.6.540 AssignDollar()

```
int AssignDollar (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 209 of file dollar.c.

4.19.6.541 WriteDollarToBuffer()

```
UBYTE * WriteDollarToBuffer (
            WORD numdollar,
            WORD par )  [extern]
```

Definition at line 596 of file dollar.c.

4.19.6.542 WriteDollarFactorToBuffer()

```
UBYTE * WriteDollarFactorToBuffer (
            WORD numdollar,
            WORD numfac,
            WORD par )  [extern]
```

Definition at line 687 of file dollar.c.

4.19.6.543 AddToDollarBuffer()

```
void AddToDollarBuffer (
            UBYTE * s )  [extern]
```

Definition at line 762 of file dollar.c.

4.19.6.544 PutTermInDollar()

```
int PutTermInDollar (
            WORD * term,
            WORD numdollar )  [extern]
```

Definition at line 863 of file dollar.c.

4.19.6.545 TermAssign()

```
void TermAssign (
            WORD * term )  [extern]
```

Definition at line 803 of file dollar.c.

4.19.6.546 WildDollars()

```
void WildDollars (
            PHEAD WORD * term )  [extern]
```

Definition at line 891 of file dollar.c.

4.19.6.547 numcommute()

```
LONG numcommute (
            WORD * terms,
            LONG * numterms )  [extern]
```

Definition at line 659 of file comtool.c.

4.19.6.548 FullRenumber()

```
int FullRenumber (
            PHEAD WORD * term,
            WORD par )  [extern]
```

Definition at line 324 of file reshuf.c.

4.19.6.549 Lus()

```
int Lus (
            WORD * term,
            WORD funnum,
            WORD loopsize,
            WORD numargs,
            WORD outfun,
            WORD mode ) [extern]
```

Definition at line 57 of file lus.c.

4.19.6.550 FindLus()

```
int FindLus (
            int from,
            int level,
            int openindex ) [extern]
```

Definition at line 515 of file lus.c.

4.19.6.551 CoReplaceLoop()

```
int CoReplaceLoop (
            UBYTE * inp ) [extern]
```

Definition at line 5135 of file compcomm.c.

4.19.6.552 CoFindLoop()

```
int CoFindLoop (
            UBYTE * inp ) [extern]
```

Definition at line 5127 of file compcomm.c.

4.19.6.553 DoFindLoop()

```
int DoFindLoop (
            UBYTE * inp,
            int mode ) [extern]
```

Definition at line 5010 of file compcomm.c.

4.19.6.554 CoFunPowers()

```
int CoFunPowers (
            UBYTE * inp ) [extern]
```

Definition at line 5155 of file compcomm.c.

4.19.6.555 SortTheList()

```
int SortTheList (
            int * slist,
            int num )  [extern]
```

Definition at line 578 of file lus.c.

4.19.6.556 MatchIsPossible()

```
int MatchIsPossible (
            WORD * pattern,
            WORD * term )  [extern]
```

Definition at line 299 of file smart.c.

4.19.6.557 StudyPattern()

```
int StudyPattern (
            WORD * lhs )  [extern]
```

Definition at line 62 of file smart.c.

4.19.6.558 DolToTensor()

```
WORD DolToTensor (
            PHEAD WORD )  [extern]
```

Definition at line 1082 of file dollar.c.

4.19.6.559 DolToFunction()

```
WORD DolToFunction (
            PHEAD WORD )  [extern]
```

Definition at line 1143 of file dollar.c.

4.19.6.560 DolToVector()

```
WORD DolToVector (
            PHEAD WORD )  [extern]
```

Definition at line 1200 of file dollar.c.

4.19.6.561 DolToNumber()

```
WORD DolToNumber (
            PHEAD WORD )  [extern]
```

Definition at line 1264 of file dollar.c.

4.19.6.562 DolToSymbol()

```
WORD DolToSymbol (
             PHEAD WORD )  [extern]
```

Definition at line 1323 of file dollar.c.

4.19.6.563 DolToIndex()

```
WORD DolToIndex (
             PHEAD WORD )  [extern]
```

Definition at line 1377 of file dollar.c.

4.19.6.564 DolToLong()

```
LONG DolToLong (
             PHEAD WORD )  [extern]
```

Definition at line 1606 of file dollar.c.

4.19.6.565 DollarFactorize()

```
int DollarFactorize (
             PHEAD WORD )  [extern]
```

Definition at line 2955 of file dollar.c.

4.19.6.566 CoPrintTable()

```
int CoPrintTable (
             UBYTE * inp )  [extern]
```

Definition at line 1748 of file comexpr.c.

4.19.6.567 CoDeallocateTable()

```
int CoDeallocateTable (
             UBYTE * inp )  [extern]
```

Definition at line 1982 of file comexpr.c.

4.19.6.568 CleanDollarFactors()

```
void CleanDollarFactors (
             DOLLARS d )  [extern]
```

Definition at line 3554 of file dollar.c.

4.19.6.569 TakeDollarContent()

```
WORD * TakeDollarContent (
            PHEAD WORD * dollarbuffer,
            WORD ** factor )  [extern]
```

Definition at line 3575 of file dollar.c.

4.19.6.570 MakeDollarInteger()

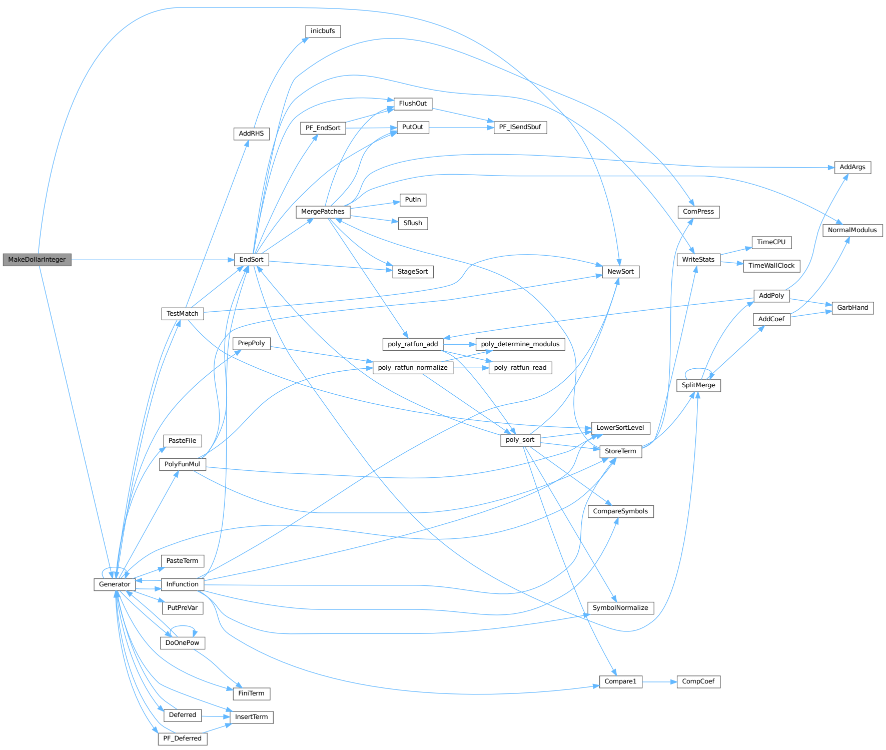
```
WORD * MakeDollarInteger (
            PHEAD WORD * bufin,
            WORD ** bufout )  [extern]
```

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in bufin. The number that comes out is the return value. The normalized argument is in bufout.

Definition at line 3627 of file dollar.c.

References EndSort(), Generator(), and NewSort().

Here is the call graph for this function:



4.19.6.571 MakeDollarMod()

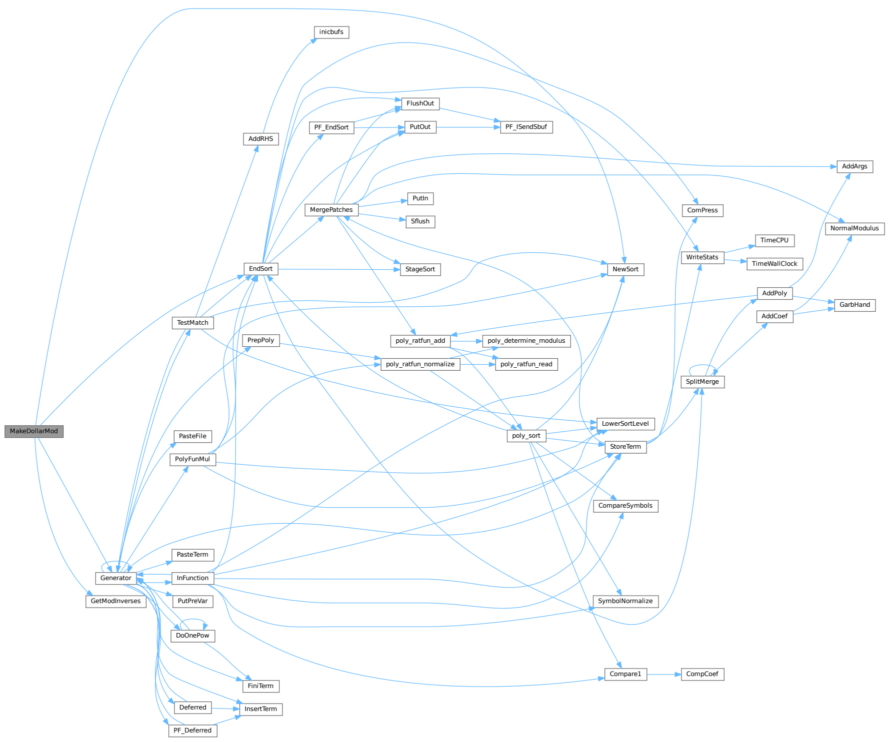
```
WORD * MakeDollarMod (
            PHEAD WORD * buffer,
            WORD ** bufout )  [extern]
```

Similar to MakeDollarInteger but now with modulus arithmetic using only a one WORD 'prime'. We make the coefficient of the first term in the argument equal to one. Already the coefficients are taken modulus AN.cmod and AN.ncmod == 1

Definition at line 3801 of file dollar.c.

References EndSort(), Generator(), GetModInverses(), and NewSort().

Here is the call graph for this function:



4.19.6.572 GetDolNum()

```
int GetDolNum (
            PHEAD WORD * t,
            WORD * tstop )  [extern]
```

Definition at line 3846 of file dollar.c.

4.19.6.573 AddPotModdollar()

```
void AddPotModdollar (
            WORD numdollar )  [extern]
```

Adds a $-variable specified by *numdollar* to the list of potentially modified $-variables unless it has already been included in the list.

Parameters

numdollar	The index of the $-variable to be added.

Definition at line 3959 of file dollar.c.

4.19.6.574 Optimize()

```
int Optimize (
            WORD exprnr,
            int do_print ) [extern]
```

Optimization of expression

4.19.7 Description

This method takes an input expression and generates optimized code to calculate its value. The following methods are called to do so:

(1) get_expression : to read to expression

(2) get_brackets : find brackets for simultaneous optimization

(3) occurrence_order or find_Horner_MCTS : to determine (the) Horner scheme(s) to use; this depends on AO.↩
optimize.horner

(4) optimize_expression_given_Horner : to do the optimizations for each Horner scheme; this method does either CSE or greedy optimizations dependings on AO.optimize.method

(5) generate_output : to format the output in Form notation and store it in a buffer

(6a) optimize_print_code : to print the expression (for "Print") or (6b) generate_expression : to modify the expression (for "#Optimize")

On ParFORM, all the processes must call this function at the same time. Then

(1) Because only the master can access to the expression to be optimized, the master broadcast the expression to all the slaves after reading the expression (PF_get_expression).

(2) get_brackets reads optimize_expr as the input and it works also on the slaves. We leave it although the bracket information is not needed on the slaves (used in (5) on the master).

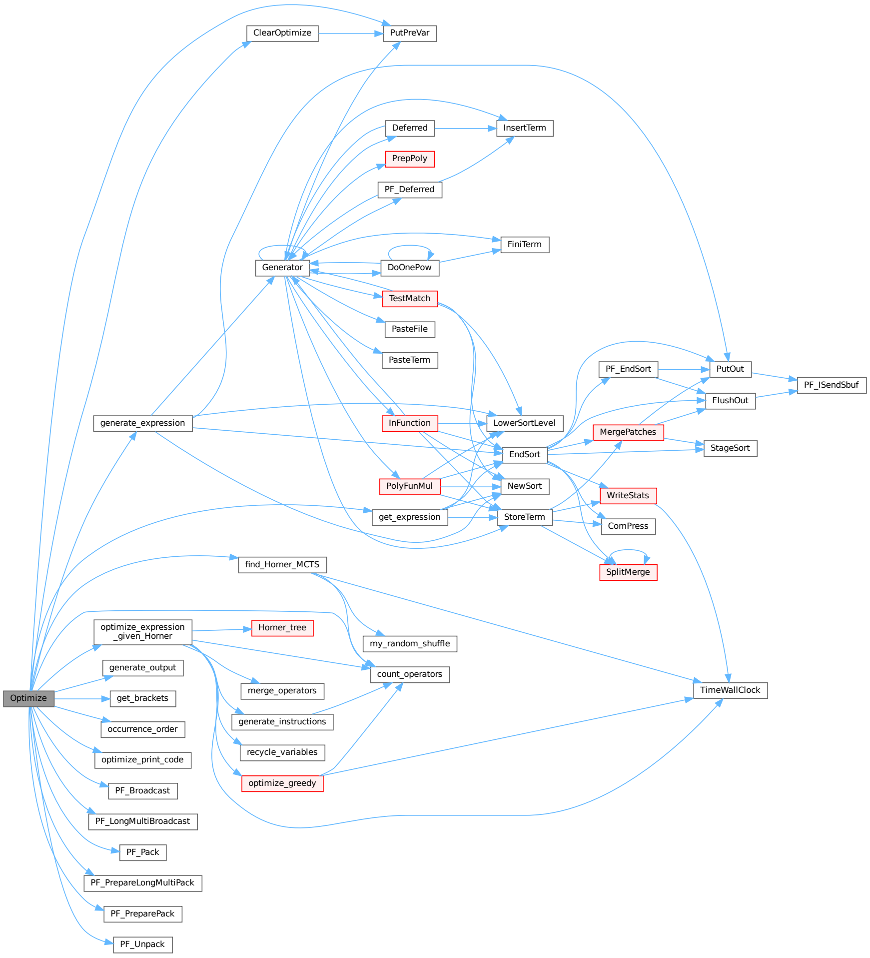
(3) and (4) find_Horner_MCTS and optimize_expression_given_Horner are parallelized.

(5), (6a) and (6b) are needed only on the master.

Definition at line 4638 of file optimize.cc.

References ClearOptimize(), count_operators(), find_Horner_MCTS(), generate_expression(), generate_output(), get_brackets(), get_expression(), occurrence_order(), optimize_expression_given_Horner(), optimize_print_code(), PF_Broadcast(), PF_LongMultiBroadcast(), PF_Pack(), PF_PrepareLongMultiPack(), PF_PreparePack(), PF_Unpack(), and PutPreVar().

Here is the call graph for this function:



4.19.7.1 ClearOptimize()

```
int ClearOptimize (
             void ) [extern]
```

Optimization of expression

4.19.8 Description

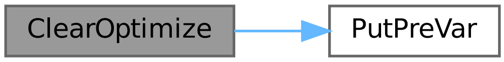
Clears the buffers that were used for optimization output. Clears the expression from the buffers (marks it to be dropped). Note: we need to use the expression by its name, because numbers may change if we drop other expressions between when we do the optimizations and clear the results (in execute.c). Also this is not 100% safe, because we could overwrite the optimized expression. But that can be done only in a Local or Global statement and hence we only have to test there that we might have to call ClearOptimize first. (in file comexpr.c)

Definition at line 4975 of file optimize.cc.

References PutPreVar().

Referenced by Optimize().

Here is the call graph for this function:



4.19.8.1 TakeLongRoot()

```
int TakeLongRoot (
            UWORD * a,
            WORD * n,
            WORD power )  [extern]
```

Definition at line 2735 of file reken.c.

4.19.8.2 TakeRatRoot()

```
int TakeRatRoot (
            UWORD * a,
            WORD * n,
            WORD power )  [extern]
```

Definition at line 680 of file reken.c.

4.19.8.3 MakeRational()

```
int MakeRational (
            WORD a,
            WORD m,
            WORD * b,
            WORD * c )  [extern]
```

Definition at line 2855 of file reken.c.

4.19.8.4 MakeLongRational()

```
int MakeLongRational (
            PHEAD UWORD * a,
            WORD na,
            UWORD * m,
            WORD nm,
            UWORD * b,
            WORD * nb )  [extern]
```

Definition at line 2903 of file reken.c.

4.19.8.5 ClearTableTree()

```
void ClearTableTree (
            TABLES T )  [extern]
```

Definition at line 72 of file tables.c.

4.19.8.6 InsTableTree()

```
int InsTableTree (
            TABLES T,
            WORD * tp )  [extern]
```

Definition at line 103 of file tables.c.

4.19.8.7 RedoTableTree()

```
void RedoTableTree (
            TABLES T,
            int newsize )  [extern]
```

Definition at line 266 of file tables.c.

4.19.8.8 FindTableTree()

```
int FindTableTree (
            TABLES T,
            WORD * tp,
            int inc )  [extern]
```

Definition at line 291 of file tables.c.

4.19.8.9 finishcbuf()

```
void finishcbuf (
            WORD num )  [extern]
```

Frees a compiler buffer.

Parameters

num	The ID number for the buffer to be freed.

Definition at line 89 of file comtool.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::BufferSize, CbUf::CanCommu, CbUf::lhs, CbUf::NumTerms, CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Referenced by PF_BroadcastCBuf().

4.19.8.10 clearcbuf()

```
void clearcbuf (
            WORD num )  [extern]
```

Clears contents in a compiler buffer.

Parameters

num	The ID number for the buffer to be cleared.

Definition at line 116 of file comtool.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::Pointer, and CbUf::rhs.

4.19.8.11 CleanUpSort()

```
void CleanUpSort (
            int num )  [extern]
```

Partially or completely frees function sort buffers.

Definition at line 4845 of file sort.c.

4.19.8.12 AllocFileHandle()

```
FILEHANDLE * AllocFileHandle (
            WORD par,
            char * name )  [extern]
```

Definition at line 1044 of file setfile.c.

4.19.8.13 DeAllocFileHandle()

```
void DeAllocFileHandle (
            FILEHANDLE * fh )  [extern]
```

Definition at line 1105 of file setfile.c.

4.19.8.14 LowerSortLevel()

```
void LowerSortLevel (
            void )  [extern]
```

Lowers the level in the sort system.

Definition at line 4945 of file sort.c.

Referenced by generate_expression(), get_expression(), InFunction(), PF_CollectModifiedDollars(), PF_Processor(), poly_factorize_expression(), poly_sort(), poly_unfactorize_expression(), PolyFunMul(), Processor(), and TestMatch().

4.19.8.15 PolyRatFunSpecial()

```
WORD * PolyRatFunSpecial (
            PHEAD WORD * t1,
            WORD * t2 )  [extern]
```

Definition at line 4962 of file sort.c.

4.19.8.16 SimpleSplitMergeRec()

```
void SimpleSplitMergeRec (
            WORD * array,
            WORD num,
            WORD * auxarray )  [extern]
```

Definition at line 5064 of file sort.c.

4.19.8.17 SimpleSplitMerge()

```
void SimpleSplitMerge (
            WORD * array,
            WORD num )  [extern]
```

Definition at line 5092 of file sort.c.

4.19.8.18 BinarySearch()

```
WORD BinarySearch (
            WORD * array,
            WORD num,
            WORD x )  [extern]
```

Definition at line 5110 of file sort.c.

4.19.8.19 InsideDollar()

```
int InsideDollar (
            PHEAD WORD * ll,
            WORD level )  [extern]
```

Definition at line 1746 of file dollar.c.

4.19.8.20 DolToTerms()

```
DOLLARS DolToTerms (
            PHEAD WORD )  [extern]
```

Definition at line 1458 of file dollar.c.

4.19.8.21 EvalDoLoopArg()

```
WORD EvalDoLoopArg (
            PHEAD WORD * arg,
            WORD par )  [extern]
```

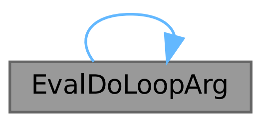
Evaluates one argument of a do loop. Such an argument is constructed from SNUMBERs DOLLAREXPRESSIONs and possibly DOLLAREXPR2s which indicate factors of the preceding dollar. Hence we have SNUMBER,num DOLLAREXPRESSION,numdollar DOLLAREXPRESSION,numdollar,DOLLAREXPR2,numfactor DOLLAREXPRESSION,numdollar,DOLLAREXPR2,numfactor,DOLLAREXPR2,numfactor etc. Because we have a do-loop at every stage we should have a number. The notation in DOLLAREXPR2 is that >= 0 is number of yet another dollar and < 0 is -n-1 with n the array element or zero. The return value is the (short) number. The routine works its way through the list in a recursive manner.

Definition at line 2651 of file dollar.c.

References EvalDoLoopArg().

Referenced by EvalDoLoopArg().

Here is the call graph for this function:



4.19.8.22 SetExprCases()

```
int SetExprCases (
            int par,
            int setunset,
            int val )  [extern]
```

Definition at line 1357 of file compcomm.c.

4.19.8.23 TestSelect()

```
int TestSelect (
            WORD * term,
            WORD * setp )  [extern]
```

Definition at line 1748 of file pattern.c.

4.19.8.24 SubsInAll()

```
void SubsInAll (
            PHEAD0 ) [extern]
```

Definition at line 1982 of file pattern.c.

4.19.8.25 TransferBuffer()

```
void TransferBuffer (
            int from,
            int to,
            int spectator ) [extern]
```

Definition at line 2143 of file pattern.c.

4.19.8.26 TakeIDfunction()

```
int TakeIDfunction (
            PHEAD WORD * term ) [extern]
```

Definition at line 2213 of file pattern.c.

4.19.8.27 MakeSetupAllocs()

```
int MakeSetupAllocs (
            void ) [extern]
```

Definition at line 1122 of file setfile.c.

4.19.8.28 TryFileSetups()

```
int TryFileSetups (
            void ) [extern]
```

Definition at line 1142 of file setfile.c.

4.19.8.29 ExchangeExpressions()

```
void ExchangeExpressions (
            int num1,
            int num2 ) [extern]
```

Definition at line 1670 of file execute.c.

4.19.8.30 ExchangeDollars()

```
void ExchangeDollars (
            int num1,
            int num2 ) [extern]
```

Definition at line 1849 of file dollar.c.

4.19.8.31 GetFirstBracket()

```
int GetFirstBracket (
            WORD * term,
            int num ) [extern]
```

Definition at line 1744 of file execute.c.

4.19.8.32 GetFirstTerm()

```
int GetFirstTerm (
            WORD * term,
            int num,
            int pre ) [extern]
```

Gets the first term of an expression. Routine should be thread safe.

Parameters

out	*term*	Pointer to the buffer where the term should be written.
in	*num*	The expression number from which to get the term.
in	*pre*	Denotes a pre-processor call (1) or not (0). Used to determine the correct source file for the expression.

Returns

First WORD of term, a negative value denotes an error.

Definition at line 1863 of file execute.c.

References FiLe::handle, VaRrEnUm::lo, and ReNuMbEr::symb.

4.19.8.33 GetContent()

```
int GetContent (
            WORD * content,
            int num ) [extern]
```

Definition at line 1971 of file execute.c.

4.19.8.34 CleanupTerm()

```
int CleanupTerm (
            WORD * term )  [extern]
```

Definition at line 2088 of file execute.c.

4.19.8.35 ContentMerge()

```
WORD ContentMerge (
            PHEAD WORD * content,
            WORD * term )  [extern]
```

Definition at line 2112 of file execute.c.

4.19.8.36 PreIfDollarEval()

```
UBYTE * PreIfDollarEval (
            UBYTE * s,
            int * value )  [extern]
```

Definition at line 1978 of file dollar.c.

4.19.8.37 TermsInDollar()

```
LONG TermsInDollar (
            WORD num )  [extern]
```

Definition at line 1867 of file dollar.c.

4.19.8.38 SizeOfDollar()

```
LONG SizeOfDollar (
            WORD num )  [extern]
```

Definition at line 1916 of file dollar.c.

4.19.8.39 TermsInExpression()

```
LONG TermsInExpression (
            WORD num )  [extern]
```

Definition at line 2376 of file execute.c.

4.19.8.40 SizeOfExpression()

```
LONG SizeOfExpression (
            WORD num )  [extern]
```

Definition at line 2388 of file execute.c.

4.19.8.41 TranslateExpression()

```
WORD * TranslateExpression (
            UBYTE * s ) [extern]
```

Definition at line 2160 of file dollar.c.

4.19.8.42 IsSetMember()

```
int IsSetMember (
            WORD * buffer,
            WORD numset ) [extern]
```

Definition at line 2217 of file dollar.c.

4.19.8.43 IsMultipleOf()

```
int IsMultipleOf (
            WORD * buf1,
            WORD * buf2 ) [extern]
```

Definition at line 2398 of file dollar.c.

4.19.8.44 TwoExprCompare()

```
int TwoExprCompare (
            WORD * buf1,
            WORD * buf2,
            int oprtr ) [extern]
```

Definition at line 2473 of file dollar.c.

4.19.8.45 UpdatePositions()

```
void UpdatePositions (
            void ) [extern]
```

Definition at line 2400 of file execute.c.

4.19.8.46 M_print()

```
void M_print (
            void ) [extern]
```

Definition at line 2517 of file tools.c.

4.19.8.47 M_check1()

```
void M_check1 (
            void ) [extern]
```

Definition at line 2516 of file tools.c.

4.19.8.48 PrintTime()

```
void PrintTime (
            UBYTE * mess ) [extern]
```

Definition at line 784 of file sch.c.

4.19.8.49 FindBracket()

```
POSITION * FindBracket (
            WORD nexp,
            WORD * bracket ) [extern]
```

Definition at line 65 of file index.c.

4.19.8.50 PutBracketInIndex()

```
void PutBracketInIndex (
            PHEAD WORD * term,
            POSITION * newpos ) [extern]
```

Definition at line 331 of file index.c.

4.19.8.51 ClearBracketIndex()

```
void ClearBracketIndex (
            WORD numexp ) [extern]
```

Definition at line 546 of file index.c.

4.19.8.52 OpenBracketIndex()

```
void OpenBracketIndex (
            WORD nexpr ) [extern]
```

Definition at line 578 of file index.c.

4.19.8.53 DoNoParallel()

```
int DoNoParallel (
            UBYTE * s ) [extern]
```

Definition at line 529 of file module.c.

4.19.8.54 DoParallel()

```
int DoParallel (
              UBYTE * s )  [extern]
```

Definition at line 544 of file module.c.

4.19.8.55 DoModSum()

```
int DoModSum (
              UBYTE * s )  [extern]
```

Definition at line 559 of file module.c.

4.19.8.56 DoModMax()

```
int DoModMax (
              UBYTE * s )  [extern]
```

Definition at line 579 of file module.c.

4.19.8.57 DoModMin()

```
int DoModMin (
              UBYTE * s )  [extern]
```

Definition at line 599 of file module.c.

4.19.8.58 DoModLocal()

```
int DoModLocal (
              UBYTE * s )  [extern]
```

Definition at line 619 of file module.c.

4.19.8.59 DoModDollar()

```
UBYTE * DoModDollar (
              UBYTE * s,
              int type )  [extern]
```

Definition at line 657 of file module.c.

4.19.8.60 DoProcessBucket()

```
int DoProcessBucket (
              UBYTE * s )  [extern]
```

Definition at line 639 of file module.c.

4.19.8.61 DoinParallel()

```
int DoinParallel (
            UBYTE * s )  [extern]
```

Definition at line 758 of file module.c.

4.19.8.62 DonotinParallel()

```
int DonotinParallel (
            UBYTE * s )  [extern]
```

Definition at line 768 of file module.c.

4.19.8.63 FlipTable()

```
int FlipTable (
            FUNCTIONS f,
            int type )  [extern]
```

Definition at line 671 of file tables.c.

4.19.8.64 ChainIn()

```
int ChainIn (
            PHEAD WORD * term,
            WORD funnum )  [extern]
```

Definition at line 691 of file function.c.

4.19.8.65 ChainOut()

```
int ChainOut (
            PHEAD WORD * term,
            WORD funnum )  [extern]
```

Definition at line 748 of file function.c.

4.19.8.66 ArgumentImplode()

```
int ArgumentImplode (
            PHEAD WORD * term,
            WORD * thelist )  [extern]
```

Definition at line 1845 of file argument.c.

4.19.8.67 ArgumentExplode()

```
int ArgumentExplode (
            PHEAD WORD * term,
            WORD * thelist )  [extern]
```

Definition at line 1949 of file argument.c.

4.19.8.68 DenToFunction()

```
int DenToFunction (
            WORD * term,
            WORD numfun )  [extern]
```

Definition at line 2556 of file wildcard.c.

4.19.8.69 HowMany()

```
WORD HowMany (
            PHEAD WORD * ifcode,
            WORD * term )  [extern]
```

Definition at line 840 of file if.c.

4.19.8.70 RemoveDollars()

```
void RemoveDollars (
            void  )  [extern]
```

Definition at line 3128 of file names.c.

4.19.8.71 CountTerms1()

```
LONG CountTerms1 (
            PHEAD0  )  [extern]
```

Definition at line 2461 of file execute.c.

4.19.8.72 TermsInBracket()

```
LONG TermsInBracket (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 2569 of file execute.c.

4.19.8.73 Crash()

```
int Crash (
            void ) [extern]
```

Definition at line 3834 of file tools.c.

4.19.8.74 str_dup()

```
char * str_dup (
            char * str ) [extern]
```

Definition at line 101 of file minos.c.

4.19.8.75 convertblock()

```
void convertblock (
            INDEXBLOCK * in,
            INDEXBLOCK * out,
            int mode ) [extern]
```

Definition at line 120 of file minos.c.

4.19.8.76 convertnamesblock()

```
void convertnamesblock (
            NAMESBLOCK * in,
            NAMESBLOCK * out,
            int mode ) [extern]
```

Definition at line 171 of file minos.c.

4.19.8.77 convertiniinfo()

```
void convertiniinfo (
            INIINFO * in,
            INIINFO * out,
            int mode ) [extern]
```

Definition at line 197 of file minos.c.

4.19.8.78 ReadIndex()

```
int ReadIndex (
            DBASE * d ) [extern]
```

Definition at line 288 of file minos.c.

4.19.8.79 WriteIndexBlock()

```
int WriteIndexBlock (
            DBASE * d,
            MLONG num )  [extern]
```

Definition at line 401 of file minos.c.

4.19.8.80 WriteNamesBlock()

```
int WriteNamesBlock (
            DBASE * d,
            MLONG num )  [extern]
```

Definition at line 422 of file minos.c.

4.19.8.81 WriteIndex()

```
int WriteIndex (
            DBASE * d )  [extern]
```

Definition at line 445 of file minos.c.

4.19.8.82 WriteIniInfo()

```
int WriteIniInfo (
            DBASE * d )  [extern]
```

Definition at line 508 of file minos.c.

4.19.8.83 ReadIniInfo()

```
int ReadIniInfo (
            DBASE * d )  [extern]
```

Definition at line 525 of file minos.c.

4.19.8.84 AddToIndex()

```
int AddToIndex (
            DBASE * d,
            MLONG number )  [extern]
```

Definition at line 1067 of file minos.c.

4.19.8.85 GetDbase()

```
DBASE * GetDbase (
            char * filename,
            MLONG rwmode )  [extern]
```

Definition at line 548 of file minos.c.

4.19.8.86 OpenDbase()

```
DBASE * OpenDbase (
            char * filename )  [extern]
```

Definition at line 822 of file minos.c.

4.19.8.87 ReadObject()

```
char * ReadObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments )  [extern]
```

Definition at line 1296 of file minos.c.

4.19.8.88 ReadijObject()

```
char * ReadijObject (
            DBASE * d,
            MLONG i,
            MLONG j,
            char * arguments )  [extern]
```

Definition at line 1377 of file minos.c.

4.19.8.89 ExistsObject()

```
int ExistsObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments )  [extern]
```

Definition at line 1429 of file minos.c.

4.19.8.90 DeleteObject()

```
int DeleteObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments )  [extern]
```

Definition at line 1461 of file minos.c.

4.19.8.91 WriteObject()

```
int WriteObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments,
            char * rhs,
            MLONG number ) [extern]
```

Definition at line 1196 of file minos.c.

4.19.8.92 AddObject()

```
MLONG AddObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments,
            char * rhs ) [extern]
```

Definition at line 1154 of file minos.c.

4.19.8.93 NewDbase()

```
DBASE * NewDbase (
            char * name,
            MLONG number ) [extern]
```

Definition at line 604 of file minos.c.

4.19.8.94 FreeTableBase()

```
void FreeTableBase (
            DBASE * d ) [extern]
```

Definition at line 741 of file minos.c.

4.19.8.95 ComposeTableNames()

```
int ComposeTableNames (
            DBASE * d ) [extern]
```

Definition at line 773 of file minos.c.

4.19.8.96 PutTableNames()

```
int PutTableNames (
            DBASE * d ) [extern]
```

Definition at line 971 of file minos.c.

4.19.8.97 AddTableName()

```
MLONG AddTableName (
            DBASE * d,
            char * name,
            TABLES T ) [extern]
```

Definition at line 856 of file minos.c.

4.19.8.98 GetTableName()

```
MLONG GetTableName (
            DBASE * d,
            char * name ) [extern]
```

Definition at line 939 of file minos.c.

4.19.8.99 FindTableNumber()

```
MLONG FindTableNumber (
            DBASE * d,
            char * name ) [extern]
```

Definition at line 1168 of file minos.c.

4.19.8.100 TryEnvironment()

```
int TryEnvironment (
            void ) [extern]
```

Definition at line 1228 of file setfile.c.

4.19.8.101 CopyExpression()

```
int CopyExpression (
            FILEHANDLE * from,
            FILEHANDLE * to ) [extern]
```

Definition at line 3308 of file store.c.

4.19.8.102 set_in()

```
int set_in (
            UBYTE ch,
            set_of_char set ) [extern]
```

Definition at line 2112 of file tools.c.

4.19.8.103 set_set()

```
one_byte set_set (
            UBYTE ch,
            set_of_char set )  [extern]
```

Definition at line 2132 of file tools.c.

4.19.8.104 set_del()

```
one_byte set_del (
            UBYTE ch,
            set_of_char set )  [extern]
```

Definition at line 2153 of file tools.c.

4.19.8.105 set_sub()

```
one_byte set_sub (
            set_of_char set,
            set_of_char set1,
            set_of_char set2 )  [extern]
```

Definition at line 2175 of file tools.c.

4.19.8.106 DoPreAddSeparator()

```
int DoPreAddSeparator (
            UBYTE * s )  [extern]
```

Definition at line 5382 of file pre.c.

4.19.8.107 DoPreRmSeparator()

```
int DoPreRmSeparator (
            UBYTE * s )  [extern]
```

Definition at line 5407 of file pre.c.

4.19.8.108 openExternalChannel()

```
int openExternalChannel (
            UBYTE * cmd,
            int daemonize,
            UBYTE * shellname,
            UBYTE * stderrname )  [extern]
```

Definition at line 230 of file extcmd.c.

4.19.8.109 initPresetExternalChannels()

```
int initPresetExternalChannels (
            UBYTE * theline,
            int thetimeout )  [extern]
```

Definition at line 232 of file extcmd.c.

4.19.8.110 closeExternalChannel()

```
int closeExternalChannel (
            int n )  [extern]
```

Definition at line 233 of file extcmd.c.

4.19.8.111 selectExternalChannel()

```
int selectExternalChannel (
            int n )  [extern]
```

Definition at line 234 of file extcmd.c.

4.19.8.112 getCurrentExternalChannel()

```
int getCurrentExternalChannel (
            void  )  [extern]
```

Definition at line 235 of file extcmd.c.

4.19.8.113 closeAllExternalChannels()

```
void closeAllExternalChannels (
            void  )  [extern]
```

Definition at line 236 of file extcmd.c.

4.19.8.114 defineChannel()

```
UBYTE * defineChannel (
            UBYTE * s,
            HANDLERS * h )  [extern]
```

Definition at line 6033 of file pre.c.

4.19.8.115 writeToChannel()

```
int writeToChannel (
            int wtype,
            UBYTE * s,
            HANDLERS * h )  [extern]
```

Definition at line 6068 of file pre.c.

4.19.8.116 ReleaseTB()

```
int ReleaseTB (
            void )  [extern]
```

Definition at line 2319 of file tables.c.

4.19.8.117 SymbolNormalize()

```
int SymbolNormalize (
            WORD * term )  [extern]
```

Routine normalizes terms that contain only symbols. Regular minimum and maximum properties are ignored.

We check whether there are negative powers in the output. This is not allowed.

Definition at line 5144 of file normal.c.

Referenced by InFunction(), and poly_sort().

4.19.8.118 TestFunFlag()

```
int TestFunFlag (
            PHEAD WORD * tfun )  [extern]
```

Definition at line 5240 of file normal.c.

4.19.8.119 CompareSymbols()

```
WORD CompareSymbols (
            PHEAD WORD * term1,
            WORD * term2,
            WORD par )  [extern]
```

Compares the terms, based on the value of AN.polysortflag. If term1 $<$ term2 the return value is -1 If term1 $>$ term2 the return value is 1 If term1 = term2 the return value is 0 The coefficients may differ. The terms contain only a single subterm of type SYMBOL. If AN.polysortflag = 0 it is a 'regular' compare. If AN.polysortflag = 1 the sum of the powers is more important par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in sort.c

Definition at line 3100 of file sort.c.

Referenced by InFunction(), and poly_sort().

4.19.8.120 CompareHSymbols()

```
WORD CompareHSymbols (
            PHEAD WORD * term1,
            WORD * term2,
            WORD par )  [extern]
```

Compares terms that can have only SYMBOL and HAAKJE subterms. If term1 < term2 the return value is -1 If term1 > term2 the return value is 1 If term1 = term2 the return value is 0 par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in sort.c

Definition at line 3143 of file sort.c.

4.19.8.121 NextPrime()

```
WORD NextPrime (
            PHEAD WORD num )  [extern]
```

Gives the next prime number in the list of prime numbers.

If the list isn't long enough we expand it. For ease in ParForm and because these lists shouldn't be very big we let each worker keep its own list.

The list is cut off at MAXPOWER, because we don't want to get into trouble that the power of a variable gets larger than the prime number.

Definition at line 3664 of file reken.c.

4.19.8.122 Moebius()

```
WORD Moebius (
            PHEAD WORD )  [extern]
```

Definition at line 3728 of file reken.c.

4.19.8.123 wranf()

```
UWORD wranf (
            PHEAD0 )  [extern]
```

Definition at line 3868 of file reken.c.

4.19.8.124 iranf()

```
UWORD iranf (
            PHEAD UWORD )  [extern]
```

Definition at line 3887 of file reken.c.

4.19.8.125 iniwranf()

```
void iniwranf (
            PHEAD0 )  [extern]
```

Definition at line 3819 of file reken.c.

4.19.8.126 PreRandom()

```
UBYTE * PreRandom (
            UBYTE * s )  [extern]
```

Definition at line 3912 of file reken.c.

4.19.8.127 TreatPolyRatFun()

```
int TreatPolyRatFun (
            PHEAD WORD * prf )  [extern]
```

Definition at line 4960 of file normal.c.

4.19.8.128 ReadSaveHeader()

```
WORD ReadSaveHeader (
            void )  [extern]
```

Reads the header in the save file and sets function pointers and flags according to the information found there. Must be called before any other ReadSave... function.

Currently works only for the exchange between 32bit and 64bit machines (WORD size must be 2 or 4 bytes)!

It is called by CoLoad().

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4065 of file store.c.

4.19.8.129 ReadSaveIndex()

```
WORD ReadSaveIndex (
            FILEINDEX * fileind )  [extern]
```

Reads a FILEINDEX from the open save file specified by AO.SaveData.Handle. Translations for adjusting endianness and data sizes are done if necessary.

Depends on the assumption that sizeof(FILEINDEX) is the same everywhere. If FILEINDEX or INDEXENTRY change, then this functions has to be adjusted.

Called by CoLoad() and FindInIndex().

Parameters

fileind	contains the read FILEINDEX after successful return. must point to allocated, big enough memory.

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4183 of file store.c.

References INFILEINDEX, FiLeInDeX::next, FiLeInDeX::number, and FuNcTiOn::number.

4.19.8.130 ReadSaveExpression()

```
WORD ReadSaveExpression (
            UBYTE * buffer,
            UBYTE * top,
            LONG * size,
            LONG * outsize )  [extern]
```

Reads an expression from the open file specified by AO.SaveData.Handle. The endianness flip and a resizing without renumbering is done in this function. Thereafter the buffer consists of chunks with a uniform maximal word size (32bit at the moment). The actual renumbering is then done by calling the function ReadSaveTerm32(). The result is returned in *buffer*.

If the translation at some point doesn't fit into the buffer anymore, the function returns and must be called again. In any case *size* returns the number of successfully read bytes, *outsize* returns the number of successfully written bytes, and the file will be positioned at the next byte after the successfully read data.

It is called by PutInStore().

Parameters

buffer	output buffer, holds the (translated) expression
top	end of buffer
size	number of read bytes
outsize	number of written bytes

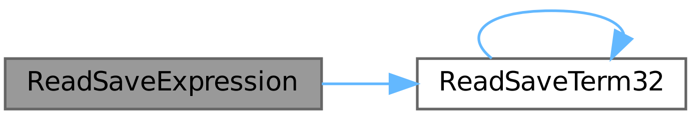
Returns

= 0 everything okay, != 0 an error occurred

Definition at line 5146 of file store.c.

References ReadSaveTerm32().

Here is the call graph for this function:



4.19.8.131 ReadSaveTerm32()

```
UBYTE * ReadSaveTerm32 (
            UBYTE * bin,
            UBYTE * binend,
            UBYTE ** bout,
            UBYTE * boutend,
            UBYTE * top,
            int terminbuf )  [extern]
```

Reads a single term from the given buffer at *bin* and write the translated term back to this buffer at *bout*.

ReadSaveTerm32() is currently the only instantiation of a ReadSaveTerm-function. It only deals with data that already has the correct endianness and that is resized to 32bit words but without being renumbered or translated in any other way. It uses the compress buffer AR.CompressBuffer.

The function is reentrant in order to cope with nested function arguments. It is called by ReadSaveExpression() and itself.

The *return value* indicates the position in the input buffer up to which the data has already been successfully processed. The parameter *bout* returns the corresponding position in the output buffer.

Parameters

bin	start of the input buffer
binend	end of the input buffer
bout	as input points to the beginning of the output buffer, as output points behind the already translated data in the output buffer
boutend	end of already decompressed data in output buffer
top	end of output buffer
terminbuf	flag whether decompressed data is already in the output buffer. used in recursive calls

Returns

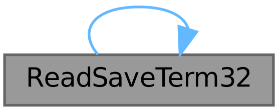
pointer to the next unprocessed data in the input buffer

Definition at line 4736 of file store.c.

References ReadSaveTerm32().

Referenced by ReadSaveExpression(), and ReadSaveTerm32().

Here is the call graph for this function:



4.19.8.132 ReadSaveVariables()

```
WORD ReadSaveVariables (
            UBYTE * buffer,
            UBYTE * top,
            LONG * size,
            LONG * outsize,
            INDEXENTRY * ind,
            LONG * stage )  [extern]
```

Reads the variables from the open file specified by AO.SaveData.Handle. It reads the ∗size bytes and writes them to the ∗buffer. It is called by PutInStore().

If translation is necessary, the data might shrink or grow in size, then ∗*size* is adjusted so that the reading and writing fits into the memory from the buffer to the top. The actual number of read bytes is returned in ∗*size*, the number of written bytes is returned in ∗*outsize*.

If the ∗size is smaller than the actual size of the variables, this function will be called several times and needs to remember the current position in the variable structure. The parameter *stage* does this job. When ReadSaveVariables() is called for the first time, this parameter should have the value -1.

The parameter *ind* is used to get the number of variables.

Parameters

buffer	read variables are written into this allocated memory
top	upper end of allocated memory
size	number of bytes to read. might return a smaller number of read bytes if translation was necessary
outsize	if translation has be done, outsize contains the number of written bytes
ind	pointer of INDEXENTRY for the current expression. read-only
stage	should be -1 for the first call, will be increased by ReadSaveVariables to memorize the position in the variable structure

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4369 of file store.c.

References InDeXeNtRy::nfunctions, InDeXeNtRy::nindices, InDeXeNtRy::nsymbols, InDeXeNtRy::nvectors, and FuNcTiOn::spec.

4.19.8.133 WriteStoreHeader()

```
WORD WriteStoreHeader (
            WORD handle )  [extern]
```

Writes header with information about system architecture and FORM revision to an open store file.

Called by SetFileIndex().

Parameters

handle	specifies open file to which header will be written

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 3972 of file store.c.

References STOREHEADER::endianness, STOREHEADER::lenLONG, STOREHEADER::lenPOINTER, STOREHEADER::lenPOS, STOREHEADER::lenWORD, STOREHEADER::maxpower, STOREHEADER::sFun, STOREHEADER::sInd, STOREHEADER::sSym, STOREHEADER::sVec, and STOREHEADER::wildoffset.

Referenced by SetFileIndex().

4.19.8.134 InitRecovery()

```
void InitRecovery (
            void )  [extern]
```

Sets up the strings for the filenames of the recovery files. This functions should only be called once to avoid memory leaks and after AM.TempDir has been initialized.

Definition at line 399 of file checkpoint.c.

4.19.8.135 CheckRecoveryFile()

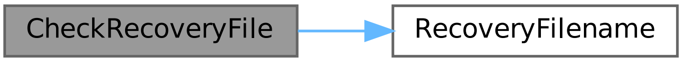
```
int CheckRecoveryFile (
            void )  [extern]
```

Checks whether a snapshot/recovery file exists. Returns 1 if it exists, 0 otherwise.

Definition at line 278 of file checkpoint.c.

References RecoveryFilename().

Here is the call graph for this function:



4.19.8.136 DeleteRecoveryFile()

```
void DeleteRecoveryFile (
            void ) [extern]
```

Deletes the recovery files. It is called by CleanUp() in the case of a successful completion.

Definition at line 333 of file checkpoint.c.

4.19.8.137 RecoveryFilename()

```
char * RecoveryFilename (
            void ) [extern]
```

Returns pointer to recovery filename.

Definition at line 364 of file checkpoint.c.

Referenced by CheckRecoveryFile(), and DoRecovery().

4.19.8.138 DoRecovery()

```
int DoRecovery (
            int * moduletype ) [extern]
```

Reads from the recovery file and restores all necessary variables and states in FORM, so that the execution can recommence in preprocessor() as if no restart of FORM had occurred.

The recovery file is read into memory as a whole. The pointer p then points into this memory at the next non-processed data. The macros by which variables are restored, like R_SET, automatically increase p appropriately.

If something goes wrong, the function returns with a non-zero value.

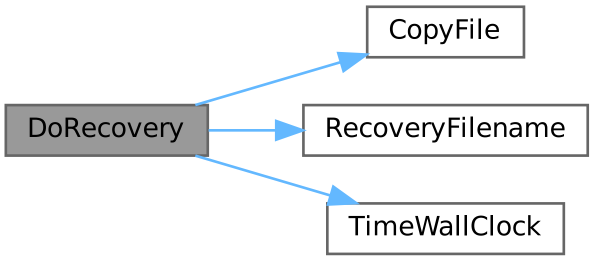
Allocated memory that would be lost when overwriting the global structs with data from the file is freed first. A major part of the code deals with the restoration of pointers. The idiom we use is to memorize the original pointer value (org), allocate new memory and copy the data from the file into this memory, calculate the offset between the old pointer value and the new allocated memory position (ofs), and then correct all affected pointers (+=ofs).

We rely on the fact that several variables (especially in AM) are already assigned the correct values by the startup functions. That means, in principle, that a change in the setup files between snapshot creation and recovery will be noticed.

Definition at line 1402 of file checkpoint.c.

References TaBlEs::argtail, TaBlEs::boomlijst, BrAcKeTiNfO::bracketbuffer, TaBlEs::buffers, TaBlEs::bufferssize, CopyFile(), TaBlEs::flags, ReNuMbEr::func, ReNuMbEr::funnum, VaRrEnUm::hi, BrAcKeTiNfO::indexbuffer, ReNuMbEr::indi, ReNuMbEr::indnum, VaRrEnUm::lo, TaBlEs::MaxTreeSize, TaBlEs::mm, TaBlEs::numind, TaBlEs::pattern, TaBlEs::prototype, TaBlEs::prototypeSize, RecoveryFilename(), ExPrEsSiOn::renumlists, TaBlEs::reserved, TaBlEs::spare, TaBlEs::sparse, VaRrEnUm::start, ReNuMbEr::symb, ReNuMbEr::symnum, TaBlEs::tablepointers, TimeWallClock(), TaBlEs::totind, ReNuMbEr::vecnum, and ReNuMbEr::vect.

Here is the call graph for this function:



4.19.8.139 DoCheckpoint()

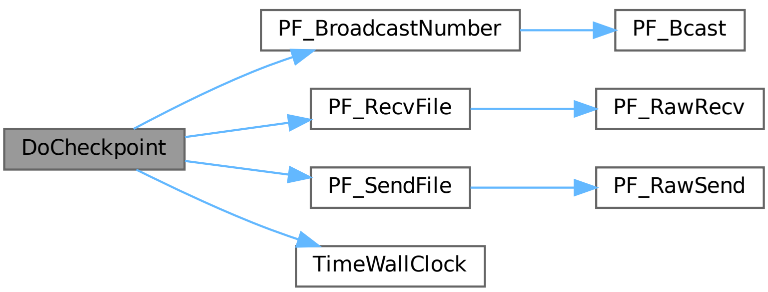
```
void DoCheckpoint (
            int moduletype )  [extern]
```

Checks whether a snapshot should be done. Calls DoSnapshot() to create the snapshot.

Definition at line 3111 of file checkpoint.c.

References PF_BroadcastNumber(), PF_RecvFile(), PF_SendFile(), and TimeWallClock().

Here is the call graph for this function:



4.19.8.140 NumberMallocAddMemory()

```
void NumberMallocAddMemory (
            PHEAD0 )  [extern]
```

Definition at line 2654 of file tools.c.

4.19.8.141 CacheNumberMallocAddMemory()

```
void CacheNumberMallocAddMemory (
            PHEAD0 )  [extern]
```

Definition at line 2728 of file tools.c.

4.19.8.142 TermMallocAddMemory()

```
void TermMallocAddMemory (
            PHEAD0 )  [extern]
```

Definition at line 2550 of file tools.c.

4.19.8.143 ExprStatus()

```
void ExprStatus (
            EXPRESSIONS e )  [extern]
```

Definition at line 66 of file bugtool.c.

4.19.8.144 iniTools()

```
void iniTools (
            void )  [extern]
```

Definition at line 2201 of file tools.c.

4.19.8.145 TestTerm()

```
int TestTerm (
            WORD * term )  [extern]
```

Tests the consistency of the term. Returns 0 when the term is OK. Any nonzero value is trouble. In the current version the testing isn't 100% complete. For instance, we don't check the validity of the symbols nor do we check the range of their powers. Etc. This should be extended when the need is there.

Parameters

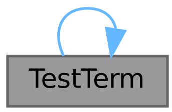
term	the term to be tested

Definition at line 3862 of file tools.c.

References TestTerm().

Referenced by TestTerm().

Here is the call graph for this function:



4.19.8.146 RunTransform()

```
WORD RunTransform (
            PHEAD WORD * term,
            WORD * params )  [extern]
```

Definition at line 732 of file transform.c.

4.19.8.147 RunEncode()

```
WORD RunEncode (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )  [extern]
```

Definition at line 982 of file transform.c.

4.19.8.148 RunDecode()

```
WORD RunDecode (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )  [extern]
```

Definition at line 1169 of file transform.c.

4.19.8.149 RunReplace()

```
WORD RunReplace (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )  [extern]
```

Definition at line 1339 of file transform.c.

4.19.8.150 RunImplode()

```
WORD RunImplode (
            WORD * fun,
            WORD * args )  [extern]
```

Definition at line 1817 of file transform.c.

4.19.8.151 RunExplode()

```
WORD RunExplode (
            PHEAD WORD * fun,
            WORD * args )  [extern]
```

Definition at line 2018 of file transform.c.

4.19.8.152 TestArgNum()

```
int TestArgNum (
            int n,
            int totarg,
            WORD * args )  [extern]
```

Definition at line 3281 of file transform.c.

4.19.8.153 PutArgInScratch()

```
WORD PutArgInScratch (
            WORD * arg,
            UWORD * scrat )  [extern]
```

Definition at line 3350 of file transform.c.

4.19.8.154 ReadRange()

```
UBYTE * ReadRange (
            UBYTE * s,
            WORD * out,
            int par )  [extern]
```

Definition at line 3386 of file transform.c.

4.19.8.155 FindRange()

```
int FindRange (
            PHEAD WORD * args,
            WORD * arg1,
            WORD * arg2,
            WORD totarg )  [extern]
```

Definition at line 3546 of file transform.c.

4.19.8.156 RunPermute()

```
WORD RunPermute (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info ) [extern]
```

Definition at line 2132 of file transform.c.

4.19.8.157 RunReverse()

```
WORD RunReverse (
            PHEAD WORD * fun,
            WORD * args ) [extern]
```

Definition at line 2359 of file transform.c.

4.19.8.158 RunCycle()

```
WORD RunCycle (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info ) [extern]
```

Definition at line 2535 of file transform.c.

4.19.8.159 RunAddArg()

```
WORD RunAddArg (
            PHEAD WORD * fun,
            WORD * args ) [extern]
```

Definition at line 2687 of file transform.c.

4.19.8.160 RunMulArg()

```
WORD RunMulArg (
            PHEAD WORD * fun,
            WORD * args ) [extern]
```

Definition at line 2776 of file transform.c.

4.19.8.161 RunIsLyndon()

```
WORD RunIsLyndon (
            PHEAD WORD * fun,
            WORD * args,
            int par ) [extern]
```

Definition at line 2917 of file transform.c.

4.19.8.162 RunToLyndon()

```
WORD RunToLyndon (
            PHEAD WORD * fun,
            WORD * args,
            int par )  [extern]
```

Definition at line 2995 of file transform.c.

4.19.8.163 RunDropArg()

```
WORD RunDropArg (
            PHEAD WORD * fun,
            WORD * args )  [extern]
```

Definition at line 3107 of file transform.c.

4.19.8.164 RunSelectArg()

```
WORD RunSelectArg (
            PHEAD WORD * fun,
            WORD * args )  [extern]
```

Definition at line 3133 of file transform.c.

4.19.8.165 RunDedup()

```
WORD RunDedup (
            PHEAD WORD * fun,
            WORD * args )  [extern]
```

Definition at line 2446 of file transform.c.

4.19.8.166 RunZtoHArg()

```
WORD RunZtoHArg (
            PHEAD WORD * fun,
            WORD * args )  [extern]
```

Definition at line 3161 of file transform.c.

4.19.8.167 RunHtoZArg()

```
WORD RunHtoZArg (
            PHEAD WORD * fun,
            WORD * args )  [extern]
```

Definition at line 3216 of file transform.c.

4.19.8.168 NormPolyTerm()

```
int NormPolyTerm (
            PHEAD WORD * term )  [extern]
```

Definition at line 53 of file notation.c.

4.19.8.169 ConvertToPoly()

```
int ConvertToPoly (
            PHEAD WORD * term,
            WORD * outterm,
            WORD * comlist,
            WORD par )  [extern]
```

Definition at line 307 of file notation.c.

4.19.8.170 LocalConvertToPoly()

```
int LocalConvertToPoly (
            PHEAD WORD * term,
            WORD * outterm,
            WORD startebuf,
            WORD par )  [extern]
```

Converts a generic term to polynomial notation in which there are only symbols and brackets. During conversion there will be only symbols. Brackets are stripped. Objects that need 'translation' are put inside a special compiler buffer and represented by a symbol. The numbering of the extra symbols is down from the maximum. In principle there can be a problem when running into the already assigned ones. This uses the FindTree for searching in the global tree and then looks further in the AT.ebufnum. This allows fully parallel processing. Hence we need no locks. Cannot be used in the same module as ConvertToPoly.

Definition at line 510 of file notation.c.

Referenced by poly_factorize_expression().

4.19.8.171 ConvertFromPoly()

```
int ConvertFromPoly (
            PHEAD WORD * term,
            WORD * outterm,
            WORD from,
            WORD to,
            WORD offset,
            WORD par )  [extern]
```

Definition at line 660 of file notation.c.

4.19.8.172 FindSubterm()

```
WORD FindSubterm (
            WORD * subterm )  [extern]
```

Definition at line 773 of file notation.c.

4.19.8.173 FindLocalSubterm()

```
WORD FindLocalSubterm (
            PHEAD WORD * subterm,
            WORD startebuf )  [extern]
```

Definition at line 842 of file notation.c.

4.19.8.174 PrintSubtermList()

```
void PrintSubtermList (
            int from,
            int to )  [extern]
```

Definition at line 895 of file notation.c.

4.19.8.175 PrintExtraSymbol()

```
void PrintExtraSymbol (
            int num,
            WORD * terms,
            int par )  [extern]
```

Definition at line 1007 of file notation.c.

4.19.8.176 FindSubexpression()

```
WORD FindSubexpression (
            WORD * subexpr )  [extern]
```

Definition at line 1094 of file notation.c.

4.19.8.177 UpdateMaxSize()

```
void UpdateMaxSize (
            void  )  [extern]
```

Definition at line 1610 of file tools.c.

4.19.8.178 CoToPolynomial()

```
int CoToPolynomial (
            UBYTE * inp )  [extern]
```

Definition at line 5925 of file compcomm.c.

4.19.8.179 CoFromPolynomial()

```
int CoFromPolynomial (
            UBYTE * inp )  [extern]
```

Definition at line 6000 of file compcomm.c.

4.19.8.180 CoArgToExtraSymbol()

```
int CoArgToExtraSymbol (
            UBYTE * s )  [extern]
```

Definition at line 6026 of file compcomm.c.

4.19.8.181 CoExtraSymbols()

```
int CoExtraSymbols (
            UBYTE * inp )  [extern]
```

Definition at line 6076 of file compcomm.c.

4.19.8.182 GetDoParam()

```
UBYTE * GetDoParam (
            UBYTE * inp,
            WORD ** wp,
            int par )  [extern]
```

Definition at line 6207 of file compcomm.c.

4.19.8.183 GetIfDollarFactor()

```
WORD * GetIfDollarFactor (
            UBYTE ** inp,
            WORD * w )  [extern]
```

Definition at line 6149 of file compcomm.c.

4.19.8.184 CoDo()

```
int CoDo (
            UBYTE * inp )  [extern]
```

Definition at line 6284 of file compcomm.c.

4.19.8.185 CoEndDo()

```
int CoEndDo (
            UBYTE * inp )  [extern]
```

Definition at line 6387 of file compcomm.c.

4.19.8.186 ExtraSymFun()

```
int ExtraSymFun (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 1146 of file notation.c.

4.19.8.187 PruneExtraSymbols()

```
int PruneExtraSymbols (
            WORD downto )  [extern]
```

Definition at line 1214 of file notation.c.

4.19.8.188 IniFbuffer()

```
int IniFbuffer (
            WORD bufnum )  [extern]
```

Initialize a factorization cache buffer. We set the size of the rhs and boomlijst buffers immediately to their final values.

Definition at line 614 of file comtool.c.

References tree::blnce, CbUf::boomlijst, CbUf::CanCommu, CbUf::dimension, tree::left, CbUf::numdum, CbUf::NumTerms, tree::parent, CbUf::rhs, tree::right, tree::usage, and tree::value.

4.19.8.189 GCDfunction()

```
int GCDfunction (
            PHEAD WORD * term,
            WORD level )  [extern]
```

Definition at line 609 of file ratio.c.

4.19.8.190 GCDfunction3()

```
WORD * GCDfunction3 (
            PHEAD WORD * in1,
            WORD * in2 )  [extern]
```

Definition at line 1141 of file ratio.c.

4.19.8.191 ReadPolyRatFun()

```
int ReadPolyRatFun (
            PHEAD WORD * term )  [extern]
```

Definition at line 2264 of file ratio.c.

4.19.8.192 FromPolyRatFun()

```
int FromPolyRatFun (
            PHEAD WORD * fun,
            WORD ** numout,
            WORD ** denout )  [extern]
```

Definition at line 2383 of file ratio.c.

4.19.8.193 PolyDiv()

```
WORD * PolyDiv (
            PHEAD WORD * a,
            WORD * b,
            char * text )  [extern]
```

Definition at line 2741 of file ratio.c.

4.19.8.194 GCDclean()

```
void GCDclean (
            PHEAD WORD * num,
            WORD * den )  [extern]
```

Definition at line 2644 of file ratio.c.

4.19.8.195 TakeSymbolContent()

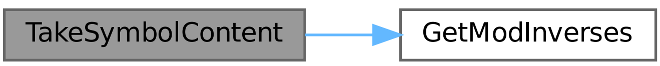
```
WORD * TakeSymbolContent (
            PHEAD WORD * in,
            WORD * term )  [extern]
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. We allow only symbols as this code is used for the polyratfun only. We have a special routine, because the generic TakeContent does too much work and speed is at a premium here. Input: in is the input expression as a sequence of terms. Output: term: the content return value: the contentfree expression. it is in new allocation, made by TermMalloc. (should be in a TermMalloc space?)

Definition at line 2434 of file ratio.c.

References GetModInverses().

Here is the call graph for this function:



4.19.8.196 GCDterms()

```
int GCDterms (
            PHEAD WORD * term1,
            WORD * term2,
            WORD * termout ) [extern]
```

Definition at line 2076 of file ratio.c.

4.19.8.197 PutExtraSymbols()

```
WORD * PutExtraSymbols (
            PHEAD WORD * in,
            WORD startebuf,
            int * actionflag ) [extern]
```

Definition at line 1244 of file ratio.c.

4.19.8.198 TakeExtraSymbols()

```
WORD * TakeExtraSymbols (
            PHEAD WORD * in,
            WORD startebuf ) [extern]
```

Definition at line 1273 of file ratio.c.

4.19.8.199 MultiplyWithTerm()

```
WORD * MultiplyWithTerm (
            PHEAD WORD * in,
            WORD * term,
            WORD par ) [extern]
```

Definition at line 1312 of file ratio.c.

4.19.8.200 TakeContent()

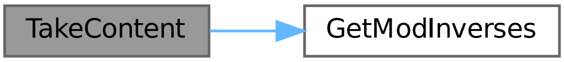
```
WORD * TakeContent (
            PHEAD WORD * in,
            WORD * term ) [extern]
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. Here the input is a sequence of terms in 'in' and the answer is a content-free sequence of terms. This sequence has been allocated by the Malloc1 routine in a call to EndSort, unless the expression was already content-free. In that case the input pointer is returned. The content is returned in term. This is supposed to be a separate allocation, made by TermMalloc in the calling routine.

Definition at line 1376 of file ratio.c.

References GetModInverses().

Here is the call graph for this function:



4.19.8.201 MergeSymbolLists()

```
int MergeSymbolLists (
             PHEAD WORD * old,
             WORD * extra,
             int par )  [extern]
```

Definition at line 1808 of file ratio.c.

4.19.8.202 MergeDotproductLists()

```
int MergeDotproductLists (
             PHEAD WORD * old,
             WORD * extra,
             int par )  [extern]
```

Definition at line 1927 of file ratio.c.

4.19.8.203 CreateExpression()

```
WORD * CreateExpression (
             PHEAD WORD )  [extern]
```

Definition at line 2020 of file ratio.c.

4.19.8.204 DIVfunction()

```
int DIVfunction (
             PHEAD WORD * term,
             WORD level,
             int par )  [extern]
```

Definition at line 2788 of file ratio.c.

4.19.8.205 MULfunc()

```
WORD * MULfunc (
            PHEAD WORD * p1,
            WORD * p2 )  [extern]
```

Definition at line 2957 of file ratio.c.

4.19.8.206 ConvertArgument()

```
WORD * ConvertArgument (
            PHEAD WORD * arg,
            int * type )  [extern]
```

Definition at line 3018 of file ratio.c.

4.19.8.207 ExpandRat()

```
int ExpandRat (
            PHEAD WORD * fun )  [extern]
```

Definition at line 3113 of file ratio.c.

4.19.8.208 InvPoly()

```
int InvPoly (
            PHEAD WORD * inpoly,
            WORD maxpow,
            WORD sym )  [extern]
```

Definition at line 3592 of file ratio.c.

4.19.8.209 TestDoLoop()

```
WORD TestDoLoop (
            PHEAD WORD * lhsbuf,
            WORD level )  [extern]
```

Definition at line 2762 of file dollar.c.

4.19.8.210 TestEndDoLoop()

```
WORD TestEndDoLoop (
            PHEAD WORD * lhsbuf,
            WORD level )  [extern]
```

Definition at line 2840 of file dollar.c.

4.19.8.211 poly_gcd()

```
WORD * poly_gcd (
            PHEAD WORD * a,
            WORD * b,
            WORD fit ) [extern]
```

Polynomial gcd

4.19.9 Description

This method calculates the greatest common divisor of two polynomials, given by two zero-terminated Form-style term lists.

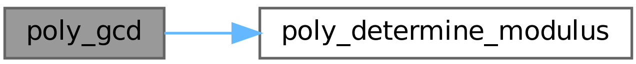
4.19.10 Notes

- The result is written at newly allocated memory
- Called from ratio.c
- Calls polygcd::gcd

Definition at line 124 of file polywrap.cc.

References poly_determine_modulus().

Here is the call graph for this function:



4.19.10.1 poly_div()

```
WORD * poly_div (
            PHEAD WORD * a,
            WORD * b,
            WORD fit ) [extern]
```

Definition at line 435 of file polywrap.cc.

4.19.10.2 poly_rem()

```
WORD * poly_rem (
            PHEAD WORD * a,
            WORD * b,
            WORD fit ) [extern]
```

Definition at line 463 of file polywrap.cc.

4.19.10.3 poly_inverse()

```
WORD * poly_inverse (
            PHEAD WORD * arga,
            WORD * argb ) [extern]
```

Definition at line 1743 of file polywrap.cc.

4.19.10.4 poly_mul()

```
WORD * poly_mul (
            PHEAD WORD * a,
            WORD * b ) [extern]
```

Definition at line 1948 of file polywrap.cc.

4.19.10.5 poly_ratfun_add()

```
WORD * poly_ratfun_add (
            PHEAD WORD * t1,
            WORD * t2 ) [extern]
```

Addition of PolyRatFuns

4.19.11 Description

This method gets two pointers to polyratfuns with up to two arguments each and calculates the sum.

4.19.12 Notes

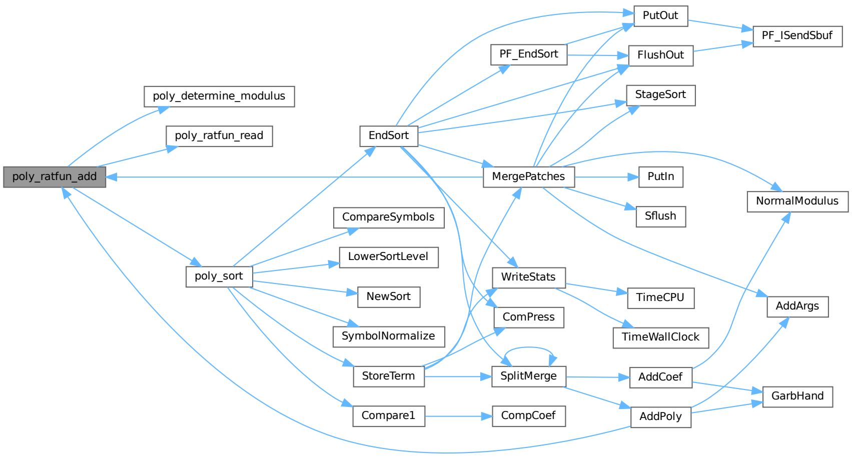
- The result is written at the workpointer
- Called from sort.c and threads.c
- Calls poly::operators and polygcd::gcd

Definition at line 633 of file polywrap.cc.

References poly_determine_modulus(), poly_ratfun_read(), and poly_sort().

Referenced by AddPoly(), and MergePatches().

Here is the call graph for this function:



4.19.12.1 poly_ratfun_normalize()

```
int poly_ratfun_normalize (
            PHEAD WORD * term )  [extern]
```

Multiplication/normalization of PolyRatFuns

4.19.13 Description

This method searches a term for multiple polyratfuns and multiplies their contents. The result is properly normalized. Normalization also works for terms with a single polyratfun.

4.19.14 Notes

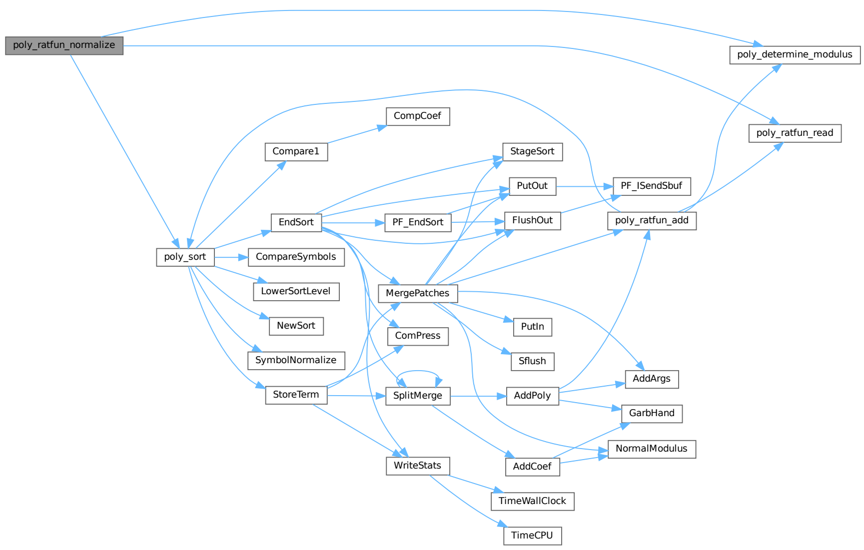
- The result overwrites the original term
- Called from proces.c
- Calls poly::operators and polygcd::gcd

Definition at line 769 of file polywrap.cc.

References poly_determine_modulus(), poly_ratfun_read(), and poly_sort().

Referenced by PolyFunMul(), and PrepPoly().

Here is the call graph for this function:



4.19.14.1 poly_factorize_argument()

```
int poly_factorize_argument (
            PHEAD WORD * argin,
            WORD * argout )  [extern]
```

Factorization of function arguments

4.19.15 Description

This method factorizes the Form-style argument argin.

4.19.16 Notes

- The result is written at argout
- Called from argument.c
- Calls poly_factorize

Definition at line 1112 of file polywrap.cc.

References poly_factorize().

Here is the call graph for this function:

4.19.16.1 poly_factorize_dollar()

```
WORD * poly_factorize_dollar (
            PHEAD WORD * argin )  [extern]
```

Factorization of dollar variables

4.19.17 Description

This method factorizes a dollar variable.

4.19.18 Notes

- The result is written at newly allocated memory.
- Called from dollar.c
- Calls poly_factorize

Definition at line 1146 of file polywrap.cc.

References poly_factorize().

Here is the call graph for this function:

4.19.18.1 poly_factorize_expression()

```
int poly_factorize_expression (
            EXPRESSIONS expr )  [extern]
```

Factorization of expressions

4.19.19 Description

This method factorizes an expression.

4.19.20 Notes

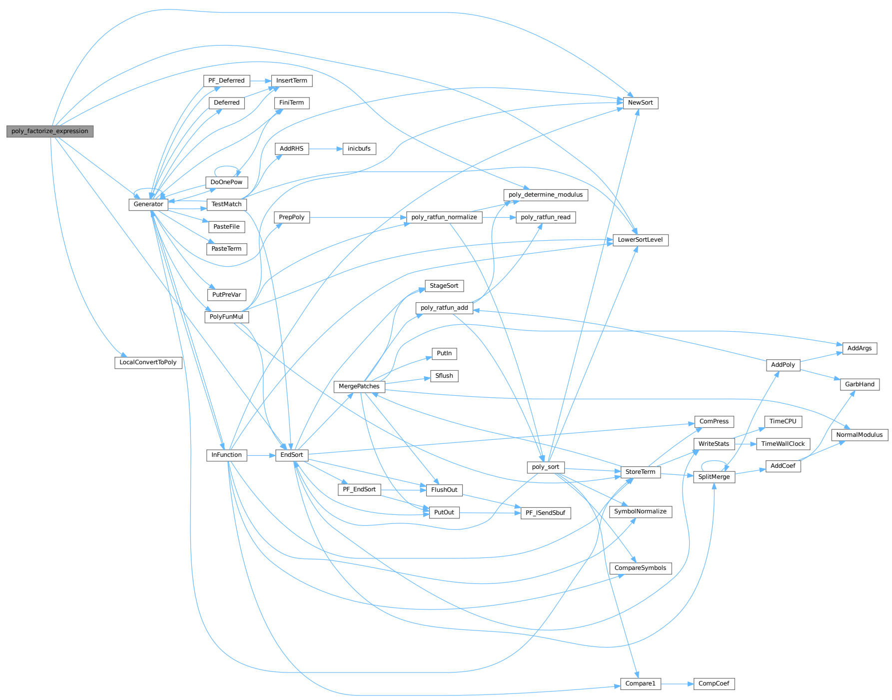
- The result overwrites the input expression
- Called from proces.c
- Calls polyfact::factorize

Definition at line 1178 of file polywrap.cc.

References EndSort(), Generator(), FiLe::handle, LocalConvertToPoly(), LowerSortLevel(), NewSort(), and poly_determine_modulus().

Referenced by PF_Processor(), and Processor().

Here is the call graph for this function:



4.19.20.1 poly_unfactorize_expression()

```
int poly_unfactorize_expression (
            EXPRESSIONS expr )  [extern]
```

Unfactorization of expressions

4.19.21 Description

This method expands a factorized expression.

4.19.22 Notes

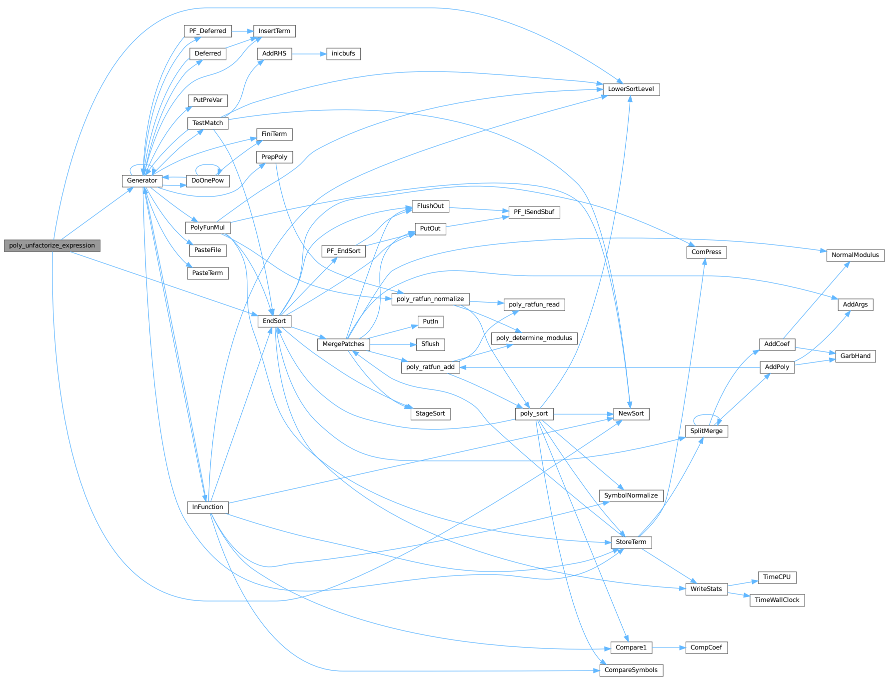
- The result overwrites the input expression
- Called from proces.c

Definition at line 1535 of file polywrap.cc.

References EndSort(), Generator(), FiLe::handle, LowerSortLevel(), and NewSort().

Referenced by PF_Processor(), and Processor().

Here is the call graph for this function:



4.19.22.1 poly_free_poly_vars()

```
void poly_free_poly_vars (
            PHEAD const char * text )  [extern]
```

Definition at line 2014 of file polywrap.cc.

4.19.22.2 optimize_print_code()

```
void optimize_print_code (
            int print_expr )  [extern]
```

Print optimized code

4.19.23 Description

This method prints the optimized code via "PrintExtraSymbol". Depending on the flag, the original expression is printed (for "Print") or not (for "#Optimize / #write "O").

Definition at line 4525 of file optimize.cc.

Referenced by Optimize().

4.19.23.1 DoPreAdd()

```
int DoPreAdd (
            UBYTE * s )  [extern]
```

Definition at line 1067 of file dict.c.

4.19.23.2 DoPreUseDictionary()

```
int DoPreUseDictionary (
            UBYTE * s )  [extern]
```

Definition at line 1022 of file dict.c.

4.19.23.3 DoPreCloseDictionary()

```
int DoPreCloseDictionary (
            UBYTE * s )  [extern]
```

Definition at line 998 of file dict.c.

4.19.23.4 DoPreOpenDictionary()

```
int DoPreOpenDictionary (
            UBYTE * s )  [extern]
```

Definition at line 959 of file dict.c.

4.19.23.5 RemoveDictionary()

```
void RemoveDictionary (
            DICTIONARY * dict )  [extern]
```

Definition at line 902 of file dict.c.

4.19.23.6 UnSetDictionary()

```
void UnSetDictionary (
            void  )  [extern]
```

Definition at line 883 of file dict.c.

4.19.23.7 SetDictionaryOptions()

```
int SetDictionaryOptions (
            UBYTE * options )  [extern]
```

Definition at line 790 of file dict.c.

4.19.23.8 AddToDictionary()

```
int AddToDictionary (
            DICTIONARY * dict,
            UBYTE * left,
            UBYTE * right )  [extern]
```

Definition at line 491 of file dict.c.

4.19.23.9 AddDictionary()

```
int AddDictionary (
            UBYTE * name )  [extern]
```

Definition at line 449 of file dict.c.

4.19.23.10 FindDictionary()

```
int FindDictionary (
            UBYTE * name )  [extern]
```

Definition at line 434 of file dict.c.

4.19.23.11 IsExponentSign()

```
UBYTE * IsExponentSign (
            void ) [extern]
```

Definition at line 238 of file dict.c.

4.19.23.12 IsMultiplySign()

```
UBYTE * IsMultiplySign (
            void ) [extern]
```

Definition at line 218 of file dict.c.

4.19.23.13 TransformRational()

```
void TransformRational (
            UWORD * a,
            WORD na ) [extern]
```

Definition at line 71 of file dict.c.

4.19.23.14 WriteDictionary()

```
void WriteDictionary (
            DICTIONARY * dict ) [extern]
```

Definition at line 1307 of file sch.c.

4.19.23.15 ShrinkDictionary()

```
void ShrinkDictionary (
            DICTIONARY * dict ) [extern]
```

Definition at line 945 of file dict.c.

4.19.23.16 MultiplyToLine()

```
void MultiplyToLine (
            void ) [extern]
```

Definition at line 671 of file sch.c.

4.19.23.17 FindSymbol()

```
UBYTE * FindSymbol (
            WORD num ) [extern]
```

Definition at line 258 of file dict.c.

4.19.23.18 FindVector()

```
UBYTE * FindVector (
            WORD num )  [extern]
```

Definition at line 279 of file dict.c.

4.19.23.19 FindIndex()

```
UBYTE * FindIndex (
            WORD num )  [extern]
```

Definition at line 301 of file dict.c.

4.19.23.20 FindFunction()

```
UBYTE * FindFunction (
            WORD num )  [extern]
```

Definition at line 323 of file dict.c.

4.19.23.21 FindFunWithArgs()

```
UBYTE * FindFunWithArgs (
            WORD * t )  [extern]
```

Definition at line 345 of file dict.c.

4.19.23.22 FindExtraSymbol()

```
UBYTE * FindExtraSymbol (
            WORD num )  [extern]
```

Definition at line 377 of file dict.c.

4.19.23.23 DictToBytes()

```
LONG DictToBytes (
            DICTIONARY * dict,
            UBYTE * buf )  [extern]
```

Definition at line 1119 of file dict.c.

4.19.23.24 DictFromBytes()

```
DICTIONARY * DictFromBytes (
            UBYTE * buf )  [extern]
```

Definition at line 1152 of file dict.c.

4.19.23.25 CoCreateSpectator()

```
int CoCreateSpectator (
            UBYTE * inp )  [extern]
```

Definition at line 89 of file spectator.c.

4.19.23.26 CoToSpectator()

```
int CoToSpectator (
            UBYTE * inp )  [extern]
```

Definition at line 181 of file spectator.c.

4.19.23.27 CoRemoveSpectator()

```
int CoRemoveSpectator (
            UBYTE * inp )  [extern]
```

Definition at line 233 of file spectator.c.

4.19.23.28 CoEmptySpectator()

```
int CoEmptySpectator (
            UBYTE * inp )  [extern]
```

Definition at line 293 of file spectator.c.

4.19.23.29 CoCopySpectator()

```
int CoCopySpectator (
            UBYTE * inp )  [extern]
```

Definition at line 459 of file spectator.c.

4.19.23.30 PutInSpectator()

```
int PutInSpectator (
            WORD * term,
            WORD specnum )  [extern]
```

Definition at line 345 of file spectator.c.

4.19.23.31 ClearSpectators()

```
void ClearSpectators (
            WORD par )  [extern]
```

Definition at line 675 of file spectator.c.

4.19.23.32 GetFromSpectator()

```
WORD GetFromSpectator (
            WORD * term,
            WORD specnum )  [extern]
```

Definition at line 601 of file spectator.c.

4.19.23.33 FlushSpectators()

```
void FlushSpectators (
            void  )  [extern]
```

Definition at line 402 of file spectator.c.

4.19.23.34 ReadFromScratch()

```
int ReadFromScratch (
            FILEHANDLE * fi,
            POSITION * pos,
            UBYTE * buffer,
            POSITION * length )  [extern]
```

Definition at line 273 of file store.c.

4.19.23.35 AddToScratch()

```
int AddToScratch (
            FILEHANDLE * fi,
            POSITION * pos,
            UBYTE * buffer,
            POSITION * length,
            int withflush )  [extern]
```

Definition at line 300 of file store.c.

4.19.23.36 DoPreAppendPath()

```
int DoPreAppendPath (
            UBYTE * s )  [extern]
```

Appends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: #appendpath <path>

Definition at line 7153 of file pre.c.

4.19.23.37 DoPrePrependPath()

```
int DoPrePrependPath (
            UBYTE * s ) [extern]
```

Prepends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: #prependpath <path>

Definition at line 7170 of file pre.c.

4.19.23.38 DoSwitch()

```
int DoSwitch (
            PHEAD WORD * term,
            WORD * lhs ) [extern]
```

Definition at line 1070 of file if.c.

4.19.23.39 DoEndSwitch()

```
int DoEndSwitch (
            PHEAD WORD * term,
            WORD * lhs ) [extern]
```

Definition at line 1086 of file if.c.

4.19.23.40 FindCase()

```
SWITCHTABLE * FindCase (
            WORD nswitch,
            WORD ncase ) [extern]
```

Definition at line 1097 of file if.c.

4.19.23.41 SwitchSplitMergeRec()

```
void SwitchSplitMergeRec (
            SWITCHTABLE * array,
            WORD num,
            SWITCHTABLE * auxarray ) [extern]
```

Definition at line 1184 of file if.c.

4.19.23.42 SwitchSplitMerge()

```
void SwitchSplitMerge (
            SWITCHTABLE * array,
            WORD num ) [extern]
```

Definition at line 1213 of file if.c.

4.19.23.43 DoubleSwitchBuffers()

```
int DoubleSwitchBuffers (
            void ) [extern]
```

Definition at line 1135 of file if.c.

4.19.23.44 DistrN()

```
int DistrN (
            int n,
            int * cpl,
            int ncpl,
            int * scratch ) [extern]
```

Definition at line 4048 of file tools.c.

4.19.23.45 DoNamespace()

```
int DoNamespace (
            UBYTE * s ) [extern]
```

Definition at line 7234 of file pre.c.

4.19.23.46 DoEndNamespace()

```
int DoEndNamespace (
            UBYTE * s ) [extern]
```

Definition at line 7282 of file pre.c.

4.19.23.47 DoUse()

```
int DoUse (
            UBYTE * s ) [extern]
```

Definition at line 7497 of file pre.c.

4.19.23.48 SkipName()

```
UBYTE * SkipName (
            UBYTE * s ) [extern]
```

Definition at line 7305 of file pre.c.

4.19.23.49 ConstructName()

```
UBYTE * ConstructName (
            UBYTE * s,
            UBYTE type )  [extern]
```

Definition at line 7405 of file pre.c.

4.19.23.50 DoSetUserFlag()

```
int DoSetUserFlag (
            UBYTE * s )  [extern]
```

Definition at line 7662 of file pre.c.

4.19.23.51 DoClearUserFlag()

```
int DoClearUserFlag (
            UBYTE * s )  [extern]
```

Definition at line 7652 of file pre.c.

4.19.23.52 CreateVertex()

```
VERTEX * CreateVertex (
            MODEL * m )
```

Definition at line 46 of file model.c.

4.19.23.53 ReadParticle()

```
UBYTE * ReadParticle (
            UBYTE * s,
            VERTEX * v,
            MODEL * m,
            int par )
```

Definition at line 76 of file model.c.

4.19.23.54 CoModel()

```
int CoModel (
            UBYTE * s )
```

Definition at line 119 of file model.c.

4.19.23.55 CoParticle()

```
int CoParticle (
            UBYTE * s )
```

Definition at line 194 of file model.c.

4.19.23.56 CoVertex()

```
int CoVertex (
            UBYTE * s )
```

Definition at line 276 of file model.c.

4.19.23.57 CoEndModel()

```
int CoEndModel (
            UBYTE * s )
```

Definition at line 379 of file model.c.

4.19.23.58 LoadModel()

```
int LoadModel (
            MODEL * m )
```

Definition at line 27 of file diawrap.cc.

4.19.23.59 CoSetUserFlag()

```
int CoSetUserFlag (
            UBYTE * s )
```

Definition at line 7384 of file compcomm.c.

4.19.23.60 CoClearUserFlag()

```
int CoClearUserFlag (
            UBYTE * s )
```

Definition at line 7412 of file compcomm.c.

4.19.23.61 CoCreateAllLoops()

```
int CoCreateAllLoops (
            UBYTE * s )
```

Definition at line 7449 of file compcomm.c.

4.19.23.62 CoCreateAllPaths()

```
int CoCreateAllPaths (
            UBYTE * s )
```

Definition at line 7569 of file compcomm.c.

4.19.23.63 CoCreateAll()

```
int CoCreateAll (
            UBYTE * s )
```

Definition at line 7697 of file compcomm.c.

4.19.23.64 AllLoops()

```
WORD AllLoops (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 650 of file lus.c.

4.19.23.65 StartLoops()

```
LONG StartLoops (
            PHEAD WORD * term,
            WORD level,
            LONG vert,
            WORD nvert,
            WORD * arglist,
            WORD nargs,
            WORD * loop,
            WORD nloop )
```

Definition at line 894 of file lus.c.

4.19.23.66 GenLoops()

```
LONG GenLoops (
            PHEAD WORD * term,
            WORD level,
            LONG vert,
            WORD nvert,
            WORD * arglist,
            WORD nargs,
            WORD * loop,
            WORD nloop )
```

Definition at line 980 of file lus.c.

4.19.23.67 LoopOutput()

```
void LoopOutput (
            PHEAD WORD * term,
            WORD level,
            WORD * loop,
            WORD nloop )
```

Definition at line 1059 of file lus.c.

4.19.23.68 AllPaths()

```
WORD AllPaths (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1123 of file lus.c.

4.19.23.69 GenPaths()

```
LONG GenPaths (
            PHEAD WORD * term,
            WORD level,
            LONG vert,
            WORD nvert,
            WORD * arglist,
            WORD nargs,
            WORD * path,
            WORD npath )
```

Definition at line 1413 of file lus.c.

4.19.23.70 PathOutput()

```
void PathOutput (
            PHEAD WORD * term,
            WORD level,
            WORD * path,
            WORD npath )
```

Definition at line 1486 of file lus.c.

4.20 declare.h

Go to the documentation of this file.

```
#ifndef __FDECLARE__
#define __FDECLARE__

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
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 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

/*
    #[ Macro's :
*/

#define MaX(x,y) ((x) > (y) ? (x): (y))
#define MiN(x,y) ((x) < (y) ? (x): (y))
#define ABS(x) ( (x) < 0 ? -(x): (x) )
#define SGN(x) ( (x) > 0 ? 1 : (x) < 0 ? -1 : 0 )
#define REDLENG(x) ((((x)<0)?((x)+1):((x)-1))/2)
#define INCLENG(x) (((x)<0)?(((x)*2)-1):(((x)*2)+1))
#define GETCOEF(x,y) x += *x;y = x[-1];x -= ABS(y);y=REDLENG(y)
#define GETSTOP(x,y) y=x+(*x)-1;y -= ABS(*y)-1
#define StuffAdd(x,y)  (((x)<0?-1:1)*(y)+((y)<0?-1:1)*(x))

#define EXCHN(t1,t2,n) { WORD a,i; for(i=0;i<n;i++){a=t1[i];t1[i]=t2[i];t2[i]=a;} }
#define EXCH(x,y) { WORD a = (x); (x) = (y); (y) = a; }

#define TOKENTOLINE(x,y) if ( AC.OutputSpaces == NOSPACEFORMAT ) { \
        TokenToLine((UBYTE *)(y)); } else { TokenToLine((UBYTE *)(x)); }

#define UngetFromStream(stream,c) ((stream)->nextchar[(stream)->isnextchar++]=c)
#ifdef WITHRETURN
#define AddLineFeed(s,n) { (s)[(n)++] = CARRIAGERETURN; (s)[(n)++] = LINEFEED; }
#else
#define AddLineFeed(s,n) { (s)[(n)++] = LINEFEED; }
#endif
#define TryRecover(x) Terminate(-1)
#define UngetChar(c) { pushbackchar = c; }
#define ParseNumber(x,s) {(x)=0;while(*(s)>='0'&&*(s)<='9')(x)=10*(x)+*(s)++ -'0';}
#define ParseSign(sgn,s) {(sgn)=0;while(*(s)=='-'||*(s)=='+'){\
          if ( *(s)++ == '-' ) sgn ^= 1;}}
#define ParseSignedNumber(x,s) { int sgn; ParseSign(sgn,s)\
          ParseNumber(x,s) if ( sgn ) x = -x; }

/* (n) is necessary here, since the macro is sometimes passed dereferenced pointers for n */
#define NCOPY(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
/*#define NCOPY(s,t,n) { memcpy(s,t,n*sizeof(WORD)); s+=n; t+=n; n = -1; }*/
#define NCOPYI(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
#define NCOPYB(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
#define NCOPYI32(s,t,n) { while ( (n)-- > 0 ) { *s++ = *t++; } }
#define WCOPY(s,t,n) { int nn=n; WORD *ss=(WORD *)s, *tt=(WORD *)t; while ( (nn)-- > 0 ) { *ss++ = *tt++; } }

#define NeedNumber(x,s,err) { int sgn = 1;                               \
        while ( *s == ' ' || *s == '\t' || *s == '-' || *s == '+' ) {      \
            if ( *s == '-' ) {sgn = -sgn;} s++; }                          \
        if ( chartype[*s] != 1 ) goto err;                                 \
        ParseNumber(x,s)                                                   \
        if ( sgn < 0 ) {(x) = -(x);} while ( *s == ' ' || *s == '\t' ) s++;\
    }
#define SKIPBLANKS(s) { while ( *(s) == ' ' || *(s) == '\t' ) (s)++; }
#define FLUSHCONSOLE if ( AP.InOutBuf > 0 ) CharOut(LINEFEED)
```

```
#define SKIPBRA1(s) { int lev1=0; s++; while(*s) { if(*s=='['){lev1++;} \
                    else {if(*s==']'&&--lev1<0){break;}} s++;} }
#define SKIPBRA2(s) { int lev2=0; s++; while(*s) { if(*s=='{'){lev2++;} \
                    else {if(*s=='}'&&--lev2<0){break;} \
                    else {if(*s=='['){SKIPBRA1(s)}}} s++;} }
#define SKIPBRA3(s) { int lev3=0; s++; while(*s) { if(*s=='('){lev3++;} \
                    else {if(*s==')'&&--lev3<0){break;} \
                    else {if(*s=='{'){SKIPBRA2(s)} \
                    else {if(*s=='['){SKIPBRA1(s)}}}} s++;} }
#define SKIPBRA4(s) { int lev4=0; s++; while(*s) { if(*s=='('){lev4++;} \
                    else {if(*s==')'&&--lev4<0){break;} \
                    else {if(*s=='['){SKIPBRA1(s)}}} s++;} }
#define SKIPBRA5(s) { int lev5=0; s++; while(*s) { if(*s=='{'){lev5++;} \
                    else {if(*s=='}'&&--lev5<0){break;} \
                    else {if(*s=='('){SKIPBRA4(s)} \
                    else {if(*s=='['){SKIPBRA1(s)}}}} s++;} }

/*
#define CYCLE1(a,i) {WORD iX,jX; iX=*a; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX;}
*/
#define CYCLE1(t,a,i) {t iX=*a; WORD jX; for(jX=1;jX<i;jX++)a[jX-1]=a[jX]; a[i-1]=iX;}

#define AddToCB(c,wx) if(c->Pointer>=c->Top) \
        {DoubleCbuffer(c-cbuf,c->Pointer,21);} \
        *(c->Pointer)++ = wx;

#define EXCHINOUT { FILEHANDLE *ffFi = AR.outfile; \
    AR.outfile = AR.infile; AR.infile = ffFi; }
#define BACKINOUT { FILEHANDLE *ffFi = AR.outfile; POSITION posi; \
    AR.outfile = AR.infile; AR.infile = ffFi; \
    SetEndScratch(AR.infile,&posi); }

#define CopyArg(to,from) { if ( *from > 0 ) { int ica = *from; NCOPY(to,from,ica) } \
        else if ( *from <= -FUNCTION ) *to++ = *from++;  \
        else { *to++ = *from++; *to++ = *from++; } }

#if ARGHEAD > 2
#define FILLARG(w) { int i = ARGHEAD-2; while ( --i >= 0 ) *w++ = 0; }
#define COPYARG(w,t) { int i = ARGHEAD-2; while ( --i >= 0 ) *w++ = *t++; }
#define ZEROARG(w) { int i; for ( i = 2; i < ARGHEAD; i++ ) w[i] = 0; }
#else
#define FILLARG(w)
#define COPYARG(w,t)
#define ZEROARG(w)
#endif

#if FUNHEAD > 2
#define FILLFUN(w) { *w++ = 0; FILLFUN3(w) }
#define COPYFUN(w,t) { *w++ = *t++; COPYFUN3(w,t) }
#else
#define FILLFUN(w)
#define COPYFUN(w,t)
#endif

#if FUNHEAD > 3
#define FILLFUN3(w) { int ie = FUNHEAD-3; while ( --ie >= 0 ) *w++ = 0; }
#define COPYFUN3(w,t) { int ie = FUNHEAD-3; while ( --ie >= 0 ) *w++ = *t++; }
#else
#define COPYFUN3(w,t)
#define FILLFUN3(w)
#endif

#if SUBEXPSIZE > 5
#define FILLSUB(w) { int ie = SUBEXPSIZE-5; while ( --ie >= 0 ) *w++ = 0; }
#define COPYSUB(w,ww) { int ie = SUBEXPSIZE-5; while ( --ie >= 0 ) *w++ = *ww++; }
#else
#define FILLSUB(w)
#define COPYSUB(w,ww)
#endif

#if EXPRHEAD > 4
#define FILLEXPR(w) { int ie = EXPRHEAD-4; while ( --ie >= 0 ) *w++ = 0; }
#else
#define FILLEXPR(w)
#endif

#define NEXTARG(x) if(*x>0) x += *x; else if(*x <= -FUNCTION)x++; else x += 2;
#define COPY1ARG(s1,t1) { int ica; if ( (ica=*t1) > 0 ) { NCOPY(s1,t1,ica) } \
        else if(*t1<=-FUNCTION){*s1++=*t1++;} else{*s1++=*t1++;*s1++=*t1++;} }

#define ZeroFillRange(w,begin,end) do { \
    int tmp_i; \
    for ( tmp_i = begin; tmp_i < end; tmp_i++ ) { (w)[tmp_i] = 0; } \
} while (0)

#define TABLESIZE(a,b) (((WORD)sizeof(a))/((WORD)sizeof(b)))
```

```
#define WORDDIF(x,y) (WORD)(x-y)
#define wsizeof(a) ((WORD)sizeof(a))
#define VARNAME(type,num) (AC.varnames->namebuffer+type[num].name)
#define DOLLARNAME(type,num) (AC.dollarnames->namebuffer+type[num].name)
#define EXPRNAME(num) (AC.exprnames->namebuffer+Expressions[num].name)

#define PREV(x) prevorder?prevorder:x

#define Terminate(x) do { TerminateImpl(x, __FILE__, __LINE__, __FUNCTION__); } while(0)
#define SETERROR(x) { Terminate(-1); return(-1); }

/* use this macro to avoid the unused parameter warning */
#define DUMMYUSE(x) (void)(x);

#ifdef _FILE_OFFSET_BITS
#if _FILE_OFFSET_BITS==64
/*:[19mar2004 mt]*/

#define ADDPOS(pp,x) (pp).p1 = ((pp).p1+(off_t)(x))
#define SETBASELENGTH(ss,x) (ss).p1 = (off_t)(x)
#define SETBASEPOSITION(pp,x) (pp).p1 = (off_t)(x)
#define ISEQUALPOSINC(pp1,pp2,x) ( (pp1).p1 == ((pp2).p1+(off_t)(x)) )
#define ISGEPOSINC(pp1,pp2,x) ( (pp1).p1 >= ((pp2).p1+(off_t)(x)) )
#define DIVPOS(pp,n) ( (pp).p1/(off_t)(n) )
#define MULPOS(pp,n) (pp).p1 *= (off_t)(n)

#else

#define ADDPOS(pp,x) (pp).p1 = ((pp).p1+(x))
#define SETBASELENGTH(ss,x) (ss).p1 = (x)
#define SETBASEPOSITION(pp,x) (pp).p1 = (x)
#define ISEQUALPOSINC(pp1,pp2,x) ( (pp1).p1 == ((pp2).p1+(LONG)(x)) )
#define ISGEPOSINC(pp1,pp2,x) ( (pp1).p1 >= ((pp2).p1+(LONG)(x)) )
#define DIVPOS(pp,n) ( (pp).p1/(n) )
#define MULPOS(pp,n) (pp).p1 *= (n)
#endif
#else

#define ADDPOS(pp,x) (pp).p1 = ((pp).p1+(LONG)(x))
#define SETBASELENGTH(ss,x) (ss).p1 = (LONG)(x)
#define SETBASEPOSITION(pp,x) (pp).p1 = (LONG)(x)
#define ISEQUALPOSINC(pp1,pp2,x) ( (pp1).p1 == ((pp2).p1+(LONG)(x)) )
#define ISGEPOSINC(pp1,pp2,x) ( (pp1).p1 >= ((pp2).p1+(LONG)(x)) )
#define DIVPOS(pp,n) ( (pp).p1/(LONG)(n) )
#define MULPOS(pp,n) (pp).p1 *= (LONG)(n)

#endif
#define DIFPOS(ss,pp1,pp2) (ss).p1 = ((pp1).p1-(pp2).p1)
#define DIFBASE(pp1,pp2) ((pp1).p1-(pp2).p1)
#define ADD2POS(pp1,pp2) (pp1).p1 += (pp2).p1
#define PUTZERO(pp) (pp).p1 = 0
#define BASEPOSITION(pp) ((pp).p1)
#define SETSTARTPOS(pp) (pp).p1 = -2
#define NOTSTARTPOS(pp) ( (pp).p1 > -2 )
#define ISMINPOS(pp) ( (pp).p1 == -1 )
#define ISEQUALPOS(pp1,pp2) ( (pp1).p1 == (pp2).p1 )
#define ISNOTEQUALPOS(pp1,pp2) ( (pp1).p1 != (pp2).p1 )
#define ISLESSPOS(pp1,pp2) ( (pp1).p1 < (pp2).p1 )
#define ISGEPOS(pp1,pp2) ( (pp1).p1 >= (pp2).p1 )
#define ISNOTZEROPOS(pp) ( (pp).p1 != 0 )
#define ISZEROPOS(pp) ( (pp).p1 == 0 )
#define ISPOSPOS(pp) ( (pp).p1 > 0 )
#define ISNEGPOS(pp) ( (pp).p1 < 0 )
extern void TELLFILE(int,POSITION *);

#define TOLONG(x) ((LONG)(x))

#define Add2Com(x) { WORD cod[2]; cod[0] = x; cod[1] = 2; AddNtoL(2,cod); }
#define Add3Com(x1,x2) { WORD cod[3]; cod[0] = x1; cod[1] = 3; cod[2] = x2; AddNtoL(3,cod); }
#define Add4Com(x1,x2,x3) { WORD cod[4]; cod[0] = x1; cod[1] = 4; \
   cod[2] = x2; cod[3] = x3; AddNtoL(4,cod); }
#define Add5Com(x1,x2,x3,x4) { WORD cod[5]; cod[0] = x1; cod[1] = 5; \
   cod[2] = x2; cod[3] = x3; cod[4] = x4; AddNtoL(5,cod); }

/*
    The temporary variable ppp is to avoid a compiler warning about strict aliassing
*/
#define WantAddPointers(x) while((AT.pWorkPointer+(x))>AR.pWorkSize){WORD ***ppp=&AT.pWorkSpace;\
    ExpandBuffer((void **)ppp,&AR.pWorkSize,(int)(sizeof(WORD *)));}
#define WantAddLongs(x) while((AT.lWorkPointer+(x))>AR.lWorkSize){LONG **ppp=&AT.lWorkSpace;\
    ExpandBuffer((void **)ppp,&AR.lWorkSize,sizeof(LONG));}
#define WantAddPositions(x) while((AT.posWorkPointer+(x))>AR.posWorkSize){POSITION **ppp=&AT.posWorkSpace;\
    ExpandBuffer((void **)ppp,&AR.posWorkSize,sizeof(POSITION));}

/* inline in form3.h (or config.h). */
#define FORM_INLINE inline
```

```
/*
    Macro's for memory management. This can be done by routines, but that
    would be slower. Inline routines could do this, but we don't want to
    leave this to the friendliness of the compiler(s).
    The routines can be found in the file tools.c
*/
#define MEMORYMACROS

#ifdef MEMORYMACROS

#define TermMalloc(x) ( (AT.TermMemTop <= 0 ) ? TermMallocAddMemory(BHEAD0), AT.TermMemHeap[--AT.TermMemTop]: AT.TermMemHeap[--AT.TermMemTop] )
#define NumberMalloc(x) ( (AT.NumberMemTop <= 0 ) ? NumberMallocAddMemory(BHEAD0), AT.NumberMemHeap[--AT.NumberMemTop]: AT.NumberMemHeap[--AT.NumberMemTop] )
#define CacheNumberMalloc(x) ( (AT.CacheNumberMemTop <= 0 ) ? CacheNumberMallocAddMemory(BHEAD0), AT.CacheNumberMemHeap[--AT.CacheNumberMemTop]: AT.CacheNumberMemHeap[--AT.CacheNumberMemTop] )
#define TermFree(TermMem,x) AT.TermMemHeap[AT.TermMemTop++] = (WORD *)(TermMem)
#define NumberFree(NumberMem,x) AT.NumberMemHeap[AT.NumberMemTop++] = (UWORD *)(NumberMem)
#define CacheNumberFree(NumberMem,x) AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = (UWORD *)(NumberMem)

#else

#define TermMalloc(x) TermMalloc2(BHEAD (char *)(x))
#define NumberMalloc(x) NumberMalloc2(BHEAD (char *)(x))
#define CacheNumberMalloc(x) CacheNumberMalloc2(BHEAD (char *)(x))
#define TermFree(x,y) TermFree2(BHEAD (WORD *)(x),(char *)(y))
#define NumberFree(x,y) NumberFree2(BHEAD (UWORD *)(x),(char *)(y))
#define CacheNumberFree(x,y) CacheNumberFree2(BHEAD (UWORD *)(x),(char *)(y))

#endif

/*
 * Macros for checking nesting levels in the compiler, used as follows:
 *
 *   AC.IfSumCheck[AC.IfLevel] = NestingChecksum();
 *   AC.IfLevel++;
 *
 *   AC.IfLevel--;
 *   if ( AC.IfSumCheck[AC.IfLevel] != NestingChecksum() ) {
 *     MesNesting();
 *   }
 *
 * Note that NestingChecksum() also contains AC.IfLevel and so in this case
 * using increment/decrement operators on it in the left-hand side may be
 * confusing.
 */
#define NestingChecksum() (AC.IfLevel + AC.RepLevel + AC.arglevel + AC.insidelevel + AC.termlevel + AC.inexprlevel + AC.dolooplevel +AC.SwitchLevel)
#define MesNesting() MesPrint("&Illegal nesting of if, repeat, argument, inside, term, inexpression and do")

#define MarkPolyRatFunDirty(T) {if(*T&&AR.PolyFunType==2){WORD *TP,*TT;TT=T+*T;TT-=ABS(TT[-1]);\
TP=T+1;while(TP<TT){if(*TP==AR.PolyFun){TP[2]|=(DIRTYFLAG|MUSTCLEANPRF);}TP+=TP[1];}}}

/*
    Macros for nesting input levels for #$name = ...; assign instructions.
    Note that the level should never go below zero.
*/
#define PUSHPREASSIGNLEVEL AP.PreAssignLevel++; { GETIDENTITY \
    if ( AP.PreAssignLevel >= AP.MaxPreAssignLevel ) { int i; \
        LONG *ap = (LONG *)Malloc1(2*AP.MaxPreAssignLevel*sizeof(LONG *),"PreAssignStack"); \
        for ( i = 0; i < AP.MaxPreAssignLevel; i++ ) ap[i] = AP.PreAssignStack[i]; \
        M_free(AP.PreAssignStack,"PreAssignStack"); \
        AP.MaxPreAssignLevel *= 2; AP.PreAssignStack = ap; \
    } \
    *AT.WorkPointer++ = AP.PreContinuation; AP.PreContinuation = 0; \
    AP.PreAssignStack[AP.PreAssignLevel] = AC.iPointer - AC.iBuffer; }

#define POPPREASSIGNLEVEL if ( AP.PreAssignLevel > 0 ) { GETIDENTITY \
    AC.iPointer = AC.iBuffer + AP.PreAssignStack[AP.PreAssignLevel--]; \
    AP.PreContinuation = *--AT.WorkPointer; \
    *AC.iPointer = 0; }

#ifdef WITHFLOAT
/*
    The following macro's are needed to avoid problems with the compilers
    and gmp.h. For the C++ files we need the Form include files to be inside
    and extern C {} environment, but then the structs.h needs gmp.h to
    recognise the mpf_t datatype. This causes no end of problems.
    Hence we collect the sensitive objects as (void *) and cast them to
    something usable with the macro's below. This way the gmp.h file
    needs to be included only in a very limited number of .c files.
*/
#define mpftab1 ((mpf_t *)(AT.mpf_tab1))
#define mpftab2 ((mpf_t *)(AT.mpf_tab2))
```

```
#define mpfaux_ ((mpf_t *)(AT.aux_))
#define aux1 (((mpf_t *)(AT.aux_))[0])
#define aux2 (((mpf_t *)(AT.aux_))[1])
#define aux3 (((mpf_t *)(AT.aux_))[2])
#define aux4 (((mpf_t *)(AT.aux_))[3])
#define aux5 (((mpf_t *)(AT.aux_))[4])
#define auxjm (((mpf_t *)(AT.aux_))[5])
#define auxjjm (((mpf_t *)(AT.aux_))[6])
#define auxsum (((mpf_t *)(AT.aux_))[7])

#define mpfdelta1 (((mpf_t *)(AS.delta_1))[0])

#endif


/*
    MesPrint("P-level popped to %d with %d",AP.PreAssignLevel,(WORD)(AC.iPointer - AC.iBuffer));

    #] Macro's :
    #[ Inline functions :
*/

/*
 * The following three functions give the unsigned absolute value of a signed
 * integer even for the most negative integer. This is beyond the scope of
 * the standard abs() function and its family, whose return-values are signed.
 * In short, we should not use the unary minus operator with signed numbers
 * unless we are sure that there are no integer overflows. Instead, we rely on
 * two well-defined operations: (i) signed-to-unsigned conversion and
 * (ii) unary minus of unsigned operands.
 *
 * See also:
 *   https://stackoverflow.com/a/4536188   (Unary minus and signed-to-unsigned conversion)
 *   https://stackoverflow.com/q/8026694   (C: unary minus operator behavior with unsigned operands)
 *   https://stackoverflow.com/q/1610947   (Why does stdlib.h's abs() family of functions return a signed value?)
 *   https://blog.regehr.org/archives/226  (A Guide to Undefined Behavior in C and C++, Part 2)
 */
static inline unsigned int IntAbs(int x)
{
    if ( x >= 0 ) return x;
    return(-((unsigned int)x));
}

static inline UWORD WordAbs(WORD x)
{
    if ( x >= 0 ) return x;
    return(-((UWORD)x));
}

static inline ULONG LongAbs(LONG x)
{
    if ( x >= 0 ) return x;
    return(-((ULONG)x));
}

/*
 * The following functions provide portable unsigned-to-signed conversions
 * (to avoid the implementation-defined behaviour), which is expected to be
 * optimized to a no-op.
 *
 * See also:
 *   https://stackoverflow.com/a/13208789  (Efficient unsigned-to-signed cast avoiding implementation-defined behavior)
 */
static inline int UnsignedToInt(unsigned int x)
{
    extern void TerminateImpl(int, const char*, int, const char*);
    if ( x <= INT_MAX ) return(x);
    if ( x >= (unsigned int)INT_MIN )
        return((int)(x - (unsigned int)INT_MIN) + INT_MIN);
    Terminate(1);
    return(0);
}

static inline WORD UWordToWord(UWORD x)
{
    extern void TerminateImpl(int, const char*, int, const char*);
    if ( x <= WORD_MAX_VALUE ) return(x);
    if ( x >= (UWORD)WORD_MIN_VALUE )
        return((WORD)(x - (UWORD)WORD_MIN_VALUE) + WORD_MIN_VALUE);
    Terminate(1);
    return(0);
}

static inline LONG ULongToLong(ULONG x)
{
```

```
    extern void TerminateImpl(int, const char*, int, const char*);
    if ( x <= LONG_MAX_VALUE ) return(x);
    if ( x >= (ULONG)LONG_MIN_VALUE )
        return((LONG)(x - (ULONG)LONG_MIN_VALUE) + LONG_MIN_VALUE);
    Terminate(1);
    return(0);
}

/*
    #] Inline functions :
    #[ Thread objects :
*/

#ifdef WITHPTHREADS

#define EXTERNLOCK(x) extern pthread_mutex_t x;
#define INILOCK(x)     pthread_mutex_t x = PTHREAD_MUTEX_INITIALIZER;
#define EXTERNRWLOCK(x) extern pthread_rwlock_t x;
#define INIRWLOCK(x)     pthread_rwlock_t x = PTHREAD_RWLOCK_INITIALIZER;
#ifdef DEBUGGINGLOCKS
#include <asm/errno.h>
#define LOCK(x)       while ( pthread_mutex_trylock(&(x)) == EBUSY ) {}
#define RWLOCKR(x)      while ( pthread_rwlock_tryrdlock(&(x)) == EBUSY ) {}
#define RWLOCKW(x)      while ( pthread_rwlock_trywrlock(&(x)) == EBUSY ) {}
#else
#define LOCK(x)       pthread_mutex_lock(&(x))
#define RWLOCKR(x)      pthread_rwlock_rdlock(&(x))
#define RWLOCKW(x)      pthread_rwlock_wrlock(&(x))
#endif
#define UNLOCK(x)     pthread_mutex_unlock(&(x))
#define UNRWLOCK(x)     pthread_rwlock_unlock(&(x))
#define MLOCK(x)      LOCK(x)
#define MUNLOCK(x)    UNLOCK(x)

#define GETBIDENTITY
#define GETIDENTITY   int identity = WhoAmI(); ALLPRIVATES *B = AB[identity];
#else

#define EXTERNLOCK(x)
#define INILOCK(x)
#define LOCK(x)
#define UNLOCK(x)
#define EXTERNRWLOCK(x)
#define INIRWLOCK(x)
#define RWLOCKR(x)
#define RWLOCKW(x)
#define UNRWLOCK(x)
#ifdef WITHMPI
#define MLOCK(x)      do { if ( PF.me != MASTER ) PF_MLock(); } while (0)
#define MUNLOCK(x)    do { if ( PF.me != MASTER ) PF_MUnlock(); } while (0)
#else
#define MLOCK(x)
#define MUNLOCK(x)
#endif
#define GETIDENTITY
#define GETBIDENTITY

#endif

/*
    #] Thread objects :
    #[ Declarations :
*/

#ifdef TERMMALLOCDEBUG
extern WORD **DebugHeap1, **DebugHeap2;
#endif

extern void   StartVariables(void);
extern void   setSignalHandlers(void);
extern UBYTE *CodeToLine(WORD,UBYTE *);
extern UBYTE *AddArrayIndex(WORD ,UBYTE *);
extern INDEXENTRY *FindInIndex(WORD,FILEDATA *,WORD,WORD);
extern INDEXENTRY *NextFileIndex(POSITION *);
extern WORD  *PasteTerm(PHEAD WORD,WORD *,WORD *,WORD,WORD);
extern UBYTE *StrCopy(UBYTE *,UBYTE *);
extern UBYTE *WrtPower(UBYTE *,WORD);
extern WORD   AccumGCD(PHEAD UWORD *,WORD *,UWORD *,WORD);
extern void   AddArgs(PHEAD WORD *,WORD *,WORD *);
extern WORD   AddCoef(PHEAD WORD **,WORD **);
extern WORD   AddLong(UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern WORD   AddPLon(UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern WORD   AddPoly(PHEAD WORD **,WORD **);
extern WORD   AddRat(PHEAD UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern void   AddToLine(UBYTE *);
extern WORD   AddWild(PHEAD WORD,WORD,WORD);
extern WORD   BigLong(UWORD *,WORD,UWORD *,WORD);
```

```
extern WORD   BinomGen(PHEAD WORD *,WORD,WORD **,WORD,WORD,WORD,WORD,WORD,UWORD *,WORD);
extern WORD   CheckWild(PHEAD WORD,WORD,WORD,WORD *);
extern WORD   Chisholm(PHEAD WORD *,WORD);
extern WORD   CleanExpr(WORD);
extern void   CleanUp(WORD);
extern void   ClearWild(PHEAD0);
extern WORD   CompareFunctions(WORD *,WORD *);
extern WORD   Commute(WORD *,WORD *);
extern WORD   DetCommu(WORD *);
extern WORD   DoesCommu(WORD *);
extern int    CompArg(WORD *,WORD *);
extern WORD   CompCoef(WORD *, WORD *);
extern WORD   CompGroup(PHEAD WORD,WORD **,WORD *,WORD *,WORD);
extern WORD   Compare1(PHEAD WORD *,WORD *,WORD);
extern WORD   CountDo(WORD *,WORD *);
extern WORD   CountFun(WORD *,WORD *);
extern WORD   DimensionSubterm(WORD *);
extern WORD   DimensionTerm(WORD *);
extern WORD   DimensionExpression(PHEAD WORD *);
extern WORD   Deferred(PHEAD WORD *,WORD);
extern WORD   DeleteStore(WORD);
extern WORD   DetCurDum(PHEAD WORD *);
extern void   DetVars(WORD *,WORD);
extern WORD   Distribute(DISTRIBUTE *,WORD);
extern WORD   DivLong(UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *,UWORD *,WORD *);
extern WORD   DivRat(PHEAD UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern WORD   Divvy(PHEAD UWORD *,WORD *,UWORD *,WORD);
extern WORD   DoDelta(WORD *);
extern WORD   DoDelta3(PHEAD WORD *,WORD);
extern WORD   TestPartitions(WORD *, PARTI *);
extern WORD   DoPartitions(PHEAD WORD *,WORD);
extern int    CoCanonicalize(UBYTE *);
extern int    DoCanonicalize(PHEAD WORD *, WORD *);
extern WORD   GenTopologies(PHEAD WORD *,WORD);
extern WORD   GenDiagrams(PHEAD WORD *,WORD);
extern int    DoTopologyCanonicalize(PHEAD WORD *,WORD,WORD,WORD *);
extern int    DoShattering(PHEAD WORD *,WORD *,WORD *,WORD);
extern WORD   GenerateTopologies(PHEAD WORD,WORD,WORD,WORD);
extern WORD   DoTableExpansion(WORD *,WORD);
extern WORD   DoDistrib(PHEAD WORD *,WORD);
extern WORD   DoShuffle(WORD *,WORD,WORD,WORD);
extern WORD   DoPermutations(PHEAD WORD *,WORD);
extern int    Shuffle(WORD *, WORD *, WORD *);
extern int    FinishShuffle(WORD *);
extern WORD   DoStuffle(WORD *,WORD,WORD,WORD);
extern int    Stuffle(WORD *, WORD *, WORD *);
extern int    FinishStuffle(WORD *);
extern WORD  *StuffRootAdd(WORD *, WORD *, WORD *);
extern WORD   TestUse(WORD *,WORD);
extern DBASE *FindTB(UBYTE *);
extern int    CheckTableDeclarations(DBASE *);
extern WORD   Apply(WORD *,WORD);
extern int    ApplyExec(WORD *,int,WORD);
extern WORD   ApplyReset(WORD);
extern WORD   TableReset(void);
extern void   ReWorkT(WORD *,WORD *,WORD);
extern WORD   GetIfDollarNum(WORD *, WORD *);
extern int    FindVar(WORD *,WORD *);
extern WORD   DoIfStatement(PHEAD WORD *,WORD *);
extern WORD   DoOnePow(PHEAD WORD *,WORD,WORD,WORD *,WORD *,WORD,WORD *);
extern void   DoRevert(WORD *,WORD *);
extern WORD   DoSumF1(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   DoSumF2(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   DoTheta(PHEAD WORD *);
extern LONG   EndSort(PHEAD WORD *,int);
extern WORD   EntVar(WORD,UBYTE *,WORD,WORD,WORD,WORD);
extern WORD   EpfCon(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   EpfFind(PHEAD WORD *,WORD *);
extern WORD   EpfGen(WORD,WORD *,WORD *,WORD *,WORD);
extern WORD   EqualArg(WORD *,WORD,WORD);
extern WORD   Evaluate(UBYTE **);
extern int    Factorial(PHEAD WORD,UWORD *,WORD *);
extern int    Bernoulli(WORD,UWORD *,WORD *);
extern int    FactorIn(PHEAD WORD *,WORD);
extern int    FactorInExpr(PHEAD WORD *,WORD);
extern WORD   FindAll(PHEAD WORD *,WORD *,WORD,WORD *);
extern WORD   FindMulti(PHEAD WORD *,WORD *);
extern WORD   FindOnce(PHEAD WORD *,WORD *);
extern WORD   FindOnly(PHEAD WORD *,WORD *);
extern WORD   FindRest(PHEAD WORD *,WORD *);
extern WORD   FindSpecial(WORD *);
extern WORD   FindrNumber(WORD,VARRENUM *);
extern void   FiniLine(void);
extern WORD   FiniTerm(PHEAD WORD *,WORD *,WORD *,WORD,WORD);
extern WORD   FlushOut(POSITION *,FILEHANDLE *,int);
extern void   FunLevel(PHEAD WORD *);
extern void   AdjustRenumScratch(PHEAD0);
```

```
extern void   GarbHand(void);
extern WORD   GcdLong(PHEAD UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern WORD   LcmLong(PHEAD UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern void   GCD(UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern ULONG  GCD2(ULONG,ULONG);
extern WORD   Generator(PHEAD WORD *,WORD);
extern WORD   GetBinom(UWORD *,WORD *,WORD,WORD);
extern WORD   GetFromStore(WORD *,POSITION *,RENUMBER,WORD *,WORD);
extern WORD   GetLong(UBYTE *,UWORD *,WORD *);
extern WORD   GetMoreTerms(WORD *);
extern WORD   GetMoreFromMem(WORD *,WORD **);
extern WORD   GetOneTerm(PHEAD WORD *,FILEHANDLE *,POSITION *,int);
extern RENUMBER GetTable(WORD,POSITION *,WORD);
extern WORD   GetTerm(PHEAD WORD *);
extern WORD   Glue(PHEAD WORD *,WORD *,WORD *,WORD);
extern WORD   InFunction(PHEAD WORD *,WORD *);
extern void   IniLine(WORD);
extern WORD   IniVars(void);
extern void   Initialize(void);
extern WORD   InsertTerm(PHEAD WORD *,WORD,WORD,WORD *,WORD *,WORD);
extern void   LongToLine(UWORD *,WORD);
extern WORD   MakeDirty(WORD *,WORD *,WORD);
extern void   MarkDirty(WORD *,WORD);
extern void   PolyFunDirty(PHEAD WORD *);
extern void   PolyFunClean(PHEAD WORD *);
extern WORD   MakeModTable(void);
extern WORD   MatchE(PHEAD WORD *,WORD *,WORD *,WORD);
extern int    MatchCy(PHEAD WORD *,WORD *,WORD *,WORD);
extern int    FunMatchCy(PHEAD WORD *,WORD *,WORD *,WORD);
extern int    FunMatchSy(PHEAD WORD *,WORD *,WORD *,WORD);
extern int    MatchArgument(PHEAD WORD *,WORD *);
extern WORD   MatchFunction(PHEAD WORD *,WORD *,WORD *);
extern WORD   MergePatches(WORD);
extern WORD   MesCerr(char *, UBYTE *);
extern WORD   MesComp(char *, UBYTE *, UBYTE *);
extern WORD   Modulus(WORD *);
extern void   MoveDummies(PHEAD WORD *,WORD);
extern WORD   MulLong(UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern WORD   MulRat(PHEAD UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern WORD   Mully(PHEAD UWORD *,WORD *,UWORD *,WORD);
extern WORD   MultDo(PHEAD WORD *,WORD *);
extern WORD   NewSort(PHEAD0);
extern WORD   ExtraSymbol(WORD,WORD,WORD,WORD *,WORD *);
extern WORD   Normalize(PHEAD WORD *);
extern WORD   BracketNormalize(PHEAD WORD *);
extern void   DropCoefficient(PHEAD WORD *);
extern void   DropSymbols(PHEAD WORD *);
extern int    PutInside(PHEAD WORD *, WORD *);
extern WORD   OpenTemp(void);
extern void   Pack(UWORD *,WORD *,UWORD *,WORD );
extern LONG   PasteFile(PHEAD WORD,WORD *,POSITION *,WORD **,RENUMBER,WORD *,WORD);
extern WORD   Permute(PERM *,WORD);
extern WORD   PermuteP(PERMP *,WORD);
extern WORD   PolyFunMul(PHEAD WORD *);
extern WORD   PopVariables(void);
extern WORD   PrepPoly(PHEAD WORD *,WORD);
extern WORD   Processor(void);
extern WORD   Product(UWORD *,WORD *,WORD);
extern void   PrtLong(UWORD *,WORD,UBYTE *);
extern void   PrtTerms(void);
extern void   PrintDeprecation(const char *,const char *);
extern void   PrintRunningTime(void);
extern LONG   GetRunningTime(void);
extern WORD   PutBracket(PHEAD WORD *);
extern LONG   PutIn(FILEHANDLE *,POSITION *,WORD *,WORD **,int);
extern WORD   PutInStore(INDEXENTRY *,WORD);
extern WORD   PutOut(PHEAD WORD *,POSITION *,FILEHANDLE *,WORD);
extern UWORD  Quotient(UWORD *,WORD *,WORD);
extern WORD   RaisPow(PHEAD UWORD *,WORD *,UWORD);
extern void   RaisPowCached (PHEAD WORD, WORD, UWORD **, WORD *);
extern WORD   RaisPowMod (WORD, WORD, WORD);
extern int    NormalModulus(UWORD *,WORD *);
extern int    MakeInverses(void);
extern int    GetModInverses(WORD,WORD,WORD *,WORD *);
extern int    GetLongModInverses(PHEAD UWORD *, WORD, UWORD *, WORD, UWORD *, WORD *, UWORD *, WORD *);
extern void   RatToLine(UWORD *,WORD);
extern WORD   RatioFind(PHEAD WORD *,WORD *);
extern WORD   RatioGen(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   ReNumber(PHEAD WORD *);
extern WORD   ReadSnum(UBYTE **);
extern WORD   Remain10(UWORD *,WORD *);
extern WORD   Remain4(UWORD *,WORD *);
extern WORD   ResetScratch(void);
extern WORD   ResolveSet(PHEAD WORD *,WORD *,WORD *);
extern WORD   RevertScratch(void);
extern WORD   ScanFunctions(PHEAD WORD *,WORD *,WORD);
```

```
extern void   SeekScratch(FILEHANDLE *,POSITION *);
extern void   SetEndScratch(FILEHANDLE *,POSITION *);
extern void   SetEndHScratch(FILEHANDLE *,POSITION *);
extern WORD   SetFileIndex(void);
extern WORD   Sflush(FILEHANDLE *);
extern WORD   Simplify(PHEAD UWORD *,WORD *,UWORD *,WORD *);
extern WORD   SortWild(WORD *,WORD);
extern FILE  *LocateBase(char **,char **,char *);
extern LONG   SplitMerge(PHEAD WORD **,LONG);
extern WORD   StoreTerm(PHEAD WORD *);
extern void   SubPLon(UWORD *,WORD,UWORD *,WORD,UWORD *,WORD *);
extern void   Substitute(PHEAD WORD *,WORD *,WORD);
extern WORD   SymFind(PHEAD WORD *,WORD *);
extern WORD   SymGen(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   Symmetrize(PHEAD WORD *,WORD *,WORD,WORD,WORD);
extern int    FullSymmetrize(PHEAD WORD *,int);
extern WORD   TakeModulus(UWORD *,WORD *,UWORD *,WORD,WORD);
extern WORD   TakeNormalModulus(UWORD *,WORD *,UWORD *,WORD,WORD);
extern void   TalToLine(UWORD);
extern WORD   TenVec(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   TenVecFind(PHEAD WORD *,WORD *);
extern WORD   TermRenumber(WORD *,RENUMBER,WORD);
extern void   TestDrop(void);
extern void   PutInVflags(WORD);
extern WORD   TestMatch(PHEAD WORD *,WORD *);
extern WORD   TestSub(PHEAD WORD *,WORD);
extern LONG   TimeCPU(WORD);
extern LONG   TimeChildren(WORD);
extern LONG   TimeWallClock(WORD);
extern LONG   Timer(int);
extern int    GetTimerInfo(LONG **,LONG **);
extern void   WriteTimerInfo(LONG *,LONG *);
extern LONG   GetWorkerTimes(void);
extern WORD   ToStorage(EXPRESSIONS,POSITION *);
extern void   TokenToLine(UBYTE *);
extern WORD   Trace4(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   Trace4Gen(PHEAD TRACES *,WORD);
extern WORD   Trace4no(WORD,WORD *,TRACES *);
extern WORD   TraceFind(PHEAD WORD *,WORD *);
extern WORD   TraceN(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   TraceNgen(PHEAD TRACES *,WORD);
extern WORD   TraceNno(WORD,WORD *,TRACES *);
extern WORD   Traces(PHEAD WORD *,WORD *,WORD,WORD);
extern WORD   Trick(WORD *,TRACES *);
extern WORD   TryDo(PHEAD WORD *,WORD *,WORD);
extern void   UnPack(UWORD *,WORD,WORD *,WORD *);
extern WORD   VarStore(UBYTE *,WORD,WORD,WORD);
extern WORD   WildFill(PHEAD WORD *,WORD *,WORD *);
extern WORD   WriteAll(void);
extern WORD   WriteOne(UBYTE *,int,int,WORD);
extern void   WriteArgument(WORD *);
extern WORD   WriteExpression(WORD *,LONG);
extern WORD   WriteInnerTerm(WORD *,WORD);
extern void   WriteLists(void);
extern void   WriteSetup(void);
extern void   WriteStats(POSITION *,WORD,WORD);
extern WORD   WriteSubTerm(WORD *,WORD);
extern WORD   WriteTerm(WORD *,WORD *,WORD,WORD,WORD);
extern WORD   execarg(PHEAD WORD *,WORD);
extern WORD   execterm(PHEAD WORD *,WORD);
extern void   SpecialCleanup(PHEAD0);
extern void   SetMods(void);
extern void   UnSetMods(void);

/*---------------------------------------------------------------------*/

extern WORD   DoExecute(WORD,WORD);
extern void   SetScratch(FILEHANDLE *,POSITION *);
extern void   Warning(char *);
extern void   HighWarning(char *);
extern int    SpareTable(TABLES);

extern UBYTE *strDup1(UBYTE *,char *);
extern void  *Malloc(LONG);
extern void  *Malloc1(LONG,const char *);
extern int    DoTail(int,UBYTE **);
extern int    OpenInput(void);
extern int    PutPreVar(UBYTE *,UBYTE *,UBYTE *,int);
extern void   Error0(char *);
extern void   Error1(char *,UBYTE *);
extern void   Error2(char *,char *,UBYTE *);
extern UBYTE  ReadFromStream(STREAM *);
extern UBYTE  GetFromStream(STREAM *);
extern UBYTE  LookInStream(STREAM *);
extern STREAM *OpenStream(UBYTE *,int,int,int);
extern int    LocateFile(UBYTE **,int);
extern STREAM *CloseStream(STREAM *);
```

```
extern void   PositionStream(STREAM *,LONG);
extern int    ReverseStatements(STREAM *);
extern int    ProcessOption(UBYTE *,UBYTE *,int);
extern int    DoSetups(void);
extern void   TerminateImpl(int, const char *,int, const char *);
extern NAMENODE *GetNode(NAMETREE *,UBYTE *);
extern int    AddName(NAMETREE *,UBYTE *,WORD,WORD,int *);
extern int    GetName(NAMETREE *,UBYTE *,WORD *,int);
extern UBYTE  *GetFunction(UBYTE *,WORD *);
extern UBYTE  *GetNumber(UBYTE *,WORD *);
extern int    GetLastExprName(UBYTE *,WORD *);
extern int    GetAutoName(UBYTE *,WORD *);
extern int    GetVar(UBYTE *,WORD *,WORD *,int,int);
extern int    MakeDubious(NAMETREE *,UBYTE *,WORD *);
extern int    GetOName(NAMETREE *,UBYTE *,WORD *,int);
extern void   DumpTree(NAMETREE *);
extern void   DumpNode(NAMETREE *,WORD,WORD);
extern void   LinkTree(NAMETREE *,WORD,WORD);
extern void   CopyTree(NAMETREE *,NAMETREE *,WORD,WORD);
extern int    CompactifyTree(NAMETREE *,WORD);
extern NAMETREE *MakeNameTree(void);
extern void   FreeNameTree(NAMETREE *);
extern int    AddExpression(UBYTE *,int,int);
extern int    AddSymbol(UBYTE *,int,int,int,int);
extern int    AddDollar(UBYTE *,WORD,WORD *,LONG);
extern int    ReplaceDollar(WORD,WORD,WORD *,LONG);
extern int    DollarRaiseLow(UBYTE *,LONG);
extern int    AddVector(UBYTE *,int,int);
extern int    AddDubious(UBYTE *);
extern int    AddIndex(UBYTE *,int,int);
extern UBYTE *DoDimension(UBYTE *,int *,int *);
extern int    AddFunction(UBYTE *,int,int,int,int,int,int,int);
extern int    CoCommuteInSet(UBYTE *);
extern int    CoFunction(UBYTE *,int,int);
extern int    TestName(UBYTE *);
extern int    AddSet(UBYTE *,WORD);
extern int    DoElements(UBYTE *,SETS,UBYTE *);
extern int    DoTempSet(UBYTE *,UBYTE *);
extern int    NameConflict(int,UBYTE *);
extern int    OpenFile(char *);
extern int    OpenAddFile(char *);
extern int    ReOpenFile(char *);
extern int    CreateFile(char *);
extern int    CreateLogFile(char *);
extern void   CloseFile(int);
extern int    CopyFile(char *, char *);
extern int    CreateHandle(void);
extern LONG   ReadFile(int,UBYTE *,LONG);
extern LONG   ReadPosFile(PHEAD FILEHANDLE *,UBYTE *,LONG,POSITION *);
extern LONG   WriteFileToFile(int,UBYTE *,LONG);
extern void   SeekFile(int,POSITION *,int);
extern LONG   TellFile(int);
extern void   FlushFile(int);
extern int    GetPosFile(int,fpos_t *);
extern int    SetPosFile(int,fpos_t *);
extern void   SynchFile(int);
extern void   TruncateFile(int);
extern int    GetChannel(char *,int);
extern int    GetAppendChannel(char *);
extern int    CloseChannel(char *);
extern void   inictable(void);
extern KEYWORD *findcommand(UBYTE *);
extern int    inicbufs(void);
extern void   StartFiles(void);
extern UBYTE *MakeDate(void);
extern void   PreProcessor(void);
extern void  *FromList(LIST *);
extern void  *From0List(LIST *);
extern void  *FromVarList(LIST *);
extern int    DoubleList(void ***,int *,int,char *);
extern int    DoubleLList(void ***,LONG *,int,char *);
extern void   DoubleBuffer(void **,void **,int,char *);
extern void   ExpandBuffer(void **,LONG *,int);
extern LONG   iexp(LONG,int);
extern int    IsLikeVector(WORD *);
extern int    AreArgsEqual(WORD *,WORD *);
extern int    CompareArgs(WORD *,WORD *);
extern UBYTE *SkipField(UBYTE *,int);
extern int    StrCmp(UBYTE *,UBYTE *);
extern int    StrICmp(UBYTE *,UBYTE *);
extern int    StrHICmp(UBYTE *,UBYTE *);
extern int    StrICont(UBYTE *,UBYTE *);
extern int    CmpArray(WORD *,WORD *,WORD);
extern int    ConWord(UBYTE *,UBYTE *);
extern int    StrLen(UBYTE *);
extern UBYTE *GetPreVar(UBYTE *,int);
extern void   ToGeneral(WORD *,WORD *,WORD);
```

```
extern WORD   ToPolyFunGeneral(PHEAD WORD *);
extern int    ToFast(WORD *,WORD *);
extern SETUPPARAMETERS *GetSetupPar(UBYTE *);
extern int    RecalcSetups(void);
extern int    AllocSetups(void);
extern SORTING *AllocSort(LONG,LONG,LONG,LONG,int,int,LONG,int);
extern void   AllocSortFileName(SORTING *);
extern UBYTE *LoadInputFile(UBYTE *,int);
extern UBYTE  GetInput(void);
extern void   ClearPushback(void);
extern UBYTE  GetChar(int);
extern void   CharOut(UBYTE);
extern void   UnsetAllowDelay(void);
extern void   PopPreVars(int);
extern void   IniModule(int);
extern void   IniSpecialModule(int);
extern int    ModuleInstruction(int *,int *);
extern int    PreProInstruction(void);
extern int    LoadInstruction(int);
extern int    LoadStatement(int);
extern KEYWORD *FindKeyWord(UBYTE *,KEYWORD *,int);
extern KEYWORD *FindInKeyWord(UBYTE *,KEYWORD *,int);
extern int    DoDefine(UBYTE *);
extern int    DoRedefine(UBYTE *);
extern int    TheDefine(UBYTE *,int);
extern int    TheUndefine(UBYTE *);
extern int    ClearMacro(UBYTE *);
extern int    DoUndefine(UBYTE *);
extern int    DoInclude(UBYTE *);
extern int    DoReverseInclude(UBYTE *);
extern int    Include(UBYTE *,int);
/*[14apr2004 mt]:*/
extern int    DoExternal(UBYTE *);
extern int    DoToExternal(UBYTE *);
extern int    DoFromExternal(UBYTE *);
extern int    DoPrompt(UBYTE *);
extern int    DoSetExternal(UBYTE *);
/*[10may2006 mt]:*/
extern int    DoSetExternalAttr(UBYTE *);
/*:[10may2006 mt]*/
extern int    DoRmExternal(UBYTE *);
/*:[14apr2004 mt]*/
extern int    DoFactDollar(UBYTE *);
extern WORD   GetDollarNumber(UBYTE **,DOLLARS);
extern int    DoSetRandom(UBYTE *);
extern int    DoOptimize(UBYTE *);
extern int    DoClearOptimize(UBYTE *);
extern int    DoSkipExtraSymbols(UBYTE *);
extern int    DoTimeOutAfter(UBYTE *);
extern int    DoMessage(UBYTE *);
extern int    DoPreOut(UBYTE *);
extern int    DoPreAppend(UBYTE *);
extern int    DoPreCreate(UBYTE *);
extern int    DoPreAssign(UBYTE *);
extern int    DoPreBreak(UBYTE *);
extern int    DoPreDefault(UBYTE *);
extern int    DoPreSwitch(UBYTE *);
extern int    DoPreEndSwitch(UBYTE *);
extern int    DoPreCase(UBYTE *);
extern int    DoPreShow(UBYTE *);
extern int    DoPreExchange(UBYTE *);
extern int    DoSystem(UBYTE *);
extern int    DoPipe(UBYTE *);
extern void   StartPrepro(void);
extern int    DoIfdef(UBYTE *,int);
extern int    DoIfydef(UBYTE *);
extern int    DoIfndef(UBYTE *);
extern int    DoElse(UBYTE *);
extern int    DoElseif(UBYTE *);
extern int    DoEndif(UBYTE *);
extern int    DoTerminate(UBYTE *);
extern int    DoIf(UBYTE *);
extern int    DoCall(UBYTE *);
extern int    DoDebug(UBYTE *);
extern int    DoContinueDo(UBYTE *);
extern int    DoDo(UBYTE *);
extern int    DoBreakDo(UBYTE *);
extern int    DoEnddo(UBYTE *);
extern int    DoEndprocedure(UBYTE *);
extern int    DoInside(UBYTE *);
extern int    DoEndInside(UBYTE *);
extern int    DoProcedure(UBYTE *);
extern int    DoPrePrintTimes(UBYTE *);
extern int    DoPreWrite(UBYTE *);
extern int    DoPreClose(UBYTE *);
extern int    DoPreRemove(UBYTE *);
extern int    DoPreSortReallocate(UBYTE *);
```

```
extern int    DoCommentChar(UBYTE *);
extern int    DoPrcExtension(UBYTE *);
extern int    DoPreReset(UBYTE *);
extern void   WriteString(int,UBYTE *,int);
extern void   WriteUnfinString(int,UBYTE *,int);
extern UBYTE *AddToString(UBYTE *,UBYTE *,int);
extern UBYTE *PreCalc(void);
extern UBYTE *PreEval(UBYTE *,LONG *);
extern void   NumToStr(UBYTE *,LONG);
extern int    PreCmp(int,int,UBYTE *,int,int,UBYTE *,int);
extern int    PreEq(int,int,UBYTE *,int,int,UBYTE *,int);
extern UBYTE *pParseObject(UBYTE *,int *,LONG *);
extern UBYTE *PreIfEval(UBYTE *,int *);
extern int    EvalPreIf(UBYTE *);
extern int    PreLoad(PRELOAD *,UBYTE *,UBYTE *,int,char *);
extern int    PreSkip(UBYTE *,UBYTE *,int);
extern UBYTE *EndOfToken(UBYTE *);
extern void   SetSpecialMode(int,int);
extern void   MakeGlobal(void);
extern int    ExecModule(int);
extern int    ExecStore(void);
extern void   FullCleanUp(void);
extern int    DoExecStatement(void);
extern int    DoPipeStatement(void);
extern int    DoPolyfun(UBYTE *);
extern int    DoPolyratfun(UBYTE *);
extern int    CompileStatement(UBYTE *);
extern UBYTE *ToToken(UBYTE *);
extern int    GetDollar(UBYTE *);
extern int    MesWork(void);
extern int    MesPrint(const char *,...);
extern int    MesCall(char *);
extern UBYTE *NumCopy(WORD,UBYTE *);
extern char  *LongCopy(LONG,char *);
extern char  *LongLongCopy(off_t *,char *);
extern void   ReserveTempFiles(int);
extern void   PrintTerm(WORD *,char *);
extern void   PrintTermC(WORD *,char *);
extern void   PrintSubTerm(WORD *,char *);
extern void   PrintWords(WORD *,LONG);
extern void   PrintSeq(WORD *,char *);
extern int    ExpandTripleDots(int);
extern LONG   ComPress(WORD **,LONG *);
extern void   StageSort(FILEHANDLE *);

#define M_alloc(x)      malloc((size_t)(x))

extern void   M_free(void *,const char *);
extern void   ClearWildcardNames(void);
extern int    AddWildcardName(UBYTE *);
extern int    GetWildcardName(UBYTE *);
extern void   Globalize(int);
extern void   ResetVariables(int);
extern void   AddToPreTypes(int);
extern void   MessPreNesting(int);
extern LONG   GetStreamPosition(STREAM *);
extern WORD  *DoubleCbuffer(int,WORD *,int);
extern WORD  *AddLHS(int);
extern WORD  *AddRHS(int,int);
extern int    AddNtoL(int,WORD *);
extern int    AddNtoC(int,int,WORD *,int);
extern void   DoubleIfBuffers(void);
extern STREAM *CreateStream(UBYTE *);

extern int    setonoff(UBYTE *,int *,int,int);
extern int    DoPrint(UBYTE *,int);
extern int    SetExpr(UBYTE *,int,int);
extern void   AddToCom(int,WORD *);
extern int    Add2ComStrings(int,WORD *,UBYTE *,UBYTE *);
extern int    DoSymmetrize(UBYTE *,int);
extern int    DoArgument(UBYTE *,int);
extern int    ArgFactorize(PHEAD WORD *,WORD *);
extern WORD  *TakeArgContent(PHEAD WORD *, WORD *);
extern WORD  *MakeInteger(PHEAD WORD *,WORD *,WORD *);
extern WORD  *MakeMod(PHEAD WORD *,WORD *,WORD *);
extern WORD   FindArg(PHEAD WORD *);
extern WORD   InsertArg(PHEAD WORD *,WORD *,int);
extern int    CleanupArgCache(PHEAD WORD);
extern int    ArgSymbolMerge(WORD *, WORD *);
extern int    ArgDotproductMerge(WORD *, WORD *);
extern void   SortWeights(LONG *,LONG *,WORD);
extern int    DoBrackets(UBYTE *,int);
extern int    DoPutInside(UBYTE *,int);
extern WORD  *CountComp(UBYTE *,WORD *);
extern int    CoAntiBracket(UBYTE *);
extern int    CoAntiSymmetrize(UBYTE *);
extern int    DoArgPlode(UBYTE *,int);
```

```
extern int     CoArgExplode(UBYTE *);
extern int     CoArgImplode(UBYTE *);
extern int     CoArgument(UBYTE *);
extern int     CoInside(UBYTE *);
extern int     ExecInside(UBYTE *);
extern int     CoInExpression(UBYTE *);
extern int     CoInParallel(UBYTE *);
extern int     CoNotInParallel(UBYTE *);
extern int     DoInParallel(UBYTE *,int);
extern int     CoEndInExpression(UBYTE *);
extern int     CoBracket(UBYTE *);
extern int     CoPutInside(UBYTE *);
extern int     CoAntiPutInside(UBYTE *);
extern int     CoMultiBracket(UBYTE *);
extern int     CoCFunction(UBYTE *);
extern int     CoCTensor(UBYTE *);
extern int     CoCollect(UBYTE *);
extern int     CoCompress(UBYTE *);
extern int     CoContract(UBYTE *);
extern int     CoCycleSymmetrize(UBYTE *);
extern int     CoDelete(UBYTE *);
extern int     CoTableBase(UBYTE *);
extern int     CoApply(UBYTE *);
extern int     CoDenominators(UBYTE *);
extern int     CoDimension(UBYTE *);
extern int     CoDiscard(UBYTE *);
extern int     CoDisorder(UBYTE *);
extern int     CoDrop(UBYTE *);
extern int     CoDropCoefficient(UBYTE *);
extern int     CoDropSymbols(UBYTE *);
extern int     CoElse(UBYTE *);
extern int     CoElseIf(UBYTE *);
extern int     CoEndArgument(UBYTE *);
extern int     CoEndInside(UBYTE *);
extern int     CoEndIf(UBYTE *);
extern int     CoEndRepeat(UBYTE *);
extern int     CoEndTerm(UBYTE *);
extern int     CoEndWhile(UBYTE *);
extern int     CoExit(UBYTE *);
extern int     CoFactArg(UBYTE *);
extern int     CoFactDollar(UBYTE *);
extern int     CoFactorize(UBYTE *);
extern int     CoNFactorize(UBYTE *);
extern int     CoUnFactorize(UBYTE *);
extern int     CoNUnFactorize(UBYTE *);
extern int     DoFactorize(UBYTE *,int);
extern int     CoFill(UBYTE *);
extern int     CoFillExpression(UBYTE *);
extern int     CoFixIndex(UBYTE *);
extern int     CoFormat(UBYTE *);
extern int     CoGlobal(UBYTE *);
extern int     CoGlobalFactorized(UBYTE *);
extern int     CoGoTo(UBYTE *);
extern int     CoId(UBYTE *);
extern int     CoIdNew(UBYTE *);
extern int     CoIdOld(UBYTE *);
extern int     CoIf(UBYTE *);
extern int     CoIfMatch(UBYTE *);
extern int     CoIfNoMatch(UBYTE *);
extern int     CoIndex(UBYTE *);
extern int     CoInsideFirst(UBYTE *);
extern int     CoKeep(UBYTE *);
extern int     CoLabel(UBYTE *);
extern int     CoLoad(UBYTE *);
extern int     CoLocal(UBYTE *);
extern int     CoLocalFactorized(UBYTE *);
extern int     CoMany(UBYTE *);
extern int     CoMerge(UBYTE *);
extern int     CoStuffle(UBYTE *);
extern int     CoMetric(UBYTE *);
extern int     CoModOption(UBYTE *);
extern int     CoModuleOption(UBYTE *);
extern int     CoModulus(UBYTE *);
extern int     CoMulti(UBYTE *);
extern int     CoMultiply(UBYTE *);
extern int     CoNFunction(UBYTE *);
extern int     CoNPrint(UBYTE *);
extern int     CoNTensor(UBYTE *);
extern int     CoNWrite(UBYTE *);
extern int     CoNoDrop(UBYTE *);
extern int     CoNoSkip(UBYTE *);
extern int     CoNormalize(UBYTE *);
extern int     CoMakeInteger(UBYTE *);
extern int     CoFlags(UBYTE *,int);
extern int     CoOff(UBYTE *);
extern int     CoOn(UBYTE *);
extern int     CoOnce(UBYTE *);
```

```
extern int    CoOnly(UBYTE *);
extern int    CoOptimizeOption(UBYTE *);
extern int    CoOptimize(UBYTE *);
extern int    CoPolyFun(UBYTE *);
extern int    CoPolyRatFun(UBYTE *);
extern int    CoPrint(UBYTE *);
extern int    CoPrintB(UBYTE *);
extern int    CoProperCount(UBYTE *);
extern int    CoUnitTrace(UBYTE *);
extern int    CoRCycleSymmetrize(UBYTE *);
extern int    CoRatio(UBYTE *);
extern int    CoRedefine(UBYTE *);
extern int    CoRenumber(UBYTE *);
extern int    CoRepeat(UBYTE *);
extern int    CoSave(UBYTE *);
extern int    CoSelect(UBYTE *);
extern int    CoSet(UBYTE *);
extern int    CoSetExitFlag(UBYTE *);
extern int    CoSkip(UBYTE *);
extern int    CoProcessBucket(UBYTE *);
extern int    CoPushHide(UBYTE *);
extern int    CoPopHide(UBYTE *);
extern int    CoHide(UBYTE *);
extern int    CoIntoHide(UBYTE *);
extern int    CoNoIntoHide(UBYTE *);
extern int    CoNoHide(UBYTE *);
extern int    CoUnHide(UBYTE *);
extern int    CoNoUnHide(UBYTE *);
extern int    CoSort(UBYTE *);
extern int    CoSplitArg(UBYTE *);
extern int    CoSplitFirstArg(UBYTE *);
extern int    CoSplitLastArg(UBYTE *);
extern int    CoSum(UBYTE *);
extern int    CoSymbol(UBYTE *);
extern int    CoSymmetrize(UBYTE *);
extern int    DoTable(UBYTE *,int);
extern int    CoTable(UBYTE *);
extern int    CoTerm(UBYTE *);
extern int    CoNTable(UBYTE *);
extern int    CoCTable(UBYTE *);
extern void   EmptyTable(TABLES);
extern int    CoToTensor(UBYTE *);
extern int    CoToVector(UBYTE *);
extern int    CoTrace4(UBYTE *);
extern int    CoTraceN(UBYTE *);
extern int    CoChisholm(UBYTE *);
extern int    CoTransform(UBYTE *);
extern int    CoClearTable(UBYTE *);
extern int    DoChain(UBYTE *,int);
extern int    CoChainin(UBYTE *);
extern int    CoChainout(UBYTE *);
extern int    CoTryReplace(UBYTE *);
extern int    CoVector(UBYTE *);
extern int    CoWhile(UBYTE *);
extern int    CoWrite(UBYTE *);
extern int    CoAuto(UBYTE *);
extern int    CoSwitch(UBYTE *);
extern int    CoCase(UBYTE *);
extern int    CoBreak(UBYTE *);
extern int    CoDefault(UBYTE *);
extern int    CoEndSwitch(UBYTE *);
extern int    CoTBaddto(UBYTE *);
extern int    CoTBaudit(UBYTE *);
extern int    CoTBcleanup(UBYTE *);
extern int    CoTBcreate(UBYTE *);
extern int    CoTBenter(UBYTE *);
extern int    CoTBhelp(UBYTE *);
extern int    CoTBload(UBYTE *);
extern int    CoTBoff(UBYTE *);
extern int    CoTBon(UBYTE *);
extern int    CoTBopen(UBYTE *);
extern int    CoTBreplace(UBYTE *);
extern int    CoTBuse(UBYTE *);
extern int    CoTestUse(UBYTE *);
extern int    CoThreadBucket(UBYTE *);
extern int    AddComString(int,WORD *,UBYTE *,int);
extern int    CompileAlgebra(UBYTE *,int,WORD *);
extern int    IsIdStatement(UBYTE *);
extern UBYTE *IsRHS(UBYTE *,UBYTE);
extern int    ParenthesesTest(UBYTE *);
extern int    tokenize(UBYTE *,WORD);
extern void   WriteTokens(SBYTE *);
extern int    simp1token(SBYTE *);
extern int    simpwtoken(SBYTE *);
extern int    simp2token(SBYTE *);
extern int    simp3atoken(SBYTE *,int);
extern int    simp3btoken(SBYTE *,int);
```

```
extern int    simp4token(SBYTE *);
extern int    simp5token(SBYTE *,int);
extern int    simp6token(SBYTE *,int);
extern UBYTE *SkipAName(UBYTE *);
extern int    TestTables(void);
extern int    GetLabel(UBYTE *);
extern int    CoIdExpression(UBYTE *,int);
extern int    CoAssign(UBYTE *);
extern int    DoExpr(UBYTE *,int,int);
extern int    CompileSubExpressions(SBYTE *);
extern int    CodeGenerator(SBYTE *);
extern int    CompleteTerm(WORD *,UWORD *,UWORD *,WORD,WORD,int);
extern int    CodeFactors(SBYTE *s);
extern WORD   GenerateFactors(WORD,WORD);
extern int    InsTree(int,int);
extern int    FindTree(int,WORD *);
extern void   RedoTree(CBUF *,int);
extern void   ClearTree(int);
extern int    CatchDollar(int);
extern int    AssignDollar(PHEAD WORD *,WORD);
extern UBYTE *WriteDollarToBuffer(WORD,WORD);
extern UBYTE *WriteDollarFactorToBuffer(WORD,WORD,WORD);
extern void   AddToDollarBuffer(UBYTE *);
extern int    PutTermInDollar(WORD *,WORD);
extern void   TermAssign(WORD *);
extern void   WildDollars(PHEAD WORD *);
extern LONG   numcommute(WORD *,LONG *);
extern int    FullRenumber(PHEAD WORD *,WORD);
extern int    Lus(WORD *,WORD,WORD,WORD,WORD,WORD);
extern int    FindLus(int,int,int);
extern int    CoReplaceLoop(UBYTE *);
extern int    CoFindLoop(UBYTE *);
extern int    DoFindLoop(UBYTE *,int);
extern int    CoFunPowers(UBYTE *);
extern int    SortTheList(int *,int);
extern int    MatchIsPossible(WORD *,WORD *);
extern int    StudyPattern(WORD *);
extern WORD   DolToTensor(PHEAD WORD);
extern WORD   DolToFunction(PHEAD WORD);
extern WORD   DolToVector(PHEAD WORD);
extern WORD   DolToNumber(PHEAD WORD);
extern WORD   DolToSymbol(PHEAD WORD);
extern WORD   DolToIndex(PHEAD WORD);
extern LONG   DolToLong(PHEAD WORD);
extern int    DollarFactorize(PHEAD WORD);
extern int    CoPrintTable(UBYTE *);
extern int    CoDeallocateTable(UBYTE *);
extern void   CleanDollarFactors(DOLLARS);
extern WORD   *TakeDollarContent(PHEAD WORD *,WORD **);
extern WORD   *MakeDollarInteger(PHEAD WORD *,WORD **);
extern WORD   *MakeDollarMod(PHEAD WORD *,WORD **);
extern int    GetDolNum(PHEAD WORD *, WORD *);
extern void   AddPotModdollar(WORD);

extern int    Optimize(WORD, int);
extern int    ClearOptimize(void);
extern int    LoadOpti(WORD);
extern int    PutObject(WORD *,int);
extern void   CleanOptiBuffer(void);
extern int    PrintOptima(WORD);
extern int    FindScratchName(void);
extern WORD   MaxPowerOpti(LONG);
extern WORD   HuntNumFactor(LONG,WORD *,int);
extern WORD   HuntFactor(LONG,WORD *,int);
extern void   HuntPairs(LONG,WORD);
extern void   HuntBrackets(LONG);
extern int    AddToOpti(WORD *,int);
extern LONG   TestNewSca(LONG,WORD *,WORD *);
extern void   NormOpti(WORD *);
extern void   SortOpti(LONG);
extern void   SplitOpti(WORD **,LONG);
extern void   CombiOpti(void);
extern int    TakeLongRoot(UWORD *,WORD *,WORD);
extern int    TakeRatRoot(UWORD *,WORD *,WORD);
extern int    MakeRational(WORD ,WORD , WORD *, WORD *);
extern int    MakeLongRational(PHEAD UWORD *,WORD ,UWORD *,WORD ,UWORD *,WORD *);
extern void   HuntPowers(LONG,WORD);
extern void   HuntNumBrackets(LONG);
extern void   ClearTableTree(TABLES);
extern int    InsTableTree(TABLES,WORD *);
extern void   RedoTableTree(TABLES,int);
extern int    FindTableTree(TABLES,WORD *,int);
extern void   finishcbuf(WORD);
extern void   clearcbuf(WORD);
extern void   CleanUpSort(int);
extern FILEHANDLE *AllocFileHandle(WORD,char *);
extern void   DeAllocFileHandle(FILEHANDLE *);
```

```
extern void   LowerSortLevel(void);
extern WORD  *PolyRatFunSpecial(PHEAD WORD *, WORD *);
extern void   SimpleSplitMergeRec(WORD *,WORD,WORD *);
extern void   SimpleSplitMerge(WORD *,WORD);
extern WORD   BinarySearch(WORD *,WORD,WORD);
extern int    InsideDollar(PHEAD WORD *,WORD);
extern DOLLARS DolToTerms(PHEAD WORD);
extern WORD   EvalDoLoopArg(PHEAD WORD *,WORD);
extern int    SetExprCases(int,int,int);
extern int    TestSelect(WORD *,WORD *);
extern void   SubsInAll(PHEAD0);
extern void   TransferBuffer(int,int,int);
extern int    TakeIDfunction(PHEAD WORD *);
extern int    MakeSetupAllocs(void);
extern int    TryFileSetups(void);
extern void   ExchangeExpressions(int,int);
extern void   ExchangeDollars(int,int);
extern int    GetFirstBracket(WORD *,int);
extern int    GetFirstTerm(WORD *,int,int);
extern int    GetContent(WORD *,int);
extern int    CleanupTerm(WORD *);
extern WORD   ContentMerge(PHEAD WORD *,WORD *);
extern UBYTE *PreIfDollarEval(UBYTE *,int *);
extern LONG   TermsInDollar(WORD);
extern LONG   SizeOfDollar(WORD);
extern LONG   TermsInExpression(WORD);
extern LONG   SizeOfExpression(WORD);
extern WORD   *TranslateExpression(UBYTE *);
extern int    IsSetMember(WORD *,WORD);
extern int    IsMultipleOf(WORD *,WORD *);
extern int    TwoExprCompare(WORD *,WORD *,int);
extern void   UpdatePositions(void);
extern void   M_check(void);
extern void   M_print(void);
extern void   M_check1(void);
extern void   PrintTime(UBYTE *);

extern POSITION *FindBracket(WORD,WORD *);
extern void   PutBracketInIndex(PHEAD WORD *,POSITION *);
extern void   ClearBracketIndex(WORD);
extern void   OpenBracketIndex(WORD);
extern int    DoNoParallel(UBYTE *);
extern int    DoParallel(UBYTE *);
extern int    DoModSum(UBYTE *);
extern int    DoModMax(UBYTE *);
extern int    DoModMin(UBYTE *);
extern int    DoModLocal(UBYTE *);
extern UBYTE *DoModDollar(UBYTE *,int);
extern int    DoProcessBucket(UBYTE *);
extern int    DoinParallel(UBYTE *);
extern int    DonotinParallel(UBYTE *);

extern int    FlipTable(FUNCTIONS,int);
extern int    ChainIn(PHEAD WORD *,WORD);
extern int    ChainOut(PHEAD WORD *,WORD);
extern int    ArgumentImplode(PHEAD WORD *,WORD *);
extern int    ArgumentExplode(PHEAD WORD *,WORD *);
extern int    DenToFunction(WORD *,WORD);

extern WORD   HowMany(PHEAD WORD *,WORD *);
extern void   RemoveDollars(void);
extern LONG   CountTerms1(PHEAD0);
extern LONG   TermsInBracket(PHEAD WORD *,WORD);
extern int    Crash(void);

extern char  *str_dup(char *);
extern void   convertblock(INDEXBLOCK *,INDEXBLOCK *,int);
extern void   convertnamesblock(NAMESBLOCK *,NAMESBLOCK *,int);
extern void   convertiniinfo(INIINFO *,INIINFO *,int);
extern int    ReadIndex(DBASE *);
extern int    WriteIndexBlock(DBASE *,MLONG);
extern int    WriteNamesBlock(DBASE *,MLONG);
extern int    WriteIndex(DBASE *);
extern int    WriteIniInfo(DBASE *);
extern int    ReadIniInfo(DBASE *);
extern int    AddToIndex(DBASE *,MLONG);
extern DBASE *GetDbase(char *,MLONG);
extern DBASE *OpenDbase(char *);
extern char  *ReadObject(DBASE *,MLONG,char *);
extern char  *ReadijObject(DBASE *,MLONG,MLONG,char *);
extern int    ExistsObject(DBASE *,MLONG,char *);
extern int    DeleteObject(DBASE *,MLONG,char *);
extern int    WriteObject(DBASE *,MLONG,char *,char *,MLONG);
extern MLONG  AddObject(DBASE *,MLONG,char *,char *);
extern int    Cleanup(DBASE *);
extern DBASE *NewDbase(char *,MLONG);
extern void   FreeTableBase(DBASE *);
```

```
extern int    ComposeTableNames(DBASE *);
extern int    PutTableNames(DBASE *);
extern MLONG  AddTableName(DBASE *,char *,TABLES);
extern MLONG  GetTableName(DBASE *,char *);
extern MLONG  FindTableNumber(DBASE *,char *);
extern int    TryEnvironment(void);

#ifdef WITHZLIB
extern int    SetupOutputGZIP(FILEHANDLE *);
extern int    PutOutputGZIP(FILEHANDLE *);
extern int    FlushOutputGZIP(FILEHANDLE *);
extern int    SetupAllInputGZIP(SORTING *);
extern LONG   FillInputGZIP(FILEHANDLE *,POSITION *,UBYTE *,LONG,int);
extern void   ClearSortGZIP(FILEHANDLE *f);
#endif

#ifdef WITHPTHREADS
extern void   BeginIdentities(void);
extern int    WhoAmI(void);
extern int    StartAllThreads(int);
extern void   StartHandleLock(void);
extern void   TerminateAllThreads(void);
extern int    GetAvailableThread(void);
extern int    ConditionalGetAvailableThread(void);
extern int    BalanceRunThread(PHEAD int,WORD *,WORD);
extern void   WakeupThread(int,int);
extern int    MasterWait(void);
extern int    InParallelProcessor(void);
extern int    ThreadsProcessor(EXPRESSIONS,WORD,WORD);
extern int    MasterMerge(void);
extern int    PutToMaster(PHEAD WORD *);
extern void   SetWorkerFiles(void);
extern int    MakeThreadBuckets(int,int);
extern int    SendOneBucket(int);
extern int    LoadOneThread(int,int,THREADBUCKET *,int);
extern void  *RunSortBot(void *);
extern void   MasterWaitAllSortBots(void);
extern int    SortBotMerge(PHEAD0);
extern int    SortBotOut(PHEAD WORD *);
extern void   DefineSortBotTree(void);
extern int    SortBotMasterMerge(void);
extern int    SortBotWait(int);
extern void   StartIdentity(void);
extern void   FinishIdentity(void *);
extern int    SetIdentity(int *);
extern ALLPRIVATES *InitializeOneThread(int);
extern void   FinalizeOneThread(int);
extern void   ClearAllThreads(void);
extern void  *RunThread(void *);
extern void   IAmAvailable(int);
extern int    ThreadWait(int);
extern int    ThreadClaimedBlock(int);
extern int    GetThread(int);
extern int    UpdateOneThread(int);
extern void   MasterWaitAll(void);
extern void   MasterWaitAllBlocks(void);
extern int    MasterWaitThread(int);
extern void   WakeupMasterFromThread(int,int);
extern int    LoadReadjusted(void);
extern int    IniSortBlocks(int);
extern int    UpdateSortBlocks(int);
extern int    TreatIndexEntry(PHEAD LONG);
extern WORD   GetTerm2(PHEAD WORD *);
extern void SetHideFiles(void);

#endif

extern int    CopyExpression(FILEHANDLE *,FILEHANDLE *);

extern int    set_in(UBYTE, set_of_char);
extern one_byte set_set(UBYTE, set_of_char);
extern one_byte set_del(UBYTE, set_of_char);
extern one_byte set_sub (set_of_char, set_of_char, set_of_char);
extern int    DoPreAddSeparator(UBYTE *);
extern int    DoPreRmSeparator(UBYTE *);

/*See the file extcmd.c*/
extern int    openExternalChannel(UBYTE *,int,UBYTE *,UBYTE *);
extern int    initPresetExternalChannels(UBYTE *, int);
extern int    closeExternalChannel(int);
extern int    selectExternalChannel(int);
extern int    getCurrentExternalChannel(void);
extern void   closeAllExternalChannels(void);

typedef int (*WRITEBUFTOEXTCHANNEL)(char *,size_t);
typedef int (*GETCFROMEXTCHANNEL)(void);
typedef int (*SETTERMINATORFOREXTERNALCHANNEL)(char *);
```

```
typedef int (*SETKILLMODEFOREXTERNALCHANNEL)(int,int);
typedef LONG (*WRITEFILE)(int,UBYTE *,LONG);
typedef WORD (*GETTERM)(PHEAD WORD *);

#define CompareTerms ((COMPARE)AR.CompareRoutine)
#define FiniShuffle AN.SHvar.finishuf
#define DoShtuffle ((DO_UFFLE)AN.SHvar.do_uffle)

extern UBYTE *defineChannel(UBYTE*, HANDLERS*);
extern int    writeToChannel(int,UBYTE *,HANDLERS*);
#ifdef WITHEXTERNALCHANNEL
extern LONG   WriteToExternalChannel(int,UBYTE *,LONG);
#endif
extern int    writeBufToExtChannelOk(char *,size_t);
extern int    getcFromExtChannelOk(void);
extern int    setKillModeForExternalChannelOk(int,int);
extern int    setTerminatorForExternalChannelOk(char *);
extern int    getcFromExtChannelFailure(void);
extern int    setKillModeForExternalChannelFailure(int,int);
extern int    setTerminatorForExternalChannelFailure(char *);
extern int    writeBufToExtChannelFailure(char *,size_t);

extern int    ReleaseTB(void);

extern int    SymbolNormalize(WORD *);
extern int    TestFunFlag(PHEAD WORD *);
extern WORD   CompareSymbols(PHEAD WORD *,WORD *,WORD);
extern WORD   CompareHSymbols(PHEAD WORD *,WORD *,WORD);
extern WORD   NextPrime(PHEAD WORD);
extern WORD   Moebius(PHEAD WORD);
extern UWORD  wranf(PHEAD0);
extern UWORD  iranf(PHEAD UWORD);
extern void   iniwranf(PHEAD0);
extern UBYTE *PreRandom(UBYTE *);

extern WORD *PolyNormPoly (PHEAD WORD);
extern WORD  *EvaluateGcd(PHEAD WORD *);
extern int TreatPolyRatFun(PHEAD WORD *);

extern WORD   ReadSaveHeader(void);
extern WORD   ReadSaveIndex(FILEINDEX *);
extern WORD   ReadSaveExpression(UBYTE *,UBYTE *,LONG *,LONG *);
extern UBYTE *ReadSaveTerm32(UBYTE *,UBYTE *,UBYTE **,UBYTE *,UBYTE *,int);
extern WORD   ReadSaveVariables(UBYTE *,UBYTE *,LONG *,LONG *,INDEXENTRY *,LONG *);
extern WORD   WriteStoreHeader(WORD);

extern void   InitRecovery(void);
extern int    CheckRecoveryFile(void);
extern void   DeleteRecoveryFile(void);
extern char  *RecoveryFilename(void);
extern int    DoRecovery(int *);
extern void   DoCheckpoint(int);

extern void NumberMallocAddMemory(PHEAD0);
extern void CacheNumberMallocAddMemory(PHEAD0);
extern void TermMallocAddMemory(PHEAD0);
#ifndef MEMORYMACROS
extern WORD *TermMalloc2(PHEAD char *text);
extern void TermFree2(PHEAD WORD *term,char *text);
extern UWORD *NumberMalloc2(PHEAD char *text);
extern UWORD *CacheNumberMalloc2(PHEAD char *text);
extern void NumberFree2(PHEAD UWORD *NumberMem,char *text);
extern void CacheNumberFree2(PHEAD UWORD *NumberMem,char *text);
#endif

extern void ExprStatus(EXPRESSIONS);
extern void iniTools(void);
extern int TestTerm(WORD *);

extern WORD RunTransform(PHEAD WORD *term, WORD *params);
extern WORD RunEncode(PHEAD WORD *fun, WORD *args, WORD *info);
extern WORD RunDecode(PHEAD WORD *fun, WORD *args, WORD *info);
extern WORD RunReplace(PHEAD WORD *fun, WORD *args, WORD *info);
extern WORD RunImplode(WORD *fun, WORD *args);
extern WORD RunExplode(PHEAD WORD *fun, WORD *args);
extern int TestArgNum(int n, int totarg, WORD *args);
extern WORD PutArgInScratch(WORD *arg,UWORD *scrat);
extern UBYTE *ReadRange(UBYTE *s, WORD *out, int par);
extern int  FindRange(PHEAD WORD *,WORD *,WORD *,WORD);
extern WORD RunPermute(PHEAD WORD *fun, WORD *args, WORD *info);
extern WORD RunReverse(PHEAD WORD *fun, WORD *args);
extern WORD RunCycle(PHEAD WORD *fun, WORD *args, WORD *info);
extern WORD RunAddArg(PHEAD WORD *fun, WORD *args);
extern WORD RunMulArg(PHEAD WORD *fun, WORD *args);
extern WORD RunIsLyndon(PHEAD WORD *fun, WORD *args, int par);
extern WORD RunToLyndon(PHEAD WORD *fun, WORD *args, int par);
extern WORD RunDropArg(PHEAD WORD *fun, WORD *args);
```

```
extern WORD RunSelectArg(PHEAD WORD *fun, WORD *args);
extern WORD RunDedup(PHEAD WORD *fun, WORD *args);
extern WORD RunZtoHArg(PHEAD WORD *fun, WORD *args);
extern WORD RunHtoZArg(PHEAD WORD *fun, WORD *args);

extern int NormPolyTerm(PHEAD WORD *);
extern WORD ComparePoly(WORD *, WORD *, WORD);
extern int ConvertToPoly(PHEAD WORD *, WORD *,WORD *,WORD);
extern int LocalConvertToPoly(PHEAD WORD *, WORD *, WORD,WORD);
extern int ConvertFromPoly(PHEAD WORD *, WORD *, WORD, WORD, WORD, WORD);
extern WORD FindSubterm(WORD *);
extern WORD FindLocalSubterm(PHEAD WORD *, WORD);
extern void PrintSubtermList(int,int);
extern void PrintExtraSymbol(int,WORD *,int);
extern WORD FindSubexpression(WORD *);

extern void UpdateMaxSize(void);

extern int CoToPolynomial(UBYTE *);
extern int CoFromPolynomial(UBYTE *);
extern int CoArgToExtraSymbol(UBYTE *);
extern int CoExtraSymbols(UBYTE *);
extern UBYTE *GetDoParam(UBYTE *, WORD **, int);
extern WORD *GetIfDollarFactor(UBYTE **, WORD *);
extern int CoDo(UBYTE *);
extern int CoEndDo(UBYTE *);
extern int ExtraSymFun(PHEAD WORD *,WORD);
extern int PruneExtraSymbols(WORD);
extern int IniFbuffer(WORD);
extern void IniFbufs(void);
extern int GCDfunction(PHEAD WORD *,WORD);
extern WORD *GCDfunction3(PHEAD WORD *,WORD *);
extern WORD *GCDfunction4(PHEAD WORD *,WORD *);
extern int ReadPolyRatFun(PHEAD WORD *);
extern int FromPolyRatFun(PHEAD WORD *, WORD **, WORD **);
extern void PRFnormalize(PHEAD WORD *);
extern WORD *PRFadd(PHEAD WORD *, WORD *);
extern WORD *PolyDiv(PHEAD WORD *,WORD *,char *);
extern WORD *PolyGCD(PHEAD WORD *,WORD *);
extern WORD *PolyAdd(PHEAD WORD *,WORD *);
extern void GCDclean(PHEAD WORD *, WORD *);
extern int RatFunNormalize(PHEAD WORD *);
extern WORD *TakeSymbolContent(PHEAD WORD *,WORD *);
extern int GCDterms(PHEAD WORD *,WORD *,WORD *);
extern WORD *PutExtraSymbols(PHEAD WORD *,WORD,int *);
extern WORD *TakeExtraSymbols(PHEAD WORD *,WORD);
extern WORD *MultiplyWithTerm(PHEAD WORD *, WORD *,WORD);
extern WORD *TakeContent(PHEAD WORD *, WORD *);
extern int MergeSymbolLists(PHEAD WORD *, WORD *, int);
extern int MergeDotproductLists(PHEAD WORD *, WORD *, int);
extern WORD *CreateExpression(PHEAD WORD);
extern int DIVfunction(PHEAD WORD *,WORD,int);
extern WORD *MULfunc(PHEAD WORD *, WORD *);
extern WORD *ConvertArgument(PHEAD WORD *,int *);
extern int ExpandRat(PHEAD WORD *);
extern int InvPoly(PHEAD WORD *,WORD,WORD);
extern WORD TestDoLoop(PHEAD WORD *,WORD);
extern WORD TestEndDoLoop(PHEAD WORD *,WORD);

extern WORD  *poly_gcd(PHEAD WORD *, WORD *, WORD);
extern WORD  *poly_div(PHEAD WORD *, WORD *, WORD);
extern WORD  *poly_rem(PHEAD WORD *, WORD *, WORD);
extern WORD  *poly_inverse(PHEAD WORD *, WORD *);
extern WORD  *poly_mul(PHEAD WORD *, WORD *);
extern WORD *poly_ratfun_add(PHEAD WORD *, WORD *);
extern int   poly_ratfun_normalize(PHEAD WORD *);
extern int   poly_factorize_argument(PHEAD WORD *, WORD *);
extern WORD *poly_factorize_dollar(PHEAD WORD *);
extern int   poly_factorize_expression(EXPRESSIONS);
extern int   poly_unfactorize_expression(EXPRESSIONS);
extern void  poly_free_poly_vars(PHEAD const char *);

#ifdef WITHFLINT
extern void  flint_final_cleanup_thread(void);
extern void  flint_final_cleanup_master(void);
extern WORD* flint_div(PHEAD WORD *, WORD *, const WORD);
extern int   flint_factorize_argument(PHEAD WORD *, WORD *);
extern WORD* flint_factorize_dollar(PHEAD WORD *);
extern WORD* flint_gcd(PHEAD WORD *, WORD *, const WORD);
extern WORD* flint_inverse(PHEAD WORD *, WORD *);
extern WORD* flint_mul(PHEAD WORD *, WORD *);
extern WORD* flint_ratfun_add(PHEAD WORD *, WORD *);
extern int   flint_ratfun_normalize(PHEAD WORD *);
extern WORD* flint_rem(PHEAD WORD *, WORD *, const WORD);
extern void  flint_startup_init(void);
#endif

```

```
extern void optimize_print_code (int);
#ifdef WITHPTHREADS
extern void find_Horner_MCTS_expand_tree(void);
extern void find_Horner_MCTS_expand_tree_threaded(void);
extern void optimize_expression_given_Horner(void);
extern void optimize_expression_given_Horner_threaded(void);
#endif
extern int DoPreAdd(UBYTE *s);
extern int DoPreUseDictionary(UBYTE *s);
extern int DoPreCloseDictionary(UBYTE *s);
extern int DoPreOpenDictionary(UBYTE *s);
extern void RemoveDictionary(DICTIONARY *dict);
extern void UnSetDictionary(void);
extern int SetDictionaryOptions(UBYTE *options);
extern int SelectDictionary(UBYTE *name,UBYTE *options);
extern int AddToDictionary(DICTIONARY *dict,UBYTE *left,UBYTE *right);
extern int AddDictionary(UBYTE *name);
extern int FindDictionary(UBYTE *name);
extern UBYTE *IsExponentSign(void);
extern UBYTE *IsMultiplySign(void);
extern void TransformRational(UWORD *a, WORD na);
extern void WriteDictionary(DICTIONARY *);
extern void ShrinkDictionary(DICTIONARY *);
extern void MultiplyToLine(void);
extern UBYTE *FindSymbol(WORD num);
extern UBYTE *FindVector(WORD num);
extern UBYTE *FindIndex(WORD num);
extern UBYTE *FindFunction(WORD num);
extern UBYTE *FindFunWithArgs(WORD *t);
extern UBYTE *FindExtraSymbol(WORD num);
extern LONG DictToBytes(DICTIONARY *dict,UBYTE *buf);
extern DICTIONARY *DictFromBytes(UBYTE *buf);
extern int CoCreateSpectator(UBYTE *inp);
extern int CoToSpectator(UBYTE *inp);
extern int CoRemoveSpectator(UBYTE *inp);
extern int CoEmptySpectator(UBYTE *inp);
extern int CoCopySpectator(UBYTE *inp);
extern int PutInSpectator(WORD *,WORD);
extern void ClearSpectators(WORD);
extern WORD GetFromSpectator(WORD *,WORD);
extern void FlushSpectators(void);
extern WORD *PreGCD(PHEAD WORD *, WORD *,int);
extern WORD *FindCommonVariables(PHEAD int,int);
extern void AddToSymbolList(PHEAD WORD);
extern int AddToListPoly(PHEAD0);
extern int InvPoly(PHEAD WORD *,WORD,WORD);
extern int ReadFromScratch(FILEHANDLE *,POSITION *,UBYTE *,POSITION *);
extern int AddToScratch(FILEHANDLE *,POSITION *,UBYTE *,POSITION *,int);
extern int DoPreAppendPath(UBYTE *);
extern int DoPrePrependPath(UBYTE *);
extern int DoSwitch(PHEAD WORD *, WORD *);
extern int DoEndSwitch(PHEAD WORD *, WORD *);
extern SWITCHTABLE *FindCase(WORD, WORD);
extern void SwitchSplitMergeRec(SWITCHTABLE *, WORD, SWITCHTABLE *);
extern void SwitchSplitMerge(SWITCHTABLE *, WORD);
extern int DoubleSwitchBuffers(void);
extern int DistrN(int, int *, int, int *);
extern int DoNamespace(UBYTE *);
extern int DoEndNamespace(UBYTE *);
extern int DoUse(UBYTE *);
extern UBYTE *SkipName(UBYTE *);
extern UBYTE *ConstructName(UBYTE *,UBYTE);
extern int DoSetUserFlag(UBYTE *);
extern int DoClearUserFlag(UBYTE *);
extern int DoUserFlag(UBYTE *,int);
VERTEX *CreateVertex(MODEL *);
UBYTE *ReadParticle(UBYTE *, VERTEX *,MODEL *, int);
int CoModel(UBYTE *);
int CoParticle(UBYTE *);
int CoVertex(UBYTE *);
int CoEndModel(UBYTE *);
int LoadModel(MODEL *);
int CoSetUserFlag(UBYTE *);
int CoClearUserFlag(UBYTE *);
int CoCreateAllLoops(UBYTE *);
int CoCreateAllPaths(UBYTE *);
int CoCreateAll(UBYTE *);
```

```
WORD AllLoops(PHEAD WORD *,WORD);
LONG StartLoops(PHEAD WORD *,WORD,LONG,WORD,WORD *,WORD,WORD *,WORD);
LONG GenLoops(PHEAD WORD *,WORD,LONG,WORD,WORD *,WORD,WORD *,WORD);
void LoopOutput(PHEAD WORD *,WORD,WORD *,WORD);
WORD AllPaths(PHEAD WORD *,WORD);
LONG GenPaths(PHEAD WORD *,WORD,LONG,WORD,WORD *,WORD,WORD *,WORD);
void PathOutput(PHEAD WORD *,WORD,WORD *,WORD);

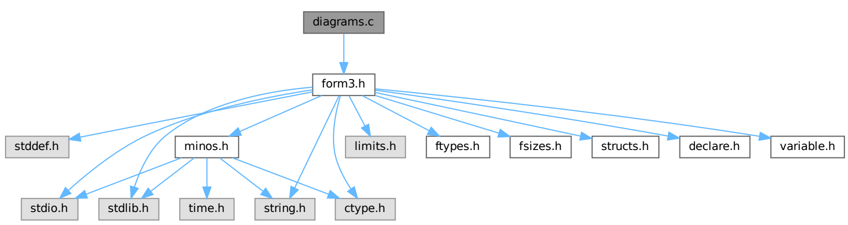
#ifdef WITHFLOAT
int DoStartFloat(UBYTE *);
int DoEndFloat(UBYTE *);
int SetFloatPrecision(WORD);
void SetupMZVTables(void);
void SetupMPFTables(void);
void ClearMZVTables(void);
int EvaluateEuler(PHEAD WORD *,WORD,WORD);
int EvaluateMZVhalf(PHEAD WORD *,WORD,WORD);
int EvaluateSqrt(PHEAD WORD *,WORD,WORD);
int CoEvaluate(UBYTE *);
int PrintFloat(WORD *fun,int numdigits);
int CoToRat(UBYTE *);
int CoToFloat(UBYTE *);
int ToRat(PHEAD WORD *,WORD);
int ToFloat(PHEAD WORD *,WORD);
WORD FloatFunToRat(PHEAD UWORD *,WORD *);
WORD AddWithFloat(PHEAD WORD **,WORD **);
WORD MergeWithFloat(PHEAD WORD **,WORD **);
void ClearfFloat(void);
int TestFloat(WORD *);
SBYTE *ReadFloat(SBYTE *);
UBYTE *CheckFloat(UBYTE *,int *);
void SetfFloatPrecision(LONG);
int EvaluateFun(PHEAD WORD *, WORD, WORD *);
#endif

/*
    #] Declarations :
*/
#endif
```

4.21 diagrams.c File Reference

`#include "form3.h"`

Include dependency graph for diagrams.c:



Functions

- int CoCanonicalize (UBYTE ∗s)
- int DoCanonicalize (PHEAD WORD ∗term, WORD ∗params)
- int DoTopologyCanonicalize (PHEAD WORD ∗term, WORD vert, WORD edge, WORD ∗args)
- int DoShattering (PHEAD WORD ∗connect, WORD ∗environ, WORD ∗partitions, WORD nvert)

4.21.1 Detailed Description

Contains the wrapper routines for diagram manipulations.

Definition in file diagrams.c.

4.21.2 Function Documentation

4.21.2.1 CoCanonicalize()

```
int CoCanonicalize (
            UBYTE * s )
```

Definition at line 64 of file diagrams.c.

4.21.2.2 DoCanonicalize()

```
int DoCanonicalize (
            PHEAD WORD * term,
            WORD * params )
```

Definition at line 185 of file diagrams.c.

4.21.2.3 DoTopologyCanonicalize()

```
int DoTopologyCanonicalize (
            PHEAD WORD * term,
            WORD vert,
            WORD edge,
            WORD * args )
```

Definition at line 417 of file diagrams.c.

4.21.2.4 DoShattering()

```
int DoShattering (
            PHEAD WORD * connect,
            WORD * environ,
            WORD * partitions,
            WORD nvert )
```

Definition at line 626 of file diagrams.c.

4.22 diagrams.c

Go to the documentation of this file.

```
00001
00005 /* #[ License : */
00006 /*
00007  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00008  *   When using this file you are requested to refer to the publication
00009  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00010  *   This is considered a matter of courtesy as the development was paid
00011  *   for by FOM the Dutch physics granting agency and we would like to
00012  *   be able to track its scientific use to convince FOM of its value
00013  *   for the community.
00014  *
00015  *   This file is part of FORM.
00016  *
00017  *   FORM is free software: you can redistribute it and/or modify it under the
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : diagrams.c
*/

#include "form3.h"

static WORD one = 1;

/*
    #] Includes :
    #[ CoCanonicalize :

    Syntax:
        Canonicalize,mainoption,...
    With the main options currently:
        Canonicalize,topology,vertexfunction,edgefunction,OutDollar,extraoptions;
        Canonicalize,polynomial,InDollar,set_or_setname_or_dollar,OutDollar,extraoptions;
    The vertex function needs to have the format (assume it is called vx):
        vx(p1,p2,-p3)  or vx(-p1,p2,p3,-p4) etc.
    The external lines have a vertex with only one line.
    All momenta that form connections should be unique.
    In principle the - signs are not relevant for the topology, but they
    may exist already in the remaining part of the diagram. They are also
    part of the canonical form.
    The edge function can be used in different ways, depending on the options.
    The extraoption(s) should be nonnegative integers or $variables that evaluate
    into nonnegative integers (integers less than 2^31).
    The Indollar variable contains the polynomial to be canonicalized.
    The OutDollar variable should be the name of a $-variable (as in $out) which
    will be filled with a replace_ function. The canonicalization can then
    be executed in the whole term with the   Multiply $out;  command.
*/

int CoCanonicalize(UBYTE *s)
{
    WORD args[10], *a, num;
    UBYTE *t, c;
    args[0] = TYPECANONICALIZE;
    a = args+2;
    while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
    t = s; while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( StrICmp(t,(UBYTE *)("topology")) == 0 ) {
        *s = c; *a++ = 0;
        while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
        s = GetFunction(s,a);
        while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
        s = GetFunction(s,a+1);
        if ( *a == 0 || a[1] == 0 ) return(1);
        a += 2;
    }
    else if ( StrICmp(t,(UBYTE *)("polynomial")) == 0 ) {
        *s = c; *a++ = 1;
        while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
/*
        Now get the name of the input $-variable
*/
        if ( *s != '$' ) {
            MesPrint("&Canonicalize statement needs a $-variable for its input.");
            return(1);
        }
        s++; t = s; while ( FG.cTable[*s] < 2 ) s++;
        c = *s; *s = 0;
        if ( GetName(AC.dollarnames,t,&num,NOAUTO) == CDOLLAR ) *a++ = num;
        else { *a++ = AddDollar(t,DOLINDEX,&one,1); }
        *s = c;
/*
        Now the set
*/
        if ( *s == '{' ) {
            t = s+1; SKIPBRA2(s)
            c = *s; *s = 0;
            *a++ = DoTempSet(t,s);
            *s++ = c;
```

```
        }
        else if ( FG.cTable[*s] == 0 || *s == '[' ) {
            t = s;
            if ( ( s = SkipAName(s) ) == 0 ) {
                MesPrint("&Illegal name for set in Canonicalize statement: %s",t);
                return(1);
            }
            c = *s; *s = 0;
            if ( GetName(AC.varnames,t,a,WITHAUTO) == CSET ) {
                if ( Sets[*a].type != CSYMBOL ) {
                    MesPrint("&In Canonicalize: %s is not a set of symbols.",t);
                    return(1);
                }
            }
            else {
                MesPrint("&In Canonicalize: %s is not a set.",t);
                return(1);
            }
            *s = c; a++;
        }
        else if ( *s == '$' ) {
            s++; t = s; while ( FG.cTable[*s] < 2 ) s++;
            c = *s; *s = 0;
            if ( GetName(AC.dollarnames,t,&num,NOAUTO) == CDOLLAR ) *a++ = -num-2;
            else {
                MesPrint("&In Canonicalize: %s is undefined.",t-1);
                return(1);
            }
            *s = c;
        }
        else {
            MesPrint("&In Canonicalize: Illegal third(=set) argument.");
            return(1);
        }
    }
    else {
        MesPrint("&Unrecognized option in Canonicalize statement: %s",t);
        return(1);
    }
/*
    Now get the name of the output $-variable
*/
    while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
    if ( *s != '$' ) {
        MesPrint("&Canonicalize statement needs a $-variable for its output.");
        return(1);
    }
    s++; t = s; while ( FG.cTable[*s] < 2 ) s++;
    c = *s; *s = 0;
    if ( GetName(AC.dollarnames,t,&num,NOAUTO) == CDOLLAR ) *a++ = num;
    else { *a++ = AddDollar(t,DOLINDEX,&one,1); }
    *s = c;
/*
    Now the options. At the moment we just do one of them.
    (the first extra option is relevant to determine the use of the edge function)
    In the future we may have to be more flexible.
*/
    a[0] = 0;  /* default value */
    while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
    if ( *s != 0 ) {
        s = GetNumber(s,a);
        if ( *a == -1 ) return(1);
        while ( *s == ',' || *s == '\t' || *s == ' ' ) s++;
        a++;
    }
/*
    Now complete the args string and put it in the compiler buffer
*/
    args[1] = a-args;
    AddNtoL(args[1],args);
    return(0);
}

/*
    #] CoCanonicalize :
    #[ DoCanonicalize :

    Does the canonicalization. The output term overwrites the input term.
*/

int DoCanonicalize(PHEAD WORD *term, WORD *params)
{
    WORD args[10];
    int i;
/*
    First check whether we need to expand dollars;
*/
```

```
    for ( i = 0; i < params[1]; i++ ) args[i] = params[i];
    if ( args[2] == 0 ) { /* topology */
        for ( i = 3; i < 5; i++ ) {
            if ( args[i] < 0 ) { /* This is a dollar */
                args[i] = DolToFunction(BHEAD -args[i]-2);
                if ( args[i] == 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Value of $-variable in Canonicalize statement should be a function.");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
            }
        }
        for ( i = 6; i < args[1]; i++ ) { /* Extra options */
            if ( args[i] < 0 ) { /* This is a dollar */
                args[i] = DolToNumber(BHEAD -args[i]-2);
                if ( args[i] < 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Value of $-variable in Canonicalize statement should be a nonnegative number < %l.",(LONG)MAXPOSITIVE);
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
            }
        }
        switch ( args[6] ) {
            case 1: {/* pass the vertex and edge functions. */
                WORD *tstop, *t, *tedge, *te;
                tstop = term + *term; tstop -= ABS(tstop[-1]);
                t = term+1;
                tedge = AT.WorkPointer; te = tedge+1;
                while ( t < tstop ) {
                    if ( *t != args[3] && *t != args[4] ) { t += t[1]; continue; }
                    for ( i = 0; i < t[1]; i++ ) te[i] = t[i];
                    te += t[1]; t += t[1];
                }
                *te++ = 1; *te++ = 1; *te++ = 3;
                tedge[0] = te-tedge;
                AT.WorkPointer = te;
/*
                DoVertexCanonicalize(BHEAD term,tedge,args[3],args[4],args[5],args[6]);
*/
                AT.WorkPointer = tedge;
            } break;
            case 2: {/* pass the edge functions only */
                WORD *tstop, *t, *tedge, *te;
                tstop = term + *term; tstop -= ABS(tstop[-1]);
                t = term+1;
                tedge = AT.WorkPointer; te = tedge+1;
                while ( t < tstop ) {
                    if ( *t != args[4] ) { t += t[1]; continue; }
                    for ( i = 0; i < t[1]; i++ ) te[i] = t[i];
                    te += t[1]; t += t[1];
                }
                *te++ = 1; *te++ = 1; *te++ = 3;
                tedge[0] = te-tedge;
                AT.WorkPointer = te;
/*
                DoEdgeCanonicalize(BHEAD term,tedge,args[5]);
*/
                AT.WorkPointer = tedge;
            } break;
            default: {
                DoTopologyCanonicalize(BHEAD term,args[3],args[4],args+5);
            } break;
        }
/*
        Call here the topology canonicalization
        We will have the arguments:
            args[3]: The function used as vertex function
            args[4]: The function used as edge function
            args[5]: The number of the dollar to be used for the output
            args[6]: Potentially other options (like saying how to use args[4]).
            term:    The term in which the topology resides.
*/

    }
    else if ( args[2] == 1 ) { /* polynomial */
        WORD *symlist, nsymlist;
        for ( i = 6; i < args[1]; i++ ) { /* Extra options */
            if ( args[i] < 0 ) { /* This is a dollar */
                args[i] = DolToNumber(BHEAD -args[i]-2);
                if ( args[i] < 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Value of $-variable in Canonicalize statement should be a nonnegative number < %l.",(LONG)MAXPOSITIVE);
                    MUNLOCK(ErrorMessageLock);
```

```
                    Terminate(-1);
                }
            }
        }
/*
        Now we sort out the set. We create a pointer to the list of set
        elements, and we determine the number of elements in the set.
*/
        symlist = AT.WorkPointer;
        if ( args[4] < -1 ) { /* Dollar that should expand into a list of symbols */
            DOLLARS d = Dollars - args[4] - 2;
            WORD *ds, *insym;
            if ( d->type != DOLWILDARGS ) {
NoWildArg:
                MLOCK(ErrorMessageLock);
                MesPrint("Value of $-variable in Canonicalize statement should be a argument wildcard
of symbol arguments.");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            insym = symlist; ds = d->where+1;
            while ( *ds ) {
                if ( *ds != -SYMBOL ) goto NoWildArg;
                *insym++ = ds[1];
                ds += 2;
            }
            nsymlist = insym-symlist;
        }
        else { /*if ( args[4] >= 0 ) */
            WORD *ss, *sy, n;
            ss = (WORD *)(AC.SetElementList.lijst)+Sets[args[4]].first;
            nsymlist = n = Sets[args[4]].last-Sets[args[4]].first;
            sy = symlist = AT.WorkPointer;
            NCOPY(sy,ss,n);
        }
        AT.WorkPointer = symlist+nsymlist;
/*
        Call here the polynomial canonicalization
        We will have the arguments:
            args[3]: The number of the dollar to be used for the input
            symlist: an array of symbols
            nsymlist: the number of symbols in symlist
            args[5]: The number of the dollar to be used for the output
            args[6]: Potentially other options.
*/
/*
        DoPolynomialCanonicalize(BHEAD args[3],symlist,nsymlist,args[5],args[6]);
*/
        AT.WorkPointer = symlist;
    }
    return(0);
}

/*
    #] DoCanonicalize :
    #[ GenTopologies :

    This function has the syntax
        topologies_(nloops,    Number of loops
            nlegs,             Number of legs
            setvertexsizes,    A set which tells which vertices are allowed like {3,4}.
            set_extmomenta,    The name of a set with the external momenta
            set_intmomenta     The name of a set with the internal momenta
            [,options])
    The output will be using the built in functions vertex_ and edge_.

    The test for whether this function can be evaluated is in TestSub (inside
    file proces.c) (search for the string TOPOLOGIES).
    This passes the code -15 in AN.TeInFun to Generator, which then calls
    the GenTopologies routine.
*/

#ifdef OLDTOPO

WORD GenTopologies(PHEAD WORD *term,WORD level)
{
    WORD *t1, *tt1, *tstop, *t, *tt;
    WORD *oldworkpointer = AT.WorkPointer;
    WORD option1 = 0, option2 = 0, setoption = -1;
    WORD retval;
/*

    We have to go through the testing procedure again, because there could
    be more than one topologies_ function and not all have to be expandable.
*/
    tstop = term+*term; tstop -= ABS(tstop[-1]);
    tt = term+1;
```

```
    while ( tt < tstop ) {
        t = tt; tt = t+t[1];
        if ( *t != TOPOLOGIES ) continue;
        tt = t + t[1]; t1 = t + FUNHEAD;
        if ( t1+10 > tt || *t1 != -SNUMBER || t1[1] < 0 ||      /* loops */
            t1[2] != -SNUMBER || ( t1[3] < 0 && t1[3] != -2 ) ||/* legs */
            t1[4] != -SETSET || Sets[t1[5]].type != CNUMBER ||  /* set vertices */
            t1[6] != -SETSET || Sets[t1[7]].type != CVECTOR ||  /* outvectors */
            t1[8] != -SETSET || Sets[t1[9]].type != CVECTOR ) continue;
        tt1 = t1 + 10;
        if ( tt1+2 <= tt && tt1[0] == -SETSET ) {
            if ( Sets[t1[5]].last-Sets[t1[5]].first !=
                 Sets[tt1[1]].last-Sets[tt1[1]].first ) continue;
            setoption = tt1[1]; tt1 += 2;
        }
        if ( tt1+2 <= tt && tt1[0] == -SNUMBER ) { option1 = tt1[1]; tt1 += 2; }
        if ( tt1+2 <= tt && tt1[0] == -SNUMBER ) { option2 = tt1[1]; tt1 += 2; }
        AT.setinterntopo = t1[9];
        AT.setexterntopo = t1[7];
        AT.TopologiesTerm = term;
        AT.TopologiesStart = t;
        AT.TopologiesLevel = level;
        AT.TopologiesOptions[0] = option1;
        AT.TopologiesOptions[1] = option2;
        retval = GenerateTopologies(BHEAD t1[1],t1[3],t1[5],setoption);
        AT.WorkPointer = oldworkpointer;
        return(retval);
    }
    MLOCK(ErrorMessageLock);
    MesPrint("Internal error: topologies_ function not encountered.");
    MUNLOCK(ErrorMessageLock);
    return(-1);

}

#endif

/*
    #] GenTopologies :
    #[ DoTopologyCanonicalize :

    term: The term
    vert: the vertex function
    edge: the edge function
    args[0]: the number of the output dollar
    args[1]: options
    return value should be zero if all is correct.

    The external lines connect to an 'external vertex' which has only one line.
    The vertices are of a type: vertex_(p1,p2,-p3)*vertex_(p3,p4,p5) etc.
    The edges indicate noninteger powers of the lines:
        edge_(p1,2) means 1/p1.p1^(2*ep)
*/

int DoTopologyCanonicalize(PHEAD WORD *term,WORD vert,WORD edge,WORD *args)
{
    int nvert = 0, nvert2, i, ii, jj, flipnames = 0, nparts/*, level, num */;
    WORD *tstop, *t, *tt, *tend, *td;
    WORD *oldworkpointer = AT.WorkPointer;
    WORD *termcopy = TermMalloc("TopologyCanonize1");
    WORD *vet= TermMalloc("TopologyCanonize2");
    WORD *partition, *environ, *connect, *pparts, *p;
/*
    WORD *pparts;
*/
    WORD momenta[150],flips[50],nmomenta = 0, nflips = 0;
/*
    Step one: the vertices should get a number. We copy the term for this.
              We need a high number for the vertex function to make sure
              that it comes after the edge function in the sorting.
*/
    if ( args[0] < args[1] ) { flipnames = 1; }
    tend = term + *term; tend -= ABS(tend[-1]); t = term+1; tt = termcopy+1;
    while ( t < tend ) {
        if ( *t == vert ) {
            for ( i = FUNHEAD; i < t[1]; i += 2 ) {
                if ( t[i] == -VECTOR || ( t[i] == -INDEX && t[i+1] < 0 ) ) {
                    momenta[nmomenta++] = -VECTOR;
                    momenta[nmomenta++] = t[i+1];
                }
                else if ( t[i] == -MINVECTOR ) {
                    momenta[nmomenta++] = -MINVECTOR;
                    momenta[nmomenta++] = t[i+1];
                }
                else goto notgoodvertex;
                momenta[nmomenta++] = nvert;
            }
```

```
            ii = FUNHEAD; i = t[1]-FUNHEAD;
            NCOPY(tt,t,ii)
            if ( flipnames ) tt[-FUNHEAD] = edge;
            tt[-FUNHEAD+1] += 2;
            *tt++ = -CNUMBER; *tt++ = nvert++;
        }
        else if ( *t == edge && flipnames ) {
            i = t[1] - 1; *tt++ = vert; t++;
        }
        else {
notgoodvertex:
            i = t[1];
        }
        NCOPY(tt,t,i)
    }
    while ( t < tend ) *tt++ = *t++;
    termcopy[0] = tt - termcopy;
    if ( flipnames ) EXCH(edge,vert)
    nvert2 = nvert*nvert;
/*
    Sort the momenta. Keep the sign order.
*/
    for ( i = 0; i < nmomenta-3; i+=3 ) {
        jj = i;
        while ( jj >= 0 && momenta[jj+4] > momenta[jj+1] ) {
            EXCH(momenta[jj+5],momenta[jj+2])
            EXCH(momenta[jj+4],momenta[jj+1])
            EXCH(momenta[jj+3],momenta[jj])
            jj -= 3;
        }
    }
/*
    Step two: make now the edge functions in the proper notation.
*/
    t = vet+1;
    for ( i = 0; i < nmomenta; i += 6 ) {
        if ( momenta[i] == -VECTOR && momenta[i+3] == -MINVECTOR
             && momenta[i+1] == momenta[i+4] ) {
        }
        else if ( momenta[i] == -MINVECTOR && momenta[i+3] == -VECTOR
             && momenta[i+1] == momenta[i+4] ) {
            flips[nflips++] = momenta[i+1];
            DUMMYUSE(flips[nflips-1]);
        }
        else { /* something wrong with the momenta */
            MLOCK(ErrorMessageLock);
            MesPrint("No momentum conservation or wrong momenta in Canonicalize statement");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        *t++ = EDGE; *t++ = FUNHEAD+10; FILLFUN(t)
        *t++ = -SNUMBER; *t++ = momenta[i+2];
        *t++ = -SNUMBER; *t++ = momenta[i+5];
        *t++ = -VECTOR; *t++ = momenta[i+1];
        *t++ = -SNUMBER; *t++ = 0; /* provisional power/color, multiple of ep */
        *t++ = -SNUMBER; *t++ = 0; /* provisional power/color, integer */
    }
    tend = t;
    *t++ = 1; *t++ = 1; *t++ = 3; vet[0] = t-vet; *t = 0;
/*
    Now the powers of the denominators
*/
    tstop = termcopy+*termcopy; tstop -= ABS(tstop[-1]); td = termcopy+1;
    while ( td < tstop ) {
        if ( *td == edge && td[1] == FUNHEAD+4 ) {
            if ( td[FUNHEAD+2] == -SNUMBER && ( td[FUNHEAD] == -VECTOR || td[FUNHEAD] == -INDEX
            || td[FUNHEAD] == -MINVECTOR ) ) {}
            else {
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal argument in edge function in Canonicalize statement");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            tt = vet+1;
            while ( tt < tend ) {
                if ( tt[FUNHEAD+5] == td[FUNHEAD+1] ) { tt[FUNHEAD+7] = td[FUNHEAD+3]; break; }
                tt += tt[1];
            }
        }
        else if ( *td == DOTPRODUCT ) break;
        td += td[1];
    }
    if ( td < tstop ) {
        tt = vet+1;
        while ( tt < tend ) {
/*
            tt[FUNHEAD+5] is a vector. We look for dotproducts with twice tt[FUNHEAD+5]
```

```
*/
            for ( i = 2; i < td[1]; i += 3 ) {
                if ( td[i] == tt[FUNHEAD+5] && td[i+1] == tt[FUNHEAD+5] ) {
                    tt[FUNHEAD+9] = td[i+2];
                    break;
                }
            }
            tt += tt[1];
        }
    }
    Normalize(BHEAD vet);
/*
    Now we have a term `vet' in the proper notation and we can start.
    To keep track of the shattering we use an array of 2*nvert.
    each entry is Number,marker
    When the marker is zero, the vertices are in the same partition.
    For the environment we need a matrix that is nvert x nvert
    At the same time we keep the connectivity matrix, because that will
    save much time later.
    The partitions are stored in a matrix as well. This allows them to be
    treated as a stack. The entries are separated by 0 if they belong to
    the same part, and by a 1 when they belong to different parts.
*/
    partition = AT.WorkPointer; AT.WorkPointer += 2*nvert2;
    for ( i = 0; i < nvert; i++ ) { partition[2*i] = i; partition[2*i+1] = 0; }
    partition[2*i-1] = -1;  /* end of the first part which is currently all vertices */
    nparts = 1;

    connect = AT.WorkPointer; AT.WorkPointer += nvert2;
    for ( i = 0; i < nvert2; i++ ) connect[i] = 0;
    tstop = vet+*vet; tstop -= ABS(tstop[-1]); t = vet+1;
    while ( t < tstop ) {
        if ( *t == EDGE ) {
            connect[t[FUNHEAD+1]*nvert+t[FUNHEAD+3]]++;
            connect[t[FUNHEAD+3]*nvert+t[FUNHEAD+1]]++;
        }
        t += t[1];
    }
for ( i = 0; i < nvert; i++ ) {
MesPrint("connectivity: %d -- %a",i,nvert,connect+i*nvert);
}
/*
    Create the environment matrix and sort it.
*/
    environ = AT.WorkPointer; AT.WorkPointer += nvert2;
/*
    And now the refinement process starts.
*/
    WantAddPointers(nvert+1);
    for ( i = 0; i < nvert2; i++ ) environ[i] = 0;
    /* level = 0; */
    pparts = partition;
    while ( nparts < nvert ) {
        nparts = DoShattering(BHEAD connect,environ,pparts,nvert);
        if ( nparts < nvert ) { /* raise level and make a copy and split a part */
            p = pparts + 2*nvert;
            /* level++; */
            for ( i = 0; i < 2*nvert; i++ ) p[i] = pparts[i];
            for ( ii = 0; ii < 2*nvert; ii += 2 ) {
                if ( p[ii+1] == 0 ) { /* found a part with more than one */
                    /* num = 2; */ i = ii+2;
                    while ( p[i+1] == 0 ) { /* num++; */ i += 2; }
                    p[ii+1] = -1; pparts = p;
                    break;
                }
            }


        }
MesPrint("partition: %d -- %a",nparts,2*nvert,pparts);

    }
/*
    Just for now
*/
    PutTermInDollar(vet,args[0]);


    TermFree(vet,"TopologyCanonize2");
    TermFree(termcopy,"TopologyCanonize1");
    AT.WorkPointer = oldworkpointer;
    return(0);
}

/*
    #] DoTopologyCanonicalize :
    #[ DoShattering :
```

```
*/

int DoShattering(PHEAD WORD *connect, WORD *environ, WORD *partitions, WORD nvert)
{
    int nparts, i, j, ii, jj, iii, jjj, newmarker;
    WORD **p = AT.pWorkSpace + AT.pWorkPointer, *part, *endpart;
    WORD *poin1, *poin2;
#ifdef SHATBUG
MesPrint("Entering DoShattering. partitions = %a",2*nvert,partitions);
#endif
restart:
/*
    Determine the number of parts
    p will be an array with pointers to the parts.
    We made space for this array in the calling routine and because this
    routine is not calling any other routines we do not need to raise
    the pointer in this stack (AT.pWorkPointer).
*/
    nparts = 0; newmarker = 0;
    part = partitions; endpart = part + 2*nvert;
    p[0] = part;
    while ( part < endpart ) {
        if ( part[1] != 0 ) { p[++nparts] = part+2; }
        part += 2;
    }
    for ( i = 0; i < nparts; i++ )
        AT.WorkPointer[i] = (p[i+1]-p[i])/2;
#ifdef SHATBUG
MesPrint("DoShattering: calculated the pointers");
MesPrint("DoShattering: sizes: %a",nparts,AT.WorkPointer);
MesPrint("DoShattering: p[0]: %a, p[1]: %a",6,p[0],6,p[1]);
#endif
    for ( i = 0; i < nparts; i++ ) {
     if ( AT.WorkPointer[i] > 1 ) {
      for ( j = 0; j < nparts; j++ ) {
/*
        Shatter part i wrt part j.
        if there is action, go to restart.
*/
        for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
          for ( jj = 0; jj < AT.WorkPointer[j]; jj++ ) {
            environ[ii*AT.WorkPointer[j]+jj] += connect[p[i][2*ii]*nvert+p[j][2*jj]];
          }
        }
#ifdef SHATBUG
for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
MesPrint("Environ(%d,%d): %a",i,j,AT.WorkPointer[j],environ+ii*AT.WorkPointer[j]);
}
#endif
/*
        Sort the rows internally, then sort the rows wrt each other
        and finally place new markers. If a new marker, we restart.
        Don't forget to clean up the environ array.
*/
        for ( ii = 0; ii < nvert; ii++ ) {
          poin1 = environ+ii*AT.WorkPointer[j];
          for ( jj = 0; jj < AT.WorkPointer[j]-1; jj++ ) {
            jjj = jj;
            while ( jjj >= 0 && poin1[jjj+1] > poin1[jjj] ) {
                EXCH(poin1[jjj+1],poin1[jjj])
                jjj--;
            }
          }
        }
#ifdef SHATBUG
for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
MesPrint("environ(%d,%d): %a",i,j,AT.WorkPointer[j],environ+ii*AT.WorkPointer[j]);
}
#endif
        for ( ii = 0; ii < AT.WorkPointer[i]-1; ii++ ) {
            poin2 = environ+ii*AT.WorkPointer[j]; poin1 = poin2+AT.WorkPointer[j];
            iii = ii;
            while ( iii >= 0 && ( CmpArray(poin1,poin2,AT.WorkPointer[j]) < 0 ) ) {
                EXCHN(poin2,poin1,AT.WorkPointer[j])
                EXCH(p[i][2*iii+2],p[i][2*iii])
                iii--; poin1 = poin2; poin2 = poin1-AT.WorkPointer[j];
            }
        }
#ifdef SHATBUG
for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
MesPrint("environ(%d,%d): %a",i,j,AT.WorkPointer[j],environ+ii*AT.WorkPointer[j]);
}
MesPrint("partitions = %a",2*nvert,partitions);
#endif
        for ( ii = 0; ii < AT.WorkPointer[i]-1; ii++ ) {
            poin2 = environ+ii*AT.WorkPointer[j]; poin1 = poin2+AT.WorkPointer[j];
            if ( CmpArray(poin1,poin2,AT.WorkPointer[j]) == 0 ) continue;
```

```
            p[i][2*ii+1] = -1; nparts++; newmarker++;
        }
#ifdef SHATBUG
MesPrint("partitions = %a",2*nvert,partitions);
#endif
/*
        Clear environ. This is probably faster than just clearing the whole array.
        Maybe in the future a test could be done on nvert to decide how to clear.
*/
        for ( ii = 0; ii < AT.WorkPointer[i]; ii++ ) {
          for ( jj = 0; jj < AT.WorkPointer[j]; jj++ ) {
            environ[ii*AT.WorkPointer[j]+jj] = 0;
          }
        }
        if ( newmarker ) { goto restart; }
      }
     }
    }
    return(nparts);
}

/*
    #] DoShattering :
*/
```

4.23 diawrap.cc

```
//  #[ Includes : diawrap.cc

extern "C" {
#include "form3.h"
}

#include "grccparam.h"
#include "grcc.h"

#define MAXPOINTS 120

typedef struct ToPoTyPe {
    WORD *vert;
    WORD *vertmax;
    Options *opt;
    int cldeg[MAXPOINTS], clnum[MAXPOINTS], clext[MAXPOINTS];
    int cmind[MAXLEGS+1],cmaxd[MAXLEGS+1];
    int ncl, nvert;
} TOPOTYPE;

static void ProcessDiagram(EGraph *eg, void *ti);
static int processVertex(TOPOTYPE *TopoInf, int pointsremaining, int level);

//  #] Includes :
//  #[ LoadModel :

int LoadModel(MODEL *m)
{
//
//  First check whether there was a model already.
//  In the Kaneko setup there can be only one at the same time.
//  Hence if there was one we need to remove it first.
//  Unless of course, it was already the model we want.
//
    if ( m->grccmodel != NULL ) return(0);

    int i,j,k;
//
//  First the information that goes into the Model struct.
//  Note that new Model takes over (does not copy) minp.cnamlist.
//
    MInput minp;
    PInput pinp;
    IInput iinp;
    if ( m->ncouplings > GRCC_MAXNCPLG ) {
        MesPrint("Too many coupling constants in model. Current limit is %d.",(WORD)GRCC_MAXNCPLG);
        MesPrint("Suggestion: recompile Form with a larger value for GRCC_MAXNCPLG");
        return(-1);
    }
    minp.defpart = GRCC_DEFBYCODE;
    minp.name = (char *)(m->name);
    minp.ncouple = m->ncouplings;
    for ( i = 0; i < GRCC_MAXNCPLG; i++ ) minp.cnamlist[i] = NULL;
    for ( i = 0; i < minp.ncouple; i++ )
        minp.cnamlist[i] = (char *)(VARNAME(symbols,m->couplings[i]));
    Model *mdl = new Model(&minp);
```

```
//
//  Now the particles
//
    for ( i = 0; i < m->nparticles; i++ ) {
        if ( minp.defpart == GRCC_DEFBYCODE ) {
            pinp.name = NULL;
            pinp.aname = NULL;
            pinp.pcode = m->vertices[i]->particles[0].number;
            pinp.acode = m->vertices[i]->particles[1].number;
            switch ( m->vertices[i]->particles[0].spin ) {
              case 1:
                pinp.ptypec = GRCC_PT_Scalar;
              break;
              case -1:
                pinp.ptypec = GRCC_PT_Ghost;
              break;
              case 2:
                if ( m->vertices[i]->particles[0].type == 0 )
                    pinp.ptypec = GRCC_PT_Majorana;
                else
                    pinp.ptypec = GRCC_PT_Undef;
              break;
              case -2:
                if ( m->vertices[i]->particles[0].type == 0 )
                    pinp.ptypec = GRCC_PT_Majorana;
                else
                    pinp.ptypec = GRCC_PT_Dirac;
              break;
              case 3:
                pinp.ptypec = GRCC_PT_Vector;
              break;
              default:
                pinp.ptypec = GRCC_PT_Undef;
              break;
            }
        }
        else {
            pinp.name   = (char *)(VARNAME(functions,m->vertices[i]->particles[0].number));
            pinp.aname  = (char *)(VARNAME(functions,m->vertices[i]->particles[1].number));
            switch ( m->vertices[i]->particles[0].spin ) {
              case 1:
                pinp.ptypen = "scalar";
              break;
              case -1:
                pinp.ptypen = "ghost";
              break;
              case 2:
                if ( m->vertices[i]->particles[0].type == 0 )
                    pinp.ptypen = "majorana";
                else
                    pinp.ptypen = "undef";
              break;
              case -2:
                if ( m->vertices[i]->particles[0].type == 0 )
                    pinp.ptypen = "majorana";
                else
                    pinp.ptypen = "dirac";
              break;
              case 3:
                pinp.ptypen = "vector";
              break;
              default:
                pinp.ptypen = "undef";
              break;
            }
        }
        pinp.extonly = m->vertices[i]->externonly;
        mdl->addParticle(&pinp);
    }
    mdl->addParticleEnd();
//
//  Now the vertices
//
    for ( i = m->nparticles; i < m->invertices; i++ ) {
        VERTEX *v = m->vertices[i];
        if ( minp.defpart == GRCC_DEFBYCODE ) {
            iinp.icode = NODEFUNCTION+i;
            iinp.name = NULL;
        }
        else {
            iinp.name = "node_";
        }
        iinp.nplistn = v->nparticles;
        for ( j = 0; j < iinp.nplistn; j++ ) {
            if ( minp.defpart == GRCC_DEFBYCODE ) {
                iinp.plistc[j] = v->particles[j].number;
            }
```

```
            else {
                iinp.plistn[j] = (char *)(VARNAME(functions,v->particles[j].number));
            }
        }
//
//      We need a properly ordered list of coupling constants
//      The ordered list is in m->couplings.
//      For each vertex they are in v->couplings.
//
//      iinp.cvallist = (int *)Malloc1(m->ncouplings*sizeof(int),"couplings");

        for ( j = 0; j < m->ncouplings; j++ ) {
            iinp.cvallist[j] = 0;
            for ( k = 0; k < v->ncouplings; k += 2 ) {
                if ( v->couplings[k] == m->couplings[j] ) {
                    iinp.cvallist[j] = v->couplings[k+1];
                    break;
                }
            }
        }
        mdl->addInteraction(&iinp);
    }
    mdl->addInteractionEnd();
    m->grccmodel = (void *)mdl;
    return(0);
}

//  #] LoadModel :
//  #[ ConvertParticle :

int ConvertParticle(Model *model,int formnum)
{
//
//  Returns the grcc number of the particle, because grcc does not convert
//
    int i;
    for ( i = 0; i < model->nParticles; i++ ) {
        if ( model->particles[i]->pcode == formnum ) { return(i); }
        else if ( model->particles[i]->acode == formnum ) { return(-i); }
    }
    MesPrint("Particle %d not found in model %s",formnum,model->name);
    Terminate(-1);
    return(0);
}

//  #] ConvertParticle :
//  #[ ReConvertParticle :

int ReConvertParticle(Model *model,int grccnum)
{
//
//  Returns the grcc number of the particle, because grcc does not convert
//
    if ( grccnum < 0 ) { return(model->particles[-grccnum]->acode); }
    else { return(model->particles[grccnum]->pcode); }
}

//  #] ReConvertParticle :
//  #[ numParticle :

int numParticle(MODEL *m,WORD n)
{
    int i;
    for ( i = 0; i < m->nparticles; i++ ) {
        if ( m->vertices[i]->particles[0].number == n ) return(i);
        if ( m->vertices[i]->particles[1].number == n ) return(i);
    }
    MesPrint("numParticle: particle %d not found in model",n);
    Terminate(-1);
    return(-1);
}

//  #] numParticle :
//  #[ ProcessDiagram :

void ProcessDiagram(EGraph *eg, void *ti)
{
//
//  This is the routine that gets a complete diagram and passes it on
//  to Form (Generator) for further algebraic manipulations.
//  The term is picked up from AT.diaterm and a new term is constructed
//  in the workspace.
//
    GETIDENTITY
    TERMINFO *info = (TERMINFO *)ti;

    if ( ( info->flags & TOPOLOGIESONLY ) == TOPOLOGIESONLY ) return;
```

```
    WORD *term = info->term, *newterm, *oldworkpointer = AT.WorkPointer;
    WORD *tdia = term + info->diaoffset;
    WORD *tail = tdia + tdia[1];
    WORD *tend = term + *term;
    WORD *fill, *startfill, *cfill, *afill;
    int i, j, intr;
    Model *model = (Model *)info->currentModel;
    MODEL *m = (MODEL *)info->currentMODEL;
    int numlegs, vect, edge;

    newterm = term + *term;
    for ( i = 1; i < info->diaoffset; i++ ) newterm[i] = term[i];
    fill = newterm + info->diaoffset;
//
//  Now get the nodes
//
    if ( ( info->flags & NONODES ) == 0 ) {
      for ( i = 0; i < eg->nNodes; i++ ) {
//
//      node_(number,coupling,particle_1(momentum_1),...,particle_n(momentum_n))
//
        numlegs = eg->nodes[i]->deg;
        startfill = fill;
        *fill++ = NODEFUNCTION;
        *fill++ = 0;
        FILLFUN(fill)
        *fill++ = -SNUMBER; *fill++ = i+1;
//
//      Now we put the coupling constants. This is done inside the
//      function for when we want to work with counterterms.
//
        if ( !eg->isExternal(i) ) {
            afill = fill; *fill++ = 0; *fill++ = 1; FILLARG(fill)
            cfill = fill; *fill++ = 0;
            intr = eg->nodes[i]->intrct;
            for ( j = 0; j < model->interacts[intr]->nclist; j++ ) {
                if ( model->interacts[intr]->clist[j] != 0 ) {
                    *fill++ = SYMBOL; *fill++ = 4;
                    *fill++ = m->couplings[j];
                    *fill++ = model->interacts[intr]->clist[j];
                }
            }
            *fill++ = 1; *fill++ = 1; *fill++ = 3;
            *cfill = fill - cfill;
            *afill = fill - afill;
        }
        else {
            *fill++ = -SNUMBER;
            *fill++ = 1;
        }
//
//      Now the particles and their momenta.
//
        for ( j = 0; j < numlegs; j++ ) {
            *fill++ = ARGHEAD+FUNHEAD+6;
            *fill++ = 0;
            FILLARG(fill)
            edge = eg->nodes[i]->edges[j];
            vect = ABS(edge)-1;
            *fill++ = 6+FUNHEAD;
            int a;
            if ( edge < 0 ) { a = ReConvertParticle(model,-eg->edges[vect]->ptcl); }
            else {            a = ReConvertParticle(model,eg->edges[vect]->ptcl); }
            *fill++ = a;
            *fill++ = FUNHEAD+2;
            FILLFUN(fill)
            *fill++ = edge < 0 ? -MINVECTOR: -VECTOR;
            if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external momenta
                *fill++ = SetElements[Sets[info->externalset].first+vect];
            }
            else { // Look up in set of internal momenta set
                *fill++ = SetElements[Sets[info->internalset].first+(vect-eg->nExtern)];
            }
            *fill++ = 1; *fill++ = 1; *fill++ = 3;
        }
        startfill[1] = fill-startfill;
      }
    }
    if ( ( info->flags & WITHEDGES ) == WITHEDGES ) {
        for ( i = 0; i < eg->nEdges; i++ ) {
            int n1 = eg->edges[i]->nodes[0];
            int n2 = eg->edges[i]->nodes[1];
//          int l1 = eg->edges[i]->nlegs[0];
//          int l2 = eg->edges[i]->nlegs[1];

            startfill = fill;
```

```
            *fill++ = EDGE;
            *fill++ = 0;
            FILLFUN(fill)
            *fill++ = -SNUMBER; *fill++ = i+1; // number of the edge
//
            *fill++ = ARGHEAD+FUNHEAD+6;
            *fill++ = 0;
            FILLARG(fill)
            *fill++ = 6+FUNHEAD;
            int a = ReConvertParticle(model,eg->edges[i]->ptcl);
            *fill++ = a;
            *fill++ = FUNHEAD+2;
            FILLFUN(fill)
                *fill++ = -VECTOR;
            // Look up in set of internal momenta set
            if ( i < info->numextern ) { // Look up in set of external momenta
                *fill++ = SetElements[Sets[info->externalset].first+i];
            }
            else { // Look up in set of internal momenta set
                *fill++ = SetElements[Sets[info->internalset].first+(i-eg->nExtern)];
            }
            *fill++ = 1; *fill++ = 1; *fill++ = 3;
//
            *fill++ = -SNUMBER; *fill++ = n1+1; // number of the node from
            *fill++ = -SNUMBER; *fill++ = n2+1; // number of the node to
            startfill[1] = fill - startfill;
        }
    }
    if ( ( info->flags & WITHBLOCKS ) == WITHBLOCKS ) {
        for ( i = 0; i < eg->econn->nblocks; i++ ) {
            startfill = fill;
            *fill++ = BLOCK;
            *fill++ = 0;
            FILLFUN(fill);
            *fill++ = -SNUMBER;
            *fill++ = i+1;
            *fill++ = -SNUMBER;
            *fill++ = eg->econn->blocks[i].loop;
//
//          Now we have to make a list of all nodes inside this block
//
            int bnodes[GRCC_MAXNODES], k;
            WORD *argfill = fill, *funfill;
            *fill++ = 0; *fill++ = 0; FILLARG(fill)
            *fill++ = 0;
            for ( k = 0; k < GRCC_MAXNODES; k++ ) bnodes[k] = 0;
            for ( k = 0; k < eg->econn->blocks[i].nmedges; k++ ) {
                bnodes[eg->econn->blocks[i].edges[k][0]] = 1;
                bnodes[eg->econn->blocks[i].edges[k][1]] = 1;
            }
            for ( k = 0; k < GRCC_MAXNODES; k++ ) {
                if ( bnodes[k] == 0 ) continue;
//
//              Now we put the node inside this argument.
//
                funfill = fill;
                *fill++ = NODEFUNCTION;
                *fill++ = 0;
                FILLFUN(fill)
                *fill++ = -SNUMBER; *fill++ = k+1;
                numlegs = eg->nodes[k]->deg;
                for ( j = 0; j < numlegs; j++ ) {
                    edge = eg->nodes[k]->edges[j];
                    vect = ABS(edge)-1;
                    *fill++ = -VECTOR;
                    if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external momenta
                        *fill++ = SetElements[Sets[info->externalset].first+vect];
                    }
                    else { // Look up in set of internal momenta set
                        *fill++ = SetElements[Sets[info->internalset].first+(vect-info->numextern)];
                    }
                }
                funfill[1] = fill-funfill;
            }
            *fill++ = 1; *fill++ = 1; *fill++ = 3;
            *argfill = fill - argfill;
            argfill[ARGHEAD] = argfill[0] - ARGHEAD;
            startfill[1] = fill-startfill;
        }
    }
    if ( ( info->flags & WITHONEPI ) == WITHONEPI ) {
        for ( i = 0; i < eg->econn->nopic; i++ ) {
            startfill = fill;
            *fill++ = ONEPI;
            *fill++ = 0;
            FILLFUN(fill);
```

```
            *fill++ = -SNUMBER;
            *fill++ = i+1;
            *fill++ = -ONEPI;
            for ( j = 0; j < eg->econn->opics[i].nnodes; j++ ) {
                *fill++ = -SNUMBER;
                *fill++ = eg->econn->opics[i].nodes[j]+1;
            }
            startfill[1] = fill-startfill;
        }
    }
//
//  Topology counter. We have exaggerated a bit with the eye on the far future.
//
    if ( info->numtopo-1 < MAXPOSITIVE ) {
        *fill++ = TOPO; *fill++ = FUNHEAD+2; FILLFUN(fill)
        *fill++ = -SNUMBER; *fill++ = (WORD)(info->numtopo-1);
    }
    else if ( info->numtopo-1 < FULLMAX-1 ) {
        *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+4; FILLFUN(fill)
        *fill++ = ARGHEAD+4; *fill++ = 0; FILLARG(fill)
        *fill++ = 4;
        *fill++ = (WORD)((info->numtopo-1) & WORDMASK);
        *fill++ = 1; *fill++ = 3;
    }
    else {  // for now: science fiction
        *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+6; FILLFUN(fill)
        *fill++ = ARGHEAD+6; *fill++ = 0; FILLARG(fill)
        *fill++ = 6; *fill++ = (WORD)((info->numtopo-1) » BITSINWORD);
        *fill++ = (WORD)((info->numtopo-1) & WORDMASK);
        *fill++ = 0; *fill++ = 1; *fill++ = 5;
    }
//
//  Symmetry factors. We let Normalize do the multiplication.
//
    if ( eg->nsym != 1 ) {
        *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->nsym; *fill++ = -1;
    }
    if ( eg->esym != 1 ) {
        *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->esym; *fill++ = -1;
    }
    if ( eg->extperm != 1 ) {
        *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->extperm; *fill++ = 1;
    }
//
//  finish it off
//
    while ( tail < tend ) *fill++ = *tail++;
    if ( eg->fsign < 0 ) fill[-1] = -fill[-1];
    *newterm = fill - newterm;
    AT.WorkPointer = fill;

//  MesPrint("<> %a",newterm[0],newterm);

    Generator(BHEAD newterm,info->level);
    AT.WorkPointer = oldworkpointer;
}

//  #] ProcessDiagram :
//  #[ fendMG :

Bool fendMG(EGraph *eg, void *ti)
{
    DUMMYUSE(eg);
    DUMMYUSE(ti);
    return True;
}

//  #] fendMG :
//  #[ ProcessTopology :

Bool ProcessTopology(EGraph *eg, void *ti)
{
//
//  This routine is called for each new topology.
//  New convention: return True;  generate the corresponding diagrams if needed
//                  return False; skip diagram generation (when asked for).
//
    TERMINFO *info = (TERMINFO *)ti;
#ifdef WITHEARLYVETO
    if ( ( ( info->flags & CHECKEXTERN ) == CHECKEXTERN ) && info->currentMODEL != NULL ) {
        int i, j;
        int numlegs, vect, edge;
        for ( i = 0; i < eg->nNodes; i++ ) {
            if ( eg->isExternal(i) ) continue;
            numlegs = eg->nodes[i]->deg;
            for ( j = 0; j < numlegs; j++ ) {
                edge = eg->nodes[i]->edges[j];
```

```
                vect = ABS(edge)-1;
                if ( vect < info->numextern && info->legcouple[vect][numlegs] == 0 ) {

//                  This cannot be.

                    info->numtopo++;
                    return False;
                }
            }
        }
    }
#endif
    if ( ( info->flags & TOPOLOGIESONLY ) == 0 ) {
        info->numtopo++;
        return True;
    }
//
//  Now we are just generating topologies.
//
    GETIDENTITY
    WORD *term = info->term, *newterm, *oldworkpointer = AT.WorkPointer;
    WORD *tdia = term + info->diaoffset;
    WORD *tail = tdia + tdia[1];
    WORD *tend = term + *term;
    WORD *fill, *startfill;
    Model *model = (Model *)info->currentModel;
    MODEL *m = (MODEL *)info->currentMODEL;
    int i, j;
    int numlegs, vect, edge;

    newterm = term + *term;
    for ( i = 1; i < info->diaoffset; i++ ) newterm[i] = term[i];
    fill = newterm + info->diaoffset;
//
//  Now get the nodes
//
    for ( i = 0; i < eg->nNodes; i++ ) {
//
//      node_(number,momentum_1,...,momentum_n)
//
        numlegs = eg->nodes[i]->deg;
        startfill = fill;
        *fill++ = NODEFUNCTION;
        *fill++ = 0;
        FILLFUN(fill)
        *fill++ = -SNUMBER; *fill++ = i+1;
        if ( model != NULL && m != NULL ) {
            if ( !eg->isExternal(i) ) {
                WORD *afill = fill; *fill++ = 0; *fill++ = 1; FILLARG(fill)
                WORD *cfill = fill; *fill++ = 0;
                int cpl = 2*eg->nodes[i]->extloop+eg->nodes[i]->deg-2;
                *fill++ = SYMBOL; *fill++ = 4;
                *fill++ = m->couplings[0];
                *fill++ = cpl;
                *fill++ = 1; *fill++ = 1; *fill++ = 3;
                *cfill = fill - cfill;
                *afill = fill - afill;
                if ( *afill == ARGHEAD+8 && afill[ARGHEAD+4] == 1 ) {
                    fill = afill; *fill++ = -SYMBOL; *fill++ = afill[ARGHEAD+3];
                }
            }
            else {
                *fill++ = -SNUMBER;
                *fill++ = 1;
            }
        }
//
//      Now the momenta.
//
        for ( j = 0; j < numlegs; j++ ) {
            edge = eg->nodes[i]->edges[j];
            vect = ABS(edge)-1;
            *fill++ = edge < 0 ? -MINVECTOR: -VECTOR;
            if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external momenta
                *fill++ = SetElements[Sets[info->externalset].first+vect];
            }
            else { // Look up in set of internal momenta set
                *fill++ = SetElements[Sets[info->internalset].first+(vect-info->numextern)];
            }
        }
        startfill[1] = fill-startfill;
    }
    if ( ( info->flags & WITHEDGES ) == WITHEDGES ) {
        for ( i = 0; i < eg->nEdges; i++ ) {
            int n1 = eg->edges[i]->nodes[0];
            int n2 = eg->edges[i]->nodes[1];
//          int l1 = eg->edges[i]->nlegs[0];
```

```
//          int l2 = eg->edges[i]->nlegs[1];
            startfill = fill;
            *fill++ = EDGE;
            *fill++ = 0;
            FILLFUN(fill)
            *fill++ = -SNUMBER; *fill++ = i+1; // number of the edge
//
            *fill++ = -VECTOR;
            if ( i < info->numextern ) { // Look up in set of external momenta
                *fill++ = SetElements[Sets[info->externalset].first+i];
            }
            else { // Look up in set of internal momenta set
                *fill++ = SetElements[Sets[info->internalset].first+(i-eg->nExtern)];
            }
//
            *fill++ = -SNUMBER; *fill++ = n1+1; // number of the node from
            *fill++ = -SNUMBER; *fill++ = n2+1; // number of the node to
            startfill[1] = fill - startfill;
        }
    }
//
//  Block information
//
    if ( ( info->flags & WITHBLOCKS ) == WITHBLOCKS ) {
        for ( i = 0; i < eg->econn->nblocks; i++ ) {
            startfill = fill;
            *fill++ = BLOCK;
            *fill++ = 0;
            FILLFUN(fill);
            *fill++ = -SNUMBER;
            *fill++ = i+1;
            *fill++ = -SNUMBER;
            *fill++ = eg->econn->blocks[i].loop;
//
//          Now we have to make a list of all nodes inside this block
//
            int bnodes[GRCC_MAXNODES], k;
            WORD *argfill = fill, *funfill;
            *fill++ = 0; *fill++ = 0; FILLARG(fill)
            *fill++ = 0;
            for ( k = 0; k < GRCC_MAXNODES; k++ ) bnodes[k] = 0;
            for ( k = 0; k < eg->econn->blocks[i].nmedges; k++ ) {
                bnodes[eg->econn->blocks[i].edges[k][0]] = 1;
                bnodes[eg->econn->blocks[i].edges[k][1]] = 1;
            }
            for ( k = 0; k < GRCC_MAXNODES; k++ ) {
                if ( bnodes[k] == 0 ) continue;
//
//              Now we put the node inside this argument.
//
                funfill = fill;
                *fill++ = NODEFUNCTION;
                *fill++ = 0;
                FILLFUN(fill)
                *fill++ = -SNUMBER; *fill++ = k+1;
                numlegs = eg->nodes[k]->deg;
                for ( j = 0; j < numlegs; j++ ) {
                    edge = eg->nodes[k]->edges[j];
                    vect = ABS(edge)-1;
                    *fill++ = -VECTOR;
                    if ( numlegs == 1 || vect < info->numextern ) { // Look up in set of external momenta
                        *fill++ = SetElements[Sets[info->externalset].first+vect];
                    }
                    else { // Look up in set of internal momenta set
                        *fill++ = SetElements[Sets[info->internalset].first+(vect-info->numextern)];
                    }
                }
                funfill[1] = fill-funfill;
            }
            *fill++ = 1; *fill++ = 1; *fill++ = 3;
            *argfill = fill - argfill;
            argfill[ARGHEAD] = argfill[0] - ARGHEAD;
            startfill[1] = fill-startfill;
        }

//      if ( eg->econn->narticuls > 0 ) {
//          startfill = fill;
//          *fill++ = BLOCK;
//          *fill++ = 0;
//          FILLFUN(fill);
//          for ( i = 0; i < eg->econn->snodes; i++ ) {
//              if ( eg->econn->articuls[i] != 0 ) {
//                  *fill++ = -SNUMBER;
//                  *fill++ = i+1;
//              }
//          }
```

```
//          startfill[1] = fill-startfill;
//      }
    }
    if ( ( info->flags & WITHONEPI ) == WITHONEPI ) {
        for ( i = 0; i < eg->econn->nopic; i++ ) {
            startfill = fill;
            *fill++ = ONEPI;
            *fill++ = 0;
            FILLFUN(fill);
            *fill++ = -SNUMBER;
            *fill++ = i+1;
            *fill++ = -ONEPI;
            for ( j = 0; j < eg->econn->opics[i].nnodes; j++ ) {
                *fill++ = -SNUMBER;
                *fill++ = eg->econn->opics[i].nodes[j]+1;
            }
            startfill[1] = fill-startfill;
        }
    }
//
//  Topology counter. We have exaggerated a bit with the eye on the far future.
//
    if ( info->numtopo < MAXPOSITIVE ) {
        *fill++ = TOPO; *fill++ = FUNHEAD+2; FILLFUN(fill)
        *fill++ = -SNUMBER; *fill++ = (WORD)(info->numtopo);
    }
    else if ( info->numtopo < FULLMAX-1 ) {
        *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+4; FILLFUN(fill)
        *fill++ = ARGHEAD+4; *fill++ = 0; FILLARG(fill)
        *fill++ = 4;
        *fill++ = (WORD)(info->numtopo & WORDMASK);
        *fill++ = 1; *fill++ = 3;
    }
    else {  // for now: science fiction
        *fill++ = TOPO; *fill++ = FUNHEAD+ARGHEAD+6; FILLFUN(fill)
        *fill++ = ARGHEAD+6; *fill++ = 0; FILLARG(fill)
        *fill++ = 6; *fill++ = (WORD)(info->numtopo » BITSINWORD);
        *fill++ = (WORD)(info->numtopo & WORDMASK);
        *fill++ = 0; *fill++ = 1; *fill++ = 5;
    }
//
//  Symmetry factors. We let Normalize do the multiplication.
//
    if ( eg->nsym != 1 ) {
        *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->nsym; *fill++ = -1;
    }
    if ( eg->esym != 1 ) {
        *fill++ = SNUMBER; *fill++ = 4; *fill++ = (WORD)eg->esym; *fill++ = -1;
    }
//
//  finish it off
//
    while ( tail < tend ) *fill++ = *tail++;
    if ( eg->fsign < 0 ) fill[-1] = -fill[-1];
    *newterm = fill - newterm;
    AT.WorkPointer = fill;

//MesPrint("<> %a",*newterm,newterm);

    Generator(BHEAD newterm,info->level);
    AT.WorkPointer = oldworkpointer;
    info->numtopo++;
    return False;
}

//  #] ProcessTopology :
//  #[ GenDiagrams :

WORD GenDiagrams(PHEAD WORD *term, WORD level)
{
    Model *model;
    MODEL *m;
    Options *opt;
    Process *proc;
    int pid = 1, x;
    int babble = 0;     // Later we may set this at the FORM code level
    TERMINFO info;
    WORD inset,outset,*coupl,setnum,optionnumber = 0;
    int i, j, cpl[GRCC_MAXNCPLG];
    int ninitl, initlPart[GRCC_MAXLEGS], nfinal, finalPart[GRCC_MAXLEGS];
    for ( i = 0; i < GRCC_MAXNCPLG; i++ ) cpl[i] = 0;
//
//  Here we create an object of type Option and load it up.
//  Next we run the diagram generation on it.
//
    info.term = term;
    info.level = level;
```

```
    info.diaoffset = AR.funoffset;
    info.externalset = term[info.diaoffset+FUNHEAD+7];
    info.internalset = term[info.diaoffset+FUNHEAD+9];
    info.flags = 0;
    inset = term[info.diaoffset+FUNHEAD+3];
    outset = term[info.diaoffset+FUNHEAD+5];
    coupl = term + info.diaoffset + FUNHEAD + 10;
    if ( *coupl < 0 ) {
        if ( term[info.diaoffset+1] > FUNHEAD + 12 ) {
            optionnumber = term[info.diaoffset+FUNHEAD+13];
        }
    }
    else {
        if ( term[info.diaoffset+1] > *coupl+FUNHEAD+10 )
            optionnumber = term[info.diaoffset+*coupl+FUNHEAD+11];
    }
    setnum = term[info.diaoffset+FUNHEAD+1];

    m = AC.models[SetElements[Sets[setnum].first]];
    LoadModel(m);
    model = (Model *)m->grccmodel;

    info.currentModel = (void *)model;
    info.currentMODEL = (void *)m;
    info.numdia = 0;
    info.numtopo = 1;
    info.flags = optionnumber;

    opt = new Options();

    opt->setOutAG(ProcessDiagram, &info);
    opt->setOutMG(ProcessTopology, &info);
//  opt->setEndMG(fendMG, &info);

    opt->values[GRCC_OPT_1PI] = ( optionnumber & ONEPARTICLEIRREDUCIBLE ) == ONEPARTICLEIRREDUCIBLE;
    opt->values[GRCC_OPT_NoTadpole] = ( optionnumber & NOTADPOLES ) == NOTADPOLES;
//
//  Next are snails:
//
    opt->values[GRCC_OPT_No1PtBlock] = ( optionnumber & NOTADPOLES ) == NOTADPOLES;
//
    if ( ( optionnumber & WITHINSERTIONS ) == WITHINSERTIONS ) {
        opt->values[GRCC_OPT_No2PtL1PI] = True;
        opt->values[GRCC_OPT_NoAdj2PtV] = True;
        opt->values[GRCC_OPT_No2PtL1PI] = True;
    }
    else {
        opt->values[GRCC_OPT_NoAdj2PtV] = True;
    }
    opt->values[GRCC_OPT_SymmInitial] = ( optionnumber & WITHSYMMETRIZE ) == WITHSYMMETRIZE;
    opt->values[GRCC_OPT_SymmFinal] = ( optionnumber & WITHSYMMETRIZE ) == WITHSYMMETRIZE;

//  opt->values[GRCC_OPT_Block] = ( optionnumber & WITHBLOCKS ) == WITHBLOCKS;

    opt->setOutputF(False,"");
    opt->setOutputP(False,"");
    opt->printLevel(babble);

//  opt->values[GRCC_OPT_Step]       = GRCC_AGraph;

//  Load the various arrays.

    ninitl = Sets[inset].last - Sets[inset].first;
    for ( i = 0; i < ninitl; i++ ) {
        x = SetElements[Sets[inset].first+i];
        initlPart[i] = ConvertParticle(model,x);
        info.legcouple[i] = m->vertices[numParticle(m,x)]->couplings;
    }
    nfinal = Sets[outset].last - Sets[outset].first;
    for ( i = 0; i < nfinal; i++ ) {
        x = SetElements[Sets[outset].first+i];
        finalPart[i] = ConvertParticle(model,x);
        info.legcouple[i+ninitl] = m->vertices[numParticle(m,x)]->couplings;
    }
    info.numextern = ninitl + nfinal;
    for ( i = 2; i <= MAXLEGS; i++ ) {
        if ( m->legcouple[i] == 1 ) {
            for ( j = 0; j < info.numextern; j++ ) {
                if ( info.legcouple[j][i] == 0 ) { info.flags |= CHECKEXTERN; goto Go_on; }
            }
        }
    }
Go_on:;
//
//  Now we have to sort out the coupling constants.
//  The argument at coupl can be of type -SNUMBER, -SYMBOL or generic
//  It has however already be tested for syntax.
```

```
//  Note that one cannot have 1 for the coupling constants.
//  In that case one should select 0 loops or something equivalent.
//
    if ( *coupl == -SNUMBER ) { // Number of loops
//
//      This is the complicated case.
//      We have to compute the number of coupling constants and then
//      generate diagrams for all combinations with the proper power.
//
        int nc = coupl[1]*2 + ninitl + nfinal - 2;
        int *scratch = (int *)Malloc1(nc*sizeof(int),"DistrN");
        scratch[0] = -2; // indicating startup cq first call.

        if ( ( info.flags & TOPOLOGIESONLY ) == 0 ) {
            while ( DistrN(nc,cpl,m->ncouplings,scratch) ) {
                proc = new Process(pid, model, opt,
                           ninitl, initlPart, nfinal, finalPart, cpl);
                delete proc;
                info.numtopo = 1;
            }
        }
        else {
            cpl[0] = nc;
            proc = new Process(pid, model, opt,
                       ninitl, initlPart, nfinal, finalPart, cpl);
            delete proc;
        }
        M_free(scratch,"DistrN");
        opt->end();
        delete opt;
        return(0);
    }
    else if ( *coupl == -SYMBOL ) { // Just a single power of one constant
        for ( i = 0; i < m->ncouplings; i++ ) {
            if ( m->couplings[i] == coupl[1] ) {
                cpl[i] = 1;
                break;
            }
        }
    }
    else {  // One term with powers of coupling constants
        WORD *t, *tstop;
        t = coupl + ARGHEAD+3;
        tstop = coupl+*coupl; tstop -= ABS(tstop[-1]);
        while ( t < tstop ) {
            for ( i = 0; i < m->ncouplings; i++ ) {
                if ( m->couplings[i] == *t ) {
                    cpl[i] = t[1];
                    break;
                }
            }
            t += 2;
        }
    }
/*
    And now the generation:
*/
    proc = new Process(pid, model, opt,
                       ninitl, initlPart, nfinal, finalPart, cpl);
    opt->end();
    delete proc;
    delete opt;
    return(0);
}

//  #] GenDiagrams :
//  #[ processVertex :

//  Routine is to be used recursively to work its way through a list
//  of possible vertices. The array of vertices is in TopoInf->vert
//  with TopoInf->nvert the number of possible vertices.
//  Currently we allow in TopoInf->vert only vertices with 3 or more edges.
//
//  We work with a point system. Each n-point vertex contributes n-2 points.
//  When all points are assigned, we can call mgraph->generate().
//
//  The number of vertices of a given number of edges is stored in
//  TopoInf->clnum[..] but the loop that determines how many there are
//  may be limited by the corresponding element in TopoInf->vertmax[level]

int processVertex(TOPOTYPE *TopoInf, int pointsremaining, int level)
{
    int i, j;

    for ( i = pointsremaining, j = 0; i >= 0; i -= TopoInf->vert[level]-2, j++ ) {
        if ( TopoInf->vertmax && TopoInf->vertmax[level] >= 0
                      && j > TopoInf->vertmax[level] ) break;
```

```
        if ( i == 0 ) { // We got one!
            TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
            TopoInf->clnum[TopoInf->ncl] = j;
            TopoInf->clext[TopoInf->ncl] = 0;
            TopoInf->ncl++;

            MGraph *mgraph = new MGraph(1, TopoInf->ncl, TopoInf->cldeg,
                                     TopoInf->clnum, TopoInf->clext,
                                     TopoInf->cmind, TopoInf->cmaxd, TopoInf->opt);

            mgraph->generate();

            delete mgraph;

            TopoInf->ncl--;

            break;
        }
        if ( level < TopoInf->nvert-1 ) {
            if ( j > 0 ) {
                TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
                TopoInf->clnum[TopoInf->ncl] = j;
                TopoInf->clext[TopoInf->ncl] = 0;
                TopoInf->ncl++;
            }
            if ( processVertex(TopoInf,i,level+1) < 0 ) return(-1);
            if ( j > 0 ) { TopoInf->ncl--; }
        }
    }
    return(0);
}

//  #] processVertex :
//  #[ GenTopologies :

#define TOPO_MAXVERT 10

WORD GenTopologies(PHEAD WORD *term, WORD level)
{
    Options *opt = new Options;
    int nlegs, nloops, i, identical;
    TERMINFO info;
    WORD *t, *t1, *tstop;
    TOPOTYPE TopoInf;
    SETS s;
//
    info.term = term;
    info.level = level;
    info.diaoffset = AR.funoffset;
    info.flags = 0;

    t = term + info.diaoffset;  // the function
    t1 = t + FUNHEAD;           // its arguments
    tstop = t + t[1];

    info.externalset = t1[7];
    info.internalset = t1[9];

    s = &(Sets[t1[5]]);
    TopoInf.nvert = s->last - s->first;
    TopoInf.vert  = &(SetElements[s->first]);

    nloops = t1[1];
    nlegs = t1[3];

    info.numextern = nlegs;

    for ( i = 0; i <= MAXLEGS; i++ ) { TopoInf.cmind[i] = TopoInf.cmaxd[i] = 0; }

    t1 += 10;
    if ( t1 < tstop && t1[0] == -SETSET ) {
        TopoInf.vertmax = &(SetElements[Sets[t1[1]].first]);
        t1 += 2;
    }
    else TopoInf.vertmax = NULL;

    info.flags |= TOPOLOGIESONLY;  // this is the topologies_ function after all.
    if ( t1 < tstop && t1[0] == -SNUMBER ) {
        if ( ( t1[1] &    NONODES ) ==    NONODES ) info.flags |=    NONODES;
        if ( ( t1[1] & WITHEDGES ) == WITHEDGES ) info.flags |= WITHEDGES;
        if ( ( t1[1] & WITHBLOCKS ) == WITHBLOCKS ) info.flags |= WITHBLOCKS;
        if ( ( t1[1] & WITHONEPI ) == WITHONEPI ) info.flags |= WITHONEPI;
        opt->values[GRCC_OPT_1PI] = ( t1[1] & ONEPARTICLEIRREDUCIBLE ) == ONEPARTICLEIRREDUCIBLE;
//      opt->values[GRCC_OPT_NoTadpole] = ( t1[1] & NOTADPOLES ) == NOTADPOLES;
        opt->values[GRCC_OPT_NoTadpole] = ( t1[1] & NOSNAILS ) == NOSNAILS;
        opt->values[GRCC_OPT_No1PtBlock] = ( t1[1] & NOTADPOLES ) == NOTADPOLES;
        opt->values[GRCC_OPT_NoExtSelf] = ( t1[1] & NOEXTSELF ) == NOEXTSELF;
```

```

        if ( ( t1[1] & WITHINSERTIONS ) == WITHINSERTIONS ) {
            opt->values[GRCC_OPT_No2PtL1PI] = True;
            opt->values[GRCC_OPT_NoAdj2PtV] = True;
            opt->values[GRCC_OPT_No2PtL1PI] = True;
        }
        opt->values[GRCC_OPT_SymmInitial] = ( t1[1] & WITHSYMMETRIZE ) == WITHSYMMETRIZE;
    }

    info.numdia = 0;
    info.numtopo = 1;

    opt->setOutAG(ProcessDiagram, &info);
    opt->setOutMG(ProcessTopology, &info);
//
//  Now we should sum over all possible vertices and run MGraph for
//  each combination. This is done by recursion in the processVertex routine
//  First load up the relevant arrays.
//

//  First the external nodes.

    if ( nlegs == -2 ) {
        nlegs = 2;
        identical = 1;
    }
    for ( i = 0; i < nlegs; i++ ) {
        TopoInf.cldeg[i] = 1; TopoInf.clnum[i] = 1; TopoInf.clext[i] = -1;
    }
    int points = 2*nloops-2+nlegs;

    if ( identical == 1 ) { /* Only propagator topologies..... */
        nlegs = 1;
        TopoInf.clnum[0] = 2;
    }
    TopoInf.ncl = nlegs;
    TopoInf.opt = opt;

    if ( points >= MAXPOINTS ) {
        MLOCK(ErrorMessageLock);
        MesPrint("GenTopologies: %d loops and %d legs considered excessive",nloops,nlegs);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( processVertex(&TopoInf,points,0) != 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Called from GenTopologies with %d loops and %d legs",nloops,nlegs);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    delete opt;
    return(0);
}

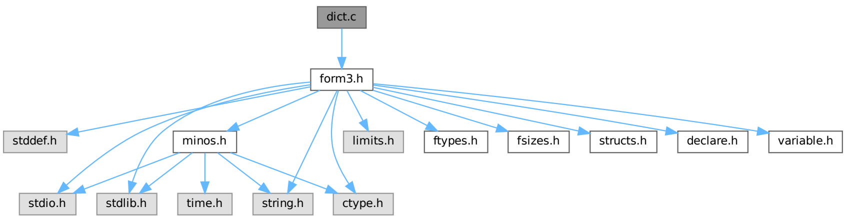
//  #] GenTopologies :

```

4.24 dict.c File Reference

`#include "form3.h"`

Include dependency graph for dict.c:



Functions

- void TransformRational (UWORD ∗a, WORD na)
- UBYTE ∗ IsMultiplySign (void)
- UBYTE ∗ IsExponentSign (void)
- UBYTE ∗ FindSymbol (WORD num)
- UBYTE ∗ FindVector (WORD num)
- UBYTE ∗ FindIndex (WORD num)
- UBYTE ∗ FindFunction (WORD num)
- UBYTE ∗ FindFunWithArgs (WORD ∗t)
- UBYTE ∗ FindExtraSymbol (WORD num)
- int FindDictionary (UBYTE ∗name)
- int AddDictionary (UBYTE ∗name)
- int AddToDictionary (DICTIONARY ∗dict, UBYTE ∗left, UBYTE ∗right)
- int UseDictionary (UBYTE ∗name, UBYTE ∗options)
- int SetDictionaryOptions (UBYTE ∗options)
- void UnSetDictionary (void)
- void RemoveDictionary (DICTIONARY ∗dict)
- void ShrinkDictionary (DICTIONARY ∗dict)
- int DoPreOpenDictionary (UBYTE ∗s)
- int DoPreCloseDictionary (UBYTE ∗s)
- int DoPreUseDictionary (UBYTE ∗s)
- int DoPreAdd (UBYTE ∗s)
- LONG DictToBytes (DICTIONARY ∗dict, UBYTE ∗buf)
- DICTIONARY ∗ DictFromBytes (UBYTE ∗buf)

4.24.1 Detailed Description

Contains the code pertaining to dictionaries Commands are: #opendictionary name #closedictionary #selectdictionary name <options> There can be several dictionaries, but only one can be active. Defining elements is done with #add object: "replacement" Replacements are strings when a dictionary is for output translation. Objects can be 1: a number (rational) 2: a variable 3: ∗ ^ 4: a function with arguments

Definition in file dict.c.

4.24.2 Function Documentation

4.24.2.1 TransformRational()

```
void TransformRational (
            UWORD * a,
            WORD na )
```

Definition at line 71 of file dict.c.

4.24.2.2 IsMultiplySign()

```
UBYTE * IsMultiplySign (
            void )
```

Definition at line 218 of file dict.c.

4.24.2.3 IsExponentSign()

```
UBYTE * IsExponentSign (
            void )
```

Definition at line 238 of file dict.c.

4.24.2.4 FindSymbol()

```
UBYTE * FindSymbol (
            WORD num )
```

Definition at line 258 of file dict.c.

4.24.2.5 FindVector()

```
UBYTE * FindVector (
            WORD num )
```

Definition at line 279 of file dict.c.

4.24.2.6 FindIndex()

```
UBYTE * FindIndex (
            WORD num )
```

Definition at line 301 of file dict.c.

4.24.2.7 FindFunction()

```
UBYTE * FindFunction (
            WORD num )
```

Definition at line 323 of file dict.c.

4.24.2.8 FindFunWithArgs()

```
UBYTE * FindFunWithArgs (
            WORD * t )
```

Definition at line 345 of file dict.c.

4.24.2.9 FindExtraSymbol()

```
UBYTE * FindExtraSymbol (
            WORD num )
```

Definition at line 377 of file dict.c.

4.24.2.10 FindDictionary()

```
int FindDictionary (
            UBYTE * name )
```

Definition at line 434 of file dict.c.

4.24.2.11 AddDictionary()

```
int AddDictionary (
            UBYTE * name )
```

Definition at line 449 of file dict.c.

4.24.2.12 AddToDictionary()

```
int AddToDictionary (
            DICTIONARY * dict,
            UBYTE * left,
            UBYTE * right )
```

Definition at line 491 of file dict.c.

4.24.2.13 UseDictionary()

```
int UseDictionary (
            UBYTE * name,
            UBYTE * options )
```

Definition at line 766 of file dict.c.

4.24.2.14 SetDictionaryOptions()

```
int SetDictionaryOptions (
            UBYTE * options )
```

Definition at line 790 of file dict.c.

4.24.2.15 UnSetDictionary()

```
void UnSetDictionary (
            void )
```

Definition at line 883 of file dict.c.

4.24.2.16 RemoveDictionary()

```
void RemoveDictionary (
            DICTIONARY * dict )
```

Definition at line 902 of file dict.c.

4.24.2.17 ShrinkDictionary()

```
void ShrinkDictionary (
            DICTIONARY * dict )
```

Definition at line 945 of file dict.c.

4.24.2.18 DoPreOpenDictionary()

```
int DoPreOpenDictionary (
            UBYTE * s )
```

Definition at line 959 of file dict.c.

4.24.2.19 DoPreCloseDictionary()

```
int DoPreCloseDictionary (
            UBYTE * s )
```

Definition at line 998 of file dict.c.

4.24.2.20 DoPreUseDictionary()

```
int DoPreUseDictionary (
            UBYTE * s )
```

Definition at line 1022 of file dict.c.

4.24.2.21 DoPreAdd()

```
int DoPreAdd (
            UBYTE * s )
```

Definition at line 1067 of file dict.c.

4.24.2.22 DictToBytes()

```
LONG DictToBytes (
            DICTIONARY * dict,
            UBYTE * buf )
```

Definition at line 1119 of file dict.c.

4.24.2.23 DictFromBytes()

```
DICTIONARY * DictFromBytes (
            UBYTE * buf )
```

Definition at line 1152 of file dict.c.

4.25 dict.c

Go to the documentation of this file.

```

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/*
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 */
/* #] License : */
/*
    #[ Includes : ratio.c

    Data setup:
        AO.Dictionaries          Array of pointers to DICTIONARY
        AO.NumDictionaries
        AO.SizeDictionaries
        AO.CurrentDictionary
        AO.CurDictNumbers
        AO.CurDictVariables
        AO.CurDictSpecials
        AP.OpenDictionary
*/

#include "form3.h"

/*
    #] Includes :
    #[ TransformRational:

    Tries to transform the rational number a according to the rules of
    the current dictionary. Whatever cannot be translated goes to the
    regular output.
    Options for AO.CurDictNumbers are:
        DICT_ALLNUMBERS, DICT_RATIONALONLY, DICT_INTEGERONLY, DICT_NONUMBERS
*/

void TransformRational(UWORD *a, WORD na)
{
    DICTIONARY *dict;
    WORD i, j, nb, i1, i2; UWORD *b;
    if ( AO.CurrentDictionary <= 0 ) goto NoAction;
    dict = AO.Dictionaries[AO.CurrentDictionary-1];
    if ( na < 0 ) na = -na;
    switch ( AO.CurDictNumbers ) {
        case DICT_NONUMBERS:
            goto NoAction;
        case DICT_INTEGERONLY:
            if ( a[na] != 1 ) goto NoAction;
            if ( na > 1 ) {
                for ( i = 1; i < na; i++ ) {
```

```
                    if ( a[na+i] != 0 ) goto NoAction;
                }
            }
Numeratoronly:;
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_INTEGERNUMBER ) {
                    if ( dict->elements[i]->size == na ) {
                        for ( j = 0; j < na; j++ ) {
                            if ( (UWORD)(dict->elements[i]->lhs[j]) != a[j] ) break;
                        }
                        if ( j == na ) {     /* Got it */
                            TokenToLine((UBYTE *)(dict->elements[i]->rhs));
                            return;
                        }
                    }
                }
            }
            goto NotFound;
        case DICT_RATIONALONLY:
            nb = 2*na;
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_RATIONALNUMBER ) {
                    if ( dict->elements[i]->size == nb+2 ) {
                        for ( j = 0; j < nb; j++ ) {
                            if ( (UWORD)(dict->elements[i]->lhs[j+1]) != a[j] ) break;
                        }
                        if ( j == nb ) {     /* Got it */
                            TokenToLine((UBYTE *)(dict->elements[i]->rhs));
                            return;
                        }
                    }
                }
            }
            goto NotFound;
        case DICT_ALLNUMBERS:
/*
            First fish for rationals
*/
            nb = 2*na;
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_RATIONALNUMBER ) {
                    if ( dict->elements[i]->size == nb+2 ) {
                        for ( j = 0; j < nb; j++ ) {
                            if ( (UWORD)(dict->elements[i]->lhs[j+1]) != a[j] ) break;
                        }
                        if ( j == nb ) {     /* Got it */
                            TokenToLine((UBYTE *)(dict->elements[i]->rhs));
                            return;
                        }
                    }
                }
            }
/*
            Now look for element[j1]/element[j2]
*/
            nb = na; b = a+na;
            while ( b[nb-1] == 0 ) nb--;
            if ( nb == 1 && b[0] == 1 ) goto Numeratoronly;
            while ( a[na-1] == 0 ) na--;
            for ( i1 = dict->numelements-1; i1 >= 0; i1-- ) {
                if ( dict->elements[i1]->type == DICT_INTEGERNUMBER ) {
                    if ( dict->elements[i1]->size == na ) {
                        for ( j = 0; j < na; j++ ) {
                            if ( (UWORD)(dict->elements[i1]->lhs[j]) != a[j] ) break;
                        }
                        if ( j == na ) break;
                    }
                }
            }
            for ( i2 = dict->numelements-1; i2 >= 0; i2-- ) {
                if ( dict->elements[i2]->type == DICT_INTEGERNUMBER ) {
                    if ( dict->elements[i2]->size == nb ) {
                        for ( j = 0; j < nb; j++ ) {
                            if ( (UWORD)(dict->elements[i2]->lhs[j]) != b[j] ) break;
                        }
                        if ( j == nb ) break;
                    }
                }
            }
            if ( i1 < 0 ) {
                if ( i2 < 0 ) goto NotFound;
                else {  /* number/replacement[i2] */
                    LongToLine(a,na);
                    if ( na > 1 || ( AO.DoubleFlag & 4 ) == 4 ) {
                        if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
                         || AC.OutputMode == CMODE ) {
                            if ( ( AO.DoubleFlag & 2 ) == 2 ) { AddToLine((UBYTE *)".Q0/"); }
```

```
                            else if ( ( AO.DoubleFlag & 1 ) == 1 ) { AddToLine((UBYTE *)".D0/"); }
                            else { AddToLine((UBYTE *)"/"); }
                        }
                    }
                    else AddToLine((UBYTE *)("/"));
                    TokenToLine((UBYTE *)(dict->elements[i2]->rhs));
                }
            }
            else if ( i2 < 0 ) { /* replacement[i1]/number */
                TokenToLine((UBYTE *)(dict->elements[i1]->rhs));
                AddToLine((UBYTE *)("/"));
                LongToLine((UWORD *)(b),nb);
                if ( nb > 1 || ( AO.DoubleFlag & 4 ) == 4 ) {
                    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
                         || AC.OutputMode == CMODE ) {
                        if ( ( AO.DoubleFlag & 2 ) == 2 ) { AddToLine((UBYTE *)".Q0"); }
                        else if ( ( AO.DoubleFlag & 1 ) == 1 ) { AddToLine((UBYTE *)".D0"); }
                    }
                }
            }
            else { /* replacement[i1]/replacement[i2] */
                TokenToLine((UBYTE *)(dict->elements[i1]->rhs));
                AddToLine((UBYTE *)("/"));
                TokenToLine((UBYTE *)(dict->elements[i2]->rhs));
            }
            break;
        default:
            MesPrint("Illegal code in TransformRational: %d",AO.CurDictNumbers);
            Terminate(-1);
    }
    return;
NotFound:
    if ( na != 1 || a[1] != 1 ) {
    if ( AO.CurDictNumberWarning ) {
        MesPrint("»»»»»Could not translate coefficient with dictionary %s«««««««",dict->name);
    } }
NoAction:
    RatToLine(a,na);
    return;
}

/*
    #] TransformRational:
    #[ IsMultiplySign:
*/

UBYTE *IsMultiplySign(void)
{
    DICTIONARY *dict;
    int i;
    if ( AO.CurrentDictionary <= 0 ) return(0);
    dict = AO.Dictionaries[AO.CurrentDictionary-1];
    if ( dict->characters == 0 ) return(0);
    for ( i = dict->numelements-1; i >= 0; i-- ) {
        if ( ( dict->elements[i]->type == DICT_SPECIALCHARACTER )
            && ( dict->elements[i]->lhs[0] == (WORD)('*') ) )
        return((UBYTE *)(dict->elements[i]->rhs));
    }
    return(0);
}

/*
    #] IsMultiplySign:
    #[ IsExponentSign:
*/

UBYTE *IsExponentSign(void)
{
    DICTIONARY *dict;
    int i;
    if ( AO.CurrentDictionary <= 0 ) return(0);
    dict = AO.Dictionaries[AO.CurrentDictionary-1];
    if ( dict->characters == 0 ) return(0);
    for ( i = dict->numelements-1; i >= 0; i-- ) {
        if ( ( dict->elements[i]->type == DICT_SPECIALCHARACTER )
            && ( dict->elements[i]->lhs[0] == (WORD)('^') ) )
        return((UBYTE *)(dict->elements[i]->rhs));
    }
    return(0);
}

/*
    #] IsExponentSign:
    #[ FindSymbol :
*/

UBYTE *FindSymbol(WORD num)
```

```
{
    if ( AO.CurrentDictionary > 0 ) {
        DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
        int i;
        if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_SYMBOL &&
                     dict->elements[i]->lhs[0] == num )
                        return((UBYTE *)(dict->elements[i]->rhs));
            }
        }
    }
    return(VARNAME(symbols,num));
}

/*
    #] FindSymbol :
    #[ FindVector :
*/

UBYTE *FindVector(WORD num)
{
    if ( AO.CurrentDictionary > 0 ) {
        DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
        int i;
        if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_VECTOR &&
                     dict->elements[i]->lhs[0] == num )
                        return((UBYTE *)(dict->elements[i]->rhs));
            }
        }
    }
    num -= AM.OffsetVector;
    return(VARNAME(vectors,num));
}

/*
    #] FindVector :
    #[ FindIndex :
*/

UBYTE *FindIndex(WORD num)
{
    if ( AO.CurrentDictionary > 0 ) {
        DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
        int i;
        if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_INDEX &&
                     dict->elements[i]->lhs[0] == num )
                        return((UBYTE *)(dict->elements[i]->rhs));
            }
        }
    }
    num -= AM.OffsetIndex;
    return(VARNAME(indices,num));
}

/*
    #] FindIndex :
    #[ FindFunction :
*/

UBYTE *FindFunction(WORD num)
{
    if ( AO.CurrentDictionary > 0 ) {
        DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
        int i;
        if ( dict->variables > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_FUNCTION &&
                     dict->elements[i]->lhs[0] == num )
                        return((UBYTE *)(dict->elements[i]->rhs));
            }
        }
    }
    num -= FUNCTION;
    return(VARNAME(functions,num));
}

/*
    #] FindFunction :
    #[ FindFunWithArgs :
*/

UBYTE *FindFunWithArgs(WORD *t)
```

```
{
    if ( AO.CurrentDictionary > 0 ) {
        DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
        int i, j;
        if ( dict->funwith > 0
                    && AO.CurDictFunWithArgs == DICT_DOFUNWITHARGS ) {
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_FUNCTION_WITH_ARGUMENTS &&
                     (WORD)(dict->elements[i]->lhs[0]) == t[0] &&
                     (WORD)(dict->elements[i]->lhs[1]) == t[1] ) {
                    for ( j = 2; j < t[1]; j++ ) {
                        if ( (WORD)(dict->elements[i]->lhs[j]) != t[j] ) break;
                    }
                    if ( j >= t[1] ) return((UBYTE *)(dict->elements[i]->rhs));
                }
            }
        }
    }
    return(0);
}

/*
    #] FindFunWithArgs :
    #[ FindExtraSymbol :

    The extra symbol is constructed in the WorkSpace. This way we do not
    have to worry about Malloc and freeing the object later.
    The input value num is already the number of the extra symbol.
    We do NOT need num = MAXVARIABLES-num;
*/

UBYTE *FindExtraSymbol(WORD num)
{
    GETIDENTITY;
    UBYTE *out = (UBYTE *)(AT.WorkPointer);
    *out = 0;
    if ( AO.CurrentDictionary > 0 ) {
        DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
        int i;
        if ( dict->ranges > 0 && AO.CurDictVariables == DICT_DOVARIABLES ) {
            for ( i = dict->numelements-1; i >= 0; i-- ) {
                if ( dict->elements[i]->type == DICT_RANGE
                     && num >= dict->elements[i]->lhs[0]
                     && num <= dict->elements[i]->lhs[1] ) {
/*
                        Now we have to translate the rhs
                        %#  gives the number
                        %@  gives the number as its position in the range
*/
                        UBYTE *r = (UBYTE *)(dict->elements[i]->rhs);
                        while ( *r ) {
                            if ( *r == (UBYTE)'%' && ( r[1] == (UBYTE)'#'
                            || r[1] == (UBYTE)'@' ) ) {
                                if ( r[1] == (UBYTE)'#' ) {
                                    out = NumCopy(num,out);
                                }
                                else {
                                    out = NumCopy(num-dict->elements[i]->lhs[0]+1,out);
                                }
                                r += 2;
                            }
                            else {
                                *out++ = *r++;
                            }
                        }
                        *out = 0;
                        return((UBYTE *)(AT.WorkPointer));
                }
            }
        }
    }

    out = StrCopy((UBYTE *)AC.extrasym,out);
    if ( AC.extrasymbols == 0 ) {
        out = NumCopy(num,out);
        out = StrCopy((UBYTE *)"_",out);
    }
    else if ( AC.extrasymbols == 1 ) {
        out = AddArrayIndex(num,out);
    }
    return((UBYTE *)(AT.WorkPointer));
}

/*
    #] FindExtraSymbol :
    #[ FindDictionary :
*/
```

```
int FindDictionary(UBYTE *name)
{
    int i;
    for ( i = 0; i < AO.NumDictionaries; i++ ) {
        if ( StrCmp(AO.Dictionaries[i]->name,name) == 0 )
            return(i+1);
    }
    return(0);
}

/*
    #] FindDictionary :
    #[ AddDictionary :
*/

int AddDictionary(UBYTE *name)
{
    DICTIONARY *dict;
/*
    First make space for the pointer in the list.
*/
    if ( AO.NumDictionaries >= AO.SizeDictionaries-1 ) {
        DICTIONARY **d;
        int i;
        if ( AO.SizeDictionaries <= 0 ) AO.SizeDictionaries = 10;
        else AO.SizeDictionaries = 2*AO.SizeDictionaries;
        d = (DICTIONARY **)Malloc1(AO.SizeDictionaries*sizeof(DICTIONARY *),"Dictionaries");
        for ( i = 0; i < AO.NumDictionaries; i++ ) d[i] = AO.Dictionaries[i];
        if ( AO.Dictionaries != 0 ) M_free(AO.Dictionaries,"Dictionaries");
        AO.Dictionaries = d;
    }
/*
    Now create an empty dictionary.
*/
    dict = (DICTIONARY *)Malloc1(sizeof(DICTIONARY),"Dictionary");
    AO.Dictionaries[AO.NumDictionaries++] = dict;
    dict->elements = 0;
    dict->name = strDup1(name,"DictionaryName");
    dict->sizeelements = 0;
    dict->numelements = 0;
    dict->numbers = 0;
    dict->variables = 0;
    dict->characters = 0;
    dict->funwith = 0;
    dict->gnumelements = 0;
    dict->ranges = 0;

    return(AO.NumDictionaries);
}

/*
    #] AddDictionary :
    #[ AddToDictionary :

        To be called from #add left:right
*/

int AddToDictionary(DICTIONARY *dict,UBYTE *left,UBYTE *right)
{
    GETIDENTITY
    CBUF *C = cbuf+AC.cbufnum;
    WORD *w = AT.WorkPointer;
    WORD *OldWork = AT.WorkPointer;
    WORD *s, oldnumrhs = C->numrhs, oldnumlhs = C->numlhs;
    WORD *ow, *ww, *mm, oldEside, *where = 0, type, number, range[3];
    LONG oldcpointer;
    int error = 0, sizelhs, sizerhs, i, retcode;
    UBYTE *r;
    DICTIONARY_ELEMENT *new;
    WORD power = (WORD)('^'), times = (WORD)('*');
    if ( ( left[0] == '^' && left[1] == 0 )
      || ( left[0] == '*' && left[1] == '*' && left[2] == 0 ) ) {
        type = DICT_SPECIALCHARACTER;
        number = 1;
        where = &power;
        goto TestDouble;
    }
    else if ( left[0] == '*' && left[1] == 0 ) {
        type = DICT_SPECIALCHARACTER;
        number = 1;
        where = &times;
        goto TestDouble;
    }
    else if ( left[0] == '(' ) {  /* range of extra symbols */
        WORD x1 = 0, x2 = 0;
        r = left+1;
```

```
        while ( FG.cTable[*r] == 1 ) x1 = 10*x1 + *r++ - '0';
        if ( *r == ',' ) {
            r++;
            while ( FG.cTable[*r] == 1 ) x2 = 10*x2 + *r++ - '0';
        }
        else x2 = x1;
        number = 2;
        if ( *r != ')' ) {
            MesPrint("&Illegal range specification in LHS of %#add instruction.");
            return(1);
        }
        type = DICT_RANGE;
        if ( x1 <= 0 || x2 <= 0 || x1 > x2 ) {
            MesPrint("&Illegal range in LHS of %#add instruction.");
            return(1);
        }
        range[0] = x1;
        range[1] = x2;
        range[2] = 0;
        where = range;
        goto TestDouble;
    }
/*
    Translate the left part. Determine type.
    We follow the code in CoIdExpression and then veto what we do not like.
    Just make sure to pop what needs to be popped in the compiler buffer.
*/
    AC.ProtoType = w;
    *w++ = SUBEXPRESSION;
    *w++ = SUBEXPSIZE;
    *w++ = C->numrhs+1;
    *w++ = 1;
    *w++ = AC.cbufnum;
    FILLSUB(w)
    AC.WildC = w;
    AC.NwildC = 0;
    AT.WorkPointer = s = w + 4*AM.MaxWildcards + 8;
/*
    Now read the LHS
*/
    oldcpointer = AddLHS(AC.cbufnum) - C->Buffer;

    if ( ( retcode = CompileAlgebra(left,LHSIDE,AC.ProtoType) ) < 0 ) { error = 1; }
    else AC.ProtoType[2] = retcode;
    AT.WorkPointer = s;
    if ( AC.NwildC && SortWild(w,AC.NwildC) ) error = 1;

    OldWork[1] = AC.WildC-OldWork;
    w = AC.WildC;
    AT.WorkPointer = w;
    s = C->rhs[C->numrhs];
/*
    We have the expression in the compiler buffers.
    The main level is at lhs[numlhs]
    The partial lhs (including ProtoType) is in OldWork (in WorkSpace)
    We need to load the result at w after the prototype
    Because these sort routines don't use the WorkSpace
    there should not be a conflict
*/
    if ( !error && *s == 0 ) {
IllLeft:MesPrint("&Illegal LHS in dictionary");
        AC.lhdollarflag = 0;
        return(1);
    }
    if ( !error && *(s+*s) != 0 ) {
        MesPrint("&LHS in dictionary should be one term only");
        return(1);
    }
    if ( error == 0 ) {
        if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) {
            if ( !error ) error = 1;
            return(error);
        }
        AN.RepPoint = AT.RepCount + 1;
        ow = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
        mm = s; ww = ow; i = *mm;
        while ( --i >= 0 ) {*ww++ = *mm++;} AT.WorkPointer = ww;
        AC.lhdollarflag = 0; oldEside = AR.Eside; AR.Eside = LHSIDE;
        AR.Cnumlhs = C->numlhs;
        if ( Generator(BHEAD ow,C->numlhs) ) {
            AR.Eside = oldEside;
            LowerSortLevel(); LowerSortLevel(); goto IllLeft;
        }
        AR.Eside = oldEside;
        AT.WorkPointer = w;
        if ( EndSort(BHEAD w,0) < 0 ) { LowerSortLevel(); goto IllLeft; }
        if ( *w == 0 || *(w+*w) != 0 ) {
```

```
            MesPrint("&LHS must be one term");
            AC.lhdollarflag = 0;
            return(1);
        }
        LowerSortLevel();
    }
    AT.WorkPointer = w + *w;
    AC.DumNum = 0;
/*
    Everything is now after OldWork. We can pop the compilerbuffer.
    Next test for illegal things like a coefficient
    At this point we have:
    w = the term of the LHS
*/
    C->Pointer = C->Buffer + oldcpointer;
    C->numrhs = oldnumrhs;
    C->numlhs = oldnumlhs;
    AC.lhdollarflag = 0;
/*
    Test for undesirables.
        1: wildcards
        2: sign
        3: more than one term
        4: composite terms
*/
    if ( AC.ProtoType[1] != SUBEXPSIZE ) {
        MesPrint("& Currently no wildcards allowed in dictionaries.");
        return(1);
    }
    if ( w[w[0]-1] < 0 ) {
        MesPrint("& Currently no sign allowed in dictionaries.");
        return(1);
    }
    if ( w[w[0]] != 0 ) {
        MesPrint("& More than one term in dictionary element.");
        return(1);
    }
    if ( w[0] == w[w[0]-1]+1 ) {    /* Only coefficient */
        WORD *numer, *denom;
        WORD nsize, dsize;
        nsize = dsize = (w[w[0]-1]-1)/2;
        numer = w+1;
        denom = numer+nsize;
        while ( numer[nsize-1] == 0 ) nsize--;
        while ( denom[dsize-1] == 0 ) dsize--;
        if ( dsize == 1 && denom[0] == 1 ) {
            type = DICT_INTEGERNUMBER;
            number = nsize;
            where = numer;
        }
        else {
            type = DICT_RATIONALNUMBER;
            number = w[0];
            where = w;
        }
    }
    else {
        s = w + w[0]-1;
        if ( s[0] != 3 || s[-1] != 1 || s[-2] != 1 ) {
Compositeness:;
            MesPrint("& Currently no composite objects allowed in dictionaries.");
            return(1);
        }
        if ( w[0] != w[2]+4 ) goto Compositeness;
        s = w+1;
        switch ( *s ) {
            case SYMBOL:
                if ( s[1] != 4 || s[3] != 1 ) goto Compositeness;
                type = DICT_SYMBOL;
                number = 1;
                where = s+2;
                break;
            case INDEX:
                if ( s[1] != 3 ) goto Compositeness;
                if ( s[2] < 0 ) type = DICT_VECTOR;
                else            type = DICT_INDEX;
                number = 1;
                where = s+2;
                break;
            default:
                if ( *s < FUNCTION ) {
                    MesPrint("& Illegal object in dictionary.");
                    return(1);
                }
                if ( s[1] == FUNHEAD ) {
                    type = DICT_FUNCTION;
                    number = 1;
```

```
                    where = s;
                    break;
                }
                else {
                    type = DICT_FUNCTION_WITH_ARGUMENTS;
                    number = s[1];
                    where = s;
                }
                break;
        }
    }
TestDouble:;
/*
    Create a new element
*/
    if ( dict->numelements >= dict->sizeelements ) {
        DICTIONARY_ELEMENT **d;
        if ( dict->sizeelements <= 0 ) dict->sizeelements = 10;
        else                           dict->sizeelements *= 2;
        d = (DICTIONARY_ELEMENT **)Malloc1(
            sizeof(DICTIONARY_ELEMENT *)*dict->sizeelements,"Dictionary elements");
        for ( i = 0; i < dict->numelements; i++ )
            d[i] = dict->elements[i];
        if ( dict->elements ) M_free(dict->elements,"Dictionary elements");
        dict->elements = d;
    }
    sizelhs = number+1;
    sizerhs = 1; r = right; while ( *r++ ) sizerhs++;
    sizerhs = (sizerhs+sizeof(WORD)-1)/sizeof(WORD)+1;
    new = (DICTIONARY_ELEMENT *)Malloc1(sizeof(DICTIONARY_ELEMENT)
                +sizeof(WORD)*(sizelhs+sizerhs),"Dictionary element");
    new->lhs = (WORD *)(new+1);
    new->rhs = new->lhs+sizelhs;
    new->type = type;
    new->size = number;
    for ( i = 0; i < number; i++ ) new->lhs[i] = where[i];
    new->lhs[i] = 0;
    r = (UBYTE *)(new->rhs);
    while ( *right ) {
        if ( *right == '\\' && ( right[1] == '`' || right[1] == '\"' ) ) right++;
        *r++ = *right++;
    }
    *r = 0;

    dict->elements[dict->numelements++] = new;

    switch ( type ) {
        case DICT_INTEGERNUMBER:
        case DICT_RATIONALNUMBER:
            dict->numbers++; break;
        case DICT_SYMBOL:
        case DICT_VECTOR:
        case DICT_INDEX:
        case DICT_FUNCTION:
            dict->variables++; break;
        case DICT_FUNCTION_WITH_ARGUMENTS:
            dict->funwith++; break;
        case DICT_SPECIALCHARACTER:
            dict->characters++; break;
        case DICT_RANGE:
            dict->ranges++; break;
    }

    AT.WorkPointer = OldWork;
    return(0);
}

/*
    #] AddToDictionary :
    #[ UseDictionary :
*/

int UseDictionary(UBYTE *name,UBYTE *options)
{
    int i;
    for ( i = 0; i < AO.NumDictionaries; i++ ) {
        if ( StrCmp(AO.Dictionaries[i]->name,name) == 0 ) {
            AO.CurrentDictionary = i+1;
            if ( SetDictionaryOptions(options) < 0 ) {
                AO.CurrentDictionary = 0;
                return(-1);
            }
            else {  /* Now test whether what is requested is really there? */
                return(0);
            }
        }
    }
```

```
    MesPrint("@There is no dictionary with the name %s",name);
    exit(-1);
}

/*
    #] UseDictionary :
    #[ SetDictionaryOptions :
*/

int SetDictionaryOptions(UBYTE *options)
{
    UBYTE *opt, *s, c;
    int retval = 0;
    s = options;
    AO.CurDictNumbers = DICT_ALLNUMBERS;
    AO.CurDictVariables = DICT_DOVARIABLES;
    AO.CurDictSpecials = DICT_DOSPECIALS;
    AO.CurDictFunWithArgs = DICT_DOFUNWITHARGS;
    AO.CurDictNumberWarning = 0;
    AO.CurDictNotInFunctions= 0;
    AO.CurDictInDollars = DICT_NOTINDOLLARS;
    while ( *s ) {
        opt = s;
        while ( *s && *s != ',' && *s != ' ' ) s++;
        c = *s; *s = 0;
        if ( opt[0] == '$' && opt[1] == 0 ) {
            AO.CurDictInDollars = DICT_INDOLLARS;
        }
        else if ( StrICmp(opt,(UBYTE *)"nonumbers") == 0 ) {
            AO.CurDictNumbers = DICT_NONUMBERS;
        }
        else if ( StrICmp(opt,(UBYTE *)"integersonly") == 0 ) {
            AO.CurDictNumbers = DICT_INTEGERONLY;
        }
        else if ( StrICmp(opt,(UBYTE *)"rationalsonly") == 0 ) {
            AO.CurDictNumbers = DICT_RATIONALONLY;
        }
        else if ( StrICmp(opt,(UBYTE *)"allnumbers") == 0 ) {
            AO.CurDictNumbers = DICT_ALLNUMBERS;
        }
        else if ( StrICmp(opt,(UBYTE *)"novariables") == 0 ) {
            AO.CurDictVariables = DICT_NOVARIABLES;
        }
        else if ( StrICmp(opt,(UBYTE *)"numbersonly") == 0 ) {
            AO.CurDictNumbers = DICT_ALLNUMBERS;
            AO.CurDictVariables = DICT_NOVARIABLES;
            AO.CurDictSpecials = DICT_NOSPECIALS;
            AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
        }
        else if ( StrICmp(opt,(UBYTE *)"variablesonly") == 0 ) {
            AO.CurDictNumbers = DICT_NONUMBERS;
            AO.CurDictVariables = DICT_DOVARIABLES;
            AO.CurDictSpecials = DICT_NOSPECIALS;
            AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
        }
        else if ( StrICmp(opt,(UBYTE *)"nospecials") == 0 ) {
            AO.CurDictSpecials = DICT_NOSPECIALS;
        }
        else if ( StrICmp(opt,(UBYTE *)"specialsonly") == 0 ) {
            AO.CurDictNumbers = DICT_NONUMBERS;
            AO.CurDictVariables = DICT_NOVARIABLES;
            AO.CurDictSpecials = DICT_DOSPECIALS;
            AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
        }
        else if ( StrICmp(opt,(UBYTE *)"nofunwithargs") == 0 ) {
            AO.CurDictFunWithArgs = DICT_NOFUNWITHARGS;
        }
        else if ( StrICmp(opt,(UBYTE *)"funwithargsonly") == 0 ) {
            AO.CurDictNumbers = DICT_NONUMBERS;
            AO.CurDictVariables = DICT_NOVARIABLES;
            AO.CurDictSpecials = DICT_NOSPECIALS;
            AO.CurDictFunWithArgs = DICT_DOFUNWITHARGS;
        }
        else if ( StrICmp(opt,(UBYTE *)"warnings") == 0
               || StrICmp(opt,(UBYTE *)"warning") == 0 ) {
            AO.CurDictNumberWarning = 1;
        }
        else if ( StrICmp(opt,(UBYTE *)"nowarnings") == 0
               || StrICmp(opt,(UBYTE *)"nowarning") == 0 ) {
            AO.CurDictNumberWarning = 0;
        }
        else if ( StrICmp(opt,(UBYTE *)"infunctions") == 0 ) {
            AO.CurDictNotInFunctions= 0;
        }
        else if ( StrICmp(opt,(UBYTE *)"notinfunctions") == 0 ) {
            AO.CurDictNotInFunctions= 1;
        }
```

```
        else {
            MesPrint("@ Unrecognized option in %#SetDictionary: %s",opt);
            retval = -1;
        }
        *s = c;
        if ( c == ',' ) s++;
    }
    return(retval);
}

/*
    #] SetDictionaryOptions :
    #[ UnSetDictionary :
*/

void UnSetDictionary(void)
{
    AO.CurrentDictionary = 0;
    AO.CurDictNumbers = -1;
    AO.CurDictVariables = -1;
    AO.CurDictSpecials = -1;
    AO.CurDictFunWithArgs = -1;
    AO.CurDictFunWithArgs = -1;
    AO.CurDictNumberWarning = -1;
    AO.CurDictNotInFunctions= -1;
}

/*
    #] UnSetDictionary :
    #[ RemoveDictionary :

    Mostly needed for .clear
*/

void RemoveDictionary(DICTIONARY *dict)
{
    int i;
    if ( dict == 0 ) return;
    for ( i = 0; i < AO.NumDictionaries; i++ ) {
        if ( AO.Dictionaries[i] == dict ) {
            for (i++; i < AO.NumDictionaries; i++ ) {
                AO.Dictionaries[i-1] = AO.Dictionaries[i];
            }
            AO.NumDictionaries--;
            goto removeit;
        }
    }
    MesPrint("@ Dictionary not found in RemoveDictionary");
    exit(-1);
removeit:;
    for ( i = 0; i < dict->numelements; i++ )
            M_free(dict->elements[i],"Dictionary element");
    for ( i = 0; i < dict->numelements; i++ ) dict->elements[i] = 0;
    if ( dict->elements ) M_free(dict->elements,"Dictionary elements");
    if ( dict->name ) {
        M_free(dict->name,"DictionaryName");
        dict->name = 0;
    }
    dict->sizeelements = 0;
    dict->numelements = 0;
    dict->numbers = 0;
    dict->variables = 0;
    dict->characters = 0;
    dict->funwith = 0;
    dict->gnumelements = 0;
    dict->ranges = 0;
}

/*
    #] RemoveDictionary :
    #[ ShrinkDictionary :

    To be called after a .store to restore the dictionary to the state
    it had at the last .global
    We do not make the elements array shorter.
*/

void ShrinkDictionary(DICTIONARY *dict)
{
    while ( dict->numelements > dict->gnumelements ) {
        dict->numelements--;
        M_free(dict->elements[dict->numelements],"Dictionary element");
        dict->elements[dict->numelements] = 0;
    }
}

/*
```

```
    #] ShrinkDictionary :
    #[ DoPreOpenDictionary :
*/

int DoPreOpenDictionary(UBYTE *s)
{
    UBYTE *name;
    int dict;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' ) s++;

    name = s; s = SkipAName(s);
    if ( *s != 0 && *s != ';' ) {
        MesPrint("@proper syntax is #opendictionary name");
        return(-1);
    }
    *s = 0;

    if ( AP.OpenDictionary > 0 ) {
        MesPrint("@you cannot nest #opendictionary instructions");
        MesPrint("@dictionary %s is open already",
                AO.Dictionaries[AP.OpenDictionary-1]->name);
        return(-1);
    }
    if ( AO.CurrentDictionary > 0 ) {
        MesPrint("@before opening a dictionary you have to first close the selected dictionary");
        return(-1);
    }
/*
    Do we have this dictionary already?
*/
    dict = FindDictionary(name);
    if ( dict == 0 ) dict = AddDictionary(name);
    AP.OpenDictionary = dict;
    return(0);
}

/*
    #] DoPreOpenDictionary :
    #[ DoPreCloseDictionary :
*/

int DoPreCloseDictionary(UBYTE *s)
{
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' ) s++;

    if ( AP.OpenDictionary == 0 && AO.CurrentDictionary == 0 ) {
        MesPrint("@you have neither an open, nor a selected dictionary");
        return(-1);
    }

    AP.OpenDictionary = 0;
    AO.CurrentDictionary = 0;

    AO.CurDictNotInFunctions = 0;

    return(0);
}

/*
    #] DoPreCloseDictionary :
    #[ DoPreUseDictionary :
*/

int DoPreUseDictionary(UBYTE *s)
{
    UBYTE *options, c, *ss, *sss, *name;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' ) s++;

    if ( AP.OpenDictionary > 0 ) {
        MesPrint("@before selecting a dictionary you have to first close the open dictionary");
        return(-1);
    }

    name = s; s = SkipAName(s);
    ss = s; while ( *s && *s != '(' ) s++;
    c = *ss; *ss = 0;
    if ( c == 0 ) {
        options = ss;
    }
    else {
        options = s+1; SKIPBRA3(s)
```

```
        if ( *s != ')' ) {
            MesPrint("@Irregular end of %#UseDictionary instruction");
            return(-1);
        }
        sss = s;
        s++; while ( *s == ' ' || *s == '\t' || *s == ';' ) s++;
        *sss = 0;
        if ( *s ) {
            MesPrint("@Irregular end of %#UseDictionary instruction");
            return(-1);
        }
    }
    return(UseDictionary(name,options));
}

/*
    #] DoPreUseDictionary :
    #[ DoPreAdd :

    Syntax:
        #add left :right
        #add left : "right"
    Adds to the currently open dictionary
*/

int DoPreAdd(UBYTE *s)
{
    UBYTE *left, *right;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' ) s++;

    if ( AP.OpenDictionary == 0 ) {
        MesPrint("@there is no open dictionary to add to");
        return(-1);
    }
/*
    Scan to the : and mark the left and right parts.
*/
    left = s;
    while ( *s && *s != ':' ) {
        if ( *s == '[' ) { SKIPBRA1(s) s++; }
        else if ( *s == '{' ) { SKIPBRA2(s) s++; }
        else if ( *s == '(' ) { SKIPBRA3(s) s++; }
        else if ( *s == ']' || *s == '}' || *s == ')' ) {
            MesPrint("@unmatched brackets in #add instruction");
            return(-1);
        }
        else s++;
    }
    if ( *s == 0 ) {
        MesPrint("@Missing : in #add instruction");
        return(-1);
    }
    *s++ = 0;
    right = s;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == '"' && s[1] ) {
        right = s+1;
        s = s+2;
        while ( *s ) s++;
        while ( s[-1] != '"' ) s--;
        if ( s <= right ) {
            MesPrint("@Irregular use of double quotes in #add instruction");
            return(-1);
        }
        s[-1] = 0;
    }
    return(AddToDictionary(AO.Dictionaries[AP.OpenDictionary-1],left,right));
}

/*
    #] DoPreAdd :
    #[ DictToBytes :
*/

LONG DictToBytes(DICTIONARY *dict,UBYTE *buf)
{
    int numelements = dict->numelements, sizeelement, i, j, x;
    UBYTE *s1, *s2 = buf;
    DICTIONARY_ELEMENT *e;
/*
    First copy the struct
*/
    s1 = (UBYTE *)dict; j = sizeof(DICTIONARY);
    NCOPY(s2,s1,j)
```

```
/*
    Now the elements. Put a size indicator in front of each of them.
*/
    for ( i = 0; i < numelements; i++ ) {
        e = dict->elements[i];
        sizeelement = sizeof(DICTIONARY_ELEMENT)+(e->size+1)*sizeof(WORD);
        s1 = (UBYTE *)e->rhs; x = 0;
        while ( *s1 ) { s1++; x++; }
        x /= sizeof(WORD);
        sizeelement += (x+1) * sizeof(WORD);
        s1 = (UBYTE *)(&sizeelement); j = sizeof(WORD); NCOPY(s2,s1,j)
        s1 = (UBYTE *)e; j = sizeof(DICTIONARY_ELEMENT); NCOPY(s2,s1,j)
        s1 = (UBYTE *)e->lhs; j = (e->size+1)*(sizeof(WORD)); NCOPY(s2,s1,j)
        s1 = (UBYTE *)e->rhs; j = (x+1)*(sizeof(WORD)); NCOPY(s2,s1,j)
    }
    return(s2-buf);
}

/*
    #] DictToBytes :
    #[ DictFromBytes :
*/

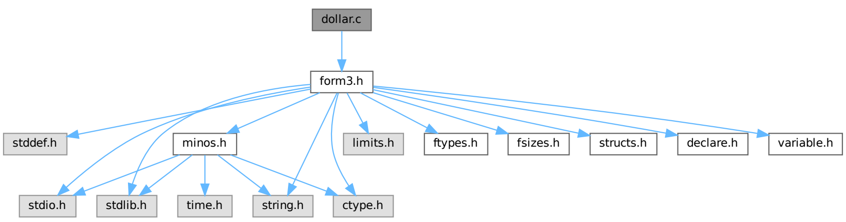
DICTIONARY *DictFromBytes(UBYTE *buf)
{
    DICTIONARY *dict = Malloc1(sizeof(DICTIONARY),"Dictionary");
    UBYTE *s1, *s2;
    int i, j, sizeelement;
    DICTIONARY_ELEMENT *e;
/*
    First read the dictionary itself
*/
    s1 = buf;
    s2 = (UBYTE *)dict; j = sizeof(DICTIONARY); NCOPY(s2,s1,j)
/*
    Allocate the elements array:
*/
    dict->elements = (DICTIONARY_ELEMENT **)Malloc1(
            sizeof(DICTIONARY_ELEMENT *)*dict->sizeelements,"dictionary elements");
    for ( i = 0; i < dict->numelements; i++ ) {
        s2 = (UBYTE *)(&sizeelement); j = sizeof(WORD); NCOPY(s2,s1,j)
        e = (DICTIONARY_ELEMENT *)Malloc1(sizeelement*sizeof(UBYTE),"dictionary element");
        dict->elements[i] = e;
        j = sizeelement; s2 = (UBYTE *)e; NCOPY(s2,s1,j)
        e->lhs = (WORD *)(e+1);
        e->rhs = e->lhs + e->size+1;
    }
    return(dict);
}

/*
    #] DictFromBytes :
*/
```

4.26 dollar.c File Reference

`#include "form3.h"`

Include dependency graph for dollar.c:



Macros

- #define STEP2

Functions

- int CatchDollar (int par)
- int AssignDollar (PHEAD WORD ∗term, WORD level)
- UBYTE ∗ WriteDollarToBuffer (WORD numdollar, WORD par)
- UBYTE ∗ WriteDollarFactorToBuffer (WORD numdollar, WORD numfac, WORD par)
- void AddToDollarBuffer (UBYTE ∗s)
- void TermAssign (WORD ∗term)
- int PutTermInDollar (WORD ∗term, WORD numdollar)
- void WildDollars (PHEAD WORD ∗term)
- WORD DolToTensor (PHEAD WORD numdollar)
- WORD DolToFunction (PHEAD WORD numdollar)
- WORD DolToVector (PHEAD WORD numdollar)
- WORD DolToNumber (PHEAD WORD numdollar)
- WORD DolToSymbol (PHEAD WORD numdollar)
- WORD DolToIndex (PHEAD WORD numdollar)
- DOLLARS DolToTerms (PHEAD WORD numdollar)
- LONG DolToLong (PHEAD WORD numdollar)
- int ExecInside (UBYTE ∗s)
- int InsideDollar (PHEAD WORD ∗ll, WORD level)
- void ExchangeDollars (int num1, int num2)
- LONG TermsInDollar (WORD num)
- LONG SizeOfDollar (WORD num)
- UBYTE ∗ PreIfDollarEval (UBYTE ∗s, int ∗value)
- WORD ∗ TranslateExpression (UBYTE ∗s)
- int IsSetMember (WORD ∗buffer, WORD numset)
- int IsMultipleOf (WORD ∗buf1, WORD ∗buf2)
- int TwoExprCompare (WORD ∗buf1, WORD ∗buf2, int oprtr)
- int DollarRaiseLow (UBYTE ∗name, LONG value)
- WORD EvalDoLoopArg (PHEAD WORD ∗arg, WORD par)
- WORD TestDoLoop (PHEAD WORD ∗lhsbuf, WORD level)
- WORD TestEndDoLoop (PHEAD WORD ∗lhsbuf, WORD level)
- int DollarFactorize (PHEAD WORD numdollar)
- void CleanDollarFactors (DOLLARS d)
- WORD ∗ TakeDollarContent (PHEAD WORD ∗dollarbuffer, WORD ∗∗factor)
- WORD ∗ MakeDollarInteger (PHEAD WORD ∗bufin, WORD ∗∗bufout)
- WORD ∗ MakeDollarMod (PHEAD WORD ∗buffer, WORD ∗∗bufout)
- int GetDolNum (PHEAD WORD ∗t, WORD ∗tstop)
- void AddPotModdollar (WORD numdollar)

4.26.1 Detailed Description

The routines that deal with the dollar variables. The name administration is to be found in the file names.c

Definition in file dollar.c.

4.26.2 Macro Definition Documentation

4.26.2.1 STEP2

```
#define STEP2
```

Factors a dollar expression. Notation: d->nfactors becomes nonzero. if the number of factors is one, we leave d->factors zero. Otherwise factors is an array of pointers to the factors. These are pointers of the type FACDOLLAR. fd->where pointer to contents in term notation fd->size size of the buffer fd->where points to fd->type DOLNUMBER or DOLTERMS fd->value value if type is DOLNUMBER and it fits in a WORD.

Definition at line 2953 of file dollar.c.

4.26.3 Function Documentation

4.26.3.1 CatchDollar()

```
int CatchDollar (
            int par )
```

Definition at line 54 of file dollar.c.

4.26.3.2 AssignDollar()

```
int AssignDollar (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 209 of file dollar.c.

4.26.3.3 WriteDollarToBuffer()

```
UBYTE * WriteDollarToBuffer (
            WORD numdollar,
            WORD par )
```

Definition at line 596 of file dollar.c.

4.26.3.4 WriteDollarFactorToBuffer()

```
UBYTE * WriteDollarFactorToBuffer (
            WORD numdollar,
            WORD numfac,
            WORD par )
```

Definition at line 687 of file dollar.c.

4.26.3.5 AddToDollarBuffer()

```
void AddToDollarBuffer (
            UBYTE * s )
```

Definition at line 762 of file dollar.c.

4.26.3.6 TermAssign()

```
void TermAssign (
            WORD * term )
```

Definition at line 803 of file dollar.c.

4.26.3.7 PutTermInDollar()

```
int PutTermInDollar (
            WORD * term,
            WORD numdollar )
```

Definition at line 863 of file dollar.c.

4.26.3.8 WildDollars()

```
void WildDollars (
            PHEAD WORD * term )
```

Definition at line 891 of file dollar.c.

4.26.3.9 DolToTensor()

```
WORD DolToTensor (
            PHEAD WORD numdollar )
```

Definition at line 1082 of file dollar.c.

4.26.3.10 DolToFunction()

```
WORD DolToFunction (
            PHEAD WORD numdollar )
```

Definition at line 1143 of file dollar.c.

4.26.3.11 DolToVector()

```
WORD DolToVector (
            PHEAD WORD numdollar )
```

Definition at line 1200 of file dollar.c.

4.26.3.12 DolToNumber()

```
WORD DolToNumber (
            PHEAD WORD numdollar )
```

Definition at line 1264 of file dollar.c.

4.26.3.13 DolToSymbol()

```
WORD DolToSymbol (
            PHEAD WORD numdollar )
```

Definition at line 1323 of file dollar.c.

4.26.3.14 DolToIndex()

```
WORD DolToIndex (
            PHEAD WORD numdollar )
```

Definition at line 1377 of file dollar.c.

4.26.3.15 DolToTerms()

```
DOLLARS DolToTerms (
            PHEAD WORD numdollar )
```

Definition at line 1458 of file dollar.c.

4.26.3.16 DolToLong()

```
LONG DolToLong (
            PHEAD WORD numdollar )
```

Definition at line 1606 of file dollar.c.

4.26.3.17 ExecInside()

```
int ExecInside (
            UBYTE * s )
```

Definition at line 1681 of file dollar.c.

4.26.3.18 InsideDollar()

```
int InsideDollar (
            PHEAD WORD * ll,
            WORD level )
```

Definition at line 1746 of file dollar.c.

4.26.3.19 ExchangeDollars()

```
void ExchangeDollars (
            int num1,
            int num2 )
```

Definition at line 1849 of file dollar.c.

4.26.3.20 TermsInDollar()

```
LONG TermsInDollar (
            WORD num )
```

Definition at line 1867 of file dollar.c.

4.26.3.21 SizeOfDollar()

```
LONG SizeOfDollar (
            WORD num )
```

Definition at line 1916 of file dollar.c.

4.26.3.22 PreIfDollarEval()

```
UBYTE * PreIfDollarEval (
            UBYTE * s,
            int * value )
```

Definition at line 1978 of file dollar.c.

4.26.3.23 TranslateExpression()

```
WORD * TranslateExpression (
            UBYTE * s )
```

Definition at line 2160 of file dollar.c.

4.26.3.24 IsSetMember()

```
int IsSetMember (
            WORD * buffer,
            WORD numset )
```

Definition at line 2217 of file dollar.c.

4.26.3.25 IsMultipleOf()

```
int IsMultipleOf (
            WORD * buf1,
            WORD * buf2 )
```

Definition at line 2398 of file dollar.c.

4.26.3.26 TwoExprCompare()

```
int TwoExprCompare (
            WORD * buf1,
            WORD * buf2,
            int oprtr )
```

Definition at line 2473 of file dollar.c.

4.26.3.27 DollarRaiseLow()

```
int DollarRaiseLow (
            UBYTE * name,
            LONG value )
```

Definition at line 2552 of file dollar.c.

4.26.3.28 EvalDoLoopArg()

```
WORD EvalDoLoopArg (
            PHEAD WORD * arg,
            WORD par )
```

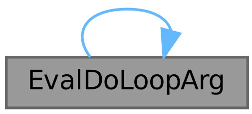
Evaluates one argument of a do loop. Such an argument is constructed from SNUMBERs DOLLAREXPRESSIONs and possibly DOLLAREXPR2s which indicate factors of the preceding dollar. Hence we have SNUMBER,num DOLLAREXPRESSION,numdollar DOLLAREXPRESSION,numdollar,DOLLAREXPR2,numfactor DOLLAREXPRESSION,numdollar,DOLLAREXPR2,numfactor,DOLLAREXPR2,numfactor etc. Because we have a do-loop at every stage we should have a number. The notation in DOLLAREXPR2 is that $>=$ 0 is number of yet another dollar and $<$ 0 is -n-1 with n the array element or zero. The return value is the (short) number. The routine works its way through the list in a recursive manner.

Definition at line 2651 of file dollar.c.

References EvalDoLoopArg().

Referenced by EvalDoLoopArg().

Here is the call graph for this function:



4.26.3.29 TestDoLoop()

```
WORD TestDoLoop (
            PHEAD WORD * lhsbuf,
            WORD level )
```

Definition at line 2762 of file dollar.c.

4.26.3.30 TestEndDoLoop()

```
WORD TestEndDoLoop (
            PHEAD WORD * lhsbuf,
            WORD level )
```

Definition at line 2840 of file dollar.c.

4.26.3.31 DollarFactorize()

```
int DollarFactorize (
            PHEAD WORD numdollar )
```

Definition at line 2955 of file dollar.c.

4.26.3.32 CleanDollarFactors()

```
void CleanDollarFactors (
            DOLLARS d )
```

Definition at line 3554 of file dollar.c.

4.26.3.33 TakeDollarContent()

```
WORD * TakeDollarContent (
            PHEAD WORD * dollarbuffer,
            WORD ** factor )
```

Definition at line 3575 of file dollar.c.

4.26.3.34 MakeDollarInteger()

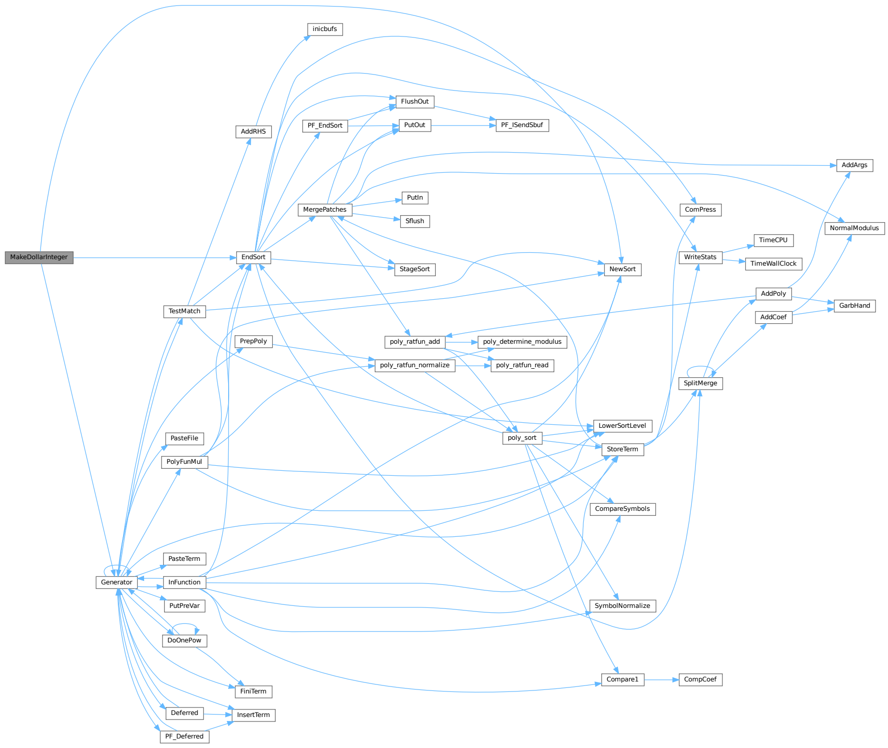
```
WORD * MakeDollarInteger (
            PHEAD WORD * bufin,
            WORD ** bufout )
```

For normalizing everything to integers we have to determine for all elements of this argument the LCM of the denominators and the GCD of the numerators. The input argument is in bufin. The number that comes out is the return value. The normalized argument is in bufout.

Definition at line 3627 of file dollar.c.

References EndSort(), Generator(), and NewSort().

Here is the call graph for this function:



4.26.3.35 MakeDollarMod()

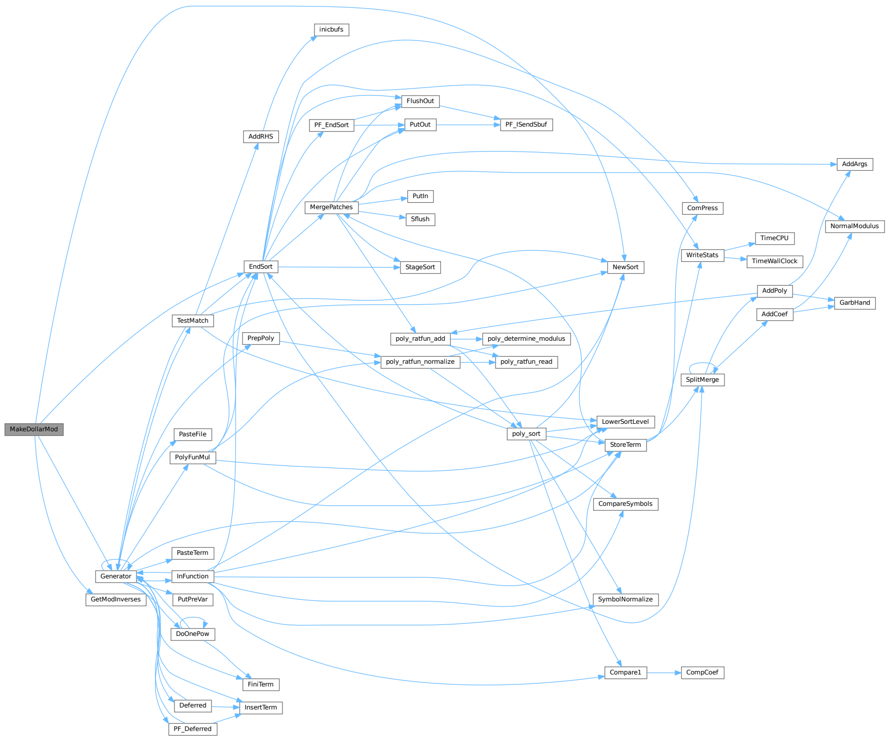
```
WORD * MakeDollarMod (
            PHEAD WORD * buffer,
            WORD ** bufout )
```

Similar to MakeDollarInteger but now with modulus arithmetic using only a one WORD 'prime'. We make the coefficient of the first term in the argument equal to one. Already the coefficients are taken modulus AN.cmod and AN.ncmod == 1

Definition at line 3801 of file dollar.c.

References EndSort(), Generator(), GetModInverses(), and NewSort().

Here is the call graph for this function:



4.26.3.36 GetDolNum()

```
int GetDolNum (
            PHEAD WORD * t,
            WORD * tstop )
```

Definition at line 3846 of file dollar.c.

4.26.3.37 AddPotModdollar()

```
void AddPotModdollar (
            WORD numdollar )
```

Adds a $-variable specified by *numdollar* to the list of potentially modified $-variables unless it has already been included in the list.

Parameters

numdollar	The index of the $-variable to be added.

Definition at line 3959 of file dollar.c.

4.27 dollar.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
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 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
*/

#include "form3.h"

/* EXTERNLOCK(dummylock) */

static UBYTE underscore[2] = {'_',0};

/*
    #] Includes :
    #[ CatchDollar :

    Works out a dollar expression during compile type.
    Steals it from the buffer and puts it in an assignment.
    At the moment we should keep this inside the small buffer.
    Later with more sort buffers we can do this better.
    Par == 0 : regular assignment
    par == -1: after error. Just make zero for now.
*/

int CatchDollar(int par)
{
    GETIDENTITY
    CBUF *C = cbuf + AC.cbufnum;
    int error = 0, numterms = 0, numdollar, resetmods = 0;
    LONG newsize, retval;
    WORD *w, *t, n, nsize, *oldwork = AT.WorkPointer, *dbuffer;
    WORD oldncmod = AN.ncmod;
    DOLLARS d;
    if ( AN.ncmod && ( ( AC.modmode & ALSODOLLARS ) == 0 ) ) AN.ncmod = 0;
    if ( AN.ncmod && AN.cmod == 0 ) { SetMods(); resetmods = 1; }

    numdollar = C->lhs[C->numlhs][2];

    d = Dollars+numdollar;
    if ( par == -1 ) {
        d->type = DOLUNDEFINED;
        cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
        cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
        if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"$-buffer old");
        d->size = 0; d->where = &(AM.dollarzero);
        cbuf[AM.dbufnum].rhs[numdollar] = d->where;
        AN.ncmod = oldncmod;
        if ( resetmods ) UnSetMods();
        return(0);
    }
#ifdef WITHMPI
    /*
     * The problem here is that only the master can make an assignment
     * like #$a=g; where g is an expression: only the master has an access to
     * the expression. So, in cases where the RHS contains expression names,
     * only the master invokes Generator() and then broadcasts the result to
     * the all slaves.
```

```
	 * Broadcasting must be performed immediately; one cannot postpone it
	 * to the end of the module because the dollar variable is visible
	 * in the current module. For the same reason, this should be done
	 * regardless of on/off parallel status.
	 * If the RHS does not contain any expression names, it can be processed
	 * in each slave.
	 */
	if ( PF.me == MASTER || !AC.RhsExprInModuleFlag ) {
#endif

	EXCHINOUT

	if ( NewSort(BHEAD0) ) { if ( !error ) error = 1; goto onerror; }
	if ( NewSort(BHEAD0) ) {
		LowerSortLevel();
		if ( !error ) error = 1;
		goto onerror;
	}
	AN.RepPoint = AT.RepCount + 1;
	w = C->rhs[C->lhs[C->numlhs][5]];
	while ( *w ) {
		n = *w; t = oldwork;
		NCOPY(t,w,n)
		AT.WorkPointer = t;
		AR.Cnumlhs = C->numlhs;
		if ( Generator(BHEAD oldwork,C->numlhs) ) { error = 1; break; }
	}
	AT.WorkPointer = oldwork;
	AN.tryterm = 0; /* for now */
	dbuffer = 0;
	if ( ( retval = EndSort(BHEAD (WORD *)((void *)(&dbuffer)),2) ) < 0 ) { error = 1; }
	LowerSortLevel();
	if ( retval <= 1 || dbuffer == 0 ) {
		d->type = DOLZERO;
		if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"$-buffer old");
		d->size = 0; d->where = &(AM.dollarzero);
		cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
		cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
		goto docopy2;
	}
	w = dbuffer;
	if ( error == 0 )
		while ( *w ) { w += *w; numterms++; }
	else
		goto onerror;
	newsize = (w-dbuffer)+1;
#ifdef WITHMPI
	}
	if ( AC.RhsExprInModuleFlag )
		/* PF_BroadcastPreDollar allocates dbuffer for slaves! */
		if ( (error = PF_BroadcastPreDollar(&dbuffer, &newsize, &numterms)) != 0 )
			goto onerror;
#endif
	if ( newsize < MINALLOC ) newsize = MINALLOC;
	newsize = ((newsize+7)/8)*8;
	if ( numterms == 0 ) {
		d->type = DOLZERO;
		goto docopy;
	}
	else if ( numterms == 1 ) {
		t = dbuffer;
		n = *t;
		nsize = t[n-1];
		if ( nsize < 0 ) { nsize = -nsize; }
		if ( nsize == (n-1) ) { /* numerical */
			nsize = (nsize-1)/2;
			w = t + 1 + nsize;
			if ( *w != 1 ) goto doterms;
			w++; while ( w < ( t + n - 1 ) ) { if ( *w ) break; w++; }
			if ( w < ( t + n - 1 ) ) goto doterms;
			d->type = DOLNUMBER;
			goto docopy;
		}
		else if ( n == 7 && t[6] == 3 && t[5] == 1 && t[4] == 1
			&& t[1] == INDEX && t[2] == 3 ) {
			d->type = DOLINDEX;
			d->index = t[3];
			goto docopy;
		}
		else goto doterms;
	}
	else {
doterms:;
		d->type = DOLTERMS;
		cbuf[AM.dbufnum].CanCommu[numdollar] = numcommute(dbuffer,
				&(cbuf[AM.dbufnum].NumTerms[numdollar]));
docopy:;
```

```
        if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"$-buffer old");
        d->size = newsize; d->where = dbuffer;
docopy2:;
        cbuf[AM.dbufnum].rhs[numdollar] = d->where;
    }
    if ( C->Pointer > C->rhs[C->numrhs] ) C->Pointer = C->rhs[C->numrhs];
    C->numlhs--; C->numrhs--;
onerror:
#ifdef WITHMPI
    if ( PF.me == MASTER || !AC.RhsExprInModuleFlag )
#endif
    BACKINOUT
    AN.ncmod = oldncmod;
    if ( resetmods ) UnSetMods();
    return(error);
}

/*
    #] CatchDollar :
    #[ AssignDollar :

    To be called from Generator. Assigns an expression to a $ variable.
    This one is slightly different from CatchDollar.
    We have no easy buffer this time.
    We will have to hack our way using what we normally use for functions.

    Note that in the threaded case we trust the user. That means that
    we are not going to recheck whether there is a maximum, minimum or sum.
    If the user says it is like that, we treat it like that.
    We only check that in this centralized version MODLOCAL isn't used.

    In a later stage dtype could be used for actually checking MODMAX
    and MODMIN cases.
*/

int AssignDollar(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    CBUF *C = cbuf+AM.rbufnum;
    int numterms = 0, numdollar = C->lhs[level][2];
    LONG newsize;
    DOLLARS d = Dollars + numdollar;
    WORD *w, *t, n, nsize, *rh = cbuf[C->lhs[level][7]].rhs[C->lhs[level][5]];
    WORD *ss, *ww;
    WORD olddefer, oldcompress, oldncmod = AN.ncmod;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1, dw;
    WORD numvalue;
    if ( AN.ncmod && ( ( AC.modmode & ALSODOLLARS ) == 0 ) ) AN.ncmod = 0;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
/*
        Here we come only when the module runs with more than one thread.
        This must be a variable with a special module option.
        For the multi-threaded version we only allow MODSUM, MODMAX and MODMIN.
*/
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt >= NumModOptdollars ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Illegal attempt to change $-variable in multi-threaded module %l",AC.CModule);
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        dtype = ModOptdollars[nummodopt].type;
        if ( dtype == MODLOCAL ) {
            d = ModOptdollars[nummodopt].dstruct+AT.identity;
        }
    }
#endif
    DUMMYUSE(term);
    w = rh;
/*
    First some shortcuts
*/
    if ( *w == 0 ) {
/*
        #[ Thread version : Zero case
*/
#ifdef WITHPTHREADS
        if ( dtype > 0 ) {
/*          LOCK(d->pthreadslockwrite); */
            LOCK(d->pthreadslockread);
NewValIsZero:;
            switch ( d->type ) {
                case DOLZERO: goto NoChangeZero;
                case DOLNUMBER:
```

```
                case DOLTERMS:
                    if ( ( dw = d->where[0] ) > 0 && d->where[dw] != 0 ) {
                        break; /* was not a single number. Trust the user */
                    }
                    if ( dtype == MODMAX && d->where[dw-1] >= 0 ) goto NoChangeZero;
                    if ( dtype == MODMIN && d->where[dw-1] <= 0 ) goto NoChangeZero;
                    break;
                default:
                    numvalue = DolToNumber(BHEAD numdollar);
                    if ( AN.ErrorInDollar != 0 ) break;
                    if ( dtype == MODMAX && numvalue >= 0 ) goto NoChangeZero;
                    if ( dtype == MODMIN && numvalue <= 0 ) goto NoChangeZero;
                    break;
            }
            d->type = DOLZERO;
            d->where[0] = 0;
            cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
            cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
NoChangeZero:;
            CleanDollarFactors(d);
/*          UNLOCK(d->pthreadslockwrite); */
            UNLOCK(d->pthreadslockread);
            AN.ncmod = oldncmod;
            return(0);
        }
#endif
/*
        #] Thread version :
*/
        d->type = DOLZERO;
        d->where[0] = 0;
        cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
        cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
        CleanDollarFactors(d);
        AN.ncmod = oldncmod;
        return(0);
    }
    else if ( *w == 4 && w[4] == 0 && w[2] == 1 ) {
/*
        #[ Thread version : New value is 'single precision'
*/
#ifdef WITHPTHREADS
        if ( dtype > 0 ) {
/*          LOCK(d->pthreadslockwrite); */
            LOCK(d->pthreadslockread);
            if ( d->size < MINALLOC ) {
                WORD oldsize, *oldwhere, i;
                oldsize = d->size; oldwhere = d->where;
                d->size = MINALLOC;
                d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"dollar contents");
                cbuf[AM.dbufnum].rhs[numdollar] = d->where;
                if ( oldsize > 0 ) {
                    for ( i = 0; i < oldsize; i++ ) d->where[i] = oldwhere[i];
                }
                else d->where[0] = 0;
                if ( oldwhere && oldwhere != &(AM.dollarzero) ) M_free(oldwhere,"dollar contents");
            }
            switch ( d->type ) {
                case DOLZERO:
HandleDolZero:;
                    if ( dtype == MODMAX && w[3] <= 0 ) goto NoChangeOne;
                    if ( dtype == MODMIN && w[3] >= 0 ) goto NoChangeOne;
                    break;
                case DOLNUMBER:
                case DOLTERMS:
                    if ( ( dw = d->where[0] ) > 0 && d->where[dw] != 0 ) {
                        break; /* was not a single number. Trust the user */
                    }
                    if ( dtype == MODMAX && CompCoef(d->where,w) >= 0 ) goto NoChangeOne;
                    if ( dtype == MODMIN && CompCoef(d->where,w) <= 0 ) goto NoChangeOne;
                    break;
                default:
                    {
/*
                        Note that we convert the type for the next time around.
*/
                        WORD extraterm[4];
                        numvalue = DolToNumber(BHEAD numdollar);
                        if ( AN.ErrorInDollar != 0 ) break;
                        if ( numvalue == 0 ) {
                            d->type = DOLZERO;
                            d->where[0] = 0;
                            cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
                            cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
                            goto HandleDolZero;
                        }
                        d->where[0] = extraterm[0] = 4;
```

```
                        d->where[1] = extraterm[1] = ABS(numvalue);
                        d->where[2] = extraterm[2] = 1;
                        d->where[3] = extraterm[3] = numvalue > 0 ? 3 : -3;
                        d->where[4] = 0;
                        d->type = DOLNUMBER;
                        if ( dtype == MODMAX && CompCoef(extraterm,w) >= 0 ) goto NoChangeOne;
                        if ( dtype == MODMIN && CompCoef(extraterm,w) <= 0 ) goto NoChangeOne;
                        break;
                    }
                }
                d->where[0] = w[0];
                d->where[1] = w[1];
                d->where[2] = w[2];
                d->where[3] = w[3];
                d->where[4] = 0;
                d->type = DOLNUMBER;
                cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
                cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
NoChangeOne:;
                CleanDollarFactors(d);
/*              UNLOCK(d->pthreadslockwrite); */
                UNLOCK(d->pthreadslockread);
                AN.ncmod = oldncmod;
                return(0);
            }
#endif
/*
            #] Thread version :
*/
            if ( d->size < MINALLOC ) {
                if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar contents");
                d->size = MINALLOC;
                d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"dollar contents");
                cbuf[AM.dbufnum].rhs[numdollar] = d->where;
            }
            d->where[0] = w[0];
            d->where[1] = w[1];
            d->where[2] = w[2];
            d->where[3] = w[3];
            d->where[4] = 0;
            d->type = DOLNUMBER;
            cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
            cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
            CleanDollarFactors(d);
            AN.ncmod = oldncmod;
            return(0);
        }
/*
        Now the real evaluation.
        In the case of threads and MODSUM this requires an immediate lock.
        Otherwise the lock could be placed later.
*/
#ifdef WITHPTHREADS
        if ( dtype == MODSUM ) {
/*          LOCK(d->pthreadslockwrite); */
            LOCK(d->pthreadslockread);
        }
#endif
        CleanDollarFactors(d);
/*
        The following case cannot occur. We treated it already

        if ( *w == 0 ) {
            ss = 0; numterms = 0; newsize = 0;
            olddefer = AR.DeferFlag; AR.DeferFlag = 0;
            oldcompress = AR.NoCompress; AR.NoCompress = 1;
        }
        else
*/
        {
/*
            New value is an expression that has to be evaluated first
            This is all generic. It won't foliate due to the sort level
*/
            if ( NewSort(BHEAD0) ) {
                AN.ncmod = oldncmod;
                return(1);
            }
            olddefer = AR.DeferFlag; AR.DeferFlag = 0;
            oldcompress = AR.NoCompress; AR.NoCompress = 1;
            while ( *w ) {
                n = *w; t = ww = AT.WorkPointer;
                NCOPY(t,w,n);
                AT.WorkPointer = t;
                if ( Generator(BHEAD ww,AR.Cnumlhs) ) {
                    AT.WorkPointer = ww;
                    LowerSortLevel();
```

```
                AR.DeferFlag = olddefer;
                AN.ncmod = oldncmod;
                return(1);
            }
            AT.WorkPointer = ww;
        }
        AN.tryterm = 0; /* for now */
        if ( ( newsize = EndSort(BHEAD (WORD *)((void *)(&ss)),2) ) < 0 ) {
            AN.ncmod = oldncmod;
            return(1);
        }
        numterms = 0; t = ss; while ( *t ) { numterms++; t += *t; }
    }
#ifdef WITHPTHREADS
    if ( dtype != MODSUM ) {
/*      LOCK(d->pthreadslockwrite); */
        LOCK(d->pthreadslockread);
    }
#endif
    if ( numterms == 0 ) {
/*
        the new value evaluates to zero
*/
#ifdef WITHPTHREADS
        if ( dtype == MODMAX || dtype == MODMIN ) {
            if ( ss ) { M_free(ss,"Sort of $"); ss = 0; }
            AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
            goto NewValIsZero;
        }
        else
#endif
        {
          if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar contents");
          d->where = &(AM.dollarzero);
          d->size = 0;
          cbuf[AM.dbufnum].rhs[numdollar] = 0;
          cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
          cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
          d->type = DOLZERO;
        }
        if ( ss ) { M_free(ss,"Sort of $"); ss = 0; }
    }
    else {
/*
        #[ Thread version :
*/
#ifdef WITHPTHREADS
        if ( dtype == MODMAX || dtype == MODMIN ) {
            if ( numterms == 1 && ( *ss-1 == ABS(ss[*ss-1]) ) ) { /* is number */
              switch ( d->type ) {
                case DOLZERO:
HandleDolZero1:;
                    if ( dtype == MODMAX && ss[*ss-1] > 0 ) break;
                    if ( dtype == MODMIN && ss[*ss-1] < 0 ) break;
                    if ( ss ) { M_free(ss,"Sort of $"); ss = 0; }
                    AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
                    goto NoChange;
                case DOLTERMS:
                case DOLNUMBER:
                    if ( ( dw = d->where[0] ) > 0 && d->where[dw] != 0 ) break;
                    if ( dtype == MODMAX && CompCoef(ss,d->where) > 0 ) break;
                    if ( dtype == MODMIN && CompCoef(ss,d->where) < 0 ) break;
                    if ( ss ) { M_free(ss,"Sort of $"); ss = 0; }
                    AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
                    goto NoChange;
                default: {
                    WORD extraterm[4];
                    numvalue = DolToNumber(BHEAD numdollar);
                    if ( AN.ErrorInDollar != 0 ) break;
                    if ( numvalue == 0 ) {
                        d->type = DOLZERO;
                        d->where[0] = 0;
                        cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
                        cbuf[AM.dbufnum].NumTerms[numdollar] = 0;
                        goto HandleDolZero1;
                    }
                    d->where[0] = extraterm[0] = 4;
                    d->where[1] = extraterm[1] = ABS(numvalue);
                    d->where[2] = extraterm[2] = 1;
                    d->where[3] = extraterm[3] = numvalue > 0 ? 3 : -3;
                    d->where[4] = 0;
                    d->type = DOLNUMBER;
                    if ( dtype == MODMAX && CompCoef(ss,extraterm) > 0 ) break;
                    if ( dtype == MODMIN && CompCoef(ss,extraterm) < 0 ) break;
                    if ( ss ) { M_free(ss,"Sort of $"); ss = 0; }
                    AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
                    goto NoChange;
```

```
                    }
                }
            }
            else {
                if ( ss ) { M_free(ss,"Sort of $"); ss = 0; }
                AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
                goto NoChange;
            }
        }
#endif
/*
        #] Thread version :
*/
        d->type = DOLTERMS;
        if ( d->where && d->where != &(AM.dollarzero) ) { M_free(d->where,"dollar contents"); d->where = 0; }
        d->size = newsize + 1;
        d->where = ss;
        cbuf[AM.dbufnum].rhs[numdollar] = w = d->where;
    }
    AR.DeferFlag = olddefer; AR.NoCompress = oldcompress;
/*
    Now find the special cases
*/
    if ( numterms == 0 ) {
        d->type = DOLZERO;
    }
    else if ( numterms == 1 ) {
        t = d->where;
        n = *t;
        nsize = t[n-1];
        if ( nsize < 0 ) { nsize = -nsize; }
        if ( nsize == (n-1) ) {
            nsize = (nsize-1)/2;
            w = t + 1 + nsize;
            if ( *w == 1 ) {
                w++; while ( w < ( t + n - 1 ) ) { if ( *w ) break; w++; }
                if ( w >= ( t + n - 1 ) ) d->type = DOLNUMBER;
            }
        }
        else if ( n == 7 && t[6] == 3 && t[5] == 1 && t[4] == 1
            && t[1] == INDEX && t[2] == 3 ) {
            d->type = DOLINDEX;
            d->index = t[3];
        }
    }
    if ( d->type == DOLTERMS ) {
        cbuf[AM.dbufnum].CanCommu[numdollar] = numcommute(d->where,
            &(cbuf[AM.dbufnum].NumTerms[numdollar]));
    }
    else {
        cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
        cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
    }
#ifdef WITHPTHREADS
NoChange:;
/*  UNLOCK(d->pthreadslockwrite); */
    UNLOCK(d->pthreadslockread);
#endif
    AN.ncmod = oldncmod;
    return(0);
}

/*
    #] AssignDollar :
    #[ WriteDollarToBuffer :

    Takes the numbered dollar expression and writes it to output.
    We catch however the output in a buffer and return its address.
    This routine is needed when we need a text representation of
    a dollar expression like for the construction `$name' in the preprocessor.
    If par==0 we leave the current printing mode.
    If par==1 we insist on normal mode
*/

UBYTE *WriteDollarToBuffer(WORD numdollar, WORD par)
{
    DOLLARS d = Dollars+numdollar;
    UBYTE *s, *oldcurbufwrt = AO.CurBufWrt;
    WORD *t, lbrac = 0, first = 0, arg[2], oldOutputMode = AC.OutputMode;
    WORD oldinfbrack = AO.InFbrack;
    int error = 0;
    int dict = AO.CurrentDictionary;

    AO.DollarOutSizeBuffer = 32;
    AO.DollarOutBuffer = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,"DollarOutBuffer");
    AO.DollarInOutBuffer = 1;
```

```
    AO.PrintType = 1;
    AO.InFbrack = 0;
    s = AO.DollarOutBuffer;
    *s = 0;
    if ( par > 0 && AO.CurDictInDollars == 0 ) {
        AC.OutputMode = NORMALFORMAT;
        AO.CurrentDictionary = 0;
    }
    else {
        AO.CurBufWrt = (UBYTE *)underscore;
    }
    AO.OutInBuffer = 1;
    switch ( d->type ) {
        case DOLARGUMENT:
            WriteArgument(d->where);
            break;
        case DOLSUBTERM:
            WriteSubTerm(d->where,1);
            break;
        case DOLNUMBER:
        case DOLTERMS:
            t = d->where;
            while ( *t ) {
                if ( WriteTerm(t,&lbrac,first,PRINTON,0) ) {
                    error = 1; break;
                }
                t += *t;
            }
            break;
        case DOLWILDARGS:
            t = d->where+1;
            while ( *t ) {
                WriteArgument(t);
                NEXTARG(t)
                if ( *t ) TokenToLine((UBYTE *)(","));
            }
            break;
        case DOLINDEX:
            arg[0] = -INDEX; arg[1] = d->index;
            WriteArgument(arg);
            break;
        case DOLZERO:
            *s++ = '0'; *s = 0;
            AO.DollarInOutBuffer = 1;
            break;
        case DOLUNDEFINED:
            *s = 0;
            AO.DollarInOutBuffer = 1;
            break;
    }
    AC.OutputMode = oldOutputMode;
    AO.OutInBuffer = 0;
    AO.InFbrack = oldinfbrack;
    AO.CurBufWrt = oldcurbufwrt;
    AO.CurrentDictionary = dict;
    if ( error ) {
        MLOCK(ErrorMessageLock);
        MesPrint("&Illegal dollar object for writing");
        MUNLOCK(ErrorMessageLock);
        M_free(AO.DollarOutBuffer,"DollarOutBuffer");
        AO.DollarOutBuffer = 0;
        AO.DollarOutSizeBuffer = 0;
        return(0);
    }
    return(AO.DollarOutBuffer);
}

/*
    #] WriteDollarToBuffer :
    #[ WriteDollarFactorToBuffer :

    Takes the numbered dollar expression and writes it to output.
    We catch however the output in a buffer and return its address.
    This routine is needed when we need a text representation of
    a dollar expression like for the construction `$name' in the preprocessor.
    If par==0 we leave the current printing mode.
    If par==1 we insist on normal mode
*/

UBYTE *WriteDollarFactorToBuffer(WORD numdollar, WORD numfac, WORD par)
{
    DOLLARS d = Dollars+numdollar;
    UBYTE *s, *oldcurbufwrt = AO.CurBufWrt;
    WORD *t, lbrac = 0, first = 0, n[5], oldOutputMode = AC.OutputMode;
    WORD oldinfbrack = AO.InFbrack;
    int error = 0;
    int dict = AO.CurrentDictionary;
```

```
    if ( numfac > d->nfactors || numfac < 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("&Illegal factor number for this dollar variable: %d",numfac);
        MesPrint("&There are %d factors",d->nfactors);
        MUNLOCK(ErrorMessageLock);
        return(0);
    }

    AO.DollarOutSizeBuffer = 32;
    AO.DollarOutBuffer = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,"DollarOutBuffer");
    AO.DollarInOutBuffer = 1;
    AO.PrintType = 1;
    AO.InFbrack = 0;
    s = AO.DollarOutBuffer;
    *s = 0;
    if ( par > 0 ) {
        AC.OutputMode = NORMALFORMAT;
        AO.CurrentDictionary = 0;
    }
    else {
        AO.CurBufWrt = (UBYTE *)underscore;
    }
    AO.OutInBuffer = 1;
    if ( numfac == 0 ) {     /* write the number d->nfactors */
        n[0] = 4; n[1] = d->nfactors; n[2] = 1; n[3] = 3; n[4] = 0; t = n;
    }
    else if ( numfac ==  1 && d->factors == 0 ) {   /* Here d->factors is zero and d->where is fine */
        t = d->where;
    }
    else if ( d->factors[numfac-1].where == 0 ) {   /* write the value */
        if ( d->factors[numfac-1].value < 0 ) {
            n[0] = 4; n[1] = -d->factors[numfac-1].value; n[2] = 1; n[3] = -3; n[4] = 0; t = n;
        }
        else {
            n[0] = 4; n[1] = d->factors[numfac-1].value; n[2] = 1; n[3] = 3; n[4] = 0; t = n;
        }
    }
    else { t = d->factors[numfac-1].where; }
    while ( *t ) {
        if ( WriteTerm(t,&lbrac,first,PRINTON,0) ) {
            error = 1; break;
        }
        t += *t;
    }
    AC.OutputMode = oldOutputMode;
    AO.OutInBuffer = 0;
    AO.InFbrack = oldinfbrack;
    AO.CurBufWrt = oldcurbufwrt;
    AO.CurrentDictionary = dict;
    if ( error ) {
        MLOCK(ErrorMessageLock);
        MesPrint("&Illegal dollar object for writing");
        MUNLOCK(ErrorMessageLock);
        M_free(AO.DollarOutBuffer,"DollarOutBuffer");
        AO.DollarOutBuffer = 0;
        AO.DollarOutSizeBuffer = 0;
        return(0);
    }
    return(AO.DollarOutBuffer);
}

/*
    #] WriteDollarFactorToBuffer :
    #[ AddToDollarBuffer :
*/

void AddToDollarBuffer(UBYTE *s)
{
    int i;
    UBYTE *t = s, *u, *newdob;
    LONG j;
    while ( *t ) { t++; }
    i = t - s;
    while ( i + AO.DollarInOutBuffer >= AO.DollarOutSizeBuffer ) {
        j = AO.DollarInOutBuffer;
        AO.DollarOutSizeBuffer *= 2;
        t = AO.DollarOutBuffer;
        newdob = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,"DollarOutBuffer");
        u = newdob;
        while ( --j >= 0 ) *u++ = *t++;
        M_free(AO.DollarOutBuffer,"DollarOutBuffer");
        AO.DollarOutBuffer = newdob;
    }
    t = AO.DollarOutBuffer + AO.DollarInOutBuffer-1;
    while ( t == AO.DollarOutBuffer && ( *s == '+' || *s == ' ' ) ) s++;
    i = 0;
```

```
    if ( AO.CurrentDictionary == 0 ) {
        while ( *s ) {
            if ( *s == ' ' ) { s++; continue; }
            *t++ = *s++; i++;
        }
    }
    else {
        while ( *s ) { *t++ = *s++; i++; }
    }
    *t = 0;
    AO.DollarInOutBuffer += i;
}

/*
    #] AddToDollarBuffer :
    #[ TermAssign :

    This routine is called from a piece of code in Normalize that has been
    commented out.
*/

void TermAssign(WORD *term)
{
    DOLLARS d;
    WORD *t, *tstop, *astop, *w, *m;
    WORD i, newsize;
    for (;;) {
        astop = term + *term;
        tstop = astop - ABS(astop[-1]);
        t = term + 1;
        while ( t < tstop ) {
            if ( *t == AM.termfunnum && t[1] == FUNHEAD+2
            && t[FUNHEAD] == -DOLLAREXPRESSION ) {
                d = Dollars + t[FUNHEAD+1];
                newsize = *term - FUNHEAD - 1;
                if ( newsize < MINALLOC ) newsize = MINALLOC;
                newsize = ((newsize+7)/8)*8;
                if ( d->size > 2*newsize && d->size > 1000 ) {
                    if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar
contents");
                    d->size = 0;
                    d->where = &(AM.dollarzero);
                }
                if ( d->size < newsize ) {
                    if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar
contents");
                    d->size = newsize;
                    d->where = (WORD *)Malloc1(newsize*sizeof(WORD),"dollar contents");
                }
                cbuf[AM.dbufnum].rhs[t[FUNHEAD+1]] = w = d->where;
                m = term;
                while ( m < t ) *w++ = *m++;
                m += t[1];
                while ( m < tstop ) {
                    if ( *m == AM.termfunnum && m[1] == FUNHEAD+2
                    && m[FUNHEAD] == -DOLLAREXPRESSION ) { m += m[1]; }
                    else {
                        i = m[1];
                        while ( --i >= 0 ) *w++ = *m++;
                    }
                }
                while ( m < astop ) *w++ = *m++;
                *(d->where) = w - d->where;
                *w = 0;
                d->type = DOLTERMS;
                w = t; m = t + t[1];
                while ( m < astop ) *w++ = *m++;
                *term = w - term;
                break;
            }
            t += t[1];
        }
        if ( t >= tstop ) return;
    }
}

/*
    #] TermAssign :
    #[ PutTermInDollar :

    We assume here that the dollar is local.
*/

int PutTermInDollar(WORD *term, WORD numdollar)
{
    DOLLARS d = Dollars+numdollar;
    WORD i;
```

```
    if ( term == 0 || *term == 0 ) {
        d->type = DOLZERO;
        return(0);
    }
    if ( d->size < *term || d->size > 2*term[0] || d->where == 0 ) {
        if ( d->size > 0 && d->where ) {
            M_free(d->where,"dollar contents");
        }
        d->where = Malloc1((term[0]+1)*sizeof(WORD),"dollar contents");
        d->size = term[0]+1;
    }
    d->type = DOLTERMS;
    for ( i = 0; i < term[0]; i++ ) d->where[i] = term[i];
    d->where[i] = 0;
    return(0);
}

/*
    #] PutTermInDollar :
    #[ WildDollars :

    Note that we cannot upload wildcards into dollar variables when WITHPTHREADS.
*/

void WildDollars(PHEAD WORD *term)
{
    GETBIDENTITY
    DOLLARS d;
    WORD *m, *t, *w, *ww, *orig = 0, *wildvalue, *wildstop;
    int numdollar;
    LONG weneed, i;
#ifdef WITHPTHREADS
    int dtype = -1;
#endif
    if ( term == 0 ) {
        m = wildvalue = AN.WildValue;
        wildstop = AN.WildStop;
    }
    else {
        ww = term + *term; ww -= ABS(ww[-1]); w = term+1;
        while ( w < ww && *w != SUBEXPRESSION ) w += w[1];
        if ( w >= ww ) return;
        wildstop = w + w[1];
        w += SUBEXPSIZE;
        wildvalue = m = w;
    }
    while ( m < wildstop ) {
        if ( *m != LOADDOLLAR ) { m += m[1]; continue; }
        t = m - 4;
        while ( *t == LOADDOLLAR || *t == FROMSET || *t == SETTONUM ) t -= 4;
        if ( t < wildvalue ) {
            MLOCK(ErrorMessageLock);
            MesPrint("&Serious bug in wildcard prototype. Found in WildDollars");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        numdollar = m[2];
        d = Dollars + numdollar;
#ifdef WITHPTHREADS
        {
            int nummodopt;
            dtype = -1;
            if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
                for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                    if ( numdollar == ModOptdollars[nummodopt].number ) break;
                }
                if ( nummodopt < NumModOptdollars ) {
                    dtype = ModOptdollars[nummodopt].type;
                    if ( dtype == MODLOCAL ) {
                        d = ModOptdollars[nummodopt].dstruct+AT.identity;
                    }
                    else {
                        MLOCK(ErrorMessageLock);
                        MesPrint("&Illegal attempt to use $-variable %s in module %l",
                            DOLLARNAME(Dollars,numdollar),AC.CModule);
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                    }
                }
            }
        }
#endif
/*
        The value of this wildcard goes into our $-variable
        First compute the space we need.
*/
        switch ( *t ) {
```

```
            case SYMTONUM:
                weneed = 5;
                break;
            case SYMTOSYM:
                weneed = 9;
                break;
            case SYMTOSUB:
            case VECTOSUB:
            case INDTOSUB:
                orig = cbuf[AT.ebufnum].rhs[t[3]];
                w = orig; while ( *w ) w += *w;
                weneed = w - orig + 1;
                break;
            case VECTOMIN:
            case VECTOVEC:
            case INDTOIND:
                weneed = 8;
                break;
            case FUNTOFUN:
                weneed = FUNHEAD+5;
                break;
            case ARGTOARG:
                orig = cbuf[AT.ebufnum].rhs[t[3]];
                if ( *orig > 0 ) weneed = *orig+2;
                else {
                    w = orig+1; while ( *w ) { NEXTARG(w) }
                    weneed = w - orig + 1;
                }
                break;
            default:
                weneed = MINALLOC;
                break;
        }
        if ( weneed < MINALLOC ) weneed = MINALLOC;
        weneed = ((weneed+7)/8)*8;
        if ( d->size > 2*weneed && d->size > 1000 ) {
            if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollarspace");
            d->where = &(AM.dollarzero);
            d->size = 0;
        }
        if ( d->size < weneed ) {
            if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollarspace");
            d->where = (WORD *)Malloc1(weneed*sizeof(WORD),"dollarspace");
            d->size = weneed;
        }
/*
        It is not clear what the following code does for TFORM

        if ( dtype != MODLOCAL ) {
*/
            cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
            cbuf[AM.dbufnum].NumTerms[numdollar] = 1;
/*          cbuf[AM.dbufnum].rhs[numdollar] = d->where; */
            cbuf[AM.dbufnum].rhs[numdollar] = (WORD *)(1);
/*

        }
        Now load up the value of the wildcard in compiler buffer format
*/
        w = d->where;
        d->type = DOLTERMS;
        switch ( *t ) {
            case SYMTONUM:
                d->where[0] = 4; d->where[2] = 1;
                if ( t[3] >= 0 ) { d->where[1] = t[3]; d->where[3] = 3; }
                else { d->where[1] = -t[3]; d->where[3] = -3; }
                if ( t[3] == 0 ) { d->type = DOLZERO; d->where[0] = 0; }
                else { d->type = DOLNUMBER; d->where[4] = 0; }
                break;
            case SYMTOSYM:
                *w++ = 8;
                *w++ = SYMBOL;
                *w++ = 4;
                *w++ = t[3];
                *w++ = 1;
                *w++ = 1;
                *w++ = 1;
                *w++ = 3;
                *w = 0;
                break;
            case SYMTOSUB:
            case VECTOSUB:
            case INDTOSUB:
                while ( *orig ) {
                    i = *orig; while ( --i >= 0 ) *w++ = *orig++;
                }
                *w = 0;
/*
```

```
                And then we have to fix up CanCommu
*/
                break;
            case VECTOMIN:
                *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[3];
                *w++ = 1; *w++ = 1; *w++ = -3; *w = 0;
                break;
            case VECTOVEC:
                *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[3];
                *w++ = 1; *w++ = 1; *w++ = 3; *w = 0;
                break;
            case INDTOIND:
                d->type = DOLINDEX; d->index = t[3]; *w = 0;
                break;
            case FUNTOFUN:
                *w++ = FUNHEAD+4; *w++ = t[3]; *w++ = FUNHEAD;
                FILLFUN(w)
                *w++ = 1; *w++ = 1; *w++ = 3; *w = 0;
                break;
            case ARGTOARG:
                if ( *orig > 0 ) ww = orig + *orig + 1;
                else {
                    ww = orig+1; while ( *ww ) { NEXTARG(ww) }
                }
                while ( orig < ww ) *w++ = *orig++;
                *w = 0;
                d->type = DOLWILDARGS;
                break;
            default:
                d->type = DOLUNDEFINED;
                break;
        }
        m += m[1];
    }
}

/*
    #] WildDollars :
    #[ DolToTensor :    with LOCK
*/

WORD DolToTensor(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
    WORD retval;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    AN.ErrorInDollar = 0;
    if ( d->type == DOLTERMS && d->where[0] == FUNHEAD+4 &&
    d->where[FUNHEAD+4] == 0 && d->where[FUNHEAD+3] == 3 &&
    d->where[FUNHEAD+2] == 1 && d->where[FUNHEAD+1] == 1 &&
    d->where[1] >= FUNCTION && d->where[1] < FUNCTION+WILDOFFSET
    && functions[d->where[1]-FUNCTION].spec >= TENSORFUNCTION ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLARGUMENT &&
    d->where[0] <= -FUNCTION && d->where[0] > -FUNCTION-WILDOFFSET
    && functions[-d->where[0]-FUNCTION].spec >= TENSORFUNCTION ) {
        retval = -d->where[0];
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] <= -FUNCTION && d->where[1] > -FUNCTION-WILDOFFSET
    && d->where[2] == 0
    && functions[-d->where[1]-FUNCTION].spec >= TENSORFUNCTION ) {
        retval = -d->where[1];
    }
    else if ( d->type == DOLSUBTERM &&
    d->where[0] >= FUNCTION && d->where[0] < FUNCTION+WILDOFFSET
    && functions[d->where[0]-FUNCTION].spec >= TENSORFUNCTION ) {
        retval = d->where[0];
    }
```

```
    else {
        AN.ErrorInDollar = 1;
        retval = 0;
    }
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(retval);
}

/*
    #] DolToTensor :
    #[ DolToFunction :  with LOCK
*/

WORD DolToFunction(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
    WORD retval;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    AN.ErrorInDollar = 0;
    if ( d->type == DOLTERMS && d->where[0] == FUNHEAD+4 &&
    d->where[FUNHEAD+4] == 0 && d->where[FUNHEAD+3] == 3 &&
    d->where[FUNHEAD+2] == 1 && d->where[FUNHEAD+1] == 1 &&
    d->where[1] >= FUNCTION && d->where[1] < FUNCTION+WILDOFFSET ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLARGUMENT &&
    d->where[0] <= -FUNCTION && d->where[0] > -FUNCTION-WILDOFFSET ) {
        retval = -d->where[0];
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] <= -FUNCTION && d->where[1] > -FUNCTION-WILDOFFSET
    && d->where[2] == 0 ) {
        retval = -d->where[1];
    }
    else if ( d->type == DOLSUBTERM &&
    d->where[0] >= FUNCTION && d->where[0] < FUNCTION+WILDOFFSET ) {
        retval = d->where[0];
    }
    else {
        AN.ErrorInDollar = 1;
        retval = 0;
    }
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(retval);
}

/*
    #] DolToFunction :
    #[ DolToVector :    with LOCK
*/

WORD DolToVector(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
    WORD retval;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
```

```
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    AN.ErrorInDollar = 0;
    if ( d->type == DOLINDEX && d->index < 0 ) {
        retval = d->index;
    }
    else if ( d->type == DOLARGUMENT && ( d->where[0] == -VECTOR
    || d->where[0] == -MINVECTOR ) ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLSUBTERM && d->where[0] == INDEX
    && d->where[1] == 3 && d->where[2] < 0 ) {
        retval = d->where[2];
    }
    else if ( d->type == DOLTERMS && d->where[0] == 7 &&
    d->where[7] == 0 && d->where[6] == 3 &&
    d->where[5] == 1 && d->where[4] == 1 &&
    d->where[1] >= INDEX && d->where[3] < 0 ) {
        retval = d->where[3];
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && ( d->where[1] == -VECTOR || d->where[1] == -MINVECTOR )
    && d->where[3] == 0 ) {
        retval = d->where[2];
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 1
    && d->where[1] < 0 ) {
        retval = d->where[1];
    }
    else {
        AN.ErrorInDollar = 1;
        retval = 0;
    }
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(retval);
}

/*
    #] DolToVector :
    #[ DolToNumber :
*/

WORD DolToNumber(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
        }
    }
#endif
    AN.ErrorInDollar = 0;
    if ( ( d->type == DOLTERMS || d->type == DOLNUMBER )
     && d->where[0] == 4 &&
    d->where[4] == 0 && ( d->where[3] == 3 || d->where[3] == -3 )
     && d->where[2] == 1 && ( d->where[1] & TOPBITONLY ) == 0 ) {
        if ( d->where[3] > 0 ) return(d->where[1]);
        else return(-d->where[1]);
    }
    else if ( d->type == DOLARGUMENT && d->where[0] == -SNUMBER ) {
        return(d->where[1]);
    }
    else if ( d->type == DOLARGUMENT && d->where[0] == -INDEX
    && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
        return(d->where[1]);
    }
    else if ( d->type == DOLZERO ) return(0);
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] == -SNUMBER && d->where[3] == 0 ) {
        return(d->where[2]);
    }
```

```
    else if ( d->type == DOLINDEX && d->index >= 0 && d->index < AM.OffsetIndex ) {
        return(d->index);
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 1
    && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
        return(d->where[1]);
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] == -INDEX && d->where[3] == 0 && d->where[2] >= 0
    && d->where[2] < AM.OffsetIndex ) {
        return(d->where[2]);
    }
    AN.ErrorInDollar = 1;
    return(0);
}

/*
    #] DolToNumber :
    #[ DolToSymbol :    with LOCK
*/

WORD DolToSymbol(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
    WORD retval;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    AN.ErrorInDollar = 0;
    if ( d->type == DOLTERMS && d->where[0] == 8 &&
    d->where[8] == 0 && d->where[7] == 3 && d->where[6] == 1
     && d->where[5] == 1 && d->where[4] == 1 && d->where[1] == SYMBOL ) {
        retval = d->where[3];
    }
    else if ( d->type == DOLARGUMENT && d->where[0] == -SYMBOL ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLSUBTERM && d->where[0] == SYMBOL
    && d->where[1] == 4 && d->where[3] == 1 ) {
        retval = d->where[2];
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] == -SYMBOL && d->where[3] == 0 ) {
        retval = d->where[2];
    }
    else {
        AN.ErrorInDollar = 1;
        retval = -1;
    }
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(retval);
}

/*
    #] DolToSymbol :
    #[ DolToIndex :     with LOCK
*/

WORD DolToIndex(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
    WORD retval;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
```

```
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    AN.ErrorInDollar = 0;
    if ( d->type == DOLTERMS && d->where[0] == 7 &&
    d->where[7] == 0 && d->where[6] == 3 && d->where[5] == 1
     && d->where[4] == 1 && d->where[1] == INDEX && d->where[3] >= 0 ) {
        retval = d->where[3];
    }
    else if ( d->type == DOLARGUMENT && d->where[0] == -SNUMBER
    && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLARGUMENT && d->where[0] == -INDEX
    && d->where[1] >= 0 ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLZERO ) return(0);
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] == -SNUMBER && d->where[3] == 0 && d->where[2] >= 0
    && d->where[2] < AM.OffsetIndex ) {
        retval = d->where[2];
    }
    else if ( d->type == DOLINDEX && d->index >= 0 ) {
        retval = d->index;
    }
    else if ( d->type == DOLNUMBER && d->where[0] == 4 && d->where[2] == 1
    && d->where[3] == 3 && d->where[4] == 0 && d->where[1] < AM.OffsetIndex ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 1
    && d->where[1] >= 0 ) {
        retval = d->where[1];
    }
    else if ( d->type == DOLSUBTERM && d->where[0] == INDEX
    && d->where[1] == 3 && d->where[2] >= 0 ) {
        retval = d->where[2];
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] == -INDEX && d->where[3] == 0 && d->where[2] >= 0 ) {
        retval = d->where[2];
    }
    else {
        AN.ErrorInDollar = 1;
        retval = 0;
    }
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(retval);
}

/*
    #] DolToIndex :
    #[ DolToTerms :

    Returns a struct of type DOLLARS which contains a copy of the
    original dollar variable, provided it can be expressed in terms of
    an expression (type = DOLTERMS). Otherwise it returns zero.
    The dollar is expressed in terms in the buffer "where"
*/

DOLLARS DolToTerms(PHEAD WORD numdollar)
{
    GETBIDENTITY
    LONG size;
    DOLLARS d = Dollars + numdollar, newd;
    WORD *t, *w, i;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
        }
```

```
    }
#endif
    AN.ErrorInDollar = 0;
    switch ( d->type ) {
        case DOLARGUMENT:
            t = d->where;
            if ( t[0] < 0 ) {
ShortArgument:
                w = AT.WorkPointer;
                if ( t[0] <= -FUNCTION ) {
                    *w++ = FUNHEAD+4; *w++ = -t[0];
                    *w++ = FUNHEAD; FILLFUN(w)
                    *w++ = 1; *w++ = 1; *w++ = 3;
                }
                else if ( t[0] == -SYMBOL ) {
                    *w++ = 8; *w++ = SYMBOL; *w++ = 4; *w++ = t[1];
                    *w++ = 1; *w++ = 1; *w++ = 1; *w++ = 3;
                }
                else if ( t[0] == -VECTOR || t[0] == -INDEX ) {
                    *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[1];
                    *w++ = 1; *w++ = 1; *w++ = 3;
                }
                else if ( t[0] == -MINVECTOR ) {
                    *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = t[1];
                    *w++ = 1; *w++ = 1; *w++ = -3;
                }
                else if ( t[0] == -SNUMBER ) {
                    *w++ = 4;
                    if ( t[1] < 0 ) {
                        *w++ = -t[1]; *w++ = 1; *w++ = -3;
                    }
                    else {
                        *w++ = t[1]; *w++ = 1; *w++ = 3;
                    }
                }
                *w = 0; size = w - AT.WorkPointer;
                w = AT.WorkPointer;
                break;
            }
            /* fall through */
        case DOLNUMBER:
        case DOLTERMS:
            t = d->where;
            while ( *t ) t += *t;
            size = t - d->where;
            w = d->where;
            break;
        case DOLSUBTERM:
            w = AT.WorkPointer;
            size = d->where[1];
            *w++ = size+4; t = d->where; NCOPY(w,t,size)
            *w++ = 1; *w++ = 1; *w++ = 3;
            w = AT.WorkPointer; size = d->where[1]+4;
            break;
        case DOLINDEX:
            w = AT.WorkPointer;
            *w++ = 7; *w++ = INDEX; *w++ = 3; *w++ = d->index;
            *w++ = 1; *w++ = 1; *w++ = 3; *w = 0;
            w = AT.WorkPointer; size = 7;
            break;
        case DOLWILDARGS:
/*
            In some cases we can make a copy
*/
            t = d->where+1;
            if ( *t == 0 ) return(0);
            NEXTARG(t);
            if ( *t ) { /* More than one argument in here */
                MLOCK(ErrorMessageLock);
                MesPrint("Trying to convert a $ with an argument field into an expression");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
/*
            Now we have a single argument
*/
            t = d->where+1;
            if ( *t < 0 ) goto ShortArgument;
            size = *t - ARGHEAD;
            w = t + ARGHEAD;
            break;
        case DOLUNDEFINED:
            MLOCK(ErrorMessageLock);
            MesPrint("Trying to use an undefined $ in an expression");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
            /* fall through */
```

```
        case DOLZERO:
            if ( d->where ) { d->where[0] = 0; }
            else d->where = &(AM.dollarzero);
            size = 0;
            w = d->where;
            break;
        default:
            return(0);
    }
    newd = (DOLLARS)Malloc1(sizeof(struct DoLlArS)+(size+1)*sizeof(WORD),
                "Copy of dollar variable");
    t = (WORD *)(newd+1);
    newd->where = t;
    newd->name = d->name;
    newd->node = d->node;
    newd->type = DOLTERMS;
    newd->size = size;
    newd->numdummies = d->numdummies;
#ifdef WITHPTHREADS
    newd->pthreadslockread  = dummylock;
    newd->pthreadslockwrite = dummylock;
#endif
    size++;
    NCOPY(t,w,size);
    newd->nfactors = d->nfactors;
    if ( d->nfactors > 1 ) {
        newd->factors = (FACDOLLAR *)Malloc1(d->nfactors*sizeof(FACDOLLAR),"Dollar factors");
        for ( i = 0; i < d->nfactors; i++ ) {
            newd->factors[i].where = 0;
            newd->factors[i].size = 0;
            newd->factors[i].type = DOLUNDEFINED;
            newd->factors[i].value = d->factors[i].value;
        }
    }
    else { newd->factors = 0; }
    return(newd);
}

/*
    #] DolToTerms :
    #[ DolToLong :
*/

LONG DolToLong(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
    LONG x;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
        }
    }
#endif
    AN.ErrorInDollar = 0;
    if ( ( d->type == DOLTERMS || d->type == DOLNUMBER )
     && d->where[0] == 4 &&
    d->where[4] == 0 && ( d->where[3] == 3 || d->where[3] == -3 )
     && d->where[2] == 1 && ( d->where[1] & TOPBITONLY ) == 0 ) {
        x = d->where[1];
        if ( d->where[3] > 0 ) return(x);
        else return(-x);
    }
    else if ( ( d->type == DOLTERMS || d->type == DOLNUMBER )
     && d->where[0] == 6 &&
    d->where[6] == 0 && ( d->where[5] == 5 || d->where[5] == -5 )
     && d->where[3] == 1 && d->where[4] == 1 && ( d->where[2] & TOPBITONLY ) == 0 ) {
        x = d->where[1] + ( (LONG)(d->where[2]) « BITSINWORD );
        if ( d->where[5] > 0 ) return(x);
        else return(-x);
    }
    else if ( d->type == DOLARGUMENT && d->where[0] == -SNUMBER ) {
        x = d->where[1];
        return(x);
    }
    else if ( d->type == DOLARGUMENT && d->where[0] == -INDEX
    && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
        x = d->where[1];
        return(x);
```

```
    }
    else if ( d->type == DOLZERO ) return(0);
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] == -SNUMBER && d->where[3] == 0 ) {
        x = d->where[2];
        return(x);
    }
    else if ( d->type == DOLINDEX && d->index >= 0 && d->index < AM.OffsetIndex ) {
        x = d->index;
        return(x);
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 1
    && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
        x = d->where[1];
        return(x);
    }
    else if ( d->type == DOLWILDARGS && d->where[0] == 0
    && d->where[1] == -INDEX && d->where[3] == 0 && d->where[2] >= 0
    && d->where[2] < AM.OffsetIndex ) {
        x = d->where[2];
        return(x);
    }
    AN.ErrorInDollar = 1;
    return(0);
}

/*
    #] DolToLong :
    #[ ExecInside :
*/

int ExecInside(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c;
    WORD *w, number;
    int error = 0;
    w = AT.WorkPointer;
    if ( AC.insidelevel >= MAXNEST ) {
        MLOCK(ErrorMessageLock);
        MesPrint("@Nesting of inside statements more than %d levels",(WORD)MAXNEST);
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AC.insidesumcheck[AC.insidelevel] = NestingChecksum();
    AC.insidestack[AC.insidelevel] = cbuf[AC.cbufnum].Pointer
                                 - cbuf[AC.cbufnum].Buffer + 2;
    AC.insidelevel++;
    *w++ = TYPEINSIDE;
    w++; w++;
    for(;;) {   /* Look for a (comma separated) list of dollar variables */
        while ( *s == ',' ) s++;
        if ( *s == 0 ) break;
        if ( *s == '$' ) {
            s++; t = s;
            if ( FG.cTable[*s] != 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal name for $ variable: %s",s-1);
                MUNLOCK(ErrorMessageLock);
                goto skipdol;
            }
            while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
            c = *s; *s = 0;
            if ( ( number = GetDollar(t) ) < 0 ) {
                number = AddDollar(t,0,0,0);
            }
            *s = c;
            *w++ = number;
            AddPotModdollar(number);
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("&Illegal object in Inside statement");
            MUNLOCK(ErrorMessageLock);
skipdol:    error = 1;
            while ( *s && *s != ',' && s[1] != '$' ) s++;
            if ( *s == 0 ) break;
        }
    }
    AT.WorkPointer[1] = w - AT.WorkPointer;
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    return(error);
}

/*
    #] ExecInside :
    #[ InsideDollar :
```

```

    Execution part of Inside $a;
    We have to take the variables one by one and then
    convert them into proper terms and call Generator for the proper levels.
    The conversion copies the whole dollar into a new buffer, making us
    insensitive to redefinitions of $a inside the Inside.
    In the end we sort and redefine $a.
*/

int InsideDollar(PHEAD WORD *ll, WORD level)
{
    GETBIDENTITY
    int numvar = (int)(ll[1]-3), j, error = 0;
    WORD numdol, *oldcterm, *oldwork = AT.WorkPointer, olddefer, *r, *m;
    WORD oldnumlhs, *dbuffer;
    DOLLARS d, newd;
    oldcterm = AN.cTerm; AN.cTerm = 0;
    oldnumlhs = AR.Cnumlhs; AR.Cnumlhs = ll[2];
    ll += 3;
    olddefer = AR.DeferFlag;
    AR.DeferFlag = 0;
    while ( --numvar >= 0 ) {
        numdol = *ll++;
        d = Dollars + numdol;
        {
#ifdef WITHPTHREADS
        int nummodopt, dtype = -1;
        if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
            for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                if ( numdol == ModOptdollars[nummodopt].number ) break;
            }
            if ( nummodopt < NumModOptdollars ) {
                dtype = ModOptdollars[nummodopt].type;
                if ( dtype == MODLOCAL ) {
                    d = ModOptdollars[nummodopt].dstruct+AT.identity;
                }
                else {
/*                  LOCK(d->pthreadslockwrite); */
                    LOCK(d->pthreadslockread);
                }
            }
        }
#endif
        newd = DolToTerms(BHEAD numdol);
        if ( newd == 0 ) {
            continue;
        }
        if ( newd->where[0] == 0 ) {
            // DolToTerms potentially allocates memory. Free it.
            // The free below is inside the while loop.
            if ( newd->factors ) M_free(newd->factors,"Dollar factors");
            M_free(newd,"Copy of dollar variable");
            continue;
        }
        r = newd->where;
        NewSort(BHEAD0);
        while ( *r ) {  /* Sum over the terms */
            m = AT.WorkPointer;
            j = *r;
            while ( --j >= 0 ) *m++ = *r++;
            AT.WorkPointer = m;
/*
            What to do with dummy indices?
*/
            if ( Generator(BHEAD oldwork,level) ) {
                LowerSortLevel();
                error = -1; goto idcall;
            }
            AT.WorkPointer = oldwork;
        }
        AN.tryterm = 0; /* for now */
        if ( EndSort(BHEAD (WORD *)((void *)(&dbuffer)),2) < 0 ) { error = 1; break; }
        if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"old buffer of dollar");
        d->where = dbuffer;
        if ( dbuffer == 0 || *dbuffer == 0 ) {
            d->type = DOLZERO;
            if ( dbuffer ) M_free(dbuffer,"buffer of dollar");
            d->where = &(AM.dollarzero); d->size = 0;
        }
        else {
            d->type = DOLTERMS;
            r = d->where; while ( *r ) r += *r;
            d->size = (r-d->where)+1;
        }
/*      cbuf[AM.dbufnum].rhs[numdol] = d->where; */
        cbuf[AM.dbufnum].rhs[numdol] = (WORD *)(1);
/*
```

```
        Now we have a little cleaning up to do
*/
#ifdef WITHPTHREADS
        if ( dtype > 0 && dtype != MODLOCAL ) {
/*          UNLOCK(d->pthreadslockwrite); */
            UNLOCK(d->pthreadslockread);
        }
#endif
        if ( newd->factors ) M_free(newd->factors,"Dollar factors");
        M_free(newd,"Copy of dollar variable");
        }
    }
idcall:;
    AR.Cnumlhs = oldnumlhs;
    AR.DeferFlag = olddefer;
    AN.cTerm = oldcterm;
    AT.WorkPointer = oldwork;
    return(error);
}

/*
    #] InsideDollar :
    #[ ExchangeDollars :
*/

void ExchangeDollars(int num1, int num2)
{
    DOLLARS d1, d2;
    WORD node1, node2;
    LONG nam;
    d1 = Dollars + num1; node1 = d1->node;
    d2 = Dollars + num2; node2 = d2->node;
    nam = d1->name; d1->name = d2->name; d2->name = nam;
    d1->node = node2; d2->node = node1;
    AC.dollarnames->namenode[node1].number = num2;
    AC.dollarnames->namenode[node2].number = num1;
}

/*
    #] ExchangeDollars :
    #[ TermsInDollar :
*/

LONG TermsInDollar(WORD num)
{
    GETIDENTITY
    DOLLARS d = Dollars + num;
    WORD *t;
    LONG n;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( num == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    if ( d->type == DOLTERMS ) {
        n = 0;
        t = d->where;
        while ( *t ) { t += *t; n++; }
    }
    else if ( d->type == DOLWILDARGS ) {
        n = 0;
        if ( d->where[0] == 0 ) {
            t = d->where+1;
            while ( *t != 0 ) { NEXTARG(t); n++; }
        }
        else if ( d->where[0] == 1 ) n = 1;
    }
    else if ( d->type == DOLZERO ) n = 0;
    else n = 1;
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(n);
}
```

```
/*
    #] TermsInDollar :
    #[ SizeOfDollar :
*/

LONG SizeOfDollar(WORD num)
{
    GETIDENTITY
    DOLLARS d = Dollars + num;
    WORD *t;
    LONG n;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( num == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    if ( d->type == DOLTERMS ) {
        t = d->where;
        while ( *t ) t += *t;
        t++;
        n = (LONG)(t - d->where);
    }
    else if ( d->type == DOLWILDARGS ) {
        n = 0;
        if ( d->where[0] == 0 ) {
            t = d->where+1;
            while ( *t != 0 ) { NEXTARG(t); n++; }
            t++;
            n = (LONG)(t - d->where);
        }
        else if ( d->where[0] == 1 ) n = 1;
    }
    else if ( d->type == DOLZERO ) n = 0;
    else n = 1;
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(n);
}

/*
    #] SizeOfDollar :
    #[ PreIfDollarEval :

    Routine is invoked in #if etc after $( is encountered.
    $(expr1 operator expr2) makes compares between expressions,
    $(expr1 operator _keyword) makes compares between expressions,
    interpreted as expressions. We are here mainly looking at $variables.
    First we look for the operator:
        >, <, ==, >=, <=, != : < means that it comes before.
    _keywords can be:
        _set(setname)    (does the expr belong to the set (only with == or !=))
        _productof(expr)
*/

UBYTE *PreIfDollarEval(UBYTE *s, int *value)
{
    GETIDENTITY
    UBYTE *s1,*s2,*s3,*s4,*s5,*t,c,c1,c2,c3;
    int oprtr, type;
    WORD *buf1 = 0, *buf2 = 0, numset, *oldwork = AT.WorkPointer;
    EXCHINOUT
/*
    Find the three composing objects (epxression, operator, expression or keyw
*/
    while ( *s == ' ' || *s == '\t' || *s == '\n' || *s == '\r' ) s++;
    s1 = t = s;
    while ( *t != '=' && *t != '!' && *t != '>' && *t != '<' ) {
        if ( *t == '[' ) { SKIPBRA1(t) }
        else if ( *t == '{' ) { SKIPBRA2(t) }
        else if ( *t == '(' ) { SKIPBRA3(t) }
        else if ( *t == ']' || *t == '}' || *t == ')' ) {
            MLOCK(ErrorMessageLock);
            MesPrint("@Improper bracketting in #if");
            MUNLOCK(ErrorMessageLock);
```

```
            goto onerror;
        }
        t++;
    }
    s2 = t;
    while ( *t == '=' || *t == '!' || *t == '>' || *t == '<' ) t++;
    s3 = t;
    while ( *t && *t != ')' ) {
        if ( *t == '[' ) { SKIPBRA1(t) }
        else if ( *t == '{' ) { SKIPBRA2(t) }
        else if ( *t == '(' ) { SKIPBRA3(t) }
        else if ( *t == ']' || *t == '}' ) {
            MLOCK(ErrorMessageLock);
            MesPrint("@Improper brackets in #if");
            MUNLOCK(ErrorMessageLock);
            goto onerror;
        }
        t++;
    }
    if ( *t == 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("@Missing ) to match $( in #if");
        MUNLOCK(ErrorMessageLock);
        goto onerror;
    }
    s4 = t; c2 = *s4; *s4 = 0;
    if ( s2+2 < s3 || s2 == s3 ) {
IllOp:;
        MLOCK(ErrorMessageLock);
        MesPrint("@Illegal operator in $( option of #if");
        MUNLOCK(ErrorMessageLock);
        goto onerror;
    }
    if ( s2+1 == s3 ) {
        if ( *s2 == '=' ) oprtr = EQUAL;
        else if ( *s2 == '>' ) oprtr = GREATER;
        else if ( *s2 == '<' ) oprtr = LESS;
        else goto IllOp;
    }
    else if ( *s2 == '!' && s2[1] == '=' ) oprtr = NOTEQUAL;
    else if ( *s2 == '=' && s2[1] == '=' ) oprtr = EQUAL;
    else if ( *s2 == '<' && s2[1] == '=' ) oprtr = LESSEQUAL;
    else if ( *s2 == '>' && s2[1] == '=' ) oprtr = GREATEREQUAL;
    else goto IllOp;
    c1 = *s2; *s2 = 0;
/*
    The two expressions are now zero terminated
    Look for the special keywords
*/
    while ( *s3 == ' ' || *s3 == '\t' || *s3 == '\n' || *s3 == '\r' ) s3++;
    t = s3;
    while ( chartype[*t] == 0 ) t++;
    if ( *t == '_' ) {
        t++; c = *t; *t = 0;
        if ( StrICmp(s3,(UBYTE *)"set_") == 0 ) {
            if ( oprtr != EQUAL && oprtr != NOTEQUAL ) {
ImpOp:;
                MLOCK(ErrorMessageLock);
                MesPrint("@Improper operator for special keyword in $( ) option");
                MUNLOCK(ErrorMessageLock);
                goto onerror;
            }
            type = 1;
        }
        else if ( StrICmp(s3,(UBYTE *)"multipleof_") == 0 ) {
            if ( oprtr != EQUAL && oprtr != NOTEQUAL ) goto ImpOp;
            type = 2;
        }
/*
        else if ( StrICmp(s3,(UBYTE *)"productof_") == 0 ) {
            if ( oprtr != EQUAL && oprtr != NOTEQUAL ) goto ImpOp;
            type = 3;
        }
*/
        else type = 0;
    }
    else { type = 0; c = *t; }
    if ( type > 0 ) {
        *t++ = c; s3 = t; s5 = s4-1;
        while ( *s5 != ')' ) {
            if ( *s5 == ' ' || *s5 == '\t' || *s5 == '\n' || *s5 == '\r' ) s5--;
            else {
                MLOCK(ErrorMessageLock);
                MesPrint("@Improper use of special keyword in $( ) option");
                MUNLOCK(ErrorMessageLock);
                goto onerror;
            }
```

```
        }
        c3 = *s5; *s5 = 0;
    }
    else { c3 = c2; s5 = s4; }
/*
    Expand the first expression.
*/
    if ( ( buf1 = TranslateExpression(s1) ) == 0 ) {
        AT.WorkPointer = oldwork;
        goto onerror;
    }
    if ( type == 1 ) {  /* determine the set */
        if ( *s3 == '{' ) {
            t = s3+1;
            SKIPBRA2(s3)
            numset = DoTempSet(t,s3);
            s3++;
            if ( numset < 0 ) {
noset:;
                MLOCK(ErrorMessageLock);
                MesPrint("@Argument of set_ is not a valid set");
                MUNLOCK(ErrorMessageLock);
                goto onerror;
            }
        }
        else {
            t = s3;
            while ( FG.cTable[*s3] == 0 || FG.cTable[*s3] == 1
                || *s3 == '_' ) s3++;
            c = *s3; *s3 = 0;
            if ( GetName(AC.varnames,t,&numset,NOAUTO) != CSET ) {
                *s3 = c; goto noset;
            }
            *s3 = c;
        }
        while ( *s3 == ' ' || *s3 == '\t' || *s3 == '\n' || *s3 == '\r' ) s3++;
        if ( s3 != s5 ) goto noset;
        *value = IsSetMember(buf1,numset);
        if ( oprtr == NOTEQUAL ) *value ^= 1;
    }
    else {
        if ( ( buf2 = TranslateExpression(s3) ) == 0 ) goto onerror;
    }
    if ( type == 0 ) {
        *value = TwoExprCompare(buf1,buf2,oprtr);
    }
    else if ( type == 2 ) {
        *value = IsMultipleOf(buf1,buf2);
        if ( oprtr == NOTEQUAL ) *value ^= 1;
    }
/*
    else if ( type == 3 ) {
        *value = IsProductOf(buf1,buf2);
        if ( oprtr == NOTEQUAL ) *value ^= 1;
    }
*/
    if ( buf1 ) M_free(buf1,"Buffer in $()");
    if ( buf2 ) M_free(buf2,"Buffer in $()");
    *s5 = c3; *s4++ = c2; *s2 = c1;
    AT.WorkPointer = oldwork;
    BACKINOUT
    return(s4);
onerror:
    if ( buf1 ) M_free(buf1,"Buffer in $()");
    if ( buf2 ) M_free(buf2,"Buffer in $()");
    AT.WorkPointer = oldwork;
    BACKINOUT
    return(0);
}

/*
    #] PreIfDollarEval :
    #[ TranslateExpression :
*/

WORD *TranslateExpression(UBYTE *s)
{
    GETIDENTITY
    CBUF *C = cbuf+AC.cbufnum;
    WORD oldnumrhs = C->numrhs;
    LONG oldcpointer = C->Pointer - C->Buffer;
    WORD *w = AT.WorkPointer;
    WORD retcode, oldEside;
    WORD *outbuffer;
    *w++ = SUBEXPSIZE + 4;
    AC.ProtoType = w;
    *w++ = SUBEXPRESSION;
```

```
    *w++ = SUBEXPSIZE;
    *w++ = C->numrhs+1;
    *w++ = 1;
    *w++ = AC.cbufnum;
    FILLSUB(w)
    *w++ = 1; *w++ = 1; *w++ = 3; *w++ = 0;
    AT.WorkPointer = w;
    if ( ( retcode = CompileAlgebra(s,RHSIDE,AC.ProtoType) ) < 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("@Error translating first expression in $( ) option");
        MUNLOCK(ErrorMessageLock);
        return(0);
    }
    else { AC.ProtoType[2] = retcode; }
/*
    Evaluate this expression
*/
    if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) { return(0); }
    AN.RepPoint = AT.RepCount + 1;
    oldEside = AR.Eside; AR.Eside = RHSIDE;
    AR.Cnumlhs = C->numlhs;
    if ( Generator(BHEAD AC.ProtoType-1,C->numlhs) ) {
        AR.Eside = oldEside;
        LowerSortLevel(); LowerSortLevel(); return(0);
    }
    AR.Eside = oldEside;
    AT.WorkPointer = w;
    AN.tryterm = 0; /* for now */
    if ( EndSort(BHEAD (WORD *)((void *)(&outbuffer)),2) < 0 ) { LowerSortLevel(); return(0); }
    LowerSortLevel();
    C->Pointer = C->Buffer + oldcpointer;
    C->numrhs = oldnumrhs;
    AT.WorkPointer = AC.ProtoType - 1;
    return(outbuffer);
}

/*
    #] TranslateExpression :
    #[ IsSetMember :

    Checks whether the expression in the buffer can be seen as an element
    of the given set.
    For the special sets: if more than one term: no match!!!
*/

int IsSetMember(WORD *buffer, WORD numset)
{
    WORD *t = buffer, *tt, num, csize, num1;
    WORD bufterm[4];
    int i, j, type;
    if ( numset < AM.NumFixedSets ) {
        if ( t[*t] != 0 ) return(0);     /* More than one term */
        if ( *t == 0 ) {
            if ( numset == POS0_ || numset == NEG0_ || numset == EVEN_
            || numset == Z_ || numset == Q_ ) return(1);
            else return(0);
        }
        if ( numset == SYMBOL_ ) {
            if ( *t == 8 && t[1] == SYMBOL && t[7] == 3 && t[6] == 1
            && t[5] == 1 && t[4] == 1 ) return(1);
            else return(0);
        }
        if ( numset == INDEX_ ) {
            if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
            && t[4] == 1 && t[3] > 0 ) return(1);
            if ( *t == 4 && t[3] == 3 && t[2] == 1 && t[1] < AM.OffsetIndex)
                return(1);
            return(0);
        }
        if ( numset == FIXED_ ) {
            if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
            && t[4] == 1 && t[3] > 0 && t[3] < AM.OffsetIndex ) return(1);
            if ( *t == 4 && t[3] == 3 && t[2] == 1 && t[1] < AM.OffsetIndex)
                return(1);
            return(0);
        }
        if ( numset == DUMMYINDEX_ ) {
            if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
            && t[4] == 1 && t[3] >= AM.IndDum && t[3] < AM.IndDum+MAXDUMMIES ) return(1);
            if ( *t == 4 && t[3] == 3 && t[2] == 1
                && t[1] >= AM.IndDum && t[1] < AM.IndDum+MAXDUMMIES ) return(1);
            return(0);
        }
        if ( numset == VECTOR_ ) {
            if ( *t == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
            && t[4] == 1 && t[3] < (AM.OffsetVector+WILDOFFSET) && t[3] >= AM.OffsetVector )
return(1);
```

```
            return(0);
        }
        tt = t + *t - 1;
        if ( ABS(tt[0]) != *t-1 ) return(0);
        if ( numset == Q_ ) return(1);
        if ( numset == POS_ || numset == POS0_ ) return(tt[0]>0);
        else if ( numset == NEG_ || numset == NEG0_ ) return(tt[0]<0);
        i = (ABS(tt[0])-1)/2;
        tt -= i;
        if ( tt[0] != 1 ) return(0);
        for ( j = 1; j < i; j++ ) { if ( tt[j] != 0 ) return(0); }
        if ( numset == Z_ ) return(1);
        if ( numset == ODD_ ) return(t[1]&1);
        if ( numset == EVEN_ ) return(1-(t[1]&1));
        return(0);
    }
    if ( t[*t] != 0 ) return(0);    /* More than one term */
    type = Sets[numset].type;
    switch ( type ) {
        case CSYMBOL:
            if ( t[0] == 8 && t[1] == SYMBOL && t[7] == 3 && t[6] == 1
            && t[5] == 1 && t[4] == 1 ) {
                num = t[3];
            }
            else if ( t[0] == 4 && t[2] == 1 && t[1] <= MAXPOWER ) {
                num = t[1];
                if ( t[3] < 0 ) num = -num;
                num += 2*MAXPOWER;
            }
            else return(0);
            break;
        case CVECTOR:
            if ( t[0] == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
            && t[4] == 1 && t[3] < 0 ) {
                num = t[3];
            }
            else return(0);
            break;
        case CINDEX:
            if ( t[0] == 7 && t[1] == INDEX && t[6] == 3 && t[5] == 1
            && t[4] == 1 && t[3] > 0 ) {
                num = t[3];
            }
            else if ( t[0] == 4 && t[3] == 3 && t[2] == 1 && t[1] < AM.OffsetIndex ) {
                num = t[1];
            }
            else return(0);
            break;
        case CFUNCTION:
            if ( t[0] == 4+FUNHEAD && t[3+FUNHEAD] == 3 && t[2+FUNHEAD] == 1
            && t[1+FUNHEAD] == 1 && t[1] >= FUNCTION ) {
                num = t[1];
            }
            else return(0);
            break;
        case CNUMBER:
            if ( t[0] == 4 && t[2] == 1 && t[1] <= AM.OffsetIndex && t[3] == 3 ) {
                num = t[1];
            }
            else return(0);
            break;
        case CRANGE:
            csize = t[t[0]-1];
            csize = ABS(csize);
            if ( csize != t[0]-1 ) return(0);
            if ( Sets[numset].first < 3*MAXPOWER ) {
                num1 = num = Sets[numset].first;
                if ( num >= MAXPOWER ) num -= 2*MAXPOWER;
                if ( num == 0 ) {
                    if ( num1 < MAXPOWER ) {
                        if ( t[t[0]-1] >= 0 ) return(0);
                    }
                    else if ( t[t[0]-1] > 0 ) return(0);
                }
                else {
                    bufterm[0] = 4; bufterm[1] = ABS(num);
                    bufterm[2] = 1;
                    if ( num < 0 ) bufterm[3] = -3;
                    else bufterm[3] = 3;
                    num = CompCoef(t,bufterm);
                    if ( num1 < MAXPOWER ) {
                        if ( num >= 0 ) return(0);
                    }
                    else if ( num > 0 ) return(0);
                }
            }
            if ( Sets[numset].last > -3*MAXPOWER ) {
```

```
                num1 = num = Sets[numset].last;
                if ( num <= -MAXPOWER ) num += 2*MAXPOWER;
                if ( num == 0 ) {
                    if ( num1 > -MAXPOWER ) {
                        if ( t[t[0]-1] <= 0 ) return(0);
                    }
                    else if ( t[t[0]-1] < 0 ) return(0);
                }
                else {
                    bufterm[0] = 4; bufterm[1] = ABS(num);
                    bufterm[2] = 1;
                    if ( num < 0 ) bufterm[3] = -3;
                    else bufterm[3] = 3;
                    num = CompCoef(t,bufterm);
                    if ( num1 > -MAXPOWER ) {
                        if ( num <= 0 ) return(0);
                    }
                    else if ( num < 0 ) return(0);
                }
            }
            return(1);
            break;
        default: return(0);
    }
    t  = SetElements + Sets[numset].first;
    tt = SetElements + Sets[numset].last;
    do {
        if ( num == *t ) return(1);
        t++;
    } while ( t < tt );
    return(0);
}

/*
    #] IsSetMember :
    #[ IsProductOf :

    Checks whether the expression in buf1 is a single term multiple of
    the expression in buf2.

int IsProductOf(WORD *buf1, WORD *buf2)
{
    return(0);
}


    #] IsProductOf :
    #[ IsMultipleOf :

    Checks whether the expression in buf1 is a numerical multiple of
    the expression in buf2.
*/

int IsMultipleOf(WORD *buf1, WORD *buf2)
{
    GETIDENTITY
    LONG num1, num2;
    WORD *t1, *t2, *m1, *m2, *r1, *r2, nc1, nc2, ni1, ni2;
    UWORD *IfScrat1, *IfScrat2;
    int i, j;
    if ( *buf1 == 0 && *buf2 == 0 ) return(1);
/*
    First count terms
*/
    t1 = buf1; t2 = buf2; num1 = 0; num2 = 0;
    while ( *t1 ) { t1 += *t1; num1++; }
    while ( *t2 ) { t2 += *t2; num2++; }
    if ( num1 != num2 ) return(0);
/*
    Test similarity of terms. Difference up to a number.
*/
    t1 = buf1; t2 = buf2;
    while ( *t1 ) {
        m1 = t1+1; m2 = t2+1; t1 += *t1; t2 += *t2;
        r1 = t1 - ABS(t1[-1]); r2 = t2 - ABS(t2[-1]);
        if ( r1-m1 != r2-m2 ) return(0);
        while ( m1 < r1 ) {
            if ( *m1 != *m2 ) return(0);
            m1++; m2++;
        }
    }
/*
    Now we have to test the constant factor
*/
    IfScrat1 = (UWORD *)(TermMalloc("IsMultipleOf")); IfScrat2 = (UWORD *)(TermMalloc("IsMultipleOf"));
    t1 = buf1; t2 = buf2;
```

```
    t1 += *t1; t2 += *t2;
    if ( *t1 == 0 && *t2 == 0 ) return(1);
    r1 = t1 - ABS(t1[-1]); r2 = t2 - ABS(t2[-1]);
    nc1 = REDLENG(t1[-1]); nc2 = REDLENG(t2[-1]);
    if ( DivRat(BHEAD (UWORD *)r1,nc1,(UWORD *)r2,nc2,IfScrat1,&ni1) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("@Called from MultipleOf in $( )");
        MUNLOCK(ErrorMessageLock);
        TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
        Terminate(-1);
    }
    while ( *t1 ) {
        t1 += *t1; t2 += *t2;
        r1 = t1 - ABS(t1[-1]); r2 = t2 - ABS(t2[-1]);
        nc1 = REDLENG(t1[-1]); nc2 = REDLENG(t2[-1]);
        if ( DivRat(BHEAD (UWORD *)r1,nc1,(UWORD *)r2,nc2,IfScrat2,&ni2) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("@Called from MultipleOf in $( )");
            MUNLOCK(ErrorMessageLock);
            TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
            Terminate(-1);
        }
        if ( ni1 != ni2 ) return(0);
        i = 2*ABS(ni1);
        for ( j = 0; j < i; j++ ) {
            if ( IfScrat1[j] != IfScrat2[j] ) {
                TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
                return(0);
            }
        }
    }
    TermFree(IfScrat1,"IsMultipleOf"); TermFree(IfScrat2,"IsMultipleOf");
    return(1);
}

/*
    #] IsMultipleOf :
    #[ TwoExprCompare :

    Compares the expressions in buf1 and buf2 according to oprtr
*/

int TwoExprCompare(WORD *buf1, WORD *buf2, int oprtr)
{
    GETIDENTITY
    WORD *t1, *t2, cond;
    t1 = buf1; t2 = buf2;
    while ( *t1 && *t2 ) {
        cond = CompareTerms(BHEAD t1,t2,1);
        if ( cond != 0 ) {
            if ( cond > 0 ) { /* t1 comes first */
                switch ( oprtr ) {  /* t1 is less */
                    case EQUAL: return(0);
                    case NOTEQUAL: return(1);
                    case GREATEREQUAL: return(0);
                    case GREATER: return(0);
                    case LESS: return(1);
                    case LESSEQUAL: return(1);
                }
            }
            else {
                switch ( oprtr ) {
                    case EQUAL: return(0);
                    case NOTEQUAL: return(1);
                    case GREATEREQUAL: return(1);
                    case GREATER: return(1);
                    case LESS: return(0);
                    case LESSEQUAL: return(0);
                }
            }
        }
        t1 += *t1; t2 += *t2;
    }
    if ( *t1 == *t2 ) { /* They are equal */
        switch ( oprtr ) {
            case EQUAL: return(1);
            case NOTEQUAL: return(0);
            case GREATEREQUAL: return(1);
            case GREATER: return(0);
            case LESS: return(0);
            case LESSEQUAL: return(1);
        }
    }
    else if ( *t1 ) {  /* t1 is greater */
        switch ( oprtr ) {
            case EQUAL: return(0);
            case NOTEQUAL: return(1);
```

```
            case GREATEREQUAL: return(1);
            case GREATER: return(1);
            case LESS: return(0);
            case LESSEQUAL: return(0);
        }
    }
    else {
        switch ( oprtr ) {  /* t1 is less */
            case EQUAL: return(0);
            case NOTEQUAL: return(1);
            case GREATEREQUAL: return(0);
            case GREATER: return(0);
            case LESS: return(1);
            case LESSEQUAL: return(1);
        }
    }
    MLOCK(ErrorMessageLock);
    MesPrint("@Internal problems with operator in $( )");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(0);
}

/*
    #] TwoExprCompare :
    #[ DollarRaiseLow :

    Raises or lowers the numerical value of a dollar variable
    Not to be used in parallel.
*/

static UWORD *dscrat = 0;
static WORD ndscrat;

int DollarRaiseLow(UBYTE *name, LONG value)
{
    GETIDENTITY
    int num;
    DOLLARS d;
    int sgn = 1;
    WORD lnum[4], nnum, *t1, *t2, i;
    UBYTE *s, c;
    s = name; while ( *s ) s++;
    if ( s[-1] == '-' && s[-2] == '-' && s > name+2 ) s -= 2;
    else if ( s[-1] == '+' && s[-2] == '+' && s > name+2 ) s -= 2;
    c = *s; *s = 0;
    num = GetDollar(name);
    *s = c;
    d = Dollars + num;
    if ( value < 0 ) { value = -value; sgn = -1; }
    if ( d->type == DOLZERO ) {
        if ( d->where ) M_free(d->where,"DollarRaiseLow");
        d->size = MINALLOC;
        d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"DollarRaiseLow");
        if ( ( value & AWORDMASK ) != 0 ) {
            d->where[0] = 6; d->where[1] = value » BITSINWORD;
            d->where[2] = (WORD)value; d->where[3] = 1; d->where[4] = 0;
            d->where[5] = 5*sgn; d->where[6] = 0;
            d->type = DOLTERMS;
        }
        else {
            d->where[0] = 4; d->where[1] = (WORD)value; d->where[2] = 1;
            d->where[3] = 3*sgn; d->where[4] = 0;
            d->type = DOLNUMBER;
        }
    }
    else if ( d->type == DOLNUMBER || ( d->type == DOLTERMS
    && d->where[d->where[0]] == 0
    && d->where[0] == ABS(d->where[d->where[0]-1])+1 ) ) {
        if ( ( value & AWORDMASK ) != 0 ) {
            lnum[0] = value » BITSINWORD;
            lnum[1] = (WORD)value; lnum[2] = 1; lnum[3] = 0;
            nnum = 2*sgn;
        }
        else {
            lnum[0] = (WORD)value; lnum[1] = 1; nnum = sgn;
        }
        i = d->where[d->where[0]-1];
        i = REDLENG(i);
        if ( dscrat == 0 ) {
            dscrat = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD),"DollarRaiseLow");
        }
        if ( AddRat(BHEAD (UWORD *)(d->where+1),i,
             (UWORD *)lnum,nnum,dscrat,&ndscrat) ) {
                MLOCK(ErrorMessageLock);
                MesCall("DollarRaiseLow");
                MUNLOCK(ErrorMessageLock);
```

```
                Terminate(-1);
        }
        ndscrat = INCLENG(ndscrat);
        i = ABS(ndscrat);
        if ( i == 0 ) {
            M_free(d->where,"DollarRaiseLow");
            d->where = 0;
            d->type = DOLZERO;
            d->size = 0;
            return(0);
        }
        if ( i+2 > d->size ) {
            M_free(d->where,"DollarRaiseLow");
            d->size = i+2;
            if ( d->size < MINALLOC ) d->size = MINALLOC;
            d->size = ((d->size+7)/8)*8;
            d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"DollarRaiseLow");
        }
        t1 = d->where; *t1++ = i+1; t2 = (WORD *)dscrat;
        while ( --i > 0 ) *t1++ = *t2++;
        *t1++ = ndscrat; *t1 = 0;
        d->type = DOLTERMS;
    }
    return(0);
}

/*
    #] DollarRaiseLow :
    #[ EvalDoLoopArg :
*/
WORD EvalDoLoopArg(PHEAD WORD *arg, WORD par)
{
    WORD num, type, *td;
    DOLLARS d;
    if ( *arg == SNUMBER ) return(arg[1]);
    if ( *arg == DOLLAREXPR2 && arg[1] < 0 ) return(-arg[1]-1);
    d = Dollars + arg[1];
#ifdef WITHPTHREADS
    {
        int nummodopt, dtype = -1;
        if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
            for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                if ( arg[1] == ModOptdollars[nummodopt].number ) break;
            }
            if ( nummodopt < NumModOptdollars ) {
                dtype = ModOptdollars[nummodopt].type;
                if ( dtype == MODLOCAL ) {
                    d = ModOptdollars[nummodopt].dstruct+AT.identity;
                }
            }
        }
    }
#endif
    if ( *arg == DOLLAREXPRESSION ) {
        if ( arg[2] != DOLLAREXPR2 ) {  /* end of chain */
endofchain:
            type = d->type;
            if ( type == DOLZERO ) {}
            else if ( type == DOLNUMBER ) {
                td = d->where;
                if ( ( td[0] != 4 ) || ( (td[1]&SPECMASK) != 0 ) || ( td[2] != 1 ) ) {
                    MLOCK(ErrorMessageLock);
                    if ( par == -1 ) {
                        MesPrint("$-variable is not a short number in print statement");
                    }
                    else {
                        MesPrint("$-variable is not a short number in do loop");
                    }
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                return( td[3] > 0 ? td[1]: -td[1] );
            }
            else {
                MLOCK(ErrorMessageLock);
                if ( par == -1 ) {
                    MesPrint("$-variable is not a number in print statement");
                }
                else {
                    MesPrint("$-variable is not a number in do loop");
                }
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            return(0);
        }
        num = EvalDoLoopArg(BHEAD arg+2,par);
```

```
    }
    else if ( *arg == DOLLAREXPR2 ) {
        if ( arg[1] < 0 ) { num = -arg[1]-1; }
        else if ( arg[2] != DOLLAREXPR2 && par == -1 ) {
            goto endofchain;
        }
        else                  { num = EvalDoLoopArg(BHEAD arg+2,par); }
    }
    else {
        MLOCK(ErrorMessageLock);
        if ( par == -1 ) {
            MesPrint("Invalid $-variable in print statement");
        }
        else {
            MesPrint("Invalid $-variable in do loop");
        }
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(0);
    }
    if ( num == 0 ) return(d->nfactors);
    if ( num > d->nfactors || num < 1 ) {
        MLOCK(ErrorMessageLock);
        if ( par == -1 ) {
            MesPrint("Not a valid factor number for $-variable in print statement");
        }
        else {
            MesPrint("Not a valid factor number for $-variable in do loop");
        }
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(0);
    }
    if ( d->factors[num].type == DOLNUMBER )
        return(d->factors[num].value);
    else {  /* If correct, type can only be DOLNUMBER or DOLTERMS */
        MLOCK(ErrorMessageLock);
        if ( par == -1 ) {
            MesPrint("$-variable in print statement is not a number");
        }
        else {
            MesPrint("$-variable in do loop is not a number");
        }
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(0);
    }
}

/*
    #] EvalDoLoopArg :
    #[ TestDoLoop :
*/

WORD TestDoLoop(PHEAD WORD *lhsbuf, WORD level)
{
    GETBIDENTITY
    WORD start,finish,incr;
    WORD *h;
    DOLLARS d;
    h = lhsbuf + 4; /* address of the start value */
    start = EvalDoLoopArg(BHEAD h,0);
    while ( ( *h == DOLLAREXPRESSION || *h == DOLLAREXPR2 )
        && ( h[2] == DOLLAREXPR2 ) ) h += 2;
    h += 2;
    finish = EvalDoLoopArg(BHEAD h,0);
    while ( ( *h == DOLLAREXPRESSION || *h == DOLLAREXPR2 )
        && ( h[2] == DOLLAREXPR2 ) ) h += 2;
    h += 2;
    incr = EvalDoLoopArg(BHEAD h,0);

    if ( ( finish == start ) || ( finish > start && incr > 0 )
    || ( finish < start && incr < 0 ) ) {}
    else { level = lhsbuf[3]; } /* skips the loop */
/*
    Put start in the dollar variable indicated by lhsbuf[2]
*/
    d = Dollars + lhsbuf[2];
#ifdef WITHPTHREADS
    {
        int nummodopt, dtype = -1;
        if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
            for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                if ( lhsbuf[2] == ModOptdollars[nummodopt].number ) break;
            }
            if ( nummodopt < NumModOptdollars ) {
                dtype = ModOptdollars[nummodopt].type;
```

```
                if ( dtype == MODLOCAL ) {
                    d = ModOptdollars[nummodopt].dstruct+AT.identity;
                }
            }
        }
    }
#endif

    if ( d->size < MINALLOC ) {
        if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar contents");
        d->size = MINALLOC;
        d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"dollar contents");
    }
    if ( start > 0 ) {
        d->where[0] = 4;
        d->where[1] = start;
        d->where[2] = 1;
        d->where[3] = 3;
        d->where[4] = 0;
        d->type = DOLNUMBER;
    }
    else if ( start < 0 ) {
        d->where[0] = 4;
        d->where[1] = -start;
        d->where[2] = 1;
        d->where[3] = -3;
        d->where[4] = 0;
        d->type = DOLNUMBER;
    }
    else
        d->type = DOLZERO;

    if ( d == Dollars + lhsbuf[2] ) {
        cbuf[AM.dbufnum].CanCommu[lhsbuf[2]] = 0;
        cbuf[AM.dbufnum].NumTerms[lhsbuf[2]] = 1;
        cbuf[AM.dbufnum].rhs[lhsbuf[2]] = d->where;
    }
    return(level);
}

/*
    #] TestDoLoop :
    #[ TestEndDoLoop :
*/

WORD TestEndDoLoop(PHEAD WORD *lhsbuf, WORD level)
{
    GETBIDENTITY
    WORD start,finish,incr,value;
    WORD *h;
    DOLLARS d;
    h = lhsbuf + 4; /* address of the start value */
    start = EvalDoLoopArg(BHEAD h,0);
    while ( ( *h == DOLLAREXPRESSION || *h == DOLLAREXPR2 )
        && ( h[2] == DOLLAREXPR2 ) ) h += 2;
    h += 2;
    finish = EvalDoLoopArg(BHEAD h,0);
    while ( ( *h == DOLLAREXPRESSION || *h == DOLLAREXPR2 )
        && ( h[2] == DOLLAREXPR2 ) ) h += 2;
    h += 2;
    incr = EvalDoLoopArg(BHEAD h,0);

    if ( ( finish == start ) || ( finish > start && incr > 0 )
    || ( finish < start && incr < 0 ) ) {}
    else { level = lhsbuf[3]; } /* skips the loop */
/*
    Put start in the dollar variable indicated by lhsbuf[2]
*/
    d = Dollars + lhsbuf[2];
#ifdef WITHPTHREADS
    {
        int nummodopt, dtype = -1;
        if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
            for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                if ( lhsbuf[2] == ModOptdollars[nummodopt].number ) break;
            }
            if ( nummodopt < NumModOptdollars ) {
                dtype = ModOptdollars[nummodopt].type;
                if ( dtype == MODLOCAL ) {
                    d = ModOptdollars[nummodopt].dstruct+AT.identity;
                }
            }
        }
    }
#endif
/*
    Get the value
```

```
*/
    if ( d->type == DOLZERO ) {
        value = 0;
    }
    else if ( ( d->type == DOLNUMBER || d->type == DOLTERMS )
    && ( d->where[4] == 0 ) && ( d->where[0] == 4 )
    && ( d->where[1] > 0 ) && ( d->where[2] == 1 ) ) {
        value = ( d->where[3] < 0 ) ? -d->where[1]: d->where[1];
    }
    else {
        MLOCK(ErrorMessageLock);
        MesPrint("Wrong type of object in do loop parameter");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(level);
    }
    value += incr;
    if ( ( finish > start && value <= finish ) ||
         ( finish < start && value >= finish ) ||
         ( finish == start && value == finish ) ) {}
    else level = lhsbuf[3];

    if ( d->size < MINALLOC ) {
        if ( d->where && d->where != &(AM.dollarzero) ) M_free(d->where,"dollar contents");
        d->size = MINALLOC;
        d->where = (WORD *)Malloc1(d->size*sizeof(WORD),"dollar contents");
    }
    if ( value > 0 ) {
        d->where[0] = 4;
        d->where[1] = value;
        d->where[2] = 1;
        d->where[3] = 3;
        d->where[4] = 0;
        d->type = DOLNUMBER;
    }
    else if ( start < 0 ) {
        d->where[0] = 4;
        d->where[1] = -value;
        d->where[2] = 1;
        d->where[3] = -3;
        d->where[4] = 0;
        d->type = DOLNUMBER;
    }
    else
        d->type = DOLZERO;

    if ( d == Dollars + lhsbuf[2] ) {
        cbuf[AM.dbufnum].CanCommu[lhsbuf[2]] = 0;
        cbuf[AM.dbufnum].NumTerms[lhsbuf[2]] = 1;
        cbuf[AM.dbufnum].rhs[lhsbuf[2]] = d->where;
    }
    return(level);
}

/*
    #] TestEndDoLoop :
    #[ DollarFactorize :
*/
/* #define STEP2 */
#define STEP2

int DollarFactorize(PHEAD WORD numdollar)
{
    GETBIDENTITY
    DOLLARS d = Dollars + numdollar;
    CBUF *C, *CC;
    WORD *oldworkpointer;
    WORD *buf1, *t, *term, *buf1content, *buf2, *termextra;
    WORD *buf3, *argextra;
#ifdef STEP2
    WORD *tstop, pow, *r;
#endif
    int i, j, jj, action = 0, sign = 1;
    LONG insize, ii;
    WORD startebuf = cbuf[AT.ebufnum].numrhs;
    WORD nfactors, factorsincontent, extrafactor = 0;
    WORD oldsorttype = AR.SortType;

#ifdef WITHPTHREADS
    int nummodopt, dtype;
    dtype = -1;
    if ( AS.MultiThreaded ) {
        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
            if ( numdollar == ModOptdollars[nummodopt].number ) break;
        }
        if ( nummodopt < NumModOptdollars ) {
            dtype = ModOptdollars[nummodopt].type;
```

```
            if ( dtype == MODLOCAL ) {
                d = ModOptdollars[nummodopt].dstruct+AT.identity;
            }
            else {
                LOCK(d->pthreadslockread);
            }
        }
    }
#endif
    CleanDollarFactors(d);
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    if ( d->type != DOLTERMS ) {    /* only one term */
        if ( d->type != DOLZERO ) d->nfactors = 1;
        return(0);
    }
    if ( d->where[d->where[0]] == 0 ) { /* only one term. easy */
    }
/*
    Here should come the code for the factorization
    We copied the routine ArgFactorize in argument.c and changed the
    memory management completely. For the actual factorization it
    calls WORD *DoFactorizeDollar(PHEAD WORD *expr) which allocates
    space for the answer. Notation:
     term,...,term,0,term,...,term,0,term,...,term,0,0

        #[ Step 1: sort the terms properly and/or make copy  --> buf1,insize
*/
    term = d->where;
    AR.SortType = SORTHIGHFIRST;
    if ( oldsorttype != AR.SortType ) {
        NewSort(BHEAD0);
        while ( *term ) {
            t = term + *term;
            if ( AN.ncmod != 0 ) {
                if ( AN.ncmod != 1 || ( (WORD)AN.cmod[0] < 0 ) ) {
                    AR.SortType = oldsorttype;
                    MLOCK(ErrorMessageLock);
                    MesPrint("Factorization modulus a number, greater than a WORD not implemented.");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                if ( Modulus(term) ) {
                    AR.SortType = oldsorttype;
                    MLOCK(ErrorMessageLock);
                    MesCall("DollarFactorize");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                if ( !*term) { term = t; continue; }
            }
            StoreTerm(BHEAD term);
            term = t;
        }
        AN.tryterm = 0; /* for now */
        EndSort(BHEAD (WORD *)((void *)(&buf1)),2);
        t = buf1; while ( *t ) t += *t;
        insize = t - buf1;
    }
    else {
        t = term; while ( *t ) t += *t;
        ii = insize = t - term;
        buf1 = (WORD *)Malloc1((insize+1)*sizeof(WORD),"DollarFactorize-1");
        t = buf1;
        NCOPY(t,term,ii);
        *t++ = 0;
    }
/*
        #] Step 1:
        #[ Step 2: take out the 'content'.
*/
#ifdef STEP2
    buf1content = TermMalloc("DollarContent");
    AN.tryterm = -1;
    if ( ( buf2 = TakeContent(BHEAD buf1,buf1content) ) == 0 ) {
        AN.tryterm = 0;
        TermFree(buf1content,"DollarContent");
        M_free(buf1,"DollarFactorize-1");
        AR.SortType = oldsorttype;
        MLOCK(ErrorMessageLock);
        MesCall("DollarFactorize");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(1);
    }
    else if ( ( buf1content[0] == 4 ) && ( buf1content[1] == 1 ) &&
```

```
            ( buf1content[2] == 1 ) && ( buf1content[3] == 3 ) ) { /* Nothing happened */
        AN.tryterm = 0;
        if ( buf2 != buf1 ) {
            M_free(buf2,"DollarFactorize-2");
            buf2 = buf1;
        }
        factorsincontent = 0;
    }
    else {
/*
        The way we took out objects is rather brutish. We have to normalize
*/
        AN.tryterm = 0;
        if ( buf2 != buf1 ) M_free(buf1,"DollarFactorize-1");
        buf1 = buf2;
        t = buf1; while ( *t ) t += *t;
        insize = t - buf1;
/*
        Now analyse how many factors there are in the content
*/
        factorsincontent = 0;
        term = buf1content;
        tstop = term + *term;
        if ( tstop[-1] < 0 ) factorsincontent++;
        if ( ABS(tstop[-1]) == 3 && tstop[-2] == 1 && tstop[-3] == 1 ) {
            tstop -= ABS(tstop[-1]);
        }
        else {
            factorsincontent++;
            tstop -= ABS(tstop[-1]);
        }
        term++;
        while ( term < tstop ) {
            switch ( *term ) {
                case SYMBOL:
                    t = term+2; i = (term[1]-2)/2;
                    while ( i > 0 ) {
                        factorsincontent += ABS(t[1]);
                        i--; t += 2;
                    }
                    break;
                case DOTPRODUCT:
                    t = term+2; i = (term[1]-2)/3;
                    while ( i > 0 ) {
                        factorsincontent += ABS(t[2]);
                        i--; t += 3;
                    }
                    break;
                case VECTOR:
                case DELTA:
                    factorsincontent += (term[1]-2)/2;
                    break;
                case INDEX:
                    factorsincontent += term[1]-2;
                    break;
                default:
                    if ( *term >= FUNCTION ) factorsincontent++;
                    break;
            }
            term += term[1];
        }
    }
#else
    factorsincontent = 0;
    buf1content = 0;
#endif
/*
        #] Step 2: take out the 'content'.
        #[ Step 3: ConvertToPoly
                if there are objects that are not SYMBOLs,
                invoke ConvertToPoly
                We keep the original in buf1 in case there are no factors
*/
    t = buf1;
    while ( *t ) {
        if ( ( t[1] != SYMBOL ) && ( *t != (ABS(t[*t-1])+1) ) ) {
            action = 1; break;
        }
        t += *t;
    }
    if ( DetCommu(buf1) > 1 ) {
        MesPrint("Cannot factorize a $-expression with more than one noncommuting object");
        AR.SortType = oldsorttype;
        M_free(buf1,"DollarFactorize-2");
        if ( buf1content ) TermFree(buf1content,"DollarContent");
        MesCall("DollarFactorize");
        Terminate(-1);
```

```
        return(-1);
    }
    if ( action ) {
        t = buf1;
        termextra = AT.WorkPointer;
        NewSort(BHEAD0);
        NewSort(BHEAD0);
        while ( *t ) {
            if ( LocalConvertToPoly(BHEAD t,termextra,startebuf,0) < 0 ) {
getout:
                AR.SortType = oldsorttype;
                M_free(buf1,"DollarFactorize-2");
                if ( buf1content ) TermFree(buf1content,"DollarContent");
                MesCall("DollarFactorize");
                Terminate(-1);
                return(-1);
            }
            StoreTerm(BHEAD termextra);
            t += *t;
        }
        AN.tryterm = 0; /* for now */
        if ( EndSort(BHEAD (WORD *)((void *)(&buf2)),2) < 0 ) { goto getout; }
        LowerSortLevel();
        t = buf2; while ( *t > 0 ) t += *t;
    }
    else {
        buf2 = buf1;
    }
/*
        #] Step 3: ConvertToPoly
        #[ Step 4: Now the hard work.
*/
    if ( ( buf3 = poly_factorize_dollar(BHEAD buf2) ) == 0 ) {
        MesCall("DollarFactorize");
        AR.SortType = oldsorttype;
        if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-3");
        M_free(buf1,"DollarFactorize-3");
        if ( buf1content ) TermFree(buf1content,"DollarContent");
        Terminate(-1);
        return(-1);
    }
    if ( buf2 != buf1 && buf2 ) {
        M_free(buf2,"DollarFactorize-3");
        buf2 = 0;
    }
    term = buf3;
    AR.SortType = oldsorttype;
/*
    Count the factors and strip a factor -1
*/
    nfactors = 0;
    while ( *term ) {
#ifdef STEP2
        if ( *term == 4 && term[4] == 0 && term[3] == -3 && term[2] == 1
            && term[1] == 1 ) {
            WORD *tt1, *tt2, *ttstop;
            sign = -sign;
            tt1 = term; tt2 = term + *term + 1;
            ttstop = tt2;
            while ( *ttstop ) {
                while ( *ttstop ) ttstop += *ttstop;
                ttstop++;
            }
            while ( tt2 < ttstop ) *tt1++ = *tt2++;
            *tt1 = 0;
            factorsincontent++;
            extrafactor++;
        }
        else
#endif
        {
            term += *term;
            while ( *term ) { term += *term; }
            nfactors++; term++;
        }
    }
/*
    We have now:
        buf1: the original before ConvertToPoly for if only one factor
        buf3: the factored expression with nfactors factors

        #] Step 4:
        #[ Step 5: ConvertFromPoly
                If ConvertToPoly was used, use now ConvertFromPoly
                Be careful: there should be more than one factor now.
*/
#ifdef WITHPTHREADS
```

```
    if ( dtype > 0 && dtype != MODLOCAL ) { LOCK(d->pthreadslockread); }
#endif
    if ( nfactors ==  1 && extrafactor == 0 ) { /* we can use the buf1 contents */
        if ( factorsincontent == 0 ) {
            d->nfactors = 1;
#ifdef WITHPTHREADS
            if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
/*
            We used here (before 3-sep-2015) the original and did not make
            provisions for having a factors struct, figuring that all info
            is identical to the full dollar. This makes things too
            complicated at later stages.
*/
            d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR),"factors in dollar");
            term = buf1; while ( *term ) term += *term;
            d->factors[0].size = i = term - buf1;
            d->factors[0].where = t = (WORD *)Malloc1(sizeof(WORD)*(i+1),"DollarFactorize-5");
            term = buf1; NCOPY(t,term,i); *t = 0;
            AR.SortType = oldsorttype;
            M_free(buf3,"DollarFactorize-4");
            if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-4");
            M_free(buf1,"DollarFactorize-4");
            if ( buf1content ) TermFree(buf1content,"DollarContent");
            return(0);
        }
        else {
            d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR)*(nfactors+factorsincontent),"factors in dollar");
            term = buf1; while ( *term ) term += *term;
            d->factors[0].size = i = term - buf1;
            d->factors[0].where = t = (WORD *)Malloc1(sizeof(WORD)*(i+1),"DollarFactorize-5");
            term = buf1; NCOPY(t,term,i); *t = 0;
            M_free(buf3,"DollarFactorize-4");
            buf3 = 0;
            if ( buf2 != buf1 && buf2 ) {
                M_free(buf2,"DollarFactorize-4");
                buf2 = 0;
            }
        }
    }
    else if ( action ) {
        C = cbuf+AC.cbufnum;
        CC = cbuf+AT.ebufnum;
        oldworkpointer = AT.WorkPointer;
        d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR)*(nfactors+factorsincontent),"factors in dollar");
        term = buf3;
        for ( i = 0; i < nfactors; i++ ) {
            argextra = AT.WorkPointer;
            NewSort(BHEAD0);
            NewSort(BHEAD0);
            while ( *term ) {
                if ( ConvertFromPoly(BHEAD term,argextra,numxsymbol,CC->numrhs-startebuf+numxsymbol
                ,startebuf-numxsymbol,1) <= 0 ) {
                    LowerSortLevel();
getout2:            AR.SortType = oldsorttype;
                    M_free(d->factors,"factors in dollar");
                    d->factors = 0;
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                    M_free(buf3,"DollarFactorize-4");
                    if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-4");
                    M_free(buf1,"DollarFactorize-4");
                    if ( buf1content ) TermFree(buf1content,"DollarContent");
                    return(-3);
                }
                AT.WorkPointer = argextra + *argextra;
/*
                ConvertFromPoly leaves terms with subexpressions. Hence:
*/
                if ( Generator(BHEAD argextra,C->numlhs+1) ) {
                    goto getout2;
                }
                term += *term;
            }
            term++;
            AT.WorkPointer = oldworkpointer;
            AN.tryterm = 0; /* for now */
            EndSort(BHEAD (WORD *)((void *)(&(d->factors[i].where))),2);
            LowerSortLevel();
            d->factors[i].type = DOLTERMS;
            t = d->factors[i].where;
            while ( *t ) t += *t;
            d->factors[i].size = t - d->factors[i].where;
        }
```

```
        CC->numrhs = startebuf;
    }
    else {
        C = cbuf+AC.cbufnum;
        oldworkpointer = AT.WorkPointer;
        d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR)*(nfactors+factorsincontent),"factors in
dollar");
        term = buf3;
        for ( i = 0; i < nfactors; i++ ) {
            NewSort(BHEAD0);
            while ( *term ) {
                argextra = oldworkpointer;
                j = *term;
                NCOPY(argextra,term,j)
                AT.WorkPointer = argextra;
                if ( Generator(BHEAD oldworkpointer,C->numlhs+1) ) {
                    goto getout2;
                }
            }
            term++;
            AT.WorkPointer = oldworkpointer;
            AN.tryterm = 0; /* for now */
            EndSort(BHEAD (WORD *)((void *)(&(d->factors[i].where))),2);
            d->factors[i].type = DOLTERMS;
            t = d->factors[i].where;
            while ( *t ) t += *t;
            d->factors[i].size = t - d->factors[i].where;
        }
    }
    d->nfactors = nfactors + factorsincontent;
/*
        #] Step 5: ConvertFromPoly
        #[ Step 6: The factors of the content
*/
    if ( buf3 ) M_free(buf3,"DollarFactorize-5");
    if ( buf2 != buf1 && buf2 ) M_free(buf2,"DollarFactorize-5");
    M_free(buf1,"DollarFactorize-5");
    j = nfactors;
#ifdef STEP2
    term = buf1content;
    tstop = term + *term;
    if ( tstop[-1] < 0 ) { tstop[-1] = -tstop[-1]; sign = -sign; }
    tstop -= tstop[-1];
    term++;
    while ( term < tstop ) {
        switch ( *term ) {
            case SYMBOL:
                t = term+2; i = (term[1]-2)/2;
                while ( i > 0 ) {
                    if ( t[1] < 0 ) { t[1] = -t[1]; pow = -1; }
                    else { pow = 1; }
                    for ( jj = 0; jj < t[1]; jj++ ) {
                        r = d->factors[j].where = (WORD *)Malloc1(9*sizeof(WORD),"factor");
                        r[0] = 8; r[1] = SYMBOL; r[2] = 4; r[3] = *t; r[4] = pow;
                        r[5] = 1; r[6] = 1; r[7] = 3; r[8] = 0;
                        d->factors[j].type = DOLTERMS;
                        d->factors[j].size = 8;
                        j++;
                    }
                    i--; t += 2;
                }
                break;
            case DOTPRODUCT:
                t = term+2; i = (term[1]-2)/3;
                while ( i > 0 ) {
                    if ( t[2] < 0 ) { t[2] = -t[2]; pow = -1; }
                    else { pow = 1; }
                    for ( jj = 0; jj < t[2]; jj++ ) {
                        r = d->factors[j].where = (WORD *)Malloc1(10*sizeof(WORD),"factor");
                        r[0] = 9; r[1] = DOTPRODUCT; r[2] = 5; r[3] = t[0]; r[4] = t[1];
                        r[5] = pow; r[6] = 1; r[7] = 1; r[8] = 3; r[9] = 0;
                        d->factors[j].type = DOLTERMS;
                        d->factors[j].size = 9;
                        j++;
                    }
                    i--; t += 3;
                }
                break;
            case VECTOR:
            case DELTA:
                t = term+2; i = (term[1]-2)/2;
                while ( i > 0 ) {
                    for ( jj = 0; jj < t[1]; jj++ ) {
                        r = d->factors[j].where = (WORD *)Malloc1(9*sizeof(WORD),"factor");
                        r[0] = 8; r[1] = *term; r[2] = 4; r[3] = *t; r[4] = t[1];
                        r[5] = 1; r[6] = 1; r[7] = 3; r[8] = 0;
                        d->factors[j].type = DOLTERMS;
```

```
                        d->factors[j].size = 8;
                        j++;
                    }
                    i--; t += 2;
                }
                break;
            case INDEX:
                t = term+2; i = term[1]-2;
                while ( i > 0 ) {
                    for ( jj = 0; jj < t[1]; jj++ ) {
                        r = d->factors[j].where = (WORD *)Malloc1(8*sizeof(WORD),"factor");
                        r[0] = 7; r[1] = *term; r[2] = 3; r[3] = *t;
                        r[4] = 1; r[5] = 1; r[6] = 3; r[7] = 0;
                        d->factors[j].type = DOLTERMS;
                        d->factors[j].size = 7;
                        j++;
                    }
                    i--; t++;
                }
                break;
            default:
                if ( *term >= FUNCTION ) {
                    r = d->factors[j].where = (WORD *)Malloc1((term[1]+5)*sizeof(WORD),"factor");
                    *r++ = d->factors[j].size = term[1]+4;
                    for ( jj = 0; jj < t[1]; jj++ ) *r++ = term[jj];
                    *r++ = 1; *r++ = 1; *r++ = 3; *r = 0;
                    j++;
                }
                break;
        }
        term += term[1];
    }
#endif
/*
        #] Step 6:
        #[ Step 7: Numerical factors
*/
#ifdef STEP2
    term = buf1content;
    tstop = term + *term;
    if ( tstop[-1] == 3 && tstop[-2] == 1 && tstop[-3] == 1 ) {}
    else if ( tstop[-1] == 3 && tstop[-2] == 1 && (UWORD)(tstop[-3]) <= MAXPOSITIVE ) {
        d->factors[j].where = 0;
        d->factors[j].size  = 0;
        d->factors[j].type  = DOLNUMBER;
        d->factors[j].value = sign*tstop[-3];
        sign = 1;
        j++;
    }
    else {
        d->factors[j].where = r = (WORD *)Malloc1((tstop[-1]+2)*sizeof(WORD),"numfactor");
        d->factors[j].size  = tstop[-1]+1;
        d->factors[j].type  = DOLTERMS;
        d->factors[j].value = 0;
        i = tstop[-1];
        t = tstop - i;
        *r++ = tstop[-1]+1;
        NCOPY(r,t,i);
        *r = 0;
        if ( sign < 0 ) {
            r = d->factors[j].where;
            while ( *r ) {
                r += *r; r[-1] = -r[-1];
            }
            sign = 1;
        }
        j++;
    }
#endif
    if ( sign < 0 ) { /* Note that this guy should come first */
        for ( jj = j; jj > 0; jj-- ) {
            d->factors[jj] = d->factors[jj-1];
        }
        d->factors[0].where = 0;
        d->factors[0].size  = 0;
        d->factors[0].type  = DOLNUMBER;
        d->factors[0].value = -1;
        j++;
    }
    d->nfactors = j;
    if ( buf1content ) TermFree(buf1content,"DollarContent");
/*
        #] Step 7:
        #[ Step 8: Sorting the factors

    There are d->nfactors factors. Look which ones have a 'where'
    Sort them by bubble sort
```

```
*/
    if ( d->nfactors > 1 ) {
        WORD ***fac, j1, j2, k, ret, *s1, *s2, *s3;
        LONG **facsize, x;
        facsize = (LONG **)Malloc1((sizeof(WORD **)+sizeof(LONG *))*d->nfactors,"SortDollarFactors");
        fac = (WORD ***)(facsize+d->nfactors);
        k = 0;
        for ( j = 0; j < d->nfactors; j++ ) {
            if ( d->factors[j].where ) {
                fac[k] = &(d->factors[j].where);
                facsize[k] = &(d->factors[j].size);
                k++;
            }
        }
        if ( k > 1 ) {
            for ( j = 1; j < k; j++ ) { /* bubble sort */
                j1 = j; j2 = j1-1;
nextj1:;
                s1 = *(fac[j1]); s2 = *(fac[j2]);
                while ( *s1 && *s2 ) {
                    if ( ( ret = CompareTerms(BHEAD s2, s1, (WORD)2) ) == 0 ) {
                        s1 += *s1; s2 += *s2;
                    }
                    else if ( ret > 0 ) goto nextj;
                    else {
exch:
                        s3 = *(fac[j1]); *(fac[j1]) = *(fac[j2]); *(fac[j2]) = s3;
                        x = *(facsize[j1]); *(facsize[j1]) = *(facsize[j2]); *(facsize[j2]) = x;
                        j1--; j2--;
                        if ( j1 > 0 ) goto nextj1;
                        goto nextj;
                    }
                }
                if ( *s1 ) goto nextj;
                if ( *s2 ) goto exch;
nextj:;
            }
        }
        M_free(facsize,"SortDollarFactors");
    }
/*
        #] Step 8:
*/
#ifdef WITHPTHREADS
    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
    return(0);
}

/*
    #] DollarFactorize :
    #[ CleanDollarFactors :
*/

void CleanDollarFactors(DOLLARS d)
{
    int i;
    if ( d->nfactors > 1 ) {
        for ( i = 0; i < d->nfactors; i++ ) {
            if ( d->factors[i].where )
                M_free(d->factors[i].where,"dollar factors");
        }
    }
    if ( d->factors ) {
        M_free(d->factors,"dollar factors");
        d->factors = 0;
    }
    d->nfactors = 0;
}

/*
    #] CleanDollarFactors :
    #[ TakeDollarContent :
*/

WORD *TakeDollarContent(PHEAD WORD *dollarbuffer, WORD **factor)
{
    WORD *remain, *t;
    int pow;
/*
    We force the sign of the first term to be positive.
*/
    t = dollarbuffer; pow = 1;
    t += *t;
    if ( t[-1] < 0 ) {
        pow = 0;
        t[-1] = -t[-1];
```

```
        while ( *t ) {
            t += *t; t[-1] = -t[-1];
        }
    }
/*
    Now the GCD of the numerators and the LCM of the denominators:
*/
    if ( AN.cmod != 0 ) {
        if ( ( *factor = MakeDollarMod(BHEAD dollarbuffer,&remain) ) == 0 ) {
            Terminate(-1);
        }
        if ( pow == 0 ) {
            (*factor)[**factor-1] = -(*factor)[**factor-1];
            (*factor)[**factor-1] += AN.cmod[0];
        }
    }
    else {
        if ( ( *factor = MakeDollarInteger(BHEAD dollarbuffer,&remain) ) == 0 ) {
            Terminate(-1);
        }
        if ( pow == 0 ) {
            (*factor)[**factor-1] = -(*factor)[**factor-1];
        }
    }
    return(remain);
}

/*
    #] TakeDollarContent :
    #[ MakeDollarInteger :
*/
WORD *MakeDollarInteger(PHEAD WORD *bufin,WORD **bufout)
{
    GETBIDENTITY
    UWORD *GCDbuffer, *GCDbuffer2, *LCMbuffer, *LCMb, *LCMc;
    WORD *r, *r1, *r2, *r3, *rnext, i, k, j, *oldworkpointer, *factor;
    WORD kGCD, kLCM, kGCD2, kkLCM, jLCM, jGCD;
    CBUF *C = cbuf+AC.cbufnum;

    GCDbuffer = NumberMalloc("MakeDollarInteger");
    GCDbuffer2 = NumberMalloc("MakeDollarInteger");
    LCMbuffer = NumberMalloc("MakeDollarInteger");
    LCMb = NumberMalloc("MakeDollarInteger");
    LCMc = NumberMalloc("MakeDollarInteger");
    r = bufin;
/*
    First take the first term to load up the LCM and the GCD
*/
    r2 = r + *r;
    j = r2[-1];
    r3 = r2 - ABS(j);
    k = REDLENG(j);
    if ( k < 0 ) k = -k;
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kGCD = 0; kGCD < k; kGCD++ ) GCDbuffer[kGCD] = r3[kGCD];
    k = REDLENG(j);
    if ( k < 0 ) k = -k;
    r3 += k;
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
    r1 = r2;
/*
    Now go through the rest of the terms in this argument.
*/
    while ( *r1 ) {
        r2 = r1 + *r1;
        j = r2[-1];
        r3 = r2 - ABS(j);
        k = REDLENG(j);
        if ( k < 0 ) k = -k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( GCDbuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
/*
            GCD is already 1
*/
        }
        else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
            if ( GcdLong(BHEAD GCDbuffer,kGCD,(UWORD *)r3,k,GCDbuffer2,&kGCD2) ) {
                goto MakeDollarIntegerErr;
            }
            kGCD = kGCD2;
            for ( i = 0; i < kGCD; i++ ) GCDbuffer[i] = GCDbuffer2[i];
        }
        else {
            kGCD = 1; GCDbuffer[0] = 1;
        }
        k = REDLENG(j);
```

```
        if ( k < 0 ) k = -k;
        r3 += k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
            for ( kLCM = 0; kLCM < k; kLCM++ )
                LCMbuffer[kLCM] = r3[kLCM];
        }
        else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
            if ( GcdLong(BHEAD LCMbuffer,kLCM,(UWORD *)r3,k,LCMb,&kkLCM) ) {
                goto MakeDollarIntegerErr;
            }
            DivLong((UWORD *)r3,k,LCMb,kkLCM,LCMb,&kkLCM,LCMc,&jLCM);
            MulLong(LCMbuffer,kLCM,LCMb,kkLCM,LCMc,&jLCM);
            for ( kLCM = 0; kLCM < jLCM; kLCM++ )
                LCMbuffer[kLCM] = LCMc[kLCM];
        }
        else {} /* LCM doesn't change */
        r1 = r2;
    }
/*
    Now put the factor together: GCD/LCM
*/
    r3 = (WORD *)(GCDbuffer);
    if ( kGCD == kLCM ) {
        for ( jGCD = 0; jGCD < kGCD; jGCD++ )
            r3[jGCD+kGCD] = LCMbuffer[jGCD];
        k = kGCD;
    }
    else if ( kGCD > kLCM ) {
        for ( jGCD = 0; jGCD < kLCM; jGCD++ )
            r3[jGCD+kGCD] = LCMbuffer[jGCD];
        for ( jGCD = kLCM; jGCD < kGCD; jGCD++ )
            r3[jGCD+kGCD] = 0;
        k = kGCD;
    }
    else {
        for ( jGCD = kGCD; jGCD < kLCM; jGCD++ )
            r3[jGCD] = 0;
        for ( jGCD = 0; jGCD < kLCM; jGCD++ )
            r3[jGCD+kLCM] = LCMbuffer[jGCD];
        k = kLCM;
    }
    j = 2*k+1;
/*
    Now we have to write this to factor
*/
    factor = r1 = (WORD *)Malloc1((j+2)*sizeof(WORD),"MakeDollarInteger");
    *r1++ = j+1; r2 = r3;
    for ( i = 0; i < k; i++ ) { *r1++ = *r2++; *r1++ = *r2++; }
    *r1++ = j;
    *r1 = 0;
/*
    Next we have to take the factor out from the argument.
    This cannot be done in location, because the denominator stuff can make
    coefficients longer.

    We do this via a sort because the things may be jumbled any way and we
    do not know in advance how much space we need.
*/
    NewSort(BHEAD0);
    r = bufin;
    oldworkpointer = AT.WorkPointer;
    while ( *r ) {
        rnext = r + *r;
        j = ABS(rnext[-1]);
        r3 = rnext - j;
        r2 = oldworkpointer;
        while ( r < r3 ) *r2++ = *r++;
        j = (j-1)/2;    /* reduced length. Remember, k is the other red length */
        if ( DivRat(BHEAD (UWORD *)r3,j,GCDbuffer,k,(UWORD *)r2,&i) ) {
            goto MakeDollarIntegerErr;
        }
        i = 2*i+1;
        r2 = r2 + i;
        if ( rnext[-1] < 0 ) r2[-1] = -i;
        else                 r2[-1] =  i;
        *oldworkpointer = r2-oldworkpointer;
        AT.WorkPointer = r2;
        if ( Generator(BHEAD oldworkpointer,C->numlhs) ) {
            goto MakeDollarIntegerErr;
        }
        r = rnext;
    }
    AT.WorkPointer = oldworkpointer;
    AN.tryterm = 0; /* for now */
    EndSort(BHEAD (WORD *)bufout,2);
/*
```

```
    Cleanup
*/
    NumberFree(LCMc,"MakeDollarInteger");
    NumberFree(LCMb,"MakeDollarInteger");
    NumberFree(LCMbuffer,"MakeDollarInteger");
    NumberFree(GCDbuffer2,"MakeDollarInteger");
    NumberFree(GCDbuffer,"MakeDollarInteger");
    return(factor);

MakeDollarIntegerErr:
    NumberFree(LCMc,"MakeDollarInteger");
    NumberFree(LCMb,"MakeDollarInteger");
    NumberFree(LCMbuffer,"MakeDollarInteger");
    NumberFree(GCDbuffer2,"MakeDollarInteger");
    NumberFree(GCDbuffer,"MakeDollarInteger");
    MesCall("MakeDollarInteger");
    Terminate(-1);
    return(0);
}

/*
    #] MakeDollarInteger :
    #[ MakeDollarMod :
*/
WORD *MakeDollarMod(PHEAD WORD *buffer, WORD **bufout)
{
    GETBIDENTITY
    WORD *r, *r1, x, xx, ix, ip;
    WORD *factor, *oldworkpointer;
    int i;
    CBUF *C = cbuf+AC.cbufnum;
    r = buffer;
    x = r[*r-3];
    if ( r[*r-1] < 0 ) x += AN.cmod[0];
    if ( GetModInverses(x,(WORD)(AN.cmod[0]),&ix,&ip) ) {
        Terminate(-1);
    }
    factor = (WORD *)Malloc1(5*sizeof(WORD),"MakeDollarMod");
    factor[0] = 4; factor[1] = x; factor[2] = 1; factor[3] = 3; factor[4] = 0;
/*
    Now we have to multiply all coefficients by ix.
    This does not make things longer, but we should keep to the conventions
    of MakeDollarInteger.
*/
    NewSort(BHEAD0);
    r = buffer;
    oldworkpointer = AT.WorkPointer;
    while ( *r ) {
        r1 = oldworkpointer; i = *r;
        NCOPY(r1,r,i);
        xx = r1[-3]; if ( r1[-1] < 0 ) xx += AN.cmod[0];
        r1[-1] = (WORD)((((LONG)xx)*ix) % AN.cmod[0]);
        *r1 = 0; AT.WorkPointer = r1;
        if ( Generator(BHEAD oldworkpointer,C->numlhs) ) {
            Terminate(-1);
        }
    }
    AT.WorkPointer = oldworkpointer;
    AN.tryterm = 0; /* for now */
    EndSort(BHEAD (WORD *)bufout,2);
    return(factor);
}
/*
    #] MakeDollarMod :
    #[ GetDolNum :

    Evaluates a chain of DOLLAREXPR2 into a number
*/

int GetDolNum(PHEAD WORD *t, WORD *tstop)
{
    DOLLARS d;
    WORD num, *w;
    if ( t+3 < tstop && t[3] == DOLLAREXPR2 ) {
        d = Dollars + t[2];
#ifdef WITHPTHREADS
        {
            int nummodopt, dtype;
            dtype = -1;
            if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
                for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                    if ( t[2] == ModOptdollars[nummodopt].number ) break;
                }
                if ( nummodopt < NumModOptdollars ) {
                    dtype = ModOptdollars[nummodopt].type;
                    if ( dtype == MODLOCAL ) {
                        d = ModOptdollars[nummodopt].dstruct+AT.identity;
```

```
                    }
                    else {
                        MLOCK(ErrorMessageLock);
                        MesPrint("&Illegal attempt to use $-variable %s in module %l",
                            DOLLARNAME(Dollars,t[2]),AC.CModule);
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                    }
                }
            }
        }
#endif
        if ( d->factors == 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Attempt to use a factor of an unfactored $-variable");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        num = GetDolNum(BHEAD t+t[1],tstop);
        if ( num == 0 ) return(d->nfactors);
        if ( num > d->nfactors ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Attempt to use an nonexisting factor %d of a $-variable",num);
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        w = d->factors[num-1].where;
        if ( w == 0 ) return(d->factors[num-1].value);
        if ( w[0] == 4 && w[4] == 0 && w[3] == 3 && w[2] == 1 && w[1] > 0
        && w[1] < MAXPOSITIVE ) return(w[1]);
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Illegal type of factor number of a $-variable");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    }
    else if ( t[2] < 0 ) {
        return(-t[2]-1);
    }
    else {
        d = Dollars + t[2];
#ifdef WITHPTHREADS
        {
            int nummodopt, dtype;
            dtype = -1;
            if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
                for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                    if ( t[2] == ModOptdollars[nummodopt].number ) break;
                }
                if ( nummodopt < NumModOptdollars ) {
                    dtype = ModOptdollars[nummodopt].type;
                    if ( dtype == MODLOCAL ) {
                        d = ModOptdollars[nummodopt].dstruct+AT.identity;
                    }
                    else {
                        MLOCK(ErrorMessageLock);
                        MesPrint("&Illegal attempt to use $-variable %s in module %l",
                            DOLLARNAME(Dollars,t[2]),AC.CModule);
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                    }
                }
            }
        }
#endif
        if ( d->type == DOLZERO ) return(0);
        if ( d->type == DOLTERMS || d->type == DOLNUMBER ) {
            if ( d->where[0] == 4 && d->where[4] == 0 && d->where[3] == 3
                && d->where[2] == 1 && d->where[1] > 0
                && d->where[1] < MAXPOSITIVE ) return(d->where[1]);
            MLOCK(ErrorMessageLock);
            MesPrint("Attempt to use an nonexisting factor of a $-variable");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal type of factor number of a $-variable");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    return(0);
}

/*
    #] GetDolNum :
    #[ AddPotModdollar :
```

```
*/

void AddPotModdollar(WORD numdollar)
{
    int i, n = NumPotModdollars;
    for ( i = 0; i < n; i++ ) {
        if ( numdollar == PotModdollars[i] ) break;
    }
    if ( i >= n ) {
        *(WORD *)FromList(&AC.PotModDolList) = numdollar;
    }
}

/*
    #] AddPotModdollar :
*/
```

4.28 evaluate.c

```
/*
    #[ includes :
*/

#include "form3.h"
#include <stdarg.h>
#include <gmp.h>
#include <mpfr.h>

#define EXTRAPRECISION 4

#define RND MPFR_RNDN

#define auxr1 (((mpfr_t *)(AT.auxr_))[0])
#define auxr2 (((mpfr_t *)(AT.auxr_))[1])
#define auxr3 (((mpfr_t *)(AT.auxr_))[2])
#define auxr4 (((mpfr_t *)(AT.auxr_))[3])
#define auxr5 (((mpfr_t *)(AT.auxr_))[4])

mpf_t ln2;

int PackFloat(WORD *,mpf_t);
int UnpackFloat(mpf_t,WORD *);
void RatToFloat(mpf_t, UWORD *, int);
void FormtoZ(mpz_t,UWORD *,WORD);

/*
    #] includes :
    #[ mpfr_ :
        #[ IntegerToFloatr :

        Converts a Form long integer to a mpfr_ float of default size.
        We assume that sizeof(unsigned long int) = 2*sizeof(UWORD).
*/

void IntegerToFloatr(mpfr_t result, UWORD *formlong, int longsize)
{
    mpz_t z;
    mpz_init(z);
    FormtoZ(z,formlong,longsize);
    mpfr_set_z(result,z,RND);
    mpz_clear(z);
}

/*
        #] IntegerToFloatr :
        #[ RatToFloatr :

        Converts a Form rational to a gmp float of default size.
*/

void RatToFloatr(mpfr_t result, UWORD *formrat, int ratsize)
{
    GETIDENTITY
    int nnum, nden;
    UWORD *num, *den;
    int sgn = 0;
    if ( ratsize < 0 ) { ratsize = -ratsize; sgn = 1; }
    nnum = nden = (ratsize-1)/2;
    num = formrat; den = formrat+nnum;
    while ( nnum > 1 && num[nnum-1] == 0 ) { nnum--; }
    while ( nden > 1 && den[nden-1] == 0 ) { nden--; }
    IntegerToFloatr(auxr4,num,nnum);
    IntegerToFloatr(auxr5,den,nden);
```

```
    mpfr_div(result,auxr4,auxr5,RND);
    if ( sgn > 0 ) mpfr_neg(result,result,RND);
}

/*
        #] RatToFloatr :
        #[ SetfFloatPrecision :

    The AC.DefaultPrecision is in bits.
    prec is in limbs.
    fprec is NOT in bits for mpfr_ which is slightly less than in mpf_
    We also set the auxr1,auxr2,auxr3,auxr4,auxr5 variables.
*/

void SetfFloatPrecision(LONG prec)
{
/*
    mpfr_prec_t fprec = prec * mp_bits_per_limb - EXTRAPRECISION;
*/
    mpfr_prec_t fprec = prec + 1;
    mpfr_set_default_prec(fprec);
#ifdef WITHPTHREADS
    int totnum = AM.totalnumberofthreads, id;
    mpfr_t *a;
  #ifdef WITHSORTBOTS
    totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
  #endif
    if ( AB[0]->T.auxr_ ) {
        for ( id = 0; id < totnum; id++ ) {
            a = (mpfr_t *)AB[id]->T.auxr_;
            mpfr_clears(a[0],a[1],a[2],a[3],a[4],(mpfr_ptr)0);
            M_free(AB[id]->T.auxr_,"AB[id]->T.auxr_");
            AB[id]->T.auxr_ = 0;
        }
    }
    for ( id = 0; id < totnum; id++ ) {
        AB[id]->T.auxr_ = (void *)Malloc1(sizeof(mpfr_t)*5,"AB[id]->T.auxr_");
        a = (mpfr_t *)AB[id]->T.auxr_;
/*
        We work here with a[0] etc because the aux1 etc contain B which
        in the current routine would be AB[0] only
*/
        mpfr_inits2(fprec,a[0],a[1],a[2],a[3],a[4],(mpfr_ptr)0);
    }
#else
    if ( AT.auxr_ ) {
        mpfr_clears(auxr1,auxr2,auxr3,auxr4,auxr5,(mpfr_ptr)0);
        M_free(AT.auxr_,"AT.auxr_");
        AT.auxr_ = 0;
    }
    AT.auxr_ = (void *)Malloc1(sizeof(mpfr_t)*5,"AT.auxr_");
    mpfr_inits2(fprec,auxr1,auxr2,auxr3,auxr4,auxr5,(mpfr_ptr)0);
#endif
}

/*
        #] SetfFloatPrecision :
        #[ ClearfFloat :
*/

void ClearfFloat(void)
{
#ifdef WITHPTHREADS
    int totnum = AM.totalnumberofthreads, id;
    mpfr_t *a;
  #ifdef WITHSORTBOTS
    totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
  #endif
    if ( AB[0]->T.auxr_ ) {
        for ( id = 0; id < totnum; id++ ) {
            a = (mpfr_t *)AB[id]->T.auxr_;
            mpfr_clears(a[0],a[1],a[2],a[3],a[4],(mpfr_ptr)0);
            M_free(AB[id]->T.auxr_,"AB[id]->T.auxr_");
            AB[id]->T.auxr_ = 0;
        }
/*
        M_free(AB[0]->T.auxr_,"AB[0]->T.auxr_");
        AB[0]->T.auxr_ = 0;
*/
    }
#else
    if ( AT.auxr_ ) {
        mpfr_clears(auxr1,auxr2,auxr3,auxr4,auxr5,(mpfr_ptr)0);
        M_free(AT.auxr_,"AT.auxr_");
        AT.auxr_ = 0;
    }
#endif
```

```
/*
    if ( AS.delta_1 ) {
        mpf_clear(ln2);
        AS.delta_1 = 0;
    }
*/
}

/*
        #] ClearFloat :
        #[ GetFloatArgument :

        Convert an argument to an mpfr float if possible.
        Return value: 0: was converted successfully.
                     -1: could not convert.
        Note: arguments with more than one term should be evaluated first
        inside an Argument environment. If there is still more than one
        term remaining we get the code: "could not convert".
        par tells which argument we need. If it is negative it should also
        be the last argument.
*/

int GetFloatArgument(PHEAD mpfr_t f_out,WORD *fun,int par)
{
    WORD *term, *tn, *t, *tstop, first, ncoef, *arg, *argp, abspar;
    arg = fun+FUNHEAD;
    abspar = ABS(par);
    while ( arg < fun+fun[1] && abspar > 1 ) { abspar--; NEXTARG(arg) }
    if ( arg >= fun+fun[1] || abspar!= 1 ) return(-1);
    if ( par < 0 ) {
        argp = arg; NEXTARG(argp); if ( argp != fun+fun[1] ) return(-1);
    }
    if ( *arg < 0 ) {
        if ( *arg == -SNUMBER ) {
            mpfr_set_si(f_out,(LONG)(arg[1]),RND);
        }
        else if ( *arg == -SYMBOL && ( arg[1] == PISYMBOL ) ) {
            mpfr_const_pi(f_out,RND);
        }
        else if ( *arg == -INDEX && arg[1] >= 0 && arg[1] < AM.OffsetIndex ) {
            mpfr_set_ui(f_out,(ULONG)(arg[1]),RND);
        }
        else { return(-1); }
        return(0);
    }
    else if ( arg[0] != arg[ARGHEAD]+ARGHEAD ) {     /* more than one term */
        return(-1);
    }
    term = arg+ARGHEAD;
    tn = term+*term;
    tstop = tn-ABS(tn[-1]);
    t = term+1;
    first = 1;
    while ( t <= tstop ) {
        if ( t == tstop ) { /* Fraction */
            if ( first ) RatToFloatr(f_out,(UWORD *)tstop,tn[-1]);
            else {
                RatToFloatr(auxr4,(UWORD *)tstop,tn[-1]);
                mpfr_mul(f_out,f_out,auxr4,RND);
            }
            return(0);
        }
        else if ( *t == FLOATFUN ) {
            if ( t+t[1] != tstop ) return(-1);
            if ( UnpackFloat(aux5,t) < 0 ) return(-1);
            if ( first ) {
                mpfr_set_f(f_out,aux5,RND);
                first = 0;
            }
            else {
                mpfr_set_f(auxr4,aux5,RND);
                mpfr_mul(f_out,f_out,auxr4,RND);
            }
            first = 0;
            if ( tn[-1] < 0 ) { /* change sign */
                ncoef = -tn[-1];
                mpfr_neg(f_out,f_out,RND);
            }
            else ncoef = tn[-1];
            if ( ncoef == 3 && tn[-2] == 1 && tn[-3] == 1 ) return(0);
/*
            If the argument was properly normalized we are not supposed to come here.
*/
            MLOCK(ErrorMessageLock);
            MesPrint("Unnormalized argument in GetFloatArgument: %a",*term,term);
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
```

```
        }
        else if ( t[1] == SYMBOL && t[2] == 4 && t[3] == PISYMBOL && t[4] == 1 ) {
            if ( first ) {
                mpfr_const_pi(f_out,RND);
                first = 0;
            }
            else {
                mpfr_const_pi(auxr5,RND);
                mpfr_mul(f_out,f_out,auxr5,RND);
            }
        }
        else {  /* This we do not / cannot do */
            return(-1);
        }
        t += t[1];
    }
    return(0);
}

/*
        #] GetFloatArgument :
        #[ GetPiArgument :

        Tests for sin,cos,tan whether the argument is a simple
        multiple of pi or can be reduced to such.
        Return value: -1: the answer is no.
        0-23: we have 0, 1/12*pi_,...,23/12*pi_
        With success most values can be worked out by simple means.
*/

int GetPiArgument(PHEAD WORD *arg)
{
    UWORD *coef, *co, *tco, twelve, *numer, *denom, *c, *d;
    WORD i, ii, i2, iflip, nc, nd, *t, *tn, *tstop;
    int rem;
/*
    One: determine whether there is a rational coefficient and a pi_:
*/
    if ( *arg == -SNUMBER && arg[1] == 0 ) return(0);
    if ( *arg == -SYMBOL && arg[1] == PISYMBOL ) return(12);
    if ( *arg < 0 ) return(-1);
    if ( arg[ARGHEAD] != *arg-ARGHEAD ) return(-1);
    t = arg+ARGHEAD+1;
    tn = arg+*arg; tstop = tn - ABS(tn[-1]);
    if ( *t != SYMBOL || t[1] != 4 || t[2] != PISYMBOL || t[3] != 1
        || t+t[1] != tstop ) return(-1);
/*
    The denominator must be a divisor of 12
    1: copy the coefficient
*/
    co = coef = NumberMalloc("GetPiArgument");
    tco = (UWORD *)tstop;
    i = tn[-1]; if ( i < 0 ) { i = -i; iflip = 1; } else iflip = 0;
    ii = (i-1)/2;
    twelve = 12;
    NCOPY(co,tco,i);
    co = coef;
    Mully(BHEAD co,&ii,&twelve,1);
/*
    Now the denominator should be 1
    i2 = i = (ii-1)/2;
*/
    i2 = i = ii;
    numer = co; denom = numer + i;
    if ( i > 1 ) {
        while ( i2 > 0 ) {
            if ( denom[i2--] != 0 ) {
                NumberFree(co,"GetPiArgument");
                return(-1);
            }
        }
    }
    if ( *denom != 1 ) {
        NumberFree(co,"GetPiArgument");
        return(-1);
    }
/*
    Now we need the numerator modulus 24.
*/
    if ( i == 1 ) {
        rem = *numer % 24;
    }
    else {
        c = NumberMalloc("GetPiArgument");
        d = NumberMalloc("GetPiArgument");
        twelve *= 2;
        DivLong(numer,i,&twelve,1,c,&nc,d,&nd);
```

```
        rem = *d % 24;
        NumberFree(d,"GetPiArgument");
        NumberFree(c,"GetPiArgument");
    }
    NumberFree(co,"GetPiArgument");
    if ( iflip && rem != 0 ) rem = 24-rem;
    return(rem);
}

/*
        #] GetPiArgument :
        #[ EvaluateFun :

        What we need to do is:
        1: look for a function to be treated.
        2: make sure its argument is treatable.
        3: call the proper mpfr_ function.
        4: accumulate the result.
        For some functions we have to insert 'smart' shortcuts as is
        the case with sin_(pi_) or sin_(pi_/6) of sqrt(4/9) etc.
        Otherwise we may have to insert a value for pi_ first.
        There are several types of arguments:
        a: (short) integers.
        b: rationals.
        c: floats.
        We accumulate the result(s) in auxr2. The argument comes in aux5
        and a result in auxr3 which then gets multiplied into auxr2.
        In the end we have to combine auxr2 with whatever coefficient
        existed already.

        When the float system is started we need for aux only 3 variables
        per thread. auxr1-auxr3. This should be done separately.
        The main problem is the conversion of mpfr_t to float_ and/or mpf_t
        and back.
*/


int EvaluateFun(PHEAD WORD *term, WORD level, WORD *pars)
{
    WORD *t, *tstop, *tt, *tnext, *newterm, i;
    WORD *oldworkpointer = AT.WorkPointer, nsize, nsgn;
    int retval = 0, first = 1, pimul;

    tstop = term + *term; tstop -= ABS(tstop[-1]);
    if ( AT.WorkPointer < term+*term ) AT.WorkPointer = term + *term;
    t = term+1;
    mpfr_set_ui(auxr2,1L,RND);
    while ( t < tstop ) {
        if ( pars[2] == *t ) {  /* have to do this one if possible */
TestArgument:
/*
            There must be a single argument, except for the AGM function
*/
            tnext = t+t[1]; tt = t+FUNHEAD; NEXTARG(tt);
            if( *t == SYMBOL) {
                for ( WORD* ti = t+2; ti < t+t[1]; ti+=2 ) {
                    if( ( *ti == PISYMBOL || *ti == EESYMBOL  || *ti == EMSYMBOL )
                     && ( pars[2] == ALLFUNCTIONS || pars[3] == *ti ) ) {
                        if ( *ti == PISYMBOL )
                            mpfr_const_pi(auxr3,RND);
                        else if ( *ti == EESYMBOL ) {
                            mpfr_set_ui(auxr3,1,RND);
                            mpfr_exp(auxr3,auxr3,RND);
                        }
                        else if ( *ti == EMSYMBOL )
                            mpfr_const_euler(auxr3,RND);
                        if ( ti[1] != 1 )
                            mpfr_pow_si(auxr3,auxr3,ti[1],RND);
                        mpfr_mul(auxr2,auxr2,auxr3,RND);
                        ti[1] = 0;
                    }
                }
                first = 0;
                goto nextfun;
            }
            if ( tt != tnext && *t != AGMFUNCTION ) goto nextfun;
            if ( *t == SINFUNCTION ) {
                pimul = GetPiArgument(BHEAD t+FUNHEAD);
                if ( pimul >= 0 && pimul < 24 ) {
                    if ( pimul == 0 || pimul == 12 ) goto getout;
                    if ( pimul > 12 ) { pimul = 24-pimul; nsgn = -1; }
                    else nsgn = 1;
                    if ( pimul > 6 ) pimul = 12-pimul;
                    switch ( pimul ) {
                        case 1:
label1:
                            mpfr_sqrt_ui(auxr3,3L,RND);
```

```
                            mpfr_sub_ui(auxr3,auxr3,1L,RND);
                            mpfr_sqrt_ui(auxr1,2L,RND);
                            mpfr_div_ui(auxr1,auxr1,4L,RND);
                            mpfr_mul(auxr3,auxr3,auxr1,RND);
                            if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                        case 2:
label2:
                            mpfr_div_ui(auxr2,auxr2,2L,RND);
                            if ( nsgn < 0 ) mpfr_neg(auxr2,auxr2,RND);
                            break;
                        case 3:
label3:
                            mpfr_sqrt_ui(auxr3,2L,RND);
                            mpfr_div_ui(auxr3,auxr3,2L,RND);
                            if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                        case 4:
label4:
                            mpfr_sqrt_ui(auxr3,3L,RND);
                            mpfr_div_ui(auxr3,auxr3,2L,RND);
                            if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                        case 5:
label5:
                            mpfr_sqrt_ui(auxr3,3L,RND);
                            mpfr_add_ui(auxr3,auxr3,1L,RND);
                            mpfr_sqrt_ui(auxr1,2L,RND);
                            mpfr_div_ui(auxr1,auxr1,4L,RND);
                            mpfr_mul(auxr3,auxr3,auxr1,RND);
                            if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                        case 6:
label6:
                            if ( nsgn < 0 ) mpfr_neg(auxr2,auxr2,RND);
                            break;
                    }
                    *t = 0; first = 0;
                    goto nextfun;
                }
            }
            else if ( *t == COSFUNCTION ) {
                pimul = GetPiArgument(BHEAD t+FUNHEAD);
                if ( pimul >= 0 && pimul < 24 ) {
                    if ( pimul > 12 ) pimul = 24-12;
                    if ( pimul > 6 ) { pimul = 12-pimul; nsgn = -1; }
                    else nsgn = 1;
                    if ( pimul == 6 ) goto getout;
                    switch ( pimul ) {
                        case 0: goto label6;
                        case 1: goto label5;
                        case 2: goto label4;
                        case 3: goto label3;
                        case 4: goto label2;
                        case 5: goto label1;
                    }
                }
            }
            else if ( *t == TANFUNCTION ) {
                pimul = GetPiArgument(BHEAD t+FUNHEAD);
                if ( pimul >= 0 && pimul < 24 ) {
                    if ( pimul == 6 || pimul == 18 ) goto nextfun;
                    if ( pimul == 0 || pimul == 12 ) goto getout;
                    if ( pimul > 12 ) { pimul = 24-pimul; nsgn = -1; }
                    else nsgn = 1;
                    if ( pimul > 6 ) { pimul = 12-pimul; nsgn = -nsgn; }
                    switch ( pimul ) {
                        case 1:
                            mpfr_sqrt_ui(auxr3,3L,RND);
                            mpfr_sub_ui(auxr3,auxr3,2L,RND);
                            if ( nsgn > 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                        case 2:
                            mpfr_sqrt_ui(auxr3,3L,RND);
                            mpfr_div_ui(auxr3,auxr3,3L,RND);
                            if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                        case 3:
                            break;
                        case 4:
                            mpfr_sqrt_ui(auxr3,3L,RND);
```

```
                            if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                        case 5:
                            mpfr_sqrt_ui(auxr3,3L,RND);
                            mpfr_add_ui(auxr3,auxr3,2L,RND);
                            if ( nsgn < 0 ) mpfr_neg(auxr3,auxr3,RND);
                            mpfr_mul(auxr2,auxr2,auxr3,RND);
                            break;
                    }
                    *t = 0;
                    goto nextfun;
                }
            }

            if ( *t == AGMFUNCTION ) {
                if ( GetFloatArgument(BHEAD auxr1,t,1) < 0 ) goto nextfun;
                if ( GetFloatArgument(BHEAD auxr3,t,-2) < 0 ) goto nextfun;
            }
            else if ( GetFloatArgument(BHEAD auxr1,t,-1) < 0 ) goto nextfun;
            nsgn = mpfr_sgn(auxr1);

            switch ( *t ) {
                case SQRTFUNCTION:
                    if ( nsgn < 0 ) goto nextfun;
                    if ( nsgn == 0 ) goto getout;
                    else mpfr_sqrt(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case LNFUNCTION:
                    if ( nsgn <= 0 ) goto nextfun;
                    else mpfr_log(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case LI2FUNCTION: /* should be between -1 and +1 */
                    mpfr_abs(auxr3,auxr1,RND);
                    if ( mpfr_cmp_ui(auxr3,1L) > 0 ) goto nextfun;
                    mpfr_li2(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case GAMMAFUN:
/*
                    We cannot do this when the argument is a non-positive integer
*/
                    if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] <= 0 ) goto nextfun;
                    if ( t[FUNHEAD] == t[1]-FUNHEAD && ABS(t[t[1]-1]) ==
                        t[FUNHEAD]-ARGHEAD-1 && t[t[1]-1] < 0 ) {
                        nsize = (-t[t[1]-1]-1)/2;
                        if ( t[t[1]-1-nsize] == 1 ) {
                            for ( i = 1; i < nsize; i++ ) {
                                if ( t[t[1]-1-nsize+i] != 0 ) break;
                            }
                            if ( i >= nsize ) goto nextfun;
                        }
                    }
                    mpfr_gamma(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case AGMFUNCTION:
                    mpfr_agm(auxr3,auxr1,auxr3,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case SINHFUNCTION:
                    mpfr_sinh(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case COSHFUNCTION:
                    mpfr_cosh(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case TANHFUNCTION:
                    mpfr_tanh(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case ASINHFUNCTION:
                    mpfr_asinh(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case ACOSHFUNCTION:
                    mpfr_acosh(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case ATANHFUNCTION:
                    if ( nsgn == 0 ) goto getout;
                    mpfr_abs(auxr3,auxr1,RND);
                    if ( mpfr_cmp_ui(auxr3,1L) > 0 ) goto nextfun;
                    mpfr_atanh(auxr3,auxr1,RND);
```

```
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case ASINFUNCTION:
                    if ( nsgn == 0 ) goto getout;
                    mpfr_abs(auxr3,auxr1,RND);
                    if ( mpfr_cmp_ui(auxr3,1L) > 0 ) goto nextfun;
                    mpfr_asin(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case ACOSFUNCTION:
                    if ( nsgn == 0 ) goto getout;
                    mpfr_abs(auxr3,auxr1,RND);
                    if ( mpfr_cmp_ui(auxr3,1L) > 0 ) goto nextfun;
                    mpfr_acos(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case ATANFUNCTION:
                    mpfr_atan(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case SINFUNCTION:
                    mpfr_sin(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case COSFUNCTION:
                    mpfr_cos(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                case TANFUNCTION:
                    mpfr_tan(auxr3,auxr1,RND);
                    mpfr_mul(auxr2,auxr2,auxr3,RND);
                break;
                default:
                    goto nextfun;
                break;
            }
            first = 0;
            *t = 0;
            goto nextfun;
        }
        else if ( pars[2] == ALLFUNCTIONS ) {
/*
            Now we have to test whether this is one of our functions
*/
            if ( t[1] == FUNHEAD ) goto nextfun;
            switch ( *t ) {
                case SQRTFUNCTION:
                case LNFUNCTION:
                case LI2FUNCTION:
                case LINFUNCTION:
                case ASINFUNCTION:
                case ACOSFUNCTION:
                case ATANFUNCTION:
                case SINHFUNCTION:
                case COSHFUNCTION:
                case TANHFUNCTION:
                case ASINHFUNCTION:
                case ACOSHFUNCTION:
                case ATANHFUNCTION:
                case SINFUNCTION:
                case COSFUNCTION:
                case TANFUNCTION:
                case AGMFUNCTION:
                case SYMBOL:
                    goto TestArgument;
                case MZV:
                case EULER:
                case MZVHALF:
                    goto nextfun;
                default:
                    MLOCK(ErrorMessageLock);
                    MesPrint("Function in evaluate statement not yet implemented.");
                    MUNLOCK(ErrorMessageLock);
                    break;
            }
        }
        else goto nextfun;
nextfun:
        t += t[1];
    }
    if ( first == 1 ) return(Generator(BHEAD term,level));
    mpfr_get_f(aux4,auxr2,RND);
/*
    Step 3:
    Now the regular coefficient, if it is not 1/1.
    We have two cases: size +- 3, or bigger.
*/
```

```
    nsize = term[*term-1];
    if ( nsize < 0 ) { nsize = -nsize; nsgn = -1; }
    else nsgn = 1;
    if ( aux4->_mp_size < 0 ) {
        aux4->_mp_size = -aux4->_mp_size;
        nsgn = -nsgn;
    }

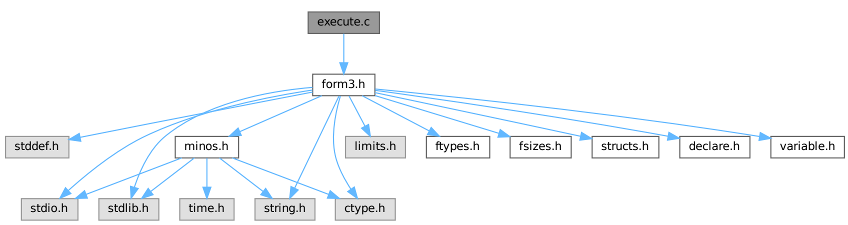
    if ( nsize == 3 ) {
        if ( tstop[0] != 1 ) {
            mpf_mul_ui(aux4,aux4,(ULONG)((UWORD)tstop[0]));
        }
        if ( tstop[1] != 1 ) {
            mpf_div_ui(aux4,aux4,(ULONG)((UWORD)tstop[1]));
        }
    }
    else {
        RatToFloat(aux5,(UWORD *)tstop,nsize);
        mpf_mul(aux4,aux4,aux5);
    }
/*
    Now we have to locate possible other float_ functions.
    Note possible incompatibilities between the mpf and mpfr formats.
*/
    t = term+1;
    while ( t < tstop ) {
        if ( *t == FLOATFUN ) {
            UnpackFloat(aux5,t);
            mpf_mul(aux4,aux4,aux5);
        }
        t += t[1];
    }
/*
    Now we should compose the new term in the WorkSpace.
*/
    t = term+1;
    newterm = AT.WorkPointer;
    tt = newterm+1;
    while ( t < tstop ) {
        if ( *t == 0 || *t == FLOATFUN ) t += t[1];
        else {
            i = t[1]; NCOPY(tt,t,i);
        }
    }
    PackFloat(tt,aux4);
    tt += tt[1];
    *tt++ = 1; *tt++ = 1; *tt++ = 3*nsgn;
    *newterm = tt-newterm;
    AT.WorkPointer = tt;
    retval = Generator(BHEAD newterm,level);
getout:
    AT.WorkPointer = oldworkpointer;
    return(retval);
}

/*
        #] EvaluateFun :
    #] mpfr_ :
*/
```

4.29 execute.c File Reference

`#include "form3.h"`

Include dependency graph for execute.c:



Macros

- #define CURRENTBRACKET 1
- #define BRACKETCURRENTEXPR 2
- #define BRACKETOTHEREXPR 3
- #define NOBRACKETACTIVE 4

Functions

- WORD CleanExpr (WORD par)
- WORD PopVariables (void)
- void MakeGlobal (void)
- void TestDrop (void)
- void PutInVflags (WORD nexpr)
- WORD DoExecute (WORD par, WORD skip)
- WORD PutBracket (PHEAD WORD ∗termin)
- void SpecialCleanup (PHEAD0)
- void SetMods (void)
- void UnSetMods (void)
- void ExchangeExpressions (int num1, int num2)
- int GetFirstBracket (WORD ∗term, int num)
- int GetFirstTerm (WORD ∗term, int num, int pre)
- int GetContent (WORD ∗content, int num)
- int CleanupTerm (WORD ∗term)
- WORD ContentMerge (PHEAD WORD ∗content, WORD ∗term)
- LONG TermsInExpression (WORD num)
- LONG SizeOfExpression (WORD num)
- void UpdatePositions (void)
- LONG CountTerms1 (PHEAD0)
- LONG TermsInBracket (PHEAD WORD ∗term, WORD level)

4.29.1 Detailed Description

The routines that start the execution phase of a module. It also contains the routines for placing the bracket subterm.

Definition in file execute.c.

4.29.2 Macro Definition Documentation

4.29.2.1 CURRENTBRACKET

```
#define CURRENTBRACKET 1
```

Definition at line 2564 of file execute.c.

4.29.2.2 BRACKETCURRENTEXPR

```
#define BRACKETCURRENTEXPR 2
```

Definition at line 2565 of file execute.c.

4.29.2.3 BRACKETOTHEREXPR

```
#define BRACKETOTHEREXPR 3
```

Definition at line 2566 of file execute.c.

4.29.2.4 NOBRACKETACTIVE

```
#define NOBRACKETACTIVE 4
```

Definition at line 2567 of file execute.c.

4.29.3 Function Documentation

4.29.3.1 CleanExpr()

```
WORD CleanExpr (
            WORD par )
```

Definition at line 47 of file execute.c.

4.29.3.2 PopVariables()

```
WORD PopVariables (
            void )
```

Definition at line 211 of file execute.c.

4.29.3.3 MakeGlobal()

```
void MakeGlobal (
            void )
```

Definition at line 382 of file execute.c.

4.29.3.4 TestDrop()

```
void TestDrop (
            void )
```

Definition at line 483 of file execute.c.

4.29.3.5 PutInVflags()

```
void PutInVflags (
            WORD nexpr )
```

Definition at line 561 of file execute.c.

4.29.3.6 DoExecute()

```
WORD DoExecute (
            WORD par,
            WORD skip )
```

Definition at line 615 of file execute.c.

4.29.3.7 PutBracket()

```
WORD PutBracket (
            PHEAD WORD * termin )
```

Definition at line 1111 of file execute.c.

4.29.3.8 SpecialCleanup()

```
void SpecialCleanup (
            PHEAD0 )
```

Definition at line 1623 of file execute.c.

4.29.3.9 SetMods()

```
void SetMods (
            void )
```

Definition at line 1637 of file execute.c.

4.29.3.10 UnSetMods()

```
void UnSetMods (
            void )
```

Definition at line 1655 of file execute.c.

4.29.3.11 ExchangeExpressions()

```
void ExchangeExpressions (
            int num1,
            int num2 )
```

Definition at line 1670 of file execute.c.

4.29.3.12 GetFirstBracket()

```
int GetFirstBracket (
            WORD * term,
            int num )
```

Definition at line 1744 of file execute.c.

4.29.3.13 GetFirstTerm()

```
int GetFirstTerm (
            WORD * term,
            int num,
            int pre )
```

Gets the first term of an expression. Routine should be thread safe.

Parameters

out	*term*	Pointer to the buffer where the term should be written.
in	*num*	The expression number from which to get the term.
in	*pre*	Denotes a pre-processor call (1) or not (0). Used to determine the correct source file for the expression.

Returns

First WORD of term, a negative value denotes an error.

Definition at line 1863 of file execute.c.

References FiLe::handle, VaRrEnUm::lo, and ReNuMbEr::symb.

4.29.3.14 GetContent()

```
int GetContent (
            WORD * content,
            int num )
```

Definition at line 1971 of file execute.c.

4.29.3.15 CleanupTerm()

```
int CleanupTerm (
            WORD * term )
```

Definition at line 2088 of file execute.c.

4.29.3.16 ContentMerge()

```
WORD ContentMerge (
            PHEAD WORD * content,
            WORD * term )
```

Definition at line 2112 of file execute.c.

4.29.3.17 TermsInExpression()

```
LONG TermsInExpression (
            WORD num )
```

Definition at line 2376 of file execute.c.

4.29.3.18 SizeOfExpression()

```
LONG SizeOfExpression (
            WORD num )
```

Definition at line 2388 of file execute.c.

4.29.3.19 UpdatePositions()

```
void UpdatePositions (
            void )
```

Definition at line 2400 of file execute.c.

4.29.3.20 CountTerms1()

```
LONG CountTerms1 (
            PHEAD0 )
```

Definition at line 2461 of file execute.c.

4.29.3.21 TermsInBracket()

```
LONG TermsInBracket (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 2569 of file execute.c.

4.30 execute.c

Go to the documentation of this file.

```
00001
00006 /* #[ License : */
00007 /*
00008  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009  *   When using this file you are requested to refer to the publication
00010  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011  *   This is considered a matter of courtesy as the development was paid
00012  *   for by FOM the Dutch physics granting agency and we would like to
00013  *   be able to track its scientific use to convince FOM of its value
00014  *   for the community.
00015  *
00016  *   This file is part of FORM.
00017  *
00018  *   FORM is free software: you can redistribute it and/or modify it under the
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00020  *   Foundation, either version 3 of the License, or (at your option) any later
00021  *   version.
00022  *
00023  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025  *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
00026  *   details.
00027  *
00028  *   You should have received a copy of the GNU General Public License along
00029  *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00030  */
00031 /* #] License : */
```

```
/*
    #[ Includes : execute.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ DoExecute :
        #[ CleanExpr :

        par == 1 after .store or .clear
        par == 0 after .sort
*/

WORD CleanExpr(WORD par)
{
    GETIDENTITY
    WORD j, n, i;
    POSITION length;
    EXPRESSIONS e_in, e_out, e;
    int numhid = 0;
    NAMENODE *node;
    n = NumExpressions;
    j = 0;
    e_in = e_out = Expressions;
    if ( n > 0 ) { do {
        e_in->vflags &= ~( TOBEFACTORED | TOBEUNFACTORED );
        if ( par ) {
            if ( e_in->renumlists ) {
                if ( e_in->renumlists != AN.dummyrenumlist )
                        M_free(e_in->renumlists,"Renumber-lists");
                e_in->renumlists = 0;
            }
            if ( e_in->renum ) {
                M_free(e_in->renum,"Renumber"); e_in->renum = 0;
            }
        }
        if ( e_in->status == HIDDENLEXPRESSION
        || e_in->status == HIDDENGEXPRESSION ) numhid++;
        switch ( e_in->status ) {
            case SPECTATOREXPRESSION:
            case LOCALEXPRESSION:
            case HIDDENLEXPRESSION:
                if ( par ) {
                    AC.exprnames->namenode[e_in->node].type = CDELETE;
                    AC.DidClean = 1;
                    if ( e_in->status != HIDDENLEXPRESSION )
                        ClearBracketIndex(e_in-Expressions);
                    break;
                }
                /* fall through */
            case GLOBALEXPRESSION:
            case HIDDENGEXPRESSION:
                if ( par ) {
#ifdef WITHMPI
                    /*
                     * Broadcast the global expression from the master to the all workers.
                     */
                    if ( PF_BroadcastExpr(e_in, e_in->status == HIDDENGEXPRESSION ? AR.hidefile : AR.outfile) ) return -1;
                    if ( PF.me == MASTER ) {
#endif
                    e = e_in;
                    i = n-1;
                    while ( --i >= 0 ) {
                        e++;
                        if ( e_in->status == HIDDENGEXPRESSION ) {
                            if ( e->status == HIDDENGEXPRESSION
                            || e->status == HIDDENLEXPRESSION ) break;
                        }
                        else {
                            if ( e->status == GLOBALEXPRESSION
                            || e->status == LOCALEXPRESSION ) break;
                        }
                    }
#ifdef WITHMPI
                    }
                    else {
                        /*
                         * On the slaves, the broadcast expression is sitting at the end of the file.
                         */
                        e = e_in;
                        i = -1;
                    }
#endif
                    if ( i >= 0 ) {
```

```
                        DIFPOS(length,e->onfile,e_in->onfile);
                    }
                    else {
                        FILEHANDLE *f = e_in->status == HIDDENGEXPRESSION ? AR.hidefile : AR.outfile;
                        if ( f->handle < 0 ) {
                            SETBASELENGTH(length,TOLONG(f->POfull)
                             - TOLONG(f->PObuffer)
                             - BASEPOSITION(e_in->onfile));
                        }
                        else {
                            SeekFile(f->handle,&(f->filesize),SEEK_SET);
                            DIFPOS(length,f->filesize,e_in->onfile);
                        }
                    }
                    if ( ToStorage(e_in,&length) ) {
                        return(MesCall("CleanExpr"));
                    }
                    e_in->status = STOREDEXPRESSION;
                    if ( e_in->status != HIDDENGEXPRESSION )
                        ClearBracketIndex(e_in-Expressions);
                }
                /* fall through */
            case SKIPLEXPRESSION:
            case DROPLEXPRESSION:
            case DROPHLEXPRESSION:
            case DROPGEXPRESSION:
            case DROPHGEXPRESSION:
            case STOREDEXPRESSION:
            case DROPSPECTATOREXPRESSION:
                if ( e_out != e_in ) {
                    node = AC.exprnames->namenode + e_in->node;
                    node->number = e_out - Expressions;

                    e_out->onfile = e_in->onfile;
                    e_out->size = e_in->size;
                    e_out->printflag = 0;
                    if ( par ) e_out->status = STOREDEXPRESSION;
                    else e_out->status = e_in->status;
                    e_out->name = e_in->name;
                    e_out->node = e_in->node;
                    e_out->renum = e_in->renum;
                    e_out->renumlists = e_in->renumlists;
                    e_out->counter = e_in->counter;
                    e_out->hidelevel = e_in->hidelevel;
                    e_out->inmem = e_in->inmem;
                    e_out->bracketinfo = e_in->bracketinfo;
                    e_out->newbracketinfo = e_in->newbracketinfo;
                    e_out->numdummies = e_in->numdummies;
                    e_out->numfactors = e_in->numfactors;
                    e_out->vflags = e_in->vflags;
                    e_out->uflags = e_in->uflags;
                    e_out->sizeprototype = e_in->sizeprototype;
                }
#ifdef PARALLELCODE
                e_out->partodo = 0;
#endif
                e_out++;
                j++;
                break;
            case DROPPEDEXPRESSION:
                break;
            default:
                AC.exprnames->namenode[e_in->node].type = CDELETE;
                AC.DidClean = 1;
                break;
        }
        e_in++;
    } while ( --n > 0 ); }
    UpdateMaxSize();
    NumExpressions = j;
    if ( numhid == 0 && AR.hidefile->PObuffer ) {
        if ( AR.hidefile->handle >= 0 ) {
            CloseFile(AR.hidefile->handle);
            remove(AR.hidefile->name);
            AR.hidefile->handle = -1;
        }
        AR.hidefile->POfull =
        AR.hidefile->POfill = AR.hidefile->PObuffer;
        PUTZERO(AR.hidefile->POposition);
    }
    FlushSpectators();
    return(0);
}

/*
        #] CleanExpr :
        #[ PopVariables :
```

```
    Pops the local variables from the tables.
    The Expressions are reprocessed and their tables are compactified.

*/

WORD PopVariables(void)
{
    GETIDENTITY
    WORD i, j, retval;
    UBYTE *s;

    retval = CleanExpr(1);
    ResetVariables(1);

    if ( AC.DidClean ) CompactifyTree(AC.exprnames,EXPRNAMES);

    AC.CodesFlag = AM.gCodesFlag;
    AC.NamesFlag = AM.gNamesFlag;
    AC.StatsFlag = AM.gStatsFlag;
#ifdef WITHFLOAT
    AC.MaxWeight = AM.gMaxWeight;
    AC.DefaultPrecision = AM.gDefaultPrecision;
#endif
    AC.OldFactArgFlag = AM.gOldFactArgFlag;
    AC.TokensWriteFlag = AM.gTokensWriteFlag;
    AC.extrasymbols = AM.gextrasymbols;
    if ( AC.extrasym ) { M_free(AC.extrasym,"extrasym"); AC.extrasym = 0; }
    i = 1; s = AM.gextrasym; while ( *s ) { s++; i++; }
    AC.extrasym = (UBYTE *)Malloc1(i*sizeof(UBYTE),"extrasym");
    for ( j = 0; j < i; j++ ) AC.extrasym[j] = AM.gextrasym[j];
    AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers;
    AO.IndentSpace = AM.gIndentSpace;
    AC.lUnitTrace = AM.gUnitTrace;
    AC.lDefDim = AM.gDefDim;
    AC.lDefDim4 = AM.gDefDim4;
    if ( AC.halfmod ) {
        if ( AC.ncmod == AM.gncmod && AC.modmode == AM.gmodmode ) {
            j = ABS(AC.ncmod);
            while ( --j >= 0 ) {
                if ( AC.cmod[j] != AM.gcmod[j] ) break;
            }
            if ( j >= 0 ) {
                M_free(AC.halfmod,"halfmod");
                AC.halfmod = 0; AC.nhalfmod = 0;
            }
        }
        else {
            M_free(AC.halfmod,"halfmod");
            AC.halfmod = 0; AC.nhalfmod = 0;
        }
    }
    if ( AC.modinverses ) {
        if ( AC.ncmod == AM.gncmod && AC.modmode == AM.gmodmode ) {
            j = ABS(AC.ncmod);
            while ( --j >= 0 ) {
                if ( AC.cmod[j] != AM.gcmod[j] ) break;
            }
            if ( j >= 0 ) {
                M_free(AC.modinverses,"modinverses");
                AC.modinverses = 0;
            }
        }
        else {
            M_free(AC.modinverses,"modinverses");
            AC.modinverses = 0;
        }
    }
    AN.ncmod = AC.ncmod = AM.gncmod;
    AC.npowmod = AM.gnpowmod;
    AC.modmode = AM.gmodmode;
    if ( ( ( AC.modmode & INVERSETABLE ) != 0 ) && ( AC.modinverses == 0 ) )
            MakeInverses();
    AC.funpowers = AM.gfunpowers;
    AC.lPolyFun = AM.gPolyFun;
    AC.lPolyFunInv = AM.gPolyFunInv;
    AC.lPolyFunType = AM.gPolyFunType;
    AC.lPolyFunExp = AM.gPolyFunExp;
    AR.PolyFunVar = AC.lPolyFunVar = AM.gPolyFunVar;
    AC.lPolyFunPow = AM.gPolyFunPow;
    AC.parallelflag = AM.gparallelflag;
    AC.ProcessBucketSize = AC.mProcessBucketSize = AM.gProcessBucketSize;
    AC.properorderflag = AM.gproperorderflag;
    AC.ThreadBucketSize = AM.gThreadBucketSize;
    AC.ThreadStats = AM.gThreadStats;
    AC.FinalStats = AM.gFinalStats;
    AC.OldGCDflag = AM.gOldGCDflag;
```

```
    AC.WTimeStatsFlag = AM.gWTimeStatsFlag;
    AC.ThreadsFlag = AM.gThreadsFlag;
    AC.ThreadBalancing = AM.gThreadBalancing;
    AC.ThreadSortFileSynch = AM.gThreadSortFileSynch;
    AC.ProcessStats = AM.gProcessStats;
    AC.OldParallelStats = AM.gOldParallelStats;
    AC.IsFortran90 = AM.gIsFortran90;
    AC.SizeCommuteInSet = AM.gSizeCommuteInSet;
    PruneExtraSymbols(AM.gnumextrasym);

    if ( AC.Fortran90Kind ) {
        M_free(AC.Fortran90Kind,"Fortran90 Kind");
        AC.Fortran90Kind = 0;
    }
    if ( AM.gFortran90Kind ) {
        AC.Fortran90Kind = strDup1(AM.gFortran90Kind,"Fortran90 Kind");
    }
    if ( AC.ThreadsFlag && AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
    {
        UWORD *p, *m;
        p = AM.gcmod;
        m = AC.cmod;
        j = ABS(AC.ncmod);
        NCOPY(m,p,j);
        p = AM.gpowmod;
        m = AC.powmod;
        j = AC.npowmod;
        NCOPY(m,p,j);
        if ( AC.DirtPow ) {
            if ( MakeModTable() ) {
                MesPrint("===No printing in powers of generator");
            }
            AC.DirtPow = 0;
        }
    }
    {
        WORD *p, *m;
        p = AM.gUniTrace;
        m = AC.lUniTrace;
        j = 4;
        NCOPY(m,p,j);
    }
    AC.Cnumpows = AM.gCnumpows;
    AC.OutputMode = AM.gOutputMode;
    AC.OutputSpaces = AM.gOutputSpaces;
    AC.OutNumberType = AM.gOutNumberType;
    AR.SortType = AC.SortType = AM.gSortType;
    AC.ShortStatsMax = AM.gShortStatsMax;
/*
    Now we have to clean up the commutation properties
*/
    for ( i = 0; i < NumFunctions; i++ ) functions[i].flags &= ~COULDCOMMUTE;
    if ( AC.CommuteInSet ) {
        WORD *g, *gg;
        g = AC.CommuteInSet;
        while ( *g ) {
            gg = g+1; g += *g;
            while ( gg < g ) {
                if ( *gg <= GAMMASEVEN && *gg >= GAMMA ) {
                    functions[GAMMA-FUNCTION].flags |= COULDCOMMUTE;
                    functions[GAMMAI-FUNCTION].flags |= COULDCOMMUTE;
                    functions[GAMMAFIVE-FUNCTION].flags |= COULDCOMMUTE;
                    functions[GAMMASIX-FUNCTION].flags |= COULDCOMMUTE;
                    functions[GAMMASEVEN-FUNCTION].flags |= COULDCOMMUTE;
                }
                else {
                    functions[*gg-FUNCTION].flags |= COULDCOMMUTE;
                }
            }
        }
    }
/*
    Clean up the dictionaries.
*/
    for ( i = AO.NumDictionaries-1; i >= AO.gNumDictionaries; i-- ) {
        RemoveDictionary(AO.Dictionaries[i]);
        M_free(AO.Dictionaries[i],"Dictionary");
    }
    for( ; i >= 0; i-- ) {
        ShrinkDictionary(AO.Dictionaries[i]);
    }
    AO.NumDictionaries = AO.gNumDictionaries;
    return(retval);
}

/*
        #] PopVariables :
```

```
        #[ MakeGlobal :
*/

void MakeGlobal(void)
{
    WORD i, j, *pp, *mm;
    UWORD *p, *m;
    UBYTE *s;
    Globalize(0);

    AM.gCodesFlag = AC.CodesFlag;
    AM.gNamesFlag = AC.NamesFlag;
    AM.gStatsFlag = AC.StatsFlag;
#ifdef WITHFLOAT
    AM.gMaxWeight = AC.MaxWeight;
    AM.gDefaultPrecision = AC.DefaultPrecision;
#endif
    AM.gOldFactArgFlag = AC.OldFactArgFlag;
    AM.gextrasymbols = AC.extrasymbols;
    if ( AM.gextrasym ) { M_free(AM.gextrasym,"extrasym"); AM.gextrasym = 0; }
    i = 1; s = AC.extrasym; while ( *s ) { s++; i++; }
    AM.gextrasym = (UBYTE *)Malloc1(i*sizeof(UBYTE),"extrasym");
    for ( j = 0; j < i; j++ ) AM.gextrasym[j] = AC.extrasym[j];
    AM.gTokensWriteFlag= AC.TokensWriteFlag;
    AM.gNoSpacesInNumbers = AO.NoSpacesInNumbers;
    AM.gIndentSpace = AO.IndentSpace;
    AM.gUnitTrace = AC.lUnitTrace;
    AM.gDefDim = AC.lDefDim;
    AM.gDefDim4 = AC.lDefDim4;
    AM.gncmod = AC.ncmod;
    AM.gnpowmod = AC.npowmod;
    AM.gmodmode = AC.modmode;
    AM.gCnumpows = AC.Cnumpows;
    AM.gOutputMode = AC.OutputMode;
    AM.gOutputSpaces = AC.OutputSpaces;
    AM.gOutNumberType = AC.OutNumberType;
    AM.gfunpowers = AC.funpowers;
    AM.gPolyFun = AC.lPolyFun;
    AM.gPolyFunInv = AC.lPolyFunInv;
    AM.gPolyFunType = AC.lPolyFunType;
    AM.gPolyFunExp = AC.lPolyFunExp;
    AM.gPolyFunVar = AC.lPolyFunVar;
    AM.gPolyFunPow = AC.lPolyFunPow;
    AM.gparallelflag = AC.parallelflag;
    AM.gProcessBucketSize = AC.ProcessBucketSize;
    AM.gproperorderflag = AC.properorderflag;
    AM.gThreadBucketSize = AC.ThreadBucketSize;
    AM.gThreadStats = AC.ThreadStats;
    AM.gFinalStats = AC.FinalStats;
    AM.gOldGCDflag = AC.OldGCDflag;
    AM.gWTimeStatsFlag = AC.WTimeStatsFlag;
    AM.gThreadsFlag = AC.ThreadsFlag;
    AM.gThreadBalancing = AC.ThreadBalancing;
    AM.gThreadSortFileSynch = AC.ThreadSortFileSynch;
    AM.gProcessStats = AC.ProcessStats;
    AM.gOldParallelStats = AC.OldParallelStats;
    AM.gIsFortran90 = AC.IsFortran90;
    AM.gSizeCommuteInSet = AC.SizeCommuteInSet;
    AM.gnumextrasym = (cbuf+AM.sbufnum)->numrhs;
    if ( AM.gFortran90Kind ) {
        M_free(AM.gFortran90Kind,"Fortran 90 Kind");
        AM.gFortran90Kind = 0;
    }
    if ( AC.Fortran90Kind ) {
        AM.gFortran90Kind = strDup1(AC.Fortran90Kind,"Fortran 90 Kind");
    }
    p = AM.gcmod;
    m = AC.cmod;
    i = ABS(AC.ncmod);
    NCOPY(p,m,i);
    p = AM.gpowmod;
    m = AC.powmod;
    i = AC.npowmod;
    NCOPY(p,m,i);
    pp = AM.gUniTrace;
    mm = AC.lUniTrace;
    i = 4;
    NCOPY(pp,mm,i);
    AM.gSortType = AC.SortType;
    AM.gShortStatsMax = AC.ShortStatsMax;

    if ( AO.CurrentDictionary > 0 || AP.OpenDictionary > 0 ) {
        Warning("You cannot have an open or selected dictionary at a .global. Dictionary closed.");
        AP.OpenDictionary = 0;
        AO.CurrentDictionary = 0;
    }

```

```
    AO.gNumDictionaries = AO.NumDictionaries;
    for ( i = 0; i < AO.NumDictionaries; i++ ) {
        AO.Dictionaries[i]->gnumelements = AO.Dictionaries[i]->numelements;
    }
    if ( AM.NumSpectatorFiles > 0 ) {
        for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
            if ( AM.SpectatorFiles[i].name != 0 )
                AM.SpectatorFiles[i].flags |= GLOBALSPECTATORFLAG;
        }
    }
}

/*
        #] MakeGlobal :
        #[ TestDrop :
*/

void TestDrop(void)
{
    EXPRESSIONS e;
    WORD j;
    for ( j = 0, e = Expressions; j < NumExpressions; j++, e++ ) {
        switch ( e->status ) {
            case SKIPLEXPRESSION:
                e->status = LOCALEXPRESSION;
                break;
            case UNHIDELEXPRESSION:
                e->status = LOCALEXPRESSION;
                ClearBracketIndex(j);
                e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
                break;
            case HIDELEXPRESSION:
                e->status = HIDDENLEXPRESSION;
                break;
            case SKIPGEXPRESSION:
                e->status = GLOBALEXPRESSION;
                break;
            case UNHIDEGEXPRESSION:
                e->status = GLOBALEXPRESSION;
                ClearBracketIndex(j);
                e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
                break;
            case HIDEGEXPRESSION:
                e->status = HIDDENGEXPRESSION;
                break;
            case DROPLEXPRESSION:
            case DROPGEXPRESSION:
            case DROPHLEXPRESSION:
            case DROPHGEXPRESSION:
            case DROPSPECTATOREXPRESSION:
                e->status = DROPPEDEXPRESSION;
                ClearBracketIndex(j);
                e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
                if ( e->replace >= 0 ) {
                    Expressions[e->replace].replace = REGULAREXPRESSION;
                    AC.exprnames->namenode[e->node].number = e->replace;
                    e->replace = REGULAREXPRESSION;
                }
                else {
                    AC.exprnames->namenode[e->node].type = CDELETE;
                    AC.DidClean = 1;
                }
                break;
            case LOCALEXPRESSION:
            case GLOBALEXPRESSION:
                ClearBracketIndex(j);
                e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
                break;
            case HIDDENLEXPRESSION:
            case HIDDENGEXPRESSION:
                break;
            case INTOHIDELEXPRESSION:
                ClearBracketIndex(j);
                e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
                e->status = HIDDENLEXPRESSION;
                break;
            case INTOHIDEGEXPRESSION:
                ClearBracketIndex(j);
                e->bracketinfo = e->newbracketinfo; e->newbracketinfo = 0;
                e->status = HIDDENGEXPRESSION;
                break;
            default:
                ClearBracketIndex(j);
                e->bracketinfo = 0;
                break;
        }
        if ( e->replace == NEWLYDEFINEDEXPRESSION ) e->replace = REGULAREXPRESSION;
```

```
    }
}

/*
        #] TestDrop :
        #[ PutInVflags :
*/

void PutInVflags(WORD nexpr)
{
    EXPRESSIONS e = Expressions + nexpr;
    POSITION *old;
    WORD *oldw;
    int i;
restart:;
    if ( AS.OldOnFile == 0 ) {
        AS.NumOldOnFile = 20;
        AS.OldOnFile = (POSITION *)Malloc1(AS.NumOldOnFile*sizeof(POSITION),"file pointers");
    }
    else if ( nexpr >= AS.NumOldOnFile ) {
        old = AS.OldOnFile;
        AS.OldOnFile = (POSITION *)Malloc1(2*AS.NumOldOnFile*sizeof(POSITION),"file pointers");
        for ( i = 0; i < AS.NumOldOnFile; i++ ) AS.OldOnFile[i] = old[i];
        AS.NumOldOnFile = 2*AS.NumOldOnFile;
        M_free(old,"process file pointers");
    }
    if ( AS.OldNumFactors == 0 ) {
        AS.NumOldNumFactors = 20;
        AS.OldNumFactors = (WORD *)Malloc1(AS.NumOldNumFactors*sizeof(WORD),"numfactors pointers");
        AS.Oldvflags = (WORD *)Malloc1(AS.NumOldNumFactors*sizeof(WORD),"vflags pointers");
        AS.Olduflags = (WORD *)Malloc1(AS.NumOldNumFactors*sizeof(WORD),"uflags pointers");
    }
    else if ( nexpr >= AS.NumOldNumFactors ) {
        oldw = AS.OldNumFactors;
        AS.OldNumFactors = (WORD *)Malloc1(2*AS.NumOldNumFactors*sizeof(WORD),"numfactors pointers");
        for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.OldNumFactors[i] = oldw[i];
        M_free(oldw,"numfactors pointers");
        oldw = AS.Oldvflags;
        AS.Oldvflags = (WORD *)Malloc1(2*AS.NumOldNumFactors*sizeof(WORD),"vflags pointers");
        for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Oldvflags[i] = oldw[i];
        M_free(oldw,"vflags pointers");
        oldw = AS.Olduflags;
        AS.Olduflags = (WORD *)Malloc1(2*AS.NumOldNumFactors*sizeof(WORD),"uflags pointers");
        for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Olduflags[i] = oldw[i];
        M_free(oldw,"uflags pointers");
        AS.NumOldNumFactors = 2*AS.NumOldNumFactors;
    }
/*
    The next is needed when we Load a .sav file with lots of expressions.
*/
    if ( nexpr >= AS.NumOldOnFile || nexpr >= AS.NumOldNumFactors ) goto restart;
    AS.OldOnFile[nexpr] = e->onfile;
    AS.OldNumFactors[nexpr] = e->numfactors;
    AS.Oldvflags[nexpr] = e->vflags;
    AS.Olduflags[nexpr] = e->uflags;
}

/*
        #] PutInVflags :
        #[ DoExecute :
*/

WORD DoExecute(WORD par, WORD skip)
{
    GETIDENTITY
    WORD RetCode = 0;
    int i, oldmultithreaded = AS.MultiThreaded;
#ifdef PARALLELCODE
    int j;
#endif

    SpecialCleanup(BHEAD0);
    if ( skip ) goto skipexec;
    if ( AC.IfLevel > 0 ) {
        MesPrint(" %d endif statement(s) missing",AC.IfLevel);
        RetCode = 1;
    }
    if ( AC.WhileLevel > 0 ) {
        MesPrint(" %d endwhile statement(s) missing",AC.WhileLevel);
        RetCode = 1;
    }
    if ( AC.arglevel > 0 ) {
        MesPrint(" %d endargument statement(s) missing",AC.arglevel);
        RetCode = 1;
    }
    if ( AC.termlevel > 0 ) {
        MesPrint(" %d endterm statement(s) missing",AC.termlevel);
```

```
        RetCode = 1;
    }
    if ( AC.insidelevel > 0 ) {
        MesPrint(" %d endinside statement(s) missing",AC.insidelevel);
        RetCode = 1;
    }
    if ( AC.inexprlevel > 0 ) {
        MesPrint(" %d endinexpression statement(s) missing",AC.inexprlevel);
        RetCode = 1;
    }
    if ( AC.NumLabels > 0 ) {
        for ( i = 0; i < AC.NumLabels; i++ ) {
            if ( AC.Labels[i] < 0 ) {
                MesPrint(" -->Label %s missing",AC.LabelNames[i]);
                RetCode = 1;
            }
        }
    }
    if ( AC.SwitchLevel > 0 ) {
        MesPrint(" %d endswitch statement(s) missing",AC.SwitchLevel);
        RetCode = 1;
    }
    if ( AC.dolooplevel > 0 ) {
        MesPrint(" %d enddo statement(s) missing",AC.dolooplevel);
        RetCode = 1;
    }
    if ( AP.OpenDictionary > 0 ) {
        MesPrint(" Dictionary %s has not been closed.",
            AO.Dictionaries[AP.OpenDictionary-1]->name);
        AP.OpenDictionary = 0;
        RetCode = 1;
    }
    if ( RetCode ) return(RetCode);
    AR.Cnumlhs = cbuf[AM.rbufnum].numlhs;

    if ( ( AS.ExecMode = par ) == GLOBALMODULE ) AS.ExecMode = 0;
#ifdef PARALLELCODE
/*
        Now check whether we have either the regular parallel flag or the
        mparallel flag set.
        Next check whether any of the expressions has partodo set.
        If any of the above we need to check what the dollar status is.
*/
    AC.partodoflag = -1;
    if ( NumPotModdollars >= 0 ) {
        for ( i = 0; i < NumExpressions; i++ ) {
            if ( Expressions[i].partodo ) { AC.partodoflag = 1; break; }
        }
    }
#ifdef WITHMPI
    if ( AC.partodoflag > 0 && PF.numtasks < 3 ) {
        AC.partodoflag = 0;
    }
#endif
    if ( AC.partodoflag > 0 || ( NumPotModdollars > 0 && AC.mparallelflag == PARALLELFLAG ) ) {
      if ( NumPotModdollars > NumModOptdollars ) {
        AC.mparallelflag |= NOPARALLEL_DOLLAR;
#ifdef WITHPTHREADS
        AS.MultiThreaded = 0;
#endif
        AC.partodoflag = 0;
      }
      else {
        for ( i = 0; i < NumPotModdollars; i++ ) {
          for ( j = 0; j < NumModOptdollars; j++ )
            if ( PotModdollars[i] == ModOptdollars[j].number ) break;
          if ( j >= NumModOptdollars ) {
            AC.mparallelflag |= NOPARALLEL_DOLLAR;
#ifdef WITHPTHREADS
            AS.MultiThreaded = 0;
#endif
            AC.partodoflag = 0;
            break;
          }
          switch ( ModOptdollars[j].type ) {
            case MODSUM:
            case MODMAX:
            case MODMIN:
            case MODLOCAL:
                break;
            default:
                AC.mparallelflag |= NOPARALLEL_DOLLAR;
                AS.MultiThreaded = 0;
                AC.partodoflag = 0;
                break;
          }
        }
```

```
        }
    }
    else if ( ( AC.mparallelflag & NOPARALLEL_USER ) != 0 ) {
#ifdef WITHPTHREADS
        AS.MultiThreaded = 0;
#endif
        AC.partodoflag = 0;
    }
    if ( AC.partodoflag == 0 ) {
        for ( i = 0; i < NumExpressions; i++ ) {
            Expressions[i].partodo = 0;
        }
    }
    else if ( AC.partodoflag == -1 ) {
        AC.partodoflag = 0;
    }
#endif
#ifdef WITHMPI
    /*
     * Check RHS expressions.
     */
    if ( AC.RhsExprInModuleFlag && (AC.mparallelflag == PARALLELFLAG || AC.partodoflag) ) {
        if (PF.rhsInParallel) {
            PF.mkSlaveInfile=1;
            if(PF.me != MASTER){
                PF.slavebuf.PObuffer=(WORD *)Malloc1(AM.ScratSize*sizeof(WORD),"PF inbuf");
                PF.slavebuf.POsize=AM.ScratSize*sizeof(WORD);
                PF.slavebuf.POfull = PF.slavebuf.POfill = PF.slavebuf.PObuffer;
                PF.slavebuf.POstop= PF.slavebuf.PObuffer+AM.ScratSize;
                PUTZERO(PF.slavebuf.POposition);
            }/*if(PF.me != MASTER)*/
        }
        else {
            AC.mparallelflag |= NOPARALLEL_RHS;
            AC.partodoflag = 0;
            for ( i = 0; i < NumExpressions; i++ ) {
                Expressions[i].partodo = 0;
            }
        }
    }
    /*
     * Set $-variables with MODSUM to zero on the slaves.
     */
    if ( (AC.mparallelflag == PARALLELFLAG || AC.partodoflag) && PF.me != MASTER ) {
        for ( i = 0; i < NumModOptdollars; i++ ) {
            if ( ModOptdollars[i].type == MODSUM ) {
                DOLLARS d = Dollars + ModOptdollars[i].number;
                d->type = DOLZERO;
                if ( d->where && d->where != &AM.dollarzero ) M_free(d->where, "old content of
dollar");
                d->where = &AM.dollarzero;
                d->size = 0;
                CleanDollarFactors(d);
            }
        }
    }
#endif
    AR.SortType = AC.SortType;
#ifdef WITHMPI
    if ( PF.me == MASTER )
#endif
    {
        if ( AC.SetupFlag ) WriteSetup();
        if ( AC.NamesFlag || AC.CodesFlag ) WriteLists();
    }
    if ( par == GLOBALMODULE ) MakeGlobal();
    if ( RevertScratch() ) return(-1);
    if ( AC.ncmod ) SetMods();
/*
    Warn if the module has to run in sequential mode due to some problems.
*/
#ifdef WITHMPI
    if ( PF.me == MASTER )
#endif
    {
        if ( !AC.ThreadsFlag || AC.mparallelflag & NOPARALLEL_USER ) {
            /* The user switched off the parallel execution explicitly. */
        }
        else if ( AC.mparallelflag & NOPARALLEL_DOLLAR ) {
            if ( AC.WarnFlag >= 2 ) {  /* HighWarning */
                int i, j, k, n;
                UBYTE *s, *s1;
                s = strDup1((UBYTE *)"","NOPARALLEL_DOLLAR s");
                n = 0;
                j = NumPotModdollars;
                for ( i = 0; i < j; i++ ) {
                    for ( k = 0; k < NumModOptdollars; k++ )
```

```
                        if ( ModOptdollars[k].number == PotModdollars[i] ) break;
                    if ( k >= NumModOptdollars ) {
                        /* global $-variable */
                        if ( n > 0 )
                            s = AddToString(s,(UBYTE *)", ",0);
                        s = AddToString(s,(UBYTE *)"$",0);
                        s = AddToString(s,DOLLARNAME(Dollars,PotModdollars[i]),0);
                        n++;
                    }
                }
                s1 = strDup1((UBYTE *)"This module is forced to run in sequential mode due to $-variable","NOPARALLEL_DOLLAR s1");
                if ( n != 1 )
                    s1 = AddToString(s1,(UBYTE *)"s",0);
                s1 = AddToString(s1,(UBYTE *)": ",0);
                s1 = AddToString(s1,s,0);
                HighWarning((char *)s1);
                M_free(s,"NOPARALLEL_DOLLAR s");
                M_free(s1,"NOPARALLEL_DOLLAR s1");
            }
        }
        else if ( AC.mparallelflag & NOPARALLEL_RHS ) {
            HighWarning("This module is forced to run in sequential mode due to RHS expression names");
        }
        else if ( AC.mparallelflag & NOPARALLEL_CONVPOLY ) {
            HighWarning("This module is forced to run in sequential mode due to conversion to extra symbols");
        }
        else if ( AC.mparallelflag & NOPARALLEL_SPECTATOR ) {
            HighWarning("This module is forced to run in sequential mode due to tospectator/copyspectator");
        }
        else if ( AC.mparallelflag & NOPARALLEL_TBLDOLLAR ) {
            HighWarning("This module is forced to run in sequential mode due to $-variable assignments in tables");
        }
        else if ( AC.mparallelflag & NOPARALLEL_NPROC ) {
            HighWarning("This module is forced to run in sequential mode because there is only one processor");
        }
    }
/*
    Now the actual execution
*/
#ifdef WITHMPI
    /*
     * Turn on AS.printflag to print runtime errors occurring on slaves.
     */
    AS.printflag = 1;
#endif
    if ( AP.preError == 0 && ( Processor() || WriteAll() ) ) RetCode = -1;
#ifdef WITHMPI
    AS.printflag = 0;
#endif
/*
    That was it. Next is cleanup.
*/
    if ( AC.ncmod ) UnSetMods();
    AS.MultiThreaded = oldmultithreaded;
    TableReset();

/*[28sep2005 mt]:*/
#ifdef WITHMPI
    /* Combine and then broadcast modified dollar variables. */
    if ( NumPotModdollars > 0 ) {
        RetCode = PF_CollectModifiedDollars();
        if ( RetCode ) return RetCode;
        RetCode = PF_BroadcastModifiedDollars();
        if ( RetCode ) return RetCode;
    }
    /* Broadcast the list of objects converted to symbols in AM.sbufnum. */
    if ( AC.topolynomialflag & TOPOLYNOMIALFLAG ) {
        RetCode = PF_BroadcastCBuf(AM.sbufnum);
        if ( RetCode ) return RetCode;
    }
    /*
     * Broadcast AR.expflags, which may be used on the slaves in the next module
     * via ZERO_ or UNCHANGED_. It also broadcasts several flags of each expression.
     */
    RetCode = PF_BroadcastExpFlags();
    if ( RetCode ) return RetCode;
    /*
     * Clean the hide file on the slaves, which was used for RHS expressions
     * broadcast from the master at the beginning of the module.
     */
    if ( PF.me != MASTER && AR.hidefile->PObuffer ) {
```

```
        if ( AR.hidefile->handle >= 0 ) {
            CloseFile(AR.hidefile->handle);
            AR.hidefile->handle = -1;
            remove(AR.hidefile->name);
        }
        AR.hidefile->POfull = AR.hidefile->POfill = AR.hidefile->PObuffer;
        PUTZERO(AR.hidefile->POposition);
    }
#endif
#ifdef WITHPTHREADS
    for ( j = 0; j < NumModOptdollars; j++ ) {
        if ( ModOptdollars[j].dstruct ) {
/*
            First clean up dollar values.
*/
            for ( i = 0; i < AM.totalnumberofthreads; i++ ) {
                if ( ModOptdollars[j].dstruct[i].size > 0 ) {
                    CleanDollarFactors(&(ModOptdollars[j].dstruct[i]));
                    M_free(ModOptdollars[j].dstruct[i].where,"Local dollar value");
                }
            }
/*
            Now clean up the whole array.
*/
            M_free(ModOptdollars[j].dstruct,"Local DOLLARS");
            ModOptdollars[j].dstruct = 0;
        }
    }
#endif
/*:[28sep2005 mt]*/

/*
    @@@@@@@@@@@@@@@
    Now follows the code to invalidate caches for all objects in the
    PotModdollars. There are NumPotModdollars of them and PotModdollars
    is an array of WORD.
*/
/*
    Cleanup:
*/
#ifdef JV_IS_WRONG
/*
    Giving back this memory gives way too much activity with Malloc1
    Better to keep it and just put the number of used objects to zero (JV)
    If you put the lijst equal to NULL, please also make maxnum = 0
*/
    if ( ModOptdollars ) M_free(ModOptdollars, "ModOptdollars pointer");
    if ( PotModdollars ) M_free(PotModdollars, "PotModdollars pointer");

    /* ModOptdollars changed to AC.ModOptDolList.lijst because AIX C compiler complained. MF 30/07/2003. */
    AC.ModOptDolList.lijst = NULL;
    /* PotModdollars changed to AC.PotModDolList.lijst because AIX C compiler complained. MF 30/07/2003. */
    AC.PotModDolList.lijst = NULL;
#endif
    NumPotModdollars = 0;
    NumModOptdollars = 0;

skipexec:
/*
    Clean up the switch information.
    We keep the switch array and heap.
*/
if ( AC.SwitchInArray > 0 ) {
    for ( i = 0; i < AC.SwitchInArray; i++ ) {
        SWITCH *sw = AC.SwitchArray + i;
        if ( sw->table ) M_free(sw->table,"Switch table");
        sw->table = 0;
        sw->defaultcase.ncase = 0;
        sw->defaultcase.value = 0;
        sw->defaultcase.compbuffer = 0;
        sw->endswitch.ncase = 0;
        sw->endswitch.value = 0;
        sw->endswitch.compbuffer = 0;
        sw->typetable = 0;
        sw->maxcase = 0;
        sw->mincase = 0;
        sw->numcases = 0;
        sw->tablesize = 0;
        sw->caseoffset = 0;
        sw->iflevel = 0;
        sw->whilelevel = 0;
        sw->nestingsum = 0;
    }
    AC.SwitchInArray = 0;
    AC.SwitchLevel = 0;
```

```
}
#ifdef PARALLELCODE
    AC.numpfirstnum = 0;
#endif
    AC.DidClean = 0;
    AC.PolyRatFunChanged = 0;
    TestDrop();
    if ( par == STOREMODULE || par == CLEARMODULE ) {
        ClearOptimize();
        if ( par == STOREMODULE && PopVariables() ) RetCode = -1;
        if ( AR.infile->handle >= 0 ) {
            CloseFile(AR.infile->handle);
            remove(AR.infile->name);
            AR.infile->handle = -1;
        }
        AR.infile->POfill = AR.infile->PObuffer;
        PUTZERO(AR.infile->POposition);
        AR.infile->POfull = AR.infile->PObuffer;
        if ( AR.outfile->handle >= 0 ) {
            CloseFile(AR.outfile->handle);
            remove(AR.outfile->name);
            AR.outfile->handle = -1;
        }
        AR.outfile->POfull =
        AR.outfile->POfill = AR.outfile->PObuffer;
        PUTZERO(AR.outfile->POposition);
        if ( AR.hidefile->handle >= 0 ) {
            CloseFile(AR.hidefile->handle);
            remove(AR.hidefile->name);
            AR.hidefile->handle = -1;
        }
        AR.hidefile->POfull =
        AR.hidefile->POfill = AR.hidefile->PObuffer;
        PUTZERO(AR.hidefile->POposition);
        AC.HideLevel = 0;
        if ( par == CLEARMODULE ) {
            if ( DeleteStore(0) < 0 ) {
                MesPrint("Cannot restart the storage file");
                RetCode = -1;
            }
            else RetCode = 0;
            CleanUp(1);
            ResetVariables(2);
            AM.gProcessBucketSize = AM.hProcessBucketSize;
            AM.gparallelflag = PARALLELFLAG;
            AM.gnumextrasym = AM.ggnumextrasym;
            PruneExtraSymbols(AM.ggnumextrasym);
            IniVars();
        }
        ClearSpectators(par);
    }
    else {
        if ( CleanExpr(0) ) RetCode = -1;
        if ( AC.DidClean ) CompactifyTree(AC.exprnames,EXPRNAMES);
        ResetVariables(0);
        CleanUpSort(-1);
    }
    clearcbuf(AC.cbufnum);
    if ( AC.MultiBracketBuf != 0 ) {
      for ( i = 0; i < MAXMULTIBRACKETLEVELS; i++ ) {
        if ( AC.MultiBracketBuf[i] ) {
            M_free(AC.MultiBracketBuf[i],"bracket buffer i");
            AC.MultiBracketBuf[i] = 0;
        }
      }
      AC.MultiBracketLevels = 0;
      M_free(AC.MultiBracketBuf,"multi bracket buffer");
      AC.MultiBracketBuf = 0;
    }

    if ( AC.SortReallocateFlag ) {
        /* Reallocate the sort buffers to reduce resident set usage */
        /* AT.SS is the same as AT.S0 here */
        SORTING* S = AT.S0;
        M_free(S->lBuffer, "SortReallocate lBuffer+sBuffer");
        S->lBuffer = Malloc1(sizeof(*(S->lBuffer))*(S->LargeSize+S->SmallEsize), "SortReallocate
lBuffer+sBuffer");
        S->lTop = S->lBuffer+S->LargeSize;
        S->sBuffer = S->lTop;
        if ( S->LargeSize == 0 ) { S->lBuffer = 0; S->lTop = 0; }
        S->sTop = S->sBuffer + S->SmallSize;
        S->sTop2 = S->sBuffer + S->SmallEsize;
        S->sHalf = S->sBuffer + (LONG)((S->SmallSize+S->SmallEsize)»1);

#ifdef WITHPTHREADS
        /* We have to re-set the pointers into master lBuffer in the SortBlocks */
        UpdateSortBlocks(AM.totalnumberofthreads-1);
```

```
        /* The SortBots do not have a real sort buffer to reallocate. */
        /* AB[0] has been reallocated above already. */
        for ( i = 1; i < AM.totalnumberofthreads; i++ ) {
            SORTING* S = AB[i]->T.S0;
            M_free(S->lBuffer, "SortReallocate lBuffer+sBuffer");
            S->lBuffer = Malloc1(sizeof(*(S->lBuffer))*(S->LargeSize+S->SmallEsize), "SortReallocate lBuffer+sBuffer");
            S->lTop = S->lBuffer+S->LargeSize;
            S->sBuffer = S->lTop;
            if ( S->LargeSize == 0 ) { S->lBuffer = 0; S->lTop = 0; }
            S->sTop = S->sBuffer + S->SmallSize;
            S->sTop2 = S->sBuffer + S->SmallEsize;
            S->sHalf = S->sBuffer + (LONG)((S->SmallSize+S->SmallEsize)»1);
        }
#endif
    }
    if ( AC.SortReallocateFlag == 2 ) {
        /* The Flag was set for a single module by the preprocessor #sortreallocate,
           so turn it off again. */
        AC.SortReallocateFlag = 0;
    }

    return(RetCode);
}

/*
        #] DoExecute :
        #[ PutBracket :

    Routine uses the bracket info to split a term into two pieces:
    1: the part outside the bracket, and
    2: the part inside the bracket.
    These parts are separated by a subterm of type HAAKJE.
    This subterm looks like: HAAKJE,3,level
    The level is used for nestings of brackets. The print routines
    cannot handle this yet (31-Mar-1988).

    The Bracket selector is in AT.BrackBuf in the form of a regular term,
    but without coefficient.
    When AR.BracketOn < 0 we have a socalled antibracket. The main effect
    is an exchange of the inner and outer part and where the coefficient goes.

    Routine recoded to facilitate b p1,p2; etc for dotproducts and tensors
    15-oct-1991
*/

WORD PutBracket(PHEAD WORD *termin)
{
    GETBIDENTITY
    WORD *t, *t1, *b, i, j, *lastfun;
    WORD *t2, *s1, *s2;
    WORD *bStop, *bb, *bf, *tStop;
    WORD *term1,*term2, *m1, *m2, *tStopa;
    WORD *bbb = 0, *bind, *binst = 0, bwild = 0, *bss = 0, *bns = 0, bset = 0;
    term1 = AT.WorkPointer+1;
    term2 = (WORD *)(((UBYTE *)(term1)) + AM.MaxTer);
    if ( ( (WORD *)(((UBYTE *)(term2)) + AM.MaxTer) ) > AT.WorkTop ) return(MesWork());
    if ( AR.BracketOn < 0 ) {
        t2 = term1; t1 = term2;     /* AntiBracket */
    }
    else {
        t1 = term1; t2 = term2;     /* Regular bracket */
    }
    b = AT.BrackBuf; bStop = b+*b; b++;
    while ( b < bStop ) {
        if ( *b == INDEX ) { bwild = 1; bbb = b+2; binst = b + b[1]; }
        if ( *b == SETSET ) { bset = 1; bss = b+2; bns = b + b[1]; }
        b += b[1];
    }

    t = termin; tStopa = t + *t; i = *(t + *t -1); i = ABS(i);
    if ( AR.PolyFun && AT.PolyAct ) tStop = termin + AT.PolyAct;
#ifdef WITHFLOAT
    else if ( AT.FloatPos ) tStop = termin + AT.FloatPos;
#endif
    else tStop = tStopa - i;
    t++;
    if ( AR.BracketOn < 0 ) {
        lastfun = 0;
        while ( t < tStop && *t >= FUNCTION
            && functions[*t-FUNCTION].commute ) {
            b = AT.BrackBuf+1;
            while ( b < bStop ) {
                if ( *b == *t ) {
                    lastfun = t;
                    while ( t < tStop && *t >= FUNCTION
```

```
                        && functions[*t-FUNCTION].commute ) t += t[1];
                    goto NextNcom1;
                }
                b += b[1];
            }
            if ( bset ) {
                b = bss;
                while ( b < bns ) {
                    if ( b[1] == CFUNCTION ) { /* Set of functions */
                        SETS set = Sets+b[0]; WORD i;
                        for ( i = set->first; i < set->last; i++ ) {
                            if ( SetElements[i] == *t ) {
                                lastfun = t;
                                while ( t < tStop && *t >= FUNCTION
                                    && functions[*t-FUNCTION].commute ) t += t[1];
                                goto NextNcom1;
                            }
                        }
                    }
                    b += 2;
                }
            }
            if ( bwild && *t >= FUNCTION && functions[*t-FUNCTION].spec ) {
                s1 = t + t[1];
                s2 = t + FUNHEAD;
                while ( s2 < s1 ) {
                    bind = bbb;
                    while ( bind < binst ) {
                        if ( *bind == *s2 ) {
                            lastfun = t;
                            while ( t < tStop && *t >= FUNCTION
                                && functions[*t-FUNCTION].commute ) t += t[1];
                            goto NextNcom1;
                        }
                        bind++;
                    }
                    s2++;
                }
            }
            t += t[1];
        }
NextNcom1:
        s1 = termin + 1;
        if ( lastfun ) {
            while ( s1 < lastfun ) *t2++ = *s1++;
            while ( s1 < t ) *t1++ = *s1++;
        }
        else {
            while ( s1 < t ) *t2++ = *s1++;
        }

    }
    else {
        lastfun = t;
        while ( t < tStop && *t >= FUNCTION
            && functions[*t-FUNCTION].commute ) {
            b = AT.BrackBuf+1;
            while ( b < bStop ) {
                if ( *b == *t ) { lastfun = t + t[1]; goto NextNcom; }
                b += b[1];
            }
            if ( bset ) {
                b = bss;
                while ( b < bns ) {
                    if ( b[1] == CFUNCTION ) { /* Set of functions */
                        SETS set = Sets+b[0]; WORD i;
                        for ( i = set->first; i < set->last; i++ ) {
                            if ( SetElements[i] == *t ) {
                                lastfun = t + t[1];
                                goto NextNcom;
                            }
                        }
                    }
                    b += 2;
                }
            }
            if ( bwild && *t >= FUNCTION && functions[*t-FUNCTION].spec ) {
                s1 = t + t[1];
                s2 = t + FUNHEAD;
                while ( s2 < s1 ) {
                    bind = bbb;
                    while ( bind < binst ) {
                        if ( *bind == *s2 ) { lastfun = t + t[1]; goto NextNcom; }
                        bind++;
                    }
                    s2++;
                }
```

```
                }
NextNcom:
                t += t[1];
            }
            s1 = termin + 1;
            while ( s1 < lastfun ) *t1++ = *s1++;
            while ( s1 < t ) *t2++ = *s1++;
        }
/*
        Now we have only commuting functions left. Move the b pointer to them.
*/
        b = AT.BrackBuf + 1;
        while ( b < bStop && *b >= FUNCTION
            && ( *b < FUNCTION || functions[*b-FUNCTION].commute ) ) {
            b += b[1];
        }
        bf = b;

        while ( t < tStop && ( bf < bStop || bwild || bset ) ) {
            b = bf;
            while ( b < bStop && *b != *t ) { b += b[1]; }
            i = t[1];
            if ( *t >= FUNCTION ) { /* We are in function territory */
                if ( b < bStop && *b == *t ) goto FunBrac;
                if ( bset ) {
                    b = bss;
                    while ( b < bns ) {
                        if ( b[1] == CFUNCTION ) { /* Set of functions */
                            SETS set = Sets+b[0]; WORD i;
                            for ( i = set->first; i < set->last; i++ ) {
                                if ( SetElements[i] == *t ) goto FunBrac;
                            }
                        }
                        b += 2;
                    }
                }
                if ( bwild && *t >= FUNCTION && functions[*t-FUNCTION].spec ) {
                    s1 = t + t[1];
                    s2 = t + FUNHEAD;
                    while ( s2 < s1 ) {
                        bind = bbb;
                        while ( bind < binst ) {
                            if ( *bind == *s2 ) goto FunBrac;
                            bind++;
                        }
                        s2++;
                    }
                }
                NCOPY(t2,t,i);
                continue;
FunBrac:        NCOPY(t1,t,i);
                continue;
            }
/*
        We have left: DELTA, INDEX, VECTOR, DOTPRODUCT, SYMBOL
*/
            if ( *t == DELTA ) {
                if ( b < bStop && *b == DELTA ) {
                    b += b[1];
                    NCOPY(t1,t,i);
                }
                else { NCOPY(t2,t,i); }
            }
            else if ( *t == INDEX ) {
                if ( bwild ) {
                    m1 = t1; m2 = t2;
                    *t1++ = *t; t1++; *t2++ = *t; t2++;
                    bind = bbb;
                    j = t[1] -2;
                    t += 2;
                    while ( --j >= 0 ) {
                        while ( *bind < *t && bind < binst ) bind++;
                        if ( *bind == *t && bind < binst ) {
                            *t1++ = *t++;
                        }
                        else if ( bset ) {
                            WORD *b3 = bss;
                            while ( b3 < bns ) {
                                if ( b3[1] == CVECTOR ) {
                                    SETS set = Sets+b3[0]; WORD i;
                                    for ( i = set->first; i < set->last; i++ ) {
                                        if ( SetElements[i] == *t ) {
                                            *t1++ = *t++;
                                            goto nextind;
                                        }
                                    }
                                }
```

```
                            b3 += 2;
                        }
                        *t2++ = *t++;
                    }
                    else *t2++ = *t++;
nextind:;
                }
                m1[1] = WORDDIF(t1,m1);
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else if ( bset ) {
                m1 = t1; m2 = t2;
                *t1++ = *t; t1++; *t2++ = *t; t2++;
                j = t[1] -2;
                t += 2;
                while ( --j >= 0 ) {
                    WORD *b3 = bss;
                    while ( b3 < bns ) {
                        if ( b3[1] == CVECTOR ) {
                            SETS set = Sets+b3[0]; WORD i;
                            for ( i = set->first; i < set->last; i++ ) {
                                if ( SetElements[i] == *t ) {
                                    *t1++ = *t++;
                                    goto nextind2;
                                }
                            }
                        }
                        b3 += 2;
                    }
                    *t2++ = *t++;
nextind2:;
                }
                m1[1] = WORDDIF(t1,m1);
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else {
                NCOPY(t2,t,i);
            }
        }
        else if ( *t == VECTOR ) {
            if ( ( b < bStop && *b == VECTOR ) || bwild ) {
                if ( b < bStop && *b == VECTOR ) {
                    bb = b + b[1]; b += 2;
                }
                else bb = b;
                j = t[1] - 2;
                m1 = t1; m2 = t2; *t1++ = *t; *t2++ = *t; t1++; t2++; t += 2;
                while ( j > 0 ) {
                    j -= 2;
                    while ( b < bb && ( *b < *t ||
                    ( *b == *t && b[1] < t[1] ) ) ) b += 2;
                    if ( b < bb && ( *t == *b && t[1] == b[1] ) ) {
                        *t1++ = *t++; *t1++ = *t++; goto nextvec;
                    }
                    else if ( bwild ) {
                        bind = bbb;
                        while ( bind < binst ) {
                            if ( *t == *bind || t[1] == *bind ) {
                                *t1++ = *t++; *t1++ = *t++;
                                goto nextvec;
                            }
                            bind++;
                        }
                    }
                    if ( bset ) {
                        WORD *b3 = bss;
                        while ( b3 < bns ) {
                            if ( b3[1] == CVECTOR ) {
                                SETS set = Sets+b3[0]; WORD i;
                                for ( i = set->first; i < set->last; i++ ) {
                                    if ( SetElements[i] == *t ) {
                                        *t1++ = *t++; *t1++ = *t++;
                                        goto nextvec;
                                    }
                                }
                            }
                            b3 += 2;
                        }
                    }
                    *t2++ = *t++; *t2++ = *t++;
nextvec:;
                }
                m1[1] = WORDDIF(t1,m1);
```

```
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else if ( bset ) {
                m1 = t1; *t1++ = *t; t1++;
                m2 = t2; *t2++ = *t; t2++;
                s2 = t + i; t += 2;
                while ( t < s2 ) {
                    WORD *b3 = bss;
                    while ( b3 < bns ) {
                        if ( b3[1] == CVECTOR ) {
                            SETS set = Sets+b3[0]; WORD i;
                            for ( i = set->first; i < set->last; i++ ) {
                                if ( SetElements[i] == *t ) {
                                    *t1++ = *t++; *t1++ = *t++;
                                    goto nextvec2;
                                }
                            }
                        }
                        b3 += 2;
                    }
                    *t2++ = *t++; *t2++ = *t++;
nextvec2:;
                }
                m1[1] = WORDDIF(t1,m1);
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else {
                NCOPY(t2,t,i);
            }
        }
        else if ( *t == DOTPRODUCT ) {
            if ( ( b < bStop && *b == *t ) || bwild ) {
                m1 = t1; *t1++ = *t; t1++;
                m2 = t2; *t2++ = *t; t2++;
                if ( b >= bStop || *b != *t ) { bb = b; s1 = b; }
                else {
                    s1 = b + b[1]; bb = b + 2;
                }
                s2 = t + i; t += 2;
                while ( t < s2 && ( bb < s1 || bwild || bset ) ) {
                    while ( bb < s1 && ( *bb < *t ||
                    ( *bb == *t && bb[1] < t[1] ) ) ) bb += 3;
                    if ( bb < s1 && *bb == *t && bb[1] == t[1] ) {
                        *t1++ = *t++; *t1++ = *t++; *t1++ = *t++; bb += 3;
                        goto nextdot;
                    }
                    else if ( bwild ) {
                        bind = bbb;
                        while ( bind < binst ) {
                            if ( *bind == *t || *bind == t[1] ) {
                                *t1++ = *t++; *t1++ = *t++; *t1++ = *t++;
                                goto nextdot;
                            }
                            bind++;
                        }
                    }
                    if ( bset ) {
                        WORD *b3 = bss;
                        while ( b3 < bns ) {
                            if ( b3[1] == CVECTOR ) {
                                SETS set = Sets+b3[0]; WORD i;
                                for ( i = set->first; i < set->last; i++ ) {
                                    if ( SetElements[i] == *t || SetElements[i] == t[1] ) {
                                        *t1++ = *t++; *t1++ = *t++; *t1++ = *t++;
                                        goto nextdot;
                                    }
                                }
                            }
                            b3 += 2;
                        }
                    }
                    *t2++ = *t++; *t2++ = *t++; *t2++ = *t++;
nextdot:;
                }
                while ( t < s2 ) *t2++ = *t++;
                m1[1] = WORDDIF(t1,m1);
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else if ( bset ) {
                m1 = t1; *t1++ = *t; t1++;
                m2 = t2; *t2++ = *t; t2++;
```

```
                s2 = t + i; t += 2;
                while ( t < s2 ) {
                    WORD *b3 = bss;
                    while ( b3 < bns ) {
                        if ( b3[1] == CVECTOR ) {
                            SETS set = Sets+b3[0]; WORD i;
                            for ( i = set->first; i < set->last; i++ ) {
                                if ( SetElements[i] == *t || SetElements[i] == t[1] ) {
                                    *t1++ = *t++; *t1++ = *t++; *t1++ = *t++;
                                    goto nextdot2;
                                }
                            }
                        }
                        b3 += 2;
                    }
                    *t2++ = *t++; *t2++ = *t++; *t2++ = *t++;
nextdot2:;
                }
                m1[1] = WORDDIF(t1,m1);
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else { NCOPY(t2,t,i); }
        }
        else if ( *t == SYMBOL ) {
            if ( b < bStop && *b == *t ) {
                m1 = t1; *t1++ = *t; t1++;
                m2 = t2; *t2++ = *t; t2++;
                s1 = b + b[1]; bb = b+2;
                s2 = t + i; t += 2;
                while ( bb < s1 && t < s2 ) {
                    while ( bb < s1 && *bb < *t ) bb += 2;
                    if ( bb >= s1 ) {
                        if ( bset ) goto TrySymbolSet;
                        break;
                    }
                    if ( *bb == *t ) { *t1++ = *t++; *t1++ = *t++; }
                    else if ( bset ) {
                        WORD *bbb;
TrySymbolSet:
                        bbb = bss;
                        while ( bbb < bns ) {
                            if ( bbb[1] == CSYMBOL ) { /* Set of symbols */
                                SETS set = Sets+bbb[0]; WORD i;
                                for ( i = set->first; i < set->last; i++ ) {
                                    if ( SetElements[i] == *t ) {
                                        *t1++ = *t++; *t1++ = *t++;
                                        goto NextSymbol;
                                    }
                                }
                            }
                            bbb += 2;
                        }
                        *t2++ = *t++; *t2++ = *t++;
                    }
                    else { *t2++ = *t++; *t2++ = *t++; }
NextSymbol:;
                }
                while ( t < s2 ) *t2++ = *t++;
                m1[1] = WORDDIF(t1,m1);
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else if ( bset ) {
                WORD *bbb;
                m1 = t1; *t1++ = *t; t1++;
                m2 = t2; *t2++ = *t; t2++;
                s2 = t + i; t += 2;
                while ( t < s2 ) {
                    bbb = bss;
                    while ( bbb < bns ) {
                        if ( bbb[1] == CSYMBOL ) { /* Set of symbols */
                            SETS set = Sets+bbb[0]; WORD i;
                            for ( i = set->first; i < set->last; i++ ) {
                                if ( SetElements[i] == *t ) {
                                    *t1++ = *t++; *t1++ = *t++;
                                    goto NextSymbol2;
                                }
                            }
                        }
                        bbb += 2;
                    }
                    *t2++ = *t++; *t2++ = *t++;
NextSymbol2:;
                }
```

```
                m1[1] = WORDDIF(t1,m1);
                if ( m1[1] == 2 ) t1 = m1;
                m2[1] = WORDDIF(t2,m2);
                if ( m2[1] == 2 ) t2 = m2;
            }
            else { NCOPY(t2,t,i); }
        }
        else {
            NCOPY(t2,t,i);
        }
    }
    if ( ( i = WORDDIF(tStop,t) ) > 0 ) NCOPY(t2,t,i);
    if ( AR.BracketOn < 0 ) {
        s1 = t1; t1 = t2; t2 = s1;
    }
    do { *t2++ = *t++; } while ( t < (WORD *)tStopa );
    t = AT.WorkPointer;
    i = WORDDIF(t1,term1);
    *t++ = 4 + i + WORDDIF(t2,term2);
    t += i;
    *t++ = HAAKJE;
    *t++ = 3;
    *t++ = 0;           /* This feature won't be used for a while */
    i = WORDDIF(t2,term2);
    t1 = term2;
    if ( i > 0 ) NCOPY(t,t1,i);

    AT.WorkPointer = t;

    return(0);
}

/*
        #] PutBracket :
        #[ SpecialCleanup :
*/

void SpecialCleanup(PHEAD0)
{
    GETBIDENTITY
    if ( AT.previousEfactor ) M_free(AT.previousEfactor,"Efactor cache");
    AT.previousEfactor = 0;
}

/*
        #] SpecialCleanup :
        #[ SetMods :
*/

#ifndef WITHPTHREADS

void SetMods(void)
{
    int i, n;
    if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
    n = ABS(AN.ncmod);
    AN.cmod = (UWORD *)Malloc1(sizeof(WORD)*n,"AN.cmod");
    for ( i = 0; i < n; i++ ) AN.cmod[i] = AC.cmod[i];
}

#endif

/*
        #] SetMods :
        #[ UnSetMods :
*/

#ifndef WITHPTHREADS

void UnSetMods(void)
{
    if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
    AN.cmod = 0;
}

#endif

/*
        #] UnSetMods :
    #] DoExecute :
    #[ Expressions :
        #[ ExchangeExpressions :
*/

void ExchangeExpressions(int num1, int num2)
{
    GETIDENTITY
```

```
    WORD node1, node2, namesize, TMproto[SUBEXPSIZE];
    INDEXENTRY *ind;
    EXPRESSIONS e1, e2;
    LONG a;
    SBYTE *s1, *s2;
    int i;
    e1 = Expressions + num1;
    e2 = Expressions + num2;
    node1 = e1->node;
    node2 = e2->node;
    AC.exprnames->namenode[node1].number = num2;
    AC.exprnames->namenode[node2].number = num1;
    a = e1->name; e1->name = e2->name; e2->name = a;
    namesize = e1->namesize; e1->namesize = e2->namesize; e2->namesize = namesize;
    e1->node = node2;
    e2->node = node1;
    if ( e1->status == STOREDEXPRESSION ) {
/*
        Find the name in the index and replace by the new name
*/
        TMproto[0] = EXPRESSION;
        TMproto[1] = SUBEXPSIZE;
        TMproto[2] = num1;
        TMproto[3] = 1;
        { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
        AT.TMaddr = TMproto;
        ind = FindInIndex(num1,&AR.StoreData,0,0);
        s1 = (SBYTE *)(AC.exprnames->namebuffer+e1->name);
        i = e1->namesize;
        s2 = ind->name;
        NCOPY(s2,s1,i);
        *s2 = 0;
        SeekFile(AR.StoreData.Handle,&(e1->onfile),SEEK_SET);
        if ( WriteFile(AR.StoreData.Handle,(UBYTE *)ind,
        (LONG)(sizeof(INDEXENTRY))) != sizeof(INDEXENTRY) ) {
            MesPrint("File error while exchanging expressions");
            Terminate(-1);
        }
        FlushFile(AR.StoreData.Handle);
    }
    if ( e2->status == STOREDEXPRESSION ) {
/*
        Find the name in the index and replace by the new name
*/
        TMproto[0] = EXPRESSION;
        TMproto[1] = SUBEXPSIZE;
        TMproto[2] = num2;
        TMproto[3] = 1;
        { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
        AT.TMaddr = TMproto;
        ind = FindInIndex(num1,&AR.StoreData,0,0);
        s1 = (SBYTE *)(AC.exprnames->namebuffer+e2->name);
        i = e2->namesize;
        s2 = ind->name;
        NCOPY(s2,s1,i);
        *s2 = 0;
        SeekFile(AR.StoreData.Handle,&(e2->onfile),SEEK_SET);
        if ( WriteFile(AR.StoreData.Handle,(UBYTE *)ind,
        (LONG)(sizeof(INDEXENTRY))) != sizeof(INDEXENTRY) ) {
            MesPrint("File error while exchanging expressions");
            Terminate(-1);
        }
        FlushFile(AR.StoreData.Handle);
    }
}

/*
        #] ExchangeExpressions :
        #[ GetFirstBracket :
*/

int GetFirstBracket(WORD *term, int num)
{
/*
        Gets the first bracket of the expression 'num'
        Puts it in term. If no brackets the answer is one.
        Routine should be thread-safe
*/
    GETIDENTITY
    POSITION position, oldposition;
    RENUMBER renumber;
    FILEHANDLE *fi;
    WORD type, *oldcomppointer, oldonefile, numword;
    WORD *t, *tstop;

    oldcomppointer = AR.CompressPointer;
    type = Expressions[num].status;
```

```
	if ( type == STOREDEXPRESSION ) {
		WORD TMproto[SUBEXPSIZE];
		TMproto[0] = EXPRESSION;
		TMproto[1] = SUBEXPSIZE;
		TMproto[2] = num;
		TMproto[3] = 1;
		{ int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
		AT.TMaddr = TMproto;
		PUTZERO(position);
		if ( ( renumber = GetTable(num,&position,0) ) == 0 ) {
			MesCall("GetFirstBracket");
			SETERROR(-1)
		}
		if ( GetFromStore(term,&position,renumber,&numword,num) < 0 ) {
			MesCall("GetFirstBracket");
			SETERROR(-1)
		}
/*
#ifdef WITHPTHREADS
*/
		if ( renumber->symb.lo != AN.dummyrenumlist )
			M_free(renumber->symb.lo,"VarSpace");
		M_free(renumber,"Renumber");
/*
#endif
*/
	}
	else {			/* Active expression */
		oldonefile = AR.GetOneFile;
		if ( type == HIDDENLEXPRESSION || type == HIDDENGEXPRESSION ) {
			AR.GetOneFile = 2; fi = AR.hidefile;
		}
		else {
			AR.GetOneFile = 0; fi = AR.infile;
		}
		if ( fi->handle >= 0 ) {
			PUTZERO(oldposition);
/*
			SeekFile(fi->handle,&oldposition,SEEK_CUR);
*/
			}
		else {
			SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
		}
		position = AS.OldOnFile[num];
		if ( GetOneTerm(BHEAD term,fi,&position,1) < 0
		|| ( GetOneTerm(BHEAD term,fi,&position,1) < 0 ) ) {
			MLOCK(ErrorMessageLock);
			MesCall("GetFirstBracket");
			MUNLOCK(ErrorMessageLock);
			SETERROR(-1)
		}
		if ( fi->handle >= 0 ) {
/*
			SeekFile(fi->handle,&oldposition,SEEK_SET);
			if ( ISNEGPOS(oldposition) ) {
				MLOCK(ErrorMessageLock);
				MesPrint("File error");
				MUNLOCK(ErrorMessageLock);
				SETERROR(-1)
			}
*/
		}
		else {
			fi->POfill = fi->PObuffer+BASEPOSITION(oldposition);
		}
		AR.GetOneFile = oldonefile;
	}
	AR.CompressPointer = oldcomppointer;
	if ( *term ) {
		tstop = term + *term; tstop -= ABS(tstop[-1]);
		t = term + 1;
		while ( t < tstop ) {
			if ( *t == HAAKJE ) break;
			t += t[1];
		}
		if ( t >= tstop ) {
			term[0] = 4; term[1] = 1; term[2] = 1; term[3] = 3;
		}
		else {
			*t++ = 1; *t++ = 1; *t++ = 3; *term = t - term;
		}
	}
	else {
		term[0] = 4; term[1] = 1; term[2] = 1; term[3] = 3;
	}
	return(*term);
```

```
}

/*
	#] GetFirstBracket :
	#[ GetFirstTerm :
*/

int GetFirstTerm(WORD *term, int num, int pre)
{
    GETIDENTITY
    POSITION position, oldposition;
    RENUMBER renumber;
    FILEHANDLE *fi;
    WORD type, *oldcomppointer, oldonefile, numword;

    oldcomppointer = AR.CompressPointer;
    type = Expressions[num].status;
    if ( type == STOREDEXPRESSION ) {
        WORD TMproto[SUBEXPSIZE];
        TMproto[0] = EXPRESSION;
        TMproto[1] = SUBEXPSIZE;
        TMproto[2] = num;
        TMproto[3] = 1;
        { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
        AT.TMaddr = TMproto;
        PUTZERO(position);
        if ( ( renumber = GetTable(num,&position,0) ) == 0 ) {
            MesCall("GetFirstTerm");
            SETERROR(-1)
        }
        if ( GetFromStore(term,&position,renumber,&numword,num) < 0 ) {
            MesCall("GetFirstTerm");
            SETERROR(-1)
        }
/*
#ifdef WITHPTHREADS
*/
        if ( renumber->symb.lo != AN.dummyrenumlist )
            M_free(renumber->symb.lo,"VarSpace");
        M_free(renumber,"Renumber");
/*
#endif
*/
    }
    else {          /* Active expression */
        oldonefile = AR.GetOneFile;
        if ( type == HIDDENLEXPRESSION || type == HIDDENGEXPRESSION ) {
            AR.GetOneFile = 2; fi = AR.hidefile;
        }
        else {
            AR.GetOneFile = 0;
            if ( Expressions[num].replace == NEWLYDEFINEDEXPRESSION ) {
                /* During execution, if the expression has already been processed it
                   will be in the outfile. If it has not, the usage is illegal according
                   to the manual, though no error is given. */
                if ( pre == 0 ) { fi = AR.outfile; }
                /* During preprocessing, the expression certainly has not been processed
                   yet. Print an error and terminate. */
                else {
                    MesPrint("&isnumerical: expression is not yet defined!");
                    SETERROR(-1);
                }
            }
            else {
                /* During execution, we should use the definition as stored at the end
                   of the previous module. This is in the infile. */
                if ( pre == 0 ) { fi = AR.infile; }
                /* During preprocessing, this function is called before the RevertScratch
                   at the beginning of this module's execution. Thus the expressions are
                   in the outfile of the previous module. */
                else { fi = AR.outfile; }
            }
        }
        if ( fi->handle >= 0 ) {
            PUTZERO(oldposition);
/*
            SeekFile(fi->handle,&oldposition,SEEK_CUR);
*/
            }
        else {
            SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
        }
        position = AS.OldOnFile[num];
        if ( GetOneTerm(BHEAD term,fi,&position,1) < 0
        || ( GetOneTerm(BHEAD term,fi,&position,1) < 0 ) ) {
            MLOCK(ErrorMessageLock);
            MesCall("GetFirstTerm");
```

```
            MUNLOCK(ErrorMessageLock);
            SETERROR(-1)
        }
        if ( fi->handle >= 0 ) {
/*
            SeekFile(fi->handle,&oldposition,SEEK_SET);
            if ( ISNEGPOS(oldposition) ) {
                MLOCK(ErrorMessageLock);
                MesPrint("File error");
                MUNLOCK(ErrorMessageLock);
                SETERROR(-1)
            }
*/
        }
        else {
            fi->POfill = fi->PObuffer+BASEPOSITION(oldposition);
        }
        AR.GetOneFile = oldonefile;
    }
    AR.CompressPointer = oldcomppointer;
    return(*term);
}

/*
        #] GetFirstTerm :
        #[ GetContent :
*/

int GetContent(WORD *content, int num)
{
/*
        Gets the content of the expression 'num'
        Puts it in content.
        Routine should be thread-safe
        The content is defined as the term that will make the expression 'num'
        with integer coefficients, no GCD and all common factors taken out,
        all negative powers removed when we divide the expression by this
        content.
*/
    GETIDENTITY
    POSITION position, oldposition;
    RENUMBER renumber;
    FILEHANDLE *fi;
    WORD type, *oldcomppointer, oldonefile, numword, *term, i;
    WORD *cbuffer = TermMalloc("GetContent");
    WORD *oldworkpointer = AT.WorkPointer;

    oldcomppointer = AR.CompressPointer;
    type = Expressions[num].status;
    if ( type == STOREDEXPRESSION ) {
        WORD TMproto[SUBEXPSIZE];
        TMproto[0] = EXPRESSION;
        TMproto[1] = SUBEXPSIZE;
        TMproto[2] = num;
        TMproto[3] = 1;
        { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
        AT.TMaddr = TMproto;
        PUTZERO(position);
        if ( ( renumber = GetTable(num,&position,0) ) == 0 ) goto CalledFrom;
        if ( GetFromStore(cbuffer,&position,renumber,&numword,num) < 0 ) goto CalledFrom;
        for(;;) {
            term = oldworkpointer;
            AR.CompressPointer = oldcomppointer;
            if ( GetFromStore(term,&position,renumber,&numword,num) < 0 ) goto CalledFrom;
            if ( *term == 0 ) break;
/*
            'merge' the two terms
*/
            if ( ContentMerge(BHEAD cbuffer,term) < 0 ) goto CalledFrom;
        }
/*
#ifdef WITHPTHREADS
*/
        if ( renumber->symb.lo != AN.dummyrenumlist )
            M_free(renumber->symb.lo,"VarSpace");
        M_free(renumber,"Renumber");
/*
#endif
*/
    }
    else {          /* Active expression */
        oldonefile = AR.GetOneFile;
        if ( type == HIDDENLEXPRESSION || type == HIDDENGEXPRESSION ) {
            AR.GetOneFile = 2; fi = AR.hidefile;
        }
        else {
            AR.GetOneFile = 0;
```

```
            if ( Expressions[num].replace == NEWLYDEFINEDEXPRESSION )
                 fi = AR.outfile;
            else fi = AR.infile;
        }
        if ( fi->handle >= 0 ) {
            PUTZERO(oldposition);
/*
            SeekFile(fi->handle,&oldposition,SEEK_CUR);
*/
            }
        else {
            SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
        }
        position = AS.OldOnFile[num];
        if ( GetOneTerm(BHEAD cbuffer,fi,&position,1) < 0 ) goto CalledFrom;
        AR.CompressPointer = oldcomppointer;
        if ( GetOneTerm(BHEAD cbuffer,fi,&position,1) < 0 ) goto CalledFrom;
/*
        Now go through the terms. For each term we have to test whether
        what is in cbuffer is also in that term. If not, we have to remove
        it from cbuffer. Additionally we have to accumulate the GCD of the
        numerators and the LCM of the denominators. This is all done in the
        routine ContentMerge.
*/
        for(;;) {
            term = oldworkpointer;
            AR.CompressPointer = oldcomppointer;
            if ( GetOneTerm(BHEAD term,fi,&position,1) < 0 ) goto CalledFrom;
            if ( *term == 0 ) break;
/*
            'merge' the two terms
*/
            if ( ContentMerge(BHEAD cbuffer,term) < 0 ) goto CalledFrom;
        }
        if ( fi->handle < 0 ) {
            fi->POfill = fi->PObuffer+BASEPOSITION(oldposition);
        }
        AR.GetOneFile = oldonefile;
    }
    AR.CompressPointer = oldcomppointer;
    for ( i = 0; i < *cbuffer; i++ ) content[i] = cbuffer[i];
    TermFree(cbuffer,"GetContent");
    AT.WorkPointer = oldworkpointer;
    return(*content);
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("GetContent");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] GetContent :
        #[ CleanupTerm :

        Removes noncommuting objects from the term
*/

int CleanupTerm(WORD *term)
{
    WORD *tstop, *t, *tfill, *tt;
    GETSTOP(term,tstop);
    t = term+1;
    while ( t < tstop ) {
        if ( *t >= FUNCTION && ( functions[*t-FUNCTION].commute || *t == DENOMINATOR ) ) {
            tfill = t; tt = t + t[1]; tstop = term + *term;
            while ( tt < tstop ) *tfill++ = *tt++;
            *term = tfill - term;
            tstop -= ABS(tfill[-1]);
        }
        else {
            t += t[1];
        }
    }
    return(0);
}

/*
        #] CleanupTerm :
        #[ ContentMerge :
*/

WORD ContentMerge(PHEAD WORD *content, WORD *term)
{
    GETBIDENTITY
    WORD *cstop, csize, crsize, sign = 1, numsize, densize, i, tnsize, tdsize;
    UWORD *num, *den, *tnum, *tden;
```

```
    WORD *outfill, *outb = TermMalloc("ContentMerge"), *ct;
    WORD *t, *tstop, tsize, trsize, *told;
    WORD *t1, *t2, *c1, *c2, i1, i2, *out1;
    WORD didsymbol = 0, diddotp = 0, tfirst;
    cstop = content + *content;
    csize = cstop[-1];
    if ( csize < 0 ) { sign = -sign; csize = -csize; }
    cstop -= csize;
    numsize = densize = crsize = (csize-1)/2;
    num = NumberMalloc("ContentMerge");
    den = NumberMalloc("ContentMerge");
    for ( i = 0; i < numsize; i++ ) num[i] = (UWORD)(cstop[i]);
    for ( i = 0; i < densize; i++ ) den[i] = (UWORD)(cstop[i+crsize]);
    while ( num[numsize-1] == 0 ) numsize--;
    while ( den[densize-1] == 0 ) densize--;
/*
    First we do the coefficient
*/
    tstop = term + *term;
    tsize = tstop[-1];
    if ( tsize < 0 ) tsize = -tsize;
/*  else { sign = 1; } */
    tstop = tstop - tsize;
    tnsize = tdsize = trsize = (tsize-1)/2;
    tnum = (UWORD *)tstop; tden = (UWORD *)(tstop + trsize);
    while ( tnum[tnsize-1] == 0 ) tnsize--;
    while ( tden[tdsize-1] == 0 ) tdsize--;
    GcdLong(BHEAD num, numsize, tnum, tnsize, num, &numsize);
    if ( LcmLong(BHEAD den, densize, tden, tdsize, den, &densize) ) goto CalledFrom;
    outfill = outb + 1;
    ct = content + 1;
    t = term + 1;
    while ( ct < cstop ) {
        switch ( *ct ) {
            case SYMBOL:
                didsymbol = 1;
                t = term+1;
                while ( t < tstop && *t != *ct ) t += t[1];
                if ( t >= tstop ) break;
                t1 = t+2; t2 = t+t[1];
                c1 = ct+2; c2 = ct+ct[1];
                out1 = outfill; *outfill++ = *ct; outfill++;
                while ( c1 < c2 && t1 < t2 ) {
                    if ( *c1 == *t1 ) {
                        if ( t1[1] <= c1[1] ) {
                            *outfill++ = *t1++; *outfill++ = *t1++;
                            c1 += 2;
                        }
                        else {
                            *outfill++ = *c1++; *outfill++ = *c1++;
                            t1 += 2;
                        }
                    }
                    else if ( *c1 < *t1 ) {
                        if ( c1[1] < 0 ) {
                            *outfill++ = *c1++; *outfill++ = *c1++;
                        }
                        else { c1 += 2; }
                    }
                    else {
                        if ( t1[1] < 0 ) {
                            *outfill++ = *t1++; *outfill++ = *t1++;
                        }
                        else t1 += 2;
                    }
                }
                while ( c1 < c2 ) {
                    if ( c1[1] < 0 ) { *outfill++ = c1[0]; *outfill++ = c1[1]; }
                    c1 += 2;
                }
                while ( t1 < t2 ) {
                    if ( t1[1] < 0 ) { *outfill++ = t1[0]; *outfill++ = t1[1]; }
                    t1 += 2;
                }
                out1[1] = outfill - out1;
                if ( out1[1] == 2 ) outfill = out1;
                break;
            case DOTPRODUCT:
                diddotp = 1;
                t = term+1;
                while ( t < tstop && *t != *ct ) t += t[1];
                if ( t >= tstop ) break;
                t1 = t+2; t2 = t+t[1];
                c1 = ct+2; c2 = ct+ct[1];
                out1 = outfill; *outfill++ = *ct; outfill++;
                while ( c1 < c2 && t1 < t2 ) {
                    if ( *c1 == *t1 && c1[1] == t1[1] ) {
```

```
                        if ( t1[2] <= c1[2] ) {
                            *outfill++ = *t1++; *outfill++ = *t1++; *outfill++ = *t1++;
                            c1 += 3;
                        }
                        else {
                            *outfill++ = *c1++; *outfill++ = *c1++; *outfill++ = *c1++;
                            t1 += 3;
                        }
                    }
                    else if ( *c1 < *t1 || ( *c1 == *t1 && c1[1] < t1[1] ) ) {
                        if ( c1[2] < 0 ) {
                            *outfill++ = *c1++; *outfill++ = *c1++; *outfill++ = *c1++;
                        }
                        else { c1 += 3; }
                    }
                    else {
                        if ( t1[2] < 0 ) {
                            *outfill++ = *t1++; *outfill++ = *t1++; *outfill++ = *t1++;
                        }
                        else t1 += 3;
                    }
                }
                while ( c1 < c2 ) {
                    if ( c1[2] < 0 ) { *outfill++ = c1[0]; *outfill++ = c1[1]; *outfill++ = c1[1]; }
                    c1 += 3;
                }
                while ( t1 < t2 ) {
                    if ( t1[2] < 0 ) { *outfill++ = t1[0]; *outfill++ = t1[1]; *outfill++ = t1[1]; }
                    t1 += 3;
                }
                out1[1] = outfill - out1;
                if ( out1[1] == 2 ) outfill = out1;
                break;
            case INDEX:
                t = term+1;
                while ( t < tstop && *t != *ct ) t += t[1];
                if ( t >= tstop ) break;
                t1 = t+2; t2 = t+t[1];
                c1 = ct+2; c2 = ct+ct[1];
                out1 = outfill; *outfill++ = *ct; outfill++;
                while ( c1 < c2 && t1 < t2 ) {
                    if ( *c1 == *t1 ) {
                        *outfill++ = *c1++;
                        t1 += 1;
                    }
                    else if ( *c1 < *t1 ) { c1 += 1; }
                    else { t1 += 1; }
                }
                out1[1] = outfill - out1;
                if ( out1[1] == 2 ) outfill = out1;
                break;
            case VECTOR:
            case DELTA:
                t = term+1;
                while ( t < tstop && *t != *ct ) t += t[1];
                if ( t >= tstop ) break;
                t1 = t+2; t2 = t+t[1];
                c1 = ct+2; c2 = ct+ct[1];
                out1 = outfill; *outfill++ = *ct; outfill++;
                while ( c1 < c2 && t1 < t2 ) {
                    if ( *c1 == *t1 && c1[1] && t1[1] ) {
                        *outfill++ = *c1++; *outfill++ = *c1++;
                        t1 += 2;
                    }
                    else if ( *c1 < *t1 || ( *c1 == *t1 && c1[1] < t1[1] ) ) {
                        c1 += 2;
                    }
                    else {
                        t1 += 2;
                    }
                }
                out1[1] = outfill - out1;
                if ( out1[1] == 2 ) outfill = out1;
                break;
            case GAMMA:
            default:            /* Functions */
                told = t;
                t = term+1;
                while ( t < tstop ) {
                    if ( *t != *ct ) { t += t[1]; continue; }
                    if ( ct[1] != t[1] ) { t += t[1]; continue; }
                    if ( ct[2] != t[2] ) { t += t[1]; continue; }
                    t1 = t; t2 = ct; i1 = t1[1]; i2 = t2[1];
                    while ( i1 > 0 ) {
                        if ( *t1 != *t2 ) break;
                        t1++; t2++; i1--;
                    }
```

```
                    if ( i1 != 0 ) { t += t[1]; continue; }
                    t1 = t;
                    for ( i = 0; i < i2; i++ ) { *outfill++ = *t++; }
/*
                    Mark as 'used'. The flags must be different!
*/
                    t1[2] |= SUBTERMUSED1;
                    ct[2] |= SUBTERMUSED2;
                    t = told;
                    break;
                }
                break;
        }
        ct += ct[1];
    }
    if ( diddotp == 0 ) {
        t = term+1; while ( t < tstop && *t != DOTPRODUCT ) t += t[1];
        if ( t < tstop ) { /* now we need the negative powers */
            tfirst = 1; told = outfill;
            for ( i = 2; i < t[1]; i += 3 ) {
                if ( t[i+2] < 0 ) {
                    if ( tfirst ) { *outfill++ = DOTPRODUCT; *outfill++ = 0; tfirst = 0; }
                    *outfill++ = t[i]; *outfill++ = t[i+1]; *outfill++ = t[i+2];
                }
            }
            if ( outfill > told ) told[1] = outfill-told;
        }
    }
    if ( didsymbol == 0 ) {
        t = term+1; while ( t < tstop && *t != SYMBOL ) t += t[1];
        if ( t < tstop ) { /* now we need the negative powers */
            tfirst = 1; told = outfill;
            for ( i = 2; i < t[1]; i += 2 ) {
                if ( t[i+1] < 0 ) {
                    if ( tfirst ) { *outfill++ = SYMBOL; *outfill++ = 0; tfirst = 0; }
                    *outfill++ = t[i]; *outfill++ = t[i+1];
                }
            }
            if ( outfill > told ) told[1] = outfill-told;
        }
    }
/*
    Now put the coefficient back.
*/
    if ( numsize < densize ) {
        for ( i = numsize; i < densize; i++ ) num[i] = 0;
        numsize = densize;
    }
    else if ( densize < numsize ) {
        for ( i = densize; i < numsize; i++ ) den[i] = 0;
        densize = numsize;
    }
    for ( i = 0; i < numsize; i++ ) *outfill++ = num[i];
    for ( i = 0; i < densize; i++ ) *outfill++ = den[i];
    csize = numsize+densize+1;
    if ( sign < 0 ) csize = -csize;
    *outfill++ = csize;
    *outb = outfill-outb;
    NumberFree(den,"ContentMerge");
    NumberFree(num,"ContentMerge");
    for ( i = 0; i < *outb; i++ ) content[i] = outb[i];
    TermFree(outb,"ContentMerge");
/*
    Now we have to 'restore' the term to its original.
    We do not restore the content, because if anything was used the
    new content overwrites the old. 6-mar-2018 JV
*/
    t = term + 1;
    while ( t < tstop ) {
        if ( *t >= FUNCTION ) t[2] &= ~SUBTERMUSED1;
        t += t[1];
    }
    return(*content);
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("GetContent");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] ContentMerge :
        #[ TermsInExpression :
*/

LONG TermsInExpression(WORD num)
{
```

```
    LONG x = Expressions[num].counter;
    if ( x >= 0 ) return(x);
    return(-1);
}

/*
        #] TermsInExpression :
        #[ SizeOfExpression :
*/

LONG SizeOfExpression(WORD num)
{
    LONG x = (LONG)(DIVPOS(Expressions[num].size,sizeof(WORD)));
    if ( x >= 0 ) return(x);
    return(-1);
}

/*
        #] SizeOfExpression :
        #[ UpdatePositions :
*/

void UpdatePositions(void)
{
    EXPRESSIONS e = Expressions;
    POSITION *old;
    WORD *oldw;
    int i;
    if ( NumExpressions > 0 &&
         ( AS.OldOnFile == 0 || AS.NumOldOnFile < NumExpressions ) ) {
        if ( AS.OldOnFile ) {
            old = AS.OldOnFile;
            AS.OldOnFile = (POSITION *)Malloc1(NumExpressions*sizeof(POSITION),"file pointers");
            for ( i = 0; i < AS.NumOldOnFile; i++ ) AS.OldOnFile[i] = old[i];
            AS.NumOldOnFile = NumExpressions;
            M_free(old,"process file pointers");
        }
        else {
            AS.OldOnFile = (POSITION *)Malloc1(NumExpressions*sizeof(POSITION),"file pointers");
            AS.NumOldOnFile = NumExpressions;
        }
    }
    if ( NumExpressions > 0 &&
         ( AS.OldNumFactors == 0 || AS.NumOldNumFactors < NumExpressions ) ) {
        if ( AS.OldNumFactors ) {
            oldw = AS.OldNumFactors;
            AS.OldNumFactors = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"numfactors pointers");
            for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.OldNumFactors[i] = oldw[i];
            M_free(oldw,"numfactors pointers");
            oldw = AS.Oldvflags;
            AS.Oldvflags = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"vflags pointers");
            for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Oldvflags[i] = oldw[i];
            AS.NumOldNumFactors = NumExpressions;
            M_free(oldw,"vflags pointers");
            oldw = AS.Olduflags;
            AS.Olduflags = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"uflags pointers");
            for ( i = 0; i < AS.NumOldNumFactors; i++ ) AS.Olduflags[i] = oldw[i];
            AS.NumOldNumFactors = NumExpressions;
            M_free(oldw,"uflags pointers");
        }
        else {
            AS.OldNumFactors = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"numfactors pointers");
            AS.Oldvflags = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"vflags pointers");
            AS.Olduflags = (WORD *)Malloc1(NumExpressions*sizeof(WORD),"uflags pointers");
            AS.NumOldNumFactors = NumExpressions;
        }
    }
    for ( i = 0; i < NumExpressions; i++ ) {
        AS.OldOnFile[i] = e[i].onfile;
        AS.OldNumFactors[i] = e[i].numfactors;
        AS.Oldvflags[i] = e[i].vflags;
        AS.Olduflags[i] = e[i].uflags;
    }
}

/*
        #] UpdatePositions :
        #[ CountTerms1 :          LONG CountTerms1()

        Counts the terms in the current deferred bracket
        Is mainly an adaptation of the routine Deferred in proces.c
*/

LONG CountTerms1(PHEAD0)
{
    GETBIDENTITY
    POSITION oldposition, startposition;
```

```
    WORD *t, *m, *mstop, decr, i, *oldwork, retval;
    WORD *oldipointer = AR.CompressPointer;
    WORD oldGetOneFile = AR.GetOneFile, olddeferflag = AR.DeferFlag;
    LONG numterms = 0;
    AR.GetOneFile = 1;
    oldwork = AT.WorkPointer;
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    AR.DeferFlag = 0;
    startposition = AR.DefPosition;
/*
        Store old position
*/
    if ( AR.infile->handle >= 0 ) {
        PUTZERO(oldposition);
/*
        SeekFile(AR.infile->handle,&oldposition,SEEK_CUR);
*/
    }
    else {
        SETBASEPOSITION(oldposition,AR.infile->POfill-AR.infile->PObuffer);
        AR.infile->POfill = (WORD *)((UBYTE *)(AR.infile->PObuffer)
                    +BASEPOSITION(startposition));
    }
/*
        Look in the CompressBuffer where the bracket contents start
*/
    t = m = AR.CompressBuffer;
    t += *t;
    mstop = t - ABS(t[-1]);
    m++;
    while ( *m != HAAKJE && m < mstop ) m += m[1];
    if ( m >= mstop ) { /* No deferred action! */
        numterms = 1;
        AR.DeferFlag = olddeferflag;
        AT.WorkPointer = oldwork;
        AR.GetOneFile = oldGetOneFile;
        return(numterms);
    }
    mstop = m + m[1];
    decr = WORDDIF(mstop,AR.CompressBuffer)-1;

    m = AR.CompressBuffer;
    t = AR.CompressPointer;
    i = *m;
    NCOPY(t,m,i);
    AR.TePos = 0;
    AN.TeSuOut = 0;
/*
        Status:
        First bracket content starts at mstop.
        Next term starts at startposition.
        Decompression information is in AR.CompressPointer.
        The outside of the bracket runs from AR.CompressBuffer+1 to mstop.
*/
    AR.CompressPointer = oldipointer;
    for(;;) {
        numterms++;
        retval = GetOneTerm(BHEAD AT.WorkPointer,AR.infile,&startposition,0);
        if ( retval >= 0 ) AR.CompressPointer = oldipointer;
        if ( retval <= 0 ) break;
        t = AR.CompressPointer;
        if ( *t < (1 + decr + ABS(*(t+*t-1))) ) break;
        t++;
        m = AR.CompressBuffer+1;
        while ( m < mstop ) {
            if ( *m != *t ) goto Thatsit;
            m++; t++;
        }
    }
Thatsit:;
/*
        Finished. Reposition the file, restore information and return.
*/
    AT.WorkPointer = oldwork;
    if ( AR.infile->handle >= 0 ) {
/*
        SeekFile(AR.infile->handle,&oldposition,SEEK_SET);
*/
    }
    else {
        AR.infile->POfill = AR.infile->PObuffer + BASEPOSITION(oldposition);
    }
    AR.DeferFlag = olddeferflag;
    AR.GetOneFile = oldGetOneFile;
    return(numterms);
}

```

```
/*
        #] CountTerms1 :
        #[ TermsInBracket :     LONG TermsInBracket(term,level)

    The function TermsInBracket_()
    Syntax:
        TermsInBracket_() : The current bracket in a Keep Brackets
        TermsInBracket_(bracket) : This bracket in the current expression
        TermsInBracket_(expression,bracket) : This bracket in the given expression
    All other specifications don't have any effect.
*/

#define CURRENTBRACKET 1
#define BRACKETCURRENTEXPR 2
#define BRACKETOTHEREXPR 3
#define NOBRACKETACTIVE 4

LONG TermsInBracket(PHEAD WORD *term, WORD level)
{
    WORD *t, *tstop, *b, *tt, *n1, *n2;
    int type = 0, i, num;
    LONG numterms = 0;
    WORD *bracketbuffer = AT.WorkPointer;
    t = term; GETSTOP(t,tstop);
    t++; b = bracketbuffer;
    while ( t < tstop ) {
        if ( *t != TERMSINBRACKET ) { t += t[1]; continue; }
        if ( t[1] == FUNHEAD || (
             t[1] == FUNHEAD+2
            && t[FUNHEAD] == -SNUMBER
            && t[FUNHEAD+1] == 0
         ) ) {
            if ( AC.ComDefer == 0 ) {
                type = NOBRACKETACTIVE;
            }
            else {
                type = CURRENTBRACKET;
            }
            *b = 0;
            break;
        }
        if ( t[FUNHEAD] == -EXPRESSION ) {
            if ( t[FUNHEAD+2] < 0 ) {
                if ( ( t[FUNHEAD+2] <= -FUNCTION ) && ( t[1] == FUNHEAD+3 ) ) {
                    type = BRACKETOTHEREXPR;
                    *b++ = FUNHEAD+4; *b++ = -t[FUNHEAD+2]; *b++ = FUNHEAD;
                    for ( i = 2; i < FUNHEAD; i++ ) *b++ = 0;
                    *b++ = 1; *b++ = 1; *b++ = 3;
                    break;
                }
                else if ( ( t[FUNHEAD+2] > -FUNCTION ) && ( t[1] == FUNHEAD+4 ) ) {
                    type = BRACKETOTHEREXPR;
                    tt = t + FUNHEAD+2;
                    switch ( *tt ) {
                        case -SYMBOL:
                            *b++ = 8; *b++ = SYMBOL; *b++ = 4; *b++ = tt[1];
                            *b++ = 1; *b++ = 1; *b++ = 1; *b++ = 3;
                            break;
                        case -SNUMBER:
                            if ( tt[1] == 1 ) {
                                *b++ = 4; *b++ = 1; *b++ = 1; *b++ = 3;
                            }
                            else goto IllBraReq;
                            break;
                        default:
                            goto IllBraReq;
                    }
                    break;
                }
            }
            else if ( ( t[FUNHEAD+2] == (t[1]-FUNHEAD-2) ) &&
                      ( t[FUNHEAD+2+ARGHEAD] == (t[FUNHEAD+2]-ARGHEAD) ) ) {
                type = BRACKETOTHEREXPR;
                tt = t + FUNHEAD + ARGHEAD; num = *tt;
                for ( i = 0; i < num; i++ ) *b++ = *tt++;
                break;
            }
        }
        else {
            if ( t[FUNHEAD] < 0 ) {
                if ( ( t[FUNHEAD] <= -FUNCTION ) && ( t[1] == FUNHEAD+1 ) ) {
                    type = BRACKETCURRENTEXPR;
                    *b++ = FUNHEAD+4; *b++ = -t[FUNHEAD+2]; *b++ = FUNHEAD;
                    for ( i = 2; i < FUNHEAD; i++ ) *b++ = 0;
                    *b++ = 1; *b++ = 1; *b++ = 3; *b = 0;
                    break;
                }
```

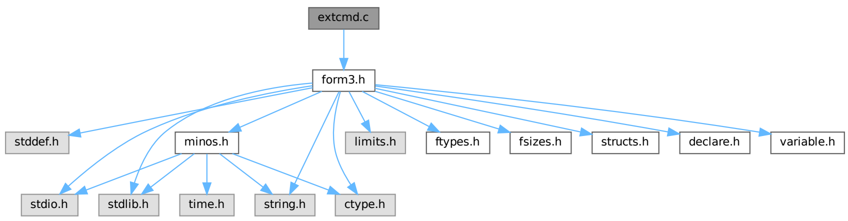
```
                else if ( ( t[FUNHEAD] > -FUNCTION ) && ( t[1] == FUNHEAD+2 ) ) {
                    type = BRACKETCURRENTEXPR;
                    tt = t + FUNHEAD+2;
                    switch ( *tt ) {
                        case -SYMBOL:
                            *b++ = 8; *b++ = SYMBOL; *b++ = 4; *b++ = tt[1];
                            *b++ = 1; *b++ = 1; *b++ = 1; *b++ = 3;
                            break;
                        case -SNUMBER:
                            if ( tt[1] == 1 ) {
                                *b++ = 4; *b++ = 1; *b++ = 1; *b++ = 3;
                            }
                            else goto IllBraReq;
                            break;
                        default:
                            goto IllBraReq;
                    }
                    break;
                }
            }
            else if ( ( t[FUNHEAD] == (t[1]-FUNHEAD) ) &&
                      ( t[FUNHEAD+ARGHEAD] == (t[FUNHEAD]-ARGHEAD) ) ) {
                type = BRACKETCURRENTEXPR;
                tt = t + FUNHEAD + ARGHEAD; num = *tt;
                for ( i = 0; i < num; i++ ) *b++ = *tt++;
                break;
            }
            else {
IllBraReq:;
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal bracket request in termsinbracket_ function.");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
        }
        t += t[1];
    }
    AT.WorkPointer = b;
    if ( AT.WorkPointer + *term +4 > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MesPrint("Called from termsinbracket_ function.");
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
/*
    We are now in the position to look for the bracket
*/
    switch ( type ) {
        case CURRENTBRACKET:
/*
            The code here should be rather similar to when we pick up
            the contents of the bracket. In our case we only count the
            terms though.
*/
            numterms = CountTerms1(BHEAD0);
            break;
        case BRACKETCURRENTEXPR:
/*
            Not implemented yet.
*/
            MLOCK(ErrorMessageLock);
            MesPrint("termsinbracket_ function currently only handles Keep Brackets.");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        case BRACKETOTHEREXPR:
            MLOCK(ErrorMessageLock);
            MesPrint("termsinbracket_ function currently only handles Keep Brackets.");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        case NOBRACKETACTIVE:
            numterms = 1;
            break;
    }
/*
    Now we have the number in numterms. We replace the function by it.
*/
    n1 = term; n2 = AT.WorkPointer; tstop = n1 + *n1;
    while ( n1 < t ) *n2++ = *n1++;
    i = numterms >> BITSINWORD;
    if ( i == 0 ) {
        *n2++ = LNUMBER; *n2++ = 4; *n2++ = 1; *n2++ = (WORD)(numterms & WORDMASK);
    }
    else {
        *n2++ = LNUMBER; *n2++ = 5; *n2++ = 2;
        *n2++ = (WORD)(numterms & WORDMASK); *n2++ = i;
    }
```

```
    n1 += n1[1];
    while ( n1 < tstop ) *n2++ = *n1++;
    AT.WorkPointer[0] = n2 - AT.WorkPointer;
    AT.WorkPointer = n2;
    if ( Generator(BHEAD n1,level) < 0 ) {
        AT.WorkPointer = bracketbuffer;
        MLOCK(ErrorMessageLock);
        MesPrint("Called from termsinbracket_ function.");
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
/*
    Finished. Reset things and return.
*/
    AT.WorkPointer = bracketbuffer;
    return(numterms);
}
/*
        #] TermsInBracket :      LONG TermsInBracket(term,level)
    #] Expressions :
*/
```

4.31 extcmd.c File Reference

`#include "form3.h"`

Include dependency graph for extcmd.c:



Functions

- int openExternalChannel (UBYTE *cmd, int daemonize, UBYTE *shellname, UBYTE *stderrname)
- int initPresetExternalChannels (UBYTE *theline, int thetimeout)
- int closeExternalChannel (int n)
- int selectExternalChannel (int n)
- int getCurrentExternalChannel (void)
- void closeAllExternalChannels (void)

4.31.1 Detailed Description

The system that takes care of communication with external programs.

Definition in file extcmd.c.

4.31.2 Function Documentation

4.31.2.1 openExternalChannel()

```
int openExternalChannel (
            UBYTE * cmd,
            int daemonize,
            UBYTE * shellname,
            UBYTE * stderrname )
```

Definition at line 230 of file extcmd.c.

4.31.2.2 initPresetExternalChannels()

```
int initPresetExternalChannels (
            UBYTE * theline,
            int thetimeout )
```

Definition at line 232 of file extcmd.c.

4.31.2.3 closeExternalChannel()

```
int closeExternalChannel (
            int n )
```

Definition at line 233 of file extcmd.c.

4.31.2.4 selectExternalChannel()

```
int selectExternalChannel (
            int n )
```

Definition at line 234 of file extcmd.c.

4.31.2.5 getCurrentExternalChannel()

```
int getCurrentExternalChannel (
            void )
```

Definition at line 235 of file extcmd.c.

4.31.2.6 closeAllExternalChannels()

```
void closeAllExternalChannels (
            void )
```

Definition at line 236 of file extcmd.c.

4.32 extcmd.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Documentation :

    This module is written by M.Tentyukov as a part of implementation of
    interaction between FORM and external processes, first release
    09.04.2004. A part of this code is copied from the DIANA project, which was
    written by M. Tentyukov and published under the GPL version 2 as
    published by the Free Software Foundation.

    This file is completely re-written by M.Tentyukov in May 2006.
    Since the interface was changed, the public function were changed,
    also. A new public functions were added: initPresetExternalChannels()
    (see comments just before this function in the present file) and
    setKillModeForExternalChannel (a pointer, not a function).

    If a macro WITHEXTERNALCHANNEL is not defined, all public functions
    are stubs returning failure.

    The idea is to start an external command  swallowing
    its stdin and stdout. This can be done by means of the function
    int openExternalChannel(cmd,daemonize,shellname,stderrname), where
    cmd is a command to run,
    daemonize: if !=0 then start the command in the "daemon" mode,
    shellname: if !=NULL, execute the command in a subshell,
    stderrname: if != NULL, redirect stderr of the command to this file.
    The function returns some small positive integer number (the
    descriptor of a newly created external channel), or -1 on failure.

    After the command is started, it becomes a _current_ opened external
    channel. The buffer can be sent to its stdin by a function
    int writeBufToExtChannel(buf, n)
    (here buf is a pointer to the buffer, n is the length in bytes; the
    function returns 0 in success, or -1 on failure),
    or one character can be read from its stdout by
    means of the function
    int getcFromExtChannel().

    The latter returns the
    character casted to integer, or something <0. This can be -2 (if there
    is no current external channel) or EOF, if the external program closes
    its stdout, or if the external program outputs a string coinciding
    with a _terminator_.

    By default, the terminator if an empty line. For the current external
    channel it can be set by means of the function
    int setTerminatorForExternalChannel(newterminator).
    The function returns 0 in success, or !0 if something is wrong (no
    current channel, too long terminator).

    After getcFromExtChannel() returns EOF, the current channel becomes
    undefined. Any further attempts to read information by
    getcFromExtChannel() result in -2. To set (re-set) a current channel,
    the function
    int selectExternalChannel(n)
    can be used. This function accepts the valid external channel
```

```
    descriptor (returned by openExternalChannel) and returns the
    descriptor of a previous current channel (0, if there was no current
    channel, or -1, if the external channel descriptor is invalid).
    If n == 0, the function undefine the current external channel.

    The function
    int closeExternalChannel(n)
    destroys the opened external channel with the descriptor n. It returns
    0 in success, or -1 on failure. If the corresponding external channel
    was the current one, the current channel becomes undefined. If n==0,
    the function closes the current external channel.

    The function
    int getCurrentExternalChannel(void)
    returns the descriptor if the current external channel, or 0 , if
    there is no current external channel.

    The function
    void closeAllExternalChannels(void)

    destroys all opened external channels.

    List of all public functions:
    int openExternalChannel(UBYTE *cmd,int daemonize,UBYTE *shellname, UBYTE * stderrname);
    int initPresetExternalChannels(UBYTE *theline, int thetimeout);
    int setTerminatorForExternalChannel(char *newterminator);
    int setKillModeForExternalChannel(int signum, int sentToWholeGroup);
    int closeExternalChannel(int n);
    int selectExternalChannel(int n);
    int writeBufToExtChannel(char *buf,int n);
    int getcFromExtChannel(void);
    int getCurrentExternalChannel(void);
    void closeAllExternalChannels(void);

    ATTENTION!

    Four of them:
    1 setTerminatorForExternalChannel
    2 setKillModeForExternalChannel
    3 writeBufToExtChannel
    4 getcFromExtChannel

    are NOT functions, but variables (pointers) of a corrsponding type.
    They are initialised by proper values to avoid repeated error checking.

    All public functions are independent of realization hidden in this module.
    All other functions may have a returned type/parameters type local w.r.t.
    this module; they are not declared outside of this file.

    #] Documentation :
    #[ Selftest initializing:
*/

/*
Uncomment to get a self-consistent program:
#define SELFTEST 1
*/


#ifdef SELFTEST
#define WITHEXTERNALCHANNEL 1
#ifdef _MSC_VER
#define FORM_INLINE __inline
#else
#define FORM_INLINE inline
#endif

/*
    From form3.h:
*/
typedef unsigned char UBYTE;

/*The following variables should be defined in variable.h:*/
extern int (*writeBufToExtChannel)(char *buffer, size_t n);
extern int (*getcFromExtChannel)();
extern int (*setTerminatorForExternalChannel)(char *buffer);
extern int (*setKillModeForExternalChannel)(int signum, int sentToWholeGroup);

#else /*ifdef SELFTEST*/
#include "form3.h"
#endif /*ifdef SELFTEST ... else*/
/*
pid_t  getExternalChannelPid(void);
*/
/*
    #] Selftest initializing:
    #[ Includes :
```

```
*/
#ifdef WITHEXTERNALCHANNEL
#include <stdio.h>
#ifndef _MSC_VER
#include <unistd.h>
#endif
#include <fcntl.h>
#include <sys/types.h>
#ifndef _MSC_VER
#include <sys/time.h>
#include <sys/wait.h>
#endif
#include <errno.h>
#include <signal.h>
#include <limits.h>
/*
    #] Includes :
    #[ FailureFunctions:
*/

/*Non-initialized variant of public functions:*/
int writeBufToExtChannelFailure(char *buf, size_t count)
{
    DUMMYUSE(buf); DUMMYUSE(count);
    return(-1);
}/*writeBufToExtChannelFailure*/

int setTerminatorForExternalChannelFailure(char *newTerminator)
{
    DUMMYUSE(newTerminator);
    return(-1);
}/*setTerminatorForExternalChannelFailure*/

int setKillModeForExternalChannelFailure(int signum, int sentToWholeGroup)
{
    DUMMYUSE(signum); DUMMYUSE(sentToWholeGroup);
    return(-1);
}/*setKillModeForExternalChannelFailure*/

int getcFromExtChannelFailure(void)
{
    return(-2);
}/*getcFromExtChannelFailure*/

int (*writeBufToExtChannel)(char *buffer, size_t n) = &writeBufToExtChannelFailure;
int (*setTerminatorForExternalChannel)(char *buffer) =
    &setTerminatorForExternalChannelFailure;
int (*setKillModeForExternalChannel)(int signum, int sentToWholeGroup) =
    &setKillModeForExternalChannelFailure;
int (*getcFromExtChannel)(void) = &getcFromExtChannelFailure;
#endif
/*
    #] FailureFunctions:
    #[ Stubs :
*/
#ifndef WITHEXTERNALCHANNEL
/*Stubs for public functions:*/
int openExternalChannel(UBYTE *cmd, int daemonize, UBYTE *shellname, UBYTE *stderrname)
{ DUMMYUSE(cmd); DUMMYUSE(daemonize); DUMMYUSE(shellname); DUMMYUSE(stderrname); return(-1); };
int initPresetExternalChannels(UBYTE *theline, int thetimeout) { DUMMYUSE(theline);
   DUMMYUSE(thetimeout); return(-1); };
int closeExternalChannel(int n) { DUMMYUSE(n); return(-1); };
int selectExternalChannel(int n) { DUMMYUSE(n); return(-1); };
int getCurrentExternalChannel(void) { return(0); };
void closeAllExternalChannels(void) {};
#else /*ifndef WITHEXTERNALCHANNEL*/
/*
    #] Stubs :
    #[ Local types :
*/
/*First argument for the function signal:*/
#ifndef INTSIGHANDLER
typedef void (*mysighandler_t)(int);
#else
/* Sometimes, this nonsense may occurs:*/
/*typedef int (*mysighandler_t)(int);*/
#endif

/*Input IO buffer size increment -- each time the buffer
is expired it will be increased by this value (in bytes):*/
#define DELTA_EXT_BUF 128

/*Re-allocatable array containing External Channel
handlers increased each time by this value:*/
#define DELTA_EXT_LIST 8

/*How many times I/O routines may attempt to continue their work
```

```
in some failures:*/
#define MAX_FAILS_IO 2

/*The external channel handler structure:*/
typedef struct ExternalChannel {
    pid_t pid;       /*PID of the external process*/
    pid_t gpid;       /*process  group ID of the external process.
                              If <=0, not used, if >0, the kill signals
                              is sent to the whole group */
    FILE *frec;      /*stdout of the external process*/
    char *INbuf;     /*External channel buffer*/
    char *IBfill;    /*Position in INbuf from which the next input character will be read*/
    char *IBfull;    /*End of read INbuf*/
    char *IBstop;    /*end of allocated space for INbuf*/
    char *terminator;/* Terminator - when extern. program outputs ONLY this string,
                              it is assumed that the answer is ready, and getcFromExtChannel
                              returns EOF. Should not be longer then the minimal buffer!*/
    /*Info fields, not changable after creating a channel:*/
    char *cmd;       /*the command*/
    char *shellname;
    char *stderrname;/*filename to redirect stderr, or NULL*/
    int fsend;       /*stdin of the external process*/
    int killSignal; /*signal to kill*/
    int daemonize;/*0 --neither setsid nor daemonize, !=0 -- full daemonization*/
    PADPOINTER(0,3,0,0);
} EXTHANDLE;


static EXTHANDLE *externalChannelsList=0;
/*Here integers are better than pointers: */
static int externalChannelsListStop=0;
static int externalChannelsListFill=0;

/*"current" external channel:*/
static EXTHANDLE *externalChannelsListTop=0;
/*
    #] Local types :
    #[ Selftest functions :
*/
#ifdef SELFTEST

/*For malloc prototype:*/
#include <stdlib.h>

/*StrLen, Malloc1, M_free and strDup1 are defined in tools.c -- here only emulation:*/
int StrLen(char *pattern)
{
register  char *p=(char*)pattern;
     while(*p)p++;
     return((int) ((p-(char*)pattern)) );
}/*StrLen*/
void *Malloc1(int l, char *c)
{
    return(malloc(l));
}
void M_free(void *p,char *c)
{
    return(free(p));
}

char *strDup1(UBYTE *instring, char *ifwrong)
{
    UBYTE *s = instring, *to;
    while ( *s ) s++;
    to = s = (UBYTE *)Malloc1((s-instring)+1,ifwrong);
    while ( *instring ) *to++ = *instring++;
    *to = 0;
    return(s);
}

/*PutPreVar from pre.c -- just ths stub:*/
int PutPreVar(UBYTE *a,UBYTE *b,UBYTE *c,int i)
{
    return(0);
}

#endif
/*
    #] Selftest functions :
    #[ Local functions :
*/

/*Initialize one cell of handler:*/
static FORM_INLINE void extHandlerInit(EXTHANDLE *h)
{
    h->pid=-1;
    h->gpid=-1;
```

```
    h->fsend=0;
    h->killSignal=SIGKILL;
    h->daemonize=1;
    h->frec=NULL;
    h->INbuf=h->IBfill=h->IBfull=h->IBstop=
    h->terminator=h->cmd=h->shellname=h->stderrname=NULL;
}/*extHandlerInit*/

/* Copies each field of handler:*/
static FORM_INLINE void extHandlerSwallowCopy(EXTHANDLE *to, EXTHANDLE *from)
{
    to->pid=from->pid;
    to->gpid=from->gpid;
    to->fsend=from->fsend;
    to->killSignal=from->killSignal;
    to->daemonize=from->daemonize;
    to->frec=from->frec;
    to->INbuf=from->INbuf;
    to->IBfill=from->IBfill;
    to->IBfull=from->IBfull;
    to->IBstop=from->IBstop;
    to->terminator=from->terminator;
    to->cmd=from->cmd;
    to->shellname=from->shellname;
    to->stderrname=from->stderrname;
}/*extHandlerSwallow*/

/*Allocates memory for fields of handler which have no fixed
 storage size and initializes some fields:*/
static FORM_INLINE void
extHandlerAlloc(EXTHANDLE *h, char *cmd, char *shellname, char *stderrname)
{
    h->IBfill=h->IBfull=h->INbuf=
                Malloc1(DELTA_EXT_BUF,"External channel buffer");
    h->IBstop=h->INbuf+DELTA_EXT_BUF;
    /*Initialize a terminator:*/
    *(h->terminator=Malloc1(DELTA_EXT_BUF,"External channel terminator"))='\n';
    (h->terminator)[1]='\0';/*By default the terminator is '\n'*/
    /*Deep copy all strings:*/
    if(cmd!=NULL)
        h->cmd=(char *)strDup1((UBYTE *)cmd,"External channel command");
    else/*cmd cannot be NULL! If this is NULL then force it to be something special*/
        h->cmd=(char *)strDup1((UBYTE *)"/","External channel command");
    if(shellname!=NULL)
        h->shellname=
             (char *)strDup1((UBYTE *)shellname,"External channel shell name");
    if(stderrname!=NULL)
        h->stderrname=
             (char *)strDup1((UBYTE *)stderrname,"External channel stderr name");
}/*extHandlerAlloc*/

/*Disallocates dynamically allocated fields of a handler:*/
static FORM_INLINE void extHandlerFree(EXTHANDLE *h)
{
    if(h->stderrname) M_free(h->stderrname,"External channel stderr name");
    if(h->shellname) M_free(h->shellname,"External channel shell name");
    if(h->cmd) M_free(h->cmd,"External channel command");
    if(h->terminator)M_free(h->terminator,"External channel terminator");
    if(h->INbuf)M_free(h->INbuf,"External channel buffer");
    extHandlerInit(h);
}/*extHandlerFree*/
/* Closes all descriptors, kills the external process, frees all internal fields,
BUT does NOT free the main container:*/
static void destroyExternalChannel(EXTHANDLE *h)
{
    /*Note, this function works in parallel mode correctly, see comments below.*/

    /*Note, for slaves in a parallel mode h->pid == 0:*/
    if( (h->pid > 0) && (h->killSignal > 0) ){
        int chstatus;
        if( h->gpid > 0)
            chstatus=kill(-h->gpid,h->killSignal);
        else
            chstatus=kill(h->pid,h->killSignal);
        if(chstatus==0)
        /*If the process will not be killed by this signal, FORM hangs up here!:*/
            waitpid(h->pid, &chstatus, 0);
    }/*if( (h->pid > 0) && (h->killSignal > 0) )*/

    /*Note, for slaves in a parallel mode h->frec == h->fsend == 0:*/
    if(h->frec) fclose(h->frec);
    if( h->fsend > 0) close(h->fsend);

    extHandlerFree(h);
    /*Does not do "free(h)"!*/
}/*destroyExternalChannel*/

```

```
/*Wrapper to the read() syscall, to handle possible interrupts by unblocked signals:*/
static FORM_INLINE ssize_t read2b(int fd, char *buf, size_t count)
{
ssize_t res;

    if( (res=read(fd,buf,count)) <1 )/*EOF or read is interrupted by a signal?:*/
         while( (errno == EINTR)&&(res <1) )
             /*The call was interrupted by  a  signal  before  any data was read, try again:*/
             res=read(fd,buf,count);
    return (res);
}/*read2b*/

/*Wrapper to the write() syscall, to handle possible interrupts by unblocked signals:*/
static FORM_INLINE ssize_t writeFromb(int fd, char *buf, size_t count)
{
ssize_t res;
    if( (res=write(fd,buf,count)) <1 )/*Is write interrupted by a signal?:*/
         while( (errno == EINTR)&&(res <1) )
             /*The call was interrupted by a signal before any data was written, try again:*/
             res=write(fd,buf,count);
    return (res);
}/*writeFromb*/

/* Read one (binary) PID from the file descriptor fd:*/
static FORM_INLINE pid_t readpid(int fd)
{
pid_t tmp;
    if(read2b(fd,(char*)&tmp,sizeof(pid_t))!=sizeof(pid_t))
      return (pid_t)-1;
    return tmp;
}/*readpid*/

/* Writeone (binary) PID to the file descriptor fd:*/
static FORM_INLINE pid_t writepid(int fd, pid_t thepid)
{
    if(writeFromb(fd,(char*)&thepid,sizeof(pid_t))!=sizeof(pid_t))
      return (pid_t)-1;
    return (pid_t)0;
}/*readpid*/

/*Wrtites exactly count bytes from the  buffer buf into the descriptor fd, independently on
  nonblocked signals and the MPU/buffer hits. Returns 0 or -1:
*/
static FORM_INLINE int writexactly(int fd, char *buf, size_t count)
{
ssize_t i;
int j=0,n=0;

    for(;;){
        if(  (i=writeFromb(fd, buf+j, count-j)) < 0 ) return(-1);
        j+=i;
        if ( ((size_t)j) == count ) break;
        if(i==0)n++;
        else n=0;
        if(n>MAX_FAILS_IO)return (-1);
    }/*for(;;)*/
    return (0);
}/*writexactly*/

/* Set the FD_CLOEXEC  flag of desc if value is nonzero,
 or clear the flag if value is 0.
 Return 0 on success, or -1 on error with errno  set. */
static int set_cloexec_flag(int desc, int value)
{
int oldflags = fcntl (desc, F_GETFD, 0);
    /* If reading the flags failed, return error indication now.*/
    if (oldflags < 0)
        return (oldflags);
    /* Set just the flag we want to set. */
    if (value != 0)
        oldflags |= FD_CLOEXEC;
    else
      oldflags &= ~FD_CLOEXEC;
    /* Store modified flag word in the descriptor. */
    return (fcntl(desc, F_SETFD, oldflags));
}/*set_cloexec_flag*/

/* Adds the integer fd to the array fifo of length top+1 so that
the array is ascendantly ordered. It is supposed that all 0 -- top-1
elements in the array are already ordered:*/
static void pushDescriptor(int *fifo, int top, int fd)
{
    if ( top == 0 ) {
        fifo[top] = fd;
    } else {
        int ins=top-1;
        if( fifo[ins]<=fd )
```

```
            fifo[top]=fd;
        else{
            /*Find the position:*/
            while( (ins>=0)&&(fifo[ins]>fd) )ins--;
            /*Move all elements starting from the position to the right:*/
            for(ins++;top>ins; top--)
                fifo[top]=fifo[top-1];
            /*Put the element:*/
            fifo[ins]=fd;
        }
    }
}/*pushDescriptor*/

/*Close all descriptors greater or equal than startFrom except those
  listed in the ascendantly ordered array usedFd of length top:*/
static FORM_INLINE void closeAllDescriptors(int startFrom, int *usedFd, int top)
{
int n,maxfd;
    for(n=0;n<top; n++){
        maxfd=usedFd[n];
        for(;startFrom<maxfd;startFrom++)/*Close all less than maxfd*/
            close(startFrom);
        startFrom++;/*skip maxfd*/
    }/*for(;startFrom<maxfd;startFrom++)*/
    /*Close all the rest:*/
    maxfd=sysconf(_SC_OPEN_MAX);
    for(;startFrom<maxfd;startFrom++)
        close(startFrom);
}/*closeAllDescriptors*/

typedef int L_APIPE[2];
/*Closes both pipe descriptors if not -1:*/
static void closepipe(L_APIPE *thepipe)
{
  if( (*thepipe)[0] != -1) close ((*thepipe)[0]);
  if( (*thepipe)[1] != -1) close ((*thepipe)[1]);
}/*closepipe*/

/*Parses the cmd line like "sh -c myprg", passes each option to the
  corresponding element of argv, ends agrv by NULL. Returns the
  number of stored argv elements, or -1 if fails:*/
static FORM_INLINE int parseline(char **argv, char *cmd)
{
int n=0;
    while(*cmd != '\0'){
        for(; (*cmd <= ' ') && (*cmd != '\0') ;cmd++);
        if(*cmd != '\0'){
            argv[n]=cmd;
            while(*++cmd > ' ');
            if(*cmd != '\0')
                *cmd++ = '\0';
            n++;
        }/*if(*cmd != '\0')*/
    }/*while(*cmd != '\0')*/
    argv[n]=NULL;
    if(n==0)return -1;
    return n;
}/*parseline*/

/*Reads positive decimal number (not bigger than maxnum)
  from the string and returns it;
  the pointer *b is set to the next non-converted character:*/
static LONG str2i(char *str, char **b, LONG maxnum)
{
    LONG n=0;
    /*Eat trailing spaces:*/
    while(*str<=' ')if(*str++ == '\0')return(-1);
    (*b)=str;
    while (*str>='0'&&*str<='9')
        if( (n=10*n + *str++ - '0')>maxnum )
            return(-1);
    if((*b)==str)/*No single number!*/
        return(-1);
    (*b)=str;
    return(n);
}

/*Converts long integer to a decimal representation.
  For portability reasons we cannot use LongCopy from tools.c
  since theoretically LONG may be smaller than pid_t:*/
static char *l2s(LONG x, char *to)
{
    char *s;
    int i = 0, j;
    s = to;
    do { *s++ = (x % 10)+'0'; i++; } while ( ( x /= 10 ) != 0 );
    *s-- = '\0';
```

```
    j = ( i - 1 ) » 1;
    while ( j >= 0 ) {
        i = to[j]; to[j] = s[-j]; s[-j] = (char)i; j--;
    }
    return(s+1);
}

/*like strcat() but returns the pointer to the end of the
resulting string:*/
static  FORM_INLINE char *addStr(char *to, char *from)
{
    while(  (*to++ = *from++)!='\0' );
    return(to-1);
}/*addStr*/


/*Try to write (atomically) short buffer (of length count) to fd.
  timeout is a timeout in millisecs. Returns number of written bytes or -1:*/
static FORM_INLINE ssize_t writeSome(int fd, char *buf, size_t count, int timeout)
{
    ssize_t res = 0;
    fd_set wfds;
    struct timeval tv;
    int nrep=5;/*five attempts it interrupted by a non-blocking signal*/
    int flags = fcntl(fd, F_GETFL,0);

    /*Add O_NONBLOCK:*/
    fcntl(fd,F_SETFL, flags | O_NONBLOCK);
    /* important -- in order to avoid blocking of short receiver buffer*/

    do{
        FD_ZERO(&wfds);
        FD_SET(fd, &wfds);
        /* Wait up to timeout. */

        tv.tv_sec =timeout /1000;
        tv.tv_usec = (timeout % 1000)*1000;
        nrep--;

        switch(select(fd+1, NULL, &wfds, NULL, &tv)){
            case -1:
                if((nrep == 0)||( errno != EINTR) ){
                    perror("select()");
                    res=-1;
                    nrep=0;
                }/*else -- A non blocked signal was caught, just repeat*/
                break;
            case 0:/*timeout*/
                res=-1;
                nrep=0;
                break;
            default:
                if( (res=write(fd,buf,count)) <0 )/*Signal?*/
                    while( (errno == EINTR)&&(res <0) )
                      res=write(fd,buf,count);
                nrep=0;
        }/*switch*/
    }while(nrep);

    /*restore the flags:*/
    fcntl(fd,F_SETFL, flags);
    return (res);
}/*writeSome*/

/*Try to read short buffer (of length not more than count)
  from fd. timeout is a timeout in millisecs. Returns number
  of written bytes or -1: */
static FORM_INLINE ssize_t readSome(int fd, char *buf, size_t count, int timeout)
{
    ssize_t res = 0;
    fd_set rfds;
    struct timeval tv;
    int nrep=5;/*five attempts it interrupted by a non-blocking signal*/

    do{
        FD_ZERO(&rfds);
        FD_SET(fd, &rfds);
        /* Wait up to timeout. */

        tv.tv_sec = timeout/1000;
        tv.tv_usec = (timeout % 1000)*1000;
        nrep--;

        switch(select(fd+1, &rfds, NULL, NULL, &tv)){
            case -1:
                if((nrep == 0)||( errno != EINTR) ){
                    perror("select()");
```

```
                    res=-1;
                    nrep=0;
                }/*else -- A non blocked signal was caught, just repeat*/
                break;
            case 0:/*timeout*/
                res=-1;
                nrep=0;
                break;
            default:
                if( (res=read(fd,buf,count)) <0 )/*Signal?*/
                    while( (errno == EINTR)&&(res <0) )
                      res=read(fd,buf,count);
                nrep=0;
        }/*switch*/
    }while(nrep);
    return (res);
}/*readSome*/

/*
    #] Local functions :
    #[ Ok functions:
*/

/*Copies (deep copy) newTerminator to thehandler->terminator. Returns 0 if
  newTerminator fits to the buffer, or !0 if it does not fit. ATT! In the
  latter case thehandler->terminator is NOT '\0' terminated! */
int setTerminatorForExternalChannelOk(char *newTerminator)
{
int i=DELTA_EXT_BUF;
/*
    No problems with externalChannelsListTop are
    possible since this function may be invoked only
    when the current channel is defined and externalChannelsListTop
  is set properly
*/
char *t=externalChannelsListTop->terminator;

    for(; i>1; i--)
        if( (*t++ = *newTerminator++)=='\0' )
            break;
    /*Add trailing '\n', if absent:*/
    if(   (i == DELTA_EXT_BUF)/*newTerminator == '\0'*/
        ||(*(t-2)!='\n')        ){
      *(t-1)='\n';*t='\0';
    }
    return(i==1);
}/*setTerminatorForExternalChannelOk*/

/*Interface to change handler fields "killSignal" and "gpid"*/
int setKillModeForExternalChannelOk(int signum, int sentToWholeGroup)
{
    if(signum<0)
        return(-1);
    /*
        No problems with externalChannelsListTop are
        possible since this function may be invoked only
        when the current channel is defined and externalChannelsListTop
        is set properly
    */
    externalChannelsListTop->killSignal=signum;
    if(sentToWholeGroup){/*gpid must be >0*/
        if(externalChannelsListTop->gpid <= 0)
            externalChannelsListTop->gpid=-externalChannelsListTop->gpid;
    }else{/*gpid must be <=0*/
        if(externalChannelsListTop->gpid>0)
            externalChannelsListTop->gpid=-externalChannelsListTop->gpid;
    }
    return(0);
}/*setKillModeForExternalChannelOk*/

/*
        #[ getcFromExtChannelOk
*/
/*Returns one character from the external channel. It the input is expired,
returns EOF. If the external process is finished completely, the function closes
the channel (and returns EOF). If the external process was finished, the function
returns EOF:*/
int getcFromExtChannelOk(void)
{
mysighandler_t oldPIPE = 0;
EXTHANDLE *h;
int ret;

    if (externalChannelsListTop->IBfill < externalChannelsListTop->IBfull)
        /*in buffer*/
        return( *(externalChannelsListTop->IBfill++) );
    /*else -- the buffer is empty*/
```

```
    ret=EOF;
    h= externalChannelsListTop;
#ifdef WITHMPI
    if ( PF.me == MASTER ){
#endif
    /* Temporary ignore this signal:*/
    /* if compiler fails here, try to change the definition of
        mysighandler_t on the beginning of this file
        (just define INTSIGHANDLER).*/
    oldPIPE=signal(SIGPIPE,SIG_IGN);
#ifdef WITHMPI
    if(  fgets(h->INbuf,h->IBstop - h->INbuf, h->frec) == 0  )/*Fail! EOF?*/
        *(h->INbuf)='\0';/*Empty line may not appear here!*/
#else
    if(  (fgets(h->INbuf,h->IBstop - h->INbuf, h->frec) == 0)/*Fail! EOF?*/
        ||( *(h->INbuf) == '\0')/*Empty line? This shouldn't be!*/
      ){
        closeExternalChannel(externalChannelsListTop-externalChannelsList+1);
        /*Note, this code is only for the sequential mode! */
        goto getcFromExtChannelReady;
        /*Here we assume that fgets is never interrupted by singals*/
    }/*if( fgets(h->INbuf,h->IBstop - h->INbuf, h->frec) == 0 )*/
#endif
#ifdef WITHMPI
    }/*if ( PF.me == MASTER */

    /*Master broadcasts result to slaves, slaves read it from the master:*/
    if( PF_BroadcastString((UBYTE *)h->INbuf) ){/*Fail!*/
          MesPrint("Fail broadcasting external channel results");
          Terminate(-1);
    }/*if( PF_BroadcastString((UBYTE *)h->INbuf) )*/

    if( *(h->INbuf) == '\0'){/*Empty line? This shouldn't be!*/
        closeExternalChannel(externalChannelsListTop-externalChannelsList+1);
        goto getcFromExtChannelReady;
    }/*if( *(h->INbuf) == '\0')*/
#endif

    {/*Block*/
        char *t=h->terminator;
        /*Move IBfull to the end of read line and compare the line with the terminator.
          Note, by construction the terminator fits to the first read line, see
          the function setTerminatorForExternalChannel.*/
        for(h->IBfull=h->INbuf; *(h->IBfull)!='\0'; (h->IBfull)++)
            if( *t== *(h->IBfull) )
                t++;
            else
                break;/*not a terminator*/
        /*Continue moving IBfullto the end of read line:*/
        while(*(h->IBfull)!='\0')(h->IBfull)++;

        if( (t-h->terminator) == (h->IBfull-h->INbuf) ){
            /*Terminator!*/
            /*Reset the channel*/
            h->IBfull=h->IBfill=h->INbuf;
            externalChannelsListTop=0;/*Undefine the current channel*/
            writeBufToExtChannel=&writeBufToExtChannelFailure;
            getcFromExtChannel=&getcFromExtChannelFailure;
            setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
            setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;
            goto getcFromExtChannelReady;
        }/*if(t == (h->IBfull-h->INbuf) )*/
    }/*Block*/
    /*Does the buffer have enough capacity?*/
    while( *(h->IBfull - 1) != '\n' ){/*Buffer is not enough!*/
        /*Extend the buffer:*/
        int l= (h->IBstop - h->INbuf)+DELTA_EXT_BUF;
        char *newbuf=Malloc1(l,"External channel buffer");
        /*We wouldn't like to use realloc.*/
        /*Copy the buffer:*/
        char *n=newbuf,*o=h->INbuf;
        while(  (*n++ = *o++)!='\0' );
        /*Att! The order of the following operators is important!:*/
        h->IBfull= newbuf+(h->IBfull-h->INbuf);
        M_free(h->INbuf,"External channel buffer");
        h->INbuf = newbuf;
        h->IBstop = h->INbuf+l;
#ifdef WITHMPI
        if ( PF.me == MASTER ){
            (h->IBfull)[1]='\0';/*Will mark (h->IBfull)[1] as '!' for failure*/
            if(  fgets(h->IBfull,h->IBstop - h->IBfull, h->frec) == 0 ){
                /*EOF! No trailing '\n'?*/
                /*Mark:*/
                (h->IBfull)[0]='\0';
                (h->IBfull)[1]='!';
                (h->IBfull)[2]='\0';
                 /*The string "\0!\0" is used as an image of NULL.*/
```

```
            }/*if(  fgets(h->IBfull,h->IBstop - h->IBfull, h->frec) == 0 )*/
        }/*if ( PF.me == MASTER )*/

        /*Master broadcasts results to slaves, slaves read it from the master:*/
        if( PF_BroadcastString((UBYTE *)h->IBfull) ){/*Fail!*/
          MesPrint("Fail broadcasting external channel results");
          Terminate(-1);
        }/*if( PF_BroadcastString(h->IBfull) )*/

        /*The string "\0!\0" is used as the image of NULL.*/
        if(
              ( (h->IBfull)[0]=='\0' )
            &&(  (h->IBfull)[1]=='!' )
            &&(  (h->IBfull)[2]=='\0' )
          )/*EOF! No trailing '\n'?*/
            break;
#else
        if(  fgets(h->IBfull,h->IBstop - h->IBfull, h->frec) == 0 )
            /*EOF! No trailing '\n'?*/
            break;
#endif
        while( *(h->IBfull)!='\0' )(h->IBfull)++;
    }/*while( *(h->IBfull - 1) != '\n' )*/
    /*In h->INbuf we have a fresh string.*/
    ret=*(h->IBfill=h->INbuf);
    h->IBfill++;/*Next time a new, isn't it?*/

    getcFromExtChannelReady:
#ifdef WITHMPI
    if ( PF.me == MASTER ){
#endif
            signal(SIGPIPE,oldPIPE);
#ifdef WITHMPI
    }/*if ( PF.me == MASTER )*/
#endif
            return(ret);
}/*getcFromExtChannelOk*/
/*
        #] getcFromExtChannelOk
*/
/*Writes exactly count bytes from the buffer buf to the external channel thehandler
  Returns 0 (on success) or -1:
*/
int writeBufToExtChannelOk(char *buf, size_t count)
{

int ret;
mysighandler_t oldPIPE;
#ifdef WITHMPI
    /*Only master communicates with the external program:*/
    if ( PF.me == MASTER ){
#endif
        /* Temporary ignore this signal:*/
        /* if compiler fails here, try to change the definition of
            mysighandler_t on the beginning of this file
            (just define INTSIGHANDLER)*/
        oldPIPE=signal(SIGPIPE,SIG_IGN);
        ret=writexactly( externalChannelsListTop->fsend, buf, count);
        signal(SIGPIPE,oldPIPE);
#ifdef WITHMPI
    }else{
        /*Do not wait the master status: this would be too slow!*/
        ret=0;
    }
#endif
    return(ret);
}/*writeBufToExtChannel*/
/*
    #] Ok functions:
    #[ do_run_cmd :
*/
/*The function returns PID of the started command*/
static FORM_INLINE pid_t do_run_cmd(
    int *fdsend,
    int *fdreceive,
    int *gpid, /*returns group process ID*/
    int ttymode,
                     /*
                      &8 - daemonizeing
                      &16 - setsid()*/
    char *cmd,
    char *argv[],
    char *stderrname
    )
{
int fdin[2]={-1,-1}, fdout[2]={-1,-1}, fdsig[2]={-1,-1};
/*initialised by -1 for possible rollback at failure, see closepipe() above*/
```

```
pid_t childpid,fatherchildpid = (pid_t)0;
mysighandler_t oldPIPE=NULL;

    if(
        (pipe(fdsig)!=0)/*This pipe will be used by a child to tell the father if fail.*/
        ||(pipe(fdin)!=0)
        ||(pipe(fdout)!=0)
    )goto fail_do_run_cmd;

    if((childpid = fork()) == -1){
         perror("fork");
         goto fail_do_run_cmd;
    }/*if((childpid = fork()) == -1)*/

    if(childpid == 0){/*Child.*/
        int fifo[3], top=0;
        /*
          To be thread safely we can't rely on ascendant order of opened
          file descriptors. So we put each of descriptor we have to
          preserve into the array fifo.
              Note, in _this_ process there are no any threads but descriptors
          were created in frame of the parent process which may have
          multiple threads.
        */
        /*Mark descriptors which will NOT be closed:*/
        pushDescriptor(fifo,top++,fdsig[1]);
        pushDescriptor(fifo,top++,fdin[0]);
        pushDescriptor(fifo,top++,fdout[1]);
        /*Close all except stdin, stdout, stderr and placed into fifo:*/
        closeAllDescriptors(3,fifo, top);

        /*Now reopen stdin and stdout.*/
        /*thread-safety is not a problem here since there are no any threads up to now:*/
        if(
            (close(0) == -1 )||/* Use fdin as stdin :*/
            (dup(fdin[0]) == -1 )||
            (close(1)==-1)||/* Use fdout as stdout:*/
            (dup(fdout[1]) == -1 )
          )
        {/*Fail!*/
            /*Signal to parent:*/
            writepid(fdsig[1],(pid_t)-2);
            _exit(1);
        }

        if(stderrname != NULL){
            if(
                (close(2) != 0 )||
                (open(stderrname,O_WRONLY)<0)
              )
            {/*Fail!*/
                writepid(fdsig[1],(pid_t)-2);
                _exit(1);
            }
        }/*if(stderrname != NULL)*/

        if( ttymode & 16 )/* create a session and sets the process group ID */
            setsid();

        /*   */
        if(set_cloexec_flag (fdsig[1], 1)!=0){/*Error?*/
            /*Signal to parent:*/
            writepid(fdsig[1],(pid_t)-2);
            _exit(1);
        }/*if(set_cloexec_flag (fdsig[1], 1)!=0)*/

        if( ttymode & 8 ){/*Daemonize*/
            int fdsig2[2];/*To check exec() success*/
            if(
                pipe(fdsig2)||
                (set_cloexec_flag (fdsig2[1], 1)!=0)
              )
            {/*Error?*/
                /*Signal to parent:*/
                writepid(fdsig[1],(pid_t)-2);
                _exit(1);
            }
            set_cloexec_flag (fdsig2[0], 1);
            switch(childpid=fork()){
                case 0:/*grandchild*/
                  /*Execute external command:*/
                  execvp(cmd, argv);
                  /* Control can  reach this point only on error!*/
                  writepid(fdsig2[1],(pid_t)-2);
                  break;
                case -1:
```

```
                  /* Control can  reach this point only on error!*/
                  /*Inform the father about the failure*/
                  writepid(fdsig[1],(pid_t)-2);
                  _exit(1);/*The child, just exit, not return*/
              default:/*Son of his father*/
                  close(fdsig2[1]);
                  /*Ignore SIGPIPE (up to the end of the process):*/
                  signal(SIGPIPE,SIG_IGN);

                  /*Wait on read() while the grandchild close the pipe
                     (on success) or send -2 (if exec() fails).*/
                  /*There are two possibilities:
                      -1 -- this is ok, the pipe was closed on exec,
                      the program was successfully executed;
                      -2 -- something is wrong, exec failed since the
                      grandchild sends -2 after exec.
                  */
                  if( readpid(fdsig2[0]) != (pid_t)-1 )/*something is wrong*/
                      writepid(fdsig[1],(pid_t)-1);
                  else/*ok, send PID of the grandchild to the father:*/
                      writepid(fdsig[1],childpid);
                  /*Die and free the life space for the grandchild:*/
                  _exit(0);/*The child, just exit, not return*/
            }/*switch(childpid=fork())*/
        }else{/*if( ttymode & 8 )*/
          execvp(cmd, argv);
          /* Control can  reach this point only on error!*/
          writepid(fdsig[1],(pid_t)-2);
          _exit(2);/*The child, just exit, not return*/
        }/*if( ttymode & 8 )...else*/
    }else{/* The (grand)father*/
        close(fdsig[1]);
        /*To prevent closing fdsig in rollback:*/
        fdsig[1]=-1;
        close(fdin[0]);
        close(fdout[1]);
        *fdsend    = fdin[1];
        *fdreceive = fdout[0];

        /*Get the process group ID.*/
        /*Avoid to use getpgid() which is non-standard.*/
        if( ttymode & 16)/*setsid() was invoked, the child is a group leader:*/
            *gpid=childpid;
        else/*the child belongs to the same process group as the this process:*/
            *gpid=getpgrp();/*if compiler fails here, try getpgrp(0) instead!*/
        /*
          Rationale: getpgrp  conform  to POSIX.1 while 4.3BSD provides a
          getpgrp() function that returns the process group ID for a
          specified process.
        */

        /* Temporary ignore this signal:*/
        /* if compiler fails here, try to change the definition of
          mysighandler_t on the beginning of this file
          (just define INTSIGHANDLER)*/
        oldPIPE=signal(SIGPIPE,SIG_IGN);

        if( ttymode & 8 ){/*Daemonize*/
            /*Read the grandchild PID from the son.*/
            fatherchildpid=childpid;
            if(  (childpid=readpid(fdsig[0]))<0  ){
                /*Daemonization process fails for some reasons!*/
                childpid=fatherchildpid;/*for rollback*/
                goto fail_do_run_cmd;
            }
        }else{
              /*fdsig[1] should be closed on exec and this read operation
                 must fail on success:*/
              if( readpid(fdsig[0])!= (pid_t)-1 )
                  goto fail_do_run_cmd;
        }/*if( ttymode & 8 ) ... else*/
    }/*if(childpid == 0)...else*/

    /*Here can be ONLY the father*/
    close(fdsig[0]);
    /*To prevent closing fdsig in rollback after goto fail_flnk_do_runcmd:*/
    fdsig[0]=-1;

    if( ttymode & 8 ){/*Daemonize*/
        int i;
        /*Wait while the father of a grandchild dies:*/
        waitpid(fatherchildpid,&i,0);
    }

    /*Restore the signal:*/
    signal(SIGPIPE,oldPIPE);

```

```
    return(childpid);

    fail_do_run_cmd:
        closepipe(&fdout);
        closepipe(&fdin);
        closepipe(&fdsig);
        return((pid_t)-1);
}/*do_run_cmd*/
/*
    #] do_run_cmd :
    #[ run_cmd :
*/
/*Starts the command cmd (directly, if shellpath is NULL, or in a subshell),
  swallowing its stdin and stdout;
  stderr will be re-directed to stderrname (if !=NULL). Returns PID of
  the started process. Stdin will be available as fdsend, and stdout
  will be available as fdreceive:*/
static FORM_INLINE pid_t run_cmd(char *cmd,
                                 int *fdsend,
                                 int *fdreceive,
                                 int *gpid,
                                 int daemonize,
                                 char *shellpath,
                                 char *stderrname )
{
char **argv;
pid_t thepid;

    cmd=(char*)strDup1((UBYTE*)cmd, "run_cmd: cmd");/*detouch cmd*/


    /* Prepare arguments for execvp:*/
    if(shellpath != NULL){/*Run in a subshell.*/
        int nopt;
        /*Allocate space which is definitely enough:*/
        argv=Malloc1(StrLen((UBYTE*)shellpath)*sizeof(char*)+2,"run_cmd:argv");
        shellpath=(char*)strDup1((UBYTE*)shellpath, "run_cmd: shellpath");/*detouch shellpath*/
        /*Parse a shell (e.g., "/bin/sh -c"):*/
        nopt=parseline(argv, shellpath);
        /* and add the command as a shell argument:*/
        argv[nopt]=cmd;
        argv[nopt+1]=NULL;
    }else{/*Run the command directly:*/
        /*Allocate space which is definitely enough:*/
        argv=Malloc1(StrLen((UBYTE*)cmd)*sizeof(char*)+1,"run_cmd:argv");
        parseline(argv, cmd);
    }

    thepid=do_run_cmd(
        fdsend,
        fdreceive,
        gpid,
        (daemonize)?(8|16):0,
        argv[0],
        argv,
        stderrname
    );

    M_free(argv,"run_cmd:argv");
    if(shellpath)
        M_free(shellpath,"run_cmd:argv");
    M_free(cmd, "run_cmd: cmd");

    return(thepid);
}/*run_cmd*/
/*
    #] run_cmd :
    #[ createExternalChannel :
*/
/*The structure to pass parameters to createExternalChannel
  and openExternalChannel in case of preset channel (instead of
  shellname):*/
typedef struct{
    int fdin;
    int fdout;
    pid_t theppid;
}ECINFOSTRUCT;

/* Creates a new external channel starting the command cmd (if cmd !=NULL)
    or using information from (ECINFOSTRUCT *)shellname, if cmd ==NULL:*/
static FORM_INLINE void *createExternalChannel(
                      EXTHANDLE *h,
                      char *cmd, /*Command to run or NULL*/
                      /*0 --neither setsid nor daemonize, !=0 -- full daemonization:*/
                      int daemonize,
                      char *shellname,/* The shell (like "/bin/sh -c") or NULL*/
                      char *stderrname/*filename to redirect stderr or NULL*/
```

```
                                                                    )
{
    int fdreceive=0;
    int gpid = 0;
    ECINFOSTRUCT *psetInfo;
#ifdef WITHMPI
    char statusbuf[2]={'\0','\0'};/*'\0' if run_cmd retuns ok, '!' otherwise.*/
#endif
    extHandlerInit(h);

    h->pid=0;

    if( cmd==NULL ){/*Instead of starting a new command, use preset channel:*/
        psetInfo=(ECINFOSTRUCT *)shellname;
        shellname=NULL;
        h->killSignal=0;
        h->daemonize=0;
    }

    /*Create a channel:*/
#ifdef WITHMPI
    if ( PF.me == MASTER ){
#endif
        if(cmd!=NULL)
            h->pid=run_cmd (cmd, &(h->fsend),
                     &fdreceive,&gpid,daemonize,shellname,stderrname);
        else{
            gpid=-psetInfo->theppid;
            h->pid=psetInfo->theppid;
            h->fsend=psetInfo->fdout;
            fdreceive=psetInfo->fdin;
        }
#ifdef WITHMPI
        if(h->pid<0)
            statusbuf[0]='!';/*Broadcast fail to slaves*/
    }
     /*else: Keep h->pid = 0 and h->fsend = 0 for slaves in parallel mode!*/

    /*Master broadcasts status to slaves, slaves read it from the master:*/
    if( PF_BroadcastString((UBYTE *)statusbuf) ){/*Fail!*/
        h->pid=-1;
    }else if( statusbuf[0]=='!')/*Master fails*/
        h->pid=-1;
#endif

    if(h->pid<0)goto createExternalChannelFails;
#ifdef WITHMPI
    if ( PF.me == MASTER ){
#endif
        h->gpid=gpid;
    /*Open stdout of a newly created program as FILE* :*/
        if( (h->frec=fdopen(fdreceive,"r")) == 0 )goto createExternalChannelFails;

#ifdef WITHMPI
    }
#endif
    /*Initialize buffers:*/
    extHandlerAlloc(h,cmd,shellname,stderrname);
    return(h);
    /*Something is wrong?*/
    createExternalChannelFails:

        destroyExternalChannel(h);
        return(NULL);
}/*createExternalChannel*/

/*
    #] createExternalChannel :
    #[ openExternalChannel :
*/
int openExternalChannel(UBYTE *cmd, int daemonize, UBYTE *shellname, UBYTE *stderrname)
{
EXTHANDLE *h=externalChannelsListTop;
int i=0;

    for(externalChannelsListTop=0;i<externalChannelsListFill;i++)
        if(externalChannelsList[i].cmd==0){
            externalChannelsListTop=externalChannelsList+i;
            break;
        }/*if(externalChannelsList[i].cmd!=0)*/

    if(externalChannelsListTop==0){
      /*No free cells, create the new one:*/
      if(externalChannelsListFill == externalChannelsListStop){
            /*The list is full, increase and reallocate it:*/
            EXTHANDLE *newbuf=Malloc1(
                                    (externalChannelsListStop+=DELTA_EXT_LIST)*sizeof(EXTHANDLE),
```

```
                                    "External channel list");
                for(i=0;i<externalChannelsListFill;i++)
                     extHandlerSwallowCopy(newbuf+i,externalChannelsList+i);
                if(externalChannelsList!=0)
                    M_free(externalChannelsList,"External channel list");
                for(;i<externalChannelsListStop;i++)
                    extHandlerInit(newbuf+i);
                externalChannelsList=newbuf;
          }/*if(externalChannelsListFill == externalChannelsListStop)*/
          externalChannelsListTop=externalChannelsList+externalChannelsListFill;
          externalChannelsListFill++;
        }/*if(externalChannelsListTop==0)*/

        if(createExternalChannel(
                                    externalChannelsListTop,
                                    (char*) cmd,
                                    daemonize,
                                    (char*)shellname,
                                    (char*)stderrname            )!=NULL){
            writeBufToExtChannel=&writeBufToExtChannelOk;
            getcFromExtChannel=&getcFromExtChannelOk;
            setTerminatorForExternalChannel=&setTerminatorForExternalChannelOk;
            setKillModeForExternalChannel=&setKillModeForExternalChannelOk;
            return(externalChannelsListTop-externalChannelsList+1);
        }
        /*else - failure!*/
        externalChannelsListTop=h;
        return(-1);
    }/*openExternalChannel*/
    /*
        #] openExternalChannel :
        #[ initPresetExternalChannels :
    */
    /*Just simple wrapper to invoke  openExternalChannel()
        from initPresetExternalChannels():*/
    static FORM_INLINE int openPresetExternalChannel(int fdin, int fdout, pid_t theppid)
    {
    ECINFOSTRUCT inf;
        inf.fdin=fdin; inf.fdout=fdout;inf.theppid=theppid;
        return( openExternalChannel(NULL,0,(UBYTE *)&inf,NULL) );
    }/*openPresetExternalChannel*/

    #define PIDTXTSIZE 23
    #define BOTHPIDSIZE 45

    #ifndef LONG_MAX
    #define LONG_MAX 0x7FFFFFFFL
    #endif

    /*There is a possibility to start FORM from another program providing
    one (or more) communication channels. These channels will be
    visible from a FORM program as ``pre-opened" external channels existing
    after FORM starts.
    Before starting FORM, the parent application must create one or more
    pairs of pipes The read-only descriptor of the first pipe in the pair
    and the write-only descriptor of the second pipe must be passed to
    FORM as an argument of a command line option -pipe in ASCII decimal
    format. The argument of the option is a comma-separated list of pairs
    r#,w# where r# is a read-only descriptor and w# is a write-only
    descriptor. Alternatively, the environment variable FORM_PIPES can be
    used.
        The following function expects as the first argument
    this comma-separated list of the  descriptor pairs and tries to
    initialize each of channel during thetimeout milliseconds:*/

    int initPresetExternalChannels(UBYTE *theline, int thetimeout)
    {
        int i, nchannels = 0;
        int pidln;            /*The length of FORM PID in pidtxt*/
        char pidtxt[PIDTXTSIZE],      /*64/Log_2[10] = 19.3, this is enough for any integer*/
             chdescriptor[PIDTXTSIZE],
             bothpidtxt[BOTHPIDSIZE],    /*"#,#\n\0"*/
             *c,*b = 0;
        int pip[2];
        pid_t ppid;
        if ( theline == NULL ) return(-1);
        /*Put into pidtxt PID\n:*/
        c = l2s((LONG)getpid(),pidtxt);
        *c++='\n';
        *c = '\0';
        pidln = c-pidtxt;

        do {
            pip[0] = (int)str2i((char*)theline,&c,0xFFFF);
            if( ( pip[0] < 0 ) || ( *c != ',' ) ) goto presetFails;

            theline = (UBYTE*)c + 1;
```

```
        pip[1] = (int)str2i((char*)theline,&c,0xFFFF);
        if ( (pip[1] < 0 ) || ( ( *c != ',' ) && ( *c != '\0' ) ) ) goto presetFails;
        theline = (UBYTE *)c + 1;
        /*Now we have two descriptors.
          According to the protocol, FORM must send to external channel
          it's PID with added '\n' and read back two comma-separated
          decimals with added '\n'. The first must be repeated FORM PID,
          the second must be the parent PID
        */
        if ( writeSome(pip[1],pidtxt,pidln,thetimeout) != pidln ) goto presetFails;
        i = readSome(pip[0],bothpidtxt,BOTHPIDSIZE,thetimeout);
        if( ( i < 4 )               /*at least 1,2\n*/
            || ( i == BOTHPIDSIZE )  /*No space for trailing '\0'*/
          ) goto presetFails;
        /*read the FORM PID:*/
        ppid = (pid_t)str2i(bothpidtxt,&b,getpid());
        if( ( *b != ',' ) || ( ppid != getpid() ) )goto presetFails;
        /*read the parent PID:*/
        /*The problem is that we do not know the the real type of pid_t.
          But long should be enough. On obsolete systems (when LONG_MAX
          is not defined) we assume pid_t is 32-bit integer.
          This can lead to problem with portability: */
        ppid = (pid_t)str2i(b+1,&b,LONG_MAX);
        if ( (*b != '\n') || (ppid<2) ) goto presetFails;
        i = openPresetExternalChannel(pip[0],pip[1],ppid);
        if ( i < 0 ) goto presetFails;
        nchannels++;
        /*Now use bothpidtxt as a buffer for preprovar, the space is enough:*/
        /*"PIPE#_" where # is ne order number of the channel:*/
        b = l2s(nchannels,addStr(bothpidtxt,"PIPE"));
        *b = '_';
        b[1] = '\0';
        *l2s(i,chdescriptor) = '\0';
        PutPreVar((UBYTE*)bothpidtxt,(UBYTE*)chdescriptor,0,0);
    } while ( *c != '\0' );
    /*Export proprovar "PIPES_":*/
    *l2s(nchannels,chdescriptor)='\0';
    PutPreVar((UBYTE*)"PIPES_",(UBYTE*)chdescriptor,0,0);

    /*success:*/
    return (nchannels);

presetFails:
        /*Here we assume the descriptors the beginning of the list!*/
        for(i=0; i<nchannels; i++)
            destroyExternalChannel(externalChannelsList+i);
        return(-1);
} /*initPresetExternalChannels*/
/*
    #] initPresetExternalChannels :
    #[ selectExternalChannel :
*/
/*
Accepts the valid external channel descriptor (returned by
openExternalChannel) and returns the descriptor of a previous current
channel (0, if there was no current channel, or -1, if the external
channel descriptor is invalid).  If n == 0, the function undefine the
current external channel:
*/
int selectExternalChannel(int n)
{
int ret=0;
    if(externalChannelsListTop!=0)
        ret=externalChannelsListTop-externalChannelsList+1;

    if(--n<0){
        if(n!=-1)
            return(-1);
        externalChannelsListTop=0;
        writeBufToExtChannel=&writeBufToExtChannelFailure;
        getcFromExtChannel=&getcFromExtChannelFailure;
        setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
        setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;
        return(ret);
    }
    if(
        (n>=externalChannelsListFill)||
        (externalChannelsList[n].cmd==0)
    )
        return(-1);

    externalChannelsListTop=externalChannelsList+n;
    writeBufToExtChannel=&writeBufToExtChannelOk;
    getcFromExtChannel=&getcFromExtChannelOk;
    setTerminatorForExternalChannel=&setTerminatorForExternalChannelOk;
    setKillModeForExternalChannel=&setKillModeForExternalChannelOk;
    return(ret);
```

```
}/*selectExternalChannel*/
/*
    #] selectExternalChannel :
    #[ closeExternalChannel :
*/

/*
Destroys the opened external channel with the descriptor n. It returns
0 in success, or -1 on failure. If the corresponding external channel
was the current one, the current channel becomes undefined. If n==0,
the function closes the current external channel.
*/
int closeExternalChannel(int n)
{
    if(n==0)
        n=externalChannelsListTop-externalChannelsList;
    else
        n--;/*Count from 0*/

    if(
        (n<0)||
        (n>=externalChannelsListFill)||
        (externalChannelsList[n].cmd==0)
    )/*No shuch a channel*/
        return(-1);

    destroyExternalChannel(externalChannelsList+n);
    /*If the current external channel was destroyed, undefine current channel:*/
    if(externalChannelsListTop==externalChannelsList+n){
        externalChannelsListTop=NULL;
        writeBufToExtChannel=&writeBufToExtChannelFailure;
        getcFromExtChannel=&getcFromExtChannelFailure;
        setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
        setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;
    }/*if(externalChannelsListTop==externalChannelsList+n)*/
    return(0);
}/*closeExternalChannel*/
/*
    #] closeExternalChannel :
    #[ closeAllExternalChannels :
*/
void closeAllExternalChannels(void)
{
int i;
    for(i=0; i<externalChannelsListFill; i++)
        destroyExternalChannel(externalChannelsList+i);
    externalChannelsListFill=externalChannelsListStop=0;
    externalChannelsListTop=NULL;

    writeBufToExtChannel=&writeBufToExtChannelFailure;
    getcFromExtChannel=&getcFromExtChannelFailure;
    setTerminatorForExternalChannel=&setTerminatorForExternalChannelFailure;
    setKillModeForExternalChannel=&setKillModeForExternalChannelFailure;

    if(externalChannelsList!=NULL){
        M_free(externalChannelsList,"External channel list");
        externalChannelsList=NULL;
    }
}/*closeAllExternalChannels*/
/*
    #] closeAllExternalChannels :
    #[ getExternalChannelPid :
*/
#ifdef SELFTEST
pid_t getExternalChannelPid(void)
{
  if(externalChannelsListTop!=0)
        return(externalChannelsListTop ->pid);
  return(-1);
}/*getExternalChannelPid*/
#endif
/*
    #] getExternalChannelPid :
    #[ getCurrentExternalChannel :
*/

int getCurrentExternalChannel(void)
{

    if ( externalChannelsListTop != 0 )
        return(externalChannelsListTop-externalChannelsList+1);
    return(0);
}/*getCurrentExternalChannel*/
/*
    #] getCurrentExternalChannel :
    #[ Selftest main :
*/
```

```

#ifdef SELFTEST

/*
    This is the example of how all these public functions may be used:
*/

char buf[1024];
char buf2[1024];

void help(void)
{
    printf("String started with a special letter is a command\n");
    printf("Known commands are:\n");
    printf("H or ? -- this help\n");
    printf("Nn<command> -- start a new command\n");
    printf("S<command> -- start a new command in a subshell,daemon,stderr>/dev/null\n");
    printf("C# -- destroy channel #\n");
    printf("R# -- set a new cahhel(number#) as a current one\n");
    printf("K#1 #2 --  set signal for kill and kill mode (0 or !=0)\n");
    printf("   ^d to quit\n");
}/*help*/

int main (void)
{
    int i, j, k,last;
    long long sum = 0;

    /*openExternalChannel(UBYTE *cmd, int daemonize, UBYTE *shellname, UBYTE *stderrname)*/

    help();

    printf("Initial channel:%d\n",last=openExternalChannel((UBYTE*)"cat",0,NULL,NULL));

    if( ( i = setTerminatorForExternalChannel("qu") ) != 0 ) return 1;
    printf("Terminator is 'qu'\n");

    while ( fgets(buf, 1024, stdin) != NULL ) {
        if ( *buf == 'N' ) {
            printf("New channel:%d\n",j=openExternalChannel((UBYTE*)buf+1,0,NULL,NULL));
            continue;
        }
        else if ( *buf == 'C' ) {
            int n;
            sscanf(buf+1,"%d",&n);
            printf("Destroy last channel:%d\n",closeExternalChannel(n));
            continue;
        }
        else if ( *buf == 'R' ) {
            int n = 0;
            sscanf(buf+1,"%d",&n);
            printf("Reopen channel %d:%d\n",n,selectExternalChannel(n));
            continue;
        }else if( *buf == 'K' ) {
            int n=0,g = 0;
            sscanf(buf+1,"%d %d",&n,&g);
            printf("setKillMode %d\n",setKillModeForExternalChannel(n,g));
            continue;
        }else if( *buf == 'S' ) {
            printf("New channel with sh&d&stderr:%d\n",
                     j=openExternalChannel((UBYTE*)buf+1,1,(UBYTE*)"/bin/sh -c",(UBYTE*)"/dev/null"));
            continue;
        }
        else if( ( *buf == 'H' )|| ( *buf == '?' )  ){
            help();
            continue;
        }

        writeBufToExtChannel(buf,k=StrLen(buf));
        sum += k;
        for ( j = 0; ( i = getcFromExtChannel() ) != '\n'; j++) {
            if ( i == EOF ) {
                printf("EOF!\n");
                break;
            }
            buf2[j] = i;
        }
        buf2[j] = '\0';
        printf("I got:'%s'; pid=%d\n",buf2,getExternalChannelPid());
    }
    printf("Total:%lld bytes\n",sum);
    closeAllExternalChannels();
    return 0;
}
#endif /*ifdef SELFTEST*/
/*
    #] Selftest main :
```

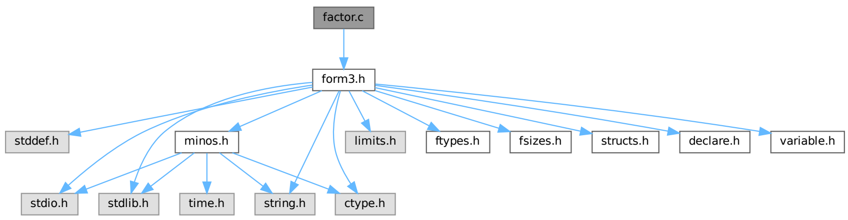
```
*/

#endif /*ifndef WITHEXTERNALCHANNEL ... else*/
```

4.33 factor.c File Reference

`#include "form3.h"`

Include dependency graph for factor.c:



Functions

- int FactorIn (PHEAD WORD ∗term, WORD level)
- int FactorInExpr (PHEAD WORD ∗term, WORD level)

4.33.1 Detailed Description

The routines for finding (one term) factors in dollars and expressions.

Definition in file factor.c.

4.33.2 Function Documentation

4.33.2.1 FactorIn()

```
int FactorIn (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 46 of file factor.c.

4.33.2.2 FactorInExpr()

```
int FactorInExpr (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 421 of file factor.c.

4.34 factor.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : factor.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ FactorIn :

    This routine tests for a factor in a dollar expression.

    Note that unlike with regular active or hidden expressions we cannot
    add memory as easily as dollars are rather volatile.
*/
int FactorIn(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD *t, *tstop, *m, *mm, *oldwork, *mstop, *n1, *n2, *n3, *n4, *n1stop, *n2stop;
    WORD *r1, *r2, *r3, *r4, j, k, kGCD, kGCD2, kLCM, jGCD, kkLCM, jLCM, size;
    UWORD *GCDbuffer, *GCDbuffer2, *LCMbuffer, *LCMb, *LCMc;
    int fromwhere = 0, i;
    DOLLARS d;
    t = term; GETSTOP(t,tstop); t++;
    while ( ( t < tstop ) && ( *t != FACTORIN || ( ( *t == FACTORIN )
     && ( t[FUNHEAD] != -DOLLAREXPRESSION || t[1] != FUNHEAD+2 ) ) ) ) t += t[1];
    if ( t >= tstop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Internal error. Could not find proper factorin_ function.");
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    oldwork = AT.WorkPointer;
    d = Dollars + t[FUNHEAD+1];
#ifdef WITHPTHREADS
    {
        int nummodopt, dtype = -1;
        if ( AS.MultiThreaded ) {
            for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                if ( t[FUNHEAD+1] == ModOptdollars[nummodopt].number ) break;
            }
            if ( nummodopt < NumModOptdollars ) {
                dtype = ModOptdollars[nummodopt].type;
                if ( dtype == MODLOCAL ) {
                    d = ModOptdollars[nummodopt].dstruct+AT.identity;
                }
            }
        }
    }
#endif

    if ( d->type == DOLTERMS ) {
        fromwhere = 1;
    }
    else if ( ( d = DolToTerms(BHEAD t[FUNHEAD+1]) ) == 0 ) {
```

```
/*
      The variable cannot convert to an expression
      We replace the function by 1.
*/
      m = oldwork; n1 = term;
      while ( n1 < t ) *m++ = *n1++;
      n1 = t + t[1]; tstop = term + *term;
      while ( n1 < tstop ) *m++ = *n1++;
      *oldwork = m - oldwork;
      AT.WorkPointer = m;
      if ( Generator(BHEAD oldwork,level) ) return(-1);
      AT.WorkPointer = oldwork;
      return(0);
   }
   if ( d->where[0] == 0 ) {
      if ( fromwhere == 0 ) {
         if ( d->factors ) M_free(d->factors,"Dollar factors");
         M_free(d,"Dollar in FactorIn_");
      }
      return(0);
   }
/*
   Now we have an expression in d->where. Find the symbolic factor that
   divides the expression and the numerical factor that makes all
   coefficients integer.

   For the symbolic factor we make a copy of the first term, and then
   go through all terms, scratching in the copy the objects that do not
   occur in the terms.
*/
   m = oldwork;
   mm = d->where;
   k = *mm - ABS((mm[*mm-1]));
   for ( j = 0; j < k; j++ ) *m++ = *mm++;
   mstop = m;
   *oldwork = k;
/*
   The copy is in place. Now search through the terms. Start at the second term
*/
   mm = d->where + d->where[0];
   while ( *mm ) {
      m = oldwork+1;
      r2 = mm+*mm;
      r2 -= ABS(r2[-1]);
      r1 = mm+1;
      while ( m < mstop ) {
         while ( r1 < r2 ) {
            if ( *r1 != *m ) {
               r1 += r1[1]; continue;
            }
/*
            Now the various cases
         #[ SYMBOL :
*/
            if ( *m == SYMBOL ) {
               n1 = m+2; n1stop = m+m[1];
               n2stop = r1+r1[1];
               while ( n1 < n1stop ) {
                  n2 = r1+2;
                  while ( n2 < n2stop ) {
                     if ( *n1 != *n2 ) { n2 += 2; continue; }
                     if ( n1[1] > 0 ) {
                        if ( n2[1] < 0 ) { n2 += 2; continue; }
                        if ( n2[1] < n1[1] ) n1[1] = n2[1];
                     }
                     else {
                        if ( n2[1] > 0 ) { n2 += 2; continue; }
                        if ( n2[1] > n1[1] ) n1[1] = n2[1];
                     }
                     break;
                  }
                  if ( n2 >= n2stop ) {   /* scratch symbol */
                     if ( m[1] == 4 ) goto scratch;
                     m[1] -= 2;
                     n3 = n1; n4 = n1+2;
                     while ( n4 < mstop ) *n3++ = *n4++;
                     *oldwork = n3 - oldwork;
                     mstop -= 2; n1stop -= 2;
                     continue;
                  }
                  n1 += 2;
               }
               break;
            }
/*
         #] SYMBOL :
         #[ DOTPRODUCT :
```

```
*/
                else if ( *m == DOTPRODUCT ) {
                    n1 = m+2; n1stop = m+m[1];
                    n2stop = r1+r1[1];
                    while ( n1 < n1stop ) {
                        n2 = r1+2;
                        while ( n2 < n2stop ) {
                            if ( *n1 != *n2 || n1[1] != n2[1] ) { n2 += 3; continue; }
                            if ( n1[2] > 0 ) {
                                if ( n2[2] < 0 ) { n2 += 3; continue; }
                                if ( n2[2] < n1[2] ) n1[2] = n2[2];
                            }
                            else {
                                if ( n2[2] > 0 ) { n2 += 3; continue; }
                                if ( n2[2] > n1[2] ) n1[2] = n2[2];
                            }
                            break;
                        }
                        if ( n2 >= n2stop ) {   /* scratch symbol */
                            if ( m[1] == 5 ) goto scratch;
                            m[1] -= 3;
                            n3 = n1; n4 = n1+3;
                            while ( n4 < mstop ) *n3++ = *n4++;
                            *oldwork = n3 - oldwork;
                            mstop -= 3; n1stop -= 3;
                            continue;
                        }
                        n1 += 3;
                    }
                    break;
                }
/*
            #] DOTPRODUCT :
            #[ VECTOR :
*/
                else if ( *m == VECTOR ) {
/*
                    Here we have to be careful if there is more than
                    one of the same
*/
                    n1 = m+2; n1stop = m+m[1];
                    n2 = r1+2;n2stop = r1+r1[1];
                    while ( n1 < n1stop ) {
                        while ( n2 < n2stop ) {
                            if ( *n1 == *n2 && n1[1] == n2[1] ) {
                                n2 += 2; goto nextn1;
                            }
                            n2 += 2;
                        }
                        if ( n2 >= n2stop ) {   /* scratch symbol */
                            if ( m[1] == 4 ) goto scratch;
                            m[1] -= 2;
                            n3 = n1; n4 = n1+2;
                            while ( n4 < mstop ) *n3++ = *n4++;
                            *oldwork = n3 - oldwork;
                            mstop -= 2; n1stop -= 2;
                            continue;
                        }
                        n2 = r1+2;
nextn1:                 n1 += 2;
                    }
                    break;
                }
/*
            #] VECTOR :
            #[ REMAINDER :
*/
                else {
/*
                    Again: watch for multiple occurrences of the same object
*/
                    if ( m[1] != r1[1] ) { r1 += r1[1]; continue; }
                    for ( j = 2; j < m[1]; j++ ) {
                        if ( m[j] != r1[j] ) break;
                    }
                    if ( j < m[1] ) { r1 += r1[1]; continue; }
                    r1 += r1[1]; /* to restart at the next potential match */
                    goto nextm;  /* match */
                }
/*
            #] REMAINDER :
*/
            }
            if ( r1 >= r2 ) { /* no factor! */
scratch:;
                r3 = m + m[1]; r4 = m;
                while ( r3 < mstop ) *r4++ = *r3++;
```

```
                *oldwork = r4 - oldwork;
                if ( *oldwork == 1 ) goto nofactor;
                mstop = r4;
                r1 = mm + 1;
                continue;
            }
            r1 = mm + 1;
nextm:      m += m[1];
        }
        mm = mm + *mm;
    }

nofactor:;
/*
    For the coefficient we have to determine the LCM of the denominators
    and the GCD of the numerators.
*/
    GCDbuffer = NumberMalloc("FactorIn"); GCDbuffer2 = NumberMalloc("FactorIn");
    LCMbuffer = NumberMalloc("FactorIn"); LCMb = NumberMalloc("FactorIn"); LCMc = NumberMalloc("FactorIn");
    r1 = d->where;
/*
    First take the first term to load up the LCM and the GCD
*/
    r2 = r1 + *r1;
    j = r2[-1];
    r3 = r2 - ABS(j);
    k = REDLENG(j);
    if ( k < 0 ) k = -k;
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kGCD = 0; kGCD < k; kGCD++ ) GCDbuffer[kGCD] = r3[kGCD];
    k = REDLENG(j);
    if ( k < 0 ) k = -k;
    r3 += k;
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
    r1 = r2;
/*
    Now go through the rest of the terms in this dollar buffer.
*/
    while ( *r1 ) {
        r2 = r1 + *r1;
        j = r2[-1];
        r3 = r2 - ABS(j);
        k = REDLENG(j);
        if ( k < 0 ) k = -k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( GCDbuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
/*
            GCD is already 1
*/
        }
        else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
            if ( GcdLong(BHEAD GCDbuffer,kGCD,(UWORD *)r3,k,GCDbuffer2,&kGCD2) ) {
                goto onerror;
            }
            kGCD = kGCD2;
            for ( i = 0; i < kGCD; i++ ) GCDbuffer[i] = GCDbuffer2[i];
        }
        else {
            kGCD = 1; GCDbuffer[0] = 1;
        }
        k = REDLENG(j);
        if ( k < 0 ) k = -k;
        r3 += k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
            for ( kLCM = 0; kLCM < k; kLCM++ )
                LCMbuffer[kLCM] = r3[kLCM];
        }
        else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
            if ( GcdLong(BHEAD LCMbuffer,kLCM,(UWORD *)r3,k,LCMb,&kkLCM) ) {
                goto onerror;
            }
            DivLong((UWORD *)r3,k,LCMb,kkLCM,LCMb,&kkLCM,LCMc,&jLCM);
            MulLong(LCMbuffer,kLCM,LCMb,kkLCM,LCMc,&jLCM);
            for ( kLCM = 0; kLCM < jLCM; kLCM++ )
                LCMbuffer[kLCM] = LCMc[kLCM];
        }
        else {} /* LCM doesn't change */
        r1 = r2;
    }
/*
    Now put the factor together: GCD/LCM
*/
    r3 = (WORD *)(GCDbuffer);
    if ( kGCD == kLCM ) {
```

```
        for ( jGCD = 0; jGCD < kGCD; jGCD++ )
            r3[jGCD+kGCD] = LCMbuffer[jGCD];
        k = kGCD;
    }
    else if ( kGCD > kLCM ) {
        for ( jGCD = 0; jGCD < kLCM; jGCD++ )
            r3[jGCD+kGCD] = LCMbuffer[jGCD];
        for ( jGCD = kLCM; jGCD < kGCD; jGCD++ )
            r3[jGCD+kGCD] = 0;
        k = kGCD;
    }
    else {
        for ( jGCD = kGCD; jGCD < kLCM; jGCD++ )
            r3[jGCD] = 0;
        for ( jGCD = 0; jGCD < kLCM; jGCD++ )
            r3[jGCD+kLCM] = LCMbuffer[jGCD];
        k = kLCM;
    }
    j = 2*k+1;
    mm = m = oldwork + oldwork[0];
/*
    Now compose the new term
*/
    n1 = term;
    while ( n1 < t ) *m++ = *n1++;
    n1 += n1[1];
    n2 = oldwork+1;
    while ( n2 < mm ) *m++ = *n2++;
    while ( n1 < tstop ) *m++ = *n1++;
/*
    And the coefficient
*/
    size = term[*term-1];
    size = REDLENG(size);
    if ( MulRat(BHEAD (UWORD *)tstop,size,(UWORD *)r3,k,
                                (UWORD *)m,&size) ) goto onerror;
    size = INCLENG(size);
    k = size < 0 ? -size: size;
    m[k-1] = size;
    m += k;
    *mm = (WORD)(m - mm);
    AT.WorkPointer = m;
    if ( Generator(BHEAD mm,level) ) goto onerror;
    AT.WorkPointer = oldwork;
    if ( fromwhere == 0 ) {
        if ( d->factors ) M_free(d->factors,"Dollar factors");
        M_free(d,"Dollar in FactorIn");
    }
    NumberFree(GCDbuffer,"FactorIn"); NumberFree(GCDbuffer2,"FactorIn");
    NumberFree(LCMbuffer,"FactorIn"); NumberFree(LCMb,"FactorIn"); NumberFree(LCMc,"FactorIn");
    return(0);
onerror:
    AT.WorkPointer = oldwork;
    MLOCK(ErrorMessageLock);
    MesCall("FactorIn");
    MUNLOCK(ErrorMessageLock);
    NumberFree(GCDbuffer,"FactorIn"); NumberFree(GCDbuffer2,"FactorIn");
    NumberFree(LCMbuffer,"FactorIn"); NumberFree(LCMb,"FactorIn"); NumberFree(LCMc,"FactorIn");
    return(-1);
}

/*
    #] FactorIn :
    #[ FactorInExpr :

    This routine tests for a factor in an active or hidden expression.

    The factor from the last call is stored in a cache.
    Main problem here is whether the cache is global or private to each thread.
    A global cache gives most likely the same traffic jam for the computation
    as a local cache. The local cache may stay valid longer.
    In the future we may make it such that threads can look at the cache
    of the others, or even whether a certain factor is under construction.
*/

int FactorInExpr(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD *t, *tstop, *m, *oldwork, *mstop, *n1, *n2, *n3, *n4, *n1stop, *n2stop;
    WORD *r1, *r2, *r3, *r4, j, k, kGCD, kGCD2, kLCM, jGCD, kkLCM, jLCM, size, sign;
    WORD *newterm, expr = 0;
    WORD olddeferflag = AR.DeferFlag, oldgetfile = AR.GetFile, oldhold = AR.KeptInHold;
    WORD newgetfile, newhold;
    int i;
    EXPRESSIONS e;
    FILEHANDLE *file = 0;
    POSITION position, oldposition, startposition;
```

```
    WORD *oldcpointer = AR.CompressPointer;
    UWORD *GCDbuffer, *GCDbuffer2, *LCMbuffer, *LCMb, *LCMc;
    GCDbuffer = NumberMalloc("FactorInExpr"); GCDbuffer2 = NumberMalloc("FactorInExpr");
    LCMbuffer = NumberMalloc("FactorInExpr"); LCMb = NumberMalloc("FactorInExpr"); LCMc = NumberMalloc("FactorInExpr");
    t = term; GETSTOP(t,tstop); t++;
    while ( t < tstop ) {
        if ( *t == FACTORIN && t[1] == FUNHEAD+2 && t[FUNHEAD] == -EXPRESSION ) {
            expr = t[FUNHEAD+1];
            break;
        }
        t += t[1];
    }
    if ( t >= tstop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Internal error. Could not find proper factorin_ function.");
        MUNLOCK(ErrorMessageLock);
        NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
        NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
        return(-1);
    }
    oldwork = AT.WorkPointer;
    if ( AT.previousEfactor && ( expr == AT.previousEfactor[0] ) ) {
/*
        We have a hit in the cache. Construct the new term.
        At the moment AT.previousEfactor[1] is reserved for future flags
*/
        goto PutTheFactor;
    }
/*
    No hit. We have to do the work. We start with constructing the factor
    in the WorkSpace. Later we will move it to the cache.
    Finally we will jump to PutTheFactor.
*/
    e = Expressions + expr;
    switch ( e->status ) {
        case LOCALEXPRESSION:
        case SKIPLEXPRESSION:
        case DROPLEXPRESSION:
        case GLOBALEXPRESSION:
        case SKIPGEXPRESSION:
        case DROPGEXPRESSION:
/*
            Expression is to be found in the input Scratch file.
            Set the file handle and the position.
            The rest is done by GetTerm.
*/
            newhold = 0;
            newgetfile = 1;
            file = AR.infile;
            break;
        case HIDDENLEXPRESSION:
        case HIDDENGEXPRESSION:
        case HIDELEXPRESSION:
        case HIDEGEXPRESSION:
        case DROPHLEXPRESSION:
        case DROPHGEXPRESSION:
        case UNHIDELEXPRESSION:
        case UNHIDEGEXPRESSION:
/*
            Expression is to be found in the hide Scratch file.
            Set the file handle and the position.
            The rest is done by GetTerm.
*/
            newhold = 0;
            newgetfile = 2;
            file = AR.hidefile;
            break;
        case STOREDEXPRESSION:
/*
            This is an 'illegal' case
*/
            MLOCK(ErrorMessageLock);
            MesPrint("Error: factorin_ cannot determine factors in stored expressions.");
            MUNLOCK(ErrorMessageLock);
            NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
            NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
            return(-1);
        case DROPPEDEXPRESSION:
/*
            We replace the function by 1.
*/
            m = oldwork; n1 = term;
            while ( n1 < t ) *m++ = *n1++;
            n1 = t + t[1]; tstop = term + *term;
```

```
            while ( n1 < tstop ) *m++ = *n1++;
            *oldwork = m - oldwork;
            AT.WorkPointer = m;
            if ( Generator(BHEAD oldwork,level) ) {
                NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
                NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
                return(-1);
            }
            AT.WorkPointer = oldwork;
            NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
            NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
            return(0);
        default:
            MLOCK(ErrorMessageLock);
            MesPrint("Error: Illegal expression in factorinexpr.");
            MUNLOCK(ErrorMessageLock);
            NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
            NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
            return(-1);
    }
/*
    Before we start with the file we set the buffers for the coefficient
    For the coefficient we have to determine the LCM of the denominators
    and the GCD of the numerators.
*/
    position = AS.OldOnFile[expr];
    AR.DeferFlag = 0; AR.KeptInHold = newhold; AR.GetFile = newgetfile;
    SeekScratch(file,&oldposition);
    SetScratch(file,&position);
    if ( GetTerm(BHEAD oldwork) <= 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("(5) Expression %d has problems in scratchfile",expr);
        MUNLOCK(ErrorMessageLock);
        NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
        NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
        return(-1);
    }
    SeekScratch(file,&startposition);
    SeekScratch(file,&position);
/*
    Load the first term in the workspace
*/
    if ( GetTerm(BHEAD oldwork) == 0 ) {
        SetScratch(file,&oldposition); /* We still need this until Processor is clean */
        AR.DeferFlag = olddeferflag;
        oldwork[0] = 4; oldwork[1] = 1; oldwork[2] = 1; oldwork[3] = 3;
        goto Complete;
    }
    SeekScratch(file,&position);
    AR.DeferFlag = olddeferflag; AR.KeptInHold = oldhold; AR.GetFile = oldgetfile;

    r2 = m = oldwork + *oldwork;
    j = m[-1];
    m -= ABS(j);
    *oldwork = (WORD)(m-oldwork);
    AT.WorkPointer = newterm = mstop = m;
/*
    Now take the coefficient of the first term to load up the LCM and the GCD
*/
    r3 = m;
    k = REDLENG(j);
    if ( k < 0 ) { k = -k; sign = -1; }
    else { sign = 1; }
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kGCD = 0; kGCD < k; kGCD++ ) GCDbuffer[kGCD] = r3[kGCD];
    k = REDLENG(j);
    if ( k < 0 ) k = -k;
    r3 += k;
    while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
    for ( kLCM = 0; kLCM < k; kLCM++ ) LCMbuffer[kLCM] = r3[kLCM];
/*
    The copy and the coefficient are in place. Now search through the terms.
*/
    for (;;) {
        AR.DeferFlag = 0; AR.KeptInHold = newhold; AR.GetFile = newgetfile;
        SetScratch(file,&position);
        size = GetTerm(BHEAD newterm);
        SeekScratch(file,&position);
        AR.DeferFlag = olddeferflag; AR.KeptInHold = oldhold; AR.GetFile = oldgetfile;
        if ( size == 0 ) break;
        m = oldwork+1;
        r2 = newterm + *newterm;
        r2 -= ABS(r2[-1]);
```

```
        r1 = newterm+1;
        while ( m < mstop ) {
            while ( r1 < r2 ) {
                if ( *r1 != *m ) {
                    r1 += r1[1]; continue;
                }
/*
                Now the various cases
            #[ SYMBOL :
*/
                if ( *m == SYMBOL ) {
                    n1 = m+2; n1stop = m+m[1];
                    n2stop = r1+r1[1];
                    while ( n1 < n1stop ) {
                        n2 = r1+2;
                        while ( n2 < n2stop ) {
                            if ( *n1 != *n2 ) { n2 += 2; continue; }
                            if ( n1[1] > 0 ) {
                                if ( n2[1] < 0 ) { n2 += 2; continue; }
                                if ( n2[1] < n1[1] ) n1[1] = n2[1];
                            }
                            else {
                                if ( n2[1] > 0 ) { n2 += 2; continue; }
                                if ( n2[1] > n1[1] ) n1[1] = n2[1];
                            }
                            break;
                        }
                        if ( n2 >= n2stop ) {   /* scratch symbol */
                            if ( m[1] == 4 ) goto scratch;
                            m[1] -= 2;
                            n3 = n1; n4 = n1+2;
                            while ( n4 < mstop ) *n3++ = *n4++;
                            *oldwork = n3 - oldwork;
                            mstop -= 2; n1stop -= 2;
                            continue;
                        }
                        n1 += 2;
                    }
                    break;
                }
/*
            #] SYMBOL :
            #[ DOTPRODUCT :
*/
                else if ( *m == DOTPRODUCT ) {
                    n1 = m+2; n1stop = m+m[1];
                    n2stop = r1+r1[1];
                    while ( n1 < n1stop ) {
                        n2 = r1+2;
                        while ( n2 < n2stop ) {
                            if ( *n1 != *n2 || n1[1] != n2[1] ) { n2 += 3; continue; }
                            if ( n1[2] > 0 ) {
                                if ( n2[2] < 0 ) { n2 += 3; continue; }
                                if ( n2[2] < n1[2] ) n1[2] = n2[2];
                            }
                            else {
                                if ( n2[2] > 0 ) { n2 += 3; continue; }
                                if ( n2[2] > n1[2] ) n1[2] = n2[2];
                            }
                            break;
                        }
                        if ( n2 >= n2stop ) {   /* scratch dotproduct */
                            if ( m[1] == 5 ) goto scratch;
                            m[1] -= 3;
                            n3 = n1; n4 = n1+3;
                            while ( n4 < mstop ) *n3++ = *n4++;
                            *oldwork = n3 - oldwork;
                            mstop -= 3; n1stop -= 3;
                            continue;
                        }
                        n1 += 3;
                    }
                    break;
                }
/*
            #] DOTPRODUCT :
            #[ VECTOR :
*/
                else if ( *m == VECTOR ) {
/*
                    Here we have to be careful if there is more than
                    one of the same
*/
                    n1 = m+2; n1stop = m+m[1];
                    n2 = r1+2;n2stop = r1+r1[1];
                    while ( n1 < n1stop ) {
                        while ( n2 < n2stop ) {
```

```
                                if ( *n1 == *n2 && n1[1] == n2[1] ) {
                                    n2 += 2; goto nextn1;
                                }
                                n2 += 2;
                            }
                            if ( n2 >= n2stop ) {    /* scratch vector */
                                if ( m[1] == 4 ) goto scratch;
                                m[1] -= 2;
                                n3 = n1; n4 = n1+2;
                                while ( n4 < mstop ) *n3++ = *n4++;
                                *oldwork = n3 - oldwork;
                                mstop -= 2; n1stop -= 2;
                                continue;
                            }
                            n2 = r1+2;
nextn1:                     n1 += 2;
                        }
                        break;
                    }
/*
            #] VECTOR :
            #[ REMAINDER :
*/
                else {
/*
                    Again: watch for multiple occurrences of the same object
*/
                    if ( m[1] != r1[1] ) { r1 += r1[1]; continue; }
                    for ( j = 2; j < m[1]; j++ ) {
                        if ( m[j] != r1[j] ) break;
                    }
                    if ( j < m[1] ) { r1 += r1[1]; continue; }
                    r1 += r1[1]; /* to restart at the next potential match */
                    goto nextm;  /* match */
                }
/*
            #] REMAINDER :
*/
            }
            if ( r1 >= r2 ) { /* no factor! */
scratch:;
                r3 = m + m[1]; r4 = m;
                while ( r3 < mstop ) *r4++ = *r3++;
                *oldwork = r4 - oldwork;
                if ( *oldwork == 1 ) goto nofactor;
                mstop = r4;
                r1 = newterm + 1;
                continue;
            }
            r1 = newterm + 1;
nextm:      m += m[1];
        }
nofactor:;
/*
        Now the coefficient
*/
        r2 = newterm + *newterm;
        j = r2[-1];
        r3 = r2 - ABS(j);
        k = REDLENG(j);
        if ( k < 0 ) k = -k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( GCDbuffer[0] == 1 ) && ( kGCD == 1 ) ) ) {
/*
            GCD is already 1
*/
        }
        else if ( ( ( k != 1 ) || ( r3[0] != 1 ) ) ) {
            if ( GcdLong(BHEAD GCDbuffer,kGCD,(UWORD *)r3,k,GCDbuffer2,&kGCD2) ) {
                goto onerror;
            }
            kGCD = kGCD2;
            for ( i = 0; i < kGCD; i++ ) GCDbuffer[i] = GCDbuffer2[i];
        }
        else {
            kGCD = 1; GCDbuffer[0] = 1;
        }
        k = REDLENG(j);
        if ( k < 0 ) k = -k;
        r3 += k;
        while ( ( k > 1 ) && ( r3[k-1] == 0 ) ) k--;
        if ( ( ( LCMbuffer[0] == 1 ) && ( kLCM == 1 ) ) ) {
            for ( kLCM = 0; kLCM < k; kLCM++ )
                LCMbuffer[kLCM] = r3[kLCM];
        }
        else if ( ( k != 1 ) || ( r3[0] != 1 ) ) {
            if ( GcdLong(BHEAD LCMbuffer,kLCM,(UWORD *)r3,k,LCMb,&kkLCM) ) {
```

```
                goto onerror;
            }
            DivLong((UWORD *)r3,k,LCMb,kkLCM,LCMb,&kkLCM,LCMc,&jLCM);
            MulLong(LCMbuffer,kLCM,LCMb,kkLCM,LCMc,&jLCM);
            for ( kLCM = 0; kLCM < jLCM; kLCM++ )
                LCMbuffer[kLCM] = LCMc[kLCM];
        }
        else {} /* LCM doesn't change */
    }
    SetScratch(file,&oldposition); /* Needed until Processor is thread safe */
    AR.DeferFlag = olddeferflag;
/*
    Now put the term together in oldwork: GCD/LCM
    We have already the algebraic contents there.
*/
    r3 = (WORD *)(GCDbuffer);
    r4 = (WORD *)(LCMbuffer);
    r1 = oldwork + *oldwork;
    if ( kGCD == kLCM ) {
        for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r3++;
        for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r4++;
        k = 2*kGCD+1;
    }
    else if ( kGCD > kLCM ) {
        for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r3++;
        for ( jGCD = 0; jGCD < kLCM; jGCD++ ) *r1++ = *r4++;
        for ( jGCD = kLCM; jGCD < kGCD; jGCD++ ) *r1++ = 0;
        k = 2*kGCD+1;
    }
    else {
        for ( jGCD = 0; jGCD < kGCD; jGCD++ ) *r1++ = *r3++;
        for ( jGCD = kGCD; jGCD < kLCM; jGCD++ ) *r1++ = 0;
        for ( jGCD = 0; jGCD < kLCM; jGCD++ ) *r1++ = *r4++;
        k = 2*kLCM+1;
    }
    if ( sign < 0 ) *r1++ = -k;
    else *r1++ = k;
    *oldwork = (WORD)(r1-oldwork);
/*
    Now put the new term in the cache
*/
Complete:;
    if ( AT.previousEfactor ) M_free(AT.previousEfactor,"Efactor cache");
    AT.previousEfactor = (WORD *)Malloc1((*oldwork+2)*sizeof(WORD),"Efactor cache");
    AT.previousEfactor[0] = expr;
    r1 = oldwork; r2 = AT.previousEfactor + 2; k = *oldwork;
    NCOPY(r2,r1,k)
    AT.previousEfactor[1] = 0;
/*
    Now we construct the new term in the workspace.
*/
PutTheFactor:;
    if ( AT.WorkPointer + AT.previousEfactor[2] >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MesPrint("Called from factorin_");
        MUNLOCK(ErrorMessageLock);
        NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
        NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
        return(-1);
    }
    n1 = oldwork; n2 = term; while ( n2 < t ) *n1++ = *n2++;
    n2 = AT.previousEfactor+2; GETSTOP(n2,n2stop); n3 = n2 + *n2;
    n2++; while ( n2 < n2stop ) *n1++ = *n2++;
    n2 = t + t[1]; while ( n2 < tstop ) *n1++ = *n2++;
    size = term[*term-1];
    size = REDLENG(size);
    k = n3[-1]; k = REDLENG(k);
    if ( MulRat(BHEAD (UWORD *)tstop,size,(UWORD *)n2stop,k,
                            (UWORD *)n1,&size) ) goto onerror;
    size = INCLENG(size);
    k = size < 0 ? -size: size;
       n1 += k; n1[-1] = size;
    *oldwork = n1 - oldwork;
    AT.WorkPointer = n1;
    if ( Generator(BHEAD oldwork,level) ) {
        NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
        NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
        return(-1);
    }
    AT.WorkPointer = oldwork;
    AR.CompressPointer = oldcpointer;
    NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
    NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr"); NumberFree(LCMc,"FactorInExpr");
```

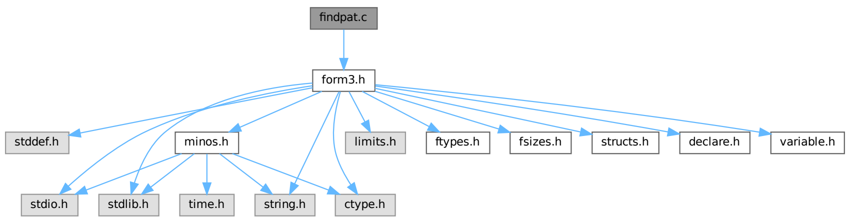
```
    return(0);
onerror:
    AT.WorkPointer = oldwork;
    AR.CompressPointer = oldcpointer;
    MLOCK(ErrorMessageLock);
    MesCall("FactorInExpr");
    MUNLOCK(ErrorMessageLock);
    NumberFree(GCDbuffer,"FactorInExpr"); NumberFree(GCDbuffer2,"FactorInExpr");
    NumberFree(LCMbuffer,"FactorInExpr"); NumberFree(LCMb,"FactorInExpr");
NumberFree(LCMc,"FactorInExpr");
    return(-1);
}

/*
    #] FactorInExpr :
*/
```

4.35 findpat.c File Reference

`#include "form3.h"`

Include dependency graph for findpat.c:



Functions

- WORD FindOnly (PHEAD WORD ∗term, WORD ∗pattern)
- WORD FindOnce (PHEAD WORD ∗term, WORD ∗pattern)
- WORD FindMulti (PHEAD WORD ∗term, WORD ∗pattern)
- WORD FindRest (PHEAD WORD ∗term, WORD ∗pattern)

4.35.1 Detailed Description

Pattern matching of symbols and dotproducts. There are various routines because of the options in the id-statements like once, only, multi and many. These are among the oldest routines in FORM and that can be noticed, because the interplay with the function matching is not complete. When we match functions and halfway we fail we can backtrack properly. With the symbols, the dotproducts and the vectors (in pattern.c) there is no proper backtracking. Hence the routines here need still quite some work or may even have to be rewritten.

Definition in file findpat.c.

4.35.2 Function Documentation

4.35.2.1 FindOnly()

```
WORD FindOnly (
            PHEAD WORD * term,
            WORD * pattern )
```

Definition at line 61 of file findpat.c.

4.35.2.2 FindOnce()

```
WORD FindOnce (
            PHEAD WORD * term,
            WORD * pattern )
```

Definition at line 419 of file findpat.c.

4.35.2.3 FindMulti()

```
WORD FindMulti (
            PHEAD WORD * term,
            WORD * pattern )
```

Definition at line 954 of file findpat.c.

4.35.2.4 FindRest()

```
WORD FindRest (
            PHEAD WORD * term,
            WORD * pattern )
```

Definition at line 1070 of file findpat.c.

4.36 findpat.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : findpat.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ Patterns :
        #[ FindOnly :          WORD FindOnly(term,pattern)
```

```

    The current version doesn't scan function arguments yet. 10-Apr-1988

    This routine searches for an exact match. This means in particular:
    1:  x^# must match exactly.
    2:  x^n? must have a single value for n that cannot be adapted.

    When setp != 0 it points to a collection of sets
    A match can occur only if no object will be left that belongs
    to any of these sets.
*/

WORD FindOnly(PHEAD WORD *term, WORD *pattern)
{
    GETBIDENTITY
    WORD *t, *m;
    WORD *tstop, *mstop;
    WORD *xstop, *ystop, *setp = AN.ForFindOnly;
    WORD n, nt, *p, nq;
    WORD older[NORMSIZE], *q, newval1, newval2, newval3;
    AN.UsedOtherFind = 0;
    m = pattern;
    mstop = m + *m;
    m++;
    t = term;
    t += *term - 1;
    tstop = t - ABS(*t) + 1;
    t = term;
    t++;
    while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
    while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
    if ( m < mstop ) { do {
/*
            #[ SYMBOLS :
*/
        if ( *m == SYMBOL ) {
            ystop = m + m[1];
            m += 2;
            n = 0;
            p = older;
            if ( t < tstop ) while ( *t != SYMBOL ) {
                t += t[1];
                if ( t >= tstop ) {
OnlyZer1:
                    do {
                        if ( *m >= 2*MAXPOWER ) return(0);
                        if ( m[1] >= 2*MAXPOWER ) nt = m[1];
                        else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
                        else return(0);
                        nt -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD nt,SYMTONUM,0,&newval3) ) return(0);
                        AddWild(BHEAD nt,SYMTONUM,0);
                        m += 2;
                    } while ( m < ystop );
                    goto EndLoop;
                }
            }
            else goto OnlyZer1;
            xstop = t + t[1];
            t += 2;
            do {
                if ( *m == *t && t < xstop ) {
                    if ( m[1] == t[1] ) { m += 2; t += 2; }
                    else if ( m[1] >= 2*MAXPOWER ) {
                        nt = t[1];
                        nq = m[1];
                        goto OnlyL2;
                    }
                    else if ( m[1] <= -2*MAXPOWER ) {
                        nt = -t[1];
                        nq = -m[1];
OnlyL2:                 nq -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD nq,SYMTONUM,nt,&newval3) ) return(0);
                        AddWild(BHEAD nq,SYMTONUM,nt);
                        m += 2;
                        t += 2;
                    }
                    else {
                        *p++ = *t++; *p++ = *t++; n += 2;
                    }
                }
                else if ( *m >= 2*MAXPOWER ) {
                    while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n += 2; }
                    nq = n;
                    p = older;
                    while ( nq > 0 ) {
                        if ( !CheckWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,*p,&newval1) ) {
```

```
                            if ( m[1] == p[1] ) {
                                AddWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,newval1);
                                break;
                            }
                            else if ( m[1] >= 2*MAXPOWER && m[1] != *m ) {
                                if ( !CheckWild(BHEAD m[1]-2*MAXPOWER,SYMTONUM,p[1],&newval3) ) {
                                    AddWild(BHEAD m[1]-2*MAXPOWER,SYMTONUM,p[1]);
                                    AddWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,newval1);
                                    break;
                                }
                            }
                            else if ( m[1] <= -2*MAXPOWER && m[1] != -(*m) ) {
                                if ( !CheckWild(BHEAD -m[1]-2*MAXPOWER,SYMTONUM,-p[1],&newval3) ) {
                                    AddWild(BHEAD -m[1]-2*MAXPOWER,SYMTONUM,-p[1]);
                                    AddWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,newval1);
                                    break;
                                }
                            }
                        }
                        nq -= 2;
                        p += 2;
                    }
                    if ( nq <= 0 ) return(0);
                    nq -= 2;
                    n -= 2;
                    q = p + 2;
                    while ( --nq >= 0 ) *p++ = *q++;
                    m += 2;
                }
                else {
                    if ( t >= xstop || *m < *t ) {
                        if ( m[1] >= 2*MAXPOWER ) nt = m[1];
                        else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
                        else return(0);
                        nt -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD nt,SYMTONUM,0,&newval3) ) return(0);
                        AddWild(BHEAD nt,SYMTONUM,0);
                        m += 2;
                    }
                    else {
                        *p++ = *t++; *p++ = *t++; n += 2;
                    }
                }
            } while ( m < ystop );
            if ( setp ) {
                while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n+= 2; }
                p = older;
                while ( n > 0 ) {
                    nq = setp[1] - 2;
                    m = setp + 2;
                    while ( --nq >= 0 ) {
                        if ( Sets[*m].type != CSYMBOL ) { m++; continue; }
                        t = SetElements + Sets[*m].first;
                        tstop = SetElements + Sets[*m].last;
                        while ( t < tstop ) {
                            if ( *t++ == *p ) return(0);
                        }
                        m++;
                    }
                    n -= 2;
                    p += 2;
                }
            }
            return(1);
        }
/*
            #] SYMBOLS :
            #[ DOTPRODUCTS :
*/
        else if ( *m == DOTPRODUCT ) {
            ystop = m + m[1];
            m += 2;
            n = 0;
            p = older;
            if ( t < tstop ) {
                if ( *t < DOTPRODUCT ) goto OnlyZer2;
                while ( *t > DOTPRODUCT ) {
                t += t[1];
                if ( t >= tstop || *t < DOTPRODUCT ) {
OnlyZer2:
                    do {
                        if ( *m >= (AM.OffsetVector+WILDOFFSET)
                        || m[1] >= (AM.OffsetVector+WILDOFFSET) ) return(0);
                        if ( m[2] >= 2*MAXPOWER ) nq = m[2];
                        else if ( m[2] <= -2*MAXPOWER ) nq = -m[2];
                        else return(0);
                        nq -= 2*MAXPOWER;
```

```
                        if ( CheckWild(BHEAD nq,SYMTONUM,0,&newval3) ) return(0);
                        AddWild(BHEAD nq,SYMTONUM,0);
                        m += 3;
                    } while ( m < ystop );
                    goto EndLoop;
                }
                }
            }
            else goto OnlyZer2;
            xstop = t + t[1];
            t += 2;
            do {
                if ( *m == *t && m[1] == t[1] && t < xstop ) {
                    if ( t[2] != m[2] ) {
                        if ( m[2] >= 2*MAXPOWER ) {
                            nq = m[2];
                            nt = t[2];
                        }
                        else if ( m[2] <= -2*MAXPOWER ) {
                            nq = -m[2];
                            nt = -t[2];
                        }
                        else return(0);
                        nq -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD nq,SYMTONUM,nt,&newval3) ) return(0);
                        AddWild(BHEAD nq,SYMTONUM,nt);
                    }
                    t += 3; m += 3;
                }
                else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
                    while ( t < xstop ) {
                        *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
                    }
                    nq = n;
                    p = older;
                    while ( nq > 0 ) {
                        if ( *m == m[1] ) {
                            if ( *p != p[1] ) goto NextInDot;
                        }
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*p,&newval1) &&
                             !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,p[1],&newval2) ) {
                            if ( p[2] == m[2] ) {
OnlyL9:                         AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval2);
                                AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newval1);
                                break;
                            }
                            if ( m[2] >= 2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,p[2],&newval3) ) {
                                    AddWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,newval3);
                                    goto OnlyL9;
                                }
                            }
                            else if ( m[2] <= -2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2],&newval3) ) {
                                    AddWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2]);
                                    goto OnlyL9;
                                }
                            }
                        }
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,p[1],&newval1) &&
                             !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,*p,&newval2) ) {
                            if ( p[2] == m[2] ) {
OnlyL10:                        AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newval1);
                                AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval2);
                                break;
                            }
                            if ( m[2] >= 2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,p[2],&newval3) ) {
                                    AddWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,p[2]);
                                    goto OnlyL10;
                                }
                            }
                            else if ( m[2] <= -2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2],&newval3) ) {
                                    AddWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2]);
                                    goto OnlyL10;
                                }
                            }
                        }
NextInDot:
                        p += 3; nq -= 3;
                    }
                    if ( nq <= 0 ) return(0);
                    q = p+3;
                    nq -= 3;
                    n -= 3;
                    while ( --nq >= 0 ) *p++ = *q++;
```

```
                    m += 3;
                }
                else if ( m[1] >= (AM.OffsetVector+WILDOFFSET) ) {
                    while ( *m >= *t && t < xstop ) {
                        *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
                    }
                    nq = n;
                    p = older;
                    while ( nq > 0 ) {
                        if ( *m == *p && !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,p[1],&newval1) ) {
                            if ( p[2] == m[2] ) {
                                AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                break;
                            }
                            else if ( m[2] >= 2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,p[2],&newval3) ) {
                                    AddWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,p[2]);
                                    AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                    break;
                                }
                            }
                            else if ( m[2] <= -2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2],&newval3) ) {
                                    AddWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2]);
                                    AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                    break;
                                }
                            }
                        }
                        if ( *m == p[1] && !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,*p,&newval1) ) {
                            if ( p[2] == m[2] ) {
                                AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                break;
                            }
                            if ( m[2] >= 2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,p[2],&newval3) ) {
                                    AddWild(BHEAD m[2]-2*MAXPOWER,SYMTONUM,p[2]);
                                    AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                    break;
                                }
                            }
                            else if ( m[2] <= -2*MAXPOWER ) {
                                if ( !CheckWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2],&newval3) ) {
                                    AddWild(BHEAD -m[2]-2*MAXPOWER,SYMTONUM,-p[2]);
                                    AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                    break;
                                }
                            }
                        }
                        p += 3; nq -= 3;
                    }
                    if ( nq <= 0 ) return(0);
                    q = p+3;
                    nq -= 3;
                    n -= 3;
                    while ( --nq >= 0 ) *p++ = *q++;
                    m += 3;
                }
                else {
                    if ( t >= xstop || *m < *t || ( *m == *t && m[1] < t[1] ) ) {
                        if ( m[2] > 2*MAXPOWER ) nt = m[2];
                        else if ( m[2] <= -2*MAXPOWER ) nt = -m[2];
                        else return(0);
                        nt -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD nt,SYMTONUM,0,&newval3) ) return(0);
                        AddWild(BHEAD nt,SYMTONUM,0);
                        m += 3;
                    }
                    else {
                        *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
                    }
                }
            } while ( m < ystop );
            t = xstop;
        }
/*
            #] DOTPRODUCTS :
*/
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Error in pattern(1)");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
EndLoop:;
    } while ( m < mstop ); }
    if ( setp ) {
```

```
/*
        while ( t < tstop && *t > SYMBOL ) t += t[1];
        if ( t < tstop && setp[1] > 2 ) return(0);
*/
                        /* There were nonempty sets */
                        /* Empty sets are rejected by the compiler */
    }
    return(1);
}

/*
        #] FindOnly :
        #[ FindOnce :           WORD FindOnce(term,pattern)

    Searches for a single match in term. The difference with multi
    lies mainly in the fact that here functions may occur.
    The functions have not been implemented yet. (10-Apr-1988)
    Wildcard powers are adjustable. The value closer to zero is taken.
    Positive and negative gives (o surprise) zero.

*/

WORD FindOnce(PHEAD WORD *term, WORD *pattern)
{
    GETBIDENTITY
    WORD *t, *m;
    WORD *tstop, *mstop;
    WORD *xstop, *ystop;
    WORD n, nt, *p, nq, mt, ch;
    WORD older[2*NORMSIZE], *q, newval1, newval2, newval3;
    AN.UsedOtherFind = 0;
    m = pattern;
    mstop = m + *m;
    m++;
    t = term;
    t += *term - 1;
    tstop = t - ABS(*t) + 1;
    t = term;
    t++;
    while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
    while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
    if ( m < mstop ) { do {
/*
            #[ SYMBOLS :
*/
        if ( *m == SYMBOL ) {
            ystop = m + m[1];
            m += 2;
            n = 0;
            p = older;
            if ( t < tstop ) while ( *t != SYMBOL ) {
                t += t[1];
                if ( t >= tstop ) {
TryZero:
                    do {
                        if ( *m >= 2*MAXPOWER ) return(0);
                        if ( m[1] >= 2*MAXPOWER ) nt = m[1];
                        else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
                        else return(0);
                        nt -= 2*MAXPOWER;
                        if ( ( ch = CheckWild(BHEAD nt,SYMTONUM,0,&newval3) ) != 0 ) {
                            if ( ch > 1 ) return(0);
                            if ( AN.oldtype != SYMTONUM ) return(0);
                            if ( *AN.MaskPointer == 2 ) return(0);
                        }
                        AddWild(BHEAD nt,SYMTONUM,0);
                        m += 2;
                    } while ( m < ystop );
                    goto EndLoop;
                }
            }
            else goto TryZero;
            xstop = t + t[1];
            t += 2;
            do {
                if ( *m == *t && t < xstop ) {
                    nt = t[1];
                    mt = m[1];
                    if ( ( mt > 0 && mt <= nt ) ||
                         ( mt < 0 && mt >= nt ) ) { m += 2; t += 2; }
                    else if ( mt >= 2*MAXPOWER ) goto OnceL2;
                    else if ( mt <= -2*MAXPOWER ) {
                        nt = -nt;
                        mt = -mt;
OnceL2:                 mt -= 2*MAXPOWER;
                        if ( ( ch = CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) != 0 ) {
                            if ( ch > 1 ) return(0);
```

```
                            if ( AN.oldtype != SYMTONUM ) return(0);
                            if ( AN.oldvalue <= 0 ) {
                                if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                else {
                                    if ( *AN.MaskPointer == 2 ) return(0);
                                    if ( nt > 0 ) nt = 0;
                                }
                            }
                            if ( AN.oldvalue >= 0 ) {
                                if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                else {
                                    if ( *AN.MaskPointer == 2 ) return(0);
                                    if ( nt < 0 ) nt = 0;
                                }
                            }
                        }
                        AddWild(BHEAD mt,SYMTONUM,nt);
                        m += 2;
                        t += 2;
                    }
                    else {
                        *p++ = *t++; *p++ = *t++; n += 2;
                    }
                }
                else if ( *m >= 2*MAXPOWER ) {
                    while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n += 2; }
                    nq = n;
                    p = older;
                    while ( nq > 0 ) {
                        nt = p[1];
                        if ( !CheckWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,*p,&newval1) ) {
                            mt = m[1];
                            if ( ( mt > 0 && mt <= nt ) ||
                                 ( mt < 0 && mt >= nt ) ) {
                                AddWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,newval1);
                                break;
                            }
                            else if ( mt >= 2*MAXPOWER && mt != *m ) {
OnceL4a:                        mt -= 2*MAXPOWER;
                                if ( ( ch = CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) != 0 ) {
                                    if ( ch > 1 ) return(0);
                                    if ( AN.oldtype == SYMTONUM ) {
                                        if ( AN.oldvalue >= 0 ) {
                                            if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt < 0 ) nt = 0;
                                            }
                                        }
                                        else {
                                            if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt > 0 ) nt = 0;
                                            }
                                        }
                                        AddWild(BHEAD mt,SYMTONUM,nt);
                                        AddWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,newval1);
                                        break;
                                    }
                                }
                                else {
                                    AddWild(BHEAD mt,SYMTONUM,nt);
                                    AddWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,newval1);
                                    break;
                                }
                            }
                            else if ( mt <= -2*MAXPOWER && mt != -(*m) ) {
                                nt = -nt;
                                mt = -mt;
                                goto OnceL4a;
                            }
                        }
                        nq -= 2;
                        p += 2;
                    }
                    if ( nq <= 0 ) return(0);
                    nq -= 2;
                    n -= 2;
                    q = p + 2;
                    while ( --nq >= 0 ) *p++ = *q++;
                    m += 2;
                }
                else {
                    if ( t >= xstop || *m < *t ) {
                        if ( m[1] >= 2*MAXPOWER ) nt = m[1];
                        else if ( m[1] <= -2*MAXPOWER ) nt = -m[1];
```

```
                        else return(0);
                        nt -= 2*MAXPOWER;
                        if ( ( ch = CheckWild(BHEAD nt,SYMTONUM,0,&newval3) ) != 0 ) {
                            if ( ch > 1 ) return(0);
                            if ( AN.oldtype != SYMTONUM ) return(0);
                            if ( *AN.MaskPointer == 2 ) return(0);
                        }
                        AddWild(BHEAD nt,SYMTONUM,0);
                        m += 2;
                    }
                    else {
                        *p++ = *t++; *p++ = *t++; n += 2;
                    }
                }
            } while ( m < ystop );
        }
/*
            #] SYMBOLS :
            #[ DOTPRODUCTS :
*/
        else if ( *m == DOTPRODUCT ) {
            ystop = m + m[1];
            m += 2;
            n = 0;
            p = older;
            if ( t < tstop ) {
                if ( *t < DOTPRODUCT ) goto OnceOp;
                while ( *t > DOTPRODUCT ) {
                t += t[1];
                if ( t >= tstop || *t < DOTPRODUCT ) {
OnceOp:
                    do {
                        if ( *m >= (AM.OffsetVector+WILDOFFSET)
                        || m[1] >= (AM.OffsetVector+WILDOFFSET) ) return(0);
                        if ( m[2] >= 2*MAXPOWER ) {
                            nq = m[2] - 2*MAXPOWER;
                        }
                        else if ( m[2] <= -2*MAXPOWER ) {
                            nq = -m[2] - 2*MAXPOWER;
                        }
                        else return(0);
                        if ( CheckWild(BHEAD nq,SYMTONUM,(WORD)0,&newval3) ) {
                            if ( AN.oldtype != SYMTONUM ) return(0);
                            if ( *AN.MaskPointer == 2 ) return(0);
                        }
                        AddWild(BHEAD nq,SYMTONUM,(WORD)0);
                        m += 3;
                    } while ( m < ystop );
                    goto EndLoop;
                }
                }
            }
            else goto OnceOp;
            xstop = t + t[1];
            t += 2;
            do {
                if ( *m == *t && m[1] == t[1] && t < xstop ) {
                    nt = t[2];
                    mt = m[2];
/*
                    if ( ( nt > 0 && nt < mt ) ||
                         ( nt < 0 && nt > mt ) ) {
                        if ( mt <= -2*MAXPOWER ) {
                            mt = -mt;
                            nt = -nt;
                        }
                        else if ( mt < 2*MAXPOWER ) return(0);
                        mt -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
                            if ( AN.oldtype != SYMTONUM ) return(0);
                            if ( AN.oldvalue <= 0 ) {
                                if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                else {
                                    if ( *AN.MaskPointer == 2 ) return(0);
                                    if ( nt > 0 ) nt = 0;
                                }
                            }
                            if ( AN.oldvalue >= 0 ) {
                                if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                else {
                                    if ( *AN.MaskPointer == 2 ) return(0);
                                    if ( nt < 0 ) nt = 0;
                                }
                            }
                        }
                        AddWild(BHEAD mt,SYMTONUM,nt);
                        m += 3; t += 3;
```

```
                    }
                    else if ( ( nt > 0 && nt >= mt && mt > -2*MAXPOWER )
                    || ( nt < 0 && nt <= mt && mt < 2*MAXPOWER ) ) {
                        m += 3; t += 3;
                    }
*/
                    if ( ( mt > 0 && mt <= nt ) ||
                         ( mt < 0 && mt >= nt ) ) { m += 3; t += 3; }
                    else if ( mt >= 2*MAXPOWER ) goto OnceL7;
                    else if ( mt <= -2*MAXPOWER ) {
                        nt = -nt;
                        mt = -mt;
OnceL7:                 mt -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
                            if ( AN.oldtype != SYMTONUM ) return(0);
                            if ( AN.oldvalue <= 0 ) {
                                if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                else {
                                    if ( *AN.MaskPointer == 2 ) return(0);
                                    if ( nt > 0 ) nt = 0;
                                }
                            }
                            if ( AN.oldvalue >= 0 ) {
                                if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                else {
                                    if ( *AN.MaskPointer == 2 ) return(0);
                                    if ( nt < 0 ) nt = 0;
                                }
                            }
                        }
                        AddWild(BHEAD mt,SYMTONUM,nt);
                        m += 3;
                        t += 3;
                    }
                    else {
                        *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
                    }
                }
                else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
                    while ( t < xstop ) {
                        *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
                    }
                    nq = n;
                    p = older;
                    while ( nq > 0 ) {
                        if ( *m == m[1] ) {
                            if ( *p != p[1] ) goto NextInDot;
                        }
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*p,&newval1) &&
                             !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,p[1],&newval2) ) {
                            nt = p[2];
                            mt = m[2];
                            if ( ( mt > 0 && nt >= mt ) ||
                                 ( mt < 0 && nt <= mt ) ) {
OnceL9:                         AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newval1);
                                AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval2);
                                break;
                            }
                            if ( mt >= 2*MAXPOWER ) {
OnceL9a:                        mt -= 2*MAXPOWER;
                                if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
                                    if ( AN.oldtype == SYMTONUM ) {
                                        if ( AN.oldvalue >= 0 ) {
                                            if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt < 0 ) nt = 0;
                                            }
                                        }
                                        else {
                                            if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt > 0 ) nt = 0;
                                            }
                                        }
                                        AddWild(BHEAD mt,SYMTONUM,nt);
                                        goto OnceL9;
                                    }
                                }
                                else {
                                    AddWild(BHEAD mt,SYMTONUM,nt);
                                    goto OnceL9;
                                }
                            }
                            else if ( mt <= -2*MAXPOWER ) {
                                mt = -mt;
```

```
                                nt = -nt;
                                goto OnceL9a;
                            }
                        }
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,p[1],&newval1) &&
                             !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,*p,&newval2) ) {
                            nt = p[2];
                            mt = m[2];
                            if ( ( mt > 0 && nt >= mt ) ||
                                 ( mt < 0 && nt <= mt ) ) {
OnceL10:                        AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newval1);
                                AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval2);
                                break;
                            }
                            if ( mt >= 2*MAXPOWER ) {
OnceL10a:                       mt -= 2*MAXPOWER;
                                if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
                                    if ( AN.oldtype == SYMTONUM ) {
                                        if ( AN.oldvalue >= 0 ) {
                                            if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt < 0 ) nt = 0;
                                            }
                                        }
                                        else {
                                            if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt > 0 ) nt = 0;
                                            }
                                        }
                                        AddWild(BHEAD mt,SYMTONUM,nt);
                                        goto OnceL10;
                                    }
                                }
                                else {
                                    AddWild(BHEAD mt,SYMTONUM,nt);
                                    goto OnceL10;
                                }
                            }
                            else if ( mt <= -2*MAXPOWER ) {
                                mt = -mt;
                                nt = -nt;
                                goto OnceL10a;
                            }
                        }
NextInDot:
                        p += 3; nq -= 3;
                    }
                    if ( nq <= 0 ) return(0);
                    else {
                        q = p+3;
                        nq -= 3;
                        n -= 3;
                        while ( --nq >= 0 ) *p++ = *q++;
                    }
                    m += 3;
                }
                else if ( m[1] >= (AM.OffsetVector+WILDOFFSET) ) {
                    while ( *m >= *t && t < xstop ) {
                        *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
                    }
                    nq = n;
                    p = older;
                    while ( nq > 0 ) {
                        if ( *m == *p && !CheckWild(BHEAD m[1]-WILDOFFSET,
                        VECTOVEC,p[1],&newval1) ) {
                            nt = p[2];
                            mt = m[2];
                            if ( ( mt > 0 && nt >= mt ) ||
                                 ( mt < 0 && nt <= mt ) ) {
                                AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                break;
                            }
                            else if ( mt >= 2*MAXPOWER ) {
OnceL7a:                        mt -= 2*MAXPOWER;
                                if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
                                    if ( AN.oldtype == SYMTONUM ) {
                                        if ( AN.oldvalue >= 0 ) {
                                            if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt < 0 ) nt = 0;
                                            }
                                        }
                                        else {
```

```
                                            if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt > 0 ) nt = 0;
                                            }
                                        }
                                        AddWild(BHEAD mt,SYMTONUM,nt);
                                        AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                        break;
                                    }
                                }
                                else {
                                    AddWild(BHEAD mt,SYMTONUM,nt);
                                    AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                    break;
                                }
                            }
                            else if ( mt <= -2*MAXPOWER ) {
                                mt = -mt;
                                nt = -nt;
                                goto OnceL7a;
                            }
                        }
                        if ( *m == p[1] && !CheckWild(BHEAD m[1]-WILDOFFSET,
                        VECTOVEC,*p,&newval1) ) {
                            nt = p[2];
                            mt = m[2];
                            if ( ( mt > 0 && nt >= mt ) ||
                                 ( mt < 0 && nt <= mt ) ) {
                                AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                break;
                            }
                            if ( mt >= 2*MAXPOWER ) {
OnceL8a:                        mt -= 2*MAXPOWER;
                                if ( CheckWild(BHEAD mt,SYMTONUM,nt,&newval3) ) {
                                    if ( AN.oldtype == SYMTONUM ) {
                                        if ( AN.oldvalue >= 0 ) {
                                            if ( nt > AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt < 0 ) nt = 0;
                                            }
                                        }
                                        else {
                                            if ( nt < AN.oldvalue ) nt = AN.oldvalue;
                                            else {
                                                if ( *AN.MaskPointer == 2 ) return(0);
                                                if ( nt > 0 ) nt = 0;
                                            }
                                        }
                                        AddWild(BHEAD mt,SYMTONUM,nt);
                                        AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                        break;
                                    }
                                }
                                else {
                                    AddWild(BHEAD mt,SYMTONUM,nt);
                                    AddWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,newval1);
                                    break;
                                }
                            }
                            else if ( mt < -2*MAXPOWER ) {
                                mt = -mt;
                                nt = -nt;
                                goto OnceL8a;
                            }
                        }
                        p += 3; nq -= 3;
                    }
                    if ( nq <= 0 ) return(0);
                    q = p+3;
                    nq -= 3;
                    n -= 3;
                    while ( --nq >= 0 ) *p++ = *q++;
                    m += 3;
                }
                else {
                    if ( t >= xstop || *m < *t || ( *m == *t && m[1] < t[1] ) ) {
                        if ( m[2] >= 2*MAXPOWER ) nt = m[2];
                        else if ( m[2] <= -2*MAXPOWER ) nt = -m[2];
                        else return(0);
                        nt -= 2*MAXPOWER;
                        if ( CheckWild(BHEAD nt,SYMTONUM,0,&newval3) ) {
                            if ( AN.oldtype != SYMTONUM ) return(0);
                            if ( *AN.MaskPointer == 2 ) return(0);
                        }
                        AddWild(BHEAD nt,SYMTONUM,0);
```

```
                        m += 3;
                    }
                    else {
                        *p++ = *t++; *p++ = *t++; *p++ = *t++; n += 3;
                    }
                }
            } while ( m < ystop );
            t = xstop;
        }
/*
            #] DOTPRODUCTS :
*/
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Error in pattern(2)");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
EndLoop:;
    } while ( m < mstop ); }
    else {
        return(-1);
    }
    return(1);
}

/*
        #] FindOnce :
        #[ FindMulti :          WORD FindMulti(term,pattern)

    Note that multi cannot deal with wildcards. Those patterns revert
    to many which gives subsequent calls to once.

*/

WORD FindMulti(PHEAD WORD *term, WORD *pattern)
{
    GETBIDENTITY
    WORD *t, *m, *p;
    WORD *tstop, *mstop;
    WORD *xstop, *ystop;
    WORD mt, power, n, nq;
    WORD older[2*NORMSIZE], *q, newval1;
    AN.UsedOtherFind = 0;
    m = pattern;
    mstop = m + *m;
    m++;
    t = term;
    t += *term - 1;
    tstop = t - ABS(*t) + 1;
    t = term;
    t++;
    while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
    while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
    power = -1;             /* No power yet */
    if ( m < mstop ) { do {
/*
            #[ SYMBOLS :
*/
        if ( *m == SYMBOL ) {
            ystop = m + m[1];
            m += 2;
            if ( t >= tstop ) return(0);
            while ( *t != SYMBOL ) { t += t[1]; if ( t >= tstop ) return(0); }
            xstop = t + t[1];
            t += 2;
            p = older;
            n = 0;
            do {
                if ( *m >= 2*MAXPOWER ) {
                    while ( t < xstop ) { *p++ = *t++; *p++ = *t++; n += 2; }
                    nq = n;
                    p = older;
                    while ( nq > 0 ) {
                        if ( !CheckWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,*p,&newval1) ) {
                            mt = p[1]/m[1];
                            if ( mt > 0 ) {
                                if ( power < 0 || mt < power ) power = mt;
                                AddWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,newval1);
                                break;
                            }
                        }
                        nq -= 2;
                        p += 2;
                    }
                    if ( nq <= 0 ) return(0);
                    nq -= 2;
```

```
                    n -= 2;
                    q = p + 2;
                    while ( --nq >= 0 ) *p++ = *q++;
                    m += 2;
                }
                else if ( t >= xstop ) return(0);
                else if ( *m == *t ) {
                    if ( ( mt = t[1]/m[1] ) <= 0 ) return(0);
                    if ( power < 0 || mt < power ) power = mt;
                    m += 2;
                    t += 2;
                }
                else if ( *m < *t ) return(0);
                else { *p++ = *t++; *p++ = *t++; n += 2; }
            } while ( m < ystop );
        }
/*
            #] SYMBOLS :
            #[ DOTPRODUCTS :
*/
        else if ( *m == DOTPRODUCT ) {
            ystop = m + m[1];
            m += 2;
            if ( t >= tstop ) return(0);
            while ( *t != DOTPRODUCT ) { t += t[1]; if ( t >= tstop ) return(0); }
            xstop = t + t[1];
            t += 2;
            do {
                if ( t >= xstop ) return(0);
                if ( *t == *m ) {
                    if ( t[1] == m[1] ) {
                        if ( ( mt = t[2]/m[2] ) <= 0 ) return(0);
                        if ( power < 0 || mt < power ) power = mt;
                        m += 3;
                    }
                    else if ( t[1] > m[1] ) return(0);
                }
                else if ( *t > *m ) return(0);
                t += 3;
            } while ( m < ystop );
            t = xstop;
        }
/*
            #] DOTPRODUCTS :
*/
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Error in pattern(3)");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    } while ( m < mstop ); }
    if ( power < 0 ) power = 0;
    return(power);
}

/*
        #] FindMulti :
        #[ FindRest :           WORD FindRest(term,pattern)

    This routine scans for anything but dotproducts and symbols.

*/

WORD FindRest(PHEAD WORD *term, WORD *pattern)
{
    GETBIDENTITY
    WORD *t, *m, *tt, wild, regular;
    WORD *tstop, *mstop;
    WORD *xstop, *ystop;
    WORD n, *p, nq;
    WORD older[NORMSIZE], *q, newval1, newval2;
    int i, ntwa;
    AN.UsedOtherFind = 0;
    AN.findTerm = term; AN.findPattern = pattern;
    m = AN.WildValue;
    i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
    ntwa = 0;
    while ( i > 0 ) {
        if ( m[0] == ARGTOARG ) ntwa++;
        m += m[1];
        i--;
    }
    t = term;
    t += *term - 1;
    tstop = t - ABS(*t) + 1;
    t = term;
```

```
    t++; p = t;
    while ( t < tstop && *t > DOTPRODUCT ) t += t[1];
    tstop = t;
    t = p;
    m = pattern;
    mstop = m + *m;
    m++;
    p = m;
    while ( m < mstop && *m > DOTPRODUCT ) m += m[1];
    mstop = m;
    m = p;
    if ( m < mstop ) {
    do {
/*
            #[ FUNCTIONS :
*/
        if ( *m >= FUNCTION ) {
            if ( *mstop > 5 && !MatchIsPossible(pattern,term) ) return(0);
            ystop = m;
            n = 0;
            do {
                m += m[1]; n++;
            } while ( m < mstop && *m >= FUNCTION );
            AT.WorkPointer += n;
            while ( t < tstop && *t == SUBEXPRESSION ) t += t[1];
            tt = xstop = t;
            nq = 0;
            while ( t < tstop && ( *t >= FUNCTION || *t == SUBEXPRESSION ) ) {
                if ( *t != SUBEXPRESSION ) {
                    nq++;
                    if ( functions[*t-FUNCTION].commute ) tt = t + t[1];
                }
                t += t[1];
            }
            if ( nq < n ) return(0);
            AN.terstart = term;
            AN.terstop = t;
            AN.terfirstcomm = tt;
            AN.patstop = m;
            AN.NumTotWildArgs = ntwa;
            if ( !ScanFunctions(BHEAD ystop,xstop,0) ) return(0);
        }
/*
            #] FUNCTIONS :
            #[ VECTORS :
*/
        else if ( *m == VECTOR ) {
            while ( t < tstop && *t != VECTOR ) t += t[1];
            if ( t >= tstop ) return(0);
            xstop = t + t[1];
            ystop = m + m[1];
            t += 2;
            m += 2;
            n = 0;
            p = older;
            do {
                if ( *m == *t && m[1] == t[1] && t < xstop ) {
                    m += 2; t += 2;
                }
                else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
                    if ( t < xstop ) {
                        p = older + n;
                        do { *p++ = *t++; n++; } while ( t < xstop );
                    }
                    p = older;
                    nq = n;
                    if ( ( m[1] < (AM.OffsetIndex+WILDOFFSET) )
                    || ( m[1] >= (AM.OffsetIndex+2*WILDOFFSET) ) ) {
                        while ( nq > 0 ) {
                            if ( m[1] == p[1] ) {
                                if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*p,&newval1) ) {
RestL11:                            AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newval1);
                                    break;
                                }
                            }
                            p += 2;
                            nq -= 2;
                        }
                    }
                    else {      /* Double wildcard */
                        while ( nq > 0 ) {
                            if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*p,&newval1) &&
                                 !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,p[1],&newval2) ) {
                                AddWild(BHEAD m[1]-WILDOFFSET,INDTOIND,newval2);
                                goto RestL11;
                            }
                            p += 2;
```

```
                            nq -= 2;
                        }
                    }
                    if ( nq > 0 ) {
                        nq -= 2; q = p + 2; n -= 2;
                        while ( --nq >= 0 ) *p++ = *q++;
                    }
                    else return(0);
                    m += 2;
                }
                else if ( ( *m <= *t )
                && ( m[1] >= (AM.OffsetIndex + WILDOFFSET) )
                && ( m[1] <  (AM.OffsetIndex + 2*WILDOFFSET) ) ) {
                    if ( *m == *t && t < xstop ) {
                        p = older;
                        p += n;
                        *p++ = *t++;
                        *p++ = *t++;
                        n += 2;
                    }
                    p = older;
                    nq = n;
                    while ( nq > 0 ) {
                        if ( *m == *p ) {
                            if ( !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,p[1],&newval1) ) {
                                AddWild(BHEAD m[1]-WILDOFFSET,INDTOIND,newval1);
                                break;
                            }
                        }
                        p += 2;
                        nq -= 2;
                    }
                    if ( nq > 0 ) {
                        nq -= 2; q = p + 2; n -= 2;
                        while ( --nq >= 0 ) *p++ = *q++;
                    }
                    else return(0);
                    m += 2;
                }
                else {
                    if ( t >= xstop ) return(0);
                    *p++ = *t++; *p++ = *t++; n += 2;
                }
            } while ( m < ystop );
        }
/*
        #] VECTORS :
        #[ INDICES :
*/
        else if ( *m == INDEX ) {
/*
            This needs only to say that there is a match, after matching
            a 'wildcard'. This has to be prepared in TestMatch. The C->rhs
            should provide the replacement inside the prototype!
            Next question: id,p=q/2+r/2
*/
            while ( *t != INDEX ) { t += t[1]; if ( t >= tstop ) return(0); }
            xstop = t + t[1];
            ystop = m + m[1];
            t += 2;
            m += 2;
            n = 0;
            p = older;
            do {
                if ( *m == *t && t < xstop && m < ystop ) {
                    t++; m++;
                }
                else if ( ( *m >= (AM.OffsetIndex+WILDOFFSET) )
                && ( *m < (AM.OffsetIndex+2*WILDOFFSET) ) ) {
                    while ( t < xstop ) {
                        *p++ = *t++; n++;
                    }
                    if ( !n ) return(0);
                    nq = n;
                    q = older;
                    do {
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*q,&newval1) ) {
                            AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newval1);
                            break;
                        }
                        q++;
                        nq--;
                    } while ( nq > 0 );
                    if ( nq <= 0 ) return (0);
                    n--;
                    nq--;
                    p = q + 1;
```

```
                    while ( nq > 0 ) { *q++ = *p++; nq--; }
                    p--;
                    m++;
                }
                else if ( ( *m >= (AM.OffsetVector+WILDOFFSET) )
                && ( *m < (AM.OffsetVector+2*WILDOFFSET) ) ) {
                    while ( t < xstop ) {
                        *p++ = *t++; n++;
                    }
                    if ( !n ) return(0);
                    nq = n;
                    q = older;
                    do {
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*q,&newval1) ) {
                            AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newval1);
                            break;
                        }
                        q++;
                        nq--;
                    } while ( nq > 0 );
                    if ( nq <= 0 ) return (0);
                    n--;
                    nq--;
                    p = q + 1;
                    while ( nq > 0 ) { *q++ = *p++; nq--; }
                    p--;
                    m++;
                }
                else {
                    if ( t >= xstop ) return(0);
                    *p++ = *t++; n++;
                }
            } while ( m < ystop );

/*
            return(0);
*/
        }
/*
            #] INDICES :
            #[ DELTAS :
*/
        else if ( *m == DELTA ) {
            while ( *t != DELTA ) { t += t[1]; if ( t >= tstop ) return(0); }
            xstop = t + t[1];
            ystop = m + m[1];
            t += 2;
            m += 2;
            n = 0;
            p = older;
            do {
                if ( *t == *m && t[1] == m[1] && t < xstop ) {
                    m += 2;
                    t += 2;
                }
                else if ( ( *m >= (AM.OffsetIndex+WILDOFFSET) )
                && ( *m < (AM.OffsetIndex+2*WILDOFFSET) )
                && ( m[1] >= (AM.OffsetIndex+WILDOFFSET) )
                && ( m[1] < (AM.OffsetIndex+2*WILDOFFSET) ) ) { /* Two dummies */
                    while ( t < xstop ) {
                        *p++ = *t++; *p++ = *t++; n += 2;
                    }
                    if ( !n ) return(0);
                    nq = n;
                    q = older;
                    do {
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*q,&newval1) &&
                             !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,q[1],&newval2) ) {
                            AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newval1);
                            AddWild(BHEAD m[1]-WILDOFFSET,INDTOIND,newval2);
                            break;
                        }
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,q[1],&newval1) &&
                             !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,*q,&newval2) ) {
                            AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newval1);
                            AddWild(BHEAD m[1]-WILDOFFSET,INDTOIND,newval2);
                            break;
                        }
                        q += 2;
                        nq -= 2;
                    } while ( nq > 0 );
                    if ( nq <= 0 ) return(0);
                    n -= 2;
                    nq -= 2;
                    p = q + 2;
                    while ( nq > 0 ) { *q++ = *p++; nq--; }
                    p -= 2;
```

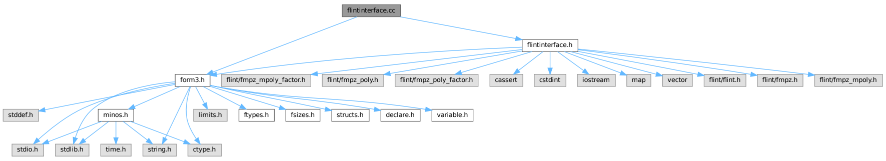
```
                    m += 2;
                }
                else if ( ( m[1] >= (AM.OffsetIndex+WILDOFFSET) )
                && ( m[1] < (AM.OffsetIndex+2*WILDOFFSET) ) ) {
                    wild = m[1]; regular = *m;
OneWild:
                    while ( ( regular == *t || regular == t[1] ) && t < xstop ) {
                        *p++ = *t++; *p++ = *t++; n += 2;
                    }
                    if ( !n ) return(0);
                    nq = n;
                    q = older;
                    do {
                        if ( regular == *q && !CheckWild(BHEAD wild-WILDOFFSET,INDTOIND,q[1],&newval1) ) {
                            AddWild(BHEAD wild-WILDOFFSET,INDTOIND,newval1);
                            break;
                        }
                        if ( regular == q[1] && !CheckWild(BHEAD wild-WILDOFFSET,INDTOIND,*q,&newval1) ) {
                            AddWild(BHEAD wild-WILDOFFSET,INDTOIND,newval1);
                            break;
                        }
                        q += 2;
                        nq -= 2;
                    } while ( nq > 0 );
                    if ( nq <= 0 ) return(0);
                    n -= 2;
                    nq -= 2;
                    p = q + 2;
                    while ( nq > 0 ) { *q++ = *p++; nq--; }
                    p -= 2;
                    m += 2;
                }
                else if ( ( *m >= (AM.OffsetIndex+WILDOFFSET) )
                && ( *m < (AM.OffsetIndex+2*WILDOFFSET) ) ) {
                    wild = *m; regular = m[1];
                    goto OneWild;
                }
                else {
                    if ( t >= tstop || *m < *t || ( *m == *t && m[1] < t[1] ) )
                        return(0);
                    *p++ = *t++; *p++ = *t++; n += 2;
                }
            } while ( m < ystop );
        }
/*
            #] DELTAS :
*/
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Pattern not yet implemented");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    } while ( m < mstop );
    return(1);
    }
    else return(-1);
}

/*
        #] FindRest :
    #] Patterns :
*/
```

4.37 flintinterface.cc File Reference

```
#include "form3.h"
#include "flintinterface.h"
```

Include dependency graph for flintinterface.cc:



Macros

- #define IFW(x) { if (write) {x;} }

4.37.1 Detailed Description

Contains functions which interface with FLINT functions to perform arithmetic with uni- and multi-variate polynomials and perform factorization.

Definition in file flintinterface.cc.

4.37.2 Macro Definition Documentation

4.37.2.1 IFW

```
#define IFW(
                x ) { if ( write ) {x;} }
```

Definition at line 1761 of file flintinterface.cc.

4.38 flintinterface.cc

Go to the documentation of this file.

```
00001
00007 #ifdef HAVE_CONFIG_H
00008 #ifndef CONFIG_H_INCLUDED
00009 #define CONFIG_H_INCLUDED
00010 #include <config.h>
00011 #endif
00012 #endif
00013
00014 extern "C" {
00015 #include "form3.h"
00016 }
00017
00018 #include "flintinterface.h"
00019
00020 /*
00021     #[ Types
00022  * Use fixed-size integer types (int32_t, uint32_t, int64_t, uint64_t) except when we refer
00023  * specifically to FORM term data, when we use WORD. This code has only been tested on 64bit
00024  * systems where WORDs are int32_t, and some functions (fmpz_get_form, fmpz_set_form) require this
00025  * to be the case. Enforce this:
00026  */
00027 static_assert(sizeof(WORD) == sizeof(int32_t), "flint interface expects 32bit WORD");
00028 static_assert(sizeof(UWORD) == sizeof(uint32_t), "flint interface expects 32bit WORD");
00029 static_assert(BITSINWORD == 32, "flint interface expects 32bit WORD");
00030 static_assert(sizeof(ulong) == sizeof(uint64_t), "flint interface expects ulong is uint64_t");
```

```
static_assert(sizeof(slong) == sizeof(int64_t), "flint interface expects slong is int64_t");

/*
 * Flint functions take arguments or return values which may be "slong" or "ulong" in its
 * documentation, and these are int64_t and uint64_t respectively.
 *
    #] Types
*/

/*
    #[ flint::cleanup :
*/
void flint::cleanup(void) {
    flint_cleanup();
}
/*
    #] flint::cleanup :
    #[ flint::cleanup_master :
*/
void flint::cleanup_master(void) {
    flint_cleanup_master();
}
/*
    #] flint::cleanup_master :

    #[ flint::divmod_mpoly :
*/
WORD* flint::divmod_mpoly(PHEAD const WORD *a, const WORD *b, const bool return_rem,
    const WORD must_fit_term, const var_map_t &var_map) {

    flint::mpoly_ctx ctx(var_map.size());
    flint::mpoly pa(ctx.d), pb(ctx.d), denpa(ctx.d), denpb(ctx.d);

    flint::from_argument_mpoly(pa.d, denpa.d, a, false, var_map, ctx.d);
    flint::from_argument_mpoly(pb.d, denpb.d, b, false, var_map, ctx.d);

    // The input won't have any symbols with negative powers, but there may be rational
    // coefficients. Verify this:
    if ( fmpz_mpoly_is_fmpz(denpa.d, ctx.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::divmod_mpoly: error: denpa is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( fmpz_mpoly_is_fmpz(denpb.d, ctx.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::divmod_mpoly: error: denpb is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }


    flint::fmpz scale;
    flint::mpoly div(ctx.d), rem(ctx.d);

    fmpz_mpoly_quasidivrem(scale.d, div.d, rem.d, pa.d, pb.d, ctx.d);

    // The quotient must be multiplied by the denominator of the divisor
    fmpz_mpoly_mul(div.d, div.d, denpb.d, ctx.d);

    // The overall denominator of both div and rem is given by scale*denpa. This we will pass to
    // to_argument_mpoly's "denscale" argument which performs the division in the result. We have
    // already checked denpa is just an fmpz.
    fmpz_mul(scale.d, scale.d, fmpz_mpoly_term_coeff_ref(denpa.d, 0, ctx.d));


    // Determine the size of the output by passing write = false.
    const bool with_arghead = false;
    bool write = false;
    const uint64_t prev_size = 0;
    const uint64_t out_size = return_rem ?
        (uint64_t)flint::to_argument_mpoly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
            rem.d, var_map, ctx.d, scale.d)
        :
        (uint64_t)flint::to_argument_mpoly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
            div.d, var_map, ctx.d, scale.d)
        ;
    WORD* res = (WORD *)Malloc1(sizeof(WORD)*out_size, "flint::divrem_mpoly");


    // Write out the result
    write = true;
    if ( return_rem ) {
        (uint64_t)flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
            rem.d, var_map, ctx.d, scale.d);
    }
    else {
```

```
        (uint64_t)flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
            div.d, var_map, ctx.d, scale.d);
    }

    return res;
}
/*
    #] flint::divmod_mpoly :
    #[ flint::divmod_poly :
*/
WORD* flint::divmod_poly(PHEAD const WORD *a, const WORD *b, const bool return_rem,
    const WORD must_fit_term, const var_map_t &var_map) {

    flint::poly pa, pb, denpa, denpb;

    flint::from_argument_poly(pa.d, denpa.d, a, false);
    flint::from_argument_poly(pb.d, denpb.d, b, false);

    // The input won't have any symbols with negative powers, but there may be rational
    // coefficients. Verify this:
    if ( fmpz_poly_length(denpa.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::divmod_poly: error: denpa is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( fmpz_poly_length(denpb.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::divmod_poly: error: denpb is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    flint::fmpz scale;
    uint64_t scale_power = 0;
    flint::poly div, rem;

    // Get the leading coefficient of pb. fmpz_poly_pseudo_divrem returns only the scaling power.
    fmpz_poly_get_coeff_fmpz(scale.d, pb.d, fmpz_poly_degree(pb.d));

    fmpz_poly_pseudo_divrem(div.d, rem.d, (ulong*)&scale_power, pa.d, pb.d);

    // The quotient must be multiplied by the denominator of the divisor
    fmpz_poly_mul(div.d, div.d, denpb.d);

    // The overall denominator of both div and rem is given by scale^scale_power*denpa. This we will
    // pass to to_argument_poly's "denscale" argument which performs the division in the result. We
    // have already checked denpa is just an fmpz.
    fmpz_pow_ui(scale.d, scale.d, scale_power);
    fmpz_mul(scale.d, scale.d, fmpz_poly_get_coeff_ptr(denpa.d, 0));


    // Determine the size of the output by passing write = false.
    const bool with_arghead = false;
    bool write = false;
    const uint64_t prev_size = 0;
    const uint64_t out_size = return_rem ?
        (uint64_t)flint::to_argument_poly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
            rem.d, var_map, scale.d)
        :
        (uint64_t)flint::to_argument_poly(BHEAD NULL, with_arghead, must_fit_term, write, prev_size,
            div.d, var_map, scale.d)
        ;
    WORD* res = (WORD *)Malloc1(sizeof(WORD)*out_size, "flint::divrem_poly");


    // Write out the result
    write = true;
    if ( return_rem ) {
        (uint64_t)flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
            rem.d, var_map, scale.d);
    }
    else {
        (uint64_t)flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size,
            div.d, var_map, scale.d);
    }

    return res;
}
/*
    #] flint::divmod_poly :

    #[ flint::factorize_mpoly :
*/
WORD* flint::factorize_mpoly(PHEAD const WORD *argin, WORD *argout, const bool with_arghead,
    const bool is_fun_arg, const var_map_t &var_map) {

```

```
    flint::mpoly_ctx ctx(var_map.size());
    flint::mpoly arg(ctx.d), den(ctx.d), base(ctx.d);

    flint::from_argument_mpoly(arg.d, den.d, argin, with_arghead, var_map, ctx.d);
    // The denominator must be 1:
    if ( fmpz_mpoly_is_one(den.d, ctx.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::factorize_mpoly error: den != 1");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }


    // Now we can factor the mpoly:
    flint::mpoly_factor arg_fac(ctx.d);
    fmpz_mpoly_factor(arg_fac.d, arg.d, ctx.d);
    const int64_t num_factors = fmpz_mpoly_factor_length(arg_fac.d, ctx.d);

    flint::fmpz overall_constant;
    fmpz_mpoly_factor_get_constant_fmpz(overall_constant.d, arg_fac.d, ctx.d);
    // FORM should always have taken the overall constant out in the content. Thus this overall
    // constant factor should be +- 1 here. Verify this:
    if ( ! ( fmpz_equal_si(overall_constant.d, 1) || fmpz_equal_si(overall_constant.d, -1) ) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::factorize_mpoly error: overall constant factor != +-1");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Construct the output. If argout is not NULL, we write the result there.
    // Otherwise, allocate memory.
    // The output is zero-terminated list of factors. If with_arghead, each has an arghead which
    // contains its size. Otherwise, the factors are zero separated.

    // We only need to determine the size of the output if we are allocating memory, but we need to
    // loop through the factors to fix their signs anyway. Do both together in one loop:

    // Initially 1, for the final trailing 0.
    uint64_t output_size = 1;

    // For finding the highest symbol, in FORM's lexicographic ordering
    var_map_t var_map_inv;
    for ( auto x: var_map ) {
        var_map_inv[x.second] = x.first;
    }

    // Store whether we should flip the factor sign in the ouput:
    vector<int32_t> base_sign(num_factors, 0);

    for ( int64_t i = 0; i < num_factors; i++ ) {
        fmpz_mpoly_factor_get_base(base.d, arg_fac.d, i, ctx.d);
        const int64_t exponent = fmpz_mpoly_factor_get_exp_si(arg_fac.d, i, ctx.d);

        // poly_factorize makes sure the highest power of the "highest symbol" (in FORM's
        // lexicographic ordering) has a positive coefficient. Check this, update overall_constant
        // of the factorization if necessary.
        // Store the sign per factor, so that we can flip the signs in the output without re-checking
        // the individual terms again.
        uint32_t max_var = 0; // FORM symbols start at 20, 0 is a good initial value.
        int32_t max_pow = -1;
        vector<int64_t> base_term_exponents(var_map.size(), 0);

        for ( int64_t j = 0; j < fmpz_mpoly_length(base.d, ctx.d); j++ ) {
            fmpz_mpoly_get_term_exp_si((slong*)base_term_exponents.data(), base.d, (slong)j, ctx.d);

            for ( size_t k = 0; k < var_map.size(); k++ ) {
                if ( base_term_exponents[k] > 0 && ( var_map_inv.at(k) > max_var ||
                    ( var_map_inv.at(k) == max_var && base_term_exponents[k] > max_pow ) ) ) {

                    max_var = var_map_inv.at(k);
                    max_pow = base_term_exponents[k];
                    base_sign[i] = fmpz_sgn(fmpz_mpoly_term_coeff_ref(base.d, j, ctx.d));
                }
            }
        }
        // If this base's sign will be flipped an odd number of times, there is a contribution to
        // the overall sign of the whole factorization:
        if ( ( base_sign[i] == -1 ) && ( exponent % 2 == 1 ) ) {
            fmpz_neg(overall_constant.d, overall_constant.d);
        }

        // Now determine the output size of the factor, if we are allocating the memory
        if ( argout == NULL ) {
            const bool write = false;
            for ( int64_t j = 0; j < exponent; j++ ) {
                output_size += (uint64_t)flint::to_argument_mpoly(BHEAD NULL, with_arghead,
                    is_fun_arg, write, 0, base.d, var_map, ctx.d);
```

```
            }
        }
    }
    if ( fmpz_sgn(overall_constant.d) == -1 && argout == NULL ) {
        // Add space for a fast-notation number or a normal-notation number and zero separator
        output_size += with_arghead ? 2 : 4+1;
    }

    // Now make the allocation if necessary:
    if ( argout == NULL ) {
        argout = (WORD*)Malloc1(sizeof(WORD)*output_size, "flint::factorize_mpoly");
    }


    // And now comes the actual output:
    WORD* old_argout = argout;

    // If the overall sign is negative, first write a full-notation -1. It will be absorbed into the
    // overall factor in the content by the caller.
    if ( fmpz_sgn(overall_constant.d) == -1 ) {
        if ( with_arghead ) {
            // poly writes in fast notation in this case. Fast notation is expected by the caller, to
            // properly merge it with the overall factor of the content.
            *argout++ = -SNUMBER;
            *argout++ = -1;
        }
        else {
            *argout++ = 4; // term size
            *argout++ = 1; // numerator
            *argout++ = 1; // denominator
            *argout++ = -3; // coeff size, negative number
            *argout++ = 0; // factor separator
        }
    }

    for ( int64_t i = 0; i < num_factors; i++ ) {
        fmpz_mpoly_factor_get_base(base.d, arg_fac.d, i, ctx.d);
        const int64_t exponent = fmpz_mpoly_factor_get_exp_si(arg_fac.d, i, ctx.d);

        if ( base_sign[i] == -1 ) {
            fmpz_mpoly_neg(base.d, base.d, ctx.d);
        }

        const bool write = true;
        for ( int64_t j = 0; j < exponent; j++ ) {
            argout += flint::to_argument_mpoly(BHEAD argout, with_arghead, is_fun_arg, write,
                argout-old_argout, base.d, var_map, ctx.d);
        }
    }
    // Final trailing zero to denote the end of the factors.
    *argout++ = 0;

    return old_argout;
}
/*
    #] flint::factorize_mpoly :
    #[ flint::factorize_poly :
*/
WORD* flint::factorize_poly(PHEAD const WORD *argin, WORD *argout, const bool with_arghead,
    const bool is_fun_arg, const var_map_t &var_map) {

    flint::poly arg, den;

    flint::from_argument_poly(arg.d, den.d, argin, with_arghead);
    // The denominator must be 1:
    if ( fmpz_poly_is_one(den.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::factorize_poly error: den != 1");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }


    // Now we can factor the poly:
    flint::poly_factor arg_fac;
    fmpz_poly_factor(arg_fac.d, arg.d);
    // fmpz_poly_factor_t lacks some convenience functions which fmpz_mpoly_factor_t has.
    // I have worked out how to get the factors by looking at how fmpz_poly_factor_print works.
    const long num_factors = (arg_fac.d)->num;


    // Construct the output. If argout is not NULL, we write the result there.
    // Otherwise, allocate memory.
    // The output is zero-terminated list of factors. If with_arghead, each has an arghead which
    // contains its size. Otherwise, the factors are zero separated.
    if ( argout == NULL ) {
        // First we need to determine the size of the output. This is the same procedure as the
```

```
        // loop below, but we don't write anything in to_argument_poly (write arg: false).
        // Initially 1, for the final trailing 0.
        uint64_t output_size = 1;

        for ( long i = 0; i < num_factors; i++ ) {
            fmpz_poly_struct* base = (arg_fac.d)->p + i;

            const long exponent = (arg_fac.d)->exp[i];

            const bool write = false;
            for ( long j = 0; j < exponent; j++ ) {
                output_size += (uint64_t)flint::to_argument_poly(BHEAD NULL, with_arghead,
                    is_fun_arg, write, 0, base, var_map);
            }
        }

        argout = (WORD*)Malloc1(sizeof(WORD)*output_size, "flint::factorize_poly");
    }

    WORD* old_argout = argout;

    for ( long i = 0; i < num_factors; i++ ) {
        fmpz_poly_struct* base = (arg_fac.d)->p + i;

        const long exponent = (arg_fac.d)->exp[i];

        const bool write = true;
        for ( long j = 0; j < exponent; j++ ) {
            argout += flint::to_argument_poly(BHEAD argout, with_arghead, is_fun_arg, write,
                argout-old_argout, base, var_map);
        }
    }
    *argout = 0;

    return old_argout;
}
/*
    #] flint::factorize_poly :

    #[ flint::form_sort :
*/
// Sort terms using form's sorting routines. Uses a custom (faster) compare routine, since here
// only symbols can appear.
// This is a modified poly_sort from polywrap.cc.
void flint::form_sort(PHEAD WORD *terms) {

    if ( terms[0] < 0 ) {
        // Fast notation, there is nothing to do
        return;
    }

    const WORD oldsorttype = AR.SortType;
    AR.SortType = SORTHIGHFIRST;

    const WORD in_size = terms[0] - ARGHEAD;
    WORD out_size;

    if ( NewSort(BHEAD0) ) {
        Terminate(-1);
    }
    AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);

    // Make sure the symbols are in the right order within the terms
    for ( WORD i = ARGHEAD; i < terms[0]; i += terms[i] ) {
        if ( SymbolNormalize(terms+i) < 0 || StoreTerm(BHEAD terms+i) ) {
            AR.SortType = oldsorttype;
            AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
            LowerSortLevel();
            Terminate(-1);
        }
    }

    if ( ( out_size = EndSort(BHEAD terms+ARGHEAD, 1) ) < 0 ) {
        AR.SortType = oldsorttype;
        AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
        Terminate(-1);
    }

    // Check the final size
    if ( in_size != out_size  ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::form_sort: error: unexpected sorted arg length change %d->%d", in_size,
            out_size);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

```

```
    AR.SortType = oldsorttype;
    AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
    terms[1] = 0; // set dirty flag to zero
}
/*
    #] flint::form_sort :

    #[ flint::from_argument_mpoly :
*/
// Convert a FORM argument (or 0-terminated list of terms: with_arghead == false) to a
// (multi-variate) fmpz_mpoly_t poly. The "denominator" is return in denpoly, and contains the
// overall negative-power numeric and symbolic factor.
uint64_t flint::from_argument_mpoly(fmpz_mpoly_t poly, fmpz_mpoly_t denpoly, const WORD *args,
    const bool with_arghead, const var_map_t &var_map, const fmpz_mpoly_ctx_t ctx) {

    // Some callers re-use their poly, denpoly to avoid calling init/clear unnecessarily.
    // Make sure they are 0 to begin.
    fmpz_mpoly_set_si(poly, 0, ctx);
    fmpz_mpoly_set_si(denpoly, 0, ctx);

    // First check for "fast notation" arguments:
    if ( *args == -SNUMBER ) {
        fmpz_mpoly_set_si(poly, *(args+1), ctx);
        fmpz_mpoly_set_si(denpoly, 1, ctx);
        return 2;
    }

    if ( *args == -SYMBOL ) {
        // A "fast notation" SYMBOL has a power and coefficient of 1:
        vector<uint64_t> exponents(var_map.size(), 0);
        exponents[var_map.at(*(args+1))] = 1;

        fmpz_mpoly_set_coeff_ui_ui(poly, (ulong)1, (ulong*)exponents.data(), ctx);
        fmpz_mpoly_set_ui(denpoly, (ulong)1, ctx);
        return 2;
    }


    // Now we can iterate through the terms of the argument. If we have
    // an ARGHEAD, we already know where to terminate. Otherwise we'll have
    // to loop until the terminating 0.
    const WORD* arg_stop = with_arghead ? args+args[0] : (WORD*)UINT64_MAX;
    uint64_t arg_size = 0;
    if ( with_arghead ) {
        arg_size = args[0];
        args += ARGHEAD;
    }


    // Search for numerical or symbol denominators to create "denpoly".
    flint::fmpz den_coeff, tmp;
    fmpz_set_si(den_coeff.d, 1);
    vector<uint64_t> neg_exponents(var_map.size(), 0);

    for ( const WORD* term = args; term < arg_stop; term += term[0] ) {
        const WORD* term_stop = term+term[0];
        const WORD  coeff_size = (term_stop)[-1];
        const WORD* symbol_stop = term_stop - ABS(coeff_size);
        const WORD* t = term;

        t++;
        if ( t == symbol_stop ) {
            // Just a number, no symbols
        }
        else {
            t++; // this entry is SYMBOL
            t++; // this entry just has the size of the symbol array, but we can use symbol_stop

            for ( const WORD* s = t; s < symbol_stop; s += 2 ) {
                if ( *(s+1) < 0 ) {
                    neg_exponents[var_map.at(*s)] =
                        MaX(neg_exponents[var_map.at(*s)], (uint64_t)(-(*(s+1))) );
                }
            }
        }

        // Now check for a denominator in the coefficient:
        if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
            flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
            // Record the LCM of the coefficient denominators:
            fmpz_lcm(den_coeff.d, den_coeff.d, tmp.d);
        }

        if ( !with_arghead && *term_stop == 0 ) {
            // + 1 for the terminating 0
            arg_size = term_stop - args + 1;
            break;
```

```
        }

    }
    // Assemble denpoly.
    fmpz_mpoly_set_coeff_fmpz_ui(denpoly, den_coeff.d, (ulong*)neg_exponents.data(), ctx);


    // For the term coefficients
    flint::fmpz coeff;

    for ( const WORD* term = args; term < arg_stop; term += term[0] ) {
        const WORD* term_stop = term+term[0];
        const WORD  coeff_size = (term_stop)[-1];
        const WORD* symbol_stop = term_stop - ABS(coeff_size);
        const WORD* t = term;

        vector<uint64_t> exponents(var_map.size(), 0);

        t++; // skip over the total size entry
        if ( t == symbol_stop ) {
            // Just a number, no symbols
        }
        else {
            t++; // this entry is SYMBOL
            t++; // this entry just has the size of the symbol array, but we can use symbol_stop
            for ( const WORD* s = t; s < symbol_stop; s += 2 ) {
                exponents[var_map.at(*s)] = *(s+1);
            }
        }
        // Now read the coefficient
        flint::fmpz_set_form(coeff.d, (UWORD*)symbol_stop, coeff_size/2);

        // Multiply by denominator LCM
        fmpz_mul(coeff.d, coeff.d, den_coeff.d);

        // Shift by neg_exponents
        for ( size_t i = 0; i < var_map.size(); i++ ) {
            exponents[i] += neg_exponents[i];
        }

        // Read the denominator if there is one, and divide it out of the coefficient
        if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
            flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
            // By construction, this is an exact division
            fmpz_divexact(coeff.d, coeff.d, tmp.d);
        }

        // Push the term to the mpoly, remember to sort when finished! This is much faster than using
        // fmpz_mpoly_set_coeff_fmpz_ui when the terms arrive in the "wrong order".
        fmpz_mpoly_push_term_fmpz_ui(poly, coeff.d, (ulong*)exponents.data(), ctx);

        if ( !with_arghead && *term_stop == 0 ) {
            break;
        }

    }
    // And now sort the mpoly
    fmpz_mpoly_sort_terms(poly, ctx);

    return arg_size;
}
/*
    #] flint::from_argument_mpoly :
    #[ flint::from_argument_poly :
*/
// Convert a FORM argument (or 0-terminated list of terms: with_arghead == false) to a
// (uni-variate) fmpz_poly_t poly. The "denominator" is return in denpoly, and contains the
// overall negative-power numeric and symbolic factor.
uint64_t flint::from_argument_poly(fmpz_poly_t poly, fmpz_poly_t denpoly, const WORD *args,
    const bool with_arghead) {

    // Some callers re-use their poly, denpoly to avoid calling init/clear unnecessarily.
    // Make sure they are 0 to begin.
    fmpz_poly_set_si(poly, 0);
    fmpz_poly_set_si(denpoly, 0);

    // First check for "fast notation" arguments:
    if ( *args == -SNUMBER ) {
        fmpz_poly_set_si(poly, *(args+1));
        fmpz_poly_set_si(denpoly, 1);
        return 2;
    }

    if ( *args == -SYMBOL ) {
        // A "fast notation" SYMBOL has a power and coefficient of 1:
        fmpz_poly_set_coeff_si(poly, 1, 1);
        fmpz_poly_set_si(denpoly, 1);
```

```
        return 2;
    }

    // Now we can iterate through the terms of the argument. If we have
    // an ARGHEAD, we already know where to terminate. Otherwise we'll have
    // to loop until the terminating 0.
    const WORD* arg_stop = with_arghead ? args+args[0] : (WORD*)UINT64_MAX;
    uint64_t arg_size = 0;
    if ( with_arghead ) {
        arg_size = args[0];
        args += ARGHEAD;
    }

    // Search for numerical or symbol denominators to create "denpoly".
    flint::fmpz den_coeff, tmp;
    fmpz_set_si(den_coeff.d, 1);
    uint64_t neg_exponent = 0;

    for ( const WORD* term = args; term < arg_stop; term += term[0] ) {

        const WORD* term_stop = term+term[0];
        const WORD  coeff_size = (term_stop)[-1];
        const WORD* symbol_stop = term_stop - ABS(coeff_size);
        const WORD* t = term;

        t++; // skip over the total size entry
        if ( t == symbol_stop ) {
            // Just a number, no symbols
        }
        else {
            t++; // this entry is SYMBOL
            t++; // this entry is the size of the symbol array
            t++; // this is the first (and only) symbol code
            if ( *t < 0 ) {
                neg_exponent = MaX(neg_exponent, (uint64_t)(-(*t)) );
            }
        }

        // Now check for a denominator in the coefficient:
        if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
            flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
            // Record the LCM of the coefficient denominators:
            fmpz_lcm(den_coeff.d, den_coeff.d, tmp.d);
        }

        if ( *term_stop == 0 ) {
            // + 1 for the terminating 0
            arg_size = term_stop - args + 1;
            break;
        }
    }
    // Assemble denpoly.
    fmpz_poly_set_coeff_fmpz(denpoly, neg_exponent, den_coeff.d);


    // For the term coefficients
    flint::fmpz coeff;

    for ( const WORD* term = args; term < arg_stop; term += term[0] ) {

        const WORD* term_stop = term+term[0];
        const WORD  coeff_size = (term_stop)[-1];
        const WORD* symbol_stop = term_stop - ABS(coeff_size);
        const WORD* t = term;

        uint64_t exponent = 0;

        t++; // skip over the total size entry
        if ( t == symbol_stop ) {
            // Just a number, no symbols
        }
        else {
            t++; // this entry is SYMBOL
            t++; // this entry is the size of the symbol array
            t++; // this is the first (and only) symbol code
            exponent = *t++;
        }

        // Now read the coefficient
        flint::fmpz_set_form(coeff.d, (UWORD*)symbol_stop, coeff_size/2);

        // Multiply by denominator LCM
        fmpz_mul(coeff.d, coeff.d, den_coeff.d);

        // Shift by neg_exponent
        exponent += neg_exponent;

```

```
        // Read the denominator if there is one, and divide it out of the coefficient
        if ( *(symbol_stop+ABS(coeff_size/2)) != 1 ) {
            flint::fmpz_set_form(tmp.d, (UWORD*)(symbol_stop+ABS(coeff_size/2)), ABS(coeff_size/2));
            // By construction, this is an exact division
            fmpz_divexact(coeff.d, coeff.d, tmp.d);
        }

        // Add the term to the poly
        fmpz_poly_set_coeff_fmpz(poly, exponent, coeff.d);

        if ( *term_stop == 0 ) {
            break;
        }
    }

    return arg_size;
}
/*
    #] flint::from_argument_poly :

    #[ flint::fmpz_get_form :
*/
// Write FORM's long integer representation of an fmpz at a, and put the number of WORDs at na.
// na carries the sign of the integer.
WORD flint::fmpz_get_form(fmpz_t z, WORD *a) {

    WORD na = 0;
    const int32_t sgn = fmpz_sgn(z);
    if ( sgn == -1 ) {
        fmpz_neg(z, z);
    }
    const int64_t nlimbs = fmpz_size(z);

    // This works but is UB?
    //fmpz_get_ui_array(reinterpret_cast<uint64_t*>(a), nlimbs, z);

    // Use fixed-size functions to get limb data where possible. These probably cover most real
    // cases.
    if ( nlimbs == 1 ) {
        const uint64_t limb = fmpz_get_ui(z);
        a[0] = (WORD)(limb & 0xFFFFFFFF);
        na++;
        a[1] = (WORD)(limb » BITSINWORD);
        if ( a[1] != 0 ) {
            na++;
        }
    }
    else if ( nlimbs == 2 ) {
        uint64_t limb_hi = 0, limb_lo = 0;
        fmpz_get_uiui((ulong*)&limb_hi, (ulong*)&limb_lo, z);
        a[0] = (WORD)(limb_lo & 0xFFFFFFFF);
            na++;
        a[1] = (WORD)(limb_lo » BITSINWORD);
            na++;
        a[2] = (WORD)(limb_hi & 0xFFFFFFFF);
            na++;
        a[3] = (WORD)(limb_hi » BITSINWORD);
        if ( a[3] != 0 ) {
            na++;
        }
    }
    else {
        vector<uint64_t> limb_data(nlimbs, 0);
        fmpz_get_ui_array((ulong*)limb_data.data(), (slong)nlimbs, z);
        for ( long i = 0; i < nlimbs; i++ ) {
            a[2*i] = (WORD)(limb_data[i] & 0xFFFFFFFF);
            na++;
            a[2*i+1] = (WORD)(limb_data[i] » BITSINWORD);
            if ( a[2*i+1] != 0 || i < (nlimbs-1) ) {
                // The final limb might fit in a single 32bit WORD. Only
                // increment na if the final WORD is non zero.
                na++;
            }
        }
    }

    // And now put the sign in the number of limbs
    if ( sgn == -1 ) {
        na = -na;
    }

    return na;
}
/*
    #] flint::fmpz_get_form :
    #[ flint::fmpz_set_form :
*/
```

```
// Create an fmpz directly from FORM's long integer representation. fmpz uses 64bit unsigned limbs,
// but FORM uses 32bit UWORDs on 64bit architectures so we can't use fmpz_set_ui_array directly.
void flint::fmpz_set_form(fmpz_t z, UWORD *a, WORD na) {

    if ( na == 0 ) {
        fmpz_zero(z);
        return;
    }

    // Negative na represenents a negative number
    int32_t sgn = 1;
    if ( na < 0 ) {
        sgn = -1;
        na = -na;
    }

    // Remove padding. FORM stores numerators and denominators with equal numbers of limbs but we
    // don't need to do this within the fmpz. It is not necessary to do this really, the fmpz
    // doesn't add zero limbs unnecessarily, but we might be able to avoid creating the limb_data
    // array below.
    while ( a[na-1] == 0 ) {
        na--;
    }

    // If the number fits in fixed-size fmpz_set functions, we don't need to use additional memory
    // to convert to uint64_t. These probably cover most real cases.
    if ( na == 1 ) {
        fmpz_set_ui(z, (uint64_t)a[0]);
    }
    else if ( na == 2 ) {
        fmpz_set_ui(z, (((uint64_t)a[1])«BITSINWORD) + (uint64_t)a[0]);
    }
    else if ( na == 3 ) {
        fmpz_set_uiui(z, (uint64_t)a[2], (((uint64_t)a[1])«BITSINWORD) + (uint64_t)a[0]);
    }
    else if ( na == 4 ) {
        fmpz_set_uiui(z, (((uint64_t)a[3])«BITSINWORD) + (uint64_t)a[2],
            (((uint64_t)a[1])«BITSINWORD) + (uint64_t)a[0]);
    }
    else {
        const int32_t nlimbs = (na+1)/2;
        vector<uint64_t> limb_data(nlimbs, 0);
        for ( int32_t i = 0; i < nlimbs; i++ ) {
            if ( 2*i+1 <= na-1 ) {
                limb_data[i] = (uint64_t)a[2*i] + (((uint64_t)a[2*i+1])«BITSINWORD);
            }
            else {
                limb_data[i] = (uint64_t)a[2*i];
            }
        }
        fmpz_set_ui_array(z, (ulong*)limb_data.data(), nlimbs);
    }

    // Finally set the sign.
    if ( sgn == -1 ) {
        fmpz_neg(z, z);
    }

    return;
}
/*
    #] flint::fmpz_set_form :

    #[ flint::gcd_mpoly :
*/
// Return a pointer to a buffer containing the GCD of the 0-terminated term lists at a and b.
// If must_fit_term, this should be a TermMalloc buffer. Otherwise Malloc1 the buffer.
// For multi-variate cases.
WORD* flint::gcd_mpoly(PHEAD const WORD *a, const WORD *b, const WORD must_fit_term,
    const var_map_t &var_map) {

    flint::mpoly_ctx ctx(var_map.size());
    flint::mpoly pa(ctx.d), pb(ctx.d), denpa(ctx.d), denpb(ctx.d), gcd(ctx.d);

    flint::from_argument_mpoly(pa.d, denpa.d, a, false, var_map, ctx.d);
    flint::from_argument_mpoly(pb.d, denpb.d, b, false, var_map, ctx.d);

    // denpa, denpb must be 1:
    if ( fmpz_mpoly_is_one(denpa.d, ctx.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::gcd_mpoly: error: denpa != 1");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( fmpz_mpoly_is_one(denpb.d, ctx.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::gcd_mpoly: error: denpb != 1");
```

```
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // poly returns pa if pa == pb, regardless of the lcoeff sign
    if ( fmpz_mpoly_equal(pa.d, pb.d, ctx.d) ) {
        fmpz_mpoly_set(gcd.d, pa.d, ctx.d);
    }
    else {
        // We need some gymnastics to have the same sign conventions as the poly class. It takes the
        // integer or univar content out of a,b, with the convention that the content sign matches
        // the lcoeff sign. Since FORM has already taken out the content, we are left with +-1. In
        // Flint, the content always has a positive sign so here we should find +1. Check this:
        flint::mpoly tmp(ctx.d);
        fmpz_mpoly_term_content(tmp.d, pa.d, ctx.d);
        if ( fmpz_mpoly_is_one(tmp.d, ctx.d) != 1 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("flint::gcd_mpoly: error: content of 1st arg != 1");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        fmpz_mpoly_term_content(tmp.d, pb.d, ctx.d);
        if ( fmpz_mpoly_is_one(tmp.d, ctx.d) != 1 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("flint::gcd_mpoly: error: content of 2nd arg != 1");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }

        // The poly class now divides the content out of a,b so that they have a positive lcoeff.
        // Then it multiplies the final gcd (which is given a positive lcoeff also) by
        // gcd(cont a, cont b). There it has gcd(1,1) = gcd(-1,1) = gcd(1,-1) = 1, and
        // gcd(-1,-1) = -1 (because of the pa==pb early return). So: if both input polys have a
        // negative lcoeff, we will flip the sign in the final result.
        bool flip_sign = 0;
        if ( ( fmpz_sgn(fmpz_mpoly_term_coeff_ref(pa.d, 0, ctx.d)) == -1 ) &&
            ( fmpz_sgn(fmpz_mpoly_term_coeff_ref(pb.d, 0, ctx.d)) == -1 ) ) {
            flip_sign = 1;
        }

        fmpz_mpoly_gcd(gcd.d, pa.d, pb.d, ctx.d);
        if ( flip_sign ) {
            fmpz_mpoly_neg(gcd.d, gcd.d, ctx.d);
        }
    }

    // This is freed by the caller
    WORD *res;
    if ( must_fit_term ) {
        res = TermMalloc("flint::gcd_mpoly");
    }
    else {
        // Determine the size of the GCD by passing write = false.
        const bool with_arghead = false;
        const bool write = false;
        const uint64_t prev_size = 0;
        const uint64_t gcd_size = (uint64_t)flint::to_argument_mpoly(BHEAD NULL,
            with_arghead, must_fit_term, write, prev_size, gcd.d, var_map, ctx.d);

        res = (WORD *)Malloc1(sizeof(WORD)*gcd_size, "flint::gcd_mpoly");
    }

    const bool with_arghead = false;
    const bool write = true;
    const uint64_t prev_size = 0;
    flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size, gcd.d,
        var_map, ctx.d);

    return res;
}
/*
    #] flint::gcd_mpoly :
    #[ flint::gcd_poly :
*/
// Return a pointer to a buffer containing the GCD of the 0-terminated term lists at a and b.
// If must_fit_term, this should be a TermMalloc buffer. Otherwise Malloc1 the buffer.
// For uni-variate cases.
WORD* flint::gcd_poly(PHEAD const WORD *a, const WORD *b, const WORD must_fit_term,
    const var_map_t &var_map) {

    flint::poly pa, pb, denpa, denpb, gcd;

    flint::from_argument_poly(pa.d, denpa.d, a, false);
    flint::from_argument_poly(pb.d, denpb.d, b, false);

    // denpa, denpb must be 1:
    if ( fmpz_poly_is_one(denpa.d) != 1 ) {
```

```
        MLOCK(ErrorMessageLock);
        MesPrint("flint::gcd_poly: error: denpa != 1");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( fmpz_poly_is_one(denpb.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::gcd_poly: error: denpb != 1");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // poly returns pa if pa == pb, regardless of the lcoeff sign
    if ( fmpz_poly_equal(pa.d, pb.d) ) {
        fmpz_poly_set(gcd.d, pa.d);
    }
    else {
        // Here, we don't have to make any sign flips like the mpoly case, because poly's
        // integer_gcd(1,1) = integer_gcd(-1,1) = integer_gcd(1,-1) = integer_gcd(-1,-1) = +1.
        // Still, verify that the content is 1:
        flint::fmpz tmp;
        fmpz_poly_content(tmp.d, pa.d);
        if ( fmpz_is_one(tmp.d) != 1 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("flint::gcd_poly: error: content of 1st arg != 1");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        fmpz_poly_content(tmp.d, pb.d);
        if ( fmpz_is_one(tmp.d) != 1 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("flint::gcd_poly: error: content of 2nd arg != 1");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }

        fmpz_poly_gcd(gcd.d, pa.d, pb.d);
    }

    // This is freed by the caller
    WORD *res;
    if ( must_fit_term ) {
        res = TermMalloc("flint::gcd_poly");
    }
    else {
        // Determine the size of the GCD by passing write = false.
        const bool with_arghead = false;
        const bool write = false;
        const uint64_t prev_size = 0;
        const uint64_t gcd_size = (uint64_t)flint::to_argument_poly(BHEAD NULL,
            with_arghead, must_fit_term, write, prev_size, gcd.d, var_map);

        res = (WORD *)Malloc1(sizeof(WORD)*gcd_size, "flint::gcd_poly");
    }

    const bool with_arghead = false;
    const uint64_t prev_size = 0;
    const bool write = true;
    flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size, gcd.d,
        var_map);

    return res;
}
/*
    #] flint::gcd_poly :

    #[ flint::get_variables :
*/
// Get the list of symbols which appear in the vector of expressions. These are polyratfun
// numerators, denominators or expressions from calls to gcd_ etc. Return this list as a map
// between indices and symbol codes.
// TODO FACTORSYMBOL last?
flint::var_map_t flint::get_variables(const vector <WORD *> &es, const bool with_arghead,
    const bool sort_vars) {

    int32_t num_vars = 0;
    // To be used if we sort by highest degree, as the polu code does.
    vector<int32_t> degrees;
    var_map_t var_map;

    // extract all variables
    for ( size_t ei = 0; ei < es.size(); ei++ ) {
        WORD *e = es[ei];

        // fast notation
        if ( *e == -SNUMBER ) {
        }
```

```
        else if ( *e == -SYMBOL ) {
            if ( !var_map.count(e[1]) ) {
                var_map[e[1]] = num_vars++;
                degrees.push_back(1);
            }
        }
        // JD: Here we need to check for non-symbol/number terms in fast notation.
        else if ( *e < 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("ERROR: polynomials and polyratfuns must contain symbols only");
            MUNLOCK(ErrorMessageLock);
            Terminate(1);
        }
        else {
            for ( WORD i = with_arghead ? ARGHEAD:0; with_arghead ? i < e[0]:e[i] != 0; i += e[i] ) {
                if ( i+1 < i+e[i]-ABS(e[i+e[i]-1]) && e[i+1] != SYMBOL ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("ERROR: polynomials and polyratfuns must contain symbols only");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(1);
                }

                for ( WORD j = i+3; j<i+e[i]-ABS(e[i+e[i]-1]); j += 2 ) {
                    if ( !var_map.count(e[j]) ) {
                        var_map[e[j]] = num_vars++;
                        degrees.push_back(e[j+1]);
                    }
                    else {
                        degrees[var_map[e[j]]] = MaX(degrees[var_map[e[j]]], e[j+1]);
                    }
                }
            }
        }
    }

    if ( sort_vars ) {
        // bubble sort variables in decreasing order of degree
        // (this seems better for factorization)
        for ( size_t i = 0; i < var_map.size(); i++ ) {
            for ( size_t j = 0; j+1 < var_map.size(); j++ ) {
                if ( degrees[j] < degrees[j+1] ) {
                    swap(degrees[j], degrees[j+1]);

                    // Find the map keys associated with the values we want to swap
                    uint32_t j0 = 0;
                    uint32_t j1 = 0;
                    for ( auto x: var_map ) {
                        if ( x.second == j ) {
                            j0 = x.first;
                        }
                        else if ( x.second == j+1 ) {
                            j1 = x.first;
                        }
                    }
                    swap(var_map.at(j0), var_map.at(j1));
                }
            }
        }
    }
    // Otherwise, sort lexicographically in FORM's ordering
    else {
        for ( size_t i = 0; i < var_map.size(); i++ ) {
            for ( size_t j = 0; j+1 < var_map.size(); j++ ) {
                uint32_t j0 = 0;
                uint32_t j1 = 0;
                for ( auto x: var_map ) {
                    if ( x.second == j ) {
                        j0 = x.first;
                    }
                    else if ( x.second == j+1 ) {
                        j1 = x.first;
                    }
                }
                if ( j0 > j1 ) {
                    swap(var_map.at(j0), var_map.at(j1));
                }
            }
        }
    }

    return var_map;
}
/*
    #] flint::get_variables :

    #[ flint::inverse_poly :
*/
```

```
WORD* flint::inverse_poly(PHEAD const WORD *a, const WORD *b, const var_map_t &var_map) {

    flint::poly pa, pb, denpa, denpb;

    flint::from_argument_poly(pa.d, denpa.d, a, false);
    flint::from_argument_poly(pb.d, denpb.d, b, false);

    // fmpz_poly_xgcd is undefined if the content of pa and pb are not 1. Take the content out.
    // fmpz_poly_content gives the non-negative content and fmpz_poly_primitive normalizes to a
    // non-negative lcoeff, so we need to add the sign to the content if the polys have a negative
    // lcoeff. We don't need to keep the content of pb, it is a numerical multiple of the modulus.
    flint::fmpz content_a, resultant;
    flint::poly inverse, tmp;
    fmpz_poly_content(content_a.d, pa.d);
    if ( fmpz_sgn(fmpz_poly_lead(pa.d)) == -1 ) {
        fmpz_neg(content_a.d, content_a.d);
    }
    fmpz_poly_primitive_part(pa.d, pa.d);
    fmpz_poly_primitive_part(pb.d, pb.d);

    // Special cases:
    // Possibly strange that we give 1 for inverse_(x1,1) but here we take MMA's convention.
    if ( fmpz_poly_is_one(pa.d) && fmpz_poly_is_one(pb.d) ) {
        fmpz_poly_one(inverse.d);
        fmpz_one(resultant.d);
    }
    else if ( fmpz_poly_is_one(pb.d) ) {
        fmpz_poly_zero(inverse.d);
        fmpz_one(resultant.d);
    }
    else {
        // Now use the extended Euclidean algorithm to find inverse, resultant, tmp of the Bezout
        // identity: inverse*pa + tmp*pb = resultant. Then inverse/resultant is the multiplicative
        // inverse of pa mod pb. We'll divide by resultant in the denscale argument of to_argument_poly.
        fmpz_poly_xgcd(resultant.d, inverse.d, tmp.d, pa.d, pb.d);

        // If the resultant is zero, the inverse does not exist:
        if ( fmpz_is_zero(resultant.d) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("flint::inverse_poly error: inverse does not exist");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    }

    // Multiply inverse by denpa. denpb is a numerical multiple of the modulus, so doesn't matter.
    // We also need to divide by content_a, which we do in the denscale argument of to_argument_poly
    // by multiplying resultant by content_a here.
    fmpz_poly_mul(inverse.d, inverse.d, denpa.d);
    fmpz_mul(resultant.d, resultant.d, content_a.d);


    WORD* res;
    // First determine the result size, and malloc. The result should have no arghead. Here we use
    // the "scale" argument of to_argument_poly since resultant might not be 1.
    const bool with_arghead = false;
    bool write = false;
    const bool must_fit_term = false;
    const uint64_t prev_size = 0;
    const uint64_t res_size = (uint64_t)flint::to_argument_poly(BHEAD NULL,
        with_arghead, must_fit_term, write, prev_size, inverse.d, var_map, resultant.d);
    res = (WORD*)Malloc1(sizeof(WORD)*res_size, "flint::inverse_poly");

    write = true;
    flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size, inverse.d,
        var_map, resultant.d);

    return res;
}
/*
    #] flint::inverse_poly :

    #[ flint::mul_mpoly :
*/
// Return a pointer to a buffer containing the product of the 0-terminated term lists at a and b.
// For multi-variate cases.
WORD* flint::mul_mpoly(PHEAD const WORD *a, const WORD *b, const var_map_t &var_map) {

    flint::mpoly_ctx ctx(var_map.size());
    flint::mpoly pa(ctx.d), pb(ctx.d), denpa(ctx.d), denpb(ctx.d);

    flint::from_argument_mpoly(pa.d, denpa.d, a, false, var_map, ctx.d);
    flint::from_argument_mpoly(pb.d, denpb.d, b, false, var_map, ctx.d);

    // denpa, denpb must be integers. Negative symbol powers have been converted to extra symbols.
    if ( fmpz_mpoly_is_fmpz(denpa.d, ctx.d) != 1 ) {
```

```
        MLOCK(ErrorMessageLock);
        MesPrint("flint::mul_mpoly: error: denpa is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( fmpz_mpoly_is_fmpz(denpb.d, ctx.d) != 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::mul_mpoly: error: denpb is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Multiply numerators, store result in pa
    fmpz_mpoly_mul(pa.d, pa.d, pb.d, ctx.d);
    // Multiply denominators, store result in denpa, and convert to an fmpz:
    fmpz_mpoly_mul(denpa.d, denpa.d, denpb.d, ctx.d);
    flint::fmpz den;
    fmpz_mpoly_get_fmpz(den.d, denpa.d, ctx.d);

    WORD* res;
    // First determine the result size, and malloc. The result should have no arghead. Here we use
    // the "scale" argument of to_argument_mpoly since den might not be 1.
    const bool with_arghead = false;
    bool write = false;
    const bool must_fit_term = false;
    const uint64_t prev_size = 0;
    const uint64_t mul_size = (uint64_t)flint::to_argument_mpoly(BHEAD NULL,
        with_arghead, must_fit_term, write, prev_size, pa.d, var_map, ctx.d, den.d);
    res = (WORD*)Malloc1(sizeof(WORD)*mul_size, "flint::mul_mpoly");

    write = true;
    flint::to_argument_mpoly(BHEAD res, with_arghead, must_fit_term, write, prev_size, pa.d,
        var_map, ctx.d, den.d);

    return res;
}
/*
    #] flint::mul_mpoly :
    #[ flint::mul_poly :
*/
// Return a pointer to a buffer containing the product of the 0-terminated term lists at a and b.
// For uni-variate cases.
WORD* flint::mul_poly(PHEAD const WORD *a, const WORD *b, const var_map_t &var_map) {

    flint::poly pa, pb, denpa, denpb;

    flint::from_argument_poly(pa.d, denpa.d, a, false);
    flint::from_argument_poly(pb.d, denpb.d, b, false);

    // denpa, denpb must be integers. Negative symbol powers have been converted to extra symbols.
    if ( fmpz_poly_degree(denpa.d) != 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::mul_poly: error: denpa is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( fmpz_poly_degree(denpb.d) != 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::mul_poly: error: denpb is non-constant");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Multiply numerators, store result in pa
    fmpz_poly_mul(pa.d, pa.d, pb.d);
    // Multiply denominators, store result in denpa, and convert to an fmpz:
    fmpz_poly_mul(denpa.d, denpa.d, denpb.d);
    flint::fmpz den;
    fmpz_poly_get_coeff_fmpz(den.d, denpa.d, 0);

    WORD* res;
    // First determine the result size, and malloc. The result should have no arghead. Here we use
    // the "scale" argument of to_argument_poly since den might not be 1.
    const bool with_arghead = false;
    bool write = false;
    const bool must_fit_term = false;
    const uint64_t prev_size = 0;
    const uint64_t mul_size = (uint64_t)flint::to_argument_poly(BHEAD NULL,
        with_arghead, must_fit_term, write, prev_size, pa.d, var_map, den.d);
    res = (WORD*)Malloc1(sizeof(WORD)*mul_size, "flint::mul_poly");

    write = true;
    flint::to_argument_poly(BHEAD res, with_arghead, must_fit_term, write, prev_size, pa.d,
        var_map, den.d);

    return res;
}
```

```
/*
    #] flint::mul_poly :

    #[ flint::ratfun_add_mpoly :
*/
// Add the multi-variate FORM rational polynomials at t1 and t2. The result is written at out.
void flint::ratfun_add_mpoly(PHEAD const WORD *t1, const WORD *t2, WORD *out,
    const var_map_t &var_map) {

    flint::mpoly_ctx ctx(var_map.size());
    flint::mpoly gcd(ctx.d), num1(ctx.d), den1(ctx.d), num2(ctx.d), den2(ctx.d);

    flint::ratfun_read_mpoly(t1, num1.d, den1.d, var_map, ctx.d);
    flint::ratfun_read_mpoly(t2, num2.d, den2.d, var_map, ctx.d);

    if ( fmpz_mpoly_cmp(den1.d, den2.d, ctx.d) != 0 ) {
        fmpz_mpoly_gcd_cofactors(gcd.d, den1.d, den2.d, den1.d, den2.d, ctx.d);

        fmpz_mpoly_mul(num1.d, num1.d, den2.d, ctx.d);
        fmpz_mpoly_mul(num2.d, num2.d, den1.d, ctx.d);

        fmpz_mpoly_add(num1.d, num1.d, num2.d, ctx.d);
        fmpz_mpoly_mul(den1.d, den1.d, den2.d, ctx.d);
        fmpz_mpoly_mul(den1.d, den1.d, gcd.d,  ctx.d);
    }
    else {
        fmpz_mpoly_add(num1.d, num1.d, num2.d, ctx.d);
    }

    // Finally divide out any common factors between the resulting num1, den1:
    fmpz_mpoly_gcd_cofactors(gcd.d, num1.d, den1.d, num1.d, den1.d, ctx.d);

    flint::util::fix_sign_fmpz_mpoly_ratfun(num1.d, den1.d, ctx.d);

    // Result in FORM notation:
    *out++ = AR.PolyFun;
    WORD* args_size = out++;
    WORD* args_flag = out++;
    *args_flag = 0; // clean prf
    FILLFUN3(out); // Remainder of funhead, if it is larger than 3

    const bool with_arghead = true;
    const bool must_fit_term = true;
    const bool write = true;
    // prev_size + 4, to account for final term size and coeff of "1/1"
    out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
        num1.d, var_map, ctx.d);
    out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
        den1.d, var_map, ctx.d);

    *args_size = out - args_size + 1; // The +1 is to include the function ID
    AT.WorkPointer = out;
}
/*
    #] flint::ratfun_add_mpoly :
    #[ flint::ratfun_add_poly :
*/
// Add the uni-variate FORM rational polynomials at t1 and t2. The result is written at out.
void flint::ratfun_add_poly(PHEAD const WORD *t1, const WORD *t2, WORD *out,
    const var_map_t &var_map) {

    flint::poly gcd, num1, den1, num2, den2;

    flint::ratfun_read_poly(t1, num1.d, den1.d);
    flint::ratfun_read_poly(t2, num2.d, den2.d);

    if ( fmpz_poly_equal(den1.d, den2.d) == 0 ) {
        flint::util::simplify_fmpz_poly(den1.d, den2.d, gcd.d);

        fmpz_poly_mul(num1.d, num1.d, den2.d);
        fmpz_poly_mul(num2.d, num2.d, den1.d);

        fmpz_poly_add(num1.d, num1.d, num2.d);
        fmpz_poly_mul(den1.d, den1.d, den2.d);
        fmpz_poly_mul(den1.d, den1.d, gcd.d);
    }
    else {
        fmpz_poly_add(num1.d, num1.d, num2.d);
    }

    // Finally divide out any common factors between the resulting num1, den1:
    flint::util::simplify_fmpz_poly(num1.d, den1.d, gcd.d);

    flint::util::fix_sign_fmpz_poly_ratfun(num1.d, den1.d);

    // Result in FORM notation:
    *out++ = AR.PolyFun;
```

```
    WORD* args_size = out++;
    WORD* args_flag = out++;
    *args_flag = 0; // clean prf
    FILLFUN3(out); // Remainder of funhead, if it is larger than 3

    const bool with_arghead = true;
    const bool must_fit_term = true;
    const bool write = true;
    // prev_size + 4, to account for final term size and coeff of "1/1"
    out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
        num1.d, var_map);
    out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-args_size+4,
        den1.d, var_map);

    *args_size = out - args_size + 1; // The +1 is to include the function ID
    AT.WorkPointer = out;
}
/*
    #] flint::ratfun_add_poly :

    #[ flint::ratfun_normalize_mpoly :
*/
// Multiply and simplify occurrences of the multi-variate FORM rational polynomials found in term.
// The final term is written in place, with the rational polynomial at the end.
void flint::ratfun_normalize_mpoly(PHEAD WORD *term, const var_map_t &var_map) {

    // The length of the coefficient
    const WORD ncoeff = (term + *term)[-1];
    // The end of the term data, before the coefficient:
    const WORD *tstop = term + *term - ABS(ncoeff);

    flint::mpoly_ctx ctx(var_map.size());
    flint::mpoly num1(ctx.d), den1(ctx.d), num2(ctx.d), den2(ctx.d), gcd(ctx.d);

    // Start with "trivial" polynomials, and multiply in the num and den of the prf which appear.
    flint::fmpz tmpNum, tmpDen;
    flint::fmpz_set_form(tmpNum.d, (UWORD*)tstop, ncoeff/2);
    flint::fmpz_set_form(tmpDen.d, (UWORD*)tstop+ABS(ncoeff/2), ABS(ncoeff/2));
    fmpz_mpoly_set_fmpz(num1.d, tmpNum.d, ctx.d);
    fmpz_mpoly_set_fmpz(den1.d, tmpDen.d, ctx.d);

    // Loop over the occurrences of PolyFun in the term, and multiply in to num1, den1.
    // s tracks where we are writing the non-PolyFun term data. The final PolyFun will
    // go at the end.
    WORD* term_size = term;
    WORD* s = term + 1;
    for ( WORD *t = term + 1; t < tstop; ) {
        if ( *t == AR.PolyFun ) {
            flint::ratfun_read_mpoly(t, num2.d, den2.d, var_map, ctx.d);

            // get gcd of num1,den2 and num2,den1 and then assemble
            fmpz_mpoly_gcd_cofactors(gcd.d, num1.d, den2.d, num1.d, den2.d, ctx.d);
            fmpz_mpoly_gcd_cofactors(gcd.d, num2.d, den1.d, num2.d, den1.d, ctx.d);

            fmpz_mpoly_mul(num1.d, num1.d, num2.d, ctx.d);
            fmpz_mpoly_mul(den1.d, den1.d, den2.d, ctx.d);

            t += t[1];
        }

        else {
            // Not a PolyFun, just copy or skip over
            WORD i = t[1];
            if ( s != t ) { NCOPY(s,t,i); }
            else { t += i; s += i; }
        }
    }

    flint::util::fix_sign_fmpz_mpoly_ratfun(num1.d, den1.d, ctx.d);

    // Result in FORM notation:
    WORD* out = s;
    *out++ = AR.PolyFun;
    WORD* args_size = out++;
    WORD* args_flag = out++;
    *args_flag &= ~MUSTCLEANPRF;

    const bool with_arghead = true;
    const bool must_fit_term = true;
    const bool write = true;
    out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
        num1.d, var_map, ctx.d);
    out += flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
        den1.d, var_map, ctx.d);

    *args_size = out - args_size + 1; // The +1 is to include the function ID

```

```
    // +3 for the coefficient of 1/1, which is added after the check
    if ( sizeof(WORD)*(out-term_size+3) > (size_t)AM.MaxTer ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::ratfun_normalize: output exceeds MaxTermSize");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    *out++ = 1;
    *out++ = 1;
    *out++ = 3; // the term's coefficient is now 1/1

    *term_size = out - term_size;
}
/*
    #] flint::ratfun_normalize_mpoly :
    #[ flint::ratfun_normalize_poly :
*/
// Multiply and simplify occurrences of the uni-variate FORM rational polynomials found in term.
// The final term is written in place, with the rational polynomial at the end.
void flint::ratfun_normalize_poly(PHEAD WORD *term, const var_map_t &var_map) {

    // The length of the coefficient
    const WORD ncoeff = (term + *term)[-1];
    // The end of the term data, before the coefficient:
    const WORD *tstop = term + *term - ABS(ncoeff);

    flint::poly num1, den1, num2, den2, gcd;

    // Start with "trivial" polynomials, and multiply in the num and den of the prf which appear.
    flint::fmpz tmpNum, tmpDen;
    flint::fmpz_set_form(tmpNum.d, (UWORD*)tstop, ncoeff/2);
    flint::fmpz_set_form(tmpDen.d, (UWORD*)tstop+ABS(ncoeff/2), ABS(ncoeff/2));
    fmpz_poly_set_fmpz(num1.d, tmpNum.d);
    fmpz_poly_set_fmpz(den1.d, tmpDen.d);

    // Loop over the occurrences of PolyFun in the term, and multiply in to num1, den1.
    // s tracks where we are writing the non-PolyFun term data. The final PolyFun will
    // go at the end.
    WORD* term_size = term;
    WORD* s = term + 1;
    for ( WORD *t = term + 1; t < tstop; ) {
        if ( *t == AR.PolyFun ) {
            flint::ratfun_read_poly(t, num2.d, den2.d);

            // get gcd of num1,den2 and num2,den1 and then assemble
            flint::util::simplify_fmpz_poly(num1.d, den2.d, gcd.d);
            flint::util::simplify_fmpz_poly(num2.d, den1.d, gcd.d);

            fmpz_poly_mul(num1.d, num1.d, num2.d);
            fmpz_poly_mul(den1.d, den1.d, den2.d);

            t += t[1];
        }

        else {
            // Not a PolyFun, just copy or skip over
            WORD i = t[1];
            if ( s != t ) { NCOPY(s,t,i); }
            else { t += i; s += i; }
        }
    }

    flint::util::fix_sign_fmpz_poly_ratfun(num1.d, den1.d);

    // Result in FORM notation:
    WORD* out = s;
    *out++ = AR.PolyFun;
    WORD* args_size = out++;
    WORD* args_flag = out++;
    *args_flag &= ~MUSTCLEANPRF;

    const bool with_arghead = true;
    const bool must_fit_term = true;
    const bool write = true;
    out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
        num1.d, var_map);
    out += flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, out-term_size,
        den1.d, var_map);

    *args_size = out - args_size + 1; // The +1 is to include the function ID

    // +3 for the coefficient of 1/1, which is added after the check
    if ( sizeof(WORD)*(out-term_size+3) > (size_t)AM.MaxTer ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::ratfun_normalize: output exceeds MaxTermSize");
        MUNLOCK(ErrorMessageLock);
```

```
        Terminate(-1);
    }

    *out++ = 1;
    *out++ = 1;
    *out++ = 3; // the term's coefficient is now 1/1

    *term_size = out - term_size;
}
/*
    #] flint::ratfun_normalize_poly :

    #[ flint::ratfun_read_mpoly :
*/
// Read the multi-variate FORM rational polynomial at a and create fmpz_mpoly_t numerator and
// denominator.
void flint::ratfun_read_mpoly(const WORD *a, fmpz_mpoly_t num, fmpz_mpoly_t den,
    const var_map_t &var_map, fmpz_mpoly_ctx_t ctx) {

    // The end of the arguments:
    const WORD* arg_stop = a+a[1];

    const bool must_normalize = (a[2] & MUSTCLEANPRF) != 0;

    a += FUNHEAD;
    if ( a >= arg_stop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("ERROR: PolyRatFun cannot have zero arguments");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Polys to collect the "den of the num" and "den of the den".
    // Input can arrive like this when enabling the PolyRatFun or moving things into it.
    flint::mpoly den_num(ctx), den_den(ctx);

    // Read the numerator
    flint::from_argument_mpoly(num, den_num.d, a, true, var_map, ctx);
    NEXTARG(a);

    if ( a < arg_stop ) {
        // Read the denominator
        flint::from_argument_mpoly(den, den_den.d, a, true, var_map, ctx);
        NEXTARG(a);
    }
    else {
        // The denominator is 1
        MLOCK(ErrorMessageLock);
        MesPrint("implement this");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( a < arg_stop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("ERROR: PolyRatFun cannot have more than two arguments");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Multiply the num by den_den and den by den_num:
    fmpz_mpoly_mul(num, num, den_den.d, ctx);
    fmpz_mpoly_mul(den, den, den_num.d, ctx);

    if ( must_normalize ) {
        flint::mpoly gcd(ctx);
        fmpz_mpoly_gcd_cofactors(gcd.d, num, den, num, den, ctx);
    }
}
/*
    #] flint::ratfun_read_mpoly :
    #[ flint::ratfun_read_poly :
*/
// Read the uni-variate FORM rational polynomial at a and create fmpz_mpoly_t numerator and
// denominator.
void flint::ratfun_read_poly(const WORD *a, fmpz_poly_t num, fmpz_poly_t den) {

    // The end of the arguments:
    const WORD* arg_stop = a+a[1];

    const bool must_normalize = (a[2] & MUSTCLEANPRF) != 0;

    a += FUNHEAD;
    if ( a >= arg_stop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("ERROR: PolyRatFun cannot have zero arguments");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
```

```
    }

    // Polys to collect the "den of the num" and "den of the den".
    // Input can arrive like this when enabling the PolyRatFun or moving things into it.
    flint::poly den_num, den_den;

    // Read the numerator
    flint::from_argument_poly(num, den_num.d, a, true);
    NEXTARG(a);

    if ( a < arg_stop ) {
        // Read the denominator
        flint::from_argument_poly(den, den_den.d, a, true);
        NEXTARG(a);
    }
    else {
        // The denominator is 1
        MLOCK(ErrorMessageLock);
        MesPrint("implement this");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( a < arg_stop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("ERROR: PolyRatFun cannot have more than two arguments");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Multiply the num by den_den and den by den_num:
    fmpz_poly_mul(num, num, den_den.d);
    fmpz_poly_mul(den, den, den_num.d);

    if ( must_normalize ) {
        flint::poly gcd;
        flint::util::simplify_fmpz_poly(num, den, gcd.d);
    }
}
/*
    #] flint::ratfun_read_poly :

    #[ flint::startup_init :
*/
// The purpose of this function is it work around threading issues in flint, reported in
// https://github.com/flintlib/flint/issues/1652 and fixed in
// https://github.com/flintlib/flint/pull/1658 . This implies versions prior to 3.1.0,
// but at least 3.0.0 (when gr was introduced).
void flint::startup_init(void) {

#if __FLINT_RELEASE >= 30000
    // Here we initialize some gr contexts so that their method tables are populated. Crashes have
    // otherwise been observed due to these two contexts in particular.
    fmpz_t dummy_fmpz;
    fmpz_init(dummy_fmpz);
    fmpz_set_si(dummy_fmpz, 19);

    gr_ctx_t dummy_gr_ctx_fmpz_mod;
    gr_ctx_init_fmpz_mod(dummy_gr_ctx_fmpz_mod, dummy_fmpz);
    gr_ctx_clear(dummy_gr_ctx_fmpz_mod);

    gr_ctx_t dummy_gr_ctx_nmod;
    gr_ctx_init_nmod(dummy_gr_ctx_nmod, 19);
    gr_ctx_clear(dummy_gr_ctx_nmod);

    fmpz_clear(dummy_fmpz);
#endif
}
/*
    #] flint::startup_init :

    #[ flint::to_argument_mpoly :
*/
// Convert a fmpz_mpoly_t to a FORM argument (or 0-terminated list of terms: with_arghead==false).
// If the caller is building an output term, prev_size contains the size of the term so far, to
// check that the output fits in AM.MaxTer if must_fit_term.
// All coefficients will be divided by denscale (which might just be 1).
// If write is false, we never write to out but only track the total would-be size. This lets this
// function be repurposed as a "size of FORM notation" function without duplicating the code.
#define IFW(x) { if ( write ) {x;} }
uint64_t flint::to_argument_mpoly(PHEAD WORD *out, const bool with_arghead,
    const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_mpoly_t poly,
    const var_map_t &var_map, const fmpz_mpoly_ctx_t ctx, const fmpz_t denscale) {

    // out is modified later, keep the pointer at entry
    const WORD* out_entry = out;

    // Track the total size written. We could do this with pointer differences, but if
```

```
    // write == false we don't write to or move out to be able to find the size that way.
    uint64_t ws = 0;

    // Check there is at least space for ARGHEAD WORDs (the arghead or fast-notation number/symbol)
    if ( write && must_fit_term && (sizeof(WORD)*(prev_size + ARGHEAD) > (size_t)AM.MaxTer) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::to_argument_mpoly: output exceeds MaxTermSize");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Create the inverse of var_map, so we don't have to search it for each symbol written
    var_map_t var_map_inv;
    for ( auto x: var_map ) {
        var_map_inv[x.second] = x.first;
    }

    vector<int64_t> exponents(var_map.size());
    const int64_t n_terms = fmpz_mpoly_length(poly, ctx);

    if ( n_terms == 0 ) {
        if ( with_arghead ) {
            IFW(*out++ = -SNUMBER); ws++;
            IFW(*out++ = 0); ws++;
            return ws;
        }
        else {
            IFW(*out++ = 0); ws++;
            return ws;
        }
    }

    // For dividing out denscale
    flint::fmpz coeff, den, gcd;

    // The mpoly might be constant or a single symbol with coeff 1. Use fast notation if possible.
    if ( with_arghead && n_terms == 1 ) {

        if ( fmpz_mpoly_is_fmpz(poly, ctx) ) {
            // The mpoly is constant. Use fast notation if the number is an integer and small enough:

            fmpz_mpoly_get_term_coeff_fmpz(coeff.d, poly, 0, ctx);
            fmpz_set(den.d, denscale);
            flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);

            if ( fmpz_is_one(den.d) && fmpz_fits_si(coeff.d) ) {
                const int64_t fast_coeff = fmpz_get_si(coeff.d);
                // While ">=", could work here, FORM does not use fast notation for INT_MIN
                if ( fast_coeff > INT32_MIN && fast_coeff <= INT32_MAX ) {
                    IFW(*out++ = -SNUMBER); ws++;
                    IFW(*out++ = (WORD)fast_coeff); ws++;
                    return ws;
                }
            }
        }

        else {
            fmpz_mpoly_get_term_coeff_fmpz(coeff.d, poly, 0, ctx);
            fmpz_set(den.d, denscale);
            flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);

            if ( fmpz_is_one(coeff.d) && fmpz_is_one(den.d) ) {
                // The coefficient is one. Now check the symbol powers:

                fmpz_mpoly_get_term_exp_si((slong*)exponents.data(), poly, 0, ctx);
                int64_t use_fast = 0;
                uint32_t fast_symbol = 0;

                for ( size_t i = 0; i < var_map.size(); i++ ) {
                    if ( exponents[i] == 1 ) fast_symbol = var_map_inv[i];
                    use_fast += exponents[i];
                }

                // use_fast has collected the total degree. If it is 1, then fast_symbol holds the code
                if ( use_fast == 1 ) {
                    IFW(*out++ = -SYMBOL); ws++;
                    IFW(*out++ = fast_symbol); ws++;
                    return ws;
                }
            }
        }
    }


    WORD *tmp_coeff = (WORD *)NumberMalloc("flint::to_argument_mpoly");
    WORD *tmp_den = (WORD *)NumberMalloc("flint::to_argument_mpoly");
```

```
    WORD* arg_size = 0;
    WORD* arg_flag = 0;
    if ( with_arghead ) {
        IFW(arg_size = out++); ws++; // total arg size
        IFW(arg_flag = out++); ws++;
        IFW(*arg_flag = 0); // clean argument
    }

    for ( int64_t i = 0; i < n_terms; i++ ) {

        fmpz_mpoly_get_term_exp_si((slong*)exponents.data(), poly, i, ctx);

        fmpz_mpoly_get_term_coeff_fmpz(coeff.d, poly, i, ctx);
        fmpz_set(den.d, denscale);
        flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);

        uint32_t num_symbols = 0;
        for ( size_t j = 0; j < var_map.size(); j++ ) {
            if ( exponents[j] != 0 ) { num_symbols += 1; }
        }

        // Convert the coefficient, write in temporary space
        const WORD num_size = flint::fmpz_get_form(coeff.d, tmp_coeff);
        const WORD den_size = flint::fmpz_get_form(den.d, tmp_den);
        const WORD coeff_size = [num_size, den_size] () -> WORD {
            WORD size = ABS(num_size) > ABS(den_size) ? ABS(num_size) : ABS(den_size);
            return size * SGN(num_size) * SGN(den_size);
        }();

        // Now we have the number of symbols and the coeff size, we can determine the output size.
        // Check it fits if necessary: term size, num,den of "coeff_size", +1 for total coeff size
        uint64_t current_size = prev_size + ws + 1 + 2*ABS(coeff_size) + 1;
        if ( num_symbols ) {
            // symbols header, code,power of each symbol:
            current_size += 2 + 2*num_symbols;
        }
        if ( write && must_fit_term && (sizeof(WORD)*current_size > (size_t)AM.MaxTer) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("flint::to_argument_mpoly: output exceeds MaxTermSize");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }

        WORD* term_size = 0;
        IFW(term_size = out++); ws++;
        if ( num_symbols ) {
            IFW(*out++ = SYMBOL); ws++;
            WORD* symbol_size = 0;
            IFW(symbol_size = out++); ws++;
            IFW(*symbol_size = 2);

            for ( size_t j = 0; j < var_map.size(); j++ ) {
                if ( exponents[j] != 0 ) {
                    IFW(*out++ = var_map_inv[j]); ws++;
                    IFW(*out++ = exponents[j]); ws++;
                    IFW(*symbol_size += 2);
                }
            }
        }

        // Copy numerator
        for ( WORD j = 0; j < ABS(num_size); j++ ) {
            IFW(*out++ = tmp_coeff[j]); ws++;
        }
        for ( WORD j = ABS(num_size); j < ABS(coeff_size); j++ ) {
            IFW(*out++ = 0); ws++;
        }
        // Copy denominator
        for ( WORD j = 0; j < ABS(den_size); j++ ) {
            IFW(*out++ = tmp_den[j]); ws++;
        }
        for ( WORD j = ABS(den_size); j < ABS(coeff_size); j++ ) {
            IFW(*out++ = 0); ws++;
        }

        IFW(*out = 2*ABS(coeff_size) + 1); // the size of the coefficient
        IFW(if ( coeff_size < 0 ) { *out = -(*out); });
        IFW(out++); ws++;

        IFW(*term_size = out - term_size);
    }

    if ( with_arghead ) {
        IFW(*arg_size = out - arg_size);
        if ( write ) {
            // Sort into form highfirst ordering
```

```
            flint::form_sort(BHEAD (WORD*)(out_entry));
        }
    }
    else {
        // with no arghead, we write a terminating zero
        IFW(*out++ = 0); ws++;
    }

    NumberFree(tmp_coeff, "flint::to_argument_mpoly");
    NumberFree(tmp_den, "flint::to_argument_mpoly");

    return ws;
}

// If no denscale argument is supplied, just set it to 1 and call the usual function
uint64_t flint::to_argument_mpoly(PHEAD WORD *out, const bool with_arghead,
    const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_mpoly_t poly,
    const var_map_t &var_map, const fmpz_mpoly_ctx_t ctx) {

    flint::fmpz tmp;
    fmpz_set_ui(tmp.d, 1);

    uint64_t ret = flint::to_argument_mpoly(BHEAD out, with_arghead, must_fit_term, write,
        prev_size, poly, var_map, ctx, tmp.d);

    return ret;
}
/*
    #] flint::to_argument_mpoly :
    #[ flint::to_argument_poly :
*/
// Convert a fmpz_poly_t to a FORM argument (or 0-terminated list of terms: with_arghead==false).
// If the caller is building an output term, prev_size contains the size of the term so far, to
// check that the output fits in AM.MaxTer if must_fit_term.
// All coefficients will be divided by denscale (which might just be 1).
// If write is false, we never write to out but only track the total would-be size. This lets this
// function be repurposed as a "size of FORM notation" function without duplicating the code.
uint64_t flint::to_argument_poly(PHEAD WORD *out, const bool with_arghead,
    const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_poly_t poly,
    const var_map_t &var_map, const fmpz_t denscale) {

    // Track the total size written. We could do this with pointer differences, but if
    // write == false we don't write to or move out to be able to find the size that way.
    uint64_t ws = 0;

    // Check there is at least space for ARGHEAD WORDs (the arghead or fast-notation number/symbol)
    if ( write && must_fit_term && (sizeof(WORD)*(prev_size + ARGHEAD) > (size_t)AM.MaxTer) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint::to_argument_poly: output exceeds MaxTermSize");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Create the inverse of var_map, so we don't have to search it for each symbol written
    var_map_t var_map_inv;
    for ( auto x: var_map ) {
        var_map_inv[x.second] = x.first;
    }

    const int64_t n_terms = fmpz_poly_length(poly);

    // The poly is zero
    if ( n_terms == 0 ) {
        if ( with_arghead ) {
            IFW(*out++ = -SNUMBER); ws++;
            IFW(*out++ = 0); ws++;
            return ws;
        }
        else {
            IFW(*out++ = 0); ws++;
            return ws;
        }
    }

    // For dividing out denscale
    flint::fmpz coeff, den, gcd;

    // The poly is constant, use fast notation if the coefficient is integer and small enough
    if ( with_arghead && n_terms == 1 ) {

        fmpz_poly_get_coeff_fmpz(coeff.d, poly, 0);
        fmpz_set(den.d, denscale);
        flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);

        if ( fmpz_is_one(den.d) && fmpz_fits_si(coeff.d) ) {
            const long fast_coeff = fmpz_get_si(coeff.d);
            // While ">=", could work here, FORM does not use fast notation for INT_MIN
```

```
                if ( fast_coeff > INT_MIN && fast_coeff <= INT_MAX ) {
                    IFW(*out++ = -SNUMBER); ws++;
                    IFW(*out++ = (WORD)fast_coeff); ws++;
                    return ws;
                }
            }
        }

        // The poly might be a single symbol with coeff 1, use fast notation if so.
        if ( with_arghead && n_terms == 2 ) {
            if ( fmpz_is_zero(fmpz_poly_get_coeff_ptr(poly, 0)) ) {
                // The constant term is zero

                fmpz_poly_get_coeff_fmpz(coeff.d, poly, 1);
                fmpz_set(den.d, denscale);
                flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);

                if ( fmpz_is_one(coeff.d) && fmpz_is_one(den.d) ) {
                    // Single symbol with coeff 1. Use fast notation:
                    IFW(*out++ = -SYMBOL); ws++;
                    IFW(*out++ = var_map_inv[0]); ws++;
                    return ws;
                }
            }
        }

        WORD *tmp_coeff = (WORD *)NumberMalloc("flint::to_argument_poly");
        WORD *tmp_den = (WORD *)NumberMalloc("flint::to_argument_mpoly");

        WORD* arg_size = 0;
        WORD* arg_flag = 0;
        if ( with_arghead ) {
            IFW(arg_size = out++); ws++; // total arg size
            IFW(arg_flag = out++); ws++;
            IFW(*arg_flag = 0); // clean argument
        }

        // In reverse, since we want a "highfirst" output
        for ( int64_t i = n_terms-1; i >= 0; i-- ) {

            // fmpz_poly is dense, there might be many zero coefficients:
            if ( !fmpz_is_zero(fmpz_poly_get_coeff_ptr(poly, i)) ) {

                fmpz_poly_get_coeff_fmpz(coeff.d, poly, i);
                fmpz_set(den.d, denscale);
                flint::util::simplify_fmpz(coeff.d, den.d, gcd.d);

                // Convert the coefficient, write in temporary space
                const WORD num_size = flint::fmpz_get_form(coeff.d, tmp_coeff);
                const WORD den_size = flint::fmpz_get_form(den.d, tmp_den);
                const WORD coeff_size = [num_size, den_size] () -> WORD {
                    WORD size = ABS(num_size) > ABS(den_size) ? ABS(num_size) : ABS(den_size);
                    return size * SGN(num_size) * SGN(den_size);
                }();

                // Now we have the coeff size, we can determine the output size
                // Check it fits if necessary: symbol code,power, num,den of "coeff_size",
                // +1 for total coeff size
                uint64_t current_size = prev_size + ws + 1 + 2*ABS(coeff_size) + 1;
                if ( i > 0 ) {
                    // and also symbols header, code,power of the symbol
                    current_size += 4;
                }
                if ( write && must_fit_term && (sizeof(WORD)*current_size > (size_t)AM.MaxTer) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("flint::to_argument_poly: output exceeds MaxTermSize");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }

                WORD* term_size = 0;
                IFW(term_size = out++); ws++;

                if ( i > 0 ) {
                    IFW(*out++ = SYMBOL); ws++;
                    IFW(*out++ = 4); ws++; // The symbol array size, it is univariate
                    IFW(*out++ = var_map_inv[0]); ws++;
                    IFW(*out++ = i); ws++;
                }

                // Copy numerator
                for ( WORD j = 0; j < ABS(num_size); j++ ) {
                    IFW(*out++ = tmp_coeff[j]); ws++;
                }
                for ( WORD j = ABS(num_size); j < ABS(coeff_size); j++ ) {
                    IFW(*out++ = 0); ws++;
                }
```

```
            // Copy denominator
            for ( WORD j = 0; j < ABS(den_size); j++ ) {
                IFW(*out++ = tmp_den[j]); ws++;
            }
            for ( WORD j = ABS(den_size); j < ABS(coeff_size); j++ ) {
                IFW(*out++ = 0); ws++;
            }

            IFW(*out = 2*ABS(coeff_size) + 1); // the size of the coefficient
            IFW(if ( coeff_size < 0 ) { *out = -(*out); });
            IFW(out++); ws++;

            IFW(*term_size = out - term_size);
        }

    }

    if ( with_arghead ) {
        IFW(*arg_size = out - arg_size);
    }
    else {
        // with no arghead, we write a terminating zero
        IFW(*out++ = 0); ws++;
    }

    NumberFree(tmp_coeff, "flint::to_argument_poly");
    NumberFree(tmp_den, "flint::to_argument_poly");

    return ws;
}

// If no denscale argument is supplied, just set it to 1 and call the usual function
uint64_t flint::to_argument_poly(PHEAD WORD *out, const bool with_arghead,
    const bool must_fit_term, const bool write, const uint64_t prev_size, const fmpz_poly_t poly,
    const var_map_t &var_map) {

    flint::fmpz tmp;
    fmpz_set_ui(tmp.d, 1);

    uint64_t ret = flint::to_argument_poly(BHEAD out, with_arghead, must_fit_term, write, prev_size,
        poly, var_map, tmp.d);

    return ret;
}
/*
    #] flint::to_argument_poly :
*/

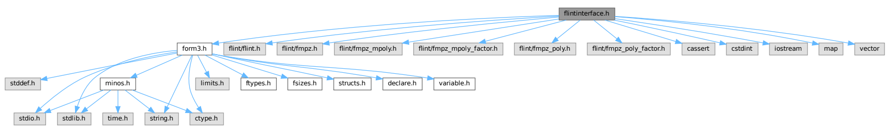
// Utility functions
/*
    #[ flint::util::simplify_fmpz :
*/
// Divide the GCD out of num and den
inline void flint::util::simplify_fmpz(fmpz_t num, fmpz_t den, fmpz_t gcd) {
    fmpz_gcd(gcd, num, den);
    if ( !fmpz_is_one(gcd) ) {
        fmpz_divexact(num, num, gcd);
        fmpz_divexact(den, den, gcd);
    }
}
/*
    #] flint::util::simplify_fmpz :
    #[ flint::util::simplify_fmpz_poly :
*/
// Divide the GCD out of num and den
inline void flint::util::simplify_fmpz_poly(fmpz_poly_t num, fmpz_poly_t den, fmpz_poly_t gcd) {
    fmpz_poly_gcd(gcd, num, den);
    if ( !fmpz_poly_is_one(gcd) ) {
#if __FLINT_RELEASE >= 30100
        // This should be faster than fmpz_poly_div, see https://github.com/flintlib/flint/pull/1766
        fmpz_poly_divexact(num, num, gcd);
        fmpz_poly_divexact(den, den, gcd);
#else
        fmpz_poly_div(num, num, gcd);
        fmpz_poly_div(den, den, gcd);
#endif
    }
}
/*
    #] flint::util::simplify_fmpz_poly :
    #[ flint::util::fix_sign_fmpz_mpoly_ratfun :
*/
inline void flint::util::fix_sign_fmpz_mpoly_ratfun(fmpz_mpoly_t num, fmpz_mpoly_t den,
    const fmpz_mpoly_ctx_t ctx) {
    // Fix sign to align with poly: the leading denominator term should have a positive coeff
    if ( fmpz_sgn(fmpz_mpoly_term_coeff_ref(den, 0, ctx)) == -1 ) {
        fmpz_mpoly_neg(num, num, ctx);
```

```
        fmpz_mpoly_neg(den, den, ctx);
    }
}
/*
    #] flint::util::fix_sign_fmpz_mpoly_ratfun :
    #[ flint::util::fix_sign_fmpz_poly_ratfun :
*/
inline void flint::util::fix_sign_fmpz_poly_ratfun(fmpz_poly_t num, fmpz_poly_t den) {
    // Fix sign to align with poly: the leading denominator term should have a positive coeff
    if ( fmpz_sgn(fmpz_poly_get_coeff_ptr(den, fmpz_poly_degree(den))) == -1 ) {
        fmpz_poly_neg(num, num);
        fmpz_poly_neg(den, den);
    }
}
/*
    #] flint::util::fix_sign_fmpz_poly_ratfun :
*/
```

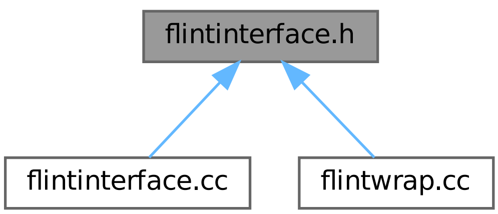
4.39 flintinterface.h File Reference

```
#include "form3.h"
#include <flint/flint.h>
#include <flint/fmpz.h>
#include <flint/fmpz_mpoly.h>
#include <flint/fmpz_mpoly_factor.h>
#include <flint/fmpz_poly.h>
#include <flint/fmpz_poly_factor.h>
#include <cassert>
#include <cstdint>
#include <iostream>
#include <map>
#include <vector>
```

Include dependency graph for flintinterface.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- class flint::fmpz
- class flint::poly
- class flint::poly_factor
- class flint::mpoly
- class flint::mpoly_factor
- class flint::mpoly_ctx

Typedefs

- typedef std::map< uint32_t, uint32_t > flint::var_map_t

Functions

- void flint::cleanup (void)
- void flint::cleanup_master (void)
- WORD * flint::divmod_mpoly (PHEAD const WORD *, const WORD *, const bool, const WORD, const var_map_t &)
- WORD * flint::divmod_poly (PHEAD const WORD *, const WORD *, const bool, const WORD, const var_map_t &)
- WORD * flint::factorize_mpoly (PHEAD const WORD *, WORD *, const bool, const bool, const var_map_t &)
- WORD * flint::factorize_poly (PHEAD const WORD *, WORD *, const bool, const bool, const var_map_t &)
- void flint::form_sort (PHEAD WORD *)
- uint64_t flint::from_argument_mpoly (fmpz_mpoly_t, fmpz_mpoly_t, const WORD *, const bool, const var_map_t &, const fmpz_mpoly_ctx_t)
- uint64_t flint::from_argument_poly (fmpz_poly_t, fmpz_poly_t, const WORD *, const bool)
- WORD flint::fmpz_get_form (fmpz_t, WORD *)
- void flint::fmpz_set_form (fmpz_t, UWORD *, WORD)
- WORD * flint::gcd_mpoly (PHEAD const WORD *, const WORD *, const WORD, const var_map_t &)
- WORD * flint::gcd_poly (PHEAD const WORD *, const WORD *, const WORD, const var_map_t &)
- var_map_t flint::get_variables (const vector< WORD * > &, const bool, const bool)
- WORD * flint::inverse_poly (PHEAD const WORD *, const WORD *, const var_map_t &)
- WORD * flint::mul_mpoly (PHEAD const WORD *, const WORD *, const var_map_t &)
- WORD * flint::mul_poly (PHEAD const WORD *, const WORD *, const var_map_t &)
- void flint::ratfun_add_mpoly (PHEAD const WORD *, const WORD *, WORD *, const var_map_t &)
- void flint::ratfun_add_poly (PHEAD const WORD *, const WORD *, WORD *, const var_map_t &)
- void flint::ratfun_normalize_mpoly (PHEAD WORD *, const var_map_t &)
- void flint::ratfun_normalize_poly (PHEAD WORD *, const var_map_t &)
- void flint::ratfun_read_mpoly (const WORD *, fmpz_mpoly_t, fmpz_mpoly_t, const var_map_t &, fmpz_mpoly_ctx_t)
- void flint::ratfun_read_poly (const WORD *, fmpz_poly_t, fmpz_poly_t)
- void flint::startup_init (void)
- uint64_t flint::to_argument_mpoly (PHEAD WORD *, const bool, const bool, const bool, const uint64_t, const fmpz_mpoly_t, const var_map_t &, const fmpz_mpoly_ctx_t)
- uint64_t flint::to_argument_mpoly (PHEAD WORD *, const bool, const bool, const bool, const uint64_t, const fmpz_mpoly_t, const var_map_t &, const fmpz_mpoly_ctx_t, const fmpz_t)
- uint64_t flint::to_argument_poly (PHEAD WORD *, const bool, const bool, const bool, const uint64_t, const fmpz_poly_t, const var_map_t &)
- uint64_t flint::to_argument_poly (PHEAD WORD *, const bool, const bool, const bool, const uint64_t, const fmpz_poly_t, const var_map_t &, const fmpz_t)
- void flint::util::simplify_fmpz (fmpz_t, fmpz_t, fmpz_t)
- void flint::util::simplify_fmpz_poly (fmpz_poly_t, fmpz_poly_t, fmpz_poly_t)
- void flint::util::fix_sign_fmpz_mpoly_ratfun (fmpz_mpoly_t, fmpz_mpoly_t, const fmpz_mpoly_ctx_t)
- void flint::util::fix_sign_fmpz_poly_ratfun (fmpz_poly_t, fmpz_poly_t)

4.39.1 Detailed Description

Prototypes for functions in flintinterface.cc

Definition in file flintinterface.h.

4.39.2 Typedef Documentation

4.39.2.1 var_map_t

```
typedef std::map<uint32_t,uint32_t> flint::var_map_t
```

Definition at line 50 of file flintinterface.h.

4.39.3 Function Documentation

4.39.3.1 cleanup()

```
void flint::cleanup (
            void )
```

Definition at line 43 of file flintinterface.cc.

4.39.3.2 cleanup_master()

```
void flint::cleanup_master (
            void )
```

Definition at line 50 of file flintinterface.cc.

4.39.3.3 divmod_mpoly()

```
WORD * flint::divmod_mpoly (
            PHEAD const WORD * a,
            const WORD * b,
            const bool return_rem,
            const WORD must_fit_term,
            const var_map_t & var_map )
```

Definition at line 58 of file flintinterface.cc.

4.39.3.4 divmod_poly()

```
WORD * flint::divmod_poly (
            PHEAD const WORD * a,
            const WORD * b,
            const bool return_rem,
            const WORD must_fit_term,
            const var_map_t & var_map )
```

Definition at line 128 of file flintinterface.cc.

4.39.3.5 factorize_mpoly()

```
WORD * flint::factorize_mpoly (
            PHEAD const WORD * argin,
            WORD * argout,
            const bool with_arghead,
            const bool is_fun_arg,
            const var_map_t & var_map )
```

Definition at line 202 of file flintinterface.cc.

4.39.3.6 factorize_poly()

```
WORD * flint::factorize_poly (
            PHEAD const WORD * argin,
            WORD * argout,
            const bool with_arghead,
            const bool is_fun_arg,
            const var_map_t & var_map )
```

Definition at line 350 of file flintinterface.cc.

4.39.3.7 form_sort()

```
void flint::form_sort (
            PHEAD WORD * terms )
```

Definition at line 423 of file flintinterface.cc.

4.39.3.8 from_argument_mpoly()

```
uint64_t flint::from_argument_mpoly (
            fmpz_mpoly_t poly,
            fmpz_mpoly_t denpoly,
            const WORD * args,
            const bool with_arghead,
            const var_map_t & var_map,
            const fmpz_mpoly_ctx_t ctx )
```

Definition at line 478 of file flintinterface.cc.

4.39.3.9 from_argument_poly()

```
uint64_t flint::from_argument_poly (
            fmpz_poly_t poly,
            fmpz_poly_t denpoly,
            const WORD * args,
            const bool with_arghead )
```

Definition at line 621 of file flintinterface.cc.

4.39.3.10 fmpz_get_form()

```
WORD flint::fmpz_get_form (
            fmpz_t z,
            WORD * a )
```

Definition at line 751 of file flintinterface.cc.

4.39.3.11 fmpz_set_form()

```
void flint::fmpz_set_form (
            fmpz_t z,
            UWORD * a,
            WORD na )
```

Definition at line 816 of file flintinterface.cc.

4.39.3.12 gcd_mpoly()

```
WORD * flint::gcd_mpoly (
            PHEAD const WORD * a,
            const WORD * b,
            const WORD must_fit_term,
            const var_map_t & var_map )
```

Definition at line 882 of file flintinterface.cc.

4.39.3.13 gcd_poly()

```
WORD * flint::gcd_poly (
            PHEAD const WORD * a,
            const WORD * b,
            const WORD must_fit_term,
            const var_map_t & var_map )
```

Definition at line 978 of file flintinterface.cc.

4.39.3.14 get_variables()

```
flint::var_map_t flint::get_variables (
            const vector< WORD * > & es,
            const bool with_arghead,
            const bool sort_vars )
```

Definition at line 1060 of file flintinterface.cc.

4.39.3.15 inverse_poly()

```
WORD * flint::inverse_poly (
            PHEAD const WORD * a,
            const WORD * b,
            const var_map_t & var_map )
```

Definition at line 1162 of file flintinterface.cc.

4.39.3.16 mul_mpoly()

```
WORD * flint::mul_mpoly (
            PHEAD const WORD * a,
            const WORD * b,
            const var_map_t & var_map )
```

Definition at line 1238 of file flintinterface.cc.

4.39.3.17 mul_poly()

```
WORD * flint::mul_poly (
            PHEAD const WORD * a,
            const WORD * b,
            const var_map_t & var_map )
```

Definition at line 1290 of file flintinterface.cc.

4.39.3.18 ratfun_add_mpoly()

```
void flint::ratfun_add_mpoly (
            PHEAD const WORD * t1,
            const WORD * t2,
            WORD * out,
            const var_map_t & var_map )
```

Definition at line 1341 of file flintinterface.cc.

4.39.3.19 ratfun_add_poly()

```
void flint::ratfun_add_poly (
            PHEAD const WORD * t1,
            const WORD * t2,
            WORD * out,
            const var_map_t & var_map )
```

Definition at line 1393 of file flintinterface.cc.

4.39.3.20 ratfun_normalize_mpoly()

```
void flint::ratfun_normalize_mpoly (
            PHEAD WORD * term,
            const var_map_t & var_map )
```

Definition at line 1446 of file flintinterface.cc.

4.39.3.21 ratfun_normalize_poly()

```
void flint::ratfun_normalize_poly (
            PHEAD WORD * term,
            const var_map_t & var_map )
```

Definition at line 1529 of file flintinterface.cc.

4.39.3.22 ratfun_read_mpoly()

```
void flint::ratfun_read_mpoly (
            const WORD * a,
            fmpz_mpoly_t num,
            fmpz_mpoly_t den,
            const var_map_t & var_map,
            fmpz_mpoly_ctx_t ctx )
```

Definition at line 1612 of file flintinterface.cc.

4.39.3.23 ratfun_read_poly()

```
void flint::ratfun_read_poly (
            const WORD * a,
            fmpz_poly_t num,
            fmpz_poly_t den )
```

Definition at line 1670 of file flintinterface.cc.

4.39.3.24 startup_init()

```
void flint::startup_init (
            void )
```

Definition at line 1730 of file flintinterface.cc.

4.39.3.25 to_argument_mpoly() [1/2]

```
uint64_t flint::to_argument_mpoly (
            PHEAD WORD * out,
            const bool with_arghead,
            const bool must_fit_term,
            const bool write,
            const uint64_t prev_size,
            const fmpz_mpoly_t poly,
            const var_map_t & var_map,
            const fmpz_mpoly_ctx_t ctx )
```

Definition at line 1958 of file flintinterface.cc.

4.39.3.26 to_argument_mpoly() [2/2]

```
uint64_t flint::to_argument_mpoly (
            PHEAD WORD * out,
            const bool with_arghead,
            const bool must_fit_term,
            const bool write,
            const uint64_t prev_size,
            const fmpz_mpoly_t poly,
            const var_map_t & var_map,
            const fmpz_mpoly_ctx_t ctx,
            const fmpz_t denscale )
```

Definition at line 1762 of file flintinterface.cc.

4.39.3.27 to_argument_poly() [1/2]

```
uint64_t flint::to_argument_poly (
            PHEAD WORD * out,
            const bool with_arghead,
            const bool must_fit_term,
            const bool write,
            const uint64_t prev_size,
            const fmpz_poly_t poly,
            const var_map_t & var_map )
```

Definition at line 2149 of file flintinterface.cc.

4.39.3.28 to_argument_poly() [2/2]

```
uint64_t flint::to_argument_poly (
            PHEAD WORD * out,
            const bool with_arghead,
            const bool must_fit_term,
            const bool write,
            const uint64_t prev_size,
            const fmpz_poly_t poly,
            const var_map_t & var_map,
            const fmpz_t denscale )
```

Definition at line 1980 of file flintinterface.cc.

4.39.3.29 simplify_fmpz()

```
void flint::util::simplify_fmpz (
            fmpz_t num,
            fmpz_t den,
            fmpz_t gcd ) [inline]
```

Definition at line 2170 of file flintinterface.cc.

4.39.3.30 simplify_fmpz_poly()

```
void flint::util::simplify_fmpz_poly (
            fmpz_poly_t num,
            fmpz_poly_t den,
            fmpz_poly_t gcd ) [inline]
```

Definition at line 2182 of file flintinterface.cc.

4.39.3.31 fix_sign_fmpz_mpoly_ratfun()

```
void flint::util::fix_sign_fmpz_mpoly_ratfun (
            fmpz_mpoly_t num,
            fmpz_mpoly_t den,
            const fmpz_mpoly_ctx_t ctx ) [inline]
```

Definition at line 2199 of file flintinterface.cc.

4.39.3.32 fix_sign_fmpz_poly_ratfun()

```
void flint::util::fix_sign_fmpz_poly_ratfun (
            fmpz_poly_t num,
            fmpz_poly_t den ) [inline]
```

Definition at line 2211 of file flintinterface.cc.

4.40 flintinterface.h

Go to the documentation of this file.

```
#pragma once
extern "C" {
#include "form3.h"
}
#if defined(WINDOWS)
// flint.h defines WORD(xx), which conflicts with the one defined in form3.h.
#undef WORD
#endif
#include <flint/flint.h>
#if __FLINT_RELEASE >= 30000
#include <flint/gr.h>
#endif
#include <flint/fmpz.h>
#include <flint/fmpz_mpoly.h>
#include <flint/fmpz_mpoly_factor.h>
#include <flint/fmpz_poly.h>
```

```
#include <flint/fmpz_poly_factor.h>

#if defined(WINDOWS)
// Redefine WORD here to match form3.h.
#undef WORD
#define WORD FORM_WORD
#endif

#include <cassert>
#include <cstdint>
#include <iostream>
#include <map>
#include <vector>


// The bits of std that are needed:
using std::cout;
using std::endl;
using std::map;
using std::swap;
using std::string;
using std::vector;


namespace flint {

    typedef std::map<uint32_t,uint32_t> var_map_t;

    // Small wrappers around the flint structs to enable RAII init and clear. "d" represents the
    // data, and this member will be passed to flint functions.
    class fmpz {
        public:
            fmpz_t d;
            fmpz() { fmpz_init(d); }
            ~fmpz() { fmpz_clear(d); }
            void print(const string& text) { cout « text; fmpz_print(d); cout « endl; }
    };
    class poly {
        public:
            fmpz_poly_t d;
            poly() { fmpz_poly_init(d); }
            ~poly() { fmpz_poly_clear(d); }
            void print(const string& text) {
                cout « text; fmpz_poly_print_pretty(d, "x"); cout « endl;
            }
    };
    class poly_factor {
        public:
            fmpz_poly_factor_t d;
            poly_factor() { fmpz_poly_factor_init(d); }
            ~poly_factor() { fmpz_poly_factor_clear(d); }
    };
    class mpoly {
        private:
            fmpz_mpoly_ctx_struct *ctx; // We need to keep a copy of the context pointer for clearing.
        public:
            fmpz_mpoly_t d;
            explicit mpoly(fmpz_mpoly_ctx_struct *ctx_in) : ctx(ctx_in) { fmpz_mpoly_init(d, ctx); }
            ~mpoly() { fmpz_mpoly_clear(d, ctx); }
            void print(const string& text) {
                cout « text;
                fmpz_mpoly_print_pretty(d, 0, ctx);
                cout « endl;
            }
    };
    class mpoly_factor {
        private:
            fmpz_mpoly_ctx_struct *ctx; // We need to keep a copy of the context pointer for clearing.
        public:
            fmpz_mpoly_factor_t d;
            explicit mpoly_factor(fmpz_mpoly_ctx_struct *ctx_in) : ctx(ctx_in) {
                fmpz_mpoly_factor_init(d, ctx);
            }
            ~mpoly_factor() { fmpz_mpoly_factor_clear(d, ctx); }
    };
    class mpoly_ctx {
        public:
            fmpz_mpoly_ctx_t d;
            explicit mpoly_ctx(int64_t nvars) { fmpz_mpoly_ctx_init(d, nvars, ORD_LEX); }
            ~mpoly_ctx() { fmpz_mpoly_ctx_clear(d); }
    };

    void cleanup(void);
    void cleanup_master(void);

    WORD* divmod_mpoly(PHEAD const WORD *, const WORD *, const bool, const WORD, const var_map_t &);
    WORD* divmod_poly(PHEAD const WORD *, const WORD *, const bool, const WORD, const var_map_t &);
```

```

    WORD* factorize_mpoly(PHEAD const WORD *, WORD *, const bool, const bool, const var_map_t &);
    WORD* factorize_poly(PHEAD const WORD *, WORD *, const bool, const bool, const var_map_t &);

    void form_sort(PHEAD WORD *);

    uint64_t from_argument_mpoly(fmpz_mpoly_t, fmpz_mpoly_t, const WORD *, const bool,
        const var_map_t &, const fmpz_mpoly_ctx_t);
    uint64_t from_argument_poly(fmpz_poly_t, fmpz_poly_t, const WORD *, const bool);

    WORD fmpz_get_form(fmpz_t, WORD *);
    void fmpz_set_form(fmpz_t, UWORD *, WORD);

    WORD* gcd_mpoly(PHEAD const WORD *, const WORD *, const WORD, const var_map_t &);
    WORD* gcd_poly(PHEAD const WORD *, const WORD *, const WORD, const var_map_t &);

    var_map_t get_variables(const vector <WORD *> &, const bool, const bool);

    WORD* inverse_poly(PHEAD const WORD *, const WORD *, const var_map_t &);

    WORD* mul_mpoly(PHEAD const WORD *, const WORD *, const var_map_t &);
    WORD* mul_poly(PHEAD const WORD *, const WORD *, const var_map_t &);

    void ratfun_add_mpoly(PHEAD const WORD *, const WORD *, WORD *, const var_map_t &);
    void ratfun_add_poly(PHEAD const WORD *, const WORD *, WORD *, const var_map_t &);

    void ratfun_normalize_mpoly(PHEAD WORD *, const var_map_t &);
    void ratfun_normalize_poly(PHEAD WORD *, const var_map_t &);

    void ratfun_read_mpoly(const WORD *, fmpz_mpoly_t, fmpz_mpoly_t, const var_map_t &,
        fmpz_mpoly_ctx_t);
    void ratfun_read_poly(const WORD *, fmpz_poly_t, fmpz_poly_t);

    void startup_init(void);

    uint64_t to_argument_mpoly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
        const fmpz_mpoly_t, const var_map_t &, const fmpz_mpoly_ctx_t);
    uint64_t to_argument_mpoly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
        const fmpz_mpoly_t, const var_map_t &, const fmpz_mpoly_ctx_t, const fmpz_t);
    uint64_t to_argument_poly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
        const fmpz_poly_t, const var_map_t &);
    uint64_t to_argument_poly(PHEAD WORD *, const bool, const bool, const bool, const uint64_t,
        const fmpz_poly_t, const var_map_t &, const fmpz_t);


    namespace util {

        void simplify_fmpz(fmpz_t, fmpz_t, fmpz_t);
        void simplify_fmpz_poly(fmpz_poly_t, fmpz_poly_t, fmpz_poly_t);

        void fix_sign_fmpz_mpoly_ratfun(fmpz_mpoly_t, fmpz_mpoly_t, const fmpz_mpoly_ctx_t);
        void fix_sign_fmpz_poly_ratfun(fmpz_poly_t, fmpz_poly_t);

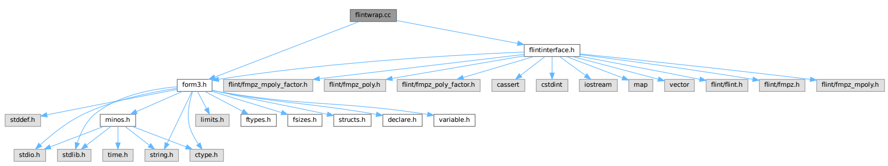
    }

}
```

4.41 flintwrap.cc File Reference

`#include "form3.h"`
`#include "flintinterface.h"`

Include dependency graph for flintwrap.cc:



Functions

- void flint_final_cleanup_thread (void)

- void flint_final_cleanup_master (void)
- WORD * flint_div (PHEAD WORD *a, WORD *b, const WORD must_fit_term)
- int flint_factorize_argument (PHEAD WORD *argin, WORD *argout)
- WORD * flint_factorize_dollar (PHEAD WORD *argin)
- WORD * flint_gcd (PHEAD WORD *a, WORD *b, const WORD must_fit_term)
- WORD * flint_inverse (PHEAD WORD *a, WORD *b)
- WORD * flint_mul (PHEAD WORD *a, WORD *b)
- WORD * flint_ratfun_add (PHEAD WORD *t1, WORD *t2)
- int flint_ratfun_normalize (PHEAD WORD *term)
- WORD * flint_rem (PHEAD WORD *a, WORD *b, const WORD must_fit_term)
- void flint_startup_init (void)

4.41.1 Detailed Description

Contains functions to call FLINT interface from the rest of FORM.

Definition in file flintwrap.cc.

4.41.2 Function Documentation

4.41.2.1 flint_final_cleanup_thread()

```
void flint_final_cleanup_thread (
            void )
```

Definition at line 16 of file flintwrap.cc.

4.41.2.2 flint_final_cleanup_master()

```
void flint_final_cleanup_master (
            void )
```

Definition at line 23 of file flintwrap.cc.

4.41.2.3 flint_div()

```
WORD * flint_div (
            PHEAD WORD * a,
            WORD * b,
            const WORD must_fit_term )
```

Definition at line 30 of file flintwrap.cc.

4.41.2.4 flint_factorize_argument()

```
int flint_factorize_argument (
            PHEAD WORD * argin,
            WORD * argout )
```

Definition at line 51 of file flintwrap.cc.

4.41.2.5 flint_factorize_dollar()

```
WORD * flint_factorize_dollar (
            PHEAD WORD * argin )
```

Definition at line 72 of file flintwrap.cc.

4.41.2.6 flint_gcd()

```
WORD * flint_gcd (
            PHEAD WORD * a,
            WORD * b,
            const WORD must_fit_term )
```

Definition at line 91 of file flintwrap.cc.

4.41.2.7 flint_inverse()

```
WORD * flint_inverse (
            PHEAD WORD * a,
            WORD * b )
```

Definition at line 111 of file flintwrap.cc.

4.41.2.8 flint_mul()

```
WORD * flint_mul (
            PHEAD WORD * a,
            WORD * b )
```

Definition at line 131 of file flintwrap.cc.

4.41.2.9 flint_ratfun_add()

```
WORD * flint_ratfun_add (
            PHEAD WORD * t1,
            WORD * t2 )
```

Definition at line 149 of file flintwrap.cc.

4.41.2.10 flint_ratfun_normalize()

```
int flint_ratfun_normalize (
            PHEAD WORD * term )
```

Definition at line 187 of file flintwrap.cc.

4.41.2.11 flint_rem()

```
WORD * flint_rem (
            PHEAD WORD * a,
            WORD * b,
            const WORD must_fit_term )
```

Definition at line 259 of file flintwrap.cc.

4.41.2.12 flint_startup_init()

```
void flint_startup_init (
            void )
```

Definition at line 280 of file flintwrap.cc.

4.42 flintwrap.cc

Go to the documentation of this file.

```

extern "C" {
#include "form3.h"
}

#include "flintinterface.h"


/*
    #[ flint_final_cleanup_thread :
*/
void flint_final_cleanup_thread(void) {
    flint::cleanup();
}
/*
    #] flint_final_cleanup_thread :
    #[ flint_final_cleanup_master :
*/
void flint_final_cleanup_master(void) {
    flint::cleanup_master();
}
/*
    #] flint_final_cleanup_master :
    #[ flint_div :
*/
WORD* flint_div(PHEAD WORD *a, WORD *b, const WORD must_fit_term) {
    // Extract expressions
    vector<WORD *> e;
    e.push_back(a);
    e.push_back(b);
    const bool with_arghead = false;
    const bool sort_vars = false;
    const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);

    const bool return_rem = false;
    if ( var_map.size() > 1 ) {
        return flint::divmod_mpoly(BHEAD a, b, return_rem, must_fit_term, var_map);
    }
    else {
        return flint::divmod_poly(BHEAD a, b, return_rem, must_fit_term, var_map);
    }
}
/*
    #] flint_div :
    #[ flint_factorize_argument :
*/
int flint_factorize_argument(PHEAD WORD *argin, WORD *argout) {

    const bool with_arghead = true;
    const bool sort_vars = true;
    const flint::var_map_t var_map = flint::get_variables(vector<WORD*>(1,argin), with_arghead,
        sort_vars);
```

```
    const bool is_fun_arg = true;
    if ( var_map.size() > 1 ) {
        flint::factorize_mpoly(BHEAD argin, argout, with_arghead, is_fun_arg, var_map);
    }
    else {
        flint::factorize_poly(BHEAD argin, argout, with_arghead, is_fun_arg, var_map);
    }

    return 0;
}
/*
    #] flint_factorize_argument :
    #[ flint_factorize_dollar :
*/
WORD* flint_factorize_dollar(PHEAD WORD *argin) {

    const bool with_arghead = false;
    const bool sort_vars = true;
    const flint::var_map_t var_map = flint::get_variables(vector<WORD*>(1,argin), with_arghead,
        sort_vars);

    const bool is_fun_arg = false;
    if ( var_map.size() > 1 ) {
        return flint::factorize_mpoly(BHEAD argin, NULL, with_arghead, is_fun_arg, var_map);
    }
    else {
        return flint::factorize_poly(BHEAD argin, NULL, with_arghead, is_fun_arg, var_map);
    }
}
/*
    #] flint_factorize_dollar :
    #[ flint_gcd :
*/
WORD* flint_gcd(PHEAD WORD *a, WORD *b, const WORD must_fit_term) {
    // Extract expressions
    vector<WORD *> e;
    e.push_back(a);
    e.push_back(b);
    const bool with_arghead = false;
    const bool sort_vars = true;
    const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);

    if ( var_map.size() > 1 ) {
        return flint::gcd_mpoly(BHEAD a, b, must_fit_term, var_map);
    }
    else {
        return flint::gcd_poly(BHEAD a, b, must_fit_term, var_map);
    }
}
/*
    #] flint_gcd :
    #[ flint_inverse :
*/
WORD* flint_inverse(PHEAD WORD *a, WORD *b) {
    // Extract expressions
    vector<WORD *> e;
    e.push_back(a);
    e.push_back(b);
    const flint::var_map_t var_map = flint::get_variables(e, false, false);

    if ( var_map.size() > 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint_inverse: error: only univariate polynomials are supported.");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    return flint::inverse_poly(BHEAD a, b, var_map);
}
/*
    #] flint_inverse :
    #[ flint_mul :
*/
WORD* flint_mul(PHEAD WORD *a, WORD *b) {
    // Extract expressions
    vector<WORD *> e;
    e.push_back(a);
    e.push_back(b);
    const flint::var_map_t var_map = flint::get_variables(e, false, false);

    if ( var_map.size() > 1 ) {
        return flint::mul_mpoly(BHEAD a, b, var_map);
    }
    else {
        return flint::mul_poly(BHEAD a, b, var_map);
    }
```

```
}
/*
    #] flint_mul :
    #[ flint_ratfun_add :
*/
WORD* flint_ratfun_add(PHEAD WORD *t1, WORD *t2) {

    if ( AR.PolyFunExp == 1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("flint_ratfun_add: PolyFunExp unimplemented.");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    WORD *oldworkpointer = AT.WorkPointer;

    // Extract expressions: the num and den of both prf
    vector<WORD *> e;
    for (WORD *t=t1+FUNHEAD; t<t1+t1[1];) {
        e.push_back(t);
        NEXTARG(t);
    }
    for (WORD *t=t2+FUNHEAD; t<t2+t2[1];) {
        e.push_back(t);
        NEXTARG(t);
    }
    const bool with_arghead = true;
    const bool sort_vars = true;
    const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);

    if ( var_map.size() > 1 ) {
        flint::ratfun_add_mpoly(BHEAD t1, t2, oldworkpointer, var_map);
    }
    else {
        flint::ratfun_add_poly(BHEAD t1, t2, oldworkpointer, var_map);
    }

    return oldworkpointer;
}
/*
    #] flint_ratfun_add :
    #[ flint_ratfun_normalize :
*/
int flint_ratfun_normalize(PHEAD WORD *term) {

    // The length of the coefficient
    const WORD ncoeff = (term + *term)[-1];
    // The end of the term data, before the coefficient:
    const WORD *tstop = term + *term - ABS(ncoeff);

    // Search the term for multiple PolyFun or one dirty one.
    unsigned num_polyratfun = 0;
    for (WORD *t = term+1; t < tstop; t += t[1]) {
        if (*t == AR.PolyFun) {
            // Found one!
            num_polyratfun++;
            if ((t[2] & MUSTCLEANPRF) != 0) {
                // Dirty, increment again to force normalisation for single PolyFun
                num_polyratfun++;
            }
            if (num_polyratfun > 1) {
                // We're not counting all occurrences, just determinining if there
                // is something to be done.
                break;
            }
        }
    }
    if (num_polyratfun <= 1) {
        // There is nothing to do, return early
        return 0;
    }

    // When there are polyratfun with only one argument: rename them temporarily to TMPPOLYFUN.
    for (WORD *t = term+1; t < tstop; t += t[1]) {
        if (*t == AR.PolyFun && (t[1] == FUNHEAD+t[FUNHEAD] || t[1] == FUNHEAD+2 ) ) {
            *t = TMPPOLYFUN;
        }
    }


    // Extract all variables in the polyfuns
    // Collect pointers to each relevant argument
    vector<WORD *> e;
    for (WORD *t=term+1; t<tstop; t+=t[1]) {
        if (*t == AR.PolyFun) {
            for (WORD *t2 = t+FUNHEAD; t2<t+t[1];) {
                e.push_back(t2);
```

```
                NEXTARG(t2);
            }
        }
    }
    const bool with_arghead = true;
    const bool sort_vars = true;
    const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);

    if ( var_map.size() > 1 ) {
        flint::ratfun_normalize_mpoly(BHEAD term, var_map);
    }
    else {
        flint::ratfun_normalize_poly(BHEAD term, var_map);
    }


    // Undo renaming of single-argument PolyFun
    const WORD *new_tstop = term + *term - ABS((term + *term)[-1]);
    for (WORD *t=term+1; t<new_tstop; t+=t[1]) {
        if (*t == TMPPOLYFUN ) *t = AR.PolyFun;
    }

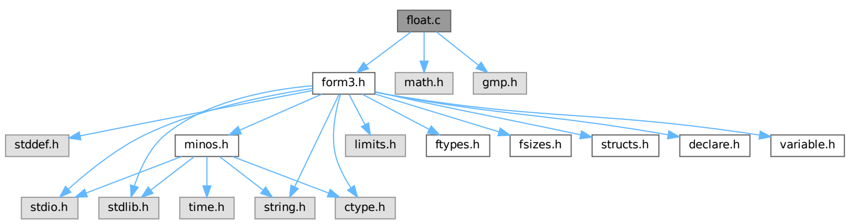
    return 0;
}
/*
    #] flint_ratfun_normalize :
    #[ flint_rem :
*/
WORD* flint_rem(PHEAD WORD *a, WORD *b, const WORD must_fit_term) {
    // Extract expressions
    vector<WORD *> e;
    e.push_back(a);
    e.push_back(b);
    const bool with_arghead = false;
    const bool sort_vars = false;
    const flint::var_map_t var_map = flint::get_variables(e, with_arghead, sort_vars);

    const bool return_rem = true;
    if ( var_map.size() > 1 ) {
        return flint::divmod_mpoly(BHEAD a, b, return_rem, must_fit_term, var_map);
    }
    else {
        return flint::divmod_poly(BHEAD a, b, return_rem, must_fit_term, var_map);
    }
}
/*
    #] flint_rem :
    #[ flint_startup_init :
*/
void flint_startup_init(void) {
    flint::startup_init();
}
/*
    #] flint_startup_init :
*/
```

4.43 float.c File Reference

```
#include "form3.h"
#include <math.h>
#include <gmp.h>
```

Include dependency graph for float.c:



Macros

- #define WITHCUTOFF
- #define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)

Functions

- void Form_mpf_init (mpf_t t)
- void Form_mpf_clear (mpf_t t)
- void Form_mpf_set_prec_raw (mpf_t t, ULONG newprec)
- void FormtoZ (mpz_t z, UWORD ∗a, WORD na)
- void ZtoForm (UWORD ∗a, WORD ∗na, mpz_t z)
- long FloatToInteger (UWORD ∗out, mpf_t floatin, long ∗bitsused)
- void IntegerToFloat (mpf_t result, UWORD ∗formlong, int longsize)
- int FloatToRat (UWORD ∗ratout, WORD ∗nratout, mpf_t floatin)
- int AddFloats (PHEAD WORD ∗fun3, WORD ∗fun1, WORD ∗fun2)
- int MulFloats (PHEAD WORD ∗fun3, WORD ∗fun1, WORD ∗fun2)
- int DivFloats (PHEAD WORD ∗fun3, WORD ∗fun1, WORD ∗fun2)
- int AddRatToFloat (PHEAD WORD ∗outfun, WORD ∗infun, UWORD ∗formrat, WORD nrat)
- int MulRatToFloat (PHEAD WORD ∗outfun, WORD ∗infun, UWORD ∗formrat, WORD nrat)
- void SimpleDelta (mpf_t sum, int m)
- void SimpleDeltaC (mpf_t sum, int m)
- void SingleTable (mpf_t ∗tabl, int N, int m, int pow)
- void DoubleTable (mpf_t ∗tabout, mpf_t ∗tabin, int N, int m, int pow)
- void EndTable (mpf_t sum, mpf_t ∗tabin, int N, int m, int pow)
- void deltaMZV (mpf_t, int ∗, int)
- void deltaEuler (mpf_t, int ∗, int)
- void deltaEulerC (mpf_t, int ∗, int)
- void CalculateMZVhalf (mpf_t, int ∗, int)
- void CalculateMZV (mpf_t, int ∗, int)
- void CalculateEuler (mpf_t, int ∗, int)
- int ExpandMZV (WORD ∗term, WORD level)
- int ExpandEuler (WORD ∗term, WORD level)
- int PackFloat (WORD ∗, mpf_t)
- int UnpackFloat (mpf_t, WORD ∗)
- void RatToFloat (mpf_t result, UWORD ∗formrat, int ratsize)
- void Form_mpf_empty (mpf_t t)
- int TestFloat (WORD ∗fun)
- WORD FloatFunToRat (PHEAD UWORD ∗ratout, WORD ∗in)
- void ZeroTable (mpf_t ∗tab, int N)
- SBYTE ∗ ReadFloat (SBYTE ∗s)
- UBYTE ∗ CheckFloat (UBYTE ∗ss, int ∗spec)
- int SetFloatPrecision (WORD prec)
- int PrintFloat (WORD ∗fun, int numdigits)
- void SetupMZVTables (void)
- void SetupMPFTables (void)
- void ClearMZVTables (void)
- int CoToFloat (UBYTE ∗s)
- int CoToRat (UBYTE ∗s)
- int ToFloat (PHEAD WORD ∗term, WORD level)
- int ToRat (PHEAD WORD ∗term, WORD level)
- WORD AddWithFloat (PHEAD WORD ∗∗ps1, WORD ∗∗ps2)
- WORD MergeWithFloat (PHEAD WORD ∗∗interm1, WORD ∗∗interm2)
- int EvaluateEuler (PHEAD WORD ∗term, WORD level, WORD par)
- int CoEvaluate (UBYTE ∗s)

4.43.1 Detailed Description

This file contains numerical routines that deal with floating point numbers. We use the GMP for arbitrary precision floating point arithmetic. The functions of the type sin, cos, ln, sqrt etc are handled by the mpfr library. The reason that for MZV's and the general notation we use the GMP (mpf_) is because the contents of the structs have been frozen and can be used for storage in a Form function float_. With mpfr_ this is not safely possible. All mpfr_ related code is in the file evaluate.c.

Definition in file float.c.

4.43.2 Macro Definition Documentation

4.43.2.1 WITHCUTOFF

```
#define WITHCUTOFF
```

Definition at line 45 of file float.c.

4.43.2.2 GMPSPREAD

```
#define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)
```

Definition at line 47 of file float.c.

4.43.3 Function Documentation

4.43.3.1 Form_mpf_init()

```
void Form_mpf_init (
            mpf_t t )
```

Definition at line 176 of file float.c.

4.43.3.2 Form_mpf_clear()

```
void Form_mpf_clear (
            mpf_t t )
```

Definition at line 200 of file float.c.

4.43.3.3 Form_mpf_set_prec_raw()

```
void Form_mpf_set_prec_raw (
            mpf_t t,
            ULONG newprec )
```

Definition at line 226 of file float.c.

4.43.3.4 FormtoZ()

```
void FormtoZ (
            mpz_t z,
            UWORD * a,
            WORD na )
```

Definition at line 511 of file float.c.

4.43.3.5 ZtoForm()

```
void ZtoForm (
            UWORD * a,
            WORD * na,
            mpz_t z )
```

Definition at line 555 of file float.c.

4.43.3.6 FloatToInteger()

```
long FloatToInteger (
            UWORD * out,
            mpf_t floatin,
            long * bitsused )
```

Definition at line 581 of file float.c.

4.43.3.7 IntegerToFloat()

```
void IntegerToFloat (
            mpf_t result,
            UWORD * formlong,
            int longsize )
```

Definition at line 604 of file float.c.

4.43.3.8 FloatToRat()

```
int FloatToRat (
            UWORD * ratout,
            WORD * nratout,
            mpf_t floatin )
```

Definition at line 668 of file float.c.

4.43.3.9 AddFloats()

```
int AddFloats (
            PHEAD WORD * fun3,
            WORD * fun1,
            WORD * fun2 )
```

Definition at line 978 of file float.c.

4.43.3.10 MulFloats()

```
int MulFloats (
            PHEAD WORD * fun3,
            WORD * fun1,
            WORD * fun2 )
```

Definition at line 994 of file float.c.

4.43.3.11 DivFloats()

```
int DivFloats (
            PHEAD WORD * fun3,
            WORD * fun1,
            WORD * fun2 )
```

Definition at line 1010 of file float.c.

4.43.3.12 AddRatToFloat()

```
int AddRatToFloat (
            PHEAD WORD * outfun,
            WORD * infun,
            UWORD * formrat,
            WORD nrat )
```

Definition at line 1028 of file float.c.

4.43.3.13 MulRatToFloat()

```
int MulRatToFloat (
            PHEAD WORD * outfun,
            WORD * infun,
            UWORD * formrat,
            WORD nrat )
```

Definition at line 1047 of file float.c.

4.43.3.14 SimpleDelta()

```
void SimpleDelta (
            mpf_t sum,
            int m )
```

Definition at line 1680 of file float.c.

4.43.3.15 SimpleDeltaC()

```
void SimpleDeltaC (
            mpf_t sum,
            int m )
```

Definition at line 1727 of file float.c.

4.43.3.16 SingleTable()

```
void SingleTable (
            mpf_t * tabl,
            int N,
            int m,
            int pow )
```

Definition at line 1774 of file float.c.

4.43.3.17 DoubleTable()

```
void DoubleTable (
            mpf_t * tabout,
            mpf_t * tabin,
            int N,
            int m,
            int pow )
```

Definition at line 1826 of file float.c.

4.43.3.18 EndTable()

```
void EndTable (
            mpf_t sum,
            mpf_t * tabin,
            int N,
            int m,
            int pow )
```

Definition at line 1877 of file float.c.

4.43.3.19 deltaMZV()

```
void deltaMZV (
            mpf_t result,
            int * indexes,
            int depth )
```

Definition at line 1922 of file float.c.

4.43.3.20 deltaEuler()

```
void deltaEuler (
            mpf_t result,
            int * indexes,
            int depth )
```

Definition at line 1989 of file float.c.

4.43.3.21 deltaEulerC()

```
void deltaEulerC (
            mpf_t result,
            int * indexes,
            int depth )
```

Definition at line 2045 of file float.c.

4.43.3.22 CalculateMZVhalf()

```
void CalculateMZVhalf (
            mpf_t result,
            int * indexes,
            int depth )
```

Definition at line 2118 of file float.c.

4.43.3.23 CalculateMZV()

```
void CalculateMZV (
            mpf_t result,
            int * indexes,
            int depth )
```

Definition at line 2139 of file float.c.

4.43.3.24 CalculateEuler()

```
void CalculateEuler (
            mpf_t result,
            int * Zindexes,
            int depth )
```

Definition at line 2204 of file float.c.

4.43.3.25 ExpandMZV()

```
int ExpandMZV (
            WORD * term,
            WORD level )
```

Definition at line 2280 of file float.c.

4.43.3.26 ExpandEuler()

```
int ExpandEuler (
            WORD * term,
            WORD level )
```

Definition at line 2292 of file float.c.

4.43.3.27 PackFloat()

```
int PackFloat (
            WORD * fun,
            mpf_t infloat )
```

Definition at line 270 of file float.c.

4.43.3.28 UnpackFloat()

```
int UnpackFloat (
            mpf_t outfloat,
            WORD * fun )
```

Definition at line 364 of file float.c.

4.43.3.29 RatToFloat()

```
void RatToFloat (
            mpf_t result,
            UWORD * formrat,
            int ratsize )
```

Definition at line 620 of file float.c.

4.43.3.30 Form_mpf_empty()

```
void Form_mpf_empty (
            mpf_t t )
```

Definition at line 213 of file float.c.

4.43.3.31 TestFloat()

```
int TestFloat (
            WORD * fun )
```

Definition at line 452 of file float.c.

4.43.3.32 FloatFunToRat()

```
WORD FloatFunToRat (
            PHEAD UWORD * ratout,
            WORD * in )
```

Definition at line 642 of file float.c.

4.43.3.33 ZeroTable()

```
void ZeroTable (
            mpf_t * tab,
            int N )
```

Definition at line 804 of file float.c.

4.43.3.34 ReadFloat()

```
SBYTE * ReadFloat (
            SBYTE * s )
```

Definition at line 822 of file float.c.

4.43.3.35 CheckFloat()

```
UBYTE * CheckFloat (
            UBYTE * ss,
            int * spec )
```

Definition at line 848 of file float.c.

4.43.3.36 SetFloatPrecision()

```
int SetFloatPrecision (
            WORD prec )
```

Definition at line 911 of file float.c.

4.43.3.37 PrintFloat()

```
int PrintFloat (
            WORD * fun,
            int numdigits )
```

Definition at line 940 of file float.c.

4.43.3.38 SetupMZVTables()

```
void SetupMZVTables (
             void )
```

Definition at line 1064 of file float.c.

4.43.3.39 SetupMPFTables()

```
void SetupMPFTables (
             void )
```

Definition at line 1161 of file float.c.

4.43.3.40 ClearMZVTables()

```
void ClearMZVTables (
             void )
```

Definition at line 1215 of file float.c.

4.43.3.41 CoToFloat()

```
int CoToFloat (
             UBYTE * s )
```

Definition at line 1292 of file float.c.

4.43.3.42 CoToRat()

```
int CoToRat (
             UBYTE * s )
```

Definition at line 1314 of file float.c.

4.43.3.43 ToFloat()

```
int ToFloat (
             PHEAD WORD * term,
             WORD level )
```

Definition at line 1338 of file float.c.

4.43.3.44 ToRat()

```
int ToRat (
             PHEAD WORD * term,
             WORD level )
```

Definition at line 1370 of file float.c.

4.43.3.45 AddWithFloat()

```
WORD AddWithFloat (
            PHEAD WORD ** ps1,
            WORD ** ps2 )
```

Definition at line 1439 of file float.c.

4.43.3.46 MergeWithFloat()

```
WORD MergeWithFloat (
            PHEAD WORD ** interm1,
            WORD ** interm2 )
```

Definition at line 1560 of file float.c.

4.43.3.47 EvaluateEuler()

```
int EvaluateEuler (
            PHEAD WORD * term,
            WORD level,
            WORD par )
```

Definition at line 2320 of file float.c.

4.43.3.48 CoEvaluate()

```
int CoEvaluate (
            UBYTE * s )
```

Definition at line 2436 of file float.c.

4.44 float.c

Go to the documentation of this file.

```

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```
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 */
/* #] License : */
/*
    #[ Includes : float.c
*/

#include "form3.h"
#include <math.h>
#include <gmp.h>

#define WITHCUTOFF

#define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)

void Form_mpf_init(mpf_t t);
void Form_mpf_clear(mpf_t t);
void Form_mpf_set_prec_raw(mpf_t t,ULONG newprec);
void FormtoZ(mpz_t z,UWORD *a,WORD na);
void ZtoForm(UWORD *a,WORD *na,mpz_t z);
long FloatToInteger(UWORD *out, mpf_t floatin, long *bitsused);
void IntegerToFloat(mpf_t result, UWORD *formlong, int longsize);
int FloatToRat(UWORD *ratout, WORD *nratout, mpf_t floatin);
int AddFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2);
int MulFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2);
int DivFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2);
int AddRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat);
int MulRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat);
void SimpleDelta(mpf_t sum, int m);
void SimpleDeltaC(mpf_t sum, int m);
void SingleTable(mpf_t *tabl, int N, int m, int pow);
void DoubleTable(mpf_t *tabout, mpf_t *tabin, int N, int m, int pow);
void EndTable(mpf_t sum, mpf_t *tabin, int N, int m, int pow);
void deltaMZV(mpf_t, int *, int);
void deltaEuler(mpf_t, int *, int);
void deltaEulerC(mpf_t, int *, int);
void CalculateMZVhalf(mpf_t, int *, int);
void CalculateMZV(mpf_t, int *, int);
void CalculateEuler(mpf_t, int *, int);
int ExpandMZV(WORD *term, WORD level);
int ExpandEuler(WORD *term, WORD level);
int PackFloat(WORD *,mpf_t);
int UnpackFloat(mpf_t, WORD *);
void RatToFloat(mpf_t result, UWORD *formrat, int ratsize);

/*
    #] Includes :
    #[ Some GMP float info :

    From gmp.h:

    typedef struct
    {
      int _mp_prec;     Max precision, in number of `mp_limb_t's.
                        Set by mpf_init and modified by
                        mpf_set_prec.  The area pointed to by the
                        _mp_d field contains `prec' + 1 limbs.
      int _mp_size;     abs(_mp_size) is the number of limbs the
                        last field points to.  If _mp_size is
                        negative this is a negative number.
      mp_exp_t _mp_exp; Exponent, in the base of `mp_limb_t'.
      mp_limb_t *_mp_d; Pointer to the limbs.
    } __mpf_struct;

    typedef __mpf_struct mpf_t[1];

    Currently:
        sizeof(int) = 4
        sizeof(mpf_t) = 24
        sizeof(mp_limb_t) = 8,
        sizeof(mp_exp_t) = 8,
        sizeof a pointer is 8.
    If any of this changes in the future we would have to change the
    PackFloat and UnpackFloat routines.

    Our float_ function is packed with the 4 arguments:
        -SNUMBER _mp_prec
        -SNUMBER _mp_size
        exponent which can be -SNUMBER or a regular term
        the limbs as n/1 in regular term format, or just an -SNUMBER.
    During normalization the sign is taken away from the second argument
    and put as the sign of the complete term. This is easier for the
    WriteInnerTerm routine in sch.c

    Wildcarding of functions excludes matches with float_
    Float_ always ends up inside the bracket, just like poly(rat)fun.
```

```

    From mpfr.h

    The formats here are different. This has, among others, to do with
    the rounding. The result is that we need different aux variables
    when working with mpfr and we need a different conversion routine
    to float. It also means that we need to treat the mzv_ etc functions
    completely separately. For now we try to develop that in the file
    Evaluate.c. At a later stage we may merge the two files.
    Originally we had the sqrt in float.c but it seems better to put
    it in Evaluate.c

    #] Some GMP float info :
    #[ Low Level :

        In the low level routines we interact directly with the content
        of the GMP structs. This can be done safely, because their
        layout is documented. We pay particular attention to the init
        and clear routines, because they involve malloc/free calls.

        #[ Explanations :

    We need a number of facilities inside Form to deal with floating point
    numbers, taking into account that they are only meant for sporadic use.
    1: We need a special function float_ who's arguments contain a limb
       representation of the mpf_t of the gmp.
    2: Some functions may return a floating point number. This is then in
       the form of an occurrence of the function float_.
    3: We need a print routine to print this float_.
    4: We need routines for conversions:
        a: (long) int to float.
        b: rat to float.
        c: float to rat (if possible)
        d: float to integer (with rounding toward zero).
    5: We need calculational operations for add, sub, mul, div, exp.
    6: We need service routines to pack and unpack the function float_.
    7: The coefficient should be pulled inside float_.
    8: Float_ cannot occur inside PolyRatFun.
    9: We need to be able to treat float_ as the coefficient during sorting.
    10: For now, we need the functions mzvf_ and eulerf_ for the evaluation
        of mzv's and euler sums.
    11: We need a setting for the default precision.
    12: We need a special flag for the dirty field of arguments to avoid
        the normalization of the argument with the limbs.
        Alternatively, the float_ should be not a regular function, but
        get a status < FUNCTION. Just like SNUMBER, LNUMBER etc.

    Note that we can keep this thread-safe. The only routine that is not
    thread-safe is the print routine, but that is in accordance with all
    other print routines in Form. When called from MesPrint it needs to
    be protected by a lock, as is done standard inside Form.

        #] Explanations :
        #[ Form_mpf_init :
*/

void Form_mpf_init(mpf_t t)
{
    mp_limb_t *d;
    int i, prec;
    if ( t->_mp_d ) { M_free(t->_mp_d,"Form_mpf_init"); t->_mp_d = 0; }
    prec = (AC.DefaultPrecision + 8*sizeof(mp_limb_t)-1)/(8*sizeof(mp_limb_t)) + 1;
    t->_mp_prec = prec;
    t->_mp_size = 0;
    t->_mp_exp = 0;
    d = (mp_limb_t *)Malloc1(prec*sizeof(mp_limb_t),"Form_mpf_init");
    if ( d == 0 ) {
        MesPrint("Fatal error in Malloc1 call in Form_mpf_init. Trying to allocate %ld bytes.",
            prec*sizeof(mp_limb_t));
        Terminate(-1);
    }
    t->_mp_d = d;
    for ( i = 0; i < prec; i++ ) d[i] = 0;
}

/*
        #] Form_mpf_init :
        #[ Form_mpf_clear :
*/

void Form_mpf_clear(mpf_t t)
{
    if ( t->_mp_d ) { M_free(t->_mp_d,"Form_mpf_init"); t->_mp_d = 0; }
    t->_mp_prec = 0;
    t->_mp_size = 0;
    t->_mp_exp = 0;
}
```

```

/*
        #] Form_mpf_clear :
        #[ Form_mpf_empty :
*/

void Form_mpf_empty(mpf_t t)
{
    int i;
    for ( i = 0; i < t->_mp_prec; i++ ) t->_mp_d[i] = 0;
    t->_mp_size = 0;
    t->_mp_exp = 0;
}

/*
        #] Form_mpf_empty :
        #[ Form_mpf_set_prec_raw :
*/

void Form_mpf_set_prec_raw(mpf_t t,ULONG newprec)
{
    ULONG newpr = (newprec + 8*sizeof(mp_limb_t)-1)/(8*sizeof(mp_limb_t)) + 1;
    if ( newpr <= (ULONG)(t->_mp_prec) ) {
        int i, used = ABS(t->_mp_size) ;
        if ( newpr > (ULONG)used ) {
            for ( i = used; i < (int)newpr; i++ ) t->_mp_d[i] = 0;
        }
        if ( t->_mp_size < 0 ) newpr = -newpr;
        t->_mp_size = newpr;
    }
    else {
/*
        We can choose here between "forget about it" or making a new malloc.
        If the program is well designed, we should never get here though.
        For now we choose the "forget it" option.
*/
        MesPrint("Trying too big a precision %ld in Form_mpf_set_prec_raw.",newprec);
        MesPrint("Old maximal precision is %ld.",
            (size_t)(t->_mp_prec-1)*sizeof(mp_limb_t)*8);
        Terminate(-1);
    }
}

/*
        #] Form_mpf_set_prec_raw :
        #[ PackFloat :

    Puts an object of type mpf_t inside a function float_
    float_(prec, nlimbs, exp, limbs);
    Complication: the limbs and the exponent are long's. That means
    two times the size of a Form (U)WORD.
    To save space we could split the limbs in two because Form stores
    coefficients as rationals with equal space for numerator and denominator.
    Hence for each limb we could put the most significant UWORD in the
    numerator and the least significant limb in the denominator.
    This gives a problem with the compilation and subsequent potential
    normalization of the 'fraction'. Hence for now we take the wasteful
    approach. At a later stage we can try to veto normalization of FLOATFUN.
    Caution: gmp/fmp is little endian and Form is big endian.

    Zero is represented by zero limbs (and zero exponent).
*/

int PackFloat(WORD *fun,mpf_t infloat)
{
    WORD *t, nlimbs, num, nnum;
    mp_limb_t *d = infloat->_mp_d; /* Pointer to the limbs.  */
    int i;
    long e = infloat->_mp_exp;

    t = fun;
    *t++ = FLOATFUN;
    t++;
    FILLFUN(t);
    *t++ = -SNUMBER;
    *t++ = infloat->_mp_prec;
    *t++ = -SNUMBER;
    *t++ = infloat->_mp_size;
/*
    Now the exponent which is a signed long
*/
    if ( e < 0 ) {
        e = -e;
        if ( ( e » (BITSINWORD-1) ) == 0 ) {
            *t++ = -SNUMBER; *t++ = -e;
        }
        else {
```

```
                *t++ = ARGHEAD+6; *t++ = 0; FILLARG(t);
                *t++ = 6;
                *t++ = (UWORD)e;
                *t++ = (UWORD)(e » BITSINWORD);
                *t++ = 1; *t++ = 0; *t++ = -5;
            }
        }
        else {
            if ( ( e » (BITSINWORD-1) ) == 0 ) {
                *t++ = -SNUMBER; *t++ = e;
            }
            else {
                *t++ = ARGHEAD+6; *t++ = 0; FILLARG(t);
                *t++ = 6;
                *t++ = (UWORD)e;
                *t++ = (UWORD)(e » BITSINWORD);
                *t++ = 1; *t++ = 0; *t++ = 5;
            }
        }
/*
        and now the limbs. Note that the number of active limbs could be less
        than prec+1 in which case we copy the smaller number.
*/
        nlimbs = infloat->_mp_size < 0 ? -infloat->_mp_size: infloat->_mp_size;
        if ( nlimbs == 0 ) {
        }
        else if ( nlimbs == 1 && (ULONG)(*d) < ((ULONG)1)«(BITSINWORD-1) ) {
            *t++ = -SNUMBER;
            *t++ = (UWORD)(*d);
        }
        else {
          if ( d[nlimbs-1] < ((ULONG)1)«BITSINWORD ) {
            num = 2*nlimbs-1; /* number of UWORDS needed */
          }
          else {
            num = 2*nlimbs;   /* number of UWORDS needed */
          }
          nnum = num;
          *t++ = 2*num+2+ARGHEAD;
          *t++ = 0;
          FILLARG(t);
          *t++ = 2*num+2;
          while ( num > 1 ) {
            *t++ = (UWORD)(*d); *t++ = (UWORD)(((ULONG)(*d))»BITSINWORD); d++;
            num -= 2;
          }
          if ( num == 1 ) { *t++ = (UWORD)(*d); }
          *t++ = 1;
          for ( i = 1; i < nnum; i++ ) *t++ = 0;
          *t++ = 2*nnum+1; /* the sign is hidden in infloat->_mp_size */
        }
        fun[1] = t-fun;
        return(fun[1]);
}

/*
            #] PackFloat :
            #[ UnpackFloat :

        Takes the arguments of a function float_ and puts it inside an mpf_t.
        For notation see commentary for PutInFloat.

        We should make a new allocation for the limbs if the precision exceeds
        the present precision of outfloat, or when outfloat has not been
        initialized yet.

        The return value is -1 if the contents of the FLOATFUN cannot be converted.
        Otherwise the return value is the number of limbs.
*/

int UnpackFloat(mpf_t outfloat,WORD *fun)
{
        UWORD *t;
        WORD *f;
        int num, i;
        mp_limb_t *d;
/*
        Very first step: check whether we can use float_:
*/
        GETIDENTITY
        if ( AT.aux_ == 0 ) {
            MesPrint("Illegal attempt at using a float_ function without proper startup.");
            MesPrint("Please use %#StartFloat <options> first.");
            Terminate(-1);
        }
/*
        Check first the number and types of the arguments
```

```
    There should be 4. -SNUMBER,-SNUMBER,-SNUMBER or a long number.
    + the limbs, either -SNUMBER or Long number in the form of a Form rational.
*/
    if ( TestFloat(fun) == 0 ) goto Incorrect;
    f = fun + FUNHEAD + 2;
    if ( ABS(f[1]) > f[-1]+1 ) goto Incorrect;
    if ( f[1] > outfloat->_mp_prec+1
      || outfloat->_mp_d == 0 ) {
        /*
            We need to make a new allocation.
            Unfortunately we cannot use Malloc1 because that is not
            recognised by gmp and hence we cannot clear with mpf_clear.
            Maybe we can do something about it by making our own
            mpf_init and mpf_clear?
        */
        if ( outfloat->_mp_d != 0 ) free(outfloat->_mp_d);
        outfloat->_mp_d = (mp_limb_t *)malloc((f[1]+1)*sizeof(mp_limb_t));
    }
    num = f[1];
    outfloat->_mp_size = num;
    if ( num < 0 ) { num = -num; }
    f += 2;
    if ( *f == -SNUMBER ) {
        outfloat->_mp_exp = (mp_exp_t)(f[1]);
        f += 2;
    }
    else {
        f += ARGHEAD+6;
        if ( f[-1] == -5 ) {
            outfloat->_mp_exp =
                -(mp_exp_t)((((ULONG)(f[-4]))<<BITSINWORD)+f[-5]);
        }
        else if ( f[-1] == 5 ) {
            outfloat->_mp_exp =
                 (mp_exp_t)((((ULONG)(f[-4]))<<BITSINWORD)+f[-5]);
        }
    }
/*
    And now the limbs if needed
*/
    d = outfloat->_mp_d;
    if ( outfloat->_mp_size == 0 ) {
        for ( i = 0; i < outfloat->_mp_prec; i++ ) *d++ = 0;
        return(0);
    }
    else if ( *f == -SNUMBER ) {
        *d++ = (mp_limb_t)f[1];
        for ( i = 0; i < outfloat->_mp_prec; i++ ) *d++ = 0;
        return(0);
    }
    num = (*f-ARGHEAD-2)/2; /* 2*number of limbs in the argument */
    t = (UWORD *)(f+ARGHEAD+1);
    while ( num > 1 ) {
        *d++ = (mp_limb_t)((((ULONG)(t[1]))<<BITSINWORD)+t[0]);
        t += 2; num -= 2;
    }
    if ( num == 1 ) *d++ = (mp_limb_t)((UWORD)(*t));
    for ( i = d-outfloat->_mp_d; i <= outfloat->_mp_prec; i++ ) *d++ = 0;
    return(0);
Incorrect:
    return(-1);
}

/*
        #] UnpackFloat :
        #[ TestFloat :

        Tells whether the function (with its arguments) is a legal float_.
        We assume, as in the whole float library that one limb is 2 WORDs.
*/

int TestFloat(WORD *fun)
{
    WORD *fstop, *f, num, nnum, i;
    fstop = fun+fun[1];
    f = fun + FUNHEAD;
    num = 0;
/*
    Count arguments
*/
    while ( f < fstop ) { num++; NEXTARG(f); }
    if ( num != 4 ) return(0);
    f = fun + FUNHEAD;
    if ( *f != -SNUMBER ) return(0);
    if ( f[1] < 0 ) return(0);
    f += 2;
    if ( *f != -SNUMBER ) return(0);
```

```
    num = ABS(f[1]); /* number of limbs */
    f += 2;
    if ( *f == -SNUMBER ) { f += 2; }
    else {
        if ( *f != ARGHEAD+6 ) return(0);
        if ( ABS(f[ARGHEAD+5]) != 5 ) return(0);
        if ( f[ARGHEAD+3] != 1 ) return(0);
        if ( f[ARGHEAD+4] != 0 ) return(0);
        f += *f;
    }
    if ( num == 0 ) return(1);
    if ( *f == -SNUMBER ) { if ( num != 1 ) return(0); }
    else {
        nnum = (ABS(f[*f-1])-1)/2;
        if ( ( *f != 4*num+2+ARGHEAD ) && ( *f != 4*num+ARGHEAD ) ) return(0);
        if ( ( nnum != 2*num ) && ( nnum != 2*num-1 ) ) return(0);
        f += ARGHEAD+nnum+1;
        if ( f[0] != 1 ) return(0);
        for ( i = 1; i < nnum; i++ ) {
            if ( f[i] != 0 ) return(0);
        }
    }
    return(1);
}

/*
        #] TestFloat :
        #[ FormtoZ :

        Converts a Form long integer to a GMP long integer.

        typedef struct
        {
          int _mp_alloc;   Number of *limbs* allocated and pointed
                           to by the _mp_d field.
          int _mp_size;    abs(_mp_size) is the number of limbs the
                           last field points to.  If _mp_size is
                           negative this is a negative number.
          mp_limb_t *_mp_d;Pointer to the limbs.
        } __mpz_struct;
        typedef __mpz_struct mpz_t[1];
*/

void FormtoZ(mpz_t z,UWORD *a,WORD na)
{
    int nlimbs, sgn = 1;
    mp_limb_t *d;
    UWORD *b = a;
    WORD nb = na;

    if ( nb == 0 ) {
        z->_mp_size = 0;
        z->_mp_d[0] = 0;
        return;
    }
    if ( nb < 0 ) { sgn = -1; nb = -nb; }
    nlimbs = (nb+1)/2;
    if ( mpz_cmp_si(z,0L) <= 1 ) {
        mpz_set_ui(z,10000000);
    }
    if ( z->_mp_alloc <= nlimbs ) {
        mpz_t zz;
        mpz_init(zz);
        while ( z->_mp_alloc <= nlimbs ) {
            mpz_mul(zz,z,z);
            mpz_set(z,zz);
        }
        mpz_clear(zz);
    }
    z->_mp_size = sgn*nlimbs;
    d = z->_mp_d;
    while ( nb > 1 ) {
        *d++ = (((mp_limb_t)(b[1]))«BITSINWORD)+(mp_limb_t)(b[0]);
        b += 2; nb -= 2;
    }
    if ( nb == 1 ) { *d = (mp_limb_t)(*b); }
}

/*
        #] FormtoZ :
        #[ ZtoForm :

        Converts a GMP long integer to a Form long integer.
        Mainly an exercise of going from little endian ULONGs.
        to big endian UWORDs
*/

```

```
void ZtoForm(UWORD *a,WORD *na,mpz_t z)
{
    mp_limb_t *d = z->_mp_d;
    int nlimbs = ABS(z->_mp_size), i;
    UWORD j;
    if ( nlimbs == 0 ) { *na = 0; return; }
    *na = 0;
    for ( i = 0; i < nlimbs-1; i++ ) {
        *a++ = (UWORD)(*d);
        *a++ = (UWORD)((*d++)»BITSINWORD);
        *na += 2;
    }
    *na += 1;
    *a++ = (UWORD)(*d);
    j = (UWORD)((*d)»BITSINWORD);
    if ( j != 0 ) { *a = j; *na += 1; }
    if ( z->_mp_size < 0 ) *na = -*na;
}

/*
        #] ZtoForm :
        #[ FloatToInteger :

        Converts a GMP float to a long Form integer.
*/

long FloatToInteger(UWORD *out, mpf_t floatin, long *bitsused)
{
    mpz_t z;
    WORD nout, nx;
    UWORD x;
    mpz_init(z);
    mpz_set_f(z,floatin);
    ZtoForm(out,&nout,z);
    mpz_clear(z);
    x = out[0]; nx = 0;
    while ( x ) { nx++; x »= 1; }
    *bitsused = (nout-1)*BITSINWORD + nx;
    return(nout);
}

/*
        #] FloatToInteger :
        #[ IntegerToFloat :

        Converts a Form long integer to a gmp float of default size.
        We assume that sizeof(unsigned long int) = 2*sizeof(UWORD).
*/

void IntegerToFloat(mpf_t result, UWORD *formlong, int longsize)
{
    mpz_t z;
    mpz_init(z);
    FormtoZ(z,formlong,longsize);
    mpf_set_z(result,z);
    mpz_clear(z);
}

/*
        #] IntegerToFloat :
        #[ RatToFloat :

        Converts a Form rational to a gmp float of default size.
*/

void RatToFloat(mpf_t result, UWORD *formrat, int ratsize)
{
    GETIDENTITY
    int nnum, nden;
    UWORD *num, *den;
    int sgn = 0;
    if ( ratsize < 0 ) { ratsize = -ratsize; sgn = 1; }
    nnum = nden = (ratsize-1)/2;
    num = formrat; den = formrat+nnum;
    while ( num[nnum-1] == 0 ) { nnum--; }
    while ( den[nden-1] == 0 ) { nden--; }
    IntegerToFloat(aux2,num,nnum);
    IntegerToFloat(aux3,den,nden);
    mpf_div(result,aux2,aux3);
    if ( sgn > 0 ) mpf_neg(result,result);
}

/*
        #] RatToFloat :
        #[ FloatFunToRat :
*/

```

```
WORD FloatFunToRat(PHEAD UWORD *ratout,WORD *in)
{
    WORD oldin = in[0], nratout;
    in[0] = FLOATFUN;
    UnpackFloat(aux4,in);
    FloatToRat(ratout,&nratout,aux4);
    in[0] = oldin;
    return(nratout);
}

/*
        #] FloatFunToRat :
        #[ FloatToRat :

    Converts a gmp float to a Form rational.
    The algorithm we use is 'repeated fractions' of the style
        n0+1/(n1+1/(n2+1/(n3+.....)))
    The moment we get an n that is very large we should have the proper fraction
    Main problem: what if the sequence keeps going?
                 (there are always rounding errors)
    We need a cutoff with an error condition.
    Basically the product n0*n1*n2*... is an underlimit on the number of
    digits in the answer. We can put a limit on this.
    A return value of zero means that everything went fine.
*/

int FloatToRat(UWORD *ratout, WORD *nratout, mpf_t floatin)
{
    GETIDENTITY
    WORD *out = AT.WorkPointer;
    WORD s; /* the sign. */
    UWORD *a, *b, *c, *d, *mc;
    WORD na, nb, nc, nd, i;
    int nout;
    LONG oldpWorkPointer = AT.pWorkPointer;
    long bitsused = 0, totalbitsused = 0, totalbits = AC.DefaultPrecision;
    int retval = 0, startnul;
    WantAddPointers(AC.DefaultPrecision);  /* Horrible overkill */
    AT.pWorkSpace[AT.pWorkPointer++] = out;
    a = NumberMalloc("FloatToRat");
    b = NumberMalloc("FloatToRat");

    s = mpf_sgn(floatin);
    if ( s < 0 ) mpf_neg(floatin,floatin);

    Form_mpf_empty(aux1);
    Form_mpf_empty(aux2);
    Form_mpf_empty(aux3);

    mpf_trunc(aux1,floatin);      /* This should now be an integer */
    mpf_sub(aux2,floatin,aux1);  /* and this >= 0 */
    if ( mpf_sgn(aux1) == 0 ) { *out++ = 0; startnul = 1; }
    else {
        nout = FloatToInteger((UWORD *)out,aux1,&totalbitsused);
        out += nout;
        startnul = 0;
    }
    AT.pWorkSpace[AT.pWorkPointer++] = out;
    if ( mpf_sgn(aux2) ) {
      for(;;) {
        mpf_ui_div(aux3,1,aux2);
        mpf_trunc(aux1,aux3);
        mpf_sub(aux2,aux3,aux1);
        if ( mpf_sgn(aux1) == 0 ) { *out++ = 0; }
        else {
            nout = FloatToInteger((UWORD *)out,aux1,&bitsused);
            out += nout;
            totalbitsused += bitsused;
        }
        if ( bitsused > (totalbits-totalbitsused)/2 ) { break; }
        if ( mpf_sgn(aux2) == 0 ) {
            /*if ( startnul == 1 )*/ AT.pWorkSpace[AT.pWorkPointer++] = out;
            break;
        }
        AT.pWorkSpace[AT.pWorkPointer++] = out;
      }
/*
      At this point we have the function with the repeated fraction.
      Now we should work out the fraction. Form code would be:
        repeat id   dum_(?a,x1?,x2?) = dum_(?a,x1+1/x2);
        id  dum_(x?) = x;
      We have to work backwards. This is why we made a list of pointers
      in AT.pWorkSpace

      Now we need the long integers a and b.
      Note that by construction there should never be a GCD!
*/
```

```
		out = (WORD *)(AT.pWorkSpace[--AT.pWorkPointer]);
		na = 1; a[0] = 1;
		c = (UWORD *)(AT.pWorkSpace[--AT.pWorkPointer]);
		nc = nb = ((UWORD *)out)-c;
		if ( nc > 10 ) {
			mc = c;
			c = (UWORD *)(AT.pWorkSpace[--AT.pWorkPointer]);
			nc = nb = ((UWORD *)mc)-c;
		}
		for ( i = 0; i < nb; i++ ) b[i] = c[i];
		mc = c = NumberMalloc("FloatToRat");
		while ( AT.pWorkPointer > oldpWorkPointer ) {
			d = (UWORD *)(AT.pWorkSpace[--AT.pWorkPointer]);
			nd = (UWORD *)(AT.pWorkSpace[AT.pWorkPointer+1])-d; /* This is the x1 in the formula */
			if ( nd == 1 && *d == 0 ) break;
			c = a; a = b; b = c; nc = na; na = nb; nb = nc; /* 1/x2 */
			MulLong(d,nd,a,na,mc,&nc);
			AddLong(mc,nc,b,nb,b,&nb);
		}
		NumberFree(mc,"FloatToRat");
		if ( startnul == 0 ) {
			c = a; a = b; b = c; nc = na; na = nb; nb = nc;
		}
	}
	else {
		c = (UWORD *)(AT.pWorkSpace[oldpWorkPointer]);
		na = (UWORD *)out - c;
		for ( i = 0; i < na; i++ ) a[i] = c[i];
		b[0] = 1; nb = 1;
	}
/*
	Now we have the fraction in a/b. Create the rational and add the sign.
*/
	d = ratout;
	if ( na == nb ) {
		*nratout = (2*na+1)*s;
		for ( i = 0; i < na; i++ ) *d++ = a[i];
		for ( i = 0; i < nb; i++ ) *d++ = b[i];
	}
	else if ( na < nb ) {
		*nratout = (2*nb+1)*s;
		for ( i = 0; i < na; i++ ) *d++ = a[i];
		for ( ; i < nb; i++ ) *d++ = 0;
		for ( i = 0; i < nb; i++ ) *d++ = b[i];
	}
	else {
		*nratout = (2*na+1)*s;
		for ( i = 0; i < na; i++ ) *d++ = a[i];
		for ( i = 0; i < nb; i++ ) *d++ = b[i];
		for ( ; i < na; i++ ) *d++ = 0;
	}
	*d = (UWORD)(*nratout);
	NumberFree(b,"FloatToRat");
	NumberFree(a,"FloatToRat");
	AT.pWorkPointer = oldpWorkPointer;
/*
	Just for check we go back to float
*/
	if ( s < 0 ) mpf_neg(floatin,floatin);
/*
	{
		WORD n = *ratout;
		RatToFloat(aux1,ratout,n);
		mpf_sub(aux2,floatin,aux1);
		gmp_printf("Diff is %.*Fe\n",40,aux2);
	}
*/
	return(retval);
}

/*
		#] FloatToRat :
		#[ ZeroTable :
*/

void ZeroTable(mpf_t *tab, int N)
{
	int i, j;
	for ( i = 0; i < N; i++ ) {
		for ( j = 0; j < tab[i]->_mp_prec; j++ ) tab[i]->_mp_d[j] = 0;
	}
}

/*
		#] ZeroTable :
		#[ ReadFloat :

```

```
    Used by the compiler. It reads a floating point number and
    outputs it at AT.WorkPointer as if it were a float_ function
    with its arguments.
    The call enters with s[-1] == TFLOAT.
*/

SBYTE *ReadFloat(SBYTE *s)
{
    GETIDENTITY
    SBYTE *ss, c;
    ss = s;
    while ( *ss > 0 ) ss++;
    c = *ss; *ss = 0;
    gmp_sscanf((char *)s,"%Ff",aux1);
    if ( AT.WorkPointer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    PackFloat(AT.WorkPointer,aux1);
    *ss = c;
    return(ss);
}

/*
        #] ReadFloat :
        #[ CheckFloat :

    For the tokenizer. Tests whether the input string can be a float.
*/

UBYTE *CheckFloat(UBYTE *ss, int *spec)
{
    GETIDENTITY
    UBYTE *s = ss;
    int zero = 1, gotdot = 0;
    while ( FG.cTable[s[-1]] == 1 ) s--;
    *spec = 0;
    if ( FG.cTable[*s] == 1 ) {
        while ( *s == '0' ) s++;
        if ( FG.cTable[*s] == 1 ) {
            s++;
            while ( FG.cTable[*s] == 1 ) s++;
            zero = 0;
        }
        if ( *s == '.' ) { goto dot; }
    }
    else if ( *s == '.' ) {
dot:
        gotdot = 1;
        s++;
        if ( FG.cTable[*s] != 1 && zero == 1 ) return(ss);
        while ( *s == '0' ) s++;
        if ( FG.cTable[*s] == 1 ) {
            s++;
            while ( FG.cTable[*s] == 1 ) s++;
            zero = 0;
        }
    }
    else return(ss);
/*
    Now we have had the mantissa part, which may be zero.
    Check for an exponent.
*/
    if ( *s == 'e' || *s == 'E' ) {
        s++;
        if ( *s == '-' || *s == '+' ) s++;
        if ( FG.cTable[*s] == 1 ) {
            s++;
            while ( FG.cTable[*s] == 1 ) s++;
        }
        else { return(ss); }
    }
    else if ( gotdot == 0 ) return(ss);
    if ( AT.aux_ == 0 ) { /* no floating point system */
        *spec = -1;
        return(s);
    }
    if ( zero ) *spec = 1;
    return(s);
}

/*
        #] CheckFloat :
    #] Low Level :
    #[ Float Routines :
```

```
        #[ SetFloatPrecision :

        We set the default precision of the floats and allocate
        space for an output string if we want to write the float.
        Space needed: exponent: up to 12 chars.
        mantissa 2+10*prec/33 + a little bit extra.
*/

int SetFloatPrecision(WORD prec)
{
    if ( prec <= 0 ) {
        MesPrint("&Illegal value for number of bits for floating point constants: %d",prec);
        return(-1);
    }
    else {
        AC.DefaultPrecision = prec;
        if ( AO.floatspace != 0 ) M_free(AO.floatspace,"floatspace");
        AO.floatsize = ((10*prec)/33+20)*sizeof(char);
        AO.floatspace = (UBYTE *)Malloc1(AO.floatsize,"floatspace");
        mpf_set_default_prec(prec);
        return(0);
    }
}

/*
        #] SetFloatPrecision :
        #[ PrintFloat :

    Print the float_ function with its arguments as a floating point
    number with numdigits digits.
    If however we use rawfloat as a format option it prints it as a
    regular Form function and it should never come here because the
    print routines in sch.c should intercept it.
    The buffer AO.floatspace is allocated when the precision of the
    floats is set in the routine SetupMZVTables.
*/

int PrintFloat(WORD *fun,int numdigits)
{
    GETIDENTITY
    int n = 0;
    int prec = (AC.DefaultPrecision-AC.MaxWeight-1)*log10(2.0);
    if ( numdigits > prec || numdigits == 0 ) {
        numdigits = prec;
    }
/*
    GMP's gmp_snprintf always prints a non-zero number before the decimal point, so we
    ask for one digit less.
*/
    if ( UnpackFloat(aux4,fun) == 0 )
        n = gmp_snprintf((char *)(AO.floatspace),AO.floatsize,"%.*Fe",numdigits-1,aux4);
    if ( numdigits == prec ) {
        UBYTE *s1, *s2;
        int n1, n2 = n;
        s1 = AO.floatspace+n;
        while ( s1 > AO.floatspace && s1[-1] != 'e'
         && s1[-1] != 'E' && s1[-1] != '.' ) { s1--; n2--; }
        if ( s1 > AO.floatspace && s1[-1] != '.' ) {
            s1--; n2--;
            s2 = s1; n1 = n2;
            while ( s1[-1] == '0' ) { s1--; n1--; }
            if ( s1[-1] == '.' ) { s1++; n1++; }
            while ( n2 < n ) { *s1++ = *s2++; n2++; }
            n -= (n2-n1);
            *s1 = 0;
        }
    }
    return(n);
}

/*
        #] PrintFloat :
        #[ AddFloats :
*/

int AddFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2)
{
    int retval = 0;
    if ( UnpackFloat(aux1,fun1) == 0 && UnpackFloat(aux2,fun2) == 0 ) {
        mpf_add(aux1,aux1,aux2);
        PackFloat(fun3,aux1);
    }
    else { retval = -1; }
    return(retval);
}

/*
```

```
        #] AddFloats :
        #[ MulFloats :
*/

int MulFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2)
{
    int retval = 0;
    if ( UnpackFloat(aux1,fun1) == 0 && UnpackFloat(aux2,fun2) == 0 ) {
        mpf_mul(aux1,aux1,aux2);
        PackFloat(fun3,aux1);
    }
    else { retval = -1; }
    return(retval);
}

/*
        #] MulFloats :
        #[ DivFloats :
*/

int DivFloats(PHEAD WORD *fun3, WORD *fun1, WORD *fun2)
{
    int retval = 0;
    if ( UnpackFloat(aux1,fun1) == 0 && UnpackFloat(aux2,fun2) == 0 ) {
        mpf_div(aux1,aux1,aux2);
        PackFloat(fun3,aux1);
    }
    else { retval = -1; }
    return(retval);
}

/*
        #] DivFloats :
        #[ AddRatToFloat :

        Note: this can be optimized, because often the rat is rather simple.
*/

int AddRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat)
{
    int retval = 0;
    if ( UnpackFloat(aux2,infun) == 0 ) {
        RatToFloat(aux1,formrat,nrat);
        mpf_add(aux2,aux2,aux1);
        PackFloat(outfun,aux2);
    }
    else retval = -1;
    return(retval);
}

/*
        #] AddRatToFloat :
        #[ MulRatToFloat :

        Note: this can be optimized, because often the rat is rather simple.
*/

int MulRatToFloat(PHEAD WORD *outfun, WORD *infun, UWORD *formrat, WORD nrat)
{
    int retval = 0;
    if ( UnpackFloat(aux2,infun) == 0 ) {
        RatToFloat(aux1,formrat,nrat);
        mpf_mul(aux2,aux2,aux1);
        PackFloat(outfun,aux2);
    }
    else retval = -1;
    return(retval);
}

/*
        #] MulRatToFloat :
        #[ SetupMZVTables :
*/

void SetupMZVTables(void)
{
/*
    Sets up a table of N+1 mpf_t floats with variable precision.
    It assumes that each next element needs one bit less.
    Initializes all of them.
    We take some extra space, to have one limb overflow everywhere.
    Because the depth of the MZV's can get close to the weight
    and each deeper sum goes one higher, we make the tablesize a bit bigger.
    This may not be needed if we fiddle with the sum boundaries.
*/
#ifdef WITHPTHREADS
    int i, Nw, id, totnum;
```

```
    size_t N;
    mpf_t *a;
    Nw = AC.DefaultPrecision;
    N = (size_t)Nw;
    totnum = AM.totalnumberofthreads;
    for ( id = 0; id < totnum; id++ ) {
        if ( AB[id]->T.mpf_tab1 ) {
            a = (mpf_t *)AB[id]->T.mpf_tab1;
            for ( i = 0; i <=Nw; i++ ) {
                mpf_clear(a[i]);
            }
            M_free(AB[id]->T.mpf_tab1,"mpftab1");
        }
        AB[id]->T.mpf_tab1 = (void *)Malloc1((N+2)*sizeof(mpf_t),"mpftab1");
        a = (mpf_t *)AB[id]->T.mpf_tab1;
        for ( i = 0; i <=Nw; i++ ) {
/*
    As explained in the comment above, we could make this variable precision
    using mpf_init2.
*/
            mpf_init(a[i]);
        }
        if ( AB[id]->T.mpf_tab2 ) {
            a = (mpf_t *)AB[id]->T.mpf_tab2;
            for ( i = 0; i <=Nw; i++ ) {
                mpf_clear(a[i]);
            }
            M_free(AB[id]->T.mpf_tab2,"mpftab2");
        }
        AB[id]->T.mpf_tab2 = (void *)Malloc1((N+2)*sizeof(mpf_t),"mpftab2");
        a = (mpf_t *)AB[id]->T.mpf_tab2;
        for ( i = 0; i <=Nw; i++ ) {
            mpf_init(a[i]);
        }
    }
#else
    int i, Nw;
    size_t N;
    Nw = AC.DefaultPrecision;
    N = (size_t)Nw;
    if ( AT.mpf_tab1 ) {
        for ( i = 0; i <= Nw; i++ ) {
            mpf_clear(mpftab1[i]);
        }
        M_free(AT.mpf_tab1,"mpftab1");
    }
    AT.mpf_tab1 = (void *)Malloc1((N+2)*sizeof(mpf_t),"mpftab1");
    for ( i = 0; i <= Nw; i++ ) {
/*
    As explained in the comment above, we could make this variable precision
    using mpf_init2.
*/
        mpf_init(mpftab1[i]);
    }
    if ( AT.mpf_tab2 ) {
        for ( i = 0; i <= Nw; i++ ) {
            mpf_clear(mpftab2[i]);
        }
        M_free(AT.mpf_tab2,"mpftab2");
    }
    AT.mpf_tab2 = (void *)Malloc1((N+2)*sizeof(mpf_t),"mpftab2");
    for ( i = 0; i <= Nw; i++ ) {
        mpf_init(mpftab2[i]);
    }
#endif
    if ( AS.delta_1 ) {
        mpf_clear(mpfdelta1);
        M_free(AS.delta_1,"delta1");
    }
    AS.delta_1 = (void *)Malloc1(sizeof(mpf_t),"delta1");
    mpf_init(mpfdelta1);
    SimpleDelta(mpfdelta1,1); /* this can speed up things. delta1 = ln(2) */
/*
    Finally the character buffer for printing
    if ( AO.floatspace ) M_free(AO.floatspace,"floatspace");
    AO.floatspace = (UBYTE *)Malloc1(((10*AC.DefaultPrecision)/33+40)*sizeof(UBYTE),"floatspace");
*/
}

/*
        #] SetupMZVTables :
        #[ SetupMPFTables :
*/

void SetupMPFTables(void)
{
#ifdef WITHPTHREADS
```

```
    int id, totnum;
    mpf_t *a;
/*
    Now the aux variables
*/
#ifdef WITHSORTBOTS
    totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
#endif
    for ( id = 0; id < totnum; id++ ) {
        if ( AB[id]->T.aux_ ) {
/*
            We work here with a[0] etc because the aux1 etc contain B which
            in the current routine would be AB[0] only
*/
            a = (mpf_t *)AB[id]->T.aux_;
            mpf_clears(a[0],a[1],a[2],a[3],a[4],a[5],a[6],a[7],(mpf_ptr)0);
            M_free(AB[id]->T.aux_,"AB[id]->T.aux_");
        }
        AB[id]->T.aux_ = (void *)Malloc1(sizeof(mpf_t)*8,"AB[id]->T.aux_");
        a = (mpf_t *)AB[id]->T.aux_;
        mpf_inits(a[0],a[1],a[2],a[3],a[4],a[5],a[6],a[7],(mpf_ptr)0);
        if ( AB[id]->T.indi1 ) M_free(AB[id]->T.indi1,"indi1");
        AB[id]->T.indi1 = (int *)Malloc1(sizeof(int)*AC.MaxWeight*2,"indi1");
        AB[id]->T.indi2 = AB[id]->T.indi1 + AC.MaxWeight;
    }
#else
/*
    Now the aux variables
*/
    if ( AT.aux_ ) {
        mpf_clears(aux1,aux2,aux3,aux4,aux5,auxjm,auxjjm,auxsum,(mpf_ptr)0);
        M_free(AT.aux_,"AT.aux");
    }
    AT.aux_ = (void *)Malloc1(sizeof(mpf_t)*8,"AT.aux_");
    mpf_inits(aux1,aux2,aux3,aux4,aux5,auxjm,auxjjm,auxsum,(mpf_ptr)0);
    if ( AT.indi1 ) M_free(AT.indi1,"indi1");
    AT.indi1 = (int *)Malloc1(sizeof(int)*AC.MaxWeight*2,"indi1");
    AT.indi2 = AT.indi1 + AC.MaxWeight;
#endif
/*
    Finally the character buffer for printing
    if ( AO.floatspace ) M_free(AO.floatspace,"floatspace");
    AO.floatspace = (UBYTE *)Malloc1(((10*AC.DefaultPrecision)/33+40)*sizeof(UBYTE),"floatspace");
*/
}

/*
        #] SetupMPFTables :
        #[ ClearMZVTables :
*/

void ClearMZVTables(void)
{
#ifdef WITHPTHREADS
    int i, id, totnum;
    mpf_t *a;
    totnum = AM.totalnumberofthreads;
    for ( id = 0; id < totnum; id++ ) {
        if ( AB[id]->T.mpf_tab1 ) {
/*
            We work here with a[0] etc because the aux1, mpftab1 etc contain B
            which in the current routine would be AB[0] only
*/
            a = (mpf_t *)AB[id]->T.mpf_tab1;
            for ( i = 0; i <=AC.DefaultPrecision; i++ ) {
                mpf_clear(a[i]);
            }
            M_free(AB[id]->T.mpf_tab1,"mpftab1");
            AB[id]->T.mpf_tab1 = 0;
        }
        if ( AB[id]->T.mpf_tab2 ) {
            a = (mpf_t *)AB[id]->T.mpf_tab2;
            for ( i = 0; i <=AC.DefaultPrecision; i++ ) {
                mpf_clear(a[i]);
            }
            M_free(AB[id]->T.mpf_tab2,"mpftab2");
            AB[id]->T.mpf_tab2 = 0;
        }
    }
#ifdef WITHSORTBOTS
    totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
#endif
    for ( id = 0; id < totnum; id++ ) {
        if ( AB[id]->T.aux_ ) {
            a = (mpf_t *)AB[id]->T.aux_;
            mpf_clears(a[0],a[1],a[2],a[3],a[4],a[5],a[6],a[7],(mpf_ptr)0);
            M_free(AB[id]->T.aux_,"AB[id]->T.aux_");
```

```
            AB[id]->T.aux_ = 0;
        }
        if ( AB[id]->T.indi1 ) { M_free(AB[id]->T.indi1,"indi1"); AB[id]->T.indi1 = 0; }
    }
#else
    int i;
    if ( AT.mpf_tab1 ) {
        for ( i = 0; i <= AC.DefaultPrecision; i++ ) {
            mpf_clear(mpftab1[i]);
        }
        M_free(AT.mpf_tab1,"mpftab1");
        AT.mpf_tab1 = 0;
    }
    if ( AT.mpf_tab2 ) {
        for ( i = 0; i <= AC.DefaultPrecision; i++ ) {
            mpf_clear(mpftab2[i]);
        }
        M_free(AT.mpf_tab2,"mpftab2");
        AT.mpf_tab2 = 0;
    }
    if ( AT.aux_ ) {
        mpf_clears(aux1,aux2,aux3,aux4,aux5,auxjm,auxjjm,auxsum,(mpf_ptr)0);
        M_free(AT.aux_,"AT.aux_");
        AT.aux_ = 0;
    }
    if ( AT.indi1 ) { M_free(AT.indi1,"indi1"); AT.indi1 = 0; }
#endif
    if ( AO.floatspace ) { M_free(AO.floatspace,"floatspace"); AO.floatspace = 0;
        AO.floatsize = 0; }
    if ( AS.delta_1 ) {
        mpf_clear(mpfdelta1);
        M_free(AS.delta_1,"delta1");
        AS.delta_1 = 0;
    }
}

/*
        #] ClearMZVTables :
        #[ CoToFloat :
*/

int CoToFloat(UBYTE *s)
{
    GETIDENTITY
    if ( AT.aux_ == 0 ) {
        MesPrint("&Illegal attempt to convert to float_ without activating floating point numbers.");
        MesPrint("&Forgotten %#startfloat instruction?");
        return(1);
    }
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( *s ) {
        MesPrint("&Illegal argument(s) in ToFloat statement: '%s'",s);
        return(1);
    }
    Add2Com(TYPETOFLOAT);
    return(0);
}

/*
        #] CoToFloat :
        #[ CoToRat :
*/

int CoToRat(UBYTE *s)
{
    GETIDENTITY
    if ( AT.aux_ == 0 ) {
        MesPrint("&Illegal attempt to convert from float_ without activating floating point
numbers.");
        MesPrint("&Forgotten %#startfloat instruction?");
        return(1);
    }
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( *s ) {
        MesPrint("&Illegal argument(s) in ToRat statement: '%s'",s);
        return(1);
    }
    Add2Com(TYPETORAT);
    return(0);
}

/*
        #] CoToRat :
        #[ ToFloat :

        Converts the coefficient to floating point if it is still a rat.
*/
```

```
int ToFloat(PHEAD WORD *term, WORD level)
{
    WORD *t, *tstop, nsize, nsign = 3;
    t = term+*term;
    nsize = ABS(t[-1]);
    tstop = t - nsize;
    if ( t[-1] < 0 ) { nsign = -nsign; }
    if ( nsize == 3 && t[-2] == 1 && t[-3] == 1 ) { /* Could be float */
        t = term + 1;
        while ( t < tstop ) {
            if ( ( *t == FLOATFUN ) && ( t+t[1] == tstop ) ) {
                return(Generator(BHEAD term,level));
            }
            t += t[1];
        }
    }
    RatToFloat(aux4,(UWORD *)tstop,nsize);
    PackFloat(tstop,aux4);
    tstop += tstop[1];
    *tstop++ = 1; *tstop++ = 1; *tstop++ = nsign;
    *term = tstop - term;
    AT.WorkPointer = tstop;
    return(Generator(BHEAD term,level));
}

/*
        #] ToFloat :
        #[ ToRat :

        Tries to convert the coefficient to rational if it is still a float.
*/

int ToRat(PHEAD WORD *term, WORD level)
{
    WORD *tstop, *t, nsize, nsign, ncoef;
/*
    1: find the float which should be at the end.
*/
    tstop = term + *term; nsize = ABS(tstop[-1]);
    nsign = tstop[-1] < 0 ? -1: 1; tstop -= nsize;
    t = term+1;
    while ( t < tstop ) {
        if ( *t == FLOATFUN && t + t[1] == tstop && TestFloat(t) &&
        nsize == 3 && tstop[0] == 1 && tstop[1] == 1 ) break;
        t += t[1];
    }
    if ( t < tstop ) {
/*
        Now t points at the float_ function and everything is correct.
        The result can go straight over the float_ function.
*/
        UnpackFloat(aux4,t);
        if ( FloatToRat((UWORD *)t,&ncoef,aux4) == 0 ) {
            t += ABS(ncoef);
            t[-1] = ncoef*nsign;
            *term = t - term;
        }
    }
    return(Generator(BHEAD term,level));
}

/*
        #] ToRat :
    #] Float Routines :
    #[ Sorting :

        We need a method to see whether two terms need addition that could
        involve floating point numbers.
        1: if PolyWise is active, we do not need anything, because
           Poly(Rat)Fun and floating point are mutually exclusive.
        2: This means that there should be only interference in AddCoef.
           and the AddRat parts in MergePatches, MasterMerge, SortBotMerge
           and PF_GetLoser.
        The compare routine(s) should recognise the float_ and give off
        a code (SortFloatMode) inside the AT struct:
        0: no float_
        1: float_ in term1 only
        2: float_ in term2 only
        3: float_ in both terms
        Be careful: we have several compare routines, because various codes
        use their own routines and we just set a variable with its address.
        Currently we have Compare1, CompareSymbols and CompareHSymbols.
        Only Compare1 should be active for this. The others should make sure
        that the proper variable is always zero.
        Remember: the sign of the coefficient is in the last word of the term.
```

```
        We need two routines:
        1: AddWithFloat for SplitMerge which sorts by pointer.
        2: MergeWithFloat for MergePatches etc which keeps terms as much
           as possible in their location.
        The routines start out the same, because AT.SortFloatMode, set in
        Compare1, tells more or less what should be done.
        The difference is in where we leave the result.

        In the future we may want to add a feature that makes the result
        zero if the sum comes within a certain distance of the numerical
        accuracy, like for instance 5% of the number of bits.

        #[ AddWithFloat :
*/

WORD AddWithFloat(PHEAD WORD **ps1, WORD **ps2)
{
    GETBIDENTITY
    SORTING *S = AT.SS;
    WORD *coef1, *coef2, size1, size2, *fun1, *fun2, *fun3;
    WORD *s1,*s2,sign3,j,jj, *t1, *t2, i;
    s1 = *ps1;
    s2 = *ps2;
    coef1 = s1+*s1; size1 = coef1[-1]; coef1 -= ABS(coef1[-1]);
    coef2 = s2+*s2; size2 = coef2[-1]; coef2 -= ABS(coef2[-1]);
    if ( AT.SortFloatMode == 3 ) {
        fun1 = s1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
        fun2 = s2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
        UnpackFloat(aux1,fun1);
        if ( size1 < 0 ) mpf_neg(aux1,aux1);
        UnpackFloat(aux2,fun2);
        if ( size2 < 0 ) mpf_neg(aux2,aux2);
    }
    else if ( AT.SortFloatMode == 1 ) {
        fun1 = s1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
        UnpackFloat(aux1,fun1);
        if ( size1 < 0 ) mpf_neg(aux1,aux1);
        fun2 = coef2;
        RatToFloat(aux2,(UWORD *)coef2,size2);
    }
    else if ( AT.SortFloatMode == 2 ) {
        fun2 = s2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
        fun1 = coef1;
        RatToFloat(aux1,(UWORD *)coef1,size1);
        UnpackFloat(aux2,fun2);
        if ( size2 < 0 ) mpf_neg(aux2,aux2);
    }
    else {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal value %d for AT.SortFloatMode in AddWithFloat.",AT.SortFloatMode);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(0);
    }
    mpf_add(aux3,aux1,aux2);
    sign3 = mpf_sgn(aux3);
    if ( sign3 == 0 ) { /* May be rare! */
        *ps1 = 0; *ps2 = 0; AT.SortFloatMode = 0; return(0);
    }
    if ( sign3 < 0 ) mpf_neg(aux3,aux3);
    fun3 = TermMalloc("AddWithFloat");
    PackFloat(fun3,aux3);
/*
    Now we have to calculate where the sumterm fits.
    If we are sloppy, we can be faster, but run the risk to need the
    extension space, even when it is not needed. At slightly lower speed
    (ie first creating the result in scribble space) we are better off.
    This is why we use TermMalloc.

    The new term will have a rational coefficient of 1,1,+-3.
    The size will be (fun1 or fun2 - term) + fun3 +3;
*/
    if ( AT.SortFloatMode == 3 ) {
        if ( fun1[1] == fun3[1]  ) {
Over1:
            i = fun3[1]; t1 = fun1; t2 = fun3; NCOPY(t1,t2,i);
            *t1++ = 1; *t1++ = 1; *t1++ = sign3 < 0 ? -3: 3;
            *s1 = t1-s1; goto Finished;
        }
        else if ( fun2[1] == fun3[1]  ) {
Over2:
            i = fun3[1]; t1 = fun2; t2 = fun3; NCOPY(t1,t2,i);
            *t1++ = 1; *t1++ = 1; *t1++ = sign3 < 0 ? -3: 3;
            *s2 = t1-s2; *ps1 = s2; goto Finished;
        }
        else if ( fun1[1] >= fun3[1]  ) goto Over1;
        else if ( fun2[1] >= fun3[1]  ) goto Over2;
```

```
    }
    else if ( AT.SortFloatMode == 1 ) {
        if ( fun1[1] >= fun3[1]  ) goto Over1;
        else if ( fun3[1]+3 <= ABS(size2) ) goto Over2;
    }
    else if ( AT.SortFloatMode == 2 ) {
        if ( fun2[1] >= fun3[1]  ) goto Over2;
        else if ( fun3[1]+3 <= ABS(size1) ) goto Over1;
    }
/*
    Does not fit. Go to extension space.
*/
    jj = fun1-s1;
    j = jj+fun3[1]+3; /* space needed */
    if ( (S->sFill + j) >= S->sTop2 ) {
        GarbHand();
        s1 = *ps1; /* new position of the term after the garbage collection */
        fun1 = s1+jj;
    }
    t1 = S->sFill;
    for ( i = 0; i < jj; i++ ) *t1++ = s1[i];
    i = fun3[1]; s1 = fun3; NCOPY(t1,s1,i);
    *t1++ = 1; *t1++ = 1; *t1++ = sign3 < 0 ? -3: 3;
    *ps1 = S->sFill;
    **ps1 = t1-*ps1;
    S->sFill = t1;
Finished:
    *ps2 = 0;
    TermFree(fun3,"AddWithFloat");
    AT.SortFloatMode = 0;
    if ( **ps1 > AM.MaxTer/((LONG)(sizeof(WORD))) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Term too complex after addition in sort. MaxTermSize = %10l",
        AM.MaxTer/sizeof(WORD));
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    return(1);
}

/*
        #] AddWithFloat :
        #[ MergeWithFloat :

        Note that we always park the result on top of term1.
        This makes life easier, because the original AddRat in MergePatches
        does this as well.
*/

WORD MergeWithFloat(PHEAD WORD **interm1, WORD **interm2)
{
    GETBIDENTITY
    WORD *coef1, *coef2, size1, size2, *fun1, *fun2, *fun3, *tt;
    WORD sign3,j,jj, *t1, *t2, i, *term1 = *interm1, *term2 = *interm2, retval = 0;
    coef1 = term1+*term1; size1 = coef1[-1]; coef1 -= ABS(size1);
    coef2 = term2+*term2; size2 = coef2[-1]; coef2 -= ABS(size2);
    if ( AT.SortFloatMode == 3 ) {
        fun1 = term1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
        fun2 = term2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
        UnpackFloat(aux1,fun1);
        if ( size1 < 0 ) mpf_neg(aux1,aux1);
        UnpackFloat(aux2,fun2);
        if ( size2 < 0 ) mpf_neg(aux2,aux2);
    }
    else if ( AT.SortFloatMode == 1 ) {
        fun1 = term1+1; while ( fun1 < coef1 && fun1[0] != FLOATFUN ) fun1 += fun1[1];
        UnpackFloat(aux1,fun1);
        if ( size1 < 0 ) mpf_neg(aux1,aux1);
        fun2 = coef2;
        RatToFloat(aux2,(UWORD *)coef2,size2);
    }
    else if ( AT.SortFloatMode == 2 ) {
        fun2 = term2+1; while ( fun2 < coef2 && fun2[0] != FLOATFUN ) fun2 += fun2[1];
        fun1 = coef1;
        RatToFloat(aux1,(UWORD *)coef1,size1);
        UnpackFloat(aux2,fun2);
        if ( size2 < 0 ) mpf_neg(aux2,aux2);
    }
    else {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal value %d for AT.SortFloatMode in MergeWithFloat.",AT.SortFloatMode);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(0);
    }
    mpf_add(aux3,aux1,aux2);
    sign3 = mpf_sgn(aux3);
```

```
    if ( sign3 == 0 ) { /* May be very rare! */
        AT.SortFloatMode = 0; return(0);
    }
/*
    Now check whether we can park the result on top of one of the input terms.
*/
    if ( sign3 < 0 ) mpf_neg(aux3,aux3);
    fun3 = TermMalloc("MergeWithFloat");
    PackFloat(fun3,aux3);
    if ( AT.SortFloatMode == 3 ) {
        if ( fun1[1] + ABS(size1) == fun3[1] + 3 ) {
OnTopOf1:   t1 = fun3; t2 = fun1;
            for ( i = 0; i < fun3[1]; i++ ) *t2++ = *t1++;
            *t2++ = 1; *t2++ = 1; *t2++ = sign3 < 0 ? -3: 3;
            retval = 1;
        }
        else if ( fun1[1] + ABS(size1) > fun3[1] + 3 ) {
Shift1:     t2 = term1 + *term1; tt = t2;
            *--t2 = sign3 < 0 ? -3: 3; *--t2 = 1; *--t2 = 1;
            t1 = fun3 + fun3[1]; for ( i = 0; i < fun3[1]; i++ ) *--t2 = *--t1;
            t1 = fun1;
            while ( t1 > term1 ) *--t2 = *--t1;
            *t2 = tt-t2; term1 = t2;
            retval = 1;
        }
        else { /* Here we have to move term1 to the left to make room. */
Over1:      jj = fun3[1]-fun1[1]+3-ABS(size1); /* This is positive */
            t2 = term1-jj; t1 = term1;
            while ( t1 < fun1 ) *t2++ = *t1++;
            term1 -= jj;
            *term1 += jj;
            for ( i = 0; i < fun3[1]; i++ ) *t2++ = fun3[i];
            *t2++ = 1; *t2++ = 1;  *t2++ = sign3 < 0 ? -3: 3;
            retval = 1;
        }
    }
    else if ( AT.SortFloatMode == 1 ) {
        if ( fun1[1] + ABS(size1) == fun3[1] + 3 ) goto OnTopOf1;
        else if ( fun1[1] + ABS(size1) > fun3[1] + 3 ) goto Shift1;
        else goto Over1;
    }
    else { /* Can only be 2, based on previous tests */
        if ( fun3[1] + 3 == ABS(size1) ) {
            t2 = coef1; t1 = fun3;
            for ( i = 0; i < fun3[1]; i++ ) *t2++ = *t1++;
            *t2++ = 1; *t2++ = 1;  *t2++ = sign3 < 0 ? -3: 3;
            retval = 1;
        }
        else if ( fun3[1] + 3 < ABS(size1) ) {
            j = ABS(size1) - fun3[1] - 3;
            t2 = term1 + *term1; tt = t2;
            *--t2 = sign3 < 0 ? -3: 3; *--t2 = 1; *--t2 = 1;
            t2 -= fun3[1]; t1 = t2-j;
            while ( t2 > term1 ) *--t2 = *--t1;
            *t2 = tt-t2; term1 = t2;
            retval = 1;
        }
        else goto Over1;
    }
    *interm1 = term1;
    TermFree(fun3,"MergeWithFloat");
    AT.SortFloatMode = 0;
    return(retval);
}

/*
        #] MergeWithFloat :
    #] Sorting :
    #[ MZV :

        The functions here allow the arbitrary precision calculation
        of MZV's and Euler sums.
        They are called when the functions mzv_ and/or euler_ are used
        and the evaluate statement is applied.
        The output is in a float_ function.
        The expand statement tries to express the functions in terms of a basis.
        The bases are the 'standard basis for the euler sums and the
        pushdown bases from the datamine, unless otherwise specified.

        #[ SimpleDelta :
*/

void SimpleDelta(mpf_t sum, int m)
{
    long s = 1;
    long prec = AC.DefaultPrecision;
    unsigned long xprec = (unsigned long)prec, x;
```

```
    int j, jmax, n;
    mpf_t jm,jjm;
    mpf_init(jm); mpf_init(jjm);
    if ( m < 0 ) { s = -1; m = -m; }

/*
    We will loop until 1/2^j/j^m is smaller than the default precision.
    Just running to prec is however overkill, specially when m is large.
    We try to estimate a better value.
    jmax = prec - ((log2(prec)-1)*m).
    Hence we need the leading bit of prec.
    We are still overshooting a bit.
*/
    n = 0; x = xprec;
    while ( x ) { x »= 1; n++; }
/*
    We have now n = floor(log2(x))+1.
*/
    n--;
    jmax = (int)((int)xprec - (n-1)*m);
    mpf_set_ui(sum,0);
    for ( j = 1; j <= jmax; j++ ) {
#ifdef WITHCUTOFF
        xprec--;
        mpf_set_prec_raw(jm,xprec);
        mpf_set_prec_raw(jjm,xprec);
#endif
        mpf_set_ui(jm,1L);
        mpf_div_ui(jjm,jm,(unsigned long)j);
        mpf_pow_ui(jm,jjm,(unsigned long)m);
        mpf_div_2exp(jjm,jm,(unsigned long)j);
        if ( s == -1 && j%2 == 1 ) mpf_sub(sum,sum,jjm);
        else                       mpf_add(sum,sum,jjm);
    }
    mpf_clear(jjm); mpf_clear(jm);
}

/*
        #] SimpleDelta :
        #[ SimpleDeltaC :
*/

void SimpleDeltaC(mpf_t sum, int m)
{
    long s = 1;
    long prec = AC.DefaultPrecision;
    unsigned long xprec = (unsigned long)prec, x;
    int j, jmax, n;
    mpf_t jm,jjm;
    mpf_init(jm); mpf_init(jjm);
    if ( m < 0 ) { s = -1; m = -m; }
/*
    We will loop until 1/2^j/j^m is smaller than the default precision.
    Just running to prec is however overkill, specially when m is large.
    We try to estimate a better value.
    jmax = prec - ((log2(prec)-1)*m).
    Hence we need the leading bit of prec.
    We are still overshooting a bit.
*/
    n = 0; x = xprec;
    while ( x ) { x »= 1; n++; }
/*
    We have now n = floor(log2(x))+1.
*/
    n--;
    jmax = (int)((int)xprec - (n-1)*m);
    if ( s < 0 ) jmax /= 2;
    mpf_set_si(sum,0L);
    for ( j = 1; j <= jmax; j++ ) {
#ifdef WITHCUTOFF
        xprec--;
        mpf_set_prec_raw(jm,xprec);
        mpf_set_prec_raw(jjm,xprec);
#endif
        mpf_set_ui(jm,1L);
        mpf_div_ui(jjm,jm,(unsigned long)j);
        mpf_pow_ui(jm,jjm,(unsigned long)m);
        if ( s == -1 ) mpf_div_2exp(jjm,jm,2*(unsigned long)j);
        else           mpf_div_2exp(jjm,jm,(unsigned long)j);
        mpf_add(sum,sum,jjm);
    }
    mpf_clear(jjm); mpf_clear(jm);
}

/*
        #] SimpleDeltaC :
        #[ SingleTable :
```

```
*/

void SingleTable(mpf_t *tabl, int N, int m, int pow)
{
/*
    Creates a table T(1,...,N) with
        T(i) = sum_(j,i,N,[sign_(j)]/2^j/j^m)
    To make this table we have two options:
    1: run the sum backwards with the potential problem that the
       precision is difficult to manage.
    2: run the sum forwards. Take sum_(j,1,i-1,...) and later subtract.
    When doing Euler sums we may need also 1/4^j
    pow: 1 -> 1/2^j
         2 -> 1/4^j
*/
    GETIDENTITY
    long s = 1;
    int j;
#ifdef WITHCUTOFF
    long prec = mpf_get_default_prec();
#endif
    mpf_t jm,jjm;
    mpf_init(jm); mpf_init(jjm);
    if ( pow < 1 || pow > 2 ) {
        printf("Wrong parameter pow in SingleTable: %d\n",pow);
        exit(-1);
    }
    if ( m < 0 ) { m = -m; s = -1; }
    mpf_set_si(auxsum,0L);
    for ( j = N; j > 0; j-- ) {
#ifdef WITHCUTOFF
        mpf_set_prec_raw(jm,prec-j);
        mpf_set_prec_raw(jjm,prec-j);
#endif
        mpf_set_ui(jm,1L);
        mpf_div_ui(jjm,jm,(unsigned long)j);
        mpf_pow_ui(jm,jjm,(unsigned long)m);
        mpf_div_2exp(jjm,jm,pow*(unsigned long)j);
        if ( pow == 2 ) mpf_add(auxsum,auxsum,jjm);
        else if ( s == -1 && j%2 == 1 ) mpf_sub(auxsum,auxsum,jjm);
        else                           mpf_add(auxsum,auxsum,jjm);
/*
        And now copy auxsum to tablelement j
*/
        mpf_set(tabl[j],auxsum);
    }
    mpf_clear(jjm); mpf_clear(jm);
}

/*
        #] SingleTable :
        #[ DoubleTable :
*/

void DoubleTable(mpf_t *tabout, mpf_t *tabin, int N, int m, int pow)
{
    GETIDENTITY
    long s = 1;
#ifdef WITHCUTOFF
    long prec = mpf_get_default_prec();
#endif
    int j;
    mpf_t jm,jjm;
    mpf_init(jm); mpf_init(jjm);
    if ( pow < -1 || pow > 2 ) {
        printf("Wrong parameter pow in SingleTable: %d\n",pow);
        exit(-1);
    }
    if ( m < 0 ) { m = -m; s = -1; }
    mpf_set_ui(auxsum,0L);
    for ( j = N; j > 0; j-- ) {
#ifdef WITHCUTOFF
        mpf_set_prec_raw(jm,prec-j);
        mpf_set_prec_raw(jjm,prec-j);
#endif
        mpf_set_ui(jm,1L);
        mpf_div_ui(jjm,jm,(unsigned long)j);
        mpf_pow_ui(jm,jjm,(unsigned long)m);
        if ( pow == -1 ) {
            mpf_mul_2exp(jjm,jm,(unsigned long)j);
            mpf_mul(jm,jjm,tabin[j+1]);
        }
        else if ( pow > 0 ) {
            mpf_div_2exp(jjm,jm,pow*(unsigned long)j);
            mpf_mul(jm,jjm,tabin[j+1]);
        }
        else {
```

```
            mpf_mul(jm,jm,tabin[j+1]);
        }
        if ( pow == 2 ) mpf_add(auxsum,auxsum,jm);
        else if ( s == -1 && j%2 == 1 ) mpf_sub(auxsum,auxsum,jm);
        else                           mpf_add(auxsum,auxsum,jm);
/*
        And now copy auxsum to tablelement j
*/
        mpf_set(tabout[j],auxsum);
    }
    mpf_clear(jjm); mpf_clear(jm);
}

/*
        #] DoubleTable :
        #[ EndTable :
*/

void EndTable(mpf_t sum, mpf_t *tabin, int N, int m, int pow)
{
    long s = 1;
#ifdef WITHCUTOFF
    long prec = mpf_get_default_prec();
#endif
    int j;
    mpf_t jm,jjm;
    mpf_init(jm); mpf_init(jjm);
    if ( pow < -1 || pow > 2 ) {
        printf("Wrong parameter pow in SingleTable: %d\n",pow);
        exit(-1);
    }
    if ( m < 0 ) { m = -m; s = -1; }
    mpf_set_si(sum,0L);
    for ( j = N; j > 0; j-- ) {
#ifdef WITHCUTOFF
        mpf_set_prec_raw(jm,prec-j);
        mpf_set_prec_raw(jjm,prec-j);
#endif
        mpf_set_ui(jm,1L);
        mpf_div_ui(jjm,jm,(unsigned long)j);
        mpf_pow_ui(jm,jjm,(unsigned long)m);
        if ( pow == -1 ) {
            mpf_mul_2exp(jjm,jm,(unsigned long)j);
            mpf_mul(jm,jjm,tabin[j+1]);
        }
        else if ( pow > 0 ) {
            mpf_div_2exp(jjm,jm,pow*(unsigned long)j);
            mpf_mul(jm,jjm,tabin[j+1]);
        }
        else {
            mpf_mul(jm,jm,tabin[j+1]);
        }
        if ( s == -1 && j%2 == 1 ) mpf_sub(sum,sum,jm);
        else                       mpf_add(sum,sum,jm);
    }
    mpf_clear(jjm); mpf_clear(jm);
}

/*
        #] EndTable :
        #[ deltaMZV :
*/

void deltaMZV(mpf_t result, int *indexes, int depth)
{
    GETIDENTITY
/*
    Because all sums go roughly like 1/2^j we need about depth steps.
*/
/*  MesPrint("deltaMZV(%a)",depth,indexes); */
    if ( depth == 1 ) {
        if ( indexes[0] == 1 ) {
            mpf_set(result,mpfdelta1);
            return;
        }
        SimpleDelta(result,indexes[0]);
    }
    else if ( depth == 2 ) {
        if ( indexes[0] == 1 && indexes[1] == 1 ) {
            mpf_pow_ui(result,mpfdelta1,2);
            mpf_div_2exp(result,result,1);
        }
        else {
            SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+1,indexes[0],1);
            EndTable(result,mpftab1,AC.DefaultPrecision-AC.MaxWeight,indexes[1],0);
        };
    }
```

```
    else if ( depth > 2 ) {
        mpf_t *mpftab3;
        int d;
/*
        Check first whether this is a power of delta1 = ln(2)
*/
        for ( d = 0; d < depth; d++ ) {
            if ( indexes[d] != 1 ) break;
        }
        if ( d == depth ) { /* divide by fac_(depth) */
            mpf_pow_ui(result,mpfdelta1,depth);
            for ( d = 2; d <= depth; d++ ) {
                mpf_div_ui(result,result,(unsigned long)d);
            }
        }
        else {
            d = depth-1;
            SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,*indexes,1);
            d--; indexes++;
            for(;;) {
                DoubleTable(mpftab2,mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,*indexes,0);
                d--; indexes++;
                if ( d == 0 ) {
                    EndTable(result,mpftab2,AC.DefaultPrecision-AC.MaxWeight,*indexes,0);
                    break;
                }
                mpftab3 = (mpf_t *)AT.mpf_tab1; AT.mpf_tab1 = AT.mpf_tab2;
                AT.mpf_tab2 = (void *)mpftab3;
            }
        }
    }
    else if ( depth == 0 ) {
        mpf_set_ui(result,1L);
    }
}

/*
        #] deltaMZV :
        #[ deltaEuler :

        Regular Euler delta with - signs, but everywhere 1/2^j
*/

void deltaEuler(mpf_t result, int *indexes, int depth)
{
    GETIDENTITY
    int m;
#ifdef DEBUG
    int i;
    printf(" deltaEuler(");
    for ( i = 0; i < depth; i++ ) {
        if ( i != 0 ) printf(",");
        printf("%d",indexes[i]);
    }
    printf(") = ");
#endif
    if ( depth == 1 ) {
        SimpleDelta(result,indexes[0]);
    }
    else if ( depth == 2 ) {
        SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+1,indexes[0],1);
        m = indexes[1]; if ( indexes[0] < 0 ) m = -m;
        EndTable(result,mpftab1,AC.DefaultPrecision,m,0);
    }
    else if ( depth > 2 ) {
        int d;
        mpf_t *mpftab3;
        d = depth-1;
        SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,*indexes,1);
        d--; indexes++;
        m = *indexes; if ( indexes[-1] < 0 ) m = -m;
        for(;;) {
            DoubleTable(mpftab2,mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,m,0);
            d--; indexes++;
            m = *indexes; if ( indexes[-1] < 0 ) m = -m;
            if ( d == 0 ) {
                EndTable(result,mpftab2,AC.DefaultPrecision-AC.MaxWeight,m,0);
                break;
            }
            mpftab3 = (mpf_t *)AT.mpf_tab1; AT.mpf_tab1 = AT.mpf_tab2;
            AT.mpf_tab2 = (void *)mpftab3;
        }
    }
    else if ( depth == 0 ) {
        mpf_set_ui(result,1L);
    }
#ifdef DEBUG
```

```
    gmp_printf("%.*Fe\n",40,result);
#endif
}

/*
        #] deltaEuler :
        #[ deltaEulerC :

        Conjugate Euler delta without - signs, but possibly 1/4^j
        When there is a - in the string we have 1/4.
*/

void deltaEulerC(mpf_t result, int *indexes, int depth)
{
    GETIDENTITY
    int m;
#ifdef DEBUG
    int i;
    printf(" deltaEulerC(");
    for ( i = 0; i < depth; i++ ) {
        if ( i != 0 ) printf(",");
        printf("%d",indexes[i]);
    }
    printf(") = ");
#endif
    mpf_set_ui(result,0);
    if ( depth == 1 ) {
        if ( indexes[0] == 0 ) {
            printf("Illegal index in depth=1 deltaEulerC: %d\n",indexes[0]);
        }
        if ( indexes[0] < 0 ) SimpleDeltaC(result,indexes[0]);
        else                  SimpleDelta(result,indexes[0]);
    }
    else if ( depth == 2 ) {
        int par;
        m = indexes[0];
        if ( m < 0 ) SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+depth+1,-m,2);
        else         SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+depth+1, m,1);
        m = indexes[1];
        if ( m < 0 ) { m = -m; par = indexes[0] < 0 ? 0: 1; }
        else { par = indexes[0] < 0 ? -1: 0; }
        EndTable(result,mpftab1,AC.DefaultPrecision-AC.MaxWeight,m,par);
    }
    else if ( depth > 2 ) {
        int d, par;
        mpf_t *mpftab3;
        d = depth-1;
        m = indexes[0];
        ZeroTable(mpftab1,AC.DefaultPrecision+1);
        if ( m < 0 ) SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+d+1,-m,2);
        else         SingleTable(mpftab1,AC.DefaultPrecision-AC.MaxWeight+d+1, m,1);
        d--; indexes++; m = indexes[0];
        if ( m < 0 ) { m = -m; par = indexes[-1] < 0 ? 0: 1; }
        else { par = indexes[-1] < 0 ? -1: 0; }
        for(;;) {
            ZeroTable(mpftab2,AC.DefaultPrecision-AC.MaxWeight+d);
            DoubleTable(mpftab2,mpftab1,AC.DefaultPrecision-AC.MaxWeight+d,m,par);
            d--; indexes++; m = indexes[0];
            if ( m < 0 ) { m = -m; par = indexes[-1] < 0 ? 0: 1; }
            else { par = indexes[-1] < 0 ? -1: 0; }
            if ( d == 0 ) {
                mpf_set_ui(result,0);
                EndTable(result,mpftab2,AC.DefaultPrecision-AC.MaxWeight,m,par);
                break;
            }
            mpftab3 = (mpf_t *)AT.mpf_tab1; AT.mpf_tab1 = AT.mpf_tab2;
            AT.mpf_tab2 = (void *)mpftab3;
        }
    }
    else if ( depth == 0 ) {
        mpf_set_ui(result,1L);
    }
#ifdef DEBUG
    gmp_printf("%.*Fe\n",40,result);
#endif
}

/*
        #] deltaEulerC :
        #[ CalculateMZVhalf :

        This routine is mainly for support when large numbers of
        MZV's have to be calculated at the same time.
*/

void CalculateMZVhalf(mpf_t result, int *indexes, int depth)
{
```

```
    int i;
    if ( depth < 0 ) {
        printf("Illegal depth in CalculateMZVhalf: %d\n",depth);
        exit(-1);
    }
    for ( i = 0; i < depth; i++ ) {
        if ( indexes[i] <= 0 ) {
            printf("Illegal index[%d] in CalculateMZVhalf: %d\n",i,indexes[i]);
            exit(-1);
        }
    }
    deltaMZV(result,indexes,depth);
}

/*
        #] CalculateMZVhalf :
        #[ CalculateMZV :
*/

void CalculateMZV(mpf_t result, int *indexes, int depth)
{
    GETIDENTITY
    int num1, num2 = 0, i, j = 0;
    if ( depth < 0 ) {
        printf("Illegal depth in CalculateMZV: %d\n",depth);
        exit(-1);
    }
    if ( indexes[0] == 1 ) {
        printf("Divergent MZV in CalculateMZV\n");
        exit(-1);
    }
/*  MesPrint("calculateMZV(%a)",depth,indexes); */
    for ( i = 0; i < depth; i++ ) {
        if ( indexes[i] <= 0 ) {
            printf("Illegal index[%d] in CalculateMZV: %d\n",i,indexes[i]);
            exit(-1);
        }
        AT.indi1[i] = indexes[i];
    }
    num1 = depth; i = 0;
    deltaMZV(result,indexes,depth);
    for(;;) {
        if ( AT.indi1[0] > 1 ) {
            AT.indi1[0] -= 1;
            for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
            AT.indi2[0] = 1; num2++;
        }
        else {
            AT.indi2[0] += 1;
            for ( j = 1; j < num1; j++ ) AT.indi1[j-1] = AT.indi1[j];
            num1--;
            if ( num1 == 0 ) break;
        }
        deltaMZV(aux1,AT.indi1,num1);
        deltaMZV(aux2,AT.indi2,num2);
        mpf_mul(aux3,aux1,aux2);
        mpf_add(result,result,aux3);
    }
    deltaMZV(aux3,AT.indi2,num2);
    mpf_add(result,result,aux3);
}

/*
        #] CalculateMZV :
        #[ CalculateEuler :

        There is a problem of notation here.
        Basically there are two notations.
        1: Z(m1,m2,m3) = sum_(j3,1,inf,sign_(m3)/j3^abs_(m3)*
                             sum_(j2,j3+1,inf,sign_(m2)/j2^abs_(m2)*
                             sum_(j1,j2+1,inf,sign_(m1)/j1^abs_(m1))))
           See Broadhurst,1996
        2: L(m1,m2,m3) = sum_(j1,1,inf,sign_(m1)*
                             sum_(j2,1,inf,sign_(m2)*
                             sum_(j3,1,inf,sign_(m3)
                                /j3^abs_(m3)
                                /(j2+j3)^abs_(m2)
                                /(j1+j2+j3)^abs_(m1) )))
           See Borwein, Bradley, Broadhurst and Lisonek, 1999
        The L corresponds with the H of the datamine up to sign_(m1*m2*m3)
        The algorithm follows 2, but the more common notation is 1.
        Hence we start with a conversion.
*/

void CalculateEuler(mpf_t result, int *Zindexes, int depth)
{
    GETIDENTITY
```

```
    int s1, num1, num2, i, j;

    int *indexes = (int *)(AT.WorkPointer);
    for ( i = 0; i < depth; i++ ) indexes[i] = Zindexes[i];
    for ( i = 0; i < depth-1; i++ ) {
        if ( Zindexes[i] < 0 ) {
            for ( j = i+1; j < depth; j++ ) indexes[j] = -indexes[j];
        }
    }

    if ( depth < 0 ) {
        printf("Illegal depth in CalculateEuler: %d\n",depth);
        exit(-1);
    }
    if ( indexes[0] == 1 ) {
        printf("Divergent Euler sum in CalculateEuler\n");
        exit(-1);
    }
    for ( i = 0, j = 0; i < depth; i++ ) {
        if ( indexes[i] == 0 ) {
            printf("Illegal index[%d] in CalculateEuler: %d\n",i,indexes[i]);
            exit(-1);
        }
        if ( indexes[i] < 0 ) j = 1;
        AT.indi1[i] = indexes[i];
    }
    if ( j == 0 ) {
        CalculateMZV(result,indexes,depth);
        return;
    }
    num1 = depth; AT.indi2[0] = 0; num2 = 0;
    deltaEuler(result,AT.indi1,depth);
    s1 = 0;
    for(;;) {
        s1++;
        if ( AT.indi1[0] > 1 ) {
            AT.indi1[0] -= 1;
            for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
            AT.indi2[0] = 1; num2++;
        }
        else if ( AT.indi1[0] < -1 ) {
            AT.indi1[0] += 1;
            for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
            AT.indi2[0] = 1; num2++;
        }
        else if ( AT.indi1[0] == -1 ) {
            for ( j = num2; j > 0; j-- ) AT.indi2[j] = AT.indi2[j-1];
            AT.indi2[0] = -1; num2++;
            for ( j = 1; j < num1; j++ ) AT.indi1[j-1] = AT.indi1[j];
            num1--;
        }
        else {
            AT.indi2[0] = AT.indi2[0] < 0 ? AT.indi2[0]-1: AT.indi2[0]+1;
            for ( j = 1; j < num1; j++ ) AT.indi1[j-1] = AT.indi1[j];
            num1--;
        }
        if ( num1 == 0 ) break;
        deltaEuler(aux1,AT.indi1,num1);
        deltaEulerC(aux2,AT.indi2,num2);
        mpf_mul(aux3,aux1,aux2);
        if ( (depth+num1+num2+s1)%2 == 0 ) mpf_add(result,result,aux3);
        else                               mpf_sub(result,result,aux3);
    }
    deltaEulerC(aux3,AT.indi2,num2);
    if ( (depth+num2+s1)%2 == 0 ) mpf_add(result,result,aux3);
    else                          mpf_sub(result,result,aux3);
}

/*
        #] CalculateEuler :
        #[ ExpandMZV :
*/

int ExpandMZV(WORD *term, WORD level)
{
    DUMMYUSE(term);
    DUMMYUSE(level);
    return(0);
}

/*
        #] ExpandMZV :
        #[ ExpandEuler :
*/

int ExpandEuler(WORD *term, WORD level)
{
```

```
    DUMMYUSE(term);
    DUMMYUSE(level);
    return(0);
}

/*
        #] ExpandEuler :
        #[ EvaluateEuler :

    There are various steps here:
    1: locate an Euler function.
    2: evaluate it into a float.
    3: convert the coefficient to a float and multiply.
    4: Put the new term together.
    5: Restart to see whether there are more Euler functions.
    The compiler should check that there is no conflict between
    the evaluate command and a potential polyfun or polyratfun.
    Yet, when reading in expressions there can be a conflict.
    Hence the Normalize routine should check as well.

    par == MZV: MZV
    par == EULER: Euler
    par == MZVHALF: MZV sums in 1/2 rather than 1. -> deltaMZV.
    par == ALLMZVFUNCTIONS: all of the above.
*/

int EvaluateEuler(PHEAD WORD *term, WORD level, WORD par)
{
    WORD *t, *tstop, *tt, *tnext, sumweight, depth, first = 1, *newterm, i;
    WORD *oldworkpointer = AT.WorkPointer, nsize, *indexes;
    int retval;
    tstop = term + *term; tstop -= ABS(tstop[-1]);
    if ( AT.WorkPointer < term+*term ) AT.WorkPointer = term + *term;
/*
    Step 1. We also need to verify that the argument we find is legal
*/
    t = term+1;
    while ( t < tstop ) {
        if ( ( *t == par ) || ( ( par == ALLMZVFUNCTIONS ) && (
            *t == MZV || *t == EULER || *t == MZVHALF ) ) ) {
    /* all arguments should be small integers */
            indexes = AT.WorkPointer;
            sumweight = 0; depth = 0;
            tnext = t + t[1]; tt = t + FUNHEAD;
            while ( tt < tnext ) {
                if ( *tt != -SNUMBER ) goto nextfun;
                if ( tt[1] == 0 ) goto nextfun;
                indexes[depth] = tt[1];
                sumweight += ABS(tt[1]); depth++;
                tt += 2;
            }
            if ( sumweight > AC.MaxWeight ) {
                MesPrint("Error: Weight of Euler/MZV sum greater than %d",sumweight);
                MesPrint("Please increase MaxWeight in form.set.");
                Terminate(-1);
            }
/*
            Step 2: evaluate:
*/
            AT.WorkPointer += depth;
            if ( first ) {
                if ( *t == MZV ) CalculateMZV(aux4,indexes,depth);
                else if ( *t == EULER ) CalculateEuler(aux4,indexes,depth);
                else if ( *t == MZVHALF ) CalculateMZVhalf(aux4,indexes,depth);
                first = 0;
            }
            else {
                if ( *t == MZV ) CalculateMZV(aux5,indexes,depth);
                else if ( *t == EULER ) CalculateEuler(aux5,indexes,depth);
                else if ( *t == MZVHALF ) CalculateMZVhalf(aux5,indexes,depth);
                mpf_mul(aux4,aux4,aux5);
            }
            *t = 0;
        }
nextfun:
        t += t[1];
    }
    if ( first == 1 ) return(Generator(BHEAD term,level));
/*
    Step 3:
    Now the regular coefficient, if it is not 1/1.
    We have two cases: size +- 3, or bigger.
*/
    nsize = term[*term-1];
    if ( nsize < 0 ) {
        mpf_neg(aux4,aux4);
        nsize = -nsize;
```

```
    }
    if ( nsize == 3 ) {
        if ( tstop[0] != 1 ) {
            mpf_mul_ui(aux4,aux4,(ULONG)((UWORD)tstop[0]));
        }
        if ( tstop[1] != 1 ) {
            mpf_div_ui(aux4,aux4,(ULONG)((UWORD)tstop[1]));
        }
    }
    else {
        RatToFloat(aux5,(UWORD *)tstop,nsize);
        mpf_mul(aux4,aux4,aux5);
    }
/*
    Now we have to locate possible other float_ functions.
*/
    t = term+1;
    while ( t < tstop ) {
        if ( *t == FLOATFUN ) {
            UnpackFloat(aux5,t);
            mpf_mul(aux4,aux4,aux5);
        }
        t += t[1];
    }
/*
    Now we should compose the new term in the WorkSpace.
*/
    t = term+1;
    newterm = AT.WorkPointer;
    tt = newterm+1;
    while ( t < tstop ) {
        if ( *t == 0 || *t == FLOATFUN ) t += t[1];
        else {
            i = t[1]; NCOPY(tt,t,i);
        }
    }
    PackFloat(tt,aux4);
    tt += tt[1];
    *tt++ = 1; *tt++ = 1; *tt++ = 3;
    *newterm = tt-newterm;
    AT.WorkPointer = tt;
    retval = Generator(BHEAD newterm,level);
    AT.WorkPointer = oldworkpointer;
    return(retval);
}

/*
        #] EvaluateEuler :
    #] MZV :
    #[ Functions :
        #[ CoEvaluate :

        Current subkeys: mzv_, euler_ and sqrt_.
*/

int CoEvaluate(UBYTE *s)
{
    UBYTE *subkey, c;
    WORD numfun, type;
    int error = 0;
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( *s == 0 ) {
/*
        No subkey, which means all functions.
*/
        Add3Com(TYPEEVALUATE,ALLFUNCTIONS);
/*
        The MZV, EULER and MZVHALF are done separately
*/
        Add3Com(TYPEEVALUATE,ALLMZVFUNCTIONS);
        return(0);
    }
    while ( *s ) {
    subkey = s;
    while ( FG.cTable[*s] == 0 ) s++;
      if ( *s == '_' ) s++;
      c = *s; *s++ = 0;
/*
        We still need provisions for pi_ and possibly other constants.
*/
      if ( ( ( type = GetName(AC.varnames,subkey,&numfun,NOAUTO) ) != CFUNCTION )
            || ( functions[numfun].spec != 0 ) ) {

        if ( type == CSYMBOL ) {
            Add4Com(TYPEEVALUATE,SYMBOL,numfun);
            break;
        }
```

```
/*
            This cannot work.
*/
        MesPrint("&%s should be a built in function that can be evaluated numerically.",s);
        error = 1;
      }
      else {
        switch ( numfun+FUNCTION ) {
            case MZV:
            case EULER:
            case MZVHALF:
            case SQRTFUNCTION:
/*
            The following functions are treated in evaluate.c

            case LNFUNCTION:
            case SINFUNCTION:
            case COSFUNCTION:
            case TANFUNCTION:
            case ASINFUNCTION:
            case ACOSFUNCTION:
            case ATANFUNCTION:
            case ATAN2FUNCTION:
            case SINHFUNCTION:
            case COSHFUNCTION:
            case TANHFUNCTION:
            case ASINHFUNCTION:
            case ACOSHFUNCTION:
            case ATANHFUNCTION:
            case LI2HFUNCTION:
            case LINHFUNCTION:
            case AGMFUNCTION:
            case GAMMAFUN:

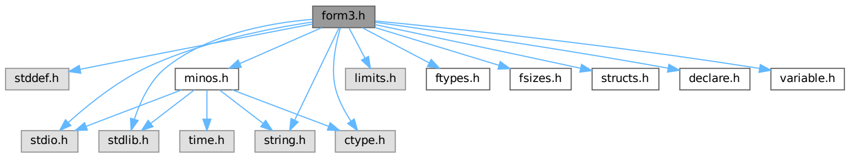
            At a later stage we can add more functions from mpfr here
                mpfr_(number,arg(s))
*/
                Add3Com(TYPEEVALUATE,numfun+FUNCTION);
                break;
            default:
                MesPrint("&%s should be a built in function that can be evaluated numerically.",s);
                error = 1;
                break;
        }
      }
      *s = c;
      while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    }
    return(error);
}

/*
        #] CoEvaluate :
    #] Functions :
*/
```

4.45 form3.h File Reference

```
#include <stddef.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <limits.h>
#include "ftypes.h"
#include "fsizes.h"
#include "minos.h"
#include "structs.h"
#include "declare.h"
#include "variable.h"
```

Include dependency graph for form3.h:



Macros

- #define MAJORVERSION 5
- #define MINORVERSION 0
- #define PRODUCTIONDATE "8-nov-2022"
- #define BETAVERSION
- #define inline
- #define NDEBUG
- #define STATIC_ASSERT(condition) STATIC_ASSERT__1(condition,__LINE__)
- #define STATIC_ASSERT__1(X, L) STATIC_ASSERT__2(X,L)
- #define STATIC_ASSERT__2(X, L) STATIC_ASSERT__3(X,L)
- #define STATIC_ASSERT__3(X, L) typedef char static_assertion_failed_##L[(!!(X))*2-1]
- #define TOPBITONLY ((ULONG)1 << (BITSINWORD - 1)) /* 0x00008000UL */
- #define TOPLONGBITONLY ((ULONG)1 << (BITSINLONG - 1)) /* 0x80000000UL */
- #define SPECMASK ((UWORD)1 << (BITSINWORD - 1)) /* 0x8000U */
- #define WILDMASK ((UWORD)1 << (BITSINWORD - 2)) /* 0x4000U */
- #define WORDMASK ((ULONG)FULLMAX - 1) /* 0x0000FFFFUL */
- #define AWORDMASK (WORDMASK << BITSINWORD) /* 0xFFFF0000UL */
- #define FULLMAX ((LONG)1 << BITSINWORD) /* 0x00010000L */
- #define MAXPOSITIVE ((LONG)(TOPBITONLY - 1)) /* 0x00007FFFL */
- #define MAXLONG ((LONG)(TOPLONGBITONLY - 1)) /* 0x7FFFFFFFL */
- #define MAXPOSITIVE2 (MAXPOSITIVE / 2) /* 0x00003FFFL */
- #define MAXPOSITIVE4 (MAXPOSITIVE / 4) /* 0x00001FFFL */
- #define form_alignof(type) offsetof(struct { char c_; type x_; }, x_)
- #define PADDUMMY(type, size) UBYTE d_u_m_m_y[form_alignof(type) - ((size) & (form_alignof(type) - 1))]
- #define PADPOSITION(ptr_, long_, int_, word_, byte_)
- #define PADPOINTER(long_, int_, word_, byte_)
- #define PADLONG(int_, word_, byte_)
- #define PADINT(word_, byte_)
- #define PADWORD(byte_)
- #define WITHSORTBOTS
- #define FILES FILE
- #define Uopen(x, y) fopen(x,y)
- #define Uflush(x) fflush(x)
- #define Uclose(x) fclose(x)
- #define Uread(x, y, z, u) fread(x,y,z,u)
- #define Uwrite(x, y, z, u) fwrite(x,y,z,u)
- #define Usetbuf(x, y) setbuf(x,y)
- #define Useek(x, y, z) fseek(x,y,z)
- #define Utell(x) ftell(x)
- #define Ugetpos(x, y) fgetpos(x,y)
- #define Usetpos(x, y) fsetpos(x,y)
- #define Usync(x) fflush(x)

- #define Utruncate(x) _chsize(_fileno(x),0)
- #define Ustdout stdout
- #define MAX_OPEN_FILES FOPEN_MAX
- #define bzero(b, len) (memset((b), 0, (len)), (void)0)
- #define GetPID() ((LONG)GetCurrentProcessId())

Typedefs

- typedef signed char SBYTE
- typedef unsigned char UBYTE
- typedef unsigned int UINT
- typedef ULONG RLONG
- typedef INT64 MLONG
- typedef char BOOL

Functions

- **STATIC_ASSERT** (sizeof(WORD) ∗8==BITSINWORD)
- **STATIC_ASSERT** (sizeof(LONG) ∗8==BITSINLONG)
- **STATIC_ASSERT** (sizeof(LONG) >=sizeof(int ∗))
- **STATIC_ASSERT** (sizeof(INT16)==2)
- **STATIC_ASSERT** (sizeof(INT32)==4)
- **STATIC_ASSERT** (sizeof(INT64)==8)

4.45.1 Detailed Description

Contains critical defines for the compilation process Also contains the inclusion of all necessary header files. There are also some system dependencies concerning file functions.

Definition in file form3.h.

4.45.2 Macro Definition Documentation

4.45.2.1 MAJORVERSION

```
#define MAJORVERSION 5
```

Definition at line 47 of file form3.h.

4.45.2.2 MINORVERSION

```
#define MINORVERSION 0
```

Definition at line 48 of file form3.h.

4.45.2.3 PRODUCTIONDATE

```
#define PRODUCTIONDATE "8-nov-2022"
```

Definition at line 53 of file form3.h.

4.45.2.4 BETAVERSION

```
#define BETAVERSION
```

Definition at line 57 of file form3.h.

4.45.2.5 inline

```
#define inline
```

Definition at line 134 of file form3.h.

4.45.2.6 NDEBUG

```
#define NDEBUG
```

Definition at line 161 of file form3.h.

4.45.2.7 STATIC_ASSERT

```
#define STATIC_ASSERT(
             condition ) STATIC_ASSERT__1(condition,__LINE__)
```

Definition at line 172 of file form3.h.

4.45.2.8 STATIC_ASSERT__1

```
#define STATIC_ASSERT__1(
             X,
             L ) STATIC_ASSERT__2(X,L)
```

Definition at line 173 of file form3.h.

4.45.2.9 STATIC_ASSERT__2

```
#define STATIC_ASSERT__2(
             X,
             L ) STATIC_ASSERT__3(X,L)
```

Definition at line 174 of file form3.h.

4.45.2.10 STATIC_ASSERT__3

```
#define STATIC_ASSERT__3(
              X,
              L )  typedef char static_assertion_failed_##L[(!!(X))*2-1]
```

Definition at line 175 of file form3.h.

4.45.2.11 TOPBITONLY

```
#define TOPBITONLY ((ULONG)1 << (BITSINWORD - 1)) /* 0x00008000UL */
```

Definition at line 310 of file form3.h.

4.45.2.12 TOPLONGBITONLY

```
#define TOPLONGBITONLY ((ULONG)1 << (BITSINLONG - 1)) /* 0x80000000UL */
```

Definition at line 311 of file form3.h.

4.45.2.13 SPECMASK

```
#define SPECMASK ((UWORD)1 << (BITSINWORD - 1)) /* 0x8000U */
```

Definition at line 312 of file form3.h.

4.45.2.14 WILDMASK

```
#define WILDMASK ((UWORD)1 << (BITSINWORD - 2)) /* 0x4000U */
```

Definition at line 313 of file form3.h.

4.45.2.15 WORDMASK

```
#define WORDMASK ((ULONG)FULLMAX - 1) /* 0x0000FFFFUL */
```

Definition at line 314 of file form3.h.

4.45.2.16 AWORDMASK

```
#define AWORDMASK (WORDMASK << BITSINWORD) /* 0xFFFF0000UL */
```

Definition at line 315 of file form3.h.

4.45.2.17 FULLMAX

```
#define FULLMAX ((LONG)1 << BITSINWORD) /* 0x00010000L */
```

Definition at line 316 of file form3.h.

4.45.2.18 MAXPOSITIVE

```
#define MAXPOSITIVE ((LONG)(TOPBITONLY - 1)) /* 0x00007FFFL */
```

Definition at line 317 of file form3.h.

4.45.2.19 MAXLONG

```
#define MAXLONG ((LONG)(TOPLONGBITONLY - 1)) /* 0x7FFFFFFFL */
```

Definition at line 318 of file form3.h.

4.45.2.20 MAXPOSITIVE2

```
#define MAXPOSITIVE2 (MAXPOSITIVE / 2) /* 0x00003FFFL */
```

Definition at line 319 of file form3.h.

4.45.2.21 MAXPOSITIVE4

```
#define MAXPOSITIVE4 (MAXPOSITIVE / 4) /* 0x00001FFFL */
```

Definition at line 320 of file form3.h.

4.45.2.22 form_alignof

```
#define form_alignof(
              type ) offsetof(struct { char c_; type x_; }, x_)
```

Definition at line 336 of file form3.h.

4.45.2.23 PADDUMMY

```
#define PADDUMMY(
              type,
              size )  UBYTE d_u_m_m_y[form_alignof(type) - ((size) & (form_alignof(type) -
1))]
```

Definition at line 391 of file form3.h.

4.45.2.24 PADPOSITION

```
#define PADPOSITION(
              ptr_,
              long_,
              int_,
              word_,
              byte_ )
```

Value:

```
PADDUMMY(off_t, \
    + sizeof(int *) * (ptr_) \
    + sizeof(LONG)  * (long_) \
    + sizeof(int)   * (int_) \
    + sizeof(WORD)  * (word_) \
    + sizeof(UBYTE) * (byte_) \
)
```

Definition at line 393 of file form3.h.

4.45.2.25 PADPOINTER

```
#define PADPOINTER(
              long_,
              int_,
              word_,
              byte_ )
```

Value:

```
PADDUMMY(int *, \
    + sizeof(LONG)  * (long_) \
    + sizeof(int)   * (int_) \
    + sizeof(WORD)  * (word_) \
    + sizeof(UBYTE) * (byte_) \
)
```

Definition at line 401 of file form3.h.

4.45.2.26 PADLONG

```
#define PADLONG(
              int_,
              word_,
              byte_ )
```

Value:

```
PADDUMMY(LONG, \
    + sizeof(int)   * (int_) \
    + sizeof(WORD)  * (word_) \
    + sizeof(UBYTE) * (byte_) \
)
```

Definition at line 408 of file form3.h.

4.45.2.27 PADINT

```
#define PADINT(
            word_,
            byte_ )
```

Value:
```
PADDUMMY(int, \
    + sizeof(WORD)  * (word_) \
    + sizeof(UBYTE) * (byte_) \
)
```

Definition at line 414 of file form3.h.

4.45.2.28 PADWORD

```
#define PADWORD(
            byte_ )
```

Value:
```
PADDUMMY(WORD, \
    + sizeof(UBYTE) * (byte_) \
)
```

Definition at line 419 of file form3.h.

4.45.2.29 WITHSORTBOTS

```
#define WITHSORTBOTS
```

Definition at line 429 of file form3.h.

4.45.2.30 FILES

```
#define FILES FILE
```

Definition at line 511 of file form3.h.

4.45.2.31 Uopen

```
#define Uopen(
            x,
            y ) fopen(x,y)
```

Definition at line 512 of file form3.h.

4.45.2.32 Uflush

```
#define Uflush(
            x ) fflush(x)
```

Definition at line 513 of file form3.h.

4.45.2.33 Uclose

```
#define Uclose(
              x ) fclose(x)
```

Definition at line 514 of file form3.h.

4.45.2.34 Uread

```
#define Uread(
              x,
              y,
              z,
              u ) fread(x,y,z,u)
```

Definition at line 515 of file form3.h.

4.45.2.35 Uwrite

```
#define Uwrite(
              x,
              y,
              z,
              u ) fwrite(x,y,z,u)
```

Definition at line 516 of file form3.h.

4.45.2.36 Usetbuf

```
#define Usetbuf(
              x,
              y ) setbuf(x,y)
```

Definition at line 517 of file form3.h.

4.45.2.37 Useek

```
#define Useek(
              x,
              y,
              z ) fseek(x,y,z)
```

Definition at line 518 of file form3.h.

4.45.2.38 Utell

```
#define Utell(
              x ) ftell(x)
```

Definition at line 519 of file form3.h.

4.45.2.39 Ugetpos

```
#define Ugetpos(
            x,
            y ) fgetpos(x,y)
```

Definition at line 520 of file form3.h.

4.45.2.40 Usetpos

```
#define Usetpos(
            x,
            y ) fsetpos(x,y)
```

Definition at line 521 of file form3.h.

4.45.2.41 Usync

```
#define Usync(
            x ) fflush(x)
```

Definition at line 522 of file form3.h.

4.45.2.42 Utruncate

```
#define Utruncate(
            x ) _chsize(_fileno(x),0)
```

Definition at line 523 of file form3.h.

4.45.2.43 Ustdout

```
#define Ustdout stdout
```

Definition at line 524 of file form3.h.

4.45.2.44 MAX_OPEN_FILES

```
#define MAX_OPEN_FILES FOPEN_MAX
```

Definition at line 525 of file form3.h.

4.45.2.45 bzero

```
#define bzero(
            b,
            len ) (memset((b), 0, (len)), (void)0)
```

Definition at line 526 of file form3.h.

4.45.2.46 GetPID

```
#define GetPID( ) ((LONG)GetCurrentProcessId())
```

Definition at line 527 of file form3.h.

4.45.3 Typedef Documentation

4.45.3.1 SBYTE

```
typedef signed char SBYTE
```

Definition at line 299 of file form3.h.

4.45.3.2 UBYTE

```
typedef unsigned char UBYTE
```

Definition at line 300 of file form3.h.

4.45.3.3 UINT

```
typedef unsigned int UINT
```

Definition at line 301 of file form3.h.

4.45.3.4 RLONG

```
typedef ULONG RLONG
```

Definition at line 302 of file form3.h.

4.45.3.5 MLONG

```
typedef INT64 MLONG
```

Definition at line 303 of file form3.h.

4.45.3.6 BOOL

```
typedef char BOOL
```

Definition at line 308 of file form3.h.

4.46 form3.h

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
#ifndef __FORM3H__
#define __FORM3H__
#ifdef HAVE_CONFIG_H
#ifndef CONFIG_H_INCLUDED
#define CONFIG_H_INCLUDED
#include <config.h>
#endif
#else  /* HAVE_CONFIG_H */
#define MAJORVERSION 5
#define MINORVERSION 0
#ifdef __DATE__
#define PRODUCTIONDATE __DATE__
#else
#define PRODUCTIONDATE "8-nov-2022"
#endif
/*#undef BETAVERSION */
#define BETAVERSION
#ifdef LINUX32
#define UNIX
#define LINUX
#define ILP32
#define SIZEOF_LONG_LONG 8
#define _FILE_OFFSET_BITS 64
#define WITHZLIB
#define WITHGMP
#define WITHPOSIXCLOCK
#endif
#ifdef LINUX64
#define UNIX
#define LINUX
#define LP64
#define WITHZLIB
#define WITHGMP
#define WITHPOSIXCLOCK
#define WITHFLOAT
#endif
#ifdef APPLE32
#define UNIX
#define ILP32
#define SIZEOF_LONG_LONG 8
#define _FILE_OFFSET_BITS 64
#define WITHZLIB
#endif
#ifdef APPLE64
```

```
#define UNIX
#define LP64
#define WITHZLIB
#define WITHGMP
#define WITHPOSIXCLOCK
#define WITHFLOAT
#define HAVE_UNORDERED_MAP
#define HAVE_UNORDERED_SET
#endif

#ifdef CYGWIN32
#define UNIX
#define ILP32
#define SIZEOF_LONG_LONG 8
#endif

#ifdef _MSC_VER
#define WINDOWS
#define _CRT_SECURE_NO_WARNINGS
#if defined(_WIN64)
#define LLP64
#elif defined(_WIN32)
#define ILP32
#define SIZEOF_LONG_LONG 8
#endif
#endif

/*
 * We must not define WITHPOSIXCLOCK in compiling the sequential FORM or ParFORM.
 */
#if !defined(WITHPTHREADS) && defined(WITHPOSIXCLOCK)
#undef WITHPOSIXCLOCK
#endif

#if !defined(__cplusplus) && !defined(inline)
#if (defined(__STDC_VERSION__) && __STDC_VERSION__ >= 199901L) || (defined(__GNUC__) && !defined(__STRICT_ANSI__))
/* "inline" is available. */
#elif defined(__GNUC__)
/* GNU C compiler has "__inline__". */
#define inline __inline__
#elif defined(_MSC_VER)
/* Microsoft C compiler has "__inline". */
#define inline __inline
#else
/* Inline functions may be not supported. Define "inline" to be empty. */
#define inline
#endif
#endif

#endif  /* HAVE_CONFIG_H */

/* Workaround for MSVC. */
#if defined(_MSC_VER)
/*
 * Old versions of MSVC didn't support C99 function `snprintf`, which is used
 * in poly.cc. On the other hand, macroizing `snprintf` gives a fatal error
 * with MSVC >= 2015.
 */
#if _MSC_VER < 1900
#define snprintf _snprintf
#endif
#endif

/*
 * Translate our dialect "DEBUGGING" to the standard "NDEBUG".
 */
#ifdef DEBUGGING
#ifdef NDEBUG
#undef NDEBUG
#endif
#else
#ifndef NDEBUG
#define NDEBUG
#endif
#endif

/*
 * STATIC_ASSERT(condition) will fail to be compiled if the given
 * condition is false.
 */
/*
#define STATIC_ASSERT(condition)
*/
#define STATIC_ASSERT(condition) STATIC_ASSERT__1(condition,__LINE__)
#define STATIC_ASSERT__1(X,L) STATIC_ASSERT__2(X,L)
#define STATIC_ASSERT__2(X,L) STATIC_ASSERT__3(X,L)
```

```
#define STATIC_ASSERT__3(X,L) \
    typedef char static_assertion_failed_##L[(!!(X))*2-1]

/*
 * UNIX or WINDOWS must be defined.
 */
#if defined(UNIX)
#define mBSD
#define ANSI
#elif defined(WINDOWS)
#define ANSI
#define WIN32_LEAN_AND_MEAN
#include <windows.h>
#include <io.h>
#include <fcntl.h>
/* Undefine/rename conflicted symbols. */
#undef MAXLONG  /* WinNT.h */
#define WORD FORM_WORD  /* WinDef.h */
#define LONG FORM_LONG /* WinNT.h */
#define ULONG FORM_ULONG  /* WinDef.h */
#define BOOL FORM_BOOL  /* WinDef.h */
#undef CreateFile  /* WinBase.h */
#undef CopyFile  /* WinBase.h */
#define OpenFile FORM_OpenFile  /* WinBase.h */
#define ReOpenFile FORM_ReOpenFile  /* WinBase.h */
#define ReadFile FORM_ReadFile  /* WinBase.h */
#define WriteFile FORM_WriteFile  /* WinBase.h */
#define DeleteObject FORM_DeleteObject  /* WinGDI.h */
#else
#error UNIX or WINDOWS must be defined!
#endif

/*
 * Data model. ILP32 or LLP64 or LP64 must be defined.
 *
 * Here we define basic types WORD, LONG and their unsigned versions
 * UWORD and ULONG. LONG must be double size of WORD. Their actual types
 * are system-dependent. BITSINWORD and BITSINLONG are also defined.
 * INT16, INT32 (also INT64 and INT128 if available) are used for
 * system independent saved expressions (store.c).
 */
#if defined(ILP32)

typedef short WORD;
typedef long LONG;
typedef unsigned short UWORD;
typedef unsigned long ULONG;
#define BITSINWORD 16
#define BITSINLONG 32
#define INT16 short
#define INT32 int
#undef INT64
#undef INT128

#ifdef SIZEOF_LONG_LONG
#if SIZEOF_LONG_LONG == 8
#define INT64 long long
#endif
#endif

#ifndef INT64
#error INT64 is not available!
#endif

#define WORD_MIN_VALUE SHRT_MIN
#define WORD_MAX_VALUE SHRT_MAX
#define LONG_MIN_VALUE LONG_MIN
#define LONG_MAX_VALUE LONG_MAX

#elif defined(LLP64)

typedef int WORD;
typedef long long LONG;
typedef unsigned int UWORD;
typedef unsigned long long ULONG;
#define BITSINWORD 32
#define BITSINLONG 64
#define INT16 short
#define INT32 int
#define INT64 long long
#undef INT128

#define WORD_MIN_VALUE INT_MIN
#define WORD_MAX_VALUE INT_MAX
#define LONG_MIN_VALUE LLONG_MIN
#define LONG_MAX_VALUE LLONG_MAX

```

```
#elif defined(LP64)

typedef int WORD;
typedef long LONG;
typedef unsigned int UWORD;
typedef unsigned long ULONG;
#define BITSINWORD 32
#define BITSINLONG 64
#define INT16 short
#define INT32 int
#define INT64 long
#undef INT128

#define WORD_MIN_VALUE INT_MIN
#define WORD_MAX_VALUE INT_MAX
#define LONG_MIN_VALUE LONG_MIN
#define LONG_MAX_VALUE LONG_MAX

#else
#error ILP32 or LLP64 or LP64 must be defined!
#endif

STATIC_ASSERT(sizeof(WORD) * 8 == BITSINWORD);
STATIC_ASSERT(sizeof(LONG) * 8 == BITSINLONG);
STATIC_ASSERT(sizeof(WORD) * 2 == sizeof(LONG));
STATIC_ASSERT(sizeof(LONG) >= sizeof(int *));
STATIC_ASSERT(sizeof(INT16) == 2);
STATIC_ASSERT(sizeof(INT32) == 4);
STATIC_ASSERT(sizeof(INT64) == 8);
#ifdef INT128
STATIC_ASSERT(sizeof(INT128) == 16);
#endif

#if BITSINWORD == 32
#define WORDSIZE32 1
#endif

typedef signed char SBYTE;
typedef unsigned char UBYTE;
typedef unsigned int UINT;
typedef ULONG RLONG;  /* Used in reken.c. */
typedef INT64 MLONG;  /* See commentary in minos.h. */
/*
 * NOTE: we don't use the standard _Bool (or C++ bool) because its size is
 * implementation-dependent and messes up the traditional PADXXX macros.
 */
typedef char BOOL;
                                                      /* E.g. in 32-bits */
#define TOPBITONLY      ((ULONG)1 « (BITSINWORD - 1))  /* 0x00008000UL */
#define TOPLONGBITONLY  ((ULONG)1 « (BITSINLONG - 1))  /* 0x80000000UL */
#define SPECMASK        ((UWORD)1 « (BITSINWORD - 1))  /*     0x8000U  */
#define WILDMASK        ((UWORD)1 « (BITSINWORD - 2))  /*     0x4000U  */
#define WORDMASK        ((ULONG)FULLMAX - 1)            /* 0x0000FFFFUL */
#define AWORDMASK       (WORDMASK « BITSINWORD)        /* 0xFFFF0000UL */
#define FULLMAX         ((LONG)1 « BITSINWORD)         /* 0x00010000L  */
#define MAXPOSITIVE     ((LONG)(TOPBITONLY - 1))        /* 0x00007FFFL  */
#define MAXLONG         ((LONG)(TOPLONGBITONLY - 1))    /* 0x7FFFFFFFL  */
#define MAXPOSITIVE2    (MAXPOSITIVE / 2)               /* 0x00003FFFL  */
#define MAXPOSITIVE4    (MAXPOSITIVE / 4)               /* 0x00001FFFL  */

/*
 * form_alignof(type) returns the number of bytes used in the alignment of
 * the type.
 */
#if !defined(form_alignof)
#if defined(__GNUC__)
/* GNU C compiler has "__alignof__". */
#define form_alignof(type) __alignof__(type)
#elif defined(_MSC_VER)
/* Microsoft C compiler has "__alignof". */
#define form_alignof(type) __alignof(type)
#elif !defined(__cplusplus)
/* Generic case in C. */
#include <stddef.h>
#define form_alignof(type) offsetof(struct { char c_; type x_; }, x_)
#else
/* Generic case in C++, at least works with a POD struct. */
#include <cstddef>
namespace alignof_impl_ {
template<typename T> struct calc {
    struct X { char c_; T x_; };
    enum { value = offsetof(X, x_) };
};
}
#define form_alignof(type) alignof_impl_::calc<type>::value
#endif
#endif
```

```

/*
 * Macros to be inserted at the end of a structure to align the whole structure.
 *
 * In the currently available systems,
 *   sizeof(POSITION) >= sizeof(pointers) == sizeof(LONG) >= sizeof(int)
 *                    >= sizeof(WORD) >= sizeof(UBYTE) = 1.
 * (POSITION is defined in struct.h and contains only an off_t variable.)
 * Thus, if we put members of a structure in this order and use those macros,
 * then we can align the data without relying on extra paddings added by
 * the compiler. For example,
 *   typedef struct {
 *     int *a;
 *     LONG b;
 *     WORD c[2];
 *     UBYTE d;
 *     PADPOINTER(1,0,2,1);
 *   } A;
 *   typedef struct {
 *     POSITION p;
 *     A a;  // aligned same as pointers
 *     int *b;
 *     LONG c;
 *     UBYTE d;
 *     PADPOSITION(1,1,0,0,1+sizeof(A));
 *   } B;
 * The cost for the use of those PADXXX macros is a padding (>= 1 byte) will
 * be always inserted even in the case that no padding is actually needed.
 *
 * Numbers for the arguments have to be calculated manually and so very
 * error-prone. Be careful!
 *
 * Note that there is a 32-bit system in which off_t is aligned on 8-byte
 * boundary, (e.g., Cygwin with large file support), but still the above
 * inequalities are satisfied.
 *
 * The legendary story of these macros--they fixed some problems in ancient
 * times when compilers were unreliable and didn't know how to correctly compute
 * structure paddings--has been handed down, though nowadays there are only
 * disadvantages for them in practice (ancient compilers most likely can't
 * compile C99 and C++98+TR1 sources anyway).
 */
#define PADDUMMY(type, size) \
    UBYTE d_u_m_m_y[form_alignof(type) - ((size) & (form_alignof(type) - 1))]
#define PADPOSITION(ptr_,long_,int_,word_,byte_) \
    PADDUMMY(off_t, \
        + sizeof(int *) * (ptr_) \
        + sizeof(LONG)  * (long_) \
        + sizeof(int)   * (int_) \
        + sizeof(WORD)  * (word_) \
        + sizeof(UBYTE) * (byte_) \
    )
#define PADPOINTER(long_,int_,word_,byte_) \
    PADDUMMY(int *, \
        + sizeof(LONG)  * (long_) \
        + sizeof(int)   * (int_) \
        + sizeof(WORD)  * (word_) \
        + sizeof(UBYTE) * (byte_) \
    )
#define PADLONG(int_,word_,byte_) \
    PADDUMMY(LONG, \
        + sizeof(int)   * (int_) \
        + sizeof(WORD)  * (word_) \
        + sizeof(UBYTE) * (byte_) \
    )
#define PADINT(word_,byte_) \
    PADDUMMY(int, \
        + sizeof(WORD)  * (word_) \
        + sizeof(UBYTE) * (byte_) \
    )
#define PADWORD(byte_) \
    PADDUMMY(WORD, \
        + sizeof(UBYTE) * (byte_) \
    )

/*
#define WITHPCOUNTER
#define DEBUGGINGLOCKS
#define WITHSTATS
*/
#define WITHSORTBOTS

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <limits.h>
```

```
#ifdef ANSI
#include <stdarg.h>
#include <time.h>
#endif
#ifdef WINDOWS
#include "fwin.h"
#endif
#ifdef UNIX
#include <unistd.h>
#include <time.h>
#include <fcntl.h>
#include <sys/file.h>
#include "unix.h"
#endif
#ifdef WITHZLIB
#ifdef WITHZSTD
#include <zstd_zlibwrapper.h>
#else
#include <zlib.h>
#endif
#endif
#ifdef WITHPTHREADS
#include <pthread.h>
#endif

/*
    PARALLELCODE indicates code that is common for TFORM and ParFORM but
    should not be there for sequential FORM.
*/
#if defined(WITHMPI) || defined(WITHPTHREADS)
#define PARALLELCODE
#endif

#include "ftypes.h"
#include "fsizes.h"
#include "minos.h"
#include "structs.h"
#include "declare.h"
#include "variable.h"

/*
 * The interface to file routines for UNIX or non-UNIX (Windows).
 */
#ifdef UNIX

#define UFILES
typedef struct FiLeS {
    int descriptor;
} FILES;
extern FILES *Uopen(char *,char *);
extern int    Uclose(FILES *);
extern size_t Uread(char *,size_t,size_t,FILES *);
extern size_t Uwrite(char *,size_t,size_t,FILES *);
extern int    Useek(FILES *,off_t,int);
extern off_t  Utell(FILES *);
extern void   Uflush(FILES *);
extern int    Ugetpos(FILES *,fpos_t *);
extern int    Usetpos(FILES *,fpos_t *);
extern void   Usetbuf(FILES *,char *);
#define Usync(f) fsync(f->descriptor)
#define Utruncate(f) { \
    if ( ftruncate(f->descriptor, 0) ) { \
        MLOCK(ErrorMessageLock); \
        MesPrint("Utruncate failed"); \
        MUNLOCK(ErrorMessageLock); \
        /* Calling Terminate() here may cause an infinite loop due to CleanUpSort(). */ \
        /* Terminate(-1); */ \
    } \
}
extern FILES *Ustdout;
#define MAX_OPEN_FILES getdtablesize()
#define GetPID() ((LONG)getpid())

#else  /* UNIX */

#define FILES FILE
#define Uopen(x,y) fopen(x,y)
#define Uflush(x) fflush(x)
#define Uclose(x) fclose(x)
#define Uread(x,y,z,u) fread(x,y,z,u)
#define Uwrite(x,y,z,u) fwrite(x,y,z,u)
#define Usetbuf(x,y) setbuf(x,y)
#define Useek(x,y,z) fseek(x,y,z)
#define Utell(x) ftell(x)
#define Ugetpos(x,y) fgetpos(x,y)
#define Usetpos(x,y) fsetpos(x,y)
#define Usync(x) fflush(x)
```

```
#define Utruncate(x) _chsize(_fileno(x),0)
#define Ustdout stdout
#define MAX_OPEN_FILES FOPEN_MAX
#define bzero(b,len) (memset((b), 0, (len)), (void)0)
#define GetPID() ((LONG)GetCurrentProcessId())

#endif  /* UNIX */

/*
 * Some system may implement the POSIX "environ" variable as a macro, e.g.,
 * https://github.com/mingw-w64/mingw-w64/blob/v10.0.0/mingw-w64-headers/crt/stdlib.h#L704
 * which breaks the definition of DoShattering() in diagrams.c that uses
 * "environ" as a formal parameter. Because FORM doesn't use the POSIX "environ"
 * or its variant anyway, we can just undefine it.
 */
#ifdef environ
#undef environ
#endif

#ifdef WITHMPI
#include "parallel.h"
#endif

#endif  /*  __FORM3H__ */
```

4.47 fsizes.h File Reference

This graph shows which files directly or indirectly include this file:

Macros

- #define MAXPRENAMESIZE 128
- #define MAXPOWER 10000
- #define MAXVARIABLES 8050
- #define MAXDOLLARVARIABLES 32000
- #define WILDOFFSET 6100
- #define MAXINNAMETREE 32768
- #define MAXDUMMIES 1000
- #define WORKBUFFER 10000000
- #define MAXTER 10000
- #define HALFMAX 0x100
- #define MAXSUBEXPRESSIONS 0x3FFF
- #define MAXFILESTREAMSIZE 1048576
- #define MAXENAME 16
- #define MAXSAVEFUNCTION 16384
- #define MAXPARLEVEL 100
- #define MAXNUMBERSIZE 200
- #define MAXREPEAT 100
- #define NORMSIZE 1000
- #define INITNODESIZE 10
- #define INITNAMESIZE 100
- #define NUMFIXED 128
- #define MAXNEST 100
- #define MAXMATCH 30
- #define MAXIF 20
- #define SIZEFACS 640L

- #define NUMFACS 50
- #define MAXLOOPS 30
- #define MAXLABELS 20
- #define COMMERCIALSIZE 24
- #define MAXFLAGS 16
- #define COMPRESSBUFFER 90000
- #define FORTRANCONTINUATIONLINES 15
- #define MAXLEVELS 2000
- #define MAXLHS 400
- #define MAXWILDC 100
- #define NUMTABLEENTRIES 1000
- #define COMPILERBUFFER 20000
- #define SMALLBUFFER 10000000L
- #define SMALLOVERFLOW 20000000L
- #define TERMSSMALL 100000L
- #define LARGEBUFFER 50000000L
- #define SCRATCHSIZE 50000000L
- #define MAXPATCHES 256
- #define MAXFPATCHES 256
- #define SORTIOSIZE 200000L
- #define SSMALLBUFFER 2560016L
- #define SSMALLOVERFLOW 3840032L
- #define STERMSSMALL 10000L
- #define SLARGEBUFFER 26880512L
- #define SMAXPATCHES 64
- #define SMAXFPATCHES 64
- #define SSORTIOSIZE 32768L
- #define SPECTATORSIZE 1048576L
- #define MAXFLEVELS 30
- #define COMPINC 2
- #define MAXNUMSIZE 10
- #define MAXBRACKETBUFFERSIZE 200000
- #define SFHSIZE 40
- #define DEFAULTPROCESSBUCKETSIZE 1000
- #define SHMWINSIZE 65536L
- #define TABLEEXTENSION 6
- #define GZIPDEFAULT 6
- #define DEFAULTTHREADS 0
- #define DEFAULTTHREADBUCKETSIZE 500
- #define DEFAULTTHREADLOADBALANCING 1
- #define THREADSCRATCHSIZE 100000L
- #define THREADSCRATCHOUTSIZE 2500000L
- #define MAXTABLECOMBUF 1000000L
- #define MAXCOMBUFRHS 32500L
- #define NUMSTORECACHES 4
- #define SIZESTORECACHE 32768
- #define INDENTSPACE 3
- #define MULTIINDENTSPACE 1
- #define MAXMULTIBRACKETLEVELS 25
- #define FBUFFERSIZE 1026
- #define NPAIR1 38
- #define NPAIR2 89
- #define MAXLINELENGTH 256
- #define MINALLOC 32
- #define JUMPRATIO 4

- #define MAXPARTICLES 20
- #define MAXCOUPLINGS 20
- #define NUMOPTIONS 20
- #define MAXLEGS 20

4.47.1 Detailed Description

The definition the default values for certain buffer sizes etc.

Definition in file fsizes.h.

4.47.2 Macro Definition Documentation

4.47.2.1 MAXPRENAMESIZE

```
#define MAXPRENAMESIZE 128
```

Definition at line 36 of file fsizes.h.

4.47.2.2 MAXPOWER

```
#define MAXPOWER 10000
```

Definition at line 58 of file fsizes.h.

4.47.2.3 MAXVARIABLES

```
#define MAXVARIABLES 8050
```

Definition at line 59 of file fsizes.h.

4.47.2.4 MAXDOLLARVARIABLES

```
#define MAXDOLLARVARIABLES 32000
```

Definition at line 60 of file fsizes.h.

4.47.2.5 WILDOFFSET

```
#define WILDOFFSET 6100
```

Definition at line 61 of file fsizes.h.

4.47.2.6 MAXINNAMETREE

```
#define MAXINNAMETREE 32768
```

Definition at line 62 of file fsizes.h.

4.47.2.7 MAXDUMMIES

```
#define MAXDUMMIES 1000
```

Definition at line 63 of file fsizes.h.

4.47.2.8 WORKBUFFER

```
#define WORKBUFFER 10000000
```

Definition at line 64 of file fsizes.h.

4.47.2.9 MAXTER

```
#define MAXTER 10000
```

Definition at line 65 of file fsizes.h.

4.47.2.10 HALFMAX

```
#define HALFMAX 0x100
```

Definition at line 66 of file fsizes.h.

4.47.2.11 MAXSUBEXPRESSIONS

```
#define MAXSUBEXPRESSIONS 0x3FFF
```

Definition at line 67 of file fsizes.h.

4.47.2.12 MAXFILESTREAMSIZE

```
#define MAXFILESTREAMSIZE 1048576
```

Definition at line 68 of file fsizes.h.

4.47.2.13 MAXENAME

```
#define MAXENAME 16
```

Definition at line 70 of file fsizes.h.

4.47.2.14 MAXSAVEFUNCTION

```
#define MAXSAVEFUNCTION 16384
```

Definition at line 71 of file fsizes.h.

4.47.2.15 MAXPARLEVEL

```
#define MAXPARLEVEL 100
```

Definition at line 73 of file fsizes.h.

4.47.2.16 MAXNUMBERSIZE

```
#define MAXNUMBERSIZE 200
```

Definition at line 74 of file fsizes.h.

4.47.2.17 MAXREPEAT

```
#define MAXREPEAT 100
```

Definition at line 76 of file fsizes.h.

4.47.2.18 NORMSIZE

```
#define NORMSIZE 1000
```

Definition at line 77 of file fsizes.h.

4.47.2.19 INITNODESIZE

```
#define INITNODESIZE 10
```

Definition at line 79 of file fsizes.h.

4.47.2.20 INITNAMESIZE

```
#define INITNAMESIZE 100
```

Definition at line 80 of file fsizes.h.

4.47.2.21 NUMFIXED

```
#define NUMFIXED 128
```

Definition at line 82 of file fsizes.h.

4.47.2.22 MAXNEST

```
#define MAXNEST 100
```

Definition at line 83 of file fsizes.h.

4.47.2.23 MAXMATCH

```
#define MAXMATCH 30
```

Definition at line 84 of file fsizes.h.

4.47.2.24 MAXIF

```
#define MAXIF 20
```

Definition at line 85 of file fsizes.h.

4.47.2.25 SIZEFACS

```
#define SIZEFACS 640L
```

Definition at line 86 of file fsizes.h.

4.47.2.26 NUMFACS

```
#define NUMFACS 50
```

Definition at line 87 of file fsizes.h.

4.47.2.27 MAXLOOPS

```
#define MAXLOOPS 30
```

Definition at line 88 of file fsizes.h.

4.47.2.28 MAXLABELS

```
#define MAXLABELS 20
```

Definition at line 89 of file fsizes.h.

4.47.2.29 COMMERCIALSIZE

```
#define COMMERCIALSIZE 24
```

Definition at line 90 of file fsizes.h.

4.47.2.30 MAXFLAGS

```
#define MAXFLAGS 16
```

Definition at line 91 of file fsizes.h.

4.47.2.31 COMPRESSBUFFER

```
#define COMPRESSBUFFER 90000
```

Definition at line 96 of file fsizes.h.

4.47.2.32 FORTRANCONTINUATIONLINES

```
#define FORTRANCONTINUATIONLINES 15
```

Definition at line 97 of file fsizes.h.

4.47.2.33 MAXLEVELS

```
#define MAXLEVELS 2000
```

Definition at line 98 of file fsizes.h.

4.47.2.34 MAXLHS

```
#define MAXLHS 400
```

Definition at line 99 of file fsizes.h.

4.47.2.35 MAXWILDC

```
#define MAXWILDC 100
```

Definition at line 100 of file fsizes.h.

4.47.2.36 NUMTABLEENTRIES

```
#define NUMTABLEENTRIES 1000
```

Definition at line 101 of file fsizes.h.

4.47.2.37 COMPILERBUFFER

```
#define COMPILERBUFFER 20000
```

Definition at line 102 of file fsizes.h.

4.47.2.38 SMALLBUFFER

```
#define SMALLBUFFER 10000000L
```

Definition at line 119 of file fsizes.h.

4.47.2.39 SMALLOVERFLOW

```
#define SMALLOVERFLOW 20000000L
```

Definition at line 120 of file fsizes.h.

4.47.2.40 TERMSSMALL

```
#define TERMSSMALL 100000L
```

Definition at line 121 of file fsizes.h.

4.47.2.41 LARGEBUFFER

```
#define LARGEBUFFER 50000000L
```

Definition at line 122 of file fsizes.h.

4.47.2.42 SCRATCHSIZE

```
#define SCRATCHSIZE 50000000L
```

Definition at line 123 of file fsizes.h.

4.47.2.43 MAXPATCHES

```
#define MAXPATCHES 256
```

Definition at line 125 of file fsizes.h.

4.47.2.44 MAXFPATCHES

```
#define MAXFPATCHES 256
```

Definition at line 126 of file fsizes.h.

4.47.2.45 SORTIOSIZE

```
#define SORTIOSIZE 200000L
```

Definition at line 127 of file fsizes.h.

4.47.2.46 SSMALLBUFFER

```
#define SSMALLBUFFER 2560016L
```

Definition at line 129 of file fsizes.h.

4.47.2.47 SSMALLOVERFLOW

```
#define SSMALLOVERFLOW 3840032L
```

Definition at line 130 of file fsizes.h.

4.47.2.48 STERMSSMALL

```
#define STERMSSMALL 10000L
```

Definition at line 131 of file fsizes.h.

4.47.2.49 SLARGEBUFFER

```
#define SLARGEBUFFER 26880512L
```

Definition at line 132 of file fsizes.h.

4.47.2.50 SMAXPATCHES

```
#define SMAXPATCHES 64
```

Definition at line 133 of file fsizes.h.

4.47.2.51 SMAXFPATCHES

```
#define SMAXFPATCHES 64
```

Definition at line 134 of file fsizes.h.

4.47.2.52 SSORTIOSIZE

```
#define SSORTIOSIZE 32768L
```

Definition at line 135 of file fsizes.h.

4.47.2.53 SPECTATORSIZE

```
#define SPECTATORSIZE 1048576L
```

Definition at line 137 of file fsizes.h.

4.47.2.54 MAXFLEVELS

```
#define MAXFLEVELS 30
```

Definition at line 139 of file fsizes.h.

4.47.2.55 COMPINC

```
#define COMPINC 2
```

Definition at line 141 of file fsizes.h.

4.47.2.56 MAXNUMSIZE

```
#define MAXNUMSIZE 10
```

Definition at line 143 of file fsizes.h.

4.47.2.57 MAXBRACKETBUFFERSIZE

```
#define MAXBRACKETBUFFERSIZE 200000
```

Definition at line 145 of file fsizes.h.

4.47.2.58 SFHSIZE

```
#define SFHSIZE 40
```

Definition at line 147 of file fsizes.h.

4.47.2.59 DEFAULTPROCESSBUCKETSIZE

```
#define DEFAULTPROCESSBUCKETSIZE 1000
```

Definition at line 149 of file fsizes.h.

4.47.2.60 SHMWINSIZE

```
#define SHMWINSIZE 65536L
```

Definition at line 150 of file fsizes.h.

4.47.2.61 TABLEEXTENSION

```
#define TABLEEXTENSION 6
```

Definition at line 152 of file fsizes.h.

4.47.2.62 GZIPDEFAULT

```
#define GZIPDEFAULT 6
```

Definition at line 154 of file fsizes.h.

4.47.2.63 DEFAULTTHREADS

```
#define DEFAULTTHREADS 0
```

Definition at line 155 of file fsizes.h.

4.47.2.64 DEFAULTTHREADBUCKETSIZE

```
#define DEFAULTTHREADBUCKETSIZE 500
```

Definition at line 156 of file fsizes.h.

4.47.2.65 DEFAULTTHREADLOADBALANCING

```
#define DEFAULTTHREADLOADBALANCING 1
```

Definition at line 157 of file fsizes.h.

4.47.2.66 THREADSCRATCHSIZE

```
#define THREADSCRATCHSIZE 100000L
```

Definition at line 158 of file fsizes.h.

4.47.2.67 THREADSCRATCHOUTSIZE

```
#define THREADSCRATCHOUTSIZE 2500000L
```

Definition at line 159 of file fsizes.h.

4.47.2.68 MAXTABLECOMBUF

```
#define MAXTABLECOMBUF 1000000L
```

Definition at line 165 of file fsizes.h.

4.47.2.69 MAXCOMBUFRHS

```
#define MAXCOMBUFRHS 32500L
```

Definition at line 166 of file fsizes.h.

4.47.2.70 NUMSTORECACHES

```
#define NUMSTORECACHES 4
```

Definition at line 169 of file fsizes.h.

4.47.2.71 SIZESTORECACHE

```
#define SIZESTORECACHE 32768
```

Definition at line 170 of file fsizes.h.

4.47.2.72 INDENTSPACE

```
#define INDENTSPACE 3
```

Definition at line 172 of file fsizes.h.

4.47.2.73 MULTIINDENTSPACE

```
#define MULTIINDENTSPACE 1
```

Definition at line 174 of file fsizes.h.

4.47.2.74 MAXMULTIBRACKETLEVELS

```
#define MAXMULTIBRACKETLEVELS 25
```

Definition at line 175 of file fsizes.h.

4.47.2.75 FBUFFERSIZE

```
#define FBUFFERSIZE 1026
```

Definition at line 177 of file fsizes.h.

4.47.2.76 NPAIR1

```
#define NPAIR1 38
```

Definition at line 181 of file fsizes.h.

4.47.2.77 NPAIR2

```
#define NPAIR2 89
```

Definition at line 182 of file fsizes.h.

4.47.2.78 MAXLINELENGTH

```
#define MAXLINELENGTH 256
```

Definition at line 184 of file fsizes.h.

4.47.2.79 MINALLOC

```
#define MINALLOC 32
```

Definition at line 185 of file fsizes.h.

4.47.2.80 JUMPRATIO

```
#define JUMPRATIO 4
```

Definition at line 187 of file fsizes.h.

4.47.2.81 MAXPARTICLES

```
#define MAXPARTICLES 20
```

Definition at line 191 of file fsizes.h.

4.47.2.82 MAXCOUPLINGS

```
#define MAXCOUPLINGS 20
```

Definition at line 192 of file fsizes.h.

4.47.2.83 NUMOPTIONS

```
#define NUMOPTIONS 20
```

Definition at line 193 of file fsizes.h.

4.47.2.84 MAXLEGS

```
#define MAXLEGS 20
```

Definition at line 194 of file fsizes.h.

4.48 fsizes.h

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

/*
    First the fixed variables
*/
#define MAXPRENAMESIZE 128
/*
    The following variables are default sizes. They can be changed
    into values read from the setup file

    Remark (21-dec-2008 JV): WILDOFFSET*3 should be larger than WILDMASK!!!!
        old value was WILDOFFSET 200000100
        be careful with old .sav files!!!
*/
#ifdef WORDSIZE32
#define MAXPOWER 500000000
#define MAXVARIABLES 200000050
#define MAXDOLLARVARIABLES 1000000000L
#define WILDOFFSET 400000100
#define MAXINNAMETREE 2000000000
#define MAXDUMMIES 100000000
#define WORKBUFFER 40000000
#define MAXTER 40000
#define HALFMAX 0x10000
#define MAXSUBEXPRESSIONS 0x1FFFFFF
#define MAXFILESTREAMSIZE 1024
#else
#define MAXPOWER 10000
#define MAXVARIABLES 8050
#define MAXDOLLARVARIABLES 32000
#define WILDOFFSET 6100
#define MAXINNAMETREE 32768
#define MAXDUMMIES 1000
#define WORKBUFFER 10000000
#define MAXTER 10000
#define HALFMAX 0x100
#define MAXSUBEXPRESSIONS 0x3FFF
#define MAXFILESTREAMSIZE 1048576
#endif
#define MAXENAME 16
#define MAXSAVEFUNCTION 16384

#define MAXPARLEVEL 100
#define MAXNUMBERSIZE 200

#define MAXREPEAT 100
#define NORMSIZE 1000

#define INITNODESIZE 10
#define INITNAMESIZE 100

#define NUMFIXED 128
#define MAXNEST 100
#define MAXMATCH 30
#define MAXIF 20
#define SIZEFACS 640L
```

```
#define NUMFACS 50
#define MAXLOOPS 30
#define MAXLABELS 20
#define COMMERCIALSIZE 24
#define MAXFLAGS 16
/*
    The next quantities should still be eliminated from the program
    This should be together with changes in setfile!
*/
#define COMPRESSBUFFER 90000
#define FORTRANCONTINUATIONLINES 15
#define MAXLEVELS 2000
#define MAXLHS 400
#define MAXWILDC 100
#define NUMTABLEENTRIES 1000
#define COMPILERBUFFER 20000

#ifdef WORDSIZE32
#ifdef WITHPTHREADS
#define SMALLBUFFER   300000000L
#define SMALLOVERFLOW 600000000L
#define TERMSSMALL      3000000L
#define LARGEBUFFER  1500000000L
#define SCRATCHSIZE   500000000L
#else
#define SMALLBUFFER   150000000L
#define SMALLOVERFLOW 300000000L
#define TERMSSMALL      2000000L
#define LARGEBUFFER   800000000L
#define SCRATCHSIZE   500000000L
#endif
#else
#define SMALLBUFFER   10000000L
#define SMALLOVERFLOW 20000000L
#define TERMSSMALL      100000L
#define LARGEBUFFER   50000000L
#define SCRATCHSIZE   50000000L
#endif
#define MAXPATCHES 256
#define MAXFPATCHES 256
#define SORTIOSIZE 200000L

#define SSMALLBUFFER 2560016L
#define SSMALLOVERFLOW 3840032L
#define STERMSSMALL 10000L
#define SLARGEBUFFER 26880512L
#define SMAXPATCHES 64
#define SMAXFPATCHES 64
#define SSORTIOSIZE 32768L

#define SPECTATORSIZE 1048576L

#define MAXFLEVELS 30

#define COMPINC 2

#define MAXNUMSIZE 10

#define MAXBRACKETBUFFERSIZE 200000

#define SFHSIZE 40

#define DEFAULTPROCESSBUCKETSIZE 1000
#define SHMWINSIZE     65536L

#define TABLEEXTENSION 6

#define GZIPDEFAULT 6
#define DEFAULTTHREADS 0
#define DEFAULTTHREADBUCKETSIZE 500
#define DEFAULTTHREADLOADBALANCING 1
#define THREADSCRATCHSIZE 100000L
#define THREADSCRATCHOUTSIZE 2500000L

#ifdef WORDSIZE32
#define MAXTABLECOMBUF 100000000000L
#define MAXCOMBUFRHS 1000000000L
#else
#define MAXTABLECOMBUF 1000000L
#define MAXCOMBUFRHS 32500L
#endif

#define NUMSTORECACHES 4
#define SIZESTORECACHE 32768

#define INDENTSPACE 3

```

```
#define MULTIINDENTSPACE 1
#define MAXMULTIBRACKETLEVELS 25

#define FBUFFERSIZE 1026
/*
    For the random number generator (see commentary there)
*/
#define NPAIR1 38
#define NPAIR2 89

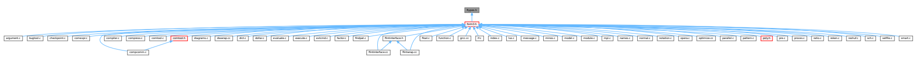
#define MAXLINELENGTH 256
#define MINALLOC 32

#define JUMPRATIO 4
/*
    Note: MAXCOUPLINGS should be at least MAXPARTICLES/2+1
*/
#define MAXPARTICLES 20
#define MAXCOUPLINGS 20
#define NUMOPTIONS 20
#define MAXLEGS 20

#ifdef WITHFLOAT
#define MAXWEIGHT 0
#define DEFAULTPRECISION 1000
#endif
```

4.49 ftypes.h File Reference

This graph shows which files directly or indirectly include this file:



Macros

- #define WITHOUTERROR 0
- #define WITHERROR 1
- #define FILESTREAM 0
- #define PREVARSTREAM 1
- #define PREREADSTREAM 2
- #define PIPESTREAM 3
- #define PRECALCSTREAM 4
- #define DOLLARSTREAM 5
- #define PREREADSTREAM2 6
- #define EXTERNALCHANNELSTREAM 7
- #define PREREADSTREAM3 8
- #define REVERSEFILESTREAM 9
- #define INPUTSTREAM 10
- #define ENDOFSTREAM 0xFF
- #define ENDOFINPUT 0xFF
- #define SUBROUTINEFILE 0
- #define PROCEDUREFILE 1
- #define HEADERFILE 2
- #define SETUPFILE 3
- #define TABLEBASEFILE 4
- #define FIRSTMODULE -1
- #define GLOBALMODULE 0
- #define SORTMODULE 1
- #define STOREMODULE 2

- #define CLEARMODULE 3
- #define ENDMODULE 4
- #define POLYFUN 0
- #define NOPARALLEL_DOLLAR 0x0001
- #define NOPARALLEL_RHS 0x0002
- #define NOPARALLEL_CONVPOLY 0x0004
- #define NOPARALLEL_SPECTATOR 0x0008
- #define NOPARALLEL_USER 0x0010
- #define NOPARALLEL_TBLDOLLAR 0x0100
- #define NOPARALLEL_NPROC 0x0200
- #define PARALLELFLAG 0x0000
- #define PRENOACTION 0
- #define PRERAISEAFTER 1
- #define PRELOWERAFTER 2
- #define WITHSEMICOLON 0
- #define WITHOUTSEMICOLON 1
- #define MODULEINSTACK 8
- #define EXECUTINGIF 0
- #define LOOKINGFORELSE 1
- #define LOOKINGFORENDIF 2
- #define NEWSTATEMENT 1
- #define OLDSTATEMENT 0
- #define EXECUTINGPRESWITCH 0
- #define SEARCHINGPRECASE 1
- #define SEARCHINGPREENDSWITCH 2
- #define PREPROONLY 1
- #define DUMPTOCOMPILER 2
- #define DUMPOUTTERMS 4
- #define DUMPINTERMS 8
- #define DUMPTOSORT 16
- #define DUMPTOPARALLEL 32
- #define THREADSDEBUG 64
- #define ERROROUT 0
- #define INPUTOUT 1
- #define STATSOUT 2
- #define EXPRSOUT 3
- #define WRITEOUT 4
- #define EXTERNALCHANNELOUT 5
- #define NUMERICALLOOP 0
- #define LISTEDLOOP 1
- #define ONEEXPRESSION 2
- #define PRETYPENONE 0
- #define PRETYPEIF 1
- #define PRETYPEDO 2
- #define PRETYPEPROCEDURE 3
- #define PRETYPESWITCH 4
- #define PRETYPEINSIDE 5
- #define DECLARATION 1
- #define SPECIFICATION 2
- #define DEFINITION 3
- #define STATEMENT 4
- #define TOOUTPUT 5
- #define ATENDOFMODULE 6
- #define MIXED2 8
- #define MIXED 9

- #define NOAUTO 0
- #define PARTEST 1
- #define WITHAUTO 2
- #define ALLVARIABLES -1
- #define SYMBOLONLY 1
- #define INDEXONLY 2
- #define VECTORONLY 4
- #define FUNCTIONONLY 8
- #define SETONLY 16
- #define EXPRESSIONONLY 32

Defines: compiler types

Type of variable found by the compiler.

- #define CDELETE -1
- #define ANYTYPE -1
- #define CSYMBOL 0
- #define CINDEX 1
- #define CVECTOR 2
- #define CFUNCTION 3
- #define CSET 4
- #define CEXPRESSION 5
- #define CDOTPRODUCT 6
- #define CNUMBER 7
- #define CSUBEXP 8
- #define CDELTA 9
- #define CDOLLAR 10
- #define CDUBIOUS 11
- #define CRANGE 12
- #define CMODEL 13
- #define CVECTOR1 21
- #define CDOUBLEDOT 22
- #define TSYMBOL -1
- #define TINDEX -2
- #define TVECTOR -3
- #define TFUNCTION -4
- #define TSET -5
- #define TEXPRESSION -6
- #define TDOTPRODUCT -7
- #define TNUMBER -8
- #define TSUBEXP -9
- #define TDELTA -10
- #define TDOLLAR -11
- #define TDUBIOUS -12
- #define LPARENTHESIS -13
- #define RPARENTHESIS -14
- #define TWILDCARD -15
- #define TWILDARG -16
- #define TDOT -17
- #define LBRACE -18
- #define RBRACE -19
- #define TCOMMA -20
- #define TFUNOPEN -21
- #define TFUNCLOSE -22
- #define TMULTIPLY -23
- #define TDIVIDE -24
- #define TPOWER -25
- #define TPLUS -26
- #define TMINUS -27
- #define TNOT -28
- #define TENDOFIT -29
- #define TSETOPEN -30

- #define TSETCLOSE -31
- #define TGENINDEX -32
- #define TCONJUGATE -33
- #define LRPARENTHESES -34
- #define TNUMBER1 -35
- #define TPOWER1 -36
- #define TEMPTY -37
- #define TSETNUM -38
- #define TSGAMMA -39
- #define TSETDOL -40
- #define TFLOAT -41
- #define TYPEISFUN 0
- #define TYPEISSUB 1
- #define TYPEISMYSTERY -1
- #define LHSIDEX 2
- #define LHSIDE 1
- #define RHSIDE 0
- #define FORTRANMODE 1
- #define REDUCEMODE 2
- #define MAPLEMODE 3
- #define MATHEMATICAMODE 4
- #define CMODE 5
- #define VORTRANMODE 6
- #define PFORTRANMODE 7
- #define DOUBLEFORTRANMODE 33
- #define DOUBLEPRECISIONFLAG 32
- #define NODOUBLEMASK 31
- #define QUADRUPLEFORTRANMODE 65
- #define QUADRUPLEPRECISIONFLAG 64
- #define ALLINTEGERDOUBLE 128
- #define NOQUADMASK 63
- #define NORMALFORMAT 0
- #define NOSPACEFORMAT 1
- #define ISNOTFORTRAN90 0
- #define ISFORTRAN90 1
- #define ALSOREVERSE 1
- #define CHISHOLM 2
- #define NOTRICK 16
- #define SORTLOWFIRST 0
- #define SORTHIGHFIRST 1
- #define SORTPOWERFIRST 2
- #define SORTANTIPOWER 3
- #define STATSSPLITMERGE 0
- #define STATSMERGETOFILE 1
- #define STATSPOSTSORT 2
- #define NOCHECKLOGTYPE 0
- #define CHECKLOGTYPE 1
- #define NMIN4SHIFT 4
- #define SYMBOL 1
- #define DOTPRODUCT 2
- #define VECTOR 3
- #define INDEX 4
- #define EXPRESSION 5
- #define SUBEXPRESSION 6
- #define DOLLAREXPRESSION 7
- #define SETSET 8
- #define ARGWILD 9
- #define MINVECTOR 10
- #define SETEXP 11
- #define DOLLAREXPR2 12
- #define HAAKJE0 9
- #define FUNCTION 20
- #define TMPPOLYFUN 14
- #define ARGFIELD 15

- #define SNUMBER 16
- #define LNUMBER 17
- #define HAAKJE 18
- #define DELTA 19
- #define EXPONENT 20
- #define DENOMINATOR 21
- #define SETFUNCTION 22
- #define GAMMA 23
- #define GAMMAI 24
- #define GAMMAFIVE 25
- #define GAMMASIX 26
- #define GAMMASEVEN 27
- #define SUMF1 28
- #define SUMF2 29
- #define DUMFUN 30
- #define REPLACEMENT 31
- #define REVERSEFUNCTION 32
- #define DISTRIBUTION 33
- #define DELTA3 34
- #define DUMMYFUN 35
- #define DUMMYTEN 36
- #define LEVICIVITA 37
- #define FACTORIAL 38
- #define INVERSEFACTORIAL 39
- #define BINOMIAL 40
- #define NUMARGSFUN 41
- #define SIGNFUN 42
- #define MODFUNCTION 43
- #define MOD2FUNCTION 44
- #define MINFUNCTION 45
- #define MAXFUNCTION 46
- #define ABSFUNCTION 47
- #define SIGFUNCTION 48
- #define INTFUNCTION 49
- #define THETA 50
- #define THETA2 51
- #define DELTA2 52
- #define DELTAP 53
- #define BERNOULLIFUNCTION 54
- #define COUNTFUNCTION 55
- #define MATCHFUNCTION 56
- #define PATTERNFUNCTION 57
- #define TERMFUNCTION 58
- #define CONJUGATION 59
- #define ROOTFUNCTION 60
- #define TABLEFUNCTION 61
- #define FIRSTBRACKET 62
- #define TERMSINEXPR 63
- #define NUMTERMSFUN 64
- #define GCDFUNCTION 65
- #define DIVFUNCTION 66
- #define REMFUNCTION 67
- #define MAXPOWEROF 68
- #define MINPOWEROF 69
- #define TABLESTUB 70
- #define FACTORIN 71
- #define TERMSINBRACKET 72
- #define WILDARGFUN 73
- #define SQRTFUNCTION 74
- #define LNFUNCTION 75
- #define SINFUNCTION 76
- #define COSFUNCTION 77
- #define TANFUNCTION 78
- #define ASINFUNCTION 79

- #define ACOSFUNCTION 80
- #define ATANFUNCTION 81
- #define ATAN2FUNCTION 82
- #define SINHFUNCTION 83
- #define COSHFUNCTION 84
- #define TANHFUNCTION 85
- #define ASINHFUNCTION 86
- #define ACOSHFUNCTION 87
- #define ATANHFUNCTION 88
- #define LI2FUNCTION 89
- #define LINFUNCTION 90
- #define EXTRASYMFUN 91
- #define RANDOMFUNCTION 92
- #define RANPERM 93
- #define NUMFACTORS 94
- #define FIRSTTERM 95
- #define CONTENTTERM 96
- #define PRIMENUMBER 97
- #define EXTEUCLIDEAN 98
- #define MAKERATIONAL 99
- #define INVERSEFUNCTION 100
- #define IDFUNCTION 101
- #define PUTFIRST 102
- #define PERMUTATIONS 103
- #define PARTITIONS 104
- #define MULFUNCTION 105
- #define MOEBIUS 106
- #define TOPOLOGIES 107
- #define DIAGRAMS 108
- #define TOPO 109
- #define NODEFUNCTION 110
- #define EDGE 111
- #define SIZEOFFUNCTION 112
- #define BLOCK 113
- #define ONEPI 114
- #define PHI 115
- #define MAXBUILTINFUNCTION 115
- #define FIRSTUSERFUNCTION 150
- #define ALLFUNCTIONS -1
- #define ALLMZVFUNCTIONS -2
- #define ISYMBOL 0
- #define PISYMBOL 1
- #define COEFFSYMBOL 2
- #define NUMERATORSYMBOL 3
- #define DENOMINATORSYMBOL 4
- #define WILDARGSYMBOL 5
- #define DIMENSIONSYMBOL 6
- #define FACTORSYMBOL 7
- #define SEPARATESYMBOL 8
- #define EESYMBOL 9
- #define EMSYMBOL 10
- #define BUILTINSYMBOLS 11
- #define FIRSTUSERSYMBOL 20
- #define BUILTININDICES 1
- #define BUILTINVECTORS 1
- #define BUILTINDOLLARS 1
- #define WILDARGVECTOR 0
- #define WILDARGINDEX 0
- #define TYPEEXPRESSION 0
- #define TYPEIDNEW 1
- #define TYPEIDOLD 2
- #define TYPEOPERATION 3
- #define TYPEREPEAT 4
- #define TYPEENDREPEAT 5

- #define TYPECOUNT 20
- #define TYPEMULT 21
- #define TYPEGOTO 22
- #define TYPEDISCARD 23
- #define TYPEIF 24
- #define TYPEELSE 25
- #define TYPEELIF 26
- #define TYPEENDIF 27
- #define TYPESUM 28
- #define TYPECHISHOLM 29
- #define TYPEREVERSE 30
- #define TYPEARG 31
- #define TYPENORM 32
- #define TYPENORM2 33
- #define TYPENORM3 34
- #define TYPEEXIT 35
- #define TYPESETEXIT 36
- #define TYPEPRINT 37
- #define TYPEFPRINT 38
- #define TYPEREDEFPRE 39
- #define TYPESPLITARG 40
- #define TYPESPLITARG2 41
- #define TYPEFACTARG 42
- #define TYPEFACTARG2 43
- #define TYPETRY 44
- #define TYPEASSIGN 45
- #define TYPERENUMBER 46
- #define TYPESUMFIX 47
- #define TYPEFINDLOOP 48
- #define TYPEUNRAVEL 49
- #define TYPEADJUSTBOUNDS 50
- #define TYPEINSIDE 51
- #define TYPETERM 52
- #define TYPESORT 53
- #define TYPEDETCURDUM 54
- #define TYPEINEXPRESSION 55
- #define TYPESPLITFIRSTARG 56
- #define TYPESPLITLASTARG 57
- #define TYPEMERGE 58
- #define TYPETESTUSE 59
- #define TYPEAPPLY 60
- #define TYPEAPPLYRESET 61
- #define TYPECHAININ 62
- #define TYPECHAINOUT 63
- #define TYPENORM4 64
- #define TYPEFACTOR 65
- #define TYPEARGIMPLODE 66
- #define TYPEARGEXPLODE 67
- #define TYPEDENOMINATORS 68
- #define TYPESTUFFLE 69
- #define TYPEDROPCOEFFICIENT 70
- #define TYPETRANSFORM 71
- #define TYPETOPOLYNOMIAL 72
- #define TYPEFROMPOLYNOMIAL 73
- #define TYPEDOLOOP 74
- #define TYPEENDDOLOOP 75
- #define TYPEDROPSYMBOLS 76
- #define TYPEPUTINSIDE 77
- #define TYPETOSPECTATOR 78
- #define TYPEARGTOEXTRASYMBOL 79
- #define TYPECANONICALIZE 80
- #define TYPESWITCH 81
- #define TYPEENDSWITCH 82
- #define TYPESETUSERFLAG 83

- #define TYPECLEARUSERFLAG 84
- #define TYPEALLLOOPS 85
- #define TYPEALLPATHS 86
- #define TAKETRACE 1
- #define CONTRACT 2
- #define RATIO 3
- #define SYMMETRIZE 4
- #define TENVEC 5
- #define SUMNUM1 6
- #define SUMNUM2 7
- #define WILDDUMMY 0
- #define SYMTONUM 1
- #define SYMTOSYM 2
- #define SYMTOSUB 3
- #define VECTOMIN 4
- #define VECTOVEC 5
- #define VECTOSUB 6
- #define INDTOIND 7
- #define INDTOSUB 8
- #define FUNTOFUN 9
- #define ARGTOARG 10
- #define ARLTOARL 11
- #define EXPTOEXP 12
- #define FROMBRAC 13
- #define FROMSET 14
- #define SETTONUM 15
- #define WILDCARDS 16
- #define SETNUMBER 17
- #define LOADDOLLAR 18
- #define NUMTONUM 20
- #define NUMTOSYM 21
- #define NUMTOIND 22
- #define NUMTOSUB 23
- #define EATTENSOR 0x2000
- #define CLEANFLAG 0
- #define DIRTYFLAG 1
- #define DIRTYSYMFLAG 2
- #define MUSTCLEANPRF 4
- #define SUBTERMUSED1 8
- #define SUBTERMUSED2 16
- #define ALLDIRTY (DIRTYFLAG|DIRTYSYMFLAG)
- #define ARGHEAD 2
- #define FUNHEAD 3
- #define SUBEXPSIZE 5
- #define EXPRHEAD 5
- #define TYPEARGHEADSIZE 6
- #define SKIP 1
- #define DROP 2
- #define HIDE 3
- #define UNHIDE 4
- #define INTOHIDE 5
- #define LOCALEXPRESSION 0
- #define SKIPLEXPRESSION 1
- #define DROPLEXPRESSION 2
- #define DROPPEDEXPRESSION 3
- #define GLOBALEXPRESSION 4
- #define SKIPGEXPRESSION 5
- #define DROPGEXPRESSION 6
- #define STOREDEXPRESSION 8
- #define HIDDENLEXPRESSION 9
- #define HIDDENGEXPRESSION 13
- #define INCEXPRESSION 9
- #define HIDELEXPRESSION 10
- #define HIDEGEXPRESSION 14

- #define DROPHLEXPRESSION 11
- #define DROPHGEXPRESSION 15
- #define UNHIDELEXPRESSION 12
- #define UNHIDEGEXPRESSION 16
- #define INTOHIDELEXPRESSION 17
- #define INTOHIDEGEXPRESSION 18
- #define SPECTATOREXPRESSION 19
- #define DROPSPECTATOREXPRESSION 20
- #define SKIPUNHIDELEXPRESSION 21
- #define SKIPUNHIDEGEXPRESSION 22
- #define PRINTOFF 0
- #define PRINTON 1
- #define PRINTCONTENTS 2
- #define PRINTCONTENT 3
- #define PRINTLFILE 4
- #define PRINTONETERM 8
- #define PRINTONEFUNCTION 16
- #define PRINTALL 32
- #define REGULAREXPRESSION -1
- #define REDEFINEDEXPRESSION -2
- #define NEWLYDEFINEDEXPRESSION -3

Defines: function specs

Function specifications.

- #define VERTEXFUNCTION -1
- #define GENERALFUNCTION 0
- #define TENSORFUNCTION 2
- #define GAMMAFUNCTION 4
- #define POS_ 0 /∗ integer > 0 ∗/
- #define POS0_ 1 /∗ integer >= 0 ∗/
- #define NEG_ 2 /∗ integer < 0 ∗/
- #define NEG0_ 3 /∗ integer <= 0 ∗/
- #define EVEN_ 4 /∗ integer (even) ∗/
- #define ODD_ 5 /∗ integer (odd) ∗/
- #define Z_ 6 /∗ integer ∗/
- #define SYMBOL_ 7 /∗ symbol only ∗/
- #define FIXED_ 8 /∗ fixed index ∗/
- #define INDEX_ 9 /∗ index only ∗/
- #define Q_ 10 /∗ rational ∗/
- #define DUMMYINDEX_ 11 /∗ dummy index only ∗/
- #define VECTOR_ 12 /∗ vector only ∗/
- #define GAMMA1 0
- #define GAMMA5 -1
- #define GAMMA6 -2
- #define GAMMA7 -3
- #define FUNNYVEC -4
- #define FUNNYWILD -5
- #define SUMMEDIND -6
- #define NOINDEX -7
- #define FUNNYDOLLAR -8
- #define EMPTYINDEX -9
- #define MINSPEC -10
- #define USEDFLAG 2
- #define DUMMYFLAG 1
- #define MAINSORT 0
- #define FUNCTIONSORT 1
- #define SUBSORT 2
- #define FLOATMODE 1
- #define RATIONALMODE 0
- #define NUMSPECSETS 10
- #define ISZERO 1
- #define ISUNMODIFIED 2

- #define ISCOMPRESSED 4
- #define ISINRHS 8
- #define ISFACTORIZED 16
- #define TOBEFACTORED 32
- #define TOBEUNFACTORED 64
- #define KEEPZERO 128
- #define VARTYPENONE 0
- #define VARTYPECOMPLEX 1
- #define VARTYPEIMAGINARY 2
- #define VARTYPEROOTOFUNITY 4
- #define VARTYPEMINUS 8
- #define CYCLESYMMETRIC 1
- #define RCYCLESYMMETRIC 2
- #define SYMMETRIC 3
- #define ANTISYMMETRIC 4
- #define REVERSEORDER 256
- #define SUBMULTI 1
- #define SUBONCE 2
- #define SUBONLY 3
- #define SUBMANY 4
- #define SUBVECTOR 5
- #define SUBSELECT 6
- #define SUBALL 7
- #define SUBMASK 15
- #define SUBDISORDER 16
- #define SUBAFTER 32
- #define SUBAFTERNOT 64
- #define IDHEAD 6
- #define DOLLARFLAG 1
- #define NORMALIZEFLAG 2
- #define GIDENT 1
- #define GFIVE 4
- #define GPLUS 3
- #define GMINUS 2
- #define LONGNUMBER 1
- #define MATCH 2
- #define COEFFI 3
- #define SUBEXPR 4
- #define MULTIPLEOF 5
- #define IFDOLLAR 6
- #define IFEXPRESSION 7
- #define IFDOLLAREXTRA 8
- #define IFISFACTORIZED 9
- #define IFOCCURS 10
- #define IFUSERFLAG 11
- #define IFFLOATNUMBER 12
- #define GREATER 0
- #define GREATEREQUAL 1
- #define LESS 2
- #define LESSEQUAL 3
- #define EQUAL 4
- #define NOTEQUAL 5
- #define ORCOND 6
- #define ANDCOND 7
- #define DUMMY 1
- #define SORT 1
- #define STORE 2
- #define END 3
- #define GLOBAL 4
- #define CLEAR 5
- #define VECTBIT 1
- #define DOTPBIT 2
- #define FUNBIT 4
- #define SETBIT 8

- #define EXTRAPARAMETER 0x4000
- #define GENCOMMUTE 0
- #define GENNONCOMMUTE 0x2000
- #define NAMENOTFOUND -9
- #define DOLUNDEFINED 0
- #define DOLNUMBER 1
- #define DOLARGUMENT 2
- #define DOLSUBTERM 3
- #define DOLTERMS 4
- #define DOLWILDARGS 5
- #define DOLINDEX 6
- #define DOLZERO 7
- #define FINDLOOP 0
- #define REPLACELOOP 1
- #define NOFUNPOWERS 0
- #define COMFUNPOWERS 1
- #define ALLFUNPOWERS 2
- #define PROPERORDERFLAG 0
- #define REGULAR 0
- #define FINISH 1
- #define POLYADD 1
- #define POLYSUB 2
- #define POLYMUL 3
- #define POLYDIV 4
- #define POLYREM 5
- #define POLYGCD 6
- #define POLYINTFAC 7
- #define POLYNORM 8
- #define MODNONE 0
- #define MODSUM 1
- #define MODMAX 2
- #define MODMIN 3
- #define MODLOCAL 4
- #define ELEMENTUSED 1
- #define ELEMENTLOADED 2
- #define POSNEG 0x1
- #define INVERSETABLE 0x2
- #define COEFFICIENTSONLY 0x4
- #define ALSOPOWERS 0x8
- #define ALSOFUNARGS 0x10
- #define ALSODOLLARS 0x20
- #define NOINVERSES 0x40
- #define POSITIVEONLY 0
- #define UNPACK 0x80
- #define NOUNPACK 0
- #define FROMFUNCTION 0x100
- #define VARNAMES 0
- #define AUTONAMES 1
- #define EXPRNAMES 2
- #define DOLLARNAMES 3
- #define DUMMYBUFFER 1
- #define ALLARGS 1
- #define NUMARG 2
- #define ARGRANGE 3
- #define MAKEARGS 4
- #define MAXRANGEINDICATOR 4
- #define REPLACEARG 5
- #define ENCODEARG 6
- #define DECODEARG 7
- #define IMPLODEARG 8
- #define EXPLODEARG 9
- #define PERMUTEARG 10
- #define REVERSEARG 11
- #define CYCLEARG 12

- #define ISLYNDON 13
- #define ISLYNDONR 14
- #define TOLYNDON 15
- #define TOLYNDONR 16
- #define ADDARG 17
- #define MULTIPLYARG 18
- #define DROPARG 19
- #define SELECTARG 20
- #define DEDUPARG 21
- #define ZTOHARG 22
- #define HTOZARG 23
- #define BASECODE 1
- #define YESLYNDON 1
- #define NOLYNDON 2
- #define TOPOLYNOMIALFLAG 1
- #define FACTARGFLAG 2
- #define OLDFACTARG 1
- #define NEWFACTARG 0
- #define FROMMODULEOPTION 0
- #define FROMPOINTINSTRUCTION 1
- #define EXTRASYMBOL 0
- #define REGULARSYMBOL 1
- #define EXPRESSIONNUMBER 2
- #define O_NONE 0
- #define O_CSE 1
- #define O_CSEGREEDY 2
- #define O_GREEDY 3
- #define O_OCCURRENCE 0
- #define O_MCTS 1
- #define O_SIMULATED_ANNEALING 2
- #define O_FORWARD 0
- #define O_BACKWARD 1
- #define O_FORWARDORBACKWARD 2
- #define O_FORWARDANDBACKWARD 3
- #define OPTHEAD 3
- #define DOALL 1
- #define ONLYFUNCTIONS 2
- #define INUSE 1
- #define COULDCOMMUTE 2
- #define DOESNOTCOMMUTE 4
- #define DICT_NONUMBERS 0
- #define DICT_INTEGERONLY 1
- #define DICT_RATIONALONLY 2
- #define DICT_ALLNUMBERS 3
- #define DICT_NOVARIABLES 0
- #define DICT_DOVARIABLES 1
- #define DICT_NOSPECIALS 0
- #define DICT_DOSPECIALS 1
- #define DICT_NOFUNWITHARGS 0
- #define DICT_DOFUNWITHARGS 1
- #define DICT_NOTINDOLLARS 0
- #define DICT_INDOLLARS 1
- #define DICT_INTEGERNUMBER 1
- #define DICT_RATIONALNUMBER 2
- #define DICT_SYMBOL 3
- #define DICT_VECTOR 4
- #define DICT_INDEX 5
- #define DICT_FUNCTION 6
- #define DICT_FUNCTION_WITH_ARGUMENTS 7
- #define DICT_SPECIALCHARACTER 8
- #define DICT_RANGE 9
- #define READSPECTATORFLAG 3
- #define GLOBALSPECTATORFLAG 1
- #define ORDEREDSET 1

- #define DENSETABLE 1
- #define SPARSETABLE 0
- #define ONEPARTICLEIRREDUCIBLE 1
- #define WITHINSERTIONS 2
- #define NOTADPOLES 4
- #define WITHSYMMETRIZE 8
- #define TOPOLOGIESONLY 16
- #define NONODES 32
- #define WITHEDGES 64
- #define CHECKEXTERN 128
- #define WITHBLOCKS 256
- #define WITHONEPI 512
- #define NOSNAILS 1024
- #define NOEXTSELF 2048

Typedefs

- typedef void(∗ PVFUNWP) ()
- typedef void(∗ PVFUNV) ()
- typedef int(∗ CFUN) ()
- typedef int(∗ TFUN) ()
- typedef int(∗ TFUN1) ()

4.49.1 Detailed Description

Contains the definitions of many internal codes Rather than using numbers directly we do this by defines, making it much easier to change things. Changing things is sometimes also a good way of testing the code.

Definition in file ftypes.h.

4.49.2 Macro Definition Documentation

4.49.2.1 WITHOUTERROR

```
#define WITHOUTERROR 0
```

The next macros were introduced when TFORM was programmed. In the case of workers, each worker may need some private data. These can in principle be accessed by some posix calls but that is unnecessarily slow. The passing of a pointer to the complete data struct with private data will be much faster. And anyway, there would have to be a macro that either makes the posix call (TFORM) or doesn't (FORM). The solution by having macro's that either pass the pointer (TFORM) or don't pass it (FORM) is seen as the best solution.

In the declarations and the calling of the functions we have to use the PHEAD or the BHEAD macro, respectively, if the pointer is to be passed. These macro's contain the comma as well. Hence we need special macro's if there are no other arguments. These are called PHEAD0 and BHEAD0.

Definition at line 51 of file ftypes.h.

4.49.2.2 WITHERROR

```
#define WITHERROR 1
```

Definition at line 52 of file ftypes.h.

4.49.2.3 FILESTREAM

```
#define FILESTREAM 0
```

Definition at line 58 of file ftypes.h.

4.49.2.4 PREVARSTREAM

```
#define PREVARSTREAM 1
```

Definition at line 59 of file ftypes.h.

4.49.2.5 PREREADSTREAM

```
#define PREREADSTREAM 2
```

Definition at line 60 of file ftypes.h.

4.49.2.6 PIPESTREAM

```
#define PIPESTREAM 3
```

Definition at line 61 of file ftypes.h.

4.49.2.7 PRECALCSTREAM

```
#define PRECALCSTREAM 4
```

Definition at line 62 of file ftypes.h.

4.49.2.8 DOLLARSTREAM

```
#define DOLLARSTREAM 5
```

Definition at line 63 of file ftypes.h.

4.49.2.9 PREREADSTREAM2

```
#define PREREADSTREAM2 6
```

Definition at line 64 of file ftypes.h.

4.49.2.10 EXTERNALCHANNELSTREAM

```
#define EXTERNALCHANNELSTREAM 7
```

Definition at line 65 of file ftypes.h.

4.49.2.11 PREREADSTREAM3

```
#define PREREADSTREAM3 8
```

Definition at line 66 of file ftypes.h.

4.49.2.12 REVERSEFILESTREAM

```
#define REVERSEFILESTREAM 9
```

Definition at line 67 of file ftypes.h.

4.49.2.13 INPUTSTREAM

```
#define INPUTSTREAM 10
```

Definition at line 68 of file ftypes.h.

4.49.2.14 ENDOFSTREAM

```
#define ENDOFSTREAM 0xFF
```

Definition at line 70 of file ftypes.h.

4.49.2.15 ENDOFINPUT

```
#define ENDOFINPUT 0xFF
```

Definition at line 71 of file ftypes.h.

4.49.2.16 SUBROUTINEFILE

```
#define SUBROUTINEFILE 0
```

Definition at line 77 of file ftypes.h.

4.49.2.17 PROCEDUREFILE

```
#define PROCEDUREFILE 1
```

Definition at line 78 of file ftypes.h.

4.49.2.18 HEADERFILE

```
#define HEADERFILE 2
```

Definition at line 79 of file ftypes.h.

4.49.2.19 SETUPFILE

```
#define SETUPFILE 3
```

Definition at line 80 of file ftypes.h.

4.49.2.20 TABLEBASEFILE

```
#define TABLEBASEFILE 4
```

Definition at line 81 of file ftypes.h.

4.49.2.21 FIRSTMODULE

```
#define FIRSTMODULE -1
```

Definition at line 87 of file ftypes.h.

4.49.2.22 GLOBALMODULE

```
#define GLOBALMODULE 0
```

Definition at line 88 of file ftypes.h.

4.49.2.23 SORTMODULE

```
#define SORTMODULE 1
```

Definition at line 89 of file ftypes.h.

4.49.2.24 STOREMODULE

```
#define STOREMODULE 2
```

Definition at line 90 of file ftypes.h.

4.49.2.25 CLEARMODULE

```
#define CLEARMODULE 3
```

Definition at line 91 of file ftypes.h.

4.49.2.26 ENDMODULE

```
#define ENDMODULE 4
```

Definition at line 92 of file ftypes.h.

4.49.2.27 POLYFUN

```
#define POLYFUN 0
```

Definition at line 94 of file ftypes.h.

4.49.2.28 NOPARALLEL_DOLLAR

```
#define NOPARALLEL_DOLLAR 0x0001
```

Definition at line 96 of file ftypes.h.

4.49.2.29 NOPARALLEL_RHS

```
#define NOPARALLEL_RHS 0x0002
```

Definition at line 97 of file ftypes.h.

4.49.2.30 NOPARALLEL_CONVPOLY

```
#define NOPARALLEL_CONVPOLY 0x0004
```

Definition at line 98 of file ftypes.h.

4.49.2.31 NOPARALLEL_SPECTATOR

```
#define NOPARALLEL_SPECTATOR 0x0008
```

Definition at line 99 of file ftypes.h.

4.49.2.32 NOPARALLEL_USER

```
#define NOPARALLEL_USER 0x0010
```

Definition at line 100 of file ftypes.h.

4.49.2.33 NOPARALLEL_TBLDOLLAR

```
#define NOPARALLEL_TBLDOLLAR 0x0100
```

Definition at line 101 of file ftypes.h.

4.49.2.34 NOPARALLEL_NPROC

```
#define NOPARALLEL_NPROC 0x0200
```

Definition at line 102 of file ftypes.h.

4.49.2.35 PARALLELFLAG

```
#define PARALLELFLAG 0x0000
```

Definition at line 103 of file ftypes.h.

4.49.2.36 PRENOACTION

```
#define PRENOACTION 0
```

Definition at line 105 of file ftypes.h.

4.49.2.37 PRERAISEAFTER

```
#define PRERAISEAFTER 1
```

Definition at line 106 of file ftypes.h.

4.49.2.38 PRELOWERAFTER

```
#define PRELOWERAFTER 2
```

Definition at line 107 of file ftypes.h.

4.49.2.39 WITHSEMICOLON

```
#define WITHSEMICOLON 0
```

Definition at line 114 of file ftypes.h.

4.49.2.40 WITHOUTSEMICOLON

```
#define WITHOUTSEMICOLON 1
```

Definition at line 115 of file ftypes.h.

4.49.2.41 MODULEINSTACK

```
#define MODULEINSTACK 8
```

Definition at line 116 of file ftypes.h.

4.49.2.42 EXECUTINGIF

```
#define EXECUTINGIF 0
```

Definition at line 117 of file ftypes.h.

4.49.2.43 LOOKINGFORELSE

```
#define LOOKINGFORELSE 1
```

Definition at line 118 of file ftypes.h.

4.49.2.44 LOOKINGFORENDIF

```
#define LOOKINGFORENDIF 2
```

Definition at line 119 of file ftypes.h.

4.49.2.45 NEWSTATEMENT

```
#define NEWSTATEMENT 1
```

Definition at line 120 of file ftypes.h.

4.49.2.46 OLDSTATEMENT

```
#define OLDSTATEMENT 0
```

Definition at line 121 of file ftypes.h.

4.49.2.47 EXECUTINGPRESWITCH

```
#define EXECUTINGPRESWITCH 0
```

Definition at line 123 of file ftypes.h.

4.49.2.48 SEARCHINGPRECASE

```
#define SEARCHINGPRECASE 1
```

Definition at line 124 of file ftypes.h.

4.49.2.49 SEARCHINGPREENDSWITCH

```
#define SEARCHINGPREENDSWITCH 2
```

Definition at line 125 of file ftypes.h.

4.49.2.50 PREPROONLY

```
#define PREPROONLY 1
```

Definition at line 127 of file ftypes.h.

4.49.2.51 DUMPTOCOMPILER

```
#define DUMPTOCOMPILER 2
```

Definition at line 128 of file ftypes.h.

4.49.2.52 DUMPOUTTERMS

```
#define DUMPOUTTERMS 4
```

Definition at line 129 of file ftypes.h.

4.49.2.53 DUMPINTERMS

```
#define DUMPINTERMS 8
```

Definition at line 130 of file ftypes.h.

4.49.2.54 DUMPTOSORT

```
#define DUMPTOSORT 16
```

Definition at line 131 of file ftypes.h.

4.49.2.55 DUMPTOPARALLEL

```
#define DUMPTOPARALLEL 32
```

Definition at line 132 of file ftypes.h.

4.49.2.56 THREADSDEBUG

```
#define THREADSDEBUG 64
```

Definition at line 133 of file ftypes.h.

4.49.2.57 ERROROUT

```
#define ERROROUT 0
```

Definition at line 135 of file ftypes.h.

4.49.2.58 INPUTOUT

```
#define INPUTOUT 1
```

Definition at line 136 of file ftypes.h.

4.49.2.59 STATSOUT

```
#define STATSOUT 2
```

Definition at line 137 of file ftypes.h.

4.49.2.60 EXPRSOUT

```
#define EXPRSOUT 3
```

Definition at line 138 of file ftypes.h.

4.49.2.61 WRITEOUT

```
#define WRITEOUT 4
```

Definition at line 139 of file ftypes.h.

4.49.2.62 EXTERNALCHANNELOUT

```
#define EXTERNALCHANNELOUT 5
```

Definition at line 141 of file ftypes.h.

4.49.2.63 NUMERICALLOOP

```
#define NUMERICALLOOP 0
```

Definition at line 143 of file ftypes.h.

4.49.2.64 LISTEDLOOP

```
#define LISTEDLOOP 1
```

Definition at line 144 of file ftypes.h.

4.49.2.65 ONEEXPRESSION

```
#define ONEEXPRESSION 2
```

Definition at line 145 of file ftypes.h.

4.49.2.66 PRETYPENONE

```
#define PRETYPENONE 0
```

Definition at line 147 of file ftypes.h.

4.49.2.67 PRETYPEIF

```
#define PRETYPEIF 1
```

Definition at line 148 of file ftypes.h.

4.49.2.68 PRETYPEDO

```
#define PRETYPEDO 2
```

Definition at line 149 of file ftypes.h.

4.49.2.69 PRETYPEPROCEDURE

```
#define PRETYPEPROCEDURE 3
```

Definition at line 150 of file ftypes.h.

4.49.2.70 PRETYPESWITCH

```
#define PRETYPESWITCH 4
```

Definition at line 151 of file ftypes.h.

4.49.2.71 PRETYPEINSIDE

```
#define PRETYPEINSIDE 5
```

Definition at line 152 of file ftypes.h.

4.49.2.72 DECLARATION

```
#define DECLARATION 1
```

Definition at line 158 of file ftypes.h.

4.49.2.73 SPECIFICATION

```
#define SPECIFICATION 2
```

Definition at line 159 of file ftypes.h.

4.49.2.74 DEFINITION

```
#define DEFINITION 3
```

Definition at line 160 of file ftypes.h.

4.49.2.75 STATEMENT

```
#define STATEMENT 4
```

Definition at line 161 of file ftypes.h.

4.49.2.76 TOOUTPUT

```
#define TOOUTPUT 5
```

Definition at line 162 of file ftypes.h.

4.49.2.77 ATENDOFMODULE

```
#define ATENDOFMODULE 6
```

Definition at line 163 of file ftypes.h.

4.49.2.78 MIXED2

```
#define MIXED2 8
```

Definition at line 164 of file ftypes.h.

4.49.2.79 MIXED

```
#define MIXED 9
```

Definition at line 165 of file ftypes.h.

4.49.2.80 NOAUTO

```
#define NOAUTO 0
```

Definition at line 191 of file ftypes.h.

4.49.2.81 PARTEST

```
#define PARTEST 1
```

Definition at line 192 of file ftypes.h.

4.49.2.82 WITHAUTO

```
#define WITHAUTO 2
```

Definition at line 193 of file ftypes.h.

4.49.2.83 ALLVARIABLES

```
#define ALLVARIABLES -1
```

Definition at line 195 of file ftypes.h.

4.49.2.84 SYMBOLONLY

```
#define SYMBOLONLY 1
```

Definition at line 196 of file ftypes.h.

4.49.2.85 INDEXONLY

```
#define INDEXONLY 2
```

Definition at line 197 of file ftypes.h.

4.49.2.86 VECTORONLY

```
#define VECTORONLY 4
```

Definition at line 198 of file ftypes.h.

4.49.2.87 FUNCTIONONLY

```
#define FUNCTIONONLY 8
```

Definition at line 199 of file ftypes.h.

4.49.2.88 SETONLY

```
#define SETONLY 16
```

Definition at line 200 of file ftypes.h.

4.49.2.89 EXPRESSIONONLY

```
#define EXPRESSIONONLY 32
```

Definition at line 201 of file ftypes.h.

4.49.2.90 CDELETE

```
#define CDELETE -1
```

Definition at line 212 of file ftypes.h.

4.49.2.91 ANYTYPE

```
#define ANYTYPE -1
```

Definition at line 213 of file ftypes.h.

4.49.2.92 CSYMBOL

```
#define CSYMBOL 0
```

Definition at line 214 of file ftypes.h.

4.49.2.93 CINDEX

```
#define CINDEX 1
```

Definition at line 215 of file ftypes.h.

4.49.2.94 CVECTOR

```
#define CVECTOR 2
```

Definition at line 216 of file ftypes.h.

4.49.2.95 CFUNCTION

```
#define CFUNCTION 3
```

Definition at line 217 of file ftypes.h.

4.49.2.96 CSET

```
#define CSET 4
```

Definition at line 218 of file ftypes.h.

4.49.2.97 CEXPRESSION

```
#define CEXPRESSION 5
```

Definition at line 219 of file ftypes.h.

4.49.2.98 CDOTPRODUCT

```
#define CDOTPRODUCT 6
```

Definition at line 220 of file ftypes.h.

4.49.2.99 CNUMBER

```
#define CNUMBER 7
```

Definition at line 221 of file ftypes.h.

4.49.2.100 CSUBEXP

```
#define CSUBEXP 8
```

Definition at line 222 of file ftypes.h.

4.49.2.101 CDELTA

```
#define CDELTA 9
```

Definition at line 223 of file ftypes.h.

4.49.2.102 CDOLLAR

```
#define CDOLLAR 10
```

Definition at line 224 of file ftypes.h.

4.49.2.103 CDUBIOUS

```
#define CDUBIOUS 11
```

Definition at line 225 of file ftypes.h.

4.49.2.104 CRANGE

```
#define CRANGE 12
```

Definition at line 226 of file ftypes.h.

4.49.2.105 CMODEL

```
#define CMODEL 13
```

Definition at line 227 of file ftypes.h.

4.49.2.106 CVECTOR1

```
#define CVECTOR1 21
```

Definition at line 228 of file ftypes.h.

4.49.2.107 CDOUBLEDOT

```
#define CDOUBLEDOT 22
```

Definition at line 229 of file ftypes.h.

4.49.2.108 TSYMBOL

```
#define TSYMBOL -1
```

Definition at line 237 of file ftypes.h.

4.49.2.109 TINDEX

```
#define TINDEX -2
```

Definition at line 238 of file ftypes.h.

4.49.2.110 TVECTOR

```
#define TVECTOR -3
```

Definition at line 239 of file ftypes.h.

4.49.2.111 TFUNCTION

```
#define TFUNCTION -4
```

Definition at line 240 of file ftypes.h.

4.49.2.112 TSET

```
#define TSET -5
```

Definition at line 241 of file ftypes.h.

4.49.2.113 TEXPRESSION

```
#define TEXPRESSION -6
```

Definition at line 242 of file ftypes.h.

4.49.2.114 TDOTPRODUCT

```
#define TDOTPRODUCT -7
```

Definition at line 243 of file ftypes.h.

4.49.2.115 TNUMBER

```
#define TNUMBER -8
```

Definition at line 244 of file ftypes.h.

4.49.2.116 TSUBEXP

```
#define TSUBEXP -9
```

Definition at line 245 of file ftypes.h.

4.49.2.117 TDELTA

```
#define TDELTA -10
```

Definition at line 246 of file ftypes.h.

4.49.2.118 TDOLLAR

```
#define TDOLLAR -11
```

Definition at line 247 of file ftypes.h.

4.49.2.119 TDUBIOUS

```
#define TDUBIOUS -12
```

Definition at line 248 of file ftypes.h.

4.49.2.120 LPARENTHESIS

```
#define LPARENTHESIS -13
```

Definition at line 249 of file ftypes.h.

4.49.2.121 RPARENTHESIS

```
#define RPARENTHESIS -14
```

Definition at line 250 of file ftypes.h.

4.49.2.122 TWILDCARD

```
#define TWILDCARD -15
```

Definition at line 251 of file ftypes.h.

4.49.2.123 TWILDARG

```
#define TWILDARG -16
```

Definition at line 252 of file ftypes.h.

4.49.2.124 TDOT

```
#define TDOT -17
```

Definition at line 253 of file ftypes.h.

4.49.2.125 LBRACE

```
#define LBRACE -18
```

Definition at line 254 of file ftypes.h.

4.49.2.126 RBRACE

```
#define RBRACE -19
```

Definition at line 255 of file ftypes.h.

4.49.2.127 TCOMMA

```
#define TCOMMA -20
```

Definition at line 256 of file ftypes.h.

4.49.2.128 TFUNOPEN

```
#define TFUNOPEN -21
```

Definition at line 257 of file ftypes.h.

4.49.2.129 TFUNCLOSE

```
#define TFUNCLOSE -22
```

Definition at line 258 of file ftypes.h.

4.49.2.130 TMULTIPLY

```
#define TMULTIPLY -23
```

Definition at line 259 of file ftypes.h.

4.49.2.131 TDIVIDE

```
#define TDIVIDE -24
```

Definition at line 260 of file ftypes.h.

4.49.2.132 TPOWER

```
#define TPOWER -25
```

Definition at line 261 of file ftypes.h.

4.49.2.133 TPLUS

```
#define TPLUS -26
```

Definition at line 262 of file ftypes.h.

4.49.2.134 TMINUS

```
#define TMINUS -27
```

Definition at line 263 of file ftypes.h.

4.49.2.135 TNOT

```
#define TNOT -28
```

Definition at line 264 of file ftypes.h.

4.49.2.136 TENDOFIT

```
#define TENDOFIT -29
```

Definition at line 265 of file ftypes.h.

4.49.2.137 TSETOPEN

```
#define TSETOPEN -30
```

Definition at line 266 of file ftypes.h.

4.49.2.138 TSETCLOSE

```
#define TSETCLOSE -31
```

Definition at line 267 of file ftypes.h.

4.49.2.139 TGENINDEX

```
#define TGENINDEX -32
```

Definition at line 268 of file ftypes.h.

4.49.2.140 TCONJUGATE

```
#define TCONJUGATE -33
```

Definition at line 269 of file ftypes.h.

4.49.2.141 LRPARENTHESES

```
#define LRPARENTHESES -34
```

Definition at line 270 of file ftypes.h.

4.49.2.142 TNUMBER1

```
#define TNUMBER1 -35
```

Definition at line 271 of file ftypes.h.

4.49.2.143 TPOWER1

```
#define TPOWER1 -36
```

Definition at line 272 of file ftypes.h.

4.49.2.144 TEMPTY

```
#define TEMPTY -37
```

Definition at line 273 of file ftypes.h.

4.49.2.145 TSETNUM

```
#define TSETNUM -38
```

Definition at line 274 of file ftypes.h.

4.49.2.146 TSGAMMA

```
#define TSGAMMA -39
```

Definition at line 275 of file ftypes.h.

4.49.2.147 TSETDOL

```
#define TSETDOL -40
```

Definition at line 276 of file ftypes.h.

4.49.2.148 TFLOAT

```
#define TFLOAT -41
```

Definition at line 277 of file ftypes.h.

4.49.2.149 TYPEISFUN

```
#define TYPEISFUN 0
```

Definition at line 279 of file ftypes.h.

4.49.2.150 TYPEISSUB

```
#define TYPEISSUB 1
```

Definition at line 280 of file ftypes.h.

4.49.2.151 TYPEISMYSTERY

```
#define TYPEISMYSTERY -1
```

Definition at line 281 of file ftypes.h.

4.49.2.152 LHSIDEX

```
#define LHSIDEX 2
```

Definition at line 283 of file ftypes.h.

4.49.2.153 LHSIDE

```
#define LHSIDE 1
```

Definition at line 284 of file ftypes.h.

4.49.2.154 RHSIDE

```
#define RHSIDE 0
```

Definition at line 285 of file ftypes.h.

4.49.2.155 FORTRANMODE

```
#define FORTRANMODE 1
```

Definition at line 291 of file ftypes.h.

4.49.2.156 REDUCEMODE

```
#define REDUCEMODE 2
```

Definition at line 292 of file ftypes.h.

4.49.2.157 MAPLEMODE

```
#define MAPLEMODE 3
```

Definition at line 293 of file ftypes.h.

4.49.2.158 MATHEMATICAMODE

```
#define MATHEMATICAMODE 4
```

Definition at line 294 of file ftypes.h.

4.49.2.159 CMODE

```
#define CMODE 5
```

Definition at line 295 of file ftypes.h.

4.49.2.160 VORTRANMODE

```
#define VORTRANMODE 6
```

Definition at line 296 of file ftypes.h.

4.49.2.161 PFORTRANMODE

```
#define PFORTRANMODE 7
```

Definition at line 297 of file ftypes.h.

4.49.2.162 DOUBLEFORTRANMODE

```
#define DOUBLEFORTRANMODE 33
```

Definition at line 298 of file ftypes.h.

4.49.2.163 DOUBLEPRECISIONFLAG

```
#define DOUBLEPRECISIONFLAG 32
```

Definition at line 299 of file ftypes.h.

4.49.2.164 NODOUBLEMASK

```
#define NODOUBLEMASK 31
```

Definition at line 300 of file ftypes.h.

4.49.2.165 QUADRUPLEFORTRANMODE

```
#define QUADRUPLEFORTRANMODE 65
```

Definition at line 301 of file ftypes.h.

4.49.2.166 QUADRUPLEPRECISIONFLAG

```
#define QUADRUPLEPRECISIONFLAG 64
```

Definition at line 302 of file ftypes.h.

4.49.2.167 ALLINTEGERDOUBLE

```
#define ALLINTEGERDOUBLE 128
```

Definition at line 303 of file ftypes.h.

4.49.2.168 NOQUADMASK

```
#define NOQUADMASK 63
```

Definition at line 304 of file ftypes.h.

4.49.2.169 NORMALFORMAT

```
#define NORMALFORMAT 0
```

Definition at line 305 of file ftypes.h.

4.49.2.170 NOSPACEFORMAT

```
#define NOSPACEFORMAT 1
```

Definition at line 306 of file ftypes.h.

4.49.2.171 ISNOTFORTRAN90

```
#define ISNOTFORTRAN90 0
```

Definition at line 308 of file ftypes.h.

4.49.2.172 ISFORTRAN90

```
#define ISFORTRAN90 1
```

Definition at line 309 of file ftypes.h.

4.49.2.173 ALSOREVERSE

```
#define ALSOREVERSE 1
```

Definition at line 311 of file ftypes.h.

4.49.2.174 CHISHOLM

```
#define CHISHOLM 2
```

Definition at line 312 of file ftypes.h.

4.49.2.175 NOTRICK

```
#define NOTRICK 16
```

Definition at line 313 of file ftypes.h.

4.49.2.176 SORTLOWFIRST

```
#define SORTLOWFIRST 0
```

Definition at line 315 of file ftypes.h.

4.49.2.177 SORTHIGHFIRST

```
#define SORTHIGHFIRST 1
```

Definition at line 316 of file ftypes.h.

4.49.2.178 SORTPOWERFIRST

```
#define SORTPOWERFIRST 2
```

Definition at line 317 of file ftypes.h.

4.49.2.179 SORTANTIPOWER

```
#define SORTANTIPOWER 3
```

Definition at line 318 of file ftypes.h.

4.49.2.180 STATSSPLITMERGE

```
#define STATSSPLITMERGE 0
```

Definition at line 324 of file ftypes.h.

4.49.2.181 STATSMERGETOFILE

```
#define STATSMERGETOFILE 1
```

Definition at line 325 of file ftypes.h.

4.49.2.182 STATSPOSTSORT

```
#define STATSPOSTSORT 2
```

Definition at line 326 of file ftypes.h.

4.49.2.183 NOCHECKLOGTYPE

```
#define NOCHECKLOGTYPE 0
```

Definition at line 329 of file ftypes.h.

4.49.2.184 CHECKLOGTYPE

```
#define CHECKLOGTYPE 1
```

Definition at line 330 of file ftypes.h.

4.49.2.185 NMIN4SHIFT

```
#define NMIN4SHIFT 4
```

Definition at line 332 of file ftypes.h.

4.49.2.186 SYMBOL

```
#define SYMBOL 1
```

Definition at line 346 of file ftypes.h.

4.49.2.187 DOTPRODUCT

```
#define DOTPRODUCT 2
```

Definition at line 347 of file ftypes.h.

4.49.2.188 VECTOR

```
#define VECTOR 3
```

Definition at line 348 of file ftypes.h.

4.49.2.189 INDEX

```
#define INDEX 4
```

Definition at line 349 of file ftypes.h.

4.49.2.190 EXPRESSION

```
#define EXPRESSION 5
```

Definition at line 350 of file ftypes.h.

4.49.2.191 SUBEXPRESSION

```
#define SUBEXPRESSION 6
```

Definition at line 351 of file ftypes.h.

4.49.2.192 DOLLAREXPRESSION

```
#define DOLLAREXPRESSION 7
```

Definition at line 352 of file ftypes.h.

4.49.2.193 SETSET

```
#define SETSET 8
```

Definition at line 353 of file ftypes.h.

4.49.2.194 ARGWILD

```
#define ARGWILD 9
```

Definition at line 354 of file ftypes.h.

4.49.2.195 MINVECTOR

```
#define MINVECTOR 10
```

Definition at line 355 of file ftypes.h.

4.49.2.196 SETEXP

```
#define SETEXP 11
```

Definition at line 356 of file ftypes.h.

4.49.2.197 DOLLAREXPR2

```
#define DOLLAREXPR2 12
```

Definition at line 357 of file ftypes.h.

4.49.2.198 HAAKJE0

```
#define HAAKJE0 9
```

Definition at line 358 of file ftypes.h.

4.49.2.199 FUNCTION

```
#define FUNCTION 20
```

Definition at line 359 of file ftypes.h.

4.49.2.200 TMPPOLYFUN

```
#define TMPPOLYFUN 14
```

Definition at line 361 of file ftypes.h.

4.49.2.201 ARGFIELD

```
#define ARGFIELD 15
```

Definition at line 362 of file ftypes.h.

4.49.2.202 SNUMBER

```
#define SNUMBER 16
```

Definition at line 363 of file ftypes.h.

4.49.2.203 LNUMBER

```
#define LNUMBER 17
```

Definition at line 364 of file ftypes.h.

4.49.2.204 HAAKJE

```
#define HAAKJE 18
```

Definition at line 365 of file ftypes.h.

4.49.2.205 DELTA

```
#define DELTA 19
```

Definition at line 366 of file ftypes.h.

4.49.2.206 EXPONENT

```
#define EXPONENT 20
```

Definition at line 367 of file ftypes.h.

4.49.2.207 DENOMINATOR

```
#define DENOMINATOR 21
```

Definition at line 368 of file ftypes.h.

4.49.2.208 SETFUNCTION

```
#define SETFUNCTION 22
```

Definition at line 369 of file ftypes.h.

4.49.2.209 GAMMA

```
#define GAMMA 23
```

Definition at line 370 of file ftypes.h.

4.49.2.210 GAMMAI

```
#define GAMMAI 24
```

Definition at line 371 of file ftypes.h.

4.49.2.211 GAMMAFIVE

```
#define GAMMAFIVE 25
```

Definition at line 372 of file ftypes.h.

4.49.2.212 GAMMASIX

```
#define GAMMASIX 26
```

Definition at line 373 of file ftypes.h.

4.49.2.213 GAMMASEVEN

```
#define GAMMASEVEN 27
```

Definition at line 374 of file ftypes.h.

4.49.2.214 SUMF1

```
#define SUMF1 28
```

Definition at line 375 of file ftypes.h.

4.49.2.215 SUMF2

```
#define SUMF2 29
```

Definition at line 376 of file ftypes.h.

4.49.2.216 DUMFUN

```
#define DUMFUN 30
```

Definition at line 377 of file ftypes.h.

4.49.2.217 REPLACEMENT

```
#define REPLACEMENT 31
```

Definition at line 378 of file ftypes.h.

4.49.2.218 REVERSEFUNCTION

```
#define REVERSEFUNCTION 32
```

Definition at line 379 of file ftypes.h.

4.49.2.219 DISTRIBUTION

```
#define DISTRIBUTION 33
```

Definition at line 380 of file ftypes.h.

4.49.2.220 DELTA3

```
#define DELTA3 34
```

Definition at line 381 of file ftypes.h.

4.49.2.221 DUMMYFUN

```
#define DUMMYFUN 35
```

Definition at line 382 of file ftypes.h.

4.49.2.222 DUMMYTEN

```
#define DUMMYTEN 36
```

Definition at line 383 of file ftypes.h.

4.49.2.223 LEVICIVITA

```
#define LEVICIVITA 37
```

Definition at line 384 of file ftypes.h.

4.49.2.224 FACTORIAL

```
#define FACTORIAL 38
```

Definition at line 385 of file ftypes.h.

4.49.2.225 INVERSEFACTORIAL

```
#define INVERSEFACTORIAL 39
```

Definition at line 386 of file ftypes.h.

4.49.2.226 BINOMIAL

```
#define BINOMIAL 40
```

Definition at line 387 of file ftypes.h.

4.49.2.227 NUMARGSFUN

```
#define NUMARGSFUN 41
```

Definition at line 388 of file ftypes.h.

4.49.2.228 SIGNFUN

```
#define SIGNFUN 42
```

Definition at line 389 of file ftypes.h.

4.49.2.229 MODFUNCTION

```
#define MODFUNCTION 43
```

Definition at line 390 of file ftypes.h.

4.49.2.230 MOD2FUNCTION

```
#define MOD2FUNCTION 44
```

Definition at line 391 of file ftypes.h.

4.49.2.231 MINFUNCTION

```
#define MINFUNCTION 45
```

Definition at line 392 of file ftypes.h.

4.49.2.232 MAXFUNCTION

```
#define MAXFUNCTION 46
```

Definition at line 393 of file ftypes.h.

4.49.2.233 ABSFUNCTION

```
#define ABSFUNCTION 47
```

Definition at line 394 of file ftypes.h.

4.49.2.234 SIGFUNCTION

```
#define SIGFUNCTION 48
```

Definition at line 395 of file ftypes.h.

4.49.2.235 INTFUNCTION

```
#define INTFUNCTION 49
```

Definition at line 396 of file ftypes.h.

4.49.2.236 THETA

```
#define THETA 50
```

Definition at line 397 of file ftypes.h.

4.49.2.237 THETA2

```
#define THETA2 51
```

Definition at line 398 of file ftypes.h.

4.49.2.238 DELTA2

```
#define DELTA2 52
```

Definition at line 399 of file ftypes.h.

4.49.2.239 DELTAP

```
#define DELTAP 53
```

Definition at line 400 of file ftypes.h.

4.49.2.240 BERNOULLIFUNCTION

```
#define BERNOULLIFUNCTION 54
```

Definition at line 401 of file ftypes.h.

4.49.2.241 COUNTFUNCTION

```
#define COUNTFUNCTION 55
```

Definition at line 402 of file ftypes.h.

4.49.2.242 MATCHFUNCTION

```
#define MATCHFUNCTION 56
```

Definition at line 403 of file ftypes.h.

4.49.2.243 PATTERNFUNCTION

```
#define PATTERNFUNCTION 57
```

Definition at line 404 of file ftypes.h.

4.49.2.244 TERMFUNCTION

```
#define TERMFUNCTION 58
```

Definition at line 405 of file ftypes.h.

4.49.2.245 CONJUGATION

```
#define CONJUGATION 59
```

Definition at line 406 of file ftypes.h.

4.49.2.246 ROOTFUNCTION

```
#define ROOTFUNCTION 60
```

Definition at line 407 of file ftypes.h.

4.49.2.247 TABLEFUNCTION

```
#define TABLEFUNCTION 61
```

Definition at line 408 of file ftypes.h.

4.49.2.248 FIRSTBRACKET

```
#define FIRSTBRACKET 62
```

Definition at line 409 of file ftypes.h.

4.49.2.249 TERMSINEXPR

```
#define TERMSINEXPR 63
```

Definition at line 410 of file ftypes.h.

4.49.2.250 NUMTERMSFUN

```
#define NUMTERMSFUN 64
```

Definition at line 411 of file ftypes.h.

4.49.2.251 GCDFUNCTION

```
#define GCDFUNCTION 65
```

Definition at line 412 of file ftypes.h.

4.49.2.252 DIVFUNCTION

```
#define DIVFUNCTION 66
```

Definition at line 413 of file ftypes.h.

4.49.2.253 REMFUNCTION

```
#define REMFUNCTION 67
```

Definition at line 414 of file ftypes.h.

4.49.2.254 MAXPOWEROF

```
#define MAXPOWEROF 68
```

Definition at line 415 of file ftypes.h.

4.49.2.255 MINPOWEROF

```
#define MINPOWEROF 69
```

Definition at line 416 of file ftypes.h.

4.49.2.256 TABLESTUB

```
#define TABLESTUB 70
```

Definition at line 417 of file ftypes.h.

4.49.2.257 FACTORIN

```
#define FACTORIN 71
```

Definition at line 418 of file ftypes.h.

4.49.2.258 TERMSINBRACKET

```
#define TERMSINBRACKET 72
```

Definition at line 419 of file ftypes.h.

4.49.2.259 WILDARGFUN

```
#define WILDARGFUN 73
```

Definition at line 420 of file ftypes.h.

4.49.2.260 SQRTFUNCTION

```
#define SQRTFUNCTION 74
```

Definition at line 428 of file ftypes.h.

4.49.2.261 LNFUNCTION

```
#define LNFUNCTION 75
```

Definition at line 429 of file ftypes.h.

4.49.2.262 SINFUNCTION

```
#define SINFUNCTION 76
```

Definition at line 430 of file ftypes.h.

4.49.2.263 COSFUNCTION

```
#define COSFUNCTION 77
```

Definition at line 431 of file ftypes.h.

4.49.2.264 TANFUNCTION

```
#define TANFUNCTION 78
```

Definition at line 432 of file ftypes.h.

4.49.2.265 ASINFUNCTION

```
#define ASINFUNCTION 79
```

Definition at line 433 of file ftypes.h.

4.49.2.266 ACOSFUNCTION

```
#define ACOSFUNCTION 80
```

Definition at line 434 of file ftypes.h.

4.49.2.267 ATANFUNCTION

```
#define ATANFUNCTION 81
```

Definition at line 435 of file ftypes.h.

4.49.2.268 ATAN2FUNCTION

```
#define ATAN2FUNCTION 82
```

Definition at line 436 of file ftypes.h.

4.49.2.269 SINHFUNCTION

```
#define SINHFUNCTION 83
```

Definition at line 437 of file ftypes.h.

4.49.2.270 COSHFUNCTION

```
#define COSHFUNCTION 84
```

Definition at line 438 of file ftypes.h.

4.49.2.271 TANHFUNCTION

```
#define TANHFUNCTION 85
```

Definition at line 439 of file ftypes.h.

4.49.2.272 ASINHFUNCTION

```
#define ASINHFUNCTION 86
```

Definition at line 440 of file ftypes.h.

4.49.2.273 ACOSHFUNCTION

```
#define ACOSHFUNCTION 87
```

Definition at line 441 of file ftypes.h.

4.49.2.274 ATANHFUNCTION

```
#define ATANHFUNCTION 88
```

Definition at line 442 of file ftypes.h.

4.49.2.275 LI2FUNCTION

```
#define LI2FUNCTION 89
```

Definition at line 443 of file ftypes.h.

4.49.2.276 LINFUNCTION

```
#define LINFUNCTION 90
```

Definition at line 444 of file ftypes.h.

4.49.2.277 EXTRASYMFUN

```
#define EXTRASYMFUN 91
```

Definition at line 446 of file ftypes.h.

4.49.2.278 RANDOMFUNCTION

```
#define RANDOMFUNCTION 92
```

Definition at line 447 of file ftypes.h.

4.49.2.279 RANPERM

```
#define RANPERM 93
```

Definition at line 448 of file ftypes.h.

4.49.2.280 NUMFACTORS

```
#define NUMFACTORS 94
```

Definition at line 449 of file ftypes.h.

4.49.2.281 FIRSTTERM

```
#define FIRSTTERM 95
```

Definition at line 450 of file ftypes.h.

4.49.2.282 CONTENTTERM

```
#define CONTENTTERM 96
```

Definition at line 451 of file ftypes.h.

4.49.2.283 PRIMENUMBER

```
#define PRIMENUMBER 97
```

Definition at line 452 of file ftypes.h.

4.49.2.284 EXTEUCLIDEAN

```
#define EXTEUCLIDEAN 98
```

Definition at line 453 of file ftypes.h.

4.49.2.285 MAKERATIONAL

```
#define MAKERATIONAL 99
```

Definition at line 454 of file ftypes.h.

4.49.2.286 INVERSEFUNCTION

```
#define INVERSEFUNCTION 100
```

Definition at line 455 of file ftypes.h.

4.49.2.287 IDFUNCTION

```
#define IDFUNCTION 101
```

Definition at line 456 of file ftypes.h.

4.49.2.288 PUTFIRST

```
#define PUTFIRST 102
```

Definition at line 457 of file ftypes.h.

4.49.2.289 PERMUTATIONS

```
#define PERMUTATIONS 103
```

Definition at line 458 of file ftypes.h.

4.49.2.290 PARTITIONS

```
#define PARTITIONS 104
```

Definition at line 459 of file ftypes.h.

4.49.2.291 MULFUNCTION

```
#define MULFUNCTION 105
```

Definition at line 460 of file ftypes.h.

4.49.2.292 MOEBIUS

```
#define MOEBIUS 106
```

Definition at line 461 of file ftypes.h.

4.49.2.293 TOPOLOGIES

```
#define TOPOLOGIES 107
```

Definition at line 462 of file ftypes.h.

4.49.2.294 DIAGRAMS

```
#define DIAGRAMS 108
```

Definition at line 463 of file ftypes.h.

4.49.2.295 TOPO

```
#define TOPO 109
```

Definition at line 464 of file ftypes.h.

4.49.2.296 NODEFUNCTION

```
#define NODEFUNCTION 110
```

Definition at line 465 of file ftypes.h.

4.49.2.297 EDGE

```
#define EDGE 111
```

Definition at line 466 of file ftypes.h.

4.49.2.298 SIZEOFFUNCTION

```
#define SIZEOFFUNCTION 112
```

Definition at line 467 of file ftypes.h.

4.49.2.299 BLOCK

```
#define BLOCK 113
```

Definition at line 468 of file ftypes.h.

4.49.2.300 ONEPI

```
#define ONEPI 114
```

Definition at line 469 of file ftypes.h.

4.49.2.301 PHI

```
#define PHI 115
```

Definition at line 470 of file ftypes.h.

4.49.2.302 MAXBUILTINFUNCTION

```
#define MAXBUILTINFUNCTION 115
```

Definition at line 482 of file ftypes.h.

4.49.2.303 FIRSTUSERFUNCTION

```
#define FIRSTUSERFUNCTION 150
```

Definition at line 485 of file ftypes.h.

4.49.2.304 ALLFUNCTIONS

```
#define ALLFUNCTIONS -1
```

Definition at line 487 of file ftypes.h.

4.49.2.305 ALLMZVFUNCTIONS

```
#define ALLMZVFUNCTIONS -2
```

Definition at line 488 of file ftypes.h.

4.49.2.306 ISYMBOL

```
#define ISYMBOL 0
```

Definition at line 496 of file ftypes.h.

4.49.2.307 PISYMBOL

```
#define PISYMBOL 1
```

Definition at line 497 of file ftypes.h.

4.49.2.308 COEFFSYMBOL

```
#define COEFFSYMBOL 2
```

Definition at line 498 of file ftypes.h.

4.49.2.309 NUMERATORSYMBOL

```
#define NUMERATORSYMBOL 3
```

Definition at line 499 of file ftypes.h.

4.49.2.310 DENOMINATORSYMBOL

```
#define DENOMINATORSYMBOL 4
```

Definition at line 500 of file ftypes.h.

4.49.2.311 WILDARGSYMBOL

```
#define WILDARGSYMBOL 5
```

Definition at line 501 of file ftypes.h.

4.49.2.312 DIMENSIONSYMBOL

```
#define DIMENSIONSYMBOL 6
```

Definition at line 502 of file ftypes.h.

4.49.2.313 FACTORSYMBOL

```
#define FACTORSYMBOL 7
```

Definition at line 503 of file ftypes.h.

4.49.2.314 SEPARATESYMBOL

```
#define SEPARATESYMBOL 8
```

Definition at line 504 of file ftypes.h.

4.49.2.315 EESYMBOL

```
#define EESYMBOL 9
```

Definition at line 505 of file ftypes.h.

4.49.2.316 EMSYMBOL

```
#define EMSYMBOL 10
```

Definition at line 506 of file ftypes.h.

4.49.2.317 BUILTINSYMBOLS

```
#define BUILTINSYMBOLS 11
```

Definition at line 508 of file ftypes.h.

4.49.2.318 FIRSTUSERSYMBOL

```
#define FIRSTUSERSYMBOL 20
```

Definition at line 509 of file ftypes.h.

4.49.2.319 BUILTININDICES

```
#define BUILTININDICES 1
```

Definition at line 511 of file ftypes.h.

4.49.2.320 BUILTINVECTORS

```
#define BUILTINVECTORS 1
```

Definition at line 512 of file ftypes.h.

4.49.2.321 BUILTINDOLLARS

```
#define BUILTINDOLLARS 1
```

Definition at line 513 of file ftypes.h.

4.49.2.322 WILDARGVECTOR

```
#define WILDARGVECTOR 0
```

Definition at line 515 of file ftypes.h.

4.49.2.323 WILDARGINDEX

```
#define WILDARGINDEX 0
```

Definition at line 516 of file ftypes.h.

4.49.2.324 TYPEEXPRESSION

```
#define TYPEEXPRESSION 0
```

Definition at line 527 of file ftypes.h.

4.49.2.325 TYPEIDNEW

```
#define TYPEIDNEW 1
```

Definition at line 528 of file ftypes.h.

4.49.2.326 TYPEIDOLD

```
#define TYPEIDOLD 2
```

Definition at line 529 of file ftypes.h.

4.49.2.327 TYPEOPERATION

```
#define TYPEOPERATION 3
```

Definition at line 530 of file ftypes.h.

4.49.2.328 TYPEREPEAT

```
#define TYPEREPEAT 4
```

Definition at line 531 of file ftypes.h.

4.49.2.329 TYPEENDREPEAT

```
#define TYPEENDREPEAT 5
```

Definition at line 532 of file ftypes.h.

4.49.2.330 TYPECOUNT

```
#define TYPECOUNT 20
```

Definition at line 536 of file ftypes.h.

4.49.2.331 TYPEMULT

```
#define TYPEMULT 21
```

Definition at line 537 of file ftypes.h.

4.49.2.332 TYPEGOTO

```
#define TYPEGOTO 22
```

Definition at line 538 of file ftypes.h.

4.49.2.333 TYPEDISCARD

```
#define TYPEDISCARD 23
```

Definition at line 539 of file ftypes.h.

4.49.2.334 TYPEIF

```
#define TYPEIF 24
```

Definition at line 540 of file ftypes.h.

4.49.2.335 TYPEELSE

```
#define TYPEELSE 25
```

Definition at line 541 of file ftypes.h.

4.49.2.336 TYPEELIF

```
#define TYPEELIF 26
```

Definition at line 542 of file ftypes.h.

4.49.2.337 TYPEENDIF

```
#define TYPEENDIF 27
```

Definition at line 543 of file ftypes.h.

4.49.2.338 TYPESUM

```
#define TYPESUM 28
```

Definition at line 544 of file ftypes.h.

4.49.2.339 TYPECHISHOLM

```
#define TYPECHISHOLM 29
```

Definition at line 545 of file ftypes.h.

4.49.2.340 TYPEREVERSE

```
#define TYPEREVERSE 30
```

Definition at line 546 of file ftypes.h.

4.49.2.341 TYPEARG

```
#define TYPEARG 31
```

Definition at line 547 of file ftypes.h.

4.49.2.342 TYPENORM

```
#define TYPENORM 32
```

Definition at line 548 of file ftypes.h.

4.49.2.343 TYPENORM2

```
#define TYPENORM2 33
```

Definition at line 549 of file ftypes.h.

4.49.2.344 TYPENORM3

```
#define TYPENORM3 34
```

Definition at line 550 of file ftypes.h.

4.49.2.345 TYPEEXIT

```
#define TYPEEXIT 35
```

Definition at line 551 of file ftypes.h.

4.49.2.346 TYPESETEXIT

```
#define TYPESETEXIT 36
```

Definition at line 552 of file ftypes.h.

4.49.2.347 TYPEPRINT

```
#define TYPEPRINT 37
```

Definition at line 553 of file ftypes.h.

4.49.2.348 TYPEFPRINT

```
#define TYPEFPRINT 38
```

Definition at line 554 of file ftypes.h.

4.49.2.349 TYPEREDEFPRE

```
#define TYPEREDEFPRE 39
```

Definition at line 555 of file ftypes.h.

4.49.2.350 TYPESPLITARG

```
#define TYPESPLITARG 40
```

Definition at line 556 of file ftypes.h.

4.49.2.351 TYPESPLITARG2

```
#define TYPESPLITARG2 41
```

Definition at line 557 of file ftypes.h.

4.49.2.352 TYPEFACTARG

```
#define TYPEFACTARG 42
```

Definition at line 558 of file ftypes.h.

4.49.2.353 TYPEFACTARG2

```
#define TYPEFACTARG2 43
```

Definition at line 559 of file ftypes.h.

4.49.2.354 TYPETRY

```
#define TYPETRY 44
```

Definition at line 560 of file ftypes.h.

4.49.2.355 TYPEASSIGN

```
#define TYPEASSIGN 45
```

Definition at line 561 of file ftypes.h.

4.49.2.356 TYPERENUMBER

```
#define TYPERENUMBER 46
```

Definition at line 562 of file ftypes.h.

4.49.2.357 TYPESUMFIX

```
#define TYPESUMFIX 47
```

Definition at line 563 of file ftypes.h.

4.49.2.358 TYPEFINDLOOP

```
#define TYPEFINDLOOP 48
```

Definition at line 564 of file ftypes.h.

4.49.2.359 TYPEUNRAVEL

```
#define TYPEUNRAVEL 49
```

Definition at line 565 of file ftypes.h.

4.49.2.360 TYPEADJUSTBOUNDS

```
#define TYPEADJUSTBOUNDS 50
```

Definition at line 566 of file ftypes.h.

4.49.2.361 TYPEINSIDE

```
#define TYPEINSIDE 51
```

Definition at line 567 of file ftypes.h.

4.49.2.362 TYPETERM

```
#define TYPETERM 52
```

Definition at line 568 of file ftypes.h.

4.49.2.363 TYPESORT

```
#define TYPESORT 53
```

Definition at line 569 of file ftypes.h.

4.49.2.364 TYPEDETCURDUM

```
#define TYPEDETCURDUM 54
```

Definition at line 570 of file ftypes.h.

4.49.2.365 TYPEINEXPRESSION

```
#define TYPEINEXPRESSION 55
```

Definition at line 571 of file ftypes.h.

4.49.2.366 TYPESPLITFIRSTARG

```
#define TYPESPLITFIRSTARG 56
```

Definition at line 572 of file ftypes.h.

4.49.2.367 TYPESPLITLASTARG

```
#define TYPESPLITLASTARG 57
```

Definition at line 573 of file ftypes.h.

4.49.2.368 TYPEMERGE

```
#define TYPEMERGE 58
```

Definition at line 574 of file ftypes.h.

4.49.2.369 TYPETESTUSE

```
#define TYPETESTUSE 59
```

Definition at line 575 of file ftypes.h.

4.49.2.370 TYPEAPPLY

```
#define TYPEAPPLY 60
```

Definition at line 576 of file ftypes.h.

4.49.2.371 TYPEAPPLYRESET

```
#define TYPEAPPLYRESET 61
```

Definition at line 577 of file ftypes.h.

4.49.2.372 TYPECHAININ

```
#define TYPECHAININ 62
```

Definition at line 578 of file ftypes.h.

4.49.2.373 TYPECHAINOUT

```
#define TYPECHAINOUT 63
```

Definition at line 579 of file ftypes.h.

4.49.2.374 TYPENORM4

```
#define TYPENORM4 64
```

Definition at line 580 of file ftypes.h.

4.49.2.375 TYPEFACTOR

```
#define TYPEFACTOR 65
```

Definition at line 581 of file ftypes.h.

4.49.2.376 TYPEARGIMPLODE

```
#define TYPEARGIMPLODE 66
```

Definition at line 582 of file ftypes.h.

4.49.2.377 TYPEARGEXPLODE

```
#define TYPEARGEXPLODE 67
```

Definition at line 583 of file ftypes.h.

4.49.2.378 TYPEDENOMINATORS

```
#define TYPEDENOMINATORS 68
```

Definition at line 584 of file ftypes.h.

4.49.2.379 TYPESTUFFLE

```
#define TYPESTUFFLE 69
```

Definition at line 585 of file ftypes.h.

4.49.2.380 TYPEDROPCOEFFICIENT

```
#define TYPEDROPCOEFFICIENT 70
```

Definition at line 586 of file ftypes.h.

4.49.2.381 TYPETRANSFORM

```
#define TYPETRANSFORM 71
```

Definition at line 587 of file ftypes.h.

4.49.2.382 TYPETOPOLYNOMIAL

```
#define TYPETOPOLYNOMIAL 72
```

Definition at line 588 of file ftypes.h.

4.49.2.383 TYPEFROMPOLYNOMIAL

```
#define TYPEFROMPOLYNOMIAL 73
```

Definition at line 589 of file ftypes.h.

4.49.2.384 TYPEDOLOOP

```
#define TYPEDOLOOP 74
```

Definition at line 590 of file ftypes.h.

4.49.2.385 TYPEENDDOLOOP

```
#define TYPEENDDOLOOP 75
```

Definition at line 591 of file ftypes.h.

4.49.2.386 TYPEDROPSYMBOLS

```
#define TYPEDROPSYMBOLS 76
```

Definition at line 592 of file ftypes.h.

4.49.2.387 TYPEPUTINSIDE

```
#define TYPEPUTINSIDE 77
```

Definition at line 593 of file ftypes.h.

4.49.2.388 TYPETOSPECTATOR

```
#define TYPETOSPECTATOR 78
```

Definition at line 594 of file ftypes.h.

4.49.2.389 TYPEARGTOEXTRASYMBOL

```
#define TYPEARGTOEXTRASYMBOL 79
```

Definition at line 595 of file ftypes.h.

4.49.2.390 TYPECANONICALIZE

```
#define TYPECANONICALIZE 80
```

Definition at line 596 of file ftypes.h.

4.49.2.391 TYPESWITCH

```
#define TYPESWITCH 81
```

Definition at line 597 of file ftypes.h.

4.49.2.392 TYPEENDSWITCH

```
#define TYPEENDSWITCH 82
```

Definition at line 598 of file ftypes.h.

4.49.2.393 TYPESETUSERFLAG

```
#define TYPESETUSERFLAG 83
```

Definition at line 599 of file ftypes.h.

4.49.2.394 TYPECLEARUSERFLAG

```
#define TYPECLEARUSERFLAG 84
```

Definition at line 600 of file ftypes.h.

4.49.2.395 TYPEALLLOOPS

```
#define TYPEALLLOOPS 85
```

Definition at line 601 of file ftypes.h.

4.49.2.396 TYPEALLPATHS

```
#define TYPEALLPATHS 86
```

Definition at line 602 of file ftypes.h.

4.49.2.397 TAKETRACE

```
#define TAKETRACE 1
```

Definition at line 614 of file ftypes.h.

4.49.2.398 CONTRACT

```
#define CONTRACT 2
```

Definition at line 615 of file ftypes.h.

4.49.2.399 RATIO

```
#define RATIO 3
```

Definition at line 616 of file ftypes.h.

4.49.2.400 SYMMETRIZE

```
#define SYMMETRIZE 4
```

Definition at line 617 of file ftypes.h.

4.49.2.401 TENVEC

```
#define TENVEC 5
```

Definition at line 618 of file ftypes.h.

4.49.2.402 SUMNUM1

```
#define SUMNUM1 6
```

Definition at line 619 of file ftypes.h.

4.49.2.403 SUMNUM2

```
#define SUMNUM2 7
```

Definition at line 620 of file ftypes.h.

4.49.2.404 WILDDUMMY

```
#define WILDDUMMY 0
```

Definition at line 626 of file ftypes.h.

4.49.2.405 SYMTONUM

```
#define SYMTONUM 1
```

Definition at line 627 of file ftypes.h.

4.49.2.406 SYMTOSYM

```
#define SYMTOSYM 2
```

Definition at line 628 of file ftypes.h.

4.49.2.407 SYMTOSUB

```
#define SYMTOSUB 3
```

Definition at line 629 of file ftypes.h.

4.49.2.408 VECTOMIN

```
#define VECTOMIN 4
```

Definition at line 630 of file ftypes.h.

4.49.2.409 VECTOVEC

```
#define VECTOVEC 5
```

Definition at line 631 of file ftypes.h.

4.49.2.410 VECTOSUB

```
#define VECTOSUB 6
```

Definition at line 632 of file ftypes.h.

4.49.2.411 INDTOIND

```
#define INDTOIND 7
```

Definition at line 633 of file ftypes.h.

4.49.2.412 INDTOSUB

```
#define INDTOSUB 8
```

Definition at line 634 of file ftypes.h.

4.49.2.413 FUNTOFUN

```
#define FUNTOFUN 9
```

Definition at line 635 of file ftypes.h.

4.49.2.414 ARGTOARG

```
#define ARGTOARG 10
```

Definition at line 636 of file ftypes.h.

4.49.2.415 ARLTOARL

```
#define ARLTOARL 11
```

Definition at line 637 of file ftypes.h.

4.49.2.416 EXPTOEXP

```
#define EXPTOEXP 12
```

Definition at line 638 of file ftypes.h.

4.49.2.417 FROMBRAC

```
#define FROMBRAC 13
```

Definition at line 639 of file ftypes.h.

4.49.2.418 FROMSET

```
#define FROMSET 14
```

Definition at line 640 of file ftypes.h.

4.49.2.419 SETTONUM

```
#define SETTONUM 15
```

Definition at line 641 of file ftypes.h.

4.49.2.420 WILDCARDS

```
#define WILDCARDS 16
```

Definition at line 642 of file ftypes.h.

4.49.2.421 SETNUMBER

```
#define SETNUMBER 17
```

Definition at line 643 of file ftypes.h.

4.49.2.422 LOADDOLLAR

```
#define LOADDOLLAR 18
```

Definition at line 644 of file ftypes.h.

4.49.2.423 NUMTONUM

```
#define NUMTONUM 20
```

Definition at line 648 of file ftypes.h.

4.49.2.424 NUMTOSYM

```
#define NUMTOSYM 21
```

Definition at line 649 of file ftypes.h.

4.49.2.425 NUMTOIND

```
#define NUMTOIND 22
```

Definition at line 650 of file ftypes.h.

4.49.2.426 NUMTOSUB

```
#define NUMTOSUB 23
```

Definition at line 651 of file ftypes.h.

4.49.2.427 EATTENSOR

```
#define EATTENSOR 0x2000
```

Definition at line 655 of file ftypes.h.

4.49.2.428 CLEANFLAG

```
#define CLEANFLAG 0
```

Definition at line 661 of file ftypes.h.

4.49.2.429 DIRTYFLAG

```
#define DIRTYFLAG 1
```

Definition at line 662 of file ftypes.h.

4.49.2.430 DIRTYSYMFLAG

```
#define DIRTYSYMFLAG 2
```

Definition at line 663 of file ftypes.h.

4.49.2.431 MUSTCLEANPRF

```
#define MUSTCLEANPRF 4
```

Definition at line 664 of file ftypes.h.

4.49.2.432 SUBTERMUSED1

```
#define SUBTERMUSED1 8
```

Definition at line 665 of file ftypes.h.

4.49.2.433 SUBTERMUSED2

```
#define SUBTERMUSED2 16
```

Definition at line 666 of file ftypes.h.

4.49.2.434 ALLDIRTY

```
#define ALLDIRTY (DIRTYFLAG|DIRTYSYMFLAG)
```

Definition at line 667 of file ftypes.h.

4.49.2.435 ARGHEAD

```
#define ARGHEAD 2
```

Definition at line 669 of file ftypes.h.

4.49.2.436 FUNHEAD

```
#define FUNHEAD 3
```

Definition at line 670 of file ftypes.h.

4.49.2.437 SUBEXPSIZE

```
#define SUBEXPSIZE 5
```

Definition at line 671 of file ftypes.h.

4.49.2.438 EXPRHEAD

```
#define EXPRHEAD 5
```

Definition at line 672 of file ftypes.h.

4.49.2.439 TYPEARGHEADSIZE

```
#define TYPEARGHEADSIZE 6
```

Definition at line 673 of file ftypes.h.

4.49.2.440 SKIP

```
#define SKIP 1
```

Definition at line 680 of file ftypes.h.

4.49.2.441 DROP

```
#define DROP 2
```

Definition at line 681 of file ftypes.h.

4.49.2.442 HIDE

```
#define HIDE 3
```

Definition at line 682 of file ftypes.h.

4.49.2.443 UNHIDE

```
#define UNHIDE 4
```

Definition at line 683 of file ftypes.h.

4.49.2.444 INTOHIDE

```
#define INTOHIDE 5
```

Definition at line 684 of file ftypes.h.

4.49.2.445 LOCALEXPRESSION

```
#define LOCALEXPRESSION 0
```

Definition at line 690 of file ftypes.h.

4.49.2.446 SKIPLEXPRESSION

```
#define SKIPLEXPRESSION 1
```

Definition at line 691 of file ftypes.h.

4.49.2.447 DROPLEXPRESSION

```
#define DROPLEXPRESSION 2
```

Definition at line 692 of file ftypes.h.

4.49.2.448 DROPPEDEXPRESSION

```
#define DROPPEDEXPRESSION 3
```

Definition at line 693 of file ftypes.h.

4.49.2.449 GLOBALEXPRESSION

```
#define GLOBALEXPRESSION 4
```

Definition at line 694 of file ftypes.h.

4.49.2.450 SKIPGEXPRESSION

```
#define SKIPGEXPRESSION 5
```

Definition at line 695 of file ftypes.h.

4.49.2.451 DROPGEXPRESSION

```
#define DROPGEXPRESSION 6
```

Definition at line 696 of file ftypes.h.

4.49.2.452 STOREDEXPRESSION

```
#define STOREDEXPRESSION 8
```

Definition at line 697 of file ftypes.h.

4.49.2.453 HIDDENLEXPRESSION

```
#define HIDDENLEXPRESSION 9
```

Definition at line 698 of file ftypes.h.

4.49.2.454 HIDDENGEXPRESSION

```
#define HIDDENGEXPRESSION 13
```

Definition at line 699 of file ftypes.h.

4.49.2.455 INCEXPRESSION

```
#define INCEXPRESSION 9
```

Definition at line 700 of file ftypes.h.

4.49.2.456 HIDELEXPRESSION

```
#define HIDELEXPRESSION 10
```

Definition at line 701 of file ftypes.h.

4.49.2.457 HIDEGEXPRESSION

```
#define HIDEGEXPRESSION 14
```

Definition at line 702 of file ftypes.h.

4.49.2.458 DROPHLEXPRESSION

```
#define DROPHLEXPRESSION 11
```

Definition at line 703 of file ftypes.h.

4.49.2.459 DROPHGEXPRESSION

```
#define DROPHGEXPRESSION 15
```

Definition at line 704 of file ftypes.h.

4.49.2.460 UNHIDELEXPRESSION

```
#define UNHIDELEXPRESSION 12
```

Definition at line 705 of file ftypes.h.

4.49.2.461 UNHIDEGEXPRESSION

```
#define UNHIDEGEXPRESSION 16
```

Definition at line 706 of file ftypes.h.

4.49.2.462 INTOHIDELEXPRESSION

```
#define INTOHIDELEXPRESSION 17
```

Definition at line 707 of file ftypes.h.

4.49.2.463 INTOHIDEGEXPRESSION

```
#define INTOHIDEGEXPRESSION 18
```

Definition at line 708 of file ftypes.h.

4.49.2.464 SPECTATOREXPRESSION

```
#define SPECTATOREXPRESSION 19
```

Definition at line 709 of file ftypes.h.

4.49.2.465 DROPSPECTATOREXPRESSION

```
#define DROPSPECTATOREXPRESSION 20
```

Definition at line 710 of file ftypes.h.

4.49.2.466 SKIPUNHIDELEXPRESSION

```
#define SKIPUNHIDELEXPRESSION 21
```

Definition at line 711 of file ftypes.h.

4.49.2.467 SKIPUNHIDEGEXPRESSION

```
#define SKIPUNHIDEGEXPRESSION 22
```

Definition at line 712 of file ftypes.h.

4.49.2.468 PRINTOFF

```
#define PRINTOFF 0
```

Definition at line 714 of file ftypes.h.

4.49.2.469 PRINTON

```
#define PRINTON 1
```

Definition at line 715 of file ftypes.h.

4.49.2.470 PRINTCONTENTS

```
#define PRINTCONTENTS 2
```

Definition at line 716 of file ftypes.h.

4.49.2.471 PRINTCONTENT

```
#define PRINTCONTENT 3
```

Definition at line 717 of file ftypes.h.

4.49.2.472 PRINTLFILE

```
#define PRINTLFILE 4
```

Definition at line 718 of file ftypes.h.

4.49.2.473 PRINTONETERM

```
#define PRINTONETERM 8
```

Definition at line 719 of file ftypes.h.

4.49.2.474 PRINTONEFUNCTION

```
#define PRINTONEFUNCTION 16
```

Definition at line 720 of file ftypes.h.

4.49.2.475 PRINTALL

```
#define PRINTALL 32
```

Definition at line 721 of file ftypes.h.

4.49.2.476 REGULAREXPRESSION

```
#define REGULAREXPRESSION -1
```

Definition at line 727 of file ftypes.h.

4.49.2.477 REDEFINEDEXPRESSION

```
#define REDEFINEDEXPRESSION -2
```

Definition at line 728 of file ftypes.h.

4.49.2.478 NEWLYDEFINEDEXPRESSION

```
#define NEWLYDEFINEDEXPRESSION -3
```

Definition at line 729 of file ftypes.h.

4.49.2.479 VERTEXFUNCTION

```
#define VERTEXFUNCTION -1
```

Definition at line 738 of file ftypes.h.

4.49.2.480 GENERALFUNCTION

```
#define GENERALFUNCTION 0
```

Definition at line 739 of file ftypes.h.

4.49.2.481 TENSORFUNCTION

```
#define TENSORFUNCTION 2
```

Definition at line 740 of file ftypes.h.

4.49.2.482 GAMMAFUNCTION

```
#define GAMMAFUNCTION 4
```

Definition at line 741 of file ftypes.h.

4.49.2.483 POS_

```
#define POS_ 0 /* integer > 0 */
```

Definition at line 748 of file ftypes.h.

4.49.2.484 POS0_

```
#define POS0_ 1 /* integer >= 0 */
```

Definition at line 749 of file ftypes.h.

4.49.2.485 NEG_

```
#define NEG_ 2 /* integer < 0 */
```

Definition at line 750 of file ftypes.h.

4.49.2.486 NEG0_

```
#define NEG0_ 3 /* integer <= 0 */
```

Definition at line 751 of file ftypes.h.

4.49.2.487 EVEN_

```
#define EVEN_ 4 /* integer (even) */
```

Definition at line 752 of file ftypes.h.

4.49.2.488 ODD_

```
#define ODD_ 5 /* integer (odd) */
```

Definition at line 753 of file ftypes.h.

4.49.2.489 Z_

```
#define Z_ 6 /* integer */
```

Definition at line 754 of file ftypes.h.

4.49.2.490 SYMBOL_

```
#define SYMBOL_ 7 /* symbol only */
```

Definition at line 755 of file ftypes.h.

4.49.2.491 FIXED_

```
#define FIXED_ 8 /* fixed index */
```

Definition at line 756 of file ftypes.h.

4.49.2.492 INDEX_

```
#define INDEX_ 9 /* index only */
```

Definition at line 757 of file ftypes.h.

4.49.2.493 Q_

```
#define Q_ 10 /* rational */
```

Definition at line 758 of file ftypes.h.

4.49.2.494 DUMMYINDEX_

```
#define DUMMYINDEX_ 11 /* dummy index only */
```

Definition at line 759 of file ftypes.h.

4.49.2.495 VECTOR_

```
#define VECTOR_ 12 /* vector only */
```

Definition at line 760 of file ftypes.h.

4.49.2.496 GAMMA1

```
#define GAMMA1 0
```

Definition at line 766 of file ftypes.h.

4.49.2.497 GAMMA5

```
#define GAMMA5 -1
```

Definition at line 767 of file ftypes.h.

4.49.2.498 GAMMA6

```
#define GAMMA6 -2
```

Definition at line 768 of file ftypes.h.

4.49.2.499 GAMMA7

```
#define GAMMA7 -3
```

Definition at line 769 of file ftypes.h.

4.49.2.500 FUNNYVEC

```
#define FUNNYVEC -4
```

Definition at line 770 of file ftypes.h.

4.49.2.501 FUNNYWILD

```
#define FUNNYWILD -5
```

Definition at line 771 of file ftypes.h.

4.49.2.502 SUMMEDIND

```
#define SUMMEDIND -6
```

Definition at line 772 of file ftypes.h.

4.49.2.503 NOINDEX

```
#define NOINDEX -7
```

Definition at line 773 of file ftypes.h.

4.49.2.504 FUNNYDOLLAR

```
#define FUNNYDOLLAR -8
```

Definition at line 774 of file ftypes.h.

4.49.2.505 EMPTYINDEX

```
#define EMPTYINDEX -9
```

Definition at line 775 of file ftypes.h.

4.49.2.506 MINSPEC

```
#define MINSPEC -10
```

Definition at line 781 of file ftypes.h.

4.49.2.507 USEDFLAG

```
#define USEDFLAG 2
```

Definition at line 783 of file ftypes.h.

4.49.2.508 DUMMYFLAG

```
#define DUMMYFLAG 1
```

Definition at line 784 of file ftypes.h.

4.49.2.509 MAINSORT

```
#define MAINSORT 0
```

Definition at line 786 of file ftypes.h.

4.49.2.510 FUNCTIONSORT

```
#define FUNCTIONSORT 1
```

Definition at line 787 of file ftypes.h.

4.49.2.511 SUBSORT

```
#define SUBSORT 2
```

Definition at line 788 of file ftypes.h.

4.49.2.512 FLOATMODE

```
#define FLOATMODE 1
```

Definition at line 790 of file ftypes.h.

4.49.2.513 RATIONALMODE

```
#define RATIONALMODE 0
```

Definition at line 791 of file ftypes.h.

4.49.2.514 NUMSPECSETS

```
#define NUMSPECSETS 10
```

Definition at line 793 of file ftypes.h.

4.49.2.515 ISZERO

```
#define ISZERO 1
```

Definition at line 795 of file ftypes.h.

4.49.2.516 ISUNMODIFIED

```
#define ISUNMODIFIED 2
```

Definition at line 796 of file ftypes.h.

4.49.2.517 ISCOMPRESSED

```
#define ISCOMPRESSED 4
```

Definition at line 797 of file ftypes.h.

4.49.2.518 ISINRHS

```
#define ISINRHS 8
```

Definition at line 798 of file ftypes.h.

4.49.2.519 ISFACTORIZED

```
#define ISFACTORIZED 16
```

Definition at line 799 of file ftypes.h.

4.49.2.520 TOBEFACTORED

```
#define TOBEFACTORED 32
```

Definition at line 800 of file ftypes.h.

4.49.2.521 TOBEUNFACTORED

```
#define TOBEUNFACTORED 64
```

Definition at line 801 of file ftypes.h.

4.49.2.522 KEEPZERO

```
#define KEEPZERO 128
```

Definition at line 802 of file ftypes.h.

4.49.2.523 VARTYPENONE

```
#define VARTYPENONE 0
```

Definition at line 804 of file ftypes.h.

4.49.2.524 VARTYPECOMPLEX

```
#define VARTYPECOMPLEX 1
```

Definition at line 805 of file ftypes.h.

4.49.2.525 VARTYPEIMAGINARY

```
#define VARTYPEIMAGINARY 2
```

Definition at line 806 of file ftypes.h.

4.49.2.526 VARTYPEROOTOFUNITY

```
#define VARTYPEROOTOFUNITY 4
```

Definition at line 807 of file ftypes.h.

4.49.2.527 VARTYPEMINUS

```
#define VARTYPEMINUS 8
```

Definition at line 808 of file ftypes.h.

4.49.2.528 CYCLESYMMETRIC

```
#define CYCLESYMMETRIC 1
```

Definition at line 809 of file ftypes.h.

4.49.2.529 RCYCLESYMMETRIC

```
#define RCYCLESYMMETRIC 2
```

Definition at line 810 of file ftypes.h.

4.49.2.530 SYMMETRIC

```
#define SYMMETRIC 3
```

Definition at line 811 of file ftypes.h.

4.49.2.531 ANTISYMMETRIC

```
#define ANTISYMMETRIC 4
```

Definition at line 812 of file ftypes.h.

4.49.2.532 REVERSEORDER

```
#define REVERSEORDER 256
```

Definition at line 813 of file ftypes.h.

4.49.2.533 SUBMULTI

```
#define SUBMULTI 1
```

Definition at line 819 of file ftypes.h.

4.49.2.534 SUBONCE

```
#define SUBONCE 2
```

Definition at line 820 of file ftypes.h.

4.49.2.535 SUBONLY

```
#define SUBONLY 3
```

Definition at line 821 of file ftypes.h.

4.49.2.536 SUBMANY

```
#define SUBMANY 4
```

Definition at line 822 of file ftypes.h.

4.49.2.537 SUBVECTOR

```
#define SUBVECTOR 5
```

Definition at line 823 of file ftypes.h.

4.49.2.538 SUBSELECT

```
#define SUBSELECT 6
```

Definition at line 824 of file ftypes.h.

4.49.2.539 SUBALL

```
#define SUBALL 7
```

Definition at line 825 of file ftypes.h.

4.49.2.540 SUBMASK

```
#define SUBMASK 15
```

Definition at line 826 of file ftypes.h.

4.49.2.541 SUBDISORDER

```
#define SUBDISORDER 16
```

Definition at line 827 of file ftypes.h.

4.49.2.542 SUBAFTER

```
#define SUBAFTER 32
```

Definition at line 828 of file ftypes.h.

4.49.2.543 SUBAFTERNOT

```
#define SUBAFTERNOT 64
```

Definition at line 829 of file ftypes.h.

4.49.2.544 IDHEAD

```
#define IDHEAD 6
```

Definition at line 831 of file ftypes.h.

4.49.2.545 DOLLARFLAG

```
#define DOLLARFLAG 1
```

Definition at line 833 of file ftypes.h.

4.49.2.546 NORMALIZEFLAG

```
#define NORMALIZEFLAG 2
```

Definition at line 834 of file ftypes.h.

4.49.2.547 GIDENT

```
#define GIDENT 1
```

Definition at line 836 of file ftypes.h.

4.49.2.548 GFIVE

```
#define GFIVE 4
```

Definition at line 837 of file ftypes.h.

4.49.2.549 GPLUS

```
#define GPLUS 3
```

Definition at line 838 of file ftypes.h.

4.49.2.550 GMINUS

```
#define GMINUS 2
```

Definition at line 839 of file ftypes.h.

4.49.2.551 LONGNUMBER

```
#define LONGNUMBER 1
```

Definition at line 845 of file ftypes.h.

4.49.2.552 MATCH

```
#define MATCH 2
```

Definition at line 846 of file ftypes.h.

4.49.2.553 COEFFI

```
#define COEFFI 3
```

Definition at line 847 of file ftypes.h.

4.49.2.554 SUBEXPR

```
#define SUBEXPR 4
```

Definition at line 848 of file ftypes.h.

4.49.2.555 MULTIPLEOF

```
#define MULTIPLEOF 5
```

Definition at line 849 of file ftypes.h.

4.49.2.556 IFDOLLAR

```
#define IFDOLLAR 6
```

Definition at line 850 of file ftypes.h.

4.49.2.557 IFEXPRESSION

```
#define IFEXPRESSION 7
```

Definition at line 851 of file ftypes.h.

4.49.2.558 IFDOLLAREXTRA

```
#define IFDOLLAREXTRA 8
```

Definition at line 852 of file ftypes.h.

4.49.2.559 IFISFACTORIZED

```
#define IFISFACTORIZED 9
```

Definition at line 853 of file ftypes.h.

4.49.2.560 IFOCCURS

```
#define IFOCCURS 10
```

Definition at line 854 of file ftypes.h.

4.49.2.561 IFUSERFLAG

```
#define IFUSERFLAG 11
```

Definition at line 855 of file ftypes.h.

4.49.2.562 IFFLOATNUMBER

```
#define IFFLOATNUMBER 12
```

Definition at line 856 of file ftypes.h.

4.49.2.563 GREATER

```
#define GREATER 0
```

Definition at line 857 of file ftypes.h.

4.49.2.564 GREATEREQUAL

```
#define GREATEREQUAL 1
```

Definition at line 858 of file ftypes.h.

4.49.2.565 LESS

```
#define LESS 2
```

Definition at line 859 of file ftypes.h.

4.49.2.566 LESSEQUAL

```
#define LESSEQUAL 3
```

Definition at line 860 of file ftypes.h.

4.49.2.567 EQUAL

```
#define EQUAL 4
```

Definition at line 861 of file ftypes.h.

4.49.2.568 NOTEQUAL

```
#define NOTEQUAL 5
```

Definition at line 862 of file ftypes.h.

4.49.2.569 ORCOND

```
#define ORCOND 6
```

Definition at line 863 of file ftypes.h.

4.49.2.570 ANDCOND

```
#define ANDCOND 7
```

Definition at line 864 of file ftypes.h.

4.49.2.571 DUMMY

```
#define DUMMY 1
```

Definition at line 865 of file ftypes.h.

4.49.2.572 SORT

```
#define SORT 1
```

Definition at line 866 of file ftypes.h.

4.49.2.573 STORE

```
#define STORE 2
```

Definition at line 867 of file ftypes.h.

4.49.2.574 END

```
#define END 3
```

Definition at line 868 of file ftypes.h.

4.49.2.575 GLOBAL

```
#define GLOBAL 4
```

Definition at line 869 of file ftypes.h.

4.49.2.576 CLEAR

```
#define CLEAR 5
```

Definition at line 870 of file ftypes.h.

4.49.2.577 VECTBIT

```
#define VECTBIT 1
```

Definition at line 872 of file ftypes.h.

4.49.2.578 DOTPBIT

```
#define DOTPBIT 2
```

Definition at line 873 of file ftypes.h.

4.49.2.579 FUNBIT

```
#define FUNBIT 4
```

Definition at line 874 of file ftypes.h.

4.49.2.580 SETBIT

```
#define SETBIT 8
```

Definition at line 875 of file ftypes.h.

4.49.2.581 EXTRAPARAMETER

```
#define EXTRAPARAMETER 0x4000
```

Definition at line 877 of file ftypes.h.

4.49.2.582 GENCOMMUTE

```
#define GENCOMMUTE 0
```

Definition at line 878 of file ftypes.h.

4.49.2.583 GENNONCOMMUTE

```
#define GENNONCOMMUTE 0x2000
```

Definition at line 879 of file ftypes.h.

4.49.2.584 NAMENOTFOUND

```
#define NAMENOTFOUND -9
```

Definition at line 881 of file ftypes.h.

4.49.2.585 DOLUNDEFINED

```
#define DOLUNDEFINED 0
```

Definition at line 887 of file ftypes.h.

4.49.2.586 DOLNUMBER

```
#define DOLNUMBER 1
```

Definition at line 888 of file ftypes.h.

4.49.2.587 DOLARGUMENT

```
#define DOLARGUMENT 2
```

Definition at line 889 of file ftypes.h.

4.49.2.588 DOLSUBTERM

```
#define DOLSUBTERM 3
```

Definition at line 890 of file ftypes.h.

4.49.2.589 DOLTERMS

```
#define DOLTERMS 4
```

Definition at line 891 of file ftypes.h.

4.49.2.590 DOLWILDARGS

```
#define DOLWILDARGS 5
```

Definition at line 892 of file ftypes.h.

4.49.2.591 DOLINDEX

```
#define DOLINDEX 6
```

Definition at line 893 of file ftypes.h.

4.49.2.592 DOLZERO

```
#define DOLZERO 7
```

Definition at line 894 of file ftypes.h.

4.49.2.593 FINDLOOP

```
#define FINDLOOP 0
```

Definition at line 896 of file ftypes.h.

4.49.2.594 REPLACELOOP

```
#define REPLACELOOP 1
```

Definition at line 897 of file ftypes.h.

4.49.2.595 NOFUNPOWERS

```
#define NOFUNPOWERS 0
```

Definition at line 899 of file ftypes.h.

4.49.2.596 COMFUNPOWERS

```
#define COMFUNPOWERS 1
```

Definition at line 900 of file ftypes.h.

4.49.2.597 ALLFUNPOWERS

```
#define ALLFUNPOWERS 2
```

Definition at line 901 of file ftypes.h.

4.49.2.598 PROPERORDERFLAG

```
#define PROPERORDERFLAG 0
```

Definition at line 903 of file ftypes.h.

4.49.2.599 REGULAR

```
#define REGULAR 0
```

Definition at line 905 of file ftypes.h.

4.49.2.600 FINISH

```
#define FINISH 1
```

Definition at line 906 of file ftypes.h.

4.49.2.601 POLYADD

```
#define POLYADD 1
```

Definition at line 908 of file ftypes.h.

4.49.2.602 POLYSUB

```
#define POLYSUB 2
```

Definition at line 909 of file ftypes.h.

4.49.2.603 POLYMUL

```
#define POLYMUL 3
```

Definition at line 910 of file ftypes.h.

4.49.2.604 POLYDIV

```
#define POLYDIV 4
```

Definition at line 911 of file ftypes.h.

4.49.2.605 POLYREM

```
#define POLYREM 5
```

Definition at line 912 of file ftypes.h.

4.49.2.606 POLYGCD

```
#define POLYGCD 6
```

Definition at line 913 of file ftypes.h.

4.49.2.607 POLYINTFAC

```
#define POLYINTFAC 7
```

Definition at line 914 of file ftypes.h.

4.49.2.608 POLYNORM

```
#define POLYNORM 8
```

Definition at line 915 of file ftypes.h.

4.49.2.609 MODNONE

```
#define MODNONE 0
```

Definition at line 917 of file ftypes.h.

4.49.2.610 MODSUM

```
#define MODSUM 1
```

Definition at line 918 of file ftypes.h.

4.49.2.611 MODMAX

```
#define MODMAX 2
```

Definition at line 919 of file ftypes.h.

4.49.2.612 MODMIN

```
#define MODMIN 3
```

Definition at line 920 of file ftypes.h.

4.49.2.613 MODLOCAL

```
#define MODLOCAL 4
```

Definition at line 921 of file ftypes.h.

4.49.2.614 ELEMENTUSED

```
#define ELEMENTUSED 1
```

Definition at line 923 of file ftypes.h.

4.49.2.615 ELEMENTLOADED

```
#define ELEMENTLOADED 2
```

Definition at line 924 of file ftypes.h.

4.49.2.616 POSNEG

```
#define POSNEG 0x1
```

Definition at line 929 of file ftypes.h.

4.49.2.617 INVERSETABLE

```
#define INVERSETABLE 0x2
```

Definition at line 930 of file ftypes.h.

4.49.2.618 COEFFICIENTSONLY

```
#define COEFFICIENTSONLY 0x4
```

Definition at line 931 of file ftypes.h.

4.49.2.619 ALSOPOWERS

```
#define ALSOPOWERS 0x8
```

Definition at line 932 of file ftypes.h.

4.49.2.620 ALSOFUNARGS

```
#define ALSOFUNARGS 0x10
```

Definition at line 933 of file ftypes.h.

4.49.2.621 ALSODOLLARS

```
#define ALSODOLLARS 0x20
```

Definition at line 934 of file ftypes.h.

4.49.2.622 NOINVERSES

```
#define NOINVERSES 0x40
```

Definition at line 935 of file ftypes.h.

4.49.2.623 POSITIVEONLY

```
#define POSITIVEONLY 0
```

Definition at line 937 of file ftypes.h.

4.49.2.624 UNPACK

```
#define UNPACK 0x80
```

Definition at line 938 of file ftypes.h.

4.49.2.625 NOUNPACK

```
#define NOUNPACK 0
```

Definition at line 939 of file ftypes.h.

4.49.2.626 FROMFUNCTION

```
#define FROMFUNCTION 0x100
```

Definition at line 940 of file ftypes.h.

4.49.2.627 VARNAMES

```
#define VARNAMES 0
```

Definition at line 942 of file ftypes.h.

4.49.2.628 AUTONAMES

```
#define AUTONAMES 1
```

Definition at line 943 of file ftypes.h.

4.49.2.629 EXPRNAMES

```
#define EXPRNAMES 2
```

Definition at line 944 of file ftypes.h.

4.49.2.630 DOLLARNAMES

```
#define DOLLARNAMES 3
```

Definition at line 945 of file ftypes.h.

4.49.2.631 DUMMYBUFFER

```
#define DUMMYBUFFER 1
```

Definition at line 1000 of file ftypes.h.

4.49.2.632 ALLARGS

```
#define ALLARGS 1
```

Definition at line 1002 of file ftypes.h.

4.49.2.633 NUMARG

```
#define NUMARG 2
```

Definition at line 1003 of file ftypes.h.

4.49.2.634 ARGRANGE

```
#define ARGRANGE 3
```

Definition at line 1004 of file ftypes.h.

4.49.2.635 MAKEARGS

```
#define MAKEARGS 4
```

Definition at line 1005 of file ftypes.h.

4.49.2.636 MAXRANGEINDICATOR

```
#define MAXRANGEINDICATOR 4
```

Definition at line 1006 of file ftypes.h.

4.49.2.637 REPLACEARG

```
#define REPLACEARG 5
```

Definition at line 1007 of file ftypes.h.

4.49.2.638 ENCODEARG

```
#define ENCODEARG 6
```

Definition at line 1008 of file ftypes.h.

4.49.2.639 DECODEARG

```
#define DECODEARG 7
```

Definition at line 1009 of file ftypes.h.

4.49.2.640 IMPLODEARG

```
#define IMPLODEARG 8
```

Definition at line 1010 of file ftypes.h.

4.49.2.641 EXPLODEARG

```
#define EXPLODEARG 9
```

Definition at line 1011 of file ftypes.h.

4.49.2.642 PERMUTEARG

```
#define PERMUTEARG 10
```

Definition at line 1012 of file ftypes.h.

4.49.2.643 REVERSEARG

```
#define REVERSEARG 11
```

Definition at line 1013 of file ftypes.h.

4.49.2.644 CYCLEARG

```
#define CYCLEARG 12
```

Definition at line 1014 of file ftypes.h.

4.49.2.645 ISLYNDON

```
#define ISLYNDON 13
```

Definition at line 1015 of file ftypes.h.

4.49.2.646 ISLYNDONR

```
#define ISLYNDONR 14
```

Definition at line 1016 of file ftypes.h.

4.49.2.647 TOLYNDON

```
#define TOLYNDON 15
```

Definition at line 1017 of file ftypes.h.

4.49.2.648 TOLYNDONR

```
#define TOLYNDONR 16
```

Definition at line 1018 of file ftypes.h.

4.49.2.649 ADDARG

```
#define ADDARG 17
```

Definition at line 1019 of file ftypes.h.

4.49.2.650 MULTIPLYARG

```
#define MULTIPLYARG 18
```

Definition at line 1020 of file ftypes.h.

4.49.2.651 DROPARG

```
#define DROPARG 19
```

Definition at line 1021 of file ftypes.h.

4.49.2.652 SELECTARG

```
#define SELECTARG 20
```

Definition at line 1022 of file ftypes.h.

4.49.2.653 DEDUPARG

```
#define DEDUPARG 21
```

Definition at line 1023 of file ftypes.h.

4.49.2.654 ZTOHARG

```
#define ZTOHARG 22
```

Definition at line 1024 of file ftypes.h.

4.49.2.655 HTOZARG

```
#define HTOZARG 23
```

Definition at line 1025 of file ftypes.h.

4.49.2.656 BASECODE

```
#define BASECODE 1
```

Definition at line 1033 of file ftypes.h.

4.49.2.657 YESLYNDON

```
#define YESLYNDON 1
```

Definition at line 1034 of file ftypes.h.

4.49.2.658 NOLYNDON

```
#define NOLYNDON 2
```

Definition at line 1035 of file ftypes.h.

4.49.2.659 TOPOLYNOMIALFLAG

```
#define TOPOLYNOMIALFLAG 1
```

Definition at line 1037 of file ftypes.h.

4.49.2.660 FACTARGFLAG

```
#define FACTARGFLAG 2
```

Definition at line 1038 of file ftypes.h.

4.49.2.661 OLDFACTARG

```
#define OLDFACTARG 1
```

Definition at line 1040 of file ftypes.h.

4.49.2.662 NEWFACTARG

```
#define NEWFACTARG 0
```

Definition at line 1041 of file ftypes.h.

4.49.2.663 FROMMODULEOPTION

```
#define FROMMODULEOPTION 0
```

Definition at line 1043 of file ftypes.h.

4.49.2.664 FROMPOINTINSTRUCTION

```
#define FROMPOINTINSTRUCTION 1
```

Definition at line 1044 of file ftypes.h.

4.49.2.665 EXTRASYMBOL

```
#define EXTRASYMBOL 0
```

Definition at line 1046 of file ftypes.h.

4.49.2.666 REGULARSYMBOL

```
#define REGULARSYMBOL 1
```

Definition at line 1047 of file ftypes.h.

4.49.2.667 EXPRESSIONNUMBER

```
#define EXPRESSIONNUMBER 2
```

Definition at line 1048 of file ftypes.h.

4.49.2.668 O_NONE

```
#define O_NONE 0
```

Definition at line 1050 of file ftypes.h.

4.49.2.669 O_CSE

```
#define O_CSE 1
```

Definition at line 1051 of file ftypes.h.

4.49.2.670 O_CSEGREEDY

```
#define O_CSEGREEDY 2
```

Definition at line 1052 of file ftypes.h.

4.49.2.671 O_GREEDY

```
#define O_GREEDY 3
```

Definition at line 1053 of file ftypes.h.

4.49.2.672 O_OCCURRENCE

```
#define O_OCCURRENCE 0
```

Definition at line 1055 of file ftypes.h.

4.49.2.673 O_MCTS

```
#define O_MCTS 1
```

Definition at line 1056 of file ftypes.h.

4.49.2.674 O_SIMULATED_ANNEALING

```
#define O_SIMULATED_ANNEALING 2
```

Definition at line 1057 of file ftypes.h.

4.49.2.675 O_FORWARD

```
#define O_FORWARD 0
```

Definition at line 1059 of file ftypes.h.

4.49.2.676 O_BACKWARD

```
#define O_BACKWARD 1
```

Definition at line 1060 of file ftypes.h.

4.49.2.677 O_FORWARDORBACKWARD

```
#define O_FORWARDORBACKWARD 2
```

Definition at line 1061 of file ftypes.h.

4.49.2.678 O_FORWARDANDBACKWARD

```
#define O_FORWARDANDBACKWARD 3
```

Definition at line 1062 of file ftypes.h.

4.49.2.679 OPTHEAD

```
#define OPTHEAD 3
```

Definition at line 1064 of file ftypes.h.

4.49.2.680 DOALL

```
#define DOALL 1
```

Definition at line 1065 of file ftypes.h.

4.49.2.681 ONLYFUNCTIONS

```
#define ONLYFUNCTIONS 2
```

Definition at line 1066 of file ftypes.h.

4.49.2.682 INUSE

```
#define INUSE 1
```

Definition at line 1068 of file ftypes.h.

4.49.2.683 COULDCOMMUTE

```
#define COULDCOMMUTE 2
```

Definition at line 1069 of file ftypes.h.

4.49.2.684 DOESNOTCOMMUTE

```
#define DOESNOTCOMMUTE 4
```

Definition at line 1070 of file ftypes.h.

4.49.2.685 DICT_NONUMBERS

```
#define DICT_NONUMBERS 0
```

Definition at line 1072 of file ftypes.h.

4.49.2.686 DICT_INTEGERONLY

```
#define DICT_INTEGERONLY 1
```

Definition at line 1073 of file ftypes.h.

4.49.2.687 DICT_RATIONALONLY

```
#define DICT_RATIONALONLY 2
```

Definition at line 1074 of file ftypes.h.

4.49.2.688 DICT_ALLNUMBERS

```
#define DICT_ALLNUMBERS 3
```

Definition at line 1075 of file ftypes.h.

4.49.2.689 DICT_NOVARIABLES

```
#define DICT_NOVARIABLES 0
```

Definition at line 1076 of file ftypes.h.

4.49.2.690 DICT_DOVARIABLES

```
#define DICT_DOVARIABLES 1
```

Definition at line 1077 of file ftypes.h.

4.49.2.691 DICT_NOSPECIALS

```
#define DICT_NOSPECIALS 0
```

Definition at line 1078 of file ftypes.h.

4.49.2.692 DICT_DOSPECIALS

```
#define DICT_DOSPECIALS 1
```

Definition at line 1079 of file ftypes.h.

4.49.2.693 DICT_NOFUNWITHARGS

```
#define DICT_NOFUNWITHARGS 0
```

Definition at line 1080 of file ftypes.h.

4.49.2.694 DICT_DOFUNWITHARGS

```
#define DICT_DOFUNWITHARGS 1
```

Definition at line 1081 of file ftypes.h.

4.49.2.695 DICT_NOTINDOLLARS

```
#define DICT_NOTINDOLLARS 0
```

Definition at line 1082 of file ftypes.h.

4.49.2.696 DICT_INDOLLARS

```
#define DICT_INDOLLARS 1
```

Definition at line 1083 of file ftypes.h.

4.49.2.697 DICT_INTEGERNUMBER

```
#define DICT_INTEGERNUMBER 1
```

Definition at line 1085 of file ftypes.h.

4.49.2.698 DICT_RATIONALNUMBER

```
#define DICT_RATIONALNUMBER 2
```

Definition at line 1086 of file ftypes.h.

4.49.2.699 DICT_SYMBOL

```
#define DICT_SYMBOL 3
```

Definition at line 1087 of file ftypes.h.

4.49.2.700 DICT_VECTOR

```
#define DICT_VECTOR 4
```

Definition at line 1088 of file ftypes.h.

4.49.2.701 DICT_INDEX

```
#define DICT_INDEX 5
```

Definition at line 1089 of file ftypes.h.

4.49.2.702 DICT_FUNCTION

```
#define DICT_FUNCTION 6
```

Definition at line 1090 of file ftypes.h.

4.49.2.703 DICT_FUNCTION_WITH_ARGUMENTS

```
#define DICT_FUNCTION_WITH_ARGUMENTS 7
```

Definition at line 1091 of file ftypes.h.

4.49.2.704 DICT_SPECIALCHARACTER

```
#define DICT_SPECIALCHARACTER 8
```

Definition at line 1092 of file ftypes.h.

4.49.2.705 DICT_RANGE

```
#define DICT_RANGE 9
```

Definition at line 1093 of file ftypes.h.

4.49.2.706 READSPECTATORFLAG

```
#define READSPECTATORFLAG 3
```

Definition at line 1095 of file ftypes.h.

4.49.2.707 GLOBALSPECTATORFLAG

```
#define GLOBALSPECTATORFLAG 1
```

Definition at line 1096 of file ftypes.h.

4.49.2.708 ORDEREDSET

```
#define ORDEREDSET 1
```

Definition at line 1098 of file ftypes.h.

4.49.2.709 DENSETABLE

```
#define DENSETABLE 1
```

Definition at line 1100 of file ftypes.h.

4.49.2.710 SPARSETABLE

```
#define SPARSETABLE 0
```

Definition at line 1101 of file ftypes.h.

4.49.2.711 ONEPARTICLEIRREDUCIBLE

```
#define ONEPARTICLEIRREDUCIBLE 1
```

Definition at line 1103 of file ftypes.h.

4.49.2.712 WITHINSERTIONS

```
#define WITHINSERTIONS 2
```

Definition at line 1104 of file ftypes.h.

4.49.2.713 NOTADPOLES

```
#define NOTADPOLES 4
```

Definition at line 1105 of file ftypes.h.

4.49.2.714 WITHSYMMETRIZE

```
#define WITHSYMMETRIZE 8
```

Definition at line 1106 of file ftypes.h.

4.49.2.715 TOPOLOGIESONLY

```
#define TOPOLOGIESONLY 16
```

Definition at line 1107 of file ftypes.h.

4.49.2.716 NONODES

```
#define NONODES 32
```

Definition at line 1108 of file ftypes.h.

4.49.2.717 WITHEDGES

```
#define WITHEDGES 64
```

Definition at line 1109 of file ftypes.h.

4.49.2.718 CHECKEXTERN

```
#define CHECKEXTERN 128
```

Definition at line 1110 of file ftypes.h.

4.49.2.719 WITHBLOCKS

```
#define WITHBLOCKS 256
```

Definition at line 1111 of file ftypes.h.

4.49.2.720 WITHONEPI

```
#define WITHONEPI 512
```

Definition at line 1112 of file ftypes.h.

4.49.2.721 NOSNAILS

```
#define NOSNAILS 1024
```

Definition at line 1113 of file ftypes.h.

4.49.2.722 NOEXTSELF

```
#define NOEXTSELF 2048
```

Definition at line 1114 of file ftypes.h.

4.49.3 Typedef Documentation

4.49.3.1 PVFUNWP

```
typedef void(* PVFUNWP) ()
```

Definition at line 183 of file ftypes.h.

4.49.3.2 PVFUNV

```
typedef void(* PVFUNV) ()
```

Definition at line 184 of file ftypes.h.

4.49.3.3 CFUN

```
typedef int(* CFUN) ()
```

Definition at line 185 of file ftypes.h.

4.49.3.4 TFUN

```
typedef int(* TFUN) ()
```

Definition at line 186 of file ftypes.h.

4.49.3.5 TFUN1

```
typedef int(* TFUN1) ()
```

Definition at line 187 of file ftypes.h.

4.50 ftypes.h

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#define WITHOUTERROR 0
#define WITHERROR 1

/*
    The various streams. (look also in tools.c)
*/

#define FILESTREAM 0
#define PREVARSTREAM 1
#define PREREADSTREAM 2
#define PIPESTREAM 3
#define PRECALCSTREAM 4
#define DOLLARSTREAM 5
#define PREREADSTREAM2 6
#define EXTERNALCHANNELSTREAM 7
#define PREREADSTREAM3 8
#define REVERSEFILESTREAM 9
#define INPUTSTREAM 10

#define ENDOFSTREAM 0xFF
#define ENDOFINPUT 0xFF

/*
    Types of files
*/

#define SUBROUTINEFILE 0
#define PROCEDUREFILE 1
#define HEADERFILE 2
#define SETUPFILE 3
#define TABLEBASEFILE 4

/*
    Types of modules
*/

#define FIRSTMODULE -1
#define GLOBALMODULE 0
#define SORTMODULE 1
#define STOREMODULE 2
#define CLEARMODULE 3
#define ENDMODULE 4

#define POLYFUN 0

#define NOPARALLEL_DOLLAR        0x0001
#define NOPARALLEL_RHS           0x0002
#define NOPARALLEL_CONVPOLY      0x0004
#define NOPARALLEL_SPECTATOR     0x0008
#define NOPARALLEL_USER          0x0010
#define NOPARALLEL_TBLDOLLAR     0x0100
#define NOPARALLEL_NPROC         0x0200
#define PARALLELFLAG             0x0000

```

```
#define PRENOACTION 0
#define PRERAISEAFTER 1
#define PRELOWERAFTER 2
/*
#define ELIUMOD 1
#define ELIZMOD 2
#define SKIUMOD 3
#define SKIZMOD 4
*/
#define WITHSEMICOLON 0
#define WITHOUTSEMICOLON 1
#define MODULEINSTACK 8
#define EXECUTINGIF 0
#define LOOKINGFORELSE 1
#define LOOKINGFORENDIF 2
#define NEWSTATEMENT 1
#define OLDSTATEMENT 0

#define EXECUTINGPRESWITCH 0
#define SEARCHINGPRECASE 1
#define SEARCHINGPREENDSWITCH 2

#define PREPROONLY 1
#define DUMPTOCOMPILER 2
#define DUMPOUTTERMS 4
#define DUMPINTERMS 8
#define DUMPTOSORT 16
#define DUMPTOPARALLEL 32
#define THREADSDEBUG 64

#define ERROROUT 0
#define INPUTOUT 1
#define STATSOUT 2
#define EXPRSOUT 3
#define WRITEOUT 4

#define EXTERNALCHANNELOUT 5

#define NUMERICALLOOP 0
#define LISTEDLOOP 1
#define ONEEXPRESSION 2

#define PRETYPENONE 0
#define PRETYPEIF 1
#define PRETYPEDO 2
#define PRETYPEPROCEDURE 3
#define PRETYPESWITCH 4
#define PRETYPEINSIDE 5

/*
    Type of statement. Used to make sure that the statements are in proper order
*/

#define DECLARATION   1
#define SPECIFICATION 2
#define DEFINITION    3
#define STATEMENT     4
#define TOOUTPUT      5
#define ATENDOFMODULE 6
#define MIXED2        8
#define MIXED         9

/*
    The typedefs are to allow the compilers to do better error checking.
*/

#ifdef ANSI
typedef void (*PVFUNWP)(WORD *);
#ifdef INTELCOMPILER
typedef void (*PVFUNV)();
typedef int (*CFUN)();
#else
typedef void (*PVFUNV)(void);
typedef int (*CFUN)(void);
#endif
typedef int (*TFUN)(UBYTE *);
typedef int (*TFUN1)(UBYTE *,int);
#else
typedef void (*PVFUNWP)();
typedef void (*PVFUNV)();
typedef int (*CFUN)();
typedef int (*TFUN)();
typedef int (*TFUN1)();
#endif


#define NOAUTO 0
```

```
#define PARTEST 1
#define WITHAUTO 2
#define ALLVARIABLES -1
#define SYMBOLONLY 1
#define INDEXONLY 2
#define VECTORONLY 4
#define FUNCTIONONLY 8
#define SETONLY 16
#define EXPRESSIONONLY 32
#define CDELETE -1
#define ANYTYPE -1
#define CSYMBOL 0
#define CINDEX 1
#define CVECTOR 2
#define CFUNCTION 3
#define CSET 4
#define CEXPRESSION 5
#define CDOTPRODUCT 6
#define CNUMBER 7
#define CSUBEXP 8
#define CDELTA 9
#define CDOLLAR 10
#define CDUBIOUS 11
#define CRANGE 12
#define CMODEL 13
#define CVECTOR1 21
#define CDOUBLEDOT 22
/*
    Types of tokens in the tokenizer.
*/
#define TSYMBOL -1
#define TINDEX -2
#define TVECTOR -3
#define TFUNCTION -4
#define TSET -5
#define TEXPRESSION -6
#define TDOTPRODUCT -7
#define TNUMBER -8
#define TSUBEXP -9
#define TDELTA -10
#define TDOLLAR -11
#define TDUBIOUS -12
#define LPARENTHESIS -13
#define RPARENTHESIS -14
#define TWILDCARD -15
#define TWILDARG -16
#define TDOT -17
#define LBRACE -18
#define RBRACE -19
#define TCOMMA -20
#define TFUNOPEN -21
#define TFUNCLOSE -22
#define TMULTIPLY -23
#define TDIVIDE -24
#define TPOWER -25
#define TPLUS -26
#define TMINUS -27
#define TNOT -28
#define TENDOFIT -29
#define TSETOPEN -30
#define TSETCLOSE -31
#define TGENINDEX -32
#define TCONJUGATE -33
#define LRPARENTHESES -34
#define TNUMBER1 -35
#define TPOWER1 -36
#define TEMPTY -37
#define TSETNUM -38
#define TSGAMMA -39
#define TSETDOL -40
#define TFLOAT -41
#define TYPEISFUN 0
#define TYPEISSUB 1
#define TYPEISMYSTERY -1
#define LHSIDEX 2
#define LHSIDE 1
#define RHSIDE 0
/*
```

```
    Output modes
*/

#define FORTRANMODE 1
#define REDUCEMODE 2
#define MAPLEMODE 3
#define MATHEMATICAMODE 4
#define CMODE 5
#define VORTRANMODE 6
#define PFORTRANMODE 7
#define DOUBLEFORTRANMODE 33
#define DOUBLEPRECISIONFLAG 32
#define NODOUBLEMASK 31
#define QUADRUPLEFORTRANMODE 65
#define QUADRUPLEPRECISIONFLAG 64
#define ALLINTEGERDOUBLE 128
#define NOQUADMASK 63
#define NORMALFORMAT 0
#define NOSPACEFORMAT 1

#define ISNOTFORTRAN90 0
#define ISFORTRAN90 1

#define ALSOREVERSE 1
#define CHISHOLM 2
#define NOTRICK 16

#define SORTLOWFIRST 0
#define SORTHIGHFIRST 1
#define SORTPOWERFIRST 2
#define SORTANTIPOWER 3

/* These control the output of WriteStats. The different sort
 * types have differently formatted stats. These variables should
 * not have arbitrary values since they are also used as array
 * indices by WriteStats. */
#define STATSSPLITMERGE 0
#define STATSMERGETOFILE 1
#define STATSPOSTSORT 2
/* CHECKLOGTYPE implies that statistics will not be printed in
 * the log file if AM.LogType = 1 (when running with -ll or -F). */
#define NOCHECKLOGTYPE 0
#define CHECKLOGTYPE 1

#define NMIN4SHIFT 4
/*
    The next are the main codes.
    Note: SETSET is not allowed to be 4*n+1
    We use those codes in CoIdExpression for function information
    after the pattern. Because SETSET also stands there we have to
    be careful!!
    Don’t forget MAXBUILTINFUNCTION when adding codes!
    The object FUNCTION is at the start of the functions that are in regular
    notation. Anything below it is in special notation.

    Remark: HAAKJE0 is for compression purposes and should only occur
    at moments that ARGWILD cannot occur.
*/
#define SYMBOL 1
#define DOTPRODUCT 2
#define VECTOR 3
#define INDEX 4
#define EXPRESSION 5
#define SUBEXPRESSION 6
#define DOLLAREXPRESSION 7
#define SETSET 8
#define ARGWILD 9
#define MINVECTOR 10
#define SETEXP 11
#define DOLLAREXPR2 12
#define HAAKJE0 9
#define FUNCTION 20

#define TMPPOLYFUN 14
#define ARGFIELD 15
#define SNUMBER 16
#define LNUMBER 17
#define HAAKJE 18
#define DELTA 19
#define EXPONENT 20
#define DENOMINATOR 21
#define SETFUNCTION 22
#define GAMMA 23
#define GAMMAI 24
#define GAMMAFIVE 25
#define GAMMASIX 26
#define GAMMASEVEN 27
```

```
#define SUMF1 28
#define SUMF2 29
#define DUMFUN 30
#define REPLACEMENT 31
#define REVERSEFUNCTION 32
#define DISTRIBUTION 33
#define DELTA3 34
#define DUMMYFUN 35
#define DUMMYTEN 36
#define LEVICIVITA 37
#define FACTORIAL 38
#define INVERSEFACTORIAL 39
#define BINOMIAL 40
#define NUMARGSFUN 41
#define SIGNFUN 42
#define MODFUNCTION 43
#define MOD2FUNCTION 44
#define MINFUNCTION 45
#define MAXFUNCTION 46
#define ABSFUNCTION 47
#define SIGFUNCTION 48
#define INTFUNCTION 49
#define THETA 50
#define THETA2 51
#define DELTA2 52
#define DELTAP 53
#define BERNOULLIFUNCTION 54
#define COUNTFUNCTION 55
#define MATCHFUNCTION 56
#define PATTERNFUNCTION 57
#define TERMFUNCTION 58
#define CONJUGATION 59
#define ROOTFUNCTION 60
#define TABLEFUNCTION 61
#define FIRSTBRACKET 62
#define TERMSINEXPR 63
#define NUMTERMSFUN 64
#define GCDFUNCTION 65
#define DIVFUNCTION 66
#define REMFUNCTION 67
#define MAXPOWEROF 68
#define MINPOWEROF 69
#define TABLESTUB 70
#define FACTORIN 71
#define TERMSINBRACKET 72
#define WILDARGFUN 73
/*
    In the past we would add new functions here and raise the numbers
    on the special reserved names. This is impractical in the light of
    the .sav files. The .sav files need a mechanism that contains the
    value of MAXBUILTINFUNCTION at the moment of writing. This allows
    form corrections if this value has changed in the mean time.
*/
#define SQRTFUNCTION 74
#define LNFUNCTION 75
#define SINFUNCTION 76
#define COSFUNCTION 77
#define TANFUNCTION 78
#define ASINFUNCTION 79
#define ACOSFUNCTION 80
#define ATANFUNCTION 81
#define ATAN2FUNCTION 82
#define SINHFUNCTION 83
#define COSHFUNCTION 84
#define TANHFUNCTION 85
#define ASINHFUNCTION 86
#define ACOSHFUNCTION 87
#define ATANHFUNCTION 88
#define LI2FUNCTION 89
#define LINFUNCTION 90

#define EXTRASYMFUN 91
#define RANDOMFUNCTION 92
#define RANPERM 93
#define NUMFACTORS 94
#define FIRSTTERM 95
#define CONTENTTERM 96
#define PRIMENUMBER 97
#define EXTEUCLIDEAN 98
#define MAKERATIONAL 99
#define INVERSEFUNCTION 100
#define IDFUNCTION 101
#define PUTFIRST 102
#define PERMUTATIONS 103
#define PARTITIONS 104
#define MULFUNCTION 105
#define MOEBIUS 106
```

```
#define TOPOLOGIES 107
#define DIAGRAMS 108
#define TOPO 109
#define NODEFUNCTION 110
#define EDGE 111
#define SIZEOFFUNCTION 112
#define BLOCK 113
#define ONEPI 114
#define PHI 115
#ifdef WITHFLOAT
#define FLOATFUN 116
#define TOFLOAT 117
#define TORAT 118
#define MZV 119
#define EULER 120
#define MZVHALF 121
#define AGMFUNCTION 122
#define GAMMAFUN 123
#define MAXBUILTINFUNCTION 123
#else
#define MAXBUILTINFUNCTION 115
#endif

#define FIRSTUSERFUNCTION 150

#define ALLFUNCTIONS -1
#define ALLMZVFUNCTIONS -2

/*
    Note: if we add a builtin table we have to look also inside names.c
    in the routine Globalize because there we assume there does not exist
    such an object
*/

#define ISYMBOL 0
#define PISYMBOL 1
#define COEFFSYMBOL 2
#define NUMERATORSYMBOL 3
#define DENOMINATORSYMBOL 4
#define WILDARGSYMBOL 5
#define DIMENSIONSYMBOL 6
#define FACTORSYMBOL 7
#define SEPARATESYMBOL 8
#define EESYMBOL 9
#define EMSYMBOL 10

#define BUILTINSYMBOLS 11
#define FIRSTUSERSYMBOL 20

#define BUILTININDICES 1
#define BUILTINVECTORS 1
#define BUILTINDOLLARS 1

#define WILDARGVECTOR 0
#define WILDARGINDEX 0

/*
    The objects that have a name that starts with TYPE are codes of statements
    made by the compiler. Each statement starts with such a code, followed by
    its size. For how most of these statements are used can be seen in the
    Generator function in the file proces.c
    TYPEOPERATION is an anachronism that remains used only for the statements
    that are executed in the file opera.c (like traces and contractions).
*/

#define TYPEEXPRESSION 0
#define TYPEIDNEW 1
#define TYPEIDOLD 2
#define TYPEOPERATION 3
#define TYPEREPEAT 4
#define TYPEENDREPEAT 5
/*
    The next counts must be higher than the ones before
*/
#define TYPECOUNT 20
#define TYPEMULT 21
#define TYPEGOTO 22
#define TYPEDISCARD 23
#define TYPEIF 24
#define TYPEELSE 25
#define TYPEELIF 26
#define TYPEENDIF 27
#define TYPESUM 28
#define TYPECHISHOLM 29
#define TYPEREVERSE 30
#define TYPEARG 31
#define TYPENORM 32
```

```
#define TYPENORM2 33
#define TYPENORM3 34
#define TYPEEXIT 35
#define TYPESETEXIT 36
#define TYPEPRINT 37
#define TYPEFPRINT 38
#define TYPEREDEFPRE 39
#define TYPESPLITARG 40
#define TYPESPLITARG2 41
#define TYPEFACTARG 42
#define TYPEFACTARG2 43
#define TYPETRY 44
#define TYPEASSIGN 45
#define TYPERENUMBER 46
#define TYPESUMFIX 47
#define TYPEFINDLOOP 48
#define TYPEUNRAVEL 49
#define TYPEADJUSTBOUNDS 50
#define TYPEINSIDE 51
#define TYPETERM 52
#define TYPESORT 53
#define TYPEDETCURDUM 54
#define TYPEINEXPRESSION 55
#define TYPESPLITFIRSTARG 56
#define TYPESPLITLASTARG 57
#define TYPEMERGE 58
#define TYPETESTUSE 59
#define TYPEAPPLY 60
#define TYPEAPPLYRESET 61
#define TYPECHAININ 62
#define TYPECHAINOUT 63
#define TYPENORM4 64
#define TYPEFACTOR 65
#define TYPEARGIMPLODE 66
#define TYPEARGEXPLODE 67
#define TYPEDENOMINATORS 68
#define TYPESTUFFLE 69
#define TYPEDROPCOEFFICIENT 70
#define TYPETRANSFORM 71
#define TYPETOPOLYNOMIAL 72
#define TYPEFROMPOLYNOMIAL 73
#define TYPEDOLOOP 74
#define TYPEENDDOLOOP 75
#define TYPEDROPSYMBOLS 76
#define TYPEPUTINSIDE 77
#define TYPETOSPECTATOR 78
#define TYPEARGTOEXTRASYMBOL 79
#define TYPECANONICALIZE 80
#define TYPESWITCH 81
#define TYPEENDSWITCH 82
#define TYPESETUSERFLAG 83
#define TYPECLEARUSERFLAG 84
#define TYPEALLLOOPS 85
#define TYPEALLPATHS 86
#ifdef WITHFLOAT
#define TYPEEVALUATE 87
#define TYPEEXPAND 88
#define TYPETOFLOAT 89
#define TYPETORAT 90
#endif

/*
    The codes for the 'operations' that are part of TYPEOPERATION.
*/

#define TAKETRACE 1
#define CONTRACT 2
#define RATIO 3
#define SYMMETRIZE 4
#define TENVEC 5
#define SUMNUM1 6
#define SUMNUM2 7

/*
    The various types of wildcards.
*/

#define WILDDUMMY 0
#define SYMTONUM 1
#define SYMTOSYM 2
#define SYMTOSUB 3
#define VECTOMIN 4
#define VECTOVEC 5
#define VECTOSUB 6
#define INDTOIND 7
#define INDTOSUB 8
#define FUNTOFUN 9
```

```
#define ARGTOARG 10
#define ARLTOARL 11
#define EXPTOEXP 12
#define FROMBRAC 13
#define FROMSET 14
#define SETTONUM 15
#define WILDCARDS 16
#define SETNUMBER 17
#define LOADDOLLAR 18
/*
    Some new types of wildcards that hold only for function arguments.
*/
#define NUMTONUM 20
#define NUMTOSYM 21
#define NUMTOIND 22
#define NUMTOSUB 23
/*
    Flag or-ed with ARGTOARG to give the number of arguments.
*/
#define EATTENSOR 0x2000

/*
    Dirty flags (introduced when functions got a field with a dirty flag)
*/

#define CLEANFLAG 0
#define DIRTYFLAG 1
#define DIRTYSYMFLAG 2
#define MUSTCLEANPRF 4
#define SUBTERMUSED1 8
#define SUBTERMUSED2 16
#define ALLDIRTY (DIRTYFLAG|DIRTYSYMFLAG)

#define ARGHEAD 2
#define FUNHEAD 3
#define SUBEXPSIZE 5
#define EXPRHEAD 5
#define TYPEARGHEADSIZE 6

/*
    Actions to be taken with expressions. They are marked with these objects
    during compilation.
*/

#define SKIP 1
#define DROP 2
#define HIDE 3
#define UNHIDE 4
#define INTOHIDE 5

/*
    Types of expressions
*/

#define LOCALEXPRESSION 0
#define SKIPLEXPRESSION 1
#define DROPLEXPRESSION 2
#define DROPPEDEXPRESSION 3
#define GLOBALEXPRESSION 4
#define SKIPGEXPRESSION 5
#define DROPGEXPRESSION 6
#define STOREDEXPRESSION 8
#define HIDDENLEXPRESSION 9
#define HIDDENGEXPRESSION 13
#define INCEXPRESSION 9
#define HIDELEXPRESSION 10
#define HIDEGEXPRESSION 14
#define DROPHLEXPRESSION 11
#define DROPHGEXPRESSION 15
#define UNHIDELEXPRESSION 12
#define UNHIDEGEXPRESSION 16
#define INTOHIDELEXPRESSION 17
#define INTOHIDEGEXPRESSION 18
#define SPECTATOREXPRESSION 19
#define DROPSPECTATOREXPRESSION 20
#define SKIPUNHIDELEXPRESSION 21
#define SKIPUNHIDEGEXPRESSION 22

#define PRINTOFF 0
#define PRINTON 1
#define PRINTCONTENTS 2
#define PRINTCONTENT 3
#define PRINTLFILE 4
#define PRINTONETERM 8
#define PRINTONEFUNCTION 16
#define PRINTALL 32

```

```
/*
    Special codes for the replace variable in the EXPRESSIONS struct
*/

#define REGULAREXPRESSION -1
#define REDEFINEDEXPRESSION -2
#define NEWLYDEFINEDEXPRESSION -3

#define VERTEXFUNCTION -1
#define GENERALFUNCTION 0
#define TENSORFUNCTION 2
#define GAMMAFUNCTION 4
/*
    Special sets
*/

#define POS_         0   /* integer > 0 */
#define POS0_        1   /* integer >= 0 */
#define NEG_         2   /* integer < 0 */
#define NEG0_        3   /* integer <= 0 */
#define EVEN_        4   /* integer (even) */
#define ODD_         5   /* integer (odd) */
#define Z_           6   /* integer */
#define SYMBOL_      7   /* symbol only */
#define FIXED_       8   /* fixed index */
#define INDEX_       9   /* index only */
#define Q_          10   /* rational */
#define DUMMYINDEX_ 11   /* dummy index only */
#define VECTOR_     12   /* vector only */

/*
    Special indices.
*/

#define GAMMA1 0
#define GAMMA5 -1
#define GAMMA6 -2
#define GAMMA7 -3
#define FUNNYVEC -4
#define FUNNYWILD -5
#define SUMMEDIND -6
#define NOINDEX -7
#define FUNNYDOLLAR -8
#define EMPTYINDEX -9

/*
    The next one should be less than all of the above special indices.
*/

#define MINSPEC -10

#define USEDFLAG 2
#define DUMMYFLAG 1

#define MAINSORT 0
#define FUNCTIONSORT 1
#define SUBSORT 2

#define FLOATMODE 1
#define RATIONALMODE 0

#define NUMSPECSETS 10

#define ISZERO 1
#define ISUNMODIFIED 2
#define ISCOMPRESSED 4
#define ISINRHS 8
#define ISFACTORIZED 16
#define TOBEFACTORED 32
#define TOBEUNFACTORED 64
#define KEEPZERO 128

#define VARTYPENONE 0
#define VARTYPECOMPLEX 1
#define VARTYPEIMAGINARY 2
#define VARTYPEROOTOFUNITY 4
#define VARTYPEMINUS 8
#define CYCLESYMMETRIC 1
#define RCYCLESYMMETRIC 2
#define SYMMETRIC 3
#define ANTISYMMETRIC 4
#define REVERSEORDER 256

/*
    Types of id statements (substitutions)
*/

```

```
#define SUBMULTI 1
#define SUBONCE 2
#define SUBONLY 3
#define SUBMANY 4
#define SUBVECTOR 5
#define SUBSELECT 6
#define SUBALL 7
#define SUBMASK 15
#define SUBDISORDER 16
#define SUBAFTER 32
#define SUBAFTERNOT 64

#define IDHEAD 6

#define DOLLARFLAG 1
#define NORMALIZEFLAG 2

#define GIDENT 1
#define GFIVE 4
#define GPLUS 3
#define GMINUS 2

/*
    Types of objects inside an if clause.
*/

#define LONGNUMBER 1
#define MATCH 2
#define COEFFI 3
#define SUBEXPR 4
#define MULTIPLEOF 5
#define IFDOLLAR 6
#define IFEXPRESSION 7
#define IFDOLLAREXTRA 8
#define IFISFACTORIZED 9
#define IFOCCURS 10
#define IFUSERFLAG 11
#define IFFLOATNUMBER 12
#define GREATER 0
#define GREATEREQUAL 1
#define LESS 2
#define LESSEQUAL 3
#define EQUAL 4
#define NOTEQUAL 5
#define ORCOND 6
#define ANDCOND 7
#define DUMMY 1
#define SORT 1
#define STORE 2
#define END 3
#define GLOBAL 4
#define CLEAR 5

#define VECTBIT 1
#define DOTPBIT 2
#define FUNBIT  4
#define SETBIT  8

#define EXTRAPARAMETER 0x4000
#define GENCOMMUTE 0
#define GENNONCOMMUTE 0x2000

#define NAMENOTFOUND -9

/*
    Types of dollar expressions.
*/

#define DOLUNDEFINED 0
#define DOLNUMBER 1
#define DOLARGUMENT 2
#define DOLSUBTERM 3
#define DOLTERMS 4
#define DOLWILDARGS 5
#define DOLINDEX 6
#define DOLZERO 7

#define FINDLOOP 0
#define REPLACELOOP 1

#define NOFUNPOWERS 0
#define COMFUNPOWERS 1
#define ALLFUNPOWERS 2

#define PROPERORDERFLAG 0

#define REGULAR 0
```

```
#define FINISH 1

#define POLYADD 1
#define POLYSUB 2
#define POLYMUL 3
#define POLYDIV 4
#define POLYREM 5
#define POLYGCD 6
#define POLYINTFAC 7
#define POLYNORM 8

#define MODNONE 0
#define MODSUM 1
#define MODMAX 2
#define MODMIN 3
#define MODLOCAL 4

#define ELEMENTUSED 1
#define ELEMENTLOADED 2
/*
    Variables for the modulus statement, flags in AC.modmode
    For explanation, see CoModulus
*/
#define POSNEG 0x1
#define INVERSETABLE 0x2
#define COEFFICIENTSONLY 0x4
#define ALSOPOWERS 0x8
#define ALSOFUNARGS 0x10
#define ALSODOLLARS 0x20
#define NOINVERSES 0x40

#define POSITIVEONLY 0
#define UNPACK 0x80
#define NOUNPACK 0
#define FROMFUNCTION 0x100

#define VARNAMES 0
#define AUTONAMES 1
#define EXPRNAMES 2
#define DOLLARNAMES 3

#ifdef WITHPTHREADS
/*
    Signals that the workers have to react to
*/

#define TERMINATETHREAD -1
#define STARTNEWEXPRESSION 1
#define LOWESTLEVELGENERATION 2
#define FINISHEXPRESSION 3
#define CLEANUPEXPRESSION 4
#define HIGHERLEVELGENERATION 5
#define STARTNEWMODULE 6
#define CLAIMOUTPUT 7
#define FINISHEXPRESSION2 8
#define INISORTBOT 7
#define RUNSORTBOT 8
#define DOONEEXPRESSION 9
#define DOBRACKETS 10
#define CLEARCLOCK 11
#define MCTSEXPANDTREE 12
#define OPTIMIZEEXPRESSION 13

#define MASTERBUFFERISFULL 1

/*
    Bucket states
*/

#define BUCKETFREE 1
#define BUCKETINUSE 0
#define BUCKETCOMINGFREE 2
#define BUCKETFILLED -1
#define BUCKETATEND -2
#define BUCKETTERMINATED 3
#define BUCKETRELEASED 4

#define NUMBEROFBLOCKSINSORT 10
#define MINIMUMNUMBEROFTERMS 10

#define BUCKETDOINGTERM 1
#define BUCKETASSIGNED -1
#define BUCKETTOBERELEASED -2
#define BUCKETPREPARINGTERM 0

#define BUCKETDOINGTERMS 0
#define BUCKETDOINGBRACKET 1
```

```
#endif

/*
    The next variable is because there is some use of cbufnum that is
    probably irrelevant. We use here DUMMYBUFNUM instead of AC.cbufnum
    just in case we run into trouble later.
*/
#define DUMMYBUFFER 1

#define ALLARGS      1
#define NUMARG       2
#define ARGRANGE     3
#define MAKEARGS     4
#define MAXRANGEINDICATOR 4
#define REPLACEARG   5
#define ENCODEARG    6
#define DECODEARG    7
#define IMPLODEARG   8
#define EXPLODEARG   9
#define PERMUTEARG  10
#define REVERSEARG  11
#define CYCLEARG    12
#define ISLYNDON    13
#define ISLYNDONR   14
#define TOLYNDON    15
#define TOLYNDONR   16
#define ADDARG      17
#define MULTIPLYARG 18
#define DROPARG     19
#define SELECTARG   20
#define DEDUPARG    21
#define ZTOHARG     22
#define HTOZARG     23
/*
#define ZTOSARG     24
#define STOZARG     25
#define HTOSARG     26
#define STOHARG     27
*/

#define BASECODE 1
#define YESLYNDON 1
#define NOLYNDON 2

#define TOPOLYNOMIALFLAG 1
#define FACTARGFLAG 2

#define OLDFACTARG 1
#define NEWFACTARG 0

#define FROMMODULEOPTION 0
#define FROMPOINTINSTRUCTION 1

#define EXTRASYMBOL 0
#define REGULARSYMBOL 1
#define EXPRESSIONNUMBER 2

#define O_NONE 0
#define O_CSE 1
#define O_CSEGREEDY 2
#define O_GREEDY 3

#define O_OCCURRENCE 0
#define O_MCTS 1
#define O_SIMULATED_ANNEALING 2

#define O_FORWARD 0
#define O_BACKWARD 1
#define O_FORWARDORBACKWARD 2
#define O_FORWARDANDBACKWARD 3

#define OPTHEAD 3
#define DOALL 1
#define ONLYFUNCTIONS 2

#define INUSE 1
#define COULDCOMMUTE 2
#define DOESNOTCOMMUTE 4

#define DICT_NONUMBERS 0
#define DICT_INTEGERONLY 1
#define DICT_RATIONALONLY 2
#define DICT_ALLNUMBERS 3
#define DICT_NOVARIABLES 0
#define DICT_DOVARIABLES 1
#define DICT_NOSPECIALS 0
#define DICT_DOSPECIALS 1
```

```
#define DICT_NOFUNWITHARGS 0
#define DICT_DOFUNWITHARGS 1
#define DICT_NOTINDOLLARS 0
#define DICT_INDOLLARS 1

#define DICT_INTEGERNUMBER 1
#define DICT_RATIONALNUMBER 2
#define DICT_SYMBOL 3
#define DICT_VECTOR 4
#define DICT_INDEX 5
#define DICT_FUNCTION 6
#define DICT_FUNCTION_WITH_ARGUMENTS 7
#define DICT_SPECIALCHARACTER 8
#define DICT_RANGE 9

#define READSPECTATORFLAG 3
#define GLOBALSPECTATORFLAG 1

#define ORDEREDSET 1

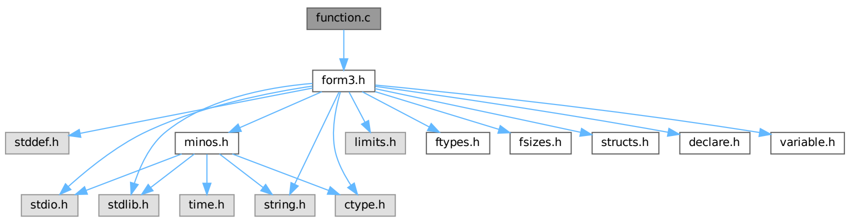
#define DENSETABLE 1
#define SPARSETABLE 0

#define ONEPARTICLEIRREDUCIBLE 1
#define WITHINSERTIONS          2
#define NOTADPOLES              4
#define WITHSYMMETRIZE          8
#define TOPOLOGIESONLY         16
#define NONODES                32
#define WITHEDGES              64
#define CHECKEXTERN           128
#define WITHBLOCKS            256
#define WITHONEPI             512
#define NOSNAILS             1024
#define NOEXTSELF            2048
```

4.51 function.c File Reference

`#include "form3.h"`

Include dependency graph for function.c:



Functions

- WORD MakeDirty (WORD ∗term, WORD ∗x, WORD par)
- void MarkDirty (WORD ∗term, WORD flags)
- void PolyFunDirty (PHEAD WORD ∗term)
- void PolyFunClean (PHEAD WORD ∗term)
- WORD Symmetrize (PHEAD WORD ∗func, WORD ∗Lijst, WORD ngroups, WORD gsize, WORD type)
- WORD CompGroup (PHEAD WORD type, WORD ∗∗args, WORD ∗a1, WORD ∗a2, WORD num)
- int FullSymmetrize (PHEAD WORD ∗fun, int type)
- WORD SymGen (PHEAD WORD ∗term, WORD ∗params, WORD num, WORD level)
- WORD SymFind (PHEAD WORD ∗term, WORD ∗params)
- int ChainIn (PHEAD WORD ∗term, WORD funnum)
- int ChainOut (PHEAD WORD ∗term, WORD funnum)
- WORD MatchFunction (PHEAD WORD ∗pattern, WORD ∗interm, WORD ∗wilds)
- WORD ScanFunctions (PHEAD WORD ∗inpat, WORD ∗inter, WORD par)

4.51.1 Detailed Description

The file with the central routines for the pattern matching of functions and their arguments. The file also contains the routines for the execution of the Symmetrize statement and its variations (like antisymmetrize etc).

Definition in file function.c.

4.51.2 Function Documentation

4.51.2.1 MakeDirty()

```
WORD MakeDirty (
            WORD * term,
            WORD * x,
            WORD par )
```

Definition at line 51 of file function.c.

4.51.2.2 MarkDirty()

```
void MarkDirty (
            WORD * term,
            WORD flags )
```

Definition at line 97 of file function.c.

4.51.2.3 PolyFunDirty()

```
void PolyFunDirty (
            PHEAD WORD * term )
```

Definition at line 139 of file function.c.

4.51.2.4 PolyFunClean()

```
void PolyFunClean (
            PHEAD WORD * term )
```

Definition at line 174 of file function.c.

4.51.2.5 Symmetrize()

```
WORD Symmetrize (
            PHEAD WORD * func,
            WORD * Lijst,
            WORD ngroups,
            WORD gsize,
            WORD type )
```

Definition at line 213 of file function.c.

4.51.2.6 CompGroup()

```
WORD CompGroup (
            PHEAD WORD type,
            WORD ** args,
            WORD * a1,
            WORD * a2,
            WORD num )
```

Definition at line 373 of file function.c.

4.51.2.7 FullSymmetrize()

```
int FullSymmetrize (
            PHEAD WORD * fun,
            int type )
```

Definition at line 473 of file function.c.

4.51.2.8 SymGen()

```
WORD SymGen (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )
```

Definition at line 518 of file function.c.

4.51.2.9 SymFind()

```
WORD SymFind (
            PHEAD WORD * term,
            WORD * params )
```

Definition at line 633 of file function.c.

4.51.2.10 ChainIn()

```
int ChainIn (
            PHEAD WORD * term,
            WORD funnum )
```

Definition at line 691 of file function.c.

4.51.2.11 ChainOut()

```
int ChainOut (
            PHEAD WORD * term,
            WORD funnum )
```

Definition at line 748 of file function.c.

4.51.2.12 MatchFunction()

```
WORD MatchFunction (
            PHEAD WORD * pattern,
            WORD * interm,
            WORD * wilds )
```

Definition at line 826 of file function.c.

4.51.2.13 ScanFunctions()

```
WORD ScanFunctions (
            PHEAD WORD * inpat,
            WORD * inter,
            WORD par )
```

Definition at line 1616 of file function.c.

4.52 function.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : function.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ Utilities :
        #[ MakeDirty :

        Routine finds the function with the address x in it
        and mark all arguments that contain x as dirty.
        if par == 0 term is a full term, else term is the start of a
        function
*/

WORD MakeDirty(WORD *term, WORD *x, WORD par)
{
    WORD *next, *n;
    if ( !par ) {
        next = term; next += *term;
```

```
        next -= ABS(next[-1]);
        term++;
        if ( x < term ) return(0);
        if ( x >= next ) return(0);
        while ( term < next ) {
            n = term + term[1];
            if ( x < n ) break;
            term = n;
        }
/*      next = n; */
    }
    else {
        next = term + term[1];
        if ( x < term || x >= next ) return(0);
    }
    if ( *term < FUNCTION ) return(0);
    if ( functions[*term-FUNCTION].spec >= TENSORFUNCTION ) return(0);
    term += FUNHEAD;
    if ( x < term ) return(0);
    next = term; NEXTARG(next)
    while ( x >= next ) { term = next; NEXTARG(next) }
    if ( *term < 0 ) return(0);
    term[1] = 1;
    term += ARGHEAD;
    if ( x < term ) return(1);
    next = term + *term;
    while ( x >= next ) { term = next; next += *next; }
    MakeDirty(term,x,0);
    return(1);
}

/*
        #] MakeDirty :
        #[ MarkDirty :

        Routine marks all functions dirty with the given flags.
        Is to be used when there is a possibility that symmetrization
        properties of functions may have changed. In that case we play
        it safe.
*/

void MarkDirty(WORD *term, WORD flags)
{
    WORD *t, *r, *m, *tstop;
    GETSTOP(term,tstop);
    t = term+1;
    while ( t < tstop ) {
        if ( *t < FUNCTION ) { t += t[1]; continue; }
        t[2] |= flags;
        if ( *t < FUNCTION+WILDOFFSET && functions[*t-FUNCTION].spec > 0 ) {
            t += t[1]; continue;
        }
        if ( *t >= FUNCTION+WILDOFFSET && functions[*t-FUNCTION-WILDOFFSET].spec > 0 ) {
            t += t[1]; continue;
        }
        r = t + FUNHEAD;
        t += t[1];
        while ( r < t ) {
            if ( *r <= 0 ) {
                if ( *r <= -FUNCTION ) r++;
                else r += 2;
                continue;
            }
            r[1] |= DIRTYFLAG;
            m = r + ARGHEAD;
            r += *r;
            while ( m < r ) {
                MarkDirty(m,flags);
                m += *m;
            }
        }
    }
}

/*
        #] MarkDirty :
        #[ PolyFunDirty :

        Routine marks the PolyFun or the PolyRatFun dirty.
        This is used when there is modular calculus and the modulus
        has changed for the current module.
*/

void PolyFunDirty(PHEAD WORD *term)
{
    GETBIDENTITY
    WORD *t, *tstop, *endarg;
```

```
    tstop = term + *term;
    tstop -= ABS(tstop[-1]);
    t = term+1;
    while ( t < tstop ) {
        if ( *t == AR.PolyFun ) {
            if ( AR.PolyFunType == 2 ) t[2] |= MUSTCLEANPRF;
            endarg = t + t[1];
            t[2] |= DIRTYFLAG;
            t += FUNHEAD;
            while ( t < endarg ) {
                if ( *t > 0 ) {
                    t[1] |= DIRTYFLAG;
                }
                NEXTARG(t);
            }
        }
        else {
            t += t[1];
        }
    }
}

/*
        #] PolyFunDirty :
        #[ PolyFunClean :

        Routine marks the PolyFun or the PolyRatFun clean.
        This is used when there is modular calculus and the modulus
        has changed for the current module.
*/

void PolyFunClean(PHEAD WORD *term)
{
    GETBIDENTITY
    WORD *t, *tstop;
    tstop = term + *term;
    tstop -= ABS(tstop[-1]);
    t = term+1;
    while ( t < tstop ) {
        if ( *t == AR.PolyFun ) {
            t[2] &= ~MUSTCLEANPRF;
        }
        t += t[1];
    }
}

/*
        #] PolyFunClean :
        #[ Symmetrize :

        (Anti)Symmetrizes the arguments of a function.
        Nlist tells of how many arguments are involved.
        Nlist == 0      All arguments must be sorted.
        Nlist > 0       Arguments mentioned are to be sorted, rest skipped.
        type = SYMMETRIC       Full symmetrization
        type = ANTISYMMETRIC:  Full symmetrization
        type = CYCLESYMMETRIC: Cyclic
        type = RCYCLESYMMETRIC:Cyclic or reverse
        Return value: OR of:
            0 even, 1 odd
            2 equal groups
            4 there was a permutation.

        The information in Lijst tells what grouping is to be applied.
        The information is:
        ngroups number of groups
        gsize size of groups
        Lijst[0]....  The groups.
*/

WORD Symmetrize(PHEAD WORD *func, WORD *Lijst, WORD ngroups, WORD gsize,
                WORD type)
{
    GETBIDENTITY
    WORD **args,**arg,nargs;
    WORD *to, *r, *fstop;
    WORD i, j, k, ff, exch, nexch, neq;
    WORD *a1, *a2, *a3;
    WORD reverseorder;
    if ( ( type & REVERSEORDER ) != 0 ) reverseorder = -1;
    else                                reverseorder = 1;
    type &= ~REVERSEORDER;

    ff = ( *func > FUNCTION ) ? functions[*func-FUNCTION].spec: 0;

    if ( 2*func[1] > AN.arglistsize ) {
        if ( AN.arglist ) M_free(AN.arglist,"Symmetrize");
```

```
        AN.arglistsize = 2*func[1] + 8;
        AN.arglist = (WORD **)Malloc1(AN.arglistsize*sizeof(WORD *),"Symmetrize");
    }
    arg = args = AN.arglist;
    to = AT.WorkPointer;
    r = func;
    fstop = r + r[1];
    r += FUNHEAD;
    nargs = 0;
    while ( r < fstop ) {    /* Make list of arguments */
        *arg++ = r;
        nargs++;
        if ( ff ) {
            if ( *r == FUNNYWILD ) r++;
            r++;
        }
        else { NEXTARG(r); }
    }
    exch = 0;
    nexch = 0;
    neq = 0;
    a1 = Lijst;
    if ( type == SYMMETRIC || type == ANTISYMMETRIC ) {
    for ( i = 1; i < ngroups; i++ ) {
        a3 = a2 = a1 + gsize;
        k = reverseorder*CompGroup(BHEAD ff,args,a1,a2,gsize);
        if ( k < 0 ) {
            j = i-1;
            for(;;) {
                for ( k = 0; k < gsize; k++ ) {
                    r = args[a1[k]]; args[a1[k]] = args[a2[k]]; args[a2[k]] = r;
                }
                exch ^= 1;
                nexch = 4;
                if ( j <= 0 ) break;
                a1 -= gsize;
                a2 -= gsize;
                k = reverseorder*CompGroup(BHEAD ff,args,a1,a2,gsize);
                if ( k == 0 ) neq = 2;
                if ( k >= 0 ) break;
                j--;
            }
        }
        else if ( k == 0 ) neq = 2;
        a1 = a3;
    }
    }
    else if ( type == CYCLESYMMETRIC || type == RCYCLESYMMETRIC ) {
        WORD rev = 0, jmin = 0, ii, iimin;
recycle:
        for ( j = 1; j < ngroups; j++ ) {
            for ( i = 0; i < ngroups; i++ ) {
                iimin = jmin + i;
                if ( iimin >= ngroups ) iimin -= ngroups;
                ii = j + i;
                if ( ii >= ngroups ) ii -= ngroups;
                k = reverseorder*CompGroup(BHEAD ff,args,Lijst+gsize*iimin,Lijst+gsize*ii,gsize);
                if ( k > 0 ) break;
                if ( k < 0 ) { jmin = j; nexch = 4; break; }
            }
        }
        if ( type == RCYCLESYMMETRIC && rev == 0 && ngroups > 1 ) {
            for ( j = 0; j < ngroups; j++ ) {
                for ( i = 0; i < ngroups; i++ ) {
                    iimin = jmin + i;
                    if ( iimin >= ngroups ) iimin -= ngroups;
                    ii = j - i;
                    if ( ii < 0 ) ii += ngroups;
                    k = reverseorder*CompGroup(BHEAD ff,args,Lijst+gsize*iimin,Lijst+gsize*ii,gsize);
                    if ( k > 0 ) break;
                    if ( k < 0 ) {
                        nexch = 4;
                        jmin = 0;
                        a1 = Lijst;
                        a2 = Lijst + gsize * (ngroups-1);
                        while ( a2 > a1 ) {
                            for ( k = 0; k < gsize; k++ ) {
                                r = args[a1[k]];
                                args[a1[k]] = args[a2[k]];
                                args[a2[k]] = r;
                            }
                            a1 += gsize; a2 -= gsize;
                        }
                        rev = 1;
                        goto recycle;
                    }
                }
```

```
            }
        }
        if ( jmin != 0 ) {
            arg = AN.arglist + func[1];
            a1 = Lijst + gsize * jmin;
            k = gsize * ngroups;
            a2 = Lijst + k;
            for ( i = 0; i < k; i++ ) {
                if ( a1 >= a2 ) a1 = Lijst;
                *arg++ = args[*a1++];
            }
            arg = AN.arglist + func[1];
            a1 = Lijst;
            for ( i = 0; i < k; i++ ) args[*a1++] = *arg++;
        }
    }
    r = func;
    i = FUNHEAD;
    NCOPY(to,r,i);
    for ( i = 0; i < nargs; i++ ) {
        if ( ff ) {
            if ( *(args[i]) == FUNNYWILD ) {
                *to++ = *(args[i]);
                *to++ = args[i][1];
            }
            else *to++ = *(args[i]);
        }
        else if ( ( j = *args[i] ) < 0 ) {
            *to++ = j;
            if ( j > -FUNCTION ) *to++ = args[i][1];
        }
        else {
            r = args[i];
            NCOPY(to,r,j);
        }
    }
    i = func[1];
    to = func;
    r = AT.WorkPointer;
    NCOPY(to,r,i);
    return ( exch | nexch | neq );
}

/*
        #] Symmetrize :
        #[ CompGroup :

            Routine compares two groups of arguments
            The arguments are in args[a1[i]] and args[a2[i]]
            for i = 0 to num
            type indicates the type of function.
            return value: -1 if there should be an exchange
            0 if they are equal
            1 if they are OK.
*/

WORD CompGroup(PHEAD WORD type, WORD **args, WORD *a1, WORD *a2, WORD num)
{
    GETBIDENTITY
    WORD *t1, *t2, i1, i2, n, k;

    for ( n = 0; n < num; n++ ) {
        t1 = args[a1[n]]; t2 = args[a2[n]];
        if ( type >= TENSORFUNCTION ) {
            if ( AR.Eside == LHSIDE || AR.Eside == LHSIDEX ) {
                if ( *t1 == FUNNYWILD ) {
                    if ( *t2 == FUNNYWILD ) {
                        if ( t1[1] < t2[1] ) return(1);
                        if ( t1[1] > t2[1] ) return(-1);
                    }
                    return(-1);
                }
                else if ( *t2 == FUNNYWILD ) {
                    return(1);
                }
                else {
                    if ( *t1 < *t2 ) return(1);
                    if ( *t1 > *t2 ) return(-1);
                }
            }
            else {
                if ( *t1 < *t2 ) return(1);
                if ( *t1 > *t2 ) return(-1);
            }
        }
        else if ( type == 0 ) {
            if ( AC.properorderflag ) {
```

```
                k = CompArg(t1,t2);
                if ( k < 0 ) return(1);
                if ( k > 0 ) return(-1);
                NEXTARG(t1)
                NEXTARG(t2)
            }
            else {
                if ( *t1 > 0 ) {
                    i1 = *t1 - ARGHEAD - 1;
                    t1 += ARGHEAD + 1;
                    if ( *t2 > 0 ) {
                        i2 = *t2 - ARGHEAD - 1;
                        t2 += ARGHEAD + 1;
                        while ( i1 > 0 && i2 > 0 ) {
                            if ( *t1 > *t2 ) return(-1);
                            else if ( *t1 < *t2 ) return(1);
                            i1--; i2--; t1++; t2++;
                        }
                        if ( i1 > 0 ) return(-1);
                        else if ( i2 > 0 ) return(1);
                    }
/*
                    This seems to be a bug. Reported by Aneesh Monahar, 28-sep-2005
                    else return(1);
*/
                    else return(-1);
                }
                else if ( *t2 > 0 ) return(1);
                else {
                    if ( *t1 != *t2 ) {
                        if ( *t1 <= -FUNCTION && *t2 <= -FUNCTION ) {
                            if ( *t1 < *t2 ) return(-1);
                            return(1);
                        }
                        else {
                            if ( *t1 < *t2 ) return(1);
                            return(-1);
                        }
                    }
                    if ( *t1 > -FUNCTION ) {
                        if ( t1[1] != t2[1] ) {
                            if ( t1[1] < t2[1] ) return(1);
                            return(-1);
                        }
                    }
                }
            }
        }
    }
    return(0);
}

/*
        #] CompGroup :
        #[ FullSymmetrize :

        Relay function for Normalize to execute a full symmetrization
        of a function fun. It hooks into Symmetrize according to the
        calling conventions for it.
        type = 0: Symmetrize
        type = 1: AntiSymmetrize
        type = 2: CycleSymmetrize
        type = 3: RCycleSymmetrize
        Return values:
        bit 0: odd permutation
        bit 1: identical arguments
        bit 2: there was a permutation.
*/

int FullSymmetrize(PHEAD WORD *fun, int type)
{
    GETBIDENTITY
    WORD *Lijst, count = 0;
    WORD *t, *funstop, i;
    int retval;

    if ( functions[*fun-FUNCTION].spec > 0 ) {
        count = fun[1] - FUNHEAD;
        for ( i = fun[1]-1; i >= FUNHEAD; i-- ) {
            if ( fun[i] == FUNNYWILD ) count--;
        }
    }
    else {
        funstop = fun + fun[1];
        t = fun + FUNHEAD;
        while ( t < funstop ) { count++; NEXTARG(t) }
    }
```

```
    if ( count < 2 ) {
        fun[2] &= ~DIRTYSYMFLAG;
        return(0);
    }
    Lijst = AT.WorkPointer;
    for ( i = 0; i < count; i++ ) Lijst[i] = i;
    AT.WorkPointer += count;
    retval = Symmetrize(BHEAD fun,Lijst,count,1,type);
    fun[2] &= ~DIRTYSYMFLAG;
    AT.WorkPointer = Lijst;
    return(retval);
}

/*
        #] FullSymmetrize :
        #[ SymGen :

        Routine does the outer work in the symmetrization.
        It locates the function(s) and loads up the parameters.
        It also studies the result.

        if params[4] = -1 and no extra -> all
                              extra -> strip groups with elements too large
                       0  -> if group with element too large: nofun
                       >0 -> must have right number of arguments
*/

WORD SymGen(PHEAD WORD *term, WORD *params, WORD num, WORD level)
{
    GETBIDENTITY
    WORD *t, *r, *m;
    WORD i, j, k, c1, c2, ngroup;
    WORD *rstop, Nlist, *inLijst, *Lijst, sign = 1, sumch = 0, count;
    DUMMYUSE(num);
    c1 = params[3];     /* function number */
    c2 = FUNCTION + WILDOFFSET;
    Nlist = params[4];
    if ( Nlist < 0 ) Nlist = 0;
    else Nlist = params[0] - 7;
    t = term;
    m = t + *t;
    m -= ABS(m[-1]);
    t++;
    while ( t < m ) {
        if ( *t == c1 || c1 > c2 ) {    /* Candidate function */
            if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
            >= TENSORFUNCTION ) {
                count = t[1] - FUNHEAD;
            }
            else {
                count = 0;
                r = t;
                rstop = t + t[1];
                r += FUNHEAD;
                while ( r < rstop ) { count++; NEXTARG(r) }
            }
            if ( ( j = params[4] ) > 0 && j != count ) goto NextFun;
            if ( j == 0 ) {
                inLijst = params+7;
                for ( i = 0; i < Nlist; i++ )
                    if ( inLijst[i] > count-1 ) goto NextFun;
            }

            if ( Nlist > (params[0] - 7) ) Nlist = params[0] - 7;
            Lijst = AT.WorkPointer;
            inLijst = params + 7;
            ngroup = params[5];
            if ( Nlist > 0 && j < 0 ) {
                k = 0;
                for ( i = 0; i < ngroup; i++ ) {
                    for ( j = 0; j < params[6]; j++ ) {
                        if ( inLijst[j] > count+1 ) {
                            inLijst += params[6];
                            goto NextGroup;
                        }
                    }
                    j = params[6];
                    NCOPY(Lijst,inLijst,j);
                    k++;
NextGroup:;
                }
                if ( k <= 1 ) goto NextFun;
                ngroup = k;
                inLijst = AT.WorkPointer;
                AT.WorkPointer = Lijst;
                Lijst = inLijst;
            }
```

```
            else if ( Nlist == 0 ) {
                for ( i = 0; i < count; i++ ) Lijst[i] = i;
                AT.WorkPointer += count;
                ngroup = count;
            }
            else {
                for ( i = 0; i < Nlist; i++ ) Lijst[i] = inLijst[i];
                AT.WorkPointer += Nlist;
            }
            j = Symmetrize(BHEAD t,Lijst,ngroup,params[6],params[2]);
            AT.WorkPointer = Lijst;
            if ( params[2] == 4 ) { /* antisymmetric */
                if ( ( j & 1 ) != 0 ) sign = -sign;
                if ( ( j & 2 ) != 0 ) return(0); /* equal arguments */
            }
            if ( ( j & 4 ) != 0 ) sumch++;
            t[2] &= ~DIRTYSYMFLAG;
        }
NextFun:
        t += t[1];
    }
    if ( sign < 0 ) {
        t = term;
        t += *t - 1;
        *t = -*t;
    }
    if ( sumch ) {
        if ( Normalize(BHEAD term) ) {
            MLOCK(ErrorMessageLock);
            MesCall("SymGen");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        if ( !*term ) return(0);
        *AN.RepPoint = 1;
        AR.expchanged = 1;
        if ( AR.CurDum > AM.IndDum && AR.sLevel <= 0 ) ReNumber(BHEAD term);
    }
    return(Generator(BHEAD term,level));
}

/*
        #] SymGen :
        #[ SymFind :

        There is a certain amount of double work here, as this routine
        finds the function to be treated, while the SymGen routine has
        to find it again. Note however that this way things remain
        uniform and simple. Moreover this avoids problems with actions
        on more than one function simultaneously.
        Output in AT.TMout:
        Number,sym/anti,fun,lenpar,ngroups,gsize,fields

*/

WORD SymFind(PHEAD WORD *term, WORD *params)
{
    GETBIDENTITY
    WORD *t, *r, *m;
    WORD j, c1, c2, count;
    WORD *rstop;
    c1 = params[4];     /* function number */
    c2 = FUNCTION + WILDOFFSET;
    t = term;
    m = t + *t;
    m -= ABS(m[-1]);
    t++;
    while ( t < m ) {
        if ( *t == c1 || c1 > c2 ) {    /* Candidate function */
            if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
                >= TENSORFUNCTION ) { count = t[1] - FUNHEAD; }
            else {
                count = 0;
                r = t;
                rstop = t + t[1];
                r += FUNHEAD;
                while ( r < rstop ) { count++; NEXTARG(r) }
            }
            if ( ( j = params[5] ) > 0 && j != count ) goto NextFun;
            if ( j == 0 ) {
                r = params + 8;
                rstop = params + params[1];
                while ( r < rstop ) {
                    if ( *r > count + 1 ) goto NextFun;
                    r++;
                }
            }
```

```

            t = AT.TMout;
            r = params;
            j = r[1] - 1;
            *t++ = j;
            *t++ = SYMMETRIZE;
            r += 3;
            j--;
            NCOPY(t,r,j);
            return(1);
        }
NextFun:
        t += t[1];
    }
    return(0);
}

/*
        #] SymFind :
        #[ ChainIn :

        Equivalent to repeat id f(?a)*f(?b) = f(?a,?b);

        This one always takes less space.
*/

int ChainIn(PHEAD WORD *term, WORD funnum)
{
    GETBIDENTITY
    WORD *t, *tend, *m, *tt, *ts;
    int action, normFlag = 0;
    if ( funnum < 0 ) { /* Dollar to be expanded */
        funnum = DolToFunction(BHEAD -funnum);
        if ( AN.ErrorInDollar || funnum <= 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Dollar variable does not evaluate to function in ChainIn statement");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
    }
    do {
        action = 0;
        tend = term+*term;
        tend -= ABS(tend[-1]);
        t = term+1;
        while ( t < tend ) {
            if ( *t != funnum ) { t += t[1]; continue; }
            m = t;
            t += t[1];
            tt = t;
            if ( t >= tend || *t != funnum ) continue;
            action = 1;
            normFlag = 1;
            while ( t < tend && *t == funnum ) {
                ts = t + t[1];
                t += FUNHEAD;
                while ( t < ts ) *tt++ = *t++;
            }
            m[1] = tt - m;
            ts = term + *term;
            while ( t < ts ) *tt++ = *t++;
            *term = tt - term;
            break;
        }
    } while ( action );

    if ( normFlag ) {
        /* We need to check the newly-constructed arguments w.r.t symmetry properties */
        MarkDirty(term, DIRTYSYMFLAG);
        AT.WorkPointer = term + *term;
        Normalize(BHEAD term);
    }

    return(0);
}

/*
        #] ChainIn :
        #[ ChainOut :

        Equivalent to repeat id f(x1?,x2?,?a) = f(x1)*f(x2,?a);
*/

int ChainOut(PHEAD WORD *term, WORD funnum)
{
    GETBIDENTITY
    WORD *t, *tend, *tt, *ts, *w, *ws;
```

```
    int flag = 0, i;
    if ( funnum < 0 ) { /* Dollar to be expanded */
        funnum = DolToFunction(BHEAD -funnum);
        if ( AN.ErrorInDollar || funnum <= 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Dollar variable does not evaluate to function in ChainOut statement");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
    }
    tend = term+*term;
    if ( AT.WorkPointer < tend ) AT.WorkPointer = tend;
    tend -= ABS(tend[-1]);
    t = term+1; tt = term; w = AT.WorkPointer;
    while ( t < tend ) {
        if ( *t != funnum || t[1] == FUNHEAD ) { t += t[1]; continue; }
        flag = 1;
        while ( tt < t ) *w++ = *tt++;
        ts = t + t[1];
        t += FUNHEAD;
        while ( t < ts ) {
            ws = w;
            for ( i = 0; i < FUNHEAD; i++ ) *w++ = tt[i];
            if ( functions[*tt-FUNCTION].spec >= TENSORFUNCTION ) {
                *w++ = *t++;
            }
            else if ( *t < 0 ) {
                if ( *t <= -FUNCTION ) *w++ = *t++;
                else { *w++ = *t++; *w++ = *t++; }
            }
            else {
                i = *t; NCOPY(w,t,i);
            }
            ws[1] = w - ws;
        }
        tt = t;
    }
    if ( flag == 1 ) {
        ts = term + *term;
        while ( tt < ts ) *w++ = *tt++;
        *AT.WorkPointer = w - AT.WorkPointer;
        t = term; w = AT.WorkPointer; i = *w;
        NCOPY(t,w,i)
        AT.WorkPointer = term + *term;
        Normalize(BHEAD term);
    }
    return(0);
}

/*
        #] ChainOut :
    #] Utilities :
    #[ Patterns :
        #[ MatchFunction :          WORD MatchFunction(pattern,interm,wilds)

        The routine assumes that the function numbers are the same.
        The contents are compared and a possible wildcard assignment
        is made. Note that it may be necessary to use a wildcard
        assignment stack to do things right.
        The routine can become arbitrarily complicated as there is
        no end to the possible wildcarding.
        Examples:
        -   a:  No wildcarding -> straight match
        -   b:  Individual arguments (object -> object)
        -   c:  whole arguments (object to subexpression)
        -   d:  any argumentlist
            e:  part of an argument (object inside subexpression)

        The ones with a minus sign in front have been implemented.

        Note: the argument wilds allows backtracking when multiple
        ?a,?b give a match that later turns out to be useless.
*/

WORD MatchFunction(PHEAD WORD *pattern, WORD *interm, WORD *wilds)
{
    GETBIDENTITY
    WORD *m, *t, *r, i;
    WORD *mstop = 0, *tstop = 0;
    WORD *argmstop, *argtstop;
    WORD *mtrmstop, *ttrmstop;
    WORD *msubstop, *mnextsub;
    WORD msizcoef, mcount, tcount, newvalue, j;
    WORD *oldm, *oldt;
    WORD *OldWork, numofwildarg;
    WORD nwstore, tobeeaten, reservevalue = 0, resernum = 0, withwild;
    WORD *wildargtaken;
```

```
    CBUF *C = cbuf+AT.ebufnum;
    int ntwa = AN.NumTotWildArgs;
    LONG oldcpointer = C->Pointer - C->Buffer;
/*
    Test first for a straight match
*/
    AN.RepFunList[AN.RepFunNum+1] = 0;
    if ( *wilds == 0 ) {
        m = pattern; t = interm;

        if ( *m != *t ) {
            if ( *m < (FUNCTION + WILDOFFSET) ) return(0);
            if ( *t < FUNCTION ) return(0);
            if ( functions[*t-FUNCTION].spec !=
            functions[*m-FUNCTION-WILDOFFSET].spec ) return(0);
        }
        i = m[1];
        if ( *m >= (FUNCTION + WILDOFFSET) ) { i--; m++; t++; }
        do { if ( *m++ != *t++ ) break; } while ( --i > 0 );
        if ( i <= 0 ) {          /* Arguments match */
            if ( AN.SignCheck && AN.ExpectedSign ) return(0);
            i = *pattern - WILDOFFSET;
            if ( i >= FUNCTION ) {
                if ( *interm != GAMMA
#ifdef WITHFLOAT
                && ( *interm != FLOATFUN )
#endif

                && !CheckWild(BHEAD i,FUNTOFUN,*interm,&newvalue) ) {
                    AddWild(BHEAD i,FUNTOFUN,newvalue);
                    return(1);
                }
                return(0);
            }
            else return(1);
        }
    }
/*
    Store the current Wildcard assignments
*/
    t = wildargtaken = OldWork = AT.WorkPointer;
    t += ntwa;
    m = AN.WildValue;
    nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
    if ( i > 0 ) {
        r = AT.WildMask;
        do {
            *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
        } while ( --i > 0 );
        *t++ = C->numrhs;
    }
    if ( t >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    AT.WorkPointer = t;

    if ( *wilds ) {
        if ( *wilds == 1 ) goto endoloop;
        else               goto enloop;         /* tensors = 2 */
    }
    m = pattern; t = interm;
/*
    Single out the specials
*/
    if ( *t == GAMMA ) {
/*
        #[ GAMMA :

        For the gamma's we need to do two things:
        a:  Find that there is a match
        b:  Find where the match occurs in the string
        This last thing cannot be stored in the current conventions,
        but once the wildcard assignments have been made it is much
        easier to find it back.
        Alternative: replace the function number in the term temporarily
        by the offset inside the string. This makes things maybe easier.
*/
        if ( *m != GAMMA ) goto NoCaseB;
        i = t[1] - m[1];
        if ( m[1] == FUNHEAD+1 ) {
            if ( i ) goto NoCaseB;
            if ( m[FUNHEAD] < (AM.OffsetIndex+WILDOFFSET) ||
            t[FUNHEAD] >= (AM.OffsetIndex+WILDOFFSET) ) goto NoCaseB;

```

```
            if ( CheckWild(BHEAD m[FUNHEAD]-WILDOFFSET,INDTOIND,t[FUNHEAD],&newvalue) ) goto NoCaseB;
            AddWild(BHEAD m[FUNHEAD]-WILDOFFSET,INDTOIND,newvalue);

            AT.WorkPointer = OldWork;
            if ( AN.SignCheck && AN.ExpectedSign ) return(0);
            return(1);      /* m was eaten. we have a match! */
        }
        if ( i < 0 ) goto NoCaseB;  /* Pattern longer than target */
        mstop = m + m[1];
        tstop = t + t[1];
        m += FUNHEAD; t += FUNHEAD;
        if ( *m >= (AM.OffsetIndex+WILDOFFSET) && *t < (AM.OffsetIndex+WILDOFFSET) ) {
            if ( CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*t,&newvalue) ) goto NoCaseB;
            reservevalue = newvalue;
            withwild = 1;
            resernum = *m-WILDOFFSET;
            AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newvalue);
        }
        else if ( *m != *t ) goto NoCaseB;
        else withwild = 0;
        m++; t++;
        oldm = m; argtstop = oldt = t;
        j = 0;                  /* No wildcard assignments yet */
        while ( i >= 0 ) {
            if ( *m == *t ) {
WithGamma:      m++; t++;
                if ( m >= mstop ) {
                    if ( t < tstop && mstop < AN.patstop ) {
                        WORD k;
                        mnextsub = pattern + pattern[1];
                        k = *mnextsub;
                        while ( k == GAMMA && mnextsub[FUNHEAD]
                        != pattern[FUNHEAD] ) {
                            mnextsub += mnextsub[1];
                            if ( mnextsub >= AN.patstop ) goto FullOK;
                            k = *mnextsub;
                        }
                        if ( k >= FUNCTION ) {
                            if ( k > (FUNCTION + WILDOFFSET) ) k -= WILDOFFSET;
                            if ( functions[k-FUNCTION].commute ) goto NoGamma;
                        }
                    }
FullOK:             if ( AN.SignCheck && AN.ExpectedSign ) goto NoGamma;
                    AN.RepFunList[AN.RepFunNum+1] = WORDDIF(oldt,argtstop);
                    return(1);
                }
                if ( t >= tstop ) goto NoCaseB;
            }
            else if ( *m >= (AM.OffsetIndex+WILDOFFSET)
            && *m < (AM.OffsetIndex + (WILDOFFSET<<1)) && ( *t >= 0 ||
            *t < MINSPEC ) ) {          /* Wildcard index */
                if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*t,&newvalue) ) {
                    AddWild(BHEAD *m-WILDOFFSET,INDTOIND,newvalue);
                    j = 1;
                    goto WithGamma;
                }
                else goto NoGamma;
            }
            else if ( *m < MINSPEC && *m >= (AM.OffsetVector+WILDOFFSET)
            && *t < MINSPEC ) {         /* Wildcard vector */
                if ( !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*t,&newvalue) ) {
                    AddWild(BHEAD *m-WILDOFFSET,VECTOVEC,newvalue);
                    j = 1;
                    goto WithGamma;
                }
                else goto NoGamma;
            }
            else {
NoGamma:
                if ( j ) {      /* Undo wildcards */
                    m = AN.WildValue;
                    t = OldWork + AN.NumTotWildArgs; r = AT.WildMask; j = nwstore;
                    if ( j > 0 ) {
                        do {
                            *m++ = *t++; *m++ = *t++;
                            *m++ = *t++; *m++ = *t++; *r++ = *t++;
                        } while ( --j > 0 );
                        C->numrhs = *t++;
                        C->Pointer = C->Buffer + oldcpointer;
                    }
                    j = 0;
                }
                m = oldm; t = ++oldt; i--;
                if ( withwild ) {
                    AddWild(BHEAD resernum,INDTOIND,reservevalue);
                }
            }
```

```
        }
        goto NoCaseB;
/*
        #] GAMMA :
        #[ Tensors :
*/
    }
    else if ( *t >= FUNCTION && functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
        mstop = m + m[1];
        tstop = t + t[1];
        mcount = 0;
        m += FUNHEAD;
        t += FUNHEAD;
        AN.WildArgs = 0;
        tcount = WORDDIF(tstop,t);
        while ( m < mstop ) {
            if ( *m == FUNNYWILD ) { m++; AN.WildArgs++; }
            m++; mcount++;
        }
        tobeeaten = tcount - mcount + AN.WildArgs;
        if ( tobeeaten ) {
            if ( tobeeaten < 0 || AN.WildArgs == 0 ) {
                AT.WorkPointer = OldWork;
                return(0);  /* Cannot match */
            }
        }
        AT.WildArgTaken[0] = AN.WildEat = tobeeaten;
        for ( i = 1; i < AN.WildArgs; i++ ) AT.WildArgTaken[i] = 0;
toploop:
        numofwildarg = 0;

        m = pattern; t = interm;
        mstop = m + m[1];
        if ( *m != *t ) {
            i = *m - WILDOFFSET;
            if ( CheckWild(BHEAD i,FUNTOFUN,*t,&newvalue) ) goto NoCaseB;
            AddWild(BHEAD i,FUNTOFUN,newvalue);
        }
        m += FUNHEAD;
        t += FUNHEAD;
        while ( m < mstop ) {
/*
            First test for an exact match
*/
            if ( *m == *t ) { m++; t++; continue; }
/*
            No exact match. Try ARGWILD
*/
            AN.argaddress = t;
            if ( *m == FUNNYWILD ) {
                tobeeaten = AT.WildArgTaken[numofwildarg++];
                if ( CheckWild(BHEAD m[1],ARGTOARG|EATTENSOR,tobeeaten,t) ) goto endloop;
                AddWild(BHEAD m[1],ARGTOARG|EATTENSOR,tobeeaten);
                m += 2;
                t += tobeeaten;
                continue;
            }
/*
            Now the various cases:
*/
            i = *m;
            if ( i < MINSPEC ) {
                if ( *t != i ) {
                    if ( *t >= MINSPEC ) goto endloop;
                    i -= WILDOFFSET;
                    if ( i < AM.OffsetVector ) goto endloop;
                    if ( CheckWild(BHEAD i,VECTOVEC,*t,&newvalue) )
                        goto endloop;
                    AddWild(BHEAD i,VECTOVEC,newvalue);
                }
            }
            else if ( i >= AM.OffsetIndex ) {           /* Index */
                if ( i < ( AM.OffsetIndex + WILDOFFSET ) ) goto endloop;
                if ( i >= ( AM.OffsetIndex + (WILDOFFSET«1) ) ) {
                                            /* Summed over index */
                    goto endloop;               /* For the moment */
                }
                i -= WILDOFFSET;
                if ( CheckWild(BHEAD i,INDTOIND,*t,&newvalue) )
                    goto endloop;       /* Assignment not allowed */
                AddWild(BHEAD i,INDTOIND,newvalue);
            }
            else goto endloop;
            m++; t++;
        }
        if ( AN.SignCheck && AN.ExpectedSign ) goto endloop;
        AT.WorkPointer = OldWork;
```

```
        if ( AN.WildArgs > 1 ) *wilds = 2;
        return(1);      /* m was eaten. we have a match! */

endloop:;
/*
    restore the current Wildcard assignments
*/
        i = nwstore;
        if ( i > 0 ) {
            m = AN.WildValue;
            t = OldWork + ntwa; r = AT.WildMask;
            do {
                *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
            } while ( --i > 0 );
            C->numrhs = *t++;
            C->Pointer = C->Buffer + oldcpointer;
        }
enloop:;
        i = AN.WildArgs - 1;
        if ( i <= 0 ) {
            AT.WorkPointer = OldWork;
            return(0);
        }
        while ( --i >= 0 ) {
            if ( AT.WildArgTaken[i] == 0 ) {
                if ( i == 0 ) {
                    AT.WorkPointer = OldWork;
                    *wilds = 0;
                    return(0);
                }
            }
            else {
                (AT.WildArgTaken[i])--;
                numofwildarg = 0;
                for ( j = 0; j <= i; j++ ) {
                    numofwildarg += AT.WildArgTaken[j];
                }
                AT.WildArgTaken[j] = AN.WildEat-numofwildarg;
                for ( j++; j < AN.WildArgs; j++ ) AT.WildArgTaken[j] = 0;
                break;
            }
        }
        goto toploop;
/*
        #] Tensors :
*/
    }
/*
    Count the number of arguments. Either equal or an argument wildcard.
*/
    mstop = m + m[1];
    tstop = t + t[1];
    mcount = 0; tcount = 0;
    m += FUNHEAD; t += FUNHEAD;
    while ( t < tstop ) { tcount++; NEXTARG(t) }
    AN.WildArgs = 0;
    while ( m < mstop ) {
        mcount++;
        if ( *m == -ARGWILD ) AN.WildArgs++;
        NEXTARG(m)
    }
    tobeeaten = tcount - mcount + AN.WildArgs;
    if ( tobeeaten ) {
        if ( tobeeaten < 0 || AN.WildArgs == 0 ) {
            AT.WorkPointer = OldWork;
            return(0);  /* Cannot match */
        }
    }
/*
    Set up the array AT.WildArgTaken for the number of arguments that each
    wildarg eats.
*/
    AT.WildArgTaken[0] = AN.WildEat = tobeeaten;
    for ( i = 1; i < AN.WildArgs; i++ ) AT.WildArgTaken[i] = 0;
topofloop:
    numofwildarg = 0;
/*
    Test for single wildcard object/argument
*/
    m = pattern; t = interm;
    if ( *m != *t ) {
        i = *m - WILDOFFSET;
        if ( CheckWild(BHEAD i,FUNTOFUN,*t,&newvalue) ) goto NoCaseB;
        AddWild(BHEAD i,FUNTOFUN,newvalue);
    }
    mstop = m + m[1];
/*  tstop = t + t[1];   */
```

```
    m += FUNHEAD;
    t += FUNHEAD;
    while ( m < mstop ) {
        argmstop = oldm = m;
        argtstop = oldt = t;
        NEXTARG(argmstop)
        NEXTARG(argtstop)
        if ( t == tstop ) { /* This concerns a very rare bug */
            if ( *m == -ARGWILD ) goto ArgAll;
            goto endofloop;
        }
        if ( *m < 0 && *t < 0 ) {
            if ( *t <= -FUNCTION ) {
                if ( *t == *m ) {}
                else if ( *m <= -FUNCTION-WILDOFFSET
                && functions[-*t-FUNCTION].spec
                == functions[-*m-FUNCTION-WILDOFFSET].spec ) {
                    i = -*m - WILDOFFSET;
                    if ( CheckWild(BHEAD i,FUNTOFUN,-*t,&newvalue) ) goto endofloop;
                    AddWild(BHEAD i,FUNTOFUN,newvalue);
                }
                else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER ) {
                    i = m[1] - 2*MAXPOWER;
                    AN.argaddress = AT.FunArg;
                    AT.FunArg[ARGHEAD+1] = -*t;
                    if ( CheckWild(BHEAD i,SYMTOSUB,1,AN.argaddress) ) goto endofloop;
                    AddWild(BHEAD i,SYMTOSUB,0);
                }
                else if ( *m == -ARGWILD ) {
ArgAll:             i = AT.WildArgTaken[numofwildarg++];
                    AN.argaddress = t;
                    if ( CheckWild(BHEAD m[1],ARGTOARG,i,t) ) goto endofloop;
                    AddWild(BHEAD m[1],ARGTOARG,i);
/*                  m += 2; */
                    while ( --i >= 0 ) { NEXTARG(t) }
                    argtstop = t;
                }
                else goto endofloop;
            }
            else if ( *t == *m ) {
                if ( t[1] == m[1] ) {}
                else if ( *t == -SYMBOL ) {
                    j = SYMTOSYM;
SymAll:
                    if ( ( i = m[1] - 2*MAXPOWER ) < 0 ) goto endofloop;
                    if ( CheckWild(BHEAD i,j,t[1],&newvalue) ) goto endofloop;
                    AddWild(BHEAD i,j,newvalue);
                }
                else if ( *t == -INDEX ) {
IndAll:             i = m[1] - WILDOFFSET;
                    if ( i < AM.OffsetIndex || i >= WILDOFFSET+AM.OffsetIndex )
                                                            goto endofloop;
                                /* We kill the summed over indices here */
                    if ( CheckWild(BHEAD i,INDTOIND,t[1],&newvalue) ) goto endofloop;
                    AddWild(BHEAD i,INDTOIND,newvalue);
                }
                else if ( *t == -VECTOR || *t == -MINVECTOR ) {
                    i = m[1] - WILDOFFSET;
                    if ( i < AM.OffsetVector ) goto endofloop;
                    if ( CheckWild(BHEAD i,VECTOVEC,t[1],&newvalue) ) goto endofloop;
                    AddWild(BHEAD i,VECTOVEC,newvalue);
                }
                else goto endofloop;
            }
            else if ( *m == -ARGWILD ) goto ArgAll;
            else if ( *m == -INDEX && m[1] >= AM.OffsetIndex+WILDOFFSET
            && m[1] < AM.OffsetIndex+(WILDOFFSET«1) ) {
                if ( *t == -VECTOR ) goto IndAll;
                if ( *t == -SNUMBER && t[1] >= 0 && t[1] < AM.OffsetIndex ) goto IndAll;
                if ( *t == -MINVECTOR ) {
                    i = m[1] - WILDOFFSET;
                    AN.argaddress = AT.MinVecArg;
                    AT.MinVecArg[ARGHEAD+3] = t[1];
                    if ( CheckWild(BHEAD i,INDTOSUB,1,AN.argaddress) ) goto endofloop;
                    AddWild(BHEAD i,INDTOSUB,(WORD)0);
                }
                else goto endofloop;
            }
            else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER && *t == -SNUMBER ) {
                j = SYMTONUM;
                goto SymAll;
            }
            else if ( *m == -VECTOR && *t == -MINVECTOR &&
            ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
/*
================================
                AN.argaddress = AT.MinVecArg;
```

```
                      AT.MinVecArg[ARGHEAD+3] = t[1];
                      if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) goto endofloop;
                      AddWild(BHEAD i,VECTOSUB,(WORD)0);
================================
*/
                      if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) goto endofloop;
                      AddWild(BHEAD i,VECTOMIN,newvalue);

                  }
                  else if ( *m == -MINVECTOR && *t == -VECTOR &&
                  ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
/*
================================
                      AN.argaddress = AT.MinVecArg;
                      AT.MinVecArg[ARGHEAD+3] = t[1];
                      if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) goto endofloop;
                      AddWild(BHEAD i,VECTOSUB,(WORD)0);
================================
*/
                      if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) goto endofloop;
                      AddWild(BHEAD i,VECTOMIN,newvalue);
                  }
                  else goto endofloop;
              }
              else if ( *t <= -FUNCTION && *m > 0 ) {
                  if ( ( m[ARGHEAD]+ARGHEAD == *m ) && m[*m-1] == 3
                  && m[*m-2] == 1 && m[*m-3] == 1 && m[ARGHEAD+1] >= FUNCTION
                  && m[ARGHEAD+2] == *m-ARGHEAD-4 ) { /* Check for f(?a) etc */
                      WORD *mmmst, *mmm;
                      if ( m[ARGHEAD+1] >= FUNCTION+WILDOFFSET ) {
/*                        i = *m - WILDOFFSET; */
                          i = m[ARGHEAD+1] - WILDOFFSET;
                          if ( CheckWild(BHEAD i,FUNTOFUN,-*t,&newvalue) ) goto endofloop;
                          AddWild(BHEAD i,FUNTOFUN,newvalue);
                      }
                      else if ( m[ARGHEAD+1] != -*t ) goto endofloop;
/*
                          Only arguments allowed are ?a etc.
*/
                      mmmst = m+*m-3;
                      mmm = m + ARGHEAD + FUNHEAD + 1;
                      while ( mmm < mmmst ) {
                          if ( *mmm != -ARGWILD ) goto endofloop;
                          i = 0;
                          AN.argaddress = t;
                          if ( CheckWild(BHEAD mmm[1],ARGTOARG,i,t) ) goto endofloop;
                          AddWild(BHEAD mmm[1],ARGTOARG,i);
                          mmm += 2;
                      }
                  }
                  else goto endofloop;
              }
              else if ( *m < 0 && *t > 0 ) {
                  if ( *m == -SYMBOL ) {           /* SYMTOSUB */
                      if ( m[1] < 2*MAXPOWER ) goto endofloop;
                      i = m[1] - 2*MAXPOWER;
                      AN.argaddress = t;
                      if ( CheckWild(BHEAD i,SYMTOSUB,1,AN.argaddress) ) goto endofloop;
                      AddWild(BHEAD i,SYMTOSUB,0);
                  }
                  else if ( *m == -VECTOR ) {
                      if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetVector )
                                                              goto endofloop;
                      AN.argaddress = t;
                      if ( CheckWild(BHEAD i,VECTOSUB,1,t) ) goto endofloop;
                      AddWild(BHEAD i,VECTOSUB,(WORD)0);
                  }
                  else if ( *m == -INDEX ) {
                      if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetIndex ) goto endofloop;
                      if ( i >= AM.OffsetIndex + WILDOFFSET ) goto endofloop;
                      AN.argaddress = t;
                      if ( CheckWild(BHEAD i,INDTOSUB,1,AN.argaddress) ) goto endofloop;
                      AddWild(BHEAD i,INDTOSUB,(WORD)0);
                  }
                  else if ( *m == -ARGWILD ) goto ArgAll;
                  else goto endofloop;
              }
              else if ( *m > 0 && *t > 0 ) {
                  WORD ii = *t-*m;
                  i = *m;
                  do { if ( *m++ != *t++ ) break; } while ( --i > 0 );
                  if ( i == 1 && ii == 0 ) {  /* sign difference */
                      goto endofloop;
                  }
                  else if ( i > 0 ) {
                      WORD *cto, *cfrom, *csav, ci;
                      WORD oRepFunNum;
```

```
                WORD *oRepFunList;
                WORD *oterstart,*oterstop,*opatstop;
                WORD oExpectedSign;
                WORD wildargs, wildeat;
/*
                Not an exact match here.
                We have to hope that the pattern contains a composite wildcard.
*/
                m = oldm; t = oldt;
                m += ARGHEAD; t += ARGHEAD;         /* Point at (first?) term */
                mtrmstop = m + *m;
                ttrmstop = t + *t;
                if ( mtrmstop < argmstop ) goto endofloop;/* More than one term */
                msizcoef = mtrmstop[-1];
                if ( msizcoef < 0 ) msizcoef = -msizcoef;
                msubstop = mtrmstop - msizcoef;
                m++;
                if ( m >= msubstop ) goto endofloop;    /* Only coefficient */
/*
                Here we have a composite term. It can match provided it
                matches the entire argument. This argument must be a
                single term also and the coefficients should match
                (more or less).
                The matching takes:
                1:  Match the functions etc. Nothing can be left.
                2:  Match dotproducts and symbols. ONLY must match
                    and nothing may be left.
                For safety it is best to take the term out and put it
                in workspace.
*/

                if ( argtstop > ttrmstop ) goto endofloop;
                m--;
                oterstart = AN.terstart;
                oterstop = AN.terstop;
                opatstop = AN.patstop;
                oRepFunList = AN.RepFunList;
                oRepFunNum = AN.RepFunNum;
                AN.RepFunNum = 0;
                AN.RepFunList = AT.WorkPointer;
                AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
                if ( AT.WorkPointer+*t+5 > AT.WorkTop ) {
                    MLOCK(ErrorMessageLock);
                    MesWork();
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                csav = cto = AT.WorkPointer;
                cfrom = t;
                ci = *t;
                while ( --ci >= 0 ) *cto++ = *cfrom++;
                AT.WorkPointer = cto;
                ci = msizcoef;
                cfrom = mtrmstop;
                --ci;
                if ( abs(*--cfrom) != abs(*--cto) ) {
                    AT.WorkPointer = csav;
                    AN.RepFunList = oRepFunList;
                    AN.RepFunNum = oRepFunNum;
                    AN.terstart = oterstart;
                    AN.terstop = oterstop;
                    AN.patstop = opatstop;
                    goto endofloop;
                }
                i = (*cfrom != *cto) ? 1 : 0; /* buffer AN.ExpectedSign until we are beyond the goto */
                while ( --ci >= 0 ) {
                    if ( *--cfrom != *--cto ) {
                        AT.WorkPointer = csav;
                        AN.RepFunList = oRepFunList;
                        AN.RepFunNum = oRepFunNum;
                        AN.terstart = oterstart;
                        AN.terstop = oterstop;
                        AN.patstop = opatstop;
                        goto endofloop;
                    }
                }
                oExpectedSign =  AN.ExpectedSign; /* buffer AN.ExpectedSign until we are beyond FindRest/FindOnly */
                AN.ExpectedSign = i;
                *m -= msizcoef;
                wildargs = AN.WildArgs;
                wildeat = AN.WildEat;
                for ( i = 0; i < wildargs; i++ ) wildargtaken[i] = AT.WildArgTaken[i];
                AN.ForFindOnly = 0; AN.UseFindOnly = 1;
                AN.nogroundlevel++;
                if ( FindRest(BHEAD csav,m) && ( AN.UsedOtherFind || FindOnly(BHEAD csav,m) ) ) {}
```

```
                else {
nomatch:
                    *m += msizcoef;
                    AT.WorkPointer = csav;
                    AN.RepFunList = oRepFunList;
                    AN.RepFunNum = oRepFunNum;
                    AN.terstart = oterstart;
                    AN.terstop = oterstop;
                    AN.patstop = opatstop;
                    AN.WildArgs = wildargs;
                    AN.WildEat = wildeat;
                    AN.ExpectedSign = oExpectedSign;
                    AN.nogroundlevel--;
                    for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
                    goto endofloop;
                }
/*              if ( *m == 1 || m[1] < FUNCTION || functions[m[1]-FUNCTION].spec >= TENSORFUNCTION ) {
*/
                if ( *m == 1 || m[1] < FUNCTION ) {
                    if ( AN.ExpectedSign ) goto nomatch;
                }
                else {
                    if ( m[1] > FUNCTION + WILDOFFSET ) {
                        if ( functions[m[1]-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION ) {
                            if ( AN.ExpectedSign != AN.RepFunList[AN.RepFunNum-1] ) goto nomatch;
                        }
                    }
                    else {
                        if ( AN.ExpectedSign != AN.RepFunList[AN.RepFunNum-1] ) goto nomatch;
/*
                        if ( functions[m[1]-FUNCTION].spec >= TENSORFUNCTION ) {
                            if ( AN.ExpectedSign != AN.RepFunList[AN.RepFunNum-1] ) goto nomatch;
                        }
*/
                    }
                }
                AN.nogroundlevel--;
                AN.ExpectedSign = oExpectedSign;
                AN.WildArgs = wildargs;
                AN.WildEat = wildeat;
                for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
                Substitute(BHEAD csav,m,1);
                cto = csav;
                cfrom = cto + *cto - msizcoef;
                cto++;
                *m += msizcoef;
                AT.WorkPointer = csav;
                AN.RepFunList = oRepFunList;
                AN.RepFunNum = oRepFunNum;
                AN.terstart = oterstart;
                AN.terstop = oterstop;
                AN.patstop = opatstop;
                if ( *cto != SUBEXPRESSION ) goto endofloop;
                cto += cto[1];
                if ( cto < cfrom ) goto endofloop;
            }
        }
        else goto endofloop;

        t = argtstop;                       /* Next argument */
        m = argmstop;
    }
    if ( AN.SignCheck && AN.ExpectedSign ) goto endofloop;
    AT.WorkPointer = OldWork;
    if ( AN.WildArgs > 1 ) *wilds = 1;
    if ( AN.SignCheck && AN.ExpectedSign ) return(0);
    return(1);      /* m was eaten. we have a match! */

endofloop:;
/*
    restore the current Wildcard assignments
*/
    i = nwstore;
    if ( i > 0 ) {
        m = AN.WildValue;
        t = OldWork + ntwa; r = AT.WildMask;
        do {
            *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
        } while ( --i > 0 );
        C->numrhs = *t++;
        C->Pointer = C->Buffer + oldcpointer;
    }

endoloop:;
    i = AN.WildArgs-1;
    if ( i <= 0 ) {
        AT.WorkPointer = OldWork;
```

```
            return(0);
        }
        while ( --i >= 0 ) {
            if ( AT.WildArgTaken[i] == 0 ) {
                if ( i == 0 ) {
                    AT.WorkPointer = OldWork;
                    return(0);
                }
            }
            else {
                (AT.WildArgTaken[i])--;
                numofwildarg = 0;
                for ( j = 0; j <= i; j++ ) {
                    numofwildarg += AT.WildArgTaken[j];
                }
                AT.WildArgTaken[j] = AN.WildEat-numofwildarg;
/* ----> bug to be replaced in other source code */
                for ( j++; j < AN.WildArgs; j++ ) AT.WildArgTaken[j] = 0;
                break;
            }
        }
        goto topofloop;
NoCaseB:
/*
        Restore the old Wildcard assignments
*/
        i = nwstore;
        if ( i > 0 ) {
            m = AN.WildValue;
            t = OldWork + ntwa; r = AT.WildMask;
            do {
                *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
            } while ( --i > 0 );
            C->numrhs = *t++;
            C->Pointer = C->Buffer + oldcpointer;
        }
        AT.WorkPointer = OldWork;
        return(0);      /* no match */
}

/*
        #] MatchFunction :
        #[ ScanFunctions :          WORD ScanFunctions(inpat,inter,par)

        Finds in which functions to look for a match.
        inpat is the start of the pattern still to be matched.
        inter is the start of the term still to be matched.
        par gives information about commutativity.
            par = 0: nothing special
            par = 1: regular noncommuting function
            par = 2: GAMMA function

        AN.patstop: end of the functions field in the search pattern
        AN.terstop: end of the functions field in the target pattern
        AN.terstart: address of entire term;

        The actual matching of the functions and their arguments is done
        in a number of different routines. Mainly MatchFunction when there
        are no symmetry properties.
        Also: MatchE
              MatchCy
              FunMatchSy
              FunMatchCy

        The main problem here is backtracking, ie continuing with wildcard
        possibilities when a first assignment doesn't work.
        Important note: this was completely forgotten in the symmetric
        functions till 6-jan-2009. As of the moment this still has to
        be fixed.  ??????21-mar-2023?????? Is this still unfixed?????

        Functions inside functions can cause problems when antisymmetric
        functions are involved. The sign of the term may be at stake.
        At the lowest level this is no problem but in f(-fas(n2,n1)) this
        plays a role. Next is when we have a product of functions inside
        an argument. The strategy must be that we test the sign only at the
        last function. Hence, when inpat+inpat[1] >= AN.patstop.
        We might relax that to the last antisymmetric function at a later stage.

    New scheme to be implemented for non-commuting objects:
    When we are matching a second (or higher) function, any match can only
    be directly after the last matched non-commuting function or a commuting
    function. This will take care of whatever happens in MatchE etc.
*/

WORD ScanFunctions(PHEAD WORD *inpat, WORD *inter, WORD par)
{
        GETBIDENTITY
```

```
    WORD i, *m, *t, *r, sym, psym;
    WORD *newpat, *newter, *instart, *oinpat = 0, *ointer = 0;
    WORD nwstore, offset, *OldWork, SetStop = 0, oRepFunNum = AN.RepFunNum;
    WORD wilds, wildargs = 0, wildeat = 0, *wildargtaken;
    WORD *Oterfirstcomm = AN.terfirstcomm;
    CBUF *C = cbuf+AT.ebufnum;
    int ntwa = AN.NumTotWildArgs;
    LONG oldcpointer = C->Pointer - C->Buffer;
    WORD oldSignCheck = AN.SignCheck;
    instart = inter;
/*
    Only active for the last function in the pattern.
    The actual test on the sign is in MatchFunction or the symmetric functions
*/
    if ( AN.nogroundlevel ) {
        AN.SignCheck = ( inpat + inpat[1] >= AN.patstop ) ? 1 : 0;
    }
    else {
        AN.SignCheck = 0;
    }
/*
            Store the current Wildcard assignments
*/
    t = wildargtaken = OldWork = AT.WorkPointer;
    t += ntwa;
    m = AN.WildValue;
    nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
    if ( i > 0 ) {
        r = AT.WildMask;
        do {
            *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
        } while ( --i > 0 );
        *t++ = C->numrhs;
    }
    if ( t >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    AT.WorkPointer = t;
    do {
#ifndef NEWCOMMUTE
/*
        Find an eligible unsubstituted function
*/
        if ( AN.RepFunNum > 0 ) {
/*
            First try a non-commuting function, just after the last
            substituted non-commuting function.
*/
            if ( *inter >= FUNCTION && functions[*inter-FUNCTION].commute ) {
                do {
                    offset = WORDDIF(inter,AN.terstart);
                    for ( i = 0; i < AN.RepFunNum; i += 2 ) {
                        if ( AN.RepFunList[i] >= offset ) break;
                    }
                    if ( i >= AN.RepFunNum ) break;
                    inter += inter[1];
                } while ( inter < AN.terfirstcomm );
                if ( inter < AN.terfirstcomm ) { /* Check that it is directly after */
                    for ( i = 0; i < AN.RepFunNum; i += 2 ) {
                        if ( functions[AN.terstart[AN.RepFunList[i]]-FUNCTION].commute
                        && AN.RepFunList[i]+AN.terstart[AN.RepFunList[i]+1] == offset ) break;
                    }
                    if ( i < AN.RepFunNum ) goto trythis;
                }
                inter = AN.terfirstcomm;
            }
/*
            Now try one of the commuting functions
*/
            while ( inter < AN.terstop ) {
                offset = WORDDIF(inter,AN.terstart);
                for ( i = 0; i < AN.RepFunNum; i += 2 ) {
                    if ( AN.RepFunList[i] == offset ) break;
                }
                if ( i >= AN.RepFunNum ) break;
                inter += inter[1];
            }
            if ( inter >= AN.terstop ) goto Failure;
trythis:;
        }
        else {
/*
            The first function can be anywhere. We have no problems.
*/
```

```
            offset = WORDDIF(inter,AN.terstart);
        }
#else
        /* first find an unsubstituted function */
        do {
            offset = WORDDIF(inter,AN.terstart);
            for ( i = 0; i < AN.RepFunNum; i += 2 ) {
                if ( AN.RepFunList[i] == offset ) break;
            }
            if ( i >= AN.RepFunNum ) break;
            inter += inter[1];
        } while ( inter < AN.terstop );
        if ( inter >= AN.terstop ) goto Failure;
#endif
        wilds = 0;
        /* We found one */
        if ( *inter >= FUNCTION && *inpat >= FUNCTION ) {
            if ( *inpat == *inter || *inpat >= FUNCTION + WILDOFFSET ) {
/*
                if ( inter[1] == FUNHEAD ) goto rewild;
*/
                if ( functions[*inter-FUNCTION].spec >= TENSORFUNCTION
                && ( *inter == *inpat ||
                functions[*inpat-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION ) ) {
                    sym = functions[*inter-FUNCTION].symmetric & ~REVERSEORDER;
                    if ( *inpat == *inter ) psym = sym;
                    else psym = functions[*inpat-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER;
                    if ( sym == ANTISYMMETRIC || sym == SYMMETRIC
                    || psym == SYMMETRIC || psym == ANTISYMMETRIC ) {
                        if ( sym == ANTISYMMETRIC && psym == SYMMETRIC ) goto rewild;
                        if ( sym == SYMMETRIC && psym == ANTISYMMETRIC ) goto rewild;
/*
                        Special function call for (anti)symmetric tensors
*/
                        if ( MatchE(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
                    }
                    else if ( sym == CYCLESYMMETRIC || sym == RCYCLESYMMETRIC
                    || psym == CYCLESYMMETRIC || psym == RCYCLESYMMETRIC ) {
/*
                        Special function call for (r)cyclic tensors
*/
                        if ( MatchCy(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
                    }
                    else goto rewild;
                }
                else if ( functions[*inter-FUNCTION].spec <= 0
                && ( *inter == *inpat ||
                functions[*inpat-FUNCTION-WILDOFFSET].spec <= 0 ) ) {
                    sym = functions[*inter-FUNCTION].symmetric & ~REVERSEORDER;
                    if ( *inpat == *inter ) psym = sym;
                    else psym = functions[*inpat-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER;
                    if ( psym == SYMMETRIC || sym == SYMMETRIC
/*
                    The next statement was commented out. Why????
                    Werkt nog niet. Teken wordt nog niet bijgehouden.
                    5-nov-2001
*/
                    || psym == ANTISYMMETRIC || sym == ANTISYMMETRIC
                    ) {
                        if ( sym == ANTISYMMETRIC && psym == SYMMETRIC ) goto rewild;
                        if ( sym == SYMMETRIC && psym == ANTISYMMETRIC ) goto rewild;
                        if ( FunMatchSy(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
                    }
                    else
                        if ( sym == CYCLESYMMETRIC || sym == RCYCLESYMMETRIC
                    || psym == CYCLESYMMETRIC || psym == RCYCLESYMMETRIC ) {
                        if ( FunMatchCy(BHEAD inpat,inter,instart,par) ) goto OnSuccess;
                    }
                    else goto rewild;
                }
                else goto rewild;
                AN.terfirstcomm = Oterfirstcomm;
            }
            else if ( par > 0 ) { SetStop = 1; goto maybenext; }
        }
        else {
rewild:
        AN.terfirstcomm = Oterfirstcomm;
        if ( *inter != SUBEXPRESSION && MatchFunction(BHEAD inpat,inter,&wilds) ) {
            AN.terfirstcomm = Oterfirstcomm;
            if ( wilds ) {
/*
                Store wildcards to continue in MatchFunction if the current
                wildcards do not work out.
*/
                wildargs = AN.WildArgs;
                wildeat = AN.WildEat;
```

```
                for ( i = 0; i < wildargs; i++ ) wildargtaken[i] = AT.WildArgTaken[i];
                oinpat = inpat; ointer = inter;
            }
            if ( par && *inter == GAMMA && AN.RepFunList[AN.RepFunNum+1] ) {
                SetStop = 1; goto NoMat;
            }
            if ( par == 2 ) {
                if ( *inter < FUNCTION || functions[*inter-FUNCTION].commute ) {
                    goto NoMat;
                }
                par = 1;
            }
            AN.RepFunList[AN.RepFunNum] = offset;
            AN.RepFunNum += 2;
            newpat = inpat + inpat[1];
            if ( newpat >= AN.patstop ) {
                if ( AN.UseFindOnly == 0 ) {
                    if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                        AN.UsedOtherFind = 1;
                        goto OnSuccess;
                    }
                    AN.RepFunNum -= 2;
                    goto NoMat;
                }
                goto OnSuccess;
            }
            if ( *inter < FUNCTION || functions[*inter-FUNCTION].commute ) {
                newter = inter + inter[1];
                if ( newter >= AN.terstop ) goto Failure;
                if ( *inter == GAMMA && inpat[1] <
                inter[1] - AN.RepFunList[AN.RepFunNum-1] ) {
                    if ( ScanFunctions(BHEAD newpat,newter,2) ) goto OnSuccess;
                    AN.terfirstcomm = Oterfirstcomm;
                }
                else if ( *newter ==  SUBEXPRESSION ) {}
                else if ( functions[*inter-FUNCTION].commute ) {
                    if ( ScanFunctions(BHEAD newpat,newter,1) ) goto OnSuccess;
                    AN.terfirstcomm = Oterfirstcomm;
                    if ( ( *newpat < (FUNCTION+WILDOFFSET)
                        && ( functions[*newpat-FUNCTION].commute == 0 ) ) ||
                        ( *newpat >= (FUNCTION+WILDOFFSET)
                        && ( functions[*newpat-FUNCTION-WILDOFFSET].commute == 0 ) ) ) {
                        newter = AN.terfirstcomm;
                        if ( newter < AN.terstop && ScanFunctions(BHEAD newpat,newter,1) ) goto OnSuccess;
                    }
                }
                else {
                    if ( ScanFunctions(BHEAD newpat,instart,1) ) goto OnSuccess;
                    AN.terfirstcomm = Oterfirstcomm;
                }
                SetStop = par;
            }
            else {
/*
                Shouldn't this be newpat instead of inpat?????
*/
                if ( par && inter > instart && ( ( *newpat < (FUNCTION+WILDOFFSET)
                && functions[*newpat-FUNCTION].commute ) ||
                ( *newpat >= (FUNCTION+WILDOFFSET)
                && functions[*newpat-FUNCTION-WILDOFFSET].commute ) ) ) {
                    SetStop = 1;
                }
                else {
                    newter = instart;
                    if ( ScanFunctions(BHEAD newpat,newter,par) ) goto OnSuccess;
                    AN.terfirstcomm = Oterfirstcomm;
                }
            }
/*
            Restore the old Wildcard assignments
*/
NoMat:
            i = nwstore;
            if ( i > 0 ) {
                m = AN.WildValue;
                t = OldWork + ntwa; r = AT.WildMask;
                do {
                    *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                } while ( --i > 0 );
                C->numrhs = *t++;
                C->Pointer = C->Buffer + oldcpointer;
            }
/*          AN.RepFunNum -= 2; */
            AN.RepFunNum = oRepFunNum;
            if ( wilds ) {
                inter = ointer; inpat = oinpat;
```

```
                AN.WildArgs = wildargs;
                AN.WildEat = wildeat;
                for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
                goto rewild;
            }
            if ( SetStop ) break;
        }
        else if ( par ) {
maybenext:
            if ( *inpat < (FUNCTION+WILDOFFSET) ) {
                if ( *inpat < FUNCTION ||
                functions[*inpat-FUNCTION].commute ) break;
            }
            else {
                if ( functions[*inpat-FUNCTION-WILDOFFSET].commute ) break;
            }
        }}
        inter += inter[1];
    } while ( inter < AN.terstop );
Failure:
    AN.SignCheck = oldSignCheck;
    AT.WorkPointer = OldWork;
    return(0);
OnSuccess:
    if ( AT.idallflag && AN.nogroundlevel <= 0 ) {
        if ( AT.idallmaxnum > 0 && AT.idallnum >= AT.idallmaxnum ) {
            AN.terfirstcomm = Oterfirstcomm;
            AN.SignCheck = oldSignCheck;
            AT.WorkPointer = OldWork;
            return(0);
        }
        SubsInAll(BHEAD0);
        AT.idallnum++;
        if ( AT.idallmaxnum == 0 || AT.idallnum < AT.idallmaxnum ) goto NoMat;
    }
    AN.terfirstcomm = Oterfirstcomm;
    AN.SignCheck = oldSignCheck;
/*
    Now the disorder test
*/
    if ( AN.DisOrderFlag && AN.RepFunNum >= 4 ) {
        WORD k, kk;
        for ( i = 2; i < AN.RepFunNum; i += 2 ) {
/*
------------> We still have to copy the code from Normalize wrt properorderflag
*/
            m = AN.terstart + AN.RepFunList[i-2];
            t = AN.terstart + AN.RepFunList[i];
            if ( *m != *t ) {
                if ( *m > *t ) continue;
                goto doesmatch;
            }
            if ( *m >= FUNCTION && functions[*m-FUNCTION].spec >=
                TENSORFUNCTION ) {
                k = m[1] - FUNHEAD;
                kk = t[1] - FUNHEAD;
                m += FUNHEAD;
                t += FUNHEAD;
            }
            else {
                k = m[1] - FUNHEAD;
                kk = t[1] - FUNHEAD;
                m += FUNHEAD;
                t += FUNHEAD;
            }
            while ( k > 0 && kk > 0 ) {
                if ( *m < *t ) goto NextFor;
                else if ( *m++ > *t++ ) goto doesmatch;
                k--; kk--;
            }
            if ( k > 0 ) goto doesmatch;
NextFor:;
        }
        SetStop = 1;
        goto NoMat;
    }
doesmatch:
    AT.WorkPointer = OldWork;
    return(1);
}

/*
        #] ScanFunctions :
    #] Patterns :
*/
```

4.53 fwin.h File Reference

Macros

- #define LINEFEED '\n'
- #define CARRIAGERETURN 0x0D
- #define WITHRETURN
- #define WITHSYSTEM
- #define P_term(code) exit((int)(code<0?-code:code))
- #define SEPARATOR '\\'
- #define ALTSEPARATOR '/'
- #define PATHSEPARATOR ';'
- #define WITH_ENV

4.53.1 Detailed Description

Settings for Windows computers.

Definition in file fwin.h.

4.53.2 Macro Definition Documentation

4.53.2.1 LINEFEED

```
#define LINEFEED '\n'
```

Definition at line 33 of file fwin.h.

4.53.2.2 CARRIAGERETURN

```
#define CARRIAGERETURN 0x0D
```

Definition at line 34 of file fwin.h.

4.53.2.3 WITHRETURN

```
#define WITHRETURN
```

Definition at line 35 of file fwin.h.

4.53.2.4 WITHSYSTEM

```
#define WITHSYSTEM
```

Definition at line 37 of file fwin.h.

4.53.2.5 P_term

```
#define P_term(
            code ) exit((int)(code<0?-code:code))
```

Definition at line 39 of file fwin.h.

4.53.2.6 SEPARATOR

```
#define SEPARATOR '\\'
```

Definition at line 41 of file fwin.h.

4.53.2.7 ALTSEPARATOR

```
#define ALTSEPARATOR '/'
```

Definition at line 42 of file fwin.h.

4.53.2.8 PATHSEPARATOR

```
#define PATHSEPARATOR ';'
```

Definition at line 43 of file fwin.h.

4.53.2.9 WITH_ENV

```
#define WITH_ENV
```

Definition at line 44 of file fwin.h.

4.54 fwin.h

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
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 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
```

```
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#define LINEFEED '\n'
#define CARRIAGERETURN 0x0D
#define WITHRETURN

#define WITHSYSTEM

#define P_term(code)    exit((int)(code<0?-code:code))

#define SEPARATOR '\\'
#define ALTSEPARATOR '/'
#define PATHSEPARATOR ';'
#define WITH_ENV
```

4.55 gentopo.cc

```
// { ( [

#ifdef HAVE_CONFIG_H
#ifndef CONFIG_H_INCLUDED
#define CONFIG_H_INCLUDED
#include <config.h>
#endif
#endif

#include <iostream>
#include <iomanip>
#include <sstream>
#include <cstdio>
#include <stdlib.h>

#include "grcc.h"
#include "gentopo.h"

#define DUMMYUSE(x)  (void)(x)

using namespace std;

//  The next limitation is imposed by the fact that the latest compilers
//  can give warnings on declarations like "int dtcl1[nNodes]"
//  With a bit of overkill there should be no real problems.
#define MAXNODES 100
#define MAXNCLASSES 100

// Generate scalar connected Feynman graphs.

//==============================================================

typedef int Bool;
const int T_True  = 1;
const int T_False = 0;

// compile options
const int CHECK    = T_True;
//const int MONITOR  = T_True;
const int MONITOR  = T_False;

const int OPTPRINT = T_False;

const int DEBUG0 = T_False;
//const int DEBUG1 = T_False;
const int DEBUG  = T_False;

// for debugging memory use
#define DEBUGM   T_False

//==============================================================
class T_MGraph;
class T_EGraph;

#define Extern
#define Global
#define Local

// External functions
Extern void toForm(T_EGraph *egraph);
Extern int  countPhiCon(int ex, int lp, int v4);

```

```
// Temporal definition: they should be replaced by FROM functions.
Local BigInt factorial(int n);
Local BigInt ipow(int n, int p);

// Local functions
Local int nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count);

Local void erEnd(const char *msg);
Local int  *newArray(int size, int val);
Local void deletArray(int *a);
Local void copyArray(int size, int *a0, int *a1);
Local int  **newMat(int n0, int n1, int val);
Local void deleteMat(int **m, int n0, int n1);
Local void printArray(int n, int *p);
Local void printMat(int n0, int n1, int **m);

#if DEBUGM
static int countNMC = 0;
static int countEG = 0;
static int countMG = 0;
#endif

//===============================================================
// class T_EGraph

T_EGraph::T_EGraph(int nnodes, int nedges, int mxdeg)
{
    int j;

    nNodes  = nnodes;
    nEdges  = nedges;
    maxdeg  = mxdeg;    // maximum value of degree of nodes
    nExtern = 0;

    nodes = new T_ENode[nNodes];
    edges = new T_EEdge[nEdges+1];
    for (j = 0; j < nNodes; j++) {
      nodes[j].deg   = 0;
      nodes[j].edges = new int[maxdeg];
    }
#if DEBUGM
    printf("+++ new    T_EGraph %d\n", ++countEG);
    if(countEG > 100) { exit(1); }
#endif
}

T_EGraph::~T_EGraph()
{
    int j;

    for (j = 0; j < nNodes; j++) {
      delete[] nodes[j].edges;
    }
    delete[] nodes;
    delete[] edges;
#if DEBUGM
    printf("+++ delete T_EGraph %d\n", countEG--);
#endif
}

// construct T_EGraph from adjacency matrix
void T_EGraph::init(int pid, long gid, int **adjmat, Bool sopi, BigInt nsm, BigInt esm)
{
    int n0, n1, ed, e, eext, eint;
//    Bool ok;

    pId  = pid;
    gId  = gid;
    opi  = sopi;
    nsym = nsm;
    esym = esm;

    for (n0 = 0; n0 < nNodes; n0++) {
        nodes[n0].deg = 0;
    }
    ed = 1;
    for (n0 = 0; n0 < nNodes; n0++) {
      for (e = 0; e < adjmat[n0][n0]/2; e++, ed++) {
        edges[ed].nodes[0] = n0;
        edges[ed].nodes[1] = n0;
        nodes[n0].edges[nodes[n0].deg++] = - ed;
        nodes[n0].edges[nodes[n0].deg++] =   ed;
        edges[ed].ext = nodes[n0].ext;
      }
      for (n1 = n0+1; n1 < nNodes; n1++) {
        for (e = 0; e < adjmat[n0][n1]; e++, ed++) {
          edges[ed].nodes[0] = n0;
```

```
          edges[ed].nodes[1] = n1;
          nodes[n0].edges[nodes[n0].deg++] = - ed;
          nodes[n1].edges[nodes[n1].deg++] =   ed;
          edges[ed].ext = (nodes[n0].ext || nodes[n1].ext);
        }
      }
    }
    if (CHECK) {
      if (ed != nEdges+1) {
        printf("*** T_EGraph::init: ed=%d != nEdges=%d\n", ed, nEdges);
        erEnd("*** T_EGraph::init: illegal connection\n");
      }
    }

    // name of momenta
    eext = 1;
    eint = 1;
    for (ed = 1; ed <= nEdges; ed++) {
      if (edges[ed].ext) {
        edges[ed].momn    = eext++;
        edges[ed].momc[0] = 'Q';
        edges[ed].momc[1] = ((char)0);
      } else {
        edges[ed].momn    = eint++;
        edges[ed].momc[0] = 'P';
        edges[ed].momc[1] = ((char)0);
      }
    }
}

// set external particle to node 'nd'
void T_EGraph::setExtern(int nd, Bool val)
{
    nodes[nd].ext = val;
}

// end of calling 'setExtern'
void T_EGraph::endSetExtern(void)
{
    int n;

    nExtern = 0;
    for (n = 0; n < nNodes; n++) {
      if (nodes[n].ext) {
        nExtern++;
      }
    }
}

// print the T_EGraph
void T_EGraph::print()
{
  int nd, lg, ed, nlp;

  nlp = nEdges - nNodes + 1;
  printf("\n");
  printf("T_EGraph: pId=%d gId=%ld nExtern=%d nLoops=%d nNodes=%d nEdges=%d\n",
         pId, gId, nExtern, nlp, nNodes, nEdges);
  printf("        sym = (%ld * %ld) maxdeg=%d\n", nsym, esym, maxdeg);
  printf("  Nodes\n");
  for (nd = 0; nd < nNodes; nd++) {
    if (nodes[nd].ext) {
      printf("    %2d Extern ", nd);
    } else {
      printf("    %2d Vertex ", nd);
    }
    printf("deg=%d [", nodes[nd].deg);
    for (lg = 0; lg < nodes[nd].deg; lg++) {
      printf(" %3d", nodes[nd].edges[lg]);
    }
    printf("]\n");
  }
  printf("  Edges\n");
  for (ed = 1; ed <= nEdges; ed++) {
    if (edges[ed].ext) {
      printf("    %2d Extern ", ed);
    } else {
      printf("    %2d Intern ", ed);
    }
    printf("%s%d", (edges[ed].ext)?"Q":"p", edges[ed].momn);
    printf(" [%3d %3d]\n", edges[ed].nodes[1], edges[ed].nodes[0]);
  }
}

//==============================================================
// class T_MNode : nodes in T_MGraph

```

```
T_MNode::T_MNode(int vid, int vdeg, Bool vext, int vclss)
{
    id     = vid;    // id of the node
    deg    = vdeg;   // degree of the node
    freelg = vdeg;   // number of free legs
    clss   = vclss;  // class to which the node belongs
    ext    = vext;   // external node or not
}

//==============================================================
//  class of node-classes for T_MGraph
//
class T_MNodeClass {
  public:
    int    nNodes;                  // the number of nodes
    int    nClasses;                // the number of classes
    int   *clist;                   // the number of nodes in each class
    int   *ndcl;                    // node --> class
    int  **clmat;                   // matrix used for classification
    int   *flist;                   // the first node in each class
    int    maxdeg;                  // maximal value of degree(node)
    int    forallignment;

    T_MNodeClass(int nnodes, int ncl);
    ~T_MNodeClass();
    void   init(int *cl, int mxdeg, int **adjmat);
    void   copy(T_MNodeClass* mnc);
    int    clCmp(int nd0, int nd1, int cn);
    void   printMat(void);

    void   mkFlist(void);
    void   mkNdCl(void);
    void   mkClMat(int **adjmat);
    void   incMat(int nd, int td, int val);
    Bool   chkOrd(int nd, int ndc, T_MNodeClass *cl, int *dtcl);
    int    cmpArray(int *a0, int *a1, int ma);
};

T_MNodeClass::T_MNodeClass(int nnodes, int ncl)
{
    nNodes   = nnodes;
    nClasses = ncl;
    clist    = new int[nClasses];
    ndcl     = new int[nNodes];
    clmat    = newMat(nNodes, nClasses, 0);
    flist    = new int[nClasses+1];

#if DEBUGM
    printf("+++ new    T_MNodeClass %d\n", ++countNMC);
    if(countNMC > 100) { exit(1); }
#endif
}

T_MNodeClass::~T_MNodeClass()
{
    delete[] clist;
    delete[] ndcl;
    deleteMat(clmat, nNodes, nClasses);
    delete[] flist;

#if DEBUGM
    printf("+++ delete T_MNodeClass %d\n", countNMC--);
#endif
}

void T_MNodeClass::init(int *cl, int mxdeg, int **adjmat)
{
    int j;

    for (j = 0; j < nClasses; j++) {
      clist[j] = cl[j];
    }
    maxdeg   = mxdeg;
    mkNdCl();
    mkClMat(adjmat);
    mkFlist();
}

void T_MNodeClass::copy(T_MNodeClass* mnc)
{
    int j, k;

    for (k = 0; k < nClasses; k++) {
      clist[k] = mnc->clist[k];
    }
    maxdeg   = mnc->maxdeg;
    for (j = 0; j < nNodes; j++) {
```

```
      ndcl[j] = mnc->ndcl[j];
      for (k = 0; k < nClasses; k++) {
        clmat[j][k] = mnc->clmat[j][k];
      }
    }
    for (k = 0; k < nClasses+1; k++) {
      flist[k] = mnc->flist[k];
    }
}

//  Construct flist
//    The set of nodes in class 'cl' is [flist[cl],...,flist[cl+1]-1]
void T_MNodeClass::mkFlist(void)
{
    int  j, f;

    f  = 0;
    for (j = 0; j < nClasses; j++) {
      flist[j] = f;
      f += clist[j];
    }
    flist[nClasses] = f;
}

//  Construct ndcl
//    ndcl[nd] = (the class id in which node 'nd' belongs)
void T_MNodeClass::mkNdCl(void)
{
    int  c, k;
    int  nd = 0;

    for (c = 0; c < nClasses; c++) {
      for (k = 0; k < clist[c]; k++) {
        ndcl[nd++] = c;
      }
    }
}

//  Comparison of two nodes 'nd0' and 'nd1'
//    Ordering is lexicographic (class, connection configuration)
int T_MNodeClass::clCmp(int nd0, int nd1, int cn)
{

    // Whether two nodes are in a same class or not.
    int cmp = ndcl[nd0] - ndcl[nd1];
    if (cmp != 0) {
      return cmp;
    }

    // Sign '-' signifies the reverse ordering
    cmp = - cmpArray(clmat[nd0], clmat[nd1], cn);
    if (cmp != 0) {
      return cmp;
    }
    // for particles ???
    return cmp;
}

//  Construct a matrix 'clmat[nd][tc]' which is the number
//  of edges connecting 'nd' and all nodes in class 'tc'.
void T_MNodeClass::mkClMat(int **adjmat)
{
    int  nd, td, tc;

    for (nd = 0; nd < nNodes; nd++) {
      for (td = 0; td < nNodes; td++) {
        tc = ndcl[td];                        // another node
        clmat[nd][tc] += adjmat[nd][td];     // the number of edges 'nd'--'td'
      }

    }
}

//  Increase the number of edges by 'val' between 'nd' and 'td'.
void T_MNodeClass::incMat(int nd, int td, int val)
{
    int tdc = ndcl[td];             // class of 'td'
    clmat[nd][tdc] += val;          // modify matrix 'clmat'.
}

//  Check whether the configuration satisfies the ordering condition or not.
Bool T_MNodeClass::chkOrd(int nd, int ndc, T_MNodeClass *cl, int *dtcl)
{
    Bool tcl[MAXNCLASSES];
//    Bool tcl[cl->nClasses];
    int  tn, tc, mxn, cmp, n;
    DUMMYUSE(nd);
```

```
    for (tc = 0; tc < cl->nClasses; tc++) {
        tcl[tc] = T_False;
    }
    for (tn = 0; tn < nNodes; tn++) {
      if (dtcl[tn] == 0) {
        tcl[cl->ndcl[tn]] = T_False;
      }
    }
    for (tc = 0; tc < cl->nClasses; tc++) {
      if (tcl[tc] && ndc != tc) {
        mxn = flist[tc+1];
        for (n = flist[tc]+1; n < mxn; n++) {
          cmp = - clmat[n-1][tc] + clmat[n][tc];
          if (cmp > 0) {
            return T_False;
          }
        }
      }
    }
    return T_True;
}

//  Print configuration matrix.
void T_MNodeClass::printMat(void)
{
    int j1, j2;

    cout « endl;

    // the first line
    cout « setw(2) « "nd" « ": " « setw(2) « "cl:   ";
    for (j2 = 0; j2 < nClasses; j2++) {
      cout « setw(2) « j2 « " ";
    }
    cout « endl;

    // print raw
    for (j1 = 0; j1 < nNodes; j1++) {
      cout « setw(2) « j1 « ": " « setw(2) « ndcl[j1] « ": [";
      for (j2 = 0; j2 < nClasses; j2++) {
        cout « " " « setw(2) « clmat[j1][j2];
      }
      cout « "] " « endl;
    }
}

int T_MNodeClass::cmpArray(int *a0, int *a1, int ma)
{
  for (int j = 0; j < ma; j++) {
    if (a0[j] < a1[j]) {
      return -1;
    } else if (a0[j] > a1[j]) {
      return 1;
    }
  }
  return 0;
}

//===============================================================
//  class T_MGraph : scalar graph expressed by matrix form

T_MGraph::T_MGraph(int pid, int ncl, int *cldeg, int *clnum, int *clext, Bool sopi)
{
    int nn, ne, j, k;

    // initial conditions
    nClasses = ncl;
    clist    = new int[ncl];

    mindeg   = -1;
    maxdeg   = -1;
    ne = 0;
    nn = 0;
    for (j = 0; j < nClasses; j++) {
      clist[j] = clnum[j];
      nn += clnum[j];
      ne += cldeg[j]*clnum[j];
      if (mindeg < 0) {
        mindeg = cldeg[j];
      } else {
        mindeg = min(mindeg, cldeg[j]);
      }
      if (maxdeg < 0) {
        maxdeg = cldeg[j];
      } else {
        maxdeg = max(maxdeg, cldeg[j]);
```

```
      }
    }
    if (ne % 2 != 0) {
        printf("Sum of degrees are not even\n");
        for (j = 0; j < nClasses; j++) {
          printf("class %2d: %2d %2d %2d\n",
                 j, cldeg[j], clnum[j], clext[j]);
        }
        erEnd("illegal degrees of nodes");
    }
    pId    = pid;
    nNodes = nn;
    nodes = new T_MNode*[nNodes];

    selOPI = sopi;

    nEdges = ne / 2;
    nLoops = nEdges - nNodes + 1;

    egraph = new T_EGraph(nNodes, nEdges, maxdeg);
    nn = 0;
    for (j = 0; j < nClasses; j++) {
      for (k = 0; k < clist[j]; k++, nn++) {
        nodes[nn] = new T_MNode(nn, cldeg[j], clext[j], j);
        egraph->setExtern(nn, clext[j]);
      }
    }
    egraph->endSetExtern();

    // generated set of graphs
    ndiag     = 0;
    n1PI      = 0;

    // the current graph
    adjMat    = newMat(nNodes, nNodes, 0);
    nsym      = ToBigInt(0);
    esym      = ToBigInt(0);
    c1PI      = 0;
    wsum   = ToFraction(0, 1);
    wsopi = ToFraction(0, 1);

    // current node classification
    curcl     = new T_MNodeClass(nNodes, nClasses);

    // measures of efficiency
    ngen      = 0;
    ngconn    = 0;

    if (MONITOR) {
      nCallRefine    = 0;
      discardOrd     = 0;
      discardRefine  = 0;
      discardDisc    = 0;
      discardIso     = 0;
    }

    // work space for isIsomorphic
    modmat = newMat(nNodes, nNodes, 0);
    permp  = newArray(nNodes, 0);
    permq  = newArray(nNodes, 0);
    permr  = newArray(nNodes, 0);

    // work space for bisearchM
    bidef = newArray(nNodes, 0);
    bilow = newArray(nNodes, 0);
    bicount = 0;
    DUMMYUSE(padding);

#if DEBUGM
    printf("+++ new    T_MGraph %d\n", ++countMG);
    if(countEG > 100) { exit(1); }
#endif
}

T_MGraph::~T_MGraph()
{
    int j;

    deletArray(bilow);
    deletArray(bidef);
    deletArray(permr);
    deletArray(permq);
    deletArray(permp);

    deleteMat(modmat, nNodes, nNodes);
    delete curcl;
    deleteMat(adjMat, nNodes, nNodes);
```

```
    delete egraph;
    for (j = 0; j < nNodes; j++) {
      delete nodes[j];
    }
    delete[] nodes;
    delete[] clist;

#if DEBUGM
    printf("+++ delete T_MGraph %d\n", countMG++);
#endif
}


void T_MGraph::printAdjMat(T_MNodeClass *cl)
{
    int j1, j2;

    cout « "      ";
    for (j2 = 0; j2 < nNodes; j2++) {
      cout « " " « setw(2) « j2;
    }
    cout « endl;
    for (j1 = 0; j1 < nNodes; j1++) {
      cout « setw(2) « j1 « ": [";
      for (j2 = 0; j2 < nNodes; j2++) {
        cout « " " « setw(2) « adjMat[j1][j2];
      }
      cout « "] " « cl->ndcl[j1] « endl;
    }
}

//---------------------------------------------------------------
//  Check graph can be a connected one.
//  If a connected component without free leg is not the whole graph then
//  return T_False, otherwise return T_True.
Bool T_MGraph::isConnected(void)
{
    int j, n, nv;

    for (j = 0; j < nNodes; j++) {
      nodes[j]->visited = -1;
    }
    if (visit(0)) {
      return T_True;
    }
    nv = 0;
    for (n = 0; n < nNodes; n++) {
      if (nodes[n]->visited >= 0) {
        nv++;
      }
    }
    return (nv == nNodes);
}

//  Visiting connected node used for 'isConnected'
//  If child nodes has free legs, then this function returns T_True.
//  otherwise it returns T_False.
Bool T_MGraph::visit(int nd)
{
  int td;

  // This node has free legs.
  if (nodes[nd]->freelg > 0) {
    return T_True;
  }
  nodes[nd]->visited = 0;
  for (td = 0; td < nNodes; td++) {
    if ((adjMat[nd][td] > 0) and (nodes[td]->visited < 0)) {
      if (visit(td)) {
        return T_True;
      }
    }
  }
  // all the child nodes has no free legs.
  return T_False;
}

//  Check whether the current graph is the Representative of a isomorphic class.
//     nsym = symmetry factor by the permutation of nodes.
//     esym = symmetry factor by the permutation of edge.
//  If this graph is not a representative, then returns T_False.
Bool T_MGraph::isIsomorphic(T_MNodeClass *cl)
{
    int j1, j2, cmp, count, nself;

    nsym = ToBigInt(0);
    esym = ToBigInt(1);
```

```
    count = 0;
    while (T_True) {
      count = nextPerm(nNodes, cl->nClasses, cl->clist,
                       permr, permq, permp, count);

      if (count < 0) {
        // calculate permutations of edges
        esym = ToBigInt(1);
        for (j1 = 0; j1 < nNodes; j1++) {
          if (adjMat[j1][j1] > 0) {
            nself = adjMat[j1][j1]/2;
            esym *= factorial(nself);
            esym *= ipow(2, nself);
          }
          for (j2 = j1+1; j2 < nNodes; j2++) {
            if (adjMat[j1][j2] > 0) {
                esym *= factorial(adjMat[j1][j2]);
            }
          }
        }
        return T_True;
      }

      permMat(nNodes, permp, adjMat, modmat);
      cmp = compMat(nNodes, adjMat, modmat);
      if (cmp < 0) {
        return T_False;
      } else if (cmp == 0) {
        // save permutation ???
        nsym = nsym + ToBigInt(1);
      }
    }
}

// apply permutation to matrix 'mat0' and obtain 'mat1'
void T_MGraph::permMat(int size, int *perm, int **mat0, int **mat1)
{
    int j1, j2;

    for (j1 = 0; j1 < size; j1++) {
        for (j2 = 0; j2 < size; j2++) {
            mat1[j1][j2] = mat0[perm[j1]][perm[j2]];
        }
    }
}

// comparison of matrix
int T_MGraph::compMat(int size, int **mat0, int **mat1)
{
    int j1, j2, cmp;

    for (j1 = 0; j1 < size; j1++) {
        for (j2 = 0; j2 < size; j2++) {
            cmp = mat0[j1][j2] - mat1[j1][j2];
            if (cmp != 0) {
                return cmp;
            }
        }
    }
    return 0;
}


//  Refine the classification
//    cl : the current 'T_MNodeClass' object
//    cn : the class number
//  Returns (the new class number corresponds to 'cn') if OK, or
//            'None' if ordering condition is not satisfied
T_MNodeClass *T_MGraph::refineClass(T_MNodeClass *cl, int cn)
{
    T_MNodeClass *ccl = cl;
    int          ccn = cn;
    T_MNodeClass *ncl = NULL;
    int          ucl[MAXNODES];
    int          nucl, nce;
    int          td, cmp;

    nucl = 0;
    while (ccl->nClasses != nucl) {    // repeat refinement.
      nce  = 0;
      nucl = 0;
      for (td = 1; td < nNodes; td++) {
        // 'td' is the next node and the current node is 'td-1'.
        // Count up the number of the elements in the current class
        // corresponding to the current node
        nce++;
```

```
        cmp = ccl->clCmp(td-1, td, ccn);

        // the ordering condition is not satisfied.
        if (cmp > 0) {
          if (DEBUG) {
            cout « "refine: cls = " « cl->clist « endl;
            cl->printMat();
            cout « "clmat" « endl;
            ccl->printMat();
            cout « "refine: discard: cls = " « ccl->clist
                 « "ucl = " « ucl « endl;
          }
          if (ccl != cl) {
            delete ccl;
          }
          return NULL;

        } else if (cmp < 0) {
          // 'td' is in the next class to the current node.
          ucl[nucl++] = nce;       // close the current class

          // start new class
          nce = 0;
        }
        // nothing to do for the case of 'cmp == 0'.

      }

      // close array 'ucl'.
      ucl[nucl++] = nce + 1;

      // class is not modified
      if (nucl == ccl->nClasses) {
        ncl = ccl;
        break;

      // inconsistent
      } else if (nucl < ccl->nClasses) {
        erEnd("refineClasses : smaller number of classes");

      // preparation of the next repetition.
      } else {
        if (cn == ccl->nClasses) {
          td = nNodes;
        } else {
          td = ccl->flist[cn+1]-1;
        }
        if (ccl != cl) {
          delete ccl;
        }
        ccl = new T_MNodeClass(nNodes, nucl);
        ccl->init(ucl, maxdeg, adjMat);
        if (td == nNodes) {
          ccn = ccl->nClasses;
        } else {
          ccn = ccl->ndcl[td];
        }
        nucl = 0;
      }
    }

    if (DEBUG) {
      cout « "refine: ucl = " « ucl « endl;
      cout « "refine: ncl = " « ncl->clist « endl;
      ncl->printMat();
    }

    return ncl;
}

//  Search biconnected component
//    visit : pd --> nd --> td
//    ne : the number of edges between pd and nd.
void T_MGraph::bisearchM(int nd, int pd, int ne)
{
    int td;

    bidef[nd] = bicount;
    bilow[nd] = bicount;
    bicount++;

    for (td = 0; td < nNodes; td++) {
      if (nodes[td]->ext) {  // ignore external node
        continue;

      } else if (td == nd) {         // pd --> nd --> nd
        continue;
```

```

      } else if (td == pd) {       // pd --> nd --> pd
        if (ne > 1) {
            bilow[nd] = min(bilow[nd], bidef[pd]);
        }

      } else if (adjMat[td][nd] < 1) {  // td is not adjacent to nd
        continue;

      } else if (bidef[td] >= 0) {  // back edge
        bilow[nd] = min(bilow[nd], bidef[td]);

      // new node
      } else {

        bisearchM(td, nd, adjMat[td][nd]);

        // ordinary case
        if (bilow[td] > bidef[nd]) {
          nBridges++;
        }
        bilow[nd] = min(bilow[nd], bilow[td]);
      }
    }

    // nd is the starting point and not an external line
    // if (pd < 0 && (!nodes[nd]->ext || nodes[nd]->deg > 1)) {
    // if (pd < 0 && !(nodes[nd]->ext && nodes[nd]->deg ==1)) {
    if (pd < 0 && nodes[nd]->deg != 1) {
      // nBridges += nodes[nd]->deg;
      nBridges++;
    }
}

//  Count the number of 1PI components.
//  The algorithm is described in
//  A.V. Aho, J.E. Hopcroft and J.D. Ullman
//   'The Design and Analysis of Computer Algorithms', Chap. 5
//   1974, Addison-Wesley.

int T_MGraph::count1PI(void)
{
    int j;

    if (nLoops < 0) {
      return 1;
    }

    // initialization
    bicount  = 0;
    nBridges = 0;
    for (j = 0; j < nNodes; j++) {
      bidef[j] = -1;
      bilow[j] = -1;
    }

    bisearchM(0, -1, 0);

    return nBridges;
}

//---------------------------------------------------------------
//  Generate graphs
//    the generation process starts from 'connectClass'.

long T_MGraph::generate(void)
{
    T_MNodeClass *cl;
    int dscl[MAXNODES];
    int n;

    for (n = 0; n < nNodes; n++) {
      dscl[n] = T_False;
    }

    // Initial classification of nodes.
    cl = new T_MNodeClass(nNodes, nClasses);
    cl->init(clist, maxdeg, adjMat);
    connectClass(cl, dscl);

    // Print the result.

    if (MONITOR) {
    cout « endl;
    cout « endl;
    cout « "* Total " « ndiag « " Graphs.";
    cout « "(" « n1PI « " 1PI)";
```

```
    // cout « " wsum = " « wsum « "(" « wsopi « "1PI)" « endl;
    cout « endl;
    }

    if (MONITOR) {
      cout « "* refine:                     " « nCallRefine « endl;
      cout « "* discard for ordering:      " « discardOrd « endl;
      cout « "* discard for refinement:  " « discardRefine « endl;
      cout « "* discard for disconnected: " « discardDisc « endl;
      cout « "* discard for duplication:  " « discardIso « endl;
    }
    delete cl;

    return ndiag;
}

int T_MGraph::findNextCl(T_MNodeClass *cl, int *dscl)
{
    int mine, cr, c, n, me;

    mine = -1;
    cr   = -1;
    for (c = 0; c < cl->nClasses; c++) {
      n = cl->flist[c];
      if (!dscl[n]) {
        me = nodes[n]->freelg;
        if (me > 0) {
          if (nodes[n]->freelg < nodes[n]->deg) {
            return c;
          }
          if (mine < 0 || mine > me) {
            mine = me;
            cr = c;
          }
        }
      }
    }
    return cr;
}

int T_MGraph::findNextTCl(T_MNodeClass *cl, int *dtcl)
{
    int c, n, mine, cr, me;

    mine = -1;
    cr   = -1;
    for (c = 0; c < cl->nClasses; c++) {
      for (n = cl->flist[c]; n < cl->flist[c+1]; n++) {
        if (!dtcl[n]) {
          me = nodes[n]->freelg;
          if (me > 0) {
            if (nodes[n]->freelg < nodes[n]->deg) {
              return c;
            }
            if (mine < 0 || mine > me) {
              mine = me;
              cr = c;
            }
          }
        }
      }
    }
    return cr;
}

// Connect nodes in a class to others
void T_MGraph::connectClass(T_MNodeClass *cl, int *dscl)
{
    int sc, sn;
    T_MNodeClass *xcl;

    if (DEBUG0) {
      printf("connectClass:begin:");
      printf(" dscl="); printArray(nNodes, dscl);
      printf("\n");
    }
    xcl = refineClass(cl, cl->nClasses);

    if (xcl == NULL) {
      if (MONITOR) {
        discardRefine++;
      }
    } else {
      sc = findNextCl(xcl, dscl);
      if (sc < 0) {
        newGraph(cl);
      } else {
```

```
        sn = xcl->flist[sc];
        connectNode(sc, sn, xcl, dscl);
      }
    }
    if (xcl != cl && xcl != NULL) {
      delete xcl;
    }
    if (DEBUG0) {
      printf("connectClass:end\n");
    }
}

//------------------------------------------------------------
void T_MGraph::connectNode(int sc, int ss, T_MNodeClass *cl, int *dscl)
{
    int sn;
    int dtcl[MAXNODES];

    if (DEBUG0) {
      printf("connectNode:begin:(%d,%d)", sc, ss);
      printf(" dscl="); printArray(nNodes, dscl);
      printf("\n");
    }
    if (ss >= cl->flist[sc+1]) {
      connectClass(cl, dscl);
      if (DEBUG0) {
        printf("connectNode:end1\n");
      }
      return;
    }

    copyArray(nNodes, dscl, dtcl);
    for (sn = ss; sn < cl->flist[sc+1]; sn++) {
      if (!dscl[sn]) {
        connectLeg(sc, sn, sc, sn, cl, dscl, dtcl);
        if (DEBUG0) {
          printf("connectNode:end2\n");
        }
        return;
      }
    }
    if (DEBUG0) {
      printf("connectNode:end3\n");
    }
}

//  Add one connection between two legs.
//    1. select another node to connect
//    2. determine multiplicity of the connection
//
//   Arguments
//     cn : the current class
//     nd : the current node to be connected.
//     nextnd : {nextnd, ...} is the possible target node of the connection.
//     cl     : the current node class

void T_MGraph::connectLeg(int sc, int sn, int tc, int ts, T_MNodeClass *cl, int *dscl, int* dtcl)
{
    int tn, maxself, nc2, nc, maxcon, ts1, wc, ncm;
    int dtcl1[MAXNODES];

    if (DEBUG0) {
      printf("connectLeg:begin:(%d,%d,%d,%d)", sc, sn, tc, ts);
      printf(" dscl="); printArray(nNodes, dscl);
      printf(" dtcl="); printArray(nNodes, dtcl);
      printf("\n");
      printAdjMat(cl);
    }

    if (sn >= cl->flist[sc+1]) {
      erEnd("*** connectLeg : illegal control");
      return;

    // There remains no free legs in the node 'sn' : move to next node.
    } else if (nodes[sn]->freelg < 1) {

      if (!isConnected()) {
        if (DEBUG0) {
          printf("connectLeg:disconnected\n");
        }
        if (MONITOR) {
          discardDisc++;
        }
      } else {
        if (DEBUG0) {
          printf("connectLeg: call conNode\n");
        }
```

```
        dscl[sn] = T_True;

        // next node in the current class.
        connectNode(sc, sn+1, cl, dscl);

        dscl[sn] = T_False;
      }

    // connect a free leg of the current node 'sn'.
    } else {
      copyArray(nNodes, dtcl, dtcl1);

      ts1 = ts;
      for (wc = 0; wc < nNodes; wc++) {
        if (ts1 >= cl->flist[tc+1]) {
          tc = findNextTCl(cl, dtcl1);
          if (DEBUG0) {
            printf("connectLeg:1:tc=%d\n", tc);
          }
          if (tc < 0) {
            if (DEBUG0) {
              printf("connectLeg:end1:(%d,%d,%d,%d)\n", sc, sn, tc, ts);
            }
            return;
          }
          if (tc != sc && !cl->chkOrd(sn, sc, cl, dtcl)) {
            if (MONITOR) {
              discardOrd++;
            }
            if (DEBUG0) {
              printf("connectLeg:end2:(%d,%d,%d,%d)\n", sc, sn, tc, ts);
            }
            return;
          }
        }

        ts1 = -1;

        // repeat for all possible target nodes
        // for all tn in tc
        if (DEBUG0) {
          printf("connectLeg:2:tc=%d, fl:%d-->%d\n", tc, cl->flist[tc], cl->flist[tc+1]);
        }
        for (tn = cl->flist[tc]; tn < cl->flist[tc+1]; tn++) {
          if (DEBUG0) {
            printf("connectLeg:2:%d=>try %d\n", sn, tn);
          }

          if (dtcl1[tn]) {
            if (DEBUG0) {
              printf("connectLeg:3\n");
            }
            continue;
          }
          dtcl1[tn] = T_True;

          if (sc == tc && sn > tn) {
            if (DEBUG0) {
              printf("connectLeg:4\n");
            }
            continue;

          // self-loop
          } else if (sn == tn) {
            if (nNodes > 1) {
              // there are two or more nodes in the graph :
              // avoid disconnected graph
              maxself = min((nodes[sn]->freelg)/2, (nodes[sn]->deg-1)/2);
            } else {
              // there is only one node the graph.
              maxself = nodes[sn]->freelg/2;
            }

            // If we can assume no tadpole, the following line can be used.
            if (selOPI and nNodes > 2) {
               maxself = min((nodes[sn]->deg-2)/2, maxself);
            }

            // vary the number of connection.
            for (nc2 = maxself; nc2 > 0; nc2--) {
              // if nNodes > 1 and ndg == nc2:
              //    // disconnected
              //    continue
              nc = 2*nc2;
              if (DEBUG0) {
                printf("connectLeg: call conLeg: (same) %d=>%d(%d)\n", sn, tn,nc);
              }
```

```
              ncm = 5*nc;
              adjMat[sn][sn] = nc;
              nodes[sn]->freelg -= nc;
              cl->incMat(sn, tn, ncm);
              ts1 = tn + 1;

              // next connection
              connectLeg(sc, sn, tc, ts1, cl, dscl, dtcl1);

              // restore the configuration
              cl->incMat(tn, sn, - ncm);
              adjMat[sn][sn] = 0;
              nodes[sn]->freelg += nc;
              if (DEBUG0) {
                printf("connectLeg: ret  conLeg: (same) %d=>%d(%d)\n", sn, tn,nc);
              }
            }

          // connections between different nodes.
          } else {
            // maximum possible connection number
            maxcon = min(nodes[sn]->freelg, nodes[tn]->freelg);

            // avoid disconnected graphs.
            if (nNodes > 2 && nodes[sn]->deg == nodes[tn]->deg) {
              maxcon = min(maxcon, nodes[sn]->deg-1);
            }

            if (CHECK) {
              if ((adjMat[sn][tn] != 0) || (adjMat[sn][tn] != adjMat[tn][sn])) {
                printf("*** inconsistent connection: sn=%d, tn=%d", sn, tn);
                printf(" dscl="); printArray(nNodes, dscl);
                printf(" dtcl="); printArray(nNodes, dtcl);
                printf("\n");
                printAdjMat(cl);
                erEnd("*** inconsistent connection ");
              }
            }

            // vary number of connections
            for (nc = maxcon; nc > 0; nc--) {
              if (DEBUG0) {
                printf("connectLeg: call conLeg: (diff) %d=>%d(%d)", sn, tn,nc);
                printf(" dtcl="); printArray(nNodes, dtcl);
                printf("\n");
              }
              adjMat[sn][tn] = nc;
              adjMat[tn][sn] = nc;
              nodes[sn]->freelg -= nc;
              nodes[tn]->freelg -= nc;
              cl->incMat(sn, tn, nc);
              cl->incMat(tn, sn, nc);
              ts1 = tn + 1;

              // next connection
              connectLeg(sc, sn, tc, ts1, cl, dscl, dtcl1);

              // restore configuration
              cl->incMat(sn, tn, - nc);
              cl->incMat(tn, sn, - nc);
              adjMat[sn][tn] = 0;
              adjMat[tn][sn] = 0;
              nodes[sn]->freelg += nc;
              nodes[tn]->freelg += nc;
              if (DEBUG0) {
                printf("connectLeg: ret  conLeg: (diff) %d=>%d(%d)\n", sn, tn,nc);
                printf(" dtcl="); printArray(nNodes, dtcl);
                printAdjMat(cl);
                printf("\n");
              }
            }
          }
        }
        if (ts1 < 0) {
          ts1 = cl->flist[tc+1];
        }
      } // for wc
    }
    if (DEBUG0) {
      printf("connectLeg:end3:(%d,%d,%d,%d)\n", sc, sn, tc, ts);
    }
}

//  A new candidate diagram is obtained.
//  It may be a disconnected graph

void T_MGraph::newGraph(T_MNodeClass *cl)
```

```
{
    int connected;
    T_MNodeClass *xcl;
    Bool sopi;

    ngen++;
    // printf("newGraph: %d\n", ngen);

    // refine class and check ordering condition
    xcl = refineClass(cl, cl->nClasses);
    if (xcl == NULL) {
      // printf("newGraph: fail refine\n");
      if (MONITOR) {
        discardRefine++;
      }

    // check whether this is a connected graph or not.
    } else {
      connected = isConnected();
      if (!connected) {
        // printf("newGraph: not connected\n");
        if (DEBUG0) {
          cout « "+++ disconnected graph" « ngen « endl;
          xcl->printMat();
        }

        if (MONITOR) {
          discardDisc++;
        }

      // connected graph : check isomorphism of the graph
      } else {
        ngconn++;
        if (!isIsomorphic(xcl)) {
          // printf("newGraph: not isomorphic\n");
          if (DEBUG) {
            cout « "+++ duplicated graph" « ngen « ngconn « endl;
            xcl->printMat();
          }

          if (MONITOR) {
            discardIso++;
          }

        // We got a new connected and a unique representation of a class.
        } else {

          c1PI =  count1PI();
          if (!selOPI || c1PI == 1) {
            wsum = wsum + ToFraction(1, nsym*esym);
            if (c1PI == 1) {
               wsopi = wsopi + ToFraction(1, nsym*esym);
            }
            curcl->copy(xcl);
            ndiag++;
            sopi = (c1PI == 1);
            if (sopi) {
              n1PI++;
            }

            if (OPTPRINT) {
              cout « endl;
              cout « "Graph :" « ndiag
                   « "(" « ngen « ")"
                   « " 1PI comp. = " « c1PI
                   « " sym. factor = (" « nsym « "*" « esym « ")"
                   « endl;
              printAdjMat(cl);
              // cl->printMat();
              if (MONITOR) {
                cout « "refine:                    " « nCallRefine « endl;
                cout « "discard for ordering:      " « discardOrd « endl;
                cout « "discard for refinement:  " « discardRefine « endl;
                cout « "discard for disconnected: " « discardDisc « endl;
                cout « "discard for duplication:  " « discardIso « endl;
              }
            }

            // go to next step
            egraph->init(pId, ndiag, adjMat, sopi, nsym, esym);
            toForm(egraph);
          }
        }
      }
    }
// printf("in newGraph cl=%p, xcl=%p\n", cl, xcl);
    if (xcl != cl && xcl != NULL) {
```

```
        delete xcl;
    }
}


//  go to next step
//  1. convert data format which treat the edges as objects.
//  2. definition of loop momenta to the edges.
//  3. analysis of loop structure in a graph.
//  4. assignment of particles to edges and vertices to nodes
//  5. recalculate symmetry factor

// Permutation
Local int nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count)
{
    int  j, k, n, e, t;
    Bool b;

    for (j = 0; j < nelem; j++) {
        p[j] = j;
    }
    if (count < 1) {
        for (j = 0; j < nelem; j++) {
            q[j] = 0;
            r[j] = 0;
        }
        j = 0;
        for (k = 0; k < nclass; k++) {
            n = cl[k];
            for (e = 0; e < n; e++) {
                r[j] = n - e - 1;
                j++;
            }
        }
        if (j != nelem) {
            erEnd("*** inconsistent # elements");
        }
        return 1;
    }
    b = T_False;
    for (j = nelem-1; j >= 0; j--) {
        if (q[j] < r[j]) {
            for (k = j+1; k < nelem; k++) {
                q[k] = 0;
            }
            q[j]++;
            b = T_True;
            break;
        }
    }
    if (!b) {
        return (-count);
    }

    for (j = 0; j < nelem; j++) {
        k = j + q[j];
        t = p[j];
        p[j] = p[k];
        p[k] = t;
    }
    return count + 1;
}

Local BigInt factorial(int n)
/* return (n < 1) ? 1 : n!
 */
{
    int r, j;

    r = 1;
    for (j = 2; j <= n; j++) {
        r *= j;
    }
    return r;
}

Local BigInt ipow(int n, int p)
{
    int r, j;
    DUMMYUSE(p);

    r = 1;
    for (j = 2; j <= n; j++) {
        r *= n;
    }
    return r;
}
```

```


Local void erEnd(const char *msg)
{
    printf("*** Error : %s\n", msg);
    exit(1);
}

/* memory allocation */
Local int   *newArray(int size, int val)
{
    int *a, j;

    a = new int[size];
    for (j = 0; j < size; j++) {
        a[j] = val;
    }
    return a;
}

Local void deletArray(int *a)
{
    delete[] a;
}

Local void copyArray(int size, int *a0, int *a1)
{
    int j;
    for (j = 0; j < size; j++) {
        a1[j] = a0[j];
    }
}

Local int  **newMat(int n0, int n1, int val)
{
    int **m, j;

    m = new int*[n0];
    for (j = 0; j < n0; j++) {
        m[j] = newArray(n1, val);
    }
    return m;
}

Local void  deleteMat(int **m, int n0, int n1)
{
    int j;
    DUMMYUSE(n1);

    for (j = 0; j < n0; j++) {
        deletArray(m[j]);
    }
    delete[] m;
}

//=============================================================
// Functions for testing the program.
//
//-------------------------------------------------------------
// Testing function nextPerm
//
Local void printArray(int n, int *p)
{
    int j;

    printf("[");
    for (j = 0; j < n; j++) {
        printf(" %2d", p[j]);
    }
    printf("]");
}

Local void printMat(int n0, int n1, int **m)
{
    int j;

    for (j = 0; j < n0; j++) {
        printArray(n1, m[j]);
        printf("\n");
    }
}

Global void testPerm()
{
    int nelem, nperm, nclist, n, count;
    int *p, *q, *r;
    int clist[] = {1, 2, 2, 3};
```

```

    nclist = sizeof(clist)/sizeof(int);
    nelem = 0;
    nperm = 1;
    for (n = 0; n < nclist; n++) {
        nelem += clist[n];
        nperm *= factorial(clist[n]);
    }

    printf("+++ clist = (%d) ", nclist);
    printArray(nclist, clist);
    printf("\n");
    printf("+++ nelem = %d, nperm = %d\n", nelem, nperm);

    p  = new int[nelem];
    q  = new int[nelem];
    r  = new int[nelem];
    count = 0;
    while (T_True) {
        count = nextPerm(nelem, nclist, clist, r, q, p, count);
        if (count < 0) {
            count = - count;
            break;
        }
        printf("%4d:", count);
        printArray(nelem, p);
        printf("\n");

        if (count > nperm) {
            break;
        }
    }
    if (count != nperm) {
        printf("*** %d != %d\n", count, nperm);
    }
    delete[] p;
    delete[] q;
    delete[] r;
}

// } ) ]

```

4.56 gentopo.h

```
#pragma once

// Temporal definition of big integer
#define BigInt long
#define ToBigInt(x)   ((BigInt) (x))
//
// Temporal definition of ratio of two big integers
#define Fraction double
#define ToFraction(x, y)   (((double) (x))/((double) (y)))

//==============================================================
// Output to FROM

//==============================================================
// class of nodes for T_EGraph

class T_ENode {
  public:
    int  deg;
    int  ext;
    int *edges;
};

class T_EEdge {
  public:
    int  nodes[2];
    int  ext;
    int  momn;
    char momc[2];        // no more used
    // momentum is printed like: ("%s%d", (enode.ext)?"Q":"p", enode.momn)
    char padding[6];
};

class T_EGraph {
  public:
    long gId;
    int  pId;
    int nNodes;
```

```
    int nEdges;
    int maxdeg;
    int nExtern;

    int opi;
    BigInt nsym, esym;

    T_ENode *nodes;
    T_EEdge *edges;

    T_EGraph(int nnodes, int nedges, int mxdeg);
    ~T_EGraph();
    void print(void);
    void init(int pid, long gid, int **adjmat, int sopi, BigInt nsym, BigInt esym);

    void setExtern(int nd, int val);
    void endSetExtern(void);
};

//===============================================================
// class of nodes for T_MGraph

class T_MNode {
  public:
    int id;                   // node id
    int deg;                  // degree(node) = the number of legs
    int clss;                 // initial class number in which the node belongs
    int ext;

    int freelg;               // the number of free legs
    int visited;

    T_MNode(int id, int deg, int ext, int clss);
};

//================================================================
//  class of node-classes for T_MGraph
class T_MNodeClass;

//================================================================
//  class of scalar graph expressed by matrix form
//
//  Input  : the classified set of nodes.
//  Output : control passed to 'T_EGraph(self)'

class T_MGraph {

  public:

    // initial conditions
    int pId;                   // process/subprocess ID
    int nNodes;                // the number of nodes
    int nEdges;                // the number of edges
    int nLoops;                // the number of loops
    T_MNode **nodes;              // table of T_MNode object
    int *clist;                 // list of initial classes
    int nClasses;              // the number of initial classes
    // int ndcl;                  // node --> initial class number
    int mindeg;                // minimum value of degree of nodes
    int maxdeg;                // maximum value of degree of nodes

    int selOPI;               // flag to select 1PI graphs

    // generated set of graphs
    long ndiag;                 // the total number of generated graphs
    long n1PI;                  // the total number of 1PI graphs
    int  nBridges;              // the number of bridges

    // the current graph
    int c1PI;                   // the number of 1PI components
    int **adjMat;               // adjacency matrix
    BigInt nsym;                   // symmetry factor from nodes
    BigInt esym;                   // symmetry factor from edges
    Fraction wsum;             // weighted sum of graphs
    Fraction wsopi;            // weighted sum of 1PI graphs
    T_MNodeClass *curcl;          // the current 'T_MNodeClass' object
    T_EGraph *egraph;

    // measures of efficiency
    long ngen;                  // generated graph before check
    long ngconn;                // generated connected graph before check

    long nCallRefine;
    long discardOrd;
    long discardRefine;
    long discardDisc;
    long discardIso;
```

```
    // functions */
    T_MGraph(int pid, int ncl, int *cldeg, int *clnum, int *clext, int sopi);
    ~T_MGraph();
    long   generate(void);

  private:
    // work space for isomorphism
    int **modmat;           // permutated adjacency matrix
    int *permp;
    int *permq;
    int *permr;

    // work space for biconnected component
    int *bidef;
    int *bilow;
    int  bicount;
    int  padding;

    void   printAdjMat(T_MNodeClass *cl);
    int    isConnected(void);
    int    visit(int nd);
    int    isIsomorphic(T_MNodeClass *cl);
    void   permMat(int size, int *perm, int **mat0, int **mat1);
    int    compMat(int size, int **mat0, int **mat1);
    T_MNodeClass *refineClass(T_MNodeClass *cl, int cn);
    void   bisearchM(int nd, int pd, int ne);
    int    count1PI(void);

    int    findNextCl(T_MNodeClass *cl, int *dscl);
    int    findNextTCl(T_MNodeClass *cl, int *dcl);
    void   connectClass(T_MNodeClass *cl, int *dscl);
    void   connectNode(int sc, int ss, T_MNodeClass *cl, int *dscl);
    void   connectLeg(int sc, int sn, int tc, int ts, T_MNodeClass *cl, int *dscl, int* dtcl);
    void   newGraph(T_MNodeClass *cl);

};
```

4.57 grcc.cc

```
// { ( [
//**************************************************************
// grcc.cc

#ifndef NOFORM
extern "C" {
#include "form3.h"
}
#endif

#include <iostream>
#include <iomanip>
#include <sstream>
#include <cstdio>
#include <stdlib.h>
#include <string.h>
#include <time.h>

// #define DEBUG9

// optimization : best combination depends on process by process
#define SIMPSEL

// optimization : lower and upper bound of deg of target node of connection.
#define MINMAXLEG

// optimization : optimization with respect to 'extonly' particles
#define OPTEXTONLY

// check unique interaction by name or by code
#define UNIQINTR

// possible extensions of the program
// #define ORBITS

// reverse direction of an edge of agraph when anti-particle flows on it.
// In this case the direction of edges will be diffrent
// from ones of original mgraph.
// #define EDGEPORDER

//-------------------------------------------------------------
#include "grcc.h"

```

```
#ifdef GRCC_NAMESPACE
using namespace Grcc;
#endif

//--------------------------------------------------------------
// Macro functions
#define CLWIGHTD(x)    (5*(x))
#define CLWIGHTO(x)    (3*(x)-2)

//--------------------------------------------------------------
// Static variables

static OptDef optDef[] = {
 {"Step",        "Generate particle assigned graphs",               GRCC_AGraph, 0},
 {"Outgrf",      "Output to file (out.grf)",                              False, 0},
 {"Outgrp",      "Output to file (out.grp)",                              False, 0},
 {"1PI",         "Only 1PI graphs",                                       True , 0},
 {"NoSelfLoop",  "Exclude graphs with loops consist of 1 edge",           True , 0},
 {"NoTadpole",   "Exclude graphs with tadpoles (2 edge connected)",       True , 0},
 {"No1PtBlock",  "Exclude graphs with tadpole blocks (2 node connected)", False, 0},
 {"No2PtL1PI",   "Exclude graphs with 2-point subgraphs",                 False, 0},
 {"NoExtSelf",   "Exclude graphs with 2-pt subgraphs at ext. particles",  False, 0},
 {"NoAdj2PtV",   "Exclude graphs with an edge connecting 2-pt vertices",  False, 0},
 {"Block",       "Exclude graphs with more than one block",               False, 0},
 {"SymmInitial","Symmetrize initial particles",                           False, 0},
 {"SymmFinal",   "Symmetrize final particles",                            False, 0},
};

static int nOptDef = sizeof(optDef)/sizeof(OptDef);

static int      prlevel = 2;
static ErExit  *erExit  = NULL;
static void    *erExitArg  = NULL;

//**************************************************************
// Utility functions
//==============================================================

static void   erEnd(const char *msg);
static Bool   nextPart(int nelem, int nclist, int *clist, int *nl, int *r);
static void   prilist(int n, const int *a, const char *msg);
static int   *newArray(int size, int val);
static int   *deleteArray(int *a);
static int  **newMat(int n0, int n1, int val);
static int  **deleteMat(int **m, int n0);
static void   prIntArray(int n, int *p, const char *msg);
static void   prIntArrayErr(int n, int *p, const char *msg);
static int   *intdup(int n, int *v);
static int   *delintdup(int *v);
static void   bsort(int n, int *ord, int *a);
static void   sorti(int n, int *a);
static int    sortb(int n, int *a);
static int    toSList(int n, int *a);
static int    intSetAdd(int n, int *a, int v, const int size);
static int    intSListAdd(int n, int *a, int v, const int size);
static int    cmpArray(int na, int *a, int nb, int *b);
static Bool   leqArray(int n, int *a, int *b);
static void   prMomStr(int mom, const char *ms, int mn);
static void   listDiff(int na, int *a, int nb, int *b, int *np, int *p, int *nq, int *q, int *nr, int *r);
static Bool   isSubSList(int na, int *a, int nb, int *b);
static int    subtrSet(int na, int *a, int nb, int *b, int *c, int size);
static int    nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count);
static BigInt ipow(int n, int p);
static BigInt factorial(int n);

// for debugging
#ifdef CHECK
static Bool   isIn(int n, int *a, int v);
#endif

//==============================================================
// class Options
//--------------------------------------------------------------
Options::Options(void)
{
    if (nOptDef != GRCC_OPT_Size) {
        fprintf(GRCC_Stderr, "*** Options: inconsistent default values\n");
        exit(1);
    }
    model  = NULL;
    proc   = NULL;
    sproc  = NULL;

    outmg  = NULL;
    endmg  = NULL;
    outag  = NULL;
```

```
    argmg  = NULL;
    argemg = NULL;
    argag  = NULL;

    setDefaultValue();

    // for output
    out    = new Output(this);

    // subrpocess
    sproc = NULL;

    // measuring time
    time0  = 0.0;
    time1  = 0.0;
}

//------------------------------------------------------------
Options::~Options(void)
{
    outmg = NULL;
    outag = NULL;

    argmg = NULL;
    argag = NULL;

    delete out;
}

//------------------------------------------------------------
void Options::setDefaultValue(void)
{
    int j;

    // default values
    for (j = 0; j < GRCC_OPT_Size; j++) {
        values[j] = optDef[j].defaultv;
    }

    values[GRCC_OPT_Step] = GRCC_AGraph;

    // print level
    prlevel = 1;
}


//------------------------------------------------------------
void Options::setOutMG(OutEGB *omg, void *pt)
{
    outmg = omg;
    argmg = pt;
}

//------------------------------------------------------------
void Options::setEndMG(OutEGB *emg, void *pt)
{
    endmg  = emg;
    argemg = pt;
}

//------------------------------------------------------------
void Options::setOutAG(OutEG *oag, void *pt)
{
    outag = oag;
    argag = pt;
}

//------------------------------------------------------------
void Options::setErExit(ErExit *ere, void *erearg)
{
    erExit    = ere;
    erExitArg = erearg;
}

//------------------------------------------------------------
void Options::printLevel(int l)
{
    prlevel = l;
}

//------------------------------------------------------------
void Options::setValue(int ind, int val)
{
    if (0 <= ind && ind < GRCC_OPT_Size) {
        values[ind] = val;
    } else {
```

```
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** Options::setValue : invalid index=%d ",
                    ind);
            fprintf(GRCC_Stderr, "(val=%d)\n", val);
        }
        erEnd("Options::setValue : invalid index");
    }
}

//--------------------------------------------------------------
int Options::getValue(int ind)
{
    if (0 <= ind && ind < GRCC_OPT_Size) {
        return values[ind];
    } else {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** Options::getValue : invalid index=%d\n", ind);
        }
        erEnd("Options::getValue : invalid index");
    }
    return -1;
}

//--------------------------------------------------------------
void Options::print(void)
{
    int j;

    printf("Options\n");
    printf("+++ GRCC_OPT_Size=%d, print level=%d: ",
           GRCC_OPT_Size, prlevel);
    printf("symbol = value (default)\n");
    for (j=0; j < GRCC_OPT_Size; j++) {
        printf("   %4d GRCC_OPT_%-15s = %2d (%2d)\n",
               j, optDef[j].name, values[j], optDef[j].defaultv);
    }
    printf("    outgrf=%s, outgrp=%s\n", out->outgrf, out->outgrp);
}

//--------------------------------------------------------------
const OptDef *Options::getDef(void)
{
    return optDef;
}


//--------------------------------------------------------------
void Options::setOutputF(Bool outf, const char *fname)
{
    values[GRCC_OPT_Outgrf] = outf;
    out->setOutgrf(fname);
}

//--------------------------------------------------------------
void Options::setOutputP(Bool outp, const char *fname)
{
    values[GRCC_OPT_Outgrp] = outp;
    out->setOutgrp(fname);
}

//--------------------------------------------------------------
void Options::printModel(void)
{
    if (prlevel > 0) {
        if (model != NULL) {
            model->prModel();
        } else {
            printf("*** model is not defined\n");
        }
    }
}

//--------------------------------------------------------------
void Options::outModel(void)
{
    if (out != NULL && model != NULL) {
        out->outModelF();
        out->outModelP();
    }
}

//--------------------------------------------------------------
void Options::begin(Model *mdl)
{
    model = mdl;
    if (out != NULL) {
        if (values[GRCC_OPT_Outgrf]) {
```

```
            out->outBeginF(model, values[GRCC_OPT_Outgrf]);
            if (model != NULL) {
                out->outModelF();
            }
        }
        if (values[GRCC_OPT_Outgrp]) {
            out->outBeginP(model, values[GRCC_OPT_Outgrp]);
            if (model != NULL) {
                out->outModelP();
            }
        }
    }
}

//---------------------------------------------------------------
void Options::end(void)
{
    if (prlevel > 0) {
        printf("Optimization: ");
#ifdef SIMPSEL
        printf("SIMPSEL=1 ");
#else
        printf("SIMPSEL=0 ");
#endif
#ifdef MINMAXLEG
        printf("MINMAXLEG=1 ");
#else
        printf("MINMAXLEG=0 ");
#endif
#ifdef OPTEXTONLY
        printf("OPTEXTONLY=1 ");
#else
        printf("OPTEXTONLY=0 ");
#endif
        printf("\n");
    }

    if (out != NULL && values[GRCC_OPT_Outgrf]) {
        out->outEndF();
    }
    if (out != NULL && values[GRCC_OPT_Outgrp]) {
        out->outEndP();
    }
    model = NULL;
}
//
//---------------------------------------------------------------
void Options::beginProc(Process *prc)
{
    proc = prc;
    if (out != NULL && values[GRCC_OPT_Outgrf]) {
        out->outProcBeginF(prc);
    }
    if (out != NULL && values[GRCC_OPT_Outgrp]) {
        out->outProcBeginP(prc);
    }
    if (proc != NULL) {
        proc->mgrcount = 0;
        proc->agrcount = 0;
    }
}

//---------------------------------------------------------------
void Options::endProc(void)
{
    int k;

    if (proc == NULL) {
        if (out != NULL && values[GRCC_OPT_Outgrf]) {
            out->outProcEndF();
        }
        if (out != NULL && values[GRCC_OPT_Outgrp]) {
            out->outProcEndP();
        }
        return;
    }
    if (prlevel > 0) {
        printf("\n");
        printf("+++ Proc %d: ext=%d, loop=%d, ",
               proc->id, proc->nExtern, proc->loop);
        if (model != NULL) {
            printf("order=");
            prIntArray(model->ncouple, proc->clist, ": ");
            model->prParticleArray(proc->ninitl, proc->initlPart, "-->");
            model->prParticleArray(proc->nfinal, proc->finalPart, "");
        }
        printf(" (%8.2f sec)\n", proc->sec);
```

```


        printf("    Proc    %d: Total M-Graphs=%ld, M-Graphs=",
               proc->id, proc->nMGraphs);
        proc->wMGraphs.print(" (Conn)\n");

        printf("    Proc    %d: Total M-Graphs=%ld, M-Graphs=",
               proc->id, proc->nMOPI);
        proc->wMOPI.print(" (1PI)\n");

        printf("    Proc    %d: Total A-Graphs=%ld, A-Graphs=",
               proc->id, proc->nAGraphs);
        proc->wAGraphs.print(" (Conn)\n");

        printf("    Proc    %d: Total A-Graphs=%ld, A-Graphs=",
               proc->id, proc->nAOPI);
        proc->wAOPI.print(" (1PI)\n");

        printf("#  { %d,{", proc->ninitl);
        for (k = 0; k < proc->ninitl; k++) {
            if (k != 0) {
                printf(", ");
            }
            if (proc->model != NULL) {
                printf("\"%s\"", proc->model->particleName(proc->initlPart[k]));
            } else {
                printf("%d", proc->initlPart[k]);
            }
        }
        printf("}, %d,{", proc->nfinal);
        for (k = 0; k < proc->nfinal; k++) {
            if (k != 0) {
                printf(", ");
            }
            if (proc->model != NULL) {
                printf("\"%s\"", proc->model->particleName(proc->finalPart[k]));
            } else {
                printf("%d", proc->initlPart[k]);
            }
        }
        printf("}, ");
        if (model != NULL) {
            printf("{");
            for (k = 0; k < model->ncouple; k++) {
                if (k != 0) {
                    printf(", ");
                }
                printf("%d", proc->clist[k]);
            }
            printf("},");
        }
        printf("},%6ldL,%6ldL,%3ldL, -1.0, %4.2f},\n",
               proc->nAOPI, proc->wAOPI.num, proc->wAOPI.den, proc->sec);
    }

    if (out != NULL && values[GRCC_OPT_Outgrf]) {
        out->outProcEndF();
    }
    if (out != NULL && values[GRCC_OPT_Outgrp]) {
        out->outProcEndP();
    }
    proc = NULL;
}

//--------------------------------------------------------------
void Options::beginSubProc(SProcess *sprc)
{
    sproc = sprc;

    // 'beginProc' is called before
    if (out != NULL && values[GRCC_OPT_Outgrf]) {
        out->outSProcBeginF(sprc);
    }
    if (out != NULL && values[GRCC_OPT_Outgrp]) {
        out->outSProcBeginP(sprc);
    }
    if (proc != NULL) {
        return;
    }
    if (sprc != NULL) {
        sproc->mgrcount = 0;
        sproc->agrcount = 0;
    }
}

//--------------------------------------------------------------
void Options::endSubProc(void)
```

```
{
    MGraph *mgraph;

    if (sproc == NULL) {
        if (out != NULL && values[GRCC_OPT_Outgrf]) {
            out->outProcEndF();
        }
        if (out != NULL && values[GRCC_OPT_Outgrp]) {
            out->outProcEndP();
        }
        return;
    }
    mgraph = sproc->mgraph;
    if (sproc != NULL) {
        sproc->resultMGraph(mgraph->cDiag, mgraph->wscon,
                            mgraph->c1PI, mgraph->wsopi);
    }

    if (prlevel > 1) {
        printf("\n");
        printf("+++ Subproc %d: ext=%d, loop=%d, nodes=%d, edges=%d\n",
               sproc->id, sproc->nExtern, sproc->loop,
               sproc->nNodes, sproc->nEdges);

        printf("    Subproc %d: Total M-Graphs=%ld, M-Wsum=",
               sproc->id, sproc->nMGraphs);
        sproc->wMGraphs.print(" (Conn)\n");

        printf("    Subproc %d: Total M-Graphs=%ld, M-Wsum=",
               sproc->id, sproc->nMOPI);
        sproc->wMOPI.print(" (1PI)\n");

        printf("    Subproc %d: Total A-Graphs=%ld, A-Wsum=",
               sproc->id, sproc->nAGraphs);
        sproc->wAGraphs.print(" (Conn)\n");

        printf("    Subproc %d: Total A-Graphs=%ld, A-Wsum=",
               sproc->id, sproc->nAOPI);
        sproc->wAOPI.print(" (1PI)\n");
    }
    if (prlevel > 0) {
        printf("\n");
        printf("* Total %ld MGraphs; %ld 1PI", mgraph->cDiag, mgraph->c1PI);
        printf(" wscon = ");
        mgraph->wscon.print(" ( ");
        mgraph->wsopi.print(" 1PI)\n");

#ifdef MONITOR
        printf("* refine:                        %ld\n", mgraph->nCallRefine);
        printf("* discarded for refinement:   %ld\n", mgraph->discardRefine);
        printf("* discarded for disconnected: %ld\n", mgraph->discardDisc);
        printf("* discarded for duplication:  %ld\n", mgraph->discardIso);
        printf("* discarded by  outMG:        %ld\n", mgraph->discardMG);
#endif
    }

    if (proc != NULL) {
        return;
    }
    if (out != NULL && values[GRCC_OPT_Outgrf]) {
        out->outProcEndF();
    }
    if (out != NULL && values[GRCC_OPT_Outgrp]) {
        out->outProcEndP();
    }
}

//--------------------------------------------------------------
void Options::newMGraph(MGraph *mgr)
{
    MGraph *mgraph = mgr;

#ifdef DEBUG
    if (proc != NULL) {
        printf("+++ New EGraph (MG): %ld\n", proc->mgrcount);
    } else if (sproc != NULL) {
        printf("+++ New EGraph (MG): %ld\n", sproc->mgrcount);
    }
#endif
    if (proc != NULL) {
        proc->mgrcount++;
        mgr->egraph->mId = proc->mgrcount;
    } else if (sproc != NULL) {
        sproc->mgrcount++;
        mgr->egraph->mId = sproc->mgrcount;
    }

```

```
    if (out != NULL
        && values[GRCC_OPT_Step] == GRCC_MGraph) {
        if (values[GRCC_OPT_Outgrf]) {
            out->outEGraphF(mgraph->egraph);
        }
        if (values[GRCC_OPT_Outgrp]) {
            out->outEGraphP(mgraph->egraph);
        }
    }
    if (sproc != NULL) {
        sproc->endMGraph(mgr);
    }
    if (endmg != NULL) {
        (*endmg)(mgr->egraph, argemg);
    }
}

//--------------------------------------------------------------
void Options::newAGraph(EGraph *egraph)
{
    Fraction sf, zr;

#ifdef DEBUG
    if (proc != NULL) {
        printf("+++ New EGraph (AG): %ld (%ld)\n",
               proc->agrcount, proc->mgrcount);
    } else if (sproc != NULL) {
        printf("+++ New EGraph (AG): %ld (%ld)\n",
               sproc->agrcount, sproc->mgrcount);
    }
#endif
    if (proc != NULL) {
        proc->agrcount++;
        egraph->sId = proc->agrcount;
    } else if (sproc != NULL) {
        sproc->agrcount++;
        egraph->sId = sproc->agrcount;
    }

    if (sproc != NULL) {
        zr = Fraction(0, 1);
        if ((egraph->nsym)*(egraph->esym) != 0) {
            sf = Fraction(egraph->extperm, egraph->nsym*egraph->esym);
        } else {
            sf = zr;
        }
        if (egraph->mgraph->opi) {
            sproc->resultAGraph(1, sf, 1, sf);
        } else {
            sproc->resultAGraph(1, sf, 0, zr);
        }
    }

    if (out != NULL && values[GRCC_OPT_Outgrf]) {
        out->outEGraphF(egraph);
    }
    if (out != NULL && values[GRCC_OPT_Outgrp]) {
        out->outEGraphP(egraph);
    }
    if (outag != NULL) {
#ifdef DEBUG1
        printf("call outag\n");
        egraph->model->prModel();
        egraph->print();
#endif
        (*outag)(egraph, argag);
    }

    sproc->endAGraph(egraph);
}

//==============================================================
Output::Output(Options *optn)
{
    opt     = optn;
    model   = NULL;
    proc    = NULL;
    sproc   = NULL;
    outgrfp = NULL;
    outgrpp = NULL;
    procId  = -1;
    outgrf  = NULL;
    outgrp  = NULL;
    outproc = False;

}

```

```
//--------------------------------------------------------------
Output::~Output(void)
{
    if (outgrfp != NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** file has not been closed : \"%s\"\n",
                    outgrf);
        }
        fclose(outgrfp);
        outgrfp = NULL;
        erEnd("file has not been closed");
    }
    if (outgrf != NULL) {
        free(outgrf);
        outgrf = NULL;
    }
    if (outgrpp != NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** file has not been closed : \"%s\"\n",
                    outgrp);
        }
        fclose(outgrpp);
        outgrpp = NULL;
        erEnd("file has not been closed");
    }
    if (outgrp != NULL) {
        free(outgrp);
        outgrp = NULL;
    }
}

//--------------------------------------------------------------
void Output::setOutgrf(const char *fname)
{
    if (outgrf != NULL && strcmp(outgrf, fname) == 0) {
        return;
    }

    if (outgrf != NULL) {
        free(outgrf);
    }
    if (fname == NULL || strlen(fname) < 1) {
        outgrf = NULL;
    } else {
        outgrf = strdup(fname);
    }
}

//--------------------------------------------------------------
void Output::setOutgrp(const char *fname)
{
    if (outgrp != NULL && strcmp(outgrp, fname) == 0) {
        return;
    }

    if (outgrp != NULL) {
        free(outgrp);
    }
    if (fname == NULL || strlen(fname) < 1) {
        outgrp = NULL;
    } else {
        outgrp = strdup(fname);
    }
}

//--------------------------------------------------------------
Bool Output::outBeginF(Model *mdl, Bool pr)
{
    model = mdl;

    if (outgrf == NULL || strlen(outgrf) < 1) {
        outgrfp = NULL;
        return True;
    } else if (outgrfp != NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** outBegin: \"%s\" is already opened\n",
                    outgrf);
        }
        erEnd("outBegin: file is already opened\n");
    }

    if (!pr) {
        return True;
    }
    if ((outgrfp = fopen(outgrf, "w")) == NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** outBegin: cannot open %s\n", outgrf);
```

```
        }
        return False;
    }
    fprintf(outgrfp, "%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\n");
    fprintf(outgrfp, "%% Generated by 'grcc'\n");
    fprintf(outgrfp, "Version={2,2,0,0};\n");
    if (model != NULL) {
        fprintf(outgrfp, "Model=\"./%s.mdl\";\n", model->name);
    } else {
        fprintf(outgrfp, "Model=\"./Phi34.mdl\";\n");
    }


    return True;
}

//--------------------------------------------------------------
Bool Output::outBeginP(Model *mdl, Bool pr)
{
    model = mdl;

    if (outgrp == NULL || strlen(outgrp) < 1) {
        outgrpp = NULL;
        return True;
    } else if (outgrpp != NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** outBegin: \"%s\" is already opened\n",
                    outgrp);
        }
        erEnd("outBegin: file is already opened\n");
    }

    if (!pr) {
        return True;
    }
    if ((outgrpp = fopen(outgrp, "w")) == NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** outBegin: cannot open %s\n", outgrp);
        }
        return False;
    }
    fprintf(outgrpp, "################################\n");
    fprintf(outgrpp, "# Generated by 'grcc'\n");

    return True;
}

//--------------------------------------------------------------
void Output::outEndF(void)
{
    if (outgrfp == NULL) {
        return;
    }
    fprintf(outgrfp, "%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\n");
    fprintf(outgrfp, "End=1;\n");
    fprintf(outgrfp, "%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\n");
    fclose(outgrfp);
    outgrfp = NULL;

    free(outgrf);
    outgrf = NULL;
}

//--------------------------------------------------------------
void Output::outEndP(void)
{
    if (outgrpp == NULL) {
        return;
    }
    fprintf(outgrpp, "################################\n");
    fprintf(outgrpp, "# End\n");
    fprintf(outgrpp, "################################\n");
    fclose(outgrpp);
    outgrpp = NULL;

    free(outgrp);
    outgrp = NULL;
}

//--------------------------------------------------------------
void Output::outProcBeginF(Process *prc)
{
    int k, ex;

    if (prc == NULL) {
        return;
    }
```

```

    proc    = prc;
    sproc   = NULL;
    procId  = proc->id;

    if (outgrfp == NULL) {
        return;
    }
    if (outproc) {
        return;
    }
    outproc = True;

    fprintf(outgrfp, "%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\n");
    fprintf(outgrfp, "Process=%d;\n", proc->id);
    fprintf(outgrfp, "External=%d;\n", proc->ninitl+proc->nfinal);
    for (k = 0, ex = 0; k < proc->ninitl; k++, ex++) {
        fprintf(outgrfp, "%4d= initial %s;\n", ex,
                proc->model->particleName(proc->initlPart[k]));
    }
    for (k = 0; k < proc->nfinal; k++, ex++) {
        fprintf(outgrfp, "%4d= final   %s;\n", ex,
                proc->model->particleName(proc->finalPart[k]));
    }
    fprintf(outgrfp, "Eend;\n");
    for (k = 0; k < proc->model->ncouple; k++) {
        fprintf(outgrfp, "%s=%d;  ", proc->model->cnlist[k], proc->clist[k]);
    }
    fprintf(outgrfp, "Loop=%d;\n", proc->loop);
    fprintf(outgrfp, "OPI=%s;\n", (opt->values[GRCC_OPT_1PI] > 0? "Yes" : "No"));
    fprintf(outgrfp, "Assign=%s;\n", (opt->values[GRCC_OPT_Step]==GRCC_AGraph ? "Yes" : "No"));
    fprintf(outgrfp, "%% Options : dummy values\n");
    fprintf(outgrfp, "Expand=Yes; ExpMin=0; ExpMax=-1;\n");
    fprintf(outgrfp, "Block=No; Tadpole=Yes; Extself=Yes;\n");
    fprintf(outgrfp, "Selfe=Yes; Countert=No; AnyCT=No;Undefp=No;\n");
}

//---------------------------------------------------------------
void Output::outProcBeginP(Process *prc)
{
    if (prc == NULL) {
        return;
    }

    proc    = prc;
    sproc   = NULL;
    procId  = proc->id;

    if (outgrpp == NULL) {
        return;
    }
    if (outproc) {
        return;
    }
    outproc = True;

    fprintf(outgrfp, "# Process=%d;\n", proc->id);
}

//---------------------------------------------------------------
void Output::outProcBegin0(int next, int couple, int loop)
{
    int k, ex;

    procId  = 0;

    if (outgrfp == NULL) {
        return;
    }
    if (outproc) {
        return;
    }
    outproc = True;

    fprintf(outgrfp, "%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\n");
    fprintf(outgrfp, "Process=%d;\n", procId);
    fprintf(outgrfp, "External=%d;\n", next);
    for (k = 0, ex = 0; k < next; k++, ex++) {
        fprintf(outgrfp, "%4d= initial undef;\n", ex);
    }
    fprintf(outgrfp, "Eend;\n");
    fprintf(outgrfp, "GRCC_PHI=%d;  ", couple);
    fprintf(outgrfp, "Loop=%d;\n", loop);
    fprintf(outgrfp, "OPI=%s;\n", (opt->values[GRCC_OPT_1PI] > 0? "Yes" : "No"));
    fprintf(outgrfp, "Assign=No;\n");
    fprintf(outgrfp, "%% Options : dummy values\n");
    fprintf(outgrfp, "Expand=Yes; ExpMin=0; ExpMax=-1;\n");
```

```
    fprintf(outgrfp, "Block=No; Tadpole=Yes; Extself=Yes;\n");
    fprintf(outgrfp, "Selfe=Yes; Countert=No; AnyCT=No;Undefp=No;\n");
}

//--------------------------------------------------------------
void Output::outSProcBeginF(SProcess *sprc)
{
    int j, k, ex;
    PNodeClass *pnc;

    if (sprc == NULL) {
        return;
    }

    sproc  = sprc;
    procId = sproc->id;
    pnc    = sproc->pnclass;

    if (outgrfp == NULL) {
        return;
    }
    if (outproc) {
        return;
    }
    outproc = True;
    fprintf(outgrfp, "%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%\n");
    fprintf(outgrfp, "Process=%d;\n", sproc->id);
    fprintf(outgrfp, "External=%d;\n", sproc->nExtern);

    ex = 0;
    for (j = 0; j < pnc->nclass; j++) {
        if (pnc->type[j] == GRCC_AT_Initial) {
            for (k = 0; k < pnc->count[j]; k++) {
                fprintf(outgrfp, "%4d= initial %s;\n", ex,
                        model->particleName(pnc->particle[j]));
                ex++;
            }
        } else if (pnc->type[j] == GRCC_AT_Final) {
            for (k = 0; k < pnc->count[j]; k++) {
                fprintf(outgrfp, "%4d= final %s;\n", ex,
                        model->particleName(-pnc->particle[j]));
                ex++;
            }
        }
    }

    fprintf(outgrfp, "Eend;\n");
    for (k = 0; k < sproc->model->ncouple; k++) {
        fprintf(outgrfp, "%s=%d;  ", sproc->model->cnlist[k], sproc->clist[k]);
    }
    fprintf(outgrfp, "Loop=%d;\n", sproc->loop);
    fprintf(outgrfp, "OPI=%s;\n", (opt->values[GRCC_OPT_1PI] > 0? "Yes" : "No"));
    fprintf(outgrfp, "Assign=%s;\n", (opt->values[GRCC_OPT_Step]==GRCC_AGraph ? "Yes" : "No"));
    fprintf(outgrfp, "%% Options : dummy values\n");
    fprintf(outgrfp, "Expand=Yes; ExpMin=0; ExpMax=-1;\n");
    fprintf(outgrfp, "Block=No; Tadpole=Yes; Extself=Yes;\n");
    fprintf(outgrfp, "Selfe=Yes; Countert=No; AnyCT=No;Undefp=No;\n");
}

//--------------------------------------------------------------
void Output::outSProcBeginP(SProcess *sprc)
{
    if (sprc == NULL) {
        return;
    }

    sproc  = sprc;
    procId = sproc->id;

    if (outgrpp == NULL) {
        return;
    }
    if (outproc) {
        return;
    }
    outproc = True;
    fprintf(outgrpp, "# SProcess=%d;\n", sproc->id);
}

//--------------------------------------------------------------
void Output::outProcEndF(void)
{
    if (outgrfp == NULL) {
        return;
    }
    if (proc != NULL) {
        fprintf(outgrfp, "%%------------------------------\n");
```

```
        fprintf(outgrfp, "Pend=%d;\n", proc->id);
    } else if (sproc != NULL) {
        fprintf(outgrfp, "%%------------------------------\n");
        fprintf(outgrfp, "Pend=%d;\n", sproc->id);
    } else {
        fprintf(outgrfp, "%%------------------------------\n");
        fprintf(outgrfp, "Pend=1;\n");
    }
    fflush(outgrfp);
}

//--------------------------------------------------------------
void Output::outProcEndP(void)
{
    if (outgrpp == NULL) {
        return;
    }
    if (proc != NULL) {
        fprintf(outgrpp, "# Pend=%d;\n", proc->id);
    }
    fflush(outgrpp);
}

//--------------------------------------------------------------
void Output::outEGraphF(EGraph *egraph)
{
    int nd, lg, ptcl, ed, intr, j, k, loop;
    Model *mdl = egraph->model;
    Bool popt;
    EFLine *fl;

#ifdef DEBUG9
    if (egraph->mgraph != NULL) {
        printf("outEGraph:sId=%ld\n", egraph->mId);
        // egraph->mgraph->print();
        egraph->mgraph->mconn->print();
    }
#endif
    if (outgrfp == NULL) {
        return;
    }
    fprintf(outgrfp, "%%------------------------------\n");
    if (egraph->assigned) {
        fprintf(outgrfp, "Graph=%ld;\n", egraph->sId);
        fprintf(outgrfp, "%% AGraph=%ld;\n", egraph->aId);
    } else {
        fprintf(outgrfp, "Graph=%ld;\n", egraph->mId);
    }
    fprintf(outgrfp, "Gtype=%ld;\n", egraph->mId);
    if (egraph->assigned) {
        fprintf(outgrfp, "Sfactor=%ld;\n",
                egraph->nsym * egraph->esym * egraph->fsign);
    } else {
        fprintf(outgrfp, "Sfactor=%ld;\n", egraph->nsym * egraph->esym);
    }
    fprintf(outgrfp, "%% ExtPerm=%ld; NSym=%ld; ESym=%ld; NSym1=%ld;"
                     " Multp=%ld;\n",
            egraph->extperm, egraph->nsym, egraph->esym, egraph->nsym1,
            egraph->multp);

    fprintf(outgrfp, "Vertex=%d;\n", egraph->nNodes - egraph->nExtern);

    for (nd = 0; nd < egraph->nNodes; nd++) {
        fprintf(outgrfp, "%4d", nd);
        if (mdl != NULL && egraph->assigned && !egraph->isExternal(nd)) {
            intr = egraph->nodes[nd]->intrct;
            loop = mdl->interacts[intr]->loop;
            popt = False;

            // not tree vertex
            if (loop > 0) {
                fprintf(outgrfp, "[loop=%d", loop);
                popt = True;
            }

            // multiple coupling constants
            if (mdl->ncouple > 1) {
                if (popt) {
                    fprintf(outgrfp, ";order={");
                } else {
                    fprintf(outgrfp, "[order={");
                    popt = True;
                }
                for (j = 0; j < mdl->interacts[intr]->nclist; j++) {
                    if (j != 0) {
                        fprintf(outgrfp, ",");
                    }
```

```
                        fprintf(outgrfp, "%d", mdl->interacts[intr]->clist[j]);
                    }
                    fprintf(outgrfp, "}");
                }
                if (popt) {
                    fprintf(outgrfp, "]");
                }
            } else if (!egraph->isExternal(nd)) {
                loop = egraph->nodes[nd]->extloop;
                // not tree vertex
                if (loop > 0) {
                    fprintf(outgrfp, "[loop=%d]", loop);
                }
            }
            fprintf(outgrfp, "={");

            // list of legs
            for (lg = 0; lg < egraph->nodes[nd]->deg; lg++) {
                if (lg != 0) {
                    fprintf(outgrfp, ", ");
                }
                ed  = V2Iedge(egraph->nodes[nd]->edges[lg]);
                fprintf(outgrfp, "%4d", egraph->nodes[nd]->edges[lg]);
                ptcl = egraph->edges[ed]->ptcl;
                if (mdl != NULL && ptcl != 0 && egraph->assigned) {
                    if (egraph->nodes[nd]->edges[lg] < 0) {
                        ptcl = mdl->normalParticle(-ptcl);
                    } else {
                        ptcl = mdl->normalParticle(ptcl);
                    }
                    fprintf(outgrfp, "[%s]", mdl->particleName(ptcl));
                } else {
                    fprintf(outgrfp, "[undef]");
                }
            }
            fprintf(outgrfp, "};\n");
        }
        // print Fermion line as comment line
        if (egraph->nflines >= 0) {
            fprintf(outgrfp, "%%  FLines=%d; FSign=%d; sId=%ld;\n",
                    egraph->nflines, egraph->fsign, egraph->sId);
        }
        for (j = 0; j < egraph->nflines; j++) {
            fl = egraph->flines[j];
            fprintf(outgrfp, "%% %4d", j);
            if (fl->ftype == FL_Open) {
                fprintf(outgrfp, "[Open]=[");
            } else if (fl->ftype == FL_Closed) {
                fprintf(outgrfp, "[Loop]=[");
            } else {
                fprintf(outgrfp, " ?%d", fl->ftype);
            }
            for (k = 0; k < fl->nlist; k++) {
                if (k != 0) {
                    fprintf(outgrfp, ", ");
                }
                fprintf(outgrfp, "%d", Abs(fl->elist[k]));
            }
            fprintf(outgrfp, "];\n");
        }
        fprintf(outgrfp, "Vend;\n");
        fprintf(outgrfp, "Gend;\n");
}

//---------------------------------------------------------------
void Output::outEGraphP(EGraph *eg)
{
    int      j, nd, lg, ed, nin, nfi;
    char     nl = '\n';
    ENode    *node;
    static char undef[] = "Undef";
    char     *s;

    // model
    fprintf(outgrpp, "egraph[%ld] = {%c", eg->sId-1, nl);
    if (model==NULL) {
        fprintf(outgrpp, "  \"Model\": [\"Undef\", 1],%c", nl);
    } else {
        fprintf(outgrpp, "  \"Model\": [\"%s\", %d],%c",
                model->name, model->ncouple, nl);
    }

    // process
    nin = nfi = 0;
    for (j = 0; j < eg->nNodes; j++) {
        if (eg->isExternal(j)) {
            if (eg->nodes[j]->extloop == GRCC_AT_Final) {
```

```
                nfi++;
            } else {
                nin++;
            }
        }
    }
    fprintf(outgrpp, "  \"Process\": [[%d, %d], %d, [",
            nin, nfi, eg->nLoops);
    if (proc != NULL && model != NULL) {
        for (j = 0; j < model->ncouple; j++) {
            if (j != 0) {
                fprintf(outgrpp, ", ");
            }
            fprintf(outgrpp, "%d", proc->clist[j]);
        }
    }
    fprintf(outgrpp, "]],%c", nl);

    // option of the process
    fprintf(outgrpp, "  \"Opt\": [");
    if (eg->opt != NULL) {
        for (j = 0; j < GRCC_OPT_Size; j++) {
            if (j != 0) {
                fprintf(outgrpp, ", ");
            }
            fprintf(outgrpp, "%d", eg->opt->values[j]);
        }
    }
    fprintf(outgrpp, "],%c", nl);

    // graph
    fprintf(outgrpp, "  \"Id\": [%ld, %ld, %ld],%c",
                     eg->mId, eg->aId, eg->sId, nl);
    fprintf(outgrpp, "  \"GStat\": [%d, %d, %d],%c",
            eg->assigned, eg->nNodes, eg->nEdges, nl);
    fprintf(outgrpp, "  \"Sym\": [%d, %ld, %ld, %ld, %ld],%c",
            eg->fsign, eg->nsym, eg->esym, eg->nsym1, eg->multp, nl);

    // nodes
    fprintf(outgrpp, "  \"Nodes\": [%c", nl);
    for (nd = 0; nd < eg->nNodes; nd++) {
        node = eg->nodes[nd];
        if (model==NULL || !eg->assigned) {
            s = undef;
        } else if (eg->isExternal(nd)) {
            s = model->particleName(node->intrct);
        } else {
            s = model->interacts[node->intrct]->name;
        }
        fprintf(outgrpp, "    [%d, \"%s\", %d, [",
                nd, s, node->extloop);
        for (lg = 0; lg < eg->nodes[nd]->deg; lg++) {
            if (lg != 0) {
                fprintf(outgrpp, ", ");
            }
            fprintf(outgrpp, "%d", eg->nodes[nd]->edges[lg]);
        }
        fprintf(outgrpp, "]],%c", nl);
    }
    fprintf(outgrpp, "  ],%c", nl);

    // edges
    fprintf(outgrpp, "  \"Edges\": [%c", nl);
    for (ed = 0; ed < eg->nEdges; ed++) {
        if (model != NULL) {
            s = model->particleName(eg->edges[ed]->ptcl);
            fprintf(outgrpp, "    [%d, \"%s\", [%d, %d]],%c",
                    ed+1, s, eg->edges[ed]->nodes[0], eg->edges[ed]->nodes[1], nl);
        } else {
            fprintf(outgrpp, "    [%d, \"Undef\", [%d, %d]],%c",
                    ed+1, eg->edges[ed]->nodes[0], eg->edges[ed]->nodes[1], nl);
        }
    }

    fprintf(outgrpp, "  ],%c", nl);

    fprintf(outgrpp, "};\n");
}

//---------------------------------------------------------------
void Output::outModelF(void)
{
    char        *mdlfn;
    FILE        *mdlfp;
    Particle    *pt;
    Interaction *vt;
    int          j, k, l, ln;
```

```
    const char        *ptn;

    if (model == NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** Output::outModel : model is not defined\n");
        }
        return;
    }
    ln = strlen(model->name)+5;
    mdlfn = new char[ln];
    snprintf(mdlfn, ln, "%s.mdl", model->name);

    if ((mdlfp = fopen(mdlfn, "w")) == NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** Output::outModel : cannot open \"%s\"\n",
                    mdlfn);
        }
        return;
    }

    for (l = 0; l < 60; l++) { putc('%', mdlfp); } putc('\n', mdlfp);
    fprintf(mdlfp, "%% File \"%s\"\n", mdlfn);
    fprintf(mdlfp, "%% Generated by graph generater\n");
    fprintf(mdlfp, " Version={2,2,0};\n");

    // copupling constants
    fprintf(mdlfp, " Order={");
    for (j = 0; j < model->ncouple; j++) {
        if (j != 0) {
            fprintf(mdlfp, ", ");
        }
        fprintf(mdlfp, "%s", model->cnlist[j]);
    }
    fprintf(mdlfp, "};\n");

    for (l = 0; l < 60; l++) { putc('%', mdlfp); } putc('\n', mdlfp);
    fprintf(mdlfp, "%% particles\n");
    for (l = 0; l < 60; l++) { putc('%', mdlfp); } putc('\n', mdlfp);
    for (j = 1; j < model->nParticles; j++) {
        pt = model->particles[j];
        fprintf(mdlfp, " Particle=%s; Antiparticle=%s;\n",
                pt->name, pt->aname);
        ptn = pt->typeGName();
        fprintf(mdlfp, " PType=%s; Charge=0; Color=1; Mass=0; Width=0;\n", ptn);
        fprintf(mdlfp, " PCode=%d; Massless; \n", pt->pcode);
        fprintf(mdlfp, " Pend;\n");
        fprintf(mdlfp, "%%\n");
    }
    for (l = 0; l < 60; l++) { putc('%', mdlfp); } putc('\n', mdlfp);
    fprintf(mdlfp, "%% interactions\n");
    for (l = 0; l < 60; l++) { putc('%', mdlfp); } putc('\n', mdlfp);
    for (j = 0; j < model->nInteracts; j++) {
        vt = model->interacts[j];
        fprintf(mdlfp, " Vertex={");
        for (k = 0; k < vt->nplist; k++) {
            if (k != 0) {
                fprintf(mdlfp, ", ");
            }
            fprintf(mdlfp, "%s", model->particleName(vt->plist[k]));
        }
        fprintf(mdlfp, "}; ");
        for (k = 0; k < model->ncouple; k++) {
            fprintf(mdlfp, "%s=%d; ", model->cnlist[k], vt->clist[k]);
        }
        fprintf(mdlfp, "FName=%s; Vend;\n", vt->name);
    }
    for (l = 0; l < 60; l++) { putc('%', mdlfp); } putc('\n', mdlfp);
    fprintf(mdlfp, " Mend;\n");
    for (l = 0; l < 60; l++) { putc('%', mdlfp); } putc('\n', mdlfp);
    fclose(mdlfp);
    delete[] mdlfn;
}

//---------------------------------------------------------------
void Output::outModelP(void)
{
    char        *mdlfn;
    FILE        *mdlfp;
    Particle    *pt;
    Interaction *vt;
    int          j, k, l, ln;
    const char        *ptn;

    if (model == NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** Output::outModel : ");
            fprintf(GRCC_Stderr, "model is not defined\n");
```

```
        }
        return;
    }
    ln = strlen(model->name)+5;
    mdlfn = new char[ln];
    snprintf(mdlfn, ln, "%s.mdp", model->name);

    if ((mdlfp = fopen(mdlfn, "w")) == NULL) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** Output::outModel : cannot open \"%s\"\n",
                    mdlfn);
        }
        return;
    }

    for (l = 0; l < 60; l++) { putc('#', mdlfp); } putc('\n', mdlfp);
    fprintf(mdlfp, "# File \"%s\"\n", mdlfn);
    fprintf(mdlfp, "# Generated by graph generater grcc\n");

    // copupling constants
    fprintf(mdlfp, "Model={\n");
    fprintf(mdlfp, "  \"Model\": [");
    fprintf(mdlfp, "\"%s\", %d, [", model->name, model->ncouple);
    for (j = 0; j < model->ncouple; j++) {
        if (j != 0) {
            fprintf(mdlfp, ", ");
        }
        fprintf(mdlfp, "%s", model->cnlist[j]);
    }
    fprintf(mdlfp, "],\n");

    fprintf(mdlfp, "  \"Particles\": [\n");
    for (j = 1; j < model->nParticles; j++) {
        pt = model->particles[j];
        ptn = pt->typeGName();
        fprintf(mdlfp, "    [\"%s\", \"%s\", \"%s\"],\n",
                pt->name, pt->aname, ptn);
    }
    fprintf(mdlfp, "  \"Interactions\": [\n");
    for (j = 0; j < model->nInteracts; j++) {
        vt = model->interacts[j];
        fprintf(mdlfp, "    [\"%s\", %d, [", vt->name, vt->nplist);
        for (k = 0; k < vt->nplist; k++) {
            if (k != 0) {
                fprintf(mdlfp, ", ");
            }
            fprintf(mdlfp, "%s", model->particleName(vt->plist[k]));
        }
        fprintf(mdlfp, "], [");
        for (k = 0; k < model->ncouple; k++) {
            if (k != 0) {
                fprintf(mdlfp, ",");
            }
            fprintf(mdlfp, "%d", vt->clist[k]);
        }
        fprintf(mdlfp, "]],\n");
    }
    fprintf(mdlfp, "  ],\n");
    fprintf(mdlfp, "],\n");
    fclose(mdlfp);
    delete[] mdlfn;
}

//****************************************************************
// model.cc

//==============================================================

static const char *ptypenames[]  = GRCC_PT_Table;
static const char *pgtypenames[] = GRCC_PT_GTable;

//==============================================================
// class Particle
//--------------------------------------------------------------
Particle::Particle(Model *modl, int pid, PInput *pinp)
{
    static char buff[100];
    static int  nbuff=100;
    int j;

    mdl      = modl;                    // the model
    id       = pid;                     // id of this particle
    if (pinp->name == NULL || strlen(pinp->name) < 1) {
        if (pinp->pcode != 0) {
            snprintf(buff, nbuff, "pa%d", pinp->pcode);
            name = strdup(buff);
        } else {
```

```
            snprintf(buff, nbuff, "pi%d", pid);
            name = strdup(buff);
        }
    } else {
        name = strdup(pinp->name);     // the name of the particle
    }
    if (pinp->aname == NULL) {
        if (pinp->acode != 0) {
            snprintf(buff, nbuff, "ap%d", Abs(pinp->acode));
            aname = strdup(buff);
        } else {
            snprintf(buff, nbuff, "ai%d", pid);
            aname = strdup(buff);
        }
    } else {
        aname   = strdup(pinp->aname); // the name of the anti-particle
    }
    pcode   = pinp->pcode;
    acode   = pinp->acode;
    if (Abs(mdl->defpart) == GRCC_DEFBYCODE) {
        neutral = (pcode == acode);
    } else if (Abs(mdl->defpart) == GRCC_DEFBYNAME) {
        neutral = ((strcmp(name, aname)==0) ? True : False);
    }

    // convert ptype string to its code.
    if (Abs(mdl->defpart) == GRCC_DEFBYCODE) {
        if (pinp->ptypec < GRCC_PT_Undef || pinp->ptypec > GRCC_PT_Size) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** particle type is not defined: %d\n",
                        pinp->ptypec);
            }
            erEnd("particle type is not defined (illegal code)");
        }
        ptype = pinp->ptypec;
    } else if (Abs(mdl->defpart) == GRCC_DEFBYNAME) {
        if (pinp->ptypen == NULL) {
            erEnd("particle type is not defined (NULL)");
        }
        ptype = -1;
        for (j = 0; j < GRCC_PT_Size; j++) {
            if (strcmp(pinp->ptypen, ptypenames[j]) == 0) {
                ptype = j;
                break;
            }
        }
        if (ptype < 0) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** particle type \"%s\" is not defined\n",
                        pinp->ptypen);
            }
            erEnd("particle type is not defined (name)");
        }
    }
    if (ptype == GRCC_PT_Undef && ptype != 0) {
        erEnd("illegal ptype");
    }

    extonly = pinp->extonly;
    cmindeg = 0;
    cmaxdeg = 0;
}

//---------------------------------------------------------------
Particle::~Particle(void)
{
    free(aname);
    free(name);
}

//---------------------------------------------------------------
char *Particle::particleName(int p)
{
    if (p < 0) {
        return aname;
    } else {
        return name;
    }
}

//---------------------------------------------------------------
int Particle::particleCode(int p)
{
    if (p < 0) {
        return acode;
    } else {
        return pcode;
```

```
    }
}

//--------------------------------------------------------------
int Particle::isNeutral(void)
{
    return neutral;
}

//--------------------------------------------------------------
const char *Particle::typeName(void)
{
    return ptypenames[ptype];
}

//--------------------------------------------------------------
const char *Particle::typeGName(void)
{
    return pgtypenames[ptype];
}

//--------------------------------------------------------------
char *Particle::aparticle(void)
{
    static char prtcl[] = "Particle";

    if (isNeutral()) {
        return prtcl;
    } else {
        return aname;
    }
}

//--------------------------------------------------------------
void Particle::prParticle(void)
{
    if (isNeutral()) {
        printf("pid=%d, name=\"%s\", real_field, ", id, name);
    } else {
        printf("pid=%d, name=\"%s\", aname=\"%s\", ", id, name, aname);
    }
    printf("ptype=%s, pcode=%d, acode=%d, extonly=%d, cdeg=(%d,%d)\n",
           ptypenames[ptype], pcode, acode, extonly, cmindeg, cmaxdeg);
}

//==============================================================
// class Interaction
//--------------------------------------------------------------
Interaction::Interaction(Model *modl, int iid, const char *nam, int icd, int *cpl, int nlgs, int *plst, int csm, int lp)
{
    static char buff[100];
    static int  nbuff=100;
    Particle   *p;
    int         j, ndir, sdir, jdir, nmaj, jmaj, ngho, sgho, jgho, ptcl;
    Bool        ok;

    mdl    = modl;         // the model
    id     = iid;          // id of this interaction
    icode  = icd;          // id of this interaction
    csum   = csm;          // the total orders of coupling constants
    nlegs  = nlgs;         // the size of plist[]
    loop   = lp;           // the number of loops

    if (nam == NULL || strlen(nam) < 1) {
        snprintf(buff, nbuff, "vt%d", icd);
        name = strdup(buff);
    } else {
        name = strdup(nam);  // the name of the interaction
    }
    nclist = mdl->ncouple;
    clist  = new int[nclist];
    icode  = icd;
    for (j = 0; j < mdl->ncouple; j++) {
        clist[j] = cpl[j];
    }
    plist  = new int[nlegs];
    nplist = nlegs;
    for (j = 0; j < nlegs; j++) {
        plist[j] = plst[j];
    }
    nslist = 0;
    slist  = NULL;

    // check fermion number conservation
    ndir = 0;
    sdir = 0;
```

```
    jdir = 0;
    nmaj = 0;
    jmaj = 0;
    ngho = 0;
    sgho = 0;
    jgho = 0;
    ok = True;
    for (j = 0; j < nplist; j++) {
        if (plist[j] > 0) {
            p = mdl->particles[plist[j]];
            ptcl = 1;
        } else {
            p = mdl->particles[-plist[j]];
            ptcl = -1;
        }

        if (p->ptype == GRCC_PT_Dirac) {
            ndir++;
            if (ptcl > 0) {
                sdir++;
            } else {
                sdir--;
            }
            if (Abs(sdir) == 1) {
                jdir = j;
            } else if ((Abs(sdir) == 0 && j != jdir + 1) || Abs(sdir) > 1) {
                fprintf(GRCC_Stderr, "*** Interaction: Dirac and anti-Dirac should "
                        "arranged in pairs in an interaction.\n");
                ok = False;
            }
        } else if (p->ptype == GRCC_PT_Ghost) {
            ngho++;
            if (ptcl > 0) {
                sgho++;
            } else {
                sgho--;
            }
            if (Abs(sgho) == 1) {
                jgho = j;
            } else if ((Abs(sgho) == 0 && j != jgho + 1) || Abs(sgho) > 1) {
                fprintf(GRCC_Stderr, "*** Interaction: Ghost and anti-Ghost should "
                        "arranged in pairs in an interaction.\n");
                ok = False;
            }
        } else if (p->ptype == GRCC_PT_Majorana) {
            nmaj++;
            if (nmaj % 2 == 1) {
                jmaj = j;
            } else if (j != jmaj + 1) {
                fprintf(GRCC_Stderr, "*** Interaction: two Majoranas should "
                        "arranged in pairs in an interaction.\n");
                ok = False;
            }
        }
    }
    if (sdir != 0) {
        fprintf(GRCC_Stderr, "*** Interaction: Dirac number is not conserved\n");
        ok = False;
    }
    if (sgho != 0) {
        fprintf(GRCC_Stderr, "*** Interaction: Ghost number is not conserved\n");
        ok = False;
    }
    if (nmaj % 2 != 0) {
        fprintf(GRCC_Stderr, "*** Interaction: odd number of Majorana particles\n");
        ok = False;
    }
    if (ndir + nmaj + ngho > 2) {
        fprintf(GRCC_Stderr, "+++ Interaction: more than 2 Dirac/Majorana/Ghost "
                             "particles are interacting.\n");
        fprintf(GRCC_Stderr, "    Interaction has %d Dirac, %d Ghost and %d Majorana "
                             "particles.\n", ndir, nmaj, ngho);
        fprintf(GRCC_Stderr, "    Sign factors related to fermions will not "
                             "be calculated correctly.\n");
        mdl->skipFLine = True;
    }
    if (!ok) {
        prInteraction();
        erEnd("illegal Dirac/Majorana/Ghost interaction");
    }
}

//--------------------------------------------------------------
Interaction::~Interaction(void)
{
    if (slist != NULL) {
        delete[] slist;
```

```
    }
    if (plist != NULL) {
        delete[] plist;
    }
    if (clist != NULL) {
        delete[] clist;
    }
    free(name);
}

//--------------------------------------------------------------
void Interaction::prInteraction(void)
{
    int j;

    printf("vid=%d, icode=%d, name=\"%s\", loop=%d, csum=%d, cpl=",
          id, icode, name, loop, csum);
    prIntArray(mdl->ncouple, clist, ", legs=[");
    for (j = 0; j < nplist; j++) {
        if (j != 0) {
            printf(", ");
        }
        printf("%s", mdl->particleName(plist[j]));
    }
    printf("];\n");

}

//==============================================================
// class Model
//--------------------------------------------------------------
Model::Model(MInput *minp)
{
    static PInput pundef = {"undef", "undef", 0, 0, "undef", 0, 0};
    int j;

    if (minp->name == NULL || strlen(minp->name) < 1) {
        erEnd("model should have a name");
    }
    name    = strdup(minp->name);
    ncouple = minp->ncouple;
    if (ncouple >= GRCC_MAXNCPLG) {
        erEnd("too many coupling constants in the model (GRCC_MAXNCPLG)");
    }
    // cnlist = minp->cnamlist;
    cnlist = new char*[ncouple];
    for (j = 0; j < ncouple; j++) {
        cnlist[j] = strdup(minp->cnamlist[j]);
    }

    nParticles = 0;
    particles  = new Particle*[GRCC_MAXMPARTICLES];
    for (j = 0; j < GRCC_MAXMPARTICLES; j++) {
        particles[j] = NULL;
    }
    pdef = False;

    nInteracts = 0;
    interacts  = new Interaction*[GRCC_MAXMINTERACT];
    for (j = 0; j < GRCC_MAXMINTERACT; j++) {
        interacts[j] = NULL;
    }
    vdef       = False;

    if (particles == NULL || interacts == NULL) {
        erEnd("memory allocation failed");
    }

#ifdef UNIQINTR
    if (minp->defpart != GRCC_DEFBYNAME
        && minp->defpart != GRCC_DEFBYCODE) {
        erEnd("illegal value of defpart");
    }
#else
    if (Abs(minp->defpart) != GRCC_DEFBYNAME
        && Abs(minp->defpart) != GRCC_DEFBYCODE) {
        erEnd("illegal value of defpart");
    }
#endif

    defpart = minp->defpart;

    maxnlegs   = 0;
    maxcpl     = 0;
    maxloop    = 0;
    ncplgcp    = 0;
    skipFLine  = False;
```

```
    addParticle(&pundef);
}

//--------------------------------------------------------------
Model::~Model(void)
{
    int j;

    for (j = ncplgcp-1; j >= 0; j--) {
        cplgvl[j] = delintdup(cplgvl[j]);
    }
    delete[] cplgvl;
    cplgnvl = delintdup(cplgnvl);
    cplglg  = delintdup(cplglg);
    cplgcp  = delintdup(cplgcp);

    for (j = GRCC_MAXMINTERACT-1; j >= 0; j--) {
        if (interacts[j] != NULL) {
            interacts[j]->slist = delintdup(interacts[j]->slist);
            delete interacts[j];
            interacts[j] = NULL;
        }
    }
    delete[] interacts;
    interacts = NULL;

    for (j = GRCC_MAXMPARTICLES-1; j >= 0; j--) {
        if (particles[j] != NULL) {
            delete particles[j];
            particles[j] = NULL;
        }
    }
    delete[] particles;
    particles = NULL;

    for (j=ncouple-1; j>=0; j--) {
        free(cnlist[j]);
    }
    delete[] cnlist;

    free(name);
}

//--------------------------------------------------------------
void Model::prModel(void)
{
    static char hdr[] = "#================================================\n";
    static char hd1[] = "#------------------------------------------------\n";
    int j;

    printf("%s", hdr);
    printf("Model=\"%s\", ", name);
    printf("coupling=[");
    for (j = 0; j < ncouple; j++) {
        if (j != 0) {
            printf(", ");
        }
        printf("\"%s\"", cnlist[j]);
    }
    printf("];\n");
    printf("%s", hd1);
    printf("Particles=%d;\n", nParticles);
    for (j = 1; j < nParticles; j++) {
        particles[j]->prParticle();
    }
    printf("EndParticle;\n");
    printf("allPart (%d) = [", nallPart);
    for (j = 0; j < nallPart; j++) {
        if (j != 0) {
            printf(", ");
        }
        printf("%d", allPart[j]);
    }
    printf("]\n");

    printf("%s", hd1);
    printf("Interactions=%d;\n", nInteracts);
    for (j = 0; j < nInteracts; j++) {
        interacts[j]->prInteraction();
    }
    printf("EndInteraction;\n");
    printf("%s", hd1);
    printf("InteractionTable;\n");
    printf("#  %d class in (total coupling, degree)\n", ncplgcp);
    for (j = 0; j < ncplgcp; j++) {
        printf("  class=%d: cp=%d, lg=%d, vl=", j, cplgcp[j], cplglg[j]);
```

```
        prIntArray(cplgnvl[j], cplgvl[j], "\n");
    }
    printf("EndInteractionTable;\n");
    printf("%s", hdr);
}

//--------------------------------------------------------------
void Model::addParticle(PInput *pinp)
{
    int  nid, aid, pid, pcd, acd;

    if (nParticles >= GRCC_MAXMPARTICLES) {
        erEnd("particle table overflow (GRCC_MAXMPARTICLES)");
    }

    // uniqueness check
    if (Abs(defpart) == GRCC_DEFBYCODE) {
        pcd = findParticleCode(pinp->pcode);
        acd = findParticleCode(pinp->acode);
        if (pcd > 0 || acd > 0) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** particle code [%d, %d] ",
                        pinp->pcode, pinp->acode);
                fprintf(GRCC_Stderr, "is already used.\n");
            }
            erEnd("particle code is already defined");
        }
    } else if (Abs(defpart) == GRCC_DEFBYNAME) {
        if (pinp->name == NULL || pinp->aname == NULL ||
            strlen(pinp->name) < 1 || strlen(pinp->aname) < 1) {
            erEnd("no name of particle or anti-particle.");
        }
        nid = findParticleName(pinp->name);
        aid = findParticleName(pinp->aname);
        if (nid > 0 || aid > 0) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** particle name [%s, %s] ",
                        pinp->name, pinp->aname);
                fprintf(GRCC_Stderr, "is already used.\n");
            }
            erEnd("particle name is already defined.");
        }
    }

    pid = nParticles++;
    particles[pid] = new Particle(this, pid, pinp);
}

//--------------------------------------------------------------
void Model::addParticleEnd(void)
{
    int p, ap;

    pdef = True;

#ifdef OPTEXTONLY
    nallPart = 0;
    for (p = nParticles-1; p >= 1; p--) {
        if (!particles[p]->extonly) {
            ap = antiParticle(p);
            if (ap != p) {
                allPart[nallPart++] = ap;
            }
        }
    }
    for (p = 1; p < nParticles; p++) {
        if (!particles[p]->extonly) {
            allPart[nallPart++] = p;
        }
    }
    sorti(nallPart, allPart);
#else
    nallPart = 0;
    for (p = nParticles-1; p >= 1; p--) {
        ap = antiParticle(p);
        if (ap != p) {
            allPart[nallPart++] = ap;
        }
    }
    for (p = 1; p < nParticles; p++) {
        allPart[nallPart++] = p;
    }
    sorti(nallPart, allPart);
#endif
}

//--------------------------------------------------------------
```

```
void Model::addInteraction(IInput *iinp)
{
    int vid, nlegs, j, k, c;
    int csum, lp2, lp;
    int plist[GRCC_MAXLEGS];

    if (!pdef) {
        erEnd("call 'addParticleEnd' before 'addInteraction'");
    }
    if (nInteracts >= GRCC_MAXMINTERACT) {
        erEnd("too many interactions in the model (GRCC_MAXMINTERACT)");
    }
    nlegs = iinp->nplistn;
    if (nlegs >= GRCC_MAXLEGS) {
        erEnd("too many legs in an interaction (GRCC_MAXLEGS)");
    }

    // check identifier
    if (defpart == GRCC_DEFBYCODE) {
        if (iinp->icode < 0) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** vertex code %d should be positive\n",
                        iinp->icode);
            }
            erEnd("vertex code should be positive");
        } else {
            vid = findInteractionCode(iinp->icode);
            if (vid >= 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** vertex code %d is already used\n",
                            iinp->icode);
                }
                erEnd("vertex code is already used");
            }
        }
    } else if (defpart == GRCC_DEFBYNAME) {
        if (iinp->name == NULL || strlen(iinp->name) < 1) {
            erEnd("no name of vertex\n");
        }
        vid = findInteractionName(iinp->name);
        if (vid >= 0) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** vertex name %s is already used\n",
                iinp->name);
                erEnd("vertex name is already used");
            }
        }
    } else {
#ifdef UNIQINTR
        // defpart =< 0
        erEnd("illegal value of defpart");
#else
        // don't care about duplicated name/code of interaction
        ;
#endif
    }
    vid = nInteracts++;

    for (j = 0; j < nlegs; j++) {
        if (Abs(defpart) == GRCC_DEFBYCODE) {
            plist[j] = findParticleCode(iinp->plistc[j]);
            if (plist[j] == 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** particle code %d ",
                            iinp->plistc[j]);
                    fprintf(GRCC_Stderr, "is not defined\n");
                }
                erEnd("particle is not defined (code)");
            }
        } else if (Abs(defpart) == GRCC_DEFBYNAME) {
            plist[j] = findParticleName(iinp->plistn[j]);
            if (plist[j] == 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** particle name %s ",
                            iinp->plistn[j]);
                    fprintf(GRCC_Stderr, "is not defined\n");
                }
                erEnd("particle is not defined (name)");
            }
        }
    }

    csum = 0;
    for (j = 0; j < ncouple; j++) {
        c = iinp->cvallist[j];
        if (c < 0) {
            if (prlevel > 0) {
```

```
                fprintf(GRCC_Stderr, "*** illegal value of c-constants \n");
                for (k = 0; k < ncouple; k++) {
                    fprintf(GRCC_Stderr, "    %s = %d\n",
                            cnlist[k], iinp->cvallist[k]);
                }
            }
            erEnd("illegal value of c-constants");
        }
        csum += c;
    }
    if (csum < 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** illegal total coupling-constants %d\n",
                    csum);
            for (k = 0; k < ncouple; k++) {
                fprintf(GRCC_Stderr, "    %s = %d\n",
                        cnlist[k], iinp->cvallist[k]);
            }
        }
        erEnd("illegal value of c-constants");
    }

    // assume : csum = nlegs - 2 + 2*loop
    lp2 = csum - nlegs + 2;
    if (lp2 % 2 != 0 || lp2 < 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** illegal coupling const : ");
            fprintf(GRCC_Stderr, "nlegs - 2 + 2*loop: 2*loop = %d\n", lp2);
        }
    }
    lp = lp2/2;

    maxnlegs = Max(maxnlegs, nlegs);
    maxloop  = Max(maxloop , lp);
    maxcpl   = Max(maxcpl  , csum);

    interacts[vid] = new Interaction(this, vid, iinp->name, iinp->icode,
                         iinp->cvallist, nlegs, plist, csum, lp);
}

//--------------------------------------------------------------
void Model::addInteractionEnd(void)
{
    int lst[GRCC_MAXMINTERACT];
    int tcp[GRCC_MAXMINTERACT], tlg[GRCC_MAXMINTERACT];
    int tnvl[GRCC_MAXMINTERACT], *tvl[GRCC_MAXMINTERACT];
    Interaction *vt;
    int cp, lg, j, k, p;

    vdef = True;
    ncplgcp = 0;
    for (cp = 0; cp <= maxcpl; cp++) {

        for (lg = 0; lg <= maxnlegs; lg++) {
            k = 0;
            for (j = 0; j < nInteracts; j++) {
                vt = interacts[j];
                if (vt->csum == cp && vt->nlegs == lg) {
                    lst[k++] = j;
                }
            }
            if (k > 0) {
                tcp[ncplgcp]  = cp;
                tlg[ncplgcp]  = lg;
                tnvl[ncplgcp] = k;
                tvl[ncplgcp]  = intdup(k, lst);
                ncplgcp++;
            }
        }
    }
    cplgcp  = intdup(ncplgcp, tcp);
    cplglg  = intdup(ncplgcp, tlg);
    cplgnvl = intdup(ncplgcp, tnvl);
    cplgvl  = new int*[ncplgcp];
    for (j = 0; j < ncplgcp; j++) {
        if (cplgnvl[j] > 0) {
            cplgvl[j] = tvl[j];
        } else {
            cplgvl[j] = NULL;
        }
    }

    for (j = 0; j < nInteracts; j++) {
        vt = interacts[j];
        if (vt->slist != NULL) {
            erEnd("addInteractionEnd: vt->nslist != NULL");
        }
```

```
        vt->slist  = intdup(vt->nplist, vt->plist);
        vt->nslist = toSList(vt->nplist, vt->slist);
    }

    for (p = 0; p < nParticles; p++) {
        particles[p]->cmindeg = 0;
        particles[p]->cmaxdeg = 0;
    }
    for (j = 0; j < nInteracts; j++) {
        vt = interacts[j];
        for (k = 0; k < vt->nplist; k++) {
            p = abs(vt->plist[k]);
            if (particles[p]->cmindeg == 0) {
                particles[p]->cmindeg = vt->nlegs;
                particles[p]->cmaxdeg = vt->nlegs;
            } else {
                particles[p]->cmindeg = Min(particles[p]->cmindeg, vt->nlegs);
                particles[p]->cmaxdeg = Max(particles[p]->cmaxdeg, vt->nlegs);
            }
        }
    }
#ifdef DEBUG
    prModel();
#endif
}

//--------------------------------------------------------------
int Model::findParticleName(const char *name)
{
    int j;

    for (j = 0; j < nParticles; j++) {
        if (strcmp(name, particles[j]->name) == 0) {
            return j;
        }
    }
    for (j = 0; j < nParticles; j++) {
        if (strcmp(name, particles[j]->aname) == 0) {
            return -j;
        }
    }
    return 0;
}

//--------------------------------------------------------------
int Model::findParticleCode(int pcd)
{
    int j;

    for (j = 0; j < nParticles; j++) {
        if (pcd == particles[j]->pcode) {
            return j;
        } else if (pcd == particles[j]->acode) {
            return -j;
        }
    }
    return 0;
}

//--------------------------------------------------------------
char *Model::particleName(int p)
{
    int q;

    if (Abs(p) >= nParticles) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "\n*** Model::particleName: ");
            fprintf(GRCC_Stderr, "illegal particle id=%d\n", p);
        }
        erEnd("Model::particleName: illegal particle");
    }
    if (p < 0) {
        q = - p;
    } else {
        q = p;
    }
    return particles[q]->particleName(p);
}

//--------------------------------------------------------------
int Model::particleCode(int p)
{
    int q;

    if (Abs(p) >= nParticles) {
        fprintf(GRCC_Stderr, "\n*** Model::particleCode: illegal particle id=%d\n",
                p);
```

```
        erEnd("Model::particleCode: illegal particle");
    }
    if (p < 0) {
        q = - p;
    } else {
        q = p;
    }
    return particles[q]->particleCode(p);
}

//---------------------------------------------------------------
void Model::prParticleArray(int n, int *a, const char *msg)
{
    int j;

    printf("[");
    for (j = 0; j < n; j++) {
        if (j != 0) {
            printf(", ");
        }
        printf("%s", particleName(a[j]));
    }
    printf("]%s", msg);
}

//---------------------------------------------------------------
int Model::findMClass(const int cpl, const int dgr)
{
    int j;

    for (j = 0; j < ncplgcp; j++) {
        if (cplgcp[j] == cpl && cplglg[j] == dgr) {
            return j;
        }
    }
    return -1;
}

//---------------------------------------------------------------
int Model::normalParticle(int pt)
{
    int ptc = Abs(pt);
    Particle *p;

    if (ptc >= nParticles) {
        return 0;
    }
    p = particles[ptc];
    if (p->isNeutral()) {
        return ptc;
    } else {
        return pt;
    }
}

//---------------------------------------------------------------
int Model::antiParticle(int pt)
{
    int ptc = Abs(pt);
    Particle *p;

    p = particles[ptc];
    if (p->isNeutral()) {
        return ptc;
    } else {
        return -pt;
    }
}

//---------------------------------------------------------------
int *Model::allParticles(int *len)
{
    *len = nallPart;
    return allPart;
}

//---------------------------------------------------------------
int Model::findInteractionName(const char *name)
{
    int j;

    for (j = 0; j < nInteracts; j++) {
        if (strcmp(name, interacts[j]->name) == 0) {
            return j;
        }
    }
    return -1;
```

```
}

//----------------------------------------------------------------
int Model::findInteractionCode(int icd)
{
    int j;

    for (j = 0; j < nInteracts; j++) {
        if (icd == interacts[j]->icode) {
            return j;
        }
    }
    return -1;
}

//**************************************************************
// proc.cc

//==============================================================
// class PNodeClass
//--------------------------------------------------------------
PNodeClass::PNodeClass(SProcess *spc, int nnods, int nclss, NCInput *cls)
{
    //  Create a class of nodes
    //    nnodes : # nodes
    //    nclss  : # classes
    //    cls    : table of class inputs
    //
    int j, k, nn, lp2;
    Bool ok = True;

    sproc    = spc;
    nnodes   = nnods;
    nclass   = nclss;
    deg      = new int[nclass];
    type     = new int[nclass];
    particle = new int[nclass];
    count    = new int[nclass];
    couple   = new int[nclass];
    cmindeg  = new int[nclass];
    cmaxdeg  = new int[nclass];
    for (j = 0; j < nclass; j++) {
        deg[j]      = cls[j].cldeg;
        count[j]    = cls[j].clnum;
        type[j]     = cls[j].cltyp;
        cmindeg[j]  = cls[j].cmind;
        cmaxdeg[j]  = cls[j].cmaxd;
        particle[j] = cls[j].ptcl;
        couple[j]   = cls[j].cple;
        lp2 = (couple[j]-deg[j]+2);
        if (!isATExternal(type[j]) && (lp2 % 2 != 0 || lp2 < 0)) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** PNodeClass: illegal loop: "
                        ": 2*loop=cpl[%d](%d)-deg[%d](%d)+2 =2*loop = %d\n",
                        j, couple[j], j, deg[j], lp2);
                for (k = 0; k < nclss; k++) {
                    fprintf(GRCC_Stderr, "k=%d, deg=%d, typ=%d, ptcl=%d, ",
                            k, deg[k], type[k], particle[k]);
                    fprintf(GRCC_Stderr, "cpl=%d, cnt=%d\n",
                            couple[k], count[k]);
                }
            }
            ok = False;
        }
    }
    if (!ok) {
        erEnd("PNodeClass: illegal loop");
    }
    cl2nd    = new int[nclass+1];
    nd2cl    = new int[nnodes];
    cl2mcl   = new int[nnodes];

    nn = 0;
    for (j = 0; j < nclass; j++) {
        cl2nd[j]   = nn;
        for (k = 0; k < count[j]; k++, nn++) {
            nd2cl[nn]   = j;
        }
        if (isATExternal(type[j])) {
            cl2mcl[j] = -1;
        } else {
            cl2mcl[j] = spc->model->findMClass(couple[j], deg[j]);
            if (cl2mcl[j] < 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** PNodeClass : no vertex : ");
                    fprintf(GRCC_Stderr, "coupling=%d, degree=%d\n",
                           couple[j], deg[j]);
```

```
                }
                erEnd("PNodeClass : no vertex");
            }
        }
    }
    cl2nd[nclass] = nnodes;
}
//--------------------------------------------------------------
PNodeClass::PNodeClass(SProcess *spc, int nnods, int nclss, int *dgs, int *typ, int *ptcl, int *cpl,
int *cnt, int *cmind, int *cmaxd)
{
    // Create a class of nodes
    //    nnodes : # nodes
    //    nclss  : # classes
    //    dgs  : degree of a node, which is common to the vertices in a class
    //    typ  : initial/final/vertex
    //    ptcl : particle code or list of possible interactions
    //    cpl  : values of coupling constants.
    //    cnt  : the number of nodes in this class
    //
    int j, k, nn, lp2;
    Bool ok = True;

    sproc    = spc;
    nnodes   = nnods;
    nclass   = nclss;
    deg      = new int[nclass];
    type     = new int[nclass];
    particle = new int[nclass];
    count    = new int[nclass];
    couple   = new int[nclass];
    cmindeg  = new int[nclass];
    cmaxdeg  = new int[nclass];
    for (j = 0; j < nclass; j++) {
        deg[j]      = dgs[j];
        type[j]     = typ[j];
        particle[j] = ptcl[j];
        count[j]    = cnt[j];
        couple[j]   = cpl[j];
        cmindeg[j]  = cmind[j];
        cmaxdeg[j]  = cmaxd[j];
        lp2 = (cpl[j]-dgs[j]+2);
        if (!isATExternal(type[j]) && (lp2 % 2 != 0 || lp2 < 0)) {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** PNodeClass: illegal loop: "
                       ": 2*loop=cpl[%d](%d)-deg[%d](%d)+2 =2*loop = %d\n",
                       j, cpl[j], j, dgs[j], lp2);
                for (k = 0; k < nclss; k++) {
                    fprintf(GRCC_Stderr, "k=%d, dgs=%d, typ=%d, ptcl=%d, ",
                           k, dgs[k], typ[k], ptcl[k]);
                    fprintf(GRCC_Stderr, "cpl=%d, cnt=%d\n", cpl[k], cnt[k]);
                }
            }
            ok = False;
        }
    }
    if (!ok) {
        erEnd("PNodeClass: illegal loop");
    }
    cl2nd   = new int[nclass+1];
    nd2cl   = new int[nnodes];
    cl2mcl  = new int[nnodes];

    nn = 0;
    for (j = 0; j < nclass; j++) {
        cl2nd[j]   = nn;
        for (k = 0; k < count[j]; k++, nn++) {
            nd2cl[nn]   = j;
        }
        if (isATExternal(type[j])) {
            cl2mcl[j] = -1;
        } else {
            cl2mcl[j] = spc->model->findMClass(couple[j], deg[j]);
            if (cl2mcl[j] < 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** PNodeClass : no vertex : ");
                    fprintf(GRCC_Stderr, "coupling=%d, degree=%d\n",
                           couple[j], deg[j]);
                }
                erEnd("PNodeClass : no vertex");
            }
        }
    }
    cl2nd[nclass] = nnodes;
}

//--------------------------------------------------------------
```

```
PNodeClass::~PNodeClass(void)
{
    delete[] cl2mcl;
    delete[] nd2cl;
    delete[] cl2nd;
    delete[] cmaxdeg;
    delete[] cmindeg;
    delete[] couple;
    delete[] count;
    delete[] particle;
    delete[] type;
    delete[] deg;
}

//---------------------------------------------------------------
void PNodeClass::prPNodeClass(void)
{
    int j;

    printf("+++ PNodeClass: nclass=%d, nnodes=%d\n", nclass, nnodes);
    for (j = 0; j < nclass; j++) {
        prElem(j);
    }
    printf("\n");
}

//---------------------------------------------------------------
void PNodeClass::prElem(int j)
{
    int tp;

    printf("%3d: %-7s(%2d), ", j, GRCC_AT_NdStr(type[j]), type[j]);
    printf("deg=%d, count=%d, nodes[%d--%d], couple=%d, cmindeg=%d, cmaxdeg=%d",
           deg[j], count[j], cl2nd[j], cl2nd[j+1], couple[j], cmindeg[j], cmaxdeg[j]);
    tp = type[j];
    if (isATExternal(tp)) {
        if (sproc->model != NULL) {
            printf(", ptcl=%s ", sproc->model->particleName(particle[j]));
        } else {
            printf(", ptcl=%d ", particle[j]);
        }
    }
    printf("\n");

}

//===============================================================
// class SProcess
//---------------------------------------------------------------
SProcess::SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, int *cdeg,
int *ctyp, int *ptcl, int *cpl, int *cnum, int *cmind, int *cmaxd)
{
    // Construct a Subprocess object
    //    mdl             : model
    //    prc             : process (may be NULL)
    //    opts            : options
    //    sid             : id of the sprocess
    //    ncls            : the number of classes
    //    cdeg[ncls]      : the table of degrees of nodes.
    //    ctyp[ncls]      : the table of nodes in the classes
    //    ptcl[ncls]      : particle(External)/interaction code(Internal)
    //    cpl[ncls]       : the table of total order of coupling constants.
    //    cnum[ncls]      : the table of nodes in the classes
    //    cmind[ncls]     : the table of min(deg of connectabl vertex)
    //    cmaxd[ncls]     : the table of max(deg of connectabl vertex)

    int j, cp, lp2, ndeg, nvrt, tcpl0;
    bool ok;

    if (ncls < 1) {
        fprintf(GRCC_Stderr, "*** SProcess::SProcess: no class %d\n", ncls);
        erEnd("SProcess::SProcess: no Class");
    }
    id        = sid;
    model     = mdl;
    proc      = prc;
    opt       = opts;
    nclass    = ncls;

    nNodes    = 0;
    nEdges    = 0;
    nExtern   = 0;
    tCouple   = 0;
    loop      = 0;
    mgraph    = NULL;
    egraph    = NULL;
    astack    = NULL;
```

```

    mgrcount = 0;
    agrcount = 0;
    extperm  = 1;

    // the results of the graph generation
    nMGraphs = 0;                 // the number of generated M-graphs
    nMOPI    = 0;                 // the number of 1PI M-graphs
    wMGraphs.setValue(0,1);    // the weighted sum of M-graphs
    wMOPI.setValue(0,1);       // the weighted sum of 1PI M-graphs

    nAGraphs = 0;                 // the number of generated A-graphs
    nAOPI    = 0;                 // the number of 1PI A-graphs
    wAGraphs.setValue(0,1);    // the weighted sum of A-graphs
    wAOPI.setValue(0,1);       // the weighted sum of 1PI A-graphs

    // check input
    nNodes  = 0;
    nExtern = 0;

    ndeg     = 0;
    nvrt     = 0;
    ok = True;

    tcpl0 = 0;
    for (j = 0; j < model->ncouple; j++) {
        clist[j] = clst[j];
        tcpl0 += clist[j];
    }
    for (j = 0; j < nclass; j++) {
        cp  = cpl[j];
        if (cp > 0) {
            lp2 = cpl[j] - cdeg[j] + 2;
            if (lp2 % 2 != 0 || lp2 < 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop:"
                           "class %d, cp=%d, deg=%d, lp2=%d\n",
                           j, cp, cdeg[j], lp2);
                }
                ok = False;
            }
        }
        tCouple +=      cp*cnum[j];
        ndeg    += cdeg[j]*cnum[j];

        if (!isATExternal(ctyp[j])) {
            nvrt += cnum[j];
        } else if (isATExternal(ctyp[j])) {
            if (model->normalParticle(ptcl[j]) == 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
                    fprintf(GRCC_Stderr, "illegal particle code: ");
                    fprintf(GRCC_Stderr, "code: class %d, ptcl=%d\n", j, ptcl[j]);
                }
                ok = False;
            } else {
                nExtern += cnum[j];
            }
            extperm *= factorial(cnum[j]);

        } else {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal type: ");
                fprintf(GRCC_Stderr, "class %d, type=%d\n", j, ctyp[j]);
            }
            ok = False;
        }
    }
    if (tcpl0 != tCouple) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
            fprintf(GRCC_Stderr, "illegal coupling constants:");
            fprintf(GRCC_Stderr, " %d != %d\n", tcpl0, tCouple);
        }
        ok = False;
    }
    if (ndeg == nExtern) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
            fprintf(GRCC_Stderr, "no vertices: ");
            fprintf(GRCC_Stderr, "nExtern=%d, ndeg=%d\n", nExtern, ndeg);
        }
        ok = False;
    }
    if (ndeg % 2 == 0) {
        nEdges = ndeg/2;
    } else {
```

```
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
            fprintf(GRCC_Stderr, "illegal total deg = %d (not even)\n", ndeg);
        }
        ok = False;
    }
    nNodes = nvrt + nExtern;
    if (nNodes >= GRCC_MAXNODES) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
            fprintf(GRCC_Stderr, "too many nodes = %d\n", nNodes);
            fprintf(GRCC_Stderr, "    nExtern=%d, nvert=%d (GRCC_MAXNODES)\n",
                    nExtern, nvert);
        }
        ok = False;
    }
    if (!ok) {
        prSProcess();
        erEnd("SProcess::SProcess: illegal input");
    }
    if (proc == NULL) {
        opt->beginProc(NULL);
    }

    lp2 = tCouple - nExtern + 2;
    if (lp2 % 2 != 0 || lp2 < 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop: "
                   "tCouple=%d, nExtern=%d, 2*loop=%d\n",
                   tCouple, nExtern, lp2);
        }
        ok = False;
    }
    loop = lp2/2;

    // stack for assignment
    if (opt->values[GRCC_OPT_Step] == GRCC_AGraph) {
        astack = new AStack(GRCC_MAXNSTACK, GRCC_MAXESTACK);
    }

    // save to PNodeClass object
    pnclass = new PNodeClass(this, nNodes, nclass, cdeg, ctyp, ptcl, cpl, cnum, cmind, cmaxd);
}

//--------------------------------------------------------------
SProcess::SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, NCInput *cls)
{
    // Construct a Subprocess object
    //    mdl            : model
    //    prc            : process (may be NULL)
    //    opts           : options
    //    sid            : id of the sprocess
    //    ncls           : the number of classes
    //    cls            : input data of classes

    int j, cp, lp2, ndeg, nvrt, tcpl0;
    bool ok;

    if (ncls < 1) {
        fprintf(GRCC_Stderr, "*** SProcess::SProcess: no class %d\n", ncls);
        erEnd("SProcess::SProcess: no Class");
    }
    id       = sid;
    model    = mdl;
    proc     = prc;
    opt      = opts;
    nclass   = ncls;

    nNodes   = 0;
    nEdges   = 0;
    nExtern  = 0;
    tCouple  = 0;
    loop     = 0;
    mgraph   = NULL;
    egraph   = NULL;
    astack   = NULL;

    mgrcount = 0;
    agrcount = 0;
    extperm  = 1;

    // the results of the graph generation
    nMGraphs = 0;                // the number of generated M-graphs
    nMOPI    = 0;                // the number of 1PI M-graphs
    wMGraphs.setValue(0,1);    // the weighted sum of M-graphs
    wMOPI.setValue(0,1);       // the weighted sum of 1PI M-graphs
```

```
    nAGraphs = 0;                // the number of generated A-graphs
    nAOPI     = 0;               // the number of 1PI A-graphs
    wAGraphs.setValue(0,1);    // the weighted sum of A-graphs
    wAOPI.setValue(0,1);       // the weighted sum of 1PI A-graphs

    // check input
    nNodes  = 0;
    nExtern = 0;

    ndeg     = 0;
    nvrt     = 0;
    ok = True;

    tcpl0 = 0;
    for (j = 0; j < model->ncouple; j++) {
        clist[j] = clst[j];
        tcpl0 += clist[j];
    }
    for (j = 0; j < nclass; j++) {
        cp  = cls[j].cple;
        if (cp > 0) {
            lp2 = cls[j].cple - cls[j].cldeg + 2;
            if (lp2 % 2 != 0 || lp2 < 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop: "
                           "class %d, cp=%d, deg=%d, lp2=%d\n",
                           j, cp, cls[j].cldeg, lp2);
                }
                ok = False;
            }
        }
        tCouple += cp*cls[j].clnum;
        ndeg    += cls[j].cldeg*cls[j].clnum;

        if (!isATExternal(cls[j].cltyp)) {
            nvrt += cls[j].clnum;
        } else if (isATExternal(cls[j].cltyp)) {
            if (model->normalParticle(cls[j].ptcl) == 0) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
                    fprintf(GRCC_Stderr, "illegal particle code: ");
                    fprintf(GRCC_Stderr, "code: class %d, ptcl=%d\n", j, cls[j].ptcl);
                }
                ok = False;
            } else {
                nExtern += cls[j].clnum;
            }
            extperm *= factorial(cls[j].clnum);

        } else {
            if (prlevel > 0) {
                fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal type: ");
                fprintf(GRCC_Stderr, "class %d, type=%d\n", j, cls[j].cltyp);
            }
            ok = False;
        }
    }
    if (tcpl0 != tCouple) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
            fprintf(GRCC_Stderr, "illegal coupling constants:");
            fprintf(GRCC_Stderr, " %d != %d\n", tcpl0, tCouple);
        }
        ok = False;
    }
    if (ndeg == nExtern) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
            fprintf(GRCC_Stderr, "no vertices: ");
            fprintf(GRCC_Stderr, "nExtern=%d, ndeg=%d\n", nExtern, ndeg);
        }
        ok = False;
    }
    if (ndeg % 2 == 0) {
        nEdges = ndeg/2;
    } else {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
            fprintf(GRCC_Stderr, "illegal total deg = %d (not even)\n", ndeg);
        }
        ok = False;
    }
    nNodes = nvrt + nExtern;
    if (nNodes >= GRCC_MAXNODES) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: ");
```

```
            fprintf(GRCC_Stderr, "too many nodes = %d\n", nNodes);
            fprintf(GRCC_Stderr, "    nExtern=%d, nvert=%d (GRCC_MAXNODES)\n",
                    nExtern, nvert);
        }
        ok = False;
    }
    if (!ok) {
        erEnd("SProcess::SProcess: illegal input");
    }
    if (proc == NULL) {
        opt->beginProc(NULL);
    }

    lp2 = tCouple - nExtern + 2;
    if (lp2 % 2 != 0 || lp2 < 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::SProcess: illegal loop: "
                   "tCouple=%d, nExtern=%d, 2*loop=%d\n",
                   tCouple, nExtern, lp2);
        }
        ok = False;
    }
    loop = lp2/2;

    // stack for assignment
    if (opt->values[GRCC_OPT_Step] == GRCC_AGraph) {
        astack = new AStack(GRCC_MAXNSTACK, GRCC_MAXESTACK);
    }

    // save to PNodeClass object
    pnclass = new PNodeClass(this, nNodes, nclass, cls);
}

//--------------------------------------------------------------
SProcess::~SProcess(void)
{
    delete pnclass;
    delete astack;
    delete mgraph;
}

//--------------------------------------------------------------
void SProcess::prSProcess(void)
{
    printf("\n");
    printf("+++ Subprocess %d, class=%d\n", id, nclass);
    if (pnclass != NULL) {
        pnclass->prPNodeClass();
    } else {
        printf("  pnclass = NULL\n");
   }
}

//--------------------------------------------------------------
BigInt SProcess::generate(void)
{
    //  Entry point of Feynman graph generation
    int  cldeg[GRCC_MAXNODES], clnum[GRCC_MAXNODES], cltyp[GRCC_MAXNODES];
    int  cmind[GRCC_MAXNODES], cmaxd[GRCC_MAXNODES];
    int  ncl;
    BigInt ng;

    ncl = toMNodeClass(cltyp, cldeg, clnum, cmind, cmaxd);

    opt->beginSubProc(this);

    mgraph = new MGraph(id, ncl, cldeg, clnum, cltyp, cmind, cmaxd, opt);

    ng = mgraph->generate();

    opt->endSubProc();

    delete mgraph;
    mgraph = NULL;

    return ng;
}

//--------------------------------------------------------------
int SProcess::toMNodeClass(int *cltyp, int *cldeg, int *clnum, int *cmind, int *cmaxd)
{
    int j;

    for (j = 0; j < nclass; j++) {
        if (isATExternal(pnclass->type[j])) {
            cltyp[j] = -1;
        } else {
```

```
            cltyp[j] = (pnclass->couple[j]-pnclass->deg[j]+2)/2;
        }
        cldeg[j] = pnclass->deg[j];
        clnum[j] = pnclass->count[j];
        cmind[j] = pnclass->cmindeg[j];
        cmaxd[j] = pnclass->cmaxdeg[j];
    }
    return nclass;
}

//-------------------------------------------------------------
void SProcess::assign(MGraph *mgr)
{
    PNodeClass *pnc;

    pnc = match(mgr);

    agraph = new Assign(this, mgr, pnc);

#ifdef CHECK
    agraph->checkAG("SProcess::assign");
#endif
    delete pnc;
    delete agraph;
    agraph = NULL;
}

//-------------------------------------------------------------
PNodeClass *SProcess::match(MGraph *mgr)
{
    PNodeClass *pnc = NULL;

    int typ[GRCC_MAXNODES], dgs[GRCC_MAXNODES];
    int cnt[GRCC_MAXNODES], ptcl[GRCC_MAXNODES];
    int cpl[GRCC_MAXNODES];
    int n2m[GRCC_MAXNODES];
    int cmind[GRCC_MAXNODES], cmaxd[GRCC_MAXNODES];
    int j, k, l, nd, mc, mcl, mclss;

    if (mgr->nNodes != nNodes) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** SProcess::match: different nNodes=%d != %d\n",
                    nNodes, mgr->nNodes);
        }
        erEnd("SProcess::match: different nNodes");
    }

    mcl = toMNodeClass(typ, dgs, cnt, cmind, cmaxd);

    nd = 0;
    for (j = 0; j < mcl; j++) {
        for (k = 0; k < cnt[j]; k++, nd++) {
            n2m[nd] = j;
        }
    }

    nd    = 0;
    mclss = mgr->curcl->nClasses;
    for (j = 0; j < mclss; j++) {
        for (k = 0; k < mgr->curcl->clist[j]; k++, nd++) {
            mc = mgr->nodes[nd]->clss;
            if (mc != n2m[nd]) {
                if (prlevel > 0) {
                    fprintf(GRCC_Stderr, "*** SProcess::match: ");
                    fprintf(GRCC_Stderr, "inconsistent class\n");
                    for (l = 0; l < nNodes; l++) {
                        fprintf(GRCC_Stderr, "    %d: %d, %d\n",
                                j, mgr->nodes[l]->clss, n2m[l]);
                    }
                }
                erEnd("SProcess::match: inconsistent class");
            }
        }
    }
    for (j = 0; j < mcl; j++) {
        dgs[j]    = pnclass->deg[j];
        typ[j]    = pnclass->type[j];
        ptcl[j]   = pnclass->particle[j];
        cnt[j]    = pnclass->count[j];
        cpl[j]    = pnclass->couple[j];
        cmind[j]  = pnclass->cmindeg[j];
        cmaxd[j]  = pnclass->cmaxdeg[j];
    }
    pnc = new PNodeClass(this, nNodes, mcl, dgs, typ, ptcl, cpl, cnt, cmind, cmaxd);

    return pnc;
}
```

```

//-----------------------------------------------------------------
void SProcess::resultMGraph(BigInt nmgraphs, Fraction mwsum, BigInt nmopi, Fraction mwopi)
{
    nMGraphs += nmgraphs;
    nMOPI    += nmopi;
    wMGraphs.add(mwsum);
    wMOPI.add(mwopi);
}

//-----------------------------------------------------------------
void SProcess::resultAGraph(BigInt nmgraphs, Fraction mwsum, BigInt nmopi, Fraction mwopi)
{
    nAGraphs += nmgraphs;
    nAOPI    += nmopi;
    wAGraphs.add(mwsum);
    wAOPI.add(mwopi);
}

//-----------------------------------------------------------------
void SProcess::endMGraph(MGraph *mgr)
{
    egraph = mgr->egraph;

    if (opt->values[GRCC_OPT_Step] != GRCC_MGraph) {
        assign(mgr);
    }
}

//-----------------------------------------------------------------
void SProcess::endAGraph(EGraph *egr)
{
    egraph = egr;
}


//=================================================================
// class Process
//-----------------------------------------------------------------
Process::Process(int pid, Model *modl, Options *optn, int nini, int *intlPrt, int nfin, int *finlPrt, int *coupling)
{
    //  Define a process and construct a set of sprocesses
    //    model      : Model object
    //    opt        : Option object
    //    initlPart  : list of particle-id of initlPart particles
    //    finalPart  : list of particle-id of final particles
    //    coupling   : list of coupling constants

    int j, lp2;
    Bool ok;

    id         = pid;
    model      = modl;
    opt        = optn;
    ninitl     = nini;
    nfinal     = nfin;
    initlPart = intdup(ninitl, intlPrt);
    finalPart = intdup(nfinal, finlPrt);

    // count total number of coupling constants.
    ctotal = 0;
    for (j = 0; j < GRCC_MAXNCPLG; j++) {
        if (j < model->ncouple) {
            clist[j] = coupling[j];
            ctotal   += coupling[j];
        } else {
            clist[j] = 0;
        }
    }
    if (ctotal < 1) {
        erEnd("Process: coupling constant = 0");
    }

    nExtern    =  0;
    loop       = -1;
    maxnlegs   =  0;
    mgrcount = 0;
    agrcount = 0;

    // table of sprocesses
    nSubproc   = 0;
    sproc    = NULL;

    // the results of the graph generation
    nMGraphs = 0;                  // the number of generated M-graphs
    nMOPI    = 0;                  // the number of 1PI M-graphs
```

```
    wMGraphs.setValue(0,1);       // the weighted sum of M-graphs
    wMOPI.setValue(0,1);          // the weighted sum of 1PI M-graphs

    nAGraphs = 0;                 // the number of generated A-graphs
    nAOPI    = 0;                 // the number of 1PI A-graphs
    wAGraphs.setValue(0,1);       // the weighted sum of A-graphs
    wAOPI.setValue(0,1);          // the weighted sum of 1PI A-graphs

    ok = True;

    // numbers
    nExtern  = ninitl + nfinal;
    maxnlegs = Max(nExtern, model->maxnlegs);

    // count loops
    lp2 = ctotal - nExtern + 2;
    if (lp2 % 2 != 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** cannot generate : 2*loop is odd : "
                    "2*loop = %d, ctotal=%d, nExtern=%d\n",
                    lp2, ctotal, nExtern);
        }
        ok = False;
    } else if (lp2 < 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** cannot make a connected graph : "
                    "2*loop=%d, ctotal=%d, nExtern=%d\n",
                    lp2, ctotal, nExtern);
        }
        ok = False;
    }

    loop = lp2 / 2;

    if (!ok) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** Process: illegal input: lp2 = %d\n", lp2);
        }
        erEnd("Process: illegal input");
    }

#ifdef DEBUG
    // print message
    prProcess();
#endif

    // construct sprocesses
    mkSProcess();
}

//--------------------------------------------------------------
Process::~Process(void)
{
    initlPart = delintdup(initlPart);
    finalPart = delintdup(finalPart);
    delete sproc;
}

//--------------------------------------------------------------
void Process::prProcess(void)
{
    printf("+++ process options : OPI = %d, (Step) = %d, coupling=%d: ",
           opt->values[GRCC_OPT_1PI], opt->values[GRCC_OPT_Step], ctotal);
    prIntArray(model->ncouple, clist, "\n");
}

//--------------------------------------------------------------
void Process::mkSProcess(void)
{
    //  Construct sprocesses
    //  Each sprocess is a set of nodes whose degree and order of
    //  coupling constants are determined.
    //  Only the total coupling constants are considered here even if
    //  two or more coupling constants are defined in the model

    NCInput cls[GRCC_MAXMINTERACT];
    int r, j, k, lin2, nvtx, nleg, nc, n0;
    int nl[GRCC_MAXMINTERACT];
    double proct0 = 0, proct1 = 0;

    // output file to start process
    opt->beginProc(this);

    // starting time
    Second(&proct0);

    // generate all possible number of (nl[i]),
```

```
    //   \sum_i nl[i]*cplgcp[i] = (total number of coupling constants)

    r = -1;
    nSubproc = 0;
    ngraphs  = 0;
    nopi     = 0;
    wgraphs  = 0;
    wopi     = 0;

    // for partitions of (leg, order)

    while (nextPart(ctotal, model->ncplgcp, model->cplgcp, nl, &r)) {
#ifdef DEBUG
        printf("nextPart:ctotal=%d, nc=%d, c=", ctotal, model->ncplgcp);
        prIntArray(model->ncplgcp, model->cplgcp, ", nl=");
        prIntArray(model->ncplgcp, nl, "");
        printf(", r=%d\n", r);
#endif
        // count the total number of vertices and legs.
        nvtx = 0;
        nleg = 0;
        for (j = 0; j < model->ncplgcp; j++) {
            nvtx += nl[j];
            nleg += nl[j]*model->cplglg[j];
        }

        // count loops
        lin2 = nExtern + nleg;
        if (lin2 % 2 != 0) {
            continue;
        }
        loop = lin2/2 - nExtern - nvtx + 1;
        if (loop < 0) {
            continue;
        }

        nc = 0;
        if (opt->values[GRCC_OPT_SymmInitial]) {
            n0 = nc;
            for (j = 0; j < ninitl; j++) {
                for (k = n0; k < nc; k++) {
                    if (cls[k].ptcl == initlPart[j]) {
                        cls[k].clnum++;
                        break;
                    }
                }
                if (k >= nc) {
                    cls[nc].ptcl  = initlPart[j];
                    cls[nc].clnum = 1;
                    cls[nc].cple  = 0;
                    cls[nc].clnum = 1;
                    cls[nc].cldeg = 1;
                    cls[nc].cltyp = GRCC_AT_Initial;
                    cls[nc].ptcl  = initlPart[j];
                    cls[nc].cmind
                        = model->particles[Abs(cls[nc].ptcl)]->cmindeg;
                    cls[nc].cmaxd
                        = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
                    if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
                        cls[nc].cmind = Max(cls[nc].cmind, 3);
                    }
                    nc++;
                }
            }
        } else {
            for (j = 0; j < ninitl; j++) {
                cls[nc].ptcl  = initlPart[j];
                cls[nc].clnum = 1;
                cls[nc].cple  = 0;
                cls[nc].cldeg = 1;
                cls[nc].cltyp = GRCC_AT_Initial;
                cls[nc].cmind = model->particles[Abs(cls[nc].ptcl)]->cmindeg;
                cls[nc].cmaxd = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
                if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
                    cls[nc].cmind = Max(cls[nc].cmind, 3);
                }
                nc++;
            }
        }
        if (opt->values[GRCC_OPT_SymmFinal]) {
            n0 = nc;
            for (j = 0; j < nfinal; j++) {
                for (k = n0; k < nc; k++) {
                    if (cls[k].ptcl == model->antiParticle(finalPart[j])) {
                        cls[k].clnum++;
                        break;
                    }
```

```
                }
                if (k >= nc) {
                    cls[nc].ptcl  = model->antiParticle(finalPart[j]);
                    cls[nc].clnum = 1;
                    cls[nc].cple  = 0;
                    cls[nc].cldeg = 1;
                    cls[nc].cltyp = GRCC_AT_Final;
                    cls[nc].cmind
                        = model->particles[Abs(cls[nc].ptcl)]->cmindeg;
                    cls[nc].cmaxd
                        = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
                    if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
                        cls[nc].cmind = Max(cls[nc].cmind, 3);
                    }
                    nc++;
                }
            }
        } else {
            for (j = 0; j < nfinal; j++) {
                cls[nc].ptcl  = model->antiParticle(finalPart[j]);
                cls[nc].cldeg = 1;
                cls[nc].cple  = 0;
                cls[nc].clnum = 1;
                cls[nc].cltyp = GRCC_AT_Final;
                cls[nc].cmind = model->particles[Abs(cls[nc].ptcl)]->cmindeg;
                cls[nc].cmaxd = model->particles[Abs(cls[nc].ptcl)]->cmaxdeg;
                if (opt->values[GRCC_OPT_NoExtSelf] > 0 && nExtern > 2) {
                    cls[nc].cmind = Max(cls[nc].cmind, 3);
                }
                nc++;
            }
        }
        for (j = 0; j < model->ncplgcp; j++) {
            if (nl[j] > 0) {
                cls[nc].ptcl  = 0;
                cls[nc].cple  = model->cplgcp[j];
                cls[nc].cldeg = model->cplglg[j];
                cls[nc].clnum = nl[j];
                // number of loops
                cls[nc].cltyp = (model->cplgcp[j] - model->cplglg[j] + 2)/2;
                cls[nc].cmind = 0;
                cls[nc].cmaxd = 0;
                nc++;
            }
        }
#ifdef DEBUG1
        for (j = 0; j < nc; j++) {
            printf("%d/%d: deg=%d, typ=%d, ptcl=%d, cple=%d, num=%d\n",
                   j, nc, cls[j].cldeg, cls[j].cltyp, cls[j].ptcl,
                   cls[j].cple, cls[j].clnum);
        }
#endif
        // create a sprocess ??? to be rewritten
        sproc = new SProcess(model, this, opt, nSubproc, clist, nc, cls);

        // list of sprocesses
        if (nSubproc >= GRCC_MAXSUBPROCS) {
            erEnd("Subclass: too many sprocesses (GRCC_MAXSUBPROCS)");
        }
        sptbl[nSubproc++] = sproc;

#ifdef DEBUG
        // print sprocesses
        sproc->prSProcess();
#endif

        // generate M-graphs
        sproc->generate();

        // summation of the numbers and weighted numbers of graphs
        nMGraphs += sproc->nMGraphs;
        nMOPI    += sproc->nMOPI;
        wMGraphs.add(sproc->wMGraphs);
        wMOPI.add(sproc->wMOPI);

        nAGraphs += sproc->nAGraphs;
        nAOPI    += sproc->nAOPI;
        wAGraphs.add(sproc->wAGraphs);
        wAOPI.add(sproc->wAOPI);

        if (nAGraphs > 0) {
            ngraphs  = nAGraphs;
        } else {
            ngraphs  = nAGraphs;
        }
        delete sproc;
        sproc = NULL;
```

```
    }

    // ending time
    Second(&proct1);
    sec = proct1-proct0;

    // output file to finish process
    opt->endProc();
}

//*****************************************************************
// mgraph.cc
//=================================================================
//-----------------------------------------------------------------
MNodeClass::MNodeClass(int nnodes, int nclasses)
{
    // Input
    //   nnodes   : possible maximal number of nodes
    //   nclasses : possible maximal number of classes

    int j, k;

    nNodes   = nnodes;
    nClasses = nclasses;
    maxdeg   = 0;
    for (j = 0; j < nNodes; j++) {
        for (k = 0; k < nNodes; k++) {
            clmat[j][k] = 0;
        }
        clist[j]   = 0;
        ndcl[j]    = 0;
        flist[j]   = 0;
        clord[j]   = 0;
        cmindeg[j] = 0;
        cmaxdeg[j] = 0;
    }
    flist[nNodes] = 0;
    flg0 = 0;
    flg1 = 0;
    flg2 = 0;
}

//-----------------------------------------------------------------
MNodeClass::~MNodeClass(void)
{
}

//=================================================================
// class MNodeClass
//-----------------------------------------------------------------
void MNodeClass::init(int *cl, int mxdeg, int **adjmat)
{
    // initialize the object
    // Argument
    //   cl[j]        : list of number of nodes in the j-th class
    //   mxdeg        : maximum degree to nodes
    //   adjmat[j][k] : adjacency matrix

    int j;

    for (j = 0; j < nClasses; j++) {
        clist[j]  = cl[j];
        clord[j]  = j;
    }
    maxdeg    = mxdeg;
    mkNdCl();
    mkClMat(adjmat);
    mkFlist();
    flg0 = 0;
    flg1 = 0;
    flg2 = 0;
}

//-----------------------------------------------------------------
void MNodeClass::copy(MNodeClass* mnc)
{
    // copy MNodeClass 'mnc' to this object

    int j, k;

    nClasses = mnc->nClasses;
    for (k = 0; k < nClasses; k++) {
        clist[k] = mnc->clist[k];
    }
    maxdeg   = mnc->maxdeg;
    for (j = 0; j < nNodes; j++) {
        ndcl[j]  = mnc->ndcl[j];
```

```
        clord[j] = mnc->clord[j];
        for (k = 0; k < nClasses; k++) {
            clmat[j][k] = mnc->clmat[j][k];
        }
    }
    for (k = 0; k < nClasses+1; k++) {
        flist[k] = mnc->flist[k];
    }
}

//-------------------------------------------------------------
void MNodeClass::mkFlist(void)
{
    //  Construct flist
    //    The set of nodes in class 'c' is [flist[c],...,flist[c+1]-1]

    int  j, f;

    f  = 0;
    for (j = 0; j < nClasses; j++) {
        flist[j] = f;
        f += clist[j];
    }
    flist[nClasses] = f;
}

//-------------------------------------------------------------
void MNodeClass::mkNdCl(void)
{
    //  Construct ndcl
    //    ndcl[nd] = (the class id in which node 'nd' belongs)

    int  c, k;
    int  nd = 0;

    for (c = 0; c < nClasses; c++) {
        for (k = 0; k < clist[c]; k++) {
            ndcl[nd++] = c;
        }
    }
}

//-------------------------------------------------------------
int MNodeClass::clCmp(int nd0, int nd1, int cn)
{
    //  Comparison of two nodes 'nd0' and 'nd1'
    //    in lexicographic ordering of [class, connection configuration]

    int cmp;
    // Wether two nodes are in a same class or not.

    cmp = ndcl[nd0] - ndcl[nd1];
    if (cmp != 0) {
        return cmp;
    }

    // Sign '-' signifies the reverse ordering
    cmp = - cmpMNCArray(clmat[nd0], clmat[nd1], cn);
    if (cmp != 0) {
        return cmp;
    }
    return cmp;
}

//-------------------------------------------------------------
void MNodeClass::mkClMat(int **adjmat)
{
    //  Construct a matrix 'clmat[nd][tc]' which is the number
    //  of edges connecting 'nd' and all nodes in class 'tc'.

    int  nd, td, tc;

    for (nd = 0; nd < nNodes; nd++) {
        for (td = 0; td < nNodes; td++) {
            tc = ndcl[td];                      // another node
            if (nd == td) {
                clmat[nd][tc] += CLWIGHTD(adjmat[nd][td]);
            } else {
                clmat[nd][tc] += CLWIGHTO(adjmat[nd][td]);
            }
        }
    }
}

//-------------------------------------------------------------
void MNodeClass::incMat(int nd, int td, int val)
{
```

```
    //  Increase the number of edges by 'val' between 'nd' and 'td'.

    int tdc = ndcl[td];             // class of 'td'
    clmat[nd][tdc] += val;          // modify matrix 'clmat'.
}

//---------------------------------------------------------------
void MNodeClass::printMat(void)
{
    //  Print configuration matrix.

    int j1, j2;

    printf("\n");

    printf("nClasses=%d", nClasses);
    printf(" clord="); prIntArray(nClasses, clord, "");
    printf(" flist="); prIntArray(nClasses, flist, "\n");
    printf("flg = (%d, %d, %d)\n", flg0, flg1, flg2);

    // the first line
    printf("nd: cl:   ");
    for (j2 = 0; j2 < nClasses; j2++) {
        printf("%2d ", j2);
    }
    printf("\n");

    // print raw
    for (j1 = 0; j1 < nNodes; j1++) {
        printf("%2d; %2d: [", j1, ndcl[j1]);
        for (j2 = 0; j2 < nClasses; j2++) {
            printf(" %2d", clmat[j1][j2]);
        }
        printf("]\n");
    }
}

//---------------------------------------------------------------
int MNodeClass::cmpMNCArray(int *a0, int *a1, int ma)
{
    int j, k;

    for (k = 0; k < ma; k++) {
        j = clord[k];
        if (a0[j] < a1[j]) {
            return -1;
        } else if (a0[j] > a1[j]) {
            return 1;
        }
    }
    return 0;
}

//---------------------------------------------------------------
void MNodeClass::reorder(MGraph *mg)
{
    int flg[GRCC_MAXNODES];
    int co, cn;
    int f0, f1, f2;

    f0 = 0;
    f1 = 0;
    f2 = 0;
    for (co = 0; co < nClasses; co++) {
        cn  = flist[co];
        if (mg->nodes[cn]->freelg < 1) {
            flg[co] = 0;
            f0++;
        } else if (mg->nodes[cn]->freelg < mg->nodes[cn]->deg) {
            flg[co] = mg->nodes[cn]->deg + maxdeg*clist[co];
            f1++;
        } else {
            flg[co] = maxdeg*(mg->nodes[cn]->deg + maxdeg*clist[co]);
            f2++;
        }
    }
    if (f0 < nClasses) {
        bsort(nClasses, clord, flg);
    }
    flg0 = f0;
    flg1 = f0 + f1;
    flg2 = f0 + f1 + f2;
}

//===============================================================
// class MNode : nodes in MGraph
//---------------------------------------------------------------
```

```
MNode::MNode(int vid, int vclss, NCInput *mgi)
{
    // Arguments
    //   vid    : identifier of the vertex
    //   vdeg   : degree of the node
    //   vextlp : external node (-1) or looped vertex
    //   vclss  : class of the node

    id      = vid;           // id of the node
    clss    = vclss;         // class to which the node belongs
    deg     = mgi->cldeg;    // degree of the node
    freelg  = mgi->cldeg;    // number of free legs
    extloop = mgi->cltyp;    // external node or not
    cmindeg = mgi->cmind;    // min(deg of connectable vertex)
    cmaxdeg = mgi->cmaxd;    // max(deg of connectable vertex)
#ifdef DEBUG3
    printf("MNode:id=%d, freelg=%d, cmind=%d, cmaxd=%d\n",
            id, freelg, cmindeg, cmaxdeg);
#endif
}
//--------------------------------------------------------------
MNode::MNode(int vid, int vdeg, int vextlp, int vclss, int cmin, int cmax)
{
    // Arguments
    //   vid    : identifier of the vertex
    //   vdeg   : degree of the node
    //   vextlp : external node (-1) or looped vertex
    //   vclss  : class of the node

    id      = vid;       // id of the node
    deg     = vdeg;      // degree of the node
    freelg  = vdeg;      // number of free legs
    clss    = vclss;     // class to which the node belongs
    extloop = vextlp;    // external node or not
    cmindeg = cmin;      // min(deg of connectable vertex)
    cmaxdeg = cmax;      // max(deg of connectable vertex)
#ifdef DEBUG3
    printf("MNode:id=%d, freelg=%d, cmind=%d, cmaxd=%d\n", id, freelg, cmindeg, cmaxdeg);
#endif
}

//===============================================================
//  class MGraph : scalar graph expressed by matrix form
//---------------------------------------------------------------
MGraph::MGraph(int pid, int ncl, int *cldeg, int *clnum, int *cltyp, int *cmind, int *cmaxd, Options
*opts)
{
    int nn, ne, j, k;

#ifdef DEBUGM
    printf("MGraph::MGraph(pid=%d, ncl=%d)\n", pid, ncl);
    for (j = 0; j < ncl; j++){
        printf("%d: deg=%d, num=%d, typ=%d, mind=%d, maxd=%d\n",
               j, cldeg[j], clnum[j], cltyp[j], cmind[j], cmaxd[j]);
    }
    opts->print();
#endif

    // initial conditions
    nClasses = ncl;
    clist    = new int[ncl];
    opt      = opts;
    mId      = -1;

    mindeg   = -1;
    maxdeg   = -1;
    ne = 0;
    nn = 0;
    for (j = 0; j < nClasses; j++) {
        clist[j] = clnum[j];
        nn += clnum[j];
        ne += cldeg[j]*clnum[j];
        if (mindeg < 0) {
            mindeg = cldeg[j];
        } else {
            mindeg = Min(mindeg, cldeg[j]);
        }
        if (maxdeg < 0) {
            maxdeg = cldeg[j];
        } else {
            maxdeg = Max(maxdeg, cldeg[j]);
        }
    }
    if (ne % 2 != 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "Sum of degrees are not even\n");
            for (j = 0; j < nClasses; j++) {
```

```
                fprintf(GRCC_Stderr, "class %2d: %2d %2d %2d\n",
                        j, cldeg[j], clnum[j], cltyp[j]);
            }
        }
        erEnd("illegal degrees of nodes");
    }
    pId    = pid;
    nNodes = nn;
    nodes  = new MNode*[nNodes];
    group  = new SGroup();
#ifdef ORBITS
    orbits = new MOrbits();
#endif

    nEdges = ne / 2;
    nLoops = nEdges - nNodes + 1;

    egraph = new EGraph(nNodes, nEdges, maxdeg);
    nn = 0;
    nExtern = 0;
#ifdef DEBUG3
    printf("MGraph::MGraph: nClasses=%d\n", nClasses);
#endif
    for (j = 0; j < nClasses; j++) {
#ifdef DEBUG3
        printf("cmind[%d]=%d, cmaxd[%d]=%d\n", j, cmind[j], j, cmaxd[j]);
#endif
        for (k = 0; k < clist[j]; k++, nn++) {
            nodes[nn] = new MNode(nn, cldeg[j], cltyp[j], j, cmind[j], cmaxd[j]);
            egraph->setExtLoop(nn, cltyp[j]);
            if (cltyp[j] < 0) {
                nExtern++;
            }
        }
    }
    egraph->endSetExtLoop();

    init();
}

//----------------------------------------------------------------
MGraph::MGraph(int pid, int ncl, NCInput *mgi, Options *opts)
{
    int nn, ne, j, k;

#ifdef DEBUGM
    printf("MGraph::MGraph(pid=%d, ncl=%d)\n", pid, ncl);
    for (j = 0; j < ncl; j++){
        printf("%d: deg=%d, num=%d, typ=%d, mind=%d, maxd=%d\n",
               j, mgi[j].cldeg, mgi[j].clnum, mgi[j].cltyp,
               mgi[j].cmind, mgi[j].cmaxd);
    }
    opts->print();
#endif

    // initial conditions
    nClasses = ncl;
    clist    = new int[ncl];
    opt      = opts;
    mId      = -1;

    mindeg   = -1;
    maxdeg   = -1;
    ne = 0;
    nn = 0;
    for (j = 0; j < nClasses; j++) {
        clist[j] = mgi[j].clnum;
        nn += mgi[j].clnum;
        ne += mgi[j].cldeg*mgi[j].clnum;
        if (mindeg < 0) {
            mindeg = mgi[j].cldeg;
        } else {
            mindeg = Min(mindeg, mgi[j].cldeg);
        }
        if (maxdeg < 0) {
            maxdeg = mgi[j].cldeg;
        } else {
            maxdeg = Max(maxdeg, mgi[j].cldeg);
        }
    }
    if (ne % 2 != 0) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "Sum of degrees are not even\n");
            for (j = 0; j < nClasses; j++) {
                fprintf(GRCC_Stderr, "class %2d: %2d %2d %2d\n",
                        j, mgi[j].cldeg, mgi[j].clnum, mgi[j].cltyp);
            }
```

```
        }
        erEnd("illegal degrees of nodes");
    }
    pId    = pid;
    nNodes = nn;
    nodes  = new MNode*[nNodes];
    group  = new SGroup();
#ifdef ORBITS
    orbits = new MOrbits();
#endif

    nEdges = ne / 2;
    nLoops = nEdges - nNodes + 1;

    egraph = new EGraph(nNodes, nEdges, maxdeg);
    nn = 0;
    nExtern = 0;
#ifdef DEBUG3
    printf("MGraph::MGraph: nClasses=%d\n", nClasses);
#endif
    for (j = 0; j < nClasses; j++) {
#ifdef DEBUG3
        printf("cmind[%d]=%d, cmaxd[%d]=%d\n",
               j, mgi[j].cmind, j, mgi[j].cmaxd);
#endif
        for (k = 0; k < clist[j]; k++, nn++) {
            nodes[nn] = new MNode(nn, j, mgi+j);
            egraph->setExtLoop(nn, mgi[j].cltyp);
            if (mgi[j].cltyp < 0) {
                nExtern++;
            }
        }
    }
    egraph->endSetExtLoop();

    init();
}

//--------------------------------------------------------------
void MGraph::init(void)
{
    // generated set of graphs
    cDiag           = 0;
    c1PI            = 0;
    cNoTadpole      = 0;
    cNoTadBlock     = 0;
    c1PINoTadBlock = 0;

    // the current graph
    adjMat     = newMat(nNodes, nNodes, 0);
    nsym       = ToBigInt(0);
    esym       = ToBigInt(0);
    wscon      = Fraction(0, 1);
    wsopi      = Fraction(0, 1);

    // current node classification
    curcl     = new MNodeClass(nNodes, nClasses);

    // table of n-edge connected components
    mconn     = new MConn(nNodes, nEdges);

    // measures of efficiency
    ngen      = 0;
    ngconn    = 0;

#ifdef MONITOR
    nCallRefine    = 0;
    discardRefine  = 0;
    discardDisc    = 0;
    discardIso     = 0;
#endif

    // work space for isIsomorphic
    modmat = newMat(nNodes, nNodes, 0);

    // work space for bisearchM
    bidef = newArray(nNodes, 0);
    bilow = newArray(nNodes, 0);
    bicount = 0;
}

//--------------------------------------------------------------
MGraph::~MGraph(void)
{
    int j;

    // group->delGroup();
```

```
    bilow = deleteArray(bilow);
    bidef = deleteArray(bidef);


    modmat = deleteMat(modmat, nNodes);
    delete mconn;
    delete curcl;
    adjMat = deleteMat(adjMat, nNodes);
#ifdef ORBITS
    delete orbits;
#endif
    delete egraph;
    delete group;
    for (j = 0; j < nNodes; j++) {
        delete nodes[j];
        nodes[j] = NULL;
    }
    delete[] nodes;
    delete[] clist;

}

//---------------------------------------------------------------
void MGraph::printAdjMat(MNodeClass *cl)
{
    int j1, j2;

    printf("      ");
    for (j2 = 0; j2 < nNodes; j2++) {
        printf(" %2d", j2);
    }
    printf("\n");
    for (j1 = 0; j1 < nNodes; j1++) {
        printf("%2d : [", j1);
        for (j2 = 0; j2 < nNodes; j2++) {
            printf(" %2d", adjMat[j1][j2]);
        }
        printf("] %2d\n", cl->ndcl[j1]);
    }
}

//---------------------------------------------------------------
void MGraph::print(void)
{
    int j;

    printf("MGraph: pId=%d, cDiag=%ld, c1PI=%ld\n", pId, cDiag, c1PI);
    printf("     nNodes=%d, nEdges=%d, nExtern=%d, nLoops=%d "
           "mindeg=%d, maxdeg=%d, sym=(%ld, %ld)\n",
           nNodes, nEdges, nExtern, nLoops, mindeg, maxdeg, nsym, esym);
    printf("  Nodes=%d\n", nNodes);
    for (j = 0; j < nNodes; j++) {
        printf("    %2d: id=%d, deg=%d, clss=%d, extloop=%d, ",
               j, nodes[j]->id, nodes[j]->deg, nodes[j]->clss,
               nodes[j]->extloop);
        printf("mind=%d, maxd=%d, freelg=%d\n",
               nodes[j]->cmindeg, nodes[j]->cmaxdeg, nodes[j]->freelg);
    }

    curcl->printMat();
    printf("\n");

    printAdjMat(curcl);
}

//---------------------------------------------------------------
Bool MGraph::isConnected(void)
{
    //  Check graph can be a connected one.
    //  If a connected component without free leg is not the whole graph then
    //  return False, otherwise return True.

    int j, n, nv;

    for (j = 0; j < nNodes; j++) {
        nodes[j]->visited = -1;
    }
    if (visit(0)) {
        return True;
    }
    nv = 0;
    for (n = 0; n < nNodes; n++) {
        if (nodes[n]->visited >= 0) {
            nv++;
        }
    }
```

```
    return (nv == nNodes);
}

//---------------------------------------------------------------
Bool MGraph::visit(int nd)
{
    //  Visiting connected node used for 'isConnected'
    //  If child nodes has free legs, then this function returns True.
    //  otherwise it returns False.

    int td;

    // This node has free legs.
    if (nodes[nd]->freelg > 0) {
        return True;
    }
    nodes[nd]->visited = 0;
    for (td = 0; td < nNodes; td++) {
        if ((adjMat[nd][td] > 0) && (nodes[td]->visited < 0)) {
            if (visit(td)) {
                return True;
            }
        }
    }
    // all the child nodes has no free legs.
    return False;
}

//---------------------------------------------------------------
Bool MGraph::isIsomorphic(MNodeClass *cl)
{
    //  Check whether the current graph is the Representative
    //  of a isomorphic class.
    //    nsym = symmetry factor by the permutation of nodes.
    //    esym = symmetry factor by the permutation of edge.
    //  If this graph is not a representative, then returns False.

    int j1, j2, cmp, nself;
    int *perm;

    nsym = ToBigInt(0);
    esym = ToBigInt(1);

    group->newGroup(nNodes, cl->nClasses, cl->clist);

#ifdef ORBITS
    orbits->initPerm(nNodes);
#endif

    while (True) {
        perm = group->genNext();

        if (perm == NULL) {
            // calculate permutations of edges
            esym = ToBigInt(1);
            for (j1 = 0; j1 < nNodes; j1++) {
                if (adjMat[j1][j1] > 0) {
                    nself = adjMat[j1][j1]/2;
                    esym *= factorial(nself);
                    esym *= ipow(2, nself);
                }
                for (j2 = j1+1; j2 < nNodes; j2++) {
                    if (adjMat[j1][j2] > 0) {
                          esym *= factorial(adjMat[j1][j2]);
                    }
                }
            }
            return True;
        }

        permMat(nNodes, perm, adjMat, modmat);
        cmp = compMat(nNodes, adjMat, modmat);
        if (cmp < 0) {
            return False;
        } else if (cmp == 0) {
            // if ngroup < 0, symmetry group is $S_{-group}$.
            group->addGroup(perm);

            nsym = nsym + ToBigInt(1);
#ifdef ORBITS
            orbits->fromPerm(perm);
#endif
        }
    }
}

//---------------------------------------------------------------
```

```
void MGraph::permMat(int size, int *perm, int **mat0, int **mat1)
{
    // apply permutation to matrix 'mat0' and obtain 'mat1'

    int j1, j2;

    for (j1 = 0; j1 < size; j1++) {
        for (j2 = 0; j2 < size; j2++) {
            mat1[j1][j2] = mat0[perm[j1]][perm[j2]];
        }
    }
}

//---------------------------------------------------------------
int MGraph::compMat(int size, int **mat0, int **mat1)
{
    // comparison of matrix

    int j1, j2, cmp;

    for (j1 = 0; j1 < size; j1++) {
        for (j2 = 0; j2 < size; j2++) {
            cmp = mat0[j1][j2] - mat1[j1][j2];
            if (cmp != 0) {
                return cmp;
            }
        }
    }
    return 0;
}

//---------------------------------------------------------------
MNodeClass *MGraph::refineClass(MNodeClass *cl)
{
    //  Refine the classification
    //    cl : the current 'MNodeClass' object
    //    cn : the class number
    //  Returns (the new class number corresponds to 'cn') if OK, or
    //            'None' if ordering condition is not satisfied

    MNodeClass *ccl = cl;
    MNodeClass *xcl = NULL;
    MNodeClass *ncl = NULL;
    int         ucl[GRCC_MAXNODES];
    int         ccn = cl->nClasses;
    int         nucl, nce;
    int         td, cmp;
    int         nccl;

#ifdef MONITOR
    nCallRefine++;
#endif
    nucl = 0;
    while (ccl->nClasses != nucl) {    // repeat refinement.
        nce  = 0;
        nucl = 0;
        for (td = 1; td < nNodes; td++) {
          // 'td' is the next node and the current node is 'td-1'.
          // Count up the number of the elements in the current class
          // corresponding to the current node
          nce++;
          cmp = ccl->clCmp(td-1, td, ccn);

          // the ordering condition is not satisfied.
          if (cmp > 0) {
            if (ccl != cl) {
                delete ccl;
            }
            return NULL;

          } else if (cmp < 0) {
              // 'td' is in the next class to the current node.
              ucl[nucl++] = nce;       // close the current class

              // start new class
              nce = 0;
          }
          // nothing to do for the case of 'cmp == 0'.

        }

        // close array 'ucl'.
        ucl[nucl++] = nce + 1;

#ifdef CHECK
        // inconsistent
        if (nucl < ccl->nClasses) {
```

```
            erEnd("refineClasses : smaller number of classes");
        }
#endif

        // preparation of the next repetition.
        xcl = new MNodeClass(nNodes, nucl);
        xcl->init(ucl, maxdeg, adjMat);
        ccn = xcl->nClasses;

        nccl = ccl->nClasses;

        if (ccl != cl) {
            delete ccl;
        }
        ccl = xcl;
        if (ccl->nClasses == nccl) {
            ccl->reorder(this);
            return ccl;
        }

        nucl = 0;
    }

    return ncl;
}

//--------------------------------------------------------------
void MGraph::biconnME(void)
{
    //  Count the number of 1PI components.

    int j, root, vr, next, nart;
    MCOpi   mopi;
    MCBlock mblk;

    // initialization
    bicount     = 0;

    mconn->init();

    for (j = 0; j < nNodes; j++) {
        bidef[j] = -1;
        bilow[j] = -1;
    }

    // find a root for the root of bisearch
    // root must be an external node
    // or an vertex when there are not external nodes.
    root = -1;
    vr   = -1;
    for (j = 0; j < nNodes; j++) {
        if (bidef[j] < 0) {
            if (isExternal(j)) {
                root = j;
                break;
            } else if (vr < 0) {
                vr = j;
            }
        }
    }
    // case (2)
    if (root < 0) {
        root = vr;
    }

    if (root >= 0) {
        bisearchME(root, -1, 0, &mopi, &mblk, &next, &nart);

        mconn->nlpopic = 0;
        mconn->nctopic = 0;
        for (j = 0; j < mconn->nopic; j++) {
            if (mconn->opics[j].loop > 0) {
                mconn->nlpopic++;
            } else if (mconn->opics[j].ctloop > 0) {
                mconn->nctopic++;
            }
        }

        mconn->ne0bridges  = 0;
        mconn->ne1bridges   = 0;
        for (j = 0; j < mconn->nbridges; j++) {
            if (mconn->bridges[j].next == 0) {
                mconn->ne0bridges++;
            }
            if (mconn->bridges[j].next == 1) {
                mconn->ne1bridges++;
            }
```

```
        }

        mconn->nselfloops = 0;
        for (j = 0; j < nNodes; j++) {
            mconn->nselfloops += adjMat[j][j]/2;
        }

        mconn->na1blocks = 0;
        for (j = 0; j < mconn->nblocks; j++) {
            if (mconn->blocks[j].nartps == 1) {
                mconn->na1blocks++;
            }
        }
        mconn->neblocks = mconn->nblocks + mconn->nctopic;

    // no vertex.
    } else {
        // no vertices
        // Connected ==> only when ex=2, loop=0
        mconn->nopic      = 0;
        mconn->ne0bridges = 0;
        mconn->ne1bridges = 0;
    }
}

//---------------------------------------------------------------
void MGraph::bisearchME(int nd, int pd, int ned, MCOpi *mopi, MCBlock *mblk, int *next, int *nart)
{
    //  Search biconnected component
    //    visit : pd --> nd --> td
    //    ned : the number of edges between pd and nd.
    //  Output
    //
    //                 |
    //                 x----+
    //                 |    |
    //             ----x pd |
    //                 |    |
    //            x----x nd |  n art. pints.
    //            |   /|\   |
    //            |  / | \  |
    //   m art p. x-x  |  x-x
    //             n1  x  n3
    //                 n2
    //
    //   output of bisearchME(n1, nd, ...) ==> npart = m
    //   output of bisearchME(n2, nd, ...) ==> npart = 1
    //   output of bisearchME(n3, nd, ...) ==> npart = n
    //
    //   output of bisearchME(nd, pd, ...) ==> npart = n+1
    //       n1 => block of (m+1) articulation points
    //       n2 => block of    2  articulation points
    //       n3 => no block
    //
    Bool newv;
    int  td, td0, lp, conn, next1, nart1, nart2, nvisit, ndart;
    int  opit, opit0, blkt;
    MCOpi   mopi1;
    MCBlock mblk1;

    mopi->init();
    mblk->init();

    *next  = 0;     // # external below 'nd' inclusive
    *nart  = 0;     // # articulation points 'nd' inclusive
    nart1  = 0;     // # articulation points below 'nd'.
    nart2  = 0;     // # 'nd' is articulation point then 1
    ndart  = 0;     // # the number of blocks attached to 'nd'

    newv = (bidef[nd] < 0);

    bidef[nd] = bicount;
    bilow[nd] = bicount;
    bicount++;

    opit0 = mconn->opistkptr;

    if (pd < 0) {
        mconn->pushNode(nd);
    }

    // external node and not the root
    if (isExternal(nd)) {
        if (pd >= 0) {
            mopi->next   = 1;
            mopi->nlegs  = 1;
            mblk->nartps = 1;
```

```
            *next        = 1;
            *nart        = 1;
            return;
        }
    }

    nvisit = 0;
    td0    = -1;
    for (td = 0; td < nNodes; td++) {
        conn = adjMat[nd][td];

        // td is not adjacent to nd
        if (conn < 1) {
            continue;
        }
        nvisit++;
        td0 = td;

        // external node
        if (isExternal(td) && bidef[td] < 0) {
            mopi->nlegs++;
            mopi->next++;
            (*next)++;
            ndart++;
            mconn->addArtic(nd, 1);
            mblk->nartps++;
            nart2 = 1;
            if (newv) {
                mconn->pushNode(td);
            }

        // self-loop : pd --> nd --> nd
        } else if (td == nd) {
            lp   = conn/2;
            mopi->loop   += lp;
            mopi->nedges += lp;
            ndart += lp;
            mconn->addArtic(nd, lp);
            mconn->addBlockSelf(nd, lp);
            mblk->nartps++;
            nart2  = 1;


        // back to the parent : pd --> nd --> pd,  pd is a vertex
        // Back edges when the connection is multi-edges (ned > 1)
        } else if (td == pd) {
            if (ned > 1) {
                bilow[nd] = Min(bilow[nd], bidef[pd]);
                mopi->loop += ned - 1;
                mblk->loop += ned - 1;
            } else {
                // cancel this visit
                nvisit--;
            }

        // back edge :td is already visited
        } else if (bidef[td] >= 0) {
            bilow[nd] = Min(bilow[nd], bidef[td]);
            if (bidef[nd] >= bidef[td]) {
                mopi->loop   += conn;
                mopi->nedges += conn;
                mblk->loop   += conn;
                mconn->pushEdge(td, nd);
            }

        // new node
        } else if (bidef[td] < 0) {

            // stack pointer
            if (pd < 0 && isExternal(nd)) {
                opit = opit0;
            } else {
                opit = mconn->opistkptr;
            }
            blkt = mconn->blkstkptr;
            mblk1.init();

            mconn->pushEdge(nd, td);

            // visit child
            bisearchME(td, nd, adjMat[td][nd], &mopi1, &mblk1, &next1, &nart1);

            // articulation point of bridge
            if (bilow[td] > bidef[nd]) {

                // new OPI component (not including 'td')
                mopi1.nlegs++;
```

```

                mconn->addOPIc(&mopi1, opit);
                mopi1.init();
                mopi1.nlegs = 1;
                if (pd >= 0 || !isExternal(nd)) {
                    // opi
                    mconn->addBridge(nd, td, next1, nExtern);

                    // block
                    ndart++;
                    mconn->addArtic(nd, 1);
                    // discard nparts from nodes below td
                    mblk1.nartps = 2;
                    mconn->addBlock(&mblk1, blkt);
                    mblk1.init();
                    mblk1.nartps = 1;

                    nart1 = 0;
                    nart2 = 1;  // nd is an articulation point
                }
            } else if (bilow[td] == bidef[nd]) {
                // articulation point of a block
                mopi1.nedges += conn;
                if (pd >= 0 || !isExternal(nd)) {
                    ndart++;
                    mconn->addArtic(nd, 1);
                    mblk1.nartps++;
                    mconn->addBlock(&mblk1, blkt);
                    mblk1.init();
                    mblk1.nartps = 1;
                    nart1 = 0;
                    nart2 = 1;  // nd is an articulation point
                }

            } else {
                // inside a block
                mopi1.nedges += conn;
            }

            bilow[nd] = Min(bilow[nd], bilow[td]);
            mopi->nlegs  += mopi1.nlegs;
            mopi->next   += mopi1.next;
            mopi->loop   += mopi1.loop;
            mopi->nedges += mopi1.nedges;
            mopi->ctloop += mopi1.ctloop;

            mblk->nartps += mblk1.nartps;
            mblk->loop   += mblk1.loop;

            *next        += next1;
            *nart        += nart1;
        }

    }  // end of for

    // no connection except for 'pd'==>'nd'.  Then 'nd' is an art. point
    if (nvisit == 0) {
        nart2 = 1;
    }
    *nart += nart2;

    // add # loop inside counter terms
    if (newv) {
        if (pd >= 0) {
            mconn->pushNode(nd);
        }
        mopi->ctloop += Max(0, nodes[nd]->extloop);
    }

    // 'nd' is the root node
    if (pd < 0) {
        if (isExternal(nd)) {
            if (td0 >= 0) {
                mconn->addArtic(td0, 1);
            }
            if (mconn->nopic > 0) {
                (mconn->opics[mconn->nopic-1].next)++;
            }
        } else {
            mconn->addOPIc(mopi, opit0);
            if (ndart == 1) {
                // 'nd' has only 1 child
                // the root is not really an art. point
                (*nart)--;
                mconn->addArtic(nd, -1);
                (mconn->blocks[mconn->nblocks-1].nartps)--;
            }
```

```
        }
    }
}

//-------------------------------------------------------------
BigInt MGraph::generate(void)
{
    //  Generate graphs
    //    the generation process starts from 'connectClass'.

    MNodeClass *cl;

#ifdef DEBUGM
    printf("MGraph::generate:\n");
    // opt->printLevel(2);
#endif
    // Initial classification of nodes.
    cl = new MNodeClass(nNodes, nClasses);
    cl->init(clist, maxdeg, adjMat);
    cl->reorder(this);
    connectClass(cl);

    delete cl;

#ifdef DEBUGM
    printf("MGraph::generate:end:%ld\n", cDiag);
#endif
    return cDiag;
}

//-------------------------------------------------------------
void MGraph::connectClass(MNodeClass *cl)
{
    // Connect nodes in a class to others

    int sc, sn;
    MNodeClass *xcl;

    xcl = refineClass(cl);

#ifdef DEBUG3
    printf("connectClass\n");
    print();
#endif
    if (xcl == NULL) {
#ifdef MONITOR
        discardRefine++;
#endif
    } else {
        if (xcl->flg0 >= xcl->nClasses) {
            newGraph(xcl);
        } else {
            sc = xcl->clord[xcl->flg0];
            sn = xcl->flist[sc];
            connectNode(xcl->flg0, sn, xcl);
        }
    }
    if (xcl != cl && xcl != NULL) {
        delete xcl;
    }
}

//----------------------------------------------------------
void MGraph::connectNode(int so, int ss, MNodeClass *cl)
{
    int sc, sn;

    sc = cl->clord[so];
    if (ss >= cl->flist[sc+1]) {
        connectClass(cl);
        return;
    }

#ifdef DEBUG3
     \printf("connectNode\n");
     print();
#endif
    for (sn = ss; sn < cl->flist[sc+1]; sn++) {
        connectLeg(so, sn, so, sn, cl);
        return;
    }
}

//----------------------------------------------------------
void MGraph::connectLeg(int so, int sn, int to, int ts, MNodeClass *cl)
{
    //  Add one connection between two legs.
```

```
    //    1. select another node to connect
    //    2. determine multiplicity of the connection

    int sc, tc, tn, maxself, nc2, nc, maxcon, ts1, ncm, to1;

    sc = cl->clord[so];

#ifdef CHECK
    if (sn >= cl->flist[sc+1]) {
        erEnd("connectLeg : illegal control");
        return;
    }
#endif
#ifdef DEBUG3
    printf("connectLeg:0\n");
    print();
#endif

    // There remains no free legs in the node 'sn' : move to next node.
    if (nodes[sn]->freelg < 1) {

        if (!isConnected()) {
#ifdef MONITOR
            discardDisc++;
#endif
        } else {
#ifdef DEBUG3
            printf("call connectNode:1\n");
#endif
            // next node in the current class.
            connectNode(so, sn+1, cl);
        }
        return;
    }

#ifdef DEBUG3
    printf("connectLeg:2\n");
    print();
#endif
    // connect a free leg of the current node 'sn'.
    for (to1 = to; to1 < cl->nClasses; to1++) {
        tc = cl->clord[to1];
        if (to1 == to) {
            ts1 = ts;
        } else {
            ts1 = cl->flist[tc];
        }
#ifdef MINMAXLEG
        if (ts1 >= nNodes) {
            continue;
        }
#   ifdef DEBUG3
        printf("connectLeg:0: sn=%d, ts1=%d, ?(%d <= %d <= %d)\n",
               sn, ts1, nodes[sn]->cmindeg, nodes[ts1]->deg, nodes[sn]->cmaxdeg);
        printf("connectLeg:0: ts1=%d, sn=%d, ?(%d <= %d <= %d)\n",
               ts1, sn, nodes[ts1]->cmindeg, nodes[sn]->deg, nodes[ts1]->cmaxdeg);
#   endif
        if ((nodes[sn]->cmindeg > 0 && nodes[ts1]->deg < nodes[sn]->cmindeg)
          ||(nodes[sn]->cmaxdeg > 0 && nodes[ts1]->deg > nodes[sn]->cmaxdeg)
          |  (nodes[ts1]->cmindeg > 0 && nodes[sn]->deg < nodes[ts1]->cmindeg)
          ||(nodes[ts1]->cmaxdeg > 0 && nodes[sn]->deg > nodes[ts1]->cmaxdeg)) {
#   ifdef DEBUG3
            printf("connectLeg:1: sn=%d, ts1=%d, !(%d <= %d <= %d)\n",
                   sn, ts1, nodes[sn]->cmindeg, nodes[ts1]->deg, nodes[sn]->cmaxdeg);
            printf("connectLeg:2: ts1=%d, sn=%d, !(%d <= %d <= %d)\n",
                   ts1, sn, nodes[ts1]->cmindeg, nodes[sn]->deg, nodes[ts1]->cmaxdeg);
#   endif
            continue;
        }
#ifdef DEBUG3
        printf("connectLeg:3: sn=%d, ts1=%d\n", sn, ts1);
#endif
#endif
        for (tn = ts1; tn < cl->flist[tc+1]; tn++) {
            if (sc == tc && sn > tn) {
                continue;

            } else if (nodes[tn]->freelg < 1) {
                continue;

            // self-loop
            } else if (sn == tn) {
                if (opt->values[GRCC_OPT_NoSelfLoop] > 0) {
                    continue;
                }
                if (nNodes > 1) {
                    // there are two or more nodes in the graph :
```

```
                    // avoid disconnected graph
                    maxself = Min((nodes[sn]->freelg)/2, (nodes[sn]->deg-1)/2);
                } else {
                    // there is only one node the graph.
                    maxself = nodes[sn]->freelg/2;
                }

                // If we can assume no tadpole, the following line can be used.
                if ((opt->values[GRCC_OPT_1PI] > 0
                     || opt->values[GRCC_OPT_NoTadpole] > 0) && nExtern > 2) {
                     maxself = Min((nodes[sn]->deg-2)/2, maxself);
                }

                // for the number of connection.
                for (nc2 = maxself; nc2 > 0; nc2--) {
                    nc = 2*nc2;
                    ncm = CLWIGHTD(nc);
                    adjMat[sn][sn] = nc;
                    nodes[sn]->freelg -= nc;
                    cl->incMat(sn, tn, ncm);

                    // next connection
#ifdef DEBUG3
                    printf("call connectLeg:1: %d--%d\n", sn, sn);
#endif
                    connectLeg(so, sn, to1, tn+1, cl);

                    // restore the configuration
                    cl->incMat(tn, sn, - ncm);
                    adjMat[sn][sn] = 0;
                    nodes[sn]->freelg += nc;
                }

            // connections between different nodes.
            } else {
                // maximum possible connection number
                maxcon = Min(nodes[sn]->freelg, nodes[tn]->freelg);

                // avoid disconnected graphs.
                if (nNodes > 2 && nodes[sn]->deg == nodes[tn]->deg) {
                    maxcon = Min(maxcon, nodes[sn]->deg-1);
                }

#ifdef CHECK
                if ((adjMat[sn][tn] != 0) ||
                     (adjMat[sn][tn] != adjMat[tn][sn])) {
                    printf("*** inconsistent connection: sn=%d, tn=%d",
                           sn, tn);
                    printAdjMat(cl);
                    erEnd("inconsistent connection ");
                }
#endif

                // vary number of connections
                for (nc = maxcon; nc > 0; nc--) {
                    ncm = CLWIGHTO(nc);
                    adjMat[sn][tn] = nc;
                    adjMat[tn][sn] = nc;
                    nodes[sn]->freelg -= nc;
                    nodes[tn]->freelg -= nc;
                    cl->incMat(sn, tn, ncm);
                    cl->incMat(tn, sn, ncm);

                    // next connection
#ifdef DEBUG3
                    printf("call connectLeg:2: %d--%d\n", sn, tn);
#endif
                    connectLeg(so, sn, to1, tn+1, cl);

                    // restore configuration
                    cl->incMat(sn, tn, - ncm);
                    cl->incMat(tn, sn, - ncm);
                    adjMat[sn][tn] = 0;
                    adjMat[tn][sn] = 0;
                    nodes[sn]->freelg += nc;
                    nodes[tn]->freelg += nc;
                }
            }
        }
    }
}

//--------------------------------------------------------------
Bool MGraph::isOptM(void)
{
    Bool ok = True;

```

```
    opi      = (mconn->nopic == 1);
    opiloop  = (mconn->nlpopic <= 1);
    tadpole  = (mconn->ne0bridges >= ((nExtern == 1) ? 0 : 1));
    selfloop = (mconn->nselfloops > 0);
    tadblock = (mconn->na1blocks > 0);
    block    = (mconn->neblocks == 1);
    extself  = (mconn->ne1bridges > 0);

    if (opt->values[GRCC_OPT_1PI] > 0) {
        ok = ok && opi;
#ifdef DEBUGM
        printf("isOptM:  1PI:%d nopic=%d\n", ok, mconn->nopic);
#endif
    } else if (opt->values[GRCC_OPT_1PI] < 0) {
        ok = ok && !opi;
#ifdef DEBUGM
        printf("isOptM: -1PI:%d nopic=%d\n", ok, mconn->nopic);
#endif
    }
    if (opt->values[GRCC_OPT_NoExtSelf] > 0) {
        ok = ok && !extself;
#ifdef DEBUGM
        printf("isOptM: NoExtSelf:%d, extself=%d\n", ok, extself);
#endif
    } else if (opt->values[GRCC_OPT_NoExtSelf] < 0) {
        ok = ok && extself;
#ifdef DEBUGM
        printf("isOptM:-NoExtSelf:%d, extself=%d\n", ok, extself);
#endif
    }
    if (opt->values[GRCC_OPT_NoTadpole] > 0) {
        ok = ok && !tadpole;
#ifdef DEBUGM
        printf("isOptM: NoTadPole:%d, tadpole=%d\n", ok, tadpole);
#endif
    } else if (opt->values[GRCC_OPT_NoTadpole] < 0) {
        ok = ok && tadpole;
#ifdef DEBUGM
        printf("isOptM: -NoTadPole:%d\n", ok);
#endif
    }
    if (opt->values[GRCC_OPT_NoSelfLoop] > 0) {
#ifdef DEBUGM
        printf("isOptM: NoSelfLoop:%d\n", ok);
#endif
    } else if (opt->values[GRCC_OPT_NoSelfLoop] < 0) {
        ok = ok && selfloop;
#ifdef DEBUGM
        printf("isOptM: -NoSelfLoop:%d\n", ok);
#endif
    }
    if (opt->values[GRCC_OPT_No1PtBlock] > 0) {
        ok = ok && !tadblock;
#ifdef DEBUGM
        printf("isOptM: No1PtBlock:%d\n", ok);
#endif
    } else if (opt->values[GRCC_OPT_No1PtBlock] < 0) {
        ok = ok && tadblock;
#ifdef DEBUGM
        printf("isOptM: -NoSelfLoop:%d\n", ok);
#endif
    }
    if (opt->values[GRCC_OPT_Block] > 0) {
        ok = ok && block;
#ifdef DEBUGM
        printf("isOptM: Block:%d\n", ok);
#endif
    } else if (opt->values[GRCC_OPT_Block] < 0) {
        ok = ok && !block;
#ifdef DEBUGM
        printf("isOptM: -Block:%d\n", ok);
#endif
    }
#ifdef DEBUGM
    printf("isOptM:%d\n", ok);
#endif
    return ok;
}

//--------------------------------------------------------------
void MGraph::newGraph(MNodeClass *cl)
{
    //  A new candidate diagram is obtained.
    //  It may be a disconnected graph
    //
    //  Things to be done in the steps followed by this function.
    //  1. convert data format which treat the edges as objects.
```

```
    //  2. definition of loop momenta to the edges.
    //  3. analysis of loop structure in a graph.
    //  4. assignment of particles to edges and vertices to nodes
    //  5. recalculate symmetry factor

    int connected;
    MNodeClass *xcl;
    Bool ok;

    ngen++;

    // refine class and check ordering condition
    xcl = refineClass(cl);
    if (xcl == NULL) {
#ifdef MONITOR
        discardRefine++;
#endif

    // check whether this is a connected graph or not.
    } else {
        connected = isConnected();
        if (!connected) {
#ifdef MONITOR
            discardDisc++;
#endif

        // connected graph : check isomorphism of the graph
        } else {
            ngconn++;
            if (!isIsomorphic(xcl)) {
#ifdef MONITOR
                discardIso++;
#endif

            // We got a new connected and a unique representation of a class.
            } else {
                biconnME();
                if (isOptM()) {
                    egraph->fromMGraph(this);
                    if (egraph->isOptE()) {
                        curcl->copy(xcl);
#ifdef ORBITS
                        orbits->toOrbits();
#endif
                        ok = True;
                        if (opt->outmg != NULL) {
                            if (opt->proc != NULL) {
                                egraph->mId = opt->proc->mgrcount + 1;
                                egraph->sId = opt->proc->agrcount + 1;
                            } else if (opt->sproc != NULL) {
                                egraph->mId = opt->sproc->mgrcount + 1;
                                egraph->sId = opt->sproc->agrcount + 1;
                            } else {
                                egraph->mId = cDiag;
                            }
#ifdef DEBUG1
                            printf("call outmg\n");
                            egraph->model->prModel();
                            opt->print();
                            egraph->print();
#endif
                            ok = (*(opt->outmg))(egraph, opt->argmg);
#ifdef DEBUG
                            printf("ok=%d\n", ok);
#endif
                        }

                        if (ok) {
                            // # generated graphs
                            cDiag++;
                            wscon.add(1, nsym*esym);

                            // # generated 1PI graphs
                            if (opi) {
                                wsopi.add(1, nsym*esym);
                                c1PI++;
                            }

                            // # no tadpoles
                            if (!tadpole) {
                                cNoTadpole++;
                            }
                            // # no tadblocks
                            if (!tadblock) {
                                cNoTadBlock++;
                                if (opi) {
                                    c1PINoTadBlock++;
```

```
                                }
                            }

#ifdef MONITOR
                            printf("\n");
                            printf("Graph : %ld (%ld) 1PIComp=%d 1PILoop=%d",
                                   cDiag, ngen, n1PIComps, n1PILoop);
                            printf(" sym. factor = (%ld*%ld)\n", nsym, esym);
                            printAdjMat(cl);
                            // cl->printMat();
#   ifdef ORBITS
                            orbits->print();
#   endif
#   ifdef DEBUGM
                            printf("refine:                    %ld\n",
                                   nCallRefine);
                            printf("discarded for refinement:   %ld\n",
                                   discardRefine);
                            printf("discarded for disconnected: %ld\n",
                                   discardDisc);
                            printf("discarded for duplication:  %ld\n",
                                   discardIso);
#   endif
#endif
                            // go to next step
#ifdef DEBUGM
                            printf("newGraph:%ld:accepted\n", ngen);
#endif
                            opt->newMGraph(this);
                        } else {
#ifdef DEBUGM
                            printf("deleted by setOutMG-function\n");
#endif
                        }
                    } else {
#ifdef DEBUGM
                        printf("newGraph:%ld:discardOptE\n", ngen);
#endif
                    }
                } else {
#ifdef DEBUGM
                    printf("newGraph:%ld:discardOptM\n", ngen);
#endif
                }
            }
        }
    }
    if (xcl != cl && xcl != NULL) {
        delete xcl;
    }
}

//===============================================================
// class MOrbits: orbits of nodes
//---------------------------------------------------------------
MOrbits::MOrbits(void)
{
    nOrbits = 0;
    nNodes  = 0;
}

//---------------------------------------------------------------
MOrbits::~MOrbits(void)
{
}

//---------------------------------------------------------------
void MOrbits::print(void)
{
    int c;

    printf("Orbits : nOrbits=%d: nd2or=", nOrbits);
    prIntArray(nNodes, nd2or, "\n");
    for (c = 0; c < nOrbits; c++) {
        printf("    %2d: (%2d -- %2d) :", c, flist[c], flist[c+1]-1);
        prIntArray(flist[c+1]-flist[c], or2nd+flist[c], "\n");
    }
}

//----------------------------------------------------------------
void MOrbits::initPerm(int nnodes)
{
    int j;

    nNodes = nnodes;
    for (j = 0; j < nNodes; j++) {
        nd2or[j] = j;
```

```
    }
}

//------------------------------------------------------------
void MOrbits::fromPerm(int *perm)
{
    // update the orbits of nodes by the permutation

    int j, j1, k, l;

    for (j = 0; j < nNodes; j++) {
        if (nd2or[j] != j) {
            // not the first element of an old orbit
            continue;
        }
        // merge orbits to orbit 'j'
        k = j;
        for (j1 = 0; j1 < nNodes && (k = perm[k]) != j; j1++) {
            if (nd2or[k] == j) {
                // 'k' is already in 'j'
                ;
            } else {
                // old orbit 'k' had several elements: merge to orbit 'j'
                for (l = 0; l < nNodes; l++) {
                    if (nd2or[l] == k) {
                        nd2or[l] = j;
                    }
                }
            }
        }
#ifdef CHECK
        if (j1 >= nNodes) {
            printf("*** fromPerm: illegal control: j=%d, k=%d\n", j, k);
            printf("perm=");
            prIntArray(nNodes, perm,  " nd2or=");
            prIntArray(nNodes, nd2or, "\n");
            erEnd("fromPerm: illegal control");
        }
#endif
    }
}

//-----------------------------------------------------------
void MOrbits::toOrbits(void)
{
    // fill elements from 'nd2or'
    int nd, nel, k, lor;

    lor = -1;
    nel = 0;
    nOrbits = 0;
    for (nd = 0; nd < nNodes; nd++) {
        if (nd2or[nd] > lor) {
            // lowest element of a new orbit
            lor = nd2or[nd];
            flist[nOrbits++] = nel;
            for (k = nd; k < nNodes; k++) {
                if (nd2or[k] == lor) {
                    or2nd[nel++] = k;
                }
            }
        }
    }
    flist[nOrbits] = nNodes;
}

//***********************************************************
// n-edge connected components
//==========================================================
// class MCOpi: one n-edge connected component
//----------------------------------------------------------
MCOpi::MCOpi(void)
{
    init();
}

//----------------------------------------------------------
MCOpi::~MCOpi(void)
{
    nodes = NULL;
}

//----------------------------------------------------------
void MCOpi::init(void)
{
    nlegs  = 0;
    nedges = 0;
```

```
    nnodes = 0;
    next   = 0;
    loop   = 0;
    ctloop = 0;
    nnodes = 0;
    nodes  = NULL;
}

//==============================================================
// class MCBridge
//--------------------------------------------------------------
MCBridge::MCBridge(void)
{
}

//--------------------------------------------------------------
MCBridge::~MCBridge(void)
{
}

//==============================================================
// class MCBlock
//--------------------------------------------------------------
MCBlock::MCBlock(void)
{
    init();
}

//--------------------------------------------------------------
MCBlock::~MCBlock(void)
{
}

//--------------------------------------------------------------
void MCBlock::init(void)
{
    edges   = NULL;
    nmedges = 0;
    nartps  = 0;
    loop    = 0;
}

//==============================================================
// class MConn: table of n-edge connected components
//--------------------------------------------------------------
MConn::MConn(int nnod, int nedg)
{
    snodes = nnod;
    sedges = nedg;

    opics    = new MCOpi[snodes];
    bridges  = new MCBridge[sedges];
    blocks   = new MCBlock[sedges];
    articuls = new int[snodes];

    // workspace
    opisp    = new int[snodes];
    opistk   = new int[snodes];
    blksp    = new Edge2n[sedges];
    blkstk   = new Edge2n[sedges];
    init();
}

//--------------------------------------------------------------
MConn::~MConn(void)
{
    delete[] blkstk;
    delete[] blksp;
    delete[] opistk;
    delete[] opisp;

    delete[] articuls;
    delete[] blocks;
    delete[] bridges;
    delete[] opics;
}

//--------------------------------------------------------------
void MConn::init(void)
{
    int j;

    nopic      = 0;
    nlpopic    = 0;
    nctopic    = 0;
    nbridges   = 0;
    ne0bridges = 0;
```

```
    ne1bridges = 0;
    nselfloops = 0;

    nblocks    = 0;
    na1blocks  = 0;
    narticuls  = 0;
    neblocks   = 0;

    opistkptr  = 0;
    nopisp     = 0;
    blkstkptr  = 0;
    nblksp     = 0;

    for (j = 0; j < snodes; j++) {
        articuls[j] = 0;
    }
}

//--------------------------------------------------------------
void MConn::pushNode(int nd)
{
    if (opistkptr >= snodes) {
        erEnd("MConn::pushNode: opistkptr >= snodes");
    }
    opistk[opistkptr++] = nd;
}

//--------------------------------------------------------------
void MConn::pushEdge(int n0, int n1)
{
    if (blkstkptr >= sedges) {
        erEnd("MConn::pushEdge: blkstkptr >= sedges");
    }
    if (n0 <= n1) {
        blkstk[blkstkptr][0] = n0;
        blkstk[blkstkptr][1] = n1;
    } else {
        blkstk[blkstkptr][0] = n1;
        blkstk[blkstkptr][1] = n0;
    }
    blkstkptr++;
}

//--------------------------------------------------------------
void MConn::addOPIc(MCOpi *mopi, int stp)
{
    int j, nn;

    if (nopic >= snodes) {
        erEnd("MConn::addOPIc: table overflow");
    }

    nn = opistkptr - stp;
    for (j = 0; j < nn; j++) {
        opisp[nopisp+j] = opistk[stp+j];
    }
    opics[nopic].nnodes = nn;
    opics[nopic].nlegs  = mopi->nlegs;
    opics[nopic].nedges = mopi->nedges;
    opics[nopic].next   = mopi->next;
    opics[nopic].loop   = mopi->loop;
    opics[nopic].ctloop = mopi->ctloop;
    opics[nopic].nodes  = opisp  + nopisp;
    nopisp   += nn;
    opistkptr = stp;

    nopic++;
}

//--------------------------------------------------------------
void MConn::addBridge(int n0, int n1, int nex, int nextot)
{
    if (nbridges >= sedges) {
        erEnd("MConn::addBridge: table overflow");
    }
    if (n0 <= n1) {
        bridges[nbridges].nodes[0] = n0;
        bridges[nbridges].nodes[1] = n1;
    } else {
        bridges[nbridges].nodes[0] = n1;
        bridges[nbridges].nodes[1] = n0;
    }
    bridges[nbridges].next = Min(nex, nextot-nex);
    nbridges++;
}

//--------------------------------------------------------------
```

```
void MConn::addArtic(int nd, int mul)
{
    articuls[nd] += mul;
    if (articuls[nd] == mul) {
        narticuls++;
    } else if (articuls[nd] == 0) {
        narticuls--;
    }
}

//------------------------------------------------------------
void MConn::addBlock(MCBlock *mblk, int stp)
{
    int j, nn;

    if (nopic >= snodes) {
        erEnd("MConn::addOPIc: table overflow");
    }

    nn = blkstkptr - stp;
    for (j = 0; j < nn; j++) {
        blksp[nblksp+j][0] = blkstk[stp+j][0];
        blksp[nblksp+j][1] = blkstk[stp+j][1];
    }
    blocks[nblocks].edges   = blksp + nblksp;
    blocks[nblocks].nmedges = nn;
    blocks[nblocks].nartps  = mblk->nartps;
    blocks[nblocks].loop    = mblk->loop;
    nblksp    += nn;
    blkstkptr = stp;

    nblocks++;
}

//------------------------------------------------------------
void MConn::addBlockSelf(int nd, int mult)
{
    MCBlock mblk;
    int j, stp;

    mblk.edges   = NULL;
    mblk.nmedges = 0;
    mblk.nartps  = 1;
    mblk.loop    = 1;

    for (j = 0; j < mult; j++) {
        stp = blkstkptr;
        pushEdge(nd, nd);
        addBlock(&mblk, stp);
    }
}

//------------------------------------------------------------
void MConn::print(void)
{
    int j, k;

    printf("+++ MConn object: snodes=%d, sedges=%d\n", snodes, sedges);
    printf("    nopic=%d, n1popic=%d, nctopic=%d, nbridges=%d, "
           "ne0bridges=%d, ne1bridges=%d\n",
           nopic, n1popic, nctopic, nbridges, ne0bridges, ne1bridges);
    printf("    nblocks=%d, neblocks=%d, na1blocks=%d, narticuls=%d, nselfloop=%d\n",
           nblocks, neblocks, na1blocks, narticuls, nselfloops);

    printf("  1PI components (%d)\n", nopic);
    for (j = 0; j < nopic; j++) {
        printf("    %d: nleg=%d, nnodes=%d, nedge=%d, ",
               j, opics[j].nlegs, opics[j].nnodes, opics[j].nedges);
        printf("next=%d, loop=%d, ctlp=%d: [",
               opics[j].next, opics[j].loop, opics[j].ctloop);
        for (k = 0; k < opics[j].nnodes; k++) {
            printf(" %d", opics[j].nodes[k]);
        }
        printf("]\n");
    }

    printf("  bridges (%d)\n", nbridges);
    if (nbridges > 0) {
        printf("    ");
        for (j = 0; j < nbridges; j++) {
            printf("(%d,%d)[ex=%d] ",
                   bridges[j].nodes[0], bridges[j].nodes[1], bridges[j].next);
        }
        printf("\n");
    }

    printf("  blocks (%d)\n", nblocks);
```

```
    for (j = 0; j < nblocks; j++) {
        printf("    %d: nmedges=%d, nartps=%d, loop=%d: [",
               j, blocks[j].nmedges, blocks[j].nartps, blocks[j].loop);
        for (k = 0; k < blocks[j].nmedges; k++) {
            printf(" (%d,%d)", blocks[j].edges[k][0], blocks[j].edges[k][1]);
        }
        printf("]\n");
    }

    printf("  articulation points (%d)\n", narticuls);
    if (narticuls > 0) {
        printf("    ");
        for (j = 0; j < snodes; j++) {
            if (articuls[j] != 0) {
#if 0
                printf("%d(%d) ", j, articuls[j]);
#else
                printf("%d ", j);
#endif
            }
        }
        printf("\n");
    }
}

//****************************************************************
// sgroup.cc
//================================================================
// compile options
//==============================================================
// table of group elements
//--------------------------------------------------------------
SGroup::SGroup(void)
{
    // should be called before graph generation
    nnodes = -1;
    nelem  = 0;
    elem   = NULL;
}

//--------------------------------------------------------------
SGroup::~SGroup(void)
{
    int j;

    if (elem != NULL) {
        for (j = GRCC_MAXGROUP-1; j >= 0; j--) {
            if (elem[j] != NULL) {
                delete[] elem[j];
                elem[j] = NULL;
            }
        }
        delete[] elem;
        elem = NULL;
    }
}

//--------------------------------------------------------------
void SGroup::print(void)
{
    int j;

    printf("SGroup : nnodes = %d, nelem=%ld, cgen=%d, csav=%d\n",
           nnodes, nelem, cgen, csav);
    printf("  eclass =");
    prIntArray(neclass, eclass, "\n");
    for (j = 0; j < nelem; j++) {
        printf("    %4d: ", j);
        prIntArray(nnodes, elem[j], "\n");
    }
}

//--------------------------------------------------------------
void SGroup::newGroup(int nnd, int nclss, int *clss)
{
    int j;

    clearGroup();
    nnodes  = nnd;
    neclass = nclss;

    if (neclass > GRCC_MAXNODES) {
        erEnd("newGroup : too many classes (GRCC_MAXNODES)");
    }
    for (j = 0; j < neclass; j++) {
        eclass[j] = clss[j];
    }
```

```
    nelem = 0;
    if (elem == NULL) {
        elem  = new int*[GRCC_MAXGROUP];
        for (j = 0; j < GRCC_MAXGROUP; j++) {
            elem[j] = NULL;
        }
    }
}

//--------------------------------------------------------------
void SGroup::clearGroup(void)
{
    nelem  = 0;
    csav   = -1;
    cgen   = -1;
}

//--------------------------------------------------------------
void SGroup::delGroup(void)
{
    int j;

    for (j = 0; j < GRCC_MAXGROUP; j++) {
        if (elem[j] != NULL) {
            delete[] elem[j];
            elem[j] = NULL;
        }
    }
    delete[] elem;
    elem = NULL;
    clearGroup();
}

//--------------------------------------------------------------
int *SGroup::genNext(void)
{
    cgen = nextPerm(nnodes, neclass, eclass, pgr, pgq, permg, cgen);
    if (cgen < 0) {
        return NULL;
    }
    return permg;
}

//--------------------------------------------------------------
void SGroup::addGroup(int *p)
{
    int j;

    if (nelem >= GRCC_MAXGROUP) {
        erEnd("addGroup : too many elements (GRCC_MAXGROUP)");
    }
    if (elem[nelem] == NULL) {
        elem[nelem] = new int[GRCC_MAXNODES];
    }
    for (j = 0; j < nnodes; j++) {
        elem[nelem][j] = p[j];
    }
    nelem++;
}

//--------------------------------------------------------------
BigInt SGroup::nElem(void)
{
    return nelem;
}

//--------------------------------------------------------------
int *SGroup::nextElem(void)
{
    if (nelem < 0) {
        cgen = nextPerm(nelem, neclass, eclass,
                        psr, psq, perms, cgen);
        return perms;
    }
    if (cgen < 0) {
        cgen = 0;
    }
    if (cgen >= nelem) {
        return NULL;
    } else {
        return elem[cgen++];
    }
}

//****************************************************************
// egraph.cc
```

```
// Generate scalar connected Feynman graphs.
//===============================================================
static const char *GRCC_ND_names[] = GRCC_ND_NAMES;
static const char *GRCC_ED_names[] = GRCC_ED_NAMES;

//===============================================================
// class ENode
//---------------------------------------------------------------
ENode::ENode(void)
{
    id      = -1;
    egraph  = NULL;
    edges   = NULL;
    maxdeg  = 0;
    deg     = 0;

    klow    = NULL;
    extloop = 0;
    ndtype  = GRCC_ND_Undef;

    intrct  = -1;
}

//---------------------------------------------------------------
ENode::ENode(EGraph *egrph, int loops, int sdeg)

{
    id      = -1;
    egraph  = egrph;
    edges   = NULL;
    maxdeg  = sdeg;
    deg     = 0;

    klow    = NULL;
    extloop = 0;
    ndtype  = GRCC_ND_Undef;

    intrct  = -1;

    edges = new int[sdeg];
    klow  = new int[loops+1];
}

//---------------------------------------------------------------
ENode::~ENode(void)
{
    if (klow != NULL) {
        delete[] klow;
        delete[] edges;
        klow  = NULL;
        edges = NULL;;
    }
}

//---------------------------------------------------------------
void ENode::copy(ENode *en)
{
    int d;

    deg = en->deg;
    extloop = en->extloop;
    ndtype  = en->ndtype;
    for (d = 0; d < deg; d++) {
        edges[d] = en->edges[d];
    }
}

//---------------------------------------------------------------
void ENode::initAss(EGraph *egrph, int nid, int sdg)
{
    id      = nid;
    egraph  = egrph;
    maxdeg  = sdg;
}

//---------------------------------------------------------------
void ENode::setId(EGraph *egrph, const int nid)
{
    id      = nid;
    egraph  = egrph;
}

//---------------------------------------------------------------
void ENode::setExtern(int typ, int pt)
{
    ndtype = typ;
    intrct = pt;
```

```
}

//---------------------------------------------------------------
void ENode::setType(int typ)
{
#ifdef CHECK
    if (ndtype != GRCC_ND_Undef && ndtype != typ) {
        fprintf(GRCC_Stderr, "*** ndtype is already defined : old=%d, new = %d\n",
               ndtype, typ);
        erEnd("ndtype is already defined");
    }
#endif
    ndtype = typ;
}

//---------------------------------------------------------------
void ENode::print(void)
{
    int j;

    printf("Enode %d deg=%d, extl=%2d, intr=%d ",
           id, deg, extloop, intrct);
    if (egraph->bicount > 0) {
        printf("(%-8s) ", GRCC_ND_names[ndtype]);
    }
    printf("edge=[");
    for (j = 0; j < deg; j++) {
        printf(" %2d", edges[j]);
    }
    printf("]\n");
}

//===============================================================
// class EEdge
//---------------------------------------------------------------
EEdge::EEdge(void)
{
    id       = -1;
    egraph   = NULL;
    nodes[0] = -1;
    nodes[1] = -1;
    // nlegs[0] = -1;
    // nlegs[1] = -1;
    ptcl     = GRCC_PT_Undef;
    deleted  = False;

    edtype   = GRCC_ED_Undef;
    cut      = False;

    emom = NULL;
    lmom = NULL;
}

//---------------------------------------------------------------
EEdge::EEdge(EGraph *egrph, int nedges, int nloops)
{
    id       = -1;
    egraph   = egrph;
    nodes[0] = -1;
    nodes[1] = -1;
    nlegs[0] = -1;
    nlegs[1] = -1;
    ptcl     = GRCC_PT_Undef;
    deleted  = False;

    edtype   = GRCC_ED_Undef;
    cut      = False;

    emom = new int[nedges];
    lmom = new int[nloops];
}

//---------------------------------------------------------------
EEdge::~EEdge(void)
{
    if (lmom != NULL) {
        delete[] lmom;
    }
    if (emom != NULL) {
        delete[] emom;
    }
}

//---------------------------------------------------------------
void EEdge::copy(EEdge *ee)
{
    nodes[0] = ee->nodes[0];
```

```
    nodes[1] = ee->nodes[1];
    id       = ee->id;
    ext      = ee->ext;
    dir      = ee->dir;
    edtype   = ee->edtype;
    cut      = ee->cut;
}

//--------------------------------------------------------------
void EEdge::setId(EGraph *egrph, const int eid)
{
    id      = eid;
    egraph  = egrph;
}

//--------------------------------------------------------------
void EEdge::setType(int typ)
{
#ifdef CHECK
    if (edtype != GRCC_ED_Undef) {
        erEnd("edtype is already defined");
    }
#endif
    edtype = typ;
}

//--------------------------------------------------------------
void EEdge::setEMom(int nedges, int *em, int dir)
{
    int e;

    for (e = 0; e < nedges; e++) {
        if (em[e] != 0) {
            emom[e] = dir;
        }
    }
}

//--------------------------------------------------------------
void EEdge::setLMom(int k, int dir)
{
    lmom[k] = dir;
}

//--------------------------------------------------------------
void EEdge::print(void)
{
    int zero, n, k;

    printf("Edge %2d ext=%d ptcl=%2d ", id, ext, ptcl);
    if (egraph == NULL || !egraph->assigned) {
        printf("[%d, %d]", nodes[0], nodes[1]);
    } else {
        printf("[(%d,%d), (%d,%d)]", nodes[0], nlegs[0], nodes[1], nlegs[1]);
    }

    if (egraph != NULL && egraph->bicount > 0) {
        if (cut) {
            printf(" %-9s ", "Cut");
        } else {
            printf(" %-9s ", GRCC_ED_names[edtype]);
        }
        printf("c%2d: ", opicomp);
        printf("%s%d =", (ext)?"Q":"p", id);
        zero = True;
        for (n = 0; n < egraph->nEdges; n++) {
            if (emom[n] != 0) {
                prMomStr(emom[n], "Q", n);
                zero = False;
            }
        }
        for (k = 0; k < egraph->nLoops; k++) {
            if (lmom[k] != 0) {
                prMomStr(lmom[k], "k", k);
                zero = False;
            }
        }
        if (zero) {
            printf(" 0");
        }
    }
    printf("\n");
}

//==============================================================
// class EFLine
//--------------------------------------------------------------
```

```
EFLine::EFLine(void)
{
    ftype = FL_Open;
    fkind = 0;
    nlist = 0;
}

//--------------------------------------------------------------
void EFLine::print(const char *msg)
{
    if (ftype == FL_Open) {
        printf(" Open");
    } else if (ftype == FL_Closed) {
        printf(" Loop");
    } else {
        printf(" ?%d", ftype);
    }
    if (fkind == GRCC_PT_Dirac) {
        printf(" Dirac");
    } else if (fkind == GRCC_PT_Majorana) {
        printf(" Major");
    } else if (fkind == GRCC_PT_Ghost) {
        printf(" Ghost");
    } else {
        printf(" ?%d", fkind);
    }
    printf(" len=%d ", nlist);
    prIntArray(nlist, elist, msg);
}

//==============================================================
// class EGraph
//--------------------------------------------------------------
EGraph::EGraph(int nnodes, int nedges, int mxdeg)
{
    // Arguments
    //   nnodes : the number of nodes
    //   nedges : the number of edges
    //   mxdeg  : the maximum number of degrees of nodes

    int j;

    opt     = NULL;
    mgraph  = NULL;
    econn   = NULL;

    sNodes  = nnodes;
    sEdges  = nedges;
    sMaxdeg = mxdeg;
    sLoops  = sEdges - sNodes + 1;   // assume connected graph

    pId     = -1;
    mId     = -1;
    aId     = -1;
    sId     =  0;
    gSubId  = -1;
    assigned = False;

    nNodes  = sNodes;
    nEdges  = sEdges;
    nLoops  = 0;
    nExtern = 0;

    nsym    = 1;
    esym    = 1;
    extperm = 1;
    nsym1   = 1;
    multp   = 1;
    totalc  = 0;

    nodes = new ENode*[sNodes];
    for (j = 0; j < sNodes; j++) {
        nodes[j] = new ENode(this, sNodes, sMaxdeg);
    }
    edges = new EEdge*[sEdges];
    for (j = 0; j < sEdges; j++) {
        edges[j] = new EEdge(this, sEdges, sLoops);
    }

    bidef   = new int[sNodes];
    bilow   = new int[sNodes];
    bicount = -1;

    extMom  = new int[sEdges+1];

    fsign   = 1;
    nflines = -1;
```

```
    for (j = 0; j < GRCC_MAXFLINES; j++) {
        flines[j] = NULL;
    }

}

//--------------------------------------------------------------
EGraph::~EGraph(void)
{
    int j;

    for (j = GRCC_MAXFLINES-1; j >= 0; j--) {
        if (flines[j] != NULL) {
            delete flines[j];
            flines[j] = NULL;
        }
    }

    delete[] extMom;

    delete[] bilow;
    delete[] bidef;

    for (j = 0; j < sEdges; j++) {
        delete edges[j];
        edges[j] = NULL;
    }
    delete[] edges;

    for (j = 0; j < sNodes; j++) {
        delete nodes[j];
        nodes[j] = NULL;
    }
    delete[] nodes;

}

//--------------------------------------------------------------
void EGraph::copy(EGraph *eg)
{
    int nd, ed;

#ifdef CHECK
    if (sNodes < eg->nNodes || sEdges < eg->nEdges ||
        sMaxdeg < eg->sMaxdeg || sLoops < eg->nLoops) {
        erEnd("EGraph::copy: sizes are too small");
    }
#endif
    model   = eg->model;
    proc    = eg->proc;
    sproc   = eg->sproc;
    mgraph  = eg->mgraph;
    opt     = mgraph->opt;
    econn   = mgraph->mconn;

    pId     = eg->pId;
    mId     = eg->mId;
    gSubId  = eg->gSubId;
    nNodes  = eg->nNodes;
    nEdges  = eg->nEdges;
    nExtern = eg->nExtern;
    nLoops  = eg->nLoops;

    for (nd = 0; nd < nNodes; nd++) {
        nodes[nd]->copy(eg->nodes[nd]);
    }
    for (ed = 0; ed < nEdges; ed++) {
        edges[ed]->copy(eg->edges[ed]);
    }

    nsym    = eg->nsym;
    esym    = eg->esym;
    extperm = eg->extperm;
    nsym1   = eg->nsym1;
    multp   = eg->multp;

    bicount = -1;
}

//--------------------------------------------------------------
void EGraph::setExtLoop(int nd, int val)
{
    // set the node 'nd' being an external node (-1) or a looped vertex

    nodes[nd]->extloop = val;
}

```

```
//-----------------------------------------------------------------
void EGraph::endSetExtLoop(void)
{
    // end of calling 'setExtLoop'.

    int n;

    nExtern = 0;
    for (n = 0; n < nNodes; n++) {
        if (isExternal(n)) {
            nExtern++;
        }
    }
}

//-----------------------------------------------------------------
void EGraph::fromDGraph(DGraph *dg)
{
    int deg[GRCC_MAXNODES];
    int maxdeg, nextern, nconn, tc;
    int n0, n1, e;

    if (dg->nnodes > GRCC_MAXNODES) {
        fprintf(GRCC_Stderr, "*** too many nodes\n");
        exit(1);
    }
    if (dg->nedges > GRCC_MAXEDGES) {
        fprintf(GRCC_Stderr, "*** too many edges\n");
        exit(1);
    }

    for (e = 0; e < dg->nedges; e++) {
        if (dg->edges[e][0] >= dg->nnodes || dg->edges[e][0] >= dg->nnodes) {
            fprintf(GRCC_Stderr, "*** undefined node:");
            fprintf(GRCC_Stderr, "edge[%d] = {%d, %d}\n",
                    e, dg->edges[e][0], dg->edges[e][0]);
            exit(1);
        }
    }

    for (n0 = 0; n0 < dg->nnodes; n0++) {
        deg[n0] = 0;
    }
    for (e = 0; e < dg->nedges; e++) {
        deg[dg->edges[e][0]]++;
        deg[dg->edges[e][1]]++;
    }
    maxdeg  = 0;
    nextern = 0;
    for (n0 = 0; n0 < dg->nnodes; n0++) {
        maxdeg = Max(maxdeg, deg[n0]);
        if (deg[n0] == 1) {
            nextern++;
        } if (deg[n0] < 0) {
            fprintf(GRCC_Stderr, "+++ node %d is isolated\n", n0);
        }
    }

    mgraph = NULL;
    model  = NULL;
    proc   = NULL;
    sproc  = NULL;
    opt    = NULL;

    nNodes  = dg->nnodes;
    nEdges  = dg->nedges;
    nLoops  = 0;
    nExtern = nextern;

    pId     = 0;
    mId     = 0;
    aId     = 0;
    gSubId  = 0;
    nsym    = 1;
    nsym1   = 1;
    esym    = 1;
    extperm = 1;
    multp   = 1;
//    maxdeg  = maxdeg;

    for (n0 = 0; n0 < dg->nnodes; n0++) {
        nodes[n0]->deg = 0;
        nodes[n0]->extloop = dg->nodes[n0].extloop;
    }
    for (e = 0; e < dg->nedges; e++) {
        n0 = dg->edges[e][0];
        n1 = dg->edges[e][1];
```

```
        edges[e]->nodes[0] = n0;
        edges[e]->nodes[1] = n1;
        edges[e]->ext = (isExternal(n0) || isExternal(n1));

        nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(e, -1);
        nodes[n1]->edges[nodes[n1]->deg++] = I2Vedge(e, +1);
    }
    for (n0 = 0; n0 < dg->nnodes; n0++) {
        nodes[n0]->setId(this, n0);
        nodes[n0]->intrct = dg->nodes[n0].intrct;
    }

    for (e = 0; e < nEdges; e++) {
        edges[e]->setId(this, e);
    }
    tc = 0;
    for (n0 = 0; n0 < nNodes; n0++) {
        if (isExternal(n0)) {
            setExtern(n0, 1, GRCC_ND_Initial);
        } else {
            tc += 2*nodes[n0]->extloop + nodes[n0]->deg - 2;
        }
    }
    totalc = tc;

    // get the number of connected components
    nconn = connComp();
    nLoops = nEdges - nNodes + nconn;

    bicount = -1;
}


//--------------------------------------------------------------
void EGraph::fromMGraph(MGraph *mg)
{
    // construct EGraph from adjacency matrix.
    // This function should be called after
    //   EGraph(), setExtLoop() and endSetExtLoop().

    int n0, n1, ed, e;
    int j, ni, nf;
    PNodeClass *pnc;
    int k;

#ifdef CHECK
    if (sNodes < mg->nNodes || sEdges < mg->nEdges || sMaxdeg < mg->maxdeg) {
        erEnd("EGraph::fromMGraph: sizes are too small");
    }
    if (sLoops < mg->nLoops) {
        printf("too small sLoops : %d < %d\n", sLoops, mg->nLoops);
        erEnd("too small sLoops");
    }
#endif
    mgraph  = mg;
    sproc   = mgraph->opt->sproc;
    if (sproc == NULL) {
        proc  = NULL;
        model = NULL;
    } else {
        proc  = sproc->proc;
        model = sproc->model;
    }
    opt     = mgraph->opt;
    econn   = mgraph->mconn;

    nNodes  = mg->nNodes;
    nEdges  = mg->nEdges;
    nLoops  = mg->nLoops;
    nExtern = mg->nExtern;
    pId     = mg->pId;
    mId     = mg->cDiag;
    aId     = -1;
    if (opt->proc != NULL) {
        mId = mg->opt->proc->mgrcount;
        sId = mg->opt->proc->agrcount;
    } else if (mg->opt->sproc != NULL) {
        mId = mg->opt->sproc->mgrcount;
        sId = mg->opt->sproc->agrcount;
    } else {
        mId = mg->cDiag;
        sId = mg->cDiag;
    }
    gSubId  = 0;
    nsym    = mg->nsym;
    esym    = mg->esym;
    extperm = 1;
```

```
    nsym1   = nsym;
    multp   = 1;
    maxdeg  = mg->maxdeg;

    totalc  = 0;
    if (proc != NULL) {
        for (j = 0; j < proc->model->ncouple; j++) {
            totalc += proc->clist[j];
        }
    }
    for (ed = 0; ed < nEdges; ed++) {
        edges[ed]->edtype = GRCC_ED_Undef;
    }

    ed = 0;
    for (n0 = 0; n0 < nNodes; n0++) {
        nodes[n0]->deg = 0;
    }
    for (n0 = 0; n0 < nNodes; n0++) {
        for (e = 0; e < mg->adjMat[n0][n0]/2; e++, ed++) {
            edges[ed]->nodes[0] = n0;
            edges[ed]->nodes[1] = n0;
            nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(ed, -1);
            nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(ed, +1);
            edges[ed]->ext = isExternal(n0);
        }
        for (n1 = n0+1; n1 < nNodes; n1++) {
            for (e = 0; e < mg->adjMat[n0][n1]; e++, ed++) {
                edges[ed]->nodes[0] = n0;
                edges[ed]->nodes[1] = n1;
                nodes[n0]->edges[nodes[n0]->deg++] = I2Vedge(ed, -1);
                nodes[n1]->edges[nodes[n1]->deg++] = I2Vedge(ed, +1);
                edges[ed]->ext = (isExternal(n0) || isExternal(n1));
            }
        }
    }
#ifdef CHECK
    if (ed != nEdges) {
        printf("*** EGraph::init: ed=%d != nEdges=%d\n",
               ed, nEdges+1);
        erEnd("EGraph::init: illegal connection");
    }
#endif

    for (j = 0; j < nNodes; j++) {
        nodes[j]->setId(this, j);
        nodes[j]->intrct = mg->nodes[j]->extloop;
    }

    for (j = 0; j < nEdges; j++) {
        edges[j]->setId(this, j);
    }

    ni = 0;
    nf = 0;
    if (proc != NULL) {
        ni = proc->ninitl;
        for (j = 0; j < ni; j++) {
            setExtern(j, proc->initlPart[j], GRCC_ND_Initial);
        }

        nf = proc->nfinal;
        for (j = 0; j <  nf; j++) {
            setExtern(j+ni, proc->finalPart[j], GRCC_ND_Final);
        }
        nExtern = ni + nf;
    } else if (sproc != NULL) {
        pnc = sproc->pnclass;
        for (j = 0; j < sproc->pnclass->nclass; j++) {
            if (pnc->type[j] == GRCC_AT_Initial || pnc->type[j] == GRCC_AT_External) {
                for (k = pnc->cl2nd[j]; k < pnc->cl2nd[j+1]; k++) {
                    setExtern(j, k, GRCC_ND_Initial);
                    ni++;
                }
            } else if (sproc->pnclass->type[j] == GRCC_AT_Final) {
                for (k = pnc->cl2nd[j]; k < pnc->cl2nd[j+1]; k++) {
                    setExtern(j, k, GRCC_ND_Final);
                    nf++;
                }
            }

        }
        nExtern = ni + nf;
    }
    totalc = 0;
    for (j = 0; j < nNodes; j++) {
        if (!isExternal(j)) {
```

```
            totalc += 2*nodes[j]->extloop + nodes[j]->deg - 2;
        }
    }

    bicount = -1;
}

//---------------------------------------------------------------
ENode *EGraph::setExtern(int n0, int pt, int ndtyp)
{
    ENode *nd;
    int npt, ept, eg;

    nd = nodes[n0];
    if (nd->deg != 1) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** illegal external particle %d, %d, %d\n",
                    n0,pt,ndtyp);
        }
        erEnd("illegal external particle");
    }

    // particle comes into the node
    if (ndtyp == GRCC_ND_Initial) {
        npt = pt;
    } else if (ndtyp == GRCC_ND_Final) {
        npt = -pt;
    } else {
        npt = 0;
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** illegal type of an external particle : "
                   "node = %d, particle = %d, type = %d", n0, pt, ndtyp);
        }
        erEnd("illegal type of an external particle");
    }

    // edge attached to the external node
    eg = n0;

    // particle flows on e from leg=0 to 1.
    if (model != NULL){
        npt = model->normalParticle(npt);
        ept = model->normalParticle(-npt);
    } else {
        npt = 0;
        ept = 0;
    }

    nd->setExtern(ndtyp, npt);
    edges[eg]->ptcl = ept;

    return nd;
}

//---------------------------------------------------------------
void EGraph::print(void)
{
    // print the EGraph

    int  nd, ed, nlp;

    nlp = nEdges - nNodes + 1;
    printf("\nEGraph\n");
    printf("    pId=%d, gSubId=%ld, mId=%ld, aId=%ld, sId=%ld\n",
           pId, gSubId, mId, aId, sId);
    printf("    sNodes=%d, sEdges=%d, sMaxdeg=%d, sLoops=%d\n",
           sNodes, sEdges, sMaxdeg, sLoops);
    printf("    nNodes=%d, nEdges=%d, nExtern=%d, nLoops=%d, totalc=%d\n",
           nNodes, nEdges, nExtern, nlp, totalc);
    printf("    ");
    if (model == NULL) {
        printf("model=NULL,");
    } else {
        printf("model=\"%s\",", model->name);
    }
    if (proc == NULL) {
        printf("proc=NULL,");
    } else {
        printf("proc=%d,", proc->id);
    }
    if (sproc == NULL) {
        printf("sproc=NULL,");
    } else {
        printf("sproc=%d,", sproc->id);
    }
    printf("\n");
    printf("    assigned=%d, sym = (%ld * %ld) ",
```

```
                assigned, nsym, esym);
    printf("extperm=%ld, nsym1=%ld, multp=%ld\n",
                extperm, nsym1, multp);

    printf("  Nodes\n");
    for (nd = 0; nd < nNodes; nd++) {
        if (isExternal(nd)) {
            printf("    %2d Extern ", nd);
        } else {
            printf("    %2d Vertex ", nd);
        }
        nodes[nd]->print();
    }
    printf("  Edges\n");
    for (ed = 0; ed < nEdges; ed++) {
        if (edges[ed]->ext) {
            printf("    %2d Extern ", ed);
        } else {
            printf("    %2d Intern ", ed);
        }
        edges[ed]->print();
    }
    if (bicount > 0) {
        printf("  Biconn: nopicomp=%d, nopi2p=%d, opi2plp=%d, nadj2ptv=%d\n",
               nopicomp, nopi2p, opi2plp, nadj2ptv);
    }
    printf("\n");

    if (nflines > 0) {
        prFLines();
    }
}
//--------------------------------------------------------------
void EGraph::prFLines(void)
{
    int j;

    printf("  Fermion lines %d, sign=%d (mId=%ld, aId=%ld)\n",
            nflines, fsign, mId, aId);
    for (j = 0; j < nflines; j++) {
        printf("%4d ", j);
        flines[j]->print("\n");
    }
}

//--------------------------------------------------------------
int EGraph::dirEdge(int n, int e)
{
    if (nodes[n]->edges[e] > 0) {
        return 1;
    } else {
        return -1;
    }
}

//--------------------------------------------------------------
int EGraph::cmpMom(int *lm0, int *em0, int *lm1, int *em1)
{
    int lp, ed, cmp, sgn = 0;

    for (lp = 0; lp < nLoops; lp++) {
        if (lm0[lp] != 0) {
            if (lm1[lp] == 0) {
                return 1;
            } else if (sgn == 0) {
                sgn = lm0[lp]*lm1[lp];    // cancel first non-zero elements
#ifdef CHECK
                if (Abs(sgn) != 1) {
                    erEnd("cmpMom : illegal element of lmom");
                }
#endif
            } else {
                cmp = lm0[lp] - sgn*lm1[lp];
                if (cmp != 0) {
                    return cmp;
                }
            }
        } else if (lm1[lp] != 0) {    // lm0[lp] == 0
            return -1;
        }
    }
    for (ed = 0; ed < nEdges; ed++) {
        if (em0[ed] != 0) {
            if (em1[ed] == 0) {
                return 1;
            } else if (sgn == 0) {
                sgn = em0[ed]*em1[ed];    // cancel first non-zero elements
```

```
#ifdef CHECK
                if (Abs(sgn) != 1) {
                    erEnd("cmpMom : illegal element of emom");
                }
#endif
            } else {
                cmp = em0[ed] - sgn*em1[ed];
                if (cmp != 0) {
                    return cmp;
                }
            }
        } else if (em1[ed] != 0) {    // em0[ed] == 0
            return -1;
        }
    }
    return 0;
}

//--------------------------------------------------------------
int EGraph::groupLMom(int *grp, int *ed2gr)
{
    //  grp[g] : the number of elements in the group g (in [0,...,(ngrp-1)])
    //  ed2gr[ed] : group number of edge ed.

    int ed0, ed1, cmp, ngrp = -1;

    for (ed0 = 0; ed0 < nEdges; ed0++) {
        ed2gr[ed0] = -1;
        grp[ed0]    = 0;
    }

    for (ed0 = 0; ed0 < nEdges; ed0++) {
        if (ed2gr[ed0] < 0) {
            ngrp++;
            ed2gr[ed0] = ngrp;
            grp[ngrp]++;
            for (ed1 = ed0+1; ed1 < nEdges; ed1++) {
                cmp = cmpMom(edges[ed0]->lmom, edges[ed0]->emom,
                             edges[ed1]->lmom, edges[ed1]->emom);
                if (cmp == 0) {
                    ed2gr[ed1] = ngrp;
                    grp[ngrp]++;
                }
            }
        }
    }
    ngrp++;

    return ngrp;
}

//--------------------------------------------------------------
Bool EGraph::isOptE(void)
{
    EGraph edupv = EGraph(sNodes, sEdges, sMaxdeg);
    EGraph *edup = &edupv;
    int    grp[GRCC_MAXEDGES], ed2gr[GRCC_MAXEDGES];
    int    g, ed, ngrp;
    Bool   ok;
    int    minopi2p;

    if (opt->values[GRCC_OPT_No2PtL1PI] == 0
        && opt->values[GRCC_OPT_NoAdj2PtV] == 0) {
        return True;
    }

    for (ed = 0; ed < nEdges; ed++) {
        edges[ed]->cut = False;
    }

    biconnE();

    if (opt->values[GRCC_OPT_NoAdj2PtV] > 0) {
        if (nadj2ptv > 0) {
            return False;
        }
    } else if (opt->values[GRCC_OPT_NoAdj2PtV] < 0) {
        if (nadj2ptv < 1) {
            return False;
        }
    }
    if (opt->values[GRCC_OPT_No2PtL1PI] == 0) {
        return True;
    }

    if (nExtern == 2 && nopicomp == 1) {
        minopi2p = 2;
```

```
    } else {
        minopi2p = 1;
    }

    if (opi2plp > 0 && nopi2p >= minopi2p) {
        if (opt->values[GRCC_OPT_No2PtL1PI] > 0) {
            return False;
        } else if (opt->values[GRCC_OPT_No2PtL1PI] < 0) {
            return True;
        }
    }

    edup->copy(this);
    ngrp = groupLMom(grp, ed2gr);

    for (g = 0; g < ngrp; g++) {
        ok = True;
        if (grp[g] < 2) {
            continue;
        }
        for (ed = 0; ed < nEdges; ed++) {
            if (ed2gr[ed] == g) {
                if (edges[ed]->ext) {
                    ok = False;
                    break;
                }
                edup->edges[ed]->cut = True;
            } else {
                edup->edges[ed]->cut = False;
            }
        }
        if (ok) {
            edup->gSubId++;
            edup->biconnE();

            if (edup->nExtern == 2 && edup->nopicomp == 1) {
                minopi2p = 2;
            } else {
                minopi2p = 1;
            }

            if (edup->opi2plp > 0 && edup->nopi2p >= minopi2p) {
                if (opt->values[GRCC_OPT_No2PtL1PI] > 0) {
                    return False;
                } else if (opt->values[GRCC_OPT_No2PtL1PI] < 0) {
                    return True;
                }
            }
        }
    }

    if (opt->values[GRCC_OPT_No2PtL1PI] > 0) {
        return True;
    } else if (opt->values[GRCC_OPT_No2PtL1PI] < 0) {
        return False;
    } else {
        erEnd("isOptE: illegal control");
        return False;
    }
}

//--------------------------------------------------------------
int EGraph::findRoot(void)
{
    int root, vr, er, e, nd0, nd1;

    root = -1;
    vr   = -1;
    er   = -1;
    for (e = 0; e < nEdges; e++) {
        if (edges[e]->cut || edges[e]->edtype == GRCC_ED_Deleted) {
            continue;
        }
        if (edges[e]->visited) {
            continue;
        }
        if (root < 0) {
            if (edges[e]->ext) {
                nd0 = edges[e]->nodes[0];
                nd1 = edges[e]->nodes[1];
                if (isExternal(nd0) && !isExternal(nd1)) {
                    root = nd1;
                    break;
                } else if (!isExternal(nd0) && isExternal(nd1)) {
                    root = nd0;
                    break;
                }
```

```
            }
        }
        if (er < 0) {
            er = edges[e]->nodes[0];
        }
        if (vr < 0) {
            vr = edges[e]->nodes[0];
        }
    }
    // if root >= 0 : vertex adjacent to an external node
    if (root < 0) {
        // if vr >= 0 : a vertex
        if (vr >= 0) {
            root = vr;

        // no vertex, only external nodes;
        } else {
            root = er;
        }
    }
    return root;
}

//---------------------------------------------------------------
int EGraph::connComp(void)
{
    //  Count the number of connected component
    //  If a connected component without free leg is not the whole graph then
    //  return False, otherwise return True.

    int j, ncc, nelem;

    for (j = 0; j < nNodes; j++) {
        nodes[j]->visited = -1;
    }
    ncc   = 0;    // # connected components
    nelem = 0;    // # visited nodes
    while (nelem < nNodes) {
        for (j = 0; j < nNodes; j++) {
            if (nodes[j]->visited < 0) {
                break;
            }
        }
        if (j >= nNodes) {
            break;
        }
        nelem += connVisit(j, ncc);
        ncc++;
    }
    if (nelem != nNodes) {
        erEnd("EGraph::connComp: illegal control");
    }
    return ncc;
}

//---------------------------------------------------------------
int EGraph::connVisit(int nd, int ncc)
{
    //  Visiting connected node used for 'connComp'

    int e, en, nn, j, nelem;

    nelem = 1;
    nodes[nd]->visited = ncc;
    for (e = 0; e < nodes[nd]->deg; e++) {
        en = V2Iedge(nodes[nd]->edges[e]);
        for (j = 0; j < 2; j++) {
            nn = edges[en]->nodes[j];
            if (nodes[nn]->visited < 0) {
                nelem += connVisit(nn, ncc);
            }
        }
    }
    return nelem;
}

//--------------------------------------------------------------
void EGraph::biconnE(void)
{
    int e, ie, root;
    int extlst[GRCC_MAXEDGES], intlst[GRCC_MAXEDGES];
    int opiext, opiloop;

    nadj2ptv = 0;
    for (e = 0; e < nEdges; e++) {
        if (nodes[edges[e]->nodes[0]]->deg == 2 &&
            nodes[edges[e]->nodes[1]]->deg == 2 &&
```

```
            edges[e]->nodes[0] != edges[e]->nodes[1]) {
            nadj2ptv++;
        }
    }

    biinitE();      // initialization
    bconn = 0;      // the number of connected components

    opiext  = 0;
    opiloop = 0;

    for (e = 0; e < nEdges; e++) {
        // root of a spanning tree to be searched.
        root = findRoot();
        if (root < 0) {
            // no root : end of search
            break;
        }

        // recursive search
        bisearchE(root, extlst, intlst, &opiext, &opiloop);

        // opi2plp = Max(opi2plp, opiloop);      // ???

        // adjust the #external for the root
        if (isExternal(root)) {
            opiext++;
        }

        // 2 point function
        if (opiext == 2) {
            opi2plp = Max(opi2plp, opiloop);
            nopi2p++;
        }
        for (ie = 0; ie < nEdges; ie++) {
            if (intlst[ie]) {
                edges[ie]->opicomp = nopicomp;
            }
        }
        nopicomp++;

        bconn++;     // the number of connected component
    }

    // momentum conservation of external particles
    extMomConsv();

#ifdef CHECK
    chkMomConsv();
#endif
#ifdef DEBUG
    printf("biconnE:opiext=%d, opiloop=%d, opi2plp=%d, nopi2p=%d, nopicomp=%d, bconn=%d\n",
           opiext, opiloop, opi2plp, nopi2p, nopicomp, bconn);
#endif

    return;
}

//--------------------------------------------------------------
void EGraph::biinitE(void)
{
    int n, j, e;

    bicount = 0;                    // counter of visiting node
    loopm   = 0;                    // # of independent loop momentum

    nopicomp = 0;
    opi2plp  = 0;
    nopi2p   = 0;

    for (n = 0; n < nNodes; n++) {
        bidef[n] = -1;
        bilow[n] = -1;
        nodes[n]->ndtype  = GRCC_ND_Undef;
        for (j = 0; j < nLoops; j++) {
            nodes[n]->klow[j] = -1;
        }
    }

    for (e = 0; e < nEdges; e++) {
        // edges[e]->cut is defined before
        edges[e]->visited = False;
        edges[e]->conid   = -1;                   // connected component
        edges[e]->edtype  = GRCC_ED_Undef;
        edges[e]->opicomp = -1;
        for (j = 0; j < nEdges; j++) {
            edges[e]->emom[j] = 0;            // coefficients of ext. mom.
```

```
        }
        for (j = 0; j < nLoops; j++) {
            edges[e]->lmom[j] = 0;            // coefficients of loop mom.
        }
    }
}

//--------------------------------------------------------------
void EGraph::bisearchE(int nd, int *extlst, int *intlst, int *opiext, int *opiloop)
{
    // Search in the spanning tree below 'nd'.
    // Arguments:
    //   nd      : input  : node to be visited
    //   extlst  : output : set of external lines in the current 1PI comp.
    //   intlst  : output : set of internal lines in the current 1PI comp.
    //   opiext  : output : no. external lines of the current 1PI component
    //   opiloop : output : no. loops of the current 1PI component
    //
    // EGraph variables
    //   bicount   : counter of visiting node
    //   loopm     : no. of independent loop momentum
    //   nopicomp  : no. of OPI components
    //   opi2plp   : maximum number loops among 2-point looped OPI components
    //   nopi2p    : no. of 2-point OPI components
    //   edges[ie]->opicomp : OPI component no. of the edge
    //

    int k, e, ie, ed, td, dir, j;
    int opiext1, opiloop1;
    int extlst1[GRCC_MAXEDGES];   // set of external nodes below
    int intlst1[GRCC_MAXEDGES];   // set of vertices in the 1PI component

    (*opiext)  = 0;
    (*opiloop) = 0;

    // visit node 'nd'
    bidef[nd] = bicount;
    bilow[nd] = bicount;
    for (k = 0; k < nLoops; k++) {
        nodes[nd]->klow[k] = bicount;
    }
    bicount++;
    for (j = 0; j < nEdges; j++) {
        extlst[j] = False;
        intlst[j] = False;
    }

    // go to children : nd --> ed --> td
    for (e = 0; e < nodes[nd]->deg; e++) {
        ed = V2Iedge(nodes[nd]->edges[e]);

        // check this edge is already visited
        if (edges[ed]->edtype == GRCC_ED_Deleted) {
            // permanently deleted edge
            continue;
        } else if (edges[ed]->visited) {
            // already visited (backword visit)
            continue;
        } else if (edges[ed]->cut) {
            // cut edge : treat as nd is an external
            (*opiext)++;
            nodes[nd]->setType(GRCC_ND_CPoint);
            continue;
        } else if (!isExternal(nd) && edges[ed]->ext) {
            // external
            edges[ed]->setType(GRCC_ED_Extern);
            edges[ed]->visited = True;
            edges[ed]->emom[ed] = 1;
            extlst[ed] = True;
            (*opiext)++;
            nodes[nd]->setType(GRCC_ND_CPoint);
            continue;
        }

        // mark visited
        edges[ed]->visited = True;

        // momentum is assigned on the backward move. (nd) 1 --> 0 (td)
        td  = edges[ed]->nodes[0];
        dir = 1;
        if (td == nd) {
            td = edges[ed]->nodes[1];
            dir = -1;
        }

        // biconnected component
        if (bidef[td] >= 0) {
```

```
            // a back edge is found. Already visited. Don't go further
            bilow[nd] = Min(bilow[nd], bilow[td]);
            edges[ed]->setType(GRCC_ED_Back);
            intlst[ed] = True;
            (*opiloop)++;

            // create new loop momentum
            k = loopm++;
            nodes[nd]->klow[k] = Min(nodes[nd]->klow[k], nodes[td]->klow[k]);
            edges[ed]->setLMom(k, dir);

            // self-loop
            if (td == nd) {
                nodes[nd]->setType(GRCC_ND_CPoint);
            }

        } else {
            // not a back edge
            // go down further
            bisearchE(td, extlst1, intlst1, &opiext1, &opiloop1);

            // the set of external particles
            for (j = 0; j < nEdges; j++) {
                extlst[j] = extlst[j] || extlst1[j];
            }

            // loop momenta
            for (k = 0; k < nLoops; k++) {
                if (nodes[td]->klow[k] <= nodes[nd]->klow[k]) {
                    // inside loop 'k'
                    edges[ed]->setLMom(k, dir);
                }
                nodes[nd]->klow[k] = Min(nodes[nd]->klow[k], nodes[td]->klow[k]);
            }

            // cut point ?
            if (bilow[td] >= bidef[nd]) {
                // a cut point is found
                if (nodes[nd]->ndtype == GRCC_ND_Undef) {
                    nodes[nd]->setType(GRCC_ND_CPoint);
#ifdef CHECK
                } else if (nodes[nd]->ndtype != GRCC_ND_CPoint) {
                    if (prlevel > 0) {
                        fprintf(GRCC_Stderr, "bisearch: node %d is a cut point ",
                                nd);
                        fprintf(GRCC_Stderr, "(not undef %d)\n",
                                nodes[nd]->ndtype);
                    }
#endif
                }
            }  // cut point

            // bridge ?
            if (bilow[td] > bidef[nd]) {
                // a bridge is found
                if (edges[ed]->edtype == GRCC_ED_Undef) {
                    edges[ed]->setType(GRCC_ED_Bridge);
                    nodes[td]->setType(GRCC_ND_CPoint);
#ifdef CHECK
                } else if (edges[ed]->edtype != GRCC_ED_Bridge) {
                    if (prlevel > 0) {
                        fprintf(GRCC_Stderr, "bisearch: edges %d is a bridge ", ed);
                        fprintf(GRCC_Stderr, "(not undef %d)\n", edges[ed]->edtype);
                    }
#endif
                }

                if (!edges[ed]->ext) {
                    // internal bridge
                    opiext1++;            // from bridge
                    for (ie = 0; ie < nEdges; ie++) {
                        if (intlst1[ie]) {
                            // OPI component No.
                            edges[ie]->opicomp = nopicomp;
                            intlst1[ie] = False;
                        }
                    }
                    // 2pt OPI component
                    if (opiext1 == 2) {
                        opi2plp = Max(opi2plp, opiloop1);

                        // the number of 2pt looped OPI components
                        nopi2p++;
                    }
                    // the number of OPI components
                    nopicomp++;
                }
```

```
                opiext1  = 1;    // from bridge
                opiloop1 = 0;

            } else {
                // not a bridge

                bilow[nd] = Min(bilow[nd], bilow[td]);

                if (edges[ed]->edtype == GRCC_ED_Undef) {
                    edges[ed]->setType(GRCC_ED_Inloop);
#ifdef CHECK
                } else if (edges[ed]->edtype != GRCC_ED_Inloop) {
                    if (prlevel > 0) {
                        fprintf(GRCC_Stderr, "bisearch: ");
                        fprintf(GRCC_Stderr, "edges %d is not undef (%d)\n",
                               ed, edges[ed]->edtype);
                    }
#endif
                }
                intlst1[ed] = True;
            }

            // merge to the current variables
            for (ie = 0; ie < nEdges; ie++) {
                intlst[ie] = intlst[ie] || intlst1[ie];
            }
            (*opiext)  += opiext1;
            (*opiloop) += opiloop1;

            // linear combination of external momenta
            edges[ed]->setEMom(nEdges, extlst1, dir);

        }  // end of a child
    }  // for (ed)

    return;
}

//--------------------------------------------------------------
void EGraph::extMomConsv(void)
{
    int e, ex, lex, rs;

    lex = -1;
    for (e = 0; e < nEdges; e++) {
        if (edges[e]->ext) {
            lex = e;
            extMom[e] = edges[e]->emom[e];
        } else {
            extMom[e] = 0;
        }
    }
    // eliminate momentum of the last external particle
    if (lex < 0) {
        return;
    }
    for (e = 0; e < nEdges; e++) {
        rs = edges[e]->emom[lex];
        if (rs != 0) {
            for (ex = 0; ex < nEdges; ex++) {
                edges[e]->emom[ex] -= rs*extMom[ex];
            }
        }
    }
}

//--------------------------------------------------------------
void EGraph::chkMomConsv(void)
{
    // check momentum conservation

    int  esum[GRCC_MAXEDGES];
    int  lsum[GRCC_MAXNODES];
    Bool ok, okn;
    int  n, ex, lk, ej, e, dir;

    if (bicount < 1) {
        return;
    }
    ok = True;
    for (n = 0; n < nNodes; n++) {
        if (isExternal(n)) {
            continue;
        }
        for (ex = 0; ex < nEdges; ex++) {
            esum[ex] = 0;
        }
```

```
        for (lk = 0; lk < nLoops; lk++) {
            lsum[lk] = 0;
        }
        for (ej = 0; ej < nodes[n]->deg; ej++) {
            e = V2Iedge(nodes[n]->edges[ej]);
            if (edges[e]->nodes[0] == edges[e]->nodes[0]) {
                continue;
            }
            dir = dirEdge(n, ej);
            for (ex = 0; ex < nEdges; ex++) {
                if (edges[e]->ext) {
                    esum[ex] += dir*edges[e]->emom[ex];
                }
            }
            for (lk = 0; lk < nLoops; lk++) {
                lsum[lk] += dir*edges[e]->lmom[lk];
            }
        }

        okn = True;
        for (ex = 0; ex < nEdges; ex++) {
            if (esum[ex] != 0) {
                okn = False;
                fprintf(GRCC_Stderr, "chkMomConsv:n=%d, esum[%d]=%d\n",
                        n, ex, esum[ex]);
            }
        }

        for (lk = 0; lk < nLoops; lk++) {
            if (lsum[lk] != 0) {
                okn = False;
                fprintf(GRCC_Stderr, "chkMomConsv:n=%d, lsum[%d]=%d\n",
                        n, lk, lsum[lk]);

            }
        }

        if (!okn) {
            ok = False;
            fprintf(GRCC_Stderr, "*** Violation of momentum conservation ");
            fprintf(GRCC_Stderr, "at node =%d\n",n);
       }
    }

    if (!ok) {
        print();
        erEnd("inconsistent momentum");
    }
}

//--------------------------------------------------------------
int  EGraph::legParticle(int ed, int el)
{
    // outgoing particle from (edge, leg) = (ed, el)

    return model->normalParticle((2*el-1)*edges[ed]->ptcl);
}

//--------------------------------------------------------------
int  EGraph::isFermion(int ed)
{
    // return if particle on the edge ed follows Fermi statistics or not.

    int ptcl, ptype;

    ptcl  = edges[ed]->ptcl;
    ptype = model->particles[Abs(ptcl)]->ptype;

    return (ptype == GRCC_PT_Dirac || ptype == GRCC_PT_Majorana || ptype == GRCC_PT_Ghost);
}

//--------------------------------------------------------------
int  EGraph::fltrace(int fk, int nd0, int *fl)
{
    int nfl, k, i, nd, nl, e, ed, el, fgcnt, fkind, lk;

    nfl = 1;
    for (k = 0; k < nEdges; k++) {
        if (k >= nfl) {
            printf("*** fltrace:illegal contorl: k=%d, nEdges=%d\n", k, nEdges);
            break;
        }
        e  = fl[k];
        ed = V2Iedge(e);
        el = V2Ileg(e);
#ifdef CHECK
        if (ed > nEdges) {
```

```
            fprintf(GRCC_Stderr, "*** fltrace: ed=%d > nEdges=%d, fl=",
                    ed, nEdges);
            prIntArray(nNodes, fl, "\n");
            erEnd("fltrace: illegal fl");
        }
#endif
        nd = edges[ed]->nodes[el];
        nl = edges[ed]->nlegs[el];

        // end of the fline
        if (isExternal(nd) || nd == nd0) {
            fgcnt = 1;
            break;
        }
        fgcnt = 0;
        ed = 0;
        lk = 0;
        for (i = 0; i < nodes[nd]->deg; i++) {
            e  = nodes[nd]->edges[i];
            ed = V2Iedge(e);
            fkind = model->particles[Abs(edges[ed]->ptcl)]->ptype;
            if (fkind == fk) {
                if (lk == 1) {
                    lk = -1;
                } else {
                    lk = 1;
                }
            } else {
                lk = 0;
            }
            if (!edges[ed]->visited) {
                if (!isFermion(ed)) {
                    edges[ed]->visited = True;
                } else {
                    if (fkind == fk) {
                        // i should be the neighbor of nl
                        // (nl, i) or (i, nl) should be a pair of fkind
                        if ((lk == 1 && nl == i+1) || (lk == -1 && nl == i-1)) {
                            edges[ed]->visited = True;
                            fgcnt++;
                            if (fgcnt == 1) {
                                fl[nfl++] = - e;
                                break;
                            }
                        }
                    }
                }
            }
        }
        if (fgcnt == 0) {
            printf("*** fline: Fermion number is not conserved\n");
            printf("    nd=%d, e=%d, ed=%d, fgcnt=%d\n",
                   nd, e, ed, fgcnt);
            // erEnd("fline: Fermion number is not conserved");
        } else if (fgcnt > 1) {
            printf("+++ fline: more than two fermions: check fsign\n");
            printf("    nd=%d, e=%d, ed=%d, fgcnt=%d\n",
                   nd, e, ed, fgcnt);
        }
    }
    if (k >= nNodes || k >= nfl) {
        printf("*** fline: illegal control\n");
        printf("    nfl=%d, ", nfl);
        prIntArray(nfl, fl, "\n");
        erEnd("fline: illegal control");
    }
    return nfl;
}

//--------------------------------------------------------------
void EGraph::getFLines(void)
{
    int fl[GRCC_MAXNODES];
    int nextn, exto[GRCC_MAXNODES];
    int e, ed, nd, floop, nswap, nfl, fkind, el, ptcl;

    nflines = 0;
    for (ed = 0; ed < nEdges; ed++) {
        edges[ed]->visited = False;
    }

    nextn = 0;
    for (ed = 0; ed < nEdges; ed++) {
        if (edges[ed]->visited) {
            continue;
        }
        if (!edges[ed]->ext) {
```

```
            continue;
        }
        el = 0;
        nd = edges[ed]->nodes[el];
        if (!isExternal(nd)) {
            el = 1;
            nd = edges[ed]->nodes[el];
            if (!isExternal(nd)) {
                erEnd("*** illegal control");
            }
        }
        if (!isFermion(ed)) {
            continue;
        }
        ptcl = legParticle(ed, 1-el);
        if (ptcl < 0) {
            continue;
        }

        e = I2Vedge(ed, el);
        fl[0] = - e;
        nfl   = 1;
        fkind = model->particles[Abs(edges[ed]->ptcl)]->ptype;
        edges[ed]->visited = True;

        nfl = fltrace(fkind, nd, fl);
        addFLine(FL_Open, fkind, nfl, fl);
        e  = fl[nfl-1];
        exto[nextn++] = edges[V2Iedge(e)]->nodes[V2Ileg(e)];
    }

    // closed Fermion line
    floop = 0;
    for (ed = 0; ed < nEdges; ed++) {
        if (edges[ed]->visited) {
            continue;
        }
        if (!isFermion(ed)) {
            continue;
        }
        el = 0;
        ptcl = legParticle(ed, 1-el);
        if (ptcl < 0) {
            el = 1;
        }
        nd = edges[ed]->nodes[el];
        e = I2Vedge(ed, 1-el);
        fl[0] = e;
        nfl   = 1;
        edges[ed]->visited = True;

        fkind = model->particles[Abs(edges[ed]->ptcl)]->ptype;
        nfl = fltrace(fkind, nd, fl);
        addFLine(FL_Closed, fkind, nfl, fl);
        floop++;
    }
    if (nextn > 0) {
        nswap = sortb(nextn, exto);
    } else {
        nswap = 0;
    }
    if ((floop+nswap) % 2 == 0) {
        fsign = 1;
    } else {
        fsign = -1;
    }
}

//--------------------------------------------------------------
void EGraph::addFLine(const FLType ft, int fk, int nfl, int *fl)
{
    int j;

    if (nflines >= GRCC_MAXFLINES) {
        erEnd("too many Fermion lines (GRCC_MAXEDGES)");
    }
    if (flines[nflines] == NULL) {
        flines[nflines] = new EFLine();
    }
    flines[nflines]->ftype = ft;
    flines[nflines]->fkind = fk;
    flines[nflines]->nlist = nfl;
    for (j = 0; j < nfl; j++) {
        flines[nflines]->elist[j] = fl[j];
    }
    nflines++;
}
```

```

//*************************************************************
// assign.cc
//==============================================================
// Assign particles to the edges and interactions to nodes.
// Completed graph data is saved in the form of EGraph.

// method : selection of assignable node

#ifdef DEBUGM
static int nordleg   = 0;
static int nopleg    = 0;
static int niso      = 0;
static int niso1     = 0;
static int niso11    = 0;
static int niso111   = 0;
static int niso112   = 0;
static int niso12    = 0;
static int niso13    = 0;
static int niso14    = 0;
static int niso2     = 0;
static int nivord    = 0;
static int nextonly  = 0;
#endif

//===============================================================
// class NCand
NCand::NCand(const NCandSt sta, const int dega, const int nilst, int *ilst)
{
    // candidate list of interactions attached to a vertex.

    int j;

    deg    = dega;
    st     = sta;
    nilist = nilst;
    for (j = 0; j < nilist; j++) {
        ilist[j] = ilst[j];
    }

#ifdef CHECK
    if (st == AS_Assigned) {
        if (nilist != 1) {
            erEnd("illegal assignment");
        }
    } else if (st == AS_AssExt) {
        if (nilist != 1) {
            erEnd("illegal assignment");
        }
    }
#endif
}

//---------------------------------------------------------------
NCand::~NCand(void)
{
}

//---------------------------------------------------------------
void NCand::prNCand(const char* msg)
{
    printf("%d %d ",  st, deg);
    prIntArray(nilist, ilist, msg);
}

//===============================================================
// class ECand
ECand::ECand(int dt, int nplst, int *plst)
{
    int j;

    det    = dt;
    nplist = nplst;
    for (j = 0; j < nplist; j++) {
        plist[j] = plst[j];
    }

    if (det && nplist != 1) {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "*** ECand : len(plist) != 1 : det=%d ", det);
            prIntArrayErr(nplist, plist, "\n");
        }
        erEnd("ECand : len(plist) != 1");
    }
}

//---------------------------------------------------------------
```

```
ECand::~ECand(void)
{
}

//-----------------------------------------------------------------
void ECand::prECand(const char *msg)
{
    prIntArray(nplist, plist, "");
    printf(" (det=%d)%s", det, msg);
}

//=================================================================
// class ANode
ANode::ANode(int dg)
{
    int j;

    deg    = dg;
    nlegs  = 0;
    anodes = new int[deg];
    aedges = new int[deg];
    aelegs = new int[deg];
    cand   = NULL;
    for (j = 0; j < deg; j++) {
        anodes[j] = -1;
        aedges[j] = -1;
        aelegs[j] = -1;
    }
}

//-----------------------------------------------------------------
ANode::~ANode(void)
{
    if (cand != NULL) {
        delete cand;
        cand = NULL;
    }
    if (aelegs != NULL) {
        delete[] aelegs;
        delete[] aedges;
        delete[] anodes;
        aelegs = NULL;
        aedges = NULL;
        anodes = NULL;
    }
}

//-----------------------------------------------------------------
int ANode::newleg(void)
{
    // add a slot for a leg

    int lg = nlegs;

    nlegs++;
#ifdef CHECK
    if (nlegs > deg) {
        fprintf(GRCC_Stderr, "*** ANode::newleg : nlegs = %d > deg = %d\n",
                nlegs, deg);
        erEnd("ANode::newleg : nlegs > deg");
    }
#endif

    return lg;
}

//=================================================================
// class AEdge
AEdge::AEdge(int n0, int l0, int n1, int l1)
{
    nodes[0]  = n0;        // node of 0 side of the edge
    nodes[1]  = n1;        // node of 1 side of the edge
    nlegs[0]  = l0;        // leg ot the node of 0 side of the edge
    nlegs[1]  = l1;        // leg ot the node of 1 side of the edge
    ptcl    = GRCC_PT_Undef;     // particle defined in the model
    cand    = NULL;        // candidate
}

//-----------------------------------------------------------------
AEdge::~AEdge(void)
{
    if (cand != NULL) {
        delete cand;
    }
}

//=================================================================
```

```
// class Assign
Assign::Assign(SProcess *sprc, MGraph *mgr, PNodeClass *pnc)
{
    Bool ok;
    int  j;

    if (sprc == NULL) {
        erEnd("Assign: sproc == NULL");
    }

    // pointers to related objects
    sproc         = sprc;
    model         = sproc->model;
    opt           = sproc->opt;
    mgraph        = mgr;
    pnclass       = pnc;

    egraph = mgraph->egraph;
    if (egraph == NULL) {
        erEnd("Assign: egraph == NULL");
    }

    astack        = sproc->astack;
    if (astack == NULL) {
        erEnd("Assign: astack == NULL");
    }

#ifdef DEBUG
    if (pnclass == NULL) {
        printf("+++ Assign::Assign : pnclass = NULL\n");
    } else {
        printf("+++ Assign::Assign : pnclass :\n");
        pnclass->prPNodeClass();
    }
#endif

    nNodes        = mgraph->nNodes;
    nEdges        = mgraph->nEdges;
    nExtern       = sproc->nExtern;
    nETotal       = 0;                // total number of edges

    // counters of graphs
    nAGraphs      = 0;
    wAGraphs      = Fraction(0,1);
    nAOPI         = 0;
    wAOPI         = Fraction(0,1);

    nodes         = new ANode*[nNodes];
    edges         = new AEdge*[nEdges];

    for (j = 0; j < nNodes; j++) {
        nodes[j] = NULL;
    }
    for (j = 0; j < nEdges; j++) {
        edges[j] = NULL;
    }
    for (j = 0; j < model->ncouple; j++) {
        cplleft[j] = sproc->clist[j];
    }

    // initialize astack
    astack->setAGraph(this);
    astack->checkPoint(checkpoint0);

    // import from mgraph
    ok = fromMGraph();
#ifdef CHECK
    checkAG("Assign::Assign");
#endif

    if (ok) {
        // for debugging
#ifdef DEBUG0
        printf("+++ End of initialization : candidate:\n");
        prCand("init");
        printf("\n");
#endif

        // start assignment
        assignAllVertices();
#ifdef DEBUGM
        printf("mId=%ld, sId=%ld, ordleg=%d, ivord=%d, extonly=%d ",
               egraph->mId, egraph->sId, nordleg, nivord, nextonly);
        printf("niso=%d [%d(%d {%d, %d}, %d, %d, %d) %d]\n",
               niso, niso1, niso11, niso111, niso112, niso12, niso13, niso14,
               niso2);
#endif
```

```
    } else {
        // cannot assign
    }

    // restore from the stack
#ifdef CHECK
    astack->restoreMsg(checkpoint0, "Assign");
#else
    astack->restore(checkpoint0);
#endif
}

//==============================================================
Assign::~Assign(void)
{
    int j;

    for (j = 0; j < nEdges; j++) {
        delete edges[j];
        edges[j] = NULL;
    }
    delete[] edges;
    edges = NULL;

    for (j = 0; j < nNodes; j++) {
        delete nodes[j];
        nodes[j] = NULL;
    }
    delete[] nodes;
    nodes = NULL;
}

//--------------------------------------------------------------
void Assign::prCand(const char *msg)
{
    //  Print candidate table
    int     n, e, ne;
    AEdge *ed;

    printf("\n");
    printf("+++ Candidate list: %s\n", msg);
    printf("  Nodes %d\n", nNodes);
    for (n = 0; n < nNodes; n++) {
       printf("%d: edges=", n);
       prIntArray(nodes[n]->deg, nodes[n]->aedges, ": cand=");
       if (nodes[n]->cand == NULL) {
           printf("NULL\n");
       } else {
           nodes[n]->cand->prNCand("\n");
       }
    }
    ne = Min(nEdges, nETotal);
    printf("  Edges %d\n", ne);
    for (e = 0; e < ne; e++) {
       ed = edges[e];
       if (ed == NULL) {
           printf("NULL_Edge\n");
       } else {
           printf("%d: %d->%d: cand=", e, ed->nodes[0], ed->nodes[1]);
           if (edges[e]->cand == NULL) {
               printf("NULL\n");
           } else {
               edges[e]->cand->prECand("\n");
           }
       }
    }
}

//==============================================================
void Assign::checkAG(const char *msg)
{
    int n, lg;
    Bool ok = True;

    for (n = 0; n < nNodes; n++) {
        for (lg = 0; lg < nodes[n]->deg; lg++) {
            if (nodes[n]->aedges[lg] < 0) {
                printf("*** checkAG:%s: n=%d, lg=%d, aedges=%d\n",
                       msg, n, lg, nodes[n]->aedges[lg]);
                ok = False;
            }
        }
    }
    if (!ok) {
        prCand("checkAG");
        erEnd("checkAG: failed");
```

```
    }
}

//===============================================================
// control of the process of particle/interaction assignment

//---------------------------------------------------------------
Bool Assign::assignAllVertices(void)
{
    //  Entry point of the assignment procedure
    Bool   ok;

#ifdef CHECK
    checkCand("assignAllVertices:1");
#endif

#ifdef DEBUG0
    printf("+++ particle assignment for '%ld'\n", mgraph->mId);
#endif

    // start main part
#ifdef SIMPSEL
    ok = selectVertexSimp(-1);
#else
    ok = selectVertex();
#endif

#ifdef DEBUG0
    printf("\n");
    printf("+++ Total %ld assigned graphs for '%ld'\n",
           nAGraphs, mgraph->mId);
    printf("result: %ld ", nAGraphs);
    wAGraphs.print(" ");
    printf("%ld ", nAOPI);
    wAOPI.print("\n");
#endif

#ifdef CHECK
    checkCand("assignAllVertices:2");
#endif

    return ok;
}

//---------------------------------------------------------------
Bool Assign::selectVertexSimp(int lastv)
{
    //  Select a vertex for assignment of particles to legs
    //
    //  Argument
    //    lastv : previously assigned vertex; Nothing when lastv<0.

    Bool ok;
    int v;

#ifdef CHECK
        checkCand("selectVertexSimp");
#endif

    // find a vertex to which interaction is assigned
    v = selUnAssVertexSimp(lastv);

    // no more vertex
    if (v < 0) {

        ok = allAssigned();
        if (ok) {
            opt->newAGraph(egraph);
        }

        return ok;
    }

    // select a leg and assign a particle to it
    ok = selectLeg(v, -1);

    return ok;
}

//---------------------------------------------------------------
Bool Assign::selectVertex(void)
{
    //  Select a vertex for assignment of particles to legs
    //
    //  Argument
    //    lastv : previously assigned vertex; Nothing when lastv<0.

```

```
    Bool ok;
    int v;

#ifdef CHECK
        checkCand("selectVertex");
#endif

    // find a vertex to which interaction is assigned
    v = selUnAssVertex();

    // no more vertex
    if (v < 0) {

        ok = allAssigned();
        if (ok) {
            opt->newAGraph(egraph);
        }

        return ok;
    }

    // select a leg and assign a particle to it
    ok = selectLeg(v, -1);

    return ok;
}

//--------------------------------------------------------------
Bool Assign::selectLeg(int v, int lastlg)
{
    //  Select a leg of vertex 'v' for assignment of particle,
    //  where the last assigned leg was 'lastlg'.

    int      pt, ln, j;
    Bool     ok;
    CheckPt  sav, sav0;
    int      nplist, plist[GRCC_MAXMPARTICLES2];

#ifdef CHECK
    checkNode(v, "selectLeg:0");
    checkCand("selectLeg");
#endif

    // find a leg to be assigned
    ln = selUnAssLeg(v, lastlg);

    // no more assignable leg for the current node
    if (ln < 0) {
        // move to next vertex
        ok = assignVertex(v);

        return ok;
    }

    // make the list of possible particles, incoming to the vertex
    astack->checkPoint(sav0);
    nplist = candPart(v, ln, plist, GRCC_MAXMPARTICLES2);

    // no candidate
    if (nplist < 1) {
        return False;
    }

    // assign each of possible particle to the leg 'lg'.
    astack->checkPoint(sav);
    for (j = 0; j < nplist; j++) {
        pt = plist[j];

        // try to assign 'pt' to (v, ln)
        if(assignPLeg(v, ln, pt)) {
            // move to next leg
            selectLeg(v, ln);
        }

#ifdef CHECK
        astack->restoreMsg(sav, "selectLeg2");
#else
        astack->restore(sav);
#endif
    }

#ifdef CHECK
    astack->restoreMsg(sav0, "selectLeg2");
#else
    astack->restore(sav0);
#endif
    return True;
```

```
}

//---------------------------------------------------------------
void Assign::saveCouple(int *sav)
{
    int j;

    if (model->ncouple > 1) {
        for (j = 0; j < model->ncouple; j++) {
            sav[j] = cplleft[j];
        }
    }
}

//---------------------------------------------------------------
void Assign::restoreCouple(int *sav)
{
    int j;

    if (model->ncouple > 1) {
        for (j = 0; j < model->ncouple; j++) {
            cplleft[j] = sav[j];
        }
    }
}

//---------------------------------------------------------------
Bool Assign::subCouple(int *cpl)
{
    int j;

    if (model->ncouple > 1) {
        for (j = 0; j < model->ncouple; j++) {
            cplleft[j] -= cpl[j];
            if (cplleft[j] < 0) {
                return False;
            }
        }
    }
    return True;
}

//---------------------------------------------------------------
Bool Assign::assignVertex(int v)
{
    //  Assign interactions to vertex 'v'

    Bool      done, ok;
    CheckPt   sav, sav1;
    NCand    *nc;
    int       ia, n, j, cl;
    NCandSt   vst;
    Bool      ok1;
    int       savc0[GRCC_MAXNCPLG], savc1[GRCC_MAXNCPLG];

#ifdef CHECK
    checkCand("assignVertex");
#endif

    // save the configuration
    astack->checkPoint(sav);
    saveCouple(savc0);

    // assign to all vertices with only one candidate
    done = False;
    for (n = 0; n < nNodes; n++) {

        // check if vertex
        cl = pnclass->nd2cl[n];
        if (!isATExternal(pnclass->type[cl])) {

            nc = nodes[n]->cand;
            if (nc->st == AS_UnAssLegs && nc->nilist == 1) {

                // bind interaction to the vertex
                ia = nc->ilist[0];
                vst = assignIVertex(n, ia);
                if (vst == AS_Impossible) {
                    // discard the current configuration
#ifdef CHECK
                    astack->restoreMsg(sav, "assignVertex");
#else
                    astack->restore(sav);
#endif
                    restoreCouple(savc0);
                    return False;

```

```
                } else if (vst == AS_Assigned) {
                    if (!subCouple(model->interacts[ia]->clist)) {
#ifdef CHECK
                        astack->restoreMsg(sav, "assignVertex");
#else
                        astack->restore(sav);
#endif
                        restoreCouple(savc0);
                        return False;
                    }
                }
                if (n == v) {
                    done = (vst == AS_Assigned);
                }
            }
        }
    }

    // uniquely determined
    ok = True;
    if (done) {
        // move to the next vertex
#ifdef SIMPSEL
        ok = selectVertexSimp(v);
#else
        ok = selectVertex();
#endif

#ifdef CHECK
        astack->restoreMsg(sav, "assignVertex");
#else
        astack->restore(sav);
#endif
        restoreCouple(savc0);

        return ok;
    }

    // there are still several possibilities.

    astack->checkPoint(sav1);
    saveCouple(savc1);
    for (j = 0; j < nodes[v]->cand->nilist; j++) {
        ia = nodes[v]->cand->ilist[j];

        // bind interaction to the vertex
        ok1 = True;
        vst = assignIVertex(v, ia);
        if (vst == AS_Assigned) {
            ok1 = subCouple(model->interacts[ia]->clist);
        }
        if (ok1 && (vst == AS_Assigned || vst == AS_Assigned0)) {
            // move to the next vertex
#ifdef SIMPSEL
            ok = selectVertexSimp(v);
#else
            ok = selectVertex();
#endif
        }

#ifdef CHECK
        astack->restoreMsg(sav1, "assignVertex");
#else
        astack->restore(sav1);
#endif
        restoreCouple(savc1);
    }

#ifdef CHECK
    astack->restoreMsg(sav, "assignVertex");
#else
    astack->restore(sav);
#endif
    restoreCouple(savc0);

    return ok;
}

//--------------------------------------------------------------
Bool Assign::allAssigned(void)
{
    // Assignment of particles and interactions is finished.
    // Check isomorphism etc. and count symmetry factor
    // The sign from Fermi statistics is calculated inf fillEGraph

    BigInt  nsym, esym, nsym1;
    MNodeClass *cl;
```

```
    Bool ok = True;
#ifdef OPTEXTONLY
#else
    Bool ext;
    int  e, p;
#endif

#ifdef CHECK
    checkCand("allAssigned");
#endif

    // check order of coupling constants
#ifdef CHECK
    if (!checkOrderCpl()) {
        erEnd("allAssigned: checkOrderCpl = False");
    }
#endif

    // check duplication by violating ordering condition
    if (!isOrdLegs()) {
#ifdef DEBUGM
        nordleg++;
#endif
        return False;
    }

    // classification of nodes
    cl = mgraph->curcl;

    // check isomorphism and calculate sfactor
    nsym = 0;
    esym = 0;

    ok = isIsomorphic(cl, &nsym, &esym, &nsym1);
    if (!ok || nsym < 1 || esym < 1) {
#ifdef DEBUGM
        niso++;
#endif
        return False;
    }

    //--------------------------
    // Now we got a new agraph.

    // Update counters
    nAGraphs++;
    wAGraphs.add(1,nsym*esym);
    if (mgraph->opi) {
        nAOPI++;
        wAOPI.add(1,nsym*esym);
    }

#ifdef DEBUG1
    for (int j = 0; j < model->ncouple; j++) {
        if (cplleft[j] != 0) {
            printf("nAgraphs=%ld: 0 != cplleft =", nAGraphs);
            prIntArray(model->ncouple, cplleft, "\n");
            break;
        }
    }
#endif

#ifdef CHECK
    checkAG("allAssigned");
#endif
    // fill information to resulting Egraph
    fillEGraph(nAGraphs, nsym, esym, nsym1);

#ifdef OPTEXTONLY
#else
    // check extonly particles
    for (e = 0; e < nEdges; e++) {
        p   = Abs(egraph->edges[e]->ptcl);
        ext = egraph->edges[e]->ext;
        if (!ext) {
            if (model->particles[p]->extonly) {
#ifdef DEBUGM
                nextonly++;
#endif
                return False;
            }
        }
    }
#endif

#ifdef DEBUG0
    printf("Assigned graph = %ld, sym = (%ld, %ld) ", nAGraphs, nsym, esym);
```

```
    prCand("allAssigned ");
#endif

    return True;
}

//==============================================================
// Interface to MGraph and EGraph
//--------------------------------------------------------------
Bool Assign::fromMGraph(void)
{
    // Import from MGraph

    int      npall, *pall;
    int      np[1];
    Bool     ok;
    int      j, k, n, nn, n1, deg, nc, ptcl, cl, typ, mcl, cl1, typ1;
    int      lcn, ia;
    int      nilist, ilist[GRCC_MAXMINTERACT];
    NCandSt st;

    // create ANodes
    for (j = 0; j < nNodes; j++) {
        nodes[j] = new ANode(mgraph->nodes[j]->deg);
    }

    // initialization of edge table
    nETotal = 0;

    // List of all particles
    pall = model->allParticles(&npall);

    // add nodes
    for (n = 0; n < mgraph->nNodes; n++) {
        cl   = pnclass->nd2cl[n];
        typ  = pnclass->type[cl];

        // external particle
        if (isATExternal(typ)) {

#ifdef CHECK
            if (mgraph->nodes[n]->deg != 1) {
                printf("*** assign:fromMGraph : "
                       "external but deg[%d] = %d != 1, type=%d\n",
                       n, mgraph->nodes[n]->deg, typ);
                mgraph->print();
                erEnd("assign:fromMGraph : illegal external particle");
            }
#endif
            np[0] = pnclass->particle[cl];
            nodes[n]->cand = new NCand(AS_AssExt, 1, 1, np);
            for (n1 = 0; n1 < mgraph->nNodes; n1++) {
                nc = mgraph->adjMat[n][n1];
                for (k = 0; k < nc; k++) {
                    addEdge(n, n1, 1, np);
                }
            }

        // vertex
        } else {

            // candidates of vertices
            deg    = mgraph->nodes[n]->deg;
            st     = AS_UnAssLegs;
            cl     = sproc->pnclass->nd2cl[n];     // class in the process
            mcl    = sproc->pnclass->cl2mcl[cl];   // class in the model

            if (nodes[n]->cand != NULL) {
                delete nodes[n]->cand;
            }
            lcn = model->ncouple;

            // multiple coupling constants are defined in the model.
            if (lcn > 1) {
                nilist = 0;
                for (j = 0; j < model->cplgnvl[mcl]; j++) {
                    ia = model->cplgvl[mcl][j];

                    // select by coupling constants
                    if (leqArray(lcn, model->interacts[ia]->clist, cplleft)) {
                        ilist[nilist++] = ia;
                    }
                }
                nodes[n]->cand = new NCand(st, deg, nilist, ilist);

            // there is only one coupling constant
            } else {
```

```
                nodes[n]->cand =
                    new NCand(st, deg, model->cplgnvl[mcl], model->cplgvl[mcl]);
            }

            // vertices
            for (n1 = n; n1 < mgraph->nNodes; n1++) {
                cl1   = pnclass->nd2cl[n1];
                typ1  = pnclass->type[cl1];
                if (!isATExternal(typ1)) {
                    nc = mgraph->adjMat[n][n1];
                    if (n == n1) {
                        nc = nc/2;
                    }
                    for (k = 0; k < nc; k++) {
                        addEdge(n, n1, npall, pall);
                    }
                }
            }
        }
    }
#ifdef CHECK
    if (nETotal != nEdges) {
        printf("*** Assign::fromMGraph nETotal=%d != nEdges=%d\n",
               nETotal, nEdges);
        erEnd("Assign::fromMGraph nETotal= != nEdges");
    }
#endif

    // assign particle of an edge next to an external particle
    for (n = 0; n < mgraph->nNodes; n++) {
        cl  = pnclass->nd2cl[n];
        typ = pnclass->type[cl];
        if (isATExternal(typ)) {

            cl   = pnclass->nd2cl[n];
            ptcl = pnclass->particle[cl];

            // particle ptcl     flows in to   the node and
            // particle (- ptcl) flows in from the edge.

#ifdef CHECK
            if (ptcl == 0) {
                erEnd("fromMGraph: ptcl=0");
            }
#endif
            ok = assignPLeg(n, 0, - ptcl);
            if (!ok) {
                // impossible config
                return False;
            }
            nn = nodes[n]->anodes[0];
            ok = updateCandNode(nn);
            if (!ok) {
                // impossible config
                return False;
            }
        }
    }

#ifdef CHECK
    checkCand("fromMGraph");
    checkAG("fromMGraph");
#endif

    return True;
}

//---------------------------------------------------------------
void Assign::addEdge(int n0, int n1, int nplist, int *plist)
{
    //  add an edge and set connection information
    //  n0 -- (new edge) -- n1, with candidates 'plist'

    int lg0, lg1, eid;
    AEdge *aed;

#ifdef CHECK
    if (n0 >= nNodes || n1 >= nNodes) {
        printf("*** Assign::addEdge : undefined nodes %d: [%d, %d]",
              nETotal, n0, n1);
        erEnd("Assign::addEdge : undefined nodes");
    }
#endif

    // legs to be connected
    lg0 = nodes[n0]->newleg();
    lg1 = nodes[n1]->newleg();
```

```
    // create edge
#ifdef CHECK
    if (nETotal >= nEdges) {
        erEnd("too many edges");
    }
#endif

    aed = new AEdge(n0, lg0, n1, lg1);
    aed->cand = new ECand(False, nplist, plist);

#ifdef CHECK
    if (edges[nETotal] != NULL) {
        erEnd("edges[nETotal] != NULL");
    }
#endif

    // registor to the edge table
    eid = nETotal;
    edges[nETotal++] = aed;

    // connect edge and nodes
    connect(n0, lg0, eid, 0, n1, lg1);
}

//---------------------------------------------------------------
void Assign::connect(int n0, int l0, int eg, int el, int n1, int l1)
{
    //  Connect (n0, l0) -- (eg, el) -- (nl)

    ANode *nd0, *nd1;
    AEdge *ed;
    int    eo;

    nd0 = nodes[n0];
    nd1 = nodes[n1];
    ed  = edges[eg];
    eo  = 1 - el;

    nd0->anodes[l0] = n1;
    nd0->aedges[l0] = eg;
    nd0->aelegs[l0] = el;

    nd1->anodes[l1] = n0;
    nd1->aedges[l1] = eg;
    nd1->aelegs[l1] = eo;

    ed->nodes[el] = n0;
    ed->nlegs[el] = l0;

    ed->nodes[eo] = n1;
    ed->nlegs[eo] = l1;
}

//---------------------------------------------------------------
Bool Assign::fillEGraph(int aid, BigInt nsym, BigInt esym, BigInt nsym1)
{
    //  Set variables of egraph with the result of particle assignment
    //  Adjust the ordering of legs of vertices in a consistent way
    //  with the interaction defined in the model.

    ANode *an;
    ENode *en;
    int    work[3][GRCC_MAXLEGS];
    int    n, lr, e;
    int   *elist;
    int    lg, ed, eg;
#ifdef CHECK
    int    cl, ok = True;
#endif

    egraph->assigned = True;

    egraph->aId    = aid;
    egraph->nsym   = nsym;
    egraph->esym   = esym;
    egraph->bicount = -1;
    if (sproc != NULL) {
        egraph->extperm = sproc->extperm;
    } else {
        egraph->extperm = 1;
    }
    egraph->nsym1 = nsym1;
    egraph->multp = (egraph->extperm * egraph->nsym1) / egraph->nsym;

#ifdef CHECK
    checkAG("fillEGraph:0");
```

```
#endif
    // external node
    for (n = 0; n < nNodes; n++) {
        // vertices
#ifdef CHECK
        cl  = pnclass->nd2cl[n];
        if (isATExternal(pnclass->type[cl])) {
            ;
        } else if (nodes[n]->cand->st != AS_Assigned) {
            printf("*** fillEGraph : node %d is not assigned", n);
            prCand("fillEGraph: node");
            erEnd("fillEGraph : node is not assigned");
        }
#endif

        an = nodes[n];
        en = egraph->nodes[n];

        en->initAss(egraph, n, an->deg);

        en->intrct = an->cand->ilist[0];
        elist = reordLeg(n, work[0], work[1], work[2]);
        for (lr = 0; lr < nodes[n]->deg; lr++) {
            if (elist == NULL) {
                lg = lr;
            } else {
                lg = elist[lr];
            }
#ifdef CHECK
            if (lg < 0 || lg >= nodes[n]->deg) {
                printf("*** fillEGraph: n=%d, lr=%d: 0 <= lg=%d < %d\n",
                       n, lr, lg, nodes[n]->deg);
                erEnd("fillEGrah: illegal reordering");
            }
#endif
            ed = an->aedges[lg];
            eg = an->aelegs[lg];

            en->edges[lr] = I2Vedge(ed, 2*eg-1);

            egraph->edges[ed]->nodes[eg] = n;
            egraph->edges[ed]->nlegs[eg] = lr;
        }
    }
    // edges
    for (e = 0; e < nEdges; e++) {
#ifdef CHECK
        if (edges[e]->cand->nplist != 1) {
            printf("*** fillEGraph : edge %d is not assigned", e);
            prCand("fillEGraph: edge");
            erEnd("fillEGraph : edge is not assigned");
        }
#endif
        egraph->edges[e]->ptcl = edges[e]->cand->plist[0];
    }
#ifdef EDGEPORDER
    for (e = 0; e < nEdges; e++) {
        if (egraph->edges[e]->ptcl < 0) {
            egraph->edges[e]->ptcl = - edges[e]->cand->plist[0];
            n  = egraph->edges[e]->nodes[0];
            lr = egraph->edges[e]->nlegs[0];
            egraph->edges[e]->nodes[0] = egraph->edges[e]->nodes[1];
            egraph->edges[e]->nlegs[0] = egraph->edges[e]->nlegs[1];
            egraph->edges[e]->nodes[1] = n;
            egraph->edges[e]->nlegs[1] = lr;

            egraph->nodes[n]->edges[lr] = - egraph->nodes[n]->edges[lr];
            n  = egraph->edges[e]->nodes[0];
            lr = egraph->edges[e]->nlegs[0];
            egraph->nodes[n]->edges[lr] = - egraph->nodes[n]->edges[lr];
        }
    }
#endif
#ifdef CHECK
    for (ed = 0; ed < nEdges; ed++) {
        n  = egraph->edges[ed]->nodes[0];
        lr = egraph->edges[ed]->nlegs[0];
        if (egraph->nodes[n]->edges[lr] != -ed-1) {
            fprintf(GRCC_Stderr, "+++ node[%d][%d]=%d != - (edge[%d][0] + 1) = %d\n",
                    n, lr, egraph->nodes[n]->edges[lr], e, -ed-1);
            ok = False;
        }
        n = egraph->edges[ed]->nodes[1];
        lr = egraph->edges[ed]->nlegs[1];
        if (egraph->nodes[n]->edges[lr] != ed+1) {
            fprintf(GRCC_Stderr, "+++ node[%d][%d]=%d != + (edge[%d][0] + 1) = %d\n",
                    n, lr, egraph->nodes[n]->edges[lr], e, ed+1);
```

```
            ok = False;
        }
    }
    if (!ok) {
        egraph->print();
        erEnd("fillEGraph: illegal connection");
    }
#endif


    // analyse fermion line and determine Fermi statistical sign factor
    if (!model->skipFLine) {
        egraph->getFLines();
    } else {
        if (prlevel > 0) {
            fprintf(GRCC_Stderr, "+++ Sign factors related to "
                    "Dirac/Majorana/Ghost particles are not calculated.\n");
        }
    }

#ifdef CHECK
    checkAG("fillEGraph:0");
#endif
    return True;

}

//---------------------------------------------------------------
int *Assign::reordLeg(int n, int *reord, int *plist, int *used)
{
    // Reorder legs of node 'n' in accordance with the definition
    // of the interaction in the model
    //
    // plist[record[j]] = model->interacts[ia]->plist[j]

    int lg, ia, lr, deg, pt, ed;
    int *ilegs;
#ifdef CHECK
    Bool found;
#endif

    // external node
    if (nodes[n]->cand->st == AS_AssExt) {
        reord[0] = 0;
        return 0;
    }

    // degree of the node
    deg = nodes[n]->deg;

    if (deg <= 1) {
        reord[0] = 0;
        return 0;
    }

    // list of particles at the legs of the node 'n'
    for (lg = 0; lg < deg; lg++) {
        ed = nodes[n]->aedges[lg];
        pt = edges[ed]->cand->plist[0];
        plist[lg] = legEdgeParticle(n, lg, pt);
        used[lg]  = 0;
    }

    // list of legs in the interaction
    ia    = nodes[n]->cand->ilist[0];
    ilegs = model->interacts[ia]->plist;

    // reorder
#ifdef CHECK
    found = False;
#endif
    for (lr = 0; lr < deg; lr++) {
        for (lg = 0; lg < deg; lg++) {
            if (used[lg] == 0 && plist[lg] == ilegs[lr]) {
                reord[lr] = lg;
                used[lg]  = 1;
#ifdef CHECK
                found = True;
#endif
                break;
            }
        }

#ifdef CHECK
        if (!found) {
            printf("*** reordLeg: illegal list of particles:"
                   "interaction %d ", ia);
```

```
            prIntArray(deg, ilegs, "; ");
            printf("vertex %d ", n);
            prIntArray(deg, plist, "\n");
            prCand("reordLeg");
            erEnd("reordLeg: illegal list of particles");
        }
#endif
    }

    return reord;
}

//==============================================================
// Adjust the direction of a particle on an edge and on a leg of
// node.
//--------------------------------------------------------------
int Assign::getLegParticle(int n,  int ln)
{
    //  Get particle code of (n, ln) when particle 'pt' runs
    //  in the direction of the edge at (n, ln).
    //
    //  Get particle code in the direction the edge at (n, ln)
    //  when particle 'pt' is at leg (n, ln).
    //
    //  These two cases are realized by the same function

    ANode *nd;
    int    elg, ed, pt;

    nd  = nodes[n];
    elg = nd->aelegs[ln];
    ed  = nd->aedges[ln];
    pt  = edges[ed]->cand->plist[0];

    // normalization of sign for neutral particle
    if (elg == 0) {
        return model->normalParticle(-pt);
    } else {
        return model->normalParticle(pt);
    }
}

//--------------------------------------------------------------
int Assign::legEdgeParticle(int n, int ln, int pt)
{
    //  Get particle code of (n, ln) when particle 'pt' runs
    //  in the direction of the edge at (n, ln).
    //
    //  Get particle code in the direction the edge at (n, nl)
    //  when particle 'pt' is at leg (n, ln).
    //
    //  These two cases are realized by the same function

    // normalization of sign for neutral particle

    if (nodes[n]->aelegs[ln] == 0) {
        return model->normalParticle(-pt);
    } else {
        return model->normalParticle(pt);
    }
}

//--------------------------------------------------------------
int Assign::legPart(int v, int lg, int nplist, int *plist, int *rlist, const int size)
{
    // Convert list of candidate particles in 'plist' at (v, lg) ==> edge
    // or edge ==> (v, lg)

    int    j, nrlist;

    nrlist = 0;
    for (j = 0; j < nplist; j++) {
        nrlist = intSetAdd(nrlist, rlist,
                           legEdgeParticle(v, lg, plist[j]), size);
    }
    return nrlist;
}

//--------------------------------------------------------------
int Assign::candPart(int v, int ln, int *plist, const int size)
{
    //  update candidate particles incoming to (v, ln)

    int      en;
    int      ntplist;
    ECand    *ec;

```

```
    en = nodes[v]->aedges[ln];

#ifdef CHECK
    if (edges[en]->cand->det) {
        printf("*** candPart : particle of leg (%d, %d) "
               "is assigned to %d\n",
               v, ln, edges[en]->cand->plist[0]);
        checkCand("candPart");
        mgraph->printAdjMat(mgraph->curcl);
        prCand("candPart");
        erEnd("candPart : particle of leg is assigned");
    }
#endif

    ec = edges[en]->cand;

    ntplist = legPart(v, ln, ec->nplist, ec->plist, plist, size);

    return  ntplist;
}

//==============================================================
// tools for assignment

//--------------------------------------------------------------
int Assign::selUnAssVertexSimp(int lastv)
{
    //  Select a vertex for assignment to its leg.
    //  We take one with minimum number of candidates
    //
    //  There may be better method.

    int v0, v;

    // select vertex one by one in the sequential order of node number
    v0 = Max(lastv+1, nExtern);
    for (v = v0; v < nNodes; v++) {
        if (nodes[v]->cand->st == AS_UnAssLegs) {
            return v;
        }
    }
    return -1;
}

//--------------------------------------------------------------
int Assign::selUnAssVertex(void)
{
    //  Select a vertex for assignment to its leg.
    //  We take one with minimum number of candidates
    //
    //  There may be better method.

    int v0, v, nl;

    // select vertex one by one in the sequential order of node number
    v0 = -1;
    nl = 0;
    for (v = 0; v < nNodes; v++) {
        if (nodes[v]->cand->st == AS_UnAssLegs) {
            if (nodes[v]->cand->nilist == 1) {
                return v;
            } else if (nodes[v]->cand->nilist > 1) {
                // select node with fewer candidates
                if (v0 < 0 || nodes[v]->cand->nilist < nl) {
                    v0 = v;
                    nl = nodes[v]->cand->nilist;
                }
            }
        }
    }
    return v0;
}

//--------------------------------------------------------------
int Assign::selUnAssLeg(int v, int lastlg)
{
    // Select a leg of vertex 'v' for the assignment.
    // The lastly selected leg was 'lastlg'.

    int lg0, lg, e;
#ifdef CHECK
    int n0, n1;
#endif

    // lg0 = Max(lastlg, lastlg + 1);
    lg0 = lastlg + 1;

```

```
    for (lg = lg0; lg < nodes[v]->deg; lg++) {

#ifdef CHECK
        if (lastlg >= 0) {
            n0 = nodes[v]->anodes[lastlg];
        } else {
            n0 = -1;
        }
        n1 = nodes[v]->anodes[lg];
        if (n0 > n1) {
            printf("*** selUnAssLeg: n0=%d > n1=%d\n", n0, n1);
            printf("*** illegal connection\n");
            erEnd("selUnAssLeg: n0 > n1");
        }
#endif

        e  = nodes[v]->aedges[lg];
        if (!edges[e]->cand->det) {
            return lg;
        }

    }
    return -1;
}

//--------------------------------------------------------------
NCandSt Assign::assignIVertex(int v, int ia)
{
    //  Try to assign interaction 'ia' to 'v'

    int      lplist[GRCC_MAXLEGS], iplist[GRCC_MAXLEGS];
    int      lg, e, pt, deg;


    if (nodes[v]->cand->st == AS_Assigned || nodes[v]->cand->st == AS_AssExt) {
        return AS_Assigned0;
    }

    deg = nodes[v]->deg;
    if (deg != model->interacts[ia]->nlegs) {
        return AS_Impossible;
    }

    // list of particles at the legs of the vertex
    for (lg = 0; lg < deg; lg++) {
        e = nodes[v]->aedges[lg];
        if (edges[e]->cand->nplist != 1) {
            return AS_UnAssLegs;
        } else {
            pt = edges[e]->cand->plist[0];
            lplist[lg] = legEdgeParticle(v, lg, pt);
        }
        iplist[lg] = model->interacts[ia]->plist[lg];
    }

    sorti(deg, lplist);
    sorti(deg, iplist);

    // from interaction
    if (cmpArray(deg, lplist, deg, iplist) == 0) {
        // orders of coupling constants should be checked ???
        astack->pushNode(v);
        nodes[v]->cand->st = AS_Assigned;
        nodes[v]->cand->nilist = 1;
        nodes[v]->cand->ilist[0] = ia;
        return AS_Assigned;
    }

    return AS_Impossible;
}

//--------------------------------------------------------------
Bool Assign::assignPLeg(int n, int ln, int pt)
{
    //  A particle 'pt' is assigned to leg 'ln' of node 'n'

    ANode *nd;
    int e, ept;
    Bool ok;

#ifdef CHECK
    checkNode(n, "assignPLeg0");
    checkCand("assignPLeg");
#endif

    // nodes and the edge
    nd  = nodes[n];
```

```
    e   = nd->aedges[ln];

    // already determined
    if (edges[e]->cand->det) {
        return True;
    }

    if (!isOrdPLeg(n, ln, pt)) {
#ifdef DEBUGM
        nopleg++;
#endif
        return False;
    }

    // particle on the edge
    ept = legEdgeParticle(n, ln, pt);

#ifdef CHECK
    if (!isIn(edges[e]->cand->nplist, edges[e]->cand->plist, ept)) {
        printf("*** assignPLeg: particle %d is not in the cand. of e=%d",
                ept, e);
        edges[e]->cand->prECand("\n");
        prCand("assignPLeg");
        erEnd("assignPLeg: particle is not in the cand.");
    }
#endif

    // save the current configuration
    astack->pushEdge(e);

    // determine the particle on the edge
    edges[e]->cand->nplist    = 1;
    edges[e]->cand->plist[0] = ept;

    ok = detEdge(e);

    return ok;
}

//--------------------------------------------------------------
Bool Assign::detEdge(int e)
{
    Bool ok0, ok1;

    if (edges[e]->cand->nplist != 1) {
        return False;
    }
    edges[e]->cand->det = True;

    // update candidate list
    ok0 = updateCandNode(edges[e]->nodes[0]);
    if (ok0) {
        ok1 = updateCandNode(edges[e]->nodes[1]);
    } else {
        ok1 = False;
    }

    return (ok0 && ok1);
}

//--------------------------------------------------------------
Bool Assign::isOrdPLeg(int n, int ln, int pt)
{
    int e0, el, ln0, ep0, pt0, nn, n0, n1, ln1, pt1, e1, ep1;

    if (nodes[n]->deg < 2) {
        return True;
    }
    nn = nodes[n]->anodes[ln];
    if (n <= nn) {
        n0  = n;
        n1  = nn;
        ln0 = ln;
        pt0 = pt;
    } else {
        n0 = nn;
        n1 = n;
        e0  = nodes[n]->aedges[ln];
        el  = nodes[n]->aelegs[ln];
        ln0 = edges[e0]->nlegs[1-el];
        ep0 = legEdgeParticle(n,  ln,  pt);
        pt0 = legEdgeParticle(n0, ln0, ep0);
    }

    for (ln1 = 0; ln1 < nodes[n0]->deg; ln1++) {
        if (ln1 == ln0 || n1 != nodes[n0]->anodes[ln1]) {
            continue;
```

```
        }
        e1  = nodes[n0]->aedges[ln1];
        if (!edges[e1]->cand->det) {
            continue;
        }
        ep1 = edges[e1]->cand->plist[0];
        pt1 = legEdgeParticle(n0, ln1, ep1);
        if ((ln0 < ln1 && pt0 > pt1) ||
            (ln0 > ln1 && pt0 < pt1)) {
            return False;
        }
    }
    return True;
}

//==============================================================
// Operations of candidate
//--------------------------------------------------------------
Bool Assign::candPartClassify(int v, int *npdass, int *pdass, int *npuass, int *puass, const int size)
{
    // Construct the classified lists of particles possible to assign to
    // the legs of node v
    //
    // pdass = (duplicated set of particles where edges has a unique
    // candidate)
    // puass = (duplicated set of other candidate particles)

    int lg, e, npl, ass, jd, ju, j, pt, pts;

    jd = 0;
    ju = 0;
    for (lg = 0; lg < nodes[v]->deg; lg++) {
        e   = nodes[v]->aedges[lg];
        npl = edges[e]->cand->nplist;
        if (npl < 1) {
            return False;
        }
        ass = (npl == 1);
        for (j = 0; j < edges[e]->cand->nplist; j++) {
            pt  = edges[e]->cand->plist[j];
            pts = legEdgeParticle(v, lg, pt);
            if (ass) {
                jd = intSListAdd(jd, pdass, pts, size);
            } else {
                ju = intSListAdd(ju, puass, pts, size);
            }
        }
    }
    *npdass = jd;
    *npuass = ju;

    return True;
}

//--------------------------------------------------------------
Bool Assign::updateCandNode(int v)
{
    //  Update the configuration
    //  - nodes[v]->cand->ilist
    //  - edges[e]->cand->plist for the adjacent edge
    //  - nodes[n]->cand->unass for the adjacent node 'n' of 'e'

    int   npdass,  pdass[GRCC_MAXPSLIST];
    int   npuass,  puass[GRCC_MAXPSLIST];
    int    nsub0,    sub0[GRCC_MAXPSLIST];
    int  niplist, iplist[GRCC_MAXPSLIST];
    int      nd0,      d0[GRCC_MAXMPARTICLES2];
    int      nd1,      d1[GRCC_MAXMPARTICLES2];
    int      ns0,      s0[GRCC_MAXMPARTICLES2];
    int  neplist, eplist[GRCC_MAXMPARTICLES2];
    int  nitlist, itlist[GRCC_MAXMINTERACT];
    int e, it, j, k, i, lg;

    // external node
    if (nodes[v]->cand->st == AS_AssExt) {
#ifdef CHECK
        e = nodes[v]->aedges[0];
        if (e < 0 || edges[e]->cand->nplist != 1) {
            printf("*** illegal external node: v=%d e=%d :", v, e);
            prCand("updateCandNode");
            erEnd("illegal external node");
        }
#endif

        return True;
    }

```

```
    // impossible configuration.
    if (nodes[v]->cand->nilist < 1) {
        return False;
    }

    // list of possible particles on the edges of the vertex.
    // pdass = (set of assigned particles)
    // puass = (list of particles of edges with two of more candidates)

    niplist = 0;
    nitlist = 0;
    if (!candPartClassify(v, &npdass, pdass, &npuass, puass, GRCC_MAXPSLIST)) {
        return False;
    }

    if (npdass < 1) {
        return False;
    }

    // from interaction : slist
    // Conditions :
    // 1. pdass \subset slist
    // 2. slist-pdass \subset puass
    // from interaction : slist
    //  conditions : pdass \subset slist \subset (pdass + puass)
    //  sub0 = slist  -   pdass,
    nitlist = 0;
    for (j = 0; j < nodes[v]->cand->nilist; j++) {
        it = nodes[v]->cand->ilist[j];
        // conditions:
        // 1. pdass \subset slist <==> pdass-slist = \emptyset
        // 2. slist-pdass \subset puass <==> (slist-pdass)-puass = \emptyset
        if (isSubSList(npdass, pdass,
                       model->interacts[it]->nslist,
                       model->interacts[it]->slist) ) {
            nsub0 = subtrSet(model->interacts[it]->nslist,
                             model->interacts[it]->slist,
                             npdass, pdass, sub0, GRCC_MAXPSLIST);
            if (nsub0 < 1) {
                // slist == pdass : no additional candidate particle
                nitlist = intSetAdd(nitlist, itlist, it, GRCC_MAXMINTERACT);

            } else {
                if (isSubSList(nsub0, sub0, npuass, puass)) {
                    nitlist = intSetAdd(nitlist, itlist, it, GRCC_MAXMINTERACT);
                    for (k = 0; k < nsub0; k++) {
                        niplist = intSetAdd(niplist,iplist,sub0[k],GRCC_MAXPSLIST);
                    }
                }
            }
        }
    }

    if (nitlist < 1) {
        return False;
    }

    if (nitlist != nodes[v]->cand->nilist) {
        astack->pushNode(v);
        nodes[v]->cand->nilist = nitlist;
        for (i = 0; i < nitlist; i++) {
            nodes[v]->cand->ilist[i] = itlist[i];
        }
#ifdef CHECK
    } else if (nitlist > nodes[v]->cand->nilist) {
        erEnd("larger ncand");
#endif
    }

    // edge
    for (lg = 0; lg < nodes[v]->deg; lg++) {
        e = nodes[v]->aedges[lg];
        if (edges[e]->cand->nplist > 1) {
            // elist = {candidate particles in the direction of the edge}
            // s0 = plist \cap elist
            neplist = legPart(v, lg, niplist, iplist, eplist, GRCC_MAXMPARTICLES2);
            listDiff(edges[e]->cand->nplist, edges[e]->cand->plist,
                     neplist, eplist,
                     &nd0, d0, &ns0, s0, &nd1, d1);

            if (ns0 < 1) {
                return False;
            }
            if (ns0 < edges[e]->cand->nplist) {
                astack->pushEdge(e);
                edges[e]->cand->nplist = ns0;
                for (i = 0; i < ns0; i++) {
```

```
                        edges[e]->cand->plist[i] = s0[i];
                    }
                }
            }
        }
        for (lg = 0; lg < nodes[v]->deg; lg++) {
            e = nodes[v]->aedges[lg];
            if (edges[e]->cand->nplist == 1 && !edges[e]->cand->det) {
                if(!detEdge(e)) {
                    return False;
                }
            }
        }

        return True;
}

//==============================================================
// Check order of coupling constants, duplication and isomorphism
//--------------------------------------------------------------
Bool Assign::checkOrderCpl(void)
{
    //  Check order of coupling constants

    int ord[GRCC_MAXNCPLG];
    int lcn, n, ia, j, cl;

    // list of coupling constants
    lcn = model->ncouple;

    if (lcn == 1) {
        return True;
    }

    // sum up for each coupling constants
    for (j = 0; j < GRCC_MAXNCPLG; j++) {
        ord[j] = 0;
    }
    for (n = 0; n < nNodes; n++) {
        cl = pnclass->nd2cl[n];
        if (!isATExternal(pnclass->type[cl])) {
            ia = nodes[n]->cand->ilist[0];
            for (j = 0; j < lcn; j++) {
                ord[j] += model->interacts[ia]->clist[j];
            }
        }
    }

    // comparison
    for (j = 0; j < lcn; j++) {
        if (sproc->clist[j] != ord[j]) {
            return False;
        }
    }
    return True;
}

//--------------------------------------------------------------
Bool Assign::isOrdLegs(void)
{
    //  Whether graph is duplicated because of the violation
    //  of ordering in the case of multiple connections
    //    v0 <-- v --> v1

    int v, l0, v0, l1, v1, e0, e1, q0, q1, p0, p1, cl;

    for (v = 0; v < nNodes; v++) {
        cl = pnclass->nd2cl[v];
        if (!isATExternal(pnclass->type[cl])) {
            for (l0 = 0; l0 < nodes[v]->deg; l0++) {
                v0 = nodes[v]->anodes[l0];
                for (l1 = l0+1; l1 < nodes[v]->deg; l1++) {
                    v1 = nodes[v]->anodes[l1];

                    if (v0 == v1 && v <= v0) {
                        // multiple connections (v, l0) and (v, l1)
                        e0 = nodes[v]->aedges[l0];
                        e1 = nodes[v]->aedges[l1];
                        q0 = edges[e0]->cand->plist[0];
                        q1 = edges[e1]->cand->plist[0];
                        p0 = legEdgeParticle(v, l0, q0);
                        p1 = legEdgeParticle(v, l1, q1);
                        if (p0 > p1) {
                            // violation of ordering
                            return False;
                        }
                    }
```

```
                }
            }
        }
    }

    return True;
}

//----------------------------------------------------------------
Bool Assign::isIsomorphic(MNodeClass *cl, BigInt *nsym, BigInt *esym, BigInt *nsym1)
{
    //  Check whether the current graph is the representive of
    //  a isomorphic class.
    //  Returns (nsym, esym)
    //    nsym = symmetry factor by the permutation of nodes.
    //    esym = symmetry factor by the permutation of edge.
    //  If this graph is not a repressentative, then returns (0,0).

    int j, cmp, n, cln;
    BigInt ngelem;
    int *p;
    Bool invext;

    ngelem = mgraph->group->nElem();
#ifdef CHECK
    if (mgraph->nsym > 1 && ngelem <= 1) {
        printf("*** isIsomorphic: illegal group: "
               "ngelem=%ld, mgraph->sym=(%ld, %ld)\n",
               ngelem, mgraph->nsym, mgraph->esym);
        erEnd("Assign::isIsomorphic: illegal group");
    }
#endif
    if (ngelem == 0) {
        *nsym  = 1;
        *nsym1 = 1;
    } else {
        *nsym  = 0;
        *nsym1 = 0;
        for (j = 0; j < ngelem; j++) {
            // check the graph is the representative and count 'nsym'.

            //p = mgraph->group->nextElem();
            p = mgraph->group->elem[j];

            cmp = cmpPermGraph(p, cl);

            if (cmp < 0) {          // duplicated graph
#ifdef DEBUGM
                niso1++;
#endif
                return False;
            } else if(cmp == 0) { // not duplicated
                (*nsym)++;

                if (opt->values[GRCC_OPT_SymmInitial] || opt->values[GRCC_OPT_SymmFinal]) {
                    invext = True;
                    for (n = 0; n < nNodes; n++) {
                        cln = pnclass->nd2cl[n];
                        if (!isATExternal(pnclass->type[cln])) {
                            continue;
                        }
                        if (p[n] != n) {
                            invext = False;
                        }
                    }
                    if (invext) {
                        (*nsym1)++;
                    }
                } else {
                    (*nsym1)++;
                }
            }
        }
    }
    if (*nsym1 == 0) {
        *nsym1 = *nsym;
    }

    // calculate permutations of edges
    *esym = edgeSym();
    if (*esym < 1) {
#ifdef DEBUGM
        niso2++;
#endif
        return False;
    }

```

```
    return True;
}

//---------------------------------------------------------------
int Assign::cmpPermGraph(int *p, MNodeClass *cl)
{
    //  compare the graph with one permutated by 'p'.

    int n, cmp, n1, n2, p1, p2, j;
    ANode *nd1, *np1;
    // ANode *nd2, *np2;
    int    njn, jn[GRCC_MAXLEGS];
    int    njp, jp[GRCC_MAXLEGS];

#ifdef CHECK
    if (p == NULL) {
        erEnd("Assign::cmpPermGraph: p==NULL");
    }
#endif
#ifdef DEBUG1
    printf("cmpPermGraph:0: p=");
    prIntArray(nNodes, p, "\n");
#endif
    for (n = 0; n < nNodes; n++) {
        if (!isATExternal(pnclass->type[pnclass->nd2cl[n]])) {
            cmp = cmpNodes(n, p[n], cl);
            if (cmp != 0) {
#ifdef DEBUGM
                if (cmp < 0) { niso11++; }
#endif
                return cmp;
            }
        }
    }

    njn = 0;
    njp = 0;
    for (n1 = 0; n1 < nNodes; n1++) {
        p1  = p[n1];
        nd1 = nodes[n1];
        np1 = nodes[p1];
        for (n2 = n1; n2 < nNodes; n2++) {
            if (mgraph->adjMat[n1][n2] == 0) {
                continue;
            }

            p2  = p[n2];
            if (p1 == n1 && p2 == n2) {
                continue;
            }
            cmp = mgraph->adjMat[n1][n2] - mgraph->adjMat[p1][p2];
            if (cmp != 0) {
#ifdef DEBUGM
                if (cmp < 0) { niso12++; }
#endif
                return cmp;
            }

            njn = 0;
            njp = 0;
            for (j = 0; j < nd1->deg; j++) {
                if (nd1->anodes[j] == n2) {
                    jn[njn++] = getLegParticle(n1, j);
                }
            }

            for (j = 0; j < np1->deg; j++) {
                if (np1->anodes[j] == p2) {
                    jp[njp++] = getLegParticle(p1, j);
                }
            }

            cmp = njn - njp;
            if (cmp != 0) {
#ifdef DEBUGM
                if (cmp < 0) { niso13++; }
#endif
                return cmp;
            }

            // ignore ordering for multiple connections
            if (mgraph->adjMat[n1][n2] >= 2) {
                sorti(njn, jn);
                sorti(njn, jp);
            }

            for (j = 0; j < njn; j++) {
```

```
                cmp = jn[j] - jp[j];
                if (cmp != 0) {
#ifdef DEBUGM
                    if (cmp < 0) { niso14++; }
#endif
                    return cmp;
                }
            }
        }
    }

    return 0;
}

//--------------------------------------------------------------
int Assign::cmpNodes(int nd0, int nd1, MNodeClass *cn)
{
    //  Comparison of two nodes 'nd0' and 'nd1'
    //    Ordering is lexicographical (class, connection configuration)
    //

    int cmp;

    // Wether two nodes are in a same class or not.
    cmp = cn->ndcl[nd0] - cn->ndcl[nd1];
    if (cmp != 0) {
#ifdef DEBUGM
        if (cmp < 0) { niso111++; }
#endif
        return cmp;
    }

    // interaction
    cmp = nodes[nd0]->cand->ilist[0] - nodes[nd1]->cand->ilist[0];
#ifdef DEBUGM
    if (cmp < 0) { niso112++; }
#endif
    return cmp;
}

//--------------------------------------------------------------
BigInt Assign::edgeSym(void)
{
    //  calculate permutations of edges

    int lg[GRCC_MAXLEGS], lt[GRCC_MAXLEGS], lc;
    ANode *nd1;
    int n1, n2, j, k, mult;
    BigInt  esym;

    esym = 1;
    for (n1 = 0; n1 < nNodes; n1++) {
        nd1 = nodes[n1];
        for (n2 = n1; n2 < nNodes; n2++) {
            if (mgraph->adjMat[n1][n2] > 1) {
                lc = 0;
                for (j = 0; j < nd1->deg; j++) {
                    if (nd1->anodes[j] == n2) {
                        lg[lc] = 1;
                        lt[lc] = getLegParticle(n1, j);
                        lc++;
                    }
                }
                for (j = 0; j < lc-1; j++) {
                    if (lg[j] > 0) {
                        for (k = lc-1; k > j; k--) {
                            if (lt[j] == lt[k]) {
                                lg[j] += lg[k];
                                lg[k]  = 0;
                            }
                        }
                    }
                }

                // calculate symmetry factor
                for (j = 0; j < lc; j++) {
                    if (lg[j] < 1) {
                        continue;
                    }
                    mult = lg[j];
                    if (mult > 1) {
                        if (n1 == n2) {
                            // self-loop
                            esym *= ipow(2,mult/2)*factorial(mult/2);
                        } else {
                            esym *= factorial(mult);
                        }
```

```
                    }
                }
            }
        }
    }

    return esym;
}

#ifdef CHECK
//===============================================================
// check
//---------------------------------------------------------------
Bool Assign::checkCand(const char *msg)
{
    //  Check the consistency of Candidate data

    Bool  ok;
    ANode *na;
    NCand *nc;
    ECand *ec;
    int   n, lg, e;

    // check 'nodes->cand'
    ok = True;
    for (n = 0; n < nNodes; n++) {
        na = nodes[n];
        nc = na->cand;

        // check unassigned vertex
        if (nc->st == AS_UnAssLegs) {

        // check assigned vertex
        } else if (nc->st == AS_Assigned) {
            if (nc->nilist < 1) {
                printf("*** checkCand:7:%s:status (%d) of node %d says"
                       " interaction is assigned to %d but ilist=",
                       msg, nc->st, n, nc->st);
                prIntArray(nc->nilist, nc->ilist, "\n");
                ok = False;
            }

            for (lg = 0; lg < nc->deg; lg++) {
                e  = na->aedges[lg];
                ec = edges[e]->cand;
                if (ec->nplist != 1) {
                    printf("*** checkCand:8:%s:status (%d) of node %d says"
                          " interaction is assigned "
                          " but unassigned edge %d is found\n",
                          msg, nc->st, n, e);
                    ok = False;
                }
            }

        // external particle
        } else if (nc->st == AS_AssExt) {
            // pt  = nc->ilist[0];
            // *** pte = legEdgeParticle(n, 0, - pt);
        } else {
            printf("*** checkCand:10:%s:illegal status of node %d : %d",
                   msg, n, nc->st);
            ok = False;
        }
    }

    // check edges

    for (e = 0; e < nEdges; e++) {
        ec = edges[e]->cand;
        if (ec != NULL && ec->nplist < 1) {
            printf("*** checkCand:12:%s:illegal edge %d\n", msg, e);
            ok = False;
        }
    }

    if (!ok) {
        printf("*** checkCand:15:%s:illegal configuration\n", msg);
        prCand("checkCand");
        printf("*** checkCand:16:illegal configuration\n");
        erEnd("checkCand:16:illegal configuration");
    }
    return ok;
}

//---------------------------------------------------------------
void Assign::checkNode(int n, const char *msg)
{
```

```
    int j, it;

    for (j = 0; j < nodes[n]->cand->nilist; j++) {
        it = nodes[n]->cand->ilist[j];
        if (Abs(it) >= GRCC_MAXMINTERACT) {
            printf("*** %s: n=%d, j=%d, it=%d\n", msg, n, j, it);
            nodes[n]->cand->prNCand(msg);
            printf("\n");
            erEnd("checkNode:illegal it");
        }
    }
}
#endif // CHECK

//**************************************************************
// astack.cc
//==============================================================
// Assign particles to the edges and interactions to nodes.
// Completed graph data is saved in the form of EGraph.

//=============================================================
// stack operations
//--------------------------------------------------------------
void NStack::print(const char *msg)
{
    printf("  node=%d, deg=%d, st=%d, ilist=", noden, deg, st);
    prilist(nilist, ilist, msg);
}

//--------------------------------------------------------------
void EStack::print(const char *msg)
{
    printf("  edge=%d, det=%d, plist=", edgen, det);
    prilist(nplist, plist, msg);
}

//--------------------------------------------------------------
AStack::AStack(int nsize, int esize)
{
    int j;

    agraph  = NULL;

    nSize   = nsize;
    eSize   = esize;
    if (nSize > 0) {
        nStack = new NStack*[nSize];
    } else {
        nStack = NULL;
    }
    if (eSize > 0) {
        eStack = new EStack*[eSize];
    } else {
        eStack = NULL;
    }

    nStackP = 0;
    eStackP = 0;

    for (j = 0; j < nSize; j++) {
        nStack[j] = new NStack();
    }

    for (j = 0; j < eSize; j++) {
        eStack[j] = new EStack();
    }
}

//--------------------------------------------------------------
AStack::~AStack(void)
{
    int j;

    if (eStack != NULL) {
        for (j = eSize-1; j >= 0; j--) {
            if (eStack[j] != NULL) {
                delete eStack[j];
                eStack[j] = NULL;
            }
        }
        delete[] eStack;
        eStack = NULL;
    }

    if (nStack != NULL) {
        for (j = nSize-1; j >= 0; j--) {
            if (nStack[j] != NULL) {
```

```
                delete nStack[j];
                nStack[j] = NULL;
            }
        }
        delete[] nStack;
        nStack = NULL;
    }
}

//--------------------------------------------------------------
void AStack::setAGraph(Assign *ag)
{
    agraph = ag;
}

//--------------------------------------------------------------
void AStack::pushNode(int n)
{
    NCand  *nc;
    NStack *ns;
    int j;

#ifdef CHECK
    if (agraph == NULL) {
        erEnd("pushNode: agraph is not defined");
    }
#endif
    if (nStackP >= GRCC_MAXNSTACK) {
        erEnd("N-stack overflow (GRCC_MAXNSTACK)");
    }
    nc = agraph->nodes[n]->cand;
    if (nc->nilist >= GRCC_MAXMINTERACT) {
        erEnd("N-stack: too long list (GRCC_MAXMINTERACT)");
    }
    ns = nStack[nStackP];
    ns->noden  = n;
    ns->deg    = nc->deg;
    ns->st     = nc->st;
    ns->nilist = nc->nilist;
    for (j = 0; j < nc->nilist; j++) {
        ns->ilist[j] = nc->ilist[j];
    }
    nStackP++;
}

//--------------------------------------------------------------
void AStack::pushEdge(int e)
{
    ECand  *ec;
    EStack *es;
    int j;

#ifdef CHECK
    if (agraph == NULL) {
        erEnd("pushEdge: agraph is not defined");
    }
#endif
    if (eStackP >= GRCC_MAXESTACK) {
        erEnd("E-stack overflow (GRCC_MAXESTACK)");
    }
    ec = agraph->edges[e]->cand;
    if (ec->nplist >= GRCC_MAXMPARTICLES2) {
        erEnd("E-stack: Too long list (GRCC_MAXMPARTICLES)");
    }
    es = eStack[eStackP];
    es->edgen  = e;
    es->det    = ec->det;
    es->nplist = ec->nplist;
    for (j = 0; j < ec->nplist; j++) {
        es->plist[j] = ec->plist[j];
    }
    eStackP++;
}

//--------------------------------------------------------------
void AStack::checkPoint(CheckPt sav)
{
    sav[0] = nStackP;
    sav[1] = eStackP;
}

//--------------------------------------------------------------
void AStack::restoreNode(int spr)
{
    NStack *stc;
    int     sp, n, j;

```

```
    for (sp = nStackP-1; sp >= spr; sp--) {
        stc = nStack[sp];
        n = stc->noden;
        agraph->nodes[n]->cand->deg     = stc->deg;
        agraph->nodes[n]->cand->st      = stc->st;
        agraph->nodes[n]->cand->nilist  = stc->nilist;
        for (j = 0; j < stc->nilist; j++) {
            agraph->nodes[stc->noden]->cand->ilist[j]  = stc->ilist[j];
        }
    }
    nStackP = spr;
}

//---------------------------------------------------------------
void AStack::restoreEdge(int spr)
{
    EStack *stc;
    int     sp, e, j;

    for (sp = eStackP-1; sp >= spr; sp--) {
        stc = eStack[sp];
        e   = eStack[sp]->edgen;
        agraph->edges[e]->cand->det = stc->det;
        agraph->edges[e]->cand->nplist = stc->nplist;
        for (j = 0; j < stc->nplist; j++) {
            agraph->edges[e]->cand->plist[j] = stc->plist[j];
        }
    }
    eStackP = spr;
}

//---------------------------------------------------------------
void AStack::restore(CheckPt sav)
{
    restoreNode(sav[0]);
    restoreEdge(sav[1]);
}

#ifdef CHECK
//---------------------------------------------------------------
void AStack::restoreMsg(CheckPt sav, const char *msg)
{
    if (agraph == NULL) {
        erEnd("restore: agraph is not defined");
    }

    restore(sav);

    if (!agraph->checkCand("restore")) {
        printf("restore is called from %s\n", msg);
    }
}
#endif

//---------------------------------------------------------------
void AStack::prStack(void)
{
    int j;

    printf("+++ prStack : (%d, %d)", nStackP, eStackP);
    for (j = 0; j < nStackP; j++) {
        printf("N:%4d ", j);
        nStack[j]->print("\n");
    }
    for (j = 0; j < eStackP; j++) {
        printf("E:%4d ", j);
        eStack[j]->print("\n");
    }
}

//***************************************************************
// class Fraction
//===============================================================
Fraction::Fraction(BigInt n, BigInt d)
{
    BigInt g;

    g     = gcd(n, d);
    num   = n/g;
    den   = d/g;
    ratio = Real(n)/Real(d);
}

//---------------------------------------------------------------
void Fraction::print(const char *msg)
{
    double err = Abs(Real(num)/Real(den) - ratio);
```

```

    if (err > GRCC_FRACERROR) {
        printf("%ld/%ld(%g)(overflow)%s", num, den, ratio, msg);
    } else {
        printf("%ld/%ld(%g)%s", num, den, ratio, msg);
    }
}

//--------------------------------------------------------------
void Fraction::setValue(BigInt n, BigInt d)
{
    num = n;
    den = d;
    ratio = Real(n)/Real(d);
}

//--------------------------------------------------------------
void  Fraction::setValue(Fraction &f)
{
    num   = f.num;
    den   = f.den;
    ratio = f.ratio;
}

//--------------------------------------------------------------
BigInt Fraction::gcd(BigInt n0, BigInt n1)
{
    BigInt nn[2], r;
    int    nc;

    nn[0] = Abs(n0);
    nn[1] = Abs(n1);
    nc    = 0;

    while (1) {
        r = nn[nc] % nn[1-nc];
        if (r == 0) {
            return nn[1-nc];
        }
        nn[nc] = r;
        nc = 1 - nc;
    }
}

//--------------------------------------------------------------
void Fraction::normal(void)
{
    BigInt g;

    if (den == 0) {
        erEnd("Fraction: den==0");
    }
    g   = gcd(num, den);
    num /= g;
    den /= g;
    if (den < 0) {
        num = - num;
        den = - den;
    }
}

//--------------------------------------------------------------
void Fraction::add(BigInt n, BigInt d)
{
    BigInt g, d1;

    if (d < 0) {
        n = -n;
        d = -d;
    }
    if (den < 0) {
        num = - num;
        den = - den;
    }
    g    = gcd(den, d);
    d1   = d/g;
    num  = num * d1 + n * (den/g);
    den  = d1*den;
    g    = gcd(num, den);
    num /= g;
    den /= g;
    ratio += Real(n)/Real(d);
}

//--------------------------------------------------------------
void Fraction::add(Fraction f)
{
```

```
    double r = ratio + f.ratio;

    add(f.num, f.den);
    ratio = r;
}

//------------------------------------------------------------
void Fraction::sub(Fraction f)
{
    double r = ratio - f.ratio;

    add(-f.num, f.den);
    ratio = r;
}

//------------------------------------------------------------
Bool Fraction::isEq(Fraction f)
{
    return (num == f.num && den == f.den);
}

//************************************************************
// common.cc
//============================================================
static void erEnd(const char *msg)
{
    if (erExit != NULL) {
        (*erExit)(msg, erExitArg);
    }
    fprintf(GRCC_Stderr, "*** Error : %s\n", msg);
    GRCC_ABORT();
}

//----------------------------------------------------------
static Bool nextPart(int nelem, int nclist, int *clist, int *nl, int *r)
{
    int rem, pn, j;

    if (*r < 0) {
        *r = 0;
        rem = nelem;
        for (j = 0; j < nclist; j++) {
            nl[j] = rem/clist[j];
            rem -= nl[j]*clist[j];
        }
        if (rem == 0) {
            return True;
        }
    } else {
        rem = 0;
    }
    for (int c = 0; c < 100; c++) {
        rem += nl[nclist-1]*clist[nclist-1];
        for (pn = nclist-2; pn >= 0 && nl[pn] == 0; pn--) {
            ;
        }
        if (pn < 0) {
            return False;
        }
        rem += clist[pn];
        nl[pn]--;
        for (j = pn+1; j < nclist; j++) {
            nl[j] = rem/clist[j];
            rem -= nl[j]*clist[j];
        }
        if (rem == 0) {
            return True;
        }
    }
    printf("*** nextPart : too many repetition\n");
    return False;
}

//----------------------------------------------------------
static int   *intdup(int n, int *a)
{
    int *r, j;

    r = new int[n];
    for (j = 0; j < n; j++) {
        r[j] = a[j];
    }
    return r;
}

//----------------------------------------------------------
static int   *delintdup(int *a)
```

```
{
    if (a != NULL) {
        delete[] a;
    }
    return NULL;
}

//-----------------------------------------------------------
static void   prilist(int n, const int *a, const char *msg)
{
    printf("[");
    for (int j = 0; j < n; j++) {
        if (j!=0) printf(", ");
        printf("%d", a[j]);
    }
    printf("]%s", msg);
}

//-----------------------------------------------------------
static int nextPerm(int nelem, int nclass, int *cl, int *r, int *q, int *p, int count)
{
    // Sequential generation of all permutations.

    int  j, k, n, e, t;
    Bool b;

    for (j = 0; j < nelem; j++) {
        p[j] = j;
    }
    if (count < 1) {
        for (j = 0; j < nelem; j++) {
            q[j] = 0;
            r[j] = 0;
        }
        j = 0;
        for (k = 0; k < nclass; k++) {
            n = cl[k];
            for (e = 0; e < n; e++) {
                r[j] = n - e - 1;
                j++;
            }
        }
#ifdef CHECK
        if (j != nelem) {
            erEnd("inconsistent # elements");
        }
#endif
        return 1;
    }
    b = False;
    for (j = nelem-1; j >= 0; j--) {
        if (q[j] < r[j]) {
            for (k = j+1; k < nelem; k++) {
                q[k] = 0;
            }
            q[j]++;
            b = True;
            break;
        }
    }
    if (!b) {
        return (-count);
    }

    for (j = 0; j < nelem; j++) {
        k = j + q[j];
        t = p[j];
        p[j] = p[k];
        p[k] = t;
    }
    return count + 1;
}

//-----------------------------------------------------------
static BigInt factorial(int n)
{
    // returns 1 for n < 1

    int r, j;

    r = 1;
    for (j = 2; j <= n; j++) {
        r *= j;
    }
    return r;
}

```

```
//----------------------------------------------------------------
static BigInt ipow(int n, int p)
{
    int r, j;

    r = 1;
    for (j = 0; j < p; j++) {
        r *= n;
    }
    return r;
}

//----------------------------------------------------------------
static int   *newArray(int size, int val)
{
    // memory allocation of an array

    int *a, j;

    a = new int[size];
    for (j = 0; j < size; j++) {
        a[j] = val;
    }
    return a;
}

//----------------------------------------------------------------
static int   *deleteArray(int *a)
{
    // memory allocation of an array

    delete[] a;
    return NULL;
}

//----------------------------------------------------------------
static int   **newMat(int n0, int n1, int val)
{
    int **m, j, k;

    m = new int*[n0];
    for (j = 0; j < n0; j++) {
        m[j] = new int[n1];
        for (k = 0; k < n1; k++) {
            m[j][k] = val;
        }
    }
    return m;
}

//----------------------------------------------------------------
static int **deleteMat(int **m, int n0)
{
    int j;

    for (j = n0-1; j >= 0; j--) {
        delete[] m[j];
    }
    delete[] m;

    return NULL;
}

//----------------------------------------------------------------
static void bsort(int n, int *ord, int *a)
{
    int i, j, t;

    for (j = n-1; j > 0; j--) {
        for (i = 0; i < j; i++) {
            if(a[ord[i+1]] < a[ord[i]]) {
                t = ord[i];
                ord[i] = ord[i+1];
                ord[i+1] = t;
            }
        }
    }
}

//------------------------------------------------------------------
static void prMomStr(int mom, const char *ms, int mn)
{
    if (mom == 0) {
        return;
    } else if (mom == 1) {
        printf(" + %s%d", ms, mn);
    } else if (mom > 0) {
```

```
        printf(" + %d*%s%d", mom, ms, mn);
    } else if (mom == -1) {
        printf(" - %s%d", ms, mn);
    } else {
        printf(" - %d*%s%d", -mom, ms, mn);
    }
}

//-------------------------------------------------------------
static void prIntArray(int n, int *p, const char *msg)
{

    int j;

    printf("[");
    for (j = 0; j < n; j++) {
        if (j!=0) printf(", ");
        printf("%2d", p[j]);
    }
    printf("]%s", msg);
}

//-------------------------------------------------------------
static void prIntArrayErr(int n, int *p, const char *msg)
{

    int j;

    fprintf(GRCC_Stderr, "[");
    for (j = 0; j < n; j++) {
        if (j!=0) fprintf(GRCC_Stderr, ", ");
        fprintf(GRCC_Stderr, "%2d", p[j]);
    }
    fprintf(GRCC_Stderr, "]%s", msg);
}

//-------------------------------------------------------------
static void sortrec(int l, int r, int *a)
{
    // quick sort : taken from N. Wirth
    int i,j, x, w;

    i = l;
    j = r;
    x = a[(l+r)/2];
    do {
        while(a[i] < x) i++;
        while(x < a[j]) j--;
        if (i <= j) {
            w = a[i];  a[i] = a[j];  a[j] = w;
            i++;  j--;
        }
    } while (i < j);
    if (l < j) sortrec(l, j, a);
    if (i < r) sortrec(i, r, a);
}

//-------------------------------------------------------------
static void sorti(int size, int *a)
{
    sortrec(0, size-1, a);
}

//-----------------------------------------------------------
static int sortb(int n, int *a)
{
    int i, j, t, nswap;

    nswap = 0;
    for (j = n-1; j > 0; j--) {
        for (i = 0; i < j; i++) {
            if(a[i+1] < a[i]) {
                t = a[i];
                a[i] = a[i+1];
                a[i+1] = t;
                nswap++;
            }
        }
    }

    // the sign of the permutation is (-1)^{nswap}
    return nswap;
}

#ifdef CHECK
//-------------------------------------------------------------
static Bool isIn(int n, int *a, int v)
```

```
{
    int j;

    for (j = 0; j < n; j++) {
        if (a[j] == v) {
            return True;
        }
    }
    return False;
}
#endif

//---------------------------------------------------------------
static int intSetAdd(int n, int *a, int v, const int size)
{
    // Add 'v' into 'a'.
    // 'n' is the current # of element.
    // 'a' is assumed sorted without duplicated elements.
    // Size of 'a' should be large enough.

    int j, k;

    if (n >= size) {
        fprintf(GRCC_Stderr, "*** intSetAdd : array out of range (>%d)\n", size);
        erEnd("intSetAdd : array out of range (GRCC_MAXPSLIST)");
    }
    for (j = 0; j < n; j++) {
        if (a[j] > v) {
            break;
        } else if (a[j] == v) {
            return n;
        }
    }

    // 'j' can be equal to 'n'
    for (k = n-1; k >= j; k--) {
        a[k+1] = a[k];
    }
    a[j] = v;
    return n+1;
}

//---------------------------------------------------------------
static int intSListAdd(int n, int *a, int v, const int size)
{
    // Add 'v' into 'a'.
    // 'n' is the current # of element.
    // 'a' is assumed sorted without duplicated elements.
    // Size of 'a' should be large enough.

    int j, k;

    if (n >= size) {
        fprintf(GRCC_Stderr, "*** intSListAdd : array out of range (>%d)\n", size);
        erEnd("intSListAdd : array out of range");
    }
    for (j = 0; j < n; j++) {
        if (a[j] >= v) {
            break;
        }
    }

    // 'j' can be equal to 'n'
    for (k = n-1; k >= j; k--) {
        a[k+1] = a[k];
    }
    a[j] = v;
    return n+1;
}

//---------------------------------------------------------------
static int    cmpArray(int na, int *a, int nb, int *b)
{
    int j, cmp;

    cmp = na - nb;
    if (cmp != 0) {
        return cmp;
    }
    for (j = 0; j < na; j++) {
        cmp = a[j] - b[j];
        if (cmp != 0) {
            return cmp;
        }
    }
    return 0;
}
```

```
//-------------------------------------------------------------
static Bool   leqArray(int n, int *a, int *b)
{
    int j;

    for (j = 0; j < n; j++) {
        if (a[j] > b[j]) {
            return False;
        }
    }
    return True;
}

//-------------------------------------------------------------
// convert list to sorted list
static int   toSList(int n, int *a)
{
    sorti(n, a);
    return n;
}

//-------------------------------------------------------------
// as set operation assuming a and b are sorted with duplication
//     p := a \ b;   q := a \cap b;   r := b \ a;
// p, q, r are sorted without duplication
static void   listDiff(int na, int *a, int nb, int *b, int *np, int *p, int *nq, int *q, int *nr, int
*r)
{
    int ja, jb;

    *np = *nq = *nr = 0;
    ja = jb = 0;
    while (ja < na && jb < nb) {
        while (ja < na && a[ja] < b[jb]) {
            p[(*np)++] = a[ja++];
        }
        while (jb < nb && b[jb] < a[ja]) {
            r[(*nr)++] = b[jb++];
        }
        if (ja < na && jb < nb && a[ja] == b[jb]) {
            q[(*nq)++] = a[ja++];
            jb++;
        }
    }
    for (; ja < na; ja++) {
        p[(*np)++] = a[ja];
    }
    for (; jb < nb; jb++) {
        r[(*nr)++] = b[jb];
    }
}

//-------------------------------------------------------------
static int isSubSList(int na, int *a, int nb, int *b)
{
    int ja, jb;

    ja = jb = 0;
    while (ja < na && jb < nb) {
        for ( ; jb < nb && b[jb] < a[ja]; jb++) {
            ;
        }
        if (jb >= nb || b[jb] > a[ja]) {
            return False;
        }
        for ( ; jb < nb && ja < na && b[jb] == a[ja]; jb++, ja++) {
            ;
        }
    }
    return (ja >= na);
}

//-------------------------------------------------------------
static int subtrSet(int na, int *a, int nb, int *b, int *c, int size)
{
    // set subtraction 'c' := 'a' - 'b'.
    // 'a' and 'b' are sorted lists (not always unique)
    // 'c' is the set (sorted and unique elements)

    int ja, jb, jc;

    jc = 0;
    ja = jb = 0;
    while (ja < na && jb < nb) {
        while (ja < na && a[ja] < b[jb]) {
            if (jc == 0 || c[jc-1] != a[ja]) {
```

```
                if (jc >= size) {
                    erEnd("subsutSet: too small size of array (GRCC_MAXPSLIST)");
                }
                c[jc++] = a[ja];
            }
            ja++;
        }
        while (jb < nb && b[jb] < a[ja]) {
            ;
        }
        if (ja < na && jb < nb && a[ja] == b[jb]) {
            ja++;
            jb++;
        }
    }
    for (; ja < na; ja++) {
        if (jc == 0 || c[jc-1] != a[ja]) {
            c[jc++] = a[ja];
        }
    }
    return jc;
}

// } ) ]
```

4.58 grcc.h

```
#pragma once

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>

//==============================================================
extern "C" {
#include "grccparam.h"
}

//==============================================================
// Name space of this library
// See 'grccparam.h' for #define
#ifdef GRCC_NAMESPACE
namespace Grcc {
#endif

//==============================================================
// For debugging
//
// #define MONITOR      // in graph elimination
// #define CHECK        // check consistency
#define DEBUGABORT
// #define DEBUG
// #define DEBUG0
// #define DEBUG1
// #define DEBUGM

//==============================================================
// Common types
//
typedef int Bool;
const int True  = 1;
const int False = 0;


//==============================================================
// Big integer (may be replaced by suitable one)

#define BigInt       long
#define ToBigInt(x)  ((BigInt) (x))

//==============================================================
// Macro functions
#define Real(x)      ((double) (x))
#define Abs(x)       (((x) >= 0)? (x): (-x))
#define Sign(x)      (((x) >= 0)? (1): (-1))
#define Max(x, y)    ((x) > (y) ? (x) : (y))
#define Min(x, y)    ((x) < (y) ? (x) : (y))
#define Second(t) ((double)(*(t)=(double)(clock()/((double)CLOCKS_PER_SEC))))
#define isATExternal(x)  ((x)==GRCC_AT_Initial||(x)==GRCC_AT_Final||(x)==GRCC_AT_External)

//==============================================================
// types and classes
```

```
typedef int  Edge2n[2];   // pair of nodes expressing an edge
class Assign;
class AStack;
class EEdge;
class EGraph;
class ENode;
class Interaction;
class MGraph;
class MNodeClass;
class MCOpi;
class MCBridge;
class MCBlock;
class MConn;
class Model;
class MOrbits;
class Options;
class Output;
class Particle;
class PNodeClass;
class Process;
class SGroup;
class SProcess;
class DCGraph;

//==============================================================
// type of output functions
typedef Bool OutEGB(EGraph *, void *);
typedef void OutEG(EGraph *, void *);
typedef void ErExit(const char *msg, void *);

//==============================================================
class Options {
  public:

    Model      *model;
    Process    *proc;
    SProcess   *sproc;
    Output     *out;         // for output
    OutEGB     *outmg;       // call back function of mgraph
    OutEGB     *endmg;       // call back function of end of mgraph
    OutEG      *outag;       // call back function of agraph
    void       *argmg;       // additional argument to outmg
    void       *argemg;      // additional argument to endmg
    void       *argag;       // additional argument to outag

    //switches for graph generation
    int values[GRCC_OPT_Size];      // array of options

    int         DUMMYPADDING;

    // measuring time
    double time0;
    double time1;

    //------------------
    Options(void);
    ~Options(void);
    void setDefaultValue(void);

    void setValue(int ind, int val);
    int  getValue(int ind);

    void print(void);

    void setOutputF(Bool outgrf, const char *fname);
    void setOutputP(Bool outgrp, const char *fname);
    void printLevel(int l);

    void printModel(void);
    void setOutMG(OutEGB *omg, void *pt);
    void setOutAG(OutEG  *oag, void *pt);
    void setEndMG(OutEGB *omg, void *pt);
    void setErExit(ErExit  *ere, void *pt);
    const OptDef *getDef(void);

    void begin(Model *mdl);
    void end(void);
    void beginProc(Process *prc);
    void endProc(void);
    void beginSubProc(SProcess *sprc);
    void endSubProc(void);
    void newMGraph(MGraph *mgr);
    void newAGraph(EGraph *egr);
    void outModel(void);
};

//==============================================================
```

```
class Output {
  public:

    Options    *opt;
    Model      *model;
    Process    *proc;
    SProcess   *sproc;
    char       *outgrf;
    FILE       *outgrfp;
    char       *outgrp;
    FILE       *outgrpp;
    int         procId;
    Bool        outproc;

    Output(Options *optn);
    ~Output(void);

    void setOutgrf(const char *fname);
    void setOutgrp(const char *fname);
    Bool outBeginF(Model *mdl, Bool pr);
    Bool outBeginP(Model *mdl, Bool pr);
    void outEndF(void);
    void outEndP(void);
    void outProcBeginF(Process *prc);
    void outProcBeginP(Process *prc);
    void outProcBegin0(int next, int couple, int loop);
    void outSProcBeginF(SProcess *sprc);
    void outSProcBeginP(SProcess *sprc);
    void outProcEndF(void);
    void outProcEndP(void);
    void outEGraphF(EGraph *egraph);
    void outEGraphP(EGraph *egraph);
    void outModelF(void);
    void outModelP(void);

};

//==============================================================
class Fraction {
  public:
    BigInt num, den;
    double ratio;

    Fraction() { num = 0;  den = 1; };
    Fraction(BigInt n, BigInt d);

    void   print(const char *msg);
    void   setValue(BigInt n, BigInt d);
    void   setValue(Fraction &f);
    void   add(BigInt n, BigInt d);
    void   add(Fraction f);
    void   sub(Fraction f);
    BigInt gcd(BigInt n0, BigInt n1);
    void   normal(void);
    Bool   isEq(Fraction f);
};

//**************************************************************
// Model
//==============================================================
//--------------------------------------------------------------
class Particle {
  public:
    Model *mdl;             // the model
    char  *name;            // the name of the particle
    char  *aname;           // the name of the anti-particle
    int    id;              // id of this particle
    int    ptype;           // the type of partice : GRCC_PT_Scalar, etc.
    int    neutral;         // True if particle==anti-particle
    int    pcode;           // Particle code
    int    acode;           // Anti-particle code
    int    cmindeg;         // min(leg of connectable interactions)
    int    cmaxdeg;         // max(leg of connectable interactions)

    int    extonly;         // can appear only as external particle

    //----------------------------
    Particle(Model *modl, int pid, PInput *pinp);
    ~Particle(void);

    char *particleName(int p);
    int   particleCode(int p);
    char *interactionName(int p);
    char *aparticle(void);
    void  prParticle(void);
    int   isNeutral(void);
    const char *typeName(void);
```

```
    const char *typeGName(void);

};

//---------------------------------------------------------------
class Interaction {
  public:
    Model *mdl;           // the model
    char  *name;          // the name of the interaction
    int   *plist;        // the list of particle codes
    int   *clist;        // the orders of coupling constants
    int   *slist;        // the sorted list of particle codes

    int    id;            // id of this interaction
    int    csum;          // the total orders of coupling constants
    int    nlegs;         // the number of legs
    int    loop;          // the number of loops
    int    icode;         // user defined interaction code

    int    nplist;        // the list of particle codes
    int    nclist;        // the orders of coupling constants
    int    nslist;        // the sorted list of particle codes

    //----------------------------
    Interaction(Model *modl, int iid, const char* nam, int icode, int *cpl, int nlgs, int *plst, int
csm, int lp);
    ~Interaction(void);

    void prInteraction(void);
};

//---------------------------------------------------------------
class Model {
  public:
    char          *name;        // the name of this model
    char         **cnlist;       // the list of coupling constant names
    int            ncouple;     // the number of coupling consants.

    int            nParticles;  // the number of particles
    Particle     **particles;   // the list of particles
    int            pdef;        // the def. of partcls is ended

    // list of particles and anti-p. without Undef.
    int            nallPart;
    int            allPart[GRCC_MAXMPARTICLES2];

    // list of particles and anti-p. without ones of extloop=True
    int            nintPart;
    int            intPart[GRCC_MAXMPARTICLES2];

    int            nInteracts;  // the number of interactions.
    Interaction **interacts;    // the list of interactions
    int            vdef;        // the definition of interaction is ended

    int            maxnlegs;    // maximum number of legs of an interaction
    int            maxcpl;      // maximum number of coupling const.
    int            maxloop;     // maximum number of loops inside an intr.

    // classification of interactions
    int           *cplgcp;      // coupling constants
    int           *cplglg;      // degree
    int           *cplgnvl;     // number of vertices
    int          **cplgvl;      // list of vertices
    int            ncplgcp;     // the number of classes

    int            defpart;     // GRCC_DEFBYNAME or GRCC_DEFBYCODE
    int            skipFLine;   // skip analysis fermion line

    int           DUMMYPADDING;

    // methods
    Model(MInput *minp);
    ~Model(void);

    void  prModel(void);
    void  addParticle(PInput *pinp);
    void  addParticleEnd(void);
    void  addInteraction(IInput *iinp);
    void  addInteractionEnd(void);
    int   findParticleName(const char *name);
    int   findParticleCode(int pcd);
    int   findInteractionName(const char *name);
    int   findInteractionCode(int icd);
    char *particleName(int p);
    int   particleCode(int p);
    int   normalParticle(int pt);
    int   antiParticle(int pt);
```

```
    int  *allParticles(int *len);
    int   findMClass(const int cpl, const int dgr);
    void  prParticleArray(int n, int *a, const char *msg);
};

//****************************************************************
// Process
//==============================================================
class PNodeClass {
  public:
    SProcess *sproc;
    int       *deg;       // The degree of each node
    int       *type;      // The type of the class : GRCC_AT_xxx
    int       *particle;  // The particle code of external node.
    int       *couple;    // The orders of coupling constans
    int       *cmindeg;   // min(deg of connectable vertex)
    int       *cmaxdeg;   // max(deg of connectable vertex)
    int       *count;     // The number of nodes in the class
    int       *cl2nd;     // cl2nd[class] <= nd < cl2nd[class+1] = set of nodes
    int       *nd2cl;     // nd2cl[node]  = class
    int       *cl2mcl;    // cl2mcl[class] = class defined in the model.
    int        nnodes;
    int        nclass;

    PNodeClass(SProcess *sprc, int nnods, int nclss, NCInput *cls);
    PNodeClass(SProcess *sprc, int nnods, int nclss, int *dgs, int *typ, int *ptcl, int *cpl, int
 *cnt, int *cmind, int *cmaxd);
    ~PNodeClass(void);

    void prPNodeClass(void);
    void prElem(int e);
};

//==============================================================
class SProcess {
  //  In sprocess the number of nodes and edges are fixed.
  //  A node is considered as
  //    (degree, total order of coupling constants),
  //  which pair of data defines a class of nodes.

  public:
    Model        *model;       // model
    Process      *proc;        // mother process
    Options      *opt;         // options
    PNodeClass   *pnclass;     // list of classes (cpl and deg is determined)
    AStack       *astack;

    MGraph       *mgraph;
    EGraph       *egraph;
    Assign       *agraph;

    int          *cl2nd;       // (code of nodes in the class[c])
                               //  = [cl2nd[c], ..., cl2nd[c+1]-1]
    int          *nd2cl;       // node nd is in the class pnclass[nd2cl[nd]]
    int           nclass;      // the number of classes

    int           ninitl;      // the number of initial particles
    int           nfinal;      // the number of final particles
    int           nvert;       // the number of vertices

    int           clist[GRCC_MAXNCPLG];  // coupling constans of the process
    int           id;          // sprocess id
    int           loop;        // the number of loops
    int           nNodes;      // the number of nodes
    int           nEdges;      // the number of edges
    int           nExtern;     // the number of external particles
    int           ncouple;     // the number of coupling constants
    int           tCouple;     // the total coupling constants

    int           DUMMYPADDING;

    BigInt        mgrcount;    // count generated mgraph
    BigInt        agrcount;    // count generated agraph
    BigInt        extperm;    // count generated agraph

    // the results of the graph generation
    BigInt   nMGraphs;         // the number of generated M-graphs
    BigInt   nMOPI;            // the number of 1PI M-graphs
    Fraction wMGraphs;         // the weighted sum of M-graphs
    Fraction wMOPI;            // the weighted sum of 1PI M-graphs

    BigInt   nAGraphs;         // the number of generated A-graphs
    BigInt   nAOPI;            // the number of 1PI A-graphs
    Fraction wAGraphs;         // the weighted sum of A-graphs
    Fraction wAOPI;            // the weighted sum of 1PI A-graphs

    // methods
```

```
    SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, NCInput *cls);
    SProcess(Model *mdl, Process *prc, Options *opts, int sid, int *clst, int ncls, int *cdeg, int
*ctyp, int *ptcl, int *cpl, int *cnum, int *cmind, int *cmaxd);

    ~SProcess(void);

    void prSProcess(void);
    BigInt generate(void);
    void assign(MGraph *mgr);

    int toMNodeClass(int *ctyp, int *cldeg, int *clnum, int *cmind, int *cmaxd);
    PNodeClass *match(MGraph *mgr);

    void endMGraph(MGraph *mgr);
    void endAGraph(EGraph *egr);

    void resultMGraph(BigInt nmgraphs, Fraction mwsum, BigInt nmopi, Fraction mwopi);
    void resultAGraph(BigInt nagraphs, Fraction awsum, BigInt naopi, Fraction awopi);

};

//===============================================================
class Process {
  public:
    Model       *model;             // model used for this process
    Options     *opt;               // options
    BigInt       mgrcount;          // count generated mgraph
    BigInt       agrcount;          // count generated agraph
    int         *initlPart;         // list of ids of initial particles
    int         *finalPart;         // list of ids of final particles

    int          id;                // process id
    int          ninitl;            // the number of initial particles
    int          nfinal;            // the number of final particles
    int          ctotal;            // the total number of coupling constants
    int          nExtern;           // the number of external particles
    int          loop;              // the number of external particles
    int          maxnlegs;          // the maximum possible degree of node
    int          clist[GRCC_MAXNCPLG];  // coupling constans of the process


    // table of sprocesses
    int        nSubproc;            // the number of sprocesses
    SProcess  *sptbl[GRCC_MAXSUBPROCS]; // the table of sprocesses
    SProcess  *sproc;               // the current sprocess

    // stack for the assignment;
    AStack      *astack;

    // the number of graphs
    BigInt       ngraphs;
    BigInt       nopi;
    BigInt       wgraphs;
    BigInt       wopi;

    // the results of the graph generation
    BigInt       nMGraphs;              // the number of generated M-graphs
    BigInt       nMOPI;                 // the number of 1PI M-graphs
    Fraction     wMGraphs;              // the weighted sum of M-graphs
    Fraction     wMOPI;                 // the weighted sum of 1PI M-graphs

    BigInt       nAGraphs;              // the number of generated A-graphs
    BigInt       nAOPI;                 // the number of 1PI A-graphs
    Fraction     wAGraphs;              // the weighted sum of A-graphs
    Fraction     wAOPI;                 // the weighted sum of 1PI A-graphs

    double       sec;

    Process(int pid, Model *model, Options *opt, int nin, int *initlPart, int nfin, int *finalPart,
int *coupling);
    Process(int pid, Model *model, Options *opt, FGInput *fgi);
    ~Process(void);

    void prProcess(void);
    void mkSProcess(void);

};

//**************************************************************
// classes for EGraph
//==============================================================
// Constants

#define GRCC_ED_Undef   0
#define GRCC_ED_Deleted 1
#define GRCC_ED_Extern  2
#define GRCC_ED_Back    3
```

```
#define GRCC_ED_Bridge  4
#define GRCC_ED_Inloop  5
#define GRCC_ED_Size    6
#define GRCC_ED_NAMES   {"Undef", "Deleted", "Extern", "Back_Edge", "Bridge", "In_Loop"}

#define GRCC_ND_Undef   0
#define GRCC_ND_Deleted 1
#define GRCC_ND_Initial 2
#define GRCC_ND_Final   3
#define GRCC_ND_CPoint  4
#define GRCC_ND_VBlock  5
#define GRCC_ND_Inblock 6
#define GRCC_ND_Size    7
#define GRCC_ND_NAMES   {"Undef", "Deleted", "Init", "Final", "CPoint", "VBlock", "In_Block"}

//---------------------------------------------------------------
class ENode {
  public:
    EGraph *egraph;
    int    *edges;           // list of edges
                             // the value is \pm [(edge index)+1]

    int     id;              // id of the enode
    int     maxdeg;          // maximum degree
    int     deg;             // degree
    int     extloop;         // loop inside this node
    int     ndtype;          // type of the node
    int     intrct;          // assigned interaction/particle (ext.)

    // used in EGraph::biconn()
    int    *klow;
    int     visited;

    int    DUMMYPADDING;

    //--------------------------------
    // functions
    ENode(void);
    ENode(EGraph *egrph, int loops, int sdeg);
    ~ENode(void);
    void initAss(EGraph *egrph, int nid, int sdg);
    void setId(EGraph *egrph, const int nid);
    void copy(ENode *en);
    void setExtern(int typ, int pt);
    void setType(int typ);
    void print(void);

};

//---------------------------------------------------------------
class EEdge {
  // momentum is printed like: ("%s%d", (enode.ext)?"Q":"p", enode.momn)
  public:
    EGraph *egraph;        // egraph

    int  id;               // id
    int  ext;
    int  ptcl;             // assigned particle (agraph)
    int  deleted;          // deleted edge, if true
    int  nodes[2];         // nodes of bothsides
    int  nlegs[2];         // nodes of both side (agraph)

    // for biconn
    int *emom;             // external momenta
    int *lmom;             // loop momenta
    int *extMom;           // set of external momenta.

    Bool cut;
    int  visited;
    int  conid;            // connected component
    int  edtype;           // type
    int  opicomp;          // id of 1PI component
    int  dir;              // direction of momentum


    //--------------------------------
    // functions
    EEdge(void);
    EEdge(EGraph *egrph, int nedges, int nloops);
    ~EEdge(void);
    void copy(EEdge *ee);
    void setId(EGraph *egrph, const int eid);
    void print(void);

    void setType(int typ);
    void setLMom(int k, int dir);
    void setEMom(int nedges, int *extn, int dir);
```

```
};

//--------------------------------------------------------------
// type of fermion lines

typedef enum {FL_Open, FL_Closed} FLType;

//--------------------------------------------------------------
class EFLine {
  public:
    int     elist[GRCC_MAXNODES];     // list of  (\pm [(edge index)+1])
    FLType ftype;
    int     fkind;
    int     nlist;

    EFLine(void);
    void print(const char *msg);
};

//--------------------------------------------------------------
class EGraph {
  public:
    Options     *opt;          // table of options
    Model       *model;
    Process     *proc;
    SProcess    *sproc;
    MGraph      *mgraph;
    MConn       *econn;

    ENode **nodes;
    EEdge **edges;          // edges[nEdges+1]: index starts from 1

    BigInt mId;             // id of mgraph
    BigInt aId;             // id of agraph in the same mgraph
    BigInt sId;             // sequential no. of agraph
    BigInt gSubId;          // ???
    Bool   assigned;        // mgraph (False) or agraph (True)

    int     fsign;
    BigInt nsym, esym;      // symmetry factor with symm. ext.
    BigInt nsym1;           // symmetry factor without symm. ext.
    BigInt extperm;         // the order of group of symm. ext.
    BigInt multp;           // multiplicity of graph in symm. ext.

    int     pId;

    int    sNodes;           // memory size
    int    sEdges;           // memory size
    int    sMaxdeg;          // memory size
    int    sLoops;           // memory size

    int     nNodes;
    int     nEdges;
    int     nExtern;
    int     maxdeg;           // maximum value of degree of nodes
    int     nLoops;
    int     totalc;           // total order of coupling constants

    // biconnect
    int     nopicomp;
    int     opi2plp;
    int     nopi2p;
    int     nadj2ptv;          // the no. of edges connecting 2point vertices

    int     DUMMYPADDING;

    int    *bidef;
    int    *bilow;
    int    *extMom;
    int     bconn;
    int     bicount;
    int     loopm;
    int     opiCount;

    // Fermion lines
    EFLine *flines[GRCC_MAXFLINES];
    int      nflines;

    int     DUMMYPADDING1;

    //--------------------------
    // functions
    EGraph(int nnodes, int nedges, int mxdeg);
    ~EGraph(void);

    void copy(EGraph *eg);
    void print(void);
```

```
    void fromDGraph(DGraph *dg);
    void fromMGraph(MGraph *mgraph);
    Bool isOptE(void);

    ENode *setExtern(int n0, int pt, int ndtp);
    Bool   isExternal(int nd)  { return (nodes[nd]->extloop < 0); };
    Bool   isFermion(int nd);

    void setExtLoop(int nd, int val);
    void endSetExtLoop(void);

    int connComp(void);
    int connVisit(int nd, int ncc);

    void biconnE(void);
    void biinitE(void);
    void bisearchE(int nd, int *extlst, int *intlst, int *opiext, int *opiloop);

    int  findRoot(void);
    int  dirEdge(int n, int e);
    void extMomConsv(void);
    int  cmpMom(int *lm0, int *em0, int *lm1, int *em1);
    int  groupLMom(int *grp, int *ed2gr);

    void chkMomConsv(void);

    void prFLines(void);
    void getFLines(void);
    int  fltrace(int fk, int nd0, int *fl);
    void addFLine(const FLType ft, int fk, int nfl, int *fl);

    int  legParticle(int ed, int lg);

};

//**************************************************************
//  Symmetry group of graphs
//===============================================================
class SGroup {
  public:
    BigInt  size;              // # of allocated elements
    BigInt  nelem;             // # of saved elements
    int    **elem;             // saved elements
    int      nnodes;           // # of nodes.
    int      neclass;          // # of classes
    int      eclass[GRCC_MAXNODES];  // table of classes
    int      cgen;             // counter for the generation
    int      csav;             // counter for the saved elements
    int      permg[GRCC_MAXNODES];    // resulting permutation
    int      perms[GRCC_MAXNODES];    // curr. elem of saved ones.

    int      pgr[GRCC_MAXNODES];  // work
    int      pgq[GRCC_MAXNODES];  // work
    int      psr[GRCC_MAXNODES];  // work
    int      psq[GRCC_MAXNODES];  // work

    //------------------------------
    // functions

    SGroup(void);
    ~SGroup(void);

    void  print(void);
    void  newGroup(int nelm, int nclss, int *clss);
    void  clearGroup(void);
    void  delGroup(void);
    int  *genNext(void);
    void  addGroup(int *p);
    BigInt  nElem(void);
    int  *nextElem(void);
};

//**************************************************************
// MGraph
//==============================================================
// class of nodes for MGraph

class MNode {
  public:
    int id;         // node id
    int deg;        // degree(node) = the number of legs
    int clss;       // initial class number in which the node belongs
    int extloop;
    int cmindeg;
    int cmaxdeg;

    int freelg;     // the number of free legs
```

```
    int visited;

    MNode(int id, int clss, NCInput *mgi);
    MNode(int id, int deg, int extloop, int clss, int cmind, int cmaxd);
};

//===============================================================
//  class of scalar graph expressed by matrix form
//
//  Input  : the classified set of nodes.
//  Output : control passed to 'EGraph(self)'

class MGraph {

  public:

    // initial conditions
    Options  *opt;          // options

    // symmetry group
    SGroup   *group;        // symmetry group
    MOrbits  *orbits;       // Orbits of nodes with respect to symmetry group

    MNode **nodes;          // table of MNode object
    BigInt mId;             // process/sprocess ID
    int    *clist;          // list of initial classes
    int     pId;            // process/sprocess ID
    int     nNodes;         // the number of nodes
    int     nEdges;         // the number of edges
    int     nLoops;         // the number of loops
    int     nExtern;        // the number of external nodes
    int     nClasses;       // the number of initial classes
    int     mindeg;         // minimum value of degree of nodes
    int     maxdeg;         // maximum value of degree of nodes

    // the current graph
    int **adjMat;           // adjacency matrix
    MNodeClass *curcl;      // the current 'MNodeClass' object
    EGraph *egraph;
    BigInt nsym;            // symmetry factor from nodes
    BigInt esym;            // symmetry factor from edges

    // generated set of graphs
    BigInt cDiag;           // the total number of generated graphs
    BigInt c1PI;            // the total number of 1PI graphs
    BigInt cNoTadpole;      // the total number of graphs without tadpoles
    BigInt cNoTadBlock;     // the total number of graphs without tad-blocks
    BigInt c1PINoTadBlock;  // the total number of 1PI graphs without tad-blocks
    Fraction wscon;          // weighted sum of graphs
    Fraction wsopi;          // weighted sum of 1PI graphs

    // measures of efficiency
    BigInt ngen;                // generated graph before check
    BigInt ngconn;              // generated connected graph before check

    BigInt nCallRefine;
    BigInt discardRefine;
    BigInt discardDisc;
    BigInt discardIso;

    // for options
    Bool opi;
    Bool opiloop;
    Bool extself;
    Bool selfloop;
    Bool tadpole;
    Bool tadblock;
    Bool block;

    int     DUMMYPADDING1;

    // table of n edge-connected components
    MConn *mconn;

    // work space for isomorphism
    int **modmat;            // permutated adjacency matrix

    // work space for biconnected component
    int *bidef;
    int *bilow;
    int  bicount;

    int     DUMMYPADDING2;

    //----------
    // functions
    MGraph(int pid, int ncl, NCInput *mgi, Options *opt);
```

```
    MGraph(int pid, int ncl, int *cldeg, int *clnum, int *clexl, int *cmind, int *cmaxd, Options *opt);
    ~MGraph(void);

    void   init(void);
    BigInt   generate(void);
    Bool   isExternal(int nd)  { return (nodes[nd]->extloop < 0); };
    void   printAdjMat(MNodeClass *cl);
    void   print(void);

    Bool   isConnected(void);
    Bool   visit(int nd);
    Bool   isIsomorphic(MNodeClass *cl);
    void   permMat(int size, int *perm, int **mat0, int **mat1);
    int    compMat(int size, int **mat0, int **mat1);
    MNodeClass *refineClass(MNodeClass *cl);
    void   bisearchME(int nd, int pd, int ned, MCOpi *mopi, MCBlock *mblk, int *next, int *nart);
    void   biconnME(void);
    Bool   isOptM(void);

    void   connectClass(MNodeClass *cl);
    void   connectNode(int sc, int ss, MNodeClass *cl);
    void   connectLeg(int sc, int sn, int tc, int ts, MNodeClass *cl);
    void   newGraph(MNodeClass *cl);

};

//==============================================================
//  class of node-classes for MGraph
//--------------------------------------------------------------
class MNodeClass {
  public:
    int   clmat[GRCC_MAXNODES][GRCC_MAXNODES];   // matrix used for classification
    int   clist[GRCC_MAXNODES];     // the number of nodes in each class
    int   ndcl[GRCC_MAXNODES];      // node --> class
    int   flist[GRCC_MAXNODES+1];   // the first node in each class
    int   clord[GRCC_MAXNODES];     // ordering of classes
    int   cmindeg[GRCC_MAXNODES];     // min(deg of connectable node)
    int   cmaxdeg[GRCC_MAXNODES];     // max(deg of connectable node)
    int   nNodes;          // the number of nodes
    int   nClasses;        // the number of classes
    int   maxdeg;          // maximal value of degree(node)

    int   flg0;
    int   flg1;
    int   flg2;

    MNodeClass(int nnodes, int nclasses);
    ~MNodeClass(void);

    void  init(int *cl, int mxdeg, int **adjmat);
    void  copy(MNodeClass* mnc);
    int   clCmp(int nd0, int nd1, int cn);
    void  printMat(void);

    void  mkFlist(void);
    void  mkNdCl(void);
    void  mkClMat(int **adjmat);
    void  incMat(int nd, int td, int val);
    int   cmpMNCArray(int *a0, int *a1, int ma);
    void  reorder(MGraph *mg);
};


//==============================================================
//  class of 1PI component
//--------------------------------------------------------------
class MCOpi {
  public:
    int *nodes;     // array of nodes in
    int  nnodes;    // # nodes in the 1PI component
    int  nlegs;     // # leg of the 1PI component
    int  next;      // # external particles of the 1PI comp.
    int  nedges;    // # edges in the 1PI comp.
    int  loop;      // # loops in the 1PI comp.
    int  ctloop;    // # loops in the counter terms in the OP comp.

    MCOpi(void);
    ~MCOpi(void);

    void init(void);
};

//==============================================================
//  class of bridge
//--------------------------------------------------------------
class MCBridge {
```

```
  public:
    Edge2n nodes;  // nodes at the both size of the bridge
    int    next;   // # momenta of ext. particles flowing on the bridge

    MCBridge(void);
    ~MCBridge(void);
};

//==============================================================
//  class of block
//--------------------------------------------------------------
class MCBlock {
  public:
    Edge2n *edges;   // array of edges in the block
    int     nmedges; // # edges in the block
    int     nartps;  // # articulation points of the block
    int     loop;    // # loop in the block

    int    DUMMYPADDING;

    MCBlock(void);
    ~MCBlock(void);
    void init(void);
};

//==============================================================
//  class of table of MConn
//--------------------------------------------------------------
class MConn {
  public:
    // 2-edge connected components
    MCOpi     *opics;       // table of n-edge-connected components
    MCBridge  *bridges;     // table of bridges
    MCBlock   *blocks;      // table of blocks
    int       *articuls;    // buffer for nodes for articulation points

    int       *opisp;       // buffer for nodes in 1PI components
    int       *opistk;      // stack of nodes for 1PI components.
    Edge2n    *blksp;       // buffer for edges in blocks
    Edge2n    *blkstk;      // stack of edges for blocks
    int        snodes;      // # nodes
    int        sedges;      // # edges

    // opi components (edge-connected)
    int        nopic;       // # 1PI components (n1PIComps)
    int        nlpopic;     // # looped 1PI components (n1PIComps)
    int        nctopic;     // # 1PI components of one counter term.
    int        nbridges;    // # bridges
    int        ne0bridges;  // # bridges whose next=0
    int        ne1bridges;  // # bridges whose next=1
    int        nselfloops;

    // blocks (node-connected)
    int        nblocks;     // # blocks
    int        na1blocks;   // # bridges whose next=0
    int        narticuls;   // # articulation points
    int        neblocks;    // # effective looped blocks

    // indices to work spaces
    int        nopisp;      // # used in opisp
    int        opistkptr;   // # stack pointer of opistk
    int        nblksp;      // # used in blksp
    int        blkstkptr;   // # stack pointer of tlkstk

    int    DUMMYPADDING;

    MConn(int nnod, int nedg);
    ~MConn(void);

    void init(void);
    void pushNode(int nd);
    void pushEdge(int n0, int n1);
    void addOPIc(MCOpi *mopi, int stp);
    void addBridge(int n0, int n1, int nex, int nextot);
    void addArtic(int nd, int mul);
    void addBlock(MCBlock *eblk, int stp);
    void addBlockSelf(int nd, int mul);
    void print(void);
};

//==============================================================
//  class of orbits of nodes
//--------------------------------------------------------------
class MOrbits {
  // Usage
  // 1. node ==> orbit
  //     (class) = nd2or[(node)]
```

```
  //
  // 2. class ==> nodes
  //    for (c = 0; c < nClass; c++) {
  //        for (j = flist[c]; j < flist[c+1]; j++) {
  //            (node) = or2nd[c];
  //        }
  //    }
  public:
    int nOrbits;
    int nNodes;
    int nd2or[GRCC_MAXNODES];
    int or2nd[GRCC_MAXNODES];
    int flist[GRCC_MAXNODES+1];

    MOrbits(void);
    ~MOrbits(void);

    void print(void);

    // construction of orbits
    void initPerm(int nnodes);
    void fromPerm(int *perm);
    void toOrbits(void);
};

//***************************************************************
// Particle assignment
//===============================================================
typedef int CheckPt[2];

typedef enum {
    AS_UnAssLegs, AS_Assigned, AS_Assigned0, AS_AssExt, AS_Impossible
} NCandSt;

//===============================================================
class NCand {
  // List of candidats for the assignment of interactions to a vertex

  public:
    int      deg;                    // degree of the node
    NCandSt  st;                     // status
    int      nilist;                 // length of the list
    int      ilist[GRCC_MAXMINTERACT];     // list of candidates

    //========
    NCand(const NCandSt sta, const int dega, const int nilst, int *ilst);
    ~NCand(void);

    // print
    void prNCand(const char* msg);
};

//===============================================================
class ECand {
  // List of candidats for the assignment of particles to an edge

  public:
    Bool  det;                      // determined or not
    int   nplist;                   // size of the list of candidates
    int   plist[GRCC_MAXMPARTICLES2];    // list of candidates

    ECand(int dt, int nplist, int *plst);
    ~ECand(void);

    // print
    void   prECand(const char *msg);
};

//===============================================================
class ANode {
  // Connection information of a node to others
  //
  //  (this node) -- (adjacent edge) -- (next node)
  //    (n0, j)             e                n1
  //
  // e  = (aedges[j], aelegs[j])
  // n1 = anodes[j]

  public:
    int     deg;               // degree of the node
    int     nlegs;             // the number of legs already assigned
    int    *anodes;            // anodes[j] = (next node of leg j)
    int    *aedges;            // aedges[j] = (next edge of leg j)
    int    *aelegs;            // aelegs[j] = (leg of the next edge)
    NCand *cand;               // candidate list

    ANode(int dg);
```

```
    ~ANode(void);

    int   newleg(void);         // get a new leg
};

//===============================================================
class AEdge {
// Connection information of an edge
// (self) ==> nodes [(nodes[0], nlegs[0]), (nodes[1], nlegs[1])]
// particle 'ptcl' flows from leg=0 to 1.

  public:

    ECand *cand;            // candidate
    int    nodes[2];        // nodes of both sides of the edge
    int    nlegs[2];        // nodes of both sides of the edge
    int    ptcl;            // particle defined in the model

    int    DUMMYPADDING;

    AEdge(int n0, int l0, int n1, int l1);
    ~AEdge(void);
};

//===============================================================
class Assign {
// Class for particle/interaction assignment.

  public:

    EGraph     *egraph;       // EGraph object
    MGraph     *mgraph;       // MGraph object
    SProcess   *sproc;        // Sub-process object
    Process    *proc;         // Process object
    Model      *model;        // Model object
    Options    *opt;          // Option object
    AStack     *astack;       // Stack for saving candidates
    PNodeClass *pnclass;      // class of nodes
    MOrbits    *orbits;       // orbits of nodes by symmetry group

    int         nNodes;       // the number of nodes
    int         nEdges;       // the number of edges
    int         nExtern;      // the number of external particles.
    int         nETotal;      // the total number of edges

    BigInt      nAGraphs;     // the number of assigned graphs
    Fraction    wAGraphs;     // the weighted sum of graphs
    BigInt      nAOPI;        // the number of assigned 1PI
    Fraction    wAOPI;        // the weighted sum of 1PI

    ANode     **nodes;        // table of nodes
    AEdge     **edges;        // table of edges

    CheckPt     checkpoint0;  // save stack pointers

    int         cplleft[GRCC_MAXNCPLG];  // coupling constants left

    //===========================================
    Assign(SProcess *sprc, MGraph *mgr, PNodeClass *pnc);
    ~Assign(void);

    //===========================================
    // print lists of candidates
    void    prCand(const char *msg);

    // check
    void checkAG(const char *msg);

    //===========================================
    // control of assignment

    // entry point of assignment
    Bool    assignAllVertices(void);

    // select a source node for assignment
    Bool    selectVertex(void);
    Bool    selectVertexSimp(int lastv);

    // select a source leg of the node for assignment
    Bool    selectLeg(int v, int lastlg);

    // assign a vertex a interaction
    Bool    assignVertex(int v);

    // assignment procedure is finished.  Needs check.
    Bool    allAssigned(void);

```

```
    //==========================================
    // Input and output

    // Input from mgraph
    Bool    fromMGraph(void);

    // add an edge
    void    addEdge(int n0, int n1, int nplist, int *plist);

    // connect nodes and edges.
    void    connect(int n0, int l0, int eg, int el, int n1, int l1);

    // Output to egraph
    Bool    fillEGraph(int aid, BigInt nsym, BigInt esym, BigInt nsym1);

    // reorder legs in accordance with the interaction definition
    int    *reordLeg(int n, int *reord, int *plist, int *used);

    //==========================================
    // direction of particle on an edge and at (node, leg)

    // convert particle code
    //   at node-leg ('n', 'ln') <==> edge at ('n', 'ln')
    int     getLegParticle(int n,  int ln);

    // convert particle 'pt' on the edge to ('n', 'ln')
    int     legEdgeParticle(int n, int ln, int pt);

    // convert candidate list at ('v', 'lg') into in-coming direction
    int     legPart(int v, int lg, int nplst, int *plst, int *rlist, const int size);

    // convert candidate list at ('v', 'lg') into in-coming direction
    int     candPart(int v, int ln, int *plist, const int size);

    //==========================================
    // assignment

    // find a vertex to be assigned
    int     selUnAssVertex(void);
    int     selUnAssVertexSimp(int lastv);

    // find a leg of the vertex to be assigned
    int     selUnAssLeg(int v, int lastlg);

    // assign the interaction 'ia' to the vertex 'v'.
    NCandSt assignIVertex(int v, int ia);

    // assign particle 'pt' the the leg 'ln' of the node 'n'.
    Bool    assignPLeg(int n, int ln, int pt);
    Bool    isOrdPLeg(int n, int ln, int pt);
    Bool    detEdge(int e);

    //==========================================
    // canndidates

    // construct lists of assigned / unassigned particles at a node
    Bool    candPartClassify(int v, int *npdass, int *pdass, int *npuass, int *puass, const int size);

    // update candidate list for a vertex 'v'.
    Bool    updateCandNode(int v);

    //==========================================
    // filter

    Bool    checkOrderCpl(void);
    Bool    isOrdLegs(void);
    Bool    isIsomorphic(MNodeClass *cl, BigInt *nsym, BigInt *esym, BigInt *nsym1);
    int     cmpPermGraph(int *p, MNodeClass *cl);
    int     cmpNodes(int nd0, int nd1, MNodeClass *cn);
    BigInt  edgeSym(void);

    void    saveCouple(int *sav);
    void    restoreCouple(int *sav);
    Bool    subCouple(int *cpl);

#ifdef CHECK
    //==========================================
    // check
    Bool    checkCand(const char *msg);

    void    checkNode(int n, const char *msg);
#endif
};

//**************************************************************
// Stack of candidate lists
//===============================================================
```

```
class NStack {
// Stack for backtracking method for a node.

  public:
    int      noden;
    int      deg;
    NCandSt  st;
    int      nilist;
    int      ilist[GRCC_MAXMINTERACT];

    NStack() { };
    ~NStack() { };

    void print(const char *msg);

};

//==============================================================
class EStack {
// Stack for backtracking method for an edge.

  public:
    int     edgen;
    int     det;
    int     nplist;
    int     plist[GRCC_MAXMPARTICLES2];

    EStack() { };
    ~EStack() { };

    void print(const char *msg);
};


//==============================================================
class AStack {
// Stack for backtracking method for an edge.

  public:
    Assign     *agraph;

    NStack    **nStack;       // stack for nodes
    int         nStackP;      // stack pointer for nodes
    int         nSize;        // stack size

    EStack    **eStack;       // stack for edges
    int         eStackP;      // stack pointer for edges
    int         eSize;        // stack size

    AStack(int nSize, int eSize);
    ~AStack(void);

    void setAGraph(Assign *ag);

    void checkPoint(CheckPt sav);
    void restore(CheckPt sav);
    void restoreMsg(CheckPt sav, const char *msg);
    void prStack(void);

    void pushNode(int n);
    void pushEdge(int e);

  private:
    void restoreNode(int spr);
    void restoreEdge(int spr);
};

//=============================================================
// end of namespace Grcc
#ifdef GRCC_NAMESPACE
}
#endif
```

4.59 grccparam.h

```
#ifdef GRCC_PARAM_H
#else
#define GRCC_PARAM_H

/*=============================================================
 * Parameters
 */
/*
```

```
#define GRCC_NAMESPACE
*/

#define GRCC_MAXNCPLG          4
#define GRCC_MAXLEGS          10
#define GRCC_MAXMPARTICLES    50
#define GRCC_MAXMINTERACT    200
#define GRCC_MAXSUBPROCS     500
#define GRCC_MAXNODES         20
#define GRCC_MAXEDGES        100
#define GRCC_MAXNSTACK        50
#define GRCC_MAXESTACK        50
#define GRCC_MAXGROUP       1000
#define GRCC_MAXPSLIST       500

#define GRCC_MAXMPARTICLES2   (2*GRCC_MAXMPARTICLES-1)
#define GRCC_MAXFLINES        GRCC_MAXEDGES

#define GRCC_FRACERROR        (1.0e-10)

/* error message */
/*
#define GRCC_Stderr           stderr
*/
#define GRCC_Stderr           stdout
/*
#define GRCC_ABORT()          exit(1)
*/
#define GRCC_ABORT()          abort()

/*---------------------------------------------------------------
 * Constants
 */

/* model definition */
#define GRCC_DEFBYNAME   1    /* define interactions by particle name */
#define GRCC_DEFBYCODE   2    /* define interactions by particle code */

/* types of particles */
#define GRCC_PT_Undef      0
#define GRCC_PT_Scalar     1
#define GRCC_PT_Dirac      2
#define GRCC_PT_Majorana   3
#define GRCC_PT_Vector     4
#define GRCC_PT_Ghost      5
#define GRCC_PT_Size       6

#define GRCC_PTS_Undef      "undef"
#define GRCC_PTS_Scalar     "scalar"
#define GRCC_PTS_Dirac      "dirac"
#define GRCC_PTS_Majorana   "majorana"
#define GRCC_PTS_Vector     "vector"
#define GRCC_PTS_Ghost      "ghost"

#define GRCC_PTO_General   0
#define GRCC_PTO_ExtOnly   1

#define GRCC_PT_Table  { GRCC_PTS_Undef, GRCC_PTS_Scalar, GRCC_PTS_Dirac, GRCC_PTS_Majorana,
      GRCC_PTS_Vector, GRCC_PTS_Ghost}

#define GRCC_PT_GTable { "Undef", "Scalar", "Fermion", "Majorana", "Vector", "Ghost"}

/* for mgraph generation */
#define GRCC_AT_Vertex     0
#define GRCC_AT_External  -1
#define GRCC_AT_Initial   -2
#define GRCC_AT_Final     -3

#define GRCC_AT_NdStr(x)    ((x)>=GRCC_AT_Vertex   ?"Vertex":       \
                            ((x)==GRCC_AT_External ?"External":     \
                            ((x)==GRCC_AT_Final    ?"Fianl":        \
                            ((x)==GRCC_AT_Initial  ?"Initial":"Undef"))))

/* graph generation */
#define GRCC_MGraph   0       /* mgraph (topology) generation */
#define GRCC_AGraph   1       /* agraph generation */

/* options for graph greneration */
#define GRCC_OPT_Step            0
#define GRCC_OPT_Outgrf          1
#define GRCC_OPT_Outgrp          2
#define GRCC_OPT_1PI             3
#define GRCC_OPT_NoSelfLoop      4
#define GRCC_OPT_NoTadpole       5
#define GRCC_OPT_No1PtBlock      6
#define GRCC_OPT_No2PtL1PI       7
#define GRCC_OPT_NoExtSelf       8
```

```
#define GRCC_OPT_NoAdj2PtV       9
#define GRCC_OPT_Block          10
#define GRCC_OPT_SymmInitial    11
#define GRCC_OPT_SymmFinal      12
#define GRCC_OPT_Size           13

typedef struct {
    const char *name;
    const char *mean;
    int         defaultv;
    int         DUMMYPADDING;
} OptDef;

/*---------------------------------------------------------------
 * Conversion of
 *     eind : index (ed) of EGraph.edges[ed]
 *     sind : value (v)  of EGraph.nodes[nd]->edges[j])
 */
#define   I2Vedge(eind, sign)     (((sign<=0)?(-1):(1))*((eind)+1))
#define   V2Iedge(sind)           (abs(sind)-1)
#define   V2Ileg(sind)            ((sind)<0 ? 0 : 1)
/*---------------------------------------------------------------
 * data types for model definition
 */
typedef struct {
    const char *name;                 /* name of the model */
          int   defpart;              /* particle is defined by name or code */
                                      /* GRCC_DEFBYNAME or GRCC_DEFBYCODE */
          int   ncouple;              /* No. of coupling constants */
    const char *cnamlist[GRCC_MAXNCPLG]; /* Names of coupling constants */
} MInput;

typedef struct {
    const char *name;                 /* name of particle */
    const char *aname;                /* name of anti-particle */
          int   pcode;                /* code of particle */
          int   acode;                /* code of anti-particle */
    const char *ptypen;               /* type : 'GRCC_PTS_Scalar' etc */
          int   ptypec;               /* type : 'GRCC_PT_Scalar' etc */
          int   extonly;              /* only as external particle */
} PInput;

typedef struct {
    const char *name;                   /* name of interaction */
          int   icode;                  /* code of interaction */
          int   nplistn;                /* the number of legs */
    const char *plistn[GRCC_MAXLEGS];      /* list of particle names */
          int   plistc[GRCC_MAXLEGS];      /* list of particle codes */
          int   cvallist[GRCC_MAXNCPLG]; /* coupling constants */
} IInput;

/*---------------------------------------------------------------
 * data type for node class input
 */
typedef struct {
    /* used as input to MGraph() and SProcess() */
    int cldeg;
    int clnum;
    int cltyp;
    int cmind;
    int cmaxd;

    /* used as input to SProcess() */
    int ptcl;
    int cple;
} NCInput;

/*---------------------------------------------------------------
 * data types for process definition
 */
typedef struct {
    long          ninitl;               /* the number of initial particles */
    const char   *initln[GRCC_MAXNODES];     /* list of initial particles (name) */
    int           initlc[GRCC_MAXNODES];     /* list of final particles (code) */
    long          nfinal;               /* the number of final particles */
    const char   *finaln[GRCC_MAXNODES];     /* list of final particles (name) */
    int           finalc[GRCC_MAXNODES];     /* list of final particles (code) */
    int           coupl[GRCC_MAXNCPLG];      /* list of orders of c. consts */
} FGInput;

/* class of node : input for SProcess */
typedef struct {
    int         cdeg;                   /* degree of each node */
    int         ctyp;                   /* typde : GRCC_AT_xxx */
    int         cnum;                   /* the number of nodes in the class */
    int         cple;                   /* total order of c.c. of each node */
                                        /* = 0 for external particle */
```

```
    const char *pname;                /* particle name defined in the model */
                                      /* = NULL for vertex */
    long       pcode;                 /* particle code defined in the model */
                                      /* = 0 for vertex */
                                      /* long = int + padding */
} PGInput;

/*---------------------------------------------------------------
 * minimum input data to EGraph
 */
typedef struct {
    int    extloop;     /* external/loop */
    int    intrct;      /* particl/interaction */
} DNode;

typedef int  DEdge[2];  /* {node0, node1} */

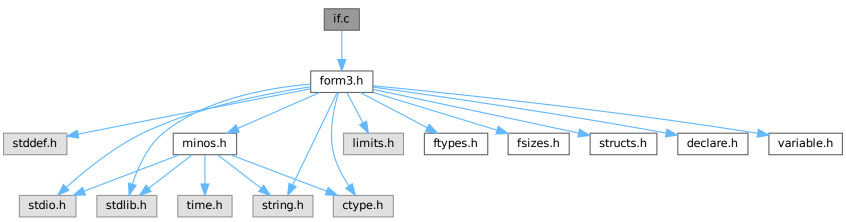
typedef struct {
    int    nnodes;
    int    nedges;
    DNode *nodes;
    DEdge *edges;
} DGraph;

/*---------------------------------------------------------------
 */
#endif
```

4.60 if.c File Reference

`#include "form3.h"`

Include dependency graph for if.c:



Functions

- WORD GetIfDollarNum (WORD ∗ifp, WORD ∗ifstop)
- int FindVar (WORD ∗v, WORD ∗term)
- WORD DoIfStatement (PHEAD WORD ∗ifcode, WORD ∗term)
- WORD HowMany (PHEAD WORD ∗ifcode, WORD ∗term)
- void DoubleIfBuffers (void)
- int DoSwitch (PHEAD WORD ∗term, WORD ∗lhs)
- int DoEndSwitch (PHEAD WORD ∗term, WORD ∗lhs)
- SWITCHTABLE ∗ FindCase (WORD nswitch, WORD ncase)
- int DoubleSwitchBuffers (void)
- void SwitchSplitMergeRec (SWITCHTABLE ∗array, WORD num, SWITCHTABLE ∗auxarray)
- void SwitchSplitMerge (SWITCHTABLE ∗array, WORD num)

4.60.1 Detailed Description

Routines for the dealing with if statements.

Definition in file if.c.

4.60.2 Function Documentation

4.60.2.1 GetIfDollarNum()

```
WORD GetIfDollarNum (
            WORD * ifp,
            WORD * ifstop )
```

Definition at line 106 of file if.c.

4.60.2.2 FindVar()

```
int FindVar (
            WORD * v,
            WORD * term )
```

Definition at line 173 of file if.c.

4.60.2.3 DoIfStatement()

```
WORD DoIfStatement (
            PHEAD WORD * ifcode,
            WORD * term )
```

Definition at line 280 of file if.c.

4.60.2.4 HowMany()

```
WORD HowMany (
            PHEAD WORD * ifcode,
            WORD * term )
```

Definition at line 840 of file if.c.

4.60.2.5 DoubleIfBuffers()

```
void DoubleIfBuffers (
            void )
```

Definition at line 1033 of file if.c.

4.60.2.6 DoSwitch()

```
int DoSwitch (
            PHEAD WORD * term,
            WORD * lhs )
```

Definition at line 1070 of file if.c.

4.60.2.7 DoEndSwitch()

```
int DoEndSwitch (
            PHEAD WORD * term,
            WORD * lhs )
```

Definition at line 1086 of file if.c.

4.60.2.8 FindCase()

```
SWITCHTABLE * FindCase (
            WORD nswitch,
            WORD ncase )
```

Definition at line 1097 of file if.c.

4.60.2.9 DoubleSwitchBuffers()

```
int DoubleSwitchBuffers (
            void )
```

Definition at line 1135 of file if.c.

4.60.2.10 SwitchSplitMergeRec()

```
void SwitchSplitMergeRec (
            SWITCHTABLE * array,
            WORD num,
            SWITCHTABLE * auxarray )
```

Definition at line 1184 of file if.c.

4.60.2.11 SwitchSplitMerge()

```
void SwitchSplitMerge (
            SWITCHTABLE * array,
            WORD num )
```

Definition at line 1213 of file if.c.

4.61 if.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : if.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ If statement :
        #[ Syntax :

        The `if' is a conglomerate of statements: if,else,endif

        The if consists in principle of:

        if ( number );
            statements
        else;
            statements
        endif;

        The first set is taken when number != 0.
        The else is not mandatory.
        TRUE = 1 and FALSE = 0

        The number can be built up via a logical expression:

                    expr1 condition expr2

        each expression can be a subexpression again. It has to be
        enclosed in parentheses in that case.
        Conditions are:
            >, >=, <, <=, ==, !=, ||, &&

        When Expressions are chained evaluation is from left to right,
        independent of whether this indicates nonsense.
        if ( a || b || c || d ); is a perfectly normal statement.
        if ( a >= b || c == d ); would be messed up. This should be:
        if ( ( a >= b ) || ( c == d ) );

        The building blocks of the Expressions are:

            Match(option,pattern)   The number of times pattern fits in term_
            Count(....)             The count value of term_
            Coeff[icient]           The coefficient of term_
            FindLoop(options)       Are there loops (as in ReplaceLoop).

        Implementation for internal notation:

        TYPEIF,length,gotolevel(if fail),EXPRTYPE,length,......

        EXPRTYPE can be:
            SHORTNUMBER         ->,4,sign,size
            LONGNUMBER          ->,|ncoef+2|,ncoef,numer,denom
            MATCH               ->,patternsiz+3,keyword,pattern
```

```
            MULTIPLEOF          ->,3,thenumber
            COUNT               ->,countsiz+2,countinfo
            TYPEFINDLOOP        ->,7 (findloop info)
            COEFFICIENT         ->,2
            IFDOLLAR            ->,3,dollarnumber
            SUBEXPR             ->,size,dummy,size1,EXPRTYPE,length,...
                                  ,2,condition1,size2,...
                                This is like functions.

        Note that there must be a restriction to the number of nestings
        of parentheses in an if statement. It has been set to 10.

        The syntax of match corresponds to the syntax of the left side
        of an id statement. The only difference is the keyword
        MATCH vs TYPEIDNEW.

        #] Syntax :
        #[ GetIfDollarNum :
*/

WORD GetIfDollarNum(WORD *ifp, WORD *ifstop)
{
    DOLLARS d;
    WORD num, *w;
    if ( ifp[2] < 0 ) { return(-ifp[2]-1); }
    d = Dollars+ifp[2];
    if ( ifp+3 < ifstop && ifp[3] == IFDOLLAREXTRA ) {
        if ( d->nfactors == 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Attempt to use a factor of an unfactored $-variable");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        num = GetIfDollarNum(ifp+3,ifstop);
        if ( num > d->nfactors ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Dollar factor number %s out of range",num);
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        if ( num == 0 ) {
            return(d->nfactors);
        }
        w = d->factors[num-1].where;
        if ( w == 0 ) return(d->factors[num].value);
getnumber:;
        if ( *w == 0 ) return(0);
        if ( *w == 4 && w[3] == 3 && w[2] == 1 && w[1] < MAXPOSITIVE && w[4] == 0 ) {
            return(w[1]);
        }
        if ( ( w[w[0]] != 0 ) || ( ABS(w[w[0]-1]) != w[0]-1 ) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Dollar factor number expected but found expression");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Dollar factor number out of range");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        return(0);
    }
/*
    Now we have just a dollar and should evaluate that into a short number
*/
    if ( d->type == DOLZERO ) {
        return(0);
    }
    else if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
        w = d->where; goto getnumber;
    }
    else {
        MLOCK(ErrorMessageLock);
        MesPrint("Dollar factor number is wrong type");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
        return(0);
    }
}

/*
        #] GetIfDollarNum :
        #[ FindVar :
*/

```

```
int FindVar(WORD *v, WORD *term)
{
    WORD *t, *tstop, *m, *mstop, *f, *fstop, *a, *astop;
    GETSTOP(term,tstop);
    t = term+1;
    while ( t < tstop ) {
        if ( *v == *t && *v < FUNCTION ) {  /* VECTOR, INDEX, SYMBOL, DOTPRODUCT */
            switch ( *v ) {
                case SYMBOL:
                    m = t+2; mstop = t+t[1];
                    while ( m < mstop ) {
                        if ( *m == v[1] ) return(1);
                        m += 2;
                    }
                    break;
                case INDEX:
                case VECTOR:
InVe:
                    m = t+2; mstop = t+t[1];
                    while ( m < mstop ) {
                        if ( *m == v[1] ) return(1);
                        m++;
                    }
                    break;
                case DOTPRODUCT:
                    m = t+2; mstop = t+t[1];
                    while ( m < mstop ) {
                        if ( *m == v[1] && m[1] == v[2] ) return(1);
                        if ( *m == v[2] && m[1] == v[1] ) return(1);
                        m += 3;
                    }
                    break;
            }
        }
        else if ( *v == VECTOR && *t == INDEX ) goto InVe;
        else if ( *v == INDEX && *t == VECTOR ) goto InVe;
        else if ( ( *v == VECTOR || *v == INDEX ) && *t == DOTPRODUCT ) {
            m = t+2; mstop = t+t[1];
            while ( m < mstop ) {
                if ( v[1] == m[0] || v[1] == m[1] ) return(1);
                m += 3;
            }
        }
        else if ( *t >= FUNCTION ) {
            if ( *v == FUNCTION && v[1] == *t ) return(1);
            if ( functions[*t-FUNCTION].spec > 0 ) {
              if ( *v == VECTOR || *v == INDEX ) { /* we need to check arguments */
                int i;
                for ( i = FUNHEAD; i < t[1]; i++ ) {
                    if ( v[1] == t[i] ) return(1);
                }
              }
            }
            else {
              fstop = t + t[1]; f = t + FUNHEAD;
              while ( f < fstop ) { /* Do the arguments one by one */
                if ( *f <= 0 ) {
                    switch ( *f ) {
                        case -SYMBOL:
                            if ( *v == SYMBOL && v[1] == f[1] ) return(1);
                            f += 2;
                            break;
                        case -VECTOR:
                        case -MINVECTOR:
                        case -INDEX:
                            if ( ( *v == VECTOR || *v == INDEX )
                             && ( v[1] == f[1] ) ) return(1);
                            f += 2;
                            break;
                        case -SNUMBER:
                            f += 2;
                            break;
                        default:
                            if ( *v == FUNCTION && v[1] == -*f && *f <= -FUNCTION ) return(1);
                            if ( *f <= -FUNCTION ) f++;
                            else f += 2;
                            break;
                    }
                }
                else {
                    a = f + ARGHEAD; astop = f + *f;
                    while ( a < astop ) {
                        if ( FindVar(v,a) == 1 ) return(1);
                        a += *a;
                    }
                    f = astop;
                }
```

```
                }
            }
        }
        t += t[1];
    }
    return(0);
}

/*
        #] FindVar :
        #[ DoIfStatement :              WORD DoIfStatement(PHEAD ifcode,term)

        The execution time part of the if-statement.
        The arguments are a pointer to the TYPEIF and a pointer to the term.
        The answer is either 1 (success) or 0 (fail).
        The calling routine can figure out where to go in case of failure
        by picking up gotolevel.
        Note that the whole setup asks for recursions.
*/

WORD DoIfStatement(PHEAD WORD *ifcode, WORD *term)
{
    GETBIDENTITY
    WORD *ifstop, *ifp;
    UWORD *coef1 = 0, *coef2, *coef3, *cc;
    WORD ncoef1, ncoef2, ncoef3, i = 0, first, *r, acoef, ismul1, ismul2, j;
    UWORD *Spac1, *Spac2;
    ifstop = ifcode + ifcode[1];
    ifp = ifcode + 3;
    if ( ifp >= ifstop ) return(1);
    if ( ( ifp + ifp[1] ) >= ifstop ) {
        switch ( *ifp ) {
            case LONGNUMBER:
                if ( ifp[2] ) return(1);
                else return(0);
            case MATCH:
            case TYPEIF:
                if ( HowMany(BHEAD ifp,term) ) return(1);
                else return(0);
            case TYPEFINDLOOP:
                if ( Lus(term,ifp[3],ifp[4],ifp[5],ifp[6],ifp[2]) ) return(1);
                else return(0);
            case TYPECOUNT:
                if ( CountDo(term,ifp) ) return(1);
                else return(0);
            case COEFFI:
            case MULTIPLEOF:
                return(1);
            case IFDOLLAR:
                {
                    DOLLARS d = Dollars + ifp[2];
#ifdef WITHPTHREADS
                    int nummodopt, dtype = -1;
                    if ( AS.MultiThreaded ) {
                        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                            if ( ifp[2] == ModOptdollars[nummodopt].number ) break;
                        }
                        if ( nummodopt < NumModOptdollars ) {
                            dtype = ModOptdollars[nummodopt].type;
                            if ( dtype == MODLOCAL ) {
                                d = ModOptdollars[nummodopt].dstruct+AT.identity;
                            }
                        }
                    }
                    dtype = d->type;
#else
                    int dtype = d->type;  /* We use dtype to make the operation atomic */
#endif
                    if ( dtype == DOLZERO ) return(0);
                    if ( dtype == DOLUNDEFINED ) {
                        if ( AC.UnsureDollarMode == 0 ) {
                            MesPrint("$%s is undefined",AC.dollarnames->namebuffer+d->name);
                            Terminate(-1);
                        }
                    }
                }
                return(1);
            case IFEXPRESSION:
                r = ifp+2; j = ifp[1] - 2;
                while ( --j >= 0 ) {
                    if ( *r == AR.CurExpr ) return(1);
                    r++;
                }
                return(0);
            case IFISFACTORIZED:
                r = ifp+2; j = ifp[1] - 2;
                if ( j == 0 ) {
```

```
                    if ( ( Expressions[AR.CurExpr].vflags & ISFACTORIZED ) != 0 )
                        return(1);
                    else
                        return(0);
                }
                while ( --j >= 0 ) {
                    if ( ( Expressions[*r].vflags & ISFACTORIZED ) == 0 ) return(0);
                    r++;
                }
                return(1);
            case IFOCCURS:
                {
                    WORD *OccStop = ifp + ifp[1];
                    ifp += 2;
                    while ( ifp < OccStop ) {
                        if ( FindVar(ifp,term) == 1 ) return(1);
                        if ( *ifp == DOTPRODUCT ) ifp += 3;
                        else ifp += 2;
                    }
                }
                return(0);
            case IFUSERFLAG:
                if ( ( Expressions[AR.CurExpr].uflags & (1 << ifp[2]) ) != 0 )
                    return(1);
                return(0);
            default:
/*
                Now we have a subexpression. Test first for one with a single item.
*/
                if ( ifp[3] == ( ifp[1] + 3 ) ) return(DoIfStatement(BHEAD ifp,term));
                ifstop = ifp + ifp[1];
                ifp += 3;
                break;
        }
    }
/*
    Here is the composite condition.
*/
    coef3 = NumberMalloc("DoIfStatement");
    Spac1 = NumberMalloc("DoIfStatement");
    Spac2 = (UWORD *)(TermMalloc("DoIfStatement"));
    ncoef1 = 0; first = 1; ismul1 = 0;
    do {
        if ( !first ) {
            ifp += 2;
            if ( ifp[-2] == ORCOND && ncoef1 ) {
                coef1 = Spac1;
                ncoef1 = 1; coef1[0] = coef1[1] = 1;
                goto SkipCond;
            }
            if ( ifp[-2] == ANDCOND && !ncoef1 ) goto SkipCond;
        }
        coef2 = Spac2;
        ncoef2 = 1;
        ismul2 = 0;
        switch ( *ifp ) {
            case LONGNUMBER:
                ncoef2 = ifp[2];
                j = 2*(ABS(ncoef2));
                cc = (UWORD *)(ifp + 3);
                for ( i = 0; i < j; i++ ) coef2[i] = cc[i];
                break;
#ifdef WITHFLOAT
            case IFFLOATNUMBER:
/*
                The sloppy solution is: Convert to rational.
                This way we can write it over coef2,ncoef2
*/
                ncoef2 = FloatFunToRat(BHEAD coef2,ifp);
                break;
#endif
            case MATCH:
            case TYPEIF:
                coef2[0] = HowMany(BHEAD ifp,term);
                coef2[1] = 1;
                if ( coef2[0] == 0 ) ncoef2 = 0;
                break;
            case TYPECOUNT:
                acoef = CountDo(term,ifp);
                coef2[0] = ABS(acoef);
                coef2[1] = 1;
                if ( acoef == 0 ) ncoef2 = 0;
                else if ( acoef < 0 ) ncoef2 = -1;
                break;
            case TYPEFINDLOOP:
                acoef = Lus(term,ifp[3],ifp[4],ifp[5],ifp[6],ifp[2]);
                coef2[0] = ABS(acoef);
```

```
                coef2[1] = 1;
                if ( acoef == 0 ) ncoef2 = 0;
                else if ( acoef < 0 ) ncoef2 = -1;
                break;
            case COEFFI:
                r = term + *term;
                ncoef2 = r[-1];
                i = ABS(ncoef2);
                cc = (UWORD *)(r - i);
                if ( ncoef2 < 0 ) ncoef2 = (ncoef2+1)»1;
                else ncoef2 = (ncoef2-1)»1;
                i--; for ( j = 0; j < i; j++ ) coef2[j] = cc[j];
                break;
            case SUBEXPR:
                ncoef2 = coef2[0] = DoIfStatement(BHEAD ifp,term);
                coef2[1] = 1;
                break;
            case MULTIPLEOF:
                ncoef2 = 1;
                coef2[0] = ifp[2];
                coef2[1] = 1;
                ismul2 = 1;
                break;
            case IFDOLLAREXTRA:
                break;
            case IFDOLLAR:
                {
/*
                    We need to abstract a long rational in coef2
                    with length ncoef2. What if that cannot be done?
*/
                    DOLLARS d = Dollars + ifp[2];
#ifdef WITHPTHREADS
                    int nummodopt, dtype = -1;
                    if ( AS.MultiThreaded ) {
                        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                            if ( ifp[2] == ModOptdollars[nummodopt].number ) break;
                        }
                        if ( nummodopt < NumModOptdollars ) {
                            dtype = ModOptdollars[nummodopt].type;
                            if ( dtype == MODLOCAL ) {
                                d = ModOptdollars[nummodopt].dstruct+AT.identity;
                            }
                            else {
                                LOCK(d->pthreadslockread);
                            }
                        }
                    }
#endif
/*
                    We have to pick up the IFDOLLAREXTRA pieces for [1], [$y] etc.
*/
                    if ( ifp+3 < ifstop && ifp[3] == IFDOLLAREXTRA ) {
                        if ( d->nfactors == 0 ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Attempt to use a factor of an unfactored $-variable");
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        } {
                        WORD num = GetIfDollarNum(ifp+3,ifstop);
                        WORD *w;
                        while ( ifp+3 < ifstop && ifp[3] == IFDOLLAREXTRA ) ifp += 3;
                        if ( num > d->nfactors ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Dollar factor number %s out of range",num);
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        }
                        if ( num == 0 ) {
                            ncoef2 = 1; coef2[0] = d->nfactors; coef2[1] = 1;
                            break;
                        }
                        w = d->factors[num-1].where;
                        if ( w == 0 ) {
                            if ( d->factors[num-1].value < 0 ) {
                                ncoef2 = -1; coef2[0] = -d->factors[num-1].value; coef2[1] = 1;
                            }
                            else {
                                ncoef2 = 1; coef2[0] = d->factors[num-1].value; coef2[1] = 1;
                            }
                            break;
                        }
                        if ( w[*w] == 0 ) {
                            r = w + *w - 1;
                            i = ABS(*r);
                            if ( i == ( *w-1 ) ) {
                                ncoef2 = (i-1)/2;
```

```
                                if ( *r < 0 ) ncoef2 = -ncoef2;
                                i--; cc = coef2; r = w + 1;
                                while ( --i >= 0 ) *cc++ = (UWORD)(*r++);
                                break;
                            }
                        }
                        goto generic;
                        }
                    }
                    else {
                      switch ( d->type ) {
                        case DOLUNDEFINED:
                            if ( AC.UnsureDollarMode == 0 ) {
#ifdef WITHPTHREADS
                                if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                MLOCK(ErrorMessageLock);
                                MesPrint("$%s is undefined",AC.dollarnames->namebuffer+d->name);
                                MUNLOCK(ErrorMessageLock);
                                Terminate(-1);
                            }
                            ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
                            break;
                        case DOLZERO:
                            ncoef2 = coef2[0] = 0; coef2[1] = 1;
                            break;
                        case DOLSUBTERM:
                            if ( d->where[0] != INDEX || d->where[1] != 3
                            || d->where[2] < 0 || d->where[2] >= AM.OffsetIndex ) {
                                if ( AC.UnsureDollarMode == 0 ) {
#ifdef WITHPTHREADS
                                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                    MLOCK(ErrorMessageLock);
                                    MesPrint("$%s is of wrong type",AC.dollarnames->namebuffer+d->name);
                                    MUNLOCK(ErrorMessageLock);
                                    Terminate(-1);
                                }
                                ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
                                break;
                            }
                            d->index = d->where[2];
                            /* fall through */
                        case DOLINDEX:
                            if ( d->index == 0 ) {
                                ncoef2 = coef2[0] = 0; coef2[1] = 1;
                            }
                            else if ( d->index > 0 && d->index < AM.OffsetIndex ) {
                                ncoef2 = 1; coef2[0] = d->index; coef2[1] = 1;
                            }
                            else if ( AC.UnsureDollarMode == 0 ) {
#ifdef WITHPTHREADS
                                if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                MLOCK(ErrorMessageLock);
                                MesPrint("$%s is of wrong type",AC.dollarnames->namebuffer+d->name);
                                MUNLOCK(ErrorMessageLock);
                                Terminate(-1);
                            }
                            ncoef2 = coef2[0] = 0; coef2[1] = 1;
                            break;
                        case DOLWILDARGS:
                            if ( d->where[0] <= -FUNCTION ||
                            ( d->where[0] < 0 && d->where[2] != 0 )
                            || ( d->where[0] > 0 && d->where[d->where[0]] != 0 )
                            ) {
                                if ( AC.UnsureDollarMode == 0 ) {
#ifdef WITHPTHREADS
                                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                    MLOCK(ErrorMessageLock);
                                    MesPrint("$%s is of wrong type",AC.dollarnames->namebuffer+d->name);
                                    MUNLOCK(ErrorMessageLock);
                                    Terminate(-1);
                                }
                                ncoef2 = coef2[0] = 0; coef2[1] = 1;
                                break;
                            }
                            /* fall through */
                        case DOLARGUMENT:
                            if ( d->where[0] == -SNUMBER ) {
                                if ( d->where[1] == 0 ) {
                                    ncoef2 = coef2[0] = 0;
```

```
                                }
                                else if ( d->where[1] < 0 ) {
                                    ncoef2 = -1;
                                    coef2[0] = -d->where[1];
                                }
                                else {
                                    ncoef2 = 1;
                                    coef2[0] = d->where[1];
                                }
                                coef2[1] = 1;
                            }
                            else if ( d->where[0] == -INDEX
                            && d->where[1] >= 0 && d->where[1] < AM.OffsetIndex ) {
                                if ( d->where[1] == 0 ) {
                                    ncoef2 = coef2[0] = 0; coef2[1] = 1;
                                }
                                else {
                                    ncoef2 = 1; coef2[0] = d->where[1];
                                    coef2[1] = 1;
                                }
                            }
                            else if ( d->where[0] > 0
                            && d->where[ARGHEAD] == (d->where[0]-ARGHEAD)
                            && ABS(d->where[d->where[0]-1]) ==
                                        (d->where[0] - ARGHEAD-1) ) {
                                i = d->where[d->where[0]-1];
                                ncoef2 = (ABS(i)-1)/2;
                                if ( i < 0 ) { ncoef2 = -ncoef2; i = -i; }
                                i--; cc = coef2; r = d->where + ARGHEAD+1;
                                while ( --i >= 0 ) *cc++ = (UWORD)(*r++);
                            }
                            else {
                                if ( AC.UnsureDollarMode == 0 ) {
#ifdef WITHPTHREADS
                                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                    MLOCK(ErrorMessageLock);
                                    MesPrint("$%s is of wrong type",AC.dollarnames->namebuffer+d->name);
                                    MUNLOCK(ErrorMessageLock);
                                    Terminate(-1);
                                }
                                ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
                            }
                            break;
                        case DOLNUMBER:
                        case DOLTERMS:
                            if ( d->where[d->where[0]] == 0 ) {
                                r = d->where + d->where[0]-1;
                                i = ABS(*r);
                                if ( i == ( d->where[0]-1 ) ) {
                                    ncoef2 = (i-1)/2;
                                    if ( *r < 0 ) ncoef2 = -ncoef2;
                                    i--; cc = coef2; r = d->where + 1;
                                    while ( --i >= 0 ) *cc++ = (UWORD)(*r++);
                                    break;
                                }
                            }
generic:;
                            if ( AC.UnsureDollarMode == 0 ) {
#ifdef WITHPTHREADS
                                if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                MLOCK(ErrorMessageLock);
                                MesPrint("$%s is of wrong type",AC.dollarnames->namebuffer+d->name);
                                MUNLOCK(ErrorMessageLock);
                                Terminate(-1);
                            }
                            ncoef2 = 0; coef2[0] = 0; coef2[1] = 1;
                            break;
                      }
                    }
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                }
                break;
            case IFEXPRESSION:
                r = ifp+2; j = ifp[1] - 2; ncoef2 = 0;
                while ( --j >= 0 ) {
                    if ( *r == AR.CurExpr ) { ncoef2 = 1; break; }
                    r++;
                }
                coef2[0] = ncoef2;
                coef2[1] = 1;
                break;
```

```
            case IFISFACTORIZED:
                r = ifp+2; j = ifp[1] - 2;
                if ( j == 0 ) {
                    ncoef2 = 0;
                    if ( ( Expressions[AR.CurExpr].vflags & ISFACTORIZED ) != 0 ) {
                        ncoef2 = 1;
                    }
                }
                else {
                    ncoef2 = 1;
                    while ( --j >= 0 ) {
                        if ( ( Expressions[*r].vflags & ISFACTORIZED ) == 0 ) {
                            ncoef2 = 0;
                            break;
                        }
                        r++;
                    }
                }
                coef2[0] = ncoef2;
                coef2[1] = 1;
                break;
            case IFOCCURS:
                {
                    WORD *OccStop = ifp + ifp[1], *ifpp = ifp+2;
                    ncoef2 = 0;
                    while ( ifpp < OccStop ) {
                        if ( FindVar(ifpp,term) == 1 ) {
                            ncoef2 = 1; break;
                        }
                        if ( *ifpp == DOTPRODUCT ) ifp += 3;
                        else ifpp += 2;
                    }
                    coef2[0] = ncoef2;
                    coef2[1] = 1;
                }
                break;
            case IFUSERFLAG:
                {
                    ncoef2 = 0;
                    if ( ( Expressions[AR.CurExpr].uflags & (1 « ifp[2]) ) != 0 )
                        ncoef2 = 1;
                    coef2[0] = ncoef2;
                    coef2[1] = 1;
                }
                break;
            default:
                break;
        }
        if ( !first ) {
            if ( ifp[-2] != ORCOND && ifp[-2] != ANDCOND ) {
                if ( ( ifp[-2] == EQUAL || ifp[-2] == NOTEQUAL ) &&
                ( ismul2 || ismul1 ) ) {
                    if ( ismul1 && ismul2 ) {
                        if ( coef1[0] == coef2[0] ) i = 1;
                        else i = 0;
                    }
                    else {
                        if ( ismul1 ) {
                            if ( ncoef2 )
                                Divvy(BHEAD coef2,&ncoef2,coef1,ncoef1);
                            cc = coef2; ncoef3 = ncoef2;
                        }
                        else {
                            if ( ncoef1 )
                                Divvy(BHEAD coef1,&ncoef1,coef2,ncoef2);
                            cc = coef1; ncoef3 = ncoef1;
                        }
                        if ( ncoef3 < 0 ) ncoef3 = -ncoef3;
                        if ( ncoef3 == 0 ) {
                            if ( ifp[-2] == EQUAL ) i = 1;
                            else i = 0;
                        }
                        else if ( cc[ncoef3] != 1 ) {
                            if ( ifp[-2] == EQUAL ) i = 0;
                            else i = 1;
                        }
                        else {
                            for ( j = 1; j < ncoef3; j++ ) {
                                if ( cc[ncoef3+j] != 0 ) break;
                            }
                            if ( j < ncoef3 ) {
                                if ( ifp[-2] == EQUAL ) i = 0;
                                else i = 1;
                            }
                            else if ( ifp[-2] == EQUAL ) i = 1;
                            else i = 0;
                        }
```

```
                    }
                    goto donemul;
                }
                else if ( AddRat(BHEAD coef1,ncoef1,coef2,-ncoef2,coef3,&ncoef3) ) {
                    NumberFree(coef3,"DoIfStatement"); NumberFree(Spac1,"DoIfStatement"); TermFree(Spac2,"DoIfStatement");
                    MesCall("DoIfStatement"); return(-1);
                }
                switch ( ifp[-2] ) {
                    case GREATER:
                        if ( ncoef3 > 0 ) i = 1;
                        else i = 0;
                        break;
                    case GREATEREQUAL:
                        if ( ncoef3 >= 0 ) i = 1;
                        else i = 0;
                        break;
                    case LESS:
                        if ( ncoef3 < 0 ) i = 1;
                        else i = 0;
                        break;
                    case LESSEQUAL:
                        if ( ncoef3 <= 0 ) i = 1;
                        else i = 0;
                        break;
                    case EQUAL:
                        if ( ncoef3 == 0 ) i = 1;
                        else i = 0;
                        break;
                    case NOTEQUAL:
                        if ( ncoef3 != 0 ) i = 1;
                        else i = 0;
                        break;
                }
donemul:        if ( i ) { ncoef2 = 1; coef2 = Spac2; coef2[0] = coef2[1] = 1; }
                else ncoef2 = 0;
                ismul1 = ismul2 = 0;
            }
        }
        else {
            first = 0;
        }
        coef1 = Spac1;
        i = 2*ABS(ncoef2);
        for ( j = 0; j < i; j++ ) coef1[j] = coef2[j];
        ncoef1 = ncoef2;
SkipCond:
        ifp += ifp[1];
    } while ( ifp < ifstop );

    NumberFree(coef3,"DoIfStatement"); NumberFree(Spac1,"DoIfStatement"); TermFree(Spac2,"DoIfStatement");
    if ( ncoef1 ) return(1);
    else return(0);
}

/*
        #] DoIfStatement :
        #[ HowMany :                    WORD HowMany(ifcode,term)

        Returns the number of times that the pattern in ifcode
        can be taken out from term. There is a subkey in ifcode[2];
        The notation is identical to the lhs of an id statement.
        Most of the code comes from TestMatch.
*/

WORD HowMany(PHEAD WORD *ifcode, WORD *term)
{
    GETBIDENTITY
    WORD *m, *t, *r, *w, power, RetVal, i, topje, *newterm;
    WORD *OldWork, *ww, *mm;
    int *RepSto, RepVal;
    int numdollars = 0;
    m = ifcode + IDHEAD;
    AN.FullProto = m;
    AN.WildValue = w = m + SUBEXPSIZE;
    m += m[1];
    AN.WildStop = m;
    OldWork = AT.WorkPointer;
    if ( ( ifcode[4] & 1 ) != 0 ) { /* We have at least one dollar in the pattern */
        AR.Eside = LHSIDEX;
        ww = AT.WorkPointer; i = m[0]; mm = m;
        NCOPY(ww,mm,i);
        *OldWork += 3;
        *ww++ = 1; *ww++ = 1; *ww++ = 3;
        AT.WorkPointer = ww;
        RepSto = AN.RepPoint;
```

```
        RepVal = *RepSto;
        NewSort(BHEAD0);
        if ( Generator(BHEAD OldWork,AR.Cnumlhs) ) {
            LowerSortLevel();
            *RepSto = RepVal;
            AN.RepPoint = RepSto;
            AT.WorkPointer = OldWork;
            return(-1);
        }
        AT.WorkPointer = ww;
        if ( EndSort(BHEAD ww,0) < 0 ) {}
        *RepSto = RepVal;
        AN.RepPoint = RepSto;
        if ( *ww == 0 || *(ww+*ww) != 0 ) {
            if ( AP.lhdollarerror == 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("&LHS must be one term");
                MUNLOCK(ErrorMessageLock);
                AP.lhdollarerror = 1;
            }
            AT.WorkPointer = OldWork;
            return(-1);
        }
        m = ww; AT.WorkPointer = ww = m + *m;
        if ( m[*m-1] < 0 ) { m[*m-1] = -m[*m-1]; }
        *m -= m[*m-1];
        AR.Eside = RHSIDE;
    }
    else {
        ww = term + *term;
        if ( AT.WorkPointer < ww ) AT.WorkPointer = ww;
    }
    ClearWild(BHEAD0);
    while ( w < AN.WildStop ) {
        if ( *w == LOADDOLLAR ) numdollars++;
        w += w[1];
    }
    AN.RepFunNum = 0;
    AN.RepFunList = AT.WorkPointer;
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    topje = cbuf[AT.ebufnum].numrhs;
    if ( AT.WorkPointer >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AN.DisOrderFlag = ifcode[2] & SUBDISORDER;
    switch ( ifcode[2] & (~SUBDISORDER) ) {
        case SUBONLY :
            /* Must be an exact match */
            AN.UseFindOnly = 1; AN.ForFindOnly = 0;
/*
        Copy the term first to scratchterm. This is needed
        because of the Substitute.
*/
            i = *term;
            t = term; newterm = r = AT.WorkPointer;
            NCOPY(r,t,i); AT.WorkPointer = r;
            RetVal = 0;
            if ( FindRest(BHEAD newterm,m) && ( AN.UsedOtherFind ||
                FindOnly(BHEAD newterm,m) ) ) {
                Substitute(BHEAD newterm,m,1);
                if ( numdollars ) {
                    WildDollars(BHEAD (WORD *)0);
                    numdollars = 0;
                }
                ClearWild(BHEAD0);
                RetVal = 1;
            }
            else RetVal = 0;
            break;
        case SUBMANY :
/*
        Copy the term first to scratchterm. This is needed
        because of the Substitute.
*/
            i = *term;
            t = term; newterm = r = AT.WorkPointer;
            NCOPY(r,t,i); AT.WorkPointer = r;
            RetVal = 0;
            AN.UseFindOnly = 0;
            if ( ( power = FindRest(BHEAD newterm,m) ) > 0 ) {
                if ( ( power = FindOnce(BHEAD newterm,m) ) > 0 ) {
                    AN.UseFindOnly = 0;
                    do {
                        Substitute(BHEAD newterm,m,1);
```

```
                        if ( numdollars ) {
                            WildDollars(BHEAD (WORD *)0);
                            numdollars = 0;
                        }
                        ClearWild(BHEAD0);
                        RetVal++;
                    } while ( FindRest(BHEAD newterm,m) && (
                        AN.UsedOtherFind || FindOnce(BHEAD newterm,m) ) );
                }
                else if ( power < 0 ) {
                    do {
                        Substitute(BHEAD newterm,m,1);
                        if ( numdollars ) {
                            WildDollars(BHEAD (WORD *)0);
                            numdollars = 0;
                        }
                        ClearWild(BHEAD0);
                        RetVal++;
                    } while ( FindRest(BHEAD newterm,m) );
                }
            }
            else if ( power < 0 ) {
                if ( FindOnce(BHEAD newterm,m) ) {
                    do {
                        Substitute(BHEAD newterm,m,1);
                        if ( numdollars ) {
                            WildDollars(BHEAD (WORD *)0);
                            numdollars = 0;
                        }
                        ClearWild(BHEAD0);
                    } while ( FindOnce(BHEAD newterm,m) );
                    RetVal = 1;
                }
            }
            break;
        case SUBONCE :
/*
        Copy the term first to scratchterm. This is needed
        because of the Substitute.
*/
            i = *term;
            t = term; newterm = r = AT.WorkPointer;
            NCOPY(r,t,i); AT.WorkPointer = r;
            RetVal = 0;
            AN.UseFindOnly = 0;
            if ( FindRest(BHEAD newterm,m) && ( AN.UsedOtherFind || FindOnce(BHEAD newterm,m) ) ) {
                Substitute(BHEAD newterm,m,1);
                if ( numdollars ) {
                    WildDollars(BHEAD (WORD *)0);
                    numdollars = 0;
                }
                ClearWild(BHEAD0);
                RetVal = 1;
            }
            else RetVal = 0;
            break;
        case SUBMULTI :
            RetVal = FindMulti(BHEAD term,m);
            break;
        case SUBVECTOR :
            RetVal = 0;
            for ( i = 0; i < *term; i++ ) ww[i] = term[i];
            while ( ( power = FindAll(BHEAD ww,m,AR.Cnumlhs,ifcode) ) != 0 ) { RetVal += power; }
            break;
        case SUBSELECT :
            ifcode += IDHEAD;   ifcode += ifcode[1];    ifcode += *ifcode;
            AN.UseFindOnly = 1; AN.ForFindOnly = ifcode;
            if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind ||
                    FindOnly(BHEAD term,m) ) ) RetVal = 1;
            else RetVal = 0;
            break;
        default :
            RetVal = 0;
            break;
    }
    AT.WorkPointer = AN.RepFunList;
    cbuf[AT.ebufnum].numrhs = topje;
    return(RetVal);
}

/*
        #] HowMany :
        #[ DoubleIfBuffers :
*/

void DoubleIfBuffers(void)
{
```

```
    int newmax, i;
    WORD *newsumcheck;
    LONG *newheap, *newifcount;
    if ( AC.MaxIf == 0 ) newmax = 10;
    else                 newmax = 2*AC.MaxIf;
    newheap = (LONG *)Malloc1(sizeof(LONG)*(newmax+1),"IfHeap");
    newsumcheck = (WORD *)Malloc1(sizeof(WORD)*(newmax+1),"IfSumCheck");
    newifcount = (LONG *)Malloc1(sizeof(LONG)*(newmax+1),"IfCount");
    if ( AC.MaxIf ) {
        for ( i = 0; i < AC.MaxIf; i++ ) {
            newheap[i] = AC.IfHeap[i];
            newsumcheck[i] = AC.IfSumCheck[i];
            newifcount[i] = AC.IfCount[i];
        }
        AC.IfStack = (AC.IfStack-AC.IfHeap) + newheap;
        M_free(AC.IfHeap,"AC.IfHeap");
        M_free(AC.IfCount,"AC.IfCount");
        M_free(AC.IfSumCheck,"AC.IfSumCheck");
    }
    else {
        AC.IfStack = newheap;
    }
    AC.IfHeap = newheap;
    AC.IfSumCheck = newsumcheck;
    AC.IfCount = newifcount;
    AC.MaxIf = newmax;
}

/*
        #] DoubleIfBuffers :
    #] If statement :
    #[ Switch statement :
        #[ DoSwitch :
*/

int DoSwitch(PHEAD WORD *term, WORD *lhs)
{
/*
    For the moment we ignore the compiler buffer problems.
*/
    WORD numdollar = lhs[2];
    WORD ncase = DolToNumber(BHEAD numdollar);
    SWITCHTABLE *swtab = FindCase(lhs[3],ncase);
    return(Generator(BHEAD term,swtab->value));
}

/*
        #] DoSwitch :
        #[ DoEndSwitch :
*/

int DoEndSwitch(PHEAD WORD *term, WORD *lhs)
{
    SWITCH *sw = AC.SwitchArray+lhs[2];
    return(Generator(BHEAD term,sw->endswitch.value+1));
}

/*
        #] DoEndSwitch :
        #[ FindCase :
*/

SWITCHTABLE *FindCase(WORD nswitch, WORD ncase)
{
/*
    First find the switch table and determine how we have to search.
*/
    SWITCH *sw = AC.SwitchArray+nswitch;
    WORD hi, lo, med;
    if ( sw->typetable == DENSETABLE ) {
        med = ncase - sw->caseoffset;
        if ( med >= sw->numcases || med < 0 ) return(&sw->defaultcase);
    }
    else {
/*
        We need a binary search in the table.
*/
        if ( ncase > sw->maxcase || ncase < sw->mincase ) return(&sw->defaultcase);
        hi = sw->numcases-1; lo = 0;
        for(;;) {
            med = (hi+lo)/2;
            if ( ncase == sw->table[med].ncase ) break;
            else if ( ncase > sw->table[med].ncase ) {
                lo = med+1;
                if ( lo > hi ) return(&sw->defaultcase);
            }
            else {
```

```
                hi = med-1;
                if ( hi < lo ) return(&sw->defaultcase);
            }
        }
    }
    return(&sw->table[med]);
}

/*
        #] FindCase :
        #[ DoubleSwitchBuffers :
*/

int DoubleSwitchBuffers(void)
{
    int newmax, i;
    SWITCH *newarray;
    WORD *newheap;
    if ( AC.MaxSwitch == 0 ) newmax = 10;
    else                     newmax = 2*AC.MaxSwitch;
    newarray = (SWITCH *)Malloc1(sizeof(SWITCH)*(newmax+1),"SwitchArray");
    newheap = (WORD *)Malloc1(sizeof(WORD)*(newmax+1),"SwitchHeap");
    if ( AC.MaxSwitch ) {
        for ( i = 0; i < AC.MaxSwitch; i++ ) {
            newarray[i] = AC.SwitchArray[i];
            newheap[i] = AC.SwitchHeap[i];
        }
        M_free(AC.SwitchHeap,"AC.SwitchHeap");
        M_free(AC.SwitchArray,"AC.SwitchArray");
    }
    for ( i = AC.MaxSwitch; i <= newmax; i++ ) {
        newarray[i].table = 0;
        newarray[i].tablesize = 0;
        newarray[i].defaultcase.ncase = 0;
        newarray[i].defaultcase.value = 0;
        newarray[i].defaultcase.compbuffer = 0;
        newarray[i].endswitch.ncase = 0;
        newarray[i].endswitch.value = 0;
        newarray[i].endswitch.compbuffer = 0;
        newarray[i].typetable = 0;
        newarray[i].mincase = 0;
        newarray[i].maxcase = 0;
        newarray[i].numcases = 0;
        newarray[i].caseoffset = 0;
        newarray[i].iflevel = 0;
        newarray[i].whilelevel = 0;
        newarray[i].nestingsum = 0;
        newheap[i] = 0;
    }
    AC.SwitchArray = newarray;
    AC.SwitchHeap = newheap;
    AC.MaxSwitch = newmax;
    return(0);
}

/*
        #] DoubleSwitchBuffers :
        #[ SwitchSplitMerge :

        Sorts an array of WORDs. No adding of equal objects.
*/

void SwitchSplitMergeRec(SWITCHTABLE *array,WORD num,SWITCHTABLE *auxarray)
{
    WORD n1,n2,i,j,k;
    SWITCHTABLE *t1,*t2, t;
    if ( num < 2 ) return;
    if ( num == 2 ) {
        if ( array[0].ncase > array[1].ncase ) {
            t = array[0]; array[0] = array[1]; array[1] = t;
        }
        return;
    }
    n1 = num/2;
    n2 = num - n1;
    SwitchSplitMergeRec(array,n1,auxarray);
    SwitchSplitMergeRec(array+n1,n2,auxarray);
    if ( array[n1-1].ncase <= array[n1].ncase ) return;

    t1 = array; t2 = auxarray; i = n1; NCOPY(t2,t1,i);
    i = 0; j = n1; k = 0;
    while ( i < n1 && j < num ) {
        if ( auxarray[i].ncase <= array[j].ncase ) { array[k++] = auxarray[i++]; }
        else { array[k++] = array[j++]; }
    }
    while ( i < n1 ) array[k++] = auxarray[i++];
/*
```

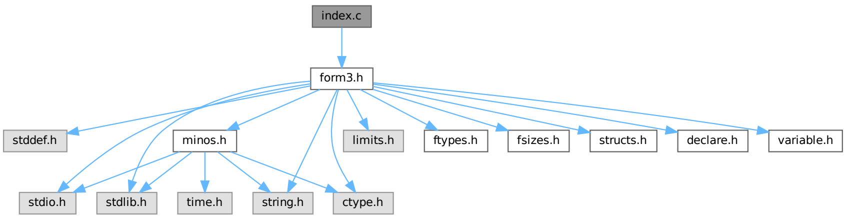
```
    Remember: remnants of j are still in place!
*/
}

void SwitchSplitMerge(SWITCHTABLE *array,WORD num)
{
    SWITCHTABLE *auxarray = (SWITCHTABLE *)Malloc1(sizeof(SWITCHTABLE)*num/2,"SwitchSplitMerge");
    SwitchSplitMergeRec(array,num,auxarray);
    M_free(auxarray,"SwitchSplitMerge");
}

/*
        #] SwitchSplitMerge :
    #] Switch statement :
*/
```

4.62 index.c File Reference

`#include "form3.h"`
Include dependency graph for index.c:



Functions

- POSITION * FindBracket (WORD nexp, WORD *bracket)
- void PutBracketInIndex (PHEAD WORD *term, POSITION *newpos)
- void ClearBracketIndex (WORD numexp)
- void OpenBracketIndex (WORD nexpr)
- int PutInside (PHEAD WORD *term, WORD *code)

4.62.1 Detailed Description

The routines that deal with bracket indexing It creates the bracket index and it can find the brackets using the index.

Definition in file index.c.

4.62.2 Function Documentation

4.62.2.1 FindBracket()

```
POSITION * FindBracket (
            WORD nexp,
            WORD * bracket )
```

Definition at line 65 of file index.c.

4.62.2.2 PutBracketInIndex()

```
void PutBracketInIndex (
            PHEAD WORD * term,
            POSITION * newpos )
```

Definition at line 331 of file index.c.

4.62.2.3 ClearBracketIndex()

```
void ClearBracketIndex (
            WORD numexp )
```

Definition at line 546 of file index.c.

4.62.2.4 OpenBracketIndex()

```
void OpenBracketIndex (
            WORD nexpr )
```

Definition at line 578 of file index.c.

4.62.2.5 PutInside()

```
int PutInside (
            PHEAD WORD * term,
            WORD * code )
```

Definition at line 609 of file index.c.

4.63 index.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 */
```

```
/* #] License : */

/*
    #[ Includes : index.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ syntax and use :

    The indexing of brackets is not automatic! It should only be used
    when one intends to use the contents of individual brackets.
    This is done with the addition of a + sign to the bracket statement:
    B+ a,b,c; or AB+ a,b,c;
    It does require resources! The index is kept in memory and is removed
    once the expression is treated and passed on to the output with different
    or no brackets.
    The index is limited to a given amount of space. Hence if there are too
    many brackets we will skip some in the index. Skipping goes by space
    occupied by the contents. We take the two adjacent bracket(s) with the
    least space together and represent them by the first one only. This gives
    a new spot.
    The expression struct has two pointers:
    bracketinfo      for using.
    newbracketinfo   for making new index.

    #] syntax and use :
    #[ FindBracket :
*/

POSITION *FindBracket(WORD nexp, WORD *bracket)
{
    GETIDENTITY
    BRACKETINDEX *bi;
    BRACKETINFO *bracketinfo;
    EXPRESSIONS e = &(Expressions[nexp]);
    LONG hi, low, med;
    int i;
    WORD oldsorttype = AR.SortType, *t1, *t2, j, bsize, *term, *p, *pstop, *pp;
    WORD *tstop, *cp, a[4], *bracketh;
    FILEHANDLE *fi;
    POSITION auxpos, toppos;
    switch ( e->status ) {
        case UNHIDELEXPRESSION:
        case UNHIDEGEXPRESSION:
        case DROPHLEXPRESSION:
        case DROPHGEXPRESSION:
        case HIDDENLEXPRESSION:
        case HIDDENGEXPRESSION:
            fi = AR.hidefile;
            break;
        default:
            fi = AR.infile;
            break;
    }
    if ( AT.bracketinfo ) bracketinfo = AT.bracketinfo;
    else                  bracketinfo = e->bracketinfo;
    hi = bracketinfo->indexfill; low = 0;
    if ( hi <= 0 ) return(0);
/*
    The next code is needed for a problem with sorting when there are
    only functions outside the bracket. This gives ordinarily the wrong
    sorting. We solve that by taking HAAKJE along with the outside while
    running the Compare. But this means that we need to copy bracket.
*/
    bracketh = TermMalloc("FindBracket");
    i = *bracket; p = bracket; pp = bracketh; NCOPY(pp,p,i)
    pp -= 3; *pp++ = HAAKJE; *pp++ = 3; *pp++ = 0; *pp++ = 1; *pp++ = 1; *pp++ = 3;
    *bracketh += 3;

    AT.fromindex = 1;
    AR.SortType = bracketinfo->SortType;
    bi = bracketinfo->indexbuffer + hi - 1;
    if ( *bracketh == 7 ) {
        if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 0;
        else i = -1;
    }
    else if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 1;
    else i = CompareTerms(BHEAD bracketh,bracketinfo->bracketbuffer+bi->bracket,0);
    if ( i < 0 ) {
        AR.SortType = oldsorttype;
        AT.fromindex = 0;
        TermFree(bracketh,"FindBracket");
        return(0);
    }
```

```
    else if ( i == 0 ) med = hi-1;
    else {
        for (;;) {
        med = (hi+low)/2;
        bi = bracketinfo->indexbuffer + med;
        if ( *bracketh == 7 ) {
            if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 0;
            else i = -1;
        }
        else if ( bracketinfo->bracketbuffer[bi->bracket] == 7 ) i = 1;
        else i = CompareTerms(BHEAD bracketh,bracketinfo->bracketbuffer+bi->bracket,0);
        if ( i == 0 ) { break; }
        if ( i > 0 ) {
            if ( low == med ) { /* no occurrence */
                AR.SortType = oldsorttype;
                AT.fromindex = 0;
                TermFree(bracketh,"FindBracket");
                return(0);
            }
            hi = med;
        }
        else if ( i < 0 ) {
            if ( low == med ) break;
            low = med;
        }
    }}
/*
    The bracket is now either bi itself or between bi and the next one
    or it is not present at all.
*/
    AN.theposition = AS.OldOnFile[nexp];
    ADD2POS(AN.theposition,bi->start);
/*
    The seek will have to move closer to the actual read so that we
    can place a lock around the two.
    if ( fi->handle >= 0 ) SeekFile(fi->handle,&AN.theposition,SEEK_SET);
    else SetScratch(fi,&AN.theposition);
*/
/*
    Put the bracket in the compress buffer as if it were the last term read.
    Have a look at AR.CompressPointer. (set it right)
*/
    term = AT.WorkPointer;
    t1 = bracketinfo->bracketbuffer+bi->bracket;
    j = *t1;
/*
    Note that in the bracketbuffer, the bracket sits with HAAKJE.

    The next is (hopefully) a bug fix. Originally the code read bsize = j
    but that overcounts one. We have the part outside the bracket and the
    coefficient which is 1,1,3. But we also have the length indicator.
    Where we use the variable bsize we do not include the length indicator,
    and we have the part outside plus 7,3,0 which is also three words.
*/
    bsize = j-1;
    t2 = AR.CompressPointer;
    NCOPY(t2,t1,j)
    if ( i == 0 ) { /* We found the proper bracket already */
        AR.SortType = oldsorttype;
        AT.fromindex = 0;
        TermFree(bracketh,"FindBracket");
        return(&AN.theposition);
    }
/*
    Here we have to skip to the bracket if it exists (!)
    Let us first look whether the expression is in memory.
    If not we have to make a buffer to increase speed..
*/
    if ( fi->handle < 0 ) {
        p = (WORD *)((UBYTE *)(fi->PObuffer)
             + BASEPOSITION(AS.OldOnFile[nexp])
             + BASEPOSITION(bi->start));
        pstop = (WORD *)((UBYTE *)(fi->PObuffer)
             + BASEPOSITION(AS.OldOnFile[nexp])
             + BASEPOSITION(bi->next));
        while ( p < pstop ) {
/*
            Check now: if size or part from previous term < size of bracket
                    we have to setup the bracket again and test.
                    Otherwise, skip immediately to the next term.
*/
            if ( *p <= -bsize ) {   /* no change of bracket */
                p++; p += *p + 1;
            }
            else if ( *p < 0 ) {    /* changes bracket */
                pp = p;
                t2 = AR.CompressPointer;
```

```
                t1 = t2 - *p++ + 1;
                j = *p++;
                NCOPY(t1,p,j)
                t2++; while ( *t2 != HAAKJE ) t2 += t2[1];
                t2 += t2[1];
                a[1] = t2[0]; a[2] = t2[1]; a[3] = t2[2];
                *t2++ = 1; *t2++ = 1; *t2++ = 3;
                *AR.CompressPointer = t2 - AR.CompressPointer;
                bsize = *AR.CompressPointer - 1;
                if ( *bracketh == 7 ) {
                    if ( AR.CompressPointer[0] == 7 ) i = 0;
                    else i = -1;
                }
                else if ( AR.CompressPointer[0] == 7 ) i = 1;
                else i = CompareTerms(BHEAD bracketh,AR.CompressPointer,0);
                t2[-3] = a[1]; t2[-2] = a[2]; t2[-1] = a[3];
                if ( i == 0 ) {
                    SETBASEPOSITION(AN.theposition,(pp-fi->PObuffer)*sizeof(WORD));
                    fi->POfill = pp;
                    goto found;
                }
                if ( i > 0 ) break; /* passed what was possible */
            }
            else {  /* no compression. We have to check! */
                WORD *oldworkpointer = AT.WorkPointer, *t3, *t4;
                t2 = p + 1; while ( *t2 != HAAKJE ) t2 += t2[1];
                t2 += t2[1];
/*
                Here we need to copy the term. Modifying has proven to
                be NOT threadsafe.
*/
                t3 = oldworkpointer; t4 = p;
                while ( t4 < t2 ) *t3++ = *t4++;
                *t3++ = 1; *t3++ = 1; *t3++ = 3;
                *oldworkpointer = t3 - oldworkpointer;
                bsize = *oldworkpointer - 1;
                AT.WorkPointer = t3;
                t3 = oldworkpointer;
                if ( *bracketh == 7 ) {
                    if ( t3[0] == 7 ) i = 0;
                    else i = -1;
                }
                else if ( t3[0] == 7 ) i = 1;
                else {
                    i = CompareTerms(BHEAD bracketh,t3,0);
                }
                AT.WorkPointer = oldworkpointer;
                if ( i == 0 ) {
                    SETBASEPOSITION(AN.theposition,(p-fi->PObuffer)*sizeof(WORD));
                    fi->POfill = p;
                    goto found;
                }
                if ( i > 0 ) break; /* passed what was possible */
                p += *p;
            }
        }
        AR.SortType = oldsorttype;
        AT.fromindex = 0;
        TermFree(bracketh,"FindBracket");
        return(0);  /* Bracket does not exist */
    }
    else {
/*
        In this case we can work with the old representation without HAAKJE.
        We stop searching when we reach toppos and we do not call Compare.
*/
        toppos = AS.OldOnFile[nexp];
        ADD2POS(toppos,bi->next);
        cp = AR.CompressPointer;
        for(;;) {
            auxpos = AN.theposition;
            GetOneTerm(BHEAD term,fi,&auxpos,0);
            if ( *term == 0 ) {
                AR.SortType = oldsorttype;
                AT.fromindex = 0;
                return(0);  /* Bracket does not exist */
            }
            tstop = term + *term;
            tstop -= ABS(tstop[-1]);
            t1 = term + 1;
            while ( *t1 != HAAKJE && t1 < tstop ) t1 += t1[1];
            i = *bracket-4;
            if ( t1-term == *bracket-3 ) {
                t1 = term + 1; t2 = bracket+1;
                while ( i > 0 && *t1 == *t2 ) { t1++; t2++; i--; }
                if ( i <= 0 ) {
                    AR.CompressPointer = cp;
```

```
                    goto found;
                }
            }
            AR.CompressPointer = cp;
            AN.theposition = auxpos;
/*
            Now check whether we passed the 'point'
*/
            if ( ISGEPOS(AN.theposition,toppos) ) {
                AR.SortType = oldsorttype;
                AR.CompressPointer = cp;
                AT.fromindex = 0;
                TermFree(bracketh,"FindBracket");
                return(0);  /* Bracket does not exist */
            }
        }
    }
found:
    AR.SortType = oldsorttype;
    AT.fromindex = 0;
    TermFree(bracketh,"FindBracket");
    return(&AN.theposition);
}

/*
    #] FindBracket :
    #[ PutBracketInIndex :

    Call via
    if ( AR.BracketOn ) PutBracketInIndex(BHEAD term);

    This means that there should be a bracket somewhere
    Note that the brackets come in in proper order.

    DON'T forget AR.SortType to be put into e->bracketinfo->SortType
*/

void PutBracketInIndex(PHEAD WORD *term, POSITION *newpos)
{
    GETBIDENTITY
    BRACKETINDEX *bi, *b1, *b2, *b3;
    BRACKETINFO *b;
    POSITION thepos;
    EXPRESSIONS e = Expressions + AR.CurExpr;
    LONG hi, i, average;
    WORD *t, *tstop, *t1, *t2, *oldt, oldsize, a[4];
    if ( ( b = e->newbracketinfo ) == 0 ) return;
    DIFPOS(thepos,*newpos,e->onfile);
    tstop = term + *term;
    tstop -= ABS(tstop[-1]);
    t = term+1;
    while ( *t != HAAKJE && t < tstop ) t += t[1];
    if ( t >= tstop ) return; /* no ticket, no laundry */
    t += t[1];  /* include HAAKJE for the sorting */
    a[0] = t[0]; a[1] = t[1]; a[2] = t[2];
    oldt = t; oldsize = *term; *t++ = 1; *t++ = 1; *t++ = 3;
    *term = t - term;
    AT.fromindex = 1;
/*
    Check now with the last bracket in the buffer.
    If it is the same we can abort.
*/
    hi = b->indexfill;
    if ( hi > 0 ) {
        bi = b->indexbuffer + hi - 1;
        bi->next = thepos;
        if ( *term == 7 ) {
            if ( b->bracketbuffer[bi->bracket] == 7 ) i = 0;
            else i = -1;
        }
        else if ( b->bracketbuffer[bi->bracket] == 7 ) i = 1;
        else i = CompareTerms(BHEAD term,b->bracketbuffer+bi->bracket,0);
        if ( i == 0 ) { /* still the same bracket */
            bi->termsinbracket++;
            goto bracketdone;
        }
        if ( i > 0 ) { /* We have a problem */
/*
            There is a special case in which we have only functions and
            term is contained completely in the bracket
*/
/*
            t = term + 1;
            tstop = term + *term - 3;
            while ( t < tstop && *t > HAAKJE ) t += t[1];
            if ( t < tstop ) goto problems;
*/
```

```
            for ( i = 1; i < *term - 3; i++ ) {
                if ( term[i] != b->bracketbuffer[bi->bracket+i] ) break;
            }
            if ( i < *term - 3 ) {
/*
problems:;
*/
                *term = oldsize; oldt[0] = a[0]; oldt[1] = a[1]; oldt[2] = a[2];
                MLOCK(ErrorMessageLock);
                MesPrint("Error!!!! Illegal bracket sequence detected in PutBracketInIndex");
#ifdef WITHPTHREADS
                MesPrint("Worker = %w");
#endif
                PrintTerm(term,"term into index");
                PrintTerm(b->bracketbuffer+bi->bracket,"Last in index");
                MUNLOCK(ErrorMessageLock);
                AT.fromindex = 0;
                Terminate(-1);
            }
            i = -1;
        }
    }
/*
    If there is room for more brackets, we add this one.
*/
    if ( b->bracketfill+*term >= b->bracketbuffersize
    && ( b->bracketbuffersize < AM.MaxBracketBufferSize
    || ( e->vflags & ISFACTORIZED ) != 0 ) ) {
/*
        Enlarge bracket buffer
*/
        WORD *oldbracketbuffer = b->bracketbuffer;
        i = MaX(b->bracketbuffersize * 2, b->bracketfill+*term+1);
        if ( i > AM.MaxBracketBufferSize && ( e->vflags & ISFACTORIZED ) == 0 )
                i = AM.MaxBracketBufferSize;
        if ( i > b->bracketfill+*term ) {
            b->bracketbuffersize = i;
            b->bracketbuffer = (WORD *)Malloc1(b->bracketbuffersize*sizeof(WORD),
                                "new bracket buffer");
            t1 = b->bracketbuffer; t2 = oldbracketbuffer;
            i = b->bracketfill;
            NCOPY(t1,t2,i)
            if ( oldbracketbuffer ) M_free(oldbracketbuffer,"old bracket buffer");
        }
    }
    if ( b->bracketfill+*term < b->bracketbuffersize ) {
        if ( b->indexfill >= b->indexbuffersize ) {
/*
            Enlarge index
*/
            BRACKETINDEX *oldindexbuffer = b->indexbuffer;
            b->indexbuffersize *= 2;
            b->indexbuffer = (BRACKETINDEX *)
                Malloc1(b->indexbuffersize*sizeof(BRACKETINDEX),"new bracket index");
            b1 = b->indexbuffer; b2 = oldindexbuffer;
            i = b->indexfill;
            NCOPY(b1,b2,i)
            if ( oldindexbuffer ) M_free(oldindexbuffer,"old bracket index");
        }
    }
    else {
/*
        We have too many brackets in the buffer. Try to improve.
        This is the interesting algorithm. We try to eliminate about 1/4 to
        1/2 of the brackets from the index. This should be done by size of
        the bracket contents to make the searching as fast as possible.
        But! Do not touch the last bracket.
        Note that we are always filling from the back.
        Algorithm: Throw away every second bracket, unless b1+b2 is much longer
        than average. How much is something we can tune.
*/
        average = DIVPOS(thepos,b->indexfill+1);
        if ( average <= 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Problems with bracket buffer. Increase MaxBracketBufferSize in form.set");
            MesPrint("Current size is %l",AM.MaxBracketBufferSize*sizeof(WORD));
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        average *= 4;  /* 2*2: one 2 for much longer, one 2 because we have pairs */
        t2 = b->bracketbuffer;
        b3 = b1 = b->indexbuffer;
        bi = b->indexbuffer + b->indexfill;
        b2 = b1+1;
        while ( b2+2 < bi ) {
            if ( DIFBASE(b2->next,b1->start) > average ) {
                t1 = b->bracketbuffer + b1->bracket;
```

```
                b1->bracket = t2 - b->bracketbuffer;
                i = *t1; NCOPY(t2,t1,i)
                *b3++ = *b1;
                t1 = b->bracketbuffer + b2->bracket;
                b2->bracket = t2 - b->bracketbuffer;
                i = *t1; NCOPY(t2,t1,i)
                *b3++ = *b2;
                if ( b3 <= b1 ) {
                    PUTZERO(b1->start);
                    PUTZERO(b1->next);
                    b1->bracket = 0;
                    b1->termsinbracket = 0;
                }
                if ( b3 <= b2 ) {
                    PUTZERO(b2->start);
                    PUTZERO(b2->next);
                    b2->bracket = 0;
                    b2->termsinbracket = 0;
                }
            }
            else {
                t1 = b->bracketbuffer + b1->bracket;
                b1->bracket = t2 - b->bracketbuffer;
                i = *t1; NCOPY(t2,t1,i)
                b1->next = b2->next;
                b1->termsinbracket += b2->termsinbracket;
                *b3++ = *b1;
                if ( b3 <= b1 ) {
                    PUTZERO(b1->start);
                    PUTZERO(b1->next);
                    b1->bracket = 0;
                    b1->termsinbracket = 0;
                }
                PUTZERO(b2->start);
                PUTZERO(b2->next);
                b2->bracket = 0;
                b2->termsinbracket = 0;
            }
            b1 += 2; b2 += 2;
        }
        while ( b1 < bi ) {
            t1 = b->bracketbuffer + b1->bracket;
            b1->bracket = t2 - b->bracketbuffer;
            i = *t1; NCOPY(t2,t1,i)
            *b3++ = *b1;
            if ( b3 <= b1 ) {
                PUTZERO(b1->start);
                PUTZERO(b1->next);
                b1->bracket = 0;
                b1->termsinbracket = 0;
            }
            b1++;
        }
        b->indexfill = b3 - b->indexbuffer;
        b->bracketfill = t2 - b->bracketbuffer;
    }
    bi = b->indexbuffer + b->indexfill;
    b->indexfill++;
    bi->bracket = b->bracketfill;
    bi->start = thepos;
    bi->next = thepos;
    bi->termsinbracket = 1;
/*
    Copy the bracket into the buffer
*/
    t1 = term; t2 = b->bracketbuffer + bi->bracket; i = *t1;
    b->bracketfill += i;
    NCOPY(t2,t1,i)
bracketdone:
    *term = oldsize; oldt[0] = a[0]; oldt[1] = a[1]; oldt[2] = a[2];
    AT.fromindex = 0;
}

/*
    #] PutBracketInIndex :
    #[ ClearBracketIndex :
*/

void ClearBracketIndex(WORD numexp)
{
    BRACKETINFO *b;
    if ( numexp >= 0 ) {
        b = Expressions[numexp].bracketinfo;
        Expressions[numexp].bracketinfo = 0;
    }
    else if ( numexp == -1 ) {
        GETIDENTITY
```

```
        b = AT.bracketinfo;
        AT.bracketinfo = 0;
    }
    else {
        numexp = -numexp-2;
        b = Expressions[numexp].newbracketinfo;
        Expressions[numexp].newbracketinfo = 0;
    }
    if ( b == 0 ) return;
    b->indexfill = b->indexbuffersize = 0;
    b->bracketfill = b->bracketbuffersize = 0;
    M_free(b->bracketbuffer,"ClearBracketBuffer");
    M_free(b->indexbuffer,"ClearIndexBuffer");
    M_free(b,"BracketInfo");
}

/*
    #] ClearBracketIndex :
    #[ OpenBracketIndex :

    Note: This routine is thread-safe
*/

void OpenBracketIndex(WORD nexpr)
{
    EXPRESSIONS e = Expressions + nexpr;
    BRACKETINFO *bi;
    LONG i;
    bi = (BRACKETINFO *)Malloc1(sizeof(BRACKETINFO),"BracketInfo");
    e->newbracketinfo = bi;
/*
    i = 20*AM.MaxTer/sizeof(WORD);
    if ( i < 1000 ) i = 1000;
*/
    i = 2000;
    bi->bracketbuffer = (WORD *)Malloc1(i*sizeof(WORD),"Bracket Buffer");
    bi->bracketbuffersize = i;
    bi->bracketfill = 0;
    i = 50;
    bi->indexbuffer = (BRACKETINDEX *)Malloc1(i*sizeof(BRACKETINDEX),"Bracket Index");
    bi->indexbuffersize = i;
    bi->indexfill = 0;
    bi->SortType = AC.SortType;
}

/*
    #] OpenBracketIndex :
    #[ PutInside :

    Puts a term, or a bracket determined part of a term inside a function.

    AT.WorkPointer points at term+*term
*/

int PutInside(PHEAD WORD *term, WORD *code)
{
    WORD *from, *to, *oldbuf, *tStop, *t, *tt, oldon, oldact, inc, argsize, *termout;
    int i, ii, error;

    if ( code[1] == 4 && ( code[2] == 0 || code[2] == 1 ) ) {
/*
        Put all inside. Move the term by 1+FUNHEAD+ARGHEAD
*/
        from = term+*term; to = from+1+ARGHEAD+FUNHEAD; i = ii = *term;
        to[0] = 1; to[1] = 1; to[2] = 3;
        while ( --i >= 0 ) *--to = *--from;
        to = term;
        *to++ = term[0]+4+ARGHEAD+FUNHEAD;
        *to++ = code[3];
        *to++ = ii+FUNHEAD+ARGHEAD;
        *to++ = 1;  /* set dirty flags, because there could be a fast notation */
        FILLFUN3(to)
        *to++ = ii+ARGHEAD;
        *to++ = 1;
        FILLARG(to)
        return(0);
    }
/*
    First we save the old bracket variables. Then we set variables to
    influence the PutBracket routine and call it.
    After that we set the values back and sort out the results by placing the
    inside of the bracket inside the function.
*/
    termout = AT.WorkPointer;
    oldbuf = AT.BrackBuf;
    oldon  = AR.BracketOn;
    oldact = AT.PolyAct;
```

```
    AR.BracketOn = -code[2];
    AT.BrackBuf  = code+4;
    AT.PolyAct = 0;
    error = PutBracket(BHEAD term);
    AT.PolyAct = oldact;
    AT.BrackBuf = oldbuf;
    AR.BracketOn = oldon;
    if ( error ) return(error);
    i = *termout; from = termout; to = term;
    NCOPY(to,from,i);
    tStop = term +*term; tStop -= tStop[-1];
    t = term+1;
    while ( t < tStop && *t != HAAKJE ) t += t[1];
    from = term + *term;
    inc = FUNHEAD+ARGHEAD-t[1]+1;
    tt = t + t[1];
    argsize = from-tt+1;
    to = from + inc;
    to[0] = 1;
    to[1] = 1;
    to[2] = 3;
    while ( from > tt ) *--to = *--from;
    *--to = argsize;
    *t++ = code[3];
    *t++ = argsize+FUNHEAD+ARGHEAD;
    *t++ = 1;
    FILLFUN3(t);
    *t++ = argsize+ARGHEAD;
    *t++ = 1;
    FILLARG(t);
    *term += inc+3;
    AT.WorkPointer = term+*term;
    if ( Normalize(BHEAD term) ) error = 1;
    return(error);
}

/*
    #] PutInside :
*/

/*
    The next routines are for indexing the local output files in a parallel
    sort. This indexing is needed to get a fast determination of the
    splitting terms needed to divide the terms evenly over the processors.
    Actually this method works well for ParFORM, but may not work well
    for TFORM.

    #[ PutTermInIndex :

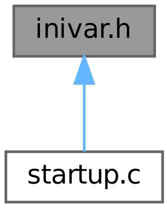
    Puts a term in the term index.
    Action:
        if the index hasn't reached its full size
            if there is room, put the term
            if there is no room: extend the buffer, put the term
        else
            check if the last term has a number of the type skip*m+1
                if no, overwrite the last term
                if yes, check whether there is room for one more term
                    yes: add the term
                    no: drop all even terms, compress the list,
                        multiply skip by 2, and add this term.

int PutTermInIndex(WORD *term,POSITION *position)
{
    return(0);
}

    #] PutTermInIndex :
*/
```

4.64 inivar.h File Reference

This graph shows which files directly or indirectly include this file:



Data Structures

- struct **fixedfun**

Variables

- FIXEDGLOBALS FG
- ALLGLOBALS A
- FIXEDSET fixedsets [ ]
- UBYTE BufferForOutput [MAXLINELENGTH+14]
- char ∗ setupfilename = "form.set"

4.64.1 Detailed Description

Contains the initialization of a number of structs at compile time This file should only be included in the file startup.c !!!

Definition in file inivar.h.

4.64.2 Variable Documentation

4.64.2.1 FG

```
FIXEDGLOBALS FG
```

Definition at line 34 of file inivar.h.

4.64.2.2 A

```
ALLGLOBALS A
```

Definition at line 133 of file inivar.h.

4.64.2.3 fixedsets

```
FIXEDSET fixedsets[]
```

Initial value:

```
= {
      {"pos_",    "integers > 0",  CSYMBOL, 0}
     ,{"pos0_",   "integers >= 0", CSYMBOL, 0}
     ,{"neg_",    "integers < 0",  CSYMBOL, 0}
     ,{"neg0_",   "integers <= 0", CSYMBOL, 0}
     ,{"even_",   "even integers", CSYMBOL, 0}
     ,{"odd_",    "odd integers",  CSYMBOL, 0}
     ,{"int_",    "all integers",  CSYMBOL, 0}
     ,{"symbol_","only symbols",  CSYMBOL, MAXPOSITIVE}
     ,{"fixed_", "fixed indices", CINDEX, 0}
     ,{"index_", "all indices",   CINDEX, 0}
     ,{"number_","all rationals", CSYMBOL, 0}
     ,{"dummyindices_", "dummy indices", CINDEX, 0}
     ,{"vector_","only vectors",  CVECTOR, 0}
}
```

Definition at line 258 of file inivar.h.

4.64.2.4 BufferForOutput

```
UBYTE BufferForOutput[MAXLINELENGTH+14]
```

Definition at line 274 of file inivar.h.

4.64.2.5 setupfilename

```
char* setupfilename = "form.set"
```

Definition at line 276 of file inivar.h.

4.65 inivar.h

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

```

```
FIXEDGLOBALS FG = {
         {Traces                /* Dummy as zeroeth argument */
        ,Traces
        ,EpfCon
        ,RatioGen
        ,SymGen
        ,TenVec
        ,DoSumF1
        ,DoSumF2}

        ,{TraceFind             /* Dummy as zeroeth argument */
        ,TraceFind
        ,EpfFind
        ,RatioFind
        ,SymFind
        ,TenVecFind}

        ,{"symbol"
        ,"index"
        ,"vector"
        ,"function"
        ,"set"
        ,"expression"
        ,"dotproduct"
        ,"number"
        ,"subexp"
        ,"delta"}

        ,{"(local)"
        ,"(skip/local)"
        ,"(drop/local)"
        ,"(dropped)"
        ,"(global)"
        ,"(skip/global)"
        ,"(drop/global)"
        ,"(dropped)"
        ,"(stored)"
        ,"(local-hidden)"
        ,"(local-to be hidden)"
        ,"(local-hidden-dropped)"
        ,"(local-to be unhidden)"
        ,"(global-hidden)"
        ,"(global-to be hidden)"
        ,"(global-hidden-dropped)"
        ,"(global-to be unhidden)"
        ,"(into-hide-local)"
        ,"(into-hide-global)"
        ,"(spectator)"
        ,"(drop/spectator)"}

        ,{" Functions"
        ," Commuting Functions"}

        ,{"left     "
        ,"active   "
        ,"in output"}

    ,(char *)0
    ,(char *)0
    ,(UBYTE *)"1"
    ,(WORD)0
    ,(WORD)0
    ,(WORD)0
    ,(WORD)0

/*  ASCII table of character types. Note that on some computers this
    table may be different from the ASCII table.

    -1  Illegal character
     0  Alphabetic character
     1  Digit
     2  . $ _ ? # or '
     3  [,]
     4  ( ) = ; or ,
     5  + - * % / ^ :
     6  blank, tab, linefeed
     7  {,|,}
     8  ! & < >
     9  "
    10  The ultimate end.
*/
    ,{    10,255,255,255,255,255,255,255,255,  6,  6,255,255,  6,255,255,
         255,255,255,255,255,255,255,255,255,255, 10,255,255,255,255,255,
           6,  8,  9,  2,  2,  5,  8,  2,  4,  4,  5,  5,  4,  5,  2,  5,
           1,  1,  1,  1,  1,  1,  1,  1,  1,  1,  5,  4,  8,  4,  8,  2,
         255,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,
           0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  3,255,  3,  5,  2,
```

```
        255,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,
          0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   0,   7,   7,   7,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,
        255,255,255,255,255,255,255,255,255,255,255,255,255,255,255,255 }
};

ALLGLOBALS A;
#ifdef WITHPTHREADS
ALLPRIVATES **AB;
#endif

static struct fixedfun {
    char *name;
    int      commu ,tensor          ,complx          ,symmetric;
} fixedfunctions[] = {
     {"exp_"     ,1 ,0               ,0                ,0} /* EXPONENT */
    ,{"denom_"   ,1 ,0               ,0                ,0} /* DENOMINATOR */
    ,{"setfun_"  ,1 ,0               ,0                ,0} /* SETFUNCTION */
    ,{"g_"       ,1 ,GAMMAFUNCTION   ,VARTYPEIMAGINARY,0} /* GAMMA */
    ,{"gi_"      ,1 ,GAMMAFUNCTION   ,VARTYPEIMAGINARY,0} /* GAMMAI */
    ,{"g5_"      ,1 ,GAMMAFUNCTION   ,VARTYPEIMAGINARY,0} /* GAMMAFIVE */
    ,{"g6_"      ,1 ,GAMMAFUNCTION   ,VARTYPEIMAGINARY,0} /* GAMMASIX */
    ,{"g7_"      ,1 ,GAMMAFUNCTION   ,VARTYPEIMAGINARY,0} /* GAMMASEVEN */
    ,{"sum_"     ,1 ,0               ,0                ,0} /* SUMF1 */
    ,{"sump_"    ,1 ,0               ,0                ,0} /* SUMF2 */
    ,{"dum_"     ,1 ,0               ,0                ,0} /* DUMFUN */
    ,{"replace_" ,0 ,0               ,0                ,0} /* REPLACEMENT */
    ,{"reverse_" ,1 ,0               ,0                ,0} /* REVERSEFUNCTION */
    ,{"distrib_" ,1 ,0               ,0                ,0} /* DISTRIBUTION */
    ,{"dd_"      ,0 ,TENSORFUNCTION  ,0                ,0} /* DELTA3 */
    ,{"dummy_"   ,0 ,0               ,0                ,0} /* DUMMYFUN */
    ,{"dummyten_",0 ,TENSORFUNCTION  ,0                ,0} /* DUMMYTEN */
    ,{"e_"       ,0 ,TENSORFUNCTION|1 ,VARTYPEIMAGINARY,ANTISYMMETRIC}
                                                         /* LEVICIVITA */
    ,{"fac_"     ,0 ,0               ,0                ,0} /* FACTORIAL */
    ,{"invfac_"  ,0 ,0               ,0                ,0} /* INVERSEFACTORIAL */
    ,{"binom_"   ,0 ,0               ,0                ,0} /* BINOMIAL */
    ,{"nargs_"   ,0 ,0               ,0                ,0} /* NUMARGSFUN */
    ,{"sign_"    ,0 ,0               ,0                ,0} /* SIGNFUN */
    ,{"mod_"     ,0 ,0               ,0                ,0} /* MODFUNCTION */
    ,{"mod2_"    ,0 ,0               ,0                ,0} /* MOD2FUNCTION */
    ,{"min_"     ,0 ,0               ,0                ,0} /* MINFUNCTION */
    ,{"max_"     ,0 ,0               ,0                ,0} /* MAXFUNCTION */
    ,{"abs_"     ,0 ,0               ,0                ,0} /* ABSFUNCTION */
    ,{"sig_"     ,0 ,0               ,0                ,0} /* SIGFUNCTION */
    ,{"integer_" ,0 ,0               ,0                ,0} /* INTFUNCTION */
    ,{"theta_"   ,0 ,0               ,0                ,0} /* THETA */
    ,{"thetap_"  ,0 ,0               ,0                ,0} /* THETA2 */
    ,{"delta_"   ,0 ,0               ,0                ,0} /* DELTA2 */
    ,{"deltap_"  ,0 ,0               ,0                ,0} /* DELTAP */
    ,{"bernoulli_",0,0               ,0                ,0} /* BERNOULLIFUNCTION */
    ,{"count_"   ,0 ,0               ,0                ,0} /* COUNTFUNCTION */
    ,{"match_"   ,0 ,0               ,0                ,0} /* MATCHFUNCTION */
    ,{"pattern_" ,0 ,0               ,VARTYPECOMPLEX   ,0} /* PATTERNFUNCTION */
    ,{"term_"    ,1 ,0               ,0                ,0} /* TERMFUNCTION */
    ,{"conjg_"   ,1 ,0               ,VARTYPECOMPLEX   ,0} /* CONJUGATEFUNCTION */
    ,{"root_"    ,0 ,0               ,0                ,0} /* ROOTFUNCTION */
    ,{"table_"   ,1 ,0               ,0                ,0} /* TABLEFUNCTION */
    ,{"firstbracket_",0 ,0           ,0                ,0} /* FIRSTBRACKET */
    ,{"termsin_" ,0 ,0               ,0                ,0} /* TERMSINEXPR */
    ,{"nterms_"  ,0 ,0               ,0                ,0} /* NUMTERMSFUN */
    ,{"gcd_"     ,0 ,0               ,0                ,0} /* GCDFUNCTION */
    ,{"div_"        ,0 ,0            ,0                ,0} /* DIVFUNCTION */
    ,{"rem_"        ,0 ,0            ,0                ,0} /* REMFUNCTION */
    ,{"maxpowerof_",0 ,0             ,0                ,0} /* MAXPOWEROF */
    ,{"minpowerof_",0 ,0             ,0                ,0} /* MINPOWEROF */
    ,{"tbl_"     ,0 ,0               ,0                ,0} /* TABLESTUB */
    ,{"factorin_",0 ,0               ,0                ,0} /* FACTORIN */
    ,{"termsinbracket_",0 ,0         ,0                ,0} /* TERMSINBRACKET */
    ,{"farg_"    ,0 ,0               ,0                ,0} /* WILDARGFUN */
/*
     The following names are reserved for the floating point library.
     As long as we have no floating point numbers they do not do anything.
*/
    ,{"sqrt_"    ,0 ,0               ,0                ,0} /* SQRTFUNCTION */
    ,{"ln_"      ,0 ,0               ,0                ,0} /* LNFUNCTION */
    ,{"sin_"     ,0 ,0               ,0                ,0} /* SINFUNCTION */
    ,{"cos_"     ,0 ,0               ,0                ,0} /* COSFUNCTION */
    ,{"tan_"     ,0 ,0               ,0                ,0} /* TANFUNCTION */
    ,{"asin_"    ,0 ,0               ,0                ,0} /* ASINFUNCTION */
    ,{"acos_"    ,0 ,0               ,0                ,0} /* ACOSFUNCTION */
```

```
    ,{"atan_"    ,0 ,0               ,0                ,0} /* ATANFUNCTION */
    ,{"atan2_"   ,0 ,0               ,0                ,0} /* ATAN2FUNCTION */
    ,{"sinh_"    ,0 ,0               ,0                ,0} /* SINHFUNCTION */
    ,{"cosh_"    ,0 ,0               ,0                ,0} /* COSHFUNCTION */
    ,{"tanh_"    ,0 ,0               ,0                ,0} /* TANHFUNCTION */
    ,{"asinh_"   ,0 ,0               ,0                ,0} /* ASINHFUNCTION */
    ,{"acosh_"   ,0 ,0               ,0                ,0} /* ACOSHFUNCTION */
    ,{"atanh_"   ,0 ,0               ,0                ,0} /* ATANHFUNCTION */
    ,{"li2_"     ,0 ,0               ,0                ,0} /* LI2FUNCTION */
    ,{"lin_"     ,0 ,0               ,0                ,0} /* LINFUNCTION */
/*
     From here on we continue with new functions (26-sep-2010)
*/
    ,{"extrasymbol_",0 ,0            ,0                ,0} /* EXTRASYMFUN */
    ,{"random_"     ,0 ,0            ,0                ,0} /* RANDOMFUNCTION */
    ,{"ranperm_"    ,1 ,0            ,0                ,0} /* RANPERM */
    ,{"numfactors_" ,0 ,0            ,0                ,0} /* NUMFACTORS */
    ,{"firstterm_"  ,0 ,0            ,0                ,0} /* FIRSTTERM */
    ,{"content_"    ,0 ,0            ,0                ,0} /* CONTENTTERM */
    ,{"prime_"      ,0 ,0            ,0                ,0} /* PRIMENUMBER */
    ,{"exteuclidean_",0 ,0           ,0                ,0} /* EXTEUCLIDEAN */
    ,{"makerational_",0 ,0           ,0                ,0} /* MAKERATIONAL */
    ,{"inverse_"    ,0 ,0            ,0                ,0} /* INVERSEFUNCTION */
    ,{"id_"         ,1 ,0            ,0                ,0} /* IDFUNCTION */
    ,{"putfirst_"   ,1 ,0            ,0                ,0} /* PUTFIRST */
    ,{"perm_"       ,1 ,0            ,0                ,0} /* PERMUTATIONS */
    ,{"partitions_" ,1 ,0            ,0                ,0} /* PARTITIONS */
    ,{"mul_"        ,0 ,0            ,0                ,0} /* MULFUNCTION */
    ,{"moebius_"    ,0 ,0            ,0                ,0} /* MOEBIUS */
    ,{"topologies_" ,0 ,0            ,0                ,0} /* TOPOLOGIES */
    ,{"diagrams_"   ,0 ,0            ,0                ,0} /* DIAGRAMS */
    ,{"topo_"       ,0 ,0            ,0                ,0} /* TOPO */
    ,{"node_"       ,0 ,0            ,0                ,0} /* VERTEX */
    ,{"edge_"       ,0 ,0            ,0                ,0} /* EDGE */
    ,{"sizeof_"     ,0 ,0            ,0                ,0} /* SIZEOFFUNCTION */
    ,{"block_"      ,0 ,0            ,0                ,0} /* BLOCK */
    ,{"onepi_"      ,0 ,0            ,0                ,0} /* ONEPI */
    ,{"phi_"        ,0 ,VERTEXFUNCTION,0               ,0} /* PHI */
#ifdef WITHFLOAT
    ,{"float_"      ,0 ,0            ,0                ,0} /* FLOATFUN */
    ,{"tofloat_"    ,0 ,0            ,0                ,0} /* TOFLOAT */
    ,{"torat_"      ,0 ,0            ,0                ,0} /* TORAT */
    ,{"mzv_"        ,0 ,0            ,0                ,0} /* MZV */
    ,{"euler_"      ,0 ,0            ,0                ,0} /* EULER */
    ,{"mzvhalf_"    ,0 ,0            ,0                ,0} /* MZVHALF */
    ,{"agm_"        ,0 ,0            ,0                ,0} /* AGMFUNCTION */
    ,{"gamma_"      ,0 ,0            ,0                ,0} /* GAMMAFUN */
#endif
};

FIXEDSET fixedsets[] = {
     {"pos_",    "integers > 0",  CSYMBOL, 0}  /* POS_    0  */
    ,{"pos0_",   "integers >= 0", CSYMBOL, 0}  /* POS0_   1  */
    ,{"neg_",    "integers < 0",  CSYMBOL, 0}  /* NEG_    2  */
    ,{"neg0_",   "integers <= 0", CSYMBOL, 0}  /* NEG0_   3  */
    ,{"even_",   "even integers", CSYMBOL, 0}  /* EVEN_   4  */
    ,{"odd_",    "odd integers",  CSYMBOL, 0}  /* ODD_    5  */
    ,{"int_",    "all integers",  CSYMBOL, 0}  /* Z_      6  */
    ,{"symbol_","only symbols",   CSYMBOL, MAXPOSITIVE}  /* SYMBOL_ 7  */
    ,{"fixed_", "fixed indices", CINDEX, 0}    /* FIXED_  8  */
    ,{"index_", "all indices",    CINDEX, 0}   /* INDEX_  9  */
    ,{"number_","all rationals", CSYMBOL, 0}   /* Q_     10  */
    ,{"dummyindices_", "dummy indices", CINDEX, 0}   /* DUMMYINDEX_ 11  */
    ,{"vector_","only vectors",   CVECTOR, 0}  /* VECTOR_ 12  */
};

UBYTE BufferForOutput[MAXLINELENGTH+14];

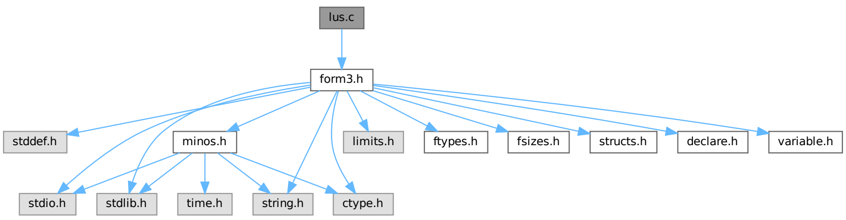
char *setupfilename = "form.set";

#ifdef WITHPTHREADS
INILOCK(ErrorMessageLock)
INILOCK(FileReadLock)
#endif
```

4.66 lus.c File Reference

`#include "form3.h"`

Include dependency graph for lus.c:



Functions

- int Lus (WORD *term, WORD funnum, WORD loopsize, WORD numargs, WORD outfun, WORD mode)
- int FindLus (int from, int level, int openindex)
- int SortTheList (int *slist, int num)
- WORD AllLoops (PHEAD WORD *term, WORD level)
- LONG StartLoops (PHEAD WORD *term, WORD level, LONG vert, WORD nvert, WORD *arglist, WORD nargs, WORD *loop, WORD nloop)
- LONG GenLoops (PHEAD WORD *term, WORD level, LONG vert, WORD nvert, WORD *arglist, WORD nargs, WORD *loop, WORD nloop)
- void LoopOutput (PHEAD WORD *term, WORD level, WORD *loop, WORD nloop)
- WORD AllPaths (PHEAD WORD *term, WORD level)
- LONG GenPaths (PHEAD WORD *term, WORD level, LONG vert, WORD nvert, WORD *arglist, WORD nargs, WORD *path, WORD npath)
- void PathOutput (PHEAD WORD *term, WORD level, WORD *path, WORD npath)
- WORD AllOnePI (WORD *term, WORD level)
- int RemoveBridges (void)
- int TakeOneLine (WORD *term, WORD level)
- int OutputOnePI (PHEAD WORD *term, WORD level)

4.66.1 Detailed Description

Routines to find loops in index contractions. These routines allow for a category of topological statements. They were originally developed for the color library.

Definition in file lus.c.

4.66.2 Function Documentation

4.66.2.1 Lus()

```
int Lus (
            WORD * term,
            WORD funnum,
            WORD loopsize,
            WORD numargs,
            WORD outfun,
            WORD mode )
```

Definition at line 57 of file lus.c.

4.66.2.2 FindLus()

```
int FindLus (
        int from,
        int level,
        int openindex )
```

Definition at line 515 of file lus.c.

4.66.2.3 SortTheList()

```
int SortTheList (
        int * slist,
        int num )
```

Definition at line 578 of file lus.c.

4.66.2.4 AllLoops()

```
WORD AllLoops (
        PHEAD WORD * term,
        WORD level )
```

Definition at line 650 of file lus.c.

4.66.2.5 StartLoops()

```
LONG StartLoops (
        PHEAD WORD * term,
        WORD level,
        LONG vert,
        WORD nvert,
        WORD * arglist,
        WORD nargs,
        WORD * loop,
        WORD nloop )
```

Definition at line 894 of file lus.c.

4.66.2.6 GenLoops()

```
LONG GenLoops (
        PHEAD WORD * term,
        WORD level,
        LONG vert,
        WORD nvert,
        WORD * arglist,
        WORD nargs,
        WORD * loop,
        WORD nloop )
```

Definition at line 980 of file lus.c.

4.66.2.7 LoopOutput()

```
void LoopOutput (
            PHEAD WORD * term,
            WORD level,
            WORD * loop,
            WORD nloop )
```

Definition at line 1059 of file lus.c.

4.66.2.8 AllPaths()

```
WORD AllPaths (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1123 of file lus.c.

4.66.2.9 GenPaths()

```
LONG GenPaths (
            PHEAD WORD * term,
            WORD level,
            LONG vert,
            WORD nvert,
            WORD * arglist,
            WORD nargs,
            WORD * path,
            WORD npath )
```

Definition at line 1413 of file lus.c.

4.66.2.10 PathOutput()

```
void PathOutput (
            PHEAD WORD * term,
            WORD level,
            WORD * path,
            WORD npath )
```

Definition at line 1486 of file lus.c.

4.66.2.11 AllOnePI()

```
WORD AllOnePI (
            WORD * term,
            WORD level )
```

Definition at line 1566 of file lus.c.

4.66.2.12 RemoveBridges()

```
int RemoveBridges (
            void )
```

Definition at line 1586 of file lus.c.

4.66.2.13 TakeOneLine()

```
int TakeOneLine (
            WORD * term,
            WORD level )
```

Definition at line 1596 of file lus.c.

4.66.2.14 OutputOnePI()

```
int OutputOnePI (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1608 of file lus.c.

4.67 lus.c

Go to the documentation of this file.

```
00001
00007 /* #[ License : */
00008 /*
00009  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00010  *   When using this file you are requested to refer to the publication
00011  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012  *   This is considered a matter of courtesy as the development was paid
00013  *   for by FOM the Dutch physics granting agency and we would like to
00014  *   be able to track its scientific use to convince FOM of its value
00015  *   for the community.
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00017  *   This file is part of FORM.
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00027  *   details.
00028  *
00029  *   You should have received a copy of the GNU General Public License along
00030  *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00031  */
00032 /* #] License : */
00033 /*
00034     #[ Includes : lus.c
00035 */
00036
00037 #include "form3.h"
00038
00039 /*
00040     #] Includes :
00041     #[ Lus :
00042
00043     Routine to find loops.
00044     Mode: 0: Just tell whether there is such a loop.
```

```
        1: Take out the functions and replace by outfun with the
           remaining arguments of the loop function
        > AM.OffsetIndex: This index must be included in the loop.
        < -AM.OffsetIndex: This index must be included in the loop. Replace.
    Return value: 0: no loop. 1: there is/was such a loop.
    funnum: the function(s) in which we look for a loop.
    numargs: the number of arguments admissible in the function.
    outfun: the output function in case of a substitution.
    loopsize: the size of the loop we are looking for.
              if < 0 we look for all loops.
*/

int Lus(WORD *term, WORD funnum, WORD loopsize, WORD numargs, WORD outfun, WORD mode)
{
    GETIDENTITY
    WORD *w, *t, *tt, *m, *r, **loc, *tstop, minloopsize;
    int nfun, i, j, jj, k, n, sign = 0, action = 0, L, ten, ten2, totnum,
    sign2, *alist, *wi, mini, maxi, medi = 0;
    if ( numargs <= 1 ) return(0);
    GETSTOP(term,tstop);
/*
    First count the number of functions with the proper number of arguments.
*/
    t = term+1; nfun = 0;
    if ( ( ten = functions[funnum-FUNCTION].spec ) >= TENSORFUNCTION ) {
        while ( t < tstop ) {
            if ( *t == funnum && t[1] == FUNHEAD+numargs ) { nfun++; }
            t += t[1];
        }
    }
    else {
        while ( t < tstop ) {
            if ( *t == funnum ) {
                i = 0; m = t+FUNHEAD; t += t[1];
                while ( m < t ) { i++; NEXTARG(m) }
                if ( i == numargs ) nfun++;
            }
            else t += t[1];
        }
    }
    if ( loopsize < 0 ) minloopsize = 2;
    else                minloopsize = loopsize;
    if ( funnum < minloopsize ) return(0); /* quick abort */
    if ( ((functions[funnum-FUNCTION].symmetric) & ~REVERSEORDER) == ANTISYMMETRIC ) sign = 1;
    if ( mode == 1 || mode < 0 ) {
        ten2 = functions[outfun-FUNCTION].spec >= TENSORFUNCTION;
    }
    else ten2 = -1;
/*
    Allocations:
*/
    if ( AN.numflocs < funnum ) {
        if ( AN.funlocs ) M_free(AN.funlocs,"Lus: AN.funlocs");
        AN.numflocs = funnum;
        AN.funlocs = (WORD **)Malloc1(sizeof(WORD *)*AN.numflocs,"Lus: AN.funlocs");
    }
    if ( AN.numfargs < funnum*numargs ) {
        if ( AN.funargs ) M_free(AN.funargs,"Lus: AN.funargs");
        AN.numfargs = funnum*numargs;
        AN.funargs = (int *)Malloc1(sizeof(int *)*AN.numfargs,"Lus: AN.funargs");
    }
/*
    Make a list of relevant indices
*/
    alist = AN.funargs; loc = AN.funlocs;
    t = term+1;
    if ( ten >= TENSORFUNCTION ) {
        while ( t < tstop ) {
            if ( *t == funnum && t[1] == FUNHEAD+numargs ) {
                *loc++ = t;
                t += FUNHEAD;
                j = i = numargs; while ( --i >= 0 ) {
                    if ( *t >= AM.OffsetIndex &&
                        ( *t >= AM.OffsetIndex+WILDOFFSET ||
                        indices[*t-AM.OffsetIndex].dimension != 0 ) ) {
                        *alist++ = *t++; j--;
                    }
                    else t++;
                }
                while ( --j >= 0 ) *alist++ = -1;
            }
            else t += t[1];
        }
    }
    else {
        nfun = 0;
        while ( t < tstop ) {
```

```
            if ( *t == funnum ) {
                w = t;
                i = 0; m = t+FUNHEAD; t += t[1];
                while ( m < t ) { i++; NEXTARG(m) }
                if ( i == numargs ) {
                    m = w + FUNHEAD;
                    while ( m < t ) {
                        if ( *m == -INDEX && m[1] >= AM.OffsetIndex &&
                        ( m[1] >= AM.OffsetIndex+WILDOFFSET ||
                        indices[m[1]-AM.OffsetIndex].dimension != 0 ) ) {
                            *alist++ = m[1]; m += 2; i--;
                        }
                        else if ( ten2 >= TENSORFUNCTION && *m != -INDEX
                         && *m != -VECTOR && *m != -MINVECTOR &&
                        ( *m != -SNUMBER || *m < 0 || *m >= AM.OffsetIndex ) ) {
                            i = numargs; break;
                        }
                        else { NEXTARG(m) }
                    }
                    if ( i < numargs ) {
                        *loc++ = w;
                        nfun++;
                        while ( --i >= 0 ) *alist++ = -1;
                    }
                }
            }
            else t += t[1];
        }
        if ( nfun < minloopsize ) return(0);
    }
/*
    We have now nfun objects. Not all indices may be usable though.
    If the list is not long, we use a quadratic algorithm to remove
    indices and vertices that cannot be used. If it becomes large we
    sort the list of available indices (and their multiplicity) and
    work with binary searches.
*/
    alist = AN.funargs; totnum = numargs*nfun;
    if ( nfun > 7 ) {
        if ( AN.funisize < totnum ) {
            if ( AN.funinds ) M_free(AN.funinds,"AN.funinds");
            AN.funisize = (totnum*3)/2;
            AN.funinds = (int *)Malloc1(AN.funisize*2*sizeof(int),"AN.funinds");
        }
        i = totnum; n = 0; wi = AN.funinds;
        while ( --i >= 0 ) {
            if ( *alist >= 0 ) { n++; *wi++ = *alist; *wi++ = 1; }
            alist++;
        }
        n = SortTheList(AN.funinds,n);
        do {
            action = 0;
            for ( i = 0; i < nfun; i++ ) {
                alist = AN.funargs + i*numargs;
                jj = numargs;
                for ( j = 0; j < jj; j++ ) {
                    if ( alist[j] < 0 ) break;
                    mini = 0; maxi = n-1;
                    while ( mini <= maxi ) {
                        medi = (mini + maxi) / 2; k = AN.funinds[2*medi];
                        if ( alist[j] > k ) mini = medi + 1;
                        else if ( alist[j] < k ) maxi = medi - 1;
                        else break;
                    }
                    if ( AN.funinds[2*medi+1] <= 1 ) {
                        (AN.funinds[2*medi+1])--;
                        jj--; k = j; while ( k < jj ) { alist[k] = alist[k+1]; k++; }
                        alist[jj] = -1; j--;
                    }
                }
                if ( jj < 2 ) {
                    if ( jj == 1 ) {
                        mini = 0; maxi = n-1;
                        while ( mini <= maxi ) {
                            medi = (mini + maxi) / 2; k = AN.funinds[2*medi];
                            if ( alist[0] > k ) mini = medi + 1;
                            else if ( alist[0] < k ) maxi = medi - 1;
                            else break;
                        }
                        (AN.funinds[2*medi+1])--;
                        if ( AN.funinds[2*medi+1] == 1 ) action++;
                    }
                    nfun--; totnum -= numargs; AN.funlocs[i] = AN.funlocs[nfun];
                    wi = AN.funargs + nfun*numargs;
                    for ( j = 0; j < numargs; j++ ) alist[j] = *wi++;
                    i--;
                }
```

```
            }
        } while ( action );
    }
    else {
        for ( i = 0; i < totnum; i++ ) {
            if ( alist[i] == -1 ) continue;
            for ( j = 0; j < totnum; j++ ) {
                if ( alist[j] == alist[i] && j != i ) break;
            }
            if ( j >= totnum ) alist[i] = -1;
        }
        do {
            action = 0;
            for ( i = 0; i < nfun; i++ ) {
                alist = AN.funargs + i*numargs;
                n = numargs;
                for ( k = 0; k < n; k++ ) {
                    if ( alist[k] < 0 ) { alist[k--] = alist[--n]; alist[n] = -1; }
                }
                if ( n <= 1 ) {
                    if ( n == 1 ) { j = alist[0]; }
                    else j = -1;
                    nfun--; totnum -= numargs; AN.funlocs[i] = AN.funlocs[nfun];
                    wi = AN.funargs + nfun * numargs;
                    for ( k = 0; k < numargs; k++ ) alist[k] = wi[k];
                    i--;
                    if ( j >= 0 ) {
                        for ( k = 0, jj = 0, wi = AN.funargs; k < totnum; k++, wi++ ) {
                            if ( *wi == j ) { jj++; if ( jj > 1 ) break; }
                        }
                        if ( jj <= 1 ) {
                            for ( k = 0, wi = AN.funargs; k < totnum; k++, wi++ ) {
                                if ( *wi == j ) { *wi = -1; action = 1; }
                            }
                        }
                    }
                }
            }
        } while ( action );
    }
    if ( nfun < minloopsize ) return(0);
/*
    Now we have nfun objects, each with at least 2 indices, each of which
    occurs at least twice in our list. There will be a loop!
*/
    if ( mode != 0 && mode != 1 ) {
        if ( mode > 0 ) AN.tohunt =  mode - 5;
        else            AN.tohunt = -mode - 5;
        AN.nargs = numargs; AN.numoffuns = nfun;
        i = 0;
        if ( loopsize < 0 ) {
            if ( loopsize == -1 ) k = nfun;
            else { k = -loopsize-1; if ( k > nfun ) k = nfun; }
            for ( L = 2; L <= k; L++ ) {
                if ( FindLus(0,L,AN.tohunt) ) goto Success;
            }
        }
        else if ( FindLus(0,loopsize,AN.tohunt) ) { L = loopsize; goto Success; }
    }
    else {
        AN.nargs = numargs; AN.numoffuns = nfun;
        if ( loopsize < 0 ) {
            jj = 2; k = nfun;
            if ( loopsize < -1 ) { k = -loopsize-1; if ( k > nfun ) k = nfun; }
        }
        else { jj = k = loopsize; }
        for ( L = jj; L <= k; L++ ) {
            for ( i = 0; i <= nfun-L; i++ ) {
                alist = AN.funargs + i * numargs;
                for ( jj = 0; jj < numargs; jj++ ) {
                    if ( alist[jj] < 0 ) continue;
                    AN.tohunt = alist[jj];
                    for ( j = jj+1; j < numargs; j++ ) {
                        if ( alist[j] < 0 ) continue;
                        if ( FindLus(i+1,L-1,alist[j]) ) {
                            alist[0] = alist[jj];
                            alist[1] = alist[j];
                            goto Success;
                        }
                    }
                }
            }
        }
    }
    return(0);
Success:;
    if ( mode == 0 || mode > 1 ) return(1);
```

```
/*
    Now we have to make the replacement and fix the potential sign
*/
    sign2 = 1;
    wi = AN.funargs + i*numargs; loc = AN.funlocs + i;
    for ( k = 0; k < L; k++ ) *(loc[k]) = -1;
    if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
    w = AT.WorkPointer + 1;
    m = t = term + 1;
    while ( t < tstop ) {
        if ( *t == -1 ) break;
        t += t[1];
    }
    while ( m < t ) *w++ = *m++;
    r = w;
    *w++ = outfun;
    w++;
    *w++ = DIRTYFLAG;
    FILLFUN3(w)
    if ( functions[outfun-FUNCTION].spec >= TENSORFUNCTION ) {
        if ( ten >= TENSORFUNCTION ) {
            for ( i = 0; i < L; i++ ) {
                alist = wi + i*numargs;
                m = loc[i] + FUNHEAD;
                for ( k = 0; k < numargs; k++ ) {
                    if ( m[k] == alist[0] ) {
                        if ( k != 0 ) {
                            jj = m[k]; m[k] = m[0]; m[0] = jj;
                            sign = -sign;
                        }
                        break;
                    }
                }
                for ( k = 1; k < numargs; k++ ) {
                    if ( m[k] == alist[1] ) {
                        if ( k != 1 ) {
                            jj = m[k]; m[k] = m[1]; m[1] = jj;
                            sign = -sign;
                        }
                        break;
                    }
                }
                m += 2;
                for ( k = 2; k < numargs; k++ ) *w++ = *m++;
            }
        }
        else {
            WORD *t1, *t2, *t3;
            for ( i = 0; i < L; i++ ) {
                alist = wi + i*numargs;
                tt = loc[i];
                m = tt + FUNHEAD;
                for ( k = 0; k < numargs; k++ ) {
                    if ( *m == -INDEX && m[1] == alist[0] ) {
                        if ( k != 0 ) {
                            if ( ( k & 1 ) != 0 ) sign = -sign;
/*
                            now move to position 0
*/
                            t2 = m+2; t1 = m; t3 = tt+FUNHEAD;
                            while ( t1 > t3 ) { *--t2 = *--t1; }
                            t3[0] = -INDEX; t3[1] = alist[0];
                        }
                        break;
                    }
                    NEXTARG(m)
                }
                m = tt + FUNHEAD + 2;
                for ( k = 1; k < numargs; k++ ) {
                    if ( *m == -INDEX && m[1] == alist[1] ) {
                        if ( k != 1 ) {
                            if ( ( k & 1 ) == 0 ) sign = -sign;
/*
                            now move to position 1
*/
                            t2 = m+2; t1 = m; t3 = tt+FUNHEAD+2;
                            while ( t1 > t3 ) { *--t2 = *--t1; }
                            t3[0] = -INDEX; t3[1] = alist[1];
                        }
                        break;
                    }
                    NEXTARG(m)
                }
/*
                now copy the remaining arguments to w
                keep in mind that the output function is a tensor!
*/
```

```
                t1 = tt + FUNHEAD + 4;
                t2 = tt + tt[1];
                while ( t1 < t2 ) {
                    if ( *t1 == -INDEX || *t1 == -VECTOR ) {
                        *w++ = t1[1]; t1 += 2;
                    }
                    else if ( *t1 == -MINVECTOR ) {
                        *w++ = t1[1]; t1 += 2; sign2 = -sign2;
                    }
                    else if ( ( *t1 == -SNUMBER ) && ( t1[1] >= 0 ) && ( t1[1] < AM.OffsetIndex ) ) {
                        *w++ = t1[1]; t1 += 2; sign2 = -sign2;
                    }
                    else {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Illegal attempt to use a non-index-like argument in a tensor in
      ReplaceLoop statement");
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                    }
                }
            }
        }
    }
    else {
        if ( ten >= TENSORFUNCTION ) {
            for ( i = 0; i < L; i++ ) {
                alist = wi + i*numargs;
                m = loc[i] + FUNHEAD;
                for ( k = 0; k < numargs; k++ ) {
                    if ( m[k] == alist[0] ) {
                        if ( k != 0 ) {
                            jj = m[k]; m[k] = m[0]; m[0] = jj;
                            sign = -sign;
                            break;
                        }
                    }
                }
                for ( k = 1; k < numargs; k++ ) {
                    if ( m[k] == alist[1] ) {
                        if ( k != 1 ) {
                            jj = m[k]; m[k] = m[1]; m[1] = jj;
                            sign = -sign;
                            break;
                        }
                    }
                }
                m += 2;
                for ( k = 2; k < numargs; k++ ) {
                    if ( *m >= AM.OffsetIndex ) { *w++ = -INDEX; }
                    else if ( *m < 0 ) { *w++ = -VECTOR; }
                    else { *w = -SNUMBER; }
                    *w++ = *m++;
                }
            }
        }
        else {
            WORD *t1, *t2, *t3;
            for ( i = 0; i < L; i++ ) {
                alist = wi + i*numargs;
                tt = loc[i];
                m = tt + FUNHEAD;
                for ( k = 0; k < numargs; k++ ) {
                    if ( *m == -INDEX && m[1] == alist[0] ) {
                        if ( k != 0 ) {
                            if ( ( k & 1 ) != 0 ) sign = -sign;
/*
                            now move to position 0
*/
                            t2 = m+2; t1 = m; t3 = tt+FUNHEAD;
                            while ( t1 > t3 ) { *--t2 = *--t1; }
                            t3[0] = -INDEX; t3[1] = alist[0];
                        }
                        break;
                    }
                    NEXTARG(m)
                }
                m = tt + FUNHEAD + 2;
                for ( k = 1; k < numargs; k++ ) {
                    if ( *m == -INDEX && m[1] == alist[1] ) {
                        if ( k != 1 ) {
                            if ( ( k & 1 ) == 0 ) sign = -sign;
/*
                            now move to position 1
*/
                            t2 = m+2; t1 = m; t3 = tt+FUNHEAD+2;
                            while ( t1 > t3 ) { *--t2 = *--t1; }
                            t3[0] = -INDEX; t3[1] = alist[1];
```

```
                            }
                            break;
                        }
                        NEXTARG(m)
                    }
/*
                    now copy the remaining arguments to w
*/
                    t1 = tt + FUNHEAD + 4;
                    t2 = tt + tt[1];
                    while ( t1 < t2 ) *w++ = *t1++;
                }
            }
        }
        r[1] = w-r;
        while ( t < tstop ) {
            if ( *t == -1 ) { t += t[1]; continue; }
            i = t[1];
            NCOPY(w,t,i)
        }
        tstop = term + *term;
        while ( t < tstop ) *w++ = *t++;
        if ( sign < 0 ) w[-1] = -w[-1];
        i = w - AT.WorkPointer;
        *AT.WorkPointer = i;
        t = term; w = AT.WorkPointer;
        NCOPY(t,w,i)
        *AN.RepPoint = 1;   /* For Repeat */
        return(1);
}

/*
        #] Lus :
        #[ FindLus :
*/

int FindLus(int from, int level, int openindex)
{
        GETIDENTITY
        int i, j, k, jj, *alist, *blist, *w, *m, partner;
        WORD **loc = AN.funlocs, *wor;
        if ( level == 1 ) {
            for ( i = from; i < AN.numoffuns; i++ ) {
                alist = AN.funargs + i*AN.nargs;
                for ( j = 0; j < AN.nargs; j++ ) {
                    if ( alist[j] == openindex ) {
                        for ( k = 0; k < AN.nargs; k++ ) {
                            if ( k == j ) continue;
                            if ( alist[k] == AN.tohunt ) {
                                loc[from] = loc[i];
                                alist = AN.funargs + from*AN.nargs;
                                alist[0] = openindex; alist[1] = AN.tohunt;
                                return(1);
                            }
                        }
                    }
                }
            }
        }
        else {
            for ( i = from; i < AN.numoffuns; i++ ) {
                alist = AN.funargs + i*AN.nargs;
                for ( j = 0; j < AN.nargs; j++ ) {
                    if ( alist[j] == openindex ) {
                        if ( from != i ) {
                            wor = loc[i]; loc[i] = loc[from]; loc[from] = wor;
                            blist = w = AN.funargs + from*AN.nargs;
                            m = alist;
                            k = AN.nargs;
                            while ( --k >= 0 ) { jj = *m; *m++ = *w; *w++ = jj; }
                        }
                        else blist = alist;
                        for ( k = 0; k < AN.nargs; k++ ) {
                            if ( k == j || blist[k] < 0 ) continue;
                            partner = blist[k];
                            if ( FindLus(from+1,level-1,partner) ) {
                                blist[0] = openindex; blist[1] = partner;
                                return(1);
                            }
                        }
                        if ( from != i ) {
                            wor = loc[i]; loc[i] = loc[from]; loc[from] = wor;
                            w = AN.funargs + from*AN.nargs;
                            m = alist;
                            k = AN.nargs;
                            while ( --k >= 0 ) { jj = *m; *m++ = *w; *w++ = jj; }
                        }
```

```
                }
            }
        }
    }
    return(0);
}

/*
    #] FindLus :
    #[ SortTheList :
*/

int SortTheList(int *slist, int num)
{
    GETIDENTITY
    int i, nleft, nright, *t1, *t2, *t3, *rlist;
    if ( num <= 2 ) {
        if ( num <=  1 ) return(num);
        if ( slist[0] < slist[2] ) return(2);
        if ( slist[0] > slist[2] ) {
            i = slist[0]; slist[0] = slist[2]; slist[2] = i;
            i = slist[1]; slist[1] = slist[3]; slist[3] = i;
            return(2);
        }
        slist[1] += slist[3];
        return(1);
    }
    else {
        nleft = num/2; rlist = slist + 2*nleft;
        nright = SortTheList(rlist,num-nleft);
        nleft = SortTheList(slist,nleft);
        if ( AN.tlistsize < nleft ) {
            if ( AN.tlistbuf ) M_free(AN.tlistbuf,"AN.tlistbuf");
            AN.tlistsize = (nleft*3)/2;
            AN.tlistbuf = (int *)Malloc1(AN.tlistsize*2*sizeof(int),"AN.tlistbuf");
        }
        i = nleft; t1 = slist; t2 = AN.tlistbuf;
        while ( --i >= 0 ) { *t2++ = *t1++; *t2++ = *t1++; }
        i = nleft+nright; t1 = AN.tlistbuf; t2 = rlist; t3 = slist;
        while ( nleft > 0 && nright > 0 ) {
            if ( *t1 < *t2 ) {
                *t3++ = *t1++; *t3++ = *t1++; nleft--;
            }
            else if ( *t1 > *t2 ) {
                *t3++ = *t2++; *t3++ = *t2++; nright--;
            }
            else {
                *t3++ = *t1++; t2++; *t3++ = (*t1++) + (*t2++); i--;
                nleft--; nright--;
            }
        }
        while ( --nleft >= 0 ) { *t3++ = *t1++; *t3++ = *t1++; }
        while ( --nright >= 0 ) { *t3++ = *t2++; *t3++ = *t2++; }
        return(i);
    }
}

/*
    #] SortTheList :
    #[ AllLoops :

    Routine finds all possible loops that can be made in the
    arguments of the symmetric commuting vertex function v and creates
    for each loop a new term which is the original term times the loop.
    The occurrences of v can have different numbers of arguments.
    Each argument that occurs twice (and in different instances of v)
    in total will be considered.

    The input parameters are in C->lhs[level] and are
        C->lhs[level][0]  TYPEALLLOOPS
        C->lhs[level][1]  6
        C->lhs[level][2]  Number of function v.
        C->lhs[level][3]  Number of the output function loop.
        C->lhs[level][4]  option1: the type of argument.
        C->lhs[level][5]  option2: 0,1: what to do if no loop.
    Eligible arguments must be of the type SYMBOL, VECTOR, INDEX or SNUMBER.
    They must all be of the same type, indicated by option1.
    Extra restriction: In a loop, each v can be visited at most once.
    If there is no loop at all, option2 determines whether the term
    remains unmodified, or is replaced by zero.
    Function v can be either a regular function or a tensor.
    To facilitate this we copy the relevant arguments into the workspace.
*/

WORD AllLoops(PHEAD WORD *term,WORD level)
{
    CBUF *C = cbuf+AM.rbufnum;
```

```
    WORD vcode = C->lhs[level][2];    /* The input function */
    WORD option1 = C->lhs[level][4];  /* type of argument */
    WORD option2 = C->lhs[level][5];  /* what to do when no loop */
    WORD *tstop, *t, *tend, *tstart, *tfrom;
    WORD i, j, jj;
    WORD *arglist, nargs, *loop, nloop;
    WORD *oldworkpointer = AT.WorkPointer;
    LONG oldpworkpointer = AT.pWorkPointer;
    LONG numgenerated = 0, vert;
    WORD *a, *a1, *a2, *a3, *v, *vv, nvert, *to, *from, *tos, action;
/*
    Search for the first occurrence of vcode.
*/
    tstop = term+*term; tstop -=  ABS(tstop[-1]);
    t = term + 1;
    while ( t < tstop && *t != vcode ) t += t[1];
    if ( t == tstop ) {
        if ( option2 == 0 ) return(0);
        else return(Generator(BHEAD term,level));
    }
    tstart = t;
    nvert = 0;
    do {
        t += t[1]; nvert++;
    } while ( t < tstop && *t == vcode );
    tend = t;
/*
    Make room for 2*nvert pointers
*/
    WantAddPointers(2*nvert);
    vert = AT.pWorkPointer;
    AT.pWorkPointer += 2*nvert;
    nvert = 0;
/*
    Next we copy these functions into the workspace, but only the arguments
    that agree with option1. Because of the difference between tensors and
    regular functions in the copy we strip the type of the argument.
*/
    to = AT.WorkPointer;
    from = tstart;
    if ( functions[vcode-FUNCTION].spec == TENSORFUNCTION ) {
        from = tstart;
        while ( from < tend ) {
            tos = to++;
            tfrom = from+from[1];
            from += FUNHEAD;
            while ( from < tfrom ) {
                if ( option1 == -INDEX && *from >= 0 ) {
                    *to++ = *from++;
                }
                else if ( option1 == -VECTOR && *from < 0 ) {
                    *to++ = *from++ - AM.OffsetVector;
                }
                else {
                    from++;
                }
            }
            *tos = to-from;
            if ( *tos < 3 ) to = tos;
            else AT.pWorkSpace[vert+nvert++] = tos;
        }
    }
    else if ( functions[vcode-FUNCTION].spec == 0 ) {
        from = tstart;
        while ( from < tend ) {
            tos = to++;
            tfrom = from + from[1];
            from += FUNHEAD;
            while ( from < tfrom ) {
                if ( option1 == -VECTOR
                  && ( from[0] == -INDEX || from[0] == -VECTOR )
                  && from[1] < 0 ) {
                    from++;
                    *to++ = *from++ - AM.OffsetVector;
                }
                else if ( option1 == *from ) {
                    from++; *to++ = *from++;
                }
                else {
                    NEXTARG(from);
                }
            }
            *tos = to-tos;
            if ( *tos < 3 ) to = tos;
            else AT.pWorkSpace[vert+nvert++] = tos;
        }
    }
```

```
    else {
        AT.WorkPointer = oldworkpointer;
        AT.pWorkPointer = oldpworkpointer;
        if ( option2 == 0 ) return(0);
        return(Generator(BHEAD term,level));
    }
    AT.WorkPointer = to;
/*
    Now make a list of all relevant arguments.
*/
    a = arglist = AT.WorkPointer;
    for ( i = 0; i < nvert; i++ ) {
        v = AT.pWorkSpace[vert+i];
        j = *v-1; v++;
        NCOPY(a,v,j);
    }
    nargs = a-arglist;
    AT.WorkPointer = a;
/*
    Now sort the list. Bubble type sort.
*/
    a1 = arglist; a2 = a1+1;
    while ( a2 < a ) {
        a3 = a2;
        while ( a3 > a1 && a3[-1] > a3[0] ) {
            EXCH(*a3,a3[-1]);
            a3--;
        }
        a2++;
    }
/*
    Now remove the elements from the list that do not occur exactly twice.
*/
    a1 = arglist; a2 = a1; a3 = a1+nargs;
    while ( a2 < a3 ) {
        if ( a2+1 == a3 ) { break; }
        else if ( a2+2 == a3 ) {
            if ( a2[0] == a2[1] ) { *a1++ = a2[0]; }
            break;
        }
        else {
            if ( a2[0] != a2[1] ) { a2++; }
            else if ( a2[0] != a2[2] ) {
                *a1++ = a2[0]; a2 += 2;
            }
            else {
                a2++; while ( a2 < a3 && a2[-1] == a2[0] ) a2++;
            }
        }
    }
    nargs = a1-arglist;
/*
    Now we need to redo the list of vertices and remove the elements
    that are not in our arglist.
*/
    do {
      action = 0;
      for ( i = 0; i < nvert; i++ ) {
        vv = v = AT.pWorkSpace[vert+i];
        v++;
        for ( j = 1; j < vv[0]; j++ ) {
            for ( jj = 0; jj < nargs; jj++ ) {
                if ( *v == arglist[jj] ) break;
            }
            if ( jj >= nargs ) {    /* was not in the list */
                vv[0] = vv[0]-1;
                for ( jj = j; jj < vv[0]; jj++ ) vv[jj] = vv[jj+1];
            }
            v++;
        }
      }
/*
      Next we need to remove vertices that have only one object remaining.
      Also, the object can be removed from arglist. After that we go back
      to clean up the list.
*/
      for ( i = 0; i < nvert; i++ ) {
        vv = AT.pWorkSpace[vert+i];
        if ( vv[0] == 1 ) {
            AT.pWorkSpace[vert+i] = AT.pWorkSpace[vert+nvert-1];
            nvert--; i--; continue;
        }
        else if ( vv[0] == 2 ) {
            for ( j = 0; j < nargs; j++ ) {
                if ( arglist[j] == vv[1] ) break;
            }
            while ( j < nargs-1 ) arglist[j] = arglist[j+1];
```

```
            nargs--;
            AT.pWorkSpace[vert+i] = AT.pWorkSpace[vert+nvert-1];
            nvert--; i--;
            action = 1;
        }
      }
    } while ( action );
/*
    Because the user can remove tadpoles rather easily as in
        id v(?a,i?,?b,i?,?c) = v(?a,?b,?c)
    there is no need to avoid them as potential loops.

    At this point we are ready to look for loops.
    We have
        arglist,nargs   list of eligible objects.
        vert,nvert      position in pWorkSpace for vertices and their number.
                        The vertices themselves are in the WorkSpace.
        loop,nloop      The buildup of the loop.
*/
    loop = AT.WorkPointer;
    AT.WorkPointer += nargs;
    nloop = 0;
    numgenerated += StartLoops(BHEAD term,level,vert,nvert,arglist,nargs,loop,nloop);
    AT.WorkPointer = oldworkpointer;
    AT.pWorkPointer = oldpworkpointer;

    if ( numgenerated == 0 && option2 != 0 ) return(Generator(BHEAD term,level));
    return(0);
}

/*
    #] AllLoops :
    #[ StartLoops :

    Algorithm.
    1: have a list of vertices and objects that can be part of the loops.
    2: starting with the first element of arglist, look for the two
       vertices that contain this object. The first is vstart.
    3: At the second, look at all possible objects to continue from here.
    4: For each, find the vertex with its partner.
    5: if the partner vertex is vstart, we have a loop. --> 9.
    6: The partner vertex is not allowed to be a vertex that we passed
       in the current build-up of the loop.
    7: Put the object in loop and increase nloop.
    8: Now sum over all allowed objects in the partner vertex. --> 4.

    9: Create the function loop with the arguments from loop,nloop.
   10: If a reverse cyclic permutation gives a smaller result, drop this
       solution and continue with the last summing over objects at a vertex.
   11: If not, output the result by adding the function loop
       to the term and call Generator(term,level)

   12: when all possibilities with the first element of arglist have been
       exhausted, raise arglist and lower nargs by one. --> 2
   13: when nargs == 1, we can stop.

    The inclusion of a new object when we sum over the possibilities at a
    vertex can best be done in a recursion, because we have no idea how
    many nested loops we would otherwise need. Of course the recursion can
    be emulated, but that makes the algorithm rather messy.

    We have two routines: StartLoops and GenLoops.
    StartLoops takes care of the first argument (and the summing over who
    is the first argument), while GenLoops treats an additional vertex until
    a loop is closed. GenLoops also sends completed loops off toe Generator.
*/

LONG StartLoops(PHEAD WORD *term,WORD level,LONG vert,WORD nvert,
                WORD *arglist,WORD nargs,WORD *loop,WORD nloop)
{
    LONG numgenerated = 0;
    WORD *v, *vv, *vstart, istart, *vpartner, ipartner, j;
    while ( nargs > 1 ) {
/*
        Look for a vertex with arglist[0]. This is our starting vertex.
        Next we look for its partner and we put arglist[0] in loop.
*/
        nloop = 0;
        loop[nloop++] = arglist[0];
        for ( istart = 0; istart < nvert; istart++ ) {
            vstart = AT.pWorkSpace[vert+istart];
            v = vstart+1; vv = vstart + *vstart;
            do {
                if ( *v == arglist[0] ) goto havestart;
                v++;
            } while ( v < vv );
        }
```

```
/*
        If we come here, we have a problem.
*/
        MesPrint("Internal error in StartLoops. Object not found.");
        Terminate(-1);
        return(-1);
havestart:
        AT.pWorkSpace[vert+nvert] = vstart;
/*
        Check for tadpole.
*/
        v++;
        while ( v < vv ) {
            if ( *v == arglist[0] ) {    /* tadpole */
                LoopOutput(BHEAD term,level,loop,nloop);
                numgenerated++;
                goto nextarg;
            }
            v++;
        }
/*
        Now the partner vertex.
*/
        for ( ipartner = istart+1; ipartner < nvert; ipartner++ ) {
            vpartner = AT.pWorkSpace[vert+ipartner];
            vv = vpartner+*vpartner; v = vpartner+1;
            do {
                if ( *v == arglist[0] ) goto havepartner;
                v++;
            } while ( v < vv );
        }
        return(numgenerated);
havepartner:
        AT.pWorkSpace[vert+nvert+1] = vpartner;
/*
        Now we run through all other possibilities at vpartner.
        vv is still OK.
*/
        v = vpartner+1;
        while ( v < vv ) {
            if ( *v != arglist[0] ) {
                for ( j = 1; j < nargs; j++ ) {
                    if ( *v == arglist[j] ) {
                        loop[nloop++] = *v;
                        numgenerated += GenLoops(BHEAD term,level,vert,nvert,
                                                arglist,nargs,loop,nloop);
                        nloop--;
                        break;
                    }
                }
            }
            v++;
        }
nextarg:
        arglist++; nargs--;
    }
    return(numgenerated);
}

/*
    #] StartLoops :
    #[ GenLoops :

    We enter with an open line in loop[nloop-1]
*/

LONG GenLoops(PHEAD WORD *term,WORD level,LONG vert,WORD nvert,
        WORD *arglist,WORD nargs,WORD *loop,WORD nloop)
{
    LONG numgenerated = 0;
    WORD *vstart, *v, *vv, i, j, *vpartner;
/*
    Start with checking whether the partner is in vstart (=vert[nvert])
*/
    vstart = AT.pWorkSpace[nvert];
    vv = vstart + *vstart; v = vstart+1;
    while ( v < vv ) {
        if ( *v == loop[nloop-1] ) {
/*
            This closes the loop.
            Now we can output it.
*/
            LoopOutput(BHEAD term,level,loop,nloop);
            numgenerated++;
            return(numgenerated);
        }
        v++;
```

```
    }
/*
    Start with finding the partner.
*/
    for ( i = 0; i < nvert; i++ ) {
        vpartner = AT.pWorkSpace[vert+i];
        if ( vpartner == vstart ) continue;
        for ( j = 0; j < nloop; j++ ) {
            if ( vpartner == AT.pWorkSpace[vert+nvert+j] ) break;
        }
        if ( j < nloop ) continue;
        v = vpartner+1; vv = vpartner + *vpartner;
        while ( v < vv ) {
            if ( *v == loop[nloop-1] ) {
/*
                Found the partner.
                Now we can sum over the remaining permitted arguments of this vertex.
*/
                v = vpartner + 1;
                while ( v < vv ) {
/*
                    *v should be in arglist.
*/
                    for ( j = 0; j < nargs; j++ ) {
                        if ( *v == arglist[j] ) break;
                    }
                    if ( j >= nargs ) { v++; continue; }
/*
                    *v should not be in loops.
*/
                    for ( j = 0; j < nloop; j++ ) {
                        if ( *v == loop[j] ) break;
                    }
                    if ( j >= nloop ) {
                        AT.pWorkSpace[vert+nvert+nloop] = vpartner;
                        loop[nloop++] = *v;
                        numgenerated += GenLoops(BHEAD term,level,vert,nvert,
                                arglist,nargs,loop,nloop);
                        nloop--;
                    }
                    v++;
                }
                return(numgenerated);
            }
            v++;
        }
    }
/*
    The partner is in a vertex we have finished before.
*/
    return(numgenerated);
}

/*
    #] GenLoops :
    #[ LoopOutput :
*/

void LoopOutput(PHEAD WORD *term, WORD level, WORD *loop, WORD nloop)
{
    CBUF *C = cbuf+AM.rbufnum;
    WORD loopfun = C->lhs[level][3];  /* The output function */
    WORD option1 = C->lhs[level][4];  /* type of argument */
    WORD *tstop, *tstop1, *t, *tt;
    WORD *outterm, *loop1;
    WORD i;
    tstop1 = term+*term; tstop = tstop1 - ABS(tstop1[-1]);
/*
    Construct the rcycle symmetrized version of loop.
*/
    if ( nloop > 2 ) {
        loop1 = AT.WorkPointer;
        loop1[0] = loop[0];
        for ( i = 1; i < nloop; i++ ) { loop1[i] = loop[nloop-i]; }
        if ( loop1[1] < loop[1] ) {
            AT.WorkPointer += nloop;
            loop = loop1;
        }
    }
    outterm = AT.WorkPointer;
    tt = outterm; t = term;
    while ( t < tstop ) *tt++ = *t++;
    *tt++ = loopfun;
    if ( functions[loopfun-FUNCTION].spec == TENSORFUNCTION ) {
        *tt++ = FUNHEAD+nloop;
        FILLFUN(tt)
        if ( option1 == -VECTOR ) {
```

```
            for ( i = 0; i < nloop; i++ ) *tt++ = loop[i]+AM.OffsetVector;
        }
        else {
            for ( i = 0; i < nloop; i++ ) *tt++ = loop[i];
        }
    }
    else {
        *tt++ = FUNHEAD+nloop*2;
        FILLFUN(tt)
        for ( i = 0; i< nloop; i++ ) {
            *tt++ = option1;
            if ( option1 == -VECTOR ) *tt++ = loop[i] + AM.OffsetVector;
            else *tt++ = loop[i];
        }
    }
    while ( t < tstop1 ) *tt++ = *t++;
    *outterm = tt - outterm;
    AT.WorkPointer = tt;
    if ( Generator(BHEAD outterm,level) ) {
        MesCall("LoopOutput");
        Terminate(-1);
    }
    AT.WorkPointer = outterm;
}

/*
    #] LoopOutput :
    #[ AllPaths :

    This routine has many similarities with AllLoops.
    In AllLoops we have startingpoint and endpoint at the same vertex.
    In AllPaths the startingpoint and the endpoints are different and
    given in advance. This endfun can occur only twice.
*/

WORD AllPaths(PHEAD WORD *term,WORD level)
{
    CBUF *C = cbuf+AM.rbufnum;
    WORD endcode = C->lhs[level][2];    /* The endpoint function */
    WORD vcode   = C->lhs[level][3];    /* The intermediate function */
    WORD option1 = C->lhs[level][5];    /* type of argument */
    WORD option2 = C->lhs[level][6];    /* what to do when no loop */
    WORD *t, *tstop, *tend1, *tend2, *tstart, *tend, numend, nvert, npass;
    WORD *tfrom, *to, *tos, *from;
    WORD *arglist, nargs, *path, npath, *a, *a1, *a2, *a3;
    WORD i, j, jj, *v, *vv, action;
    LONG vert,vert1; /* ,vert2; */
    WORD numgenerated = 0;
    WORD *oldworkpointer = AT.WorkPointer;
    LONG oldpworkpointer = AT.pWorkPointer;
/*
    Search for the two occurrences of endcode.
*/
    tstop = term+*term; tstop -=  ABS(tstop[-1]);
    t = term + 1;
    while ( t < tstop && *t != endcode ) t += t[1];
    if ( t == tstop ) { /* no endpoints */
        if ( option2 == 0 ) return(0);
        else return(Generator(BHEAD term,level));
    }
    tend1 = t;
    numend = 0;
    while ( t < tstop && *t == endcode ) { tend2 = t; t += t[1]; numend++; }
    if ( numend != 2 ) { /* not 2 endpoints -> no path */
        if ( option2 == 0 ) return(0);
        else return(Generator(BHEAD term,level));
    }
/*
    Search for the intermediate functions.
*/
    t = term + 1;
    nvert = 0;
    while ( t < tstop && *t != vcode ) t += t[1];
    tstart = t;
    while ( t < tstop && *t == vcode ) {
        t += t[1]; nvert++;
    }
    tend = t;
/*
    Next we copy the relevant content of the various functions into the
    WorkSpace. We keep pointers to the functions in pWorkSpace.
    First the two endpoints and then the intermediate ones.
*/
    WantAddPointers((2*nvert+8));
    vert = AT.pWorkPointer+2;
    vert1 = AT.pWorkPointer;
/*  vert2 = AT.pWorkPointer+1; */
```

```
    AT.pWorkPointer += 2*nvert+8;
    nvert = 0;
/*
    Copy the endpoints.
*/
    to = AT.WorkPointer;

    if ( functions[endcode-FUNCTION].spec == TENSORFUNCTION ) {
        from = tend1; npass = 0;
redo1:
        tos = to++;
        tfrom = from+from[1];
        from += FUNHEAD;
        while ( from < tfrom ) {
            if ( option1 == -INDEX && *from >= 0 ) {
                *to++ = *from++;
            }
            else if ( option1 == -VECTOR && *from < 0 ) {
                *to++ = *from++ - AM.OffsetVector;
            }
            else {
                from++;
            }
        }
        *tos = to-tos;
        if ( *tos < 2 ) to = tos;
        else AT.pWorkSpace[vert1+npass] = tos;
        npass++;
        if ( from == tend2 ) goto redo1;
    }
    else if ( functions[endcode-FUNCTION].spec == 0 ) {
        from = tend1;
        npass = 0;
redo2:
        tos = to++;
        tfrom = from + from[1];
        from += FUNHEAD;
        while ( from < tfrom ) {
            if ( option1 == -VECTOR
             && ( from[0] == -INDEX || from[0] == -VECTOR )
             && from[1] < 0 ) {
                from++;
                *to++ = *from++ - AM.OffsetVector;
            }
            else if ( option1 == *from ) {
                from++; *to++ = *from++;
            }
            else {
                NEXTARG(from);
            }
        }
        *tos = to-tos;
        if ( *tos < 2 ) to = tos;
        else AT.pWorkSpace[vert1+npass] = tos;
        npass++;
        if ( from == tend2 ) goto redo2;
    }
    else {
        AT.WorkPointer = oldworkpointer;
        AT.pWorkPointer = oldpworkpointer;
        if ( option2 == 0 ) return(0);
        return(Generator(BHEAD term,level));
    }
/*
    And now the intermediate functions
*/
    from = tstart;
    if ( functions[vcode-FUNCTION].spec == TENSORFUNCTION ) {
        from = tstart;
        while ( from < tend ) {
            tos = to++;
            tfrom = from+from[1];
            from += FUNHEAD;
            while ( from < tfrom ) {
                if ( option1 == -INDEX && *from >= 0 ) {
                    *to++ = *from++;
                }
                else if ( option1 == -VECTOR && *from < 0 ) {
                    *to++ = *from++ - AM.OffsetVector;
                }
                else {
                    from++;
                }
            }
            *tos = to-tos;
            if ( *tos < 3 ) to = tos;
            else AT.pWorkSpace[vert+nvert++] = tos;
```

```
        }
    }
    else if ( functions[vcode-FUNCTION].spec == 0 ) {
        from = tstart;
        while ( from < tend ) {
            tos = to++;
            tfrom = from + from[1];
            from += FUNHEAD;
            while ( from < tfrom ) {
                if ( option1 == -VECTOR
                     && ( from[0] == -INDEX || from[0] == -VECTOR )
                     && from[1] < 0 ) {
                    from++;
                    *to++ = *from++ - AM.OffsetVector;
                }
                else if ( option1 == *from ) {
                    from++; *to++ = *from++;
                }
                else {
                    NEXTARG(from);
                }
            }
            *tos = to-tos;
            if ( *tos < 3 ) to = tos;
            else AT.pWorkSpace[vert+nvert++] = tos;
        }
    }
    else {
        AT.WorkPointer = oldworkpointer;
        AT.pWorkPointer = oldpworkpointer;
        if ( option2 == 0 ) return(0);
        return(Generator(BHEAD term,level));
    }
    AT.WorkPointer = to;
/*
    Now make a list of all relevant arguments.
*/
    a = arglist = AT.WorkPointer;
    for ( i = -2; i < nvert; i++ ) {
        v = AT.pWorkSpace[vert+i];
        j = *v-1; v++;
        NCOPY(a,v,j);
    }
    nargs = a-arglist;
    AT.WorkPointer = a;
/*
    Now sort the list. Bubble type sort.
*/
    a1 = arglist; a2 = a1+1;
    while ( a2 < a ) {
        a3 = a2;
        while ( a3 > a1 && a3[-1] > a3[0] ) {
            EXCH(*a3,a3[-1]);
            a3--;
        }
        a2++;
    }
/*
    Now remove the elements from the list that do not occur exactly twice.
*/
    a1 = arglist; a2 = a1; a3 = a1+nargs;
    while ( a2 < a3 ) {
        if ( a2+1 == a3 ) { break; }
        else if ( a2+2 == a3 ) {
            if ( a2[0] == a2[1] ) { *a1++ = a2[0]; }
            break;
        }
        else {
            if ( a2[0] != a2[1] ) { a2++; }
            else if ( a2[0] != a2[2] ) {
                *a1++ = a2[0]; a2 += 2;
            }
            else {
                a2++; while ( a2 < a3 && a2[-1] == a2[0] ) a2++;
            }
        }
    }
    nargs = a1-arglist;
/*
    Now we need to redo the list of vertices and remove the elements
    that are not in our arglist.
*/
    do {
      action = 0;
      for ( i = -2; i < nvert; i++ ) {
        vv = v = AT.pWorkSpace[vert+i];
        v++;
```

```
            for ( j = 1; j < vv[0]; j++ ) {
                for ( jj = 0; jj < nargs; jj++ ) {
                    if ( *v == arglist[jj] ) break;
                }
                if ( jj >= nargs ) {     /* was not in the list */
                    vv[0] = vv[0]-1;
                    for ( jj = j; jj < vv[0]; jj++ ) vv[jj] = vv[jj+1];
                }
                v++;
            }
        }
/*
        Next we need to remove vertices that have only one object remaining.
        Also, the object can be removed from arglist. After that we go back
        to clean up the list.
*/
        for ( i = -2; i < nvert; i++ ) {
            vv = AT.pWorkSpace[vert+i];
            if ( vv[0] == 1 ) {
                AT.pWorkSpace[vert+i] = AT.pWorkSpace[vert+nvert-1];
                nvert--; i--; continue;
            }
            else if ( vv[0] == 2 && i >= 0 ) {
                for ( j = 0; j < nargs; j++ ) {
                    if ( arglist[j] == vv[1] ) break;
                }
                while ( j < nargs-1 ) arglist[j] = arglist[j+1];
                nargs--;
                AT.pWorkSpace[vert+i] = AT.pWorkSpace[vert+nvert-1];
                nvert--; i--;
                action = 1;
            }
        }
    } while ( action );
/*
    Now we have a clean list of connections and we can start building
    the path. We start with summing over all objects in vert1.
*/
    path = AT.WorkPointer; npath = 0;
    AT.WorkPointer += nvert+8;

    t = AT.pWorkSpace[vert1];
    if ( *t >= 2 ) {
        for ( i = 1; i < *t; i++ ) {
            AT.pWorkSpace[vert+nvert] = t;
            path[npath++] = t[i];
            numgenerated += GenPaths(BHEAD term,level,vert,nvert,arglist,nargs,path,npath);
            npath--;
        }
    }
    AT.WorkPointer = oldworkpointer;
    AT.pWorkPointer = oldpworkpointer;
    if ( numgenerated == 0 && option2 != 0 ) return(Generator(BHEAD term,level));
    return(0);
}

/*
    #] AllPaths :
    #[ GenPaths :

    We enter with an open line in path[npath-1]
    We traverse though the vertices until we reach vert-1, the endpoint.
*/

LONG GenPaths(PHEAD WORD *term, WORD level, LONG vert, WORD nvert,
        WORD *arglist, WORD nargs, WORD *path, WORD npath)
{
    LONG numgenerated = 0;
    WORD *t, *vpartner, *v, *vv;
    WORD i, j;
/*
    Check whether path[npath-1] is part of the endpoint.
*/
    t = AT.pWorkSpace[vert-1];
    for ( i = 1; i < *t; i++ ) {
        if ( t[i] == path[npath-1] ) { /* Got a path! */
            PathOutput(BHEAD term,level,path,npath);
            numgenerated++;
            return(numgenerated);
        }
    }
/*
    Start with finding the partner.
*/
    for ( i = 0; i < nvert; i++ ) {
        vpartner = AT.pWorkSpace[vert+i];
        for ( j = 0; j < npath; j++ ) {
```

```
            if ( vpartner == AT.pWorkSpace[vert+nvert+j] ) break;
        }
        if ( j < npath ) continue;
        v = vpartner+1; vv = vpartner + *vpartner;
        while ( v < vv ) {
            if ( *v == path[npath-1] ) {
/*
                Found the partner.
                Now we can sum over the remaining permitted arguments of this vertex.
*/
                v = vpartner + 1;
                while ( v < vv ) {
/*
                    *v should be in arglist.
*/
                    for ( j = 0; j < nargs; j++ ) {
                        if ( *v == arglist[j] ) break;
                    }
                    if ( j >= nargs ) { v++; continue; }
/*
                    *v should not be in path.
*/
                    for ( j = 0; j < npath; j++ ) {
                        if ( *v == path[j] ) break;
                    }
                    if ( j >= npath ) {
                        AT.pWorkSpace[vert+nvert+npath] = vpartner;
                        path[npath++] = *v;
                        numgenerated += GenPaths(BHEAD term,level,vert,nvert,
                                arglist,nargs,path,npath);
                        npath--;
                    }
                    v++;
                }
                return(numgenerated);
            }
            v++;
        }
    }
/*
    The partner is in a vertex we have finished before.
*/
    return(numgenerated);
}

/*
    #] GenPaths :
    #[ PathOutput :
*/

void PathOutput(PHEAD WORD *term, WORD level, WORD *path, WORD npath)
{
    CBUF *C = cbuf+AM.rbufnum;
    WORD pathfun = C->lhs[level][4];  /* The output function */
    WORD option1 = C->lhs[level][5];  /* type of argument */
    WORD *tstop, *tstop1, *t, *tt;
    WORD *outterm;
    WORD i;
    tstop1 = term+*term; tstop = tstop1 - ABS(tstop1[-1]);
    outterm = AT.WorkPointer;
    tt = outterm; t = term;
    while ( t < tstop ) *tt++ = *t++;
    *tt++ = pathfun;
    if ( functions[pathfun-FUNCTION].spec == TENSORFUNCTION ) {
        *tt++ = FUNHEAD+npath;
        FILLFUN(tt)
        if ( option1 == -VECTOR ) {
            for ( i = 0; i < npath; i++ ) *tt++ = path[i]+AM.OffsetVector;
        }
        else {
            for ( i = 0; i < npath; i++ ) *tt++ = path[i];
        }
    }
    else {
        *tt++ = FUNHEAD+npath*2;
        FILLFUN(tt)
        for ( i = 0; i< npath; i++ ) {
            *tt++ = option1;
            if ( option1 == -VECTOR ) *tt++ = path[i] + AM.OffsetVector;
            else *tt++ = path[i];
        }
    }
    while ( t < tstop1 ) *tt++ = *t++;
    *outterm = tt - outterm;
    AT.WorkPointer = tt;
    if ( Generator(BHEAD outterm,level) ) {
        MesCall("PathOutput");
```

```
        Terminate(-1);
    }
    AT.WorkPointer = outterm;
}



/*
    #] PathOutput :
    #[ AllOnePI :

    We assume a graph that has loops and is OnePI.
    This routine initializes the recursion that creates all onePI subgraphs.

    Algorithm:
        a: have a routine that removes all bridges: RemoveBridges
        b: output an empty diagram.
        c: output the diagram itself
        d: cut a line, followed by removing all bridges.
        e: if the diagram still has lines, go to c, else go to the next line in d.
    In order to avoid factorial blowup, we need to prune the tree as fast as possible.

    1: list of lines to be cut.
    2: once we have tried a line and removed its bridges, we do not have to
       try this line deeper in the tree, neither its bridges.


     1 2 3
        cut 1.                        x
            cut 2 -> bridge 3          x
        cut 2.                        x
            cut 3 -> bridge 1   ????   mag niet
        cut 3.                        x
    Hence  1,2,3,4,5,6,7
        1 still to go 2,3,4,5,6,7
        2 still to go 3,4,5,6,7
        3 still to go 4,5,6,7
        etc.
    bridges: if x is bridge, we take x from the list_to_go.
    If we have a bridge that is a number lower than the list_to_go we skip this possibility.
    We reach the end when the list_to_go is empty.
*/

WORD AllOnePI(WORD *term,WORD level)
{
    CBUF *C = cbuf+AM.rbufnum;
    WORD vcode   = C->lhs[level][2];     /* The vertex function */
    WORD option1 = C->lhs[level][4];     /* type of argument */
/*
    First we have to collect all relevant information about the diagram.
    We should start with removing bridges to make the real starting point onePI.
*/
    DUMMYUSE(term)
    DUMMYUSE(vcode)
    DUMMYUSE(option1)
    return(0);
}

/*
    #] AllOnePI :
    #[ RemoveBridges :
*/

int RemoveBridges(void)
{
    return(0);
}

/*
    #] RemoveBridges :
    #[ TakeOneLine :
*/

int TakeOneLine(WORD*term,WORD level)
{
    DUMMYUSE(term)
    DUMMYUSE(level)
    return(0);
}

/*
    #] TakeOneLine :
    #[ OutputOnePI :
*/

int OutputOnePI(PHEAD WORD *term,WORD level)
{
```

```
    return(Generator(BHEAD term,level));
}

/*
    #] OutputOnePI :
*/

```

4.68 mallocprotect.h

```
#ifndef MALLOCPROTECT_H
#define MALLOCPROTECT_H

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

/*
    #[ Documentation :

The enhanced malloc debugger. It intercepts access (both write AND
read) to the memory outside the allocated chunk in the following
manner: it prints out (to the stderr) the information about the event
and goes to the spilock. The user is able to attach the debugger to
the running process, investigate the problem and continue evaluation.

In case of error, the debugger will print to the stderr the following
information:
"***** PID: ###, Addr: ###, signal ### (code ###)"
"Attach gdb -p ### and set var loopForever=0 in a corresponding frame to continue"
or for TFORM:
"Attach gdb -p ### and set var loopForever=0 in a corresponding frame
   of a thread with LPW id ### to continue"

and go to the spilock. The user may attach gdb by the command
gdb -p ### and type "where" (in case of TFORM, the user should first
switch to the proper thread).  Then investigation of the corresponding
frames should clarify the situation. In order to continue the program,
the user might set (in the corresponding frame of the corresponding
thread) the variable loopForever to 0. The debugger will try to remove
local problem lead to the exception.

To activate the debugger, the macro MALLOCPROTECT should be
defined. There are three possible values:
#define MALLOCPROTECT -1 (any integer <0)
#define MALLOCPROTECT 0
#define MALLOCPROTECT 1 (any integer>0)

If MALLOCPROTECT < 0, the debugger will intercept any access to the
memory before the allocated chunk, even one byte.

If MALLOCPROTECT == 0, the debugger will intercept any access to the
memory before the allocated chunk, even one byte, and access to the
memory page next to the allocated one.

If MALLOCPROTECT > 0, the debugger will intercept any access to the
memory after the allocated chunk, even one byte, and access to the
memory page before the allocated one.

The original FORM malloc debugger is able to catch errors when the
```

```
allocated memory is freed improper, or if some small portion OUT of
the allocated chunk is written. This debugger is a complementary
one. It permits to catch situation when some small portion of memory
before or after the allocated chunk is written or even just read.

The idea is to protect the beginning or the end of the allocated chunk
for any kind of access and install the SIGSEGV handler in order to
intercept the access to this memory. Moreover, the allocated memory is
always immediately returned to the system so if the user tries to
access the memory after it is freed then the handler is triggered,
also.

The problem here is that we are able to allocate / protect only the
whole page. So the user has to run the debugger twice: one time with
the left alignment (#define MALLOCPROTECT <0), and then with the right
alignment (#define MALLOCPROTECT >0). During the first run the possible
errors like x[-1] will be caught, and the second run will manifest
reading over the allocated ares.

The leftmost extra page is always allocated and protected. The size
of the allocated chunk is stored in the beginning of this page.

    #] Documentation :
    #[ Includes :
*/

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/mman.h>

#ifdef WITHPTHREADS
#ifdef LINUX
#include <sys/syscall.h>
#endif
#endif

/*
    #] Includes :
*/

static int pageSize=4096;/*the default value*/


/*
    #[ segv_handler :
*/

/*The handler will be invoked on some signal (usually SIGSEGV). It
will print some diagnostics to stderr and hands up in infinite loop
waiting the user for debugging:*/
static void segv_handler(int sig, siginfo_t *sip, void *xxx) {
    char *vadr;
    char *errStr;
    int actionBeforeExit=0;/* < 0 - unprotect, > 0 - map */
    switch(sip->si_signo){/* All POSIX signals for which the field siginfo_t.si_addr is defined:*/
        case SIGILL:
            switch(sip->si_code){
            case ILL_ILLOPC:
                errStr="SIGILL: Illegal opcode";
                break;
            case ILL_ILLOPN:
                errStr="SIGILL: Illegal operand";
                break;
            case ILL_ILLADR:
                errStr="SIGILL: Illegal addressing mode";
                break;
            case ILL_ILLTRP:
                errStr="SIGILL: Illegal trap";
                break;
            case ILL_PRVOPC:
                errStr="SIGILL: Privileged opcode";
                break;
            case ILL_PRVREG:
                errStr="SIGILL: Privileged register";
                break;
            case ILL_COPROC:
                errStr="SIGILL: Coprocessor error";
                break;
            case ILL_BADSTK:
                errStr="SIGILL: Internal stack error";
                break;
            default:
                errStr="SIGILL: Unknown signal code";
        }/*case SIGILL:*/
        break;
```

```
        case SIGFPE:
            switch(sip->si_code){
                case FPE_INTDIV:
                    errStr="SIGFPE: Integer divide-by-zero";
                    break;
                case FPE_INTOVF:
                    errStr="SIGFPE: Integer overflow";
                    break;
                case FPE_FLTDIV:
                    errStr="SIGFPE: Floating point divide-by-zero";
                    break;
                case FPE_FLTOVF:
                    errStr="SIGFPE: Floating point overflow";
                    break;
                case FPE_FLTUND:
                    errStr="SIGFPE: Floating point underflow";
                    break;
                case FPE_FLTRES:
                    errStr="SIGFPE: Floating point inexact result";
                    break;
                case FPE_FLTINV:
                    errStr="SIGFPE: Invalid floating point operation";
                    break;
                case FPE_FLTSUB:
                    errStr="SIGFPE: Subscript out of range";
                    break;
                default:
                    errStr="SIGFPE: Unknown signal code";
            }/*switch(sip->si_code)*/
            break;
        case SIGSEGV:
            switch(sip->si_code){
                case SEGV_MAPERR:
                    errStr="SIGSEGV: Address not mapped";
                    actionBeforeExit = 1;
                    break;
                case SEGV_ACCERR:
                    errStr="SIGSEGV: Invalid permissions";
                    actionBeforeExit = -1;
                    break;
                default:
                    errStr="SIGSEGV: Unknown signal code";
            }/*switch(sip->si_code)*/
            break;
        case SIGBUS:
            switch(sip->si_code){
                case BUS_ADRALN:
                    errStr="SIGBUS: Invalid address alignment";
                    break;
                case BUS_ADRERR:
                    errStr="SIGBUS: Non-existent physical address";
                    break;
                case BUS_OBJERR:
                    errStr="SIGBUS: Object-specific hardware error";
                    break;
                default:
                    errStr="SIGBUS: Unknown signal code";
            }/*switch(sip->si_code)*/
            break;
        default:
            errStr="Unknown signal";
    }/*switch(sip->si_signo)*/

    vadr = (caddr_t)sip->si_addr;
    fprintf(stderr, "\n***** PID: %ld, Addr: %p, signal %s (code %d)\n",
        (long)getpid(), vadr, errStr,sip->si_code);

    {/*Block*/
        /*The process hangs up at this block.
        Attach gdb to investigate and continue:
        */
        volatile int loopForever=1;
        size_t alignedAdr=((size_t)vadr) & (~(pageSize-1));
        fprintf(stderr, "   Attach gdb -p %ld and set var loopForever=0 in a corresponding frame",
            (long)getpid());
#ifdef CORRECTCODE
#ifdef WITHPTHREADS
#ifdef LINUX
        fprintf(stderr, "\n   of a thread with LPW id %ld",(long)syscall(SYS_gettid));
#else
        /*If the compiler fails here, just remove the next line:*/
        fprintf(stderr, "\n   of thread with LPW id %ld",(long)gettid());
#endif
#endif
#endif
        fprintf(stderr, " to continue\n");

```

```
        while(loopForever)
            sleep(1);

        if(actionBeforeExit<0)/*After changing loopForever=0, unprotect the page to continue:*/
            mprotect((char*)alignedAdr, pageSize, PROT_READ | PROT_WRITE);
        if(actionBeforeExit>0)/*After changing loopForever=0, map the page to continue:*/
/*          mmap((void*)alignedAdr,pageSize,PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, 0, 0); */
            mmap((void*)alignedAdr,pageSize,PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANON, 0, 0);
    }/*Block*/
}/*segv_handler*/

/*
    #] segv_handler :
*/

/*
    #[ mprotectInit :
*/

static FORM_INLINE int mprotectInit(void)
{
    struct sigaction sa;
    pageSize = getpagesize();
    sa.sa_sigaction = &segv_handler;

    sigemptyset(&sa.sa_mask);
    sigaddset(&sa.sa_mask, SIGIO);
    sigaddset(&sa.sa_mask, SIGALRM);

    sa.sa_flags = SA_SIGINFO;

    if (sigaction(SIGILL, &sa, NULL)) {
        fprintf(stderr,"Error on assigning %s.\n","SIGILL");
        return SIGILL;
    }
    if (sigaction(SIGFPE, &sa, NULL)) {
        fprintf(stderr,"Error on assigning %s.\n","SIGFPE");
        return SIGFPE;
    }
    if (sigaction(SIGSEGV, &sa, NULL)) {
        fprintf(stderr,"Error on assigning %s.\n","SIGSEGV");
        return SIGSEGV;
    }
    if (sigaction(SIGBUS, &sa, NULL)) {
        fprintf(stderr,"Error on assigning %s.\n","SIGBUS");
        return SIGBUS;
    }
    return 0;
}/*mprotectInit*/

/*
    #] mprotectInit :
*/


/*
    #[ mprotectMalloc :
*/

static void *mprotectMalloc(size_t theSize)
{
#if MALLOCPROTECT < 0
    /*Only one side is protected*/
    size_t nPages=1;
#else
    /*Both sides are protected*/
    size_t nPages=2;
#if MALLOCPROTECT > 0
    /*will need the original theSize*/
    size_t requestedSize=theSize;
#endif
#endif

    char *ret=NULL;
    size_t *ptr;



    /*Align required size to the pagesize:*/
    if(theSize % pageSize)
        nPages++;
    theSize= (theSize/pageSize+nPages)*pageSize;

/*  ret=(char*)mmap(0,theSize,PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, 0, 0); */
    ret=(char*)mmap(0,theSize,PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANON, 0, 0);
    if(ret == MAP_FAILED)
```

```
    return NULL;
    ptr=(size_t *)ret;
    *ptr=theSize;
    if(mprotect(ret, pageSize, PROT_NONE))return NULL;
#if MALLOCPROTECT < 0
    return ret+pageSize;
#else
    if(mprotect(ret+(theSize-pageSize), pageSize, PROT_NONE))return NULL;
#if MALLOCPROTECT ==0
    return ret+pageSize;
#endif
#endif
    /*MALLOCPROTECT > 0, but we need conditional compilation
    since the variable requestedSize:*/
#if MALLOCPROTECT > 0

    /*Potential problems with alignment if the requested size is not
    a multiple of items. But no problems on x86-64:*/
    return ret+ (theSize-pageSize-requestedSize);
#endif
}/*mprotectMalloc*/
/*
    #] mprotectMalloc :
*/

/*
    #[ mprotectFree :
*/

static void mprotectFree(char *ptr)
{
    size_t theSize;
    if(ptr==NULL) return;
#if MALLOCPROTECT > 0
    /*The memory block was moved to the right, find the left page boundary:*/
    {/*Block*/
        size_t alignedPtr=((size_t)ptr) & (~(pageSize-1));
        ptr=(char*)alignedPtr;
    }/*Block*/
#endif

    ptr-=pageSize;
    mprotect(ptr, pageSize, PROT_READ);
    theSize=*((size_t*)ptr);
    munmap(ptr,theSize);
}/*mprotectFree*/

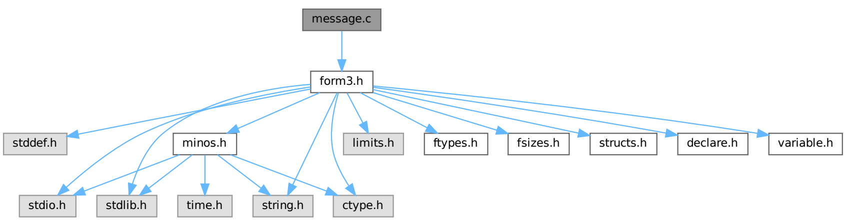
/*
    #] mprotectFree :
*/

#endif
```

4.69 message.c File Reference

`#include "form3.h"`

Include dependency graph for message.c:



Functions

- void Error0 (char *s)

- void Error1 (char ∗s, UBYTE ∗t)
- void Error2 (char ∗s1, char ∗s2, UBYTE ∗t)
- int MesWork (void)
- int MesPrint (va_alist)
- void Warning (char ∗s)
- void HighWarning (char ∗s)
- int MesCall (char ∗s)
- WORD MesCerr (char ∗s, UBYTE ∗t)
- WORD MesComp (char ∗s, UBYTE ∗p, UBYTE ∗q)
- void PrintTerm (WORD ∗term, char ∗where)
- void PrintTermC (WORD ∗term, char ∗where)
- void PrintSubTerm (WORD ∗term, char ∗where)
- void PrintWords (WORD ∗buffer, LONG number)
- void PrintSeq (WORD ∗a, char ∗text)

4.69.1 Detailed Description

Contains the routines that write messages. This includes the very important routine MesPrint which is the FORM equivalent of printf but then with escape sequences that are relevant for symbolic manipulation. The FORM statement Print "...." is passed almost literally to MesPrint.

Definition in file message.c.

4.69.2 Function Documentation

4.69.2.1 Error0()

```
void Error0 (
            char * s )
```

Definition at line 56 of file message.c.

4.69.2.2 Error1()

```
void Error1 (
            char * s,
            UBYTE * t )
```

Definition at line 67 of file message.c.

4.69.2.3 Error2()

```
void Error2 (
            char * s1,
            char * s2,
            UBYTE * t )
```

Definition at line 78 of file message.c.

4.69.2.4 MesWork()

```
int MesWork (
              void )
```

Definition at line 89 of file message.c.

4.69.2.5 MesPrint()

```
int MesPrint (
              va_alist )
```

Definition at line 140 of file message.c.

4.69.2.6 Warning()

```
void Warning (
              char * s )
```

Definition at line 790 of file message.c.

4.69.2.7 HighWarning()

```
void HighWarning (
              char * s )
```

Definition at line 802 of file message.c.

4.69.2.8 MesCall()

```
int MesCall (
              char * s )
```

Definition at line 814 of file message.c.

4.69.2.9 MesCerr()

```
WORD MesCerr (
              char * s,
              UBYTE * t )
```

Definition at line 824 of file message.c.

4.69.2.10 MesComp()

```
WORD MesComp (
              char * s,
              UBYTE * p,
              UBYTE * q )
```

Definition at line 843 of file message.c.

4.69.2.11 PrintTerm()

```
void PrintTerm (
             WORD * term,
             char * where )
```

Definition at line 857 of file message.c.

4.69.2.12 PrintTermC()

```
void PrintTermC (
             WORD * term,
             char * where )
```

Definition at line 887 of file message.c.

4.69.2.13 PrintSubTerm()

```
void PrintSubTerm (
             WORD * term,
             char * where )
```

Definition at line 921 of file message.c.

4.69.2.14 PrintWords()

```
void PrintWords (
             WORD * buffer,
             LONG number )
```

Definition at line 943 of file message.c.

4.69.2.15 PrintSeq()

```
void PrintSeq (
             WORD * a,
             char * text )
```

Definition at line 961 of file message.c.

4.70 message.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 */
/* #] License : */
/*
    #[ Includes :

    The static variables for the messages can remain as such also for
    the parallel version as messages are to be locked to avoid problems
    with simultaneous messages.
*/

#include "form3.h"

static int iswarning = 0;

static char hex[] = {'0','1','2','3','4','5','6','7','8','9',
                     'A','B','C','D','E','F'};

/*
    #] Includes :
    #[ exit :
        #[ Error0 :
*/

void Error0(char *s)
{
    MesPrint("=== %s",s);
    Terminate(-1);
}

/*
        #] Error0 :
        #[ Error1 :
*/

void Error1(char *s, UBYTE *t)
{
    MesPrint("@%s %s",s,t);
    Terminate(-1);
}

/*
        #] Error1 :
        #[ Error2 :
*/

void Error2(char *s1, char *s2, UBYTE *t)
{
    MesPrint("@%s%s %s",s1,s2,t);
    Terminate(-1);
}

/*
        #] Error2 :
        #[ MesWork :
*/

int MesWork(void)
```

```
{
    MesPrint("=== Workspace overflow. %l bytes is not enough.",AM.WorkSize);
    MesPrint("=== Change parameter WorkSpace in %s",setupfilename);
    Terminate(-1);
    return(-1);
}

/*
        #] MesWork :
        #[ MesPrint :

    Kind of a printf function for simple messages.
    The main concern is getting the arguments in a portable way.
    Note: many compilers have errors when sizeof(WORD) < sizeof(int)
    %a  array of size n WORDs (two parameters, first is int, second WORD *)
    %b  array of size n UBYTEs (two parameters, first is int, second UBYTE *)
    %C  array of size n chars (two parameters, first is int, second char *)
    %d  word;
    %l  long;
    %L  long long *;
    %s  string;
    %#i unsigned word filled
    %#d word positioned
    %#l long word positioned.
    %#L long long word * positioned.
    %#s string positioned.
    %#p position in file.
    %r  The current term in raw format (internal representation)
    %t  The current term (AN.currentTerm)
    %T  The current term (AN.currentTerm) with its sign
    %w  Number of the thread(worker)
    %$  The next $ in AN.listinprint
    %x  hexadecimal. Takes 8 places. Mainly for debugging.
    %%  %
    %#  #
    #   " ==> "
    @   " ==> "   Preprocessor error
    &   ' --> '   Regular compiler error
    Each call is terminated with a new line.
    Put a % at the end of the string to suppress the new line.

    New feature (7-dec-2011): The & will only work when we do not block it
    from the execution of the print statement because we need the & also for
    the tabulator in the print "" statement.
*/

int
#ifdef ANSI
MesPrint(const char *fmt, ... )
#else
MesPrint(va_alist)
va_dcl
#endif
{
    GETIDENTITY
    char Out[MAXLINELENGTH+14], *stopper, *t, *s, *u, c, *carray;
    UBYTE extrabuffer[MAXLINELENGTH+14];
    int w, x, i, specialerror = 0;
    LONG num, y;
    WORD *array;
    UBYTE *oldoutfill = AO.OutputLine, *barray;
    /*[19apr2004 mt]:*/
    LONG (*OldWrite)(int handle, UBYTE *buffer, LONG size) = WriteFile;
    /*:[19apr2004 mt]*/
    va_list ap;
#ifdef ANSI
    va_start(ap,fmt);
    s = (char *)fmt;
#else
    va_start(ap);
    s = va_arg(ap,char *);
#endif
#ifdef WITHMPI
    /*
     * On slaves, if AS.printflag is
     *   = 0 : print nothing.
     *   > 0 : synchronized output. All text will be sent to the master
     *         in the next MUNLOCK().
     *   < 0 : normal output.
     */
    if ( PF.me != MASTER && AS.printflag == 0 ) return(0);
    if ( PF.me == MASTER || AS.printflag < 0 )
#endif
    FLUSHCONSOLE;
    /*
     * MesPrints() never prints a message to an external channel even if
     * WriteFile is set to &WriteToExternalChannel.
```

```
     */
#ifdef WITHMPI
    WriteFile = PF.me == MASTER || AS.printflag > 0 ? &PF_WriteFileToFile : &WriteFileToFile;
#else
    WriteFile = &WriteFileToFile;
#endif
    AO.OutputLine = extrabuffer;
    t = Out;
    stopper = Out + AC.LineLength;
    while ( *s ) {
        if ( ( ( *s == '&' && AO.ErrorBlock == 0 ) || *s == '@' || *s == '#' ) && AC.CurrentStream !=
0 ) {
            u = (char *)AC.CurrentStream->name;
            while ( *u ) {
                *t++ = *u++;
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    num = 0; t = Out;
                }
            }
            *t++ = ' ';
            if ( t+20 >= stopper ) {
                num = t - Out;
                WriteString(ERROROUT,(UBYTE *)Out,num);
                num = 0; t = Out;
            }
            *t++ = 'L'; *t++ = 'i'; *t++ = 'n'; *t++ = 'e'; *t++ = ' ';
            if ( *s == '&' ) y = AC.CurrentStream->prevline;
            else             y = AC.CurrentStream->linenumber;
            t = LongCopy(y,t);
            if ( !iswarning && ( *s == '&' || *s == '@' ) ) {
                for ( i = 0; i < NumDoLoops; i++ ) DoLoops[i].errorsinloop = 1;
            }
        }
        if ( ( *s == '&' && AO.ErrorBlock == 0 ) ) {
            *t++ = ' '; *t++ = '-'; *t++ = '-'; *t++ = '>'; *t++ = ' '; s++;
        }
        else if ( *s == '@' || *s == '#' ) {
            *t++ = ' '; *t++ = '='; *t++ = '='; *t++ = '>'; *t++ = ' '; s++;
        }
/*
        else if ( *s == '&' && AO.ErrorBlock == 1 ) {

        }
*/
        else if ( *s != '%' ) {
            *t++ = *s++;
            if ( t >= stopper ) {
                num = t - Out;
                WriteString(ERROROUT,(UBYTE *)Out,num);
                num = 0; t = Out;
            }
        }
        else {
            s++;
            if ( *s == 'd' ) {
                if ( ( w = va_arg(ap, int) ) < 0 ) { *t++ = '-'; w = -w; }
                t = (char *)NumCopy(w,(UBYTE *)t);
            }
            else if ( *s == 'l' ) {
                if ( ( y = va_arg(ap, LONG) ) < 0 ) { *t++ = '-'; y = -y; }
                t = LongCopy(y,t);
            }
/*  #ifdef __GLIBC_HAVE_LONG_LONG */
            else if ( *s == 'p' ) {
                POSITION *pp;
                off_t ly;
                pp = va_arg(ap, POSITION *);
                ly = BASEPOSITION(*pp);
                if ( ly < 0 ) { *t++ = '-'; ly = -ly; }
/*----change 10-feb-2003 did not have & */
                t = LongLongCopy(&(ly),t);
            }
/*  #endif  */
            else if ( *s == 'c' ) {
                c = (char)(va_arg(ap, int));
                *t++ = c; *t = 0;
            }
            else if ( *s == 'a' ) {
                w = va_arg(ap, int);
                array = va_arg(ap,WORD *);
                while ( w > 0 ) {
                    t = (char *)NumCopy(*array,(UBYTE *)t);
                    if ( t >= stopper ) {
                        num = t - Out;
                        WriteString(ERROROUT,(UBYTE *)Out,num);
```

```
                    t = Out;
                    *t++ = ' ';
                }
                *t++ = ' ';
                w--; array++;
            }
        }
        else if ( *s == 'b' ) {
            w = va_arg(ap, int);
            barray = va_arg(ap,UBYTE *);
            while ( w > 0 ) {
                *t++ = hex[((*barray)»4)&0xF];
                *t++ = hex[(*barray)&0xF];
                *t = 0;
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    t = Out;
                    *t++ = ' ';
                }
                *t++ = ' ';
                w--; barray++;
            }
        }
        else if ( *s == 'C' ) {
            w = va_arg(ap, int);
            carray = va_arg(ap,char *);
            while ( w > 0 ) {
                if ( *carray < 32 ) *t++ = '^';
                else *t++ = *carray;
                *t = 0;
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    t = Out;
                    *t++ = ' ';
                }
                w--; carray++;
            }
        }
        else if ( *s == 'I' ) {
            int *iarray;
            w = va_arg(ap, int);
            iarray = va_arg(ap,int *);
            while ( w > 0 ) {
                t = (char *)LongCopy((LONG)(*iarray),(char *)t);
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    t = Out;
                    *t++ = ' ';
                }
                *t++ = ' ';
                w--; array++;
            }
        }
        else if ( *s == 'E' ) {
            LONG *larray;
            w = va_arg(ap, int);
            larray = va_arg(ap,LONG *);
            while ( w > 0 ) {
                t = (char *)LongCopy(*larray,(char *)t);
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    t = Out;
                    *t++ = ' ';
                }
                *t++ = ' ';
                w--; array++;
            }
        }
        else if ( *s == 's' ) {
            u = va_arg(ap,char *);
            while ( *u ) {
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    t = Out;
                }
                *t++ = *u++;
            }
            *t = 0;
        }
        else if ( *s == 't' || *s == 'T' ) {
            WORD oldskip = AO.OutSkip, noleadsign;
            WORD oldmode = AC.OutputMode;
```

```
                WORD oldbracket = AO.IsBracket;
                WORD oldlength = AC.LineLength;
                UBYTE *oldStop = AO.OutStop;
                if ( AN.currentTerm ) {
                    if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
                    AO.IsBracket = 0;
                    AO.OutSkip = 1;
                    AC.OutputMode = 0;
                    AO.OutFill = AO.OutputLine;
                    AO.OutStop = AO.OutputLine + AC.LineLength;
                    *t = 0;
                    AddToLine((UBYTE *)Out);
                    if ( *s == 'T' ) noleadsign = 1;
                    else noleadsign = 0;
                    if ( WriteInnerTerm(AN.currentTerm,noleadsign) ) Terminate(-1);
                    t = Out;
                    u = (char *)AO.OutputLine;
                    *(AO.OutFill) = 0;
                    while ( u < (char *)(AO.OutFill) ) *t++ = *u++;
                    *t = 0;
                    AO.OutSkip = oldskip;
                    AC.OutputMode = oldmode;
                    AO.IsBracket = oldbracket;
                    AC.LineLength = oldlength;
                    AO.OutStop = oldStop;
                }
            }
            else if ( *s == 'r' ) {
                WORD oldskip = AO.OutSkip;
                WORD oldmode = AC.OutputMode;
                WORD oldbracket = AO.IsBracket;
                WORD oldlength = AC.LineLength;
                UBYTE *oldStop = AO.OutStop;
                if ( AN.currentTerm ) {
                    WORD *tt = AN.currentTerm;
                    if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
                    AO.IsBracket = 0;
                    AO.OutSkip = 1;
                    AC.OutputMode = 0;
                    AO.OutFill = AO.OutputLine;
                    AO.OutStop = AO.OutputLine + AC.LineLength;
                    *t = 0;
                    i = *tt;
                    while ( --i >= 0 ) {
                        t = (char *)NumCopy(*tt,(UBYTE *)t);
                        tt++;
                        if ( t >= stopper ) {
                            num = t - Out;
                            WriteString(ERROROUT,(UBYTE *)Out,num);
                            num = 0; t = Out;
                        }
                        *t++ = ' '; *t++ = ' ';
                    }
                    *t = 0;
                    AO.OutSkip = oldskip;
                    AC.OutputMode = oldmode;
                    AO.IsBracket = oldbracket;
                    AC.LineLength = oldlength;
                    AO.OutStop = oldStop;
                }
            }
            else if ( *s == '$' ) {
/*
            #[ dollars :
*/
                WORD oldskip = AO.OutSkip;
                WORD oldmode = AC.OutputMode;
                WORD oldbracket = AO.IsBracket;
                WORD oldlength = AC.LineLength;
                UBYTE *oldStop = AO.OutStop;
                WORD *term, indsubterm[3], *tt;
                WORD value[5], first, num;
                if ( *AN.listinprint != DOLLAREXPRESSION ) {
                    specialerror = 1;
                }
                else {
                    DOLLARS d = Dollars + AN.listinprint[1];
#ifdef WITHPTHREADS
                    int nummodopt, dtype;
                    dtype = -1;
                    if ( AS.MultiThreaded ) {
                        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                            if ( AN.listinprint[1] == ModOptdollars[nummodopt].number ) break;
                        }
                        if ( nummodopt < NumModOptdollars ) {
                            dtype = ModOptdollars[nummodopt].type;
                            if ( dtype == MODLOCAL ) {
```

```
                                d = ModOptdollars[nummodopt].dstruct+AT.identity;
                            }
                            else {
                                LOCK(d->pthreadslockread);
                            }
                        }
                    }
#endif
                    AO.IsBracket = 0;
                    AO.OutSkip = 0;
                    AC.OutputMode = 0;
                    AO.OutFill = AO.OutputLine;
                    AO.OutStop = AO.OutputLine + AC.LineLength;
                    *t = 0;
                    AddToLine((UBYTE *)Out);
                    if ( d->nfactors >= 1 && AN.listinprint[2] == DOLLAREXPR2 ) {
                        if ( d->type == 0 ||
                         ( d->factors == 0 && d->nfactors != 1 ) ) goto dollarzero;
                        num = EvalDoLoopArg(BHEAD AN.listinprint+2,-1);
                        if ( num == 0 ) {
                            value[0] = 4; value[1] = d->nfactors; value[2] = 1; value[3] = 3; value[4] = 0;
                            term = value; goto printterms;
                        }
                        if ( num == 1 && d->nfactors == 1 ) {
                            term = d->where;
                            if ( *term == 0 ) goto dollarzero;
                            goto printterms;
                        }
                        if ( num > d->nfactors ) {
                            MesPrint("\nFactor number for dollar is too large.");
                            Terminate(-1);
                        }
                        term = d->factors[num-1].where;
                        if ( term == 0 ) {
                            if ( d->factors[num-1].value < 0 ) {
                                value[0] = 4; value[1] = -d->factors[num-1].value; value[2] = 1; value[3] = -3; value[4] = 0;
                            }
                            else {
                                value[0] = 4; value[1] = d->factors[num-1].value; value[2] = 1; value[3] = 3; value[4] = 0;
                            }
                            term = value;
                        }
                        goto printterms;
                    }
                    if ( d->type == DOLTERMS || d->type == DOLNUMBER ) {
                        term = d->where;
printterms:             first = 1;
                        do {
                            if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
                            AO.IsBracket = 0;
                            AO.OutSkip = 1;
                            AC.OutputMode = 0;
                            AO.OutFill = AO.OutputLine;
                            AO.OutStop = AO.OutputLine + AC.LineLength;
                            *t = 0;
                            AddToLine((UBYTE *)Out);
                            if ( WriteInnerTerm(term,first) ) {
#ifdef WITHPTHREADS
                                if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                Terminate(-1);
                            }
                            first = 0;
                            t = Out;
                            u = (char *)AO.OutputLine;
                            *(AO.OutFill) = 0;
                            while ( u < (char *)(AO.OutFill) ) *t++ = *u++;
                            *t = 0;
                            AO.OutSkip = oldskip;
                            AC.OutputMode = oldmode;
                            AO.IsBracket = oldbracket;
                            AC.LineLength = oldlength;
                            AO.OutStop = oldStop;
                            term += *term;
                        } while ( *term );
                        AO.OutSkip = oldskip;
                    }
                    else if ( d->type == DOLSUBTERM ) {
                        tt = d->where;
dosubterm:              if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
                        AO.IsBracket = 0;
                        AO.OutSkip = 1;
                        AC.OutputMode = 0;
                        AO.OutFill = AO.OutputLine;
```

```
                        AO.OutStop = AO.OutputLine + AC.LineLength;
                        *t = 0;
                        AddToLine((UBYTE *)Out);
                        if ( WriteSubTerm(tt,1) ) {
#ifdef WITHPTHREADS
                            if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                            Terminate(-1);
                        }
                        t = Out;
                        u = (char *)AO.OutputLine;
                        *(AO.OutFill) = 0;
                        while ( u < (char *)(AO.OutFill) ) *t++ = *u++;
                        *t = 0;
                        AO.OutSkip = oldskip;
                        AC.OutputMode = oldmode;
                        AO.IsBracket = oldbracket;
                        AC.LineLength = oldlength;
                        AO.OutStop = oldStop;
                    }
                    else if ( d->type == DOLUNDEFINED ) {
                        *t++ = '*'; *t++ = '*'; *t++ = '*'; *t = 0;
                    }
                    else if ( d->type == DOLZERO ) {
dollarzero:             *t++ = '0'; *t = 0;
                    }
                    else if ( d->type == DOLINDEX ) {
                        tt = indsubterm; *tt = INDEX;
                        tt[1] = 3; tt[2] = d->index;
                        goto dosubterm;
                    }
                    else if ( d->type == DOLARGUMENT ) {
                        if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
                        AO.IsBracket = 0;
                        AO.OutSkip = 1;
                        AC.OutputMode = 0;
                        AO.OutFill = AO.OutputLine;
                        AO.OutStop = AO.OutputLine + AC.LineLength;
                        *t = 0;
                        AddToLine((UBYTE *)Out);
                        WriteArgument(d->where);
                        t = Out;
                        u = (char *)AO.OutputLine;
                        *(AO.OutFill) = 0;
                        while ( u < (char *)(AO.OutFill) ) *t++ = *u++;
                        *t = 0;
                        AO.OutSkip = oldskip;
                        AC.OutputMode = oldmode;
                        AO.IsBracket = oldbracket;
                        AC.LineLength = oldlength;
                        AO.OutStop = oldStop;
                    }
                    else if ( d->type == DOLWILDARGS ) {
                        tt = d->where;
                        if ( *tt == 0 ) { tt++;
                         while ( *tt ) {
                          if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
                          AO.IsBracket = 0;
                          AO.OutSkip = 1;
                          AC.OutputMode = 0;
                          AO.OutFill = AO.OutputLine;
                          AO.OutStop = AO.OutputLine + AC.LineLength;
                          *t = 0;
                          AddToLine((UBYTE *)Out);
                          WriteArgument(tt);
                          NEXTARG(tt);
                          if ( *tt ) TokenToLine((UBYTE *)",");
                          t = Out;
                          u = (char *)AO.OutputLine;
                          *(AO.OutFill) = 0;
                          while ( u < (char *)(AO.OutFill) ) *t++ = *u++;
                          *t = 0;
                          AO.OutSkip = oldskip;
                          AC.OutputMode = oldmode;
                          AO.IsBracket = oldbracket;
                          AC.LineLength = oldlength;
                          AO.OutStop = oldStop;
                         }
                        }
                        else if ( *tt > 0 ) {   /* Tensor arguments */
                            i = *tt++;
                            while ( --i >= 0 ) {
                                indsubterm[0] = INDEX;
                                indsubterm[1] = 3;
                                indsubterm[2] = *tt++;
                                if ( AC.LineLength > MAXLINELENGTH ) AC.LineLength = MAXLINELENGTH;
                                AO.IsBracket = 0;
```

```
                                AO.OutSkip = 1;
                                AC.OutputMode = 0;
                                AO.OutFill = AO.OutputLine;
                                AO.OutStop = AO.OutputLine + AC.LineLength;
                                *t = 0;
                                AddToLine((UBYTE *)Out);
                                if ( WriteSubTerm(indsubterm,1) ) Terminate(-1);
                                if ( i > 0 ) TokenToLine((UBYTE *)",");
                                t = Out;
                                u = (char *)AO.OutputLine;
                                *(AO.OutFill) = 0;
                                while ( u < (char *)(AO.OutFill) ) *t++ = *u++;
                                *t = 0;
                                AO.OutSkip = oldskip;
                                AC.OutputMode = oldmode;
                                AO.IsBracket = oldbracket;
                                AC.LineLength = oldlength;
                                AO.OutStop = oldStop;
                            }
                        }
                    }
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                    AN.listinprint += 2;
                    while ( AN.listinprint[0] == DOLLAREXPR2 ) AN.listinprint += 2;
                }
/*
            #] dollars :
*/
            }
#ifdef WITHPTHREADS
            else if ( *s == 'W' ) { /* number of the thread with time */
                LONG millitime;
                WORD timepart;
                t = (char *)NumCopy(identity,(UBYTE *)t);
                millitime = TimeCPU(1);
                timepart = (WORD)(millitime%1000);
                millitime /= 1000;
                timepart /= 10;
                *t++ = '('; *t = 0;
                t = (char *)LongCopy(millitime,(char *)t);
                *t++ = '.'; *t = 0;
                t = (char *)NumCopy(timepart,(UBYTE *)t);
                *t++ = ')'; *t = 0;
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    num = 0; t = Out;
                }
            }
            else if ( *s == 'w' ) { /* number of the thread */
                t = (char *)NumCopy(identity,(UBYTE *)t);
            }
#elif defined(WITHMPI)
            else if ( *s == 'W' ) { /* number of the thread with time */
                LONG millitime;
                WORD timepart;
                t = (char *)NumCopy(PF.me,(UBYTE *)t);
                millitime = TimeCPU(1);
                timepart = (WORD)(millitime%1000);
                millitime /= 1000;
                timepart /= 10;
                *t++ = '('; *t = 0;
                t = (char *)LongCopy(millitime,(char *)t);
                *t++ = '.'; *t = 0;
                t = (char *)NumCopy(timepart,(UBYTE *)t);
                *t++ = ')'; *t = 0;
                if ( t >= stopper ) {
                    num = t - Out;
                    WriteString(ERROROUT,(UBYTE *)Out,num);
                    num = 0; t = Out;
                }
            }
            else if ( *s == 'w' ) { /* number of the thread */
                t = (char *)NumCopy(PF.me,(UBYTE *)t);
            }
#else
            else if ( *s == 'w' ) { }
            else if ( *s == 'W' ) { }
#endif
            else if ( FG.cTable[(int)*s] == 1 ) {
                x = *s++ - '0';
                while ( FG.cTable[(int)*s] == 1 )
                    x = 10 * x + *s++ - '0';

                if ( *s == 'l' || *s == 'd' ) {
```

```
                    if ( *s == 'l' ) { y = va_arg(ap,LONG); }
                    else { y = va_arg(ap,int); }
                    if ( y < 0 ) { y = -y; w = 1; }
                    else w = 0;
                    u = t + x;
                    do { *--u = y%10+'0'; y /= 10; } while ( y && u > t );
                    if ( w && u > t ) *--u = '-';
                    while ( --u >= t ) *u = ' ';
                    t += x;
                }
                else if ( *s == 's' ) {
                    u = va_arg(ap,char *);
                    i = 0;
                    while ( *u ) { i++; u++; }
                    if ( i > x ) i = x;
                    while ( x > i ) { *t++ = ' '; x--; }
                    t += x;
                    while ( --i >= 0 ) { *--t = *--u; }
                    t += x;
                }
                else if ( *s == 'p' ) {
                    POSITION *pp;
/*#ifdef __GLIBC_HAVE_LONG_LONG */
                    off_t ly;
/*
#else
                    LONG ly;
#endif
*/
                    pp = va_arg(ap,POSITION *);
                    ly = BASEPOSITION(*pp);
                    u = t + x;
                    do { *--u = ly%10+'0'; ly /= 10; } while ( ly && u > t );
                    while ( --u >= t ) *u = ' ';
                    t += x;
                }
                else if ( *s == 'i' ) {
                    w = va_arg(ap, int);
                    u = t + x;
                    do { *--u = (char)(w%10+'0'); w /= 10; } while ( u > t );
                    t += x;
                }
                else {
                    w = va_arg(ap, int);
                    u = t + x;
                    do { *--u = (char )(w%10+'0'); w /= 10; } while ( w && u > t );
                    while ( --u >= t ) *u = ' ';
                    t += x;
                }
            }
            else if ( *s == 'x' ) {
                char ccc;
                y = va_arg(ap, LONG);
                i = 2*sizeof(LONG);
                while ( --i > 0 ) {
                    ccc = ( y » (i*4) ) & 0xF;
                    if ( ccc ) break;
                }
                do {
                    ccc = ( y » (i*4) ) & 0xF;
                    *t++ = hex[(int)ccc];
                } while ( --i >= 0 );
            }
            else if ( *s == '#' ) *t++ = *s;
            else if ( *s == '%' ) *t++ = *s;
            else if ( *s == 0 ) { *t++ = 0; break; }
            else if ( *s == '&' ) {
                *t++ = *s;
            }
            else {
                *t++ = '%';
                s--;
            }
            s++;
        }
    }
    num = t - Out;
    WriteString(ERROROUT,(UBYTE *)Out,num);
    va_end(ap);
    if ( specialerror == 1 ) {
        MesPrint("!!!Wrong object in Print statement!!!");
        MesPrint("!!!Object encountered is of a different type as in the format specifier");
    }
    AO.OutputLine = oldoutfill;
    /*[19apr2004 mt]:*/
    WriteFile=OldWrite;
    /*:[19apr2004 mt]*/
```

```
    return(-1);
}

/*
        #] MesPrint :
        #[ Warning :
*/

void Warning(char *s)
{
    iswarning = 1;
    if ( AC.WarnFlag ) MesPrint("&Warning: %s",s);
    iswarning = 0;
}

/*
        #] Warning :
        #[ HighWarning :
*/

void HighWarning(char *s)
{
    iswarning = 1;
    if ( AC.WarnFlag >= 2 ) MesPrint("&Warning: %s",s);
    iswarning = 0;
}

/*
        #] HighWarning :
        #[ MesCall :
*/

int MesCall(char *s)
{
    return(MesPrint((char *)"Called from %s",s));
}

/*
        #] MesCall :
        #[ MesCerr :
*/

WORD MesCerr(char *s, UBYTE *t)
{
    UBYTE *u, c;
    WORD i = 11;
    u = t;
    while ( *u && --i >= 0 ) u--;
    u++;
    c = *++t;
    *t = 0;
    MesPrint("&Illegal %s: %s",s,u);
    *t = c;
    return(-1);
}

/*
        #] MesCerr :
        #[ MesComp :
*/

WORD MesComp(char *s, UBYTE *p, UBYTE *q)
{
    UBYTE c;
    c = *++q; *q = 0;
    MesPrint("&%s: %s",s,p);
    *q = c;
    return(-1);
}

/*
        #] MesComp :
        #[ PrintTerm :
*/

void PrintTerm(WORD *term, char *where)
{
    UBYTE OutBuf[140];
    WORD *t, x;
    int i;
    AO.OutFill = AO.OutputLine = OutBuf;
    t = term;
    AO.OutSkip = 3;
    FiniLine();
    TokenToLine((UBYTE *)where);
    TokenToLine((UBYTE *)": ");
    i = *t;
```

```
    while ( --i >= 0 ) {
        x = *t++;
        if ( x < 0 ) {
            x = -x;
            TokenToLine((UBYTE *)"-");
        }
        TalToLine((UWORD)(x));
        TokenToLine((UBYTE *)"  ");
    }
    AO.OutSkip = 0;
    FiniLine();
}

/*
        #] PrintTerm :
        #[ PrintTermC :
*/

void PrintTermC(WORD *term, char *where)
{
    UBYTE OutBuf[140];
    WORD *t, x;
    int i;
    if ( *term >= 0 ) {
        PrintTerm(term,where);
        return;
    }
    AO.OutFill = AO.OutputLine = OutBuf;
    t = term;
    AO.OutSkip = 3;
    FiniLine();
    TokenToLine((UBYTE *)where);
    TokenToLine((UBYTE *)": ");
    i = t[1]+2;
    while ( --i >= 0 ) {
        x = *t++;
        if ( x < 0 ) {
            x = -x;
            TokenToLine((UBYTE *)"-");
        }
        TalToLine((UWORD)(x));
        TokenToLine((UBYTE *)"  ");
    }
    AO.OutSkip = 0;
    FiniLine();
}

/*
        #] PrintTermC :
        #[ PrintSubTerm :
*/

void PrintSubTerm(WORD *term, char *where)
{
    UBYTE OutBuf[140];
    WORD *t;
    int i;
    AO.OutFill = AO.OutputLine = OutBuf;
    t = term;
    AO.OutSkip = 3;
    FiniLine();
    TokenToLine((UBYTE *)where);
    TokenToLine((UBYTE *)": ");
    i = t[1];
    while ( --i >= 0 ) { TalToLine((UWORD)(*t++)); TokenToLine((UBYTE *)"  "); }
    AO.OutSkip = 0;
    FiniLine();
}

/*
        #] PrintSubTerm :
        #[ PrintWords :
*/

void PrintWords(WORD *buffer, LONG number)
{
    UBYTE OutBuf[140];
    WORD *t;
    AO.OutFill = AO.OutputLine = OutBuf;
    t = buffer;
    AO.OutSkip = 3;
    FiniLine();
    while ( --number >= 0 ) { TalToLine((UWORD)(*t++)); TokenToLine((UBYTE *)"  "); }
    AO.OutSkip = 0;
    FiniLine();
}

```

```
/*
        #] PrintWords :
        #[ PrintSeq :
*/

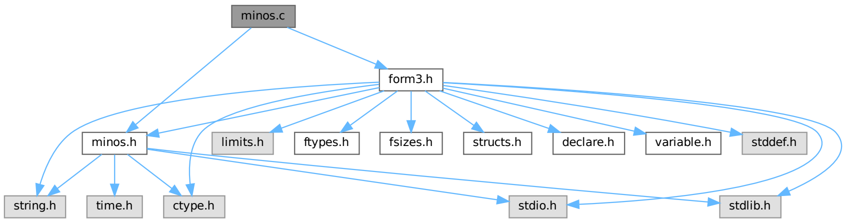
void PrintSeq(WORD *a,char *text)
{
    MesPrint(" %s:",text);
    while ( *a ) {
        MesPrint("     %a",a[0],a);
        a += *a;
    }
}

/*
        #] PrintSeq :
    #] exit :
*/
```

4.71 minos.c File Reference

```
#include "form3.h"
#include "minos.h"
```

Include dependency graph for minos.c:



Macros

- #define CFD(y, s, type, x, j)
- #define CTD(y, s, type, x, j)

Functions

- int minosread (FILE *f, char *buffer, MLONG size)
- int minoswrite (FILE *f, char *buffer, MLONG size)
- char * str_dup (char *str)
- void convertblock (INDEXBLOCK *in, INDEXBLOCK *out, int mode)
- void convertnamesblock (NAMESBLOCK *in, NAMESBLOCK *out, int mode)
- void convertiniinfo (INIINFO *in, INIINFO *out, int mode)
- FILE * LocateBase (char **name, char **newname, char *iomode)
- int ReadIndex (DBASE *d)
- int WriteIndexBlock (DBASE *d, MLONG num)
- int WriteNamesBlock (DBASE *d, MLONG num)
- int WriteIndex (DBASE *d)
- int WriteIniInfo (DBASE *d)
- int ReadIniInfo (DBASE *d)
- DBASE * GetDbase (char *filename, MLONG rwmode)

- DBASE ∗ NewDbase (char ∗name, MLONG number)
- void FreeTableBase (DBASE ∗d)
- int ComposeTableNames (DBASE ∗d)
- DBASE ∗ OpenDbase (char ∗filename)
- MLONG AddTableName (DBASE ∗d, char ∗name, TABLES T)
- MLONG GetTableName (DBASE ∗d, char ∗name)
- int PutTableNames (DBASE ∗d)
- int AddToIndex (DBASE ∗d, MLONG number)
- MLONG AddObject (DBASE ∗d, MLONG tablenumber, char ∗arguments, char ∗rhs)
- MLONG FindTableNumber (DBASE ∗d, char ∗name)
- int WriteObject (DBASE ∗d, MLONG tablenumber, char ∗arguments, char ∗rhs, MLONG number)
- char ∗ ReadObject (DBASE ∗d, MLONG tablenumber, char ∗arguments)
- char ∗ ReadijObject (DBASE ∗d, MLONG i, MLONG j, char ∗arguments)
- int ExistsObject (DBASE ∗d, MLONG tablenumber, char ∗arguments)
- int DeleteObject (DBASE ∗d, MLONG tablenumber, char ∗arguments)

Variables

- int withoutflush = 0

4.71.1 Detailed Description

These are the low level functions for the database part of the tablebases. These routines have been copied (and then adapted) from the minos database program. This file goes together with minos.h

Definition in file minos.c.

4.71.2 Macro Definition Documentation

4.71.2.1 CFD

```
#define CFD(
        y,
        s,
        type,
        x,
        j )
```

Value:

```
for(x=0,j=0;j<((int)sizeof(type));j++) \
{x=(x«8)+((*s++)&0x00FF);} y=x;
```

Definition at line 55 of file minos.c.

4.71.2.2 CTD

```
#define CTD(
        y,
        s,
        type,
        x,
        j )
```

Value:

```
x=y;for(j=sizeof(type)-1;j>=0;j--){s[j]=x&0xFF; \
x»=8;} s += sizeof(type);
```

Definition at line 57 of file minos.c.

4.71.3 Function Documentation

4.71.3.1 minosread()

```
int minosread (
            FILE * f,
            char * buffer,
            MLONG size )
```

Definition at line 66 of file minos.c.

4.71.3.2 minoswrite()

```
int minoswrite (
            FILE * f,
            char * buffer,
            MLONG size )
```

Definition at line 83 of file minos.c.

4.71.3.3 str_dup()

```
char * str_dup (
            char * str )
```

Definition at line 101 of file minos.c.

4.71.3.4 convertblock()

```
void convertblock (
            INDEXBLOCK * in,
            INDEXBLOCK * out,
            int mode )
```

Definition at line 120 of file minos.c.

4.71.3.5 convertnamesblock()

```
void convertnamesblock (
            NAMESBLOCK * in,
            NAMESBLOCK * out,
            int mode )
```

Definition at line 171 of file minos.c.

4.71.3.6 convertiniinfo()

```
void convertiniinfo (
            INIINFO * in,
            INIINFO * out,
            int mode )
```

Definition at line 197 of file minos.c.

4.71.3.7 LocateBase()

```
FILE * LocateBase (
            char ** name,
            char ** newname,
            char * iomode )
```

Definition at line 225 of file minos.c.

4.71.3.8 ReadIndex()

```
int ReadIndex (
            DBASE * d )
```

Definition at line 288 of file minos.c.

4.71.3.9 WriteIndexBlock()

```
int WriteIndexBlock (
            DBASE * d,
            MLONG num )
```

Definition at line 401 of file minos.c.

4.71.3.10 WriteNamesBlock()

```
int WriteNamesBlock (
            DBASE * d,
            MLONG num )
```

Definition at line 422 of file minos.c.

4.71.3.11 WriteIndex()

```
int WriteIndex (
            DBASE * d )
```

Definition at line 445 of file minos.c.

4.71.3.12 WriteIniInfo()

```
int WriteIniInfo (
            DBASE * d )
```

Definition at line 508 of file minos.c.

4.71.3.13 ReadIniInfo()

```
int ReadIniInfo (
            DBASE * d )
```

Definition at line 525 of file minos.c.

4.71.3.14 GetDbase()

```
DBASE * GetDbase (
            char * filename,
            MLONG rwmode )
```

Definition at line 548 of file minos.c.

4.71.3.15 NewDbase()

```
DBASE * NewDbase (
            char * name,
            MLONG number )
```

Definition at line 604 of file minos.c.

4.71.3.16 FreeTableBase()

```
void FreeTableBase (
            DBASE * d )
```

Definition at line 741 of file minos.c.

4.71.3.17 ComposeTableNames()

```
int ComposeTableNames (
            DBASE * d )
```

Definition at line 773 of file minos.c.

4.71.3.18 OpenDbase()

```
DBASE * OpenDbase (
            char * filename )
```

Definition at line 822 of file minos.c.

4.71.3.19 AddTableName()

```
MLONG AddTableName (
            DBASE * d,
            char * name,
            TABLES T )
```

Definition at line 856 of file minos.c.

4.71.3.20 GetTableName()

```
MLONG GetTableName (
            DBASE * d,
            char * name )
```

Definition at line 939 of file minos.c.

4.71.3.21 PutTableNames()

```
int PutTableNames (
            DBASE * d )
```

Definition at line 971 of file minos.c.

4.71.3.22 AddToIndex()

```
int AddToIndex (
            DBASE * d,
            MLONG number )
```

Definition at line 1067 of file minos.c.

4.71.3.23 AddObject()

```
MLONG AddObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments,
            char * rhs )
```

Definition at line 1154 of file minos.c.

4.71.3.24 FindTableNumber()

```
MLONG FindTableNumber (
            DBASE * d,
            char * name )
```

Definition at line 1168 of file minos.c.

4.71.3.25 WriteObject()

```
int WriteObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments,
            char * rhs,
            MLONG number )
```

Definition at line 1196 of file minos.c.

4.71.3.26 ReadObject()

```
char * ReadObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments )
```

Definition at line 1296 of file minos.c.

4.71.3.27 ReadijObject()

```
char * ReadijObject (
            DBASE * d,
            MLONG i,
            MLONG j,
            char * arguments )
```

Definition at line 1377 of file minos.c.

4.71.3.28 ExistsObject()

```
int ExistsObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments )
```

Definition at line 1429 of file minos.c.

4.71.3.29 DeleteObject()

```
int DeleteObject (
            DBASE * d,
            MLONG tablenumber,
            char * arguments )
```

Definition at line 1461 of file minos.c.

4.71.4 Variable Documentation

4.71.4.1 withoutflush

```
int withoutflush = 0
```

Definition at line 45 of file minos.c.

4.72 minos.c

Go to the documentation of this file.

```
00001
00007 /* #[ License : */
00008 /*
00009  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00010  *   When using this file you are requested to refer to the publication
00011  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00012  *   This is considered a matter of courtesy as the development was paid
00013  *   for by FOM the Dutch physics granting agency and we would like to
00014  *   be able to track its scientific use to convince FOM of its value
00015  *   for the community.
00016  *
00017  *   This file is part of FORM.
00018  *
00019  *   FORM is free software: you can redistribute it and/or modify it under the
00020  *   terms of the GNU General Public License as published by the Free Software
00021  *   Foundation, either version 3 of the License, or (at your option) any later
00022  *   version.
00023  *
00024  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00025  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00026  *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
00027  *   details.
00028  *
00029  *   You should have received a copy of the GNU General Public License along
00030  *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00031  */
00032 /* #] License : */
00033 /*
00034     #[ Includes :
00035 
00036     File contains the lowlevel routines for the management of primitive
00037     database-like files. The structures are explained in the file manage.h
00038     Original file for minos made by J.Vermaseren, april-1994.
00039 
00040     Note: the minos primitives for writing are never invoked in parallel.
00041 */
00042 
00043 #include "form3.h"
00044 #include "minos.h"
00045 int withoutflush = 0;
00046 
00047 /*
00048     #] Includes :
00049     #[ Variables :
00050 */
00051 
00052 static INDEXBLOCK scratchblock;
00053 static NAMESBLOCK scratchnamesblock;
00054 
00055 #define CFD(y,s,type,x,j) for(x=0,j=0;j<((int)sizeof(type));j++) \
00056     {x=(x«8)+((*s++)&0x00FF);} y=x;
00057 #define CTD(y,s,type,x,j) x=y;for(j=sizeof(type)-1;j>=0;j--){s[j]=x&0xFF; \
00058                       x»=8;} s += sizeof(type);
00059 
00060 /*
00061     #] Variables :
00062     #[ Utilities :
00063         #[ minosread :
00064 */
00065 
00066 int minosread(FILE *f,char *buffer,MLONG size)
00067 {
00068     MLONG x;
00069     while ( size > 0 ) {
00070         x = fread(buffer,sizeof(char),size,f);
00071         if ( x <= 0 ) return(-1);
```

```
        buffer += x;
        size -= x;
    }
    return(0);
}

/*
        #] minosread :
        #[ minoswrite :
*/

int minoswrite(FILE *f,char *buffer,MLONG size)
{
    MLONG x;
    while ( size > 0 ) {
        x = fwrite(buffer,sizeof(char),size,f);
        if ( x <= 0 ) return(-1);
        buffer += x;
        size -= x;
    }
    if ( withoutflush == 0 ) fflush(f);
    return(0);
}

/*
        #] minoswrite :
        #[ str_dup :
*/

char *str_dup(char *str)
{
    char *s, *t;
    int i;
    s = str;
    while ( *s ) s++;
    i = s - str + 1;
    if ( ( s = (char *)Malloc1((size_t)i,"a string copy") ) == 0 ) return(0);
    t = s;
    while ( *str ) *t++ = *str++;
    *t = 0;
    return(s);
}

/*
        #] str_dup :
        #[ convertblock :
*/

void convertblock(INDEXBLOCK *in,INDEXBLOCK *out,int mode)
{
    char *s,*t;
    MLONG i, x;
    int j;
    OBJECTS *obj;
    switch ( mode ) {
    case TODISK:
        s = (char *)out;
        CTD(in->flags,s,MLONG,x,j)
        CTD(in->previousblock,s,MLONG,x,j)
        CTD(in->position,s,MLONG,x,j)
        for ( i = 0, obj = in->objects; i < NUMOBJECTS; i++, obj++ ) {
            CTD(obj->position,s,MLONG,x,j)
            CTD(obj->size,s,MLONG,x,j)
            CTD(obj->date,s,MLONG,x,j)
            CTD(obj->tablenumber,s,MLONG,x,j)
            CTD(obj->uncompressed,s,MLONG,x,j)
            CTD(obj->spare1,s,MLONG,x,j)
            CTD(obj->spare2,s,MLONG,x,j)
            CTD(obj->spare3,s,MLONG,x,j)
            t = obj->element;
            for ( j = 0; j < ELEMENTSIZE; j++ ) *s++ = *t++;
        }
        break;
    case FROMDISK:
        s = (char *)in;
        CFD(out->flags,s,MLONG,x,j)
        CFD(out->previousblock,s,MLONG,x,j)
        CFD(out->position,s,MLONG,x,j)
        for ( i = 0, obj = out->objects; i < NUMOBJECTS; i++, obj++ ) {
            CFD(obj->position,s,MLONG,x,j)
            CFD(obj->size,s,MLONG,x,j)
            CFD(obj->date,s,MLONG,x,j)
            CFD(obj->tablenumber,s,MLONG,x,j)
            CFD(obj->uncompressed,s,MLONG,x,j)
            CFD(obj->spare1,s,MLONG,x,j)
            CFD(obj->spare2,s,MLONG,x,j)
            CFD(obj->spare3,s,MLONG,x,j)
```

```
            t = obj->element;
            for ( j = 0; j < ELEMENTSIZE; j++ ) *t++ = *s++;
        }
        break;
    }
}

/*
        #] convertblock :
        #[ convertnamesblock :
*/

void convertnamesblock(NAMESBLOCK *in,NAMESBLOCK *out,int mode)
{
    char *s;
    MLONG x;
    int j;
    switch ( mode ) {
    case TODISK:
        s = (char *)out;
        CTD(in->previousblock,s,MLONG,x,j)
        CTD(in->position,s,MLONG,x,j)
        for ( j = 0; j < NAMETABLESIZE; j++ ) out->names[j] = in->names[j];
        break;
    case FROMDISK:
        s = (char *)in;
        CFD(out->previousblock,s,MLONG,x,j)
        CFD(out->position,s,MLONG,x,j)
        for ( j = 0; j < NAMETABLESIZE; j++ ) out->names[j] = in->names[j];
        break;
    }
}

/*
        #] convertnamesblock :
        #[ convertiniinfo :
*/

void convertiniinfo(INIINFO *in,INIINFO *out,int mode)
{
    char *s;
    MLONG i, x, *y;
    int j;
    switch ( mode ) {
    case TODISK:
        s = (char *)out; y = (MLONG *)in;
        for ( i = sizeof(INIINFO)/sizeof(MLONG); i > 0; i-- ) {
            CTD(*y,s,MLONG,x,j)
            y++;
        }
        break;
    case FROMDISK:
        s = (char *)in; y = (MLONG *)out;
        for ( i = sizeof(INIINFO)/sizeof(MLONG); i > 0; i-- ) {
            CFD(*y,s,MLONG,x,j)
            y++;
        }
        break;
    }
}

/*
        #] convertiniinfo :
        #[ LocateBase :
*/

FILE *LocateBase(char **name, char **newname, char *iomode)
{
    FILE *handle;
    int  namesize, i;
    UBYTE *s, *to, *u1, *u2, *indir;

    if ( ( handle = fopen(*name,iomode) ) != 0 ) {
        *newname = (char *)strDup1((UBYTE *)(*name),"LocateBase");
        return(handle);
    }
    namesize = 2; s = (UBYTE *)(*name);
    while ( *s ) { s++; namesize++; }
    indir = AM.IncDir;
    if ( indir ) {
        s = indir; i = 0;
        while ( *s ) { s++; i++; }
        *newname = (char *)Malloc1(namesize+i,"LocateBase");
        s = indir; to = (UBYTE *)(*newname);
        while ( *s ) *to++ = *s++;
        if ( to > (UBYTE *)(*newname) && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
        s = (UBYTE *)(*name);
```

```
        while ( *s ) *to++ = *s++;
        *to = 0;
        if ( ( handle = fopen(*newname,iomode) ) != 0 ) {
            return(handle);
        }
        M_free(*newname,"LocateBase, incdir/file");
    }
    if ( AM.Path ) {
        u1 = AM.Path;
        while ( *u1 ) {
            u2 = u1; i = 0;
            while ( *u1 && *u1 != ':' ) {
                if ( *u1 == '\\' ) u1++;
                u1++; i++;
            }
            *newname = (char *)Malloc1(namesize+i,"LocateBase");
            s = u2; to = (UBYTE *)(*newname);
            while ( s < u1 ) {
                if ( *s == '\\' ) s++;
                *to++ = *s++;
            }
            if ( to > (UBYTE *)(*newname) && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
            s = (UBYTE *)(*name);
            while ( *s ) *to++ = *s++;
            *to = 0;
            if ( ( handle = fopen(*newname,iomode) ) != 0 ) {
                return(handle);
            }
            M_free(*newname,"LocateBase Path/file");
            if ( *u1 ) u1++;
        }
    }
/*  Error1("LocateBase: Cannot find file",*name); */
    return(0);
}

/*
        #] LocateBase :
    #] Utilities :
    #[ ReadIndex :
*/

int ReadIndex(DBASE *d)
{
    MLONG i;
    INDEXBLOCK **ib;
    NAMESBLOCK **ina;
    MLONG position, size;
/*
    Allocate the pieces one by one (makes it easier to give it back)
*/
    if ( d->info.numberofindexblocks <= 0 ) return(0);
#ifndef WORDSIZE32
    if ( sizeof(INDEXBLOCK)*d->info.numberofindexblocks > MAXINDEXSIZE ) {
        MesPrint("We need more than %ld bytes for the index.\n",MAXINDEXSIZE);
        MesPrint("The file %s may not be a proper database\n",d->name);
        return(-1);
    }
#endif
    size = sizeof(INDEXBLOCK *)*d->info.numberofindexblocks;
    if ( ( ib = (INDEXBLOCK **)Malloc1(size,"tb,index") ) == 0 ) return(-1);
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
        if ( ( ib[i] = (INDEXBLOCK *)Malloc1(sizeof(INDEXBLOCK),"index block") ) == 0 ) {
            for ( --i; i >= 0; i-- ) M_free(ib[i],"tb,indexblock");
            M_free(ib,"tb,index");
            return(-1);
        }
    }
    size = sizeof(NAMESBLOCK *)*d->info.numberofnamesblocks;
    if ( ( ina = (NAMESBLOCK **)Malloc1(size,"tb,indexnames") ) == 0 ) return(-1);
    for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
        if ( ( ina[i] = (NAMESBLOCK *)Malloc1(sizeof(NAMESBLOCK),"index names block") ) == 0 ) {
            for ( --i; i >= 0; i-- ) M_free(ina[i],"index names block");
            M_free(ina,"tb,indexnames");
            for ( i = 0; i < d->info.numberofindexblocks; i++ ) M_free(ib[i],"tb,indexblock");
            M_free(ib,"tb,index");
            return(-1);
        }
    }
/*
    Read the index blocks, from the back to the front. The links are only
    reliable that way.
*/
    position = d->info.lastindexblock;
    for ( i = d->info.numberofindexblocks - 1; i >= 0; i-- ) {
        fseek(d->handle,position,SEEK_SET);
        if ( minosread(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
```

```
            MesPrint("Error while reading file %s\n",d->name);
thisiswrong:
            for ( i = 0; i < d->info.numberofnamesblocks; i++ ) M_free(ina[i],"index names block");
            M_free(ina,"tb,indexnames");
            for ( i = 0; i < d->info.numberofindexblocks; i++ ) M_free(ib[i],"tb,indexblock");
            M_free(ib,"tb,index");
            return(-1);
        }
        convertblock(&scratchblock,ib[i],FROMDISK);
        if ( ib[i]->position != position ||
        ( ib[i]->previousblock <= 0 && i > 0 ) ) {
            MesPrint("File %s has inconsistent contents\n",d->name);
            goto thisiswrong;
        }
        position = ib[i]->previousblock;
    }
    d->info.firstindexblock = ib[0]->position;
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
        ib[i]->flags &= MCLEANFLAG;
    }
/*
    Read the names blocks, from the back to the front. The links are only
    reliable that way.
*/
    position = d->info.lastnameblock;
    for ( i = d->info.numberofnamesblocks - 1; i >= 0; i-- ) {
        fseek(d->handle,position,SEEK_SET);
        if ( minosread(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
            MesPrint("Error while reading file %s\n",d->name);
            goto thisiswrong;
        }
        convertnamesblock(&scratchnamesblock,ina[i],FROMDISK);
        if ( ina[i]->position != position ||
        ( ina[i]->previousblock <= 0 && i > 0 ) ) {
            MesPrint("File %s has inconsistent contents\n",d->name);
            goto thisiswrong;
        }
        position = ina[i]->previousblock;
    }
    d->info.firstnameblock = ina[0]->position;
/*
    Give the old info back to the system.
*/
    if ( d->iblocks ) {
        for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
            if ( d->iblocks[i] ) M_free(d->iblocks[i],"d->iblocks[i]");
        }
        M_free(d->iblocks,"d->iblocks");
    }
    if ( d->nblocks ) {
        for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
            if ( d->nblocks[i] ) M_free(d->nblocks[i],"d->nblocks[i]");
        }
        M_free(d->nblocks,"d->nblocks");
    }
/*
    And substitute the new blocks
*/
    d->iblocks = ib;
    d->nblocks = ina;
    return(0);
}

/*
    #] ReadIndex :
    #[ WriteIndexBlock :
*/

int WriteIndexBlock(DBASE *d,MLONG num)
{
    if ( num >= d->info.numberofindexblocks ) {
        MesPrint("Illegal number specified for number of index blocks\n");
        return(-1);
    }
    fseek(d->handle,d->iblocks[num]->position,SEEK_SET);
    convertblock(d->iblocks[num],&scratchblock,TODISK);
    if ( minoswrite(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
        MesPrint("Error while writing an index block in file %s\n",d->name);
        MesPrint("File may be unreliable now\n");
        return(-1);
    }
    return(0);
}

/*
    #] WriteIndexBlock :
    #[ WriteNamesBlock :
```

```
*/

int WriteNamesBlock(DBASE *d,MLONG num)
{
    if ( num >= d->info.numberofnamesblocks ) {
        MesPrint("Illegal number specified for number of names blocks\n");
        return(-1);
    }
    fseek(d->handle,d->nblocks[num]->position,SEEK_SET);
    convertnamesblock(d->nblocks[num],&scratchnamesblock,TODISK);
    if ( minoswrite(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
        MesPrint("Error while writing a names block in file %s\n",d->name);
        MesPrint("File may be unreliable now\n");
        return(-1);
    }
    return(0);
}

/*
    #] WriteNamesBlock :
    #[ WriteIndex :

    Problem here is to get the links right.
*/

int WriteIndex(DBASE *d)
{
    MLONG i, position;
    if ( d->iblocks == 0 ) return(0);
    if ( d->nblocks == 0 ) return(0);
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
        if ( d->iblocks[i] == 0 ) {
            MesPrint("Error: unassigned index blocks. Cannot write\n");
            return(-1);
        }
    }
    for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
        if ( d->nblocks[i] == 0 ) {
            MesPrint("Error: unassigned names blocks. Cannot write\n");
            return(-1);
        }
    }
    d->info.lastindexblock = -1;
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
        position = d->iblocks[i]->position;
        if ( position <= 0 ) {
            fseek(d->handle,0,SEEK_END);
            position = ftell(d->handle);
            d->iblocks[i]->position = position;
            if ( i <= 0 ) d->iblocks[i]->previousblock = -1;
            else d->iblocks[i]->previousblock = d->iblocks[i-1]->position;
        }
        else fseek(d->handle,position,SEEK_SET);
        convertblock(d->iblocks[i],&scratchblock,TODISK);
        if ( minoswrite(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
            MesPrint("Error while writing index of file %s",d->name);
            d->iblocks[i]->position = -1;
            return(-1);
        }
        d->info.lastindexblock = position;
    }
    d->info.lastnameblock = -1;
    for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
        position = d->nblocks[i]->position;
        if ( position <= 0 ) {
            fseek(d->handle,0,SEEK_END);
            position = ftell(d->handle);
            d->nblocks[i]->position = position;
            if ( i <= 0 ) d->nblocks[i]->previousblock = -1;
            else d->nblocks[i]->previousblock = d->nblocks[i-1]->position;
        }
        else fseek(d->handle,position,SEEK_SET);
        convertnamesblock(d->nblocks[i],&scratchnamesblock,TODISK);
        if ( minoswrite(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
            MesPrint("Error while writing index of file %s",d->name);
            d->nblocks[i]->position = -1;
            return(-1);
        }
        d->info.lastnameblock = position;
    }
    return(0);
}

/*
    #] WriteIndex :
    #[ WriteIniInfo :
*/
```

```
int WriteIniInfo(DBASE *d)
{
    INIINFO inf;
    fseek(d->handle,0,SEEK_SET);
    convertiniinfo(&(d->info),&inf,TODISK);
    if ( minoswrite(d->handle,(char *)(&inf),sizeof(INIINFO)) ) {
        MesPrint("Error while writing masterindex of file %s",d->name);
        return(-1);
    }
    return(0);
}

/*
    #] WriteIniInfo :
    #[ ReadIniInfo :
*/

int ReadIniInfo(DBASE *d)
{
    INIINFO inf;
    fseek(d->handle,0,SEEK_SET);
    if ( minosread(d->handle,(char *)(&inf),sizeof(INIINFO)) ) {
        MesPrint("Error while reading masterindex of file %s",d->name);
        return(-1);
    }
    convertiniinfo(&inf,&(d->info),FROMDISK);
    if ( d->info.entriesinindex < 0
    || d->info.numberofindexblocks < 0
    || d->info.lastindexblock < 0 ) {
        MesPrint("The file %s is not a proper database\n",d->name);
        return(-1);
    }
    return(0);
}

/*
    #] ReadIniInfo :
    #[ GetDbase :
*/

DBASE *GetDbase(char *filename, MLONG rwmode)
{
    FILE *f;
    DBASE *d;
    char *newname;
    if ( rwmode == 0 ) {
        if ( ( f = LocateBase(&filename,&newname,"rb") ) == 0 ) {

            MesPrint("&Trying to open non-existent TableBase in readonly mode: %s", filename);
            Terminate(-1);
        }
    } else {
        if ( ( f = LocateBase(&filename,&newname,"r+b") ) == 0 ) {

            return(NewDbase(filename,0));
        }
    }

/*  setbuf(f,0); */
    d = (DBASE *)From0List(&(AC.TableBaseList));
    d->mode = 0;
    d->tablenamessize = 0;
    d->topnumber = 0;
    d->tablenamefill = 0;
    d->iblocks = 0;
    d->nblocks = 0;
    d->tablenames = 0;
    d->rwmode = rwmode;

    d->info.entriesinindex = 0;
    d->info.numberofindexblocks = 0;
    d->info.firstindexblock = 0;
    d->info.lastindexblock = 0;
    d->info.numberoftables = 0;
    d->info.numberofnamesblocks = 0;
    d->info.firstnameblock = 0;
    d->info.lastnameblock = 0;

    d->name = str_dup(filename); /* For the moment just for the error messages */
    d->handle = f;
    if ( ReadIniInfo(d) || ReadIndex(d) ) { M_free(d,"index-d"); fclose(f); return(0); }
    if ( ComposeTableNames(d) < 0 ) { FreeTableBase(d); fclose(f); return(0); }
    // free allocation from previous str_dup
    M_free(d->name, "from str_dup");
    d->name = str_dup(filename);
    d->fullname = newname;
```

```
    return(d);
}

/*
    #] GetDbase :
    #[ NewDbase :

    Creates a new database with 'number' entries in the index.
*/

DBASE *NewDbase(char *name,MLONG number)
{
    FILE *f;
    DBASE *d;
    MLONG numblocks, numnameblocks, i;
    char *s;
/*----------change 10-feb-2003 */
    int j, jj;
    MLONG t = (MLONG)(time(0));
/*-------------------------------*/
    if ( number < 0 ) number = 0;
    if ( ( f = fopen(name,"w+b") ) == 0 ) {
        MesPrint("Could not create a new file with name %s\n",name);
        return(0);
    }
    numblocks = (number+NUMOBJECTS-1)/NUMOBJECTS;
    numnameblocks = 1;
    if ( numblocks <= 0 ) numblocks = 1;
    if ( numnameblocks <= 0 ) numnameblocks = 1;
    d = (DBASE *)FromOList(&(AC.TableBaseList));
    if ( ( d->iblocks = (INDEXBLOCK **)Malloc1(numblocks*sizeof(INDEXBLOCK *),
    "new database") ) == 0 ) {
        NumTableBases--;
        return(0);
    }
    d->tablenames = 0;
    d->tablenamessize = 0;
    d->topnumber = 0;
    d->tablenamefill = 0;
    d->rwmode = 1;

    d->mode = 0;
    if ( ( d->nblocks = (NAMESBLOCK **)Malloc1(sizeof(NAMESBLOCK *)*numnameblocks,
    "new database") ) == 0 ) {
        M_free(d->iblocks,"new database");
        NumTableBases--;
        return(0);
    }
    if ( ( f = fopen(name,"w+b") ) == 0 ) {
        MesPrint("Could not create new file %s\n",name);
        NumTableBases--;
        return(0);
    }
/*  setbuf(f,0); */
    d->name = str_dup(name);
    d->fullname = str_dup(name);
    d->handle = f;

    d->info.entriesinindex = number;
    d->info.numberofindexblocks = numblocks;
    d->info.numberofnamesblocks = numnameblocks;
    d->info.firstindexblock = 0;
    d->info.lastindexblock = 0;
    d->info.numberoftables = 0;
    d->info.firstnameblock = 0;
    d->info.lastnameblock = 0;

    if ( WriteIniInfo(d) ) {
getout:
        fclose(f);
        remove(d->fullname);
        if ( d->name ) { M_free(d->name,"name tablebase"); d->name = 0; }
        if ( d->fullname ) { M_free(d->fullname,"fullname tablebase"); d->fullname = 0; }
        M_free(d->nblocks,"new database");
        M_free(d->iblocks,"new database");
        NumTableBases--;
        return(0);
    }
    for ( i = 0; i < numblocks; i++ ) {
        if ( ( d->iblocks[i] = (INDEXBLOCK *)Malloc1(sizeof(INDEXBLOCK),
        "index blocks of new database") ) == 0 ) {
            while ( --i >= 0 ) M_free(d->iblocks[i],"index blocks of new database");
            goto getout;
        }
        if ( i > 0 ) d->iblocks[i]->previousblock = d->iblocks[i-1]->position;
        else d->iblocks[i]->previousblock = -1;
        d->iblocks[i]->position = ftell(f);
```

```
        // Initialise, to keep valgrind happy
        d->iblocks[i]->flags = -1;
/*----------change 10-feb-2003 */
/*
            Zero things properly. We don't want garbage in the file.
*/
            for ( j = 0; j < NUMOBJECTS; j++ ) {
                d->iblocks[i]->objects[j].date = t;
                d->iblocks[i]->objects[j].size = 0;
                d->iblocks[i]->objects[j].position = -1;
                d->iblocks[i]->objects[j].tablenumber = 0;
                d->iblocks[i]->objects[j].uncompressed = 0;
                d->iblocks[i]->objects[j].spare1 = 0;
                d->iblocks[i]->objects[j].spare2 = 0;
                d->iblocks[i]->objects[j].spare3 = 0;
                for ( jj = 0; jj < ELEMENTSIZE; jj++ ) d->iblocks[i]->objects[j].element[jj] = 0;
            }
        convertblock(d->iblocks[i],&scratchblock,TODISK);
        if ( minoswrite(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
            MesPrint("Error while writing new index blocks\n");
            goto getout;
        }
    }
    for ( i = 0; i < numnameblocks; i++ ) {
        if ( ( d->nblocks[i] = (NAMESBLOCK *)Malloc1(sizeof(NAMESBLOCK),
        "names blocks of new database") ) == 0 ) {
            while ( --i >= 0 ) { M_free(d->nblocks[i],"names blocks of new database"); }
            for ( i = 0; i < numblocks; i++ ) M_free(d->iblocks[i],"index blocks of new database");
            goto getout;
        }
        if ( i > 0 ) d->nblocks[i]->previousblock = d->nblocks[i-1]->position;
        else d->nblocks[i]->previousblock = -1;
        d->nblocks[i]->position = ftell(f);
        s = d->nblocks[i]->names;
        for ( j = 0; j < NAMETABLESIZE; j++ ) *s++ = 0;
        convertnamesblock(d->nblocks[i],&scratchnamesblock,TODISK);
        if ( minoswrite(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
            MesPrint("Error while writing new names blocks\n");
            for ( i = 0; i < numnameblocks; i++ ) M_free(d->nblocks[i],"names blocks of new database");
            for ( i = 0; i < numblocks; i++ ) M_free(d->iblocks[i],"index blocks of new database");
            goto getout;
        }
    }
    d->info.firstindexblock = d->iblocks[0]->position;
    d->info.lastindexblock = d->iblocks[numblocks-1]->position;
    d->info.firstnameblock = d->nblocks[0]->position;
    d->info.lastnameblock = d->nblocks[numnameblocks-1]->position;
    if ( WriteIniInfo(d) ) {
        for ( i = 0; i < numnameblocks; i++ ) M_free(d->nblocks[i],"names blocks of new database");
        for ( i = 0; i < numblocks; i++ ) M_free(d->iblocks[i],"index blocks of new database");
        goto getout;
    }
    return(d);
}

/*
    #] NewDbase :
    #[ FreeTableBase :
*/

void FreeTableBase(DBASE *d)
{
    int i, j, *old, *newL;
    LIST *L;
    for ( i = 0; i < d->info.numberofnamesblocks; i++ ) M_free(d->nblocks[i],"nblocks[i]");
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) M_free(d->iblocks[i],"iblocks[i]");
    M_free(d->nblocks,"nblocks");
    M_free(d->iblocks,"iblocks");
    if ( d->tablenames ) M_free(d->tablenames,"d->tablenames");
    if ( d->name ) M_free(d->name,"d->name");
    if ( d->fullname ) M_free(d->fullname,"d->fullname");
    i = d - tablebases;
    if ( i < ( NumTableBases - 1 ) ) {
        L = &(AC.TableBaseList);
        j = ( ( NumTableBases - i - 1 ) * L->size ) / sizeof(int);
        old = (int *)d; newL = (int *)(d+1);
        while ( --j >= 0 ) *newL++ = *old++;
        j = L->size / sizeof(int);
        while ( --j >= 0 ) *newL++ = 0;
    }
    NumTableBases--;
    M_free(d,"tb,d");
}

/*
    #] FreeTableBase :
```

```
    #[ ComposeTableNames :

        The nameblocks are supposed to be in memory.
        Hence we have to go through them
*/

int ComposeTableNames(DBASE *d)
{
    MLONG nsize = 0;
    int i, j, k;
    char *s, *t, *ss;
    d->topnumber = 0;
    i = 0; s = d->nblocks[i]->names; j = NAMETABLESIZE;
    while ( *s ) {
      if ( *s ) d->topnumber++;
      for ( k = 0; k < 2; k++ ) { /* name and argtail */
        while ( *s ) {
            j--;
            if ( j <= 0 ) {
                i++; if ( i >= d->info.numberofnamesblocks ) goto gotall;
                s = d->nblocks[i]->names; j = NAMETABLESIZE;
            }
            else s++;
        }
        j--;
        if ( j <= 0 ) {
            i++; if ( i >= d->info.numberofnamesblocks ) goto gotall;
            s = d->nblocks[i]->names; j = NAMETABLESIZE;
        }
        else s++;
      }
    }
gotall:;
    nsize = (d->info.numberofnamesblocks-1)*NAMETABLESIZE +
                (s-d->nblocks[i]->names)+1;
    if ( ( d->tablenames = (char *)Malloc1((2*nsize+30)*sizeof(char),"tablenames") )
        == 0 ) { return(-1); }
    t = d->tablenames;
    d->tablenamessize = 2*nsize+30;
    d->tablenamefill = nsize-1;
    for ( k = 0; k < i; k++ ) {
        ss = d->nblocks[k]->names;
        for ( j = 0; j < NAMETABLESIZE; j++ ) *t++ = *ss++;
    }
    ss = d->nblocks[i]->names;
    while ( ss < s ) *t++ = *ss++;
    *t = 0;
    return(0);
}

/*
    #] ComposeTableNames :
    #[ OpenDbase :
*/

DBASE *OpenDbase(char *filename)
{
    FILE *f;
    DBASE *d;
    char *newname;
    if ( ( f = LocateBase(&filename,&newname,"r+b") ) == 0 ) {
        MesPrint("Cannot open file %s\n",filename);
        return(0);
    }
/*  setbuf(f,0); */
    d = (DBASE *)From0List(&(AC.TableBaseList));
    d->name = filename; /* For the moment just for the error messages */
    d->handle = f;
    if ( ReadIniInfo(d) || ReadIndex(d) ) { M_free(d,"OpenDbase"); fclose(f); return(0); }
    if ( ComposeTableNames(d) ) {
        FreeTableBase(d);
        fclose(f);
        return(0);
    }
    d->name = str_dup(filename);
    d->fullname = newname;
    return(d);
}

/*
    #] OpenDbase :
    #[ AddTableName :

    Adds a name of a table. Writes the namelist to disk.
    Returns the number of this tablename in the database.
    If the name was already in the table we return its value in negative.
    Zero is an error!
```

```
*/

MLONG AddTableName(DBASE *d,char *name,TABLES T)
{
    char *s, *t, *tt;
    int namesize, tailsize;
    MLONG newsize, i, num;
/*
    First search for the name in what we have already
*/
    if ( d->tablenames ) {
        num = 0;
        s = d->tablenames;
        while ( *s ) {
            num++;
            t = name;
            while ( ( *s == *t ) && *t ) { s++; t++; }
            if ( *s == *t ) { return(-num); }
            while ( *s ) s++;
            s++;
            while ( *s ) s++;
            s++;
        }
    }
/*
    This name has to be added
*/
    MesPrint("We add the name %s\n",name);
    t = name;
    while ( *t ) { t++; }
    namesize = t-name;
    if ( ( t = (char *)(T->argtail) ) != 0 ) {
        while ( *t ) { t++; }
        tailsize = t - (char *)(T->argtail);
    }
    else { tailsize = 0; }
    if ( d->tablenames == 0 ) {
        if ( ComposeTableNames(d) ) {
            FreeTableBase(d);
            M_free(d,"AddTableName");
            return(0);
        }
    }
    d->info.numberoftables++;
    while ( ( d->tablenamefill+namesize+tailsize+3 > d->tablenamessize )
                  || ( d->tablenames == 0 ) ) {
        newsize = 2*d->tablenamessize + 2*namesize + 2*tailsize + 6;
        if ( ( t = (char *)Malloc1(newsize*sizeof(char),"AddTableName") ) == 0 )
            return(0);
        tt = t;
        if ( d->tablenames ) {
            s = d->tablenames;
            for ( i = 0; i < d->tablenamefill; i++ ) *t++ = *s++;
            *t = 0;
            M_free(d->tablenames,"d->tablenames");
        }
        d->tablenames = tt;
        d->tablenamessize = newsize;
    }
    s = d->tablenames + d->tablenamefill;
    t = name;
    while ( *t ) *s++ = *t++;
    *s++ = 0;
    t = (char *)(T->argtail);
    while ( *t ) *s++ = *t++;
    *s++ = 0;
    *s = 0;
    d->tablenamefill = s - d->tablenames;
    d->topnumber++;
/*
    Now we have to synchronize
*/
    if ( PutTableNames(d) ) return(0);
    return(d->topnumber);
}

/*
    #] AddTableName :
    #[ GetTableName :

    Gets a name of a table.
    Returns the number of this tablename in the database.
    Zero -> error
*/

MLONG GetTableName(DBASE *d,char *name)
{
```

```
    char *s, *t;
    MLONG num;
/*
    search for the name in what we have
*/
    if ( d->tablenames ) {
        num = 0;
        s = d->tablenames;
        while ( *s ) {
            num++;
            t = name;
            while ( ( *s == *t ) && *t ) { s++; t++; }
            if ( *s == *t ) { return(num); }
            while ( *s ) s++;
            s++;
            while ( *s ) s++;
            s++;
        }
    }
    return(0);
}

/*
    #] GetTableName :
    #[ PutTableNames :

    Takes the names string in d->tablenames and puts it in the nblocks
    pieces. Writes what has been changed to disk.
*/

int PutTableNames(DBASE *d)
{
    NAMESBLOCK **nnew;
    int i, j, firstdif;
    MLONG m;
    char *s, *t;
/*
    Determine how many blocks are needed.
*/
    MLONG numblocks = d->tablenamefill/NAMETABLESIZE + 1;
    if ( d->info.numberofnamesblocks < numblocks ) {
/*
        We need more blocks. First make sure of the space for nblocks.
*/
        if ( ( nnew = (NAMESBLOCK **)Malloc1(sizeof(NAMESBLOCK *)*numblocks,
            "new names block") ) == 0 ) {
            return(-1);
        }
        for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
            nnew[i] = d->nblocks[i];
        }
        free(d->nblocks);
        d->nblocks = nnew;
        for ( ; i < numblocks; i++ ) {
            if ( ( d->nblocks[i] = (NAMESBLOCK *)Malloc1(sizeof(NAMESBLOCK),
            "additional names blocks ") ) == 0 ) {
                FreeTableBase(d);
                return(-1);
            }
            d->nblocks[i]->previousblock = -1;
            d->nblocks[i]->position = -1;
            s = d->nblocks[i]->names;
            for ( j = 0; j < NAMETABLESIZE; j++ ) *s++ = 0;
        }
        d->info.numberofnamesblocks = numblocks;
    }
/*
    Now look till where the new contents agree with the old.
*/
    firstdif = 0;
    i = 0; t = d->nblocks[i]->names; j = 0; s = d->tablenames;
    for ( m = 0; m < d->tablenamefill; m++ ) {
        if ( *s == *t ) {
            s++; t++; j++;
            if ( j >= NAMETABLESIZE ) {
                i++;
                t = d->nblocks[i]->names;
                j = 0;
            }
        }
        else {
            firstdif = i;
            for ( ; m < d->tablenamefill; m++ ) {
                *t++ = *s++; j++;
                if ( j >= NAMETABLESIZE ) {
                    i++;
                    t = d->nblocks[i]->names;
```

```
                    j = 0;
                }
            }
            *t = 0;
            break;
        }
    }
    for ( i = 0; i < d->info.numberofnamesblocks; i++ ) {
        if ( i == firstdif ) break;
        if ( d->nblocks[i]->position < 0 ) { firstdif = i; break; }
    }
/*
    Now we have to (re)write the blocks, starting at firstdif.
*/
    for ( i = firstdif; i < d->info.numberofnamesblocks; i++ ) {
        if ( i > 0 ) d->nblocks[i]->previousblock = d->nblocks[i-1]->position;
        else d->nblocks[i]->previousblock = -1;
        if ( d->nblocks[i]->position < 0 ) {
            fseek(d->handle,0,SEEK_END);
            d->nblocks[i]->position = ftell(d->handle);
        }
        else fseek(d->handle,d->nblocks[i]->position,SEEK_SET);
        convertnamesblock(d->nblocks[i],&scratchnamesblock,TODISK);
        if ( minoswrite(d->handle,(char *)(&scratchnamesblock),sizeof(NAMESBLOCK)) ) {
            MesPrint("Error while writing names blocks\n");
            FreeTableBase(d);
            return(-1);
        }
    }
    d->info.lastnameblock = d->nblocks[d->info.numberofnamesblocks-1]->position;
    d->info.firstnameblock = d->nblocks[0]->position;
    return(WriteIniInfo(d));
}

/*
    #] PutTableNames :
    #[ AddToIndex :
*/

int AddToIndex(DBASE *d,MLONG number)
{
    MLONG i, oldnumofindexblocks = d->info.numberofindexblocks;
    MLONG j, newnumofindexblocks, jj;
    INDEXBLOCK **ib;
    MLONG t = (MLONG)(time(0));
    if ( number == 0 ) return(0);
    else if ( number < 0 ) {
        if ( d->info.entriesinindex < -number ) {
            MesPrint("There are only %ld entries in the index of file %s\n",
            d->info.entriesinindex,d->name);
            return(-1);
        }
        d->info.entriesinindex += number;
dowrite:
        if ( WriteIniInfo(d) ) {
            d->info.entriesinindex -= number;
            MesPrint("File may be corrupted\n");
            return(-1);
        }
    }
    else if ( d->info.entriesinindex+number <=
    NUMOBJECTS*d->info.numberofindexblocks ) {
        d->info.entriesinindex += number;
        goto dowrite;
    }
    else {
        d->info.entriesinindex += number;
        newnumofindexblocks = d->info.numberofindexblocks + ((number -
            (NUMOBJECTS*d->info.numberofindexblocks - d->info.entriesinindex))
            +NUMOBJECTS-1)/NUMOBJECTS;
        if ( ( ib = (INDEXBLOCK **)Malloc1(sizeof(INDEXBLOCK *)*newnumofindexblocks,
        "index") ) == 0 ) return(-1);
        for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
            ib[i] = d->iblocks[i];
        }
        for ( i = d->info.numberofindexblocks; i < newnumofindexblocks; i++ ) {
            if ( ( ib[i] = (INDEXBLOCK *)Malloc1(sizeof(INDEXBLOCK),"index block") ) == 0 ) {
                FreeTableBase(d);
                return(-1);
            }
            if ( i > 0 ) ib[i]->previousblock = ib[i-1]->position;
            else ib[i]->previousblock = -1;
/*
            Zero things properly. We don't want garbage in the file.
*/
            for ( j = 0; j < NUMOBJECTS; j++ ) {
                ib[i]->objects[j].date = t;
```

```
                ib[i]->objects[j].size = 0;
                ib[i]->objects[j].position = -1;
                ib[i]->objects[j].tablenumber = 0;
                ib[i]->objects[j].uncompressed = 0;
                ib[i]->objects[j].spare1 = 0;
                ib[i]->objects[j].spare2 = 0;
                ib[i]->objects[j].spare3 = 0;
                for ( jj = 0; jj < ELEMENTSIZE; jj++ ) ib[i]->objects[j].element[jj] = 0;
            }
            fseek(d->handle,0,SEEK_END);
            ib[i]->position = ftell(d->handle);
            convertblock(ib[i],&scratchblock,TODISK);
            if ( minoswrite(d->handle,(char *)(&scratchblock),sizeof(INDEXBLOCK)) ) {
                MesPrint("Error while writing new index of file %s",d->name);
                FreeTableBase(d);
                return(-1);
            }
        }
        d->info.lastindexblock = ib[newnumofindexblocks-1]->position;
        d->info.firstindexblock = ib[0]->position;
        d->info.numberofindexblocks = newnumofindexblocks;
        if ( WriteIniInfo(d) ) {
            d->info.numberofindexblocks = oldnumofindexblocks;
            d->info.entriesinindex -= number;
            MesPrint("File may be corrupted\n");
            FreeTableBase(d);
            return(-1);
        }
        M_free(d->iblocks,"AddToIndex");
        d->iblocks = ib;
    }
    return(0);
}

/*
    #] AddToIndex :
    #[ AddObject :
*/

MLONG AddObject(DBASE *d,MLONG tablenumber,char *arguments,char *rhs)
{
    MLONG number;
    number = d->info.entriesinindex;
    if ( AddToIndex(d,1) ) return(-1);
    if ( WriteObject(d,tablenumber,arguments,rhs,number) ) return(-1);
    return(number);
}

/*
    #] AddObject :
    #[ FindTableNumber :
*/

MLONG FindTableNumber(DBASE *d,char *name)
{
    char *s = d->tablenames, *t, *ss;
    MLONG num = 0;
    ss = d->tablenames + d->tablenamefill;
    while ( s < ss ) {
        num++;
        t = name;
        while ( *s == *t && *t ) {
            s++; t++;
        }
        if ( *s == 0 && *t == 0 ) return(num);
        while ( *s ) s++;
        s++;
/*
        Skip also the argument tail
*/
        while ( *s ) s++;
        s++;
    }
    return(-1);      /* Name not found */
}

/*
    #] FindTableNumber :
    #[ WriteObject :
*/

int WriteObject(DBASE *d,MLONG tablenumber,char *arguments,char *rhs,MLONG number)
{
    char *s, *a;
#ifdef WITHZLIB
    char *buffer = 0;
    uLongf newsize = 0, oldsize = 0;
```

```
    uLong ssize;
    int error = 0;
#endif
    MLONG i, j, position, size, n;
    OBJECTS *obj;
    if ( ( d->mode & INPUTONLY ) == INPUTONLY ) {
        MesPrint("Not allowed to write to input\n");
        return(-1);
    }
    if ( number >= d->info.entriesinindex ) {
        MesPrint("Reference to non-existing object number %ld\n",number+1);
        return(0);
    }
    j = number/NUMOBJECTS;
    i = number%NUMOBJECTS;
    obj = &(d->iblocks[j]->objects[i]);
    a = arguments;
    while ( *a ) a++;
    a++; n = a - arguments;
    if ( n > ELEMENTSIZE ) {
        MesPrint("Table element %s has more than %ld characters.\n",arguments,
            (MLONG)ELEMENTSIZE);
        return(-1);
    }
    s = obj->element;
    a = arguments;
    while ( *a ) *s++ = *a++;
    *s++ = 0;
    while ( n < ELEMENTSIZE ) { *s++ = 0; n++; }
    obj->spare1 = obj->spare2 = obj->spare3 = 0;

    fseek(d->handle,0,SEEK_END);
    position = ftell(d->handle);
    s = rhs;
    while ( *s ) s++;
    s++;
    size = s - rhs;
#ifdef WITHZLIB
    if ( ( d->mode & NOCOMPRESS ) == 0 ) {
        newsize = size + size/1000 + 20;
        if ( ( buffer = (char *)Malloc1(newsize*sizeof(char),"compress buffer") )
            == 0 ) {
            MesPrint("No compress used for element %s in file %s\n",arguments,d->name);
        }
    }
    else buffer = 0;
    if ( buffer ) {
        ssize = size;
#ifdef WITHZSTD
        // Force the use of zlib for compressed Tablebase entries, so that tablebases created
        // with zstd-supported FORM builds can be used by zstd-unsupported FORM builds.
        const int old_isUsingZSTDcompression = ZWRAP_isUsingZSTDcompression();
        ZWRAP_useZSTDcompression(0);
#endif
        if ( ( error = compress((Bytef *)buffer,&newsize,(Bytef *)rhs,ssize) ) != Z_OK ) {
            MesPrint("Error = %d\n",error);
            MesPrint("Due to error no compress used for element %s in file %s\n",arguments,d->name);
            M_free(buffer,"tb,WriteObject");
            buffer = 0;
        }
#ifdef WITHZSTD
        ZWRAP_useZSTDcompression(old_isUsingZSTDcompression);
#endif
    }
    if ( buffer ) {
        rhs = buffer;
        oldsize = size;
        size = newsize;
    }
#endif
    if ( minoswrite(d->handle,rhs,size) ) {
        MesPrint("Error while writing rhs\n");
        return(-1);
    }
    obj->position = position;
    obj->size = size;
    obj->date = (MLONG)(time(0));
    obj->tablenumber = tablenumber;
#ifdef WITHZLIB
    obj->uncompressed = oldsize;
    if ( buffer ) M_free(buffer,"tb,WriteObject");
#else
    obj->uncompressed = 0;
#endif
    return(WriteIndexBlock(d,j));
}

```

```
/*
    #] WriteObject :
    #[ ReadObject :

    Returns a pointer to the proper rhs
*/

char *ReadObject(DBASE *d,MLONG tablenumber,char *arguments)
{
    OBJECTS *obj;
    MLONG i, j;
    char *buffer1, *s, *t;
#ifdef WITHZLIB
    char *buffer2 = 0;
    uLongf finallength = 0;
#endif
    if ( tablenumber > d->topnumber ) {
        MesPrint("Reference to non-existing table number in tablebase %s: %ld\n",
                d->name,tablenumber);
        return(0);
    }
/*
    Start looking for the object
*/
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
        for ( j = 0; j < NUMOBJECTS; j++ ) {
            if ( d->iblocks[i]->objects[j].tablenumber != tablenumber ) continue;
            s = arguments; t = d->iblocks[i]->objects[j].element;
            while ( *s == *t && *s ) { s++; t++; }
            if ( *t == 0 && *s == 0 ) goto foundelement;
        }
    }
    s = d->tablenames; i = 1;
    while ( *s ) {
        if ( i == tablenumber ) break;
        while ( *s ) s++;
        s++;
        while ( *s ) s++;
        s++;
        i++;
    }
    MesPrint("%s(%s) not found in tablebase %s\n",s,arguments,d->name);
    return(0);

foundelement:;
    obj = &(d->iblocks[i]->objects[j]);
    fseek(d->handle,obj->position,SEEK_SET);
    if ( ( buffer1 = (char *)Malloc1(obj->size,"reading rhs buffer1") ) == 0 ) {
        return(0);
    }
#ifdef WITHZLIB
    if ( obj->uncompressed > 0 ) {
        if ( ( buffer2 = (char *)Malloc1(obj->uncompressed,"reading rhs buffer2") ) == 0 ) {
            return(0);
        }
    }
    else buffer2 = 0;
#endif
    if ( minosread(d->handle,buffer1,obj->size) ) {
        MesPrint("Could not read rhs %s in file %s\n",arguments,d->name);
        M_free(buffer1,"tb,ReadObject");
#ifdef WITHZLIB
        if ( buffer2 ) M_free(buffer2,"tb,ReadObject");
#endif
        return(0);
    }
#ifdef WITHZLIB
    if ( buffer2 == 0 ) return(buffer1);
    finallength = obj->uncompressed;
    if ( uncompress((Bytef *)buffer2,&finallength,(Bytef *)buffer1,obj->size) != Z_OK ) {
        MesPrint("Cannot uncompress element %s in file %s\n",arguments,d->name);
        M_free(buffer1,"tb,ReadObject"); M_free(buffer2,"tb,ReadObject");
        return(0);
    }
    M_free(buffer1,"tb,ReadObject");
    return(buffer2);
#else
    return(buffer1);
#endif
}

/*
    #] ReadObject :
    #[ ReadijObject :

    Returns a pointer to the proper rhs
*/
```

```
char *ReadijObject(DBASE *d,MLONG i,MLONG j,char *arguments)
{
    OBJECTS *obj;
    char *buffer1;
#ifdef WITHZLIB
    char *buffer2 = 0;
    uLongf finallength = 0;
#endif
    obj = &(d->iblocks[i]->objects[j]);
    fseek(d->handle,obj->position,SEEK_SET);
    if ( ( buffer1 = (char *)Malloc1(obj->size,"reading rhs buffer1") ) == 0 ) {
        return(0);
    }
#ifdef WITHZLIB
    if ( obj->uncompressed > 0 ) {
        if ( ( buffer2 = (char *)Malloc1(obj->uncompressed,"reading rhs buffer2") ) == 0 ) {
            return(0);
        }
    }
    else buffer2 = 0;
#endif
    if ( minosread(d->handle,buffer1,obj->size) ) {
        MesPrint("Could not read rhs %s in file %s\n",arguments,d->name);
        if ( buffer1 ) M_free(buffer1,"rhs buffer1");
#ifdef WITHZLIB
        if ( buffer2 ) M_free(buffer2,"rhs buffer2");
#endif
        return(0);
    }
#ifdef WITHZLIB
    if ( buffer2 == 0 ) return(buffer1);
    finallength = obj->uncompressed;
    if ( uncompress((Bytef *)buffer2,&finallength,(Bytef *)buffer1,obj->size) != Z_OK ) {
        MesPrint("Cannot uncompress element %s in file %s\n",arguments,d->name);
        if ( buffer1 ) M_free(buffer1,"rhs buffer1");
        if ( buffer2 ) M_free(buffer2,"rhs buffer2");
        return(0);
    }
    M_free(buffer1,"rhs buffer1");
    return(buffer2);
#else
    return(buffer1);
#endif
}

/*
    #] ReadijObject :
    #[ ExistsObject :

    Returns 1 if Object exists
*/

int ExistsObject(DBASE *d,MLONG tablenumber,char *arguments)
{
    MLONG i, j;
    char *s, *t;
    if ( tablenumber > d->topnumber ) {
        MesPrint("Reference to non-existing table number in tablebase %s: %ld\n",
                d->name,tablenumber);
        return(0);
    }
/*
    Start looking for the object
*/
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
        for ( j = 0; j < NUMOBJECTS; j++ ) {
            if ( d->iblocks[i]->objects[j].tablenumber != tablenumber ) continue;
            s = arguments; t = d->iblocks[i]->objects[j].element;
            while ( *s == *t && *s ) { s++; t++; }
            if ( *t == 0 && *s == 0 ) return(1);
        }
    }
    return(0);
}

/*
    #] ExistsObject :
    #[ DeleteObject :

    Returns 1 if Object has been deleted.
    We leave a hole. Actually the object is still there but has been
    inactivated. It can be reactivated by calling this routine again.
*/

int DeleteObject(DBASE *d,MLONG tablenumber,char *arguments)
{
```

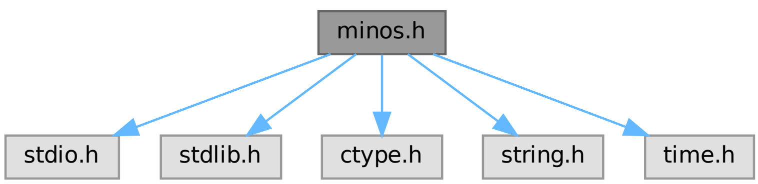
```
    MLONG i, j;
    char *s, *t;
    if ( tablenumber > d->topnumber ) {
        MesPrint("Reference to non-existing table number in tablebase %s: %ld\n",
                d->name,tablenumber);
        return(0);
    }
/*
    Start looking for the object
*/
    for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
        for ( j = 0; j < NUMOBJECTS; j++ ) {
            if ( d->iblocks[i]->objects[j].tablenumber != tablenumber ) continue;
            s = arguments; t = d->iblocks[i]->objects[j].element;
            while ( *s == *t && *s ) { s++; t++; }
            if ( *t == 0 && *s == 0 ) {
                d->iblocks[i]->objects[j].tablenumber =
                    -d->iblocks[i]->objects[j].tablenumber - 1;
                return(1);
            }
        }
    }
    return(0);
}

/*
    #] DeleteObject :
*/
```

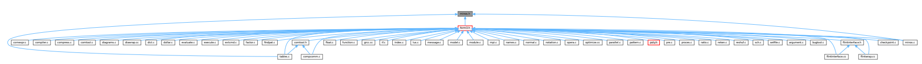
4.73 minos.h File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <ctype.h>
#include <string.h>
#include <time.h>
```

Include dependency graph for minos.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct iniinfo
- struct objects
- struct indexblock
- struct nameblock
- struct dbase

Macros

- #define MAXBASES 16
- #define NUMOBJECTS 100
- #define MAXINDEXSIZE 33000000L
- #define NAMETABLESIZE 1008
- #define ELEMENTSIZE 128
- #define TODISK 0
- #define FROMDISK 1
- #define MDIRTYFLAG 1
- #define MCLEANFLAG (∼MDIRTYFLAG)
- #define INANDOUT 0
- #define INPUTONLY 1
- #define OUTPUTONLY 2
- #define NOCOMPRESS 4

Typedefs

- typedef struct iniinfo **INIINFO**
- typedef struct objects **OBJECTS**
- typedef struct indexblock **INDEXBLOCK**
- typedef struct nameblock **NAMESBLOCK**
- typedef struct dbase **DBASE**

Functions

- int minosread (FILE ∗f, char ∗buffer, MLONG size)
- int minoswrite (FILE ∗f, char ∗buffer, MLONG size)

Variables

- int withoutflush

4.73.1 Detailed Description

Contains all needed declarations and definitions for the tablebase low level file routines. These have been taken from the minos database system and modified somewhat.

!!!CAUTION!!! Changes in this file will most likely have consequences for the recovery mechanism (see checkpoint.c). You need to care for the code in checkpoint.c as well and modify the code there accordingly!

Definition in file minos.h.

4.73.2 Macro Definition Documentation

4.73.2.1 MAXBASES

```
#define MAXBASES 16
```

Definition at line 62 of file minos.h.

4.73.2.2 NUMOBJECTS

```
#define NUMOBJECTS 100
```

Definition at line 69 of file minos.h.

4.73.2.3 MAXINDEXSIZE

```
#define MAXINDEXSIZE 33000000L
```

Definition at line 70 of file minos.h.

4.73.2.4 NAMETABLESIZE

```
#define NAMETABLESIZE 1008
```

Definition at line 71 of file minos.h.

4.73.2.5 ELEMENTSIZE

```
#define ELEMENTSIZE 128
```

Definition at line 72 of file minos.h.

4.73.2.6 TODISK

```
#define TODISK 0
```

Definition at line 143 of file minos.h.

4.73.2.7 FROMDISK

```
#define FROMDISK 1
```

Definition at line 144 of file minos.h.

4.73.2.8 MDIRTYFLAG

```
#define MDIRTYFLAG 1
```

Definition at line 145 of file minos.h.

4.73.2.9 MCLEANFLAG

```
#define MCLEANFLAG (~MDIRTYFLAG)
```

Definition at line 146 of file minos.h.

4.73.2.10 INANDOUT

```
#define INANDOUT 0
```

Definition at line 147 of file minos.h.

4.73.2.11 INPUTONLY

```
#define INPUTONLY 1
```

Definition at line 148 of file minos.h.

4.73.2.12 OUTPUTONLY

```
#define OUTPUTONLY 2
```

Definition at line 149 of file minos.h.

4.73.2.13 NOCOMPRESS

```
#define NOCOMPRESS 4
```

Definition at line 150 of file minos.h.

4.73.3 Function Documentation

4.73.3.1 minosread()

```
int minosread (
            FILE * f,
            char * buffer,
            MLONG size )
```

Definition at line 66 of file minos.c.

4.73.3.2 minoswrite()

```
int minoswrite (
            FILE * f,
            char * buffer,
            MLONG size )
```

Definition at line 83 of file minos.c.

4.73.4 Variable Documentation

4.73.4.1 withoutflush

```
int withoutflush [extern]
```

Definition at line 45 of file minos.c.

4.74 minos.h

Go to the documentation of this file.

```
#ifndef __MANAGE_H__
#define __MANAGE_H__
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
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 *   terms of the GNU General Public License as published by the Free Software
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 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
#include <stdio.h>
#include <stdlib.h>
#include <ctype.h>
#include <string.h>
#include <time.h>
/*
    The following typedef has been moved to form3.h where all the sizes
    are defined for the various memory models.
    We want MLONG to have a more or less fixed size.
    In form3.h we try to fix it at 8 bytes.
    This should make files exchangeable
    between various 32-bits and 64-bits systems. At 4 bytes it might have
    problems with files of more than 2 Gbytes.
typedef long MLONG;
*/
#define MAXBASES 16
#ifdef WORDSIZE32
#define NUMOBJECTS 1024
#define MAXINDEXSIZE 1000000000L
#define NAMETABLESIZE 1008
#define ELEMENTSIZE 256
#else
#define NUMOBJECTS 100
#define MAXINDEXSIZE 33000000L
#define NAMETABLESIZE 1008
#define ELEMENTSIZE 128
#endif
int minosread(FILE *f,char *buffer,MLONG size);
int minoswrite(FILE *f,char *buffer,MLONG size);
/*
```

```
    ELEMENTSIZE should make a nice number of sizeof(OBJECTS)
    Usually this will be much too much, but there are cases.....
*/

typedef struct iniinfo {
/* should contains only MLONG variables or convertiniinfo should be modified */
    MLONG   entriesinindex;
    MLONG   numberofindexblocks;
    MLONG   firstindexblock;
    MLONG   lastindexblock;
    MLONG   numberoftables;
    MLONG   numberofnamesblocks;
    MLONG   firstnameblock;
    MLONG   lastnameblock;
} INIINFO;

typedef struct objects {
/* if any changes, convertblock should be adapted too!!!! */
    MLONG position;             /* position of RHS= */
    MLONG size;                 /* size on disk (could be compressed) */
    MLONG date;                 /* Time stamp */
    MLONG tablenumber;          /* Number of table. Refers to name in special index */
    MLONG uncompressed;         /* uncompressed size if compressed. If not: 0 */
    MLONG spare1;
    MLONG spare2;
    MLONG spare3;
    char element[ELEMENTSIZE];  /* table element in character form */
} OBJECTS;

typedef struct indexblock {
    MLONG   flags;
    MLONG   previousblock;
    MLONG   position;
    OBJECTS  objects[NUMOBJECTS];
} INDEXBLOCK;

typedef struct nameblock {
    MLONG   previousblock;
    MLONG   position;
    char    names[NAMETABLESIZE];
} NAMESBLOCK;

typedef struct dbase {
    INIINFO      info;
    MLONG       mode;
    MLONG       tablenamessize;
    MLONG       topnumber;
    MLONG       tablenamefill;
    MLONG       rwmode;
    INDEXBLOCK    **iblocks;
    NAMESBLOCK  **nblocks;
    FILE        *handle;
    char        *name;
    char        *fullname;
    char        *tablenames;
} DBASE;

/*
typedef int (*SFUN)(char *);
typedef struct compile {
    char *keyword;
    SFUN func;
} MCFUNCTION;
 */
#define TODISK 0
#define FROMDISK 1
#define MDIRTYFLAG 1
#define MCLEANFLAG (~MDIRTYFLAG)
#define INANDOUT 0
#define INPUTONLY 1
#define OUTPUTONLY 2
#define NOCOMPRESS 4

extern int withoutflush;

#endif
```

4.75 model.c

```
/*
    #[ Explanations :

    This is a second generation much simplified version of model.c
```

```
    Syntax:
        Model QED;
            Particle electron,positron,-2;
            Particle photon,+3;
            Vertex electron,positron,photon: g;
        EndModel;
    or:
        Model QCD;
            Particle quark,antiquark,-2;
            Particle ghost,antighost,-1;
            Particle gluon,+3;
            Vertex quark,antiquark,gluon: g;
            Vertex qhost,antighost,gluon: g;
            Vertex gluon,gluon,gluon: g;
            Vertex gluon,gluon,gluon,gluon: g^2;
        EndModel;
    Remarks:
    1: if no second particle is given, a particle and its antiparticle are the same
    2: Spin statistics is by dimension of SU(2) representation with a sign.
    3: In a vertex we need to mention the coupling constants.
    4: Internally a particle gives a vertex with only two lines.
    5: All particles must have been declared before any vertex.

    The diagrams should be provided as a collection of nodes and edges
    with momenta with a direction substituted. The other parameters (like
    Lorentz indices, color indices etc.) can then be provided in a procedure
    that should go with this mode definition.
    In principle the edges are not needed if there are no 'insertions', but
    because we would like to have the insertions as well we need the edges.

    #] Explanations :
    #[ Includes : model.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ CreateVertex :
*/

VERTEX *CreateVertex(MODEL *m)
{
    VERTEX *v, **new;
    WORD newsize;
    int i;
    if ( m->invertices >= m->sizevertices ) {
        if ( m->sizevertices == 0 ) newsize = 20;
        else newsize = 2*m->sizevertices;
        new = (VERTEX **)Malloc1(newsize*sizeof(VERTEX *),"m->vertices");
        for ( i = 0; i < m->sizevertices; i++ ) new[i] = m->vertices[i];
        if ( m->sizevertices > 0 ) M_free(m->vertices,"m->vertices");
        m->vertices = new;
        m->sizevertices = newsize;
    }
    v = (VERTEX *)Malloc1(sizeof(VERTEX),"VERTEX");
    m->vertices[m->invertices++] = v;
    v->nparticles = 0;
    v->ncouplings = 0;
    v->type = 0;
    v->error = 0;
    v->externonly = 0;
    v->spare = 0;
    return(v);
}

/*
    #] CreateVertex :
    #[ ReadParticle :
*/

UBYTE *ReadParticle(UBYTE *s, VERTEX *v, MODEL *m, int par)
{
    UBYTE *name, c;
    PARTICLE *p = v->particles + v->nparticles++;
    WORD type, funnum;
    int i, j;
    SKIPBLANKS(s)
    name = s; s = SkipName(s); c = *s; *s = 0;
    if ( GetVar(name,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND ) {
        p->number = AddFunction(name,0,VERTEXFUNCTION,0,0,0,-1,-1) + FUNCTION;
    }
    else if ( par == 1 && type == CFUNCTION && functions[funnum].spec == VERTEXFUNCTION ) {
        p->number = funnum+FUNCTION;
    }
    else if ( par == 0 && type == CFUNCTION && functions[funnum].spec == VERTEXFUNCTION ) {
/*
```

```
        We should check whether this particle exists already in this model.
        If so, this will be an error.
*/
        for ( i = 0; i < m->nparticles-1; i++ ) {
          for ( j = 0; j < m->vertices[i]->nparticles; j++ ) {
            if ( m->vertices[i]->particles[j].number == funnum ) {
                MesPrint("&Illegal attempt to redefine a particle in the same model: %s",name);
                v->error = 1;
            }
          }
        }
        p->number = funnum+FUNCTION;
    }
    else {
        MesPrint("&Name of particle previously declared as another variable: %s",name);
        v->error = 1;
    }
    *s = c;
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    return(s);
}

/*
    #] ReadParticle :
    #[ CoModel :
*/

int CoModel(UBYTE *s)
{
    UBYTE *name, c;
    MODEL *m = (MODEL *)Malloc1(sizeof(MODEL),"Model");
    WORD numberofset, *e;
    SETS set;
    SKIPBLANKS(s)
    AC.ModelLevel++;
    int i;
/*
    We now have to add the model to the list of models.
*/
    if ( AC.modelspace == 0 || AC.nummodels >= AC.modelspace ) {
        int newspace = 2*AC.modelspace+2, i;
        MODEL **models = (MODEL **)Malloc1((sizeof(MODEL *))*newspace,"AC.models");
        for ( i = 0; i < AC.modelspace; i++ ) models[i] = AC.models[i];
        if ( AC.models ) M_free(AC.models,"AC.models");
        AC.modelspace = newspace;
        AC.models = models;
    }

    AC.models[AC.nummodels++] = m;

    m->vertices = 0;
    m->couplings = 0;
    m->nparticles = m->nvertices = m->sizevertices = m->invertices = 0;
    m->sizecouplings = 0;
    m->error = 0;
    m->grccmodel = NULL;

    name = s;
    if ( FG.cTable[*s] == 0 ) {
        while ( FG.cTable[*s] <= 1 ) s++;
        c = *s; *s = 0;
        m->name = strDup1(name,"Model name");
        *s = c;
        SKIPBLANKS(s)
        if ( *s != 0 ) {
            MesPrint("&Illegal option in model statement: %s",s);
            return(1);
        }
    }
    else {
        m->name = strDup1((UBYTE *)"---","Model name");
        m->error = 1;
        MesPrint("&Illegal name for model: %s",name);
        return(1);
    }
/*
    Now make it a set with one element. The type of the set is rather special
*/
    if ( GetName(AC.varnames,m->name,&numberofset,NOAUTO) != NAMENOTFOUND ) {
        MesPrint("&Name conflict with name %s of model",m->name);
        return(1);
    }
    numberofset = AddSet(name,0);
    set = Sets + numberofset;
    set->type = CMODEL;
    e = (WORD *)FromVarList(&AC.SetElementList);
    *e = AC.nummodels-1;
```

```
    set->last++;
    AC.SetList.numtemp = AC.SetList.num;
    AC.SetElementList.numtemp = AC.SetElementList.num;

    for ( i = 0; i <= MAXLEGS; i++ ) m->legcouple[i] = 0;
    m->legcouple[2] = 1;

    return(0);
}

/*
    #] CoModel :
    #[ CoParticle :
*/

int CoParticle(UBYTE *s)
{
    MODEL *m;
    VERTEX *v;
    int i;
    if ( AC.nummodels <= 0 ) {
        MesPrint("&No open model for particle statement");
        return(1);
    }
    m = AC.models[AC.nummodels-1];
    if ( m->nvertices > 0 ) {
        MesPrint("&In a model description Particle statements should be before Vertex statements.");
        m->error = 1;
        return(1);
    }
    v = CreateVertex(m);
    m->nparticles++;
    s = ReadParticle(s,v,m,0);
    v->particles[0].type = 1;
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    if ( v->error == 0 ) {
        if ( FG.cTable[*s] == 0 ) {
            s = ReadParticle(s,v,m,0);
            v->particles[1].type = -1;
            if ( v->particles[0].number == v->particles[1].number ) {
                v->particles[0].type = 0;
                v->particles[1].type = 0;
            }
        }
        else {  /* Particle = antiparticle */
            v->particles[1] = v->particles[0];
            v->nparticles++;
            v->particles[0].type = 0;
            v->particles[1].type = 0;
        }
    }
    if ( v->error == 0 && ( *s == '+' || *s == '-' ) ) {
        int x = 0, sign = ( *s ==  '-' ) ? -1: 1;
        s++;
        while ( FG.cTable[*s] == 1 ) { x = 10*x + *s++ - '0'; }
        if ( x == 0 ) {
            v->error = 1;
            MesPrint("&Spin goes by dimension of SU(2) representation. Zero is not allowed.");
        }
        else { v->particles[0].spin = v->particles[1].spin = sign*x; }
        SKIPBLANKS(s)
    }
    else if ( v->error == 0 ) {
        v->particles[0].spin = v->particles[1].spin = 1;
        v->particles[0].type = 0;
        v->particles[1].type = 0;
    }
/*
    Here will come future options
*/
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    while ( v->error == 0 && *s != 0 && FG.cTable[*s] == 0 ) {
        UBYTE *opt = s, c;
        while ( FG.cTable[*s] == 0 ) s++;
        c = *s; *s = 0;
        if ( StrICont(opt,(UBYTE *)"external") == 0 ) { v->externonly = 1; }
        else {
            MesPrint("&Unrecognized option %s in particle statement.",opt);
            v->error = 1;
        }
        *s = c;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    }
/*
    The couplings array will be used for registering in what kind of vertices the
    particle is involved. (example: in QCD the quarks only in 2 and 3-point vertices)
*/
```

```
    for ( i = 0; i < MAXPARTICLES; i++ ) v->couplings[i] = 0;
    v->couplings[2] = 1;
    return(v->error);
}

/*
    #] CoParticle :
    #[ CoVertex :
*/

int CoVertex(UBYTE *s)
{
    UBYTE *ss, *name, c;
    WORD type;
    MODEL *m;
    VERTEX *v;
    if ( AC.ModelLevel <= 0 ) {
        MesPrint("&No open model for vertex statement");
        return(1);
    }
    m = AC.models[AC.nummodels-1];
/*
    Get an object of type VERTEX
*/
    v = CreateVertex(m);
    m->nvertices++;
    SKIPBLANKS(s);
    while ( *s && *s != ':' ) {
        if ( v->error == 0 ) {
            s = ReadParticle(s,v,m,1);
            if ( v->error ) m->error = 1;
        }
        else {
            while ( *s && *s != ':' ) {
                if ( *s == '[' ) { SKIPBRA1(s) }
                else if ( *s == '(' ) { SKIPBRA3(s) }
                else if ( *s == 0 ) {
                    v->error = 1;
                    MesPrint("&No coupling constant in vertex statement.");
                    break;
                }
            }
            break;
        }
    }
    if ( v->error == 0 ) {
      if ( *s == ':' ) { /* read symbols and powers */
        s++; SKIPBLANKS(s);
        while ( *s ) {
            if ( v->ncouplings >= 2*MAXCOUPLINGS ) {
                MesPrint("&More than %d coupling constants in vertex.",(WORD)MAXCOUPLINGS);
                MesPrint("    Recompile with a larger value for MAXCOUPLINGS.");
                return(1);
            }
            ss = s; s = SkipName(s); c = *s; *s = 0; name = ConstructName(ss,0);
            if ( GetVar(name,&type,&v->couplings[v->ncouplings],CSYMBOL,WITHAUTO)
                 == NAMENOTFOUND ) {
                WORD minpow = -MAXPOWER;
                WORD maxpow =  MAXPOWER;
                WORD cplx = 0, dim = 0;
                v->couplings[v->ncouplings] = AddSymbol(name,minpow,maxpow,cplx,dim);
            }
            *s = c;
            v->ncouplings++;
/*
            Now possibly a power
*/
            if ( *s == '^' ) { /* we need an integer */
                WORD x = 0; WORD sign = 1;
                s++;
                while ( *s == '-' || *s == '+' ) {
                    if ( *s == '-' ) sign = -sign;
                    s++;
                }
                while ( FG.cTable[*s] == 1 ) x = 10*x + *s++ - '0';
                v->couplings[v->ncouplings++] = x;
            }
            else {
                v->couplings[v->ncouplings++] = 1;
            }
            SKIPBLANKS(s);
            if ( *s == '*' ) { s++; continue; }
            if ( *s != ',' ) break;
            s++;
            SKIPBLANKS(s);
        }
      }
```

```
        else {
            v->error = 1;
            MesPrint("&A vertex statement needs at least one coupling constant.");
        }
    }
    if ( v->error == 0 ) {
/*
        Register which types of vertices are possible for each particle.
*/
        int i, j;
        for ( i = 0; i < v->nparticles; i++ ) {
            for ( j = 0; j < m->nparticles; j++ ) {
                if ( m->vertices[j]->particles[0].number == v->particles[i].number ) break;
                if ( m->vertices[j]->particles[1].number == v->particles[i].number ) break;
            }
            m->vertices[j]->couplings[v->nparticles] = 1;
        }
    }
    return(v->error);
}

/*
    #] CoVertex :
    #[ CoEndModel :
*/

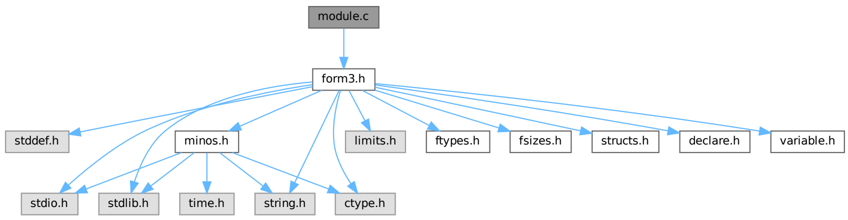
int CoEndModel(UBYTE *s)
{
    int i, j, k, kk;
    WORD csize = 0, *newcouplings, newsize;
    MODEL *m;
    VERTEX *v;
    if ( AC.ModelLevel <= 0 ) {
        MesPrint("&EndModel statement without matching Model statement");
        return(1);
    }
    m = AC.models[AC.nummodels-1];
/*
    Now create a list of all coupling constants.
    Note that we do not expect an astronomical number of them and hence
    we use a simple insertion algorithm.
*/
    m->ncouplings = 0;
    for ( i = 0; i < m->nvertices; i++ ) {
        v = m->vertices[i+m->nparticles];
        m->legcouple[v->nparticles] = 1;
        if ( m->ncouplings + v->ncouplings > csize ) {
            if ( csize == 0 ) newsize = v->ncouplings + 10;
            else newsize = 2*csize;
            newcouplings = (WORD *)Malloc1(newsize*sizeof(WORD),"m->couplings");
            for ( k = 0; k < m->ncouplings; k++ ) newcouplings[k] = m->couplings[k];
            if ( csize > 0 ) M_free(m->couplings,"m->couplings");
            m->couplings = newcouplings;
            csize = newsize;
        }
        for ( j = 0; j < v->ncouplings; j += 2 ) {
            WORD sym = v->couplings[j];
            for ( k = 0; k < m->ncouplings; k++ ) {
                if ( sym == m->couplings[k] ) break;
                if ( sym < m->couplings[k] ) {
                    for ( kk = m->ncouplings; kk > k; kk-- )
                        m->couplings[kk] = m->couplings[kk-1];
                    m->couplings[k] = sym;
                    m->ncouplings++;
                    break;
                }
            }
            if ( k >= m->ncouplings ) m->couplings[m->ncouplings++] = sym;
        }
    }
    AC.ModelLevel--;
    DUMMYUSE(s)
    return(LoadModel(m));
}

/*
    #] CoEndModel :
*/
```

4.76 module.c File Reference

`#include "form3.h"`

Include dependency graph for module.c:



Functions

- int ModuleInstruction (int ∗moduletype, int ∗specialtype)
- int CoModuleOption (UBYTE ∗s)
- int CoModOption (UBYTE ∗s)
- void SetSpecialMode (int moduletype, int specialtype)
- int ExecModule (int moduletype)
- int ExecStore (void)
- void FullCleanUp (void)
- int DoPolyfun (UBYTE ∗s)
- int DoPolyratfun (UBYTE ∗s)
- int DoNoParallel (UBYTE ∗s)
- int DoParallel (UBYTE ∗s)
- int DoModSum (UBYTE ∗s)
- int DoModMax (UBYTE ∗s)
- int DoModMin (UBYTE ∗s)
- int DoModLocal (UBYTE ∗s)
- int DoProcessBucket (UBYTE ∗s)
- UBYTE ∗ DoModDollar (UBYTE ∗s, int type)
- int DoinParallel (UBYTE ∗s)
- int DonotinParallel (UBYTE ∗s)
- int DoExecStatement (void)
- int DoPipeStatement (void)

4.76.1 Detailed Description

A number of routines that deal with the moduleoption statement and the execution of modules. Additionally there are the execution of the exec and pipe instructions.

Definition in file module.c.

4.76.2 Function Documentation

4.76.2.1 ModuleInstruction()

```
int ModuleInstruction (
            int * moduletype,
            int * specialtype )
```

Definition at line 76 of file module.c.

4.76.2.2 CoModuleOption()

```
int CoModuleOption (
            UBYTE * s )
```

Definition at line 143 of file module.c.

4.76.2.3 CoModOption()

```
int CoModOption (
            UBYTE * s )
```

Definition at line 212 of file module.c.

4.76.2.4 SetSpecialMode()

```
void SetSpecialMode (
            int moduletype,
            int specialtype )
```

Definition at line 257 of file module.c.

4.76.2.5 ExecModule()

```
int ExecModule (
            int moduletype )
```

Definition at line 274 of file module.c.

4.76.2.6 ExecStore()

```
int ExecStore (
            void )
```

Definition at line 299 of file module.c.

4.76.2.7 FullCleanUp()

```
void FullCleanUp (
            void )
```

Definition at line 314 of file module.c.

4.76.2.8 DoPolyfun()

```
int DoPolyfun (
            UBYTE * s )
```

Definition at line 395 of file module.c.

4.76.2.9 DoPolyratfun()

```
int DoPolyratfun (
            UBYTE * s )
```

Definition at line 458 of file module.c.

4.76.2.10 DoNoParallel()

```
int DoNoParallel (
            UBYTE * s )
```

Definition at line 529 of file module.c.

4.76.2.11 DoParallel()

```
int DoParallel (
            UBYTE * s )
```

Definition at line 544 of file module.c.

4.76.2.12 DoModSum()

```
int DoModSum (
            UBYTE * s )
```

Definition at line 559 of file module.c.

4.76.2.13 DoModMax()

```
int DoModMax (
            UBYTE * s )
```

Definition at line 579 of file module.c.

4.76.2.14 DoModMin()

```
int DoModMin (
            UBYTE * s )
```

Definition at line 599 of file module.c.

4.76.2.15 DoModLocal()

```
int DoModLocal (
            UBYTE * s )
```

Definition at line 619 of file module.c.

4.76.2.16 DoProcessBucket()

```
int DoProcessBucket (
            UBYTE * s )
```

Definition at line 639 of file module.c.

4.76.2.17 DoModDollar()

```
UBYTE * DoModDollar (
            UBYTE * s,
            int type )
```

Definition at line 657 of file module.c.

4.76.2.18 DoinParallel()

```
int DoinParallel (
            UBYTE * s )
```

Definition at line 758 of file module.c.

4.76.2.19 DonotinParallel()

```
int DonotinParallel (
            UBYTE * s )
```

Definition at line 768 of file module.c.

4.76.2.20 DoExecStatement()

```
int DoExecStatement (
            void )
```

Definition at line 780 of file module.c.

4.76.2.21 DoPipeStatement()

```
int DoPipeStatement (
            void )
```

Definition at line 797 of file module.c.

4.77 module.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
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 *
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
*/

#include "form3.h"

/*
    #] Includes :
    #[ Modules :
        #[ ModuleInstruction :

        Enters after the . of .sort etc
        We have word[[(options)][:commentary];]
        Word is one of 'clear end global sort store'
        Options are 'polyfun, endloopifunchanged, endloopifzero'
        Additions for solving equations can be added to 'word'

        The flag in the moduleoptions is for telling whether the current
        option can be followed by others. If it is > 0 it cannot.
*/

static KEYWORD ModuleWords[] = {
     {"clear",   (TFUN)0,    CLEARMODULE,    0}
    ,{"end",     (TFUN)0,    ENDMODULE,      0}
    ,{"global",  (TFUN)0,    GLOBALMODULE,   0}
    ,{"sort",    (TFUN)0,    SORTMODULE,     0}
    ,{"store",   (TFUN)0,    STOREMODULE,    0}
};

static KEYWORD ModuleOptions[] = {
     {"inparallel",          DoinParallel,   1,              1}
    ,{"local",               DoModLocal,     MODLOCAL,       0}
    ,{"maximum",             DoModMax,       MODMAX,         0}
    ,{"minimum",             DoModMin,       MODMIN,         0}
    ,{"noparallel",          DoNoParallel,   NOPARALLEL_USER,0}
    ,{"notinparallel",       DonotinParallel,0,              1}
    ,{"parallel",            DoParallel,     PARALLELFLAG,   0}
    ,{"polyfun",             DoPolyfun,      POLYFUN,        0}
    ,{"polyratfun",          DoPolyratfun,   POLYFUN,        0}
    ,{"processbucketsize",   DoProcessBucket,0,              0}
    ,{"sum",                 DoModSum,       MODSUM,         0}
```

```c
};

int ModuleInstruction(int *moduletype, int *specialtype)
{
    UBYTE *t, *s, *u, c;
    KEYWORD *key;
    int addit = 0, error = 0, i, j;
    DUMMYUSE(specialtype);
    LoadInstruction(0);
    AC.firstctypemessage = 0;
    s = AP.preStart; SKIPBLANKS(s)
    t = EndOfToken(s); c = *t; *t = 0;
    AC.origin = FROMPOINTINSTRUCTION;
    key = FindKeyWord(AP.preStart,ModuleWords,sizeof(ModuleWords)/sizeof(KEYWORD));
    if ( key == 0 ) {
        MesPrint("@Unrecognized module terminator: %s",s);
        error = 1;
        key = ModuleWords;
        while ( StrCmp((UBYTE *)key->name,(UBYTE *)"end") ) key++;
    }
    *t = c;
    *moduletype = key->type;
    SKIPBLANKS(t);
    while ( *t == '(' ) {   /* There are options */
        s = t+1; SKIPBRA3(t)
        if ( *t == 0 ) {
            MesPrint("@Improper options field in . instruction");
            error = 1;
        }
        else {
            *t = 0;
            if ( CoModOption(s) ) error = 1;
            *t++ = ')';
        }
    }
    if ( *t == ':' ) {  /* There is an 'advertisement' */
        t++;
        SKIPBLANKS(t)
        s = t; i = 0;
        while ( *t && *t != ';' ) {
            if ( *t == '\\' ) t++;
            t++; i++;
        }
        u = t;
        while ( u > s && u[-1] == ' ' ) { u--; i--; }
        if ( *u == '\\' ) { u++; i++; }
        for ( j = COMMERCIALSIZE-1; j >= 0; j-- ) {
            if ( i <= 0 ) break;
            AC.Commercial[j] = *--u; i--;
            if ( u > s && u[-1] == '\\' ) u--;
        }
        for ( ; j >= 0; j-- ) AC.Commercial[j] = ' ';
        AC.Commercial[COMMERCIALSIZE] = 0;
        addit += 2;
    }
    if ( addit && *t != ';' ) {
        MesPrint("@Improper ending of . instruction");
        error = -1;
    }
    return(error);
}

/*
        #] ModuleInstruction :
        #[ CoModuleOption :

    ModuleOption, options;
*/

int CoModuleOption(UBYTE *s)
{
    UBYTE *t,*tt,c;
    KEYWORD *option;
    int error = 0, polyflag = 0;
    AC.origin = FROMMODULEOPTION;
    if ( *s ) do {
        s = ToToken(s);
        t = EndOfToken(s);
        c = *t; *t = 0;
        option = FindKeyWord(s,ModuleOptions,
            sizeof(ModuleOptions)/sizeof(KEYWORD));
        if ( option == 0 ) {
            if ( polyflag ) {
                *t = c; t++; s = SkipAName(t);
                polyflag = 0;
                continue;
            }
```

```
            else {
                MesPrint("@Unrecognized module option: %s",s);
                error = 1;
                polyflag = 0;
                *t = c;
            }
        }
        else {
            *t = c;
            SKIPBLANKS(t)
            if ( (option->func)(t) ) error = 1;
        }
        if ( StrCmp((UBYTE *)(option->name),(UBYTE *)("polyfun")) == 0
         || StrCmp((UBYTE *)(option->name),(UBYTE *)("polyratfun")) == 0 ) {
            polyflag = 1;
            t++;
            t = SkipAName(t);
        }
        else polyflag = 0;
        if ( option->flags > 0 ) return(error);
        while ( *t ) {
            if ( *t == ',' ) {
                tt = t+1;
                while ( *tt == ',' ) tt++;
                if ( *tt != '$' ) break;
                t = tt+1;
            }
            if ( *t == ')' ) break;
            if ( *t == '(' ) SKIPBRA3(t)
            else if ( *t == '{' ) SKIPBRA2(t)
            else if ( *t == '[' ) SKIPBRA1(t)
            t++;
        }
        s = t;
    } while ( *s == ',' );
    if ( *s ) {
        MesPrint("@Unrecognized module option: %s",s);
        error = 1;
    }
    return(error);
}

/*
        #] CoModuleOption :
        #[ CoModOption :

    To be called from a .instruction.
    Only recognizes polyfun. The newer ones should be via the
    ModuleOption statement.
*/

int CoModOption(UBYTE *s)
{
    UBYTE *t,c;
    int error = 0;
    AC.origin = FROMPOINTINSTRUCTION;
    if ( *s ) do {
        s = ToToken(s);
        t = EndOfToken(s);
        c = *t; *t = 0;
        if ( StrICmp(s,(UBYTE *)"polyfun") == 0 ) {
            *t = c;
            SKIPBLANKS(t)
            if ( DoPolyfun(t) ) error = 1;
        }
        else if ( StrICmp(s,(UBYTE *)"polyratfun") == 0 ) {
            *t = c;
            SKIPBLANKS(t)
            if ( DoPolyratfun(t) ) error = 1;
        }
        else {
            MesPrint("@Unrecognized module option in .instruction: %s",s);
            error = 1;
            *t = c;
        }
        while ( *t ) {
            if ( *t == ',' || *t == ')' ) break;
            if ( *t == '(' ) SKIPBRA3(t)
            else if ( *t == '{' ) SKIPBRA2(t)
            else if ( *t == '[' ) SKIPBRA1(t)
            t++;
        }
        s = t;
    } while ( *s == ',' );
    if ( *s ) {
        MesPrint("@Unrecognized module option in .instruction: %s",s);
        error = 1;
```

```
    }
    return(error);
}

/*
        #] CoModOption :
        #[ SetSpecialMode :
*/

void SetSpecialMode(int moduletype, int specialtype)
{
    DUMMYUSE(moduletype); DUMMYUSE(specialtype);
}

/*
        #] SetSpecialMode :
        #[ MakeGlobal :

void MakeGlobal()
{
}

        #] MakeGlobal :
        #[ ExecModule :
*/

int ExecModule(int moduletype)
{
/*
    Here we check module options and other things that should
    be done at the start of a module.
*/
#ifdef WITHFLOAT
    if ( ( AC.tMaxWeight != 0 && AC.tMaxWeight != AC.MaxWeight )
      || ( AC.tDefaultPrecision != 0 && AC.tDefaultPrecision != AC.DefaultPrecision ) ) {
        AC.DefaultPrecision = AC.tDefaultPrecision;
        AC.tDefaultPrecision = 0;
        AC.MaxWeight = AC.tMaxWeight;
        AC.tMaxWeight = 0;
        ClearMZVTables();
        SetupMZVTables();
    }
#endif
    return(DoExecute(moduletype,0));
}

/*
        #] ExecModule :
        #[ ExecStore :
*/

int ExecStore(void)
{
    return(0);
}

/*
        #] ExecStore :
        #[ FullCleanUp :

        Remark 27-oct-2005 by JV
        This routine (and CleanUp in startup.c) may still need some work:
            What to do with preprocessor variables
            What to do with files we write to
*/

void FullCleanUp(void)
{
    int j;

    while ( AC.CurrentStream->previous >= 0 )
        AC.CurrentStream = CloseStream(AC.CurrentStream);
    AP.PreSwitchLevel = AP.PreIfLevel = 0;

    for ( j = NumProcedures-1; j >= 0; j-- ) {
        if ( Procedures[j].name ) M_free(Procedures[j].name,"name of procedure");
        if ( Procedures[j].p.buffer ) M_free(Procedures[j].p.buffer,"buffer of procedure");
    }
    NumProcedures = 0;

    while ( NumPre > AP.gNumPre ) {
        NumPre--;
        M_free(PreVar[NumPre].name,"PreVar[NumPre].name");
        PreVar[NumPre].name = PreVar[NumPre].value = 0;
    }

    AC.DidClean = 0;
```

```
    for ( j = 0; j < NumExpressions; j++ ) {
        AC.exprnames->namenode[Expressions[j].node].type = CDELETE;
        AC.DidClean = 1;
    }

    CompactifyTree(AC.exprnames,EXPRNAMES);

    for ( j = AO.NumDictionaries-1; j >= 0; j-- ) {
        RemoveDictionary(AO.Dictionaries[j]);
        M_free(AO.Dictionaries[j],"Dictionary");
    }
    AO.NumDictionaries = AO.gNumDictionaries = 0;
    M_free(AO.Dictionaries,"Dictionaries");
    AO.Dictionaries = 0;
    AO.SizeDictionaries = 0;
    AP.OpenDictionary = 0;
    AO.CurrentDictionary = 0;

    AP.ComChar = AP.cComChar;
    if ( AP.procedureExtension ) M_free(AP.procedureExtension,"procedureextension");
    AP.procedureExtension = strDup1(AP.cprocedureExtension,"procedureextension");

    AC.StatsFlag = AM.gStatsFlag = AM.ggStatsFlag;
    AC.extrasymbols = AM.gextrasymbols = AM.ggextrasymbols;
    AC.extrasym[0] = AM.gextrasym[0] = AM.ggextrasym[0] = 'Z';
    AC.extrasym[1] = AM.gextrasym[1] = AM.ggextrasym[1] = 0;
    AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers = AM.ggNoSpacesInNumbers;
    AO.IndentSpace = AM.gIndentSpace = AM.ggIndentSpace;
    AC.ThreadStats = AM.gThreadStats = AM.ggThreadStats;
    AC.OldFactArgFlag = AM.gOldFactArgFlag = AM.ggOldFactArgFlag;
    AC.FinalStats = AM.gFinalStats = AM.ggFinalStats;
    AC.OldGCDflag = AM.gOldGCDflag = AM.ggOldGCDflag;
    AC.ThreadsFlag = AM.gThreadsFlag = AM.ggThreadsFlag;
    if ( AC.ThreadsFlag && AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
    AC.ThreadBucketSize = AM.gThreadBucketSize = AM.ggThreadBucketSize;
    AC.ThreadBalancing = AM.gThreadBalancing = AM.ggThreadBalancing;
    AC.ThreadSortFileSynch = AM.gThreadSortFileSynch = AM.ggThreadSortFileSynch;
    AC.ShortStatsMax = AM.gShortStatsMax = AM.ggShortStatsMax;
    AC.SizeCommuteInSet = AM.gSizeCommuteInSet = 0;
#ifdef WITHFLOAT
    AC.MaxWeight = AM.gMaxWeight = AM.ggMaxWeight;
    AC.DefaultPrecision = AM.gDefaultPrecision = AM.ggDefaultPrecision;
#endif

    NumExpressions = 0;
    if ( DeleteStore(0) < 0 ) {
        MesPrint("@Cannot restart the storage file");
        Terminate(-1);
    }
    RemoveDollars();
    CleanUp(1);
    ResetVariables(2);
    IniVars();
}

/*
        #] FullCleanUp :
        #[ DoPolyfun :
*/

int DoPolyfun(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c;
    WORD funnum, eqsign = 0;
    if ( AC.origin == FROMPOINTINSTRUCTION ) {
        if ( *s == 0 || *s == ',' || *s == ')' ) {
            AR.PolyFun = 0; AR.PolyFunType = 0;
            return(0);
        }
        if ( *s != '=' ) {
            MesPrint("@Proper use in point instructions is: PolyFun[=functionname]");
            return(-1);
        }
        eqsign = 1;
    }
    else {
        if ( *s == 0 ) {
            AR.PolyFun = 0; AR.PolyFunType = 0;
            return(0);
        }
        if ( *s != '=' && *s != ',' ) {
            MesPrint("@Proper use is: PolyFun[{ ,=}functionname]");
            return(-1);
        }
        if ( *s == '=' ) eqsign = 1;
    }
```

```
    s++;
    SKIPBLANKS(s)
    t = EndOfToken(s);
    c = *t; *t = 0;

    if ( GetName(AC.varnames,s,&funnum,WITHAUTO) != CFUNCTION ) {
        if ( AC.origin != FROMPOINTINSTRUCTION && eqsign == 0 ) {
            AR.PolyFun = 0; AR.PolyFunType = 0;
            return(0);
        }
        MesPrint("@ %s is not a properly declared function",s);
        *t = c;
        return(-1);
    }
    if ( functions[funnum].spec > 0 || functions[funnum].commute != 0 ) {
        MesPrint("@The PolyFun must be a regular commuting function!");
        *t = c;
        return(-1);
    }
    AR.PolyFun = funnum+FUNCTION; AR.PolyFunType = 1;
    *t = c;
    SKIPBLANKS(t)
    if ( *t && *t != ',' && *t != ')' ) {
        t++; c = *t; *t = 0;
        MesPrint("@Improper ending of end-of-module instruction: %s",s);
        *t = c;
        return(-1);
    }
    return(0);
}

/*
        #] DoPolyfun :
        #[ DoPolyratfun :
*/

int DoPolyratfun(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c;
    WORD funnum;
    if ( AC.origin == FROMPOINTINSTRUCTION ) {
        if ( *s == 0 || *s == ',' || *s == ')' ) {
            AR.PolyFun = 0; AR.PolyFunType = 0; AR.PolyFunInv = 0; AR.PolyFunExp = 0;
            return(0);
        }
        if ( *s != '=' ) {
            MesPrint("@Proper use in point instructions is:
PolyRatFun[=functionname[+functionname]]");
            return(-1);
        }
    }
    else {
        if ( *s == 0 ) {
            AR.PolyFun = 0; AR.PolyFunType = 0; AR.PolyFunInv = 0; AR.PolyFunExp = 0;
            return(0);
        }
        if ( *s != '=' && *s != ',' ) {
            MesPrint("@Proper use is: PolyRatFun[{ ,=}functionname[+functionname]]");
            return(-1);
        }
    }
    s++;
    SKIPBLANKS(s)
    t = EndOfToken(s);
    c = *t; *t = 0;

    if ( GetName(AC.varnames,s,&funnum,WITHAUTO) != CFUNCTION ) {
Error1:;
        MesPrint("@ %s is not a properly declared function",s);
        *t = c;
        return(-1);
    }
    if ( functions[funnum].spec > 0 || functions[funnum].commute != 0 ) {
Error2:;
        MesPrint("@The PolyRatFun must be a regular commuting function!");
        *t = c;
        return(-1);
    }
    AR.PolyFun = funnum+FUNCTION; AR.PolyFunType = 2;
    AR.PolyFunInv = 0;
    AR.PolyFunExp = 0;
    AC.PolyRatFunChanged = 1;
    *t = c;
    if ( *t == '+' ) {
        t++; s = t;
        t = EndOfToken(s);
```

```
        c = *t; *t = 0;
        if ( GetName(AC.varnames,s,&funnum,WITHAUTO) != CFUNCTION ) goto Error1;
        if ( functions[funnum].spec > 0 || functions[funnum].commute != 0 ) goto Error2;
        AR.PolyFunInv = funnum+FUNCTION;
        *t = c;
    }
    SKIPBLANKS(t)
    if ( *t && *t != ',' && *t != ')' ) {
        t++; c = *t; *t = 0;
        MesPrint("@Improper ending of end-of-module instruction: %s",s);
        *t = c;
        return(-1);
    }
    return(0);
}

/*
        #] DoPolyratfun :
        #[ DoNoParallel :
*/

int DoNoParallel(UBYTE *s)
{
    if ( *s == 0 || *s == ',' || *s == ')' ) {
        AC.mparallelflag |= NOPARALLEL_USER;
        return(0);
    }
    MesPrint("@NoParallel should not have extra parameters");
    return(-1);
}

/*
        #] DoNoParallel :
        #[ DoParallel :
*/

int DoParallel(UBYTE *s)
{
    if ( *s == 0 || *s == ',' || *s == ')' ) {
        AC.mparallelflag &= ~NOPARALLEL_USER;
        return(0);
    }
    MesPrint("@Parallel should not have extra parameters");
    return(-1);
}

/*
        #] DoParallel :
        #[ DoModSum :
*/

int DoModSum(UBYTE *s)
{
    while ( *s == ',' ) s++;
    if ( *s != '$' ) {
        MesPrint("@Module Sum should mention which $-variables");
        return(-1);
    }
    s = DoModDollar(s,MODSUM);
    if ( s && *s != 0 && *s != ')' ) {
        MesPrint("@Irregular end of Sum option of Module statement");
        return(-1);
    }
    return(0);
}

/*
        #] DoModSum :
        #[ DoModMax :
*/

int DoModMax(UBYTE *s)
{
    while ( *s == ',' ) s++;
    if ( *s != '$' ) {
        MesPrint("@Module Maximum should mention which $-variables");
        return(-1);
    }
    s = DoModDollar(s,MODMAX);
    if ( s && *s != 0 ) {
        MesPrint("@Irregular end of Maximum option of Module statement");
        return(-1);
    }
    return(0);
}

/*
```

```
        #] DoModMax :
        #[ DoModMin :
*/

int DoModMin(UBYTE *s)
{
    while ( *s == ',' ) s++;
    if ( *s != '$' ) {
        MesPrint("@Module Minimum should mention which $-variables");
        return(-1);
    }
    s = DoModDollar(s,MODMIN);
    if ( s && *s != 0 ) {
        MesPrint("@Irregular end of Minimum option of Module statement");
        return(-1);
    }
    return(0);
}

/*
        #] DoModMin :
        #[ DoModLocal :
*/

int DoModLocal(UBYTE *s)
{
    while ( *s == ',' ) s++;
    if ( *s != '$' ) {
        MesPrint("@ModuleOption Local should mention which $-variables");
        return(-1);
    }
    s = DoModDollar(s,MODLOCAL);
    if ( s && *s != 0 ) {
        MesPrint("@Irregular end of Local option of ModuleOption statement");
        return(-1);
    }
    return(0);
}

/*
        #] DoModLocal :
        #[ DoProcessBucket :
*/

int DoProcessBucket(UBYTE *s)
{
    LONG x;
    while ( *s == ',' || *s == '=' ) s++;
    ParseNumber(x,s)
    if ( *s && *s != ' ' && *s != '\t' ) {
        MesPrint("&Numerical value expected for ProcessBucketSize");
        return(1);
    }
    AC.mProcessBucketSize = x;
    return(0);
}

/*
        #] DoProcessBucket :
        #[ DoModDollar :
*/

UBYTE * DoModDollar(UBYTE *s, int type)
{
    UBYTE *name, c;
    WORD number;
    MODOPTDOLLAR *md;
    while ( *s == '$' ) {
/*
        Read the name of the dollar
        Mark the type
*/
        s++;
        name = s;
        if ( FG.cTable[*s] == 0 ) {
            while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
            c = *s; *s = 0;
            number = GetDollar(name);
            if ( number < 0 ) {
                number = AddDollar(s,0,0,0);
                Warning("&Undefined $-variable in module statement");
            }
            md = (MODOPTDOLLAR *)FromList(&AC.ModOptDolList);
            md->number = number;
            md->type = type;
#ifdef WITHPTHREADS
            if ( type == MODLOCAL ) {
```

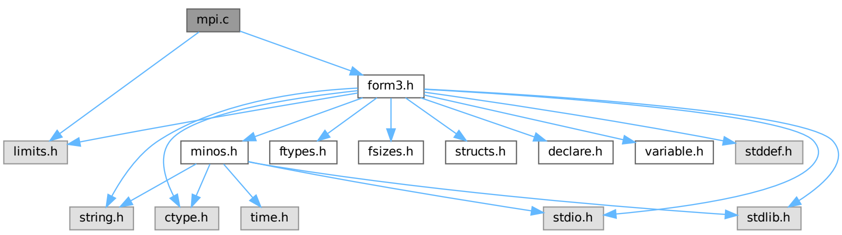
```
                int j, i;
                DOLLARS dglobal, dlocal;
                md->dstruct = (DOLLARS)Malloc1(
                    sizeof(struct DoLlArS)*AM.totalnumberofthreads,"Local DOLLARS");
/*
                Now copy the global dollar into the local copies.
                This can be nontrivial if the value needs an allocation.
                We don't really need the locks.
*/
                dglobal = Dollars + number;
                for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
                    dlocal = md->dstruct + j;
                    dlocal->index = dglobal->index;
                    dlocal->node = dglobal->node;
                    dlocal->type = dglobal->type;
                    dlocal->name = dglobal->name;
                    dlocal->size = dglobal->size;
                    dlocal->where = dglobal->where;
                    if ( dlocal->size > 0 ) {
                        dlocal->where = (WORD *)Malloc1((dlocal->size+1)*sizeof(WORD),"Local dollar value");
                        for ( i = 0; i < dlocal->size; i++ )
                            dlocal->where[i] = dglobal->where[i];
                        dlocal->where[dlocal->size] = 0;
                    }
                    dlocal->pthreadslockread = dummylock;
                    dlocal->pthreadslockwrite = dummylock;
                    dlocal->nfactors = dglobal->nfactors;
                    if ( dglobal->nfactors > 1 ) {
                        int nsize;
                        WORD *t, *m;
                        dlocal->factors = (FACDOLLAR *)Malloc1(dglobal->nfactors*sizeof(FACDOLLAR),"Dollar factors");
                        for ( i = 0; i < dglobal->nfactors; i++ ) {
                            nsize = dglobal->factors[i].size;
                            dlocal->factors[i].type = DOLUNDEFINED;
                            dlocal->factors[i].value = dglobal->factors[i].value;
                            if ( ( dlocal->factors[i].size = nsize ) > 0 ) {
                                dlocal->factors[i].where = t = (WORD *)Malloc1(sizeof(WORD)*(nsize+1),"DollarCopyFactor");
                                m = dglobal->factors[i].where;
                                NCOPY(t,m,nsize);
                                *t = 0;
                            }
                            else {
                                dlocal->factors[i].where = 0;
                            }
                        }
                    }
                    else { dlocal->factors = 0; }
                }
            }
            else {
                md->dstruct = 0;
            }
#endif
            *s = c;
        }
        else {
            MesPrint("&Illegal name for $-variable in module option");
            while ( *s != ',' && *s != 0 && *s != ')' ) s++;
        }
        while ( *s == ',' ) s++;
    }
    return(s);
}

/*
        #] DoModDollar :
        #[ DoinParallel :

        The idea is that we should have the commands
        ModuleOption,InParallel;
        ModuleOption,InParallel,name1,name2,...,namen;
        ModuleOption,NotInParallel,name1,name2,...,namen;
        The advantage over the InParallel statement is that this statement
        comes after the definition of the expressions.
*/

int DoinParallel(UBYTE *s)
{
    return(DoInParallel(s,1));
}

/*
        #] DoinParallel :
        #[ DonotinParallel :
```

```
*/

int DonotinParallel(UBYTE *s)
{
    return(DoInParallel(s,0));
}

/*
        #] DonotinParallel :
    #] Modules :
    #[ External :
        #[ DoExecStatement :
*/

int DoExecStatement(void)
{
#ifdef WITHSYSTEM
    FLUSHCONSOLE;
    if ( system((char *)(AP.preStart)) ) return(-1);
    return(0);
#else
    Error0("External programs not implemented on this computer/system");
    return(-1);
#endif
}

/*
        #] DoExecStatement :
        #[ DoPipeStatement :
*/

int DoPipeStatement(void)
{
#ifdef WITHPIPE
    FLUSHCONSOLE;
    if ( OpenStream(AP.preStart,PIPESTREAM,0,PRENOACTION) == 0 ) return(-1);
    return(0);
#else
    Error0("Pipes not implemented on this computer/system");
    return(-1);
#endif
}

/*
        #] DoPipeStatement :
    #] External :
*/
```

4.78 mpi.c File Reference

```
#include <limits.h>
#include "form3.h"
```

Include dependency graph for mpi.c:



Data Structures

- struct longMultiStruct

Macros

- #define PF_PACKSIZE 1600
- #define MPI_ERRCODE_CHECK(err)

Typedefs

- typedef struct longMultiStruct **PF_LONGMULTI**

Functions

- LONG PF_RealTime (int i)
- int PF_LibInit (int ∗argcp, char ∗∗∗argvp)
- int PF_LibTerminate (int error)
- int PF_Probe (int ∗src)
- int PF_ISendSbuf (int to, int tag)
- int PF_RecvWbuf (WORD ∗b, LONG ∗s, int ∗src)
- int PF_IRecvRbuf (PF_BUFFER ∗r, int bn, int from)
- int PF_WaitRbuf (PF_BUFFER ∗r, int bn, LONG ∗size)
- int PF_Bcast (void ∗buffer, int count)
- int PF_RawSend (int dest, void ∗buf, LONG l, int tag)
- LONG PF_RawRecv (int ∗src, void ∗buf, LONG thesize, int ∗tag)
- int PF_RawProbe (int ∗src, int ∗tag, int ∗bytesize)
- int PF_PrintPackBuf (char ∗s, int size)
- int PF_PreparePack (void)
- int PF_Pack (const void ∗buffer, size_t count, MPI_Datatype type)
- int PF_Unpack (void ∗buffer, size_t count, MPI_Datatype type)
- int PF_PackString (const UBYTE ∗str)
- int PF_UnpackString (UBYTE ∗str)
- int PF_Send (int to, int tag)
- int PF_Receive (int src, int tag, int ∗psrc, int ∗ptag)
- int PF_Broadcast (void)
- int PF_PrepareLongSinglePack (void)
- int PF_LongSinglePack (const void ∗buffer, size_t count, MPI_Datatype type)
- int PF_LongSingleUnpack (void ∗buffer, size_t count, MPI_Datatype type)
- int PF_LongSingleSend (int to, int tag)
- int PF_LongSingleReceive (int src, int tag, int ∗psrc, int ∗ptag)
- int PF_PrepareLongMultiPack (void)
- int PF_LongMultiPackImpl (const void ∗buffer, size_t count, size_t eSize, MPI_Datatype type)
- int PF_LongMultiUnpackImpl (void ∗buffer, size_t count, size_t eSize, MPI_Datatype type)
- int PF_LongMultiBroadcast (void)

Variables

- LONG PF_maxDollarChunkSize = 0

4.78.1 Detailed Description

MPI dependent functions of parform

This file contains all the functions for the parallel version of form3 that explicitly need to call mpi routines. This is the only file that really needs to be linked to the mpi-library!

Definition in file mpi.c.

4.78.2 Macro Definition Documentation

4.78.2.1 PF_PACKSIZE

```
#define PF_PACKSIZE 1600
```

Definition at line 58 of file mpi.c.

4.78.2.2 MPI_ERRCODE_CHECK

```
#define MPI_ERRCODE_CHECK(
              err )
```

Value:

```
    do { \
        int _tmp_err = (err); \
        if ( _tmp_err != MPI_SUCCESS ) return _tmp_err != 0 ? _tmp_err : -1; \
    } while (0)
```

A macro which exits the caller with a non-zero return value if *err* is not MPI_SUCCESS.

Parameters

err	The return value of a MPI function to be checked.

Remarks

The MPI standard defines MPI_SUCCESS == 0. Then (_tmp_err == 0) appears twice and we can expect the second evaluation will be eliminated by the compiler optimization.

Definition at line 84 of file mpi.c.

4.78.3 Function Documentation

4.78.3.1 PF_RealTime()

```
LONG PF_RealTime (
             int i )
```

Returns the realtime in 1/100 sec. as a LONG.

Parameters

i	the timer will be reset if i == 0.

Returns

the real elapsed time in 1/100 second.

Definition at line 101 of file mpi.c.

Referenced by PF_Init().

4.78.3.2 PF_LibInit()

```
int PF_LibInit (
            int * argcp,
            char *** argvp )
```

Performs all library dependent initializations.

Parameters

argcp	pointer to the number of arguments.
argvp	pointer to the arguments.

Returns

0 if OK, nonzero on error.

Definition at line 123 of file mpi.c.

Referenced by PF_Init().

4.78.3.3 PF_LibTerminate()

```
int PF_LibTerminate (
            int error )
```

Exits mpi, when there is an error either indicated or happening, returnvalue is 1, else returnvalue is 0.

Parameters

error	an error code.

Returns

0 if OK, nonzero on error.

Definition at line 209 of file mpi.c.

Referenced by PF_Terminate().

4.78.3.4 PF_Probe()

```
int PF_Probe (
            int * src )
```

Probes the next incoming message. If src == PF_ANY_SOURCE this operation is blocking, otherwise nonblocking.

Parameters

`in,out`	*src*	the source process number. In output, the process number of actual found source.

Returns

the tag value of the next incoming message if found, 0 if a nonblocking probe (input src != PF_ANY_SOURCE) did not find any messages. A negative returned value indicates an error.

Definition at line 230 of file mpi.c.

4.78.3.5 PF_ISendSbuf()

```
int PF_ISendSbuf (
            int to,
            int tag )
```

Nonblocking send operation. It sends everything from `buff` to `fill` of the active buffer. Depending on *tag* it also can do waiting for other sends to finish or set the active buffer to the next one.

Parameters

to	the destination process number.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 261 of file mpi.c.

Referenced by FlushOut(), PF_Processor(), and PutOut().

4.78.3.6 PF_RecvWbuf()

```
int PF_RecvWbuf (
            WORD * b,
            LONG * s,
            int * src )
```

Blocking receive of a `WORD` buffer.

Parameters

`out`	*b*	the buffer to store the received data.
`in,out`	*s*	the size of the buffer. The output value is the actual size of the received data.
`in,out`	*src*	the source process number. The output value is the process number of actual source.

Returns

the received message tag. A negative value indicates an error.

Definition at line 337 of file mpi.c.

4.78.3.7 PF_IRecvRbuf()

```
int PF_IRecvRbuf (
            PF_BUFFER * r,
            int bn,
            int from )
```

Posts nonblocking receive for the active receive buffer. The buffer is filled from `full` to `stop`.

Parameters

r	the `PF_BUFFER` struct for the nonblocking receive.
bn	the index of the cyclic buffer.
from	the source process number.

Returns

0 if OK, nonzero on error.

Definition at line 366 of file mpi.c.

4.78.3.8 PF_WaitRbuf()

```
int PF_WaitRbuf (
            PF_BUFFER * r,
            int bn,
            LONG * size )
```

Waits for the buffer *bn* to finish a pending nonblocking receive. It returns the received tag and in `*size` the number of field received. If the receive is already finished, just return the flag and size, else wait for it to finish, but also check for other pending receives.

Parameters

	r	the `PF_BUFFER` struct for the pending nonblocking receive.
	bn	the index of the cyclic buffer.
`out`	*size*	the actual size of received data.

Returns

the received message tag. A negative value indicates an error.

Definition at line 400 of file mpi.c.

4.78.3.9 PF_Bcast()

```
int PF_Bcast (
            void * buffer,
            int count )
```

Broadcasts a message from the master to slaves.

Parameters

in,out	*buffer*	the starting address of buffer. The contents in this buffer on the master will be transferred to those on the slaves.
	count	the length of the buffer in bytes.

Returns

0 if OK, nonzero on error.

Definition at line 440 of file mpi.c.

Referenced by PF_BroadcastBuffer(), and PF_BroadcastNumber().

4.78.3.10 PF_RawSend()

```
int PF_RawSend (
            int dest,
            void * buf,
            LONG l,
            int tag )
```

Sends *l* bytes from *buf* to *dest*. Returns 0 on success, or -1.

Parameters

dest	the destination process number.
buf	the send buffer.
l	the size of data in the send buffer in bytes.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 463 of file mpi.c.

Referenced by PF_MUnlock(), and PF_SendFile().

4.78.3.11 PF_RawRecv()

```
LONG PF_RawRecv (
            int * src,
```

```
            void * buf,
            LONG thesize,
            int * tag )
```

Receives not more than *thesize* bytes from *src*, returns the actual number of received bytes, or -1 on failure.

Parameters

in,out	*src*	the source process number. In output, that of the actual received message.
out	*buf*	the receive buffer.
	thesize	the size of the receive buffer in bytes.
out	*tag*	the message tag of the actual received message.

Returns

the actual sizeof received data in bytes, or -1 on failure.

Definition at line 484 of file mpi.c.

Referenced by PF_RecvFile().

4.78.3.12 PF_RawProbe()

```
int PF_RawProbe (
            int * src,
            int * tag,
            int * bytesize )
```

Probes an incoming message.

Parameters

in,out	*src*	the source process number. In output, that of the actual received message.
in,out	*tag*	the message tag. In output, that of the actual received message.
out	*bytesize*	the size of incoming data in bytes.

Returns

0 if OK, nonzero on error.

Definition at line 508 of file mpi.c.

4.78.3.13 PF_PrintPackBuf()

```
int PF_PrintPackBuf (
            char * s,
            int size )
```

Prints the contents in the pack buffer.

Parameters

s	a message to be shown.
size	the length of the buffer to be shown.

Returns

0 if OK, nonzero on error.

Definition at line 594 of file mpi.c.

4.78.3.14 PF_PreparePack()

```
int PF_PreparePack (
            void )
```

Prepares for the next pack operations on the sender.

Returns

0 if OK, nonzero on error.

Definition at line 624 of file mpi.c.

Referenced by Optimize(), PF_BroadcastPreDollar(), PF_BroadcastString(), and PF_Init().

4.78.3.15 PF_Pack()

```
int PF_Pack (
            const void * buffer,
            size_t count,
            MPI_Datatype type )
```

Adds data into the pack buffer.

Parameters

buffer	the pointer to the buffer storing the data to be packed.
count	the number of elements in the buffer.
type	the data type of elements in the buffer.

Returns

0 if OK, nonzero on error.

Definition at line 642 of file mpi.c.

References MPI_ERRCODE_CHECK.

Referenced by Optimize(), PF_BroadcastPreDollar(), and PF_Init().

4.78.3.16 PF_Unpack()

```
int PF_Unpack (
            void * buffer,
            size_t count,
            MPI_Datatype type )
```

Retrieves the next data in the pack buffer.

Parameters

out	*buffer*	the pointer to the buffer to store the unpacked data.
	count	the number of elements of data to be received.
	type	the data type of elements of data to be received.

Returns

0 if OK, nonzero on error.

Definition at line 671 of file mpi.c.

References MPI_ERRCODE_CHECK.

Referenced by Optimize(), PF_BroadcastPreDollar(), and PF_Init().

4.78.3.17 PF_PackString()

```
int PF_PackString (
            const UBYTE * str )
```

Packs a string *str* into the packed buffer PF_packbuf, including the trailing zero.

The first element (PF_INT) is the length of the packed portion of the string. If the string does not fit to the buffer PF←_packbuf, the function packs only the initial portion. It returns the number of packed bytes, so if (str[length-1]=='\0') then the whole string fits to the buffer, if not, then the rest (str+length) must be packed and send again. On error, the function returns the negative error code.

One exception: the string "\0!\0" is used as an image of the NULL, so all 3 characters will be packed.

Parameters

str	a string to be packed.

Returns

the number of packed bytes, or a negative value on failure.

Definition at line 706 of file mpi.c.

Referenced by PF_BroadcastString().

4.78.3.18 PF_UnpackString()

```
int PF_UnpackString (
            UBYTE * str )
```

Unpacks a string to *str* from the packed buffer PF_packbuf, including the trailing zero.

It returns the number of unpacked bytes, so if (str[length-1]=='\0') then the whole string was unpacked, if not, then the rest must be appended to (str+length). On error, the function returns the negative error code.

Parameters

out	*str*	the buffer to store the unpacked string

Returns

the number of unpacked bytes, or a negative value on failure.

Definition at line 774 of file mpi.c.

Referenced by PF_BroadcastString().

4.78.3.19 PF_Send()

```
int PF_Send (
            int to,
            int tag )
```

Sends the contents in the pack buffer to the process specified by *to*.

Example:
```
if ( PF.me == SRC ) {
  PF_PreparePack();
  // Packing operations here...
  PF_Send(DEST, TAG);
}
else if ( PF.me == DEST ) {
  PF_Receive(SRC, TAG, &actual_src, &actual_tag);
  // Unpacking operations here...
}
```

Parameters

to	the destination process number.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 822 of file mpi.c.

References MPI_ERRCODE_CHECK.

4.78.3.20 PF_Receive()

```
int PF_Receive (
            int src,
            int tag,
            int * psrc,
            int * ptag )
```

Receives data into the pack buffer from the process specified by *src*. This function allows &src == psrc or &tag == ptag. Either *psrc* or *ptag* can be NULL.

See the example of PF_Send().

Parameters

	src	the source process number (can be PF_ANY_SOURCE).
	tag	the source message tag (can be PF_ANY_TAG).
out	*psrc*	the actual source process number of received message.
out	*ptag*	the received message tag.

Returns

0 if OK, nonzero on error.

Definition at line 848 of file mpi.c.

References MPI_ERRCODE_CHECK.

4.78.3.21 PF_Broadcast()

```
int PF_Broadcast (
            void )
```

Broadcasts the contents in the pack buffer on the master to those on the slaves.

Example:

```
if ( PF.me == MASTER ) {
  PF_PreparePack();
  // Packing operations here...
}
PF_Broadcast();
if ( PF.me != MASTER ) {
  // Unpacking operations here...
}
```

Returns

0 if OK, nonzero on error.

Definition at line 883 of file mpi.c.

References MPI_ERRCODE_CHECK.

Referenced by Optimize(), PF_BroadcastPreDollar(), PF_BroadcastString(), and PF_Init().

4.78.3.22 PF_PrepareLongSinglePack()

```
int PF_PrepareLongSinglePack (
            void )
```

Prepares for the next long-single-pack operations on the sender.

Returns

0 if OK, nonzero on error.

Definition at line 1451 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.78.3.23 PF_LongSinglePack()

```
int PF_LongSinglePack (
            const void * buffer,
            size_t count,
            MPI_Datatype type )
```

Adds data into the "long single" pack buffer.

Parameters

buffer	the pointer to the buffer storing the data to be packed.
count	the number of elements in the buffer.
type	the data type of elements in the buffer.

Returns

0 if OK, nonzero on error.

Definition at line 1469 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.78.3.24 PF_LongSingleUnpack()

```
int PF_LongSingleUnpack (
            void * buffer,
            size_t count,
            MPI_Datatype type )
```

Retrieves the next data in the "long single" pack buffer.

Parameters

out	*buffer*	the pointer to the buffer to store the unpacked data.
	count	the number of elements of data to be received.
	type	the data type of elements of data to be received.

Returns

0 if OK, nonzero on error.

Definition at line 1503 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.78.3.25 PF_LongSingleSend()

```
int PF_LongSingleSend (
        int to,
        int tag )
```

Sends the contents in the "long single" pack buffer to the process specified by *to*.

Example:

```
if ( PF.me == SRC ) {
  PF_PrepareLongSinglePack();
  // Packing operations here...
  PF_LongSingleSend(DEST, TAG);
}
else if ( PF.me == DEST ) {
  PF_LongSingleReceive(SRC, TAG, &actual_src, &actual_tag);
  // Unpacking operations here...
}
```

Parameters

to	the destination process number.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 1540 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.78.3.26 PF_LongSingleReceive()

```
int PF_LongSingleReceive (
        int src,
        int tag,
        int * psrc,
        int * ptag )
```

Receives data into the "long single" pack buffer from the process specified by *src*. This function allows &src == psrc or &tag == ptag. Either *psrc* or *ptag* can be NULL.

See the example of PF_LongSingleSend().

Parameters

	src	the source process number (can be PF_ANY_SOURCE).
	tag	the source message tag (can be PF_ANY_TAG).
out	*psrc*	the actual source process number of received message.
out	*ptag*	the received message tag.

Returns

0 if OK, nonzero on error.

Definition at line 1583 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.78.3.27 PF_PrepareLongMultiPack()

```
int PF_PrepareLongMultiPack (
            void )
```

Prepares for the next long-multi-pack operations on the sender.

Returns

0 if OK, nonzero on error.

Definition at line 1643 of file mpi.c.

Referenced by Optimize(), PF_BroadcastCBuf(), PF_BroadcastExpFlags(), PF_BroadcastModifiedDollars(), and PF_BroadcastRedefinedPreVars().

4.78.3.28 PF_LongMultiPackImpl()

```
int PF_LongMultiPackImpl (
            const void * buffer,
            size_t count,
            size_t eSize,
            MPI_Datatype type )
```

Adds data into the "long multi" pack buffer.

Parameters

buffer	the pointer to the buffer storing the data to be packed.
count	the number of elements in the buffer.
eSize	the byte size of each element of data.
type	the data type of elements in the buffer.

Returns

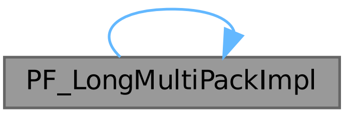
0 if OK, nonzero on error.

Definition at line 1662 of file mpi.c.

References PF_LongMultiPackImpl().

Referenced by PF_LongMultiPackImpl().

Here is the call graph for this function:



4.78.3.29 PF_LongMultiUnpackImpl()

```
int PF_LongMultiUnpackImpl (
            void * buffer,
            size_t count,
            size_t eSize,
            MPI_Datatype type )
```

Retrieves the next data in the "long multi" pack buffer.

Parameters

out	*buffer*	the pointer to the buffer to store the unpacked data.
	count	the number of elements of data to be received.
	eSize	the byte size of each element of data.
	type	the data type of elements of data to be received.

Returns

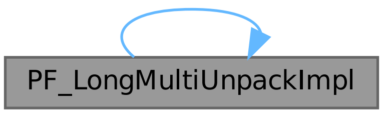
0 if OK, nonzero on error.

Definition at line 1721 of file mpi.c.

References PF_LongMultiUnpackImpl().

Referenced by PF_LongMultiUnpackImpl().

Here is the call graph for this function:



4.78.3.30 PF_LongMultiBroadcast()

```
int PF_LongMultiBroadcast (
             void )
```

Broadcasts the contents in the "long multi" pack buffer on the master to those on the slaves.

Example:

```
if ( PF.me == MASTER ) {
  PF_PrepareLongMultiPack();
  // Packing operations here...
}
PF_LongMultiBroadcast();
if ( PF.me != MASTER ) {
  // Unpacking operations here...
}
```

Returns

0 if OK, nonzero on error.

Definition at line 1807 of file mpi.c.

Referenced by Optimize(), PF_BroadcastCBuf(), PF_BroadcastExpFlags(), PF_BroadcastModifiedDollars(), and PF_BroadcastRedefinedPreVars().

4.78.4 Variable Documentation

4.78.4.1 PF_maxDollarChunkSize

```
LONG PF_maxDollarChunkSize = 0
```

Definition at line 69 of file mpi.c.

4.79 mpi.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   version.
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 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes and variables :
*/

#include <limits.h>
#include "form3.h"

#ifdef MPICH_PROFILING
# include "mpe.h"
#endif

#ifdef MPIDEBUGGING
#include "mpidbg.h"
#endif

/*[12oct2005 mt]:*/
/*
    Today there was some cleanup, some stuff is moved into another place
    in this file, and PF.packsize is removed and PF_packsize is used
    instead. It is rather difficult to proper comment it, so not all these
    changing are marked by "[12oct2005 mt]"
*/

#define PF_PACKSIZE 1600

/*
    Size in bytes, will be initialized soon as
    PF_packsize=PF_PACKSIZE/sizeof(int)*sizeof(int); for possible
    future developing we prefer to do this initialization not here,
    but in PF_LibInit:
*/

static int PF_packsize = 0;
static MPI_Status PF_status;
LONG PF_maxDollarChunkSize = 0;      /*:[04oct2005 mt]*/

static int PF_ShortPackInit(void);
static int PF_longPackInit(void);      /*:[12oct2005 mt]*/

#define MPI_ERRCODE_CHECK(err) \
    do { \
        int _tmp_err = (err); \
        if ( _tmp_err != MPI_SUCCESS ) return _tmp_err != 0 ? _tmp_err : -1; \
    } while (0)

/*
    #] Includes and variables :
    #[ PF_RealTime :
*/

LONG PF_RealTime(int i)
{
    static double starttime;
    if ( i == PF_RESET ) {
        starttime = MPI_Wtime();
```

```
            return((LONG)0);
        }
        return((LONG)( 100. * (MPI_Wtime() - starttime) ) );
    }

    /*
        #] PF_RealTime :
        #[ PF_LibInit :
    */

    int PF_LibInit(int *argcp, char ***argvp)
    {
        int ret;
        ret = MPI_Init(argcp,argvp);
        if ( ret != MPI_SUCCESS ) return(ret);
        ret = MPI_Comm_rank(PF_COMM,&PF.me);
        if ( ret != MPI_SUCCESS ) return(ret);
        ret = MPI_Comm_size(PF_COMM,&PF.numtasks);
        if ( ret != MPI_SUCCESS ) return(ret);

        /* Initialization of packed communications. */
        PF_packsize = PF_PACKSIZE/sizeof(int)*sizeof(int);
        if ( PF_ShortPackInit() ) return -1;
        if ( PF_longPackInit() ) return -1;

        {/*Block*/
            int bytes, totalbytes=0;
    /*
                There is one problem with maximal possible packing: there is no API to
                convert bytes to the record number. So, here we calculate the buffer
                size needed for storing dollarvars:

                LONG PF_maxDollarChunkSize is the size for the portion of the dollar
                variable buffer suitable for broadcasting. This variable should be
                visible from parallel.c

                Evaluate PF_Pack(numterms,1,PF_INT):
    */
            if ( ( ret = MPI_Pack_size(1,PF_INT,PF_COMM,&bytes) )!=MPI_SUCCESS )
                return(ret);

            totalbytes+=bytes;
    /*
                Evaluate PF_Pack( newsize,1,PF_LONG):
    */
            if ( ( ret = MPI_Pack_size(1,PF_LONG,PF_COMM,&bytes) )!=MPI_SUCCESS )
                return(ret);

            totalbytes += bytes;
    /*
                Now available room is PF_packsize-totalbytes
    */
            totalbytes = PF_packsize-totalbytes;
    /*
                Now totalbytes is the size of chunk in bytes.
                Evaluate this size in number of records:

                Rough estimate:
    */
            PF_maxDollarChunkSize=totalbytes/sizeof(WORD);
    /*
                Go to the up limit:
    */
            do {
                if ( ( ret = MPI_Pack_size(
                            ++PF_maxDollarChunkSize,PF_WORD,PF_COMM,&bytes) )!=MPI_SUCCESS )
                    return(ret);
            } while ( bytes<totalbytes );
    /*
                Now the chunk size is too large
                And now evaluate the exact value:
    */
            do {
                if ( ( ret = MPI_Pack_size(
                            --PF_maxDollarChunkSize,PF_WORD,PF_COMM,&bytes) )!=MPI_SUCCESS )
                    return(ret);
            } while ( bytes>totalbytes );
    /*
                Now PF_maxDollarChunkSize is the size of chunk of PF_WORD fitting the
                buffer <= (PF_packsize-PF_INT-PF_LONG)
    */
        }/*Block*/
        return(0);
    }
    /*
        #] PF_LibInit :
        #[ PF_LibTerminate :
```

```
*/

int PF_LibTerminate(int error)
{
    DUMMYUSE(error);
    return(MPI_Finalize());
}

/*
    #] PF_LibTerminate :
    #[ PF_Probe :
*/

int PF_Probe(int *src)
{
    int ret, flag;
    if ( *src == PF_ANY_SOURCE ) { /*Blocking call*/
        ret = MPI_Probe(*src,MPI_ANY_TAG,PF_COMM,&PF_status);
        flag = 1;
    }
    else { /*Non-blocking call*/
        ret = MPI_Iprobe(*src,MPI_ANY_TAG,PF_COMM,&flag,&PF_status);
    }
    *src = PF_status.MPI_SOURCE;
    if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
    if ( !flag ) return(0);
    return(PF_status.MPI_TAG);
}

/*
    #] PF_Probe :
    #[ PF_ISendSbuf :
*/

int PF_ISendSbuf(int to, int tag)
{
    PF_BUFFER *s = PF.sbuf;
    int a = s->active;
    int size = s->fill[a] - s->buff[a];
    int r = 0;

    static int finished;

    s->fill[a] = s->buff[a];
    if ( s->numbufs == 1 ) {
        r = MPI_Ssend(s->buff[a],size,PF_WORD,MASTER,tag,PF_COMM);
        if ( r != MPI_SUCCESS ) {
            fprintf(stderr,"[%d|%d] PF_ISendSbuf: MPI_Ssend returns: %d \n",
                    PF.me,(int)AC.CModule,r);
            fflush(stderr);
            return(r);
        }
        return(0);
    }

    switch ( tag ) { /* things to do before sending */
        case PF_TERM_MSGTAG:
            if ( PF.sbuf->request[to] != MPI_REQUEST_NULL)
                r = MPI_Wait(&PF.sbuf->request[to],&PF.sbuf->retstat[to]);
            if ( r != MPI_SUCCESS ) return(r);
            break;
        default:
            break;
    }

    r = MPI_Isend(s->buff[a],size,PF_WORD,to,tag,PF_COMM,&s->request[a]);

    if ( r != MPI_SUCCESS ) return(r);

    switch ( tag ) { /* things to do after initialising sending */
        case PF_TERM_MSGTAG:
            finished = 0;
            break;
        case PF_ENDSORT_MSGTAG:
            if ( ++finished == PF.numtasks - 1 )
                r = MPI_Waitall(s->numbufs,s->request,s->status);
            if ( r != MPI_SUCCESS ) return(r);
            break;
        case PF_BUFFER_MSGTAG:
            if ( ++s->active >= s->numbufs ) s->active = 0;
            while ( s->request[s->active] != MPI_REQUEST_NULL ) {
                r = MPI_Waitsome(s->numbufs,s->request,&size,s->index,s->retstat);
                if ( r != MPI_SUCCESS ) return(r);
            }
            break;
        case PF_ENDBUFFER_MSGTAG:
            if ( ++s->active >= s->numbufs ) s->active = 0;
```

```
            r = MPI_Waitall(s->numbufs,s->request,s->status);
            if ( r != MPI_SUCCESS ) return(r);
            break;
        default:
            return(-99);
            break;
    }
    return(0);
}

/*
    #] PF_ISendSbuf :
    #[ PF_RecvWbuf :
*/

int PF_RecvWbuf(WORD *b, LONG *s, int *src)
{
    int i, r = 0;

    r = MPI_Recv(b,(int)*s,PF_WORD,*src,PF_ANY_MSGTAG,PF_COMM,&PF_status);
    if ( r != MPI_SUCCESS ) { if ( r > 0 ) r *= -1; return(r); }

    r = MPI_Get_count(&PF_status,PF_WORD,&i);
    if ( r != MPI_SUCCESS ) { if ( r > 0 ) r *= -1; return(r); }

    *s = (LONG)i;
    *src = PF_status.MPI_SOURCE;
    return(PF_status.MPI_TAG);
}

/*
    #] PF_RecvWbuf :
    #[ PF_IRecvRbuf :
*/

int PF_IRecvRbuf(PF_BUFFER *r, int bn, int from)
{
    int ret;
    r->type[bn] = PF_WORD;

    if ( r->numbufs == 1 ) {
        r->tag[bn] = MPI_ANY_TAG;
        r->from[bn] = from;
    }
    else {
        ret = MPI_Irecv(r->full[bn],(int)(r->stop[bn] - r->full[bn]),PF_WORD,from,
                        MPI_ANY_TAG,PF_COMM,&r->request[bn]);
        if (ret != MPI_SUCCESS) { if(ret > 0) ret *= -1; return(ret); }
    }
    return(0);
}

/*
    #] PF_IRecvRbuf :
    #[ PF_WaitRbuf :
*/

int PF_WaitRbuf(PF_BUFFER *r, int bn, LONG *size)
{
    int ret, rsize;

    if ( r->numbufs == 1 ) {
        *size = r->stop[bn] - r->full[bn];
        ret = MPI_Recv(r->full[bn],(int)*size,r->type[bn],r->from[bn],r->tag[bn],
                       PF_COMM,&(r->status[bn]));
        if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
        ret = MPI_Get_count(&(r->status[bn]),r->type[bn],&rsize);
        if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
        if ( rsize > *size ) return(-99);
        *size = (LONG)rsize;
    }
    else {
        while ( r->request[bn] != MPI_REQUEST_NULL ) {
            ret = MPI_Waitsome(r->numbufs,r->request,&rsize,r->index,r->retstat);
            if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
            while ( --rsize >= 0 ) r->status[r->index[rsize]] = r->retstat[rsize];
        }
        ret = MPI_Get_count(&(r->status[bn]),r->type[bn],&rsize);
        if ( ret != MPI_SUCCESS ) { if ( ret > 0 ) ret *= -1; return(ret); }
        *size = (LONG)rsize;
    }
    return(r->status[bn].MPI_TAG);
}

/*
    #] PF_WaitRbuf :
    #[ PF_Bcast :
```

```
*/

int PF_Bcast(void *buffer, int count)
{
    if ( MPI_Bcast(buffer,count,MPI_BYTE,MASTER,PF_COMM) != MPI_SUCCESS )
        return(-1);
    return(0);
}

/*
    #] PF_Bcast :
    #[ PF_RawSend :
*/

int PF_RawSend(int dest, void *buf, LONG l, int tag)
{
    int ret=MPI_Ssend(buf,(int)l,MPI_BYTE,dest,tag,PF_COMM);
    if ( ret != MPI_SUCCESS ) return(-1);
    return(0);
}
/*
    #] PF_RawSend :
    #[ PF_RawRecv :
*/

LONG PF_RawRecv(int *src,void *buf,LONG thesize,int *tag)
{
    MPI_Status stat;
    int ret=MPI_Recv(buf,(int)thesize,MPI_BYTE,*src,MPI_ANY_TAG,PF_COMM,&stat);
    if ( ret != MPI_SUCCESS ) return(-1);
    if ( MPI_Get_count(&stat,MPI_BYTE,&ret) != MPI_SUCCESS ) return(-1);
    *tag = stat.MPI_TAG;
    *src = stat.MPI_SOURCE;
    return(ret);
}

/*
    #] PF_RawRecv :
    #[ PF_RawProbe :
*/

int PF_RawProbe(int *src, int *tag, int *bytesize)
{
    MPI_Status stat;
    int srcval = src != NULL ? *src : PF_ANY_SOURCE;
    int tagval = tag != NULL ? *tag : PF_ANY_MSGTAG;
    int ret = MPI_Probe(srcval, tagval, PF_COMM, &stat);
    if ( ret != MPI_SUCCESS ) return -1;
    if ( src != NULL ) *src = stat.MPI_SOURCE;
    if ( tag != NULL ) *tag = stat.MPI_TAG;
    if ( bytesize != NULL ) {
        ret = MPI_Get_count(&stat, MPI_BYTE, bytesize);
        if ( ret != MPI_SUCCESS ) return -1;
    }
    return 0;
}

/*
    #] PF_RawProbe :
    #[ The pack buffer :
        #[ Variables :
*/

/*
 * The pack buffer with the fixed size (= PF_packsize).
 */
static UBYTE *PF_packbuf  = NULL;
static UBYTE *PF_packstop = NULL;
static int    PF_packpos  = 0;

/*
        #] Variables :
        #[ PF_ShortPackInit :
*/

static int PF_ShortPackInit(void)
{
    PF_packbuf = (UBYTE *)Malloc1(sizeof(UBYTE) * PF_packsize, "PF_ShortPackInit");
    if ( PF_packbuf == NULL ) return -1;
    PF_packstop = PF_packbuf + PF_packsize;
    return 0;
}

/*
        #] PF_ShortPackInit :
        #[ PF_InitPackBuf :
*/
```

```
static inline int PF_InitPackBuf(void)
{
/*
        This is definitely not the best place for allocating the
        buffer! Moved to PF_LibInit():

    if ( PF_packbuf == 0 ) {
        PF_packbuf = (UBYTE *)Malloc1(sizeof(UBYTE)*PF.packsize,"PF_InitPackBuf");
        if ( PF_packbuf == 0 ) return(-1);
        PF_packstop = PF_packbuf + PF.packsize;
    }
*/
    PF_packpos = 0;
    return(0);
}

/*
        #] PF_InitPackBuf :
        #[ PF_PrintPackBuf :
*/

int PF_PrintPackBuf(char *s, int size)
{
#ifdef NOMESPRINTYET
/*
        The use of printf should be discouraged. The results are flushed to
        the output at unpredictable moments. We should use printf only
        during startup when MesPrint doesn't have its buffers and output
        channels initialized.
*/
    int i;
    printf("[%d] %s: ",PF.me,s);
    for(i=0;i<size;i++) printf("%d ",PF_packbuf[i]);
    printf("\n");
#else
    MesPrint("[%d] %s: %a",PF.me,s,size,(WORD *)(PF_packbuf));
#endif
    return(0);
}

/*
        #] PF_PrintPackBuf :
        #[ PF_PreparePack :
*/

int PF_PreparePack(void)
{
    return PF_InitPackBuf();
}

/*
        #] PF_PreparePack :
        #[ PF_Pack :
*/

int PF_Pack(const void *buffer, size_t count, MPI_Datatype type)
{
    int err, bytes;

    if ( count > INT_MAX ) return -99;

    err = MPI_Pack_size((int)count, type, PF_COMM, &bytes);
    MPI_ERRCODE_CHECK(err);
    if ( PF_packpos + bytes > PF_packstop - PF_packbuf ) return -99;

    err = MPI_Pack((void *)buffer, (int)count, type, PF_packbuf, PF_packsize, &PF_packpos, PF_COMM);
    MPI_ERRCODE_CHECK(err);

    return 0;
}

/*
        #] PF_Pack :
        #[ PF_Unpack :
*/

int PF_Unpack(void *buffer, size_t count, MPI_Datatype type)
{
    int err;

    if ( count > INT_MAX ) return -99;

    err = MPI_Unpack(PF_packbuf, PF_packsize, &PF_packpos, buffer, (int)count, type, PF_COMM);
    MPI_ERRCODE_CHECK(err);

    return 0;
```

```
}

/*
        #] PF_Unpack :
        #[ PF_PackString :
*/

int PF_PackString(const UBYTE *str)
{
    int ret,buflength,bytes,length;
/*
        length will be packed in the beginning.
        Decrement buffer size by the length of the field "length":
*/
    if ( ( ret = MPI_Pack_size(1,PF_INT,PF_COMM,&bytes) ) != MPI_SUCCESS )
        return(ret);
    buflength = PF_packsize - bytes;
/*
        Calculate the string length (INCLUDING the trailing zero!):
*/
    for ( length = 0; length < buflength; length++ ) {
        if ( str[length] == '\0' ) {
            length++; /* since the trailing zero must be accounted */
            break;
        }
    }
/*
        The string "\0!\0" is used as an image of the NULL.
*/
    if ( ( str[0] == '\0' ) /* empty string */
      && ( str[1] == '!' )  /* Special case? */
      && ( str[2] == '\0' ) /* Yes, pass 3 initial symbols */
            ) length += 2;  /* all 3 characters will be packed */
    length++;               /* Will be decremented in the following loop */
/*
        The problem: packed size of byte may be not equal 1! So first, suppose
        it is 1, and if this is not the case decrease the length of the string
        until it fits the buffer:
*/
    do {
        if ( ( ret = MPI_Pack_size(--length,PF_BYTE,PF_COMM,&bytes) )
            != MPI_SUCCESS ) return(ret);
    } while ( bytes > buflength );
/*
        Note, now if str[length-1] == '\0' then the string fits to the buffer
        (INCLUDING the trailing zero!);if not, the rest must be packed further!

        Pack the length to PF_packbuf:
*/
    if ( ( ret = MPI_Pack(&length,1,PF_INT,PF_packbuf,PF_packsize,
            &PF_packpos,PF_COMM) ) != MPI_SUCCESS ) return(ret);
/*
        Pack the string to PF_packbuf:
*/
    if ( ( ret = MPI_Pack((UBYTE *)str,length,PF_BYTE,PF_packbuf,PF_packsize,
            &PF_packpos,PF_COMM) ) != MPI_SUCCESS ) return(ret);
    return(length);
}

/*
        #] PF_PackString :
        #[ PF_UnpackString :
*/

int PF_UnpackString(UBYTE *str)
{
    int ret,length;
/*
        Unpack the length:
*/
    if(  (ret = MPI_Unpack(PF_packbuf,PF_packsize,&PF_packpos,
            &length,1,PF_INT,PF_COMM))!= MPI_SUCCESS )
                return(ret);
/*
        Unpack the string:
*/
    if ( ( ret = MPI_Unpack(PF_packbuf,PF_packsize,&PF_packpos,
            str,length,PF_BYTE,PF_COMM) ) != MPI_SUCCESS ) return(ret);
/*
        Now if str[length-1]=='\0' then the whole string
        (INCLUDING the trailing zero!) was unpacked ;if not, the rest
        must be unpacked to str+length.
*/
    return(length);
}

/*
```

```
        #] PF_UnpackString :
        #[ PF_Send :
*/

int PF_Send(int to, int tag)
{
    int err;
    err = MPI_Ssend(PF_packbuf, PF_packpos, MPI_PACKED, to, tag, PF_COMM);
    MPI_ERRCODE_CHECK(err);
    return 0;
}

/*
        #] PF_Send :
        #[ PF_Receive :
*/

int PF_Receive(int src, int tag, int *psrc, int *ptag)
{
    int err;
    MPI_Status status;
    PF_InitPackBuf();
    err = MPI_Recv(PF_packbuf, PF_packsize, MPI_PACKED, src, tag, PF_COMM, &status);
    MPI_ERRCODE_CHECK(err);
    if ( psrc ) *psrc = status.MPI_SOURCE;
    if ( ptag ) *ptag = status.MPI_TAG;
    return 0;
}

/*
        #] PF_Receive :
        #[ PF_Broadcast :
*/

int PF_Broadcast(void)
{
    int err;
/*
 * If PF_SHORTBROADCAST is defined, then the broadcasting will be performed in
 * 2 steps. First, the size of the buffer will be broadcast, then the buffer of
 * exactly used size. This should be faster with slow connections, but slower on
 * SMP shmem MPI because of the latency.
 */
#ifdef PF_SHORTBROADCAST
    int pos = PF_packpos;
#endif
    if ( PF.me != MASTER ) {
        err = PF_InitPackBuf();
        if ( err ) return err;
    }
#ifdef PF_SHORTBROADCAST
    err = MPI_Bcast(&pos, 1, MPI_INT, MASTER, PF_COMM);
    MPI_ERRCODE_CHECK(err);
    err = MPI_Bcast(PF_packbuf, pos, MPI_PACKED, MASTER, PF_COMM);
#else
    err = MPI_Bcast(PF_packbuf, PF_packsize, MPI_PACKED, MASTER, PF_COMM);
#endif
    MPI_ERRCODE_CHECK(err);
    return 0;
}

/*
        #] PF_Broadcast :
    #] The pack buffer :
    #[ Long pack stuff :
        #[ Explanations :

    The problems here are:
        1. We need to send/receive long dollar variables. For
    preprocessor-defined dollarvars we used multiply
    packing/broadcasting  (see parallel.c:PF_BroadcastPreDollar())
    since each variable must be broadcast immediately. For run-time
    the changed dollar variables, collecting and broadcasting are
    performed at the end of the module and all modified dollarvars
    are transferred "at once", that is why the size of packed and
    transferred buffers may be really very large.
        2. There is some strange feature of MPI_Bcast() on Altix MPI
    implementation, namely, sometimes it silently fails with big
    buffers. For better performance, it would be useful to send one
    big buffer instead of several small ones (since the latency is more
    important than the bandwidth). That is why we need two different
    sets of routines: for long point-to-point communication we collect
    big re-allocatable buffer, the corresponding routines have the
    prefix PF_longSingle, and for broadcasting we pack data into
    several smaller buffers, the corresponding routines have the
    prefix PF_longMulti.
        Note, from portability reasons we cannot split large packed
```

```
    buffer into small chunks, send them and collect back on the other
    side, see "Advice to users" on page 180 MPI--The Complete Reference
    Volume1, second edition.
        OPTIMIZING:
        We assume, for most communications, the single buffer of size
    PF_packsize is enough.

        How does it work:
        For point-to-point, there is one big re-allocatable
    buffer PF_longPackBuf with two integer positions: PF_longPackPos
    and PF_longPackTop (due to re-allocatable character of the buffer,
    it is better to use integers rather than pointers).
        Each time of re-allocation, the size of the buffer
    PF_longPackBuf is incremented by the same size of a "standard" chunk
    PF_packsize.
        For broadcasting there is one linked list (PF_longMultiRoot),
    which contains either positions of a chunk of PF_longPackBuf, or
    it's own buffer. This is done for better memory utilisation:
    longSingle and longMulti are never used simultaneously.
        When a new cell is needed for LongMulti packing, we increment
    the counter PF_longPackN and just follow the list. If it is not
    possible, we allocate the cell's own buffer and link it to the end
    of the list PF_longMultiRoot.
        When PF_longPackPos is reallocated, we link new chunks into
    existing PF_longMultiRoot list before the first longMulti allocated
    cell's own buffer. The pointer PF_longMultiLastChunk points to the last
    cell of PF_longMultiRoot containing the pointer to the chunk of
    PF_longPackBuf.
        Initialization PF_longPackBuf is made by the function
    PF_longSingleReset(). In the begin of the PF_longPackBuf it packs
    the size of the last sent buffer. Upon sending, the program checks,
    whether there was at list one re-allocation (PF_longPackN>1) .
    If so, the sender first packs and sends small buffer
    (PF_longPackSmallBuf) containing one integer number -- the
    _negative_ new size of the send buffer. Getting the buffer, a
    receiver unpacks one integer and checks whether it is <0 . If so,
    the receiver will repeat receiving, but first it checks whether
    it has enough buffer and increase it, if necessary.
        Initialization PF_longMultiRoot is made by the function
    PF_longMultiReset(). In the begin of the first chunk it packs
    one integer -- the number 1. Upon sending, the program checks,
    how many cells were packed (PF_longPackN). If more than 1, the
    sender packs to the next cell the integer PF_longPackN, than
    packs PF_longPackN pairs of integers -- the information about how many
    times chunk on each cell was accessed by the packing procedure,
    this information is contained by the nPacks field of the cell
    structure, and how many non-complete items was at the end of this
    chunk the structure field lastLen. Then the sender sends first
    this auxiliary chunk.
    The receiver unpacks the integer from obtained chunk and, if this
    integer is more than 1, it gets more chunks, unpacking information
    from the first auxiliary chunk into the corresponding nPacks
    fields. Unpacking information from multiple chunks, the receiver
    knows, when the chunk is expired and it must switch to the next cell,
    successively decrementing corresponding nPacks field.

    XXX: There are still some flaws:
    PF_LongSingleSend/PF_LongSingleReceive may fail, for example, for data
    transfers from the master to many slaves. Suppose that the master sends big
    data to slaves, which needs an increase of the buffer of the receivers. For
    the first data transfer, the master sends the new buffer size as the first
    message, and then sends the data as the second message, because
    PF_LongSinglePack records the increase of the buffer size on the master. For
    the next time, however, the master sends the data without sending the new
    buffer size, and then MPI_Recv fails due to the data overflow.
    In parallel.c, they are used for the communication from slaves to the
    master. In this case, this problem does not occur because the master always
    has enough buffer.
    The maximum size that PF_LongMultiBroadcast can broadcast is limited to
    around 320kB because the current implementation tries to pack all
    information of chained buffers into one buffer, whose size is PF_packsize
    = 1600B.

        #] Explanations :
        #[ Variables :
*/

typedef struct longMultiStruct {
    UBYTE *buffer; /* NULL if */
    int bufpos;    /* if >=0, PF_longPackBuf+bufpos is the chunk start */
    int packpos;   /* the current position */
    int nPacks;    /* How many times PF_longPack operates on this cell */
    int lastLen;   /* if > 0, the last packing didn't fit completely to this
                      chunk, only lastLen items was packed, the rest is in
                      the next cell. */
    struct longMultiStruct *next;  /* next linked cell, or NULL */
} PF_LONGMULTI;
```

```

static UBYTE *PF_longPackBuf = NULL;
static void *PF_longPackSmallBuf = NULL;
static int PF_longPackPos = 0;
static int PF_longPackTop = 0;
static PF_LONGMULTI *PF_longMultiRoot = NULL;
static PF_LONGMULTI *PF_longMultiTop = NULL;
static PF_LONGMULTI *PF_longMultiLastChunk = NULL;
static int PF_longPackN = 0;

/*
        #] Variables :
        #[ Long pack private functions :
        #[ PF_longMultiNewCell :
*/

static inline int PF_longMultiNewCell(void)
{
/*
        Allocate a new cell:
*/
    PF_longMultiTop->next = (PF_LONGMULTI *)
            Malloc1(sizeof(PF_LONGMULTI),"PF_longMultiCell");
    if ( PF_longMultiTop->next == NULL ) return(-1);
/*
        Allocate a private buffer:
*/
    PF_longMultiTop->next->buffer=(UBYTE*)
            Malloc1(sizeof(UBYTE)*PF_packsize,"PF_longMultiChunk");
    if ( PF_longMultiTop->next->buffer == NULL ) return(-1);
/*
        For the private buffer position is -1:
*/
    PF_longMultiTop->next->bufpos = -1;
/*
        This is the last cell in the chain:
*/
    PF_longMultiTop->next->next = NULL;
/*
        packpos and nPacks are not initialized!
*/
    return(0);
}

/*
        #] PF_longMultiNewCell :
        #[ PF_longMultiPack2NextCell :
*/
static inline int PF_longMultiPack2NextCell(void)
{
/*
        Is there a free cell in the chain?
*/
    if ( PF_longMultiTop->next == NULL ) {
/*
            No, allocate the new cell with a private buffer:
*/
        if ( PF_longMultiNewCell() ) return(-1);
    }
/*
        Move to the next cell in the chain:
*/
    PF_longMultiTop = PF_longMultiTop->next;
/*
        if >=0, the cell buffer is the chunk of PF_longPackBuf, initialize it:
*/
    if ( PF_longMultiTop->bufpos > -1 )
        PF_longMultiTop->buffer = PF_longPackBuf+PF_longMultiTop->bufpos;
/*
    else -- the cell has it's own private buffer.
    Initialize the cell fields:
*/
    PF_longMultiTop->nPacks  = 0;
    PF_longMultiTop->lastLen = 0;
    PF_longMultiTop->packpos = 0;
    return(0);
}

/*
        #] PF_longMultiPack2NextCell :
        #[ PF_longMultiNewChunkAdded :
*/

static inline int PF_longMultiNewChunkAdded(int n)
{
/*
        Store the list tail:
```

```
*/
    PF_LONGMULTI *MemCell = PF_longMultiLastChunk->next;
    int pos = PF_longPackTop;

    while ( n-- > 0 ) {
/*
            Allocate a new cell:
*/
        PF_longMultiLastChunk->next = (PF_LONGMULTI *)
                Malloc1(sizeof(PF_LONGMULTI),"PF_longMultiCell");
        if ( PF_longMultiLastChunk->next == NULL ) return(-1);
/*
            Update the Last Chunk Pointer:
*/
        PF_longMultiLastChunk = PF_longMultiLastChunk->next;
/*
            Initialize the new cell:
*/
        PF_longMultiLastChunk->bufpos = pos;
        pos += PF_packsize;
        PF_longMultiLastChunk->buffer = NULL;
        PF_longMultiLastChunk->packpos = 0;
        PF_longMultiLastChunk->nPacks  = 0;
        PF_longMultiLastChunk->lastLen = 0;
    }
/*
        Hitch the tail:
*/
    PF_longMultiLastChunk->next = MemCell;
    return(0);
}

/*
        #] PF_longMultiNewChunkAdded :
        #[ PF_longCopyChunk :
*/

static inline void PF_longCopyChunk(int *to, int *from, int n)
{
    NCOPYI(to,from,n)
/*  for ( ; n > 0; n-- ) *to++ = *from++; */
}

/*
        #] PF_longCopyChunk :
        #[ PF_longAddChunk :

    The chunk must be increased by n*PF_packsize.
*/

static int PF_longAddChunk(int n, int mustRealloc)
{
    UBYTE *newbuf;
    if ( ( newbuf = (UBYTE*)Malloc1(sizeof(UBYTE)*(PF_longPackTop+n*PF_packsize),
                "PF_longPackBuf") ) == NULL ) return(-1);
/*
        Allocate and chain a new cell for longMulti:
*/
    if ( PF_longMultiNewChunkAdded(n) ) return(-1);
/*
        Copy the content to the new buffer:
*/
    if ( mustRealloc ) {
        PF_longCopyChunk((int*)newbuf,(int*)PF_longPackBuf,PF_longPackTop/sizeof(int));
    }
/*
        Note, PF_packsize is multiple by sizeof(int) by construction!
*/
    PF_longPackTop += (n*PF_packsize);
/*
        Free the old buffer and store the new one:
*/
    M_free(PF_longPackBuf,"PF_longPackBuf");
    PF_longPackBuf = newbuf;
/*
        Count number of re-allocs:
*/
    PF_longPackN += n;
    return(0);
}

/*
        #] PF_longAddChunk :
        #[ PF_longMultiHowSplit :

    "count" of "type" elements in an input buffer occupy "bytes" bytes.
    We know from the algorithm, that it is too many. How to split
```

```
    the buffer so that the head fits to rest of a storage buffer?*/
static inline int PF_longMultiHowSplit(int count, MPI_Datatype type, int bytes)
{
    int ret, items, totalbytes;

    if ( count < 2 ) return(0); /* Nothing to split */
/*
        A rest of a storage buffer:
*/
    totalbytes = PF_packsize - PF_longMultiTop->packpos;
/*
    Rough estimate:
*/
    items = (int)((double)totalbytes*count/bytes);
/*
        Go to the up limit:
*/
    do {
        if ( ( ret = MPI_Pack_size(++items,type,PF_COMM,&bytes) )
            !=MPI_SUCCESS ) return(ret);
    } while ( bytes < totalbytes );
/*
        Now the value of "items" is too large
        And now evaluate the exact value:
*/
    do {
        if ( ( ret = MPI_Pack_size(--items,type,PF_COMM,&bytes) )
            !=MPI_SUCCESS ) return(ret);
        if ( items == 0 ) /* Nothing about MPI_Pack_size(0) == 0 in standards! */
            return(0);
    } while ( bytes > totalbytes );
    return(items);
}
/*
        #] PF_longMultiHowSplit :
        #[ PF_longPackInit :
*/

static int PF_longPackInit(void)
{
    int ret;
    PF_longPackBuf = (UBYTE*)Malloc1(sizeof(UBYTE)*PF_packsize,"PF_longPackBuf");
    if ( PF_longPackBuf == NULL ) return(-1);
/*
        PF_longPackTop is not initialized yet, use in as a return value:
*/
    ret = MPI_Pack_size(1,MPI_INT,PF_COMM,&PF_longPackTop);
    if ( ret != MPI_SUCCESS ) return(ret);

    PF_longPackSmallBuf =
            (void*)Malloc1(sizeof(UBYTE)*PF_longPackTop,"PF_longPackSmallBuf");

    PF_longPackTop = PF_packsize;
    PF_longMultiRoot =
            (PF_LONGMULTI *)Malloc1(sizeof(PF_LONGMULTI),"PF_longMultiRoot");
    if ( PF_longMultiRoot == NULL ) return(-1);
    PF_longMultiRoot->bufpos = 0;
    PF_longMultiRoot->buffer = NULL;
    PF_longMultiRoot->next   = NULL;
    PF_longMultiLastChunk    = PF_longMultiRoot;

    PF_longPackPos = 0;
    PF_longMultiRoot->packpos = 0;
    PF_longMultiTop = PF_longMultiRoot;
    PF_longPackN = 1;
    return(0);
}

/*
        #] PF_longPackInit :
        #[ PF_longMultiPreparePrefix :
*/

static inline int PF_longMultiPreparePrefix(void)
{
    int ret;
    PF_LONGMULTI *thePrefix;
    int i = PF_longPackN;
/*
        Here we have PF_longPackN>1!
        New cell (at the list end) to create the auxiliary chunk:
*/
    if ( PF_longMultiPack2NextCell() ) return(-1);
/*
        Store the pointer to the chunk we will proceed:
*/
    thePrefix = PF_longMultiTop;
```

```
/*
      Pack PF_longPackN:
*/
    ret = MPI_Pack(&(PF_longPackN),
                   1,
                   MPI_INT,
                   thePrefix->buffer,
                   PF_packsize,
                   &(thePrefix->packpos),
                   PF_COMM);
    if ( ret != MPI_SUCCESS ) return(ret);
/*
      And start from the beginning:
*/
    for ( PF_longMultiTop = PF_longMultiRoot; i > 0; i-- ) {
/*
          Pack number of Pack hits:
*/
        ret = MPI_Pack(&(PF_longMultiTop->nPacks),
                       1,
                       MPI_INT,
                       thePrefix->buffer,
                       PF_packsize,
                       &(thePrefix->packpos),
                       PF_COMM);
/*
              Pack the length of the last fit portion:
*/
        ret |= MPI_Pack(&(PF_longMultiTop->lastLen),
                        1,
                        MPI_INT,
                        thePrefix->buffer,
                        PF_packsize,
                        &(thePrefix->packpos),
                        PF_COMM);
/*
              Check the size -- not necessary, MPI_Pack did it.
*/
        if (  ret != MPI_SUCCESS ) return(ret);
/*
          Go to the next cell:
*/
        PF_longMultiTop = PF_longMultiTop->next;
    }

    PF_longMultiTop = thePrefix;
/*
      PF_longMultiTop is ready!
*/
    return(0);
}

/*
      #] PF_longMultiPreparePrefix :
      #[ PF_longMultiProcessPrefix :
*/

static inline int PF_longMultiProcessPrefix(void)
{
    int ret,i;
/*
      We have PF_longPackN records packed in PF_longMultiRoot->buffer,
      pairs nPacks and lastLen. Loop through PF_longPackN cells,
      unpacking these integers into proper fields:
*/
    for ( PF_longMultiTop = PF_longMultiRoot, i = 0; i < PF_longPackN; i++ ) {
/*
          Go to th next cell, allocating, when necessary:
*/
        if ( PF_longMultiPack2NextCell() ) return(-1);
/*
          Unpack the number of Pack hits:
*/
        ret = MPI_Unpack(PF_longMultiRoot->buffer,
                         PF_packsize,
                         &(  PF_longMultiRoot->packpos),
                         &(PF_longMultiTop->nPacks),
                         1,
                         MPI_INT,
                         PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
/*
          Unpack the length of the last fit portion:
*/
        ret = MPI_Unpack(PF_longMultiRoot->buffer,
                         PF_packsize,
                         &(  PF_longMultiRoot->packpos),
```

```
                            &(PF_longMultiTop->lastLen),
                            1,
                            MPI_INT,
                            PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
    }
    return(0);
}

/*
        #] PF_longMultiProcessPrefix :
        #[ PF_longSingleReset :
*/

static inline int PF_longSingleReset(int is_sender)
{
    int ret;
    PF_longPackPos=0;
    if ( is_sender ) {
        ret = MPI_Pack(&PF_longPackTop,1,MPI_INT,
            PF_longPackBuf,PF_longPackTop,&PF_longPackPos,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
        PF_longPackN = 1;
    }
    else {
        PF_longPackN=0;
    }
    return(0);
}

/*
        #] PF_longSingleReset :
        #[ PF_longMultiReset :
*/

static inline int PF_longMultiReset(int is_sender)
{
    int ret = 0, theone = 1;
    PF_longMultiRoot->packpos = 0;
    if ( is_sender ) {
        ret = MPI_Pack(&theone,1,MPI_INT,
            PF_longPackBuf,PF_longPackTop,&(PF_longMultiRoot->packpos),PF_COMM);
        PF_longPackN = 1;
    }
    else {
        PF_longPackN = 0;
    }
    PF_longMultiRoot->nPacks = 0;   /* The auxiliary field is not counted */
    PF_longMultiRoot->lastLen = 0;
    PF_longMultiTop = PF_longMultiRoot;
    PF_longMultiRoot->buffer = PF_longPackBuf;
    return ret;
}

/*
        #] PF_longMultiReset :
        #] Long pack private functions :
        #[ PF_PrepareLongSinglePack :
*/

int PF_PrepareLongSinglePack(void)
{
    return PF_longSingleReset(1);
}

/*
        #] PF_PrepareLongSinglePack :
        #[ PF_LongSinglePack :
*/

int PF_LongSinglePack(const void *buffer, size_t count, MPI_Datatype type)
{
    int ret, bytes;
    /* XXX: Limited by int size. */
    if ( count > INT_MAX ) return -99;
    ret = MPI_Pack_size((int)count,type,PF_COMM,&bytes);
    if ( ret != MPI_SUCCESS ) return(ret);

    while ( PF_longPackPos+bytes > PF_longPackTop ) {
        if ( PF_longAddChunk(1, 1) ) return(-1);
    }
/*
        PF_longAddChunk(1, 1) means, the chunk must
        be increased by 1 and re-allocated
*/
    ret = MPI_Pack((void *)buffer,(int)count,type,
                   PF_longPackBuf,PF_longPackTop,&PF_longPackPos,PF_COMM);
```

```
    if ( ret != MPI_SUCCESS ) return(ret);
    return(0);
}

/*
        #] PF_LongSinglePack :
        #[ PF_LongSingleUnpack :
*/

int PF_LongSingleUnpack(void *buffer, size_t count, MPI_Datatype type)
{
    int ret;
    /* XXX: Limited by int size. */
    if ( count > INT_MAX ) return -99;
    ret = MPI_Unpack(PF_longPackBuf,PF_longPackTop,&PF_longPackPos,
                     buffer,(int)count,type,PF_COMM);
    if ( ret != MPI_SUCCESS ) return(ret);
    return(0);
}

/*
        #] PF_LongSingleUnpack :
        #[ PF_LongSingleSend :
*/

int PF_LongSingleSend(int to, int tag)
{
    int ret, pos = 0;
/*
        Note, here we assume that this function couldn't be used
        with to == PF_ANY_SOURCE!
*/
    if ( PF_longPackN > 1 ) {
        /* The buffer was incremented, pack send the new size first: */
        int tmp = -PF_longPackTop;
/*
            Negative value means there will be the second buffer
*/
        ret = MPI_Pack(&tmp, 1,PF_INT,
                       PF_longPackSmallBuf,PF_longPackTop,&pos,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
        ret = MPI_Ssend(PF_longPackSmallBuf,pos,MPI_PACKED,to,tag,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
    }
    ret = MPI_Ssend(PF_longPackBuf,PF_longPackPos,MPI_PACKED,to,tag,PF_COMM);
    if ( ret != MPI_SUCCESS ) return(ret);
    return(0);
}

/*
        #] PF_LongSingleSend :
        #[ PF_LongSingleReceive :
*/

int PF_LongSingleReceive(int src, int tag, int *psrc, int *ptag)
{
    int ret, missed, oncemore;
    MPI_Status status;
    PF_longSingleReset(0);
    do {
        ret = MPI_Recv(PF_longPackBuf,PF_longPackTop,MPI_PACKED,src,tag,
                       PF_COMM,&status);
        if ( ret != MPI_SUCCESS ) return(ret);
/*
            The source and tag must be specified here for the case if
            MPI_Recv is performed more than once:
*/
        src = status.MPI_SOURCE;
        tag = status.MPI_TAG;
        if ( psrc ) *psrc = status.MPI_SOURCE;
        if ( ptag ) *ptag = status.MPI_TAG;
/*
            Now we got either small buffer with the new PF_longPackTop,
            or just a regular chunk.
*/
        ret = MPI_Unpack(PF_longPackBuf,PF_longPackTop,&PF_longPackPos,
                         &missed,1,MPI_INT,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);

        if ( missed < 0 ) { /* The small buffer was received. */
            oncemore = 1; /* repeat receiving afterwards */
                          /* Reallocate the buffer and get the data */
            missed = -missed;
/*
                restore after unpacking small from buffer:
*/
            PF_longPackPos = 0;
```

```
        }
        else {
            oncemore = 0;   /* That's all, no repetition */
        }
        if ( missed > PF_longPackTop ) {
            /*
             * The room must be increased. We need a re-allocation for the
             * case that there is no repetition.
             */
            if ( PF_longAddChunk( (missed-PF_longPackTop)/PF_packsize, !oncemore ) )
                return(-1);
        }
    } while ( oncemore );
    return(0);
}

/*
        #] PF_LongSingleReceive :
        #[ PF_PrepareLongMultiPack :
*/

int PF_PrepareLongMultiPack(void)
{
    return PF_longMultiReset(1);
}

/*
        #] PF_PrepareLongMultiPack :
        #[ PF_LongMultiPackImpl :
*/

int PF_LongMultiPackImpl(const void*buffer, size_t count, size_t eSize, MPI_Datatype type)
{
    int ret, items;

    /* XXX: Limited by int size. */
    if ( count > INT_MAX ) return -99;

    ret = MPI_Pack_size((int)count,type,PF_COMM,&items);
    if ( ret != MPI_SUCCESS ) return(ret);

    if ( PF_longMultiTop->packpos + items <= PF_packsize ) {
        ret = MPI_Pack((void *)buffer,(int)count,type,PF_longMultiTop->buffer,
                       PF_packsize,&(PF_longMultiTop->packpos),PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
        PF_longMultiTop->nPacks++;
        return(0);
    }
/*
        The data do not fit to the rest of the buffer.
        There are two possibilities here: go to the next cell
        immediately, or first try to pack some portion. The function
        PF_longMultiHowSplit() returns the number of items could be
        packed in the end of the current cell:
*/
    if ( ( items = PF_longMultiHowSplit((int)count,type,items) ) < 0 ) return(items);

    if ( items > 0 ) {    /* store the head */
        ret = MPI_Pack((void *)buffer,items,type,PF_longMultiTop->buffer,
                       PF_packsize,&(PF_longMultiTop->packpos),PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
        PF_longMultiTop->nPacks++;
        PF_longMultiTop->lastLen = items;
    }
/*
        Now the rest should be packed to the new cell.
        Slide to the new cell:
*/
    if ( PF_longMultiPack2NextCell() ) return(-1);
    PF_longPackN++;
/*
        Pack the rest to the next cell:
*/
    return(PF_LongMultiPackImpl((char *)buffer+items*eSize,count-items,eSize,type));
}

/*
        #] PF_LongMultiPackImpl :
        #[ PF_LongMultiUnpackImpl :
*/

int PF_LongMultiUnpackImpl(void *buffer, size_t count, size_t eSize, MPI_Datatype type)
{
    int ret;

    /* XXX: Limited by int size. */
    if ( count > INT_MAX ) return -99;
```

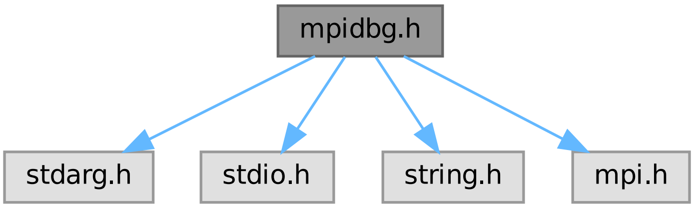
```

    if ( PF_longPackN < 2 ) { /* Just unpack the buffer from the single cell */
        ret = MPI_Unpack(
                    PF_longMultiTop->buffer,
                    PF_packsize,
                    &(PF_longMultiTop->packpos),
                    buffer,
                    count,type,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
        return(0);
    }
/*
        More than one cell is in use.
*/
    if ( ( PF_longMultiTop->nPacks > 1 )     /* the cell is not expired */
        ||          /* The last cell contains exactly required portion: */
        ( ( PF_longMultiTop->nPacks == 1 ) && ( PF_longMultiTop->lastLen == 0 ) )
    ) {    /* Just unpack the buffer from the current cell */
        ret = MPI_Unpack(
                    PF_longMultiTop->buffer,
                    PF_packsize,
                    &(PF_longMultiTop->packpos),
                    buffer,
                    count,type,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
        (PF_longMultiTop->nPacks)--;
        return(0);
    }
    if ( ( PF_longMultiTop->nPacks == 1 ) && ( PF_longMultiTop->lastLen != 0 ) ) {
/*
            Unpack the head:
*/
        ret = MPI_Unpack(
                    PF_longMultiTop->buffer,
                    PF_packsize,
                    &(PF_longMultiTop->packpos),
                    buffer,
                    PF_longMultiTop->lastLen,type,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
/*
            Decrement the counter by read items:
*/
        count -= PF_longMultiTop->lastLen;
        if ( count <= 0 ) return(-1);  /*Something is wrong! */
/*
            Shift the output buffer position:
*/
        buffer = (char *)buffer + PF_longMultiTop->lastLen * eSize;
        (PF_longMultiTop->nPacks)--;
    }
/*
        Here PF_longMultiTop->nPacks == 0
*/
    if ( ( PF_longMultiTop = PF_longMultiTop->next ) == NULL ) return(-1);
    return(PF_LongMultiUnpackImpl(buffer,count,eSize,type));
}

/*
        #] PF_LongMultiUnpackImpl :
        #[ PF_LongMultiBroadcast :
*/

int PF_LongMultiBroadcast(void)
{
    int ret, i;

    if ( PF.me == MASTER ) {
/*
            PF_longPackN is the number of packed chunks. If it is more
            than 1, we have to pack a new one and send it first
*/
        if ( PF_longPackN > 1 ) {
            if ( PF_longMultiPreparePrefix() ) return(-1);
            ret = MPI_Bcast((void*)PF_longMultiTop->buffer,
                                PF_packsize,MPI_PACKED,MASTER,PF_COMM);
            if ( ret != MPI_SUCCESS ) return(ret);
/*
                PF_longPackN was not incremented by PF_longMultiPreparePrefix()!
*/
        }
/*
            Now we start from the beginning:
*/
        PF_longMultiTop = PF_longMultiRoot;
/*
            Just broadcast all the chunks:
*/
```

```
        for ( i = 0; i < PF_longPackN; i++ ) {
            ret = MPI_Bcast((void*)PF_longMultiTop->buffer,
                            PF_packsize,MPI_PACKED,MASTER,PF_COMM);
            if ( ret != MPI_SUCCESS ) return(ret);
            PF_longMultiTop = PF_longMultiTop->next;
        }
        return(0);
    }
/*
        else - the slave
*/
    PF_longMultiReset(0);
/*
        Get the first chunk; it can be either the only data chunk, or
        an auxiliary chunk, if the data do not fit the single chunk:
*/
    ret = MPI_Bcast((void*)PF_longMultiRoot->buffer,
                    PF_packsize,MPI_PACKED,MASTER,PF_COMM);
    if ( ret != MPI_SUCCESS ) return(ret);

    ret = MPI_Unpack((void*)PF_longMultiRoot->buffer,
                     PF_packsize,
                     &(PF_longMultiRoot->packpos),
                     &PF_longPackN,1,MPI_INT,PF_COMM);
    if ( ret != MPI_SUCCESS ) return(ret);
/*
        Now in PF_longPackN we have the number of cells used
        for broadcasting. If it is >1, then we have to allocate
        enough cells, initialize them and receive all the chunks.
*/
    if ( PF_longPackN < 2 ) /* That's all, the single chunk is received. */
        return(0);
/*
        Here we have to get PF_longPackN chunks. But, first,
        initialize cells by info from the received auxiliary chunk.
*/
    if ( PF_longMultiProcessPrefix() ) return(-1);
/*
        Now we have free PF_longPackN cells, starting
        from PF_longMultiRoot->next,  with properly initialized
        nPacks and lastLen fields. Get chunks:
*/
    for ( PF_longMultiTop = PF_longMultiRoot->next, i = 0; i < PF_longPackN; i++ ) {
        ret = MPI_Bcast((void*)PF_longMultiTop->buffer,
                        PF_packsize,MPI_PACKED,MASTER,PF_COMM);
        if ( ret != MPI_SUCCESS ) return(ret);
        if ( i == 0 ) {   /* The first chunk, it contains extra "1". */
            int tmp;
/*
                Extract this 1 into tmp and forget about it.
*/
            ret = MPI_Unpack((void*)PF_longMultiTop->buffer,
                             PF_packsize,
                             &(PF_longMultiTop->packpos),
                             &tmp,1,MPI_INT,PF_COMM);
            if ( ret != MPI_SUCCESS ) return(ret);
        }
        PF_longMultiTop = PF_longMultiTop->next;
    }
/*
        multiUnPack starts with PF_longMultiTop, skip auxiliary chunk in
        PF_longMultiRoot:
*/
    PF_longMultiTop = PF_longMultiRoot->next;
    return(0);
}

/*
        #] PF_LongMultiBroadcast :
    #] Long pack stuff :
*/
```

4.80 mpidbg.h File Reference

```
#include <stdarg.h>
#include <stdio.h>
#include <string.h>
```

`#include <mpi.h>`
Include dependency graph for mpidbg.h:



Macros

- #define MPIDBG_RANK MPIDBG_Get_rank()
- #define MPIDBG_Out(...) MPIDBG_Out(file, line, func, __VA_ARGS__)
- #define MPIDBG_EXTARG const char *file, int line, const char *func
- #define MPI_Send(...) MPIDBG_Send(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Recv(...) MPIDBG_Recv(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Bsend(...) MPIDBG_Bsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Ssend(...) MPIDBG_Ssend(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Rsend(...) MPIDBG_Rsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Isend(...) MPIDBG_Isend(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Ibsend(...) MPIDBG_Ibsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Issend(...) MPIDBG_Issend(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Irsend(...) MPIDBG_Irsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Irecv(...) MPIDBG_Irecv(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Wait(...) MPIDBG_Wait(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Test(...) MPIDBG_Test(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Waitany(...) MPIDBG_Waitany(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Testany(...) MPIDBG_Testany(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Waitall(...) MPIDBG_Waitall(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Testall(...) MPIDBG_Testall(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Waitsome(...) MPIDBG_Waitsome(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Testsome(...) MPIDBG_Testsome(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Iprobe(...) MPIDBG_Iprobe(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Probe(...) MPIDBG_Probe(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Cancel(...) MPIDBG_Cancel(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Test_cancelled(...) MPIDBG_Test_cancelled(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Barrier(...) MPIDBG_Barrier(__VA_ARGS__, __FILE__, __LINE__, __func__)
- #define MPI_Bcast(...) MPIDBG_Bcast(__VA_ARGS__, __FILE__, __LINE__, __func__)

4.80.1 Detailed Description

MPI APIs with the logging feature. NOTE: This file needs C99.

Definition in file mpidbg.h.

4.80.2 Macro Definition Documentation

4.80.2.1 MPIDBG_RANK

```
#define MPIDBG_RANK MPIDBG_Get_rank()
```

Definition at line 55 of file mpidbg.h.

4.80.2.2 MPIDBG_Out

```
#define MPIDBG_Out(
              ... ) MPIDBG_Out(file, line, func, __VA_ARGS__)
```

Definition at line 72 of file mpidbg.h.

4.80.2.3 MPIDBG_EXTARG

```
#define MPIDBG_EXTARG const char *file, int line, const char *func
```

Definition at line 139 of file mpidbg.h.

4.80.2.4 MPI_Send

```
#define MPI_Send(
              ... ) MPIDBG_Send(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 157 of file mpidbg.h.

4.80.2.5 MPI_Recv

```
#define MPI_Recv(
              ... ) MPIDBG_Recv(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 178 of file mpidbg.h.

4.80.2.6 MPI_Bsend

```
#define MPI_Bsend(
              ... ) MPIDBG_Bsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 197 of file mpidbg.h.

4.80.2.7 MPI_Ssend

```
#define MPI_Ssend(
              ... ) MPIDBG_Ssend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 216 of file mpidbg.h.

4.80.2.8 MPI_Rsend

```
#define MPI_Rsend(
              ... ) MPIDBG_Rsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 235 of file mpidbg.h.

4.80.2.9 MPI_Isend

```
#define MPI_Isend(
              ... ) MPIDBG_Isend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 254 of file mpidbg.h.

4.80.2.10 MPI_Ibsend

```
#define MPI_Ibsend(
              ... ) MPIDBG_Ibsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 273 of file mpidbg.h.

4.80.2.11 MPI_Issend

```
#define MPI_Issend(
              ... ) MPIDBG_Issend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 292 of file mpidbg.h.

4.80.2.12 MPI_Irsend

```
#define MPI_Irsend(
              ... ) MPIDBG_Irsend(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 311 of file mpidbg.h.

4.80.2.13 MPI_Irecv

```
#define MPI_Irecv(
              ... ) MPIDBG_Irecv(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 330 of file mpidbg.h.

4.80.2.14 MPI_Wait

```
#define MPI_Wait(
              ... ) MPIDBG_Wait(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 353 of file mpidbg.h.

4.80.2.15 MPI_Test

```
#define MPI_Test(
                ... ) MPIDBG_Test(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 381 of file mpidbg.h.

4.80.2.16 MPI_Waitany

```
#define MPI_Waitany(
                ... ) MPIDBG_Waitany(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 407 of file mpidbg.h.

4.80.2.17 MPI_Testany

```
#define MPI_Testany(
                ... ) MPIDBG_Testany(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 438 of file mpidbg.h.

4.80.2.18 MPI_Waitall

```
#define MPI_Waitall(
                ... ) MPIDBG_Waitall(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 462 of file mpidbg.h.

4.80.2.19 MPI_Testall

```
#define MPI_Testall(
                ... ) MPIDBG_Testall(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 491 of file mpidbg.h.

4.80.2.20 MPI_Waitsome

```
#define MPI_Waitsome(
                ... ) MPIDBG_Waitsome(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 515 of file mpidbg.h.

4.80.2.21 MPI_Testsome

```
#define MPI_Testsome(
                ... ) MPIDBG_Testsome(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 539 of file mpidbg.h.

4.80.2.22 MPI_Iprobe

```
#define MPI_Iprobe(
              ... ) MPIDBG_Iprobe(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 565 of file mpidbg.h.

4.80.2.23 MPI_Probe

```
#define MPI_Probe(
              ... ) MPIDBG_Probe(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 586 of file mpidbg.h.

4.80.2.24 MPI_Cancel

```
#define MPI_Cancel(
              ... ) MPIDBG_Cancel(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 605 of file mpidbg.h.

4.80.2.25 MPI_Test_cancelled

```
#define MPI_Test_cancelled(
              ... ) MPIDBG_Test_cancelled(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 629 of file mpidbg.h.

4.80.2.26 MPI_Barrier

```
#define MPI_Barrier(
              ... ) MPIDBG_Barrier(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 648 of file mpidbg.h.

4.80.2.27 MPI_Bcast

```
#define MPI_Bcast(
              ... ) MPIDBG_Bcast(__VA_ARGS__, __FILE__, __LINE__, __func__)
```

Definition at line 667 of file mpidbg.h.

4.81 mpidbg.h

Go to the documentation of this file.

```
#ifndef MPIDBG_H
#define MPIDBG_H

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

/*
    #[ Includes :
*/

#include <stdarg.h>
#include <stdio.h>
#include <string.h>
#include <mpi.h>

/*
    #] Includes :
    #[ Utilities :
        #[ MPIDBG_RANK :
*/

static inline int MPIDBG_Get_rank(void) {
    return PF.me;  /* Assume we are working with ParFORM. */
}
#define MPIDBG_RANK MPIDBG_Get_rank()

/*
        #] MPIDBG_RANK :
        #[ MPIDBG_Out :
*/

static inline void MPIDBG_Out(const char *file, int line, const char *func, const char *fmt, ...) {
    char buf[1024];  /* Enough. */
    va_list ap;
    va_start(ap, fmt);
    snprintf(buf,1024, "*** [%d] %10s %4d @ %-16s: ", MPIDBG_RANK, file, line, func);
    vsnprintf(buf + strlen(buf),1024-strlen(buf), fmt, ap);
    va_end(ap);
    /* Assume fprintf with a line will work well even in multi-processes. */
    fprintf(stderr, "%s\n", buf);
}
#define MPIDBG_Out(...) MPIDBG_Out(file, line, func, __VA_ARGS__)

/*
        #] MPIDBG_Out :
        #[ MPIDBG_sprint_requests :
*/

static inline void MPIDBG_sprint_requests(char *buf, int count, const MPI_Request *requests)
{
    /* Assume sprintf never fail and returns >= 0... */
    buf += sprintf(buf, "(");
    int i, first = 1;
    for ( i = 0; i < count; i++ ) {
        if ( first ) {
            first = 0;
        }
        else {
```

```
            buf += sprintf(buf, ",");
        }
        if ( requests[i] != MPI_REQUEST_NULL ) {
            buf += sprintf(buf, "%d", i);
        }
        else {
            buf += sprintf(buf, "*");
        }
    }
    buf += sprintf(buf, ")");
}

/*
        #] MPIDBG_sprint_requests :
        #[ MPIDBG_sprint_statuses :
*/

static inline void MPIDBG_sprint_statuses(char *buf, int count, const MPI_Request *old_requests, const MPI_Request *new_requests, const MPI_Status *statuses)
{
    /* Assume sprintf never fail and returns >= 0... */
    buf += sprintf(buf, "(");
    int i, first = 1;
    for ( i = 0; i < count; i++ ) {
        if ( first ) {
            first = 0;
        }
        else {
            buf += sprintf(buf, ",");
        }
        if ( old_requests[i] != MPI_REQUEST_NULL && new_requests[i] == MPI_REQUEST_NULL ) {
            int ret_size = 0;
            MPI_Get_count((MPI_Status *)&statuses[i], MPI_BYTE, &ret_size);
            buf += sprintf(buf, "(source=%d,tag=%d,size=%d)", statuses[i].MPI_SOURCE, statuses[i].MPI_TAG, ret_size);
        } else {
            buf += sprintf(buf, "*");
        }
    }
    buf += sprintf(buf, ")");
}

/*
        #] MPIDBG_sprint_statuses :
    #] Utilities :
    #[ MPI APIs :
*/

/*
 * The followings are the MPI APIs with the logging.
 */

#define MPIDBG_EXTARG const char *file, int line, const char *func

/*
        #[ MPI_Send :
*/

static inline int MPIDBG_Send(void *buf, int count, MPI_Datatype datatype, int dest, int tag, MPI_Comm comm, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Send: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Send(buf, count, datatype, dest, tag, comm);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Send: OK");
    }
    else {
        MPIDBG_Out("MPI_Send: Failed");
    }
    return ret;
}
#define MPI_Send(...) MPIDBG_Send(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Send :
        #[ MPI_Recv :
*/

static inline int MPIDBG_Recv(void* buf, int count, MPI_Datatype datatype, int source, int tag, MPI_Comm comm, MPI_Status *status, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Recv: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
    int ret = MPI_Recv(buf, count, datatype, source, tag, comm, status);
    if ( ret == MPI_SUCCESS ) {
        int ret_count = 0;
        MPI_Get_count(status, datatype, &ret_count);
        MPIDBG_Out("MPI_Recv: OK src=%d dest=%d tag=%d count=%d", status->MPI_SOURCE, MPIDBG_RANK,
```

```
status->MPI_TAG, ret_count);
    }
    else {
        MPIDBG_Out("MPI_Recv: Failed");
    }
    return ret;
}
#define MPI_Recv(...) MPIDBG_Recv(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Recv :
        #[ MPI_Bsend :
*/

static inline int MPIDBG_Bsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
    MPI_Comm comm, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Bsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Bsend(buf, count, datatype, dest, tag, comm);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Bsend: OK");
    }
    else {
        MPIDBG_Out("MPI_Bsend: Failed");
    }
    return ret;
}
#define MPI_Bsend(...) MPIDBG_Bsend(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Bsend :
        #[ MPI_Ssend :
*/

static inline int MPIDBG_Ssend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
    MPI_Comm comm, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Ssend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Ssend(buf, count, datatype, dest, tag, comm);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Ssend: OK");
    }
    else {
        MPIDBG_Out("MPI_Ssend: Failed");
    }
    return ret;
}
#define MPI_Ssend(...) MPIDBG_Ssend(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Ssend :
        #[ MPI_Rsend :
*/

static inline int MPIDBG_Rsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
    MPI_Comm comm, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Rsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Rsend(buf, count, datatype, dest, tag, comm);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Rsend: OK");
    }
    else {
        MPIDBG_Out("MPI_Rsend: Failed");
    }
    return ret;
}
#define MPI_Rsend(...) MPIDBG_Rsend(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Rsend :
        #[ MPI_Isend :
*/

static inline int MPIDBG_Isend(void* buf, int count, MPI_Datatype datatype, int dest, int tag,
    MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Isend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Isend(buf, count, datatype, dest, tag, comm, request);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Isend: OK");
    }
    else {
        MPIDBG_Out("MPI_Isend: Failed");
    }
    return ret;
}
```

```
#define MPI_Isend(...) MPIDBG_Isend(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Isend :
        #[ MPI_Ibsend :
*/

static inline int MPIDBG_Ibsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag, MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Ibsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Ibsend(buf, count, datatype, dest, tag, comm, request);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Ibsend: OK");
    }
    else {
        MPIDBG_Out("MPI_Ibsend: Failed");
    }
    return ret;
}
#define MPI_Ibsend(...) MPIDBG_Ibsend(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Ibsend :
        #[ MPI_Issend :
*/

static inline int MPIDBG_Issend(void* buf, int count, MPI_Datatype datatype, int dest, int tag, MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Issend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Issend(buf, count, datatype, dest, tag, comm, request);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Issend: OK");
    }
    else {
        MPIDBG_Out("MPI_Issend: Failed");
    }
    return ret;
}
#define MPI_Issend(...) MPIDBG_Issend(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Issend :
        #[ MPI_Irsend :
*/

static inline int MPIDBG_Irsend(void* buf, int count, MPI_Datatype datatype, int dest, int tag, MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Irsend: src=%d dest=%d tag=%d count=%d", MPIDBG_RANK, dest, tag, count);
    int ret = MPI_Irsend(buf, count, datatype, dest, tag, comm, request);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Irsend: OK");
    }
    else {
        MPIDBG_Out("MPI_Irsend: Failed");
    }
    return ret;
}
#define MPI_Irsend(...) MPIDBG_Irsend(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Irsend :
        #[ MPI_Irecv :
*/

static inline int MPIDBG_Irecv(void* buf, int count, MPI_Datatype datatype, int source, int tag, MPI_Comm comm, MPI_Request *request, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Irecv: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
    int ret = MPI_Irecv(buf, count, datatype, source, tag, comm, request);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Irecv: OK dest=%d", MPIDBG_RANK);
    }
    else {
        MPIDBG_Out("MPI_Irecv: Failed");
    }
    return ret;
}
#define MPI_Irecv(...) MPIDBG_Irecv(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Irecv :
        #[ MPI_Wait :
*/

```

```
static inline int MPIDBG_Wait(MPI_Request *request, MPI_Status *status, MPIDBG_EXTARG)
{
    char buf[256 * 1];  /* Enough. */
    MPI_Request old_request = *request;
    MPIDBG_sprint_requests(buf, 1, request);
    MPIDBG_Out("MPI_Wait: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Wait(request, status);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_sprint_statuses(buf, 1, request, &old_request, status);
        MPIDBG_Out("MPI_Wait: OK rank=%d result=%s", MPIDBG_RANK, buf);
    }
    else {
        MPIDBG_Out("MPI_Wait: Failed");
    }
    return ret;
}
#define MPI_Wait(...) MPIDBG_Wait(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Wait :
        #[ MPI_Test :
*/

static inline int MPIDBG_Test(MPI_Request *request, int *flag, MPI_Status *status, MPIDBG_EXTARG)
{
    char buf[256 * 1];  /* Enough. */
    MPI_Request old_request = *request;
    MPIDBG_sprint_requests(buf, 1, request);
    MPIDBG_Out("MPI_Test: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Test(request, flag, status);
    if ( ret == MPI_SUCCESS ) {
        if ( *flag ) {
            MPIDBG_sprint_statuses(buf, 1, request, &old_request, status);
            MPIDBG_Out("MPI_Test: OK rank=%d result=%s", MPIDBG_RANK, buf);
        }
        else {
            MPIDBG_Out("MPI_Test: OK flag=false");
        }
    }
    else {
        MPIDBG_Out("MPI_Test: Failed");
    }
    return ret;
}
#define MPI_Test(...) MPIDBG_Test(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Test :
        #[ MPI_Waitany :
*/

static inline int MPIDBG_Waitany(int count, MPI_Request *array_of_requests, int *index, MPI_Status *status, MPIDBG_EXTARG)
{
    char buf[256];  /* Enough. */
    MPI_Request old_requests[count];
    memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
    MPIDBG_sprint_requests(buf, count, array_of_requests);
    MPIDBG_Out("MPI_Waitany: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Waitany(count, array_of_requests, index, status);
    if ( ret == MPI_SUCCESS ) {
        MPI_Status statuses[count];
        statuses[*index] = *status;
        MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, statuses);
        MPIDBG_Out("MPI_Waitany: OK rank=%d result=%s", MPIDBG_RANK, buf);
    }
    else {
        MPIDBG_Out("MPI_Waitany: Failed");
    }
    return ret;
}
#define MPI_Waitany(...) MPIDBG_Waitany(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Waitany :
        #[ MPI_Testany :
*/

static inline int MPIDBG_Testany(int count, MPI_Request *array_of_requests, int *index, int *flag, MPI_Status *status, MPIDBG_EXTARG)
{
    char buf[256];  /* Enough. */
    MPI_Request old_requests[count];
    memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
    MPIDBG_sprint_requests(buf, count, array_of_requests);
    MPIDBG_Out("MPI_Testany: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Testany(count, array_of_requests, index, flag, status);
```

```
    if ( ret == MPI_SUCCESS ) {
        if ( *flag ) {
            MPI_Status statuses[count];
            statuses[*index] = *status;
            MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, statuses);
            MPIDBG_Out("MPI_Testany: OK rank=%d result=%s", MPIDBG_RANK, buf);
        }
        else {
            MPIDBG_Out("MPI_Testany: OK flag=false");
        }
    }
    else {
        MPIDBG_Out("MPI_Testany: Failed");
    }
    return ret;
}
#define MPI_Testany(...) MPIDBG_Testany(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Testany :
        #[ MPI_Waitall :
*/

static inline int MPIDBG_Waitall(int count, MPI_Request *array_of_requests, MPI_Status
*array_of_statuses, MPIDBG_EXTARG)
{
    char buf[256 * count];  /* Enough. */
    MPI_Request old_requests[count];
    memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
    MPIDBG_sprint_requests(buf, count, array_of_requests);
    MPIDBG_Out("MPI_Waitall: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Waitall(count, array_of_requests, array_of_statuses);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, array_of_statuses);
        MPIDBG_Out("MPI_Waitall: OK rank=%d result=%s", MPIDBG_RANK, buf);
    }
    else {
        MPIDBG_Out("MPI_Waitall: Failed");
    }
    return ret;
}
#define MPI_Waitall(...) MPIDBG_Waitall(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Waitall :
        #[ MPI_Testall :
*/

static inline int MPIDBG_Testall(int count, MPI_Request *array_of_requests, int *flag, MPI_Status
*array_of_statuses, MPIDBG_EXTARG)
{
    char buf[256 * count];  /* Enough. */
    MPI_Request old_requests[count];
    memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * count);
    MPIDBG_sprint_requests(buf, count, array_of_requests);
    MPIDBG_Out("MPI_Testall: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Testall(count, array_of_requests, flag, array_of_statuses);
    if ( ret == MPI_SUCCESS ) {
        if ( *flag ) {
            MPIDBG_sprint_statuses(buf, count, old_requests, array_of_requests, array_of_statuses);
            MPIDBG_Out("MPI_Testall: OK rank=%d result=%s", MPIDBG_RANK, buf);
        }
        else {
            MPIDBG_Out("MPI_Testall: OK flag=false");
        }
    }
    else {
        MPIDBG_Out("MPI_Testall: Failed");
    }
    return ret;
}
#define MPI_Testall(...) MPIDBG_Testall(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Testall :
        #[ MPI_Waitsome :
*/

static inline int MPIDBG_Waitsome(int incount, MPI_Request *array_of_requests, int *outcount, int
*array_of_indices, MPI_Status *array_of_statuses, MPIDBG_EXTARG)
{
    char buf[256 * incount];  /* Enough. */
    MPI_Request old_requests[incount];
    memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * incount);
    MPIDBG_sprint_requests(buf, incount, array_of_requests);
    MPIDBG_Out("MPI_Waitsome: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Waitsome(incount, array_of_requests, outcount, array_of_indices, array_of_statuses);
```

```
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_sprint_statuses(buf, incount, old_requests, array_of_requests, array_of_statuses);
        MPIDBG_Out("MPI_Waitsome: OK rank=%d result=%s", MPIDBG_RANK, buf);
    }
    else {
        MPIDBG_Out("MPI_Waitsome: Failed");
    }
    return ret;
}
#define MPI_Waitsome(...) MPIDBG_Waitsome(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Waitsome :
        #[ MPI_Testsome :
*/

static inline int MPIDBG_Testsome(int incount, MPI_Request *array_of_requests, int *outcount, int *array_of_indices, MPI_Status *array_of_statuses, MPIDBG_EXTARG)
{
    char buf[256 * incount];  /* Enough. */
    MPI_Request old_requests[incount];
    memcpy(old_requests, array_of_requests, sizeof(MPI_Request) * incount);
    MPIDBG_sprint_requests(buf, incount, array_of_requests);
    MPIDBG_Out("MPI_Testsome: rank=%d request=%s", MPIDBG_RANK, buf);
    int ret = MPI_Testsome(incount, array_of_requests, outcount, array_of_indices, array_of_statuses);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_sprint_statuses(buf, incount, old_requests, array_of_requests, array_of_statuses);
        MPIDBG_Out("MPI_Testsome: OK rank=%d result=%s", MPIDBG_RANK, buf);
    }
    else {
        MPIDBG_Out("MPI_Testsome: Failed");
    }
    return ret;
}
#define MPI_Testsome(...) MPIDBG_Testsome(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Testsome :
        #[ MPI_Iprobe :
*/

static inline int MPIDBG_Iprobe(int source, int tag, MPI_Comm comm, int *flag, MPI_Status *status, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Iprobe: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
    int ret = MPI_Iprobe(source, tag, comm, flag, status);
    if ( ret == MPI_SUCCESS ) {
        if ( *flag ) {
            int ret_size = 0;
            MPI_Get_count(status, MPI_BYTE, &ret_size);
            MPIDBG_Out("MPI_Iprobe: OK src=%d dest=%d tag=%d size=%d", status->MPI_SOURCE, MPIDBG_RANK, status->MPI_TAG, ret_size);
        }
        else {
            MPIDBG_Out("MPI_Iprobe: OK flag=false");
        }
    }
    else {
        MPIDBG_Out("MPI_Iprobe: Failed");
    }
    return ret;
}
#define MPI_Iprobe(...) MPIDBG_Iprobe(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Iprobe :
        #[ MPI_Probe :
*/

static inline int MPIDBG_Probe(int source, int tag, MPI_Comm comm, MPI_Status *status, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Probe: src=%d dest=%d tag=%d", source, MPIDBG_RANK, tag);
    int ret = MPI_Probe(source, tag, comm, status);
    if ( ret == MPI_SUCCESS ) {
        int ret_size = 0;
        MPI_Get_count(status, MPI_BYTE, &ret_size);
        MPIDBG_Out("MPI_Probe: OK src=%d dest=%d tag=%d size=%d", status->MPI_SOURCE, MPIDBG_RANK, status->MPI_TAG, ret_size);
    }
    else {
        MPIDBG_Out("MPI_Probe: Failed");
    }
    return ret;
}
#define MPI_Probe(...) MPIDBG_Probe(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
```

```
        #] MPI_Probe :
        #[ MPI_Cancel :
*/

static inline int MPIDBG_Cancel(MPI_Request *request, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Cancel: rank=%d", MPIDBG_RANK);
    int ret = MPI_Cancel(request);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Cancel: OK");
    }
    else {
        MPIDBG_Out("MPI_Cancel: Failed");
    }
    return ret;
}
#define MPI_Cancel(...) MPIDBG_Cancel(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Cancel :
        #[ MPI_Test_cancelled :
*/

static inline int MPIDBG_Test_cancelled(MPI_Status *status, int *flag, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Test_cancelled: rank=%d", MPIDBG_RANK);
    int ret = MPI_Test_cancelled(status, flag);
    if ( ret == MPI_SUCCESS ) {
        if ( *flag ) {
            MPIDBG_Out("MPI_Test_cancelled: OK flag=true");
        }
        else {
            MPIDBG_Out("MPI_Test_cancelled: OK flag=false");
        }
    }
    else {
        MPIDBG_Out("MPI_Test_cancelled: Failed");
    }
    return ret;
}
#define MPI_Test_cancelled(...) MPIDBG_Test_cancelled(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Test_cancelled :
        #[ MPI_Barrier :
*/

static inline int MPIDBG_Barrier(MPI_Comm comm, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Barrier: rank=%d", MPIDBG_RANK);
    int ret = MPI_Barrier(comm);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Barrier: OK");
    }
    else {
        MPIDBG_Out("MPI_Barrier: Failed");
    }
    return ret;
}
#define MPI_Barrier(...) MPIDBG_Barrier(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Barrier :
        #[ MPI_Bcast :
*/

static inline int MPIDBG_Bcast(void* buffer, int count, MPI_Datatype datatype, int root, MPI_Comm
comm, MPIDBG_EXTARG)
{
    MPIDBG_Out("MPI_Bcast: root=%d count=%d", root, count);
    int ret = MPI_Bcast(buffer, count, datatype, root, comm);
    if ( ret == MPI_SUCCESS ) {
        MPIDBG_Out("MPI_Bcast: OK");
    }
    else {
        MPIDBG_Out("MPI_Bcast: Failed");
    }
    return ret;
}
#define MPI_Bcast(...) MPIDBG_Bcast(__VA_ARGS__, __FILE__, __LINE__, __func__)

/*
        #] MPI_Bcast :
    #] MPI APIs :
*/

#endif  /* MPIDBG_H */
```

4.82 mytime.cc

```
#ifdef HAVE_CONFIG_H
#include <config.h>
#endif

// A timing routine for debugging. Only on Unix (where sys/time.h is available).
#ifdef UNIX

#include <sys/time.h>
#include <cstdlib>
#include <cstdio>
#include <string>

#ifndef timersub
/* timersub is not in POSIX, but presents on most BSD derivatives.
   This implementation is borrowed from glibc. (TU 23 Oct 2011) */
#define timersub(a, b, result) \
    do { \
        (result)->tv_sec = (a)->tv_sec - (b)->tv_sec; \
        (result)->tv_usec = (a)->tv_usec - (b)->tv_usec; \
        if ((result)->tv_usec < 0) { \
            --(result)->tv_sec; \
            (result)->tv_usec += 1000000; \
        } \
    } while (0)
#endif

bool starttime_set = false;
timeval starttime;

double thetime () {
    if (!starttime_set) {
        gettimeofday(&starttime,NULL);
        starttime_set=true;
    }

  timeval now,diff;
  gettimeofday(&now,NULL);
  timersub(&now,&starttime,&diff);
  return diff.tv_sec+diff.tv_usec/1000000.0;
}

std::string thetime_str() {
    char res[10];
    snprintf (res,10,"%.4lf", thetime());
    return res;
}

#endif  // UNIX
```

4.83 mytime.h

```
#pragma once
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 *   be able to track its scientific use to convince FOM of its value
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 *
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

double thetime();
std::string thetime_str();
```
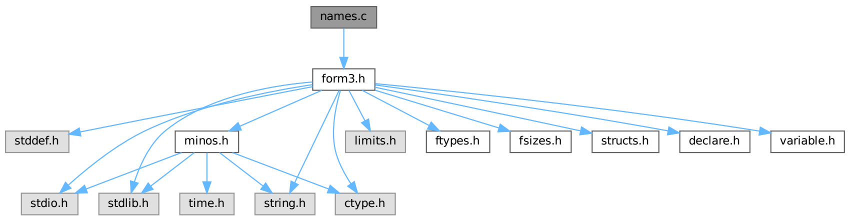

4.84 names.c File Reference

`#include "form3.h"`

Include dependency graph for names.c:



Functions

- NAMENODE * GetNode (NAMETREE *nametree, UBYTE *name)
- int AddName (NAMETREE *nametree, UBYTE *name, WORD type, WORD number, int *nodenum)
- int GetName (NAMETREE *nametree, UBYTE *namein, WORD *number, int par)
- UBYTE * GetFunction (UBYTE *s, WORD *funnum)
- UBYTE * GetNumber (UBYTE *s, WORD *num)
- int GetLastExprName (UBYTE *name, WORD *number)
- int GetOName (NAMETREE *nametree, UBYTE *name, WORD *number, int par)
- int GetAutoName (UBYTE *name, WORD *number)
- int GetVar (UBYTE *name, WORD *type, WORD *number, int wantedtype, int par)
- WORD EntVar (WORD type, UBYTE *name, WORD x, WORD y, WORD z, WORD d)
- int GetDollar (UBYTE *name)
- void DumpTree (NAMETREE *nametree)
- void DumpNode (NAMETREE *nametree, WORD node, WORD depth)
- int CompactifyTree (NAMETREE *nametree, WORD par)
- void CopyTree (NAMETREE *newtree, NAMETREE *oldtree, WORD node, WORD par)
- void LinkTree (NAMETREE *tree, WORD offset, WORD numnodes)
- NAMETREE * MakeNameTree (void)
- void FreeNameTree (NAMETREE *n)
- void ClearWildcardNames (void)
- int AddWildcardName (UBYTE *name)
- int GetWildcardName (UBYTE *name)
- int AddSymbol (UBYTE *name, int minpow, int maxpow, int cplx, int dim)
- int CoSymbol (UBYTE *s)
- int AddIndex (UBYTE *name, int dim, int dim4)
- int CoIndex (UBYTE *s)
- UBYTE * DoDimension (UBYTE *s, int *dim, int *dim4)
- int CoDimension (UBYTE *s)
- int AddVector (UBYTE *name, int cplx, int dim)
- int CoVector (UBYTE *s)
- int AddFunction (UBYTE *name, int comm, int istensor, int cplx, int symprop, int dim, int argmax, int argmin)
- int CoCommuteInSet (UBYTE *s)
- int CoFunction (UBYTE *s, int comm, int istensor)
- int CoNFunction (UBYTE *s)
- int CoCFunction (UBYTE *s)
- int CoNTensor (UBYTE *s)

- int CoCTensor (UBYTE *s)
- int DoTable (UBYTE *s, int par)
- int CoTable (UBYTE *s)
- int CoNTable (UBYTE *s)
- int CoCTable (UBYTE *s)
- void EmptyTable (TABLES T)
- int AddSet (UBYTE *name, WORD dim)
- int DoElements (UBYTE *s, SETS set, UBYTE *name)
- int CoSet (UBYTE *s)
- int DoTempSet (UBYTE *from, UBYTE *to)
- int CoAuto (UBYTE *inp)
- int AddDollar (UBYTE *name, WORD type, WORD *start, LONG size)
- int ReplaceDollar (WORD number, WORD newtype, WORD *newstart, LONG newsize)
- int AddDubious (UBYTE *name)
- int MakeDubious (NAMETREE *nametree, UBYTE *name, WORD *number)
- int NameConflict (int type, UBYTE *name)
- int AddExpression (UBYTE *name, int x, int y)
- int GetLabel (UBYTE *name)
- void ResetVariables (int par)
- void RemoveDollars (void)
- void Globalize (int par)
- int TestName (UBYTE *name)

4.84.1 Detailed Description

The complete names administration. All variables with a name have to pass here to be properly registered, have structs of the proper type assigned to them etc. The file also contains the utility routines for maintaining the balanced trees that make searching for names rather fast.

Definition in file names.c.

4.84.2 Function Documentation

4.84.2.1 GetNode()

```
NAMENODE * GetNode (
            NAMETREE * nametree,
            UBYTE * name )
```

Definition at line 49 of file names.c.

4.84.2.2 AddName()

```
int AddName (
            NAMETREE * nametree,
            UBYTE * name,
            WORD type,
            WORD number,
            int * nodenum )
```

Definition at line 71 of file names.c.

4.84.2.3 GetName()

```
int GetName (
            NAMETREE * nametree,
            UBYTE * namein,
            WORD * number,
            int par )
```

Definition at line 252 of file names.c.

4.84.2.4 GetFunction()

```
UBYTE * GetFunction (
            UBYTE * s,
            WORD * funnum )
```

Definition at line 342 of file names.c.

4.84.2.5 GetNumber()

```
UBYTE * GetNumber (
            UBYTE * s,
            WORD * num )
```

Definition at line 388 of file names.c.

4.84.2.6 GetLastExprName()

```
int GetLastExprName (
            UBYTE * name,
            WORD * number )
```

Definition at line 438 of file names.c.

4.84.2.7 GetOName()

```
int GetOName (
            NAMETREE * nametree,
            UBYTE * name,
            WORD * number,
            int par )
```

Definition at line 460 of file names.c.

4.84.2.8 GetAutoName()

```
int GetAutoName (
            UBYTE * name,
            WORD * number )
```

Definition at line 479 of file names.c.

4.84.2.9 GetVar()

```
int GetVar (
            UBYTE * name,
            WORD * type,
            WORD * number,
            int wantedtype,
            int par )
```

Definition at line 524 of file names.c.

4.84.2.10 EntVar()

```
WORD EntVar (
            WORD type,
            UBYTE * name,
            WORD x,
            WORD y,
            WORD z,
            WORD d )
```

Definition at line 557 of file names.c.

4.84.2.11 GetDollar()

```
int GetDollar (
            UBYTE * name )
```

Definition at line 590 of file names.c.

4.84.2.12 DumpTree()

```
void DumpTree (
            NAMETREE * nametree )
```

Definition at line 602 of file names.c.

4.84.2.13 DumpNode()

```
void DumpNode (
            NAMETREE * nametree,
            WORD node,
            WORD depth )
```

Definition at line 615 of file names.c.

4.84.2.14 CompactifyTree()

```
int CompactifyTree (
            NAMETREE * nametree,
            WORD par )
```

Definition at line 634 of file names.c.

4.84.2.15 CopyTree()

```
void CopyTree (
            NAMETREE * newtree,
            NAMETREE * oldtree,
            WORD node,
            WORD par )
```

Definition at line 696 of file names.c.

4.84.2.16 LinkTree()

```
void LinkTree (
            NAMETREE * tree,
            WORD offset,
            WORD numnodes )
```

Definition at line 784 of file names.c.

4.84.2.17 MakeNameTree()

```
NAMETREE * MakeNameTree (
            void )
```

Definition at line 815 of file names.c.

4.84.2.18 FreeNameTree()

```
void FreeNameTree (
            NAMETREE * n )
```

Definition at line 834 of file names.c.

4.84.2.19 ClearWildcardNames()

```
void ClearWildcardNames (
            void )
```

Definition at line 849 of file names.c.

4.84.2.20 AddWildcardName()

```
int AddWildcardName (
            UBYTE * name )
```

Definition at line 854 of file names.c.

4.84.2.21 GetWildcardName()

```
int GetWildcardName (
            UBYTE * name )
```

Definition at line 898 of file names.c.

4.84.2.22 AddSymbol()

```
int AddSymbol (
            UBYTE * name,
            int minpow,
            int maxpow,
            int cplx,
            int dim )
```

Definition at line 920 of file names.c.

4.84.2.23 CoSymbol()

```
int CoSymbol (
            UBYTE * s )
```

Definition at line 946 of file names.c.

4.84.2.24 AddIndex()

```
int AddIndex (
            UBYTE * name,
            int dim,
            int dim4 )
```

Definition at line 1088 of file names.c.

4.84.2.25 CoIndex()

```
int CoIndex (
            UBYTE * s )
```

Definition at line 1112 of file names.c.

4.84.2.26 DoDimension()

```
UBYTE * DoDimension (
            UBYTE * s,
            int * dim,
            int * dim4 )
```

Definition at line 1160 of file names.c.

4.84.2.27 CoDimension()

```
int CoDimension (
            UBYTE * s )
```

Definition at line 1226 of file names.c.

4.84.2.28 AddVector()

```
int AddVector (
            UBYTE * name,
            int cplx,
            int dim )
```

Definition at line 1244 of file names.c.

4.84.2.29 CoVector()

```
int CoVector (
            UBYTE * s )
```

Definition at line 1267 of file names.c.

4.84.2.30 AddFunction()

```
int AddFunction (
            UBYTE * name,
            int comm,
            int istensor,
            int cplx,
            int symprop,
            int dim,
            int argmax,
            int argmin )
```

Definition at line 1326 of file names.c.

4.84.2.31 CoCommuteInSet()

```
int CoCommuteInSet (
            UBYTE * s )
```

Definition at line 1356 of file names.c.

4.84.2.32 CoFunction()

```
int CoFunction (
            UBYTE * s,
            int comm,
            int istensor )
```

Definition at line 1446 of file names.c.

4.84.2.33 CoNFunction()

```
int CoNFunction (
            UBYTE * s )
```

Definition at line 1624 of file names.c.

4.84.2.34 CoCFunction()

```
int CoCFunction (
            UBYTE * s )
```

Definition at line 1625 of file names.c.

4.84.2.35 CoNTensor()

```
int CoNTensor (
            UBYTE * s )
```

Definition at line 1626 of file names.c.

4.84.2.36 CoCTensor()

```
int CoCTensor (
            UBYTE * s )
```

Definition at line 1627 of file names.c.

4.84.2.37 DoTable()

```
int DoTable (
            UBYTE * s,
            int par )
```

Definition at line 1666 of file names.c.

4.84.2.38 CoTable()

```
int CoTable (
            UBYTE * s )
```

Definition at line 2048 of file names.c.

4.84.2.39 CoNTable()

```
int CoNTable (
            UBYTE * s )
```

Definition at line 2058 of file names.c.

4.84.2.40 CoCTable()

```
int CoCTable (
            UBYTE * s )
```

Definition at line 2068 of file names.c.

4.84.2.41 EmptyTable()

```
void EmptyTable (
            TABLES T )
```

Definition at line 2078 of file names.c.

4.84.2.42 AddSet()

```
int AddSet (
            UBYTE * name,
            WORD dim )
```

Definition at line 2122 of file names.c.

4.84.2.43 DoElements()

```
int DoElements (
            UBYTE * s,
            SETS set,
            UBYTE * name )
```

Definition at line 2155 of file names.c.

4.84.2.44 CoSet()

```
int CoSet (
            UBYTE * s )
```

Definition at line 2355 of file names.c.

4.84.2.45 DoTempSet()

```
int DoTempSet (
            UBYTE * from,
            UBYTE * to )
```

Definition at line 2480 of file names.c.

4.84.2.46 CoAuto()

```
int CoAuto (
            UBYTE * inp )
```

Definition at line 2572 of file names.c.

4.84.2.47 AddDollar()

```
int AddDollar (
            UBYTE * name,
            WORD type,
            WORD * start,
            LONG size )
```

Definition at line 2602 of file names.c.

4.84.2.48 ReplaceDollar()

```
int ReplaceDollar (
            WORD number,
            WORD newtype,
            WORD * newstart,
            LONG newsize )
```

Definition at line 2646 of file names.c.

4.84.2.49 AddDubious()

```
int AddDubious (
            UBYTE * name )
```

Definition at line 2679 of file names.c.

4.84.2.50 MakeDubious()

```
int MakeDubious (
            NAMETREE * nametree,
            UBYTE * name,
            WORD * number )
```

Definition at line 2693 of file names.c.

4.84.2.51 NameConflict()

```
int NameConflict (
            int type,
            UBYTE * name )
```

Definition at line 2728 of file names.c.

4.84.2.52 AddExpression()

```
int AddExpression (
            UBYTE * name,
            int x,
            int y )
```

Definition at line 2744 of file names.c.

4.84.2.53 GetLabel()

```
int GetLabel (
            UBYTE * name )
```

Definition at line 2787 of file names.c.

4.84.2.54 ResetVariables()

```
void ResetVariables (
            int par )
```

Definition at line 2832 of file names.c.

4.84.2.55 RemoveDollars()

```
void RemoveDollars (
            void )
```

Definition at line 3128 of file names.c.

4.84.2.56 Globalize()

```
void Globalize (
            int par )
```

Definition at line 3155 of file names.c.

4.84.2.57 TestName()

```
int TestName (
            UBYTE * name )
```

Definition at line 3254 of file names.c.

4.85 names.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
*/

#include "form3.h"

/* EXTERNLOCK(dummylock) */

/*
    #] Includes :

    #[ GetNode :
*/

NAMENODE *GetNode(NAMETREE *nametree, UBYTE *name)
{
    NAMENODE *n;
    int node, newnode, i;
    if ( nametree->namenode == 0 ) return(0);
    newnode = nametree->headnode;
    do {
        node = newnode;
        n = nametree->namenode+node;
        if ( ( i = StrCmp(name,nametree->namebuffer+n->name) ) < 0 )
            newnode = n->left;
        else if ( i > 0 ) newnode = n->right;
        else { return(n); }
    } while ( newnode >= 0 );
```

```
    return(0);
}

/*
    #] GetNode :
    #[ AddName :
*/

int AddName(NAMETREE *nametree, UBYTE *name, WORD type, WORD number, int *nodenum)
{
    NAMENODE *n, *nn, *nnn;
    UBYTE *s, *ss, *sss;
    LONG *c1,*c2, j, newsize;
    int node, newnode, node3, r, rr = 0, i, retval = 0;
    if ( nametree->namenode == 0 ) {
        s = name; i = 1; while ( *s ) { i++; s++; }
        j = INITNAMESIZE;
        if ( i > j ) j = i;
        nametree->namenode = (NAMENODE *)Malloc1(INITNODESIZE*sizeof(NAMENODE),
            "new nametree in AddName");
        nametree->namebuffer = (UBYTE *)Malloc1(j,
            "new namebuffer in AddName");
        nametree->nodesize = INITNODESIZE;
        nametree->namesize = j;
        nametree->namefill = i;
        nametree->nodefill = 1;
        nametree->headnode = 0;
        n = nametree->namenode;
        n->parent = n->left = n->right = -1;
        n->balance = 0;
        n->type = type;
        n->number = number;
        n->name = 0;
        s = name;
        ss = nametree->namebuffer;
        while ( *s ) *ss++ = *s++;
        *ss = 0;
        *nodenum = 0;
        return(retval);
    }
    newnode = nametree->headnode;
    do {
        node = newnode;
        n = nametree->namenode+node;
        if ( StrCmp(name,nametree->namebuffer+n->name) < 0 ) {
            newnode = n->left; r = -1;
        }
        else {
            newnode = n->right; r = 1;
        }
    } while ( newnode >= 0 );
/*
    We are at the insertion point. Add the node.
*/
    if ( nametree->nodefill >= nametree->nodesize ) {    /* Double allocation */
        newsize = nametree->nodesize * 2;
        if ( newsize > MAXINNAMETREE ) newsize = MAXINNAMETREE;
        if ( nametree->nodefill >= MAXINNAMETREE ) {
            MesPrint("!!!More than %l names in one object",(LONG)MAXINNAMETREE);
            Terminate(-1);
        }
        nnn = (NAMENODE *)Malloc1(2*((LONG)newsize*sizeof(NAMENODE)),
            "extra names in AddName");
        c1 = (LONG *)nnn; c2 = (LONG *)nametree->namenode;
        i = (nametree->nodefill * sizeof(NAMENODE))/sizeof(LONG);
        while ( --i >= 0 ) *c1++ = *c2++;
        M_free(nametree->namenode,"nametree->namenode");
        nametree->namenode = nnn;
        nametree->nodesize = newsize;
        n = nametree->namenode+node;
    }
    *nodenum = newnode = nametree->nodefill++;
    nn = nametree->namenode+newnode;
    nn->parent = node;
    if ( r < 0 ) n->left = newnode; else n->right = newnode;
    nn->left = nn->right = -1;
    nn->type = type;
    nn->number = number;
    nn->balance = 0;
    i = 1; s = name; while ( *s ) { i++; s++; }
    while ( nametree->namefill + i >= nametree->namesize ) { /* Double alloc */
        sss = (UBYTE *)Malloc1(2*nametree->namesize,
            "extra names in AddName");
        s = sss; ss = nametree->namebuffer; j = nametree->namefill;
        while ( --j >= 0 ) *s++ = *ss++;
        M_free(nametree->namebuffer,"nametree->namebuffer");
        nametree->namebuffer = sss;
```

```
        nametree->namesize *= 2;
    }
    s = nametree->namebuffer+nametree->namefill;
    nn->name = nametree->namefill;
    retval = nametree->namefill;
    nametree->namefill += i;
    while ( *name ) *s++ = *name++;
    *s = 0;
/*
    Adjust the balance factors
*/
    while ( node >= 0 ) {
        n = nametree->namenode + node;
        if ( newnode == n->left ) rr = -1;
        else rr = 1;
        if ( n->balance == -rr ) { n->balance = 0; return(retval); }
        else if ( n->balance == rr ) break;
        n->balance = rr;
        newnode = node;
        node = n->parent;
    }
    if ( node < 0 ) return(retval);
/*
    We have to rebalance the tree. There are two basic operations.
    n/node is the unbalanced node. newnode is its child.
    rr is the old balance of n/node.
*/
    nn = nametree->namenode + newnode;
    if ( nn->balance == -rr ) {  /* The difficult case */
        if ( rr > 0 ) {
            node3 = nn->left;
            nnn = nametree->namenode + node3;
            nnn->parent = n->parent;
            n->parent = nn->parent = node3;
            if ( nnn->right >= 0 ) nametree->namenode[nnn->right].parent = newnode;
            if ( nnn->left >= 0 ) nametree->namenode[nnn->left].parent = node;
            n->right = nnn->left; nnn->left = node;
            nn->left = nnn->right; nnn->right = newnode;
            if ( nnn->balance > 0 ) { n->balance = -1; nn->balance = 0; }
            else if ( nnn->balance == 0 ) { n->balance = nn->balance = 0; }
            else                         { nn->balance = 1; n->balance = 0; }
        }
        else {
            node3 = nn->right;
            nnn = nametree->namenode + node3;
            nnn->parent = n->parent;
            n->parent = nn->parent = node3;
            if ( nnn->right >= 0 ) nametree->namenode[nnn->right].parent = node;
            if ( nnn->left >= 0 ) nametree->namenode[nnn->left].parent = newnode;
            n->left = nnn->right; nnn->right = node;
            nn->right = nnn->left; nnn->left = newnode;
            if ( nnn->balance < 0 ) { n->balance = 1; nn->balance = 0; }
            else if ( nnn->balance == 0 ) { n->balance = nn->balance = 0; }
            else                         { nn->balance = -1; n->balance = 0; }
        }
        nnn->balance = 0;
        if ( nnn->parent >= 0 ) {
            nn = nametree->namenode + nnn->parent;
            if ( node == nn->left ) nn->left = node3;
            else nn->right = node3;
        }
        if ( node == nametree->headnode ) nametree->headnode = node3;
    }
    else if ( nn->balance == rr ) {  /* The easy case */
        nn->parent = n->parent; n->parent = newnode;
        if ( rr > 0 ) {
            if ( nn->left >= 0 ) nametree->namenode[nn->left].parent = node;
            n->right = nn->left; nn->left = node;
        }
        else {
            if ( nn->right >= 0 ) nametree->namenode[nn->right].parent = node;
            n->left = nn->right; nn->right = node;
        }
        if ( nn->parent >= 0 ) {
            nnn = nametree->namenode + nn->parent;
            if ( node == nnn->left ) nnn->left = newnode;
            else nnn->right = newnode;
        }
        nn->balance = n->balance = 0;
        if ( node == nametree->headnode ) nametree->headnode = newnode;
    }
#ifdef DEBUGON
    else {  /* Cannot be. Code here for debugging only */
        MesPrint("We ran into an impossible case in AddName\n");
        DumpTree(nametree);
        Terminate(-1);
    }
```

```
#endif
    return(retval);
}

/*
    #] AddName :
    #[ GetName :

    When AutoDeclare is an active statement.
    If par == WITHAUTO and the variable is not found we have to check:
    1: that nametree != AC.exprnames && nametree != AC.dollarnames
    2: check that the variable is not in AC.exprnames after all.
    3: call GetAutoName and return its values.
*/

int GetName(NAMETREE *nametree, UBYTE *namein, WORD *number, int par)
{
    NAMENODE *n;
    int node, newnode, i;
    UBYTE *s, *t, *u, *name;
/*  name = ConstructName(namein,0); */
    name = namein;
    if ( nametree->namenode == 0 || nametree->namefill == 0 ) goto NotFound;
    newnode = nametree->headnode;
    do {
        node = newnode;
        n = nametree->namenode+node;
        if ( ( i = StrCmp(name,nametree->namebuffer+n->name) ) < 0 )
            newnode = n->left;
        else if ( i > 0 ) newnode = n->right;
        else {
            *number = n->number;
            return(n->type);
        }
    } while ( newnode >= 0 );
    s = name;
    while ( *s ) s++;
    if ( s > name && s[-1] == '_' && nametree == AC.varnames ) {
/*
            The Kronecker delta d_ is very special. It is not really a function.
*/
        if ( s == name+2 && ( *name == 'd' || *name == 'D' ) ) {
            *number = DELTA-FUNCTION;
            return(CDELTA);
        }
/*
            Test for N#_? type variables (summed indices)
*/
        if ( s > name+2 && *name == 'N' ) {
            t = name+1; i = 0;
            while ( FG.cTable[*t] == 1 ) i = 10*i + *t++ -'0';
            if ( s == t+1 ) {
                *number = i + AM.IndDum - AM.OffsetIndex;
                return(CINDEX);
            }
        }
/*
            Now test for any built in object
*/
        newnode = nametree->headnode;
        do {
            node = newnode;
            n = nametree->namenode+node;
            if ( ( i = StrHICmp(name,nametree->namebuffer+n->name) ) < 0 )
                newnode = n->left;
            else if ( i > 0 ) newnode = n->right;
            else {
                *number = n->number; return(n->type);
            }
        } while ( newnode >= 0 );
/*
            Now we test for the extra symbols of the type STR###_
            The string sits in AC.extrasym and is followed by digits.
            The name is only legal if the number is in the
            range 1,...,cbuf[AM.sbufnum].numrhs
*/
        t = name; u = AC.extrasym;
        while ( *t == *u ) { t++; u++; }
        if ( *u == 0 && *t != 0 ) { /* potential hit */
            WORD x = 0;
            while ( FG.cTable[*t] == 1 ) {
                x = 10*x + (*t++ - '0');
            }
            if ( *t == '_' && x > 0 && x <= cbuf[AM.sbufnum].numrhs ) { /* Hit */
                *number = MAXVARIABLES-x;
                return(CSYMBOL);
            }
```

```
        }
    }
NotFound:;
    if ( par != WITHAUTO || nametree == AC.autonames ) return(NAMENOTFOUND);
    return(GetAutoName(name,number));
}

/*
    #] GetName :
    #[ GetFunction :

    Gets either a function or a $ that should expand into a function
    during runtime. In the case of the $ the value in funnum is -dolnum-1.
    The return value is the position after the name of the function or the $.
*/

static WORD one = 1;

UBYTE *GetFunction(UBYTE *s,WORD *funnum)
{
    int type;
    WORD numfun;
    UBYTE *t1, c;
    if ( *s == '$' ) {
        t1 = s+1; while ( FG.cTable[*t1] < 2 ) t1++;
        c = *t1; *t1 = 0;
        if ( ( type = GetName(AC.dollarnames,s+1,&numfun,NOAUTO) ) == CDOLLAR ) {
            *funnum = -numfun-2;
        }
        else {
            MesPrint("&%s is undefined",s);
            numfun = AddDollar(s+1,DOLINDEX,&one,1);
            *funnum = 0;
        }
    }
    else {
        t1 = SkipAName(s);
        c = *t1; *t1 = 0;
        if ( ( ( type = GetName(AC.varnames,s,&numfun,WITHAUTO) ) != CFUNCTION )
            || ( functions[numfun].spec > 0 ) ) {
            MesPrint("&%s should be a regular function",s);
            *funnum = 0;
            if ( type < 0 ) {
                if ( GetName(AC.exprnames,s,&numfun,NOAUTO) == NAMENOTFOUND )
                AddFunction(s,0,0,0,0,0,-1,-1);
            }
            *t1 = c;
            return(t1);
        }
        *funnum = numfun+FUNCTION;
    }
    *t1 = c;
    return(t1);
}

/*
    #] GetFunction :
    #[ GetNumber :

    Gets either a number or a $ that should expand into a number
    during runtime. In the case of the $ the value in num is -dolnum-2.
    The return value is the position after the number or the $.
*/

UBYTE *GetNumber(UBYTE *s,WORD *num)
{
    int type;
    WORD numfun;
    UBYTE *t1, c;
    while ( *s == '+' ) s++;
    if ( *s == '$' ) {
        t1 = s+1; while ( FG.cTable[*t1] < 2 ) t1++;
        c = *t1; *t1 = 0;
        if ( ( type = GetName(AC.dollarnames,s+1,&numfun,NOAUTO) ) == CDOLLAR ) {
            *num = -numfun-2;
        }
        else {
            MesPrint("&%s is undefined",s);
            numfun = AddDollar(s+1,DOLINDEX,&one,1);
            *num = -1;
        }
    }
    else if ( *s >= '0' && *s <= '9' ) {
        ULONG x = *s++ - '0';
        while ( *s >= '0' && *s <= '9' ) { x = 10*x + (*s++-'0'); }
        t1 = s;
        if ( x >= MAXPOSITIVE ) goto illegal;
```

```
            *num = (WORD)x;
            return(t1);
        }
        else {
            if ( *s == '-' ) { s++; }
            if ( *s >= '0' && *s <= '9' ) { while ( *s >= '0' && *s <= '9' ) s++; t1 = s; }
            else { t1 = SkipAName(s); }
illegal:
            *num = -1;
            MesPrint("&Illegal option in Canonicalize statement. Should be a nonnegative number or $ variable.");
            return(t1);
        }
        *t1 = c;
        return(t1);
}

/*
    #] GetNumber :
    #[ GetLastExprName :

    When AutoDeclare is an active statement.
    If par == WITHAUTO and the variable is not found we have to check:
    1: that nametree != AC.exprnames && nametree != AC.dollarnames
    2: check that the variable is not in AC.exprnames after all.
    3: call GetAutoName and return its values.
*/

int GetLastExprName(UBYTE *name, WORD *number)
{
    int i;
    EXPRESSIONS e;
    for ( i = NumExpressions; i > 0; i-- ) {
        e = Expressions+i-1;
        if ( StrCmp(AC.exprnames->namebuffer+e->name,name) == 0 ) {
            *number = i-1;
            return(1);
        }
    }
    return(0);
}

/*
    #] GetLastExprName :
    #[ GetOName :

    Adds the proper offsets, so we do not have to do that in the calling
    routine.
*/

int GetOName(NAMETREE *nametree, UBYTE *name, WORD *number, int par)
{
    int retval = GetName(nametree,name,number,par);
    switch ( retval ) {
        case CVECTOR: *number += AM.OffsetVector; break;
        case CINDEX:  *number += AM.OffsetIndex;  break;
        case CFUNCTION: *number += FUNCTION; break;
        default: break;
    }
    return(retval);
}

/*
    #] GetOName :
    #[ GetAutoName :

    This routine gets the automatic declarations
*/

int GetAutoName(UBYTE *name, WORD *number)
{
    UBYTE *s, c;
    int type;
    if ( GetName(AC.exprnames,name,number,NOAUTO) != NAMENOTFOUND )
                return(NAMENOTFOUND);
    s = name;
    while ( *s ) { s++; }
    if ( s[-1] == '_' ) {
        return(NAMENOTFOUND);
    }
    while ( s > name ) {
        c = *s; *s = 0;
        type = GetName(AC.autonames,name,number,NOAUTO);
        *s = c;
        switch(type) {
            case CSYMBOL: {
                SYMBOLS sym = ((SYMBOLS)(AC.AutoSymbolList.lijst)) + *number;
```

```
                *number = AddSymbol(name,sym->minpower,sym->maxpower,sym->complex,sym->dimension);
                return(type); }
            case CVECTOR: {
                VECTORS vec = ((VECTORS)(AC.AutoVectorList.lijst)) + *number;
                *number = AddVector(name,vec->complex,vec->dimension);
                return(type); }
            case CINDEX: {
                INDICES ind = ((INDICES)(AC.AutoIndexList.lijst)) + *number;
                *number = AddIndex(name,ind->dimension,ind->nmin4);
                return(type); }
            case CFUNCTION: {
                FUNCTIONS fun = ((FUNCTIONS)(AC.AutoFunctionList.lijst)) + *number;
                *number =
    AddFunction(name,fun->commute,fun->spec,fun->complex,fun->symmetric,fun->dimension,fun->maxnumargs,fun->minnumargs);
                return(type); }
            default:
                break;
        }
        s--;
    }
    return(NAMENOTFOUND);
}

/*
    #] GetAutoName :
    #[ GetVar :
*/

int GetVar(UBYTE *name, WORD *type, WORD *number, int wantedtype, int par)
{
    WORD funnum;
    int typ;
    if ( ( typ = GetName(AC.varnames,name,number,par) ) != wantedtype ) {
        if ( typ != NAMENOTFOUND ) {
            if ( wantedtype == -1 ) {
                *type = typ;
                return(1);
            }
            NameConflict(typ,name);
            MakeDubious(AC.varnames,name,&funnum);
            return(-1);
        }
        if ( ( typ = GetName(AC.exprnames,name,&funnum,par) ) != NAMENOTFOUND ) {
            if ( typ == wantedtype || wantedtype == -1 ) {
                *number = funnum; *type = typ; return(1);
            }
            NameConflict(typ,name);
            return(-1);
        }
        return(NAMENOTFOUND);
    }
    if ( typ == -1 ) { return(0); }
    *type = typ;
    return(1);
}

/*
    #] GetVar :
    #[ EntVar :
*/

WORD EntVar(WORD type, UBYTE *name, WORD x, WORD y, WORD z, WORD d)
{
    switch ( type ) {
        case CSYMBOL:
            return(AddSymbol(name,y,z,x,d));
            break;
        case CINDEX:
            return(AddIndex(name,x,z));
            break;
        case CVECTOR:
            return(AddVector(name,x,d));
            break;
        case CFUNCTION:
            return(AddFunction(name,y,z,x,0,d,-1,-1));
            break;
        case CSET:
            AC.SetList.numtemp++;
            return(AddSet(name,d));
            break;
        case CEXPRESSION:
            return(AddExpression(name,x,y));
            break;
        default:
            break;
    }
    return(-1);
```

```
}

/*
    #] EntVar :
    #[ GetDollar :
*/

int GetDollar(UBYTE *name)
{
    WORD number;
    if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) return(-1);
    return((int)number);
}

/*
    #] GetDollar :
    #[ DumpTree :
*/

void DumpTree(NAMETREE *nametree)
{
    if ( nametree->headnode >= 0
        && nametree->namebuffer && nametree->namenode ) {
        DumpNode(nametree,nametree->headnode,0);
    }
}

/*
    #] DumpTree :
    #[ DumpNode :
*/

void DumpNode(NAMETREE *nametree, WORD node, WORD depth)
{
    NAMENODE *n;
    int i;
    char *name;
    n = nametree->namenode + node;
    if ( n->left >= 0 ) DumpNode(nametree,n->left,depth+1);
    for ( i = 0; i < depth; i++ ) printf(" ");
    name = (char *)(nametree->namebuffer+n->name);
    printf("%s(%d): {%d}(%d)(%d)[%d]\n",
        name,node,n->parent,n->left,n->right,n->balance);
    if ( n->right >= 0 ) DumpNode(nametree,n->right,depth+1);
}

/*
    #] DumpNode :
    #[ CompactifyTree :
*/

int CompactifyTree(NAMETREE *nametree,WORD par)
{
    NAMETREE newtree;
    NAMENODE *n;
    LONG i, j, ns, k;
    UBYTE *s;

    for ( i = 0, j = 0, k = 0, n = nametree->namenode, ns = 0;
    i < nametree->nodefill; i++, n++ ) {
        if ( n->type != CDELETE ) {
            s = nametree->namebuffer+n->name;
            while ( *s ) { s++; ns++; }
            j++;
        }
        else k++;
    }
    if ( k == 0 ) return(0);
    if ( j == 0 ) {
        if ( nametree->namebuffer ) M_free(nametree->namebuffer,"nametree->namebuffer");
        if ( nametree->namenode ) M_free(nametree->namenode,"nametree->namenode");
        nametree->namebuffer = 0;
        nametree->namenode = 0;
        nametree->namesize = nametree->namefill =
        nametree->nodesize = nametree->nodefill =
        nametree->oldnamefill = nametree->oldnodefill = 0;
        nametree->globalnamefill = nametree->globalnodefill =
        nametree->clearnamefill = nametree->clearnodefill = 0;
        nametree->headnode = -1;
        return(0);
    }
    ns += j;
    if ( j < 10 ) j = 10;
    if ( ns < 100 ) ns = 100;
    newtree.namenode = (NAMENODE *)Malloc1(2*j*sizeof(NAMENODE),"compactify namestree");
    newtree.nodefill = 0; newtree.nodesize = 2*j;
    newtree.namebuffer = (UBYTE *)Malloc1(2*ns,"compactify namestree");
```

```
    newtree.namefill = 0; newtree.namesize = 2*ns;
    CopyTree(&newtree,nametree,nametree->headnode,par);
    newtree.namenode[newtree.nodefill»1].parent = -1;
    LinkTree(&newtree,(WORD)0,newtree.nodefill);
    newtree.headnode = newtree.nodefill » 1;
    M_free(nametree->namebuffer,"nametree->namebuffer");
    M_free(nametree->namenode,"nametree->namenode");
    nametree->namebuffer  = newtree.namebuffer;
    nametree->namenode    = newtree.namenode;
    nametree->namesize    = newtree.namesize;
    nametree->namefill    = newtree.namefill;
    nametree->nodesize    = newtree.nodesize;
    nametree->nodefill    = newtree.nodefill;
    nametree->oldnamefill = newtree.namefill;
    nametree->oldnodefill = newtree.nodefill;
    nametree->headnode    = newtree.headnode;

/*  DumpTree(nametree); */
    return(0);
}

/*
    #] CompactifyTree :
    #[ CopyTree :
*/

void CopyTree(NAMETREE *newtree, NAMETREE *oldtree, WORD node, WORD par)
{
    NAMENODE *n, *m;
    UBYTE *s, *t;
    n = oldtree->namenode+node;
    if ( n->left >= 0 ) CopyTree(newtree,oldtree,n->left,par);
    if ( n->type != CDELETE ) {
        m = newtree->namenode+newtree->nodefill;
        m->type = n->type;
        m->number = n->number;
        m->name = newtree->namefill;
        m->left = m->right = -1;
        m->balance = 0;
        switch ( n->type ) {
            case CSYMBOL:
                if ( par == AUTONAMES ) {
                    autosymbols[n->number].name = newtree->namefill;
                    autosymbols[n->number].node = newtree->nodefill;
                }
                else {
                    symbols[n->number].name = newtree->namefill;
                    symbols[n->number].node = newtree->nodefill;
                }
                break;
            case CINDEX :
                if ( par == AUTONAMES ) {
                    autoindices[n->number].name = newtree->namefill;
                    autoindices[n->number].node = newtree->nodefill;
                }
                else {
                    indices[n->number].name = newtree->namefill;
                    indices[n->number].node = newtree->nodefill;
                }
                break;
            case CVECTOR:
                if ( par == AUTONAMES ) {
                    autovectors[n->number].name = newtree->namefill;
                    autovectors[n->number].node = newtree->nodefill;
                }
                else {
                    vectors[n->number].name = newtree->namefill;
                    vectors[n->number].node = newtree->nodefill;
                }
                break;
            case CFUNCTION:
                if ( par == AUTONAMES ) {
                    autofunctions[n->number].name = newtree->namefill;
                    autofunctions[n->number].node = newtree->nodefill;
                }
                else {
                    functions[n->number].name = newtree->namefill;
                    functions[n->number].node = newtree->nodefill;
                }
                break;
            case CSET:
                Sets[n->number].name = newtree->namefill;
                Sets[n->number].node = newtree->nodefill;
                break;
            case CEXPRESSION:
                Expressions[n->number].name = newtree->namefill;
                Expressions[n->number].node = newtree->nodefill;
```

```
                break;
            case CDUBIOUS:
                Dubious[n->number].name = newtree->namefill;
                Dubious[n->number].node = newtree->nodefill;
                break;
            case CDOLLAR:
                Dollars[n->number].name = newtree->namefill;
                Dollars[n->number].node = newtree->nodefill;
                break;
            default:
                MesPrint("Illegal variable type in CopyTree: %d",n->type);
                break;
        }
        newtree->nodefill++;
        s = newtree->namebuffer + newtree->namefill;
        t = oldtree->namebuffer + n->name;
        while ( *t ) { *s++ = *t++; newtree->namefill++; }
        *s = 0; newtree->namefill++;
    }
    if ( n->right >= 0 ) CopyTree(newtree,oldtree,n->right,par);
}

/*
    #] CopyTree :
    #[ LinkTree :
*/

void LinkTree(NAMETREE *tree, WORD offset, WORD numnodes)
{
/*
    Makes the tree into a binary tree
*/
    int med,numleft,numright,medleft,medright;
    med = numnodes >> 1;
    numleft = med;
    numright = numnodes - med - 1;
    medleft = numleft >> 1;
    medright = ( numright >> 1 ) + med + 1;
    if ( numleft > 0 ) {
        tree->namenode[offset+med].left = offset+medleft;
        tree->namenode[offset+medleft].parent = offset+med;
    }
    if ( numright > 0 ) {
        tree->namenode[offset+med].right = offset+medright;
        tree->namenode[offset+medright].parent = offset+med;
    }
    if ( numleft  > 0 ) LinkTree(tree,offset,numleft);
    if ( numright > 0 ) LinkTree(tree,offset+med+1,numright);
    while ( numleft && numright ) { numleft >>= 1; numright >>= 1; }
    if ( numleft ) tree->namenode[offset+med].balance = -1;
    else if ( numright ) tree->namenode[offset+med].balance = 1;
}

/*
    #] LinkTree :
    #[ MakeNameTree :
*/

NAMETREE *MakeNameTree(void)
{
    NAMETREE *n;
    n = (NAMETREE *)Malloc1(sizeof(NAMETREE),"new nametree");
    n->namebuffer = 0;
    n->namenode = 0;
    n->namesize = n->namefill = n->nodesize = n->nodefill =
    n->oldnamefill = n->oldnodefill = 0;
    n->globalnamefill = n->globalnodefill =
    n->clearnamefill = n->clearnodefill = 0;
    n->headnode = -1;
    return(n);
}

/*
    #] MakeNameTree :
    #[ FreeNameTree :
*/

void FreeNameTree(NAMETREE *n)
{
    if ( n ) {
        if ( n->namebuffer ) M_free(n->namebuffer,"nametree->namebuffer");
        if ( n->namenode ) M_free(n->namenode,"nametree->namenode");
        M_free(n,"nametree");
    }
}

/*
```

```
    #] FreeNameTree :

    #[ WildcardNames :
*/

void ClearWildcardNames(void)
{
    AC.NumWildcardNames = 0;
}

int AddWildcardName(UBYTE *name)
{
    GETIDENTITY
    int size = 0, tocopy, i;
    UBYTE *s = name, *t, *newbuffer;
    while ( *s ) { s++; size++; }
    for ( i = 0, t = AC.WildcardNames; i < AC.NumWildcardNames; i++ ) {
        s = name;
        while ( ( *s == *t ) && *s ) { s++; t++; }
        if ( *s == 0 && *t == 0 ) return(i+1);
        while ( *t ) t++;
        t++;
    }
    tocopy = t - AC.WildcardNames;
    if ( tocopy + size + 1 > AC.WildcardBufferSize ) {
        if ( AC.WildcardBufferSize == 0 ) {
            AC.WildcardBufferSize = size+1;
            if ( AC.WildcardBufferSize < 100 ) AC.WildcardBufferSize = 100;
        }
        else if ( size+1 >= AC.WildcardBufferSize ) {
            AC.WildcardBufferSize += size+1;
        }
        else {
            AC.WildcardBufferSize *= 2;
        }
        newbuffer = (UBYTE *)Malloc1((LONG)AC.WildcardBufferSize,"argument list names");
        t = newbuffer;
        if ( AC.WildcardNames ) {
            s = AC.WildcardNames;
            while ( tocopy > 0 ) { *t++ = *s++; tocopy--; }
            M_free(AC.WildcardNames,"AC.WildcardNames");
        }
        AC.WildcardNames = newbuffer;
        M_free(AT.WildArgTaken,"AT.WildArgTaken");
        AT.WildArgTaken = (WORD *)Malloc1((LONG)AC.WildcardBufferSize*sizeof(WORD)/2
                ,"argument list names");
    }
    s = name;
    while ( *s ) *t++ = *s++;
    *t = 0;
    AC.NumWildcardNames++;
    return(AC.NumWildcardNames);
}

int GetWildcardName(UBYTE *name)
{
    UBYTE *s, *t;
    int i;
    for ( i = 0, t = AC.WildcardNames; i < AC.NumWildcardNames; i++ ) {
        s = name;
        while ( ( *s == *t ) && *s ) { s++; t++; }
        if ( *s == 0 && *t == 0 ) return(i+1);
        while ( *t ) t++;
        t++;
    }
    return(0);
}

/*
    #] WildcardNames :

    #[ AddSymbol :

    The actual addition. Special routine for additions 'on the fly'
*/

int AddSymbol(UBYTE *name, int minpow, int maxpow, int cplx, int dim)
{
    int nodenum, numsymbol = AC.Symbols->num;
    UBYTE *s = name;
    SYMBOLS sym = (SYMBOLS)FromVarList(AC.Symbols);
    bzero(sym,sizeof(struct SymbOl));
    sym->name = AddName(*AC.activenames,name,CSYMBOL,numsymbol,&nodenum);
    sym->minpower = minpow;
    sym->maxpower = maxpow;
    sym->complex  = cplx;
    sym->flags    = 0;
```

```
    sym->node      = nodenum;
    sym->dimension= dim;
    while ( *s ) s++;
    sym->namesize = (s-name)+1;
    return(numsymbol);
}

/*
    #] AddSymbol :
    #[ CoSymbol :

    Symbol declarations.   name[#{R|I|C}][([min]:[max])]
    Note that we know already that the parentheses match properly
*/

int CoSymbol(UBYTE *s)
{
    int type, error = 0, minpow, maxpow, cplx, sgn, dim;
    WORD numsymbol;
    UBYTE *name, *oldc, c, cc;
    do {
        minpow = -MAXPOWER;
        maxpow =  MAXPOWER;
        cplx = 0;
        dim = 0;
        name = s;
        if ( ( s = SkipAName(s) ) == 0 ) {
IllForm:    MesPrint("&Illegally formed name in symbol statement");
            error = 1;
            s = SkipField(name,0);
            goto eol;
        }
        oldc = s; cc = c = *s; *s = 0;
        if ( TestName(name) ) { *s = c; goto IllForm; }
        if ( cc == '#' ) {
            s++;
                 if ( tolower(*s) == 'r' ) cplx = VARTYPENONE;
            else if ( tolower(*s) == 'c' ) cplx = VARTYPECOMPLEX;
            else if ( tolower(*s) == 'i' ) cplx = VARTYPEIMAGINARY;
            else if ( ( ( *s == '-' || *s == '+' || *s == '=' )
                 && ( s[1] >= '0' && s[1] <= '9' ) )
                 || ( *s >= '0' && *s <= '9' ) ) {
                LONG x;
                sgn = 0;
                if ( *s == '-' ) { sgn = VARTYPEMINUS; s++; }
                else if ( *s == '+' || *s == '=' ) { sgn = 0; s++; }
                x = *s -'0';
                while ( s[1] >= '0' && s[1] <= '9' ) {
                    x = 10*x + (s[1] - '0'); s++;
                }
                if ( x >= MAXPOWER || x <= 1 ) {
                    MesPrint("&Illegal value for root of unity %s",name);
                    error = 1;
                }
                else {
                    maxpow = x;
                }
                cplx = VARTYPEROOTOFUNITY | sgn;
            }
            else {
                MesPrint("&Illegal specification for complexity of symbol %s",name);
                *oldc = c;
                error = 1;
                s = SkipField(s,0);
                goto eol;
            }
            s++; cc = *s;
        }
        if ( cc == '{' ) {
            s++;
            if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
                s += 2;
                if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
                    ParseSignedNumber(dim,s)
                    if ( dim < -HALFMAX || dim > HALFMAX ) {
                        MesPrint("&Warning: dimension of %s (%d) out of range"
                        ,name,dim);
                    }
                }
                if ( *s != '}' ) goto IllDim;
                else s++;
            }
            else {
IllDim:         MesPrint("&Error: Illegal dimension field for variable %s",name);
                error = 1;
                s = SkipField(s,0);
                goto eol;
```

```
            }
            cc = *s;
        }
        if ( cc == '(' ) {
            if ( ( cplx & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
                MesPrint("&Root of unity property for %s cannot be combined with power restrictions",name);
                error = 1;
            }
            s++;
            if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
                ParseSignedNumber(minpow,s)
                if ( minpow < -MAXPOWER ) {
                    minpow = -MAXPOWER;
                    if ( AC.WarnFlag )
                    MesPrint("&Warning: minimum power of %s corrected to %d"
                    ,name,-MAXPOWER);
                }
            }
            if ( *s != ':' ) {
skippar:        error = 1;
                s = SkipField(s,1);
                goto eol;
            }
            else s++;
            if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
                ParseSignedNumber(maxpow,s)
                if ( maxpow > MAXPOWER ) {
                    maxpow = MAXPOWER;
                    if ( AC.WarnFlag )
                    MesPrint("&Warning: maximum power of %s corrected to %d"
                    ,name,MAXPOWER);
                }
            }
            if ( *s != ')' ) goto skippar;
            s++;
        }
        if ( ( AC.AutoDeclareFlag == 0 &&
         ( ( type = GetName(AC.exprnames,name,&numsymbol,NOAUTO) )
                 != NAMENOTFOUND ) )
        || ( ( type = GetName(*(AC.activenames),name,&numsymbol,NOAUTO) ) != NAMENOTFOUND ) ) {
            if ( type != CSYMBOL ) error = NameConflict(type,name);
            else {
                SYMBOLS sym = (SYMBOLS)(AC.Symbols->lijst) + numsymbol;
                if ( ( numsymbol == AC.lPolyFunVar ) && ( AC.lPolyFunType > 0 )
                     && ( AC.lPolyFun != 0 ) && ( minpow > -MAXPOWER || maxpow < MAXPOWER ) ) {
                    MesPrint("&The symbol %s is used by power expansions in the PolyRatFun!",name);
                    error = 1;
                }
                sym->complex  = cplx;
                sym->minpower = minpow;
                sym->maxpower = maxpow;
                sym->dimension= dim;
            }
        }
        else {
            AddSymbol(name,minpow,maxpow,cplx,dim);
        }
        *oldc = c;
eol:    while ( *s == ',' ) s++;
    } while ( *s );
    return(error);
}

/*
    #] CoSymbol :
    #[ AddIndex :

    The actual addition. Special routine for additions 'on the fly'
*/

int AddIndex(UBYTE *name, int dim, int dim4)
{
    int nodenum, numindex = AC.Indices->num;
    INDICES ind = (INDICES)FromVarList(AC.Indices);
    UBYTE *s = name;
    bzero(ind,sizeof(struct InDeX));
    ind->name      = AddName(*AC.activenames,name,CINDEX,numindex,&nodenum);
    ind->type      = 0;
    ind->dimension = dim;
    ind->flags     = 0;
    ind->nmin4     = dim4;
    ind->node      = nodenum;
    while ( *s ) s++;
    ind->namesize = (s-name)+1;
    return(numindex);
}
```

```
/*
    #] AddIndex :
    #[ CoIndex :

    Index declarations. name[={number|symbol[:othersymbol]}]
*/

int CoIndex(UBYTE *s)
{
    int type, error = 0, dim, dim4;
    WORD numindex;
    UBYTE *name, *oldc, c;
    do {
        dim = AC.lDefDim;
        dim4 = AC.lDefDim4;
        name = s;
        if ( ( s = SkipAName(s) ) == 0 ) {
IllForm:    MesPrint("&Illegally formed name in index statement");
            error = 1;
            s = SkipField(name,0);
            goto eol;
        }
        oldc = s; c = *s; *s = 0;
        if ( TestName(name) ) { *s = c; goto IllForm; }
        if ( c == '=' ) {
            s++;
            if ( ( s = DoDimension(s,&dim,&dim4) ) == 0 ) {
                *oldc = c;
                error = 1;
                s = SkipField(name,0);
                goto eol;
            }
        }
        if ( ( AC.AutoDeclareFlag == 0 &&
        ( ( type = GetName(AC.exprnames,name,&numindex,NOAUTO) )
         != NAMENOTFOUND ) )
        || ( ( type = GetName(*(AC.activenames),name,&numindex,NOAUTO) ) != NAMENOTFOUND ) ) {
            if ( type != CINDEX ) error = NameConflict(type,name);
            else { /* reset the dimensions */
                indices[numindex].dimension = dim;
                indices[numindex].nmin4     = dim4;
            }
        }
        else AddIndex(name,dim,dim4);
        *oldc = c;
eol:    while ( *s == ',' ) s++;
    } while ( *s );
    return(error);
}

/*
    #] CoIndex :
    #[ DoDimension :
*/

UBYTE *DoDimension(UBYTE *s, int *dim, int *dim4)
{
    UBYTE c, *t = s;
    int type, error = 0;
    WORD numsymbol;
    NAMETREE **oldtree = AC.activenames;
    LIST* oldsymbols = AC.Symbols;
    *dim4 = -NMIN4SHIFT;
    if ( FG.cTable[*s] == 1 ) {
retry:
        ParseNumber(*dim,s)
#if ( BITSINWORD/8 < 4 )
        if ( *dim >= (1 « (BITSINWORD-1)) ) goto illeg;
#endif
        *dim4 = *dim - 4;
        return(s);
    }
    else if ( ( (FG.cTable[*s] == 0 ) || ( *s == '[' ) )
        && ( s = SkipAName(s) ) != 0 ) {
        AC.activenames = &(AC.varnames);
        AC.Symbols = &(AC.SymbolList);
        c = *s; *s = 0;
        if ( ( ( type = GetName(AC.exprnames,t,&numsymbol,NOAUTO) ) != NAMENOTFOUND )
        || ( ( type = GetName(AC.varnames,t,&numsymbol,WITHAUTO) ) != NAMENOTFOUND ) ) {
            if ( type != CSYMBOL ) error = NameConflict(type,t);
        }
        else {
            numsymbol = AddSymbol(t,-MAXPOWER,MAXPOWER,0,0);
            if ( AC.WarnFlag )
            MesPrint("&Warning: Implicit declaration of %s as a symbol",t);
        }
```

```
        *dim = -numsymbol;
        if ( ( *s = c ) == ':' ) {
            s++;
            t = s;
            if ( ( s = SkipAName(s) ) == 0 ) goto illeg;
            if ( ( ( type = GetName(AC.exprnames,t,&numsymbol,NOAUTO) ) != NAMENOTFOUND )
            || ( ( type = GetName(AC.varnames,t,&numsymbol,WITHAUTO) ) != NAMENOTFOUND ) ) {
                if ( type != CSYMBOL ) error = NameConflict(type,t);
            }
            else {
                numsymbol = AddSymbol(t,-MAXPOWER,MAXPOWER,0,0);
                if ( AC.WarnFlag )
                MesPrint("&Warning: Implicit declaration of %s as a symbol",t);
            }
            *dim4 = -numsymbol-NMIN4SHIFT;
        }
    }
    else if ( *s == '+' && FG.cTable[s[1]] == 1 ) {
        s++; goto retry;
    }
    else {
illeg:  MesPrint("&Illegal dimension specification. Should be number >= 0, symbol or symbol:symbol");
        return(0);
    }
    AC.Symbols = oldsymbols;
    AC.activenames = oldtree;
    if ( error ) return(0);
    return(s);
}

/*
    #] DoDimension :
    #[ CoDimension :
*/

int CoDimension(UBYTE *s)
{
    s = DoDimension(s,&AC.lDefDim,&AC.lDefDim4);
    if ( s == 0 ) return(1);
    if ( *s != 0 ) {
        MesPrint("&Argument of dimension statement should be number >= 0, symbol or symbol:symbol");
        return(1);
    }
    return(0);
}

/*
    #] CoDimension :
    #[ AddVector :

    The actual addition. Special routine for additions 'on the fly'
*/

int AddVector(UBYTE *name, int cplx, int dim)
{
    int nodenum, numvector = AC.Vectors->num;
    VECTORS v = (VECTORS)FromVarList(AC.Vectors);
    UBYTE *s = name;
    bzero(v,sizeof(struct VeCtOr));
    v->name = AddName(*AC.activenames,name,CVECTOR,numvector,&nodenum);
    v->complex = cplx;
    v->node    = nodenum;
    v->dimension = dim;
    v->flags = 0;
    while ( *s ) s++;
    v->namesize = (s-name)+1;
    return(numvector);
}

/*
    #] AddVector :
    #[ CoVector :

    Vector declarations. The descriptor string is "(,%n)"
*/

int CoVector(UBYTE *s)
{
    int type, error = 0, dim;
    WORD numvector;
    UBYTE *name, c, *endname;
    do {
        name = s;
        dim = 0;
        if ( ( s = SkipAName(s) ) == 0 ) {
IllForm:    MesPrint("&Illegally formed name in vector statement");
            error = 1;
```

```
                s = SkipField(s,0);
            }
            else {
                c = *s; *s = 0, endname = s;
                if ( TestName(name) ) { *s = c; goto IllForm; }
                if ( c == '{' ) {
                    s++;
                    if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
                        s += 2;
                        if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
                            ParseSignedNumber(dim,s)
                            if ( dim < -HALFMAX || dim > HALFMAX ) {
                                MesPrint("&Warning: dimension of %s (%d) out of range"
                                ,name,dim);
                            }
                        }
                        if ( *s != '}' ) goto IllDim;
                        else s++;
                    }
                    else {
IllDim:                 MesPrint("&Error: Illegal dimension field for variable %s",name);
                        error = 1;
                        s = SkipField(s,0);
                        while ( *s == ',' ) s++;
                        continue;
                    }
                }
                if ( ( AC.AutoDeclareFlag == 0 &&
                 ( ( type = GetName(AC.exprnames,name,&numvector,NOAUTO) )
                 != NAMENOTFOUND ) )
                || ( ( type = GetName(*(AC.activenames),name,&numvector,NOAUTO) ) != NAMENOTFOUND ) ) {
                    if ( type != CVECTOR ) error = NameConflict(type,name);
                }
                else AddVector(name,0,dim);
                *endname = c;
            }
            while ( *s == ',' ) s++;
        } while ( *s );
        return(error);
}

/*
    #] CoVector :
    #[ AddFunction :

    The actual addition. Special routine for additions 'on the fly'
*/

int AddFunction(UBYTE *name, int comm, int istensor, int cplx, int symprop, int dim, int argmax, int
argmin)
{
    int nodenum, numfunction = AC.Functions->num;
    FUNCTIONS fun = (FUNCTIONS)FromVarList(AC.Functions);
    UBYTE *s = name;
    bzero(fun,sizeof(struct FuNcTiOn));
    fun->name = AddName(*AC.activenames,name,CFUNCTION,numfunction,&nodenum);
    fun->commute = comm;
    fun->spec = istensor;
    fun->complex  = cplx;
    fun->tabl = 0;
    fun->flags = 0;
    fun->node = nodenum;
    fun->symminfo = 0;
    fun->symmetric = symprop;
    fun->dimension = dim;
    fun->maxnumargs = argmax;
    fun->minnumargs = argmin;
    while ( *s ) s++;
    fun->namesize = (s-name)+1;
    return(numfunction);
}

/*
    #] AddFunction :
    #[ CoCommuteInSet :

    Commuting,f1,...,fn;
*/

int CoCommuteInSet(UBYTE *s)
{
    UBYTE *name, *ss, c, *start = s;
    WORD number, type, *g, *gg;
    int error = 0, i, len = StrLen(s), len2 = 0;
    if ( AC.CommuteInSet != 0 ) {
        g = AC.CommuteInSet;
        while ( *g ) g += *g;
```

```
        len2 = g - AC.CommuteInSet;
        if ( len2+len+3 > AC.SizeCommuteInSet ) {
            gg = (WORD *)Malloc1((len2+len+3)*sizeof(WORD),"CommuteInSet");
            for ( i = 0; i < len2; i++ ) gg[i] = AC.CommuteInSet[i];
            gg[len2] = 0;
            M_free(AC.CommuteInSet,"CommuteInSet");
            AC.CommuteInSet = gg;
            AC.SizeCommuteInSet = len+len2+3;
            g = AC.CommuteInSet+len2;
        }
    }
    else {
        AC.SizeCommuteInSet = len+2;
        g = AC.CommuteInSet = (WORD *)Malloc1((len+3)*sizeof(WORD),"CommuteInSet");
        *g = 0;
    }
    gg = g++;
    ss = s-1;
    for(;;) {
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
        if ( *s == 0 ) {
            if ( s - start >= len ) break;
            *s = '}'; s++;
            *g = 0;
            *gg = g-gg;
            if ( *gg < 2 ) {
                MesPrint("&There should be at least two noncommuting functions or tensors in a
commuting statement.");
                error = 1;
            }
            else if ( *gg == 2 ) {
                gg[2] = gg[1]; gg[3] = 0; gg[0] = 3;
            }
            gg = g++;
            continue;
        }
        if ( s > ss ) {
            if ( *s != '{' ) {
                MesPrint("&The CommuteInSet statement should have sets enclosed in {}.");
                error = 1;
                break;
            }
            ss = s;
            SKIPBRA2(ss) /* Note that parentheses were tested before */
            *ss = 0;
            s++;
        }
        name = s;
        s = SkipAName(s);
        c = *s; *s = 0;
        if ( ( type = GetName(AC.varnames,name,&number,NOAUTO) ) != CFUNCTION ) {
            MesPrint("&%s is not a function or tensor",name);
            error = 1;
        }
        else if ( functions[number].commute == 0 ){
            MesPrint("&%s is not a noncommuting function or tensor",name);
            error = 1;
        }
        else {
            *g++ = number+FUNCTION;
            functions[number].flags |= COULDCOMMUTE;
            if ( number+FUNCTION >= GAMMA && number+FUNCTION <= GAMMASEVEN ) {
                functions[GAMMA-FUNCTION].flags |= COULDCOMMUTE;
                functions[GAMMAI-FUNCTION].flags |= COULDCOMMUTE;
                functions[GAMMAFIVE-FUNCTION].flags |= COULDCOMMUTE;
                functions[GAMMASIX-FUNCTION].flags |= COULDCOMMUTE;
                functions[GAMMASEVEN-FUNCTION].flags |= COULDCOMMUTE;
            }
        }
        *s = c;
    }
    return(error);
}

/*
    #] CoCommuteInSet :
    #[ CoFunction + ...:

    Function declarations.
    The second parameter indicates commutation properties.
    The third parameter tells whether we have a tensor.
*/

int CoFunction(UBYTE *s, int comm, int istensor)
{
    int type, error = 0, cplx, symtype, dim, argmax, argmin;
    WORD numfunction, reverseorder = 0, addone;
```

```
    UBYTE *name, *oldc, *par, c, cc;
    do {
        symtype = cplx = 0, argmin = argmax = -1;
        dim = 0;
        name = s;
        if ( ( s = SkipAName(s) ) == 0 ) {
IllForm:    MesPrint("&Illegally formed function/tensor name");
            error = 1;
            s = SkipField(name,0);
            goto eol;
        }
        oldc = s; cc = c = *s; *s = 0;
        if ( TestName(name) ) { *s = c; goto IllForm; }
        if ( c == '#' ) {
            s++;
                 if ( tolower(*s) == 'r' ) cplx = VARTYPENONE;
            else if ( tolower(*s) == 'c' ) cplx = VARTYPECOMPLEX;
            else if ( tolower(*s) == 'i' ) cplx = VARTYPEIMAGINARY;
            else {
                MesPrint("&Illegal specification for complexity of %s",name);
                *oldc = c;
                error = 1;
                s = SkipField(s,0);
                goto eol;
            }
            s++; cc = *s;
        }
        if ( cc == '{' ) {
            s++;
            if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
                s += 2;
                if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
                    ParseSignedNumber(dim,s)
                    if ( dim < -HALFMAX || dim > HALFMAX ) {
                        MesPrint("&Warning: dimension of %s (%d) out of range"
                        ,name,dim);
                    }
                }
                if ( *s != '}' ) goto IllDim;
                else s++;
            }
            else {
IllDim:         MesPrint("&Error: Illegal dimension field for variable %s",name);
                error = 1;
                s = SkipField(s,0);
                goto eol;
            }
            cc = *s;
        }
        if ( cc == '(' ) {
            s++;
            if ( *s == '-' ) {
                reverseorder = REVERSEORDER;
                s++;
            }
            else {
                reverseorder = 0;
            }
            par = s;
            while ( FG.cTable[*s] == 0 ) s++;
            cc = *s; *s = 0;
            if ( s <= par ) {
illegsym:       *s = cc;
                MesPrint("&Illegal specification for symmetry of %s",name);
                *oldc = c;
                error = 1;
                s = SkipField(s,1);
                goto eol;
            }
            if ( StrICont(par,(UBYTE *)"symmetric") == 0 ) symtype = SYMMETRIC;
            else if ( StrICont(par,(UBYTE *)"antisymmetric") == 0 ) symtype = ANTISYMMETRIC;
            else if ( ( StrICont(par,(UBYTE *)"cyclesymmetric") == 0 )
            || ( StrICont(par,(UBYTE *)"cyclic") == 0 ) ) symtype = CYCLESYMMETRIC;
            else if ( ( StrICont(par,(UBYTE *)"rcyclesymmetric") == 0 )
            || ( StrICont(par,(UBYTE *)"rcyclic") == 0 )
            || ( StrICont(par,(UBYTE *)"reversecyclic") == 0 ) ) symtype = RCYCLESYMMETRIC;
            else goto illegsym;
            *s = cc;
            if ( *s != ')' || ( s[1] && s[1] != ',' && s[1] != '<' ) ) {
                Warning("&Excess information in symmetric properties currently ignored");
                s = SkipField(s,1);
            }
            else s++;
            symtype |= reverseorder;
            cc = *s;
        }
retry:;
```

```
        if ( cc == '<' ) {
            s++; addone = 0;
            if ( *s == '=' ) { addone++; s++; }
            argmax = 0;
            while ( FG.cTable[*s] == 1 ) { argmax = 10*argmax + *s++ - '0'; }
            argmax += addone;
            par = s;
            while ( FG.cTable[*s] == 0 ) s++;
            if ( s > par ) {
                cc = *s; *s = 0;
                if ( ( StrICont(par,(UBYTE *)"arguments") == 0 )
                || ( StrICont(par,(UBYTE *)"args") == 0 ) ) {}
                else {
                    Warning("&Illegal information in number of arguments properties currently ignored");
                    error = 1;
                }
                *s = cc;
            }
            if ( argmax <= 0 ) {
                MesPrint("&Error: Cannot have fewer than 0 arguments for variable %s",name);
                error = 1;
            }
            cc = *s;
        }
        if ( cc == '>' ) {
            s++; addone = 1;
            if ( *s == '=' ) { addone = 0; s++; }
            argmin = 0;
            while ( FG.cTable[*s] == 1 ) { argmin = 10*argmin + *s++ - '0'; }
            argmin += addone;
            par = s;
            while ( FG.cTable[*s] == 0 ) s++;
            if ( s > par ) {
                cc = *s; *s = 0;
                if ( ( StrICont(par,(UBYTE *)"arguments") == 0 )
                || ( StrICont(par,(UBYTE *)"args") == 0 ) ) {}
                else {
                    Warning("&Illegal information in number of arguments properties currently ignored");
                    error = 1;
                }
                *s = cc;
            }
            cc = *s;
        }
        if ( cc == '<' ) goto retry;
        if ( ( AC.AutoDeclareFlag == 0 &&
         ( ( type = GetName(AC.exprnames,name,&numfunction,NOAUTO) )
         != NAMENOTFOUND ) )
        || ( ( type = GetName(*(AC.activenames),name,&numfunction,NOAUTO) ) != NAMENOTFOUND ) ) {
            if ( type != CFUNCTION ) error = NameConflict(type,name);
            else {
/*              FUNCTIONS fun = (FUNCTIONS)(AC.Functions->lijst) + numfunction-FUNCTION; */
                FUNCTIONS fun = (FUNCTIONS)(AC.Functions->lijst) + numfunction;

                if ( fun->tabl != 0 ) {
                    MesPrint("&Illegal attempt to change table into function");
                    error = 1;
                }

                fun->complex = cplx;
                fun->commute = comm;
                if ( istensor && fun->spec == 0 ) {
                    MesPrint("&Function %s changed to tensor",name);
                    error = 1;
                }
                else if ( istensor == 0 && fun->spec > 0 ) {
                    MesPrint("&Tensor %s changed to function",name);
                    error = 1;
                }
                fun->spec = istensor;
                if ( fun->symmetric != symtype ) {
                    fun->symmetric = symtype;
                    AC.SymChangeFlag = 1;
                }
                fun->maxnumargs = argmax;
                fun->minnumargs = argmin;
            }
        }
        else {
            AddFunction(name,comm,istensor,cplx,symtype,dim,argmax,argmin);
        }
        *oldc = c;
eol:    while ( *s == ',' ) s++;
    } while ( *s );
    return(error);
```

```
}

int CoNFunction(UBYTE *s) { return(CoFunction(s,1,0)); }
int CoCFunction(UBYTE *s) { return(CoFunction(s,0,0)); }
int CoNTensor(UBYTE *s) { return(CoFunction(s,1,2)); }
int CoCTensor(UBYTE *s) { return(CoFunction(s,0,2)); }

/*
    #] CoFunction + ...:
    #[ DoTable :

        Syntax:
        Table [check] [strict|relax] [zerofill] name(:1:2,...,regular arguments);
        name must be the name of a regular function.
        the table indices must be the first arguments.
        The parenthesis indicates 'name' as opposed to the options.

        We leave behind:
        a struct tabl in the FUNCTION struct
        Regular table:
            an array tablepointers for the pointers to elements of rhs
            in the compiler struct cbuf[T->bufnum]
            an array MINMAX T->mm with the minima and maxima
            a prototype array
            an offset in the compiler buffer for the pattern to be matched
        Sparse table:
            Just the number of dimensions
            We will keep track of the number of defined elements in totind
            and in tablepointers we will have numind+1 positions for each
            element. The first numind elements for the indices and the
            last one for the element in cbuf[T->bufnum].rhs

            If the number of dimensions is *<number>, there is not a fixed
            number of dimensions. Just a maximum. In that case the first
            index should be the number of other dimensions. This first index
            does not count in <number>.

        Complication: to preserve speed we need a prototype and a pattern
        for each thread when we use WITHPTHREADS. This is because we write
        into those when looking for the pattern.
*/

static int nwarntab = 1;

int DoTable(UBYTE *s, int par)
{
    GETIDENTITY
    UBYTE *name, *p, *inp, c;
    int i, j, k, sparseflag = 0, rflag = 0, checkflag = 0;
    int error = 0, ret, oldcbufnum, oldEside;
    WORD funnum, type, *OldWork, *w, *ww, *t, *tt, *flags1, oldnumrhs,oldnumlhs;
    LONG oldcpointer;
    MINMAX *mm, *mm1;
    LONG x, y;
    TABLES T;
    CBUF *C;

    while ( *s == ',' ) s++;
    do {
        name = s;
        if ( ( s = SkipAName(s) ) == 0 ) {
IllForm:    MesPrint("&Illegal name or option in table declaration");
            return(1);
        }
        c = *s; *s = 0;
        if ( TestName(name) ) { *s = c; goto IllForm; }
        *s = c;
        if ( *s == '(' ) break;
        if ( *s != ',' ) {
            MesPrint("&Illegal definition of table");
            return(1);
        }
        *s = 0;
/*
        Secondary options
*/
        if      ( StrICmp(name,(UBYTE *)("check" )) == 0 ) checkflag = 1;
        else if ( StrICmp(name,(UBYTE *)("zero"  )) == 0 ) checkflag = 2;
        else if ( StrICmp(name,(UBYTE *)("one"   )) == 0 ) checkflag = 3;
        else if ( StrICmp(name,(UBYTE *)("strict")) == 0 ) rflag = 1;
        else if ( StrICmp(name,(UBYTE *)("relax" )) == 0 ) rflag = -1;
        else if ( StrICmp(name,(UBYTE *)("zerofill" )) == 0 ) { rflag = -2; checkflag = 2; }
        else if ( StrICmp(name,(UBYTE *)("onefill" )) == 0 ) { rflag = -3; checkflag = 3; }
        else if ( StrICmp(name,(UBYTE *)("sparse")) == 0 ) sparseflag |= 1;
        else if ( StrICmp(name,(UBYTE *)("base")) == 0 ) sparseflag |= 3;
        else if ( StrICmp(name,(UBYTE *)("tablebase")) == 0 ) sparseflag |= 3;
        else {
```

```
            MesPrint("&Illegal option in table definition: '%s'",name);
            error = 1;
        }
        *s++ = ',';
        while ( *s == ',' ) s++;
    } while ( *s );
    if ( name == s || *s == 0 ) {
        MesPrint("&Illegal name or option in table declaration");
        return(1);
    }
    *s = 0;     /* *s could only have been a parenthesis */
    if ( sparseflag ) {
        if ( checkflag == 1 ) rflag = 0;
        else if ( checkflag == 2 ) rflag = -2;
        else if ( checkflag == 3 ) rflag = -3;
        else rflag = -1;
    }
    if ( ( ret = GetVar(name,&type,&funnum,CFUNCTION,NOAUTO) ) ==
                    NAMENOTFOUND ) {
        if ( par == 0 ) {
            funnum = EntVar(CFUNCTION,name,0,1,0,0);
        }
        else if ( par == 1 || par == 2 ) {
            funnum = EntVar(CFUNCTION,name,0,0,0,0);
        }
    }
    else if ( ret <= 0 ) {
        funnum = EntVar(CFUNCTION,name,0,0,0,0);
        error = 1;
    }
    else {
        if ( par == 2 ) {
            if ( nwarntab ) {
                Warning("Table now declares its (commuting) function.");
                Warning("Earlier definition in Function statement obsolete. Please remove.");
                nwarntab = 0;
            }
        }
        else {
            error = 1;
            MesPrint("&(N)(C)Tables should not be declared previously");
        }
    }
    if ( functions[funnum].spec > 0 ) {
        MesPrint("&Tensors cannot become tables");
        return(1);
    }
    if ( functions[funnum].symmetric > 0 ) {
        MesPrint("&Functions with nontrivial symmetrization properties cannot become tables");
        return(1);
    }
    if ( functions[funnum].tabl ) {
        MesPrint("&Redefinition of an existing table is not allowed.");
        return(1);
    }
    functions[funnum].tabl = T = (TABLES)Malloc1(sizeof(struct TaBlEs),"table");
/*
    Next we find the size of the table (if it is not sparse)
*/
    T->defined = T->mdefined = 0; T->sparse = sparseflag; T->mm = 0; T->flags = 0;
    T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
    T->boomlijst = 0;
    T->strict = rflag;
    T->bounds = checkflag;
    T->bufnum = inicbufs();
    T->argtail = 0;
    T->spare = 0;
    T->bufferssize = 8;
    T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->bufferssize,"Table buffers");
    T->buffersfill = 0;
    T->buffers[T->buffersfill++] = T->bufnum;
    T->mode = 0;
    T->numdummies = 0;
    mm = T->mm;
    T->numind = 0;
    if ( rflag > 0 ) AC.MustTestTable++;
    T->totind = 0;      /* Table hasn't been checked */

    p = s; *s = '(';
    if ( sparseflag ) {
/*
        First copy the tail, just in case we will construct a tablebase
        Note that we keep the ( to indicate a tail
        The actual arguments can be found after the comma. Before we have
        the dimension which the tablebase will need for consistency checking.
*/
        inp = p+1;
```

```
        SKIPBRA3(inp)
        c = *inp; *inp = 0;
        T->argtail = strDup1(p,"argtail");
        *inp = c;
/*
        Now the regular compilation
*/
        inp = p++;
        if ( *p == '<' ) {
            WORD inc = 1;
            p++;
            if ( *p == '=' ) { inc++; p++; }
/*
            We will use one extra number for telling how many there really are.
*/
            x = 0;
            while ( *p <= '9' && *p >= '0' ) x = 10*x + (*p++-'0');
            x = -x-inc;
            if ( x == -1 ) {
                MesPrint("&Maximum number of dimensions in *-table should be at least one.");
                error = 1;
                goto FinishUp2;
            }
        }
        else {
            ParseNumber(x,p)
            if ( FG.cTable[p[-1]] != 1 || ( *p != ',' && *p != ')' ) ) {
                p = inp;
                MesPrint("&First argument in a sparse table must be a number of dimensions");
                error = 1;
                x = 1;
            }
        }
        T->numind = x;
        T->mm = (MINMAX *)Malloc1(ABS(x)*sizeof(MINMAX),"table dimensions");
        T->flags = (WORD *)Malloc1(ABS(x)*sizeof(WORD),"table flags");
        mm = T->mm;
        inp = p;
        if ( *inp != ')' ) inp++;
        T->totind = 0;            /* At the moment there are this many */
        T->tablepointers = 0;
        T->reserved = 0;
    }
    else {
        T->numind = 0;
        T->totind = 1;
        for(;;) {   /* Read the dimensions as far as they can be recognized */
            inp = ++p;
            if ( FG.cTable[*p] != 1 && *p != '+' && *p != '-' ) break;
            ParseSignedNumber(x,p)
            if ( FG.cTable[p[-1]] != 1 || *p != ':' ) break;
            p++;
            ParseSignedNumber(y,p)
            if ( FG.cTable[p[-1]] != 1 || ( *p != ',' && *p != ')' ) ) {
                MesPrint("&Illegal dimension field in table declaration");
                return(1);
            }
            mm1 = (MINMAX *)Malloc1((T->numind+1)*sizeof(MINMAX),"table dimensions");
            flags1 = (WORD *)Malloc1((T->numind+1)*sizeof(WORD),"table flags");
            for ( i = 0; i < T->numind; i++ ) { mm1[i] = T->mm[i]; flags1[i] = T->flags[i]; }
            if ( T->mm ) M_free(T->mm,"table dimensions");
            if ( T->flags ) M_free(T->flags,"table flags");
            T->mm = mm1;
            T->flags = flags1;
            mm = T->mm + T->numind;
            mm->mini = x; mm->maxi = y;
            T->totind *= mm->maxi-mm->mini+1;
            T->numind++;
            if ( *p == ')' ) { inp = p; break; }
        }
        w = T->tablepointers
          = (WORD *)Malloc1(TABLEEXTENSION*sizeof(WORD)*(T->totind),"table pointers");
        i = T->totind;
        for ( i = TABLEEXTENSION*T->totind; i > 0; i-- ) *w++ = -1; /* means: undefined */
        for ( i = T->numind-1, x = 1; i >= 0; i-- ) {
            T->mm[i].size = x;     /* Defines increment in this dimension */
            x *= T->mm[i].maxi - T->mm[i].mini + 1;
        }
    }
/*
    Now we redo the 'function part' and send it to the compiler.
    The prototype has to be picked up properly.
*/
    AT.WorkPointer++; /* We need one extra word later */
    OldWork = AT.WorkPointer;
    oldcbufnum = AC.cbufnum;
    AC.cbufnum = T->bufnum;
```

```
    C = cbuf+AC.cbufnum;
    oldcpointer = C->Pointer - C->Buffer;
    oldnumlhs = C->numlhs;
    oldnumrhs = C->numrhs;
    AddLHS(AC.cbufnum);
    while ( s >= name ) *--inp = *s--;
    w = AT.WorkPointer;
    AC.ProtoType = w;
    *w++ = SUBEXPRESSION;
    *w++ = SUBEXPSIZE;
    *w++ = 0;
    *w++ = 1;
    *w++ = AC.cbufnum;
    FILLSUB(w)
    AC.WildC = w;
    AC.NwildC = 0;
    AT.WorkPointer = w + 4*AM.MaxWildcards;
    if ( ( ret = CompileAlgebra(inp,LHSIDE,AC.ProtoType) ) < 0 ) {
        error = 1; goto FinishUp;
    }
    if ( AC.NwildC && SortWild(w,AC.NwildC) ) error = 1;
    w += AC.NwildC;
    i = w-OldWork;
    OldWork[1] = i;
/*
    Basically we have to pull this pattern through Generator in case
    there are functions inside functions, or parentheses.
    We have to temporarily disable the .tabl to avoid problems with
    TestSub.
    Essential: we need to start NewSort twice to avoid the PutOut routines.
    The ground pattern is sitting in C->numrhs, but it could be that it
    has subexpressions in it. Hence it has to be worked out as the lhs in
    id statements (in comexpr.c).
*/
    OldWork[2] = C->numrhs;
    *w++ = 1; *w++ = 1; *w++ = 3;
    OldWork[-1] = w-OldWork+1;
    AT.WorkPointer = w;
    ww = C->rhs[C->numrhs];
    for ( j = 0; j < *ww; j++ ) w[j] = ww[j];
    AT.WorkPointer = w+*w;
    if ( *ww == 0 || ww[*ww] != 0 ) {
        MesPrint("&Illegal table pattern definition");
        AC.lhdollarflag = 0;
        error = 1;
    }
    if ( error ) goto FinishUp;

    if ( NewSort(BHEAD0) || NewSort(BHEAD0) ) { error = 1; goto FinishUp; }
    AN.RepPoint = AT.RepCount + 1;
    AC.lhdollarflag = 0; oldEside = AR.Eside; AR.Eside = LHSIDE;
    AR.Cnumlhs = C->numlhs;
    functions[funnum].tabl = 0;
    if ( Generator(BHEAD w,C->numlhs) ) {
        functions[funnum].tabl = T;
        AR.Eside = oldEside;
        LowerSortLevel(); LowerSortLevel(); goto FinishUp;
    }
    functions[funnum].tabl = T;
    AR.Eside = oldEside;
    AT.WorkPointer = w;
    if ( EndSort(BHEAD w,0) < 0 ) { LowerSortLevel(); goto FinishUp; }
    if ( *w == 0 || *(w+*w) != 0 ) {
        MesPrint("&Irregular pattern in table definition");
        error = 1;
        goto FinishUp;
    }
    LowerSortLevel();
    if ( AC.lhdollarflag ) {
        MesPrint("&Unexpanded dollar variables are not allowed in table definition");
        error = 1;
        goto FinishUp;
    }
    AT.WorkPointer = ww = w + *w;
    if ( ww[-1] != 3 || ww[-2] != 1 || ww[-3] != 1 ) {
        MesPrint("&Coefficient of pattern in table definition should be 1.");
        error = 1;
        goto FinishUp;
    }
    AC.DumNum = 0;
/*
    Now we have to allocate space for prototype+pattern
    In the case of TFORM we need extra pointers, because each worker has its own
*/
    j = *w + ABS(T->numind)*2-3;
#ifdef WITHPTHREADS
    { int n;
```

```
    T->prototypeSize = ((i+j)*sizeof(WORD)+2*sizeof(WORD *)) * AM.totalnumberofthreads;
    T->prototype = (WORD **)Malloc1(T->prototypeSize,"table prototype");
    T->pattern = T->prototype + AM.totalnumberofthreads;
    t = (WORD *)(T->pattern + AM.totalnumberofthreads);
    for ( n = 0; n < AM.totalnumberofthreads; n++ ) {
        T->prototype[n] = t;
        for ( k = 0; k < i; k++ ) *t++ = OldWork[k];
    }
    T->pattern[0] = t;
    j--; w++;
    w[1] += ABS(T->numind)*2;
    for ( k = 0; k < FUNHEAD; k++ ) *t++ = *w++;
    j -= FUNHEAD;
    for ( k = 0; k < ABS(T->numind); k++ ) { *t++ = -SNUMBER; *t++ = 0; j -= 2; }
    for ( k = 0; k < j; k++ ) *t++ = *w++;
    if ( sparseflag ) T->pattern[0][1] = t - T->pattern[0];
    k = t - T->pattern[0];
    for ( n = 1; n < AM.totalnumberofthreads; n++ ) {
        T->pattern[n] = t; tt = T->pattern[0];
        for ( i = 0; i < k; i++ ) *t++ = *tt++;
    }
    }
#else
    T->prototypeSize = (i+j)*sizeof(WORD);
    T->prototype = (WORD *)Malloc1(T->prototypeSize, "table prototype");
    T->pattern = T->prototype + i;
    for ( k = 0; k < i; k++ ) T->prototype[k] = OldWork[k];
    t = T->pattern;
    j--; w++;
    w[1] += ABS(T->numind)*2;
    for ( k = 0; k < FUNHEAD; k++ ) *t++ = *w++;
    j -= FUNHEAD;
    for ( k = 0; k < ABS(T->numind); k++ ) { *t++ = -SNUMBER; *t++ = 0; j -= 2; }
    for ( k = 0; k < j; k++ ) *t++ = *w++;
    if ( sparseflag ) T->pattern[1] = t - T->pattern;
#endif
/*
    At this point we can pop the compilerbuffer.
*/
    C->Pointer = C->Buffer + oldcpointer;
    C->numrhs = oldnumrhs;
    C->numlhs = oldnumlhs;
/*
    Now check whether wildcards get converted to dollars (for PARALLEL)
    We give a warning!
*/
#ifdef WITHPTHREADS
    t = T->prototype[0];
#else
    t = T->prototype;
#endif
    tt = t + t[1]; t += SUBEXPSIZE;
    while ( t < tt ) {
        if ( *t == LOADDOLLAR ) {
            Warning("The use of $-variable assignments in tables disables parallel\
 execution for the whole program.");
            AM.hparallelflag |= NOPARALLEL_TBLDOLLAR;
            AC.mparallelflag |= NOPARALLEL_TBLDOLLAR;
            AddPotModdollar(t[2]);
        }
        t += t[1];
    }
FinishUp:;
    AT.WorkPointer = OldWork - 1;
    AC.cbufnum = oldcbufnum;
FinishUp2:;
    if ( T->sparse ) ClearTableTree(T);
    if ( ( sparseflag & 2 ) != 0 ) {
        if ( T->spare == 0 ) { SpareTable(T); }
    }
    return(error);
}

/*
    #] DoTable :
    #[ CoTable :
*/

int CoTable(UBYTE *s)
{
    return(DoTable(s,2));
}

/*
    #] CoTable :
    #[ CoNTable :
*/
```

```
int CoNTable(UBYTE *s)
{
    return(DoTable(s,0));
}

/*
    #] CoNTable :
    #[ CoCTable :
*/

int CoCTable(UBYTE *s)
{
    return(DoTable(s,1));
}

/*
    #] CoCTable :
    #[ EmptyTable :
*/

void EmptyTable(TABLES T)
{
    int j;
    if ( T->sparse ) ClearTableTree(T);
    if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
    T->boomlijst = 0;
    for (j = 0; j < T->buffersfill; j++ ) { /* was <= */
        finishcbuf(T->buffers[j]);
    }
    if ( T->buffers ) M_free(T->buffers,"Table buffers");
    finishcbuf(T->bufnum);
    T->bufnum = inicbufs();
    T->bufferssize = 8;
    T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->bufferssize,"Table buffers");
    T->buffersfill = 0;
    T->buffers[T->buffersfill++] = T->bufnum;
    T->defined = T->mdefined = 0; T->flags = 0;
    T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
    T->spare = 0; T->reserved = 0;
    if ( T->spare ) {
        TABLES TT = T->spare;
        if ( TT->mm ) M_free(TT->mm,"tableminmax");
        if ( TT->flags ) M_free(TT->flags,"tableflags");
        if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
        for (j = 0; j < TT->buffersfill; j++ ) {
            finishcbuf(TT->buffers[j]);
        }
        if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
        if ( TT->buffers ) M_free(TT->buffers,"Table buffers");
        M_free(TT,"table");
        SpareTable(T);
    }
    else {
        WORD *w = T->tablepointers;
        j = T->totind;
        for ( j = TABLEEXTENSION*T->totind; j > 0; j-- ) *w++ = -1; /* means: undefined */
    }
}

/*
    #] EmptyTable :
    #[ AddSet :
*/

int AddSet(UBYTE *name, WORD dim)
{
    int nodenum, numset = AC.SetList.num;
    SETS set = (SETS)FromVarList(&AC.SetList);
    UBYTE *s;
    if ( name ) {
        set->name = AddName(AC.varnames,name,CSET,numset,&nodenum);
        s = name;
        while ( *s ) s++;
        set->namesize = (s-name)+1;
        set->node  = nodenum;
    }
    else {
        set->name = 0;
        set->namesize = 0;
        set->node  = -1;
    }
    set->first =
    set->last  = AC.SetElementList.num; /* set has no elements yet */
    set->type  = -1;                    /* undefined as of yet */
    set->dimension = dim;
    set->flags = 0;
```

```
    return(numset);
}

/*
    #] AddSet :
    #[ DoElements :

    Remark (25-mar-2011): If the dimension has been set (dim != MAXPOSITIVE)
    we want to test dimensions. Numbers count as dimension zero?
*/

int DoElements(UBYTE *s, SETS set, UBYTE *name)
{
    int type, error = 0, x, sgn, i;
    WORD numset, *e;
    UBYTE c, *cname;
    while ( *s ) {
        if ( *s == ',' ) { s++; continue; }
        sgn = 0;
        while ( *s == '-' || *s == '+' ) {
            if ( *s == '-' ) sgn ^= 1;
            s++;
        }
        cname = s;
        if ( FG.cTable[*s] == 0 || *s == '_' || *s == '[' ) {
            if ( ( s = SkipAName(s) ) == 0 ) {
                MesPrint("&Illegal name in set definition");
                return(1);
            }
            c = *s; *s = 0;
            if ( ( ( type = GetName(AC.exprnames,cname,&numset,NOAUTO) ) == NAMENOTFOUND )
            && ( ( type = GetOName(AC.varnames,cname,&numset,WITHAUTO) ) == NAMENOTFOUND ) ) {
                DUBIOUSV dv;
                int nodenum;
                MesPrint("&%s has not been declared",cname);
/*
                We enter a 'dubious' declaration to cut down on errors
*/
                numset = AC.DubiousList.num;
                dv = (DUBIOUSV)FromVarList(&AC.DubiousList);
                dv->name = AddName(AC.varnames,cname,CDUBIOUS,numset,&nodenum);
                dv->node = nodenum;
                set->type = type = CDUBIOUS;
                set->dimension = 0;
                error = 1;
            }
            if ( set->type == -1 ) {
                if ( type == CSYMBOL ) {
                    for ( i = set->first; i < set->last; i++ ) {
                        SetElements[i] += 2*MAXPOWER;
                    }
                }
                set->type = type;
            }
            if ( set->type != type && set->type != CDUBIOUS
            && type != CDUBIOUS ) {
                if ( set->type != CNUMBER || ( type != CSYMBOL
                && type != CINDEX ) ) {
                    MesPrint(
                    "&%s has not the same type as the other members of the set"
                    ,cname);
                    error = 1;
                    set->type = CDUBIOUS;
                }
                else {
                    if ( type == CSYMBOL ) {
                        for ( i = set->first; i < set->last; i++ ) {
                            SetElements[i] += 2*MAXPOWER;
                        }
                    }
                    set->type = type;
                }
            }
            if ( set->dimension != MAXPOSITIVE ) { /* Dimension check */
                switch ( set->type ) {
                    case CSYMBOL:
                        if ( symbols[numset].dimension != set->dimension ) {
                            MesPrint("&Dimension check failed in set %s, symbol %s",
                                VARNAME(Sets,(set-Sets)),
                                VARNAME(symbols,numset));
                            error = 1;
                            set->dimension = MAXPOSITIVE;
                        }
                        break;
                    case CVECTOR:
                        if ( vectors[numset-AM.OffsetVector].dimension != set->dimension ) {
                            MesPrint("&Dimension check failed in set %s, vector %s",
```

```
                            VARNAME(Sets,(set-Sets)),
                            VARNAME(vectors,(numset-AM.OffsetVector)));
                        error = 1;
                        set->dimension = MAXPOSITIVE;
                    }
                    break;
                case CFUNCTION:
                    if ( functions[numset-FUNCTION].dimension != set->dimension ) {
                        MesPrint("&Dimension check failed in set %s, function %s",
                            VARNAME(Sets,(set-Sets)),
                            VARNAME(functions,(numset-FUNCTION)));
                        error = 1;
                    }
                    break;
                        set->dimension = MAXPOSITIVE;
            }
        }
        if ( sgn ) {
            if ( type != CVECTOR ) {
                MesPrint("&Illegal use of - sign in set. Can use only with vector or number");
                error = 1;
            }
/*
            numset = AM.OffsetVector - numset;
            numset |= SPECMASK;
            numset = AM.OffsetVector - numset;
*/
            numset -= WILDMASK;
        }
        *s = c;
        if ( name == 0 && *s == '?' ) {
            s++;
            switch ( set->type ) {
                case CSYMBOL:
                    numset = -numset; break;
                case CVECTOR:
                    numset += WILDOFFSET; break;
                case CINDEX:
                    numset |= WILDMASK; break;
                case CFUNCTION:
                    numset |= WILDMASK; break;
            }
            AC.wildflag = 1;
        }
/*
        Now add the element to the set.
*/
        e = (WORD *)FromVarList(&AC.SetElementList);
        *e = numset;
        (set->last)++;
    }
    else if ( FG.cTable[*s] == 1 ) {
        ParseNumber(x,s)
        if ( sgn ) x = -x;
        if ( x >= MAXPOWER || x <= -MAXPOWER ||
        ( set->type == CINDEX && ( x < 0 || x >= AM.OffsetIndex ) ) ) {
            MesPrint("&Illegal value for set element: %d",x);
            if ( AC.firstconstindex ) {
            MesPrint("&0 <= Fixed indices < ConstIndex(which is %d)",
                AM.OffsetIndex-1);
            MesPrint("&For setting ConstIndex, read the chapter on the setup file");
            AC.firstconstindex = 0;
            }
            error = 1;
            x = 0;
        }
/*
        Check what is allowed with the type.
*/
        if ( set->type == -1 ) {
            if ( x < 0 || x >= AM.OffsetIndex ) {
                for ( i = set->first; i < set->last; i++ ) {
                    SetElements[i] += 2*MAXPOWER;
                }
                set->type = CSYMBOL;
            }
            else set->type = CNUMBER;
        }
        else if ( set->type == CDUBIOUS ) {}
        else if ( set->type == CNUMBER && x < 0 ) {
            for ( i = set->first; i < set->last; i++ ) {
                SetElements[i] += 2*MAXPOWER;
            }
            set->type = CSYMBOL;
        }
        else if ( set->type != CSYMBOL && ( x < 0 ||
        ( set->type != CINDEX && set->type != CNUMBER ) ) ) {
```

```
                MesPrint("&Illegal mixture of element types in set");
                error = 1;
                set->type = CDUBIOUS;
            }
/*
            Allocate an element
*/
            e = (WORD *)FromVarList(&AC.SetElementList);
            (set->last)++;
            if ( set->type == CSYMBOL ) *e = x + 2*MAXPOWER;
/*          else if ( set->type == CINDEX ) *e = x; */
            else *e = x;
        }
        else {
            MesPrint("&Illegal object in list of set elements");
            return(1);
        }
    }
    if ( error == 0 && ( ( set->flags & ORDEREDSET ) == ORDEREDSET ) ) {
/*
        The set->last-set->first list of numbers must be sorted.
        Because we plan here potentially thousands of elements we use
        a simple version of splitmerge. In ordered sets we can search
        later with a binary search.
*/
        SimpleSplitMerge(SetElements+set->first,set->last-set->first);
    }
    return(error);
}

/*
    #] DoElements :
    #[ CoSet :

    Set declarations.
*/

int CoSet(UBYTE *s)
{
    int type, error = 0, ordered = 0;
    UBYTE *name = s, c, *ss;
    SETS set;
    WORD numberofset, dim = MAXPOSITIVE;
#ifdef WITHFLOAT
/*-----------------------------------------------------------------*/
    {
        WORD numeq = 0;
        LONG x;
        ss = s;
        while ( *ss && *ss != ':' ) { if ( *ss == '=' ) numeq++; ss++; }
        if ( *ss == 0 && numeq == 1 ) { /* We have the Set var = value; variety */
            while ( FG.cTable[*s] == 0 ) s++;
            ss = s; c = *s; *s = 0;
            if ( c != '=' ) {
Proper:
                MesPrint("&Proper syntax for value-set is `Set name = value'");
                return(1);
            }
            x = 0; s++;
            while ( *s >= '0' && *s <= '9' ) x = 10*x + (*s++-'0');
            if ( *s ) goto Proper;
            if ( StrICmp(name,(UBYTE *)"maxweight") == 0 ) {
                AC.tMaxWeight = x;    /* Temporary. Made permanent later */
            }
            else if ( StrICmp(name,(UBYTE *)"defaultprecision") == 0 ) {
                AC.tDefaultPrecision = x;    /* Temporary. Made permanent later */
            }
            else {
                MesPrint("&Illegal subkey in value set: %s",name);
                return(1);
            }
        }
    }
/*-----------------------------------------------------------------*/
#endif
    if ( ( s = SkipAName(s) ) == 0 ) {
IllForm:MesPrint("&Illegal name for set");
        return(1);
    }
    c = *s; *s = 0;
    if ( TestName(name) ) goto IllForm;
    if ( ( ( type = GetName(AC.exprnames,name,&numberofset,NOAUTO) ) != NAMENOTFOUND )
    || ( ( type = GetName(AC.varnames,name,&numberofset,NOAUTO) ) != NAMENOTFOUND ) ) {
        if ( type != CSET ) NameConflict(type,name);
        else {
            MesPrint("&There is already a set with the name %s",name);
        }
```

```
        return(1);
    }
    if ( c == 0 ) {
        numberofset = AddSet(name,0);
        set = Sets + numberofset;
        return(0);      /* empty set */
    }
    *s = c; ss = s;     /* ss marks the end of the name */
    if ( *s == '(' ) {
        UBYTE *sss, cc;
        s++; sss = s;   /* Beginning of option */
        while ( *s != ',' && *s != ')' && *s ) s++;
        cc = *s; *s = 0;
        if ( StrICont(sss,(UBYTE *)"ordered") == 0 ) {
            ordered = ORDEREDSET;
        }
        else {
            MesPrint("&Error: Illegal option in set definition: %s",sss);
            error = 1;
        }
        *s = cc;
        if ( *s != ')' ) {
            MesPrint("&Error: Currently only one option allowed in set definition.");
            error = 1;
            while ( *s && *s != ')' ) s++;
        }
        s++;
    }
    if ( *s == '{' ) {
        s++;
        if ( ( *s == 'd' || *s == 'D' ) && s[1] == '=' ) {
            s += 2;
            if ( *s == '-' || *s == '+' || FG.cTable[*s] == 1 ) {
                ParseSignedNumber(dim,s)
                if ( dim < -HALFMAX || dim > HALFMAX ) {
                    MesPrint("&Warning: dimension of %s (%d) out of range"
                    ,name,dim);
                }
            }
            if ( *s != '}' ) goto IllDim;
            else s++;
        }
        else {
IllDim:     MesPrint("&Error: Illegal dimension field for set %s",name);
            error = 1;
            s = SkipField(s,0);
        }
        while ( *s == ',' ) s++;
    }
    c = *ss; *ss = 0;
    numberofset = AddSet(name,dim);
    *ss = c;
    set = Sets + numberofset;
    set->flags |= ordered;
    if ( *s != ':' ) {
        MesPrint("&Proper syntax is `Set name:elements'");
        return(1);
    }
    s++;
    error = DoElements(s,set,name);
    AC.SetList.numtemp = AC.SetList.num;
    AC.SetElementList.numtemp = AC.SetElementList.num;
    return(error);
}

/*
    #] CoSet :
    #[ DoTempSet :

        Gets a {} set definition and returns a set number if the set is
        properly structured. This number refers either to an already
        existing set, or to a set that is defined here.
        From and to refer to the contents. They exclude the {}.
*/

int DoTempSet(UBYTE *from, UBYTE *to)
{
    int i, num, j, sgn;
    WORD *e, *ep;
    UBYTE c;
    int setnum = AddSet(0,MAXPOSITIVE);
    SETS set = Sets + setnum, setp;
    set->name = -1;
    set->type = -1;
    c = *to; *to = 0;
    AC.wildflag = 0;
    while ( *from == ',' ) from++;
```

```
    if ( *from == '<' || *from == '>' ) {
        set->type = CRANGE;
        set->first = 3*MAXPOWER;
        set->last = -3*MAXPOWER;
        while ( *from == '<' || *from == '>' ) {
            if ( *from == '<' ) {
                j = 1; from++;
                if ( *from == '=' ) { from++; j++; }
            }
            else {
                j = -1; from++;
                if ( *from == '=' ) { from++; j--; }
            }
            sgn = 1;
            while ( *from == '-' || *from == '+' ) {
                if ( *from == '-' ) sgn = -sgn;
                from++;
            }
            ParseNumber(num,from)
            if ( *from && *from != ',' ) {
                MesPrint("&Illegal number in ranged set definition");
                return(-1);
            }
            if ( sgn < 0 ) num = -num;
            if ( num >= MAXPOWER || num <= -MAXPOWER ) {
                Warning("Value in ranged set too big. Adjusted to infinity.");
                if ( num > 0 ) num =  3*MAXPOWER;
                else           num = -3*MAXPOWER;
            }
            else if ( j ==  2 ) num += 2*MAXPOWER;
            else if ( j == -2 ) num -= 2*MAXPOWER;
            if ( j > 0 ) set->first = num;
            else set->last = num;
            while ( *from == ',' ) from++;
        }
        if ( *from ) {
            MesPrint("&Definition of ranged set contains illegal objects");
            return(-1);
        }
    }
    else if ( DoElements(from,set,(UBYTE *)0) != 0 ) {
        AC.SetElementList.num = set->first;
        AC.SetList.num--; *to = c;
        return(-1);
    }
    *to = c;
/*
    Now we have to test whether this set exists already.
*/
    num = set->last - set->first;
    for ( setp = Sets, i = 0; i < AC.SetList.num-1; i++, setp++ ) {
        if ( num != setp->last - setp->first ) continue;
        if ( set->type != setp->type ) continue;
        if ( set->type == CRANGE ) {
            if ( set->first == setp->first ) return(setp-Sets);
        }
        else {
            e = SetElements + set->first;
            ep = SetElements + setp->first;
            j = num;
            while ( --j >= 0 ) if ( *e++ != *ep++ ) break;
            if ( j < 0 ) {
                AC.SetElementList.num = set->first;
                AC.SetList.num--;
                return(setp - Sets);
            }
        }
    }
    return(setnum);
}

/*
    #] DoTempSet :
    #[ CoAuto :

    To prepare first:
        Use of the proper pointers in the various declaration routines
        Proper action in .store and .clear
*/

int CoAuto(UBYTE *inp)
{
    int retval;

    AC.Symbols = &(AC.AutoSymbolList);
    AC.Vectors = &(AC.AutoVectorList);
    AC.Indices = &(AC.AutoIndexList);
```

```
    AC.Functions = &(AC.AutoFunctionList);
    AC.activenames = &(AC.autonames);
    AC.AutoDeclareFlag = WITHAUTO;

    while ( *inp == ',' ) inp++;
    retval = CompileStatement(inp);

    AC.AutoDeclareFlag = 0;
    AC.Symbols = &(AC.SymbolList);
    AC.Vectors = &(AC.VectorList);
    AC.Indices = &(AC.IndexList);
    AC.Functions = &(AC.FunctionList);
    AC.activenames = &(AC.varnames);
    return(retval);
}

/*
    #] CoAuto :
    #[ AddDollar :

    The actual addition. Special routine for additions 'on the fly'
*/

int AddDollar(UBYTE *name, WORD type, WORD *start, LONG size)
{
    int nodenum, numdollar = AP.DollarList.num;
    WORD *s, *t;
    DOLLARS dol = (DOLLARS)FromVarList(&AP.DollarList);
    dol->name = AddName(AC.dollarnames,name,CDOLLAR,numdollar,&nodenum);
    dol->type = type;
    dol->node = nodenum;
    dol->zero = 0;
    dol->numdummies = 0;
#ifdef WITHPTHREADS
    dol->pthreadslockread = dummylock;
    dol->pthreadslockwrite = dummylock;
#endif
    dol->nfactors = 0;
    dol->factors = 0;
    AddRHS(AM.dbufnum,1);
    AddLHS(AM.dbufnum);
    if ( start && size > 0 ) {
        dol->size = size;
        dol->where =
        s = (WORD *)Malloc1((size+1)*sizeof(WORD),"$-variable contents");
        t = start;
        while ( --size >= 0 ) *s++ = *t++;
        *s = 0;
    }
    else { dol->where = &(AM.dollarzero); dol->size = 0; }
    cbuf[AM.dbufnum].rhs[numdollar] = dol->where;
    cbuf[AM.dbufnum].CanCommu[numdollar] = 0;
    cbuf[AM.dbufnum].NumTerms[numdollar] = 0;

    return(numdollar);
}

/*
    #] AddDollar :
    #[ ReplaceDollar :

    Replacements of dollar variables can happen at any time.
    For debugging purposes we should have a tracing facility.

    Not in use????
*/

int ReplaceDollar(WORD number, WORD newtype, WORD *newstart, LONG newsize)
{
    int error = 0;
    DOLLARS dol = Dollars + number;
    WORD *s, *t;
    LONG i;
    dol->type = newtype;
    if ( dol->size == newsize && newsize > 0 && newstart ) {
        s = dol->where; t = newstart; i = newsize;
        while ( --i >= 0 ) { if ( *s++ != *t++ ) break; }
        if ( i < 0 ) return(0);
    }
    if ( dol->where && dol->where != &(dol->zero) ) {
        M_free(dol->where,"dollar->where"); dol->where = &(dol->zero); dol->size = 0;
    }
    if ( newstart && newsize > 0 ) {
        dol->size = newsize;
        dol->where =
        s = (WORD *)Malloc1((newsize+1)*sizeof(WORD),"$-variable contents");
        t = newstart; i = newsize;
```

```
        while ( --i >= 0 ) *s++ = *t++;
        *s = 0;
    }
    return(error);
}

/*
    #] ReplaceDollar :
    #[ AddDubious :

    This adds a variable of which we do not know the proper type.
*/

int AddDubious(UBYTE *name)
{
    int nodenum, numdubious = AC.DubiousList.num;
    DUBIOUSV dub = (DUBIOUSV)FromVarList(&AC.DubiousList);
    dub->name = AddName(AC.varnames,name,CDUBIOUS,numdubious,&nodenum);
    dub->node = nodenum;
    return(numdubious);
}

/*
    #] AddDubious :
    #[ MakeDubious :
*/

int MakeDubious(NAMETREE *nametree, UBYTE *name, WORD *number)
{
    NAMENODE *n;
    int node, newnode, i;
    if ( nametree->namenode == 0 ) return(-1);
    newnode = nametree->headnode;
    do {
        node = newnode;
        n = nametree->namenode+node;
        if ( ( i = StrCmp(name,nametree->namebuffer+n->name) ) < 0 )
            newnode = n->left;
        else if ( i > 0 ) newnode = n->right;
        else {
            if ( n->type != CDUBIOUS ) {
                int numdubious = AC.DubiousList.num;
                FUNCTIONS dub = (FUNCTIONS)FromVarList(&AC.DubiousList);
                dub->name = n->name;
                n->number = numdubious;
            }
            *number = n->number;
            return(CDUBIOUS);
        }
    } while ( newnode >= 0 );
    return(-1);
}

/*
    #] MakeDubious :
    #[ NameConflict :
*/

static char *nametype[] = { "symbol", "index", "vector", "function",
        "set", "expression" };
static char *plural[] = { "","n","","","","n" };

int NameConflict(int type, UBYTE *name)
{
    if ( type == NAMENOTFOUND ) {
        MesPrint("&%s has not been declared",name);
    }
    else if ( type != CDUBIOUS )
        MesPrint("&%s has been declared as a%s %s already"
            ,name,plural[type],nametype[type]);
    return(1);
}

/*
    #] NameConflict :
    #[ AddExpression :
*/

int AddExpression(UBYTE *name, int x, int y)
{
    int nodenum, numexpr = AC.ExpressionList.num;
    EXPRESSIONS expr = (EXPRESSIONS)FromVarList(&AC.ExpressionList);
    UBYTE *s;
    expr->status = x;
    expr->printflag = y;
    PUTZERO(expr->onfile);
    PUTZERO(expr->size);
```

```
    expr->renum = 0;
    expr->renumlists = 0;
    expr->hidelevel = 0;
    expr->inmem = 0;
    expr->bracketinfo = expr->newbracketinfo = 0;
    if ( name ) {
        expr->name = AddName(AC.exprnames,name,CEXPRESSION,numexpr,&nodenum);
        expr->node = nodenum;
        expr->replace = NEWLYDEFINEDEXPRESSION ;
        s = name;
        while ( *s ) s++;
        expr->namesize = (s-name)+1;
    }
    else {
        expr->replace = REDEFINEDEXPRESSION;
        expr->name = AC.TransEname;
        expr->node = -1;
        expr->namesize = 0;
    }
    expr->vflags = 0;
    expr->numdummies = 0;
    expr->numfactors = 0;
#ifdef PARALLELCODE
    expr->partodo = 0;
#endif
    expr->uflags = 0;
    return(numexpr);
}

/*
    #] AddExpression :
    #[ GetLabel :
*/

int GetLabel(UBYTE *name)
{
    int i;
    LONG newnum;
    UBYTE **NewLabelNames;
    int *NewLabel;
    for ( i = 0; i < AC.NumLabels; i++ ) {
        if ( StrCmp(name,AC.LabelNames[i]) == 0 ) return(i);
    }
    if ( AC.NumLabels >= AC.MaxLabels ) {
        newnum = 2*AC.MaxLabels;
        if ( newnum == 0 ) newnum = 10;
        if ( newnum > 32765 ) newnum = 32765;
        if ( newnum == AC.MaxLabels ) {
            MesPrint("&More than 32765 labels in one module. Please simplify.");
            Terminate(-1);
        }
        NewLabelNames = (UBYTE **)Malloc1((sizeof(UBYTE *)+sizeof(int))
                *newnum,"Labels");
        NewLabel = (int *)(NewLabelNames+newnum);
        for ( i = 0; i< AC.MaxLabels; i++ ) {
            NewLabelNames[i] = AC.LabelNames[i];
            NewLabel[i] = AC.Labels[i];
        }
        if ( AC.LabelNames ) M_free(AC.LabelNames,"Labels");
        AC.LabelNames = NewLabelNames;
        AC.Labels = NewLabel;
        AC.MaxLabels = newnum;
    }
    i = AC.NumLabels++;
    AC.LabelNames[i] = strDup1(name,"Labels");
    AC.Labels[i] = -1;
    return(i);
}

/*
    #] GetLabel :
    #[ ResetVariables :

    Resets the variables.
    par = 0  The list of temporary sets (after each .sort)
    par = 1  The list of local variables (after each .store)
    par = 2  All variables (after each .clear)
*/

void ResetVariables(int par)
{
    int i, j;
    TABLES T;
    switch ( par ) {
    case 0 : /* Only the sets without a name */
        AC.SetList.num        = AC.SetList.numtemp;
        AC.SetElementList.num = AC.SetElementList.numtemp;
```

```
            break;
        case 2 :
            for ( i = AC.SymbolList.numclear; i < AC.SymbolList.num; i++ )
                AC.varnames->namenode[symbols[i].node].type = CDELETE;
            AC.SymbolList.num = AC.SymbolList.numglobal = AC.SymbolList.numclear;
            for ( i = AC.VectorList.numclear; i < AC.VectorList.num; i++ )
                AC.varnames->namenode[vectors[i].node].type = CDELETE;
            AC.VectorList.num = AC.VectorList.numglobal = AC.VectorList.numclear;
            for ( i = AC.IndexList.numclear; i < AC.IndexList.num; i++ )
                AC.varnames->namenode[indices[i].node].type = CDELETE;
            AC.IndexList.num = AC.IndexList.numglobal = AC.IndexList.numclear;
            for ( i = AC.FunctionList.numclear; i < AC.FunctionList.num; i++ ) {
                AC.varnames->namenode[functions[i].node].type = CDELETE;
                if ( ( T = functions[i].tabl ) != 0 ) {
                    if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
                    if ( T->prototype ) M_free(T->prototype,"tableprototype");
                    if ( T->mm ) M_free(T->mm,"tableminmax");
                    if ( T->flags ) M_free(T->flags,"tableflags");
                    if ( T->argtail ) M_free(T->argtail,"table arguments");
                    if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
                    for (j = 0; j < T->buffersfill; j++ ) { /* was <= */
                        finishcbuf(T->buffers[j]);
                    }
                    /*[07apr2004 mt]:*/ /*memory leak*/
                    if ( T->buffers ) M_free(T->buffers,"Table buffers");
                    /*:[07apr2004 mt]*/
                    finishcbuf(T->bufnum);
                    if ( T->spare ) {
                        TABLES TT = T->spare;
                        if ( TT->mm ) M_free(TT->mm,"tableminmax");
                        if ( TT->flags ) M_free(TT->flags,"tableflags");
                        if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
                        for (j = 0; j < TT->buffersfill; j++ ) { /* was <= */
                            finishcbuf(TT->buffers[j]);
                        }
                        if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
                        /*[07apr2004 mt]:*/ /*memory leak*/
                        if ( TT->buffers )M_free(TT->buffers,"Table buffers");
                        /*:[07apr2004 mt]*/
                        M_free(TT,"table");
                    }
                    M_free(T,"table");
                }
            }
            AC.FunctionList.num = AC.FunctionList.numglobal = AC.FunctionList.numclear;
            for ( i = AC.SetList.numclear; i < AC.SetList.num; i++ ) {
                if ( Sets[i].node >= 0 )
                    AC.varnames->namenode[Sets[i].node].type = CDELETE;
            }
            AC.SetList.numtemp = AC.SetList.num = AC.SetList.numglobal = AC.SetList.numclear;
            for ( i = AC.DubiousList.numclear; i < AC.DubiousList.num; i++ )
                AC.varnames->namenode[Dubious[i].node].type = CDELETE;
            AC.DubiousList.num = AC.DubiousList.numglobal = AC.DubiousList.numclear;
            AC.SetElementList.numtemp = AC.SetElementList.num =
                AC.SetElementList.numglobal = AC.SetElementList.numclear;
            CompactifyTree(AC.varnames,VARNAMES);
            AC.varnames->namefill = AC.varnames->globalnamefill = AC.varnames->clearnamefill;
            AC.varnames->nodefill = AC.varnames->globalnodefill = AC.varnames->clearnodefill;

            for ( i = AC.AutoSymbolList.numclear; i < AC.AutoSymbolList.num; i++ )
                AC.autonames->namenode[
                        ((SYMBOLS)(AC.AutoSymbolList.lijst))[i].node].type = CDELETE;
            AC.AutoSymbolList.num = AC.AutoSymbolList.numglobal
                                  = AC.AutoSymbolList.numclear;
            for ( i = AC.AutoVectorList.numclear; i < AC.AutoVectorList.num; i++ )
                AC.autonames->namenode[
                        ((VECTORS)(AC.AutoVectorList.lijst))[i].node].type = CDELETE;
            AC.AutoVectorList.num = AC.AutoVectorList.numglobal
                                  = AC.AutoVectorList.numclear;
            for ( i = AC.AutoIndexList.numclear; i < AC.AutoIndexList.num; i++ )
                AC.autonames->namenode[
                        ((INDICES)(AC.AutoIndexList.lijst))[i].node].type = CDELETE;
            AC.AutoIndexList.num = AC.AutoIndexList.numglobal
                                 = AC.AutoIndexList.numclear;
            for ( i = AC.AutoFunctionList.numclear; i < AC.AutoFunctionList.num; i++ ) {
                AC.autonames->namenode[
                        ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].node].type = CDELETE;
                if ( ( T = ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].tabl ) != 0 ) {
                    if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
                    if ( T->prototype ) M_free(T->prototype,"tableprototype");
                    if ( T->mm ) M_free(T->mm,"tableminmax");
                    if ( T->flags ) M_free(T->flags,"tableflags");
                    if ( T->argtail ) M_free(T->argtail,"table arguments");
                    if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
                    for (j = 0; j < T->buffersfill; j++ ) { /* was <= */
                        finishcbuf(T->buffers[j]);
                    }
```

```
                if ( T->spare ) {
                    TABLES TT = T->spare;
                    if ( TT->mm ) M_free(TT->mm,"tableminmax");
                    if ( TT->flags ) M_free(TT->flags,"tableflags");
                    if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
                    for (j = 0; j < TT->buffersfill; j++ ) { /* was <= */
                        finishcbuf(TT->buffers[j]);
                    }
                    if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
                    M_free(TT,"table");
                }
                M_free(T,"table");
            }
        }
        AC.AutoFunctionList.num = AC.AutoFunctionList.numglobal
                                = AC.AutoFunctionList.numclear;
        CompactifyTree(AC.autonames,AUTONAMES);
        AC.autonames->namefill = AC.autonames->globalnamefill
                               = AC.autonames->clearnamefill;
        AC.autonames->nodefill = AC.autonames->globalnodefill
                               = AC.autonames->clearnodefill;
        ReleaseTB();
        break;
    case 1 :
        for ( i = AC.SymbolList.numglobal; i < AC.SymbolList.num; i++ )
            AC.varnames->namenode[symbols[i].node].type = CDELETE;
        AC.SymbolList.num = AC.SymbolList.numglobal;
        for ( i = AC.VectorList.numglobal; i < AC.VectorList.num; i++ )
            AC.varnames->namenode[vectors[i].node].type = CDELETE;
        AC.VectorList.num = AC.VectorList.numglobal;
        for ( i = AC.IndexList.numglobal; i < AC.IndexList.num; i++ )
            AC.varnames->namenode[indices[i].node].type = CDELETE;
        AC.IndexList.num = AC.IndexList.numglobal;
        for ( i = AC.FunctionList.numglobal; i < AC.FunctionList.num; i++ ) {
            AC.varnames->namenode[functions[i].node].type = CDELETE;
            if ( ( T = functions[i].tabl ) != 0 ) {
                if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
                if ( T->prototype ) M_free(T->prototype,"tableprototype");
                if ( T->mm ) M_free(T->mm,"tableminmax");
                if ( T->flags ) M_free(T->flags,"tableflags");
                if ( T->argtail ) M_free(T->argtail,"table arguments");
                if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
                for (j = 0; j < T->buffersfill; j++ ) { /* was <= */
                    finishcbuf(T->buffers[j]);
                }
                /*[07apr2004 mt]:*/  /*memory leak*/
                if ( T->buffers ) M_free(T->buffers,"Table buffers");
                /*:[07apr2004 mt]*/
                finishcbuf(T->bufnum);
                if ( T->spare ) {
                    TABLES TT = T->spare;
                    if ( TT->mm ) M_free(TT->mm,"tableminmax");
                    if ( TT->flags ) M_free(TT->flags,"tableflags");
                    if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
                    for (j = 0; j < TT->buffersfill; j++ ) { /* was <= */
                        finishcbuf(TT->buffers[j]);
                    }
                    if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
                    /*[07apr2004 mt]:*/ /*memory leak*/
                    if ( TT->buffers ) M_free(TT->buffers,"Table buffers");
                    /*:[07apr2004 mt]*/
                    M_free(TT,"table");
                }
                M_free(T,"table");
            }
        }
#ifdef TABLECLEANUP
        {
        int j;
        WORD *tp;
        for ( i = 0; i < AC.FunctionList.numglobal; i++ ) {
/*
            Now, if the table definition is from after the .global
            while the function is from before, there is a problem.
            This could be resolved by defining CTable (=Table), Ntable
            and do away with the previous function definition.
*/
            if ( ( T = functions[i].tabl ) != 0 ) {
/*
                First restore overwritten definitions.
*/
                if ( T->sparse ) {
                    T->totind = T->mdefined;
                    for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {
                        tp += ABS(T->numind);
#if TABLEEXTENSION == 2
                        tp[0] = tp[1];
```

```
#else
                        tp[0] = tp[2];
                        tp[1] = tp[3];
                        tp[4] = tp[5];
#endif
                        tp += TABLEEXTENSION;
                    }
                    RedoTableTree(T,T->totind);
                    if ( T->spare ) {
                        TABLES TT = T->spare;
                        TT->totind = TT->mdefined;
                        for ( j = 0, tp = TT->tablepointers; j < TT->totind; j++ ) {
                            tp += ABS(TT->numind);
#if TABLEEXTENSION == 2
                            tp[0] = tp[1];
#else
                            tp[0] = tp[2];
                            tp[1] = tp[3];
                            tp[4] = tp[5];
#endif
                            tp += TABLEEXTENSION;
                        }
                        RedoTableTree(TT,TT->totind);
                        cbuf[TT->bufnum].numlhs = cbuf[TT->bufnum].mnumlhs;
                        cbuf[TT->bufnum].numrhs = cbuf[TT->bufnum].mnumrhs;
                    }
                }
                else {
                    for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {
#if TABLEEXTENSION == 2
                        tp[0] = tp[1];
#else
                        tp[0] = tp[2];
                        tp[1] = tp[3];
                        tp[4] = tp[5];
#endif
                    }
                    T->defined = T->mdefined;
                }
                cbuf[T->bufnum].numlhs = cbuf[T->bufnum].mnumlhs;
                cbuf[T->bufnum].numrhs = cbuf[T->bufnum].mnumrhs;
            }
        }
        }
#endif
        AC.FunctionList.num = AC.FunctionList.numglobal;
        for ( i = AC.SetList.numglobal; i < AC.SetList.num; i++ ) {
            if ( Sets[i].node >= 0 )
                AC.varnames->namenode[Sets[i].node].type = CDELETE;
        }
        AC.SetList.numtemp = AC.SetList.num = AC.SetList.numglobal;
        for ( i = AC.DubiousList.numglobal; i < AC.DubiousList.num; i++ )
            AC.varnames->namenode[Dubious[i].node].type = CDELETE;
        AC.DubiousList.num = AC.DubiousList.numglobal;
        AC.SetElementList.numtemp = AC.SetElementList.num =
            AC.SetElementList.numglobal;
        CompactifyTree(AC.varnames,VARNAMES);
        AC.varnames->namefill = AC.varnames->globalnamefill;
        AC.varnames->nodefill = AC.varnames->globalnodefill;

        for ( i = AC.AutoSymbolList.numglobal; i < AC.AutoSymbolList.num; i++ )
            AC.autonames->namenode[
                    ((SYMBOLS)(AC.AutoSymbolList.lijst))[i].node].type = CDELETE;
        AC.AutoSymbolList.num = AC.AutoSymbolList.numglobal;
        for ( i = AC.AutoVectorList.numglobal; i < AC.AutoVectorList.num; i++ )
            AC.autonames->namenode[
                    ((VECTORS)(AC.AutoVectorList.lijst))[i].node].type = CDELETE;
        AC.AutoVectorList.num = AC.AutoVectorList.numglobal;
        for ( i = AC.AutoIndexList.numglobal; i < AC.AutoIndexList.num; i++ )
            AC.autonames->namenode[
                    ((INDICES)(AC.AutoIndexList.lijst))[i].node].type = CDELETE;
        AC.AutoIndexList.num = AC.AutoIndexList.numglobal;
        for ( i = AC.AutoFunctionList.numglobal; i < AC.AutoFunctionList.num; i++ ) {
            AC.autonames->namenode[
                    ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].node].type = CDELETE;
            if ( ( T = ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].tabl ) != 0 ) {
                if ( T->tablepointers ) M_free(T->tablepointers,"tablepointers");
                if ( T->prototype ) M_free(T->prototype,"tableprototype");
                if ( T->mm ) M_free(T->mm,"tableminmax");
                if ( T->flags ) M_free(T->flags,"tableflags");
                if ( T->argtail ) M_free(T->argtail,"table arguments");
                if ( T->boomlijst ) M_free(T->boomlijst,"TableTree");
                for (j = 0; j < T->buffersfill; j++ ) { /* was <= */
                    finishcbuf(T->buffers[j]);
                }
                if ( T->spare ) {
                    TABLES TT = T->spare;
```

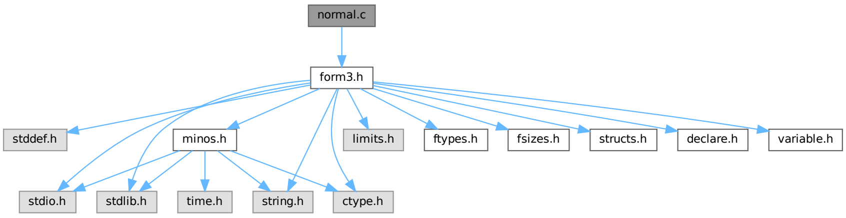
```
                        if ( TT->mm ) M_free(TT->mm,"tableminmax");
                        if ( TT->flags ) M_free(TT->flags,"tableflags");
                        if ( TT->tablepointers ) M_free(TT->tablepointers,"tablepointers");
                        for (j = 0; j < TT->buffersfill; j++ ) { /* was <= */
                            finishcbuf(TT->buffers[j]);
                        }
                        if ( TT->boomlijst ) M_free(TT->boomlijst,"TableTree");
                        M_free(TT,"table");
                    }
                    M_free(T,"table");
                }
            }
            AC.AutoFunctionList.num = AC.AutoFunctionList.numglobal;

            CompactifyTree(AC.autonames,AUTONAMES);

            AC.autonames->namefill = AC.autonames->globalnamefill;
            AC.autonames->nodefill = AC.autonames->globalnodefill;
            break;
    }
}

/*
    #] ResetVariables :
    #[ RemoveDollars :
*/

void RemoveDollars(void)
{
    DOLLARS d;
    CBUF *C = cbuf + AM.dbufnum;
    int numdollar = AP.DollarList.num;
    if ( numdollar > 0 ) {
      while ( numdollar > AM.gcNumDollars ) {
        numdollar--;
        d = Dollars + numdollar;
        if ( d->where && d->where != &(d->zero) && d->where != &(AM.dollarzero) ) {
            M_free(d->where,"dollar->where"); d->where = &(d->zero); d->size = 0;
        }
        AC.dollarnames->namenode[d->node].type = CDELETE;
      }
      AP.DollarList.num = AM.gcNumDollars;
      CompactifyTree(AC.dollarnames,DOLLARNAMES);

      C->numrhs = C->mnumrhs;
      C->numlhs = C->mnumlhs;
    }
}

/*
    #] RemoveDollars :
    #[ Globalize :
*/

void Globalize(int par)
{
    int i, j;
    WORD *tp;
    if ( par == 1 ) {
        AC.SymbolList.numclear     = AC.SymbolList.num;
        AC.VectorList.numclear     = AC.VectorList.num;
        AC.IndexList.numclear      = AC.IndexList.num;
        AC.FunctionList.numclear   = AC.FunctionList.num;
        AC.SetList.numclear        = AC.SetList.num;
        AC.DubiousList.numclear    = AC.DubiousList.num;
        AC.SetElementList.numclear = AC.SetElementList.num;
        AC.varnames->clearnamefill = AC.varnames->namefill;
        AC.varnames->clearnodefill = AC.varnames->nodefill;

        AC.AutoSymbolList.numclear   = AC.AutoSymbolList.num;
        AC.AutoVectorList.numclear   = AC.AutoVectorList.num;
        AC.AutoIndexList.numclear    = AC.AutoIndexList.num;
        AC.AutoFunctionList.numclear = AC.AutoFunctionList.num;
        AC.autonames->clearnamefill  = AC.autonames->namefill;
        AC.autonames->clearnodefill  = AC.autonames->nodefill;
    }
/*  for ( i = AC.FunctionList.numglobal; i < AC.FunctionList.num; i++ ) { */
    for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
/*
        We need here not only the not-yet-global functions. The already
        global ones may have obtained extra elements.
*/
        if ( functions[i].tabl ) {
            TABLES T = functions[i].tabl;
            if ( T->sparse ) {
                T->mdefined = T->totind;
                for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {
```

```
                    tp += ABS(T->numind);
#if TABLEEXTENSION == 2
                    tp[1] = tp[0];
#else
                    tp[2] = tp[0]; tp[3] = tp[1]; tp[5] = tp[4] & (~ELEMENTUSED);
#endif
                    tp += TABLEEXTENSION;
                }
                if ( T->spare ) {
                    TABLES TT = T->spare;
                    TT->mdefined = TT->totind;
                    for ( j = 0, tp = TT->tablepointers; j < TT->totind; j++ ) {
                        tp += ABS(TT->numind);
#if TABLEEXTENSION == 2
                        tp[1] = tp[0];
#else
                        tp[2] = tp[0]; tp[3] = tp[1]; tp[5] = tp[4] & (~ELEMENTUSED);
#endif
                        tp += TABLEEXTENSION;
                    }
                    cbuf[TT->bufnum].mnumlhs = cbuf[TT->bufnum].numlhs;
                    cbuf[TT->bufnum].mnumrhs = cbuf[TT->bufnum].numrhs;
                }
            }
            else {
                T->mdefined = T->defined;
                for ( j = 0, tp = T->tablepointers; j < T->totind; j++ ) {
#if TABLEEXTENSION == 2
                    tp[1] = tp[0];
#else
                    tp[2] = tp[0]; tp[3] = tp[1]; tp[5] = tp[4] & (~ELEMENTUSED);
#endif
                }
            }
            cbuf[T->bufnum].mnumlhs = cbuf[T->bufnum].numlhs;
            cbuf[T->bufnum].mnumrhs = cbuf[T->bufnum].numrhs;
        }
    }
    for ( i = AC.AutoFunctionList.numglobal; i < AC.AutoFunctionList.num; i++ ) {
        if ( ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].tabl )
            ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].tabl->mdefined =
                ((FUNCTIONS)(AC.AutoFunctionList.lijst))[i].tabl->defined;
    }
    AC.SymbolList.numglobal     = AC.SymbolList.num;
    AC.VectorList.numglobal     = AC.VectorList.num;
    AC.IndexList.numglobal      = AC.IndexList.num;
    AC.FunctionList.numglobal   = AC.FunctionList.num;
    AC.SetList.numglobal        = AC.SetList.num;
    AC.DubiousList.numglobal    = AC.DubiousList.num;
    AC.SetElementList.numglobal = AC.SetElementList.num;
    AC.varnames->globalnamefill = AC.varnames->namefill;
    AC.varnames->globalnodefill = AC.varnames->nodefill;

    AC.AutoSymbolList.numglobal   = AC.AutoSymbolList.num;
    AC.AutoVectorList.numglobal   = AC.AutoVectorList.num;
    AC.AutoIndexList.numglobal    = AC.AutoIndexList.num;
    AC.AutoFunctionList.numglobal = AC.AutoFunctionList.num;
    AC.autonames->globalnamefill  = AC.autonames->namefill;
    AC.autonames->globalnodefill  = AC.autonames->nodefill;
}

/*
    #] Globalize :
    #[ TestName :
*/

int TestName(UBYTE *name)
{
    if ( *name == '[' ) {
        while ( *name ) name++;
        if ( name[-1] == ']' ) return(0);
        return(-1);
    }
    while ( *name ) {
        if ( *name == '_' ) return(-1);
        name++;
    }
    return(0);
}

/*
    #] TestName :
*/
```

4.86 normal.c File Reference

```
#include "form3.h"
```

Include dependency graph for normal.c:



Macros

- #define MAXNUMBEROFNONCOMTERMS 2

Functions

- WORD CompareFunctions (WORD ∗fleft, WORD ∗fright)
- WORD Commute (WORD ∗fleft, WORD ∗fright)
- WORD Normalize (PHEAD WORD ∗term)
- WORD ExtraSymbol (WORD sym, WORD pow, WORD nsym, WORD ∗ppsym, WORD ∗ncoef)
- WORD DoTheta (PHEAD WORD ∗t)
- WORD DoDelta (WORD ∗t)
- void DoRevert (WORD ∗fun, WORD ∗tmp)
- WORD DetCommu (WORD ∗terms)
- WORD DoesCommu (WORD ∗term)
- int TreatPolyRatFun (PHEAD WORD ∗prf)
- void DropCoefficient (PHEAD WORD ∗term)
- void DropSymbols (PHEAD WORD ∗term)
- int SymbolNormalize (WORD ∗term)
- int TestFunFlag (PHEAD WORD ∗tfun)
- WORD BracketNormalize (PHEAD WORD ∗term)

4.86.1 Detailed Description

Mainly the routine Normalize. This routine brings terms to standard FORM. Currently it has one serious drawback. Its buffers are all in the stack. This means these buffers have a fixed size (NORMSIZE). In the past this has caused problems and NORMSIZE had to be increased.

It is not clear whether Normalize can be called recursively.

Definition in file normal.c.

4.86.2 Macro Definition Documentation

4.86.2.1 MAXNUMBEROFNONCOMTERMS

```
#define MAXNUMBEROFNONCOMTERMS 2
```

Definition at line 4508 of file normal.c.

4.86.3 Function Documentation

4.86.3.1 CompareFunctions()

```
WORD CompareFunctions (
            WORD * fleft,
            WORD * fright )
```

Definition at line 54 of file normal.c.

4.86.3.2 Commute()

```
WORD Commute (
            WORD * fleft,
            WORD * fright )
```

Definition at line 118 of file normal.c.

4.86.3.3 Normalize()

```
WORD Normalize (
            PHEAD WORD * term )
```

Definition at line 193 of file normal.c.

4.86.3.4 ExtraSymbol()

```
WORD ExtraSymbol (
            WORD sym,
            WORD pow,
            WORD nsym,
            WORD * ppsym,
            WORD * ncoef )
```

Definition at line 4196 of file normal.c.

4.86.3.5 DoTheta()

```
WORD DoTheta (
            PHEAD WORD * t )
```

Definition at line 4258 of file normal.c.

4.86.3.6 DoDelta()

```
WORD DoDelta (
            WORD * t )
```

Definition at line 4355 of file normal.c.

4.86.3.7 DoRevert()

```
void DoRevert (
            WORD * fun,
            WORD * tmp )
```

Definition at line 4422 of file normal.c.

4.86.3.8 DetCommu()

```
WORD DetCommu (
            WORD * terms )
```

Definition at line 4510 of file normal.c.

4.86.3.9 DoesCommu()

```
WORD DoesCommu (
            WORD * term )
```

Definition at line 4569 of file normal.c.

4.86.3.10 TreatPolyRatFun()

```
int TreatPolyRatFun (
            PHEAD WORD * prf )
```

Definition at line 4960 of file normal.c.

4.86.3.11 DropCoefficient()

```
void DropCoefficient (
            PHEAD WORD * term )
```

Definition at line 5096 of file normal.c.

4.86.3.12 DropSymbols()

```
void DropSymbols (
            PHEAD WORD * term )
```

Definition at line 5114 of file normal.c.

4.86.3.13 SymbolNormalize()

```
int SymbolNormalize (
            WORD * term )
```

Routine normalizes terms that contain only symbols. Regular minimum and maximum properties are ignored.

We check whether there are negative powers in the output. This is not allowed.

Definition at line 5144 of file normal.c.

Referenced by InFunction(), and poly_sort().

4.86.3.14 TestFunFlag()

```
int TestFunFlag (
            PHEAD WORD * tfun )
```

Definition at line 5240 of file normal.c.

4.86.3.15 BracketNormalize()

```
WORD BracketNormalize (
            PHEAD WORD * term )
```

Definition at line 5271 of file normal.c.

4.87 normal.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : normal.c
*/

#include "form3.h"
#ifdef WITHFLOAT
```

```
#include <gmp.h>

int PackFloat(WORD *,mpf_t);
int UnpackFloat(mpf_t, WORD *);
void RatToFloat(mpf_t result, UWORD *formrat, int ratsize);
#endif
/*
    #] Includes :
    #[ Normalize :
        #[ CompareFunctions :
*/

WORD CompareFunctions(WORD *fleft,WORD *fright)
{
    WORD k, kk;
    if ( AC.properorderflag ) {
        if ( ( *fleft >= (FUNCTION+WILDOFFSET)
        && functions[*fleft-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION )
        || ( *fleft >= FUNCTION && *fleft < (FUNCTION + WILDOFFSET)
        && functions[*fleft-FUNCTION].spec >= TENSORFUNCTION ) ) {}
        else {
            WORD *s1, *s2, *ss1, *ss2;
            s1 = fleft+FUNHEAD; s2 = fright+FUNHEAD;
            ss1 = fleft + fleft[1]; ss2 = fright + fright[1];
            while ( s1 < ss1 && s2 < ss2 ) {
                k = CompArg(s1,s2);
                if ( k > 0 ) return(1);
                if ( k < 0 ) return(0);
                NEXTARG(s1)
                NEXTARG(s2)
            }
            if ( s1 < ss1 ) return(1);
            return(0);
        }
        k = fleft[1] - FUNHEAD;
        kk = fright[1] - FUNHEAD;
        fleft += FUNHEAD;
        fright += FUNHEAD;
        while ( k > 0 && kk > 0 ) {
            if ( *fleft < *fright ) return(0);
            else if ( *fleft++ > *fright++ ) return(1);
            k--; kk--;
        }
        if ( k > 0 ) return(1);
        return(0);
    }
    else {
        k = fleft[1] - FUNHEAD;
        kk = fright[1] - FUNHEAD;
        fleft += FUNHEAD;
        fright += FUNHEAD;
        while ( k > 0 && kk > 0 ) {
            if ( *fleft < *fright ) return(0);
            else if ( *fleft++ > *fright++ ) return(1);
            k--; kk--;
        }
        if ( k > 0 ) return(1);
        return(0);
    }
}

/*
        #] CompareFunctions :
        #[ Commute :

    This function gets two adjacent function pointers and decides
    whether these two functions should be exchanged to obtain a
    natural ordering.

    Currently there is only an ordering of gamma matrices belonging
    to different spin lines.

    Note that we skip for now the cases of (F)^(3/2) or 1/F and a few more
    of such funny functions.
*/

WORD Commute(WORD *fleft, WORD *fright)
{
    WORD fun1, fun2;
    if ( *fleft == DOLLAREXPRESSION || *fright == DOLLAREXPRESSION ) return(0);
    fun1 = ABS(*fleft); fun2 = ABS(*fright);
    if ( *fleft >= GAMMA && *fleft <= GAMMASEVEN
        && *fright >= GAMMA && *fright <= GAMMASEVEN ) {
        if ( fleft[FUNHEAD] < AM.OffsetIndex && fleft[FUNHEAD] > fright[FUNHEAD] )
            return(1);
        return(0);
    }
```

```
    if ( fun1 >= WILDOFFSET ) fun1 -= WILDOFFSET;
    if ( fun2 >= WILDOFFSET ) fun2 -= WILDOFFSET;
    if ( ( ( functions[fun1-FUNCTION].flags & COULDCOMMUTE ) == 0 )
      || ( ( functions[fun2-FUNCTION].flags & COULDCOMMUTE ) == 0 ) ) return(0);
/*
    if other conditions will come here, keep in mind that if *fleft < 0
    or *fright < 0 they are arguments in the exponent function as in f^(3/2)
*/
    if ( AC.CommuteInSet == 0 ) return(0);
/*
    The code for CompareFunctions can be stolen from the commuting case.

    We need the syntax:
        Commute Fun1,Fun2,...,Fun'n';
    For this Fun1,...,Fun'n' need to be noncommuting functions.
    These functions will commute with all members of the group.
    In the AC.paircommute buffer the representation is
        `n'+1,element1,...,element'n','m'+1,element1,...,element'm',0
    A function can belong to more than one group.
    If a function commutes with itself, it is most efficient to make a separate
    group of two elements for it as in
        Commute T,T;   -> 3,T,T
*/
    if ( fun1 >= fun2 ) {
        WORD *group = AC.CommuteInSet, *g1, *g2, *g3;
        while ( *group > 0 ) {
            g3 = group + *group;
            g1 = group+1;
            while ( g1 < g3 ) {
                if ( *g1 == fun1 || ( fun1 <= GAMMASEVEN && fun1 >= GAMMA
                 && *g1 <= GAMMASEVEN && *g1 >= GAMMA ) ) {
                    g2 = group+1;
                    while ( g2 < g3 ) {
                        if ( g1 != g2 && ( *g2 == fun2 ||
                         ( fun2 <= GAMMASEVEN && fun2 >= GAMMA
                         && *g2 <= GAMMASEVEN && *g2 >= GAMMA ) ) ) {
                            if ( fun1 != fun2 ) return(1);
                            if ( *fleft < 0 ) return(0);
                            if ( *fright < 0 ) return(1);
                            return(CompareFunctions(fleft,fright));
                        }
                        g2++;
                    }
                    break;
                }
                g1++;
            }
            group = g3;
        }
    }
    return(0);
}

/*
        #] Commute :
        #[ Normalize :

    This is the big normalization routine. It has a great need
    to be economical.
    There is a fixed limit to the number of objects coming in.
    Something should be done about it.

*/

WORD Normalize(PHEAD WORD *term)
{
/*
    #[ Declarations :
*/
    GETBIDENTITY
    WORD *t, *m, *r, i, j, k, l, nsym, *ss, *tt, *u;
    WORD shortnum, stype;
    WORD *stop, *to = 0, *from = 0;
/*
    The next variables could be better off in the AT.WorkSpace (?)
    Now they make stackallocations rather bothersome.
*/
    WORD psym[7*NORMSIZE],*ppsym;
    WORD pvec[NORMSIZE],*ppvec,nvec;
    WORD pdot[3*NORMSIZE],*ppdot,ndot;
    WORD pdel[2*NORMSIZE],*ppdel,ndel;
    WORD pind[NORMSIZE],nind;
    WORD *peps[NORMSIZE/3],neps;
    WORD *pden[NORMSIZE/3],nden;
    WORD *pcom[NORMSIZE],ncom;
    WORD *pnco[NORMSIZE],nnco;
    WORD *pcon[2*NORMSIZE],ncon;        /* Pointer to contractable indices */
```

```
    WORD *n_coef, ncoef;                /* Accumulator for the coefficient */
    WORD *n_llnum, *lnum, nnum;
    WORD *termout, oldtoprhs = 0, subtype;
    WORD ReplaceType, ReplaceVeto = 0, didcontr, regval = 0;
    WORD *ReplaceSub;
    WORD *fillsetexp;
    CBUF *C = cbuf+AT.ebufnum;
    WORD *ANsc = 0, *ANsm = 0, *ANsr = 0, PolyFunMode;
#ifdef WITHFLOAT
    WORD withfloat = 0;
    WORD *firstfloat = 0;
#endif
    LONG oldcpointer = 0, x;
    n_coef = TermMalloc("NormCoef");
    n_llnum = TermMalloc("n_llnum");
    lnum = n_llnum+1;
/*
    int termflag;
*/
/*
    #] Declarations :
    #[ Setup :
PrintTerm(term,"Normalize");
*/

Restart:
    didcontr = 0;
    ReplaceType = -1;
    t = term;
    if ( !*t ) { TermFree(n_coef,"NormCoef"); TermFree(n_llnum,"n_llnum"); return(regval); }
    r = t + *t;
    ncoef = r[-1];
    i = ABS(ncoef);
    r -= i;
    m = r;
    t = n_coef;
    NCOPY(t,r,i);
    termout = AT.WorkPointer;
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    fillsetexp = termout+1;
    AN.PolyNormFlag = 0; PolyFunMode = AN.PolyFunTodo;
/*
    termflag = 0;
*/
/*
    #] Setup :
    #[ First scan :
*/
    nsym = nvec = ndot = ndel = neps = nden =
    nind = ncom = nnco = ncon = 0;
    ppsym = psym;
    ppvec = pvec;
    ppdot = pdot;
    ppdel = pdel;
    t = term + 1;
conscan:;
    if ( t < m ) do {
        r = t + t[1];
        switch ( *t ) {
            case SYMBOL :
                t += 2;
                from = m;
                do {
                    if ( t[1] == 0 ) {
/*                      if ( *t == 0 || *t == MAXPOWER ) goto NormZZ; */
                        t += 2;
                        goto NextSymbol;
                    }
                    if ( *t <= DENOMINATORSYMBOL && *t >= COEFFSYMBOL ) {
/*
                        if ( AN.NoScrat2 == 0 ) {
                            AN.NoScrat2 = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD),"Normalize");
                        }
*/
                        if ( AN.cTerm ) m = AN.cTerm;
                        else m = term;
                        m += *m;
                        ncoef = REDLENG(ncoef);
                        if ( *t == COEFFSYMBOL ) {
                          i = t[1];
                          nnum = REDLENG(m[-1]);
                          m -= ABS(m[-1]);
                          if ( i > 0 ) {
                            while ( i > 0 ) {
                                if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)m,nnum,
                                (UWORD *)n_coef,&ncoef) ) goto FromNorm;
                                i--;
```

```
                            }
                          }
                          else if ( i < 0 ) {
                            while ( i < 0 ) {
                                if ( DivRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)m,nnum,
                                (UWORD *)n_coef,&ncoef) ) goto FromNorm;
                                i++;
                            }
                          }
                        }
                        else {
                          i = m[-1];
                          nnum = (ABS(i)-1)/2;
                          if ( *t == NUMERATORSYMBOL ) { m -= nnum + 1; }
                          else { m--; }
                          while ( *m == 0 && nnum > 1 ) { m--; nnum--; }
                          m -= nnum;
                          if ( i < 0 && *t == NUMERATORSYMBOL ) nnum = -nnum;
                          i = t[1];
                          if ( i > 0 ) {
                            while ( i > 0 ) {
                                if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)m,nnum) )
                                        goto FromNorm;
                                i--;
                            }
                          }
                          else if ( i < 0 ) {
                            while ( i < 0 ) {
                                if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)m,nnum) )
                                        goto FromNorm;
                                i++;
                            }
                          }
                        }
                        ncoef = INCLENG(ncoef);
                        t += 2;
                        goto NextSymbol;
                    }
                    else if ( *t == DIMENSIONSYMBOL ) {
                        if ( AN.cTerm ) m = AN.cTerm;
                        else m = term;
                        k = DimensionTerm(m);
                        if ( k == 0 ) goto NormZero;
                        if ( k == MAXPOSITIVE ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Dimension_ is undefined in term %t");
                            MUNLOCK(ErrorMessageLock);
                            goto NormMin;
                        }
                        if ( k == -MAXPOSITIVE ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Dimension_ out of range in term %t");
                            MUNLOCK(ErrorMessageLock);
                            goto NormMin;
                        }
                        if ( k > 0 ) { *((UWORD *)lnum) = k; nnum = 1; }
                        else { *((UWORD *)lnum) = -k; nnum = -1; }
                        ncoef = REDLENG(ncoef);
                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) ) goto FromNorm;
                        ncoef = INCLENG(ncoef);
                        t += 2;
                        goto NextSymbol;
                    }
                    if ( ( *t >= MAXPOWER && *t < 2*MAXPOWER )
                        || ( *t < -MAXPOWER && *t > -2*MAXPOWER ) ) {
/*
            #[ TO SNUMBER :
*/
                if ( *t < 0 ) {
                    *t += MAXPOWER;
                    *t = -*t;
                    if ( t[1] & 1 ) ncoef = -ncoef;
                }
                else if ( *t == MAXPOWER ) {
                    if ( t[1] > 0 ) goto NormZero;
                    goto NormInf;
                }
                else {
                    *t -= MAXPOWER;
                }
                lnum[0] = *t;
                nnum = 1;
                if ( t[1] && RaisPow(BHEAD (UWORD *)lnum,&nnum,(UWORD)(ABS(t[1]))) )
                    goto FromNorm;
                ncoef = REDLENG(ncoef);
                if ( t[1] < 0 ) {
                    if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
```

```
                        goto FromNorm;
                }
                else if ( t[1] > 0 ) {
                    if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
                        goto FromNorm;
                }
                ncoef = INCLENG(ncoef);
/*
            #] TO SNUMBER :
*/
                        t += 2;
                        goto NextSymbol;
                    }
                    if ( ( *t <= NumSymbols && *t > -MAXPOWER )
                     && ( symbols[*t].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
                        if ( t[1] <= 2*MAXPOWER && t[1] >= -2*MAXPOWER ) {
                        t[1] %= symbols[*t].maxpower;
                        if ( t[1] < 0 ) t[1] += symbols[*t].maxpower;
                        if ( ( symbols[*t].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
                            if ( ( ( symbols[*t].maxpower & 1 ) == 0 ) &&
                            ( t[1] >= symbols[*t].maxpower/2 ) ) {
                                t[1] -= symbols[*t].maxpower/2; ncoef = -ncoef;
                            }
                        }
                        if ( t[1] == 0 ) { t += 2; goto NextSymbol; }
                        }
                    }
                    i = nsym;
                    m = ppsym;
                    if ( i > 0 ) do {
                        m -= 2;
                        if  ( *t == *m ) {
                            t++; m++;
                            if  ( *t > 2*MAXPOWER || *t < -2*MAXPOWER
                            ||  *m > 2*MAXPOWER || *m < -2*MAXPOWER ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Illegal wildcard power combination.");
                                MUNLOCK(ErrorMessageLock);
                                goto NormMin;
                            }
                            *m += *t;
                            if ( ( t[-1] <= NumSymbols && t[-1] > -MAXPOWER )
                             && ( symbols[t[-1]].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY
) {
                                *m %= symbols[t[-1]].maxpower;
                                if ( *m < 0 ) *m += symbols[t[-1]].maxpower;
                                if ( ( symbols[t[-1]].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
                                    if ( ( ( symbols[t[-1]].maxpower & 1 ) == 0 ) &&
                                    ( *m >= symbols[t[-1]].maxpower/2 ) ) {
                                        *m -= symbols[t[-1]].maxpower/2; ncoef = -ncoef;
                                    }
                                }
                            }
                            if  ( *m >= 2*MAXPOWER || *m <= -2*MAXPOWER ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Power overflow during normalization");
                                MUNLOCK(ErrorMessageLock);
                                goto NormMin;
                            }
                            if ( !*m ) {
                                m--;
                                while ( i < nsym )
                                    { *m = m[2]; m++; *m = m[2]; m++; i++; }
                                ppsym -= 2;
                                nsym--;
                            }
                            t++;
                            goto NextSymbol;
                        }
                    } while ( *t < *m && --i > 0 );
                    m = ppsym;
                    while ( i < nsym )
                        { m--; m[2] = *m; m--; m[2] = *m; i++; }
                    *m++ = *t++;
                    *m = *t++;
                    ppsym += 2;
                    nsym++;
NextSymbol:;
                } while ( t < r );
                m = from;
                break;
            case VECTOR :
                t += 2;
                do {
                    if ( t[0] == AM.vectorzero ) goto NormZero;
                    if ( t[1] == FUNNYVEC ) {
                        pind[nind++] = *t;
```

```
                        t += 2;
                    }
                    else if ( t[1] < 0 ) {
                        if ( *t == NOINDEX && t[1] == NOINDEX ) t += 2;
                        else {
                            if ( t[1] == AM.vectorzero ) goto NormZero;
                            *ppdot++ = *t++; *ppdot++ = *t++; *ppdot++ = 1; ndot++;
                        }
                    }
                    else { *ppvec++ = *t++; *ppvec++ = *t++; nvec += 2; }
                } while ( t < r );
                break;
            case DOTPRODUCT :
                t += 2;
                do {
                    if ( t[2] == 0 ) t += 3;
                    else if ( ndot > 0 && t[0] == ppdot[-3]
                        && t[1] == ppdot[-2] ) {
                        ppdot[-1] += t[2];
                        t += 3;
                        if ( ppdot[-1] == 0 ) { ppdot -= 3; ndot--; }
                    }
                    else if ( t[0] == AM.vectorzero || t[1] == AM.vectorzero ) {
                        if ( t[2] > 0 ) goto NormZero;
                        goto NormInf;
                    }
                    else {
                        *ppdot++ = *t++; *ppdot++ = *t++;
                        *ppdot++ = *t++; ndot++;
                    }
                } while ( t < r );
                break;
            case HAAKJE :
                break;
            case SETSET:
                if ( WildFill(BHEAD termout,term,AT.dummysubexp) < 0 ) goto FromNorm;
                i = *termout;
                t = termout; m = term;
                NCOPY(m,t,i);
                goto Restart;
            case DOLLAREXPRESSION :
/*
                We have DOLLAREXPRESSION,4,number,power
                Replace by SUBEXPRESSION and exit elegantly to let
                TestSub pick it up. Of course look for special cases first.
                Note that we have a special compiler buffer for the values.
*/
                if ( AR.Eside != LHSIDE ) {
                DOLLARS d = Dollars + t[2];
#ifdef WITHPTHREADS
                int nummodopt, ptype = -1;
                if ( AS.MultiThreaded ) {
                    for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                        if ( t[2] == ModOptdollars[nummodopt].number ) break;
                    }
                    if ( nummodopt < NumModOptdollars ) {
                        ptype = ModOptdollars[nummodopt].type;
                        if ( ptype == MODLOCAL ) {
                            d = ModOptdollars[nummodopt].dstruct+AT.identity;
                        }
                        else {
                            LOCK(d->pthreadslockread);
                        }
                    }
                }
#endif
                if ( d->type == DOLZERO ) {
#ifdef WITHPTHREADS
                    if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                    if ( t[3] == 0 ) goto NormZZ;
                    if ( t[3] < 0 ) goto NormInf;
                    goto NormZero;
                }
                else if ( d->type == DOLNUMBER ) {
                    nnum = d->where[0];
                    if ( nnum > 0 ) {
                        nnum = d->where[nnum-1];
                        if ( nnum < 0 ) { ncoef = -ncoef; nnum = -nnum; }
                        nnum = (nnum-1)/2;
                        for ( i = 1; i <= nnum; i++ ) lnum[i-1] = d->where[i];
                    }
                    if ( nnum == 0 || ( nnum == 1 && lnum[0] == 0 ) ) {
#ifdef WITHPTHREADS
                        if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                        if ( t[3] < 0 ) goto NormInf;
```

```
                        else if ( t[3] == 0 ) goto NormZZ;
                        goto NormZero;
                    }
                    if ( t[3] && RaisPow(BHEAD (UWORD *)lnum,&nnum,(UWORD)(ABS(t[3]))) ) goto FromNorm;
                    ncoef = REDLENG(ncoef);
                    if ( t[3] < 0 ) {
                        if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) ) {
#ifdef WITHPTHREADS
                            if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                            goto FromNorm;
                        }
                    }
                    else if ( t[3] > 0 ) {
                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) ) {
#ifdef WITHPTHREADS
                            if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                            goto FromNorm;
                        }
                    }
                    ncoef = INCLENG(ncoef);
                }
                else if ( d->type == DOLINDEX ) {
                    if ( d->index == 0 ) {
#ifdef WITHPTHREADS
                        if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                        goto NormZero;
                    }
                    if ( d->index != NOINDEX ) pind[nind++] = d->index;
                }
                else if ( d->type == DOLTERMS ) {
                    if ( t[3] >= MAXPOWER || t[3] <= -MAXPOWER ) {
                        if ( d->where[0] == 0 ) goto NormZero;
                        if ( d->where[d->where[0]] != 0 ) {
IllDollarExp:
                            MLOCK(ErrorMessageLock);
                            MesPrint("!!!Illegal $ expansion with wildcard power!!!");
                            MUNLOCK(ErrorMessageLock);
                            goto FromNorm;
                        }
/*
                        At this point we should only admit symbols and dotproducts
                        We expand the dollar directly and do not send it back.
*/
                        {   WORD *td, *tdstop, dj;
                            td = d->where+1;
                            tdstop = d->where+d->where[0];
                            if ( tdstop[-1] != 3 || tdstop[-2] != 1
                                || tdstop[-3] != 1 ) goto IllDollarExp;
                            tdstop -= 3;
                            if ( td >= tdstop ) goto IllDollarExp;
                            while ( td < tdstop ) {
                                if ( *td == SYMBOL ) {
                                    for ( dj = 2; dj < td[1]; dj += 2 ) {
                                        if ( td[dj+1] == 1 ) {
                                            *ppsym++ = td[dj];
                                            *ppsym++ = t[3];
                                            nsym++;
                                        }
                                        else if ( td[dj+1] == -1 ) {
                                            *ppsym++ = td[dj];
                                            *ppsym++ = -t[3];
                                            nsym++;
                                        }
                                        else goto IllDollarExp;
                                    }
                                }
                                else if ( *td == DOTPRODUCT ) {
                                    for ( dj = 2; dj < td[1]; dj += 3 ) {
                                        if ( td[dj+2] == 1 ) {
                                            *ppdot++ = td[dj];
                                            *ppdot++ = td[dj+1];
                                            *ppdot++ = t[3];
                                            ndot++;
                                        }
                                        else if ( td[dj+2] == -1 ) {
                                            *ppdot++ = td[dj];
                                            *ppdot++ = td[dj+1];
                                            *ppdot++ = -t[3];
                                            ndot++;
                                        }
                                        else goto IllDollarExp;
                                    }
                                }
```

```
                                else goto IllDollarExp;
                                td += td[1];
                            }
                            regval = 2;
                            break;
                        }
                    }
                    t[0] = SUBEXPRESSION;
                    t[4] = AM.dbufnum;
                    if ( t[3] == 0 ) {
#ifdef WITHPTHREADS
                        if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                        break;
                    }
                    regval = 2;
                    t = r;
                    while ( t < m ) {
                        if ( *t == DOLLAREXPRESSION ) {
#ifdef WITHPTHREADS
                            if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                            d = Dollars + t[2];
#ifdef WITHPTHREADS
                            if ( AS.MultiThreaded ) {
                                for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                                    if ( t[2] == ModOptdollars[nummodopt].number ) break;
                                }
                                if ( nummodopt < NumModOptdollars ) {
                                    ptype = ModOptdollars[nummodopt].type;
                                    if ( ptype == MODLOCAL ) {
                                        d = ModOptdollars[nummodopt].dstruct+AT.identity;
                                    }
                                    else {
                                        LOCK(d->pthreadslockread);
                                    }
                                }
                            }
#endif
                            if ( d->type == DOLTERMS ) {
                                t[0] = SUBEXPRESSION;
                                t[4] = AM.dbufnum;
                            }
                        }
                        t += t[1];
                    }
#ifdef WITHPTHREADS
                    if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                    goto RegEnd;
                }
                else {
#ifdef WITHPTHREADS
                    if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                    MLOCK(ErrorMessageLock);
                    MesPrint("!!!This $ variation has not been implemented yet!!!");
                    MUNLOCK(ErrorMessageLock);
                    goto NormMin;
                }
#ifdef WITHPTHREADS
                if ( ptype > 0 && ptype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                }
                else {
                    pnco[nnco++] = t;
/*
                    The next statement should be safe as the value is used
                    only by the compiler (ie the master).
*/
                    AC.lhdollarflag = 1;
                }
                break;
            case DELTA :
                t += 2;
                do {
                    if ( *t < 0 ) {
                        if ( *t == SUMMEDIND ) {
                            if ( t[1] < -NMIN4SHIFT ) {
                                k = -t[1]-NMIN4SHIFT;
                                k = ExtraSymbol(k,1,nsym,ppsym,&ncoef);
                                nsym += k;
                                ppsym += (k * 2);
                            }
                            else if ( t[1] == 0 ) goto NormZero;
                            else {
                                if ( t[1] < 0 ) {
```

```
                                             lnum[0] = -t[1];
                                             nnum = -1;
                                        }
                                        else {
                                             lnum[0] = t[1];
                                             nnum = 1;
                                        }
                                        ncoef = REDLENG(ncoef);
                                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
                                             goto FromNorm;
                                        ncoef = INCLENG(ncoef);
                                   }
                                   t += 2;
                              }
                              else if ( *t == NOINDEX && t[1] == NOINDEX ) t += 2;
                              else if ( *t == EMPTYINDEX && t[1] == EMPTYINDEX ) {
                                   *ppdel++ = *t++; *ppdel++ = *t++; ndel += 2;
                              }
                              else
                              if ( t[1] < 0 ) {
                                   *ppdot++ = *t++; *ppdot++ = *t++; *ppdot++ = 1; ndot++;
                              }
                              else {
                                   *ppvec++ = *t++; *ppvec++ = *t++; nvec += 2;
                              }
                         }
                         else {
                              if ( t[1] < 0 ) {
                                   *ppvec++ = t[1]; *ppvec++ = *t; t+=2; nvec += 2;
                              }
                              else { *ppdel++ = *t++; *ppdel++ = *t++; ndel += 2; }
                         }
                    } while ( t < r );
                    break;
               case FACTORIAL :
/*
                    (FACTORIAL,FUNHEAD+2,..,-SNUMBER,number)
*/
                    if ( t[FUNHEAD] == -SNUMBER && t[1] == FUNHEAD+2
                                                       && t[FUNHEAD+1] >= 0 ) {
                         if ( Factorial(BHEAD t[FUNHEAD+1],(UWORD *)lnum,&nnum) )
                              goto FromNorm;
MulIn:
                         ncoef = REDLENG(ncoef);
                         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) ) goto FromNorm;
                         ncoef = INCLENG(ncoef);
                    }
                    else pcom[ncom++] = t;
                    break;
               case BERNOULLIFUNCTION :
/*
                    (BERNOULLIFUNCTION,FUNHEAD+2,..,-SNUMBER,number)
*/
                    if ( ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] >= 0 )
                    && ( t[1] == FUNHEAD+2 || ( t[1] == FUNHEAD+4 &&
                    t[FUNHEAD+2] == -SNUMBER && ABS(t[FUNHEAD+3]) == 1 ) ) ) {
                         WORD inum, mnum;
                         if ( Bernoulli(t[FUNHEAD+1],(UWORD *)lnum,&nnum) )
                              goto FromNorm;
                         if ( nnum == 0 ) goto NormZero;
                         inum = nnum; if ( inum < 0 ) inum = -inum;
                         inum--; inum /= 2;
                         mnum = inum;
                         while ( lnum[mnum-1] == 0 ) mnum--;
                         if ( nnum < 0 ) mnum = -mnum;
                         ncoef = REDLENG(ncoef);
                         if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,mnum) ) goto FromNorm;
                         mnum = inum;
                         while ( lnum[inum+mnum-1] == 0 ) mnum--;
                         if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)(lnum+inum),mnum) ) goto FromNorm;
                         ncoef = INCLENG(ncoef);
                         if ( t[1] == FUNHEAD+4 && t[FUNHEAD+1] == 1
                          && t[FUNHEAD+3] == -1 ) ncoef = -ncoef;
                    }
                    else pcom[ncom++] = t;
                    break;
               case NUMARGSFUN:
/*
                    Numerical function giving the number of arguments.
*/
                    k = 0;
                    t += FUNHEAD;
                    while ( t < r ) {
                         k++;
                         NEXTARG(t);
                    }
```

```
                if ( k == 0 ) goto NormZero;
                *((UWORD *)lnum) = k;
                nnum = 1;
                goto MulIn;
            case NUMFACTORS:
/*
                Numerical function giving the number of factors in an expression.
*/
                t += FUNHEAD;
                if ( *t == -EXPRESSION ) {
                    k = AS.OldNumFactors[t[1]];
                }
                else if ( *t == -DOLLAREXPRESSION ) {
                    k = Dollars[t[1]].nfactors;
                }
                else {
                    pcom[ncom++] = t;
                    break;
                }
                if ( k == 0 ) goto NormZero;
                *((UWORD *)lnum) = k;
                nnum = 1;
                goto MulIn;
            case NUMTERMSFUN:
/*
                Numerical function giving the number of terms in the single argument.
*/
                if ( t[FUNHEAD] < 0 ) {
                    if ( t[FUNHEAD] <= -FUNCTION && t[1] == FUNHEAD+1 ) break;
                    if ( t[FUNHEAD] > -FUNCTION && t[1] == FUNHEAD+2 ) {
                        if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) goto NormZero;
                        break;
                    }
                    pcom[ncom++] = t;
                    break;
                }
                if ( t[FUNHEAD] > 0 && t[FUNHEAD] == t[1]-FUNHEAD ) {
                    k = 0;
                    t += FUNHEAD+ARGHEAD;
                    while ( t < r ) {
                        k++;
                        t += *t;
                    }
                    if ( k == 0 ) goto NormZero;
                    *((UWORD *)lnum) = k;
                    nnum = 1;
                    goto MulIn;
                }
                else pcom[ncom++] = t;
                break;
            case COUNTFUNCTION:
                if ( AN.cTerm ) {
                    k = CountFun(AN.cTerm,t);
                }
                else {
                    k = CountFun(term,t);
                }
                if ( k == 0 ) goto NormZero;
                if ( k > 0 ) { *((UWORD *)lnum) = k; nnum = 1; }
                else { *((UWORD *)lnum) = -k; nnum = -1; }
                goto MulIn;
                break;
            case MAKERATIONAL:
                if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+2] == -SNUMBER
                    && t[1] == FUNHEAD+4 && t[FUNHEAD+3] > 1 ) {
                    WORD x1[2], sgn;
                    if ( t[FUNHEAD+1] == 0 ) goto NormZero;
                    if ( t[FUNHEAD+1] < 0 ) { t[FUNHEAD+1] = -t[FUNHEAD+1]; sgn = -1; }
                    else sgn = 1;
                    if ( MakeRational(t[FUNHEAD+1],t[FUNHEAD+3],x1,x1+1) ) {
                        static int warnflag = 1;
                        if ( warnflag ) {
                            MesPrint("%w Warning: fraction could not be reconstructed in MakeRational_");
                            warnflag = 0;
                        }
                        x1[0] = t[FUNHEAD+1]; x1[1] = 1;
                    }
                    if ( sgn < 0 ) { t[FUNHEAD+1] = -t[FUNHEAD+1]; x1[0] = -x1[0]; }
                    if ( x1[0] < 0 ) { sgn = -1; x1[0] = -x1[0]; }
                    else sgn = 1;
                    ncoef = REDLENG(ncoef);
                    if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)x1,sgn,
                    (UWORD *)n_coef,&ncoef) ) goto FromNorm;
                    ncoef = INCLENG(ncoef);
                }
                else {
```

```
				WORD narg = 0, *tt, *ttstop, *arg1 = 0, *arg2 = 0;
				UWORD *x1, *x2, *xx;
				WORD nx1,nx2,nxx;
				ttstop = t + t[1]; tt = t+FUNHEAD;
				while ( tt < ttstop ) {
					narg++;
					if ( narg == 1 ) arg1 = tt;
					else arg2 = tt;
					NEXTARG(tt);
				}
				if ( narg != 2 ) goto defaultcase;
				if ( *arg2 == -SNUMBER && arg2[1] <= 1 ) goto defaultcase;
				else if ( *arg2 > 0 && ttstop[-1] < 0 ) goto defaultcase;
				x1 = NumberMalloc("Norm-MakeRational");
				if ( *arg1 == -SNUMBER ) {
					if ( arg1[1] == 0 ) {
						NumberFree(x1,"Norm-MakeRational");
						goto NormZero;
					}
					if ( arg1[1] < 0 ) { x1[0] = -arg1[1]; nx1 = -1; }
					else { x1[0] = arg1[1]; nx1 = 1; }
				}
				else if ( *arg1 > 0 ) {
					WORD *tc;
					nx1 = (ABS(arg2[-1])-1)/2;
					tc = arg1+ARGHEAD+1+nx1;
					if ( tc[0] != 1 ) {
						NumberFree(x1,"Norm-MakeRational");
						goto defaultcase;
					}
					for ( i = 1; i < nx1; i++ ) if ( tc[i] != 0 ) {
						NumberFree(x1,"Norm-MakeRational");
						goto defaultcase;
					}
					tc = arg1+ARGHEAD+1;
					for ( i = 0; i < nx1; i++ ) x1[i] = tc[i];
					if ( arg2[-1] < 0 ) nx1 = -nx1;
				}
				else {
					NumberFree(x1,"Norm-MakeRational");
					goto defaultcase;
				}
				x2 = NumberMalloc("Norm-MakeRational");
				if ( *arg2 == -SNUMBER ) {
					if ( arg2[1] <= 1 ) {
						NumberFree(x2,"Norm-MakeRational");
						NumberFree(x1,"Norm-MakeRational");
						goto defaultcase;
					}
					else { x2[0] = arg2[1]; nx2 = 1; }
				}
				else if ( *arg2 > 0 ) {
					WORD *tc;
					nx2 = (ttstop[-1]-1)/2;
					tc = arg2+ARGHEAD+1+nx2;
					if ( tc[0] != 1 ) {
						NumberFree(x2,"Norm-MakeRational");
						NumberFree(x1,"Norm-MakeRational");
						goto defaultcase;
					}
					for ( i = 1; i < nx2; i++ ) if ( tc[i] != 0 ) {
						NumberFree(x2,"Norm-MakeRational");
						NumberFree(x1,"Norm-MakeRational");
						goto defaultcase;
					}
					tc = arg2+ARGHEAD+1;
					for ( i = 0; i < nx2; i++ ) x2[i] = tc[i];
				}
				else {
					NumberFree(x2,"Norm-MakeRational");
					NumberFree(x1,"Norm-MakeRational");
					goto defaultcase;
				}
				if ( BigLong(x1,ABS(nx1),x2,nx2) >= 0 ) {
					UWORD *x3 = NumberMalloc("Norm-MakeRational");
					UWORD *x4 = NumberMalloc("Norm-MakeRational");
					WORD nx3, nx4;
					DivLong(x1,nx1,x2,nx2,x3,&nx3,x4,&nx4);
					for ( i = 0; i < ABS(nx4); i++ ) x1[i] = x4[i];
					nx1 = nx4;
					NumberFree(x4,"Norm-MakeRational");
					NumberFree(x3,"Norm-MakeRational");
				}
				xx = (UWORD *)(TermMalloc("Norm-MakeRational"));
				if ( MakeLongRational(BHEAD x1,nx1,x2,nx2,xx,&nxx) ) {
					static int warnflag = 1;
					if ( warnflag ) {
```

```
                            MesPrint("%w Warning: fraction could not be reconstructed in MakeRational_");
                            warnflag = 0;
                        }
                        ncoef = REDLENG(ncoef);
                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,x1,nx1) )
                            goto FromNorm;
                    }
                    else {
                        ncoef = REDLENG(ncoef);
                        if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,xx,nxx,
                        (UWORD *)n_coef,&ncoef) ) goto FromNorm;
                    }
                    ncoef = INCLENG(ncoef);
                    TermFree(xx,"Norm-MakeRational");
                    NumberFree(x2,"Norm-MakeRational");
                    NumberFree(x1,"Norm-MakeRational");
                }
                break;
            case TERMFUNCTION:
                if ( t[1] == FUNHEAD && AN.cTerm ) {
                    ANsr = r; ANsm = m; ANsc = AN.cTerm;
                    AN.cTerm = 0;
                    t = ANsc + 1;
                    m = ANsc + *ANsc;
                    ncoef = REDLENG(ncoef);
                    nnum = REDLENG(m[-1]);
                    m -= ABS(m[-1]);
                    if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)m,nnum,
                    (UWORD *)n_coef,&ncoef) ) goto FromNorm;
                    ncoef = INCLENG(ncoef);
                    r = t;
                }
                break;
            case FIRSTBRACKET:
                if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
                    if ( GetFirstBracket(termout,t[FUNHEAD+1]) < 0 ) goto FromNorm;
                    if ( *termout == 0 ) goto NormZero;
                    if ( *termout > 4 ) {
                        WORD *r1, *r2, *r3;
                        while ( r < m ) *t++ = *r++;
                        r1 = term + *term;
                        r2 = termout + *termout; r2 -= ABS(r2[-1]);
                        while ( r < r1 ) *r2++ = *r++;
                        r3 = termout + 1;
                        while ( r3 < r2 ) *t++ = *r3++;
                        *term = t - term;
                        if ( AT.WorkPointer > term && AT.WorkPointer < t )
                            AT.WorkPointer = t;
                        goto Restart;
                    }
                }
                break;
            case FIRSTTERM:
            case CONTENTTERM:
                if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
                    {
                        EXPRESSIONS e = Expressions+t[FUNHEAD+1];
                        POSITION oldondisk = AS.OldOnFile[t[FUNHEAD+1]];
                        if ( e->replace == NEWLYDEFINEDEXPRESSION ) {
                            AS.OldOnFile[t[FUNHEAD+1]] = e->onfile;
                        }
                        if ( *t == FIRSTTERM ) {
                            if ( GetFirstTerm(termout,t[FUNHEAD+1],0) < 0 ) goto FromNorm;
                        }
                        else if ( *t == CONTENTTERM ) {
                            if ( GetContent(termout,t[FUNHEAD+1]) < 0 ) goto FromNorm;
                        }
                        AS.OldOnFile[t[FUNHEAD+1]] = oldondisk;
                        if ( *termout == 0 ) goto NormZero;
                    }
PasteIn:;
                    {
                        WORD *r1, *r2, *r3, *r4, *r5, nr1, *rterm;
                        r2 = termout + *termout; lnum = r2 - ABS(r2[-1]);
                        nnum = REDLENG(r2[-1]);

                        r1 = term + *term; r3 = r1 - ABS(r1[-1]);
                        nr1 = REDLENG(r1[-1]);
                        if ( Mully(BHEAD (UWORD *)lnum,&nnum,(UWORD *)r3,nr1) ) goto FromNorm;
                        nnum = INCLENG(nnum); nr1 = ABS(nnum); lnum[nr1-1] = nnum;
                        rterm = TermMalloc("FirstTerm/ContentTerm");
                        r4 = rterm+1; r5 = term+1; while ( r5 < t ) *r4++ = *r5++;
                        r5 = termout+1; while ( r5 < lnum ) *r4++ = *r5++;
                        r5 = r; while ( r5 < r3 ) *r4++ = *r5++;
                        r5 = lnum; NCOPY(r4,r5,nr1);
                        *rterm = r4-rterm;
```

```
                        nr1 = *rterm; r1 = term; r2 = rterm; NCOPY(r1,r2,nr1);
                        TermFree(rterm,"FirstTerm/ContentTerm");
                        if ( AT.WorkPointer > term && AT.WorkPointer < r1 )
                            AT.WorkPointer = r1;
                        goto Restart;
                    }
                }
                else if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -DOLLAREXPRESSION ) {
                    DOLLARS d = Dollars + t[FUNHEAD+1], newd = 0;
                    int idol, ido;
#ifdef WITHPTHREADS
                    int nummodopt, dtype = -1;
                    if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
                        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                            if ( t[FUNHEAD+1] == ModOptdollars[nummodopt].number ) break;
                        }
                        if ( nummodopt < NumModOptdollars ) {
                            dtype = ModOptdollars[nummodopt].type;
                            if ( dtype == MODLOCAL ) {
                                d = ModOptdollars[nummodopt].dstruct+AT.identity;
                            }
                        }
                    }
#endif
                    if ( d->where && ( d->type == DOLTERMS || d->type == DOLNUMBER ) ) {
                        newd = d;
                    }
                    else {
                        if ( ( newd = DolToTerms(BHEAD t[FUNHEAD+1]) ) == 0 )
                            goto NormZero;
                    }
                    if ( newd->where[0] == 0 ) {
                        M_free(newd,"Copy of dollar variable");
                        goto NormZero;
                    }
                    if ( *t == FIRSTTERM ) {
                        idol = newd->where[0];
                        for ( ido = 0; ido < idol; ido++ ) termout[ido] = newd->where[ido];
                    }
                    else if ( *t == CONTENTTERM ) {
                        WORD *tterm;
                        tterm = newd->where;
                        idol = tterm[0];
                        for ( ido = 0; ido < idol; ido++ ) termout[ido] = tterm[ido];
                        tterm += *tterm;
                        while ( *tterm ) {
                            if ( ContentMerge(BHEAD termout,tterm) < 0 ) goto FromNorm;
                            tterm += *tterm;
                        }
                    }
                    if ( newd != d ) {
                        if ( newd->factors ) M_free(newd->factors,"Dollar factors");
                        M_free(newd,"Copy of dollar variable");
                        newd = 0;
                    }
                    goto PasteIn;
                }
                break;
            case TERMSINEXPR:
                {
                if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
                    x = TermsInExpression(t[FUNHEAD+1]);
multermnum:         if ( x == 0 ) goto NormZero;
                    if ( x < 0 ) {
                        x = -x;
                        if ( x > (LONG)WORDMASK ) { lnum[0] = x & WORDMASK;
                            lnum[1] = x » BITSINWORD; nnum = -2;
                        }
                        else { lnum[0] = x; nnum = -1; }
                    }
                    else if ( x > (LONG)WORDMASK ) {
                        lnum[0] = x & WORDMASK;
                        lnum[1] = x » BITSINWORD;
                        nnum = 2;
                    }
                    else { lnum[0] = x; nnum = 1; }
                    ncoef = REDLENG(ncoef);
                    if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
                        goto FromNorm;
                    ncoef = INCLENG(ncoef);
                }
                else if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -DOLLAREXPRESSION ) {
                    x = TermsInDollar(t[FUNHEAD+1]);
                    goto multermnum;
                }
                else { pcom[ncom++] = t; }
                }
```

```
                    break;
                case SIZEOFFUNCTION:
                    {
                    if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -EXPRESSION ) {
                        x = SizeOfExpression(t[FUNHEAD+1]);
                        goto multermnum;
                    }
                    else if ( ( t[1] == FUNHEAD+2 ) && t[FUNHEAD] == -DOLLAREXPRESSION ) {
                        x = SizeOfDollar(t[FUNHEAD+1]);
                        goto multermnum;
                    }
                    else { pcom[ncom++] = t; }
                    }
                    break;
                case MATCHFUNCTION:
                case PATTERNFUNCTION:
                    break;
                case BINOMIAL:
/*
                    Binomial function for internal use for the moment.
                    The routine in reken.c should be more efficient.
*/
                    if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SNUMBER
                    && t[FUNHEAD+1] >= 0 && t[FUNHEAD+2] == -SNUMBER
                    && t[FUNHEAD+3] >= 0 && t[FUNHEAD+1] >= t[FUNHEAD+3] ) {
                        if ( t[FUNHEAD+1] > t[FUNHEAD+3] ) {
                            if ( GetBinom((UWORD *)lnum,&nnum,
                            t[FUNHEAD+1],t[FUNHEAD+3]) ) goto FromNorm;
                            if ( nnum == 0 ) goto NormZero;
                            goto MulIn;
                        }
                    }
                    else pcom[ncom++] = t;
                    break;
                case SIGNFUN:
/*
                    Numerical function giving (-1)^arg
*/
                    if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
                        if ( ( t[FUNHEAD+1] & 1 ) != 0 ) ncoef = -ncoef;
                    }
                    else if ( ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+t[FUNHEAD] )
                    && ( t[FUNHEAD] == ARGHEAD+1+abs(t[t[1]-1]) ) ) {
                        UWORD *numer1,*denom1;
                        WORD nsize = abs(t[t[1]-1]), nnsize, isize;
                        nnsize = (nsize-1)/2;
                        numer1 = (UWORD *)(t + FUNHEAD+ARGHEAD+1);
                        denom1 = numer1 + nnsize;
                        for ( isize = 1; isize < nnsize; isize++ ) {
                            if ( denom1[isize] ) break;
                        }
                        if ( ( denom1[0] != 1 ) || isize < nnsize ) {
                            pcom[ncom++] = t;
                        }
                        else {
                            if ( ( numer1[0] & 1 ) != 0 ) ncoef = -ncoef;
                        }
                    }
                    else {
                        goto doflags;
/*                      pcom[ncom++] = t; */
                    }
                    break;
                case SIGFUNCTION:
/*
                    Numerical function giving the sign of the numerical argument
                    The sign of zero is 1.
                    If there are roots of unity they are part of the sign.
*/
                    if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
                        if ( t[FUNHEAD+1] < 0 ) ncoef = -ncoef;
                    }
                    else if ( ( t[1] == FUNHEAD+2 ) && ( t[FUNHEAD] == -SYMBOL )
                        && ( ( t[FUNHEAD+1] <= NumSymbols && t[FUNHEAD+1] > -MAXPOWER )
                        && ( symbols[t[FUNHEAD+1]].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY )
      ) {
                        k = t[FUNHEAD+1];
                        from = m;
                        i = nsym;
                        m = ppsym;
                        if ( i > 0 ) do {
                            m -= 2;
                            if  ( k == *m ) {
                                m++;
                                *m = *m + 1;
                                *m %= symbols[k].maxpower;
                                if ( ( symbols[k].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
```

```
                                if ( ( ( symbols[k].maxpower & 1 ) == 0 ) &&
                                ( *m >= symbols[k].maxpower/2 ) ) {
                                    *m -= symbols[k].maxpower/2; ncoef = -ncoef;
                                }
                            }
                            if ( !*m ) {
                                m--;
                                while ( i < nsym )
                                    { *m = m[2]; m++; *m = m[2]; m++; i++; }
                                ppsym -= 2;
                                nsym--;
                            }
                            goto sigDoneSymbol;
                        }
                    } while ( k < *m && --i > 0 );
                    m = ppsym;
                    while ( i < nsym )
                        { m--; m[2] = *m; m--; m[2] = *m; i++; }
                    *m++ = k;
                    *m = 1;
                    ppsym += 2;
                    nsym++;
sigDoneSymbol:;
                    m = from;
                }
                else if ( ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+t[FUNHEAD] ) ) {
                    if ( t[FUNHEAD] == ARGHEAD+1+abs(t[t[1]-1]) ) {
                        if ( t[t[1]-1] < 0 ) ncoef = -ncoef;
                    }
/*
                    Now we should fish out the roots of unity
*/
                    else if ( ( t[FUNHEAD+ARGHEAD]+FUNHEAD+ARGHEAD == t[1] )
                    && ( t[FUNHEAD+ARGHEAD+1] == SYMBOL ) ) {
                        WORD *ts = t + FUNHEAD+ARGHEAD+3;
                        WORD its = ts[-1]-2;
                        while ( its > 0 ) {
                            if ( ( *ts != 0 ) && ( ( *ts > NumSymbols || *ts <= -MAXPOWER )
                             || ( symbols[*ts].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
) {
                                goto signogood;
                            }
                            ts += 2; its -= 2;
                        }
/*
                        Now we have only roots of unity which should be
                        registered in the list of symbols.
*/
                        if ( t[t[1]-1] < 0 ) ncoef = -ncoef;
                        ts = t + FUNHEAD+ARGHEAD+3;
                        its = ts[-1]-2;
                        from = m;
                        while ( its > 0 ) {
                            i = nsym;
                            m = ppsym;
                            if ( i > 0 ) do {
                                m -= 2;
                                if  ( *ts == *m ) {
                                    ts++; m++;
                                    *m += *ts;
                                    if ( ( ts[-1] <= NumSymbols && ts[-1] > -MAXPOWER ) &&
                                     ( symbols[ts[-1]].complex & VARTYPEROOTOFUNITY ) ==
VARTYPEROOTOFUNITY ) {
                                        *m %= symbols[ts[-1]].maxpower;
                                        if ( *m < 0 ) *m += symbols[ts[-1]].maxpower;
                                        if ( ( symbols[ts[-1]].complex & VARTYPEMINUS ) ==
VARTYPEMINUS ) {
                                            if ( ( ( symbols[ts[-1]].maxpower & 1 ) == 0 ) &&
                                            ( *m >= symbols[ts[-1]].maxpower/2 ) ) {
                                                *m -= symbols[ts[-1]].maxpower/2; ncoef = -ncoef;
                                            }
                                        }
                                    }
                                    if ( !*m ) {
                                        m--;
                                        while ( i < nsym )
                                            { *m = m[2]; m++; *m = m[2]; m++; i++; }
                                        ppsym -= 2;
                                        nsym--;
                                    }
                                    ts++; its -= 2;
                                    goto sigNextSymbol;
                                }
                            } while ( *ts < *m && --i > 0 );
                            m = ppsym;
                            while ( i < nsym )
                                { m--; m[2] = *m; m--; m[2] = *m; i++; }
```

```
                            *m++ = *ts++;
                            *m = *ts++;
                            ppsym += 2;
                            nsym++; its -= 2;
sigNextSymbol:;
                        }
                        m = from;
                    }
                    else {
signogood:              pcom[ncom++] = t;
                    }
                }
                else pcom[ncom++] = t;
                break;
            case ABSFUNCTION:
/*
                Numerical function giving the absolute value of the
                numerical argument. Or roots of unity.
*/
                if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
                    k = t[FUNHEAD+1];
                    if ( k < 0 ) k = -k;
                    if ( k == 0 ) goto NormZero;
                    *((UWORD *)lnum) = k; nnum = 1;
                    goto MulIn;

                }
                else if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SYMBOL ) {
                    k = t[FUNHEAD+1];
                    if ( ( k > NumSymbols || k <= -MAXPOWER )
                     || ( symbols[k].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
                        goto absnogood;
                }
                else if ( ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+t[FUNHEAD] )
                && ( t[1] == FUNHEAD+ARGHEAD+t[FUNHEAD+ARGHEAD] ) ) {
                    if ( t[FUNHEAD] == ARGHEAD+1+abs(t[t[1]-1]) ) {
                        WORD *ts;
absnosymbols:           ts = t + t[1] -1;
                        ncoef = REDLENG(ncoef);
                        nnum = REDLENG(*ts);
                        if ( nnum < 0 ) nnum = -nnum;
                        if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,
                        (UWORD *)(ts-ABS(*ts)+1),nnum,
                        (UWORD *)n_coef,&ncoef) ) goto FromNorm;
                        ncoef = INCLENG(ncoef);
                    }
/*
                    Now get rid of the roots of unity. This includes i_
*/
                    else if ( t[FUNHEAD+ARGHEAD+1] == SYMBOL ) {
                        WORD *ts = t+FUNHEAD+ARGHEAD+1;
                        WORD its = ts[1] - 2;
                        ts += 2;
                        while ( its > 0 ) {
                            if ( *ts == 0 ) { }
                            else if ( ( *ts > NumSymbols || *ts <= -MAXPOWER )
                             || ( symbols[*ts].complex & VARTYPEROOTOFUNITY )
                                != VARTYPEROOTOFUNITY ) goto absnogood;
                            its -= 2; ts += 2;
                        }
                        goto absnosymbols;
                    }
                    else {
absnogood:                  pcom[ncom++] = t;
                    }
                }
                else pcom[ncom++] = t;
                break;
            case MODFUNCTION:
            case MOD2FUNCTION:
/*
                Mod function. Does work if two arguments and the
                second argument is a positive short number
*/
                if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SNUMBER
                    && t[FUNHEAD+2] == -SNUMBER && t[FUNHEAD+3] > 1 ) {
                    WORD tmod;
                    tmod = (t[FUNHEAD+1]%t[FUNHEAD+3]);
                    if ( tmod < 0 ) tmod += t[FUNHEAD+3];
                    if ( *t == MOD2FUNCTION && tmod > t[FUNHEAD+3]/2 )
                            tmod -= t[FUNHEAD+3];
                    if ( tmod < 0 ) {
                        *((UWORD *)lnum) = -tmod;
                        nnum = -1;
                    }
                    else if ( tmod > 0 ) {
                        *((UWORD *)lnum) = tmod;
```

```
                        nnum = 1;
                    }
                    else goto NormZero;
                    goto MulIn;
                }
                else if ( t[1] > t[FUNHEAD+2] && t[FUNHEAD] > 0
                && t[FUNHEAD+t[FUNHEAD]] == -SNUMBER
                && t[FUNHEAD+t[FUNHEAD]+1] > 1
                && t[1] == FUNHEAD+2+t[FUNHEAD] ) {
                    WORD *ttt = t+FUNHEAD, iii;
                    iii = ttt[*ttt-1];
                    if ( *ttt == ttt[ARGHEAD]+ARGHEAD &&
                        ttt[ARGHEAD] == ABS(iii)+1 ) {
                        WORD ncmod = 1;
                        WORD cmod = ttt[*ttt+1];
                        iii = REDLENG(iii);
                        if ( *t == MODFUNCTION ) {
                            if ( TakeModulus((UWORD *)(ttt+ARGHEAD+1)
                            ,&iii,(UWORD *)(&cmod),ncmod,UNPACK|NOINVERSES) )
                                goto FromNorm;
                        }
                        else {
                            if ( TakeModulus((UWORD *)(ttt+ARGHEAD+1)
                            ,&iii,(UWORD *)(&cmod),ncmod,UNPACK|POSNEG|NOINVERSES) )
                                goto FromNorm;
                        }
                        if ( *t == MOD2FUNCTION && ttt[ARGHEAD+1] > cmod/2 && iii > 0 ) {
                            ttt[ARGHEAD+1] -= cmod;
                        }
                        if ( ttt[ARGHEAD+1] < 0 ) {
                            *((UWORD *)lnum) = -ttt[ARGHEAD+1];
                            nnum = -1;
                        }
                        else if ( ttt[ARGHEAD+1] > 0 ) {
                            *((UWORD *)lnum) = ttt[ARGHEAD+1];
                            nnum = 1;
                        }
                        else goto NormZero;
                        goto MulIn;
                    }
                }
                else if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER ) {
                    *((UWORD *)lnum) = t[FUNHEAD+1];
                    if ( *lnum == 0 ) goto NormZero;
                    nnum = 1;
                    goto MulIn;
                }
                else if ( ( ( t[FUNHEAD] < 0 ) && ( t[FUNHEAD] == -SNUMBER )
                && ( t[1] >= ( FUNHEAD+6+ARGHEAD ) )
                && ( t[FUNHEAD+2] >= 4+ARGHEAD )
                && ( t[t[1]-1] == t[FUNHEAD+2+ARGHEAD]-1 ) ) ||
                ( ( t[FUNHEAD] > 0 )
                && ( t[FUNHEAD]-ARGHEAD-1 == ABS(t[FUNHEAD+t[FUNHEAD]-1]) )
                && ( t[FUNHEAD+t[FUNHEAD]]-ARGHEAD-1 == t[t[1]-1] ) ) ) {
/*
                    Check that the last (long) number is integer
*/
                    WORD *ttt = t + t[1], iii, iii1;
                    UWORD coefbuf[2], *coef2, ncoef2;
                    iii = (ttt[-1]-1)/2;
                    ttt -= iii;
                    if ( ttt[-1] != 1 ) {
exitfromhere:
                        pcom[ncom++] = t;
                        break;
                    }
                    iii--;
                    for ( iii1 = 0; iii1 < iii; iii1++ ) {
                        if ( ttt[iii1] != 0 ) goto exitfromhere;
                    }
/*
                    Now we have a hit!
                    The first argument will be put in lnum.
                    It will be a rational.
                    The second argument will be a long integer in coef2.
*/
                    ttt = t + FUNHEAD;
                    if ( *ttt < 0 ) {
                        if ( ttt[1] < 0 ) {
                            nnum = -1; lnum[0] = -ttt[1]; lnum[1] = 1;
                        }
                        else {
                            nnum = 1; lnum[0] = ttt[1]; lnum[1] = 1;
                        }
                    }
                    else {
                        nnum = ABS(ttt[ttt[0]-1] - 1);
```

```
                        for ( iii = 0; iii < nnum; iii++ ) {
                            lnum[iii] = ttt[ARGHEAD+1+iii];
                        }
                        nnum = nnum/2;
                        if ( ttt[ttt[0]-1] < 0 ) nnum = -nnum;
                    }
                    NEXTARG(ttt);
                    if ( *ttt < 0 ) {
                        coef2 = coefbuf;
                        ncoef2 = 3; *coef2 = (UWORD)(ttt[1]);
                        coef2[1] = 1;
                    }
                    else {
                        coef2 = (UWORD *)(ttt+ARGHEAD+1);
                        ncoef2 = (ttt[ttt[0]-1]-1)/2;
                    }
                    if ( TakeModulus((UWORD *)lnum,&nnum,(UWORD *)coef2,ncoef2,
                                    UNPACK|NOINVERSES|FROMFUNCTION) ) {
                        goto FromNorm;
                    }
                    if ( *t == MOD2FUNCTION && nnum > 0 ) {
                        UWORD *coef3 = NumberMalloc("Mod2Function"), two = 2;
                        WORD ncoef3;
                        if ( MulLong((UWORD *)lnum,nnum,&two,1,coef3,&ncoef3) )
                            goto FromNorm;
                        if ( BigLong(coef3,ncoef3,(UWORD *)coef2,ncoef2) > 0 ) {
                            nnum = -nnum;
                            AddLong((UWORD *)lnum,nnum,(UWORD *)coef2,ncoef2
                                    ,(UWORD *)lnum,&nnum);
                            nnum = -nnum;
                        }
                        NumberFree(coef3,"Mod2Function");
                    }
/*
                    Do we have to pack? No, because the answer is not a fraction
*/
                    ncoef = REDLENG(ncoef);
                    if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
                            goto FromNorm;
                    ncoef = INCLENG(ncoef);
                }
                else pcom[ncom++] = t;
                break;
            case EXTEUCLIDEAN:
                {
                    WORD argcount = 0, *tc, *ts, xc, xs, *tcc;
                    UWORD *Num1, *Num2, *Num3, *Num4;
                    WORD size1, size2, size3, size4, space;
                    tc = t+FUNHEAD; ts = t + t[1];
                    while ( argcount < 3 && tc < ts ) { NEXTARG(tc); argcount++; }
                    if ( argcount != 2 ) goto defaultcase;
                    if ( t[FUNHEAD] == -SNUMBER ) {
                        if ( t[FUNHEAD+1] <= 1 ) goto defaultcase;
                        if ( t[FUNHEAD+2] == -SNUMBER ) {
                            if ( t[FUNHEAD+3] <= 1 ) goto defaultcase;
                            Num2 = NumberMalloc("modinverses");
                            *Num2 = t[FUNHEAD+3]; size2 = 1;
                        }
                        else {
                            if ( ts[-1] < 0 ) goto defaultcase;
                            if ( ts[-1] != t[FUNHEAD+2]-ARGHEAD-1 ) goto defaultcase;
                            xs = (ts[-1]-1)/2;
                            tcc = ts-xs-1;
                            if ( *tcc != 1 ) goto defaultcase;
                            for ( i = 1; i < xs; i++ ) {
                                if ( tcc[i] != 0 ) goto defaultcase;
                            }
                            Num2 = NumberMalloc("modinverses");
                            size2 = xs;
                            for ( i = 0; i < xs; i++ ) Num2[i] = t[FUNHEAD+ARGHEAD+3+i];
                        }
                        Num1 = NumberMalloc("modinverses");
                        *Num1 = t[FUNHEAD+1]; size1 = 1;
                    }
                    else {
                        tc = t + FUNHEAD + t[FUNHEAD];
                        if ( tc[-1] < 0 ) goto defaultcase;
                        if ( tc[-1] != t[FUNHEAD]-ARGHEAD-1 ) goto defaultcase;
                        xc = (tc[-1]-1)/2;
                        tcc = tc-xc-1;
                        if ( *tcc != 1 ) goto defaultcase;
                        for ( i = 1; i < xc; i++ ) {
                            if ( tcc[i] != 0 ) goto defaultcase;
                        }
                        if ( *tc == -SNUMBER ) {
                            if ( tc[1] <= 1 ) goto defaultcase;
                            Num2 = NumberMalloc("modinverses");
```

```
                            *Num2 = tc[1]; size2 = 1;
                        }
                        else {
                            if ( ts[-1] < 0 ) goto defaultcase;
                            if ( ts[-1] != t[FUNHEAD+2]-ARGHEAD-1 ) goto defaultcase;
                            xs = (ts[-1]-1)/2;
                            tcc = ts-xs-1;
                            if ( *tcc != 1 ) goto defaultcase;
                            for ( i = 1; i < xs; i++ ) {
                                if ( tcc[i] != 0 ) goto defaultcase;
                            }
                            Num2 = NumberMalloc("modinverses");
                            size2 = xs;
                            for ( i = 0; i < xs; i++ ) Num2[i] = tc[ARGHEAD+1+i];
                        }
                        Num1 = NumberMalloc("modinverses");
                        size1 = xc;
                        for ( i = 0; i < xc; i++ ) Num1[i] = t[FUNHEAD+ARGHEAD+1+i];
                    }
                    Num3 = NumberMalloc("modinverses");
                    Num4 = NumberMalloc("modinverses");
                    GetLongModInverses(BHEAD Num1,size1,Num2,size2
                                ,Num3,&size3,Num4,&size4);
/*
                    Now we have to compose the answer. This needs more space
                    and hence we have to put this inside the term.
                    Compute first how much extra space we need.
                    Then move the trailing part of the term upwards.
                    Do not forget relevant pointers!!! (r, m, termout, AT.WorkPointer)
*/
                    space = 0;
                    if ( ( size3 == 1 || size3 == -1 ) && (*Num3&TOPBITONLY) == 0 ) space += 2;
                    else space += ARGHEAD + 2*ABS(size3) + 2;
                    if ( ( size4 == 1 || size4 == -1 ) && (*Num4&TOPBITONLY) == 0 ) space += 2;
                    else space += ARGHEAD + 2*ABS(size4) + 2;
                    tt = term + *term; u = tt + space;
                    while ( tt >= ts ) *--u = *--tt;
                    m += space; r += space;
                    *term += space;
                    t[1] += space;
                    if ( ( size3 == 1 || size3 == -1 ) && (*Num3&TOPBITONLY) == 0 ) {
                        *ts++ = -SNUMBER; *ts = (WORD)(*Num3);
                        if ( size3 < 0 ) *ts = -*ts;
                        ts++;
                    }
                    else {
                        *ts++ = 2*ABS(size3)+ARGHEAD+2;
                        *ts++ = 0; FILLARG(ts)
                        *ts++ = 2*ABS(size3)+1;
                        for ( i = 0; i < ABS(size3); i++ ) *ts++ = Num3[i];
                        *ts++ = 1;
                        for ( i = 1; i < ABS(size3); i++ ) *ts++ = 0;
                        if ( size3 < 0 ) *ts++ = 2*size3-1;
                        else             *ts++ = 2*size3+1;
                    }
                    if ( ( size4 == 1 || size4 == -1 ) && (*Num4&TOPBITONLY) == 0 ) {
                        *ts++ = -SNUMBER; *ts = *Num4;
                        if ( size4 < 0 ) *ts = -*ts;
                        ts++;
                    }
                    else {
                        *ts++ = 2*ABS(size4)+ARGHEAD+2;
                        *ts++ = 0; FILLARG(ts)
                        *ts++ = 2*ABS(size4)+2;
                        for ( i = 0; i < ABS(size4); i++ ) *ts++ = Num4[i];
                        *ts++ = 1;
                        for ( i = 1; i < ABS(size4); i++ ) *ts++ = 0;
                        if ( size4 < 0 ) *ts++ = 2*size4-1;
                        else             *ts++ = 2*size4+1;
                    }
                    NumberFree(Num4,"modinverses");
                    NumberFree(Num3,"modinverses");
                    NumberFree(Num1,"modinverses");
                    NumberFree(Num2,"modinverses");
                    t[2] = 0; /* mark function as clean. */
                    goto Restart;
                }
                break;
            case GCDFUNCTION:
#ifdef EVALUATEGCD
#ifdef NEWGCDFUNCTION
                {
/*
                Has two integer arguments
                Four cases: S,S, S,L, L,S, L,L
*/
                WORD *num1, *num2, size1, size2, stor1, stor2, *ttt, ti;
```

```
                if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SNUMBER
                && t[FUNHEAD+2] == -SNUMBER && t[FUNHEAD+1] != 0
                && t[FUNHEAD+3] != 0 ) {     /* Short,Short */
                    stor1 = t[FUNHEAD+1];
                    stor2 = t[FUNHEAD+3];
                    if ( stor1 < 0 ) stor1 = -stor1;
                    if ( stor2 < 0 ) stor2 = -stor2;
                    num1 = &stor1; num2 = &stor2;
                    size1 = size2 = 1;
                    goto gcdcalc;
                }
                else if ( t[1] > FUNHEAD+4 ) {
                    if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] != 0
                    && t[FUNHEAD+2] == t[1]-FUNHEAD-2 &&
                    ABS(t[t[1]-1]) == t[FUNHEAD+2]-1-ARGHEAD ) { /* Short,Long */
                        num2 = t + t[1];
                        size2 = ABS(num2[-1]);
                        ttt = num2-1;
                        num2 -= size2;
                        size2 = (size2-1)/2;
                        ti = size2;
                        while ( ti > 1 && ttt[-1] == 0 ) { ttt--; ti--; }
                        if ( ti == 1 && ttt[-1] == 1 ) {
                            stor1 = t[FUNHEAD+1];
                            if ( stor1 < 0 ) stor1 = -stor1;
                            num1 = &stor1;
                            size1 = 1;
                            goto gcdcalc;
                        }
                        else pcom[ncom++] = t;
                    }
                    else if ( t[FUNHEAD] > 0 &&
                    t[FUNHEAD]-1-ARGHEAD == ABS(t[t[FUNHEAD]+FUNHEAD-1]) ) {
                        num1 = t + FUNHEAD + t[FUNHEAD];
                        size1 = ABS(num1[-1]);
                        ttt = num1-1;
                        num1 -= size1;
                        size1 = (size1-1)/2;
                        ti = size1;
                        while ( ti > 1 && ttt[-1] == 0 ) { ttt--; ti--; }
                        if ( ti == 1 && ttt[-1] == 1 ) {
                         if ( t[1]-FUNHEAD == t[FUNHEAD]+2 && t[t[1]-2] == -SNUMBER
                         && t[t[1]-1] != 0 ) { /* Long,Short */
                            stor2 = t[t[1]-1];
                            if ( stor2 < 0 ) stor2 = -stor2;
                            num2 = &stor2;
                            size2 = 1;
                            goto gcdcalc;
                         }
                         else if ( t[1]-FUNHEAD == t[FUNHEAD]+t[FUNHEAD+t[FUNHEAD]]
                         && ABS(t[t[1]-1]) == t[FUNHEAD+t[FUNHEAD]] - ARGHEAD-1 ) {
                          num2 = t + t[1];
                          size2 = ABS(num2[-1]);
                          ttt = num2-1;
                          num2 -= size2;
                          size2 = (size2-1)/2;
                          ti = size2;
                          while ( ti > 1 && ttt[-1] == 0 ) { ttt--; ti--; }
                          if ( ti == 1 && ttt[-1] == 1 ) {
gcdcalc:                       if ( GcdLong(BHEAD (UWORD *)num1,size1,(UWORD *)num2,size2
                                ,(UWORD *)lnum,&nnum) ) goto FromNorm;
                            goto MulIn;
                          }
                          else pcom[ncom++] = t;
                         }
                         else pcom[ncom++] = t;
                        }
                        else pcom[ncom++] = t;
                    }
                    else pcom[ncom++] = t;
                }
                else pcom[ncom++] = t;
                }
#else
                {
                    WORD *gcd = AT.WorkPointer;
                    if ( ( gcd = EvaluateGcd(BHEAD t) ) == 0 ) goto FromNorm;
                    if ( *gcd == 4 && gcd[1] == 1 && gcd[2] == 1 && gcd[4] == 0 ) {
                        AT.WorkPointer = gcd;
                    }
                    else if ( gcd[*gcd] == 0 ) {
                        WORD *t1, iii, change, *num, *den, numsize, densize;
                        if ( gcd[*gcd-1] < *gcd-1 ) {
                            t1 = gcd+1;
                            for ( iii = 2; iii < t1[1]; iii += 2 ) {
                                change = ExtraSymbol(t1[iii],t1[iii+1],nsym,ppsym,&ncoef);
                                nsym += change;
```

```
                                ppsym += change * 2;
                            }
                        }
                        t1 = gcd + *gcd;
                        iii = t1[-1]; num = t1-iii; numsize = (iii-1)/2;
                        den = num + numsize; densize = numsize;
                        while ( numsize > 1 && num[numsize-1] == 0 ) numsize--;
                        while ( densize > 1 && den[densize-1] == 0 ) densize--;
                        if ( numsize > 1 || num[0] != 1 ) {
                            ncoef = REDLENG(ncoef);
                            if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)num,numsize) ) goto
      FromNorm;
                            ncoef = INCLENG(ncoef);
                        }
                        if ( densize > 1 || den[0] != 1 ) {
                            ncoef = REDLENG(ncoef);
                            if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)den,densize) ) goto
      FromNorm;
                            ncoef = INCLENG(ncoef);
                        }
                        AT.WorkPointer = gcd;
                    }
                    else {  /* a whole expression */
/*
                        Action: Put the expression in a compiler buffer.
                        Insert a SUBEXPRESSION subterm
                        Set the return value of the routine such that in
                        Generator the term gets sent again to TestSub.

                        1: put in C (ebufnum)
                        2: after that the WorkSpace is free again.
                        3: insert the SUBEXPRESSION
                        4: copy the top part of the term down
*/
                        LONG size = AT.WorkPointer - gcd;

                        ss = AddRHS(AT.ebufnum,1);
                        while ( (ss + size + 10) > C->Top ) ss = DoubleCbuffer(AT.ebufnum,ss,13);
                        tt = gcd;
                        NCOPY(ss,tt,size);
                        C->rhs[C->numrhs+1] = ss;
                        C->Pointer = ss;

                        t[0] = SUBEXPRESSION;
                        t[1] = SUBEXPSIZE;
                        t[2] = C->numrhs;
                        t[3] = 1;
                        t[4] = AT.ebufnum;
                        t += 5;
                        tt = term + *term;
                        while ( r < tt ) *t++ = *r++;
                        *term = t - term;

                        regval = 1;
                        goto RegEnd;
                    }
                }
#endif
#else
                MesPrint(" Unexpected call to EvaluateGCD");
                Terminate(-1);
#endif
                break;
            case MINFUNCTION:
            case MAXFUNCTION:
                if ( t[1] == FUNHEAD ) break;
                {
                    WORD *ttt = t + FUNHEAD;
                    WORD *tttstop = t + t[1];
                    WORD tterm[4], iii;
                    while ( ttt < tttstop ) {
                        if ( *ttt > 0 ) {
                            if ( ttt[ARGHEAD]-1 > ABS(ttt[*ttt-1]) ) goto nospec;
                            ttt += *ttt;
                        }
                        else {
                            if ( *ttt != -SNUMBER ) goto nospec;
                            ttt += 2;
                        }
                    }
/*
                    Function has only numerical arguments
                    Pick up the first argument.
*/
                    ttt = t + FUNHEAD;
                    if ( *ttt > 0 ) {
loadnew1:
```

```
                        for ( iii = 0; iii < ttt[ARGHEAD]; iii++ )
                            n_llnum[iii] = ttt[ARGHEAD+iii];
                        ttt += *ttt;
                    }
                    else {
loadnew2:
                        if ( ttt[1] == 0 ) {
                            n_llnum[0] = n_llnum[1] = n_llnum[2] = n_llnum[3] = 0;
                        }
                        else {
                            n_llnum[0] = 4;
                            if ( ttt[1] > 0 ) { n_llnum[1] = ttt[1]; n_llnum[3] = 3; }
                            else { n_llnum[1] = -ttt[1]; n_llnum[3] = -3; }
                            n_llnum[2] = 1;
                        }
                        ttt += 2;
                    }
/*
                    Now loop over the other arguments
*/
                    while ( ttt < tttstop ) {
                        if ( *ttt > 0 ) {
                            if ( n_llnum[0] == 0 ) {
                                if ( ( *t == MINFUNCTION && ttt[*ttt-1] < 0 )
                                || ( *t == MAXFUNCTION && ttt[*ttt-1] > 0 ) )
                                    goto loadnew1;
                            }
                            else {
                                ttt += ARGHEAD;
                                iii = CompCoef(n_llnum,ttt);
                                if ( ( iii > 0 && *t == MINFUNCTION )
                                || ( iii < 0 && *t == MAXFUNCTION ) ) {
                                    for ( iii = 0; iii < ttt[0]; iii++ )
                                        n_llnum[iii] = ttt[iii];
                                }
                            }
                            ttt += *ttt;
                        }
                        else {
                            if ( n_llnum[0] == 0 ) {
                                if ( ( *t == MINFUNCTION && ttt[1] < 0 )
                                || ( *t == MAXFUNCTION && ttt[1] > 0 ) )
                                    goto loadnew2;
                            }
                            else if ( ttt[1] == 0 ) {
                                if ( ( *t == MINFUNCTION && n_llnum[*n_llnum-1] > 0 )
                                || ( *t == MAXFUNCTION && n_llnum[*n_llnum-1] < 0 ) ) {
                                    n_llnum[0] = 0;
                                }
                            }
                            else {
                                tterm[0] = 4; tterm[2] = 1;
                                if ( ttt[1] < 0 ) { tterm[1] = -ttt[1]; tterm[3] = -3; }
                                else { tterm[1] = ttt[1]; tterm[3] = 3; }
                                iii = CompCoef(n_llnum,tterm);
                                if ( ( iii > 0 && *t == MINFUNCTION )
                                || ( iii < 0 && *t == MAXFUNCTION ) ) {
                                    for ( iii = 0; iii < 4; iii++ )
                                        n_llnum[iii] = tterm[iii];
                                }
                            }
                            ttt += 2;
                        }
                    }
                    if ( n_llnum[0] == 0 ) goto NormZero;
                    ncoef = REDLENG(ncoef);
                    nnum = REDLENG(n_llnum[*n_llnum-1]);
                    if ( MulRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)lnum,nnum,
                    (UWORD *)n_coef,&ncoef) ) goto FromNorm;
                    ncoef = INCLENG(ncoef);
                }
                break;
            case INVERSEFACTORIAL:
                if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] >= 0 ) {
                    if ( Factorial(BHEAD t[FUNHEAD+1],(UWORD *)lnum,&nnum) )
                        goto FromNorm;
                    ncoef = REDLENG(ncoef);
                    if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) ) goto FromNorm;
                    ncoef = INCLENG(ncoef);
                }
                else {
nospec:             pcom[ncom++] = t;
                }
                break;
            case MAXPOWEROF:
                if ( ( t[FUNHEAD] == -SYMBOL )
                     && ( t[FUNHEAD+1] > 0 ) && ( t[1] == FUNHEAD+2 ) ) {
```

```
                        *((UWORD *)lnum) = symbols[t[FUNHEAD+1]].maxpower;
                        nnum = 1;
                        goto MulIn;
                    }
                    else { pcom[ncom++] = t; }
                    break;
                case MINPOWEROF:
                    if ( ( t[FUNHEAD] == -SYMBOL )
                         && ( t[FUNHEAD] > 0 ) && ( t[1] == FUNHEAD+2 ) ) {
                        *((UWORD *)lnum) = symbols[t[FUNHEAD+1]].minpower;
                        nnum = 1;
                        goto MulIn;
                    }
                    else { pcom[ncom++] = t; }
                    break;
                case PRIMENUMBER :
                    if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER
                         && t[FUNHEAD+1] > 0 ) {
                        UWORD xp = (UWORD)(NextPrime(BHEAD t[FUNHEAD+1]));
                        ncoef = REDLENG(ncoef);
                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,&xp,1) ) goto FromNorm;
                        ncoef = INCLENG(ncoef);
                    }
                    else goto defaultcase;
                    break;
                case MOEBIUS:
/*
                    Numerical function giving -1,0,1 or no evaluation
*/
                    if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER
                        && t[FUNHEAD+1] > 0 ) {
                        WORD val = Moebius(BHEAD t[FUNHEAD+1]);
                        if ( val == 0 ) goto NormZero;
                        if ( val < 0 ) ncoef = -ncoef;
                    }
                    else {
                        pcom[ncom++] = t;
                    }
                    break;
                case LNUMBER :
                    ncoef = REDLENG(ncoef);
                    if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)(t+3),t[2]) ) goto FromNorm;
                    ncoef = INCLENG(ncoef);
                    break;
                case SNUMBER :
                    if ( t[2] < 0 ) {
                        t[2] = -t[2];
                        if ( t[3] & 1 ) ncoef = -ncoef;
                    }
                    else if ( t[2] == 0 ) {
                        if ( t[3] < 0 ) goto NormInf;
                        goto NormZero;
                    }
                    lnum[0] = t[2];
                    nnum = 1;
                    if ( t[3] && RaisPow(BHEAD (UWORD *)lnum,&nnum,(UWORD)(ABS(t[3]))) ) goto FromNorm;
                    ncoef = REDLENG(ncoef);
                    if ( t[3] < 0 ) {
                        if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
                            goto FromNorm;
                    }
                    else if ( t[3] > 0 ) {
                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
                            goto FromNorm;
                    }
                    ncoef = INCLENG(ncoef);
                    break;
                case GAMMA :
                case GAMMAI :
                case GAMMAFIVE :
                case GAMMASIX :
                case GAMMASEVEN :
                    if ( t[1] == FUNHEAD ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Gamma matrix without spin line encountered.");
                        MUNLOCK(ErrorMessageLock);
                        goto NormMin;
                    }
                    pnco[nnco++] = t;
                    t += FUNHEAD+1;
                    goto ScanCont;
                case LEVICIVITA :
                    peps[neps++] = t;
                    if ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) {
                        t[2] &= ~DIRTYFLAG;
                        t[2] |= DIRTYSYMFLAG;
                    }
```

```
                t += FUNHEAD;
ScanCont:       while ( t < r ) {
                    if ( *t >= AM.OffsetIndex &&
                        ( *t >= AM.DumInd || ( *t < AM.WilInd &&
                        indices[*t-AM.OffsetIndex].dimension ) ) )
                        pcon[ncon++] = t;
                    t++;
                }
                break;
            case EXPONENT :
                {
                    WORD *rr;
                    k = 1;
                    rr = t + FUNHEAD;
                    if ( *rr == ARGHEAD || ( *rr == -SNUMBER && rr[1] == 0 ) )
                        k = 0;
                    if ( *rr == -SNUMBER && rr[1] == 1 ) break;
                    if ( *rr <= -FUNCTION ) k = *rr;
                    NEXTARG(rr)
                    if ( *rr == ARGHEAD || ( *rr == -SNUMBER && rr[1] == 0 ) ) {
                        if ( k == 0 ) goto NormZZ;
                        break;
                    }
                    if ( *rr == -SNUMBER && rr[1] > 0 && rr[1] < MAXPOWER && k < 0 ) {
                        k = -k;
                        if ( functions[k-FUNCTION].commute ) {
                            for ( i = 0; i < rr[1]; i++ ) pnco[nnco++] = rr-1;
                        }
                        else {
                            for ( i = 0; i < rr[1]; i++ ) pcom[ncom++] = rr-1;
                        }
                        break;
                    }
                    if ( k == 0 ) goto NormZero;
                    if ( t[FUNHEAD] == -SYMBOL && *rr == -SNUMBER && t[1] == FUNHEAD+4 ) {
                        if ( rr[1] < MAXPOWER ) {
                            t[FUNHEAD+2] = t[FUNHEAD+1]; t += FUNHEAD+2;
                            from = m;
                            goto NextSymbol;
                        }
                    }
/*
                    if ( ( t[FUNHEAD] > 0 && t[FUNHEAD+1] != 0 )
                    || ( *rr > 0 && rr[1] != 0 ) ) {}
                    else
*/
                    t[2] &= ~DIRTYSYMFLAG;

                    pnco[nnco++] = t;
                }
                break;
            case DENOMINATOR :
                t[2] &= ~DIRTYSYMFLAG;
                pden[nden++] = t;
                pnco[nnco++] = t;
                break;
            case INDEX :
                t += 2;
                do {
                    if ( *t == 0 || *t == AM.vectorzero ) goto NormZero;
                    if ( *t > 0 && *t < AM.OffsetIndex ) {
                        lnum[0] = *t++;
                        nnum = 1;
                        ncoef = REDLENG(ncoef);
                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)lnum,nnum) )
                            goto FromNorm;
                        ncoef = INCLENG(ncoef);
                    }
                    else if ( *t == NOINDEX ) t++;
                    else pind[nind++] = *t++;
                } while ( t < r );
                break;
            case SUBEXPRESSION :
                if ( t[3] == 0 ) break;
            case EXPRESSION :
                goto RegEnd;
            case ROOTFUNCTION :
/*
                Tries to take the n-th root inside the rationals
                If this is not possible, it clears all flags and
                hence tries no more.
                Notation:
                root_(power(=integer),(rational)number)
*/
                { WORD nc;
                if ( t[2] == 0 ) goto defaultcase;
                if ( t[FUNHEAD] != -SNUMBER || t[FUNHEAD+1] < 0 ) goto defaultcase;
```

```
				if ( t[FUNHEAD+2] == -SNUMBER ) {
					if ( t[FUNHEAD+1] == 0 && t[FUNHEAD+3] == 0 ) goto NormZZ;
					if ( t[FUNHEAD+1] == 0 ) break;
					if ( t[FUNHEAD+3] < 0 ) {
						AT.WorkPointer[0] = -t[FUNHEAD+3];
						nc = -1;
					}
					else {
						AT.WorkPointer[0] = t[FUNHEAD+3];
						nc = 1;
					}
					AT.WorkPointer[1] = 1;
				}
				else if ( t[FUNHEAD+2] == t[1]-FUNHEAD-2
				&& t[FUNHEAD+2] == t[FUNHEAD+2+ARGHEAD]+ARGHEAD
				&& ABS(t[t[1]-1]) == t[FUNHEAD+2+ARGHEAD] - 1 ) {
					WORD *r1, *r2;
					if ( t[FUNHEAD+1] == 0 ) break;
					i = t[t[1]-1]; r1 = t + FUNHEAD+ARGHEAD+3;
					nc = REDLENG(i);
					i = ABS(i) - 1;
					r2 = AT.WorkPointer;
					while ( --i >= 0 ) *r2++ = *r1++;
				}
				else goto defaultcase;
				if ( TakeRatRoot((UWORD *)AT.WorkPointer,&nc,t[FUNHEAD+1]) ) {
					t[2] = 0;
					goto defaultcase;
				}
				ncoef = REDLENG(ncoef);
				if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)AT.WorkPointer,nc) )
						goto FromNorm;
				if ( nc < 0 ) nc = -nc;
				if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)(AT.WorkPointer+nc),nc) )
						goto FromNorm;
				ncoef = INCLENG(ncoef);
				}
				break;
			case RANDOMFUNCTION :
				{
				WORD nnc, nc, nca, nr;
				UWORD xx;
/*
				Needs one positive integer argument.
				returns (wranf()%argument)+1.
				We may call wranf several times to paste UWORDS together
				when we need long numbers.
				We make little errors when taking the % operator
				(not 100% uniform). We correct for that by redoing the
				calculation in the (unlikely) case that we are in leftover area
*/
				if ( t[1] == FUNHEAD ) goto defaultcase;
				if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -SNUMBER &&
					t[FUNHEAD+1] > 0 ) {
					if ( t[FUNHEAD+1] == 1 ) break;
redoshort:
					((UWORD *)AT.WorkPointer)[0] = wranf(BHEAD0);
					((UWORD *)AT.WorkPointer)[1] = wranf(BHEAD0);
					nr = 2;
					if ( ((UWORD *)AT.WorkPointer)[1] == 0 ) {
						nr = 1;
						if ( ((UWORD *)AT.WorkPointer)[0] == 0 ) {
							nr = 0;
						}
					}
					xx = (UWORD)(t[FUNHEAD+1]);
					if ( nr ) {
					DivLong((UWORD *)AT.WorkPointer,nr
							,&xx,1
							,((UWORD *)AT.WorkPointer)+4,&nnc
							,((UWORD *)AT.WorkPointer)+2,&nc);
					((UWORD *)AT.WorkPointer)[4] = 0;
					((UWORD *)AT.WorkPointer)[5] = 0;
					((UWORD *)AT.WorkPointer)[6] = 1;
					DivLong((UWORD *)AT.WorkPointer+4,3
							,&xx,1
							,((UWORD *)AT.WorkPointer)+9,&nnc
							,((UWORD *)AT.WorkPointer)+7,&nca);
					AddLong((UWORD *)AT.WorkPointer+4,3
							,((UWORD *)AT.WorkPointer)+7,-nca
							,((UWORD *)AT.WorkPointer)+9,&nnc);
					if ( BigLong((UWORD *)AT.WorkPointer,nr
							,((UWORD *)AT.WorkPointer)+9,nnc) >= 0 ) goto redoshort;
					}
					else nc = 0;
					if ( nc == 0 ) {
						AT.WorkPointer[2] = (WORD)xx;
```

```
                        nc = 1;
                    }
                    ncoef = REDLENG(ncoef);
                    if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,((UWORD *)(AT.WorkPointer))+2,nc) )
                            goto FromNorm;
                    ncoef = INCLENG(ncoef);
                }
                else if ( t[FUNHEAD] > 0 && t[1] == t[FUNHEAD]+FUNHEAD
                && ABS(t[t[1]-1]) == t[FUNHEAD]-1-ARGHEAD && t[t[1]-1] > 0 ) {
                    WORD nna, nnb, nni, nnb2, nnb2a;
                    UWORD *nnt;
                    nna = t[t[1]-1];
                    nnb2 = nna-1;
                    nnb = nnb2/2;
                    nnt = (UWORD *)(t+t[1]-1-nnb);  /* start of denominator */
                    if ( *nnt != 1 ) goto defaultcase;
                    for ( nni = 1; nni < nnb; nni++ ) {
                        if ( nnt[nni] != 0 ) goto defaultcase;
                    }
                    nnt = (UWORD *)(t + FUNHEAD + ARGHEAD + 1);

                    for ( nni = 0; nni < nnb2; nni++ ) {
                        ((UWORD *)AT.WorkPointer)[nni] = wranf(BHEAD0);
                    }
                    nnb2a = nnb2;
                    while ( nnb2a > 0 && ((UWORD *)AT.WorkPointer)[nnb2a-1] == 0 ) nnb2a--;
                    if ( nnb2a > 0 ) {
                    DivLong((UWORD *)AT.WorkPointer,nnb2a
                            ,nnt,nnb
                            ,((UWORD *)AT.WorkPointer)+2*nnb2,&nnc
                            ,((UWORD *)AT.WorkPointer)+nnb2,&nc);
                    for ( nni = 0; nni < nnb2; nni++ ) {
                        ((UWORD *)AT.WorkPointer)[nni+2*nnb2] = 0;
                    }
                    ((UWORD *)AT.WorkPointer)[3*nnb2] = 1;
                    DivLong((UWORD *)AT.WorkPointer+2*nnb2,nnb2+1
                            ,nnt,nnb
                            ,((UWORD *)AT.WorkPointer)+4*nnb2+1,&nnc
                            ,((UWORD *)AT.WorkPointer)+3*nnb2+1,&nca);
                    AddLong((UWORD *)AT.WorkPointer+2*nnb2,nnb2+1
                            ,((UWORD *)AT.WorkPointer)+3*nnb2+1,-nca
                            ,((UWORD *)AT.WorkPointer)+4*nnb2+1,&nnc);
                    if ( BigLong((UWORD *)AT.WorkPointer,nnb2a
                            ,((UWORD *)AT.WorkPointer)+4*nnb2+1,nnc) >= 0 ) goto redoshort;
                    }
                    else nc = 0;
                    if ( nc == 0 ) {
                        for ( nni = 0; nni < nnb; nni++ ) {
                            ((UWORD *)AT.WorkPointer)[nnb2+nni] = nnt[nni];
                        }
                        nc = nnb;
                    }
                    ncoef = REDLENG(ncoef);
                    if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,((UWORD *)(AT.WorkPointer))+nnb2,nc) )
                            goto FromNorm;
                    ncoef = INCLENG(ncoef);
                }
                else goto defaultcase;
                }
                break;
            case RANPERM :
                if ( *t == RANPERM && t[1] > FUNHEAD && t[FUNHEAD] <= -FUNCTION ) {
                    WORD **pwork;
                    WORD *mm, *ww, *ow = AT.WorkPointer;
                    WORD *Array, *targ, *argstop, narg = 0, itot;
                    int ie;
                    argstop = t+t[1];
                    targ = t+FUNHEAD+1;
                    while ( targ < argstop ) {
                        narg++; NEXTARG(targ);
                    }
                    WantAddPointers(narg);
                    pwork = AT.pWorkSpace + AT.pWorkPointer;
                    targ = t+FUNHEAD+1; narg = 0;
                    while ( targ < argstop ) {
                        pwork[narg++] = targ;
                        NEXTARG(targ);
                    }
/*
                    Make a random permutation of the numbers 0,...,narg-1
                    The following code works also for narg == 0 and narg == 1
*/
                    ow = AT.WorkPointer;
                    Array = AT.WorkPointer;
                    AT.WorkPointer += narg;
                    for ( i = 0; i < narg; i++ ) Array[i] = i;
                    for ( i = 2; i <= narg; i++ ) {
```

```
                        itot = (WORD)(iranf(BHEAD i));
                        for ( j = 0; j < itot; j++ ) CYCLE1(WORD,Array,i)
                    }
                    mm = AT.WorkPointer;
                    *mm++ = -t[FUNHEAD];
                    *mm++ = t[1] - 1;
                    for ( ie = 2; ie < FUNHEAD; ie++ ) *mm++ = t[ie];
                    for ( i = 0; i < narg; i++ ) {
                        ww = pwork[Array[i]];
                        CopyArg(mm,ww);
                    }
                    mm = AT.WorkPointer; t++; ww = t;
                    i = mm[1]; NCOPY(ww,mm,i)
                    AT.WorkPointer = ow;
                    goto TryAgain;
                }
                pnco[nnco++] = t;
                break;
            case PUTFIRST : /* First argument should be a function, second a number */
                if ( ( t[2] & DIRTYFLAG ) != 0 && t[FUNHEAD] <= -FUNCTION
                && t[FUNHEAD+1] == -SNUMBER && t[FUNHEAD+2] > 0 ) {
                    WORD *rr = t+t[1], *mm = t+FUNHEAD+3, *tt, *tt1, *tt2, num = 0;
/*
                    now count the arguments. If not enough: no action.
*/
                    while ( mm < rr ) { num++; NEXTARG(mm); }
                    if ( num < t[FUNHEAD+2] ) { pnco[nnco++] = t; break; }
                    /* Replace "putfirst_" with the resulting function code. mm goes to arg start. */
                    *t = -t[FUNHEAD]; mm = t+FUNHEAD+3;
                    /* Set i to the arg number we are putting first, then move mm to its start. */
                    i = t[FUNHEAD+2];
                    while ( --i > 0 ) { NEXTARG(mm); }
                    /* Keep a pointer to the arguments trailing the selected: */
                    WORD *argTail = mm;
                    NEXTARG(argTail);
                    tt = TermMalloc("Select_"); /* Move selected out of the way into tmp space */
                    tt1 = tt;
                    /* Normal argument: */
                    if ( *mm > 0 ) {
                        for ( i = 0; i < *mm; i++ ) *tt1++ = mm[i];
                    }
                    /* Fast-notation function: single word */
                    else if ( *mm <= -FUNCTION ) { *tt1++ = *mm; }
                    /* Fast-notation symbol, number etc: two words */
                    else { *tt1++ = mm[0]; *tt1++ = mm[1]; }
                    /* Put tt2 at the start of the original arguments */
                    tt2 = t+FUNHEAD+3;
                    /* Copy leading original arguments after the new first, in the tmp space. */
                    while ( tt2 < mm ) *tt1++ = *tt2++;
                    /* i contains the size so far. tt1 goes to the start of the tmp space.
                        tt2 to the final argument location (we overwrite "putfirst_") */
                    i = tt1-tt; tt1 = tt; tt2 = t+FUNHEAD;
                    /* Copy everything so far to its final place */
                    NCOPY(tt2,tt1,i);
                    /* We are finished with the tmp space */
                    TermFree(tt,"Select_");
                    /* Now copy the trailing args. Use the stored pointer, *mm has been edited
                        during the copy to tt2 above! NEXTARG(mm) would produce nonsense. */
                    while ( argTail < rr ) *tt2++ = *argTail++;
                    /* Set the size of all function args */
                    t[1] = tt2 - t;
                    /* Now copy the rest of the term, and set the final term size */
                    rr = term + *term;
                    while ( argTail < rr ) *tt2++ = *argTail++;
                    *term = tt2-term;
                    goto Restart;
                }
                else pnco[nnco++] = t;
                break;
#ifdef WITHFLOAT
            case FLOATFUN :
/*
                If it is a proper float_ we give it special treatment.
                If it is not proper, we treat it as a regular commuting function.
*/
                if ( withfloat == 0 ) {
                    if ( TestFloat(t) == 0 ) goto defaultcase;
                    firstfloat = t;
                    withfloat = 1;
                }
                else { /* There are now at least two float_'s. Time for action. */
                    if ( TestFloat(t) == 0 ) goto defaultcase;
                    if ( withfloat == 1 ) UnpackFloat(aux4,firstfloat);
                    withfloat++;
                    UnpackFloat(aux5,t);
                    mpf_mul(aux4,aux4,aux5);
                }
```

```
                break;
#endif
            case INTFUNCTION :
/*
                Can be resolved if the first argument is a number
                and the second argument either doesn't exist or has
                the value +1, 0, -1
                +1 : rounding up
                0  : rounding towards zero
                -1 : rounding down (same as no argument)
*/
                if ( t[1] <= FUNHEAD ) break;
                {
                    WORD *rr, den, num;
                    to = t + FUNHEAD;
                    if ( *to > 0 ) {
                        if ( *to == ARGHEAD ) goto NormZero;
                        rr = to + *to;
                        i = rr[-1];
                        j = ABS(i);
                        if ( to[ARGHEAD] != j+1 ) goto NoInteg;
                        if ( rr >= r ) k = -1;
                        else if ( *rr == ARGHEAD ) { k = 0; rr += ARGHEAD; }
                        else if ( *rr == -SNUMBER ) { k = rr[1]; rr += 2; }
                        else goto NoInteg;
                        if ( rr != r ) goto NoInteg;
                        if ( k > 1 || k < -1 ) goto NoInteg;
                        to += ARGHEAD+1;
                        j = (j-1) » 1;
                        i = ( i < 0 ) ? -j: j;
                        UnPack((UWORD *)to,i,&den,&num);
/*
                        Potentially the use of NoScrat2 is unsafe.
                        It makes the routine not reentrant, but because it is
                        used only locally and because we only call the
                        low level routines DivLong and AddLong which never
                        make calls involving Normalize, things are OK after all
*/
                        if ( AN.NoScrat2 == 0 ) {
                            AN.NoScrat2 = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD),"Normalize");
                        }
                        if ( DivLong((UWORD *)to,num,(UWORD *)(to+j),den
                        ,(UWORD *)AT.WorkPointer,&num,AN.NoScrat2,&den) ) goto FromNorm;
                        if ( k < 0 && den < 0 ) {
                            *AN.NoScrat2 = 1;
                            den = -1;
                            if ( AddLong((UWORD *)AT.WorkPointer,num
                            ,AN.NoScrat2,den,(UWORD *)AT.WorkPointer,&num) )
                                goto FromNorm;
                        }
                        else if ( k > 0 && den > 0 ) {
                            *AN.NoScrat2 = 1;
                            den = 1;
                            if ( AddLong((UWORD *)AT.WorkPointer,num,
                            AN.NoScrat2,den,(UWORD *)AT.WorkPointer,&num) )
                                goto FromNorm;
                        }

                    }
                    else if ( *to == -SNUMBER ) {    /* No rounding needed */
                        if ( to[1] < 0 ) { *AT.WorkPointer = -to[1]; num = -1; }
                        else if ( to[1] == 0 ) goto NormZero;
                        else { *AT.WorkPointer = to[1]; num = 1; }
                    }
                    else goto NoInteg;
                    if ( num == 0 ) goto NormZero;
                    ncoef = REDLENG(ncoef);
                    if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)AT.WorkPointer,num) )
                        goto FromNorm;
                    ncoef = INCLENG(ncoef);
                    break;
                }
NoInteg:;
/*
                Fall through if it cannot be resolved
*/
            default :
defaultcase:;
                if ( *t < FUNCTION ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Illegal code in Norm");
#ifdef DEBUGON
                    {
                        UBYTE OutBuf[140];
                        AO.OutFill = AO.OutputLine = OutBuf;
                        t = term;
                        AO.OutSkip = 3;
```

```
                        FiniLine();
                        i = *t;
                        while ( --i >= 0 ) {
                            TalToLine((UWORD)(*t++));
                            TokenToLine((UBYTE *)"  ");
                        }
                        AO.OutSkip = 0;
                        FiniLine();
                    }
#endif
                    MUNLOCK(ErrorMessageLock);
                    goto NormMin;
                }
                if ( *t == REPLACEMENT ) {
                    if ( AR.Eside != LHSIDE ) ReplaceVeto--;
                    pcom[ncom++] = t;
                    break;
                }
/*
                if ( *t == AM.termfunnum && t[1] == FUNHEAD+2
                && t[FUNHEAD] == -DOLLAREXPRESSION ) termflag++;
*/
                if ( *t == DUMMYFUN || *t == DUMMYTEN ) {}
                else {
                    if ( *t < (FUNCTION + WILDOFFSET) ) {
                        if ( ( ( functions[*t-FUNCTION].maxnumargs > 0 )
                        || ( functions[*t-FUNCTION].minnumargs > 0 ) )
                         && ( ( t[2] & DIRTYFLAG ) != 0 ) ) {
/*
                            Number of arguments is bounded. And we have not checked.
*/
                            WORD *ta = t + FUNHEAD, *tb = t + t[1];
                            int numarg = 0;
                            while ( ta < tb ) { numarg++; NEXTARG(ta) }
                            if ( ( functions[*t-FUNCTION].maxnumargs > 0 )
                            && ( numarg >= functions[*t-FUNCTION].maxnumargs ) )
                                goto NormZero;
                            if ( ( functions[*t-FUNCTION].minnumargs > 0 )
                            && ( numarg < functions[*t-FUNCTION].minnumargs ) )
                                goto NormZero;
                        }
doflags:
                        if ( ( ( t[2] & DIRTYFLAG ) != 0 ) && ( functions[*t-FUNCTION].tabl == 0 ) ) {
                            t[2] &= ~DIRTYFLAG;
                            t[2] |= DIRTYSYMFLAG;
                        }
                        if ( functions[*t-FUNCTION].commute ) { pnco[nnco++] = t; }
                        else { pcom[ncom++] = t; }
                    }
                    else {
                        if ( ( ( t[2] & DIRTYFLAG ) != 0 ) && ( functions[*t-FUNCTION-WILDOFFSET].tabl
== 0 ) ) {
                            t[2] &= ~DIRTYFLAG;
                            t[2] |= DIRTYSYMFLAG;
                        }
                        if ( functions[*t-FUNCTION-WILDOFFSET].commute ) {
                            pnco[nnco++] = t;
                        }
                        else { pcom[ncom++] = t; }
                    }
                }

                /* Now hunt for contractible indices */

                if ( ( *t < (FUNCTION + WILDOFFSET)
                 && functions[*t-FUNCTION].spec >= TENSORFUNCTION ) || (
                 *t >= (FUNCTION + WILDOFFSET)
                 && functions[*t-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION ) ) {
                    if ( *t >= GAMMA && *t <= GAMMASEVEN ) t++;
                    t += FUNHEAD;
                    while ( t < r ) {
                        if ( *t == AM.vectorzero ) goto NormZero;
                        if ( *t >= AM.OffsetIndex && ( *t >= AM.DumInd
                        || ( *t < AM.WilInd && indices[*t-AM.OffsetIndex].dimension ) ) ) {
                            pcon[ncon++] = t;
                        }
                        else if ( *t == FUNNYWILD ) { t++; }
                        t++;
                    }
                }
                else {
                    t += FUNHEAD;
                    while ( t < r ) {
                        if ( *t > 0 ) {
/*
                            Here we should worry about a recursion
                            A problem is the possibility of a construct
```

```
                            like  f(mu+nu)
*/
                            t += *t;
                        }
                        else if ( *t <= -FUNCTION ) t++;
                        else if ( *t == -INDEX ) {
                            if ( t[1] >= AM.OffsetIndex &&
                                ( t[1] >= AM.DumInd || ( t[1] < AM.WilInd
                                && indices[t[1]-AM.OffsetIndex].dimension ) ) )
                                pcon[ncon++] = t+1;
                            t += 2;
                        }
                        else if ( *t == -SYMBOL ) {
                            if ( t[1] >= MAXPOWER && t[1] < 2*MAXPOWER ) {
                                *t = -SNUMBER;
                                t[1] -= MAXPOWER;
                            }
                            else if ( t[1] < -MAXPOWER && t[1] > -2*MAXPOWER  ) {
                                *t = -SNUMBER;
                                t[1] += MAXPOWER;
                            }
                            else t += 2;
                        }
                        else t += 2;
                    }
                }
                break;
        }
        t = r;
TryAgain:;
    } while ( t < m );
    if ( ANsc ) {
        AN.cTerm = ANsc;
        r = t = ANsr; m = ANsm;
        ANsc = ANsm = ANsr = 0;
        goto conscan;
    }
/*
    #] First scan :
    #[ Easy denominators :

    Easy denominators are denominators that can be replaced by
    negative powers of individual subterms. This may add to all
    our sublists.

*/
    if ( nden ) {
        for ( k = 0, i = 0; i < nden; i++ ) {
            t = pden[i];
            if ( ( t[2] & DIRTYFLAG ) == 0 ) continue;
            r = t + t[1]; m = t + FUNHEAD;
            if ( m >= r ) {
                for ( j = i+1; j < nden; j++ ) pden[j-1] = pden[j];
                nden--;
                for ( j = 0; j < nnco; j++ ) if ( pnco[j] == t ) break;
                for ( j++; j < nnco; j++ ) pnco[j-1] = pnco[j];
                nnco--;
                i--;
            }
            else {
                NEXTARG(m);
                if ( m >= r ) continue;
/*
                We have more than one argument. Split the function.
*/
                if ( k == 0 ) {
                    k = 1; to = termout; from = term;
                }
                while ( from < t ) *to++ = *from++;
                m = t + FUNHEAD;
                while ( m < r ) {
                    stop = to;
                    *to++ = DENOMINATOR;
                    for ( j = 1; j < FUNHEAD; j++ ) *to++ = 0;
                    if ( *m < -FUNCTION ) *to++ = *m++;
                    else if ( *m < 0 ) { *to++ = *m++; *to++ = *m++; }
                    else {
                        j = *m; while ( --j >= 0 ) *to++ = *m++;
                    }
                    stop[1] = WORDDIF(to,stop);
                }
                from = r;
                if ( i == nden - 1 ) {
                    stop = term + *term;
                    while ( from < stop ) *to++ = *from++;
                    i = *termout = WORDDIF(to,termout);
                    to = term; from = termout;
```

```
                    while ( --i >= 0 ) *to++ = *from++;
                    goto Restart;
                }
            }
        }
        for ( i = 0; i < nden; i++ ) {
            t = pden[i];
            if ( ( t[2] & DIRTYFLAG ) == 0 ) continue;
            t[2] = 0;
            if ( t[FUNHEAD] == -SYMBOL ) {
                WORD change;
                t += FUNHEAD+1;
                change = ExtraSymbol(*t,-1,nsym,ppsym,&ncoef);
                nsym += change;
                ppsym += change * 2;
                goto DropDen;
            }
            else if ( t[FUNHEAD] == -SNUMBER ) {
                t += FUNHEAD+1;
                if ( *t == 0 ) goto NormInf;
                if ( *t < 0 ) { *AT.WorkPointer = -*t; j = -1; }
                else { *AT.WorkPointer = *t; j = 1; }
                ncoef = REDLENG(ncoef);
                if ( Divvy(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)AT.WorkPointer,j) )
                    goto FromNorm;
                ncoef = INCLENG(ncoef);
                goto DropDen;
            }
            else if ( t[FUNHEAD] == ARGHEAD ) goto NormInf;
            else if ( t[FUNHEAD] > 0 && t[FUNHEAD+ARGHEAD] ==
            t[FUNHEAD]-ARGHEAD ) {
                /* Only one term */
                r = t + t[1] - 1;
                t += FUNHEAD + ARGHEAD + 1;
                j = *r;
                m = r - ABS(*r) + 1;
                if ( j != 3 || ( ( *m != 1 ) || ( m[1] != 1 ) ) ) {
                    ncoef = REDLENG(ncoef);
                    if ( DivRat(BHEAD (UWORD *)n_coef,ncoef,(UWORD *)m,REDLENG(j),(UWORD
*)n_coef,&ncoef) ) goto FromNorm;
                    ncoef = INCLENG(ncoef);
                    j = ABS(j) - 3;
                    t[-FUNHEAD-ARGHEAD] -= j;
                    t[-ARGHEAD-1] -= j;
                    t[-1] -= j;
                    m[0] = m[1] = 1;
                    m[2] = 3;
                }
                while ( t < m ) {
                    r = t + t[1];
                    if ( *t == SYMBOL || *t == DOTPRODUCT ) {
                        k = t[1];
                        pden[i][1] -= k;
                        pden[i][FUNHEAD] -= k;
                        pden[i][FUNHEAD+ARGHEAD] -= k;
                        m -= k;
                        stop = m + 3;
                        tt = to = t;
                        from = r;
                        if ( *t == SYMBOL ) {
                            t += 2;
                            while ( t < r ) {
                                WORD change;
                                change = ExtraSymbol(*t,-t[1],nsym,ppsym,&ncoef);
                                nsym += change;
                                ppsym += change * 2;
                                t += 2;
                            }
                        }
                        else {
                            t += 2;
                            while ( t < r ) {
                                *ppdot++ = *t++;
                                *ppdot++ = *t++;
                                *ppdot++ = -*t++;
                                ndot++;
                            }
                        }
                        while ( to < stop ) *to++ = *from++;
                        r = tt;
                    }
#ifdef WITHFLOAT
                    else if ( *t == FLOATFUN && TestFloat(t) ) {
                        k = t[1];
                        pden[i][1] -= k;
                        pden[i][FUNHEAD] -= k;
                        pden[i][FUNHEAD+ARGHEAD] -= k;
```

```
                        UnpackFloat(aux5,t);
                        if ( withfloat == 0 ) {
                            mpf_ui_div(aux4,1,aux5);
                            withfloat = 2;
                        }
                        else {
                            if ( withfloat == 1 ) UnpackFloat(aux4,firstfloat);
                            mpf_div(aux4,aux4,aux5);
                            withfloat++;
                        }
                        tt = to = t;
                        while ( r < m ) *to++ = *r++;
                        *to++ = 1; *to++ = 1; *to++ = 3;
                        if ( ncoef < 0 ) t[-1] = -t[-1];
                        m -= k;
                        stop = m + 3;
                        r = tt;
                    }
#endif
                    t = r;
                }
                if ( pden[i][1] == 4+FUNHEAD+ARGHEAD ) {
DropDen:
                    for ( j = 0; j < nnco; j++ ) {
                        if ( pden[i] == pnco[j] ) {
                            --nnco;
                            while ( j < nnco ) {
                                pnco[j] = pnco[j+1];
                                j++;
                            }
                            break;
                        }
                    }
                    pden[i--] = pden[--nden];
                }
            }
        }
    }
/*
    #] Easy denominators :
    #[ Index Contractions :
*/
    if ( ndel ) {
        t = pdel;
        for ( i = 0; i < ndel; i += 2 ) {
            if ( t[0] == t[1] ) {
                if ( t[0] == EMPTYINDEX ) {}
                else if ( *t < AM.OffsetIndex ) {
                    k = AC.FixIndices[*t];
                    if ( k < 0 ) { j = -1; k = -k; }
                    else if ( k > 0 ) j = 1;
                    else goto NormZero;
                    goto WithFix;
                }
                else if ( *t >= AM.DumInd ) {
                    k = AC.lDefDim;
                    if ( k ) goto docontract;
                }
                else if ( *t >= AM.WilInd ) {
                    k = indices[*t-AM.OffsetIndex-WILDOFFSET].dimension;
                    if ( k ) goto docontract;
                }
                else if ( ( k = indices[*t-AM.OffsetIndex].dimension ) != 0 ) {
docontract:
                    if ( k > 0 ) {
                        j = 1;
WithFix:                shortnum = k;
                        ncoef = REDLENG(ncoef);
                        if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)(&shortnum),j) )
                            goto FromNorm;
                        ncoef = INCLENG(ncoef);
                    }
                    else {
                        WORD change;
                        change = ExtraSymbol((WORD)(-k),(WORD)1,nsym,ppsym,&ncoef);
                        nsym += change;
                        ppsym += change * 2;
                    }
                    t[1] = pdel[ndel-1];
                    t[0] = pdel[ndel-2];
HaveCon:
                    ndel -= 2;
                    i -= 2;
                }
            }
            else {
                if ( *t < AM.OffsetIndex && t[1] < AM.OffsetIndex ) goto NormZero;
```

```
                j = *t - AM.OffsetIndex;
                if ( j >= 0 && ( ( *t >= AM.DumInd && AC.lDefDim )
                || ( *t < AM.WilInd && indices[j].dimension ) ) ) {
                    for ( j = i + 2, m = pdel+j; j < ndel; j += 2, m += 2 ) {
                        if ( *t == *m ) {
                            *t = m[1];
                            *m++ = pdel[ndel-2];
                            *m = pdel[ndel-1];
                            goto HaveCon;
                        }
                        else if ( *t == m[1] ) {
                            *t = *m;
                            *m++ = pdel[ndel-2];
                            *m   = pdel[ndel-1];
                            goto HaveCon;
                        }
                    }
                }
                j = t[1]-AM.OffsetIndex;
                if ( j >= 0 && ( ( t[1] >= AM.DumInd && AC.lDefDim )
                || ( t[1] < AM.WilInd && indices[j].dimension ) ) ) {
                    for ( j = i + 2, m = pdel+j; j < ndel; j += 2, m += 2 ) {
                        if ( t[1] == *m ) {
                            t[1] = m[1];
                            *m++ = pdel[ndel-2];
                            *m   = pdel[ndel-1];
                            goto HaveCon;
                        }
                        else if ( t[1] == m[1] ) {
                            t[1] = *m;
                            *m++ = pdel[ndel-2];
                            *m   = pdel[ndel-1];
                            goto HaveCon;
                        }
                    }
                }
                t += 2;
            }
        }
        if ( ndel > 0 ) {
            if ( nvec ) {
                t = pdel;
                for ( i = 0; i < ndel; i++ ) {
                    if ( *t >= AM.OffsetIndex && ( ( *t >= AM.DumInd && AC.lDefDim ) ||
                    ( *t < AM.WilInd && indices[*t-AM.OffsetIndex].dimension ) ) ) {
                        r = pvec + 1;
                        for ( j = 1; j < nvec; j += 2 ) {
                            if ( *r == *t ) {
                                if ( i & 1 ) {
                                    *r = t[-1];
                                    *t-- = pdel[--ndel];
                                    i -= 2;
                                }
                                else {
                                    *r = t[1];
                                    t[1] = pdel[--ndel];
                                    i--;
                                }
                                *t-- = pdel[--ndel];
                                break;
                            }
                            r += 2;
                        }
                    }
                    t++;
                }
            }
            if ( ndel > 0 && ncon ) {
                t = pdel;
                for ( i = 0; i < ndel; i++ ) {
                    if ( *t >= AM.OffsetIndex && ( ( *t >= AM.DumInd && AC.lDefDim ) ||
                    ( *t < AM.WilInd && indices[*t-AM.OffsetIndex].dimension ) ) ) {
                        for ( j = 0; j < ncon; j++ ) {
                            if ( *pcon[j] == *t ) {
                                if ( i & 1 ) {
                                    *pcon[j] = t[-1];
                                    *t-- = pdel[--ndel];
                                    i -= 2;
                                }
                                else {
                                    *pcon[j] = t[1];
                                    t[1] = pdel[--ndel];
                                    i--;
                                }
                                *t-- = pdel[--ndel];
                                didcontr++;
                                r = pcon[j];
```

```
                                for ( j = 0; j < nnco; j++ ) {
                                    m = pnco[j];
                                    if ( r > m && r < m+m[1] ) {
                                        m[2] |= DIRTYSYMFLAG;
                                        break;
                                    }
                                }
                                for ( j = 0; j < ncom; j++ ) {
                                    m = pcom[j];
                                    if ( r > m && r < m+m[1] ) {
                                        m[2] |= DIRTYSYMFLAG;
                                        break;
                                    }
                                }
                                for ( j = 0; j < neps; j++ ) {
                                    m = peps[j];
                                    if ( r > m && r < m+m[1] ) {
                                        m[2] |= DIRTYSYMFLAG;
                                        break;
                                    }
                                }
                                break;
                            }
                        }
                    }
                    t++;
                }
            }
        }
    }
    if ( nvec ) {
        t = pvec + 1;
        for ( i = 3; i < nvec; i += 2 ) {
            k = *t - AM.OffsetIndex;
            if ( k >= 0 && ( ( *t > AM.DumInd && AC.lDefDim )
            || ( *t < AM.WilInd && indices[k].dimension ) ) ) {
                r = t + 2;
                for ( j = i; j < nvec; j += 2 ) {
                    if ( *r == *t ) {    /* Another dotproduct */
                        *ppdot++ = t[-1];
                        *ppdot++ = r[-1];
                        *ppdot++ = 1;
                        ndot++;
                        *r-- = pvec[--nvec];
                        *r   = pvec[--nvec];
                        *t-- = pvec[--nvec];
                        *t-- = pvec[--nvec];
                        i -= 2;
                        break;
                    }
                    r += 2;
                }
            }
            t += 2;
        }
        if ( nvec > 0 && ncon ) {
            t = pvec + 1;
            for ( i = 1; i < nvec; i += 2 ) {
                k = *t - AM.OffsetIndex;
                if ( k >= 0 && ( ( *t >= AM.DumInd && AC.lDefDim )
                || ( *t < AM.WilInd && indices[k].dimension ) ) ) {
                    for ( j = 0; j < ncon; j++ ) {
                        if ( *pcon[j] == *t ) {
                            *pcon[j] = t[-1];
                            *t-- = pvec[--nvec];
                            *t-- = pvec[--nvec];
                            r = pcon[j];
                            pcon[j] = pcon[--ncon];
                            i -= 2;
                            for ( j = 0; j < nnco; j++ ) {
                                m = pnco[j];
                                if ( r > m && r < m+m[1] ) {
                                    m[2] |= DIRTYSYMFLAG;
                                    break;
                                }
                            }
                            for ( j = 0; j < ncom; j++ ) {
                                m = pcom[j];
                                if ( r > m && r < m+m[1] ) {
                                    m[2] |= DIRTYSYMFLAG;
                                    break;
                                }
                            }
                            for ( j = 0; j < neps; j++ ) {
                                m = peps[j];
                                if ( r > m && r < m+m[1] ) {
                                    m[2] |= DIRTYSYMFLAG;
```

```
                                        break;
                                    }
                                }
                                break;
                            }
                        }
                    }
                    t += 2;
                }
            }
        }
/*
    #] Index Contractions :
    #[ NonCommuting Functions :
*/
    m = fillsetexp;
    if ( nnco ) {
        for ( i = 0; i < nnco; i++ ) {
            t = pnco[i];
            if ( ( *t >= (FUNCTION+WILDOFFSET)
            && functions[*t-FUNCTION-WILDOFFSET].spec <= 0 )
            || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
            && functions[*t-FUNCTION].spec <= 0 ) ) {
                DoRevert(t,m);
                if ( didcontr ) {
                    r = t + FUNHEAD;
                    t += t[1];
                    while ( r < t ) {
                        if ( *r == -INDEX && r[1] >= 0 && r[1] < AM.OffsetIndex ) {
                            *r = -SNUMBER;
                            didcontr--;
                            pnco[i][2] |= DIRTYSYMFLAG;
                        }
                        NEXTARG(r)
                    }
                }
            }
        }

        /* First should come the code for function properties. */

        /* First we test for symmetric properties and the DIRTYSYMFLAG */

        for ( i = 0; i < nnco; i++ ) {
            t = pnco[i];
            if ( *t > 0 && ( t[2] & DIRTYSYMFLAG ) && *t != DOLLAREXPRESSION ) {
                l = 0; /* to make the compiler happy */
                if ( ( *t >= (FUNCTION+WILDOFFSET)
                && ( l = functions[*t-FUNCTION-WILDOFFSET].symmetric ) > 0 )
                || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
                && ( l = functions[*t-FUNCTION].symmetric ) > 0 ) ) {
                    if ( *t >= (FUNCTION+WILDOFFSET) ) {
                        *t -= WILDOFFSET;
                        j = FullSymmetrize(BHEAD t,l);
                        *t += WILDOFFSET;
                    }
                    else j = FullSymmetrize(BHEAD t,l);
                    if ( (l & ~REVERSEORDER) == ANTISYMMETRIC ) {
                        if ( ( j & 2 ) != 0 ) goto NormZero;
                        if ( ( j & 1 ) != 0 ) ncoef = -ncoef;
                    }
                }
                else t[2] &= ~DIRTYSYMFLAG;
            }
        }

        /* Non commuting functions are then tested for commutation
           rules. If needed their order is exchanged. */

        k = nnco - 1;
        for ( i = 0; i < k; i++ ) {
            j = i;
            while ( Commute(pnco[j],pnco[j+1]) ) {
                t = pnco[j]; pnco[j] = pnco[j+1]; pnco[j+1] = t;
                l = j-1;
                while ( l >= 0 && Commute(pnco[l],pnco[l+1]) ) {
                    t = pnco[l]; pnco[l] = pnco[l+1]; pnco[l+1] = t;
                    l--;
                }
                if ( ++j >= k ) break;
            }
        }

        /* Finally they are written to output. gamma matrices
           are bundled if possible */

        for ( i = 0; i < nnco; i++ ) {
```

```
            t = pnco[i];
            if ( *t == IDFUNCTION ) AN.idfunctionflag = 1;
            if ( *t >= GAMMA && *t <= GAMMASEVEN ) {
                WORD gtype;
                to = m;
                *m++ = GAMMA;
                m++;
                FILLFUN(m)
                *m++ = stype = t[FUNHEAD];      /* type of string */
                j = 0;
                nnum = 0;
                do {
                    r = t + t[1];
                    if ( *t == GAMMAFIVE ) {
                        gtype = GAMMA5; t += FUNHEAD; goto onegammamatrix; }
                    else if ( *t == GAMMASIX ) {
                        gtype = GAMMA6; t += FUNHEAD; goto onegammamatrix; }
                    else if ( *t == GAMMASEVEN ) {
                        gtype = GAMMA7; t += FUNHEAD; goto onegammamatrix; }
                    t += FUNHEAD+1;
                    while ( t < r ) {
                        gtype = *t;
onegammamatrix:
                        if ( gtype == GAMMA5 ) {
                            if ( j == GAMMA1 ) j = GAMMA5;
                            else if ( j == GAMMA5 ) j = GAMMA1;
                            else if ( j == GAMMA7 ) ncoef = -ncoef;
                            if ( nnum & 1 ) ncoef = -ncoef;
                        }
                        else if ( gtype == GAMMA6 || gtype == GAMMA7 ) {
                            if ( nnum & 1 ) {
                                if ( gtype == GAMMA6 ) gtype = GAMMA7;
                                else                   gtype = GAMMA6;
                            }
                            if ( j == GAMMA1 ) j = gtype;
                            else if ( j == GAMMA5 ) {
                                j = gtype;
                                if ( j == GAMMA7 ) ncoef = -ncoef;
                            }
                            else if ( j != gtype ) goto NormZero;
                            else {
                                shortnum = 2;
                                ncoef = REDLENG(ncoef);
                                if ( Mully(BHEAD (UWORD *)n_coef,&ncoef,(UWORD *)(&shortnum),1) ) goto FromNorm;
                                ncoef = INCLENG(ncoef);
                            }
                        }
                        else {
                            *m++ = gtype; nnum++;
                        }
                        t++;
                    }

                } while ( ( ++i < nnco ) && ( *(t = pnco[i]) >= GAMMA
                && *t <= GAMMASEVEN ) && ( t[FUNHEAD] == stype ) );
                i--;
                if ( j ) {
                    k = WORDDIF(m,to) - FUNHEAD-1;
                    r = m;
                    from = m++;
                    while ( --k >= 0 ) *from-- = *--r;
                    *from = j;
                }
                to[1] = WORDDIF(m,to);
            }
            else if ( *t < 0 ) {
                *m++ = -*t; *m++ = FUNHEAD; *m++ = 0;
                FILLFUN3(m)
            }
            else {
                if ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG
                        && *t != REPLACEMENT && *t != DOLLAREXPRESSION
                        && TestFunFlag(BHEAD t) ) ReplaceVeto = 1;
                k = t[1];
                NCOPY(m,t,k);
            }
        }

    }
/*
    #] NonCommuting Functions :
    #[ Commuting Functions :
*/
    if ( ncom ) {
        for ( i = 0; i < ncom; i++ ) {
            t = pcom[i];
```

```
            if ( ( *t >= (FUNCTION+WILDOFFSET)
            && functions[*t-FUNCTION-WILDOFFSET].spec <= 0 )
            || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
            && functions[*t-FUNCTION].spec <= 0 ) ) {
                DoRevert(t,m);
                if ( didcontr ) {
                    r = t + FUNHEAD;
                    t += t[1];
                    while ( r < t ) {
                        if ( *r == -INDEX && r[1] >= 0 && r[1] < AM.OffsetIndex ) {
                            *r = -SNUMBER;
                            didcontr--;
                            pcom[i][2] |= DIRTYSYMFLAG;
                        }
                        NEXTARG(r)
                    }
                }
            }
        }

        /* Now we test for symmetric properties and the DIRTYSYMFLAG */

        for ( i = 0; i < ncom; i++ ) {
            t = pcom[i];
            if ( *t > 0 && ( t[2] & DIRTYSYMFLAG ) ) {
                l = 0; /* to make the compiler happy */
                if ( ( *t >= (FUNCTION+WILDOFFSET)
                && ( l = functions[*t-FUNCTION-WILDOFFSET].symmetric ) > 0 )
                || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
                && ( l = functions[*t-FUNCTION].symmetric ) > 0 ) ) {
                    if ( *t >= (FUNCTION+WILDOFFSET) ) {
                        *t -= WILDOFFSET;
                        j = FullSymmetrize(BHEAD t,l);
                        *t += WILDOFFSET;
                    }
                    else j = FullSymmetrize(BHEAD t,l);
                    if ( (l & ~REVERSEORDER) == ANTISYMMETRIC ) {
                        if ( ( j & 2 ) != 0 ) goto NormZero;
                        if ( ( j & 1 ) != 0 ) ncoef = -ncoef;
                    }
                }
                else t[2] &= ~DIRTYSYMFLAG;
            }
        }
/*
        Sort the functions
        From a purists point of view this can be improved.
        There are slow and fast arguments and no conversions are
        taken into account here.
*/
        for ( i = 1; i < ncom; i++ ) {
            for ( j = i; j > 0; j-- ) {
                WORD jj,kk;
                jj = j-1;
                t = pcom[jj];
                r = pcom[j];
                if ( *t < 0 ) {
                    if ( *r < 0 ) { if ( *t >= *r ) goto NextI; }
                    else { if ( -*t <= *r ) goto NextI; }
                    goto jexch;
                }
                else if ( *r < 0 ) {
                    if ( *t < -*r ) goto NextI;
                    goto jexch;
                }
                else if ( *t != *r ) {
                    if ( *t < *r ) goto NextI;
jexch:              t = pcom[j]; pcom[j] = pcom[jj]; pcom[jj] = t;
                    continue;
                }
                if ( AC.properorderflag ) {
                    if ( ( *t >= (FUNCTION+WILDOFFSET)
                    && functions[*t-FUNCTION-WILDOFFSET].spec >= TENSORFUNCTION )
                    || ( *t >= FUNCTION && *t < (FUNCTION + WILDOFFSET)
                    && functions[*t-FUNCTION].spec >= TENSORFUNCTION ) ) {}
                    else {
                        WORD *s1, *s2, *ss1, *ss2;
                        s1 = t+FUNHEAD; s2 = r+FUNHEAD;
                        ss1 = t + t[1]; ss2 = r + r[1];
                        while ( s1 < ss1 && s2 < ss2 ) {
                            k = CompArg(s1,s2);
                            if ( k > 0 ) goto jexch;
                            if ( k < 0 ) goto NextI;
                            NEXTARG(s1)
                            NEXTARG(s2)
                        }
                        if ( s1 < ss1 ) goto jexch;
```

```
                        goto NextI;
                    }
                    k = t[1] - FUNHEAD;
                    kk = r[1] - FUNHEAD;
                    t += FUNHEAD;
                    r += FUNHEAD;
                    while ( k > 0 && kk > 0 ) {
                        if ( *t < *r ) goto NextI;
                        else if ( *t++ > *r++ ) goto jexch;
                        k--; kk--;
                    }
                    if ( k > 0 ) goto jexch;
                    goto NextI;
                }
                else
                {
                    k = t[1] - FUNHEAD;
                    kk = r[1] - FUNHEAD;
                    t += FUNHEAD;
                    r += FUNHEAD;
                    while ( k > 0 && kk > 0 ) {
                        if ( *t < *r ) goto NextI;
                        else if ( *t++ > *r++ ) goto jexch;
                        k--; kk--;
                    }
                    if ( k > 0 ) goto jexch;
                    goto NextI;
                }
            }
NextI:;
        }
        for ( i = 0; i < ncom; i++ ) {
            t = pcom[i];
            if ( *t == THETA || *t == THETA2 ) {
                if ( ( k = DoTheta(BHEAD t) ) == 0 ) goto NormZero;
                else if ( k < 0 ) {
                    k = t[1];
                    NCOPY(m,t,k);
                }
            }
            else if ( *t == DELTA2 || *t == DELTAP ) {
                if ( ( k = DoDelta(t) ) == 0 ) goto NormZero;
                else if ( k < 0 ) {
                    k = t[1];
                    NCOPY(m,t,k);
                }
            }
            else if ( *t == AR.PolyFunInv && AR.PolyFunType == 2 ) {
/*
                If there are two arguments, exchange them, change the
                name of the function and go to dealing with PolyRatFun.
*/
                WORD *mm, *tt = t, numt = 0;
                tt += FUNHEAD;
                while ( tt < t+t[1] ) { numt++; NEXTARG(tt) }
                if ( numt == 2 ) {
                    tt = t; mm = m; k = t[1];
                    NCOPY(mm,tt,k)
                    mm = m+FUNHEAD;
                    NEXTARG(mm);
                    tt = t+FUNHEAD;
                    if ( *mm < 0 ) {
                        if ( *mm <= -FUNCTION ) { *tt++ = *mm++; }
                        else { *tt++ = *mm++;  *tt++ = *mm++; }
                    }
                    else {
                        k = *mm; NCOPY(tt,mm,k)
                    }
                    mm = m+FUNHEAD;
                    if ( *mm < 0 ) {
                        if ( *mm <= -FUNCTION ) { *tt++ = *mm++; }
                        else { *tt++ = *mm++;  *tt++ = *mm++; }
                    }
                    else {
                        k = *mm; NCOPY(tt,mm,k)
                    }
                    *t = AR.PolyFun;
                    t[2] |= MUSTCLEANPRF;
                    goto regularratfun;
                }
            }
            else if ( *t == AR.PolyFun ) {
              if ( AR.PolyFunType == 1 ) { /* Regular PolyFun with one argument */
                if ( t[FUNHEAD+1] == 0 && AR.Eside != LHSIDE &&
                t[1] == FUNHEAD + 2 && t[FUNHEAD] == -SNUMBER ) goto NormZero;
                if ( i > 0 && pcom[i-1][0] == AR.PolyFun ) {
                    if ( AN.PolyNormFlag == 0 ) {
```

```
                            AN.PolyNormFlag = 1;
                            AN.PolyFunTodo = 0;
                        }
                    }
                    k = t[1];
                    NCOPY(m,t,k);
                }
                else if ( AR.PolyFunType == 2 ) {
/*
                    PolyRatFun.
                        Regular type: Two arguments
                        Power expanded: One argument. Here to be treated as
                            AR.PolyFunType == 1, but with power cutoff.
*/
regularratfun:;
/*
                    First check for zeroes.
*/
                    if ( t[FUNHEAD+1] == 0 && AR.Eside != LHSIDE &&
                    t[1] > FUNHEAD + 2 && t[FUNHEAD] == -SNUMBER ) {
                        u = t + FUNHEAD + 2;
                        if ( *u < 0 ) {
                            if ( *u <= -FUNCTION ) {}
                            else if ( t[1] == FUNHEAD+4 && t[FUNHEAD+2] == -SNUMBER
                                && t[FUNHEAD+3] == 0 ) goto NormPRF;
                            else if ( t[1] == FUNHEAD+4 ) goto NormZero;
                        }
                        else if ( t[1] == *u+FUNHEAD+2 ) goto NormZero;
                    }
                    else {
                        u = t+FUNHEAD; NEXTARG(u);
                        if ( *u == -SNUMBER && u[1] == 0 ) goto NormInf;
                    }
                    if ( i > 0 && pcom[i-1][0] == AR.PolyFun ) AN.PolyNormFlag = 1;
                    else if ( i < ncom-1 && pcom[i+1][0] == AR.PolyFun ) AN.PolyNormFlag = 1;
                    k = t[1];
                    if ( AN.PolyNormFlag ) {
                        if ( AR.PolyFunExp == 0 ) {
                            AN.PolyFunTodo = 0;
                            NCOPY(m,t,k);
                        }
                        else if ( AR.PolyFunExp == 1 ) { /* get highest divergence */
                            if ( PolyFunMode == 0 ) {
                                NCOPY(m,t,k);
                                AN.PolyFunTodo = 1;
                            }
                            else {
                                WORD *mmm = m;
                                NCOPY(m,t,k);
                                if ( TreatPolyRatFun(BHEAD mmm) != 0 )
                                        goto FromNorm;
                                m = mmm+mmm[1];
                            }
                        }
                        else {
                            if ( PolyFunMode == 0 ) {
                                NCOPY(m,t,k);
                                    AN.PolyFunTodo = 1;
                            }
                            else {
                                WORD *mmm = m;
                                NCOPY(m,t,k);
                                if ( ExpandRat(BHEAD mmm) != 0 )
                                        goto FromNorm;
                                m = mmm+mmm[1];
                            }
                        }
                    }
                    else {
                        if ( AR.PolyFunExp == 0 ) {
                            AN.PolyFunTodo = 0;
                            NCOPY(m,t,k);
                        }
                        else if ( AR.PolyFunExp == 1 ) { /* get highest divergence */
                            WORD *mmm = m;
                            NCOPY(m,t,k);
                            if ( TreatPolyRatFun(BHEAD mmm) != 0 )
                                    goto FromNorm;
                            m = mmm+mmm[1];
                        }
                        else {
                            WORD *mmm = m;
                            NCOPY(m,t,k);
                            if ( ExpandRat(BHEAD mmm) != 0 )
                                    goto FromNorm;
                            m = mmm+mmm[1];
                        }
```

```
                }
              }
            }
            else if ( *t > 0 ) {
                if ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG
                    && *t != REPLACEMENT && TestFunFlag(BHEAD t) ) ReplaceVeto = 1;
                k = t[1];
                NCOPY(m,t,k);
            }
            else {
                *m++ = -*t; *m++ = FUNHEAD; *m++ = 0;
                FILLFUN3(m)
            }
        }
    }
/*
    #] Commuting Functions :
    #[ Track Replace_ :
*/
    if ( ReplaceVeto < 0 ) {
/*
        We found one (or more) replace_ functions and all other
        functions are 'clean' (no dirty flag).
        Now we check whether one of these functions can be used.
        Thus far the functions go from fillsetexp to m.
        Somewhere in there there are -ReplaceVeto occurrences of REPLACEMENT.
        Hunt for the first one that fits the bill.
        Note that replace_ is a commuting function.
*/
        WORD *ma = fillsetexp, *mb, *mc;
        while ( ma < m ) {
            mb = ma + ma[1];
            if ( *ma != REPLACEMENT ) {
                ma = mb;
                continue;
            }
            if ( *ma == REPLACEMENT && ReplaceType == -1 ) {
                mc = ma;
                ReplaceType = 0;
                if ( AN.RSsize < 2*ma[1]+SUBEXPSIZE ) {
                    if ( AN.ReplaceScrat ) M_free(AN.ReplaceScrat,"AN.ReplaceScrat");
                    AN.RSsize = 2*ma[1]+SUBEXPSIZE+40;
                    AN.ReplaceScrat = (WORD *)Malloc1((AN.RSsize+1)*sizeof(WORD),"AN.ReplaceScrat");
                }
                ma += FUNHEAD;
                ReplaceSub = AN.ReplaceScrat;
                ReplaceSub += SUBEXPSIZE;
                while ( ma < mb ) {
                    if ( *ma > 0 ) goto NoRep;
                    if ( *ma <= -FUNCTION ) {
                        *ReplaceSub++ = FUNTOFUN;
                        *ReplaceSub++ = 4;
                        *ReplaceSub++ = -*ma++;
                        if ( *ma > -FUNCTION ) goto NoRep;
                        *ReplaceSub++ = -*ma++;
                    }
                    else if ( ma+4 > mb ) goto NoRep;
                    else {
                        if ( *ma == -SYMBOL ) {
                            if ( ma[2] == -SYMBOL && ma+4 <= mb )
                                *ReplaceSub++ = SYMTOSYM;
                            else if ( ma[2] == -SNUMBER && ma+4 <= mb ) {
                                *ReplaceSub++ = SYMTONUM;
                                if ( ReplaceType == 0 ) {
                                    oldtoprhs = C->numrhs;
                                    oldcpointer = C->Pointer - C->Buffer;
                                }
                                ReplaceType = 1;
                            }
                            else if ( ma[2] == ARGHEAD && ma+2+ARGHEAD <= mb ) {
                                *ReplaceSub++ = SYMTONUM;
                                *ReplaceSub++ = 4;
                                *ReplaceSub++ = ma[1];
                                *ReplaceSub++ = 0;
                                ma += 2+ARGHEAD;
                                continue;
                            }
/*
                            Next is the subexpression. We have to test that
                            it isn't vector-like or index-like
*/
                            else if ( ma[2] > 0 ) {
                                WORD *sstop, *ttstop, n;
                                ss = ma+2;
                                sstop = ss + *ss;
                                ss += ARGHEAD;
                                while ( ss < sstop ) {
```

```
                                    tt = ss + *ss;
                                    ttstop = tt - ABS(tt[-1]);
                                    ss++;
                                    while ( ss < ttstop ) {
                                        if ( *ss == INDEX ) goto NoRep;
                                        ss += ss[1];
                                    }
                                    ss = tt;
                                }
                                subtype = SYMTOSUB;
                                if ( ReplaceType == 0 ) {
                                    oldtoprhs = C->numrhs;
                                    oldcpointer = C->Pointer - C->Buffer;
                                }
                                ReplaceType = 1;
                                ss = AddRHS(AT.ebufnum,1);
                                tt = ma+2;
                                n = *tt - ARGHEAD;
                                tt += ARGHEAD;
                                while ( (ss + n + 10) > C->Top ) ss = DoubleCbuffer(AT.ebufnum,ss,14);
                                while ( --n >= 0 ) *ss++ = *tt++;
                                *ss++ = 0;
                                C->rhs[C->numrhs+1] = ss;
                                C->Pointer = ss;
                                *ReplaceSub++ = subtype;
                                *ReplaceSub++ = 4;
                                *ReplaceSub++ = ma[1];
                                *ReplaceSub++ = C->numrhs;
                                ma += 2 + ma[2];
                                continue;
                            }
                            else goto NoRep;
                        }
                        else if ( ( *ma == -VECTOR || *ma == -MINVECTOR ) && ma+4 <= mb ) {
                            if ( ma[2] == -VECTOR ) {
                                if ( *ma == -VECTOR ) *ReplaceSub++ = VECTOVEC;
                                else *ReplaceSub++ = VECTOMIN;
                            }
                            else if ( ma[2] == -MINVECTOR ) {
                                if ( *ma == -VECTOR ) *ReplaceSub++ = VECTOMIN;
                                else *ReplaceSub++ = VECTOVEC;
                            }
/*
                            Next is a vector-like subexpression
                            Search for vector nature first
*/
                            else if ( ma[2] > 0 ) {
                                WORD *sstop, *ttstop, *w, *mm, n, count;
                                WORD *v1, *v2 = 0;
                                if ( *ma == -MINVECTOR ) {
                                    ss = ma+2;
                                    sstop = ss + *ss;
                                    ss += ARGHEAD;
                                    while ( ss < sstop ) {
                                        ss += *ss;
                                        ss[-1] = -ss[-1];
                                    }
                                    *ma = -VECTOR;
                                }
                                ss = ma+2;
                                sstop = ss + *ss;
                                ss += ARGHEAD;
                                while ( ss < sstop ) {
                                    tt = ss + *ss;
                                    ttstop = tt - ABS(tt[-1]);
                                    ss++;
                                    count = 0;
                                    while ( ss < ttstop ) {
                                        if ( *ss == INDEX ) {
                                            n = ss[1] - 2; ss += 2;
                                            while ( --n >= 0 ) {
                                                if ( *ss < MINSPEC ) count++;
                                                ss++;
                                            }
                                        }
                                        else ss += ss[1];
                                    }
                                    if ( count != 1 ) goto NoRep;
                                    ss = tt;
                                }
                                subtype = VECTOSUB;
                                if ( ReplaceType == 0 ) {
                                    oldtoprhs = C->numrhs;
                                    oldcpointer = C->Pointer - C->Buffer;
                                }
                                ReplaceType = 1;
                                mm = AddRHS(AT.ebufnum,1);
```

```
                *ReplaceSub++ = subtype;
                *ReplaceSub++ = 4;
                *ReplaceSub++ = ma[1];
                *ReplaceSub++ = C->numrhs;
                w = ma+2;
                n = *w - ARGHEAD;
                w += ARGHEAD;
                while ( (mm + n + 10) > C->Top )
                    mm = DoubleCbuffer(AT.ebufnum,mm,15);
                while ( --n >= 0 ) *mm++ = *w++;
                *mm++ = 0;
                C->rhs[C->numrhs+1] = mm;
                C->Pointer = mm;
                mm = AddRHS(AT.ebufnum,1);
                w = ma+2;
                n = *w - ARGHEAD;
                w += ARGHEAD;
                while ( (mm + n + 13) > C->Top )
                    mm = DoubleCbuffer(AT.ebufnum,mm,16);
                sstop = w + n;
                while ( w < sstop ) {
                    tt = w + *w; ttstop = tt - ABS(tt[-1]);
                    ss = mm; mm++; w++;
                    while ( w < ttstop ) {      /* Subterms */
                        if ( *w != INDEX ) {
                            n = w[1];
                            NCOPY(mm,w,n);
                        }
                        else {
                            v1 = mm;
                            *mm++ = *w++;
                            *mm++ = n = *w++;
                            n -= 2;
                            while ( --n >= 0 ) {
                                if ( *w >= MINSPEC ) *mm++ = *w++;
                                else v2 = w++;
                            }
                            n = WORDDIF(mm,v1);
                            if ( n != v1[1] ) {
                                if ( n <= 2 ) mm -= 2;
                                else v1[1] = n;
                                *mm++ = VECTOR;
                                *mm++ = 4;
                                *mm++ = *v2;
                                *mm++ = FUNNYVEC;
                            }
                        }
                    }
                    while ( w < tt ) *mm++ = *w++;
                    *ss = WORDDIF(mm,ss);
                }
                *mm++ = 0;
                C->rhs[C->numrhs+1] = mm;
                C->Pointer = mm;
                if ( mm > C->Top ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Internal error in Normalize with extra compiler buffer");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                ma += 2 + ma[2];
                continue;
            }
            else goto NoRep;
        }
        else if ( *ma == -INDEX ) {
            if ( ( ma[2] == -INDEX || ma[2] == -VECTOR )
            && ma+4 <= mb )
                *ReplaceSub++ = INDTOIND;
            else if ( ma[1] >= AM.OffsetIndex ) {
                if ( ma[2] == -SNUMBER && ma+4 <= mb
                && ma[3] >= 0 && ma[3] < AM.OffsetIndex )
                    *ReplaceSub++ = INDTOIND;
                else if ( ma[2] == ARGHEAD && ma+2+ARGHEAD <= mb ) {
                    *ReplaceSub++ = INDTOIND;
                    *ReplaceSub++ = 4;
                    *ReplaceSub++ = ma[1];
                    *ReplaceSub++ = 0;
                    ma += 2+ARGHEAD;
                    continue;
                }
                else goto NoRep;
            }
            else goto NoRep;
        }
        else goto NoRep;
```

```
                        *ReplaceSub++ = 4;
                        *ReplaceSub++ = ma[1];
                        *ReplaceSub++ = ma[3];
                        ma += 4;
                    }

                }
                AN.ReplaceScrat[1] = ReplaceSub-AN.ReplaceScrat;
/*
                Success. This means that we have to remove the replace_
                from the functions. It starts at mc and end at mb.
*/
                while ( mb < m ) *mc++ = *mb++;
                m = mc;
                break;
NoRep:
                if ( ReplaceType > 0 ) {
                    C->numrhs = oldtoprhs;
                    C->Pointer = C->Buffer + oldcpointer;
                }
                ReplaceType = -1;
                if ( ++ReplaceVeto >= 0 ) break;
            }
            ma = mb;
        }
    }
/*
    #] Track Replace_ :
    #[ LeviCivita tensors :
*/
    if ( neps ) {
        to = m;
        for ( i = 0; i < neps; i++ ) {  /* Put the indices in order */
            t = peps[i];
            if ( ( t[2] & DIRTYSYMFLAG ) != DIRTYSYMFLAG ) continue;
            t[2] &= ~DIRTYSYMFLAG;
            if ( AR.Eside == LHSIDE || AR.Eside == LHSIDEX ) {
                        /* Potential problems with FUNNYWILD */
/*
                First make sure all FUNNIES are at the end.
                Then sort separately
*/
                r = t + FUNHEAD;
                m = tt = t + t[1];
                while ( r < m ) {
                    if ( *r != FUNNYWILD ) { r++; continue; }
                    k = r[1]; u = r + 2;
                    while ( u < tt ) {
                        u[-2] = *u;
                        if ( *u != FUNNYWILD ) ncoef = -ncoef;
                        u++;
                    }
                    tt[-2] = FUNNYWILD; tt[-1] = k; m -= 2;
                }
                t += FUNHEAD;
                do {
                    for ( r = t + 1; r < m; r++ ) {
                        if ( *r < *t ) { k = *r; *r = *t; *t = k; ncoef = -ncoef; }
                        else if ( *r == *t ) goto NormZero;
                    }
                    t++;
                } while ( t < m );
                do {
                    for ( r = t + 2; r < tt; r += 2 ) {
                        if ( r[1] < t[1] ) {
                            k = r[1]; r[1] = t[1]; t[1] = k; ncoef = -ncoef; }
                        else if ( r[1] == t[1] ) goto NormZero;
                    }
                    t += 2;
                } while ( t < tt );
            }
            else {
                m = t + t[1];
                t += FUNHEAD;
                do {
                    for ( r = t + 1; r < m; r++ ) {
                        if ( *r < *t ) { k = *r; *r = *t; *t = k; ncoef = -ncoef; }
                        else if ( *r == *t ) goto NormZero;
                    }
                    t++;
                } while ( t < m );
            }
        }

        /* Sort the tensors */

        for ( i = 0; i < (neps-1); i++ ) {
```

```
            t = peps[i];
            for ( j = i+1; j < neps; j++ ) {
                r = peps[j];
                if ( t[1] > r[1] ) {
                    peps[i] = m = r; peps[j] = r = t; t = m;
                }
                else if ( t[1] == r[1] ) {
                    k = t[1] - FUNHEAD;
                    m = t + FUNHEAD;
                    r += FUNHEAD;
                    do {
                        if ( *r < *m ) {
                            m = peps[j]; peps[j] = t; peps[i] = t = m;
                            break;
                        }
                        else if ( *r++ > *m++ ) break;
                    } while ( --k > 0 );
                }
            }
        }
        m = to;
        for ( i = 0; i < neps; i++ ) {
            t = peps[i];
            k = t[1];
            NCOPY(m,t,k);
        }
    }
/*
    #] LeviCivita tensors :
    #[ Delta :
*/
    if ( ndel ) {
        r = t = pdel;
        for ( i = 0; i < ndel; i += 2, r += 2 ) {
            if ( r[1] < r[0] ) { k = *r; *r = r[1]; r[1] = k; }
        }
        for ( i = 2; i < ndel; i += 2, t += 2 ) {
            r = t + 2;
            for ( j = i; j < ndel; j += 2 ) {
                if ( *r > *t ) { r += 2; }
                else if ( *r < *t ) {
                    k = *r; *r++ = *t; *t++ = k;
                    k = *r; *r++ = *t; *t-- = k;
                }
                else {
                    if ( *++r < t[1] ) {
                        k = *r; *r = t[1]; t[1] = k;
                    }
                    r++;
                }
            }
        }
        t = pdel;
        *m++ = DELTA;
        *m++ = ndel + 2;
        i = ndel;
        NCOPY(m,t,i);
    }
/*
    #] Delta :
    #[ Loose Vectors/Indices :
*/
    if ( nind ) {
        t = pind;
        for ( i = 0; i < nind; i++ ) {
            r = t + 1;
            for ( j = i+1; j < nind; j++ ) {
                if ( *r < *t ) {
                    k = *r; *r = *t; *t = k;
                }
                r++;
            }
            t++;
        }
        t = pind;
        *m++ = INDEX;
        *m++ = nind + 2;
        i = nind;
        NCOPY(m,t,i);
    }
/*
    #] Loose Vectors/Indices :
    #[ Vectors :
*/
    if ( nvec ) {
        t = pvec;
        for ( i = 2; i < nvec; i += 2 ) {
```

```
            r = t + 2;
            for ( j = i; j < nvec; j += 2 ) {
                if ( *r == *t ) {
                    if ( *++r < t[1] ) {
                        k = *r; *r = t[1]; t[1] = k;
                    }
                    r++;
                }
                else if ( *r < *t ) {
                    k = *r; *r++ = *t; *t++ = k;
                    k = *r; *r++ = *t; *t-- = k;
                }
                else { r += 2; }
            }
            t += 2;
        }
        t = pvec;
        *m++ = VECTOR;
        *m++ = nvec + 2;
        i = nvec;
        NCOPY(m,t,i);
    }
/*
    #] Vectors :
    #[ Dotproducts :
*/
    if ( ndot ) {
        to = m;
        m = t = pdot;
        i = ndot;
        while ( --i >= 0 ) {
            if ( *t > t[1] ) { j = *t; *t = t[1]; t[1] = j; }
            t += 3;
        }
        t = m;
        ndot *= 3;
        m += ndot;
        while ( t < (m-3) ) {
            r = t + 3;
            do {
                if ( *r == *t ) {
                    if ( *++r == *++t ) {
                        r++;
                        if ( ( *r < MAXPOWER && t[1] < MAXPOWER )
                        || ( *r > -MAXPOWER && t[1] > -MAXPOWER ) ) {
                            t++;
                            *t += *r;
                            if ( *t > MAXPOWER || *t < -MAXPOWER ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Exponent of dotproduct out of range: %d",*t);
                                MUNLOCK(ErrorMessageLock);
                                goto NormMin;
                            }
                            ndot -= 3;
                            *r-- = *--m;
                            *r-- = *--m;
                            *r   = *--m;
                            if ( !*t ) {
                                ndot -= 3;
                                *t-- = *--m;
                                *t-- = *--m;
                                *t   = *--m;
                                t -= 3;
                                break;
                            }
                        }
                        else if ( *r < *++t ) {
                            k = *r; *r++ = *t; *t = k;
                        }
                        else r++;
                        t -= 2;
                    }
                    else if ( *r < *t ) {
                        k = *r; *r++ = *t; *t++ = k;
                        k = *r; *r++ = *t; *t = k;
                        t -= 2;
                    }
                    else { r += 2; t--; }
                }
                else if ( *r < *t ) {
                    k = *r; *r++ = *t; *t++ = k;
                    k = *r; *r++ = *t; *t++ = k;
                    k = *r; *r++ = *t; *t   = k;
                    t -= 2;
                }
                else { r += 3; }
            } while ( r < m );
```

```
            t += 3;
        }
        m = to;
        t = pdot;
        if ( ( i = ndot ) > 0 ) {
            *m++ = DOTPRODUCT;
            *m++ = i + 2;
            NCOPY(m,t,i);
        }
    }
/*
    #] Dotproducts :
    #[ Symbols :
*/
    if ( nsym ) {
        nsym «= 1;
        t = psym;
        *m++ = SYMBOL;
        r = m;
        *m++ = ( i = nsym ) + 2;
        if ( i ) { do {
            if ( !*t ) {
                if ( t[1] < (2*MAXPOWER) ) {         /* powers of i */
                    if ( t[1] & 1 ) { *m++ = 0; *m++ = 1; }
                    else *r -= 2;
                    if ( *++t & 2 ) ncoef = -ncoef;
                    t++;
                }
            }
            else if ( *t <= NumSymbols && *t > -2*MAXPOWER ) {  /* Put powers in range */
                if ( ( ( ( t[1] > symbols[*t].maxpower ) && ( symbols[*t].maxpower < MAXPOWER ) ) ||
                       ( ( t[1] < symbols[*t].minpower ) && ( symbols[*t].minpower > -MAXPOWER ) ) )
&&
                       ( t[1] < 2*MAXPOWER ) && ( t[1] > -2*MAXPOWER ) ) {
                    if ( i <= 2 || t[2] != *t ) goto NormZero;
                }
                if ( AN.ncmod == 1 && ( AC.modmode & ALSOPOWERS ) != 0 ) {
                    if ( AC.cmod[0] == 1 ) t[1] = 0;
                    else if ( t[1] >= 0 ) t[1] = 1 + (t[1]-1)%(AC.cmod[0]-1);
                    else {
                        t[1] = -1 - (-t[1]-1)%(AC.cmod[0]-1);
                        if ( t[1] < 0 ) t[1] += (AC.cmod[0]-1);
                    }
                }
                if ( ( t[1] < (2*MAXPOWER) && t[1] >= MAXPOWER )
                || ( t[1] > -(2*MAXPOWER) && t[1] <= -MAXPOWER ) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Exponent out of range: %d",t[1]);
                    MUNLOCK(ErrorMessageLock);
                    goto NormMin;
                }
                if ( AT.TrimPower && AR.PolyFunVar == *t && t[1] > AR.PolyFunPow ) {
                    goto NormZero;
                }
                else if ( t[1] ) {
                    *m++ = *t++;
                    *m++ = *t++;
                }
                else { *r -= 2; t += 2; }
            }
            else {
                *m++ = *t++; *m++ = *t++;
            }
        } while ( (i-=2) > 0 ); }
        if ( *r <= 2 ) m = r-1;
    }
/*
    #] Symbols :
    #[ float_ :

    Here we treat float_ functions and combined them with the regular
    coefficient.
*/

#ifdef WITHFLOAT
    if ( withfloat ) {
        WORD floatsign = 3;
/*
        First check whether the coefficient is already 1/1
*/
        if ( ABS(ncoef) == 3 && n_coef[0] == 1 && n_coef[1] == 1 ) {
            if ( withfloat == 1 ) {
                t = firstfloat;
/*
                We only interfere by making the sign positive.
                The float_ function has already been tested.
*/
```

```
                if ( t[FUNHEAD+3] < 0 ) {
                    t[FUNHEAD+3] = -t[FUNHEAD+3];
                    floatsign = -floatsign;
                }
                if ( ncoef < 0 ) floatsign = -floatsign;
/*
                Copy to output;
*/
                AT.FloatPos = m-termout;
                i = t[1]; NCOPY(m,t,i)
            }
            else {
                PackFloat(m,aux4);
                if ( m[FUNHEAD+3] < 0 ) {
                    m[FUNHEAD+3] = -m[FUNHEAD+3];
                    floatsign = -floatsign;
                }
                if ( ncoef < 0 ) floatsign = -floatsign;
                AT.FloatPos = m-termout;
                m += m[1];
            }
        }
        else {
            if ( withfloat == 1 ) UnpackFloat(aux4,firstfloat);
            RatToFloat(aux5,(UWORD *)n_coef,ncoef);
            mpf_mul(aux4,aux4,aux5);
            PackFloat(m,aux4);
            AT.FloatPos = m-termout;
            if ( m[FUNHEAD+3] < 0 ) {
                m[FUNHEAD+3] = -m[FUNHEAD+3];
                floatsign = -floatsign;
            }
            m += m[1];
        }
        n_coef[0] = 1; n_coef[1] = 1; ncoef = floatsign;
    }
    else AT.FloatPos = 0;
#endif

/*
    #] float_ :
    #[ Do Replace_ :
*/
    stop = (WORD *)(((UBYTE *)(termout)) + AM.MaxTer);
    i = ABS(ncoef);
    if ( ( m + i ) > stop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Term too complex during normalization");
        MUNLOCK(ErrorMessageLock);
        goto NormMin;
    }
    if ( ReplaceType >= 0 ) {
        t = n_coef;
        i--;
        NCOPY(m,t,i);
        *m++ = ncoef;
        t = termout;
        *t = WORDDIF(m,t);
        if ( ReplaceType == 0 ) {
            AT.WorkPointer = termout+*termout;
            WildFill(BHEAD term,termout,AN.ReplaceScrat);
            termout = term + *term;
        }
        else {
            AT.WorkPointer = r = termout + *termout;
            WildFill(BHEAD r,termout,AN.ReplaceScrat);
            i = *r; m = term;
            NCOPY(m,r,i);
            termout = m;


            r = m = term;
            r += *term; r -= ABS(r[-1]);
            m++;
            while ( m < r ) {
                if ( *m >= FUNCTION && m[1] > FUNHEAD &&
                functions[*m-FUNCTION].spec != TENSORFUNCTION )
                    m[2] |= DIRTYFLAG;
                m += m[1];
            }
        }
/*
        The next 'reset' cannot be done. We still need the expression
        in the buffer. Note though that this may cause a runaway pointer
        if we are not very careful.

        C->numrhs = oldtoprhs;
```

```
        C->Pointer = C->Buffer + oldcpointer;
*/
        TermFree(n_llnum,"n_llnum");
        TermFree(n_coef,"NormCoef");
        return(1);
    }
    else {
        t = termout;
        k = WORDDIF(m,t);
        *t = k + i;
        m = term;
        NCOPY(m,t,k);
        i--;
        t = n_coef;
        NCOPY(m,t,i);
        *m++ = ncoef;
    }
/*
    #] Do Replace_ :
    #[ Errors and Finish :
*/
RegEnd:
    if ( termout < term + *term && termout >= term ) AT.WorkPointer = term + *term;
    else AT.WorkPointer = termout;
/*
    if ( termflag ) {    We have to assign the term to $variable(s)
        TermAssign(term);
    }
*/
    TermFree(n_llnum,"n_llnum");
    TermFree(n_coef,"NormCoef");
    return(regval);

NormInf:
    MLOCK(ErrorMessageLock);
    MesPrint("Division by zero during normalization");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);

NormZZ:
    MLOCK(ErrorMessageLock);
    MesPrint("0^0 during normalization of term");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);

NormPRF:
    MLOCK(ErrorMessageLock);
    MesPrint("0/0 in polyratfun during normalization of term");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);

NormZero:
    *term = 0;
    AT.WorkPointer = termout;
    TermFree(n_llnum,"n_llnum");
    TermFree(n_coef,"NormCoef");
    return(regval);

NormMin:
    TermFree(n_llnum,"n_llnum");
    TermFree(n_coef,"NormCoef");
    return(-1);

FromNorm:
    MLOCK(ErrorMessageLock);
    MesCall("Norm");
    MUNLOCK(ErrorMessageLock);
    TermFree(n_llnum,"n_llnum");
    TermFree(n_coef,"NormCoef");
    return(-1);

/*
    #] Errors and Finish :
*/
}

/*
        #] Normalize :
        #[ ExtraSymbol :
*/

WORD ExtraSymbol(WORD sym, WORD pow, WORD nsym, WORD *ppsym, WORD *ncoef)
{
    WORD *m, i;
    i = nsym;
    m = ppsym - 2;
    while  ( i > 0 ) {
```

```
        if ( sym == *m ) {
            m++;
            if  ( pow > 2*MAXPOWER || pow < -2*MAXPOWER
            ||  *m > 2*MAXPOWER || *m < -2*MAXPOWER ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal wildcard power combination.");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            *m += pow;

            if ( ( sym <= NumSymbols && sym > -MAXPOWER )
             && ( symbols[sym].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
                *m %= symbols[sym].maxpower;
                if ( *m < 0 ) *m += symbols[sym].maxpower;
                if ( ( symbols[sym].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
                    if ( ( ( symbols[sym].maxpower & 1 ) == 0 ) &&
                        ( *m >= symbols[sym].maxpower/2 ) ) {
                        *m -= symbols[sym].maxpower/2; *ncoef = -*ncoef;
                    }
                }
            }

            if  ( *m >= 2*MAXPOWER || *m <= -2*MAXPOWER ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Power overflow during normalization");
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            if ( !*m ) {
                m--;
                while ( i < nsym )
                    { *m = m[2]; m++; *m = m[2]; m++; i++; }
                return(-1);
            }
            return(0);
        }
        else if ( sym < *m ) {
            m -= 2;
            i--;
        }
        else break;
    }
    m = ppsym;
    while ( i < nsym )
        { m--; m[2] = *m; m--; m[2] = *m; i++; }
    *m++ = sym;
    *m = pow;
    return(1);
}

/*
        #] ExtraSymbol :
        #[ DoTheta :
*/

WORD DoTheta(PHEAD WORD *t)
{
    GETBIDENTITY
    WORD k, *r1, *r2, *tstop, type;
    WORD ia, *ta, *tb, *stopa, *stopb;
    if ( AC.BracketNormalize ) return(-1);
    type = *t;
    k = t[1];
    tstop = t + k;
    t += FUNHEAD;
    if ( k <= FUNHEAD ) return(1);
    r1 = t;
    NEXTARG(r1)
    if ( r1 == tstop ) {
/*
        One argument
*/
        if ( *t == ARGHEAD ) {
            if ( type == THETA ) return(1);
            else return(0);   /* THETA2 */
        }
        if ( *t < 0 ) {
            if ( *t == -SNUMBER ) {
                if ( t[1] < 0 ) return(0);
                else {
                    if ( type == THETA2 && t[1] == 0 ) return(0);
                    else return(1);
                }
            }
            return(-1);
        }
```

```
        k = t[*t-1];
        if ( *t == ABS(k)+1+ARGHEAD ) {
            if ( k > 0 ) return(1);
            else return(0);
        }
        return(-1);
    }
/*
    At least two arguments
*/
    r2 = r1;
    NEXTARG(r2)
    if ( r2 < tstop ) return(-1);    /* More than 2 arguments */
/*
    Note now that zero has to be treated specially
    We take the criteria from the symmetrize routine
*/
    if ( *t == -SNUMBER && *r1 == -SNUMBER ) {
        if ( t[1] > r1[1] ) return(0);
        else if ( t[1] < r1[1] ) {
            return(1);
        }
        else if ( type == THETA ) return(1);
        else return(0);    /* THETA2 */
    }
    else if ( t[1] == 0 && *t == -SNUMBER ) {
        if ( *r1 > 0 ) { }
        else if ( *t < *r1 ) return(1);
        else if ( *t > *r1 ) return(0);
    }
    else if ( r1[1] == 0 && *r1 == -SNUMBER ) {
        if ( *t > 0 ) { }
        else if ( *t < *r1 ) return(1);
        else if ( *t > *r1 ) return(0);
    }
    r2 = AT.WorkPointer;
    if ( *t < 0 ) {
        ta = r2;
        ToGeneral(t,ta,0);
        r2 += *r2;
    }
    else ta = t;
    if ( *r1 < 0 ) {
        tb = r2;
        ToGeneral(r1,tb,0);
    }
    else tb = r1;
    stopa = ta + *ta;
    stopb = tb + *tb;
    ta += ARGHEAD; tb += ARGHEAD;
    while ( ta < stopa ) {
        if ( tb >= stopb ) return(0);
        if ( ( ia = CompareTerms(BHEAD ta,tb,(WORD)1) ) < 0 ) return(0);
        if ( ia > 0 ) return(1);
        ta += *ta;
        tb += *tb;
    }
    if ( type == THETA ) return(1);
    else return(0); /* THETA2 */
}

/*
        #] DoTheta :
        #[ DoDelta :
*/

WORD DoDelta(WORD *t)
{
    WORD k, *r1, *r2, *tstop, isnum, isnum2, type = *t;
    if ( AC.BracketNormalize ) return(-1);
    k = t[1];
    if ( k <= FUNHEAD ) goto argzero;
    if ( k == FUNHEAD+ARGHEAD && t[FUNHEAD] == ARGHEAD ) goto argzero;
    tstop = t + k;
    t += FUNHEAD;
    r1 = t;
    NEXTARG(r1)
    if ( *t < 0 ) {
        k = 1;
        if ( *t == -SNUMBER ) { isnum = 1; k = t[1]; }
        else isnum = 0;
    }
    else {
        k = t[*t-1];
        k = ABS(k);
        if ( k == *t-ARGHEAD-1 ) isnum = 1;
        else isnum = 0;
```

```
            k = 1;
        }
        if ( r1 >= tstop ) {         /* Single argument */
            if ( !isnum ) return(-1);
            if ( k == 0 ) goto argzero;
            goto argnonzero;
        }
        r2 = r1;
        NEXTARG(r2)
        if ( r2 < tstop ) return(-1);
        if ( *r1 < 0 ) {
            if ( *r1 == -SNUMBER ) { isnum2 = 1; }
            else isnum2 = 0;
        }
        else {
            k = r1[*r1-1];
            k = ABS(k);
            if ( k == *r1-ARGHEAD-1 ) isnum2 = 1;
            else isnum2 = 0;
        }
        if ( isnum != isnum2 ) return(-1);
        tstop = r1;
        while ( t < tstop && r1 < r2 ) {
            if ( *t != *r1 ) {
                if ( !isnum ) return(-1);
                goto argnonzero;
            }
            t++; r1++;
        }
        if ( t != tstop || r1 != r2 ) {
            if ( !isnum ) return(-1);
            goto argnonzero;
        }
    argzero:
        if ( type == DELTA2 ) return(1);
        else return(0);
    argnonzero:
        if ( type == DELTA2 ) return(0);
        else return(1);
    }

    /*
            #] DoDelta :
            #[ DoRevert :
    */

    void DoRevert(WORD *fun, WORD *tmp)
    {
        WORD *t, *r, *m, *to, *tt, *mm, i, j;
        to = fun + fun[1];
        r = fun + FUNHEAD;
        while ( r < to ) {
            if ( *r <= 0 ) {
                if ( *r == -REVERSEFUNCTION ) {
                    m = r; mm = m+1;
                    while ( mm < to ) *m++ = *mm++;
                    to--;
                    (fun[1])--;
                    fun[2] |= DIRTYSYMFLAG;
                }
                else if ( *r <= -FUNCTION ) r++;
                else {
                    if ( *r == -INDEX && r[1] < MINSPEC ) *r = -VECTOR;
                    r += 2;
                }
            }
            else {
                if ( ( *r > ARGHEAD )
                && ( r[ARGHEAD+1] == REVERSEFUNCTION )
                && ( *r == (r[ARGHEAD]+ARGHEAD) )
                && ( r[ARGHEAD] == (r[ARGHEAD+2]+4) )
                && ( *(r+*r-3) == 1 )
                && ( *(r+*r-2) == 1 )
                && ( *(r+*r-1) == 3 ) ) {
                    mm = r;
                    r += ARGHEAD + 1;
                    tt = r + r[1];
                    r += FUNHEAD;
                    m = tmp;
                    t = r;
                    j = 0;
                    while ( t < tt ) {
                        NEXTARG(t)
                        j++;
                    }
                    while ( --j >= 0 ) {
                        i = j;
```

```
                    t = r;
                    while ( --i >= 0 ) {
                        NEXTARG(t)
                    }
                    if ( *t > 0 ) {
                        i = *t;
                        NCOPY(m,t,i);
                    }
                    else if ( *t <= -FUNCTION ) *m++ = *t++;
                    else { *m++ = *t++; *m++ = *t++; }
                }
                i = WORDDIF(m,tmp);
                m = tmp;
                t = mm;
                r = t + *t;
                NCOPY(t,m,i);
                m = r;
                r = t;
                i = WORDDIF(to,m);
                NCOPY(t,m,i);
                fun[1] = WORDDIF(t,fun);
                to = t;
                fun[2] |= DIRTYSYMFLAG;
            }
            else r += *r;
        }
    }
}

/*
        #] DoRevert :
    #] Normalize :
    #[ DetCommu :

    Determines the number of terms in an expression that contain
    noncommuting objects. This can be used to see whether products of
    this expression can be evaluated with binomial coefficients.

    We don't try to be fancy. If a term contains noncommuting objects
    we are not looking whether they can commute with complete other
    terms.

    If the number gets too large we cut it off.
*/

#define MAXNUMBEROFNONCOMTERMS 2

WORD DetCommu(WORD *terms)
{
    WORD *t, *tnext, *tstop;
    WORD num = 0;
    if ( *terms == 0 ) return(0);
    if ( terms[*terms] == 0 ) return(0);
    t = terms;
    while ( *t ) {
        tnext = t + *t;
        tstop = tnext - ABS(tnext[-1]);
        t++;
        while ( t < tstop ) {
            if ( *t >= FUNCTION ) {
                if ( functions[*t-FUNCTION].commute ) {
                    num++;
                    if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
                    break;
                }
            }
            else if ( *t == SUBEXPRESSION ) {
                if ( cbuf[t[4]].CanCommu[t[2]] ) {
                    num++;
                    if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
                    break;
                }
            }
            else if ( *t == EXPRESSION ) {
                num++;
                if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
                break;
            }
            else if ( *t == DOLLAREXPRESSION ) {
/*
                Technically this is not correct. We have to test first
                whether this is MODLOCAL (in TFORM) and if so, use the
                local version. Anyway, this should be rare to never
                occurring because dollars should be replaced.
*/
                if ( cbuf[AM.dbufnum].CanCommu[t[2]] ) {
                    num++;
```

```
                        if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
                        break;
                    }
                }
                t += t[1];
            }
            t = tnext;
        }
        return(num);
}

/*
        #] DetCommu :
        #[ DoesCommu :

        Determines the number of noncommuting objects in a term.
        If the number gets too large we cut it off.
*/

WORD DoesCommu(WORD *term)
{
        WORD *tstop;
        WORD num = 0;
        if ( *term == 0 ) return(0);
        tstop = term + *term;
        tstop = tstop - ABS(tstop[-1]);
        term++;
        while ( term < tstop ) {
            if ( ( *term >= FUNCTION ) && ( functions[*term-FUNCTION].commute ) ) {
                num++;
                if ( num >= MAXNUMBEROFNONCOMTERMS ) return(num);
            }
            term += term[1];
        }
        return(num);
}

/*
        #] DoesCommu :
        #[ PolyNormPoly :

            Normalizes a polynomial
*/

#ifdef EVALUATEGCD
WORD *PolyNormPoly (PHEAD WORD *Poly) {

        GETBIDENTITY;
        WORD *buffer = AT.WorkPointer;
        WORD *p;
        if ( NewSort(BHEAD0) ) { Terminate(-1); }
        AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
        while ( *Poly ) {
            p = Poly + *Poly;
            if ( SymbolNormalize(Poly) < 0 ) return(0);
            if ( StoreTerm(BHEAD Poly) ) {
                AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
                LowerSortLevel();
                Terminate(-1);
            }
            Poly = p;
        }
        if ( EndSort(BHEAD buffer,1) < 0 ) {
            AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
            Terminate(-1);
        }
        p = buffer;
        while ( *p ) p += *p;
        AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
        AT.WorkPointer = p + 1;
        return(buffer);
}
#endif

/*
        #] PolyNormPoly :
        #[ EvaluateGcd :

        Try to evaluate the GCDFUNCTION gcd_.
        This function can have a number of arguments which can be numbers
        and/or polynomials. If there are objects that aren't SYMBOLS or numbers
        it cannot work currently.

        To make this work properly we have to intervene in proces.c
         proces.c:                        if ( Normalize(BHEAD m) ) {
1060 proces.c:                            if ( Normalize(BHEAD r) ) {
1126?proces.c:                            if ( Normalize(BHEAD term) ) {
```

```
     proces.c:              if ( Normalize(BHEAD AT.WorkPointer) ) goto PasErr;
2308!proces.c:      if ( ( retnorm = Normalize(BHEAD term) ) != 0 ) {
     proces.c:                  ReNumber(BHEAD term); Normalize(BHEAD term);
     proces.c:              if ( Normalize(BHEAD v) ) Terminate(-1);
     proces.c:      if ( Normalize(BHEAD w) ) { LowerSortLevel(); goto PolyCall; }
     proces.c:      if ( Normalize(BHEAD term) ) goto PolyCall;
*/
#ifdef EVALUATEGCD

WORD *EvaluateGcd(PHEAD WORD *subterm)
{
    GETBIDENTITY
    WORD *oldworkpointer = AT.WorkPointer, *work1, *work2, *work3;
    WORD *t, *tt, *ttt, *t1, *t2, *t3, *t4, *tstop;
    WORD ct, nnum;
    UWORD gcdnum, stor;
    WORD *lnum=n_llnum+1;
    WORD *num1, *num2, *num3, *den1, *den2, *den3;
    WORD sizenum1, sizenum2, sizenum3, sizeden1, sizeden2, sizeden3;
    int i, isnumeric = 0, numarg = 0 /*, sizearg */;
    LONG size;
/*
    Step 1: Look for -SNUMBER or -SYMBOL arguments.
            If encountered, treat everybody with it.
*/
    tt = subterm + subterm[1]; t = subterm + FUNHEAD;

    while ( t < tt ) {
        numarg++;
        if ( *t == -SNUMBER ) {
            if ( t[1] == 0 ) {
gcdzero:;
                MLOCK(ErrorMessageLock);
                MesPrint("Trying to take the GCD involving a zero term.");
                MUNLOCK(ErrorMessageLock);
                return(0);
            }
            gcdnum = ABS(t[1]);
            t1 = subterm + FUNHEAD;
            while ( gcdnum > 1 && t1 < tt ) {
                if ( *t1 == -SNUMBER ) {
                    stor = ABS(t1[1]);
                    if ( stor == 0 ) goto gcdzero;
                    if ( GcdLong(BHEAD (UWORD *)&stor,1,(UWORD *)&gcdnum,1,
                                    (UWORD *)lnum,&nnum) ) goto FromGCD;
                    gcdnum = lnum[0];
                    t1 += 2;
                    continue;
                }
                else if ( *t1 == -SYMBOL ) goto gcdisone;
                else if ( *t1 < 0 ) goto gcdillegal;
/*
                Now we have to go through all the terms in the argument.
                This includes long numbers.
*/
                ttt = t1 + *t1;
                ct = *ttt; *ttt = 0;
                if ( t1[1] != 0 ) { /* First normalize the argument */
                    t1 = PolyNormPoly(BHEAD t1+ARGHEAD);
                }
                else t1 += ARGHEAD;
                while ( *t1 ) {
                    t1 += *t1;
                    i = ABS(t1[-1]);
                    t2 = t1 - i;
                    i = (i-1)/2;
                    t3 = t2+i-1;
                    while ( t3 > t2 && *t3 == 0 ) { t3--; i--; }
                    if ( GcdLong(BHEAD (UWORD *)t2,(WORD)i,(UWORD *)&gcdnum,1,
                                    (UWORD *)lnum,&nnum) ) {
                        *ttt = ct;
                        goto FromGCD;
                    }
                    gcdnum = lnum[0];
                    if ( gcdnum == 1 ) {
                        *ttt = ct;
                        goto gcdisone;
                    }
                }
                *ttt = ct;
                t1 = ttt;
                AT.WorkPointer = oldworkpointer;
            }
            if ( gcdnum == 1 ) goto gcdisone;
            oldworkpointer[0] = 4;
            oldworkpointer[1] = gcdnum;
            oldworkpointer[2] = 1;
```

```
            oldworkpointer[3] = 3;
            oldworkpointer[4] = 0;
            AT.WorkPointer = oldworkpointer + 5;
            return(oldworkpointer);
        }
        else if ( *t == -SYMBOL ) {
            t1 = subterm + FUNHEAD;
            i = t[1];
            while ( t1 < tt ) {
                if ( *t1 == -SNUMBER ) goto gcdisone;
                if ( *t1 == -SYMBOL ) {
                    if ( t1[1] != i ) goto gcdisone;
                    t1 += 2; continue;
                }
                if ( *t1 < 0 ) goto gcdillegal;
                ttt = t1 + *t1;
                ct = *ttt; *ttt = 0;
                if ( t1[1] != 0 ) { /* First normalize the argument */
                    t2 = PolyNormPoly(BHEAD t1+ARGHEAD);
                }
                else t2 = t1 + ARGHEAD;
                while ( *t2 ) {
                    t3 = t2+1;
                    t2 = t2 + *t2;
                    tstop = t2 - ABS(t2[-1]);
                    while ( t3 < tstop ) {
                        if ( *t3 != SYMBOL ) {
                            *ttt = ct;
                            goto gcdillegal;
                        }
                        t4 = t3 + 2;
                        t3 += t3[1];
                        while ( t4 < t3 ) {
                            if ( *t4 == i && t4[1] > 0 ) goto nextterminarg;
                            t4 += 2;
                        }
                    }
                    *ttt = ct;
                    goto gcdisone;
nextterminarg:;
                }
                *ttt = ct;
                t1 = ttt;
                AT.WorkPointer = oldworkpointer;
            }
            oldworkpointer[0] = 8;
            oldworkpointer[1] = SYMBOL;
            oldworkpointer[2] = 4;
            oldworkpointer[3] = t[1];
            oldworkpointer[4] = 1;
            oldworkpointer[5] = 1;
            oldworkpointer[6] = 1;
            oldworkpointer[7] = 3;
            oldworkpointer[8] = 0;
            AT.WorkPointer = oldworkpointer+9;
            return(oldworkpointer);
        }
        else if ( *t < 0 ) {
gcdillegal:;
            MLOCK(ErrorMessageLock);
            MesPrint("Illegal object in gcd_ function. Object not a number or a symbol.");
            MUNLOCK(ErrorMessageLock);
            goto FromGCD;
        }
        else if ( ABS(t[*t-1]) == *t-ARGHEAD-1 ) isnumeric = numarg;
        else if ( t[1] != 0 ) {
            ttt = t + *t; ct = *ttt; *ttt = 0;
            t = PolyNormPoly(BHEAD t+ARGHEAD);
            *ttt = ct;
            if ( t[*t] == 0 && ABS(t[*t-1]) == *t-ARGHEAD-1 ) isnumeric = numarg;
            AT.WorkPointer = oldworkpointer;
            t = ttt;
        }
        t += *t;
    }
/*
    At this point there are only generic arguments.
    There are however still two cases:
    1: There is an argument that is purely numerical
       In that case we have to take the gcd of all coefficients
    2: All arguments are nontrivial polynomials.
       Here we don't worry so much about the factor. (???)
    We know whether case 1 occurs when isnumeric > 0.
    We can look up numarg to get a good starting value.
*/
    AT.WorkPointer = oldworkpointer;
    if ( isnumeric ) {
```

```
        t = subterm + FUNHEAD;
        for ( i = 1; i < isnumeric; i++ ) {
            NEXTARG(t);
        }
        if ( t[1] != 0 ) {  /* First normalize the argument */
            ttt = t + *t; ct = *ttt; *ttt = 0;
            t = PolyNormPoly(BHEAD t+ARGHEAD);
            *ttt = ct;
        }
        t += *t;
        i = (ABS(t[-1])-1)/2;
        den1 = t - 1 - i;
        num1 = den1 - i;
        sizenum1 = sizeden1 = i;
        while ( sizenum1 > 1 && num1[sizenum1-1] == 0 ) sizenum1--;
        while ( sizeden1 > 1 && den1[sizeden1-1] == 0 ) sizeden1--;
        work1 = AT.WorkPointer+1; work2 = work1+sizenum1;
        for ( i = 0; i < sizenum1; i++ ) work1[i] = num1[i];
        for ( i = 0; i < sizeden1; i++ ) work2[i] = den1[i];
        num1 = work1; den1 = work2;
        AT.WorkPointer = work2 = work2 + sizeden1;
        t = subterm + FUNHEAD;
        while ( t < tt ) {
            ttt = t + *t; ct = *ttt; *ttt = 0;
            if ( t[1] != 0 ) {
                t = PolyNormPoly(BHEAD t+ARGHEAD);
            }
            else t += ARGHEAD;
            while ( *t ) {
                t += *t;
                i = (ABS(t[-1])-1)/2;
                den2 = t - 1 - i;
                num2 = den2 - i;
                sizenum2 = sizeden2 = i;
                while ( sizenum2 > 1 && num2[sizenum2-1] == 0 ) sizenum2--;
                while ( sizeden2 > 1 && den2[sizeden2-1] == 0 ) sizeden2--;
                num3 = AT.WorkPointer;
                if ( GcdLong(BHEAD (UWORD *)num2,sizenum2,(UWORD *)num1,sizenum1,
                                    (UWORD *)num3,&sizenum3) ) goto FromGCD;
                sizenum1 = sizenum3;
                for ( i = 0; i < sizenum1; i++ ) num1[i] = num3[i];
                den3 = AT.WorkPointer;
                if ( GcdLong(BHEAD (UWORD *)den2,sizeden2,(UWORD *)den1,sizeden1,
                                    (UWORD *)den3,&sizeden3) ) goto FromGCD;
                sizeden1 = sizeden3;
                for ( i = 0; i < sizeden1; i++ ) den1[i] = den3[i];
                if ( sizenum1 == 1 && num1[0] == 1 && sizeden1 == 1 && den1[1] == 1 )
                    goto gcdisone;
            }
            *ttt = ct;
            t = ttt;
            AT.WorkPointer = work2;
        }
        AT.WorkPointer = oldworkpointer;
/*
        Now copy the GCD to the 'output'
*/
        if ( sizenum1 > sizeden1 ) {
            while ( sizenum1 > sizeden1 ) den1[sizeden1++] = 0;
        }
        else if ( sizenum1 < sizeden1 ) {
            while ( sizenum1 < sizeden1 ) num1[sizenum1++] = 0;
        }
        t = oldworkpointer;
        i = 2*sizenum1+1;
        *t++ = i+1;
        if ( num1 != t ) { NCOPY(t,num1,sizenum1); }
        else t += sizenum1;
        if ( den1 != t ) { NCOPY(t,den1,sizeden1); }
        else t += sizeden1;
        *t++ = i;
        *t++ = 0;
        AT.WorkPointer = t;
        return(oldworkpointer);
    }
/*
    Now the real stuff with only polynomials.
    Pick up the shortest term to start.
    We are a bit brutish about this.
*/
    t = subterm + FUNHEAD;
    AT.WorkPointer += AM.MaxTer/sizeof(WORD);
    work2 = AT.WorkPointer;
/*
    sizearg = subterm[1];
*/
    i = 0; work3 = 0;
```

```
    while ( t < tt ) {
        i++;
        work1 = AT.WorkPointer;
        ttt = t + *t; ct = *ttt; *ttt = 0;
        t = PolyNormPoly(BHEAD t+ARGHEAD);
        if ( *work1 < AT.WorkPointer-work1 ) {
/*
            sizearg = AT.WorkPointer-work1;
*/
            numarg = i;
            work3 = work1;
        }
        *ttt = ct; t = ttt;
    }
    *AT.WorkPointer++ = 0;
/*
    We have properly normalized arguments and the shortest is indicated in work3
*/
    work1 = work3;
    while ( *work2 ) {
        if ( work2 != work3 ) {
            work1 = PolyGCD2(BHEAD work1,work2);
        }
        while ( *work2 ) work2 += *work2;
        work2++;
    }
    work2 = work1;
    while ( *work2 ) work2 += *work2;
    size = work2 - work1 + 1;
    t = oldworkpointer;
    NCOPY(t,work1,size);
    AT.WorkPointer = t;
    return(oldworkpointer);

gcdisone:;
    oldworkpointer[0] = 4;
    oldworkpointer[1] = 1;
    oldworkpointer[2] = 1;
    oldworkpointer[3] = 3;
    oldworkpointer[4] = 0;
    AT.WorkPointer = oldworkpointer+5;
    return(oldworkpointer);
FromGCD:
    MLOCK(ErrorMessageLock);
    MesCall("EvaluateGcd");
    MUNLOCK(ErrorMessageLock);
    return(0);
}

#endif

/*
    #] EvaluateGcd :
    #[ TreatPolyRatFun :

    if ( AR.PolyFunExp == 1 ) we have to trim the contents of the polyratfun
    down to its most divergent term and give it coefficient +1. This is done
    by taking the terms with the least power in the variable in the numerator
    and in the denominator and then combine them.
    Answer is either PolyRatFun(ep^n,1) or PolyRatFun(1,1) or PolyRatFun(1,ep^n)
*/

int TreatPolyRatFun(PHEAD WORD *prf)
{
    WORD *t, *tstop, *r, *rstop, *m, *mstop;
    WORD exp1 = MAXPOWER, exp2 = MAXPOWER;
    t = prf+FUNHEAD;
    if ( *t < 0 ) {
        if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) {
            if ( exp1 > 1 ) exp1 = 1;
            t += 2;
        }
        else {
            if ( exp1 > 0 ) exp1 = 0;
            NEXTARG(t)
        }
    }
    else {
        tstop = t + *t;
        t += ARGHEAD;
        while ( t < tstop ) {
/*
            Now look for the minimum power of AR.PolyFunVar
*/
            r = t+1;
            t += *t;
            rstop = t - ABS(t[-1]);
```

```
            while ( r < rstop ) {
                if ( *r != SYMBOL ) { r += r[1]; continue; }
                m = r;
                mstop = m + m[1];
                m += 2;
                while ( m < mstop ) {
                    if ( *m == AR.PolyFunVar ) {
                        if ( m[1] < exp1 ) exp1 = m[1];
                        break;
                    }
                    m += 2;
                }
                if ( m == mstop ) {
                    if ( exp1 > 0 ) exp1 = 0;
                }
                break;
            }
            if ( r == rstop ) {
                if ( exp1 > 0 ) exp1 = 0;
            }
        }
        t = tstop;
    }
    if ( *t < 0 ) {
        if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) {
            if ( exp2 > 1 ) exp2 = 1;
        }
        else {
            if ( exp2 > 0 ) exp2 = 0;
        }
    }
    else {
        tstop = t + *t;
        t += ARGHEAD;
        while ( t < tstop ) {
/*
            Now look for the minimum power of AR.PolyFunVar
*/
            r = t+1;
            t += *t;
            rstop = t - ABS(t[-1]);
            while ( r < rstop ) {
                if ( *r != SYMBOL ) { r += r[1]; continue; }
                m = r;
                mstop = m + m[1];
                m += 2;
                while ( m < mstop ) {
                    if ( *m == AR.PolyFunVar ) {
                        if ( m[1] < exp2 ) exp2 = m[1];
                        break;
                    }
                    m += 2;
                }
                if ( m == mstop ) {
                    if ( exp2 > 0 ) exp2 = 0;
                }
                break;
            }
            if ( r == rstop ) {
                if ( exp2 > 0 ) exp2 = 0;
            }
        }
    }
/*
    Now we can compose the output.
    Notice that the output can never be longer than the input provided
    we never can have arguments that consist of just a function.
*/
    exp1 = exp1-exp2;
/*  if ( exp1 > 0 ) exp1 = 0; */
    t = prf+FUNHEAD;
    if ( exp1 == 0 ) {
        *t++ = -SNUMBER; *t++ = 1;
        *t++ = -SNUMBER; *t++ = 1;
    }
    else if ( exp1 > 0 ) {
        if ( exp1 == 1 ) {
            *t++ = -SYMBOL; *t++ = AR.PolyFunVar;
        }
        else {
            *t++ = 8+ARGHEAD;
            *t++ = 0;
            FILLARG(t);
            *t++ = 8; *t++ = SYMBOL; *t++ = 4; *t++ = AR.PolyFunVar;
            *t++ = exp1; *t++ = 1; *t++ = 1; *t++ = 3;
        }
        *t++ = -SNUMBER; *t++ = 1;
```

```
    }
    else {
        *t++ = -SNUMBER; *t++ = 1;
        if ( exp1 == -1 ) {
            *t++ = -SYMBOL; *t++ = AR.PolyFunVar;
        }
        else {
            *t++ = 8+ARGHEAD;
            *t++ = 0;
            FILLARG(t);
            *t++ = 8; *t++ = SYMBOL; *t++ = 4; *t++ = AR.PolyFunVar;
            *t++ = -exp1; *t++ = 1; *t++ = 1; *t++ = 3;
        }
    }
    prf[2] = 0;  /* Clean */
    prf[1] = t - prf;
    return(0);
}

/*
    #] TreatPolyRatFun :
    #[ DropCoefficient :
*/

void DropCoefficient(PHEAD WORD *term)
{
    GETBIDENTITY
    WORD *t = term + *term;
    WORD n, na;
    n = t[-1]; na = ABS(n);
    t -= na;
    if ( n == 3 && t[0] == 1 && t[1] == 1 ) return;
    *AN.RepPoint = 1;
    t[0] = 1; t[1] = 1; t[2] = 3;
    *term -= (na-3);
}

/*
    #] DropCoefficient :
    #[ DropSymbols :
*/

void DropSymbols(PHEAD WORD *term)
{
    GETBIDENTITY
    WORD *tend = term + *term, *t1, *t2, *tstop;
    tstop = tend - ABS(tend[-1]);
    t1 = term+1;
    while ( t1 < tstop ) {
        if ( *t1 == SYMBOL ) {
            *AN.RepPoint = 1;
            t2 = t1+t1[1];
            while ( t2 < tend ) *t1++ = *t2++;
            *term = t1 - term;
            break;
        }
        t1 += t1[1];
    }
}

/*
    #] DropSymbols :
    #[ SymbolNormalize :
*/
int SymbolNormalize(WORD *term)
{
    GETIDENTITY
    WORD buffer[7*NORMSIZE], *t, *b, *bb, *tt, *m, *tstop;
    int i;
    b = buffer;
    *b++ = SYMBOL; *b++ = 2;
    t = term + *term;
    tstop = t - ABS(t[-1]);
    t = term + 1;
    while ( t < tstop ) {   /* Step 1: collect symbols */
      if ( *t == SYMBOL && t < tstop ) {
        for ( i = 2; i < t[1]; i += 2 ) {
            bb = buffer+2;
            while ( bb < b ) {
                if ( bb[0] == t[i] ) {  /* add powers */
                    bb[1] += t[i+1];
                    if ( bb[1] > MAXPOWER || bb[1] < -MAXPOWER ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Power in SymbolNormalize out of range");
                        MUNLOCK(ErrorMessageLock);
                        return(-1);
                    }
```

```
                        if ( bb[1] == 0 ) {
                            b -= 2;
                            while ( bb < b ) {
                                bb[0] = bb[2]; bb[1] = bb[3]; bb += 2;
                            }
                        }
                        goto Nexti;
                    }
                    else if ( bb[0] > t[i] ) { /* insert it */
                        m = b;
                        while ( m > bb ) { m[1] = m[-1]; m[0] = m[-2]; m -= 2; }
                        b += 2;
                        bb[0] = t[i];
                        bb[1] = t[i+1];
                        goto Nexti;
                    }
                    bb += 2;
                }
                if ( bb >= b ) { /* add it to the end */
                    *b++ = t[i]; *b++ = t[i+1];
                }
Nexti:;
            }
        }
        else {
          MLOCK(ErrorMessageLock);
          MesPrint("Illegal term in SymbolNormalize");
          MUNLOCK(ErrorMessageLock);
          return(-1);
        }
        t += t[1];
    }
    buffer[1] = b - buffer;
/*
    Veto negative powers
*/
    if ( AT.LeaveNegative == 0 ) {
        b = buffer; bb = b + b[1]; b += 3;
        while ( b < bb ) {
            if ( *b < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Negative power in SymbolNormalize");
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            b += 2;
        }
    }
/*
    Now we use the fact that the new term will not be longer than the old one
    Actually it should be shorter when there is more than one subterm!
    Copy back.
*/
    i = buffer[1];
    b = buffer; tt = term + 1;
    if ( i > 2 ) { NCOPY(tt,b,i) }
    if ( tt < tstop ) {
        i = term[*term-1];
        if ( i < 0 ) i = -i;
        *term -= (tstop-tt);
        NCOPY(tt,tstop,i)
    }
    return(0);
}

/*
    #] SymbolNormalize :
    #[ TestFunFlag :

    Tests whether a function still has unsubstituted subexpressions
    This function has its dirtyflag on!
*/

int TestFunFlag(PHEAD WORD *tfun)
{
    WORD *t, *tstop, *r, *rstop, *m, *mstop;
    if ( functions[*tfun-FUNCTION].spec <= 0 ) return(0);
    tstop = tfun + tfun[1];
    t = tfun + FUNHEAD;
    while ( t < tstop ) {
        if ( *t < 0 ) { NEXTARG(t); continue; }
        rstop = t + *t;
        if ( t[1] == 0 ) { t = rstop; continue; }
        r = t + ARGHEAD;
        while ( r < rstop ) { /* Here we loop over terms */
            m = r+1; mstop = r+*r; mstop -= ABS(mstop[-1]);
            while ( m < mstop ) {    /* Loop over the subterms */
```

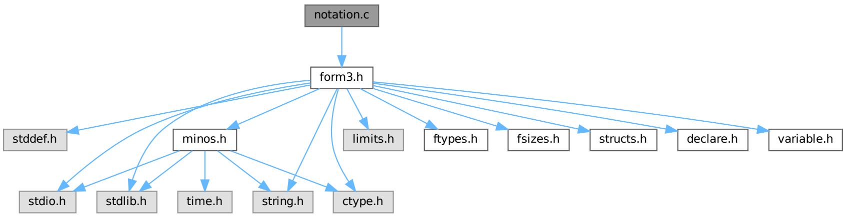
```
                if ( *m == SUBEXPRESSION || *m == EXPRESSION || *m == DOLLAREXPRESSION ) return(1);
                if ( ( *m >= FUNCTION ) && ( ( m[2] & DIRTYFLAG ) == DIRTYFLAG )
                    && ( *m != REPLACEMENT ) && TestFunFlag(BHEAD m) ) return(1);
                m += m[1];
            }
            r += *r;
        }
        t += *t;
    }
    return(0);
}

/*
    #] TestFunFlag :
    #[ BracketNormalize :
*/

WORD BracketNormalize(PHEAD WORD *term)
{
    WORD *stop = term+*term-3, *t, *tt, *tstart, *r;
    WORD *oldwork = AT.WorkPointer;
    WORD *termout;
    WORD i, ii, j;
    termout = AT.WorkPointer = term+*term;
/*
    First collect all functions and sort them
*/
    tt = termout+1; t = term+1;
    while ( t < stop ) {
        if ( *t >= FUNCTION ) { i = t[1]; NCOPY(tt,t,i); }
        else t += t[1];
    }
    if ( tt > termout+1 && tt-termout-1 > termout[2] ) { /* sorting */
        r = termout+1; ii = tt-r;
        for ( i = 0; i < ii-FUNHEAD; i += FUNHEAD ) {   /* Bubble sort */
            for ( j = i+FUNHEAD; j > 0; j -= FUNHEAD ) {
                if ( functions[r[j-FUNHEAD]-FUNCTION].commute
                  && functions[r[j]-FUNCTION].commute == 0 ) break;
                if ( r[j-FUNHEAD] > r[j] ) EXCH(r[j-FUNHEAD],r[j])
                else break;
            }
        }
    }

    tstart = tt; t = term + 1; *tt++ = DELTA; *tt++ = 2;
    while ( t < stop ) {
        if ( *t == DELTA ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
        else t += t[1];
    }
    if ( tstart[1] > 2 ) {
        for ( r = tstart+2; r < tstart+tstart[1]; r += 2 ) {
            if ( r[0] > r[1] ) EXCH(r[0],r[1])
        }
    }
    if ( tstart[1] > 4 ) { /* sorting */
        r = tstart+2; ii = tstart[1]-2;
        for ( i = 0; i < ii-2; i += 2 ) {   /* Bubble sort */
            for ( j = i+2; j > 0; j -= 2 ) {
                if ( r[j-2] > r[j] ) {
                    EXCH(r[j-2],r[j])
                    EXCH(r[j-1],r[j+1])
                }
                else if ( r[j-2] < r[j] ) break;
                else {
                    if ( r[j-1] > r[j+1] ) EXCH(r[j-1],r[j+1])
                    else break;
                }
            }
        }
        tt = tstart+tstart[1];
    }
    else if ( tstart[1] == 2 ) { tt = tstart; }
    else tt = tstart+4;

    tstart = tt; t = term + 1; *tt++ = INDEX; *tt++ = 2;
    while ( t < stop ) {
        if ( *t == INDEX ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
        else t += t[1];
    }
    if ( tstart[1] >= 4 ) { /* sorting */
        r = tstart+2; ii = tstart[1]-2;
        for ( i = 0; i < ii-1; i += 1 ) {   /* Bubble sort */
            for ( j = i+1; j > 0; j -= 1 ) {
                if ( r[j-1] > r[j] ) EXCH(r[j-1],r[j])
                else break;
            }
        }
```

```
        tt = tstart+tstart[1];
    }
    else if ( tstart[1] == 2 ) { tt = tstart; }
    else tt = tstart+3;

    tstart = tt; t = term + 1; *tt++ = DOTPRODUCT; *tt++ = 2;
    while ( t < stop ) {
        if ( *t == DOTPRODUCT ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
        else t += t[1];
    }
    if ( tstart[1] > 5 ) { /* sorting */
        r = tstart+2; ii = tstart[1]-2;
        for ( i = 0; i < ii; i += 3 ) {
            if ( r[i] > r[i+1] ) EXCH(r[i],r[i+1])
        }
        for ( i = 0; i < ii-3; i += 3 ) {   /* Bubble sort */
            for ( j = i+3; j > 0; j -= 3 ) {
                if ( r[j-3] < r[j] ) break;
                if ( r[j-3] > r[j] ) {
                    EXCH(r[j-3],r[j])
                    EXCH(r[j-2],r[j+1])
                }
                else {
                    if ( r[j-2] > r[j+1] ) EXCH(r[j-2],r[j+1])
                    else break;
                }
            }
        }
        tt = tstart+tstart[1];
    }
    else if ( tstart[1] == 2 ) { tt = tstart; }
    else {
        if ( tstart[2] > tstart[3] ) EXCH(tstart[2],tstart[3])
        tt = tstart+5;
    }

    tstart = tt; t = term + 1; *tt++ = SYMBOL; *tt++ = 2;
    while ( t < stop ) {
        if ( *t == SYMBOL ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
        else t += t[1];
    }
    if ( tstart[1] > 4 ) { /* sorting */
        r = tstart+2; ii = tstart[1]-2;
        for ( i = 0; i < ii-2; i += 2 ) {   /* Bubble sort */
            for ( j = i+2; j > 0; j -= 2 ) {
                if ( r[j-2] > r[j] ) EXCH(r[j-2],r[j])
                else break;
            }
        }
        tt = tstart+tstart[1];
    }
    else if ( tstart[1] == 2 ) { tt = tstart; }
    else tt = tstart+4;

    tstart = tt; t = term + 1; *tt++ = SETSET; *tt++ = 2;
    while ( t < stop ) {
        if ( *t == SETSET ) { i = t[1]-2; t += 2; tstart[1] += i; NCOPY(tt,t,i); }
        else t += t[1];
    }
    if ( tstart[1] > 4 ) { /* sorting */
        r = tstart+2; ii = tstart[1]-2;
        for ( i = 0; i < ii-2; i += 2 ) {   /* Bubble sort */
            for ( j = i+2; j > 0; j -= 2 ) {
                if ( r[j-2] > r[j] ) {
                    EXCH(r[j-2],r[j])
                    EXCH(r[j-1],r[j+1])
                }
                else break;
            }
        }
        tt = tstart+tstart[1];
    }
    else if ( tstart[1] == 2 ) { tt = tstart; }
    else tt = tstart+4;
    *tt++ = 1; *tt++ = 1; *tt++ = 3;
    t = term; i = *termout = tt - termout; tt = termout;
    NCOPY(t,tt,i);
    AT.WorkPointer = oldwork;
    return(0);
}

/*
    #] BracketNormalize :
*/
```

4.88 notation.c File Reference

```
#include "form3.h"
```

Include dependency graph for notation.c:



Functions

- int NormPolyTerm (PHEAD WORD ∗term)
- int ConvertToPoly (PHEAD WORD ∗term, WORD ∗outterm, WORD ∗comlist, WORD par)
- int LocalConvertToPoly (PHEAD WORD ∗term, WORD ∗outterm, WORD startebuf, WORD par)
- int ConvertFromPoly (PHEAD WORD ∗term, WORD ∗outterm, WORD from, WORD to, WORD offset, WORD par)
- WORD FindSubterm (WORD ∗subterm)
- WORD FindLocalSubterm (PHEAD WORD ∗subterm, WORD startebuf)
- void PrintSubtermList (int from, int to)
- void PrintExtraSymbol (int num, WORD ∗terms, int par)
- WORD FindSubexpression (WORD ∗subexpr)
- int ExtraSymFun (PHEAD WORD ∗term, WORD level)
- int PruneExtraSymbols (WORD downto)

4.88.1 Detailed Description

Contains the functions that deal with the rewriting and manipulation of expressions/terms in polynomial representation.

Definition in file notation.c.

4.88.2 Function Documentation

4.88.2.1 NormPolyTerm()

```
int NormPolyTerm (
            PHEAD WORD ∗ term )
```

Definition at line 53 of file notation.c.

4.88.2.2 ConvertToPoly()

```
int ConvertToPoly (
            PHEAD WORD * term,
            WORD * outterm,
            WORD * comlist,
            WORD par )
```

Definition at line 307 of file notation.c.

4.88.2.3 LocalConvertToPoly()

```
int LocalConvertToPoly (
            PHEAD WORD * term,
            WORD * outterm,
            WORD startebuf,
            WORD par )
```

Converts a generic term to polynomial notation in which there are only symbols and brackets. During conversion there will be only symbols. Brackets are stripped. Objects that need 'translation' are put inside a special compiler buffer and represented by a symbol. The numbering of the extra symbols is down from the maximum. In principle there can be a problem when running into the already assigned ones. This uses the FindTree for searching in the global tree and then looks further in the AT.ebufnum. This allows fully parallel processing. Hence we need no locks. Cannot be used in the same module as ConvertToPoly.

Definition at line 510 of file notation.c.

Referenced by poly_factorize_expression().

4.88.2.4 ConvertFromPoly()

```
int ConvertFromPoly (
            PHEAD WORD * term,
            WORD * outterm,
            WORD from,
            WORD to,
            WORD offset,
            WORD par )
```

Definition at line 660 of file notation.c.

4.88.2.5 FindSubterm()

```
WORD FindSubterm (
            WORD * subterm )
```

Definition at line 773 of file notation.c.

4.88.2.6 FindLocalSubterm()

```
WORD FindLocalSubterm (
            PHEAD WORD * subterm,
            WORD startebuf )
```

Definition at line 842 of file notation.c.

4.88.2.7 PrintSubtermList()

```
void PrintSubtermList (
            int from,
            int to )
```

Definition at line 895 of file notation.c.

4.88.2.8 PrintExtraSymbol()

```
void PrintExtraSymbol (
            int num,
            WORD * terms,
            int par )
```

Definition at line 1007 of file notation.c.

4.88.2.9 FindSubexpression()

```
WORD FindSubexpression (
            WORD * subexpr )
```

Definition at line 1094 of file notation.c.

4.88.2.10 ExtraSymFun()

```
int ExtraSymFun (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1146 of file notation.c.

4.88.2.11 PruneExtraSymbols()

```
int PruneExtraSymbols (
            WORD downto )
```

Definition at line 1214 of file notation.c.

4.89 notation.c

Go to the documentation of this file.

```
/* #[ License : */
/*
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 */
/* #] License : */
/*
    #[ Includes :
*/

#include "form3.h"

/*
    #] Includes :
        #[ NormPolyTerm :

        Brings a term to normal form.

        This routine knows objects of the following types:
            SYMBOL
            HAAKJE
            SNUMBER
            LNUMBER
        The SNUMBER and LNUMBER are worked into the coefficient.
        One of the essences here is that everything can be done in place.
*/

int NormPolyTerm(PHEAD WORD *term)
{
    WORD *tcoef, ncoef, *tstop, *tfill, *t, *tt;
    int equal, i;
    WORD *r1, *r2, *r3, *r4, *r5, *rfirst, rv;
    WORD *lnum, nnum;       /* Scratch, originally for factorials */
/*
    One: find the coefficient
*/
    tcoef = term+*term;
    ncoef = tcoef[-1];
    tstop = tcoef - ABS(tcoef[-1]);
    tfill = t = term + 1;
    rfirst = 0;
    if ( t >= tstop ) { return(*term); }
    while ( t < tstop ) {
        switch ( *t ) {
            case SYMBOL:
                if ( rfirst == 0 ) {
/*
                    Here we only need to sort
                    1: assume no equals. Bubble.
*/
                    rfirst = t;
                    r2 = rfirst+4; tt = r3 = t + t[1]; equal = 0;
                    while ( r2 < r3 ) {
                        r1 = r2 - 2;
                        if ( *r2 > *r1 ) { r2 += 2; continue; }
                        if ( *r2 == *r1 ) { r2 += 2; equal = 1; continue; }
                        rv = *r1; *r1 = *r2; *r2 = rv;
                        r1 -= 2; r2 -= 2; r4 = r2 + 2;
                        while ( r1 > t ) {
                            if ( *r2 >= *r1 ) { r2 = r4; break; }
                            rv = *r1; *r1 = *r2; *r2 = rv;
```

```
                            r1 -= 2; r2 -= 2;
                        }
                    }
/*
                    2: hunt down the equal objects
                       postpone eliminating zero powers.
*/
                    if ( equal ) {
                        r1 = t+2; r2 = r1+2;
                        while ( r2 < r3 ) {
                            if ( *r1 == *r2 ) {
                                r1[1] += r2[1];
                                r4 = r2+2;
                                while ( r4 < r3 ) *r2++ = *r4++;
                                t[1] -= 2;
                                r2 = r1 + 2; r3 -= 2;
                            }
                        }
                    }
                }
                else {
/*
                    Here we only need to insert
*/
                    r1 = t + 2; tt = r3 = t + t[1];
                    while ( r1 < r3 ) {
                        r2 = rfirst+2; r4 = rfirst + rfirst[1];
                        while ( r2 < r4 ) {
                            if ( *r1 == *r2 ) {
                                r2[1] += r1[1];
                                break;
                            }
                            else if ( *r2 > *r1 ) {
                                r5 = r4;
                                while ( r5 > r2 ) { r5[1] = r5[-1]; r5[0] = r5[-2]; r5 -= 2; }
                                rfirst[1] += 2;
                                *r2 = *r1; r2[1] = r1[1];
                                break;
                            }
                            r2 += 2;
                        }
                        if ( r2 == r4 ) {
                            rfirst[1] += 2;
                            *r2++ = *r1++; *r2++ = *r1++;
                        }
                        else r1 += 2;
                    }
                }
                t = tt;
                break;
            case HAAKJE:    /* Here we skip brackets */
                t += t[1];
                break;
            case SNUMBER:
                if ( t[2] < 0 ) {
                    t[2] = -t[2];
                    if ( t[3] & 1 ) ncoef = -ncoef;
                }
                else if ( t[2] == 0 ) {
                    if ( t[3] < 0 ) goto NormInf;
                    goto NormZero;
                }
                lnum = TermMalloc("lnum");
                lnum[0] = t[2];
                nnum = 1;
                if ( t[3] && RaisPow(BHEAD (UWORD *)lnum,&nnum,(UWORD)(ABS(t[3]))) ) goto FromNorm;
                ncoef = REDLENG(ncoef);
                if ( t[3] < 0 ) {
                    if ( Divvy(BHEAD (UWORD *)tstop,&ncoef,(UWORD *)lnum,nnum) )
                        goto FromNorm;
                }
                else if ( t[3] > 0 ) {
                    if ( Mully(BHEAD (UWORD *)tstop,&ncoef,(UWORD *)lnum,nnum) )
                        goto FromNorm;
                }
                ncoef = INCLENG(ncoef);
                t += t[1];
                TermFree(lnum,"lnum");
                break;
            case LNUMBER:
                ncoef = REDLENG(ncoef);
                if ( Mully(BHEAD (UWORD *)tstop,&ncoef,(UWORD *)(t+3),t[2]) ) goto FromNorm;
                ncoef = INCLENG(ncoef);
                t += t[1];
                break;
            default:
                MLOCK(ErrorMessageLock);
```

```
                MesPrint("Illegal code in NormPolyTerm");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
                break;
        }
    }
/*
    Now we try to eliminate objects to the power zero.
*/
    if ( rfirst ) {
        r2 = rfirst+2;
        r3 = rfirst + rfirst[1];
        while ( r2 < r3 ) {
            if ( r2[1] == 0 ) {
                r1 = r2 + 2;
                while ( r1 < r3 ) { r1[-2] = r1[0]; r1[-1] = r1[1]; r1 += 2; }
                r3 -= 2;
                rfirst[1] -= 2;
            }
            else { r2 += 2; }
        }
        if ( rfirst[1] < 4 ) rfirst = 0;
    }
/*
    Finally we put the term together
*/
    if ( rfirst ) {
        i = rfirst[1];
        NCOPY(tfill,rfirst,i)
    }
    i = ABS(ncoef)-1;
    NCOPY(tfill,tstop,i)
    *tfill++ = ncoef;
    *term = tfill - term;
    return(*term);
NormZero:
    *term = 0;
    return(0);
NormInf:
    MLOCK(ErrorMessageLock);
    MesPrint("0^0 in NormPolyTerm");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(-1);
FromNorm:
    MLOCK(ErrorMessageLock);
    MesCall("NormPolyTerm");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(-1);
}

/*
        #] NormPolyTerm :
        #[ ComparePoly :
*/
#ifdef WITHCOMPAREPOLY

WORD ComparePoly(WORD *term1, WORD *term2, WORD level)
{
    WORD *t1, *t2, *t3, *t4, *tstop1, *tstop2;
    tstop1 = term1 + *term1;
    tstop1 -= ABS(tstop1[-1]);
    tstop2 = term2 + *term2;
    tstop2 -= ABS(tstop2[-1]);
    t1 = term1+1;
    t2 = term2+1;
    while ( t1 < tstop1 && t2 < tstop2 ) {
        if ( *t1 == *t2 ) {
            if ( *t1 == HAAKJE ) {
                if ( t1[2] != t2[2] ) return(t2[2]-t1[2]);
                t1 += t1[1]; t2 += t2[1];
            }
            else {  /* must be type SYMBOL */
                t3 = t1 + t1[1]; t4 = t2 + t2[1];
                t1 += 2; t2 += 2;
                while ( t1 < t3 && t2 < t4 ) {
                    if ( *t1 != *t2 ) return(*t2-*t1);
                    if ( t1[1] != t2[1] ) return(t2[1]-t1[1]);
                    t1 += 2; t2 += 2;
                }
                if ( t1 < t3 ) return(-1);
                if ( t2 < t4 ) return(1);
            }
        }
        else return(*t2-*t1);
    }
```

```
    if ( t1 < tstop1 ) return(-1);
    if ( t2 < tstop2 ) return(1);
    return(0);
}

#endif

/*
        #] ComparePoly :
        #[ ConvertToPoly :
*/
static int FirstWarnConvertToPoly = 1;

int ConvertToPoly(PHEAD WORD *term, WORD *outterm, WORD *comlist, WORD par)
{
    WORD *tout, *tstop, ncoef, *t, *r, *tt, *ttwo = 0;
    int i, action = 0;
    tt = term + *term;
    ncoef = ABS(tt[-1]);
    tstop = tt - ncoef;
    tout = outterm+1;
    t = term + 1;
    if ( comlist[2] == DOALL ) {
      while ( t < tstop ) {
        if ( *t == SYMBOL ) {
            r = t+2;
            t += t[1];
            while ( r < t ) {
                if ( r[1] > 0 ) {
                    *tout++ = SYMBOL;
                    *tout++ = 4;
                    *tout++ = r[0];
                    *tout++ = r[1];
                }
                else {
                    tout[1] = SYMBOL;
                    tout[2] = 4;
                    tout[3] = r[0];
                    tout[4] = -1;
                    i = FindSubterm(tout+1);
                    *tout++ = SYMBOL;
                    *tout++ = 4;
                    *tout++ = MAXVARIABLES-i;
                    *tout++ = -r[1];
                    action = 1;
                }
                r += 2;
            }
        }
        else if ( *t == DOTPRODUCT ) {
            r = t + 2;
            t += t[1];
            while ( r < t ) {
                tout[1] = DOTPRODUCT;
                tout[2] = 5;
                tout[3] = r[0];
                tout[4] = r[1];
                if ( r[2] < 0 ) {
                    tout[5] = -1;
                }
                else {
                    tout[5] = 1;
                }
                i = FindSubterm(tout+1);
                *tout++ = SYMBOL;
                *tout++ = 4;
                *tout++ = MAXVARIABLES-i;
                *tout++ = ABS(r[2]);
                r += 3;
                action = 1;
            }
        }
        else if ( *t == VECTOR ) {
            r = t + 2;
            t += t[1];
            while ( r < t ) {
                tout[1] = VECTOR;
                tout[2] = 4;
                tout[3] = r[0];
                tout[4] = r[1];
                i = FindSubterm(tout+1);
                *tout++ = SYMBOL;
                *tout++ = 4;
                *tout++ = MAXVARIABLES-i;
                *tout++ = 1;
                r += 2;
                action = 1;
```

```
                }
            }
            else if ( *t == INDEX ) {
                r = t + 2;
                t += t[1];
                while ( r < t ) {
                    tout[1] = INDEX;
                    tout[2] = 3;
                    tout[3] = r[0];
                    i = FindSubterm(tout+1);
                    *tout++ = SYMBOL;
                    *tout++ = 4;
                    *tout++ = MAXVARIABLES-i;
                    *tout++ = 1;
                    r++;
                    action = 1;
                }
            }
            else if ( *t == HAAKJE) {
                if ( par ) {
                    tout[0] = 1; tout[1] = 1; tout[2] = 3;
                    *outterm = (tout+3)-outterm;
                    if ( NormPolyTerm(BHEAD outterm) < 0 ) return(-1);
                    tout = outterm + *outterm;
                    tout -= 3;
                    i = t[1]; NCOPY(tout,t,i);
                    ttwo = tout-1;
                }
                else { t += t[1]; }
            }
            else if ( *t >= FUNCTION ) {
                i = FindSubterm(t);
                t += t[1];
                *tout++ = SYMBOL;
                *tout++ = 4;
                *tout++ = MAXVARIABLES-i;
                *tout++ = 1;
                action = 1;
            }
            else {
                if ( FirstWarnConvertToPoly ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Illegal object in conversion to polynomial notation");
                    MUNLOCK(ErrorMessageLock);
                    FirstWarnConvertToPoly = 0;
                }
                return(-1);
            }
          }
          NCOPY(tout,tstop,ncoef)
          if ( ttwo ) {
            WORD hh = *ttwo;
            *ttwo = tout-ttwo;
            if ( ( i = NormPolyTerm(BHEAD ttwo) ) >= 0 ) i = action;
            tout = ttwo + *ttwo;
            *ttwo = hh;
            *outterm = tout - outterm;
          }
          else {
            *outterm = tout-outterm;
            if ( ( i = NormPolyTerm(BHEAD outterm) ) >= 0 ) i = action;
          }
        }
        else if ( comlist[2] == ONLYFUNCTIONS ) {
          while ( t < tstop ) {
            if ( *t >= FUNCTION ) {
                if ( comlist[1] == 3 ) {
                    i = FindSubterm(t);
                    t += t[1];
                    *tout++ = SYMBOL;
                    *tout++ = 4;
                    *tout++ = MAXVARIABLES-i;
                    *tout++ = 1;
                    action = 1;
                }
                else {
                    for ( i = 3; i < comlist[1]; i++ ) {
                        if ( *t == comlist[i] ) break;
                    }
                    if ( i < comlist[1] ) {
                        i = FindSubterm(t);
                        t += t[1];
                        *tout++ = SYMBOL;
                        *tout++ = 4;
                        *tout++ = MAXVARIABLES-i;
                        *tout++ = 1;
                        action = 1;
```

```
                }
                else {
                    i = t[1]; NCOPY(tout,t,i);
                }
            }
        }
        else {
            i = t[1]; NCOPY(tout,t,i);
        }
      }
      NCOPY(tout,tstop,ncoef)
      *outterm = tout-outterm;
      Normalize(BHEAD outterm);
      i = action;
    }
    else {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal internal code in conversion to polynomial notation");
        MUNLOCK(ErrorMessageLock);
        i = -1;
    }
    return(i);
}

/*
        #] ConvertToPoly :
        #[ LocalConvertToPoly :
*/
int LocalConvertToPoly(PHEAD WORD *term, WORD *outterm, WORD startebuf, WORD par)
{
    WORD *tout, *tstop, ncoef, *t, *r, *tt, *ttwo = 0;
    int i, action = 0;
    tt = term + *term;
    ncoef = ABS(tt[-1]);
    tstop = tt - ncoef;
    tout = outterm+1;
    t = term + 1;
    while ( t < tstop ) {
        if ( *t == SYMBOL ) {
            r = t+2;
            t += t[1];
            while ( r < t ) {
                if ( r[1] > 0 ) {
                    *tout++ = SYMBOL;
                    *tout++ = 4;
                    *tout++ = r[0];
                    *tout++ = r[1];
                }
                else {
                    tout[1] = SYMBOL;
                    tout[2] = 4;
                    tout[3] = r[0];
                    tout[4] = -1;
                    i = FindLocalSubterm(BHEAD tout+1,startebuf);
                    *tout++ = SYMBOL;
                    *tout++ = 4;
                    *tout++ = MAXVARIABLES-i;
                    *tout++ = -r[1];
                    action = 1;
                }
                r += 2;
            }
        }
        else if ( *t == DOTPRODUCT ) {
            r = t + 2;
            t += t[1];
            while ( r < t ) {
                tout[1] = DOTPRODUCT;
                tout[2] = 5;
                tout[3] = r[0];
                tout[4] = r[1];
                if ( r[2] < 0 ) {
                    tout[5] = -1;
                }
                else {
                    tout[5] = 1;
                }
                i = FindLocalSubterm(BHEAD tout+1,startebuf);
                *tout++ = SYMBOL;
                *tout++ = 4;
                *tout++ = MAXVARIABLES-i;
                *tout++ = ABS(r[2]);
                r += 3;
                action = 1;
            }
        }
        else if ( *t == VECTOR ) {
```

```
            r = t + 2;
            t += t[1];
            while ( r < t ) {
                tout[1] = VECTOR;
                tout[2] = 4;
                tout[3] = r[0];
                tout[4] = r[1];
                i = FindLocalSubterm(BHEAD tout+1,startebuf);
                *tout++ = SYMBOL;
                *tout++ = 4;
                *tout++ = MAXVARIABLES-i;
                *tout++ = 1;
                r += 2;
                action = 1;
            }
        }
        else if ( *t == INDEX ) {
            r = t + 2;
            t += t[1];
            while ( r < t ) {
                tout[1] = INDEX;
                tout[2] = 3;
                tout[3] = r[0];
                i = FindLocalSubterm(BHEAD tout+1,startebuf);
                *tout++ = SYMBOL;
                *tout++ = 4;
                *tout++ = MAXVARIABLES-i;
                *tout++ = 1;
                r++;
                action = 1;
            }
        }
        else if ( *t == HAAKJE) {
            if ( par ) {
                tout[0] = 1; tout[1] = 1; tout[2] = 3;
                *outterm = (tout+3)-outterm;
                if ( NormPolyTerm(BHEAD outterm) < 0 ) return(-1);
                tout = outterm + *outterm;
                tout -= 3;
                i = t[1]; NCOPY(tout,t,i);
                ttwo = tout-1;
            }
            else { t += t[1]; }
        }
        else if ( *t >= FUNCTION ) {
            i = FindLocalSubterm(BHEAD t,startebuf);
            t += t[1];
            *tout++ = SYMBOL;
            *tout++ = 4;
            *tout++ = MAXVARIABLES-i;
            *tout++ = 1;
            action = 1;
        }
        else {
            if ( FirstWarnConvertToPoly ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal object in conversion to polynomial notation");
                MUNLOCK(ErrorMessageLock);
                FirstWarnConvertToPoly = 0;
            }
            return(-1);
        }
    }
    NCOPY(tout,tstop,ncoef)
    if ( ttwo ) {
        WORD hh = *ttwo;
        *ttwo = tout-ttwo;
        if ( ( i = NormPolyTerm(BHEAD ttwo) ) >= 0 ) i = action;
        tout = ttwo + *ttwo;
        *ttwo = hh;
        *outterm = tout - outterm;
    }
    else {
        *outterm = tout-outterm;
        if ( ( i = NormPolyTerm(BHEAD outterm) ) >= 0 ) i = action;
    }
    return(i);
}

/*
        #] LocalConvertToPoly :
        #[ ConvertFromPoly :

        Converts a generic term from polynomial notation to the original
        in which the extra symbols have been replaced by their values.
        The output is in outterm.
        We only deal with the extra symbols in the range from < i <= to
```

```
        The output has to be sent to TestSub because it may contain
        subexpressions when extra symbols have been replaced.
*/

int ConvertFromPoly(PHEAD WORD *term, WORD *outterm, WORD from, WORD to, WORD offset, WORD par)
{
    WORD *tout, *tstop, *tstop1, ncoef, *t, *r, *tt;
    int i;
/*  first = 1; */
    tt = term + *term;
    tout = outterm+1;
    ncoef = ABS(tt[-1]);
    tstop = tt - ncoef;
/*
    r = t = term + 1;
    while ( t < tstop ) {
        if ( *t == SYMBOL ) {
            tstop1 = t + t[1];
            tt = t + 2;
            while ( tt < tstop1 ) {
                if ( ( *tt < MAXVARIABLES - to )
                  || ( *tt >= MAXVARIABLES - from ) ) {
                    tt += 2;
                }
                else break;
            }
            if ( tt >= tstop1 ) { t = tstop1; continue; }
            while ( r < t ) *tout++ = *r++;
            t += 2;
            first = 0;
            while ( t < tstop1 ) {
                if ( ( *t < MAXVARIABLES - to )
                  || ( *t >= MAXVARIABLES - from ) ) {
                    *tout++ = SYMBOL;
                    *tout++ = 4;
                    *tout++ = *t++;
                    *tout++ = *t++;
                }
                else {
                    *tout++ = SUBEXPRESSION;
                    *tout++ = SUBEXPSIZE;
                    *tout++ = MAXVARIABLES - *t++ + offset;
                    *tout++ = *t++;
                    if ( par ) *tout++ = AT.ebufnum;
                    else       *tout++ = AM.sbufnum;
                    FILLSUB(tout)
                }
            }
            r = t;
        }
        else {
            t += t[1];
        }
    }
    if ( first ) {
        i = *term; t = term;
        NCOPY(outterm,t,i);
        return(*term);
    }
    while ( r < t ) *tout++ = *r++;
    NCOPY(tout,tstop,ncoef)
    *outterm = tout-outterm;
*/
    t = term + 1;
    while ( t < tstop ) {
        if ( *t == SYMBOL ) {
            tstop1 = t + t[1];
            tt = t + 2;
            while ( tt < tstop1 ) {
                if ( ( *tt < MAXVARIABLES - to )
                  || ( *tt >= MAXVARIABLES - from ) ) {
                    tt += 2;
                }
                else {
                    *tout++ = SUBEXPRESSION;
                    *tout++ = SUBEXPSIZE;
                    *tout++ = MAXVARIABLES - *tt++ + offset;
                    *tout++ = *tt++;
                    if ( par ) *tout++ = AT.ebufnum;
                    else       *tout++ = AM.sbufnum;
                    FILLSUB(tout)
                }
            }
            r = tout; t += 2;
            *tout++ = SYMBOL; *tout++ = 0;
            while ( t < tstop1 ) {
                if ( ( *t < MAXVARIABLES - to )
```

```
                  || ( *t >= MAXVARIABLES - from ) ) {
                    *tout++ = *t++;
                    *tout++ = *t++;
                }
                else { t += 2; }
            }
            r[1] = tout - r;
            if ( r[1] <= 2 ) tout = r;
        }
        else {
            i = t[1]; NCOPY(tout,t,i)
        }
    }
    NCOPY(tout,tstop,ncoef)
    *outterm = tout-outterm;
    return(*outterm);
}

/*
        #] ConvertFromPoly :
        #[ FindSubterm :

        In this routine we look up a variable.
        If we don't find it we will enter it in the subterm compiler buffer
        Searching is by tree structure.
        Adding changes the tree.

        Notice that in TFORM we should be in sequential mode.
*/

WORD FindSubterm(WORD *subterm)
{
    WORD old[5], *ss, *term, number;
    CBUF *C = cbuf + AM.sbufnum;
    LONG oldCpointer;
    term = subterm-1;
    ss = subterm+subterm[1];
/*
        Convert to proper term
*/
    old[0] = *term; old[1] = ss[0]; old[2] = ss[1]; old[3] = ss[2]; old[4] = ss[3];
    ss[0] = 1; ss[1] = 1; ss[2] = 3; ss[3] = 0; *term = subterm[1]+4;
/*
        We may have to add the term to the compiler
        buffer and then to the tree. This cannot be done in parallel and
        hence we have to set a lock.
*/
    LOCK(AM.sbuflock);

    oldCpointer = C->Pointer-C->Buffer; /* Offset of course !!!!!*/
    AddRHS(AM.sbufnum,1);
    AddNtoC(AM.sbufnum,*term,term,8);
    AddToCB(C,0)
/*
        See whether we have this one already. If not, insert it in the tree.
*/
    number = InsTree(AM.sbufnum,C->numrhs);
/*
        Restore old values and return what is needed.
*/
    if ( number < (C->numrhs) ) {   /* It existed already */
        C->Pointer = oldCpointer + C->Buffer;
        C->numrhs--;
    }
    else {
        GETIDENTITY
        WORD dim = DimensionSubterm(subterm);

        if ( dim == -MAXPOSITIVE ) {    /* Give error message but continue */
            WORD *old = AN.currentTerm;
            AN.currentTerm = term;
            MLOCK(ErrorMessageLock);
            MesPrint("Dimension out of range in %t");
            MUNLOCK(ErrorMessageLock);
            AN.currentTerm = old;
        }
/*
        Store the dimension
*/
        C->dimension[number] = dim;
    }
    UNLOCK(AM.sbuflock);

    *term = old[0]; ss[0] = old[1]; ss[1] = old[2]; ss[2] = old[3]; ss[3] = old[4];
    return(number);
}
```

```
/*
        #] FindSubterm :
        #[ FindLocalSubterm :

        In this routine we look up a variable.
        If we don't find it we will enter it in the subterm compiler buffer
        Searching is by tree structure.
        Adding changes the tree.

        Notice that in TFORM we should be in sequential mode.
*/

WORD FindLocalSubterm(PHEAD WORD *subterm, WORD startebuf)
{
    WORD old[5], *ss, *term, number, i, j, *t1, *t2;
    CBUF *C = cbuf + AT.ebufnum;
    term = subterm-1;
    ss = subterm+subterm[1];
/*
        Convert to proper term
*/
    old[0] = *term; old[1] = ss[0]; old[2] = ss[1]; old[3] = ss[2]; old[4] = ss[3];
    ss[0] = 1; ss[1] = 1; ss[2] = 3; ss[3] = 0; *term = subterm[1]+4;
/*
        First see whether we have this one already in the global buffer.
*/
    number = FindTree(AM.sbufnum,term);
    if ( number > 0 ) goto wearehappy;
/*
    Now look whether it is in the ebufnum between startebuf and numrhs
    Note however that we need an offset of (numxsymbol-startebuf)
*/
    for ( i = startebuf+1; i <= C->numrhs; i++ ) {
        t1 = C->rhs[i]; t2 = term;
        if ( *t1 == *t2 ) {
            j = *t1;
            while ( *t1 == *t2 && j > 0 ) { t1++; t2++; j--; }
            if ( j <= 0 ) {
                number = i-startebuf+numxsymbol;
                goto wearehappy;
            }
        }
    }
/*
    Now we have to add it to cbuf[AT.ebufnum]
*/
    AddRHS(AT.ebufnum,1);
    AddNtoC(AT.ebufnum,*term,term,9);
    AddToCB(C,0)
    number = C->numrhs-startebuf+numxsymbol;
wearehappy:
    *term = old[0]; ss[0] = old[1]; ss[1] = old[2]; ss[2] = old[3]; ss[3] = old[4];
    return(number);
}

/*
        #] FindLocalSubterm :
        #[ PrintSubtermList :

        Prints all the expressions in the subterm compiler buffer.
        The format is such that they give definitions of the temporary
        variables of which the contents are stored in this buffer.
        These variables have the names Z_123 etc.
*/

void PrintSubtermList(int from,int to)
{
    UBYTE buffer[80], *out, outbuffer[300];
    int first, i, ii, inc = 1;
    WORD *term;
    CBUF *C = cbuf + AM.sbufnum;
/*
    if ( to < from ) inc = -1;
    if ( to == from ) inc = 0;
*/
    if ( from <= to ) {
        inc = 1; to += inc;
    }
    else {
        inc = -1; to += inc;
    }
    AO.OutFill = AO.OutputLine = outbuffer;
    AO.OutStop = AO.OutputLine+AC.LineLength;
    AO.IsBracket = 0;
    AO.OutSkip = 3;

    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
```

```
        TokenToLine((UBYTE *)"      ");
        AO.OutSkip = 7;
    }
    else if ( ( AO.Optimize.debugflags & 1 ) == 1 ) {}
    else if ( AO.OutSkip > 0 ) {
        for ( i = 0; i < AO.OutSkip; i++ ) TokenToLine((UBYTE *)" ");
    }
    i = from;
    do {
        if ( ( AO.Optimize.debugflags & 1 ) == 1 ) {
            TokenToLine((UBYTE *)"id ");
            for ( ii = 3; ii < AO.OutSkip; ii++ ) TokenToLine((UBYTE *)" ");
        }
/*
        if ( AC.OutputMode == NORMALFORMAT ) {
            TokenToLine((UBYTE *)"id ");
        }
*/
        else if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {}
        else { TokenToLine((UBYTE *)" "); }

        out = StrCopy((UBYTE *)AC.extrasym,buffer);
        if ( AC.extrasymbols == 0 ) {
            out = NumCopy(i,out);
            out = StrCopy((UBYTE *)"_",out);
        }
        else if ( AC.extrasymbols == 1 ) {
            out = AddArrayIndex(i,out);
        }
        out = StrCopy((UBYTE *)"=",out);
        TokenToLine(buffer);
        term = C->rhs[i];
        first = 1;
        if ( *term == 0 ) {
            out = StrCopy((UBYTE *)"0",buffer);
            if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE ) {
                out = StrCopy((UBYTE *)";",out);
            }
            TokenToLine(buffer);
        }
        else {
            while ( *term ) {
                if ( WriteInnerTerm(term,first) ) Terminate(-1);
                term += *term;
                first = 0;
            }
            if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE ) {
                out = StrCopy((UBYTE *)";",buffer);
                TokenToLine(buffer);
            }
        }
/*
        There is a problem with FiniLine because it prepares for a
        continuation line in fortran mode.
        But the next statement should start on a blank line.
*/
/*
        FiniLine();
        if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
            AO.OutFill = AO.OutputLine;
            TokenToLine((UBYTE *)"      ");
            AO.OutSkip = 7;
        }
*/
        if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
            AO.OutSkip = 6;
            FiniLine();
            AO.OutSkip = 7;
        }
        else {
            FiniLine();
        }
        i += inc;
    } while ( i != to );
}

/*

        #] PrintSubtermList :
        #[ PrintExtraSymbol :

        Prints the definition of extra symbol num as the contents
        of the expression in terms.
        The parameter par has three options:
            EXTRASYMBOL      num is interpreted as the number of an extra symbol
            REGULARSYMBOL    num is interpreted as the number of a symbol.
                             It could still be an extra symbol.
            EXPRESSIONNUMBER num is the number of an expression.
```

```
        terms contains the rhs expression.
*/

void PrintExtraSymbol(int num, WORD *terms,int par)
{
    UBYTE buffer[80], *out, outbuffer[300];
    int first, i;
    WORD *term;

    AO.OutFill = AO.OutputLine = outbuffer;
    AO.OutStop = AO.OutputLine+AC.LineLength;
    AO.IsBracket = 0;

    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
        TokenToLine((UBYTE *)"      ");
        AO.OutSkip = 7;
    }
    else if ( ( AO.Optimize.debugflags & 1 ) == 1 ) {
        TokenToLine((UBYTE *)"id ");
        for ( i = 3; i < AO.OutSkip; i++ ) TokenToLine((UBYTE *)" ");
    }
    else if ( AO.OutSkip > 0 ) {
        for ( i = 0; i < AO.OutSkip; i++ ) TokenToLine((UBYTE *)" ");
    }
    out = buffer;
    switch ( par ) {
        case REGULARSYMBOL:
            if ( num >= MAXVARIABLES-cbuf[AM.sbufnum].numrhs ) {
                num = MAXVARIABLES-num;
            }
            else {
                out = StrCopy(FindSymbol(num),out);
/*              out = StrCopy(VARNAME(symbols,num),out); */
                break;
            }
            /* fall through */
        case EXTRASYMBOL:
            out = StrCopy(FindExtraSymbol(num),out);
/*
            out = StrCopy((UBYTE *)AC.extrasym,out);
            if ( AC.extrasymbols == 0 ) {
                out = NumCopy(num,out);
                out = StrCopy((UBYTE *)"_",out);
            }
            else if ( AC.extrasymbols == 1 ) {
                out = AddArrayIndex(num,out);
            }
*/
            break;
        case EXPRESSIONNUMBER:
            out = StrCopy(EXPRNAME(num),out);
            break;
        default:
            MesPrint("Illegal option in PrintExtraSymbol");
            Terminate(-1);
    }
    out = StrCopy((UBYTE *)"=",out);
    TokenToLine(buffer);
    term = terms;
    first = 1;
    if ( *term == 0 ) {
        out = StrCopy((UBYTE *)"0",buffer);
        TokenToLine(buffer);
    }
    else {
        while ( *term ) {
            if ( WriteInnerTerm(term,first) ) Terminate(-1);
            term += *term;
            first = 0;
        }
    }
    if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE ) {
        out = StrCopy((UBYTE *)";",buffer);
        TokenToLine(buffer);
    }
    FiniLine();
}

/*
        #] PrintExtraSymbol :
        #[ FindSubexpression :

        In this routine we look up a subexpression.
        If we don't find it we will enter it in the subterm compiler buffer
        Searching is by tree structure.
        Adding changes the tree.

```

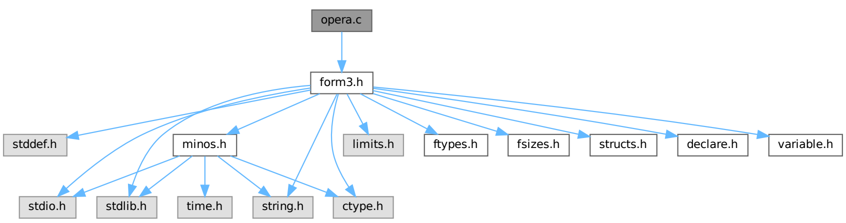
```
        Notice that in TFORM we should be in sequential mode.
*/

WORD FindSubexpression(WORD *subexpr)
{
    WORD *term, number;
    CBUF *C = cbuf + AM.sbufnum;
    LONG oldCpointer;

    term = subexpr;
    while ( *term ) term += *term;
    number = term - subexpr;
/*
        We may have to add the subexpression to the tree.
        This requires a lock.
*/
    LOCK(AM.sbuflock);

    oldCpointer = C->Pointer-C->Buffer; /* Offset of course !!!!!*/
    AddRHS(AM.sbufnum,1);
/*
        Add the terms to the compiler buffer. Paste on a zero.
*/
    AddNtoC(AM.sbufnum,number,subexpr,10);
    AddToCB(C,0)
/*
        See whether we have this one already. If not, insert it in the tree.
*/
    number = InsTree(AM.sbufnum,C->numrhs);
/*
        Restore old values and return what is needed.
*/
    if ( number < (C->numrhs) ) {   /* It existed already */
        C->Pointer = oldCpointer + C->Buffer;
        C->numrhs--;
    }
    else {
        GETIDENTITY
        WORD dim = DimensionExpression(BHEAD subexpr);
/*
        Store the dimension
*/
        C->dimension[number] = dim;
    }

    UNLOCK(AM.sbuflock);

    return(number);
}

/*
        #] FindSubexpression :
        #[ ExtraSymFun :
*/

int ExtraSymFun(PHEAD WORD *term,WORD level)
{
    WORD *oldworkpointer = AT.WorkPointer;
    WORD *termout, *t1, *t2, *t3, *tstop, *tend, i;
    int retval = 0;
    tend = termout = term + *term;
    tstop = tend - ABS(tend[-1]);
    t3 = t1 = term+1; t2 = termout+1;
/*
    First refind the function(s). There is at least one.
*/
    while ( t1 < tstop ) {
        if ( *t1 == EXTRASYMFUN && t1[1] == FUNHEAD+2 ) {
            if ( t1[FUNHEAD] == -SNUMBER && t1[FUNHEAD+1] <= numxsymbol
                                && t1[FUNHEAD+1] > 0 ) {
                i = t1[FUNHEAD+1];
            }
            else if ( t1[FUNHEAD] == -SYMBOL && t1[FUNHEAD+1] < MAXVARIABLES
                                && t1[FUNHEAD+1] >= MAXVARIABLES-numxsymbol ) {
                i = MAXVARIABLES - t1[FUNHEAD+1];
            }
            else goto nocase;
            while ( t3 < t1 ) *t2++ = *t3++;
/*
                Now inset the rhs pointer
*/
            *t2++ = SUBEXPRESSION;
            *t2++ = SUBEXPSIZE;
            *t2++ = i;
            *t2++ = 1;
            *t2++ = AM.sbufnum;
            FILLSUB(t2)
```

```
            t3 = t1 = t1 + t1[1];
        }
        else if ( *t1 == EXTRASYMFUN && t1[1] == FUNHEAD ) {
            while ( t3 < t1 ) *t2++ = *t3++;
            t3 = t1 = t1 + t1[1];
        }
        else {
nocase:;
            t1 = t1 + t1[1];
        }
    }
    while ( t3 < tend ) *t2++ = *t3++;
    *termout = t2 - termout;
    AT.WorkPointer = t2;
    if ( AT.WorkPointer >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        AT.WorkPointer = oldworkpointer;
        return(-1);
    }
    retval = Generator(BHEAD termout,level);
    AT.WorkPointer = oldworkpointer;
    if ( retval < 0 ) {
        MLOCK(ErrorMessageLock);
        MesCall("ExtraSymFun");
        MUNLOCK(ErrorMessageLock);
    }
    return(retval);
}

/*
        #] ExtraSymFun :
        #[ PruneExtraSymbols :
*/

int PruneExtraSymbols(WORD downto)
{
    CBUF *C = cbuf + AM.sbufnum;
    if ( downto < C->numrhs && downto >= 0 ) {  /* !!!!! */
        ClearTree(AM.sbufnum);
        C->numrhs = downto;
        if ( downto == 0 ) {
            C->Pointer = C->Buffer;
        }
        else {
            WORD *w = C->rhs[downto], i;
            while ( *w ) w += *w;
            C->Pointer = w+1;
            for ( i = 1; i <= downto; i++ ) {
                InsTree(AM.sbufnum,i);
            }
        }
    }
    return(0);
}

/*
        #] PruneExtraSymbols :
*/
```

4.90 opera.c File Reference

```
#include "form3.h"
```

Include dependency graph for opera.c:



Functions

- WORD EpfFind (PHEAD WORD ∗term, WORD ∗params)
- WORD EpfCon (PHEAD WORD ∗term, WORD ∗params, WORD num, WORD level)
- WORD EpfGen (WORD number, WORD ∗inlist, WORD ∗kron, WORD ∗perm, WORD sgn)
- WORD Trick (WORD ∗in, TRACES ∗t)
- WORD Trace4no (WORD number, WORD ∗kron, TRACES ∗t)
- WORD Trace4 (PHEAD WORD ∗term, WORD ∗params, WORD num, WORD level)
- WORD Trace4Gen (PHEAD TRACES ∗t, WORD number)
- WORD TraceNno (WORD number, WORD ∗kron, TRACES ∗t)
- WORD TraceN (PHEAD WORD ∗term, WORD ∗params, WORD num, WORD level)
- WORD TraceNgen (PHEAD TRACES ∗t, WORD number)
- WORD Traces (PHEAD WORD ∗term, WORD ∗params, WORD num, WORD level)
- WORD TraceFind (PHEAD WORD ∗term, WORD ∗params)
- WORD Chisholm (PHEAD WORD ∗term, WORD level)
- WORD TenVecFind (PHEAD WORD ∗term, WORD ∗params)
- WORD TenVec (PHEAD WORD ∗term, WORD ∗params, WORD num, WORD level)

4.90.1 Detailed Description

Contains the 'operations' These are the trace routines, the contractions of the Levi-Civita tensors and the tensor to vector/vector to tensor routines. The trace and contraction routines are done in a special way (see commentary with the FIXEDGLOBALS struct)

Definition in file opera.c.

4.90.2 Function Documentation

4.90.2.1 EpfFind()

```
WORD EpfFind (
            PHEAD WORD * term,
            WORD * params )
```

Definition at line 58 of file opera.c.

4.90.2.2 EpfCon()

```
WORD EpfCon (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )
```

Definition at line 212 of file opera.c.

4.90.2.3 EpfGen()

```
WORD EpfGen (
            WORD number,
            WORD * inlist,
            WORD * kron,
            WORD * perm,
            WORD sgn )
```

Definition at line 282 of file opera.c.

4.90.2.4 Trick()

```
WORD Trick (
            WORD * in,
            TRACES * t )
```

Definition at line 331 of file opera.c.

4.90.2.5 Trace4no()

```
WORD Trace4no (
            WORD number,
            WORD * kron,
            TRACES * t )
```

Definition at line 418 of file opera.c.

4.90.2.6 Trace4()

```
WORD Trace4 (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )
```

Definition at line 694 of file opera.c.

4.90.2.7 Trace4Gen()

```
WORD Trace4Gen (
            PHEAD TRACES * t,
            WORD number )
```

Definition at line 856 of file opera.c.

4.90.2.8 TraceNno()

```
WORD TraceNno (
            WORD number,
            WORD * kron,
            TRACES * t )
```

Definition at line 1341 of file opera.c.

4.90.2.9 TraceN()

```
WORD TraceN (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )
```

Definition at line 1382 of file opera.c.

4.90.2.10 TraceNgen()

```
WORD TraceNgen (
            PHEAD TRACES * t,
            WORD number )
```

Definition at line 1446 of file opera.c.

4.90.2.11 Traces()

```
WORD Traces (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )
```

Definition at line 1831 of file opera.c.

4.90.2.12 TraceFind()

```
WORD TraceFind (
            PHEAD WORD * term,
            WORD * params )
```

Definition at line 1853 of file opera.c.

4.90.2.13 Chisholm()

```
WORD Chisholm (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1963 of file opera.c.

4.90.2.14 TenVecFind()

```
WORD TenVecFind (
            PHEAD WORD * term,
            WORD * params )
```

Definition at line 2178 of file opera.c.

4.90.2.15 TenVec()

```
WORD TenVec (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )
```

Definition at line 2312 of file opera.c.

4.91 opera.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : opera.c
*/

#include "form3.h"
/*
    int hulp;
*/

/*
    #] Includes :
    #[ Operations :
        #[ EpfFind :            WORD EpfFind(term,params)

        Searches for a pair of Levi-Civita tensors that should be
        contracted.
        If a match is found its settings are recorded in AT.TMout.
        type indicates the number of indices that is searched for,
        unless all are searched for (type = 0).
        number is the number of tensors that should survive contraction.

*/
```

```
WORD EpfFind(PHEAD WORD *term, WORD *params)
{
    GETBIDENTITY
    WORD *t, *m, *r, n1 = 0, n2, min = -1, count, fac;
    WORD *c1 = 0, *c2 = 0, sgn = 1;
    WORD *tstop, *mstop;
    UWORD *facto = (UWORD *)AT.WorkPointer;
    WORD ncoef,nfac;
    WORD number, type;
    if ( ( AT.WorkPointer = (WORD *)(facto + AM.MaxTal) ) > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    number = params[3];
    type = params[4];
    t = term;
    GETSTOP(t,tstop);
    t++;
    if ( !type ) {
        while ( *t != LEVICIVITA && t < tstop ) t += t[1];
        if ( t >= tstop ) return(0);
        m = t;
        while ( *m == LEVICIVITA && m < tstop ) { n1++; m += m[1]; }
AllLev:
        if ( n1 <= (number+1) || n1 <= 1 ) return(0);
        mstop = m;
        m = t + t[1];
        do {
            while ( m[1] == t[1] ) {
                m += FUNHEAD;
                r = t+FUNHEAD;
                count = fac = n1 = n2 = t[1] - FUNHEAD;
                while ( n1 && n2 ) {
                    if ( *m > *r ) {
                        r++; n2--;
                    }
                    else if ( *m < *r ) {
                        m++; n1--;
                    }
                    else {
                        if ( *m >= AM.OffsetIndex &&
                        ( ( *m >= AM.IndDum && AC.lDefDim == fac ) ||
                        ( *m < AM.IndDum &&
                        indices[*m-AM.OffsetIndex].dimension == fac ) ) ) {
                            count--;
                        }
                        r++; m++; n1--; n2--;
                    }
                }
                m += n1;
                if ( min < 0 || count < min ) {
                    c1 = t;
                    c2 = m - fac - FUNHEAD;
                    min = count;
                }
                if ( m >= mstop ) break;
            }
            t += t[1];
        } while ( ( m = t + t[1] ) < mstop );
    }
    else {
        fac = type + FUNHEAD;
        while ( *t != LEVICIVITA && t < tstop ) t += t[1];
        while ( *t == LEVICIVITA && t < tstop && t[1] != fac ) t += t[1];
        if ( t >= tstop ) return(0);
        m = t;
        while ( *m == LEVICIVITA && m < tstop && m[1] == fac ) { n1++; m += m[1]; }
        goto AllLev;
    }
/*
    We have now the two tensors that give the minimum contraction
    in c1 and c2.
    Prepare the AT.TMout array;
*/
    if ( min < 0 ) return(0);    /* No matching pair! */
    t = c1;
    mstop = c2;
    fac = t[1] - FUNHEAD;
    m = AT.TMout;
    *m++ = 3 + (min*2);      /* The full length */
    *m++ = CONTRACT;
    if ( number < 0 ) *m++ = 1;
    else *m++ = 0;
    n1 = n2 = t[1] - FUNHEAD;
```

```
    r = c1 + FUNHEAD;
    c1 = m;
    m = c2 + FUNHEAD;
    c2 = c1 + min;
    while ( n1 && n2 ) {
        if ( *m > *r ) { *c1++ = *r++; n2--; }
        else if ( *m < *r ) { *c2++ = *m++; n1--; }
        else {
            if ( *m < AM.OffsetIndex || ( *m < AM.IndDum &&
            ( indices[*m-AM.OffsetIndex].dimension != fac ) ) ||
            ( *m >= AM.IndDum && AC.lDefDim != fac ) ) {
                *c1++ = *r++; *c2++ = *m++;
            }
            else { if ( ( n1 ^ n2 ) & 1 ) sgn = -sgn; r++; m++; }
            n1--; n2--;
        }
    }
    if ( n1 ) { NCOPY(c2,m,n1); }
    else if ( n2 ) { NCOPY(c1,r,n2); }
    fac -= min;
    m = t + t[1];
    while ( m < mstop ) *t++ = *m++;
    m += m[1];
    while ( m < tstop ) *t++ = *m++;
    *t++ = SUBEXPRESSION;
    *t++ = SUBEXPSIZE;
    *t++ = -1;
    *t++ = 1;
    *t++ = DUMMYBUFFER;
    FILLSUB(t)
    r = term;
    r += *r - 1;
    mstop = r;
    ncoef = REDLENG(*r);
    tstop = t;
    while ( m < mstop ) *t++ = *m++;
    if ( Factorial(BHEAD fac,facto,&nfac) || Mully(BHEAD (UWORD *)tstop,&ncoef,facto,nfac) ) {
        MLOCK(ErrorMessageLock);
        MesCall("EpfFind");
        MUNLOCK(ErrorMessageLock);
        SETERROR(-1)
    }
    tstop += (ABS(ncoef))*2;
    if ( sgn < 0 ) ncoef = -ncoef;
    ncoef *= 2;
    *tstop++ = (ncoef<0)?(ncoef-1):(ncoef+1);
    *term = WORDDIF(tstop,term);
    return(1);
}

/*
        #] EpfFind :
        #[ EpfCon :              WORD EpfCon(term,params,num,level)

        Contraction of two strings of indices/vectors. They come
        from LeviCivita tensors that are being contracted.
        term is the term with the subterm to be replaced.
        params is the full indicator:
            Length, number, raise, parameters.
        Length is the length of the parameter field.
        number is the number of the operation.
        raise tells whether level should be raised afterwards.
        the parameters are the two strings.
        level is the id level.
        The factorial has been multiplied in already.

*/

WORD EpfCon(PHEAD WORD *term, WORD *params, WORD num, WORD level)
{
    GETBIDENTITY
    WORD *kron, *perm, *termout, *tstop, size2;
    WORD *m, *t, sizes, sgn = 0, i;
    sizes = *params - 3;
    kron = AT.WorkPointer;
    perm = (AT.WorkPointer += sizes);
    termout = (AT.WorkPointer += sizes);
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    if ( AT.WorkPointer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    params += 2;
    if ( !(*params++) ) level--;
    size2 = sizes»1;
```

```
    if ( !size2 ) goto DoOnce;
    while ( ( sgn = EpfGen(size2,params,kron,perm,sgn) ) != 0 ) {
DoOnce:
        t = term;
        GETSTOP(t,tstop);
        m = termout;
        tstop -= SUBEXPSIZE;
        while ( t < tstop ) *m++ = *t++;
        if ( t[2] != num || *t != SUBEXPRESSION ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Serious error in EpfCon");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        tstop += SUBEXPSIZE;
        if ( sizes ) {
            *m++ = DELTA;
            *m++ = sizes + 2;
            t = kron;
            i = sizes;
            if ( i ) { NCOPY(m,t,i); }
        }
        t = tstop;
        tstop = term + *term;
        while ( t < tstop ) *m++ = *t++;
        *termout = WORDDIF(m,termout);
        m--;
        if ( sgn < 0 ) *m = - *m;
        if ( *termout ) {
            *AN.RepPoint = 1;
            AR.expchanged = 1;
            AT.WorkPointer = termout + *termout;
            if ( Generator(BHEAD termout,level) < 0 ) goto EpfCall;
        }
    }
    AT.WorkPointer = kron;
    return(0);
EpfCall:
    if ( AM.tracebackflag ) {
        MLOCK(ErrorMessageLock);
        MesCall("EpfCon");
        MUNLOCK(ErrorMessageLock);
    }
    return(-1);
}

/*
        #] EpfCon :
        #[ EpfGen :            WORD EpfGen(number,inlist,kron,perm,sgn)
*/

WORD EpfGen(WORD number, WORD *inlist, WORD *kron, WORD *perm, WORD sgn)
{
    WORD i, *in2, k, a;
    if ( !sgn ) {
        in2 = inlist + number;
        number *= 2;
        for ( i = 1; i < number; i += 2 ) {
            *perm++ = i;
            *perm++ = i;
            *kron++ = *inlist++;
            *kron++ = *in2++;
        }
        if ( number <= 0 ) return(0);
        else return(1);
    }
    number *= 2;
    i = number - 1;
    while ( ( i -= 2 ) >= 0 ) {
        if ( ( k = perm[i] ) != i ) {
            sgn = -sgn;
            a = kron[i];
            kron[i] = kron[k];
            kron[k] = a;
        }
        if ( ( k = ( perm[i] += 2 ) ) < number ) {
            a = kron[i];
            kron[i] = kron[k];
            kron[k] = a;
            sgn = - sgn;
            for ( k = i + 2; k < number; k += 2 ) perm[k] = k;
            return(sgn);
        }
    }
    return(0);
}
```

```
/*
        #] EpfGen :
        #[ Trick :              WORD Trick(in,t)

        This routine implements the identity:
        g_(j,mu)*g_(j,nu)*g_(j,ro)=e_(mu,nu,ro,si)*g5_(j)*g_(j,si)
        +d_(mu,nu)*g_(j,ro)-d_(mu,ro)*g_(j,nu)+d_(nu,ro)*g_(j,mu)
        which is for 4 dimensions only!

        Note that z->gamm = 1 if there is no g5 present.

*/

WORD Trick(WORD *in, TRACES *t)
{
    WORD n, n1;
    n = t->stap;
    n1 = t->step1;
    switch ( t->eers[n] ) {
        case 0: {
            WORD *p;
            p = t->pepf + t->mepf[n];
            *p++ = *in++;
            *p++ = *in++;
            *p++ = *in;
            *p   = ++(t->mdum);
            (t->mepf[n1]) += 4;
            *in  = t->mdum;
            t->gamm = - t->gamm;
            t->eers[n] = 5;
            break;
        }
        case 4: {
            WORD *p;
            p = t->pdel + t->mdel[n];
            (t->mepf[n1]) -= 4;
            (t->mdum)--;
            *p++ = *in++;
            *p   = *in++;
            *in  = *(t->pepf + t->mepf[n] + 2);
            (t->mdel[n1]) += 2;
            t->gamm = - t->gamm;
            break;
        }
        case 3: {
            t->sign1 = - t->sign1;
            *(t->pdel + t->mdel[n] + 1) = in[2];
            in[2] = in[1];
            break;
        }
        case 2: {
            t->sign1 = - t->sign1;
            in[2] = in[0];
            *(t->pdel + t->mdel[n]) = in[1];
            break;
        }
        case 1: {
            in[2] = *(t->pdel + t->mdel[n] + 1);
            (t->mdel[n1]) -= 2;
            break;
        }
        default: {
            return(0);
        }
    }
    return(--(t->eers[n]));
}

/*
        #] Trick :
        #[ Trace4no :           WORD Trace4no(number,kron,t)

        Takes the trace of a string of gamma matrices in 4 dimensions.
        There is no test for indices or vectors that are the same.
        The four dimensions refer to the contraction in the algebra:
        g_(i,a,b,c) = e_(a,b,c,d)*g_(i,5_,d) + g_(i,a)*d_(b,c)
                    - g_(i,b)*d_(a,c) + g_(i,c)*d_(a,b)
        This simplifies life very much and leads to shorter expressions
        than in the n dimensional case.

        Parameters:
        number: the number of vectors/indices in inlist.
        inlist: the indices/vectors in the string.
        kron:   the output delta's.
        gamma5: the potential gamma5 in front.
        t:      the struct for scratch manipulations.
        stack:  the space to put all scratch arrays in.
```

```

        The return value is zero if there are no more terms, 1 if a
        term was generated with a positive sign and -1 if a term was
        generated with a negative sign.
        The value of one is increased to two if the first 4 values
        in kron should be interpreted as a Levi-Civita tensor.

        Note that kron should have more places reserved than the number
        of indices in inlist, because it will contain dummy indices
        temporarily. In principle there can be ’number*1/4’ extra dummies.

*/

WORD Trace4no(WORD number, WORD *kron, TRACES *t)
{
    WORD i;
    WORD *p, *m;
    WORD retval, *stop, oldsign;
    if ( !t->sgn ) {        /* Startup */
        if ( ( number < 0 ) || ( number & 1 ) ) return(0);
        if ( number <= 2 ) {
            if ( t->gamma5 == GAMMA5 ) return(0);
            if ( number == 2 ) {
                *kron++ = *t->inlist;
                *kron++ = t->inlist[1];
            }
            return(1);
        }
        t->sgn = 1;
        {
            WORD nhalf = number » 1;
            WORD ndouble = number * 2;
            p = t->eers;
            t->eers = p;    p += nhalf;
            t->mepf = p;    p += nhalf;
            t->mdel = p;    p += nhalf;
            t->pdel = p;    p += number + nhalf;
            t->pepf = p;    p += ndouble;
            t->e4 = p;      p += number;
            t->e3 = p;      p += ndouble;
            t->nt3 = p;     p += nhalf;
            t->nt4 = p;     p += nhalf;
            t->j3 = p;      p += ndouble;
            t->j4 = p;
        }
        t->mepf[0] = 0;
        t->mdel[0] = 0;
        t->mdum = AM.mTraceDum;
        t->kstep = -2;
        t->step1 = 0;
        t->sign1 = 1;
        t->lc3 = -1;
        t->lc4 = -1;
        t->gamm = 1;

        do {
            t->stap = (t->step1)++;
            t->kstep += 2;
            t->eers[t->stap] = 0;
            t->mepf[t->step1] = t->mepf[t->stap];
            t->mdel[t->step1] = t->mdel[t->stap];
CallTrick:  while ( !Trick(t->inlist+t->kstep,t) ) {
                t->kstep -= 2;
                t->step1 = (t->stap)--;
                if ( t->stap < 0 ) {
                    return(0);
                }
            }
        } while ( t->kstep < (number-4) );
/*
        Take now the trace of the leftover matrices.
        If gamma5 causes the term to vanish there will be a
        renewed call to Trick for its next term.
*/
        t->sign2 = t->sign1;
        if ( ( t->gamma5 == GAMMA7 ) && ( t->gamm == -1 ) ) {
            t->sign2 = - t->sign2;
        }
        else if ( ( t->gamma5 == GAMMA5 ) && ( t->gamm == 1 ) ) {
            goto CallTrick;
        }
        else if ( ( t->gamma5 == GAMMA1 ) && ( t->gamm == -1 ) ) {
            goto CallTrick;
        }
        p = t->pdel + t->mdel[t->step1];
        *p++ = t->inlist[t->kstep+2];
        *p++ = t->inlist[t->kstep+3];
```

```
/*
        Now the trace has been expressed in terms of Levi-Civita tensors
        and Kronecker delta's.
        The Levi-Civita tensors are in t->pepf
        and there are t->mepf[step1] elements in this array.
        The Kronecker delta's are in t->pdel
        and there are t->mdel[step1] elements in this array.

        Next we rake the Levi-Civita tensors together such that there
        is an optimal use of the contractions.
*/
        {
            WORD ae;
            ae = t->mepf[t->step1];
            t->ad = t->mdel[t->step1]+2;
            t->a4 = 0;
            t->a3 = 0;
            while ( ( ae -= 4 ) >= 0 ) {
                if ( t->pepf[ae] > AM.mTraceDum && t->pepf[ae] <= t->mdum ) {
                    p = t->e3 + t->a3;
                    m = t->pepf + ae;
                    for ( i = 0; i < 3; i++ ) {
                        p[3] = m[3-i];
                        *p++ = m[i-4];
                    }
                    t->a3 += 6;
                    ae -= 4;
                }
                else {
                    p = t->e4 + t->a4;
                    m = t->pepf + ae;
                    for ( i = 0; i < 4; i++ ) *p++ = *m++;
                    t->a4 += 4;
                }
            }
        }
/*
        Now e3 contains pairs of LeviCivita tensors that have
        three indices each and a3 is the total number of indices.
        e4 contains individual tensors with 4 indices.
        Some indices may be contracted with Kronecker delta's.

        Contract the e3 tensors first.
*/

        while ( t->a3 > 0 ) {
            t->nt3[++(t->lc3)] = 0;
            while ( ( t->nt3[t->lc3] = EpfGen(3,t->e3+t->a3-6,
            t->pdel+t->ad,t->j3+6*t->lc3,oldsign = t->nt3[t->lc3]) ) == 0 ) {
                if ( oldsign < 0 ) t->sign2 = - t->sign2;
                (t->lc3)--;
NextE3:         if ( t->lc3 < 0 ) goto CallTrick;
                t->ad -= 6;
                t->a3 += 6;
            }
            if ( oldsign ) {
                if ( oldsign != t->nt3[t->lc3] ) t->sign2 = - t->sign2;
            }
            else if ( t->nt3[t->lc3] < 0 ) t->sign2 = - t->sign2;
            t->a3 -= 6;
            t->ad += 6;
        }
/*
        Contract the e4 tensors.
*/
        while ( t->a4 > 4 ) {
            t->nt4[++(t->lc4)] = 0;
            while ( ( t->nt4[t->lc4] = EpfGen(4,t->e4+t->a4-8,
            t->pdel+t->ad,t->j4+8*t->lc4,oldsign = t->nt4[t->lc4]) ) == 0 ) {
                if ( oldsign < 0 ) t->sign2 = - t->sign2;
                (t->lc4)--;
NextE4:         if ( t->lc4 < 0 ) goto NextE3;
                t->ad -= 8;
                t->a4 += 8;
            }
            if ( oldsign ) {
                if ( oldsign != t->nt4[t->lc4] ) t->sign2 = - t->sign2;
            }
            else if ( t->nt4[t->lc4] < 0 ) t->sign2 = - t->sign2;
            t->a4 -= 8;
            t->ad += 8;
        }
/*
        Finally the extra dummy indices can be eliminated.
        Note that there are currently t->ad words in t->pdel forming
        t->ad / 2 Kronecker delta's. We are however not allowed to
        alter anything in these arrays, so the results should be
```

```
        copied to kron.
*/
        m = kron;
        if ( t->a4 == 4 ) {
            p = t->e4;
            *m++ = *p++; *m++ = *p++; *m++ = *p++; *m++ = *p++;
            retval = 2;
        }
        else retval = 1;
        if ( t->sign2 < 0 ) retval = - retval;
        p = t->pdel;
        for ( i = 0; i < t->ad; i++ ) *m++ = *p++;
        p = kron;
        if ( t->a4 == 4 ) {
/*
            Test for dummies in the last position of the e_.
*/
            stop = p + t->ad + 4;
            p += 3;
            while ( *p >= AM.mTraceDum && *p <= t->mdum ) {
                m = p + 1;
                do {
                    if ( *m == *p ) {
                        *p = m[1];
                        *m = *--stop;
                        m[1] = *--stop;
                        break;
                    }
                    else if ( m[1] == *p ) {
                        *p = *m;
                        *m = *--stop;
                        m[1] = *--stop;
                        break;
                    }
                    else m += 2;
                } while ( m < stop );
            }
            p++;
        }
        else stop = p + t->ad;
        while ( p < (stop-2) ) {
            while ( *p >= AM.mTraceDum && *p <= t->mdum ) {
                m = p + 2;
                do {
                    if ( *m == *p ) {
                        *p = m[1];
                        *m = *--stop;
                        m[1] = *--stop;
                        break;
                    }
                    else if ( m[1] == *p ) {
                        *p = *m;
                        *m = *--stop;
                        m[1] = *--stop;
                        break;
                    }
                    else m += 2;
                } while ( m < stop );
            }
            p++;
            while ( *p >= AM.mTraceDum && *p <= t->mdum ) {
                m = p + 1;
                do {
                    if ( *m == *p ) {
                        *p = m[1];
                        *m = *--stop;
                        m[1] = *--stop;
                        break;
                    }
                    else if ( m[1] == *p ) {
                        *p = *m;
                        *m = *--stop;
                        m[1] = *--stop;
                        break;
                    }
                    else m += 2;
                } while ( m < stop );
            }
            p++;
        }
        return(retval);
    }
    if ( number <= 2 ) return(0);
    else { goto NextE4; }
}

/*
```

```
        #] Trace4no :
        #[ Trace4 :             WORD Trace4(term,params,num,level)

        Generates traces of the string of gamma matrices in 'instring'.

        The difference with the routine tracen ( for n dimensions )
        lies in the treatment of gamma 5 and the specific form
        of the Chisholm identities. The identities used here are
        g(mu)*g(a1)*...*g(an)*g(mu)=
        n=odd:  -2*g(an)*...*g(a1)    ( reversed order )
        n=even: 2*g(an)*g(a1)*...*g(a(n-1))
            +2*g(a(n-1))*...*g(a1)*g(an)
        There is a special case for n=2 : 4*d(a1,a2)*gi

        The main difference with the old fortran version lies in
        the recursion that is used here. That cleans up the variables
        very much.

        The contents of the AT.TMout array are:
        length,type,gamma5,gamma's

        The space for the vectors in t is at most 14 * number.

        The condition params[5] == 0 corresponds to finding gamma6*gamma7
        during the pick up of the matrices.

*/

WORD Trace4(PHEAD WORD *term, WORD *params, WORD num, WORD level)
{
    GETBIDENTITY
    TRACES *t;
    WORD *p, *m, number, i;
    WORD *OldW;
    WORD j, minimum, minimum2, *min, *stopper;
    OldW = AT.WorkPointer;
    if ( AN.numtracesctack >= AN.intracestack ) {
        number = AN.intracestack + 2;
        t = (TRACES *)Malloc1(number*sizeof(TRACES),"TRACES-struct");
        if ( AN.tracestack ) {
            for ( i = 0; i < AN.intracestack; i++ ) { t[i] = AN.tracestack[i]; }
            M_free(AN.tracestack,"TRACES-struct");
        }
        AN.tracestack = t;
        AN.intracestack = number;
    }

    number = *params - 6;
    if ( number < 0 || ( number & 1 ) || !params[5] ) return(0);

    t = AN.tracestack + AN.numtracesctack;
    AN.numtracesctack++;

    t->finalstep = ( params[2] & 16 ) ? 1 : 0;
    t->gamma5 = params[3];
    if ( t->finalstep && t->gamma5 != GAMMA1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Gamma5 not allowed in this option of the trace command");
        MUNLOCK(ErrorMessageLock);
        AN.numtracesctack--;
        SETERROR(-1)
    }
    t->inlist = AT.WorkPointer;
    t->accup = t->accu = t->inlist + number;
    t->perm = t->accu + (number*2);
    t->eers = t->perm + number;
    if ( ( AT.WorkPointer += 19 * number ) >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    t->num = num;
    t->level = level;
    p = t->inlist;
    m = params+6;
    for ( i = 0; i < number; i++ ) *p++ = *m++;
    t->termp = term;
    t->factor = params[4];
    t->allsign = params[5];
    if ( number >= 10 || ( t->gamma5 != GAMMA1 && number > 4 ) ) {
/*
        The next code should `normal order' the string
        We need the lexicographic smallest string, taking also the
        reverse strings into account.
*/
        minimum = 0; min = t->inlist;
```

```
        stopper = min + number;
        for ( i = 1; i < number; i++ ) {
            p = min;
            m = t->inlist + i;
            for ( j = 0; j < number; j++ ) {
                if ( *p < *m ) break;
                if ( *p > *m ) {
                    min = t->inlist+i;
                    minimum = i;
                    break;
                }
                p++; m++;
                if ( m >= stopper ) m = t->inlist;
                if ( p >= stopper ) p = t->inlist;
            }
        }
        p = min;
        min = m = AT.WorkPointer;
        i = number;
        while ( --i >= 0 ) {
            *m++ = *p++;
            if ( p >= stopper ) p = t->inlist;
        }
        p = t->inlist;
        m = min;
        i = number;
        while ( --i >= 0 ) *p++ = *m++;
        p = t->inlist;
        m = stopper;
        while ( p < m ) {   /* reverse string */
            i = *p; *p++ = *--m; *m = i;
        }
        minimum2 = 0;
        for ( i = 0; i < number; i++ ) {
            p = min;
            m = t->inlist + i;
            for ( j = 0; j < number; j++ ) {
                if ( *p < *m ) break;
                if ( *p > *m ) {
                    m = t->inlist + i;
                    p = min;
                    j = number;
                    while ( --j >= 0 ) {
                        *p++ = *m++;
                        if ( m >= stopper ) m = t->inlist;
                    }
                    minimum2 = i;
                    break;
                }
                p++; m++;
                if ( m >= stopper ) m = t->inlist;
            }
        }
        minimum ^= minimum2;
        if ( ( minimum & 1 ) != 0 ) {
            if ( t->gamma5 == GAMMA5 ) t->allsign = - t->allsign;
            else if ( t->gamma5 != GAMMA1 )
                t->gamma5 = GAMMA6 + GAMMA7 - t->gamma5;
        }
        p = min; m = t->inlist; i = number;
        while ( --i >= 0 ) *m++ = *p++;
/*
        Now the trace is in normal order
*/
    }
    number = Trace4Gen(BHEAD t,number);
    AT.WorkPointer = OldW;
    AN.numtracesctack--;
    return(number);
}

/*
        #] Trace4 :
        #[ Trace4Gen :          WORD Trace4Gen(t,number)

        The recursive breakdown of a trace in 4 dimensions.
        We test first whether the trace has zero or two gamma's left.
        This case can be done quickly.
        Next we test whether we can eliminate adjacent objects that are
        the same.
        Then we test for Chisholm identities (I). First for identities
        with an odd number of gamma matrices (II), then for those with an
        even number of matrices (III). The special thing here is the demand
        that the contraction be between indices with 4 dimensions only.
        Then there is a scan for objects that are the same, not regarding
        their type (IV). This is exactly the same as in n dimensions.
        Finally we have a string left in which all objects are different (V).
```

```
        This case is treated by the routine Trace4no (no stands for no
        objects are the same).

        In case I we have one d_ of which the result of the contraction
        has not yet been fixed.
        Case II gives just a reordering of the matrices and a factor -2.
        Case III gives two terms: one for the anti commutation, such that
        the number of intermediate matrices becomes odd and the other
        from the Chisholm rule for an odd number of matrices. Both have
        a factor 2.
        Case IV gives m+1 terms when m is the number of matrices inbetween.
        We take the shortest path. The sign alternates and all terms have
        a factor two, except for the last one.

*/

WORD Trace4Gen(PHEAD TRACES *t, WORD number)
{
    GETBIDENTITY
    WORD *termout, *stop;
    WORD *p, *m, oldval;
    WORD *pold, *mold, diff, *oldstring, cp;
/*
            #[ Special cases :
*/
    if ( number <= 2 ) {    /* Special cases */
        if ( t->gamma5 == GAMMA5 ) return(0);
        termout = AT.WorkPointer;
        p = t->termp;
        stop = p + *p;
        m = termout;
        p++;
        if ( p < stop ) do {
            if ( *p == SUBEXPRESSION && p[2] == t->num ) {
                oldstring = p;
                p = t->termp;
                do { *m++ = *p++; } while ( p < oldstring );
                p += p[1];
                *m++ = AC.lUniTrace[0];
                *m++ = AC.lUniTrace[1];
                *m++ = AC.lUniTrace[2];
                *m++ = AC.lUniTrace[3];
                if ( number == 2 || t->accup > t->accu ) {
                    oldstring = m;
                    *m++ = DELTA;
                    *m++ = 4;
                    if ( number == 2 ) {
                        *m++ = t->inlist[0];
                        *m++ = t->inlist[1];
                    }
                    if ( t->accup > t->accu ) {
                        pold = p;
                        p = t->accu;
                        while ( p < t->accup ) *m++ = *p++;
                        oldstring[1] = WORDDIF(m,oldstring);
                        p = pold;
                    }
                }
                if ( t->factor ) {
                    *m++ = SNUMBER;
                    *m++ = 4;
                    *m++ = 2;
                    *m++ = t->factor;
                }
                do { *m++ = *p++; } while ( p < stop );
                *termout = WORDDIF(m,termout);
                if ( t->allsign < 0 ) m[-1] = -m[-1];
                if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
                    MLOCK(ErrorMessageLock);
                    MesWork();
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                *AN.RepPoint = 1;
                AR.expchanged = 1;
                if ( *termout ) {
                    *AN.RepPoint = 1;
                    AR.expchanged = 1;
                    if ( Generator(BHEAD termout,t->level) ) goto TracCall;
                }
                AT.WorkPointer= termout;
                return(0);
            }
            p += p[1];
        } while ( p < stop );
        return(0);
    }
```

```
/*
        #] Special cases :
        #[ Adjacent objects :
*/
    p = t->inlist;
    stop = p + number - 1;
    if ( *p == *stop ) {          /* First and last of string */
        oldval = *p;
        *(t->accup)++ = *p;
        *(t->accup)++ = *p;
        m = p+1;
        while ( m < stop ) *p++ = *m++;
        if ( t->gamma5 != GAMMA1 ) {
            if ( t->gamma5 == GAMMA5 ) t->allsign = - t->allsign;
            else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
            else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
        }
        if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
        t = AN.tracestack + AN.numtracesctack - 1;
        if ( t->gamma5 != GAMMA1 ) {
            if ( t->gamma5 == GAMMA5 ) t->allsign = - t->allsign;
            else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
            else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
        }
        while ( p > t->inlist ) *--m = *--p;
        *p = *stop = oldval;
        t->accup -= 2;
        return(0);
    }
    do {
        if ( *p == p[1] ) {       /* Adjacent in string */
            oldval = *p;
            pold = p;
            m = p+2;
            *(t->accup)++ = *p;
            *(t->accup)++ = *p;
            while ( m <= stop ) *p++ = *m++;
            if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
            t = AN.tracestack + AN.numtracesctack - 1;
            while ( p > pold ) *--m = *--p;
            *p++ = oldval;
            *p++ = oldval;
            t->accup -= 2;
            return(0);
        }
        p++;
    } while ( p < stop );
/*
        #] Adjacent objects :
        #[ Odd Contraction :
*/
    p = t->inlist;
    stop = p + number;
    do {
        if ( *p >= AM.OffsetIndex && (
        ( *p < WILDOFFSET + AM.OffsetIndex &&
         indices[*p-AM.OffsetIndex].dimension == 4 )
        || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim == 4 ) ) ) {
            m = p+2;
            while ( m < stop ) {
                if ( *p == *m ) {
                    pold = p;
                    mold = m;
                    oldval = *p;
                    (t->factor)++;
                    t->allsign = - t->allsign;
                    *p++ = *--m;
                    m--;
                    while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
                    p = mold - 1;
                    m = mold + 1;
                    while ( m < stop ) *p++ = *m++;
                    if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
                    t = AN.tracestack + AN.numtracesctack - 1;
                    m--;
                    while ( m > mold ) *m-- = *--p;
                    p = pold;
                    *m-- = oldval;
                    *m-- = *p;
                    *p++ = oldval;
                    while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
                    t->allsign = - t->allsign;
                    (t->factor)--;
                    return(0);
                }
                m += 2;
            }
```

```
        }
        p++;
    } while ( p < stop );
/*
            #] Odd Contraction :
            #[ Even Contraction :
        First the case with two matrices inbetween.
*/
    p = t->inlist;
    stop = p + number;
    do {
        if ( *p >= AM.OffsetIndex && (
        ( *p < WILDOFFSET + AM.OffsetIndex &&
         indices[*p-AM.OffsetIndex].dimension == 4 )
        || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim == 4 ) ) ) {
            m = p+3;
            if ( m >= stop ) m -= number;
            if ( *p == *m ) {
                WORD oldfactor, old5;
                oldstring = AT.WorkPointer;
                AT.WorkPointer += number;
                oldfactor = t->allsign;
                old5 = t->gamma5;
                if ( m < p ) cp = (WORDDIF(m,t->inlist) + 1 ) & 1;
                else cp = 0;
                if ( cp && ( t->gamma5 != GAMMA1 ) ) {
                    if ( t->gamma5 == GAMMA5 ) t->allsign = -t->allsign;
                    else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
                    else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
                }
                mold = m;
                p = oldstring;
                m = t->inlist;
                while ( m < stop ) *p++ = *m++;
/*
        Rotate the string
*/
                m = mold + 1;
                p = t->inlist;
                while ( m < stop ) *p++ = *m++;
                m = oldstring;
                if ( !cp && ((WORDDIF(stop,p))&1) != 0 && ( t->gamma5 != GAMMA1 ) ) {
                    if ( t->gamma5 == GAMMA5 ) t->allsign = -t->allsign;
                    else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
                    else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
                }
                while ( p < stop ) *p++ = *m++;
                t->factor += 2;
                m = p - 3;
                p = t->inlist;
                oldval = number - 4;
                while ( oldval > 0 ) {
                    if ( *p >= AM.OffsetIndex && (
                    ( *p < WILDOFFSET + AM.OffsetIndex &&
                     indices[*p-AM.OffsetIndex].dimension )
                    || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim ) ) ) {
                        if ( *p == *m ) {
                            *p = m[1];
                            break;
                        }
                        else if ( *p == m[1] ) {
                            *p = *m;
                            break;
                        }
                    }
                    p++; oldval--;
                }
                if ( oldval <= 0 ) {
                    *(t->accup)++ = *m++;
                    *(t->accup)++ = *m++;
                }
                if ( Trace4Gen(BHEAD t,number-4) ) goto TracCall;
                t = AN.tracestack + AN.numtracesctack - 1;
                t->factor -= 2;
                if ( oldval <= 0 ) t->accup -= 2;
                t->gamma5 = old5;
                t->allsign = oldfactor;
                AT.WorkPointer = p = oldstring;
                m = t->inlist;
                while ( m < stop ) *m++ = *p++;
                return(0);
            }
        }
        p++;
    } while ( p < stop );
/*
        The case with at least 4 matrices inbetween.
```

```
*/
    p = t->inlist;
    stop = p + number;
    do {
        if ( *p >= AM.OffsetIndex && (
        ( *p < WILDOFFSET + AM.OffsetIndex &&
         indices[*p-AM.OffsetIndex].dimension == 4 )
        || ( *p >= WILDOFFSET + AM.OffsetIndex && AC.lDefDim == 4 ) ) ) {
            m = p+5;
            while ( m < stop ) {
                if ( *p == *m ) {
                    WORD *pex, *mex;
                    pold = p;
                    mold = m;
                    oldval = *p;
/*
            g_(1,mu)*g_(1,a1)*...*g_(1,aj)*g_(1,an)*g_(1,mu) ->
            first:
            2*g_(1,an)*g_(1,a1)*...*g_(1,aj)
*/
                    (t->factor)++;
/*
                    The variable hulp seems unnecessary
                    *p = hulp = m[-1];
*/
                    *p = m[-1];
                    p = m - 1;
                    m++;
                    while ( m < stop ) *p++ = *m++;
                    if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
                    t = AN.tracestack + AN.numtracesctack - 1;
                    pex = p; mex = m;
                    p = pold;
                    m = mold - 2;
                    while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
/*
            and then:
            2*g_(1,aj)*...*g_(1,a1)*g_(1,an)
*/
                    if ( Trace4Gen(BHEAD t,number-2) ) goto TracCall;
                    t = AN.tracestack + AN.numtracesctack - 1;
                    p = pold;
                    m = mold - 2;
                    while ( m > p ) { diff = *p; *p++ = *m; *m-- = diff; }
                    m = mex;
                    p = pex;
                    m--;
                    while ( m > mold ) *m-- = *--p;
                    m = mold;
                    *m-- = oldval;
                    p = pold;
                    *m = *p;
                    *p = oldval;
                    (t->factor)--;
                    return(0);
                }
                m += 2;
            }
        }
        p++;
    } while ( p < stop );
/*
            #] Even Contraction :
            #[ Same Objects :
*/
    p = t->inlist;
    stop = p + number - 1;
    diff = 2;
    do {
        p = t->inlist;
        while ( p <= stop ) {
            m = p + diff;
            if ( m > stop ) m -= number;
            if ( *p == *m ) {
                WORD oldfactor, c, old5;
                oldfactor = t->allsign;
                old5 = t->gamma5;
                cp = (WORDDIF(m,t->inlist)) & 1;
                if ( !cp && ( t->gamma5 != GAMMA1 ) ) {
                    if ( t->gamma5 == GAMMA5 ) t->allsign = -t->allsign;
                    else if ( t->gamma5 == GAMMA6 ) t->gamma5 = GAMMA7;
                    else if ( t->gamma5 == GAMMA7 ) t->gamma5 = GAMMA6;
                }
                oldstring = AT.WorkPointer;
                AT.WorkPointer += number;
                mold = m;
                oldval = *p;
```

```
                p = oldstring;
                m = t->inlist;
                while ( m <= stop ) *p++ = *m++;
/*
        Rotate the string
*/
                m = mold + 1;
                p = t->inlist;
                while ( m <= stop ) *p++ = *m++;
                m = oldstring;
                while ( p <= stop ) *p++ = *m++;
                (t->factor)++;
                p -= diff + 1;
                m = stop;
                *(t->accup) = oldval;
                t->accup += 2;
                m--;
                while ( m > p ) {
                    c = t->accup[-1];
                    t->accup[-1] = *m;
                    *m = c;
                    if ( Trace4Gen(BHEAD t,number-2) ) goto Trac4Call;
                    t = AN.tracestack + AN.numtracesctack - 1;
                    m--;
                    t->allsign = - t->allsign;
                }
                c = t->accup[-1];
                t->accup[-1] = *m;
                *m = c;
                (t->factor)--;
                if ( Trace4Gen(BHEAD t,number-2) ) goto Trac4Call;
                t = AN.tracestack + AN.numtracesctack - 1;
                t->accup -= 2;
                t->allsign = oldfactor;
                AT.WorkPointer = p = oldstring;
                m = t->inlist;
                while ( m <= stop ) *m++ = *p++;
                t->gamma5 = old5;
                return(0);
            }
            p++;
        }
    } while ( ++diff <= (number»1) );
/*
            #] Same Objects :
            #[ All Different :

        Here we have a string with all different objects.

*/
    t->sgn = 0;
    termout = AT.WorkPointer;
    for(;;) {
        if ( t->finalstep == 0 ) diff = Trace4no(number,t->accup,t);
        else                     diff = TraceNno(number,t->accup,t);
/*  while ( ( diff = Trace4no(number,t->accup,t) ) != 0 ) */
        if ( diff == 0 ) break;
        p = t->termp;
        stop = p + *p;
        m = termout;
        p++;
        if ( p < stop ) do {
            if ( *p == SUBEXPRESSION && p[2] == t->num ) {
                oldstring = p;
                p = t->termp;
                do { *m++ = *p++; } while ( p < oldstring );
                p += p[1];
                pold = p;
                *m++ = AC.lUniTrace[0];
                *m++ = AC.lUniTrace[1];
                *m++ = AC.lUniTrace[2];
                *m++ = AC.lUniTrace[3];
                *m++ = SNUMBER;
                *m++ = 4;
                *m++ = 2;
                *m++ = t->factor;
                p = t->accup;
                oldval = number;
                if ( diff == 2 || diff == -2 ) {
                    *m++ = LEVICIVITA;
                    *m++ = 4+FUNHEAD;
                    *m++ = DIRTYFLAG;
                    FILLFUN3(m)
                    *m++ = *p++; *m++ = *p++; *m++ = *p++; *m++ = *p++;
                    oldval -= 4;
                }
                if ( oldval > 0 || t->accup > t->accu ) {
```

```
                    oldstring = m;
                    *m++ = DELTA;
                    *m++ = oldval + 2;
                    if ( oldval > 0 ) NCOPY(m,p,oldval);
                    if ( t->accup > t->accu ) {
                        p = t->accu;
                        while ( p < t->accup ) *m++ = *p++;
                        oldstring[1] = WORDDIF(m,oldstring);
                    }
                }
                p = pold;
                do { *m++ = *p++; } while ( p < stop );
                *termout = WORDDIF(m,termout);
                if ( ( diff ^ t->allsign ) < 0 ) m[-1] = - m[-1];
                if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
                    MLOCK(ErrorMessageLock);
                    MesWork();
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                if ( *termout ) {
                    *AN.RepPoint = 1;
                    AR.expchanged = 1;
                    if ( Generator(BHEAD termout,t->level) ) {
                        AT.WorkPointer = termout;
                        goto TracCall;
                    }
                    t = AN.tracestack + AN.numtracesctack - 1;
                }
                break;
            }
            p += p[1];
        } while ( p < stop );
    }
    AT.WorkPointer = termout;
    return(0);

/*
            #] All Different :
*/
Trac4Call:
    AT.WorkPointer = oldstring;
TracCall:
    if ( AM.tracebackflag ) {
        MLOCK(ErrorMessageLock);
        MesCall("Trace4Gen");
        MUNLOCK(ErrorMessageLock);
    }
    return(-1);
}

/*
        #] Trace4Gen :
        #[ TraceNno :           WORD TraceNno(number,kron,t)

        Routine takes the trace in N dimensions of a string
        of gamma matrices. It is assumed that there are no
        contractions and no vectors that are the same. For
        the treatment of those cases there are special routines,
        that call this routine as a final stage.
        The calling routine must reserve 'number' WORDs for perm
        and kron each.
        kron and perm may not be altered during the generation!

*/

WORD TraceNno(WORD number, WORD *kron, TRACES *t)
{
    WORD i, j, a, *p;
    if ( !t->sgn ) {
        if ( !number || ( number & 1 ) ) return(0);
        p = t->inlist;
        for ( i = 0; i < number; i++ ) {
            t->perm[i] = i;
            *kron++ = *p++;
        }
        t->sgn = 1;
        return(1);
    }
    else {
        i = number - 3;
        while ( i > 0 ) {
            a = kron[i];
            p = t->perm + i;
            for ( j = i + 1; j <= *p; j++ ) kron[j-1] = kron[j];
            kron[(*p)++] = a;
            if ( *p < number ) {
```

```
                a = kron[*p];
                j = *p;
                while ( j >= (i+1) ) { kron[j] = kron[j-1]; j--; }
                kron[i] = a;
                number -= 2;
                for ( j = i+2; j < number; j += 2 ) t->perm[j] = j;
                t->sgn = - t->sgn;
                return(t->sgn);
            }
            i -= 2;
        }
    }
    return(0);
}

/*
        #] TraceNno :
        #[ TraceN :             WORD TraceN(term,params,num,level)
*/

WORD TraceN(PHEAD WORD *term, WORD *params, WORD num, WORD level)
{
    GETBIDENTITY
    TRACES *t;
    WORD *p, *m, number, i;
    WORD *OldW;
    if ( params[3] != GAMMA1 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Gamma5 not allowed in n-trace");
        MUNLOCK(ErrorMessageLock);
        SETERROR(-1)
    }
    OldW = AT.WorkPointer;
    if ( AN.numtracesctack >= AN.intracestack ) {
        number = AN.intracestack + 2;
        t = (TRACES *)Malloc1(number*sizeof(TRACES),"TRACES-struct");
        if ( AN.tracestack ) {
            for ( i = 0; i < AN.intracestack; i++ ) { t[i] = AN.tracestack[i]; }
            M_free(AN.tracestack,"TRACES-struct");
        }
        AN.tracestack = t;
        AN.intracestack = number;
    }
    number = *params - 6;
    if ( number < 0 || ( number & 1 ) || !params[5] ) return(0);

    t = AN.tracestack + AN.numtracesctack;
    AN.numtracesctack++;

    t->inlist = AT.WorkPointer;
    t->accup = t->accu = t->inlist + number;
    t->perm = t->accu + number;
    if ( ( AT.WorkPointer += 3 * number ) >= AT.WorkTop ) {
        AN.numtracesctack--;
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    t->num = num;
    t->level = level;
    p = t->inlist;
    m = params+6;
    for ( i = 0; i < number; i++ ) *p++ = *m++;
    t->termp = term;
    t->factor = params[4];
    t->allsign = params[5];
    number = TraceNgen(BHEAD t,number);
    AT.WorkPointer = OldW;
    AN.numtracesctack--;
    return(number);
}

/*
        #] TraceN :
        #[ TraceNgen :          WORD TraceNgen(t,number)

        This routine is a simplified version of Trace4Gen. We know here
        only three cases: Adjacent objects, same objects and all different.
        The other difference lies of course in the struct which is now
        not of type TRACES, but of type TRACES.

*/

WORD TraceNgen(PHEAD TRACES *t, WORD number)
{
    GETBIDENTITY
```

```
    WORD *termout, *stop;
    WORD *p, *m, oldval;
    WORD *pold, *mold, diff, *oldstring;
/*
            #[ Special cases :
*/
    if ( number <= 2 ) {     /* Special cases */
        termout = AT.WorkPointer;
        p = t->termp;
        stop = p + *p;
        m = termout;
        p++;
        if ( p < stop ) do {
            if ( *p == SUBEXPRESSION && p[2] == t->num ) {
                oldstring = p;
                p = t->termp;
                do { *m++ = *p++; } while ( p < oldstring );
                p += p[1];
                *m++ = AC.lUniTrace[0];
                *m++ = AC.lUniTrace[1];
                *m++ = AC.lUniTrace[2];
                *m++ = AC.lUniTrace[3];
                if ( number == 2 || t->accup > t->accu ) {
                    oldstring = m;
                    *m++ = DELTA;
                    *m++ = 4;
                    if ( number == 2 ) {
                        *m++ = t->inlist[0];
                        *m++ = t->inlist[1];
                    }
                    if ( t->accup > t->accu ) {
                        pold = p;
                        p = t->accu;
                        while ( p < t->accup ) *m++ = *p++;
                        oldstring[1] = WORDDIF(m,oldstring);
                        p = pold;
                    }
                }
                if ( t->factor ) {
                    *m++ = SNUMBER;
                    *m++ = 4;
                    *m++ = 2;
                    *m++ = t->factor;
                }
                do { *m++ = *p++; } while ( p < stop );
                *termout = WORDDIF(m,termout);
                if ( t->allsign < 0 ) m[-1] = -m[-1];
                if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
                    MLOCK(ErrorMessageLock);
                    MesWork();
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                if ( *termout ) {
                    *AN.RepPoint = 1;
                    AR.expchanged = 1;
                    if ( Generator(BHEAD termout,t->level) ) goto TracCall;
                }
                AT.WorkPointer= termout;
                return(0);
            }
            p += p[1];
        } while ( p < stop );
        return(0);
    }
/*
            #] Special cases :
            #[ Adjacent objects :
*/
    p = t->inlist;
    stop = p + number - 1;
    if ( *p == *stop ) {           /* First and last of string */
        oldval = *p;
        *(t->accup)++ = *p;
        *(t->accup)++ = *p;
        m = p+1;
        while ( m < stop ) *p++ = *m++;
        if ( TraceNgen(BHEAD t,number-2) ) goto TracCall;
        t = AN.tracestack + AN.numtracesctack - 1;
        while ( p > t->inlist ) *--m = *--p;
        *p = *stop = oldval;
        t->accup -= 2;
        return(0);
    }
    do {
        if ( *p == p[1] ) {      /* Adjacent in string */
            oldval = *p;
```

```
            pold = p;
            m = p+2;
            *(t->accup)++ = *p;
            *(t->accup)++ = *p;
            while ( m <= stop ) *p++ = *m++;
            if ( TraceNgen(BHEAD t,number-2) ) goto TracCall;
            t = AN.tracestack + AN.numtracesctack - 1;
            while ( p > pold ) *--m = *--p;
            *p++ = oldval;
            *p++ = oldval;
            t->accup -= 2;
            return(0);
        }
        p++;
    } while ( p < stop );
/*
            #] Adjacent objects :
            #[ Same Objects :
*/
    p = t->inlist;
    stop = p + number - 1;
    diff = 2;
    do {
        p = t->inlist;
        while ( p <= stop ) {
            m = p + diff;
            if ( m > stop ) m -= number;
            if ( *p == *m ) {
                WORD oldfactor, c;
                oldstring = AT.WorkPointer;
                AT.WorkPointer += number;
                mold = m;
                oldval = *p;
                p = oldstring;
                m = t->inlist;
                while ( m <= stop ) *p++ = *m++;
/*
        Rotate the string
*/
                {
                    m = mold + 1;
                    p = t->inlist;
                    while ( m <= stop ) *p++ = *m++;
                    m = oldstring;
                    while ( p <= stop ) *p++ = *m++;
                }
                oldfactor = t->allsign;
                (t->factor)++;
                p -= diff + 1;
                m = stop;
                if ( oldval >= ( AM.OffsetIndex + WILDOFFSET ) ||
                    ( oldval >= AM.OffsetIndex
                    && indices[oldval-AM.OffsetIndex].dimension ) ) {
                    m--;
/*
        We distinguish 4 cases:
        m-p=1   Use g_(1,mu,a,mu)  = (2-d_(mu,mu))*g_(1,a)
        m-p=2   Use g_(1,mu,a,b,mu) = 4*d_(a,b)+(d_(mu,mu)-4)*g_(1,a,b)
        m-p=3   Use g_(1,mu,a,b,c,mu) = -2*g_(1,c,b,a)
                        -(d_(mu,mu)-4)*g_(1,a,b,c)
        m-p>3   Reduce down to m-p=3 with the old technique
*/
                    while ( m > (p+3) ) {
                        c = *p;
                        *p = *m;
                        *m = c;
                        if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                        t = AN.tracestack + AN.numtracesctack - 1;
                        m--;
                        t->allsign = - t->allsign;
                    }
                    switch ( WORDDIF(m,p) ) {
                        case 1:
                            c = *p;
                            *p = *m;
                            *m = c;
                            if ( oldval < ( AM.OffsetIndex + WILDOFFSET )
                            && indices[oldval-AM.OffsetIndex].nmin4
                            != -NMIN4SHIFT ) {
                                t->allsign = - t->allsign;
                                if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                                t = AN.tracestack + AN.numtracesctack - 1;
                                (t->factor)--;
                                *(t->accup)++ = SUMMEDIND;
                                *(t->accup)++ =
                                    indices[oldval-AM.OffsetIndex].nmin4;
                            }
```

```
                        else
                        {
                            if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                            t = AN.tracestack + AN.numtracesctack - 1;
                            t->allsign = - t->allsign;
                            (t->factor)--;
                            *(t->accup)++ = oldval;
                            *(t->accup)++ = oldval;
                        }
                        if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                        t = AN.tracestack + AN.numtracesctack - 1;
                        t->accup -= 2;
                        break;
                    case 2:
                        { WORD one, two;
                        one = *p = p[1];
                        two = p[1] = *m;
                        (t->factor)++;            /* 4 */
                        *(t->accup)++ = *p;       /* d_(a,b) */
                        *(t->accup)++ = *m;
                        if ( TraceNgen(BHEAD t,number-4) ) goto TracnCall;
                        t = AN.tracestack + AN.numtracesctack - 1;
                        *p = one; p[1] = two;
                        t->accup -= 2;
                        if ( oldval < ( AM.OffsetIndex + WILDOFFSET )
                        && indices[oldval-AM.OffsetIndex].nmin4
                        != -NMIN4SHIFT ) {
                            t->factor -= 2;
                            *(t->accup)++ = SUMMEDIND;
                            *(t->accup)++ =
                                indices[oldval-AM.OffsetIndex].nmin4;
                        }
                        else {
                            t->allsign = - t->allsign;
                            if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                            t = AN.tracestack + AN.numtracesctack - 1;
                            t->allsign = - t->allsign;
                            t->factor -= 2;
                            *(t->accup)++ = oldval;
                            *(t->accup)++ = oldval;
                        }
                        if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                        t = AN.tracestack + AN.numtracesctack - 1;
                        t->accup -= 2;
                        }
                        break;
                    default:
                        c = *p;
                        *p = *m;
                        *m = c;
                        c = m[-1]; m[-1] = m[-2]; m[-2] = c;
                        t->allsign = - t->allsign;
                        if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                        t = AN.tracestack + AN.numtracesctack - 1;
                        m--;
                        c = *p;
                        *p = *m;
                        *m = c;
                        (t->factor)--;
                        if ( oldval < ( AM.OffsetIndex + WILDOFFSET )
                        && indices[oldval-AM.OffsetIndex].nmin4
                        != -NMIN4SHIFT ) {
                            *(t->accup)++ = SUMMEDIND;
                            *(t->accup)++ =
                                indices[oldval-AM.OffsetIndex].nmin4;
                            if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                            t = AN.tracestack + AN.numtracesctack - 1;
                            t->accup -= 2;
                            t->allsign = - t->allsign;
                        }
                        else
                        {
                            *(t->accup)++ = oldval;
                            *(t->accup)++ = oldval;
                            if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                            t = AN.tracestack + AN.numtracesctack - 1;
                            t->accup -= 2;
                            t->allsign = - t->allsign;
                            t->factor += 2;
                            if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                            t = AN.tracestack + AN.numtracesctack - 1;
                            t->factor -= 2;
                        }
                        break;
                }
            }
            else {
```

```
                    *(t->accup) = oldval;
                    t->accup += 2;
                    m--;
                    while ( m > p ) {
                        c = t->accup[-1];
                        t->accup[-1] = *m;
                        *m = c;
                        if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                        t = AN.tracestack + AN.numtracesctack - 1;
                        m--;
                        t->allsign = - t->allsign;
                    }
                    c = t->accup[-1];
                    t->accup[-1] = *m;
                    *m = c;
                    (t->factor)--;
                    if ( TraceNgen(BHEAD t,number-2) ) goto TracnCall;
                    t = AN.tracestack + AN.numtracesctack - 1;
                    t->accup -= 2;
                }
                t->allsign = oldfactor;
                p = oldstring;
                m = t->inlist;
                while ( m <= stop ) *m++ = *p++;
                AT.WorkPointer = oldstring;
                return(0);
            }
            p++;
        }
        diff++;
    } while ( diff <= (number»1) );
/*
            #] Same Objects :
            #[ All Different :

        Here we have a string with all different objects.

*/
    t->sgn = 0;
    termout = AT.WorkPointer;
    while ( ( diff = TraceNno(number,t->accup,t) ) != 0 ) {
        p = t->termp;
        stop = p + *p;
        m = termout;
        p++;
        if ( p < stop ) do {
            if ( *p == SUBEXPRESSION && p[2] == t->num ) {
                oldstring = p;
                p = t->termp;
                do { *m++ = *p++; } while ( p < oldstring );
                p += p[1];
                pold = p;
                *m++ = AC.lUniTrace[0];
                *m++ = AC.lUniTrace[1];
                *m++ = AC.lUniTrace[2];
                *m++ = AC.lUniTrace[3];
                *m++ = SNUMBER;
                *m++ = 4;
                *m++ = 2;
                *m++ = t->factor;
                p = t->accup;
                oldval = number;
                oldstring = m;
                *m++ = DELTA;
                *m++ = oldval + 2;
                NCOPY(m,p,oldval);
                if ( t->accup > t->accu ) {
                    p = t->accu;
                    while ( p < t->accup ) *m++ = *p++;
                    oldstring[1] = WORDDIF(m,oldstring);
                }
                p = pold;
                do { *m++ = *p++; } while ( p < stop );
                *termout = WORDDIF(m,termout);
                if ( ( diff ^ t->allsign ) < 0 ) m[-1] = - m[-1];
                if ( ( AT.WorkPointer = m ) > AT.WorkTop ) {
                    MLOCK(ErrorMessageLock);
                    MesWork();
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                if ( *termout ) {
                    *AN.RepPoint = 1;
                    AR.expchanged = 1;
                    if ( Generator(BHEAD termout,t->level) ) {
                        AT.WorkPointer = termout;
                        goto TracCall;
```

```
                    }
                    t = AN.tracestack + AN.numtracesctack - 1;
                }
                break;
            }
            p += p[1];
        } while ( p < stop );
    }
    AT.WorkPointer = termout;
    return(0);

/*
            #] All Different :
*/
TracnCall:
    AT.WorkPointer = oldstring;
TracCall:
    if ( AM.tracebackflag ) {
        MLOCK(ErrorMessageLock);
        MesCall("TraceNGen");
        MUNLOCK(ErrorMessageLock);
    }
    return(-1);
}

/*
        #] TraceNgen :
        #[ Traces :             WORD Traces(term,params,num,level)

        The contents of the AT.TMout array are:
        length,type,subtype,gamma5,factor,sign,gamma's

*/

WORD Traces(PHEAD WORD *term, WORD *params, WORD num, WORD level)
{
    GETBIDENTITY
    switch ( AT.TMout[2] ) {    /* Subtype gives dimension */
        case 0:
            return(TraceN(BHEAD term,params,num,level));
        case 4:
            return(Trace4(BHEAD term,params,num,level));
        case 12:
            return(Trace4(BHEAD term,params,num,level));
        case 20:
            return(Trace4(BHEAD term,params,num,level));
        default:
            return(0);
    }
}

/*
        #] Traces :
        #[ TraceFind :          WORD TraceFind(term,params)
*/

WORD TraceFind(PHEAD WORD *term, WORD *params)
{
    GETBIDENTITY
    WORD *p, *m, *to;
    WORD *termout, *stop, *stop2, number = 0;
    WORD first = 1;
    WORD type, spinline, sp;
    type = params[3];
    spinline = params[4];
    if ( spinline < 0 ) {   /* $ variable. Evaluate */
        sp = DolToIndex(BHEAD -spinline);
        if ( AN.ErrorInDollar || sp < 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("$%s does not have an index value in trace statement in module %l",
                DOLLARNAME(Dollars,-spinline),AC.CModule);
            MUNLOCK(ErrorMessageLock);
            return(0);
        }
        spinline = sp;
    }
    to = AT.TMout;
    to++;
    *to++ = TAKETRACE;
    *to++ = type;
    *to++ = GAMMA1;
    *to++ = 0;              /* Powers of two */
    *to++ = 1;              /* sign */
    p = term;
    m = p + *p - 1;
    stop = m - ABS(*m);
    termout = m = AT.WorkPointer;
```

```
    m++;
    p++;
    while ( p < stop ) {
        stop2 = p + p[1];
        if ( *p == GAMMA && p[FUNHEAD] == spinline ) {
            if ( first ) {
                *m++ = SUBEXPRESSION;
                *m++ = SUBEXPSIZE;
                *m++ = -1;
                *m++ = 1;
                *m++ = DUMMYBUFFER;
                FILLSUB(m)
                first = 0;
            }
            p += FUNHEAD+1;
            while ( p < stop2 ) {
                if ( *p == GAMMA5 ) {
                    if ( AT.TMout[3] == GAMMA5 ) AT.TMout[3] = GAMMA1;
                    else if ( AT.TMout[3] == GAMMA1 ) AT.TMout[3] = GAMMA5;
                    else if ( AT.TMout[3] == GAMMA7 ) AT.TMout[5] = -AT.TMout[5];
                    if ( number & 1 ) AT.TMout[5] = - AT.TMout[5];
                    p++;
                }
                else if ( *p == GAMMA6 ) {
                    if ( number & 1 ) goto F7;
F6:
                    if ( AT.TMout[3] == GAMMA6 ) (AT.TMout[4])++;
                    else if ( AT.TMout[3] == GAMMA1 ) AT.TMout[3] = GAMMA6;
                    else if ( AT.TMout[3] == GAMMA5 ) AT.TMout[3] = GAMMA6;
                    else if ( AT.TMout[3] == GAMMA7 ) AT.TMout[5] = 0;
                    p++;
                }
                else if ( *p == GAMMA7 ) {
                    if ( number & 1 ) goto F6;
F7:
                    if ( AT.TMout[3] == GAMMA7 ) (AT.TMout[4])++;
                    else if ( AT.TMout[3] == GAMMA1 ) AT.TMout[3] = GAMMA7;
                    else if ( AT.TMout[3] == GAMMA5 ) {
                        AT.TMout[3] = GAMMA7;
                        AT.TMout[5] = -AT.TMout[5];
                    }
                    else if ( AT.TMout[3] == GAMMA6 ) AT.TMout[5] = 0;
                    p++;
                }
                else {
                    *to++ = *p++;
                    number++;
                }
            }
        }
        else {
            while ( p < stop2 ) *m++ = *p++;
        }
    }
    if ( first ) return(0);
    AT.TMout[0] = WORDDIF(to,AT.TMout);
    to = term;
    to += *to;
    while ( p < to ) *m++ = *p++;
    *termout = WORDDIF(m,termout);
    to = term;
    p = termout;
    do { *to++ = *p++; } while ( p < m );
    AT.WorkPointer = term + *term;
    return(1);
}

/*
        #] TraceFind :
        #[ Chisholm :           WORD Chisholm(term,level,num)

        Routines for reorganizing traces.
        The command
            Chisholm,1;
        will collect the gamma matrices in spinline 1 and see whether
        they have an index in common with another gamma matrix. If this
        is the case the identity
            g_(2,mu)*Tr[g_(1,mu)*S(2)] = S(2)+SR(2)
        is applied (SR is the reversed string).
*/

WORD Chisholm(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD *t, *r, *m, *s, *tt, *rr;
    WORD *mat, *matpoint, *termout, *rdo;
    CBUF *C = cbuf+AM.rbufnum;
    WORD i, j, num = C->lhs[level][2], gam5;
    WORD norm = 0, k, *matp;
```

```
/*
    #[ Find : Find possible matrices
*/
    mat = matpoint = AT.WorkPointer;
    t = term;
    r = t + *t - 1; r -= ABS(*r);
    t++;
    i = 0;
    gam5 = GAMMA1;
    while ( t < r ) {
        if ( *t == GAMMA && t[FUNHEAD] == num ) {
            m = t + t[1];
            t += FUNHEAD+1;
            while ( t < m ) {
                if ( *t >= 0 || *t < MINSPEC ) i++;
                else {
                    if ( gam5 == GAMMA1 ) gam5 = *t;
                    else if ( gam5 == GAMMA5 ) {
                        if ( *t == GAMMA5 ) gam5 = GAMMA1;
                        else if ( *t != GAMMA1 ) gam5 = *t;
                    }
                }
                *matpoint++ = *t++;
            }
        }
        else t += t[1];
    }
    if ( ( i & 1 ) != 0 ) return(0);     /* odd trace */
/*
    #] Find :
    #[ Test : Test for contracted index

    This code should be modified.

    We have to check for all possible matches if C->lhs[level][3] == 1
    and the trace contains a gamma5, gamma6 or gamma7.
    Then we normalize by the number of possible contractions (norm) and
    do all of them. This way the Levi-Civita tensors have a maximum
    chance of cancelling each other. This option is activated with
    `contract' and `symmetrize'. Defaults are that they are on, but
    they can be switched off with nocontract and nosymmetrize.
*/
    s = mat;
    while ( s < matpoint ) {
/*
        if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
        indices[*s-AM.OffsetIndex].dimension == 0 ) ) {
*/
        if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
        indices[*s-AM.OffsetIndex].dimension != 4 )
        || ( ( AC.lDefDim != 4 ) && ( *s >= ( AM.OffsetIndex + WILDOFFSET ) ) ) ) {
            s++; continue;
        }
        t = term+1;
        while ( t < r ) {
            if ( *t == GAMMA && t[FUNHEAD] != num ) {
                m = t + t[1];
                t += FUNHEAD+1;
                while ( t < m ) {
                    if ( *t == *s ) {
                        norm++;
                    }
                    t++;
                }
            }
            else t += t[1];
        }
        s++;
    }
    if ( norm == 0 ) return(Generator(BHEAD term,level));  /* No Action */
/*
    #] Test :
    #[ Do : Process the string

    tt: The subterm
    t:  The matrix
    s:  The matrix in the relevant string

    Cycle the string in mat so that s is at the end.
    Copy the part till the critical GAMMA.
    Copy inside the critical string, copy S, copy tail inside string.
    Important to remember where S is so that we can reverse it later.
    Add term UnitTrace/2/norm.
    Copy rest of term.
    Continue execution with S.
    Reverse S.
    Continue execution with SR.
```

```
*/

    if ( C->lhs[level][3] == 0 /* || gam5 == GAMMA1 */ ) norm = 1;

    matp = matpoint;
    for ( k = 0; k < norm; k++ ) {
        matpoint = matp;
        s = mat;
        while ( s < matpoint ) {
/*
            if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
            indices[*s-AM.OffsetIndex].dimension == 0 ) ) {
*/
            if ( *s < AM.OffsetIndex || ( *s < ( AM.OffsetIndex + WILDOFFSET ) &&
            indices[*s-AM.OffsetIndex].dimension != 4 ) ) {
                s++; continue;
            }
            t = term+1;
            while ( t < r ) {
                if ( *t == GAMMA && t[FUNHEAD] != num ) {
                    tt = t;
                    m = t + t[1];
                    t += FUNHEAD+1;
                    while ( t < m ) {
                        if ( *t == *s ) {
                            i = WORDDIF(t,tt);
                            m = mat;
                            while ( m <= s ) *matpoint++ = *m++;
                            t = mat;
                            while ( m < matpoint ) *t++ = *m++;
                            termout = t;
                            m = termout + 1;
                            t = term + 1;
                            while ( t < tt ) {
                                if ( *t != GAMMA || t[FUNHEAD] != num ) {
                                    j = t[1];
                                    NCOPY(m,t,j);
                                }
                                else t += t[1];
                            }

                            tt += tt[1];
                            rdo = m;
                            j = i;
                            while ( --j >= 0 ) *m++ = *t++;
                            matpoint = m;
                            s = mat;
                            while ( s < termout ) *m++ = *s++;
                            m--;
                            t++;
                            while ( t < tt ) *m++ = *t++;
                            rdo[1] = WORDDIF(m,rdo);

                            *m++ = AC.lUniTrace[0];
                            *m++ = AC.lUniTrace[1];
                            *m++ = AC.lUniTrace[2];
                            *m++ = AC.lUniTrace[3];
                            *m++ = SNUMBER;
                            *m++ = 4;
                            *m++ = 2*norm;
                            *m++ = -1;

                            while ( t < r ) {
                                if ( *t != GAMMA || t[FUNHEAD] != num ) {
                                    j = t[1];
                                    NCOPY(m,t,j);
                                }
                                else t += t[1];
                            }
                            rr = term + *term;
                            while ( t < rr ) *m++ = *t++;

                            *termout = WORDDIF(m,termout);
                            rr = m;
                            t = termout;
                            j = *termout;
                            NCOPY(m,t,j);
                            AT.WorkPointer = m;
                            if ( Generator(BHEAD t,level) ) goto ChisCall;

                            j = WORDDIF(termout,mat)-1;
                            t = matpoint;
                            m = t + j;
                            AT.WorkPointer = rr;
                            while ( m > t ) {
                                i = *--m; *m = *t; *t++ = i;
                            }
```

```
                            if ( Generator(BHEAD termout,level) ) goto ChisCall;
                            AT.WorkPointer = mat;

                            goto NextK;
                        }
                        t++;
                    }
                }
                else t += t[1];
            }
            s++;
        }
NextK:;
    }
    return(0);
/*
    #] Do :
*/
ChisCall:
    if ( AM.tracebackflag ) {
        MLOCK(ErrorMessageLock);
        MesCall("Chisholm");
        MUNLOCK(ErrorMessageLock);
    }
    return(-1);
}

/*
        #] Chisholm :
        #[ TenVecFind :         WORD TenVecFind(term,params)
*/

WORD TenVecFind(PHEAD WORD *term, WORD *params)
{
    GETBIDENTITY
    WORD *t, *w, *m, *tstop;
    WORD i, mode, thevector, thetensor, spectator;
    thetensor = params[3];
    thevector = params[4];
    mode = params[5];
    if ( thetensor < 0 ) {  /* $-expression */
        thetensor = DolToTensor(BHEAD -thetensor);
        if ( thetensor < FUNCTION ) {
            if ( thevector > 0 ) {
                thetensor = DolToTensor(BHEAD thevector);
                if ( thetensor < FUNCTION ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("$%s should have been a tensor in module %l"
                        ,DOLLARNAME(Dollars,params[4]),AC.CModule);
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                thevector = DolToVector(BHEAD -params[3]);
                if ( thevector >= 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("$%s should have been a vector in module %l"
                        ,DOLLARNAME(Dollars,-params[3]),AC.CModule);
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
            }
            else {
                MLOCK(ErrorMessageLock);
                MesPrint("$%s should have been a tensor in module %l"
                    ,DOLLARNAME(Dollars,-params[3]),AC.CModule);
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
        }
    }
    if ( thevector > 0 ) {  /* $-expression */
        thevector = DolToVector(BHEAD thevector);
        if ( thevector >= 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("$%s should have been a vector in module %l"
                ,DOLLARNAME(Dollars,params[4]),AC.CModule);
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
    }
    if ( ( mode & 1 ) != 0 ) {  /* Vector to tensor */
        GETSTOP(term,tstop);
        t = term + 1;
        while ( t < tstop ) {
            if ( *t == DOTPRODUCT ) {
                i = t[1] - 2; t += 2;
```

```
                    while ( i > 0 ) {
                        spectator = 0;
                        if ( t[2] < 0 ) {}
                        else if ( *t == thevector && t[1] == thevector ) {
                            if ( ( mode & 2 ) == 0 ) spectator = thevector;
                        }
                        else if ( *t == thevector ) spectator = t[1];
                        else if ( t[1] == thevector ) spectator = *t;
                        if ( spectator ) {
                            if ( ( mode & 8 ) == 0 ) goto match;
                            w = SetElements + Sets[params[6]].first;
                            m = SetElements + Sets[params[6]].last;
                            while ( w < m ) {
                                if ( *w == spectator ) break;
                                w++;
                            }
                            if ( w >= m ) goto match;
                        }
                        t += 3;
                        i -= 3;
                    }
                }
                else if ( *t == VECTOR ) {
                    i = t[1] - 2; t += 2;
                    while ( i > 0 ) {
                        if ( *t == thevector ) goto match;
                        t += 2;
                        i -= 2;
                    }
                }
                else if ( *t == thetensor ) t += t[1];
                else if ( *t >= FUNCTION ) {
                    if ( functions[*t-FUNCTION].spec > 0 ) {
                        w = t + t[1];
                        t += FUNHEAD;
                        while ( t < w ) {
                            if ( *t == thevector ) goto match;
                            t++;
                        }
                    }
                    else if ( ( mode & 4 ) != 0 ) {
                        w = t + t[1];
                        t += FUNHEAD;
                        while ( t < w ) {
                            if ( *t == -VECTOR && t[1] == thevector ) goto match;
                            else if ( *t > 0 ) t += *t;
                            else if ( *t <= -FUNCTION ) t++;
                            else t += 2;
                        }
                    }
                    else t += t[1];
                }
                else t += t[1];
            }
        }
        else {                          /* Tensor to Vector */
            GETSTOP(term,tstop);
            t = term+1;
            while ( t < tstop ) {
                if ( *t == thetensor ) goto match;
                t += t[1];
            }
        }
        return(0);
match:
        AT.TMout[0] = 5;
        AT.TMout[1] = TENVEC;
        AT.TMout[2] = thetensor;
        AT.TMout[3] = thevector;
        AT.TMout[4] = mode;
        if ( ( mode & 8 ) != 0 ) { AT.TMout[0] = 6; AT.TMout[5] = params[6]; }
        return(1);

}

/*
        #] TenVecFind :
        #[ TenVec :             WORD TenVec(term,params,num,level)
*/

WORD TenVec(PHEAD WORD *term, WORD *params, WORD num, WORD level)
{
        GETBIDENTITY
        WORD *t, *m, *w, *termout, *tstop, *outlist, *ou, *ww, *mm;
        WORD i, j, k, x, mode, thevector, thetensor, DumNow, spectator;
        DUMMYUSE(num);
        thetensor = params[2];
```

```
    thevector = params[3];
    mode = params[4];
    termout = AT.WorkPointer;
    DumNow = AR.CurDum = DetCurDum(BHEAD term);
    if ( ( mode & 1 ) != 0 ) {  /* Vector to tensor */
        AT.WorkPointer += *term;
        ou = outlist = AT.WorkPointer;
        GETSTOP(term,tstop);
        t = term + 1;
        m = termout + 1;
        while ( t < tstop ) {
            if ( *t == DOTPRODUCT ) {
                i = t[1] - 2;
                w = m;
                *m++ = *t++; *m++ = *t++;
                while ( i > 0 ) {
                    spectator = 0;
                    if ( t[2] < 0 ) {
                        *m++ = *t++; *m++ = *t++; *m++ = *t++;
                    }
                    else if ( *t == thevector && t[1] == thevector ) {
                        if ( ( mode & 2 ) == 0 ) spectator = thevector;
                        else {
                            *m++ = *t++; *m++ = *t++; *m++ = *t++;
                        }
                    }
                    else if ( *t == thevector ) spectator = t[1];
                    else if ( t[1] == thevector ) spectator = *t;
                    else {
                        *m++ = *t++; *m++ = *t++; *m++ = *t++;
                    }
                    if ( spectator ) {
                        if ( ( mode & 8 ) == 0 ) goto noveto;
                        ww = SetElements + Sets[params[5]].first;
                        mm = SetElements + Sets[params[5]].last;
                        while ( ww < mm ) {
                            if ( *ww == spectator ) break;
                            ww++;
                        }
                        if ( ww < mm ) {
                            *m++ = *t++; *m++ = *t++; *m++ = *t++;
                        }
                        else {
noveto:                 if ( spectator == thevector ) {
                            for ( j = 0; j < t[2]; j++ ) {
                                *ou++ = ++AR.CurDum;
                                *ou++ = AR.CurDum;
                            }
                            t += 3;
                        }
                        else {
                            for ( j = 0; j < t[2]; j++ ) *ou++ = spectator;
                            t += 3;
                        }}
                    }
                    i -= 3;
                }
                w[1] = WORDDIF(m,w);
                if ( w[1] == 2 ) m = w;
            }
            else if ( *t == VECTOR ) {
                i = t[1] - 2; w = m;
                *m++ = *t++; *m++ = *t++;
                while ( i > 0 ) {
                    if ( *t == thevector ) {
                        *ou++ = t[1];
                        t += 2;
                    }
                    else { *m++ = *t++; *m++ = *t++; }
                    i -= 2;
                }
                w[1] = WORDDIF(m,w);
                if ( w[1] == 2 ) m = w;
            }
            else if ( *t == thetensor ) {
                i = t[1] - FUNHEAD;
                t += FUNHEAD;
                NCOPY(ou,t,i);
            }
            else if ( *t >= FUNCTION ) {
                if ( functions[*t-FUNCTION].spec > 0 ) {
                    w = t + t[1];
                    i = FUNHEAD;
                    NCOPY(m,t,i);
                    while ( t < w ) {
                        if ( *t == thevector ) {
                            *m++ = ++AR.CurDum;
```

```
                            *ou++ = AR.CurDum;
                            t++;
                        }
                        else *m++ = *t++;
                    }
                }
                else if ( ( mode & 4 ) != 0 ) {
                    w = t + t[1];
                    i = FUNHEAD;
                    NCOPY(m,t,i);
                    while ( t < w ) {
                        if ( *t == -VECTOR && t[1] == thevector ) {
                            *m++ = -INDEX;
                            *m++ = ++AR.CurDum;
                            *ou++ = AR.CurDum;
                            t += 2;
                        }
                        else if ( *t > 0 ) {
                            i = *t;
                            NCOPY(m,t,i);
                        }
                        else if ( *t <= -FUNCTION ) *m++ = *t++;
                        else { *m++ = *t++; *m++ = *t++; }
                    }
                }
                else goto docopy;
            }
            else {
docopy:
                i = t[1];
                NCOPY(m,t,i);
            }
        }
        i = WORDDIF(ou,outlist);
        if ( i > 0 ) {
            for ( j = 1; j < i; j++ ) {
                if ( outlist[j-1] > outlist[j] ) {
                    x = outlist[j-1]; outlist[j-1] = outlist[j]; outlist[j] = x;
                    for ( k = j-1; k > 0; k-- ) {
                        if ( outlist[k-1] <= outlist[k] ) break;
                        x = outlist[k-1]; outlist[k-1] = outlist[k]; outlist[k] = x;
                    }
                }
            }

            *m++ = thetensor;
            *m++ = FUNHEAD + i;
            *m++ = DIRTYSYMFLAG;
            FILLFUN3(m)
            ou = outlist;
            NCOPY(m,ou,i);
        }
        w = term + *term;
        while ( t < w ) *m++ = *t++;
    }
    else {                      /* Tensor to Vector */
        GETSTOP(term,tstop);
        t = term+1;
        m = termout+1;
        while ( t < tstop ) {
            if ( *t != thetensor ) {
                i = t[1];
                NCOPY(m,t,i);
            }
            else {
                i = t[1] - FUNHEAD;
                t += FUNHEAD;
                if ( i > 0 ) {
                    w = m; m += 2;
                    while ( --i >= 0 ) {
                        *m++ = thevector;
                        *m++ = *t++;
                    }
                    *w = DELTA;
                    w[1] = WORDDIF(m,w);
                }
            }
        }
        w = term + *term;
        while ( t < w ) *m++ = *t++;
    }
    *termout = WORDDIF(m,termout);
    AT.WorkPointer = m;
    *AT.TMout = 0;
    if ( Generator(BHEAD termout,level) ) goto fromTenVec;
    AR.CurDum = DumNow;
    AT.WorkPointer = termout;
```

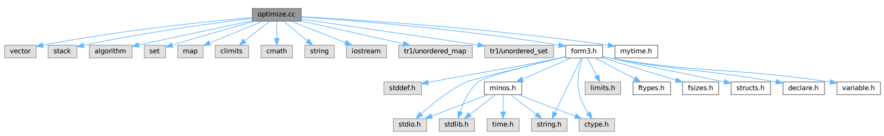
```
    return(0);
fromTenVec:
    if ( AM.tracebackflag ) {
        MLOCK(ErrorMessageLock);
        MesCall("TenVec");
        MUNLOCK(ErrorMessageLock);
    }
    return(-1);
}

/*
        #] TenVec :
    #] Operations :
*/
```

4.92 optimize.cc File Reference

```
#include <vector>
#include <stack>
#include <algorithm>
#include <set>
#include <map>
#include <climits>
#include <cmath>
#include <string>
#include <iostream>
#include <tr1/unordered_map>
#include <tr1/unordered_set>
#include "form3.h"
#include "mytime.h"
```

Include dependency graph for optimize.cc:



Data Structures

- class tree_node
- struct CSEHash
- struct CSEEq
- struct node
- struct NodeHash
- struct NodeEq
- class optimization

Typedefs

- typedef struct node **NODE**

Functions

- void print_instr (const vector< WORD > &instr, WORD num)
- template<class RandomAccessIterator >
  void my_random_shuffle (PHEAD RandomAccessIterator fr, RandomAccessIterator to)
- LONG get_expression (int exprnr)
- vector< vector< WORD > > get_brackets ()
- int count_operators (const WORD *expr, bool print=false)
- int count_operators (const vector< WORD > &instr, bool print=false)
- vector< WORD > occurrence_order (const WORD *expr, bool rev)
- WORD getpower (const WORD *term, int var, int pos)
- void fixarg (UWORD *t, WORD &n)
- void GcdLong_fix_args (PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
- void DivLong_fix_args (UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc, UWORD *d, WORD *nd)
- void build_Horner_tree (const WORD **terms, int numterms, int var, int maxvar, int pos, vector< WORD > *res)
- bool term_compare (const WORD *a, const WORD *b)
- vector< WORD > Horner_tree (const WORD *expr, const vector< WORD > &order)
- void print_tree (const vector< WORD > &tree)
- template<typename T >
  size_t hash_range (T *array, int size)
- vector< WORD > generate_instructions (const vector< WORD > &tree, bool do_CSE)
- int count_operators_cse (const vector< WORD > &tree)
- NODE * buildTree (vector< WORD > &tree)
- int count_operators_cse_topdown (vector< WORD > &tree)
- vector< WORD > simulated_annealing ()
- void find_Horner_MCTS_expand_tree ()
- void find_Horner_MCTS ()
- vector< WORD > merge_operators (const vector< WORD > &all_instr, bool move_coeff)
- vector< optimization > find_optimizations (const vector< WORD > &instr)
- bool do_optimization (const optimization optim, vector< WORD > &instr, int newid)
- int partial_factorize (vector< WORD > &instr, int n, int improve)
- vector< WORD > optimize_greedy (vector< WORD > instr, LONG time_limit)
- vector< WORD > recycle_variables (const vector< WORD > &all_instr)
- void optimize_expression_given_Horner ()
- void generate_output (const vector< WORD > &instr, int exprnr, int extraoffset, const vector< vector< WORD > > &brackets)
- WORD generate_expression (WORD exprnr)
- void optimize_print_code (int print_expr)
- int Optimize (WORD exprnr, int do_print)
- int ClearOptimize ()

Variables

- const WORD OPER_ADD = -1
- const WORD OPER_MUL = -2
- const WORD OPER_COMMA = -3
- WORD * optimize_expr
- vector< vector< WORD > > optimize_best_Horner_schemes
- int optimize_num_vars
- int optimize_best_num_oper
- vector< WORD > optimize_best_instr
- vector< WORD > optimize_best_vars

- bool mcts_factorized
- bool mcts_separated
- vector< WORD > mcts_vars
- tree_node mcts_root
- int mcts_expr_score
- set< pair< int, vector< WORD > > > mcts_best_schemes

4.92.1 Detailed Description

experimental routines for the optimization of FORTRAN or C output.

Definition in file optimize.cc.

4.92.2 Function Documentation

4.92.2.1 print_instr()

```
void print_instr (
            const vector< WORD > & instr,
            WORD num )
```

Definition at line 180 of file optimize.cc.

4.92.2.2 my_random_shuffle()

```
template<class RandomAccessIterator >
void my_random_shuffle (
            PHEAD RandomAccessIterator fr,
            RandomAccessIterator to )
```

Random shuffle

4.92.3 Description

Randomly permutes elements in the range [fr,to). Functionality is the same as C++'s "random_shuffle", but it uses Form's "wranf".

Definition at line 202 of file optimize.cc.

Referenced by find_Horner_MCTS().

4.92.3.1 get_expression()

```
LONG get_expression (
            int exprnr )
```

Get expression

4.92.4 Description

Reads an expression from the input file into a buffer (called optimize_expr). This buffer is used during the optimization process. Non-symbols are removed by ConvertToPoly and are put in temporary symbols.

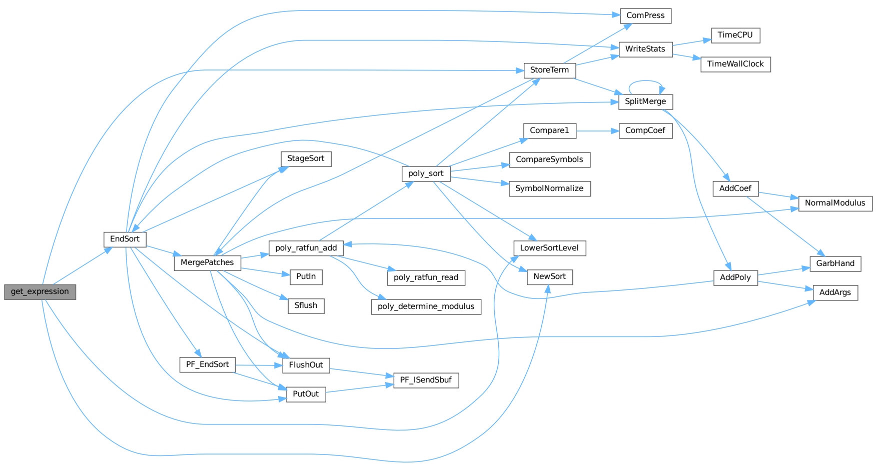
The return value is the length of the expression in WORDs, or a negative number if it failed.

Definition at line 226 of file optimize.cc.

References EndSort(), LowerSortLevel(), NewSort(), and StoreTerm().

Referenced by Optimize().

Here is the call graph for this function:



4.92.4.1 get_brackets()

```
vector< vector< WORD > > get_brackets ( )
```

Get brackets

4.92.5 Description

Checks whether the input expression (stored in optimize_expr) contains brackets. If so, this method replaces terms outside brackets by powers of SEPERATESYMBOL (equal brackets have equal powers) and the brackets are returned. If not, the result is empty.

Brackets are used for simultaneous optimization. The symbol SEPARATESYMBOL is always the first one used in a Horner scheme.

Definition at line 303 of file optimize.cc.

Referenced by Optimize().

4.92.5.1 count_operators() [1/2]

```
int count_operators (
            const WORD * expr,
            bool print = false )
```

Count operators

4.92.6 Description

Counts the number of operators in a Form-style expression.

Definition at line 423 of file optimize.cc.

Referenced by find_Horner_MCTS(), generate_instructions(), Optimize(), optimize_expression_given_Horner(), and optimize_greedy().

4.92.6.1 count_operators() [2/2]

```
int count_operators (
            const vector< WORD > & instr,
            bool print = false )
```

Count operators

4.92.7 Description

Counts the number of operators in a vector of instructions

Definition at line 477 of file optimize.cc.

4.92.7.1 occurrence_order()

```
vector< WORD > occurrence_order (
            const WORD * expr,
            bool rev )
```

Occurrence order

4.92.8 Description

Extracts all variables from an expression and sorts them with most occurring first (or last, with rev=true)

Definition at line 520 of file optimize.cc.

Referenced by Optimize().

4.92.8.1 getpower()

```
WORD getpower (
            const WORD * term,
            int var,
            int pos )
```

Horner tree building

4.92.9 Description

Given a Form-style expression (in a buffer in memory), this builds an expression tree. The tree is determined by a multivariate Horner scheme, i.e., something of the form:

1+y+x∗(2+y∗(1+y)+x∗(3-y∗(...)))

The order of the variables is given to the method "Horner_tree", which renumbers ad reorders the terms of the expression. Next, the recursive method "build_Horner_tree" does the actual tree construction.

The tree is represented in postfix notation. Tokens are of the following forms:

- SNUMBER tokenlength num den coefflength
- SYMBOL tokenlength variable power
- OPER_ADD or OPER_MUL

4.92.10 Note

Sets AN.poly_num_vars and allocates AN.poly_vars. The latter should be freed later. Get power of variable (helper function for build_Horner_tree)

4.92.11 Description

Returns the power of the variable "var", which is at position "pos" in this term, if it is present.

Definition at line 601 of file optimize.cc.

Referenced by build_Horner_tree().

4.92.11.1 fixarg()

```
void fixarg (
            UWORD * t,
            WORD & n )
```

Call GcdLong/DivLong with leading zeroes

4.92.12 Description

These method remove leading zeroes, so that GcdLong and DivLong can safely be called.

Definition at line 615 of file optimize.cc.

4.92.12.1 GcdLong_fix_args()

```
void GcdLong_fix_args (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 621 of file optimize.cc.

4.92.12.2 DivLong_fix_args()

```
void DivLong_fix_args (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc,
            UWORD * d,
            WORD * nd )
```

Definition at line 627 of file optimize.cc.

4.92.12.3 build_Horner_tree()

```
void build_Horner_tree (
            const WORD ** terms,
            int numterms,
            int var,
            int maxvar,
            int pos,
            vector< WORD > * res )
```

Build the Horner tree

4.92.13 Description

Constructs the Horner tree. The method processes one variable and continues recursively until the Horner scheme is completed.

"terms" is a pointer to the starts of the terms. "numterms" is the number of terms to be processed. "var" is the next variable to be processed (index between 0 and #maxvar) and "maxvar" is the last variable to be processed, so that partial Horner trees can also be constructed. "pos" is the position that the power of "var" should be in (one level further in the recursion, "pos" has increased by 0 or 1 depending on whether the previous power was 0 or not). The result is written at the pointer "res".

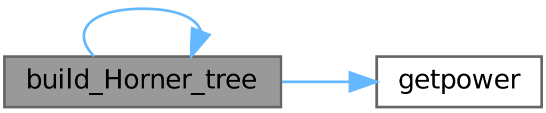
This method also factors out gcds of the coefficients. The result should end with "gcd OPER_MUL" at all times, so that one level higher gcds can be extracted again.

Definition at line 653 of file optimize.cc.

References build_Horner_tree(), and getpower().

Referenced by build_Horner_tree(), and Horner_tree().

Here is the call graph for this function:



4.92.13.1 term_compare()

```
bool term_compare (
            const WORD * a,
            const WORD * b )
```

Term compare (helper function for Horner_tree)

4.92.14 Description

Compares two terms of the form "L SYM 4 x n coeff" or "L coeff". Lower powers of lower-indexed symbols come first. This is used to sort the terms in correct order.

Definition at line 853 of file optimize.cc.

Referenced by Horner_tree().

4.92.14.1 Horner_tree()

```
vector< WORD > Horner_tree (
            const WORD * expr,
            const vector< WORD > & order )
```

Prepare Horner tree building

4.92.15 Description

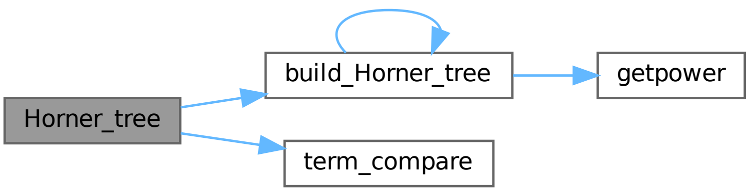
This method renumbers the variables to 0...#vars-1 according to the specified order. Next, it stored pointer to individual terms and sorts the terms with higher power first. Then the sorted lists of power is used for the construction of the Horner tree.

Definition at line 874 of file optimize.cc.

References build_Horner_tree(), and term_compare().

Referenced by optimize_expression_given_Horner().

Here is the call graph for this function:



4.92.15.1 print_tree()

```
void print_tree (
            const vector< WORD > & tree )
```

Definition at line 1002 of file optimize.cc.

4.92.15.2 hash_range()

```
template<typename T >
size_t hash_range (
            T * array,
            int size )
```

Definition at line 1071 of file optimize.cc.

4.92.15.3 generate_instructions()

```
vector< WORD > generate_instructions (
            const vector< WORD > & tree,
            bool do_CSE )
```

Generate instructions

4.92.16 Description

Converts the expression tree to a list of instructions that directly translate to code. Instructions are of the form:

expr.nr operator length [operands]+ trailing.zero

The operands are of the form:

length [(EXTRA)SYMBOL length variable power] coeff

This method only generates binary operators. Merging is done later. The method also checks for common subexpressions and eliminates them and the flag "do_CSE" is set.

4.92.17 Implementation details

The map "ID" keeps track of which subexpressions already exist. The key is formatted as one of the following:

SYMBOL x n SNUMBER coeff OPERATOR LHS RHS

with LHS/RHS formatted as one of the following:

SNUMBER idx 0 (EXTRA)SYMBOL x n

ID[symbol] or ID[operator] equals a subexpression number. ID[coeff] equals the position of the number in the input.

The stack s is used the process the postfix expression tree. Three-word tokens of the form:

SNUMBER idx.of.coeff 0 SYMBOL x n EXTRASYMBOL x 1

are pushed onto it. Operators pop two operands and push the resulting expression.

(Extra)symbols are 1-indexed, because -X is also needed to represent -1 times this term.

There is currently a bug. The notation cannot tell if there is a single bracket and then ignores the bracket.

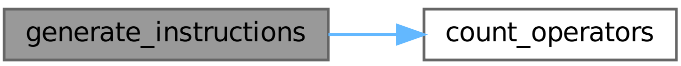
TODO: check if this method performs properly if do_CSE=false

Definition at line 1133 of file optimize.cc.

References count_operators().

Referenced by optimize_expression_given_Horner().

Here is the call graph for this function:



4.92.17.1 count_operators_cse()

```
int count_operators_cse (
            const vector< WORD > & tree )
```

Count number of operators in a binary tree, while removing CSEs on the fly. The instruction set is not created, which makes this method slightly faster.

A hash is created on the fly and is passed through the stack. TODO: find better hash functions

Definition at line 1342 of file optimize.cc.

4.92.17.2 buildTree()

```
NODE * buildTree (
            vector< WORD > & tree )
```

Definition at line 1564 of file optimize.cc.

4.92.17.3 count_operators_cse_topdown()

```
int count_operators_cse_topdown (
            vector< WORD > & tree )
```

Definition at line 1626 of file optimize.cc.

4.92.17.4 simulated_annealing()

```
vector< WORD > simulated_annealing ( )
```

Definition at line 1681 of file optimize.cc.

4.92.17.5 find_Horner_MCTS_expand_tree()

```
void find_Horner_MCTS_expand_tree ( )
```

Definition at line 2054 of file optimize.cc.

4.92.17.6 find_Horner_MCTS()

```
void find_Horner_MCTS ( )
```

Find best Horner schemes using MCTS

4.92.18 Description

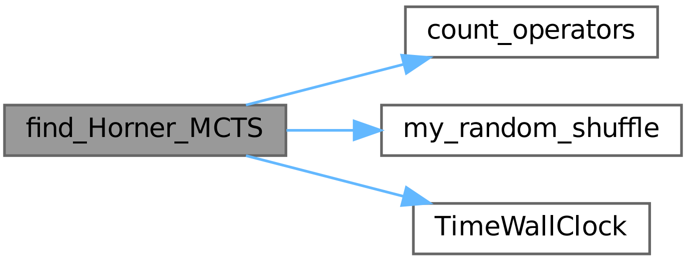
The method governs the MCTS for the best Horner schemes. It does some pre-processing, calls "find_Horner_↩
MCTS_expand_tree" a number of times and does some post-processing.

Definition at line 2255 of file optimize.cc.

References count_operators(), my_random_shuffle(), and TimeWallClock().

Referenced by Optimize().

Here is the call graph for this function:



4.92.18.1 merge_operators()

```
vector< WORD > merge_operators (
            const vector< WORD > & all_instr,
            bool move_coeff )
```

Merge operators

4.92.19 Description

The input instructions form a binary DAG. This method merges expressions like

Z1 = a+b; Z2 = Z1+c;

into

Z2 = a+b+c;

An instruction is merged iff it only has one parent and the operator equals its parent's operator.

This still leaves some freedom: where should the coefficients end up in cases as:

Z1 = Z2 + x <=> Z1 = 2∗Z2 + x Z2 = 2∗x∗y Z2 = x∗y

Both are relevant, e.g. for CSE of the form "2∗x" and "2∗Z2". The flag "move_coeff" moves coefficients from LHS-like expressions to RHS-like expressions.

Furthermore, this method removes empty equation (Z1=0), that are introduced by some "optimize_greedy" substitutions.

4.92.20 Implementation details

Expressions are mostly traversed via a stack, so that parents are evaluated before their children.

With "move_coeff" set coefficients are moved, but this leads to some tricky cases, e.g.

Z1 = Z2 + x Z2 = 2∗y

Here Z2 reduces to the trivial equation Z2=y, which should be eliminated. Here the array skip[i] comes in.

Furthermore in the case

Z1 = Z2 + x Z2 = 2∗Z3 Z3 = x∗Z4 Z4 = y∗z

after substituting Z1 = 2∗Z3 + x, the parent expression for Z4 becomes Z3 instead of Z2. This is where renum_par[i] comes in.

Finally, once a coefficient has been moved, skip_coeff[i] is set and this coefficient is copied into the new expression anymore.

Definition at line 2403 of file optimize.cc.

Referenced by optimize_expression_given_Horner().

4.92.20.1 find_optimizations()

```
vector< optimization > find_optimizations (
            const vector< WORD > & instr )
```

Find optimizations

4.92.21 Description

This method find all optimization of the form described in "class Optimization". It process every equation, looking for possible optimizations and stores them in a fast-access data structure to count the total improvement of an optimization.

Definition at line 2698 of file optimize.cc.

Referenced by optimize_greedy().

4.92.21.1 do_optimization()

```
bool do_optimization (
            const optimization optim,
            vector< WORD > & instr,
            int newid )
```

Do optimization

4.92.22 Description

This method performs an optimization. It scans through the equations of "optim.eqnidxs" and looks in which this optimization can still be performed (due to other performed optimizations this isn't always the case). If possible, it substitutes the common subexpression by a new extra symbol numbered "newid". Finally, the new extrasymbol is defined accordingly.

Substitutions may lead to trivial equations of the form "Zi=Zj", but these are removed in the end of the method. The method returns whether the substitution has been done once or more (or not).

Definition at line 2931 of file optimize.cc.

Referenced by optimize_greedy().

4.92.22.1 partial_factorize()

```
int partial_factorize (
            vector< WORD > & instr,
            int n,
            int improve )
```

Partial factorization of instructions

4.92.23 Description

This method performs partial factorization of instructions. In particular the following instructions

Z1 = x∗a∗b Z2 = x∗c∗d∗e Z3 = 2∗x + Z1 + Z2 + more

are replaced by

Z1 = a∗b Z2 = c∗d∗e Z3 = Zj + more Zi = 2 + Z1 + Z2 Zj = x∗Zi

Here it is necessary that no other equations refer to Z1 and Z2. The generation of trivial instructions (Zi=Zj or Zi=x) is prevented.

Definition at line 3480 of file optimize.cc.

Referenced by optimize_greedy().

4.92.23.1 optimize_greedy()

```
vector< WORD > optimize_greedy (
            vector< WORD > instr,
            LONG time_limit )
```

Optimize instructions greedily

4.92.24 Description

This method optimizes an expression greedily. It calls "find_optimizations" to obtain candidates and performs the best one(s) by calling "do_optimization".

How many different optimization are done, before "find_optimization" is called again, is determined by the settings "greedyminnum" and "greedymaxperc".

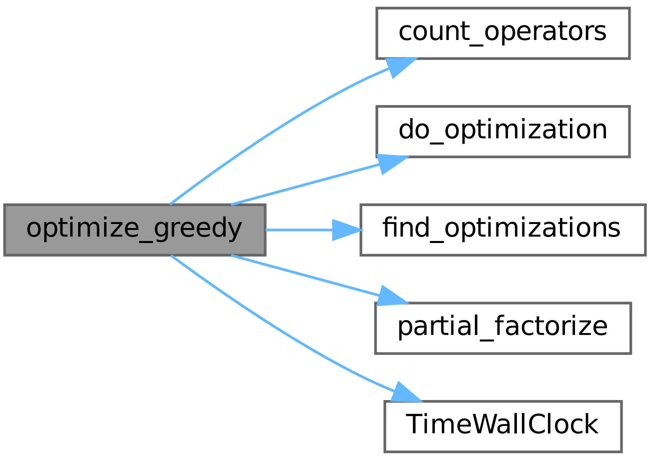
During the optimization process, sequences of zeroes are introduced in the instructions, since moving all instructions when one gets optimized, is very costly. Therefore, in the end, the instructions are "compressed" again to remove these extra zeroes.

Definition at line 3774 of file optimize.cc.

References count_operators(), do_optimization(), find_optimizations(), partial_factorize(), and TimeWallClock().

Referenced by optimize_expression_given_Horner().

Here is the call graph for this function:



4.92.24.1 recycle_variables()

```
vector< WORD > recycle_variables (
            const vector< WORD > & all_instr )
```

Recycle variables

4.92.25 Description

The current input uses many temporary variables. Many of them become obsolete at some point during the evaluation of the code, so can be recycled. This method renumbers the temporary variables, so that they are recycled. Furthermore, the input is order in depth-first order, so that the instructions can be performed consecutively.

4.92.26 Implementation details

First, for each subDAG, an estimate for the number of variables needed is made. This is done by the following recursive formula:

#vars(x) = max(#vars(ch_i(x)) + i),

with ch_i(x) the i-th child of x, where the childs are ordered w.r.t. #vars(ch_i). This formula is exact if the input forms a tree, and otherwise gives a reasonable estimate.

Then, the instructions are reordered in a depth-first order with childs ordered w.r.t. #vars. Next, the times that variables become obsolete are found. Each LHS of an instruction is renumbered to the lowest-numbered temporary variable that is available at that time.

Definition at line 3914 of file optimize.cc.

Referenced by optimize_expression_given_Horner().

4.92.26.1 optimize_expression_given_Horner()

```
void optimize_expression_given_Horner ( )
```

Optimize expression given a Horner scheme

4.92.27 Description

This method picks one Horner scheme from the list of best Horner schemes, applies this scheme to the expression and then, depending on optimize.settings, does a common subexpression elimination (CSE) or performs greedy optimizations.

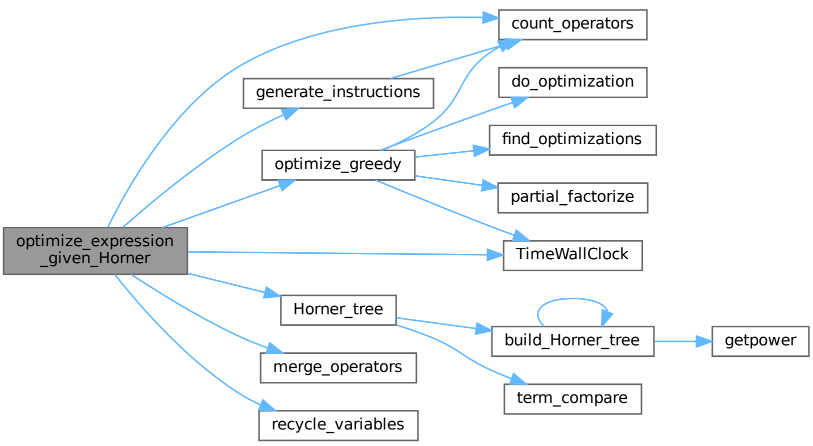
CSE is fast, while greedy might be slow. CSE followed by greedy is faster than greedy alone, but typically results in slightly worse code (not proven; just observed). eventually do greedy optimizations

Definition at line 4061 of file optimize.cc.

References count_operators(), generate_instructions(), Horner_tree(), merge_operators(), optimize_greedy(), recycle_variables(), and TimeWallClock().

Referenced by Optimize().

Here is the call graph for this function:



4.92.27.1 generate_output()

```
void generate_output (
            const vector< WORD > & instr,
            int exprnr,
            int extraoffset,
            const vector< vector< WORD > > & brackets )
```

Generate output

4.92.28 Description

This method prepares the instructions for printing. They are stored in Form format, so that they can be printed by "PrintExtraSymbol". The results are stored in the buffer AO.OptimizeResult.

Definition at line 4292 of file optimize.cc.

Referenced by Optimize().

4.92.28.1 generate_expression()

```
WORD generate_expression (
            WORD exprnr )
```

Generate expression

4.92.29 Description

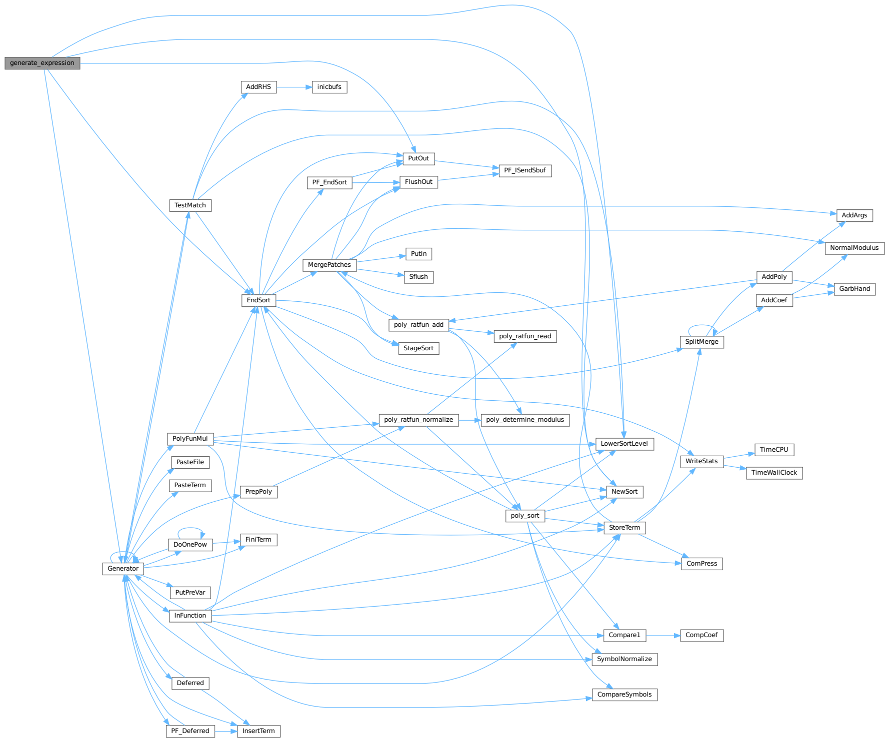
This method modifies the original optimized expression by an expression with extra symbols. This is used for "#Optimize".

Definition at line 4441 of file optimize.cc.

References EndSort(), Generator(), LowerSortLevel(), NewSort(), and PutOut().

Referenced by Optimize().

Here is the call graph for this function:



4.92.29.1 optimize_print_code()

```
void optimize_print_code (
             int print_expr )
```

Print optimized code

4.92.30 Description

This method prints the optimized code via "PrintExtraSymbol". Depending on the flag, the original expression is printed (for "Print") or not (for "#Optimize / #write "O").

Definition at line 4525 of file optimize.cc.

Referenced by Optimize().

4.92.30.1 Optimize()

```
int Optimize (
             WORD exprnr,
             int do_print )
```

Optimization of expression

4.92.31 Description

This method takes an input expression and generates optimized code to calculate its value. The following methods are called to do so:

(1) get_expression : to read to expression

(2) get_brackets : find brackets for simultaneous optimization

(3) occurrence_order or find_Horner_MCTS : to determine (the) Horner scheme(s) to use; this depends on AO.↩ optimize.horner

(4) optimize_expression_given_Horner : to do the optimizations for each Horner scheme; this method does either CSE or greedy optimizations dependings on AO.optimize.method

(5) generate_output : to format the output in Form notation and store it in a buffer

(6a) optimize_print_code : to print the expression (for "Print") or (6b) generate_expression : to modify the expression (for "#Optimize")

On ParFORM, all the processes must call this function at the same time. Then

(1) Because only the master can access to the expression to be optimized, the master broadcast the expression to all the slaves after reading the expression (PF_get_expression).

(2) get_brackets reads optimize_expr as the input and it works also on the slaves. We leave it although the bracket information is not needed on the slaves (used in (5) on the master).

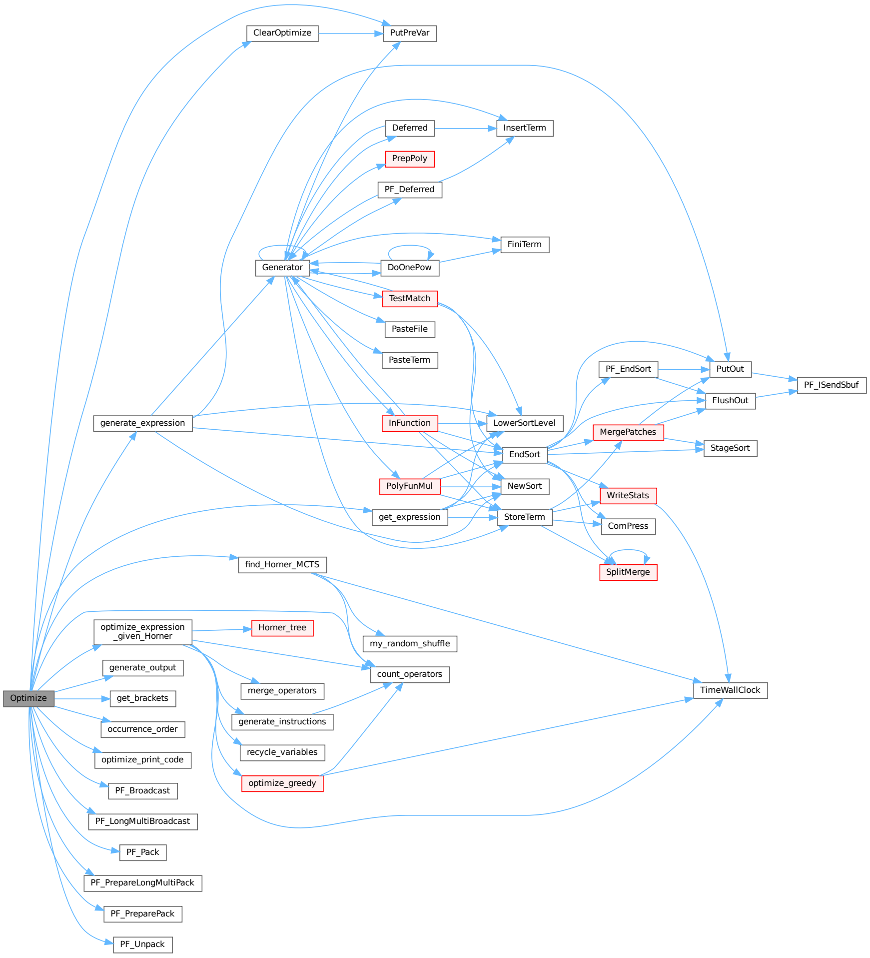
(3) and (4) find_Horner_MCTS and optimize_expression_given_Horner are parallelized.

(5), (6a) and (6b) are needed only on the master.

Definition at line 4638 of file optimize.cc.

References ClearOptimize(), count_operators(), find_Horner_MCTS(), generate_expression(), generate_output(), get_brackets(), get_expression(), occurrence_order(), optimize_expression_given_Horner(), optimize_print_code(), PF_Broadcast(), PF_LongMultiBroadcast(), PF_Pack(), PF_PrepareLongMultiPack(), PF_PreparePack(), PF_Unpack(), and PutPreVar().

Here is the call graph for this function:



4.92.31.1 ClearOptimize()

```
int ClearOptimize (
            void  )
```

Optimization of expression

4.92.32 Description

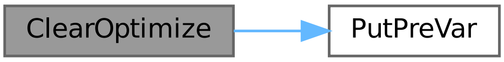
Clears the buffers that were used for optimization output. Clears the expression from the buffers (marks it to be dropped). Note: we need to use the expression by its name, because numbers may change if we drop other expressions between when we do the optimizations and clear the results (in execute.c). Also this is not 100% safe, because we could overwrite the optimized expression. But that can be done only in a Local or Global statement and hence we only have to test there that we might have to call ClearOptimize first. (in file comexpr.c)

Definition at line 4975 of file optimize.cc.

References PutPreVar().

Referenced by Optimize().

Here is the call graph for this function:



4.92.33 Variable Documentation

4.92.33.1 OPER_ADD

```
const WORD OPER_ADD = -1
```

Definition at line 120 of file optimize.cc.

4.92.33.2 OPER_MUL

```
const WORD OPER_MUL = -2
```

Definition at line 121 of file optimize.cc.

4.92.33.3 OPER_COMMA

```
const WORD OPER_COMMA = -3
```

Definition at line 122 of file optimize.cc.

4.92.33.4 optimize_expr

```
WORD* optimize_expr
```

Definition at line 157 of file optimize.cc.

4.92.33.5 optimize_best_Horner_schemes

```
vector<vector<WORD> > optimize_best_Horner_schemes
```

Definition at line 158 of file optimize.cc.

4.92.33.6 optimize_num_vars

```
int optimize_num_vars
```

Definition at line 159 of file optimize.cc.

4.92.33.7 optimize_best_num_oper

```
int optimize_best_num_oper
```

Definition at line 160 of file optimize.cc.

4.92.33.8 optimize_best_instr

```
vector<WORD> optimize_best_instr
```

Definition at line 161 of file optimize.cc.

4.92.33.9 optimize_best_vars

```
vector<WORD> optimize_best_vars
```

Definition at line 162 of file optimize.cc.

4.92.33.10 mcts_factorized

```
bool mcts_factorized
```

Definition at line 165 of file optimize.cc.

4.92.33.11 mcts_separated

```
bool mcts_separated
```

Definition at line 165 of file optimize.cc.

4.92.33.12 mcts_vars

```
vector<WORD> mcts_vars
```

Definition at line 166 of file optimize.cc.

4.92.33.13 mcts_root

```
tree_node mcts_root
```

Definition at line 167 of file optimize.cc.

4.92.33.14 mcts_expr_score

```
int mcts_expr_score
```

Definition at line 168 of file optimize.cc.

4.92.33.15 mcts_best_schemes

```
set<pair<int,vector<WORD> > > mcts_best_schemes
```

Definition at line 169 of file optimize.cc.

4.93 optimize.cc

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/*
    #] License :
    #[ includes :
*/

//#define DEBUG
//#define DEBUG_MORE
//#define DEBUG_MCTS
//#define DEBUG_GREEDY

#ifdef HAVE_CONFIG_H
#ifndef CONFIG_H_INCLUDED
#define CONFIG_H_INCLUDED
#include <config.h>
#endif
#endif

#include <vector>
#include <stack>
#include <algorithm>
#include <set>
#include <map>
#include <climits>
#include <cmath>
#include <string>
#include <algorithm>
#include <iostream>

#ifdef APPLE64
#ifndef HAVE_UNORDERED_MAP
#define HAVE_UNORDERED_MAP
#endif
```

```
#ifndef HAVE_UNORDERED_SET
#define HAVE_UNORDERED_SET
#endif
#endif

#ifdef HAVE_UNORDERED_MAP
    #include <unordered_map>
    using std::unordered_map;
#elif !defined(HAVE_TR1_UNORDERED_MAP) && defined(HAVE_BOOST_UNORDERED_MAP_HPP)
    #include <boost/unordered_map.hpp>
    using boost::unordered_map;
#else
    #include <tr1/unordered_map>
    using std::tr1::unordered_map;
#endif
#ifdef HAVE_UNORDERED_SET
    #include <unordered_set>
    using std::unordered_set;
#elif !defined(HAVE_TR1_UNORDERED_SET) && defined(HAVE_BOOST_UNORDERED_SET_HPP)
    #include <boost/unordered_set.hpp>
    using boost::unordered_set;
#else
    #include <tr1/unordered_set>
    using std::tr1::unordered_set;
#endif

#if defined(HAVE_BUILTIN_POPCOUNT)
    static inline int popcount(unsigned int x) {
        return __builtin_popcount(x);
    }
#elif defined(HAVE_POPCNT)
    #include <intrin.h>
    static inline int popcount(unsigned int x) {
        return __popcnt(x);
    }
#else
    static inline int popcount(unsigned int x) {
        int count = 0;
        while (x > 0) {
            if ((x & 1) == 1) count++;
            x »= 1;
        }
        return count;
    }
#endif

extern "C" {
#include "form3.h"
}

//#ifdef DEBUG
#include "mytime.h"
//#endif

using namespace std;

// operators
const WORD OPER_ADD = -1;
const WORD OPER_MUL = -2;
const WORD OPER_COMMA = -3;

// class for a node of the MCTS tree
class tree_node {
public:
    vector<tree_node> childs;
    double sum_results;
    int num_visits;
    WORD var;
    bool finished;
    PADPOINTER(1,1,1,1);

    tree_node (int _var=0):
        sum_results(0), num_visits(0), var(_var), finished(false) {}

    tree_node(const tree_node& other):
        childs(other.childs),
        sum_results(other.sum_results),
        num_visits(other.num_visits),
        var(other.var),
        finished(other.finished) {}

    tree_node& operator=(const tree_node& other) {
        if (this != &other) {
            childs = other.childs;
            sum_results = other.sum_results;
            num_visits = other.num_visits;
            var = other.var;
```

```
            finished = other.finished;
        }
        return *this;
    }
};

// global variables for multithreading
WORD *optimize_expr;
vector<vector<WORD> > optimize_best_Horner_schemes;
int optimize_num_vars;
int optimize_best_num_oper;
vector<WORD> optimize_best_instr;
vector<WORD> optimize_best_vars;

// global variables for MCTS
bool mcts_factorized, mcts_separated;
vector<WORD> mcts_vars;
tree_node mcts_root;
int mcts_expr_score;
set<pair<int,vector<WORD> > > mcts_best_schemes;

#ifdef WITHPTHREADS
pthread_mutex_t optimize_lock;
#endif

/*
    #] includes :
    #[ print_instr :
*/

void print_instr (const vector<WORD> &instr, WORD num)
{
    const WORD *tbegin = &*instr.begin();
    const WORD *tend = tbegin+instr.size();
    for (const WORD *t=tbegin; t!=tend; t+=*(t+2)) {
        MesPrint("<%d> %a",num,t[2],t);
    }
}

/*
    #] print_instr :
    #[ my_random_shuffle :
*/

template <class RandomAccessIterator>
void my_random_shuffle (PHEAD RandomAccessIterator fr, RandomAccessIterator to) {
  for (int i=to-fr-1; i>0; --i)
        swap (fr[i],fr[wranf(BHEAD0) % (i+1)]);
}

/*
    #] my_random_shuffle :
    #[ get_expression :
*/

static WORD comlist[3] = {TYPETOPOLYNOMIAL,3,DOALL};

LONG get_expression (int exprnr) {

    GETIDENTITY;

  AR.NoCompress = 1;

  NewSort(BHEAD0);
  EXPRESSIONS e = Expressions+exprnr;
  SetScratch(AR.infile,&(e->onfile));

  // get header term
  WORD *term = AT.WorkPointer;
    GetTerm(BHEAD term);

  NewSort(BHEAD0);

  // get terms
  while (GetTerm(BHEAD term) > 0) {
    AT.WorkPointer = term + *term;
    WORD *t1 = term;
    WORD *t2 = term + *term;
    if (ConvertToPoly(BHEAD t1,t2,comlist,1) < 0) return -1;
    int n = *t2;
    NCOPY(t1,t2,n);
    AT.WorkPointer = term + *term;
    if (StoreTerm(BHEAD term)) return -1;
  }

  // sort and store in buffer
  LONG len = EndSort(BHEAD (WORD *)((void *)(&optimize_expr)),2);
```

```
  LowerSortLevel();
  AT.WorkPointer = term;

    return len;
}

/*
    #] get_expression :
    #[ PF_get_expression :
*/
#ifdef WITHMPI

// get_expression for ParFORM
LONG PF_get_expression (int exprnr) {
    LONG len;
    if (PF.me == MASTER) {
        len = get_expression(exprnr);
    }
    if (PF.numtasks > 1) {
        PF_BroadcastBuffer(&optimize_expr, &len);
    }
    return len;
}

// replace get_expression called in Optimize
#define get_expression PF_get_expression

#endif
/*
    #] PF_get_expression :
    #[ get_brackets :
*/

vector<vector<WORD> > get_brackets () {

  // check for brackets in expression
  bool has_brackets = false;
  for (WORD *t=optimize_expr; *t!=0; t+=*t) {
    WORD *tend=t+*t; tend-=ABS(*(tend-1));
    for (WORD *t1=t+1; t1<tend; t1+=*(t1+1))
      if (*t1 == HAAKJE)
        has_brackets=true;
  }

  // replace brackets by SEPARATESYMBOL
  vector<vector<WORD> > brackets;

  if (has_brackets) {
        int exprlen=10; // we need potential space for an empty bracket
        for (WORD *t=optimize_expr; *t!=0; t+=*t)
            exprlen += *t;
    WORD *newexpr = (WORD *)Malloc1(exprlen*sizeof(WORD), "optimize newexpr");

    int i=0;
    int sep_power = 0;

    for (WORD *t=optimize_expr; *t!=0; t+=*t) {
      WORD *t1 = t+1;

            // scan for bracket
      vector<WORD> bracket;
      for (; *t1!=HAAKJE; t1+=*(t1+1))
        bracket.insert(bracket.end(), t1, t1+*(t1+1));

      if (brackets.size()==0 || bracket!=brackets.back()) {
        sep_power++;
        brackets.push_back(bracket);
      }
      t1+=*(t1+1);

      WORD left = t + *t - t1;
      bool more_symbols = (left != ABS(*(t+*t-1)));

            // add power of SEPARATESYMBOL
      newexpr[i++] = 1 + left + (more_symbols ? 2 : 4);
      newexpr[i++] = SYMBOL;
      newexpr[i++] = (more_symbols ? *(t1+1) + 2 : 4);
      newexpr[i++] = SEPARATESYMBOL;
      newexpr[i++] = sep_power;

            // add remaining terms
      if (more_symbols) {
        t1+=2;
        left-=2;
      }
      while (left-->0)
        newexpr[i++] = *(t1++);
```

```
    }
/*
    We insert here an empty bracket that is zero.
    It is used for the case that there is only a single bracket which is
    outside the notation for trees at a later stage.

    There may be a problem now in that in the case of sep_power==1
    newexpr is bigger than optimize_expr. We have to check that.
*/
    if ( sep_power == 1 )
    {
      WORD *t;
      vector<WORD> bracket;
      bracket.push_back(0);
      bracket.push_back(0);
      bracket.push_back(0);
      bracket.push_back(0);
      sep_power++;
      brackets.push_back(bracket);
      newexpr[i++] = 8;
      newexpr[i++] = SYMBOL;
      newexpr[i++] = 4;
      newexpr[i++] = SEPARATESYMBOL;
      newexpr[i++] = sep_power;
      newexpr[i++] = 1;
      newexpr[i++] = 1;
      newexpr[i++] = 3;
      newexpr[i++] = 0;
      for (t=optimize_expr; *t!=0; t+=*t) {}
      if ( t-optimize_expr+1 < i ) {  // We have to redo this
        M_free(optimize_expr,"$-sort space");
        optimize_expr = (WORD *)Malloc1(i*sizeof(WORD),"$-sort space");
      }
    }
    else {
        newexpr[i++] = 0;
    }
    memcpy(optimize_expr, newexpr, i*sizeof(WORD));
    M_free(newexpr,"optimize newexpr");

    // if factorized, replace SEP by FAC and remove brackets
    if (brackets[0].size()>0 && brackets[0][2]==FACTORSYMBOL) {
        for (WORD *t=optimize_expr; *t!=0; t+=*t) {
            if (*t == ABS(*(t+*t-1))+1) continue;
            if (t[1]==SYMBOL)
                for (int i=3; i<t[2]; i+=2)
                    if (t[i]==SEPARATESYMBOL) t[i]=FACTORSYMBOL;
        }
        return vector<vector<WORD> >();
    }
  }

  return brackets;
}

/*
    #] get_brackets :
    #[ count_operators :
*/

int count_operators (const WORD *expr, bool print=false) {

    int n=0;
    while (*(expr+n)!=0) n+=*(expr+n);

    int cntpow=0, cntmul=0, cntadd=0, sumpow=0;
    WORD maxpowfac=1, maxpowsep=1;

    for (const WORD *t=expr; *t!=0; t+=*t) {
        if (t!=expr) cntadd++;                // new term
        if (*t==ABS(*(t+*t-1))+1) continue; // only coefficient

        int cntsym=0;

        if (t[1]==SYMBOL)
            for (int i=3; i<t[2]; i+=2) {
                if (t[i]==FACTORSYMBOL) {
                    maxpowfac = max(maxpowfac, t[i+1]);
                    continue;
                }
                if (t[i]==SEPARATESYMBOL) {
                    maxpowsep = max(maxpowsep, t[i+1]);
                    continue;
                }
                if (t[i+1]>2) {           // (extra)symbol power>2
                    cntpow++;
                    sumpow += (int)floor(log(t[i+1])/log(2.0)) + popcount(t[i+1]) - 1;
```

```
                }
                if (t[i+1]==2) cntmul++; // (extra)symbol squared
                cntsym++;
            }

        if (ABS(*(t+*t-1))!=3 || *(t+*t-2)!=1 || *(t+*t-3)!=1) cntsym++; // non +/-1 coefficient

        if (cntsym > 0) cntmul+=cntsym-1;
    }

    cntadd -= maxpowfac-1;
    cntmul += maxpowfac-1;

    cntadd -= maxpowsep-1;

    if (print)
        MesPrint ("*** STATS: original  %lP %lM %lA : %l", cntpow,cntmul,cntadd,sumpow+cntmul+cntadd);

    return sumpow+cntmul+cntadd;
}

int count_operators (const vector<WORD> &instr, bool print=false) {

    int cntpow=0, cntmul=0, cntadd=0, sumpow=0;

    const WORD *ebegin = &*instr.begin();
    const WORD *eend = ebegin+instr.size();

    for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
        for (const WORD *t=e+3; *t!=0; t+=*t) {
            if (t!=e+3) {
                if (*(e+1)==OPER_ADD) cntadd++;      // new term
                if (*(e+1)==OPER_MUL) cntmul++;      // new term
            }
            if (*t == ABS(*(t+*t-1))+1) continue;    // only coefficient
            if (*(t+1)==SYMBOL || *(t+1)==EXTRASYMBOL) {
                if (*(t+4)==2) cntmul++;             // (extra)symbol squared
                if (*(t+4)>2) {                      // (extra)symbol power>2
                    cntpow++;
                    sumpow += (int)floor(log(*(t+4))/log(2.0)) + popcount(*(t+4)) - 1;
                }
            }
            if (ABS(*(t+*t-1))!=3 || *(t+*t-2)!=1 || *(t+*t-3)!=1) cntmul++; // non +/-1 coefficient
        }
    }

    if (print)
        MesPrint ("*** STATS: optimized %lP %lM %lA : %l", cntpow,cntmul,cntadd,sumpow+cntmul+cntadd);

    return sumpow+cntmul+cntadd;
}

/*
    #] count_operators :
    #[ occurrence_order :
*/

vector<WORD> occurrence_order (const WORD *expr, bool rev) {

    // count the number of occurrences of variables
    map<WORD,int> cnt;
    for (const WORD *t=expr; *t!=0; t+=*t) {
        if (*t == ABS(*(t+*t-1))+1) continue; // skip constant term
        if (t[1] == SYMBOL)
            for (int i=3; i<t[2]; i+=2)
                cnt[t[i]]++;
    }

    bool is_fac=false, is_sep=false;
    if (cnt.count(FACTORSYMBOL)) {
        is_fac=true;
        cnt.erase(FACTORSYMBOL);
    }
    if (cnt.count(SEPARATESYMBOL)) {
        is_sep=true;
        cnt.erase(SEPARATESYMBOL);
    }

    // determine the order of the variables
    vector<pair<int,WORD> > cnt_order;
    for (map<WORD,int>::iterator i=cnt.begin(); i!=cnt.end(); i++)
        cnt_order.push_back(make_pair(i->second, i->first));
    sort(cnt_order.rbegin(), cnt_order.rend());

    // create resulting order
    vector<WORD> order;
    for (int i=0; i<(int)cnt_order.size(); i++)
```

```
        order.push_back(cnt_order[i].second);

    if (rev) reverse(order.begin(),order.end());

    // add FACTORSYMBOL/SEPARATESYMBOL
    if (is_fac) order.insert(order.begin(), FACTORSYMBOL);
    if (is_sep) order.insert(order.begin(), SEPARATESYMBOL);

    return order;
}

/*
    #] occurrence_order :
    #[ Horner_tree :
*/

WORD getpower (const WORD *term, int var, int pos) {
    if (*term == ABS(*(term+*term-1))+1) return 0; // constant term
    if (2*pos+2 >= term[2]) return 0;              // too few symbols
    if (term[2*pos+3] != var) return 0;            // incorrect symbol
    return term[2*pos+4];                          // return power
}

void fixarg (UWORD *t, WORD &n) {
    int an=ABS(n), sn=SGN(n);
    while (*(t+an-1)==0) an--;
    n=an*sn;
}

void GcdLong_fix_args (PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc) {
    fixarg(a,na);
    fixarg(b,nb);
    GcdLong(BHEAD a,na,b,nb,c,nc);
}

void DivLong_fix_args(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c,   WORD *nc, UWORD *d, WORD *nd) {
    fixarg(a,na);
    fixarg(b,nb);
    DivLong(a,na,b,nb,c,nc,d,nd);
}

void build_Horner_tree (const WORD **terms, int numterms, int var, int maxvar, int pos, vector<WORD> *res) {

    GETIDENTITY;

    if (var == maxvar) {
        // Horner tree is finished, so add remaining terms unfactorized
        // (note: since only complete Horner schemes seem to be useful, numterms=1 here

        for (int fr=0; fr<numterms; fr++) {

            bool empty = true;
            const WORD *t = terms[fr];

            // add symbols
            if (*t != ABS(*(t+*t-1))+1)
                for (int i=3+2*pos; i<t[2]; i+=2) {
                    res->push_back(SYMBOL);
                    res->push_back(4);
                    res->push_back(t[i]);
                    res->push_back(t[i+1]);
                    if (!empty) res->push_back(OPER_MUL);
                    empty = false;
                }

            // if empty, add a 1, since the result should look like "... coeff *"
            if (empty) {
                res->push_back(SNUMBER);
                res->push_back(5);
                res->push_back(1);
                res->push_back(1);
                res->push_back(3);
            }

            // add coefficient
            res->push_back(SNUMBER);
            WORD n = ABS(*(t+*t-1));
            res->push_back(n+2);
            for (int i=0; i<n; i++)
                res->push_back(*(t+*t-n+i));
            res->push_back(OPER_MUL);

            if (fr>0) res->push_back(OPER_ADD);
        }

```

```
        // result should end with gcd of the terms; right now this never
        // triggers, but if one would allow for incomplete Horner schemes,
        // one should extract the gcd here
        if (numterms > 1) {
            res->push_back(SNUMBER);
            res->push_back(5);
            res->push_back(1);
            res->push_back(1);
            res->push_back(3);
            res->push_back(OPER_MUL);
        }
    }
    else {
        // extract variable "var" and the gcd and proceed recursively

        WORD nnum = 0, nden = 0, ntmp = 0, ndum = 0;
        UWORD *num = NumberMalloc("build_Horner_tree");
        UWORD *den = NumberMalloc("build_Horner_tree");
        UWORD *tmp = NumberMalloc("build_Horner_tree");
        UWORD *dum = NumberMalloc("build_Horner_tree");

        // previous coefficient for gcd extraction or coefficient multiplication
        int prev_coeff_idx = -1;

        for (int fr=0; fr<numterms;) {

            // find power of current term
            WORD pow = getpower(terms[fr], var, pos);

            // find all terms with that power
            int to=fr+1;
            while (to<numterms && getpower(terms[to], var, pos) == pow) to++;

            // recursively build Horner tree of all terms proportional to var^pow
            build_Horner_tree (terms+fr, to-fr, var+1, maxvar, pow==0?pos:pos+1, res);

            if (AN.poly_vars[var] != FACTORSYMBOL && AN.poly_vars[var] != SEPARATESYMBOL) {
                // if normal symbol, find gcd(numerators) and gcd(denominators)
                WORD  n1 = res->at(res->size()-2) / 2;
                WORD *t1 = &res->at(res->size()-2-2*ABS(n1));

                WORD *t2 = fr==0 ? t1 : &res->at(prev_coeff_idx);
                WORD  n2 = fr==0 ? n1 : *(t2+*(t2+1)-1) / 2;
                if (fr>0) t2+=2;

                GcdLong_fix_args(BHEAD (UWORD *)t1,n1,(UWORD *)t2,n2,num,&nnum);
                GcdLong_fix_args(BHEAD (UWORD *)t1+ABS(n1),ABS(n1),(UWORD *)t2+ABS(n2),ABS(n2),den,&nden);

                // divide out gcds; note: leading zeroes can be added here
                for (int i=0; i<2; i++) {
                    if (i==1 && fr==0) break;

                    WORD *t = i==0 ? t1 : t2;
                    WORD n = i==0 ? n1 : n2;

                    DivLong_fix_args((UWORD *)t, n, num, nnum, tmp, &ntmp, dum, &ndum);
                    for (int j=0; j<ABS(ntmp); j++) *t++ = tmp[j];
                    for (int j=0; j<ABS(n)-ABS(ntmp); j++) *t++ = 0;

                    if (SGN(ntmp) != SGN(n)) n=-n;

                    DivLong_fix_args((UWORD *)t, n, den, nden, tmp, &ntmp, dum, &ndum);
                    for (int j=0; j<ABS(ntmp); j++) *t++ = tmp[j];
                    for (int j=0; j<ABS(n)-ABS(ntmp); j++) *t++ = 0;

                    *t++ = SGN(n) * (2*ABS(n)+1);
                }

                // add the addition operator
                if (fr>0) res->push_back(OPER_ADD);

                // add the power of "var"
                WORD nextpow = (to==numterms ? 0 : getpower(terms[to], var, pos));

                if (pow-nextpow > 0) {
                    res->push_back(SYMBOL);
                    res->push_back(4);
                    res->push_back(var);
                    res->push_back(pow-nextpow);
                    res->push_back(OPER_MUL);
                }

                // add the extracted gcd
                res->push_back(SNUMBER);
                WORD n = MaX(ABS(nnum),nden);
                res->push_back(n*2+3);
```

```
                for (int i=0; i<ABS(nnum); i++) res->push_back(num[i]);
                for (int i=0; i<n-ABS(nnum); i++) res->push_back(0);
                for (int i=0; i<nden; i++) res->push_back(den[i]);
                for (int i=0; i<n-ABS(nden); i++) res->push_back(0);
                res->push_back(SGN(nnum)*(2*n+1));
                res->push_back(OPER_MUL);

                prev_coeff_idx = res->size() - ABS(res->at(res->size()-2)) - 3;
            }
            else if (AN.poly_vars[var]==FACTORSYMBOL) {

                // if factorsymbol, multiply overall integer contents

                if (fr>0) {
                    WORD  n1 = res->at(res->size()-2) / 2;
                    WORD *t1 = &res->at(res->size()-2-2*ABS(n1));
                    WORD *t2 = &res->at(prev_coeff_idx);
                    WORD  n2 = *(t2+*(t2+1)-1) / 2;
                    t2+=2;

                    MulRat(BHEAD (UWORD *)t1,n1,(UWORD *)t2,n2,tmp,&ntmp);

                    // replace previous coefficient with 1
                    n2=ABS(n2);
                    for (int i=0; i<ABS(n2); i++)
                        t2[i] = t2[n2+i] = i==0 ? 1 : 0;
                    t2[2*n2] = 2*n2+1;

                    // remove this coefficient
                    for (int i=0; i<2*ABS(n1)+2; i++)
                        res->pop_back();

                    // add product
                    res->back() = 2*ABS(ntmp)+3;                      // adjust size of term
                    res->insert(res->end(), tmp, tmp+2*ABS(ntmp)); // num/den coefficient
                    res->push_back(SGN(ntmp)*(2*ABS(ntmp)+1));      // size of coefficient
                    res->push_back(OPER_MUL);                         // operator
                }

                prev_coeff_idx = res->size() - ABS(res->at(res->size()-2)) - 3;

                // multiply previous factors with this factor
                if (fr>0)
                    res->push_back(OPER_MUL);
            }
            else { // AN.poly_vars[var]==SEPARATESYMBOL
                if (fr>0)
                    res->push_back(OPER_COMMA);
                prev_coeff_idx = -1;
            }

            fr=to;
        }

        // cleanup
        NumberFree(dum,"build_Horner_tree");
        NumberFree(tmp,"build_Horner_tree");
        NumberFree(den,"build_Horner_tree");
        NumberFree(num,"build_Horner_tree");
    }
}

bool term_compare (const WORD *a, const WORD *b) {
    if (*a == ABS(*(a+*a-1))+1) return true; // coefficient comes first
    if (*b == ABS(*(b+*b-1))+1) return false;
    if (a[1]!=SYMBOL) return true;
    if (b[1]!=SYMBOL) return false;
    for (int i=3; i<a[2] && i<b[2]; i+=2) {
        if (a[i]  !=b[i]  ) return a[i]  >b[i];
        if (a[i+1]!=b[i+1]) return a[i+1]<b[i+1];
    }
    return a[2]<b[2];
}

vector<WORD> Horner_tree (const WORD *expr, const vector<WORD> &order) {
#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: Horner_tree(%a)", thetime_str().c_str(), order.size(), &order[0]);
#endif

    GETIDENTITY;

    // find the renumbering scheme (new numbers are 0,1,...,#vars-1)
    map<WORD,WORD> renum;
    AN.poly_num_vars = order.size();
    AN.poly_vars = (WORD *)Malloc1(AN.poly_num_vars*sizeof(WORD), "AN.poly_vars");
    for (int i=0; i<AN.poly_num_vars; i++) {
        AN.poly_vars[i] = order[i];
```

```
        renum[order[i]] = i;
    }

    // sort variables in individual terms using bubble sort
    WORD* sorted;
    WORD* dynamicAlloc = 0;
    LONG sumsize = 0;

    for (const WORD *t=expr; *t!=0; t+=*t) {
        sumsize += *t;
    }

    // We actually need sumsize + 1 WORDS available, due to the "*sorted = 0;"
    // at the end of the sort loop.
    if ( AT.WorkPointer + sumsize + 1 > AT.WorkTop ) {
        dynamicAlloc = (WORD*)Malloc1(sizeof(*dynamicAlloc)*(sumsize+1), "Horner_tree alloc");
        sorted = dynamicAlloc;
    }
    else {
        sorted = AT.WorkPointer;
    }

    for (const WORD *t=expr; *t!=0; t+=*t) {
        memcpy(sorted, t, *t*sizeof(WORD));

        if (*t != ABS(*(t+*t-1))+1) {
            // non-constant term

            // renumber variables, adding new variables if necessary
            for (int i=3; i<sorted[2]; i+=2) {
                if (!renum.count(sorted[i])) {
                    renum[sorted[i]] = AN.poly_num_vars;

                    WORD *new_poly_vars = (WORD *)Malloc1((AN.poly_num_vars+1)*sizeof(WORD),
"AN.poly_vars");
                    memcpy(new_poly_vars, AN.poly_vars, AN.poly_num_vars*sizeof(WORD));
                    M_free(AN.poly_vars,"poly_vars");
                    AN.poly_vars = new_poly_vars;
                    AN.poly_vars[AN.poly_num_vars] = sorted[i];
                    AN.poly_num_vars++;
                }
                sorted[i] = renum[sorted[i]];
            }
            // order variables
            for (int i=0; i<sorted[2]/2; i++)
                for (int j=3; j+2<sorted[2]; j+=2)
                    if (sorted[j] > sorted[j+2]) {
                        swap(sorted[j]  , sorted[j+2]);
                        swap(sorted[j+1], sorted[j+3]);
                    }
        }

        sorted += *sorted;
    }

    *sorted = 0;
    if ( dynamicAlloc ) {
        sorted = dynamicAlloc;
    }
    else {
        sorted = AT.WorkPointer;
    }

    // find pointers to all terms and sort them efficiently
    vector<const WORD *> terms;
    for (const WORD *t=sorted; *t!=0; t+=*t)
        terms.push_back(t);
    sort(terms.rbegin(),terms.rend(),term_compare);

    // construct the Horner tree
    vector<WORD> res;
    build_Horner_tree(&terms[0], terms.size(), 0, AN.poly_num_vars, 0, &res);

    // remove leading zeroes in coefficients
    int j=0;
    for (int i=0; i<(int)res.size();) {
        if (res[i]==OPER_ADD || res[i]==OPER_MUL || res[i]==OPER_COMMA)
            res[j++] = res[i++];
        else if (res[i]==SYMBOL) {
            memmove(&res[j], &res[i], res[i+1]*sizeof(WORD));
            i+=res[j+1];
            j+=res[j+1];
        }
        else if (res[i]==SNUMBER) {
            int n = (res[i+1]-2)/2;
            int dn = 0;
            while (res[i+1+n-dn]==0 && res[i+1+2*n-dn]==0) dn++;
```

```
            res[j++] = SNUMBER;
            res[j++] = 2*(n-dn) + 3;
            memmove(&res[j], &res[i+2], (n-dn)*sizeof(WORD));
            j+=n-dn;
            memmove(&res[j], &res[i+n+2], (n-dn)*sizeof(WORD));
            j+=n-dn;
            res[j++] = SGN(res[i+2*n+2])*(2*(n-dn)+1);
            i+=2*n+3;
        }
    }
    res.resize(j);

    if ( dynamicAlloc ) {
        M_free(dynamicAlloc, "Horner_tree alloc");
    }

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: Horner_tree(%a)", thetime_str().c_str(), order.size(), &order[0]);
#endif

    return res;
}

/*
    #] Horner_tree :
    #[ print_tree :
*/

// print Horner tree (for debugging)
void print_tree (const vector<WORD> &tree) {

    GETIDENTITY;

    for (int i=0; i<(int)tree.size();) {
        if (tree[i]==OPER_ADD) {
            MesPrint("+%");
            i++;
        }
        else if (tree[i]==OPER_MUL) {
            MesPrint("*%");
            i++;
        }
        else if (tree[i]==OPER_COMMA) {
            MesPrint(",%");
            i++;
        }
        else if (tree[i]==SNUMBER) {
            UBYTE buf[100];
            int n = tree[i+tree[i+1]-1]/2;
            PrtLong((UWORD *)&tree[i+2], n, buf);
            int l = strlen((char *)buf);
            buf[l]='/';
            n=ABS(n);
            PrtLong((UWORD *)&tree[i+2+n], n, buf+l+1);
            MesPrint("%s%",buf);
            i+=tree[i+1];
        }
        else if (tree[i]==SYMBOL) {
            if (AN.poly_vars[tree[i+2]] < 10000)
                MesPrint("%s^%d%", VARNAME(symbols,AN.poly_vars[tree[i+2]]), tree[i+3]);
            else
                MesPrint("Z%d^%d%", MAXVARIABLES-AN.poly_vars[tree[i+2]], tree[i+3]);
            i+=tree[i+1];
        }
        else {
            MesPrint("error");
            exit(1);
        }

        MesPrint(" %");
    }

    MesPrint("");
}

/*
    #] print_tree :
    #[ generate_instructions :
*/


struct CSEHash {
    size_t operator()(const vector<WORD>& n) const {
        return n[0];
    }
};

```

```
struct CSEEq {
    bool operator()(const vector<WORD>& lhs, const vector<WORD>& rhs) const {
            if (lhs.size() != rhs.size()) return false;
            for (unsigned int i = 1; i < lhs.size(); i++) {
                if (lhs[i] != rhs[i]) return false;
            }
            return true;
    }
};


template<typename T> size_t hash_range(T* array, int size) {
    size_t hash = 0;

    for (int i = 0; i < size; i++) {
        hash ^= array[i] + 0x9e3779b9 + (hash « 6) + (hash » 2);
    }

    return hash;
}

vector<WORD> generate_instructions (const vector<WORD> &tree, bool do_CSE) {
#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: generate_instructions(cse=%d)",
                        thetime_str().c_str(), do_CSE?1:0);
#endif

    typedef unordered_map<vector<WORD>, int, CSEHash, CSEEq> csemap;
    csemap ID;

    // reserve lots of space, to prevent later rehashes
    // TODO: what if this is too large? make a parameter?
    if (do_CSE) {
            ID.rehash(mcts_expr_score * 2);
    }

    // s is a stack of operands to process when you encounter operators
    // in the postfix expression tree. Operands consist of three WORDs,
    // formatted as the LHS/RHS of the keys in ID.
    stack<int> s;
    vector<WORD> instr;
    WORD numinstr = 0;
    vector<WORD> x;

    // process the expression tree
    for (int i=0; i<(int)tree.size();) {
        x.clear();

        if (tree[i]==SNUMBER) {
            WORD hash = hash_range(&tree[i], tree[i + 1]);
            x.push_back(hash);
            x.push_back(SNUMBER);
            x.insert(x.end(),&tree[i],&tree[i]+tree[i+1]);
            int sign = SGN(x.back());
            x.back() *= sign;

            std::pair<csemap::iterator, bool> suc = ID.insert(csemap::value_type(x, i + 1));
            s.push(0);
            s.push(sign * suc.first->second);
            s.push(SNUMBER);
            s.push(hash);
            i+=tree[i+1];
        }
        else if (tree[i]==SYMBOL) {
            WORD hash = hash_range(&tree[i], tree[i + 1]);
            if (tree[i+3]>1) {
                x.push_back(hash);
                x.push_back(SYMBOL);
                x.push_back(tree[i+2]+1); // variable (1-indexed)
                x.push_back(tree[i+3]);   // power

                csemap::iterator it = ID.find(x);
                if (do_CSE && it != ID.end()) {
                    // already-seen power of a symbol
                    s.push(1);
                    s.push(it->second);
                    s.push(EXTRASYMBOL);
                } else {
                    //MesPrint("strange");
                    if (numinstr == MAXPOSITIVE) {
                        MesPrint((char *)"ERROR: too many temporary variables needed in optimization");
                        Terminate(-1);
                    }

                    // new power greater than 1 of a symbol
                    instr.push_back(numinstr);   // expr.nr
```

```
                    instr.push_back(OPER_MUL);   // operator
                    instr.push_back(9+OPTHEAD);  // length total
                    instr.push_back(8);          // length operand
                    instr.push_back(SYMBOL);     // SYMBOL
                    instr.push_back(4);          // length symbol
                    instr.push_back(tree[i+2]);  // variable
                    instr.push_back(tree[i+3]);  // power
                    instr.push_back(1);          // numerator
                    instr.push_back(1);          // denominator
                    instr.push_back(3);          // length coeff
                    instr.push_back(0);          // trailing 0

                    ID[x] = ++numinstr;
                    s.push(1);
                    s.push(numinstr);
                    s.push(EXTRASYMBOL);
                }
            }
            else {
                // power of 1
                s.push(tree[i+3]);    // power
                s.push(tree[i+2]+1); // variable (1-indexed)
                s.push(SYMBOL);
            }

            s.push(hash); // push hash
            i+=tree[i+1];
        }
        else { // tree[i]==OPERATOR
            int oper = tree[i];
            i++;

            x.push_back(0); // placeholder for hash
            x.push_back(oper);
            vector<WORD> hash;
            hash.push_back(oper);

            // get two operands from the stack
            for (int operand=0; operand<2; operand++) {
                hash.push_back(s.top()); s.pop();
                x.push_back(s.top()); s.pop();
                x.push_back(s.top()); s.pop();
                x.push_back(s.top()); s.pop();
            }

            x[0] = hash_range(&hash[0], 3);

            // get rid of multiplications by +/-1
            if (oper==OPER_MUL) {
                bool do_continue = false;

                for (int operand=0; operand<2; operand++) {
                    int idx_oper1 = operand==0 ? 2 : 5;
                    int idx_oper2 = operand==0 ? 5 : 2;

                    // check whether operand 1 equals +/-1
                    if (x[idx_oper1]==SNUMBER) {

                        int idx = ABS(x[idx_oper1+1])-1;
                        if (tree[idx+2]==1 && tree[idx+3]==1 && ABS(tree[idx+4])==3) {
                            // push +/- other operand and continue
                            s.push(x[idx_oper2+2]);
                            s.push(x[idx_oper2+1]*SGN(x[idx_oper1+1]));
                            s.push(x[idx_oper2]);
                            s.push(hash[1 + (operand + 1) % 2]);
                            do_continue = true;
                            break;
                        }
                    }
                }

                if (do_continue) continue;
            }

            // check whether this subexpression has been seen before
            // if not, generate instruction to define it
            csemap::iterator it = ID.find(x);
            if (!do_CSE || it == ID.end()) {
                if (numinstr == MAXPOSITIVE) {
                    MesPrint((char *)"ERROR: too many temporary variables needed in optimization");
                    Terminate(-1);
                }

                instr.push_back(numinstr); // expr.nr.
                instr.push_back(x[1]);     // operator
                instr.push_back(3);        // length

```

```
                ID[x] = ++numinstr;

                int lenidx = instr.size()-1;

                for (int j=0; j<2; j++)
                    if (x[3*j+2]==SYMBOL || x[3*j+2]==EXTRASYMBOL) {
                        instr.push_back(8);                         // length total
                        instr.push_back(x[3*j+2]);                  // (extra)symbol
                        instr.push_back(4);                         // length (extra)symbol
                        instr.push_back(ABS(x[3*j+3])-1);           // variable (0-indexed)
                        instr.push_back(x[3*j+4]);                  // power
                        instr.push_back(1);                         // numerator
                        instr.push_back(1);                         // denominator
                        instr.push_back(3*SGN(x[3*j+3]));           // length coeff
                        instr[lenidx] += 8;
                    }
                    else { // x[3*j+1]==SNUMBER
                        int t = ABS(x[3*j+3])-1;
                        instr.push_back(tree[t+1]-1);                               // length number
                        instr.insert(instr.end(), &tree[t+2], &tree[t]+tree[t+1]); // digits
                        instr.back() *= SGN(instr.back()) * SGN(x[3*j+3]);
                        instr[lenidx] += tree[t+1]-1;
                    }

                instr.push_back(0); // trailing 0
                instr[lenidx]++;
            }

            // push new expression on the stack
            s.push(1);
            s.push(ID[x]);
            s.push(EXTRASYMBOL);
            s.push(x[0]); // push hash
        }
    }

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: generate_instructions(cse=%d) : numoper=%d",
                        thetime_str().c_str(), do_CSE?1:0, count_operators(instr));
#endif

    return instr;
}

/*
    #] generate_instructions :
    #[ count_operators_cse :
*/


int count_operators_cse (const vector<WORD> &tree) {
    //MesPrint ("*** [%s] Starting CSEE", thetime_str().c_str());

    typedef unordered_map<vector<WORD>, int, CSEHash, CSEEq> csemap;
    csemap ID;

    // reserve lots of space, to prevent later rehashes
    // TODO: what if this is too large? make a parameter?
    ID.rehash(mcts_expr_score * 2);

    // s is a stack of operands to process when you encounter operators
    // in the postfix expression tree. Operands consist of three WORDs,
    // formatted as the LHS/RHS of the keys in ID.
    stack<int> s;
    int numinstr = 0, numcommas = 0;
    vector<WORD> x;

    // process the expression tree
    for (int i=0; i<(int)tree.size();) {
        x.clear();

        if (tree[i]==SNUMBER) {
            WORD hash = hash_range(&tree[i], tree[i + 1]);
            x.push_back(hash);
            x.push_back(SNUMBER);
            x.insert(x.end(),&tree[i],&tree[i]+tree[i+1]);
            int sign = SGN(x.back());
            x.back() *= sign;

            std::pair<csemap::iterator, bool> suc = ID.insert(csemap::value_type(x, i + 1));
            s.push(0);
            s.push(sign * suc.first->second);
            s.push(SNUMBER);
            s.push(hash);
            i+=tree[i+1];
        }
        else if (tree[i]==SYMBOL) {
```

```
            WORD hash = hash_range(&tree[i], tree[i + 1]);
            if (tree[i+3]>1) {
                x.push_back(hash);
                x.push_back(SYMBOL);
                x.push_back(tree[i+2]+1); // variable (1-indexed)
                x.push_back(tree[i+3]);   // power

                csemap::iterator it = ID.find(x);
                if (it != ID.end()) {
                    // already-seen power of a symbol
                    s.push(1);
                    s.push(it->second);
                    s.push(EXTRASYMBOL);
                } else {
                    if (tree[i + 3] == 2)
                        numinstr++;
                    else
                        numinstr += (int)floor(log(tree[i+3])/log(2.0)) + popcount(tree[i+3]) - 1;

                    ID[x] = numinstr;
                    s.push(1);
                    s.push(numinstr);
                    s.push(EXTRASYMBOL);
                }
            }
            else {
                // power of 1
                s.push(tree[i+3]);   // power
                s.push(tree[i+2]+1); // variable (1-indexed)
                s.push(SYMBOL);
            }

            s.push(hash); // push hash

            i+=tree[i+1];
        }
        else { // tree[i]==OPERATOR
            int oper = tree[i];
            i++;

            x.push_back(0); // placeholder for hash
            x.push_back(oper);
            vector<WORD> hash;
            hash.push_back(oper);

            // get two operands from the stack
            for (int operand=0; operand<2; operand++) {
                hash.push_back(s.top()); s.pop();
                x.push_back(s.top()); s.pop();
                x.push_back(s.top()); s.pop();
                x.push_back(s.top()); s.pop();
            }


            x[0] = hash_range(&hash[0], 3);

            // get rid of multiplications by +/-1
            if (oper==OPER_MUL) {
                bool do_continue = false;

                for (int operand=0; operand<2; operand++) {
                    int idx_oper1 = operand==0 ? 2 : 5;
                    int idx_oper2 = operand==0 ? 5 : 2;

                    // check whether operand 1 equals +/-1
                    if (x[idx_oper1]==SNUMBER) {

                        int idx = ABS(x[idx_oper1+1])-1;
                        if (tree[idx+2]==1 && tree[idx+3]==1 && ABS(tree[idx+4])==3) {
                            // push +/- other operand and continue
                            s.push(x[idx_oper2+2]);
                            s.push(x[idx_oper2+1]*SGN(x[idx_oper1+1]));
                            s.push(x[idx_oper2]);
                            s.push(hash[1 + (operand + 1) % 2]);
                            do_continue = true;
                            break;
                        }
                    }
                }

                if (do_continue) continue;
            }

            // check whether this subexpression has been seen before
            // if not, generate instruction to define it
            csemap::iterator it = ID.find(x);
            if (it == ID.end()) {
```

```
                if (numinstr == MAXPOSITIVE) {
                    MesPrint((char *)"ERROR: too many temporary variables needed in optimization");
                    Terminate(-1);
                }

                if (oper == OPER_COMMA) numcommas++;
                ID[x] = ++numinstr;
                s.push(1);
                s.push(numinstr);
                s.push(EXTRASYMBOL);
            } else {
                // push new expression on the stack
                s.push(1);
                s.push(it->second);
                s.push(EXTRASYMBOL);
            }

            s.push(x[0]); // push hash
        }
    }

    //MesPrint ("*** [%s] Stopping CSEE", thetime_str().c_str());
    return numinstr - numcommas;
}

/*
    #] count_operators_cse :
    #[ count_operators_cse_topdown :
*/

typedef struct node {
    const WORD* data;
    struct node* l;
    struct node* r; // TODO: add l,r to data?
    WORD sign; // TODO: use data for this?
    UWORD hash;

    node() : l(NULL), r(NULL), sign(1), hash(0) {};
    node(const WORD* data) : data(data), l(NULL), r(NULL), sign(1), hash(0) {};

    // a minus sign in the tree should only count as a different entry if
    // it is a compound expression: a = -a, but T+-V != T+V
    int cmp(const struct node* rhs) const {
        if (this == rhs) return 0;

        if (data[0] != rhs->data[0]) return data[0] < rhs->data[0] ? -1 : 1;
        int mod = data[0] == SNUMBER ? -1 : 0; // don't check sign, for numbers
        if (data[0] == SYMBOL || data[0] == SNUMBER) {
            for (int i = 0; i < data[1] + mod; i++) {
                if (data[i] != rhs->data[i]) return data[i] < rhs->data[i] ? -1 : 1;
                }
        } else {
            int lv = l->cmp(rhs->l);
            if (lv != 0) return lv;
            int rv = r->cmp(rhs->r);
            if (rv != 0) return rv;

            // TODO: only for ADD operation
            if (l->sign != rhs->l->sign) return l->sign < rhs->l->sign ? -1 : 1;
            if (r->sign != rhs->r->sign) return r->sign < rhs->r->sign ? -1 : 1;
        }

        return 0;
    }

        // less than operator
    bool operator() (const struct node* lhs, const struct node* rhs) const
    {
        return lhs->cmp(rhs) < 0;
    }

    void calcHash() {
        int mod = data[0] == SNUMBER ? -1 : 0; // don't check sign, for numbers
        if (data[0] == SYMBOL || data[0] == SNUMBER) {
            hash = hash_range(data, data[1] + mod);
        } else {
            if (l->hash == 0) l->calcHash();
            if (r->hash == 0) r->calcHash();

            // signs only matter for compound expressions
            size_t newr[] = {(size_t)data[0], l->hash, (size_t)l->sign, r->hash, (size_t)r->sign};
            hash = hash_range(newr, 5);
        }
    }
} NODE;

struct NodeHash {
```

```
    size_t operator()(const NODE* n) const {
        return n->hash; // already computed
    }
};

struct NodeEq {
    bool operator()(const NODE* lhs, const NODE* rhs) const {
        return lhs->cmp(rhs) == 0;
    }
};

NODE* buildTree(vector<WORD> &tree) {
    //MesPrint ("*** [%s] Starting CSEE topdown", thetime_str().c_str());

    // allocate spaces for the tree, cannot be more nodes than tree size
    NODE* ar = (NODE*)Malloc1(tree.size() * sizeof(NODE), "CSE tree");
    NODE* c = 0;
    unsigned int curIndex = 0;

    stack<NODE*> st;
    for (int i=0; i<(int)tree.size();) {
        c = ar + curIndex;
        new (c) NODE(&tree[i]); // placement new
        curIndex++;

        if (tree[i]==SYMBOL || tree[i] == SNUMBER) {
            // extract the sign to a new class member
            if (tree[i] == SNUMBER) {
                c->sign = SGN(tree[i + tree[i + 1] -1]);
            }

            c->calcHash();
            st.push(c);
            i+=tree[i+1];
        } else {
            c->r = st.top(); st.pop();
            c->l = st.top(); st.pop();

            // filter *1 and *-1
            // TODO: also multiply if there are two numbers?
            if (c->data[0] == OPER_MUL) {
                NODE* ch[] = {c->r, c->l};
                for (int j = 0; j < 2; j++)
                    if (ch[j]->data[0] == SNUMBER && ch[j]->data[1] == 5 && ch[j]->data[2]==1 &&
ch[j]->data[3]==1) {
                        ch[(j+1)%2]->sign *= ch[j]->sign; // transfer sign
                        c = ch[(j+1)%2];
                        break;
                    }
            }

            c->calcHash();
            st.push(c);
            i++;
        }
    }

    // TODO: reallocate to smaller size? Could save memory
    //MesPrint("Memory difference: %d vs %d", curIndex, tree.size());

    // we want to make the root of the tree the first element
    // so that we can easily free the array.
    // we swap the first element with the root
    // we need to change the pointer in the operator node that has this element as a child
    // TODO: check performance
    for (unsigned int i = 0; i < curIndex; i++) {
        if (ar[i].l == ar) ar[i].l = st.top();
        if (ar[i].r == ar) ar[i].r = st.top();
    }

    swap(ar[0], *st.top());
    return ar;
}

int count_operators_cse_topdown (vector<WORD> &tree) {
    typedef unordered_set<NODE*, NodeHash, NodeEq> nodeset;
    nodeset ID;

    // reserve lots of space, to prevent later rehashes
    // TODO: what if this is too large? make a parameter?
    ID.rehash(mcts_expr_score * 2);

    int numinstr = 0;

    NODE* root = buildTree(tree);

    stack<NODE*> stack;
```

```
    stack.push(root);
    while (!stack.empty())
    {
        NODE* c = stack.top();
        stack.pop();

        if (c->data[0] == SYMBOL) {
            if (c->data[3] > 1) {
                std::pair<nodeset::iterator, bool> suc = ID.insert(c);
                if (suc.second) { // new
                    if (c->data[3] == 2)
                        numinstr++;
                    else
                        numinstr += (int)floor(log(c->data[3])/log(2.0)) + popcount(c->data[3]) - 1;
                }
            }
        } else {
            if (c->data[0] != SNUMBER) {
                // operator
                std::pair<nodeset::iterator, bool> suc = ID.insert(c);
                if (suc.second) {
                    stack.push(c->r);
                    stack.push(c->l);

                    // ignore OPER_COMMA
                    if (c->data[0] == OPER_MUL || c->data[0] == OPER_ADD)
                        numinstr++;
                }
            }
        }
    }

    //MesPrint ("*** [%s] Stopping CSEE", thetime_str().c_str());
    M_free(root, "CSE tree");

    return numinstr;
}

/*
    #] count_operators_cse_topdown :
    #[ simulated_annealing :
*/
vector<WORD> simulated_annealing() {
    float minT = AO.Optimize.saMinT.fval;
    float maxT = AO.Optimize.saMaxT.fval;
    float T = maxT;
    float coolrate = pow(minT / maxT, 1 / (float)AO.Optimize.saIter);

    GETIDENTITY;

    // create a valid state where FACTORSYMBOL/SEPARATESYMBOL remains first
    vector<WORD> state = occurrence_order(optimize_expr, false);
    int startindex = 0;
    if ((state.size() > 0 && state[0] == SEPARATESYMBOL) || (state.size() > 1 && state[1] == FACTORSYMBOL)) startindex++;
    if (state.size() > 1 && state[1] == FACTORSYMBOL) startindex++;

    my_random_shuffle(BHEAD state.begin() + startindex, state.end()); // start from random scheme

    vector<WORD> tree = Horner_tree(optimize_expr, state);
    int curscore = count_operators_cse_topdown(tree);

    // clean poly_vars, that are allocated by Horner_tree
    AN.poly_num_vars = 0;
    M_free(AN.poly_vars,"poly_vars");

    std::vector<WORD> best = state; // best state
    int bestscore = curscore;

    if (startindex == (int)state.size() || (int)state.size() - startindex < 2) {
        return best;
    }

    for (int o = 0; o < AO.Optimize.saIter; o++) {
        int inda = iranf(BHEAD state.size() - startindex) + startindex;
        int indb = iranf(BHEAD state.size() - startindex) + startindex;

        if (inda == indb) {
            continue;
        }

        swap(state[inda], state[indb]); // swap works best for Horner

        vector<WORD> tree = Horner_tree(optimize_expr, state);
        int newscore = count_operators_cse_topdown(tree);

        // clean poly_vars, that are allocated by Horner_tree
```

```
        AN.poly_num_vars = 0;
        M_free(AN.poly_vars,"poly_vars");

        if (newscore <= curscore || 2.0 * wranf(BHEAD0) / (float)(UWORD)(-1) < exp((curscore - newscore) / T)) {
            curscore = newscore;

            if (curscore < bestscore) {
                bestscore = curscore;
                best = state;
            }
        } else {
            swap(state[inda], state[indb]);
        }

#ifdef DEBUG_SA
    MesPrint("Score at step %d: %d", o, curscore);
#endif
        T *= coolrate;
    }

#ifdef DEBUG_SA
    MesPrint("Simulated annealing score: %d", bestscore);
#endif

    return best;
}

/*
    #] simulated_annealing :
    #[ printpstree :
*/

/*
// print MCTS tree with LaTeX/pstricks (for analysis)
void printpstree_rec (tree_node x, string pre="") {

    if (x.num_visits==1) {
        MesPrint("%s\\TR{%d}",pre.c_str(),x.var);
    }
    else {
        MesPrint("%s\\pstree%s{\\TR{%d}}{",pre.c_str(),
                                pre==" "?"[nodesep=0, levelsep=40]":"",
                                x.var);
        for (int i=0; i<(int)x.childs.size(); i++)
            if (x.childs[i].num_visits>0)
                printpstree_rec(x.childs[i], pre+"  ");
        MesPrint("%s}",pre.c_str());
    }
}

void printpstree () {
    // draw tree with pstricks
    MesPrint ("\\documentclass{article}");
    MesPrint ("\\usepackage{pstricks,pst-node,pst-tree,graphicx}");
    MesPrint ("\\begin{document}");
    MesPrint ("\\scalebox{0.02}{");
    printpstree_rec(mcts_root," ");
    MesPrint ("}");
    MesPrint ("\\end{document}");
}
*/

/*
    #] printpstree :
    #[ find_Horner_MCTS_expand_tree :
*/

/*
        #[ next_MCTS_scheme :
*/

// find a Horner scheme to be used for the next simulation
inline static void next_MCTS_scheme (PHEAD vector<WORD> *porder, vector<WORD> *pscheme, vector<tree_node *> *ppath) {

    vector<WORD> &order = *porder;
    vector<WORD> &schemev = *pscheme;
    vector<tree_node *> &path = *ppath;
    int depth = 0, nchild0;
    float slide_down_factor = 1.0;

    order.clear();
    path.clear();

    // MCTS step I: select
    tree_node *select = &mcts_root;
```

```
    path.push_back(select);
    nchild0 = select->childs.size();
    while (select->childs.size() > 0) {
        // add virtual loss
        select->num_visits++;
        select->sum_results+=mcts_expr_score;

//-------------------------------------------------------------------
                switch ( AO.Optimize.mctsdecaymode ) {
                    case 1:  // Based on http://arxiv.org/abs/arXiv:1312.0841
                        slide_down_factor = 1.0-(1.0*AT.optimtimes)/(1.0*AO.Optimize.mctsnumexpand);
                        break;
                    case 2:  // This gives a bit more cleanup time at the end.
                        if ( 2*AT.optimtimes < AO.Optimize.mctsnumexpand ) {
                            slide_down_factor = 1.0*(AO.Optimize.mctsnumexpand-2*AT.optimtimes);
                            slide_down_factor /= 1.0*AO.Optimize.mctsnumexpand;
                        }
                        else {
                            slide_down_factor = 0.0001;
                        }
                        break;
                    case 3:  // depth dependent factor combined with case 1
                        float dd = 1.0-(1.0*depth)/(1.0*nchild0);
                        slide_down_factor = 1.0-(1.0*AT.optimtimes)/(1.0*AO.Optimize.mctsnumexpand);
                        if ( dd <= 0.000001 ) slide_down_factor = 1.0;
                        else slide_down_factor /= dd;
                        if ( slide_down_factor > 1.0 ) slide_down_factor = 1.0;
                        break;
                }
//-------------------------------------------------------------------

#ifdef DEBUG_MCTS
        MesPrint("select %d",select->var);
#endif

        // find most-promising node
        double best=0;
        tree_node *next=NULL;
        for (vector<tree_node>::iterator p=select->childs.begin(); p<select->childs.end(); p++) {
            double score;
            if (p->num_visits >= 1) {

                // there are results calculated, so select with the UCT formula
                score = mcts_expr_score / (p->sum_results/p->num_visits) +
//-------------------------------------------------------------------------
                    slide_down_factor *
//-------------------------------------------------------------------------
                    2 * AO.Optimize.mctsconstant.fval * sqrt(2*log(select->num_visits) /
p->num_visits);

#ifdef DEBUG_MCTS
                printf("%d: %.2lf [x=%.2lf n=%d fin=%i]\n",p->var,score,mcts_expr_score /
(p->sum_results/p->num_visits),
                             p->num_visits,p->finished?1:0);
                fflush(stdout);
#endif
            }
            else {
        // no results yet, so select this node by setting score=infinite
                score = 1e100;

#ifdef DEBUG_MCTS
                printf("%d: inf\n",p->var); fflush(stdout);
#endif
            }

            // update best candidate
            if (!p->finished && score>best) {
                best=score;
                next=&*p;
            }
        }

        // if no node is found, this node is finished
        if (next==NULL) {
            select->finished=true;
            break;
        }

        // traverse down the tree
        select = next;
        path.push_back(select);
        order.push_back(select->var);
        depth++;
    }

    // MCTS step II: expand
```

```

#ifdef DEBUG_MCTS
    MesPrint("expand %d",select->var);
#endif

    // variables used so far
    set<WORD> var_used;

    for (int i=0; i<(int)order.size(); i++)
        var_used.insert(ABS(order[i])-1);

    // if this a new node, create node and add children
    if (!select->finished && select->childs.size()==0) {
        tree_node new_node(select->var);
        int sign = (order.size() == 0) ? 1 : SGN(order.back());
        for (int i=0; i<(int)mcts_vars.size(); i++)
            if (!var_used.count(mcts_vars[i])) {
                new_node.childs.push_back(tree_node(sign*(mcts_vars[i]+1)));
                if (AO.Optimize.hornerdirection==O_FORWARDANDBACKWARD)
                    new_node.childs.push_back(tree_node(-sign*(mcts_vars[i]+1)));
            }
        my_random_shuffle(BHEAD new_node.childs.begin(), new_node.childs.end());

        // here locking is necessary, since operator=(tree_node) is a
        // non-atomic operation (using pointers makes this lock obsolete)
        LOCK(optimize_lock);
        *select = new_node;
        UNLOCK(optimize_lock);
    }
    // set finished if necessary
    if (select->childs.size()==0)
        select->finished = true;

    // add virtual loss of number of operators in original expression
    select->num_visits++;
    select->sum_results+=mcts_expr_score;

    // MCTS step III: simulation

    // create complete Horner scheme
    deque<WORD> scheme;

    for (int i=0; i<(int)mcts_vars.size(); i++)
        if (!var_used.count(mcts_vars[i]))
            scheme.push_back(mcts_vars[i]);
    my_random_shuffle(BHEAD scheme.begin(), scheme.end());

    for (int i=(int)order.size()-1; i>=0; i--) {
        if (order[i] > 0)
            scheme.push_front(order[i]-1);
        else
            scheme.push_back(-order[i]-1);
    }

    // add FACTORSYMBOL/SEPARATESYMBOL is necessary
    if (mcts_factorized)
        scheme.push_front(FACTORSYMBOL);
    if (mcts_separated)
        scheme.push_front(SEPARATESYMBOL);

    // Horner scheme as a vector
    schemev = vector<WORD>(scheme.begin(), scheme.end());

}

/*
        #] next_MCTS_scheme :
        #[ try_MCTS_scheme :
*/

// count the number of operators in the given Horner scheme
inline static void try_MCTS_scheme (PHEAD const vector<WORD> &scheme, int *pnum_oper) {

    // do Horner, CSE and count the number of operators
    vector<WORD> tree = Horner_tree(optimize_expr, scheme);
    //vector<WORD> instr = generate_instructions(tree, true);
    //int num_oper = count_operators(instr);
    //int num_oper2 = count_operators_cse(tree);
    //int num_oper2 = count_operators_cse_topdown(tree);
    //MesPrint("%d %d", num_oper, num_oper2);

    int num_oper = count_operators_cse_topdown(tree);

    // clean poly_vars, that is allocated by Horner_tree
    AN.poly_num_vars = 0;
    M_free(AN.poly_vars,"poly_vars");

```

```
    *pnum_oper = num_oper;

}

/*
        #] try_MCTS_scheme :
        #[ update_MCTS_scheme :
*/

// update the best score and the search tree
inline static void update_MCTS_scheme (int num_oper, const vector<WORD> &scheme, vector<tree_node *> *ppath) {

    vector<tree_node *> &path = *ppath;

    // update the (global) list of best Horner scheme
    if ((int)mcts_best_schemes.size() < AO.Optimize.mctsnumkeep ||
            (--mcts_best_schemes.end())->first > num_oper) {
        // here locking is necessary, for otherwise best_schemes may
        // become corrupted; lock can be prevented if each thread keeps
        // track of it's own list and those lists are merged in the end,
        // but this seems not useful to implement
        LOCK(optimize_lock);
        mcts_best_schemes.insert(make_pair(num_oper,scheme));
        if ((int)mcts_best_schemes.size() > AO.Optimize.mctsnumkeep)
            mcts_best_schemes.erase(--mcts_best_schemes.end());
        UNLOCK(optimize_lock);
    }

    // MCTS step IV: backpropagate

    // add number of operator and subtract mcts_expr_score, which
    // behaves as a "virtual loss"
    for (vector<tree_node *>::iterator p=path.begin(); p<path.end(); p++)
        (*p)->sum_results += num_oper - mcts_expr_score;

}

/*
        #] update_MCTS_scheme :
*/

void find_Horner_MCTS_expand_tree () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: find_Horner_MCTS_expand_tree", thetime_str().c_str());
#endif

    GETIDENTITY;

    // the order for the Horner scheme up to the selected node, with signs
    // indicating forward or backward.
    vector<WORD> order;

    // complete Horner scheme
    vector<WORD> scheme;

    // path to the selected node
    vector<tree_node *> path;

    // the number of operations obtained by the simulation
    int num_oper;

    next_MCTS_scheme(BHEAD &order, &scheme, &path);
    try_MCTS_scheme(BHEAD scheme, &num_oper);
#ifdef DEBUG_MCTS
    // Actually "order" is needed only for this debug output.
    MesPrint ("{%a} -> {%a} -> %d", order.size(), &order[0], scheme.size(), &scheme[0], num_oper);
#endif
    update_MCTS_scheme(num_oper, scheme, &path);

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: find_Horner_MCTS_expand_tree(%a-> %d)",
                        thetime_str().c_str(), scheme.size(), &scheme[0], num_oper);
#endif

}

/*
    #] find_Horner_MCTS_expand_tree :
    #[ PF_find_Horner_MCTS_expand_tree :
*/
#ifdef WITHMPI

// To remember which task is assigned to each slave. A task is represented by
// a pair of a Horner scheme and the selected path for the scheme.
// The index range is from 1 to PF.numtasks-1.
```

```
vector<pair<vector<WORD>, vector<tree_node *> > > PF_opt_MCTS_tasks;

// Initialization.
void PF_find_Horner_MCTS_expand_tree_master_init () {

    PF_opt_MCTS_tasks.resize(PF.numtasks);
    for (int i = 1; i < PF.numtasks; i++) {
        pair<vector<WORD>, vector<tree_node *> > &p = PF_opt_MCTS_tasks[i];
        p.first.clear();
        p.second.clear();
    }

}

// Wait for an idle slave and return the process number.
int PF_find_Horner_MCTS_expand_tree_master_next () {

    // Find an idle slave.
    int next;
    PF_Receive(PF_ANY_SOURCE, PF_OPT_MCTS_MSGTAG, &next, NULL);

    // Check if the slave had a task.
    pair<vector<WORD>, vector<tree_node *> > &p = PF_opt_MCTS_tasks[next];
    if (!p.first.empty()) {
        // If so, update the result.
        int num_oper;
        PF_Unpack(&num_oper, 1, PF_INT);
        update_MCTS_scheme(num_oper, p.first, &p.second);

        // Clear the task.
        p.first.clear();
        p.second.clear();
    }

    return next;

}

// The main function on the master.
void PF_find_Horner_MCTS_expand_tree_master () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: PF_find_Horner_MCTS_expand_tree_master", thetime_str().c_str());
#endif

    vector<WORD> order;
    vector<WORD> scheme;
    vector<tree_node *> path;

    next_MCTS_scheme(BHEAD &order, &scheme, &path);

    // Find an idle slave.
    int next = PF_find_Horner_MCTS_expand_tree_master_next();

    // Send a new task to the slave.
    PF_PrepareLongSinglePack();
    int len = scheme.size();
    PF_LongSinglePack(&len, 1, PF_INT);
    PF_LongSinglePack(&scheme[0], len, PF_WORD);
    PF_LongSingleSend(next, PF_OPT_MCTS_MSGTAG);

    // Remember the task.
    pair<vector<WORD>, vector<tree_node *> > &p = PF_opt_MCTS_tasks[next];
    p.first = scheme;
    p.second = path;

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: PF_find_Horner_MCTS_expand_tree_master", thetime_str().c_str());
#endif

}

// Wait for all the slaves to finish their tasks.
void PF_find_Horner_MCTS_expand_tree_master_wait () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: PF_find_Horner_MCTS_expand_tree_master_wait",
thetime_str().c_str());
#endif

    // Wait for all the slaves.
    for (int i = 1; i < PF.numtasks; i++) {
        int next = PF_find_Horner_MCTS_expand_tree_master_next();
        // Send a null task.
        PF_PrepareLongSinglePack();
        int len = 0;
        PF_LongSinglePack(&len, 1, PF_INT);
```

```
        PF_LongSingleSend(next, PF_OPT_MCTS_MSGTAG);
    }

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: PF_find_Horner_MCTS_expand_tree_master_wait",
thetime_str().c_str());
#endif

}

// The main function on the slaves.
void PF_find_Horner_MCTS_expand_tree_slave () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: PF_find_Horner_MCTS_expand_tree_slave", thetime_str().c_str());
#endif

    vector<WORD> scheme;

    {
        // Send the first message to the master, which indicates I am idle.
        PF_PreparePack();
        int dummy = 0;
        PF_Pack(&dummy, 1, PF_INT);
        PF_Send(MASTER, PF_OPT_MCTS_MSGTAG);
    }

    for (;;) {
        // Get a task from the master.
        PF_LongSingleReceive(MASTER, PF_OPT_MCTS_MSGTAG, NULL, NULL);

        // Length of the task.
        int len;
        PF_LongSingleUnpack(&len, 1, PF_INT);

        // No task remains.
        if (len == 0) break;

        // Perform the given task.
        scheme.resize(len);
        PF_LongSingleUnpack(&scheme[0], len, PF_WORD);
        int num_oper;
        try_MCTS_scheme(scheme, &num_oper);

        // Send the result to the master.
        PF_PreparePack();
        PF_Pack(&num_oper, 1, PF_INT);
        PF_Send(MASTER, PF_OPT_MCTS_MSGTAG);
    }

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: PF_find_Horner_MCTS_expand_tree_slave", thetime_str().c_str());
#endif

}

#endif
/*
    #] PF_find_Horner_MCTS_expand_tree :
    #[ find_Horner_MCTS :
*/

//vector<vector<WORD> > find_Horner_MCTS () {
void find_Horner_MCTS () {

#ifdef WITHMPI
    if (PF.me != MASTER) {
        if (PF.numtasks <= 1) return;
        PF_find_Horner_MCTS_expand_tree_slave();
        return;
    }
#endif

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: find_Horner_MCTS", thetime_str().c_str());
#endif

    GETIDENTITY;

    LONG start_time = TimeWallClock(1);

    // initialize the used global variables
    mcts_expr_score = count_operators(optimize_expr);
    mcts_root = tree_node();

    // extract all symbols from the expression
    set<WORD> var_set;
```

```
    for (WORD *t=optimize_expr; *t!=0; t+=*t)
        if (t[1] == SYMBOL)
            for (int i=3; i<t[2]; i+=2)
                var_set.insert(t[i]);

    // check for factorized/separated expression and make sure that
    // FACTORSYMBOL/SEPARATESYMBOL isn't included in the MCTS
    mcts_factorized = var_set.count(FACTORSYMBOL);
    if (mcts_factorized) var_set.erase(FACTORSYMBOL);
    mcts_separated = var_set.count(SEPARATESYMBOL);
    if (mcts_separated) var_set.erase(SEPARATESYMBOL);

    mcts_vars = vector<WORD>(var_set.begin(), var_set.end());
    optimize_num_vars = (int)mcts_vars.size();
    // initialize MCTS tree root
    for (int i=0; i<(int)mcts_vars.size(); i++) {
        if (AO.Optimize.hornerdirection != O_BACKWARD)
            mcts_root.childs.push_back(tree_node(+(mcts_vars[i]+1)));
        if (AO.Optimize.hornerdirection != O_FORWARD)
            mcts_root.childs.push_back(tree_node(-(mcts_vars[i]+1)));
    }
    my_random_shuffle(BHEAD mcts_root.childs.begin(), mcts_root.childs.end());

#if defined(WITHMPI)
    PF_find_Horner_MCTS_expand_tree_master_init();
#endif

    // initialize a potential variable mctsconstant scheme.
    AT.optimtimes = 0;

    // call expand_tree until it is called "mctsnumexpand" times, the
    // time limit is reached or the tree is fully finished
    for (int times=0; times<AO.Optimize.mctsnumexpand && !mcts_root.finished &&
                 (AO.Optimize.mctstimelimit==0 ||   (TimeWallClock(1)-start_time)/100 <
AO.Optimize.mctstimelimit);
              times++) {
        AT.optimtimes = times;
    // call expand_tree routine depending on threading mode
#if defined(WITHPTHREADS)
        if (AM.totalnumberofthreads > 1)
            find_Horner_MCTS_expand_tree_threaded();
        else
#elif defined(WITHMPI)
        if (PF.numtasks > 1)
            PF_find_Horner_MCTS_expand_tree_master();
        else
#endif
            find_Horner_MCTS_expand_tree();
    }

    // if TForm or ParForm, wait for everyone to finish
#ifdef WITHPTHREADS
    MasterWaitAll();
#endif
#ifdef WITHMPI
    PF_find_Horner_MCTS_expand_tree_master_wait();
#endif
#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: find_Horner_MCTS", thetime_str().c_str());
#endif

}

/*
    #] find_Horner_MCTS :
    #[ merge_operators :
*/

vector<WORD> merge_operators (const vector<WORD> &all_instr, bool move_coeff) {

#ifdef DEBUG_MORE
    MesPrint ("*** [%s, w=%w] CALL: merge_operators", thetime_str().c_str());
#endif

    // get starting positions of instructions
    vector<const WORD *> instr;
    const WORD *tbegin = &*all_instr.begin();
    const WORD *tend = tbegin+all_instr.size();
    // copy all instructions to temp space. There will be n of them in instr.
    for (const WORD *t=tbegin; t!=tend; t+=*(t+2)) {
        instr.push_back(t);
    }
    int n = instr.size();

    // find parents and number of parents of instructions
    vector<int> par(n), numpar(n,0);
    for (int i=0; i<n; i++) par[i]=i;
```

```
    for (int i=0; i<n; i++) {
        for (const WORD *t=instr[i]+OPTHEAD; *t!=0; t+=*t) {
            if ( *(t+1)==EXTRASYMBOL && *t!=1+ABS(*(t+*t-1)) ) {
                // extra symbol t[3] is referred to in instr i.
                par[*(t+3)]=i;
                numpar[*(t+3)]++;

                // if coefficient isn't +/-1 or power > 1, increase numpar,
                // so that this is not merged
                if (*(t+*t-3)!=1 || *(t+*t-2)!=1 || ABS(*(t+*t-1))!=3) numpar[*(t+3)]++;
                if (*(t+4)>1) numpar[*(t+3)]++;
            }
        }
    }
    // determine which instructions to merge
    stack<int> s;
    s.push(n-1);
    vector<bool> vis(n,false);

    while (!s.empty()) {

        int i=s.top(); s.pop();
        if (vis[i]) continue;
        vis[i]=true;

        for (const WORD *t=instr[i]+OPTHEAD; *t!=0; t+=*t)
            if ( *(t+1)==EXTRASYMBOL && *t!=1+ABS(*(t+*t-1)) )
                s.push(*(t+3));

        // condition: one parent and equal operator as parent
        if (numpar[i]==1 && *(instr[i]+1)==*(instr[par[i]]+1))
            par[i] = par[par[i]]; // The expr into which we subst par[i] to get i
        else
            par[i] = i;
    }

    // merge instructions into new instructions
    vector<WORD> newinstr;

    // stack of new expressions, all 0-indexed
    stack<WORD> new_expr;
    new_expr.push(n-1);
    vis = vector<bool>(n,false);

    // skip empty equations (might be introduced by greedy optimizations)
    vector<bool> skip(n,false), skipcoeff(n,false);
    for (int i=0; i<n; i++)
        if (*(instr[i]+OPTHEAD) == 0) skip[i]=skipcoeff[i]=true;

    // for renumbering merged parents
    vector<int> renum_par(n);
    for (int i=0; i<n; i++)
        renum_par[i]=i;

    while (!new_expr.empty()) {
        int x = new_expr.top(); new_expr.pop();
        if (vis[x]) continue;
        vis[x] = true;

        // find all instructions with parent=x and copy their arguments
        // into a new expression
        bool first_copy=true;
        int lenidx = newinstr.size()+2;

        // 1-indexed, since signs may occur
        stack<WORD> this_expr;
        this_expr.push(x+1);

        while (!this_expr.empty()) {
            // pop from stack, determine expr.nr and sign
            int i = this_expr.top(); this_expr.pop();
            int sign = SGN(i);
            i = ABS(i)-1;
            for (const WORD *t=instr[i]+OPTHEAD; *t!=0; t+=*t) { // terms in i
                // don't copy a term if:
                // (1) skip=true, since then it's merged into the parent
                // (2) extrasymbol with parent=x, because its children should be copied
                // (3) coefficient with skipcoeff=true, since it's already copied
                bool copy = !skip[i];
                if (*t!=1+ABS(*(t+*t-1)) && *(t+1)==EXTRASYMBOL) {
                    if (par[*(t+3)] == x) {
                        // parent of term refers to x. we push it with its sign if no skip is true
                        // and the sign of the expr.
                        this_expr.push(sign * (skip[i]||skipcoeff[i] ? 1 : SGN(*(t+*t-1))) *
(1+*(t+3)));
                        if (*(instr[i]+1) == OPER_MUL) sign=1;
```

```
                        copy=false;
                    }
                    else {
                        new_expr.push(*(t+3));
                    }
                }

                if (*t == 1+ABS(*(t+*t-1)) && skipcoeff[i]) {
                    copy=false;
                }

                if (copy) {
                    // first term, so add header
                    if (first_copy) {
                        newinstr.push_back(renum_par[x]);  // expr.nr.
                        newinstr.push_back(*(instr[x]+1)); // operator
                        newinstr.push_back(3);             // length   OPTHEAD?
                        first_copy=false;
                    }

                    // copy term and adjust sign
                    int thislenidx = newinstr.size();
                    newinstr.insert(newinstr.end(), t, t+*t); // Put the whole term in newinstr
                    newinstr.back() *= sign;
                    if (*(instr[i]+1) == OPER_MUL) sign=1;
                    newinstr[lenidx] += *t;

                    // check for moving coefficients up
                    // necessary condition: MUL-expression with 1 parent
                    if (move_coeff && *t!=1+ABS(*(t+*t-1)) && *(instr[i]+1)!=OPER_COMMA &&
                            *(t+1)==EXTRASYMBOL && numpar[*(t+3)]==1 && *(instr[*(t+3)]+1)==OPER_MUL)
{

                        // coefficient is always the first term (that's how Horner+generate works)
                        const WORD *t1 = instr[*(t+3)]+OPTHEAD;
                        const WORD *t2 = t1+*t1;

                        if (*t1 == 1+ABS(*(t1+*t1-1))) {
                            // t1 pointer to a coefficient, so move it

                            // remove old coefficient of 1
                            WORD *t3 = &*newinstr.end();                     //
                            int sign2 = SGN(t3[-1]);                          //
                            newinstr.erase(newinstr.end()-3, newinstr.end());
                            // count number of arguments; iff it is 2 move the (extra)symbol too
                            int numargs=0;
                            for (const WORD *tt=t1; *tt!=0; tt+=*tt) {
                                numargs++;
                            }
                            if (numargs==2 && *(t2+4)==1) {
                                // replace (extra)symbol
                                newinstr[newinstr.size()-4] = *(t2+1);
                                newinstr[newinstr.size()-3] = *(t2+2);
                                newinstr[newinstr.size()-2] = *(t2+3);
                                newinstr[newinstr.size()-1] = *(t2+4);
                                sign2 *= SGN(*(t2+*t2-1));          // was t2[7]

                                // ignore this expression from now on
                                skip[*(t+3)]=true;
                                if (*(t2+1)==EXTRASYMBOL)
                                    renum_par[*(t+3)] = *(t2+3);
                            }
                            else {
                                // otherwise, ignore coefficient from now on
                                // we need to collect the signs of the terms
                                // first and set them to one. This was forgotten
                                // before. Gave occasional errors.
                                if ( numargs > 2 || ( numargs == 2 && t2[4] > 1 ) ) {
                                    for (WORD *tt=(WORD *)t2; *tt!=0; tt+=*tt) {
                                        if ( tt[*tt-1] < 0 ) {
                                            tt[*tt-1] = -tt[*tt-1];
                                            sign2 = -sign2;
                                        }
                                    }
                                }
                                skipcoeff[*(t+3)]=true;
                            }

                            // add new coefficient
                            newinstr.insert(newinstr.end(), t1+1, t1+*t1);
                            newinstr.back() *= sign2;
                            newinstr[thislenidx] += ABS(newinstr.back()) - 3;
                            newinstr[lenidx] += ABS(newinstr.back()) - 3;
                        }
                    }
                }
            }
```

```
        }

        // if something has been copied, add trailing zero
        if (!first_copy) {
            newinstr.push_back(0);
            newinstr[lenidx]++;
        }
    }

    // renumber the expressions to 0,1,2,..,; only keep expressions with
    // skip=false which are their own parent after a renumbering in case
    // of moved coefficients

    // find renumber scheme
    vector<int> renum(n,-1);
    int next=0;
    for (int i=0; i<n; i++)
        if (!skip[i] && renum_par[par[i]]==i) renum[renum_par[i]]=next++;

    // find new instruction index
    tbegin = &*newinstr.begin();
    tend = tbegin+newinstr.size();
    for (const WORD *t=tbegin; t!=tend; t+=*(t+2))
        instr[renum[*t]] = t;

    // renumbering expressions and sort them lexicographically (in
    // new_instr they are in the preorder of the expression tree)
    vector<WORD> sortinstr;
    for (int i=0; i<next; i++) {
        int idx = sortinstr.size();
        sortinstr.insert(sortinstr.end(), instr[i], instr[i]+*(instr[i]+2));

        sortinstr[idx] = renum[sortinstr[idx]];

        // renumber content of an expression
        for (WORD *t2=&sortinstr[idx]+3; *t2!=0; t2+=*t2)
            if (*t2!=1+ABS(*(t2+*t2-1)) && *(t2+1)==EXTRASYMBOL)
                *(t2+3) = renum[*(t2+3)];
    }

#ifdef DEBUG_MORE
    MesPrint ("*** [%s, w=%w] DONE: merge_operators", thetime_str().c_str());
#endif

    return sortinstr;
}

/*
    #] merge_operators :
    #[ class Optimization :
*/

class optimization {
public:
    int type, arg1, arg2, improve;
    vector<WORD> coeff;
    vector<int> eqnidxs;

    bool operator< (const optimization &a) const {
        if (arg1 != a.arg1) return arg1 < a.arg1;
        if (arg2 != a.arg2) return arg2 < a.arg2;
        if (type != a.type) return type < a.type;
        return coeff < a.coeff;
    }
};

/*
    #] class Optimization :
    #[ find_optimizations :
*/

vector<optimization> find_optimizations (const vector<WORD> &instr) {

#ifdef DEBUG_GREEDY
    MesPrint ("*** [%s, w=%w] CALL: find_optimizations", thetime_str().c_str());
#endif

//  #[ Startup :
    // the resulting vector of optimizations
    vector<optimization> res;

    // a map to count the improvement of an optimization; the
    // improvement is stored as a vector<int> with equation numbers
    map<optimization, vector<int> > cnt;

    // a map to identify coefficients
    map<vector<int>,int> idx_coeff;
```

```
    const WORD *ebegin = &*instr.begin();
    const WORD *eend = ebegin+instr.size();

    for (int optim_type=0; optim_type<=4; optim_type++) {

        cnt.clear();
        idx_coeff.clear();

      optimization optim;
        optim.type = optim_type;
//  #] Startup :

//  #[ type 0 :   find optimizations of the form z=x^n (optim.type==0)
        if (optim_type == 0) {
            for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
                if (*(e+1) != OPER_MUL) continue;
                for (const WORD *t=e+3; *t!=0; t+=*t) {
                    if (*t == ABS(*(t+*t-1))+1) continue;
                    if (*(t+4) > 1) {
                        optim.arg1 = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3) + 1);
                        optim.arg2 = *(t+4);
                        cnt[optim].push_back(e-ebegin);
                    }
                }
            }
        }
//  #] type 0 :
//  #[ type 1 :  find optimizations of the form z=x*y (optim.type==1)
        if (optim_type == 1) {
            for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
                if (*(e+1) != OPER_MUL) continue;

                for (const WORD *t1=e+3; *t1!=0; t1+=*t1) {
                    if (*t1 == ABS(*(t1+*t1-1))+1) continue;
                    int x1 = (*(t1+1)==SYMBOL ? 1 : -1) * (*(t1+3) + 1);

                    for (const WORD *t2=t1+*t1; *t2!=0; t2+=*t2) {
                        if (*t2 == ABS(*(t2+*t2-1))+1) continue;
                        int x2 = (*(t2+1)==SYMBOL ? 1 : -1) * (*(t2+3) + 1);

                        if (*(t1+4) == *(t2+4)) {
                            optim.arg1 = x1;
                            optim.arg2 = x2;
                            if (optim.arg1 > optim.arg2)
                                swap(optim.arg1, optim.arg2);

                            if (*(t1+4) == 1)
                                cnt[optim].push_back(e-ebegin);
                            else {
                                // E=x^n*y^n -> z=x*y; E=z^n is double improvement
                                cnt[optim].push_back(e-ebegin);
                                cnt[optim].push_back(e-ebegin);
                            }
                        }
                    }
                }
            }
        }
//  #] type 1 :
//  #[ type 2 :  find optimizations of the form z=c*x (optim.type==2)
        if (optim_type == 2) {
            for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {

                if (*(e+1) == OPER_ADD) {
                    // in ADD-equation

                    for (const WORD *t=e+3; *t!=0; t+=*t) {
                        if (*t == ABS(*(t+*t-1))+1) continue;

                        if (*(t+4)==1) {
                            if (ABS(*(t+*t-1))==3 && *(t+*t-2)==1 && *(t+*t-3)==1) continue;

                            optim.coeff = vector<WORD>(t+*t-ABS(*(t+*t-1)), t+*t);
                            optim.coeff.back() = ABS(optim.coeff.back());

                            optim.arg1 = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3) + 1);
                            optim.arg2 = 0;

                            cnt[optim].push_back(e-ebegin);
                        }
                    }
                }
                else if (*(e+1) == OPER_MUL) {
                    // in MUL-equation
                    optim.coeff.clear();

```

```
                    for (const WORD *t=e+3; *t!=0; t+=*t)
                        if (*t == ABS(*(t+*t-1))+1) {
                            if (ABS(*(t+*t-1))==3 && *(t+*t-2)==1 && *(t+*t-3)==1) continue;
                            optim.coeff = vector<WORD>(t+*t-ABS(*(t+*t-1)), t+*t);
                            optim.coeff.back() = ABS(optim.coeff.back());
                        }

                    if (!optim.coeff.empty())
                        for (const WORD *t=e+3; *t!=0; t+=*t) {
                            if (*t == ABS(*(t+*t-1))+1) continue;
                            if (*(t+4) != 1) continue;
                            optim.arg1 = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3) + 1);
                            optim.arg2 = 0;
                            cnt[optim].push_back(e-ebegin);
                        }
                }
            }
        }
//  #] type 2 :
//  #[ type 3 :  find optimizations of the form z=x+c (optim.type==3)
        if (optim_type == 3) {
            for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {

                if (*(e+1) != OPER_ADD) continue;

                optim.coeff.clear();

                for (const WORD *t=e+3; *t!=0; t+=*t)
                    if (*t == ABS(*(t+*t-1))+1) {
                        optim.coeff = vector<WORD>(t+*t-ABS(*(t+*t-1)), t+*t);
                    }

                if (!optim.coeff.empty()) {
                    for (const WORD *t=e+3; *t!=0; t+=*t) {
                        if (*t == ABS(*(t+*t-1))+1) continue;
                        if (ABS(*(t+*t-1))!=3 || *(t+*t-2)!=1 || *(t+*t-3)!=1) continue;
                        if (*(t+*t-1)==-3) optim.coeff.back() *= -1;

                        optim.arg1 = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3) + 1);
                        optim.arg2 = 0;
                        cnt[optim].push_back(e-ebegin);
                        if (*(t+*t-1)==-3) optim.coeff.back() *= -1;
                    }
                }
            }
        }
//  #] type 3 :
//  #[ type 4,5 : find optimizations of the form z=x+y or z=x-y (optim.type==4 or 5)
        if (optim_type == 4) {
            for (const WORD *e=ebegin; e!=eend; e+=*(e+2)) {
                if (*(e+1) != OPER_ADD) continue;

                for (const WORD *t1=e+3; *t1!=0; t1+=*t1) {
                    if (*t1 == ABS(*(t1+*t1-1))+1) continue;
                    int x1 = (*(t1+1)==SYMBOL ? 1 : -1) * (*(t1+3) + 1);

                    for (const WORD *t2=t1+*t1; *t2!=0; t2+=*t2) {
                        if (*t2 == ABS(*(t2+*t2-1))+1) continue;
                        int x2 = (*(t2+1)==SYMBOL ? 1 : -1) * (*(t2+3) + 1);

                        int sign1 = SGN(*(t1+*t1-1));
                        int sign2 = SGN(*(t2+*t2-1));

                        if (BigLong((UWORD *)t1+5, ABS(*(t1+*t1-1))-1, (UWORD *)t2+5,
ABS(*(t2+*t2-1))-1) == 0) {

                            optim.type = (sign1 * sign2 == 1 ? 4 : 5); // optimization type
                            optim.arg1 = x1;
                            optim.arg2 = x2;
                            if (optim.arg1 > optim.arg2) {
                                swap(optim.arg1, optim.arg2);
                            }

                            if (ABS(*(t1+*t1-1))==3 && *(t1+*t1-2)==1 && *(t1+*t1-3)==1)
                                cnt[optim].push_back(e-ebegin);
                            else {
                                // E=2x+2y -> z=x+y; E=2z is improvement bby itself
                                cnt[optim].push_back(e-ebegin);
                                cnt[optim].push_back(e-ebegin);
                            }
                        }
                    }
                }
            }
        }
//  #] type 4,5 :
//  #[ add :
```

```
        // add optimizations with positive improvement to the result
        for (map<optimization, vector<int> >::iterator i=cnt.begin(); i!=cnt.end(); i++) {
            int improve = i->second.size() - 1;
            if (improve > 0) {
                res.push_back(i->first);
                res.back().improve = improve;
                res.back().eqnidxs = i->second;

                // remove duplicates, that were add to get the correct improvement
                res.back().eqnidxs.erase(unique(res.back().eqnidxs.begin(), res.back().eqnidxs.end()), res.back().eqnidxs.end());
            }
        }
    }
//  #] add :

#ifdef DEBUG_GREEDY
    MesPrint ("*** [%s, w=%w] DONE: find_optimizations",thetime_str().c_str());
#endif

    return res;
}

/*
    #] find_optimizations :
    #[ do_optimization :
*/

bool do_optimization (const optimization optim, vector<WORD> &instr, int newid) {

//  #[ Debug code :
#ifdef DEBUG_GREEDY
    if (optim.type==0)
        MesPrint ("*** [%s, w=%w] CALL: do_optimization(improve=%d, %c%d^%d)",
          thetime_str().c_str(), optim.improve,
        optim.arg1>0?'x':'Z', ABS(optim.arg1)-1, optim.arg2);
    else if (optim.type==1 || optim.type>=4)
        MesPrint ("*** [%s, w=%w] CALL: do_optimization(improve=%d, %c%d%c%c%d)",
        thetime_str().c_str(), optim.improve,
        optim.arg1>0?'x':'Z', ABS(optim.arg1)-1,
        optim.type==1 ? '*' : optim.type==4 ? '+' : '-',
        optim.arg2>0?'x':'Z', ABS(optim.arg2)-1);
    else {
    WORD n = optim.coeff.back()/2;
    UBYTE num[BITSINWORD*ABS(n)], den[BITSINWORD*ABS(n)];
    PrtLong((UWORD *)&optim.coeff[0], n, num);
    PrtLong((UWORD *)&optim.coeff[ABS(n)], ABS(n), den);
        MesPrint ("*** [%s, w=%w] CALL: do_optimization(improve=%d, %c%d%c%s/%s)",
        thetime_str().c_str(), optim.improve,
        optim.arg1>0?'x':'Z', ABS(optim.arg1)-1,
        optim.type==2 ? '*' : '+', num,den);
  }
#endif
//  #] Debug code :

    bool substituted = false;
    WORD *ebegin = &*instr.begin();

//  #[ type 0 : substitution of the form z=x^n (optim.type==0)
    if (optim.type == 0) {

        int vartypex = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
        int varnumx  = ABS(optim.arg1) - 1;
        int n = optim.arg2;

        for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
            WORD *e = ebegin + optim.eqnidxs[i];
            if (*(e+1) != OPER_MUL) continue;

            // scan through equation
            for (WORD *t=e+3; *t!=0; t+=*t) {
                if (*t == ABS(*(t+*t-1))+1) continue;
                if (*(t+1)==vartypex &&
                        *(t+3)==varnumx &&
                        *(t+4) % n == 0) {

                    // substitute
                    *(t+1) = EXTRASYMBOL;
                    *(t+3) = newid;
                    *(t+4) /= n;

                    substituted = true;
                }
            }
        }

```

```
        if (!substituted) {
#ifdef DEBUG_GREEDY
            MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(), optim.improve);
#endif
            return false;
        }

        // add extra equation (Tnew = x^n)
        instr.push_back(newid);     // eqn.nr
        instr.push_back(OPER_MUL); // operator
        instr.push_back(12);        // total length
        instr.push_back(8);         // term length
        instr.push_back(vartypex); // (extra)symbol
        instr.push_back(4);         // symbol length
        instr.push_back(varnumx);  // symbol id
        instr.push_back(n);         // power
        instr.push_back(1);
        instr.push_back(1);         // coefficient 1
        instr.push_back(3);
        instr.push_back(0);         // trailing 0
    }
//  #] type 0 :
//  #[ type 1 : substitution of the form z=x*y (optim.type==1)
    if (optim.type == 1) {

        int vartypex = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
        int varnumx  = ABS(optim.arg1) - 1;
        int vartypey = optim.arg2>0 ? SYMBOL : EXTRASYMBOL;
        int varnumy  = ABS(optim.arg2) - 1;

        for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
            WORD *e = ebegin + optim.eqnidxs[i];
            if (*(e+1) != OPER_MUL) continue;

            // scan through equation
            int powx=0, powy=0;
            for (WORD *t=e+3; *t!=0; t+=*t) {
                if (*t == ABS(*(t+*t-1))+1) continue;
                if (*(t+1)==vartypex && *(t+3)==varnumx) powx = *(t+4);
                if (*(t+1)==vartypey && *(t+3)==varnumy) powy = *(t+4);
            }

            // substitute if found
            if (powx>0 && powy>0 && powx==powy) {

                WORD sign = 1;
                WORD *newt = e+3;

                for (WORD *t=e+3; *t!=0;) {
                    int dt=*t;

                    if (*t == ABS(*(t+*t-1))+1 ||
                            (!(*(t+1)==vartypex && *(t+3)==varnumx) &&
                             !(*(t+1)==vartypey && *(t+3)==varnumy))) {
                        memmove(newt, t, *t*sizeof(WORD));
                        newt += dt;
                    }
                    else {
                        sign *= SGN(*(t+*t-1));
                    }

                    t+=dt;
                }

                *newt++ = 8;            // term length
                *newt++ = EXTRASYMBOL; // extrasymbol
                *newt++ = 4;            // symbol length
                *newt++ = newid;        // symbol id
                *newt++ = powx;         // power
                *newt++ = 1;
                *newt++ = 1;            // coefficient +/-1
                *newt++ = 3*sign;
                *newt++ = 0;            // trailing 0

                substituted = true;
            }
        }

        if (!substituted) {
#ifdef DEBUG_GREEDY
            MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(), optim.improve);
#endif
            return false;
        }

```

```
        // add extra equation (Tnew = x*y)
        instr.push_back(newid);    // eqn.nr
        instr.push_back(OPER_MUL); // operator
        instr.push_back(20);       // total length
        instr.push_back(8);        // LHS length
        instr.push_back(vartypex); // (extra)symbol
        instr.push_back(4);        // symbol length
        instr.push_back(varnumx);  // symbol id
        instr.push_back(1);        // power 1
        instr.push_back(1);
        instr.push_back(1);        // coefficient 1
        instr.push_back(3);
        instr.push_back(8);        // RHS length
        instr.push_back(vartypey); // (extra)symbol
        instr.push_back(4);        // symbol length
        instr.push_back(varnumy);  // symbol id
        instr.push_back(1);        // power 1
        instr.push_back(1);
        instr.push_back(1);        // coefficient 1
        instr.push_back(3);
        instr.push_back(0);        // trailing 0
    }
//  #] type 1 :
//  #[ type 2 : substitution of the form z=c*x (optim.type==2)

    if (optim.type == 2) {
        int vartype = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
        int varnum  = ABS(optim.arg1) - 1;

        WORD ncoeff = optim.coeff.back();

        // scan through equations
        for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
            WORD *e = ebegin + optim.eqnidxs[i];

            if (*(e+1) == OPER_ADD) {
                // scan through ADD-equation
                for (WORD *t=e+3; *t!=0; t+=*t) {

                    if (*t == ABS(*(t+*t-1))+1) continue;

                    if (*(t+1)==vartype && ABS(*(t+3))==varnum && *(t+4)==1 &&
                            BigLong((UWORD *)&optim.coeff[0],ncoeff-1,
                                    (UWORD *)t+*t-ABS(*(t+*t-1)),ABS(*(t+*t-1))-1) == 0) {
                        // substitute

                        int sign = SGN(*(t+*t-1));

                        WORD *tend = t;
                        while (*tend!=0) tend+=*tend;
                        WORD nmove = tend - t - *t;
                        memmove(t, t+*t, nmove*sizeof(WORD));
                        t += nmove;

                        *t++ = 8;           // term length
                        *t++ = EXTRASYMBOL; // (extra)symbol
                        *t++ = 4;           // symbol length
                        *t++ = newid;       // symbol id
                        *t++ = 1;           // power of 1
                        *t++ = 1;
                        *t++ = 1;           // coefficient of +/-1
                        *t++ = 3 * sign;
                        *t++ = 0;           // trailing 0

                        substituted = true;
                        break;
                    }
                }
            }
            else if (*(e+1) == OPER_MUL) {

                bool coeff_match=false, var_match=false;
                int sign = 1;

                // scan through MUL-equation
                for (WORD *t=e+3; *t!=0; t+=*t) {
                    if (*t == ABS(*(t+*t-1))+1 && BigLong((UWORD *)&optim.coeff[0],ncoeff-1,                                    (UWORD
*)t+*t-ABS(*(t+*t-1)),ABS(*(t+*t-1))-1) == 0) {
                        coeff_match = true;
                        sign *= SGN(*(t+*t-1));
                    }
                    else if (*(t+1)==vartype && ABS(*(t+3))==varnum && *(t+4)==1) {
                        var_match = true;
                        sign *= SGN(*(t+*t-1));
                    }
                }
```

```

                // substitute if found
                if (coeff_match && var_match) {

                    WORD *newt = e+3;

                    for (WORD *t=e+3; *t!=0;) {

                        int dt=*t;

                        if (*t!=ABS(*(t+*t-1))+1 && !(*(t+1)==vartype && ABS(*(t+3))==varnum && *(t+4)==1)) {
                            memmove(newt, t, dt*sizeof(WORD));
                            newt += dt;
                        }

                        t+=dt;
                    }

                    *newt++ = 8;           // term length
                    *newt++ = EXTRASYMBOL; // extrasymbol
                    *newt++ = 4;           // symbol length
                    *newt++ = newid;       // symbol id
                    *newt++ = 1;           // power of 1
                    *newt++ = 1;
                    *newt++ = 1;           // coefficient of +/-1
                    *newt++ = 3 * sign;
                    *newt++ = 0;           // trailing 0

                    substituted = true;
                }
            }
        }

        if (!substituted) {
#ifdef DEBUG_GREEDY
            MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(), optim.improve);
#endif
            return false;
        }

        // add extra equation (Tnew = c*y)
        instr.push_back(newid);               // eqn.nr
        instr.push_back(OPER_ADD);            // operator
        instr.push_back(9+ABS(ncoeff));       // total length
        instr.push_back(5+ABS(ncoeff));       // term length
        instr.push_back(vartype);             // (extra)symbol
        instr.push_back(4);                   // symbol length
        instr.push_back(varnum);              // symbol id
        instr.push_back(1);                   // power of 1
        for (int i=0; i<ABS(ncoeff); i++)     // coefficient
            instr.push_back(optim.coeff[i]);
        instr.push_back(0);                   // trailing 0
    }
//   #] type 2 :
//   #[ type 3 : substitution of the form z=x+c (optim.type==3)
    if (optim.type == 3) {
        int vartype = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
        int varnum  = ABS(optim.arg1) - 1;

        WORD ncoeff = optim.coeff.back();

        // scan through equation
        for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
            WORD *e = ebegin + optim.eqnidxs[i];

            if (*(e+1) != OPER_ADD) continue;

            int coeff_match=0, var_match=0;

            for (WORD *t=e+3; *t!=0; t+=*t) {
                if (*t == ABS(*(t+*t-1))+1 && BigLong((UWORD *)&optim.coeff[0],ABS(ncoeff)-1,
                                                                                   (UWORD *)t+*t-ABS(*(t+*t-1)),ABS(*(t+*t-1))-1) == 0)
                    coeff_match = SGN(ncoeff) * SGN(*(t+*t-1));
                else if (*(t+1)==vartype && ABS(*(t+3))==varnum && *(t+4)==1)
                    var_match = SGN(*(t+7));
            }

            // substitute if found (x+c and -x-c and matches)
            if (coeff_match * var_match == 1) {

                WORD *newt = e+3;

                for (WORD *t=e+3; *t!=0;) {
                    int dt=*t;
```

```
                    if (*t!=ABS(*(t+*t-1))+1 && !(*(t+1)==vartype && ABS(*(t+3))==varnum && *(t+4)==1)) {
                        memmove(newt, t, dt*sizeof(WORD));
                        newt += dt;
                    }
                    t+=dt;
                }

                *newt++ = 8;            // term length
                *newt++ = EXTRASYMBOL;  // extrasymbol
                *newt++ = 4;            // symbol length
                *newt++ = newid;        // symbol id
                *newt++ = 1;            // power of 1
                *newt++ = 1;
                *newt++ = 1;            // coefficient of +/-1
                *newt++ = 3*coeff_match;
                *newt++ = 0;            // trailing zero

                substituted = true;
            }
        }

        if (!substituted) {
#ifdef DEBUG_GREEDY
            MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(), optim.improve);
#endif
            return false;
        }

        // add extra equation (Tnew = x+c)
        instr.push_back(newid);           // eqn.nr
        instr.push_back(OPER_ADD);        // operator
        instr.push_back(13+ABS(ncoeff));  // total length
        instr.push_back(8);               // x-term length
        instr.push_back(vartype);         // (extra)symbol
        instr.push_back(4);               // symbol length
        instr.push_back(varnum);          // symbol id
        instr.push_back(1);               // power of 1
        instr.push_back(1);
        instr.push_back(1);               // coefficient of 1
        instr.push_back(3);
        instr.push_back(ABS(ncoeff)+1);   // c-term length
        for (int i=0; i<ABS(ncoeff); i++) // coefficient
            instr.push_back(optim.coeff[i]);
        instr.push_back(0);               // trailing zero
    }
//  #] type 3 :
//  #[ type 4,5 : substitution of the form z=x+y or z=x-y (optim.type=4 or 5)
    if (optim.type >= 4) {

        int vartypex = optim.arg1>0 ? SYMBOL : EXTRASYMBOL;
        int varnumx  = ABS(optim.arg1) - 1;
        int vartypey = optim.arg2>0 ? SYMBOL : EXTRASYMBOL;
        int varnumy  = ABS(optim.arg2) - 1;

        // scan through equations
        for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
            WORD *e = ebegin + optim.eqnidxs[i];

            if (*(e+1) != OPER_ADD) continue;

            const WORD *coeffx=NULL, *coeffy=NULL;
            WORD ncoeffx=0,ncoeffy=0;

            // looks for terms
            for (WORD *t=e+3; *t!=0; t+=*t) {
                if (*t == ABS(*(t+*t-1))+1) continue; // constant
                if (*(t+1)==vartypex && *(t+3)==varnumx && *(t+4)==1) {
                    coeffx = t+5;
                    ncoeffx = *(t+*t-1);
                }
                if (*(t+1)==vartypey && *(t+3)==varnumy && *(t+4)==1) {
                    coeffy = t+5;
                    ncoeffy = *(t+*t-1);
                }
            }

            // check signs (type=4: x+y and -x-y, type=5: x-y and -x+y) ??????
            // check signs (type=4: x+y, type=5: x-y) !!!!!!!!!!
            if (SGN(ncoeffx) * SGN(ncoeffy) * (optim.type==4 ? 1 : -1) == 1) {
                // check absolute value of coefficients
                if (BigLong((UWORD *)coeffx, ABS(ncoeffx)-1, (UWORD *)coeffy, ABS(ncoeffy)-1) == 0) {
                    // substitute
                    vector<WORD> coeff(coeffx, coeffx+ABS(ncoeffx));

                    WORD *newt = e+3;
```

```
/*
if ( optim.type == 5 ) {
    while ( *newt ) newt+=*newt;
    int i = (newt - e) - 3;
    MesPrint("  < %a",i,e+3);
    newt = e+3;
}
*/
                    for (WORD *t=e+3; *t!=0;) {
                        int dt=*t;
                        if (*t == ABS(*(t+*t-1))+1 ||
                                (!(*(t+1)==vartypex && *(t+3)==varnumx) &&
                                 !(*(t+1)==vartypey && *(t+3)==varnumy))) {
                            memmove(newt, t, dt*sizeof(WORD));
                            newt += dt;
                        }
                        t+=dt;
                    }

                    *newt++ = 5 + ABS(ncoeffx);       // term length
                    *newt++ = EXTRASYMBOL;            // extrasymbol
                    *newt++ = 4;                      // symbol length
                    *newt++ = newid;                  // symbol id
                    *newt++ = 1;                      // power of 1
                    for (int j=0; j<ABS(ncoeffx); j++)// coefficient
                        *newt++ = coeff[j];
                    *newt++ = 0;                      // trailing 0
                    substituted = true;
/*
if ( optim.type == 5 ) {
    int i = (newt - e) - 4;
    MesPrint("  > %a",i,e+3);
}
*/
                }
            }
        }

        if (!substituted) {
#ifdef DEBUG_GREEDY
            MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=false", thetime_str().c_str(),
optim.improve);
#endif
            return false;
        }
/*
if ( optim.type == 5 )
MesPrint ("improve=%d, %c%d%c%c%d)", optim.improve,
    optim.arg1>0?'x':'Z', ABS(optim.arg1)-1,
    optim.type==1 ? '*' : optim.type==4 ? '+' : '-',
    optim.arg2>0?'x':'Z', ABS(optim.arg2)-1);
*/
        // add extra equation (Tnew = x+/-y)
        instr.push_back(newid);    // eqn.nr
        instr.push_back(OPER_ADD); // operator
        instr.push_back(20);       // total length
        instr.push_back(8);        // term length
        instr.push_back(vartypex); // (extra)symbol
        instr.push_back(4);        // symbol length
        instr.push_back(varnumx);  // symbol id
        instr.push_back(1);        // power of 1
        instr.push_back(1);
        instr.push_back(1);        // coefficient of 1
        instr.push_back(3);
        instr.push_back(8);        // term length
        instr.push_back(vartypey); // (extra)symbol
        instr.push_back(4);        // symbol length
        instr.push_back(varnumy);  // symbol id
        instr.push_back(1);        // power of 1
        instr.push_back(1);
        instr.push_back(1);        // coefficient of +/-1
        instr.push_back(3*(optim.type==4?1:-1));
        instr.push_back(0);        // trailing 0
    }
//  #] type 4,5 :
//  #[ trivial :  remove trivial equations of the form Zi = +/-Zj
    vector<int> renum(newid+1, 0);
    bool do_renum=false;

    // vector may be moved when it is extended
    ebegin = &*instr.begin();

    for (int i=0; i<(int)optim.eqnidxs.size(); i++) {
        WORD *e = ebegin + optim.eqnidxs[i];
        WORD *t = e+3;
        if (*t==0) continue;      // empty (removed) equation
        if (*(t+*t)!=0) continue; // more than 1 term
```

```
        if (*t!=8) continue;       // term not of correct form
        if (*(t+4)!=1) continue;  // power != 1
        if (*(t+5)!=1 || *(t+6)!=1) continue; // coeff != +/-1

        // trivial term, so renumber this one
        renum[*e] = SGN(*(t+7)) * (*(t+3) + 1);
        do_renum = true;

        // remove equation
        *t=0;
    }
//  #] trivial :
//  #[ renumbering :

    // there are renumberings to be done, so loop through all equations
    if (do_renum) {
        WORD *eend = ebegin+instr.size();

        for (WORD *e=ebegin; e!=eend; e+=*(e+2)) {
            for (WORD *t=e+3; *t!=0; t+=*t) {
                if (*t == ABS(*(t+*t-1))+1) continue;
                if (*(t+1)==EXTRASYMBOL && renum[*(t+3)]!=0) {
                    *(t+*t-1) *= SGN(renum[*(t+3)]);
                    *(t+3) = ABS(renum[*(t+3)]) - 1;
                }
            }
        }
    }
//  #] renumbering :

#ifdef DEBUG_GREEDY
    MesPrint ("*** [%s, w=%w] DONE: do_optimization : res=true", thetime_str().c_str(), optim.improve);
#endif

    return true;
}

/*
    #] do_optimization :
    #[ partial_factorize :
*/

int partial_factorize (vector<WORD> &instr, int n, int improve) {

#ifdef DEBUG_GREEDY
    MesPrint ("*** [%s, w=%w] CALL: partial_factorize (n=%d)", thetime_str().c_str(), n);
#endif

    GETIDENTITY;

    // get starting positions of instructions
    vector<int> instr_idx(n);
    WORD *ebegin = &*instr.begin();
    WORD *eend = ebegin+instr.size();
    for (WORD *e=ebegin; e!=eend; e+=*(e+2)) {
        instr_idx[*e] = e - ebegin;
    }

    // get reference counts
/*
 *  The next construction replaces the vector construction which is
 *  rather costly for valgrind (and maybe also in normal running)
 */
    int nmax = 2*n;
    WORD *numpar = (WORD *)Malloc1(nmax*sizeof(WORD),"numpar");
    for ( int i = 0; i < nmax; i++ ) numpar[i] = 0;
//  vector<WORD> numpar(n);
    for (WORD *e=ebegin; e!=eend; e+=*(e+2))
        for (WORD *t=e+3; *t!=0; t+=*t) {
            if (*t == ABS(*(t+*t-1))+1) continue;
            if (*(t+1) == EXTRASYMBOL) numpar[*(t+3)]++;
        }

    // find factorizable expressions
    for (int i=0; i<n; i++) {
        WORD *e = &*instr.begin() + instr_idx[i];
        if (*(e+1) != OPER_ADD) continue;

        // count symbol occurrences
        map<WORD,WORD> cnt; // 1-indexed, <0:EXTRASYMBOL, >0:SYMBOL

        for (WORD *t=e+3; *t!=0; t+=*t) {
            if (*t==ABS(*(t+*t-1))+1) continue;

            // count symbols in t
            if (*(t+4)==1)
```

```
                cnt[(*(t+1)==SYMBOL ? 1 : -1) * (*(t+3)+1)]++;

            // count symbols in extrasymbols of t
            if (*(t+1)==EXTRASYMBOL && *(t+4)==1 && numpar[*(t+3)]==1) {
                WORD *t2 = &*instr.begin() + instr_idx[*(t+3)];
                if (*(t2+1) != OPER_MUL) continue;
                for (t2+=3; *t2!=0; t2+=*t2) {
                    if (*t2 == ABS(*(t2+*t2-1))+1) continue;
                    if (*(t2+4)==1)
                        cnt[(*(t2+1)==SYMBOL ? 1 : -1) * (*(t2+3)+1)]++;
                }
            }
        }

        // find most-occurring symbol
        WORD x=0, best=0;
        for (map<WORD,WORD>::iterator it=cnt.begin(); it!=cnt.end(); it++)
            if (it->second > best) { x=it->first; best=it->second; }

        // occurrence>=2 and occurrence>improve, so factorize

        if (best>=2 && best>improve) {
            // initialize new equation (Zi from example above)
            vector<WORD> new_eqn;
            new_eqn.push_back(n);
            new_eqn.push_back(OPER_ADD);
            new_eqn.push_back(0); // length

            WORD dt;
            WORD *newt=e+3;
            for (WORD *t=e+3; *t!=0; t+=dt) {
                dt = *t;
                bool keep=true;

                if (*t!=ABS(*(t+*t-1))+1) {

                    // factorized symbol is in t itself
                    if (*(t+4)==1) {
                        WORD y = (*(t+1)==SYMBOL ? 1 : -1) * (*(t+3)+1);
                        if (y==x) {
                            new_eqn.push_back(*t-4);
                            new_eqn.insert(new_eqn.end(), t+5, t+dt);
                            keep=false;
                        }
                    }

                    // look in extrasymbol of t with ref.count=1
                    if (*(t+1)==EXTRASYMBOL && *(t+4)==1 && numpar[*(t+3)]==1) {
                        WORD *t2 = &*instr.begin() + instr_idx[*(t+3)];
                        if (*(t2+1) == OPER_MUL) {
                            bool has_x=false;
                            for (t2+=3; *t2!=0; t2+=*t2) {
                                if (*t2 == ABS(*(t2+*t2-1))+1) continue;
                                WORD y = (*(t2+1)==SYMBOL ? 1 : -1) * (*(t2+3)+1);
                                // extrasymbol has factorized symbol
                                if (y==x && *(t2+4)==1) {
                                    has_x=true;
                                    // copy remaining part
                                    WORD *tend=t2+*t2;
                                    WORD sign = SGN(*(tend-1));
                                    while (*tend!=0) tend+=*tend;
                                    int dt2 = tend - (t2+*t2);
                                    memmove(t2, t2+*t2, (dt2+1)*sizeof(WORD));
                                    t2 += dt2;
                                    *(t2-1) *= sign;
                                    break;
                                }
                            }
                            if (has_x) {
                                // extrasymbol has x, so add it to new equation
                                keep=false;
                                int thisidx=new_eqn.size();
                                new_eqn.insert(new_eqn.end(), t, t+dt);
                                t2 = &*instr.begin() + instr_idx[*(t+3)] + 3;
                                // if becomes trivial, substitute the term
                                if (*(t2+*t2)==0) {
                                    // it's a number
                                    if (*t2 == ABS(*(t2+*t2-1))+1) {
                                        if (ABS(new_eqn[new_eqn.size()-1])==3 && new_eqn[new_eqn.size()-2]==1 && new_eqn[new_eqn.size()-3]==1) {
                                            // original equation has coefficient of +/-1, so replace it
                                            WORD sign = SGN(new_eqn.back());
                                            new_eqn.erase(new_eqn.begin()+thisidx, new_eqn.end());
                                            new_eqn.insert(new_eqn.end(), t2, t2+*t2);
                                            new_eqn.back() *= sign;
                                            *t2 = 0;
```

```
                                        }
                                        else {
                                            // two non-trivial coefficients, so multiply them
                                            // note: untested code (found no way to trigger it)
                                            UWORD *tmp = NumberMalloc("partial_factorize");
                                            WORD ntmp=0;
                                            MulRat(BHEAD (UWORD *)t2+*t2-ABS(*(t2+*t2-1)),
*(t2+*t2-1),
                                                          (UWORD
*)&*(new_eqn.end()-ABS(new_eqn.back())), new_eqn.back(),
                                                          tmp, &ntmp);
                                            new_eqn.erase(new_eqn.begin()+thisidx, new_eqn.end());
                                            new_eqn.push_back(ABS(ntmp)+1);
                                            new_eqn.insert(new_eqn.end(), tmp, tmp+ABS(ntmp));
                                            NumberFree(tmp,"partial_factorize");
                                            *t2 = 0;
                                        }
                                    }
                                    else if (*(t2+4)==1) {
                                        // it's a variable
                                        new_eqn.back() *= SGN(*(t2+*t2-1));
                                        new_eqn[thisidx+1] = *(t2+1);
                                        new_eqn[thisidx+2] = *(t2+2);
                                        new_eqn[thisidx+3] = *(t2+3);
                                        new_eqn[thisidx+4] = *(t2+4);
                                        *t2 = 0;
                                    }
                                }
                            }
                        }
                    }
                }

                // no x, so copy it
                if (keep) {
                    memmove(newt, t, dt*sizeof(WORD));
                    newt += dt;
                }
            }

            // finalize new equation
            new_eqn.push_back(0);
            new_eqn[2] = new_eqn.size();

            bool empty = newt == e+3;
            if ( n+1 >= nmax ) {
                int i, newnmax = nmax*2;
                WORD *newnumpar = (WORD *)Malloc1(newnmax*sizeof(WORD),"newnumpar");
                for ( i = 0; i < n; i++ ) newnumpar[i] = numpar[i];
                for ( ; i < newnmax; i++ ) newnumpar[i] = 0;
                M_free(numpar,"numpar");
                numpar = newnumpar;
                nmax = newnmax;
            }
//          numpar.push_back(0);
            n++;

            // if original is not empty, add new equation (Zj) to it
            // otherwise replace it later
            if (!empty) {
                *newt++ = 8;
                *newt++ = EXTRASYMBOL;
                *newt++ = 4;
                *newt++ = n;
                *newt++ = 1;
                *newt++ = 1;
                *newt++ = 1;
                *newt++ = 3;
                *newt++ = 0;
            }

            // add new equation to instructions
            instr_idx.push_back(instr.size());
            instr.insert(instr.end(), new_eqn.begin(), new_eqn.end());

            // generate another new equation (Zj=x*Zi)
            new_eqn.clear();
            new_eqn.push_back(n);
            new_eqn.push_back(OPER_MUL);
            new_eqn.push_back(20);
            new_eqn.push_back(8);

            // add factorized symbol
            if (x>0) {
                new_eqn.push_back(SYMBOL);
                new_eqn.push_back(4);
                new_eqn.push_back(x-1);
```

```
                new_eqn.push_back(1);
            }
            else {
                new_eqn.push_back(EXTRASYMBOL);
                new_eqn.push_back(4);
                new_eqn.push_back(-x-1);
                new_eqn.push_back(1);
            }
            new_eqn.push_back(1);
            new_eqn.push_back(1);
            new_eqn.push_back(3);
            new_eqn.push_back(8);
            // add new equation (Zi)
            new_eqn.push_back(EXTRASYMBOL);
            new_eqn.push_back(4);
            new_eqn.push_back(n-1);
            new_eqn.push_back(1);
            new_eqn.push_back(1);
            new_eqn.push_back(1);
            new_eqn.push_back(3);
            new_eqn.push_back(0);

            if (!empty) {
                // add new equation (Zj) to instructions
                instr_idx.push_back(instr.size());
                instr.insert(instr.end(), new_eqn.begin(), new_eqn.end());
                if ( n+1 >= nmax ) {
                    int i, newnmax = nmax*2;
                    WORD *newnumpar = (WORD *)Malloc1(newnmax*sizeof(WORD),"newnumpar");
                    for ( i = 0; i < n; i++ ) newnumpar[i] = numpar[i];
                    for ( ; i < newnmax; i++ ) newnumpar[i] = 0;
                    M_free(numpar,"numpar");
                    numpar = newnumpar;
                    nmax = newnmax;
                }
//              numpar.push_back(0);
                n++;
            }
            else {
                // replace e with Zj
                e = &*instr.begin() + instr_idx[i];
                e[1] = OPER_MUL;
                memcpy(e+3, &new_eqn[3], (new_eqn.size()-3)*sizeof(WORD));
            }

            // decrease i, so this expression is factorized again if possible
            i--;
        }
    }

#ifdef DEBUG_GREEDY
    MesPrint ("*** [%s, w=%w] DONE: partial_factorize (n=%d)", thetime_str().c_str(), n);
#endif
    M_free(numpar,"numpar");
    return n;
}

/*
    #] partial_factorize :
    #[ optimize_greedy :
*/

vector<WORD> optimize_greedy (vector<WORD> instr, LONG time_limit) {

#ifdef DEBUG
    int old_num_oper = count_operators(instr);
    MesPrint ("*** [%s, w=%w] CALL: optimize_greedy(numoper=%d)",
                        thetime_str().c_str(), old_num_oper);
#endif

    LONG start_time = TimeWallClock(1);

    WORD *ebegin = &*instr.begin();
    WORD *eend = ebegin+instr.size();

    // store final equation, since it must be the last equation later
    int final_eqn_idx = 0;
    int next_eqn = 0;

    for (WORD *e=ebegin; e!=eend; e+=*(e+2)) {
        next_eqn = *e + 1;
        final_eqn_idx = e-ebegin;
    }
    // optimize instructions
    while (TimeWallClock(1)-start_time < time_limit) {
        int old_next_eqn = next_eqn;

```

```
        // find optimizations
        vector<optimization> optim = find_optimizations(instr);

        // add_eqnidxs contains modified equations, which might have to be updated later again
        vector<int> add_eqnidxs;

        // number of optimizations to do
        int num_do_optims = max(AO.Optimize.greedyminnum,
    (int)optim.size()*AO.Optimize.greedymaxperc/100);

        // if best improvement is one, do all optimizations
        int best_improve=0;
        for (int i=0; i<(int)optim.size(); i++)
            best_improve = max(best_improve, optim[i].improve);
        if (best_improve <= 1)
            num_do_optims = MAXPOSITIVE;
        // do a number of optimizations
        while (optim.size() > 0 && num_do_optims-- > 0) {

            // find best optimization
            int best=0;
            best_improve=0;
            for (int i=0; i<(int)optim.size(); i++)
                if (optim[i].improve > best_improve) {
                    best=i;
                    best_improve=optim[i].improve;
                }

            // add extra equations
            for (int i=0; i<(int)add_eqnidxs.size(); i++)
                optim[best].eqnidxs.push_back(add_eqnidxs[i]);

            // do optimization, update next_eqn if successful
            int next_idx = instr.size();
            if (do_optimization(optim[best], instr, next_eqn)) {
                next_eqn++;
                add_eqnidxs.push_back(next_idx);
            }

            optim.erase(optim.begin()+best);
        }

        // partially factorize with improve >= best_improve
        next_eqn = partial_factorize(instr, next_eqn, best_improve);

        // check whether nothing has changed
        if (next_eqn == old_next_eqn) break;
    }

    // add final equation to the back (must be by definition)
    instr.push_back(next_eqn);
    instr.insert(instr.end(), instr.begin()+final_eqn_idx+1,
instr.begin()+final_eqn_idx+instr[final_eqn_idx+2]);

    // removed original final equation
    instr[final_eqn_idx+3] = 0;

    // remove extra zeroes
    WORD *t = &instr[0];

    ebegin = &*instr.begin();
    eend = ebegin+instr.size();
    int de=0;

    for (WORD *e=ebegin; e!=eend; e+=de) {
        de = *(e+2);
        int n=3;
        while (*(e+n) != 0) n+=*(e+n);
        n++;
        memmove (t, e, n*sizeof(WORD));
        *(t+2) = n;
        t += n;
    }

    instr.resize(t - &instr[0]);

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: optimize_greedy(numoper=%d) : numoper=%d",
                        thetime_str().c_str(), old_num_oper, count_operators(instr));
#endif

    return instr;
}

/*
    #] optimize_greedy :
    #[ recycle_variables :
```

```
*/

vector<WORD> recycle_variables (const vector<WORD> &all_instr) {

#ifdef DEBUG_MORE
    MesPrint ("*** [%s, w=%w] CALL: recycle_variables", thetime_str().c_str());
#endif

    // get starting positions of instructions
    vector<const WORD *> instr;
    const WORD *tbegin = &*all_instr.begin();
    const WORD *tend = tbegin+all_instr.size();
    for (const WORD *t=tbegin; t!=tend; t+=*(t+2))
        instr.push_back(t);
    int n = instr.size();

    // determine with expressions are connected, how many intermediate
    // are needed (assuming it's a expression tree instead of a DAG) and
    // sort the leaves such that you need a minimal number of variables
    vector<int> vars_needed(n);
    vector<bool> vis(n,false);
    vector<vector<int> > conn(n);

    stack<int> s;
    s.push(n);

    while (!s.empty()) {
        int i=s.top(); s.pop();
        if (i>0) {
            i--;
            if (vis[i]) continue;
            vis[i]=true;
            s.push(-(i+1));

            // find all connections
            for (const WORD *t=instr[i]+3; *t!=0; t+=*t)
                if (*t!=1+ABS(*(t+*t-1)) && *(t+1)==EXTRASYMBOL) {
                    int k = *(t+3);
                    conn[i].push_back(k);
                    s.push(k+1);
                }
        }
        else {
            i=-i-1;

            // sort the childs w.r.t. needed variables
            vector<pair<int,int> > need;
            for (int j=0; j<(int)conn[i].size(); j++)
                need.push_back(make_pair(vars_needed[conn[i][j]], conn[i][j]));

            // keep the comma expression in proper order
            if (*(instr[i]+1) != OPER_COMMA)
                sort(need.rbegin(), need.rend());

            vars_needed[i] = 1;
            for (int j=0; j<(int)need.size(); j++) {
                vars_needed[i] = max(vars_needed[i], need[j].first+j);
                conn[i][j] = need[j].second;
            }
        }
    }

    // order the instructions in depth-first order and determine the first
    // and last occurrences of variables
    vector<int> order, first(n,0), last(n,0);
    vis = vector<bool>(n,false);
    s.push(n);

    while (!s.empty()) {

        int i=s.top(); s.pop();

        if (i>0) {
            i--;
            if (vis[i]) continue;
            vis[i]=true;
            s.push(-(i+1));
            for (int j=(int)conn[i].size()-1; j>=0; j--)
                s.push(conn[i][j]+1);
        }
        else {
            i=-i-1;
            first[i] = last[i] = order.size();
            order.push_back(i);
            for (int j=0; j<(int)conn[i].size(); j++) {
                int k = conn[i][j];
                last[k] = max(last[k], first[i]);
```

```
            }
        }
    }

    // find the renumbering to recycled variables, where at any time the
    // lowest-indexed variable that can be used is chosen
    int numvar=0;
    set<int> var;
    vector<int> renum(n);

    for (int i=0; i<(int)order.size(); i++) {
        for (int j=0; j<(int)conn[order[i]].size(); j++) {
            int k = conn[order[i]][j];
            if (last[k] == i) var.insert(renum[k]);
        }

        if (var.empty()) var.insert(numvar++);
        renum[order[i]] = *var.begin(); var.erase(var.begin());
    }

    // put the number of variables used in a preprocessor variable

    // generate new instructions with the renumbering
    vector<WORD> newinstr;

    for (int i=0; i<(int)order.size(); i++) {
        int x = order[i];
        int j = newinstr.size();
        newinstr.insert(newinstr.end(), instr[x], instr[x]+*(instr[x]+2));

        newinstr[j] = renum[newinstr[j]];

        for (WORD *t=&newinstr[j+3]; *t!=0; t+=*t)
            if (*t!=1+ABS(*(t+*t-1)) && *(t+1)==EXTRASYMBOL)
                *(t+3) = renum[*(t+3)];
    }

#ifdef DEBUG_MORE
    MesPrint ("*** [%s, w=%w] DONE: recycle_variables", thetime_str().c_str());
#endif

    return newinstr;
}

/*
    #] recycle_variables :
    #[ optimize_expression_given_Horner :
*/
void optimize_expression_given_Horner () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: optimize_expression_given_Horner", thetime_str().c_str());
#endif

    GETIDENTITY;

    // initialize timer
    LONG start_time = TimeWallClock(1);
    LONG time_limit = 100 * AO.Optimize.greedytimelimit / (AO.Optimize.horner == O_MCTS ?
AO.Optimize.mctsnumkeep : 1);
    if (time_limit == 0) time_limit=MAXPOSITIVE;

    // pick a Horner scheme from the list
    LOCK(optimize_lock);
    vector<WORD> Horner_scheme = optimize_best_Horner_schemes.back();
    optimize_best_Horner_schemes.pop_back();
    UNLOCK(optimize_lock);

//  if ( ( AO.Optimize.debugflags&2 ) == 2 ) {
//      MesPrint ("Scheme: %a",Horner_scheme.size(),&(Horner_scheme[0]));
//  }

    // apply Horner scheme
    vector<WORD> tree = Horner_tree(optimize_expr, Horner_scheme);

    // generate instructions, eventually with CSE
    vector<WORD> instr;

    if (AO.Optimize.method == O_CSE || AO.Optimize.method == O_CSEGREEDY)
        instr = generate_instructions(tree, true);
    else
        instr = generate_instructions(tree, false);
    if (AO.Optimize.method == O_CSEGREEDY || AO.Optimize.method == O_GREEDY) {
        instr = merge_operators(instr, false);
        instr = optimize_greedy(instr, time_limit-(TimeWallClock(1)-start_time));
        instr = merge_operators(instr, true);
```

```
        instr = optimize_greedy(instr, time_limit-(TimeWallClock(1)-start_time));
    }
    instr = merge_operators(instr, true);

    // recycle the temporary variables
    instr = recycle_variables(instr);

    // determine the quality of the code and possibly update the best code
    int num_oper = count_operators(instr);

    LOCK(optimize_lock);
    if (num_oper < optimize_best_num_oper) {
        optimize_num_vars = Horner_scheme.size();
        optimize_best_num_oper = num_oper;
        optimize_best_instr = instr;
        optimize_best_vars = vector<WORD>(AN.poly_vars, AN.poly_vars+AN.poly_num_vars);
    }
    UNLOCK(optimize_lock);

  // clean poly_vars, that are allocated by Horner_tree
    AN.poly_num_vars = 0;
    M_free(AN.poly_vars,"poly_vars");

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: optimize_expression_given_Horner", thetime_str().c_str());
#endif
}

/*
    #] optimize_expression_given_Horner :
    #[ PF_optimize_expression_given_Horner :
*/
#ifdef WITHMPI

// Initialization.
void PF_optimize_expression_given_Horner_master_init () {
    // Nothing to do for now.
}

// Wait for an idle slave and return the process number.
int PF_optimize_expression_given_Horner_master_next() {

    // Find an idle slave.
    int next;
    PF_LongSingleReceive(PF_ANY_SOURCE, PF_OPT_HORNER_MSGTAG, &next, NULL);
    return next;

}

// The main function on the master.
void PF_optimize_expression_given_Horner_master () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: PF_optimize_expression_given_Horner_master",
thetime_str().c_str());
#endif

    // pick a Horner scheme from the list
    vector<WORD> Horner_scheme = optimize_best_Horner_schemes.back();
    optimize_best_Horner_schemes.pop_back();

    // Find an idle slave.
    int next = PF_optimize_expression_given_Horner_master_next();

    // Send a new task to the slave.
    PF_PrepareLongSinglePack();
    int len = Horner_scheme.size();
    PF_LongSinglePack(&len, 1, PF_INT);
    PF_LongSinglePack(&Horner_scheme[0], len, PF_WORD);
    PF_LongSingleSend(next, PF_OPT_HORNER_MSGTAG);

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: PF_optimize_expression_given_Horner_master",
thetime_str().c_str());
#endif

}

// Wait for all the slaves to finish their tasks.
void PF_optimize_expression_given_Horner_master_wait () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: PF_optimize_expression_given_Horner_master_wait",
thetime_str().c_str());
#endif

    // Wait for all the slaves.
```

```
    for (int i = 1; i < PF.numtasks; i++) {
        int next = PF_optimize_expression_given_Horner_master_next();
        // Send a null task.
        PF_PrepareLongSinglePack();
        int len = 0;
        PF_LongSinglePack(&len, 1, PF_INT);
        PF_LongSingleSend(next, PF_OPT_HORNER_MSGTAG);
    }

    // Combine the result from all the slaves.
    optimize_best_num_oper = INT_MAX;
    for (int i = 1; i < PF.numtasks; i++) {
        PF_LongSingleReceive(PF_ANY_SOURCE, PF_OPT_COLLECT_MSGTAG, NULL, NULL);

        int len;

        // The first integer is the number of operations.
        PF_LongSingleUnpack(&len, 1, PF_INT);

        if (len >= optimize_best_num_oper) continue;

        // Update the best result.
        optimize_best_num_oper = len;
        PF_LongSingleUnpack(&len, 1, PF_INT);
        optimize_best_instr.resize(len);
        PF_LongSingleUnpack(&optimize_best_instr[0], len, PF_WORD);
        PF_LongSingleUnpack(&len, 1, PF_INT);
        optimize_best_vars.resize(len);
        PF_LongSingleUnpack(&optimize_best_vars[0], len, PF_WORD);

        optimize_num_vars = optimize_best_vars.size();  // TODO
    }

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: PF_optimize_expression_given_Horner_master_wait",
thetime_str().c_str());
#endif

}

// The main function on the slaves.
void PF_optimize_expression_given_Horner_slave () {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: PF_optimize_expression_given_Horner_slave",
thetime_str().c_str());
#endif

    optimize_best_Horner_schemes.clear();
    optimize_best_num_oper = INT_MAX;

    int dummy = 0;
    int len;

    for (;;) {
        // Ask the master for the next task.
        PF_PrepareLongSinglePack();
        PF_LongSinglePack(&dummy, 1, PF_INT);
        PF_LongSingleSend(MASTER, PF_OPT_HORNER_MSGTAG);

        // Get a task from the master.
        PF_LongSingleReceive(MASTER, PF_OPT_HORNER_MSGTAG, NULL, NULL);

        // Length of the task.
        PF_LongSingleUnpack(&len, 1, PF_INT);

        // No task remains.
        if (len == 0) break;

        // Perform the given task.
        optimize_best_Horner_schemes.push_back(vector<WORD>());
        vector<WORD> &Horner_scheme = optimize_best_Horner_schemes.back();
        Horner_scheme.resize(len);
        PF_LongSingleUnpack(&Horner_scheme[0], len, PF_WORD);
        optimize_expression_given_Horner ();
    }

    // Send the result to the master.
    PF_PrepareLongSinglePack();
    PF_LongSinglePack(&optimize_best_num_oper, 1, PF_INT);
    if (optimize_best_num_oper != INT_MAX) {
        len = optimize_best_instr.size();
        PF_LongSinglePack(&len, 1, PF_INT);
        PF_LongSinglePack(&optimize_best_instr[0], len, PF_WORD);
        len = optimize_best_vars.size();
        PF_LongSinglePack(&len, 1, PF_INT);
        PF_LongSinglePack(&optimize_best_vars[0], len, PF_WORD);
```

```
    }
    PF_LongSingleSend(MASTER, PF_OPT_COLLECT_MSGTAG);

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: PF_optimize_expression_given_Horner_slave", thetime_str().c_str());
#endif

}

#endif
/*
    #] PF_optimize_expression_given_Horner :
    #[ generate_output :
*/

void generate_output (const vector<WORD> &instr, int exprnr, int extraoffset, const vector<vector<WORD> > &brackets) {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: generate_output", thetime_str().c_str());
#endif

    GETIDENTITY;
    vector<WORD> output;

    // one-indexed instead of zero-indexed
    extraoffset++;
    int num = 0;
    int maxsize = (int)instr.size();
    for (int i=0; i<maxsize; i+=instr[i+2]) num++;
    int *tstart = (int *)Malloc1(num*sizeof(int),"nplaces");
    num = 0;
    for (int i=0; i<maxsize; i+=instr[i+2]) tstart[num++] = i;
    for (int j=0; j<num; j++) {
        int i = tstart[j];

        // copy arguments
        WCOPY(AT.WorkPointer, &instr[i+3], (instr[i+2]-3));

        // update maximal variable number
        AO.OptimizeResult.maxvar = MaX(AO.OptimizeResult.maxvar, instr[i]+extraoffset);

        // renumber symbols and extrasymbols to correct values
        for (WORD *t=AT.WorkPointer; *t!=0; t+=*t) {
            if (*t == ABS(*(t+*t-1))+1) continue;
            if (*(t+1)==SYMBOL) *(t+3) = optimize_best_vars[*(t+3)];
            if (*(t+1)==EXTRASYMBOL) {
                *(t+1) = SYMBOL;
                *(t+3) = MAXVARIABLES-*(t+3)-extraoffset;
            }
        }

        // reformat multiplication instructions, since their current
        // format is "expr.nr OPER_MUL length arguments", which differs
        // from Form's format for a product of symbols
        if (instr[i+1]==OPER_MUL) {

            WORD *now=AT.WorkPointer+1;
            int dt;
            bool coeff=false;
            int coeff_sign=1;

            for (WORD *t=AT.WorkPointer; *t!=0; t+=dt) {
                dt = *t;
                if (*t == ABS(*(t+*t-1))+1) {
                    // copy coefficient
                    memmove(AT.WorkPointer+instr[i+2], t, *t*sizeof(WORD));
                    coeff = true;
                }
                else {
                    // move symbol
                    int n = *(t+2);
                    memmove(now, t+1, n*sizeof(WORD));
                    now += n;
                    coeff_sign *= SGN(*(t+dt-1));
                }
            }

            if (coeff) {
                // add existing coefficient
                int n = *(AT.WorkPointer + instr[i+2]) - 1;
                memmove(now, AT.WorkPointer+instr[i+2]+1, n*sizeof(WORD));
                now += n;
            }
            else {
                // add coefficient of one
```

```
                *now++=1;
                *now++=1;
                *now++=3;
            }

            *(now-1) *= coeff_sign;

            *AT.WorkPointer = now - AT.WorkPointer;
            *now++ = 0;
        }

        // in the case of simultaneous optimization of expressions, add the
        // brackets to the final expression
        if (instr[i+1]==OPER_COMMA) {
            WORD *start = AT.WorkPointer + instr[i+2];
            WORD *now = start;
            int b=0;
            for (const WORD *t=AT.WorkPointer; *t!=0; t+=*t) {
                if ( ( brackets[b].size() != 0 ) && ( brackets[b][0] == 0 ) ) break;
                *now++ = *t + brackets[b].size();
                if (brackets[b].size() != 0) {
                    memcpy(now, &brackets[b][0], brackets[b].size()*sizeof(WORD));
                    now += brackets[b].size();
                }
                memcpy(now, t+1, (*t-1)*sizeof(WORD));
                now += *t-1;
                b++;
            }
            *now++ = 0;
            memmove(AT.WorkPointer, start, (now-start)*sizeof(WORD));
        }

        // add the number of the extra symbol; if it is the last one,
        // replace the extra symbol number with the expression number
        if (i+instr[i+2]<(int)instr.size())
            output.push_back(instr[i]+extraoffset);
        else {
            output.push_back(-(exprnr+1));
        }

        // add code for this symbol
        int n=0;
        while (*(AT.WorkPointer+n)!=0)
            n += *(AT.WorkPointer+n);
        n++;
        output.insert(output.end(), AT.WorkPointer, AT.WorkPointer+n);
    }

    // trailing zero
    output.push_back(0);

    // clear buffer
    if (AO.OptimizeResult.code != NULL)
        M_free(AO.OptimizeResult.code, "optimize output");

    M_free(tstart,"nplaces");

    // copy buffer
    AO.OptimizeResult.codesize = output.size();
    AO.OptimizeResult.code = (WORD *)Malloc1(output.size()*sizeof(WORD), "optimize output");
    memcpy(AO.OptimizeResult.code, &output[0], output.size()*sizeof(WORD));

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: generate_output", thetime_str().c_str());
#endif
}

/*
    #] generate_output :
    #[ generate_expression :
*/

WORD generate_expression (WORD exprnr) {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: generate_expression", thetime_str().c_str());
#endif

    GETIDENTITY;

    WORD *oldWorkPointer = AT.WorkPointer;

    CBUF *C = cbuf+AC.cbufnum;
    WORD *term = AT.WorkPointer, oldcurexpr = AR.CurExpr;
    POSITION position;
    EXPRESSIONS e = Expressions+exprnr;
    SetScratch(AR.infile,&(e->onfile));
```

```
    if ( GetTerm(BHEAD term) <= 0 ) {
        MesPrint("Expression %d has problems in scratchfile",exprnr);
        Terminate(-1);
    }
    SeekScratch(AR.outfile,&position);
    e->onfile = position;
    if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) {
        MesPrint("Expression %d has problems in output scratchfile",exprnr);
        Terminate(-1);
    }

    AR.CurExpr = exprnr;
    NewSort(BHEAD0);

    // scan for the original expression (marked by *t<0) and give the
    // terms to Generator
    WORD *t = AO.OptimizeResult.code;
    {
        WORD old = AR.Cnumlhs; AR.Cnumlhs = 0;
        // We can use the remaining part of the WorkSpace for every term that
        // goes through Generator. We have WorkSpace problems here if we use
        // the (modified) AT.WorkPointer every time in the loop.
        WORD* currentWorkPointer = AT.WorkPointer;
        while (*t!=0) {
            bool is_expr = *t < 0;
            t++;
            while (*t!=0) {
                if (is_expr) {
                    memcpy(currentWorkPointer, t, *t*sizeof(WORD));
                    Generator(BHEAD currentWorkPointer, C->numlhs);
                }
                t+=*t;
            }
            t++;
        }
        AR.Cnumlhs = old;
    }

    // final sorting
    if (EndSort(BHEAD NULL,0) < 0) {
        LowerSortLevel();
        Terminate(-1);
    }

    AT.WorkPointer = oldWorkPointer;
    AR.CurExpr = oldcurexpr;

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: generate_expression", thetime_str().c_str());
#endif

    return 0;
}

/*
    #] generate_expression :
    #[ optimize_print_code :
*/

void optimize_print_code (int print_expr) {

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: optimize_print_code", thetime_str().c_str());
#endif
    if ( ( AO.Optimize.debugflags & 1 ) != 0 ) {
/*
 *      The next code is for debugging purposes. We may want the statements
 *      in reverse order to substitute them all back.
 *      Jan used a Mathematica program to do this. Here we make that
 *          Format Ox,debugflag=1;
 *      Creates reverse order during printing.
 *      All we have to do is put id in front of the statements. This is done
 *      in PrintExtraSymbol.
 */
        WORD *t = AO.OptimizeResult.code;
        WORD num = 0;   // First we count the number of objects.
        while (*t!=0) {
            num++;
            t++; while (*t!=0) t+=*t; t++;
        }
        WORD **tstart = (WORD **)Malloc1(num*sizeof(WORD *),"print objects");
        t = AO.OptimizeResult.code; num = 0; // Now we get the addresses
        while (*t!=0) {
            tstart[num++] = t;
            t++; while (*t!=0) t+=*t; t++;
        }
```

```
        // Flip the addresses
        int halfnum = num/2;
        for (int i=0; i<halfnum; i++) { swap(tstart[i],tstart[num-1-i]); }
        for ( int i = 0; i < num; i++ ) {
            t = tstart[i];
            if (*t > 0)
                PrintExtraSymbol(*t, t+1, EXTRASYMBOL);
            else if (print_expr)
                PrintExtraSymbol(-*t-1, t+1, EXPRESSIONNUMBER);
        }
        CBUF *C = cbuf + AM.sbufnum;
        if (C->numrhs >= AO.OptimizeResult.minvar)
            PrintSubtermList(AO.OptimizeResult.minvar, C->numrhs);
    }
    else {
        // print extra symbols from ConvertToPoly in optimization
        CBUF *C = cbuf + AM.sbufnum;
        if (C->numrhs >= AO.OptimizeResult.minvar)
            PrintSubtermList(AO.OptimizeResult.minvar, C->numrhs);

        WORD *t = AO.OptimizeResult.code;
        while (*t!=0) {
            if (*t > 0) {
                PrintExtraSymbol(*t, t+1, EXTRASYMBOL);
            }
            else if (print_expr)
                PrintExtraSymbol(-*t-1, t+1, EXPRESSIONNUMBER);
            t++;
            while (*t!=0) t+=*t;
            t++;
        }
    }

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: optimize_print_code", thetime_str().c_str());
#endif
}

/*
    #] optimize_print_code :
    #[ Optimize :
*/

int Optimize (WORD exprnr, int do_print) {

#if defined(WITHMPI) && (defined(DEBUG) || defined(DEBUG_MORE) || defined(DEBUG_MCTS) || defined(DEBUG_GREEDY))
    // set AS.printflag negative temporary.
    struct save_printflag {
        save_printflag() {
            oldprintflag = AS.printflag;
            AS.printflag = -1;
        }
        ~save_printflag() {
            AS.printflag = oldprintflag;
        }
        int oldprintflag;
    } save_printflag_;
#endif

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] CALL: Optimize", thetime_str().c_str());
    MesPrint ("*** %"); PrintRunningTime();
#endif

#ifdef WITHPTHREADS
    optimize_lock = dummylock;
#endif

    AO.OptimizeResult.minvar = (cbuf + AM.sbufnum)->numrhs + 1;

    if (get_expression(exprnr) < 0) return -1;
    vector<vector<WORD> > brackets = get_brackets();

#ifdef DEBUG
#ifdef WITHMPI
        if (PF.me == MASTER)
#endif
        MesPrint ("*** runtime after preparing the expression: %"); PrintRunningTime();
#endif

    if (optimize_expr[0]==0 ||
            (optimize_expr[optimize_expr[0]]==0 && optimize_expr[0]==ABS(optimize_expr[optimize_expr[0]-1])+1) ||
            (optimize_expr[optimize_expr[0]]==0 && optimize_expr[0]==8 &&
             optimize_expr[5]==1 && optimize_expr[6]==1 && ABS(optimize_expr[7])==3)) {
        if (AO.OptimizeResult.code != NULL)
```

```
            M_free(AO.OptimizeResult.code, "optimize output");

        // zero terms or one trivial term (number or +/-variable), so no
        // optimization, so copy expression; special case because without
        // operators the optimization crashes
        AO.OptimizeResult.code = (WORD *)Malloc1((optimize_expr[0]+3)*sizeof(WORD), "optimize output");
        AO.OptimizeResult.code[0] = -(exprnr+1);
        memcpy(AO.OptimizeResult.code+1, optimize_expr, (optimize_expr[0]+1)*sizeof(WORD));
        AO.OptimizeResult.code[optimize_expr[0]+2] = 0;
    }
    else {
        // find Horner scheme(s)
        optimize_best_Horner_schemes.clear();
        if (AO.Optimize.horner == O_OCCURRENCE) {
            if (AO.Optimize.hornerdirection != O_BACKWARD)
                optimize_best_Horner_schemes.push_back(occurrence_order(optimize_expr, false));
            if (AO.Optimize.hornerdirection != O_FORWARD)
                optimize_best_Horner_schemes.push_back(occurrence_order(optimize_expr, true));
        }
        else {
            if (AO.Optimize.horner == O_SIMULATED_ANNEALING) {
                optimize_best_Horner_schemes.push_back(simulated_annealing());
            }
            else {
                mcts_best_schemes.clear();
                if ( AO.inscheme ) {
                    optimize_best_Horner_schemes.push_back(vector<WORD>(AO.schemenum));
                    for ( int i=0; i < AO.schemenum; i++ )
                        optimize_best_Horner_schemes[0][i] = AO.inscheme[i];
                }
                else {
                    for ( int i = 0; i < AO.Optimize.mctsnumrepeat; i++ )
                        find_Horner_MCTS();
                    // generate results
                    for (set<pair<int,vector<WORD> > >::iterator i=mcts_best_schemes.begin(); i!=mcts_best_schemes.end(); i++) {
                        optimize_best_Horner_schemes.push_back(i->second);
#ifdef DEBUG_MCTS
                        MesPrint ("{%a} -> %d",i->second.size(), &i->second[0], i->first);
#endif
                    }
                }
                // clear the tree by making a new empty one.
                mcts_root = tree_node();
            }
        }
#ifdef DEBUG
#ifdef WITHMPI
        if (PF.me == MASTER)
#endif
        MesPrint ("*** runtime after Horner: %"); PrintRunningTime();
#endif

#ifdef WITHMPI
        if (PF.me == MASTER ) {
            PF_optimize_expression_given_Horner_master_init();
#endif

        // find best Horner scheme and results
        optimize_best_num_oper = INT_MAX;

        int imax = (int)optimize_best_Horner_schemes.size();

        for (int i=0; i<imax; i++) {
#if defined(WITHPTHREADS)
            if (AM.totalnumberofthreads > 1)
                optimize_expression_given_Horner_threaded();
            else
#elif defined(WITHMPI)
            if (PF.numtasks > 1)
                PF_optimize_expression_given_Horner_master();
            else
#endif
                optimize_expression_given_Horner();
        }

#ifdef WITHMPI
            PF_optimize_expression_given_Horner_master_wait();
        }
        else {
            if (PF.numtasks > 1)
                PF_optimize_expression_given_Horner_slave();
        }
#endif

#ifdef WITHPTHREADS
```

```
        MasterWaitAll();
#endif
        // format results, then print it (for "Print") or modify
        // expression (for "#Optimize")
#ifdef WITHMPI
        if (PF.me == MASTER)
#endif
        generate_output(optimize_best_instr, exprnr, cbuf[AM.sbufnum].numrhs, brackets);
#ifdef WITHMPI
        else {
            // non-null dummy code for slaves
            if (AO.OptimizeResult.code == NULL) {
                AO.OptimizeResult.code = (WORD *)Malloc1(sizeof(WORD), "optimize output");
            }
        }
#endif
    }

#ifdef WITHMPI
    if (PF.me == MASTER) {
        PF_PreparePack();
        PF_Pack(&AO.OptimizeResult.minvar, 1, PF_WORD);
        PF_Pack(&AO.OptimizeResult.maxvar, 1, PF_WORD);
    }
    PF_Broadcast();
    if (PF.me != MASTER) {
        PF_Unpack(&AO.OptimizeResult.minvar, 1, PF_WORD);
        PF_Unpack(&AO.OptimizeResult.maxvar, 1, PF_WORD);
    }
#endif

    // set preprocessor variables
    char str[100];
    snprintf (str,100,"%d",AO.OptimizeResult.minvar);
    PutPreVar((UBYTE *)"optimminvar_",(UBYTE *)str,0,1);
    snprintf (str,100,"%d",AO.OptimizeResult.maxvar);
    PutPreVar((UBYTE *)"optimmaxvar_",(UBYTE *)str,0,1);

    if (do_print) {
#ifdef WITHMPI
        if (PF.me == MASTER)
#endif
        optimize_print_code(1);
        ClearOptimize();
    }
    else {
#ifdef WITHMPI
        if (PF.me == MASTER)
#endif
        generate_expression(exprnr);
    }

#ifdef WITHMPI
    if (PF.me == MASTER) {
#endif

    if ( AO.Optimize.printstats > 0 ) {
        char str[20];
        MesPrint("");
        count_operators(optimize_expr,true);
        int numop = count_operators(optimize_best_instr,true);
        snprintf(str,20,"%d",numop);
        PutPreVar((UBYTE *)"optimvalue_",(UBYTE *)str,0,1);
    }
    else {
        char str[20];
        int numop = count_operators(optimize_best_instr,false);
        snprintf(str,20,"%d",numop);
        PutPreVar((UBYTE *)"optimvalue_",(UBYTE *)str,0,1);
    }

    if ( ( AO.Optimize.schemeflags&1 ) == 1 ) {
        GETIDENTITY
        UBYTE *OutScr, *Out, *old1 = AO.OutputLine, *old2 = AO.OutFill;
        int i, sym;
        AO.OutputLine = AO.OutFill = (UBYTE *)AT.WorkPointer;
        FiniLine();
        OutScr = (UBYTE *)AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) /2;
        TokenToLine((UBYTE *)" Scheme selected: ");
        for ( i = 0; i < optimize_num_vars; i++ ) {
            Out = OutScr;
            sym = optimize_best_vars[i];
            if ( i > 0 ) TokenToLine((UBYTE *)",");
            if ( sym < NumSymbols ) {
                StrCopy(FindSymbol(sym),OutScr);
/*              StrCopy(VARNAME(symbols,sym),OutScr); */
            }
```

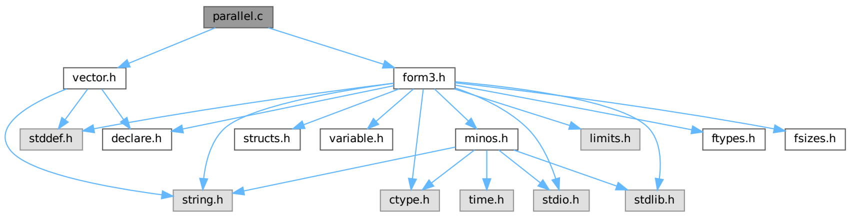
```
            else {
                Out = StrCopy((UBYTE *)AC.extrasym,Out);
                if ( AC.extrasymbols == 0 ) {
                    Out = NumCopy((MAXVARIABLES-sym),Out);
                    Out = StrCopy((UBYTE *)"_",Out);
                }
                else if ( AC.extrasymbols == 1 ) {
                    Out = AddArrayIndex((MAXVARIABLES-sym),Out);
                }
            }
            TokenToLine(OutScr);
        }
        TokenToLine((UBYTE *)";");
        FiniLine();
        AO.OutFill = old2;
        AO.OutputLine = old1;
    }

    {
        GETIDENTITY
        UBYTE *OutScr, *Out, *outstring = 0;
        int i, sym;
        AO.OutputLine = AO.OutFill = (UBYTE *)AT.WorkPointer;
        OutScr = (UBYTE *)AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) /2;
        for ( i = 0; i < optimize_num_vars; i++ ) {
            Out = OutScr;
            sym = optimize_best_vars[i];
            if ( sym < NumSymbols ) {
                StrCopy(FindSymbol(sym),OutScr);
/*              StrCopy(VARNAME(symbols,sym),OutScr); */
            }
            else {
                Out = StrCopy((UBYTE *)AC.extrasym,Out);
                if ( AC.extrasymbols == 0 ) {
                    Out = NumCopy((MAXVARIABLES-sym),Out);
                    Out = StrCopy((UBYTE *)"_",Out);
                }
                else if ( AC.extrasymbols == 1 ) {
                    Out = AddArrayIndex((MAXVARIABLES-sym),Out);
                }
            }
            outstring = AddToString(outstring,OutScr,1);
        }
        if ( outstring == 0 ) {
            PutPreVar((UBYTE *)"optimscheme_",(UBYTE *)"",0,1);
        }
        else {
            PutPreVar((UBYTE *)"optimscheme_",(UBYTE *)outstring,0,1);
            M_free(outstring,"AddToString");
        }
    }

#ifdef WITHMPI
    }

    // synchronize optimvalue_ and optimscheme_
    if ( PF.me == MASTER ) {
        UBYTE *value;
        int bytes;

        PF_PrepareLongMultiPack();

        value = GetPreVar((UBYTE *)"optimvalue_", WITHERROR);
        bytes = strlen((char *)value);
        PF_LongMultiPack(&bytes, 1, PF_INT);
        PF_LongMultiPack(value, bytes, PF_BYTE);

        value = GetPreVar((UBYTE *)"optimscheme_", WITHERROR);
        bytes = strlen((char *)value);
        PF_LongMultiPack(&bytes, 1, PF_INT);
        PF_LongMultiPack(value, bytes, PF_BYTE);
    }
    PF_LongMultiBroadcast();
    if ( PF.me != MASTER ) {
        static vector<UBYTE> prevarbuf;
        UBYTE *value;
        int bytes;

        PF_LongMultiUnpack(&bytes, 1, PF_INT);
        prevarbuf.reserve(bytes + 1);
        value = &prevarbuf[0];
        PF_LongMultiUnpack(value, bytes, PF_BYTE);
        value[bytes] = '\0';  // null terminator
        PutPreVar((UBYTE *)"optimvalue_", value, NULL, 1);

        PF_LongMultiUnpack(&bytes, 1, PF_INT);
        prevarbuf.reserve(bytes + 1);
```

```
        value = &prevarbuf[0];
        PF_LongMultiUnpack(value, bytes, PF_BYTE);
        value[bytes] = '\0';  // null terminator
        PutPreVar((UBYTE *)"optimscheme_", value, NULL, 1);
    }
#endif

    // cleanup
    M_free(optimize_expr,"LoadOptim");

#ifdef DEBUG
    MesPrint ("*** [%s, w=%w] DONE: Optimize", thetime_str().c_str());
#endif

    return 0;
}

/*
    #] Optimize :
    #[ ClearOptimize :
*/

int ClearOptimize()
{
    char str[20];
    WORD numexpr, *w;
    int error = 0;
    if ( AO.OptimizeResult.code != NULL ) {
        M_free(AO.OptimizeResult.code, "optimize output");
        AO.OptimizeResult.code = NULL;
        AO.OptimizeResult.codesize = 0;
        PutPreVar((UBYTE *)"optimminvar_",(UBYTE *)("0"),0,1);
        PutPreVar((UBYTE *)"optimmaxvar_",(UBYTE *)("0"),0,1);
        PruneExtraSymbols(AO.OptimizeResult.minvar-1);
        cbuf[AM.sbufnum].numrhs = AO.OptimizeResult.minvar-1;
        AO.OptimizeResult.minvar = AO.OptimizeResult.maxvar = 0;
    }
    if ( AO.OptimizeResult.nameofexpr != NULL ) {
/*
        We have to pick up the expression by its name. Numbers may change.
        Note that this requires that when we overwrite an expression, we
        check that it is not an optimized expression. See execute.c and
        comexpr.c
*/
        if ( GetName(AC.exprnames,AO.OptimizeResult.nameofexpr,&numexpr,NOAUTO) != CEXPRESSION ) {
            MesPrint("@Internal error while clearing optimized expression %s
",AO.OptimizeResult.nameofexpr);
            Terminate(-1);
        }
        M_free(AO.OptimizeResult.nameofexpr, "optimize expression name");
        AO.OptimizeResult.nameofexpr = NULL;
        w = &(Expressions[numexpr].status);
        *w = SetExprCases(DROP,1,*w);
        if ( *w < 0 ) error = 1;
    }
    snprintf (str,20,"%d",cbuf[AM.sbufnum].numrhs);
    PutPreVar(AM.oldnumextrasymbols,(UBYTE *)str,0,1);
    PutPreVar((UBYTE *)"optimvalue_",(UBYTE *)("0"),0,1);
    PutPreVar((UBYTE *)"optimscheme_",(UBYTE *)("0"),0,1);
    return(error);
}

/*
    #] ClearOptimize :
*/
```

4.94 parallel.c File Reference

```
#include "form3.h"
#include "vector.h"
```

Include dependency graph for parallel.c:



Data Structures

- struct NoDe
- struct dollar_buf
- struct bufIPstruct

Macros

- #define PRINTFBUF(TEXT, TERM, SIZE) {}
- #define SWAP(x, y)
- #define PACK_LONG(p, n)
- #define UNPACK_LONG(p, n)
- #define CHECK(condition) _CHECK(condition, __FILE__, __LINE__)
- #define _CHECK(condition, file, line) __CHECK(condition, file, line)
- #define __CHECK(condition, file, line)
- #define DBGOUT(lv1, lv2, a) do { if (lv1 >= lv2) { printf a; fflush(stdout); } } while (0)
- #define DBGOUT_NINTERMS(lv, a)
- #define PF_STATS_SIZE 5
- #define PF_SNDFILEBUFSIZE 4096
- #define recvBuffer logBuffer /∗ (master) The buffer for receiving messages. ∗/

Typedefs

- typedef struct NoDe NODE
- typedef struct bufIPstruct **bufIPstruct_t**

Functions

- LONG PF_RealTime (int)
- int PF_LibInit (int ∗, char ∗∗∗)
- int PF_LibTerminate (int)
- int PF_Probe (int ∗)
- int PF_RecvWbuf (WORD ∗, LONG ∗, int ∗)
- int PF_IRecvRbuf (PF_BUFFER ∗, int, int)
- int PF_WaitRbuf (PF_BUFFER ∗, int, LONG ∗)
- int PF_RawSend (int dest, void ∗buf, LONG l, int tag)
- LONG PF_RawRecv (int ∗src, void ∗buf, LONG thesize, int ∗tag)
- int PF_RawProbe (int ∗src, int ∗tag, int ∗bytesize)
- int PF_EndSort (void)

- WORD PF_Deferred (WORD ∗term, WORD level)
- int PF_Processor (EXPRESSIONS e, WORD i, WORD LastExpression)
- int PF_Init (int ∗argc, char ∗∗∗argv)
- int PF_Terminate (int errorcode)
- LONG PF_GetSlaveTimes (void)
- LONG PF_BroadcastNumber (LONG x)
- void PF_BroadcastBuffer (WORD ∗∗buffer, LONG ∗length)
- int PF_BroadcastString (UBYTE ∗str)
- int PF_BroadcastPreDollar (WORD ∗∗dbuffer, LONG ∗newsize, int ∗numterms)
- int PF_CollectModifiedDollars (void)
- int PF_BroadcastModifiedDollars (void)
- int PF_BroadcastRedefinedPreVars (void)
- int PF_StoreInsideInfo (void)
- int PF_RestoreInsideInfo (void)
- int PF_BroadcastCBuf (int bufnum)
- int PF_BroadcastExpFlags (void)
- int PF_BroadcastExpr (EXPRESSIONS e, FILEHANDLE ∗file)
- int PF_BroadcastRHS (void)
- int PF_InParallelProcessor (void)
- int PF_SendFile (int to, FILE ∗fd)
- int PF_RecvFile (int from, FILE ∗fd)
- void PF_MLock (void)
- void PF_MUnlock (void)
- LONG PF_WriteFileToFile (int handle, UBYTE ∗buffer, LONG size)
- void PF_FlushStdOutBuffer (void)
- void PF_FreeErrorMessageBuffers (void)

Variables

- PARALLELVARS PF

4.94.1 Detailed Description

Message passing library independent functions of parform

This file contains functions needed for the parallel version of form3 these functions need no real link to the message passing libraries, they only need some interface dependent preprocessor definitions (check parallel.h). So there still need two different objectfiles to be compiled for mpi and pvm!

Definition in file parallel.c.

4.94.2 Macro Definition Documentation

4.94.2.1 PRINTFBUF

```
#define PRINTFBUF(
              TEXT,
              TERM,
              SIZE ) {}
```

Definition at line 117 of file parallel.c.

4.94.2.2 SWAP

```
#define SWAP(
            x,
            y )
```

Value:
```
do { \
    char swap_tmp__[sizeof(x) == sizeof(y) ? (int)sizeof(x) : -1]; \
    memcpy(swap_tmp__, &y, sizeof(x)); \
    memcpy(&y, &x, sizeof(x)); \
    memcpy(&x, swap_tmp__, sizeof(x)); \
} while (0)
```

Swaps the variables *x* and *y*. If sizeof(x) != sizeof(y) then a compilation error will occur. A set of memcpy calls with constant sizes is expected to be inlined by the optimisation.

Definition at line 124 of file parallel.c.

4.94.2.3 PACK_LONG

```
#define PACK_LONG(
            p,
            n )
```

Value:
```
do { \
    *(p)++ = (UWORD)((ULONG)(n) & (ULONG)WORDMASK); \
    *(p)++ = (UWORD)(((ULONG)(n) » BITSINWORD) & (ULONG)WORDMASK); \
} while (0)
```

Packs a LONG value *n* to a WORD buffer *p*.

Definition at line 135 of file parallel.c.

4.94.2.4 UNPACK_LONG

```
#define UNPACK_LONG(
            p,
            n )
```

Value:
```
do { \
    (n) = (LONG)((((ULONG)(p)[1] & (ULONG)WORDMASK) « BITSINWORD) | ((ULONG)(p)[0] & (ULONG)WORDMASK));
  \
    (p) += 2; \
} while (0)
```

Unpacks a LONG value *n* from a WORD buffer *p*.

Definition at line 144 of file parallel.c.

4.94.2.5 CHECK

```
#define CHECK(
            condition ) _CHECK(condition, __FILE__, __LINE__)
```

A simple check for unrecoverable errors.

Definition at line 153 of file parallel.c.

4.94.2.6 _CHECK

```
#define _CHECK(
        condition,
        file,
        line ) __CHECK(condition, file, line)
```

Definition at line 154 of file parallel.c.

4.94.2.7 __CHECK

```
#define __CHECK(
        condition,
        file,
        line )
```

Value:

```
do { \
    if ( !(condition) ) { \
        Error0("Fatal error at " file ":" #line); \
        Terminate(-1); \
    } \
} while (0)
```

Definition at line 155 of file parallel.c.

4.94.2.8 DBGOUT

```
#define DBGOUT(
        lv1,
        lv2,
        a ) do { if ( lv1 >= lv2 ) { printf a; fflush(stdout); } } while (0)
```

Definition at line 166 of file parallel.c.

4.94.2.9 DBGOUT_NINTERMS

```
#define DBGOUT_NINTERMS(
        lv,
        a )
```

Definition at line 169 of file parallel.c.

4.94.2.10 PF_STATS_SIZE

```
#define PF_STATS_SIZE 5
```

Definition at line 178 of file parallel.c.

4.94.2.11 PF_SNDFILEBUFSIZE

```
#define PF_SNDFILEBUFSIZE 4096
```

Definition at line 4212 of file parallel.c.

4.94.2.12 recvBuffer

```
#define recvBuffer logBuffer /* (master) The buffer for receiving messages. */
```

Definition at line 4320 of file parallel.c.

4.94.3 Typedef Documentation

4.94.3.1 NODE

```
typedef struct NoDe NODE
```

A node for the tree of losers in the final sorting on the master.

4.94.4 Function Documentation

4.94.4.1 PF_RealTime()

```
LONG PF_RealTime (
            int i )
```

Returns the realtime in 1/100 sec. as a LONG.

Parameters

i	the timer will be reset if i == 0.

Returns

the real elapsed time in 1/100 second.

Definition at line 101 of file mpi.c.

Referenced by PF_Init().

4.94.4.2 PF_LibInit()

```
int PF_LibInit (
            int * argcp,
            char *** argvp )
```

Performs all library dependent initializations.

Parameters

argcp	pointer to the number of arguments.
argvp	pointer to the arguments.

Returns

0 if OK, nonzero on error.

Definition at line 123 of file mpi.c.

Referenced by PF_Init().

4.94.4.3 PF_LibTerminate()

```
int PF_LibTerminate (
            int error )
```

Exits mpi, when there is an error either indicated or happening, returnvalue is 1, else returnvalue is 0.

Parameters

error	an error code.

Returns

0 if OK, nonzero on error.

Definition at line 209 of file mpi.c.

Referenced by PF_Terminate().

4.94.4.4 PF_Probe()

```
int PF_Probe (
            int * src )
```

Probes the next incoming message. If src == PF_ANY_SOURCE this operation is blocking, otherwise nonblocking.

Parameters

in,out	*src*	the source process number. In output, the process number of actual found source.

Returns

the tag value of the next incoming message if found, 0 if a nonblocking probe (input src != PF_ANY_SOURCE) did not find any messages. A negative returned value indicates an error.

Definition at line 230 of file mpi.c.

4.94.4.5 PF_RecvWbuf()

```
int PF_RecvWbuf (
            WORD * b,
            LONG * s,
            int * src )
```

Blocking receive of a `WORD` buffer.

Parameters

out	*b*	the buffer to store the received data.
in,out	*s*	the size of the buffer. The output value is the actual size of the received data.
in,out	*src*	the source process number. The output value is the process number of actual source.

Returns

the received message tag. A negative value indicates an error.

Definition at line 337 of file mpi.c.

4.94.4.6 PF_IRecvRbuf()

```
int PF_IRecvRbuf (
            PF_BUFFER * r,
            int bn,
            int from )
```

Posts nonblocking receive for the active receive buffer. The buffer is filled from `full` to `stop`.

Parameters

r	the `PF_BUFFER` struct for the nonblocking receive.
bn	the index of the cyclic buffer.
from	the source process number.

Returns

0 if OK, nonzero on error.

Definition at line 366 of file mpi.c.

4.94.4.7 PF_WaitRbuf()

```
int PF_WaitRbuf (
            PF_BUFFER * r,
            int bn,
            LONG * size )
```

Waits for the buffer *bn* to finish a pending nonblocking receive. It returns the received tag and in `*size` the number of field received. If the receive is already finished, just return the flag and size, else wait for it to finish, but also check for other pending receives.

Parameters

	r	the `PF_BUFFER` struct for the pending nonblocking receive.
	bn	the index of the cyclic buffer.
out	*size*	the actual size of received data.

Returns

the received message tag. A negative value indicates an error.

Definition at line 400 of file mpi.c.

4.94.4.8 PF_RawSend()

```
int PF_RawSend (
            int dest,
            void * buf,
            LONG l,
            int tag )
```

Sends *l* bytes from *buf* to *dest*. Returns 0 on success, or -1.

Parameters

dest	the destination process number.
buf	the send buffer.
l	the size of data in the send buffer in bytes.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 463 of file mpi.c.

Referenced by PF_MUnlock(), and PF_SendFile().

4.94.4.9 PF_RawRecv()

```
LONG PF_RawRecv (
            int * src,
            void * buf,
            LONG thesize,
            int * tag )
```

Receives not more than *thesize* bytes from *src*, returns the actual number of received bytes, or -1 on failure.

Parameters

in,out	*src*	the source process number. In output, that of the actual received message.
out	*buf*	the receive buffer.
	thesize	the size of the receive buffer in bytes.
out	*tag*	the message tag of the actual received message.

Returns

the actual sizeof received data in bytes, or -1 on failure.

Definition at line 484 of file mpi.c.

Referenced by PF_RecvFile().

4.94.4.10 PF_RawProbe()

```
int PF_RawProbe (
            int * src,
            int * tag,
            int * bytesize )
```

Probes an incoming message.

Parameters

in,out	*src*	the source process number. In output, that of the actual received message.
in,out	*tag*	the message tag. In output, that of the actual received message.
out	*bytesize*	the size of incoming data in bytes.

Returns

0 if OK, nonzero on error.

Definition at line 508 of file mpi.c.

4.94.4.11 PF_EndSort()

```
int PF_EndSort (
            void )
```

Finishes a master sorting with collecting terms from slaves. Called by EndSort().

If this is not the masterprocess, just initialize the sendbuffers and return 0, else PF_EndSort() sends the rest of the terms in the sendbuffer to the next slave and a dummy message to all slaves with tag PF_ENDSORT_MSGTAG. Then it receives the sorted terms, sorts them using a recursive 'tree of losers' (PF_GetLoser()) and writes them to the outputfile.

Returns

1 if the sorting on the master was done. 0 if EndSort() still must perform a regular sorting because it is not at the ground level or not on the master or in the sequential mode or in the InParallel mode. -1 if an error occurred.

Remarks

The slaves will send the sorted terms back to the master in the regular sorting (after the initialization of the send buffer in PF_EndSort()). See PutOut() and FlushOut().

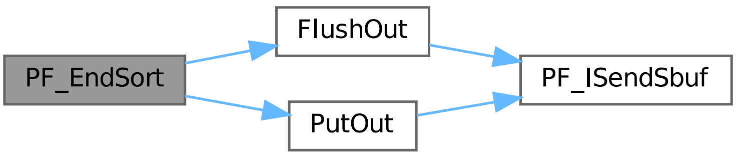
This function has been changed such that when it returns 1, AM.S0->TermsLeft is set correctly. But AM.S0->GenTerms is not set: it will be set after collecting the statistics from the slaves at the end of PF_Processor(). (TU 30 Jun 2011)

Definition at line 864 of file parallel.c.

References FlushOut(), and PutOut().

Referenced by EndSort().

Here is the call graph for this function:



4.94.4.12 PF_Deferred()

```
WORD PF_Deferred (
            WORD * term,
            WORD level )
```

Replaces Deferred() on the slaves.

Parameters

term	the term that must be multiplied by the contents of the current bracket.
level	the compiler level.

Returns

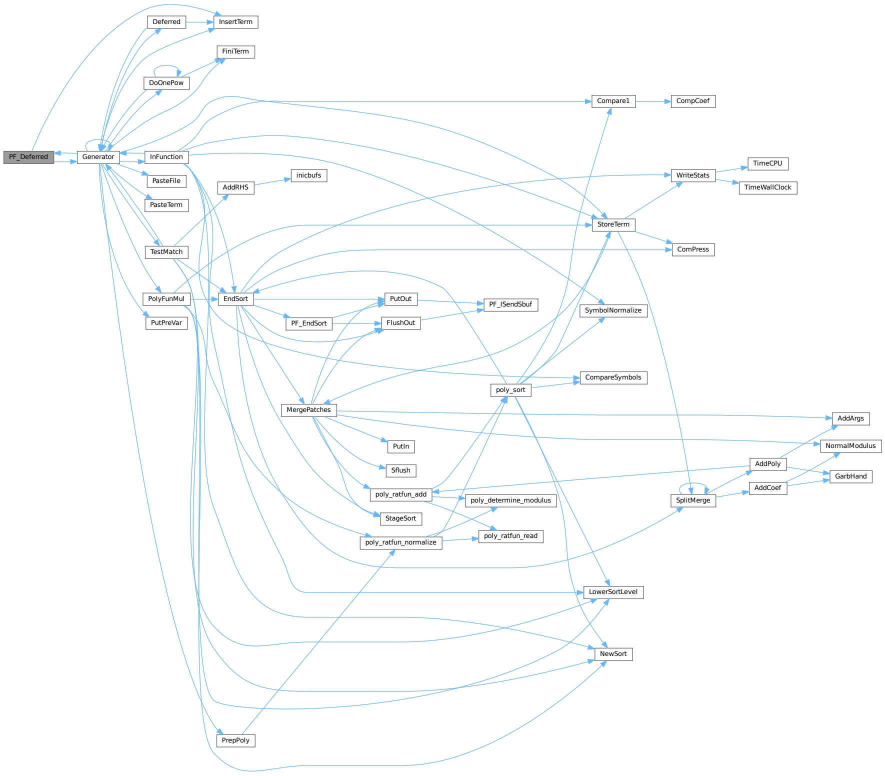
0 if OK, nonzero on error.

Definition at line 1208 of file parallel.c.

References Generator(), and InsertTerm().

Referenced by Generator().

Here is the call graph for this function:



4.94.4.13 PF_Processor()

```
int PF_Processor (
            EXPRESSIONS e,
            WORD i,
            WORD LastExpression )
```

Replaces parts of Processor() on the masters and slaves. On the master PF_Processor() is responsible for proper distribution of terms from the input file to the slaves. On the slaves it calls Generator() for all the terms that this process gets, but PF_GetTerm() gets terms from the master (not directly from infile).

Parameters

e	The pointer to the current expression.
i	The index for the current expression.
LastExpression	The flag indicating whether it is the last expression.

Returns

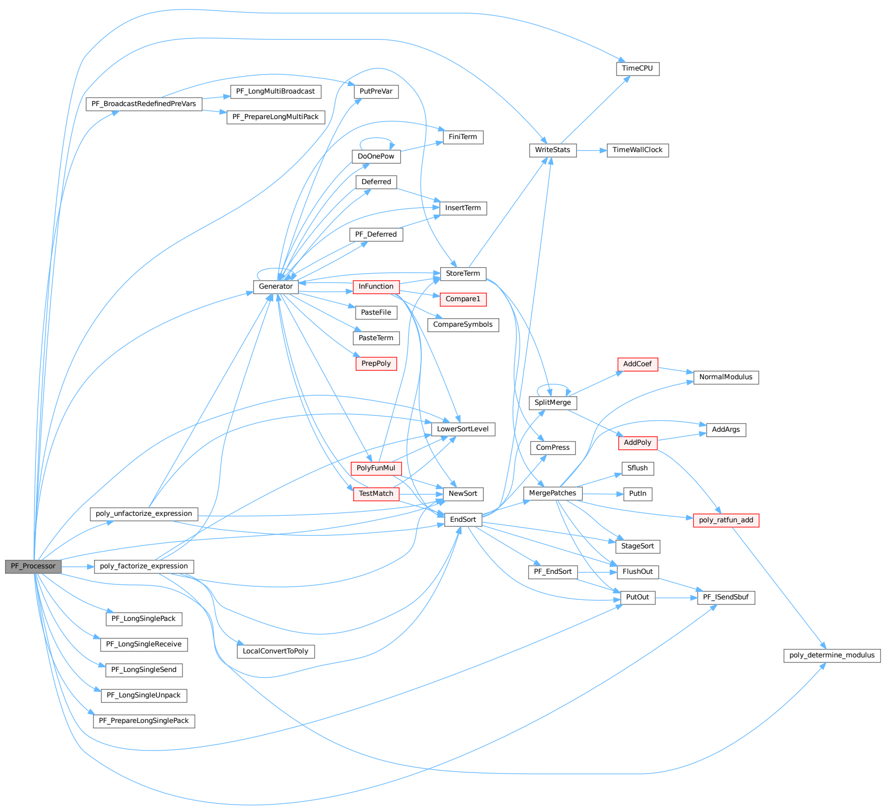
0 if OK, nonzero on error.

Definition at line 1540 of file parallel.c.

References EndSort(), Generator(), LowerSortLevel(), NewSort(), PACK_LONG, PF_BroadcastRedefinedPreVars(), PF_ISendSbuf(), PF_LongSinglePack(), PF_LongSingleReceive(), PF_LongSingleSend(), PF_LongSingleUnpack(), PF_PrepareLongSinglePack(), poly_factorize_expression(), poly_unfactorize_expression(), PutOut(), StoreTerm(), SWAP, TimeCPU(), and WriteStats().

Referenced by Processor().

Here is the call graph for this function:



4.94.4.14 PF_Init()

```
int PF_Init (
            int * argc,
            char *** argv )
```

All the library independent stuff. PF_LibInit() should do all library dependent initializations.

Parameters

argc	pointer to the number of arguments.
argv	pointer to the arguments.

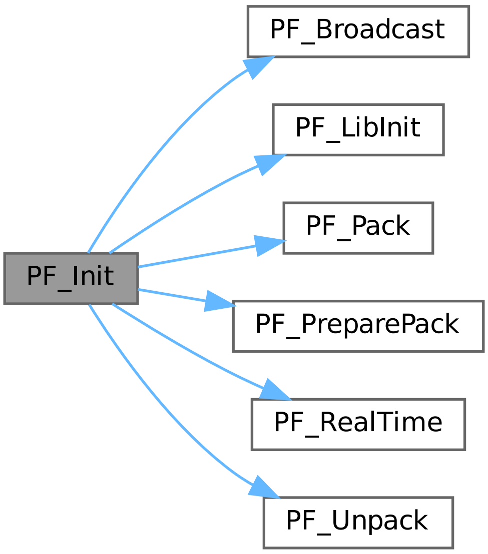
Returns

0 if OK, nonzero on error.

Definition at line 1963 of file parallel.c.

References PF_Broadcast(), PF_LibInit(), PF_Pack(), PF_PreparePack(), PF_RealTime(), and PF_Unpack().

Here is the call graph for this function:



4.94.4.15 PF_Terminate()

```
int PF_Terminate (
        int errorcode )
```

Performs the finalization of ParFORM. To be called by Terminate().

Parameters

error	an error code.

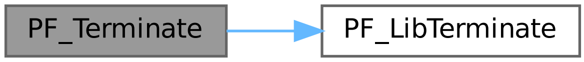
Returns

0 if OK, nonzero on error.

Definition at line 2058 of file parallel.c.

References PF_LibTerminate().

Here is the call graph for this function:



4.94.4.16 PF_GetSlaveTimes()

```
LONG PF_GetSlaveTimes (
            void )
```

Returns the total CPU time of all slaves together. This function must be called on the master and all slaves.

Returns

on the master, the sum of CPU times on all slaves.

Definition at line 2074 of file parallel.c.

References TimeCPU().

Here is the call graph for this function:

4.94.4.17 PF_BroadcastNumber()

```
LONG PF_BroadcastNumber (
            LONG x )
```

Broadcasts a LONG value from the master to the all slaves.

Parameters

x	the number to be broadcast (set on the master).

Returns

the synchronised result.

Definition at line 2094 of file parallel.c.

References PF_Bcast().

Referenced by DoCheckpoint(), PF_BroadcastBuffer(), and StartVariables().

Here is the call graph for this function:

4.94.4.18 PF_BroadcastBuffer()

```
void PF_BroadcastBuffer (
            WORD ** buffer,
            LONG * length )
```

Broadcasts a buffer from the master to all the slaves.

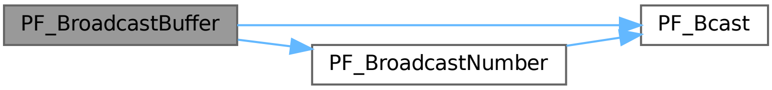
Parameters

in,out	*buffer*	on the master, the buffer to be broadcast. On the slaves, the buffer will be allocated if the length is greater than 0. The caller must free it.
in,out	*length*	on the master, the length of the buffer to be broadcast. On the slaves, it receives the length of transferred buffer. The actual transfer occurs only if the length is greater than 0.

Definition at line 2121 of file parallel.c.

References PF_Bcast(), and PF_BroadcastNumber().

Here is the call graph for this function:



4.94.4.19 PF_BroadcastString()

```
int PF_BroadcastString (
            UBYTE * str )
```

Broadcasts a string from the master to all slaves.

Parameters

in,out	*str*	The pointer to a null-terminated string.

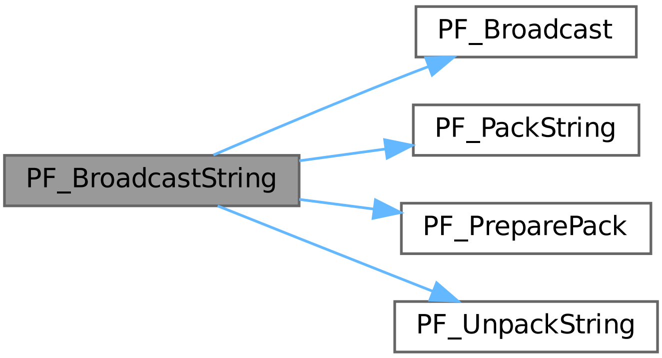
Returns

0 if OK, nonzero on error.

Definition at line 2163 of file parallel.c.

References PF_Broadcast(), PF_PackString(), PF_PreparePack(), and PF_UnpackString().

Here is the call graph for this function:



4.94.4.20 PF_BroadcastPreDollar()

```
int PF_BroadcastPreDollar (
            WORD ** dbuffer,
            LONG * newsize,
            int * numterms )
```

Broadcasts dollar variables set as a preprocessor variables. Only the master is able to make an assignment like #$a=g; where g is an expression: only the master has an access to the expression. So, the master broadcasts the result to slaves.

The result is in ∗dbuffer of the size is ∗newsize (in number of WORDs), +1 for trailing zero. For slave newsize and numterms are output parameters.

Parameters

in,out	*dbuffer*	the buffer for a dollar variable.
in,out	*newsize*	the size of the dollar variable in WORDs.
in,out	*numterms*	the number of terms in the dollar variable.

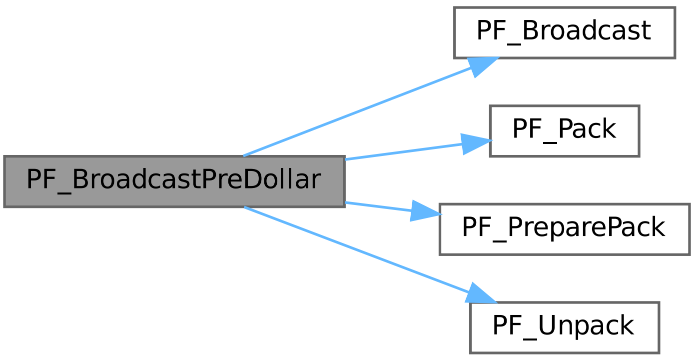
Returns

0 if OK, nonzero on error.

Definition at line 2218 of file parallel.c.

References PF_Broadcast(), PF_Pack(), PF_PreparePack(), and PF_Unpack().

Here is the call graph for this function:



4.94.4.21 PF_CollectModifiedDollars()

```
int PF_CollectModifiedDollars (
            void )
```

Combines modified dollar variables on the all slaves, and store them into those on the master.

The potentially modified dollar variables are given in PotModdollars, and the number of them is given by NumPot←Moddollars.

The current module could be executed in parallel only if all potentially modified variables are listed in ModOptdollars, otherwise the module was switched to the sequential mode.

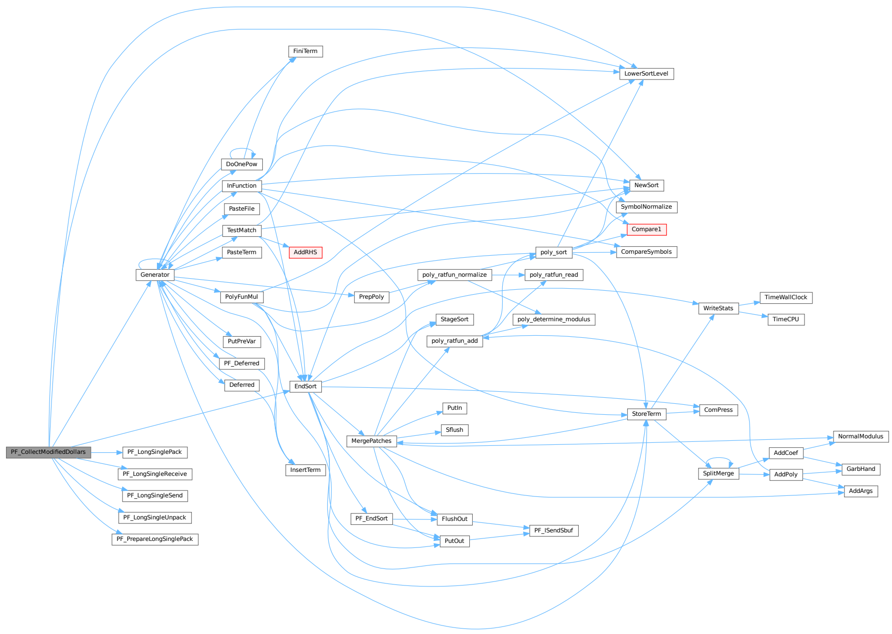
Returns

0 if OK, nonzero on error.

Definition at line 2506 of file parallel.c.

References EndSort(), Generator(), LowerSortLevel(), NewSort(), PF_LongSinglePack(), PF_LongSingleReceive(), PF_LongSingleSend(), PF_LongSingleUnpack(), PF_PrepareLongSinglePack(), VectorInit, VectorPtr, VectorReserve, and VectorSize.

Here is the call graph for this function:



4.94.4.22 PF_BroadcastModifiedDollars()

```
int PF_BroadcastModifiedDollars (
            void )
```

Broadcasts modified dollar variables on the master to the all slaves.

The potentially modified dollar variables are given in PotModdollars, and the number of them is given by NumPot↩ Moddollars.

The current module could be executed in parallel only if all potentially modified variables are listed in ModOptdollars, otherwise the module was switched to the sequential mode. In either cases, we need to broadcast them.

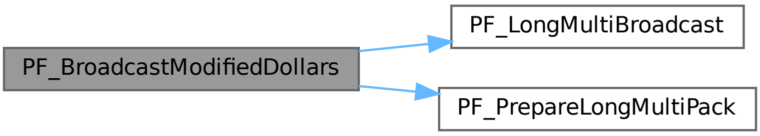
Returns

0 if OK, nonzero on error.

Definition at line 2785 of file parallel.c.

References PF_LongMultiBroadcast(), and PF_PrepareLongMultiPack().

Here is the call graph for this function:



4.94.4.23 PF_BroadcastRedefinedPreVars()

```
int PF_BroadcastRedefinedPreVars (
            void  )
```

Broadcasts preprocessor variables, which were changed by the Redefine statements in the current module, from the master to the all slaves.

The potentially redefined preprocessor variables are given in AC.pfirstnum, and the number of them is given by AC.numpfirstnum. For an actually redefined variable, the corresponding value in AC.inputnumbers is non-negative.

Returns

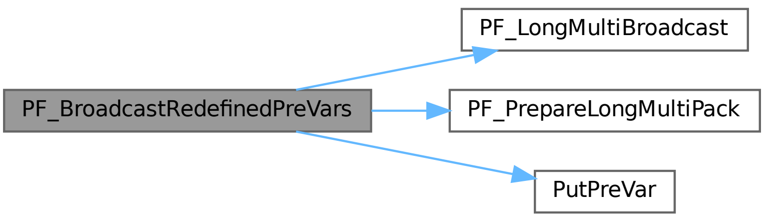
0 if OK, nonzero on error.

Definition at line 3002 of file parallel.c.

References PF_LongMultiBroadcast(), PF_PrepareLongMultiPack(), PutPreVar(), VectorPtr, and VectorReserve.

Referenced by PF_Processor().

Here is the call graph for this function:



4.94.4.24 PF_StoreInsideInfo()

```
int PF_StoreInsideInfo (
            void )
```

Definition at line 3092 of file parallel.c.

4.94.4.25 PF_RestoreInsideInfo()

```
int PF_RestoreInsideInfo (
            void )
```

Definition at line 3118 of file parallel.c.

4.94.4.26 PF_BroadcastCBuf()

```
int PF_BroadcastCBuf (
            int bufnum )
```

Broadcasts a compiler buffer specified by *bufnum* from the master to the all slaves.

Parameters

bufnum	The index of the compiler buffer to be broadcast.

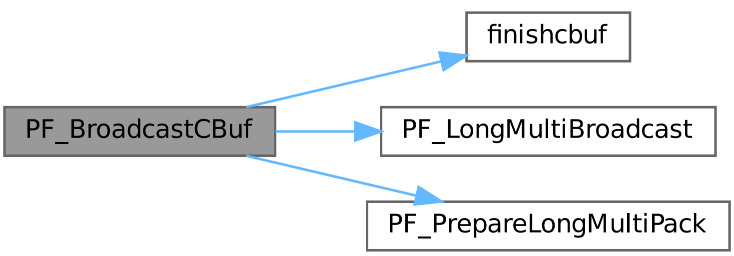
Returns

0 if OK, nonzero on error.

Definition at line 3144 of file parallel.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::BufferSize, CbUf::CanCommu, CbUf::dimension, finishcbuf(), CbUf::lhs, CbUf::numdum, CbUf::NumTerms, PF_LongMultiBroadcast(), PF_PrepareLongMultiPack(), CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Here is the call graph for this function:



4.94.4.27 PF_BroadcastExpFlags()

```
int PF_BroadcastExpFlags (
            void )
```

Broadcasts AR.expflags and several properties of each expression, e.g., e->vflags, from the master to all slaves.

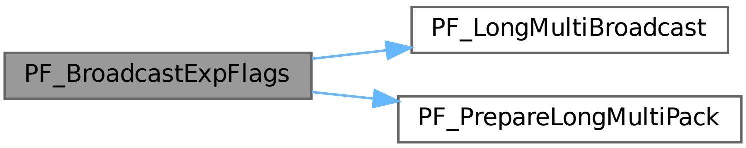
Returns

0 if OK, nonzero on error.

Definition at line 3255 of file parallel.c.

References PF_LongMultiBroadcast(), and PF_PrepareLongMultiPack().

Here is the call graph for this function:



4.94.4.28 PF_BroadcastExpr()

```
int PF_BroadcastExpr (
            EXPRESSIONS e,
            FILEHANDLE * file )
```

Broadcasts an expression from the master to the all slaves.

Parameters

e	The expression to be broadcast.
file	The file in which the expression is sitting.

Returns

0 if OK, nonzero on error.

Definition at line 3549 of file parallel.c.

Referenced by PF_BroadcastRHS().

4.94.4.29 PF_BroadcastRHS()

```
int PF_BroadcastRHS (
            void )
```

Broadcasts expressions appearing in the right-hand side from the master to the all slaves.

Returns

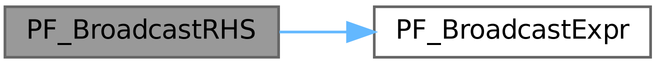
0 if OK, nonzero on error.

Definition at line 3577 of file parallel.c.

References PF_BroadcastExpr().

Referenced by Processor().

Here is the call graph for this function:



4.94.4.30 PF_InParallelProcessor()

```
int PF_InParallelProcessor (
            void )
```

Processes expressions in the InParallel mode, i.e., dividing expressions marked by partodo over the slaves.

Returns

0 if OK, nonzero on error.

Definition at line 3624 of file parallel.c.

Referenced by Processor().

4.94.4.31 PF_SendFile()

```
int PF_SendFile (
            int to,
            FILE * fd )
```

Sends a file to the process specified by *to*.

Parameters

to	the destination process number.
fd	the file to be sent.

Returns

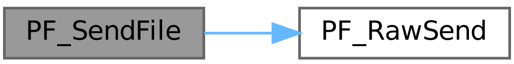
the size of sent data in bytes, or -1 on error.

Definition at line 4221 of file parallel.c.

References PF_RawSend().

Referenced by DoCheckpoint().

Here is the call graph for this function:



4.94.4.32 PF_RecvFile()

```
int PF_RecvFile (
            int from,
            FILE * fd )
```

Receives a file from the process specified by *from*.

Parameters

from	the source process number.
fd	the file to save the received data.

Returns

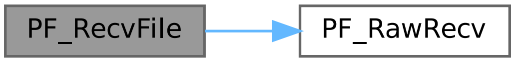
the size of received data in bytes, or -1 on error.

Definition at line 4259 of file parallel.c.

References PF_RawRecv().

Referenced by DoCheckpoint().

Here is the call graph for this function:



4.94.4.33 PF_MLock()

```
void PF_MLock (
            void  )
```

A function called by MLOCK(ErrorMessageLock) for slaves.

Definition at line 4340 of file parallel.c.

References VectorClear.

4.94.4.34 PF_MUnlock()

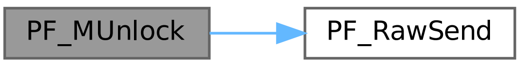
```
void PF_MUnlock (
            void  )
```

A function called by MUNLOCK(ErrorMessageLock) for slaves.

Definition at line 4356 of file parallel.c.

References PF_RawSend(), VectorEmpty, VectorPtr, and VectorSize.

Here is the call graph for this function:



4.94.4.35 PF_WriteFileToFile()

```
LONG PF_WriteFileToFile (
            int handle,
            UBYTE * buffer,
            LONG size )
```

Replaces WriteFileToFile() on the master and slaves.

It copies the given buffer into internal buffers if called between MLOCK(ErrorMessageLock) and MUNLOCK(Error↵ MessageLock) for slaves and handle is StdOut or LogHandle, otherwise calls WriteFileToFile().

Parameters

handle	a file handle that specifies the output.
buffer	a pointer to the source buffer containing the data to be written.
size	the size of data to be written in bytes.

Returns

the actual size of data written to the output in bytes.

Definition at line 4385 of file parallel.c.

References VectorClear, VectorPtr, VectorPushBacks, and VectorSize.

4.94.4.36 PF_FlushStdOutBuffer()

```
void PF_FlushStdOutBuffer (
            void )
```

Explicitly Flushes the buffer for the standard output on the master, which is used if PF_ENABLE_STDOUT_↩
BUFFERING is defined.

Definition at line 4479 of file parallel.c.

References VectorClear, VectorPtr, and VectorSize.

4.94.4.37 PF_FreeErrorMessageBuffers()

```
void PF_FreeErrorMessageBuffers (
            void )
```

Frees the buffers allocated for the synchronized output.

Currently, not used anywhere, but could be used in PF_Terminate().

Definition at line 4615 of file parallel.c.

References VectorFree.

4.94.5 Variable Documentation

4.94.5.1 PF

```
PARALLELVARS PF
```

Definition at line 90 of file parallel.c.

4.95 parallel.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 *   for the community.
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ includes :
*/
#include "form3.h"
#include "vector.h"

/*
#define PF_DEBUG_BCAST_LONG
#define PF_DEBUG_BCAST_BUF
#define PF_DEBUG_BCAST_PREDOLLAR
#define PF_DEBUG_BCAST_RHSEXPR
#define PF_DEBUG_BCAST_DOLLAR
#define PF_DEBUG_BCAST_PREVAR
#define PF_DEBUG_BCAST_CBUF
#define PF_DEBUG_BCAST_EXPRFLAGS
#define PF_DEBUG_REDUCE_DOLLAR
*/

/* mpi.c */
LONG PF_RealTime(int);
int PF_LibInit(int*, char***);
int PF_LibTerminate(int);
int PF_Probe(int*);
int PF_RecvWbuf(WORD*,LONG*,int*);
int PF_IRecvRbuf(PF_BUFFER*,int,int);
int PF_WaitRbuf(PF_BUFFER *,int,LONG *);
int PF_RawSend(int dest, void *buf, LONG l, int tag);
LONG PF_RawRecv(int *src,void *buf,LONG thesize,int *tag);
int PF_RawProbe(int *src, int *tag, int *bytesize);

/* Private functions */

static int PF_WaitAllSlaves(void);

static void PF_PackRedefinedPreVars(void);
static void PF_UnpackRedefinedPreVars(void);

static int PF_Wait4MasterIP(int tag);
static int PF_DoOneExpr(void);
static int PF_ReadMaster(void);/*reads directly to its scratch!*/
static int PF_Slave2MasterIP(int src);/*both master and slave*/
static int PF_Master2SlaveIP(int dest, EXPRESSIONS e);
static int PF_WalkThrough(WORD *t, LONG l, LONG chunk, LONG *count);
static int PF_SendChunkIP(FILEHANDLE *curfile,  POSITION *position, int to, LONG thesize);
static int PF_RecvChunkIP(FILEHANDLE *curfile, int from, LONG thesize);

static void PF_ReceiveErrorMessage(int src, int tag);
static void PF_CatchErrorMessages(int *src, int *tag);
static void PF_CatchErrorMessagesForAll(void);
static int PF_ProbeWithCatchingErrorMessages(int *src);

/* Variables */

PARALLELVARS PF;
#ifdef MPI2
```

```
 WORD *PF_shared_buff;
#endif

static LONG     PF_goutterms;  /* (master) Total out terms at PF_EndSort(), used in PF_Statistics(). */
static POSITION PF_exprsize;   /* (master) The size of the expression at PF_EndSort(), used in PF_Processor(). */

/*
    This will work well only under Linux, see
        #ifdef PF_WITH_SCHED_YIELD
    below in PF_WaitAllSlaves().
*/
#ifdef PF_WITH_SCHED_YIELD
 #include <sched.h>
#endif

#ifdef PF_WITHLOG
 #define PRINTFBUF(TEXT,TERM,SIZE)  { UBYTE lbuf[24]; if(PF.log){ WORD iii;\
  NumToStr(lbuf,AC.CModule); \
  fprintf(stderr,"[%d|%s] %s : ",PF.me,lbuf,(char*)TEXT);\
  if(TERM){ fprintf(stderr,"[%d] ",(int)(*TERM));\
    if((SIZE)<500 && (SIZE)>0) for(iii=1;iii<(SIZE);iii++)\
      fprintf(stderr,"%d ",TERM[iii]); }\
  fprintf(stderr,"\n");\
  fflush(stderr); } }
#else
 #define PRINTFBUF(TEXT,TERM,SIZE) {}
#endif

#define SWAP(x, y) \
    do { \
        char swap_tmp__[sizeof(x) == sizeof(y) ? (int)sizeof(x) : -1]; \
        memcpy(swap_tmp__, &y, sizeof(x)); \
        memcpy(&y, &x, sizeof(x)); \
        memcpy(&x, swap_tmp__, sizeof(x)); \
    } while (0)

#define PACK_LONG(p, n) \
    do { \
        *(p)++ = (UWORD)((ULONG)(n) & (ULONG)WORDMASK); \
        *(p)++ = (UWORD)(((ULONG)(n) » BITSINWORD) & (ULONG)WORDMASK); \
    } while (0)

#define UNPACK_LONG(p, n) \
    do { \
         (n) = (LONG)((((ULONG)(p)[1] & (ULONG)WORDMASK) « BITSINWORD) | ((ULONG)(p)[0] & (ULONG)WORDMASK)); \
         (p) += 2; \
    } while (0)

#define CHECK(condition) _CHECK(condition, __FILE__, __LINE__)
#define _CHECK(condition, file, line) __CHECK(condition, file, line)
#define __CHECK(condition, file, line) \
    do { \
        if ( !(condition) ) { \
            Error0("Fatal error at " file ":" #line); \
            Terminate(-1); \
        } \
    } while (0)

/*
 * For debugging.
 */
#define DBGOUT(lv1, lv2, a) do { if ( lv1 >= lv2 ) { printf a; fflush(stdout); } } while (0)

/* (AN.ninterms of master) == max(AN.ninterms of slaves) == sum(PF_linterms of slaves) at EndSort(). */
#define DBGOUT_NINTERMS(lv, a)
/* #define DBGOUT_NINTERMS(lv, a) DBGOUT(1, lv, a) */

/*
    #] includes :
    #[ statistics :
        #[ variables : (should be part of a struct?)
*/
static LONG PF_linterms;     /* local interms on this proces: PF_Proces */
#define PF_STATS_SIZE 5
static LONG **PF_stats = NULL;/* space for collecting statistics of all procs */
static LONG PF_laststat;     /* last realtime when statistics were printed */
static LONG PF_statsinterval;/* timeinterval for printing statistics */
/*
        #] variables :
        #[ PF_Statistics :
*/

static int PF_Statistics(LONG **stats, int proc)
```

```
{
    GETIDENTITY
    LONG real, cpu;
    WORD rpart, cpart;
    int i, j;

    if ( AT.SS == AM.S0 && PF.me == MASTER ) {
        real = PF_RealTime(PF_TIME); rpart = (WORD)(real%100); real /= 100;

        if ( PF_stats == NULL ) {
            PF_stats = (LONG**)Malloc1(PF.numtasks*sizeof(LONG*),"PF_stats 1");
            for ( i = 0; i < PF.numtasks; i++ ) {
                PF_stats[i] = (LONG*)Malloc1(PF_STATS_SIZE*sizeof(LONG),"PF_stats 2");
                for ( j = 0; j < PF_STATS_SIZE; j++ ) PF_stats[i][j] = 0;
            }
        }
        if ( proc > 0 ) for ( i = 0; i < PF_STATS_SIZE; i++ ) PF_stats[proc][i] = stats[0][i];

        if ( real >= PF_laststat + PF_statsinterval || proc == 0 ) {
            LONG sum[PF_STATS_SIZE];

            for ( i = 0; i < PF_STATS_SIZE; i++ ) sum[i] = 0;
            sum[0] = cpu = TimeCPU(1);
            cpart = (WORD)(cpu%1000);
            cpu /= 1000;
            cpart /= 10;
            if ( AC.OldParallelStats ) MesPrint("");
            if ( proc > 0 && AC.StatsFlag && AC.OldParallelStats ) {
                MesPrint("proc          CPU         in        gen       left       byte");
                MesPrint("%3d  : %7l.%2i %10l",0,cpu,cpart,AN.ninterms);
            }
            else if ( AC.StatsFlag && AC.OldParallelStats ) {
                MesPrint("proc          CPU         in        gen       out        byte");
                MesPrint("%3d  : %7l.%2i %10l %10l %10l",0,cpu,cpart,AN.ninterms,0,PF_goutterms);
            }

            for ( i = 1; i < PF.numtasks; i++ ) {
                cpart = (WORD)(PF_stats[i][0]%1000);
                cpu = PF_stats[i][0] / 1000;
                cpart /= 10;
                if ( AC.StatsFlag && AC.OldParallelStats )
                    MesPrint("%3d  : %7l.%2i %10l %10l %10l",i,cpu,cpart,
                            PF_stats[i][2],PF_stats[i][3],PF_stats[i][4]);
                for ( j = 0; j < PF_STATS_SIZE; j++ ) sum[j] += PF_stats[i][j];
            }
            cpart = (WORD)(sum[0]%1000);
            cpu = sum[0] / 1000;
            cpart /= 10;
            if ( AC.StatsFlag && AC.OldParallelStats ) {
                MesPrint("Sum  = %7l.%2i %10l %10l %10l",cpu,cpart,sum[2],sum[3],sum[4]);
                MesPrint("Real = %7l.%2i %20s (%l) %16s",
                        real,rpart,AC.Commercial,AC.CModule,EXPRNAME(AR.CurExpr));
                MesPrint("");
            }
            PF_laststat = real;
        }
    }
    return(0);
}
/*
        #] PF_Statistics :
    #] statistics :
    #[ sort.c :
        #[ sort variables :
*/

typedef struct NoDe {
    struct NoDe *left;
    struct NoDe *rght;
    int lloser;
    int rloser;
    int lsrc;
    int rsrc;
} NODE;

/*
    should/could be put in one struct
*/
static  NODE *PF_root;          /* root of tree of losers */
static  WORD PF_loser;          /* this is the last loser */
static  WORD **PF_term;         /* these point to the active terms */
static  WORD **PF_newcpos;      /* new coeffs of merged terms */
static  WORD *PF_newclen;       /* length of new coefficients */

/*
    preliminary: could also write somewhere else?
*/
```

```
static  WORD *PF_WorkSpace;     /* used in PF_EndSort() */
static  UWORD *PF_ScratchSpace; /* used in PF_GetLoser() */

/*
        #] sort variables :
        #[ PF_AllocBuf :
*/

static PF_BUFFER *PF_AllocBuf(int nbufs, LONG bsize, WORD free)
{
    PF_BUFFER *buf;
    UBYTE *p, *stop;
    LONG allocsize;
    int i;

    allocsize =
        (LONG)(sizeof(PF_BUFFER) + 4*nbufs*sizeof(WORD*) + (nbufs-free)*bsize);

    allocsize +=
        (LONG)( nbufs * (   2 * sizeof(MPI_Status)
                         +      sizeof(MPI_Request)
                         +      sizeof(MPI_Datatype)
                        )  );
    allocsize += (LONG)( nbufs * 3 * sizeof(int) );

    if ( ( buf = (PF_BUFFER*)Malloc1(allocsize,"PF_AllocBuf") ) == NULL ) return(NULL);

    p = ((UBYTE *)buf) + sizeof(PF_BUFFER);
    stop = ((UBYTE *)buf) + allocsize;

    buf->numbufs = nbufs;
    buf->active = 0;

    buf->buff    = (WORD**)p;         p += buf->numbufs*sizeof(WORD*);
    buf->fill    = (WORD**)p;         p += buf->numbufs*sizeof(WORD*);
    buf->full    = (WORD**)p;         p += buf->numbufs*sizeof(WORD*);
    buf->stop    = (WORD**)p;         p += buf->numbufs*sizeof(WORD*);
    buf->status  = (MPI_Status *)p;   p += buf->numbufs*sizeof(MPI_Status);
    buf->retstat = (MPI_Status *)p;   p += buf->numbufs*sizeof(MPI_Status);
    buf->request = (MPI_Request *)p;  p += buf->numbufs*sizeof(MPI_Request);
    buf->type    = (MPI_Datatype *)p; p += buf->numbufs*sizeof(MPI_Datatype);
    buf->index   = (int *)p;          p += buf->numbufs*sizeof(int);

    for ( i = 0; i < buf->numbufs; i++ ) buf->request[i] = MPI_REQUEST_NULL;
    buf->tag     = (int *)p;          p += buf->numbufs*sizeof(int);
    buf->from    = (int *)p;          p += buf->numbufs*sizeof(int);
/*
        and finally the real bufferspace
*/
    for ( i = free; i < buf->numbufs; i++ ) {
        buf->buff[i] = (WORD*)p; p += bsize;
        buf->stop[i] = (WORD*)p;
        buf->fill[i] = buf->full[i] = buf->buff[i];
    }
    if ( p != stop ) {
        MesPrint("Error in PF_AllocBuf p = %x stop = %x\n",p,stop);
        return(NULL);
    }
    return(buf);
}

/*
        #] PF_AllocBuf :
        #[ PF_InitTree :
*/

static int PF_InitTree(void)
{
    GETIDENTITY
    PF_BUFFER **rbuf = PF.rbufs;
    UBYTE *p, *stop;
    int numrbufs,numtasks = PF.numtasks;
    int i, j, src, numnodes;
    int numslaves = numtasks - 1;
    LONG size;
/*
        #[ the buffers : for the new coefficients and the terms
           we need one for each slave
*/
    if ( PF_term == NULL ) {
        size =  2*numtasks*sizeof(WORD*) + sizeof(WORD)*
             ( numtasks*(1 + AM.MaxTal) + (AM.MaxTer/sizeof(WORD)+1) + 2*(AM.MaxTal+2));

        PF_term = (WORD **)Malloc1(size,"PF_term");
        stop = ((UBYTE*)PF_term) + size;
        p = ((UBYTE*)PF_term) + numtasks*sizeof(WORD*);
```

```
 
        PF_newcpos = (WORD **)p;  p += sizeof(WORD*) * numtasks;
        PF_newclen =  (WORD *)p;  p += sizeof(WORD)  * numtasks;
        for ( i = 0; i < numtasks; i++ ) {
            PF_newcpos[i] = (WORD *)p; p += sizeof(WORD)*AM.MaxTal;
            PF_newclen[i] = 0;
        }
        PF_WorkSpace = (WORD *)p;     p += AM.MaxTer+sizeof(WORD);
        PF_ScratchSpace = (UWORD*)p; p += 2*(AM.MaxTal+2)*sizeof(UWORD);

        if ( p != stop ) { MesPrint("error in PF_InitTree"); return(-1); }
    }
/*
        #] the buffers :
        #[ the receive buffers :
*/
    numrbufs = PF.numrbufs;
/*
        this is the size we have in the combined sortbufs for one slave
*/
    size = (AT.SS->sTop2 - AT.SS->lBuffer - 1)/(PF.numtasks - 1);

    if ( rbuf == NULL ) {
        if ( ( rbuf = (PF_BUFFER**)Malloc1(numtasks*sizeof(PF_BUFFER*), "Master: rbufs") ) == NULL )
return(-1);
        if ( (rbuf[0] = PF_AllocBuf(1,0,1) ) == NULL ) return(-1);
        for ( i = 1; i < numtasks; i++ ) {
            if (!(rbuf[i] = PF_AllocBuf(numrbufs,sizeof(WORD)*size,1))) return(-1);
        }
    }
    rbuf[0]->buff[0] = AT.SS->lBuffer;
    rbuf[0]->full[0] = rbuf[0]->fill[0] = rbuf[0]->buff[0];
    rbuf[0]->stop[0] = rbuf[1]->buff[0] = rbuf[0]->buff[0] + 1;
    rbuf[1]->full[0] = rbuf[1]->fill[0] = rbuf[1]->buff[0];
    for ( i = 2; i < numtasks; i++ ) {
        rbuf[i-1]->stop[0] = rbuf[i]->buff[0] = rbuf[i-1]->buff[0] + size;
        rbuf[i]->full[0] = rbuf[i]->fill[0] = rbuf[i]->buff[0];
    }
    rbuf[numtasks-1]->stop[0] = rbuf[numtasks-1]->buff[0] + size;

    for ( i = 1; i < numtasks; i++ ) {
        for ( j = 0; j < rbuf[i]->numbufs; j++ ) {
            rbuf[i]->full[j] = rbuf[i]->fill[j] = rbuf[i]->buff[j] + AM.MaxTer/sizeof(WORD) + 2;
        }
        PF_term[i] = rbuf[i]->fill[rbuf[i]->active];
        *PF_term[i] = 0;
        PF_IRecvRbuf(rbuf[i],rbuf[i]->active,i);
    }
    rbuf[0]->active = 0;
    PF_term[0] = rbuf[0]->buff[0];
    PF_term[0][0] = 0;  /* PF_term[0] is used for a zero term. */
    PF.rbufs = rbuf;
/*
        #] the receive buffers :
        #[ the actual tree :

     calculate number of nodes in mergetree and allocate space for them
*/
    if ( numslaves < 3 ) numnodes = 1;
    else {
        numnodes = 2;
        while ( numnodes < numslaves ) numnodes *= 2;
        numnodes -= 1;
    }

    if ( PF_root == NULL )
    if ( ( PF_root = (NODE*)Malloc1(sizeof(NODE)*numnodes,"nodes in mergetree") ) == NULL )
        return(-1);
/*
        then initialize all the nodes
*/
    src = 1;
    for ( i = 0; i < numnodes; i++ ) {
        if ( 2*(i+1) <= numnodes ) {
            PF_root[i].left = &(PF_root[2*(i+1)-1]);
            PF_root[i].lsrc = 0;
        }
        else {
            PF_root[i].left = 0;
            if ( src < numtasks ) PF_root[i].lsrc = src++;
            else                  PF_root[i].lsrc = 0;
        }
        PF_root[i].lloser = 0;
    }
    for ( i = 0; i < numnodes; i++ ) {
        if ( 2*(i+1)+1 <= numnodes ) {
            PF_root[i].rght = &(PF_root[2*(i+1)]);
```

```
            PF_root[i].rsrc = 0;
        }
        else {
            PF_root[i].rght = 0;
            if (src<numtasks) PF_root[i].rsrc = src++;
            else              PF_root[i].rsrc = 0;
        }
        PF_root[i].rloser = 0;
    }
/*
        #] the actual tree :
*/
    return(numnodes);
}

/*
        #] PF_InitTree :
        #[ PF_PutIn :
*/

static WORD *PF_PutIn(int src)
{
    int tag;
    WORD im, r;
    WORD *m1, *m2;
    LONG size;
    PF_BUFFER *rbuf = PF.rbufs[src];
    int a = rbuf->active;
    int next = a+1 >= rbuf->numbufs ? 0 : a+1 ;
    WORD *lastterm = PF_term[src];
    WORD *term = rbuf->fill[a];

    if ( src <= 0 ) return(PF_term[0]);

    if ( rbuf->full[a] == rbuf->buff[a] + AM.MaxTer/sizeof(WORD) + 2 ) {
/*
            very first term from this src
*/
        tag = PF_WaitRbuf(rbuf,a,&size);
        rbuf->full[a] += size;
        if ( tag == PF_ENDBUFFER_MSGTAG ) *rbuf->full[a]++ = 0;
        else if ( rbuf->numbufs > 1 ) {
/*
                post a nonblock. recv. for the next buffer
*/
            rbuf->full[next] = rbuf->buff[next] + AM.MaxTer/sizeof(WORD) + 2;
            size = (LONG)(rbuf->stop[next] - rbuf->full[next]);
            PF_IRecvRbuf(rbuf,next,src);
        }
    }
    if ( *term == 0 && term != rbuf->full[a] ) return(PF_term[0]);
/*
        exception is for rare cases when the terms fitted exactly into buffer
*/
    if ( term + *term > rbuf->full[a] || term + 1 >= rbuf->full[a] ) {
newterms:
        m1 = rbuf->buff[next] + AM.MaxTer/sizeof(WORD) + 1;
        if ( *term < 0 || term == rbuf->full[a] ) {
/*
            copy term and lastterm to the new buffer, so that they end at m1
*/
            m2 = rbuf->full[a] - 1;
            while ( m2 >= term ) *m1-- = *m2--;
            rbuf->fill[next] = term = m1 + 1;
            m2 = lastterm + *lastterm - 1;
            while ( m2 >= lastterm ) *m1-- = *m2--;
            lastterm = m1 + 1;
        }
        else {
/*
            copy beginning of term to the next buffer so that it ends at m1
*/
            m2 = rbuf->full[a] - 1;
            while ( m2 >= term ) *m1-- = *m2--;
            rbuf->fill[next] = term = m1 + 1;
        }
        if ( rbuf->numbufs == 1 ) {
            rbuf->full[a] = rbuf->buff[a] + AM.MaxTer/sizeof(WORD) + 2;
            size = (LONG)(rbuf->stop[a] - rbuf->full[a]);
            PF_IRecvRbuf(rbuf,a,src);
        }
/*
            wait for new terms in the next buffer
*/
        rbuf->full[next] = rbuf->buff[next] + AM.MaxTer/sizeof(WORD) + 2;
        tag = PF_WaitRbuf(rbuf,next,&size);
        rbuf->full[next] += size;
```

```
        if ( tag == PF_ENDBUFFER_MSGTAG ) {
            *rbuf->full[next]++ = 0;
        }
        else if ( rbuf->numbufs > 1 ) {
/*
            post a nonblock. recv. for active buffer, it is not needed anymore
*/
            rbuf->full[a] = rbuf->buff[a] + AM.MaxTer/sizeof(WORD) + 2;
            size = (LONG)(rbuf->stop[a] - rbuf->full[a]);
            PF_IRecvRbuf(rbuf,a,src);
        }
/*
            now safely make next buffer active
*/
        a = rbuf->active = next;
    }

    if ( *term < 0 ) {
/*
            We need to decompress the term
*/
        im = *term;
        r = term[1] - im + 1;
        m1 = term + 2;
        m2 = lastterm - im + 1;
        while ( ++im <= 0 ) *--m1 = *--m2;
        *--m1 = r;
        rbuf->fill[a] = term = m1;
        if ( term + *term > rbuf->full[a] ) goto newterms;
    }
    rbuf->fill[a] += *term;
    return(term);
}

/*
        #] PF_PutIn :
        #[ PF_GetLoser :
*/

static int PF_GetLoser(NODE *n)
{
    GETIDENTITY
    WORD comp;

    if ( PF_loser == 0 ) {
/*
            this is for the right initialization of the tree only
*/
        if ( n->left ) n->lloser = PF_GetLoser(n->left);
        else {
            n->lloser = n->lsrc;
            if ( *(PF_term[n->lsrc] = PF_PutIn(n->lsrc)) == 0) n->lloser = 0;
        }
        PF_loser = 0;
        if ( n->rght ) n->rloser = PF_GetLoser(n->rght);
        else{
            n->rloser = n->rsrc;
            if ( *(PF_term[n->rsrc] = PF_PutIn(n->rsrc)) == 0 ) n->rloser = 0;
        }
        PF_loser = 0;
    }
    else if ( PF_loser == n->lloser ) {
        if ( n->left ) n->lloser = PF_GetLoser(n->left);
        else {
            n->lloser = n->lsrc;
            if ( *(PF_term[n->lsrc] = PF_PutIn(n->lsrc)) == 0 ) n->lloser = 0;
        }
    }
    else if ( PF_loser == n->rloser ) {
newright:
        if ( n->rght ) n->rloser = PF_GetLoser(n->rght);
        else {
            n->rloser = n->rsrc;
            if ( *(PF_term[n->rsrc] = PF_PutIn(n->rsrc)) == 0 ) n->rloser = 0;
        }
    }
    if ( n->lloser > 0 && n->rloser > 0 ) {
        comp = CompareTerms(BHEAD PF_term[n->lloser],PF_term[n->rloser],(WORD)0);
        if ( comp > 0 )      return(n->lloser);
        else if (comp < 0 ) return(n->rloser);
        else {
/*
        #[ terms are equal :
*/
            WORD *lcpos, *rcpos;
            UWORD *newcpos;
            WORD lclen, rclen, newclen, newnlen;
```

```
            SORTING *S = AT.SS;

            if ( S->PolyWise ) {
/*
            #[ Here we work with PolyFun :
*/
                WORD *tt1, *w;
                WORD r1,r2;
                WORD *ml = PF_term[n->lloser];
                WORD *mr = PF_term[n->rloser];

                if ( ( r1 = (int)*PF_term[n->lloser] ) <= 0 ) r1 = 20;
                if ( ( r2 = (int)*PF_term[n->rloser] ) <= 0 ) r2 = 20;
                tt1 = ml;
                ml += S->PolyWise;
                mr += S->PolyWise;
                if ( S->PolyFlag == 2 ) {
                    w = poly_ratfun_add(BHEAD ml,mr);
                    if ( *tt1 + w[1] - ml[1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
                        MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l is too
      small",AM.MaxTer);
                        Terminate(-1);
                    }
                    AT.WorkPointer = w;
                }
                else {
                    w = AT.WorkPointer;
                    if ( w + ml[1] + mr[1] > AT.WorkTop ) {
                        MesPrint("A WorkSpace of %10l is too small",AM.WorkSize);
                        Terminate(-1);
                    }
                    AddArgs(BHEAD ml,mr,w);
                }
                r1 = w[1];
                if ( r1 <= FUNHEAD || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) ) {
                    goto cancelled;
                }
                if ( r1 == ml[1] ) {
                    NCOPY(ml,w,r1);
                }
                else if ( r1 < ml[1] ) {
                    r2 = ml[1] - r1;
                    mr = w + r1;
                    ml += ml[1];
                    while ( --r1 >= 0 ) *--ml = *--mr;
                    mr = ml - r2;
                    r1 = S->PolyWise;
                    while ( --r1 >= 0 ) *--ml = *--mr;
                    *ml -= r2;
                    PF_term[n->lloser] = ml;
                }
                else {
                    r2 = r1 - ml[1];
                    if ( r2 > 2*AM.MaxTal )
                        MesPrint("warning: new term in polyfun is large");
                    mr = tt1 - r2;
                    r1 = S->PolyWise;
                    ml = tt1;
                    *ml += r2;
                    PF_term[n->lloser] = mr;
                    NCOPY(mr,ml,r1);
                    r1 = w[1];
                    NCOPY(mr,w,r1);
                }
                PF_newclen[n->rloser] = 0;
                PF_loser = n->rloser;
                goto newright;
/*
            #] Here we work with PolyFun :
*/
            }
            if ( ( lclen = PF_newclen[n->lloser] ) != 0 ) lcpos = PF_newcpos[n->lloser];
            else {
                lcpos = PF_term[n->lloser];
                lclen = *(lcpos += *lcpos - 1);
                lcpos -= ABS(lclen) - 1;
            }
            if ( ( rclen = PF_newclen[n->rloser] ) != 0 ) rcpos = PF_newcpos[n->rloser];
            else {
                rcpos = PF_term[n->rloser];
                rclen = *(rcpos += *rcpos - 1);
                rcpos -= ABS(rclen) -1;
            }
            lclen = ( (lclen > 0) ? (lclen-1) : (lclen+1) ) » 1;
            rclen = ( (rclen > 0) ? (rclen-1) : (rclen+1) ) » 1;
            newcpos = PF_ScratchSpace;
            if ( AddRat(BHEAD (UWORD *)lcpos,lclen,(UWORD *)rcpos,rclen,newcpos,&newnlen) )
```

```
return(-1);
            if ( AN.ncmod != 0 ) {
                if ( ( AC.modmode & POSNEG ) != 0 ) {
                    NormalModulus(newcpos,&newnlen);
                }
                if ( BigLong(newcpos,newnlen,(UWORD *)AC.cmod,ABS(AN.ncmod)) >=0 ) {
                    WORD ii;
                    SubPLon(newcpos,newnlen,(UWORD *)AC.cmod,ABS(AN.ncmod),newcpos,&newnlen);
                    newcpos[newnlen] = 1;
                    for ( ii = 1; ii < newnlen; ii++ ) newcpos[newnlen+ii] = 0;
                }
            }
            if ( newnlen == 0 ) {
/*
                    terms cancel, get loser of left subtree and then of right subtree
*/
cancelled:
                PF_loser = n->lloser;
                PF_newclen[n->lloser] = 0;
                if ( n->left ) n->lloser = PF_GetLoser(n->left);
                else {
                    n->lloser = n->lsrc;
                    if ( *(PF_term[n->lsrc] = PF_PutIn(n->lsrc)) == 0 ) n->lloser = 0;
                }
                PF_loser = n->rloser;
                PF_newclen[n->rloser] = 0;
                goto newright;
            }
            else {
/*
                    keep the left term and get the loser of right subtree
*/
                newnlen *= 2;
                newclen = ( newnlen > 0 ) ? ( newnlen + 1 ) : ( newnlen - 1 );
                if ( newnlen < 0 ) newnlen = -newnlen;
                PF_newclen[n->lloser] = newclen;
                lcpos = PF_newcpos[n->lloser];
                if ( newclen < 0 ) newclen = -newclen;
                while ( newclen-- ) *lcpos++ = *newcpos++;
                PF_loser = n->rloser;
                PF_newclen[n->rloser] = 0;
                goto newright;
            }
/*
        #] terms are equal :
*/
        }
    }
    if (n->lloser > 0) return(n->lloser);
    if (n->rloser > 0) return(n->rloser);
    return(0);
}
/*
        #] PF_GetLoser :
        #[ PF_EndSort :
*/

int PF_EndSort(void)
{
    GETIDENTITY
    FILEHANDLE *fout = AR.outfile;
    PF_BUFFER *sbuf=PF.sbuf;
    SORTING *S = AT.SS;
    WORD *outterm,*pp;
    LONG size, noutterms;
    POSITION position, oldposition;
    WORD i,cc;
    int oldgzipCompress;

    if ( AT.SS != AT.S0 || !PF.parallel ) return 0;

    if ( PF.me != MASTER ) {
/*
        #[ the slaves have to initialize their sendbuffer :

        this is a slave and it's PObuffer should be the minimum of the
        sortiosize on the master and the POsize of our file.
        First save the original PObuffer and POstop of the outfile
*/
        size = (S->sTop2 - S->lBuffer - 1)/(PF.numtasks - 1);
        size -= (AM.MaxTer/sizeof(WORD) + 2);
        if ( fout->POsize < (LONG)(size*sizeof(WORD)) ) size = fout->POsize/sizeof(WORD);
        if ( sbuf == NULL ) {
            if ( (sbuf = PF_AllocBuf(PF.numsbufs, size*sizeof(WORD), 1)) == NULL ) return -1;
            sbuf->active = 0;
            PF.sbuf = sbuf;
        }
```

```
        sbuf->buff[0] = fout->PObuffer;
        sbuf->stop[0] = fout->PObuffer+size;
        if ( sbuf->stop[0] > fout->POstop ) return -1;
        for ( i = 0; i < PF.numsbufs; i++ )
            sbuf->fill[i] = sbuf->full[i] = sbuf->buff[i];

        fout->PObuffer = sbuf->buff[sbuf->active];
        fout->POstop = sbuf->stop[sbuf->active];
        fout->POsize = size*sizeof(WORD);
        fout->POfill = fout->POfull = fout->PObuffer;
/*
        #] the slaves have to initialize their sendbuffer :
*/
        return(0);
    }
/*
        this waits for all slaves to be ready to send terms back
*/
    PF_WaitAllSlaves(); /* Note, the returned value should be 0 on success. */
/*
        Now collect the terms of all slaves and merge them.
        PF_GetLoser gives the position of the smallest term, which is the real
        work. The smallest term needs to be copied to the outbuf: use PutOut.
*/
    PF_InitTree();
    if ( AR.PolyFun == 0 ) { S->PolyFlag = 0; }
    else if ( AR.PolyFunType == 1 ) { S->PolyFlag = 1; }
    else if ( AR.PolyFunType == 2 ) {
        if ( AR.PolyFunExp == 2
          || AR.PolyFunExp == 3 ) S->PolyFlag = 1;
        else                      S->PolyFlag = 2;
    }
    *AR.CompressPointer = 0;
    SeekScratch(fout, &position);
    oldposition = position;
    oldgzipCompress = AR.gzipCompress;
    AR.gzipCompress = 0;

    noutterms = 0;

    while ( PF_loser >= 0 ) {
        if ( (PF_loser = PF_GetLoser(PF_root)) == 0 ) break;
        outterm = PF_term[PF_loser];
        noutterms++;

        if ( PF_newclen[PF_loser] != 0 ) {
/*
            #[ this is only when new coeff was too long :
*/
            outterm = PF_WorkSpace;
            pp = PF_term[PF_loser];
            cc = *pp;
            while ( cc-- ) *outterm++ = *pp++;
            outterm = (outterm[-1] > 0) ? outterm-outterm[-1] : outterm+outterm[-1];
            if ( PF_newclen[PF_loser] > 0 ) cc =  (WORD)PF_newclen[PF_loser] - 1;
            else                             cc = -(WORD)PF_newclen[PF_loser] - 1;
            pp =  PF_newcpos[PF_loser];
            while ( cc-- ) *outterm++ = *pp++;
            *outterm++ = PF_newclen[PF_loser];
            *PF_WorkSpace = outterm - PF_WorkSpace;
            outterm = PF_WorkSpace;
            *PF_newcpos[PF_loser] = 0;
            PF_newclen[PF_loser] = 0;
/*
            #] this is only when new coeff was too long :
*/
        }
        PRINTFBUF("PF_EndSort to PutOut: ",outterm,*outterm);
        PutOut(BHEAD outterm,&position,fout,1);
    }
    if ( FlushOut(&position,fout,0) ) {
        AR.gzipCompress = oldgzipCompress;
        return(-1);
    }
    S->TermsLeft = PF_goutterms = noutterms;
    DIFPOS(PF_exprsize, position, oldposition);
    AR.gzipCompress = oldgzipCompress;
    return(1);
}

/*
        #] PF_EndSort :
    #] sort.c :
    #[ proces.c :
        #[ variables :
*/

```

```
static  WORD *PF_CurrentBracket;

/*
        #] variables :
        #[ PF_GetTerm :
*/

static WORD PF_GetTerm(WORD *term)
{
    GETIDENTITY
    FILEHANDLE *fi = AC.RhsExprInModuleFlag && PF.rhsInParallel ? &PF.slavebuf : AR.infile;
    WORD i;
    WORD *next, *np, *last, *lp = 0, *nextstop, *tp=term;

    /* Only on the slaves. */

    AN.deferskipped = 0;
    if ( fi->POfill >= fi->POfull || fi->POfull == fi->PObuffer ) {
ReceiveNew:
      {
/*
        #[ receive new terms from master :
*/
        int src = MASTER, tag;
        int follow = 0;
        LONG size,cpu,space = 0;

        if ( PF.log ) {
            fprintf(stderr,"[%d] Starting to send to Master\n",PF.me);
            fflush(stderr);
        }

        cpu = TimeCPU(1);
        PF_PreparePack();
        PF_Pack(&cpu              ,1,PF_LONG);
        PF_Pack(&space            ,1,PF_LONG);
        PF_Pack(&PF_linterms      ,1,PF_LONG);
        PF_Pack(&(AM.S0->GenTerms) ,1,PF_LONG);
        PF_Pack(&(AM.S0->TermsLeft),1,PF_LONG);
        PF_Pack(&follow           ,1,PF_INT );

        if ( PF.log ) {
            fprintf(stderr,"[%d] Now sending with tag = %d\n",PF.me,PF_READY_MSGTAG);
            fflush(stderr);
        }

        PF_Send(MASTER, PF_READY_MSGTAG);

        if ( PF.log ) {
            fprintf(stderr,"[%d] returning from send\n",PF.me);
            fflush(stderr);
        }

        size = fi->POstop - fi->PObuffer - 1;
#ifdef AbsolutelyExtra
        PF_Receive(MASTER,PF_ANY_MSGTAG,&src,&tag);
#ifdef MPI2
        if ( tag == PF_TERM_MSGTAG ) {
            PF_Unpack(&size, 1, PF_LONG);
            if ( PF_Put_target(src) == 0 ) {
                printf("PF_Put_target error ...\n");
            }
        }
        else {
            PF_RecvWbuf(fi->PObuffer,&size,&src);
        }
#else
        PF_RecvWbuf(fi->PObuffer,&size,&src);
#endif
#endif
        tag=PF_RecvWbuf(fi->PObuffer,&size,&src);

        fi->POfill = fi->PObuffer;
        /* Get AN.ninterms which sits in the first 2 WORDs. */
        {
            LONG ninterms;
            UNPACK_LONG(fi->POfill, ninterms);
            if ( fi->POfill < fi->POfull ) {
                DBGOUT_NINTERMS(2, ("PF.me=%d AN.ninterms=%d PF_linterms=%d ninterms=%d GET\n",
 (int)PF.me, (int)AN.ninterms, (int)PF_linterms, (int)ninterms));
                AN.ninterms = ninterms - 1;
            } else {
                DBGOUT_NINTERMS(2, ("PF.me=%d AN.ninterms=%d PF_linterms=%d ninterms=%d GETEND\n",
 (int)PF.me, (int)AN.ninterms, (int)PF_linterms, (int)ninterms));
            }
        }
        fi->POfull = fi->PObuffer + size;
```

```
        if ( tag == PF_ENDSORT_MSGTAG ) *fi->POfull++ = 0;
/*
        #] receive new terms from master :
*/
      }
      if ( PF_CurrentBracket ) *PF_CurrentBracket = 0;
    }
    if ( *fi->POfill == 0 ) {
        fi->POfill = fi->POfull = fi->PObuffer;
        *term = 0;
        goto RegRet;
    }
    if ( AR.DeferFlag ) {
        if ( !PF_CurrentBracket ) {
/*
        #[ alloc space :
*/
            PF_CurrentBracket =
                    (WORD*)Malloc1(AM.MaxTer,"PF_CurrentBracket");
            *PF_CurrentBracket = 0;
/*
        #] alloc space :
*/
        }
        while ( *PF_CurrentBracket ) {  /* "for each term in the buffer" */
/*
        #[ test : bracket & skip if it's equal to the last in PF_CurrentBracket
*/
            next = fi->POfill;
            nextstop = next + *next; nextstop -= ABS(nextstop[-1]);
            next++;
            last = PF_CurrentBracket+1;
            while ( next < nextstop ) {
/*
                    scan the next term and PF_CurrentBracket
*/
                if ( *last == HAAKJE && *next == HAAKJE ) {
/*
                    the part outside brackets is equal => skip this term
*/
                    PRINTFBUF("PF_GetTerm skips",fi->POfill,*fi->POfill);
                    break;
                }
/*
                    check if the current subterms are equal
*/
                np = next; next += next[1];
                lp = last; last += last[1];
                while ( np < next ) if ( *lp++ != *np++ ) goto strip;
            }
/*
                go on to next term
*/
            fi->POfill += *fi->POfill;
            AN.deferskipped++;
/*
                the usual checks
*/
            if ( fi->POfill >= fi->POfull || fi->POfull == fi->PObuffer )
                goto ReceiveNew;
            if ( *fi->POfill == 0 ) {
                fi->POfill = fi->POfull = fi->PObuffer;
                *term = 0;
                goto RegRet;
            }
/*
        #] test :
*/
        }
/*
        #[ copy :

        this term to CurrentBracket and the part outside of bracket
        to WorkSpace at term
*/
strip:
        next = fi->POfill;
        nextstop = next + *next; nextstop -= ABS(nextstop[-1]);
        next++;
        tp++;
        lp = PF_CurrentBracket + 1;
        while ( next < nextstop ) {
            if ( *next == HAAKJE ) {
                fi->POfill += *fi->POfill;
                while ( next < fi->POfill ) *lp++ = *next++;
                *PF_CurrentBracket = lp - PF_CurrentBracket;
                *lp = 0;
```

```
                *tp++ = 1;
                *tp++ = 1;
                *tp++ = 3;
                *term = WORDDIF(tp,term);
                PRINTFBUF("PF_GetTerm new brack",PF_CurrentBracket,*PF_CurrentBracket);
                PRINTFBUF("PF_GetTerm POfill",fi->POfill,*fi->POfill);
                goto RegRet;
            }
            np = next; next += next[1];
            while ( np < next ) *tp++ = *lp++ = *np++;
        }
        tp = term;
/*
        #] copy :
*/
    }

    i = *fi->POfill;
    while ( i-- ) *tp++ = *fi->POfill++;
RegRet:
    PRINTFBUF("PF_GetTerm returns",term,*term);
    return(*term);
}

/*
        #] PF_GetTerm :
        #[ PF_Deferred :
*/

WORD PF_Deferred(WORD *term, WORD level)
{
    GETIDENTITY
    WORD *bra, *bstop;
    WORD *tstart;
    FILEHANDLE *fi = AC.RhsExprInModuleFlag && PF.rhsInParallel ? &PF.slavebuf : AR.infile;
    WORD *next = fi->POfill;
    WORD *termout = AT.WorkPointer;
    WORD *oldwork = AT.WorkPointer;

    AT.WorkPointer = (WORD *)((UBYTE *)(AT.WorkPointer) + AM.MaxTer);
    AR.DeferFlag = 0;

    PRINTFBUF("PF_Deferred (Term)    ",term,*term);
    PRINTFBUF("PF_Deferred (Bracket)",PF_CurrentBracket,*PF_CurrentBracket);

    bra = bstop = PF_CurrentBracket;
    if ( *bstop > 0 ) {
        bstop += *bstop;
        bstop -= ABS(bstop[-1]);
    }
    bra++;
    while ( *bra != HAAKJE && bra < bstop ) bra += bra[1];
    if ( bra >= bstop ) {    /* No deferred action! */
        AT.WorkPointer = term + *term;
        if ( Generator(BHEAD term,level) ) goto DefCall;
        AR.DeferFlag = 1;
        AT.WorkPointer = oldwork;
        return(0);
    }
    bstop = bra;
    tstart = bra + bra[1];
    bra = PF_CurrentBracket;
    tstart--;
    *tstart = bra + *bra - tstart;
    bra++;
/*
        Status of affairs:
        First bracket content starts at tstart.
        Next term starts at next.
        The outside of the bracket runs from bra = PF_CurrentBracket to bstop.
*/
    for(;;) {
        if ( InsertTerm(BHEAD term,0,AM.rbufnum,tstart,termout,0) < 0 ) {
            goto DefCall;
        }
/*
            call Generator with new composed term
*/
        AT.WorkPointer = termout + *termout;
        if ( Generator(BHEAD termout,level) ) goto DefCall;
        AT.WorkPointer = termout;
        tstart = next + 1;
        if ( tstart >= fi->POfull ) goto ThatsIt;
        next += *next;
/*
            compare with current bracket
*/
```

```
        while ( bra <= bstop ) {
            if ( *bra != *tstart ) goto ThatsIt;
            bra++; tstart++;
        }
/*
            now bra and tstart should both be a HAAKJE
*/
        bra--; tstart--;
        if ( *bra != HAAKJE || *tstart != HAAKJE ) goto ThatsIt;
        tstart += tstart[1];
        tstart--;
        *tstart = next - tstart;
        bra = PF_CurrentBracket + 1;
    }

ThatsIt:
/*
    AT.WorkPointer = oldwork;
*/
    AR.DeferFlag = 1;
    return(0);
DefCall:
    MesCall("PF_Deferred");
    SETERROR(-1);
}

/*
        #] PF_Deferred :
        #[ PF_Wait4Slave :
*/

static LONG **PF_W4Sstats = 0;

static int PF_Wait4Slave(int src)
{
    int j, tag, next;

    tag = PF_ANY_MSGTAG;
    PF_CatchErrorMessages(&src, &tag);
    PF_Receive(src, tag, &next, &tag);

    if ( tag != PF_READY_MSGTAG ) {
        MesPrint("[%d] PF_Wait4Slave: received MSGTAG %d",(WORD)PF.me,(WORD)tag);
        return(-1);
    }
    if ( PF_W4Sstats == 0 ) {
        PF_W4Sstats = (LONG**)Malloc1(sizeof(LONG*),"");
        PF_W4Sstats[0] = (LONG*)Malloc1(PF_STATS_SIZE*sizeof(LONG),"");
    }
    PF_Unpack(PF_W4Sstats[0],PF_STATS_SIZE,PF_LONG);
    PF_Statistics(PF_W4Sstats,next);

    PF_Unpack(&j,1,PF_INT);

    if ( j ) {
/*
        actions depending on rest of information in last message
*/
    }
    return(next);
}

/*
        #] PF_Wait4Slave :
        #[ PF_Wait4SlaveIP :
*/
/*
    array of expression numbers for PF_InParallel processor.
    Each time the master sends expression "i" to the slave
    "next" it sets partodoexr[next]=i:
*/
static WORD *partodoexr=NULL;

static int PF_Wait4SlaveIP(int *src)
{
    int j,tag,next;

    tag = PF_ANY_MSGTAG;
    PF_CatchErrorMessages(src, &tag);
    PF_Receive(*src, tag, &next, &tag);
    *src=tag;
    if ( PF_W4Sstats == 0 ) {
        PF_W4Sstats = (LONG**)Malloc1(sizeof(LONG*),"");
        PF_W4Sstats[0] = (LONG*)Malloc1(PF_STATS_SIZE*sizeof(LONG),"");
    }

    PF_Unpack(PF_W4Sstats[0],PF_STATS_SIZE,PF_LONG);
```

```
    if ( tag == PF_DATA_MSGTAG )
        AR.CurExpr = partodoexr[next];
    PF_Statistics(PF_W4Sstats,next);

    PF_Unpack(&j,1,PF_INT);

    if ( j ) {
    /* actions depending on rest of information in last message */
    }

    return(next);
}
/*
        #] PF_Wait4SlaveIP :
        #[ PF_WaitAllSlaves :
*/

static int PF_WaitAllSlaves(void)
{
    int i, readySlaves, tag, next = PF_ANY_SOURCE;
    UBYTE *has_sent = 0;

    has_sent = (UBYTE*)Malloc1(sizeof(UBYTE)*(PF.numtasks + 1),"PF_WaitAllSlaves");
    for ( i = 0; i < PF.numtasks; i++ ) has_sent[i] = 0;

    for ( readySlaves = 1; readySlaves < PF.numtasks; ) {
        if ( next != PF_ANY_SOURCE) { /*Go to the next slave:*/
            do{ /*Note, here readySlaves<PF.numtasks, so this loop can't be infinite*/
                if ( ++next >= PF.numtasks ) next = 1;
            } while ( has_sent[next] == 1 );
        }
/*
            Here PF_ProbeWithCatchingErrorMessages() is BLOCKING function if next = PF_ANY_SOURCE:
*/
        tag = PF_ProbeWithCatchingErrorMessages(&next);
/*
            Here next != PF_ANY_SOURCE
*/
        switch ( tag ) {
            case PF_BUFFER_MSGTAG:
            case PF_ENDBUFFER_MSGTAG:
/*
                    Slaves are ready to send their results back
*/
                if ( has_sent[next] == 0 ) {
                    has_sent[next] = 1;
                    readySlaves++;
                }
                else {  /*error?*/
                    fprintf(stderr,"ERROR next=%d tag=%d\n",next,tag);
                }
/*
                    Note, we do NOT read results here! Messages from these slaves will be read
                    only after all slaves are ready, further in caller function
*/
                break;
            case 0:
/*
                    The slave is not ready. Just go to the next slave.
                    It may appear that there are no more ready slaves, and the master
                    will wait them in infinite loop. Stupid situation - the master can
                    receive buffers from ready slaves!
*/
#ifdef PF_WITH_SCHED_YIELD
/*
                        Relinquish the processor:
*/
                    sched_yield();
#endif
                break;
            case PF_DATA_MSGTAG:
                tag=next;
                next=PF_Wait4SlaveIP(&tag);
/*
    tag must be == PF_DATA_MSGTAG!
*/
                PF_Statistics(PF_stats,0);
                PF_Slave2MasterIP(next);
                PF_Master2SlaveIP(next,NULL);
                if ( has_sent[next] == 0 ) {
                    has_sent[next]=1;
                    readySlaves++;
                }else{
                    /*error?*/
                    fprintf(stderr,"ERROR next=%d tag=%d\n",next,tag);
                }/*if ( has_sent[next] == 0 )*/
                break;
```

```
            case PF_EMPTY_MSGTAG:
                tag=next;
                next=PF_Wait4SlaveIP(&tag);
/*
    tag must be == PF_EMPTY_MSGTAG!
*/
                PF_Master2SlaveIP(next,NULL);
                if ( has_sent[next] == 0 ) {
                    has_sent[next]=1;
                    readySlaves++;
                }else{
                    /*error?*/
                    fprintf(stderr,"ERROR next=%d tag=%d\n",next,tag);
                }/*if ( has_sent[next] == 0 )*/
                break;
            case PF_READY_MSGTAG:
/*
                    idle slave
                    May be only PF_READY_MSGTAG:
*/
                next = PF_Wait4Slave(next);
                if ( next == -1 ) return(next); /*Cannot be!*/
                if ( has_sent[0] == 0 ) {  /*Send the last chunk to the slave*/
                    PF.sbuf->active = 0;
                    has_sent[0] = 1;
                }
                else {
/*
                        Last chunk was sent, so just send to slave ENDSORT
                        AN.ninterms must be sent because the slave expects it:
*/
                    PACK_LONG(PF.sbuf->fill[next], AN.ninterms);
/*
                        This will tell to the slave that there are no more terms:
*/
                    *(PF.sbuf->fill[next])++ = 0;
                    PF.sbuf->active = next;
                }
/*
                    Send ENDSORT
*/
                PF_ISendSbuf(next,PF_ENDSORT_MSGTAG);
                break;
            default:
/*
                    Error?
                    Indicates the error. This will force exit from the main loop:
*/
                MesPrint("!!!Unexpected MPI message src=%d tag=%d.", next, tag);
                readySlaves = PF.numtasks+1;
                break;
        }
    }

    if ( has_sent ) M_free(has_sent,"PF_WaitAllSlaves");
/*
        0 on success (exit from the main loop by loop condition), or -1 if fails
        (exit from the main loop since readySlaves=PF.numtasks+1):
*/
    return(PF.numtasks-readySlaves);
}

/*
        #] PF_WaitAllSlaves :
        #[ PF_Processor :
*/

int PF_Processor(EXPRESSIONS e, WORD i, WORD LastExpression)
{
    GETIDENTITY
    WORD *term = AT.WorkPointer;
    LONG dd = 0;
    PF_BUFFER *sb = PF.sbuf;
    WORD j, *s, next;
    LONG size, cpu;
    POSITION position;
    int k, src, tag;
    FILEHANDLE *oldoutfile = AR.outfile;

#ifdef MPI2
    if ( PF_shared_buff == NULL ) {
        if ( PF_SMWin_Init() == 0 ) {
            MesPrint("PF_SMWin_Init error");
            exit(-1);
        }
    }
#endif
```

```
    if ( ( (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer ) ) > AT.WorkTop ) return(MesWork());

    /* For redefine statements. */
    if ( AC.numpfirstnum > 0 ) {
        for ( j = 0; j < AC.numpfirstnum; j++ ) {
            AC.inputnumbers[j] = -1;
        }
    }

    if ( AC.mparallelflag != PARALLELFLAG ) return(0);

    if ( PF.me == MASTER ) {
/*
        #[ Master:
            #[ write prototype to outfile:
*/
        WORD oldBracketOn = AR.BracketOn;
        WORD *oldBrackBuf = AT.BrackBuf;
        WORD oldbracketindexflag = AT.bracketindexflag;

        LONG maxinterms;     /* the maximum number of terms in the bucket */
        int cmaxinterms;     /* a variable controling the transition of maxinterms */
        LONG termsinbucket;  /* the number of filled terms in the bucket */
        LONG ProcessBucketSize = AC.mProcessBucketSize;

        if ( PF.log && AC.CModule >= PF.log )
            MesPrint("[%d] working on expression %s in module %l",PF.me,EXPRNAME(i),AC.CModule);
        if ( GetTerm(BHEAD term) <= 0 ) {
            MesPrint("[%d] Expression %d has problems in scratchfile",PF.me,i);
            return(-1);
        }
        term[3] = i;
        if ( AR.outtohide ) {
            SeekScratch(AR.hidefile,&position);
            e->onfile = position;
            if ( PutOut(BHEAD term,&position,AR.hidefile,0) < 0 ) return(-1);
        }
        else {
            SeekScratch(AR.outfile,&position);
            e->onfile = position;
            if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) return(-1);
        }
        AR.DeferFlag = 0;  /* The master leave the brackets!!! */
        AR.Eside = RHSIDE;
        if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
            AR.BracketOn = 1;
            AR.BrackBuf = AM.BracketFactors;
            AT.bracketindexflag = 1;
        }
        if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
/*
            #] write prototype to outfile:
            #[ initialize sendbuffer if necessary:

            the size of the sendbufs is:
            MIN(1/PF.numtasks*(AT.SS->sBufsize+AT.SS->lBufsize),AR.infile->POsize)
            No allocation for extra buffers necessary, just make sb->buf... point
            to the right places in the sortbuffers.
*/
        NewSort(BHEAD0);   /* we need AT.SS to be set for this!!! */
        if ( sb == 0 || sb->buff[0] != AT.SS->lBuffer ) {
            size = (LONG)((AT.SS->sTop2 - AT.SS->lBuffer)/(PF.numtasks));
            if ( size > (LONG)(AR.infile->POsize/sizeof(WORD) - 1) )
                size = AR.infile->POsize/sizeof(WORD) - 1;
            if ( sb == 0 ) {
                if ( ( sb = PF_AllocBuf(PF.numtasks,size*sizeof(WORD),PF.numtasks) ) == NULL )
                    return(-1);
            }
            sb->buff[0] = AT.SS->lBuffer;
            sb->full[0] = sb->fill[0] = sb->buff[0];
            for ( j = 1; j < PF.numtasks; j++ ) {
                sb->stop[j-1] = sb->buff[j] = sb->buff[j-1] + size;
            }
            sb->stop[PF.numtasks-1] = sb->buff[PF.numtasks-1] + size;
            PF.sbuf = sb;
        }
        for ( j = 0; j < PF.numtasks; j++ ) {
            sb->full[j] = sb->fill[j] = sb->buff[j];
        }
/*
            #] initialize sendbuffer if necessary:
            #[ loop for all terms in infile:
*/
        /*
         * The initial value of maxinterms is determined by the user given
         * ProcessBucketSize and the number of terms in the current expression.
```

```
         * We make the initial maxinterms smaller, so that we get the all
         * workers busy as soon as possible.
         */
        maxinterms = ProcessBucketSize / 100;
        if ( maxinterms > e->counter / (PF.numtasks - 1) / 4 )
            maxinterms = e->counter / (PF.numtasks - 1) / 4;
        if ( maxinterms < 1 ) maxinterms = 1;
        cmaxinterms = 0;
        /*
         * Copy them always to sb->buff[0]. When that is full, wait for
         * the next slave to accept terms, exchange sb->buff[0] and
         * sb->buff[next], send sb->buff[next] to next slave and go on
         * filling the now empty sb->buff[0].
         */
        AN.ninterms = 0;
        termsinbucket = 0;
        PACK_LONG(sb->fill[0], 1);
        while ( GetTerm(BHEAD term) ) {
            AN.ninterms++; dd = AN.deferskipped;
            if ( AC.CollectFun && *term <= (LONG)(AM.MaxTer/(2*sizeof(WORD))) ) {
                if ( GetMoreTerms(term) < 0 ) {
                    LowerSortLevel(); return(-1);
                }
            }
            PRINTFBUF("PF_Processor gets",term,*term);
            if ( termsinbucket >= maxinterms || sb->fill[0] + *term >= sb->stop[0] ) {
                next = PF_Wait4Slave(PF_ANY_SOURCE);

                sb->fill[next] = sb->fill[0];
                sb->full[next] = sb->full[0];
                SWAP(sb->stop[next], sb->stop[0]);
                SWAP(sb->buff[next], sb->buff[0]);
                sb->fill[0] = sb->full[0] = sb->buff[0];
                sb->active = next;

#ifdef MPI2
                if ( PF_Put_origin(next) == 0 ) {
                    printf("PF_Put_origin error...\n");
                }
#else
                PF_ISendSbuf(next,PF_TERM_MSGTAG);
#endif
                /* Initialize the next bucket. */
                termsinbucket = 0;
                PACK_LONG(sb->fill[0], AN.ninterms);
                /*
                 * For the "slow startup". We double maxinterms up to ProcessBucketSize
                 * after (hopefully) the all workers got some terms.
                 */
                if ( cmaxinterms >= PF.numtasks - 2 ) {
                    maxinterms *= 2;
                    if ( maxinterms >= ProcessBucketSize ) {
                        cmaxinterms = -1;
                        maxinterms = ProcessBucketSize;
                    }
                }
                else if ( cmaxinterms >= 0 ) {
                    cmaxinterms++;
                }
            }
            j = *(s = term);
            NCOPY(sb->fill[0], s, j);
            termsinbucket++;
        }
        /* NOTE: The last chunk will be sent to a slave at EndSort() => PF_EndSort()
         *       => PF_WaitAllSlaves(). */
        AN.ninterms += dd;
/*
            #] loop for all terms in infile:
            #[ Clean up & EndSort:
*/
        if ( LastExpression ) {
            UpdateMaxSize();
            if ( AR.infile->handle >= 0 ) {
                CloseFile(AR.infile->handle);
                AR.infile->handle = -1;
                remove(AR.infile->name);
                PUTZERO(AR.infile->POposition);
            }
            AR.infile->POfill = AR.infile->POfull = AR.infile->PObuffer;
        }
        if ( AR.outtohide ) AR.outfile = AR.hidefile;
        PF.parallel = 1;
        if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) return(-1);
        PF.parallel = 0;
        if ( AR.outtohide ) {
            AR.outfile = oldoutfile;
```

```
            AR.hidefile->POfull = AR.hidefile->POfill;
        }
        UpdateMaxSize();
        AR.BracketOn = oldBracketOn;
        AT.BrackBuf = oldBrackBuf;
        if ( ( e->vflags & TOBEFACTORED ) != 0 )
            poly_factorize_expression(e);
        else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
         && ( ( e->vflags & ISFACTORIZED ) != 0 ) )
            poly_unfactorize_expression(e);
        AT.bracketindexflag = oldbracketindexflag;
        AR.GetFile = 0;
        AR.outtohide = 0;
        /*
         * NOTE: e->numdummies, e->vflags and AR.exprflags will be updated
         *       after gathering the information from all slaves.
         */
/*
            #] Clean up & EndSort:
            #[ Collect (stats,prepro,...):
*/
        DBGOUT_NINTERMS(1, ("PF.me=%d AN.ninterms=%d ENDSORT\n", (int)PF.me, (int)AN.ninterms));
        PF_CatchErrorMessagesForAll();
        e->numdummies = 0;
        for ( k = 1; k < PF.numtasks; k++ ) {
            PF_LongSingleReceive(PF_ANY_SOURCE, PF_ENDSORT_MSGTAG, &src, &tag);
            PF_LongSingleUnpack(PF_stats[src], PF_STATS_SIZE, PF_LONG);
            {
                WORD numdummies, expchanged;
                PF_LongSingleUnpack(&numdummies, 1, PF_WORD);
                PF_LongSingleUnpack(&expchanged, 1, PF_WORD);
                if ( e->numdummies < numdummies ) e->numdummies = numdummies;
                AR.expchanged |= expchanged;
            }
            /* Now handle redefined preprocessor variables. */
            if ( AC.numpfirstnum > 0 ) PF_UnpackRedefinedPreVars();
        }
        /* Broadcast redefined preprocessor variables. */
        if ( AC.numpfirstnum > 0 ) {
            int RetCode = PF_BroadcastRedefinedPreVars();
            if ( RetCode ) return RetCode;
        }
        if ( ! AC.OldParallelStats ) {
            /* Now we can calculate AT.SS->GenTerms from the statistics of the slaves. */
            LONG genterms = 0;
            for ( k = 1; k < PF.numtasks; k++ ) {
                genterms += PF_stats[k][3];
            }
            AT.SS->GenTerms = genterms;
            WriteStats(&PF_exprsize, STATSPOSTSORT, NOCHECKLOGTYPE);
            Expressions[AR.CurExpr].size = PF_exprsize;
        }
        PF_Statistics(PF_stats,0);
/*
            #] Collect (stats,prepro,...):
            #[ Update flags :
*/
        if ( AM.S0->TermsLeft ) e->vflags &= ~ISZERO;
        else e->vflags |= ISZERO;
        if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
        if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
        if ( AR.expchanged ) AR.expflags |= ISUNMODIFIED;
/*
            #] Update flags :
        #] Master:
*/
    }
    else {
/*
        #[ Slave :
*/
/*
            #[ Generator Loop & EndSort :

            loop for all terms to get from master, call Generator for each of them
            then call EndSort and do cleanup (to be implemented)
*/
        WORD oldBracketOn = AR.BracketOn;
        WORD *oldBrackBuf = AT.BrackBuf;
        WORD oldbracketindexflag = AT.bracketindexflag;

        /* For redefine statements. */
        if ( AC.numpfirstnum > 0 ) {
            for ( j = 0; j < AC.numpfirstnum; j++ ) {
                AC.inputnumbers[j] = -1;
            }
        }
```

```
        SeekScratch(AR.outfile,&position);
        e->onfile = position;
        AR.DeferFlag = AC.ComDefer;
        AR.Eside = RHSIDE;
        if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
            AR.BracketOn = 1;
            AT.BrackBuf = AM.BracketFactors;
            AT.bracketindexflag = 1;
        }
        NewSort(BHEAD0);
        AR.MaxDum = AM.IndDum;
        AN.ninterms = 0;
        PF_linterms = 0;
        PF.parallel = 1;
#ifdef MPI2
        AR.infile->POfull = AR.infile->POfill = AR.infile->PObuffer = PF_shared_buff;
#endif
        {
            FILEHANDLE *fi = AC.RhsExprInModuleFlag && PF.rhsInParallel ? &PF.slavebuf : AR.infile;
            fi->POfull = fi->POfill = fi->PObuffer;
        }
        /* FIXME: AN.ninterms is still broken when AN.deferskipped is non-zero.
         *        It still needs some work, also in PF_GetTerm(). (TU 30 Aug 2011) */
        while ( PF_GetTerm(term) ) {
            PF_linterms++; AN.ninterms++; dd = AN.deferskipped;
            AT.WorkPointer = term + *term;
            AN.RepPoint = AT.RepCount + 1;
            if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
                StoreTerm(BHEAD term);
                continue;
            }
            if ( AR.DeferFlag ) {
                AR.CurDum = AN.IndDum = Expressions[AR.CurExpr].numdummies + AM.IndDum;
            }
            else {
                AN.IndDum = AM.IndDum;
                AR.CurDum = ReNumber(BHEAD term);
            }
            if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
            if ( AN.ncmod ) {
                if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
                else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
            }
            else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
            if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
                && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
                PolyFunClean(BHEAD term);
            }
            if ( Generator(BHEAD term,0) ) {
                MesPrint("[%d] PF_Processor: Error in Generator",PF.me);
                LowerSortLevel(); return(-1);
            }
            PF_linterms += dd; AN.ninterms += dd;
        }
        PF_linterms += dd; AN.ninterms += dd;
        {
            /*
             * EndSort() overrides AR.outfile->PObuffer etc. (See also PF_EndSort()),
             * but it causes a problem because
             *  (1) PF_EndSort() sets AR.outfile->PObuffer to a send-buffer.
             *  (2) RevertScratch() clears AR.infile, but then swaps buffers of AR.infile
             *      and AR.outfile.
             *  (3) RHS expressions are stored to AR.infile->PObuffer.
             *  (4) Again, PF_EndSort() sets AR.outfile->PObuffer, but now AR.outfile->PObuffer
             *      == AR.infile->PObuffer because of (1) and (2).
             *  (5) The result goes to AR.outfile. This breaks the RHS expressions,
             *      which may be needed for the next expression.
             * Solution: backup & restore AR.outfile->PObuffer etc. (TU 14 Sep 2011)
             */
            FILEHANDLE *fout = AR.outfile;
            WORD *oldbuff = fout->PObuffer;
            WORD *oldstop = fout->POstop;
            LONG  oldsize = fout->POsize;
            if ( EndSort(BHEAD AM.S0->sBuffer, 0) < 0 ) return -1;
            fout->PObuffer = oldbuff;
            fout->POstop   = oldstop;
            fout->POsize   = oldsize;
            fout->POfill = fout->POfull = fout->PObuffer;
        }
        AR.BracketOn = oldBracketOn;
        AT.BrackBuf = oldBrackBuf;
        AT.bracketindexflag = oldbracketindexflag;
/*
            #] Generator Loop & EndSort :
            #[ Collect (stats,prepro...) :
*/
```

```
        DBGOUT_NINTERMS(1, ("PF.me=%d AN.ninterms=%d PF_linterms=%d ENDSORT\n", (int)PF.me,
 (int)AN.ninterms, (int)PF_linterms));
        PF_PrepareLongSinglePack();
        cpu = TimeCPU(1);
        size = 0;
        PF_LongSinglePack(&cpu,               1, PF_LONG);
        PF_LongSinglePack(&size,              1, PF_LONG);
        PF_LongSinglePack(&PF_linterms,       1, PF_LONG);
        PF_LongSinglePack(&AM.S0->GenTerms,   1, PF_LONG);
        PF_LongSinglePack(&AM.S0->TermsLeft, 1, PF_LONG);
        {
            WORD numdummies = AR.MaxDum - AM.IndDum;
            PF_LongSinglePack(&numdummies,     1, PF_WORD);
            PF_LongSinglePack(&AR.expchanged, 1, PF_WORD);
        }
        /* Now handle redefined preprocessor variables. */
        if ( AC.numpfirstnum > 0 ) PF_PackRedefinedPreVars();
        PF_LongSingleSend(MASTER, PF_ENDSORT_MSGTAG);
        /* Broadcast redefined preprocessor variables. */
        if ( AC.numpfirstnum > 0 ) {
            int RetCode = PF_BroadcastRedefinedPreVars();
            if ( RetCode ) return RetCode;
        }
/*
            #] Collect (stats,prepro...) :

        This operation is moved to the beginning of each block, see PreProcessor
        in pre.c.

        #] Slave :
*/
        if ( PF.log ) {
            UBYTE lbuf[24];
            NumToStr(lbuf,AC.CModule);
            fprintf(stderr,"[%d|%s] Endsort,Collect,Broadcast done\n",PF.me,lbuf);
            fflush(stderr);
        }
    }
    return(0);
}

/*
        #] PF_Processor :
    #] proces.c :
    #[ startup :, prepro & compile
        #[ PF_Init :
*/

int PF_Init(int *argc, char ***argv)
{
/*
        this should definitely be somewhere else ...
*/
    PF_CurrentBracket = 0;

    PF.numtasks = 0; /* number of tasks, is determined in PF_LibInit ! */
    PF.numsbufs = 2; /* might be changed by the environment variable on the master ! */
    PF.numrbufs = 2; /* might be changed by the environment variable on the master ! */

    int ret = PF_LibInit(argc,argv);
    if (ret) { return ret; }
    PF_RealTime(PF_RESET);

    PF.log = 0;
    PF.parallel = 0;
    PF_statsinterval = 10;
    PF.rhsInParallel=1;
    PF.exprbufsize=4096;/*in WORDs*/

#ifdef PF_WITHGETENV
    if ( PF.me == MASTER ) {
        char *c;
/*
            get these from the environment at the moment should be in setfile/tail
*/
        if ( ( c = getenv("PF_LOG") ) != 0 ) {
            if ( *c ) PF.log = (int)atoi(c);
            else PF.log = 1;
            fprintf(stderr,"[%d] changing PF.log to %d\n",PF.me,PF.log);
            fflush(stderr);
        }
        if ( ( c = (char*)getenv("PF_RBUFS") ) != 0 ) {
            PF.numrbufs = (int)atoi(c);
            fprintf(stderr,"[%d] changing numrbufs to: %d\n",PF.me,PF.numrbufs);
            fflush(stderr);
        }
        if ( ( c = (char*)getenv("PF_SBUFS") ) != 0 ) {
```

```
            PF.numsbufs = (int)atoi(c);
            fprintf(stderr,"[%d] changing numsbufs to: %d\n",PF.me,PF.numsbufs);
            fflush(stderr);
        }
        if ( PF.numsbufs > 10 ) PF.numsbufs = 10;
        if ( PF.numsbufs <  1 ) PF.numsbufs = 1;
        if ( PF.numrbufs >  2 ) PF.numrbufs = 2;
        if ( PF.numrbufs <  1 ) PF.numrbufs = 1;

        if ( ( c = getenv("PF_STATS") ) ) {
            UBYTE lbuf[24];
            PF_statsinterval = (int)atoi(c);
            NumToStr(lbuf,PF_statsinterval);
            fprintf(stderr,"[%d] changing PF_statsinterval to %s\n",PF.me,lbuf);
            fflush(stderr);
            if ( PF_statsinterval < 1 ) PF_statsinterval = 10;
        }
    }
#endif
/*
    #[ Broadcast settings from getenv: could also be done in PF_DoSetup
*/
    if ( PF.me == MASTER ) {
        PF_PreparePack();
        PF_Pack(&PF.log,1,PF_INT);
        PF_Pack(&PF.numrbufs,1,PF_WORD);
        PF_Pack(&PF.numsbufs,1,PF_WORD);
    }
    PF_Broadcast();
    if ( PF.me != MASTER ) {
        PF_Unpack(&PF.log,1,PF_INT);
        PF_Unpack(&PF.numrbufs,1,PF_WORD);
        PF_Unpack(&PF.numsbufs,1,PF_WORD);
        if ( PF.log ) {
            fprintf(stderr, "[%d] log=%d rbufs=%d sbufs=%d\n",
                    PF.me, PF.log, PF.numrbufs, PF.numsbufs);
            fflush(stderr);
        }
    }
/*
    #] Broadcast settings from getenv:
*/
    return(0);
}
/*
        #] PF_Init :
        #[ PF_Terminate :
*/

int PF_Terminate(int errorcode)
{
    return PF_LibTerminate(errorcode);
}

/*
        #] PF_Terminate :
        #[ PF_GetSlaveTimes :
*/

LONG PF_GetSlaveTimes(void)
{
    LONG slavetimes = 0;
    LONG t = PF.me == MASTER ? 0 : AM.SumTime + TimeCPU(1);
    MPI_Reduce(&t, &slavetimes, 1, PF_LONG, MPI_SUM, MASTER, PF_COMM);
    return slavetimes;
}

/*
        #] PF_GetSlaveTimes :
    #] startup :
    #[ PF_BroadcastNumber :
*/

LONG PF_BroadcastNumber(LONG x)
{
#ifdef PF_DEBUG_BCAST_LONG
    if ( PF.me == MASTER ) {
        MesPrint("» Broadcast LONG: %l", x);
    }
#endif
    PF_Bcast(&x, sizeof(LONG));
    return x;
}

/*
    #] PF_BroadcastNumber :
    #[ PF_BroadcastBuffer :
```

```
*/

void PF_BroadcastBuffer(WORD **buffer, LONG *length)
{
    WORD *p;
    LONG rest;
#ifdef PF_DEBUG_BCAST_BUF
    if ( PF.me == MASTER ) {
        MesPrint(">> Broadcast Buffer: length=%l", *length);
    }
#endif
    /* Initialize the buffer on the slaves. */
    if ( PF.me != MASTER ) {
        *buffer = NULL;
    }
    /* Broadcast the length of the buffer. */
    *length = PF_BroadcastNumber(*length);
    if ( *length <= 0 ) return;
    /* Allocate the buffer on the slaves. */
    if ( PF.me != MASTER ) {
        *buffer = (WORD *)Malloc1(*length * sizeof(WORD), "PF_BroadcastBuffer");
    }
    /* Broadcast the data in the buffer. */
    p = *buffer;
    rest = *length;
    while ( rest > 0 ) {
        int l = rest < (LONG)PF.exprbufsize ? (int)rest : PF.exprbufsize;
        PF_Bcast(p, l * sizeof(WORD));
        p += l;
        rest -= l;
    }
}

/*
    #] PF_BroadcastBuffer :
    #[ PF_BroadcastString :
*/

int PF_BroadcastString(UBYTE *str)
{
    int clength = 0;
/*
        If string does not fit to the PF_buffer, it
        will be split into chunks. Next chunk is started at  str+clength
*/
        UBYTE *cstr=str;
/*
        Note, compilation is performed INDEPENDENTLY on AC.mparallelflag!
        No if ( AC.mparallelflag == PARALLELFLAG ) !!
*/
    do {
        cstr += clength; /*at each step for all slaves and master */

        if ( MASTER == PF.me ) { /*Pack str*/
/*
                initialize buffers
*/
            if ( PF_PreparePack() != 0 ) Terminate(-1);
            if ( ( clength = PF_PackString(cstr) ) <0  ) Terminate(-1);
        }
        PF_Broadcast();

        if ( MASTER != PF.me ) {
/*
                Slave - unpack received string
                For slaves buffers are initialised automatically.
*/
            if ( ( clength = PF_UnpackString(cstr) ) < 0 ) Terminate(-1);
        }
    } while ( cstr[clength-1] != '\0' );
    return (0);
}

/*
    #] PF_BroadcastString :
    #[ PF_BroadcastPreDollar :
*/

int PF_BroadcastPreDollar(WORD **dbuffer, LONG *newsize, int *numterms)
{
    int err = 0;
    LONG i;
/*
        Note, compilation is performed INDEPENDENTLY on AC.mparallelflag!
        No if(AC.mparallelflag==PARALLELFLAG) !!
*/
    if ( MASTER == PF.me ) {
```

```
/*
            The problem is that sometimes dollar variables are longer
            than PF_packbuf! So we split long expression into chunks.
            There are n filled chunks and one partially filled chunk:
*/
        LONG n = ((*newsize)+1)/PF_maxDollarChunkSize;
/*
            ...and one more chunk for the rest; if the expression fits to
            the buffer without splitting, the latter will be the only one.

            PF_maxDollarChunkSize is the maximal number of items fitted to
            the buffer. It is calculated in PF_LibInit() in mpi.c.
            PF_maxDollarChunkSize is calculated for the first step, when
            two fields (numterms and newsize, see below) are already packed.
            For simplicity, this value is used also for all steps, in
            despite  of it is   a bit less than maximally available space.
*/
        WORD *thechunk = *dbuffer;

        err = PF_PreparePack();             /* initialize buffers */
        err |= PF_Pack(numterms,1,PF_INT);
        err |= PF_Pack(newsize,1,PF_LONG); /* pack the size */
/*
            Pack and broadcast completely filled chunks.
            It may happen, this loop is not entered at all:
*/
        for ( i = 0; i < n; i++ ) {
            err |= PF_Pack(thechunk,PF_maxDollarChunkSize,PF_WORD);
            err |= PF_Broadcast();
            thechunk +=PF_maxDollarChunkSize;
            PF_PreparePack();
        }
/*
            Pack and broadcast the rest:
*/
        if ( ( n = ( (*newsize)+1)%PF_maxDollarChunkSize ) != 0 ) {
            err |= PF_Pack(thechunk,n,PF_WORD);
            err |= PF_Broadcast();
        }
#ifdef PF_DEBUG_BCAST_PREDOLLAR
        MesPrint("» Broadcast PreDollar: newsize=%d numterms=%d", (int)*newsize, *numterms);
#endif
    }
    if ( MASTER != PF.me ) {  /* Slave - unpack received buffer */
        WORD *thechunk;
        LONG n, therest, thesize;
        err |= PF_Broadcast();
        err |=PF_Unpack(numterms,1,PF_INT);
        err |=PF_Unpack(newsize,1,PF_LONG);
/*
            Now we know the buffer size.
*/
        thesize = (*newsize)+1;
/*
            Evaluate the number of completely filled chunks. The last step must be
            treated separately, so -1:
*/
        n = (thesize/PF_maxDollarChunkSize) - 1;
/*
            Note, here n can be <0, this is ok.
*/
        therest = thesize % PF_maxDollarChunkSize;
        thechunk = *dbuffer =
            (WORD*)Malloc1( thesize * sizeof(WORD),"$-buffer slave");
        if ( thechunk == NULL ) return(err|4);
/*
            Unpack completely filled chunks and receive the next portion.
            It may happen, this loop is not entered at all:
*/
        for ( i = 0; i < n; i++ ) {
            err |= PF_Unpack(thechunk,PF_maxDollarChunkSize,PF_WORD);
            thechunk += PF_maxDollarChunkSize;
            err |= PF_Broadcast();
        }
/*
            Now the last completely filled chunk:
*/
        if ( n >= 0 ) {
            err |= PF_Unpack(thechunk,PF_maxDollarChunkSize,PF_WORD);
            thechunk += PF_maxDollarChunkSize;
            if ( therest != 0 ) err |= PF_Broadcast();
        }
/*
            Unpack the rest (it is already received!):
*/
        if ( therest != 0 ) err |= PF_Unpack(thechunk,therest,PF_WORD);
    }
```

```
    return (err);
}

/*
    #] PF_BroadcastPreDollar :
    #[ Synchronization of modified dollar variables :
        #[ Helper functions :
            #[ dollarlen :
*/

static inline LONG dollarlen(const WORD *terms)
{
    const WORD *p = terms;
    while ( *p ) p += *p;
    return p - terms;  /* Not including the null terminator. */
}

/*
            #] dollarlen :
            #[ dollar_mod_type :
*/

static inline WORD dollar_mod_type(WORD index)
{
    int i;
    for ( i = 0; i < NumModOptdollars; i++ )
        if ( ModOptdollars[i].number == index ) break;
    if ( i >= NumModOptdollars ) return -1;
    return ModOptdollars[i].type;
}


/*
            #] dollar_mod_type :
        #] Helper functions :
        #[ PF_CollectModifiedDollars :
*/

/*
            #[ dollar_to_be_collected :
*/

static inline int dollar_to_be_collected(WORD index)
{
    switch ( dollar_mod_type(index) ) {
        case MODSUM:
        case MODMAX:
        case MODMIN:
            return 1;
        default:
            return 0;
    }
}

/*
            #] dollar_to_be_collected :
            #[ copy_dollar :
*/

static inline void copy_dollar(WORD index, WORD type, const WORD *where, LONG size)
{
    DOLLARS d = Dollars + index;

    CleanDollarFactors(d);

    if ( type != DOLZERO && where != NULL && where != &AM.dollarzero && where[0] != 0 && size > 0 ) {
        if ( size > d->size || size < d->size / 4 ) {  /* Reallocate if not enough or too much. */
            if ( d->where && d->where != &AM.dollarzero )
                M_free(d->where, "old content of dollar");
            d->where = Malloc1(sizeof(WORD) * size, "copy buffer to dollar");
            d->size  = size;
        }
        d->type  = type;
        WCOPY(d->where, where, size);
    }
    else {
        if ( d->where && d->where != &AM.dollarzero )
            M_free(d->where, "old content of dollar");
        d->type  = DOLZERO;
        d->where = &AM.dollarzero;
        d->size  = 0;
    }
}

/*
            #] copy_dollar :
            #[ compare_two_expressions :
```

```
*/

static inline int compare_two_expressions(const WORD *e1, const WORD *e2)
{
    GETIDENTITY
    /*
     * We consider the cases that
     *   (1) the expression has no term,
     *   (2) the expression has only one term and it is a number,
     *   (3) otherwise.
     * Assume that the expressions are sorted and all terms are normalized.
     * The numerators of the coefficients must never be zero.
     *
     * Note that TwoExprCompare() is not adequate for our purpose
     * (as of 6 Aug. 2013), e.g., TwoExprCompare({0}, {4, 1, 1, -1}, LESS)
     * returns TRUE.
     */
    if ( e1[0] == 0 ) {
        if ( e2[0] == 0 ) {
            return(0);
        }
        else if ( e2[e2[0]] == 0 && e2[0] == ABS(e2[e2[0] - 1]) + 1 ) {
            if ( e2[e2[0] - 1] > 0 )
                return(-1);
            else
                return(+1);
        }
    }
    else if ( e1[e1[0]] == 0 && e1[0] == ABS(e1[e1[0] - 1]) + 1 ) {
        if ( e2[0] == 0 ) {
            if ( e1[e1[0] - 1] > 0 )
                return(+1);
            else
                return(-1);
        }
        else if ( e2[e2[0]] == 0 && e2[0] == ABS(e2[e2[0] - 1]) + 1 ) {
            return(CompCoef((WORD *)e1, (WORD *)e2));
        }
    }
    /* The expressions are not so simple. Define the order by each term. */
    while ( e1[0] && e2[0] ) {
        int c = CompareTerms(BHEAD (WORD *)e1, (WORD *)e2, 1);
        if ( c < 0 )
            return(-1);
        else if ( c > 0 )
            return(+1);
        e1 += e1[0];
        e2 += e2[0];
    }
    if ( e1[0] ) return(+1);
    if ( e2[0] ) return(-1);
    return(0);
}

/*
            #] compare_two_expressions :
            #[ Variables :
*/

typedef struct {
    VectorStruct(WORD) buf;
    LONG size;
    WORD type;
    PADPOINTER(1,0,1,0);
} dollar_buf;

/* Buffers used to store data for each variable from each slave. */
static Vector(dollar_buf, dollar_slave_bufs);

/*
            #] Variables :
*/

int PF_CollectModifiedDollars(void)
{
    int i, j, ndollars;
    /*
     * If the current module was executed in the sequential mode,
     * there are no modified module on the slaves.
     */
    if ( AC.mparallelflag != PARALLELFLAG && !AC.partodoflag ) return 0;
    /*
     * Count the number of (potentially) modified dollar variables, which we need to collect.
     * Here we need to collect all max/min/sum variables.
     */
    ndollars = 0;
    for ( i = 0; i < NumPotModdollars; i++ ) {
```

```
        WORD index = PotModdollars[i];
        if ( dollar_to_be_collected(index) ) ndollars++;
    }
    if ( ndollars == 0 ) return 0;  /* No dollars to be collected. */

    if ( PF.me == MASTER ) {
/*
            #[ Master :
*/
        int nslaves, nvars;
        /* Prepare receive buffers. We need ndollars*(PF.numtasks-1) buffers. */
        int nbufs = ndollars * (PF.numtasks - 1);
        VectorReserve(dollar_slave_bufs, nbufs);
        for ( i = VectorSize(dollar_slave_bufs); i < nbufs; i++ ) {
            VectorInit(VectorPtr(dollar_slave_bufs)[i].buf);
        }
        VectorSize(dollar_slave_bufs) = nbufs;
        /* Receive data from each slave. */
        for ( nslaves = 1; nslaves < PF.numtasks; nslaves++ ) {
            int src;
            PF_LongSingleReceive(PF_ANY_SOURCE, PF_DOLLAR_MSGTAG, &src, NULL);
            nvars = 0;
            for ( i = 0; i < NumPotModdollars; i++ ) {
                WORD index = PotModdollars[i];
                dollar_buf *b;
                if ( !dollar_to_be_collected(index) ) continue;
                b = &VectorPtr(dollar_slave_bufs)[(PF.numtasks - 1) * nvars + (src - 1)];
                PF_LongSingleUnpack(&b->type, 1, PF_WORD);
                if ( b->type != DOLZERO ) {
                    LONG size;
                    WORD *where;
                    PF_LongSingleUnpack(&size, 1, PF_LONG);
                    VectorReserve(b->buf, size + 1);
                    where = VectorPtr(b->buf);
                    PF_LongSingleUnpack(where, size, PF_WORD);
                    where[size] = 0;  /* The null terminator is needed. */
                    b->size = size + 1;  /* Including the null terminator. */
                    /* Note that we don't collect factored stuff for max/min/sum variables. */
                }
                else {
                    VectorReserve(b->buf, 1);
                    VectorPtr(b->buf)[0] = 0;
                    b->size = 0;
                }
                nvars++;
            }
        }
        /*
         * Combine received dollars. The FORM reference manual says maximum/minimum/sum
         * $-variables must have a numerical value, however, this routine should work also
         * for non-numerical cases, although the maximum/minimum value for non-numerical
         * terms has ambiguity.
         */
        nvars = 0;
        for ( i = 0; i < NumPotModdollars; i++ ) {
            WORD index = PotModdollars[i];
            WORD dtype;
            DOLLARS d;
            dollar_buf *b;
            if ( !dollar_to_be_collected(index) ) continue;
            d = Dollars + index;
            b = &VectorPtr(dollar_slave_bufs)[(PF.numtasks - 1) * nvars];
            dtype = dollar_mod_type(index);
            switch ( dtype ) {
                case MODMAX:
                case MODMIN: {
/*
            #[ MODMAX & MODMIN :
*/
                    int selected = 0;
                    for ( j = 1; j < PF.numtasks - 1; j++ ) {
                        int c = compare_two_expressions(VectorPtr(b[j].buf),
VectorPtr(b[selected].buf));
                        if ( (dtype == MODMAX && c > 0) || (dtype == MODMIN && c < 0) )
                            selected = j;
                    }
                    b = b + selected;
                    copy_dollar(index, b->type, VectorPtr(b->buf), b->size);
/*
            #] MODMAX & MODMIN :
*/
                    break;
                }
                case MODSUM: {
/*
            #[ MODSUM :
*/
```

```
        GETIDENTITY
        int err = 0;

        CBUF *C = cbuf + AM.rbufnum;
        WORD *oldwork = AT.WorkPointer, *oldcterm = AN.cTerm;
        WORD olddefer = AR.DeferFlag, oldnumlhs = AR.Cnumlhs, oldnumrhs = C->numrhs;

        LONG size;
        WORD type, *dbuf;

        AN.cTerm = 0;
        AR.DeferFlag = 0;

        if ( ((WORD *)((UBYTE *)AT.WorkPointer + AM.MaxTer)) > AT.WorkTop ) {
            err = -1;
            goto cleanup;
            MesWork();
        }

        if ( NewSort(BHEAD0) ) {
            err = -1;
            goto cleanup;
        }
        if ( NewSort(BHEAD0) ) {
            LowerSortLevel();
            err = -1;
            goto cleanup;
        }

        /*
         * Sum up the original $-variable in the master and $-variables on all slaves.
         * Note that $-variables on the slaves are set to zero at the beginning of
         * the module (See also DoExecute()).
         */
        for ( j = 0; j < PF.numtasks; j++ ) {
            const WORD *r;
            for ( r = j == 0 ? Dollars[index].where : VectorPtr(b[j - 1].buf); *r; r += *r ) {
                WCOPY(AT.WorkPointer, r, *r);
                AT.WorkPointer += *r;
                AR.Cnumlhs = 0;
                if ( Generator(BHEAD oldwork, 0) ) {
                    LowerSortLevel(); LowerSortLevel();
                    err = -1;
                    goto cleanup;
                }
                AT.WorkPointer = oldwork;
            }
        }

        size = EndSort(BHEAD (WORD *)&dbuf, 2);
        if ( size < 0 ) {
            LowerSortLevel();
            err = -1;
            goto cleanup;
        }
        LowerSortLevel();

        /* Find special cases. */
        type = DOLTERMS;
        if ( dbuf[0] == 0 ) {
            type = DOLZERO;
        }
        else if ( dbuf[dbuf[0]] == 0 ) {
            const WORD *t = dbuf, *w;
            WORD n, nsize;
            n = *t;
            nsize = t[n - 1];
            if ( nsize < 0 ) nsize = -nsize;
            if ( nsize == n - 1 ) {
                nsize = (nsize - 1) / 2;
                w = t + 1 + nsize;
                if ( *w == 1 ) {
                    w++; while ( w < t + n - 1 ) { if ( *w ) break; w++; }
                    if ( w >= t + n - 1 ) type =  DOLNUMBER;
                }
                else if ( n == 7 && t[6] == 3 && t[5] == 1 && t[4] == 1 && t[1] == INDEX && t[2] == 3 ) {
                    type = DOLINDEX;
                    d->index = t[3];
                }
            }
        }
        copy_dollar(index, type, dbuf, dollarlen(dbuf) + 1);
        M_free(dbuf, "temporary dollar buffer");
cleanup:
        AR.Cnumlhs = oldnumlhs;
```

```
                    C->numrhs = oldnumrhs;
                    AR.DeferFlag = olddefer;
                    AN.cTerm = oldcterm;
                    AT.WorkPointer = oldwork;

                    if ( err ) return err;
/*
            #] MODSUM :
*/
                    break;
                }
            }
            if ( d->type == DOLTERMS )
                cbuf[AM.dbufnum].CanCommu[index] = numcommute(d->where,
&cbuf[AM.dbufnum].NumTerms[index]);
            cbuf[AM.dbufnum].rhs[index] = d->where;
            nvars++;
#ifdef PF_DEBUG_REDUCE_DOLLAR
            MesPrint("« Reduce $-var: %s", AC.dollarnames->namebuffer + d->name);
#endif
        }
/*
            #] Master :
*/
    }
    else {
/*
            #[ Slave :
*/
        PF_PrepareLongSinglePack();
        /* Pack each variable. */
        for ( i = 0; i < NumPotModdollars; i++ ) {
            WORD index = PotModdollars[i];
            DOLLARS d;
            if ( !dollar_to_be_collected(index) ) continue;
            d = Dollars + index;
            PF_LongSinglePack(&d->type, 1, PF_WORD);
            if ( d->type != DOLZERO ) {
                /*
                 * NOTE: d->size is the allocated buffer size for d->where in WORDs.
                 *       So dollarlen(d->where) can be < d->size-1. (TU 15 Dec 2011)
                 */
                LONG size = dollarlen(d->where);
                PF_LongSinglePack(&size, 1, PF_LONG);
                PF_LongSinglePack(d->where, size, PF_WORD);
                /* Note that we don't collect factored stuff for max/min/sum variables. */
            }
        }
        PF_LongSingleSend(MASTER, PF_DOLLAR_MSGTAG);
/*
            #] Slave :
*/
    }
    return 0;
}

/*
        #] PF_CollectModifiedDollars :
        #[ PF_BroadcastModifiedDollars :
*/

/*
            #[ dollar_to_be_broadcast :
*/

static inline int dollar_to_be_broadcast(WORD index)
{
    switch ( dollar_mod_type(index) ) {
        case MODLOCAL:
            return 0;
        default:
            return 1;
    }
}

/*
            #] dollar_to_be_broadcast :
*/

int PF_BroadcastModifiedDollars(void)
{
    int i, j, ndollars;
    /*
     * Count the number of (potentially) modified dollar variables, which we need to broadcast.
     * Here we need to broadcast all non-local variables.
     */
    ndollars = 0;
```

```
    for ( i = 0; i < NumPotModdollars; i++ ) {
        WORD index = PotModdollars[i];
        if ( dollar_to_be_broadcast(index) ) ndollars++;
    }
    if ( ndollars == 0 ) return 0;  /* No dollars to be broadcast. */

    if ( PF.me == MASTER ) {
/*
            #[ Master :
*/
        PF_PrepareLongMultiPack();
        /* Pack each variable. */
        for ( i = 0; i < NumPotModdollars; i++ ) {
            WORD index = PotModdollars[i];
            DOLLARS d;
            if ( !dollar_to_be_broadcast(index) ) continue;
            d = Dollars + index;
            PF_LongMultiPack(&d->type, 1, PF_WORD);
            if ( d->type != DOLZERO ) {
                /*
                 * NOTE: d->size is the allocated buffer size for d->where in WORDs.
                 *       So dollarlen(d->where) can be < d->size-1. (TU 15 Dec 2011)
                 */
                LONG size = dollarlen(d->where);
                PF_LongMultiPack(&size, 1, PF_LONG);
                PF_LongMultiPack(d->where, size, PF_WORD);
                /* ...and the factored stuff. */
                PF_LongMultiPack(&d->nfactors, 1, PF_WORD);
                if ( d->nfactors > 1 ) {
                    for ( j = 0; j < d->nfactors; j++ ) {
                        FACDOLLAR *f = &d->factors[j];
                        PF_LongMultiPack(&f->type, 1, PF_WORD);
                        PF_LongMultiPack(&f->size, 1, PF_LONG);
                        if ( f->size > 0 )
                            PF_LongMultiPack(f->where, f->size, PF_WORD);
                        else
                            PF_LongMultiPack(&f->value, 1, PF_WORD);
                    }
                }
            }
#ifdef PF_DEBUG_BCAST_DOLLAR
            MesPrint(">> Broadcast $-var: %s", AC.dollarnames->namebuffer + d->name);
#endif
        }
/*
            #] Master :
*/
    }
    if ( PF_LongMultiBroadcast() ) return -1;
    if ( PF.me != MASTER ) {
/*
            #[ Slave :
*/
        for ( i = 0; i < NumPotModdollars; i++ ) {
            WORD index = PotModdollars[i];
            DOLLARS d;
            if ( !dollar_to_be_broadcast(index) ) continue;
            d = Dollars + index;
            /* Clear the contents of the dollar variable. */
            if ( d->where && d->where != &AM.dollarzero )
                M_free(d->where, "old content of dollar");
            d->where = &AM.dollarzero;
            d->size = 0;
            CleanDollarFactors(d);
            /* Unpack and store the contents. */
            PF_LongMultiUnpack(&d->type, 1, PF_WORD);
            if ( d->type != DOLZERO ) {
                LONG size;
                PF_LongMultiUnpack(&size, 1, PF_LONG);
                d->size = size + 1;
                d->where = (WORD *)Malloc1(sizeof(WORD) * d->size, "dollar content");
                PF_LongMultiUnpack(d->where, size, PF_WORD);
                d->where[size] = 0;  /* The null terminator is needed. */
                /* ...and the factored stuff. */
                PF_LongMultiUnpack(&d->nfactors, 1, PF_WORD);
                if ( d->nfactors > 1 ) {
                    d->factors = (FACDOLLAR *)Malloc1(sizeof(FACDOLLAR) * d->nfactors, "dollar
factored stuff");
                    for ( j = 0; j < d->nfactors; j++ ) {
                        FACDOLLAR *f = &d->factors[j];
                        PF_LongMultiUnpack(&f->type, 1, PF_WORD);
                        PF_LongMultiUnpack(&f->size, 1, PF_LONG);
                        if ( f->size > 0 ) {
                            f->where = (WORD *)Malloc1(sizeof(WORD) * (f->size + 1), "dollar factor
content");
                            PF_LongMultiUnpack(f->where, f->size, PF_WORD);
                            f->where[f->size] = 0;  /* The null terminator is needed. */
```

```
                            f->value = 0;
                        }
                        else {
                            f->where = NULL;
                            PF_LongMultiUnpack(&f->value, 1, PF_WORD);
                        }
                    }
                }
            }
            if ( d->type == DOLTERMS )
                cbuf[AM.dbufnum].CanCommu[index] = numcommute(d->where,
&cbuf[AM.dbufnum].NumTerms[index]);
            cbuf[AM.dbufnum].rhs[index] = d->where;
        }
/*
            #] Slave :
*/
    }
    return 0;
}

/*
        #] PF_BroadcastModifiedDollars :
    #] Synchronization of modified dollar variables :
    #[ Synchronization of redefined preprocessor variables :
        #[ Variables :
*/

/* A buffer used in receivers. */
static Vector(UBYTE, prevarbuf);

/*
        #] Variables :
        #[ PF_PackRedefinedPreVars :
*/

static void PF_PackRedefinedPreVars(void)
{
    int i;
    /* First, pack the number of redefined preprocessor variables. */
    int nredefs = 0;
    for ( i = 0; i < AC.numpfirstnum; i++ )
        if ( AC.inputnumbers[i] >= 0 ) nredefs++;
    PF_LongSinglePack(&nredefs, 1, PF_INT);
    /* Then, pack each variable. */
    for ( i = 0; i < AC.numpfirstnum; i++ )
        if ( AC.inputnumbers[i] >= 0) {
            WORD index = AC.pfirstnum[i];
            UBYTE *value = PreVar[index].value;
            int bytes = strlen((char *)value);
            PF_LongSinglePack(&index, 1, PF_WORD);
            PF_LongSinglePack(&bytes, 1, PF_INT);
            PF_LongSinglePack(value, bytes, PF_BYTE);
            PF_LongSinglePack(&AC.inputnumbers[i], 1, PF_LONG);
    }
}

/*
        #] PF_PackRedefinedPreVars :
        #[ PF_UnpackRedefinedPreVars :
*/

static void PF_UnpackRedefinedPreVars(void)
{
    int i, j;
    /* Unpack the number of redefined preprocessor variables. */
    int nredefs;
    PF_LongSingleUnpack(&nredefs, 1, PF_INT);
    if ( nredefs > 0 ) {
        /* Then unpack each variable. */
        for ( i = 0; i < nredefs; i++ ) {
            WORD index;
            int bytes;
            UBYTE *value;
            LONG inputnumber;
            PF_LongSingleUnpack(&index, 1, PF_WORD);
            PF_LongSingleUnpack(&bytes, 1, PF_INT);
            VectorReserve(prevarbuf, bytes + 1);
            value = VectorPtr(prevarbuf);
            PF_LongSingleUnpack(value, bytes, PF_BYTE);
            value[bytes] = '\0';  /* The null terminator is needed. */
            PF_LongSingleUnpack(&inputnumber, 1, PF_LONG);
            /* Put this variable if it must be updated. */
            for ( j = 0; j < AC.numpfirstnum; j++ )
                if ( AC.pfirstnum[j] == index ) break;
            if ( AC.inputnumbers[j] < inputnumber ) {
                AC.inputnumbers[j] = inputnumber;
```

```
                PutPreVar(PreVar[index].name, value, NULL, 1);
            }
        }
    }
}

/*
        #] PF_UnpackRedefinedPreVars :
        #[ PF_BroadcastRedefinedPreVars :
*/

int PF_BroadcastRedefinedPreVars(void)
{
    /*
     * NOTE: Because the compilation is performed on the all processes
     * independently on AC.mparallelflag, we always have to broadcast redefined
     * preprocessor variables from the master to the all slaves.
     */
    if ( PF.me == MASTER ) {
/*
            #[ Master :
*/
        int i, nredefs;
        PF_PrepareLongMultiPack();
        /* First, pack the number of redefined preprocessor variables. */
        nredefs = 0;
        for ( i = 0; i < AC.numpfirstnum; i++ )
            if ( AC.inputnumbers[i] >= 0 ) nredefs++;
        PF_LongMultiPack(&nredefs, 1, PF_INT);
        /* Then, pack each variable. */
        for ( i = 0; i < AC.numpfirstnum; i++ )
            if ( AC.inputnumbers[i] >= 0) {
                WORD index = AC.pfirstnum[i];
                UBYTE *value = PreVar[index].value;
                int bytes = strlen((char *)value);
                PF_LongMultiPack(&index, 1, PF_WORD);
                PF_LongMultiPack(&bytes, 1, PF_INT);
                PF_LongMultiPack(value, bytes, PF_BYTE);
#ifdef PF_DEBUG_BCAST_PREVAR
                MesPrint("» Broadcast PreVar: %s = \"%s\"", PreVar[index].name, value);
#endif
            }
/*
            #] Master :
*/
    }
    if ( PF_LongMultiBroadcast() ) return -1;
    if ( PF.me != MASTER ) {
/*
            #[ Slave :
*/
        int i, nredefs;
        /* Unpack the number of redefined preprocessor variables. */
        PF_LongMultiUnpack(&nredefs, 1, PF_INT);
        if ( nredefs > 0 ) {
            /* Then unpack each variable and put it. */
            for ( i = 0; i < nredefs; i++ ) {
                WORD index;
                int bytes;
                UBYTE *value;
                PF_LongMultiUnpack(&index, 1, PF_WORD);
                PF_LongMultiUnpack(&bytes, 1, PF_INT);
                VectorReserve(prevarbuf, bytes + 1);
                value = VectorPtr(prevarbuf);
                PF_LongMultiUnpack(value, bytes, PF_BYTE);
                value[bytes] = '\0';  /* The null terminator is needed. */
                PutPreVar(PreVar[index].name, value, NULL, 1);
            }
        }
/*
            #] Slave :
*/
    }
    return 0;
}

/*
        #] PF_BroadcastRedefinedPreVars :
    #] Synchronization of redefined preprocessor variables :
    #[ Preprocessor Inside instruction :
        #[ Variables :
*/

/* Saved values of AC.RhsExprInModuleFlag, PotModdollars and AC.pfirstnum. */
static WORD oldRhsExprInModuleFlag;
static Vector(WORD, oldPotModdollars);
static Vector(WORD, oldpfirstnum);
```

```

/*
        #] Variables :
        #[ PF_StoreInsideInfo :
*/

/*
 * Saves the current values of AC.RhsExprInModuleFlag, PotModdollars
 * and AC.pfirstnum.
 *
 * Called by DoInside().
 *
 * @return  0  if OK, nonzero on error.
 */
int PF_StoreInsideInfo(void)
{
    int i;
    oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
    VectorClear(oldPotModdollars);
    for ( i = 0; i < NumPotModdollars; i++ )
        VectorPushBack(oldPotModdollars, PotModdollars[i]);
    VectorClear(oldpfirstnum);
    for ( i = 0; i < AC.numpfirstnum; i++ )
        VectorPushBack(oldpfirstnum, AC.pfirstnum[i]);
    return 0;
}

/*
        #] PF_StoreInsideInfo :
        #[ PF_RestoreInsideInfo :
*/

/*
 * Restores the saved values of AC.RhsExprInModuleFlag, PotModdollars
 * and AC.pfirstnum.
 *
 * Called by DoEndInside().
 *
 * @return  0  if OK, nonzero on error.
 */
int PF_RestoreInsideInfo(void)
{
    int i;
    AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
    NumPotModdollars = VectorSize(oldPotModdollars);
    for ( i = 0; i < NumPotModdollars; i++ )
        PotModdollars[i] = VectorPtr(oldPotModdollars)[i];
    AC.numpfirstnum = VectorSize(oldpfirstnum);
    for ( i = 0; i < AC.numpfirstnum; i++ )
        AC.pfirstnum[i] = VectorPtr(oldpfirstnum)[i];
    return 0;
}

/*
        #] PF_RestoreInsideInfo :
    #] Preprocessor Inside instruction :
    #[ PF_BroadcastCBuf :
*/

int PF_BroadcastCBuf(int bufnum)
{
    CBUF *C = cbuf + bufnum;
    int i;
    LONG l;
    if ( PF.me == MASTER ) {
/*
        #[ Master :
*/
        PF_PrepareLongMultiPack();
        /* Pack CBUF struct except pointers. */
        PF_LongMultiPack(&C->BufferSize, 1, PF_LONG);
        PF_LongMultiPack(&C->numlhs, 1, PF_INT);
        PF_LongMultiPack(&C->numrhs, 1, PF_INT);
        PF_LongMultiPack(&C->maxlhs, 1, PF_INT);
        PF_LongMultiPack(&C->maxrhs, 1, PF_INT);
        PF_LongMultiPack(&C->mnumlhs, 1, PF_INT);
        PF_LongMultiPack(&C->mnumrhs, 1, PF_INT);
        PF_LongMultiPack(&C->numtree, 1, PF_INT);
        PF_LongMultiPack(&C->rootnum, 1, PF_INT);
        PF_LongMultiPack(&C->MaxTreeSize, 1, PF_INT);
        /* Now pointers. Pointer, lhs and rhs are packed as offsets. We don't pack Top. */
        l = C->Pointer - C->Buffer;
        PF_LongMultiPack(&l, 1, PF_LONG);
        PF_LongMultiPack(C->Buffer, l, PF_WORD);
        for ( i = 0; i < C->numlhs + 1; i++ ) {
            l = C->lhs[i] - C->Buffer;
            PF_LongMultiPack(&l, 1, PF_LONG);
```

```
        }
        for ( i = 0; i < C->numrhs + 1; i++ ) {
            l = C->rhs[i] - C->Buffer;
            PF_LongMultiPack(&l, 1, PF_LONG);
        }
        PF_LongMultiPack(C->CanCommu, C->numrhs + 1, PF_LONG);
        PF_LongMultiPack(C->NumTerms, C->numrhs + 1, PF_LONG);
        PF_LongMultiPack(C->numdum, C->numrhs + 1, PF_WORD);
        PF_LongMultiPack(C->dimension, C->numrhs + 1, PF_WORD);
        if ( C->MaxTreeSize > 0 )
            PF_LongMultiPack(C->boomlijst, (C->numtree + 1) * (sizeof(COMPTREE) / sizeof(int)),
PF_INT);
#ifdef PF_DEBUG_BCAST_CBUF
        MesPrint(">> Broadcast CBuf %d", bufnum);
#endif
/*
        #] Master :
*/
    }
    if ( PF_LongMultiBroadcast() ) return -1;
    if ( PF.me != MASTER ) {
/*
        #[ Slave :
*/
        /* First, free already allocated buffers. */
        finishcbuf(bufnum);
        /* Unpack CBUF struct except pointers. */
        PF_LongMultiUnpack(&C->BufferSize, 1, PF_LONG);
        PF_LongMultiUnpack(&C->numlhs, 1, PF_INT);
        PF_LongMultiUnpack(&C->numrhs, 1, PF_INT);
        PF_LongMultiUnpack(&C->maxlhs, 1, PF_INT);
        PF_LongMultiUnpack(&C->maxrhs, 1, PF_INT);
        PF_LongMultiUnpack(&C->mnumlhs, 1, PF_INT);
        PF_LongMultiUnpack(&C->mnumrhs, 1, PF_INT);
        PF_LongMultiUnpack(&C->numtree, 1, PF_INT);
        PF_LongMultiUnpack(&C->rootnum, 1, PF_INT);
        PF_LongMultiUnpack(&C->MaxTreeSize, 1, PF_INT);
        /* Allocate new buffers. */
        C->Buffer = (WORD *)Malloc1(C->BufferSize * sizeof(WORD), "compiler buffer");
        C->Top = C->Buffer + C->BufferSize;
        C->lhs = (WORD **)Malloc1(C->maxlhs * sizeof(WORD *), "compiler buffer");
        C->rhs = (WORD **)Malloc1(C->maxrhs * (sizeof(WORD *) + 2 * sizeof(LONG) + 2 * sizeof(WORD)),
"compiler buffer");
        C->CanCommu = (LONG *)(C->rhs + C->maxrhs);
        C->NumTerms = C->CanCommu + C->maxrhs;
        C->numdum = (WORD *)(C->NumTerms + C->maxrhs);
        C->dimension = C->numdum + C->maxrhs;
        if ( C->MaxTreeSize > 0 )
            C->boomlijst = (COMPTREE *)Malloc1(C->MaxTreeSize * sizeof(COMPTREE), "compiler buffer");
        /* Unpack buffers. */
        PF_LongMultiUnpack(&l, 1, PF_LONG);
        PF_LongMultiUnpack(C->Buffer, l, PF_WORD);
        C->Pointer = C->Buffer + l;
        for ( i = 0; i < C->numlhs + 1; i++ ) {
            PF_LongMultiUnpack(&l, 1, PF_LONG);
            C->lhs[i] = C->Buffer + l;
        }
        for ( i = 0; i < C->numrhs + 1; i++ ) {
            PF_LongMultiUnpack(&l, 1, PF_LONG);
            C->rhs[i] = C->Buffer + l;
        }
        PF_LongMultiUnpack(C->CanCommu, C->numrhs + 1, PF_LONG);
        PF_LongMultiUnpack(C->NumTerms, C->numrhs + 1, PF_LONG);
        PF_LongMultiUnpack(C->numdum, C->numrhs + 1, PF_WORD);
        PF_LongMultiUnpack(C->dimension, C->numrhs + 1, PF_WORD);
        if ( C->MaxTreeSize > 0 )
            PF_LongMultiUnpack(C->boomlijst, (C->numtree + 1) * (sizeof(COMPTREE) / sizeof(int)),
PF_INT);
/*
        #] Slave :
*/
    }
    return 0;
}

/*
    #] PF_BroadcastCBuf :
    #[ PF_BroadcastExpFlags :
*/

int PF_BroadcastExpFlags(void)
{
    WORD i;
    EXPRESSIONS e;
    if ( PF.me == MASTER ) {
/*
        #[ Master :
```

```
*/
        PF_PrepareLongMultiPack();
        PF_LongMultiPack(&AR.expflags, 1, PF_WORD);
        for ( i = 0; i < NumExpressions; i++ ) {
            e = &Expressions[i];
            PF_LongMultiPack(&e->counter,    1, PF_WORD);
            PF_LongMultiPack(&e->vflags,     1, PF_WORD);
            PF_LongMultiPack(&e->uflags,     1, PF_WORD);
            PF_LongMultiPack(&e->numdummies, 1, PF_WORD);
            PF_LongMultiPack(&e->numfactors, 1, PF_WORD);
#ifdef PF_DEBUG_BCAST_EXPRFLAGS
            MesPrint(">> Broadcast ExprFlags: %s", AC.exprnames->namebuffer + e->name);
#endif
        }
/*
        #] Master :
*/
    }
    if ( PF_LongMultiBroadcast() ) return -1;
    if ( PF.me != MASTER ) {
/*
        #[ Slave :
*/
        PF_LongMultiUnpack(&AR.expflags, 1, PF_WORD);
        for ( i = 0; i < NumExpressions; i++ ) {
            e = &Expressions[i];
            PF_LongMultiUnpack(&e->counter,    1, PF_WORD);
            PF_LongMultiUnpack(&e->vflags,     1, PF_WORD);
            PF_LongMultiUnpack(&e->uflags,     1, PF_WORD);
            PF_LongMultiUnpack(&e->numdummies, 1, PF_WORD);
            PF_LongMultiUnpack(&e->numfactors, 1, PF_WORD);
        }
/*
        #] Slave :
*/
    }
    return 0;
}

/*
    #] PF_BroadcastExpFlags :
    #[ PF_SetScratch :
*/

static void PF_SetScratch(FILEHANDLE *f,POSITION *position)
{
    if(
                ( f->handle >= 0) && ISGEPOS(*position,f->POposition) &&
                ( ISGEPOSINC(*position,f->POposition,(f->POfull-f->PObuffer)*sizeof(WORD)) ==0 )
        )/*position is inside the buffer! SetScratch() will do nothing.*/
                f->POfull=f->PObuffer;/*force SetScratch() to re-read the position from the beginning:*/
    SetScratch(f,position);
}

/*
    #] PF_SetScratch :
    #[ PF_pushScratch :
*/

static int PF_pushScratch(FILEHANDLE *f)
{
    LONG size,RetCode;
    if ( f->handle < 0){
        /*Create the file*/
        if ( ( RetCode = CreateFile(f->name) ) >= 0 ) {
            f->handle = (WORD)RetCode;
            PUTZERO(f->filesize);
            PUTZERO(f->POposition);
        }
        else{
            MesPrint("Cannot create scratch file %s",f->name);
            return(-1);
        }
    }/*if ( f->handle < 0)*/
    size = (f->POfill-f->PObuffer)*sizeof(WORD);
    if( size > 0 ){
        SeekFile(f->handle,&(f->POposition),SEEK_SET);
        if ( WriteFile(f->handle,(UBYTE *)(f->PObuffer),size) != size ){
            MesPrint("Error while writing to disk. Disk full?");
            return(-1);
        }
        ADDPOS(f->filesize,size);
        ADDPOS(f->POposition,size);
        f->POfill = f->POfull=f->PObuffer;
    }/*if( size > 0 )*/
    return(0);
}
```

```

/*
    #] PF_pushScratch :
    #[ Broadcasting RHS expressions :
        #[ PF_WalkThroughExprMaster :
    Returns <=0 if the expression is ready, or dl+1;
*/

static int PF_WalkThroughExprMaster(FILEHANDLE *curfile, int dl)
{
    LONG l=0;
    for(;;){
        if(curfile->POfull-curfile->POfill < dl){
            POSITION pos;
            SeekScratch(curfile,&pos);
            PF_SetScratch(curfile,&pos);
        }/*if(curfile->POfull-curfile->POfill < dl)*/
        curfile->POfill+=dl;
        l+=dl;
        if( l >= PF.exprbufsize){
            if( l == PF.exprbufsize){
                if( *(curfile->POfill) == 0)/*expression is ready*/
                    return(0);
                }
            l-=PF.exprbufsize;
            curfile->POfill-=l;
            return l+1;
        }

        dl=*(curfile->POfill);
        if(dl == 0)
            return l-PF.exprbufsize;

        if(dl<0){/*compressed term*/
            if(curfile->POfull-curfile->POfill < 1){
                POSITION pos;
                SeekScratch(curfile,&pos);
                PF_SetScratch(curfile,&pos);
            }/*if(curfile->POfull-curfile->POfill < 1)*/
            dl=*(curfile->POfill+1)+2;
        }/*if(*(curfile->POfill)<0)*/
    }/*for(;;)*/
}

/*
        #] PF_WalkThroughExprMaster :
        #[ PF_WalkThroughExprSlave :
    Returns <=0 if the expression is ready, or dl+1;
*/

static int PF_WalkThroughExprSlave(FILEHANDLE *curfile, LONG *counter, int dl)
{
    LONG l=0;
    for(;;){
        if(curfile->POstop-curfile->POfill < dl){
            if(PF_pushScratch(curfile))
                return(-PF.exprbufsize-1);
        }
        curfile->POfill+=dl;
        curfile->POfull=curfile->POfill;
        l+=dl;
        if( l >= PF.exprbufsize){
            if( l == PF.exprbufsize){
                /*
                 * This access is valid because PF.exprbufsize+1 WORDs are
                 * broadcasted, this shortcut is not mandatory though. (TU 15 Sep 2011)
                 */
                if( *(curfile->POfill) == 0)/*expression is ready*/
                    return(0);
                }
            l-=PF.exprbufsize;
            curfile->POfill-=l;
            curfile->POfull=curfile->POfill;
            return l+1;
        }

        dl=*(curfile->POfill);
        if(dl == 0)
            return l-PF.exprbufsize;
        (*counter)++;
        if(dl<0){/*compressed term*/
            if(curfile->POstop-curfile->POfill < 1){
                if(PF_pushScratch(curfile))
                    return(-PF.exprbufsize-1);
            }
            /*
             * This access is always valid because PF.exprbufsize+1 WORDs are
```

```
            * broadcasted. (TU 15 Sep 2011)
            */
           dl=*(curfile->POfill+1)+2;
       }/*if(*(curfile->POfill)<0)*/
   }/*for(;;)*/
}

/*
       #] PF_WalkThroughExprSlave :
       #[ PF_rhsBCastMaster :
*/

static int PF_rhsBCastMaster(FILEHANDLE *curfile, EXPRESSIONS e)
{
    LONG l=1;/*PF_WalkThroughExpr returns length + 1*/
    SetScratch(curfile,&(e->onfile));
    do{
        /*
         * We need to broadcast PF.exprbufsize+1 WORDs because PF_WalkThroughExprSlave
         * may access to an additional 1 WORD. It is better to rewrite the routines
         * in such a way as to broadcast only PF.exprbufsize WORDs. (TU 15 Sep 2011)
         */
        if ( curfile->POfull - curfile->POfill < PF.exprbufsize + 1 ) {
            POSITION pos;
            SeekScratch(curfile,&pos);
            PF_SetScratch(curfile,&pos);
        }
        if ( PF_Bcast(curfile->POfill, (PF.exprbufsize + 1) * sizeof(WORD)) )
            return -1;
        l=PF_WalkThroughExprMaster(curfile,l-1);
    }while(l>0);
    if(l<0)/*The tail is extra, decrease POfill*/
        curfile->POfill-=l;
    return(0);
}

/*
       #] PF_rhsBCastMaster :
       #[ PF_rhsBCastSlave :
*/

static int PF_rhsBCastSlave(FILEHANDLE *curfile, EXPRESSIONS e)
{
    LONG l=1;/*PF_WalkThroughExpr returns length + 1*/
    LONG counter = 0;
    do{
        /*
         * We need to broadcast PF.exprbufsize+1 WORDs because PF_WalkThroughExprSlave
         * may access to an additional 1 WORD. It is better to rewrite the routines
         * in such a way as to broadcast only PF.exprbufsize WORDs. (TU 15 Sep 2011)
         */
        if ( curfile->POstop - curfile->POfill < PF.exprbufsize + 1 ) {
            if(PF_pushScratch(curfile))
                return(-1);
        }
        if ( PF_Bcast(curfile->POfill, (PF.exprbufsize + 1) * sizeof(WORD)) )
            return(-1);
        l = PF_WalkThroughExprSlave(curfile, &counter, l - 1);
    }while(l>0);
    if(l<0){/*The tail is extra, decrease POfill*/
        if(l<-PF.exprbufsize)/*error due to a PF_pushScratch() failure */
            return(-1);
        curfile->POfill-=l;
    }
    if ( curfile->handle >= 0 ) {
        if ( PF_pushScratch(curfile) ) return -1;
    }
    curfile->POfull=curfile->POfill;
    if ( curfile != AR.hidefile ) AR.InInBuf = curfile->POfull-curfile->PObuffer;
    else                          AR.InHiBuf = curfile->POfull-curfile->PObuffer;
    CHECK(counter == e->counter + 1);  /* The first term is the prototype. */
    return(0);
}

/*
       #] PF_rhsBCastSlave :
       #[ PF_BroadcastExpr :
*/

int PF_BroadcastExpr(EXPRESSIONS e, FILEHANDLE *file)
{
    if ( PF.me == MASTER ) {
        if ( PF_rhsBCastMaster(file, e) ) return -1;
#ifdef PF_DEBUG_BCAST_RHSEXPR
        MesPrint("» Broadcast RhsExpr: %s", AC.exprnames->namebuffer + e->name);
#endif
    }
```

```
    else {
        POSITION pos;
        SetEndHScratch(file, &pos);
        e->onfile = pos;
        if ( PF_rhsBCastSlave(file, e) ) return -1;
    }
    return 0;
}

/*
        #] PF_BroadcastExpr :
        #[ PF_BroadcastRHS :
*/

int PF_BroadcastRHS(void)
{
    int i;
    for ( i = 0; i < NumExpressions; i++ ) {
        EXPRESSIONS e = &Expressions[i];
        if ( !(e->vflags & ISINRHS) ) continue;
        switch ( e->status ) {
            case LOCALEXPRESSION:
            case SKIPLEXPRESSION:
            case DROPLEXPRESSION:
            case GLOBALEXPRESSION:
            case SKIPGEXPRESSION:
            case DROPGEXPRESSION:
            case HIDELEXPRESSION:
            case HIDEGEXPRESSION:
            case INTOHIDELEXPRESSION:
            case INTOHIDEGEXPRESSION:
                if ( PF_BroadcastExpr(e, AR.infile) ) return -1;
                break;
            case HIDDENLEXPRESSION:
            case HIDDENGEXPRESSION:
            case DROPHLEXPRESSION:
            case DROPHGEXPRESSION:
            case UNHIDELEXPRESSION:
            case UNHIDEGEXPRESSION:
                if ( PF_BroadcastExpr(e, AR.hidefile) ) return -1;
                break;
        }
    }
    if ( PF.me != MASTER )
        UpdatePositions();
    return 0;
}

/*
        #] PF_BroadcastRHS :
    #] Broadcasting RHS expressions :
    #[ InParallel mode :
        #[ PF_InParallelProcessor :
*/

int PF_InParallelProcessor(void)
{
    GETIDENTITY
    int i, next,tag;
    EXPRESSIONS e;
    /*
     * Skip expressions with zero terms. All the master and slaves need to
     * change the "partodo" flag.
     */
    if ( PF.numtasks >= 3 ) {
        for ( i = 0; i < NumExpressions; i++ ) {
            e = Expressions + i;
            if ( e->partodo > 0 && e->counter == 0 ) {
                e->partodo = 0;
            }
        }
    }
    if(PF.me == MASTER){
        if ( PF.numtasks >= 3 ) {
            partodoexr = (WORD*)Malloc1(sizeof(WORD)*(PF.numtasks+1),"PF_InParallelProcessor");
            for ( i = 0; i < NumExpressions; i++ ) {
                e = Expressions+i;
                if ( e->partodo <= 0 ) continue;
                switch(e->status){
                    case LOCALEXPRESSION:
                    case GLOBALEXPRESSION:
                    case UNHIDELEXPRESSION:
                    case UNHIDEGEXPRESSION:
                    case INTOHIDELEXPRESSION:
                    case INTOHIDEGEXPRESSION:
                        tag=PF_ANY_SOURCE;
                        next=PF_Wait4SlaveIP(&tag);
```

```
                        if(next<0)
                            return(-1);
                        if(tag == PF_DATA_MSGTAG){
                            PF_Statistics(PF_stats,0);
                            if(PF_Slave2MasterIP(next))
                                return(-1);
                        }
                        if(PF_Master2SlaveIP(next,e))
                            return(-1);
                        partodoexr[next]=i;
                        break;
                    default:
                        e->partodo = 0;
                        continue;
                }/*switch(e->status)*/
            }/*for ( i = 0; i < NumExpressions; i++ )*/
            /*Here some slaves are working, other are waiting on PF_Send.
                Wait all of them.*/
            /*At this point no new slaves may be launched so PF_WaitAllSlaves()
                does not modify partodoexr[].*/
            if(PF_WaitAllSlaves())
                return(-1);

            if ( AC.CollectFun ) AR.DeferFlag = 0;
            if(partodoexr){
                M_free(partodoexr,"PF_InParallelProcessor");
                partodoexr=NULL;
            }/*if(partodoexr)*/
        }/*if ( PF.numtasks >= 3 ) */
        else {
            for ( i = 0; i < NumExpressions; i++ ) {
                Expressions[i].partodo = 0;
            }
        }
        return(0);
    }/*if(PF.me == MASTER)*/
    /*Slave:*/
    if(PF_Wait4MasterIP(PF_EMPTY_MSGTAG))
        return(-1);
    /*master is ready to listen to me*/
    do{
        WORD *oldwork= AT.WorkPointer;
        tag=PF_ReadMaster();/*reads directly to its scratch!*/
        if(tag<0)
            return(-1);
        if(tag == PF_DATA_MSGTAG){
            oldwork = AT.WorkPointer;

            /* For redefine statements. */
            if ( AC.numpfirstnum > 0 ) {
                int j;
                for ( j = 0; j < AC.numpfirstnum; j++ ) {
                    AC.inputnumbers[j] = -1;
                }
            }

            if(PF_DoOneExpr())/*the processor*/
                return(-1);
            if(PF_Wait4MasterIP(PF_DATA_MSGTAG))
                return(-1);
            if(PF_Slave2MasterIP(PF.me))/*both master and slave*/
                return(-1);
            AT.WorkPointer=oldwork;
        }/*if(tag == PF_DATA_MSGTAG)*/
    }while(tag!=PF_EMPTY_MSGTAG);
    PF.exprtodo=-1;
    return(0);
}/*PF_InParallelProcessor*/

/*
        #] PF_InParallelProcessor :
        #[ PF_Wait4MasterIP :
*/

static int PF_Wait4MasterIP(int tag)
{
    int follow = 0;
    LONG cpu,space = 0;

    if(PF.log){
        fprintf(stderr,"[%d] Starting to send to Master\n",PF.me);
        fflush(stderr);
    }

    PF_PreparePack();
    cpu = TimeCPU(1);
    PF_Pack(&cpu               ,1,PF_LONG);
```

```
    PF_Pack(&space              ,1,PF_LONG);
    PF_Pack(&PF_linterms        ,1,PF_LONG);
    PF_Pack(&(AM.S0->GenTerms)  ,1,PF_LONG);
    PF_Pack(&(AM.S0->TermsLeft),1,PF_LONG);
    PF_Pack(&follow             ,1,PF_INT );

    if(PF.log){
        fprintf(stderr,"[%d] Now sending with tag = %d\n",PF.me,tag);
        fflush(stderr);
    }

    PF_Send(MASTER, tag);

    if(PF.log){
        fprintf(stderr,"[%d] returning from send\n",PF.me);
        fflush(stderr);
    }
    return(0);
}
/*
        #] PF_Wait4MasterIP :
        #[ PF_DoOneExpr :
*/

static int PF_DoOneExpr(void)/*the processor*/
{
                GETIDENTITY
                EXPRESSIONS e;
                int i;
                WORD *term;
                POSITION position, outposition;
                FILEHANDLE *fi, *fout;
                LONG dd = 0;
                WORD oldBracketOn = AR.BracketOn;
                WORD *oldBrackBuf = AT.BrackBuf;
                WORD oldbracketindexflag = AT.bracketindexflag;
                e = Expressions + PF.exprtodo;
                i = PF.exprtodo;
                AR.CurExpr = i;
                AR.SortType = AC.SortType;
                AR.expchanged = 0;
                if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
                    AR.BracketOn = 1;
                    AT.BrackBuf = AM.BracketFactors;
                    AT.bracketindexflag = 1;
                }

                position = AS.OldOnFile[i];
                if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENGEXPRESSION ) {
                    AR.GetFile = 2; fi = AR.hidefile;
                }
                else {
                    AR.GetFile = 0; fi = AR.infile;
                }
/*
                PUTZERO(fi->POposition);
                if ( fi->handle >= 0 ) {
                    fi->POfill = fi->POfull = fi->PObuffer;
                }
*/
                SetScratch(fi,&position);
                term = AT.WorkPointer;
                AR.CompressPointer = AR.CompressBuffer;
                AR.CompressPointer[0] = 0;
                AR.KeptInHold = 0;
                if ( GetTerm(BHEAD term) <= 0 ) {
                    MesPrint("Expression %d has problems in scratchfile",i);
                    Terminate(-1);
                }
                if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
                term[3] = i;
                PUTZERO(outposition);
                fout = AR.outfile;
                fout->POfill = fout->POfull = fout->PObuffer;
                fout->POposition = outposition;
                if ( fout->handle >= 0 ) {
                    fout->POposition = outposition;
                }
/*
                The next statement is needed because we need the system
                to believe that the expression is at position zero for
                the moment. In this worker, with no memory of other expressions,
                it is. This is needed for when a bracket index is made
                because there e->onfile is an offset. Afterwards, when the
                expression is written to its final location in the masters
                output e->onfile will get its real value.
*/
```

```
                PUTZERO(e->onfile);
                if ( PutOut(BHEAD term,&outposition,fout,0) < 0 ) return -1;

                AR.DeferFlag = AC.ComDefer;

/*              AR.sLevel = AB[0]->R.sLevel;*/
                term = AT.WorkPointer;
                NewSort(BHEAD0);
                AR.MaxDum = AM.IndDum;
                AN.ninterms = 0;
                while ( GetTerm(BHEAD term) ) {
                  SeekScratch(fi,&position);
                  AN.ninterms++; dd = AN.deferskipped;
                  if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
                      StoreTerm(BHEAD term);
                  }
                  else {
                  if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)sizeof(WORD))) ) {
                    if ( GetMoreTerms(term) < 0 ) {
                      LowerSortLevel(); return(-1);
                    }
                    SeekScratch(fi,&position);
                  }
                  AT.WorkPointer = term + *term;
                  AN.RepPoint = AT.RepCount + 1;
                  if ( AR.DeferFlag ) {
                    AR.CurDum = AN.IndDum = Expressions[PF.exprtodo].numdummies;
                  }
                  else {
                    AN.IndDum = AM.IndDum;
                    AR.CurDum = ReNumber(BHEAD term);
                  }
                  if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
                  if ( AN.ncmod ) {
                    if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
                    else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
                  }
                  else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
                  if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
                        && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
                        PolyFunClean(BHEAD term);
                  }
                  if ( Generator(BHEAD term,0) ) {
                    LowerSortLevel(); return(-1);
                  }
                  AN.ninterms += dd;
                  }
                  SetScratch(fi,&position);
                  if ( fi == AR.hidefile ) {
                    AR.InHiBuf = (fi->POfull-fi->PObuffer)
                        -DIFBASE(position,fi->POposition)/sizeof(WORD);
                  }
                  else {
                    AR.InInBuf = (fi->POfull-fi->PObuffer)
                        -DIFBASE(position,fi->POposition)/sizeof(WORD);
                  }
                }
                AN.ninterms += dd;
                if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) return(-1);
                e->numdummies = AR.MaxDum - AM.IndDum;
                AR.BracketOn = oldBracketOn;
                AT.BrackBuf = oldBrackBuf;
                if ( ( e->vflags & TOBEFACTORED ) != 0 )
                        poly_factorize_expression(e);
                else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
                 && ( ( e->vflags & ISFACTORIZED ) != 0 ) )
                        poly_unfactorize_expression(e);
                if ( AM.S0->TermsLeft )   e->vflags &= ~ISZERO;
                else                      e->vflags |= ISZERO;
                if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
/*              if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
                if ( AR.expchanged )    AR.expflags |= ISUNMODIFIED;*/
                AR.GetFile = 0;
                AT.bracketindexflag = oldbracketindexflag;

                fout->POfull = fout->POfill;
    return(0);
}

/*
        #] PF_DoOneExpr :
        #[ PF_Slave2MasterIP :
*/

typedef struct bufIPstruct {
    LONG i;
    struct ExPrEsSiOn e;
```

```c
} bufIPstruct_t;

static int PF_Slave2MasterIP(int src)/*both master and slave*/
{
    EXPRESSIONS e;
    bufIPstruct_t exprData;
    int i,l;
    FILEHANDLE *fout=AR.outfile;
    POSITION pos;
    /*Here we know the length of data to send in advance:
        slave has the only one expression in its scratch file, and it sends
        this information to the master.*/
    if(PF.me != MASTER){/*slave*/
        e = Expressions + PF.exprtodo;
        /*Fill in the expression data:*/
        memcpy(&(exprData.e), e, sizeof(struct ExPrEsSiOn));
        SeekScratch(fout,&pos);
        exprData.i=BASEPOSITION(pos);
        /*Send the metadata:*/
        if(PF_RawSend(MASTER,&exprData,sizeof(bufIPstruct_t),0))
            return(-1);
        i=exprData.i;
        SETBASEPOSITION(pos,0);
        do{
            int blen=PF.exprbufsize*sizeof(WORD);
            if(i<blen)
                blen=i;
            l=PF_SendChunkIP(fout,&pos, MASTER, blen);
            /*Here always l == blen!*/
            if(l<0)
                return(-1);
            ADDPOS(pos,l);
            i-=l;
        }while(i>0);
        if ( fout->handle >= 0 ) { /* Now get rid of the file */
            CloseFile(fout->handle);
            fout->handle = -1;
            remove(fout->name);
            PUTZERO(fout->POposition);
            PUTZERO(fout->filesize);
            fout->POfill = fout->POfull = fout->PObuffer;
        }
        /* Now handle redefined preprocessor variables. */
        if ( AC.numpfirstnum > 0 ) {
            PF_PrepareLongSinglePack();
            PF_PackRedefinedPreVars();
            PF_LongSingleSend(MASTER, PF_MISC_MSGTAG);
        }
        return(0);
    }/*if(PF.me != MASTER)*/
    /*Master*/
    /*partodoexr[src] is the number of expression.*/
    e = Expressions +partodoexr[src];
    /*Get metadata:*/
    if (PF_RawRecv(&src, &exprData,sizeof(bufIPstruct_t),&i)!= sizeof(bufIPstruct_t))
        return(-1);
    /*Fill in the expression data:*/
/*  memcpy(e, &(exprData.e), sizeof(struct ExPrEsSiOn)); */
    e->counter    = exprData.e.counter;
    e->vflags     = exprData.e.vflags;
    e->uflags     = exprData.e.uflags;
    e->numdummies = exprData.e.numdummies;
    e->numfactors = exprData.e.numfactors;
    if ( !(e->vflags & ISZERO) )       AR.expflags |= ISZERO;
    if ( !(e->vflags & ISUNMODIFIED) ) AR.expflags |= ISUNMODIFIED;
    SeekScratch(fout,&pos);
    e->onfile = pos;
    i=exprData.i;
    while(i>0){
        int blen=PF.exprbufsize*sizeof(WORD);
        if(i<blen)
            blen=i;
        l=PF_RecvChunkIP(fout,src,blen);
        /*Here always l == blen!*/
        if(l<0)
            return(-1);
        i-=l;
    }
    /* Now handle redefined preprocessor variables. */
    if ( AC.numpfirstnum > 0 ) {
        PF_LongSingleReceive(src, PF_MISC_MSGTAG, NULL, NULL);
        PF_UnpackRedefinedPreVars();
    }
    return(0);
}

/*
```

```
        #] PF_Slave2MasterIP :
        #[ PF_Master2SlaveIP :
*/

static int PF_Master2SlaveIP(int dest, EXPRESSIONS e)
{
    bufIPstruct_t exprData;
    FILEHANDLE *fi;
    POSITION pos;
    int l;
    LONG ll=0,count=0;
    WORD *t;
    if(e==NULL){/*Say to the slave that no more job:*/
        if(PF_RawSend(dest,&exprData,sizeof(bufIPstruct_t),PF_EMPTY_MSGTAG))
            return(-1);
        return(0);
    }
    memcpy(&(exprData.e), e, sizeof(struct ExPrEsSiOn));
    exprData.i=e-Expressions;
    if ( AC.StatsFlag && AC.OldParallelStats ) {
        MesPrint("");
        MesPrint(" Sending expression %s to slave %d",EXPRNAME(exprData.i),dest);
    }
    if(PF_RawSend(dest,&exprData,sizeof(bufIPstruct_t),PF_DATA_MSGTAG))
        return(-1);
    if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENGEXPRESSION )
        fi = AR.hidefile;
    else
        fi = AR.infile;
    pos=e->onfile;
    SetScratch(fi,&pos);
    do{
        l=PF_SendChunkIP(fi, &pos, dest, PF.exprbufsize*sizeof(WORD));
        if(l<0)
            return(-1);
        t=fi->PObuffer+ (DIFBASE(pos,fi->POposition))/sizeof(WORD);
        ll=PF_WalkThrough(t,ll,l/sizeof(WORD),&count);
        ADDPOS(pos,l);
    }while(ll>-2);
    return(0);
}

/*
        #] PF_Master2SlaveIP :
        #[ PF_ReadMaster :
*/

static int PF_ReadMaster(void)/*reads directly to its scratch!*/
{
    bufIPstruct_t exprData;
    int tag,m=MASTER;
    EXPRESSIONS e;
    FILEHANDLE *fi;
    POSITION pos;
    LONG count=0;
    WORD *t;
    LONG ll=0;
    int l;
    /*Get metadata:*/
    if (PF_RawRecv(&m, &exprData,sizeof(bufIPstruct_t),&tag)!= sizeof(bufIPstruct_t))
        return(-1);

    if(tag == PF_EMPTY_MSGTAG)/*No data, no job*/
        return(tag);

    /*data expected, tag must be == PF_DATA_MSTAG!*/
    PF.exprtodo=exprData.i;
    e=Expressions + PF.exprtodo;
    /*Fill in the expression data:*/
/*  memcpy(e, &(exprData.e), sizeof(struct ExPrEsSiOn)); */
    if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENGEXPRESSION )
        fi = AR.hidefile;
    else
        fi = AR.infile;
    SetEndHScratch(fi,&pos);
    e->onfile=AS.OldOnFile[PF.exprtodo]=pos;

    do{
        l=PF_RecvChunkIP(fi,MASTER,PF.exprbufsize*sizeof(WORD));
        if(l<0)
            return(-1);
        t=fi->POfull-l/sizeof(WORD);
        ll=PF_WalkThrough(t,ll,l/sizeof(WORD),&count);
    }while(ll>-2);
    /*Now -ll-2 is the number of "extra" elements transferred from the master.*/
    fi->POfull-=-ll-2;
    fi->POfill=fi->POfull;
```

```
    return(PF_DATA_MSGTAG);
}

/*
        #] PF_ReadMaster :
        #[ PF_SendChunkIP :
    thesize is in bytes. Returns the number of sent bytes or <0 on error:
*/

static int PF_SendChunkIP(FILEHANDLE *curfile, POSITION *position, int to, LONG thesize)
{
    LONG l=thesize;
    if(
        ISLESSPOS(*position,curfile->POposition) ||
        ISGEPOSINC(*position,curfile->POposition,
        ((curfile->POfull-curfile->PObuffer)*sizeof(WORD)-thesize) )
    ){
        if(curfile->handle< 0)
            l=(curfile->POfull-curfile->PObuffer)*sizeof(WORD) - (LONG)(position->p1);
        else{
            PF_SetScratch(curfile,position);
            if(
                ISGEPOSINC(*position,curfile->POposition,
                ((curfile->POfull-curfile->PObuffer)*sizeof(WORD)-thesize) )
                )
            l=(curfile->POfull-curfile->PObuffer)*sizeof(WORD) - (LONG)position->p1;
        }
    }
    /*Now we are able to sent l bytes from the
        curfile->PObuffer[position-curfile->POposition]*/
    if(PF_RawSend(to,curfile->PObuffer+ (DIFBASE(*position,curfile->POposition))/sizeof(WORD),l,0))
        return(-1);
    return(l);
}

/*
        #] PF_SendChunkIP :
        #[ PF_RecvChunkIP :
    thesize is in bytes. Returns the number of sent bytes or <0 on error:
*/

static int PF_RecvChunkIP(FILEHANDLE *curfile, int from, LONG thesize)
{
    LONG receivedBytes;

    if( (LONG)((curfile->POstop - curfile->POfull)*sizeof(WORD)) < thesize )
        if(PF_pushScratch(curfile))
            return(-1);
    /*Now there is enough space from curfile->POfill to curfile->POstop*/
    {/*Block:*/
        int tag=0;
        receivedBytes=PF_RawRecv(&from,curfile->POfull,thesize,&tag);
    }/*:Block*/
    if(receivedBytes >= 0 ){
        curfile->POfull+=receivedBytes/sizeof(WORD);
        curfile->POfill=curfile->POfull;
    }/*if(receivedBytes >= 0 )*/
    return(receivedBytes);
}

/*
        #] PF_RecvChunkIP :
        #[ PF_WalkThrough :
    Returns:
    >=  0 -- initial offset,
        -1 -- the first element of t contains the length of the tail of compressed term,
    <= -2 -- -(d+2), where d is the number of extra transferred elements.
    Expects:
    l -- initial offset or -1,
    chunk -- number of transferred elements (not bytes!)
    *count -- incremented each time a new term is found
*/

static int PF_WalkThrough(WORD *t, LONG l, LONG chunk, LONG *count)
{
    if(l<0) /*==-1!*/
        l=(*t)+1;/*the first element of t contains the length of
                        the tail of compressed term*/
    else{
        if(l>=chunk)/*next term is out of the chunk*/
            return(l-chunk);
        t+=l;
        chunk-=l;/*note, l was less than chunk so chunk >0!*/
        l=*t;
    }
    /*Main loop:*/
    while(l!=0){
```

```
        if(l>0){/*an offset to the next term*/
            if(l<chunk){
                t+=l;
                chunk-=l;/*note, l was less than chunk so chunk >0!*/
                l=*t;
                (*count)++;
            }/*if(l<chunk)*/
            else
                return(l-chunk);
        }/*if(l>0)*/
        else{ /* l<0 */
            if(chunk < 2)/*i.e., chunk == 1*/
                return(-1);/*the first WORD in the next chunk is length of the tail of the compressed term*/
            l=*(t+1)+2;/**2 since
                    1. t points to the length field -1,
                    2. the size of a tail of compressed term is equal to the number of WORDs in this tail*/
        }
    }/*while(l!=0)*/
    return(-1-chunk);/* -(2+(chunk-1)), chunk>0 ! */
}

/*
        #] PF_WalkThrough :
    #] InParallel mode :
    #[ PF_SendFile :
*/

#define PF_SNDFILEBUFSIZE 4096

int PF_SendFile(int to, FILE *fd)
{
    size_t len=0;
    if(fd == NULL){
        if(PF_RawSend(to,&to,sizeof(int),PF_EMPTY_MSGTAG))
            return(-1);
        return(0);
    }
    for(;;){
        char buf[PF_SNDFILEBUFSIZE];
        size_t l;
        l=fread(buf, 1, PF_SNDFILEBUFSIZE, fd);
        len+=l;
        if(l==PF_SNDFILEBUFSIZE){
            if(PF_RawSend(to,buf,PF_SNDFILEBUFSIZE,PF_BUFFER_MSGTAG))
                return(-1);
        }
        else{
            if(PF_RawSend(to,buf,l,PF_ENDBUFFER_MSGTAG))
                return(-1);
            break;
        }
    }/*for(;;)*/
    return(len);
}

/*
    #] PF_SendFile :
    #[ PF_RecvFile :
*/

int PF_RecvFile(int from, FILE *fd)
{
    size_t len=0;
    int tag;
    do{
        char buf[PF_SNDFILEBUFSIZE];
        int l;
            l=PF_RawRecv(&from,buf,PF_SNDFILEBUFSIZE,&tag);
            if(l<0)
                return(-1);
            if(tag == PF_EMPTY_MSGTAG)
                return(-1);

            if( fwrite(buf,l,1,fd)!=1 )
                return(-1);
            len+=l;
    }while(tag!=PF_ENDBUFFER_MSGTAG);
    return(len);
}

/*
    #] PF_RecvFile :
    #[ Synchronised output :
        #[ Explanations :
*/
```

```

/*
 * If the master and slaves output statistics or error messages to the same stream
 * or file (e.g., the standard output or the log file) simultaneously, then
 * a mixing of their outputs can occur. To avoid this, TFORM uses a lock of
 * ErrorMessageLock, but there is no locking functionality in the original MPI
 * specification. We need to synchronise the output from the master and slaves.
 *
 * The idea of the synchronised output (by, e.g., MesPrint()) implemented here is
 *   Slaves:
 *     1. Save the output by WriteFile() (set to PF_WriteFileToFile())
 *        into some buffers between MLOCK(ErrorMessageLock) and
 *        MUNLOCK(ErrorMessageLock), which call PF_MLock() and PF_MUnlock(),
 *        respectively. The output for AM.StdOut and AC.LogHandle are saved to
 *        the buffers.
 *     2. At MUNLOCK(ErrorMessageLock), send the output in the buffer to the master,
 *        with PF_STDOUT_MSGTAG or PF_LOG_MSGTAG.
 *   Master:
 *     1. Receive the buffered output from slaves, and write them by
 *        WriteFileToFile().
 *   The main problem is how and where the master receives messages from
 *   the slaves (PF_ReceiveErrorMessage()). For this purpose there are three
 *   helper functions: PF_CatchErrorMessages() and PF_CatchErrorMessagesForAll()
 *   which remove messages with PF_STDOUT_MSGTAG or PF_LOG_MSGTAG from the top
 *   of the message queue, and PF_ProbeWithCatchingErrorMessages() which is same as
 *   PF_Probe() except removing these messages.
 */

/*
        #] Explanations :
        #[ Variables :
*/

static int errorMessageLock = 0;     /* (slaves) The lock count. See PF_MLock() and PF_MUnlock(). */
static Vector(UBYTE, stdoutBuffer);  /* (slaves) The buffer for AM.StdOut. */
static Vector(UBYTE, logBuffer);     /* (slaves) The buffer for AC.LogHandle. */
#define recvBuffer logBuffer         /* (master) The buffer for receiving messages. */

/*
 * If PF_ENABLE_STDOUT_BUFFERING is defined, the master performs the line buffering
 * (using stdoutBuffer) at PF_WriteFileToFile().
 */
#ifndef PF_ENABLE_STDOUT_BUFFERING
#ifdef UNIX
#define PF_ENABLE_STDOUT_BUFFERING
#endif
#endif

/*
        #] Variables :
        #[ PF_MLock :
*/

void PF_MLock(void)
{
    /* Only on slaves. */
    if ( errorMessageLock++ > 0 ) return;
    VectorClear(stdoutBuffer);
    VectorClear(logBuffer);
}

/*
        #] PF_MLock :
        #[ PF_MUnlock :
*/

void PF_MUnlock(void)
{
    /* Only on slaves. */
    if ( --errorMessageLock > 0 ) return;
    if ( !VectorEmpty(stdoutBuffer) ) {
        PF_RawSend(MASTER, VectorPtr(stdoutBuffer), VectorSize(stdoutBuffer), PF_STDOUT_MSGTAG);
    }
    if ( !VectorEmpty(logBuffer) ) {
        PF_RawSend(MASTER, VectorPtr(logBuffer), VectorSize(logBuffer), PF_LOG_MSGTAG);
    }
}

/*
        #] PF_MUnlock :
        #[ PF_WriteFileToFile :
*/

LONG PF_WriteFileToFile(int handle, UBYTE *buffer, LONG size)
{
    if ( PF.me != MASTER && errorMessageLock > 0 ) {
        if ( handle == AM.StdOut ) {
```

```
            VectorPushBacks(stdoutBuffer, buffer, size);
            return size;
        }
        else if ( handle == AC.LogHandle ) {
            VectorPushBacks(logBuffer, buffer, size);
            return size;
        }
    }
#ifdef PF_ENABLE_STDOUT_BUFFERING
    /*
     * On my computer, sometimes a single linefeed "\n" sent to the standard
     * output is ignored on the execution of mpiexec. A typical example is:
     *   $ cat foo.c
     *     #include <unistd.h>
     *     int main() {
     *       write(1, "    ", 4);
     *       write(1, "\n", 1);
     *       write(1, "    ", 4);
     *       write(1, "123\n", 4);
     *       return 0;
     *     }
     * or even as a shell script:
     *   $ cat foo.sh
     *     #! bin/sh
     *     printf "    "
     *     printf "\n"
     *     printf "    "
     *     printf "123\n"
     * When I ran it on mpiexec
     *   $ while :; do mpiexec -np 1 ./foo.sh; done
     * I observed the single linefeed (printf "\n") was sometimes ignored. Even
     * though this phenomenon might be specific to my environment, I added this
     * code because someone may encounter a similar phenomenon and feel it
     * frustrating. (TU 16 Jun 2011)
     *
     * Phenomenon:
     *   A single linefeed sent to the standard output occasionally ignored
     *   on mpiexec.
     *
     * Environment:
     *   openSUSE 11.4 (x86_64)
     *   kernel: 2.6.37.6-0.5-desktop
     *   gcc: 4.5.1 20101208
     *   mpich2-1.3.2p1 configured with '--enable-shared --with-pm=smpd'
     *
     * Solution:
     *   In Unix (in which Uwrite() calls write() system call without any buffering),
     *   we perform the line buffering here. A single linefeed is also buffered.
     *
     * XXX:
     *   At the end of the program the buffered output (text without LF) will not be flushed,
     *   i.e., will not be written to the standard output. This is not problematic at a normal run.
     *   The buffer can be explicitly flushed by PF_FlushStdOutBuffer().
     */
    if ( PF.me == MASTER && handle == AM.StdOut ) {
        size_t oldsize;
        /* Assume the newline character is LF (when UNIX is defined). */
        if ( (size > 0 && buffer[size - 1] != LINEFEED) || (size == 1 && buffer[0] == LINEFEED) ) {
            VectorPushBacks(stdoutBuffer, buffer, size);
            return size;
        }
        if ( (oldsize = VectorSize(stdoutBuffer)) > 0 ) {
            LONG ret;
            VectorPushBacks(stdoutBuffer, buffer, size);
            ret = WriteFileToFile(handle, VectorPtr(stdoutBuffer), VectorSize(stdoutBuffer));
            VectorClear(stdoutBuffer);
            if ( ret < 0 ) {
                return ret;
            }
            else if ( ret < (LONG)oldsize ) {
                return 0;  /* This means the buffered output in previous calls is lost. */
            }
            else {
                return ret - (LONG)oldsize;
            }
        }
    }
#endif
    return WriteFileToFile(handle, buffer, size);
}

/*
        #] PF_WriteFileToFile :
        #[ PF_FlushStdOutBuffer :
*/

void PF_FlushStdOutBuffer(void)
```

```
{
#ifdef PF_ENABLE_STDOUT_BUFFERING
    if ( PF.me == MASTER && VectorSize(stdoutBuffer) > 0 ) {
        WriteFileToFile(AM.StdOut, VectorPtr(stdoutBuffer), VectorSize(stdoutBuffer));
        VectorClear(stdoutBuffer);
    }
#endif
}

/*
        #] PF_FlushStdOutBuffer :
        #[ PF_ReceiveErrorMessage :
*/

static void PF_ReceiveErrorMessage(int src, int tag)
{
    /* Only on the master. */
    int size;
    int ret = PF_RawProbe(&src, &tag, &size);
    CHECK(ret == 0);
    switch ( tag ) {
        case PF_STDOUT_MSGTAG:
        case PF_LOG_MSGTAG:
            VectorReserve(recvBuffer, size);
            ret = PF_RawRecv(&src, VectorPtr(recvBuffer), size, &tag);
            CHECK(ret == size);
            if ( size > 0 ) {
                int handle = (tag == PF_STDOUT_MSGTAG) ? AM.StdOut : AC.LogHandle;
#ifdef PF_ENABLE_STDOUT_BUFFERING
                if ( handle == AM.StdOut ) PF_WriteFileToFile(handle, VectorPtr(recvBuffer), size);
                else
#endif
                WriteFileToFile(handle, VectorPtr(recvBuffer), size);
            }
            break;
    }
}

/*
        #] PF_ReceiveErrorMessage :
        #[ PF_CatchErrorMessages :
*/

static void PF_CatchErrorMessages(int *src, int *tag)
{
    /* Only on the master. */
    for (;;) {
        int asrc = *src;
        int atag = *tag;
        int ret = PF_RawProbe(&asrc, &atag, NULL);
        CHECK(ret == 0);
        if ( atag == PF_STDOUT_MSGTAG || atag == PF_LOG_MSGTAG ) {
            PF_ReceiveErrorMessage(asrc, atag);
            continue;
        }
        *src = asrc;
        *tag = atag;
        break;
    }
}

/*
        #] PF_CatchErrorMessages :
        #[ PF_CatchErrorMessagesForAll :
*/

static void PF_CatchErrorMessagesForAll(void)
{
    /* Only on the master. */
    int i;
    for ( i = 1; i < PF.numtasks; i++ ) {
        int src = i;
        int tag = PF_ANY_MSGTAG;
        PF_CatchErrorMessages(&src, &tag);
    }
}

/*
        #] PF_CatchErrorMessagesForAll :
        #[ PF_ProbeWithCatchingErrorMessages :
*/

static int PF_ProbeWithCatchingErrorMessages(int *src)
{
    for (;;) {
        int newsrc = *src;
        int tag = PF_Probe(&newsrc);
```

```
        if ( tag == PF_STDOUT_MSGTAG || tag == PF_LOG_MSGTAG ) {
            PF_ReceiveErrorMessage(newsrc, tag);
            continue;
        }
        if ( tag > 0 ) *src = newsrc;
        return tag;
    }
}

/*
        #] PF_ProbeWithCatchingErrorMessages :
        #[ PF_FreeErrorMessageBuffers :
*/

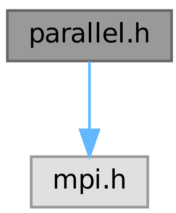
void PF_FreeErrorMessageBuffers(void)
{
    VectorFree(stdoutBuffer);
    VectorFree(logBuffer);
}

/*
        #] PF_FreeErrorMessageBuffers :
    #] Synchronised output :
*/
```

4.96 parallel.h File Reference

`#include <mpi.h>`

Include dependency graph for parallel.h:



Data Structures

- struct PF_BUFFER
- struct ParallelVars

Macros

- #define MASTER 0
- #define PF_RESET 0
- #define PF_TIME 1
- #define PF_TERM_MSGTAG 10 /* master -> slave: sending terms */
- #define PF_ENDSORT_MSGTAG 11 /* master -> slave: no more terms to be distributed, slave -> master: after EndSort() */
- #define PF_DOLLAR_MSGTAG 12 /* slave -> master: sending $-variables */
- #define PF_BUFFER_MSGTAG 20 /* slave -> master: sending sorted terms, or in PF_SendFile()/PF_RecvFile() */

- #define PF_ENDBUFFER_MSGTAG 21 /∗ same as PF_BUFFER_MSGTAG, but indicates the end of operation ∗/
- #define PF_READY_MSGTAG 30 /∗ slave -> master: slave is idle and can accept terms ∗/
- #define PF_DATA_MSGTAG 50 /∗ InParallel, DoCheckpoint() ∗/
- #define PF_EMPTY_MSGTAG 52 /∗ InParallel, DoCheckpoint(), PF_SendFile(), PF_RecvFile() ∗/
- #define PF_STDOUT_MSGTAG 60 /∗ slave -> master: sending text to the stdout ∗/
- #define PF_LOG_MSGTAG 61 /∗ slave -> master: sending text to the log file ∗/
- #define PF_OPT_MCTS_MSGTAG 70 /∗ master <-> slave: optimization ∗/
- #define PF_OPT_HORNER_MSGTAG 71 /∗ master <-> slave: optimization ∗/
- #define PF_OPT_COLLECT_MSGTAG 72 /∗ slave -> master: optimization ∗/
- #define PF_MISC_MSGTAG 100
- #define GNUC_PREREQ(major, minor, patchlevel) 0
- #define OMPI_SKIP_MPICXX 1
- #define indices ((INDICES)(AC.IndexList.lijst))
- #define PF_ANY_SOURCE MPI_ANY_SOURCE
- #define PF_ANY_MSGTAG MPI_ANY_TAG
- #define PF_COMM MPI_COMM_WORLD
- #define PF_BYTE MPI_BYTE
- #define PF_INT MPI_INT
- #define PF_LongMultiPack(buffer, count, type) PF_LongMultiPackImpl(buffer, count, sizeof_datatype(type), type)
- #define PF_LongMultiUnpack(buffer, count, type) PF_LongMultiUnpackImpl(buffer, count, sizeof_←datatype(type), type)

Typedefs

- typedef struct ParallelVars **PARALLELVARS**

Functions

- int PF_ISendSbuf (int, int)
- int PF_Bcast (void ∗buffer, int count)
- int PF_RawSend (int, void ∗, LONG, int)
- LONG PF_RawRecv (int ∗, void ∗, LONG, int ∗)
- int PF_PreparePack (void)
- int PF_Pack (const void ∗buffer, size_t count, MPI_Datatype type)
- int PF_Unpack (void ∗buffer, size_t count, MPI_Datatype type)
- int PF_PackString (const UBYTE ∗str)
- int PF_UnpackString (UBYTE ∗str)
- int PF_Send (int to, int tag)
- int PF_Receive (int src, int tag, int ∗psrc, int ∗ptag)
- int PF_Broadcast (void)
- int PF_PrepareLongSinglePack (void)
- int PF_LongSinglePack (const void ∗buffer, size_t count, MPI_Datatype type)
- int PF_LongSingleUnpack (void ∗buffer, size_t count, MPI_Datatype type)
- int PF_LongSingleSend (int to, int tag)
- int PF_LongSingleReceive (int src, int tag, int ∗psrc, int ∗ptag)
- int PF_PrepareLongMultiPack (void)
- int PF_LongMultiPackImpl (const void ∗buffer, size_t count, size_t eSize, MPI_Datatype type)
- int PF_LongMultiUnpackImpl (void ∗buffer, size_t count, size_t eSize, MPI_Datatype type)
- int PF_LongMultiBroadcast (void)
- int PF_EndSort (void)
- WORD PF_Deferred (WORD ∗, WORD)

- int PF_Processor (EXPRESSIONS, WORD, WORD)
- int PF_Init (int ∗, char ∗∗∗)
- int PF_Terminate (int)
- LONG PF_GetSlaveTimes (void)
- LONG PF_BroadcastNumber (LONG)
- void PF_BroadcastBuffer (WORD ∗∗buffer, LONG ∗length)
- int PF_BroadcastString (UBYTE ∗)
- int PF_BroadcastPreDollar (WORD ∗∗, LONG ∗, int ∗)
- int PF_CollectModifiedDollars (void)
- int PF_BroadcastModifiedDollars (void)
- int PF_BroadcastRedefinedPreVars (void)
- int PF_BroadcastCBuf (int bufnum)
- int PF_BroadcastExpFlags (void)
- int PF_StoreInsideInfo (void)
- int PF_RestoreInsideInfo (void)
- int PF_BroadcastExpr (EXPRESSIONS e, FILEHANDLE ∗file)
- int PF_BroadcastRHS (void)
- int PF_InParallelProcessor (void)
- int PF_SendFile (int to, FILE ∗fd)
- int PF_RecvFile (int from, FILE ∗fd)
- void PF_MLock (void)
- void PF_MUnlock (void)
- LONG PF_WriteFileToFile (int, UBYTE ∗, LONG)
- void PF_FlushStdOutBuffer (void)

Variables

- PARALLELVARS PF
- LONG PF_maxDollarChunkSize

4.96.1 Detailed Description

Header file with things relevant to ParForm.

Definition in file parallel.h.

4.96.2 Macro Definition Documentation

4.96.2.1 MASTER

```
#define MASTER 0
```

Definition at line 39 of file parallel.h.

4.96.2.2 PF_RESET

```
#define PF_RESET 0
```

Definition at line 41 of file parallel.h.

4.96.2.3 PF_TIME

```
#define PF_TIME 1
```

Definition at line 42 of file parallel.h.

4.96.2.4 PF_TERM_MSGTAG

```
#define PF_TERM_MSGTAG 10 /* master -> slave:  sending terms */
```

Definition at line 44 of file parallel.h.

4.96.2.5 PF_ENDSORT_MSGTAG

```
#define PF_ENDSORT_MSGTAG 11 /* master -> slave:  no more terms to be distributed, slave ->
master:  after EndSort() */
```

Definition at line 45 of file parallel.h.

4.96.2.6 PF_DOLLAR_MSGTAG

```
#define PF_DOLLAR_MSGTAG 12 /* slave -> master:  sending $-variables */
```

Definition at line 46 of file parallel.h.

4.96.2.7 PF_BUFFER_MSGTAG

```
#define PF_BUFFER_MSGTAG 20 /* slave -> master:  sending sorted terms, or in PF_SendFile()/PF_RecvFile()
*/
```

Definition at line 47 of file parallel.h.

4.96.2.8 PF_ENDBUFFER_MSGTAG

```
#define PF_ENDBUFFER_MSGTAG 21 /* same as PF_BUFFER_MSGTAG, but indicates the end of operation
*/
```

Definition at line 48 of file parallel.h.

4.96.2.9 PF_READY_MSGTAG

```
#define PF_READY_MSGTAG 30 /* slave -> master:  slave is idle and can accept terms */
```

Definition at line 49 of file parallel.h.

4.96.2.10 PF_DATA_MSGTAG

```
#define PF_DATA_MSGTAG 50 /* InParallel, DoCheckpoint() */
```

Definition at line 50 of file parallel.h.

4.96.2.11 PF_EMPTY_MSGTAG

```
#define PF_EMPTY_MSGTAG 52 /* InParallel, DoCheckpoint(), PF_SendFile(), PF_RecvFile() */
```

Definition at line 51 of file parallel.h.

4.96.2.12 PF_STDOUT_MSGTAG

```
#define PF_STDOUT_MSGTAG 60 /* slave -> master:  sending text to the stdout */
```

Definition at line 52 of file parallel.h.

4.96.2.13 PF_LOG_MSGTAG

```
#define PF_LOG_MSGTAG 61 /* slave -> master:  sending text to the log file */
```

Definition at line 53 of file parallel.h.

4.96.2.14 PF_OPT_MCTS_MSGTAG

```
#define PF_OPT_MCTS_MSGTAG 70 /* master <-> slave:  optimization */
```

Definition at line 54 of file parallel.h.

4.96.2.15 PF_OPT_HORNER_MSGTAG

```
#define PF_OPT_HORNER_MSGTAG 71 /* master <-> slave:  optimization */
```

Definition at line 55 of file parallel.h.

4.96.2.16 PF_OPT_COLLECT_MSGTAG

```
#define PF_OPT_COLLECT_MSGTAG 72 /* slave -> master:  optimization */
```

Definition at line 56 of file parallel.h.

4.96.2.17 PF_MISC_MSGTAG

```
#define PF_MISC_MSGTAG 100
```

Definition at line 57 of file parallel.h.

4.96.2.18 GNUC_PREREQ

```
#define GNUC_PREREQ(
              major,
              minor,
              patchlevel ) 0
```

Definition at line 67 of file parallel.h.

4.96.2.19 OMPI_SKIP_MPICXX

```
#define OMPI_SKIP_MPICXX 1
```

Definition at line 108 of file parallel.h.

4.96.2.20 indices

```
#define indices ((INDICES)(AC.IndexList.lijst))
```

Definition at line 113 of file parallel.h.

4.96.2.21 PF_ANY_SOURCE

```
#define PF_ANY_SOURCE MPI_ANY_SOURCE
```

Definition at line 123 of file parallel.h.

4.96.2.22 PF_ANY_MSGTAG

```
#define PF_ANY_MSGTAG MPI_ANY_TAG
```

Definition at line 124 of file parallel.h.

4.96.2.23 PF_COMM

```
#define PF_COMM MPI_COMM_WORLD
```

Definition at line 125 of file parallel.h.

4.96.2.24 PF_BYTE

```
#define PF_BYTE MPI_BYTE
```

Definition at line 126 of file parallel.h.

4.96.2.25 PF_INT

```
#define PF_INT MPI_INT
```

Definition at line 127 of file parallel.h.

4.96.2.26 PF_LongMultiPack

```
#define PF_LongMultiPack(
              buffer,
              count,
              type ) PF_LongMultiPackImpl(buffer, count, sizeof_datatype(type), type)
```

Definition at line 235 of file parallel.h.

4.96.2.27 PF_LongMultiUnpack

```
#define PF_LongMultiUnpack(
              buffer,
              count,
              type ) PF_LongMultiUnpackImpl(buffer, count, sizeof_datatype(type), type)
```

Definition at line 236 of file parallel.h.

4.96.3 Function Documentation

4.96.3.1 PF_ISendSbuf()

```
int PF_ISendSbuf (
             int to,
             int tag )  [extern]
```

Nonblocking send operation. It sends everything from `buff` to `fill` of the active buffer. Depending on *tag* it also can do waiting for other sends to finish or set the active buffer to the next one.

Parameters

to	the destination process number.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 261 of file mpi.c.

Referenced by FlushOut(), PF_Processor(), and PutOut().

4.96.3.2 PF_Bcast()

```
int PF_Bcast (
            void * buffer,
            int count )  [extern]
```

Broadcasts a message from the master to slaves.

Parameters

in,out	*buffer*	the starting address of buffer. The contents in this buffer on the master will be transferred to those on the slaves.
	count	the length of the buffer in bytes.

Returns

0 if OK, nonzero on error.

Definition at line 440 of file mpi.c.

Referenced by PF_BroadcastBuffer(), and PF_BroadcastNumber().

4.96.3.3 PF_RawSend()

```
int PF_RawSend (
            int dest,
            void * buf,
            LONG l,
            int tag )  [extern]
```

Sends *l* bytes from *buf* to *dest*. Returns 0 on success, or -1.

Parameters

dest	the destination process number.
buf	the send buffer.
l	the size of data in the send buffer in bytes.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 463 of file mpi.c.

4.96.3.4 PF_RawRecv()

```
LONG PF_RawRecv (
            int * src,
```

```
            void * buf,
            LONG thesize,
            int * tag )  [extern]
```

Receives not more than *thesize* bytes from *src*, returns the actual number of received bytes, or -1 on failure.

Parameters

in,out	*src*	the source process number. In output, that of the actual received message.
out	*buf*	the receive buffer.
	thesize	the size of the receive buffer in bytes.
out	*tag*	the message tag of the actual received message.

Returns

the actual sizeof received data in bytes, or -1 on failure.

Definition at line 484 of file mpi.c.

4.96.3.5 PF_PreparePack()

```
int PF_PreparePack (
            void )  [extern]
```

Prepares for the next pack operations on the sender.

Returns

0 if OK, nonzero on error.

Definition at line 624 of file mpi.c.

Referenced by Optimize(), PF_BroadcastPreDollar(), PF_BroadcastString(), and PF_Init().

4.96.3.6 PF_Pack()

```
int PF_Pack (
            const void * buffer,
            size_t count,
            MPI_Datatype type )  [extern]
```

Adds data into the pack buffer.

Parameters

buffer	the pointer to the buffer storing the data to be packed.
count	the number of elements in the buffer.
type	the data type of elements in the buffer.

Returns

0 if OK, nonzero on error.

Definition at line 642 of file mpi.c.

References MPI_ERRCODE_CHECK.

Referenced by Optimize(), PF_BroadcastPreDollar(), and PF_Init().

4.96.3.7 PF_Unpack()

```
int PF_Unpack (
            void * buffer,
            size_t count,
            MPI_Datatype type ) [extern]
```

Retrieves the next data in the pack buffer.

Parameters

out	*buffer*	the pointer to the buffer to store the unpacked data.
	count	the number of elements of data to be received.
	type	the data type of elements of data to be received.

Returns

0 if OK, nonzero on error.

Definition at line 671 of file mpi.c.

References MPI_ERRCODE_CHECK.

Referenced by Optimize(), PF_BroadcastPreDollar(), and PF_Init().

4.96.3.8 PF_PackString()

```
int PF_PackString (
            const UBYTE * str ) [extern]
```

Packs a string *str* into the packed buffer PF_packbuf, including the trailing zero.

The first element (PF_INT) is the length of the packed portion of the string. If the string does not fit to the buffer PF←_packbuf, the function packs only the initial portion. It returns the number of packed bytes, so if (str[length-1]=='\0') then the whole string fits to the buffer, if not, then the rest (str+length) must be packed and send again. On error, the function returns the negative error code.

One exception: the string "\0!\0" is used as an image of the NULL, so all 3 characters will be packed.

Parameters

str	a string to be packed.

Returns

the number of packed bytes, or a negative value on failure.

Definition at line 706 of file mpi.c.

Referenced by PF_BroadcastString().

4.96.3.9 PF_UnpackString()

```
int PF_UnpackString (
            UBYTE * str )  [extern]
```

Unpacks a string to *str* from the packed buffer PF_packbuf, including the trailing zero.

It returns the number of unpacked bytes, so if (str[length-1]=='\0') then the whole string was unpacked, if not, then the rest must be appended to (str+length). On error, the function returns the negative error code.

Parameters

out	*str*	the buffer to store the unpacked string

Returns

the number of unpacked bytes, or a negative value on failure.

Definition at line 774 of file mpi.c.

Referenced by PF_BroadcastString().

4.96.3.10 PF_Send()

```
int PF_Send (
            int to,
            int tag )  [extern]
```

Sends the contents in the pack buffer to the process specified by *to*.

Example:
```
if ( PF.me == SRC ) {
  PF_PreparePack();
  // Packing operations here...
  PF_Send(DEST, TAG);
}
else if ( PF.me == DEST ) {
  PF_Receive(SRC, TAG, &actual_src, &actual_tag);
  // Unpacking operations here...
}
```

Parameters

to	the destination process number.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 822 of file mpi.c.

References MPI_ERRCODE_CHECK.

4.96.3.11 PF_Receive()

```
int PF_Receive (
        int src,
        int tag,
        int * psrc,
        int * ptag )  [extern]
```

Receives data into the pack buffer from the process specified by *src*. This function allows &src == psrc or &tag == ptag. Either *psrc* or *ptag* can be NULL.

See the example of PF_Send().

Parameters

	src	the source process number (can be PF_ANY_SOURCE).
	tag	the source message tag (can be PF_ANY_TAG).
out	*psrc*	the actual source process number of received message.
out	*ptag*	the received message tag.

Returns

0 if OK, nonzero on error.

Definition at line 848 of file mpi.c.

References MPI_ERRCODE_CHECK.

4.96.3.12 PF_Broadcast()

```
int PF_Broadcast (
        void )  [extern]
```

Broadcasts the contents in the pack buffer on the master to those on the slaves.

Example:

```
if ( PF.me == MASTER ) {
  PF_PreparePack();
  // Packing operations here...
}
PF_Broadcast();
if ( PF.me != MASTER ) {
  // Unpacking operations here...
}
```

Returns

0 if OK, nonzero on error.

Definition at line 883 of file mpi.c.

References MPI_ERRCODE_CHECK.

Referenced by Optimize(), PF_BroadcastPreDollar(), PF_BroadcastString(), and PF_Init().

4.96.3.13 PF_PrepareLongSinglePack()

```
int PF_PrepareLongSinglePack (
            void ) [extern]
```

Prepares for the next long-single-pack operations on the sender.

Returns

0 if OK, nonzero on error.

Definition at line 1451 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.96.3.14 PF_LongSinglePack()

```
int PF_LongSinglePack (
            const void * buffer,
            size_t count,
            MPI_Datatype type ) [extern]
```

Adds data into the "long single" pack buffer.

Parameters

buffer	the pointer to the buffer storing the data to be packed.
count	the number of elements in the buffer.
type	the data type of elements in the buffer.

Returns

0 if OK, nonzero on error.

Definition at line 1469 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.96.3.15 PF_LongSingleUnpack()

```
int PF_LongSingleUnpack (
            void * buffer,
            size_t count,
            MPI_Datatype type ) [extern]
```

Retrieves the next data in the "long single" pack buffer.

Parameters

out	*buffer*	the pointer to the buffer to store the unpacked data.
	count	the number of elements of data to be received.
	type	the data type of elements of data to be received.

Returns

0 if OK, nonzero on error.

Definition at line 1503 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.96.3.16 PF_LongSingleSend()

```
int PF_LongSingleSend (
              int to,
              int tag )   [extern]
```

Sends the contents in the "long single" pack buffer to the process specified by *to*.

Example:
```
if ( PF.me == SRC ) {
  PF_PrepareLongSinglePack();
  // Packing operations here...
  PF_LongSingleSend(DEST, TAG);
}
else if ( PF.me == DEST ) {
  PF_LongSingleReceive(SRC, TAG, &actual_src, &actual_tag);
  // Unpacking operations here...
}
```

Parameters

to	the destination process number.
tag	the message tag.

Returns

0 if OK, nonzero on error.

Definition at line 1540 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.96.3.17 PF_LongSingleReceive()

```
int PF_LongSingleReceive (
              int src,
              int tag,
              int * psrc,
              int * ptag )   [extern]
```

Receives data into the "long single" pack buffer from the process specified by *src*. This function allows &src == psrc or &tag == ptag. Either *psrc* or *ptag* can be NULL.

See the example of PF_LongSingleSend().

Parameters

	src	the source process number (can be PF_ANY_SOURCE).
	tag	the source message tag (can be PF_ANY_TAG).
out	*psrc*	the actual source process number of received message.
out	*ptag*	the received message tag.

Returns

0 if OK, nonzero on error.

Definition at line 1583 of file mpi.c.

Referenced by PF_CollectModifiedDollars(), and PF_Processor().

4.96.3.18 PF_PrepareLongMultiPack()

```
int PF_PrepareLongMultiPack (
            void ) [extern]
```

Prepares for the next long-multi-pack operations on the sender.

Returns

0 if OK, nonzero on error.

Definition at line 1643 of file mpi.c.

Referenced by Optimize(), PF_BroadcastCBuf(), PF_BroadcastExpFlags(), PF_BroadcastModifiedDollars(), and PF_BroadcastRedefinedPreVars().

4.96.3.19 PF_LongMultiPackImpl()

```
int PF_LongMultiPackImpl (
            const void * buffer,
            size_t count,
            size_t eSize,
            MPI_Datatype type ) [extern]
```

Adds data into the "long multi" pack buffer.

Parameters

buffer	the pointer to the buffer storing the data to be packed.
count	the number of elements in the buffer.
eSize	the byte size of each element of data.
type	the data type of elements in the buffer.

Returns

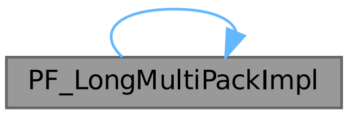
0 if OK, nonzero on error.

Definition at line 1662 of file mpi.c.

References PF_LongMultiPackImpl().

Referenced by PF_LongMultiPackImpl().

Here is the call graph for this function:



4.96.3.20 PF_LongMultiUnpackImpl()

```
int PF_LongMultiUnpackImpl (
            void * buffer,
            size_t count,
            size_t eSize,
            MPI_Datatype type )  [extern]
```

Retrieves the next data in the "long multi" pack buffer.

Parameters

out	*buffer*	the pointer to the buffer to store the unpacked data.
	count	the number of elements of data to be received.
	eSize	the byte size of each element of data.
	type	the data type of elements of data to be received.

Returns

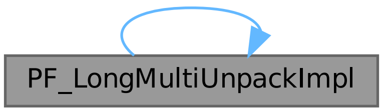
0 if OK, nonzero on error.

Definition at line 1721 of file mpi.c.

References PF_LongMultiUnpackImpl().

Referenced by PF_LongMultiUnpackImpl().

Here is the call graph for this function:



4.96.3.21 PF_LongMultiBroadcast()

```
int PF_LongMultiBroadcast (
              void ) [extern]
```

Broadcasts the contents in the "long multi" pack buffer on the master to those on the slaves.

Example:

```
if ( PF.me == MASTER ) {
  PF_PrepareLongMultiPack();
  // Packing operations here...
}
PF_LongMultiBroadcast();
if ( PF.me != MASTER ) {
  // Unpacking operations here...
}
```

Returns

0 if OK, nonzero on error.

Definition at line 1807 of file mpi.c.

Referenced by Optimize(), PF_BroadcastCBuf(), PF_BroadcastExpFlags(), PF_BroadcastModifiedDollars(), and PF_BroadcastRedefinedPreVars().

4.96.3.22 PF_EndSort()

```
int PF_EndSort (
              void ) [extern]
```

Finishes a master sorting with collecting terms from slaves. Called by EndSort().

If this is not the masterprocess, just initialize the sendbuffers and return 0, else PF_EndSort() sends the rest of the terms in the sendbuffer to the next slave and a dummy message to all slaves with tag PF_ENDSORT_MSGTAG. Then it receives the sorted terms, sorts them using a recursive 'tree of losers' (PF_GetLoser()) and writes them to the outputfile.

Returns

1 if the sorting on the master was done. 0 if EndSort() still must perform a regular sorting because it is not at the ground level or not on the master or in the sequential mode or in the InParallel mode. -1 if an error occurred.

Remarks

The slaves will send the sorted terms back to the master in the regular sorting (after the initialization of the send buffer in PF_EndSort()). See PutOut() and FlushOut().

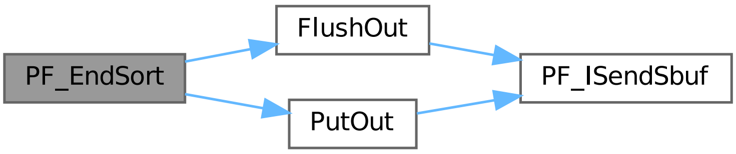
This function has been changed such that when it returns 1, AM.S0->TermsLeft is set correctly. But AM.S0->GenTerms is not set: it will be set after collecting the statistics from the slaves at the end of PF_Processor(). (TU 30 Jun 2011)

Definition at line 864 of file parallel.c.

References FlushOut(), and PutOut().

Referenced by EndSort().

Here is the call graph for this function:



4.96.3.23 PF_Deferred()

```
WORD PF_Deferred (
            WORD * term,
            WORD level )  [extern]
```

Replaces Deferred() on the slaves.

Parameters

term	the term that must be multiplied by the contents of the current bracket.
level	the compiler level.

Returns

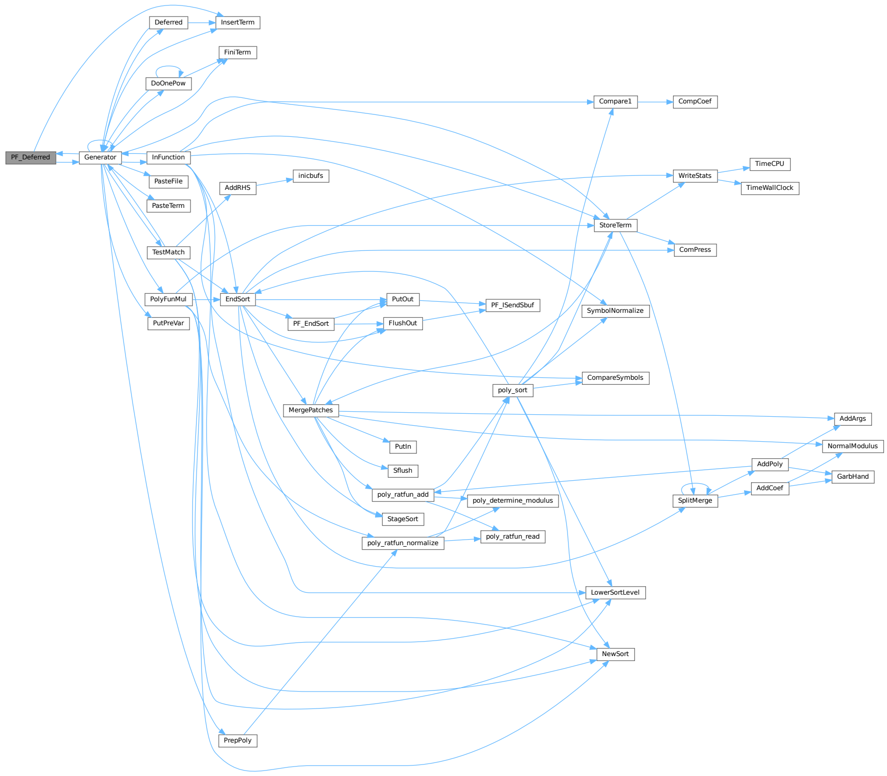
0 if OK, nonzero on error.

Definition at line 1208 of file parallel.c.

References Generator(), and InsertTerm().

Referenced by Generator().

Here is the call graph for this function:



4.96.3.24 PF_Processor()

```
int PF_Processor (
            EXPRESSIONS e,
            WORD i,
            WORD LastExpression )  [extern]
```

Replaces parts of Processor() on the masters and slaves. On the master PF_Processor() is responsible for proper distribution of terms from the input file to the slaves. On the slaves it calls Generator() for all the terms that this process gets, but PF_GetTerm() gets terms from the master (not directly from infile).

Parameters

e	The pointer to the current expression.
i	The index for the current expression.
LastExpression	The flag indicating whether it is the last expression.

Returns

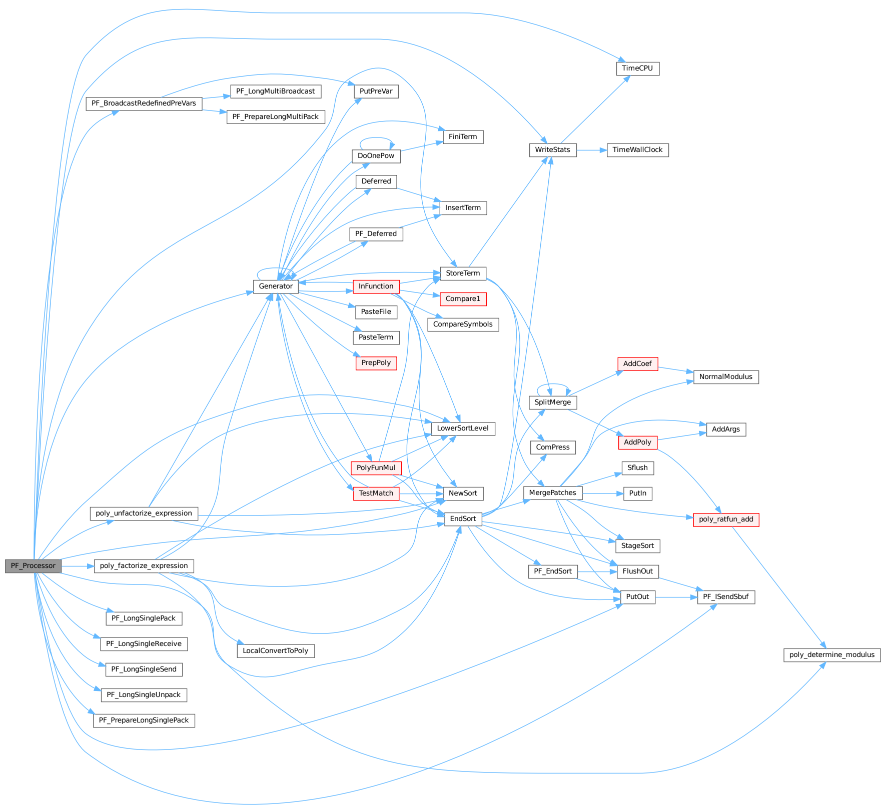
0 if OK, nonzero on error.

Definition at line 1540 of file parallel.c.

References EndSort(), Generator(), LowerSortLevel(), NewSort(), PACK_LONG, PF_BroadcastRedefinedPreVars(), PF_ISendSbuf(), PF_LongSinglePack(), PF_LongSingleReceive(), PF_LongSingleSend(), PF_LongSingleUnpack(), PF_PrepareLongSinglePack(), poly_factorize_expression(), poly_unfactorize_expression(), PutOut(), StoreTerm(), SWAP, TimeCPU(), and WriteStats().

Referenced by Processor().

Here is the call graph for this function:



4.96.3.25 PF_Init()

```
int PF_Init (
            int * argc,
            char *** argv )  [extern]
```

All the library independent stuff. PF_LibInit() should do all library dependent initializations.

Parameters

argc	pointer to the number of arguments.
argv	pointer to the arguments.

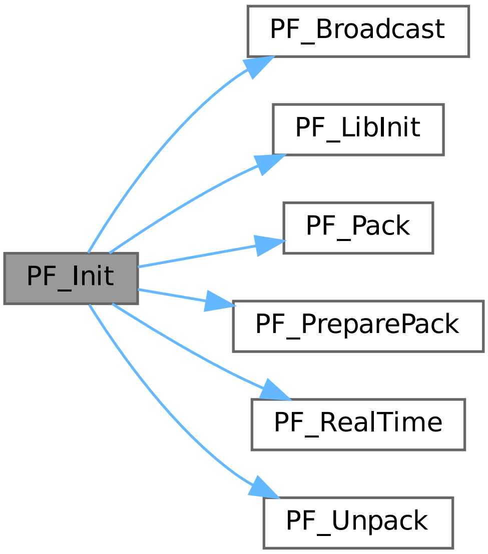
Returns

0 if OK, nonzero on error.

Definition at line 1963 of file parallel.c.

References PF_Broadcast(), PF_LibInit(), PF_Pack(), PF_PreparePack(), PF_RealTime(), and PF_Unpack().

Here is the call graph for this function:



4.96.3.26 PF_Terminate()

```
int PF_Terminate (
            int errorcode )  [extern]
```

Performs the finalization of ParFORM. To be called by Terminate().

Parameters

error	an error code.

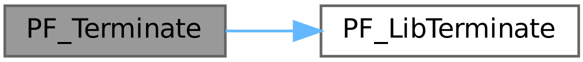
Returns

0 if OK, nonzero on error.

Definition at line 2058 of file parallel.c.

References PF_LibTerminate().

Here is the call graph for this function:



4.96.3.27 PF_GetSlaveTimes()

```
LONG PF_GetSlaveTimes (
            void ) [extern]
```

Returns the total CPU time of all slaves together. This function must be called on the master and all slaves.

Returns

on the master, the sum of CPU times on all slaves.

Definition at line 2074 of file parallel.c.

References TimeCPU().

Here is the call graph for this function:

4.96.3.28 PF_BroadcastNumber()

```
LONG PF_BroadcastNumber (
            LONG x ) [extern]
```

Broadcasts a LONG value from the master to the all slaves.

Parameters

x	the number to be broadcast (set on the master).

Returns

the synchronised result.

Definition at line 2094 of file parallel.c.

References PF_Bcast().

Referenced by DoCheckpoint(), PF_BroadcastBuffer(), and StartVariables().

Here is the call graph for this function:

4.96.3.29 PF_BroadcastBuffer()

```
void PF_BroadcastBuffer (
            WORD ** buffer,
            LONG * length )  [extern]
```

Broadcasts a buffer from the master to all the slaves.

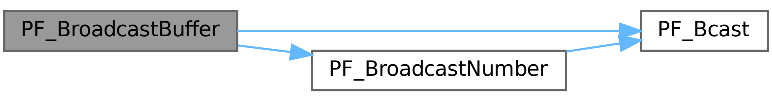
Parameters

in,out	*buffer*	on the master, the buffer to be broadcast. On the slaves, the buffer will be allocated if the length is greater than 0. The caller must free it.
in,out	*length*	on the master, the length of the buffer to be broadcast. On the slaves, it receives the length of transferred buffer. The actual transfer occurs only if the length is greater than 0.

Definition at line 2121 of file parallel.c.

References PF_Bcast(), and PF_BroadcastNumber().

Here is the call graph for this function:



4.96.3.30 PF_BroadcastString()

```
int PF_BroadcastString (
            UBYTE * str )  [extern]
```

Broadcasts a string from the master to all slaves.

Parameters

in,out	*str*	The pointer to a null-terminated string.

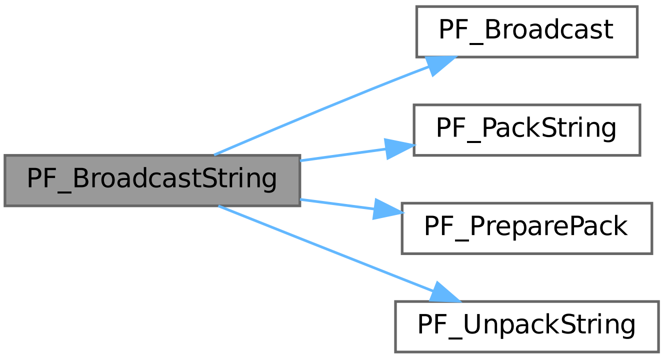
Returns

0 if OK, nonzero on error.

Definition at line 2163 of file parallel.c.

References PF_Broadcast(), PF_PackString(), PF_PreparePack(), and PF_UnpackString().

Here is the call graph for this function:



4.96.3.31 PF_BroadcastPreDollar()

```
int PF_BroadcastPreDollar (
            WORD ** dbuffer,
            LONG * newsize,
            int * numterms )  [extern]
```

Broadcasts dollar variables set as a preprocessor variables. Only the master is able to make an assignment like #$a=g; where g is an expression: only the master has an access to the expression. So, the master broadcasts the result to slaves.

The result is in ∗dbuffer of the size is ∗newsize (in number of WORDs), +1 for trailing zero. For slave newsize and numterms are output parameters.

Parameters

in,out	*dbuffer*	the buffer for a dollar variable.
in,out	*newsize*	the size of the dollar variable in WORDs.
in,out	*numterms*	the number of terms in the dollar variable.

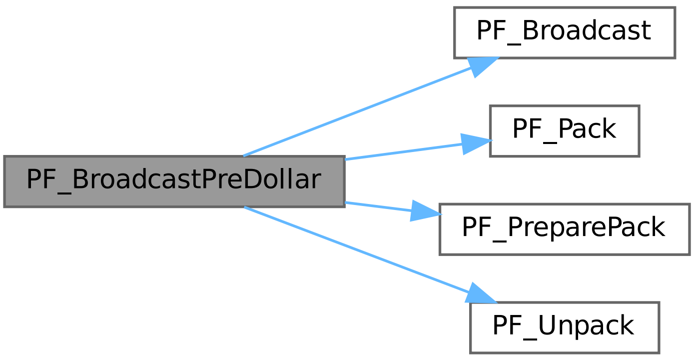
Returns

0 if OK, nonzero on error.

Definition at line 2218 of file parallel.c.

References PF_Broadcast(), PF_Pack(), PF_PreparePack(), and PF_Unpack().

Here is the call graph for this function:



4.96.3.32 PF_CollectModifiedDollars()

```
int PF_CollectModifiedDollars (
            void ) [extern]
```

Combines modified dollar variables on the all slaves, and store them into those on the master.

The potentially modified dollar variables are given in PotModdollars, and the number of them is given by NumPot←Moddollars.

The current module could be executed in parallel only if all potentially modified variables are listed in ModOptdollars, otherwise the module was switched to the sequential mode.

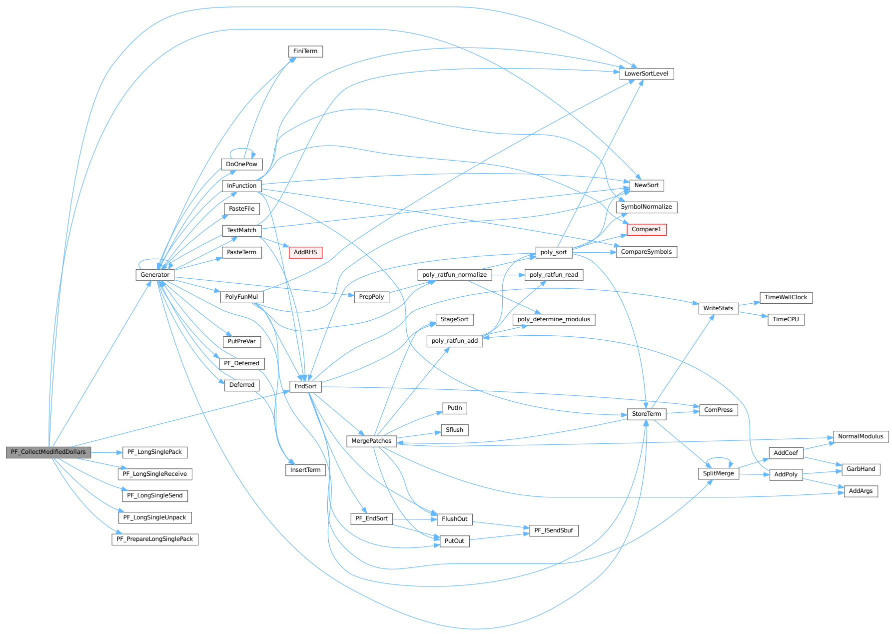
Returns

0 if OK, nonzero on error.

Definition at line 2506 of file parallel.c.

References EndSort(), Generator(), LowerSortLevel(), NewSort(), PF_LongSinglePack(), PF_LongSingleReceive(), PF_LongSingleSend(), PF_LongSingleUnpack(), PF_PrepareLongSinglePack(), VectorInit, VectorPtr, VectorReserve, and VectorSize.

Here is the call graph for this function:



4.96.3.33 PF_BroadcastModifiedDollars()

```
int PF_BroadcastModifiedDollars (
             void ) [extern]
```

Broadcasts modified dollar variables on the master to the all slaves.

The potentially modified dollar variables are given in PotModdollars, and the number of them is given by NumPot←Moddollars.

The current module could be executed in parallel only if all potentially modified variables are listed in ModOptdollars, otherwise the module was switched to the sequential mode. In either cases, we need to broadcast them.

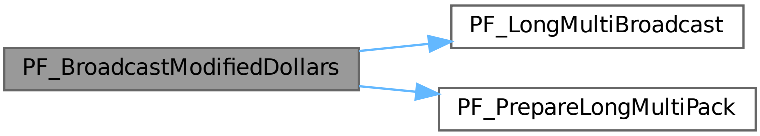
Returns

0 if OK, nonzero on error.

Definition at line 2785 of file parallel.c.

References PF_LongMultiBroadcast(), and PF_PrepareLongMultiPack().

Here is the call graph for this function:



4.96.3.34 PF_BroadcastRedefinedPreVars()

```
int PF_BroadcastRedefinedPreVars (
            void ) [extern]
```

Broadcasts preprocessor variables, which were changed by the Redefine statements in the current module, from the master to the all slaves.

The potentially redefined preprocessor variables are given in AC.pfirstnum, and the number of them is given by AC.numpfirstnum. For an actually redefined variable, the corresponding value in AC.inputnumbers is non-negative.

Returns

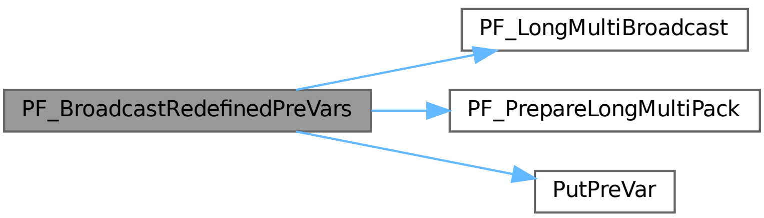
0 if OK, nonzero on error.

Definition at line 3002 of file parallel.c.

References PF_LongMultiBroadcast(), PF_PrepareLongMultiPack(), PutPreVar(), VectorPtr, and VectorReserve.

Referenced by PF_Processor().

Here is the call graph for this function:



4.96.3.35 PF_BroadcastCBuf()

```
int PF_BroadcastCBuf (
            int bufnum )  [extern]
```

Broadcasts a compiler buffer specified by *bufnum* from the master to the all slaves.

Parameters

bufnum	The index of the compiler buffer to be broadcast.

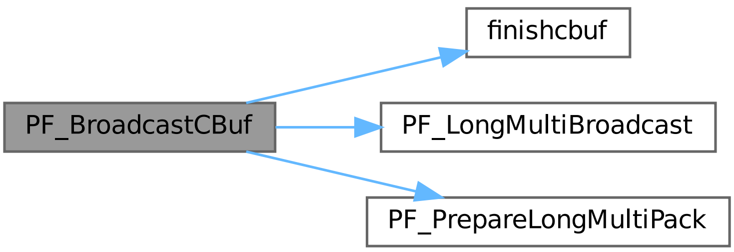
Returns

0 if OK, nonzero on error.

Definition at line 3144 of file parallel.c.

References CbUf::boomlijst, CbUf::Buffer, CbUf::BufferSize, CbUf::CanCommu, CbUf::dimension, finishcbuf(), CbUf::lhs, CbUf::numdum, CbUf::NumTerms, PF_LongMultiBroadcast(), PF_PrepareLongMultiPack(), CbUf::Pointer, CbUf::rhs, and CbUf::Top.

Here is the call graph for this function:



4.96.3.36 PF_BroadcastExpFlags()

```
int PF_BroadcastExpFlags (
            void )  [extern]
```

Broadcasts AR.expflags and several properties of each expression, e.g., e->vflags, from the master to all slaves.

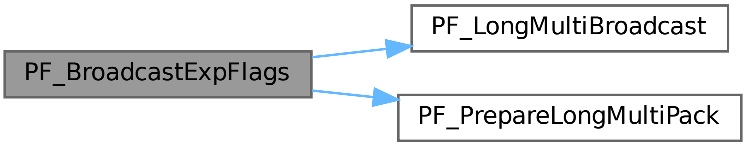
Returns

0 if OK, nonzero on error.

Definition at line 3255 of file parallel.c.

References PF_LongMultiBroadcast(), and PF_PrepareLongMultiPack().

Here is the call graph for this function:



4.96.3.37 PF_StoreInsideInfo()

```
int PF_StoreInsideInfo (
            void )  [extern]
```

Definition at line 3092 of file parallel.c.

4.96.3.38 PF_RestoreInsideInfo()

```
int PF_RestoreInsideInfo (
            void )  [extern]
```

Definition at line 3118 of file parallel.c.

4.96.3.39 PF_BroadcastExpr()

```
int PF_BroadcastExpr (
            EXPRESSIONS e,
            FILEHANDLE * file )  [extern]
```

Broadcasts an expression from the master to the all slaves.

Parameters

e	The expression to be broadcast.
file	The file in which the expression is sitting.

Returns

0 if OK, nonzero on error.

Definition at line 3549 of file parallel.c.

Referenced by PF_BroadcastRHS().

4.96.3.40 PF_BroadcastRHS()

```
int PF_BroadcastRHS (
            void ) [extern]
```

Broadcasts expressions appearing in the right-hand side from the master to the all slaves.

Returns

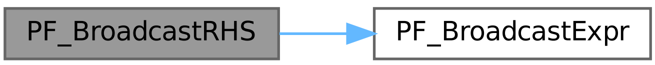
0 if OK, nonzero on error.

Definition at line 3577 of file parallel.c.

References PF_BroadcastExpr().

Referenced by Processor().

Here is the call graph for this function:



4.96.3.41 PF_InParallelProcessor()

```
int PF_InParallelProcessor (
            void ) [extern]
```

Processes expressions in the InParallel mode, i.e., dividing expressions marked by partodo over the slaves.

Returns

0 if OK, nonzero on error.

Definition at line 3624 of file parallel.c.

Referenced by Processor().

4.96.3.42 PF_SendFile()

```
int PF_SendFile (
            int to,
            FILE * fd ) [extern]
```

Sends a file to the process specified by *to*.

Parameters

to	the destination process number.
fd	the file to be sent.

Returns

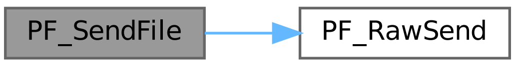
the size of sent data in bytes, or -1 on error.

Definition at line 4221 of file parallel.c.

References PF_RawSend().

Referenced by DoCheckpoint().

Here is the call graph for this function:



4.96.3.43 PF_RecvFile()

```
int PF_RecvFile (
            int from,
            FILE * fd ) [extern]
```

Receives a file from the process specified by *from*.

Parameters

from	the source process number.
fd	the file to save the received data.

Returns

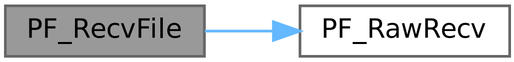
the size of received data in bytes, or -1 on error.

Definition at line 4259 of file parallel.c.

References PF_RawRecv().

Referenced by DoCheckpoint().

Here is the call graph for this function:



4.96.3.44 PF_MLock()

```
void PF_MLock (
            void ) [extern]
```

A function called by MLOCK(ErrorMessageLock) for slaves.

Definition at line 4340 of file parallel.c.

References VectorClear.

4.96.3.45 PF_MUnlock()

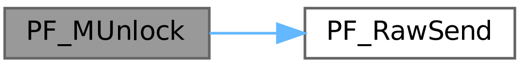
```
void PF_MUnlock (
            void ) [extern]
```

A function called by MUNLOCK(ErrorMessageLock) for slaves.

Definition at line 4356 of file parallel.c.

References PF_RawSend(), VectorEmpty, VectorPtr, and VectorSize.

Here is the call graph for this function:



4.96.3.46 PF_WriteFileToFile()

```
LONG PF_WriteFileToFile (
            int handle,
            UBYTE * buffer,
            LONG size ) [extern]
```

Replaces WriteFileToFile() on the master and slaves.

It copies the given buffer into internal buffers if called between MLOCK(ErrorMessageLock) and MUNLOCK(Error↩ MessageLock) for slaves and handle is StdOut or LogHandle, otherwise calls WriteFileToFile().

Parameters

handle	a file handle that specifies the output.
buffer	a pointer to the source buffer containing the data to be written.
size	the size of data to be written in bytes.

Returns

the actual size of data written to the output in bytes.

Definition at line 4385 of file parallel.c.

References VectorClear, VectorPtr, VectorPushBacks, and VectorSize.

4.96.3.47 PF_FlushStdOutBuffer()

```
void PF_FlushStdOutBuffer (
            void ) [extern]
```

Explicitly Flushes the buffer for the standard output on the master, which is used if PF_ENABLE_STDOUT_↩ BUFFERING is defined.

Definition at line 4479 of file parallel.c.

References VectorClear, VectorPtr, and VectorSize.

4.96.4 Variable Documentation

4.96.4.1 PF

```
PARALLELVARS PF [extern]
```

Definition at line 90 of file parallel.c.

4.96.4.2 PF_maxDollarChunkSize

```
LONG PF_maxDollarChunkSize [extern]
```

Definition at line 69 of file mpi.c.

4.97 parallel.h

Go to the documentation of this file.

```
#ifndef __PARALLEL__
#define __PARALLEL__

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

/*
    #[ macros & definitions :
*/
#define MASTER 0

#define PF_RESET 0
#define PF_TIME  1

#define PF_TERM_MSGTAG          10  /* master -> slave: sending terms */
#define PF_ENDSORT_MSGTAG       11  /* master -> slave: no more terms to be distributed, slave -> master: after EndSort() */
#define PF_DOLLAR_MSGTAG        12  /* slave -> master: sending $-variables */
#define PF_BUFFER_MSGTAG        20  /* slave -> master: sending sorted terms, or in PF_SendFile()/PF_RecvFile() */
#define PF_ENDBUFFER_MSGTAG     21  /* same as PF_BUFFER_MSGTAG, but indicates the end of operation */
#define PF_READY_MSGTAG         30  /* slave -> master: slave is idle and can accept terms */
#define PF_DATA_MSGTAG          50  /* InParallel, DoCheckpoint() */
#define PF_EMPTY_MSGTAG         52  /* InParallel, DoCheckpoint(), PF_SendFile(), PF_RecvFile() */
#define PF_STDOUT_MSGTAG        60  /* slave -> master: sending text to the stdout */
#define PF_LOG_MSGTAG           61  /* slave -> master: sending text to the log file */
#define PF_OPT_MCTS_MSGTAG      70  /* master <-> slave: optimization */
#define PF_OPT_HORNER_MSGTAG    71  /* master <-> slave: optimization */
#define PF_OPT_COLLECT_MSGTAG   72  /* slave -> master: optimization */
#define PF_MISC_MSGTAG         100

/*
 * A macro for checking the version of gcc.
 */
#if defined(__GNUC__) && defined(__GNUC_MINOR__) && defined(__GNUC_PATCHLEVEL__)
#  define GNUC_PREREQ(major, minor, patchlevel) \
      ((__GNUC__ « 16) + (__GNUC_MINOR__ « 8) + __GNUC_PATCHLEVEL__ >= \
      ((major) « 16) + ((minor) « 8) + (patchlevel))
#else
#  define GNUC_PREREQ(major, minor, patchlevel) 0
#endif

/*
 * The macro "indices" defined in variable.h collides with some function
 * argument names in the MPI-3.0 standard.
 */
#undef indices

/* Avoid messy padding warnings which may appear in mpi.h. */
#if GNUC_PREREQ(4, 6, 0)
#  pragma GCC diagnostic push
#  pragma GCC diagnostic ignored "-Wpadded"
#  pragma GCC diagnostic ignored "-Wunused-parameter"
#endif
#if defined(__clang__) && defined(__has_warning)
#  pragma clang diagnostic push
#  if __has_warning("-Wpadded")
#    pragma clang diagnostic ignored "-Wpadded"
```

```
#  endif
#  if __has_warning("-Wunused-parameter")
#    pragma clang diagnostic ignored "-Wunused-parameter"
#  endif
#endif

#  ifdef __cplusplus
     /*
      * form3.h (which includes parallel.h) is included from newpoly.h as
      *   extern "C" {
      *   #include "form3.h"
      *   }
      * On the other hand, C++ interfaces to MPI are defined in mpi.h if it is
      * included from C++ sources. We first leave from the C-linkage, include
      * mpi.h, and then go back to the C-linkage.
      * (TU 7 Jun 2011)
      */
}
#    define OMPI_SKIP_MPICXX 1
#    include <mpi.h>
extern "C" {
#  else
#    define OMPI_SKIP_MPICXX 1
#    include <mpi.h>
#  endif

/* Now redefine "indices" in the same way as in variable.h. */
#define indices ((INDICES)(AC.IndexList.lijst))

/* Restore the warning settings. */
#if GNUC_PREREQ(4, 6, 0)
#  pragma GCC diagnostic pop
#endif
#if defined(__clang__) && defined(__has_warning)
#  pragma clang diagnostic pop
#endif

#  define PF_ANY_SOURCE MPI_ANY_SOURCE
#  define PF_ANY_MSGTAG MPI_ANY_TAG
#  define PF_COMM MPI_COMM_WORLD
#  define PF_BYTE MPI_BYTE
#  define PF_INT  MPI_INT
#  if defined(ILP32)
#    define PF_WORD MPI_SHORT
#    define PF_LONG MPI_LONG
#  elif defined(LLP64)
#    define PF_WORD MPI_INT
#    define PF_LONG MPI_LONG_LONG_INT
#  elif defined(LP64)
#    define PF_WORD MPI_INT
#    define PF_LONG MPI_LONG
#  endif

/*
    #] macros & definitions :
    #[ s/r-bufs :
*/

typedef struct {
    WORD **buff;
    WORD **fill;
    WORD **full;
    WORD **stop;
    MPI_Status *status;
    MPI_Status *retstat;
    MPI_Request *request;
    MPI_Datatype *type;    /* this is needed in PF_Wait for Get_count */
    int *index;            /* dummies for returnvalues */
    int *tag;              /* for the version with blocking send/receives */
    int *from;
    int numbufs;           /* number of cyclic buffers */
    int active;            /* flag telling which buffer is active */
    PADPOINTER(0,2,0,0);
} PF_BUFFER;

/*
    #] s/r-bufs :
    #[ global variables used by the PF_functions : need to be known everywhere
*/

typedef struct ParallelVars {
    FILEHANDLE  slavebuf;       /* (slave) allocated if there are RHS expressions */
    /* special buffers for nonblocking, unbuffered send/receives */
    PF_BUFFER  *sbuf;           /* set of cyclic send buffers for master _and_ slave */
    PF_BUFFER **rbufs;          /* array of sets of cyclic receive buffers for master */
    int         me;             /* Internal number of task: master is 0 */
    int         numtasks;       /* total number of tasks */
```

```
    int         parallel;       /* flags telling the master and slaves to do the sorting parallel */
                                /* [05nov2003 mt] This flag must be set to 0 in iniModule! */
    int         rhsInParallel;  /* flag for parallel executing even if there are RHS expressions */
    int         mkSlaveInfile;  /* flag tells that slavebuf is used on the slaves */
    int         exprbufsize;    /* buffer size in WORDs to be used for transferring expressions */
    int         exprtodo;       /* >= 0: the expression to do in InParallel, -1: otherwise */
    int         log;            /* flag for logging mode */
    WORD        numsbufs;       /* number of cyclic send buffers (PF.sbuf->numbufs) */
    WORD        numrbufs;       /* number of cyclic receive buffers (PF.rbufs[i]->numbufs, i=1,...numtasks-1) */
    PADPOSITION(2,0,8,2,0);
} PARALLELVARS;

extern PARALLELVARS PF;
/*[04oct2005 mt]:*/
/*for broadcasting dollarvars, see parallel.c:PF_BroadcastPreDollar():*/
extern LONG PF_maxDollarChunkSize;
/*:[04oct2005 mt]*/

/*
    #] global variables used by the PF_functions :
    #[ Function prototypes :
*/

/* mpi.c */
extern int    PF_ISendSbuf(int,int);
extern int    PF_Bcast(void *buffer, int count);
extern int    PF_RawSend(int,void *,LONG,int);
extern LONG   PF_RawRecv(int *,void *,LONG,int *);

extern int    PF_PreparePack(void);
extern int    PF_Pack(const void *buffer, size_t count, MPI_Datatype type);
extern int    PF_Unpack(void *buffer, size_t count, MPI_Datatype type);
extern int    PF_PackString(const UBYTE *str);
extern int    PF_UnpackString(UBYTE *str);
extern int    PF_Send(int to, int tag);
extern int    PF_Receive(int src, int tag, int *psrc, int *ptag);
extern int    PF_Broadcast(void);

extern int    PF_PrepareLongSinglePack(void);
extern int    PF_LongSinglePack(const void *buffer, size_t count, MPI_Datatype type);
extern int    PF_LongSingleUnpack(void *buffer, size_t count, MPI_Datatype type);
extern int    PF_LongSingleSend(int to, int tag);
extern int    PF_LongSingleReceive(int src, int tag, int *psrc, int *ptag);

extern int    PF_PrepareLongMultiPack(void);
extern int    PF_LongMultiPackImpl(const void *buffer, size_t count, size_t eSize, MPI_Datatype type);
extern int    PF_LongMultiUnpackImpl(void *buffer, size_t count, size_t eSize, MPI_Datatype type);
extern int    PF_LongMultiBroadcast(void);

static inline size_t sizeof_datatype(MPI_Datatype type)
{
    if ( type == PF_BYTE ) return sizeof(char);
    if ( type == PF_INT  ) return sizeof(int);
    if ( type == PF_WORD ) return sizeof(WORD);
    if ( type == PF_LONG ) return sizeof(LONG);
    return(0);
}

#define PF_LongMultiPack(buffer, count, type) PF_LongMultiPackImpl(buffer, count, sizeof_datatype(type), type)
#define PF_LongMultiUnpack(buffer, count, type) PF_LongMultiUnpackImpl(buffer, count, sizeof_datatype(type), type)

/* parallel.c */
extern int    PF_EndSort(void);
extern WORD   PF_Deferred(WORD *,WORD);
extern int    PF_Processor(EXPRESSIONS,WORD,WORD);
extern int    PF_Init(int*,char ***);
extern int    PF_Terminate(int);
extern LONG   PF_GetSlaveTimes(void);
extern LONG   PF_BroadcastNumber(LONG);
extern void   PF_BroadcastBuffer(WORD **buffer, LONG *length);
extern int    PF_BroadcastString(UBYTE *);
extern int    PF_BroadcastPreDollar(WORD **, LONG *,int *);
extern int    PF_CollectModifiedDollars(void);
extern int    PF_BroadcastModifiedDollars(void);
extern int    PF_BroadcastRedefinedPreVars(void);
extern int    PF_BroadcastCBuf(int bufnum);
extern int    PF_BroadcastExpFlags(void);
extern int    PF_StoreInsideInfo(void);
extern int    PF_RestoreInsideInfo(void);
extern int    PF_BroadcastExpr(EXPRESSIONS e, FILEHANDLE *file);
extern int    PF_BroadcastRHS(void);
extern int    PF_InParallelProcessor(void);
extern int    PF_SendFile(int to, FILE *fd);
extern int    PF_RecvFile(int from, FILE *fd);
```

```
extern void   PF_MLock(void);
extern void   PF_MUnlock(void);
extern LONG   PF_WriteFileToFile(int,UBYTE *,LONG);
extern void   PF_FlushStdOutBuffer(void);

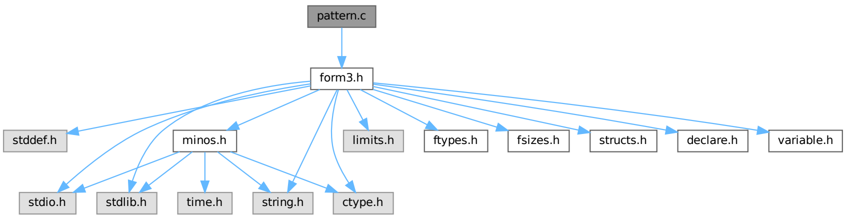
/*
    #] Function prototypes :
*/

#endif
```

4.98 pattern.c File Reference

`#include "form3.h"`

Include dependency graph for pattern.c:



Macros

- #define PutInBuffers(pow)

Functions

- WORD TestMatch (PHEAD WORD ∗term, WORD ∗level)
- void Substitute (PHEAD WORD ∗term, WORD ∗pattern, WORD power)
- WORD FindAll (PHEAD WORD ∗term, WORD ∗pattern, WORD level, WORD ∗par)
- int TestSelect (WORD ∗term, WORD ∗setp)
- void SubsInAll (PHEAD0)
- void TransferBuffer (int from, int to, int spectator)
- int TakeIDfunction (PHEAD WORD ∗term)

4.98.1 Detailed Description

Top level pattern matching routines. More pattern matching is found in findpat.c, function.c, symmetr.c and smart.c. The last three files contain the matching inside functions. The file pattern.c contains also the very important routine Substitute. All regular pattern matching is just the finding of the pattern and indicating what are the wildcards etc. The routine Substitute does the actual removal of the pattern and replaces it by a subterm of the type SUBEXPRESSION.

Definition in file pattern.c.

4.98.2 Macro Definition Documentation

4.98.2.1 PutInBuffers

```
#define PutInBuffers(
              pow )
```

Value:

```
AddRHS(AT.ebufnum,1); \
*out++ = SUBEXPRESSION; \
*out++ = SUBEXPSIZE; \
*out++ = C->numrhs; \
*out++ = pow; \
*out++ = AT.ebufnum; \
FILLSUB(out) \
r = AT.pWorkSpace[rhs+i]; \
if ( *r > 0 ) { \
    oldinr = r[*r]; r[*r] = 0; \
    AddNtoC(AT.ebufnum,(*r+1-ARGHEAD),(r+ARGHEAD),14); \
    r[*r] = oldinr; \
} \
else { \
    ToGeneral(r,buffer,1); \
    buffer[buffer[0]] = 0; \
    AddNtoC(AT.ebufnum,buffer[0]+1,buffer,15); \
}
```

Definition at line 2193 of file pattern.c.

4.98.3 Function Documentation

4.98.3.1 TestMatch()

```
WORD TestMatch (
            PHEAD WORD * term,
            WORD * level )
```

This routine governs the pattern matching. If it decides that a substitution should be made, this can be either the insertion of a right hand side (C->rhs) or the automatic generation of terms as a result of an operation (like trace). The object to be replaced is removed from term and a subexpression pointer is inserted. If the substitution is made more than once there can be more subexpression pointers. Its number is positive as it corresponds to the level at which the C->rhs can be found in the compiler output. The subexpression pointer contains the wildcard substitution information. The power is found in ∗AT.TMout. For operations the subexpression pointer is negative and corresponds to an address in the array AT.TMout. In this array are then the instructions for the routine to be called and its number in the array 'Operations' The format is here: length,functionnumber,length-2 parameters

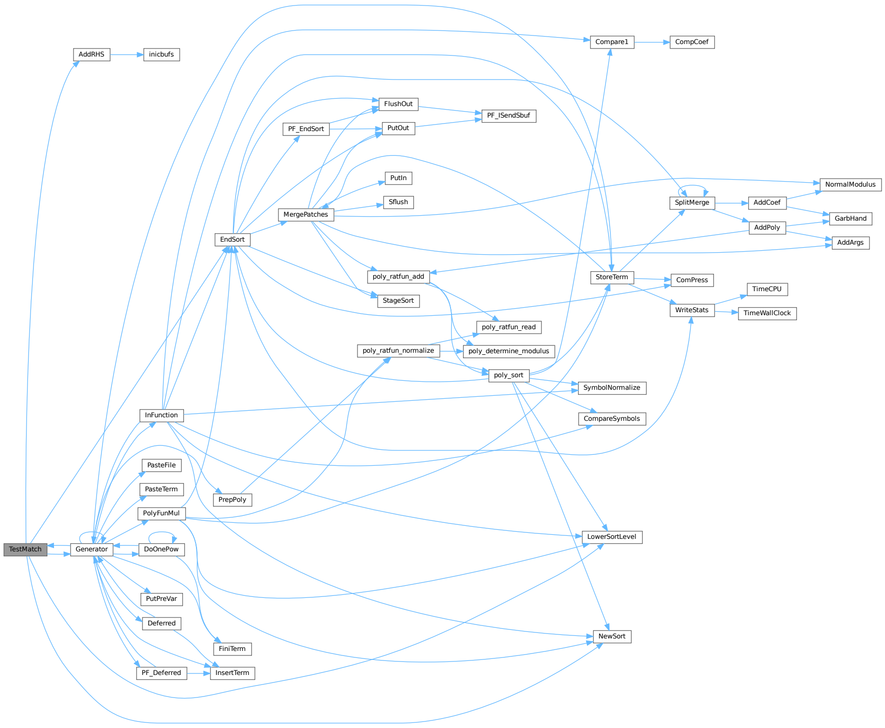
There is a certain complexity wrt repeat levels. Another complication is the poking of the wildcard values in the subexpression prototype in the compiler buffer. This was how things were done in the past with sequential FORM, but with the advent of TFORM this cannot be maintained. Now, for TFORM we make a copy of it. 7-may-2008 (JV): We cannot yet guarantee that this has been done 100% correctly. There are errors that occur in TFORM only and that may indicate problems.

Definition at line 97 of file pattern.c.

References AddRHS(), EndSort(), Generator(), CbUf::lhs, LowerSortLevel(), and NewSort().

Referenced by Generator().

Here is the call graph for this function:



4.98.3.2 Substitute()

```
void Substitute (
            PHEAD WORD * term,
            WORD * pattern,
            WORD power )
```

Definition at line 640 of file pattern.c.

4.98.3.3 FindAll()

```
WORD FindAll (
            PHEAD WORD * term,
            WORD * pattern,
            WORD level,
            WORD * par )
```

Definition at line 1197 of file pattern.c.

4.98.3.4 TestSelect()

```
int TestSelect (
            WORD * term,
            WORD * setp )
```

Definition at line 1748 of file pattern.c.

4.98.3.5 SubsInAll()

```
void SubsInAll (
            PHEAD0 )
```

Definition at line 1982 of file pattern.c.

4.98.3.6 TransferBuffer()

```
void TransferBuffer (
            int from,
            int to,
            int spectator )
```

Definition at line 2143 of file pattern.c.

4.98.3.7 TakeIDfunction()

```
int TakeIDfunction (
            PHEAD WORD * term )
```

Definition at line 2213 of file pattern.c.

4.99 pattern.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
!!! Notice the change in OnePV in FindAll (7-may-2008 JV).

    #[ Includes : pattern.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ Patterns :
        #[ Rules :
```

```

        There are several rules governing the allowable replacements.
        1:  Multi with anything but symbols or dotproducts reverts
            to many.
        2:  Each symbol can have only one (wildcard) power, so
            x^2*x^n? is illegal.
        3:  when a single vector is used it replaces all occurrences
            of the vector. Therefore q*q(mu) or q*q(mu) cannot occur.
            Also q*q cannot be done.
        4:  Loose vector elements are replaced with p(mu), dotproducts
            with p?.q.
        5:  p?.q? is allowed.
        6:  x^n? can revert to n = 0 if there is no power of x.
        7:  x?^n? must match some x. There could be an ambiguity otherwise.

        #] Rules :
        #[ TestMatch :          WORD TestMatch(term,level)
*/

WORD TestMatch(PHEAD WORD *term, WORD *level)
{
    GETBIDENTITY
    WORD *ll, *m, *w, *llf, *OldWork, *StartWork, *ww, *mm, *t, *OldTermBuffer = 0;
    WORD power = 0, match = 0, i, msign = 0, ll2;
    int numdollars = 0, protosize, oldallnumrhs;
    CBUF *C = cbuf+AM.rbufnum, *CC;
    AT.idallflag = 0;
    do {
/*
        #[ Preliminaries :
*/
    ll = C->lhs[*level];
    if ( *ll == TYPEEXPRESSION ) {
/*
        Expressions are not subject to anything.
*/
        return(0);
    }
    else if ( *ll == TYPEREPEAT ) {
        *++AN.RepPoint = 0;
        return(0);          /* Will force the next level */
    }
    else if ( *ll == TYPEENDREPEAT ) {
        if ( *AN.RepPoint ) {
            AN.RepPoint[-1] = 1;        /* Mark the higher level as dirty */
            *AN.RepPoint = 0;
            *level = ll[2];         /* Level to jump back to */
        }
        else {
            AN.RepPoint--;
            if ( AN.RepPoint < AT.RepCount ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Internal problems with REPEAT count");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
        }
        return(0);          /* Force the next level */
    }
    else if ( *ll == TYPEOPERATION ) {
/*
        Operations have always their own level.
*/
        if ( (*(FG.OperaFind[ll[2]]))(BHEAD term,ll) ) return(-1);
        else return(0);
    }
/*
        #] Preliminaries :
*/
    OldWork = AT.WorkPointer;
    if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
    ww = AT.WorkPointer;
/*
        Here we need to make a copy of the subexpression object because we
        will be writing the values of the wildcards in it.
        Originally we copied it into the private version of the compiler buffer
        that is used for scratch space (ebufnum). This caused errors in the
        routines like ScanFunctions when the ebufnum Buffer was expanded
        and inpat was still pointing at the old Buffer. This expansion
        could be done in AddWild and hence cannot be fixed at > 100 places.
        The solution is to use AN.patternbuffer (JV 16-mar-2009).
*/
    {
        WORD *ta = ll, *ma;
        int ja = ta[1];
/*
        New code (16-mar-2009) JV
```

```
*/
        if ( ( ja + 2 ) > AN.patternbuffersize ) {
            if ( AN.patternbuffer ) M_free(AN.patternbuffer,"AN.patternbuffer");
            AN.patternbuffersize = 2 * ja + 2;
            AN.patternbuffer = (WORD *)Malloc1(AN.patternbuffersize * sizeof(WORD),
                    "AN.patternbuffer");
        }
        ma = AN.patternbuffer;
        m = ma + IDHEAD;
        NCOPY(ma,ta,ja);
        *ma = 0;
    }
    AN.FullProto = m;
    AN.WildValue = w = m + SUBEXPSIZE;
    protosize = IDHEAD + m[1];
    m += m[1];
    AN.WildStop = m;
    StartWork = ww;
    ll2 = ll[2];
/*
        #[ Expand dollars :
*/
    if ( ( ll[4] & DOLLARFLAG ) != 0 ) {     /* We have at least one dollar in the pattern */
        WORD oldRepPoint = *AN.RepPoint, olddefer = AR.DeferFlag;
        AR.Eside = LHSIDEX;
/*
        Copy into WorkSpace. This means that AN.patternbuffer will be free.
*/
        ww = AT.WorkPointer; i = m[0]; mm = m;
        NCOPY(ww,mm,i);
        *StartWork += 3;
        *ww++ = 1; *ww++ = 1; *ww++ = 3;
        AT.WorkPointer = ww;
        AR.DeferFlag = 0;
        NewSort(BHEAD0);
        if ( Generator(BHEAD StartWork,AR.Cnumlhs) ) {
            LowerSortLevel();
            AT.WorkPointer = OldWork;
            AR.DeferFlag = olddefer;
            return(-1);
        }
        AT.WorkPointer = ww;
        if ( EndSort(BHEAD ww,0) < 0 ) {}
        AR.DeferFlag = olddefer;
        if ( *ww == 0 || *(ww+*ww) != 0 ) {
            if ( AP.lhdollarerror == 0 ) {
/*
                If race condition we just get more error messages
*/
                MLOCK(ErrorMessageLock);
                MesPrint("&LHS must be one term");
                MUNLOCK(ErrorMessageLock);
                AP.lhdollarerror = 1;
            }
            AT.WorkPointer = OldWork;
            return(-1);
        }
        m = ww; ww = m + *m;
        if ( m[*m-1] < 0 ) { msign = 1; m[*m-1] = -m[*m-1]; }
        if ( *ww || m[*m-1] != 3 || m[*m-2] != 1 || m[*m-3] != 1 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Dollar variable develops into an illegal pattern in id-statement");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        *m -= m[*m-1];
        if ( ( *m + 1 + protosize ) > AN.patternbuffersize ) {
            if ( AN.patternbuffer ) M_free(AN.patternbuffer,"AN.patternbuffer");
            AN.patternbuffersize = 2 * (*m) + 2 + protosize;
            AN.patternbuffer = (WORD *)Malloc1(AN.patternbuffersize * sizeof(WORD),
                    "AN.patternbuffer");
            mm = ll; ww = AN.patternbuffer; i = protosize;
            NCOPY(ww,mm,i);
            AN.FullProto = AN.patternbuffer + IDHEAD;
            AN.WildValue = w = AN.FullProto + SUBEXPSIZE;
            AN.WildStop = AN.patternbuffer + protosize;
        }
        mm = AN.patternbuffer + protosize;
        i = *m;
        NCOPY(mm,m,i);
        m = AN.patternbuffer + protosize;
        AR.Eside = RHSIDE;
        *mm = 0;
/*
        Test the pattern. If only wildcard powers -> SUBONCE
*/
        {
```

```
            WORD *mmm = m + *m, *m1 = m+1, jm, noveto = 0;
            while ( m1 < mmm ) {
                if ( *m1 == SYMBOL ) {
                    for ( jm = 2; jm < m1[1]; jm+=2 ) {
                        if ( m1[jm+1] < MAXPOWER && m1[jm+1] > -MAXPOWER ) break;
                    }
                    if ( jm < m1[1] ) { noveto = 1; break; }
                }
                else if ( *m1 == DOTPRODUCT ) {
                    for ( jm = 2; jm < m1[1]; jm+=3 ) {
                        if ( m1[jm+2] < MAXPOWER && m1[jm+2] > -MAXPOWER ) break;
                    }
                    if ( jm < m1[1] ) { noveto = 1; break; }
                }
                else { noveto = 1; break; }
                m1 += m1[1];
            }
            if ( noveto == 0 ) {
                ll2 = ll2 & ~SUBMASK;
                ll2 |= SUBONCE;
            }
        }
        AT.WorkPointer = ww = StartWork;
        *AN.RepPoint = oldRepPoint;
    }
/*
        #] Expand dollars :

    In case of id,all we have to check at this point that there are only
    functions in the pattern.
*/
    if ( ( ll2 & SUBMASK ) == SUBALL ) {
        WORD *t = AN.patternbuffer+IDHEAD, *tt;
        WORD *tstop, *ttstop, ii;
        t += t[1]; tstop = t + *t; t++;
        while ( t < tstop ) {
            if ( *t < FUNCTION ) break;
            t += t[1];
        }
        if ( t < tstop ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Error: id,all can only be used with (products of) functions and/or tensors.");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        OldTermBuffer = AN.termbuffer;
        AN.termbuffer = TermMalloc("id,all");
/*
        Now make sure that only regular functions and tensors can take part.
*/
        tt = term; ttstop = tt+*tt; ttstop -= ABS(ttstop[-1]); tt++;
        t = AN.termbuffer+1;
        while ( tt < ttstop ) {
            if ( *tt >= FUNCTION && *tt != AR.PolyFun && *tt != AR.PolyFunInv ) {
                ii = tt[1]; NCOPY(t,tt,ii);
            }
            else tt += tt[1];
        }
        *t++ = 1; *t++ = 1; *t++ = 3; AN.termbuffer[0] = t-AN.termbuffer;
    }
/*
    To be puristic, we need to check that all wildcards in the prototype
    are actually present. If the LHS contained a replace_ this may not be
    the case.
*/
    ClearWild(BHEAD0);
    while ( w < AN.WildStop ) {
        if ( *w == LOADDOLLAR ) numdollars++;
        w += w[1];
    }
    AN.RepFunNum = 0;
    /* rep = */ AN.RepFunList = AT.WorkPointer;
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer/2);
    if ( AT.WorkPointer >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AN.DisOrderFlag = ll2 & SUBDISORDER;
    AN.nogroundlevel = 0;
    switch ( ll2 & SUBMASK ) {
        case SUBONLY :
            /* Must be an exact match */
            AN.UseFindOnly = 1; AN.ForFindOnly = 0;
            if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind ||
                FindOnly(BHEAD term,m) ) ) {
```

```
                power = 1;
                if ( msign ) term[term[0]-1] = -term[term[0]-1];
            }
            else power = 0;
            break;
        case SUBMANY :
            AN.UseFindOnly = -1;
            if ( ( power = FindRest(BHEAD term,m) ) > 0 ) {
                if ( ( power = FindOnce(BHEAD term,m) ) > 0 ) {
                    AN.UseFindOnly = 0;
                    do {
                        if ( msign ) term[term[0]-1] = -term[term[0]-1];
                        Substitute(BHEAD term,m,1);
                        if ( numdollars ) {
                            WildDollars(BHEAD (WORD *)0);
                            numdollars = 0;
                        }
                        if ( ww < term+term[0] ) ww = term+term[0];
                        ClearWild(BHEAD0);
                        AT.WorkPointer = ww;
/*                      if ( rep < ww ) {*/
                            AN.RepFunNum = 0;
                            /* rep = */ AN.RepFunList = ww;
                            AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer/2);
                            if ( AT.WorkPointer >= AT.WorkTop ) {
                                MLOCK(ErrorMessageLock);
                                MesWork();
                                MUNLOCK(ErrorMessageLock);
                                return(-1);
                            }
/*
                        }
                        else {
                            AN.RepFunList = rep;
                            AN.RepFunNum = 0;
                        }
*/
                        AN.nogroundlevel = 0;
                    } while ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind ||
                            FindOnce(BHEAD term,m) ) );
                    match = 1;
                }
                else if ( power < 0 ) {
                    do {
                        if ( msign ) term[term[0]-1] = -term[term[0]-1];
                        Substitute(BHEAD term,m,1);
                        if ( numdollars ) {
                            WildDollars(BHEAD (WORD *)0);
                            numdollars = 0;
                        }
                        if ( ww < term+term[0] ) ww = term+term[0];
                        ClearWild(BHEAD0);
                        AT.WorkPointer = ww;
/*                      if ( rep < ww ) { */
                            AN.RepFunNum = 0;
                            /* rep = */ AN.RepFunList = ww;
                            AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer/2);
                            if ( AT.WorkPointer >= AT.WorkTop ) {
                                MLOCK(ErrorMessageLock);
                                MesWork();
                                MUNLOCK(ErrorMessageLock);
                                return(-1);
                            }
/*
                        }
                        else {
                            AN.RepFunList = rep;
                            AN.RepFunNum = 0;
                        }
*/
                    } while ( FindRest(BHEAD term,m) );
                    match = 1;
                }
            }
            else if ( power < 0 ) {
                if ( FindOnce(BHEAD term,m) ) {
                    do {
                        if ( msign ) term[term[0]-1] = -term[term[0]-1];
                        Substitute(BHEAD term,m,1);
                        if ( numdollars ) {
                            WildDollars(BHEAD (WORD *)0);
                            numdollars = 0;
                        }
                        if ( ww < term+term[0] ) ww = term+term[0];
                        ClearWild(BHEAD0);
                        AT.WorkPointer = ww;
/*                      if ( rep < ww ) { */
```

```
                            AN.RepFunNum = 0;
                            /* rep = */ AN.RepFunList = ww;
                            AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer/2);
                            if ( AT.WorkPointer >= AT.WorkTop ) {
                                MLOCK(ErrorMessageLock);
                                MesWork();
                                MUNLOCK(ErrorMessageLock);
                                return(-1);
                            }
/*
                        }
                        else {
                            AN.RepFunList = rep;
                            AN.RepFunNum = 0;
                        }
*/
                    } while ( FindOnce(BHEAD term,m) );
                    match = 1;
                }
            }
            if ( match ) {
                if ( ( ll2 & SUBAFTER ) != 0 ) *level = AC.Labels[ll[3]];
            }
            else {
                if ( ( ll2 & SUBAFTERNOT ) != 0 ) *level = AC.Labels[ll[3]];
            }
            goto nextlevel;
        case SUBONCE :
            AN.UseFindOnly = 0;
            if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind || FindOnce(BHEAD term,m) ) ) {
                power = 1;
                if ( msign ) term[term[0]-1] = -term[term[0]-1];
            }
            else power = 0;
            break;
        case SUBMULTI :
            power = FindMulti(BHEAD term,m);
            if ( ( power & 1 ) != 0 && msign ) term[term[0]-1] = -term[term[0]-1];
            break;
        case SUBVECTOR :
            while ( ( power = FindAll(BHEAD term,m,*level,(WORD *)0) ) != 0 ) {
                if ( ( power & 1 ) != 0 && msign ) term[term[0]-1] = -term[term[0]-1];
                match = 1;
            }
            break;
        case SUBSELECT :
            llf = ll + IDHEAD;  llf += llf[1];  llf += *llf;
            AN.UseFindOnly = 1; AN.ForFindOnly = llf;
            if ( FindRest(BHEAD term,m) && ( AN.UsedOtherFind || FindOnly(BHEAD term,m) ) ) {
                if ( msign ) term[term[0]-1] = -term[term[0]-1];
/*
                The following code needs to be hacked a bit to allow for
                all types of sets and for occurrence anywhere in the term
                The code at the end of FindOnly is a bit mysterious.
*/
                if ( llf[1] > 2 ) {
                    WORD *t1, *t2;
                    if ( *term > AN.sizeselecttermundo ) {
                        if ( AN.selecttermundo ) M_free(AN.selecttermundo,"AN.selecttermundo");
                        AN.sizeselecttermundo = *term +10;
                        AN.selecttermundo = (WORD *)Malloc1(
                            AN.sizeselecttermundo*sizeof(WORD),"AN.selecttermundo");
                    }
                    t1 = term; t2 = AN.selecttermundo; i = *term;
                    NCOPY(t2,t1,i);
                }
                power = 1;
                Substitute(BHEAD term,m,power);
                if ( llf[1] > 2 ) {
                    if ( TestSelect(term,llf) ) {
                        WORD *t1, *t2;
                        power = 0;
                        t1 = term; t2 = AN.selecttermundo; i = *t2;
                        NCOPY(t1,t2,i);
#if IDHEAD > 3
                        if ( ( ll2 & SUBAFTERNOT ) != 0 ) {
                            *level = AC.Labels[ll[3]];
                        }
#endif
                        goto nextlevel;
                    }
                }
                if ( numdollars ) {
                    WildDollars(BHEAD (WORD *)0);
                    numdollars = 0;
                }
                match = 1;
```

```
                if ( ( ll2 & SUBAFTER ) != 0 ) {
                    *level = AC.Labels[ll[3]];
                }
            }
            else {
                if ( ( ll2 & SUBAFTERNOT ) != 0 ) {
                    *level = AC.Labels[ll[3]];
                }
                power = 0;
            }
            goto nextlevel;
        case SUBALL:
            AN.UseFindOnly = 0;
            CC = cbuf+AT.allbufnum;
            oldallnumrhs = CC->numrhs;
            t = AddRHS(AT.allbufnum,1);
            *t = 0;
            AT.idallflag = 1;
            AT.idallmaxnum = ll[5];
            AT.idallnum = 0;
            if ( FindRest(BHEAD AN.termbuffer,m) || AT.idallflag > 1 ) {
                WORD *t, *tstop, *tt, first = 1, ii;
                power = 1;
                *CC->Pointer++ = 0;
                if ( msign ) term[term[0]-1] = -term[term[0]-1];
/*
                If we come here the matches are all already in the
                compiler buffer. All we need to do is take out all
                functions and replace them by a SUBEXPRESSION that
                points to this buffer.
                Note: the PolyFun/PolyRatFun should be excluded from this.
                This works because each match writes incrementally to
                the buffer using the routine SubsInAll.

                The call to WildDollars should be made in Generator.....
*/
                t = term; tstop = t + *t; ii = ABS(tstop[-1]); tstop -= ii;
                tt = AT.WorkPointer+1;
                t++;
                while ( t < tstop ) {
                    if ( *t >= FUNCTION && *t != AR.PolyFun && *t != AR.PolyFunInv ) {
                        if ( first ) { /* SUBEXPRESSION */
                            *tt++ = SUBEXPRESSION;
                            *tt++ = SUBEXPSIZE;
                            *tt++ = CC->numrhs;
                            *tt++ = 1;
                            *tt++ = AT.allbufnum;
                            FILLSUB(tt)
                            first = 0;
                        }
                        t += t[1];
                    }
                    else {
                        i = t[1]; NCOPY(tt,t,i);
                    }
                }
                if ( ( ll[4] & NORMALIZEFLAG ) != 0 ) {
/*
                    In case of the normalization option, we have to divide
                    by AT.idallnum;
*/
                    WORD na = t[ii-1];
                    na = REDLENG(na);
                    for ( i = 0; i < ii; i++ ) tt[i] = t[i];
                    Divvy(BHEAD (UWORD *)tt,&na,(UWORD *)(&(AT.idallnum)),1);
                    na = INCLENG(na);
                    ii = ABS(na);
                    tt[ii-1] = na;
                    tt += ii;
                }
                else {
                    NCOPY(tt,t,ii);
                }
                ii = tt-AT.WorkPointer;
                *(AT.WorkPointer) = ii;
                tt = AT.WorkPointer; t = term;
                NCOPY(t,tt,ii);

                if ( ( ll2 & SUBAFTER ) != 0 ) { /* ifmatch -> */
                    *level = AC.Labels[ll[3]];
                }
                TermFree(AN.termbuffer,"id,all");
                AN.termbuffer = OldTermBuffer;
                AT.WorkPointer = AN.RepFunList;
                AT.idallflag = 0;
                CC->Pointer[0] = 0;
                TransferBuffer(AT.aebufnum,AT.ebufnum,AT.allbufnum);
```

```
                return(1);
            }
            AT.idallflag = 0;
            power = 0;
            CC->numrhs = oldallnumrhs;
            TermFree(AN.termbuffer,"id,all");
            AN.termbuffer = OldTermBuffer;
            break;
        default :
            break;
    }
    if ( power ) {
        Substitute(BHEAD term,m,power);
        if ( numdollars ) {
            WildDollars(BHEAD (WORD *)0);
            numdollars = 0;
        }
        match = 1;
        if ( ( ll2 & SUBAFTER ) != 0 ) { /* ifmatch -> */
            *level = AC.Labels[ll[3]];
        }
    }
    else {
        AT.WorkPointer = AN.RepFunList;
        if ( ( ll2 & SUBAFTERNOT ) != 0 ) { /* ifnomatch -> */
            *level = AC.Labels[ll[3]];
        }
    }
nextlevel:;
    } while ( (*level)++ < AR.Cnumlhs && C->lhs[*level][0] == TYPEIDOLD );
    (*level)--;
    AT.WorkPointer = AN.RepFunList;
    return(match);
}

/*
        #] TestMatch :
        #[ Substitute :         void Substitute(term,pattern,power)
*/

void Substitute(PHEAD WORD *term, WORD *pattern, WORD power)
{
    GETBIDENTITY
    WORD *TemTerm;
    WORD *t, *m;
    WORD *tstop, *mstop;
    WORD *xstop, *ystop;
    WORD nt, *fill, nq, mt;
    WORD *q, *subterm, *tcoef, oldval1 = 0, newval3, i = 0;
    WORD PutExpr = 0, sign = 0;
    TemTerm = AT.WorkPointer;
    if ( ( (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer*2) ) > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    m = pattern;
    mstop = m + *m;
    m++;
    t = term;
    t += *term - 1;
    tcoef = t;
    tstop = t - ABS(*t) + 1;
    t = term;
    t++;
    fill = TemTerm;
    fill++;
    if ( m < mstop ) { do {
/*
            #[ SYMBOLS :
*/
        if ( *m == SYMBOL ) {
            ystop = m + m[1];
            m += 2;
            while ( *t != SYMBOL && t < tstop ) {
                nq = t[1];
                NCOPY(fill,t,nq);
            }
            if ( t >= tstop ) goto SubCoef;
            *fill++ = SYMBOL;
            fill++;
            subterm = fill;
            xstop = t + t[1];
            t += 2;
            do {
                if ( *m == *t && t < xstop ) {
```

```
                    nt = t[1];
                    mt = m[1];
                    if ( mt >= 2*MAXPOWER ) {
                        if ( CheckWild(BHEAD mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                            nt -= AN.oldvalue;
                            goto SubsL1;
                        }
                    }
                    else if ( mt <= -2*MAXPOWER ) {
                        if ( CheckWild(BHEAD -mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                            nt += AN.oldvalue;
                            goto SubsL1;
                        }
                    }
                    else {
                        nt -= mt * power;
SubsL1:                 if ( nt ) {
                            *fill++ = *t;
                            *fill++ = nt;
                        }
                    }
                    m += 2; t+= 2;
                }
                else if ( *m >= 2*MAXPOWER ) {
                    while ( t < xstop ) { *fill++ = *t++; *fill++ = *t++; }
                    nq = WORDDIF(fill,subterm);
                    fill = subterm;
                    while ( nq > 0 ) {
                        if ( !CheckWild(BHEAD *m-2*MAXPOWER,SYMTOSYM,*fill,&newval3) ) {
                            mt = m[1];
                            if ( mt >= 2*MAXPOWER ) {
                                if ( CheckWild(BHEAD mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                                    if ( fill[1] -= AN.oldvalue ) goto SubsL2;
                                }
                            }
                            else if ( mt <= -2*MAXPOWER ) {
                                if ( CheckWild(BHEAD -mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                                    if ( fill[1] += AN.oldvalue ) goto SubsL2;
                                }
                            }
                            else {
                                if ( fill[1] -= mt * power ) {
SubsL2:                             fill += nq;
                                    nq = 0;
                                }
                            }
                            break;
                        }
                        nq -= 2;
                        fill += 2;
                    }
                    if ( nq ) {
                        nq -= 2;
                        q = fill + 2;
                        while ( --nq >= 0 ) *fill++ = *q++;
                    }
                    m += 2;
                }
                else if ( *m < *t || t >= xstop ) { m += 2; }
                else { *fill++ = *t++; *fill++ = *t++; }
            } while ( m < ystop );
            while ( t < xstop ) *fill++ = *t++;
            nq = WORDDIF(fill,subterm);
            if ( nq > 0 ) {
                nq += 2;
                subterm[-1] = nq;
            }
            else { fill = subterm; fill -= 2; }
        }
/*
            #] SYMBOLS :
            #[ DOTPRODUCTS :
*/
        else if ( *m == DOTPRODUCT ) {
            ystop = m + m[1];
            m += 2;
            while ( *t > DOTPRODUCT && t < tstop ) {
                nq = t[1];
                NCOPY(fill,t,nq);
            }
            if ( t >= tstop ) goto SubCoef;
            if ( *t != DOTPRODUCT ) {
                m = ystop;
                goto EndLoop;
            }
            *fill++ = DOTPRODUCT;
            fill++;
```

```
            subterm = fill;
            xstop = t + t[1];
            t += 2;
            do {
                if ( *m == *t && m[1] == t[1] && t < xstop ) {
                    nt = t[2];
                    mt = m[2];
                    if ( mt >= 2*MAXPOWER ) {
                        if ( CheckWild(BHEAD mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                            nt -= AN.oldvalue;
                            goto SubsL3;
                        }
                    }
                    else if ( mt <= -2*MAXPOWER ) {
                        if ( CheckWild(BHEAD -mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                            nt += AN.oldvalue;
                            goto SubsL3;
                        }
                    }
                    else {
                        nt -= mt * power;
SubsL3:                 if ( nt ) {
                            *fill++ = *t++;
                            *fill++ = *t;
                            *fill++ = nt;
                            t += 2;
                        }
                        else t += 3;
                    }
                    m += 3;
                }
                else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
                    while ( t < xstop ) {
                        *fill++ = *t++; *fill++ = *t++; *fill++ = *t++;
                    }
                    oldval1 = 1;
                    goto SubsL4;
                }
                else if ( m[1] >= (AM.OffsetVector+WILDOFFSET) ) {
                    while ( *m >= *t && t < xstop ) {
                        *fill++ = *t++; *fill++ = *t++; *fill++ = *t++;
                    }
                    oldval1 = 0;
SubsL4:             nq = WORDDIF(fill,subterm);
                    fill = subterm;
                    while ( nq > 0 ) {
                        if ( ( oldval1 && ( (
                           !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*fill,&newval3)
                        && !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,fill[1],&newval3)
                        ) || (
                           !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,*fill,&newval3)
                        && !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,fill[1],&newval3)
                        ) ) ) || ( !oldval1 && ( (
                           *m == *fill
                        && !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,fill[1],&newval3)
                        ) || (
                           !CheckWild(BHEAD m[1]-WILDOFFSET,VECTOVEC,*fill,&newval3)
                        && *m == fill[1] ) ) ) ) {
                            mt = m[2];
                            if ( mt >= 2*MAXPOWER ) {
                                if ( CheckWild(BHEAD mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                                    if ( fill[2] -= AN.oldvalue )
                                            goto SubsL5;
                                }
                            }
                            else if ( mt <= -2*MAXPOWER ) {
                                if ( CheckWild(BHEAD -mt-2*MAXPOWER,SYMTONUM,-MAXPOWER,&newval3) ) {
                                    if ( fill[2] += AN.oldvalue )
                                            goto SubsL5;
                                }
                            }
                            else {
                                if ( fill[2] -= mt * power ) {
SubsL5:                             fill += nq;
                                    nq = 0;
                                }
                            }
                            m += 3;
                            break;
                        }
                        fill += 3; nq -= 3;
                    }
                    if ( nq ) {
                        nq -= 3;
                        q = fill + 3;
                        while ( --nq >= 0 ) *fill++ = *q++;
                    }
```

```
                }
                else if ( t >= xstop || *m < *t || ( *m == *t && m[1] < t[1] ) )
                    { m += 3; }
                else {
                    *fill++ = *t++; *fill++ = *t++; *fill++ = *t++;
                }
            } while ( m < ystop );
            while ( t < xstop ) *fill++ = *t++;
            nq = WORDDIF(fill,subterm);
            if ( nq > 0 ) {
                nq += 2;
                subterm[-1] = nq;
            }
            else { fill = subterm; fill -= 2; }
        }
/*
            #] DOTPRODUCTS :
            #[ FUNCTIONS :
*/
        else if ( *m >= FUNCTION ) {
            while ( *t >= FUNCTION || *t == SUBEXPRESSION ) {
                nt = WORDDIF(t,term);
                for ( mt = 0; mt < AN.RepFunNum; mt += 2 ) {
                    if ( nt == AN.RepFunList[mt] ) break;
                }
                if ( mt >= AN.RepFunNum ) {
                    nq = t[1];
                    NCOPY(fill,t,nq);
                }
                else {
                    WORD *oldt = 0;
                    if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
                        oldt = t;
                        if ( ( i = AN.RepFunList[mt+1] ) > 0 ) {
                            *fill++ = GAMMA;
                            *fill++ = i + FUNHEAD+1;
                            FILLFUN(fill)
                            nq = i + 1;
                            t += FUNHEAD;
                            NCOPY(fill,t,nq);
                        }
                        t = oldt;
                    }
                    else if ( ( *t == LEVICIVITA ) || ( *t >= FUNCTION
                    && (functions[*t-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
                                ) sign += AN.RepFunList[mt+1];
                    else if ( *m >= FUNCTION+WILDOFFSET
                    && (functions[*m-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER) == ANTISYMMETRIC
                                ) sign += AN.RepFunList[mt+1];
                    if ( !PutExpr ) {
                        xstop = t + t[1];
                        t = AN.FullProto;
                        nq = t[1];
                        t[3] = power;
                        NCOPY(fill,t,nq);
                        t = xstop;
                        PutExpr = 1;
                    }
                    else t += t[1];
                    if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
                        i = oldt[1] - m[1] - i;
                        if ( i > 0 ) {
                            *fill++ = GAMMA;
                            *fill++ = i + FUNHEAD+1;
                            FILLFUN(fill)
                            *fill++ = oldt[FUNHEAD];
                            t = t - i;
                            NCOPY(fill,t,i);
                        }
                    }
                    break;
                }
            }
            m += m[1];
        }
/*
            #] FUNCTIONS :
            #[ VECTORS :
*/
        else if ( *m == VECTOR ) {
            while ( *t > VECTOR ) {
                nq = t[1];
                NCOPY(fill,t,nq);
            }
            xstop = t + t[1];
            ystop = m + m[1];
            t += 2;
```

```
            m += 2;
            *fill++ = VECTOR;
            fill++;
            subterm = fill;
            do {
                if ( *m == *t && m[1] == t[1] ) {
                    m += 2; t += 2;
                }
                else if ( *m >= (AM.OffsetVector+WILDOFFSET) ) {
                    while ( t < xstop ) *fill++ = *t++;
                    nq = WORDDIF(fill,subterm);
                    fill = subterm;
                    if ( m[1] < (AM.OffsetIndex+WILDOFFSET) ) {
                        do {
                            if ( m[1] == fill[1] &&
                            !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*fill,&newval3) )
                                break;
                            fill += 2;
                            nq -= 2;
                        } while ( nq > 0 );
                    }
                    else {      /* Double wildcard */
                        do {
                            if ( !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3)
                            && !CheckWild(BHEAD *m-WILDOFFSET,VECTOVEC,*fill,&newval3) )
                                break;
                            if ( *fill == oldval1 && fill[1] == AN.oldvalue ) break;
                            fill += 2;
                            nq -= 2;
                        } while ( nq > 0 );
                    }
                    nq -= 2;
                    q = fill + 2;
                    if ( nq > 0 ) { NCOPY(fill,q,nq); }
                    m += 2;
                }
                else if ( *m <= *t &&
                m[1] >= (AM.OffsetIndex + WILDOFFSET) ) {
                    while ( *m == *t && t < xstop )
                        { *fill++ = *t++; *fill++ = *t++; }
                    nq = WORDDIF(fill,subterm);
                    fill = subterm;
                    do {
                        if ( *m == *fill &&
                        !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3) )
                            break;
                        nq -= 2;
                        fill += 2;
                    } while ( nq > 0 );
                    nq -= 2;
                    q = fill + 2;
                    if ( nq > 0 ) { NCOPY(fill,q,nq); }
                    m += 2;
                }
                else { *fill++ = *t++; *fill++ = *t++; }
            } while ( m < ystop );
            while ( t < xstop ) *fill++ = *t++;
            nq = WORDDIF(fill,subterm);
            if ( nq > 0 ) {
                nq += 2;
                subterm[-1] = nq;
            }
            else { fill = subterm; fill -= 2; }
        }
/*
            #] VECTORS :
            #[ INDICES :

            Currently without wildcards
*/
        else if ( *m == INDEX ) {
            while ( *t > INDEX ) {
                nq = t[1];
                NCOPY(fill,t,nq);
            }
            xstop = t + t[1];
            ystop = m + m[1];
            t += 2;
            m += 2;
            *fill++ = INDEX;
            fill++;
            subterm = fill;
            do {
                if ( *m == *t ) {
                    m += 1; t += 1;
                }
                else if ( *m >= (AM.OffsetIndex+WILDOFFSET) ) {
```

```
                    while ( t < xstop ) *fill++ = *t++;
                    nq = WORDDIF(fill, subterm);
                    fill = subterm;
                    do {
                        if ( !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*fill,&newval3) ) {
                            break;
                        }
                        fill += 1;
                        nq -= 1;
                    } while ( nq > 0 );
                    nq -= 1;
                    if ( nq > 0 ) {
                        q = fill + 1;
                        NCOPY(fill,q,nq);
                    }
                    m += 1;
                }
                else {
                    *fill++ = *t++;
                }
            } while ( m < ystop );
            while ( t < xstop ) *fill++ = *t++;
            nq = WORDDIF(fill,subterm);
            if ( nq > 0 ) {
                nq += 2;
                subterm[-1] = nq;
            }
            else { fill = subterm; fill -= 2; }
        }
/*
            #] INDICES :
            #[ DELTAS :
*/
        else if ( *m == DELTA ) {
            while ( *t > DELTA ) {
                nq = t[1];
                NCOPY(fill,t,nq);
            }
            xstop = t + t[1];
            ystop = m + m[1];
            t += 2;
            m += 2;
            *fill++ = DELTA;
            fill++;
            subterm = fill;
            do {
                if ( *t == *m && t[1] == m[1] ) { m += 2; t += 2; }
                else if ( *m >= (AM.OffsetIndex+WILDOFFSET) ) { /* Two dummies */
                    while ( t < xstop ) *fill++ = *t++;
/*                  fill = subterm; */
                    oldval1 = 1;
                    goto SubsL6;
                }
                else if ( m[1] >= (AM.OffsetIndex+WILDOFFSET) ) {
                    while ( (*m == *t || *m == t[1] ) && ( t < xstop ) ) {
                        *fill++ = *t++; *fill++ = *t++;
                    }
                    oldval1 = 0;
SubsL6:             nq = WORDDIF(fill,subterm);
                    fill = subterm;
                    do {
                        if ( ( oldval1 && ( (
                           !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,*fill,&newval3)
                        && !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3)
                        ) || (
                           !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,*fill,&newval3)
                        && !CheckWild(BHEAD *m-WILDOFFSET,INDTOIND,fill[1],&newval3)
                        ) ) ) || ( !oldval1 && ( (
                           *m == *fill
                        && !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,fill[1],&newval3)
                        ) || (
                           *m == fill[1]
                        && !CheckWild(BHEAD m[1]-WILDOFFSET,INDTOIND,*fill,&newval3)
                        ) ) ) ) break;
                        fill += 2;
                        nq -= 2;
                    } while ( nq > 0 );
                    nq -= 2;
                    if ( nq > 0 ) {
                        q = fill + 2;
                        NCOPY(fill,q,nq);
                    }
                    m += 2;
                }
                else {
                    *fill++ = *t++; *fill++ = *t++;
                }
```

```
            } while ( m < ystop );
            while ( t < xstop ) *fill++ = *t++;
            nq = WORDDIF(fill,subterm);
            if ( nq > 0 ) {
                nq += 2;
                subterm[-1] = nq;
            }
            else { fill = subterm; fill -= 2; }
        }
/*
            #] DELTAS :
*/
EndLoop:;
    } while ( m < mstop ); }
    while ( t < tstop ) *fill++ = *t++;
SubCoef:
    if ( !PutExpr ) {
        t = AN.FullProto;
        nq = t[1];
        t[3] = power;
        NCOPY(fill,t,nq);
    }
    t = tcoef;
    nq = ABS(*t);
    t = tstop;
    NCOPY(fill,t,nq);
    nq = WORDDIF(fill,TemTerm);
    fill = term;
    t = TemTerm;
    *fill++ = nq--;
    t++;
    NCOPY(fill,t,nq);
    if ( sign ) {
        if ( ( sign & 1 ) != 0 ) fill[-1] = -fill[-1];
    }
    if ( AT.WorkPointer < fill ) AT.WorkPointer = fill;
    AN.RepFunNum = 0;
}

/*
        #] Substitute :
        #[ FindSpecial :        WORD FindSpecial(term)

    Routine to detect simplifications regarding the special functions
    exponent, denominator.


WORD FindSpecial(WORD *term)
{
    WORD *t;
    WORD *tstop;
    t = term; t += *t - 1; tstop = t - ABS(*t) + 1; t = term;
    t++;
    if ( t < tstop ) { do {
        if ( *t == EXPONENT ) {
            Exponents can become simpler when:
            a: the exponent of an expression becomes an integer.
            b: The expression becomes zero.
        }
        else if ( *t == DENOMINATOR ) {
            Denominators can become simpler when:
            a: The denominator is a single term without functions.
            b: An overall coefficient can be removed.
            c: An overall object can be removed.
            The task is here to bring the denominator in an unique form.
        }
        t += *t;
    } while ( t < tstop ); }
    return(0);
}

        #] FindSpecial :
        #[ FindAll :            WORD FindAll(term,pattern,level,par)
*/

WORD FindAll(PHEAD WORD *term, WORD *pattern, WORD level, WORD *par)
{
    GETBIDENTITY
    WORD *t, *m, *r, *mm, rnum;
    WORD *tstop, *mstop, *TwoProto, *vwhere = 0, oldv, oldvv, vv, level2;
    WORD v, nq, OffNum = AM.OffsetVector + WILDOFFSET, i, ii = 0, jj;
    WORD fromindex, *intens, notflag1 = 0, notflag2 = 0;
    CBUF *C;
    C = cbuf+AM.rbufnum;
    v = pattern[3];      /* The vector to be found */
    m = t = term;
    m += *m;
```

```
    m -= ABS(m[-1]);
    t++;
    if ( t < m ) do {
        tstop = t + t[1];
        fromindex = 2;
/*
            #[ VECTOR :
*/
        if ( *t == VECTOR ) {
            r = t;
            r += 2;
InVect:
            while ( r < tstop ) {
                oldv = *r;
                if ( v >= OffNum ) {
                    vwhere = AN.FullProto + 3 + SUBEXPSIZE;
                    if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
                        WORD *afirst, *alast, j;
                        j = vwhere[3];
                        if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
                        else { notflag1 = 0; }
                        afirst = SetElements + Sets[j].first;
                        alast  = SetElements + Sets[j].last;
                        ii = 1;
                        if ( notflag1 == 0 ) {
                          do {
                            if ( *afirst == *r ) {
                                if ( vwhere[1] == SETTONUM ) {
                                    AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
                                    AN.FullProto[11+SUBEXPSIZE] = ii;
                                }
                                else if ( vwhere[4] >= 0 ) {
                                    oldv = *(afirst - Sets[j].first
                                    + Sets[vwhere[4]].first);
                                }
                                goto DoVect;
                            }
                            ii++;
                          } while ( ++afirst < alast );
                        }
                        else {
                          do {
                            if ( *afirst == *r ) break;
                          } while ( ++afirst < alast );
                          if ( afirst >= alast ) goto DoVect;
                        }
                    }
                    else goto DoVect;
                }
                else if ( v == *r ) {
DoVect:             m = AT.WorkPointer;
                    tstop = t;
                    t = term;
                    mstop = t + *t;
                    do { *m++ = *t++; } while ( t < tstop );
                    vwhere = m;
                    t = AN.FullProto;
                    nq = t[1];
                    t[3] = 1;
                    NCOPY(m,t,nq);
                    t = tstop;
                    if ( fromindex == 1 ) m[-1] = FUNNYVEC;
                    else m[-1] = r[1];       /* The index is always here! */
                    if ( v >= OffNum ) vwhere[3+SUBEXPSIZE] = oldv;
                    if ( vwhere[1] > 12+SUBEXPSIZE ) {
                        vwhere[11+SUBEXPSIZE] = ii;
                        vwhere[8+SUBEXPSIZE] = SYMTONUM;
                    }
                    if ( t[1] > fromindex+2 ) {
                        *m++ = *t++;
                        *m++ = *t++ - fromindex;
                        while ( t < r ) *m++ = *t++;
                        t += fromindex;
                    }
                    else t += t[1];
                    do { *m++ = *t++; } while ( t < mstop );
                    *AT.WorkPointer = nq = WORDDIF(m,AT.WorkPointer);
                    m = AT.WorkPointer;
                    t = term;
                    NCOPY(t,m,nq);
                    AT.WorkPointer = t;
                    return(1);
                }
                r += fromindex;
            }
        }
/*
```

```
            #] VECTOR :
            #[ DOTPRODUCT :
*/
        else if ( *t == DOTPRODUCT ) {
            r = t;
            r += 2;
            do {
                if ( ( i = r[2] ) < 0 ) goto NextDot;
                if ( *r == r[1] ) {     /* p.p */
                    oldv = *r;
                    if ( v == *r ) {    /* v.v */
TwoVec:
                        m = AT.WorkPointer;
                        tstop = t;
                        t = term;
                        mstop = t + *t;
                        do { *m++ = *t++; } while ( t < tstop );
                        do {
                            vwhere = m;
                            t = AN.FullProto;
                            nq = t[1];
                            t[3] = 2;
                            NCOPY(m,t,nq);
                            m[-1] = ++AR.CurDum;
                            if ( v >= OffNum ) vwhere[3+SUBEXPSIZE] = oldv;
                        } while ( --i > 0 );
CopRest:
                        t = tstop;
                        if ( t[1] > 5 ) {
                            *m++ = *t++;
                            *m++ = *t++ - 3;
                            while ( t < r ) *m++ = *t++;
                            t += 3;
                        }
                        else t += t[1];
                        do { *m++ = *t++; } while ( t < mstop );
                        *AT.WorkPointer = nq = WORDDIF(m,AT.WorkPointer);
                        m = AT.WorkPointer;
                        t = term;
                        NCOPY(t,m,nq);
                        AT.WorkPointer = t;
                        return(1);
                    }
                    else if ( v >= OffNum ) {   /* v?.v? */
                        vwhere = AN.FullProto + 3+SUBEXPSIZE;
                        if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
                            WORD *afirst, *alast, j;
                            j = vwhere[3];
                            if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
                            else { notflag1 = 0; }
                            afirst = SetElements + Sets[j].first;
                            alast  = SetElements + Sets[j].last;
                            ii = 1;
                            if ( notflag1 == 0 ) {
                              do {
                                if ( *afirst == *r ) {
                                    if ( vwhere[1] == SETTONUM ) {
                                        AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
                                        AN.FullProto[11+SUBEXPSIZE] = ii;
                                    }
                                    else if ( vwhere[4] >= 0 ) {
                                        oldv = *(afirst - Sets[j].first
                                        + Sets[vwhere[4]].first);
                                    }
                                    goto TwoVec;
                                }
                                ii++;
                              } while ( ++afirst < alast );
                            }
                            else {
                              do {
                                if ( *afirst == *r ) break;
                              } while ( ++afirst < alast );
                              if ( afirst >= alast ) goto TwoVec;
                            }
                        }
                        else goto TwoVec;
                    }
                }
                else {
                    if ( v == r[1] ) { r[1] = *r; *r = v; }
                    oldv = *r;
                    oldvv = r[1];
                    if ( v == *r ) {
                        if ( !par ) { while ( ++level <= AR.Cnumlhs
                        && C->lhs[level][0] == TYPEIDOLD ) {
                            m = C->lhs[level];
                            m += IDHEAD;
                            if ( m[-IDHEAD+2] == SUBVECTOR ) {
```

```
                            if ( ( vv = m[m[1]+3] ) == r[1] ) {
OnePV:                          TwoProto = AN.FullProto;
TwoPV:                          m = AT.WorkPointer;
                                tstop = t;
                                t = term;
                                mstop = t + *t;
                                do { *m++ = *t++; } while ( t < tstop );
                                do {
                                    t = AN.FullProto;
                                    vwhere = m + 3 +SUBEXPSIZE;
                                    nq = t[1];
                                    t[3] = 1;
                                    NCOPY(m,t,nq);
                                    m[-1] = ++AR.CurDum;
                                    if ( v >= OffNum ) *vwhere = oldv;
                                    if ( vwhere[-2-SUBEXPSIZE] > 12+SUBEXPSIZE ) {
                                        vwhere[8] = ii;
                                        vwhere[5] = SYMTONUM;
                                    }
                                    t = TwoProto;
                                    vwhere = m + 3+SUBEXPSIZE;
                                    mm = m;
                                    nq = t[1];
                                    t[3] = 1;
                                    NCOPY(m,t,nq);
/*
        The next two lines repair a bug. without them it takes twice
        the rhs of the first vector.
*/
                                    mm[2] = C->lhs[level][IDHEAD+2];
                                    mm[4] = C->lhs[level][IDHEAD+4];
                                    m[-1] = AR.CurDum;
                                    if ( vv >= OffNum ) *vwhere = oldvv;
                                } while ( --i > 0 );
                                goto CopRest;
                            }
                            else if ( vv > OffNum ) {
                                vwhere = AN.FullProto + 3+SUBEXPSIZE;
                                if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
                                    WORD *afirst, *alast, j;
                                    j = vwhere[3];
                                    if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
                                    else { notflag1 = 0; }
                                    afirst = SetElements + Sets[j].first;
                                    alast  = SetElements + Sets[j].last;
                                    if ( notflag1 == 0 ) {
                                      ii = 1;
                                      do {
                                        if ( *afirst == r[1] ) {
                                            if ( vwhere[1] == SETTONUM ) {
                                                AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
                                                AN.FullProto[11+SUBEXPSIZE] = ii;
                                            }
                                            else if ( vwhere[4] >= 0 ) {
                                                oldvv = *(afirst - Sets[j].first
                                                + Sets[vwhere[4]].first);
                                            }
                                            goto OnePV;
                                        }
                                        ii++;
                                      } while ( ++afirst < alast );
                                    }
                                    else {
                                      do {
                                        if ( *afirst == *r ) break;
                                      } while ( ++afirst < alast );
                                      if ( afirst >= alast ) goto OnePV;
                                    }
                                }
                                else goto OnePV;
                            }
                            }
                        }}
/*
            v.q with v matching and no match for the q, also
            not in following idold statements.
            Notice that a following q.p? cannot match.
*/
                        rnum = r[1];
OneOnly:                m = AT.WorkPointer;
                        tstop = t;
                        t = term;
                        mstop = t + *t;
                        do { *m++ = *t++; } while ( t < tstop );
                        vwhere = m;
                        t = AN.FullProto;
                        nq = t[1];
```

```
                t[3] = i;
                NCOPY(m,t,nq);
                m[-4] = INDTOIND;
                m[-1] = rnum;
                if ( v >= OffNum ) vwhere[3+SUBEXPSIZE] = oldv;
                goto CopRest;
            }
            else if ( v >= OffNum ) {
                vwhere = AN.FullProto + 3+SUBEXPSIZE;
                if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
                    WORD *afirst, *alast, *bfirst, *blast, j;
                    j = vwhere[3];
                    if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
                    else { notflag1 = 0; }
                    afirst = SetElements + Sets[j].first;
                    alast  = SetElements + Sets[j].last;
                    ii = 1;
                    if ( notflag1 == 0 ) {
                      do {
                        if ( *afirst == *r ) {
                            if ( vwhere[1] == SETTONUM ) {
                                AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
                                AN.FullProto[11+SUBEXPSIZE] = ii;
                            }
                            else if ( vwhere[4] >= 0 ) {
                                oldv = *(afirst - Sets[j].first
                                + Sets[vwhere[4]].first);
                            }
Hitlevel1:                  level2 = level;
                            do {
                                if ( !par ) m = C->lhs[level2];
                                else m = par;
                                m += IDHEAD;
                                if ( m[-IDHEAD+2] == SUBVECTOR ) {
                                if ( ( vv = m[m[1]+3] ) == r[1] )
                                    goto OnePV;
                                else if ( vv >= OffNum ) {
                                    if ( m[SUBEXPSIZE+4] != FROMSET &&
                                    m[SUBEXPSIZE+4] != SETTONUM ) goto OnePV;
                                    j = m[SUBEXPSIZE+6];
                                    if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag2 = 1; }
                                    else { notflag2 = 0; }
                                    bfirst = SetElements + Sets[j].first;
                                    blast  = SetElements + Sets[j].last;
                                    jj = 1;
                                    if ( notflag2 == 0 ) {
                                      do {
                                        if ( *bfirst == r[1] ) {
                                            if ( m[SUBEXPSIZE+4] == SETTONUM ) {
                                                m[SUBEXPSIZE+8] = SYMTONUM;
                                                m[SUBEXPSIZE+11] = jj;
                                            }
                                            else if ( m[SUBEXPSIZE+7] >= 0 ) {
                                                oldvv = *(bfirst - Sets[j].first
                                                + Sets[m[SUBEXPSIZE+7]].first);
                                            }
                                            goto OnePV;
                                        }
                                        jj++;
                                      } while ( ++bfirst < blast );
                                    }
                                    else {
                                      do {
                                        if ( *bfirst == r[1] ) break;
                                      } while ( ++bfirst < blast );
                                      if ( bfirst >= blast ) goto OnePV;
                                    }
                                }
                                    }
                            } while ( ++level2 < AR.Cnumlhs &&
                            C->lhs[level2][0] == TYPEIDOLD );
                            rnum = r[1];
                            goto OneOnly;
                        }
                        else if ( *afirst == r[1] ) {
                            if ( vwhere[1] == SETTONUM ) {
                                AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
                                AN.FullProto[11+SUBEXPSIZE] = ii;
                            }
                            else if ( vwhere[4] >= 0 ) {
                                oldv = *(afirst - Sets[j].first
                                + Sets[vwhere[4]].first);
                            }
Hitlevel2:                  level2 = level;
                            while ( ++level2 < AR.Cnumlhs &&
                            C->lhs[level2][0] == TYPEIDOLD ) {
                                if ( !par ) m = C->lhs[level2];
```

```
                                        else m = par;
                                        m += IDHEAD;
                                        if ( m[-IDHEAD+2] == SUBVECTOR ) {
                                        if ( ( vv = m[6] ) == *r )
                                            goto OnePV;
                                        else if ( vv >= OffNum ) {
                                            if ( m[SUBEXPSIZE+4] != FROMSET && m[SUBEXPSIZE+4]
                                            != SETTONUM ) {
                                                j = *r;
                                                *r = r[1];
                                                r[1] = j;
                                                goto OnePV;
                                            }
                                            j = m[SUBEXPSIZE+6];
                                            bfirst = SetElements + Sets[j].first;
                                            blast  = SetElements + Sets[j].last;
                                            jj = 1;
                                            do {
                                                if ( *bfirst == *r ) {
                                                    if ( m[SUBEXPSIZE+4] == SETTONUM ) {
                                                        m[SUBEXPSIZE+8] = SYMTONUM;
                                                        m[SUBEXPSIZE+11] = jj;
                                                    }
                                                    else if ( m[SUBEXPSIZE+7] >= 0 ) {
                                                        oldvv = *(bfirst - Sets[j].first
                                                        + Sets[m[SUBEXPSIZE+7]].first);
                                                    }
                                                    j = *r;
                                                    *r = r[1];
                                                    r[1] = j;
                                                    j = oldv; oldv = oldvv; oldvv = j;
                                                    goto OnePV;
                                                }
                                                jj++;
                                            } while ( ++bfirst < blast );
                                        }
                                            }
                                    }
                                    jj = *r; *r = r[1]; r[1] = jj;
                                    jj = oldv; oldv = oldvv; oldvv = j;
                                    rnum = r[1];
                                    goto OneOnly;
                                }
                                ii++;
                              } while ( ++afirst < alast );
                            }
                            else {
                              do {
                                if ( *afirst == *r ) break;
                              } while ( ++afirst < alast );
                              if ( afirst >= alast ) goto Hitlevel1;
                              do {
                                if ( *afirst == r[1] ) break;
                              } while ( ++afirst < alast );
                              if ( afirst >= alast ) goto Hitlevel2;
                            }
                        }
                        else { /* Matches twice */
                            vv = v;
                            TwoProto = AN.FullProto;
                            goto TwoPV;
                        }
                    }
                }
NextDot:            r += 3;
            } while ( r < tstop );
        }
/*
            #] DOTPRODUCT :
            #[ LEVICIVITA :
*/
        else if ( *t == LEVICIVITA ) {
            intens = 0;
            r = t;
            r += FUNHEAD;
OneVect:;
            while ( r < tstop ) {
                oldv = *r;
                if ( v >= OffNum && *r < -10 ) {
                    vwhere = AN.FullProto + 3+SUBEXPSIZE;
                    if ( vwhere[1] == FROMSET || vwhere[1] == SETTONUM ) {
                        WORD *afirst, *alast, j;
                        j = vwhere[3];
                        if ( j > WILDOFFSET ) { j -= 2*WILDOFFSET; notflag1 = 1; }
                        else { notflag1 = 0; }
                        afirst = SetElements + Sets[j].first;
                        alast  = SetElements + Sets[j].last;
```

```
                        ii = 1;
                        if ( notflag1 == 0 ) {
                          do {
                            if ( *afirst == *r ) {
                                if ( vwhere[1] == SETTONUM ) {
                                    AN.FullProto[8+SUBEXPSIZE] = SYMTONUM;
                                    AN.FullProto[11+SUBEXPSIZE] = ii;
                                }
                                else if ( vwhere[4] >= 0 ) {
                                    oldv = *(afirst - Sets[j].first
                                    + Sets[vwhere[4]].first);
                                }
                                goto DoVect;
                            }
                            ii++;
                          } while ( ++afirst < alast );
                        }
                        else {
                          do {
                            if ( *afirst == *r ) break;
                          } while ( ++afirst < alast );
                          if ( afirst >= alast ) goto DoVect;
                        }
                    }
                    else goto LeVect;
                }
                else if ( v == *r ) {
LeVect:             m = AT.WorkPointer;
                    mstop = term + *term;
                    t = term;
                    *r = ++AR.CurDum;
                    if ( intens ) *intens = DIRTYSYMFLAG;
                    do { *m++ = *t++; } while ( t < tstop );
                    t = AN.FullProto;
                    nq = t[1];
                    t[3] = 1;
                    if ( v >= OffNum ) *vwhere = oldv;
                    NCOPY(m,t,nq);
                    m[-1] = AR.CurDum;
                    t = tstop;
                    do { *m++ = *t++; } while ( t < mstop );
                    *AT.WorkPointer = nq = WORDDIF(m,AT.WorkPointer);
                    m = AT.WorkPointer;
                    t = term;
                    NCOPY(t,m,nq);
                    AT.WorkPointer = t;
                    return(1);
                }
                r++;
            }
        }
/*
        #] LEVICIVITA :
        #[ GAMMA :
*/
        else if ( *t == GAMMA ) {
            intens = 0;
            r = t;
            r += FUNHEAD+1;
            if ( r < tstop ) goto OneVect;
        }
/*
        #] GAMMA :
        #[ INDEX :
*/
        else if ( *t == INDEX ) {   /* The 'forgotten' part */
            r = t;
            r += 2;
            fromindex = 1;
            goto InVect;
        }
/*
        #] INDEX :
        #[ FUNCTION :
*/
        else if ( *t >= FUNCTION ) {
            if ( *t >= FUNCTION
             && functions[*t-FUNCTION].spec >= TENSORFUNCTION
             && t[1] > FUNHEAD ) {
/*
                Tensors are linear in their vectors!
*/
                r = t;
                r += FUNHEAD;
                intens = t+2;
                goto OneVect;
            }
```

```
        }
/*
            #] FUNCTION :
*/
        t += t[1];
    } while ( t < m );
    return(0);
}

/*
        #] FindAll :
        #[ TestSelect :

        Returns 1 if any of the objects in any of the sets in setp
        occur anywhere in the term
*/

int TestSelect(WORD *term, WORD *setp)
{
    WORD *tstop, *t, *s, *el, *elstop, *termstop, *tt, n, ns;
    GETSTOP(term,tstop);
    term += 1;
    while ( term < tstop ) {
    switch ( *term ) {
        case SYMBOL:
            n = term[1] - 2;
            t = term + 2;
            while ( n > 0 ) {
                ns = setp[1] - 2;
                s = setp + 2;
                while ( --ns >= 0 ) {
                    if ( Sets[*s].type != CSYMBOL ) { s++; continue; }
                    el = SetElements + Sets[*s].first;
                    elstop = SetElements + Sets[*s].last;
                    while ( el < elstop ) {
                        if ( *el++ == *t ) return(1);
                    }
                    s++;
                }
                n -= 2;
                t += 2;
            }
            break;
        case VECTOR:
            n = term[1] - 2;
            t = term + 2;
            while ( n > 0 ) {
                ns = setp[1] - 2;
                s = setp + 2;
                while ( --ns >= 0 ) {
                    if ( Sets[*s].type != CVECTOR ) { s++; continue; }
                    el = SetElements + Sets[*s].first;
                    elstop = SetElements + Sets[*s].last;
                    while ( el < elstop ) {
                        if ( *el++ == *t ) return(1);
                    }
                    s++;
                }
                t++;
                ns = setp[1] - 2;
                s = setp + 2;
                while ( --ns >= 0 ) {
                    if ( Sets[*s].type != CINDEX
                    && Sets[*s].type != CNUMBER ) { s++; continue; }
                    el = SetElements + Sets[*s].first;
                    elstop = SetElements + Sets[*s].last;
                    while ( el < elstop ) {
                        if ( *el++ == *t ) return(1);
                    }
                    s++;
                }
                n -= 2;
                t++;
            }
            break;
        case INDEX:
            n = term[1] - 2;
            t = term + 2;
            goto dotensor;
        case DOTPRODUCT:
            n = term[1] - 2;
            t = term + 2;
            while ( n > 0 ) {
                ns = setp[1] - 2;
                s = setp + 2;
                while ( --ns >= 0 ) {
                    if ( Sets[*s].type != CVECTOR ) { s++; continue; }
```

```
                    el = SetElements + Sets[*s].first;
                    elstop = SetElements + Sets[*s].last;
                    while ( el < elstop ) {
                        if ( *el++ == *t ) return(1);
                    }
                    s++;
                }
                t++;
                ns = setp[1] - 2;
                s = setp + 2;
                while ( --ns >= 0 ) {
                    if ( Sets[*s].type != CVECTOR ) { s++; continue; }
                    el = SetElements + Sets[*s].first;
                    elstop = SetElements + Sets[*s].last;
                    while ( el < elstop ) {
                        if ( *el++ == *t ) return(1);
                    }
                    s++;
                }
                n -= 3;
                t += 2;
            }
            break;
        case DELTA:
            n = term[1] - 2;
            t = term + 2;
            goto dotensor;
        default:
            if ( *term < FUNCTION ) break;
            ns = setp[1] - 2;
            s = setp + 2;
            while ( --ns >= 0 ) {
                if ( Sets[*s].type != CFUNCTION ) { s++; continue; }
                el = SetElements + Sets[*s].first;
                elstop = SetElements + Sets[*s].last;
                while ( el < elstop ) {
                    if ( *el++ == *term ) return(1);
                }
                s++;
            }
            if ( functions[*term-FUNCTION].spec > 0 ) {
                n = term[1] - FUNHEAD;
                t = term + FUNHEAD;
dotensor:
                while ( n > 0 ) {
                    ns = setp[1] - 2;
                    s = setp + 2;
                    while ( --ns >= 0 ) {
                        if ( *t < MINSPEC ) {
                            if ( Sets[*s].type != CVECTOR ) { s++; continue; }
                        }
                        else if ( *t >= 0 ) {
                            if ( Sets[*s].type != CINDEX
                            && Sets[*s].type != CNUMBER ) { s++; continue; }
                        }
                        else { s++; continue; }
                        el = SetElements + Sets[*s].first;
                        elstop = SetElements + Sets[*s].last;
                        while ( el < elstop ) {
                            if ( *el++ == *t ) return(1);
                        }
                        s++;
                    }
                    t++;
                    n--;
                }
            }
            else {
                termstop = term + term[1];
                tt = term + FUNHEAD;
                while ( tt < termstop ) {
                    if ( *tt < 0 ) {
                        if ( *tt == -SYMBOL ) {
                            ns = setp[1] - 2;
                            s = setp + 2;
                            while ( --ns >= 0 ) {
                                if ( Sets[*s].type != CSYMBOL ) { s++; continue; }
                                el = SetElements + Sets[*s].first;
                                elstop = SetElements + Sets[*s].last;
                                while ( el < elstop ) {
                                    if ( *el++ == tt[1] ) return(1);
                                }
                                s++;
                            }
                            tt += 2;
                        }
                        else if ( *tt == -VECTOR || *tt == -MINVECTOR ) {
```

```
                                ns = setp[1] - 2;
                                s = setp + 2;
                                while ( --ns >= 0 ) {
                                    if ( Sets[*s].type != CVECTOR ) { s++; continue; }
                                    el = SetElements + Sets[*s].first;
                                    elstop = SetElements + Sets[*s].last;
                                    while ( el < elstop ) {
                                        if ( *el++ == tt[1] ) return(1);
                                    }
                                    s++;
                                }
                                tt += 2;
                            }
                            else if ( *tt == -INDEX ) {
                                ns = setp[1] - 2;
                                s = setp + 2;
                                while ( --ns >= 0 ) {
                                    if ( Sets[*s].type != CINDEX
                                    && Sets[*s].type != CNUMBER ) { s++; continue; }
                                    el = SetElements + Sets[*s].first;
                                    elstop = SetElements + Sets[*s].last;
                                    while ( el < elstop ) {
                                        if ( *el++ == tt[1] ) return(1);
                                    }
                                    s++;
                                }
                                tt += 2;
                            }
                            else if ( *tt <= -FUNCTION ) {
                                ns = setp[1] - 2;
                                s = setp + 2;
                                while ( --ns >= 0 ) {
                                    if ( Sets[*s].type != CFUNCTION ) { s++; continue; }
                                    el = SetElements + Sets[*s].first;
                                    elstop = SetElements + Sets[*s].last;
                                    while ( el < elstop ) {
                                        if ( *el++ == -(*tt) ) return(1);
                                    }
                                    s++;
                                }
                                tt++;
                            }
                            else tt += 2;
                        }
                        else {
                            t = tt + ARGHEAD;
                            tt += *tt;
                            while ( t < tt ) {
                                if ( TestSelect(t,setp) ) return(1);
                                t += *t;
                            }
                        }
                    }
                }
                break;
        }
        term += term[1];
    }
    return(0);
}

/*
        #] TestSelect :
        #[ SubsInAll :          void SubsInAll()

        This routine takes a match in id,all and stores it away in
        the AT.allbufnum 'compiler' buffer, after taking out the pattern.
        The main problem here is that id,all usually has (lots of) wildcards
        and their assignments are on stack and the difficult ones are in
        AT.ebufnum. Popping the stack while looking for more matches would
        loose those. Hence we have to copy them into yet another compiler
        buffer: AT.aebufnum. Because this may involve many matches and
        because the original term has only a limited number of arguments,
        it will pay to look for already existing ones in this buffer.
        (to be done later).
*/

void SubsInAll(PHEAD0)
{
    GETBIDENTITY
    WORD *TemTerm;
    WORD *t, *m, *term;
    WORD *tstop, *mstop, *xstop;
    WORD nt, *fill, nq, mt;
    WORD *tcoef, i = 0;
    WORD PutExpr = 0, sign = 0;
/*
```

```
    We start with building the term in the WorkSpace.
    Afterwards we will transfer it to AT.allbufnum.
    We have to make sure there is room in the WorkSpace.
*/
    AT.idallflag = 2;
    TemTerm = AT.WorkPointer;
    if ( ( (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer*2) ) > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    m = AN.patternbuffer + IDHEAD; m += m[1];
    mstop = m + *m;
    m++;
    term = AN.termbuffer;
    tstop = term + *term; tcoef = tstop-1; tstop -= ABS(tstop[-1]);
    t = term;
    t++;
    fill = TemTerm;
    fill++;
    while ( m < mstop ) {
        while ( t < tstop ) {
            nt = WORDDIF(t,term);
            for ( mt = 0; mt < AN.RepFunNum; mt += 2 ) {
                if ( nt == AN.RepFunList[mt] ) break;
            }
            if ( mt >= AN.RepFunNum ) {
                nq = t[1];
                NCOPY(fill,t,nq);
            }
            else {
                WORD *oldt = 0;
                if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
                    oldt = t;
                    if ( ( i = AN.RepFunList[mt+1] ) > 0 ) {
                        *fill++ = GAMMA;
                        *fill++ = i + FUNHEAD+1;
                        FILLFUN(fill)
                        nq = i + 1;
                        t += FUNHEAD;
                        NCOPY(fill,t,nq);
                    }
                    t = oldt;
                }
                else if ( ( *t == LEVICIVITA ) || ( *t >= FUNCTION
                && (functions[*t-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
                                 ) sign += AN.RepFunList[mt+1];
                else if ( *m >= FUNCTION+WILDOFFSET
                && (functions[*m-FUNCTION-WILDOFFSET].symmetric & ~REVERSEORDER) == ANTISYMMETRIC
                                 ) sign += AN.RepFunList[mt+1];
                if ( !PutExpr ) {
                    WORD *pstart = fill, *p, *w, *ww;
                    xstop = t + t[1];
                    t = AN.FullProto;
                    nq = t[1];
                    t[3] = 1;
                    NCOPY(fill,t,nq);
                    t = xstop;
                    PutExpr = 1;
/*
                    Here we need provisions for keeping wildcard matches
                    that reside in AT.ebufnum. We will move them to
                    AT.aebufnum.
                    Problem: the SUBEXPRESSION assumes automatically
                    that the compiler buffer is AT.ebufnum. We have to
                    correct that in TransferBuffer.
*/
                    p = pstart + SUBEXPSIZE;
                    while ( p < fill ) {
                        switch ( *p ) {
                            case SYMTOSUB:
                            case VECTOSUB:
                            case INDTOSUB:
                            case ARGTOARG:
                            case ARLTOARL:
                                w = cbuf[AT.ebufnum].rhs[p[3]];
                                ww = cbuf[AT.ebufnum].rhs[p[3]+1];
/*
                                Here we could search for whether this
                                object sits in the buffer already.
                                To be done later.
                                By the way: ww-w fits inside a WORD.
*/
                                AddRHS(AT.aebufnum,1);
                                AddNtoC(AT.aebufnum,ww-w,w,11);
                                p[3] = cbuf[AT.aebufnum].numrhs;
```

```
                                cbuf[AT.aebufnum].rhs[p[3]+1] = cbuf[AT.aebufnum].Pointer;
                                p += p[1];
                                break;
                            case FROMSET:
                            case SETTONUM:
                            case LOADDOLLAR:
                                p += p[1];
                                break;
                            default:
                                p += p[1];
                                break;
                        }

                    }
                }
                else t += t[1];
                if ( *m == GAMMA && m[1] != FUNHEAD+1 ) {
                    i = oldt[1] - m[1] - i;
                    if ( i > 0 ) {
                        *fill++ = GAMMA;
                        *fill++ = i + FUNHEAD+1;
                        FILLFUN(fill)
                        *fill++ = oldt[FUNHEAD];
                        t = t - i;
                        NCOPY(fill,t,i);
                    }
                }
                break;
            }
        }
        m += m[1];
    }
    while ( t < tstop ) *fill++ = *t++;
    if ( !PutExpr ) {
        t = AN.FullProto;
        nq = t[1];
        t[3] = 1;
        NCOPY(fill,t,nq);
    }
    t = tcoef;
    nq = ABS(*t);
    t = tstop;
    NCOPY(fill,t,nq);
    if ( sign ) {
        if ( ( sign & 1 ) != 0 ) fill[-1] = -fill[-1];
    }
    *TemTerm = fill-TemTerm;
/*
    And now we copy this to AT.allbufnum
*/
    AddNtoC(AT.allbufnum,TemTerm[0],TemTerm,12);
    cbuf[AT.allbufnum].Pointer[0] = 0;
    AN.RepFunNum = 0;
}

/*
        #] SubsInAll :
        #[ TransferBuffer :

        Adds the whole content of a (compiler)buffer to another buffer.
        In spectator we have an expression in the RHS that needs the
        wildcard resolutions adapted by an offset.
*/

void TransferBuffer(int from,int to,int spectator)
{
    CBUF *C  = cbuf + spectator;
    CBUF *Cf = cbuf + from;
    CBUF *Ct = cbuf + to;
    int offset = Ct->numrhs;
    LONG i;
    WORD *t, *tt, *ttt, *tstop, size;
    for ( i = 1; i <= Cf->numrhs; i++ ) {
        size = Cf->rhs[i+1]-Cf->rhs[i];
        AddRHS(to,1);
        AddNtoC(to,size,Cf->rhs[i],13);
    }
    Ct->rhs[Ct->numrhs+1] = Ct->Pointer;
    Cf->numrhs = 0;
/*
    Now we have to update the 'pointers' in the spectator.
*/
    t = C->rhs[C->numrhs];
    while ( *t ) {
        tt = t+1; t += *t;
        tstop = t-ABS(t[-1]);
        while ( tt < tstop ) {
```

```
            if ( *tt == SUBEXPRESSION ) {
                ttt = tt+SUBEXPSIZE; tt += tt[1];
                while ( ttt < tt ) {
                    switch ( *ttt ) {
                        case SYMTOSUB:
                        case VECTOSUB:
                        case INDTOSUB:
                        case ARGTOARG:
                        case ARLTOARL:
                            ttt[3] += offset;
                            break;
                        default:
                            break;
                    }
                    ttt += 4;
                }
            }
            else tt += tt[1];
        }
    }
}

/*
        #] TransferBuffer :
        #[ TakeIDfunction :
*/

#define PutInBuffers(pow) \
    AddRHS(AT.ebufnum,1); \
    *out++ = SUBEXPRESSION; \
    *out++ = SUBEXPSIZE; \
    *out++ = C->numrhs; \
    *out++ = pow; \
    *out++ = AT.ebufnum; \
    FILLSUB(out) \
    r = AT.pWorkSpace[rhs+i]; \
    if ( *r > 0 ) { \
        oldinr = r[*r]; r[*r] = 0; \
        AddNtoC(AT.ebufnum,(*r+1-ARGHEAD),(r+ARGHEAD),14); \
        r[*r] = oldinr; \
    } \
    else { \
        ToGeneral(r,buffer,1); \
        buffer[buffer[0]] = 0; \
        AddNtoC(AT.ebufnum,buffer[0]+1,buffer,15); \
    }

int TakeIDfunction(PHEAD WORD *term)
{
    WORD *tstop, *t, *r, *m, *f, *nextf, *funstop, *left, *l, *newterm;
    WORD *out, oldinr, pow;
    WORD buffer[20];
    int i, ii, j, numsub, numfound = 0, first;
    LONG lhs,rhs;
    CBUF *C;
    GETSTOP(term,tstop);
    for ( t = term+1; t < tstop; t += t[1] ) { if ( *t == IDFUNCTION ) break; }
    if ( t >= tstop ) return(0);
/*
    Step 1: test validity
*/
    funstop = t + t[1]; f = t + FUNHEAD;
    left = term + *term;
    l = left+1; numsub = 0;
    while ( f < funstop ) {
        nextf = f; NEXTARG(nextf)
        if ( nextf >= funstop ) { return(0); } /* odd number of arguments */
        if ( *f == -SYMBOL ) { *l++ = SYMBOL; *l++ = 4; *l++ = f[1]; *l++ = 1; }
        else if ( *f < -FUNCTION ) { *l++ = *f; *l++ = FUNHEAD; FILLFUN(l) }
        else if ( *f > 0 ) {
            if ( *f != f[ARGHEAD]+ARGHEAD ) goto noaction;
            if ( nextf[-1] != 3 || nextf[-2] != 1 || nextf[-3] != 1 ) goto noaction;
            if ( f[ARGHEAD] <= 4 ) goto noaction;
            if ( f[ARGHEAD] != f[ARGHEAD+2]+4 ) goto noaction;
            if ( f[ARGHEAD] == 8 && f[ARGHEAD+1] == SYMBOL ) {
                for ( i = 0; i < 4; i++ ) *l++ = f[ARGHEAD+1+i];
            }
            else if ( f[ARGHEAD] == 9 && f[ARGHEAD+1] == DOTPRODUCT ) {
                for ( i = 0; i < 5; i++ ) *l++ = f[ARGHEAD+1+i];
            }
            else if ( f[ARGHEAD+1] >= FUNCTION ) {
                for ( i = 0; i < f[ARGHEAD+1]-4; i++ ) *l++ = f[ARGHEAD+1+i];
            }
            else goto noaction;
        }
        else goto noaction;
        numsub++;
```

```
        f = nextf;
        NEXTARG(f)
    }
    C = cbuf+AT.ebufnum;
    AT.WorkPointer = l;
    *left = l-left;
/*
    Put the pointers to the lhs and the rhs in the pointer workspace
*/
    WantAddPointers(2*numsub);
    lhs = AT.pWorkPointer;
    rhs = lhs+numsub;
    AT.pWorkPointer = rhs+numsub;
    f = t + FUNHEAD; l = left+1;
    for ( i = 0; i < numsub; i++ ) {
        AT.pWorkSpace[lhs+i] = l; l += l[1];
        NEXTARG(f);
        AT.pWorkSpace[rhs+i] = f;
        NEXTARG(f);
    }
/*
    Take out the patterns and replace them by SUBEXPRESSIONs pointing at
    the e buffer. We put the resulting term above the left sides.
    Note that we take out only the first id_ if there is more than one!
*/
    first = 1;
    t = term+1; newterm = AT.WorkPointer; out = newterm+1;
    while ( t < tstop ) {
        if ( *t == IDFUNCTION && first ) { first = 0; t += t[1]; continue; }
        if ( *t >= FUNCTION ) {
            for ( i = 0; i < numsub; i++ ) {
                m = AT.pWorkSpace[lhs+i];
                if ( *m != *t ) continue;
                for ( j = 1; j < t[1]; j++ ) {
                    if ( m[j] != t[j] ) break;
                }
                if ( j != t[1] ) continue;
                numfound++;
/*
                We have a match! Set up a SUBEXPRESSION subterm and put the
                corresponding rhs in the eBuffer.
*/
                PutInBuffers(1)
                t += t[1];
            }
            if ( i == numsub ) {    /* no match. Just copy to output. */
                j = t[1]; NCOPY(out,t,j)
            }
        }
        else if ( *t == SYMBOL ) {
            for ( i = 0; i < numsub; i++ ) {
                m = AT.pWorkSpace[lhs+i];
                if ( *m != SYMBOL ) continue;
                for ( ii = 2; ii < t[1]; ii += 2 ) {
                    if ( m[2] != t[ii] ) continue;
                    pow = t[ii+1]/m[3];
                    if ( pow <= 0 ) continue;
                    t[ii+1] = t[ii+1]%m[3];
                    numfound++;
/*
                    Create the proper rhs in the eBuffer and set up a
                    SUBEXPRESSION subterm.
*/
                    PutInBuffers(pow)
                }
            }
/*
            Now we copy whatever remains of the SYMBOL subterm to the output
*/
            m = out; *out++ = t[0]; *out++ = t[1];
            for ( ii = 2; ii < t[1]; ii += 2 ) {
                if ( t[ii+1] ) { *out++ = t[ii]; *out++ = t[ii+1]; }
            }
            m[1] = out-m;
            if ( m[1] == 2 ) out = m;
            t += t[1];
        }
        else if ( *t == DOTPRODUCT ) {
            for ( i = 0; i < numsub; i++ ) {
                m = AT.pWorkSpace[lhs+i];
                if ( *m != DOTPRODUCT ) continue;
                for ( ii = 2; ii < t[1]; ii += 3 ) {
                    if ( m[2] != t[ii] || m[3] != t[ii+1] ) continue;
                    pow = t[ii+2]/m[4];
                    if ( pow <= 0 ) continue;
                    t[ii+2] = t[ii+2]%m[4];
                    numfound++;
```

```
/*
                    Create the proper rhs in the eBuffer and set up a
                    SUBEXPRESSION subterm.
*/
                    PutInBuffers(pow)
                }
            }
/*
            Now we copy whatever remains of the DOTPRODUCT subterm to the output
*/
            m = out; *out++ = t[0]; *out++ = t[1];
            for ( ii = 2; ii < t[1]; ii += 3 ) {
                if ( t[ii+2] ) { *out++ = t[ii]; *out++ = t[ii+1]; *out++ = t[ii+2]; }
            }
            m[1] = out-m;
            if ( m[1] == 2 ) out = m;
            t += t[1];
        }
        else {
            j = t[1]; NCOPY(out,t,j)
        }
    }
/*
    Copy the coefficient and set the size.
*/
    t = tstop; r = term+*term; while ( t < r ) *out++ = *t++;
    *newterm = out-newterm;
/*
    Finally we move the new term over the original term.
*/
    i = *newterm;
    t = term; r = newterm; NCOPY(t,r,i)
/*
    At this point we can return and if the calling Generator jumps back to
    its start, TestSub can take care of the expansions of SUBEXPRESSIONs.
*/
    AT.pWorkPointer = lhs;
    AT.WorkPointer = t;
    return(numfound);
noaction:
    return(0);
}

/*
        #] TakeIDfunction :
    #] Patterns :
*/

```

4.100 poly.cc

```
/* @file poly.cc
 *
 *  Contains the class for representing sparse multivariate
 *  polynomials with integer coefficients
 */
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 *   details.
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
```

```
    #[ includes :
*/

#include "poly.h"
#include "polygcd.h"

#include <cassert>
#include <cctype>
#include <cmath>
#include <algorithm>
#include <map>
#include <iostream>

using namespace std;

/*
    #] includes :
    #[ constructor (small constant polynomial) :
*/

// constructor for a small constant polynomial
poly::poly (PHEAD int a, WORD modp, WORD modn):
    size_of_terms(AM.MaxTer/(LONG)sizeof(WORD)),
    modp(0),
    modn(1)
{
    POLY_STOREIDENTITY;

    terms = TermMalloc("poly::poly(int)");

    if (a == 0) {
        terms[0] = 1; // length
    }
    else {
        terms[0] = 4 + AN.poly_num_vars;                         // length
        terms[1] = 3 + AN.poly_num_vars;                         // length
        for (int i=0; i<AN.poly_num_vars; i++) terms[2+i] = 0; // powers
        terms[2+AN.poly_num_vars] = ABS(a);                      // coefficient
        terms[3+AN.poly_num_vars] = a>0 ? 1 : -1;               // length coefficient
    }

    if (modp!=-1) setmod(modp,modn);
}

/*
    #] constructor (small constant polynomial) :
    #[ constructor (large constant polynomial) :
*/

// constructor for a large constant polynomial
poly::poly (PHEAD const UWORD *a, WORD na, WORD modp, WORD modn):
    size_of_terms(AM.MaxTer/(LONG)sizeof(WORD)),
    modp(0),
    modn(1)
{
    POLY_STOREIDENTITY;

    terms = TermMalloc("poly::poly(UWORD*,WORD)");

    // remove leading zeros
    while (*(a+ABS(na)-1)==0)
        na -= SGN(na);

    terms[0] = 3 + AN.poly_num_vars + ABS(na);                // length
    terms[1] = terms[0] - 1;                                  // length
    for (int i=0; i<AN.poly_num_vars; i++) terms[2+i] = 0;   // powers
    termscopy((WORD *)a, 2+AN.poly_num_vars, ABS(na));       // coefficient
    terms[2+AN.poly_num_vars+ABS(na)] = na;                  // length coefficient

    if (modp!=-1) setmod(modp,modn);
}

/*
    #] constructor (large constant polynomial) :
    #[ constructor (deep copy polynomial) :
*/

// copy constructor: makes a deep copy of a polynomial
poly::poly (const poly &a, WORD modp, WORD modn):
    size_of_terms(AM.MaxTer/(LONG)sizeof(WORD))
{
#ifdef POLY_MOVE_DEBUG
    std::cout « "poly copy ctor" « std::endl;
#endif
    POLY_GETIDENTITY(a);
    POLY_STOREIDENTITY;

```

```
    terms = TermMalloc("poly::poly(poly&)");

    *this = a;

    if (modp!=-1) setmod(modp,modn);
}

/*
    #] constructor (deep copy polynomial) :
    #[ destructor :
*/

// destructor
poly::~poly () {

    POLY_GETIDENTITY(*this);

    if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD))
        TermFree(terms, "poly::~poly");
    else
        M_free(terms, "poly::~poly");
}

/*
    #] destructor :
    #[ expand_memory :
*/

// expands the memory allocated for terms to at least twice its size
void poly::expand_memory (int i) {

    POLY_GETIDENTITY(*this);

    LONG new_size_of_terms = MaX(2*size_of_terms, i);
    if (new_size_of_terms > MAXPOSITIVE) {
        MLOCK(ErrorMessageLock);
        MesPrint ((char*)"ERROR: polynomials too large (> WORDSIZE)");
        MUNLOCK(ErrorMessageLock);
        Terminate(1);
    }

    WORD *newterms = (WORD *)Malloc1(new_size_of_terms * sizeof(WORD), "poly::expand_memory");
    WCOPY(newterms, terms, size_of_terms);

    if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD))
        TermFree(terms, "poly::expand_memory");
    else
        M_free(terms, "poly::expand_memory");

    terms = newterms;
    size_of_terms = new_size_of_terms;
}

/*
    #] expand_memory :
    #[ setmod :
*/

// sets the coefficient space to ZZ/p^n
void poly::setmod(WORD _modp, WORD _modn) {

    POLY_GETIDENTITY(*this);

    if (_modp>0 && (_modp!=modp || _modn<modn)) {
        modp = _modp;
        modn = _modn;

        WORD nmodq=0;
        UWORD *modq=NULL;
        bool smallq;

        if (modn == 1) {
            modq = (UWORD *)&modp;
            nmodq = 1;
            smallq = true;
        }
        else {
            RaisPowCached(BHEAD modp,modn,&modq,&nmodq);
            smallq = false;
        }

        coefficients_modulo(modq,nmodq,smallq);
    }
    else {
        modp = _modp;
        modn = _modn;
    }
```

```
}

/*
    #] setmod :
    #[ coefficients_modulo :
*/

// reduces all coefficients of the polynomial modulo a
void poly::coefficients_modulo (UWORD *a, WORD na, bool small) {

    POLY_GETIDENTITY(*this);

    int j=1;
    for (int i=1, di; i<terms[0]; i+=di) {
        di = terms[i];

        if (i!=j)
            for (int k=0; k<di; k++)
                terms[j+k] = terms[i+k];

        WORD n = terms[j+terms[j]-1];

        if (ABS(n)==1 && small) {
            // small number modulo small number
            terms[j+1+AN.poly_num_vars] = n*((UWORD)terms[j+1+AN.poly_num_vars] % a[0]);
            if (terms[j+1+AN.poly_num_vars] == 0)
                n=0;
            else {
                if (terms[j+1+AN.poly_num_vars] > +(WORD)a[0]/2) terms[j+1+AN.poly_num_vars]-=a[0];
                if (terms[j+1+AN.poly_num_vars] < -(WORD)a[0]/2) terms[j+1+AN.poly_num_vars]+=a[0];
                n = SGN(terms[j+1+AN.poly_num_vars]);
                terms[j+1+AN.poly_num_vars] = ABS(terms[j+1+AN.poly_num_vars]);
            }
        }
        else
            TakeNormalModulus((UWORD *)&terms[j+1+AN.poly_num_vars], &n, a, na, NOUNPACK);

        if (n!=0) {
            terms[j] = 2+AN.poly_num_vars+ABS(n);
            terms[j+terms[j]-1] = n;
            j += terms[j];
        }
    }

    terms[0] = j;
}

/*
    #] coefficients_modulo :
    #[ to_string :
*/

// converts an integer to a string
const string int_to_string (WORD x) {
    char res[41];
    snprintf (res, 41, "%i",x);
    return res;
}

// converts a polynomial to a string
const string poly::to_string() const {

    POLY_GETIDENTITY(*this);

    string res;

    int printtimes;
    UBYTE *scoeff = (UBYTE *)NumberMalloc("poly::to_string");

    if (terms[0]==1)
        // zero
        res = "0";
    else {
        for (int i=1; i<terms[0]; i+=terms[i]) {

            // sign
            WORD ncoeff = terms[i+terms[i]-1];
            if (ncoeff < 0) {
                ncoeff*=-1;
                res += "-";
            }
            else {
                if (i>1) res += "+";
            }

            if (ncoeff==1 && terms[i+terms[i]-1-ncoeff]==1) {
                // coeff=1, so don't print coefficient and '*'
```

```
                printtimes = 0;
            }
            else {
                // print coefficient
                PrtLong((UWORD*)&terms[i+terms[i]-1-ncoeff], ncoeff, scoeff);
                res += string((char *)scoeff);
                printtimes=1;
            }

            // print variables
            for (int j=0; j<AN.poly_num_vars; j++) {
                if (terms[i+1+j] > 0) {
                    if (printtimes) res += "*";
                    res += string(1,'a'+j);
                    if (terms[i+1+j] > 1) res += "^" + int_to_string(terms[i+1+j]);
                    printtimes = 1;
                }
            }

            // iff coeff=1 and all power=0, print '1' after all
            if (!printtimes) res += "1";
        }
    }

    // eventual modulo
    if (modp>0) {
        res += " (mod ";
        res += int_to_string(modp);
        if (modn>1) {
            res += "^";
            res += int_to_string(modn);
        }
        res += ")";
    }

    NumberFree(scoeff,"poly::to_string");

    return res;
}

/*
    #] to_string :
    #[ ostream operator :
*/

// output stream operator
ostream& operator« (ostream &out, const poly &a) {
    return out « a.to_string();
}

/*
    #] ostream operator :
    #[ monomial_compare :
*/

// compares two monomials (0:equal, <0:a smaller, >0:b smaller)
int poly::monomial_compare (PHEAD const WORD *a, const WORD *b) {
    for (int i=0; i<AN.poly_num_vars; i++)
        if (a[i+1]!=b[i+1]) return a[i+1]-b[i+1];
    return 0;
}

/*
    #] monomial_compare :
    #[ normalize :
*/

// brings a polynomial to normal form
const poly & poly::normalize() {

    POLY_GETIDENTITY(*this);

    WORD nmodq=0;
    UWORD *modq=NULL;

    if (modp!=0) {
        if (modn == 1) {
            modq = (UWORD *)&modp;
            nmodq = 1;
        }
        else {
            RaisPowCached(BHEAD modp,modn,&modq,&nmodq);
        }
    }

    // find and sort all monomials
    // terms[0]/num_vars+3 is an upper bound for number of terms in a
```

```

    LONG poffset = AT.pWorkPointer;
    WantAddPointers((terms[0]/(AN.poly_num_vars+3)));
    AT.pWorkPointer += terms[0]/(AN.poly_num_vars+3);
    #define p (&(AT.pWorkSpace[poffset]))

//  WORD **p = (WORD **)Malloc1(terms[0]/(AN.poly_num_vars+3) * sizeof(WORD*), "poly::normalize");

    int nterms = 0;
    for (int i=1; i<terms[0]; i+=terms[i])
        p[nterms++] = terms + i;

    sort(p, p + nterms, monomial_larger(BHEAD0));

    WORD *tmp;
    if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD)) {
        tmp = (WORD *)TermMalloc("poly::normalize");
    }
    else {
        tmp = (WORD *)Malloc1(size_of_terms * sizeof(WORD), "poly::normalize");
    }
    int j=1;
    int prevj=0;
    tmp[0]=0;
    tmp[1]=0;

    for (int i=0; i<nterms; i++) {
        if (i>0 && monomial_compare(BHEAD &tmp[j], p[i])==0) {
            // duplicate term, so add coefficients
            WORD ncoeff = tmp[j+tmp[j]-1];
            AddLong((UWORD *)&tmp[j+1+AN.poly_num_vars], ncoeff,
                            (UWORD *)&p[i][1+AN.poly_num_vars], p[i][p[i][0]-1],
                            (UWORD *)&tmp[j+1+AN.poly_num_vars], &ncoeff);

            tmp[j+1+AN.poly_num_vars+ABS(ncoeff)] = ncoeff;
            tmp[j] = 2+AN.poly_num_vars+ABS(ncoeff);
        }
        else {
            // new term
            prevj = j;
            j += tmp[j];
            WCOPY(&tmp[j],p[i],p[i][0]);
        }

        if (modp!=0) {
            // bring coefficient to normal form mod p^n
            WORD ntmp = tmp[j+tmp[j]-1];
            TakeNormalModulus((UWORD *)&tmp[j+1+AN.poly_num_vars], &ntmp,
                                                    modq,nmodq, NOUNPACK);
            tmp[j] = 2+AN.poly_num_vars+ABS(ntmp);
            tmp[j+tmp[j]-1] = ntmp;
        }

        // add terms to polynomial
        if (tmp[j+tmp[j]-1]==0) {
            tmp[j]=0;
            j=prevj;
        }
    }

    j+=tmp[j];

    tmp[0] = j;
    WCOPY(terms,tmp,tmp[0]);

//  M_free(p, "poly::normalize");

    if (size_of_terms == AM.MaxTer/(LONG)sizeof(WORD))
        TermFree(tmp, "poly::normalize");
    else
        M_free(tmp, "poly::normalize");

    AT.pWorkPointer = poffset;
    #undef p

    return *this;
}

/*
    #] normalize :
    #[ last_monomial_index :
*/

// index of the last monomial, i.e., the constant term
WORD poly::last_monomial_index () const {
    POLY_GETIDENTITY(*this);
    return terms[0] - ABS(terms[terms[0]-1]) - AN.poly_num_vars - 2;
```

```
}


// pointer to the last monomial (the constant term)
WORD * poly::last_monomial () const {
    return &terms[last_monomial_index()];
}

/*
    #] last_monomial_index :
    #[ compare_degree_vector :
*/

int poly::compare_degree_vector(const poly & b) const {
    POLY_GETIDENTITY(*this);

    // special cases if one or both are 0
    if (terms[0] == 1 &&  b[0] == 1) return 0;
    if (terms[0] == 1) return -1;
    if (b[0] == 1) return 1;

    for (int i = 0; i < AN.poly_num_vars; i++) {
        int d1 = degree(i);
        int d2 = b.degree(i);
        if (d1 != d2) return d1 - d2;
    }

    return 0;
}

/*
    #] compare_degree_vector :
    #[ degree_vector :
*/

std::vector<int> poly::degree_vector() const {
    POLY_GETIDENTITY(*this);

    std::vector<int> degrees(AN.poly_num_vars);
    for (int i = 0; i < AN.poly_num_vars; i++) {
        degrees[i] = degree(i);
    }

    return degrees;
}

/*
    #] degree_vector :
    #[ compare_degree_vector :
*/

int poly::compare_degree_vector(const vector<int> & b) const {
    POLY_GETIDENTITY(*this);

    if (terms[0] == 1) return -1;

    for (int i = 0; i < AN.poly_num_vars; i++) {
        int d1 = degree(i);
        if (d1 != b[i]) return d1 - b[i];
    }

    return 0;
}

/*
    #] compare_degree_vector :
    #[ add :
*/

// addition of polynomials by merging
void poly::add (const poly &a, const poly &b, poly &c) {

    POLY_GETIDENTITY(a);

  c.modp = a.modp;
  c.modn = a.modn;

    WORD nmodq=0;
    UWORD *modq=NULL;

    bool both_mod_small=false;

    if (c.modp!=0) {
        if (c.modn == 1) {
            modq = (UWORD *)&c.modp;
            nmodq = 1;
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
```

```
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD c.modp,c.modn,&modq,&nmodq);
        }
    }

    int ai=1,bi=1,ci=1;

    while (ai<a[0] || bi<b[0]) {

        c.check_memory(ci);

        int cmp = ai<a[0] && bi<b[0] ? monomial_compare(BHEAD &a[ai],&b[bi]) : 0;

        if (bi==b[0] || cmp>0) {
            // insert term from a
            c.termscopy(&a[ai],ci,a[ai]);
            ci+=a[ai];
            ai+=a[ai];
        }
        else if (ai==a[0] || cmp<0) {
            // insert term from b
            c.termscopy(&b[bi],ci,b[bi]);
            ci+=b[bi];
            bi+=b[bi];
        }
        else {
            // insert term from a+b
            c.termscopy(&a[ai],ci,MaX(a[ai],b[bi]));
            WORD nc=0;
            if (both_mod_small) {
                c[ci+1+AN.poly_num_vars] = ((LONG)a[ai+1+AN.poly_num_vars]*a[ai+a[ai]-1]+

(LONG)b[bi+1+AN.poly_num_vars]*b[bi+b[bi]-1]) % c.modp;
                if ((WORD)c[ci+1+AN.poly_num_vars] > +c.modp/2) c[ci+1+AN.poly_num_vars] -= c.modp;
                if ((WORD)c[ci+1+AN.poly_num_vars] < -c.modp/2) c[ci+1+AN.poly_num_vars] += c.modp;
                nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : SGN((WORD)c[ci+1+AN.poly_num_vars]));
                c[ci+1+AN.poly_num_vars] = ABS((WORD)c[ci+1+AN.poly_num_vars]);

            }
            else {
                AddLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
                                (UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
                                (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);
                if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
                                                                                 modq, nmodq,
NOUNPACK);
            }

            if (nc!=0) {
                c[ci] = 2+AN.poly_num_vars+ABS(nc);
                c[ci+c[ci]-1] = nc;
                ci += c[ci];
            }

            ai+=a[ai];
            bi+=b[bi];
        }
    }

    c[0]=ci;
}

/*
    #] add :
    #[ sub :
*/

// subtraction of polynomials by merging
void poly::sub (const poly &a, const poly &b, poly &c) {

    POLY_GETIDENTITY(a);

  c.modp = a.modp;
  c.modn = a.modn;

    WORD nmodq=0;
    UWORD *modq=NULL;

    bool both_mod_small=false;

    if (c.modp!=0) {
        if (c.modn == 1) {
            modq = (UWORD *)&c.modp;
            nmodq = 1;
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
```

```
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD c.modp,c.modn,&modq,&nmodq);
        }
    }

    int ai=1,bi=1,ci=1;

    while (ai<a[0] || bi<b[0]) {

        c.check_memory(ci);

        int cmp = ai<a[0] && bi<b[0] ? monomial_compare(BHEAD &a[ai],&b[bi]) : 0;

        if (bi==b[0] || cmp>0) {
            // insert term from a
            c.termscopy(&a[ai],ci,a[ai]);
            ci+=a[ai];
            ai+=a[ai];
        }
        else if (ai==a[0] || cmp<0) {
            // insert term from b
            c.termscopy(&b[bi],ci,b[bi]);
            ci+=b[bi];
            bi+=b[bi];
            c[ci-1]*=-1;
        }
        else {
            // insert term from a+b
            c.termscopy(&a[ai],ci,MaX(a[ai],b[bi]));
            WORD nc=0;
            if (both_mod_small) {
                c[ci+1+AN.poly_num_vars] = ((LONG)a[ai+1+AN.poly_num_vars]*a[ai+a[ai]-1]-

(LONG)b[bi+1+AN.poly_num_vars]*b[bi+b[bi]-1]) % c.modp;
                if ((WORD)c[ci+1+AN.poly_num_vars] > +c.modp/2) c[ci+1+AN.poly_num_vars] -= c.modp;
                if ((WORD)c[ci+1+AN.poly_num_vars] < -c.modp/2) c[ci+1+AN.poly_num_vars] += c.modp;
                nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : SGN((WORD)c[ci+1+AN.poly_num_vars]));
                c[ci+1+AN.poly_num_vars] = ABS((WORD)c[ci+1+AN.poly_num_vars]);

            }
            else {
                AddLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
                                (UWORD *)&b[bi+1+AN.poly_num_vars],-b[bi+b[bi]-1], // -b[...] causes
subtraction
                                (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);
                if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
                                                                        modq, nmodq,
NOUNPACK);
            }

            if (nc!=0) {
                c[ci] = 2+AN.poly_num_vars+ABS(nc);
                c[ci+c[ci]-1] = nc;
                ci += c[ci];
            }

            ai+=a[ai];
            bi+=b[bi];
        }
    }

    c[0]=ci;
}

/*
    #] sub :
    #[ pop_heap :
*/

// pops the largest monomial from the heap and stores it in heap[n]
void poly::pop_heap (PHEAD WORD **heap, int n) {

    WORD *old = heap[0];

    heap[0] = heap[--n];

    int i=0;
    while (2*i+2<n && (monomial_compare(BHEAD heap[2*i+1]+3, heap[i]+3)>0 ||
                                        monomial_compare(BHEAD heap[2*i+2]+3, heap[i]+3)>0)) {

        if (monomial_compare(BHEAD heap[2*i+1]+3, heap[2*i+2]+3)>0) {
            swap(heap[i], heap[2*i+1]);
            i=2*i+1;
        }
        else {
```

```
            swap(heap[i], heap[2*i+2]);
            i=2*i+2;
        }
    }

    if (2*i+1<n && monomial_compare(BHEAD heap[2*i+1]+3, heap[i]+3)>0)
        swap(heap[i], heap[2*i+1]);

    heap[n] = old;
}

/*
    #] pop_heap :
    #[ push_heap :
*/

// pushes the monomial in heap[n] onto the heap
void poly::push_heap (PHEAD WORD **heap, int n)  {

    int i=n-1;

    while (i>0 && monomial_compare(BHEAD heap[i]+3, heap[(i-1)/2]+3) > 0) {
        swap(heap[(i-1)/2], heap[i]);
        i=(i-1)/2;
    }
}

/*
    #] push_heap :
    #[ mul_one_term :
*/

// multiplies a polynomial with a monomial
void poly::mul_one_term (const poly &a, const poly &b, poly &c) {

  POLY_GETIDENTITY(a);

  int ci=1;

    WORD nmodq=0;
    UWORD *modq=NULL;

    bool both_mod_small=false;

    if (c.modp!=0) {
        if (c.modn == 1) {
            modq = (UWORD *)&c.modp;
            nmodq = 1;
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD c.modp,c.modn,&modq,&nmodq);
        }
    }

  for (int ai=1; ai<a[0]; ai+=a[ai])
    for (int bi=1; bi<b[0]; bi+=b[bi]) {

            c.check_memory(ci);

      for (int i=0; i<AN.poly_num_vars; i++)
        c[ci+1+i] = a[ai+1+i] + b[bi+1+i];
      WORD nc;

            // if both polynomials are modulo p^1, use integer calculus
            if (both_mod_small) {
                c[ci+1+AN.poly_num_vars] =
                    ((LONG)a[ai+1+AN.poly_num_vars] * a[ai+2+AN.poly_num_vars] *
                     b[bi+1+AN.poly_num_vars] * b[bi+2+AN.poly_num_vars]) % c.modp;
                nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : 1);
            }
            else {
                // otherwise, use form long calculus
                MulLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
                                (UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
                                (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);
                if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
                                                                                 modq, nmodq,
NOUNPACK);
            }

            if (nc!=0) {
                c[ci] = 2+AN.poly_num_vars+ABS(nc);
                ci += c[ci];

                if (both_mod_small) {
```

```
                        if (c[ci-2] > +c.modp/2) c[ci-2] -= c.modp;
                        if (c[ci-2] < -c.modp/2) c[ci-2] += c.modp;
                        c[ci-1] = SGN(c[ci-2]);
                        c[ci-2] = ABS(c[ci-2]);
                    }
                    else {
                        c[ci-1] = nc;
                    }
                }
    }

  c[0]=ci;
}

/*
    #] mul_one_term :
    #[ mul_univar :
*/

// dense univariate multiplication, i.e., for each power find all
// pairs of monomials that result in that power
void poly::mul_univar (const poly &a, const poly &b, poly &c, int var) {

    POLY_GETIDENTITY(a);

    WORD nmodq=0;
    UWORD *modq=NULL;

    bool both_mod_small=false;

    if (c.modp!=0) {
        if (c.modn == 1) {
            modq = (UWORD *)&c.modp;
            nmodq = 1;
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD c.modp,c.modn,&modq,&nmodq);
        }
    }

    poly t(BHEAD 0);
    WORD nt;

    int ci=1;

    // bounds on the powers in a*b
    int minpow = AN.poly_num_vars==0 ? 0 : a.last_monomial()[1+var] + b.last_monomial()[1+var];
    int maxpow = AN.poly_num_vars==0 ? 0 : a[2+var]+b[2+var];
    int afirst=1, blast=1;

    for (int pow=maxpow; pow>=minpow; pow--) {

        c.check_memory(ci);

        WORD nc=0;

        // adjust range in a or b
        if (a[afirst+1+var] + b[blast+1+var] > pow) {
            if (blast+b[blast] < b[0])
                blast+=b[blast];
            else
                afirst+=a[afirst];
        }

        // find terms that result in the correct power
        for (int ai=afirst, bi=blast; ai<a[0] && bi>=1;) {

            int thispow = AN.poly_num_vars==0 ? 0 : a[ai+1+var] + b[bi+1+var];

            if (thispow == pow) {

                // if both polynomials are modulo p^1, use integer calculus
                if (both_mod_small) {
                    c[ci+1+AN.poly_num_vars] =
                        ((nc==0 ? 0 : (LONG)c[ci+1+AN.poly_num_vars] * nc) +
                        (LONG)a[ai+1+AN.poly_num_vars] * a[ai+2+AN.poly_num_vars] *
                         b[bi+1+AN.poly_num_vars] * b[bi+2+AN.poly_num_vars]) % c.modp;
                    nc = (c[ci+1+AN.poly_num_vars]==0 ? 0 : 1);
                }
                else {
                    // otherwise, use form long calculus

                    MulLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
                                    (UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
                                    (UWORD *)&t[0], &nt);
```

```
                    AddLong ((UWORD *)&t[0], nt,
                                  (UWORD *)&c[ci+1+AN.poly_num_vars], nc,
                                  (UWORD *)&c[ci+1+AN.poly_num_vars], &nc);

                    if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
                                                                                       modq, nmodq,
NOUNPACK);
                }

                ai += a[ai];
                bi -= ABS(b[bi-1]) + 2 + AN.poly_num_vars;
            }
            else if (thispow > pow)
                ai += a[ai];
            else
                bi -= ABS(b[bi-1]) + 2 + AN.poly_num_vars;
        }

        // add term to result
        if (nc != 0) {
            for (int j=0; j<AN.poly_num_vars; j++)
                c[ci+1+j] = 0;
            if (AN.poly_num_vars > 0)
                c[ci+1+var] = pow;

            c[ci] = 2+AN.poly_num_vars+ABS(nc);
            ci += c[ci];

            // if necessary, adjust to range [-p/2,p/2]
            if (both_mod_small) {
                if (c[ci-2] > +c.modp/2) c[ci-2] -= c.modp;
                if (c[ci-2] < -c.modp/2) c[ci-2] += c.modp;
                c[ci-1] = SGN(c[ci-2]);
                c[ci-2] = ABS(c[ci-2]);
            }
            else {
                c[ci-1] = nc;
            }
        }
    }

    c[0] = ci;
}

/*
    #] mul_univar :
    #[ mul_heap :
*/

void poly::mul_heap (const poly &a, const poly &b, poly &c) {

    POLY_GETIDENTITY(a);

    WORD nmodq=0;
    UWORD *modq=NULL;

    bool both_mod_small=false;

    if (c.modp!=0) {
        if (c.modn == 1) {
            modq = (UWORD *)&c.modp;
            nmodq = 1;
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD c.modp,c.modn,&modq,&nmodq);
        }
    }

    // find maximum powers in different variables
    WORD *maxpower  = AT.WorkPointer;
    AT.WorkPointer += AN.poly_num_vars;
    WORD *maxpowera = AT.WorkPointer;
    AT.WorkPointer += AN.poly_num_vars;
    WORD *maxpowerb = AT.WorkPointer;
    AT.WorkPointer += AN.poly_num_vars;

    for (int i=0; i<AN.poly_num_vars; i++)
        maxpowera[i] = maxpowerb[i] = 0;

    for (int ai=1; ai<a[0]; ai+=a[ai])
        for (int j=0; j<AN.poly_num_vars; j++)
            maxpowera[j] = MaX(maxpowera[j], a[ai+1+j]);

    for (int bi=1; bi<b[0]; bi+=b[bi])
```

```
        for (int j=0; j<AN.poly_num_vars; j++)
            maxpowerb[j] = MaX(maxpowerb[j], b[bi+1+j]);

    for (int i=0; i<AN.poly_num_vars; i++)
        maxpower[i] = maxpowera[i] + maxpowerb[i];

    // if PROD(maxpower) small, use hash array
    bool use_hash = true;
    int nhash = 1;

    for (int i=0; i<AN.poly_num_vars; i++) {
        if (nhash > POLY_MAX_HASH_SIZE / (maxpower[i]+1)) {
            nhash = 1;
            use_hash = false;
            break;
        }
        nhash *= maxpower[i]+1;
    }

    // allocate heap and hash
    int nheap=a.number_of_terms();

    WantAddPointers(nheap+nhash);
    WORD **heap = AT.pWorkSpace + AT.pWorkPointer;

    for (int ai=1, i=0; ai<a[0]; ai+=a[ai], i++) {
        heap[i] = (WORD *) NumberMalloc("poly::mul_heap");
        heap[i][0] = ai;
        heap[i][1] = 1;
        heap[i][2] = -1;
        heap[i][3] = 0;
        for (int j=0; j<AN.poly_num_vars; j++)
            heap[i][4+j] = MAXPOSITIVE;
    }

    WORD **hash = AT.pWorkSpace + AT.pWorkPointer + nheap;
    for (int i=0; i<nhash; i++)
        hash[i] = NULL;

    int ci = 1;

    // multiply
    while (nheap > 0) {

        c.check_memory(ci);

        pop_heap(BHEAD heap, nheap--);
        WORD *p = heap[nheap];

        // if non-zero
        if (p[3] != 0) {
            if (use_hash) hash[p[2]] = NULL;

            c[0] = ci;

            // append this term to the result
            if (use_hash || ci==1 || monomial_compare(BHEAD p+3, c.last_monomial())!=0) {
                p[4 + AN.poly_num_vars + ABS(p[3])] = p[3];
                p[3] = 2 + AN.poly_num_vars + ABS(p[3]);
                c.termscopy(&p[3],ci,p[3]);
                ci += c[ci];
            }
            else {
                // add this term to the last term of the result
                ci = c.last_monomial_index();
                WORD nc = c[ci+c[ci]-1];

                // if both polynomials are modulo p^1, use integer calculus
                if (both_mod_small) {
                    c[ci+AN.poly_num_vars+1] = ((LONG)c[ci+AN.poly_num_vars+1]*nc +
p[4+AN.poly_num_vars]*p[3]) % c.modp;
                    if (c[ci+1+AN.poly_num_vars]==0)
                        nc = 0;
                    else {
                        if (c[ci+1+AN.poly_num_vars] > +c.modp/2) c[ci+1+AN.poly_num_vars] -= c.modp;
                        if (c[ci+1+AN.poly_num_vars] < -c.modp/2) c[ci+1+AN.poly_num_vars] += c.modp;
                        nc = SGN(c[ci+1+AN.poly_num_vars]);
                        c[ci+1+AN.poly_num_vars] = ABS(c[ci+1+AN.poly_num_vars]);
                    }
                }
                else {
                    // otherwise, use form long calculus
                    AddLong ((UWORD *)&p[4+AN.poly_num_vars], p[3],
                                     (UWORD *)&c[ci+AN.poly_num_vars+1], nc,
                                     (UWORD *)&c[ci+AN.poly_num_vars+1],&nc);

                    if (c.modp!=0) TakeNormalModulus((UWORD *)&c[ci+1+AN.poly_num_vars], &nc,
```

```
                                                                                      modq, nmodq,
NOUNPACK);
                }

                if (nc!=0) {
                    c[ci] = 2 + AN.poly_num_vars + ABS(nc);
                    ci += c[ci];
                    c[ci-1] = nc;
                }
            }
        }

        // add new term to the heap (ai, bi+1)
        while (p[1] < b[0]) {

            for (int j=0; j<AN.poly_num_vars; j++)
                p[4+j] = a[p[0]+1+j] + b[p[1]+1+j];

            // if both polynomials are modulo p^1, use integer calculus
            if (both_mod_small) {
                p[4+AN.poly_num_vars] = ((LONG)a[p[0]+1+AN.poly_num_vars]*a[p[0]+a[p[0]]-1]*
b[p[1]+1+AN.poly_num_vars]*b[p[1]+b[p[1]]-1]) % c.modp;
                if (p[4+AN.poly_num_vars]==0)
                    p[3]=0;
                else {
                    if (p[4+AN.poly_num_vars] > +c.modp/2) p[4+AN.poly_num_vars] -= c.modp;
                    if (p[4+AN.poly_num_vars] < -c.modp/2) p[4+AN.poly_num_vars] += c.modp;
                    p[3] = SGN(p[4+AN.poly_num_vars]);
                    p[4+AN.poly_num_vars] = ABS(p[4+AN.poly_num_vars]);
                }
            }
            else {
                // otherwise, use form long calculus
                MulLong((UWORD *)&a[p[0]+1+AN.poly_num_vars], a[p[0]+a[p[0]]-1],
                                (UWORD *)&b[p[1]+1+AN.poly_num_vars], b[p[1]+b[p[1]]-1],
                                (UWORD *)&p[4+AN.poly_num_vars], &p[3]);

                if (c.modp!=0) TakeNormalModulus((UWORD *)&p[4+AN.poly_num_vars], &p[3],
                                                                                  modq, nmodq,
NOUNPACK);
            }

            p[1] += b[p[1]];

            if (use_hash) {
                int ID = 0;
                for (int i=0; i<AN.poly_num_vars; i++)
                    ID = (maxpower[i]+1)*ID + p[4+i];

                // if hash and unused, push onto heap
                if (hash[ID] == NULL) {
                    p[2] = ID;
                    hash[ID] = p;
                    push_heap(BHEAD heap, ++nheap);
                    break;
                }
                else {
                    // if hash and used, add to heap element
                    WORD *h = hash[ID];
                    // if both polynomials are modulo p^1, use integer calculus
                    if (both_mod_small) {
                        h[4+AN.poly_num_vars] = ((LONG)p[4+AN.poly_num_vars]*p[3] +
h[4+AN.poly_num_vars]*h[3]) % c.modp;
                        if (h[4+AN.poly_num_vars]==0)
                            h[3]=0;
                        else {
                            if (h[4+AN.poly_num_vars] > +c.modp/2) h[4+AN.poly_num_vars] -= c.modp;
                            if (h[4+AN.poly_num_vars] < -c.modp/2) h[4+AN.poly_num_vars] += c.modp;
                            h[3] = SGN(h[4+AN.poly_num_vars]);
                            h[4+AN.poly_num_vars] = ABS(h[4+AN.poly_num_vars]);
                        }
                    }
                    else {
                        // otherwise, use form long calculus
                        AddLong ((UWORD *)&p[4+AN.poly_num_vars],  p[3],
                                         (UWORD *)&h[4+AN.poly_num_vars],  h[3],
                                         (UWORD *)&h[4+AN.poly_num_vars], &h[3]);

                        if (c.modp!=0) TakeNormalModulus((UWORD *)&h[4+AN.poly_num_vars], &h[3],
                                                                                          modq, nmodq,
NOUNPACK);
                    }
                }
            }
            else {
                // if no hash, push onto heap
```

```
                p[2] = -1;
                push_heap(BHEAD heap, ++nheap);
                break;
            }
        }
    }

    c[0] = ci;

    for (int ai=1, i=0; ai<a[0]; ai+=a[ai], i++)
        NumberFree(heap[i],"poly::mul_heap");
    AT.WorkPointer -= 3 * AN.poly_num_vars;
}

/*
    #] mul_heap :
    #[ mul :
*/

void poly::mul (const poly &a, const poly &b, poly &c) {

  c.modp = a.modp;
  c.modn = a.modn;

    if (a.is_zero() || b.is_zero()) { c[0]=1; return; }
    if (a.is_one()) {
        c.check_memory(b[0]);
        c.termscopy(b.terms,0,b[0]);
        // normalize if necessary
        if (a.modp!=b.modp || a.modn!=b.modn) {
            c.modp=0;
            c.setmod(a.modp,a.modn);
        }
        return;
    }
    if (b.is_one()) {
        c.check_memory(a[0]);
        c.termscopy(a.terms,0,a[0]);
        return;
    }

    int na=a.number_of_terms();
    int nb=b.number_of_terms();

    if (na==1 || nb==1) {
        mul_one_term(a,b,c);
        return;
    }

    int vara = a.is_dense_univariate();
    int varb = b.is_dense_univariate();

    if (vara!=-2 && varb!=-2 && vara==varb) {
        mul_univar(a,b,c,vara);
        return;
    }

    if (na < nb)
        mul_heap(a,b,c);
    else
        mul_heap(b,a,c);
}

/*
    #] mul :
    #[ divmod_one_term : :
*/

// divides a polynomial by a monomial
void poly::divmod_one_term (const poly &a, const poly &b, poly &q, poly &r, bool only_divides) {

    POLY_GETIDENTITY(a);

    int qi=1, ri=1;

    WORD nmodq=0;
    UWORD *modq=NULL;

    WORD nltbinv=0;
    UWORD *ltbinv=NULL;

    bool both_mod_small=false;

    if (q.modp!=0) {
        if (q.modn == 1) {
            modq = (UWORD *)&q.modp;
            nmodq = 1;
```

```
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD q.modp,q.modn,&modq,&nmodq);
        }

        ltbinv = NumberMalloc("poly::div_one_term");

        if (both_mod_small) {
            WORD ltb = b[b[1]]*b[2+AN.poly_num_vars];
            GetModInverses(ltb + (ltb<0?q.modp:0), q.modp, (WORD*)ltbinv, NULL);
            nltbinv = 1;
        }
        else
            GetLongModInverses(BHEAD (UWORD *)&b[2+AN.poly_num_vars], b[b[1]], modq, nmodq, ltbinv, &nltbinv, NULL, NULL);
    }

    for (int ai=1; ai<a[0]; ai+=a[ai]) {

        q.check_memory(qi);
        r.check_memory(ri);

        // check divisibility of powers
        bool div=true;
        for (int j=0; j<AN.poly_num_vars; j++) {
            q[qi+1+j] = a[ai+1+j]-b[2+j];
            r[ri+1+j] = a[ai+1+j];
            if (q[qi+1+j] < 0) div=false;
        }

        WORD nq,nr;

        if (div) {
            // if variables are divisible, divide coefficient
            if (q.modp==0) {
                DivLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
                                (UWORD *)&b[2+AN.poly_num_vars], b[b[1]],
                                (UWORD *)&q[qi+1+AN.poly_num_vars], &nq,
                                (UWORD *)&r[ri+1+AN.poly_num_vars], &nr);
            }
            else {
                // if both polynomials are modulo p^1, use integer calculus
                if (both_mod_small) {
                    q[qi+1+AN.poly_num_vars] =
                        ((LONG)a[ai+1+AN.poly_num_vars] * a[ai+a[ai]-1] * ltbinv[0] * nltbinv) % q.modp;
                    nq = (q[qi+1+AN.poly_num_vars]==0 ? 0 : 1);
                }
                else {
                    // otherwise, use form long calculus
                    MulLong((UWORD *)&a[ai+1+AN.poly_num_vars], a[ai+a[ai]-1],
                                    ltbinv, nltbinv,
                                    (UWORD *)&q[qi+1+AN.poly_num_vars], &nq);
                    TakeNormalModulus((UWORD *)&q[qi+1+AN.poly_num_vars], &nq,
                                                        modq,nmodq, NOUNPACK);
                }

                nr=0;
            }
        }
        else {
            // if not, term becomes part of the remainder
            nq=0;
            nr=a[ai+a[ai]-1];
            r.termscopy(&a[ai+1+AN.poly_num_vars], ri+1+AN.poly_num_vars, ABS(nr));
        }

        // add terms to quotient/remainder
        if (nq!=0) {
            q[qi] = 2+AN.poly_num_vars+ABS(nq);
            qi += q[qi];

            // if necessary, adjust to range [-p/2,p/2]
            if (both_mod_small) {
                if (q[qi-2] > +q.modp/2) q[qi-2] -= q.modp;
                if (q[qi-2] < -q.modp/2) q[qi-2] += q.modp;
                q[qi-1] = SGN(q[qi-2]);
                q[qi-2] = ABS(q[qi-2]);
            }
            else {
                q[qi-1] = nq;
            }
        }

        if (nr != 0) {
```

```
            if (only_divides) { r = poly(BHEAD 1); ri=r[0]; break; }
            r[ri] = 2+AN.poly_num_vars+ABS(nr);
            ri += r[ri];
            r[ri-1] = nr;
        }
    }

    q[0]=qi;
    r[0]=ri;

    if (q.modp!=0 || ltbinv != NULL) NumberFree(ltbinv,"poly::div_one_term");
}

/*
    #] divmod_one_term :
    #[ divmod_univar : :
*/
void poly::divmod_univar (const poly &a, const poly &b, poly &q, poly &r, int var, bool only_divides)
{

    POLY_GETIDENTITY(a);

    WORD nmodq=0;
    UWORD *modq=NULL;

    WORD nltbinv=0;
    UWORD *ltbinv=NULL;

    bool both_mod_small=false;

    if (q.modp!=0) {
        if (q.modn == 1) {
            modq = (UWORD *)&q.modp;
            nmodq = 1;
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD q.modp,q.modn,&modq,&nmodq);
        }
        ltbinv = NumberMalloc("poly::div_univar");

        if (both_mod_small) {
            WORD ltb = b[b[1]]*b[2+AN.poly_num_vars];
            GetModInverses(ltb + (ltb<0?q.modp:0), q.modp, (WORD*)ltbinv, NULL);
            nltbinv = 1;
        }
        else
            GetLongModInverses(BHEAD (UWORD *)&b[2+AN.poly_num_vars], b[b[1]], modq, nmodq, ltbinv,
&nltbinv, NULL, NULL);
    }

    WORD ns=0;
    WORD nt;
    UWORD *s = NumberMalloc("poly::div_univar");
    UWORD *t = NumberMalloc("poly::div_univar");
    s[0]=0;

    int bpow = b[2+var];

    int ai=1, qi=1, ri=1;

    for (int pow=a[2+var]; pow>=0; pow--) {

        q.check_memory(qi);
        r.check_memory(ri);

        // look for the correct power in a
        while (ai<a[0] && a[ai+1+var] > pow)
            ai+=a[ai];

        // first term of the r.h.s. of the above equation
        if (ai<a[0] && a[ai+1+var] == pow) {
            ns = a[ai+a[ai]-1];
            WCOPY(s, &a[ai+1+AN.poly_num_vars], ABS(ns));
        }
        else {
            ns = 0;
        }

        int bi=1, qj=qi;

        // second term(s) of the r.h.s. of the above equation
        while (qj>1 && bi<b[0]) {

            qj -= 2 + AN.poly_num_vars + ABS(q[qj-1]);
```

```

            while (bi<b[0] && b[bi+1+var]+q[qj+1+var] > pow)
                bi += b[bi];

            if (bi<b[0] && b[bi+1+var]+q[qj+1+var] == pow) {
                // if both polynomials are modulo p^1, use integer calculus

                if (both_mod_small) {
                    s[0] = ((WORD)s[0]*ns - (LONG)b[bi+1+AN.poly_num_vars] * b[bi+b[bi]-1] *
                                    q[qj+1+AN.poly_num_vars] * q[qj+q[qj]-1]) % q.modp;
                    ns = (s[0]==0 ? 0 : 1);
                }
                else {
                    // otherwise, use form long calculus
                    MulLong((UWORD *)&b[bi+1+AN.poly_num_vars], b[bi+b[bi]-1],
                                    (UWORD *)&q[qj+1+AN.poly_num_vars], q[qj+q[qj]-1],
                                    t, &nt);
                    nt *= -1;
                    AddLong(t,nt,s,ns,s,&ns);
                    if (q.modp!=0) TakeNormalModulus((UWORD *)s,&ns,modq, nmodq, NOUNPACK);
                }
            }
        }

        if (ns != 0) {
            // if necessary, adjust to range [-p/2,p/2]
            if (both_mod_small) {
                s[0]*=ns;
                if ((WORD)s[0] > +q.modp/2) s[0] -= q.modp;
                if ((WORD)s[0] < -q.modp/2) s[0] += q.modp;
                ns = SGN((WORD)s[0]);
                s[0] = ABS((WORD)s[0]);
            }

            if (pow >= bpow) {
                // large power, so divide by b
                if (q.modp == 0) {
                    DivLong(s, ns,
                                    (UWORD *)&b[2+AN.poly_num_vars], b[b[1]],
                                    (UWORD *)&q[qi+1+AN.poly_num_vars], &ns, t, &nt);
                }
                else {
                    if (both_mod_small) {
                        q[qi+1+AN.poly_num_vars] = ((LONG)s[0]*ns*ltbinv[0]*nltbinv) % q.modp;
                        if ((WORD)q[qi+1+AN.poly_num_vars] > +q.modp/2) q[qi+1+AN.poly_num_vars] -= q.modp;
                        if ((WORD)q[qi+1+AN.poly_num_vars] < -q.modp/2) q[qi+1+AN.poly_num_vars] += q.modp;
                        ns = (q[qi+1+AN.poly_num_vars]==0 ? 0 : SGN((WORD)q[qi+1+AN.poly_num_vars]));
                        q[qi+1+AN.poly_num_vars] = ABS((WORD)q[qi+1+AN.poly_num_vars]);

                    }
                    else {
                        MulLong(s, ns, ltbinv, nltbinv, (UWORD *)&q[qi+1+AN.poly_num_vars], &ns);
                        TakeNormalModulus((UWORD *)&q[qi+1+AN.poly_num_vars], &ns,
                                                            modq,nmodq, NOUNPACK);
                    }
                    nt=0;
                }
            }
            else {
                // small power, so remainder
                WCOPY(t,s,ABS(ns));
                nt = ns;
                ns = 0;
            }

            // add terms to quotient/remainder
            if (ns!=0) {
                for (int i=0; i<AN.poly_num_vars; i++)
                    q[qi+1+i] = 0;
                q[qi+1+var] = pow-bpow;

                q[qi] = 2+AN.poly_num_vars+ABS(ns);
                qi += q[qi];
                q[qi-1] = ns;
            }

            if (nt != 0) {
                if (only_divides) { r=poly(BHEAD 1); ri=r[0]; break; }
                for (int i=0; i<AN.poly_num_vars; i++)
                    r[ri+1+i] = 0;
                r[ri+1+var] = pow;

                for (int i=0; i<ABS(nt); i++)
                    r[ri+1+AN.poly_num_vars+i] = t[i];

```

```
                r[ri] = 2+AN.poly_num_vars+ABS(nt);
                ri += r[ri];
                r[ri-1] = nt;
            }
        }
    }

    q[0] = qi;
    r[0] = ri;

    NumberFree(s,"poly::div_univar");
    NumberFree(t,"poly::div_univar");

    if (q.modp!=0 || ltbinv!=NULL) NumberFree(ltbinv,"poly::div_univar");
}

/*
    #] divmod_univar :
    #[ divmod_heap :
*/

void poly::divmod_heap (const poly &a, const poly &b, poly &q, poly &r, bool only_divides, bool
check_div, bool &div_fail) {

    POLY_GETIDENTITY(a);

    div_fail = false;
    q[0] = r[0] = 1;

    WORD nmodq=0;
    UWORD *modq=NULL;

    WORD nltbinv=0;
    UWORD *ltbinv=NULL;
    LONG oldpWorkPointer = AT.pWorkPointer;

    bool both_mod_small=false;

    if (q.modp!=0) {
        if (q.modn == 1) {
            modq = (UWORD *)&q.modp;
            nmodq = 1;
            if (a.modp>0 && b.modp>0 && a.modn==1 && b.modn==1)
                both_mod_small=true;
        }
        else {
            RaisPowCached(BHEAD q.modp,q.modn,&modq,&nmodq);
        }
        ltbinv = NumberMalloc("poly::div_heap-a");

        if (both_mod_small) {
            WORD ltb = b[b[1]]*b[2+AN.poly_num_vars];
            GetModInverses(ltb + (ltb<0?q.modp:0), q.modp, (WORD*)ltbinv, NULL);
            nltbinv = 1;
        }
        else
            GetLongModInverses(BHEAD (UWORD *)&b[2+AN.poly_num_vars], b[b[1]], modq, nmodq, ltbinv,
&nltbinv, NULL, NULL);
    }

    // allocate heap
    int nb=b.number_of_terms();
    WantAddPointers(nb);
    WORD **heap = AT.pWorkSpace + AT.pWorkPointer;
    AT.pWorkPointer += nb;

    for (int i=0; i<nb; i++)
        heap[i] = (WORD *) NumberMalloc("poly::div_heap-b");
    int nheap = 1;
    heap[0][0] = 1;
    heap[0][1] = 0;
    heap[0][2] = -1;
    WCOPY(&heap[0][3], &a[1], a[1]);
    heap[0][3] = a[a[1]];

    int qi=1, ri=1;

    int s = nb;
    WORD *t = (WORD *) NumberMalloc("poly::div_heap-c");

    // insert contains element that still have to be inserted to the heap
    // (exists to avoid code duplication).
    vector<pair<int,int> > insert;

    while (insert.size()>0 || nheap>0) {

        q.check_memory(qi);
```

```
        r.check_memory(ri);

        // collect a term t for the quotient/remainder
        t[0] = -1;

        do {

            WORD *p = heap[nheap];
            bool this_insert;

            if (insert.empty()) {
                // extract element from the heap and prepare adding new ones
                this_insert = false;

                pop_heap(BHEAD heap, nheap--);
                p = heap[nheap];

                if (t[0] == -1) {
                    WCOPY(t, p, (5+ABS(p[3])+AN.poly_num_vars));
                }
                else {
                    // if both polynomials are modulo p^1, use integer calculus
                    if (both_mod_small) {
                        t[4+AN.poly_num_vars] = ((LONG)t[4+AN.poly_num_vars]*t[3] + p[4+AN.poly_num_vars]*p[3]) % q.modp;
                        if (t[4+AN.poly_num_vars]==0)
                            t[3]=0;
                        else {
                            if (t[4+AN.poly_num_vars] > +q.modp/2) t[4+AN.poly_num_vars] -= q.modp;
                            if (t[4+AN.poly_num_vars] < -q.modp/2) t[4+AN.poly_num_vars] += q.modp;
                            t[3] = SGN(t[4+AN.poly_num_vars]);
                            t[4+AN.poly_num_vars] = ABS(t[4+AN.poly_num_vars]);
                        }
                    }
                    else {
                        // otherwise, use form long calculus
                        AddLong ((UWORD *)&p[4+AN.poly_num_vars],  p[3],
                                 (UWORD *)&t[4+AN.poly_num_vars],  t[3],
                                 (UWORD *)&t[4+AN.poly_num_vars], &t[3]);
                        if (q.modp!=0) TakeNormalModulus((UWORD *)&t[4+AN.poly_num_vars], &t[3],
                                                                           modq, nmodq, NOUNPACK);
                    }
                }
            }
            else {
                // prepare adding an element of insert to the heap
                this_insert = true;

                p[0] = insert.back().first;
                p[1] = insert.back().second;
                insert.pop_back();
            }

            // add elements to the heap
            while (true) {
                // prepare the element
                if (p[1]==0) {
                    p[0] += a[p[0]];
                    if (p[0]==a[0]) break;
                    WCOPY(&p[3], &a[p[0]], a[p[0]]);
                    p[3] = p[2+p[3]];
                }
                else {
                    if (!this_insert)
                        p[1] += q[p[1]];
                    this_insert = false;

                    if (p[1]==qi) { s++; break; }

                    for (int i=0; i<AN.poly_num_vars; i++)
                        p[4+i] = b[p[0]+1+i] + q[p[1]+1+i];

                    // if both polynomials are modulo p^1, use integer calculus
                    if (both_mod_small) {
                        p[4+AN.poly_num_vars] = ((LONG)b[p[0]+1+AN.poly_num_vars]*b[p[0]+b[p[0]]-1]*
                            q[p[1]+1+AN.poly_num_vars]*q[p[1]+q[p[1]]-1]) % q.modp;
                        if (p[4+AN.poly_num_vars]==0)
                            p[3]=0;
                        else {
                            if (p[4+AN.poly_num_vars] > +q.modp/2) p[4+AN.poly_num_vars] -= q.modp;
                            if (p[4+AN.poly_num_vars] < -q.modp/2) p[4+AN.poly_num_vars] += q.modp;
                            p[3] = SGN(p[4+AN.poly_num_vars]);
                            p[4+AN.poly_num_vars] = ABS(p[4+AN.poly_num_vars]);
                        }
                    }
```

```
                    else {
                        // otherwise, use form long calculus
                        MulLong((UWORD *)&b[p[0]+1+AN.poly_num_vars], b[p[0]+b[p[0]]-1],
                                        (UWORD *)&q[p[1]+1+AN.poly_num_vars], q[p[1]+q[p[1]]-1],
                                        (UWORD *)&p[4+AN.poly_num_vars], &p[3]);
                        if (q.modp!=0) TakeNormalModulus((UWORD *)&p[4+AN.poly_num_vars], &p[3],
                                                                                 modq, nmodq, NOUNPACK);
                    }

                    p[3] *= -1;
                }

                // no hashing
                p[2] = -1;

                // add it to a heap element
                swap (heap[nheap],p);
                push_heap(BHEAD heap, ++nheap);
                break;
            }
        }
        while (t[0]==-1 || (nheap>0 && monomial_compare(BHEAD heap[0]+3, t+3)==0));

        if (t[3] == 0) continue;

        // check divisibility
        bool div = true;
        for (int i=0; i<AN.poly_num_vars; i++)
            if (t[4+i] < b[2+i]) div=false;

        if (!div) {
            // not divisible, so append it to the remainder
            if (only_divides) { r=poly(BHEAD 1); ri=r[0]; break; }
            t[4 + AN.poly_num_vars + ABS(t[3])] = t[3];
            t[3] = 2 + AN.poly_num_vars + ABS(t[3]);
            r.termscopy(&t[3], ri, t[3]);
            ri += t[3];
        }
        else {
            // divisible, so divide coefficient as well
            WORD nq, nr;

            if (q.modp==0) {
                DivLong((UWORD *)&t[4+AN.poly_num_vars], t[3],
                                (UWORD *)&b[2+AN.poly_num_vars], b[b[1]],
                                (UWORD *)&q[qi+1+AN.poly_num_vars], &nq,
                                (UWORD *)&r[ri+1+AN.poly_num_vars], &nr);
                if(check_div && nr != 0)
                {
                    // non-zero remainder from coefficient division => result not expressible in integers
                    div_fail = true;
                    break;
                }
            }
            else {
                // if both polynomials are modulo p^1, use integer calculus
                if (both_mod_small) {
                    q[qi+1+AN.poly_num_vars] = ((LONG)t[4+AN.poly_num_vars]*t[3]*ltbinv[0]*nltbinv) % q.modp;
                    if (q[qi+1+AN.poly_num_vars]==0)
                        nq=0;
                    else {
                        if (q[qi+1+AN.poly_num_vars] > +q.modp/2) q[qi+1+AN.poly_num_vars] -= q.modp;
                        if (q[qi+1+AN.poly_num_vars] < -q.modp/2) q[qi+1+AN.poly_num_vars] += q.modp;
                        nq = SGN(q[qi+1+AN.poly_num_vars]);
                        q[qi+1+AN.poly_num_vars] = ABS(q[qi+1+AN.poly_num_vars]);
                    }
                }
                else {
                    // otherwise, use form long calculus
                    MulLong((UWORD *)&t[4+AN.poly_num_vars], t[3], ltbinv, nltbinv, (UWORD *)&q[qi+1+AN.poly_num_vars], &nq);
                    TakeNormalModulus((UWORD *)&q[qi+1+AN.poly_num_vars], &nq, modq, nmodq, NOUNPACK);
                }

                nr=0;
            }

            // add terms to quotient and remainder
            if (nq != 0) {
                int bi = 1;
                for (int j=1; j<s; j++) {
                    bi += b[bi];
                    insert.push_back(make_pair(bi,qi));
                }
```

```
                s=1;

                q[qi] = 2+AN.poly_num_vars+ABS(nq);
                for (int i=0; i<AN.poly_num_vars; i++)
                    q[qi+1+i] = t[4+i] - b[2+i];
                qi += q[qi];
                q[qi-1] = nq;
            }

            if (nr != 0) {
                r[ri] = 2+AN.poly_num_vars+ABS(nr);
                for (int i=0; i<AN.poly_num_vars; i++)
                    r[ri+1+i] = t[4+i];
                ri += r[ri];
                r[ri-1] = nr;
            }
        }
    }

    q[0] = qi;
    r[0] = ri;

    for (int i=0; i<nb; i++)
        NumberFree(heap[i],"poly::div_heap-b");

    NumberFree(t,"poly::div_heap-c");

    if (q.modp!=0||ltbinv!=NULL) NumberFree(ltbinv,"poly::div_heap-a");
    AT.pWorkPointer = oldpWorkPointer;
}

/*
    #] divmod_heap :
    #[ divmod :
*/

void poly::divmod (const poly &a, const poly &b, poly &q, poly &r, bool only_divides) {

    q.modp = r.modp = a.modp;
    q.modn = r.modn = a.modn;

    if (a.is_zero()) {
        q[0]=1;
        r[0]=1;
        return;
    }
    if (b.is_zero()) {
        MLOCK(ErrorMessageLock);
        MesPrint ((char*)"ERROR: division by zero polynomial");
        MUNLOCK(ErrorMessageLock);
        Terminate(1);
    }
    if (b.is_one()) {
        q.check_memory(a[0]);
        q.termscopy(a.terms,0,a[0]);
        r[0]=1;
        return;
    }

    if (b.is_monomial()) {
        divmod_one_term(a,b,q,r,only_divides);
        return;
    }

    int vara = a.is_dense_univariate();
    int varb = b.is_dense_univariate();

    if (vara!=-2 && varb!=-2 && (vara==-1 || varb==-1 || vara==varb)) {
        divmod_univar(a,b,q,r,MaX(vara,varb),only_divides);
    }
    else {
        bool div_fail;
        divmod_heap(a,b,q,r,only_divides,false,div_fail);
    }
}

/*
    #] divmod :
    #[ divides :
*/

// returns whether a exactly divides b
bool poly::divides (const poly &a, const poly &b) {
    POLY_GETIDENTITY(a);
    poly q(BHEAD 0), r(BHEAD 0);
    divmod(b,a,q,r,true);
    return r.is_zero();
```

```
}

/*
    #] divides :
    #[ div :
*/

// the quotient of two polynomials
void poly::div (const poly &a, const poly &b, poly &c) {
    POLY_GETIDENTITY(a);
    poly d(BHEAD 0);
    divmod(a,b,c,d,false);
}

/*
    #] div :
    #[ mod :
*/

// the remainder of division of two polynomials
void poly::mod (const poly &a, const poly &b, poly &c) {
    POLY_GETIDENTITY(a);
    poly d(BHEAD 0);
    divmod(a,b,d,c,false);
}

/*
    #] mod :
    #[ copy operator :
*/

// copy operator
poly & poly::operator= (const poly &a) {
#ifdef POLY_MOVE_DEBUG
    std::cout « "poly copy assign" « std::endl;
#endif
    if (&a != this) {
        modp = a.modp;
        modn = a.modn;
        check_memory(a[0]);
        termscopy(a.terms,0,a[0]);
    }

    return *this;
}

/*
    #] copy operator :
    #[ operator overloads :
*/

// binary operators for polynomial arithmetic
const poly poly::operator+ (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0); add(*this,a,b); return b; }
const poly poly::operator- (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0); sub(*this,a,b); return b; }
const poly poly::operator* (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0); mul(*this,a,b); return b; }
const poly poly::operator/ (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0); div(*this,a,b); return b; }
const poly poly::operator% (const poly &a) const { POLY_GETIDENTITY(a); poly b(BHEAD 0); mod(*this,a,b); return b; }

// assignment operators for polynomial arithmetic
poly& poly::operator+= (const poly &a) { return *this = *this + a; }
poly& poly::operator-= (const poly &a) { return *this = *this - a; }
poly& poly::operator*= (const poly &a) { return *this = *this * a; }
poly& poly::operator/= (const poly &a) { return *this = *this / a; }
poly& poly::operator%= (const poly &a) { return *this = *this % a; }

// comparison operators
bool poly::operator== (const poly &a) const {
    for (int i=0; i<terms[0]; i++)
        if (terms[i] != a[i]) return 0;
    return 1;
}

bool poly::operator!= (const poly &a) const {   return !(*this == a); }

/*
    #] operator overloads :
    #[ num_terms :
*/

int poly::number_of_terms () const {

    int res=0;
```

```
    for (int i=1; i<terms[0]; i+=terms[i])
        res++;
    return res;
}

/*
    #] num_terms :
    #[ first_variable :
*/

// returns the lexicographically first variable of a polynomial
int poly::first_variable () const {

    POLY_GETIDENTITY(*this);

    int var = AN.poly_num_vars;
    for (int j=0; j<var; j++)
        if (terms[2+j]>0) return j;
    return var;
}

/*
    #] first_variable :
    #[ all_variables :
*/

// returns a list of all variables of a polynomial
const vector<int> poly::all_variables () const {

    POLY_GETIDENTITY(*this);

    vector<bool> used(AN.poly_num_vars, false);

    for (int i=1; i<terms[0]; i+=terms[i])
        for (int j=0; j<AN.poly_num_vars; j++)
            if (terms[i+1+j]>0) used[j] = true;

    vector<int> vars;
    for (int i=0; i<AN.poly_num_vars; i++)
        if (used[i]) vars.push_back(i);

    return vars;
}

/*
    #] all_variables :
    #[ sign :
*/

// returns the sign of the leading coefficient
int poly::sign () const {
    if (terms[0]==1) return 0;
    return terms[terms[1]] > 0 ? 1 : -1;
}

/*
    #] sign :
    #[ degree :
*/

// returns the degree of x of a polynomial (deg=-1 iff a=0)
int poly::degree (int x) const {
    int deg = -1;
    for (int i=1; i<terms[0]; i+=terms[i])
        deg = MaX(deg, terms[i+1+x]);
    return deg;
}

/*
    #] degree :
    #[ total_degree :
*/

// returns the total degree of a polynomial (deg=-1 iff a=0)
int poly::total_degree () const {

    POLY_GETIDENTITY(*this);

    int tot_deg = -1;
    for (int i=1; i<terms[0]; i+=terms[i]) {
        int deg=0;
        for (int j=0; j<AN.poly_num_vars; j++)
            deg += terms[i+1+j];
        tot_deg = MaX(deg, tot_deg);
    }
    return tot_deg;
}
```

```
/*
    #] total_degree :
    #[ integer_lcoeff :
*/

// returns the integer coefficient of the leading monomial
const poly poly::integer_lcoeff () const {

    POLY_GETIDENTITY(*this);

    poly res(BHEAD 0);
    res.modp = modp;
    res.modn = modn;

    res.termscopy(&terms[1],1,terms[1]);
    res[0] = res[1] + 1; // length
    for (int i=0; i<AN.poly_num_vars; i++)
        res[2+i] = 0; // powers

    return res;
}

/*
    #] integer_lcoeff :
    #[ coefficient :
*/

// returns the polynomial coefficient of x^n
const poly poly::coefficient (int x, int n) const {

    POLY_GETIDENTITY(*this);

    poly res(BHEAD 0);
    res.modp = modp;
    res.modn = modn;
    res[0] = 1;

    for (int i=1; i<terms[0]; i+=terms[i])
        if (terms[i+1+x] == n) {
            res.check_memory(res[0]+terms[i]);
            res.termscopy(&terms[i], res[0], terms[i]);
            res[res[0]+1+x] -= n;  // power of x
            res[0] += res[res[0]]; // length
        }

    return res;
}

/*
    #] coefficient :
    #[ lcoeff_multivar :
*/

// returns the leading coefficient with the polynomial viewed as a
// polynomial in x, so that the result is a polynomial in the
// remaining variables
const poly poly::lcoeff_univar (int x) const {
    return coefficient(x, degree(x));
}

// returns the leading coefficient with the polynomial viewed as a
// polynomial with coefficients in ZZ[x], so that the result
// polynomial is in ZZ[x] too
const poly poly::lcoeff_multivar (int x) const {

    POLY_GETIDENTITY(*this);

    poly res(BHEAD 0,modp,modn);

    for (int i=1; i<terms[0]; i+=terms[i]) {
        bool same_powers = true;
        for (int j=0; j<AN.poly_num_vars; j++)
            if (j!=x && terms[i+1+j]!=terms[2+j]) {
                same_powers = false;
                break;
            }
        if (!same_powers) break;

        res.check_memory(res[0]+terms[i]);
        res.termscopy(&terms[i],res[0],terms[i]);
        for (int k=0; k<AN.poly_num_vars; k++)
            if (k!=x) res[res[0]+1+k]=0;

        res[0] += terms[i];
    }

```

```
    return res;
}

/*
    #] lcoeff_multivar :
    #[ derivative :
*/

// returns the derivative with respect to x
const poly poly::derivative (int x) const {

    POLY_GETIDENTITY(*this);

    poly b(BHEAD 0);
    int bi=1;

    for (int i=1; i<terms[0]; i+=terms[i]) {

        int power = terms[i+1+x];

        if (power > 0) {
            b.check_memory(bi);
            b.termscopy(&terms[i], bi, terms[i]);
            b[bi+1+x]--;

            WORD nb = b[bi+b[bi]-1];
            Product((UWORD *)&b[bi+1+AN.poly_num_vars], &nb, power);

            b[bi] = 2 + AN.poly_num_vars + ABS(nb);
            b[bi+b[bi]-1] = nb;

            bi += b[bi];
        }
    }

    b[0] = bi;
    b.setmod(modp, modn);
    return b;
}

/*
    #] derivative :
    #[ is_zero :
*/

// returns whether the polynomial is zero
bool poly::is_zero () const {
    return terms[0] == 1;
}

/*
    #] is_zero :
    #[ is_one :
*/

// returns whether the polynomial is one
bool poly::is_one () const {

    POLY_GETIDENTITY(*this);

    if (terms[0] != 4+AN.poly_num_vars) return false;
    if (terms[1] != 3+AN.poly_num_vars) return false;
    for (int i=0; i<AN.poly_num_vars; i++)
        if (terms[2+i] != 0) return false;
    if (terms[2+AN.poly_num_vars]!=1) return false;
    if (terms[3+AN.poly_num_vars]!=1) return false;

    return true;
}

/*
    #] is_one :
    #[ is_integer :
*/

// returns whether the polynomial is an integer
bool poly::is_integer () const {

    POLY_GETIDENTITY(*this);

    if (terms[0] == 1) return true;
    if (terms[0] != terms[1]+1) return false;

    for (int j=0; j<AN.poly_num_vars; j++)
        if (terms[2+j] != 0)
            return false;

```

```
    return true;
}

/*
    #] is_integer :
    #[ is_monomial :
*/

// returns whether the polynomial consist of one term
bool poly::is_monomial () const {
    return terms[0]>1 && terms[0]==terms[1]+1;
}

/*
    #] is_monomial :
    #[ is_dense_univariate :
*/

int poly::is_dense_univariate () const {

    POLY_GETIDENTITY(*this);

    int num_terms=0, res=-1;

    // test univariate
    for (int i=1; i<terms[0]; i+=terms[i]) {
        for (int j=0; j<AN.poly_num_vars; j++)
            if (terms[i+1+j] > 0) {
                if (res == -1) res = j;
                if (res != j) return -2;
            }

        num_terms++;
    }

    // constant polynomial
    if (res == -1) return -1;

    // test density
    int deg = terms[2+res];
    if (2*num_terms < deg+1) return -2;

    return res;
}

/*
    #] is_dense_univariate :
    #[ simple_poly (small) :
*/

// returns the polynomial (x-a)^b mod p^n with a small
const poly poly::simple_poly (PHEAD int x, int a, int b, int p, int n) {

    poly tmp(BHEAD 0,p,n);

    int idx=1;
    tmp[idx++] = 3 + AN.poly_num_vars;                          // length
    for (int i=0; i<AN.poly_num_vars; i++)
        tmp[idx++] = i==x ? 1 : 0;                                // powers
    tmp[idx++] = 1;                                             // coefficient
    tmp[idx++] = 1;                                             // length coefficient

    if (a != 0) {
        tmp[idx++] = 3 + AN.poly_num_vars;                        // length
        for (int i=0; i<AN.poly_num_vars; i++) tmp[idx++] = 0;   // powers
        tmp[idx++] = ABS(a);                                      // coefficient
        tmp[idx++] = -SGN(a);                                     // length coefficient
    }

    tmp[0] = idx;                                               // length

    if (b == 1) return tmp;

    poly res(BHEAD 1,p,n);

    while (b!=0) {
        if (b&1) res*=tmp;
        tmp*=tmp;
        b»=1;
    }

    return res;
}

/*
    #] simple_poly (small) :
    #[ simple_poly (large) :
```

```
*/

// returns the polynomial (x-a)^b mod p^n with a large
const poly poly::simple_poly (PHEAD int x, const poly &a, int b, int p, int n) {

    poly res(BHEAD 1,p,n);
    poly tmp(BHEAD 0,p,n);

    int idx=1;

    tmp[idx++] = 3 + AN.poly_num_vars;                                // length
    for (int i=0; i<AN.poly_num_vars; i++)
        tmp[idx++] = i==x ? 1 : 0;                                    // powers
    tmp[idx++] = 1;                                                   // coefficient
    tmp[idx++] = 1;                                                   // length coefficient

    if (!a.is_zero()) {
        tmp[idx++] = 2 + AN.poly_num_vars + ABS(a[a[0]-1]); // length
        for (int i=0; i<AN.poly_num_vars; i++) tmp[idx++] = 0;        // powers
        for (int i=0; i<ABS(a[a[0]-1]); i++)
            tmp[idx++] = a[2 + AN.poly_num_vars + i];                  // coefficient
        tmp[idx++] = -a[a[0]-1];                                      // length coefficient
    }

    tmp[0] = idx;                                                     // length

    while (b!=0) {
        if (b&1) res*=tmp;
        tmp*=tmp;
        b»=1;
    }

    return res;
}

/*
    #] simple_poly (large) :
    #[ get_variables :
*/

// gets all variables in the expressions and stores them in AN.poly_vars
// (it is assumed that AN.poly_vars=NULL)
void poly::get_variables (PHEAD const vector<WORD *> &es, bool with_arghead, bool sort_vars) {
    assert(AN.poly_vars == NULL);

    AN.poly_num_vars = 0;

    vector<int> vars;
    vector<int> degrees;
    map<int,int> var_to_idx;

    // extract all variables
    for (int ei=0; ei<(int)es.size(); ei++) {
        const WORD *e = es[ei];

        // fast notation
        if (*e == -SNUMBER) {
        }
        else if (*e == -SYMBOL) {
            if (!var_to_idx.count(e[1])) {
                vars.push_back(e[1]);
                var_to_idx[e[1]] = AN.poly_num_vars++;
                degrees.push_back(1);
            }
        }
        // JD: Here we need to check for non-symbol/number terms in fast notation.
        else if (*e < 0) {
            MLOCK(ErrorMessageLock);
            MesPrint ((char*)"ERROR: polynomials and polyratfuns must contain symbols only");
            MUNLOCK(ErrorMessageLock);
            Terminate(1);
        }
        else {
            for (int i=with_arghead ? ARGHEAD : 0; with_arghead ? i<e[0] : e[i]!=0; i+=e[i]) {
                if (i+1<i+e[i]-ABS(e[i+e[i]-1]) && e[i+1]!=SYMBOL) {
                    MLOCK(ErrorMessageLock);
                    MesPrint ((char*)"ERROR: polynomials and polyratfuns must contain symbols only");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(1);
                }

                for (int j=i+3; j<i+e[i]-ABS(e[i+e[i]-1]); j+=2) {
                    if (!var_to_idx.count(e[j])) {
                        vars.push_back(e[j]);
                        var_to_idx[e[j]] = AN.poly_num_vars++;
                        degrees.push_back(e[j+1]);
                    }
```

```
                    else {
                        degrees[var_to_idx[e[j]]] = MaX(degrees[var_to_idx[e[j]]], e[j+1]);
                    }
                }
            }
        }
    }

    // make sure an eventual FACTORSYMBOL appear as last
    if (var_to_idx.count(FACTORSYMBOL)) {
        int i = var_to_idx[FACTORSYMBOL];
        while (i+1<AN.poly_num_vars) {
            swap(vars[i], vars[i+1]);
            swap(degrees[i], degrees[i+1]);
            i++;
        }
        degrees[i] = -1; // makes sure it stays last
    }

    // AN.poly_vars will be deleted in calling functions from polywrap.c
    // For that we use the function poly_free_poly_vars
    // We add one to the allocation to avoid copying in poly_divmod

    if ( (AN.poly_num_vars+1)*sizeof(WORD) > (size_t)(AM.MaxTer) ) {
        // This can happen only in expressions with excessively many variables.
        AN.poly_vars = (WORD *)Malloc1((AN.poly_num_vars+1)*sizeof(WORD), "AN.poly_vars");
        AN.poly_vars_type = 1;
    }
    else {
        AN.poly_vars = TermMalloc("AN.poly_vars");
        AN.poly_vars_type = 0;
    }

    for (int i=0; i<AN.poly_num_vars; i++)
        AN.poly_vars[i] = vars[i];

    if (sort_vars) {
        // bubble sort variables in decreasing order of degree
        // (this seems better for factorization)
        for (int i=0; i<AN.poly_num_vars; i++)
            for (int j=0; j+1<AN.poly_num_vars; j++)
                if (degrees[j] < degrees[j+1]) {
                    swap(degrees[j], degrees[j+1]);
                    swap(AN.poly_vars[j], AN.poly_vars[j+1]);
                }
    }
    else {
        // sort lexicographically
        sort(AN.poly_vars, AN.poly_vars+AN.poly_num_vars - var_to_idx.count(FACTORSYMBOL));
    }
}

/*
    #] get_variables :
    #[ argument_to_poly :
*/

// converts a form expression to a polynomial class "poly"
const poly poly::argument_to_poly (PHEAD WORD *e, bool with_arghead, bool sort_univar, poly *denpoly)
{

    map<int,int> var_to_idx;
    for (int i=0; i<AN.poly_num_vars; i++)
        var_to_idx[AN.poly_vars[i]] = i;

    poly res(BHEAD 0);

    // fast notation
    if (*e == -SNUMBER) {
        if (denpoly!=NULL) *denpoly = poly(BHEAD 1);

        if (e[1] == 0) {
            res[0] = 1;
            return res;
        }
        else {
            res[0] = 4 + AN.poly_num_vars;
            res[1] = 3 + AN.poly_num_vars;
            for (int i=0; i<AN.poly_num_vars; i++)
                res[2+i] = 0;
            res[2+AN.poly_num_vars] = ABS(e[1]);
            res[3+AN.poly_num_vars] = SGN(e[1]);

            return res;
        }
    }

```

```
    if (*e == -SYMBOL) {
        if (denpoly!=NULL) *denpoly = poly(BHEAD 1);

        res[0] = 4 + AN.poly_num_vars;
        res[1] = 3 + AN.poly_num_vars;
        for (int i=0; i<AN.poly_num_vars; i++)
            res[2+i] = 0;
        res[2+var_to_idx.find(e[1])->second] = 1;
        res[2+AN.poly_num_vars] = 1;
        res[3+AN.poly_num_vars] = 1;
        return res;
    }

    // find LCM of denominators of all terms
    WORD nden=1, npro=0, ngcd=0, ndum=0;
    UWORD *den = NumberMalloc("poly::argument_to_poly");
    UWORD *pro = NumberMalloc("poly::argument_to_poly");
    UWORD *gcd = NumberMalloc("poly::argument_to_poly");
    UWORD *dum = NumberMalloc("poly::argument_to_poly");
    den[0]=1;

    for (int i=with_arghead ? ARGHEAD : 0; with_arghead ? i<e[0] : e[i]!=0; i+=e[i]) {
        int ncoe = ABS(e[i+e[i]-1]/2);
        UWORD *coe = (UWORD *)&e[i+e[i]-ncoe-1];
        while (ncoe>0 && coe[ncoe-1]==0) ncoe--;
        MulLong(den,nden,coe,ncoe,pro,&npro);
        GcdLong(BHEAD den,nden,coe,ncoe,gcd,&ngcd);
        DivLong(pro,npro,gcd,ngcd,den,&nden,dum,&ndum);
    }

    if (denpoly!=NULL) *denpoly = poly(BHEAD den, nden);

    int ri=1;

    // ordinary notation
    for (int i=with_arghead ? ARGHEAD : 0; with_arghead ? i<e[0] : e[i]!=0; i+=e[i]) {
        res.check_memory(ri);
        WORD nc = e[i+e[i]-1];                                            // length coefficient
        for (int j=0; j<AN.poly_num_vars; j++)
            res[ri+1+j]=0;                                                // powers=0
        WCOPY(dum, &e[i+e[i]-ABS(nc)], ABS(nc));                          // coefficient to dummy
        nc /= 2;                                                          // remove denominator
        Mully(BHEAD dum, &nc, den, nden);                                 // multiply with
  overall den
        res.termscopy((WORD *)dum, ri+1+AN.poly_num_vars, ABS(nc));       // coefficient to res
        res[ri] = ABS(nc) + AN.poly_num_vars + 2;                         // length
        res[ri+res[ri]-1] = nc;                                           // length coefficient
        for (int j=i+3; j<i+e[i]-ABS(e[i+e[i]-1]); j+=2)
            res[ri+1+var_to_idx.find(e[j])->second] = e[j+1];             // powers
        ri += res[ri];                                                    // length
    }

    res[0] = ri;

    // JD: For univariate cases, check whether the ordering is correct or not.
    // There are various ways to arrive here, but with an incorrect ordering, such
    // as interactions with collect, moving a polyratfun out of another function
    // argument, etc.
    // If the ordering is not correct, make sure we normalize. In multivariate cases,
    // always normalize as before.
    if (sort_univar == false && AN.poly_num_vars == 1) {
        if (res.number_of_terms() >= 2) {
            if(res[2] < res.last_monomial()[2]) {
                sort_univar = true;
            }
        }
    }

    if (sort_univar || AN.poly_num_vars>1) {
        res.normalize();
    }

    NumberFree(den,"poly::argument_to_poly");
    NumberFree(pro,"poly::argument_to_poly");
    NumberFree(gcd,"poly::argument_to_poly");
    NumberFree(dum,"poly::argument_to_poly");

    return res;
}

/*
    #] argument_to_poly :
    #[ poly_to_argument :
*/

// converts a polynomial class "poly" to a form expression
void poly::poly_to_argument (const poly &a, WORD *res, bool with_arghead) {
```

```
    POLY_GETIDENTITY(a);

    // special case: a=0
    if (a[0]==1) {
        if (with_arghead) {
            res[0] = -SNUMBER;
            res[1] = 0;
        }
        else {
            res[0] = 0;
        }
        return;
    }

    if (with_arghead) {
        res[1] = AN.poly_num_vars>1 ? DIRTYFLAG : 0; // dirty flag
        for (int i=2; i<ARGHEAD; i++)
            res[i] = 0;                               // remainder of arghead
    }

    int L = with_arghead ? ARGHEAD : 0;

    for (int i=1; i!=a[0]; i+=a[i]) {

        res[L]=1; // length

        bool first=true;

        for (int j=0; j<AN.poly_num_vars; j++)
            if (a[i+1+j] > 0) {
                if (first) {
                    first=false;
                    res[L+1] = 1; // symbols
                    res[L+2] = 2; // length
                }
                res[L+1+res[L+2]++] = AN.poly_vars[j]; // symbol
                res[L+1+res[L+2]++] = a[i+1+j];  // power
            }

        if (!first) res[L] += res[L+2]; // fix length

        WORD nc = a[i+a[i]-1];
        WCOPY(&res[L+res[L]], &a[i+a[i]-1-ABS(nc)], ABS(nc)); // numerator

        res[L] += ABS(nc);                                // fix length
        memset(&res[L+res[L]], 0, ABS(nc)*sizeof(WORD)); // denominator one
        res[L+res[L]] = 1;                               // denominator one
        res[L] += ABS(nc);                               // fix length
        res[L+res[L]] = SGN(nc) * (2*ABS(nc)+1);         // length of coefficient
        res[L]++;                                        // fix length
        L += res[L];                                     // fix length
    }

    if (with_arghead) {
        res[0] = L;
        // convert to fast notation if possible
        ToFast(res,res);
    }
    else {
        res[L] = 0;
    }
}

/*
    #] poly_to_argument :
    #[ poly_to_argument_with_den :
*/

// converts a polynomial class "poly" divided by a number (nden, den) to a form expression
// cf. poly::poly_to_argument()
void poly::poly_to_argument_with_den (const poly &a, WORD nden, const UWORD *den, WORD *res, bool
with_arghead) {

    POLY_GETIDENTITY(a);

    // special case: a=0
    if (a[0]==1) {
        if (with_arghead) {
            res[0] = -SNUMBER;
            res[1] = 0;
        }
        else {
            res[0] = 0;
        }
        return;
    }
```

```

    if (with_arghead) {
        res[1] = AN.poly_num_vars>1 ? DIRTYFLAG : 0; // dirty flag
        for (int i=2; i<ARGHEAD; i++)
            res[i] = 0;                               // remainder of arghead
    }

    WORD nden1;
    UWORD *den1 = (UWORD *)NumberMalloc("poly_to_argument_with_den");

    int L = with_arghead ? ARGHEAD : 0;

    for (int i=1; i!=a[0]; i+=a[i]) {

        res[L]=1; // length

        bool first=true;

        for (int j=0; j<AN.poly_num_vars; j++)
            if (a[i+1+j] > 0) {
                if (first) {
                    first=false;
                    res[L+1] = 1; // symbols
                    res[L+2] = 2; // length
                }
                res[L+1+res[L+2]++] = AN.poly_vars[j]; // symbol
                res[L+1+res[L+2]++] = a[i+1+j];  // power
            }

        if (!first) res[L] += res[L+2]; // fix length

        // numerator
        WORD nc = a[i+a[i]-1];
        WCOPY(&res[L+res[L]], &a[i+a[i]-1-ABS(nc)], ABS(nc));

        // denominator
        nden1 = nden;
        WCOPY(den1, den, ABS(nden));

        if (nden != 1 || den[0] != 1) {
            // remove gcd(num,den)
            Simplify(BHEAD (UWORD *)&res[L+res[L]], &nc, den1, &nden1);
        }

        Pack((UWORD *)&res[L+res[L]], &nc, den1, nden1);  // format
        res[L] += 2*ABS(nc)+1;                            // fix length
        res[L+res[L]-1] = SGN(nc)*(2*ABS(nc)+1);          // length of coefficient
        L += res[L];                                      // fix length
    }

    NumberFree(den1, "poly_to_argument_with_den");

    if (with_arghead) {
        res[0] = L;
        // convert to fast notation if possible
        ToFast(res,res);
    }
    else {
        res[L] = 0;
    }
}

/*
    #] poly_to_argument_with_den :
    #[ size_of_form_notation :
*/

// the size of the polynomial in form notation (without argheads and fast notation)
int poly::size_of_form_notation () const {

    POLY_GETIDENTITY(*this);

    // special case: a=0
    if (terms[0]==1) return 0;

    int len = 0;

    for (int i=1; i!=terms[0]; i+=terms[i]) {
        len++;
        int npow = 0;
        for (int j=0; j<AN.poly_num_vars; j++)
            if (terms[i+1+j] > 0) npow++;
        if (npow > 0) len += 2*npow + 2;
        len += 2 * ABS(terms[i+terms[i]-1]) + 1;
    }

    return len;
```

```
}

/*
    #] size_of_form_notation :
    #[ size_of_form_notation_with_den :
*/

// the size of the polynomial divided by a number (its size is given by nden)
// in form notation (without argheads and fast notation)
// cf. poly::size_of_form_notation()
int poly::size_of_form_notation_with_den (WORD nden) const {

    POLY_GETIDENTITY(*this);

    // special case: a=0
    if (terms[0]==1) return 0;

    nden = ABS(nden);
    int len = 0;

    for (int i=1; i!=terms[0]; i+=terms[i]) {
        len++;
        int npow = 0;
        for (int j=0; j<AN.poly_num_vars; j++)
            if (terms[i+1+j] > 0) npow++;
        if (npow > 0) len += 2*npow + 2;
        len += 2 * MaX(ABS(terms[i+terms[i]-1]), nden) + 1;
    }

    return len;
}

/*
    #] size_of_form_notation_with_den :
    #[ to_coefficient_list :
*/

// returns the coefficient list of a univariate polynomial
const vector<WORD> poly::to_coefficient_list (const poly &a) {

    POLY_GETIDENTITY(a);

    if (a.is_zero()) return vector<WORD>();

    int x = a.first_variable();
    if (x == AN.poly_num_vars) x=0;

    vector<WORD> res(1+a[2+x],0);

    for (int i=1; i<a[0]; i+=a[i])
        res[a[i+1+x]] = (a[i+a[i]-1] * a[i+a[i]-2]) % a.modp;

    return res;
}

/*
    #] to_coefficient_list :
    #[ coefficient_list_divmod :
*/

// divides two polynomials represented by coefficient lists
const vector<WORD> poly::coefficient_list_divmod (const vector<WORD> &a, const vector<WORD> &b, WORD
p, int divmod) {

    int bsize = (int)b.size();
    while (b[bsize-1]==0) bsize--;

    WORD inv;
    GetModInverses(b[bsize-1] + (b[bsize-1]<0?p:0), p, &inv, NULL);

    vector<WORD> q(a.size(),0);
    vector<WORD> r(a);

    while ((int)r.size() >= bsize) {
        LONG mul = ((LONG)inv * r.back()) % p;
        int off = r.size()-bsize;
        q[off] = mul;
        for (int i=0; i<bsize; i++)
            r[off+i] = (r[off+i] - mul*b[i]) % p;
        while (r.size()>0 && r.back()==0)
            r.pop_back();
    }

    if (divmod==0) {
        while (q.size()>0 && q.back()==0)
            q.pop_back();
        return q;
```

```
    }
    else {
        while (r.size()>0 && r.back()==0)
            r.pop_back();
        return r;
    }
}

/*
    #] coefficient_list_divmod :
    #[ from_coefficient_list :
*/

// converts a coefficient list to a "poly"
const poly poly::from_coefficient_list (PHEAD const vector<WORD> &a, int x, WORD p) {

    poly res(BHEAD 0);
    int ri=1;

    for (int i=(int)a.size()-1; i>=0; i--)
        if (a[i] != 0) {
            res.check_memory(ri);
            res[ri] = AN.poly_num_vars+3;               // length
            for (int j=0; j<AN.poly_num_vars; j++)
                res[ri+1+j]=0;                           // powers
            res[ri+1+x] = i;                            // power of x
            res[ri+1+AN.poly_num_vars] = ABS(a[i]);     // coefficient
            res[ri+1+AN.poly_num_vars+1] = SGN(a[i]);   // sign
            ri += res[ri];
        }

    res[0]=ri;                                      // total length
    res.setmod(p,1);

    return res;
}

/*
    #] from_coefficient_list :
*/
```

4.101 poly.h

```
#pragma once
/* #[ License : */
/*
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

extern "C" {
#include "form3.h"
}

#include <iostream>
#include <string>
#include <vector>

// for debugging
// #define POLY_MOVE_DEBUG

// macros for tform
```

```
#ifndef WITHPTHREADS
#define POLY_GETIDENTITY(X)
#define POLY_STOREIDENTITY
#else
#define POLY_GETIDENTITY(X) ALLPRIVATES *B = (X).Bpointer
#define POLY_STOREIDENTITY Bpointer = B
#endif

// maximum size of the hash table used for multiplication and division

const int POLY_MAX_HASH_SIZE = MiN(1«20, MAXPOSITIVE);

class poly {

public:

    // variables
#ifdef WITHPTHREADS
    ALLPRIVATES *Bpointer;
#endif

    WORD *terms;
    LONG size_of_terms;
    WORD modp,modn;

    // constructors/destructor
    poly (PHEAD int, WORD=-1, WORD=1);
    poly (PHEAD const UWORD *, WORD, WORD=-1, WORD=1);
    poly (const poly &, WORD=-1, WORD=1);
    ~poly ();

    // operators
    poly& operator+= (const poly&);
    poly& operator-= (const poly&);
    poly& operator*= (const poly&);
    poly& operator/= (const poly&);
    poly& operator%= (const poly&);

    const poly operator+ (const poly&) const;
    const poly operator- (const poly&) const;
    const poly operator* (const poly&) const;
    const poly operator/ (const poly&) const;
    const poly operator% (const poly&) const;

    bool operator== (const poly&) const;
    bool operator!= (const poly&) const;
    poly& operator= (const poly &);
    WORD& operator[] (int);
    const WORD& operator[] (int) const;

#if __cplusplus >= 201103L
    // support move semantics
    poly (poly &&, WORD=-1, WORD=1) noexcept;
    poly& operator= (poly &&) noexcept;
#endif

    // memory management
    void termscopy (const WORD *, int, int);
    void check_memory(int);
    void expand_memory(int);

    // check type of polynomial
    bool is_zero () const;
    bool is_one () const;
    bool is_integer () const;
    bool is_monomial () const;
    int is_dense_univariate () const;

    // properties
    int sign () const;
    int degree (int) const;
    int total_degree () const;
    int first_variable () const;
    int number_of_terms () const;
    const std::vector<int> all_variables () const;
    const poly integer_lcoeff () const;
    const poly lcoeff_univar (int) const;
    const poly lcoeff_multivar (int) const;
    const poly coefficient (int, int) const;
    const poly derivative (int) const;

    // modulo calculus
    void setmod(WORD, WORD=1);
    void coefficients_modulo (UWORD *, WORD, bool);

    // simple polynomials
    static const poly simple_poly (PHEAD int, int=0, int=1, int=0, int=1);
```

```
    static const poly simple_poly (PHEAD int, const poly&, int=1, int=0, int=1);

    // conversion from/to form notation
    static void get_variables (PHEAD const std::vector<WORD *> &, bool, bool);
    static const poly argument_to_poly (PHEAD WORD *, bool, bool, poly *den=NULL);
    static void poly_to_argument (const poly &, WORD *, bool);
    static void poly_to_argument_with_den (const poly &, WORD, const UWORD *, WORD *, bool);
    int size_of_form_notation () const;
    int size_of_form_notation_with_den (WORD) const;
    const poly & normalize ();

    // operations for coefficient lists
    static const poly from_coefficient_list (PHEAD const std::vector<WORD> &, int, WORD);
    static const std::vector<WORD> to_coefficient_list (const poly &);
    static const std::vector<WORD> coefficient_list_divmod (const std::vector<WORD> &, const std::vector<WORD> &, WORD, int);

    // string output for debugging
    const std::string to_string() const;

    // monomials
    static int monomial_compare (PHEAD const WORD *, const WORD *);
    WORD last_monomial_index () const;
    WORD* last_monomial () const;
    int compare_degree_vector(const poly &) const;
    std::vector<int> degree_vector() const;
    int compare_degree_vector(const std::vector<int> &) const;

    // (internal) mathematical operations
    static void add (const poly &, const poly &, poly &);
    static void sub (const poly &, const poly &, poly &);
    static void mul (const poly &, const poly &, poly &);
    static void div (const poly &, const poly &, poly &);
    static void mod (const poly &, const poly &, poly &);
    static void divmod (const poly &, const poly &, poly &, poly &, bool only_divides);
    static bool divides (const poly &, const poly &);

    static void mul_one_term (const poly &, const poly &, poly &);
    static void mul_univar (const poly &, const poly &, poly &, int);
    static void mul_heap (const poly &, const poly &, poly &);

    static void divmod_one_term (const poly &, const poly &, poly &, poly &, bool);
    static void divmod_univar (const poly &, const poly &, poly &, poly &, int, bool);
    static void divmod_heap (const poly &, const poly &, poly &, poly &, bool, bool, bool&);

    static void push_heap (PHEAD WORD **, int);
    static void pop_heap (PHEAD WORD **, int);

    PADPOINTER(1,0,2,0);
};

// comparison class for monomials (for std::sort)
class monomial_larger {

public:

#ifndef WITHPTHREADS
    monomial_larger() {}
#else
    ALLPRIVATES *B;
    monomial_larger(ALLPRIVATES *b): B(b) {}
#endif

    bool operator()(const WORD *a, const WORD *b) {
        return poly::monomial_compare(BHEAD a, b) > 0;
    }
};

// stream operator
std::ostream& operator« (std::ostream &, const poly &);

// inline function definitions

#if __cplusplus >= 201103L

inline poly::poly (poly &&a, WORD modp, WORD modn) noexcept {
#ifdef POLY_MOVE_DEBUG
    std::cout « "poly move ctor" « std::endl;
#endif
    POLY_GETIDENTITY(a);
    POLY_STOREIDENTITY;
    terms = a.terms;
    size_of_terms = a.size_of_terms;
    this->modp = a.modp;
    this->modn = a.modn;
    if (modp != -1) {
        setmod(modp,modn);
```

```
    }
    a.terms = nullptr;
    a.size_of_terms = -1;
}

inline poly& poly::operator= (poly &&a) noexcept {
#ifdef POLY_MOVE_DEBUG
    std::cout « "poly move assign" « std::endl;
#endif
    if (this != &a) {
        WORD *old_terms = terms;
        LONG old_size_of_terms = size_of_terms;
        terms = a.terms;
        size_of_terms = a.size_of_terms;
        modp = a.modp;
        modn = a.modn;
        a.terms = old_terms;
        a.size_of_terms = old_size_of_terms;
    }
    return *this;
}

#endif

/*  Checks whether the terms array is large enough to add another
 *  term (of size AM.MaxTal) to the polynomials. In case not, it is
 *  expanded.
 */
inline void poly::check_memory (int i) {
    POLY_GETIDENTITY(*this);
    if (i + 3 + AN.poly_num_vars + AM.MaxTal >= size_of_terms) expand_memory(i + AM.MaxTal);
//  Used to be i+2 but there should also be space for a trailing zero
}

// indexing operators
inline WORD& poly::operator[] (int i) {
    return terms[i];
}

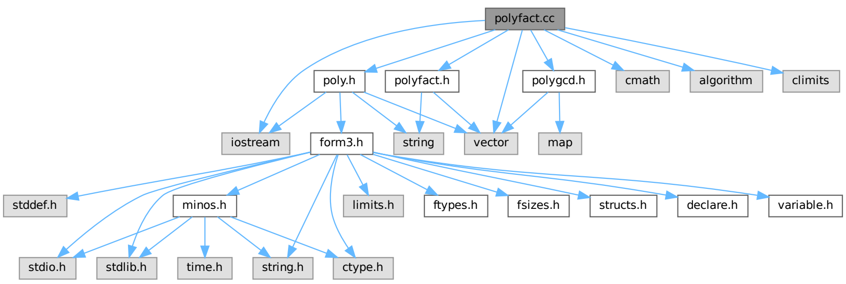
inline const WORD& poly::operator[] (int i) const {
    return terms[i];
}

/*  Copies "num" WORD-sized terms from the pointer "source" to the
 *  current polynomial at index "dest"
 */
inline void poly::termscopy (const WORD *source, int dest, int num) {
#ifdef POLY_MOVE_DEBUG
    std::cout « "poly termscopy: num=" « num « std::endl;
#endif
    memcpy (terms+dest, source, num*sizeof(WORD));
}
```

4.102 polyfact.cc File Reference

```
#include "poly.h"
#include "polygcd.h"
#include "polyfact.h"
#include <cmath>
#include <vector>
#include <iostream>
#include <algorithm>
#include <climits>
```

Include dependency graph for polyfact.cc:



Functions

- ostream & operator<< (ostream &out, const factorized_poly &a)
- template<class T >
  ostream & operator<< (ostream &out, const vector< T > &v)

4.102.1 Detailed Description

Contains the routines for factorizing multivariate polynomials

Definition in file polyfact.cc.

4.102.2 Function Documentation

4.102.2.1 operator<<() [1/2]

```
ostream & operator<< (
            ostream & out,
            const factorized_poly & a )
```

Definition at line 104 of file polyfact.cc.

4.102.2.2 operator<<() [2/2]

```
template<class T >
ostream & operator<< (
            ostream & out,
            const vector< T > & v )
```

Definition at line 109 of file polyfact.cc.

4.103 polyfact.cc

Go to the documentation of this file.

```
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ include :
*/

#include "poly.h"
#include "polygcd.h"
#include "polyfact.h"

#include <cmath>
#include <vector>
#include <iostream>
#include <algorithm>
#include <climits>

//#define DEBUG

#ifdef DEBUG
#include "mytime.h"
#endif

using namespace std;

/*
    #] include :
    #[ tostring :
*/

// Turns a factorized_poly into a readable string
const string factorized_poly::tostring () const {

    // empty
    if (factor.size()==0)
        return "no_factors";

    string res;

    // polynomial
    for (int i=0; i<(int)factor.size(); i++) {
        if (i>0) res += "*";
        res += "(";
        res += poly(factor[i],0,1).to_string();
        res += ")";
        if (power[i]>1) {
            res += "^";
            char tmp[100];
            snprintf (tmp,100,"%i",power[i]);
            res += tmp;
        }
    }

    // modulo p^n
    if (factor[0].modp>0) {
        res += " (mod ";
        char tmp[12];
        snprintf (tmp,12,"%i",factor[0].modp);
```

```
        res += tmp;
        if (factor[0].modn > 1) {
            snprintf (tmp,12,"%i",factor[0].modn);
            res += "^";
            res += tmp;
        }
        res += ")";
    }

    return res;
}

/*
    #] tostring :
    #[ ostream operator :
*/

// ostream operator for outputting a factorized_poly
ostream& operator« (ostream &out, const factorized_poly &a) {
    return out « a.tostring();
}

// ostream operator for outputting a vector<T>
template<class T> ostream& operator« (ostream &out, const vector<T> &v) {
    out«"{";
    for (int i=0; i<(int)v.size(); i++) {
        if (i>0) out«",";
        out«v[i];
    }
    out«"}";
    return out;
}

/*
    #] ostream operator :
    #[ add_factor :
*/

// adds a factor f^p to a factorization
void factorized_poly::add_factor(const poly &f, int p) {
    factor.push_back(f);
    power.push_back(p);
}

/*
    #] add_factor :
    #[ extended_gcd_Euclidean_lifted :
*/

const vector<poly> polyfact::extended_gcd_Euclidean_lifted (const poly &a, const poly &b) {

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: extended_Euclidean_lifted("«a«","«b«")"«endl;
#endif

    POLY_GETIDENTITY(a);

    // Calculate s,t,gcd (mod p) with the Extended Euclidean Algorithm.
    poly s(a,a.modp,1);
    poly t(b,b.modp,1);
    poly sa(BHEAD 1,a.modp,1);
    poly sb(BHEAD 0,b.modp,1);
    poly ta(BHEAD 0,a.modp,1);
    poly tb(BHEAD 1,b.modp,1);

    while (!t.is_zero()) {
        poly x(s/t);
        swap(s -=x*t , t);
        swap(sa-=x*ta, ta);
        swap(sb-=x*tb, tb);
    }

    // Normalize the result.
    sa /= s.integer_lcoeff();
    sb /= s.integer_lcoeff();

    // Lift the result to modulo p^n with p-adic Newton's iteration.
    poly samodp(sa);
    poly sbmodp(sb);
    poly term(BHEAD 1);

    sa.setmod(0,1);
    sb.setmod(0,1);

    poly amodp(a,a.modp,1);
    poly bmodp(b,a.modp,1);
    poly error(poly(BHEAD 1) - sa*a - sb*b);
```

```
    poly p(BHEAD a.modp);

    for (int n=1; n<a.modn && !error.is_zero(); n++) {
        error /= p;
        term *= p;
        poly errormodp(error,a.modp,1);
        poly dsa((samodp * errormodp) % bmodp);
        // equivalent, but faster than the symmetric
        // poly dsb((sbmodp * errormodp) % amodp);
        poly dsb((errormodp - dsa*amodp) / bmodp);
        sa += term * dsa;
        sb += term * dsb;
        error -= a*dsa + b*dsb;
    }

    sa.setmod(a.modp,a.modn);
    sb.setmod(a.modp,a.modn);

    // Output the result
    vector<poly> res;
    res.push_back(sa);
    res.push_back(sb);

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : extended_Euclidean_lifted("«a«","«b«") = "«res«endl;
#endif

    return res;
}

/*
    #] extended_gcd_Euclidean_lifted :
    #[ solve_Diophantine_univariate :
*/

const vector<poly> polyfact::solve_Diophantine_univariate (const vector<poly> &a, const poly &b) {

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: solve_Diophantine_univariate(" «a«","«b«")"«endl;
#endif

    POLY_GETIDENTITY(b);

    vector<poly> s(1,b);

    for (int i=0; i+1<(int)a.size(); i++) {
        poly A(BHEAD 1,b.modp,b.modn);
        for (int j=i+1; j<(int)a.size(); j++) A *= a[j];

        vector<poly> t(extended_gcd_Euclidean_lifted(a[i],A));
        poly prev(s.back());
        s.back() = t[1] * prev % a[i];
        s.push_back(t[0] * prev % A);
    }

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : solve_Diophantine_univariate(" «a«","«b«") = "«s«endl;
#endif

    return s;
}

/*
    #] solve_Diophantine_univariate :
    #[ solve_Diophantine_multivariate :
*/

const vector<poly> polyfact::solve_Diophantine_multivariate (const vector<poly> &a, const poly &b,
const vector<int> &x, const vector<int> &c, int d) {

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: solve_Diophantine_multivariate("
«a«","«b«","«x«","«c«","«d«")"«endl;
#endif

    POLY_GETIDENTITY(b);

    if (b.is_zero()) return vector<poly>(a.size(),poly(BHEAD 0));

    if (x.size() == 1) return solve_Diophantine_univariate(a,b);

    // Reduce the polynomial with the homomorphism <xm-c{m-1}>
    poly simple(poly::simple_poly(BHEAD x.back(),c.back()));

    vector<poly> ared (a);
    for (int i=0; i<(int)ared.size(); i++)
        ared[i] %= simple;
```

```
    poly bred(b % simple);
    vector<int> xred(x.begin(),x.end()-1);
    vector<int> cred(c.begin(),c.end()-1);

    // Solve the equation in one less variable
    vector<poly> s(solve_Diophantine_multivariate(ared,bred,xred,cred,d));
    if (s == vector<poly>()) return vector<poly>();

    // Cache the Ai = product(aj | j!=i).
    vector<poly> A(a.size(), poly(BHEAD 1,b.modp,b.modn));
    for (int i=0; i<(int)a.size(); i++)
        for (int j=0; j<(int)a.size(); j++)
            if (i!=j) A[i] *= a[j];

    // Add the powers (xm-c{m-1})^k with ideal-adic Newton iteration.
    poly term(BHEAD 1,b.modp,b.modn);

    poly error(b);
    for (int i=0; i<(int)A.size(); i++)
        error -= s[i] * A[i];

    for (int deg=1; deg<=d; deg++) {

        if (error.is_zero()) break;

        error /= simple;
        term *= simple;

        vector<poly> ds(solve_Diophantine_multivariate(ared, error%simple, xred, cred, d));
        if (ds == vector<poly>()) return vector<poly>();

        for (int i=0; i<(int)s.size(); i++) {
            s[i] += ds[i] * term;
            error -= ds[i] * A[i];
        }
    }

    if (!error.is_zero()) return vector<poly>();

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : solve_Diophantine_multivariate("
«a«","«b«","«x«","«c«","«d«") = "«s«endl;
#endif

    return s;
}

/*
    #] solve_Diophantine_multivariate :
    #[ lift_coefficients :
*/

const vector<poly> polyfact::lift_coefficients (const poly &_A, const vector<poly> &_a) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: lift_coefficients("«_A«","«_a«")"«endl;
#endif

    POLY_GETIDENTITY(_A);

    poly A(_A);
    vector<poly> a(_a);
    poly term(BHEAD 1);

    int x = A.first_variable();

    // Replace the leading term of all factors with lterm(A) mod p
    poly lead(A.integer_lcoeff());
    for (int i=0; i<(int)a.size(); i++) {
        a[i] *= lead / a[i].integer_lcoeff();
        if (i>0) A*=lead;
    }

    // Solve Diophantine equation
    vector<poly> s(solve_Diophantine_univariate(a,poly(BHEAD 1,A.modp,1)));

    // Replace the leading term of all factors with lterm(A) mod p^n
    for (int i=0; i<(int)a.size(); i++) {
        a[i].setmod(A.modp,A.modn);
        a[i] += (lead - a[i].integer_lcoeff()) * poly::simple_poly(BHEAD x,0,a[i].degree(x));
    }

    // Calculate the error, express it in terms of ai and add corrections.
    for (int k=2; k<=A.modn; k++) {
        term *= poly(BHEAD A.modp);
```

```

        poly error(BHEAD -1);
        for (int i=0; i<(int)a.size(); i++) error *= a[i];
        error += A;
        if (error.is_zero()) break;

        error /= term;
        error.setmod(A.modp,1);

        for (int i=0; i<(int)a.size(); i++)
            a[i] += term * (error * s[i] % a[i]);
    }

    // Fix leading coefficients by dividing out integer contents.
    for (int i=0; i<(int)a.size(); i++)
        a[i] /= polygcd::integer_content(poly(a[i],0,1));

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : lift_coefficients("«_A«","«_a«") = "«a«endl;
#endif

    return a;
}

/*
    #] lift_coefficients :
        #[ predetermine :
*/

void polyfact::predetermine (int dep, const vector<vector<int> > &state, vector<vector<vector<int> > > &terms, vector<int> &term, int sumdeg) {
    // store the term
    if (dep == (int)state.size()) {
        terms[sumdeg].push_back(term);
        return;
    }

    // recursively create new terms
    term.push_back(0);

    for (int deg=0; sumdeg+deg<(int)state[dep].size(); deg++)
        if (state[dep][deg] > 0) {
            term.back() = deg;
            predetermine(dep+1, state, terms, term, sumdeg+deg);
        }

    term.pop_back();
}

/*
        #] predetermine :
    #[ lift_variables :
*/

const vector<poly> polyfact::lift_variables (const poly &A, const vector<poly> &_a, const vector<int> &x, const vector<int> &c, const vector<poly> &lc) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: lift_variables("«A«","«_a«","«x«","«c«","«lc«")\n";
#endif

    // If univariate, don't lift
    if (x.size()<=1) return _a;

    POLY_GETIDENTITY(A);

    vector<poly> a(_a);

    // First method: predetermine coefficients

    // check feasibility, otherwise it tries too many possibilities
    int cnt = POLYFACT_MAX_PREDETERMINATION;
    for (int i=0; i<(int)a.size(); i++) {
        if (a[i].number_of_terms() == 0) return vector<poly>();
        cnt /= a[i].number_of_terms();
    }

    if (cnt>0) {
        // state[n][d]: coefficient of x^d in a[n] is
        // 0: non-existent, 1: undetermined, 2: determined
        int D = A.degree(x[0]);
        vector<vector<int> > state(a.size(), vector<int>(D+1, 0));

        for (int i=0; i<(int)a.size(); i++)
            for (int j=1; j<a[i][0]; j+=a[i][j])
                state[i][a[i][j+1+x[0]]] = j==1 ? 2 : 1;

```

```
        // collect all products of terms
        vector<vector<vector<int> > > terms(D+1);
        vector<int> term;
        predetermine(0,state,terms,term);

        // count the number of undetermined coefficients
        vector<int> num_undet(terms.size(),0);
        for (int i=0; i<(int)terms.size(); i++)
            for (int j=0; j<(int)terms[i].size(); j++)
                for (int k=0; k<(int)terms[i][j].size(); k++)
                    if (state[k][terms[i][j][k]] == 1) num_undet[i]++;

        // replace the current leading coefficients by the correct ones
        for (int i=0; i<(int)a.size(); i++)
            a[i] += (lc[i] - a[i].lcoeff_univar(x[0])) * poly::simple_poly(BHEAD x[0],0,a[i].degree(x[0]));

        bool changed;
        do {
            changed = false;

            for (int i=0; i<(int)terms.size(); i++) {
                // is only one coefficient in a equation is undetermined, solve
                // the equation to determine this coefficient
                if (num_undet[i] == 1) {
                    // generate equation
                    poly lhs(BHEAD 0), rhs(A.coefficient(x[0],i), A.modp, A.modn);
                    int which_idx=-1, which_deg=-1;
                    for (int j=0; j<(int)terms[i].size(); j++) {
                        poly coeff(BHEAD 1, A.modp, A.modn);
                        bool undet=false;
                        for (int k=0; k<(int)terms[i][j].size(); k++) {
                            if (state[k][terms[i][j][k]] == 1) {
                                undet = true;
                                which_idx=k;
                                which_deg=terms[i][j][k];
                            }
                            else
                                coeff *= a[k].coefficient(x[0], terms[i][j][k]);
                        }
                        if (undet)
                            lhs = coeff;
                        else
                            rhs -= coeff;
                    }
                    // solve equation
                    if (A.modn > 1) rhs.setmod(0,1);
                    if (lhs.is_zero() || !(rhs%lhs).is_zero()) return vector<poly>();
                    a[which_idx] += (rhs / lhs - a[which_idx].coefficient(x[0],which_deg)) * poly::simple_poly(BHEAD x[0],0,which_deg);
                    state[which_idx][which_deg] = 2;

                    // update number of undetermined coefficients
                    for (int j=0; j<(int)terms.size(); j++)
                        for (int k=0; k<(int)terms[j].size(); k++)
                            if (terms[j][k][which_idx] == which_deg)
                                num_undet[j]--;

                    changed = true;
                }
            }
        }
        while (changed);

        // if this is the complete result, skip lifting
        poly check(BHEAD 1, A.modn>1?0:A.modp, 1);
        for (int i=0; i<(int)a.size(); i++)
            check *= a[i];

        if (check == A) return a;
    }

    // Second method: Hensel lifting

    // Calculate A and lc's modulo Ii = <xi-c{i-1],...,xm-c{m-1}> (for i=2,...,m)
    vector<poly> simple(x.size(), poly(BHEAD 0));
    for (int i=(int)x.size()-2; i>=0; i--)
        simple[i] = poly::simple_poly(BHEAD x[i+1],c[i],1);

    // Calculate the maximum degree of A in x2,...,xm
    int maxdegA=0;
    for (int i=1; i<(int)x.size(); i++)
        maxdegA = MaX(maxdegA, A.degree(x[i]));

    // Iteratively add the variables x2,...,xm
    for (int xi=1; xi<(int)x.size(); xi++) {
        // replace the current leading coefficients by the correct ones
```

```
        for (int i=0; i<(int)a.size(); i++)
            a[i] += (lc[i] - a[i].lcoeff_univar(x[0])) * poly::simple_poly(BHEAD x[0],0,a[i].degree(x[0]));

        vector<poly> anew(a);
        for (int i=0; i<(int)anew.size(); i++)
            for (int j=xi-1; j<(int)c.size(); j++)
                anew[i] %= simple[j];

        vector<int> xnew(x.begin(), x.begin()+xi);
        vector<int> cnew(c.begin(), c.begin()+xi-1);
        poly term(BHEAD 1,A.modp,A.modn);

        // Iteratively add the powers xi^k
        for (int deg=1, maxdeg=A.degree(x[xi]); deg<=maxdeg; deg++) {

            term *= simple[xi-1];

            // Calculate the error, express it in terms of ai and add corrections.
            poly error(BHEAD -1,A.modp,A.modn);
            for (int i=0; i<(int)a.size(); i++) error *= a[i];
            error += A;
            for (int i=xi; i<(int)c.size(); i++) error %= simple[i];

            if (error.is_zero()) break;

            error /= term;
            error %= simple[xi-1];

            vector<poly> s(solve_Diophantine_multivariate(anew,error,xnew,cnew,maxdegA));
            if (s == vector<poly>()) return vector<poly>();

            for (int i=0; i<(int)a.size(); i++)
                a[i] += s[i] * term;
        }

        // check whether PRODUCT(a[i]) = A mod <xi-c{i-1},...,xm-c{m-1]> over the integers or ZZ/p
        poly check(BHEAD -1, A.modn>1?0:A.modp, 1);
        for (int i=0; i<(int)a.size(); i++) check *= a[i];
        check += A;
        for (int i=xi; i<(int)c.size(); i++) check %= simple[i];
        if (!check.is_zero()) return vector<poly>();
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : lift_variables("«A«","«_a«","«x«","«c«","«lc«") = " « a « endl;
#endif

    return a;
}

/*
    #] lift_variables :
    #[ choose_prime :
*/

WORD polyfact::choose_prime (const poly &a, const vector<int> &x, WORD p) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: choose_prime("«a«","«x«","«p«")"«endl;
#endif

    POLY_GETIDENTITY(a);

    poly icont_lcoeff(polygcd::integer_content(a.lcoeff_univar(x[0])));

    if (p==0) p = POLYFACT_FIRST_PRIME;

    poly icont_lcoeff_modp(BHEAD 0);

    do {

        bool is_prime;

        do {
            p += 2;
            is_prime = true;
            for (int d=2; d*d<=p; d++)
                if (p%d==0) { is_prime=false; break; }
        }
        while (!is_prime);

        icont_lcoeff_modp = icont_lcoeff;
        icont_lcoeff_modp.setmod(p,1);
    }
    while (icont_lcoeff_modp.is_zero());
```

```
#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : choose_prime("«a«","«x«",?) = "«p«endl;
#endif

    return p;
}

/*
    #] choose_prime :
    #[ choose_prime_power :
*/
WORD polyfact::choose_prime_power (const poly &a, WORD p) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: choose_prime_power("«a«","«p«")"«endl;
#endif

    POLY_GETIDENTITY(a);

    // analyse the polynomial for calculating the bound
    double maxdegree=0, maxlogcoeff=0, numterms=0;

    for (int i=1; i<a[0]; i+=a[i]) {
        for (int j=0; j<AN.poly_num_vars; j++)
            maxdegree = MaX(maxdegree, a[i+1+j]);

        maxlogcoeff = MaX(maxlogcoeff,
                                          log(1.0+(UWORD)a[i+a[i]-2]) +            // most significant digit + 1
                                          BITSINWORD*log(2.0)*(ABS(a[i+a[i]-1])-1)); // number of digits
        numterms++;
    }

    WORD res = (WORD)ceil((log((sqrt(5.0)+1)/2)*maxdegree + maxlogcoeff + 0.5*log(numterms)) / log((double)p));

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: choose_prime_power("«a«","«p«") = "«res«endl;
#endif

    return res;
}

/*
    #] choose_prime_power :
    #[ choose_ideal :
*/
const vector<int> polyfact::choose_ideal (const poly &a, int p, const factorized_poly &lc, const vector<int> &x) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: polyfact::choose_ideal("
             «a«","«p«","«lc«","«x«")"«endl;
#endif

    if (x.size()==1) return vector<int>();

    POLY_GETIDENTITY(a);

    vector<int> c(x.size()-1);

    int dega = a.degree(x[0]);
    poly amodI(a);

    // choose random c
    for (int i=0; i<(int)c.size(); i++) {
        c[i] = 1 + wranf(BHEAD0) % ((p-1) / POLYFACT_IDEAL_FRACTION);
        amodI %= poly::simple_poly(BHEAD x[i+1],c[i],1);
    }

    poly amodIp(amodI);
    amodIp.setmod(p,1);

    // check if leading coefficient is non-zero [equivalent to degree=old_degree]
    if (amodIp.degree(x[0]) != dega)
        return c = vector<int>();

    // check if leading coefficient is squarefree [equivalent to gcd(a,a')==const]
    if (!polygcd::gcd_Euclidean(amodIp, amodIp.derivative(x[0])).is_integer())
        return c = vector<int>();

    if (a.modp>0 && a.modn==1) return c;

```

```
    // check for unique prime factors in each factor lc[i] of the leading coefficient
    vector<poly> d(1, polygcd::integer_content(amodI));

    for (int i=0; i<(int)lc.factor.size(); i++) {
        // constant factor
        if (i==0 && lc.factor[i].is_integer()) {
            d[0] *= lc.factor[i];
            continue;
        }

        // factor modulo I
        poly q(lc.factor[i]);
        for (int j=0; j<(int)c.size(); j++)
            q %= poly::simple_poly(BHEAD x[j+1],c[j]);
        if (q.sign() == -1) q *= poly(BHEAD -1);

        // divide out common factors
        for (int j=(int)d.size()-1; j>=0; j--) {
            poly r(d[j]);
            while (!r.is_one()) {
                r = polygcd::integer_gcd(r,q);
                q /= r;
            }
        }

        // check whether there is some factor left
        if (q.is_one()) return vector<int>();
        d.push_back(q);
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : polyfact::choose_ideal("
             «a«","«p«","«lc«","«x«") = "«c«endl;
#endif

    return c;
}

/*
    #] choose_ideal :
    #[ squarefree_factors_Yun :
*/

const factorized_poly polyfact::squarefree_factors_Yun (const poly &_a) {

    factorized_poly res;
    poly a(_a);

    int pow = 1;
    int x = a.first_variable();

    poly b(a.derivative(x));
    poly c(polygcd::gcd(a,b));

    while (true) {
        a /= c;
        b /= c;
        b -= a.derivative(x);
        if (b.is_zero()) break;
        c = polygcd::gcd(a,b);
        if (!c.is_one()) res.add_factor(c,pow);
        pow++;
    }

    if (!a.is_one()) res.add_factor(a,pow);
    return res;
}

/*
    #] squarefree_factors_Yun :
    #[ squarefree_factors_modp :
*/

const factorized_poly polyfact::squarefree_factors_modp (const poly &_a) {

    factorized_poly res;
    poly a(_a);

    int pow = 1;
    int x = a.first_variable();
    poly b(a.derivative(x));

    // poly contains terms of the form c(x)^n (n!=c*p)
    if (!b.is_zero()) {
        poly c(polygcd::gcd(a,b));
        a /= c;

```

```
        while (!a.is_one()) {
            b = polygcd::gcd(a,c);
            a /= b;
            if (!a.is_one()) res.add_factor(a,pow);
            pow++;
            a = b;
            c /= a;
        }

        a = c;
    }

    // polynomial contains terms of the form c(x)^p
    if (!a.is_one()) {
        for (int i=1; i<a[1]; i+=a[i])
            a[i+1+x] /= a.modp;
        factorized_poly res2(squarefree_factors(a));
        for (int i=0; i<(int)res2.factor.size(); i++) {
            res.factor.push_back(res2.factor[i]);
            res.power.push_back(a.modp*res2.power[i]);
        }
    }

    return res;
}

/*
    #] squarefree_factors_modp :
    #[ squarefree_factors :
*/
const factorized_poly polyfact::squarefree_factors (const poly &a) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: squarefree_factors("«a«")\n";
#endif

    if (a.is_one()) return factorized_poly();

    factorized_poly res;

    if (a.modp==0)
        res = squarefree_factors_Yun(a);
    else
        res = squarefree_factors_modp(a);

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : squarefree_factors("«a«") = "«res«"\n";
#endif

    return res;
}

/*
    #] squarefree_factors :
    #[ Berlekamp_Qmatrix :
*/
const vector<vector<WORD> > polyfact::Berlekamp_Qmatrix (const poly &_a) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: Berlekamp_Qmatrix("«_a«")\n";
#endif

    if (_a.all_variables() == vector<int>())
        return vector<vector<WORD> >(0);

    POLY_GETIDENTITY(_a);

    poly a(_a);
    int x = a.first_variable();
    int n = a.degree(x);
    int p = a.modp;

    poly lc(a.integer_lcoeff());
    a /= lc;

    vector<vector<WORD> > Q(n, vector<WORD>(n));

    // c is the vector of coefficients of the polynomial a
    vector<WORD> c(n+1,0);
    for (int j=1; j<a[0]; j+=a[j])
        c[a[j+1+x]] = a[j+1+AN.poly_num_vars] * a[j+2+AN.poly_num_vars];

    // d is the vector of coefficients of x^i mod a, starting with i=0
    vector<WORD> d(n,0);
    d[0]=1;
```

```

    for (int i=0; i<=(n-1)*p; i++) {
        // store the coefficients of x^(i*p) mod a
        if (i%p==0) Q[i/p] = d;

        // transform d=x^i mod a into d=x^(i+1) mod a
        vector<WORD> e(n);
        for (int j=0; j<n; j++) {
            e[j] = (-(LONG)d[n-1]*c[j] + (j>0?d[j-1]:0)) % p;
            if (e[j]<0) e[j]+=p;
        }

        d=e;
    }

    // Q = Q - I
    for (int i=0; i<n; i++)
        Q[i][i] = (Q[i][i] - 1 + p) % p;

    // Gaussian elimination
    for (int i=0; i<n; i++) {
        // Find pivot
        int ii=i; while (ii<n && Q[i][ii]==0) ii++;
        if (ii==n) continue;

        for (int k=0; k<n; k++)
            swap(Q[k][ii],Q[k][i]);

        // normalize row i, which becomes the pivot
        WORD mul;
        GetModInverses (Q[i][i], p, &mul, NULL);
        vector<int> idx;

        for (int k=0; k<n; k++) if (Q[k][i] != 0) {
            // store indices of non-zero elements for sparse matrices
            idx.push_back(k);
            Q[k][i] = ((LONG)Q[k][i] * mul) % p;
        }

        // reduce
        for (int j=0; j<n; j++)
            if (j!=i && Q[i][j]!=0) {
                LONG mul = Q[i][j];
                for (int k=0; k<(int)idx.size(); k++) {
                    Q[idx[k]][j] = (Q[idx[k]][j] - mul*Q[idx[k]][i]) % p;
                    if (Q[idx[k]][j] < 0) Q[idx[k]][j]+=p;
                }
            }
    }

    for (int i=0; i<n; i++) {
        // Q = Q - I
        Q[i][i] = Q[i][i]-1;

        // reduce all coefficients in the range 0,1,...,p-1
        for (int j=0; j<n; j++) {
            Q[i][j] = -Q[i][j]%p;
            if (Q[i][j]<0) Q[i][j]+=p;
        }
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : Berlekamp_Qmatrix("«_a«") = "«Q«"\n";
#endif

    return Q;
}

/*
    #] Berlekamp_Qmatrix :
    #[ Berlekamp_find_factors :
*/

const vector<poly> polyfact::Berlekamp_find_factors (const poly &_a, const vector<vector<WORD> > &_q)
{

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: Berlekamp_find_factors("«_a«","«_q«")\n";
#endif

    if (_a.all_variables() == vector<int>()) return vector<poly>(1,_a);

    POLY_GETIDENTITY(_a);

    vector<vector<WORD> > q=_q;

    int rank=0;
```

```
    for (int i=0; i<(int)q.size(); i++)
        if (q[i]!=vector<WORD>(q[i].size(),0)) rank++;

    poly a(_a);
    int x = a.first_variable();
    int n = a.degree(x);
    int p = a.modp;

    a /= a.integer_lcoeff();

    // Vector of factors, represented as dense polynomials mod p
    vector<vector<WORD> > fac(1, vector<WORD>(n+1,0));

    fac[0] = poly::to_coefficient_list(a);
    bool finished=false;

    // Loop over the columns of q + constant, i.e., an exhaustive list of possible factors
    for (int i=1; i<n && !finished; i++) {
        if (q[i] == vector<WORD>(n,0)) continue;

        for (int s=0; s<p && !finished; s++) {
            for (int j=0; j<(int)fac.size() && !finished; j++) {
                vector<WORD> c = polygcd::coefficient_list_gcd(fac[j], q[i], p);

                // If a non-trivial factor is found, add it to the list
                if (c.size()!=1 && c.size()!=fac[j].size()) {
                    fac.push_back(c);
                    fac[j] = poly::coefficient_list_divmod(fac[j], c, p, 0);
                    if ((int)fac.size() == rank) finished=true;
                }
            }

            // Increase the constant term by one
            q[i][0] = (q[i][0]+1) % p;
        }
    }

    // Convert the densely represented polynomials to sparse ones
    vector<poly> res(fac.size(),poly(BHEAD 0, p));
    for (int i=0; i<(int)fac.size(); i++)
        res[i] = poly::from_coefficient_list(BHEAD fac[i],x,p);

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : Berlekamp_find_factors("«_a«","«_q«") = "«res«"\n";
#endif

    return res;
}

/*
    #] Berlekamp_find_factors :
    #[ combine_factors :
*/

const vector<poly> polyfact::combine_factors (const poly &a, const vector<poly> &f) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: combine_factors("«a«","«f«")\n";
#endif

    POLY_GETIDENTITY(a);

    poly a0(a,0,1);
    vector<poly> res;

    int num_used = 0;
    vector<bool> used(f.size(), false);

    // Loop over all bitmasks with num=1,2,...,size(factors)/2 bits
    // set, that contain only unused factors
    for (int num=1; num<=(int)(f.size() - num_used)/2; num++) {
        vector<int> next(f.size() - num_used, 0);
        for (int i=0; i<num; i++) next[next.size()-1-i] = 1;

        do {
            poly fac(BHEAD 1,a.modp,a.modn);
            for (int i=0, j=0; i<(int)f.size(); i++)
                if (!used[i] && next[j++]) fac *= f[i];
            fac /= fac.integer_lcoeff();
            fac *= a.integer_lcoeff();
            fac /= polygcd::integer_content(poly(fac,0,1));

            if ((a0 % fac).is_zero()) {
                res.push_back(fac);
                for (int i=0, j=0; i<(int)f.size(); i++)
                    if (!used[i]) used[i] = next[j++];
                num_used += num;
```

```
                num--;
                break;
            }
        }
        while (next_permutation(next.begin(), next.end()));
    }

    // All unused factors together form one more factor
    if (num_used != (int)f.size()) {
        poly fac(BHEAD 1,a.modp,a.modn);
        for (int i=0; i<(int)f.size(); i++)
            if (!used[i]) fac *= f[i];
        fac /= fac.integer_lcoeff();
        fac *= a.integer_lcoeff();
        fac /= polygcd::integer_content(poly(fac,0,1));
        res.push_back(fac);
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : combine_factors("«a«","«f«") = "«res«"\n";
#endif

    return res;
}

/*
    #] combine_factors :
    #[ factorize_squarefree :
*/

const vector<poly> polyfact::factorize_squarefree (const poly &a, const vector<int> &x) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: factorize_squarefree("«a«")\n";
#endif

    POLY_GETIDENTITY(a);

    WORD p=a.modp, n=a.modn;

 try_again:

    int bestp=0;
    int min_factors = INT_MAX;
    poly amodI(BHEAD 0), bestamodI(BHEAD 0);
    vector<int> c,d,bestc,bestd;
    vector<vector<WORD> > q,bestq;

    // Factorize leading coefficient
    factorized_poly lc(factorize(a.lcoeff_univar(x[0])));

    // Try a number of primes
    int prime_tries = 0;

    while (prime_tries<POLYFACT_NUM_CONFIRMATIONS && min_factors>1) {
        if (a.modp == 0) {
            p = choose_prime(a,x,p);
            n = 0;
            if (a.degree(x[0]) % p == 0) continue;

            // Univariate case: check whether the polynomial mod p is squarefree
            // Multivariate case: this check is done after choosing I (for efficiency)
            if (x.size()==1) {
                poly amodp(a,p,1);
                if (polygcd::gcd_Euclidean(amodp, amodp.derivative(x[0])).degree(x[0]) != 0)
                    continue;
            }
        }

        // Try a number of ideals
        if (x.size()>1)
            for (int ideal_tries=0; ideal_tries<POLYFACT_MAX_IDEAL_TRIES; ideal_tries++) {
                c = choose_ideal(a,p,lc,x);
                if (c.size()>0) break;
            }

        if (x.size()==1 || c.size()>0) {
            amodI = a;
            for (int i=0; i<(int)c.size(); i++)
                amodI %= poly::simple_poly(BHEAD x[i+1],c[i]);

            // Determine Q-matrix and its rank. Smaller rank is better.
            q = Berlekamp_Qmatrix(poly(amodI,p,1));
            int rank=0;
            for (int i=0; i<(int)q.size(); i++)
                if (q[i]!=vector<WORD>(q[i].size(),0)) rank++;

```

```
            if (rank<min_factors) {
                bestp=p;
                bestc=c;
                bestq=q;
                bestamodI=amodI;
                min_factors = rank;
                prime_tries = 0;
            }

            if (rank==min_factors)
                prime_tries++;

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  (A) : factorize_squarefree("«a«
        ") try p=" « p « " #factors=" « rank « " (min="«min_factors«"x"«prime_tries«")" « endl;
#endif
        }
    }

    p=bestp;
    c=bestc;
    q=bestq;
    amodI=bestamodI;

    // Determine to which power of p to lift
    if (n==0) {
        n = choose_prime_power(amodI,p);
        n = MaX(n, choose_prime_power(a,p));
    }

    amodI.setmod(p,n);

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  (B) : factorize_squarefree("«a«
        ") chosen c = " « c « ", p^n = "«p«"^"«n«endl;
    cout « "*** [" « thetime() « "]  (C) : factorize_squarefree("«a«
        ") #factors = " « min_factors « endl;
#endif

    // Find factors
    vector<poly> f(Berlekamp_find_factors(poly(amodI,p,1),q));

    // Lift coefficients
    if (f.size() > 1) {
        f = lift_coefficients(amodI,f);
        if (f==vector<poly>()) {
#ifdef DEBUG
            cout « "factor_squarefree failed (lift_coeff step) : " « endl;
#endif
            goto try_again;
        }

        // Combine factors
        if (a.modp==0) f = combine_factors(amodI,f);
    }

    // Lift variables
    if (f.size() == 1)
        f = vector<poly>(1, a);

    if (x.size() > 1 && f.size() > 1) {

        // The correct leading coefficients of the factors can be
        // reconstructed from prime number factors of the leading
        // coefficients modulo I. This is possible since all factors of
        // the leading coefficient have unique prime factors for the ideal
        // I is chosen as such.

        poly amodI(a);
        for (int i=0; i<(int)c.size(); i++)
            amodI %= poly::simple_poly(BHEAD x[i+1],c[i]);
        poly delta(polygcd::integer_content(amodI));

        vector<poly> lcmodI(lc.factor.size(), poly(BHEAD 0));
        for (int i=0; i<(int)lc.factor.size(); i++) {
            lcmodI[i] = lc.factor[i];
            for (int j=0; j<(int)c.size(); j++)
                lcmodI[i] %= poly::simple_poly(BHEAD x[j+1],c[j]);
        }

        vector<poly> correct_lc(f.size(), poly(BHEAD 1,p,n));

        for (int j=0; j<(int)f.size(); j++) {
            poly lc_f(f[j].integer_lcoeff() * delta);
            WORD nlc_f = lc_f[lc_f[1]];
            poly quo(BHEAD 0),rem(BHEAD 0);
            WORD nquo,nrem;
```

```
        
            for (int i=(int)lcmodI.size()-1; i>=0; i--) {

                if (i==0 && lc.factor[i].is_integer()) continue;

                do {
                    DivLong((UWORD *)&lc_f[2+AN.poly_num_vars], nlc_f,
                                    (UWORD *)&lcmodI[i][2+AN.poly_num_vars], lcmodI[i][lcmodI[i][1]],
                                    (UWORD *)&quo[0], &nquo,
                                    (UWORD *)&rem[0], &nrem);

                    if (nrem == 0) {
                        correct_lc[j] *= lc.factor[i];
                        lc_f.termscopy(&quo[0], 2+AN.poly_num_vars, ABS(nquo));
                        nlc_f = nquo;
                    }
                }
                while (nrem == 0);
            }
        }

        for (int i=0; i<(int)correct_lc.size(); i++) {
            poly correct_modI(correct_lc[i]);
            for (int j=0; j<(int)c.size(); j++)
                correct_modI %= poly::simple_poly(BHEAD x[j+1],c[j]);

            poly d(polygcd::integer_gcd(correct_modI, f[i].integer_lcoeff()));
            correct_lc[i] *= f[i].integer_lcoeff() / d;
            delta /= correct_modI / d;
            f[i] *= correct_modI / d;
        }

        // increase n, because of multiplying with delta
        if (!delta.is_one()) {
            poly deltapow(BHEAD 1);
            for (int i=1; i<(int)correct_lc.size(); i++)
                    deltapow *= delta;
            while (!deltapow.is_zero()) {
                deltapow /= poly(BHEAD p);
                n++;
            }

            for (int i=0; i<(int)f.size(); i++) {
                f[i].modn = n;
                correct_lc[i].modn = n;
            }
        }

        poly aa(a,p,n);

        for (int i=0; i<(int)correct_lc.size(); i++) {
            correct_lc[i] *= delta;
            f[i] *= delta;
            if (i>0) aa *= delta;
        }

        f = lift_variables(aa,f,x,c,correct_lc);

        for (int i=0; i<(int)f.size(); i++)
            if (a.modp == 0)
                f[i] /= polygcd::integer_content(poly(f[i],0,1));
            else
                f[i] /= polygcd::content_univar(f[i], x[0]);

        if (f==vector<poly>()) {
#ifdef DEBUG
            cout « "factor_squarefree failed (lift_var step)" « endl;
#endif
            goto try_again;
        }
    }

    // set modulus of the factors correctly
    if (a.modp==0)
        for (int i=0; i<(int)f.size(); i++)
            f[i].setmod(0,1);

    // Final check (not sure if this is necessary, but it doesn't hurt)
    poly check(BHEAD 1,a.modp,a.modn);
    for (int i=0; i<(int)f.size(); i++)
        check *= f[i];

    if (check != a) {
#ifdef DEBUG
        cout « "factor_squarefree failed (final check) : " « f « endl;
#endif
        goto try_again;
```

```
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : factorize_squarefree("«a«","«x«","«c«") = "«f«"\n";
#endif

    return f;
}

/*
    #] factorize_squarefree :
    #[ factorize :
*/

const factorized_poly polyfact::factorize (const poly &a) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: factorize("«a«")\n";
#endif
    vector<int> x = a.all_variables();

    // No variables, so just one factor
    if (x.size() == 0) {
        factorized_poly res;
        if (a.is_one()) return res;
        res.add_factor(a,1);
        return res;
    }

    // Remove content
    poly conta(polygcd::content_univar(a,x[0]));

    factorized_poly faca(factorize(conta));

    poly ppa(a / conta);

    // Find a squarefree factorization
    factorized_poly b(squarefree_factors(ppa));

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  ... : factorize("«a«") : SFF = "«b«"\n";
#endif

    // Factorize each squarefree factor and build the "factorized_poly"
    for (int i=0; i<(int)b.factor.size(); i++) {

        vector<poly> c(factorize_squarefree(b.factor[i], x));

        for (int j=0; j<(int)c.size(); j++) {
            faca.factor.push_back(c[j]);
            faca.power.push_back(b.power[i]);
        }
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : factorize("«a«") = "«faca«"\n";
#endif

    return faca;
}

/*
    #] factorize :
*/
```

4.104 polyfact.h

```
#pragma once
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
```

```
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#include <vector>
#include <string>

// First prime modulo which factorization is tried. Too small results
// in more unsuccessful attempts; too large is slower.
const int POLYFACT_FIRST_PRIME = 17;

// Fraction of [1,p) that is used for substitutions of variables. Too
// small results in more unsuccessful attempts; too large is slower.
const int POLYFACT_IDEAL_FRACTION = 5;

// Number of ideals that are tried before failure due to unlucky
// choices is accepted.
const int POLYFACT_MAX_IDEAL_TRIES = 3;

// Number of confirmations for the minimal number of factors before
// Hensel lifting is started.
const int POLYFACT_NUM_CONFIRMATIONS = 3;

// Maximum number of equations for predetermination of coefficients
// for multivariate Hensel lifting
const int POLYFACT_MAX_PREDETERMINATION = 10000;

class poly;

class factorized_poly {
    /*   Class for representing a factorized polynomial
     *   The polynomial is given by: PRODUCT(factor[i] ^ power[i])
     */
public:
    std::vector<poly> factor;
    std::vector<int> power;

    void add_factor(const poly &f, int p=1);
    const std::string tostring () const;
    friend std::ostream& operator« (std::ostream &out, const poly &p);
};

std::ostream& operator« (std::ostream &out, const factorized_poly &a);

namespace polyfact {

    // factorization routine
    const factorized_poly factorize (const poly &a);

    // methods for squarefree factorization
    const factorized_poly squarefree_factors (const poly &a);
    const factorized_poly squarefree_factors_Yun (const poly &a);
    const factorized_poly squarefree_factors_modp (const poly &a);

    // methods for choosing suitable reductions
    const std::vector<poly> factorize_squarefree (const poly &a, const std::vector<int> &x);
    WORD choose_prime (const poly &a, const std::vector<int> &x, WORD p=0);
    WORD choose_prime_power (const poly &a, WORD p);
    const std::vector<int> choose_ideal (const poly &a, int p, const factorized_poly &lc, const std::vector<int> &x);

    // methods for univariate factorization
    const std::vector<std::vector<WORD> > Berlekamp_Qmatrix (const poly &a);
    const std::vector<poly> Berlekamp_find_factors (const poly &a, const std::vector<std::vector<WORD> > &Q);
    const std::vector<poly> combine_factors (const poly &a, const std::vector<poly> &f);

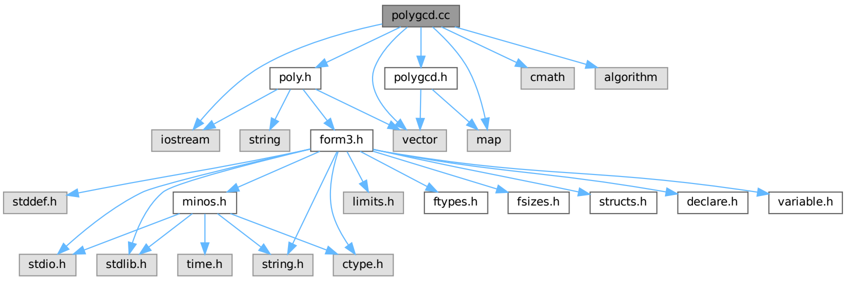
    // methods for Hensel lifting
    const std::vector<poly> extended_gcd_Euclidean_lifted (const poly &a, const poly &b);
    const std::vector<poly> solve_Diophantine_univariate (const std::vector<poly> &a, const poly &b);
    const std::vector<poly> solve_Diophantine_multivariate (const std::vector<poly> &a, const poly &b, const std::vector<int> &x, const std::vector<int> &c, int d);
    const std::vector<poly> lift_coefficients (const poly &a, const std::vector<poly> &f);
    const std::vector<poly> lift_variables (const poly &a, const std::vector<poly> &f, const std::vector<int> &x, const std::vector<int> &c, const std::vector<poly> &lc);
    void predetermine (int dep, const std::vector<std::vector<int> > &state, std::vector<std::vector<std::vector<int> > > &terms, std::vector<int> &term, int sumdeg=0);

}
```

4.105 polygcd.cc File Reference

```
#include "poly.h"
#include "polygcd.h"
#include <iostream>
#include <vector>
#include <cmath>
#include <map>
#include <algorithm>
```

Include dependency graph for polygcd.cc:



Data Structures

- struct BracketInfo

Functions

- bool gcd_heuristic_possible (const poly &a)
- const poly gcd_linear_helper (const poly &a, const poly &b)

4.105.1 Detailed Description

Contains the routines for calculating greatest commons divisors of multivariate polynomials

Definition in file polygcd.cc.

4.105.2 Function Documentation

4.105.2.1 gcd_heuristic_possible()

```
bool gcd_heuristic_possible (
            const poly & a )
```

Heuristic greatest common divisor of multivariate polynomials

4.105.3 Description

Checks whether the heuristic seems possible by estimating

MAX_{terms} (coeff ^ PROD_{i=1..#vars} (pow_i+1))

and comparing this with GCD_HEURISTIC_MAX_DIGITS.

4.105.4 Notes

- For small polynomials, this consumes time and never triggers.

Definition at line 1142 of file polygcd.cc.

4.105.4.1 gcd_linear_helper()

```
const poly gcd_linear_helper (
            const poly & a,
            const poly & b )
```

Definition at line 1403 of file polygcd.cc.

4.106 polygcd.cc

Go to the documentation of this file.

```

/* #[ License : */
/*
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 */
/* #] License : */
/*
    #[ include :
*/

#include "poly.h"
#include "polygcd.h"

#include <iostream>
#include <vector>
#include <cmath>
#include <map>
#include <algorithm>
```

```
//#define DEBUG
//#define DEBUGALL

#ifdef DEBUG
#include "mytime.h"
#endif

using namespace std;

/*
    #] include :
    #[ ostream operator :
*/

#ifdef DEBUG
// ostream operator for outputting vector<T>s   for debugging purposes
template<class T> ostream& operator« (ostream &out, const vector<T> &x) {
    out«"{";
    for (int i=0; i<(int)x.size(); i++) {
        if (i>0) out«",";
        out«x[i];
    }
    out«"}";
    return out;
}
#endif

/*
    #] ostream operator :
    #[ integer_gcd :
*/

const poly polygcd::integer_gcd (const poly &a, const poly &b) {

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: integer_gcd(" « a « "," « b « ")" « endl;
#endif

    POLY_GETIDENTITY(a);

    if (a.is_zero()) return b;
    if (b.is_zero()) return a;

    poly c(BHEAD 0, a.modp, a.modn);
    WORD nc;

    GcdLong(BHEAD
                        (UWORD *)&a[AN.poly_num_vars+2],a[a[0]-1],
                        (UWORD *)&b[AN.poly_num_vars+2],b[b[0]-1],
                        (UWORD *)&c[AN.poly_num_vars+2],&nc);

    WORD x = 2 + AN.poly_num_vars + ABS(nc);
    c[1] = x;      // term length
    c[0] = x+1;    // total length
    c[x] = nc;     // coefficient length

    for (int i=0; i<AN.poly_num_vars; i++)
        c[2+i] = 0; // powers

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : integer_gcd(" « a « "," « b « ") = " « c « endl;
#endif

    return c;
}

/*
    #] integer_gcd :
    #[ integer_content :
*/

const poly polygcd::integer_content (const poly &a) {

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: integer_content(" « a « ")" « endl;
#endif

    POLY_GETIDENTITY(a);

    if (a.modp>0) return a.integer_lcoeff();

    poly c(BHEAD 0, 0, 1);
    WORD *d = (WORD *)NumberMalloc("polygcd::integer_content");
    WORD nc=0;

    for (int i=0; i<AN.poly_num_vars; i++)
```

```
        c[2+i] = 0;

    for (int i=1; i<a[0]; i+=a[i]) {

        WCOPY(d,&c[2+AN.poly_num_vars],nc);

        GcdLong(BHEAD (UWORD *)d, nc,
                        (UWORD *)&a[i+1+AN.poly_num_vars], a[i+a[i]-1],
                        (UWORD *)&c[2+AN.poly_num_vars], &nc);

        WORD x = 2 + AN.poly_num_vars + ABS(nc);
        c[1] = x;    // term length
        c[0] = x+1; // total length
        c[x] = nc;  // coefficient length
    }

    if (a.sign() != c.sign()) c *= poly(BHEAD -1);

    NumberFree(d,"polygcd::integer_content");

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : integer_content(" « a « ") = " « c « endl;
#endif

    return c;
}

/*
    #] integer_content :
    #[ content_univar :
*/

const poly polygcd::content_univar (const poly &a, int x) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: content_univar(" « a « "," « string(1,'a'+x) « ")" « endl;
#endif

    POLY_GETIDENTITY(a);

    poly res(BHEAD 0, a.modp, a.modn);

    for (int i=1; i<a[0];) {
        poly b(BHEAD 0, a.modp, a.modn);
        int deg = a[i+1+x];

        for (; i<a[0] && a[i+1+x]==deg; i+=a[i]) {
            b.check_memory(b[0]+a[i]);
            b.termscopy(&a[i],b[0],a[i]);
            b[b[0]+1+x] = 0;
            b[0] += a[i];
        }

        res = gcd(res, b);

        if (res.is_integer()) {
            res = integer_content(a);
            break;
        }
    }

    if (a.sign() != res.sign()) res *= poly(BHEAD -1);

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : content_univar(" « a « "," « string(1,'a'+x) « ") = " « res
« endl;
#endif

    return res;
}

/*
    #] content_univar :
        #[ content_multivar :
*/

const poly polygcd::content_multivar (const poly &a, int x) {

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: content_multivar(" « a « "," « string(1,'a'+x) « ")" «
endl;
#endif

    POLY_GETIDENTITY(a);

    poly res(BHEAD 0, a.modp, a.modn);

```

```
    for (int i=1,j; i<a[0]; i=j) {
        poly b(BHEAD 0, a.modp, a.modn);

        for (j=i; j<a[0]; j+=a[j]) {
            bool same_powers = true;
            for (int k=0; k<AN.poly_num_vars; k++)
                if (k!=x && a[i+1+k]!=a[j+1+k]) {
                    same_powers = false;
                    break;
                }
            if (!same_powers) break;

            b.check_memory(b[0]+a[j]);
            b.termscopy(&a[j],b[0],a[j]);
            for (int k=0; k<AN.poly_num_vars; k++)
                if (k!=x) b[b[0]+1+k]=0;

            b[0] += a[j];
        }

        res = gcd_Euclidean(res, b);

        if (res.is_integer()) {
            res = poly(BHEAD a.sign(),a.modp,a.modn);
            break;
        }
    }

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : content_multivar(" « a « "," « string(1,'a'+x) « ") = " « res « endl;
#endif

    return res;
}

/*
    #] content_multivar :
        #[ coefficient_list_gcd :
*/

const vector<WORD> polygcd::coefficient_list_gcd (const vector<WORD> &_a, const vector<WORD> &_b, WORD p) {

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: coefficient_list_gcd("«_a«","«_b«","«p«")"«endl;
#endif

    vector<WORD> a(_a), b(_b);

    while (b.size() != 0) {
      a = poly::coefficient_list_divmod(a,b,p,1);
        swap(a,b);
    }

    while (a.back()==0) a.pop_back();

    WORD inv;
    GetModInverses(a.back() + (a.back()<0?p:0), p, &inv, NULL);

    for (int i=0; i<(int)a.size(); i++)
        a[i] = (LONG)inv*a[i] % p;

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : coefficient_list_gcd("«_a«","«_b«","«p«") = "«a«endl;
#endif

    return a;
}

/*
        #] coefficient_list_gcd :
    #[ gcd_Euclidean :
*/

const poly polygcd::gcd_Euclidean (const poly &a, const poly &b) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: gcd_Euclidean("«a«","«b«")"«endl;
#endif

    POLY_GETIDENTITY(a);

    if (a.is_zero()) return b;
    if (b.is_zero()) return a;
    if (a.is_integer() || b.is_integer())   return integer_gcd(a,b);

```

```
    poly res(BHEAD 0);

    if (a.is_dense_univariate()>=-1 && b.is_dense_univariate()>=-1) {
        vector<WORD> coeff = coefficient_list_gcd(poly::to_coefficient_list(a),

poly::to_coefficient_list(b), a.modp);
        res = poly::from_coefficient_list(BHEAD coeff, a.first_variable(), a.modp);
    }
    else {
        res = a;
        poly rem(b);
        while (!rem.is_zero())
            swap(res%=rem, rem);
        res /= res.integer_lcoeff();
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : gcd_Euclidean("«a«","«b«") = "«res«endl;
#endif

    return res;
}

/*
    #] gcd_Euclidean :
        #[ chinese_remainder :
*/

const poly polygcd::chinese_remainder (const poly &a1, const poly &m1, const poly &a2, const poly &m2)
{

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  CALL: chinese_remainder(" « a1 « "," « m1 « "," « a2 « "," « m2 «
")" « endl;
#endif

    POLY_GETIDENTITY(a1);

    WORD nx,ny,nz;
    UWORD *x = (UWORD *)NumberMalloc("polygcd::chinese_remainder");
    UWORD *y = (UWORD *)NumberMalloc("polygcd::chinese_remainder");
    UWORD *z = (UWORD *)NumberMalloc("polygcd::chinese_remainder");

    GetLongModInverses(BHEAD (UWORD *)&m1[2+AN.poly_num_vars], m1[m1[1]],
                                          (UWORD *)&m2[2+AN.poly_num_vars], m2[m2[1]],
                                          (UWORD *)x, &nx, NULL, NULL);

    AddLong((UWORD *)&a2[2+AN.poly_num_vars], a2.is_zero() ? 0 :  a2[a2[1]],
                    (UWORD *)&a1[2+AN.poly_num_vars], a1.is_zero() ? 0 : -a1[a1[1]],
                    y, &ny);

    MulLong (x,nx,y,ny,z,&nz);
    MulLong (z,nz,(UWORD *)&m1[2+AN.poly_num_vars],m1[m1[1]],x,&nx);

    AddLong (x,nx,(UWORD *)&a1[2+AN.poly_num_vars], a1.is_zero() ? 0 : a1[a1[1]],y,&ny);

    MulLong ((UWORD *)&m1[2+AN.poly_num_vars], m1[m1[1]],
                     (UWORD *)&m2[2+AN.poly_num_vars], m2[m2[1]],
                     (UWORD *)z,&nz);

    TakeNormalModulus (y,&ny,(UWORD *)z,nz,NOUNPACK);

    poly res(BHEAD y,ny);

    NumberFree(x,"polygcd::chinese_remainder");
    NumberFree(y,"polygcd::chinese_remainder");
    NumberFree(z,"polygcd::chinese_remainder");

#ifdef DEBUGALL
    cout « "*** [" « thetime() « "]  RES : chinese_remainder(" « a1 « "," « m1 « "," « a2 « "," « m2 «
") = " « res « endl;
#endif

    return res;
}

/*
    #] chinese_remainder :
        #[ substitute :
*/

const poly polygcd::substitute(const poly &a, int x, int c) {

    POLY_GETIDENTITY(a);

    poly b(BHEAD 0);

```

```
    if (a.is_zero()) {
        return b;
    }

    bool zero=true;
    int bi=1;

    // cache size is bounded by the degree in x, twice the number of terms of
    // the polynomial and a constant
    vector<WORD> cache(min(a.degree(x)+1,min(2*a.number_of_terms(),
POLYGCD_RAISPOWMOD_CACHE_SIZE)), 0);

    for (int ai=1; ai<=a[0]; ai+=a[ai]) {
        // last term or different power, then add term to b iff non-zero
        if (!zero) {
            bool add=false;
            if (ai==a[0])
                add=true;
            else {
                for (int i=0; i<AN.poly_num_vars; i++)
                    if (i!=x && a[ai+1+i]!=b[bi+1+i]) {
                        zero=true;
                        add=true;
                        break;
                    }
            }

            if (add) {
                if (b[bi+AN.poly_num_vars+1] < 0)
                    b[bi+AN.poly_num_vars+1] += a.modp;
                bi+=b[bi];
            }

            if (ai==a[0]) break;
        }

        b.check_memory(bi);

        // create new term in b
        if (zero) {
            b[bi] = 3+AN.poly_num_vars;
            for (int i=0; i<AN.poly_num_vars; i++)
                b[bi+1+i] = a[ai+1+i];
            b[bi+1+x] = 0;
            b[bi+AN.poly_num_vars+1] = 0;
            b[bi+AN.poly_num_vars+2] = 1;
        }

        // add term of a to the current term in b
        LONG coeff = a[ai+1+AN.poly_num_vars] * a[ai+2+AN.poly_num_vars];
        int pow = a[ai+1+x];

        if (pow<(int)cache.size()) {
            if (cache[pow]==0)
                cache[pow] = RaisPowMod(c, pow, a.modp);
            coeff = (coeff * cache[pow]) % a.modp;
        }
        else {
            coeff = (coeff * RaisPowMod(c, pow, a.modp)) % a.modp;
        }

        b[bi+AN.poly_num_vars+1] = (coeff + b[bi+AN.poly_num_vars+1]) % a.modp;
        if (b[bi+AN.poly_num_vars+1] != 0) zero=false;
    }

    b[0]=bi;
    b.setmod(a.modp);

    return b;
}

/*
        #] substitute :
        #[ sparse_interpolation helper functions :
*/

// Returns a list of size #terms(a) with entries PROD(ci^powi, i=2..n)
const vector<int> polygcd::sparse_interpolation_get_mul_list (const poly &a, const vector<int> &x,
const vector<int> &c) {
    // cache size for variable x is bounded by the degree in x, twice
    // the number of terms of the polynomial and a constant
    vector<vector<WORD> > cache(c.size());
    int max_cache_size = min(2*a.number_of_terms(),POLYGCD_RAISPOWMOD_CACHE_SIZE);
    for (int i=0; i<(int)c.size(); i++)
        cache[i] = vector<WORD>(min(a.degree(x[i+1])+1,max_cache_size), 0);

```

```
    vector<int> res;
    for (int i=1; i<a[0]; i+=a[i]) {
        LONG coeff=1;
        for (int j=0; j<(int)c.size(); j++) {
            int pow = a[i+1+x[j+1]];
            if (pow<(int)cache[j].size()) {
                if (cache[j][pow]==0)
                    cache[j][pow] = RaisPowMod(c[j], pow, a.modp);
                coeff = (coeff * cache[j][pow]) % a.modp;
            }
            else {
                coeff = (coeff * RaisPowMod(c[j], pow, a.modp)) % a.modp;
            }
        }
        res.push_back(coeff);
    }
    return res;
}

// Multiplies the coefficients of a with the entries of mul
void polygcd::sparse_interpolation_mul_poly (poly &a, const vector<int> &mul) {
    for (int i=1,j=0; i<a[0]; i+=a[i],j++)
        a[i+a[i]-2] = ((LONG)a[i+a[i]-2]*mul[j]) % a.modp;
}

// Sets all coefficients to the range 0..modp-1 and the powers of x2...xn to 0
const poly polygcd::sparse_interpolation_reduce_poly (const poly &a, const vector<int> &x) {
    poly res(a);
    for (int i=1; i<a[0]; i+=a[i]) {
        for (int j=1; j<(int)x.size(); j++)
            res[i+1+x[j]]=0;
        if (res[i+a[i]-1]==-1) {
            res[i+a[i]-1]=1;
            res[i+a[i]-2]=a.modp-res[i+a[i]-2];
        }
    }
    return res;
}

// Collects entries with equal powers, so that the result is a proper polynomial
const poly polygcd::sparse_interpolation_fix_poly (const poly &a, int x) {

    POLY_GETIDENTITY(a);
    poly res(BHEAD 0,a.modp,1);

    int j=1;
    bool newterm=true;

    for (int i=1; i<a[0]; i+=a[i]) {
        if (newterm)
            res.termscopy(&a[i], j, a[i]);
        else
            res[j+res[j]-2] = ((LONG)res[j+res[j]-2] + a[i+a[i]-2]) % a.modp;

        newterm = i+a[i] == a[0] || res[j+1+x] != a[i+a[i]+1+x];
        if (newterm && res[j+res[j]-2]!=0) j += res[j];
    }

    res[0]=j;
    return res;
}

/*
        #] sparse_interpolation helper functions :
        #[ gcd_modular_sparse_interpolation :
*/

const poly polygcd::gcd_modular_sparse_interpolation (const poly &origa, const poly &origb, const
vector<int> &x, const poly &s) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: gcd_modular_sparse_interpolation("
             « origa « "," « origb « "," « x « "," « "," « s «")" « endl;
#endif

    POLY_GETIDENTITY(origa);

    // strip multivariate content
    poly conta(content_multivar(origa,x.back()));
    poly contb(content_multivar(origb,x.back()));
    poly gcdconts(gcd_Euclidean(conta,contb));
    const poly& a = conta.is_one() ? origa : origa/conta;
    const poly& b = contb.is_one() ? origb : origb/contb;

    // for non-monic cases, we need to normalize with the gcd of the lcoeffs of a poly in x[0]
    // or else the shape fitting does not work.
    // FIXME: the current implementation still rejects some valid shapes.
```

```
    poly lcgcd(BHEAD 1, a.modp);
    if (!s.lcoeff_univar(x[0]).is_integer()) {
        lcgcd = gcd_modular_dense_interpolation(a.lcoeff_univar(x[0]), b.lcoeff_univar(x[0]), x,
poly(BHEAD 0));
    }

    // reduce polynomials
    poly ared(sparse_interpolation_reduce_poly(a,x));
    poly bred(sparse_interpolation_reduce_poly(b,x));
    poly sred(sparse_interpolation_reduce_poly(s,x));
    poly lred(sparse_interpolation_reduce_poly(lcgcd,x));

    // set all coefficients to 1
    for (int i=1; i<sred[0]; i+=sred[i]) {
        sred[i+sred[i]-2] = sred[i+sred[i]-1] = 1;
    }

    // generate random numbers and check there this set doesn't result
    // in a singular matrix
    vector<int> c(x.size()-1);
    vector<int> smul;

    bool duplicates;
    do {
        for (int i=0; i<(int)c.size(); i++)
            c[i] = 1 + wranf(BHEAD0) % (a.modp-1);
        smul = sparse_interpolation_get_mul_list(s,x,c);

        duplicates = false;

        int fr=0,to=0;
        for (int i=1; i<s[0];) {
            int pow = s[i+1+x[0]];
            while (i<s[0] && s[i+1+x[0]]==pow) i+=s[i], to++;
            for (int j=fr; j<to; j++)
                for (int k=fr; k<j; k++)
                    if (smul[j] == smul[k])
                        duplicates = true;
            fr=to;
        }
    }
    while (duplicates);

    // get the lists to multiply the polynomials with every iteration
    vector<int> amul(sparse_interpolation_get_mul_list(a,x,c));
    vector<int> bmul(sparse_interpolation_get_mul_list(b,x,c));
    vector<int> lmul(sparse_interpolation_get_mul_list(lcgcd,x,c));

    vector<vector<vector<LONG> > > M;
    vector<vector<LONG> > V;

    int maxMsize=0;

    // create (empty) matrices
    for (int i=1; i<s[0]; i+=s[i]) {
        if (i==1 || s[i+1+x[0]]!=s[i+1+x[0]-s[i]]) {
            M.push_back(vector<vector<LONG> >());
            V.push_back(vector<LONG>());
        }
        M.back().push_back(vector<LONG>());
        V.back().push_back(0);
        maxMsize = max(maxMsize, (int)M.back().size());
    }

    // generate linear equations
    for (int numg=0; numg<maxMsize; numg++) {

        poly amodI(sparse_interpolation_fix_poly(ared,x[0]));
        poly bmodI(sparse_interpolation_fix_poly(bred,x[0]));
        poly lmodI(sparse_interpolation_fix_poly(lred,x[0]));

        // A fix for non-monic gcds. This could be slow if lmodI has many terms,
        // since it overfits the gcd now. Another gcd has to be run to remove the
        // extra terms.
        poly gcd(lmodI * gcd_Euclidean(amodI,bmodI));

        // if correct gcd
        if (!gcd.is_zero() && gcd[2+x[0]]==sred[2+x[0]]) {

            // for each power in the gcd, generate an equation if needed
            int gi=1, midx=0;

            for (int si=1; si<s[0];) {
                // if the term exists, set Vi=coeff, otherwise Vi remains 0
                if (gi<gcd[0] && gcd[gi+1+x[0]]==sred[si+1+x[0]]) {
                    if (numg < (int)V[midx].size())
                        V[midx][numg] = gcd[gi+gcd[gi]-1]*gcd[gi+gcd[gi]-2];
```

```
                    gi += gcd[gi];
                }

                // add the coefficients of s to the matrix M
                for (int i=0; i<(int)M[midx].size(); i++) {
                    if (numg < (int)M[midx].size())
                        M[midx][numg].push_back(sred[si+1+AN.poly_num_vars]);
                    si += s[si];
                }

                midx++;
            }
        }
        else {
            // incorrect gcd
            if (!gcd.is_zero() && gcd[2+x[0]]<sred[2+x[0]])
                return poly(BHEAD 0);
            numg--;
        }

        // multiply polynomials by the lists to obtain new ones
        sparse_interpolation_mul_poly(ared,amul);
        sparse_interpolation_mul_poly(bred,bmul);
        sparse_interpolation_mul_poly(sred,smul);
        sparse_interpolation_mul_poly(lred,lmul);
    }

    // solve the linear equations
    for (int i=0; i<(int)M.size(); i++) {
        int n = M[i].size();

        // Gaussian elimination
        for (int j=0; j<n; j++) {
            for (int k=0; k<j; k++) {
                LONG x = M[i][j][k];
                for (int l=k; l<n; l++)
                    M[i][j][l] = (M[i][j][l] - M[i][k][l]*x) % a.modp;
                V[i][j] = (V[i][j] - V[i][k]*x) % a.modp;
            }

            // normalize row
            WORD x = M[i][j][j]; // WORD for GetModInverses
            GetModInverses(x + (x<0?a.modp:0), a.modp, &x, NULL);
            for (int k=0; k<n; k++)
                M[i][j][k] = (M[i][j][k]*x) % a.modp;
            V[i][j] = (V[i][j]*x) % a.modp;
        }

        // solve
        for (int j=n-1; j>=0; j--)
            for (int k=j+1; k<n; k++)
                V[i][j] = (V[i][j] - M[i][j][k]*V[i][k]) % a.modp;
    }

    // create coefficient list
    vector<LONG> coeff;
    for (int i=0; i<(int)V.size(); i++)
        for (int j=0; j<(int)V[i].size(); j++)
            coeff.push_back(V[i][j]);

    // create resulting polynomial
    poly res(BHEAD 0);
    int ri=1, i=0;
    for (int si=1; si<s[0]; si+=s[si]) {
        res.check_memory(ri);
        res[ri] = 3 + AN.poly_num_vars;                   // term length
        for (int j=0; j<AN.poly_num_vars; j++)
            res[ri+1+j] = s[si+1+j];                      // powers
        res[ri+1+AN.poly_num_vars] = ABS(coeff[i]); // coefficient
        res[ri+2+AN.poly_num_vars] = SGN(coeff[i]); // coefficient length
        i++;
        ri += res[ri];
    }
    res[0]=ri;                                          // total length
    res.setmod(a.modp,1);

    if (!poly::divides(res, lcgcd * a) || !poly::divides(res, lcgcd * b)) {
        return poly(BHEAD 0); // bad shape
    } else {
        // refine gcd
        if (!poly::divides(res, a))
            res = gcd_modular_dense_interpolation(res, a, x, poly(BHEAD 0));
        if (!poly::divides(res, b))
            res = gcd_modular_dense_interpolation(res, b, x, poly(BHEAD 0));
    }

#ifdef DEBUG
```

```
    cout « "*** [" « thetime() « "]  RES : gcd_modular_sparse_interpolation("
             « a « "," « b « "," « x « "," « "," « s «") = " « res « endl;
#endif

    return gcdconts * res;
}

/*
    #] gcd_modular_sparse_interpolation :
        #[ gcd_modular_dense_interpolation :
*/

const poly polygcd::gcd_modular_dense_interpolation (const poly &a, const poly &b, const vector<int> &x, const poly &s) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: gcd_modular_dense_interpolation(" « a « "," « b « "," « x « "," « s «")" « endl;
#endif

    POLY_GETIDENTITY(a);

    // if univariate, then use Euclidean algorithm
    if (x.size() == 1) {
        return gcd_Euclidean(a,b);
    }

    // if shape is known, use sparse interpolation
    if (!s.is_zero()) {
        poly res = gcd_modular_sparse_interpolation (a,b,x,s);
        if (!res.is_zero()) return res;
        // apparently the shape was not correct. continue.
    }

    // divide out multivariate content in last variable
    int X = x.back();

    poly conta(content_multivar(a,X));
    poly contb(content_multivar(b,X));
    poly gcdconts(gcd_Euclidean(conta,contb));
    const poly& ppa = conta.is_one() ? a : poly(a/conta);
    const poly& ppb = contb.is_one() ? b : poly(b/contb);

    // gcd of leading coefficients
    poly lcoeffa(ppa.lcoeff_multivar(X));
    poly lcoeffb(ppb.lcoeff_multivar(X));
    poly gcdlcoeffs(gcd_Euclidean(lcoeffa,lcoeffb));

    // calculate the degree bound for each variable
    int m = MiN(ppa.degree(x[x.size() - 2]),ppb.degree(x[x.size() - 2]));

    poly res(BHEAD 0);
    poly oldres(BHEAD 0);
    poly newshape(BHEAD 0);
    poly modpoly(BHEAD 1,a.modp);

    while (true) {
        // generate random constants and substitute it
        int c = 1 + wranf(BHEAD0) % (a.modp-1);
        if (substitute(gcdlcoeffs,X,c).is_zero()) continue;
        if (substitute(modpoly,X,c).is_zero()) continue;

        poly amodc(substitute(ppa,X,c));
        poly bmodc(substitute(ppb,X,c));

        // calculate gcd recursively
        poly gcdmodc(gcd_modular_dense_interpolation(amodc,bmodc,vector<int>(x.begin(),x.end()-1), newshape));
        int n = gcdmodc.degree(x[x.size() - 2]);

        // normalize
        gcdmodc = (gcdmodc * substitute(gcdlcoeffs,X,c)) / gcdmodc.integer_lcoeff();
        poly simple(poly::simple_poly(BHEAD X,c,1,a.modp)); // (X-c) mod p

        // if power is smaller, the old one was wrong
        if ((res.is_zero() && n == m) || n < m) {
            m = n;
            res = gcdmodc;
            newshape = gcdmodc; // set a new shape (interpolation does not change it)
            modpoly = simple;
        }
        else if (n == m) {
            oldres = res;
            // equal powers, so interpolate results
            poly coeff_poly(substitute(modpoly,X,c));
            WORD coeff_word = coeff_poly[2+AN.poly_num_vars] * coeff_poly[3+AN.poly_num_vars];
            if (coeff_word < 0) coeff_word += a.modp;
```

```
            GetModInverses(coeff_word, a.modp, &coeff_word, NULL);

            res.setmod(a.modp); // make sure the mod is set before substituting
            res += poly(BHEAD coeff_word, a.modp, 1) * modpoly * (gcdmodc - substitute(res,X,c));
            modpoly *= simple;
        }

        // check whether this is the complete gcd
        if (!res.is_zero() && res == oldres) {
            poly nres = res / content_multivar(res, X);
            if (poly::divides(nres,ppa) && poly::divides(nres,ppb)) {
#ifdef DEBUG
                cout « "*** [" « thetime() « "]  RES : gcd_modular_dense_interpolation(" « a « "," « b
« ","
                         « x « "," « "," « s «") = " « gcdconts * nres « endl;
#endif
                return gcdconts * nres;
            }

            // At this point, the gcd may be too large due to bad luck
            // TODO: create an efficient fail state that tries to find a smaller
            // polynomial without interpolating bad ones?
            newshape = poly(BHEAD 0); // reset the shape, important!
        }
    }
}

/*
    #] gcd_modular_dense_interpolation : :
        #[ gcd_modular :
*/

const poly polygcd::gcd_modular (const poly &origa, const poly &origb, const vector<int> &x) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: gcd_modular(" « origa « "," « origb « "," « x « ")" « endl;
#endif

    POLY_GETIDENTITY(origa);

    if (origa.is_zero()) return origa.modp==0 ? origb : origb / origb.integer_lcoeff();
    if (origb.is_zero()) return origa.modp==0 ? origa : origa / origa.integer_lcoeff();
    if (origa==origb) return origa.modp==0 ? origa : origa / origa.integer_lcoeff();

    poly ac = integer_content(origa);
    poly bc = integer_content(origb);
    const poly& a = ac.is_one() ? origa : poly(origa/ac);
    const poly& b = bc.is_one() ? origb : poly(origb/bc);
    poly ic = integer_gcd(ac, bc);
    poly g = integer_gcd(a.integer_lcoeff(), b.integer_lcoeff());

    int pnum=0;

    poly d(BHEAD 0);
    poly m1(BHEAD 1);
    int mindeg=MAXPOSITIVE;

    while (true) {
        // choose a prime and solve modulo the prime
        WORD p = NextPrime(BHEAD pnum++);
        if (poly(a.integer_lcoeff(),p).is_zero()) continue;
        if (poly(b.integer_lcoeff(),p).is_zero()) continue;

        poly c(gcd_modular_dense_interpolation(poly(a,p),poly(b,p),x,poly(d,p)));
        c = (c * poly(g,p)) / c.integer_lcoeff(); // normalize so that lcoeff(c) = g mod p

        if (c.is_zero()) {
            // unlucky choices somewhere, so start all over again
            d = poly(BHEAD 0);
            m1 = poly(BHEAD 1);
            mindeg = MAXPOSITIVE;
            continue;
        }

        if (!(poly(a,p)%c).is_zero()) continue;
        if (!(poly(b,p)%c).is_zero()) continue;

        int deg = c.degree(x[0]);

        if (deg < mindeg) {
            // small degree, so the old one is wrong
            d=c;
            d.modp=a.modp;
            d.modn=a.modn;
            m1 = poly(BHEAD p);
            mindeg=deg;
```

```
        }
        else if (deg == mindeg) {
            // same degree, so use Chinese Remainder Algorithm
            poly newd(BHEAD 0);

            for (int ci=1,di=1; ci<c[0]||di<d[0]; ) {
                int comp = ci==c[0] ? -1 : di==d[0] ? +1 : poly::monomial_compare(BHEAD &c[ci],&d[di]);
                poly a1(BHEAD 0),a2(BHEAD 0);

                newd.check_memory(newd[0]);

                if (comp <= 0) {
                    newd.termscopy(&d[di],newd[0],1+AN.poly_num_vars);
                    a1 = poly(BHEAD (UWORD *)&d[di+1+AN.poly_num_vars],d[di+d[di]-1]);
                    di+=d[di];
                }
                if (comp >= 0) {
                    newd.termscopy(&c[ci],newd[0],1+AN.poly_num_vars);
                    a2 = poly(BHEAD (UWORD *)&c[ci+1+AN.poly_num_vars],c[ci+c[ci]-1]);
                    ci+=c[ci];
                }

                poly e(chinese_remainder(a1,m1,a2,poly(BHEAD p)));
                newd.termscopy(&e[2+AN.poly_num_vars], newd[0]+1+AN.poly_num_vars, ABS(e[e[1]])+1);
                newd[newd[0]] = 2 + AN.poly_num_vars + ABS(e[e[1]]);
                newd[0] += newd[newd[0]];
            }

            m1 *= poly(BHEAD p);
            d=newd;
        }

        // divide out spurious integer content
        poly ppd(d / integer_content(d));

        // check whether this is the complete gcd
        if (poly::divides(ppd,a) && poly::divides(ppd,b)) {
#ifdef DEBUG
            cout « "*** [" « thetime() « "]  RES : gcd_modular(" « origa « "," « origb « "," « x « ") = "
                     « ic * ppd « endl;
#endif
            return ic * ppd;
        }
#ifdef DEBUG
        cout « "*** [" « thetime() « "] Retrying modular_gcd with new prime" « endl;
#endif
    }
}

/*
    #] gcd_modular :
    #[ gcd_heuristic_possible :
*/

bool gcd_heuristic_possible (const poly &a) {

    POLY_GETIDENTITY(a);

    double prod_deg = 1;
    for (int j=0; j<AN.poly_num_vars; j++)
        prod_deg *= a[2+j]+1;

    double digits = ABS(a[1+a[1]-1]);
  double lead = a[1+1+AN.poly_num_vars];

    return prod_deg*(digits-1+log(2*ABS(lead))/log(2.0)/(BITSINWORD/2)) < POLYGCD_HEURISTIC_MAX_DIGITS;
}

/*
    #] gcd_heuristic_possible :
    #[ gcd_heuristic :
*/


const poly polygcd::gcd_heuristic (const poly &a, const poly &b, const vector<int> &x, int max_tries) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: gcd_heuristic("«a«","«b«","«x«")\n";
#endif

    if (a.is_integer()) return integer_gcd(a,integer_content(b));
    if (b.is_integer()) return integer_gcd(integer_content(a),b);

```

```
    POLY_GETIDENTITY(a);

    // Calculate xi = 2*min(max(coefficients a),max(coefficients b))+2

    UWORD *pxi=NULL;
    WORD nxi=0;

    for (int i=1; i<a[0]; i+=a[i]) {
        WORD na = ABS(a[i+a[i]-1]);
        if (BigLong((UWORD *)&a[i+a[i]-1-na], na, pxi, nxi) > 0) {
            pxi = (UWORD *)&a[i+a[i]-1-na];
            nxi = na;
        }
    }

    for (int i=1; i<b[0]; i+=b[i]) {
        WORD nb = ABS(b[i+b[i]-1]);
        if (BigLong((UWORD *)&b[i+b[i]-1-nb], nb, pxi, nxi) > 0) {
            pxi = (UWORD *)&b[i+b[i]-1-nb];
            nxi = nb;
        }
    }

    poly xi(BHEAD pxi,nxi);

    // Addition of another random factor gives better performance
    xi = xi*poly(BHEAD 2) + poly(BHEAD 2 + wranf(BHEAD0)%POLYGCD_HEURISTIC_MAX_ADD_RANDOM);

    // If degree*digits(xi) is too large, throw exception
    if (max(a.degree(x[0]),b.degree(x[0])) * xi[xi[1]] >= MiN(AM.MaxTal,
POLYGCD_HEURISTIC_MAX_DIGITS)) {
#ifdef DEBUG
        cout « "*** [" « thetime() « "]  RES : gcd_heuristic("«a«","«b«","«x«") = overflow\n";
#endif
        throw(gcd_heuristic_failed());
    }

    for (int times=0; times<max_tries; times++) {

        // Recursively calculate the gcd

        poly gamma(gcd_heuristic(a % poly::simple_poly(BHEAD x[0],xi),
                                                                 b % poly::simple_poly(BHEAD x[0],xi),
                                                                 vector<int>(x.begin()+1,x.end()),1));

        // If a gcd is found, reconstruct the powers of x
        if (!gamma.is_zero()) {
            // res is construct is reverse order. idx/len are for reversing
            // it in the correct order
            poly res(BHEAD 0), c(BHEAD 0);
            vector<int> idx, len;

            for (int power=0; !gamma.is_zero(); power++) {

                // calculate c = gamma % xi (c and gamma are polynomials, xi is integer)
                c = gamma;
                c.coefficients_modulo((UWORD *)&xi[2+AN.poly_num_vars], xi[xi[0]-1], false);

                // Add the terms c * x^power to res
                res.check_memory(res[0]+c[0]);
                res.termscopy(&c[1],res[0],c[0]-1);
                for (int i=1; i<c[0]; i+=c[i])
                    res[res[0]-1+i+1+x[0]] = power;

                // Store idx/len for reversing
                if (!c.is_zero()) {
                    idx.push_back(res[0]);
                    len.push_back(c[0]-1);
                }

                res[0] += c[0]-1;

                // Divide gamma by xi
                gamma = (gamma - c) / xi;
            }

            // Reverse the resulting polynomial
            poly rev(BHEAD 0);
            rev.check_memory(res[0]);

            rev[0] = 1;
            for (int i=idx.size()-1; i>=0; i--) {
                rev.termscopy(&res[idx[i]], rev[0], len[i]);
                rev[0] += len[i];
            }
            res = rev;

```

```
            poly ppres = res / integer_content(res);

            if ((a%ppres).is_zero() && (b%ppres).is_zero()) {
#ifdef DEBUG
                cout « "*** [" « thetime() « "]  RES : gcd_heuristic("«a«","«b«","«x«") = "«res«"\n";
#endif
                return res;
            }
        }

        // Next xi by multiplying with the golden ratio to avoid correlated errors
        xi = xi * poly(BHEAD 28657) / poly(BHEAD 17711) + poly(BHEAD wranf(BHEAD0) %
POLYGCD_HEURISTIC_MAX_ADD_RANDOM);
    }

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : gcd_heuristic("«a«","«b«","«x«") = failed\n";
#endif

    return poly(BHEAD 0);
}

/*
    #] gcd_heuristic :
    #[ bracket :
*/

const map<vector<int>,poly> polygcd::full_bracket(const poly &a, const vector<int>& filter) {
    POLY_GETIDENTITY(a);

    map<vector<int>,poly> bracket;
    for (int ai=1; ai<a[0]; ai+=a[ai]) {
        vector<int> varpattern(AN.poly_num_vars);
        for (int i=0; i<AN.poly_num_vars; i++)
            if (filter[i] == 1 && a[ai + i + 1] > 0)
                varpattern[i] = a[ai + i + 1];

        // create monomial
        poly mon(BHEAD 1);
        mon.setmod(a.modp);
        mon[0] = a[ai] + 1;
        for (int i=0; i<a[ai]; i++)
            if (i > 0 && i <= AN.poly_num_vars && varpattern[i - 1])
                mon[1+i] = 0;
            else
                mon[1+i] = a[ai+i];

        map<vector<int>,poly>::iterator i = bracket.find(varpattern);
        if (i == bracket.end()) {
            bracket.insert(std::make_pair(varpattern, mon));
        } else {
            i->second += mon;
        }
    }

    return bracket;
}

const poly polygcd::bracket(const poly &a, const vector<int>& pattern, const vector<int>& filter) {
    POLY_GETIDENTITY(a);

    poly bracket(BHEAD 0);
    for (int ai=1; ai<a[0]; ai+=a[ai]) {
        bool ispat = true;
        for (int i=0; i<AN.poly_num_vars; i++)
            if (filter[i] == 1 && pattern[i] != a[ai + i + 1]) {
                ispat = false;
                break;
            }

        if (ispat) {
            poly mon(BHEAD 1);
            mon.setmod(a.modp);
            mon[0] = a[ai] + 1;
            for (int i=0; i<a[ai]; i++)
                if (i > 0 && i <= AN.poly_num_vars && pattern[i - 1])
                    mon[1+i] = 0;
                else
                    mon[1+i] = a[ai+i];
            bracket += mon;
        }
    }

    return bracket;
}

const map<vector<int>,int> polygcd::bracket_count(const poly &a, const vector<int>& filter) {
```

```
    POLY_GETIDENTITY(a);

    map<vector<int>,int> bracket;
    for (int ai=1; ai<a[0]; ai+=a[ai]) {
        vector<int> varpattern(AN.poly_num_vars);
        for (int i=0; i<AN.poly_num_vars; i++)
            if (filter[i] == 1 && a[ai + i + 1] > 0)
                varpattern[i] = a[ai + i + 1];

        map<vector<int>,int>::iterator i = bracket.find(varpattern);
        if (i == bracket.end()) {
            bracket.insert(std::make_pair(varpattern, 0));
        } else {
            i->second++;
        }
    }

    return bracket;
}

struct BracketInfo {
    std::vector<int> pattern;
    int num_terms, dummy = 0;
    const poly* p;

    BracketInfo(const std::vector<int>& pattern, int num_terms, const poly* p) : pattern(pattern), num_terms(num_terms), p(p) {}
    bool operator<(const BracketInfo& rhs) const { return num_terms > rhs.num_terms; } // biggest should be first!
};

/*
    #] bracket :
    #[ gcd_linear:
*/

const poly gcd_linear_helper (const poly &a, const poly &b) {
    POLY_GETIDENTITY(a);

    for (int i = 0; i < AN.poly_num_vars; i++)
        if (a.degree(i) == 1) {
            vector<int> filter(AN.poly_num_vars);
            filter[i] = 1;

            // bracket the linear variable
            map<vector<int>,poly> ba = polygcd::full_bracket(a, filter);

            poly subgcd(BHEAD 1);
            if (ba.size() == 2)
                subgcd = polygcd::gcd_linear(ba.begin()->second, (++ba.begin())->second);
            else
                subgcd = ba.begin()->second;

            poly linfac = a / subgcd;
            if (poly::divides(linfac,b))
                return linfac * polygcd::gcd_linear(subgcd, b / linfac);

            return polygcd::gcd_linear(subgcd, b);
        }

    return poly(BHEAD 0);
}

const poly polygcd::gcd_linear (const poly &a, const poly &b) {
#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: gcd_linear("«a«","«b«")\n";
#endif

    POLY_GETIDENTITY(a);

    if (a.is_zero()) return a.modp==0 ? b : b / b.integer_lcoeff();
    if (b.is_zero()) return a.modp==0 ? a : a / a.integer_lcoeff();
    if (a==b) return a.modp==0 ? a : a / a.integer_lcoeff();

    if (a.is_integer() || b.is_integer()) {
        if (a.modp > 0) return poly(BHEAD 1,a.modp,a.modn);
        return poly(integer_gcd(integer_content(a),integer_content(b)),0,1);
    }

    poly h = gcd_linear_helper(a, b);
    if (!h.is_zero()) return h;

    h = gcd_linear_helper(b, a);
    if (!h.is_zero()) return h;

    vector<int> xa = a.all_variables();
    vector<int> xb = b.all_variables();
```

```

    vector<int> used(AN.poly_num_vars,0);
    for (int i=0; i<(int)xa.size(); i++) used[xa[i]]++;
    for (int i=0; i<(int)xb.size(); i++) used[xb[i]]++;
    vector<int> x;
    for (int i=0; i<AN.poly_num_vars; i++)
        if (used[i]) x.push_back(i);

    return gcd_modular(a,b,x);
}

/*
    #] gcd_linear:
    #[ gcd :
*/

const poly polygcd::gcd (const poly &a, const poly &b) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL: gcd("«a«","«b«")\n";
#endif

    POLY_GETIDENTITY(a);

    if (a.is_zero()) return a.modp==0 ? b : b / b.integer_lcoeff();
    if (b.is_zero()) return a.modp==0 ? a : a / a.integer_lcoeff();
    if (a==b) return a.modp==0 ? a : a / a.integer_lcoeff();

    if (a.is_integer() || b.is_integer()) {
        if (a.modp > 0) return poly(BHEAD 1,a.modp,a.modn);
        return poly(integer_gcd(integer_content(a),integer_content(b)),0,1);
    }

    // Generate a list of variables of a and b
    vector<int> xa = a.all_variables();
    vector<int> xb = b.all_variables();

    vector<int> used(AN.poly_num_vars,0);
    for (int i=0; i<(int)xa.size(); i++) used[xa[i]]++;
    for (int i=0; i<(int)xb.size(); i++) used[xb[i]]++;
    vector<int> x;
    for (int i=0; i<AN.poly_num_vars; i++)
        if (used[i]) x.push_back(i);

    if (a.is_monomial() || b.is_monomial()) {

        poly res(BHEAD 1,a.modp,a.modn);
        if (a.modp == 0) res = integer_gcd(integer_content(a),integer_content(b));

        for (int i=0; i<(int)x.size(); i++)
            res[2+x[i]] = 1«(BITSINWORD-2);

        for (int i=1; i<a[0]; i+=a[i])
            for (int j=0; j<(int)x.size(); j++)
                res[2+x[j]] = MiN(res[2+x[j]], a[i+1+x[j]]);

        for (int i=1; i<b[0]; i+=b[i])
            for (int j=0; j<(int)x.size(); j++)
                res[2+x[j]] = MiN(res[2+x[j]], b[i+1+x[j]]);

        return res;
    }

    // Calculate the contents, their gcd and the primitive parts
    poly conta(x.size()==1 ? integer_content(a) : content_univar(a,x[0]));
    poly contb(x.size()==1 ? integer_content(b) : content_univar(b,x[0]));
    poly gcdconts(x.size()==1 ? integer_gcd(conta,contb) : gcd(conta,contb));
    const poly& ppa = conta.is_one() ? a : poly(a/conta);
    const poly& ppb = contb.is_one() ? b : poly(b/contb);

    if (ppa == ppb)
        return ppa * gcdconts;

    poly res(BHEAD 0);

#ifdef POLYGCD_USE_HEURISTIC
    // Try the heuristic gcd algorithm
    if (a.modp==0 && gcd_heuristic_possible(a) && gcd_heuristic_possible(b)) {
        try {
            res = gcd_heuristic(ppa,ppb,x);
            if (!res.is_zero()) res /= integer_content(res);
        }
        catch (gcd_heuristic_failed) {}
    }
#endif

    // If gcd==0, the heuristic algorithm failed, so we do more extensive checks.
```

```
    // First, we filter out variables that appear in only one of the expressions.
    if (res.is_zero()) {
        bool unusedVars = false;
        for (unsigned int i = 0; i < used.size(); i++) {
            if (used[i] == 1) {
                unusedVars = true;
                break;
            }
        }

        // if there are no unused variables, go to the linear routine directly
        if (!unusedVars) {
            res = gcd_linear(ppa,ppb);
#ifdef DEBUG
            cout « "New GCD attempt (unused vars): " « res « endl;
#endif
        }

        // if res is not the gcd, it is 0 or larger than the gcd.
        // we bracket the expression in all the variables that appear only in one expr.
        // and we refine the gcd.
        bool diva = !res.is_zero() && poly::divides(res,ppa);
        bool divb = !res.is_zero() && poly::divides(res,ppb);
        if (!diva || !divb) {
            vector<BracketInfo> bracketinfo;

            if (!diva) {
                map<vector<int>,int> ba = bracket_count(ppa, used);
                for(map<vector<int>,int>::iterator it = ba.begin(); it != ba.end(); it++)
                    bracketinfo.push_back(BracketInfo(it->first, it->second, &ppa));
            }

            if (!divb) {
                map<vector<int>,int> bb = bracket_count(ppb, used);
                for(map<vector<int>,int>::iterator it = bb.begin(); it != bb.end(); it++)
                    bracketinfo.push_back(BracketInfo(it->first, it->second, &ppb));
            }

            // sort so that the smallest bracket will be last
            sort(bracketinfo.begin(), bracketinfo.end());

            if (res.is_zero()) {
                res = bracket(*bracketinfo.back().p, bracketinfo.back().pattern, used);
                bracketinfo.pop_back();
            }

            while (bracketinfo.size() > 0) {
                poly subpoly(bracket(*bracketinfo.back().p, bracketinfo.back().pattern, used));
                if (!poly::divides(res,subpoly)) {
                    // if we can filter out more variables, call gcd again
                    if (res.all_variables() != subpoly.all_variables())
                        res = gcd(subpoly,res);
                    else
                        res = gcd_linear(subpoly,res);
                }

                bracketinfo.pop_back();
            }
        }

        if (res.is_zero() || !poly::divides(res,ppa) || !poly::divides(res,ppb)) {
            MesPrint("Bad gcd found.");
            std::cout « "Bad gcd:" « res « " for " « ppa « " " « ppb « std::endl;
            Terminate(1);
        }
    }

    res *= gcdconts * poly(BHEAD res.sign());

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  RES : gcd("«a«","«b«") = "«res«endl;
#endif

    return res;
}

/*
    #] gcd :
*/
```

4.107 polygcd.h

```
#pragma once
```

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 */
/* #] License : */

#include <vector>
#include <map>

class poly; // polynomial class
class gcd_heuristic_failed {}; // class for throwing exceptions

// whether or not to use the heuristic before Zippel's algorithm
#define POLYGCD_USE_HEURISTIC

// maximum number of words in a coefficient for gcd_heuristic to continue
const int POLYGCD_HEURISTIC_MAX_DIGITS = 1000;

// maximum number of retries after the heuristic has failed
const int POLYGCD_HEURISTIC_MAX_TRIES = 10;

// a fudge factor, which improves efficiency
const int POLYGCD_HEURISTIC_MAX_ADD_RANDOM = 10;

// maximum cached power in substitute_last and sparse_interpolation_get_mul_list
const int POLYGCD_RAISPOWMOD_CACHE_SIZE = 1000;

namespace polygcd {

    // functions to call the gcd routines
    const poly integer_gcd (const poly &a, const poly &b);
    const poly integer_content (const poly &a);

    const poly gcd (const poly &a, const poly &b);
    const poly content_univar (const poly &a, int x);
    const poly content_multivar (const poly &a, int x);

    const std::vector<WORD> coefficient_list_gcd (const std::vector<WORD> &a, const std::vector<WORD> &b, WORD p);

    // internal functions
    const poly gcd_heuristic (const poly &a, const poly &b, const std::vector<int> &x, int max_tries=POLYGCD_HEURISTIC_MAX_TRIES);
    const poly gcd_Euclidean (const poly &a, const poly &b);
    const poly gcd_modular (const poly &a, const poly &b, const std::vector<int> &x);
    const poly gcd_modular_dense_interpolation (const poly &a, const poly &b, const std::vector<int> &x, const poly &s);
    const poly gcd_modular_sparse_interpolation (const poly &a, const poly &b, const std::vector<int> &x, const poly &s);

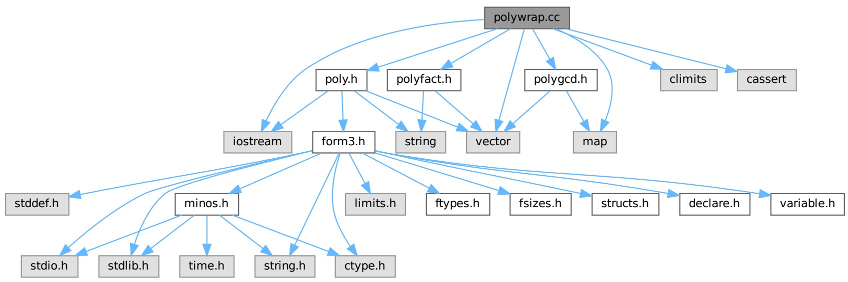
    const std::vector<int> sparse_interpolation_get_mul_list (const poly &a, const std::vector<int> &x, const std::vector<int> &c);
    void sparse_interpolation_mul_poly (poly &a, const std::vector<int> &m);
    const poly sparse_interpolation_reduce_poly (const poly &a, const std::vector<int> &x);
    const poly sparse_interpolation_fix_poly (const poly &a, int x);

    const poly chinese_remainder (const poly &a1, const poly &m1, const poly &a2, const poly &m2);
    const poly substitute(const poly &a, int x, int c);
    const std::map<std::vector<int>,poly> full_bracket(const poly &a, const std::vector<int>& filter);
    const poly bracket(const poly &a, const std::vector<int>& pattern, const std::vector<int>& filter);
    const std::map<std::vector<int>,int> bracket_count(const poly &a, const std::vector<int>& filter);
    const poly gcd_linear (const poly &a, const poly &b);
}
```

4.108 polywrap.cc File Reference

```
#include "poly.h"
#include "polygcd.h"
#include "polyfact.h"
#include <iostream>
#include <vector>
#include <map>
#include <climits>
#include <cassert>
```

Include dependency graph for polywrap.cc:



Functions

- WORD poly_determine_modulus (PHEAD bool multi_error, bool is_fun_arg, string message)
- WORD ∗ poly_gcd (PHEAD WORD ∗a, WORD ∗b, WORD fit)
- WORD ∗ poly_divmod (PHEAD WORD ∗a, WORD ∗b, int divmod, WORD fit)
- WORD ∗ poly_div (PHEAD WORD ∗a, WORD ∗b, WORD fit)
- WORD ∗ poly_rem (PHEAD WORD ∗a, WORD ∗b, WORD fit)
- void poly_ratfun_read (WORD ∗a, poly &num, poly &den)
- void poly_sort (PHEAD WORD ∗a)
- WORD ∗ poly_ratfun_add (PHEAD WORD ∗t1, WORD ∗t2)
- int poly_ratfun_normalize (PHEAD WORD ∗term)
- void poly_fix_minus_signs (factorized_poly &a)
- WORD ∗ poly_factorize (PHEAD WORD ∗argin, WORD ∗argout, bool with_arghead, bool is_fun_arg)
- int poly_factorize_argument (PHEAD WORD ∗argin, WORD ∗argout)
- WORD ∗ poly_factorize_dollar (PHEAD WORD ∗argin)
- int poly_factorize_expression (EXPRESSIONS expr)
- int poly_unfactorize_expression (EXPRESSIONS expr)
- WORD ∗ poly_inverse (PHEAD WORD ∗arga, WORD ∗argb)
- WORD ∗ poly_mul (PHEAD WORD ∗a, WORD ∗b)
- void poly_free_poly_vars (PHEAD const char ∗text)

Variables

- const int POLYWRAP_DENOMPOWER_INCREASE_FACTOR = 2

4.108.1 Detailed Description

Contains methods to call the polynomial methods (written in C++) from the rest of Form (written in C). These include polynomial gcd computation, factorization and polyratfuns.

Definition in file polywrap.cc.

4.108.2 Function Documentation

4.108.2.1 poly_determine_modulus()

```
WORD poly_determine_modulus (
            PHEAD bool multi_error,
            bool is_fun_arg,
            string message )
```

Modulus for polynomial algebra

4.108.3 Description

This method determines whether polynomial algebra is done with a modulus or not. This depends on AC.ncmod. If only_funargs is set it also depends on (AC.modmode & ALSOFUNARGS).

The program terminates if the feature is not implemented. Polynomial algebra modulo M > WORDSIZE in not implemented. If multi_error is set, multivariate algebra mod M is not implemented.

4.108.4 Notes

- If AC.ncmod>0 and only_funargs=true and AC.modmode&ALSOFUNARGS=false, AN.ncmod is set to zero, for otherwise RaisPow calculates mod M.

Definition at line 79 of file polywrap.cc.

Referenced by poly_factorize(), poly_factorize_expression(), poly_gcd(), poly_ratfun_add(), and poly_ratfun_normalize().

4.108.4.1 poly_gcd()

```
WORD * poly_gcd (
            PHEAD WORD * a,
            WORD * b,
            WORD fit )
```

Polynomial gcd

4.108.5 Description

This method calculates the greatest common divisor of two polynomials, given by two zero-terminated Form-style term lists.

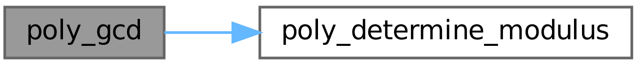
4.108.6 Notes

- The result is written at newly allocated memory
- Called from ratio.c
- Calls polygcd::gcd

Definition at line 124 of file polywrap.cc.

References poly_determine_modulus().

Here is the call graph for this function:



4.108.6.1 poly_divmod()

```
WORD * poly_divmod (
            PHEAD WORD * a,
            WORD * b,
            int divmod,
            WORD fit )
```

Definition at line 203 of file polywrap.cc.

4.108.6.2 poly_div()

```
WORD * poly_div (
            PHEAD WORD * a,
            WORD * b,
            WORD fit )
```

Definition at line 435 of file polywrap.cc.

4.108.6.3 poly_rem()

```
WORD * poly_rem (
            PHEAD WORD * a,
            WORD * b,
            WORD fit )
```

Definition at line 463 of file polywrap.cc.

4.108.6.4 poly_ratfun_read()

```
void poly_ratfun_read (
            WORD * a,
            poly & num,
            poly & den )
```

Read a PolyRatFun

4.108.7 Description

This method reads a polyratfun starting at the pointer a. The resulting numerator and denominator are written in num and den. If MUSTCLEANPRF, the result is normalized.

4.108.8 Notes

- Calls polygcd::gcd

Definition at line 496 of file polywrap.cc.

Referenced by poly_ratfun_add(), and poly_ratfun_normalize().

4.108.8.1 poly_sort()

```
void poly_sort (
            PHEAD WORD * a )
```

Sort the polynomial terms

4.108.9 Description

Sorts the terms of a polynomial in Form poly(rat)fun order, i.e. lexicographical order with highest degree first.

4.108.10 Notes

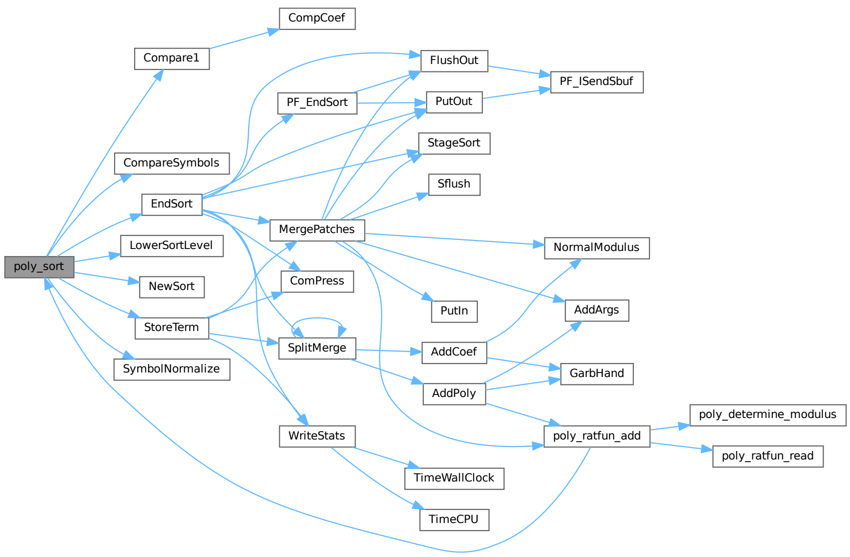
- Uses Form sort routines with custom compare

Definition at line 590 of file polywrap.cc.

References Compare1(), CompareSymbols(), EndSort(), LowerSortLevel(), NewSort(), StoreTerm(), and SymbolNormalize().

Referenced by poly_ratfun_add(), and poly_ratfun_normalize().

Here is the call graph for this function:



4.108.10.1 poly_ratfun_add()

```
WORD * poly_ratfun_add (
            PHEAD WORD * t1,
            WORD * t2 )
```

Addition of PolyRatFuns

4.108.11 Description

This method gets two pointers to polyratfuns with up to two arguments each and calculates the sum.

4.108.12 Notes

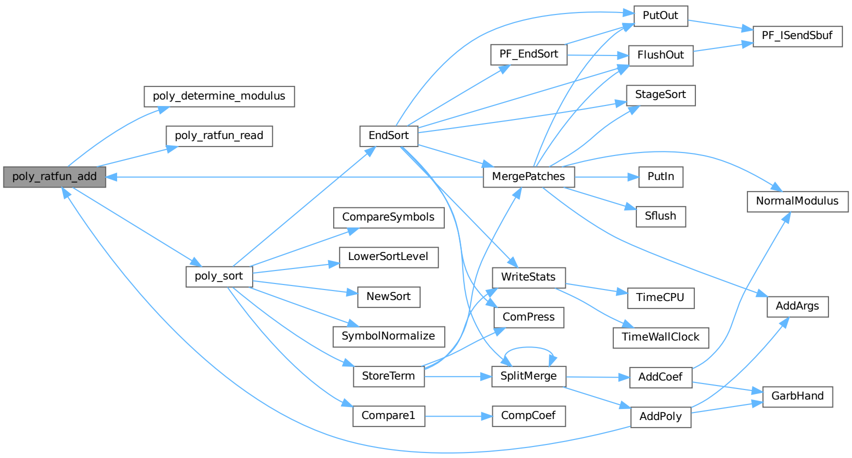
- The result is written at the workpointer
- Called from sort.c and threads.c
- Calls poly::operators and polygcd::gcd

Definition at line 633 of file polywrap.cc.

References poly_determine_modulus(), poly_ratfun_read(), and poly_sort().

Referenced by AddPoly(), and MergePatches().

Here is the call graph for this function:



4.108.12.1 poly_ratfun_normalize()

```
int poly_ratfun_normalize (
            PHEAD WORD * term )
```

Multiplication/normalization of PolyRatFuns

4.108.13 Description

This method searches a term for multiple polyratfuns and multiplies their contents. The result is properly normalized. Normalization also works for terms with a single polyratfun.

4.108.14 Notes

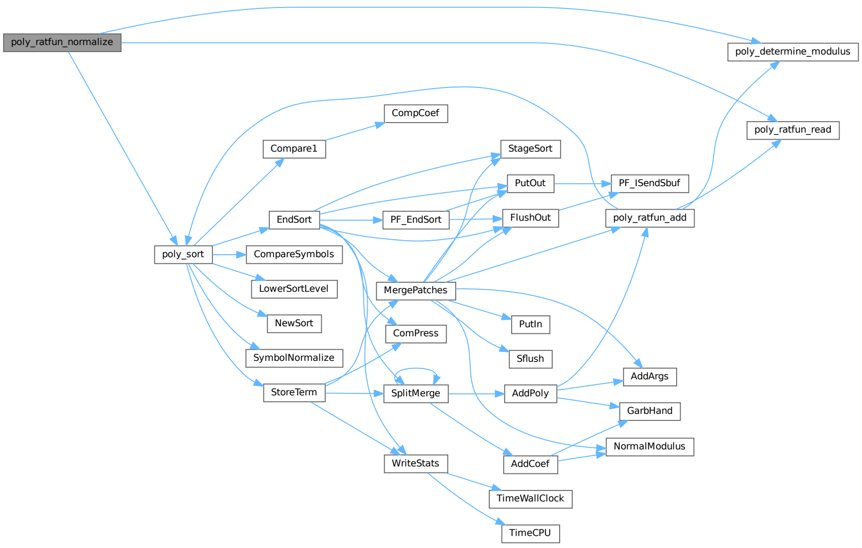
- The result overwrites the original term
- Called from proces.c
- Calls poly::operators and polygcd::gcd

Definition at line 769 of file polywrap.cc.

References poly_determine_modulus(), poly_ratfun_read(), and poly_sort().

Referenced by PolyFunMul(), and PrepPoly().

Here is the call graph for this function:



4.108.14.1 poly_fix_minus_signs()

```
void poly_fix_minus_signs (
            factorized_poly & a )
```

Definition at line 921 of file polywrap.cc.

4.108.14.2 poly_factorize()

```
WORD * poly_factorize (
            PHEAD WORD * argin,
            WORD * argout,
            bool with_arghead,
            bool is_fun_arg )
```

Factorization of function arguments / dollars

4.108.15 Description

This method factorizes a Form style argument or zero-terminated term list.

4.108.16 Notes

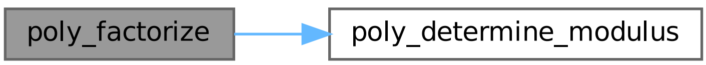
- Called from poly_factorize_{argument,dollar}
- Calls polyfact::factorize

Definition at line 986 of file polywrap.cc.

References poly_determine_modulus().

Referenced by poly_factorize_argument(), and poly_factorize_dollar().

Here is the call graph for this function:



4.108.16.1 poly_factorize_argument()

```
int poly_factorize_argument (
            PHEAD WORD * argin,
            WORD * argout )
```

Factorization of function arguments

4.108.17 Description

This method factorizes the Form-style argument argin.

4.108.18 Notes

- The result is written at argout
- Called from argument.c
- Calls poly_factorize

Definition at line 1112 of file polywrap.cc.

References poly_factorize().

Here is the call graph for this function:

4.108.18.1 poly_factorize_dollar()

```
WORD * poly_factorize_dollar (
            PHEAD WORD * argin )
```

Factorization of dollar variables

4.108.19 Description

This method factorizes a dollar variable.

4.108.20 Notes

- The result is written at newly allocated memory.
- Called from dollar.c
- Calls poly_factorize

Definition at line 1146 of file polywrap.cc.

References poly_factorize().

Here is the call graph for this function:

4.108.20.1 poly_factorize_expression()

```
int poly_factorize_expression (
            EXPRESSIONS expr )
```

Factorization of expressions

4.108.21 Description

This method factorizes an expression.

4.108.22 Notes

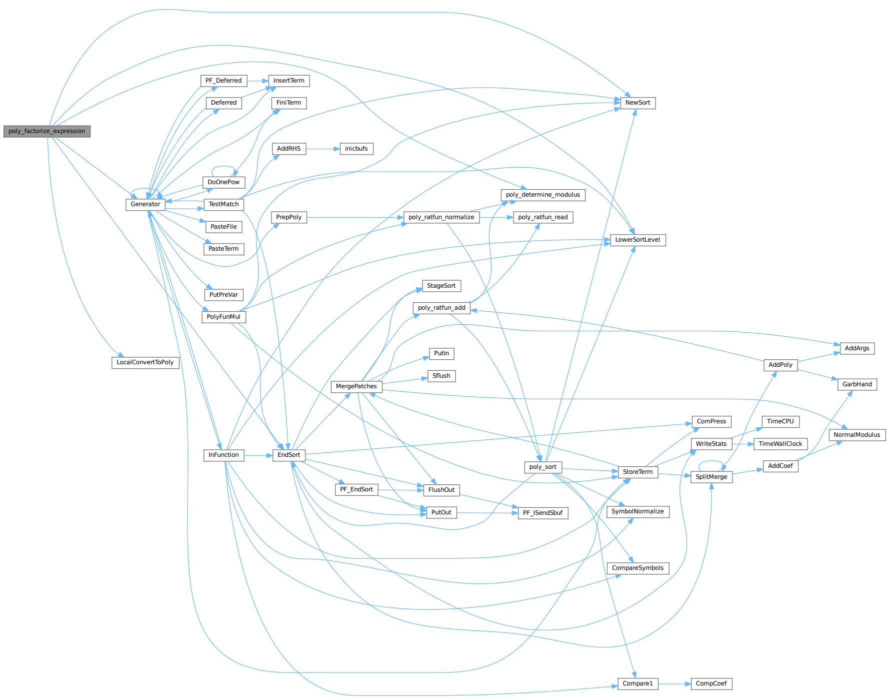
- The result overwrites the input expression
- Called from proces.c
- Calls polyfact::factorize

Definition at line 1178 of file polywrap.cc.

References EndSort(), Generator(), FiLe::handle, LocalConvertToPoly(), LowerSortLevel(), NewSort(), and poly_determine_modulus().

Referenced by PF_Processor(), and Processor().

Here is the call graph for this function:



4.108.22.1 poly_unfactorize_expression()

```
int poly_unfactorize_expression (
            EXPRESSIONS expr )
```

Unfactorization of expressions

4.108.23 Description

This method expands a factorized expression.

4.108.24 Notes

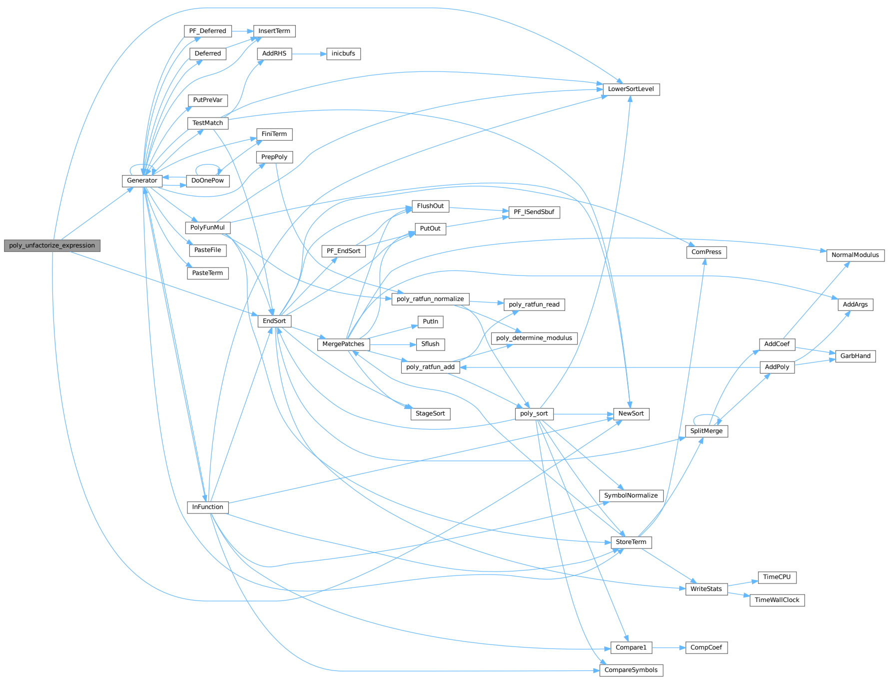
- The result overwrites the input expression
- Called from proces.c

Definition at line 1535 of file polywrap.cc.

References EndSort(), Generator(), FiLe::handle, LowerSortLevel(), and NewSort().

Referenced by PF_Processor(), and Processor().

Here is the call graph for this function:



4.108.24.1 poly_inverse()

```
WORD * poly_inverse (
            PHEAD WORD * arga,
            WORD * argb )
```

Definition at line 1743 of file polywrap.cc.

4.108.24.2 poly_mul()

```
WORD * poly_mul (
            PHEAD WORD * a,
            WORD * b )
```

Definition at line 1948 of file polywrap.cc.

4.108.24.3 poly_free_poly_vars()

```
void poly_free_poly_vars (
            PHEAD const char * text )
```

Definition at line 2014 of file polywrap.cc.

4.108.25 Variable Documentation

4.108.25.1 POLYWRAP_DENOMPOWER_INCREASE_FACTOR

```
const int POLYWRAP_DENOMPOWER_INCREASE_FACTOR = 2
```

Definition at line 52 of file polywrap.cc.

4.109 polywrap.cc

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *
```

```
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#include "poly.h"
#include "polygcd.h"
#include "polyfact.h"

#include <iostream>
#include <vector>
#include <map>
#include <climits>
#include <cassert>

//#define DEBUG

#ifdef DEBUG
#include "mytime.h"
#endif

// denompower is increased by this factor when divmod_heap fails
const int POLYWRAP_DENOMPOWER_INCREASE_FACTOR = 2;

using namespace std;

/*
    #[ poly_determine_modulus :
*/

WORD poly_determine_modulus (PHEAD bool multi_error, bool is_fun_arg, string message) {

    if (AC.ncmod==0) return 0;

    if (!is_fun_arg || (AC.modmode & ALSOFUNARGS)) {

        if (ABS(AC.ncmod)>1) {
            MLOCK(ErrorMessageLock);
            MesPrint ((char*)"ERROR: %s with modulus > WORDSIZE not implemented",message.c_str());
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }

        if (multi_error && AN.poly_num_vars>1) {
            MLOCK(ErrorMessageLock);
            MesPrint ((char*)"ERROR: multivariate %s with modulus not implemented",message.c_str());
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }

        return *AC.cmod;
    }

    AN.ncmod = 0;
    return 0;
}

/*
    #] poly_determine_modulus :
    #[ poly_gcd :
*/

WORD *poly_gcd(PHEAD WORD *a, WORD *b, WORD fit) {

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        WORD *ret = flint_gcd(BHEAD a, b, fit);
        return ret;
    }
#endif

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_gcd" « endl;
#endif

//
//MesPrint("Calling poly_gcd with:");
//{
//  WORD *at = a;
//  MesPrint("      a:");
//  while ( *at ) {
//      MesPrint("   %a",*at,at);
//      at += *at;
//  }
//  MesPrint("      b:");
//  at = b;
//  while ( *at ) {
```

```
//      MesPrint("   %a",*at,at);
//      at += *at;
//  }
//}

    // Extract variables
    vector<WORD *> e;
    e.reserve(2);
    e.push_back(a);
    e.push_back(b);
    poly::get_variables(BHEAD e, false, true);

    // Check for modulus calculus
    WORD modp=poly_determine_modulus(BHEAD true, true, "polynomial GCD");

    // Convert to polynomials
    poly pa(poly::argument_to_poly(BHEAD a, false, true), modp, 1);
    poly pb(poly::argument_to_poly(BHEAD b, false, true), modp, 1);

    // Calculate gcd
    poly gcd(polygcd::gcd(pa,pb));

    // Allocate new memory and convert to Form notation
    int newsize = (gcd.size_of_form_notation()+1);
    WORD *res;
    if ( fit ) {
        if ( newsize*sizeof(WORD) >= (size_t)(AM.MaxTer) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("poly_gcd: Term too complex (%d words). Maybe increasing MaxTermSize (%d words) can help",
                newsize, AM.MaxTer/sizeof(WORD));
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        res = TermMalloc("poly_gcd");
    }
    else {
        res = (WORD *)Malloc1(newsize*sizeof(WORD), "poly_gcd");
    }
    poly::poly_to_argument(gcd, res, false);

    poly_free_poly_vars(BHEAD "AN.poly_vars_qcd");

    // reset modulo calculation
    AN.ncmod = AC.ncmod;
    return res;
}

/*
    #] poly_gcd :
    #[ poly_divmod :

    if fit == 1 the answer must fit inside a term.
*/

WORD *poly_divmod(PHEAD WORD *a, WORD *b, int divmod, WORD fit) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_divmod" « endl;
#endif

    // check for modulus calculus
    WORD modp=poly_determine_modulus(BHEAD false, true, "polynomial division");

    // get variables
    vector<WORD *> e;
    e.reserve(2);
    e.push_back(a);
    e.push_back(b);
    poly::get_variables(BHEAD e, false, false);

    // add extra variables to keep track of denominators
    const int DENOMSYMBOL = MAXPOSITIVE;

//  WORD *new_poly_vars = (WORD *)Malloc1((AN.poly_num_vars+1)*sizeof(WORD), "AN.poly_vars");
//  WCOPY(new_poly_vars, AN.poly_vars, AN.poly_num_vars);
//  new_poly_vars[AN.poly_num_vars] = DENOMSYMBOL;
//  if (AN.poly_num_vars > 0)
//      M_free(AN.poly_vars, "AN.poly_vars");
//  AN.poly_num_vars++;
//  AN.poly_vars = new_poly_vars;
    AN.poly_vars[AN.poly_num_vars++] = DENOMSYMBOL;

    // convert to polynomials
    poly dena(BHEAD 0);
    poly denb(BHEAD 0);
    poly pa(poly::argument_to_poly(BHEAD a, false, true, &dena), modp, 1);
```

```
    poly pb(poly::argument_to_poly(BHEAD b, false, true, &denb), modp, 1);

    // remove contents
    poly numres(polygcd::integer_content(pa));
    poly denres(polygcd::integer_content(pb));
    pa /= numres;
    pb /= denres;

    if (divmod==0) {
        numres *= denb;
        denres *= dena;
    }
    else {
        denres = dena;
    }

    poly gcdres(polygcd::integer_gcd(numres,denres));
    numres /= gcdres;
    denres /= gcdres;

    // determine lcoeff(b)
    poly lcoeffb(pb.integer_lcoeff());

    poly pres(BHEAD 0);

    if (!lcoeffb.is_one()) {

        if (AN.poly_num_vars > 2) {
            // the original polynomial is multivariate (one dummy variable has
            // been added), so it is not trivial to determine which power of
            // lcoeff(b) can be in the answer

            int denompower = 0, prevdenompower = 0;
            poly pq(BHEAD 0);
            poly pr(BHEAD 0);

            // try denompower = 0, if this fails increase denompower until division succeeds
            bool div_fail = true;
            do
            {
                if(denompower < prevdenompower)
                {
                    // denompower increased beyond INT_MAX
                    MLOCK(ErrorMessageLock);
                    MesPrint ((char*)"ERROR: pseudo-division failed in poly_divmod (denompower > INT_MAX)");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(1);
                }

                if(denompower != 0)
                {
                    // multiply a by lcoeffb^(denompower-prevdenompower)
                    WORD n = lcoeffb[lcoeffb[1]];
                    RaisPow(BHEAD (UWORD *)&lcoeffb[2+AN.poly_num_vars], &n, denompower-prevdenompower);
                    lcoeffb[1] = 2 + AN.poly_num_vars + ABS(n);
                    lcoeffb[0] = 1 + lcoeffb[1];
                    lcoeffb[lcoeffb[1]] = n;

                    pa *= lcoeffb;
                    denres *= lcoeffb;
                }

                // try division
                poly ppow(BHEAD 0);
                poly::divmod_heap(pa,pb,pq,pr,false,true,div_fail); // sets div_fail

                // increase denompower for next iteration
                prevdenompower = denompower;
                denompower = (denompower==0 ? 1 : denompower*POLYWRAP_DENOMPOWER_INCREASE_FACTOR+1 ); // generates 2^n-1 when POLYWRAP_DENOMPOWER_INCREASE_FACTOR = 2
            }
            while(div_fail);

            pres = (divmod==0 ? pq : pr);

        }
        else {
            // one variable, so the power is the difference of the degrees

            int denompower = MaX(0, pa.degree(0) - pb.degree(0) + 1);

            // multiply a by that power
            WORD n = lcoeffb[lcoeffb[1]];
            RaisPow(BHEAD (UWORD *)&lcoeffb[2+AN.poly_num_vars], &n, denompower);
            lcoeffb[1] = 2 + AN.poly_num_vars + ABS(n);
```

```
            lcoeffb[0] = 1 + lcoeffb[1];
            lcoeffb[lcoeffb[1]] = n;

            pa *= lcoeffb;
            denres *= lcoeffb;

            pres = (divmod==0 ? pa/pb : pa%pb);

        }

    }
    else {

        pres = (divmod==0 ? pa/pb : pa%pb);

    }

    // convert to Form notation
    // NOTE: this part can be rewritten with poly::size_of_form_notation_with_den()
    // and poly::poly_to_argument_with_den().
    WORD *res;

    // special case: a=0
    if (pres[0]==1) {
        if ( fit ) {
            res = TermMalloc("poly_divmod");
        }
        else {
            res = (WORD *)Malloc1(sizeof(WORD), "poly_divmod");
        }
        res[0] = 0;
    }
    else {
        pres *= numres;

        WORD nden;
        UWORD *den = (UWORD *)NumberMalloc("poly_divmod");

        // allocate the memory; note that this overestimates the size,
        // since the estimated denominators are too large
        int ressize = pres.size_of_form_notation() + pres.number_of_terms()*2*ABS(denres[denres[1]]) + 1;

        if ( fit ) {
            if ( ressize*sizeof(WORD) > (size_t)(AM.MaxTer) ) {
                MLOCK(ErrorMessageLock);
                MesPrint("poly_divmod: Term too complex (%d words). Maybe increasing MaxTermSize (%d words) can help",
                    ressize, AM.MaxTer/sizeof(WORD));
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            res = TermMalloc("poly_divmod");
        }
        else {
            res = (WORD *)Malloc1(ressize*sizeof(WORD), "poly_divmod");
        }
        int L=0;

        for (int i=1; i!=pres[0]; i+=pres[i]) {

            res[L]=1; // length
            bool first = true;

            for (int j=0; j<AN.poly_num_vars; j++)
                if (pres[i+1+j] > 0) {
                    if (first) {
                        first = false;
                        res[L+1] = 1; // symbols
                        res[L+2] = 2; // length
                    }
                    res[L+1+res[L+2]++] = AN.poly_vars[j]; // symbol
                    res[L+1+res[L+2]++] = pres[i+1+j];     // power
                }

            if (!first) res[L] += res[L+2]; // fix length

            // numerator
            WORD nnum = pres[i+pres[i]-1];
            WCOPY(&res[L+res[L]], &pres[i+pres[i]-1-ABS(nnum)], ABS(nnum));

            // calculate denominator
            nden = denres[denres[1]];
            WCOPY(den, &denres[2+AN.poly_num_vars], ABS(nden));

            if (nden!=1 || den[0]!=1)
                Simplify(BHEAD (UWORD *)&res[L+res[L]], &nnum, den, &nden); // gcd(num,den)
```

```
            Pack((UWORD *)&res[L+res[L]], &nnum, den, nden);              // format
            res[L] += 2*ABS(nnum)+1;                                      // fix length
            res[L+res[L]-1] = SGN(nnum)*(2*ABS(nnum)+1);                  // length of coefficient
            L += res[L];                                                  // fix length
        }

        res[L] = 0;

        NumberFree(den,"poly_divmod");
    }

    // clean up
    poly_free_poly_vars(BHEAD "AN.poly_vars_divmod");

    // reset modulo calculation
    AN.ncmod = AC.ncmod;

    return res;
}

/*
    #] poly_divmod :
    #[ poly_div :

    Routine divides the expression in arg1 by the expression in arg2.
    We did not take out special cases.
    The arguments are zero terminated sequences of term(s).
    The action is to divide arg1 by arg2: [arg1/arg2].
    The answer should be a buffer (allocated by Malloc1) with a zero
    terminated sequence of terms (or just zero).
*/
WORD *poly_div(PHEAD WORD *a, WORD *b, WORD fit) {

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        WORD *ret = flint_div(BHEAD a, b, fit);
        return ret;
    }
#endif

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_div" « endl;
#endif

    return poly_divmod(BHEAD a, b, 0, fit);
}

/*
    #] poly_div :
    #[ poly_rem :

    Routine divides the expression in arg1 by the expression in arg2
    and takes the remainder.
    We did not take out special cases.
    The arguments are zero terminated sequences of term(s).
    The action is to divide arg1 by arg2 and take the remainder: [arg1%arg2].
    The answer should be a buffer (allocated by Malloc1) with a zero
    terminated sequence of terms (or just zero).
*/
WORD *poly_rem(PHEAD WORD *a, WORD *b, WORD fit) {

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        WORD *ret = flint_rem(BHEAD a, b, fit);
        return ret;
    }
#endif

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_rem" « endl;
#endif

    return poly_divmod(BHEAD a, b, 1, fit);
}

/*
    #] poly_rem :
    #[ poly_ratfun_read :
*/

void poly_ratfun_read (WORD *a, poly &num, poly &den) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_ratfun_read" « endl;
#endif

    POLY_GETIDENTITY(num);
```

```

    int modp = num.modp;

    WORD *astop = a+a[1];

    bool clean = (a[2] & MUSTCLEANPRF) == 0;

    a += FUNHEAD;
    if (a >= astop) {
        MLOCK(ErrorMessageLock);
        MesPrint ((char*)"ERROR: PolyRatFun cannot have zero arguments");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    poly den_num(BHEAD 1),den_den(BHEAD 1);

    num = poly::argument_to_poly(BHEAD a, true, !clean, &den_num);
    num.setmod(modp,1);
    NEXTARG(a);

    if (a < astop) {
        den = poly::argument_to_poly(BHEAD a, true, !clean, &den_den);
        den.setmod(modp,1);
        NEXTARG(a);
    }
    else {
        den = poly(BHEAD 1, modp, 1);
    }

    if (a < astop) {
        MLOCK(ErrorMessageLock);
        MesPrint ((char*)"ERROR: PolyRatFun cannot have more than two arguments");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // JD: At this point, num and den are certainly sorted into the correct order by
    // poly::argument_to_poly, but we can't rely on the clean flag to know if there
    // are any negative powers. Check for them, and set clean = false if there are any.
    vector<WORD> minpower(AN.poly_num_vars, MAXPOSITIVE);

    for (int i=1; i<num[0]; i+=num[i]) {
        for (int j=0; j<AN.poly_num_vars; j++) {
            minpower[j] = MiN(minpower[j], num[i+1+j]);
        }
    }
    for (int i=1; i<den[0]; i+=den[i]) {
        for (int j=0; j<AN.poly_num_vars; j++) {
            minpower[j] = MiN(minpower[j], den[i+1+j]);
            if ( minpower[j] < 0 ) clean = false;
        }
    }

    if (!clean) {
        for (int i=1; i<num[0]; i+=num[i])
            for (int j=0; j<AN.poly_num_vars; j++)
                num[i+1+j] -= minpower[j];
        for (int i=1; i<den[0]; i+=den[i])
            for (int j=0; j<AN.poly_num_vars; j++)
                den[i+1+j] -= minpower[j];

        num *= den_den;
        den *= den_num;

        poly gcd = polygcd::gcd(num,den);
        num /= gcd;
        den /= gcd;
    }
}

/*
    #] poly_ratfun_read :
    #[ poly_sort :
*/

void poly_sort(PHEAD WORD *a) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_sort" « endl;
#endif
    if (NewSort(BHEAD0)) { Terminate(-1); }
    AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);

    for (int i=ARGHEAD; i<a[0]; i+=a[i]) {
        if (SymbolNormalize(a+i)<0 || StoreTerm(BHEAD a+i)) {
            AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
```

```
            LowerSortLevel();
            Terminate(-1);
        }
    }

    if (EndSort(BHEAD a+ARGHEAD,1) < 0) {
        AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
        Terminate(-1);
    }

    AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
    a[1] = 0; // set dirty flag to zero
}

/*
    #] poly_sort :
    #[ poly_ratfun_add :
*/

WORD *poly_ratfun_add (PHEAD WORD *t1, WORD *t2) {

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        WORD *ret = flint_ratfun_add(BHEAD t1, t2);
        return ret;
    }
#endif

    if ( AR.PolyFunExp == 1 ) return PolyRatFunSpecial(BHEAD t1, t2);

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_ratfun_add" « endl;
#endif

    WORD *oldworkpointer = AT.WorkPointer;

    // Extract variables
    vector<WORD *> e;
    e.reserve(4);

    for (WORD *t=t1+FUNHEAD; t<t1+t1[1];) {
        e.push_back(t);
        NEXTARG(t);
    }
    for (WORD *t=t2+FUNHEAD; t<t2+t2[1];) {
        e.push_back(t);
        NEXTARG(t);
    }

    assert(e.size() == 4);

    poly::get_variables(BHEAD e, true, true);

    // Check for modulus calculus
    WORD modp=poly_determine_modulus(BHEAD true, true, "PolyRatFun");

    // Find numerators / denominators
    poly num1(BHEAD 0,modp,1), den1(BHEAD 0,modp,1), num2(BHEAD 0,modp,1), den2(BHEAD 0,modp,1);

    poly_ratfun_read(t1, num1, den1);
    poly_ratfun_read(t2, num2, den2);

    poly num(BHEAD 0),den(BHEAD 0),gcd(BHEAD 0);

    // Calculate result
    if (den1 != den2) {
        gcd = polygcd::gcd(den1,den2);
#ifdef OLDADDITION
        num = num1*(den2/gcd) + num2*(den1/gcd);
        den = (den1/gcd)*den2;
        gcd = polygcd::gcd(num,den);
#else
        den = den1/gcd;
        num = num1*(den2/gcd) + num2*den;
        den = den*den2;
        gcd = polygcd::gcd(num,gcd);
#endif
    }
    else {
        num = num1 + num2;
        den = den1;
        gcd = polygcd::gcd(num,den);
    }

    num /= gcd;
    den /= gcd;

```

```
    // Fix sign
    if (den.sign() == -1) { num*=poly(BHEAD -1); den*=poly(BHEAD -1); }

    // Check size: include FUNHEAD for the prf itself, an ARGHEAD each for num and den,
    // and 3 for the final coeff "1/1". We don't know here what the rest of the term looks like,
    // but it certainly has at least its total size (so +1):
    if ((num.size_of_form_notation() + den.size_of_form_notation() + FUNHEAD + 2*ARGHEAD + 3 + 1)
        > AM.MaxTer/(int)sizeof(WORD)) {

        MLOCK(ErrorMessageLock);
        MesPrint ("ERROR: PolyRatFun doesn't fit in a term");
        MesPrint ("(1) num size = %d, den size = %d, rest = %d, MaxTermSize = %d words",
                num.size_of_form_notation()+ARGHEAD,
                den.size_of_form_notation()+ARGHEAD,
                FUNHEAD + 3 + 1,
                AM.MaxTer/sizeof(WORD));
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Format result in Form notation
    WORD *t = oldworkpointer;

    *t++ = AR.PolyFun;                      // function
    *t++ = 0;                               // length (to be determined)
//  *t++ &= ~MUSTCLEANPRF;                  // clean polyratfun
    *t++ = 0;
    FILLFUN3(t);                            // header
    poly::poly_to_argument(num,t, true); // argument 1 (numerator)
    if (*t>0 && t[1]==DIRTYFLAG)            // to Form order
        poly_sort(BHEAD t);
    t += (*t>0 ? *t : 2);
    poly::poly_to_argument(den,t, true); // argument 2 (denominator)
    if (*t>0 && t[1]==DIRTYFLAG)            // to Form order
        poly_sort(BHEAD t);
    t += (*t>0 ? *t : 2);

    oldworkpointer[1] = t - oldworkpointer; // length
    AT.WorkPointer = t;

    poly_free_poly_vars(BHEAD "AN.poly_vars_ratfun_add");

    // reset modulo calculation
    AN.ncmod = AC.ncmod;

    return oldworkpointer;
}

/*
    #] poly_ratfun_add :
    #[ poly_ratfun_normalize :
*/

int poly_ratfun_normalize (PHEAD WORD *term) {

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        flint_ratfun_normalize(BHEAD term);
        return 0;
    }
#endif

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_ratfun_normalize" « endl;
#endif

    // Strip coefficient
    WORD *tstop = term + *term;
    int ncoeff = tstop[-1];
    tstop -= ABS(ncoeff);

    // if only one clean polyratfun, return immediately
    int num_polyratfun = 0;

    for (WORD *t=term+1; t<tstop; t+=t[1])
        if (*t == AR.PolyFun) {
            num_polyratfun++;
            if ((t[2] & MUSTCLEANPRF) != 0)
                num_polyratfun = INT_MAX;
            if (num_polyratfun > 1) break;
        }

    if (num_polyratfun <= 1) return 0;

    WORD oldsorttype = AR.SortType;
    AR.SortType = SORTHIGHFIRST;

```

```
/*
    When there are polyratfun's with only one variable: rename them
    temporarily to TMPPOLYFUN.
*/
    for (WORD *t=term+1; t<tstop; t+=t[1]) {
        if (*t == AR.PolyFun && (t[1] == FUNHEAD+t[FUNHEAD]
            || t[1] == FUNHEAD+2 ) ) { *t = TMPPOLYFUN; }
    }


    // Extract all variables in the polyfuns
    vector<WORD *> e;

    for (WORD *t=term+1; t<tstop; t+=t[1]) {
        if (*t == AR.PolyFun)
            for (WORD *t2 = t+FUNHEAD; t2<t+t[1];) {
                e.push_back(t2);
                NEXTARG(t2);
            }
    }
    poly::get_variables(BHEAD e, true, true);

    // Check for modulus calculus
    WORD modp=poly_determine_modulus(BHEAD true, true, "PolyRatFun");

    // Accumulate total denominator/numerator and copy the remaining terms
    // We start with 'trivial' polynomials
    poly num1(BHEAD (UWORD *)tstop, ncoeff/2, modp, 1);
    poly den1(BHEAD (UWORD *)tstop+ABS(ncoeff/2), ABS(ncoeff)/2, modp, 1);

    WORD *s = term+1;

    for (WORD *t=term+1; t<tstop;)
        if (*t == AR.PolyFun) {

            poly num2(BHEAD 0,modp,1);
            poly den2(BHEAD 0,modp,1);
            poly_ratfun_read(t,num2,den2);

            if ((t[2] & MUSTCLEANPRF) != 0) { // first normalize
                poly gcd1(polygcd::gcd(num2,den2));
                num2 = num2/gcd1;
                den2 = den2/gcd1;
            }
            t += t[1];
            poly gcd1(polygcd::gcd(num1,den2));
            poly gcd2(polygcd::gcd(num2,den1));

            num1 = (num1 / gcd1) * (num2 / gcd2);
            den1 = (den1 / gcd2) * (den2 / gcd1);
        }
        else {
            int i = t[1];
            if ( s != t ) { NCOPY(s,t,i) }
            else { t += i; s += i; }
        }

    // Fix sign
    if (den1.sign() == -1) { num1*=poly(BHEAD -1); den1*=poly(BHEAD -1); }

    // Check size: include FUNHEAD for the prf itself, an ARGHEAD each for num and den,
    // s-term for the copied term so far, and 3 for final coeff "1/1"
    if ((num1.size_of_form_notation() + den1.size_of_form_notation() + FUNHEAD + 2*ARGHEAD
        + s-term + 3) > AM.MaxTer/(int)sizeof(WORD)) {

        MLOCK(ErrorMessageLock);
        MesPrint ("ERROR: PolyRatFun doesn't fit in a term");
        MesPrint ("(2) num size = %d, den size = %d, rest = %d, MaxTermSize = %d words",
                num1.size_of_form_notation()+ARGHEAD,
                den1.size_of_form_notation()+ARGHEAD,
                FUNHEAD + s-term + 3,
                AM.MaxTer/sizeof(WORD));
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    // Format result in Form notation
    WORD *t = s;
    *t++ = AR.PolyFun;                    // function
    *t++ = 0;                             // size (to be determined)
    *t++ &= ~MUSTCLEANPRF;                    // clean polyratfun
    FILLFUN3(t);                          // header
    poly::poly_to_argument(num1,t,true); // argument 1 (numerator)
    if (*t>0 && t[1]==DIRTYFLAG)          // to Form order
        poly_sort(BHEAD t);
    t += (*t>0 ? *t : 2);
    poly::poly_to_argument(den1,t,true); // argument 2 (denominator)
```

```
    if (*t>0 && t[1]==DIRTYFLAG)          // to Form order
        poly_sort(BHEAD t);
    t += (*t>0 ? *t : 2);

    s[1] = t - s;                         // function length

    *t++ = 1;                             // term coefficient
    *t++ = 1;
    *t++ = 3;

    term[0] = t-term;                     // term length

    poly_free_poly_vars(BHEAD "AN.poly_vars_ratfun_normalize");

    // reset modulo calculation
    AN.ncmod = AC.ncmod;

    tstop = term + *term; tstop -= ABS(tstop[-1]);
    for (WORD *t=term+1; t<tstop; t+=t[1]) {
        if (*t == TMPPOLYFUN ) *t = AR.PolyFun;
    }

    AR.SortType = oldsorttype;
    return 0;
}

/*
    #] poly_ratfun_normalize :
    #[ poly_fix_minus_signs :
*/

void poly_fix_minus_signs (factorized_poly &a) {

    if ( a.factor.empty() ) return;

    POLY_GETIDENTITY(a.factor[0]);

    int overall_sign = +1;

    // find term with maximum power of highest symbol
    for (int i=0; i<(int)a.factor.size(); i++) {

        int maxvar = -1;
        int maxpow = -1;
        int sign = +1;

        WORD *tstop = a.factor[i].terms; tstop += *tstop;
        for (WORD *t=a.factor[i].terms+1; t<tstop; t+=*t)
            for (int j=0; j<AN.poly_num_vars; j++) {
                int var = AN.poly_vars[j];
                int pow = t[1+j];
                if (pow>0 && (var>maxvar || (var==maxvar && pow>maxpow))) {
                    maxvar = var;
                    maxpow = pow;
                    sign = SGN(*(t+*t-1));
                }
            }

        // if negative coefficient, multiply by -1
        if (sign==-1) {
            a.factor[i] *= poly(BHEAD sign);
            if (a.power[i] % 2 == 1) overall_sign*=-1;
        }
    }

    // if overall minus sign
    if (overall_sign == -1) {
        // look at constant factor and multiply by -1
        for (int i=0; i<(int)a.factor.size(); i++)
            if (a.factor[i].is_integer()) {
                a.factor[i] *= poly(BHEAD -1);
                return;
            }

        // otherwise, add a factor of -1
        a.add_factor(poly(BHEAD -1), 1);
    }
}

/*
    #] poly_fix_minus_signs :
    #[ poly_factorize :
*/

WORD *poly_factorize (PHEAD WORD *argin, WORD *argout, bool with_arghead, bool is_fun_arg) {

#ifdef DEBUG
```

```
    cout « "*** [" « thetime() « "]  CALL : poly_factorize" « endl;
#endif

    poly::get_variables(BHEAD vector<WORD*>(1,argin), with_arghead, true);
    poly den(BHEAD 0);
    poly a(poly::argument_to_poly(BHEAD argin, with_arghead, true, &den));

    // check for modulus calculus
    WORD modp=poly_determine_modulus(BHEAD true, is_fun_arg, "polynomial factorization");
    a.setmod(modp,1);

    // factorize
    factorized_poly f(polyfact::factorize(a));
    poly_fix_minus_signs(f);

    poly num(BHEAD 1);
    for (int i=0; i<(int)f.factor.size(); i++)
        if (f.factor[i].is_integer())
            num = f.factor[i];

    // determine size
    int len = with_arghead ? ARGHEAD : 0;

    if (!num.is_one() || !den.is_one()) {
        len++;
        len += MaX(ABS(num[num[1]]), den[den[1]])*2+1;
        len += with_arghead ? ARGHEAD : 1;
    }

    for (int i=0; i<(int)f.factor.size(); i++) {
        if (!f.factor[i].is_integer()) {
            len += f.power[i] * f.factor[i].size_of_form_notation();
            len += f.power[i] * (with_arghead ? ARGHEAD : 1);
        }
    }

    len++;

    if (argout != NULL) {
        // check size
        if (len >= AM.MaxTer) {
            MLOCK(ErrorMessageLock);
            MesPrint ("ERROR: factorization doesn't fit in a term (len = %d, MaxTermSize = %d words)",
                len/sizeof(WORD), AM.MaxTer/sizeof(WORD));
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    }
    else {
        // allocate size
        argout = (WORD*) Malloc1(len*sizeof(WORD), "poly_factorize");
    }

    WORD *old_argout = argout;

    // constant factor
    if (!num.is_one() || !den.is_one()) {
        int n = max(ABS(num[num[1]]), ABS(den[den[1]]));

        if (with_arghead) {
            *argout++ = ARGHEAD + 2 + 2*n;
            for (int i=1; i<ARGHEAD; i++)
                *argout++ = 0;
        }

        *argout++ = 2 + 2*n;

        for (int i=0; i<n; i++)
            *argout++ = i<ABS(num[num[1]]) ? num[2+AN.poly_num_vars+i] : 0;
        for (int i=0; i<n; i++)
            *argout++ = i<ABS(den[den[1]]) ? den[2+AN.poly_num_vars+i] : 0;

        *argout++ = SGN(num[num[1]]) * (2*n+1);

        if (!with_arghead)
            *argout++ = 0;
        else {
            if (ToFast(old_argout, old_argout))
                argout = old_argout+2;
        }
    }

    // non-constant factors
    for (int i=0; i<(int)f.factor.size(); i++)
        if (!f.factor[i].is_integer())
            for (int j=0; j<f.power[i]; j++) {
                poly::poly_to_argument(f.factor[i],argout,with_arghead);
```

```

                if (with_arghead)
                    argout += *argout > 0 ? *argout : 2;
                else {
                    while (*argout!=0) argout+=*argout;
                    argout++;
                }
            }

    *argout=0;

    poly_free_poly_vars(BHEAD "AN.poly_vars_factorize");

    // reset modulo calculation
    AN.ncmod = AC.ncmod;

    return old_argout;
}

/*
    #] poly_factorize :
    #[ poly_factorize_argument :
*/

int poly_factorize_argument(PHEAD WORD *argin, WORD *argout) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_factorize_argument" « endl;
#endif

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        flint_factorize_argument(BHEAD argin, argout);
        return 0;
    }
#endif

    poly_factorize(BHEAD argin,argout,true,true);
    return 0;
}

/*
    #] poly_factorize_argument :
    #[ poly_factorize_dollar :
*/

WORD *poly_factorize_dollar (PHEAD WORD *argin) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_factorize_dollar" « endl;
#endif

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        return flint_factorize_dollar(BHEAD argin);
    }
#endif

    return poly_factorize(BHEAD argin,NULL,false,false);
}

/*
    #] poly_factorize_dollar :
    #[ poly_factorize_expression :
*/

int poly_factorize_expression(EXPRESSIONS expr) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_factorize_expression" « endl;
#endif

    GETIDENTITY;

    if (AT.WorkPointer + AM.MaxTer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    WORD *term = AT.WorkPointer;
    WORD startebuf = cbuf[AT.ebufnum].numrhs;
    FILEHANDLE *file;
    POSITION pos;

    FILEHANDLE *oldinfile = AR.infile;
```

```
    FILEHANDLE *oldoutfile = AR.outfile;
    WORD oldBracketOn = AR.BracketOn;
    WORD *oldBrackBuf = AT.BrackBuf;
    WORD oldbracketindexflag = AT.bracketindexflag;
    char oldCommercial[COMMERCIALSIZE+2];

    strcpy(oldCommercial, (char*)AC.Commercial);
    strcpy((char*)AC.Commercial, "factorize");

    // locate is the input
    if (expr->status == HIDDENGEXPRESSION || expr->status == HIDDENLEXPRESSION ||
            expr->status == INTOHIDEGEXPRESSION || expr->status == INTOHIDELEXPRESSION) {
        AR.InHiBuf = 0; file = AR.hidefile; AR.GetFile = 2;
    }
    else {
        AR.InInBuf = 0; file = AR.outfile; AR.GetFile = 0;
    }

    // read and write to expression file
    AR.infile = AR.outfile = file;

    // dummy indices are not allowed
    if (expr->numdummies > 0) {
        MesPrint("ERROR: factorization with dummy indices not implemented");
        Terminate(-1);
    }

    // determine whether the expression in on file or in memory
    if (file->handle >= 0) {
        pos = expr->onfile;
        SeekFile(file->handle,&pos,SEEK_SET);
        if (ISNOTEQUALPOS(pos,expr->onfile)) {
            MesPrint("ERROR: something wrong in scratch file [poly_factorize_expression]");
            Terminate(-1);
        }
        file->POposition = expr->onfile;
        file->POfull = file->PObuffer;
        if (expr->status == HIDDENGEXPRESSION)
            AR.InHiBuf = 0;
        else
            AR.InInBuf = 0;
    }
    else {
        file->POfill = (WORD *)((UBYTE *)(file->PObuffer)+BASEPOSITION(expr->onfile));
    }

    SetScratch(AR.infile, &(expr->onfile));

    // read the first header term
    WORD size = GetTerm(BHEAD term);
    if (size <= 0) {
        MesPrint ("ERROR: something wrong with expression [poly_factorize_expression]");
        Terminate(-1);
    }

    // store position: this is where the output will go
    pos = expr->onfile;
    ADDPOS(pos, size*sizeof(WORD));

    // use polynomial as buffer, because it is easy to extend
    poly buffer(BHEAD 0);
    int bufpos = 0;
    int sumcommu = 0;

    // read all terms
    while (GetTerm(BHEAD term)) {
        // substitute non-symbols by extra symbols
        sumcommu += DoesCommu(term);
        if ( sumcommu > 1 ) {
            MesPrint("ERROR: Cannot factorize an expression with more than one noncommuting object");
            Terminate(-1);
        }
        buffer.check_memory(bufpos);
        if (LocalConvertToPoly(BHEAD term, buffer.terms + bufpos, startebuf,0) < 0) {
            MesPrint("ERROR: in LocalConvertToPoly [factorize_expression]");
            Terminate(-1);
        }
        bufpos += *(buffer.terms + bufpos);
    }
    buffer[bufpos] = 0;

    // parse the polynomial

    AN.poly_num_vars = 0;
    poly::get_variables(BHEAD vector<WORD*>(1,buffer.terms), false, true);
    poly den(BHEAD 0);
    poly a(poly::argument_to_poly(BHEAD buffer.terms, false, true, &den));
```

```
    // check for modulus calculus
    WORD modp=poly_determine_modulus(BHEAD true, false, "polynomial factorization");
    a.setmod(modp,1);

    // create output
    SetScratch(file, &pos);
    NewSort(BHEAD0);

    CBUF *C = cbuf+AC.cbufnum;
    CBUF *CC = cbuf+AT.ebufnum;

    // turn brackets on. We force the existence of a bracket index.
    WORD nexpr = expr - Expressions;
    AR.BracketOn = 1;
    AT.BrackBuf = AM.BracketFactors;
    AT.bracketindexflag = 1;
    ClearBracketIndex(-nexpr-2); // Clears the index made during primary generation
    OpenBracketIndex(nexpr);     // Set up a new index

    if (a.is_zero()) {
        expr->numfactors = 1;
    }
    else if (a.is_one() && den.is_one()) {
        expr->numfactors = 1;

        term[0] = 8;
        term[1] = SYMBOL;
        term[2] = 4;
        term[3] = FACTORSYMBOL;
        term[4] = 1;
        term[5] = 1;
        term[6] = 1;
        term[7] = 3;

        AT.WorkPointer += *term;
        Generator(BHEAD term, C->numlhs);
        AT.WorkPointer = term;
    }
    else {
        factorized_poly fac;
        bool iszero = false;

        if (!(expr->vflags & ISFACTORIZED)) {
            // factorize the polynomial
            fac = polyfact::factorize(a);
            poly_fix_minus_signs(fac);
        }
        else {
            int factorsymbol=-1;
            for (int i=0; i<AN.poly_num_vars; i++)
                if (AN.poly_vars[i] == FACTORSYMBOL)
                    factorsymbol = i;

            poly denpow(BHEAD 1);

            // already factorized, so factorize the factors
            for (int i=1; i<=expr->numfactors; i++) {
                poly origfac(a.coefficient(factorsymbol, i));
                factorized_poly fac2;
                if (origfac.is_zero())
                    iszero=true;
                else {
                    fac2 = polyfact::factorize(origfac);
                    poly_fix_minus_signs(fac2);
                    denpow *= den;
                }
                for (int j=0; j<(int)fac2.power.size(); j++)
                    fac.add_factor(fac2.factor[j], fac2.power[j]);
            }

            // update denominator, since each factor was scaled
            den=denpow;
        }

        expr->numfactors = 0;

        // coefficient
        poly num(BHEAD 1);
        for (int i=0; i<(int)fac.factor.size(); i++)
            if (fac.factor[i].is_integer())
                num *= fac.factor[i];

        poly gcd(polygcd::integer_gcd(num,den));
        den/=gcd;
        num/=gcd;

```

```
        if (iszero)
            expr->numfactors++;

        if (!iszero || (expr->vflags & KEEPZERO)) {
            if (!num.is_one() || !den.is_one()) {
                expr->numfactors++;

                int n = max(ABS(num[num[1]]), ABS(den[den[1]]));

                term[0] = 6 + 2*n;
                term[1] = SYMBOL;
                term[2] = 4;
                term[3] = FACTORSYMBOL;
                term[4] = expr->numfactors;
                for (int i=0; i<n; i++) {
                    term[5+i]   = i<ABS(num[num[1]]) ? num[2+AN.poly_num_vars+i] : 0;
                    term[5+n+i] = i<ABS(den[den[1]]) ? den[2+AN.poly_num_vars+i] : 0;
                }
                term[5+2*n] = SGN(num[num[1]]) * (2*n+1);
                AT.WorkPointer += *term;
                Generator(BHEAD term, C->numlhs);
                AT.WorkPointer = term;
            }

            vector<poly> fac_arg(fac.factor.size(), poly(BHEAD 0));

            // convert the non-constant factors to Form-style arguments
            for (int i=0; i<(int)fac.factor.size(); i++)
                if (!fac.factor[i].is_integer()) {
                    buffer.check_memory(fac.factor[i].size_of_form_notation()+1);
                    poly::poly_to_argument(fac.factor[i], buffer.terms, false);
                    NewSort(BHEAD0);

                    for (WORD *t=buffer.terms; *t!=0; t+=*t) {
                        // substitute extra symbols
                        if (ConvertFromPoly(BHEAD t, term, numxsymbol,
CC->numrhs-startebuf+numxsymbol,
                                                                startebuf-numxsymbol, 1) <= 0 ) {
                            MesPrint("ERROR: in ConvertFromPoly [factorize_expression]");
                            Terminate(-1);
                            return(-1);
                        }

                        // store term
                        AT.WorkPointer += *term;
                        Generator(BHEAD term, C->numlhs);
                        AT.WorkPointer = term;
                    }

                    // sort and store in buffer
                    WORD *buffer;
                    if (EndSort(BHEAD (WORD *)((void *)(&buffer)),2) < 0) return -1;

                    LONG bufsize=0;
                    for (WORD *t=buffer; *t!=0; t+=*t)
                        bufsize+=*t;

                    fac_arg[i].check_memory(bufsize+ARGHEAD+1);

                    for (int j=0; j<ARGHEAD; j++)
                        fac_arg[i].terms[j] = 0;
                    fac_arg[i].terms[0] = ARGHEAD + bufsize;
                    WCOPY(fac_arg[i].terms+ARGHEAD, buffer, bufsize);
                    M_free(buffer, "polynomial factorization");
                }

            // compare and sort the factors in Form notation
            vector<int> order;
            vector<vector<int> > comp(fac.factor.size(), vector<int>(fac.factor.size(), 0));

            for (int i=0; i<(int)fac.factor.size(); i++)
                if (!fac.factor[i].is_integer()) {
                    order.push_back(i);

                    for (int j=i+1; j<(int)fac.factor.size(); j++)
                        if (!fac.factor[j].is_integer()) {
                            comp[i][j] = CompArg(fac_arg[j].terms, fac_arg[i].terms);
                            comp[j][i] = -comp[i][j];
                        }
                }

            for (int i=0; i<(int)order.size(); i++)
                for (int j=0; j+1<(int)order.size(); j++)
                    if (comp[order[i]][order[j]] == 1)
                        swap(order[i],order[j]);

            // create the final expression
```

```
            for (int i=0; i<(int)order.size(); i++)
                for (int j=0; j<fac.power[order[i]]; j++) {

                    expr->numfactors++;

                    WORD *tstop = fac_arg[order[i]].terms + *fac_arg[order[i]].terms;
                    for (WORD *t=fac_arg[order[i]].terms+ARGHEAD; t<tstop; t+=*t) {

                        WCOPY(term+4, t, *t);

                        // add special symbol "factor_"
                        *term = *(term+4) + 4;
                        *(term+1) = SYMBOL;
                        *(term+2) = 4;
                        *(term+3) = FACTORSYMBOL;
                        *(term+4) = expr->numfactors;

                        // store term
                        AT.WorkPointer += *term;
                        Generator(BHEAD term, C->numlhs);
                        AT.WorkPointer = term;
                    }
                }
        }
    }

    // final sorting
    if (EndSort(BHEAD NULL,0) < 0) {
        LowerSortLevel();
        Terminate(-1);
    }

    // set factorized flag
    if (expr->numfactors > 0)
        expr->vflags |= ISFACTORIZED;

    // clean up
    AR.infile = oldinfile;
    AR.outfile = oldoutfile;
    AR.BracketOn = oldBracketOn;
    AT.BrackBuf = oldBrackBuf;
    AT.bracketindexflag = oldbracketindexflag;
    strcpy((char*)AC.Commercial, oldCommercial);

    poly_free_poly_vars(BHEAD "AN.poly_vars_factorize_expression");

    return 0;
}

/*
    #] poly_factorize_expression :
    #[ poly_unfactorize_expression :
*/

#if ( SUBEXPSIZE == 5 )
static WORD genericterm[] = {38,1,4,FACTORSYMBOL,0
    ,EXPRESSION,15,0,1,0,13,10,8,1,4,FACTORSYMBOL,0,1,1,3
    ,EXPRESSION,15,0,1,0,13,10,8,1,4,FACTORSYMBOL,0,1,1,3
    ,1,1,3,0};
static WORD genericterm2[] = {23,1,4,FACTORSYMBOL,0
    ,EXPRESSION,15,0,1,0,13,10,8,1,4,FACTORSYMBOL,0,1,1,3
    ,1,1,3,0};
#endif

int poly_unfactorize_expression(EXPRESSIONS expr)
{
    GETIDENTITY;
    int i, j, nfac = expr->numfactors, nfacp, nexpr = expr - Expressions;
    int expriszero = 0;

    FILEHANDLE *oldinfile = AR.infile;
    FILEHANDLE *oldoutfile = AR.outfile;
    char oldCommercial[COMMERCIALSIZE+2];

    WORD *oldworkpointer = AT.WorkPointer;
    WORD *term = AT.WorkPointer, *t, *w, size;

    FILEHANDLE *file;
    POSITION pos, oldpos;

    WORD oldBracketOn = AR.BracketOn;
    WORD *oldBrackBuf = AT.BrackBuf;
    CBUF *C = cbuf+AC.cbufnum;

    if ( ( expr->vflags & ISFACTORIZED ) == 0 ) return(0);

    if ( AT.WorkPointer + AM.MaxTer > AT.WorkTop ) {
```

```
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    oldpos = AS.OldOnFile[nexpr];
    AS.OldOnFile[nexpr] = expr->onfile;

    strcpy(oldCommercial, (char*)AC.Commercial);
    strcpy((char*)AC.Commercial, "unfactorize");
/*
    locate the input
*/
    if ( expr->status == HIDDENGEXPRESSION || expr->status == HIDDENLEXPRESSION ||
            expr->status == INTOHIDEGEXPRESSION || expr->status == INTOHIDELEXPRESSION ) {
        AR.InHiBuf = 0; file = AR.hidefile; AR.GetFile = 2;
    }
    else {
        AR.InInBuf = 0; file = AR.outfile; AR.GetFile = 0;
    }
/*
    read and write to expression file
*/
    AR.infile = AR.outfile = file;
/*
    set the input file to the correct position
*/
    if ( file->handle >= 0 ) {
        pos = expr->onfile;
        SeekFile(file->handle,&pos,SEEK_SET);
        if (ISNOTEQUALPOS(pos,expr->onfile)) {
            MesPrint("ERROR: something wrong in scratch file unfactorize_expression");
            Terminate(-1);
        }
        file->POposition = expr->onfile;
        file->POfull = file->PObuffer;
        if ( expr->status == HIDDENGEXPRESSION )
            AR.InHiBuf = 0;
        else
            AR.InInBuf = 0;
    }
    else {
        file->POfill = (WORD *)((UBYTE *)(file->PObuffer)+BASEPOSITION(expr->onfile));
    }
    SetScratch(AR.infile, &(expr->onfile));
/*
    Test for whether the first factor is zero.
*/
    if ( GetFirstBracket(term,nexpr) < 0 ) Terminate(-1);
    if ( term[4] != 1 || *term != 8 || term[1] != SYMBOL || term[3] != FACTORSYMBOL || term[4] != 1 )
{
        expriszero = 1;
    }
    SetScratch(AR.infile, &(expr->onfile));
/*
    Read the prototype. After this we have the file ready for the output at pos.
*/
    size = GetTerm(BHEAD term);
    if ( size <= 0 ) {
        MesPrint ("ERROR: something wrong with expression unfactorize_expression");
        Terminate(-1);
    }
    pos = expr->onfile;
    ADDPOS(pos, size*sizeof(WORD));
/*
    Set the brackets straight
*/
    AR.BracketOn = 1;
    AT.BrackBuf = AM.BracketFactors;
    AT.bracketinfo = 0;
    while ( nfac > 2 ) {
        nfacp = nfac - nfac%2;
/*
        Prepare the bracket index. We have:
            e->bracketinfo:    the old input bracket index
            e->newbracketinfo: the bracket index made for our current input
        We need to keep e->bracketinfo in case other workers need it (InParallel)
        Hence we work with AT.bracketinfo which takes priority.
        Note that in Processor we forced a newbracketinfo to be made.
*/
        if ( AT.bracketinfo != 0 ) ClearBracketIndex(-1);
        AT.bracketinfo = expr->newbracketinfo;
        OpenBracketIndex(nexpr);
/*
        Now emulate the terms:
            sum_(i,0,nfacp,2,factor_^(i/2+1)*F[factor_^(i+1)]*F[factor_^(i+2)])
```

```
            +factor_^(nfacp/2+1)*F[factor_^nfac]
*/
        NewSort(BHEAD0);
        if ( expriszero == 0 ) {
            for ( i = 0; i < nfacp; i += 2 ) {
                t = genericterm; w = term = oldworkpointer;
                j = *t; NCOPY(w,t,j);
                term[4] = i/2+1;
                term[7] = nexpr;
                term[16] = i+1;
                term[22] = nexpr;
                term[31] = i+2;
                AT.WorkPointer = term + *term;
                Generator(BHEAD term, C->numlhs);
            }
            if ( nfac > nfacp ) {
                t = genericterm2; w = term = oldworkpointer;
                j = *t; NCOPY(w,t,j);
                term[4] = i/2+1;
                term[7] = nexpr;
                term[16] = nfac;
                AT.WorkPointer = term + *term;
                Generator(BHEAD term, C->numlhs);
            }
        }
        if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) {
            LowerSortLevel();
            Terminate(-1);
        }
/*
        Set the file back into reading position
*/
        SetScratch(file, &pos);
        nfac = (nfac+1)/2;
        if ( expriszero ) { nfac = 1; }
    }
    if ( AT.bracketinfo != 0 ) ClearBracketIndex(-1);
    AT.bracketinfo = expr->newbracketinfo;
    expr->newbracketinfo = 0;
/*
    Reset the brackets to make them ready for the final pass
*/
    AR.BracketOn = oldBracketOn;
    AT.BrackBuf = oldBrackBuf;
    if ( AR.BracketOn ) OpenBracketIndex(nexpr);
/*
    We distinguish two cases: nfac == 2 and nfac == 1
    After preparing the term we skip the factor_ part.
*/
    NewSort(BHEAD0);
    if ( expriszero == 0 ) {
        if ( nfac == 1 ) {
            t = genericterm2; w = term = oldworkpointer;
            j = *t; NCOPY(w,t,j);
            term[7] = nexpr;
            term[16] = nfac;
        }
        else if ( nfac == 2 ) {
            t = genericterm; w = term = oldworkpointer;
            j = *t; NCOPY(w,t,j);
            term[7] = nexpr;
            term[16] = 1;
            term[22] = nexpr;
            term[31] = 2;
        }
        else {
            AS.OldOnFile[nexpr] = oldpos;
            return(-1);
        }
        term[4] = term[0]-4;
        term += 4;
        AT.WorkPointer = term + *term;
        Generator(BHEAD term, C->numlhs);
    }
    if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) {
        LowerSortLevel();
        Terminate(-1);
    }
/*
    Final Cleanup
*/
    expr->numfactors = 0;
    expr->vflags &= ~ISFACTORIZED;
    if ( AT.bracketinfo != 0 ) ClearBracketIndex(-1);

    AR.infile = oldinfile;
    AR.outfile = oldoutfile;
```

```
    strcpy((char*)AC.Commercial, oldCommercial);
    AT.WorkPointer = oldworkpointer;
    AS.OldOnFile[nexpr] = oldpos;

    return(0);
}

/*
    #] poly_unfactorize_expression :
    #[ poly_inverse :
*/

WORD *poly_inverse(PHEAD WORD *arga, WORD *argb) {

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        return flint_inverse(BHEAD arga, argb);
    }
#endif

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_inverse" « endl;
#endif

    // Extract variables
    vector<WORD *> e;
    e.reserve(2);
    e.push_back(arga);
    e.push_back(argb);
    poly::get_variables(BHEAD e, false, true);

    if (AN.poly_num_vars > 1) {
        MLOCK(ErrorMessageLock);
        MesPrint ((char*)"ERROR: multivariate polynomial inverse is generally impossible");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    poly finalden(BHEAD 1), finalres(BHEAD 1);
    int ressize = 0;
    WORD *res = NULL;

    // Convert to polynomials
    poly dena(BHEAD 0); // We need to keep the overall denominator of arga, to multiply the result
    poly a(poly::argument_to_poly(BHEAD arga, false, true, &dena));
    poly b(poly::argument_to_poly(BHEAD argb, false, true));

    // Divide out the integer content, FORM has not already done this.
    poly content_a(BHEAD 0), content_b(BHEAD 0);
    content_a = polygcd::integer_content(a);
    content_b = polygcd::integer_content(b);
    a /= content_a;
    b /= content_b;

    poly invamodp(BHEAD 0), invbmodp(BHEAD 0);
    WORD modp = 0;

    // Special cases:
    // Possibly strange that we give 1 for inverse_(x1,1) but here we take MMA's convention.
    if ((a.is_one() && b.is_one()) || a.is_one()) {
        finalres = poly(BHEAD 1);
    }
    else if (b.is_one()) {
        finalres = poly(BHEAD 0);
    }
    else {
        // Check for modulus calculus
        modp=poly_determine_modulus(BHEAD true, true, "polynomial inverse");
        const bool mod_calc = modp==0 ? false : true;
        a.setmod(modp,1);
        b.setmod(modp,1);

        // Check the gcd of a,b: if it is != 1, the inverse does not exist.
        poly gcd(polygcd::gcd(a,b));
        if (!gcd.is_one()) {
            MLOCK(ErrorMessageLock);
            if (mod_calc) {
                MesPrint ((char*)"ERROR: polynomial inverse does not exist (mod %d)", modp);
            }
            else {
                MesPrint ((char*)"ERROR: polynomial inverse does not exist");
            }
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }

        bool inv_exists = true;
```

```
        do {
            // If we are not using modulus calculus, find a suitable prime for xgcd:
            if (!mod_calc) {
                vector<int> x(1,0);
                modp = polyfact::choose_prime(a.integer_lcoeff()*b.integer_lcoeff(), x, modp);
            }

            poly amodp(a,modp,1);
            poly bmodp(b,modp,1);

            // Calculate gcd
            vector<poly> xgcd(polyfact::extended_gcd_Euclidean_lifted(amodp,bmodp));
            invamodp = poly(xgcd[0]);
            invbmodp = poly(xgcd[1]);

            inv_exists = ((invamodp * amodp) % bmodp).is_one();
            if (!inv_exists && mod_calc) {
                // Control should not reach here!
                MLOCK(ErrorMessageLock);
                MesPrint ((char*)"ERROR: polynomial inverse does not exist (mod %d) B", modp);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }

        // If the inverse does not exist and we are not working in modulus calculus,
        // choose a new prime and try again. This loop should always terminate, as
        // we have already checked that gcd(a,b) == 1.
        } while (!inv_exists);

        // estimate of the size of the Form notation; might be extended later
        ressize = invamodp.size_of_form_notation()+1;
        res = (WORD *)Malloc1(ressize*sizeof(WORD), "poly_inverse");

        // initialize polynomials to store the result
        poly primepower(BHEAD modp);
        poly inva(invamodp,modp,1);
        poly invb(invbmodp,modp,1);

        while (true) {
            // convert to Form notation
            int j=0;
            WORD n=0;
            for (int i=1; i<inva[0]; i+=inva[i]) {

                // check whether res should be extended
                while (ressize < j + 2*ABS(inva[i+inva[i]-1]) + (inva[i+1]>0?4:0) + 3) {
                    int newressize = 2*ressize;

                    WORD *newres = (WORD *)Malloc1(newressize*sizeof(WORD), "poly_inverse");
                    WCOPY(newres, res, ressize);
                    M_free(res, "poly_inverse");
                    res = newres;
                    ressize = newressize;
                }

                res[j] = 1;
                if (inva[i+1]>0) {
                    res[j+res[j]++] = SYMBOL;
                    res[j+res[j]++] = 4;
                    res[j+res[j]++] = AN.poly_vars[0];
                    res[j+res[j]++] = inva[i+1];
                }
                MakeLongRational(BHEAD (UWORD *)&inva[i+2], inva[i+inva[i]-1],
                                                 (UWORD*)&primepower.terms[3],
primepower.terms[primepower.terms[1]],
                                                 (UWORD *)&res[j+res[j]], &n);
                res[j] += 2*ABS(n);
                res[j+res[j]++] = SGN(n)*(2*ABS(n)+1);
                j += res[j];
            }
            res[j]=0;

            // if modulus calculus is set, this is the answer
            if (a.modp != 0) break;

            // otherwise check over integers
            poly den(BHEAD 0);
            poly check(poly::argument_to_poly(BHEAD res, false, true, &den));
            // Shortcut: if b's lcoeff doesn't divide the lcoeff of check*a,
            // b certainly doesn't divide check*a:
            if (poly::divides(b.integer_lcoeff(), check.integer_lcoeff()*a.integer_lcoeff())) {
                check = check*a - den;
                if (poly::divides(b, check)) break;
            }

            // if incorrect, lift with quadratic p-adic Newton's iteration.
            poly error((poly(BHEAD 1) - a*inva - b*invb) / primepower);
```

```
            poly errormodpp(error, modp, inva.modn);

            inva.modn *= 2;
            invb.modn *= 2;

            poly dinva((inva * errormodpp) % b);
            poly dinvb((invb * errormodpp) % a);

            inva += dinva * primepower;
            invb += dinvb * primepower;

            primepower *= primepower;
        }

        // One more round trip from form -> poly -> form, to multiply by dena in an easy way
        finalres = poly(poly::argument_to_poly(BHEAD res, false, true, &finalden));
    }

    finalres *= dena;
    // The overall denominator additionally needs to be multiplied by content_a:
    finalden *= content_a;
    const WORD finalden_size = finalden.terms[finalden.terms[1]];
    const int finalsize = finalres.size_of_form_notation_with_den(finalden_size)+1;
    if (ressize < finalsize) {
        if (res != NULL) {
            M_free(res, "poly_inverse");
        }
        res = (WORD *)Malloc1(finalsize*sizeof(WORD), "poly_inverse");
    }
    poly::poly_to_argument_with_den(finalres, finalden_size,
        (UWORD*)&(finalden.terms[finalden.terms[1] - ABS(finalden_size)]), res, false);

    // clean up and reset modulo calculation
    poly_free_poly_vars(BHEAD "AN.poly_vars_inverse");

    AN.ncmod = AC.ncmod;
    return res;
}

/*
    #] poly_inverse :
    #[ poly_mul :
*/

WORD *poly_mul(PHEAD WORD *a, WORD *b) {

#ifdef DEBUG
    cout « "*** [" « thetime() « "]  CALL : poly_mul" « endl;
#endif

#ifdef WITHFLINT
    if ( AC.FlintPolyFlag && AC.ncmod==0 ) {
        return flint_mul(BHEAD a, b);
    }
#endif

    // Extract variables
    vector<WORD *> e;
    e.reserve(2);
    e.push_back(a);
    e.push_back(b);
    poly::get_variables(BHEAD e, false, false);  // TODO: any performance effect by sort_vars=true?

    // Convert to polynomials
    poly dena(BHEAD 0);
    poly denb(BHEAD 0);
    poly pa(poly::argument_to_poly(BHEAD a, false, true, &dena));
    poly pb(poly::argument_to_poly(BHEAD b, false, true, &denb));

    // Check for modulus calculus
    WORD modp = poly_determine_modulus(BHEAD true, true, "polynomial multiplication");
    pa.setmod(modp, 1);

    // NOTE: mul_ is currently implemented by translating negative powers of
    // symbols to extra symbols. For future improvement, it may be better to
    // compute
    //   (content(a) * content(b)) * (a/content(a)) * (b/content(b))
    // to avoid introducing extra symbols for "mixed" cases, e.g.,
    // (1+x) * (1/x) -> (1+x) * (1+Z1_).
    assert(dena.is_integer());
    assert(denb.is_integer());
    assert(modp == 0 || dena.is_one());
    assert(modp == 0 || denb.is_one());

    // multiplication
    pa *= pb;

```

```
    // convert to Form notation
    WORD *res;
    if (dena.is_one() && denb.is_one()) {
        res = (WORD *)Malloc1((pa.size_of_form_notation() + 1) * sizeof(WORD), "poly_mul");
        poly::poly_to_argument(pa, res, false);
    } else {
        dena *= denb;
        res = (WORD *)Malloc1((pa.size_of_form_notation_with_den(dena[dena[1]]) + 1) * sizeof(WORD), "poly_mul");
        poly::poly_to_argument_with_den(pa, dena[dena[1]], (const UWORD *)&dena[2+AN.poly_num_vars], res, false);
    }

    // clean up and reset modulo calculation
    poly_free_poly_vars(BHEAD "AN.poly_vars_mul");
    AN.ncmod = AC.ncmod;

    return res;
}

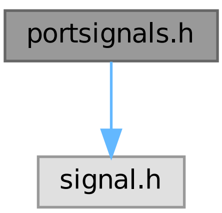
/*
    #] poly_mul :
    #[ poly_free_poly_vars :
*/

void poly_free_poly_vars(PHEAD const char *text)
{
    if ( AN.poly_vars_type == 0 ) {
        TermFree(AN.poly_vars, text);
    }
    else {
        M_free(AN.poly_vars, text);
    }
    AN.poly_num_vars = 0;
    AN.poly_vars = 0;
}

/*
    #] poly_free_poly_vars :
*/
```

4.110 portsignals.h File Reference

`#include <signal.h>`
Include dependency graph for portsignals.h:



Macros

- #define FATAL_SIG_ERROR 4
- #define NSIG (1024)
- #define SIGSEGV (NSIG+1)
- #define SIGFPE (NSIG+2)

- #define SIGILL (NSIG+3)
- #define SIGEMT (NSIG+4)
- #define SIGSYS (NSIG+5)
- #define SIGPIPE (NSIG+6)
- #define SIGLOST (NSIG+7)
- #define SIGXCPU (NSIG+8)
- #define SIGXFSZ (NSIG+9)
- #define SIGTERM (NSIG+10)
- #define SIGINT (NSIG+11)
- #define SIGQUIT (NSIG+12)
- #define SIGHUP (NSIG+13)
- #define SIGALRM (NSIG+14)
- #define SIGVTALRM (NSIG+15)
- #define SIGPROF (NSIG+16)

4.110.1 Detailed Description

Contains definitions for signals used/intercepted in FORM.

Some systems (especially LINUX) have not enough signals available so some of the (!documented!) signals are not defined. This file contains the definition of all signals used in the program. If the signal is not defined we define it as unused (>NSIG).

The include of signal.h must be first, before we try to define undefined signals.

Definition in file portsignals.h.

4.110.2 Macro Definition Documentation

4.110.2.1 FATAL_SIG_ERROR

```
#define FATAL_SIG_ERROR 4
```

Definition at line 47 of file portsignals.h.

4.110.2.2 NSIG

```
#define NSIG (1024)
```

Definition at line 53 of file portsignals.h.

4.110.2.3 SIGSEGV

```
#define SIGSEGV (NSIG+1)
```

Definition at line 57 of file portsignals.h.

4.110.2.4 SIGFPE

```
#define SIGFPE (NSIG+2)
```

Definition at line 60 of file portsignals.h.

4.110.2.5 SIGILL

```
#define SIGILL (NSIG+3)
```

Definition at line 63 of file portsignals.h.

4.110.2.6 SIGEMT

```
#define SIGEMT (NSIG+4)
```

Definition at line 66 of file portsignals.h.

4.110.2.7 SIGSYS

```
#define SIGSYS (NSIG+5)
```

Definition at line 69 of file portsignals.h.

4.110.2.8 SIGPIPE

```
#define SIGPIPE (NSIG+6)
```

Definition at line 72 of file portsignals.h.

4.110.2.9 SIGLOST

```
#define SIGLOST (NSIG+7)
```

Definition at line 75 of file portsignals.h.

4.110.2.10 SIGXCPU

```
#define SIGXCPU (NSIG+8)
```

Definition at line 78 of file portsignals.h.

4.110.2.11 SIGXFSZ

```
#define SIGXFSZ (NSIG+9)
```

Definition at line 81 of file portsignals.h.

4.110.2.12 SIGTERM

```
#define SIGTERM (NSIG+10)
```

Definition at line 84 of file portsignals.h.

4.110.2.13 SIGINT

```
#define SIGINT (NSIG+11)
```

Definition at line 87 of file portsignals.h.

4.110.2.14 SIGQUIT

```
#define SIGQUIT (NSIG+12)
```

Definition at line 90 of file portsignals.h.

4.110.2.15 SIGHUP

```
#define SIGHUP (NSIG+13)
```

Definition at line 93 of file portsignals.h.

4.110.2.16 SIGALRM

```
#define SIGALRM (NSIG+14)
```

Definition at line 96 of file portsignals.h.

4.110.2.17 SIGVTALRM

```
#define SIGVTALRM (NSIG+15)
```

Definition at line 99 of file portsignals.h.

4.110.2.18 SIGPROF

```
#define SIGPROF (NSIG+16)
```

Definition at line 102 of file portsignals.h.

4.111 portsignals.h

Go to the documentation of this file.

```
#ifndef PORTSIGNAL_H
#define PORTSIGNAL_H

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#include <signal.h>

#define FATAL_SIG_ERROR 4

#ifndef NSIG
/*
    The value of NSIG must be enough to fall outside the range of defined signals
*/
#define NSIG (1024)
#endif

#ifndef SIGSEGV
#define SIGSEGV (NSIG+1)
#endif
#ifndef SIGFPE
#define SIGFPE (NSIG+2)
#endif
#ifndef SIGILL
#define SIGILL (NSIG+3)
#endif
#ifndef SIGEMT
#define SIGEMT (NSIG+4)
#endif
#ifndef SIGSYS
#define SIGSYS (NSIG+5)
#endif
#ifndef SIGPIPE
#define SIGPIPE  (NSIG+6)
#endif
#ifndef SIGLOST
#define SIGLOST (NSIG+7)
#endif
#ifndef SIGXCPU
#define SIGXCPU  (NSIG+8)
#endif
#ifndef SIGXFSZ
#define SIGXFSZ (NSIG+9)
#endif
#ifndef SIGTERM
#define SIGTERM (NSIG+10)
#endif
#ifndef SIGINT
#define SIGINT (NSIG+11)
#endif
#ifndef SIGQUIT
#define SIGQUIT (NSIG+12)
#endif
#ifndef SIGHUP
#define SIGHUP (NSIG+13)
#endif
#ifndef SIGALRM
#define SIGALRM (NSIG+14)
```

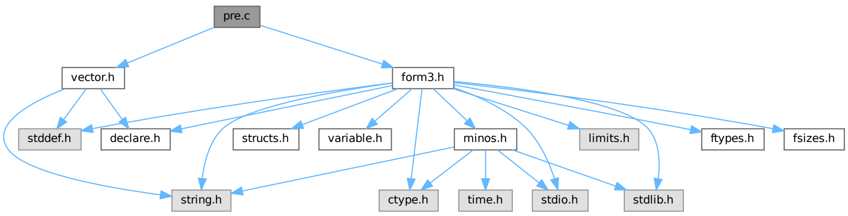
```
#endif
#ifndef SIGVTALRM
#define SIGVTALRM (NSIG+15)
#endif
#ifndef SIGPROF
#define SIGPROF (NSIG+16)
#endif

#endif
```

4.112 pre.c File Reference

```
#include "form3.h"
#include "vector.h"
```

Include dependency graph for pre.c:



Macros

- #define STRINGIFY(x) STRINGIFY__(x)
- #define STRINGIFY__(x) #x
- #define SKIPBUFSIZE 20
- #define KILL "kill"
- #define KILLALL "killall"
- #define DAEMON "daemon"
- #define SHELL "shell"
- #define STDERR "stderr"
- #define TRUE_EXPR "true"
- #define FALSE_EXPR "false"
- #define NOSHELL "noshell"
- #define TERMINAL "terminal"

Functions

- UBYTE GetInput (void)
- void ClearPushback (void)
- UBYTE GetChar (int level)
- void CharOut (UBYTE c)
- void UnsetAllowDelay (void)
- UBYTE ∗ GetPreVar (UBYTE ∗name, int flag)
- int PutPreVar (UBYTE ∗name, UBYTE ∗value, UBYTE ∗args, int mode)
- void PopPreVars (int tonumber)
- void IniModule (int type)
- void IniSpecialModule (int type)

- void PreProcessor (void)
- int PreProInstruction (void)
- int LoadInstruction (int mode)
- int LoadStatement (int type)
- int ExpandTripleDots (int par)
- KEYWORD * FindKeyWord (UBYTE *theword, KEYWORD *table, int size)
- KEYWORD * FindInKeyWord (UBYTE *theword, KEYWORD *table, int size)
- int TheDefine (UBYTE *s, int mode)
- int DoCommentChar (UBYTE *s)
- int DoPreAssign (UBYTE *s)
- int DoDefine (UBYTE *s)
- int DoRedefine (UBYTE *s)
- int ClearMacro (UBYTE *name)
- int TheUndefine (UBYTE *name)
- int DoUndefine (UBYTE *s)
- int DoInclude (UBYTE *s)
- int DoReverseInclude (UBYTE *s)
- int Include (UBYTE *s, int type)
- int DoPreExchange (UBYTE *s)
- int DoCall (UBYTE *s)
- int DoDebug (UBYTE *s)
- int DoTerminate (UBYTE *s)
- int DoContinueDo (UBYTE *s)
- int DoDo (UBYTE *s)
- int DoBreakDo (UBYTE *s)
- int DoElse (UBYTE *s)
- int DoElseif (UBYTE *s)
- int DoEnddo (UBYTE *s)
- int DoEndif (UBYTE *s)
- int DoEndprocedure (UBYTE *s)
- int DoIf (UBYTE *s)
- int DoIfdef (UBYTE *s, int par)
- int DoIfydef (UBYTE *s)
- int DoIfndef (UBYTE *s)
- int DoInside (UBYTE *s)
- int DoEndInside (UBYTE *s)
- int DoMessage (UBYTE *s)
- int DoPipe (UBYTE *s)
- int DoPrcExtension (UBYTE *s)
- int DoPreOut (UBYTE *s)
- int DoPrePrintTimes (UBYTE *s)
- int DoPreSortReallocate (UBYTE *s)
- int DoPreAppend (UBYTE *s)
- int DoPreCreate (UBYTE *s)
- int DoPreRemove (UBYTE *s)
- int DoPreClose (UBYTE *s)
- int DoPreWrite (UBYTE *s)
- int DoProcedure (UBYTE *s)
- int DoPreBreak (UBYTE *s)
- int DoPreCase (UBYTE *s)
- int DoPreDefault (UBYTE *s)
- int DoPreEndSwitch (UBYTE *s)
- int DoPreSwitch (UBYTE *s)
- int DoPreShow (UBYTE *s)
- int DoSystem (UBYTE *s)

- int PreLoad (PRELOAD *p, UBYTE *start, UBYTE *stop, int mode, char *message)
- int PreSkip (UBYTE *start, UBYTE *stop, int mode)
- void StartPrepro (void)
- int EvalPreIf (UBYTE *s)
- UBYTE * PreIfEval (UBYTE *s, int *value)
- int PreCmp (int type, int val, UBYTE *t, int type2, int val2, UBYTE *t2, int cmpop)
- int PreEq (int type, int val, UBYTE *t, int type2, int val2, UBYTE *t2, int eqop)
- UBYTE * pParseObject (UBYTE *s, int *type, LONG *val2)
- UBYTE * PreCalc (void)
- UBYTE * PreEval (UBYTE *s, LONG *x)
- void AddToPreTypes (int type)
- void MessPreNesting (int par)
- int DoPreAddSeparator (UBYTE *s)
- int DoPreRmSeparator (UBYTE *s)
- int DoExternal (UBYTE *s)
- int DoPrompt (UBYTE *s)
- int DoSetExternal (UBYTE *s)
- int DoSetExternalAttr (UBYTE *s)
- int DoRmExternal (UBYTE *s)
- int DoFromExternal (UBYTE *s)
- int DoToExternal (UBYTE *s)
- UBYTE * defineChannel (UBYTE *s, HANDLERS *h)
- int writeToChannel (int wtype, UBYTE *s, HANDLERS *h)
- int DoFactDollar (UBYTE *s)
- WORD GetDollarNumber (UBYTE **inp, DOLLARS d)
- int DoSetRandom (UBYTE *s)
- int DoOptimize (UBYTE *s)
- int DoClearOptimize (UBYTE *s)
- int DoSkipExtraSymbols (UBYTE *s)
- int DoPreReset (UBYTE *s)
- int DoPreAppendPath (UBYTE *s)
- int DoPrePrependPath (UBYTE *s)
- int DoTimeOutAfter (UBYTE *s)
- int DoNamespace (UBYTE *s)
- int DoEndNamespace (UBYTE *s)
- UBYTE * SkipName (UBYTE *s)
- UBYTE * ConstructName (UBYTE *s, UBYTE type)
- int DoUse (UBYTE *s)
- int UserFlags (UBYTE *s, int par)
- int DoClearUserFlag (UBYTE *s)
- int DoSetUserFlag (UBYTE *s)

4.112.1 Detailed Description

This is the preprocessor and all its routines.

Definition in file pre.c.

4.112.2 Macro Definition Documentation

4.112.2.1 STRINGIFY

```
#define STRINGIFY(
              x ) STRINGIFY__(x)
```

Definition at line 4251 of file pre.c.

4.112.2.2 STRINGIFY__

```
#define STRINGIFY__(
              x ) #x
```

Definition at line 4252 of file pre.c.

4.112.2.3 SKIPBUFSIZE

```
#define SKIPBUFSIZE 20
```

Definition at line 4375 of file pre.c.

4.112.2.4 KILL

```
#define KILL "kill"
```

Definition at line 5595 of file pre.c.

4.112.2.5 KILLALL

```
#define KILLALL "killall"
```

Definition at line 5596 of file pre.c.

4.112.2.6 DAEMON

```
#define DAEMON "daemon"
```

Definition at line 5597 of file pre.c.

4.112.2.7 SHELL

```
#define SHELL "shell"
```

Definition at line 5598 of file pre.c.

4.112.2.8 STDERR

```
#define STDERR "stderr"
```

Definition at line 5599 of file pre.c.

4.112.2.9 TRUE_EXPR

```
#define TRUE_EXPR "true"
```

Definition at line 5601 of file pre.c.

4.112.2.10 FALSE_EXPR

```
#define FALSE_EXPR "false"
```

Definition at line 5602 of file pre.c.

4.112.2.11 NOSHELL

```
#define NOSHELL "noshell"
```

Definition at line 5603 of file pre.c.

4.112.2.12 TERMINAL

```
#define TERMINAL "terminal"
```

Definition at line 5604 of file pre.c.

4.112.3 Function Documentation

4.112.3.1 GetInput()

```
UBYTE GetInput (
             void )
```

Definition at line 132 of file pre.c.

4.112.3.2 ClearPushback()

```
void ClearPushback (
             void )
```

Definition at line 161 of file pre.c.

4.112.3.3 GetChar()

```
UBYTE GetChar (
            int level )
```

Definition at line 179 of file pre.c.

4.112.3.4 CharOut()

```
void CharOut (
            UBYTE c )
```

Definition at line 489 of file pre.c.

4.112.3.5 UnsetAllowDelay()

```
void UnsetAllowDelay (
            void )
```

Definition at line 510 of file pre.c.

4.112.3.6 GetPreVar()

```
UBYTE * GetPreVar (
            UBYTE * name,
            int flag )
```

Definition at line 536 of file pre.c.

4.112.3.7 PutPreVar()

```
int PutPreVar (
            UBYTE * name,
            UBYTE * value,
            UBYTE * args,
            int mode )
```

Inserts/Updates a preprocessor variable in the name administration.

Parameters

name	Character string with the variable name.
value	Character string with a possible value. Special case: if this argument is zero, then we have no value. Note: This is different from having an empty argument! This should only occur when the name starts with a ?
args	Character string with possible arguments.
mode	=0: always create a new name entry, =1: try to do a redefinition if possible.

Returns

Index of used entry in name list.

Definition at line 718 of file pre.c.

Referenced by ClearOptimize(), Generator(), Optimize(), PF_BroadcastRedefinedPreVars(), StartVariables(), and TheDefine().

4.112.3.8 PopPreVars()

```
void PopPreVars (
            int tonumber )
```

Definition at line 820 of file pre.c.

4.112.3.9 IniModule()

```
void IniModule (
            int type )
```

Definition at line 835 of file pre.c.

4.112.3.10 IniSpecialModule()

```
void IniSpecialModule (
            int type )
```

Definition at line 937 of file pre.c.

4.112.3.11 PreProcessor()

```
void PreProcessor (
            void )
```

Definition at line 947 of file pre.c.

4.112.3.12 PreProInstruction()

```
int PreProInstruction (
            void )
```

Definition at line 1166 of file pre.c.

4.112.3.13 LoadInstruction()

```
int LoadInstruction (
            int mode )
```

Definition at line 1254 of file pre.c.

4.112.3.14 LoadStatement()

```
int LoadStatement (
            int type )
```

Definition at line 1457 of file pre.c.

4.112.3.15 ExpandTripleDots()

```
int ExpandTripleDots (
            int par )
```

Definition at line 1595 of file pre.c.

4.112.3.16 FindKeyWord()

```
KEYWORD * FindKeyWord (
            UBYTE * theword,
            KEYWORD * table,
            int size )
```

Definition at line 1963 of file pre.c.

4.112.3.17 FindInKeyWord()

```
KEYWORD * FindInKeyWord (
            UBYTE * theword,
            KEYWORD * table,
            int size )
```

Definition at line 1994 of file pre.c.

4.112.3.18 TheDefine()

```
int TheDefine (
            UBYTE * s,
            int mode )
```

Preprocessor assignment. Possible arguments and values are treated and the new preprocessor variable is put into the name administration.

Parameters

s	Pointer to the character string following the preprocessor command.
mode	Bitmask. 0-bit clear: always create a new name entry, 0-bit set: try to redefine an existing name, 1-bit set: ignore preprocessor if/switch status.

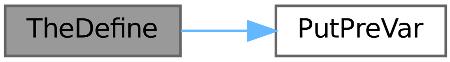
Returns

zero: no errors, negative number: errors.

Definition at line 2024 of file pre.c.

References PutPreVar().

Here is the call graph for this function:



4.112.3.19 DoCommentChar()

```
int DoCommentChar (
            UBYTE * s )
```

Definition at line 2097 of file pre.c.

4.112.3.20 DoPreAssign()

```
int DoPreAssign (
            UBYTE * s )
```

Definition at line 2128 of file pre.c.

4.112.3.21 DoDefine()

```
int DoDefine (
            UBYTE * s )
```

Definition at line 2159 of file pre.c.

4.112.3.22 DoRedefine()

```
int DoRedefine (
            UBYTE * s )
```

Definition at line 2169 of file pre.c.

4.112.3.23 ClearMacro()

```
int ClearMacro (
             UBYTE * name )
```

Definition at line 2181 of file pre.c.

4.112.3.24 TheUndefine()

```
int TheUndefine (
             UBYTE * name )
```

Definition at line 2209 of file pre.c.

4.112.3.25 DoUndefine()

```
int DoUndefine (
             UBYTE * s )
```

Definition at line 2263 of file pre.c.

4.112.3.26 DoInclude()

```
int DoInclude (
             UBYTE * s )
```

Definition at line 2315 of file pre.c.

4.112.3.27 DoReverseInclude()

```
int DoReverseInclude (
             UBYTE * s )
```

Definition at line 2322 of file pre.c.

4.112.3.28 Include()

```
int Include (
             UBYTE * s,
             int type )
```

Definition at line 2329 of file pre.c.

4.112.3.29 DoPreExchange()

```
int DoPreExchange (
             UBYTE * s )
```

Definition at line 2490 of file pre.c.

4.112.3.30 DoCall()

```
int DoCall (
        UBYTE * s )
```

Definition at line 2551 of file pre.c.

4.112.3.31 DoDebug()

```
int DoDebug (
        UBYTE * s )
```

Definition at line 2720 of file pre.c.

4.112.3.32 DoTerminate()

```
int DoTerminate (
        UBYTE * s )
```

Definition at line 2757 of file pre.c.

4.112.3.33 DoContinueDo()

```
int DoContinueDo (
        UBYTE * s )
```

Definition at line 2780 of file pre.c.

4.112.3.34 DoDo()

```
int DoDo (
        UBYTE * s )
```

Definition at line 2811 of file pre.c.

4.112.3.35 DoBreakDo()

```
int DoBreakDo (
        UBYTE * s )
```

Definition at line 3026 of file pre.c.

4.112.3.36 DoElse()

```
int DoElse (
        UBYTE * s )
```

Definition at line 3095 of file pre.c.

4.112.3.37 DoElseif()

```
int DoElseif (
            UBYTE * s )
```

Definition at line 3132 of file pre.c.

4.112.3.38 DoEnddo()

```
int DoEnddo (
            UBYTE * s )
```

Definition at line 3167 of file pre.c.

4.112.3.39 DoEndif()

```
int DoEndif (
            UBYTE * s )
```

Definition at line 3313 of file pre.c.

4.112.3.40 DoEndprocedure()

```
int DoEndprocedure (
            UBYTE * s )
```

Definition at line 3341 of file pre.c.

4.112.3.41 DoIf()

```
int DoIf (
            UBYTE * s )
```

Definition at line 3367 of file pre.c.

4.112.3.42 DoIfdef()

```
int DoIfdef (
            UBYTE * s,
            int par )
```

Definition at line 3391 of file pre.c.

4.112.3.43 DoIfydef()

```
int DoIfydef (
            UBYTE * s )
```

Definition at line 3416 of file pre.c.

4.112.3.44 DoIfndef()

```
int DoIfndef (
            UBYTE * s )
```

Definition at line 3426 of file pre.c.

4.112.3.45 DoInside()

```
int DoInside (
            UBYTE * s )
```

Definition at line 3451 of file pre.c.

4.112.3.46 DoEndInside()

```
int DoEndInside (
            UBYTE * s )
```

Definition at line 3544 of file pre.c.

4.112.3.47 DoMessage()

```
int DoMessage (
            UBYTE * s )
```

Definition at line 3676 of file pre.c.

4.112.3.48 DoPipe()

```
int DoPipe (
            UBYTE * s )
```

Definition at line 3690 of file pre.c.

4.112.3.49 DoPrcExtension()

```
int DoPrcExtension (
            UBYTE * s )
```

Definition at line 3713 of file pre.c.

4.112.3.50 DoPreOut()

```
int DoPreOut (
            UBYTE * s )
```

Definition at line 3747 of file pre.c.

4.112.3.51 DoPrePrintTimes()

```
int DoPrePrintTimes (
            UBYTE * s )
```

Definition at line 3770 of file pre.c.

4.112.3.52 DoPreSortReallocate()

```
int DoPreSortReallocate (
            UBYTE * s )
```

Definition at line 3784 of file pre.c.

4.112.3.53 DoPreAppend()

```
int DoPreAppend (
            UBYTE * s )
```

Definition at line 3804 of file pre.c.

4.112.3.54 DoPreCreate()

```
int DoPreCreate (
            UBYTE * s )
```

Definition at line 3847 of file pre.c.

4.112.3.55 DoPreRemove()

```
int DoPreRemove (
            UBYTE * s )
```

Definition at line 3887 of file pre.c.

4.112.3.56 DoPreClose()

```
int DoPreClose (
            UBYTE * s )
```

Definition at line 3920 of file pre.c.

4.112.3.57 DoPreWrite()

```
int DoPreWrite (
            UBYTE * s )
```

Definition at line 3964 of file pre.c.

4.112.3.58 DoProcedure()

```
int DoProcedure (
              UBYTE * s )
```

Definition at line 4005 of file pre.c.

4.112.3.59 DoPreBreak()

```
int DoPreBreak (
              UBYTE * s )
```

Definition at line 4047 of file pre.c.

4.112.3.60 DoPreCase()

```
int DoPreCase (
              UBYTE * s )
```

Definition at line 4071 of file pre.c.

4.112.3.61 DoPreDefault()

```
int DoPreDefault (
              UBYTE * s )
```

Definition at line 4110 of file pre.c.

4.112.3.62 DoPreEndSwitch()

```
int DoPreEndSwitch (
              UBYTE * s )
```

Definition at line 4134 of file pre.c.

4.112.3.63 DoPreSwitch()

```
int DoPreSwitch (
              UBYTE * s )
```

Definition at line 4161 of file pre.c.

4.112.3.64 DoPreShow()

```
int DoPreShow (
              UBYTE * s )
```

Definition at line 4215 of file pre.c.

4.112.3.65 DoSystem()

```
int DoSystem (
            UBYTE * s )
```

Definition at line 4254 of file pre.c.

4.112.3.66 PreLoad()

```
int PreLoad (
            PRELOAD * p,
            UBYTE * start,
            UBYTE * stop,
            int mode,
            char * message )
```

Definition at line 4294 of file pre.c.

4.112.3.67 PreSkip()

```
int PreSkip (
            UBYTE * start,
            UBYTE * stop,
            int mode )
```

Definition at line 4377 of file pre.c.

4.112.3.68 StartPrepro()

```
void StartPrepro (
            void )
```

Definition at line 4432 of file pre.c.

4.112.3.69 EvalPreIf()

```
int EvalPreIf (
            UBYTE * s )
```

Definition at line 4460 of file pre.c.

4.112.3.70 PreIfEval()

```
UBYTE * PreIfEval (
            UBYTE * s,
            int * value )
```

Definition at line 4495 of file pre.c.

4.112.3.71 PreCmp()

```
int PreCmp (
            int type,
            int val,
            UBYTE * t,
            int type2,
            int val2,
            UBYTE * t2,
            int cmpop )
```

Definition at line 4624 of file pre.c.

4.112.3.72 PreEq()

```
int PreEq (
            int type,
            int val,
            UBYTE * t,
            int type2,
            int val2,
            UBYTE * t2,
            int eqop )
```

Definition at line 4648 of file pre.c.

4.112.3.73 pParseObject()

```
UBYTE * pParseObject (
            UBYTE * s,
            int * type,
            LONG * val2 )
```

Definition at line 4677 of file pre.c.

4.112.3.74 PreCalc()

```
UBYTE * PreCalc (
            void )
```

Definition at line 5113 of file pre.c.

4.112.3.75 PreEval()

```
UBYTE * PreEval (
            UBYTE * s,
            LONG * x )
```

Definition at line 5191 of file pre.c.

4.112.3.76 AddToPreTypes()

```
void AddToPreTypes (
             int type )
```

Definition at line 5341 of file pre.c.

4.112.3.77 MessPreNesting()

```
void MessPreNesting (
             int par )
```

Definition at line 5359 of file pre.c.

4.112.3.78 DoPreAddSeparator()

```
int DoPreAddSeparator (
             UBYTE * s )
```

Definition at line 5382 of file pre.c.

4.112.3.79 DoPreRmSeparator()

```
int DoPreRmSeparator (
             UBYTE * s )
```

Definition at line 5407 of file pre.c.

4.112.3.80 DoExternal()

```
int DoExternal (
             UBYTE * s )
```

Definition at line 5424 of file pre.c.

4.112.3.81 DoPrompt()

```
int DoPrompt (
             UBYTE * s )
```

Definition at line 5506 of file pre.c.

4.112.3.82 DoSetExternal()

```
int DoSetExternal (
             UBYTE * s )
```

Definition at line 5539 of file pre.c.

4.112.3.83 DoSetExternalAttr()

```
int DoSetExternalAttr (
              UBYTE * s )
```

Definition at line 5609 of file pre.c.

4.112.3.84 DoRmExternal()

```
int DoRmExternal (
              UBYTE * s )
```

Definition at line 5726 of file pre.c.

4.112.3.85 DoFromExternal()

```
int DoFromExternal (
              UBYTE * s )
```

Definition at line 5791 of file pre.c.

4.112.3.86 DoToExternal()

```
int DoToExternal (
              UBYTE * s )
```

Definition at line 5986 of file pre.c.

4.112.3.87 defineChannel()

```
UBYTE * defineChannel (
              UBYTE * s,
              HANDLERS * h )
```

Definition at line 6033 of file pre.c.

4.112.3.88 writeToChannel()

```
int writeToChannel (
              int wtype,
              UBYTE * s,
              HANDLERS * h )
```

Definition at line 6068 of file pre.c.

4.112.3.89 DoFactDollar()

```
int DoFactDollar (
            UBYTE * s )
```

Definition at line 6633 of file pre.c.

4.112.3.90 GetDollarNumber()

```
WORD GetDollarNumber (
            UBYTE ** inp,
            DOLLARS d )
```

Definition at line 6670 of file pre.c.

4.112.3.91 DoSetRandom()

```
int DoSetRandom (
            UBYTE * s )
```

Definition at line 6760 of file pre.c.

4.112.3.92 DoOptimize()

```
int DoOptimize (
            UBYTE * s )
```

Definition at line 6803 of file pre.c.

4.112.3.93 DoClearOptimize()

```
int DoClearOptimize (
            UBYTE * s )
```

Definition at line 6936 of file pre.c.

4.112.3.94 DoSkipExtraSymbols()

```
int DoSkipExtraSymbols (
            UBYTE * s )
```

Definition at line 6956 of file pre.c.

4.112.3.95 DoPreReset()

```
int DoPreReset (
            UBYTE * s )
```

Definition at line 6995 of file pre.c.

4.112.3.96 DoPreAppendPath()

```
int DoPreAppendPath (
            UBYTE * s )
```

Appends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: #appendpath <path>

Definition at line 7153 of file pre.c.

4.112.3.97 DoPrePrependPath()

```
int DoPrePrependPath (
            UBYTE * s )
```

Prepends the given path (absolute or relative to the current file directory) to the FORM path.

Syntax: #prependpath <path>

Definition at line 7170 of file pre.c.

4.112.3.98 DoTimeOutAfter()

```
int DoTimeOutAfter (
            UBYTE * s )
```

Definition at line 7182 of file pre.c.

4.112.3.99 DoNamespace()

```
int DoNamespace (
            UBYTE * s )
```

Definition at line 7234 of file pre.c.

4.112.3.100 DoEndNamespace()

```
int DoEndNamespace (
            UBYTE * s )
```

Definition at line 7282 of file pre.c.

4.112.3.101 SkipName()

```
UBYTE * SkipName (
            UBYTE * s )
```

Definition at line 7305 of file pre.c.

4.112.3.102 ConstructName()

```
UBYTE * ConstructName (
            UBYTE * s,
            UBYTE type )
```

Definition at line 7405 of file pre.c.

4.112.3.103 DoUse()

```
int DoUse (
            UBYTE * s )
```

Definition at line 7497 of file pre.c.

4.112.3.104 UserFlags()

```
int UserFlags (
            UBYTE * s,
            int par )
```

Definition at line 7545 of file pre.c.

4.112.3.105 DoClearUserFlag()

```
int DoClearUserFlag (
            UBYTE * s )
```

Definition at line 7652 of file pre.c.

4.112.3.106 DoSetUserFlag()

```
int DoSetUserFlag (
            UBYTE * s )
```

Definition at line 7662 of file pre.c.

4.113 pre.c

Go to the documentation of this file.

```
/* #[ License : */
/*
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
*/
#include "form3.h"

static UBYTE pushbackchar = 0;
static int oldmode = 0;
static int stopdelay = 0;
static STREAM *oldstream = 0;
static UBYTE underscore[2] = {'_',0};
static PREVAR *ThePreVar = 0;

static KEYWORD precommands[] = {
     {"add"          , DoPreAdd       , 0, 0}
    ,{"addseparator" , DoPreAddSeparator,0,0}
    ,{"append"       , DoPreAppend    , 0, 0}
    ,{"appendpath"   , DoPreAppendPath, 0, 0}
    ,{"assign"       , DoPreAssign    , 0, 0}
    ,{"break"        , DoPreBreak     , 0, 0}
    ,{"breakdo"      , DoBreakDo      , 0, 0}
    ,{"call"         , DoCall         , 0, 0}
    ,{"case"         , DoPreCase      , 0, 0}
    ,{"clearflag"    , DoClearUserFlag, 0, 0}
    ,{"clearoptimize", DoClearOptimize, 0, 0}
    ,{"close"        , DoPreClose     , 0, 0}
    ,{"closedictionary", DoPreCloseDictionary,0,0}
    ,{"commentchar"  , DoCommentChar  , 0, 0}
    ,{"continuedo"  ,  DoContinueDo   , 0, 0}
    ,{"create"       , DoPreCreate    , 0, 0}
    ,{"debug"        , DoDebug        , 0, 0}
    ,{"default"      , DoPreDefault   , 0, 0}
    ,{"define"       , DoDefine       , 0, 0}
    ,{"do"           , DoDo           , 0, 0}
    ,{"else"         , DoElse         , 0, 0}
    ,{"elseif"       , DoElseif       , 0, 0}
    ,{"enddo"        , DoEnddo        , 0, 0}
#ifdef WITHFLOAT
    ,{"endfloat"     , DoEndFloat     , 0, 0}
#endif
    ,{"endif"        , DoEndif        , 0, 0}
    ,{"endinside"    , DoEndInside    , 0, 0}
    ,{"endnamespace" , DoEndNamespace , 0, 0}
    ,{"endprocedure" , DoEndprocedure , 0, 0}
    ,{"endswitch"    , DoPreEndSwitch , 0, 0}
    ,{"exchange"     , DoPreExchange  , 0, 0}
    ,{"external"     , DoExternal     , 0, 0}
    ,{"factdollar"   , DoFactDollar   , 0, 0}
    ,{"fromexternal" , DoFromExternal , 0, 0}
    ,{"if"           , DoIf           , 0, 0}
    ,{"ifdef"        , DoIfydef       , 0, 0}
    ,{"ifndef"       , DoIfndef       , 0, 0}
    ,{"include"      , DoInclude      , 0, 0}
    ,{"inside"       , DoInside       , 0, 0}
    ,{"message"      , DoMessage      , 0, 0}
    ,{"namespace"    , DoNamespace    , 0, 0}
```

```
    ,{"opendictionary", DoPreOpenDictionary,0,0}
    ,{"optimize"      , DoOptimize      , 0, 0}
    ,{"pipe"          , DoPipe          , 0, 0}
    ,{"preout"        , DoPreOut        , 0, 0}
    ,{"prependpath"   , DoPrePrependPath,0, 0}
    ,{"printtimes"    , DoPrePrintTimes, 0, 0}
    ,{"procedure"     , DoProcedure     , 0, 0}
    ,{"procedureextension" , DoPrcExtension   , 0, 0}
    ,{"prompt"        , DoPrompt        , 0, 0}
    ,{"redefine"      , DoRedefine      , 0, 0}
    ,{"remove"        , DoPreRemove     , 0, 0}
    ,{"reset"         , DoPreReset      , 0, 0}
    ,{"reverseinclude"    , DoReverseInclude   , 0, 0}
    ,{"rmexternal"    , DoRmExternal    , 0, 0}
    ,{"rmseparator"   , DoPreRmSeparator,0, 0}
    ,{"setexternal"   , DoSetExternal   , 0, 0}
    ,{"setexternalattr"  , DoSetExternalAttr  , 0, 0}
    ,{"setflag"       , DoSetUserFlag   , 0, 0}
    ,{"setrandom"     , DoSetRandom     , 0, 0}
    ,{"show"          , DoPreShow       , 0, 0}
    ,{"skipextrasymbols" , DoSkipExtraSymbols , 0, 0}
    ,{"sortreallocate", DoPreSortReallocate , 0, 0}
#ifdef WITHFLOAT
    ,{"startfloat"    , DoStartFloat    , 0, 0}
#endif
    ,{"switch"        , DoPreSwitch     , 0, 0}
    ,{"system"        , DoSystem        , 0, 0}
    ,{"terminate"     , DoTerminate     , 0, 0}
    ,{"timeoutafter"  , DoTimeOutAfter  , 0, 0}
    ,{"toexternal"    , DoToExternal    , 0, 0}
    ,{"undefine"      , DoUndefine      , 0, 0}
    ,{"use"           , DoUse           , 0, 0}
    ,{"usedictionary", DoPreUseDictionary,0,0}
    ,{"write"         , DoPreWrite      , 0, 0}
};

/*
    #] Includes :
    # [ PreProcessor :
        #[ GetInput :

        Gets one input character. If we reach the end of a stream
        we pop to the previous stream and try again.
        If there are no more streams we let this be known.
*/

UBYTE GetInput(void)
{
    UBYTE c;
    while ( AC.CurrentStream ) {
        c = GetFromStream(AC.CurrentStream);
        if ( c != ENDOFSTREAM ) {
#ifdef WITHMPI
            if ( PF.me == MASTER
                 && AC.NoShowInput <= 0
                 && AC.CurrentStream->type != PREVARSTREAM )
#else
            if ( AC.NoShowInput <= 0 && AC.CurrentStream->type != PREVARSTREAM )
#endif
                CharOut(c);
            return(c);
        }
        AC.CurrentStream = CloseStream(AC.CurrentStream);
        if ( stopdelay && AC.CurrentStream == oldstream ) {
            stopdelay = 0; AP.AllowDelay = 1;
        }
    }
    return(ENDOFINPUT);
}

/*
        #] GetInput :
        #[ ClearPushback :
*/

void ClearPushback(void)
{
    pushbackchar = 0;
}

/*
        #] ClearPushback :
        #[ GetChar :

        Reads one character. If it encounters a quote it immediately
        takes the whole preprocessor variable and opens a stream
        for it and starts reading the stream.
```

```
        Note that we have to take special precautions for escaped quotes.
        That is why we remember the previous character. We allow the
        (dubious?) construction of ending a stream with a backslash and
        then using it to escape an object in the parent stream.
*/

UBYTE GetChar(int level)
{
    UBYTE namebuf[MAXPRENAMESIZE+2], c, *s, *t;
    static UBYTE lastchar, charinbuf = 0;
    int i, j, raiselow, olddelay;
    STREAM *stream;
    if ( level > 0 ) {
        lastchar = '`';
        goto higherlevel;
    }
    if ( pushbackchar ) { c = pushbackchar; pushbackchar = 0; return(c); }
    if ( charinbuf ) { c = charinbuf; charinbuf = 0; return(c); }
    c = GetInput();
    for(;;) {
        if ( c == '\\' ) {
            charinbuf = GetInput();
            if ( charinbuf != LINEFEED ) {
                pushbackchar = charinbuf;
                charinbuf = 0;
                break;
            }
            charinbuf = 0;  /* Escaped linefeed -> skip leading blanks */
            while ( ( c = GetInput() ) == ' ' || c == '\t' ) {}
        }
        else if ( c == '\"' || c == '`' ) {
            if ( AP.DelayPrevar == 1 && c == '\"' ) {
                AP.DelayPrevar = 0;
                break;
            }
            lastchar = c;
higherlevel:
            c = GetInput();
            if ( c == '!' && lastchar == '`' ) {
                if ( stopdelay == 0 ) oldstream = AC.CurrentStream;
                AP.AllowDelay = 0;
                stopdelay = 1;
                c = GetInput();
            }
            if ( c == '~' && lastchar == '`' ) {
                if ( AP.AllowDelay ) {
                    pushbackchar = c;
                    c = lastchar;
                    AP.DelayPrevar = 1;
                    break;
                }
            }
            else {
                pushbackchar = c;
            }
            olddelay = AP.DelayPrevar;
            AP.DelayPrevar = 0;
            i = 0; lastchar = 0;
            for (;;) {
                if ( pushbackchar ) { c = pushbackchar; pushbackchar = 0; }
                else { c = GetInput(); }
                if ( c == ENDOFINPUT || ( ( c == '\"' || c == LINEFEED )
                && lastchar != '\\' ) ) {
                    break;
                }
                if ( c == '{' ) { /* Try the preprocessor calculator */
                    if ( PreCalc() == 0 ) Terminate(-1);
                    c = GetInput();     /* This is either a { or a number */
                    if ( c == '{' ) {
                        MesPrint("@Illegal set inside preprocessor variable name");
                        Terminate(-1);
                    }
                }
                if ( c == '`' && lastchar != '\\' ) {
                    c = GetChar(1);
                    if ( c == ENDOFINPUT || ( ( c == '\"' || c == LINEFEED )
                    && lastchar != '\\' ) ) {
                        break;
                    }
                }
                if ( lastchar == '\\' ) { i--; lastchar = 0; }
                else lastchar = c;
                namebuf[i++] = c;
                if ( i > MAXPRENAMESIZE ) {
                    namebuf[i] = 0;
                    Error1("Preprocessor variable name too long: ",namebuf);
                }
```

```
            }
            namebuf[i++] = 0;
            if ( c != '\"' ) {
                Error1("Unmatched quotes for preprocessor variable",namebuf);
            }
            AP.DelayPrevar = olddelay;
            if ( namebuf[0] == '$' ) {
                raiselow = PRENOACTION;
                if ( AP.PreproFlag && *AP.preStart) {
                    s = EndOfToken(AP.preStart);
                    c = *s; *s = 0;
                    if ( ( StrICmp(AP.preStart,(UBYTE *)"ifdef") == 0
                    || StrICmp(AP.preStart,(UBYTE *)"ifndef") == 0 )
                    && GetDollar(namebuf+1) < 0 ) {
                        *s = c; c = ' ';
                        break;
                    }
                    *s = c;
                }
                else {
                    s = EndOfToken(namebuf+1);
                    if ( *s == '[' ) { while ( *s ) s++; }
                }
                if ( *s == '-' && s[1] == '-' && s[2] == 0 )
                    raiselow = PRELOWERAFTER;
                else if ( *s == '+' && s[1] == '+' && s[2] == 0 )
                    raiselow = PRERAISEAFTER;
                c = *s; *s = 0;
                if ( OpenStream(namebuf+1,DOLLARSTREAM,0,raiselow) == 0 ) {
                    *s = c;
                    MesPrint("@Undefined variable %s used as preprocessor variable",
                        namebuf);
                    Terminate(-1);
                }
                *s = c;
            }
            else {
                raiselow = PRENOACTION;
                if ( AP.PreproFlag && *AP.preStart) {
                    s = EndOfToken(AP.preStart);
                    c = *s; *s = 0;
                    if ( ( StrICmp(AP.preStart,(UBYTE *)"ifdef") == 0
                    || StrICmp(AP.preStart,(UBYTE *)"ifndef") == 0 )
                    && GetPreVar(namebuf,WITHOUTERROR) == 0 ) {
                        *s = c; c = ' ';
                        break;
                    }
                    *s = c;
                }
                s = EndOfToken(namebuf);
                if ( *s == '_' ) s++;
                if ( *s == '-' && s[1] == '-' && s[2] == 0 )
                    raiselow = PRELOWERAFTER;
                else if ( *s == '+' && s[1] == '+' && s[2] == 0 )
                    raiselow = PRERAISEAFTER;
                else if ( *s == '(' && namebuf[i-2] == ')' ) {
/*
                    Now count the arguments and separate them by zeroes
                    Check on the ?var construction and if present, reset
                    some comma's.
                    Make the assignments of the variables
                    Run the macro.
                    Undefine the variables
*/
                    int nargs = 1;
                    PREVAR *p;
                    size_t p_offset;
                    *s++ = 0; namebuf[i-2] = 0;
                    if ( StrICmp(namebuf,(UBYTE *)"random_") == 0 ) {
                        UBYTE *ranvalue;
                        ranvalue = PreRandom(s);
                        PutPreVar(namebuf,ranvalue,(UBYTE *)"?a",1);
                        M_free(ranvalue,"PreRandom");
                        goto dostream;
                    }
                    else if ( StrICmp(namebuf,(UBYTE *)"tolower_") == 0 ) {
                        UBYTE *ss = s;
                        while ( *ss ) { *ss = (UBYTE)(tolower(*ss)); ss++; }
                        PutPreVar(namebuf,s,(UBYTE *)"?a",1);
                        goto dostream;
                    }
                    else if ( StrICmp(namebuf,(UBYTE *)"toupper_") == 0 ) {
                        UBYTE *ss = s;
                        while ( *ss ) { *ss = (UBYTE)(toupper(*ss)); ss++; }
                        PutPreVar(namebuf,s,(UBYTE *)"?a",1);
                        goto dostream;
                    }
```

```
                    else if ( StrICmp(namebuf,(UBYTE *)"takeleft_") == 0 ) {
                        UBYTE *ss = s;
                        int x = 0, nsize;
                        while ( *ss != ',' && *ss ) ss++;
                        nsize = ss-s;
                        if ( *ss ) {
                            *ss++ = 0;
                            while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
                            if ( x > nsize ) x = nsize;
                        }
                        else x = 0;
                        PutPreVar(namebuf,s+x,(UBYTE *)"?a",1);
                        goto dostream;
                    }
                    else if ( StrICmp(namebuf,(UBYTE *)"takeright_") == 0 ) {
                        UBYTE *ss = s;
                        int x = 0, nsize;
                        while ( *ss != ',' && *ss ) ss++;
                        nsize = ss-s;
                        if ( *ss ) {
                            *ss++ = 0;
                            while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
                            if ( x > nsize ) x = nsize;
                        }
                        else x = 0;
                        x = nsize - x;
                        s[x] = 0;
                        PutPreVar(namebuf,s,(UBYTE *)"?a",1);
                        goto dostream;
                    }
                    else if ( StrICmp(namebuf,(UBYTE *)"keepleft_") == 0 ) {
                        UBYTE *ss = s;
                        int x = 0, nsize;
                        while ( *ss != ',' && *ss ) ss++;
                        nsize = ss-s;
                        if ( *ss ) {
                            *ss++ = 0;
                            while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
                            if ( x > nsize ) x = nsize;
                        }
                        else x = nsize;
                        s[x] = 0;
                        PutPreVar(namebuf,s,(UBYTE *)"?a",1);
                        goto dostream;
                    }
                    else if ( StrICmp(namebuf,(UBYTE *)"keepright_") == 0 ) {
                        UBYTE *ss = s;
                        int x = 0, nsize;
                        while ( *ss != ',' && *ss ) ss++;
                        nsize = ss-s;
                        if ( *ss ) {
                            *ss++ = 0;
                            while ( FG.cTable[*ss] == 1 ) x = 10*x + (*ss++ - '0');
                            if ( x > nsize ) x = nsize;
                        }
                        else x = nsize;
                        x = nsize-x;
                        PutPreVar(namebuf,s+x,(UBYTE *)"?a",1);
                        goto dostream;
                    }
                    while ( *s ) {
                        if ( *s == '\\' ) s++;
                        if ( *s == ',' ) { *s = 0; nargs++; }
                        s++;
                    }
                    GetPreVar(namebuf,WITHERROR);
                    p = ThePreVar;
                    if ( p == 0 ) {
                        MesPrint("@Illegal use of arguments in preprocessor variable %s",namebuf);
                        Terminate(-1);
                    }
                    if ( p->nargs <= 0 || ( p->wildarg == 0 && nargs != p->nargs )
                        || ( p->wildarg > 0 && nargs < p->nargs-1 ) ) {
                        MesPrint("@Arguments of macro %s do not match",namebuf);
                        Terminate(-1);
                    }
                    if ( p->wildarg > 0 ) {
/*
                        Change some zeroes into commas
*/
                        s = namebuf;
                        for ( j = 0; j < p->wildarg; j++ ) {
                            while ( *s ) s++;
                            s++;
                        }
                        for ( j = 0; j < nargs-p->nargs; j++ ) {
                            while ( *s ) s++;
```

```
                            *s++ = ',';
                        }
                    }
/*
                        Now we can make the assignments
*/
                    s = namebuf;
                    while ( *s ) s++;
                    s++;
                    t = p->argnames;
                    p_offset = p - PreVar;
                    for ( j = 0; j < p->nargs; j++ ) {
                        if ( ( nargs == p->nargs-1 ) && ( *t == '?' ) ) {
                            PutPreVar(t,0,0,0);
                        }
                        else {
                            PutPreVar(t,s,0,0);
                            while ( *s ) s++;
                            s++;
                        }
                        p = PreVar + p_offset;
                        while ( *t ) t++;
                        t++;
                    }
                }
dostream:;
                if ( ( stream = OpenStream(namebuf,PREVARSTREAM,0,raiselow) ) == 0 ) {
/*
                    Eat comma before or after. This is `no value'
*/
                }
                else if ( stream->inbuffer == 0 ) {
                    c = GetInput();
                    if ( level > 0 && c == '\" ) return(c);
                    goto endofloop;
                }
            }
            c = GetInput();
        }
        else if ( c == '{' ) { /* Try the preprocessor calculator */
            if ( PreCalc() == 0 ) Terminate(-1);
            c = GetInput();    /* This is either a { or a number */
            break;
        }
        else break;
endofloop:;
    }
    return(c);
}

/*
        #] GetChar :
        #[ CharOut :
*/

void CharOut(UBYTE c)
{
    if ( c == LINEFEED ) {
        AM.OutBuffer[AP.InOutBuf++] = c;
        WriteString(INPUTOUT,AM.OutBuffer,AP.InOutBuf);
        AP.InOutBuf = 0;
    }
    else {
        if ( AP.InOutBuf >= AM.OutBufSize || c == LINEFEED ) {
            WriteString(INPUTOUT,AM.OutBuffer,AP.InOutBuf);
            AP.InOutBuf = 0;
        }
        AM.OutBuffer[AP.InOutBuf++] = c;
    }
}

/*
        #] CharOut :
        #[ UnsetAllowDelay :
*/

void UnsetAllowDelay(void)
{
    if ( ThePreVar != 0 ) {
        if ( ThePreVar->nargs > 0 ) AP.AllowDelay = 0;
    }
}

/*
        #] UnsetAllowDelay :
        #[ GetPreVar :

```

```
        We use the model of a heap. If the same name has been used more
        than once the last definition is used. This gives the impression
        of local variables.

        There are two types: The regular ones and the expression variables.
        The last ones are like UNCHANGED_exprname and ZERO_exprname or
        UNCHANGED_ and ZERO_.
*/

static UBYTE *yes = (UBYTE *)"1";
static UBYTE *no  = (UBYTE *)"0";
static UBYTE numintopolynomial[12];
#include "vector.h"
static Vector(UBYTE, exprstr);  /* Used for numactiveexprs_ and activeexprnames_. */

UBYTE *GetPreVar(UBYTE *name, int flag)
{
    GETIDENTITY
    int i, mode;
    WORD number;
    UBYTE *t, c = 0, *tt = 0;
    t = name; while ( *t ) t++;
    if ( t[-1] == '-' && t[-2] == '-' && t-2 > name && t[-3] != '_' ) {
        t -= 2; c = *t; *t = 0; tt = t;
    }
    else if ( t[-1] == '+' && t[-2] == '+' && t-2 > name && t[-3] != '_' ) {
        t -= 2; c = *t; *t = 0; tt = t;
    }
    else if ( StrICmp(name,(UBYTE *)"time_") == 0 ) {
        UBYTE millibuf[24];
        LONG millitime, timepart;
        int timepart1, timepart2;
        static char timestring[40];
/*      millitime = TimeCPU(1); */
        millitime = GetRunningTime();
        timepart = millitime%1000;
        millitime /= 1000;
        timepart /= 10;
        timepart1 = timepart / 10;
        timepart2 = timepart % 10;
        NumToStr(millibuf,millitime);
        snprintf(timestring,40,"%s.%1d%1d",millibuf,timepart1,timepart2);
        return((UBYTE *)timestring);
    }
    else if ( ( StrICmp(name,(UBYTE *)"timer_") == 0 )
           || ( StrICmp(name,(UBYTE *)"stopwatch_") == 0 ) ) {
        static char timestring[40];
        snprintf(timestring,40,"%ld",(GetRunningTime() - AP.StopWatchZero));
        return((UBYTE *)timestring);
    }
    else if ( StrICmp(name, (UBYTE *)"numactiveexprs_") == 0 ) {
        /* the number of active expressions */
        int n = 0;
        for ( i = 0; i < NumExpressions; i++ ) {
            EXPRESSIONS e = Expressions + i;
            switch ( e->status ) {
                case LOCALEXPRESSION:
                case GLOBALEXPRESSION:
                case UNHIDELEXPRESSION:
                case UNHIDEGEXPRESSION:
                case INTOHIDELEXPRESSION:
                case INTOHIDEGEXPRESSION:
                    n++;
                    break;
            }
        }
        VectorReserve(exprstr, 41);  /* up to 128-bit */
        LongCopy(n, (char *)VectorPtr(exprstr));
        return VectorPtr(exprstr);
    }
    else if ( StrICmp(name, (UBYTE *)"activeexprnames_") == 0 ) {
        /* the list of active expressions separated by commas */
        int j = 0;
        VectorReserve(exprstr, 16);  /* at least 1 character for '\0' */
        for ( i = 0; i < NumExpressions; i++ ) {
            UBYTE *p, *s;
            int len, k;
            EXPRESSIONS e = Expressions + i;
            switch ( e->status ) {
                case LOCALEXPRESSION:
                case GLOBALEXPRESSION:
                case UNHIDELEXPRESSION:
                case UNHIDEGEXPRESSION:
                case INTOHIDELEXPRESSION:
                case INTOHIDEGEXPRESSION:
                    s = AC.exprnames->namebuffer + e->name;
                    len = StrLen(s);
```

```
                    VectorSize(exprstr) = j;  /* j bytes must be copied in extending the buffer. */
                    VectorReserve(exprstr, j + len * 2 + 1);
                    p = VectorPtr(exprstr);
                    if ( j > 0 ) p[j++] = ',';
                    for ( k = 0; k < len; k++ ) {
                        if ( s[k] == ',' || s[k] == '|' ) p[j++] = '\\';
                        p[j++] = s[k];
                    }
                    break;
            }
        }
        VectorPtr(exprstr)[j] = '\0';
        return VectorPtr(exprstr);
    }
    else if ( StrICmp(name, (UBYTE *)"path_") == 0 ) {
        /* the current FORM path (for debugging both in .c and .frm) */
        if ( AM.Path ) {
            return(AM.Path);
        }
        else {
            return((UBYTE *)"");
        }
    }
    t = name;
    while ( *t && *t != '_' ) t++;
    for ( i = NumPre-1; i >= 0; i-- ) {
        if ( *t == '_' && ( StrICmp(name,PreVar[i].name) == 0 ) ) {
            if ( c ) *tt = c;
            ThePreVar = PreVar+i;
            return(PreVar[i].value);
        }
        else if ( StrCmp(name,PreVar[i].name) == 0 ) {
            if ( c ) *tt = c;
            ThePreVar = PreVar+i;
            return(PreVar[i].value);
        }
    }
    if ( *t == '_' ) {
        if ( StrICmp(name,(UBYTE *)"EXTRASYMBOLS_") == 0 ) goto extrashort;
        *t = 0;
        if ( StrICmp(name,(UBYTE *)"UNCHANGED") == 0 ) mode = 1;
        else if ( StrICmp(name,(UBYTE *)"ZERO") == 0 ) mode = 0;
        else if ( StrICmp(name,(UBYTE *)"SHOWINPUT") == 0 ) {
            *t++ = '_';
            if ( c ) *tt = c;
            if ( AC.NoShowInput > 0 ) return(no);
            else return(yes);
        }
        else if ( StrICmp(name,(UBYTE *)"EXTRASYMBOLS") == 0 ) {
            *t++ = '_';
extrashort:;
            number = cbuf[AM.sbufnum].numrhs;
            t = numintopolynomial;
            NumCopy(number,t);
            return(numintopolynomial);
        }
        else mode = -1;
        *t++ = '_';
        if ( mode >= 0 ) {
            ThePreVar = 0;
            if ( *t ) {
                if ( GetName(AC.exprnames,t,&number,NOAUTO) == CEXPRESSION ) {
                    if ( c ) *tt = c;
                    if ( ( Expressions[number].vflags & ( 1 << mode ) ) != 0 )
                         return(yes);
                    else return(no);
                }
            }
            else {
/*
                Here we have to test all active results.
                These are in `negative' so the flags have to be zero.
*/
                if ( c ) *tt = c;
                if ( ( AR.expflags & ( 1 << mode ) ) == 0 ) return(yes);
                else return(no);
            }
        }
    }
    if ( ( t = (UBYTE *)(getenv((char *)(name))) ) != 0 ) {
        if ( c ) *tt = c;
        ThePreVar = 0;
        return(t);
    }
    if ( c ) *tt = c;
    if ( flag == WITHERROR ) {
        Error1("Undefined preprocessor variable",name);
```

```
    }
    return(0);
}

/*
       #] GetPreVar :
       #[ PutPreVar :
*/

int PutPreVar(UBYTE *name, UBYTE *value, UBYTE *args, int mode)
{
    int i, ii, num = 2, nnum = 2, numargs = 0;
    UBYTE *s, *t, *u = 0;
    PREVAR *p;
    if ( value == 0 && name[0] != '?' ) {
        MesPrint("@Illegal empty value for preprocessor variable %s",name);
        Terminate(-1);
    }
    if ( args ) {
        s = args; num++;
        while ( *s ) {
            if ( *s != ' ' && *s != '\t' ) num++;
            s++;
        }
    }
    if ( mode == 1 ) {
        i = NumPre;
        while ( --i >= 0 ) {
            if ( StrCmp(name,PreVar[i].name) == 0 ) {
                u = PreVar[i].name;
                break;
            }
        }
    }
    else i = -1;
    if ( i < 0 ) { p = (PREVAR *)FromList(&AP.PreVarList); ii = p - PreVar; }
    else         { p = &(PreVar[i]);            ii = i; }
    if ( value ) {
        s = value; while ( *s ) { s++; num++; }
    }
    else num = 1;
    if ( i >= 0 ) {
        if ( p->value ) {
            s = p->value;
            while ( *s ) { s++; nnum++; }
        }
        else nnum = 1;
        if ( nnum >= num ) {
/*
            We can keep this in place
*/
            if ( value && p->value ) {
                s = value;
                t = p->value;
                while ( *s ) *t++ = *s++;
                *t = 0;
            }
            else p->value = 0;
            return(i);
        }
    }
    s = name;  while ( *s ) { s++; num++; }
    t = (UBYTE *)Malloc1(num,"PreVariable");
    p->name = t;
    s = name;  while ( *s ) *t++ = *s++; *t++ = 0;
    if ( value ) {
        p->value = t;
        s = value; while ( *s ) *t++ = *s++; *t = 0;
        if ( AM.atstartup && t[-1] == '\n' ) t[-1] = 0;
    }
    else p->value = 0;
    p->wildarg = 0;
    if ( args ) {
        int first = 1;
        t++; p->argnames = t;
        s = args;
        while ( *s ) {
            if ( *s == ' ' || *s == '\t' ) { s++; continue; }
            if ( *s == ',' ) {
                s++; *t++ = 0; numargs++;
                while ( *s == ' ' || *s == '\t' ) s++;
                if ( *s == '?' ) {
                    if ( p->wildarg > 0 ) {
                        Error0("More than one ?var in #define");
                    }
                    p->wildarg = numargs;
                }
```

```
            }
            else if ( *s == '?' && first ) {
                p->wildarg = 1; *t++ = *s++;
            }
            else { *t++ = *s++; }
            first = 0;
        }
        *t = 0;
        numargs++;
        p->nargs = numargs;
    }
    else {
        p->nargs = 0;
        p->argnames = 0;
    }
    if ( u ) M_free(u,"replace PreVar value");
    return(ii);
}

/*
        #] PutPreVar :
        #[ PopPreVars :
*/

void PopPreVars(int tonumber)
{
    PREVAR *p = &(PreVar[NumPre]);
    while ( NumPre > tonumber ) {
        NumPre--; p--;
        M_free(p->name,"popping PreVar");
        p->name = p->value = 0;
    }
}

/*
        #] PopPreVars :
        #[ IniModule :
*/

void IniModule(int type)
{
    GETIDENTITY
    WORD **w, i;
    CBUF *C = cbuf+AC.cbufnum;
    /*[05nov2003 mt]:*/
#ifdef WITHMPI
    /* To prevent
     *    (1) FlushOut() and PutOut() on the slaves to send a mess to the master
     *        compiling a module,
     *    (2) EndSort() called from poly_factorize_expression() on the master
     *        waits for the slaves.
     */
    PF.parallel=0;
    /*BTW, this was the bug preventing usage of more than 1 expression!*/
#endif

    AR.BracketOn = 0;
    AR.StoreData.dirtyflag = 0;
    AC.bracketindexflag = 0;
    AT.bracketindexflag = 0;

/*[06nov2003 mt]:*/
#ifdef WITHMPI
    /* This flag may be set in the procedure tokenize(). */
    AC.RhsExprInModuleFlag = 0;
/*[20oct2009 mt]:*/
    PF.mkSlaveInfile=0;
    PF.slavebuf.PObuffer=NULL;
    for(i=0; i<NumExpressions; i++)
        Expressions[i].vflags &= ~ISINRHS;
/*:[20oct2009 mt]*/
#endif
/*:[06nov2003 mt]*/

    /*[19nov2003 mt]:*/
    /*The module counter:*/
    (AC.CModule)++;
    /*:[19nov2003 mt]*/

    if ( !type ) {
        if ( C->rhs ) {
            w = C->rhs; i = C->maxrhs;
            do { *w++ = 0; } while ( --i > 0 );
        }
        if ( C->lhs ) {
            w = C->lhs; i = C->maxlhs;
            do { *w++ = 0; } while ( --i > 0 );
```

```
        }
    }
    C->numlhs = C->numrhs = 0;
    ClearTree(AC.cbufnum);
    while ( AC.NumLabels > 0 ) {
        AC.NumLabels--;
        if ( AC.LabelNames[AC.NumLabels] ) M_free(AC.LabelNames[AC.NumLabels],"LabelName");
    }

    C->Pointer = C->Buffer;

    AC.Commercial[0] = 0;

    AC.IfStack = AC.IfHeap;
    AC.arglevel = 0;
    AC.termlevel = 0;
    AC.IfLevel = 0;
    AC.WhileLevel = 0;
    AC.RepLevel = 0;
    AC.insidelevel = 0;
    AC.dolooplevel = 0;
    AC.MustTestTable = 0;
    AO.PrintType = 0;                /* Otherwise statistics can get spoiled */
    AC.ComDefer = 0;
    AC.CollectFun = 0;
    AM.S0->PolyWise = 0;
    AC.SymChangeFlag = 0;
    AP.lhdollarerror = 0;
    AR.PolyFun = AC.lPolyFun;
    AR.PolyFunInv = AC.lPolyFunInv;
    AR.PolyFunType = AC.lPolyFunType;
    AR.PolyFunExp = AC.lPolyFunExp;
    AR.PolyFunVar = AC.lPolyFunVar;
    AR.PolyFunPow = AC.lPolyFunPow;
    AC.mparallelflag = AC.parallelflag | AM.hparallelflag;
    AC.inparallelflag = 0;
    AC.mProcessBucketSize = AC.ProcessBucketSize;
    NumPotModdollars = 0;
    AC.topolynomialflag = 0;
#ifdef WITHPTHREADS
    if ( AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
    else AS.MultiThreaded = 0;
    for ( i = 1; i < AM.totalnumberofthreads; i++ ) {
        AB[i]->T.S0->PolyWise = 0;
    }
#endif
    OpenTemp();
}

/*
        #] IniModule :
        #[ IniSpecialModule :
*/

void IniSpecialModule(int type)
{
    DUMMYUSE(type);
}

/*
        #] IniSpecialModule :
        #[ PreProcessor :
*/

void PreProcessor(void)
{
    int moduletype = FIRSTMODULE;
    int specialtype = 0;
    int error1 = 0, error2 = 0, retcode, retval;
    UBYTE c, *t, *s;
    AP.StopWatchZero = GetRunningTime();
    AC.compiletype = 0;
    AP.PreContinuation = 0;
    AP.PreAssignLevel = 0;
    AP.gNumPre = NumPre;
    AC.iPointer = AC.iBuffer;
    AC.iPointer[0] = 0;

    if ( AC.CheckpointFlag == -1 ) DoRecovery(&moduletype);
    AC.CheckpointStamp = Timer(0);

    for(;;) {
/*      if ( A.StatisticsFlag ) CharOut(LINEFEED); */

        IniModule(moduletype);

        /*Re-define preprocessor variable CMODULE_ as a current module number, starting from 1*/
```

```
        /*The module counter is AC.CModule, it is incremented in IniModule*/
        {
            UBYTE buf[24];/*64/Log_2[10] = 19.3, this is enough for any integer*/
            NumToStr(buf,AC.CModule);
            PutPreVar((UBYTE *)"CMODULE_",buf,0,1);
        }

        if ( specialtype ) IniSpecialModule(specialtype);

        for(;;) {   /* Read a single line/statement */
            c = GetChar(0);
            if ( c == AP.ComChar ) {  /* This line is commentary */
                LoadInstruction(5);
                if ( AC.CurrentStream->FoldName ) {
                    t = AP.preStart;
                    if ( *t && t[1] && t[2] == '#' && t[3] == ']' ) {
                        t += 4;
                        while ( *t == ' ' || *t == '\t' ) t++;
                        s = AC.CurrentStream->FoldName;
                        while ( *s == *t ) { s++; t++; }
                        if ( *s == 0 && ( *t == ' ' || *t == '\t'
                        || *t == ':' ) ) {
                            while ( *t == ' ' || *t == '\t' ) t++;
                            if ( *t == ':' ) {
                                AC.CurrentStream = CloseStream(AC.CurrentStream);
                            }
                        }
                    }
                }
                *AP.preStart = 0;
                continue;
            }
            while ( c == ' ' || c == '\t' ) c = GetChar(0);
            if ( c == LINEFEED ) continue;
            if ( c == ENDOFINPUT ) {
/*              CharOut(LINEFEED); */
                Warning(".end instruction generated");
                moduletype = ENDMODULE; specialtype = 0;
                goto endmodule; /* Fake one */
            }
            if ( c == '#' ) {
                if ( PreProInstruction() ) { error1++; error2++; AP.preError++; }
                *AP.preStart = 0;
            }
            else if ( c == '.' ) {
                if ( ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ||
                ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) ) {
                    LoadInstruction(1);
                    continue;
                }
                if ( ModuleInstruction(&moduletype,&specialtype) ) { error2++; AP.preError++; }
                if ( specialtype ) SetSpecialMode(moduletype,specialtype);
                if ( AP.PreInsideLevel != 0 ) {
                    MesPrint("@end of module instructions may not be used inside");
                    MesPrint("@the scope of a %#inside %#endinside construction.");
                    Terminate(-1);
                }
                if ( AC.RepLevel > 0 ) {
                    MesPrint("&EndRepeat statement(s) missing");
                    error2++; AP.preError++;
                }
                if ( AC.tablecheck == 0 ) {
                    AC.tablecheck = 1;
                    if ( TestTables() ) { error2++; AP.preError++; }
                }
                if ( AP.PreContinuation ) {
                    error1++; error2++;
                    MesPrint("&Unfinished statement. Missing ;?");
                }
                if ( moduletype == GLOBALMODULE ) MakeGlobal();
                else {
endmodule:          if ( error2 == 0 && AM.qError == 0 ) {
                        retcode = ExecModule(moduletype);
#ifdef WITHMPI
                        if(PF.slavebuf.PObuffer!=NULL){
                            M_free(PF.slavebuf.PObuffer,"PF inbuf");
                            PF.slavebuf.PObuffer=NULL;
                        }
#endif
                        UpdatePositions();
                        if ( retcode < 0 ) error1++;
                        if ( retcode ) { error2++; AP.preError++; }
                    }
                    else {
                        EXPRESSIONS e;
                        WORD j;
                        for ( j = 0, e = Expressions; j < NumExpressions; j++, e++ ) {
```

```
                            if ( e->replace == NEWLYDEFINEDEXPRESSION ) e->replace = REGULAREXPRESSION;
                        }
                    }
                    switch ( moduletype ) {
                        case STOREMODULE:
                            if ( ExecStore() ) error1++;
                            break;
                        case CLEARMODULE:
                            FullCleanUp();
                            error1 = error2 = AP.preError = 0;
                            AM.atstartup = 1;
                            PutPreVar((UBYTE *)"DATE_",(UBYTE *)MakeDate(),0,1);
                            AM.atstartup = 0;
                            if ( AM.resetTimeOnClear ) {
#ifdef WITHPTHREADS
                                ClearAllThreads();
#endif
                                AM.SumTime += TimeCPU(1);
                                TimeCPU(0);
                            }
                            AP.StopWatchZero = GetRunningTime();
                            break;
                        case ENDMODULE:
                            Terminate( -( error1 | error2 ) );
                    }
                }
                AC.tablecheck = 0;
                AC.compiletype = 0;
                if ( AC.exprfillwarning > 0 ) {
                    AC.exprfillwarning = 0;
                }
                if ( AC.CheckpointFlag && error1 == 0 && error2 == 0 ) DoCheckpoint(moduletype);
                break;  /* start a new module */
            }
            else {
                if ( ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ||
                ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) ) {
                    pushbackchar = c;
                    LoadInstruction(5);
                    continue;
                }
                UngetChar(c);
                if ( AP.PreContinuation ) {
                    retval = LoadStatement(OLDSTATEMENT);
                }
                else {
                    AC.CurrentStream->prevline = AC.CurrentStream->linenumber;
                    retval = LoadStatement(NEWSTATEMENT);
                }
                if ( retval < 0 ) {
                    error1++;
                    if ( retval == -1 ) AP.PreContinuation = 0;
                    else AP.PreContinuation = 1;
                    TryRecover(0);
                }
                else if ( retval > 0 ) AP.PreContinuation = 0;
                else AP.PreContinuation = 1;
                if ( error1 == 0 && !AP.PreContinuation ) {
                    if ( ( AP.PreDebug & PREPROONLY ) == 0 ) {
                        int onpmd = NumPotModdollars;
#ifdef WITHMPI
                        WORD oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
                        if ( AP.PreAssignFlag ) AC.RhsExprInModuleFlag = 0;
#endif
                        if ( AP.PreOut || ( AP.PreDebug & DUMPTOCOMPILER )
                                == DUMPTOCOMPILER )
                                    MesPrint(" %s",AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]);
                        retcode = CompileStatement(AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]);
                        if ( retcode < 0 ) error1++;
                        if ( retcode ) { error2++; AP.preError++; }
                        if ( AP.PreAssignFlag ) {
                            if ( retcode == 0 ) {
                                if ( ( retcode = CatchDollar(0) ) < 0 ) error1++;
                                else if ( retcode > 0 ) { error2++; AP.preError++; }
                            }
                            else CatchDollar(-1);
                            POPPREASSIGNLEVEL;
                            if ( AP.PreAssignLevel <=0 )
                                AP.PreAssignFlag = 0;
                            NumPotModdollars = onpmd;
#ifdef WITHMPI
                            AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
#endif
                        }
                    }
                    else {
```

```
                        MesPrint(" %s",AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]);
                    }
                }
                else if ( !AP.PreContinuation ) {
                    if ( AP.PreAssignLevel > 0 ) {
                        POPPREASSIGNLEVEL;
                        if ( AP.PreAssignLevel <=0 )
                            AP.PreAssignFlag = 0;
                    }
                }
/*
                if ( !AP.PreContinuation ) AP.PreAssignFlag = 0;
*/
            }
        }
    }
}

/*
        #] PreProcessor :
        #[ PreProInstruction :
*/

int PreProInstruction(void)
{
    UBYTE *s, *t;
    KEYWORD *key;
    AP.PreproFlag = 1;
    AP.preFill = 0;
    AP.AllowDelay = 0;
    AP.DelayPrevar = 0;

    oldmode = 0;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) {
        LoadInstruction(3);
        if ( ( StrICmp(AP.preStart,(UBYTE *)"case") == 0
        || StrICmp(AP.preStart,(UBYTE *)"default") == 0 )
        && AP.PreSwitchModes[AP.PreSwitchLevel] == SEARCHINGPRECASE ) {
            LoadInstruction(0);
        }
        else if ( StrICmp(AP.preStart,(UBYTE *)"assign ") == 0 ) {}
        else { LoadInstruction(1); }
    }
    else if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) {
        LoadInstruction(3);
        if ( ( StrICmp(AP.preStart,(UBYTE *)"else") == 0
        || StrICmp(AP.preStart,(UBYTE *)"elseif") == 0 )
        && AP.PreIfStack[AP.PreIfLevel] == LOOKINGFORELSE ) {
            LoadInstruction(0);
        }
        else if ( StrICmp(AP.preStart,(UBYTE *)"assign ") == 0 ) {}
        else {
            LoadInstruction(1);
        }
    }
    else {
        LoadInstruction(0);
    }
    AP.PreproFlag = 0;
    t = AP.preStart;
    if ( *t == '-' ) {
        if ( AP.PreSwitchModes[AP.PreSwitchLevel] == EXECUTINGPRESWITCH
            && AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF )
            AC.NoShowInput = 1;
    }
    else if ( *t == '+' ) {
        if ( AP.PreSwitchModes[AP.PreSwitchLevel] == EXECUTINGPRESWITCH
            && AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF )
            AC.NoShowInput = 0;
    }
    else if ( *t == ':' ) {}
    else {
retry:;
        key = FindKeyWord(t,precommands,sizeof(precommands)/sizeof(KEYWORD));
        s = EndOfToken(t);
        if ( key == 0 ) {
            if ( *s == ';' ) {
                *s = 0; goto retry;
            }
            else {
                *s = 0;
                MesPrint("@Unrecognized preprocessor instruction: %s",t);
                return(-1);
            }
        }
        while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
        t = s;
```

```
        while ( *t ) t++;
        while ( ( t[-1] == ';' ) && ( t[-2] != '\\' ) ) {
            t--; *t = 0;
        }
        return((key->func)(s));
    }
    return(0);
}

/*
        #] PreProInstruction :
        #[ LoadInstruction :

        0:  preprocessor instruction that may involve matching of brackets
        1:  runs straight to end-of-line
        2:  runs to ;
        3:  only gets one word without `' interpretation.
        5:  with pushbackchar, but inside commentary. -> 1

To be added:
    In define, redefine, call and listed do we may have delayed substitution
    of preprocessor variables.
*/

int LoadInstruction(int mode)
{
    UBYTE *s, *sstart, *t, c, cp;
    LONG position, fillpos = 0;
    int bralevel = 0, parlevel = 0, first = 1;
    int quotelevel = 0;
    if ( AP.preFill ) {
        s = AP.preFill;
        AP.preFill = 0;
        if ( s[1] != LINEFEED && s[1] != ENDOFINPUT ) {
            s[0] = s[1]; s++;
        }
        else { oldmode = mode; return(0); }
    }
    else { s = AP.preStart; }
    sstart = s; *s = 0;
    for(;;) {
        if ( ( mode & 1 ) == 1 ) {
            if ( pushbackchar && ( mode == 3 || mode == 5 ) ) {
                c = pushbackchar; pushbackchar = 0;
            }
            else c = GetInput();
        }
        else {
            c = GetChar(0);
        }

        if ( mode == 2 && c == ';' ) break;
        if ( ( mode == 1 || mode == 5 ) && c == LINEFEED ) break;
        if ( mode == 3 && FG.cTable[c] != 0 ) {
            if ( c == '$' ) {
                pushbackchar = '$';
                *s++ = 'a'; *s++ = 's'; *s++ = 's'; *s++ = 'i';
                *s++ = 'g'; *s++ = 'n'; *s++ = ' '; *s = 0;
            }
            if ( c == '\"' || c == '`' ) { /* we do not expand preprocessor variables */
                mode = 1;
            }
            else {
                AP.preFill = s; *s++ = 0; *s = c;
                oldmode = mode;
                return(0);
            }
        }
        if ( mode == 0 && first ) {
            if ( c == '$' ) {
dodollar:       s = sstart;
                *s++ = 'a'; *s++ = 's'; *s++ = 's'; *s++ = 'i';
                *s++ = 'g'; *s++ = 'n'; *s = 0;
                pushbackchar = c;
                oldmode = mode;
                return(0);
            }
            if ( c == ' ' || c == '\t' || c == ',' ) {}
            else first = 0;
        }
        else if ( mode == 1 && first && c == '$' && oldmode == 3 ) goto dodollar;
        if ( c == ENDOFINPUT || ( c == LINEFEED
/*      && bralevel == 0  */
        && quotelevel == 0 ) ) {
            if ( mode == 2 && c == ENDOFINPUT ) {
                MesPrint("@Unexpected end of instruction");
                oldmode = mode;
```

```
                return(-1);
            }
/*
            if ( mode == 0 && bralevel ) {
                MesPrint("@Unmatched brackets");
                oldmode = mode;
                return(-1);
            }
*/
            if ( mode != 2 ) break;
        }
        if ( quotelevel ) {
            if ( c == '\\' ) {
                if ( ( mode == 1 ) || ( mode == 5 ) ) c = GetInput();
                else {
                    c = GetChar(0);
                }
                if ( c == ENDOFINPUT ) {
                    MesPrint("@Unmatched \"");
                    if ( mode == 2 && c == ENDOFINPUT ) {
                        MesPrint("@Unexpected end of instruction");
                    }
/*
                    if ( mode == 0 && bralevel ) {
                        MesPrint("@Unmatched brackets");
                    }
*/
                    oldmode = mode;
                    return(-1);
                }
                else if ( c == LINEFEED ) {}
                else if ( c == '"' ) { *s++ = '\\'; }
                else {
                    *s++ = '\\';
                }
            }
            else if ( c == '"' ) {
                quotelevel = 0;
                AP.AllowDelay = 0;
            }
        }
        else if ( c == '\\' ) {
            if ( ( mode == 1 ) || ( mode == 5 ) ) cp = GetInput();
            else {
                cp = GetChar(0);
            }
            if ( cp == LINEFEED ) continue;
            if ( mode != 2 || cp != ';' ) *s++ = c;
            c = cp;
        }
        else if ( c == '"' ) {
/*
            Now look back in the buffer and determine what the keyword is.
            If it is define or redefine, put AllowDelay to 1.
*/
            t = AP.preStart;
            while ( FG.cTable[*t] <= 1 ) t++;
            cp = *t; *t = 0;
            if ( ( StrICmp(AP.preStart,(UBYTE *)"define") == 0 )
                || ( StrICmp(AP.preStart,(UBYTE *)"redefine") == 0 ) ) {
                AP.AllowDelay = 1;
                oldstream = AC.CurrentStream;
            }
            *t = cp;
            quotelevel = 1;
        }
        else if ( quotelevel == 0 && bralevel == 0 && c == '(' ) {
            t = AP.preStart;
            while ( FG.cTable[*t] <= 1 ) t++;
            cp = *t; *t = 0;
            if ( ( parlevel == 0 )
                 && ( StrICmp(AP.preStart,(UBYTE *)"call") == 0 ) ) {
                AP.AllowDelay = 1;
                oldstream = AC.CurrentStream;
            }
            *t = cp;
            parlevel++;
        }
        else if ( quotelevel == 0 && bralevel == 0 && c == ')' ) {
            parlevel--;
        }
        else if ( quotelevel == 0 && parlevel == 0 && c == '{' ) {
            t = AP.preStart;
            while ( FG.cTable[*t] <= 1 ) t++;
            cp = *t; *t = 0;
            if ( ( bralevel == 0 )
                && ( ( StrICmp(AP.preStart,(UBYTE *)"call") == 0 )
```

```
                || ( StrICmp(AP.preStart,(UBYTE *)"do") == 0 ) ) ) {
                AP.AllowDelay = 1;
                oldstream = AC.CurrentStream;
            }
            *t = cp;
            bralevel++;
        }
        else if ( quotelevel == 0 && parlevel == 0 && c == '}' ) {
            bralevel--;
            if ( bralevel < 0 ) {
                if ( mode != 5 ) {
                    MesPrint("@Unmatched brackets");
                    oldmode = mode;
                    return(-1);
                }
                bralevel = 0;
            }
        }
        if ( s >= (AP.preStop-1) ) {
            UBYTE **ppp;
            position = s - AP.preStart;
            if ( AP.preFill ) fillpos = AP.preFill - AP.preStart;
            ppp = &(AP.preStart); /* to avoid a compiler warning */
            if ( DoubleLList((void ***)ppp,&AP.pSize,sizeof(UBYTE),
            "instruction buffer") ) { *s = 0; oldmode = mode; return(-1); }
            AP.preStop = AP.preStart + AP.pSize-3;
            s = AP.preStart + position;
            if ( AP.preFill ) AP.preFill = fillpos + AP.preStart;
        }
        *s++ = c;
    }
    *s = 0;
    oldmode = mode;
    if ( mode == 0 ) {
        if ( ExpandTripleDots(1) < 0 ) return(-1);
    }
    return(0);
}

/*
        #] LoadInstruction :
        #[ LoadStatement :

        Puts the current string together in the input buffer.
        Does things like placing comma's where needed and expand ...
        We force a comma after the keyword. Before 8-sep-2009 the program might
        not put a comma if a + or - followed. And then the compiler ate
        the + or - and we needed repair code in the routines that used the
        + or - (Print, modulus, multiply and (a)bracket). This worked but
        the problem was with statements like  Dimension -4; which then would
        be processed as Dimension 4; (JV)
*/

int LoadStatement(int type)
{
    UBYTE *s, c, cp;
    int retval = 0, stringlevel = 0, newstatement = 0;
    if ( type == NEWSTATEMENT ) { AP.eat = 1; newstatement = 1;
        s = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; }
    else { s = AC.iPointer; *s = 0; c = ' '; goto blank; }
    *s = 0;
    for(;;) {
        c = GetChar(0);
        if ( c == ENDOFINPUT ) { retval = -1; break; }
        if ( stringlevel == 0 ) {
            if ( c == LINEFEED ) {
                if ( AP.eat < 0 ) { s--; AP.eat = 0; }
                retval = 0; break;
            }
            if ( c == ';' ) {
                if ( AP.eat < 0 ) { s--; AP.eat = 0; }
                while ( ( c = GetChar(0) ) == ' ' || c == '\t' ) {}
                if ( c != LINEFEED ) UngetChar(c);
                retval = 1;
                break;
            }
        }
        if ( c == '\\' ) {
            cp = GetChar(0);
            if ( cp == LINEFEED ) continue;
            *s++ = c;
            c = cp;
        }
        if ( c == '"' ) {
            if ( stringlevel == 0 ) stringlevel = 1;
            else stringlevel = 0;
            AP.eat = 0;
```

```
        }
        else if ( stringlevel == 0 ) {
            if ( c == '\t' ) c = ' ';
            if ( c == ' ' ) {
blank:          if ( newstatement < 0 ) newstatement = 0;
                if ( AP.eat && ( newstatement == 0 ) ) continue;
                c = ',';
                AP.eat = -2;
                if ( newstatement > 0 ) newstatement = -1;
            }
            else if ( chartype[c] <= 3 ) {
                AP.eat = 0;
                if ( newstatement < 0 ) newstatement = 0;
            }
            else if ( c == ',' ) {
                if ( newstatement > 0 ) {
                    newstatement = -1;
                    AP.eat = -2;
                }
/*              else if ( AP.eat == -2 ) { s--; } */
                else if ( AP.eat == -2 ) { AP.eat = 1; continue; }
                else { goto doall; }
            }
            else {
doall:;         if ( AP.eat < 0 ) {
                    if ( newstatement == 0 ) s--;
                    else { newstatement = 0; }
                }
                else if ( newstatement == 1 ) newstatement = 0;
                AP.eat = 1;
                if ( c == '*' && s > AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel] && s[-1] == '*' )
    {
                    s[-1] = '^';
                    continue;
                }
            }
        }
        if ( s >= AC.iStop ) {
            if ( !AP.iBufError ) {
                LONG position = s - AC.iBuffer;
                LONG position2 = AC.iPointer - AC.iBuffer;
                UBYTE **ppp = &(AC.iBuffer); /* to avoid a compiler warning */
                if ( DoubleLList((void ***)ppp,&AC.iBufferSize
                ,sizeof(UBYTE),"statement buffer") ) {
                    *s = 0; retval = -1; AP.iBufError = 1;
                }
                AC.iPointer = AC.iBuffer + position2;
                AC.iStop = AC.iBuffer + AC.iBufferSize-2;
                s = AC.iBuffer + position;
            }
            if ( AP.iBufError ) {
                for(;;){
                    c = GetChar(0);
                    if ( c == ENDOFINPUT ) { retval = -1; break; }
                    if ( c == '"' ) {
                        if ( stringlevel > 0 ) stringlevel = 0;
                        else stringlevel = 1;
                    }
                    else if ( c == LINEFEED && !stringlevel ) { retval = -2; break; }
                    else if ( c == ';' && !stringlevel ) {
                        while ( ( c = GetChar(0) ) == ' ' || c == '\t' ) {}
                        if ( c != LINEFEED ) UngetChar(c);
                        retval = -1;
                        break;
                    }
                    else if ( c == '\\' ) c = GetChar(0);
                }
                break;
            }
        }
        *s++ = c;
    }
    AC.iPointer = s;
    *s = 0;
    if ( stringlevel > 0 ) {
        MesPrint("@Unbalanced \". Runaway string");
        retval = -1;
    }
    if ( retval == 1 ) {
        if ( ExpandTripleDots(0) < 0 ) retval = -1;
    }
    return(retval);
}

/*
        #] LoadStatement :
        #[ ExpandTripleDots :
```

```
*/

static inline int IsSignChar(UBYTE c)
{
    return c == '+' || c == '-';
}

static inline int IsAlphanumericChar(UBYTE c)
{
    return FG.cTable[c] == 0 || FG.cTable[c] == 1;
}

static inline int CanParseSignedNumber(const UBYTE *s)
{
    while ( IsSignChar(*s) ) s++;
    return FG.cTable[*s] == 1;
}

int ExpandTripleDots(int par)
{
    UBYTE *s, *s1, *s2, *n1, *n2, *t1, *t2, *startp, operator1, operator2, c, cc;
    UBYTE *nBuffer, *strngs, *Buffer, *Stop;
    LONG withquestion, x1, x2, y1, y2, number, inc, newsize, pow, fullsize;
    int i, error = 0, i1 ,i2, ii, *nums = 0;

    if ( par == 0 ) {
        Buffer = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; Stop = AC.iStop;
    }
    else {
        Buffer = AP.preStart; Stop = AP.preStop;
    }
    s = Buffer; while ( *s ) s++;
    fullsize = s - Buffer;
    if ( fullsize < 7 ) return(error);

    s = Buffer+2;
    while ( *s ) {
        if ( *s != '.' || ( s[-1] != ',' && FG.cTable[s[-1]] != 5 ) )
            { s++; continue; }
        if ( s[-1] == '%' || s[-1] == '^' || s[1] != '.' || s[2] != '.' )
            { s++; continue; }
        s1 = s - 2;
        s += 3;
        if ( *s != s[-4] && ( *s != '+' || s[-4] != '-' )
         && ( *s != '-' || s[-4] != '+' ) ) {
            MesPrint("&Improper operators for ...");
            error = -1;
        }
        operator1 = s[-4];
        operator2 = *s++;
        if ( operator1 == ':' ) operator1 = '.';
        if ( operator2 == ':' ) operator2 = '.';
/*
        We have now O1...O2 (O stands for operator)
        Full syntax is
        [str]#1[?]O1...O2[str]#2[?]   (Special case)
        in which both strings are identical and if one ? then also the other.
        <pattern1>O1...O2<pattern2>   (General case)
        in which the difference in the patterns is just numerical.
*/
        s2 = s;          /* the beginning of the second string */
        if ( *s2 != '<' || *s1 != '>' ) {   /* Special case */
            startp = s1+1;
            withquestion = ( *s1 == '?' ); s1--;
            while ( FG.cTable[*s1] == 1 && s1 >= Buffer ) s1--;
            n1 = s1+1;       /* Beginning of first number */
            if ( FG.cTable[*n1] != 1 ) {
                MesPrint("&No first number in ... operator");
                error = -1;
            }
            while ( FG.cTable[*s1] <= 1 && s1 >= Buffer ) s1--;
            s1++;
/*
            We have now the first string from s1 to n1, number from n1
*/
            t1 = s1; t2 = s2;
            while ( t1 < n1 && *t1 == *t2 ) { t1++; t2++; }
            n2 = t2;
            if ( FG.cTable[*t2] != 1 ) {
                MesPrint("&No second number in ... operator");
                error = -1;
            }
            x2 = 0;
            while ( FG.cTable[*t2] == 1 ) x2 = 10*x2 + *t2++ - '0';
            x1 = 0;
            while ( FG.cTable[*t1] == 1 ) x1 = 10*x1 + *t1++ - '0';
            if ( withquestion != ( *t2 == '?' ) ) {
```

```
                MesPrint("&Improper use of ? in ... operator");
                if ( *t2 == '?' ) t2++;
                error = -1;
            }
            else if ( withquestion ) t2++;
            if ( FG.cTable[*t2] <= 2 ) {
                MesPrint("&Illegal object after ... construction");
                error = -1;
            }
            c = *n1; *n1 = 0; s = t2;
            if ( error ) continue;
/*
            At this point the syntax has been fulfilled. We have
            str in s1.
            x1,x2 are #1,#2
            operator1,operator2 are the two operators.
            s points at whatever comes after.
            Expansion will have to be computed.
*/
            if ( x2 < x1 ) { number = x1-x2; inc = -1; y1 = x2; y2 = x1; }
            else           { number = x2-x1; inc =  1; y1 = x1; y2 = x2; }
            newsize = (number+1)*(n1-s1)            /* the strings   */
                     + number                       /* the operators */
                     +(number+1)*(withquestion?1:0) /* questionmarks */
                     +(number+1);                   /* last digits   */
            pow = 10;
            for ( i = 1; i < 10; i++, pow *= 10 ) {
                if ( y1 >= pow )      newsize += number+1;
                else if ( y2 >= pow ) newsize += y2-pow+1;
                else break;
            }
            while ( Buffer+(fullsize+newsize-(s-s1)) >= Stop ) {
                LONG strpos = s1-Buffer;
                LONG endstr = n1-Buffer;
                LONG startq = startp - Buffer;
                LONG position = s - Buffer;
                UBYTE **ppp;
                if ( par == 0 ) {
                    LONG position2 = AC.iPointer - AC.iBuffer;
                    ppp = &(AC.iBuffer); /* to avoid a compiler warning */
                    if ( DoubleLList((void ***)ppp,&AC.iBufferSize
                        ,sizeof(UBYTE),"statement buffer") ) {
                            Terminate(-1);
                    }
                    AC.iPointer = AC.iBuffer + position2;
                    AC.iStop = AC.iBuffer + AC.iBufferSize-2;
                    Buffer = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; Stop = AC.iStop;
                }
                else {
                    LONG fillpos = 0;
                    if ( AP.preFill ) fillpos = AP.preFill - AP.preStart;
                    ppp = &(AP.preStart); /* to avoid a compiler warning */
                    if ( DoubleLList((void ***)ppp,&AP.pSize,sizeof(UBYTE),
                        "instruction buffer") ) {
                            Terminate(-1);
                    }
                    AP.preStop = AP.preStart + AP.pSize-3;
                    if ( AP.preFill ) AP.preFill = fillpos + AP.preStart;
                    Buffer = AP.preStart; Stop = AP.preStop;
                }
                s  = Buffer + position;
                n1 = Buffer + endstr;
                s1 = Buffer + strpos;
                startp = Buffer + startq;
            }
/*
            We have space for the expansion in the buffer.
            There are two cases: new size >  old size
                                 old size >= new size
            Note that whereever we move things, it will be at least startp.
*/
            if ( newsize > (s-s1) ) {
                t2 = Buffer + fullsize;
                t1 = t2 + (newsize - (s-s1));
                *t1 = 0;
                while ( t2 > s ) { *--t1 = *--t2; }
            }
            else if ( newsize < (s-s1) ) {
                t1 = s1 + newsize; t2 = s; s = t1;
                while ( *t2 ) *t1++ = *t2++;
                *t1 = 0;
            }
            for ( x1 += inc, t1 = startp; number > 0; number--, x1 += inc ) {
                *t1++ = operator1;
                cc = operator1; operator1 = operator2; operator2 = cc;
                t2 = s1; while ( *t2 ) *t1++ = *t2++;
                x2 = x1; n2 = t1;
```

```
                do {
                    *t1++ = '0' + x2 % 10;
                    x2 /= 10;
                } while ( x2 );
                s2 = t1 - 1;
                while ( s2 > n2 ) { cc = *s2; *s2 = *n2; *n2++ = cc; s2--; }
                if ( withquestion ) *t1++ = '?';
            }
            fullsize += newsize - ( s - s1 );
            *n1 = c;
        }
        else {      /* General case. Find the patterns first */
            t1 = s1; s1--;
            while ( s1 > Buffer ) {
                if ( *s1 == '<' ) break;
                s1--;
            }
            t2 = s2;
            while ( *t2 ) {
                if ( *t2 == '>' ) break;
                t2++;
            }
            if ( *s1 != '<' || *t2 != '>' ) {
                MesPrint("&Illegal attempt to use ... operator");
                return(-1);
            }
            s1++; s2++;     /* Pointers to the patterns */
            nums = (int *)Malloc1((t1-s1)*2*(sizeof(int)+sizeof(UBYTE))
                            ,"Expand ...");
            strngs = (UBYTE *)(nums + 2*(t1-s1));
            n1 = s1; n2 = s2; ii = -1; i = 0;
            s = strngs;
            while ( n1 < t1 || n2 < t2 ) {
                /* Check the next characters can be parsed as numbers including signs. */
                if ( CanParseSignedNumber(n1) && CanParseSignedNumber(n2) ) {
                    /*
                     * Don't allow the cases that one has the sign and the other doesn't,
                     * and the meaning changes without the sign. For example,
                     *   <f(1)>+...+<f(3)>      Allowed
                     *   <f(-2)>+...+<f(2)>     Allowed
                     *   <f(x-2)>+...+<f(x+2)>  Allowed
                     *   <f(x-2)>+...+<f(x2)>   Not allowed
                     */
                    int sign1 = IsSignChar(*n1);
                    int sign2 = IsSignChar(*n2);
                    int inword1 = s1 < n1 && IsAlphanumericChar(n1[-1]);
                    int inword2 = s2 < n2 && IsAlphanumericChar(n2[-1]);
                    if ( ( sign1 ^ sign2 ) && ( inword1 || inword2 ) ) break;  /* Not allowed. */
                    if ( sign1 || sign2 ) {
                        *s++ = '+';  /* Marker indicating we need the sign. */
                    }
                } else {
                    /* If they are not numbers, they should be same. */
                    if ( *n1 == *n2 ) { *s++ = *n1++; n2++; continue; }
                    else break;
                }
                ParseSignedNumber(x1,n1)
                ParseSignedNumber(x2,n2)
                if ( x1 == x2 ) {
                    if ( s != strngs && ( s[-1] == '+' || s[-1] == '-' ) ) {
                        /* We need the sign. */
                        s--;
                        if ( x1 >= 0 ) {
                            *s++ = '+';
                        }
                    }
                    s = NumCopy(x1, s);
                }
                else {
                    nums[2*i] = x1; nums[2*i+1] = x2;
                    i++; *s++ = 0;
                }
            }
            if ( n1 < t1 || n2 < t2 ) {
                MesPrint("&Improper use of ... operator.");
theend:         M_free(nums,"Expand ...");
                return(-1);
            }
            *s = 0;
            if ( i == 0 ) ii = 0;
            else {
                ii = nums[0] - nums[1];
                if ( ii < 0 ) ii = -ii;
                for ( x1 = 1; x1 < i; x1++ ) {
                    x2 = nums[2*x1]-nums[2*x1+1];
                    if ( x2 < 0 ) x2 = -x2;
                    if ( x2 != ii ) {
```

```
                        MesPrint("&Improper synchronization of numbers in ... operator");
                        goto theend;
                    }
                }
            }
            ii++;
/*
            We have now proper syntax.
            There are i+1 strings in strngs and i pairs of numbers
            in nums. Each time a start value and a finish value.
            We have ii steps. If ii <= 2, it will fit in the existing
            allocation. But this is hardly useful.
            We make a new allocation and copy from the old.
            Compute space.
*/
            x2 = s - strngs - i;  /* -1 for end-of-string and +1 for the operator*/
            for ( i1 = 0; i1 < i; i1++ ) {
                i2 = nums[2*i1];
                x1 = nums[2*i1+1];
                if ( i2 < 0 ) i2 = -i2;
                if ( x1 < 0 ) x1 = -x1;
                if ( x1 > i2 ) i2 = x1;
                x1 = 2;
                while ( i2 > 0 ) { i2 /= 10; x1++; }
                x2 += x1;
            }
            x2 *= ii;   /* Space for the expanded string (a bit more) */
            x2 += fullsize;
            x2 += 5;        /* This will definitely hold everything */
            x2 += sizeof(UBYTE *);
            x2 = x2 - (x2 & (sizeof(UBYTE *)-1));

            nBuffer = (UBYTE *)Malloc1(x2,"input buffer");
            n1 = nBuffer; s = Buffer; s1--;
            while ( s < s1 ) *n1++ = *s++;
/*
            Solution of the special case that no comma was generated
            due to the presence of < to start the pattern.
            We get a comma when the word before ends in an alphanumeric
            character, a _ or a ] and the word inside starts with an
            alphanumeric character, a [ (or an _ (for future considerations))
*/
            if ( ( ( n1 > nBuffer ) && ( ( FG.cTable[n1[-1]] <= 1 )
            || ( n1[-1] == '_' ) || ( n1[-1] == ']' ) ) ) &&
            ( ( FG.cTable[strngs[0]] <= 1 ) || ( strngs[0] == '[' )
             || ( strngs[0] == '_' ) ) ) *n1++ = ',';

            for ( i1 = 0; i1 < ii; i1++ ) {
                s = strngs; while ( *s ) *n1++ = *s++;
                for ( i2 = 0; i2 < i; i2++ ) {
                    if ( n1 > nBuffer && IsSignChar(n1[-1]) ) {
                        /* We need the sign of counters. */
                        n1--;
                        if ( nums[2*i2] >= 0 ) {
                            *n1++ = '+';
                        }
                    }
                    n1 = NumCopy((WORD)(nums[2*i2]),n1);
                    if ( nums[2*i2] > nums[2*i2+1] ) nums[2*i2]--;
                    else nums[2*i2]++;
                    s++; while ( *s ) *n1++ = *s++;
                }
                if ( ( i1 & 1 ) == 0 ) *n1++ = operator1;
                else *n1++ = operator2;
            }
            n1--;   /* drop the trailing operator */
            s = t2 + 1; n2 = n1;
/*
            Similar extra comma
*/
            if ( ( ( ( FG.cTable[n1[-1]] <= 1 )
            || ( n1[-1] == '_' ) || ( n1[-1] == ']' ) ) ) &&
            ( ( FG.cTable[s[0]] <= 1 ) || ( s[0] == '[' )
             || ( s[0] == '_' ) ) ) *n1++ = ',';

            while ( *s ) *n1++ = *s++;
            *n1 = 0;
            if ( par == 0 ) {
                LONG nnn1 = n1-nBuffer;
                LONG nnn2 = n2-nBuffer;
                LONG nnn3;
                while ( AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel] + x2 >= AC.iStop ) {
                    LONG position = s-Buffer;
                    LONG position2 = AC.iPointer - AC.iBuffer;
                    UBYTE **ppp;
                    ppp = &(AC.iBuffer); /* to avoid a compiler warning */
                    if ( DoubleLList((void ***)ppp,&AC.iBufferSize
```

```
                        ,sizeof(UBYTE),"statement buffer") ) {
                            Terminate(-1);
                    }
                    AC.iPointer = AC.iBuffer + position2;
                    AC.iStop = AC.iBuffer + AC.iBufferSize-2;
                    Buffer = AC.iBuffer+AP.PreAssignStack[AP.PreAssignLevel]; Stop = AC.iStop;
                    s  = Buffer + position;
                }
/*
                This can be improved. We only have to start from the first term.
*/
                for ( nnn3 = 0; nnn3 < nnn1; nnn3++ ) Buffer[nnn3] = nBuffer[nnn3];
                Buffer[nnn3] = 0;
                n1 = Buffer + nnn1;
                n2 = Buffer + nnn2;
                M_free(nBuffer,"input buffer");
                M_free(nums,"Expand ...");
            }
            else { /* Comes here only inside a real preprocessor instruction */
                AP.preStop = nBuffer + x2 - 2;
                AP.pSize = x2;
                M_free(AP.preStart,"input buffer");
                M_free(nums,"Expand ...");
                AP.preStart = nBuffer;
                Buffer = AP.preStart; Stop = AP.preStop;
            }
            fullsize = n1 - Buffer;
            s = n2;
        }
    }
    return(error);
}

/*
        #] ExpandTripleDots :
        #[ FindKeyWord :
*/

KEYWORD *FindKeyWord(UBYTE *theword, KEYWORD *table, int size)
{
    int low,med,hi;
    UBYTE *s1, *s2;
    low = 0;
    hi = size-1;
    while ( hi >= low ) {
        med = (hi+low)/2;
        s1 = (UBYTE *)(table[med].name);
        s2 = theword;
        while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
        if ( *s1 == 0 &&
/*[30apr2004 mt]:*/
/* The bug!:
                FG.cTable[*s2] != 1 && FG.cTable[*s2] != 2
*/
                FG.cTable[*s2] != 0 && FG.cTable[*s2] != 1
/*      ( *s2 == ' ' || *s2 == '\t' || *s2 == 0 || *s2 == ',' || *s2 == '(' ) */
             )
            return(table+med);
        if ( tolower(*s2) > tolower(*s1) ) low = med+1;
        else hi = med - 1;
    }
    return(0);
}

/*
        #] FindKeyWord :
        #[ FindInKeyWord :
*/

KEYWORD *FindInKeyWord(UBYTE *theword, KEYWORD *table, int size)
{
    int i;
    UBYTE *s1, *s2;
    for ( i = 0; i < size; i++ ) {
        s1 = (UBYTE *)(table[i].name);
        s2 = theword;
        while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
        if ( *s2 == 0 || *s2 == ' ' || *s2 == ',' || *s2 == '\t' )
            return(table+i);
    }
    return(0);
}

/*
        #] FindInKeyWord :
        #[ TheDefine :
*/
```

```

int TheDefine(UBYTE *s, int mode)
{
    UBYTE *name, *value, *valpoin, *args = 0, c;
    if ( ( mode & 2 ) == 0 ) {
        if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
        if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    }
    else { mode &= ~2; }
    name = s;
    if ( chartype[*s] != 0 ) goto illname;
    s++;
    while ( chartype[*s] <= 1 ) s++;
    value = s;
    while ( *s == ' ' || *s == '\t' ) s++;
    c = *s; *value = 0;
    if ( c == 0 ) {
        if ( PutPreVar(name,(UBYTE *)"1",0,mode) < 0 ) return(-1);
        return(0);
    }
    if ( c == '(' ) {   /* arguments. scan for correctness */
        s++; args = s;
        for (;;) {
            if ( chartype[*s] != 0 ) goto illarg;
            s++;
            while ( chartype[*s] <= 1 ) s++;
            while ( *s == ' ' || *s == '\t' ) s++;
            if ( *s == ')' ) break;
            if ( *s != ',' ) goto illargs;
            s++;
            while ( *s == ' ' || *s == '\t' ) s++;
        }
        *s++ = 0;
        while ( *s == ' ' || *s == '\t' ) s++;
        c = *s;
    }
    if ( c == '"' ) {
        s++; valpoin = value = s;
        while ( *s != '"' ) {
            if ( *s == '\\' ) {
                if ( s[1] == 'n' ) { *valpoin++ = LINEFEED; s += 2; }
                else if ( s[1] == '"' ) { *valpoin++ = '"'; s += 2; }
                else if ( s[1] == 0 ) goto illval;
                else { *valpoin++ = *s++; *valpoin++ = *s++; }
            }
            else *valpoin++ = *s++;
        }
        *valpoin = 0;
        if ( PutPreVar(name,value,args,mode) < 0 ) return(-1);
    }
    else {
        MesPrint("@Illegal string for preprocessor variable %s. Forgotten double quotes (\") ?",name);
        return(-1);
    }
    return(0);
illname:;
    MesPrint("@Illegally formed name of preprocessor variable");
    return(-1);
illarg:;
    MesPrint("@Illegally formed name of argument of preprocessor definition");
    return(-1);
illargs:;
    MesPrint("@Illegally formed arguments of preprocessor definition");
    return(-1);
illval:;
    MesPrint("@Illegal valpoin for preprocessor variable %s",name);
    return(-1);
}

/*
        #] TheDefine :
        #[ DoCommentChar :
*/

int DoCommentChar(UBYTE *s)
{
    UBYTE c;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == 0 || *s == '\n' ) {
        MesPrint("@No valid comment character specified");
        return(-1);
    }
    c = *s++;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s != 0 && *s != '\n' ) {
```

```
        MesPrint("@Comment character should be a single valid character");
        return(-1);
    }
    AP.ComChar = c;
    return(0);
}

/*
        #] DoCommentChar :
        #[ DoPreAssign :

        Routine assigns a 'value' to a $variable.
        Syntax: #assign
            next line(s) a statement of the type
            $name = expression;
        Note: at the moment of the assign there cannot be an 'open' statement.
*/

int DoPreAssign(UBYTE *s)
{
    int error = 0;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) {
        return(0);
    }
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) {
        return(0);
    }
    if ( *s ) {
        MesPrint("@Illegal characters in %#assign instruction");
        error = 1;
    }
    PUSHPREASSIGNLEVEL;
    AP.PreAssignFlag = 1;
/*
    if ( AP.PreContinuation ) {
        MesPrint("@Assign instructions cannot occur inside statements");
        MesPrint("@Missing ; ?");
        AP.PreContinuation = 0;
        error = 1;
    }
*/
    return(error);
}

/*
        #] DoPreAssign :
        #[ DoDefine :
*/

int DoDefine(UBYTE *s)
{
    return(TheDefine(s,0));
}

/*
        #] DoDefine :
        #[ DoRedefine :
*/

int DoRedefine(UBYTE *s)
{
    return(TheDefine(s,1));
}

/*
        #] DoRedefine :
        #[ ClearMacro :

        Undefines the arguments of a macro after its use.
*/

int ClearMacro(UBYTE *name)
{
    int i;
    PREVAR *p;
    UBYTE *s;
    for ( i = NumPre-1, p = &(PreVar[NumPre-1]); i >= 0; i--, p-- ) {
        if ( StrCmp(name,p->name) == 0 ) break;
    }
    if ( i < 0 ) return(-1);
    if ( p->nargs <= 0 ) return(0);
    s = p->argnames;
    for ( i = 0; i < p->nargs; i++ ) {
        TheUndefine(s);
        while ( *s ) s++;
        s++;
    }
```

```
    return(0);
}

/*
        #] ClearMacro :
        #[ TheUndefine :

        There is a complication here. If there are redefine statements
        they will be pointing at the wrong variable if their number is
        greater than the number of the variable we pop.
*/

int TheUndefine(UBYTE *name)
{
    int i, inum, error = 0;
    PREVAR *p;
    for ( i = NumPre-1, p = &(PreVar[NumPre-1]); i >= 0; i--, p-- ) {
        if ( StrCmp(name,p->name) == 0 ) {
            M_free(p->name,"undefining PreVar");
            NumPre--;
            inum = i;
            while ( i < NumPre ) {
                p->name  = p[1].name;
                p->value = p[1].value;
                p++; i++;
            }
            p->name = 0; p->value = 0;
            {
                CBUF *CC = cbuf + AC.cbufnum;
                int j, k;
                for ( j = 1; j <= CC->numlhs; j++ ) {
                    if ( CC->lhs[j][0] == TYPEREDEFPRE ) {
                        if ( CC->lhs[j][2] > inum ) CC->lhs[j][2]--;
                        else if ( CC->lhs[j][2] == inum ) {
                            for ( k = inum - 1; k >= 0; k-- )
                                if ( StrCmp(name, PreVar[k].name) == 0 ) break;
                            if ( k >= 0 ) CC->lhs[j][2] = k;
                            else {
                                MesPrint("@Conflict between undefining a preprocessor variable and a
redefine statement");
                                error = 1;
                            }
                        }
                    }
                }
#ifdef PARALLELCODE
                for ( j = 0; j < AC.numpfirstnum; j++ ) {
                    if ( AC.pfirstnum[j] > inum ) AC.pfirstnum[j]--;
                    else if ( AC.pfirstnum[j] == inum ) {
                        for ( k = inum - 1; k >= 0; k-- )
                            if ( StrCmp(name, PreVar[k].name) == 0 ) break;
                        if ( k >= 0 ) AC.pfirstnum[j] = k;
                    }
                }
#endif
            }
            break;
        }
    }
    return(error);
}

/*
        #] TheUndefine :
        #[ DoUndefine :
*/

int DoUndefine(UBYTE *s)
{
    UBYTE *name, *t;
    int error = 0, retval;
/*
    int i;
    PREVAR *p;
*/
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    name = s;
    if ( chartype[*s] != 0 ) goto illname;
    s++;
    while ( chartype[*s] <= 1 ) s++;
    t = s;
    if ( *s && *s != ' ' && *s != '\t' ) goto illname;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s ) {
        MesPrint("@Undefine should just have a variable name");
        error = -1;
```

```
    }
    *t = 0;
    if ( ( retval = TheUndefine(name) ) != 0 ) {
        if ( error == 0 ) return(retval);
        if ( error > 0 ) error = retval;
    }
/*
    for ( i = NumPre-1, p = &(PreVar[NumPre-1]); i >= 0; i--, p-- ) {
        if ( StrCmp(name,p->name) == 0 ) {
            M_free(p->name,"undefining PreVar");
            NumPre--;
            while ( i < NumPre ) {
                p->name  = p[1].name;
                p->value = p[1].value;
                p++; i++;
            }
            p->name = 0; p->value = 0;
            break;
        }
    }
*/
    return(error);
illname:;
    MesPrint("@Illegally formed name of preprocessor variable");
    return(-1);
}

/*
        #] DoUndefine :
        #[ DoInclude :
*/

int DoInclude(UBYTE *s) { return(Include(s,FILESTREAM)); }

/*
        #] DoInclude :
        #[ DoReverseInclude :
*/

int DoReverseInclude(UBYTE *s) { return(Include(s,REVERSEFILESTREAM)); }

/*
        #] DoReverseInclude :
        #[ Include :
*/

int Include(UBYTE *s, int type)
{
    UBYTE *name = s, *fold, *t, c, c1 = 0, c2 = 0, c3 = 0;
    int str1offset, withnolist = AC.NoShowInput;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( *s == '-' || *s == '+' ) {
        if ( *s == '-' ) withnolist = 1;
        else             withnolist = 0;
        s++;
        while ( *s == ' ' || *s == '\t' ) s++;
        name = s;
    }
    if ( *s == '"' ) {
        while ( *s && *s != '"' ) {
            if ( *s == '\\' ) s++;
            s++;
        }
        t = s++;
    }
    else {
        while ( *s && *s != ' ' && *s != '\t' ) {
            if ( *s == '\\' ) s++;
            s++;
        }
        t = s;
    }
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == '#' ) {
        *t = 0;
        s++;
        while ( *s == ' ' || *s == '\t' ) s++;
        fold = s;
        if ( *s == 0 ) {
            MesPrint("@Empty fold name");
            return(-1);
        }
continue_fold:
        while ( *s && *s != ' ' && *s != '\t' ) {
            if ( *s == '\\' ) s++;
            s++;
```

```
        }
        t = s;
        while ( *s == ' ' || *s == '\t' ) s++;
        if ( *s ) {
            /*
             * A non-whitespace character is found. Continue parsing the fold.
             */
            goto continue_fold;
        }
    }
    else if ( *s == 0 ) {
        fold = 0;
    }
    else {
        MesPrint("@Improper syntax for file name");
        return(-1);
    }
    *t = 0;
    if ( fold ) {
        fold = strDup1(fold,"foldname");
    }
/*
    We have the name of the file in 'name' and the fold in 'fold' (or NULL)
*/
    if ( OpenStream(name,type,0,PRENOACTION) == 0 ) {
        if ( fold ) { M_free(fold,"foldname"); fold = 0; }
        return(-1);
    }
    if ( fold ) {
        LONG position = -1;
        int foldopen = 0;
        LONG linenum = 0, prevline = 0;
        name = strDup1(name,"name of include file");
        AC.CurrentStream->FoldName = strDup1(fold,"name of fold");
        AC.NoShowInput++;
        for(;;) {
            c = GetFromStream(AC.CurrentStream);
            if ( c == ENDOFSTREAM ) {
                AC.CurrentStream = CloseStream(AC.CurrentStream);
                goto nofold;
            }
            if ( c == AP.ComChar ) {
                str1offset = AC.CurrentStream-AC.Streams;
                LoadInstruction(1);
                if ( AC.CurrentStream != str1offset+AC.Streams ) {
                    c = ENDOFSTREAM;
                }
                else {
                    t = AP.preStart;
                    if ( t[2] == '#' && ( ( t[3] == '[' && !foldopen )
                    || ( t[3] == ']' && foldopen ) ) ) {
                        t += 4;
                        while ( *t == ' ' || *t == '\t' ) t++;
                        s = AC.CurrentStream->FoldName;
                        while ( *s == *t ) { s++; t++; }
                        if ( *s == 0 && ( *t == ' ' || *t == '\t'
                        || *t == ':' ) ) {
                            while ( *t == ' ' || *t == '\t' ) t++;
                            if ( *t == ':' ) {
                                if ( foldopen == 0 ) {
                                    foldopen = 1;
                                    position = GetStreamPosition(AC.CurrentStream);
                                    linenum = AC.CurrentStream->linenumber;
                                    prevline = AC.CurrentStream->prevline;
                                    c3 = AC.CurrentStream->isnextchar;
                                    c1 = AC.CurrentStream->nextchar[0];
                                    c2 = AC.CurrentStream->nextchar[1];
                                }
                                else {
                                    foldopen = 0;
                                    PositionStream(AC.CurrentStream,position);
                                    AC.CurrentStream->linenumber = linenum;
                                    AC.CurrentStream->prevline = prevline;
                                    AC.CurrentStream->eqnum = 1;
                                    AC.NoShowInput--;
                                    AC.CurrentStream->isnextchar = c3;
                                    AC.CurrentStream->nextchar[0] = c1;
                                    AC.CurrentStream->nextchar[1] = c2;
                                    break;
                                }
                            }
                        }
                    }
                }
            }
            else {
                while ( c != LINEFEED && c != ENDOFSTREAM ) {
```

```
                    c = GetFromStream(AC.CurrentStream);
                    if ( c == ENDOFSTREAM ) {
                        AC.CurrentStream = CloseStream(AC.CurrentStream);
                        break;
                    }
                }
            }
            if ( c == ENDOFSTREAM ) {
nofold:
                MesPrint("@Cannot find fold %s in file %s",fold,name);
                UngetChar(c);
                AC.NoShowInput--;
                M_free(name,"name of include file");
                Terminate(-1);
            }
        }
        M_free(name,"name of include file");
    }
    AC.NoShowInput = withnolist;
    if ( fold ) { M_free(fold,"foldname"); fold = 0; }
    return(0);
}

/*
        #] Include :
        #[ DoPreExchange :

        Exchanges the names of expressions or the contents of dollars
        Syntax:
            #exchange expr1,expr2
            #exchange $var1,$var2
*/

int DoPreExchange(UBYTE *s)
{
    int error = 0;
    UBYTE *s1, *s2;
    WORD num1, num2;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( *s == '$' ) {
        s++; s1 = s; while ( FG.cTable[*s] <= 1 ) s++;
        if ( *s != ',' && *s != ' ' && *s != '\t' ) goto syntax;
        *s++ = 0;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
        if ( *s != '$' ) goto syntax;
        s++; s2 = s; while ( FG.cTable[*s] <= 1 ) s++;
        if ( *s != 0 && *s != ';' ) goto syntax;
        *s = 0;
        if ( ( num1 = GetDollar(s1) ) <= 0 ) {
            MesPrint("@$%s has not been defined (yet)",s1);
            error = 1;
        }
        if ( ( num2 = GetDollar(s2) ) <= 0 ) {
            MesPrint("@$%s has not been defined (yet)",s2);
            error = 1;
        }
        if ( error == 0 ) {
            ExchangeDollars((int)num1,(int)num2);
        }
    }
    else {
        s1 = s; s = SkipAName(s);
        if ( *s != ',' && *s != ' ' && *s != '\t' ) goto syntax;
        *s++ = 0;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
        if ( FG.cTable[*s] != 0 && *s != '[' ) goto syntax;
        s2 = s; s = SkipAName(s);
        if ( *s != 0 && *s != ';' ) goto syntax;
        *s = 0;
        if ( GetName(AC.exprnames,s1,&num1,NOAUTO) != CEXPRESSION ) {
            MesPrint("@%s is not an expression",s1);
            error = 1;
        }
        if ( GetName(AC.exprnames,s2,&num2,NOAUTO) != CEXPRESSION ) {
            MesPrint("@%s is not an expression",s2);
            error = 1;
        }
        if ( error == 0 ) {
            ExchangeExpressions((int)num1,(int)num2);
        }
    }
    return(error);
syntax:
    MesPrint("@Proper syntax: %#exchange expr1,expr2 or %#exchange $var1,$var2");
    return(1);
```

```
}

/*
 	#] DoPreExchange :
 	#[ DoCall :
*/

int DoCall(UBYTE *s)
{
	UBYTE *t, *u, *v, *name, c, cp, *args1, *args2, *t1, *t2, *wild = 0;
	int bratype = 0, wildargs = 0, inwildargs = 0, nwildargs = 0;
	PROCEDURE *p;
	int streamoffset;
	int i, namesize, narg1, narg2, bralevel, numpre;
	LONG i1, i2;
	if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
	if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
/*
	1:	Get the name of the procedure.
	2:	Locate the procedure.
*/
	name = s; s = EndOfToken(s); c = *s; *s = 0;
	for ( i = NumProcedures-1; i >= 0; i-- ) {
		if ( StrCmp(Procedures[i].name,name) == 0 ) break;
	}
	p = (PROCEDURE *)FromList(&AP.ProcList);
	if ( i < 0 ) {	/* Try to find a file */
		namesize = 0;
		t = name;
		while ( *t ) { t++; namesize++; }
		t = AP.procedureExtension;
		while ( *t ) { t++; namesize++; }
		t = p->name = (UBYTE *)Malloc1(namesize+2,"procedure");
		u = name;
		while ( *u ) *t++ = *u++;
		*t++ = '.';
		v = AP.procedureExtension;
		while ( *v ) *t++ = *v++;
		*t = 0;
		p->loadmode = 0;	/* buffer should be freed at end */
		p->p.buffer = LoadInputFile(p->name,PROCEDUREFILE);
		if ( p->p.buffer == 0 ) return(-1);
		t[-4] = 0;
	}
	else {
		p->p.buffer = Procedures[i].p.buffer;
		p->name = Procedures[i].name;
		p->loadmode = 1;
	}
	t = p->p.buffer;
	SKIPBLANKS(t)
	if ( *t++ != '#' ) goto wrongfile;
	SKIPBLANKS(t)
	t += 9;
	SKIPBLANKS(t)
	u = EndOfToken(t);
	cp = *u; *u = 0;
	if ( StrCmp(t,name) != 0 ) goto wrongfile;
	*u = cp;
	*s = c;
/*
	The pointer p points to the contents of the procedure (in memory)
	Now we have to match the arguments. u points to after the name
	in the 'file', s to after the name in the call statement.
*/
	bralevel = narg1 = narg2 = 0; args2 = u;
	SKIPBLANKS(u)
	if ( *u == '(' ) {
		u++; SKIPBLANKS(u)
		args2 = u;
		while ( *u != ')' ) {
			if ( *u == '?' ) { wildargs++; u++; nwildargs = narg2+1; }
			narg2++; u = EndOfToken(u); SKIPBLANKS(u)
			if ( *u == ',' ) { u++; SKIPBLANKS(u) }
			else if ( *u != ')' || ( wildargs > 1 ) ) {
				MesPrint("@Illegal argument field in procedure %s",p->name);
				return(-1);
			}
		}
	}
	while ( *u != LINEFEED ) u++;
	SKIPBLANKS(s)
	args1 = s+1;
	if ( *s == '(' ) bratype = 1;
	do {
		if ( *s == '{' && bratype == 0 ) bralevel++;
		else if ( *s == '(' && bratype == 1 ) bralevel++;
```

```
        else if ( *s == '}' && bratype == 0 ) {
            bralevel--;
            if ( bralevel == 0 ) {
                *s = 0; narg1++;
                if ( wildargs && narg1 == nwildargs ) wild = s;
            }
        }
        else if ( *s == ')' && bratype == 1 ) {
            bralevel--;
            if ( bralevel == 0 ) {
                *s = 0; narg1++;
                if ( wildargs && narg1 == nwildargs ) wild = s;
            }
        }
        /*[12dec2003 mt]:*/
        /*else if ( *s == ',' || *s == '|' ) {*/
        else if (set_in(*s,AC.separators)) {/*Function set_in see in
                                                    file tools.c*/
        /*:[12dec2003 mt]*/
            *s = 0; narg1++;
            if ( wildargs && narg1 == nwildargs ) wild = s;
        }
        else if ( *s == '\\' ) s++;
        s++;
    } while ( bralevel > 0 );
    if ( wildargs && narg1 >= narg2-1 ) {
        inwildargs = narg1-narg2+1;
        if ( inwildargs == 0 ) nwildargs = 0;
        else {
            while ( inwildargs > 1 ) {
                *wild = ',';
                while ( *wild ) wild++;
                inwildargs--;
            }
        }
    }
    else if ( narg1 != narg2 && ( narg2 != 0 || narg1 != 1 || *args1 != 0 ) ) {
        MesPrint("@Arguments of procedure %s are not matching",p->name);
        return(-1);
    }
    numpre = -NumPre-1; /* For the stream */
    for ( i = 0; i < narg2; i++ ) {
        t = args2;
        if ( *t == '?' ) {
            args2++;
        }
        if ( *t == '?' && inwildargs == 0 ) {
            args2 = EndOfToken(args2); c = *args2; *args2 = 0;
            if ( PutPreVar(t,(UBYTE *)"",0,0) < 0 ) return(-1);
        }
        else {
            args2 = EndOfToken(args2); c = *args2; *args2 = 0;
            t1 = t2 = args1;
            while ( *t1 ) {
                if ( *t1 == '\\' ) t1++;
                if ( t1 != t2 ) *t2 = *t1;
                t2++; t1++;
            }
            *t2 = 0;
            if ( PutPreVar(t,args1,0,0) < 0 ) return(-1);
            args1 = t1+1;                   /* Next argument */
        }
        *args2 = c; SKIPBLANKS(args2)  /* skip to next name */
        args2++; SKIPBLANKS(args2)
    }
    streamoffset = AC.CurrentStream - AC.Streams;
    args1 = AC.CurrentStream->name;
    AC.CurrentStream->name = p->name;
    i1 = AC.CurrentStream->linenumber;
    i2 = AC.CurrentStream->prevline;
    AC.CurrentStream->prevline   =
    AC.CurrentStream->linenumber = 2;
    OpenStream(u+1,PREREADSTREAM3,numpre,PRENOACTION);
    AC.Streams[streamoffset].name = args1;
    AC.Streams[streamoffset].linenumber = i1;
    AC.Streams[streamoffset].prevline   = i2;
    AddToPreTypes(PRETYPEPROCEDURE);
    return(0);
wrongfile:;
    if ( i < 0 ) MesPrint("@File %s is not a proper procedure",p->name);
    else MesPrint("!!!Internal error with procedure names: %s",name);
    return(-1);
}

/*
        #] DoCall :
        #[ DoDebug :
```

```
*/

int DoDebug(UBYTE *s)
{
    int x;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    NeedNumber(x,s,nonumber)
    if ( x < 0 || x >(PREPROONLY
                    | DUMPTOCOMPILER
                    | DUMPOUTTERMS
                    | DUMPINTERMS
                    | DUMPTOSORT
                    | DUMPTOPARALLEL
#ifdef WITHPTHREADS
                    | THREADSDEBUG
#endif
             ) ) goto nonumber;
    AP.PreDebug = 0;
    if ( ( x & PREPROONLY     ) != 0 ) AP.PreDebug |= PREPROONLY;     /* 1  */
    if ( ( x & DUMPTOCOMPILER ) != 0 ) AP.PreDebug |= DUMPTOCOMPILER; /* 2  */
    if ( ( x & DUMPOUTTERMS   ) != 0 ) AP.PreDebug |= DUMPOUTTERMS;   /* 4  */
    if ( ( x & DUMPINTERMS    ) != 0 ) AP.PreDebug |= DUMPINTERMS;    /* 8  */
    if ( ( x & DUMPTOSORT     ) != 0 ) AP.PreDebug |= DUMPTOSORT;     /* 16 */
    if ( ( x & DUMPTOPARALLEL ) != 0 ) AP.PreDebug |= DUMPTOPARALLEL; /* 32 */
#ifdef WITHPTHREADS
    if ( ( x & THREADSDEBUG   ) != 0 ) AP.PreDebug |= THREADSDEBUG;   /* 64 */
#endif
    return(0);
nonumber:
    MesPrint("@Illegal argument for debug instruction");
    return(1);
}

/*
        #] DoDebug :
        #[ DoTerminate :
*/

int DoTerminate(UBYTE *s)
{
    int x;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( *s ) {
        NeedNumber(x,s,nonumber)
        Terminate(x);
    }
    else {
        Terminate(-1);
    }
    return(0);
nonumber:
    MesPrint("@Illegal argument for terminate instruction");
    return(1);
}

/*
        #] DoTerminate :
        #[ DoContinueDo :
*/

int DoContinueDo(UBYTE *s)
{
    DOLOOP *loop;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);

    if ( NumDoLoops <= 0 ) {
        MesPrint("@%#continuedo without %#do");
        return(1);
    }

    loop = &(DoLoops[NumDoLoops-1]);
    AP.NumPreTypes = loop->NumPreTypes+1;
    AP.PreIfLevel = loop->PreIfLevel;
    AP.PreSwitchLevel = loop->PreSwitchLevel;

    return(DoEnddo(s));
}

/*
        #] DoContinueDo :
        #[ DoDo :

        The do loop has three varieties:
```

```
        #do i = num1,num2 [,num3]
        #do i = {string1,string2,....,stringn}
            The | as separator is also allowed for backwards compatibility
        #do i = expression      One by one all terms of the expression
*/

int DoDo(UBYTE *s)
{
    GETIDENTITY
    UBYTE *t, c, *u, *uu;
    DOLOOP *loop;
    WORD expnum;
    LONG linenum  = AC.CurrentStream->linenumber;
    int oldNoShowInput = AC.NoShowInput, i, oldpreassignflag;

    if ( ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH )
    || ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ) {
        if ( PreSkip((UBYTE *)"do",(UBYTE *)"enddo",1) ) return(-1);
        return(0);
    }

/*
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
*/
    AddToPreTypes(PRETYPEDO);

    loop = (DOLOOP *)FromList(&AP.LoopList);
    loop->firstdollar = loop->lastdollar = loop->incdollar = -1;
    loop->NumPreTypes = AP.NumPreTypes-1;
    loop->PreIfLevel = AP.PreIfLevel;
    loop->PreSwitchLevel = AP.PreSwitchLevel;
    AC.NoShowInput = 1;
    if ( PreLoad(&(loop->p),(UBYTE *)"do",(UBYTE *)"enddo",1,"doloop") ) return(-1);
    AC.NoShowInput = oldNoShowInput;
    loop->NoShowInput = AC.NoShowInput;
/*
    Get now the name. We have to take great care when the name is terminated!
*/
    s = loop->p.buffer + (s - AP.preStart);
    SKIPBLANKS(s)
    loop->name = s;
    if ( chartype[*s] != 0 ) goto illname;
    s++;
    while ( chartype[*s] <= 1 ) s++;
    t = s;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s != '=' ) goto illdo;
    s++;
    while ( *s == ' ' || *s == '\t' ) s++;
    *t = 0;

    if ( *s == '{' ) {
        loop->type = LISTEDLOOP;
        s++; loop->vars = s;
        loop->lastnum = 0;
        while ( *s != '}' && *s != 0 ) {
            if ( set_in(*s,AC.separators) ) { *s = 0; loop->lastnum++; }
            else if ( *s == '\\' ) s++;
            s++;
        }
        if ( *s == 0 ) goto illdo;
        *s++ = 0;
        loop->lastnum++;
        loop->firstnum = 0;
        loop->contents = s;
    }
    else if ( *s == '-' || *s == '+' || chartype[*s] == 1 || *s == '$' ) {
        loop->type = NUMERICALLOOP;
        t = s;
        while ( *s && *s != ',' ) s++;
        if ( *s == 0 ) goto illdo;
        if ( *t == '$' ) {
            c = *s; *s = 0;
            if ( GetName(AC.dollarnames,t+1,&loop->firstdollar,NOAUTO) != CDOLLAR ) {
                MesPrint("@%s is undefined in first parameter in %#do instruction",t);
                return(-1);
            }
            loop->firstnum = DolToLong(BHEAD loop->firstdollar);
            if ( AN.ErrorInDollar ) {
                MesPrint("@%s does not evaluate into a valid loop parameter",t);
                return(-1);
            }
            *s++ = c;
        }
        else {
            *s = '}';
```

```
            if ( PreEval(t,&loop->firstnum) == 0 ) goto illdo;
            *s++ = ',';
        }
        t = s;
        while ( *s && *s != ',' && *s != ';' && *s != LINEFEED ) s++;
        c = *s;
        if ( *t == '$' ) {
            *s = 0;
            if ( GetName(AC.dollarnames,t+1,&loop->lastdollar,NOAUTO) != CDOLLAR ) {
                MesPrint("@%s is undefined in second parameter in %#do instruction",t);
                return(-1);
            }
            loop->lastnum = DolToLong(BHEAD loop->lastdollar);
            if ( AN.ErrorInDollar ) {
                MesPrint("@%s does not evaluate into a valid loop parameter",t);
                return(-1);
            }
            *s++ = c;
        }
        else {
            *s = '}';
            if ( PreEval(t,&loop->lastnum) == 0 ) goto illdo;
            *s++ = c;
        }
        if ( c == ',' ) {
            t = s;
            while ( *s && *s != ';' && *s != LINEFEED ) s++;
            if ( *t == '$' ) {
                c = *s; *s = 0;
                if ( GetName(AC.dollarnames,t+1,&loop->incdollar,NOAUTO) != CDOLLAR ) {
                    MesPrint("@%s is undefined in third parameter in %#do instruction",t);
                    return(-1);
                }
                loop->incnum = DolToLong(BHEAD loop->incdollar);
                if ( AN.ErrorInDollar ) {
                    MesPrint("@%s does not evaluate into a valid loop parameter",t);
                    return(-1);
                }
                *s++ = c;
            }
            else {
                c = *s; *s = '}';
                if ( PreEval(t,&loop->incnum) == 0 ) goto illdo;
                *s++ = c;
            }
        }
        else loop->incnum = 1;
        loop->contents = s;
    }
    else if ( ( chartype[*s] == 0 ) || ( *s == '[' ) ) {
        int oldNumPotModdollars = NumPotModdollars;
#ifdef WITHMPI
        WORD oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
        AC.RhsExprInModuleFlag = 0;
#endif
        t = s;
        if ( ( s = SkipAName(s) ) == 0 ) goto illdo;
        c = *s; *s = 0;
        if ( GetName(AC.exprnames,t,&expnum,NOAUTO) == CEXPRESSION ) {
            loop->type = ONEEXPRESSION;
/*
            We should remember the expression by name for when it gets
            renumbered!!! If it gets deleted there will be a crash or at
            least the loop terminates.
*/
            loop->vars = t;
        }
        else goto illdo;
        if ( c == ',' || c == '\t' || c == ';' ) { s++; }
        else if ( c != 0 && c != '\n' ) goto illdo;
        while ( *s == ',' || *s == '\t' || *s == ';' ) s++;
        if ( *s != 0 && *s != '\n' ) goto illdo;
        loop->firstnum = 0;
        s++;
        loop->contents = s;
        loop->incnum = 0;
/*
        Next determine size of statement and allocate space
*/
        while ( *t ) t++;
        i = t - loop->vars;
        t = loop->name;
        while ( *t ) { t++; i++; }
        i += 4;
        loop->dollarname = Malloc1((LONG)i,"do-loop instruction");
/*
        Construct the statement
```

```
*/
        u = loop->dollarname;
        *u++ = '$'; t = loop->name; while ( *t ) *u++ = *t++;
        *u++ = '_'; uu = u; *u++ = '='; t = loop->vars;
        while ( *t ) *u++ = *t++;
        *t = 0; *u = 0;
/*
        Compile and put in dollar variable.
        Note that we remember the dollar by name and that this name ends in _
*/
        oldpreassignflag = AP.PreAssignFlag;
        AP.PreAssignFlag = 2;
        CompileStatement(loop->dollarname);
        if ( CatchDollar(0) ) {
            MesPrint("@Cannot load expression in do loop");
            return(-1);
        }
        AP.PreAssignFlag = oldpreassignflag;
        NumPotModdollars = oldNumPotModdollars;
#ifdef WITHMPI
        AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
#endif
        *uu = 0;
    }
    else goto illdo; /* Syntax problems */
    loop->errorsinloop = 0;
/*  loop->startlinenumber = linenum+1; 5-oct-2000 One too much? */
    loop->startlinenumber = linenum;
    PutPreVar(loop->name,(UBYTE *)"0",0,0);
    loop->firstloopcall = 1;
    return(DoEnddo(s));
illname:;
    MesPrint("@Improper name for do loop variable");
    return(-1);
illdo:;
    MesPrint("@Improper syntax in do loop instruction");
    return(-1);
}

/*
        #] DoDo :
        #[ DoBreakDo :

        #breakdo [num]
        jumps out of num #do-loops (if there are that many) (default is 1)
*/

int DoBreakDo(UBYTE *s)
{
    DOLOOP *loop;
    WORD levels;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);

    if ( NumDoLoops <= 0 ) {
        MesPrint("@%#breakdo without %#do");
        return(1);
    }
/*
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEDO ) { MessPreNesting(4); return(-1); }
*/
    while ( *s && ( *s == ',' || *s == ' ' || *s == '\t' ) ) s++;
    if ( *s == 0 ) {
        levels = 1;
    }
    else if ( FG.cTable[*s] == 1 ) {
        levels = 0;
        while ( *s >= '0' && *s <= '9' ) { levels = 10*levels + *s++ - '0'; }
        if ( *s != 0 ) goto improper;
    }
    else {
improper:
        MesPrint("@Improper syntax of %#breakdo instruction");
        return(1);
    }
    if ( levels > NumDoLoops ) {
        MesPrint("@Too many loop levels requested in %#breakdo instruction");
        Terminate(-1);
    }
    while ( levels > 0 ) {
        while ( AC.CurrentStream->type != PREREADSTREAM
          && AC.CurrentStream->type != PREREADSTREAM2
          && AC.CurrentStream->type != PREREADSTREAM3 ) {
            AC.CurrentStream = CloseStream(AC.CurrentStream);
        }
        while ( AP.PreTypes[AP.NumPreTypes] != PRETYPEDO
```

```
        && AP.PreTypes[AP.NumPreTypes] != PRETYPEPROCEDURE ) AP.NumPreTypes--;
        if ( AC.CurrentStream->type == PREREADSTREAM3
        || AP.PreTypes[AP.NumPreTypes] == PRETYPEPROCEDURE ) {
            MesPrint("@Trying to jump out of a procedure with a %#breakdo instruction");
            Terminate(-1);
        }
        loop = &(DoLoops[NumDoLoops-1]);
        AP.NumPreTypes = loop->NumPreTypes;
        AP.PreIfLevel = loop->PreIfLevel;
        AP.PreSwitchLevel = loop->PreSwitchLevel;
/*
        AP.NumPreTypes--;
*/
        NumDoLoops--;
        DoUndefine(loop->name);
        M_free(loop->p.buffer,"loop->p.buffer");
        loop->firstloopcall = 0;

        AC.CurrentStream = CloseStream(AC.CurrentStream);
        levels--;
    }
    return(0);
}

/*
        #] DoBreakDo :
        #[ DoElse :
*/

int DoElse(UBYTE *s)
{
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEIF ) {
        if ( AP.PreIfLevel <= 0 ) MesPrint("@%#else without corresponding %#if");
        else MessPreNesting(1);
        return(-1);
    }
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    while ( *s == ' ' ) s++;
    if ( tolower(*s) == 'i' && tolower(s[1]) == 'f' && s[2]
        && FG.cTable[s[2]] > 1 && s[2] != '_' ) {
        s += 2;
        while ( *s == ' ' ) s++;
        return(DoElseif(s));
    }
    if ( AP.PreIfLevel <= 0 ) {
        MesPrint("@%#else without corresponding %#if");
        return(-1);
    }
    switch ( AP.PreIfStack[AP.PreIfLevel] ) {
        case EXECUTINGIF:
            AP.PreIfStack[AP.PreIfLevel] = LOOKINGFORENDIF;
            break;
        case LOOKINGFORELSE:
            AP.PreIfStack[AP.PreIfLevel] = EXECUTINGIF;
            break;
        case LOOKINGFORENDIF:
            break;
    }
    return(0);
}

/*
        #] DoElse :
        #[ DoElseif :
*/

int DoElseif(UBYTE *s)
{
    int condition;
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEIF ) {
        if ( AP.PreIfLevel <= 0 ) MesPrint("@%#elseif without corresponding %#if");
        else MessPreNesting(2);
        return(-1);
    }
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfLevel <= 0 ) {
        MesPrint("@%#elseif without corresponding %#if");
        return(-1);
    }
    switch ( AP.PreIfStack[AP.PreIfLevel] ) {
        case EXECUTINGIF:
            AP.PreIfStack[AP.PreIfLevel] = LOOKINGFORENDIF;
            break;
        case LOOKINGFORELSE:
            if ( ( condition = EvalPreIf(s) ) < 0 ) return(-1);
            AP.PreIfStack[AP.PreIfLevel] = condition;
            break;
```

```
        case LOOKINGFORENDIF:
            break;
    }
    return(0);
}

/*
        #] DoElseif :
        #[ DoEnddo :

        At the first call there is no stream yet.
        After that we have to close the stream and start a new one.
*/

int DoEnddo(UBYTE *s)
{
    GETIDENTITY
    DOLOOP *loop;
    UBYTE *t, *tt, *value, numstr[16];
    LONG xval;
    int xsign, retval;
    DUMMYUSE(s);
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
/*
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ||
        AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) {
        if ( AP.PreTypes[AP.NumPreTypes] == PRETYPEDO ) AP.NumPreTypes--;
        else { MessPreNesting(3); return(-1); }
        return(0);
    }
*/
    if ( NumDoLoops <= 0 ) {
        MesPrint("@%#enddo without %#do");
        return(1);
    }
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEDO ) { MessPreNesting(4); return(-1); }
    loop = &(DoLoops[NumDoLoops-1]);
    if ( !loop->firstloopcall ) AC.CurrentStream = CloseStream(AC.CurrentStream);

    if ( loop->errorsinloop ) {
        MesPrint("++++Errors in Loop");
        goto finish;
    }
    if ( loop->type == LISTEDLOOP ) {
        if ( loop->firstnum >= loop->lastnum ) goto finish;
        loop->firstnum++;
        t = value = loop->vars;
        while ( *value ) value++;
        value++;
        loop->vars = value;
        value = tt = t;
        while ( *value ) {
            if ( *value == '\\' ) value++;
            *tt++ = *value++;
        }
        *tt = 0;
        PutPreVar(loop->name,t,0,1);    /* We overwrite the definition */
    }
    else if ( loop->type == NUMERICALLOOP ) {

        if ( !loop->firstloopcall ) {
/*
            Test whether the variable was changed inside the loop into
            a different numerical value. If so, adjust.
*/
            t = GetPreVar(loop->name,WITHOUTERROR);
            if ( t ) {
                value = t;
                xsign = 1;
                while ( *value && ( *value == ' '
                    || *value == '-' || *value == '+' ) ) {
                    if ( *value == '-' ) xsign = -xsign;
                    value++;
                }
                t = value; xval = 0;
                while ( *value >= '0' && *value <= '9' ) xval = 10*xval + *value++ - '0';
                while ( *value && *value == ' ' ) value++;
                if ( *value == 0 ) {
/*
                    Now we may substitute the loopvalue.
*/
                    if ( xsign < 0 ) xval = -xval;
                    if ( loop->incdollar >= 0 ) {
                        loop->incnum = DolToLong(BHEAD loop->incdollar);
                        if ( AN.ErrorInDollar ) {
                            MesPrint("@%s does not evaluate into a valid third loop
```

```
parameter",DOLLARNAME(Dollars,loop->incdollar));
                            return(-1);
                        }
                    }
                    loop->firstnum = xval + loop->incnum;
                }
            }
            if ( loop->lastdollar >= 0 ) {
                loop->lastnum = DolToLong(BHEAD loop->lastdollar);
                if ( AN.ErrorInDollar ) {
                    MesPrint("@%s does not evaluate into a valid second loop
parameter",DOLLARNAME(Dollars,loop->lastdollar));
                    return(-1);
                }
            }
        }
        if ( ( loop->incnum > 0 && loop->firstnum > loop->lastnum )
        || ( loop->incnum < 0 && loop->firstnum < loop->lastnum ) ) goto finish;
        NumToStr(numstr,loop->firstnum);
        t = numstr;
        loop->firstnum += loop->incnum;
        PutPreVar(loop->name,t,0,1);   /* We overwrite the definition */
    }
    else if ( loop->type == ONEEXPRESSION ) {
/*
        Find the dollar expression
*/
        WORD numdollar = GetDollar(loop->dollarname+1);
        DOLLARS d = Dollars + numdollar;
        WORD *w, *dw, v, *ww;
        if ( (d->where) == 0 ) {
            d->type = DOLUNDEFINED;
            M_free(loop->dollarname,"do-loop instruction");
            goto finish;
        }
        w = d->where + loop->incnum;
        if ( *w == 0 ) {
            M_free(d->where,"dollar");
            d->where = 0;
            d->type = DOLUNDEFINED;
            M_free(loop->dollarname,"do-loop instruction");
            goto finish;
        }
        loop->incnum += *w;
/*
        Now the term has to be converted to text.
*/
        ww = w + *w; v = *ww; *ww = 0;
        dw = d->where; d->where = w;
        t = WriteDollarToBuffer(numdollar,1);
        d->where = dw; *ww = v;
        PutPreVar(loop->name,t,0,1);   /* We overwrite the definition */
        M_free(t,"dollar");
    }
    if ( loop->firstloopcall ) OpenStream(loop->contents,PREREADSTREAM2,0,PRENOACTION);
    else OpenStream(loop->contents,PREREADSTREAM,0,PRENOACTION);
    AC.CurrentStream->prevline   =
    AC.CurrentStream->linenumber = loop->startlinenumber;
    AC.CurrentStream->eqnum = 0;
    loop->firstloopcall = 0;
    return(0);
finish:;
    NumDoLoops--;
    retval = DoUndefine(loop->name);
    M_free(loop->p.buffer,"loop->p.buffer");
    loop->firstloopcall = 0;
    AP.NumPreTypes--;
    return(retval);
}

/*
        #] DoEnddo :
        #[ DoEndif :
*/

int DoEndif(UBYTE *s)
{
    DUMMYUSE(s);
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEIF ) {
        if ( AP.PreIfLevel <= 0 ) MesPrint("@%#endif without corresponding %#if");
        else MessPreNesting(5);
        return(-1);
    }
    AP.NumPreTypes--;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfLevel <= 0 ) {
        MesPrint("@%#endif without corresponding %#if");
```

```
        return(-1);
    }
    AP.PreIfLevel--;
    return(0);
}

/*
        #] DoEndif : 
        #[ DoEndprocedure :

        Action is simple: close the current stream if it is still
        the stream from which the statement came.
        Then pop the current procedure and all its local derivatives.
        if loadmode > 1 the procedure was defined locally.
*/

int DoEndprocedure(UBYTE *s)
{
    DUMMYUSE(s);
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEPROCEDURE ) {
        MessPreNesting(6);
        return(-1);
    }
    AP.NumPreTypes--;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    AC.CurrentStream = CloseStream(AC.CurrentStream);
    do {
        NumProcedures--;
        if ( Procedures[NumProcedures].loadmode == 0 ) {
            M_free(Procedures[NumProcedures].p.buffer,"procedures buffer");
            M_free(Procedures[NumProcedures].name,"procedures name");
        }
    } while ( Procedures[NumProcedures].loadmode > 1 );
    return(0);
}

/*
        #] DoEndprocedure : 
        #[ DoIf :
*/

int DoIf(UBYTE *s)
{
    int condition;
    AddToPreTypes(PRETYPEIF);
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF ) {
        condition = EvalPreIf(s);
        if ( condition < 0 ) return(-1);
    }
    else condition = LOOKINGFORENDIF;
    if ( AP.PreIfLevel+1 >= AP.MaxPreIfLevel ) {
        int **ppp = &AP.PreIfStack; /* To avoid a compiler warning */
        if ( DoubleList((void ***)ppp,&AP.MaxPreIfLevel,sizeof(int),
            "PreIfLevels") ) return(-1);
    }
    AP.PreIfStack[++AP.PreIfLevel] = condition;
    return(0);
}

/*
        #] DoIf : 
        #[ DoIfdef :
*/

int DoIfdef(UBYTE *s, int par)
{
    int condition;
    AddToPreTypes(PRETYPEIF);
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] == EXECUTINGIF ) {
        while ( *s == ' ' || *s == '\t' ) s++;
        if ( ( *s == 0 ) == ( par == 1 ) ) condition = LOOKINGFORELSE;
        else                               condition = EXECUTINGIF;
    }
    else condition = LOOKINGFORENDIF;
    if ( AP.PreIfLevel+1 >= AP.MaxPreIfLevel ) {
        int **ppp = &AP.PreIfStack; /* to avoid a compiler warning */
        if ( DoubleList((void ***)ppp,&AP.MaxPreIfLevel,sizeof(int),
            "PreIfLevels") ) return(-1);
    }
    AP.PreIfStack[++AP.PreIfLevel] = condition;
    return(0);
}

/*
```

```
        #] DoIfdef :
        #[ DoIfydef :
*/

int DoIfydef(UBYTE *s)
{
    return DoIfdef(s,1);
}

/*
        #] DoIfydef :
        #[ DoIfndef :
*/

int DoIfndef(UBYTE *s)
{
    return DoIfdef(s,2);
}

/*
        #] DoIfndef :
        #[ DoInside :

    #inside $var1,...,$varn
        statements without .sort
    #endinside

    executes the statements on the contents of the $ variables as if they
    are a module. The results are put back in the dollar variables.
    To do this right we need a struct with
        old compiler buffer
        list of numbers of dollars
        length of the list
        length of the array containing the list
    Because we need to compose statements, the statement buffer must be
    empty. This means that we have to test for that. Same at the end. We
    must have a completed statement.
*/

int DoInside(UBYTE *s)
{
    GETIDENTITY
    int numdol, error = 0;
    WORD *nb, newsize, i;
    UBYTE *name, c;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.PreInsideLevel != 0 ) {
        MesPrint("@Illegal nesting of %#inside/%#endinside instructions");
        return(-1);
    }
/*
    if ( AP.PreContinuation ) {
        error = -1;
        MesPrint("@%#inside cannot be inside a regular statement");
    }
*/
    PUSHPREASSIGNLEVEL
/*
    Now the dollars to do
*/
    AP.inside.numdollars = 0;
    for(;;) {
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
        if ( *s == 0 ) break;
        if ( *s != '$' ) {
            MesPrint("@%#inside instruction can have only $ variables for parameters");
            return(-1);
        }
        s++;
        name = s;
        while (chartype[*s] <= 1 ) s++;
        c = *s; *s = 0;
        if ( ( numdol = GetDollar(name) ) < 0 ) {
            MesPrint("@%#inside: $%s has not (yet) been defined",name);
            *s = c;
            error = -1;
        }
        else {
            *s = c;
            if ( AP.inside.numdollars >= AP.inside.size ) {
                if ( AP.inside.buffer == 0 ) newsize = 20;
                else                         newsize = 2*AP.inside.size;
                nb = (WORD *)Malloc1(newsize*sizeof(WORD),"insidebuffer");
                if ( AP.inside.buffer ) {
                    for ( i = 0; i < AP.inside.size; i++ ) nb[i] = AP.inside.buffer[i];
                    M_free(AP.inside.buffer,"insidebuffer");
```

```
                }
                AP.inside.buffer = nb;
                AP.inside.size = newsize;
            }
            AP.inside.buffer[AP.inside.numdollars++] = numdol;
        }
    }
/*
    We have to store the configuration of the compiler buffer, so that
    we know where to start executing and how to reset the buffer.
*/
    AP.inside.oldcompiletype = AC.compiletype;
    AP.inside.oldparallelflag = AC.mparallelflag;
    AP.inside.oldnumpotmoddollars = NumPotModdollars;
    AP.inside.oldcbuf = AC.cbufnum;
    AP.inside.oldrbuf = AM.rbufnum;
    AP.inside.oldcnumlhs = AR.Cnumlhs,
    AddToPreTypes(PRETYPEINSIDE);
    AP.PreInsideLevel = 1;
    AC.cbufnum = AP.inside.inscbuf;
    AM.rbufnum = AP.inside.inscbuf;
    clearcbuf(AC.cbufnum);
    AC.compiletype = 0;
    AC.mparallelflag = PARALLELFLAG;
#ifdef WITHMPI
    /*
     * We use AC.RhsExprInModuleFlag, PotModdollars, and AC.pfirstnum
     * in order to check (1) whether there are expression names in RHS,
     * (2) which dollar variables can be modified, and (3) which
     * preprocessor variables can be redefined, in #inside.
     * We store the current values of them, and then reset them.
     */
    PF_StoreInsideInfo();
    AC.RhsExprInModuleFlag = 0;
    NumPotModdollars = 0;
    AC.numpfirstnum = 0;
#endif
    return(error);
}

/*
        #] DoInside :
        #[ DoEndInside :
*/

int DoEndInside(UBYTE *s)
{
    GETIDENTITY
    WORD numdol, *oldworkpointer = AT.WorkPointer, *term, *t, j, i;
    DOLLARS d, nd;
    WORD oldbracketon = AR.BracketOn;
    WORD *oldcompresspointer = AR.CompressPointer;
    int oldmultithreaded = AS.MultiThreaded;
    /* int oldmparallelflag = AC.mparallelflag; */
    FILEHANDLE *f;
#ifdef WITHMPI
    int error = 0;
#endif
    DUMMYUSE(s);
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPEINSIDE ) {
        if ( AP.PreInsideLevel != 1 ) MesPrint("@%#endinside without corresponding %#inside");
        else MessPreNesting(11);
        return(-1);
    }
    AP.NumPreTypes--;
    if ( AP.PreInsideLevel != 1 ) {
        MesPrint("@%#endinside without corresponding %#inside");
        return(-1);
    }
    if ( AP.PreContinuation ) {
        MesPrint("@%#endinside: previous statement not terminated.");
        Terminate(-1);
    }
    AC.compiletype = AP.inside.oldcompiletype;
    AR.Cnumlhs = cbuf[AM.rbufnum].numlhs;
#ifdef WITHMPI
    /*
     * If the #inside...#endinside contains expressions in RHS, only the master executes it
     * and then broadcasts the result to the all slaves. If not, the all processes execute
     * it and in this case no MPI interactions are needed.
     */
    if ( PF.me == MASTER || !AC.RhsExprInModuleFlag ) {
#endif
    AR.BracketOn = 0;
    AS.MultiThreaded = 0;
```

```
    /* AC.mparallelflag = PARALLELFLAG; */
    if ( AR.CompressPointer == 0 ) AR.CompressPointer = AR.CompressBuffer;
    f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
/*
    Now we have to execute the statements on the proper dollars.
*/
    for ( i = 0; i < AP.inside.numdollars; i++ ) {
        numdol = AP.inside.buffer[i];
        nd = d = Dollars + numdol;
        if ( d->type != DOLZERO ) {
            if ( d->type != DOLTERMS ) nd = DolToTerms(BHEAD numdol);
            term = nd->where;
            NewSort(BHEAD0);
            NewSort(BHEAD0);
            AR.MaxDum = AM.IndDum;
            while ( *term ) {
                t = oldworkpointer; j = *term;
                NCOPY(t,term,j);
                AT.WorkPointer = t;
                AN.IndDum = AM.IndDum;
                AR.CurDum = ReNumber(BHEAD term);
                if ( Generator(BHEAD oldworkpointer,0) ) {
                    MesPrint("@Called from %#endinside");
                    MesPrint("@Evaluating variable $%s",DOLLARNAME(Dollars,numdol));
                    Terminate(-1);
                }
            }
            AT.WorkPointer = oldworkpointer;
            CleanDollarFactors(d);
            if ( d->where ) { M_free(d->where,"dollar contents"); d->where = 0; }
            EndSort(BHEAD (WORD *)((void *)(&(d->where))),2);
            LowerSortLevel();
            term = d->where; while ( *term ) term += *term;
            d->size = term - d->where;
            if ( nd != d ) M_free(nd,"Copy of dollar variable");
            if ( d->where[0] == 0 ) {
                M_free(d->where,"dollar contents"); d->where = 0;
                d->type = DOLZERO;
            }
        }
    }
#ifdef WITHMPI
    }
    if ( AC.RhsExprInModuleFlag ) {
        /*
         * The only master executed the statements in #inside.
         * We need to broadcast the result to the all slaves.
         */
        for ( i = 0; i < AP.inside.numdollars; i++ ) {
            /*
             * Mark $-variables specified in the #inside instruction as modified
             * such that they will be broadcast.
             */
            AddPotModdollar(AP.inside.buffer[i]);
        }
        /* Now actual broadcast of modified variables. */
        if ( NumPotModdollars > 0 ) {
            error = PF_BroadcastModifiedDollars();
            if ( error ) goto cleanup;
        }
        if ( AC.numpfirstnum > 0 ) {
            error = PF_BroadcastRedefinedPreVars();
            if ( error ) goto cleanup;
        }
    }
cleanup:
#endif
    f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
    AC.cbufnum = AP.inside.oldcbuf;
    AM.rbufnum = AP.inside.oldrbuf;
    AR.Cnumlhs = AP.inside.oldcnumlhs;
    AR.BracketOn = oldbracketon;
    AP.PreInsideLevel = 0;
    AR.CompressPointer = oldcompresspointer;
    AS.MultiThreaded = oldmultithreaded;
    AC.mparallelflag = AP.inside.oldparallelflag;
    NumPotModdollars = AP.inside.oldnumpotmoddollars;
    POPPREASSIGNLEVEL
#ifdef WITHMPI
    PF_RestoreInsideInfo();
    if ( error ) return error;
#endif
    return(0);
}

/*
        #] DoEndInside :
```

```
        #[ DoMessage :
*/

int DoMessage(UBYTE *s)
{
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    MesPrint("~~~%s",s);
    return(0);
}

/*
        #] DoMessage :
        #[ DoPipe :
*/

int DoPipe(UBYTE *s)
{
#ifndef WITHPIPE
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
#ifdef WITHPIPE
    FLUSHCONSOLE;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( OpenStream(s,PIPESTREAM,0,PRENOACTION) == 0 ) return(-1);
    return(0);
#else
    Error0("Pipes not implemented on this computer/system");
    return(-1);
#endif
}

/*
        #] DoPipe :
        #[ DoPrcExtension :
*/

int DoPrcExtension(UBYTE *s)
{
    UBYTE *t, *u, c;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == 0 || *s == '\n' ) {
        MesPrint("@No valid procedure extension specified");
        return(-1);
    }
    if ( FG.cTable[*s] != 0 ) {
        MesPrint("@Procedure extension should be a string starting with an alphabetic character. No whitespace.");
        return(-1);
    }
    t = s;
    while ( *s && *s != '\n' && *s != ' ' && *s != '\t' ) s++;
    u = s;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s != 0 && *s != '\n' ) {
        MesPrint("@Too many parameters in ProcedureExtension instruction");
        return(-1);
    }
    c = *u; *u = 0;
    if ( AP.procedureExtension ) M_free(AP.procedureExtension,"ProcedureExtension");
    AP.procedureExtension = strDup1(t,"ProcedureExtension");
    *u = c;
    return(0);
}

/*
        #] DoPrcExtension :
        #[ DoPreOut :
*/

int DoPreOut(UBYTE *s)
{
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( tolower(*s) == 'o' ) {
        if ( tolower(s[1]) == 'n' && s[2] == 0 ) {
            AP.PreOut = 1;
            return(0);
        }
        if ( tolower(s[1]) == 'f' && tolower(s[2]) == 'f' && s[3] == 0 ) {
            AP.PreOut = 0;
            return(0);
```

```
        }
    }
    MesPrint("@Illegal option in PreOut instruction");
    return(-1);
}

/*
        #] DoPreOut :
        #[ DoPrePrintTimes :
*/

int DoPrePrintTimes(UBYTE *s)
{
    DUMMYUSE(s);
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    PrintRunningTime();
    return(0);
}

/*
        #] DoPrePrintTimes :
        #[ DoPreSortReallocate :
*/

int DoPreSortReallocate(UBYTE *s)
{
    DUMMYUSE(s);
    if ( AC.SortReallocateFlag == 0 ) {
        /* Currently off, so set to 2. Then the reallocation code knows the flag was
           set here, since "On sortreallocate;" sets it to 1. */
        AC.SortReallocateFlag = 2;
    }
    /* If the flag is already on, do nothing. */
    return(0);
}

/*
        #] DoPreSortReallocate :
        #[ DoPreAppend :

        Syntax:
        #append <filename>
*/

int DoPreAppend(UBYTE *s)
{
    UBYTE *name, *to;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
/*
    Determine where to write
*/
    if ( *s == '<' ) {
        s++;
        name = to = s;
        while ( *s && *s != '>' ) {
            if ( *s == '\\' ) s++;
            *to++ = *s++;
        }
        if ( *s == 0 ) {
            MesPrint("@Improper termination of filename");
            return(-1);
        }
        s++;
        *to = 0;
        if ( *name ) { GetAppendChannel((char *)name); }
        else goto improper;
    }
    else {
improper:
        MesPrint("@Proper syntax is: %#append <filename>");
        return(-1);
    }
    return(0);
}

/*
        #] DoPreAppend :
        #[ DoPreCreate :

        Syntax:
        #create <filename>
*/
```

```
int DoPreCreate(UBYTE *s)
{
    UBYTE *name, *to;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
/*
    Determine where to write
*/
    if ( *s == '<' ) {
        s++;
        name = to = s;
        while ( *s && *s != '>' ) {
            if ( *s == '\\' ) s++;
            *to++ = *s++;
        }
        if ( *s == 0 ) {
            MesPrint("@Improper termination of filename");
            return(-1);
        }
        s++;
        *to = 0;
        if ( *name ) { GetChannel((char *)name,0); }
        else goto improper;
    }
    else {
improper:
        MesPrint("@Proper syntax is: %#create <filename>");
        return(-1);
    }
    return(0);
}

/*
        #] DoPreCreate :
        #[ DoPreRemove :
*/

int DoPreRemove(UBYTE *s)
{
    UBYTE *name, *to;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == '<' ) { s++; }
    else {
        MesPrint("@Proper syntax is: %#remove <filename>");
        return(-1);
    }
    name = to = s;
    while ( *s && *s != '>' ) {
        if ( *s == '\\' ) s++;
        *to++ = *s++;
    }
    if ( *s == 0 ) {
        MesPrint("@Improper filename");
        return(-1);
    }
    s++;
    *to = 0;
    CloseChannel((char *)name);
    remove((char *)name);
    return(0);
}

/*
        #] DoPreRemove :
        #[ DoPreClose :
*/

int DoPreClose(UBYTE *s)
{
    UBYTE *name, *to;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == '<' ) { s++; }
    else {
        MesPrint("@Proper syntax is: %#close <filename>");
        return(-1);
    }
    name = to = s;
```

```
    while ( *s && *s != '>' ) {
        if ( *s == '\\' ) s++;
        *to++ = *s++;
    }
    if ( *s == 0 ) {
        MesPrint("@Improper filename");
        return(-1);
    }
    s++;
    *to = 0;
    return(CloseChannel((char *)name));
}

/*
        #] DoPreClose :
        #[ DoPreWrite :

        Syntax:
        #write [<filename>] "formatstring" [,objects]
        The format string can contain the following special objects/codes
        \n  newline
        \t  tab
        \!  if last entry in string: no linefeed at end
        \b  put \ in output
        %$  $-variable (to be found among the objects)
        %e  expression (name to be found among the objects)
        %E  expression without ; (name to be found among the objects)
        %s  string (to be found among the objects) (with or without "")
        %S  subterms (see PrintSubtermList)
*/

int DoPreWrite(UBYTE *s)
{
    HANDLERS h;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);

#ifdef WITHMPI
    if ( PF.me != MASTER ) return 0;
#endif

    h.oldsilent    = AM.silent;
    h.newlogonly   = h.oldlogonly = AM.FileOnlyFlag;
    h.newhandle    = h.oldhandle  = AC.LogHandle;
    h.oldprinttype = AO.PrintType;

    while ( *s == ' ' || *s == '\t' ) s++;
/*
    Determine where to write
*/
    if( (s=defineChannel(s,&h))==0 ) return(-1);

    return(writeToChannel(WRITEOUT,s,&h));
}

/*
        #] DoPreWrite :
        #[ DoProcedure :

        We have to read this procedure into a buffer.
        The only complications are:
        1: we have to seek through the file to do this efficiently
           the file operations under VMS cannot do this properly
           (unless we use the proper ANSI structs?)
           This is the reason why we read whole input files under VMS.
        2: what to do when the same name is used twice.
        Note that we have to do the reading without substitution of
        preprocessor variables.
*/

int DoProcedure(UBYTE *s)
{
    UBYTE c;
    PROCEDURE *p;
    LONG i;
    if ( ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH )
    || ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) ) {
        if ( PreSkip((UBYTE *)"procedure",(UBYTE *)"endprocedure",1) ) return(-1);
        return(0);
    }
    p = (PROCEDURE *)FromList(&AP.ProcList);
    if ( PreLoad(&(p->p),(UBYTE *)"procedure",(UBYTE *)"endprocedure"
        ,1,(char *)"procedure") ) return(-1);

    p->loadmode = 2;
```

```
    s = p->p.buffer + 10;
    while ( *s == ' ' || *s == LINEFEED ) s++;
    if ( chartype[*s] ) {
        MesPrint("@Illegal name for procedure");
        return(-1);
    }
    p->name = s++;
    while ( chartype[*s] == 0 || chartype[*s] == 1 ) s++;
    c = *s; *s = 0;
    p->name = strDup1(p->name,"procedure");
    *s = c;
/*
    Check for double names
*/
    for ( i = NumProcedures-2; i >= 0; i-- ) {
        if ( StrCmp(Procedures[i].name,p->name) == 0 ) {
            Error1("Multiple occurrence of procedure name ",p->name);
        }
    }
    return(0);
}

/*
        #] DoProcedure :
        #[ DoPreBreak :
*/

int DoPreBreak(UBYTE *s)
{
    DUMMYUSE(s);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {
        if ( AP.PreSwitchLevel <= 0 )
             MesPrint("@Break without corresponding Switch");
        else MessPreNesting(7);
        return(-1);
    }
    if ( AP.PreSwitchLevel <= 0 ) {
        MesPrint("@Break without corresponding Switch");
        return(-1);
    }
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] == EXECUTINGPRESWITCH )
        AP.PreSwitchModes[AP.PreSwitchLevel] = SEARCHINGPREENDSWITCH;
    return(0);
}

/*
        #] DoPreBreak :
        #[ DoPreCase :
*/

int DoPreCase(UBYTE *s)
{
    UBYTE *t;
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {
        if ( AP.PreSwitchLevel <= 0 )
             MesPrint("@Case without corresponding Switch");
        else MessPreNesting(8);
        return(-1);
    }
    if ( AP.PreSwitchLevel <= 0 ) {
        MesPrint("@Case without corresponding Switch");
        return(-1);
    }
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != SEARCHINGPRECASE ) return(0);

    SKIPBLANKS(s)
    t = s;
    while ( *s ) { if ( *s == '\\' ) s++; s++; }
    while ( s > t && ( s[-1] == ' ' || s[-1] == '\t' ) && s[-2] != '\\' ) {
        if ( s[-2] == '\\' ) s--;
        s--;
    }
    if ( *t == '"' && s > t+1 && s[-1] == '"' && s[-2] != '\\' ) {
        t++; s--; *s = 0;
    }
    else *s = 0;
    s = AP.PreSwitchStrings[AP.PreSwitchLevel];
    while ( *t == *s && *t ) { s++; t++; }
    if ( *t || *s ) return(0);  /* case did not match */
    AP.PreSwitchModes[AP.PreSwitchLevel] = EXECUTINGPRESWITCH;
    return(0);
}

/*
        #] DoPreCase :
```

```
        #[ DoPreDefault :
*/

int DoPreDefault(UBYTE *s)
{
    DUMMYUSE(s);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {
        if ( AP.PreSwitchLevel <= 0 )
             MesPrint("@Default without corresponding Switch");
        else MessPreNesting(9);
        return(-1);
    }
    if ( AP.PreSwitchLevel <= 0 ) {
        MesPrint("@Default without corresponding Switch");
        return(-1);
    }
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != SEARCHINGPRECASE ) return(0);
    AP.PreSwitchModes[AP.PreSwitchLevel] = EXECUTINGPRESWITCH;
    return(0);
}

/*
        #] DoPreDefault :
        #[ DoPreEndSwitch :
*/

int DoPreEndSwitch(UBYTE *s)
{
    DUMMYUSE(s);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.PreTypes[AP.NumPreTypes] != PRETYPESWITCH ) {
        if ( AP.PreSwitchLevel <= 0 )
             MesPrint("@EndSwitch without corresponding Switch");
        else MessPreNesting(10);
        return(-1);
    }
    AP.NumPreTypes--;
    if ( AP.PreSwitchLevel <= 0 ) {
        MesPrint("@EndSwitch without corresponding Switch");
        return(-1);
    }
    M_free(AP.PreSwitchStrings[AP.PreSwitchLevel--],"pre switch string");
    return(0);
}

/*
        #] DoPreEndSwitch :
        #[ DoPreSwitch :

        There should be a string after this.
        We have to store it somewhere.
*/

int DoPreSwitch(UBYTE *s)
{
    UBYTE *t, *switchstring, **newstrings;
    int newnum, i, *newmodes;
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    SKIPBLANKS(s)
    t = s;
    while ( *s ) { if ( *s == '\\' ) s++; s++; }
    while ( s > t && ( s[-1] == ' ' || s[-1] == '\t' ) && s[-2] != '\\' ) {
        if ( s[-2] == '\\' ) s--;
        s--;
    }
    if ( *t == '"' && s > t+1 && s[-1] == '"' && s[-2] != '\\' ) {
        t++; s--; *s = 0;
    }
    else *s = 0;
    switchstring = (UBYTE *)Malloc1((s-t)+1,"case string");
    s = switchstring;
    while ( *t ) {
        if ( *t == '\\' ) t++;
        *s++ = *t++;
    }
    *s = 0;
    if ( AP.PreSwitchLevel >= AP.NumPreSwitchStrings ) {
        newnum = 2*AP.NumPreSwitchStrings;
        newstrings = (UBYTE **)Malloc1(sizeof(UBYTE *)*(newnum+1),"case strings");
        newmodes   = (int *)Malloc1(sizeof(int)*(newnum+1),"case strings");
        for ( i = 0; i < AP.NumPreSwitchStrings; i++ )
            newstrings[i] = AP.PreSwitchStrings[i];
        M_free(AP.PreSwitchStrings,"AP.PreSwitchStrings");
        for ( i = 0; i <= AP.NumPreSwitchStrings; i++ )
            newmodes[i] = AP.PreSwitchModes[i];
        M_free(AP.PreSwitchModes,"AP.PreSwitchModes");
```

```
        AP.PreSwitchStrings = newstrings;
        AP.PreSwitchModes   = newmodes;
        AP.NumPreSwitchStrings = newnum;
    }
    AP.PreSwitchStrings[++AP.PreSwitchLevel] = switchstring;
    if ( ( AP.PreSwitchLevel > 1 )
     && ( AP.PreSwitchModes[AP.PreSwitchLevel-1] != EXECUTINGPRESWITCH ) )
        AP.PreSwitchModes[AP.PreSwitchLevel] = SEARCHINGPREENDSWITCH;
    else
        AP.PreSwitchModes[AP.PreSwitchLevel] = SEARCHINGPRECASE;
    AddToPreTypes(PRETYPESWITCH);
    return(0);
}

/*
        #] DoPreSwitch :
        #[ DoPreShow :

        Print the contents of the preprocessor variables
*/

int DoPreShow(UBYTE *s)
{
    int i;
    UBYTE *name, c;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == 0 ) {
        MesPrint("%#The preprocessor variables:");
        for ( i = 0; i < NumPre; i++ ) {
            MesPrint("%d: %s = \"%s\"",i,PreVar[i].name,PreVar[i].value);
        }
    }
    else {
        while ( *s ) {
            name = s; while ( *s && *s != ' ' && *s != '\t' && *s != ',' ) s++;
            c = *s; *s = 0;
            for ( i = 0; i < NumPre; i++ ) {
                if ( StrCmp(PreVar[i].name,name) == 0 )
                    MesPrint("%d: %s = \"%s\"",i,PreVar[i].name,PreVar[i].value);
            }
            *s = c;
            while ( *s == ' ' || *s == '\t' ) s++;
        }
    }
    return(0);
}

/*
        #] DoPreShow :
        #[ DoSystem :
*/

/*
 * A macro for translating the contents of `x' into a string after expanding.
 */
#define STRINGIFY(x)  STRINGIFY__(x)
#define STRINGIFY__(x) #x

int DoSystem(UBYTE *s)
{
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);
#ifdef WITHSYSTEM
    FLUSHCONSOLE;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == '-' && s[1] == 'e' ) {
        LONG err;
        UBYTE str[24];
        s += 2;
        if ( *s != ' ' ) {
            MesPrint("@Syntax error in #system command.");
            return(-1);
        }
        while ( *s == ' ' || *s == '\t' ) s++;
        err = system((char *)s);
        NumToStr(str,err);
        PutPreVar((UBYTE *)"SYSTEMERROR_",str,0,1);
    }
    else if ( system((char *)s) ) {
        MesPrint("@System call returned with error condition");
        Terminate(-1);
    }
    return(0);
#else
```

```
    Error0("External programs not implemented on this computer/system");
    return(-1);
#endif
}

/*
        #] DoSystem :
        #[ PreLoad :

        Loads a loop or procedure into a special buffer.
        Note: The current instruction is already in the preStart buffer
*/

int PreLoad(PRELOAD *p, UBYTE *start, UBYTE *stop, int mode, char *message)
{
    UBYTE *s, *t, *top, *newbuffer, c;
    LONG i, ppsize, linenum = AC.CurrentStream->linenumber;
    int size1, size2, level, com=0, last=1, strng = 0;
    p->size = AP.pSize;
    p->buffer = (UBYTE *)Malloc1(p->size+1,message);
    top = p->buffer + p->size - 2;
    t = p->buffer; *t++ = '#';
    s = start; size1 = size2 = 0;
    while ( *s ) { s++; size1++; }
    s = stop; while ( *s ) { s++; size2++; }
    s = AP.preStart; while ( *s ) *t++ = *s++; *t++ = LINEFEED;
    level = 1;
    i = 100;
    for (;;) {
        c = GetInput();
        if ( c == ENDOFINPUT ) {
            MesPrint("@Missing %#%s, Should match line %l",stop,linenum);
            return(-1);
        }
        if ( c == AP.ComChar && last == 1 ) com = 1;
        if ( c == LINEFEED ) { last = 1; com = 0; }
        else last = 0;

        if ( ( c == '"' ) && ( com == 0 ) ) { strng ^= 1; }

        if ( ( c == '#' ) && ( com == 0 ) ) i = 0;
        else i++;

        if ( t >= top ) {
            ppsize = t - p->buffer;
            p->size *= 2;
            newbuffer = (UBYTE *)Malloc1(p->size,message);
            t = newbuffer; s = p->buffer;
            while ( --ppsize >= 0 ) *t++ = *s++;
            M_free(p->buffer,"loading do loop");
            p->buffer = newbuffer;
            top = p->buffer + p->size - 2;
        }
        *t++ = c;
        if ( strng == 0 ) {
          if ( ( i == size2 ) && ( com == 0 ) ) {
            *t = 0;
            if ( StrICmp(t-size2,(UBYTE *)(stop)) == 0 ) {
                while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
                level--;
                if ( level <= 0 ) break;
                if ( c == ENDOFINPUT ) Error1("Missing #",stop);
                *t++ = LINEFEED; *t = 0; last = 1;
            }
          }
          if ( ( i == size1 ) && mode && ( com == 0 ) ) {
            *t = 0;
            if ( StrICmp(t-size1,(UBYTE *)(start)) == 0 ) {
/*
                while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
                if ( c == ENDOFINPUT ) Error1("Missing #",stop);
*/
                level++;
            }
          }
          if ( i == 1 && t[-2] == LINEFEED ) {
            if ( c == '-' )       AC.NoShowInput = 1;
            else if ( c == '+' ) AC.NoShowInput = 0;
          }
        }
    }
    *t++ = LINEFEED;
    *t = 0;
    return(0);
}

/*
```

```
        #] PreLoad :
        #[ PreSkip :

        Skips a loop or procedure.
        Note: The current instruction is already in the preStart buffer
*/

#define SKIPBUFSIZE 20

int PreSkip(UBYTE *start, UBYTE *stop, int mode)
{
    UBYTE *s, *t, buffer[SKIPBUFSIZE+2], c;
    LONG i, linenum = AC.CurrentStream->linenumber;
    int size1, size2, level, com=0, last=1;

    t = buffer; *t++ = '#';
    s = start; size1 = size2 = 0;
    while ( *s ) { s++; size1++; }
    s = stop; while ( *s ) { s++; size2++; }
    level = 1;
    i = 0;
    for (;;) {
        c = GetInput();
        if ( c == ENDOFINPUT ) {
            MesPrint("@Missing %#%s, Should match line %l",stop,linenum);
            return(-1);
        }
        if ( c == AP.ComChar && last == 1 ) com = 1;
        if ( c == LINEFEED ) { last = 1; com = 0; i = 0; t = buffer; }
        else last = 0;
        if ( ( c == '#' ) && ( com == 0 ) ) { i = 0; t = buffer; }
        else i++;

        if ( i < SKIPBUFSIZE ) *t++ = c;
        if ( ( i == size2 ) && ( com == 0 ) ) {
            *t = 0;
            if ( StrICmp(t-size2,(UBYTE *)(stop)) == 0 ) {
                while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
                level--;
                if ( level <= 0 ) {
                    pushbackchar = LINEFEED;
                    break;
                }
                if ( c == ENDOFINPUT ) Error1("Missing #",stop);
                i = 0; t = buffer;
            }
        }
        if ( ( i == size1 ) && mode && ( com == 0 ) ) {
            *t = 0;
            if ( StrICmp(t-size1,(UBYTE *)(start)) == 0 ) {
                while ( ( c = GetInput() ) != LINEFEED && c != ENDOFINPUT ) {}
                level++;
                i = 0; t = buffer;
            }
        }
    }
    return(0);
}

/*
        #] PreSkip :
        #[ StartPrepro :
*/

void StartPrepro(void)
{
    int **ppp;
    AP.MaxPreIfLevel = 2;
    ppp = &AP.PreIfStack;
    if ( DoubleList((void ***)ppp,&AP.MaxPreIfLevel,sizeof(int),
            "PreIfLevels") ) Terminate(-1);
    AP.PreIfLevel = 0; AP.PreIfStack[0] = EXECUTINGIF;

    AP.NumPreSwitchStrings = 10;
    AP.PreSwitchStrings = (UBYTE **)Malloc1(sizeof(UBYTE *)*
                                (AP.NumPreSwitchStrings+1),"case strings");
    AP.PreSwitchModes   = (int *)Malloc1(sizeof(int)*
                                (AP.NumPreSwitchStrings+1),"case strings");
    AP.PreSwitchModes[0] = EXECUTINGPRESWITCH;
    AP.PreSwitchLevel = 0;
}

/*
        #] StartPrepro :
        #[ EvalPreIf :

        Evaluates the condition in an if instruction.
```

```
        The return value is EXECUTINGIF if the condition is true.
        If it is false the returnvalue is LOOKINGFORELSE.
        An error gives a return value of -1
*/

int EvalPreIf(UBYTE *s)
{
    UBYTE *t, *u;
    int val;
    t = s;
    while ( *t ) t++;
    *t++ = ')';
    *t = 0;
    if ( ( u = PreIfEval(s,&val) ) == 0 ) return(-1);
    if ( u < t ) {
        MesPrint("@Unmatched parentheses in condition");
        return(-1);
    }
    if ( val ) return(EXECUTINGIF);
    else       return(LOOKINGFORELSE);
}

/*
        #] EvalPreIf :
        #[ PreIfEval :

        Used for recursions in the evaluation of a preprocessor if-condition.
        It determines whether the contents of () is true or false
        (or in error).
        The return value is the address of the first character after the
        closing parenthesis or null if there is an error.
        In value we find true(1) or false(0)
        We enter after the opening parenthesis.
        There are levels:
            0: orlevel:  a || b
            1: andlevel: a && b
            2: eqlevel:  a == b or a != b or a = b
            3: cmplevel: a > b or a >= b or a < b or a <= b or a >~ b etc
*/

UBYTE *PreIfEval(UBYTE *s, int *value)
{
    int orlevel = 0, andlevel = 0, eqlevel = 0, cmplevel = 0;
    int type, val;
    LONG val2;
    int ortype, orval, cmptype, cmpval, eqtype, eqval, andtype, andval;
    UBYTE *t, *eqt, *cmpt, c;
    int eqop, cmpop;
    ortype = orval = cmptype = cmpval = eqtype = eqval = andtype = andval = 0;
    eqop = cmpop = 0;
    eqt = cmpt = 0;
    *value = 0;
    while ( *s != ')' ) {
        while ( *s == ' ' || *s == '\t' || *s == '\n' || *s == '\r' ) s++;
        t = s;
        s = pParseObject(s,&type,&val2);
        if ( s == 0 ) return(0);
        val = val2;
        c = *s;
        *s++ = 0;     /* in case the object is a string without " */
        while ( c == ' ' || c == '\t' || c == '\n' || c == '\r' ) {
            c = *s; *s++ = 0;
        }
        if ( *t == '"' ) t++;
        switch(c) {
            case '|':
                if ( *s != '|' ) goto illoper;
                s++;
                /* fall through */
            case ')':
                if ( cmplevel ) {
                    if ( type == 0 || cmptype == 0 ) goto illobject;
                    val = PreCmp(type,val,t,cmptype,cmpval,cmpt,cmpop);
                    type = 0;
                    cmplevel = 0;
                }
                if ( eqlevel ) {
                    val = PreEq(type,val,t,eqtype,eqval,eqt,eqop);
                    type = 0;
                    eqlevel = 0;
                }
                if ( andlevel ) {
                    if ( andtype != 0 || type != 0 ) goto illobject;
                    val &= andval;
                    andlevel = 0;
                }
                if ( orlevel ) {
```

```
                    if ( ortype != 0 || type != 0 ) goto illobject;
                    val |= orval;
                }
                if ( c == ')' ) {
                    *value = val;
                    return(s);
                }
                orlevel = 1;
                orval = val;
                ortype = type;
                break;
            case '&':
                if ( *s != '&' ) goto illoper;
                s++;
                if ( cmplevel ) {
                    if ( type == 0 || cmptype == 0 ) goto illobject;
                    val = PreCmp(type,val,t,cmptype,cmpval,cmpt,cmpop);
                    type = 0;
                    cmplevel = 0;
                }
                if ( eqlevel ) {
                    val = PreEq(type,val,t,eqtype,eqval,eqt,eqop);
                    type = 0;
                    eqlevel = 0;
                }
                if ( andlevel ) {
                    if ( andtype != 0 || type != 0 ) goto illobject;
                    val &= andval;
                }
                andlevel = 1;
                andval = val;
                andtype = type;
                break;
            case '!':
            case '=':
                if ( eqlevel ) goto illorder;
                if ( cmplevel ) {
                    if ( type == 0 || cmptype == 0 ) goto illobject;
                    val = PreCmp(type,val,t,cmptype,cmpval,cmpt,cmpop);
                    type = 0;
                    cmplevel = 0;
                }
                if ( c == '!' && *s != '=' ) goto illoper;
                if ( *s == '=' ) s++;
                if ( c == '!' ) eqop = 1;
                else eqop = 0;
                eqlevel = 1; eqt = t; eqval = val; eqtype = type;
                break;
            case '>':
            case '<':
                if ( cmplevel ) goto illorder;
                if ( c == '<' ) cmpop = -1;
                else            cmpop = 1;
                cmplevel = 1; cmpt = t; cmpval = val; cmptype = type;
                if ( *s == '=' ) {
                    s++;
                    if ( *s == '~' ) { s++; cmpop *= 4; }
                    else cmpop *= 2;
                }
                else if ( *s == '~' ) { s++; cmpop *= 3; }
                break;
            default:
                goto illoper;
        }
    }
    return(s);
illorder:
    MesPrint("@illegal order of operators");
    return(0);
illobject:
    MesPrint("@illegal object for this operator");
    return(0);
illoper:
    MesPrint("@illegal operator");
    return(0);
}

/*
        #] PreIfEval :
        #[ PreCmp :
*/

int PreCmp(int type, int val, UBYTE *t, int type2, int val2, UBYTE *t2, int cmpop)
{
    if ( type == 2 || type2 == 2 || cmpop < -2 || cmpop > 2 ) {
        if ( cmpop < 0 && cmpop > -3 ) cmpop -= 2;
        if ( cmpop > 0 && cmpop <  3 ) cmpop += 2;
```

```
            if ( cmpop ==  3 ) val = StrCmp(t2,t) >  0;
        else if ( cmpop ==  4 ) val = StrCmp(t2,t) >= 0;
        else if ( cmpop == -3 ) val = StrCmp(t2,t) <  0;
        else if ( cmpop == -4 ) val = StrCmp(t2,t) <= 0;
    }
    else {
            if ( cmpop ==  1 ) val = ( val2 >  val );
        else if ( cmpop ==  2 ) val = ( val2 >= val );
        else if ( cmpop == -1 ) val = ( val2 <  val );
        else if ( cmpop == -2 ) val = ( val2 <= val );
    }
    return(val);
}

/*
        #] PreCmp :
        #[ PreEq :
*/

int PreEq(int type, int val, UBYTE *t, int type2, int val2, UBYTE *t2, int eqop)
{
    UBYTE str[20];
    if ( type == 2 || type2 == 2 ) {
        if ( type  != 2 ) { NumToStr(str,val ); t  = str; }
        if ( type2 != 2 ) { NumToStr(str,val2); t2 = str; }
        if ( eqop == 1 ) val = StrCmp(t,t2) != 0;
        else             val = StrCmp(t,t2) == 0;
    }
    else {
        if ( eqop ) val = val != val2;
        else        val = val == val2;
    }
    return(val);
}

/*
        #] PreEq :
        #[ pParseObject :

        Parses a preprocessor object. We can have:
        1: a number   (type = 1)
        2: a string   (type = 2)
        3: an expression between parentheses (type = 0)
        4: a special function (type = 3)
        If the object is not a number, an expression or a special operator
        we try to interpret it as a string.
*/

UBYTE *pParseObject(UBYTE *s, int *type, LONG *val2)
{
    UBYTE *t, c;
    int sign, val = 0;
    LONG x;
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == '(' ) {
        s++;
        while ( *s == ' ' || *s == '\t' || *s == '\n' || *s == '\r' ) s++;
        s = PreIfEval(s,&val);
        *type = 0;
        *val2 = val;
        return(s);
    }
    else if ( *s == '$' && s[1] == '(' ) {
        s += 2;
        while ( *s == ' ' || *s == '\t' || *s == '\n' || *s == '\r' ) s++;
        s = PreIfDollarEval(s,&val);
        *type = 0; *val2 = val;
        return(s);
    }
    if ( *s == 0 ) {
illend:
        MesPrint("@illegal end of condition");
        return(0);
    }
    if ( *s == '"' ) {
        s++;
        while ( *s && *s != '"' ) {
            if ( *s == '\\' ) s++;
            s++;
        }
        if ( *s == 0 ) goto illend;
        else *s = 0;
        *type = 2;
        s++;

        while ( *s == ' ' || *s == '\t' || *s == '\n' || *s == '\r' ) s++;

```

```
            return(s);
        }
        t = s; sign = 1; x = 0;
        if ( chartype[*t] == 0 ) {  /* Special operators and strings without "" */
            do { t++; } while ( chartype[*t] <= 1 );
            if ( *t == '(' ) {
                WORD ttype;
                c = *t; *t = 0;
                if ( StrICmp(s,(UBYTE *)"termsin") == 0 ) {
                    UBYTE *tt;
                    WORD numdol, numexp;
                    ttype = 0;
together:
                    *t++ = c;
                    while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
                    if ( *t == '$' ) {
                        t++; tt = t; while (chartype[*tt] <= 1 ) tt++;
                        c = *tt; *tt = 0;
                        if ( ( numdol = GetDollar(t) ) > 0 ) {
                            *tt = c;
                            if ( ttype == 1 ) {
                                x = SizeOfDollar(numdol);
                            }
                            else {
                                x = TermsInDollar(numdol);
                            }
                        }
                        else {
                            MesPrint("@$%s has not (yet) been defined",t);
                            *tt = c;
                            Terminate(-1);
                        }
                    }
                    else {
                        tt = SkipAName(t);
                        c = *tt; *tt = 0;
                        if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) {
                            MesPrint("@%s has not (yet) been defined",t);
                            *tt = c;
                            Terminate(-1);
                        }
                        else {
                            *tt = c;
                            if ( ttype == 1 ) {
                                x = SizeOfExpression(numexp);
                            }
                            else {
                                x = TermsInExpression(numexp);
                            }
                        }
                    }
                    while ( *tt == ' ' || *tt == '\t'
                          || *tt == '\n' || *tt == '\r' ) tt++;
                    if ( *tt != ')' ) {
                        MesPrint("@Improper use of terms($var) or terms(expr)");
                        Terminate(-1);
                    }
                    *type = 3;
                    s = tt+1;
                    *val2 = x;
                    return(s);
                }
                else if ( StrICmp(s,(UBYTE *)"sizeof") == 0 ) {
                    ttype = 1;
                    goto together;
                }
                else if ( StrICmp(s,(UBYTE *)"exists") == 0 ) {
                    UBYTE *tt;
                    WORD numdol, numexp;
                    *t++ = c;
                    while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
                    if ( *t == '$' ) {
                        t++; tt = t; while (chartype[*tt] <= 1 ) tt++;
                        c = *tt; *tt = 0;
                        if ( ( numdol = GetDollar(t) ) >= 0 ) { x = 1; }
                        else                                  { x = 0; }
                        *tt = c;
                    }
                    else if ( *t == '"' ) { /* see whether a file exists */
/*                      UBYTE *name, *oldname; */
                        t++; tt = t;
                        for (;;) {
                            if ( *tt == '\\' ) tt++;
                            else if ( *tt == '"' ) break;
                            tt++;
                        }
                        c = *tt; *tt = 0;
```

```
/*
                    Try to open the file. If possible, return 1.
                    Afterwards close it.
                    We do have to run through the FORMPATH. Hence we use LocateFile.
                    This routine may change the name to the full name.

                    oldname = name = strDup1(t,"name in exists");
                    x = LocateFile(&name,-1);
*/
                    x = OpenFile((char *)t);
                    if ( x >= 0 ) {
                        CloseFile(x);
                        x = 1;
/*
                        if ( name != oldname ) M_free(name,"name from LocateFile");
*/
                    }
                    else x = 0;
/*
                    M_free(oldname,"name in exists");
*/
                    *tt++ = c;
                }
                else {
                    tt = SkipAName(t);
                    c = *tt; *tt = 0;
                    if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) { x = 0; }
                    else { x = 1; }
                    *tt = c;
                }
                while ( *tt == ' ' || *tt == '\t'
                        || *tt == '\n' || *tt == '\r' ) tt++;
                if ( *tt != ')' ) {
                    MesPrint("@Improper use of exists($var) or exists(expr)");
                    Terminate(-1);
                }
                *type = 3;
                s = tt+1;
                *val2 = x;
                return(s);
            }
            else if ( StrICmp(s,(UBYTE *)"isnumerical") == 0 ) {
                GETIDENTITY
                UBYTE *tt;
                WORD numdol, numexp;
                *t++ = c;
                while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
                if ( *t == '$' ) {
                    t++; tt = t; while (chartype[*tt] <= 1 ) tt++;
                    c = *tt; *tt = 0;
                    if ( ( numdol = GetDollar(t) ) < 0 ) {
                        MesPrint("@$ variable in isnumerical(%s) does not exist",t);
                        Terminate(-1);
                    }
                    x = DolToLong(BHEAD numdol);
                    if ( AN.ErrorInDollar ) {
                        DOLLARS d = Dollars + numdol;
                        x = 0;
                        if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
                            if ( d->where[0] == 0 ) x = 1;
                            else if ( d->where[d->where[0]] == 0 ) {
                                if ( ABS(d->where[d->where[0]-1]) == d->where[0]-1 )
                                    x = 1;
                            }
                        }
                    }
                    else x = 1;
                    *tt = c;
                }
                else {
                    tt = SkipAName(t);
                    c = *tt; *tt = 0;
                    if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) {
                        MesPrint("@expression in isnumerical(%s) does not exist",t);
                        Terminate(-1);
                    }
                    x = TermsInExpression(numexp);
                    if ( x != 1 ) x = 0;
                    else {
                        WORD *term = AT.WorkPointer;
                        if ( GetFirstTerm(term,numexp,1) < 0 ) {
                            MesPrint("@error reading expression in isnumerical(%s)",t);
                            Terminate(-1);
                        }
                        if ( *term == ABS(term[*term-1])+1 ) x = 1;
                        else                                  x = 0;
                    }
```

```
                    *tt = c;
                }
                while ( *tt == ' ' || *tt == '\t'
                        || *tt == '\n' || *tt == '\r' ) tt++;
                if ( *tt != ')' ) {
                    MesPrint("@Improper use of isnumerical($var) or numerical(expr)");
                    Terminate(-1);
                }
                *type = 3;
                s = tt+1;
                *val2 = x;
                return(s);
            }
            else if ( StrICmp(s,(UBYTE *)("maxpowerof")) == 0 ) {
                UBYTE *tt;
                WORD numsym;
                int stype;
                *t++ = c;
                while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
                tt = SkipAName(t);
                c = *tt; *tt = 0;
                if ( ( stype = GetName(AC.varnames,t,&numsym,NOAUTO) ) == NAMENOTFOUND ) {
                    MesPrint("@%s has not (yet) been defined",t);
                    *tt = c;
                    Terminate(-1);
                }
                else if ( stype != CSYMBOL ) {
                    MesPrint("@%s should be a symbol",t);
                    *tt = c;
                    Terminate(-1);
                }
                else {
                    *tt = c;
                    x = symbols[numsym].maxpower;
                }
                while ( *tt == ' ' || *tt == '\t'
                        || *tt == '\n' || *tt == '\r' ) tt++;
                if ( *tt != ')' ) {
                    MesPrint("@Improper use of maxpowerof(symbol)");
                    Terminate(-1);
                }
                *type = 3;
                s = tt+1;
                *val2 = x;
                return(s);
            }
            else if ( StrICmp(s,(UBYTE *)("minpowerof")) == 0 ) {
                UBYTE *tt;
                WORD numsym;
                int stype;
                *t++ = c;
                while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
                tt = SkipAName(t);
                c = *tt; *tt = 0;
                if ( ( stype = GetName(AC.varnames,t,&numsym,NOAUTO) ) == NAMENOTFOUND ) {
                    MesPrint("@%s has not (yet) been defined",t);
                    *tt = c;
                    Terminate(-1);
                }
                else if ( stype != CSYMBOL ) {
                    MesPrint("@%s should be a symbol",t);
                    *tt = c;
                    Terminate(-1);
                }
                else {
                    *tt = c;
                    x = symbols[numsym].minpower;
                }
                while ( *tt == ' ' || *tt == '\t'
                        || *tt == '\n' || *tt == '\r' ) tt++;
                if ( *tt != ')' ) {
                    MesPrint("@Improper use of minpowerof(symbol)");
                    Terminate(-1);
                }
                *type = 3;
                s = tt+1;
                *val2 = x;
                return(s);
            }
            else if ( StrICmp(s,(UBYTE *)"isfactorized") == 0 ) {
                UBYTE *tt;
                WORD numdol, numexp;
                *t++ = c;
                while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
                if ( *t == '$' ) {
                    t++; tt = t; while (chartype[*tt] <= 1 ) tt++;
                    c = *tt; *tt = 0;
```

```
                        if ( ( numdol = GetDollar(t) ) > 0 ) {
                            if ( Dollars[numdol].factors != 0 ) x = 1;
                            else x = 0;
                        }
                        else {
                            MesPrint("@ %s should be the name of an expression or a $ variable",t-1);
                            Terminate(-1);
                        }
                        *tt = c;
                    }
                    else {
                        tt = SkipAName(t);
                        c = *tt; *tt = 0;
                        if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) {
                            MesPrint("@ %s should be the name of an expression or a $ variable",t);
                            Terminate(-1);
                        }
                        else {
                            if ( ( Expressions[numexp].vflags & ISFACTORIZED ) != 0 ) x = 1;
                            else x = 0;
                        }
                        *tt = c;
                    }
                    while ( *tt == ' ' || *tt == '\t'
                           || *tt == '\n' || *tt == '\r' ) tt++;
                    if ( *tt != ')' ) {
                        MesPrint("@Improper use of isfactorized($var) or isfactorized(expr)");
                        Terminate(-1);
                    }
                    *type = 3;
                    s = tt+1;
                    *val2 = x;
                    return(s);
                }
                else if ( StrICmp(s,(UBYTE *)"isdefined") == 0 ) {
                    UBYTE *tt;
                    *t++ = c;
                    while ( *t == ' ' || *t == '\t' || *t == '\n' || *t == '\r' ) t++;
                    tt = SkipAName(t);
                    c = *tt; *tt = 0;
                    if ( GetPreVar(t,WITHOUTERROR) != 0 ) x = 1;
                    else x = 0;
                    *tt = c;
                    while ( *tt == ' ' || *tt == '\t'
                           || *tt == '\n' || *tt == '\r' ) tt++;
                    if ( *tt != ')' ) {
                        MesPrint("@Improper use of isdefined(var)");
                        Terminate(-1);
                    }
                    *type = 3;
                    s = tt+1;
                    *val2 = x;
                    return(s);
                }
                else if ( StrICmp(s,(UBYTE *)"flag") == 0 ) {
                    UBYTE *tt;
                    WORD x = 0, numexp;
                    {
                    *t++ = c;
                    while ( *t == ' ' || *t == '\t' ) t++;
                    if ( FG.cTable[*t] != 1 ) goto flagerror;
                    while ( FG.cTable[*t] == 1 ) x = 10*x + (*t++ - '0');
                    if ( x < 1 || x > BITSINWORD ) {
                        MesPrint("@Illegal number %d for flag in flag condition",x);
                        goto flagerror;
                    }
                    while ( *t == ' ' || *t == '\t' ) t++;
                    if ( *t != ',' ) goto flagerror;
                    t++;
                    while ( *t == ' ' || *t == '\t' ) t++;
                    tt = SkipAName(t);
                    c = *tt; *tt = 0;
                    if ( GetName(AC.exprnames,t,&numexp,NOAUTO) == NAMENOTFOUND ) {
                        MesPrint("@ %s should be the name of an expression",t);
                        goto flagerror;
                    }
                    *tt = c;
                    while ( *t == ' ' || *t == '\t' ) t++;
                    if ( *tt != ')' ) {
flagerror:
                        MesPrint("@Improper use of flag(num,expr)");
                        Terminate(-1);
                        return(0);
                    }
                    if ( ( Expressions[numexp].uflags & ( 1 << (x-1) ) ) != 0 )
                        *val2 = 1;
                    else *val2 = 0;
```

```
                *type = 3;
                s = tt+1;
                return(s);
                }
            }
            else *t = c;
        }
        else if ( *t == '=' || *t == '<' || *t == '>' || *t == '!'
        || *t == ')' || *t == ' ' || *t == '\t' || *t == 0 || *t == '\n' ) {
            *val2 = 0;
            *type = 2;
            return(t);
        }
        else {
            MesPrint("@Illegal use of string in preprocessor condition: %s",s);
            Terminate(-1);
        }
    }
    while ( *t == '-' || *t == '+' || *t == ' ' || *t == '\t' ) {
        if ( *t == '-' ) sign = -sign;
        t++;
    }
    while ( chartype[*t] == 1 ) { x = 10*x + *t++ - '0'; }
    while ( *t == ' ' || *t == '\t' ) t++;
    if ( chartype[*t] == 8 || *t == ')' || *t == '=' || *t == 0 ) {
        *val2 = sign > 0 ? x: -x;
        *type = 1;
        return(t);
    }
    while ( chartype[*t] != 8 && *t != ')' && *t != '=' && *t ) t++;
    while ( ( t > s ) && ( t[-1] == ' ' || t[-1] == '\t' ) ) t--;
    *type = 2;
    *val2 = val;
    return(t);
}

/*
        #] pParseObject :
        #[ PreCalc :

        To be called when a { is encountered.
        Action: read first till matching }. This is to be stored.
        Next we look whether this is a set or whether it can be
        evaluated. If it is a set we consider it as a new stream.
        The stream will have to be deallocated when read completely.
        If it is to be evaluated we do that and put the result in
        a stream.
*/

UBYTE *PreCalc(void)
{
    UBYTE *buff, *s = 0, *t, *newb, c;
    int size, i, n, parlevel = 0, bralevel = 0;
    LONG answer;
    ULONG uanswer;
    size = n = 0;
    buff = 0; c = '{';
    for (;;) {
        if ( n >= size ) {
            if ( size == 0 ) size = 72;
            else size *= 2;
            if ( ( newb = (UBYTE *)Malloc1(size+2,"{}") ) == 0 ) return(0);
            s = newb;
            if ( buff ) {
                i = n;
                t = buff;
                NCOPYB(s,t,i);
                M_free(buff,"pre calc buffer");
            }
            else s = newb;
            buff = newb;
        }
        *s++ = c; n++;
        c = GetChar(0);
        if ( c == 0 ) {
            Error0("Unmatched {}");
            M_free(buff,"precalc buffer");
            return(0);
        }
        else if ( c == '{' ) { bralevel++; }
        else if ( c == '}' ) {
            if ( --bralevel < 0 ) { *s++ = c; *s = 0; break; }
        }
        else if ( c == '(' ) { parlevel++; }
        else if ( c == ')' ) {
            if ( --parlevel < 0 ) { *s++ = c; *s = 0; goto setstring; }
        }
```

```
        else if ( chartype[c] != 1 && chartype[c] != 5
        && chartype[c] != 6 && c != '!' && c != '&'
        && c != '|' && c != '\\' ) { *s++ = c; *s = 0; goto setstring; }
    }
    if ( parlevel > 0 ) goto setstring;
/*
    Try now to evaluate the string.
    If it works, copy the resulting value back into buff as a string.
*/
    answer = 0;
    if ( PreEval(buff+1,&answer) == 0 ) goto setstring;
    t = buff + size;
    s = buff;
    if ( answer < 0 ) { *s++ = '-'; }
    uanswer = LongAbs(answer);
    n = 0;
    do {
        *--t = ( uanswer % 10 ) + '0';
        uanswer /= 10;
        n++;
    } while ( uanswer > 0 );
    NCOPYB(s,t,n);
    *s = 0;
setstring:;
/*
    Open a stream that contains the current string.
    Mark it to be removed after termination.
*/
    if ( OpenStream(buff,PRECALCSTREAM,0,PRENOACTION) == 0 ) return(0);
    return(buff);
}

/*
        #] PreCalc :
        #[ PreEval :

        Operations are:
        +, -, *, /, %, &, |, ^, !,  ^% (postfix 2log), ^/ (postfix sqrt)
*/

UBYTE *PreEval(UBYTE *s, LONG *x)
{
    LONG y, z, a;
    int tobemultiplied, tobeadded = 1, expsign, i;
    UBYTE *t;
    *x = 0; a = 1;
    while ( *s == ' ' || *s == '\t' ) s++;
    for(;;){
        if ( *s == '+' || *s == '-' ) {
            if ( *s == '-' ) tobeadded = -1;
            else tobeadded = 1;
            s++;
            while ( *s == '-' || *s == '+' || *s == ' ' || *s == '\t' ) {
                if ( *s == '-' ) tobeadded = -tobeadded;
                s++;
            }
        }
        tobemultiplied = 0;
        for(;;){
            while ( *s == ' ' || *s == '\t' ) s++;
            if ( *s <= '9' && *s >= '0' ) {
                ULONG uy;
                ParseNumber(uy,s)
                y = uy;  /* may cause an implementation-defined behaviour */
            }
            else if ( *s == '(' || *s == '{' ) {
                if ( ( t = PreEval(s+1,&y) ) == 0 ) return(0);
                s = t;
            }
            else return(0);
            while ( *s == ' ' || *s == '\t' ) s++;
            expsign = 1;
            while ( *s == '^' || *s == '!' ) {
                s++;
                if ( s[-1] == '!' ) {   /* factorial of course */
                    while ( *s == ' ' || *s == '\t' ) s++;
                    if ( y < 0 ) {
                        MesPrint("@Negative value in preprocessor factorial: %l",y);
                        return(0);
                    }
                    else if ( y == 0 ) y = 1;
                    else if ( y > 1 ) {
                        z = y-1;
                        while ( z > 0 ) { y = y*z; z--; }
                    }
                    continue;
                }
```

```
                else if ( *s == '%' ) { /* ^% is postfix 2log */
                    s++;
                    while ( *s == ' ' || *s == '\t' ) s++;
                    z = y;
                    if ( z <= 0 ) {
                        MesPrint("@Illegal value in preprocessor logarithm: %l",z);
                        return(0);
                    }
                    y = 0; z >>= 1;
                    while ( z ) { y++; z >>= 1; }
                    continue;
                }
                else if ( *s == '/' ) { /* ^/ is postfix sqrt */
                    LONG yy, zz;
                    s++;
                    while ( *s == ' ' || *s == '\t' ) s++;
                    z = y;
                    if ( z <= 0 ) {
                        MesPrint("@Illegal value in preprocessor square root: %l",z);
                        return(0);
                    }
                    if ( z > 8 ) {      /* Very crude integer square root */
                        zz = z;
                        yy = 0; zz >>= 1;
                        while ( zz ) { yy++; zz >>= 1; }
                        zz = z >> (yy/2); i = 10; y = 0;
                        do {
                            yy = zz/2 + z/(2*zz); i--;
                            if ( y == yy ) break;
                            y = zz; zz = yy;
                        } while ( y != yy && i > 0 );
                        while ( y*y < z ) y++;
                        while ( y*y > z ) y--;
                    }
                    else if ( z >= 4 ) y = 2;
                    else if ( z == 0 ) y = 0;
                    else y = 1;
                    continue;
                }
                while ( *s == ' ' || *s == '\t' ) s++;
                while ( *s == '-' || *s == '+' || *s == ' ' || *s == '\t' ) {
                    if ( *s == '-' ) expsign = -expsign;
                }
                if ( *s <= '9' && *s >= '0' ) {
                    ParseNumber(z,s)
                }
                else if ( *s == '(' || *s == '{' ) {
                    if ( ( t = PreEval(s+1,&z) ) == 0 ) return(0);
                    s = t;
                }
                else return(0);
                while ( *s == ' ' || *s == '\t' ) s++;
                y = iexp(y,(int)z);
            }
            if ( tobemultiplied == 0 ) {
                if ( expsign < 0 ) a = 1/y;
                else a = y;
            }
            else {
                if ( tobemultiplied > 2 && expsign != 1 ) {
                    MesPrint("&Incorrect use of ^ with & or |. Use brackets!");
                    Terminate(-1);
                }
                tobemultiplied *= expsign;
                if ( tobemultiplied == 1 ) a *= y;
                else if ( tobemultiplied == 3 ) a &= y;
                else if ( tobemultiplied == 4 ) a |= y;
                else {
                    if ( y == 0 || tobemultiplied == -2 ) {
                        MesPrint("@Division by zero in preprocessor calculator");
                        Terminate(-1);
                    }
                    if ( tobemultiplied == 2 ) a %= y;
                    else a /= y;
                }
            }
            if ( *s == '%' ) tobemultiplied = 2;
            else if ( *s == '*' ) tobemultiplied = 1;
            else if ( *s == '/' ) tobemultiplied = -1;
            else if ( *s == '&' ) tobemultiplied = 3;
            else if ( *s == '|' ) tobemultiplied = 4;
            else {
                ULONG ux, ua;
                ux = *x;
                ua = a;
                if ( tobeadded >= 0 ) ux += ua;
                else ux -= ua;
```

```
                *x = ULongToLong(ux);
                if ( *s == ')' || *s == '}' ) return(s+1);
                else if ( *s == '-' || *s == '+' ) { tobeadded = 1; break; }
                else return(0);
            }
            s++;
        }
    }
/*  return(0); */
}

/*
        #] PreEval :
        #[ AddToPreTypes :
*/

void AddToPreTypes(int type)
{
    if ( AP.NumPreTypes >= AP.MaxPreTypes ) {
        int i, *newlist = (int *)Malloc1(sizeof(int)*(2*AP.MaxPreTypes+1)
                        ,"preprocessor type lists");
        for ( i = 0; i <= AP.MaxPreTypes; i++ ) newlist[i] = AP.PreTypes[i];
        M_free(AP.PreTypes,"preprocessor type lists");
        AP.PreTypes = newlist;
        AP.MaxPreTypes = 2*AP.MaxPreTypes;
    }
    AP.PreTypes[++AP.NumPreTypes] = type;
}

/*
        #] AddToPreTypes :
        #[ MessPreNesting :
*/

void MessPreNesting(int par)
{
    MesPrint("@(%d)Illegal nesting of %#if, %#do, %#procedure and/or %#switch",par);
}

/*
        #] MessPreNesting :
        #[ DoPreAddSeparator :

        Preprocessor directives "addseparator" and "rmseparator" add/remove
        separator characters used to separate function arguments.
        Example:

            #define QQ "a|g|a"
            #addseparator %
            *Comma must be quoted!:
            #rmseparator ","
            #rmseparator |
            #call H(a,a%'QQ')

        Characters ' ', '\t' and '"' are ignored!
*/

int DoPreAddSeparator(UBYTE *s)
{
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    for(;*s != '\0';s++){
        while ( *s == ' ' || *s == '\t' || *s == '"') s++;
        /* Todo:
        if ( set_in(*s,invalidseparators) ) {
            MesPrint("@Invalid separator specified");
            return(-1);
        }
        */
        set_set(*s,AC.separators);
    }
    return(0);
}

/*
        #] DoPreAddSeparator :
        #[ DoPreRmSeparator :

        See commentary with DoPreAddSeparator

        Characters ' ', '\t' and '"' are ignored!
*/
int DoPreRmSeparator(UBYTE *s)
{
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    for(;*s != '\0';s++){
```

```
        while ( *s == ' ' || *s == '\t' || *s == '"') s++;
        set_del(*s,AC.separators);
    }
    return(0);
}

/*
        #] DoPreRmSeparator :
        #[ DoExternal:

        #external ["prevar"] command
*/
int DoExternal(UBYTE *s)
{
#ifdef WITHEXTERNALCHANNEL
    UBYTE *prevar=0;
    int externalD= 0;
#else
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);

#ifdef WITHEXTERNALCHANNEL
    while ( *s == ' ' || *s == '\t' ) s++;
    if(*s == '"'){/*prevar to store the descriptor is defined*/
        prevar=++s;

        if ( chartype[*s] == 0 )for(;*s != '"'; s++)switch(chartype[*s]){
            case 10:/*'\0' fits here*/
                MesPrint("@Can't finde closing \"");
                Terminate(-1);
            case 0:case 1: continue;
            default:
                break;
        }
        if(*s != '"'){
                MesPrint("@Illegal name of preprocessor variable to store external channel");
                return(-1);
      }
      *s='\0';
        for(s++; *s == ' ' || *s == '\t'; s++);
    }

    if(*s == '\0'){
        MesPrint("@Illegal external command");
        return(-1);
    }
    /*here s is a command*/
   /*See the file extcmd.c*/
    /*[08may2006 mt]:*/
    externalD=openExternalChannel(
                s,
                AX.daemonize,
                AX.shellname,
                AX.stderrname);
    /*:[08may2006 mt]*/
    if(externalD<1){/*error?*/
        /*Not quite correct - terminate the program on error:*/
        Error1("Can't start external program",s);
        return(-1);
    }
   /*Now external command runs.*/

   if(prevar){/*Store the external channel descriptor in the provided variable:*/
        UBYTE buf[21];/* 64/Log_2[10] = 19.3, so this is enough forever...*/
        NumToStr(buf,externalD);
        if ( PutPreVar(prevar,buf,0,1) < 0 ) return(-1);
    }

    AX.currentExternalChannel=externalD;
    /*[08may2006 mt]:*/
    if(AX.currentPrompt!=0){/*Change default terminator*/
        if(setTerminatorForExternalChannel(  (char *)AX.currentPrompt)){
            MesPrint("@Prompt is too long");
            return(-1);
        }
    }
    setKillModeForExternalChannel(AX.killSignal,AX.killWholeGroup);
    /*:[08may2006 mt]*/
    return(0);
#else /*ifdef WITHEXTERNALCHANNEL*/
    Error0("External channel: not implemented on this computer/system");
    return(-1);
#endif /*ifdef WITHEXTERNALCHANNEL ... else*/
}
```

```
/*
        #] DoExternal:
        #[ DoPrompt:
            #prompt string
*/

int DoPrompt(UBYTE *s)
{
#ifndef WITHEXTERNALCHANNEL
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);

#ifdef WITHEXTERNALCHANNEL
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( AX.currentPrompt )
            M_free(AX.currentPrompt,"external channel prompt");
    if ( *s == '\0' )
        AX.currentPrompt = (UBYTE *)strDup1((UBYTE *)"","external channel prompt");
    else
        AX.currentPrompt = strDup1(s,"external channel prompt");
    if(  setTerminatorForExternalChannel( (char *)AX.currentPrompt) > 0  ){
        MesPrint("@Prompt is too long");
        return(-1);
    }
    /*else: if 0, ok; if -1, there is no current channel-ok, just prompt is stored.*/
    return(0);
#else /*ifdef WITHEXTERNALCHANNEL*/
    Error0("External channel: not implemented on this computer/system");
    return(-1);
#endif /*ifdef WITHEXTERNALCHANNEL ... else*/
}
/*
        #] DoPrompt:
        #[ DoSetExternal:
            #setexternal n
*/

int DoSetExternal(UBYTE *s)
{
#ifdef WITHEXTERNALCHANNEL
    int n=0;
#else
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);

#ifdef WITHEXTERNALCHANNEL
    while ( *s == ' ' || *s == '\t' ) s++;
    while ( chartype[*s] == 1 ) { n = 10*n + *s++ - '0'; }
    while ( *s == ' ' || *s == '\t' ) s++;
    if(*s!='\0'){
        MesPrint("@setexternal: number expected");
        return(-1);
    }
    if(selectExternalChannel(n)<0){
        MesPrint("@setexternal: invalid number");
        return(-1);
    }
    AX.currentExternalChannel=n;
    return(0);
#else /*ifdef WITHEXTERNALCHANNEL*/
    Error0("External channel: not implemented on this computer/system");
    return(-1);
#endif /*ifdef WITHEXTERNALCHANNEL ... else*/
}
/*
        #] DoSetExternal:
        #[ DoSetExternalAttr:
*/

static FORM_INLINE UBYTE *pickupword(UBYTE *s)
{

    for(;*s>' ';s++)switch(*s){
        case '=':
        case ',':
        case ';':
            return(s);
    }/*for(;*s>' ';s++)switch(*s)*/
    return(s);
}
/*Returns 0 if the first string (case insensitively) equal to
```

```
  the beginning of the second string (of length n):
*/
static inline int strINCmp(UBYTE *a, UBYTE *b, int n)
{
    for(;n>0;n--)if(tolower(*a++)!=tolower(*b++))
        return(1);
    return(*a != '\0');
}

#define KILL "kill"
#define KILLALL "killall"
#define DAEMON "daemon"
#define SHELL "shell"
#define STDERR "stderr"

#define TRUE_EXPR "true"
#define FALSE_EXPR "false"
#define NOSHELL "noshell"
#define TERMINAL "terminal"

/*
    Expects comma-separated list of pairs name=value
*/
int DoSetExternalAttr(UBYTE *s)
{
#ifdef WITHEXTERNALCHANNEL
    int lnam,lval;
    UBYTE *nam,*val;
#else
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);

#ifdef WITHEXTERNALCHANNEL
    do{
        /*Read the name:*/
        while ( *s == ' ' || *s == '\t' ) s++;
        s=pickupword(nam=s);
        lnam=s-nam;
        while ( *s == ' ' || *s == '\t' ) s++;
        if(*s++!='='){
            MesPrint("@External channel:'=' expected instead of %s",s-1);
            return(-1);
        }
        /*Read the value:*/
        while ( *s == ' ' || *s == '\t' ) s++;
        val=s;

        for(;;){
            UBYTE *m;
            s=pickupword(s);
            m=s;
            while ( *s == ' ' || *s == '\t' ) s++;
         if( (*s == ',')||(*s == '\n')||(*s == ';')||(*s == '\0') ){
                s=m;
                break;
            }
        }/*for(;;)*/

        lval=s-val;
        while ( *s == ' ' || *s == '\t' ) s++;

        if(strINCmp((UBYTE *)SHELL,nam,lnam)==0){
            if(AX.shellname!=NULL)
                M_free(AX.shellname,"external channel shellname");
            if(strINCmp((UBYTE *)NOSHELL,val,lval)==0)
                AX.shellname=NULL;
            else{
                UBYTE *ch,*b;
                b=ch=AX.shellname=Malloc1(lval+1,"external channel shellname");
                while(ch-b<lval)
                    *ch++=*val++;
                *ch='\0';
            }
        }else if(strINCmp((UBYTE *)DAEMON,nam,lnam)==0){
            if(strINCmp((UBYTE *)TRUE_EXPR,val,lval)==0)
                AX.daemonize = 1;
            else if(strINCmp((UBYTE *)FALSE_EXPR,val,lval)==0)
                AX.daemonize = 0;
            else{
                MesPrint("@External channel:true or false expected for %s",DAEMON);
                return(-1);
            }
        }else  if(strINCmp((UBYTE *)KILLALL,nam,lnam)==0){
            if(strINCmp((UBYTE *)TRUE_EXPR,val,lval)==0)
```

```
                AX.killWholeGroup = 1;
            else if(strINCmp((UBYTE *)FALSE_EXPR,val,lval)==0)
                AX.killWholeGroup = 0;
            else{
                MesPrint("@External channel: true or false expected for %s",KILLALL);
                return(-1);
            }
        }else   if(strINCmp((UBYTE *)KILL,nam,lnam)==0){
            int i,n=0;
            for(i=0;i<lval;i++) {
                if( *val>='0' && *val<= '9' )
                    n = 10*n + *val++  - '0';
                else{
                    MesPrint("@External channel: number expected for %s",KILL);
                    return(-1);
                }
            }
            AX.killSignal=n;
        }else   if(strINCmp((UBYTE *)STDERR,nam,lnam)==0){
            if( AX.stderrname != NULL ) {
                M_free(AX.stderrname,"external channel stderrname");
            }
            if(strINCmp((UBYTE *)TERMINAL,val,lval)==0)
                AX.stderrname = NULL;
            else{
                UBYTE *ch,*b;
                b=ch=AX.stderrname=Malloc1(lval+1,"external channel stderrname");
                while(ch-b<lval)
                    *ch++=*val++;
                *ch='\0';
            }
        }else{
            nam[lnam+1]='\0';
            MesPrint("@External channel: unrecognized attribute",nam);
            return(-1);
        }
    }while(*s++ == ',');
    if(  (*(s-1)>' ')&&(*(s-1)!=';')  ){
        MesPrint("@External channel: syntax error: %s",s-1);
        return(-1);
    }
   return(0);
#else /*ifdef WITHEXTERNALCHANNEL*/
    Error0("External channel: not implemented on this computer/system");
    return(-1);
#endif /*ifdef WITHEXTERNALCHANNEL ... else*/
}
/*
        #] DoSetExternalAttr:
        #[ DoRmExternal:
            #rmexternal [n] (if 0, close all)
*/

int DoRmExternal(UBYTE *s)
{
#ifdef WITHEXTERNALCHANNEL
    int n = -1;
#else
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);

#ifdef WITHEXTERNALCHANNEL
    while ( *s == ' ' || *s == '\t' ) s++;
    if( chartype[*s] == 1 ){
        for(n=0; chartype[*s] == 1 ; s++) { n = 10*n + *s - '0'; }
        while ( *s == ' ' || *s == '\t' ) s++;
    }
    if(*s!='\0'){
        MesPrint("@rmexternal: invalid number");
        return(-1);
    }
    switch(n){
        case 0:/*Close all opened channels*/
            closeAllExternalChannels();
            AX.currentExternalChannel=0;
            /*Do not clean AX.currentPrompt!*/
            return(0);
        case -1:/*number is not specified - try current*/
            n=AX.currentExternalChannel;
            /* fall through */
        default:
            closeExternalChannel(n);/*No reaction for possible error*/
    }
    if (n == AX.currentExternalChannel)/*cleaned up by closeExternalChannel()*/
```

```
        AX.currentExternalChannel=0;
    return(0);
#else /*ifdef WITHEXTERNALCHANNEL*/
    Error0("External channel: not implemented on this computer/system");
    return(-1);
#endif /*ifdef WITHEXTERNALCHANNEL ... else*/

}
/*
        #] DoRmExternal:
        #[ DoFromExternal :
                #fromexternal
                    is used to read the text from the running external
                    program, the syntax is similar to the #include
                    directive.
                #fromexternal "varname"
                    is used to read the text from the running external
                    program into the preprocessor variable varname.
                    directive.
                #fromexternal "varname" maxlength
                    is used to read the text from the running external
                    program into the preprocessor variable varname.
                    directive. Only first maxlength characters are
                    stored.

                    FORM continues to read the running external
                    program output until the external program outputs a
                    prompt.

*/

int DoFromExternal(UBYTE *s)
{
#ifdef WITHEXTERNALCHANNEL
    UBYTE *prevar=0;
    int lbuf=-1;
    int withNoList=AC.NoShowInput;
    int oldpreassignflag;
#else
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);
#ifdef WITHEXTERNALCHANNEL

    FLUSHCONSOLE;

    while ( *s == ' ' || *s == '\t' ) s++;
    /*[17may2006 mt]:*/
    if ( *s == '-' || *s == '+' ) {
        if ( *s == '-' )
            withNoList = 1;
        else
            withNoList = 0;
        s++;
        while ( *s == ' ' || *s == '\t' ) s++;
    }/*if ( *s == '-' || *s == '+' )*/
    /*:[17may2006 mt]*/
    /*[02feb2006 mt]:*/
    if(*s == '"'){/*prevar to store the output is defined*/
        prevar=++s;

        if ( *s=='$' || chartype[*s] == 0 )for(;*s != '"'; s++)switch(chartype[*s]){
            case 10:/*'\0' fits here*/
                MesPrint("@Can't finde closing \"");
                Terminate(-1);
            case 0:case 1: continue;
            default:
                break;
        }
        if(*s != '"'){
                MesPrint("@Illegal name to store output of external channel");
                return(-1);
      }
      *s='\0';
        for(s++; *s == ' ' || *s == '\t'; s++);
    }/*if(*s == '"')*/

    if(*s != '\0'){
        if( chartype[*s] == 1 ){
            for(lbuf=0; chartype[*s] == 1 ; s++) { lbuf = 10*lbuf + *s - '0'; }
            while ( *s == ' ' || *s == '\t' ) s++;
        }
        if( (*s!='\0')||(lbuf<0) ){
            MesPrint("@Illegal buffer length in fromexternal");
            return(-1);
```

```
        }
    }/*if(*s != '\0')*/
    /*:[02feb20006 mt]*/
    if(getCurrentExternalChannel()!=AX.currentExternalChannel)
        /*[08may20006 mt]:*/
        /*selectExternalChannel(AX.currentExternalChannel);*/
        if(selectExternalChannel(AX.currentExternalChannel)){
            MesPrint("@No current external channel");
            return(-1);
        }
        /*:[08may20006 mt]*/

    /*[02feb2006 mt]:*/
    if(prevar!=0){/*The result must be stored into preprovar*/
      UBYTE *buf;
        int cc = 0;
        if(lbuf == -1){/*Unlimited buffer, everything must be stored*/
            int i;
            buf=Malloc1( (lbuf=255)+1,"Fromexternal");
            /*[18may20006 mt]:*/
            /*for(i=0;(cc=getcFromExtChannel())!=EOF;i++){*/
            /* May 2006: now getcFromExtChannelOk returns EOF while
                getcFromExtChannelFailure returns -2 (see comments in
                exctcmd.c):*/
            for(i=0;(cc=getcFromExtChannel())>0;i++){
            /*:[18may20006 mt]*/
                if(i==lbuf){
                    int j;
                    UBYTE *tmp=Malloc1( (lbuf*=2)+1,"Fromexternal");
                    for(j=0;j<i;j++)tmp[j]=buf[j];
                    M_free(buf,"Fromexternal");
                    buf=tmp;
                }
                buf[i]=(UBYTE)(cc);
            }/*for(i=0;(cc=getcFromExtChannel())>0;i++)*/
            /*[18may20006 mt]:*/
         if(cc == -2){
                MesPrint("@No current external channel");
                return(-1);
            }
            lbuf=i;
            /*:[18may20006 mt]*/
            buf[i]='\0';
        }else{/*Fixed buffer, only lbuf chars must be stored*/
            int i;
            buf=Malloc1(lbuf+1,"Fromexternal");
            for(i=0; i<lbuf;i++){
            /*[18may20006 mt]:*/
                /*if( (cc=getcFromExtChannel())==EOF )*/
                /* May 2006: now getcFromExtChannelOk returns EOF while
                    getcFromExtChannelFailure returns -2 (see comments in
                    exctcmd.c):*/
                if( (cc=getcFromExtChannel())<1 )
            /*:[18may20006 mt]*/
                    break;
                buf[i]=(UBYTE)(cc);
            }
            buf[i]='\0';
            /*[18may20006 mt]:*/
            /*if(cc!=EOF)
                while(getcFromExtChannel()!=EOF);*//*Eat the rest*/
            /* May 2006: now getcFromExtChannelOk returns EOF while
                getcFromExtChannelFailure returns -2 (see comments in
                exctcmd.c):*/
            if(cc>0)
                while(getcFromExtChannel()>0);/*Eat the rest*/
            else if(cc == -2){
                MesPrint("@No current external channel");
                return(-1);
            }
            /*:[18may20006 mt]*/
        }
        /*[18may20006 mt]:*/
        if(*prevar == '$'){/*Put the answer to the dollar variable*/
            int oldNumPotModdollars = NumPotModdollars;
#ifdef WITHMPI
            WORD oldRhsExprInModuleFlag = AC.RhsExprInModuleFlag;
            AC.RhsExprInModuleFlag = 0;
#endif
            /*Here lbuf is the actual length of buf!*/
            /*"prevar=buf'\0'":*/
            UBYTE *pbuf=Malloc1(StrLen(prevar)+1+lbuf+1,"Fromexternal to dollar");
            UBYTE *c=pbuf;
            UBYTE *b=prevar;
            while(*b!='\0'){*c++ = *b++;}
            *c++='=';
            b=buf;
```

```
            while(  (*c++=*b++)!='\0'  );
            oldpreassignflag = AP.PreAssignFlag;
            AP.PreAssignFlag = 1;
            if ( ( cc = CompileStatement(pbuf) ) || ( cc = CatchDollar(0) ) ) {
                Error1("External channel: can't asign output to dollar variable ",prevar);
            }
            AP.PreAssignFlag = oldpreassignflag;
            NumPotModdollars = oldNumPotModdollars;
#ifdef WITHMPI
            AC.RhsExprInModuleFlag = oldRhsExprInModuleFlag;
#endif
            M_free(pbuf,"Fromexternal to dollar");
        }else{
            cc = PutPreVar(prevar, buf, 0, 1) < 0;
        }
        /*:[18may20006 mt]*/
        M_free(buf,"Fromexternal");
        if ( cc ) return(-1);
        return(0);
    }
    /*:[02feb2006 mt]*/
    if ( OpenStream(s,EXTERNALCHANNELSTREAM,0,PRENOACTION) == 0 ) return(-1);
    /*[17may2006 mt]:*/
    AC.NoShowInput = withNoList;
    /*:[17may2006 mt]*/
    return(0);
#else
    Error0("External channel: not implemented on this computer/system");
    return(-1);
#endif
}

/*
        #] DoFromExternal :
        #[ DoToExternal :
            #toexetrnal
*/

#ifdef WITHEXTERNALCHANNEL

/*A wrapper to writeBufToExtChannel, see the file extcmd.c:*/
LONG WriteToExternalChannel(int handle, UBYTE *buffer, LONG size)
{
    /*ATT! handle is not used! Actual output is performed to
        the current external channel, see extcmd.c!*/
    DUMMYUSE(handle);
    if(writeBufToExtChannel((char*)buffer,size))
        return(-1);
    return(size);
}
#endif /*ifdef WITHEXTERNALCHANNEL*/

int DoToExternal(UBYTE *s)
{
#ifdef WITHEXTERNALCHANNEL
   HANDLERS h;
   LONG (*OldWrite)(int handle, UBYTE *buffer, LONG size) = WriteFile;
    int ret=-1;
#else
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.preError ) return(0);
#ifdef WITHEXTERNALCHANNEL

    h.oldsilent=AM.silent;
    h.newlogonly = h.oldlogonly = AM.FileOnlyFlag;
    h.newhandle = h.oldhandle = AC.LogHandle;
    h.oldprinttype = AO.PrintType;

    WriteFile=&WriteToExternalChannel;

    while ( *s == ' ' || *s == '\t' ) s++;

    if(AX.currentExternalChannel==0){
        MesPrint("@No current external channel");
        goto DoToExternalReady;
    }

    if(getCurrentExternalChannel()!=AX.currentExternalChannel)
        selectExternalChannel(AX.currentExternalChannel);

    ret=writeToChannel(EXTERNALCHANNELOUT,s,&h);
    DoToExternalReady:
        WriteFile=OldWrite;
        return(ret);
```

```
#else /*ifdef WITHEXTERNALCHANNEL*/
    Error0("External channel: not implemented on this computer/system");
    return(-1);
#endif /*ifdef WITHEXTERNALCHANNEL ... else*/

}

/*
        #] DoToExternal :
        #[ defineChannel :
*/

UBYTE *defineChannel(UBYTE *s, HANDLERS *h)
{
    UBYTE *name,*to;

    if ( *s != '<' )
        return(s);

    s++;
    name = to = s;
    while ( *s && *s != '>' ) {
        if ( *s == '\\' ) s++;
        *to++ = *s++;
    }
    if ( *s == 0 ) {
        MesPrint("@Improper termination of filename");
        return(0);
    }
    s++;
    *to = 0;
    if ( *name ) {
        h->newhandle = GetChannel((char *)name,0);
        h->newlogonly = 1;
    }
    else if ( AC.LogHandle >= 0 ) {
        h->newhandle = AC.LogHandle;
        h->newlogonly = 1;
    }
    return(s);
}

/*
        #] defineChannel :
        #[ writeToChannel :
*/

int writeToChannel(int wtype, UBYTE *s, HANDLERS *h)
{
    UBYTE *to, *fstring, *ss, *sss, *s1, c, c1;
    WORD  num, number, nfac;
    WORD oldOptimizationLevel;
    UBYTE Out[MAXLINELENGTH+14], *stopper;
    int nosemi, i;
    int plus = 0;

/*
    Now determine the format string
*/
    while ( *s == ',' || *s == ' ' ) s++;
    if ( *s != '"' ) {
        MesPrint("@No format string present");
        return(-1);
    }
    s++; fstring = to = s;
    while ( *s ) {
        if ( *s == '\\' ) {
            s++;
            if ( *s == '\\' ) {
                *to++ = *s++;
                if ( *s == '\\' ) *to++ = *s++;
            }
            else if ( *s == '"' ) *to++ = *s++;
            else { *to++ = '\\'; *to++ = *s++; }
        }
        else if ( *s == '"' ) break;
        else *to++ = *s++;
    }
    if ( *s != '"' ) {
        MesPrint("@No closing \" in format string");
        return(-1);
    }
    *to = 0; s++;
    if ( AC.LineLength > 20 && AC.LineLength <= MAXLINELENGTH ) stopper = Out + AC.LineLength;
    else stopper = Out + MAXLINELENGTH;
    to = Out;
/*
```

```
    s points now at the list of objects (if any)
    we can start executing the format string.
*/
    AM.silent = 0;
    AC.LogHandle = h->newhandle;
    AM.FileOnlyFlag = h->newlogonly;
    if ( h->newhandle >= 0 ) {
        AO.PrintType |= PRINTLFILE;
    }
    while ( *fstring ) {
        if ( to >= stopper ) {
            if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                *to++ = '&';
            }
            num = to - Out;
            WriteString(wtype,Out,num);
            to = Out;
            if ( AC.OutputMode == FORTRANMODE
             || AC.OutputMode == PFORTRANMODE ) {
                number = 7;
                for ( i = 0; i < number; i++ ) *to++ = ' ';
                to[-2] = '&';
            }
        }
        if ( *fstring == '\\' ) {
            fstring++;
            if ( *fstring == 'n' ) {
                num = to - Out;
                WriteString(wtype,Out,num);
                to = Out;
                fstring++;
            }
            else if ( *fstring == 't' ) { *to++ = '\t'; fstring++; }
            else if ( *fstring == 'b' ) { *to++ = '\\'; fstring++; }
            else *to++ = *fstring++;
        }
        else if ( *fstring == '%' ) {
            plus = 0;
retry:
            fstring++;
            if ( *fstring == 'd' ) {
                int sign,dig;
                number = -1;
donumber:
                while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
                sign = 1;
                while ( *s == '+' || *s == '-' ) {
                    if ( *s == '-' ) sign = -sign;
                    s++;
                }
                dig = 0; ss = s; if ( sign < 0 ) { ss--; *ss = '-'; dig++; }
                while ( *s >= '0' && *s <= '9' ) { s++; dig++; }
                if ( number < 0 ) {
                    while ( ss < s ) {
                        if ( to >= stopper ) {
                            num = to - Out;
                            WriteString(wtype,Out,num);
                            to = Out;
                        }
                        if ( *ss == '\\' ) ss++;
                        *to++ = *ss++;
                    }
                }
                else {
                    if ( number < dig ) { dig = number; ss = s - dig; }
                    while ( number > dig ) {
                        if ( to >= stopper ) {
                            num = to - Out;
                            WriteString(wtype,Out,num);
                            to = Out;
                        }
                        *to++ = ' '; number--;
                    }
                    while ( ss < s ) {
                        if ( to >= stopper ) {
                            num = to - Out;
                            WriteString(wtype,Out,num);
                            to = Out;
                        }
                        if ( *ss == '\\' ) ss++;
                        *to++ = *ss++;
                    }
                }
                fstring++;
            }
            else if ( *fstring == '$' ) {
                UBYTE *dolalloc;
```

```
                    number = AO.OutSkip;
dodollar:
                    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
                    if ( AC.OutputMode == FORTRANMODE
                     || AC.OutputMode == PFORTRANMODE ) {
                        number = 7;
                    }
                    if ( *s != '$' ) {
nodollar:               MesPrint("@$-variable expected in #write instruction");
                        AM.FileOnlyFlag = h->oldlogonly;
                        AC.LogHandle = h->oldhandle;
                        AO.PrintType = h->oldprinttype;
                        AM.silent = h->oldsilent;
                        return(-1);
                    }
                    s++; ss = s;
                    while ( chartype[*s] <= 1 ) s++;
                    if ( s == ss ) goto nodollar;
                    c = *s; *s = 0;
                    num = GetDollar(ss);
                    if ( num < 0 ) {
                        MesPrint("@#write instruction: $%s has not been defined",ss);
                        AM.FileOnlyFlag = h->oldlogonly;
                        AC.LogHandle = h->oldhandle;
                        AO.PrintType = h->oldprinttype;
                        AM.silent = h->oldsilent;
                        return(-1);
                    }
                    *s = c;
                    if ( *s == '[' ) {
                        if ( Dollars[num].nfactors <= 0 ) {
                            *s = 0;
                            MesPrint("@#write instruction: $%s has not been factorized",ss);
                            AM.FileOnlyFlag = h->oldlogonly;
                            AC.LogHandle = h->oldhandle;
                            AO.PrintType = h->oldprinttype;
                            AM.silent = h->oldsilent;
                            return(-1);
                        }
/*
                        Now get the number between the []
*/
                        nfac = GetDollarNumber(&s,Dollars+num);

                        if ( Dollars[num].nfactors == 1 && nfac == 1 ) goto writewhole;

                        if ( ( dolalloc = WriteDollarFactorToBuffer(num,nfac,0) ) == 0 ) {
                            AM.FileOnlyFlag = h->oldlogonly;
                            AC.LogHandle = h->oldhandle;
                            AO.PrintType = h->oldprinttype;
                            AM.silent = h->oldsilent;
                            return(-1);
                        }
                        goto writealloc;
                    }
                    else if ( *s && *s != ' ' && *s != ',' && *s != '\t' ) {
                        MesPrint("@#write instruction: illegal characters after $-variable");
                        AM.FileOnlyFlag = h->oldlogonly;
                        AC.LogHandle = h->oldhandle;
                        AO.PrintType = h->oldprinttype;
                        AM.silent = h->oldsilent;
                        return(-1);
                    }
                    else {
writewhole:
                      if ( ( dolalloc = WriteDollarToBuffer(num,0) ) == 0 ) {
                        AM.FileOnlyFlag = h->oldlogonly;
                        AC.LogHandle = h->oldhandle;
                        AO.PrintType = h->oldprinttype;
                        AM.silent = h->oldsilent;
                        return(-1);
                      }
                      else {
writealloc:
                        ss = dolalloc;
                        while ( *ss ) {
                            if ( to >= stopper ) {
                                if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                                    *to++ = '&';
                                }
                                num = to - Out;
                                WriteString(wtype,Out,num);
                                to = Out;
                                for ( i = 0; i < number; i++ ) *to++ = ' ';
                                if ( AC.OutputMode == FORTRANMODE
                                 || AC.OutputMode == PFORTRANMODE ) to[-2] = '&';
                            }
```

```
                        if ( chartype[*ss] > 3 ) { *to++ = *ss++; }
                        else {
                            sss = ss; while ( chartype[*ss] <= 3 ) ss++;
                            if ( ( to + (ss-sss) ) >= stopper ) {
                                if ( (ss-sss) >= (stopper-Out) ) {
                                    if ( ( to - stopper ) < 10 ) {
                                        if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                                            *to++ = '&';
                                        }
                                        num = to - Out;
                                        WriteString(wtype,Out,num);
                                        to = Out;
                                        for ( i = 0; i < number; i++ ) *to++ = ' ';
                                        if ( AC.OutputMode == FORTRANMODE
                                         || AC.OutputMode == PFORTRANMODE ) to[-2] = '&';
                                    }
                                    while ( (ss-sss) >= (stopper-Out) ) {
                                        while ( to < stopper-1 ) {
                                            *to++ = *sss++;
                                        }
                                        if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                                            *to++ = '&';
                                        }
                                        else {
                                            *to++ = '\\';
                                        }
                                        num = to - Out;
                                        WriteString(wtype,Out,num);
                                        to = Out;
                                        if ( AC.OutputMode == FORTRANMODE
                                         || AC.OutputMode == PFORTRANMODE ) {
                                            for ( i = 0; i < number; i++ ) *to++ = ' ';
                                            to[-2] = '&';
                                        }
                                    }
                                }
                                else {
                                    if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                                        *to++ = '&';
                                    }
                                    num = to - Out;
                                    WriteString(wtype,Out,num);
                                    to = Out;
                                    for ( i = 0; i < number; i++ ) *to++ = ' ';
                                    if ( AC.OutputMode == FORTRANMODE
                                     || AC.OutputMode == PFORTRANMODE ) to[-2] = '&';
                                }
                            }
                            while ( sss < ss ) *to++ = *sss++;
                        }
                    }
                  }
                  M_free(dolalloc,"written dollar");
                  fstring++;
                }
            }
            else if ( *fstring == 's' ) {
                fstring++;
                while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
                if ( *s == '"' ) {
                    s++; ss = s;
                    while ( *s ) {
                        if ( *s == '\\' ) s++;
                        else if ( *s == '"' ) break;
                        s++;
                    }
                    if ( *s == 0 ) {
                        MesPrint("@#write instruction: Missing \" in string");
                        AM.FileOnlyFlag = h->oldlogonly;
                        AC.LogHandle = h->oldhandle;
                        AO.PrintType = h->oldprinttype;
                        AM.silent = h->oldsilent;
                        return(-1);
                    }
                    while ( ss < s ) {
                        if ( to >= stopper ) {
                            num = to - Out;
                            WriteString(wtype,Out,num);
                            to = Out;
                        }
                        if ( *ss == '\\' ) ss++;
                        *to++ = *ss++;
                    }
                    s++;
```

```
                }
                else {
                    sss = ss = s;
                    while ( *s && *s != ',' ) {
                        if ( *s == '\\' ) { s++; sss = s+1; }
                        s++;
                    }
                    while ( s > sss+1 && ( s[-1] == ' ' || s[-1] == '\t' ) ) s--;
                    while ( ss < s ) {
                        if ( to >= stopper ) {
                            num = to - Out;
                            WriteString(wtype,Out,num);
                            to = Out;
                        }
                        if ( *ss == '\\' ) ss++;
                        *to++ = *ss++;
                    }
                }
            }
            else if ( *fstring == 'X' ) {
                fstring++;
                if ( cbuf[AM.sbufnum].numrhs > 0 ) {
/*
                    This should be only to the value of AM.oldnumextrasymbols
*/
                    UBYTE *s = GetPreVar(AM.oldnumextrasymbols,0);
                    WORD x = 0;
                    while ( *s >= '0' && *s <= '9' ) x = 10*x + *s++ - '0';
                    if ( x > 0 )
                        PrintSubtermList(1,x);
                    else
                        PrintSubtermList(1,cbuf[AM.sbufnum].numrhs);
                }
            }
            else if ( *fstring == 'O' ) {
                number = AO.OutSkip;
dooptim:
                fstring++;
/*
                First test whether there is an optimization buffer
*/
                if ( AO.OptimizeResult.code == NULL && AO.OptimizationLevel != 0 ) {
                    MesPrint("@In #write instruction: no optimization results available!");
                    return(-1);
                }
                num = to - Out;
                WriteString(wtype,Out,num);
                to = Out;
                if ( AO.OptimizationLevel != 0 ) {
                    WORD oldoutskip = AO.OutSkip;
                    AO.OutSkip = number;
                    optimize_print_code(0);
                    AO.OutSkip = oldoutskip;
                }
            }
            else if ( *fstring == 'e' || *fstring == 'E' ) {
                if ( *fstring == 'E'
                    || AC.OutputMode == FORTRANMODE
                    || AC.OutputMode == PFORTRANMODE ) nosemi = 1;
                else nosemi = 0;
                fstring++;
                while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
                if ( chartype[*s] != 0 && *s != '[' ) {
noexpr:             MesPrint("@expression name expected in #write instruction");
                    AM.FileOnlyFlag = h->oldlogonly;
                    AC.LogHandle = h->oldhandle;
                    AO.PrintType = h->oldprinttype;
                    AM.silent = h->oldsilent;
                    return(-1);
                }
                ss = s;
                if ( ( s = SkipAName(ss) ) == 0 || s[-1] == '_' ) goto noexpr;
                s1 = s; c = c1 = *s1;
                if ( c1 == '(' ) {
                    SKIPBRA3(s)
                    if ( *s == ')' ) {
                        AO.CurBufWrt = s1+1;
                        c = *s; *s = 0;
                    }
                    else {
                        MesPrint("@Illegal () specifier in expression name in #write");
                        AM.FileOnlyFlag = h->oldlogonly;
                        AC.LogHandle = h->oldhandle;
                        AO.PrintType = h->oldprinttype;
                        AM.silent = h->oldsilent;
                        return(-1);
                    }
```

```
            }
            else AO.CurBufWrt = (UBYTE *)underscore;
            *s1 = 0;
            num = to - Out;
            if ( num > 0 ) WriteUnfinString(wtype,Out,num);
            to = Out;
            oldOptimizationLevel = AO.OptimizationLevel;
            AO.OptimizationLevel = 0;
            if ( WriteOne(ss,(int)num,nosemi,plus) < 0 ) {
                AM.FileOnlyFlag = h->oldlogonly;
                AC.LogHandle = h->oldhandle;
                AO.PrintType = h->oldprinttype;
                AM.silent = h->oldsilent;
                return(-1);
            }
            AO.OptimizationLevel = oldOptimizationLevel;
            *s1 = c1;
            if ( s > s1 ) *s++ = c;
        }
/*
        File content
*/
        else if ( ( *fstring == 'f' ) || ( *fstring == 'F' ) ) {
            LONG n;
            while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
            ss = s;
            while ( *s && *s != ',' ) {
                if ( *s == '\\' ) s++;
                s++;
            }
            c = *s; *s = 0;
            s1 = LoadInputFile(ss,HEADERFILE);
            *s = c;
/*
            There should have been a way to pass the file size.
            Also there should be conversions for \r\n etc.
*/
            if ( s1 ) {
                ss = s1; while ( *ss ) ss++;
                n = ss-s1;
                WriteString(wtype,s1,n);
                M_free(s1,"copy file");
            }
            else if ( *fstring == 'F' ) {
                *s = 0;
                MesPrint("@Error in #write: could not open file %s",ss);
                *s = c;
                goto ReturnWithError;
            }
            fstring++;
        }
        else if ( *fstring == '%' ) {
            *to++ = *fstring++;
        }
        else if ( FG.cTable[*fstring] == 1 ) {  /* %#S */
            number = 0;
            while ( FG.cTable[*fstring] == 1 ) {
                number = 10*number + *fstring++ - '0';
            }
            if ( *fstring == 'O' ) goto dooptim;
            else if ( *fstring == 'd' ) goto donumber;
            else if ( *fstring == '$' ) goto dodollar;
            else if ( *fstring == 't' ) {   /* `tab' position */
                if ( number < (WORD)(stopper-Out) ) {
                    while ( (WORD)(to-Out) < number ) *to++ = ' ';
                }
                fstring++;
            }
            else if ( *fstring == 'X' || *fstring == 'x' ) {
                if ( number > 0 && number <= cbuf[AM.sbufnum].numrhs ) {
                    UBYTE buffer[80], *out, *old1, *old2, *old3;
                    WORD *term, first;
                    if ( *fstring == 'X' ) {
                        out = StrCopy((UBYTE *)AC.extrasym,buffer);
                        if ( AC.extrasymbols == 0 ) {
                            out = NumCopy(number,out);
                            out = StrCopy((UBYTE *)"_",out);
                        }
                        else if ( AC.extrasymbols == 1 ) {
                            if ( AC.OutputMode == CMODE ) {
                                out = StrCopy((UBYTE *)"[",out);
                                out = NumCopy(number,out);
                                out = StrCopy((UBYTE *)"]",out);
                            }
                            else {
                                out = StrCopy((UBYTE *)"(",out);
                                out = NumCopy(number,out);
```

```
                                    out = StrCopy((UBYTE *)")",out);
                                }
                            }
                            out = StrCopy((UBYTE *)"=",out);
                            ss = buffer;
                            while ( ss < out ) {
                                if ( to >= stopper ) {
                                    num = to - Out;
                                    WriteString(wtype,Out,num);
                                    to = Out;
                                }
                                *to++ = *ss++;
                            }
                        }
                        term = cbuf[AM.sbufnum].rhs[number];
                        first = 1;
                        if ( *term == 0 ) {
                            *to++ = '0';
                        }
                        else {
                            old1 = AO.OutFill;
                            old2 = AO.OutputLine;
                            old3 = AO.OutStop;
                            AO.OutFill = to;
                            AO.OutputLine = Out;
                            AO.OutStop = Out + AC.LineLength;
                            while ( *term ) {
                                if ( WriteInnerTerm(term,first) ) Terminate(-1);
                                term += *term;
                                first = 0;
                            }
                            to = Out + (AO.OutFill-AO.OutputLine);
                            AO.OutFill = old1;
                            AO.OutputLine = old2;
                            AO.OutStop = old3;
                        }
                    }
                    fstring++;
                }
                else {
                    goto IllegControlSequence;
                }
            }
            else if ( *fstring == '+' ) {
                plus = 1; goto retry;
            }
            else if ( *fstring == 0 ) {
                *to++ = 0;
            }
            else {
IllegControlSequence:
                MesPrint("@Illegal control sequence in format string in #write instruction");
ReturnWithError:
                AM.FileOnlyFlag = h->oldlogonly;
                AC.LogHandle = h->oldhandle;
                AO.PrintType = h->oldprinttype;
                AM.silent = h->oldsilent;
                return(-1);
            }
        }
        else {
            *to++ = *fstring++;
        }
    }
/*
    Now flush the output
*/
    num = to - Out;
    /*[15apr2004 mt]:*/
    if(wtype==EXTERNALCHANNELOUT){
        if(num!=0)
            WriteUnfinString(wtype,Out,num);
    }else
    /*:[15apr2004 mt]*/
    WriteString(wtype,Out,num);
/*
    and restore original parameters
*/
    AM.FileOnlyFlag = h->oldlogonly;
    AC.LogHandle = h->oldhandle;
    AO.PrintType = h->oldprinttype;
    AM.silent = h->oldsilent;
    return(0);
}

/*
        #] writeToChannel :
```

```
        #[ DoFactDollar :

        Executes the #factdollar $var
              instruction
*/

int DoFactDollar(UBYTE *s)
{
    GETIDENTITY
    WORD numdollar, *oldworkpointer;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == '$' ) {
        if ( GetName(AC.dollarnames,s+1,&numdollar,NOAUTO) != CDOLLAR ) {
            MesPrint("@%s is undefined",s);
            return(-1);
        }
        s = SkipAName(s+1);
        if ( *s != 0 ) {
            MesPrint("@#FactDollar should have a single $variable for its argument");
            return(-1);
        }
        NewSort(BHEAD0);
        oldworkpointer = AT.WorkPointer;
        if ( DollarFactorize(BHEAD numdollar) ) return(-1);
        AT.WorkPointer = oldworkpointer;
        LowerSortLevel();
        return(0);
    }
    else if ( ParenthesesTest(s) ) return(-1);
    else {
        MesPrint("@#FactDollar should have a single $variable for its argument");
        return -1;
    }
}

/*
        #] DoFactDollar :
        #[ GetDollarNumber :
*/

WORD GetDollarNumber(UBYTE **inp, DOLLARS d)
{
    UBYTE *s = *inp, c, *name;
    WORD number, nfac, *w;
    DOLLARS dd;
    s++;
    if ( *s == '$' ) {
        s++; name = s;
        while ( FG.cTable[*s] < 2 ) s++;
        c = *s; *s = 0;
        if ( GetName(AC.dollarnames,name,&number,NOAUTO) == NAMENOTFOUND ) {
            MesPrint("@dollar in #write should have been defined previously");
            Terminate(-1);
        }
        *s = c;
        dd = Dollars + number;
        if ( c == '[' ) {
            *inp = s;
            nfac = GetDollarNumber(inp,dd);
            s = *inp;
            if ( *s != ']' ) {
                MesPrint("@Illegal factor for dollar variable");
                Terminate(-1);
            }
            *inp = s+1;
            if ( nfac == 0 ) {
                if ( dd->nfactors > d->nfactors ) {
TooBig:
                    MesPrint("@Factor number for dollar variable too large");
                    Terminate(-1);
                }
                return(dd->nfactors);
            }
            w = dd->factors[nfac-1].where;
            if ( w == 0 ) {
                if ( dd->factors[nfac-1].value > d->nfactors ||
                     dd->factors[nfac-1].value < 0 ) goto TooBig;
                return(dd->factors[nfac-1].value);
            }
            if ( *w == 4 && w[4] == 0 && w[3] == 3 && w[2] == 1
              && w[1] <= d->nfactors ) return(w[1]);
            if ( w[*w] == 0 && w[*w-1] == *w-1 ) goto TooBig;
IllNum:
            MesPrint("@Illegal factor number for dollar variable");
```

```
            Terminate(-1);
        }
        else {  /* The dollar should be a number */
            if ( dd->type == DOLZERO ) {
                return(0);
            }
            else if ( dd->type == DOLTERMS || dd->type == DOLNUMBER ) {
                w = dd->where;
                if ( *w == 4 && w[4] == 0 && w[3] == 3 && w[2] == 1
                  && w[1] <= d->nfactors ) return(w[1]);
                if ( w[*w] == 0 && w[*w-1] == *w-1 ) goto TooBig;
                goto IllNum;
            }
            else goto IllNum;
        }
    }
    else if ( FG.cTable[*s] == 1 ) {
        WORD x = *s++ - '0';
        while ( FG.cTable[*s] == 1 ) {
            x = 10*x + *s++ - '0';
            if ( x > d->nfactors ) {
                MesPrint("@Factor number %d for dollar variable too large",x);
                Terminate(-1);
            }
        }
        if ( *s != ']' ) {
            MesPrint("@Illegal factor number for dollar variable");
            Terminate(-1);
        }
        s++; *inp = s;
        return(x);
    }
    else {
        MesPrint("@Illegal factor indicator for dollar variable");
        Terminate(-1);
    }
    return(-1);
}

/*
        #] GetDollarNumber :
        #[ DoSetRandom :

        Executes the #SetRandom number
*/

int DoSetRandom(UBYTE *s)
{
    ULONG x;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    x = 0;
    while ( FG.cTable[*s] == 1 ) {
        x = 10*x + (*s++-'0');
    }
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s  == 0 ) {
#ifdef WITHPTHREADS
#ifdef WITHSORTBOTS
        int id, totnum = MaX(2*AM.totalnumberofthreads-3,AM.totalnumberofthreads);
#else
        int id, totnum = AM.totalnumberofthreads;
#endif
        for ( id = 0; id < totnum; id++ ) {
            AB[id]->R.wranfseed = x;
            if ( AB[id]->R.wranfia ) M_free(AB[id]->R.wranfia,"wranf");
            AB[id]->R.wranfia = 0;
        }
#else
        AR.wranfseed = x;
        if ( AR.wranfia ) M_free(AR.wranfia,"wranf");
        AR.wranfia = 0;
#endif
        return(0);
    }
    else {
        MesPrint("@proper syntax is #SetRandom number");
        return(-1);
    }
}

/*
        #] DoSetRandom :
        #[ DoOptimize :

        Executes the #Optimize(expr) instruction.
```

```
*/

int DoOptimize(UBYTE *s)
{
    GETIDENTITY
    UBYTE *exprname;
    WORD numexpr;
    int error = 0, i;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    DUMMYUSE(*s)
    exprname = s; s = SkipAName(s);
    if ( *s != 0 && *s != ';' ) {
        MesPrint("@proper syntax is #Optimize,expression");
        return(-1);
    }
    *s = 0;
    if ( GetName(AC.exprnames,exprname,&numexpr,NOAUTO) != CEXPRESSION ) {
        MesPrint("@%s is not an expression",exprname);
        error = 1;
    }
    else if ( AP.preError == 0 ) {
        EXPRESSIONS e = Expressions + numexpr;
        POSITION position;
        int firstterm;
        WORD *term = AT.WorkPointer;
        ClearOptimize();
        if ( AO.OptimizationLevel == 0 ) return(0);
        switch ( e->status ) {
            case LOCALEXPRESSION:
            case GLOBALEXPRESSION:
                break;
            default:
                MesPrint("@Expression %s is not an active unhidden local or global
expression.",exprname);
                Terminate(-1);
                break;
        }
#ifdef WITHMPI
        if ( PF.me == MASTER )
#endif
        RevertScratch();
        for ( i = NumExpressions-1; i >= 0; i-- ) {
            AS.OldOnFile[i] = Expressions[i].onfile;
            AS.OldNumFactors[i] = Expressions[i].numfactors;
            AS.Oldvflags[i] = Expressions[i].vflags;
            Expressions[i].vflags &= ~(ISUNMODIFIED|ISZERO);
        }
        for ( i = 0; i < NumExpressions; i++ ) {
            if ( i == numexpr ) {
                PutPreVar(AM.oldnumextrasymbols,
                        GetPreVar((UBYTE *)"EXTRASYMBOLS_",0),0,1);
                Optimize(numexpr, 0);
                AO.OptimizeResult.nameofexpr = strDup1(exprname,"optimize expression name");
                continue;
            }
#ifdef WITHMPI
            if ( PF.me == MASTER ) {
#endif
            e = Expressions + i;
            switch ( e->status ) {
                case LOCALEXPRESSION:
                case SKIPLEXPRESSION:
                case DROPLEXPRESSION:
                case DROPPEDEXPRESSION:
                case GLOBALEXPRESSION:
                case SKIPGEXPRESSION:
                case DROPGEXPRESSION:
                case HIDELEXPRESSION:
                case HIDEGEXPRESSION:
                case DROPHLEXPRESSION:
                case DROPHGEXPRESSION:
                case INTOHIDELEXPRESSION:
                case INTOHIDEGEXPRESSION:
                    break;
                default:
                    continue;
            }
            AR.GetFile = 0;
            SetScratch(AR.infile,&(e->onfile));
            if ( GetTerm(BHEAD term) <= 0 ) {
                MesPrint("@Expression %d has problems reading from scratchfile",i);
                Terminate(-1);
            }
            term[3] = i;
            AR.DeferFlag = 0;
            SeekScratch(AR.outfile,&position);
```

```
            e->onfile = position;
            *AM.S0->sBuffer = 0; firstterm = -1;
            do {
                WORD *oldipointer = AR.CompressPointer;
                WORD *comprtop = AR.ComprTop;
                AR.ComprTop = AM.S0->sTop;
                AR.CompressPointer = AM.S0->sBuffer;
                if ( firstterm > 0 ) {
                    if ( PutOut(BHEAD term,&position,AR.outfile,1) < 0 ) goto DoSerr;
                }
                else if ( firstterm < 0 ) {
                    if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) goto DoSerr;
                    firstterm++;
                }
                else {
                    if ( PutOut(BHEAD term,&position,AR.outfile,-1) < 0 ) goto DoSerr;
                    firstterm++;
                }
                AR.CompressPointer = oldipointer;
                AR.ComprTop = comprtop;
            } while ( GetTerm(BHEAD term) );
            if ( FlushOut(&position,AR.outfile,1) ) {
DoSerr:
                MesPrint("@Expression %d has problems writing to scratchfile",i);
                Terminate(-1);
            }
#ifdef WITHMPI
            }
#endif
        }
/*
        Now some administration and we are done
*/
        UpdateMaxSize();
    }
    else {
        ClearOptimize();
    }
    return(error);

}

/*
        #] DoOptimize :
        #[ DoClearOptimize :

        Clears all relevant buffers of the output optimization
*/

int DoClearOptimize(UBYTE *s)
{
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    DUMMYUSE(*s);
    return(ClearOptimize());
}

/*
        #] DoClearOptimize :
        #[ DoSkipExtraSymbols :

        Adds the intermediate variables of the previous optimization
        to the list of extra symbols, provided it has not yet been erased
        by a #clearoptimize
        To remove them again one needs to use the 'delete extrasymbols;'
        or the 'delete extrasymbols>num;' statement in which num is the
        old number of extra symbols.
*/

int DoSkipExtraSymbols(UBYTE *s)
{
    CBUF *C = cbuf + AM.sbufnum;
    WORD tt = 0, j = 0, oldval = AO.OptimizeResult.minvar;
    if ( AO.OptimizeResult.code == NULL ) return(0);
    if ( AO.OptimizationLevel == 0 ) return(0);
    while ( *s == ',' ) s++;
    if ( *s == 0 ) {
        AO.OptimizeResult.minvar = AO.OptimizeResult.maxvar+1;
    }
    else {
        while ( *s <= '9' && *s >= '0' ) j = 10*j + *s++ - '0';
        if ( *s ) {
            MesPrint("@Illegal use of #SkipExtraSymbols instruction");
            Terminate(-1);
        }
        AO.OptimizeResult.minvar += j;
        if ( AO.OptimizeResult.minvar > AO.OptimizeResult.maxvar )
```

```
            AO.OptimizeResult.minvar = AO.OptimizeResult.maxvar+1;
    }
    j = AO.OptimizeResult.minvar - oldval;
    while ( j > 0 ) {
        AddRHS(AM.sbufnum,1);
        AddNtoC(AM.sbufnum,1,&tt,16);
        AddToCB(C,0)
        InsTree(AM.sbufnum,C->numrhs);
        j--;
    }
    return(0);
}

/*
        #] DoSkipExtraSymbols :
        #[ DoPreReset :

        Does a reset of variables.
        Currently only the timer (stopwatch) of `timer_'
*/

int DoPreReset(UBYTE *s)
{
    UBYTE *ss, c;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s == 0 ) {
        MesPrint("@proper syntax is #Reset variable");
        return(-1);
    }
    ss = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    if ( ( StrICmp(ss,(UBYTE *)"timer") == 0 )
      || ( StrICmp(ss,(UBYTE *)"stopwatch") == 0 ) ) {
        *s = c;
        AP.StopWatchZero = GetRunningTime();
        return(0);
    }
    else {
        *s = c;
        MesPrint("@proper syntax is #Reset variable");
        return(-1);
    }
}

/*
        #] DoPreReset :
        #[ DoPreAppendPath :
*/

static int DoAddPath(UBYTE *s, int bPrepend)
{
    /* NOTE: this doesn't support some file systems, e.g., 0x5c with CP932. */

    UBYTE *path, *path_end, *current_dir, *current_dir_end, *NewPath, *t;
    int bRelative, n;

    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);

    /* Parse the path in the input. */
    while ( *s == ' ' || *s == '\t' ) s++;  /* skip spaces */
    if ( *s == '"' ) {  /* the path is given by "..." */
        path = ++s;
        while ( *s && *s != '"' ) {
            if ( SEPARATOR != '\\' && *s == '\\' ) {  /* escape character, e.g., "\\"" */
                if ( !s[1] ) goto ImproperPath;
                s++;
            }
            s++;
        }
        if ( *s != '"' ) goto ImproperPath;
        path_end = s++;
    }
    else {
        path = s;
        while ( *s && *s != ' ' && *s != '\t' ) {
            if ( SEPARATOR != '\\' && *s == '\\' ) {  /* escape character, e.g., "\\ " */
                if ( !s[1] ) goto ImproperPath;
                s++;
            }
            s++;
        }
        path_end = s;
    }
```

```
    if ( path == path_end ) goto ImproperPath;  /* empty path */
    while ( *s == ' ' || *s == '\t' ) s++;  /* skip spaces */
    if ( *s ) goto ImproperPath;  /* extra tokens found */

    /* Check if the path is an absolute path. */
    bRelative = 1;
    if ( path[0] == SEPARATOR ) {  /* starts with the directory separator */
        bRelative = 0;
    }
#ifdef WINDOWS
    else if ( chartype[path[0]] == 0 && path[1] == ':' ) {  /* starts with (drive letter): */
        bRelative = 0;
    }
#endif

    /* Get the current file directory when a relative path is given. */
    if ( bRelative ) {
        if ( !AC.CurrentStream ) goto FileNameUnavailable;
        if ( AC.CurrentStream->type != FILESTREAM && AC.CurrentStream->type != REVERSEFILESTREAM ) goto FileNameUnavailable;
        if ( !AC.CurrentStream->name ) goto FileNameUnavailable;
        s = current_dir = current_dir_end = AC.CurrentStream->name;
        while ( *s ) {
            if ( SEPARATOR != '\\' && *s == '\\' && s[1] ) {  /* escape character, e.g., "\\"" */
                s += 2;
                continue;
            }
            if ( *s == SEPARATOR ) {
                current_dir_end = s;
            }
            s++;
        }
    }
    else {
        current_dir = current_dir_end = NULL;
    }

    /* Allocate a buffer for new AM.Path. */
    n = path_end - path;
    if ( AM.Path ) n += StrLen(AM.Path) + 1;
    if ( current_dir != current_dir_end ) n+= current_dir_end - current_dir + 1;
    s = NewPath = (UBYTE *)Malloc1(n + 1,"add path");

    /* Construct new FORM path. */
    if ( bPrepend ) {
        if ( current_dir != current_dir_end ) {
            t = current_dir;
            while ( t != current_dir_end ) *s++ = *t++;
            *s++ = SEPARATOR;
        }
        t = path;
        while ( t != path_end ) *s++ = *t++;
        if ( AM.Path ) *s++ = PATHSEPARATOR;
    }
    if ( AM.Path ) {
        t = AM.Path;
        while ( *t ) *s++ = *t++;
    }
    if ( !bPrepend ) {
        if ( AM.Path ) *s++ = PATHSEPARATOR;
        if ( current_dir != current_dir_end ) {
            t = current_dir;
            while ( t != current_dir_end ) *s++ = *t++;
            *s++ = SEPARATOR;
        }
        t = path;
        while ( t != path_end ) *s++ = *t++;
    }
    *s = '\0';

    /* Update AM.Path. */
    if ( AM.Path ) M_free(AM.Path,"add path");
    AM.Path = NewPath;

    return(0);

ImproperPath:
    MesPrint("@Improper syntax for %#%sPath", bPrepend ? "Prepend" : "Append");
    return(-1);

FileNameUnavailable:
    /* This may be improved in future. */
    MesPrint("@Sorry, %#%sPath can't resolve the current file name from here", bPrepend ? "Prepend" : "Append");
    return(-1);
}
```

```
int DoPreAppendPath(UBYTE *s)
{
    return DoAddPath(s, 0);
}

/*
        #] DoPreAppendPath :
        #[ DoPrePrependPath :
*/

int DoPrePrependPath(UBYTE *s)
{
    return DoAddPath(s, 1);
}

/*
        #] DoPrePrependPath :
        #[ DoTimeOutAfter :

        Executes the #timeoutafter number
*/

int DoTimeOutAfter(UBYTE *s)
{
#ifdef WITH_ALARM
    ULONG x;
#else
    DUMMYUSE(s);
#endif
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
#ifdef WITH_ALARM
    while ( *s == ' ' || *s == '\t' ) s++;
    x = 0;
    while ( FG.cTable[*s] == 1 ) {
        x = 10*x + (*s++-'0');
    }
    while ( *s == ' ' || *s == '\t' ) s++;
    if ( *s  == 0 ) {
        alarm(x);
        return(0);
    }
    else {
        MesPrint("@proper syntax is #TimeoutAfter number");
        return(-1);
    }
#else
    Error0("#timeoutafter not implemented on this computer/system");
    return(-1);
#endif
}

/*
        #] DoTimeOutAfter :
        #[ DoNamespace :

        Syntax:
            #Namespace name
                .....
            #use variables
                .....
            #EndNamespace
        Effect:
            All variables/expressions defined inside the range of the
            namespace get name_ prepended.
            This holds also for $-variables, names of procedures and
            names of files.
        Namespaces can be used in a nested way. cf this_is_deep_x
        A leading _ takes the role of what is super:: in some other languages.
        Remarks:
            Names of preprocessor variables are excluded!
            Names of built in objects are excluded! (like sum_, d_ etc.)
*/

int DoNamespace(UBYTE *s)
{
    UBYTE *s1, *s2, c;
    NAMESPACE *namespace;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( FG.cTable[*s] != 0 ) {
        MesPrint("@Illegal name in #namespace instruction: %s",s);
        return(-1);
    }
    s1 = s;
    while ( FG.cTable[*s1] <= 1 ) s1++;
```

```
    s2 = s1;
    while ( *s2 == ' ' || *s2 == ',' || *s2 == '\t' ) s2++;
    if ( *s2 != 0 ) {
        MesPrint("@A #namespace instruction can only have one name with only alphanumeric
characters.");
        return(-1);
    }
    c = *s1; *s1 = 0;
/*
    Now we have the name and the statement is legal.
    We can proceed creating the namespace and its use tree.
*/
    namespace = (NAMESPACE *)Malloc1(sizeof(NAMESPACE),"namespace");
    namespace->name = strDup1(s,"namespace_name");
    namespace->usenames = MakeNameTree();
    if ( AP.firstnamespace == 0 ) {
        namespace->previous = 0;
        namespace->next = 0;
        AP.firstnamespace = namespace;
        AP.lastnamespace = namespace;
    }
    else {
        AP.lastnamespace->next = namespace;
        namespace->next = 0;
        namespace->previous = AP.lastnamespace;
        AP.lastnamespace = namespace;
    }
    *s1 = c;
    return(0);
}

/*
        #] DoNamespace :
        #[ DoEndNamespace :
*/

int DoEndNamespace(UBYTE *s)
{
    NAMESPACE *namespace;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    if ( *s != 0 ) {
        MesPrint("@Illegal #endnamespace instruction");
        return(-1);
    }
    namespace = AP.lastnamespace;
    AP.lastnamespace = namespace->previous;
    M_free(namespace->name,"namespace_name");
    FreeNameTree(namespace->usenames);
    M_free(namespace,"namespace");
    return(0);
}

/*
        #] DoEndNamespace :
        #[ SkipName :
*/

UBYTE *SkipName(UBYTE *s)
{
    UBYTE *t = s, *s1, c;
    int num = 0, block = 0;
    if ( *s == '[' ) {
straight:
        SKIPBRA1(s)
        if ( *s == 0 ) {
            MesPrint("&Illegal name: '%s'",t);
            return(0);
        }
        s++; s1 = s;
        while ( FG.cTable[*s] <= 1 || *s == '_' ) s++;
        if ( s1 != s ) goto witherror;
    }
    else if ( *s == '$' ) {
        s++;
        while ( *s == '_' ) { s++; num++; }
        block = 1;
        while ( FG.cTable[*s] <= 1 || *s == '_' ) {
            if ( FG.cTable[*s] != 0 && block == 1 ) {
blocked:
                MesPrint("&Illegally formed name: %s",t);
                return(0);
            }
            if ( *s == '_' ) { num++; block = 1; }
            else block = 0;
            s++;
```

```
        }
        if ( s[-1] == '_' && num > 1 ) goto built;
    }
    else if ( FG.cTable[*s] == 0 ) {
regular:
        while ( FG.cTable[*s] <= 1 || *s == '_' ) {
            if ( FG.cTable[*s] != 0 && block == 1 ) goto blocked;
            if ( *s == '_' ) { block = 1; num++; }
            else block = 0;
            s++;
        }
        if ( *s == '[' ) goto straight;
        if ( s[-1] == '_' && num > 1 ) {
built:
            c = *s; *s = 0;
            MesPrint("&Built in objects cannot be part of namespaces: %s",t);
            *s = c;
            return(0);
        }
    }
    else if ( *s == '_' ) {
        while ( *s == '_' ) { s++; num++; }
        if ( FG.cTable[*s] == 0 ) { block = 0; goto regular; }
        goto witherror;
    }
    else if ( *s == '@' ) {
        s++; block = 1;
        if ( *s == '_' || FG.cTable[*s] == 1 ) {
            MesPrint("@Illegally formed name: %s",s-1);
        }
        goto regular;
    }
    else if ( *s == '#' ) { /* name of a procedure */
        s++;
        while ( *s == '_' ) { s++; num++; }
        block = 1;
        while ( FG.cTable[*s] <= 1 || *s == '_' ) {
            if ( FG.cTable[*s] != 0 && block == 1 ) goto blocked;
            if ( *s == '_' ) { num++; block = 1; }
            else block = 0;
            s++;
        }
        if ( s[-1] == '_' ) {
witherror:
            c = *s; *s = 0;
            MesPrint("&Illegally formed name: %s",t);
            *s = c;
            return(0);
        }
    }
    else if ( *s == '<' ) { /* name of a file. Can be anything (more or less) */
        s++;
        while ( *s && *s != '>' ) s++;
        if ( *s != '>' ) goto witherror;
        s++;
    }
    return(s);
}

/*
        #] SkipName :
        #[ ConstructName :

        Routine gets a 'raw' name and modifies it if the namespace
        settings ask for it. It puts the new name in a buffer that
        may be expanded if the names become rather long.
        Note that eventually that name needs to be copied, because
        we do not allocate new buffers for each name.

        type tells what kind of name we look for
*/

UBYTE *ConstructName(UBYTE *s,UBYTE type)
{
    int len;
    UBYTE *t, *u;
    WORD number;
    NAMESPACE *namespace;
    if ( AP.lastnamespace == 0 ) return(s);
    if ( *s == '@' ) return(s+1);
    if ( GetName(AP.lastnamespace->usenames,s,&number,NOAUTO) !=
            NAMENOTFOUND ) return(s);
/*
    Now the real stuff
    First we have to compute the size of the new name.
*/
    len = StrLen(s) + 1;
```

```
    namespace = AP.firstnamespace;
    while ( namespace ) {
        len += StrLen(namespace->name)+1;
        namespace = namespace->previous;
    }
    if ( len > AP.fullnamesize ) {
        while ( len > AP.fullnamesize ) AP.fullnamesize *= 2;
        M_free(AP.fullname,"AP.fullname");
        AP.fullname = (UBYTE *)Malloc1(AP.fullnamesize*sizeof(UBYTE *),"AP.fullname");
    }
    namespace = AP.firstnamespace;
    t = AP.fullname;
    switch ( type ) {
        case ' ':
        case 0:
            while ( namespace ) {
                u = namespace->name;
                while ( *u ) *t++ = *u++;
                *t++ = '_';
                namespace = namespace->previous;
            }
            while ( *s ) *t++ = *s++;
            *t = 0;
            break;
        case '$':
        case '#':
            *t++ = type;
            while ( namespace ) {
                u = namespace->name;
                while ( *u ) *t++ = *u++;
                *t++ = '_';
                namespace = namespace->previous;
            }
            if ( type == '$' ) s++;
            while ( *s ) *t++ = *s++;
            *t = 0;
            break;
        case '<':
            while ( namespace ) {
                u = namespace->name;
                while ( *u ) *t++ = *u++;
                *t++ = '_';
                namespace = namespace->previous;
            }
            s++;
            while ( *s ) *t++ = *s++;
            t--;  /* strip the '>' */
            *t = 0;
            break;
        default:
            MesPrint("&Unrecognized datatype in ConstructName");
            *t = 0;
            break;
    }
    return(AP.fullname);
}

/*
        #] ConstructName :
        #[ DoUse :

        Routine makes (inside the confines of the current namespace)
        a list of variables that are excluded from the namespace.
        Once the namespace is ended, the list is removed.
        The list can include names of variables, dollars and procedures.
        Preprocessor variables are excluded from the namespace. Their
        inclusion would be too complicated for the input streams.
        Note that the preprocessor variables that are arguments in a #do
        or in a procedure are on a stack and should not cause problems.
        Names of variables can be just that.
        Names of $-variables are also straightforward.
        Names of procedures should be preceeded by a # character.
        Names of files (like in #include) should be enclosed by <>.
        The names are stored in a balanced tree. Each namespace may have
        its own tree. The toplevel (no namespace) does not allow a #use.
*/

int DoUse(UBYTE *s)
{
    NAMESPACE *namespace;
    UBYTE *t, c;
    int number;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    if ( AP.lastnamespace == 0 ) {
        MesPrint("@It is not allowed to use #use outside the scope of a namespace.");
        return(-1);
```

```
    }
    namespace = AP.lastnamespace;
    while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    while ( *s ) {
        t = s;
        if ( ( s = SkipName(t) ) == 0 ) return(-1);
        if ( s == t ) {
            MesPrint("@Unrecognized object in #use instruction: %s",t);
            return(-1);
        }
        c = *s; *s = 0;
/*
        In usenames we only need the names to know whether they are 'protected'.
        We need to keep the $, # and <> to avoid potential double names.
*/
        AddName(namespace->usenames,t,0,0,&number);
        *s = c;
        while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    }
    return(0);
}

/*
        #] DoUse :
        #[ UserFlags :

    Syntax:
        #ClearFlag number(s),expression(s)
        #ClearFlag number(s)
        #ClearFlag expression(s)
        #ClearFlag
        #SetFlag number(s),expression(s)
        #SetFlag number(s)
        #SetFlag expression(s)
        #SetFlag
        par == 0: Clear, par == 1: Set.
*/

int UserFlags(UBYTE *s,int par)
{
    int mask = 0, error = 0, i;
    if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
    if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    if ( *s == 0 ) {    /* Treat all flags in all active expressions */
allexpr:
        for ( i = 0; i < NumExpressions; i++ ) {
            switch ( Expressions[i].status ) {
                case UNHIDELEXPRESSION:
                case UNHIDEGEXPRESSION:
                case INTOHIDELEXPRESSION:
                case INTOHIDEGEXPRESSION:
                case LOCALEXPRESSION:
                case GLOBALEXPRESSION:
                case SKIPLEXPRESSION:
                case SKIPGEXPRESSION:
                case HIDELEXPRESSION:
                case HIDEGEXPRESSION:
                    if ( par == 1 ) Expressions[i].uflags |= ~mask;
                    else            Expressions[i].uflags &= mask;
                    break;
                case DROPPEDEXPRESSION:
                case DROPLEXPRESSION:
                case DROPGEXPRESSION:
                case DROPHLEXPRESSION:
                case DROPHGEXPRESSION:
                case STOREDEXPRESSION:
                case HIDDENLEXPRESSION:
                case HIDDENGEXPRESSION:
                case SPECTATOREXPRESSION:
                default:
                    break;
            }
        }
    }
    else if ( FG.cTable[*s] == 1 ) {
        mask = (int)WORDMASK;
        while ( FG.cTable[*s] == 1 ) {
            int x = 0;
            while ( FG.cTable[*s] == 1 ) { x = 10*x + (*s++-'0'); }
            if ( x < 1 || x > BITSINWORD ) {
                MesPrint("@Illegal number %d for flag in #...Flag instruction",x);
                return(1);
            }
            mask ^= (1«(x-1));
            while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
        }
```

```
        if ( *s == 0 ) goto allexpr;
    }
    else {  /* Clear all flags in all expressions that are specified */
        mask = (int)WORDMASK;
    }
    while ( *s ) { /* now read the expressions */
        UBYTE *s1, c;
        WORD num1;
        if ( FG.cTable[*s] != 0 && *s != '[' ) goto syntax;
        s1 = s; s = SkipAName(s);
        c = *s; *s = 0;
        if ( GetName(AC.exprnames,s1,&num1,NOAUTO) != CEXPRESSION ) {
            MesPrint("@%s is not an active expression",s1);
            error = 1;
            return(error);
        }
        switch ( Expressions[num1].status ) {
            case UNHIDELEXPRESSION:
            case UNHIDEGEXPRESSION:
            case INTOHIDELEXPRESSION:
            case INTOHIDEGEXPRESSION:
            case LOCALEXPRESSION:
            case GLOBALEXPRESSION:
            case SKIPLEXPRESSION:
            case SKIPGEXPRESSION:
            case HIDELEXPRESSION:
            case HIDEGEXPRESSION:
                if ( par == 1 ) Expressions[num1].uflags |= ~mask;
                else            Expressions[num1].uflags &= mask;
                break;
            case DROPPEDEXPRESSION:
            case DROPLEXPRESSION:
            case DROPGEXPRESSION:
            case DROPHLEXPRESSION:
            case DROPHGEXPRESSION:
            case STOREDEXPRESSION:
            case HIDDENLEXPRESSION:
            case HIDDENGEXPRESSION:
            case SPECTATOREXPRESSION:
            default:
                MesPrint("@%s is not an active expression",s1);
                error = 1;
                break;
        }
        *s = c;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    }
    return(error);
syntax:
    MesPrint("@Illegal name in #...Flag instruction.");
    return(1);
}

/*
        #] UserFlags :
        #[ DoClearUserFlag :
*/

int DoClearUserFlag(UBYTE *s)
{
    return(UserFlags(s,0));
}

/*
        #] DoClearUserFlag :
        #[ DoSetUserFlag :
*/

int DoSetUserFlag(UBYTE *s)
{
    return(UserFlags(s,1));
}

/*
        #] DoSetUserFlag :
        #[ DoStartFloat :

        If there is a number follwing, it will be the new default precision.
        If float has been started before, the old one will be removed first.
        If there are two numbers, the second one is the maximum weight for
        MZV's.
*/
#ifdef WITHFLOAT

int DoStartFloat(UBYTE *s)
{
    GETIDENTITY
```

```
	int error = 0;
	LONG x;
	if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
	if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
	if ( AR.PolyFun != 0 ) {
		MesPrint("@Simultaneous use of Poly(Rat)Fun and float_ is not allowed.");
		error = 1;
	}
	if ( AC.ncmod != 0 ) {
		MesPrint("@Simultaneous use of floating point and modulus arithmetic makes no sense.");
		error = 1;
	}
	while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
	if ( *s >= '0' && *s <= '9' ) {
		x = 0;
		do {
			x = 10*x + (*s++-'0');
		} while ( *s >= '0' && *s <= '9' );
		AC.tDefaultPrecision = x;
		while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
		if ( *s >= '0' && *s <= '9' ) {
			x = 0;
			do {
				x = 10*x + (*s++ - '0');
			} while ( *s >= '0' && *s <= '9' );
			AC.tMaxWeight = x;
			while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
		}
		else {
			AC.tMaxWeight = 0;
		}
		if ( *s ) goto IllPar;
	}
	else if ( *s != 0 ) {
IllPar:
		MesPrint("@Illegal parameter in %#StartFloat instruction: %s ",s);
		error = 1;
	}
	if ( error == 0 ) {
		if ( AC.tDefaultPrecision && ( AC.tDefaultPrecision != AC.DefaultPrecision
		|| AT.aux_ == 0 ) ) {
			AC.DefaultPrecision = AC.tDefaultPrecision;
			AC.tDefaultPrecision = 0;
		}
		if ( AC.tMaxWeight && ( AC.tMaxWeight != AC.MaxWeight
		|| AT.aux_ == 0 ) ) {
			AC.MaxWeight = AC.tMaxWeight;
			AC.tMaxWeight = 0;
		}
		SetFloatPrecision(AC.DefaultPrecision+AC.MaxWeight+1);
		SetupMPFTables();
		if ( AC.MaxWeight > 0 ) SetupMZVTables();
		SetfFloatPrecision(AC.DefaultPrecision);
	}
	else {
		AC.tDefaultPrecision = 0;
		AC.tMaxWeight = 0;
	}
	return(error);
}

#endif
/*
		#] DoStartFloat :
		#[ DoEndFloat :
*/
#ifdef WITHFLOAT

int DoEndFloat(UBYTE *s)
{
	int error = 0;
	if ( AP.PreSwitchModes[AP.PreSwitchLevel] != EXECUTINGPRESWITCH ) return(0);
	if ( AP.PreIfStack[AP.PreIfLevel] != EXECUTINGIF ) return(0);
	while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
	if ( *s != 0 ) {
		MesPrint("@Illegal parameter in %#EndFloat instruction: %s ",s);
		error = 1;
	}
	if ( error == 0 ) {
		ClearfFloat();
		ClearMZVTables();
	}
	return(error);
}

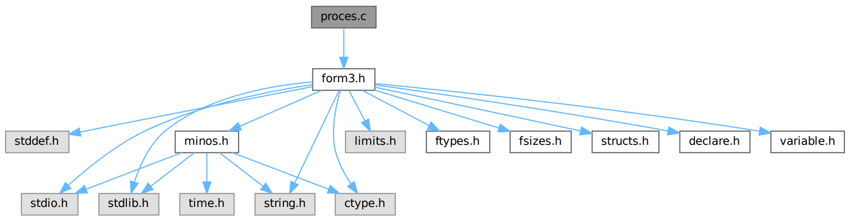
#endif
/*
```

```
          #] DoEndFloat :
      # ] PreProcessor :
*/
```

4.114 proces.c File Reference

`#include "form3.h"`

Include dependency graph for proces.c:



Macros

- #define DONE(x) { retvalue = x; goto Done; }

Functions

- WORD Processor (void)
- WORD TestSub (PHEAD WORD ∗term, WORD level)
- WORD InFunction (PHEAD WORD ∗term, WORD ∗termout)
- WORD InsertTerm (PHEAD WORD ∗term, WORD replac, WORD extractbuff, WORD ∗position, WORD ∗termout, WORD tepos)
- LONG PasteFile (PHEAD WORD number, WORD ∗accum, POSITION ∗position, WORD ∗∗accfill, RENUMBER renumber, WORD ∗freeze, WORD nexpr)
- WORD ∗ PasteTerm (PHEAD WORD number, WORD ∗accum, WORD ∗position, WORD times, WORD divby)
- WORD FiniTerm (PHEAD WORD ∗term, WORD ∗accum, WORD ∗termout, WORD number, WORD tepos)
- WORD Generator (PHEAD WORD ∗term, WORD level)
- WORD DoOnePow (PHEAD WORD ∗term, WORD power, WORD nexp, WORD ∗accum, WORD ∗aa, WORD level, WORD ∗freeze)
- WORD Deferred (PHEAD WORD ∗term, WORD level)
- WORD PrepPoly (PHEAD WORD ∗term, WORD par)
- WORD PolyFunMul (PHEAD WORD ∗term)

Variables

- WORD printscratch [2]

4.114.1 Detailed Description

Contains the central terms processor routines. This is the core of the virtual machine. All other files are to help these routines.

Definition in file proces.c.

4.114.2 Macro Definition Documentation

4.114.2.1 DONE

```
#define DONE(
             x ) { retvalue = x; goto Done; }
```

TestSub hunts for subexpression pointers. If one is found its power is given in AN.TeSuOut. and the returnvalue is 'expressionnumber'. If the expression number is negative it is an expression on disk.

In addition this routine tries to locate subexpression pointers in functions. It also notices that action must be taken with any of the special functions.

Parameters

term	The term in which TestSub hunts for potential action
level	The number of the 'level' in the compiler buffer.

Returns

The number of the (sub)expression that was encountered.

Other values that are returned are in AN.TeSuOut, AR.TePos, AT.TMbuff, AN.TeInFun, AN.Frozen, AT.TMaddr

The level in the compiler buffer is more or less the number of the statement in the module. Hence it refers to the element in the lhs array.

This routine is one of the most important routines in FORM.

Definition at line 682 of file proces.c.

4.114.3 Function Documentation

4.114.3.1 Processor()

```
WORD Processor (
            void )
```

This is the central processor. It accepts a stream of Expressions which is accessed by calls to GetTerm. The expressions reside either in AR.infile or AR.hidefile The definitions of an expression are seen as an id-statement, so the primary Expressions should be written to the system of scratch files as single terms with an expression pointer. Each expression is terminated with a zero and the whole is terminated by two zeroes.

The routine DoExecute should determine whether results are to be printed, should revert the scratch I/O directions etc. In principle it is DoExecute that calls Processor.

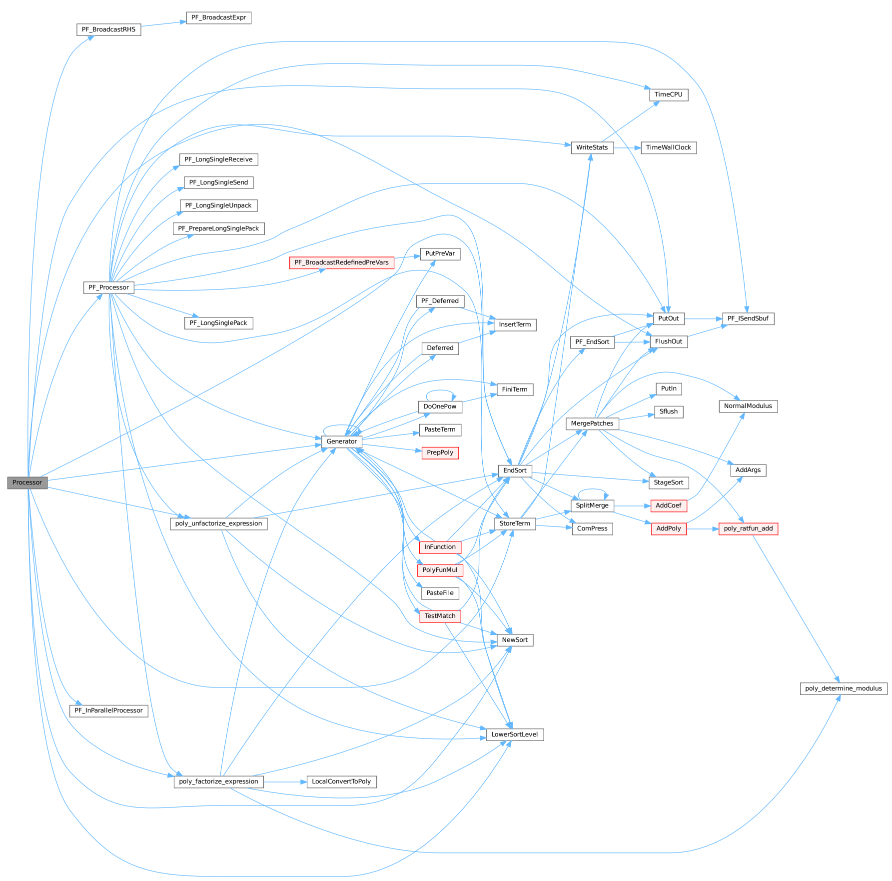
Returns

if everything OK: 0. Otherwise error. The preprocessor may continue with compilation though. Really fatal errors should return on the spot by calling Terminate.

Definition at line 64 of file proces.c.

References CbUf::Buffer, EndSort(), FlushOut(), Generator(), CbUf::lhs, LowerSortLevel(), NewSort(), PF_BroadcastRHS(), PF_InParallelProcessor(), PF_Processor(), CbUf::Pointer, poly_factorize_expression(), poly_unfactorize_expression(), PutOut(), CbUf::rhs, and StoreTerm().

Here is the call graph for this function:



4.114.3.2 TestSub()

```
WORD TestSub (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 684 of file proces.c.

4.114.3.3 InFunction()

```
WORD InFunction (
            PHEAD WORD * term,
            WORD * termout )
```

Makes the replacement of the subexpression with the number 'replac' in a function argument. Additional information is passed in some of the AR, AN, AT variables.

Parameters

term	The input term
termout	The output term

Returns

0: everything is fine, Negative: fatal, Positive: error.

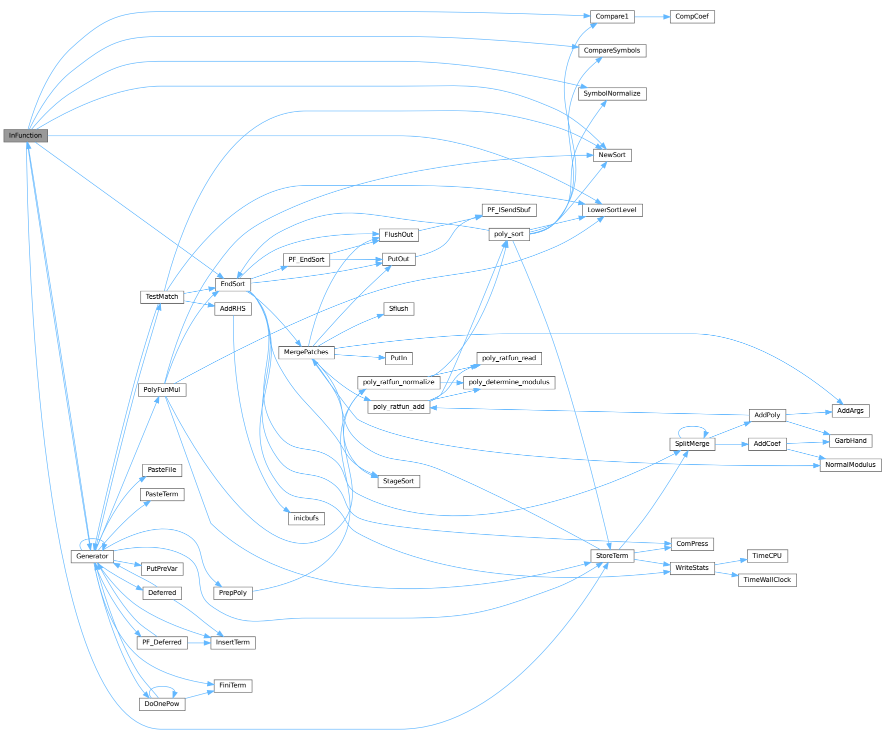
Special attention should be given to nested functions!

Definition at line 2174 of file proces.c.

References Compare1(), CompareSymbols(), EndSort(), Generator(), LowerSortLevel(), NewSort(), StoreTerm(), and SymbolNormalize().

Referenced by Generator().

Here is the call graph for this function:



4.114.3.4 InsertTerm()

```
WORD InsertTerm (
            PHEAD WORD * term,
            WORD replac,
            WORD extractbuff,
            WORD * position,
```

```
WORD * termout,
WORD tepos )
```

Puts the terms 'term' and 'position' together into a single legal term in termout. replac is the number of the subexpression that should be replaced. It must be a positive term. When action is needed in the argument of a function all terms in that argument are dealt with recursively. The subexpression is sorted. Only one subexpression is done at a time this way.

Parameters

term	the input term
replac	number of the subexpression pointer to replace
extractbuff	number of the compiler buffer replac refers to
position	position from where to take the term in the compiler buffer
termout	the output term
tepos	offset in term where the subexpression is.

Returns

Normal conventions (OK = 0).

Definition at line 2744 of file proces.c.

Referenced by Deferred(), Generator(), and PF_Deferred().

4.114.3.5 PasteFile()

```
LONG PasteFile (
            PHEAD WORD number,
            WORD * accum,
            POSITION * position,
            WORD ** accfill,
            RENUMBER renumber,
            WORD * freeze,
            WORD nexpr )
```

Gets a term from stored expression expr and puts it in the accumulator at position number. It returns the length of the term that came from file.

Parameters

number	number of partial terms to skip in accum
accum	the accumulator
position	file position from where to get the stored term
accfill	returns tail position in accum
renumber	the renumber struct for the variables in the stored expression
freeze	information about if we need only the contents of a bracket
nexpr	the number of the stored expression

Returns

Normal conventions (OK = 0).

Definition at line 2880 of file proces.c.

Referenced by Generator().

4.114.3.6 PasteTerm()

```
WORD * PasteTerm (
            PHEAD WORD number,
            WORD * accum,
            WORD * position,
            WORD times,
            WORD divby )
```

Puts the term at position in the accumulator accum at position 'number+1'. if times $>$ 0 the coefficient of this term is multiplied by times/divby.

Parameters

number	The number of term fragments in accum that should be skipped
accum	The accumulator of term fragments
position	A position in (typically) a compiler buffer from where a (piece of a) term comes.
times	Multiply the result by this
divby	Divide the result by this.

This routine is typically used when we have to replace a (sub)expression pointer by a power of a (sub)expression. This uses mostly a binomial expansion and the new term is the old term multiplied one by one by terms of the new expression. The factors times and divby keep track of the binomial coefficient. Once this is complete, the routine FiniTerm will make the contents of the accumulator into a proper term that still needs to be normalized.

Definition at line 3002 of file proces.c.

Referenced by Generator().

4.114.3.7 FiniTerm()

```
WORD FiniTerm (
            PHEAD WORD * term,
            WORD * accum,
            WORD * termout,
            WORD number,
            WORD tepos )
```

Concatenates the contents of the accumulator into a single legal term, which replaces the subexpression pointer

Parameters

term	the input term with the (sub)expression subterm
accum	the accumulator with the term fragments
termout	the location where the output should be written
number	the number of term fragments in the accumulator
tepos	the position of the subterm in term to be replaced

Definition at line 3067 of file proces.c.

Referenced by DoOnePow(), and Generator().

4.114.3.8 Generator()

```
WORD Generator (
            PHEAD WORD * term,
            WORD level )
```

The heart of the program. Here the expansion tree is set up in one giant recursion

Parameters

term	the input term. may be overwritten
level	the level in the compiler buffer (number of statement)

Returns

Normal conventions (OK = 0).

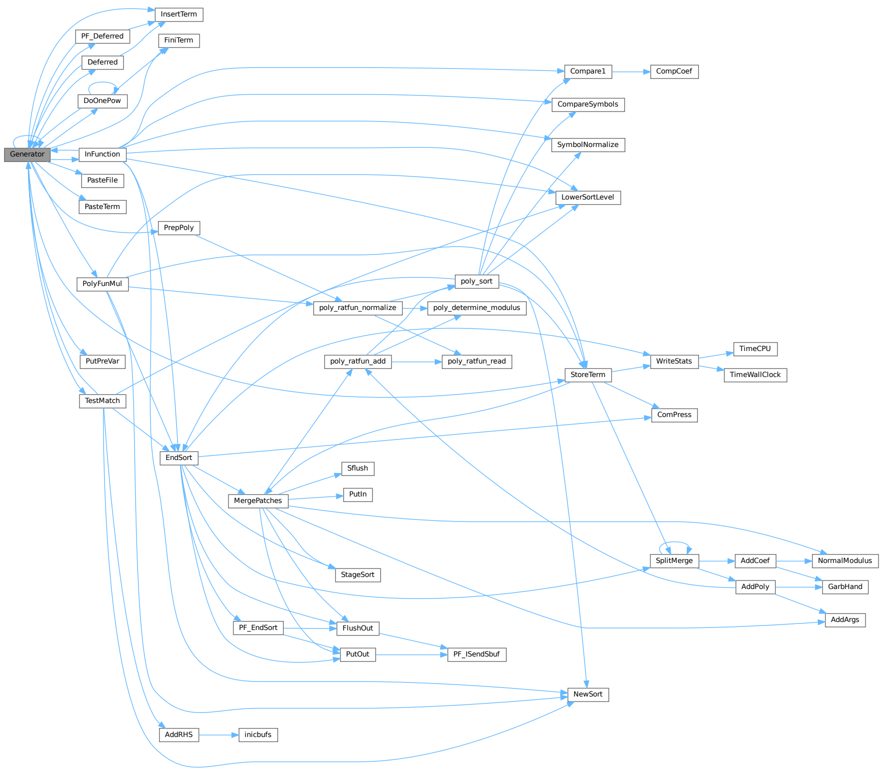
The routine looks first whether there are unsubstituted (sub)expressions. If so, one of them gets inserted term by term and the new term is used in a renewed call to Generator. If there are no (sub)expressions, the term is normalized, the compiler level is raised (next statement) and the program looks what type of statement this is. If this is a special statement it is either treated on the spot or the appropriate routine is called. If it is a substitution, the pattern matcher is called (TestMatch) which tells whether there was a match. If so we need to call TestSub again to test for (sub)expressions. If we run out of levels, the term receives a final treatment for modulus calculus and/or brackets and is then sent off to the sorting routines.

Definition at line 3266 of file proces.c.

References Deferred(), DoOnePow(), FiniTerm(), Generator(), InFunction(), InsertTerm(), CbUf::lhs, VaRrEnUm::lo, PasteFile(), PasteTerm(), PF_Deferred(), CbUf::Pointer, PolyFunMul(), PrepPoly(), PutPreVar(), CbUf::rhs, StoreTerm(), ReNuMbEr::symb, and TestMatch().

Referenced by Deferred(), DoOnePow(), generate_expression(), Generator(), InFunction(), MakeDollarInteger(), MakeDollarMod(), PF_CollectModifiedDollars(), PF_Deferred(), PF_Processor(), poly_factorize_expression(), poly_unfactorize_expression(), Processor(), and TestMatch().

Here is the call graph for this function:



4.114.3.9 DoOnePow()

```
WORD DoOnePow (
            PHEAD WORD * term,
            WORD power,
            WORD nexp,
            WORD * accum,
            WORD * aa,
            WORD level,
            WORD * freeze )
```

Routine gets one power of an expression in the scratch system. If there are more powers needed there will be a recursion.

No attempt is made to use binomials because we have no information about commuting properties.

There is a searching for the contents of brackets if needed. This searching may be rather slow because of the single links.

Parameters

term	is the term we are adding to.
power	is the power of the expression that we need.
nexp	is the number of the expression.
accum	is the accumulator of terms. It accepts the termfragments that are made into a proper term in FiniTerm

Parameters

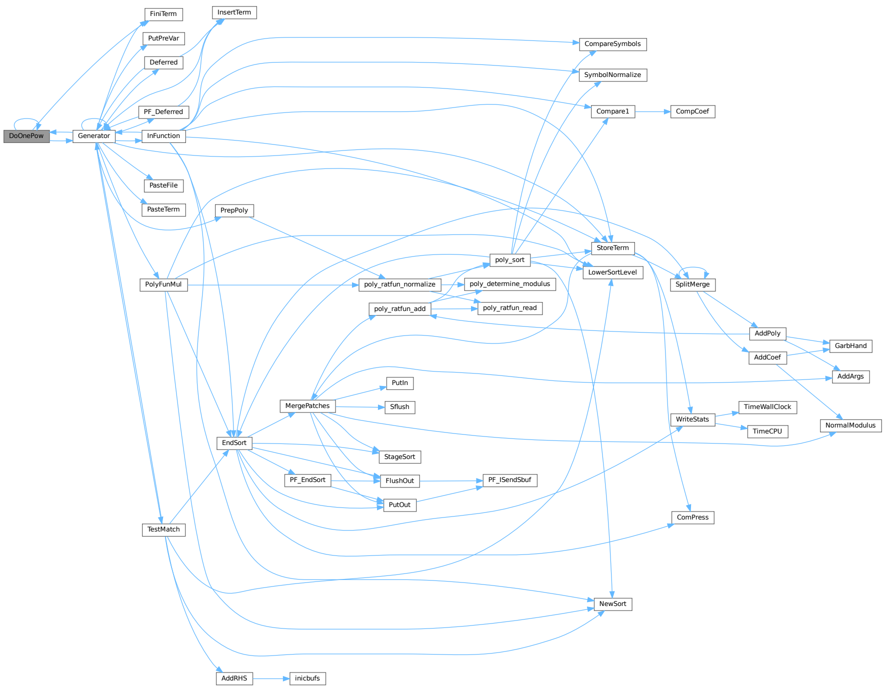
aa	points to the start of the entire accumulator. In ∗aa we store the number of term fragments that are in the accumulator.
level	is the current depth in the tree of statements. It is needed to continue to the next operation/substitution with each generated term
freeze	is the pointer to the bracket information that should be matched.

Definition at line 4623 of file proces.c.

References DoOnePow(), FiniTerm(), Generator(), and FiLe::handle.

Referenced by DoOnePow(), and Generator().

Here is the call graph for this function:



4.114.3.10 Deferred()

```
WORD Deferred (
            PHEAD WORD * term,
            WORD level )
```

Picks up the deferred brackets. These are the bracket contents of which we postpone the reading when we use the 'Keep Brackets' statement. These contents are multiplying the terms just before they are sent to the sorting system. Special attention goes to having it thread-safe We have to lock positioning the file and reading it in a thread specific buffer.

Parameters

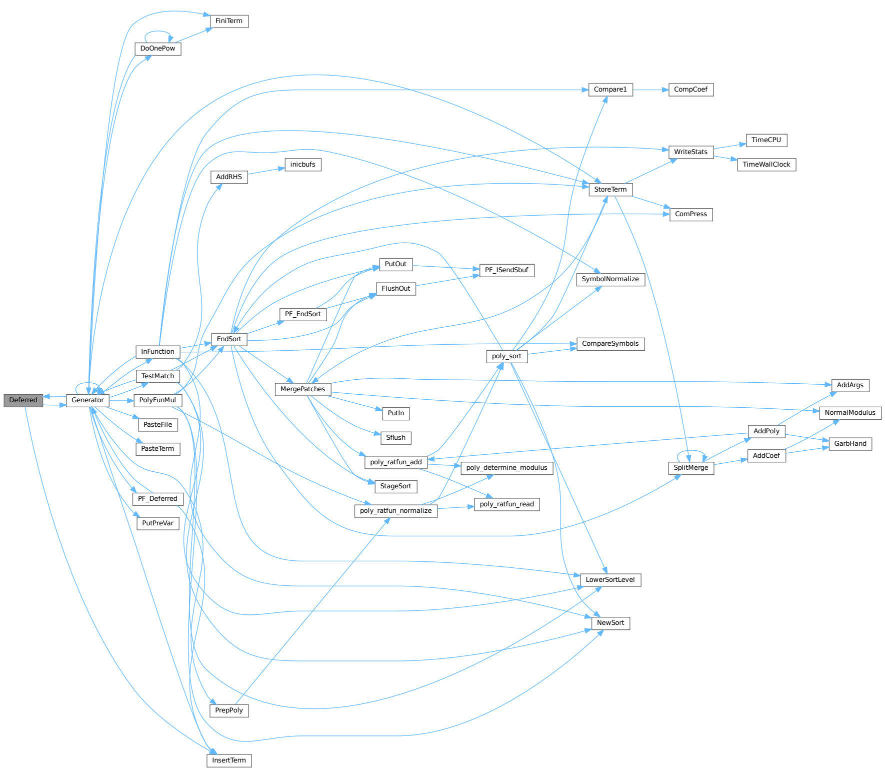
term	The term that must be multiplied by the contents of the current bracket
level	The compiler level. This is needed because after multiplying term by term we call Generator again.

Definition at line 4844 of file proces.c.

References Generator(), and InsertTerm().

Referenced by Generator().

Here is the call graph for this function:



4.114.3.11 PrepPoly()

```
WORD PrepPoly (
            PHEAD WORD * term,
            WORD par )
```

Routine checks whether the count of function AR.PolyFun is zero or one. If it is one and it has one scalarlike argument the coefficient of the term is pulled inside the argument. If the count is zero a new function is made with the coefficient as its only argument. The function should be placed at its proper position.

When this function is active it places the PolyFun as last object before the coefficient. This is needed because otherwise the compress algorithm has problems in MergePatches.

The bracket routine should also place the PolyFun at a comparable spot. The compression should then stop at the PolyFun. It doesn't really have to stop when writing the final result but this may be too complicated.

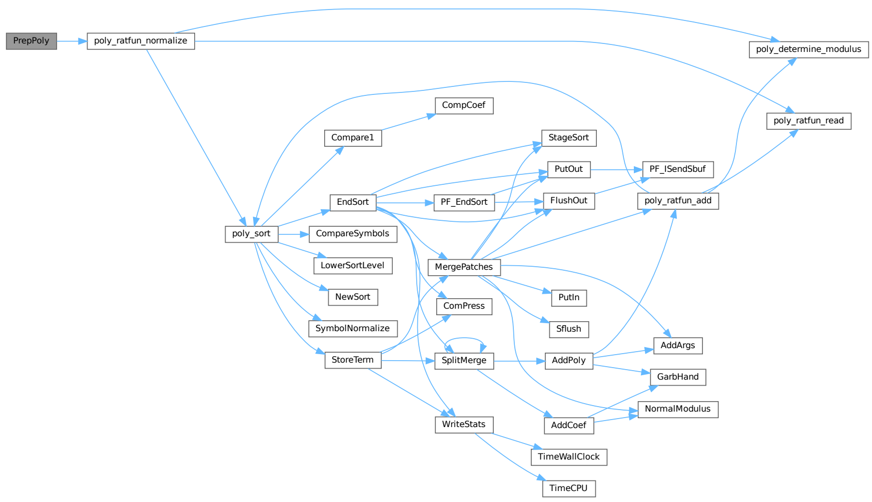
The parameter par tells whether we are at groundlevel or inside a function or dollar variable.

Definition at line 4972 of file proces.c.

References poly_ratfun_normalize().

Referenced by Generator().

Here is the call graph for this function:



4.114.3.12 PolyFunMul()

```
WORD PolyFunMul (
            PHEAD WORD * term )
```

Multiplies the arguments of multiple occurrences of the polyfun. In this routine we do the original PolyFun with one argument only. The PolyRatFun (PolyFunType = 2) is done in a dedicated routine in the file polywrap.cc The new result is written over the old result.

Parameters

term	It contains the input term and later the output.

Returns

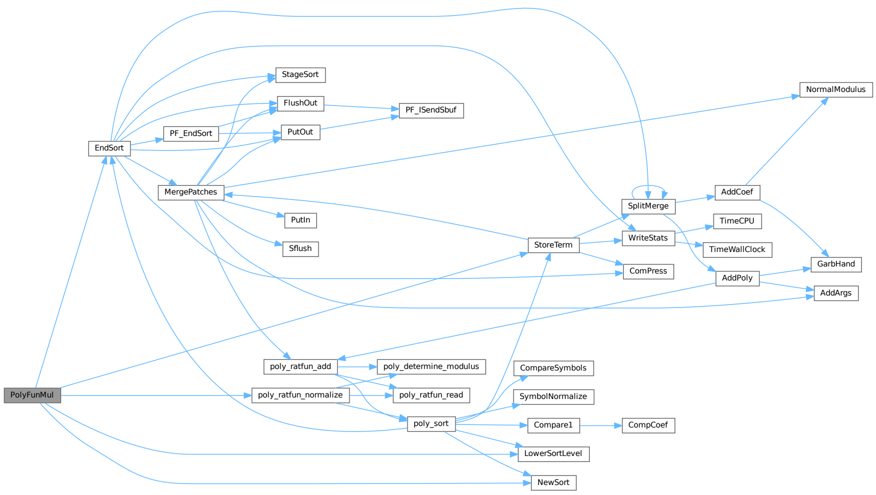
Normal conventions (OK = 0).

Definition at line 5360 of file proces.c.

References EndSort(), LowerSortLevel(), NewSort(), poly_ratfun_normalize(), and StoreTerm().

Referenced by Generator().

Here is the call graph for this function:



4.114.4 Variable Documentation

4.114.4.1 printscratch

```
WORD printscratch[2]
```

Definition at line 39 of file proces.c.

4.115 proces.c

Go to the documentation of this file.

```
00001
00006 /* #[ License : */
00007 /*
00008  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009  *   When using this file you are requested to refer to the publication
00010  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011  *   This is considered a matter of courtesy as the development was paid
00012  *   for by FOM the Dutch physics granting agency and we would like to
00013  *   be able to track its scientific use to convince FOM of its value
00014  *   for the community.
00015  *
00016  *   This file is part of FORM.
00017  *
00018  *   FORM is free software: you can redistribute it and/or modify it under the
00019  *   terms of the GNU General Public License as published by the Free Software
00020  *   Foundation, either version 3 of the License, or (at your option) any later
00021  *   version.
00022  *
00023  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025  *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
00026  *   details.
00027  *
00028  *   You should have received a copy of the GNU General Public License along
00029  *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00030  */
00031 /* #] License : */
00032 /*
00033 #define HIDEDEBUG
```

```
    #[ Includes : proces.c
*/

#include "form3.h"

WORD printscratch[2];

/*
    #] Includes :
    #[ Processor :
        #[ Processor :          WORD Processor()
*/
WORD Processor(void)
{
    GETIDENTITY
    WORD *term, *t, i, retval = 0, size;
    EXPRESSIONS e;
    POSITION position;
    WORD last, LastExpression, fromspectator;
    LONG dd = 0;
    CBUF *C = cbuf+AC.cbufnum;
    int firstterm;
    CBUF *CC = cbuf+AT.ebufnum;
    WORD **w, *cpo, *cbo;
    FILEHANDLE *curfile, *oldoutfile = AR.outfile;
    WORD oldBracketOn = AR.BracketOn;
    WORD *oldBrackBuf = AT.BrackBuf;
    WORD oldbracketindexflag = AT.bracketindexflag;
#ifdef WITHPTHREADS
    int OldMultiThreaded = AS.MultiThreaded, Oldmparallelflag = AC.mparallelflag;
#endif
    if ( CC->numrhs > 0 || CC->numlhs > 0 ) {
        if ( CC->rhs ) {
            w = CC->rhs; i = CC->numrhs;
            do { *w++ = 0; } while ( --i > 0 );
        }
        if ( CC->lhs ) {
            w = CC->lhs; i = CC->numlhs;
            do { *w++ = 0; } while ( --i > 0 );
        }
        CC->numlhs = CC->numrhs = 0;
        ClearTree(AT.ebufnum);
        CC->Pointer = CC->Buffer;
    }

    if ( NumExpressions == 0 ) return(0);
    AR.expflags = 0;
    AR.CompressPointer = AR.CompressBuffer;
    AR.NoCompress = AC.NoCompress;
    term = AT.WorkPointer;
    if ( ( (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer) ) > AT.WorkTop ) return(MesWork());
    UpdatePositions();
    C->rhs[C->numrhs+1] = C->Pointer;
    AR.KeptInHold = 0;
    if ( AC.CollectFun ) AR.DeferFlag = 0;
    AR.outtohide = 0;
    AN.PolyFunTodo = 0;
#ifdef HIDEDEBUG
    MesPrint("Status at the start of Processor (HideLevel = %d)",AC.HideLevel);
    for ( i = 0; i < NumExpressions; i++ ) {
        e = Expressions+i;
        ExprStatus(e);
    }
#endif
/*
        Next determine the last expression. This is used for removing the
        input file when the final stage of the sort of this expression is
        reached. That can save up to 1/3 in disk space.
*/
    for ( i = NumExpressions-1; i >= 0; i-- ) {
        e = Expressions+i;
        if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
        || e->status == HIDELEXPRESSION || e->status == HIDEGEXPRESSION
        || e->status == SKIPLEXPRESSION || e->status == SKIPGEXPRESSION
        || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
        || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION
        ) break;
    }
    last = i;
    for ( i = NumExpressions-1; i >= 0; i-- ) {
        AS.OldOnFile[i] = Expressions[i].onfile;
        AS.OldNumFactors[i] = Expressions[i].numfactors;
/*      AS.Oldvflags[i] = e[i].vflags; */
        AS.Oldvflags[i] = Expressions[i].vflags;
        AS.Olduflags[i] = Expressions[i].uflags;
        Expressions[i].vflags &= ~(ISUNMODIFIED|ISZERO);
    }
```

```
#ifdef WITHPTHREADS
/*
        When we run with threads we have to make sure that all local input
        buffers are pointed correctly. Of course this isn't needed if we
        run on a single thread only.
*/
    if ( AC.partodoflag && AM.totalnumberofthreads > 1 ) {
        AS.MultiThreaded = 1; AC.mparallelflag = PARALLELFLAG;
    }
    if ( AS.MultiThreaded && AC.mparallelflag == PARALLELFLAG ) {
        SetWorkerFiles();
    }
/*
        We start with running the expressions with expr->partodo in parallel.
        The current model is: give each worker an expression. Wait for
        workers to finish and tell them where to write.
        Then give them a new expression. Workers may have to wait for each
        other. This is also the case with the last one.
*/
    if ( AS.MultiThreaded && AC.mparallelflag == PARALLELFLAG ) {
        if ( InParallelProcessor() ) {
            retval = 1;
        }
        AS.MultiThreaded = OldMultiThreaded;
        AC.mparallelflag = Oldmparallelflag;
    }
#endif
#ifdef WITHMPI
    if ( AC.RhsExprInModuleFlag && PF.rhsInParallel && (AC.mparallelflag == PARALLELFLAG ||
AC.partodoflag) ) {
        if ( PF_BroadcastRHS() ) {
            retval = -1;
        }
    }
    PF.exprtodo = -1; /* This means, the slave does not perform inparallel */
    if ( AC.partodoflag > 0 ) {
        if ( PF_InParallelProcessor() ) {
            retval = -1;
        }
    }
#endif
    for ( i = 0; i < NumExpressions; i++ ) {
#ifdef INNERTEST
        if ( AC.InnerTest ) {
            if ( StrCmp(AC.TestValue,(UBYTE *)INNERTEST) == 0 ) {
                MesPrint("Testing(Processor): value = %s",AC.TestValue);
            }
        }
#endif
        e = Expressions+i;
#ifdef WITHPTHREADS
        if ( AC.partodoflag > 0 && e->partodo > 0 && AM.totalnumberofthreads > 2 ) {
            e->partodo = 0;
            continue;
        }
#endif
#ifdef WITHMPI
        if ( AC.partodoflag > 0 && e->partodo > 0 && PF.numtasks > 2 ) {
            e->partodo = 0;
            continue;
        }
#endif
        AS.CollectOverFlag = 0;
        AR.expchanged = 0;
        if ( i == last ) LastExpression = 1;
        else             LastExpression = 0;
        if ( e->inmem ) {
/*
            #[ in memory : Memory allocated by poly.c only thusfar.
                           Here GetTerm cannot work.
                           For the moment we ignore this for parallelization.
*/
            WORD j;

            AR.GetFile = 0;
            SetScratch(AR.infile,&(e->onfile));
            if ( GetTerm(BHEAD term) <= 0 ) {
                MesPrint("(1) Expression %d has problems in scratchfile",i);
                retval = -1;
                break;
            }
            term[3] = i;
            AR.CurExpr = i;
            SeekScratch(AR.outfile,&position);
            e->onfile = position;
            if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) goto ProcErr;
            AR.DeferFlag = AC.ComDefer;
```

```
            NewSort(BHEAD0);
            AN.ninterms = 0;
            t = e->inmem;
            while ( *t ) {
                for ( j = 0; j < *t; j++ ) term[j] = t[j];
                t += *t;
                AN.ninterms++; dd = AN.deferskipped;
                if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)(sizeof(WORD)))) ) {
                    if ( GetMoreFromMem(term,&t) ) {
                        LowerSortLevel(); goto ProcErr;
                    }
                }
                AT.WorkPointer = term + *term;
                AN.RepPoint = AT.RepCount + 1;
                AN.IndDum = AM.IndDum;
                AR.CurDum = ReNumber(BHEAD term);
                if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
                if ( AN.ncmod ) {
                    if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
                    else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
                }
                else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
                if ( Generator(BHEAD term,0) ) {
                    LowerSortLevel(); goto ProcErr;
                }
                AN.ninterms += dd;
            }
            AN.ninterms += dd;
            if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) goto ProcErr;
            if ( AM.S0->TermsLeft ) e->vflags &= ~ISZERO;
            else e->vflags |= ISZERO;
            if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
                if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
            if ( AR.expchanged ) AR.expflags |= ISUNMODIFIED;
            AR.GetFile = 0;
/*
            #] in memory :
*/
        }
        else {
            AR.CurExpr = i;
            switch ( e->status ) {
            case UNHIDELEXPRESSION:
            case UNHIDEGEXPRESSION:
                AR.GetFile = 2;
#ifdef WITHMPI
                if ( PF.me == MASTER ) SetScratch(AR.hidefile,&(e->onfile));
#else
                SetScratch(AR.hidefile,&(e->onfile));
                AR.InHiBuf = AR.hidefile->POfull-AR.hidefile->POfill;
#ifdef HIDEDEBUG
                MesPrint("Hidefile: onfile: %15p, POposition: %15p, filesize: %15p",&(e->onfile)
                ,&(AR.hidefile->POposition),&(AR.hidefile->filesize));
                MesPrint("Set hidefile to buffer position %l/%l; AR.InHiBuf = %l"
                    ,(AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD)
                    ,(AR.hidefile->POfull-AR.hidefile->PObuffer)*sizeof(WORD)
                    ,AR.InHiBuf
                );
#endif
#endif
                curfile = AR.hidefile;
                goto commonread;
            case INTOHIDELEXPRESSION:
            case INTOHIDEGEXPRESSION:
                AR.outtohide = 1;
/*
                BugFix 12-feb-2016
                This may not work when the file is open and we move around.
                AR.hidefile->POfill = AR.hidefile->POfull;
*/
                SetEndHScratch(AR.hidefile,&position);
                /* fall through */
            case LOCALEXPRESSION:
            case GLOBALEXPRESSION:
                AR.GetFile = 0;
/*[20oct2009 mt]:*/
#ifdef WITHMPI
                if( ( PF.me == MASTER ) || (PF.mkSlaveInfile) )
#endif
                SetScratch(AR.infile,&(e->onfile));
/*:[20oct2009 mt]*/
                curfile = AR.infile;
commonread:;
#ifdef WITHMPI
                if ( PF_Processor(e,i,LastExpression) ) {
                    MesPrint("Error in PF_Processor");
                    goto ProcErr;
```

```
                }
/*[20oct2009 mt]:*/
                if ( AC.mparallelflag != PARALLELFLAG ){
                    if(PF.me != MASTER)
                        break;
#endif
/*:[20oct2009 mt]*/
                if ( GetTerm(BHEAD term) <= 0 ) {
#ifdef HIDEDEBUG
                    MesPrint("Error condition 1a");
                    ExprStatus(e);
#endif
                    MesPrint("(2) Expression %d has problems in scratchfile(process)",i);
                    retval = -1;
                    break;
                }
                term[3] = i;
                if ( term[5] < 0 ) {    /* Fill with spectator */
                    fromspectator = -term[5];
                    PUTZERO(AM.SpectatorFiles[fromspectator-1].readpos);
                    term[5] = AC.cbufnum;
                }
                else fromspectator = 0;
                if ( AR.outtohide ) {
                    SeekScratch(AR.hidefile,&position);
                    e->onfile = position;
                    if ( PutOut(BHEAD term,&position,AR.hidefile,0) < 0 ) goto ProcErr;
                }
                else {
                    SeekScratch(AR.outfile,&position);
                    e->onfile = position;
                    if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) goto ProcErr;
                }
                AR.DeferFlag = AC.ComDefer;
                AR.Eside = RHSIDE;
                if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
                    AR.BracketOn = 1;
                    AT.BrackBuf = AM.BracketFactors;
                    AT.bracketindexflag = 1;
                }
                if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
#ifdef WITHPTHREADS
                if ( AS.MultiThreaded && AC.mparallelflag == PARALLELFLAG ) {
                    if ( ThreadsProcessor(e,LastExpression,fromspectator) ) {
                        MesPrint("Error in ThreadsProcessor");
                        goto ProcErr;
                    }
                    if ( AR.outtohide ) {
                        AR.outfile = oldoutfile;
                        AR.hidefile->POfull = AR.hidefile->POfill;
                    }
                }
                else
#endif
                {
                    NewSort(BHEAD0);
                    AR.MaxDum = AM.IndDum;
                    AN.ninterms = 0;
                    for(;;) {
                     if ( fromspectator ) size = GetFromSpectator(term,fromspectator-1);
                     else                 size = GetTerm(BHEAD term);
                     if ( size <= 0 ) break;
                     SeekScratch(curfile,&position);
                     if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
                      StoreTerm(BHEAD term);
                     }
                     else {
                      AN.ninterms++; dd = AN.deferskipped;
                      if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)(sizeof(WORD)))) ) {
                        if ( GetMoreTerms(term) < 0 ) {
                          LowerSortLevel(); goto ProcErr;
                        }
                        SeekScratch(curfile,&position);
                      }
                      AT.WorkPointer = term + *term;
                      AN.RepPoint = AT.RepCount + 1;
                      if ( AR.DeferFlag ) {
                        AN.IndDum = Expressions[AR.CurExpr].numdummies + AM.IndDum;
                        AR.CurDum = AN.IndDum;
                      }
                      else {
                        AN.IndDum = AM.IndDum;
                        AR.CurDum = ReNumber(BHEAD term);
                      }
                      if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
                      if ( AN.ncmod ) {
                        if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
```

```
                        else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
                      }
                      else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
                      if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
                        && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
                        PolyFunClean(BHEAD term);
                      }
                      if ( Generator(BHEAD term,0) ) {
                        LowerSortLevel(); goto ProcErr;
                      }
                      AN.ninterms += dd;
                     }
                     SetScratch(curfile,&position);
                     if ( AR.GetFile == 2 ) {
                        AR.InHiBuf = (curfile->POfull-curfile->PObuffer)
                            -DIFBASE(position,curfile->POposition)/sizeof(WORD);
                     }
                     else {
                        AR.InInBuf = (curfile->POfull-curfile->PObuffer)
                            -DIFBASE(position,curfile->POposition)/sizeof(WORD);
                     }
                    }
                    AN.ninterms += dd;
                    if ( LastExpression ) {
                        UpdateMaxSize();
                        if ( AR.infile->handle >= 0 ) {
                            CloseFile(AR.infile->handle);
                            AR.infile->handle = -1;
                            remove(AR.infile->name);
                            PUTZERO(AR.infile->POposition);
                        }
                        AR.infile->POfill = AR.infile->POfull = AR.infile->PObuffer;
                    }
                    if ( AR.outtohide ) AR.outfile = AR.hidefile;
                    if ( EndSort(BHEAD AM.S0->sBuffer,0) < 0 ) goto ProcErr;
                    if ( AR.outtohide ) {
                        AR.outfile = oldoutfile;
                        AR.hidefile->POfull = AR.hidefile->POfill;
                    }
                    e->numdummies = AR.MaxDum - AM.IndDum;
                    UpdateMaxSize();
                }
                AR.BracketOn = oldBracketOn;
                AT.BrackBuf = oldBrackBuf;
                if ( ( e->vflags & TOBEFACTORED ) != 0 ) {
                    poly_factorize_expression(e);
                }
                else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
                 && ( ( e->vflags & ISFACTORIZED ) != 0 ) ) {
                    poly_unfactorize_expression(e);
                }
                AT.bracketindexflag = oldbracketindexflag;
                if ( AM.S0->TermsLeft )   e->vflags &= ~ISZERO;
                else                      e->vflags |= ISZERO;
                if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
                if ( AM.S0->TermsLeft ) AR.expflags |= ISZERO;
                if ( AR.expchanged )    AR.expflags |= ISUNMODIFIED;
                AR.GetFile = 0;
                AR.outtohide = 0;
/*[20oct2009 mt]:*/
#ifdef WITHMPI
                }
#endif
#ifdef WITHPTHREADS
                if ( e->status == INTOHIDELEXPRESSION ||
                     e->status == INTOHIDEGEXPRESSION ) {
                    SetHideFiles();
                }
#endif
                break;
            case SKIPLEXPRESSION:
            case SKIPGEXPRESSION:
/*
                This can be greatly improved of course by file-to-file copy.
*/
#ifdef WITHMPI
                if ( PF.me != MASTER ) break;
#endif
                AR.GetFile = 0;
                SetScratch(AR.infile,&(e->onfile));
                if ( GetTerm(BHEAD term) <= 0 ) {
#ifdef HIDEDEBUG
                    MesPrint("Error condition 1b");
                    ExprStatus(e);
#endif
                    MesPrint("(3) Expression %d has problems in scratchfile",i);
                    retval = -1;
```

```
                        break;
                    }
                    term[3] = i;
                    AR.DeferFlag = 0;
                    SeekScratch(AR.outfile,&position);
                    e->onfile = position;
                    *AM.S0->sBuffer = 0; firstterm = -1;
                    do {
                        WORD *oldipointer = AR.CompressPointer;
                        WORD *comprtop = AR.ComprTop;
                        AR.ComprTop = AM.S0->sTop;
                        AR.CompressPointer = AM.S0->sBuffer;
                        if ( firstterm > 0 ) {
                            if ( PutOut(BHEAD term,&position,AR.outfile,1) < 0 ) goto ProcErr;
                        }
                        else if ( firstterm < 0 ) {
                            if ( PutOut(BHEAD term,&position,AR.outfile,0) < 0 ) goto ProcErr;
                            firstterm++;
                        }
                        else {
                            if ( PutOut(BHEAD term,&position,AR.outfile,-1) < 0 ) goto ProcErr;
                            firstterm++;
                        }
                        AR.CompressPointer = oldipointer;
                        AR.ComprTop = comprtop;
                    } while ( GetTerm(BHEAD term) );
                    if ( FlushOut(&position,AR.outfile,1) ) goto ProcErr;
                    UpdateMaxSize();
                    break;
                case HIDELEXPRESSION:
                case HIDEGEXPRESSION:
#ifdef WITHMPI
                    if ( PF.me != MASTER ) break;
#endif
                    AR.GetFile = 0;
                    SetScratch(AR.infile,&(e->onfile));
                    if ( GetTerm(BHEAD term) <= 0 ) {
#ifdef HIDEDEBUG
                        MesPrint("Error condition 1c");
                        ExprStatus(e);
#endif
                        MesPrint("(4) Expression %d has problems in scratchfile",i);
                        retval = -1;
                        break;
                    }
                    term[3] = i;
                    AR.DeferFlag = 0;
                    SetEndHScratch(AR.hidefile,&position);
                    e->onfile = position;
#ifdef HIDEDEBUG
                    if ( AR.hidefile->handle >= 0 ) {
                        POSITION possize,pos;
                        PUTZERO(possize);
                        PUTZERO(pos);
                        SeekFile(AR.hidefile->handle,&pos,SEEK_CUR);
                        SeekFile(AR.hidefile->handle,&possize,SEEK_END);
                        SeekFile(AR.hidefile->handle,&pos,SEEK_SET);
                        MesPrint("Processor Hide1: filesize(th) = %12p, filesize(ex) = %12p",&(position),
                                &(possize));
                        MesPrint("     in buffer: %l",(AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD));
                    }
#endif
                    *AM.S0->sBuffer = 0; firstterm = -1;
                    cbo = cpo = AM.S0->sBuffer;
                    do {
                        WORD *oldipointer = AR.CompressPointer;
                        WORD *oldibuffer = AR.CompressBuffer;
                        WORD *comprtop = AR.ComprTop;
                        AR.ComprTop = AM.S0->sTop;
                        AR.CompressPointer = cpo;
                        AR.CompressBuffer = cbo;
                        if ( firstterm > 0 ) {
                            if ( PutOut(BHEAD term,&position,AR.hidefile,1) < 0 ) goto ProcErr;
                        }
                        else if ( firstterm < 0 ) {
                            if ( PutOut(BHEAD term,&position,AR.hidefile,0) < 0 ) goto ProcErr;
                            firstterm++;
                        }
                        else {
                            if ( PutOut(BHEAD term,&position,AR.hidefile,-1) < 0 ) goto ProcErr;
                            firstterm++;
                        }
                        cpo = AR.CompressPointer;
                        cbo = AR.CompressBuffer;
                        AR.CompressPointer = oldipointer;
                        AR.CompressBuffer = oldibuffer;
```

```
                    AR.ComprTop = comprtop;
                } while ( GetTerm(BHEAD term) );
#ifdef HIDEDEBUG
                if ( AR.hidefile->handle >= 0 ) {
                    POSITION possize,pos;
                    PUTZERO(possize);
                    PUTZERO(pos);
                    SeekFile(AR.hidefile->handle,&pos,SEEK_CUR);
                    SeekFile(AR.hidefile->handle,&possize,SEEK_END);
                    SeekFile(AR.hidefile->handle,&pos,SEEK_SET);
                    MesPrint("Processor Hide2: filesize(th) = %12p, filesize(ex) = %12p",&(position),
                            &(possize));
                    MesPrint("    in buffer:
%l",(AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD));
                }
#endif
                if ( FlushOut(&position,AR.hidefile,1) ) goto ProcErr;
                AR.hidefile->POfull = AR.hidefile->POfill;
#ifdef HIDEDEBUG
                if ( AR.hidefile->handle >= 0 ) {
                    POSITION possize,pos;
                    PUTZERO(possize);
                    PUTZERO(pos);
                    SeekFile(AR.hidefile->handle,&pos,SEEK_CUR);
                    SeekFile(AR.hidefile->handle,&possize,SEEK_END);
                    SeekFile(AR.hidefile->handle,&pos,SEEK_SET);
                    MesPrint("Processor Hide3: filesize(th) = %12p, filesize(ex) = %12p",&(position),
                            &(possize));
                    MesPrint("    in buffer:
%l",(AR.hidefile->POfill-AR.hidefile->PObuffer)*sizeof(WORD));
                }
#endif
/*
                Because we direct the e->onfile already to the hide file, we
                need to change the status of the expression. Otherwise the use
                of parts (or the whole) of the expression looks in the infile
                while the position is that of the hide file.
                We choose to get everything from the hide file. On average that
                should give least file activity.
*/
                if ( e->status == HIDELEXPRESSION ) {
                    e->status = HIDDENLEXPRESSION;
                    AS.OldOnFile[i] = e->onfile;
                    AS.OldNumFactors[i] = Expressions[i].numfactors;
                }
                if ( e->status == HIDEGEXPRESSION ) {
                    e->status = HIDDENGEXPRESSION;
                    AS.OldOnFile[i] = e->onfile;
                    AS.OldNumFactors[i] = Expressions[i].numfactors;
                }
#ifdef WITHPTHREADS
                SetHideFiles();
#endif
                UpdateMaxSize();
                break;
            case DROPPEDEXPRESSION:
            case DROPLEXPRESSION:
            case DROPGEXPRESSION:
            case DROPHLEXPRESSION:
            case DROPHGEXPRESSION:
            case STOREDEXPRESSION:
            case HIDDENLEXPRESSION:
            case HIDDENGEXPRESSION:
            case SPECTATOREXPRESSION:
            default:
                break;
        }
        }
        AR.KeptInHold = 0;
    }
    AR.DeferFlag = 0;
    AT.WorkPointer = term;
#ifdef HIDEDEBUG
    MesPrint("Status at the end of Processor (HideLevel = %d)",AC.HideLevel);
    for ( i = 0; i < NumExpressions; i++ ) {
        e = Expressions+i;
        ExprStatus(e);
    }
#endif
    return(retval);
ProcErr:
    AT.WorkPointer = term;
    if ( AM.tracebackflag ) MesCall("Processor");
    return(-1);
}
/*
        #] Processor :
```

```
        #[ TestSub :            WORD TestSub(term,level)
*/
#define DONE(x) { retvalue = x; goto Done; }

WORD TestSub(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD *m, *t, *r, retvalue = 0, funflag, j, oldncmod, nexpr, *Tpattern = 0;
    WORD *stop, *t1, *t2, funnum, wilds, tbufnum, stilldirty = 0;
    NESTING n;
    CBUF *C = cbuf+AT.ebufnum;
    LONG isp, i;
    TABLES T;
    COMPARE oldcompareroutine = (COMPARE)(AR.CompareRoutine);
    WORD oldsorttype = AR.SortType;
ReStart:
    tbufnum = 0; i = 0; retvalue = 0;
    AT.TMbuff = AM.rbufnum;
    funflag = 0;
    t = term;
    r = t + *t - 1;
    m = r - ABS(*r) + 1;
    t++;
    if ( t < m ) do {
        if ( *t == SUBEXPRESSION ) {
            /*
                Subexpression encountered
                There may be more than one.
                The old strategy was to take the last.
                A newer strategy was to take the lowest power first.
                The current strategy is that we compute the number of terms
                generated by this subexpression and take the minimum of that.
            */

#ifdef WHICHSUBEXPRESSION

            WORD *tmin = t, AN.nbino;
/*          LONG minval = MAXLONG; */
            LONG minval = -1;
            LONG mm, mnum1 = 1;
            if ( AN.BinoScrat == 0 ) {
                AN.BinoScrat = (UWORD *)Malloc1((AM.MaxTal+2)*sizeof(UWORD),"GetBinoScrat");
            }
#endif
            if ( t[3] ) {
                r = t + t[1];
                while ( AN.subsubveto == 0 &&
                            *r == SUBEXPRESSION && r < m && r[3] ) {
#ifdef WHICHSUBEXPRESSION
                    mnum1++;
#endif
                    if ( r[1] == t[1] && r[2] == t[2] && r[4] == t[4] ) {
                        j = t[1] - SUBEXPSIZE;
                        t1 = t + SUBEXPSIZE;
                        t2 = r + SUBEXPSIZE;
                        while ( j > 0 && *t1++ == *t2++ ) j--;
                        if ( j <= 0 ) {
                            t[3] += r[3];
                            if ( t[3] == 0 ) {
                                t1 = r + r[1];
                                t2 = term + *term;
                                *term -= r[1]+t[1];
                                r = t;
                                while ( t1 < t2 ) *r++ = *t1++;
                                goto ReStart;
                            }
                            else {
                                t1 = r + r[1];
                                t2 = term + *term;
                                *term -= r[1];
                                m -= r[1];
                                while ( t1 < t2 ) *r++ = *t1++;
                                r = t;
                            }
                        }
                    }
#ifdef WHICHSUBEXPRESSION

                    else if ( t[2] >= 0 ) {
/*
                        Compute Binom(numterms+power-1,power-1)
                        We need potentially long arithmetic.
                        That is why we had to allocate AN.BinoScrat
*/
                        if ( AN.last1 == t[3] && AN.last2 == cbuf[t[4]].NumTerms[t[2]] + t[3] - 1 ) {
                            if ( AN.last3 > minval ) {
                                minval = AN.last3; tmin = t;
```

```
                            }
                        }
                        else {
                        AN.last1 = t[3]; mm = AN.last2 = cbuf[t[4]].NumTerms[t[2]] + t[3] - 1;
                        if ( t[3] == 1 ) {
                            if ( mm > minval ) {
                                minval = mm; tmin = t;
                            }
                        }
                        else if ( t[3] > 0 ) {
                            if ( mm > MAXPOSITIVE ) goto TooMuch;
                            GetBinom(AN.BinoScrat,&AN.nbino,(WORD)mm,t[3]);
                            if ( AN.nbino > 2 ) goto TooMuch;
                            if ( AN.nbino == 2 ) {
                                mm = AN.BinoScrat[1];
                                mm = ( mm « BITSINWORD ) + AN.BinoScrat[0];
                            }
                            else if ( AN.nbino == 1 ) mm = AN.BinoScrat[0];
                            else mm = 0;
                            if ( mm > minval ) {
                                minval = mm; tmin = t;
                            }
                        }
                        AN.last3 = mm;
                        }
                    }
#endif
                    t = r;
                    r += r[1];
                }
#ifdef WHICHSUBEXPRESSION
                if ( mnum1 > 1 && t[2] >= 0 ) {
/*
                    To keep the flowcontrol simple we duplicate some code here
*/
                    if ( AN.last1 == t[3] && AN.last2 == cbuf[t[4]].NumTerms[t[2]] + t[3] - 1 ) {
                        if ( AN.last3 > minval ) {
                            minval = AN.last3; tmin = t;
                        }
                    }
                    else {
                    AN.last1 = t[3]; mm = AN.last2 = cbuf[t[4]].NumTerms[t[2]] + t[3] - 1;
                    if ( t[3] == 1 ) {
                        if ( mm > minval ) {
                            minval = mm; tmin = t;
                        }
                    }
                    else if ( t[3] > 0 ) {
                        if ( mm > MAXPOSITIVE ) {
/*
                            We will generate more terms than we can count
*/
TooMuch:;
                            MLOCK(ErrorMessageLock);
                            MesPrint("Attempt to generate more terms than FORM can count");
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        }
                        GetBinom(AN.BinoScrat,&AN.nbino,(WORD)mm,t[3]);
                        if ( AN.nbino > 2 ) goto TooMuch;
                        if ( AN.nbino == 2 ) {
                            mm = AN.BinoScrat[1];
                            mm = ( mm « BITSINWORD ) + AN.BinoScrat[0];
                        }
                        else if ( AN.nbino == 1 ) mm = AN.BinoScrat[0];
                        else mm = 0;
                        if ( mm > minval ) {
                            minval = mm; tmin = t;
                        }
                    }
                    AN.last3 = mm;
                    }
                }
                t = tmin;
#endif
/*              AR.TePos = 0;  */
                AR.TePos = WORDDIF(t,term);
                AT.TMbuff = t[4];
                if ( t[4] == AM.dbufnum && (t+t[1]) < m && t[t[1]] == DOLLAREXPR2 ) {
                    if ( t[t[1]+2] < 0 ) AT.TMdolfac = -t[t[1]+2];
                    else {  /* resolve the element number */
                        AT.TMdolfac = GetDolNum(BHEAD t+t[1],m)+1;
                    }
                }
                else AT.TMdolfac = 0;
                if ( t[3] < 0 ) {
                    AN.TeInFun = 1;
```

```
                    AR.TePos = WORDDIF(t,term);
                    DONE(t[2])
                }
                else {
                    AN.TeInFun = 0;
                    AN.TeSuOut = t[3];
                }
                if ( t[2] < 0 ) {
                    AN.TeSuOut = -t[3];
                    DONE(-t[2])
                }
                DONE(t[2])
            }
        }
        else if ( *t == EXPRESSION ) {
            WORD *toTMaddr;
            i = -t[2] - 1;
            if ( t[3] < 0 ) {
                AN.TeInFun = 1;
                AR.TePos = WORDDIF(t,term);
                DONE(i)
            }
            nexpr = t[3];
            toTMaddr = m = AT.WorkPointer;
            AN.Frozen = 0;
/*
            We have to be very careful with respect to setting variables
            like AN.TeInFun, because we may still call Generator and that
            may change those variables. That is why we set them at the
            last moment only.
*/
            j = t[1];
            AT.WorkPointer += j;
            r = t;
            NCOPY(m,r,j);
            r = t + t[1];
            t += SUBEXPSIZE;
            while ( t < r ) {
                if ( *t == FROMBRAC ) {
                    WORD *ttstop,*tttstop;
/*
                    Note: Convention is that wildcards are done
                    after the expression has been picked up. So
                    no wildcard substitutions are needed here.
*/
                    t += 2;
                    AN.Frozen = m = AT.WorkPointer;
/*
                    We should check now for subexpressions and if necessary
                    we substitute them. Keep in mind: only one term allowed!

                    In retrospect (26-jan-2010): take also functions that
                    have a dirty flag on
*/
                    j = *t; tttstop = t + j;
                    GETSTOP(t,ttstop);
                    *m++ = j; t++;
                    while ( t < ttstop ) {
                        if ( *t == SUBEXPRESSION ) break;
                        if ( *t >= FUNCTION && ( ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) ) break;
                        j = t[1]; NCOPY(m,t,j);
                    }
                    if ( t < ttstop ) {
/*
                        We ran into a subexpression or a function with a
                        'dirty' argument. It could also be a $ or
                        just e[(a^2)*b]. In all cases we should evaluate
*/
                        while ( t < tttstop ) *m++ = *t++;
                        *AT.WorkPointer = m-AT.WorkPointer;
                        m = AT.WorkPointer;
                        AT.WorkPointer = m + *m;
                        NewSort(BHEAD0);
                        if ( Generator(BHEAD m,AR.Cnumlhs) ) {
                            LowerSortLevel(); goto EndTest;
                        }
                        if ( EndSort(BHEAD m,0) < 0 ) goto EndTest;
                        AN.Frozen = m;
                        if ( *m == 0 ) {
                            *m++ = 4; *m++ = 1; *m++ = 1; *m++ = 3;
                        }
                        else if ( m[*m] != 0 ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Bracket specification in expression should be one single term");
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        }
```

```
                        else {
                            m += *m;
                            m -= ABS(m[-1]);
                            *m++ = 1; *m++ = 1; *m++ = 3;
                            *AN.Frozen = m - AN.Frozen;
                        }
                    }
                    else {
                        while ( t < tttstop ) *m++ = *t++;
                        *AT.WorkPointer = m-AT.WorkPointer;
                        m = AT.WorkPointer;
                        AT.WorkPointer = m + *m;
                        if ( Normalize(BHEAD m) ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Error while picking up contents of bracket");
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        }
                        if ( !*m ) {
                            *m++ = 4; *m++ = 1; *m++ = 1; *m++ = 3;
                        }
                        else m += *m;
                    }
                    AT.WorkPointer = m;
                    break;
                }
                t += t[1];
            }
            AN.TeInFun = 0;
            AR.TePos = 0;
            AN.TeSuOut = nexpr;
            AT.TMaddr = toTMaddr;
            DONE(i)
        }
        else if ( *t >= FUNCTION ) {
            if ( t[0] == EXPONENT ) {
                if ( t[1] == FUNHEAD+4 && t[FUNHEAD] == -SYMBOL &&
                t[FUNHEAD+2] == -SNUMBER && t[FUNHEAD+3] < MAXPOWER
                && t[FUNHEAD+3] > -MAXPOWER ) {
                    t[0] = SYMBOL;
                    t[1] = 4;
                    t[2] = t[FUNHEAD+1];
                    t[3] = t[FUNHEAD+3];
                    r = term + *term;
                    m = t + FUNHEAD+4;
                    t += 4;
                    while ( m < r ) *t++ = *m++;
                    *term = WORDDIF(t,term);
                    goto ReStart;
                }
                else if ( t[1] == FUNHEAD+ARGHEAD+11 && t[FUNHEAD] == ARGHEAD+9
                 && t[FUNHEAD+ARGHEAD] == 9 && t[FUNHEAD+ARGHEAD+1] == DOTPRODUCT
                 && t[FUNHEAD+ARGHEAD+8] == 3
                 && t[FUNHEAD+ARGHEAD+7] == 1
                 && t[FUNHEAD+ARGHEAD+6] == 1
                 && t[FUNHEAD+ARGHEAD+5] == 1
                 && t[FUNHEAD+ARGHEAD+9] == -SNUMBER
                 && t[FUNHEAD+ARGHEAD+10] < MAXPOWER
                 && t[FUNHEAD+ARGHEAD+10] > -MAXPOWER ) {
                    t[0] = DOTPRODUCT;
                    t[1] = 5;
                    t[2] = t[FUNHEAD+ARGHEAD+3];
                    t[3] = t[FUNHEAD+ARGHEAD+4];
                    t[4] = t[FUNHEAD+ARGHEAD+10];
                    r = term + *term;
                    m = t + FUNHEAD+ARGHEAD+11;
                    t += 5;
                    while ( m < r ) *t++ = *m++;
                    *term = WORDDIF(t,term);
                    goto ReStart;
                }
            }
            funnum = *t;
            if ( *t >= FUNCTION + WILDOFFSET ) funnum -= WILDOFFSET;
            if ( *t == EXPONENT ) {
/*
                Test whether the second argument is an integer
*/
                r = t+FUNHEAD;
                NEXTARG(r)
                if ( *r == -SNUMBER && r[1] < MAXPOWER && r+2 == t+t[1] &&
                t[FUNHEAD] > -FUNCTION && ( t[FUNHEAD] != -SNUMBER
                || t[FUNHEAD+1] != 0 ) && t[FUNHEAD] != ARGHEAD ) {
                  if ( r[1] == 0 ) {
                    if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Encountered 0^0. Fatal error.");
```

```
                        MUNLOCK(ErrorMessageLock);
                        SETERROR(-1);
                    }
                    *t = DUMMYFUN;
/*
                        Now mark it clean to avoid further interference.
                        Normalize will remove this object.
*/
                    t[2] = 0;
                  }
                  else {
                    /* Note that the case 0^ is treated in Normalize */

                    t1 = AddRHS(AT.ebufnum,1);
                    m = t + FUNHEAD;
                    if ( *m > 0 ) {
                        m += ARGHEAD;
                        i = t[FUNHEAD] - ARGHEAD;
                        while ( (t1 + i + 10) > C->Top )
                            t1 = DoubleCbuffer(AT.ebufnum,t1,9);
                        while ( --i >= 0 ) *t1++ = *m++;
                    }
                    else {
                        if ( (t1 + 20) > C->Top )
                            t1 = DoubleCbuffer(AT.ebufnum,t1,10);
                        ToGeneral(m,t1,1);
                        t1 += *t1;
                    }
                    *t1++ = 0;
                    C->rhs[C->numrhs+1] = t1;
                    C->Pointer = t1;

                    /* No provisions yet for commuting objects */

                    C->CanCommu[C->numrhs] = 1;
                    *t++ = SUBEXPRESSION;
                    *t++ = SUBEXPSIZE;
                    *t++ = C->numrhs;
                    *t++ = r[1];
                    *t++ = AT.ebufnum;
#if SUBEXPSIZE > 5
Important: we may not have enough spots here
#endif
                    FILLSUB(t)  /* Important: We have maybe only 5 spots! */
                    r += 2;
                    m = term + *term;
                    do { *t++ = *r++; } while ( r < m );
                    *term -= WORDDIF(r,t);
                    goto ReStart;
                  }
                }
            }
            else if ( *t == SUMF1 || *t == SUMF2 ) {
/*
                What we are looking for is:
                1-st argument:  Single symbol or index.
                2-nd argument:  Number.
                3-rd argument:  Number.
                (4-th argument):Number.
                One more argument.
                This would activate the summation procedure.
                Note that the initiated recursion here can be done
                without upsetting the regular procedures.
*/
                WORD *tstop, lcounter, lcmin, lcmax, lcinc;
                tstop = t + t[1];
                r = t+FUNHEAD;
                if ( r+6 < tstop && r[2] == -SNUMBER && r[4] == -SNUMBER
                && ( ( r[0] == -SYMBOL )
                || ( r[0] == -INDEX && r[1] >= AM.OffsetIndex
                && r[3] >= 0 && r[3] < AM.OffsetIndex
                && r[5] >= 0 && r[5] < AM.OffsetIndex ) ) ) {
                    lcounter = r[0] == -INDEX ? -r[1]: r[1]; /* The loop counter */
                    lcmin = r[3];
                    lcmax = r[5];
                    r += 6;
                    if ( *r == -SNUMBER && r+2 < tstop ) {
                        lcinc = r[1];
                        r += 2;
                    }
                    else lcinc = 1;
                    if ( r < tstop && ( ( *r > 0 && (r+*r) == tstop )
                    || ( *r <= -FUNCTION && r+1 == tstop )
                    || ( *r > -FUNCTION && *r < 0 && r+2 == tstop ) ) ) {
                        m = AddRHS(AT.ebufnum,1);
                        if ( *r > 0 ) {
                            i = *r - ARGHEAD;
```

```
                        r += ARGHEAD;
                        while ( (m + i + 10) > C->Top )
                            m = DoubleCbuffer(AT.ebufnum,m,11);
                        while ( --i >= 0 ) *m++ = *r++;
                    }
                    else {
                        while ( (m + 20) > C->Top )
                            m = DoubleCbuffer(AT.ebufnum,m,12);
                        ToGeneral(r,m,1);
                        m += *m;
                    }
                    *m++ = 0;
                    C->rhs[C->numrhs+1] = m;
                    C->Pointer = m;
                    m = AT.TMout;
                    *m++ = 6;
                    if ( *t == SUMF1 ) *m++ = SUMNUM1;
                    else               *m++ = SUMNUM2;
                    *m++ = lcounter;
                    *m++ = lcmin;
                    *m++ = lcmax;
                    *m++ = lcinc;
                    m = t + t[1];
                    r = C->rhs[C->numrhs];
/*
                    Test now if the argument was already evaluated.
                    In that case it needs a new subexpression prototype.
                    In either case we replace the function now by a
                    subexpression prototype.
*/
                    if ( *r >= (SUBEXPSIZE+4)
                    && ABS(*(r+*r-1)) < (*r - 1)
                    && r[1] == SUBEXPRESSION ) {
                        r++;
                        i = r[1] - 5;
                        *t++ = *r++; *t++ = *r++; *t++ = C->numrhs;
                        r++; *t++ = *r++; *t++ = AT.ebufnum; r++;
                        while ( --i >= 0 ) *t++ = *r++;
                    }
                    else {
                        *t++ = SUBEXPRESSION;
                        *t++ = 4+SUBEXPSIZE;
                        *t++ = C->numrhs;
                        *t++ = 1;
                        *t++ = AT.ebufnum;
                        FILLSUB(t)
                        if ( lcounter < 0 ) {
                            *t++ = INDTOIND;
                            *t++ = 4;
                            *t++ = -lcounter;
                        }
                        else {
                            *t++ = SYMTONUM;
                            *t++ = 4;
                            *t++ = lcounter;
                        }
                        *t++ = lcmin;
                    }
                    t2 = term + *term;
                    while ( m < t2 ) *t++ = *m++;
                    *term = WORDDIF(t,term);
                    AN.TeInFun = -C->numrhs;
                    AR.TePos = 0;
                    AN.TeSuOut = 0;
                    AT.TMbuff = AT.ebufnum;
                    DONE(C->numrhs)
                }
            }
        }
        else if ( *t == TOPOLOGIES ) {
/*
            Syntax:
            topologies_(nloops,nlegs,setvertexsizes,setext,setint[,options])
*/
            t1 = t+FUNHEAD; t2 = t+t[1];
            if ( *t1 == -SNUMBER && t1[1] >= 0 &&
                t1[2] == -SNUMBER && ( t1[3] >= 0 || t1[3] == -2 ) &&
                t1[4] == -SETSET && Sets[t1[5]].type == CNUMBER &&
                t1[6] == -SETSET && Sets[t1[7]].type == CVECTOR &&
                t1[8] == -SETSET && Sets[t1[9]].type == CVECTOR &&
                t1+10 <= t2 ) {
                if ( t1+10 == t2 || ( t1+12 <= t2 && ( t1[10] == -SNUMBER ||
                    ( t1[10] == -SETSET &&
                        Sets[t1[5]].last-Sets[t1[5]].first ==
                        Sets[t1[11]].last-Sets[t1[11]].first ) ) ) ) {
                    AN.TeInFun = -15;
                    AN.TeSuOut = 0;
```

```
                        AR.TePos = -1;
                        AR.funoffset = t - term;
                        DONE(1)
                    }
                }
            }
            else if ( *t == DIAGRAMS ) {
/*
                Syntax:
                diagrams_(model,setinparticles,setoutparticles,
                    setextmomenta,setintmomenta,couplings or loops)
*/
                if ( AC.nummodels > 0 ) { /* No model, no diagrams */
                  t1 = t+FUNHEAD; t2 = t+t[1];
                  if (
                  t1[0] == -SETSET && Sets[t1[1]].type == CMODEL &&
                  t1[2] == -SETSET && ( Sets[t1[3]].type == CFUNCTION
                    || ( Sets[t1[3]].type == ANYTYPE && ( Sets[t1[3]].first == Sets[t1[3]].last ) ) ) &&
                  t1[4] == -SETSET && ( Sets[t1[5]].type == CFUNCTION
                    || ( Sets[t1[5]].type == ANYTYPE && ( Sets[t1[5]].first == Sets[t1[5]].last ) ) ) &&
                  t1[6] == -SETSET && Sets[t1[7]].type == CVECTOR &&
                  t1[8] == -SETSET && Sets[t1[9]].type == CVECTOR &&
                  t1+12 <= t2 ) {
/*
                    Test that the sets of particles correspond to particles
                    of the set model.
*/
                    MODEL *m = AC.models[SetElements[Sets[t1[1]].first]];
                    int nn0,nn1,nn2;
                    for ( nn0 = 3; nn0 <= 5; nn0 += 2 ) {
                      for ( nn1 = Sets[t1[nn0]].first; nn1 < Sets[t1[nn0]].last; nn1++ ) {
                        for ( nn2 = 0; nn2 < m->nparticles; nn2++ ) {
                            if ( m->vertices[nn2]->particles[0].number == SetElements[nn1]
                              || m->vertices[nn2]->particles[1].number == SetElements[nn1] ) break;
                        }
                        if ( nn2 >= m->nparticles ) goto doesnotwork;
                      }
                    }
/*
                    Now test for a single argument indicating the order
                    in perturbation theory.
*/
                    if ( ( t1[10] == -SNUMBER && t1[11] >= 0 && t1+12 == t2 )
                    || ( t1[10] == -SYMBOL && t1+12 == t2 )
                    || ( t1+10+t1[10] == t2 && t1+10+ARGHEAD+t1[10+ARGHEAD] == t2
                    && t1[11+ARGHEAD] == SYMBOL ) ) {
/*
                        Now test that all symbols are valid coupling constants.
*/
                        if ( t1+12 > t2 ) {
                            WORD *tt1 = t1+13+ARGHEAD, im;
                            t2 -=  ABS(t2[-1]);
                            while ( tt1 < t2 ) {
                                for ( im = 0; im < m->ncouplings; im++ ) {
                                    if ( *tt1 == m->couplings[im] ) break;
                                }
                                if ( im >= m->ncouplings ) goto doesnotwork;
                                tt1 += 2;
                            }
                        }
                        AN.TeInFun = -16;
                        AN.TeSuOut = 0;
                        AR.TePos = -1;
                        AR.funoffset = t - term;
                        DONE(1)
                    }
                    else if ( ( ( t1[10] == -SNUMBER && t1[11] >= 0 && t1+12 == t2-2 )
                    || ( t1[10] == -SYMBOL && t1+12 == t2-2 )
                    || ( t1+10+t1[10] == t2-2 && t1+10+ARGHEAD+t1[10+ARGHEAD] == t2-2
                    && t1[11+ARGHEAD] == SYMBOL ) ) && t2[-2] == -SNUMBER ) {
/*
                        With options at t2[-2],t2[-1]
                        Now test that all symbols are valid coupling constants.
*/
                        t2 -= 2;
                        if ( t1+12 > t2 ) {
                            WORD *tt1 = t1+13+ARGHEAD, im;
                            t2 -=  ABS(t2[-1]);
                            while ( tt1 < t2 ) {
                                for ( im = 0; im < m->ncouplings; im++ ) {
                                    if ( *tt1 == m->couplings[im] ) break;
                                }
                                if ( im >= m->ncouplings ) goto doesnotwork;
                                tt1 += 2;
                            }
```

```
                        }
                        AN.TeInFun = -16;
                        AN.TeSuOut = 0;
                        AR.TePos = -1;
                        AR.funoffset = t - term;
                        DONE(1)
                    }
doesnotwork:;
                  }
                }
            }
            if ( functions[funnum-FUNCTION].spec <= 0
                || ( t[2] & (DIRTYFLAG|MUSTCLEANPRF) ) != 0 ) {
                funflag = 1;
            }
            if ( *t <= MAXBUILTINFUNCTION ) {
              if ( *t <= DELTAP && *t >= THETA ) { /* Speeds up by 2 or 3 compares */
              if ( *t == THETA || *t == THETA2 ) {
                WORD *tstop, *tt2, kk;
                tstop = t + t[1];
                tt2 = t + FUNHEAD;
                while ( tt2 < tstop ) {
                    if ( *tt2 > 0 && tt2[1] != 0 ) goto DoSpec;
                    NEXTARG(tt2)
                }
                if ( !AT.RecFlag ) {
                    if ( ( kk = DoTheta(BHEAD t) ) == 0 ) {
                        *term = 0;
                        DONE(0)
                    }
                    else if ( kk > 0 ) {
                        m = t + t[1];
                        r = term + *term;
                        while ( m < r ) *t++ = *m++;
                        *term = WORDDIF(t,term);
                        goto ReStart;
                    }
                }
              }
              else if ( *t == DELTA2 || *t == DELTAP ) {
                WORD *tstop, *tt2, kk;
                tstop = t + t[1];
                tt2 = t + FUNHEAD;
                while ( tt2 < tstop ) {
                    if ( *tt2 > 0 && tt2[1] != 0 ) goto DoSpec;
                    NEXTARG(tt2)
                }
                if ( !AT.RecFlag ) {
                    if ( ( kk = DoDelta(t) ) == 0 ) {
                        *term = 0;
                        DONE(0)
                    }
                    else if ( kk > 0 ) {
                        m = t + t[1];
                        r = term + *term;
                        while ( m < r ) *t++ = *m++;
                        *term = WORDDIF(t,term);
                        goto ReStart;
                    }
                }
              } }
              else if ( *t == DISTRIBUTION && t[FUNHEAD] == -SNUMBER
              && t[FUNHEAD+1] >= -2 && t[FUNHEAD+1] <= 2
              && t[FUNHEAD+2] == -SNUMBER
              && t[FUNHEAD+4] <= -FUNCTION
              && t[FUNHEAD+5] <= -FUNCTION ) {
                WORD *ttt = t+FUNHEAD+6, *tttstop = t+t[1];
                while ( ttt < tttstop ) {
                    if ( *ttt == -DOLLAREXPRESSION ) break;
                    NEXTARG(ttt);
                }
                if ( ttt >= tttstop ) {
                    AN.TeInFun = -1;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
              }
              else if ( *t == DELTA3 && ((t[1]-FUNHEAD) & 1 ) == 0 ) {
                AN.TeInFun = -2;
                AN.TeSuOut = 0;
                AR.TePos = -1;
                DONE(1)
              }
              else if ( ( *t == TABLEFUNCTION ) && ( t[FUNHEAD] <= -FUNCTION )
              && ( T = functions[-t[FUNHEAD]-FUNCTION].tabl ) != 0
              && ( t[1] >= FUNHEAD+1+2*ABS(T->numind) )
```

```
              && ( t[FUNHEAD+1] == -SYMBOL ) ) {
/*
                The case of table_(tab,sym1,...,symn)
*/
                for ( isp = 0; isp < ABS(T->numind); isp++ ) {
                    if ( t[FUNHEAD+1+2*isp] != -SYMBOL ) break;
                }
                if ( isp >= ABS(T->numind) ) {
                    AN.TeInFun = -3;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
              }
              else if ( *t == TABLEFUNCTION && t[FUNHEAD] <= -FUNCTION
              && ( T = functions[-t[FUNHEAD]-FUNCTION].tabl ) != 0
              && ( t[1] == FUNHEAD+2 )
              && ( t[FUNHEAD+1] <= -FUNCTION ) ) {
/*
                The case of table_(tab,fun)
*/
                AN.TeInFun = -3;
                AN.TeSuOut = 0;
                AR.TePos = -1;
                DONE(1)
              }
              else if ( *t == FACTORIN ) {
                if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -DOLLAREXPRESSION ) {
                    AN.TeInFun = -4;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
                else if ( t[1] == FUNHEAD+2 && t[FUNHEAD] == -EXPRESSION ) {
                    AN.TeInFun = -5;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
              }
              else if ( *t == TERMSINBRACKET ) {
                if ( t[1] == FUNHEAD || (
                     t[1] == FUNHEAD+2
                    && t[FUNHEAD] == -SNUMBER
                    && t[FUNHEAD+1] == 0
                 ) ) {
                    AN.TeInFun = -6;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
/*
                The other cases have not yet been implemented
                We still have to add the case of short arguments
                First the different bracket in same expression

                else if ( t[1] > FUNHEAD+ARGHEAD
                        && t[FUNHEAD] == t[1]-FUNHEAD
                        && t[FUNHEAD+ARGHEAD] == t[1]-FUNHEAD-ARGHEAD
                        && t[t[1]-1] == 3
                        && t[t[1]-2] == 1
                        && t[t[1]-3] == 1 ) {
                    AN.TeInFun = -6;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }

                Next the bracket in an other expression

                else if ( t[1] > FUNHEAD+ARGHEAD+2
                        && t[FUNHEAD] == -EXPRESSION
                        && t[FUNHEAD+2] == t[1]-FUNHEAD-2
                        && t[FUNHEAD+ARGHEAD+2] == t[1]-FUNHEAD-ARGHEAD-2
                        && t[t[1]-1] == 3
                        && t[t[1]-2] == 1
                        && t[t[1]-3] == 1 ) {
                    AN.TeInFun = -6;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
*/
              }
              else if ( *t == EXTRASYMFUN ) {
                if ( t[1] == FUNHEAD+2 && (
                    ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] <= cbuf[AM.sbufnum].numrhs
```

```
                    && t[FUNHEAD+1] > 0 ) ||
                ( t[FUNHEAD] == -SYMBOL && t[FUNHEAD+1] < MAXVARIABLES
                    && t[FUNHEAD+1] >= MAXVARIABLES-cbuf[AM.sbufnum].numrhs ) ) ) {
                AN.TeInFun = -7;
                AN.TeSuOut = 0;
                AR.TePos = -1;
                DONE(1)
            }
            else if ( t[1] == FUNHEAD ) {
                AN.TeInFun = -7;
                AN.TeSuOut = 0;
                AR.TePos = -1;
                DONE(1)
            }
        }
        else if ( *t == DIVFUNCTION || *t == REMFUNCTION
                    || *t == INVERSEFUNCTION || *t == MULFUNCTION
                    || *t == GCDFUNCTION ) {
            WORD *tf;
            int todo = 1, numargs = 0;
            tf = t + FUNHEAD;
            while ( tf < t + t[1] ) {
                DOLLARS d;
                if ( *tf == -DOLLAREXPRESSION ) {
                    d = Dollars + tf[1];
                    if ( d->type == DOLWILDARGS ) {
                        WORD *tterm = AT.WorkPointer, *tw;
                        WORD *ta = term, *tb = tterm, *tc, *td = term + *term;
                        while ( ta < t ) *tb++ = *ta++;
                        tc = tb;
                        while ( ta < tf ) *tb++ = *ta++;
                        tw = d->where+1;
                        while ( *tw ) {
                            if ( *tw < 0 ) {
                                if ( *tw > -FUNCTION ) *tb++ = *tw++;
                                *tb++ = *tw++;
                            }
                            else {
                                int ia;
                                for ( ia = 0; ia < *tw; ia++ ) *tb++ = *tw++;
                            }
                        }
                        NEXTARG(ta)
                        while ( ta < t+t[1] ) *tb++ = *ta++;
                        tc[1] = tb-tc;
                        while ( ta < td ) *tb++ = *ta++;
                        *tterm = tb - tterm;
                        {
                            int ia, na = *tterm;
                            ta = tterm; tb  = term;
                            for ( ia = 0; ia < na; ia++ ) *tb++ = *ta++;
                        }
                        if ( tb > AT.WorkTop ) {
                            MLOCK(ErrorMessageLock);
                            MesWork();
                            goto EndTest2;
                        }
                        AT.WorkPointer = tb;
                        goto ReStart;
                    }
                }
                NEXTARG(tf);
            }
            tf = t + FUNHEAD;
            while ( tf < t + t[1] ) {
                numargs++;
                if ( *tf > 0 && tf[1] != 0 ) todo = 0;
                NEXTARG(tf);
            }
            if ( todo && numargs == 2 ) {
                if ( *t == DIVFUNCTION ) AN.TeInFun = -9;
                else if ( *t == REMFUNCTION ) AN.TeInFun = -10;
                else if ( *t == INVERSEFUNCTION ) AN.TeInFun = -11;
                else if ( *t == MULFUNCTION ) AN.TeInFun = -14;
                else if ( *t == GCDFUNCTION ) AN.TeInFun = -8;
                AN.TeSuOut = 0;
                AR.TePos = -1;
                DONE(1)
            }
            else if ( todo && numargs == 3 ) {
                if ( *t == DIVFUNCTION ) AN.TeInFun = -9;
                else if ( *t == REMFUNCTION ) AN.TeInFun = -10;
                else if ( *t == GCDFUNCTION ) AN.TeInFun = -8;
                AN.TeSuOut = 0;
                AR.TePos = -1;
                DONE(1)
            }
```

```
                else if ( todo && *t == GCDFUNCTION ) {
                    AN.TeInFun = -8;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
              }
              else if ( *t == PERMUTATIONS && ( ( t[1] >= FUNHEAD+1
                && t[FUNHEAD] <= -FUNCTION ) || ( t[1] >= FUNHEAD+3
                && t[FUNHEAD] == -SNUMBER && t[FUNHEAD+2] <= -FUNCTION ) ) ) {
                AN.TeInFun = -12;
                AN.TeSuOut = 0;
                AR.TePos = -1;
                DONE(1)
              }
              else if ( *t == PARTITIONS ) {
                if ( TestPartitions(t,&(AT.partitions)) ) {
                    AT.partitions.where = t-term;
                    AN.TeInFun = -13;
                    AN.TeSuOut = 0;
                    AR.TePos = -1;
                    DONE(1)
                }
              }
            }
        }
        t += t[1];
    } while ( t < m );
    if ( funflag ) {    /* Search in functions */
DoSpec:
        t = term;
        AT.NestPoin->termsize = t;
        if ( AT.NestPoin == AT.Nest ) AN.EndNest = t + *t;
        t++;
        oldncmod = AN.ncmod;
        if ( t < m ) do {
            if ( *t < FUNCTION ) {
                t += t[1]; continue;
            }
            if ( AN.ncmod && ( ( AC.modmode & ALSOFUNARGS ) == 0 ) ) {
                if ( *t != AR.PolyFun ) AN.ncmod = 0;
                else                    AN.ncmod = oldncmod;
            }
            r = t + t[1];
            funnum = *t;
            if ( *t >= FUNCTION + WILDOFFSET ) funnum -= WILDOFFSET;
            if ( ( *t == NUMFACTORS || *t == FIRSTTERM || *t == CONTENTTERM )
             && t[1] == FUNHEAD+2 &&
            ( t[FUNHEAD] == -EXPRESSION || t[FUNHEAD] == -DOLLAREXPRESSION ) ) {
/*
                if ( *t == NUMFACTORS ) {
                    This we leave for Normalize
                }
*/
            }
            else if ( functions[funnum-FUNCTION].spec <= 0 ) {
                AT.NestPoin->funsize = t + 1;
                t1 = t;
                t += FUNHEAD;
                while ( t < r ) {   /* Sum over arguments */
                    if ( *t > 0 && t[1] ) { /* Argument is dirty  */
                        AT.NestPoin->argsize = t;
                        AT.NestPoin++;
/*                      stop = t + *t; */
                        t2 = t;
                        t += ARGHEAD;
                        while ( t < AT.NestPoin[-1].argsize+*(AT.NestPoin[-1].argsize) ) {
                                            /* Sum over terms */
                            AT.RecFlag++;
                            i = *t;
                            AN.subsubveto = 1;
/*
                            AN.subsubveto repairs a bug that became apparent
                            in an example by York Schroeder:
                                f(k1.k1)*replace_(k1,2*k2)
                            Is it possible to repair the counting of the various
                            length indicators? (JV 1-jun-2010)
*/
                            if ( ( retvalue = TestSub(BHEAD t,level) ) != 0 ) {
/*
                                Possible size changes:
                                Note defs at 471,467,460,400,425,328
*/
redosize:
                                if ( i > *t ) {
/*
                                    i -= *t;
```

```
                                    *t2 -= i;
                                    t1[1] -= i;
                                    t += *t;
                                    r = t + i;
                                    m = term + *term;
                                    while ( r < m ) *t++ = *r++;
                                    *term -= i;
*/
                                    i -= *t;
                                    t += *t;
                                    r = t + i;
                                    m = AN.EndNest;
                                    while ( r < m ) *t++ = *r++;
                                    t = AT.NestPoin[-1].argsize + ARGHEAD;
                                    n = AT.Nest;
                                    while ( n < AT.NestPoin ) {
                                        *(n->argsize) -= i;
                                        *(n->funsize) -= i;
                                        *(n->termsize) -= i;
                                        n++;
                                    }
                                    AN.EndNest -= i;
                                }
                                AN.subsubveto = 0;
                                t1[2] = 1;
                                if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 )
                                    t1[2] |= MUSTCLEANPRF;
                                AT.RecFlag--;
                                AT.NestPoin--;
                                AN.TeInFun++;
                                AR.TePos = 0;
                                AN.ncmod = oldncmod;
                                DONE(retvalue)
                            }
                            else {
                                /*
                                 * Somehow the next line fixes Issue #106.
                                 */
                                i = *t;
                                Normalize(BHEAD t);
/*
                                if ( i > *t ) { retvalue = 1; goto redosize; } */
                                /*
                                 * Experimentally, the next line fixes Issue #105.
                                 */
                                if ( *t == 0 ) { retvalue = 1; goto redosize; }
                                {
                                    WORD *tend = t + *t, *tt = t+1;
                                    stilldirty = 0;
                                    tend -= ABS(tend[-1]);
                                    while ( tt < tend ) {
                                        if ( *tt == SUBEXPRESSION || *tt == EXPRESSION ) {
                                            stilldirty = 1; break;
                                        }
                                        tt += tt[1];
                                    }
                                }
                                if ( i > *t ) {
/*

                                    We should not forget to correct the Nest
                                    stack. That caused trouble in the past.
*/

                                    retvalue = 1;
                                    i -= *t;
                                    t += *t;
                                    r = t + i;
                                    m = AN.EndNest;
                                    while ( r < m ) *t++ = *r++;
                                    t = AT.NestPoin[-1].argsize + ARGHEAD;
                                    n = AT.Nest;
                                    while ( n < AT.NestPoin ) {
                                        *(n->argsize) -= i;
                                        *(n->funsize) -= i;
                                        *(n->termsize) -= i;
                                        n++;
                                    }
                                    AN.EndNest -= i;
                                }
                            }
                            AN.subsubveto = 0;
                            AT.RecFlag--;
                            t += *t;
                        }
                        AT.NestPoin--;
/*

                        Argument contains no subexpressions.
                        It should be normalized and sorted.
                        The main problem is the storage.
```

```
*/
                        t = AT.NestPoin->argsize;
                        j = *t;
                        t += ARGHEAD;
                        NewSort(BHEAD0);
                        if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
                            AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
                            AR.SortType = SORTHIGHFIRST;
                        }
                        if ( AT.WorkPointer < term + *term )
                            AT.WorkPointer = term + *term;

                        while ( t < AT.NestPoin->argsize+*(AT.NestPoin->argsize) ) {
                            m = AT.WorkPointer;
                            r = t + *t;
                            do { *m++ = *t++; } while ( t < r );
                            r = AT.WorkPointer;
                            AT.WorkPointer = r + *r;
                            if ( Normalize(BHEAD r) ) {
                                if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
                                    AR.SortType = oldsorttype;
                                    AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
                                    t1[2] |= MUSTCLEANPRF;
                                }
                                LowerSortLevel(); goto EndTest;
                            }
                            if ( AN.ncmod != 0 ) {
                                if ( *r ) {
                                    if ( Modulus(r) ) {
                                        LowerSortLevel();
                                        AT.WorkPointer = r;
                                        if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
                                            AR.SortType = oldsorttype;
                                            AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
                                            t1[2] |= MUSTCLEANPRF;
                                        }
                                        goto EndTest;
                                    }
                                }
                            }
                            if ( AR.PolyFun > 0 ) {
                                if ( PrepPoly(BHEAD r,1) != 0 ) goto EndTest;
                            }
                            if ( *r ) StoreTerm(BHEAD r);
                            AT.WorkPointer = r;
                        }
                        if ( EndSort(BHEAD AT.WorkPointer+ARGHEAD,1) < 0 ) goto EndTest;
                        m = AT.WorkPointer+ARGHEAD;
                        if ( *t1 == AR.PolyFun && AR.PolyFunType == 2 ) {
                            AR.SortType = oldsorttype;
                            AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
                            t1[2] |= MUSTCLEANPRF;
                        }
                        while ( *m ) m += *m;
                        i = WORDDIF(m,AT.WorkPointer);
                        *AT.WorkPointer = i;
                        AT.WorkPointer[1] = stilldirty;
                        if ( ToFast(AT.WorkPointer,AT.WorkPointer) ) {
                            m = AT.WorkPointer;
                            if ( *m <= -FUNCTION ) { m++; i = 1; }
                            else { m += 2; i = 2; }
                        }
                        j = i - j;
                        if ( j > 0 ) {
                            r = m + j;
                            if ( r > AT.WorkTop ) {
                                MLOCK(ErrorMessageLock);
                                MesWork();
                                goto EndTest2;
                            }
                            do { *--r = *--m; } while ( m > AT.WorkPointer );
                            AT.WorkPointer = r;
                            m = AN.EndNest;
                            r = m + j;
                            stop = AT.NestPoin->argsize+*(AT.NestPoin->argsize);
                            do { *--r = *--m; } while ( m >= stop );
                        }
                        else if ( j < 0 ) {
                            m = AT.NestPoin->argsize+*(AT.NestPoin->argsize);
                            r = m + j;
                            do { *r++ = *m++; } while ( m < AN.EndNest );
                        }
                        m = AT.NestPoin->argsize;
                        r = AT.WorkPointer;
                        while ( --i >= 0 ) *m++ = *r++;
                        n = AT.Nest;
                        while ( n <= AT.NestPoin ) {
```

```
                            if ( *(n->argsize) > 0 && n != AT.NestPoin )
                                *(n->argsize) += j;
                            *(n->funsize) += j;
                            *(n->termsize) += j;
                            n++;
                        }
                        AN.EndNest += j;
/*                      (AT.NestPoin->argsize)[1] = 0;  */
                        if ( funnum == DENOMINATOR || funnum == EXPONENT ) {
                            if ( Normalize(BHEAD term) ) {
/*
                                In this case something has been substituted
                                Either a $ or a replace_?????
                                Originally we had here:

                                goto EndTest;

                                It seems better to restart.
*/
                                AN.ncmod = oldncmod;
                                goto ReStart;
                            }
/*
                            And size changes here?????
*/
                        }
                        AN.ncmod = oldncmod;
                        goto ReStart;
                    }
                    else if ( *t == -DOLLAREXPRESSION ) {
                        if ( ( *t1 == TERMSINEXPR || *t1 == SIZEOFFUNCTION )
                         && t1[1] == FUNHEAD+2 ) {}
                        else {
                            if ( AR.Eside != LHSIDE ) {
                                AN.TeInFun = 1; AR.TePos = 0;
                                AT.TMbuff = AM.dbufnum; t1[2] |= DIRTYFLAG;
                                AN.ncmod = oldncmod;
                                DONE(1)
                            }
                            AC.lhdollarflag = 1;
                        }
                    }
                    else if ( *t == -TERMSINBRACKET ) {
                        if ( AR.Eside != LHSIDE ) {
                            AN.TeInFun = 1; AR.TePos = 0;
                            t1[2] |= DIRTYFLAG;
                            AN.ncmod = oldncmod;
                            DONE(1)
                        }
                    }
                    else if ( AN.ncmod != 0 && *t == -SNUMBER ) {
                        if ( AN.ncmod == 1 || AN.ncmod == -1 ) {
                            isp = (UWORD)(AC.cmod[0]);
                            isp = t[1] % isp;
                            if ( ( AC.modmode & POSNEG ) != 0 ) {
                                if ( isp > (UWORD)(AC.cmod[0])/2 ) isp = isp - (UWORD)(AC.cmod[0]);
                                else if ( -isp > (UWORD)(AC.cmod[0])/2 ) isp = isp + (UWORD)(AC.cmod[0]);
                            }
                            else {
                                if ( isp < 0 ) isp += (UWORD)(AC.cmod[0]);
                            }
                            if ( isp <= MAXPOSITIVE && isp >= -MAXPOSITIVE ) {
                                t[1] = isp;
                            }
                        }
                    }
                    NEXTARG(t)
                }
                if ( funnum >= FUNCTION && functions[funnum-FUNCTION].tabl ) {
/*
                    Test whether the table catches
                    Test 1: index arguments and range. i will be the number
                        of the element in the table.
*/
                    WORD rhsnumber, *oldwork = AT.WorkPointer;
                    WORD ii, *p, *pp, *ppstop;
                    MINMAX *mm;
                    T = functions[funnum-FUNCTION].tabl;
/*
                    Because of tables with a variable number of indices
                    we need to make a copy of the pattern.
                    If we do this in the WorkSpace we get problems with EndNest.
                    This is why we use TermMalloc.
                    Now Tpattern is a copy that can be modified.
*/
#ifdef WITHPTHREADS
```

```
                    pp = T->pattern[AT.identity];
#else
                    pp = T->pattern;
#endif
                    if ( Tpattern == 0 ) Tpattern = TermMalloc("Tpattern");
                    p = Tpattern;
                    ii = pp[1];
                    for ( i = 0; i < ii; i++ ) *p++ = *pp++;

                    p = Tpattern + FUNHEAD+1;
                    mm = T->mm;
                    if ( T->sparse ) {
                        WORD xx;
                        t = t1+FUNHEAD;
                        if ( T->numind == 0 ) { isp = 0; xx = 0; }
                        else {
                            if ( T->numind < 0 ) {
                              if ( *t != -SNUMBER && t[2] != -SNUMBER ) {
                                xx = ABS(T->numind);
                                goto teststrict;
                              }
                              xx = t[1]+1;
                              if ( xx < 2 || xx > -T->numind ) goto teststrict;
                            }
                            else xx = T->numind;
                            for ( i = 0; i < xx; i++, t += 2 ) {
                                if ( *t != -SNUMBER ) break;
                            }
                            if ( i < xx ) goto teststrict;
                            isp = FindTableTree(T,t1+FUNHEAD,2);
                        }
                        if ( isp < 0 ) {
teststrict:                 if ( T->strict == -2 ) {
                                rhsnumber = AM.zerorhs;
                                tbufnum = AM.zbufnum;
                            }
                            else if ( T->strict == -3 ) {
                                rhsnumber = AM.onerhs;
                                tbufnum = AM.zbufnum;
                            }
                            else if ( T->strict < 0 ) goto NextFun;
                            else {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Element in table is undefined");
                                if ( Tpattern ) {
                                    TermFree(Tpattern,"Tpattern");
                                    Tpattern = 0;
                                }
                                goto showtable;
                            }
/*
                            Copy the indices;
*/
                            t = t1+FUNHEAD+1;
                            for ( i = 0; i < xx; i++ ) {
                                *p = *t; p+=2; t+=2;
                            }
                        }
                        else {
                            rhsnumber = T->tablepointers[isp+ABS(T->numind)];
#if ( TABLEEXTENSION == 2 )
                            tbufnum = T->bufnum;
#else
                            tbufnum = T->tablepointers[isp+ABS(T->numind)+1];
#endif
                            t = t1+FUNHEAD+1;
                            ii = xx;
                            while ( --ii >= 0 ) {
                                *p = *t; t += 2; p += 2;
                            }
                        }
                        if ( xx < ABS(T->numind) ) {
                            p--; ppstop = Tpattern+Tpattern[1];
                            pp = p+2*(-T->numind-xx);
                            while ( pp < ppstop ) *p++ = *pp++;
                            Tpattern[1] = p - Tpattern;
                        }
                        goto caughttable;
                    }
                    else {
                        i = 0;
                        t = t1 + FUNHEAD;
                        j = T->numind;
                        while ( --j >= 0 ) {
                            if ( *t != -SNUMBER ) goto NextFun;
                            t++;
                            if ( *t < mm->mini || *t > mm->maxi ) {
```

```
                                if ( T->bounds ) {
                                    MLOCK(ErrorMessageLock);
                                    MesPrint("Table boundary check. Argument %d",
                                    T->numind-j);
showtable:                          AO.OutFill = AO.OutputLine = (UBYTE *)m;
                                    AO.OutSkip = 8;
                                    IniLine(0);
                                    WriteSubTerm(t1,1);
                                    FiniLine();
                                    MUNLOCK(ErrorMessageLock);
                                    if ( Tpattern ) {
                                        TermFree(Tpattern,"Tpattern");
                                        Tpattern = 0;
                                    }
                                    SETERROR(-1)
                                }
                                if ( Tpattern ) {
                                    TermFree(Tpattern,"Tpattern");
                                    Tpattern = 0;
                                }
                                goto NextFun;
                            }
                            i += ( *t - mm->mini ) * (LONG)(mm->size);
                            *p = *t++;
                            p += 2;
                            mm++;
                        }
/*
                        Test now whether the entry exists.
*/
                        i *= TABLEEXTENSION;
                        if ( T->tablepointers[i] == -1 ) {
                            if ( T->strict == -2 ) {
                                rhsnumber = AM.zerorhs;
                                tbufnum = AM.zbufnum;
                            }
                            else if ( T->strict == -3 ) {
                                rhsnumber = AM.onerhs;
                                tbufnum = AM.zbufnum;
                            }
                            else if ( T->strict < 0 ) {
                                if ( Tpattern ) {
                                    TermFree(Tpattern,"Tpattern");
                                    Tpattern = 0;
                                }
                                goto NextFun;
                            }
                            else {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Element in table is undefined");
                                if ( Tpattern ) {
                                    TermFree(Tpattern,"Tpattern");
                                    Tpattern = 0;
                                }
                                goto showtable;
                            }
                        }
                        else {
                            rhsnumber = T->tablepointers[i];
#if ( TABLEEXTENSION == 2 )
                            tbufnum = T->bufnum;
#else
                            tbufnum = T->tablepointers[i+1];
#endif
                        }
                    }
/*
                    If there are more arguments we have to do some
                    pattern matching. This should be easy. We adapted the
                    pattern, so that the array indices match already.
                    Note that if there is no match the program will become
                    very slow.
*/
caughttable:
#ifdef WITHPTHREADS
                    AN.FullProto = T->prototype[AT.identity];
#else
                    AN.FullProto = T->prototype;
#endif
                    AN.WildValue = AN.FullProto + SUBEXPSIZE;
                    AN.WildStop = AN.FullProto+AN.FullProto[1];
                    ClearWild(BHEAD0);
                    AN.RepFunNum = 0;
                    AN.RepFunList = AN.EndNest;
                    AT.WorkPointer = (WORD *)(((UBYTE *)(AN.EndNest)) + AM.MaxTer/2);
                    if ( AT.WorkPointer >= AT.WorkTop ) {
                        MLOCK(ErrorMessageLock);
```

```
                        MesWork();
                        MUNLOCK(ErrorMessageLock);
                    }
                    wilds = 0;
                    if ( MatchFunction(BHEAD Tpattern,t1,&wilds) > 0 ) {
                        AT.WorkPointer = oldwork;
                        if ( AT.NestPoin != AT.Nest ) {
                            AN.ncmod = oldncmod;
                            if ( Tpattern ) {
                                TermFree(Tpattern,"Tpattern");
                                Tpattern = 0;
                            }
                            DONE(1)
                        }

                        m = AN.FullProto;
                        retvalue = m[2] = rhsnumber;
                        m[4] = tbufnum;
                        t = t1;
                        j = t[1];
                        i = m[1];
                        if ( j > i ) {
                            j = i - j;
                            NCOPY(t,m,i);
                            m = term + *term;
                            while ( r < m ) *t++ = *r++;
                            *term += j;
                        }
                        else if ( j < i ) {
                            j = i-j;
                            t = term + *term;
                            while ( t >= r ) { t[j] = *t; t--; }
                            t = t1;
                            NCOPY(t,m,i);
                            *term += j;
                        }
                        else {
                            NCOPY(t,m,j);
                        }
                        AN.TeInFun = 0;
                        AR.TePos = 0;
                        AN.TeSuOut = -1;
                        if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
                        AT.TMbuff = tbufnum;
                        AN.ncmod = oldncmod;
                        DONE(retvalue);
                    }
                    AT.WorkPointer = oldwork;
                    if ( Tpattern ) {
                        TermFree(Tpattern,"Tpattern");
                        Tpattern = 0;
                    }
                }
NextFun:;
            }
            else if ( ( t[2] & DIRTYFLAG ) != 0 ) {
                t += FUNHEAD;
                while ( t < r ) {
                    if ( *t == FUNNYDOLLAR ) {
                        if ( AR.Eside != LHSIDE ) {
                            AN.TeInFun = 1;
                            AR.TePos = 0;
                            AT.TMbuff = AM.dbufnum;
                            AN.ncmod = oldncmod;
                            DONE(1)
                        }
                        AC.lhdollarflag = 1;
                    }
                    t++;
                }
            }
            t = r;
            AN.ncmod = oldncmod;
        } while ( t < m );
    }
Done:
    if ( Tpattern ) {
        TermFree(Tpattern,"Tpattern");
        Tpattern = 0;
    }
    return(retvalue);
EndTest:;
    MLOCK(ErrorMessageLock);
EndTest2:;
    MesCall("TestSub");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
```

```
}

/*
        #] TestSub :
        #[ InFunction :         WORD InFunction(term,termout)
*/
WORD InFunction(PHEAD WORD *term, WORD *termout)
{
    GETBIDENTITY
    WORD *m, *t, *r, *rr, sign = 1, oldncmod;
    WORD *u, *v, *w, *from, *to,
        ipp, olddefer = AR.DeferFlag, oldPolyFun = AR.PolyFun, i, j;
    LONG numterms;
    from = t = term;
    r = t + *t - 1;
    m = r - ABS(*r) + 1;
    t++;
    while ( t < m ) {
        if ( *t >= FUNCTION+WILDOFFSET ) ipp = *t - WILDOFFSET;
        else ipp = *t;
        if ( AR.TePos ) {
            if ( ( term + AR.TePos ) == t ) {
                m = termout;
                while ( from < t ) *m++ = *from++;
                *m++ = DENOMINATOR;
                *m++ = t[1] + 4 + FUNHEAD + ARGHEAD;
                *m++ = DIRTYFLAG;
                FILLFUN3(m)
                *m++ = t[1] + 4 + ARGHEAD;
                *m++ = 1;
                FILLARG(m)
                *m++ = t[1] + 4;
                t[3] = -t[3];
                v = t + t[1];
                while ( t < v ) *m++ = *t++;
                from[3] = -from[3];
                *m++ = 1;
                *m++ = 1;
                *m++ = 3;
                r = term + *term;
                while ( t < r ) *m++ = *t++;
                if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Output term too large (%d words) (MaxTermSize: %d words)", m-termout, AM.MaxTer/sizeof(WORD));
                    MUNLOCK(ErrorMessageLock);
                    goto TooLarge;
                }
                *termout = WORDDIF(m,termout);
                return(0);
            }
        }
        else if ( ( *t >= FUNCTION && functions[ipp-FUNCTION].spec <= 0 )
        && ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) {
            m = termout;
            r = t + t[1];
            u = t;
            t += FUNHEAD;
            oldncmod = AN.ncmod;
            while ( t < r ) {   /* t points at an argument */
                if ( *t > 0 && t[1] ) { /* Argument has been modified */
                    WORD oldsorttype = AR.SortType;
                    /* This whole argument must be redone */

                    if ( ( AN.ncmod != 0 )
                        && ( ( AC.modmode & ALSOFUNARGS ) == 0 )
                        && ( *u != AR.PolyFun ) ) { AN.ncmod = 0; }
                    AR.DeferFlag = 0;
                    v = t + *t;
                    t += ARGHEAD;       /* First term */
                    w = 0;  /* to appease the compilers warning devices */
                    while ( from < t ) {
                        if ( from == u ) w = m;
                        *m++ = *from++;
                    }
                    to = m;
                    NewSort(BHEAD0);
                    if ( *u == AR.PolyFun && AR.PolyFunType == 2 ) {
                        AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
                        AR.SortType = SORTHIGHFIRST;
                    }
/*
                    AR.PolyFun = 0;
*/
                    while ( t < v ) {
                        i = *t;
                        NCOPY(m,t,i);
```

```
                        m = to;
                        if ( AT.WorkPointer < m+*m ) AT.WorkPointer = m + *m;
                        if ( Generator(BHEAD m,AR.Cnumlhs) ) {
                            AN.ncmod = oldncmod;
                            LowerSortLevel(); goto InFunc;
                        }
                    }
                    /* w = the function */
                    /* v = the next argument */
                    /* u = the function */
                    /* to is new argument */

                    to -= ARGHEAD;
                    if ( EndSort(BHEAD m,1) < 0 ) {
                        AN.ncmod = oldncmod;
                        goto InFunc;
                    }
                    AR.PolyFun = oldPolyFun;
                    if ( *u == AR.PolyFun && AR.PolyFunType == 2 ) {
                        AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
                        AR.SortType = oldsorttype;
                    }
                    while ( *m ) m += *m;
                    *to = WORDDIF(m,to);
                    to[1] = 1;  /* ??????? or rather 0?. 24-mar-2006 JV */
                    if ( ToFast(to,to) ) {
                        if ( *to <= -FUNCTION ) m = to+1;
                        else m = to+2;
                    }
                    w[1] = WORDDIF(m,w) + WORDDIF(r,v);
                    r = term + *term;
                    t = v;
                    while ( t < r ) *m++ = *t++;
                    if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
      m-termout, AM.MaxTer/sizeof(WORD));
                        MUNLOCK(ErrorMessageLock);
                        goto TooLarge;
                    }
                    *termout = WORDDIF(m,termout);
                    AR.DeferFlag = olddefer;
                    AN.ncmod = oldncmod;
                    return(0);
                }
                else if ( *t == -DOLLAREXPRESSION ) {
                    if ( AR.Eside == LHSIDE ) {
                        NEXTARG(t)
                        AC.lhdollarflag = 1;
                    }
                    else {
/*
                        This whole argument must be redone
*/
                    DOLLARS d = Dollars + t[1];
#ifdef WITHPTHREADS
                    int nummodopt, dtype = -1;
                    if ( AS.MultiThreaded ) {
                        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                            if ( t[1] == ModOptdollars[nummodopt].number ) break;
                        }
                        if ( nummodopt < NumModOptdollars ) {
                            dtype = ModOptdollars[nummodopt].type;
                            if ( dtype == MODLOCAL ) {
                                d = ModOptdollars[nummodopt].dstruct+AT.identity;
                            }
                            else {
                                LOCK(d->pthreadslockread);
                            }
                        }
                    }
#endif
                    oldncmod = AN.ncmod;
                    if ( ( AN.ncmod != 0 )
                        && ( ( AC.modmode & ALSOFUNARGS ) == 0 )
                        && ( *u != AR.PolyFun ) ) { AN.ncmod = 0; }
                    AR.DeferFlag = 0;
                    v = t + 2;
                    w = 0;  /* to appease the compilers warning devices */
                    while ( from < t ) {
                        if ( from == u ) w = m;
                        *m++ = *from++;
                    }
                    to = m;
                    switch ( d->type ) {
                        case DOLINDEX:
                            if ( d->index >= 0 && d->index < AM.OffsetIndex ) {
```

```
                                *m++ = -SNUMBER; *m++ = d->index;
                            }
                            else { *m++ = -INDEX; *m++ = d->index; }
                            break;
                        case DOLZERO:
                            *m++ = -SNUMBER; *m++ = 0; break;
                        case DOLNUMBER:
                            if ( d->where[0] == 4 &&
                            ( d->where[1] & MAXPOSITIVE ) == d->where[1] ) {
                                *m++ = -SNUMBER;
                                if ( d->where[3] >= 0 ) *m++ = d->where[1];
                                else *m++ = -d->where[1];
                                break;
                            }
                            /* fall through */
                        case DOLTERMS:
/*
                            Here we have the special case of the PolyRatFun
                            That function may have a different sort of the
                            terms in the argument.
*/
                            to = m; r = d->where;
                            *m++ = 0; *m++ = 1;
                            FILLARG(m)
                            while ( *r ) {
                                i = *r; NCOPY(m,r,i)
                            }
                            *to = m-to;
                            if ( ToFast(to,to) ) {
                                if ( *to <= -FUNCTION ) m = to+1;
                                else m = to+2;
                            }
                            else if ( *u == AR.PolyFun && AR.PolyFunType == 2 ) {
                                AR.PolyFun = 0;
                                NewSort(BHEAD0);
                                AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
                                r = to + ARGHEAD;
                                while ( r < m ) {
                                    rr = r; r += *r;
                                    if ( SymbolNormalize(rr) ) goto InFunc;
                                    if ( StoreTerm(BHEAD rr) ) {
                                        AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
                                        LowerSortLevel();
                                        Terminate(-1);
                                    }
                                }
                                if ( EndSort(BHEAD to+ARGHEAD,1) < 0 ) goto InFunc;
                                AR.PolyFun = oldPolyFun;
                                AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
                                m = to+ARGHEAD;
                                if ( *m == 0 ) {
                                    *to = -SNUMBER;
                                    to[1] = 0;
                                    m = to + 2;
                                }
                                else {
                                    while ( *m ) m += *m;
                                    *t = m - to;
                                    if ( ToFast(to,to) ) {
                                        if ( *to <= -FUNCTION ) m = to+1;
                                        else m = to+2;
                                    }
                                }
                            }
                            w[1] = w[1] - 2 + (m-to);
                            break;
                        case DOLSUBTERM:
                            to = m; r = d->where;
                            i = r[1];
                            *m++ = i+4+ARGHEAD; *m++ = 1;
                            FILLARG(m)
                            *m++ = i+4;
                            while ( --i >= 0 ) *m++ = *r++;
                            *m++ = 1; *m++ = 1; *m++ = 3;
                            if ( ToFast(to,to) ) {
                                if ( *to <= -FUNCTION ) m = to+1;
                                else m = to+2;
                            }
                            w[1] = w[1] - 2 + (m-to);
                            break;
                        case DOLARGUMENT:
                            to = m; r = d->where;
                            if ( *r > 0 ) {
                                i = *r - 2;
                                *m++ = *r++; *m++ = 1; r++;
                                while ( --i >= 0 ) *m++ = *r++;
                            }
```

```
                            else if ( *r <= -FUNCTION ) *m++ = *r++;
                            else { *m++ = *r++; *m++ = *r++; }
                            w[1] = w[1] - 2 + (m-to);
                            break;
                        case DOLWILDARGS:
                            to = m; r = d->where;
                            if ( *r > 0 ) { /* Tensor arguments */
                                i = *r++;
                                while ( --i >= 0 ) {
                                    if ( *r < 0 ) {
                                        *m++ = -VECTOR; *m++ = *r++;
                                    }
                                    else if ( *r >= AM.OffsetIndex ) {
                                        *m++ = -INDEX; *m++ = *r++;
                                    }
                                    else { *m++ = -SNUMBER; *m++ = *r++; }
                                }
                            }
                            else { /* Regular arguments */
                                r++;
                                while ( *r ) {
                                    if ( *r > 0 ) {
                                        i = *r - 2;
                                        *m++ = *r++; *m++ = 1; r++;
                                        while ( --i >= 0 ) *m++ = *r++;
                                    }
                                    else if ( *r <= -FUNCTION ) *m++ = *r++;
                                    else { *m++ = *r++; *m++ = *r++; }
                                }
                            }
                            w[1] = w[1] - 2 + (m-to);
                            break;
                        case DOLUNDEFINED:
                        default:
                            MLOCK(ErrorMessageLock);
                            MesPrint("!!!Undefined $-variable: $%s!!!",
                            AC.dollarnames->namebuffer+d->name);
                            MUNLOCK(ErrorMessageLock);
#ifdef WITHPTHREADS
                            if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                            Terminate(-1);
                    }
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                    r = term + *term;
                    t = v;
                    while ( t < r ) *m++ = *t++;
                    if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
m-termout, AM.MaxTer/sizeof(WORD));
                        MUNLOCK(ErrorMessageLock);
                        goto TooLarge;
                    }
                    *termout = WORDDIF(m,termout);
                    AR.DeferFlag = olddefer;
                    AN.ncmod = oldncmod;
                    return(0);
                }
                }
                else if ( *t == -TERMSINBRACKET ) {
                    if ( AC.ComDefer ) numterms = CountTerms1(BHEAD0);
                    else               numterms = 1;
/*
                    Compose the output term
                    First copy the part till this function argument
                    m points at the output term space
                    u points at the start of the function
                    t points at the start of the argument
*/
                    w = 0;
                    while ( from < t ) {
                        if ( from == u ) w = m;
                        *m++ = *from++;
                    }
                    if ( ( numterms & MAXPOSITIVE ) == numterms ) {
                        *m++ = -SNUMBER; *m++ =  numterms & MAXPOSITIVE;
                        w[1] += 1;
                    }
                    else if ( ( i = numterms » BITSINWORD ) == 0 ) {
                        *m++ = ARGHEAD+4;
                        for ( j = 1; j < ARGHEAD; j++ ) *m++ = 0;
                        *m++ = 4; *m++ = numterms & WORDMASK; *m++ = 1; *m++ = 3;
                        w[1] += ARGHEAD+3;
                    }
```

```
                    else {
                        *m++ = ARGHEAD+6;
                        for ( j = 1; j < ARGHEAD; j++ ) *m++ = 0;
                        *m++ = 6; *m++ = numterms & WORDMASK;
                        *m++ = i; *m++ = 1; *m++ = 0; *m++ = 5;
                        w[1] += ARGHEAD+5;
                    }
                    from++;  /* Skip our function */
                    r = term + *term;
                    while ( from < r ) *m++ = *from++;
                    if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
m-termout, AM.MaxTer/sizeof(WORD));
                        MUNLOCK(ErrorMessageLock);
                        goto TooLarge;
                    }
                    *termout = WORDDIF(m,termout);
                    return(0);
                }
                else { NEXTARG(t) }
            }
            t = u;
        }
        else if ( ( *t >= FUNCTION && functions[ipp-FUNCTION].spec > 0 )
        && ( t[2] & DIRTYFLAG ) == DIRTYFLAG ) { /* Could be FUNNYDOLLAR */
            u = t; v = t + t[1];
            t += FUNHEAD;
            while ( t < v ) {
                if ( *t == FUNNYDOLLAR ) {
                    if ( AR.Eside != LHSIDE ) {
                    DOLLARS d = Dollars + t[1];
#ifdef WITHPTHREADS
                    int nummodopt, dtype = -1;
                    if ( AS.MultiThreaded ) {
                        for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                            if ( t[1] == ModOptdollars[nummodopt].number ) break;
                        }
                        if ( nummodopt < NumModOptdollars ) {
                            dtype = ModOptdollars[nummodopt].type;
                            if ( dtype == MODLOCAL ) {
                                d = ModOptdollars[nummodopt].dstruct+AT.identity;
                            }
                            else {
                                LOCK(d->pthreadslockread);
                            }
                        }
                    }
#endif
                    oldncmod = AN.ncmod;
                    if ( ( AN.ncmod != 0 )
                        && ( ( AC.modmode & ALSOFUNARGS ) == 0 )
                        && ( *u != AR.PolyFun ) ) { AN.ncmod = 0; }
                    m = termout; w = 0;
                    while ( from < t ) {
                        if ( from == u ) w = m;
                        *m++ = *from++;
                    }
                    to = m;
                    switch ( d->type ) {
                        case DOLINDEX:
                            *m++ = d->index; break;
                        case DOLZERO:
                            *m++ = 0; break;
                        case DOLNUMBER:
                        case DOLTERMS:
                            if ( d->where[0] == 4 && d->where[4] == 0
                            && d->where[3] == 3 && d->where[2] == 1
                            && d->where[1] < AM.OffsetIndex ) {
                                *m++ = d->where[1];
                            }
                            else {
wrongtype:;
#ifdef WITHPTHREADS
                                if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                MLOCK(ErrorMessageLock);
                                MesPrint("$%s has wrong type for tensor substitution",
                                AC.dollarnames->namebuffer+d->name);
                                MUNLOCK(ErrorMessageLock);
                                AN.ncmod = oldncmod;
                                return(-1);
                            }
                            break;
                        case DOLARGUMENT:
                            if ( d->where[0] == -INDEX ) {
                                *m++ = d->where[1]; break;
```

```
                            }
                            else if ( d->where[0] == -VECTOR ) {
                                *m++ = d->where[1]; break;
                            }
                            else if ( d->where[0] == -MINVECTOR ) {
                                *m++ = d->where[1];
                                sign = -sign;
                                break;
                            }
                            else if ( d->where[0] == -SNUMBER ) {
                                if ( d->where[1] >= 0
                                && d->where[1] < AM.OffsetIndex ) {
                                    *m++ = d->where[1]; break;
                                }
                            }
                            goto wrongtype;
                        case DOLWILDARGS:
                            if ( d->where[0] > 0 ) {
                                r = d->where; i = *r++;
                                while ( --i >= 0 ) *m++ = *r++;
                            }
                            else {
                                r = d->where + 1;
                                while ( *r ) {
                                    if ( *r == -INDEX ) {
                                        *m++ = r[1]; r += 2; continue;
                                    }
                                    else if ( *r == -VECTOR ) {
                                        *m++ = r[1]; r += 2; continue;
                                    }
                                    else if ( *r == -MINVECTOR ) {
                                        *m++ = r[1]; r += 2;
                                        sign = -sign; continue;
                                    }
                                    else if ( *r == -SNUMBER ) {
                                        if ( r[1] >= 0
                                        && r[1] < AM.OffsetIndex ) {
                                            *m++ = r[1]; r += 2; continue;
                                        }
                                    }
                                    goto wrongtype;
                                }
                            }
                            break;
                        case DOLSUBTERM:
                            r = d->where;
                            if ( *r == INDEX && r[1] == 3 ) {
                                *m++ = r[2];
                            }
                            else goto wrongtype;
                            break;
                        case DOLUNDEFINED:
#ifdef WITHPTHREADS
                            if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                            MLOCK(ErrorMessageLock);
                            MesPrint("$%s is undefined in tensor substitution",
                            AC.dollarnames->namebuffer+d->name);
                            MUNLOCK(ErrorMessageLock);
                            AN.ncmod = oldncmod;
                            return(-1);
                    }
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                    w[1] = w[1] - 2 + (m-to);
                    from += 2;
                    term += *term;
                    while ( from < term ) *m++ = *from++;
                    if ( sign < 0 ) m[-1] = -m[-1];
                    if ( (m-termout) > (LONG)(AM.MaxTer/sizeof(WORD)) ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Output term too large (%d words) (MaxTermSize: %d words)",
      m-termout, AM.MaxTer/sizeof(WORD));
                        MUNLOCK(ErrorMessageLock);
                        goto TooLarge;
                    }
                    *termout = m - termout;
                    AN.ncmod = oldncmod;
                    return(0);
                }
                else {
                    AC.lhdollarflag = 1;
                }
                }
                t++;
            }
```

```
            t = u;
        }
        t += t[1];
    }
    MLOCK(ErrorMessageLock);
    MesPrint("Internal error in InFunction: Function not encountered.");
    if ( AM.tracebackflag ) {
        MesPrint("%w: AR.TePos = %d",AR.TePos);
        MesPrint("%w: AN.TeInFun = %d",AN.TeInFun);
        termout = term;
        AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer + AM.MaxTer;
        AO.OutSkip = 3;
        FiniLine();
        i = *termout;
        while ( --i >= 0 ) {
            TalToLine((UWORD)(*termout++));
            TokenToLine((UBYTE *)"  ");
        }
        AO.OutSkip = 0;
        FiniLine();
        MesCall("InFunction");
    }
    MUNLOCK(ErrorMessageLock);
    return(1);

InFunc:
    MLOCK(ErrorMessageLock);
    MesCall("InFunction");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)

TooLarge:
    MLOCK(ErrorMessageLock);
    MesCall("InFunction");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] InFunction :
        #[ InsertTerm :         WORD InsertTerm(term,replac,extractbuff,position,termout)
*/
WORD InsertTerm(PHEAD WORD *term, WORD replac, WORD extractbuff, WORD *position, WORD *termout,
                WORD tepos)
{
    GETBIDENTITY
    WORD *m, *t, *r, i, l2, j;
    WORD *u, *v, l1, *coef;
    coef = AT.WorkPointer;
    if ( ( AT.WorkPointer = coef + 2*AM.MaxTal ) > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    t = term;
    r = t + *t;
    l1 = l2 = r[-1];
    m = r - ABS(l2);
    if ( tepos > 0 ) {
        t = term + tepos;
        goto foundit;
    }
    t++;
    while ( t < m ) {
        if ( *t == SUBEXPRESSION && t[2] == replac && t[3] && t[4] == extractbuff ) {
            r = t + t[1];
            while ( *r == SUBEXPRESSION && r[2] == replac && r[3] && r < m && r[4] == extractbuff ) {
                t = r; r += r[1];
            }
foundit:;
            u = m;
            r = term;
            m = termout;
            do { *m++ = *r++; } while ( r < t );
            if ( t[1] > SUBEXPSIZE ) {
/*
                if this is a dollar expression there are no wildcards
*/
                i = *--m;
                if ( ( l2 = WildFill(BHEAD m,position,t) ) < 0 ) goto InsCall;
                *m = i;
                m += l2-1;
                l2 = *m;
                i = ( j = ABS(l2) ) - 1;
                r = coef + i;
                do { *--r = *--m; } while ( --i > 0 );
```

```
            }
            else {
                v = t;
                t = position;
                r = t + *t;
                l2 = r[-1];
                r -= ( j = ABS(l2) );
                t++;
                if ( t < r ) do { *m++ = *t++; } while ( t < r );
                t = v;
            }
            t += t[1];
            while ( t < u && *t == DOLLAREXPR2 ) t += t[1];
ComAct:     if ( t < u ) do { *m++ = *t++; } while ( t < u );
            if ( *r == 1 && r[1] == 1 && j == 3 ) {
                if ( l2 < 0 ) l1 = -l1;
                i = ABS(l1)-1;
                NCOPY(m,t,i);
                *m++ = l1;
            }
            else {
                if ( MulRat(BHEAD (UWORD *)u,REDLENG(l1),(UWORD *)r,REDLENG(l2),
                (UWORD *)m,&l1) ) goto InsCall;
                l2 = l1;
                l2 *= 2;
                if ( l2 < 0 ) {
                    m -= l2;
                    *m++ = l2-1;
                }
                else {
                    m += l2;
                    *m++ = l2+1;
                }
            }
            *termout = WORDDIF(m,termout);
            if ( (*termout)*((LONG)sizeof(WORD)) > AM.MaxTer ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Term too complex during substitution (%d words). MaxTermSize (%l words) is
too small.", *termout, AM.MaxTer/(LONG)sizeof(WORD) );
                goto InsCall2;
            }
            AT.WorkPointer = coef;
            return(0);
        }
        t += t[1];
    }
/*
    The next action is for when there is no subexpression pointer.
    We append the extra term. Effectively the routine becomes now a
    merge routine for two terms.
*/
    v = t;
    u = m;
    r = term;
    m = termout;
    do { *m++ = *r++; } while ( r < t );
    t = position;
    r = t + *t;
    l2 = r[-1];
    r -= ( j = ABS(l2) );
    t++;
    if ( t < r ) do { *m++ = *t++; } while ( t < r );
    t = v;
    goto ComAct;

InsCall:
    MLOCK(ErrorMessageLock);
InsCall2:
    MesCall("InsertTerm");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] InsertTerm :
        #[ PasteFile :          WORD PasteFile(num,acc,pos,accf,renum,freeze,nexpr)
*/
LONG PasteFile(PHEAD WORD number, WORD *accum, POSITION *position, WORD **accfill,
               RENUMBER renumber, WORD *freeze, WORD nexpr)
{
    GETBIDENTITY
    WORD *r, l, *m, i;
    WORD *stop, *s1, *s2;
/*  POSITION AccPos; bug 12-apr-2008 JV */
    WORD InCompState;
    WORD *oldipointer;
    LONG retlength;
```

```
    stop = (WORD *)(((UBYTE *)(accum)) + 2*AM.MaxTer);
    *accum++ = number;
    while ( --number >= 0 ) accum += *accum;
    if ( freeze ) {
/*      AccPos = *position; bug 12-apr-2008 JV */
        oldipointer = AR.CompressPointer;
        do {
            AR.CompressPointer = oldipointer;
/*          if ( ( l = GetFromStore(accum,&AccPos,renumber,&InCompState,nexpr) ) < 0 ) bug 12-apr-2008 JV */
            if ( ( l = GetFromStore(accum,position,renumber,&InCompState,nexpr) ) < 0 )
                goto PasErr;
            if ( !l ) { *accum = 0; return(0); }
            r = accum;
            m = r + *r;
            m -= ABS(m[-1]);
            r++;
            while ( r < m && *r != HAAKJE ) r += r[1];
            if ( r >= m ) {
                if ( *freeze != 4 ) l = -1;
            }
            else {
/*
                The algorithm for accepting terms with a given (freeze)
                representation outside brackets is rather crude. A refinement
                would be to store the part outside the bracket and skip the
                term when this part doesn't alter (and is unacceptable).
                Once accepting one can keep accepting till the bracket alters
                and then one may stop the generation. It is necessary to
                set up a struct to remember the bracket and the progress
                status.
*/
                m = AT.WorkPointer;
                s2 = r;
                r = accum;
                *m++ = WORDDIF(s2,r) + 3;
                r++;
                while ( r < s2 ) *m++ = *r++;
                *m++ = 1; *m++ = 1; *m++ = 3;
                m = AT.WorkPointer;
                if ( Normalize(BHEAD AT.WorkPointer) ) goto PasErr;
                r = freeze;
                i = *m;
                while ( --i >= 0 && *m++ == *r++ ) {}
                if ( i > 0 ) {
                    l = -1;
                }
                else {  /* Term to be accepted */
                    r = accum;
                    s1 = r + *r;
                    r++;
                    m = s2;
                    m += m[1];
                    do { *r++ = *m++; } while ( m < s1 );
                    *accum = l = WORDDIF(r,accum);
                }
            }
        } while ( l < 0 );
        retlength = InCompState;
/*      retlength = DIFBASE(AccPos,*position) / sizeof(WORD);  bug 12-apr-2008 JV */
    }
    else {
        if ( ( l = GetFromStore(accum,position,renumber,&InCompState,nexpr) ) < 0 ) {
            MLOCK(ErrorMessageLock);
            MesCall("PasteFile");
            MUNLOCK(ErrorMessageLock);
            SETERROR(-1)
        }
        if ( l == 0 ) { *accum = 0; return(0); }
        retlength = InCompState;
    }
    accum += l;
    if ( accum > stop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Buffer too small in PasteFile");
        MUNLOCK(ErrorMessageLock);
        SETERROR(-1)
    }
    *accum = 0;
    *accfill = accum;
    return(retlength);
PasErr:
    MLOCK(ErrorMessageLock);
    MesCall("PasteFile");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}
```

```
/*
        #] PasteFile :
        #[ PasteTerm :          WORD PasteTerm(number,accum,position,times,divby)
*/
WORD *PasteTerm(PHEAD WORD number, WORD *accum, WORD *position, WORD times, WORD divby)
{
    GETBIDENTITY
    WORD *t, *r, x, y, z;
    WORD *m, *u, l1, a[2];
    m = (WORD *)(((UBYTE *)(accum)) + AM.MaxTer);
/*  m = (WORD *)(((UBYTE *)(accum)) + 2*AM.MaxTer); */
    *accum++ = number;
    while ( --number >= 0 ) accum += *accum;
    if ( times == divby ) {
        t = position;
        r = t + *t;
        if ( t < r ) do { *accum++ = *t++; } while ( t < r );
    }
    else {
        u = accum;
        t = position;
        r = t + *t - 1;
        l1 = *r;
        r -= ABS(*r) - 1;
        if ( t < r ) do { *accum++ = *t++; } while ( t < r );
        if ( divby > times ) { x = divby; y = times; }
        else { x = times; y = divby; }
        z = x%y;
        while ( z ) { x = y; y = z; z = x%y; }
        if ( y != 1 ) { divby /= y; times /= y; }
        a[1] = divby;
        a[0] = times;
        if ( MulRat(BHEAD (UWORD *)t,REDLENG(l1),(UWORD *)a,1,(UWORD *)accum,&l1) ) {
            MLOCK(ErrorMessageLock);
            MesCall("PasteTerm");
            MUNLOCK(ErrorMessageLock);
            return(0);
        }
        x = l1;
        x *= 2;
        if ( x < 0 ) { accum -= x; *accum++ = x - 1; }
        else         { accum += x; *accum++ = x + 1; }
        *u = WORDDIF(accum,u);
    }
    if ( accum >= m ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Buffer too small in PasteTerm");
        MUNLOCK(ErrorMessageLock);
        return(0);
    }
    *accum = 0;
    return(accum);
}

/*
        #] PasteTerm :
        #[ FiniTerm :          WORD FiniTerm(term,accum,termout,number)
*/
WORD FiniTerm(PHEAD WORD *term, WORD *accum, WORD *termout, WORD number, WORD tepos)
{
    GETBIDENTITY
    WORD *m, *t, *r, i, numacc, l2, ipp;
    WORD *u, *v, l1, *coef = AT.WorkPointer, *oldaccum;
    if ( ( AT.WorkPointer = coef + 2*AM.MaxTal ) > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    oldaccum = accum;
    t = term;
    m = t + *t - 1;
    l1 = REDLENG(*m);
    i = ABS(*m) - 1;
    r = coef + i;
    do { *--r = *--m; } while ( --i > 0 ); /* Copies coefficient */
    if ( tepos > 0 ) {
        t = term + tepos;
        goto foundit;
    }
    t++;
    if ( t < m ) do {
        if ( ( ( *t == SUBEXPRESSION && ( *(r=t+t[1]) != SUBEXPRESSION
        || r >= m || !r[3] ) ) || *t == EXPRESSION ) && t[2] == number && t[3] ) {
foundit:;
            u = m;
```

```
            r = term;
            m = termout;
            if ( r < t ) do { *m++ = *r++; } while ( r < t );
            numacc = *accum++;
            if ( numacc >= 0 ) do {
                if ( *t == EXPRESSION ) {
                    v = t + t[1];
                    r = t + SUBEXPSIZE;
                    while ( r < v ) {
                        if ( *r == WILDCARDS ) {
                            r += 2;
                            i = *--m;
                            if ( ( l2 = WildFill(BHEAD m,accum,r) ) < 0 ) goto FiniCall;
                            goto AllWild;
                        }
                        r += r[1];
                    }
                    goto NoWild;
                }
                else if ( t[1] > SUBEXPSIZE && t[SUBEXPSIZE] != FROMBRAC ) {
                    i = *--m;
                    if ( ( l2 = WildFill(BHEAD m,accum,t) ) < 0 ) goto FiniCall;
AllWild:            *m = i;
                    m += l2-1;
                    l2 = *m;
                    m -= ABS(l2) - 1;
                    r = m;
                }
                else {
NoWild:             r = accum;
                    v = r + *r - 1;
                    l2 = *v;
                    v -= ABS(l2) - 1;
                    r++;
                    if ( r < v ) do { *m++ = *r++; } while ( r < v );
                }
                if ( *r == 1 && r[1] == 1 && ABS(l2) == 3 ) {
                    if ( l2 < 0 ) l1 = -l1;
                }
                else {
                    l2 = REDLENG(l2);
                    if ( l2 == 0 ) {
                        t = oldaccum;
                        numacc = *t++;
                        AO.OutSkip = 3;
                        FiniLine();
                        while ( --numacc >= 0 ) {
                            i = *t;
                            while ( --i >= 0 ) {
                                TalToLine((UWORD)(*t++));
                                TokenToLine((UBYTE *)"  ");
                            }
                        }
                        AO.OutSkip = 0;
                        FiniLine();
                        goto FiniCall;
                    }
                    if ( MulRat(BHEAD (UWORD *)coef,l1,(UWORD *)r,l2,(UWORD *)coef,&l1) ) goto FiniCall;
                    if ( AN.ncmod != 0 && TakeModulus((UWORD *)coef,&l1,AC.cmod,AN.ncmod,UNPACK|AC.modmode) ) goto FiniCall;
                }
                accum += *accum;
            } while ( --numacc >= 0 );
            if ( *t == SUBEXPRESSION ) {
                 while ( t+t[1] < u && t[t[1]] == DOLLAREXPR2 ) t += t[1];
            }
            t += t[1];
            if ( t < u ) do { *m++ = *t++; } while ( t < u );
            l2 = l1;
/*
            Code to economize when taking x = (a+b)/2
*/
            r = termout+1;
            while ( r < m ) {
                if ( *r == SUBEXPRESSION ) {
                    t = r + r[1];
                    l1 = (WORD)(cbuf[r[4]].CanCommu[r[2]]);
                    while ( t < m ) {
                        if ( *t == SUBEXPRESSION &&
                            t[1] == r[1] && t[2] == r[2] && t[4] == r[4] ) {
                                i = t[1] - SUBEXPSIZE;
                                u = r + SUBEXPSIZE; v = t + SUBEXPSIZE;
                                while ( i > 0 ) {
                                    if ( *v++ != *u++ ) break;
                                    i--;
                                }
```

```
                                if ( i <= 0 ) {
                                    u = r;
                                    r[3] += t[3];
                                    r = t + t[1];
                                    while ( r < m ) *t++ = *r++;
                                    m = t;
                                    r = u;
                                    goto Nextr;
                                }
                                if ( l1 && cbuf[t[4]].CanCommu[t[2]] ) break;
                                while ( t+t[1] < m && t[t[1]] == DOLLAREXPR2 ) t += t[1];
                        }
                        else if ( l1 ) {
                            if ( *t == SUBEXPRESSION && cbuf[t[4]].CanCommu[t[2]] )
                                break;
                            if ( *t >= FUNCTION+WILDOFFSET )
                                ipp = *t - WILDOFFSET;
                            else ipp = *t;
                            if ( *t >= FUNCTION
                             && functions[ipp-FUNCTION].commute && l1 ) break;
                            if ( *t == EXPRESSION ) break;
                        }
                        t += t[1];
                    }
                    r += r[1];
                }
                else r += r[1];
Nextr:;
            }

            i = ABS(l2);
            i *= 2;
            i++;
            l2 = ( l2 >= 0 ) ? i: -i;
            r = coef;
            while ( --i > 0 ) *m++ = *r++;
            *m++ = l2;
            *termout = WORDDIF(m,termout);
            AT.WorkPointer = coef;
            return(0);
        }
        t += t[1];
    } while ( t < m );
    AT.WorkPointer = coef;
    return(1);

FiniCall:
    MLOCK(ErrorMessageLock);
    MesCall("FiniTerm");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] FiniTerm :
        #[ Generator :          WORD Generator(BHEAD term,level)
*/

static WORD zeroDollar[] = { 0, 0 };
/*
static LONG debugcounter = 0;
*/

WORD Generator(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD replac, *accum, *termout, *t, i, j, tepos, applyflag = 0, *StartBuf;
    WORD *a, power, power1, DumNow = AR.CurDum, oldtoprhs, oldatoprhs, retnorm, extractbuff;
    int *RepSto = AN.RepPoint, iscopy = 0;
    CBUF *C = cbuf+AM.rbufnum, *CC = cbuf + AT.ebufnum, *CCC = cbuf + AT.aebufnum;
    LONG posisub, oldcpointer, oldacpointer;
    DOLLARS d = 0;
    WORD numfac[5], idfunctionflag;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1, id;
#endif
    oldtoprhs = CC->numrhs;
    oldcpointer = CC->Pointer - CC->Buffer;
    oldatoprhs = CCC->numrhs;
    oldacpointer = CCC->Pointer - CCC->Buffer;
ReStart:
    if ( ( replac = TestSub(BHEAD term,level) ) == 0 ) {
        if ( applyflag ) { TableReset(); applyflag = 0; }
/*
        if ( AN.PolyNormFlag > 1 ) {
            if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
            AN.PolyNormFlag = 0;
```

```
            if ( !*term ) goto Return0;
        }
*/
Renormalize:
        AN.PolyNormFlag = 0;
        AN.idfunctionflag = 0;
        if ( ( retnorm = Normalize(BHEAD term) ) != 0 ) {
            if ( retnorm > 0 ) {
                if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
                goto ReStart;
            }
            goto GenCall;
        }
        idfunctionflag = AN.idfunctionflag;
        if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }

        if ( AN.PolyNormFlag ) {
            if ( AN.PolyFunTodo == 0 ) {
                if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
                if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
            }
            else {
                WORD oldPolyFunExp = AR.PolyFunExp;
                AR.PolyFunExp = 0;
                if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
                AT.WorkPointer = term+*term;
                AR.PolyFunExp = oldPolyFunExp;
                if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
                if ( Normalize(BHEAD term) < 0 ) goto GenCall;
                if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
                AT.WorkPointer = term+*term;
                if ( AN.PolyNormFlag ) {
                    if ( PolyFunMul(BHEAD term) < 0 ) goto GenCall;
                    if ( !*term ) { AN.PolyNormFlag = 0; goto Return0; }
                    AT.WorkPointer = term+*term;
                }
                AN.PolyFunTodo = 0;
            }
        }
        if ( idfunctionflag > 0 ) {
            if ( TakeIDfunction(BHEAD term) ) {
                AT.WorkPointer = term + *term;
                goto ReStart;
            }
        }
        if ( AT.WorkPointer < (WORD *)(((UBYTE *)(term)) + AM.MaxTer) )
             AT.WorkPointer = (WORD *)(((UBYTE *)(term)) + AM.MaxTer);
        do {
SkipCount:  level++;
            if ( level > AR.Cnumlhs ) {
                if ( AR.DeferFlag && AR.sLevel <= 0 ) {
#ifdef WITHMPI
                    if ( PF.me != MASTER && AC.mparallelflag == PARALLELFLAG && PF.exprtodo < 0 ) {
                        if ( PF_Deferred(term,level) ) goto GenCall;
                    }
                    else
#endif
                    {
                        if ( Deferred(BHEAD term,level) ) goto GenCall;
                    }
                    goto Return0;
                }
                if ( AN.ncmod != 0 ) {
                    if ( Modulus(term) ) goto GenCall;
                    if ( !*term ) goto Return0;
                }
                if ( AR.CurDum > AM.IndDum && AR.sLevel <= 0 ) {
                    WORD olddummies = AN.IndDum;
                    AN.IndDum = AM.IndDum;
                    ReNumber(BHEAD term);
                    Normalize(BHEAD term);
                    AN.IndDum = olddummies;
                    if ( !*term ) goto Return0;
                    olddummies = DetCurDum(BHEAD term);
                    if ( olddummies > AR.MaxDum ) AR.MaxDum = olddummies;
                }
                if ( AR.PolyFun > 0 && ( AR.sLevel <= 0 || AN.FunSorts[AR.sLevel]->PolyFlag > 0 ) ) {
                    if ( PrepPoly(BHEAD term,0) != 0 ) goto Return0;
                }
                else if ( AR.PolyFun > 0 ) {
                    if ( PrepPoly(BHEAD term,1) != 0 ) goto Return0;
                }
                if ( AR.sLevel <= 0 && AR.BracketOn ) {
                    if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
                    termout = AT.WorkPointer;
                    if ( AT.WorkPointer + *term + 3 > AT.WorkTop ) goto OverWork;
                    if ( PutBracket(BHEAD term) ) return(-1);
```

```
                    AN.RepPoint = RepSto;
                    *AT.WorkPointer = 0;
                    i = StoreTerm(BHEAD termout);
                    AT.WorkPointer = termout;
                    CC->numrhs = oldtoprhs;
                    CC->Pointer = CC->Buffer + oldcpointer;
                    CCC->numrhs = oldatoprhs;
                    CCC->Pointer = CCC->Buffer + oldacpointer;
                    return(i);
                }
                else {
                    if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
                    if ( AT.WorkPointer >= AT.WorkTop ) goto OverWork;
                    *AT.WorkPointer = 0;
                    AN.RepPoint = RepSto;
                    i = StoreTerm(BHEAD term);
                    CC->numrhs = oldtoprhs;
                    CC->Pointer = CC->Buffer + oldcpointer;
                    CCC->numrhs = oldatoprhs;
                    CCC->Pointer = CCC->Buffer + oldacpointer;
                    return(i);
                }
            }
            i = C->lhs[level][0];
            if ( i >= TYPECOUNT ) {
/*
            #[ Special action :
*/
                switch ( i ) {
                  case TYPECOUNT:
                    if ( CountDo(term,C->lhs[level]) < C->lhs[level][2] ) {
                        AT.WorkPointer = term + *term;
                        goto Return0;
                    }
                    break;
                  case TYPEMULT:
                    if ( MultDo(BHEAD term,C->lhs[level]) ) goto GenCall;
                    goto ReStart;
                  case TYPEGOTO:
                    level = AC.Labels[C->lhs[level][2]];
                    break;
                  case TYPEDISCARD:
                    AT.WorkPointer = term + *term;
                    goto Return0;
                  case TYPEIF:
#ifdef WITHPTHREADS
                    {
/*
                        We may be writing in the space here when wildcards
                        are involved in a match(). Hence we have to make
                        a private copy here!!!!
*/
                        WORD ic, jc, *ifcode, *jfcode;
                        jfcode = C->lhs[level]; jc = jfcode[1];
                        ifcode = AT.WorkPointer; AT.WorkPointer += jc;
                        for ( ic = 0; ic < jc; ic++ ) ifcode[ic] = jfcode[ic];
                        while ( !DoIfStatement(BHEAD ifcode,term) ) {
                            level = C->lhs[level][2];
                            if ( C->lhs[level][0] != TYPEELIF ) break;
                        }
                        AT.WorkPointer = ifcode;
                    }
#else
                    while ( !DoIfStatement(BHEAD C->lhs[level],term) ) {
                        level = C->lhs[level][2];
                        if ( C->lhs[level][0] != TYPEELIF ) break;
                    }
#endif
                    break;
                  case TYPEELIF:
                    do {
                        level = C->lhs[level][2];
                    } while ( C->lhs[level][0] == TYPEELIF );
                    break;
                  case TYPEELSE:
                  case TYPEENDIF:
                    level = C->lhs[level][2];
                    break;
                  case TYPESUMFIX:
                    {
                        WORD *cp = AR.CompressPointer, *op = AR.CompressPointer;
                        WORD *tlhs = C->lhs[level] + 3, *m, jlhs;
                        WORD theindex = C->lhs[level][2];
                        if ( theindex < 0 ) {   /* $-variable */
#ifdef WITHPTHREADS
                            int ddtype = -1;
                            theindex = -theindex;
```

```
                            d = Dollars + theindex;
                            if ( AS.MultiThreaded ) {
                                for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                                    if ( theindex == ModOptdollars[nummodopt].number ) break;
                                }
                                if ( nummodopt < NumModOptdollars ) {
                                    ddtype = ModOptdollars[nummodopt].type;
                                    if ( ddtype == MODLOCAL ) {
                                        d = ModOptdollars[nummodopt].dstruct+AT.identity;
                                    }
                                    else {
                                        LOCK(d->pthreadslockread);
                                    }
                                }
                            }
#else
                            theindex = -theindex;
                            d = Dollars + theindex;
#endif

                            if ( d->type != DOLINDEX
                            || d->index < AM.OffsetIndex
                            || d->index >= AM.OffsetIndex + WILDOFFSET ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("$%s should have been an index"
                                ,AC.dollarnames->namebuffer+d->name);
                                AN.currentTerm = term;
                                MesPrint("Current term: %t");
                                AN.listinprint = printscratch;
                                printscratch[0] = DOLLAREXPRESSION;
                                printscratch[1] = theindex;
                                MesPrint("$%s = %$"
                                ,AC.dollarnames->namebuffer+d->name);
                                MUNLOCK(ErrorMessageLock);
#ifdef WITHPTHREADS
                            if ( ddtype > 0 && ddtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                                goto GenCall;
                            }
                            theindex = d->index;
#ifdef WITHPTHREADS
                            if ( ddtype > 0 && ddtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
                        }
                        cp[1] = SUBEXPSIZE+4;
                        cp += SUBEXPSIZE;
                        *cp++ = INDTOIND;
                        *cp++ = 4;
                        *cp++ = theindex;
                        i = C->lhs[level][1] - 3;
                        cp++;
                        AR.CompressPointer = cp;
                        while ( --i >= 0 ) {
                            cp[-1] = *tlhs++;
                            termout = AT.WorkPointer;
                            if ( ( jlhs = WildFill(BHEAD termout,term,op)) < 0 )
                                goto GenCall;
                            m = term;
                            jlhs = *m;
                            while ( --jlhs >= 0 ) {
                                if ( *m++ != *termout++ ) break;
                            }
                            if ( jlhs >= 0 ) {
                                termout = AT.WorkPointer;
                                AT.WorkPointer = termout + *termout;
                                if ( Generator(BHEAD termout,level) ) goto GenCall;
                                AT.WorkPointer = termout;
                            }
                            else {
                                AR.CompressPointer = op;
                                goto SkipCount;
                            }
                        }
                        AR.CompressPointer = op;
                        goto CommonEnd;
                    }
                  case TYPESUM:
                    {
                        WORD *wp, *cp = AR.CompressPointer, *op = AR.CompressPointer;
                        WORD theindex;
                        WORD *ow;
/*
                        At this point it is safest to determine CurDum
*/
                        AR.CurDum = DetCurDum(BHEAD term);
                        i = C->lhs[level][1]-2;
                        wp = C->lhs[level] + 2;
```

```
                        cp[1] = SUBEXPSIZE+4*i;
                        cp += SUBEXPSIZE;
                        while ( --i >= 0 ) {
                            theindex = *wp++;
                            if ( theindex < 0 ) {   /* $-variable */
#ifdef WITHPTHREADS
                                int ddtype = -1;
                                theindex = -theindex;
                                d = Dollars + theindex;
                                if ( AS.MultiThreaded ) {
                                    for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                                        if ( theindex == ModOptdollars[nummodopt].number ) break;
                                    }
                                    if ( nummodopt < NumModOptdollars ) {
                                        ddtype = ModOptdollars[nummodopt].type;
                                        if ( ddtype == MODLOCAL ) {
                                            d = ModOptdollars[nummodopt].dstruct+AT.identity;
                                        }
                                        else {
                                            LOCK(d->pthreadslockread);
                                        }
                                    }
                                }
#else
                                theindex = -theindex;
                                d = Dollars + theindex;
#endif
                                if ( d->type != DOLINDEX
                                || d->index < AM.OffsetIndex
                                || d->index >= AM.OffsetIndex + WILDOFFSET ) {
                                    MLOCK(ErrorMessageLock);
                                    MesPrint("$%s should have been an index"
                                    ,AC.dollarnames->namebuffer+d->name);
                                    AN.currentTerm = term;
                                    MesPrint("Current term: %t");
                                    AN.listinprint = printscratch;
                                    printscratch[0] = DOLLAREXPRESSION;
                                    printscratch[1] = theindex;
                                    MesPrint("$%s = %$"
                                    ,AC.dollarnames->namebuffer+d->name);
                                    MUNLOCK(ErrorMessageLock);
#ifdef WITHPTHREADS
                                    if ( ddtype > 0 && ddtype != MODLOCAL ) {
UNLOCK(d->pthreadslockread); }
#endif
                                    goto GenCall;
                                }
                                theindex = d->index;
#ifdef WITHPTHREADS
                                if ( ddtype > 0 && ddtype != MODLOCAL ) { UNLOCK(d->pthreadslockread);
}
#endif
                            }
                            *cp++ = INDTOIND;
                            *cp++ = 4;
                            *cp++ = theindex;
                            *cp++ = ++AR.CurDum;
                        }
                        ow = AT.WorkPointer;
                        AR.CompressPointer = cp;
                        if ( WildFill(BHEAD ow,term,op) < 0 ) goto GenCall;
                        AR.CompressPointer = op;
                        i = ow[0];
                        WORD term_changed = 0;
                        for ( j = 0; j < i; j++ ) {
                            if ( term[j] != ow[j] ) term_changed = 1;
                            term[j] = ow[j];
                        }
                        // If the term was modified by WildFill, set RepCount.
                        if ( term_changed ) *AN.RepPoint = 1;
                        AT.WorkPointer = ow;
                        ReNumber(BHEAD term);
                        goto Renormalize;
                    }
                  case TYPECHISHOLM:
                    if ( Chisholm(BHEAD term,level) ) goto GenCall;
CommonEnd:
                    AT.WorkPointer = term + *term;
                    goto Return0;
                  case TYPEARG:
                    if ( ( i = execarg(BHEAD term,level) ) < 0 ) goto GenCall;
                    level = C->lhs[level][2];
                    if ( i > 0 ) goto ReStart;
                    break;
                  case TYPENORM:
                  case TYPENORM2:
                  case TYPENORM3:
```

```
                    case TYPENORM4:
                    case TYPESPLITARG:
                    case TYPESPLITARG2:
                    case TYPESPLITFIRSTARG:
                    case TYPESPLITLASTARG:
                    case TYPEARGTOEXTRASYMBOL:
                      if ( execarg(BHEAD term,level) < 0 ) goto GenCall;
                      level = C->lhs[level][2];
                      break;
                    case TYPEFACTARG:
                    case TYPEFACTARG2:
                      { WORD jjj;
                      if ( ( jjj = execarg(BHEAD term,level) ) < 0 ) goto GenCall;
                      if ( jjj > 0 ) goto ReStart;
                      level = C->lhs[level][2];
                      break; }
                    case TYPEEXIT:
                      if ( C->lhs[level][2] > 0 ) {
                          MLOCK(ErrorMessageLock);
                          MesPrint("%s",C->lhs[level]+3);
                          MUNLOCK(ErrorMessageLock);
                      }
                      Terminate(-1);
                      goto GenCall;
                    case TYPESETEXIT:
                      AM.exitflag = 1; /* no danger of race conditions */
                      break;
                    case TYPEPRINT:
                      AN.currentTerm = term;
                      AN.numlistinprint = (C->lhs[level][1] - C->lhs[level][4] - 5)/2;
                      AN.listinprint = C->lhs[level]+5+C->lhs[level][4];
                      MLOCK(ErrorMessageLock);
                      AO.ErrorBlock = 1;
                      MesPrint((char *)(C->lhs[level]+5));
                      AO.ErrorBlock = 0;
                      MUNLOCK(ErrorMessageLock);
                      break;
                    case TYPEFPRINT:
                      {
                      int oldFOflag;
                      WORD oldPrintType, oldLogHandle = AC.LogHandle;
                      AC.LogHandle = C->lhs[level][2];
                      MLOCK(ErrorMessageLock);
                      oldFOflag = AM.FileOnlyFlag;
                      oldPrintType = AO.PrintType;
                      if ( AC.LogHandle >= 0 ) {
                          AM.FileOnlyFlag = 1;
                          AO.PrintType |= PRINTLFILE;
                      }
                      AO.PrintType |= C->lhs[level][3];
                      AN.currentTerm = term;
                      AN.numlistinprint = (C->lhs[level][1] - C->lhs[level][4] - 5)/2;
                      AN.listinprint = C->lhs[level]+5+C->lhs[level][4];
                      MesPrint((char *)(C->lhs[level]+5));
                      AO.PrintType = oldPrintType;
                      AM.FileOnlyFlag = oldFOflag;
                      MUNLOCK(ErrorMessageLock);
                      AC.LogHandle = oldLogHandle;
                      }
                      break;
                    case TYPEREDEFPRE:
                      j = C->lhs[level][2];
#ifdef WITHMPI
                      {
                          /*
                           * Regardless of parallel/nonparallel switch, we need to set
                           * AC.inputnumbers[ii], which indicates that the corresponding
                           * preprocessor variable is redefined and so we need to
                           * send/broadcast it.
                           */
                          int ii;
                          for ( ii = 0; ii < AC.numpfirstnum; ii++ ) {
                              if ( AC.pfirstnum[ii] == j ) break;
                          }
                          AC.inputnumbers[ii] = AN.ninterms;
                      }
#endif
#ifdef WITHPTHREADS
                      if ( AS.MultiThreaded ) {
                          int ii;
                          for ( ii = 0; ii < AC.numpfirstnum; ii++ ) {
                              if ( AC.pfirstnum[ii] == j ) break;
                          }
                          if ( AN.inputnumber < AC.inputnumbers[ii] ) break;
                          LOCK(AP.PreVarLock);
                          if ( AN.inputnumber >= AC.inputnumbers[ii] ) {
                              a = C->lhs[level]+4;
```

```
                            if ( a[a[-1]] == 0 )
                                PutPreVar(PreVar[j].name,(UBYTE *)(a),0,1);
                            else
                                PutPreVar(PreVar[j].name,(UBYTE *)(a)
                                    ,(UBYTE *)(a+a[-1]+1),1);
/*
                            PutPreVar(PreVar[j].name,(UBYTE *)(C->lhs[level]+4),0,1);
*/
                            AC.inputnumbers[ii] = AN.inputnumber;
                        }
                        UNLOCK(AP.PreVarLock);
                    }
                    else
#endif
                    {
                        a = C->lhs[level]+4;
                        LOCK(AP.PreVarLock);
                        if ( a[a[-1]] == 0 )
                            PutPreVar(PreVar[j].name,(UBYTE *)(a),0,1);
                        else
                            PutPreVar(PreVar[j].name,(UBYTE *)(a)
                                ,(UBYTE *)(a+a[-1]+1),1);
                        UNLOCK(AP.PreVarLock);
                    }
                    break;
                  case TYPERENUMBER:
                    AT.WorkPointer = term + *term;
                    if ( FullRenumber(BHEAD term,C->lhs[level][2]) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    if ( *term == 0 ) goto Return0;
                    break;
                  case TYPETRY:
                    if ( TryDo(BHEAD term,C->lhs[level],level) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    goto Return0;
                  case TYPEASSIGN:
                    { WORD onc = AR.NoCompress, oldEside = AR.Eside;
                    WORD oldrepeat = *AN.RepPoint;
/*
                    Here we have to assign an expression to a $ variable.
*/
                    AR.Eside = RHSIDE;
                    AR.NoCompress = 1;
                    AN.cTerm = AN.currentTerm = term;
                    AT.WorkPointer = term + *term;
                    *AT.WorkPointer++ = 0;
                    if ( AssignDollar(BHEAD term,level) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    AN.cTerm = 0;
                    *AN.RepPoint = oldrepeat;
                    AR.NoCompress = onc;
                    AR.Eside = oldEside;
                    break;
                    }
                  case TYPEFINDLOOP:
                    if ( Lus(term,C->lhs[level][3],C->lhs[level][4],
                    C->lhs[level][5],C->lhs[level][6],C->lhs[level][2]) ) {
                        AT.WorkPointer = term + *term;
                        goto Renormalize;
                    }
                    break;
                  case TYPEINSIDE:
                    if ( InsideDollar(BHEAD C->lhs[level],level) < 0 ) goto GenCall;
                    level = C->lhs[level][2];
                    break;
                  case TYPETERM:
                    retnorm = execterm(BHEAD term,level);
                    AN.RepPoint = RepSto;
                    AR.CurDum = DumNow;
                    CC->numrhs = oldtoprhs;
                    CC->Pointer = CC->Buffer + oldcpointer;
                    CCC->numrhs = oldatoprhs;
                    CCC->Pointer = CCC->Buffer + oldacpointer;
                    return(retnorm);
                  case TYPEDETCURDUM:
                    AT.WorkPointer = term + *term;
                    AR.CurDum = DetCurDum(BHEAD term);
                    break;
                  case TYPEINEXPRESSION:
                    {WORD *ll = C->lhs[level];
                    int numexprs = (int)(ll[1]-3);
                    ll += 3;
                    while ( numexprs-- >= 0 ) {
                        if ( *ll == AR.CurExpr ) break;
                        ll++;
                    }
                    if ( numexprs < 0 ) level = C->lhs[level][2];
```

```
                    }
                    break;
                  case TYPEMERGE:
                    AT.WorkPointer = term + *term;
                    if ( DoShuffle(term,level,C->lhs[level][2],C->lhs[level][3]) )
                        goto GenCall;
                    AT.WorkPointer = term + *term;
                    goto Return0;
                  case TYPESTUFFLE:
                    AT.WorkPointer = term + *term;
                    if ( DoStuffle(term,level,C->lhs[level][2],C->lhs[level][3]) )
                        goto GenCall;
                    AT.WorkPointer = term + *term;
                    goto Return0;
                  case TYPETESTUSE:
                    AT.WorkPointer = term + *term;
                    if ( TestUse(term,level) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    break;
                  case TYPEAPPLY:
                    AT.WorkPointer = term + *term;
                    if ( ApplyExec(term,C->lhs[level][2],level) < C->lhs[level][2] ) {
                        AT.WorkPointer = term + *term;
                        *AN.RepPoint = 1;
                        goto ReStart;
                    }
                    AT.WorkPointer = term + *term;
                    break;
/*
                  case TYPEAPPLYRESET:
                    AT.WorkPointer = term + *term;
                    if ( ApplyReset(level) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    break;
*/
                  case TYPECHAININ:
                    { int iter = *term;
                    AT.WorkPointer = term + *term;
                    if ( ChainIn(BHEAD term,C->lhs[level][2]) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    /* Symmetry properties might mean the term has vanished */
                    if ( *term == 0 ) goto Return0;
                    if ( *term != iter ) *AN.RepPoint = 1;
                    }
                    break;
                  case TYPECHAINOUT:
                    { int iter = *term;
                    AT.WorkPointer = term + *term;
                    if ( ChainOut(BHEAD term,C->lhs[level][2]) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    if ( *term != iter ) *AN.RepPoint = 1;
                    }
                    break;
                  case TYPEFACTOR:
                    AT.WorkPointer = term + *term;
                    if ( DollarFactorize(BHEAD C->lhs[level][2]) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    break;
                  case TYPEARGIMPLODE:
                    AT.WorkPointer = term + *term;
                    if ( ArgumentImplode(BHEAD term,C->lhs[level]) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    break;
                  case TYPEARGEXPLODE:
                    AT.WorkPointer = term + *term;
                    if ( ArgumentExplode(BHEAD term,C->lhs[level]) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    break;
                  case TYPEDENOMINATORS:
                    if ( DenToFunction(term,C->lhs[level][2]) ) goto ReStart;
                    break;
                  case TYPEDROPCOEFFICIENT:
                    DropCoefficient(BHEAD term);
                    break;
                  case TYPETRANSFORM:
                    AT.WorkPointer = term + *term;
                    if ( RunTransform(BHEAD term,C->lhs[level]+2) ) goto GenCall;
                    AT.WorkPointer = term + *term;
                    if ( *term == 0 ) goto Return0;
                    goto ReStart;
                  case TYPETOPOLYNOMIAL:
                    AT.WorkPointer = term + *term;
                    termout = AT.WorkPointer;
                    if ( ConvertToPoly(BHEAD term,termout,C->lhs[level],0) < 0 ) goto GenCall;
                    if ( *termout == 0 ) goto Return0;
                    i = termout[0]; t = term; NCOPY(t,termout,i);
                    AT.WorkPointer = term + *term;
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                        break;
                    case TYPEFROMPOLYNOMIAL:
                        AT.WorkPointer = term + *term;
                        termout = AT.WorkPointer;
                        if ( ConvertFromPoly(BHEAD term,termout,0,numxsymbol,0,0) < 0 ) goto GenCall;
                        if ( *term == 0 ) goto Return0;
                        i = termout[0]; t = term; NCOPY(t,termout,i);
                        AT.WorkPointer = term + *term;
                        goto ReStart;
                    case TYPEDOLOOP:
                        level = TestDoLoop(BHEAD C->lhs[level],level);
                        if ( level < 0 ) goto GenCall;
                        break;
                    case TYPEENDDOLOOP:
                        level = TestEndDoLoop(BHEAD C->lhs[C->lhs[level][2]],C->lhs[level][2]);
                        if ( level < 0 ) goto GenCall;
                        break;
                    case TYPEDROPSYMBOLS:
                        DropSymbols(BHEAD term);
                        break;
                    case TYPEPUTINSIDE:
                        AT.WorkPointer = term + *term;
                        if ( PutInside(BHEAD term,C->lhs[level]) < 0 ) goto GenCall;
                        AT.WorkPointer = term + *term;
                        /*
                         * We need to call Generator() to convert slow notation to
                         * fast notation, which fixes Issue #30.
                         */
                        if ( Generator(BHEAD term,level) < 0 ) goto GenCall;
                        goto Return0;
                    case TYPETOSPECTATOR:
                        if ( PutInSpectator(term,C->lhs[level][2]) < 0 ) goto GenCall;
                        goto Return0;
                    case TYPECANONICALIZE:
                        AT.WorkPointer = term + *term;
                        if ( DoCanonicalize(BHEAD term,C->lhs[level]) ) goto GenCall;
                        AT.WorkPointer = term + *term;
                        if ( *term == 0 ) goto Return0;
                        break;
                    case TYPESWITCH:
                        AT.WorkPointer = term + *term;
                        if ( DoSwitch(BHEAD term,C->lhs[level]) ) goto GenCall;
                        goto Return0;
                    case TYPEENDSWITCH:
                        AT.WorkPointer = term + *term;
                        if ( DoEndSwitch(BHEAD term,C->lhs[level]) ) goto GenCall;
                        goto Return0;
                    case TYPESETUSERFLAG:
                        Expressions[AR.CurExpr].uflags |= 1 « (C->lhs[level][2]);
                        break;
                    case TYPECLEARUSERFLAG:
                        Expressions[AR.CurExpr].uflags &= ~(1 « (C->lhs[level][2]));
                        break;
                    case TYPEALLLOOPS:
                        AT.WorkPointer = term + *term;
                        if ( AllLoops(BHEAD term,level) ) goto GenCall;
                        goto Return0;
                    case TYPEALLPATHS:
                        AT.WorkPointer = term + *term;
                        if ( AllPaths(BHEAD term,level) ) goto GenCall;
                        goto Return0;
#ifdef WITHFLOAT
                    case TYPEEVALUATE:
                        AT.WorkPointer = term + *term;
                        if ( C->lhs[level][2] == MZV
                             || C->lhs[level][2] == EULER
                             || C->lhs[level][2] == MZVHALF
                             || C->lhs[level][2] == ALLMZVFUNCTIONS
                             ) {
                            if ( EvaluateEuler(BHEAD term,level,C->lhs[level][2]) ) goto GenCall;
                        }
                        else {
                            if ( EvaluateFun(BHEAD term,level,C->lhs[level]) ) goto GenCall;
                        }
/*
                        else if ( C->lhs[level][2] == SQRTFUNCTION ) {
                            if ( EvaluateSqrt(BHEAD term,level,C->lhs[level][2]) ) goto GenCall;
                        }
                        else {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Illegal function %d in evaluate statement.",C->lhs[level][2]);
                            MUNLOCK(ErrorMessageLock);
                            goto GenCall;
                        }
*/
                        goto Return0;
                    case TYPETOFLOAT:
```

```
                        AT.WorkPointer = term + *term;
                        if ( ToFloat(BHEAD term,level) ) goto GenCall;
                        goto Return0;
                      case TYPETORAT:
                        AT.WorkPointer = term + *term;
                        if ( ToRat(BHEAD term,level) ) goto GenCall;
                        goto Return0;
#endif
                    }
                    goto SkipCount;
/*
            #] Special action :
*/
                }
            } while ( ( i = TestMatch(BHEAD term,&level) ) == 0 );
            if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
            if ( i > 0 ) replac = TestSub(BHEAD term,level);
            else replac = i;
            if ( replac >= 0 || AT.TMout[1] != SYMMETRIZE ) {
                *AN.RepPoint = 1;
                AR.expchanged = 1;
            }
            if ( replac < 0 ) {      /* Terms come from automatic generation */
AutoGen:        i = *AT.TMout;
                t = termout = AT.WorkPointer;
                if ( ( AT.WorkPointer += i ) > AT.WorkTop ) goto OverWork;
                accum = AT.TMout;
                while ( --i >= 0 ) *t++ = *accum++;
                if ( (*(FG.Operation[termout[1]]))(BHEAD term,termout,replac,level) ) goto GenCall;
                AT.WorkPointer = termout;
                goto Return0;
            }
        }
        if ( applyflag ) { TableReset(); applyflag = 0; }

        if ( AN.TeInFun ) { /* Match in function argument */
            if ( AN.TeInFun < 0 && !AN.TeSuOut ) {

                if ( AR.TePos >= 0 ) goto AutoGen;
                switch ( AN.TeInFun ) {
                    case -1:
                        if ( DoDistrib(BHEAD term,level) ) goto GenCall;
                        break;
                    case -2:
                        if ( DoDelta3(BHEAD term,level) ) goto GenCall;
                        break;
                    case -3:
                        if ( DoTableExpansion(term,level) ) goto GenCall;
                        break;
                    case -4:
                        if ( FactorIn(BHEAD term,level) ) goto GenCall;
                        break;
                    case -5:
                        if ( FactorInExpr(BHEAD term,level) ) goto GenCall;
                        break;
                    case -6:
                        if ( TermsInBracket(BHEAD term,level) < 0 ) goto GenCall;
                        break;
                    case -7:
                        if ( ExtraSymFun(BHEAD term,level) < 0 ) goto GenCall;
                        break;
                    case -8:
                        if ( GCDfunction(BHEAD term,level) < 0 ) goto GenCall;
                        break;
                    case -9:
                        if ( DIVfunction(BHEAD term,level,0) < 0 ) goto GenCall;
                        break;
                    case -10:
                        if ( DIVfunction(BHEAD term,level,1) < 0 ) goto GenCall;
                        break;
                    case -11:
                        if ( DIVfunction(BHEAD term,level,2) < 0 ) goto GenCall;
                        break;
                    case -12:
                        if ( DoPermutations(BHEAD term,level) ) goto GenCall;
                        break;
                    case -13:
                        if ( DoPartitions(BHEAD term,level) ) goto GenCall;
                        break;
                    case -14:
                        if ( DIVfunction(BHEAD term,level,3) < 0 ) goto GenCall;
                        break;
                    case -15:
                        if ( GenTopologies(BHEAD term,level) < 0 ) goto GenCall;
                        break;
                    case -16:
                        if ( GenDiagrams(BHEAD term,level) < 0 ) goto GenCall;
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                    break;
                }
            }
            else {
                termout = AT.WorkPointer;
                AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
                if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
                if ( InFunction(BHEAD term,termout) ) goto GenCall;
                AT.WorkPointer = termout + *termout;
                *AN.RepPoint = 1;
                AR.expchanged = 1;
                if ( *termout && Generator(BHEAD termout,level) < 0 ) goto GenCall;
                AT.WorkPointer = termout;
            }
        }
        else if ( replac > 0 ) {
            power = AN.TeSuOut;
            tepos = AR.TePos;
            if ( power < 0 ) {  /* Table expansion */
                power = -power; tepos = 0;
            }
            extractbuff = AT.TMbuff;
            if ( extractbuff == AM.dbufnum ) {
                d = DolToTerms(BHEAD replac);
                if ( d && d->where != 0 ) {
                    iscopy = 1;
                    if ( AT.TMdolfac > 0 ) {    /* We need a factor */
                      if ( AT.TMdolfac == 1 ) {
                        if ( d->nfactors ) {
                            numfac[0] = 4;
                            numfac[1] = d->nfactors;
                            numfac[2] = 1;
                            numfac[3] = 3;
                            numfac[4] = 0;
                        }
                        else {
                            numfac[0] = 0;
                        }
                        StartBuf = numfac;
                      }
                      else {
                        if ( (AT.TMdolfac-1) > d->nfactors && d->nfactors > 0 ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Attempt to use an nonexisting factor %d of a $-variable",(WORD)(AT.TMdolfac-1));
                            if ( d->nfactors == 1 )
                                MesPrint("There is only one factor");
                            else
                                MesPrint("There are only %d factors",(WORD)(d->nfactors));
                            MUNLOCK(ErrorMessageLock);
                            goto GenCall;
                        }
                        if ( d->nfactors > 1 ) {
                            DOLLARS dd;
                            LONG dsize;
                            WORD *td1, *td2;
                            dd = Dollars + replac;
#ifdef WITHPTHREADS
                            {
                                int nummodopt, dtype = -1;
                                if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
                                    for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                                        if ( replac == ModOptdollars[nummodopt].number ) break;
                                    }
                                    if ( nummodopt < NumModOptdollars ) {
                                        dtype = ModOptdollars[nummodopt].type;
                                        if ( dtype == MODLOCAL ) {
                                            dd = ModOptdollars[nummodopt].dstruct+AT.identity;
                                        }
                                    }
                                }
                            }
#endif
                            dsize = dd->factors[AT.TMdolfac-2].size;
/*
                            We copy only the factor we need
*/
                            if ( dsize == 0 ) {
                                numfac[0] = 4;
                                numfac[1] = d->factors[AT.TMdolfac-2].value;
                                numfac[2] = 1;
                                numfac[3] = 3;
                                numfac[4] = 0;
                                StartBuf = numfac;
                                if ( numfac[1] < 0 ) {
                                    numfac[1] = -numfac[1];
                                    numfac[3] = -numfac[3];
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                            }
                        }
                        else {
                        d->factors[AT.TMdolfac-2].where = td2 = (WORD *)Malloc1(
                            (dsize+1)*sizeof(WORD),"Copy of factor");
                        td1 = dd->factors[AT.TMdolfac-2].where;
                        StartBuf = td2;
                        d->size = dsize; d->type = DOLTERMS;
                        NCOPY(td2,td1,dsize);
                        *td2 = 0;
                        }
                    }
                    else if ( d->nfactors == 1 ) {
                        StartBuf = d->where;
                    }
                    else {
                        MLOCK(ErrorMessageLock);
                        if ( d->nfactors == 0 ) {
                            MesPrint("Attempt to use factor %d of an unfactored $-variable",(WORD)(AT.TMdolfac-1));
                        }
                        else {
                            MesPrint("Internal error. Illegal number of factors for $-variable");
                        }
                        MUNLOCK(ErrorMessageLock);
                        goto GenCall;
                    }
                  }
                }
                else StartBuf = d->where;
            }
            else {
                d = Dollars + replac;
                StartBuf = zeroDollar;
            }
            posisub = 0;
            i = DetCommu(d->where);
#ifdef WITHPTHREADS
            if ( AS.MultiThreaded ) {
                for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                    if ( replac == ModOptdollars[nummodopt].number ) break;
                }
                if ( nummodopt < NumModOptdollars ) {
                    dtype = ModOptdollars[nummodopt].type;
                    if ( dtype != MODLOCAL && dtype != MODSUM ) {
                        if ( StartBuf[0] && StartBuf[StartBuf[0]] ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("A dollar variable with modoption max or min can have only one term");
                            MUNLOCK(ErrorMessageLock);
                            goto GenCall;
                        }
                        LOCK(d->pthreadslockread);
                    }
                }
            }
#endif
        }
        else {
            StartBuf = cbuf[extractbuff].Buffer;
            posisub = cbuf[extractbuff].rhs[replac] - StartBuf;
            i = (WORD)cbuf[extractbuff].CanCommu[replac];
        }
        if ( power == 1 ) {      /* Just a single power */
            termout = AT.WorkPointer;
            AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
            if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
            while ( StartBuf[posisub] ) {
                if ( extractbuff == AT.allbufnum ) WildDollars(BHEAD &(StartBuf[posisub]));
                AT.WorkPointer = (WORD *)(((UBYTE *)(termout)) + AM.MaxTer);
                if ( InsertTerm(BHEAD term,replac,extractbuff,
                    &(StartBuf[posisub]),termout,tepos) < 0 ) goto GenCall;
                AT.WorkPointer = termout + *termout;
                *AN.RepPoint = 1;
                AR.expchanged = 1;
                posisub += StartBuf[posisub];
/*
                    For multiple table substitutions it may be better to
                    do modulus arithmetic right here
                    Turns out to be not very effective.

                if ( AN.ncmod != 0 ) {
                    if ( Modulus(termout) ) goto GenCall;
                    if ( !*termout ) goto Return0;
                }
*/
#ifdef WITHPTHREADS
```

```
                    if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadslockread); }
                    if ( ( AS.Balancing && CC->numrhs == 0 ) && StartBuf[posisub] ) {
                        if ( ( id = ConditionalGetAvailableThread() ) >= 0 ) {
                            if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;
                        }
                    }
                    else
#endif
                    if ( Generator(BHEAD termout,level) < 0 ) goto GenCall;
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL ) { dtype = 0; break; }
#endif
                    if ( iscopy == 0 && ( extractbuff != AM.dbufnum ) ) {
/*
                        There are cases in which a bigger buffer is created
                        on the fly, like with wildcard buffers.
                        We play it safe here. Maybe we can be more selective
                        in some distant future?
*/
                        StartBuf = cbuf[extractbuff].Buffer;
                    }
                }
                if ( extractbuff == AT.allbufnum ) {
                    CBUF *Ce = cbuf + extractbuff;
                    Ce->Pointer = Ce->rhs[Ce->numrhs--];
                }
#ifdef WITHPTHREADS
                if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadslockread); dtype = 0; }
#endif
                if ( iscopy ) {
                    if ( d->nfactors > 1 ) {
                        int j;
                        for ( j = 0; j < d->nfactors; j++ ) {
                            if ( d->factors[j].where ) M_free(d->factors[j].where,"Copy of factor");
                        }
                        M_free(d->factors,"Dollar factors");
                    }
                    M_free(d,"Copy of dollar variable");
                    d = 0; iscopy = 0;
                }
                AT.WorkPointer = termout;
            }
            else if ( i <= 1 ) {        /* Use binomials */
                LONG posit, olw;
                WORD *same, *ow = AT.WorkPointer;
                LONG olpw = AT.posWorkPointer;
                power1 = power+1;
                WantAddLongs(power1);
                olw = posit = AT.lWorkPointer; AT.lWorkPointer += power1;
                same = ++AT.WorkPointer;
                a = accum = ( AT.WorkPointer += power1+1 );
                AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + 2*AM.MaxTer);
                if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
                AT.lWorkSpace[posit] = posisub;
                same[-1] = 0;
                *same = 1;
                *accum = 0;
                tepos = AR.TePos;
                i = 1;
                do {
                    if ( StartBuf[AT.lWorkSpace[posit]] ) {
                        if ( ( a = PasteTerm(BHEAD i-1,accum,
                            &(StartBuf[AT.lWorkSpace[posit]]),i,*same) ) == 0 )
                            goto GenCall;
                        AT.lWorkSpace[posit+1] = AT.lWorkSpace[posit];
                        same[1] = *same + 1;
                        if ( i > 1 && AT.lWorkSpace[posit] < AT.lWorkSpace[posit-1] ) *same = 1;
                        AT.lWorkSpace[posit] += StartBuf[AT.lWorkSpace[posit]];
                        i++;
                        posit++;
                        same++;
                    }
                    else {
                        i--; posit--; same--;
                    }
                    if ( i > power ) {
                        termout = AT.WorkPointer = a;
                        AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + 2*AM.MaxTer);
                        if ( AT.WorkPointer > AT.WorkTop )
                            goto OverWork;
                        if ( FiniTerm(BHEAD term,accum,termout,replac,tepos) ) goto GenCall;
                        AT.WorkPointer = termout + *termout;
                        *AN.RepPoint = 1;
                        AR.expchanged = 1;
#ifdef WITHPTHREADS
```

```
                    if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadslockread); }
                    if ( ( AS.Balancing && CC->numrhs == 0 ) && ( i > 0 )
                    && ( id = ConditionalGetAvailableThread() ) >= 0 ) {
                        if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;
                    }
                    else
#endif
                    if ( Generator(BHEAD termout,level) ) goto GenCall;
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL ) { dtype = 0; break; }
#endif
                    if ( iscopy == 0 && ( extractbuff != AM.dbufnum ) )
                            StartBuf = cbuf[extractbuff].Buffer;
                    i--; posit--; same--;
                }
            } while ( i > 0 );
#ifdef WITHPTHREADS
            if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadslockread); dtype = 0; }
#endif
            if ( iscopy ) {
                if ( d->nfactors > 1 ) {
                    int j;
                    for ( j = 0; j < d->nfactors; j++ ) {
                        if ( d->factors[j].where ) M_free(d->factors[j].where,"Copy of factor");
                    }
                    M_free(d->factors,"Dollar factors");
                }
                M_free(d,"Copy of dollar variable");
                d = 0; iscopy = 0;
            }
            AT.WorkPointer = ow; AT.lWorkPointer = olw; AT.posWorkPointer = olpw;
        }
        else {                          /* No binomials */
            LONG posit, olw, olpw = AT.posWorkPointer;
            WantAddLongs(power);
            posit = olw = AT.lWorkPointer; AT.lWorkPointer += power;
            a = accum = AT.WorkPointer;
            AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + 2*AM.MaxTer);
            if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
            for ( i = 0; i < power; i++ ) AT.lWorkSpace[posit++] = posisub;
            posit = olw;
            *accum = 0;
            tepos = AR.TePos;
            i = 0;
            while ( i >= 0 ) {
                if ( StartBuf[AT.lWorkSpace[posit]] ) {
                    if ( ( a = PasteTerm(BHEAD i,accum,
                        &(StartBuf[AT.lWorkSpace[posit]]),1,1) ) == 0 ) goto GenCall;
                    AT.lWorkSpace[posit] += StartBuf[AT.lWorkSpace[posit]];
                    i++; posit++;
                }
                else {
                    AT.lWorkSpace[posit--] = posisub;
                    i--;
                }
                if ( i >= power ) {
                    termout = AT.WorkPointer = a;
                    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + 2*AM.MaxTer);
                    if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
                    if ( FiniTerm(BHEAD term,accum,termout,replac,tepos) ) goto GenCall;
                    AT.WorkPointer = termout + *termout;
                    *AN.RepPoint = 1;
                    AR.expchanged = 1;
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadslockread); }
                    if ( ( AS.Balancing && CC->numrhs == 0 ) && ( i > 0 ) && ( id = ConditionalGetAvailableThread() ) >= 0 ) {
                        if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;
                    }
                    else
#endif
                    if ( Generator(BHEAD termout,level) ) goto GenCall;
#ifdef WITHPTHREADS
                    if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { dtype = 0; break; }
#endif
                    if ( iscopy == 0 && ( extractbuff != AM.dbufnum ) )
                            StartBuf = cbuf[extractbuff].Buffer;
                    i--; posit--;
                }
            }
#ifdef WITHPTHREADS
            if ( dtype > 0 && dtype != MODLOCAL && dtype != MODSUM ) { UNLOCK(d->pthreadslockread); dtype = 0; }
#endif
```

```
            if ( iscopy ) {
                if ( d->nfactors > 1 ) {
                    int j;
                    for ( j = 0; j < d->nfactors; j++ ) {
                        if ( d->factors[j].where ) M_free(d->factors[j].where,"Copy of factor");
                    }
                    M_free(d->factors,"Dollar factors");
                }
                M_free(d,"Copy of dollar variable");
                d = 0; iscopy = 0;
            }
            AT.WorkPointer = accum;
            AT.lWorkPointer = olw;
            AT.posWorkPointer = olpw;
        }
    }
    else {                                  /* Expression from disk */
        POSITION StartPos;
        LONG position, olpw, opw, comprev, extra;
        RENUMBER renumber;
        WORD *Freeze, *aa, *dummies;
        replac = -replac-1;
        power = AN.TeSuOut;
        Freeze = AN.Frozen;
        if ( Expressions[replac].status == STOREDEXPRESSION ) {
            POSITION firstpos;
            SETSTARTPOS(firstpos);

/*          Note that AT.TMaddr is needed for GetTable just once! */
/*

            We need space for the previous term in the compression
            This is made available in AR.CompressBuffer, although we may get
            problems with this sooner or later. Hence we need to keep
            a set of pointers in AR.CompressBuffer
            Note that after the last call there has been no use made
            of AR.CompressPointer, so it points automatically at its original
            position!
*/
            WantAddPointers(power+1);
            comprev = opw = AT.pWorkPointer;
            AT.pWorkPointer += power+1;
            WantAddPositions(power+1);
            position = olpw = AT.posWorkPointer;
            AT.posWorkPointer += power + 1;

            AT.pWorkSpace[comprev++] = AR.CompressPointer;

            for ( i = 0; i < power; i++ ) {
                PUTZERO(AT.posWorkSpace[position]); position++;
            }
            position = olpw;
            if ( ( renumber = GetTable(replac,&(AT.posWorkSpace[position]),1) ) == 0 ) goto GenCall;
            dummies = AT.WorkPointer;
            *dummies++ = AR.CurDum;
            AT.WorkPointer += power+2;
            accum = AT.WorkPointer;
            AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + 2*AM.MaxTer);
            if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
            aa = AT.WorkPointer;
            *accum = 0;
            i = 0; StartPos = AT.posWorkSpace[position];
            dummies[i] = AR.CurDum;
            while ( i >= 0 ) {
skippedfirst:
                AR.CompressPointer = AT.pWorkSpace[comprev-1];
                if ( ( extra = PasteFile(BHEAD i,accum,&(AT.posWorkSpace[position])
                        ,&a,renumber,Freeze,replac) ) < 0 ) goto GenCall;
                if ( Expressions[replac].numdummies > 0 ) {
                    AR.CurDum = dummies[i] + Expressions[replac].numdummies;
                }
                if ( NOTSTARTPOS(firstpos) ) {
                    if ( ISMINPOS(firstpos) || ISEQUALPOS(firstpos,AT.posWorkSpace[position]) ) {
                        firstpos = AT.posWorkSpace[position];
/*
                        ADDPOS(AT.posWorkSpace[position],extra * sizeof(WORD));
*/
                        goto skippedfirst;
                    }
                }
                if ( extra ) {
/*
                    ADDPOS(AT.posWorkSpace[position],extra * sizeof(WORD));
*/
                    i++; AT.posWorkSpace[++position] = StartPos;
                    AT.pWorkSpace[comprev++] = AR.CompressPointer;
                    dummies[i] = AR.CurDum;
                }
```

```
                else {
                    PUTZERO(AT.posWorkSpace[position]); position--; i--;
                    AR.CurDum = dummies[i];
                    comprev--;
                }
                if ( i >= power ) {
                    termout = AT.WorkPointer = a;
                    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + 2*AM.MaxTer);
                    if ( AT.WorkPointer > AT.WorkTop ) goto OverWork;
                    if ( FiniTerm(BHEAD term,accum,termout,replac,0) ) goto GenCall;
                    if ( *termout ) {
                        AT.WorkPointer = termout + *termout;
                        *AN.RepPoint = 1;
                        AR.expchanged = 1;
#ifdef WITHPTHREADS
                        if ( ( AS.Balancing && CC->numrhs == 0 ) && ( i > 0 ) && ( id = ConditionalGetAvailableThread() ) >= 0 ) {
                            if ( BalanceRunThread(BHEAD id,termout,level) < 0 ) goto GenCall;

                        }
                        else
#endif
                        if ( Generator(BHEAD termout,level) ) goto GenCall;
                    }
                    i--; position--;
                    AR.CurDum = dummies[i];
                    comprev--;
                }
                AT.WorkPointer = aa;
            }
            AT.WorkPointer = accum;
            AT.posWorkPointer = olpw;
            AT.pWorkPointer = opw;
/*
        Bug fix. See also GetTable
#ifdef WITHPTHREADS
            M_free(renumber->symb.lo,"VarSpace");
            M_free(renumber,"Renumber");
#endif
*/
            if ( renumber->symb.lo != AN.dummyrenumlist )
                M_free(renumber->symb.lo,"VarSpace");
            M_free(renumber,"Renumber");

        }
        else {          /* Active expression */
            aa = accum = AT.WorkPointer;
            if ( ( (WORD *)(((UBYTE *)(AT.WorkPointer)) + 2 * AM.MaxTer + sizeof(WORD)) ) > AT.WorkTop )
                    goto OverWork;
            *accum++ = -1; AT.WorkPointer++;
            if ( DoOnePow(BHEAD term,power,replac,accum,aa,level,Freeze) ) goto GenCall;
            AT.WorkPointer = aa;
        }
    }
Return0:
    AR.CurDum = DumNow;
    AN.RepPoint = RepSto;
    CC->numrhs = oldtoprhs;
    CC->Pointer = CC->Buffer + oldcpointer;
    CCC->numrhs = oldatoprhs;
    CCC->Pointer = CCC->Buffer + oldacpointer;
    return(0);

GenCall:
    if ( AM.tracebackflag ) {
        termout = term;
        MLOCK(ErrorMessageLock);
        AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer;
        AO.OutSkip = 3;
        FiniLine();
        i = *termout;
        while ( --i >= 0 ) {
            TalToLine((UWORD)(*termout++));
            TokenToLine((UBYTE *)"  ");
        }
        AO.OutSkip = 0;
        FiniLine();
        MesCall("Generator");
        MUNLOCK(ErrorMessageLock);
    }
    CC->numrhs = oldtoprhs;
    CC->Pointer = CC->Buffer + oldcpointer;
    CCC->numrhs = oldatoprhs;
    CCC->Pointer = CCC->Buffer + oldacpointer;
    return(-1);
OverWork:
```

```
    CC->numrhs = oldtoprhs;
    CC->Pointer = CC->Buffer + oldcpointer;
    CCC->numrhs = oldatoprhs;
    CCC->Pointer = CCC->Buffer + oldacpointer;
    MLOCK(ErrorMessageLock);
    MesWork();
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] Generator :
        #[ DoOnePow :           WORD DoOnePow(term,power,nexp,accum,aa,level,freeze)
*/
#ifdef WITHPTHREADS
char freezestring[] = "freeze<-xxxx";
#endif

WORD DoOnePow(PHEAD WORD *term, WORD power, WORD nexp, WORD * accum,
              WORD *aa, WORD level, WORD *freeze)
{
    GETBIDENTITY
    POSITION oldposition, startposition;
    WORD *acc, *termout, fromfreeze = 0;
    WORD *oldipointer = AR.CompressPointer;
    FILEHANDLE *fi;
    WORD type, retval;
    WORD oldGetOneFile = AR.GetOneFile;
    WORD olddummies = AR.CurDum;
    WORD extradummies = Expressions[nexp].numdummies;
/*
    The next code is for some tricky debugging. (5-jan-2010 JV)
    Normally it should be disabled.
*/
/*
#ifdef WITHPTHREADS
    if ( freeze ) {
        MLOCK(ErrorMessageLock);
        if ( AT.identity < 10 ) {
            freezestring[8] = '0'+AT.identity;
            freezestring[9] = '>';
            freezestring[10] = 0;
        }
        else if ( AT.identity < 100 ) {
            freezestring[8] = '0'+AT.identity/10;
            freezestring[9] = '0'+AT.identity%10;
            freezestring[10] = '>';
            freezestring[11] = 0;
        }
        else {
            freezestring[8] = 0;
        }
        PrintTerm(freeze,freezestring);
        MUNLOCK(ErrorMessageLock);
    }
#else
    if ( freeze ) PrintTerm(freeze,"freeze");
#endif
*/
    type = Expressions[nexp].status;
    if ( type == HIDDENLEXPRESSION || type == HIDDENGEXPRESSION
     || type == DROPHLEXPRESSION || type == DROPHGEXPRESSION
     || type == UNHIDELEXPRESSION || type == UNHIDEGEXPRESSION ) {
        AR.GetOneFile = 2; fi = AR.hidefile;
    }
    else {
        AR.GetOneFile = 0; fi = AR.infile;
    }
    if ( fi->handle >= 0 ) {
        PUTZERO(oldposition);
#ifdef WITHSEEK
        LOCK(AS.inputslock);
        SeekFile(fi->handle,&oldposition,SEEK_CUR);
        UNLOCK(AS.inputslock);
#endif
    }
    else {
        SETBASEPOSITION(oldposition,fi->POfill-fi->PObuffer);
    }
    if ( freeze && ( Expressions[nexp].bracketinfo != 0 ) ) {
        POSITION *brapos;
/*
        There is a bracket index
        AR.CompressPointer = oldipointer;
*/
        (*aa)++;
        power--;
```

```
        if ( ( brapos = FindBracket(nexp,freeze) ) == 0 )
            goto EndExpr;
        startposition = *brapos;
        goto doterms;
    }
    startposition = AS.OldOnFile[nexp];
    retval = GetOneTerm(BHEAD accum,fi,&startposition,0);
    if ( retval > 0 ) {         /* Skip prototype */
        (*aa)++;
        power--;
doterms:
        AR.CompressPointer = oldipointer;
        for (;;) {
            retval = GetOneTerm(BHEAD accum,fi,&startposition,0);
            if ( retval <= 0 ) break;
/*
            Here should come the code to test for [].
*/
            if ( freeze ) {
                WORD *t, *m, *r, *mstop;
                WORD *tset;
                t = accum;
                m = freeze;
                m += *m;
                m -= ABS(m[-1]);
                mstop = m;
                m = freeze + 1;
                r = t;
                r += *t;
                r -= ABS(r[-1]);
                t++;
                tset = t;
                while ( t < r && *t != HAAKJE ) t += t[1];
                if ( t >= r ) {
                    if ( m < mstop ) {
                        if ( fromfreeze ) goto EndExpr;
                        goto NextTerm;
                    }
                    t = tset;
                }
                else {
                    r = tset;
                    while ( r < t && m < mstop ) {
                        if ( *r == *m ) { m++; r++; }
                        else {
                            if ( fromfreeze ) goto EndExpr;
                            goto NextTerm;
                        }
                    }
                    if ( r < t || m < mstop ) {
                        if ( fromfreeze ) goto EndExpr;
                        goto NextTerm;
                    }
                }
                fromfreeze = 1;
                r = tset;
                m = accum;
                m += *m;
                while ( t < m ) *r++ = *t++;
                *accum = WORDDIF(r,accum);
            }
            if ( extradummies > 0 ) {
                if ( olddummies > AM.IndDum ) {
                    MoveDummies(BHEAD accum,olddummies-AM.IndDum);
                }
                AR.CurDum = olddummies+extradummies;
            }
            acc = accum;
            acc += *acc;
            if ( power <= 0 ) {
                termout = acc;
                AT.WorkPointer = (WORD *)(((UBYTE *)(acc)) + 2*AM.MaxTer);
                if ( AT.WorkPointer > AT.WorkTop ) {
                    MLOCK(ErrorMessageLock);
                    MesWork();
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                if ( FiniTerm(BHEAD term,aa,termout,nexp,0) ) goto PowCall;
                if ( *termout ) {
                    MarkPolyRatFunDirty(termout)
/*                  PolyFunDirty(BHEAD termout); */
                    AT.WorkPointer = termout + *termout;
                    *AN.RepPoint = 1;
                    AR.expchanged = 1;
                    if ( Generator(BHEAD termout,level) ) goto PowCall;
                }
```

```
                }
                else {
                    if ( acc > AT.WorkTop ) {
                        MLOCK(ErrorMessageLock);
                        MesWork();
                        MUNLOCK(ErrorMessageLock);
                        return(-1);
                    }
                    if ( DoOnePow(BHEAD term,power,nexp,acc,aa,level,freeze) ) goto PowCall;
                }
NextTerm:;
                AR.CompressPointer = oldipointer;
            }
EndExpr:
            (*aa)--;
        }
        AR.CompressPointer = oldipointer;
        if ( fi->handle >= 0 ) {
#ifdef WITHSEEK
            LOCK(AS.inputslock);
            SeekFile(fi->handle,&oldposition,SEEK_SET);
            UNLOCK(AS.inputslock);
            if ( ISNEGPOS(oldposition) ) {
                MLOCK(ErrorMessageLock);
                MesPrint("File error");
                goto PowCall2;
            }
#endif
        }
        else {
            fi->POfill = fi->PObuffer + BASEPOSITION(oldposition);
        }
        AR.GetOneFile = oldGetOneFile;
        AR.CurDum = olddummies;
        return(0);
PowCall:;
        MLOCK(ErrorMessageLock);
#ifdef WITHSEEK
PowCall2:;
#endif
        MesCall("DoOnePow");
        MUNLOCK(ErrorMessageLock);
        SETERROR(-1)
}

/*
            #] DoOnePow :
            #[ Deferred :           WORD Deferred(term,level)
*/
WORD Deferred(PHEAD WORD *term, WORD level)
{
        GETBIDENTITY
        POSITION startposition;
        WORD *t, *m, *mstop, *tstart, decr, oldb, *termout, i, *oldwork, retval;
        WORD *oldipointer = AR.CompressPointer, *oldPOfill = AR.infile->POfill;
        WORD oldGetOneFile = AR.GetOneFile;
        AR.GetOneFile = 1;
        oldwork = AT.WorkPointer;
        AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
        termout = AT.WorkPointer;
        AR.DeferFlag = 0;
        startposition = AR.DefPosition;
/*
            Store old position
*/
        if ( AR.infile->handle >= 0 ) {
/*
            PUTZERO(oldposition);
            SeekFile(AR.infile->handle,&oldposition,SEEK_CUR);
*/
        }
        else {
/*
            SETBASEPOSITION(oldposition,AR.infile->POfill-AR.infile->PObuffer);
*/
            AR.infile->POfill = (WORD *)((UBYTE *)(AR.infile->PObuffer)
                        +BASEPOSITION(startposition));
        }
/*
            Look in the CompressBuffer where the bracket contents start
*/
        t = m = AR.CompressBuffer;
        t += *t;
        mstop = t - ABS(t[-1]);
        m++;
        while ( *m != HAAKJE && m < mstop ) m += m[1];
        if ( m >= mstop ) { /* No deferred action! */
```

```
        AT.WorkPointer = term + *term;
        if ( Generator(BHEAD term,level) ) goto DefCall;
        AR.DeferFlag = 1;
        AT.WorkPointer = oldwork;
        AR.GetOneFile = oldGetOneFile;
        return(0);
    }
    mstop = m + m[1];
    decr = WORDDIF(mstop,AR.CompressBuffer)-1;
    tstart = AR.CompressPointer + decr;

    m = AR.CompressBuffer;
    t = AR.CompressPointer;
    i = *m;
    NCOPY(t,m,i);
    oldb = *tstart;
    AR.TePos = 0;
    AN.TeSuOut = 0;
/*
        Status:
        First bracket content starts at mstop.
        Next term starts at startposition.
        Decompression information is in AR.CompressPointer.
        The outside of the bracket runs from AR.CompressBuffer+1 to mstop.
*/
    for(;;) {
        *tstart = *(AR.CompressPointer)-decr;
        AR.CompressPointer = AR.CompressPointer+AR.CompressPointer[0];
        if ( InsertTerm(BHEAD term,0,AM.rbufnum,tstart,termout,0) < 0 ) {
            goto DefCall;
        }
        *tstart = oldb;
        AT.WorkPointer = termout + *termout;
        if ( Generator(BHEAD termout,level) ) goto DefCall;
        AR.CompressPointer = oldipointer;
        AT.WorkPointer = termout;
        retval = GetOneTerm(BHEAD AT.WorkPointer,AR.infile,&startposition,0);
        if ( retval >= 0 ) AR.CompressPointer = oldipointer;
        if ( retval <= 0 ) break;
        t = AR.CompressPointer;
        if ( *t < (1 + decr + ABS(*(t+*t-1))) ) break;
        t++;
        m = AR.CompressBuffer+1;
        while ( m < mstop ) {
            if ( *m != *t ) goto Thatsit;
            m++; t++;
        }
    }
Thatsit:;
/*
        Finished. Reposition the file, restore information and return.
*/
    if ( AR.infile->handle < 0 ) AR.infile->POfill = oldPOfill;
    AR.DeferFlag = 1;
    AR.GetOneFile = oldGetOneFile;
    AT.WorkPointer = oldwork;
    return(0);
DefCall:;
    MLOCK(ErrorMessageLock);
    MesCall("Deferred");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] Deferred :
        #[ PrepPoly :           WORD PrepPoly(term,par)
*/
WORD PrepPoly(PHEAD WORD *term,WORD par)
{
    GETBIDENTITY
    WORD count = 0, i, jcoef, ncoef;
    WORD *t, *m, *r, *tstop, *poly = 0, *v, *w, *vv, *ww;
    WORD *oldworkpointer = AT.WorkPointer;
/*
    The problem here is that the function will be forced into 'long'
    notation. After this -SNUMBER,1 becomes 6,0,4,1,1,3 and the
    pattern matcher cannot match a short 1 with a long 1.
    But because this is an undocumented feature for very special
    purposes, we don't do anything about it. (30-aug-2011)
*/
    if ( AR.PolyFunType == 2 && AR.PolyFunExp != 2 ) {
        WORD oldtype = AR.SortType;
        AR.SortType = SORTHIGHFIRST;
        if ( poly_ratfun_normalize(BHEAD term) != 0 ) Terminate(-1);
/*      if ( ReadPolyRatFun(BHEAD term) != 0 ) Terminate(-1); */
        oldworkpointer = AT.WorkPointer;
```

```
        AR.SortType = oldtype;
    }
    AT.PolyAct = 0;
    t = term;
    GETSTOP(t,tstop);
    t++;
    while ( t < tstop ) {
        if ( *t == AR.PolyFun ) {
            if ( count > 0 ) return(0);
            poly = t;
            count++;
        }
        t += t[1];
    }
    r = m = term + *term;
    i = ABS(m[-1]);
    if ( par > 0 ) {
        if ( count == 0 ) return(0);
        else if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) )
            goto DoOne;
        else if ( AR.PolyFunType == 2 )
            goto DoTwo;
        else
            goto DoError;
    }
    else if ( count == 0 ) {
/*
        #[ Create a PolyFun :
*/
        poly = t = tstop;
        if ( i == 3 && m[-2] == 1 && (m[-3]&MAXPOSITIVE) == m[-3] ) {
            *m++ = AR.PolyFun;
            if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) ) {
                *m++ = FUNHEAD+2;
                FILLFUN(m)
                *m++ = -SNUMBER;
                *m = m[-2-FUNHEAD] < 0 ? -m[-4-FUNHEAD]: m[-4-FUNHEAD];
                m++;
            }
            else if ( AR.PolyFunType == 2 ) {
                *m++ = FUNHEAD+4;
                FILLFUN(m)
                *m++ = -SNUMBER;
                *m = m[-2-FUNHEAD] < 0 ? -m[-4-FUNHEAD]: m[-4-FUNHEAD];
                m++;
                *m++ = -SNUMBER;
                *m++ = 1;
            }
        }
        else {
            WORD *vm;
            r = tstop;
            if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) ) {
                *m++ = AR.PolyFun;
                *m++ = FUNHEAD+ARGHEAD+i+1;
                FILLFUN(m)
                *m++ = ARGHEAD+i+1;
                *m++ = 0;
                FILLARG(m)
                *m++ = i+1;
                NCOPY(m,r,i);
            }
            else if ( AR.PolyFunType == 2 ) {
                WORD *num, *den, size, sign, sizenum, sizeden;
                if ( m[-1] < 0 ) { sign = -1; size = -m[-1]; }
                else             { sign =  1; size =  m[-1]; }
                num = m - size; size = (size-1)/2; den = num + size;
                sizenum = size; while ( num[sizenum-1] == 0 ) sizenum--;
                sizeden = size; while ( den[sizeden-1] == 0 ) sizeden--;
                v = m;
                AT.PolyAct = WORDDIF(v,term);
                *v++ = AR.PolyFun;
                v++;
                FILLFUN(v);
                vm = v;
                *v++ = ARGHEAD+2*sizenum+2;
                *v++ = 0;
                FILLARG(v);
                *v++ = 2*sizenum+2;
                for ( i = 0; i < sizenum; i++ ) *v++ = num[i];
                *v++ = 1;
                for ( i = 1; i < sizenum; i++ ) *v++ = 0;
                *v++ = sign*(2*sizenum+1);
                if ( ToFast(vm,vm) ) v = vm+2;
                vm = v;
                *v++ = ARGHEAD+2*sizeden+2;
                *v++ = 0;
```

```
                FILLARG(v);
                *v++ = 2*sizeden+2;
                for ( i = 0; i < sizeden; i++ ) *v++ = den[i];
                *v++ = 1;
                for ( i = 1; i < sizeden; i++ ) *v++ = 0;
                *v++ = 2*sizeden+1;
                if ( ToFast(vm,vm) ) v = vm+2;
                i = v-m;
                m[1] = i;
                w = num;
                NCOPY(w,m,i);
                *w++ = 1; *w++ = 1; *w++ = 3; *term = w - term;
                return(0);
            }
        }
/*
        #] Create a PolyFun :
*/
    }
    else if ( AR.PolyFunType == 1 || (AR.PolyFunType == 2 && AR.PolyFunExp == 2) ) {
        DoOne:;
/*
        #[ One argument :
*/
        m = term + *term;
        r = poly + poly[1];
        if ( ( poly[1] == FUNHEAD+2 && poly[FUNHEAD+1] == 0
        && poly[FUNHEAD] == -SNUMBER ) || poly[1] == FUNHEAD ) return(1);
        t = poly + FUNHEAD;
        if ( t >= r ) return(0);
        if ( m[-1] == 3 && *tstop == 1 && tstop[1] == 1 ) {
            i = poly[1];
            t = poly;
            NCOPY(m,t,i);
        }
        else if ( *t <= -FUNCTION ) {
            if ( t+1 < r ) return(0);   /* More than one argument */
            r = tstop;
            *m++ = AR.PolyFun;
            *m++ = FUNHEAD*2+ARGHEAD+i+1;
            FILLFUN(m)
            *m++ = FUNHEAD+ARGHEAD+i+1;
            *m++ = 0;
            FILLARG(m)
            *m++ = FUNHEAD+i+1;
            *m++ = -*t++;
            *m++ = FUNHEAD;
            FILLFUN(m)
            NCOPY(m,r,i);
        }
        else if ( *t < 0 ) {
            if ( t+2 < r ) return(0);   /* More than one argument */
            r = tstop;
            if ( *t == -SNUMBER ) {
                if ( t[1] == 0 ) return(1); /* Term should be zero now */
                *m = AR.PolyFun;
                w = m+1;
                m += FUNHEAD+ARGHEAD;
                v = m;
                *m++ = 5+i;
                *m++ = SNUMBER;
                *m++ = 4;
                *m++ = t[1];
                *m++ = 1;
                NCOPY(m,r,i);
                if ( m >= AT.WorkSpace && m < AT.WorkTop )
                    AT.WorkPointer = m;
                if ( Normalize(BHEAD v) ) Terminate(-1);
                AT.WorkPointer = oldworkpointer;
                m = w;
                if ( *v == 4 && v[2] == 1 && (v[1]&MAXPOSITIVE) == v[1] ) {
                    *m++ = FUNHEAD+2;
                    FILLFUN(m)
                    *m++ = -SNUMBER;
                    *m++ = v[3] < 0 ? -v[1] : v[1];
                }
                else if ( *v == 0 ) return(1);
                else {
                    *m++ = FUNHEAD+ARGHEAD+*v;
                    FILLFUN(m)
                    *m++ = ARGHEAD+*v;
                    *m++ = 0;
                    FILLARG(m)
                    m = v + *v;
                }
            }
            else if ( *t == -SYMBOL ) {
```

```
                *m++ = AR.PolyFun;
                *m++ = FUNHEAD+ARGHEAD+5+i;
                FILLFUN(m)
                *m++ = ARGHEAD+5+i;
                *m++ = 0;
                FILLARG(m)
                *m++ = 5+i;
                *m++ = SYMBOL;
                *m++ = 4;
                *m++ = t[1];
                *m++ = 1;
                NCOPY(m,r,i);
            }
            else return(0);         /* Not symbol-like */
        }
        else {
            if ( t + *t < r ) return(0); /* More than one argument */
            i = m[-1];
            *m++ = AR.PolyFun;
            w = m;
            m += ARGHEAD+FUNHEAD-1;
            t += ARGHEAD;
            jcoef = i < 0 ? (i+1)>>1:(i-1)>>1;
            v = t;
/*
            Test now the scalar nature of the argument.
            No indices allowed.
*/
            while ( t < r ) {
                WORD *vstop;
                vv = t + *t;
                vstop = vv - ABS(vv[-1]);
                t++;
                while( t < vstop ) {
                    if ( *t == INDEX ) return(0);
                    t += t[1];
                }
                t = vv;
            }
/*
            Now multiply each term by the coefficient.
*/
            t = v;
            while ( t < r ) {
                ww = m;
                v = t + *t;
                ncoef = v[-1];
                vv = v - ABS(ncoef);
                if ( ncoef < 0 ) ncoef++;
                else ncoef--;
                ncoef >>= 1;
                while ( t < vv ) *m++ = *t++;
                if ( MulRat(BHEAD (UWORD *)vv,ncoef,(UWORD *)tstop,jcoef,
                    (UWORD *)m,&ncoef) ) Terminate(-1);
                ncoef *= 2;
                m += ABS(ncoef);
                if ( ncoef < 0 ) ncoef--;
                else ncoef++;
                *m++ = ncoef;
                *ww = WORDDIF(m,ww);
                if ( AN.ncmod != 0 ) {
                    if ( Modulus(ww) ) Terminate(-1);
                    if ( *ww == 0 ) return(1);
                    m = ww + *ww;
                }
                t = v;
            }
            *w = (WORDDIF(m,w))+1;
            w[FUNHEAD-1] = w[0] - FUNHEAD;
            w[FUNHEAD] = 0;
            w[1] = 0; /* omission survived for years. 23-mar-2006 JV */
            w += FUNHEAD-1;
            if ( ToFast(w,w) ) {
                if ( *w <= -FUNCTION ) { w[-FUNHEAD+1] = FUNHEAD+1; m = w+1; }
                else { w[-FUNHEAD+1] = FUNHEAD+2; m = w+2; }

            }
        }
        t = poly + poly[1];
        while ( t < tstop ) *poly++ = *t++;
/*
        #] One argument :
*/
    }
    else if ( AR.PolyFunType == 2 ) {
        DoTwo:;
/*
```

```
        #[ Two arguments :
*/
        WORD *num, *den, size, sign, sizenum, sizeden;
/*
        First make sure that the PolyFun is last
*/
        m = term + *term;
        if ( poly + poly[1] < tstop ) {
            for ( i = 0; i < poly[1]; i++ ) m[i] = poly[i];
            t = poly; v = poly + poly[1];
            while ( v < tstop ) *t++ = *v++;
            poly = t;
            for ( i = 0; i < m[1]; i++ ) t[i] = m[i];
            t += m[1];
        }
        AT.PolyAct = WORDDIF(poly,term);
/*
        If needed we convert the coefficient into a PolyRatFun and then
        we call poly_ratfun_normalize
*/
        if ( m[-1] == 3 && m[-2] == 1 && m[-3] == 1 ) return(0);
        if ( AR.PolyFunExp != 1 ) {
        if ( m[-1] < 0 ) { sign = -1; size = -m[-1]; } else { sign = 1; size = m[-1]; }
        num = m - size; size = (size-1)/2; den = num + size;
        sizenum = size; while ( num[sizenum-1] == 0 ) sizenum--;
        sizeden = size; while ( den[sizeden-1] == 0 ) sizeden--;
        v = m;
        *v++ = AR.PolyFun;
        *v++ = FUNHEAD + 2*(ARGHEAD+sizenum+sizeden+2);
/*      *v++ = MUSTCLEANPRF; */
        *v++ = 0;
        FILLFUN3(v);
        *v++ = ARGHEAD+2*sizenum+2;
        *v++ = 0;
        FILLARG(v);
        *v++ = 2*sizenum+2;
        for ( i = 0; i < sizenum; i++ ) *v++ = num[i];
        *v++ = 1;
        for ( i = 1; i < sizenum; i++ ) *v++ = 0;
        *v++ = sign*(2*sizenum+1);
        *v++ = ARGHEAD+2*sizeden+2;
        *v++ = 0;
        FILLARG(v);
        *v++ = 2*sizeden+2;
        for ( i = 0; i < sizeden; i++ ) *v++ = den[i];
        *v++ = 1;
        for ( i = 1; i < sizeden; i++ ) *v++ = 0;
        *v++ = 2*sizeden+1;
        w = num;
        i = v - m;
        NCOPY(w,m,i);
        }
        else {
            w = m-ABS(m[-1]);
        }
        *w++ = 1; *w++ = 1; *w++ = 3; *term = w - term;
        {
            WORD oldtype = AR.SortType;
            AR.SortType = SORTHIGHFIRST;
/*
            if ( count > 0 )
                poly_ratfun_normalize(BHEAD term);
            else
                ReadPolyRatFun(BHEAD term);
*/
            poly_ratfun_normalize(BHEAD term);

/*          oldworkpointer = AT.WorkPointer; */
            AR.SortType = oldtype;
        }
        goto endofit;
/*
        #] Two arguments :
*/
    }
    else {
        DoError:;
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal value for PolyFunType in PrepPoly");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    r = term + *term;
    AT.PolyAct = WORDDIF(poly,term);
    while ( r < m ) *poly++ = *r++;
    *poly++ = 1;
    *poly++ = 1;
```

```
    *poly++ = 3;
    *term = WORDDIF(poly,term);
endofit:;
    return(0);
}

/*
        #] PrepPoly :
        #[ PolyFunMul :         WORD PolyFunMul(term)
*/
WORD PolyFunMul(PHEAD WORD *term)
{
    GETBIDENTITY
    WORD *t, *fun1, *fun2, *t1, *t2, *m, *w, *ww, *tt1, *tt2, *tt4, *arg1, *arg2;
    WORD *tstop, i, dirty = 0, OldPolyFunPow = AR.PolyFunPow, minp1, minp2;
    WORD n1, n2, i1, i2, l1, l2, l3, l4, action = 0, noac = 0, retval = 0;
    if ( AR.PolyFunType == 2 && AR.PolyFunExp == 1 ) {
        WORD pow = 0, pow1;
        t = term + 1; t1 = term + *term; t1 -= ABS(t1[-1]);
        w = t;
        while ( t < t1 ) {
            if ( *t != AR.PolyFun ) {
SkipFun:
                if ( t == w ) { t += t[1]; w = t; }
                else { i = t[1]; NCOPY(w,t,i) }
                continue;
            }
            pow1 = 0;
            t2 = t + t[1]; t += FUNHEAD;
            if ( *t < 0 ) {
                if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) pow1++;
                else if ( *t != -SNUMBER ) goto NoLegal;
                t += 2;
            }
            else if ( t[0] == ARGHEAD+8 && t[ARGHEAD] == 8
             && t[ARGHEAD+1] == SYMBOL && t[ARGHEAD+3] == AR.PolyFunVar
             && t[ARGHEAD+5] == 1 && t[ARGHEAD+6] == 1 && t[ARGHEAD+7] == 3 ) {
                pow1 += t[ARGHEAD+4];
                t += *t;
            }
            else {
NoLegal:
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal term with divergence in PolyRatFun");
                MesCall("PolyFunMul");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            if ( *t < 0 ) {
                if ( *t == -SYMBOL && t[1] == AR.PolyFunVar ) pow1--;
                else if ( *t != -SNUMBER ) goto NoLegal;
                t += 2;
            }
            else if ( t[0] == ARGHEAD+8 && t[ARGHEAD] == 8
             && t[ARGHEAD+1] == SYMBOL && t[ARGHEAD+3] == AR.PolyFunVar
             && t[ARGHEAD+5] == 1 && t[ARGHEAD+6] == 1 && t[ARGHEAD+7] == 3 ) {
                pow1 -= t[ARGHEAD+4];
                t += *t;
            }
            else goto NoLegal;
            if ( t == t2 ) pow += pow1;
            else goto SkipFun;
        }
        m = w;
        *w++ = AR.PolyFun; *w++ = 0; FILLFUN(w);
        if ( pow > 1 ) {
            *w++ = 8+ARGHEAD; *w++ = 0; FILLARG(w);
            *w++ = 8; *w++ = SYMBOL; *w++ = 4; *w++ = AR.PolyFunVar; *w++ = pow;
            *w++ = 1; *w++ = 1; *w++ = 3; *w++ = -SNUMBER; *w++ = 1;
        }
        else if ( pow == 1 ) {
            *w++ = -SYMBOL; *w++ = AR.PolyFunVar; *w++ = -SNUMBER; *w++ = 1;
        }
        else if ( pow < -1 ) {
            *w++ = -SNUMBER; *w++ = 1; *w++ = 8+ARGHEAD; *w++ = 0; FILLARG(w);
            *w++ = 8; *w++ = SYMBOL; *w++ = 4; *w++ = AR.PolyFunVar; *w++ = -pow;
            *w++ = 1; *w++ = 1; *w++ = 3;
        }
        else if ( pow == -1 ) {
            *w++ = -SNUMBER; *w++ = 1; *w++ = -SYMBOL; *w++ = AR.PolyFunVar;
        }
        else {
            *w++ = -SNUMBER; *w++ = 1; *w++ = -SNUMBER; *w++ = 1;
        }
        m[1] = w - m;
        *w++ = 1; *w++ = 1; *w++ = 3;
        *term = w - term;
```

```
        if ( w > AT.WorkSpace && w < AT.WorkTop ) AT.WorkPointer = w;
        return(0);
    }
ReStart:
    if ( AR.PolyFunType == 2 && ( ( AR.PolyFunExp != 2 )
     || ( AR.PolyFunExp == 2 && AN.PolyNormFlag > 1 ) ) ) {
        WORD count1 = 0, count2 = 0, count3;
        WORD oldtype = AR.SortType;
        t = term + 1; t1 = term + *term; t1 -= ABS(t1[-1]);
        while ( t < t1 ) {
            if ( *t == AR.PolyFun ) {
              if ( t[2] && dirty == 0 ) { /* Any dirty flag on? */
                dirty = 1;
/*              ReadPolyRatFun(BHEAD term); */
/*              ToPolyFunGeneral(BHEAD term); */
                poly_ratfun_normalize(BHEAD term);
                if ( term[0] == 0 ) return(0);
                count1 = 0;
                action++;
                goto ReStart;
              }
              t2 = t + t[1]; tt2 = t+FUNHEAD; count3 = 0;
              while ( tt2 < t2 ) { count3++; NEXTARG(tt2); }
              if ( count3 == 2 ) {
                count1++;
                if ( ( t[2] & MUSTCLEANPRF ) != 0 ) {   /* Better civilize this guy */
                    action++;
                    w = AT.WorkPointer;
                    AR.SortType = SORTHIGHFIRST;
                    t2 = t + t[1]; tt2 = t+FUNHEAD;
                    while ( tt2 < t2 ) {
                        if ( *tt2 > 0 ) {
                            tt4 = tt2; tt1 = tt2 + ARGHEAD; tt2 += *tt2;
                            NewSort(BHEAD0);
                            while ( tt1 < tt2 ) {
                                i = *tt1; ww = w; NCOPY(ww,tt1,i);
                                AT.WorkPointer = ww;
                                Normalize(BHEAD w);
                                StoreTerm(BHEAD w);
                            }
                            EndSort(BHEAD w,1);
                            ww = w; while ( *ww ) ww += *ww;
                            if ( ww-w != *tt4-ARGHEAD ) { /* Little problem */
/*
                                Solution: brute force copy
                                Maybe it will never come here????
*/
                                WORD *r1 = TermMalloc("PolyFunMul");
                                WORD ii = (ww-w)-(*tt4-ARGHEAD); /* increment */
                                WORD *r2 = tt4+ARGHEAD, *r3, *r4 = r1;
                                i = r2 - term; r3 = term; NCOPY(r4,r3,i);
                                i = ww-w; ww = w; NCOPY(r4,ww,i);
                                r3 = tt2; i = term+*term-tt2; NCOPY(r4,r3,i);
                                *r1 = i = r4-r1; r4 = term; r3 = r1;
                                NCOPY(r4,r3,i);
                                t[1] += ii; t1 += ii; *tt4 += ii;
                                tt2 = tt4 + *tt4;
                                TermFree(r1,"PolyFunMul");
                            }
                            else {
                                i = ww-w; ww = w; tt1 = tt4+ARGHEAD;
                                NCOPY(tt1,ww,i);
                                AT.WorkPointer = w;
                            }
                        }
                        else if ( *tt2 <= -FUNCTION ) tt2++;
                        else tt2 += 2;
                    }
                    AR.SortType = oldtype;
                }
              }
            }
            t += t[1];
        }
        if ( count1 <= 1 ) { goto checkaction; }
        if ( AR.PolyFunExp == 1 ) {
            t = term + *term; t -= ABS(t[-1]);
            *t++ = 1; *t++ = 1; *t++ = 3; *term = t - term;
        }
        {
            AR.SortType = SORTHIGHFIRST;
/*          retval = ReadPolyRatFun(BHEAD term); */
/*          ToPolyFunGeneral(BHEAD term); */
            retval = poly_ratfun_normalize(BHEAD term);
            if ( *term == 0 ) return(retval);
            AR.SortType = oldtype;
        }
```

```
        t = term + 1; t1 = term + *term; t1 -= ABS(t1[-1]);
        while ( t < t1 ) {
            if ( *t == AR.PolyFun ) {
              t2 = t + t[1]; tt2 = t+FUNHEAD; count3 = 0;
              while ( tt2 < t2 ) { count3++; NEXTARG(tt2); }
              if ( count3 == 2 ) {
                count2++;
              }
            }
            t += t[1];
        }
        if ( count1 >= count2 ) {
            t = term + 1;
            while ( t < t1 ) {
                if ( *t == AR.PolyFun ) {
                    t2 = t;
                    t = t + t[1];
                    t2[2] |= (DIRTYFLAG|MUSTCLEANPRF);
                    t2 += FUNHEAD;
                    while ( t2 < t ) {
                        if ( *t2 > 0 ) t2[1] = DIRTYFLAG;
                        NEXTARG(t2);
                    }
                }
                else t += t[1];
            }
        }

        w = term + *term;
        if ( w > AT.WorkSpace && w < AT.WorkTop ) AT.WorkPointer = w;
checkaction:
        if ( action ) retval = action;
        return(retval);
    }
retry:
    if ( term >= AT.WorkSpace && term+*term < AT.WorkTop )
        AT.WorkPointer = term + *term;
    GETSTOP(term,tstop);
    t = term+1;
    while ( *t != AR.PolyFun && t < tstop ) t += t[1];
    while ( t < tstop && *t == AR.PolyFun ) {
        if ( t[1] > FUNHEAD ) {
            if ( t[FUNHEAD] < 0 ) {
                if ( t[FUNHEAD] <= -FUNCTION && t[1] == FUNHEAD+1 ) break;
                if ( t[FUNHEAD]  > -FUNCTION && t[1] == FUNHEAD+2 ) {
                    if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) {
                        *term = 0;
                        return(0);
                    }
                    break;
                }
            }
            else if ( t[FUNHEAD] == t[1] - FUNHEAD ) break;
        }
        noac = 1;
        t += t[1];
    }
    if ( *t != AR.PolyFun || t >= tstop ) goto done;
    fun1 = t;
    t += t[1];
    while ( t < tstop && *t == AR.PolyFun ) {
        if ( t[1] > FUNHEAD ) {
            if ( t[FUNHEAD] < 0 ) {
                if ( t[FUNHEAD] <= -FUNCTION && t[1] == FUNHEAD+1 ) break;
                if ( t[FUNHEAD]  > -FUNCTION && t[1] == FUNHEAD+2 ) {
                    if ( t[FUNHEAD] == -SNUMBER && t[FUNHEAD+1] == 0 ) {
                        *term = 0;
                        return(0);
                    }
                    break;
                }
            }
            else if ( t[FUNHEAD] == t[1] - FUNHEAD ) break;
        }
        noac = 1;
        t += t[1];
    }
    if ( *t != AR.PolyFun || t >= tstop ) goto done;
    fun2 = t;
/*
    We have two functions of the proper type.
    Count terms (needed for the specials)
*/
    t = fun1 + FUNHEAD;
    if ( *t < 0 ) {
        n1 = 1; arg1 = AT.WorkPointer;
```

```
            ToGeneral(t,arg1,1);
            AT.WorkPointer = arg1 + *arg1;
        }
        else {
            t += ARGHEAD;
            n1 = 0; t1 = fun1 + fun1[1]; arg1 = t;
            while ( t < t1 ) { n1++; t += *t; }
        }
        t = fun2 + FUNHEAD;
        if ( *t < 0 ) {
            n2 = 1; arg2 = AT.WorkPointer;
            ToGeneral(t,arg2,1);
            AT.WorkPointer = arg2 + *arg2;
        }
        else {
            t += ARGHEAD;
            n2 = 0; t2 = fun2 + fun2[1]; arg2 = t;
            while ( t < t2 ) { n2++; t += *t; }
        }
/*
        Now we can start the multiplications. We first multiply the terms
        without coefficients, then normalize, and finally put the coefficients
        in place. This is because one has often truncated series and the
        high powers may get killed, while their coefficients are the most
        expensive ones.
        Note: We may run into fun(-SNUMBER,value)
*/
        w = AT.WorkPointer;
        NewSort(BHEAD0);
        if ( AR.PolyFunType == 2 && AR.PolyFunExp == 2 ) {
            AT.TrimPower = 1;
/*
            We have to find the lowest power in both polynomials.
            This will be needed to temporarily correct the AR.PolyFunPow
*/
            minp1 = MAXPOWER;
            for ( t1 = arg1, i1 = 0; i1 < n1; i1++, t1 += *t1 ) {
                if ( *t1 == 4 ) {
                    if ( minp1 > 0 ) minp1 = 0;
                }
                else if ( ABS(t1[*t1-1]) == (*t1-1) ) {
                    if ( minp1 > 0 ) minp1 = 0;
                }
                else {
                    if ( t1[1] == SYMBOL && t1[2] == 4 && t1[3] == AR.PolyFunVar ) {
                        if ( t1[4] < minp1 ) minp1 = t1[4];
                    }
                    else {
                        MesPrint("Illegal term in expanded polyratfun.");
                        goto PolyCall;
                    }
                }
            }
            minp2 = MAXPOWER;
            for ( t2 = arg2, i2 = 0; i2 < n2; i2++, t2 += *t2 ) {
                if ( *t2 == 4 ) {
                    if ( minp2 > 0 ) minp2 = 0;
                }
                else if ( ABS(t2[*t2-1]) == (*t2-1) ) {
                    if ( minp2 > 0 ) minp2 = 0;
                }
                else {
                    if ( t2[1] == SYMBOL && t2[2] == 4 && t2[3] == AR.PolyFunVar ) {
                        if ( t2[4] < minp2 ) minp2 = t2[4];
                    }
                    else {
                        MesPrint("Illegal term in expanded polyratfun.");
                        goto PolyCall;
                    }
                }
            }
            AR.PolyFunPow += minp1+minp2;
        }
        for ( t1 = arg1, i1 = 0; i1 < n1; i1++, t1 += *t1 ) {
        for ( t2 = arg2, i2 = 0; i2 < n2; i2++, t2 += *t2 ) {
            m = w;
            m++;
            GETSTOP(t1,tt1);
            t = t1 + 1;
            while ( t < tt1 ) *m++ = *t++;
            GETSTOP(t2,tt2);
            t = t2+1;
            while ( t < tt2 ) *m++ = *t++;
            *m++ = 1; *m++ = 1; *m++ = 3; *w = WORDDIF(m,w);
            AT.WorkPointer = m;
            if ( Normalize(BHEAD w) ) { LowerSortLevel(); goto PolyCall; }
            if ( *w ) {
```

```
            m = w + *w;
            if ( m[-1] != 3 || m[-2] != 1 || m[-3] != 1 ) {
                l3 = REDLENG(m[-1]);
                m -= ABS(m[-1]);
                t = t1 + *t1 - 1;
                l1 = REDLENG(*t);
                if ( MulRat(BHEAD (UWORD *)m,l3,(UWORD *)tt1,l1,(UWORD *)m,&l4) ) {
                    LowerSortLevel(); goto PolyCall; }
                if ( AN.ncmod != 0 && TakeModulus((UWORD *)m,&l4,AC.cmod,AN.ncmod,UNPACK|AC.modmode) ) {
                    LowerSortLevel(); goto PolyCall; }
                if ( l4 == 0 ) continue;
                t = t2 + *t2 - 1;
                l2 = REDLENG(*t);
                if ( MulRat(BHEAD (UWORD *)m,l4,(UWORD *)tt2,l2,(UWORD *)m,&l3) ) {
                    LowerSortLevel(); goto PolyCall; }
                if ( AN.ncmod != 0 && TakeModulus((UWORD *)m,&l3,AC.cmod,AN.ncmod,UNPACK|AC.modmode) ) {
                    LowerSortLevel(); goto PolyCall; }
            }
            else {
                m -= 3;
                t = t1 + *t1 - 1;
                l1 = REDLENG(*t);
                t = t2 + *t2 - 1;
                l2 = REDLENG(*t);
                if ( MulRat(BHEAD (UWORD *)tt1,l1,(UWORD *)tt2,l2,(UWORD *)m,&l3) ) {
                    LowerSortLevel(); goto PolyCall; }
                if ( AN.ncmod != 0 && TakeModulus((UWORD *)m,&l3,AC.cmod,AN.ncmod,UNPACK|AC.modmode) ) {
                    LowerSortLevel(); goto PolyCall; }
            }
            if ( l3 == 0 ) continue;
            l3 = INCLENG(l3);
            m += ABS(l3);
            m[-1] = l3;
            *w = WORDDIF(m,w);
            AT.WorkPointer = m;
            if ( StoreTerm(BHEAD w) ) { LowerSortLevel(); goto PolyCall; }
        }
    }
    }
    if ( EndSort(BHEAD w,0) < 0 ) goto PolyCall;
    AR.PolyFunPow = OldPolyFunPow;
    AT.TrimPower = 0;
    if ( *w == 0 ) {
        *term = 0;
        return(0);
    }
    t = w;
    while ( *t ) t += *t;
    AT.WorkPointer = t;
    n1 = WORDDIF(t,w);
    t1 = term;
    while ( t1 < fun1 ) *t++ = *t1++;
    t2 = t;
    *t++ = AR.PolyFun;
    *t++ = FUNHEAD+ARGHEAD+n1;
    *t++ = 0;
    FILLFUN3(t)
    *t++ = ARGHEAD+n1;
    *t++ = 0;
    FILLARG(t)
    NCOPY(t,w,n1);
    if ( ToFast(t2+FUNHEAD,t2+FUNHEAD) ) {
        if ( t2[FUNHEAD] > -FUNCTION ) t2[1] = FUNHEAD+2;
        else t2[FUNHEAD] = FUNHEAD+1;
        t = t2 + t2[1];
    }
    t1 = fun1 + fun1[1];
    while ( t1 < fun2 ) *t++ = *t1++;
    t1 = fun2 + fun2[1];
    t2 = term + *term;
    while ( t1 < t2 ) *t++ = *t1++;
    *AT.WorkPointer = n1 = WORDDIF(t,AT.WorkPointer);
    if ( n1*((LONG)sizeof(WORD)) > AM.MaxTer ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Term too complex (%d words). MaxTermSize (%l words) is too small.", n1,
AM.MaxTer/(LONG)sizeof(WORD) );
        goto PolyCall2;
    }
    m = term; t = AT.WorkPointer;
    NCOPY(m,t,n1);
    action++;
    goto retry;
done:
    AT.WorkPointer = term + *term;
```

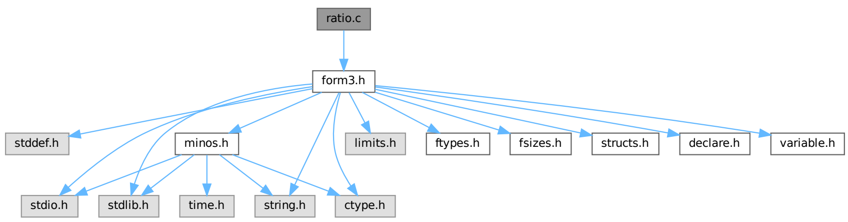
```
    if ( action && noac ) {
        if ( Normalize(BHEAD term) ) goto PolyCall;
        AT.WorkPointer = term + *term;
    }
    return(0);
PolyCall:;
    MLOCK(ErrorMessageLock);
PolyCall2:;
    AR.PolyFunPow = OldPolyFunPow;
    MesCall("PolyFunMul");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] PolyFunMul :
    #] Processor :
*/
```

4.116 ratio.c File Reference

`#include "form3.h"`

Include dependency graph for ratio.c:



Data Structures

- struct ARGBUFFER

Functions

- WORD RatioFind (PHEAD WORD ∗term, WORD ∗params)
- WORD RatioGen (PHEAD WORD ∗term, WORD ∗params, WORD num, WORD level)
- WORD BinomGen (PHEAD WORD ∗term, WORD level, WORD ∗∗tstops, WORD x1, WORD x2, WORD pow1, WORD pow2, WORD sign, UWORD ∗coef, WORD ncoef)
- WORD DoSumF1 (PHEAD WORD ∗term, WORD ∗params, WORD replac, WORD level)
- WORD Glue (PHEAD WORD ∗term1, WORD ∗term2, WORD ∗sub, WORD insert)
- WORD DoSumF2 (PHEAD WORD ∗term, WORD ∗params, WORD replac, WORD level)
- int GCDfunction (PHEAD WORD ∗term, WORD level)
- WORD ∗ GCDfunction3 (PHEAD WORD ∗in1, WORD ∗in2)
- WORD ∗ PutExtraSymbols (PHEAD WORD ∗in, WORD startebuf, int ∗actionflag)
- WORD ∗ TakeExtraSymbols (PHEAD WORD ∗in, WORD startebuf)
- WORD ∗ MultiplyWithTerm (PHEAD WORD ∗in, WORD ∗term, WORD par)
- WORD ∗ TakeContent (PHEAD WORD ∗in, WORD ∗term)
- int MergeSymbolLists (PHEAD WORD ∗old, WORD ∗extra, int par)
- int MergeDotproductLists (PHEAD WORD ∗old, WORD ∗extra, int par)
- WORD ∗ CreateExpression (PHEAD WORD nexp)

- int GCDterms (PHEAD WORD ∗term1, WORD ∗term2, WORD ∗termout)
- int ReadPolyRatFun (PHEAD WORD ∗term)
- int FromPolyRatFun (PHEAD WORD ∗fun, WORD ∗∗numout, WORD ∗∗denout)
- WORD ∗ TakeSymbolContent (PHEAD WORD ∗in, WORD ∗term)
- void GCDclean (PHEAD WORD ∗num, WORD ∗den)
- WORD ∗ PolyDiv (PHEAD WORD ∗a, WORD ∗b, char ∗text)
- int DIVfunction (PHEAD WORD ∗term, WORD level, int par)
- WORD ∗ MULfunc (PHEAD WORD ∗p1, WORD ∗p2)
- WORD ∗ ConvertArgument (PHEAD WORD ∗arg, int ∗type)
- int ExpandRat (PHEAD WORD ∗fun)
- int InvPoly (PHEAD WORD ∗inpoly, WORD maxpow, WORD sym)

Variables

- WORD divrem [4] = { DIVFUNCTION, REMFUNCTION, INVERSEFUNCTION, MULFUNCTION }
- char ∗ TheErrorMessage [ ]

4.116.1 Detailed Description

A variety of routines: The ratio command for partial fractioning (rather old. Schoonschip inheritance) The sum routines.

Definition in file ratio.c.

4.116.2 Function Documentation

4.116.2.1 RatioFind()

```
WORD RatioFind (
            PHEAD WORD * term,
            WORD * params )
```

Definition at line 62 of file ratio.c.

4.116.2.2 RatioGen()

```
WORD RatioGen (
            PHEAD WORD * term,
            WORD * params,
            WORD num,
            WORD level )
```

Definition at line 170 of file ratio.c.

4.116.2.3 BinomGen()

```
WORD BinomGen (
            PHEAD WORD * term,
            WORD level,
            WORD ** tstops,
            WORD x1,
            WORD x2,
            WORD pow1,
            WORD pow2,
            WORD sign,
            UWORD * coef,
            WORD ncoef )
```

Definition at line 320 of file ratio.c.

4.116.2.4 DoSumF1()

```
WORD DoSumF1 (
            PHEAD WORD * term,
            WORD * params,
            WORD replac,
            WORD level )
```

Definition at line 395 of file ratio.c.

4.116.2.5 Glue()

```
WORD Glue (
            PHEAD WORD * term1,
            WORD * term2,
            WORD * sub,
            WORD insert )
```

Definition at line 461 of file ratio.c.

4.116.2.6 DoSumF2()

```
WORD DoSumF2 (
            PHEAD WORD * term,
            WORD * params,
            WORD replac,
            WORD level )
```

Definition at line 520 of file ratio.c.

4.116.2.7 GCDfunction()

```
int GCDfunction (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 609 of file ratio.c.

4.116.2.8 GCDfunction3()

```
WORD * GCDfunction3 (
            PHEAD WORD * in1,
            WORD * in2 )
```

Definition at line 1141 of file ratio.c.

4.116.2.9 PutExtraSymbols()

```
WORD * PutExtraSymbols (
            PHEAD WORD * in,
            WORD startebuf,
            int * actionflag )
```

Definition at line 1244 of file ratio.c.

4.116.2.10 TakeExtraSymbols()

```
WORD * TakeExtraSymbols (
            PHEAD WORD * in,
            WORD startebuf )
```

Definition at line 1273 of file ratio.c.

4.116.2.11 MultiplyWithTerm()

```
WORD * MultiplyWithTerm (
            PHEAD WORD * in,
            WORD * term,
            WORD par )
```

Definition at line 1312 of file ratio.c.

4.116.2.12 TakeContent()

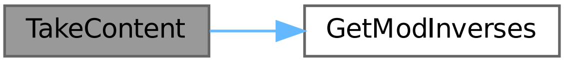
```
WORD * TakeContent (
            PHEAD WORD * in,
            WORD * term )
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. Here the input is a sequence of terms in 'in' and the answer is a content-free sequence of terms. This sequence has been allocated by the Malloc1 routine in a call to EndSort, unless the expression was already content-free. In that case the input pointer is returned. The content is returned in term. This is supposed to be a separate allocation, made by TermMalloc in the calling routine.

Definition at line 1376 of file ratio.c.

References GetModInverses().

Here is the call graph for this function:



4.116.2.13 MergeSymbolLists()

```
int MergeSymbolLists (
            PHEAD WORD * old,
            WORD * extra,
            int par )
```

Definition at line 1808 of file ratio.c.

4.116.2.14 MergeDotproductLists()

```
int MergeDotproductLists (
            PHEAD WORD * old,
            WORD * extra,
            int par )
```

Definition at line 1927 of file ratio.c.

4.116.2.15 CreateExpression()

```
WORD * CreateExpression (
            PHEAD WORD nexp )
```

Definition at line 2020 of file ratio.c.

4.116.2.16 GCDterms()

```
int GCDterms (
            PHEAD WORD * term1,
            WORD * term2,
            WORD * termout )
```

Definition at line 2076 of file ratio.c.

4.116.2.17 ReadPolyRatFun()

```
int ReadPolyRatFun (
            PHEAD WORD * term )
```

Definition at line 2264 of file ratio.c.

4.116.2.18 FromPolyRatFun()

```
int FromPolyRatFun (
            PHEAD WORD * fun,
            WORD ** numout,
            WORD ** denout )
```

Definition at line 2383 of file ratio.c.

4.116.2.19 TakeSymbolContent()

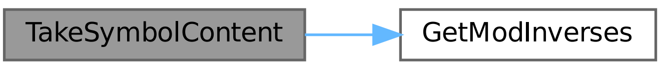
```
WORD * TakeSymbolContent (
            PHEAD WORD * in,
            WORD * term )
```

Implements part of the old ExecArg in which we take common factors from arguments with more than one term. We allow only symbols as this code is used for the polyratfun only. We have a special routine, because the generic TakeContent does too much work and speed is at a premium here. Input: in is the input expression as a sequence of terms. Output: term: the content return value: the contentfree expression. it is in new allocation, made by TermMalloc. (should be in a TermMalloc space?)

Definition at line 2434 of file ratio.c.

References GetModInverses().

Here is the call graph for this function:



4.116.2.20 GCDclean()

```
void GCDclean (
            PHEAD WORD * num,
            WORD * den )
```

Definition at line 2644 of file ratio.c.

4.116.2.21 PolyDiv()

```
WORD * PolyDiv (
            PHEAD WORD * a,
            WORD * b,
            char * text )
```

Definition at line 2741 of file ratio.c.

4.116.2.22 DIVfunction()

```
int DIVfunction (
            PHEAD WORD * term,
            WORD level,
            int par )
```

Definition at line 2788 of file ratio.c.

4.116.2.23 MULfunc()

```
WORD * MULfunc (
            PHEAD WORD * p1,
            WORD * p2 )
```

Definition at line 2957 of file ratio.c.

4.116.2.24 ConvertArgument()

```
WORD * ConvertArgument (
            PHEAD WORD * arg,
            int * type )
```

Definition at line 3018 of file ratio.c.

4.116.2.25 ExpandRat()

```
int ExpandRat (
            PHEAD WORD * fun )
```

Definition at line 3113 of file ratio.c.

4.116.2.26 InvPoly()

```
int InvPoly (
            PHEAD WORD * inpoly,
            WORD maxpow,
            WORD sym )
```

Definition at line 3592 of file ratio.c.

4.116.3 Variable Documentation

4.116.3.1 divrem

```
WORD divrem[4] = { DIVFUNCTION, REMFUNCTION, INVERSEFUNCTION, MULFUNCTION }
```

Definition at line 2786 of file ratio.c.

4.116.3.2 TheErrorMessage

```
char* TheErrorMessage[]
```

Initial value:

```
= {
     "PolyRatFun not of a type that FORM will expand: incorrect variable inside."
    ,"Division by zero in PolyRatFun encountered in ExpandRat."
    ,"Irregular code in PolyRatFun encountered in ExpandRat."
    ,"Called from ExpandRat."
    ,"WorkSpace overflow. Change parameter WorkSpace in setup file?"
    }
```

Definition at line 3105 of file ratio.c.

4.117 ratio.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : ratio.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ Ratio :

    These are the special operations regarding simple polynomials.
    The first and most needed is the partial fractioning expansion.
    Ratio,x1,x2,x3

    The files belonging to the ratio command serve also as a good example
    of how to implement a new operation.

```

```
        #[ RatioFind :

        The routine that should locate the need for a ratio command.
        If located the corresponding symbols are removed and the
        operational parameters are loaded. A subexpression pointer
        is inserted and the code for success is returned.

        params points at the compiler output block defined in RatioComp.

*/

WORD RatioFind(PHEAD WORD *term, WORD *params)
{
    GETBIDENTITY
    WORD *t, *m, *r;
    WORD x1, x2, i;
    WORD *y1, *y2, n1 = 0, n2 = 0;
    x1 = params[3];
    x2 = params[4];
    m = t = term;
    m += *m;
    m -= ABS(m[-1]);
    t++;
    if ( t < m ) do {
        if ( *t == SYMBOL ) {
            y1 = 0;
            y2 = 0;
            r = t + t[1];
            m = t + 2;
            do {
                if ( *m == x1 ) { y1 = m; n1 = m[1]; }
                else if ( *m == x2 ) { y2 = m; n2 = m[1]; }
                m += 2;
            } while ( m < r );
            if ( !y1 || !y2 || ( n1 > 0 && n2 > 0 ) ) return(0);
            m -= 2;
            if ( y1 > y2 ) { r = y1; y1 = y2; y2 = r; }
            *y2 = *m; y2[1] = m[1];
            m -= 2;
            *y1 = *m; y1[1] = m[1];
            i = WORDDIF(m,t);
#if SUBEXPSIZE > 6
We have to revise the code for the second case.
#endif
            if ( i > 2 ) {      /* Subexpression fits exactly */
                t[1] = i;
                y1 = term+*term;
                y2 = y1+SUBEXPSIZE-4;
                r = m+4;
                while ( y1 > r ) *--y2 = *--y1;
                *m++ = SUBEXPRESSION;
                *m++ = SUBEXPSIZE;
                *m++ = -1;
                *m++ = 1;
                *m++ = DUMMYBUFFER;
                FILLSUB(m)
                *term += SUBEXPSIZE-4;
            }
            else {              /* All symbols are gone. Rest has to be moved */
                m -= 2;
                *m++ = SUBEXPRESSION;
                *m++ = SUBEXPSIZE;
                *m++ = -1;
                *m++ = 1;
                *m++ = DUMMYBUFFER;
                FILLSUB(m)
                t = term;
                t += *t;
                *term += SUBEXPSIZE-6;
                r = m + 6-SUBEXPSIZE;
                do { *m++ = *r++; } while ( r < t );
            }
            t = AT.TMout;           /* Load up the TM out array for the generator */
            *t++ = 7;
            *t++ = RATIO;
            *t++ = x1;
            *t++ = x2;
            *t++ = params[5];
            *t++ = n1;
            *t++ = n2;
            return(1);
        }
        t += t[1];
    } while ( t < m );
    return(0);
}
```

```
/*
        #] RatioFind :
        #[ RatioGen :

        The algorithm:
        x1^-n1*x2^n2    ==>  x2 --> x1 + x3
        x1^n1*x2^-n2    ==>  x1 --> x2 - x3
        x1^-n1*x2^-n2   ==>

           +sum(i=0,n1-1){(-1)^i*binom(n2-1+i,n2-1)
                    *x3^-(n2+i)*x1^-(n1-i)}
           +sum(i=0,n2-1){(-1)^(n1)*binom(n1-1+i,n1-1)
                    *x3^-(n1+i)*x2^-(n2-i)}

        Actually there is an amount of arbitrariness in the first two
        formulae and the replacement x2 -> x1 + x3 could be made 'by hand'.
        It is better to use the nontrivial 'minimal change' formula:

        x1^-n1*x2^n2:   if ( n1 >= n2 ) {
                           +sum(i=0,n2){x3^i*x1^-(n1-n2+i)*binom(n2,i)}
                        }
                        else {
                            sum(i=0,n2-n1){x2^(n2-n1-i)*x3^i*binom(n1-1+i,n1-1)}
                           +sum(i=0,n1-1){x3^(n2-i)*x1^-(n1-i)*binom(n2,i)}
                        }
        x1^n1*x2^-n2:   Same but x3 -> -x3.

        The contents of the AT.TMout/params array are:
        length,type,x1,x2,x3,n1,n2

*/

WORD RatioGen(PHEAD WORD *term, WORD *params, WORD num, WORD level)
{
    GETBIDENTITY
    WORD *t, *m;
    WORD *tstops[3];
    WORD n1, n2, i, j;
    WORD x1,x2,x3;
    UWORD *coef;
    WORD ncoef, sign = 0;
    coef = (UWORD *)AT.WorkPointer;
    t = term;
    tstops[2] = m = t + *t;
    m -= ABS(m[-1]);
    t++;
    do {
        if ( *t == SUBEXPRESSION && t[2] == num ) break;
        t += t[1];
    } while ( t < m );
    tstops[0] = t;
    tstops[1] = t + t[1];
/*
    Copying to termout will be from term to tstop1, then the induced part
    and finally from tstop2 to tstop3

    Now separate the various cases:

*/
    t = params + 2;
    x1 = *t++;
    x2 = *t++;
    x3 = *t++;
    n1 = *t++;
    n2 = *t++;
    if ( n1 > 0 ) {     /* Flip the variables and indicate -x3 */
        n2 = -n2;
        sign = 1;
        i = n1; n1 = n2; n2 = i;
        i = x1; x1 = x2; x2 = i;
        goto PosNeg;
    }
    else if ( n2 > 0 ) {
        n1 = -n1;
PosNeg:
        if ( n2 <= n1 ) {   /* x1 -> x2 + x3 */
            *coef = 1;
            ncoef = 1;
            AT.WorkPointer = (WORD *)(coef + 1);
            j = n2;
            for ( i = 0; i <= n2; i++ ) {
                if ( BinomGen(BHEAD term,level,tstops,x1,x3,n2-n1-i,i,sign&i
                ,coef,ncoef) ) goto RatioCall;
                if ( i < n2 ) {
                    if ( Product(coef,&ncoef,j) ) goto RatioCall;
                    if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
                    j--;
```

```
                    AT.WorkPointer = (WORD *)(coef + ABS(ncoef));
                }
            }
            AT.WorkPointer = (WORD *)(coef);
            return(0);
        }
        else {
/*
            sum(i=0,n2-n1){x2^(n2-n1-i)*x3^i*binom(n1-1+i,n1-1)}
           +sum(i=0,n1-1){x3^(n2-i)*x1^-(n1-i)*binom(n2,i)}
*/
            *coef = 1;
            ncoef = 1;
            AT.WorkPointer = (WORD *)(coef + 1);
            j = n2 - n1;
            for ( i = 0; i <= j; i++ ) {
                if ( BinomGen(BHEAD term,level,tstops,x2,x3,n2-n1-i,i,sign&i
                ,coef,ncoef) ) goto RatioCall;
                if ( i < j ) {
                    if ( Product(coef,&ncoef,n1+i) ) goto RatioCall;
                    if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
                    AT.WorkPointer = (WORD *)(coef + ABS(ncoef));
                }
            }
            *coef = 1;
            ncoef = 1;
            AT.WorkPointer = (WORD *)(coef + 1);
            j = n1-1;
            for ( i = 0; i <= j; i++ ) {
                if ( BinomGen(BHEAD term,level,tstops,x1,x3,i-n1,n2-i,sign&(n2-i)
                ,coef,ncoef) ) goto RatioCall;
                if ( i < j ) {
                    if ( Product(coef,&ncoef,n2-i) ) goto RatioCall;
                    if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
                    AT.WorkPointer = (WORD *)(coef + ABS(ncoef));
                }
            }
            AT.WorkPointer = (WORD *)(coef);
            return(0);
        }
    }
    else {
        n2 = -n2;
        n1 = -n1;
/*
           +sum(i=0,n1-1){(-1)^i*binom(n2-1+i,n2-1)
                    *x3^-(n2+i)*x1^-(n1-i)}
           +sum(i=0,n2-1){(-1)^(n1)*binom(n1-1+i,n1-1)
                    *x3^-(n1+i)*x2^-(n2-i)}
*/
        *coef = 1;
        ncoef = 1;
        AT.WorkPointer = (WORD *)(coef + 1);
        j = n1-1;
        for ( i = 0; i <= j; i++ ) {
            if ( BinomGen(BHEAD term,level,tstops,x1,x3,i-n1,-n2-i,i&1
            ,coef,ncoef) ) goto RatioCall;
            if ( i < j ) {
                if ( Product(coef,&ncoef,n2+i) ) goto RatioCall;
                if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
                AT.WorkPointer = (WORD *)(coef + ABS(ncoef));
            }
        }
        *coef = 1;
        ncoef = 1;
        AT.WorkPointer = (WORD *)(coef + 1);
        j = n2-1;
        for ( i = 0; i <= j; i++ ) {
            if ( BinomGen(BHEAD term,level,tstops,x2,x3,i-n2,-n1-i,n1&1
            ,coef,ncoef) ) goto RatioCall;
            if ( i < j ) {
                if ( Product(coef,&ncoef,n1+i) ) goto RatioCall;
                if ( Quotient(coef,&ncoef,i+1) ) goto RatioCall;
                AT.WorkPointer = (WORD *)(coef + ABS(ncoef));
            }
        }
        AT.WorkPointer = (WORD *)(coef);
        return(0);
    }

RatioCall:
    MLOCK(ErrorMessageLock);
    MesCall("RatioGen");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}
```

```
/*
        #] RatioGen :
        #[ BinomGen :

        Routine for the generation of terms in a binomialtype expansion.

*/

WORD BinomGen(PHEAD WORD *term, WORD level, WORD **tstops, WORD x1, WORD x2,
              WORD pow1, WORD pow2, WORD sign, UWORD *coef, WORD ncoef)
{
    GETBIDENTITY
    WORD *t, *r;
    WORD *termout;
    WORD k;
    termout = AT.WorkPointer;
    t = termout;
    r = term;
    do { *t++ = *r++; } while ( r < tstops[0] );
    *t++ = SYMBOL;
    if ( pow2 == 0 ) {
        if ( pow1 == 0 ) t--;
        else { *t++ = 4; *t++ = x1; *t++ = pow1; }
    }
    else if ( pow1 == 0 ) {
        *t++ = 4; *t++ = x2; *t++ = pow2;
    }
    else {
        *t++ = 6; *t++ = x1; *t++ = pow1; *t++ = x2; *t++ = pow2;
    }
    *t++ = LNUMBER;
    *t++ = ABS(ncoef) + 3;
    *t = ncoef;
    if ( sign ) *t = -*t;
    t++;
    ncoef = ABS(ncoef);
    for ( k = 0; k < ncoef; k++ ) *t++ = coef[k];
    r = tstops[1];
    do { *t++ = *r++; } while ( r < tstops[2] );
    *termout = WORDDIF(t,termout);
    AT.WorkPointer = t;
    if ( AT.WorkPointer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    *AN.RepPoint = 1;
    AR.expchanged = 1;
    if ( Generator(BHEAD termout,level) ) {
        MLOCK(ErrorMessageLock);
        MesCall("BinomGen");
        MUNLOCK(ErrorMessageLock);
        SETERROR(-1)
    }
    AT.WorkPointer = termout;
    return(0);
}

/*
        #] BinomGen :
    #] Ratio :
    #[ Sum :
        #[ DoSumF1 :

        Routine expands a sum_ function.
        Its arguments are:
        The term in which the function occurs.
        The parameter list:
            length of parameter field
            function number (SUMNUM1)
            number of the symbol that is loop parameter
            min value
            max value
            increment
        the number of the subexpression to be removed
        the level in the generation tree.

        Note that the insertion of the loop parameter in the argument
        is done via the regular wildcard substitution mechanism.

*/

WORD DoSumF1(PHEAD WORD *term, WORD *params, WORD replac, WORD level)
{
    GETBIDENTITY
    WORD *termout, *t, extractbuff = AT.TMbuff;
```

```
    WORD isum, ival, iinc;
    LONG from;
    CBUF *C;
    ival = params[3];
    iinc = params[5];
    if ( ( iinc > 0 && params[4] >= ival )
      || ( iinc < 0 && params[4] <= ival ) ) {
        isum = (params[4] - ival)/iinc + 1;
    }
    else return(0);
    termout = AT.WorkPointer;
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    if ( AT.WorkPointer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    t = term + 1;
    while ( *t != SUBEXPRESSION || t[2] != replac || t[4] != extractbuff )
                    t += t[1];
    C = cbuf+t[4];
    t += SUBEXPSIZE;
    if ( params[2] < 0 ) {
        while ( *t != INDTOIND || t[2] != -params[2] ) t += t[1];
        *t = INDTOIND;
    }
    else {
        while ( *t > SYMTOSUB || t[2] != params[2] ) t += t[1];
        *t = SYMTONUM;
    }
    do {
        t[3] = ival;
        from = C->rhs[replac] - C->Buffer;
        while ( C->Buffer[from] ) {
            if ( InsertTerm(BHEAD term,replac,extractbuff,C->Buffer+from,termout,0) < 0 ) goto SumF1Call;
            AT.WorkPointer = termout + *termout;
            if ( Generator(BHEAD termout,level) < 0 ) goto SumF1Call;
            from += C->Buffer[from];
        }
        ival += iinc;
    } while ( --isum > 0 );
    AT.WorkPointer = termout;
    return(0);
SumF1Call:
    MLOCK(ErrorMessageLock);
    MesCall("DoSumF1");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] DoSumF1 :
        #[ Glue :

        Routine multiplies two terms. The second term is subject
        to the wildcard substitutions in sub.
        Output in the first term. This routine is a variation on
        the routine InsertTerm.

*/

WORD Glue(PHEAD WORD *term1, WORD *term2, WORD *sub, WORD insert)
{
    GETBIDENTITY
    UWORD *coef;
    WORD ncoef, *t, *t1, *t2, i, nc2, nc3, old, newer;
    coef = (UWORD *)(TermMalloc("Glue"));
    t = term1;
    t += *t;
    i = t[-1];
    t -= ABS(i);
    old = WORDDIF(t,term1);
    ncoef = REDLENG(i);
    if ( i < 0 ) i = -i;
    i--;
    t1 = t;
    t2 = (WORD *)coef;
    while ( --i >= 0 ) *t2++ = *t1++;
    i = *--t;
    nc2 = WildFill(BHEAD t,term2,sub);
    *t = i;
    t += nc2;
    nc2 = t[-1];
    t -= ABS(nc2);
    newer = WORDDIF(t,term1);
```

```
    if ( MulRat(BHEAD (UWORD *)t,REDLENG(nc2),coef,ncoef,(UWORD *)t,&nc3) ) {
        MLOCK(ErrorMessageLock);
        MesCall("Glue");
        MUNLOCK(ErrorMessageLock);
        TermFree(coef,"Glue");
        SETERROR(-1)
    }
    i = (ABS(nc3))*2;
    t += i++;
    *t++ = (nc3 >= 0)?i:-i;
    *term1 = WORDDIF(t,term1);
/*
    Switch the new piece with the old tail, so that noncommuting
    variables get into their proper spot.
*/
    i = old - insert;
    t1 = t;
    t2 = term1+insert;
    NCOPY(t1,t2,i);
    i = newer - old;
    t1 = term1+insert;
    t2 = term1+old;
    NCOPY(t1,t2,i);
    t2 = t;
    i = old - insert;
    NCOPY(t1,t2,i);
    TermFree(coef,"Glue");
    return(0);
}

/*
        #] Glue :
        #[ DoSumF2 :
*/

WORD DoSumF2(PHEAD WORD *term, WORD *params, WORD replac, WORD level)
{
    GETBIDENTITY
    WORD *termout, *t, *from, *sub, *to, extractbuff = AT.TMbuff;
    WORD isum, ival, iinc, insert, i;
    CBUF *C;
    ival = params[3];
    iinc = params[5];
    if ( ( iinc > 0 && params[4] >= ival )
      || ( iinc < 0 && params[4] <= ival ) ) {
        isum = (params[4] - ival)/iinc + 1;
    }
    else return(0);
    termout = AT.WorkPointer;
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    if ( AT.WorkPointer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    t = term + 1;
    while ( *t != SUBEXPRESSION || t[2] != replac || t[4] != extractbuff ) t += t[1];
    insert = WORDDIF(t,term);

    from = term;
    to = termout;
    while ( from < t ) *to++ = *from++;
    from += t[1];
    sub = term + *term;
    while ( from < sub ) *to++ = *from++;
    *termout -= t[1];

    sub = t;
    C = cbuf+t[4];
    t += SUBEXPSIZE;
    if ( params[2] < 0 ) {
        while ( *t != INDTOIND || t[2] != -params[2] ) t += t[1];
        *t = INDTOIND;
    }
    else {
        while ( *t > SYMTOSUB || t[2] != params[2] ) t += t[1];
        *t = SYMTONUM;
    }
    t[3] = ival;
    for(;;) {
        AT.WorkPointer = termout + *termout;
        to = AT.WorkPointer;
        if ( ( to + *termout ) > AT.WorkTop ) {
            MLOCK(ErrorMessageLock);
            MesWork();
            MUNLOCK(ErrorMessageLock);
```

```
            return(-1);
        }
        from = termout;
        i = *termout;
        NCOPY(to,from,i);
        from = AT.WorkPointer;
        AT.WorkPointer = to;
        if ( Generator(BHEAD from,level) < 0 ) goto SumF2Call;
        if ( --isum <= 0 ) break;
        ival += iinc;
        t[3] = ival;
        if ( Glue(BHEAD termout,C->rhs[replac],sub,insert) < 0 ) goto SumF2Call;
    }
    AT.WorkPointer = termout;
    return(0);
SumF2Call:
    MLOCK(ErrorMessageLock);
    MesCall("DoSumF2");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] DoSumF2 :
    #] Sum :
    #[ GCDfunction :
        #[ GCDfunction :
*/

typedef struct {
    WORD *buffer;
    DOLLARS dollar;
    LONG size;
    int type;
    int dummy;
} ARGBUFFER;

int GCDfunction(PHEAD WORD *term,WORD level)
{
    GETBIDENTITY
    WORD *t, *tstop, *tf, *termout, *tin, *tout, *m, *mnext, *mstop, *mm;
    int todo, i, ii, j, istart, sign = 1, action = 0;
    WORD firstshort = 0, firstvalue = 0, gcdisone = 0, mlength, tlength, newlength;
    WORD totargs = 0, numargs, argsdone = 0, *mh, oldval1, *g, *gcdout = 0;
    WORD *arg1, *arg2;
    UWORD x1,x2,x3;
    LONG args;
#if ( FUNHEAD > 4 )
    WORD sh[FUNHEAD+5];
#else
    WORD sh[9];
#endif
    DOLLARS d;
    ARGBUFFER *abuf = 0, ab;
/*
    #[ Find Function. Count arguments :

    First find the proper function
*/
    t = term + *term; tlength = t[-1];
    tstop = t - ABS(tlength);
    t = term + 1;
    while ( t < tstop ) {
        if ( *t != GCDFUNCTION ) { t += t[1]; continue; }
        todo = 1; totargs = 0;
        tf = t + FUNHEAD;
        while ( tf < t + t[1] ) {
            totargs++;
            if ( *tf > 0 && tf[1] != 0 ) todo = 0;
            NEXTARG(tf);
        }
        if ( todo ) break;
        t += t[1];
    }
    if ( t >= tstop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Internal error. Indicated gcd_ function not encountered.");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    WantAddPointers(totargs);
    args = AT.pWorkPointer; AT.pWorkPointer += totargs;
/*
    #] Find Function. Count arguments :
    #[ Do short arguments :

    The function we need, in agreement with TestSub, is now in t
```

```
    Make first a compilation of the short arguments (except $-s and expressions)
    to see whether we need to do much work.
    This means that after this scan we can ignore all short arguments with
    the exception of unevaluated $-s and expressions.
*/
    numargs = 0;
    firstshort = 0;
    tf = t + FUNHEAD;
    while ( tf < t + t[1] ) {
        if ( *tf == -SNUMBER && tf[1] == 0 ) { NEXTARG(tf); continue; }
        if ( *tf > 0 || *tf == -DOLLAREXPRESSION || *tf == -EXPRESSION ) {
            AT.pWorkSpace[args+numargs++] = tf;
            NEXTARG(tf); continue;
        }
        if ( firstshort == 0 ) {
            firstshort = *tf;
            if ( *tf <= -FUNCTION ) { firstvalue = -(*tf); }
            else                    { firstvalue = tf[1]; }
            NEXTARG(tf);
            argsdone++;
            continue;
        }
        else if ( *tf != firstshort ) {
            if ( *tf != -INDEX && *tf != -VECTOR && *tf != -MINVECTOR ) {
                argsdone++; gcdisone = 1; break;
            }
            if ( firstshort != -INDEX && firstshort != -VECTOR && firstshort != -MINVECTOR ) {
                argsdone++; gcdisone = 1; break;
            }
            if ( tf[1] != firstvalue ) {
                argsdone++; gcdisone = 1; break;
            }
            if ( *t == -MINVECTOR ) { firstshort = -VECTOR; }
            if ( firstshort == -MINVECTOR ) { firstshort = -VECTOR; }
        }
        else if ( *tf > -FUNCTION && *tf != -SNUMBER && tf[1] != firstvalue ) {
            argsdone++; gcdisone = 1; break;
        }
        if ( *tf == -SNUMBER && firstvalue != tf[1] ) {
/*
            make a new firstvalue which is gcd_(firstvalue,tf[1])
*/
            if ( firstvalue == 1 || tf[1] == 1 ) { gcdisone = 1; break; }
            if ( firstvalue < 0 && tf[1] < 0 ) {
                x1 = -firstvalue; x2 = -tf[1]; sign = -1;
            }
            else {
                x1 = ABS(firstvalue); x2 = ABS(tf[1]); sign = 1;
            }
            while ( ( x3 = x1%x2 ) != 0 ) { x1 = x2; x2 = x3; }
            firstvalue = ((WORD)x2)*sign;
            argsdone++;
            if ( firstvalue == 1 ) { gcdisone = 1; break; }
        }
        NEXTARG(tf);
    }
    termout = AT.WorkPointer;
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    if ( AT.WorkPointer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
/*
    #] Do short arguments :
    #[ Do trivial GCD :

    Copy head
*/
    i = t - term; tin = term; tout = termout;
    NCOPY(tout,tin,i);
    if ( gcdisone || ( firstshort == -SNUMBER && firstvalue == 1 ) ) {
        sign = 1;
gcdone:
        tin += t[1]; tstop = term + *term;
        while ( tin < tstop ) *tout++ = *tin++;
        *termout = tout - termout;
        if ( sign < 0 ) tout[-1] = -tout[-1];
        AT.WorkPointer = tout;
        if ( argsdone && Generator(BHEAD termout,level) < 0 ) goto CalledFrom;
        AT.WorkPointer = termout;
        AT.pWorkPointer = args;
        return(0);
    }
/*
    #] Do trivial GCD :
```

```
    #[ Do short argument GCD :
*/
    if ( numargs == 0 ) {   /* basically we are done */
doshort:
        sign = 1;
        if ( firstshort == 0 ) goto gcdone;
        if ( firstshort == -SNUMBER ) {
            *tout++ = SNUMBER; *tout++ = 4; *tout++ = firstvalue; *tout++ = 1;
            goto gcdone;
        }
        else if ( firstshort == -SYMBOL ) {
            *tout++ = SYMBOL; *tout++ = 4; *tout++ = firstvalue; *tout++ = 1;
            goto gcdone;
        }
        else if ( firstshort == -VECTOR || firstshort == -INDEX ) {
            *tout++ = INDEX; *tout++ = 3; *tout++ = firstvalue; goto gcdone;
        }
        else if ( firstshort == -MINVECTOR ) {
            sign = -1;
            *tout++ = INDEX; *tout++ = 3; *tout++ = firstvalue; goto gcdone;
        }
        else if ( firstshort <= -FUNCTION ) {
            *tout++ = firstvalue; *tout++ = FUNHEAD; FILLFUN(tout);
            goto gcdone;
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Internal error. Illegal short argument in GCDfunction.");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    }
/*
    #] Do short argument GCD :
    #[ Convert short argument :

    Now we allocate space for the arguments in general notation.
    First the special one if there were short arguments
*/
    if ( firstshort ) {
      switch ( firstshort ) {
        case -SNUMBER:
            sh[0] = 4; sh[1] = ABS(firstvalue); sh[2] = 1;
            if ( firstvalue < 0 ) sh[3] = -3;
            else sh[3] = 3;
            sh[4] = 0;
            break;
        case -MINVECTOR:
        case -VECTOR:
        case -INDEX:
            sh[0] = 8; sh[1] = INDEX; sh[2] = 3; sh[3] = firstvalue;
            sh[4] = 1; sh[5] = 1;
            if ( firstshort == -MINVECTOR ) sh[6] = -3;
            else sh[6] = 3;
            sh[7] = 0;
            break;
        case -SYMBOL:
            sh[0] = 8; sh[1] = SYMBOL; sh[2] = 4; sh[3] = firstvalue; sh[4] = 1;
            sh[5] = 1; sh[6] = 1; sh[7] = 3; sh[8] = 0;
            break;
        default:
            sh[0] = FUNHEAD+4; sh[1] = firstshort; sh[2] = FUNHEAD;
            for ( i = 2; i < FUNHEAD; i++ ) sh[i+1] = 0;
            sh[FUNHEAD+1] = 1; sh[FUNHEAD+2] = 1; sh[FUNHEAD+3] = 3; sh[FUNHEAD+4] = 0;
        break;
      }
    }
/*
    #] Convert short argument :
    #[ Sort arguments :

    Now we should sort the arguments in a way that the dollars and the
    expressions come last. That way we may never need them.
*/
    for ( i = 1; i < numargs; i++ ) {
        for ( ii = i; ii > 0; ii-- ) {
            arg1 = AT.pWorkSpace[args+ii];
            arg2 = AT.pWorkSpace[args+ii-1];
            if ( *arg1 < 0 ) {
                if ( *arg2 < 0 ) {
                    if ( *arg1 == -EXPRESSION ) break;
                    if ( *arg2 == -DOLLAREXPRESSION ) break;
                    AT.pWorkSpace[args+ii] = arg2;
                    AT.pWorkSpace[args+ii-1] = arg1;
                }
                else break;
            }
```

```
            else if ( *arg2 < 0 ) {
                AT.pWorkSpace[args+ii] = arg2;
                AT.pWorkSpace[args+ii-1] = arg1;
            }
            else {
                if ( *arg1 > *arg2 ) {
                    AT.pWorkSpace[args+ii] = arg2;
                    AT.pWorkSpace[args+ii-1] = arg1;
                }
                else break;
            }
        }
    }
/*
    #] Sort arguments :
    #[ There is a single term argument :
*/
    if ( firstshort ) {
        mh = sh; istart = 0;
oneterm:;
        for ( i = istart; i < numargs; i++ ) {
            arg1 = AT.pWorkSpace[args+i];
            if ( *arg1 > 0 ) {
                oldval1 = arg1[*arg1]; arg1[*arg1] = 0;
                m = arg1+ARGHEAD;
                while ( *m ) {
                    GCDterms(BHEAD mh,m,mh); m += *m;
                    if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
                        gcdisone = 1; sign = 1; arg1[*arg1] = oldval1; goto gcdone;
                    }
                }
                arg1[*arg1] = oldval1;
            }
            else if ( *arg1 == -DOLLAREXPRESSION ) {
                if ( ( d = DolToTerms(BHEAD arg1[1]) ) != 0 ) {
                    m = d->where;
                    while ( *m ) {
                        GCDterms(BHEAD mh,m,mh); m += *m;
                        argsdone++;
                        if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
                            gcdisone = 1; sign = 1;
                            if ( d->factors ) M_free(d->factors,"Dollar factors");
                            M_free(d,"Copy of dollar variable"); goto gcdone;
                        }
                    }
                    if ( d->factors ) M_free(d->factors,"Dollar factors");
                    M_free(d,"Copy of dollar variable");
                }
            }
            else {
                mm = CreateExpression(BHEAD arg1[1]);
                m = mm;
                while ( *m ) {
                    GCDterms(BHEAD mh,m,mh); m += *m;
                    argsdone++;
                    if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
                        gcdisone = 1; sign = 1; M_free(mm,"CreateExpression"); goto gcdone;
                    }
                }
                M_free(mm,"CreateExpression");
            }
        }
        if ( firstshort ) {
            if ( mh[0] == 4 ) {
                firstshort = -SNUMBER; firstvalue = mh[1] * (mh[3]/3);
            }
            else if ( mh[1] == SYMBOL ) {
                firstshort = -SYMBOL; firstvalue = mh[3];
            }
            else if ( mh[1] == INDEX ) {
                firstshort = -INDEX; firstvalue = mh[3];
                if ( mh[6] == -3 ) firstshort = -MINVECTOR;
            }
            else if ( mh[1] >= FUNCTION ) {
                firstshort = -mh[1]; firstvalue = mh[1];
            }
            goto doshort;
        }
        else {
/*
            We have a GCD that is only a single term.
            Paste it in and combine the coefficients.
*/
            mh[mh[0]] = 0;
            mm = mh;
            ii = 0;
            goto multiterms;
```

```
        }
    }
/*
    Now we have only regular arguments.
    But some have not yet been expanded.
    Check whether there are proper long arguments and if so if there is
    one with just a single term
*/
    for ( i = 0; i < numargs; i++ ) {
        arg1 = AT.pWorkSpace[args+i];
        if ( *arg1 > 0 && arg1[ARGHEAD]+ARGHEAD == *arg1 ) {
/*
            We have an argument with a single term
*/
            if ( i != 0 ) {
                arg2 = AT.pWorkSpace[args];
                AT.pWorkSpace[args] = arg1;
                AT.pWorkSpace[args+1] = arg2;
            }
            m = mh = AT.WorkPointer;
            mm = arg1+ARGHEAD; i = *mm;
            NCOPY(m,mm,i);
            AT.WorkPointer = m;
            istart = 1;
            argsdone++;
            goto oneterm;
        }
    }
/*
    #] There is a single term argument :
    #[ Expand $ and expr :

    We have: 1: regular multiterm arguments
             2: dollars
             3: expressions.
    The sum of them is numargs. Their addresses are in args. The problem is
    that expansion will lead to allocations that we have to return and all
    these allocations are different in nature.
*/
    action = 1;
    abuf = (ARGBUFFER *)Malloc1(numargs*sizeof(ARGBUFFER),"argbuffer");
    for ( i = 0; i < numargs; i++ ) {
        arg1 = AT.pWorkSpace[args+i];
        if ( *arg1 > 0 ) {
            m = (WORD *)Malloc1(*arg1*sizeof(WORD),"argbuffer type 0");
            abuf[i].buffer = m;
            abuf[i].type = 0;
            mm = arg1+ARGHEAD;
            j = *arg1-ARGHEAD;
            abuf[i].size = j;
            if ( j ) argsdone++;
            NCOPY(m,mm,j);
            *m = 0;
        }
        else if ( *arg1 == -DOLLAREXPRESSION ) {
            d = DolToTerms(BHEAD arg1[1]);
            abuf[i].buffer = d->where;
            abuf[i].type = 1;
            abuf[i].dollar = d;
            m = abuf[i].buffer;
            if ( *m ) argsdone++;
            while ( *m ) m+= *m;
            abuf[i].size = m-abuf[i].buffer;
        }
        else if ( *arg1 == -EXPRESSION ) {
            abuf[i].buffer = CreateExpression(BHEAD arg1[1]);
            abuf[i].type = 2;
            m = abuf[i].buffer;
            if ( *m ) argsdone++;
            while ( *m ) m+= *m;
            abuf[i].size = m-abuf[i].buffer;
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("What argument is this?");
            MUNLOCK(ErrorMessageLock);
            goto CalledFrom;
        }
    }
    for ( i = 0; i < numargs; i++ ) {
        arg1 = abuf[i].buffer;
        if ( *arg1 == 0 ) {}
        else if ( arg1[*arg1] == 0 ) {
/*
            After expansion there is an argument with a single term
*/
            ab = abuf[i]; abuf[i] = abuf[0]; abuf[0] = ab;
```

```
            mh = abuf[0].buffer;
            for ( j = 1; j < numargs; j++ ) {
                m = abuf[j].buffer;
                while ( *m ) {
                    GCDterms(BHEAD mh,m,mh); m += *m;
                    argsdone++;
                    if ( mh[0] == 4 && mh[1] == 1 && mh[2] == 1 && mh[3] == 3 ) {
                        gcdisone = 1; sign = 1; break;
                    }
                }
                if ( *m ) break;
            }
            mm = mh + *mh; if ( mm[-1] < 0 ) { sign = -1; mm[-1] = -mm[-1]; }
            mstop = mm - mm[-1]; m = mh+1; mlength = mm[-1];
            while ( tin < t ) *tout++ = *tin++;
            while ( m < mstop ) *tout++ = *m++;
            tin += tin[1];
            while ( tin < tstop ) *tout++ = *tin++;
            tlength = REDLENG(tlength);
            mlength = REDLENG(mlength);
            if ( MulRat(BHEAD (UWORD *)tstop,tlength,(UWORD *)mstop,mlength,
                    (UWORD *)tout,&newlength) < 0 ) goto CalledFrom;
            mlength = INCLENG(newlength);
            tout += ABS(mlength);
            tout[-1] = mlength*sign;
            *termout = tout - termout;
            AT.WorkPointer = tout;
            if ( argsdone && Generator(BHEAD termout,level) < 0 ) goto CalledFrom;
            goto cleanup;
        }
    }
/*
    There are only arguments with more than one term.
    We order them by size to make the computations as easy as possible.
*/
    for ( i = 1; i < numargs; i++ ) {
        for ( ii = i; ii > 0; ii-- ) {
            if ( abuf[ii-1].size <= abuf[ii].size ) break;
            ab = abuf[ii-1]; abuf[ii-1] = abuf[ii]; abuf[ii] = ab;
        }
    }
/*
    #] Expand $ and expr :
    #[ Multiterm subexpressions :
*/
    ii = 0;
    gcdout = abuf[ii].buffer;
    for ( i = 0; i < numargs; i++ ) {
        if ( abuf[i].buffer[0] ) { gcdout = abuf[i].buffer; ii = i; i++; argsdone++; break; }
    }
    for ( ; i < numargs; i++ ) {
      if ( abuf[i].buffer[0] ) {
        g = GCDfunction3(BHEAD gcdout,abuf[i].buffer);
        argsdone++;
        if ( gcdout != abuf[ii].buffer ) M_free(gcdout,"gcdout");
        gcdout = g;
        if ( gcdout[*gcdout] == 0 && gcdout[0] == 4 && gcdout[1] == 1
        && gcdout[2] == 1 && gcdout[3] == 3 ) break;
      }
    }
    mm = gcdout;
multiterms:;
    tlength = REDLENG(tlength);
    while ( *mm ) {
        tin = term; tout = termout; while ( tin < t ) *tout++ = *tin++;
        tin += t[1];
        mnext = mm + *mm; mlength = mnext[-1]; mstop = mnext - ABS(mlength);
        mm++;
        while ( mm < mstop ) *tout++ = *mm++;
        while ( tin < tstop ) *tout++ = *tin++;
        mlength = REDLENG(mlength);
        if ( MulRat(BHEAD (UWORD *)tstop,tlength,(UWORD *)mm,mlength,
                (UWORD *)tout,&newlength) < 0 ) goto CalledFrom;
        mlength = INCLENG(newlength);
        tout += ABS(mlength);
        tout[-1] = mlength;
        *termout = tout - termout;
        AT.WorkPointer = tout;
        if ( argsdone && Generator(BHEAD termout,level) < 0 ) goto CalledFrom;
        mm = mnext; /* next term */
    }
    if ( action && ( gcdout != abuf[ii].buffer ) ) M_free(gcdout,"gcdout");
/*
    #] Multiterm subexpressions :
    #[ Cleanup :
*/
cleanup:;
```

```
    if ( action ) {
        for ( i = 0; i < numargs; i++ ) {
            if ( abuf[i].type == 0 ) { M_free(abuf[i].buffer,"argbuffer type 0"); }
            else if ( abuf[i].type == 1 ) {
                d = abuf[i].dollar;
                if ( d->factors ) M_free(d->factors,"Dollar factors");
                M_free(d,"Copy of dollar variable");
            }
            else if ( abuf[i].type == 2 ) { M_free(abuf[i].buffer,"CreateExpression"); }
        }
        M_free(abuf,"argbuffer");
    }
/*
    #] Cleanup :
*/
    AT.pWorkPointer = args;
    AT.WorkPointer = termout;
    return(0);

CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("GCDfunction");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
    return(-1);
}

/*
        #] GCDfunction :
        #[ GCDfunction3 :

    Finds the GCD of the two arguments which are buffers with terms.
    In principle the first buffer can have only one term.

    If both buffers have more than one term, we need to replace all
    non-symbolic objects by generated symbols and substitute that back
    afterwards. The rest we leave to the powerful routines.
    Philosophical problem: What do we do with GCD_(x/z+y,x+y*z) ?

    Method:
    If we have only negative powers of z and no positive powers we let
    the EXTRASYMBOLS do their job. When mixed, multiply the arguments with
    the negative powers with enough powers of z to eliminate the negative powers.
    The DENOMINATOR function is always eliminated with the mechanism as we
    cannot tell whether there are positive powers of its contents.
*/

WORD *GCDfunction3(PHEAD WORD *in1, WORD *in2)
{
    GETBIDENTITY
    WORD oldsorttype = AR.SortType, *ow = AT.WorkPointer;;
    WORD *t, *tt, *gcdout, *term1, *term2, *confree1, *confree2, *gcdout1, *proper1, *proper2;
    int i, actionflag1, actionflag2;
    WORD startebuf = cbuf[AT.ebufnum].numrhs;
    WORD tryterm1, tryterm2;
    if ( in2[*in2] == 0 ) { t = in1; in1 = in2; in2 = t; }
    if ( in1[*in1] == 0 ) { /* First input with only one term */
        gcdout = (WORD *)Malloc1((*in1+1)*sizeof(WORD),"gcdout");
        i = *in1; t = gcdout; tt = in1; NCOPY(t,tt,i); *t = 0;
        t = in2;
        while ( *t ) {
            GCDterms(BHEAD gcdout,t,gcdout);
            if ( gcdout[0] == 4 && gcdout[1] == 1
             && gcdout[2] == 1 && gcdout[3] == 3 ) break;
            t += *t;
        }
        AT.WorkPointer = ow;
        return(gcdout);
    }
/*
    We need to take out the content from the two expressions
    and determine their GCD. This plays with the negative powers!
*/
    AR.SortType = SORTHIGHFIRST;
    term1 = TermMalloc("GCDfunction3-a");
    term2 = TermMalloc("GCDfunction3-b");

    confree1 = TakeContent(BHEAD in1,term1);
    tryterm1 = AN.tryterm; AN.tryterm = 0;
    confree2 = TakeContent(BHEAD in2,term2);
    tryterm2 = AN.tryterm; AN.tryterm = 0;
/*
    confree1 = TakeSymbolContent(BHEAD in1,term1);
    confree2 = TakeSymbolContent(BHEAD in2,term2);
*/
    GCDterms(BHEAD term1,term2,term1);
    TermFree(term2,"GCDfunction3-b");
```

```
/*
    Now we have to replace all non-symbols and symbols to a negative power
    by extra symbols.
*/
    if ( ( proper1 = PutExtraSymbols(BHEAD confree1,startebuf,&actionflag1) ) == 0 ) goto CalledFrom;
    if ( confree1 != in1 ) {
        if ( tryterm1 ) { TermFree(confree1,"TakeContent"); }
        else { M_free(confree1,"TakeContent"); }
    }
/*
    TermFree(confree1,"TakeSymbolContent");
*/
    if ( ( proper2 = PutExtraSymbols(BHEAD confree2,startebuf,&actionflag2) ) == 0 ) goto CalledFrom;
    if ( confree2 != in2 ) {
        if ( tryterm2 ) { TermFree(confree2,"TakeContent"); }
        else { M_free(confree2,"TakeContent"); }
    }
/*
    TermFree(confree2,"TakeSymbolContent");
*/
/*
    And now the real work:
*/
    gcdout1 = poly_gcd(BHEAD proper1,proper2,0);
    M_free(proper1,"PutExtraSymbols");
    M_free(proper2,"PutExtraSymbols");

    AR.SortType = oldsorttype;
    if ( actionflag1 || actionflag2 ) {
        if ( ( gcdout = TakeExtraSymbols(BHEAD gcdout1,startebuf) ) == 0 ) goto CalledFrom;
        M_free(gcdout1,"gcdout");
    }
    else {
        gcdout = gcdout1;
    }

    cbuf[AT.ebufnum].numrhs = startebuf;
/*
    Now multiply gcdout by term1
*/
    if ( term1[0] != 4 || term1[3] != 3 || term1[1] != 1 || term1[2] != 1 ) {
        AN.tryterm = -1;
        if ( ( gcdout1 = MultiplyWithTerm(BHEAD gcdout,term1,2) ) == 0 ) goto CalledFrom;
        AN.tryterm = 0;
        M_free(gcdout,"gcdout");
        gcdout = gcdout1;
    }
    TermFree(term1,"GCDfunction3-a");
    AT.WorkPointer = ow;
    return(gcdout);
CalledFrom:
    AN.tryterm = 0;
    MLOCK(ErrorMessageLock);
    MesCall("GCDfunction3");
    MUNLOCK(ErrorMessageLock);
    return(0);
}

/*
        #] GCDfunction3 :
        #[ PutExtraSymbols :
*/

WORD *PutExtraSymbols(PHEAD WORD *in,WORD startebuf,int *actionflag)
{
    WORD *termout = AT.WorkPointer;
    int action;
    *actionflag = 0;
    NewSort(BHEAD0);
    while ( *in ) {
        if ( ( action = LocalConvertToPoly(BHEAD in,termout,startebuf,0) ) < 0 ) {
            LowerSortLevel();
            goto CalledFrom;
        }
        if ( action > 0 ) *actionflag = 1;
        StoreTerm(BHEAD termout);
        in += *in;
    }
    if ( EndSort(BHEAD (WORD *)((void *)(&termout)),2) < 0 ) goto CalledFrom;
    return(termout);
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("PutExtraSymbols");
    MUNLOCK(ErrorMessageLock);
    return(0);
}
```

```
/*
        #] PutExtraSymbols :
        #[ TakeExtraSymbols :
*/

WORD *TakeExtraSymbols(PHEAD WORD *in,WORD startebuf)
{
    CBUF *C = cbuf+AC.cbufnum;
    CBUF *CC = cbuf+AT.ebufnum;
    WORD *oldworkpointer = AT.WorkPointer, *termout;

    termout = AT.WorkPointer;
    NewSort(BHEAD0);
    while ( *in ) {
        if ( ConvertFromPoly(BHEAD in,termout,numxsymbol,CC->numrhs-startebuf+numxsymbol,startebuf-numxsymbol,1) <= 0 ) {
            LowerSortLevel();
            goto CalledFrom;
        }
        in += *in;
        AT.WorkPointer = termout + *termout;
/*
        ConvertFromPoly leaves terms with subexpressions. Hence:
*/
        if ( Generator(BHEAD termout,C->numlhs) ) {
            LowerSortLevel();
            goto CalledFrom;
        }
    }
    AT.WorkPointer = oldworkpointer;
    if ( EndSort(BHEAD (WORD *)((void *)(&termout)),2) < 0 ) goto CalledFrom;
    return(termout);

CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("TakeExtraSymbols");
    MUNLOCK(ErrorMessageLock);
    return(0);
}

/*
        #] TakeExtraSymbols :
        #[ MultiplyWithTerm :
*/

WORD *MultiplyWithTerm(PHEAD WORD *in, WORD *term, WORD par)
{
    WORD *termout, *t, *tt, *tstop, *ttstop;
    WORD length, length1, length2;
    WORD oldsorttype = AR.SortType;
    COMPARE oldcompareroutine = (COMPARE)(AR.CompareRoutine);
    AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);

    if ( par == 0 || par == 2 ) AR.SortType = SORTHIGHFIRST;
    else            AR.SortType = SORTLOWFIRST;
    termout = AT.WorkPointer;
    NewSort(BHEAD0);
    while ( *in ) {
        tt = termout + 1;
        tstop = in + *in; tstop -= ABS(tstop[-1]); t = in + 1;
        while ( t < tstop ) *tt++ = *t++;
        ttstop = term + *term; ttstop -= ABS(ttstop[-1]); t = term + 1;
        while ( t < ttstop ) *tt++ = *t++;
        length1 = REDLENG(in[*in-1]); length2 = REDLENG(term[*term-1]);
        if ( MulRat(BHEAD (UWORD *)tstop,length1,
                (UWORD *)ttstop,length2,(UWORD *)tt,&length) ) goto CalledFrom;
        length = INCLENG(length);
        tt += ABS(length); tt[-1] = length;
        *termout = tt - termout;
        SymbolNormalize(termout);
        StoreTerm(BHEAD termout);
        in += *in;
    }
    if ( par == 2 ) {
/*      if ( AN.tryterm == 0 ) AN.tryterm = 1; */
        AN.tryterm = 0; /* For now */
        if ( EndSort(BHEAD (WORD *)((void *)(&termout)),2) < 0 ) goto CalledFrom;
    }
    else {
        if ( EndSort(BHEAD termout,1) < 0 ) goto CalledFrom;
    }

    AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;

    AR.SortType = oldsorttype;
    return(termout);

```

```
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("MultiplyWithTerm");
    MUNLOCK(ErrorMessageLock);
    return(0);
}

/*
        #] MultiplyWithTerm :
        #[ TakeContent :
*/
WORD *TakeContent(PHEAD WORD *in, WORD *term)
{
    GETBIDENTITY
    WORD *t, *tstop, *tcom, *tout, *tstore, *r, *rstop, *m, *mm, *w, *ww, *wterm;
    WORD *tnext, *tt, *tterm, code[2];
    WORD *inp, a, *den;
    int i, j, k, action = 0, sign;
    UWORD *GCDbuffer, *GCDbuffer2, *LCMbuffer, *LCMbuffer2, *ap;
    WORD GCDlen, GCDlen2, LCMlen, LCMlen2, length, redlength, len1, len2;
    tout = tstore = term+1;
/*
    #[ INDEX :
*/
    t = in;
    tnext = t + *t;
    tstop = tnext-ABS(tnext[-1]);
    t++;
    while ( t < tstop ) {
        if ( *t == INDEX ) {
            i = t[1]; NCOPY(tout,t,i); break;
        }
        else t += t[1];
    }
    if ( tout > tstore ) { /* There are indices in the first term */
        t = tnext;
        while ( *t ) {
            tnext = t + *t;
            rstop = tnext - ABS(tnext[-1]);
            r = t+1;
            if ( r == rstop ) goto noindices;
            while ( r < rstop ) {
                if ( *r != INDEX ) { r += r[1]; continue; }
                m = tstore+2;
                while ( m < tout ) {
                    for ( i = 2; i < r[1]; i++ ) {
                        if ( *m == r[i] ) break;
                        if ( *m > r[i] ) continue;
                        mm = m+1;
                        while ( mm < tout ) { mm[-1] = mm[0]; mm++; }
                        tout--; tstore[1]--; m--;
                        break;
                    }
                    m++;
                }
            }
            if ( r >= rstop || tout <= tstore+2 ) {
                tout = tstore; break;
            }
        }
        if ( tout > tstore+2 ) { /* Now we have to take out what is in tstore */
            t = in; w = in;
            while ( *t ) {
                wterm = w;
                tnext = t + *t; t++; w++;
                while ( *t != INDEX ) { i = t[1]; NCOPY(w,t,i); }
                tt = t + t[1]; t += 2; r = tstore+2; ww = w; *w++ = INDEX; w++;
                while ( r < tout && t < tt ) {
                    if ( *r > *t ) { *w++ = *t++; }
                    else if ( *r == *t ) { r++; t++; }
                    else goto CalledFrom;
                }
                if ( r < tout ) goto CalledFrom;
                while (  t < tt ) *w++ = *t++;
                ww[1] = w - ww;
                if ( ww[1] == 2 ) w = ww;
                while ( t < tnext ) *w++ = *t++;
                *wterm = w - wterm;
            }
            *w = 0;
        }
noindices:
        tstore = tout;
    }
/*
    #] INDEX :
    #[ VECTOR/DELTA :
```

```
*/
    code[0] = VECTOR; code[1] = DELTA;
    for ( k = 0; k < 2; k++ ) {
      t = in;
      tnext = t + *t;
      tstop = tnext-ABS(tnext[-1]);
      t++;
      while ( t < tstop ) {
        if ( *t == code[k] ) {
            i = t[1]; NCOPY(tout,t,i); break;
        }
        else t += t[1];
      }
      if ( tout > tstore ) { /* There are vectors in the first term */
        t = tnext;
        while ( *t ) {
            tnext = t + *t;
            rstop = tnext - ABS(tnext[-1]);
            r = t+1;
            if ( r == rstop ) { tstore = tout; goto novectors; }
            while ( r < rstop ) {
                if ( *r != code[k] ) { r += r[1]; continue; }
                m = tstore+2;
                while ( m < tout ) {
                    for ( i = 2; i < r[1]; i += 2 ) {
                        if ( *m == r[i] && m[1] == r[i+1] ) break;
                        if ( *m > r[i] || ( *m == r[i] && m[1] > r[i+1] ) ) continue;
                        mm = m+2;
                        while ( mm < tout ) { mm[-2] = mm[0]; mm[-1] = mm[1]; mm += 2; }
                        tout -= 2; tstore[1] -= 2; m -= 2;
                        break;
                    }
                    m += 2;
                }
            }
            if ( r >= rstop || tout <= tstore+2 ) {
                tout = tstore; break;
            }
        }
        if ( tout > tstore+2 ) { /* Now we have to take out what is in tstore */
            t = in; w = in;
            while ( *t ) {
                wterm = w;
                tnext = t + *t; t++; w++;
                while ( *t != code[k] ) { i = t[1]; NCOPY(w,t,i); }
                tt = t + t[1]; t += 2; r = tstore+2; ww = w; *w++ = code[k]; w++;
                while ( r < tout && t < tt ) {
                    if ( ( *r > *t ) || ( *r == *t && r[1] > t[1] ) )
                        { *w++ = *t++; *w++ = *t++; }
                    else if ( *r == *t && r[1] == t[1] ) { r += 2; t += 2; }
                    else goto CalledFrom;
                }
                if ( r < tout ) goto CalledFrom;
                while (  t < tt ) *w++ = *t++;
                ww[1] = w - ww;
                if ( ww[1] == 2 ) w = ww;
                while ( t < tnext ) *w++ = *t++;
                *wterm = w - wterm;
            }
            *w = 0;
        }
        tstore = tout;
      }
    }
novectors:;
/*
    #] VECTOR/DELTA :
    #[ FUNCTIONS :
*/
    t = in;
    tnext = t + *t;
    tstop = tnext-ABS(tnext[-1]);
    t++;
    tcom = 0;
    while ( t < tstop ) {
        if ( *t >= FUNCTION ) {
            if ( functions[*t-FUNCTION].commute ) {
                if ( tcom == 0 ) { tcom = tstore; }
                else {
                    for ( i = 0; i < t[1]; i++ ) {
                        if ( t[i] != tcom[i] ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("GCD or factorization of more than one noncommuting object not
allowed");
                            MUNLOCK(ErrorMessageLock);
                            goto CalledFrom;
                        }
```

```
                        }
                    }
                }
                i = t[1]; NCOPY(tstore,t,i);
            }
            else t += t[1];
        }
        if ( tout > tstore ) {
            t = tnext;
            while ( *t ) {
                tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
                if ( t == tstop ) goto nofunctions;
                r = tstore;
                while ( r < tout ) {
                    tt = t;
                    while ( tt < tstop ) {
                        for ( i = 0; i < r[1]; i++ ) {
                            if ( r[i] != tt[i] ) break;
                        }
                        if ( i == r[1] ) { r += r[1]; goto nextr1; }
                    }
/*
                    Not encountered in this term. Scratch from list
*/
                    m = r; mm = r + r[1];
                    while ( mm < tout ) *m++ = *mm++;
                    tout = m;
nextr1:;
                }
                if ( tout <= tstore ) break;
                t += *t;
            }
        }
        if ( tout > tstore ) {
/*
            Now we have one or more functions left that are common in all terms.
            Take them out. We do this one by one.
*/
            r = tstore;
            while ( r < tout ) {
                t = in; ww = in; w = ww+1;
                while ( *t ) {
                    tnext = t + *t;
                    t++;
                    for(;;) {
                        for ( i = 0; i < r[1]; i++ ) {
                            if ( t[i] != r[i] ) {
                                j = t[1]; NCOPY(w,t,j);
                                break;
                            }
                        }
                        if ( i == r[1] ) {
                            t += t[1];
                            while ( t < tnext ) *w++ = *t++;
                            *ww = w - ww;
                            break;
                        }
                    }
                }
                r += r[1];
                *w = 0;
            }
nofunctions:
            tstore = tout;
        }
/*
        #] FUNCTIONS :
        #[ SYMBOL :

        We make a list of symbols and their minimal powers.
        This includes negative powers. In the end we have to multiply by the
        inverse of this list. That takes out all negative powers and leaves
        things ready for further processing.
*/
        tterm = AT.WorkPointer; tt = tterm+1;
        tout[0] = SYMBOL; tout[1] = 2;
        t = in;
        tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
        while ( t < tstop ) {
            if ( *t == SYMBOL ) {
                for ( i = 0; i < t[1]; i++ ) tout[i] = t[i];
                break;
            }
            t += t[1];
        }
        t = tnext;
        while ( *t ) {
```

```
        tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
        if ( t == tstop ) {
            tout[1] = 2;
            break;
        }
        else {
            while ( t < tstop ) {
                if ( *t == SYMBOL ) {
                    MergeSymbolLists(BHEAD tout,t,-1);
                    break;
                }
                t += t[1];
            }
            t = tnext;
        }
    }
    if ( tout[1] > 2 ) {
        t = tout;
        tt[0] = t[0]; tt[1] = t[1];
        for ( i = 2; i < t[1]; i += 2 ) {
            tt[i] = t[i]; tt[i+1] = -t[i+1];
        }
        tt += tt[1];
        tout += tout[1];
        action++;
    }
/*
    #] SYMBOL :
    #[ DOTPRODUCT :

    We make a list of dotproducts and their minimal powers.
    This includes negative powers. In the end we have to multiply by the
    inverse of this list. That takes out all negative powers and leaves
    things ready for further processing.
*/
    tout[0] = DOTPRODUCT; tout[1] = 2;
    t = in;
    while ( *t ) {
        tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
        if ( t == tstop ) {
            tout[1] = 2;
            break;
        }
        while ( t < tstop ) {
            if ( *t == DOTPRODUCT ) {
                MergeDotproductLists(BHEAD tout,t,-1);
                break;
            }
            t += t[1];
        }
        t = tnext;
    }
    if ( tout[1] > 2 ) {
        t = tout;
        tt[0] = t[0]; tt[1] = t[1];
        for ( i = 2; i < t[1]; i += 3 ) {
            tt[i] = t[i]; tt[i+1] = t[i+1]; tt[i+2] = -t[i+2];
        }
        tt += tt[1];
        tout += tout[1];
        action++;
    }
/*
    #] DOTPRODUCT :
    #[ Coefficient :

    Now we have to collect the GCD of the numerators
    and the LCM of the denominators.
*/
    AT.WorkPointer = tt;
    if ( AN.cmod != 0 ) {
      WORD x, ix, ip;
      t = in; tnext = t + *t; tstop = tnext - ABS(tnext[-1]);
      x = tstop[0];
      if ( tnext[-1] < 0 ) x += AC.cmod[0];
      if ( GetModInverses(x,(WORD)(AN.cmod[0]),&ix,&ip) ) goto CalledFrom;
      *tout++ = x; *tout++ = 1; *tout++ = 3;
      *tt++ = ix; *tt++ = 1; *tt++ = 3;
    }
    else {
      GCDbuffer  = NumberMalloc("MakeInteger");
      GCDbuffer2 = NumberMalloc("MakeInteger");
      LCMbuffer  = NumberMalloc("MakeInteger");
      LCMbuffer2 = NumberMalloc("MakeInteger");
      t = in;
      tnext = t + *t; length = tnext[-1];
      if ( length < 0 ) { sign = -1; length = -length; }
```

```
        else              { sign = 1; }
        tstop = tnext - length;
        redlength = (length-1)/2;
        for ( i = 0; i < redlength; i++ ) {
          GCDbuffer[i] = (UWORD)(tstop[i]);
          LCMbuffer[i] = (UWORD)(tstop[redlength+i]);
        }
        GCDlen = LCMlen = redlength;
        while ( GCDbuffer[GCDlen-1] == 0 ) GCDlen--;
        while ( LCMbuffer[LCMlen-1] == 0 ) LCMlen--;
        t = tnext;
        while ( *t ) {
          tnext = t + *t; length = ABS(tnext[-1]);
          tstop = tnext - length; redlength = (length-1)/2;
          len1 = len2 = redlength;
          den = tstop + redlength;
          while ( tstop[len1-1] == 0 ) len1--;
          while ( den[len2-1] == 0 ) len2--;
          if ( GCDlen == 1 && GCDbuffer[0] == 1 ) {}
          else {
              GcdLong(BHEAD (UWORD *)tstop,len1,GCDbuffer,GCDlen,GCDbuffer2,&GCDlen2);
              ap = GCDbuffer; GCDbuffer = GCDbuffer2; GCDbuffer2 = ap;
              a = GCDlen; GCDlen = GCDlen2; GCDlen2 = a;
          }
          if ( len2 == 1 && den[0] == 1 ) {}
          else {
              GcdLong(BHEAD LCMbuffer,LCMlen,(UWORD *)den,len2,LCMbuffer2,&LCMlen2);
              DivLong((UWORD *)den,len2,LCMbuffer2,LCMlen2,
                  GCDbuffer2,&GCDlen2,(UWORD *)AT.WorkPointer,&a);
              MulLong(LCMbuffer,LCMlen,GCDbuffer2,GCDlen2,LCMbuffer2,&LCMlen2);
              ap = LCMbuffer; LCMbuffer = LCMbuffer2; LCMbuffer2 = ap;
              a = LCMlen; LCMlen = LCMlen2; LCMlen2 = a;
          }
          t = tnext;
        }
        if ( GCDlen != 1 || GCDbuffer[0] != 1 || LCMlen != 1 || LCMbuffer[0] != 1 ) {
          redlength = GCDlen; if ( LCMlen > GCDlen ) redlength = LCMlen;
          for ( i = 0; i < GCDlen; i++ ) *tout++ = (WORD)(GCDbuffer[i]);
          for ( ; i < redlength; i++ ) *tout++ = 0;
          for ( i = 0; i < LCMlen; i++ ) *tout++ = (WORD)(LCMbuffer[i]);
          for ( ; i < redlength; i++ ) *tout++ = 0;
          *tout++ = (2*redlength+1)*sign;
          for ( i = 0; i < LCMlen; i++ ) *tt++ = (WORD)(LCMbuffer[i]);
          for ( ; i < redlength; i++ ) *tt++ = 0;
          for ( i = 0; i < GCDlen; i++ ) *tt++ = (WORD)(GCDbuffer[i]);
          for ( ; i < redlength; i++ ) *tt++ = 0;
          *tt++ = (2*redlength+1)*sign;
          action++;
        }
        else {
          *tout++ = 1; *tout++ = 1; *tout++ = 3*sign;
          *tt++ = 1; *tt++ = 1; *tt++ = 3*sign;
          if ( sign != 1 ) action++;
        }
        *tout = 0;
        NumberFree(LCMbuffer2,"MakeInteger");
        NumberFree(LCMbuffer ,"MakeInteger");
        NumberFree(GCDbuffer2,"MakeInteger");
        NumberFree(GCDbuffer ,"MakeInteger");
      }
/*
    #] Coefficient :
    #[ Multiply by the inverse content :
*/
    if ( action ) {
        *tterm = tt - tterm;
        AT.WorkPointer = tt;
        inp = MultiplyWithTerm(BHEAD in,tterm,2);
        AT.WorkPointer = tterm;
        in = inp;
    }
/*
    #] Multiply by the inverse content :
*/
    *term = tout - term;
    AT.WorkPointer = tterm;
    return(in);
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("TakeContent");
    MUNLOCK(ErrorMessageLock);
    return(0);
}

/*
        #] TakeContent :
        #[ MergeSymbolLists :
```

```

        Merges the extra list into the old.
        If par == -1 we take minimum powers
        If par ==  1 we take maximum powers
        If par ==  0 we take minimum of the absolute value of the powers
                    if one is positive and the other negative we get zero.
        We assume that the symbols are in order in both lists
*/

int MergeSymbolLists(PHEAD WORD *old, WORD *extra, int par)
{
    GETBIDENTITY
    WORD *new = TermMalloc("MergeSymbolLists");
    WORD *t1, *t2, *fill;
    int i1,i2;
    fill = new + 2; *new = SYMBOL;
    i1 = old[1] - 2; i2 = extra[1] - 2;
    t1 = old + 2; t2 = extra + 2;
    switch ( par ) {
        case -1:
            while ( i1 > 0 && i2 > 0 ) {
                if ( *t1 > *t2 ) {
                    if ( t2[1] < 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
                    else t2 += 2;
                    i2 -= 2;
                }
                else if ( *t1 < *t2 ) {
                    if ( t1[1] < 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
                    else t1 += 2;
                    i1 -= 2;
                }
                else if ( t1[1] < t2[1] ) {
                    *fill++ = *t1++; *fill++ = *t1++; t2 += 2;
                    i1 -= 2; i2 -=2;
                }
                else {
                    *fill++ = *t2++; *fill++ = *t2++; t1 += 2;
                    i1 -= 2; i2 -=2;
                }
            }
            for ( ; i1 > 0; i1 -= 2 ) {
                if ( t1[1] < 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
                else t1 += 2;
            }
            for ( ; i2 > 0; i2 -= 2 ) {
                if ( t2[1] < 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
                else t2 += 2;
            }
            break;
        case 1:
            while ( i1 > 0 && i2 > 0 ) {
                if ( *t1 > *t2 ) {
                    if ( t2[1] > 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
                    else t2 += 2;
                    i2 -=2;
                }
                else if ( *t1 < *t2 ) {
                    if ( t1[1] > 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
                    else t1 += 2;
                    i1 -= 2;
                }
                else if ( t1[1] > t2[1] ) {
                    *fill++ = *t1++; *fill++ = *t1++; t2 += 2;
                    i1 -= 2; i2 -=2;
                }
                else {
                    *fill++ = *t2++; *fill++ = *t2++; t1 += 2;
                    i1 -= 2; i2 -=2;
                }
            }
            for ( ; i1 > 0; i1 -= 2 ) {
                if ( t1[1] > 0 ) { *fill++ = *t1++; *fill++ = *t1++; }
                else t1 += 2;
            }
            for ( ; i2 > 0; i2 -= 2 ) {
                if ( t2[1] > 0 ) { *fill++ = *t2++; *fill++ = *t2++; }
                else t2 += 2;
            }
            break;
        case 0:
            while ( i1 > 0 && i2 > 0 ) {
                if ( *t1 > *t2 ) {
                    t2 += 2; i2 -= 2;
                }
                else if ( *t1 < *t2 ) {
                    t1 += 2; i1 -= 2;
                }
```

```
                else if ( ( t1[1] > 0 ) && ( t2[1] < 0 ) ) { t1 += 2; t2 += 2; i1 -= 2; i2 -= 2; }
                else if ( ( t1[1] < 0 ) && ( t2[1] > 0 ) ) { t1 += 2; t2 += 2; i1 -= 2; i2 -= 2; }
                else if ( t1[1] > 0 ) {
                    if ( t1[1] < t2[1] ) {
                        *fill++ = *t1++; *fill++ = *t1++; t2 += 2; i2 -= 2;
                    }
                    else {
                        *fill++ = *t2++; *fill++ = *t2++; t1 += 2; i1 -= 2;
                    }
                }
                else {
                    if ( t2[1] < t1[1] ) {
                        *fill++ = *t2++; *fill++ = *t2++; t1 += 2; i1 -= 2; i2 -= 2;
                    }
                    else {
                        *fill++ = *t1++; *fill++ = *t1++; t2 += 2; i1 -= 2; i2 -= 2;
                    }
                }
            }
            for ( ; i1 > 0; i1-- ) *fill++ = *t1++;
            for ( ; i2 > 0; i2-- ) *fill++ = *t2++;
            break;
    }
    i1 = new[1] = fill - new;
    t2 = new; t1 = old; NCOPY(t1,t2,i1);
    TermFree(new,"MergeSymbolLists");
    return(0);
}

/*
        #] MergeSymbolLists :
        #[ MergeDotproductLists :

        Merges the extra list into the old.
        If par == -1 we take minimum powers
        If par ==  1 we take maximum powers
        If par ==  0 we take minimum of the absolute value of the powers
                     if one is positive and the other negative we get zero.
        We assume that the dotproducts are in order in both lists
*/

int MergeDotproductLists(PHEAD WORD *old, WORD *extra, int par)
{
    GETBIDENTITY
    WORD *new = TermMalloc("MergeDotproductLists");
    WORD *t1, *t2, *fill;
    int i1,i2;
    fill = new + 2;
    i1 = old[1] - 2; i2 = extra[1] - 2;
    t1 = old + 2; t2 = extra + 2;
    switch ( par ) {
        case -1:
            while ( i1 > 0 && i2 > 0 ) {
                if ( ( *t1 > *t2 ) || ( *t1 == *t2 && t1[1] > t2[1] ) ) {
                    if ( t2[2] < 0 ) { *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; }
                    else t2 += 3;
                }
                else if ( ( *t1 < *t2 ) || ( *t1 == *t2 && t1[1] < t2[1] ) ) {
                    if ( t1[2] < 0 ) { *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; }
                    else t1 += 3;
                }
                else if ( t1[2] < t2[2] ) {
                    *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
                }
                else {
                    *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
                }
            }
            break;
        case 1:
            while ( i1 > 0 && i2 > 0 ) {
                if ( ( *t1 > *t2 ) || ( *t1 == *t2 && t1[1] > t2[1] ) ) {
                    if ( t2[2] > 0 ) { *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; }
                    else t2 += 3;
                }
                else if ( ( *t1 < *t2 ) || ( *t1 == *t2 && t1[1] < t2[1] ) ) {
                    if ( t1[2] > 0 ) { *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; }
                    else t1 += 3;
                }
                else if ( t1[2] > t2[2] ) {
                    *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
                }
                else {
                    *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
                }
            }
            break;
```

```
        case 0:
            while ( i1 > 0 && i2 > 0 ) {
                if ( ( *t1 > *t2 ) || ( *t1 == *t2 && t1[1] > t2[1] ) ) {
                    t2 += 3;
                }
                else if ( ( *t1 < *t2 ) || ( *t1 == *t2 && t1[1] < t2[1] ) ) {
                    t1 += 3;
                }
                else if ( ( t1[2] > 0 ) && ( t2[2] < 0 ) ) { t1 += 3; t2 += 3; }
                else if ( ( t1[2] < 0 ) && ( t2[2] > 0 ) ) { t1 += 3; t2 += 3; }
                else if ( t1[2] > 0 ) {
                    if ( t1[2] < t2[2] ) {
                        *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
                    }
                    else {
                        *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
                    }
                }
                else {
                    if ( t2[2] < t1[2] ) {
                        *fill++ = *t2++; *fill++ = *t2++; *fill++ = *t2++; t1 += 3;
                    }
                    else {
                        *fill++ = *t1++; *fill++ = *t1++; *fill++ = *t1++; t2 += 3;
                    }
                }
            }
            break;
    }
    i1 = new[1] = fill - new;
    t2 = new; t1 = old; NCOPY(t1,t2,i1);
    TermFree(new,"MergeDotproductLists");
    return(0);
}

/*
        #] MergeDotproductLists :
        #[ CreateExpression :

        Looks for the expression in the argument, reads it and puts it
        in a buffer. Returns the address of the buffer.
        We send the expression through the Generator system, because there
        may be unsubstituted (sub)expressions as in
        Local F = (a+b);
        Local G = gcd_(F,...);
*/

WORD *CreateExpression(PHEAD WORD nexp)
{
    GETBIDENTITY
    CBUF *C = cbuf+AC.cbufnum;
    POSITION startposition, oldposition;
    FILEHANDLE *fi;
    WORD *term, *oldipointer = AR.CompressPointer;
;
    switch ( Expressions[nexp].status ) {
        case HIDDENLEXPRESSION:
        case HIDDENGEXPRESSION:
        case DROPHLEXPRESSION:
        case DROPHGEXPRESSION:
        case UNHIDELEXPRESSION:
        case UNHIDEGEXPRESSION:
            AR.GetOneFile = 2; fi = AR.hidefile;
            break;
        default:
            AR.GetOneFile = 0; fi = AR.infile;
            break;
    }
    SeekScratch(fi,&oldposition);
    startposition = AS.OldOnFile[nexp];
    term = AT.WorkPointer;
    if ( GetOneTerm(BHEAD term,fi,&startposition,0) <= 0 ) goto CalledFrom;
    NewSort(BHEAD0);
    AR.CompressPointer = oldipointer;
    while ( GetOneTerm(BHEAD term,fi,&startposition,0) > 0 ) {
        AT.WorkPointer = term + *term;
        if ( Generator(BHEAD term,C->numlhs) ) {
            LowerSortLevel();
            goto CalledFrom;
        }
        AR.CompressPointer = oldipointer;
    }
    AT.WorkPointer = term;
    if ( EndSort(BHEAD (WORD *)((void *)(&term)),2) < 0 ) goto CalledFrom;
    SetScratch(fi,&oldposition);
    return(term);
CalledFrom:
```

```
    MLOCK(ErrorMessageLock);
    MesCall("CreateExpression");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(0);
}

/*
        #] CreateExpression :
        #[ GCDterms :       GCD of two terms

    Computes the GCD of two terms.
    Output in termout.
    termout may overlap with term1.
*/

int GCDterms(PHEAD WORD *term1, WORD *term2, WORD *termout)
{
    GETBIDENTITY
    WORD *t1, *t1stop, *t1next, *t2, *t2stop, *t2next, *tout, *tt1, *tt2;
    int count1, count2, i, ii, x1, sign;
    WORD length1, length2;
    t1 = term1 + *term1; t1stop = t1 - ABS(t1[-1]); t1 = term1+1;
    t2 = term2 + *term2; t2stop = t2 - ABS(t2[-1]); t2 = term2+1;
    tout = termout+1;
    while ( t1 < t1stop ) {
        t1next = t1 + t1[1];
        t2 = term2+1;
        if ( *t1 == SYMBOL ) {
            while ( t2 < t2stop && *t2 != SYMBOL ) t2 += t2[1];
            if ( t2 < t2stop && *t2 == SYMBOL ) {
                t2next = t2+t2[1];
                tt1 = t1+2; tt2 = t2+2; count1 = 0;
                while ( tt1 < t1next && tt2 < t2next ) {
                    if ( *tt1 < *tt2 ) tt1 += 2;
                    else if ( *tt1 > *tt2 ) tt2 += 2;
                    else if ( ( tt1[1] > 0 && tt2[1] < 0 ) ||
                              ( tt2[1] > 0 && tt1[1] < 0 ) ) {
                        tt1 += 2; tt2 += 2;
                    }
                    else {
                        x1 = tt1[1];
                        if ( tt1[1] < 0 ) { if ( tt2[1] > x1 ) x1 = tt2[1]; }
                        else              { if ( tt2[1] < x1 ) x1 = tt2[1]; }
                        tout[count1+2] = *tt1;
                        tout[count1+3] = x1;
                        tt1 += 2; tt2 += 2;
                        count1 += 2;
                    }
                }
                if ( count1 > 0 ) {
                    *tout = SYMBOL; tout[1] = count1+2; tout += tout[1];
                }
            }
        }
        else if ( *t1 == DOTPRODUCT ) {
            while ( t2 < t2stop && *t2 != DOTPRODUCT ) t2 += t2[1];
            if ( t2 < t2stop && *t2 == DOTPRODUCT ) {
                t2next = t2+t2[1];
                tt1 = t1+2; tt2 = t2+2; count1 = 0;
                while ( tt1 < t1next && tt2 < t2next ) {
                    if ( *tt1 < *tt2 || ( *tt1 == *tt2 && tt1[1] < tt2[1] ) ) tt1 += 3;
                    else if ( *tt1 > *tt2 || ( *tt1 == *tt2 && tt1[1] > tt2[1] ) ) tt2 += 3;
                    else if ( ( tt1[2] > 0 && tt2[2] < 0 ) ||
                              ( tt2[2] > 0 && tt1[2] < 0 ) ) {
                        tt1 += 3; tt2 += 3;
                    }
                    else {
                        x1 = tt1[2];
                        if ( tt1[2] < 0 ) { if ( tt2[2] > x1 ) x1 = tt2[2]; }
                        else              { if ( tt2[2] < x1 ) x1 = tt2[2]; }
                        tout[count1+2] = *tt1;
                        tout[count1+3] = tt1[1];
                        tout[count1+4] = x1;
                        tt1 += 3; tt2 += 3;
                        count1 += 3;
                    }
                }
                if ( count1 > 0 ) {
                    *tout = DOTPRODUCT; tout[1] = count1+2; tout += tout[1];
                }
            }
        }
        else if ( *t1 == VECTOR ) {
            while ( t2 < t2stop && *t2 != VECTOR ) t2 += t2[1];
            if ( t2 < t2stop && *t2 == VECTOR ) {
                t2next = t2+t2[1];
```

```
                tt1 = t1+2; tt2 = t2+2; count1 = 0;
                while ( tt1 < t1next && tt2 < t2next ) {
                    if ( *tt1 < *tt2 || ( *tt1 == *tt2 && tt1[1] < tt2[1] ) ) tt1 += 2;
                    else if ( *tt1 > *tt2 || ( *tt1 == *tt2 && tt1[1] > tt2[1] ) ) tt2 += 2;
                    else {
                        tout[count1+2] = *tt1;
                        tout[count1+3] = tt1[1];
                        tt1 += 2; tt2 += 2;
                        count1 += 2;
                    }
                }
                if ( count1 > 0 ) {
                    *tout = VECTOR; tout[1] = count1+2; tout += tout[1];
                }
            }
        }
        else if ( *t1 == INDEX ) {
            while ( t2 < t2stop && *t2 != INDEX ) t2 += t2[1];
            if ( t2 < t2stop && *t2 == INDEX ) {
                t2next = t2+t2[1];
                tt1 = t1+2; tt2 = t2+2; count1 = 0;
                while ( tt1 < t1next && tt2 < t2next ) {
                    if ( *tt1 < *tt2 ) tt1 += 1;
                    else if ( *tt1 > *tt2 ) tt2 += 1;
                    else {
                        tout[count1+2] = *tt1;
                        tt1 += 1; tt2 += 1;
                        count1 += 1;
                    }
                }
                if ( count1 > 0 ) {
                    *tout = INDEX; tout[1] = count1+2; tout += tout[1];
                }
            }
        }
        else if ( *t1 == DELTA ) {
            while ( t2 < t2stop && *t2 != DELTA ) t2 += t2[1];
            if ( t2 < t2stop && *t2 == DELTA ) {
                t2next = t2+t2[1];
                tt1 = t1+2; tt2 = t2+2; count1 = 0;
                while ( tt1 < t1next && tt2 < t2next ) {
                    if ( *tt1 < *tt2 || ( *tt1 == *tt2 && tt1[1] < tt2[1] ) ) tt1 += 2;
                    else if ( *tt1 > *tt2 || ( *tt1 == *tt2 && tt1[1] > tt2[1] ) ) tt2 += 2;
                    else {
                        tout[count1+2] = *tt1;
                        tout[count1+3] = tt1[1];
                        tt1 += 2; tt2 += 2;
                        count1 += 2;
                    }
                }
                if ( count1 > 0 ) {
                    *tout = DELTA; tout[1] = count1+2; tout += tout[1];
                }
            }
        }
        else if ( *t1 >= FUNCTION ) { /* noncommuting functions? Forbidden! */
/*
            Count how many times this function occurs.
            Then count how many times it is in term2.
*/
            count1 = 1;
            while ( t1next < t1stop && *t1 == *t1next && t1[1] == t1next[1] ) {
                for ( i = 2; i < t1[1]; i++ ) {
                    if ( t1[i] != t1next[i] ) break;
                }
                if ( i < t1[1] ) break;
                count1++;
                t1next += t1next[1];
            }
            count2 = 0;
            while ( t2 < t2stop ) {
                if ( *t2 == *t1 && t2[1] == t1[1] ) {
                    for ( i = 2; i < t1[1]; i++ ) {
                        if ( t2[i] != t1[i] ) break;
                    }
                    if ( i >= t1[1] ) count2++;
                }
                t2 += t2[1];
            }
            if ( count1 < count2 ) count2 = count1; /* number of common occurrences */
            if ( count2 > 0 ) {
                if ( tout == t1 ) {
                    while ( count2 > 0 ) { tout += tout[1]; count2--; }
                }
                else {
                    i = t1[1]*count2;
                    NCOPY(tout,t1,i);
```

```
                }
            }
        }
        t1 = t1next;
    }
/*
        Now the coefficients. They are in t1stop and t2stop. Should go to tout.
*/
    sign = 1;
    length1 = term1[*term1-1]; ii = i = ABS(length1); t1 = t1stop;
    if ( t1 != tout ) { NCOPY(tout,t1,i); tout -= ii; }
    length2 = term2[*term2-1];
    if ( length1 < 0 && length2 < 0 ) sign = -1;
    if ( AccumGCD(BHEAD (UWORD *)tout,&length1,(UWORD *)t2stop,length2) ) {
        MLOCK(ErrorMessageLock);
        MesCall("GCDterms");
        MUNLOCK(ErrorMessageLock);
        SETERROR(-1)
    }
    if ( sign < 0 && length1 > 0 ) length1 = -length1;
    tout += ABS(length1); tout[-1] = length1;
    *termout = tout - termout; *tout = 0;
    return(0);
}

/*
        #] GCDterms :
        #[ ReadPolyRatFun :
*/

int ReadPolyRatFun(PHEAD WORD *term)
{
    WORD *oldworkpointer = AT.WorkPointer;
    int flag, i;
    WORD *t, *fun, *nextt, *num, *den, *t1, *t2, size, numsize, densize;
    WORD *term1, *term2, *confree1, *confree2, *gcd, *num1, *den1, move, *newnum, *newden;
    WORD *tstop, *m1, *m2;
    WORD oldsorttype = AR.SortType;
    COMPARE oldcompareroutine = (COMPARE)(AR.CompareRoutine);
    AR.SortType = SORTHIGHFIRST;
    AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);

    tstop = term + *term; tstop -= ABS(tstop[-1]);
    if ( term + *term == AT.WorkPointer ) flag = 1;
    else flag = 0;
    t = term+1;
    while ( t < tstop ) {
        if ( *t != AR.PolyFun ) { t += t[1]; continue; }
        if ( ( t[2] & MUSTCLEANPRF ) == 0 ) { t += t[1]; continue; }
        fun = t;
        nextt = t + t[1];
        if ( fun[1] > FUNHEAD && fun[FUNHEAD] == -SNUMBER && fun[FUNHEAD+1] == 0 )
            { *term = 0; break; }
        if ( FromPolyRatFun(BHEAD fun, &num, &den) > 0 ) { t = nextt; continue; }
        if ( *num == ARGHEAD ) { *term = 0; break; }
/*
        Now we have num and den. Both are in general argument notation,
        but can also be used as expressions as in num+ARGHEAD, den+ARGHEAD.
        We need the gcd. For this we have to take out the contents
        because PreGCD does not like contents.
*/
        term1 = TermMalloc("ReadPolyRatFun");
        term2 = TermMalloc("ReadPolyRatFun");
        confree1 = TakeSymbolContent(BHEAD num+ARGHEAD,term1);
        confree2 = TakeSymbolContent(BHEAD den+ARGHEAD,term2);
        GCDclean(BHEAD term1,term2);
/*      gcd = PreGCD(BHEAD confree1,confree2,1); */
        gcd = poly_gcd(BHEAD confree1,confree2,1);
        newnum = PolyDiv(BHEAD confree1,gcd,"ReadPolyRatFun");
        newden = PolyDiv(BHEAD confree2,gcd,"ReadPolyRatFun");
        TermFree(confree2,"ReadPolyRatFun");
        TermFree(confree1,"ReadPolyRatFun");
        num1 = MULfunc(BHEAD term1,newnum);
        den1 = MULfunc(BHEAD term2,newden);
        TermFree(newnum,"ReadPolyRatFun");
        TermFree(newden,"ReadPolyRatFun");
/*      M_free(gcd,"poly_gcd"); */
        TermFree(gcd,"poly_gcd");
        TermFree(term1,"ReadPolyRatFun");
        TermFree(term2,"ReadPolyRatFun");
/*
        Now we can put the function back together.
        Notice that we cannot use ToFast, because there is no reservation
        for the header of the argument. Fortunately there are only two
        types of fast arguments.
*/
        if ( num1[0] == 4 && num1[4] == 0 && num1[2] == 1 && num1[1] > 0 ) {
```

```
            numsize = 2; num1[0] = -SNUMBER;
            if ( num1[3] < 0 ) num1[1] = -num1[1];
        }
        else if ( num1[0] == 8 && num1[8] == 0 && num1[7] == 3 && num1[6] == 1
            && num1[5] == 1 && num1[1] == SYMBOL && num1[4] == 1 ) {
            numsize = 2; num1[0] = -SYMBOL; num1[1] = num1[3];
        }
        else { m1 = num1; while ( *m1 ) m1 += *m1; numsize = (m1-num1)+ARGHEAD; }
        if ( den1[0] == 4 && den1[4] == 0 && den1[2] == 1 && den1[1] > 0 ) {
            densize = 2; den1[0] = -SNUMBER;
            if ( den1[3] < 0 ) den1[1] = -den1[1];
        }
        else if ( den1[0] == 8 && den1[8] == 0 && den1[7] == 3 && den1[6] == 1
            && den1[5] == 1 && den1[1] == SYMBOL && den1[4] == 1 ) {
            densize = 2; den1[0] = -SYMBOL; den1[1] = den1[3];
        }
        else { m2 = den1; while ( *m2 ) m2 += *m2; densize = (m2-den1)+ARGHEAD; }
        size = FUNHEAD+numsize+densize;

        if ( size > fun[1] ) {
            move = size - fun[1];
            t1 = term+*term; t2 = t1+move;
            while ( t1 > nextt ) *--t2 = *--t1;
            tstop += move; nextt += move;
            *term += move;
        }
        else if ( size < fun[1] ) {
            move = fun[1]-size;
            t2 = fun+size; t1 = nextt;
            tstop -= move; nextt -= move;
            t = term+*term;
            while ( t1 < t ) *t2++ = *t1++;
            *term -= move;
        }
        else { /* no need to move anything */ }
        fun[1] = size; fun[2] = 0;
        t2 = fun+FUNHEAD; t1 = num1;
        if ( *num1 < 0 ) { *t2++ = num1[0]; *t2++ = num1[1]; }
        else { *t2++ = numsize; *t2++ = 0; FILLARG(t2);
            i = numsize-ARGHEAD; NCOPY(t2,t1,i) }
        t1 = den1;
        if ( *den1 < 0 ) { *t2++ = den1[0]; *t2++ = den1[1]; }
        else { *t2++ = densize; *t2++ = 0; FILLARG(t2);
            i = densize-ARGHEAD; NCOPY(t2,t1,i) }

        TermFree(num1,"MULfunc");
        TermFree(den1,"MULfunc");
        t = nextt;
    }

    if ( flag ) AT.WorkPointer = term +*term;
    else AT.WorkPointer = oldworkpointer;
    AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
    AR.SortType = oldsorttype;
    return(0);
}

/*
        #] ReadPolyRatFun :
        #[ FromPolyRatFun :
*/

int FromPolyRatFun(PHEAD WORD *fun, WORD **numout, WORD **denout)
{
    WORD *nextfun, *tt, *num, *den;
    int i;
    nextfun = fun + fun[1];
    fun += FUNHEAD;
    num = AT.WorkPointer;
    if ( *fun < 0 ) {
        if ( *fun != -SNUMBER && *fun != -SYMBOL ) goto Improper;
        ToGeneral(fun,num,0);
        tt = num + *num; *tt++ = 0;
        fun += 2;
    }
    else { i = *fun; tt = num; NCOPY(tt,fun,i); *tt++ = 0; }
    den = tt;
    if ( *fun < 0 ) {
        if ( *fun != -SNUMBER && *fun != -SYMBOL ) goto Improper;
        ToGeneral(fun,den,0);
        tt = den + *den; *tt++ = 0;
        fun += 2;
    }
    else { i = *fun; tt = den; NCOPY(tt,fun,i); *tt++ = 0; }
    *numout = num; *denout = den;
    if ( fun != nextfun ) { return(1); }
    AT.WorkPointer = tt;
```

```
    return(0);
Improper:
    MLOCK(ErrorMessageLock);
    MesPrint("Improper use of PolyRatFun");
    MesCall("FromPolyRatFun");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1);
}

/*
        #] FromPolyRatFun :
        #[ TakeSymbolContent :
*/
WORD *TakeSymbolContent(PHEAD WORD *in, WORD *term)
{
    GETBIDENTITY
    WORD *t, *tstop, *tout, *tstore;
    WORD *tnext, *tt, *tterm;
    WORD *inp, a, *den, *oldworkpointer = AT.WorkPointer;
    int i, action = 0, sign, first;
    UWORD *GCDbuffer, *GCDbuffer2, *LCMbuffer, *LCMbuffer2, *ap;
    WORD GCDlen, GCDlen2, LCMlen, LCMlen2, length, redlength, len1, len2;
    LONG j;
    tout = tstore = term+1;
/*
    #[ SYMBOL :

    We make a list of symbols and their minimal powers.
    This includes negative powers. In the end we have to multiply by the
    inverse of this list. That takes out all negative powers and leaves
    things ready for further processing.
*/
    tterm = AT.WorkPointer; tt = tterm+1;
    tout[0] = SYMBOL; tout[1] = 2;
    t = in; first = 1;
    while ( *t ) {
        tnext = t + *t; tstop = tnext - ABS(tnext[-1]); t++;
        while ( t < tstop ) {
            if ( first ) {
                if ( *t == SYMBOL ) {
                    for ( i = 0; i < t[1]; i++ ) tout[i] = t[i];
                    goto didwork;
                }
                else {
                    MLOCK(ErrorMessageLock);
                    MesPrint ((char*)"ERROR: polynomials and polyratfuns must contain symbols only");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(1);
                }
            }
            else if ( *t == SYMBOL ) {
                MergeSymbolLists(BHEAD tout,t,-1);
                goto didwork;
            }
            else {
                t += t[1];
            }
        }
/*
        Here we come when there were no symbols. Only keep the negative ones.
*/
        if ( first == 0 ) {
            int j = 2;
            for ( i = 2; i < tout[1]; i += 2 ) {
                if ( tout[i+1] < 0 ) {
                    if ( i == j ) { j += 2; }
                    else { tout[j] = tout[i]; tout[j+1] = tout[i+1]; j += 2; }
                }
            }
            tout[1] = j;
        }
didwork:;
        first = 0;
        t = tnext;
    }
    if ( tout[1] > 2 ) {
        t = tout;
        tt[0] = t[0]; tt[1] = t[1];
        for ( i = 2; i < t[1]; i += 2 ) {
            tt[i] = t[i]; tt[i+1] = -t[i+1];
        }
        tt += tt[1];
        tout += tout[1];
        action++;
    }
/*
    #] SYMBOL :
```

```
	#[ Coefficient :

	Now we have to collect the GCD of the numerators
	and the LCM of the denominators.
*/
	AT.WorkPointer = tt;
	if ( AN.cmod != 0 ) {
	  WORD x, ix, ip;
	  t = in; tnext = t + *t; tstop = tnext - ABS(tnext[-1]);
	  x = tstop[0];
	  if ( tnext[-1] < 0 ) x += AC.cmod[0];
	  if ( GetModInverses(x,(WORD)(AN.cmod[0]),&ix,&ip) ) goto CalledFrom;
	  *tout++ = x; *tout++ = 1; *tout++ = 3;
	  *tt++ = ix; *tt++ = 1; *tt++ = 3;
	}
	else {
	  GCDbuffer  = NumberMalloc("MakeInteger");
	  GCDbuffer2 = NumberMalloc("MakeInteger");
	  LCMbuffer  = NumberMalloc("MakeInteger");
	  LCMbuffer2 = NumberMalloc("MakeInteger");
	  t = in;
	  tnext = t + *t; length = tnext[-1];
	  if ( length < 0 ) { sign = -1; length = -length; }
	  else              { sign = 1; }
	  tstop = tnext - length;
	  redlength = (length-1)/2;
	  for ( i = 0; i < redlength; i++ ) {
	    GCDbuffer[i] = (UWORD)(tstop[i]);
	    LCMbuffer[i] = (UWORD)(tstop[redlength+i]);
	  }
	  GCDlen = LCMlen = redlength;
	  while ( GCDbuffer[GCDlen-1] == 0 ) GCDlen--;
	  while ( LCMbuffer[LCMlen-1] == 0 ) LCMlen--;
	  t = tnext;
	  while ( *t ) {
	    tnext = t + *t; length = ABS(tnext[-1]);
	    tstop = tnext - length; redlength = (length-1)/2;
	    len1 = len2 = redlength;
	    den = tstop + redlength;
	    while ( tstop[len1-1] == 0 ) len1--;
	    while ( den[len2-1] == 0 ) len2--;
	    if ( GCDlen == 1 && GCDbuffer[0] == 1 ) {}
	    else {
	        GcdLong(BHEAD (UWORD *)tstop,len1,GCDbuffer,GCDlen,GCDbuffer2,&GCDlen2);
	        ap = GCDbuffer; GCDbuffer = GCDbuffer2; GCDbuffer2 = ap;
	        a = GCDlen; GCDlen = GCDlen2; GCDlen2 = a;
	    }
	    if ( len2 == 1 && den[0] == 1 ) {}
	    else {
	        GcdLong(BHEAD LCMbuffer,LCMlen,(UWORD *)den,len2,LCMbuffer2,&LCMlen2);
	        DivLong((UWORD *)den,len2,LCMbuffer2,LCMlen2,
	            GCDbuffer2,&GCDlen2,(UWORD *)AT.WorkPointer,&a);
	        MulLong(LCMbuffer,LCMlen,GCDbuffer2,GCDlen2,LCMbuffer2,&LCMlen2);
	        ap = LCMbuffer; LCMbuffer = LCMbuffer2; LCMbuffer2 = ap;
	        a = LCMlen; LCMlen = LCMlen2; LCMlen2 = a;
	    }
	    t = tnext;
	  }
	  if ( GCDlen != 1 || GCDbuffer[0] != 1 || LCMlen != 1 || LCMbuffer[0] != 1 ) {
	    redlength = GCDlen; if ( LCMlen > GCDlen ) redlength = LCMlen;
	    for ( i = 0; i < GCDlen; i++ ) *tout++ = (WORD)(GCDbuffer[i]);
	    for ( ; i < redlength; i++ ) *tout++ = 0;
	    for ( i = 0; i < LCMlen; i++ ) *tout++ = (WORD)(LCMbuffer[i]);
	    for ( ; i < redlength; i++ ) *tout++ = 0;
	    *tout++ = (2*redlength+1)*sign;
	    for ( i = 0; i < LCMlen; i++ ) *tt++ = (WORD)(LCMbuffer[i]);
	    for ( ; i < redlength; i++ ) *tt++ = 0;
	    for ( i = 0; i < GCDlen; i++ ) *tt++ = (WORD)(GCDbuffer[i]);
	    for ( ; i < redlength; i++ ) *tt++ = 0;
	    *tt++ = (2*redlength+1)*sign;
	    action++;
	  }
	  else {
	    *tout++ = 1; *tout++ = 1; *tout++ = 3*sign;
	    *tt++ = 1; *tt++ = 1; *tt++ = 3*sign;
	    if ( sign != 1 ) action++;
	  }
	  NumberFree(LCMbuffer2,"MakeInteger");
	  NumberFree(LCMbuffer ,"MakeInteger");
	  NumberFree(GCDbuffer2,"MakeInteger");
	  NumberFree(GCDbuffer ,"MakeInteger");
	}
/*
	#] Coefficient :
	#[ Multiply by the inverse content :
*/
	if ( action ) {
```

```
        *term = tout - term; *tout = 0;
        *tterm = tt - tterm; *tt = 0;
        AT.WorkPointer = tt;
        inp = MultiplyWithTerm(BHEAD in,tterm,2);
        AT.WorkPointer = tterm;
        t = inp; while ( *t ) t += *t;
        j = (t-inp); t = inp;
        if ( j*sizeof(WORD) > (size_t)(AM.MaxTer) ) goto OverWork;
        in = tout = TermMalloc("TakeSymbolContent");
        NCOPY(tout,t,j); *tout = 0;
        if ( AN.tryterm > 0 ) { TermFree(inp,"MultiplyWithTerm"); AN.tryterm = 0; }
        else { M_free(inp,"MultiplyWithTerm"); }
    }
    else {
        t = in; while ( *t ) t += *t;
        j = (t-in); t = in;
        if ( j*sizeof(WORD) > (size_t)(AM.MaxTer) ) goto OverWork;
        in = tout = TermMalloc("TakeSymbolContent");
        NCOPY(tout,t,j); *tout = 0;
        term[0] = 4; term[1] = 1; term[2] = 1; term[3] = 3; term[4] = 0;
    }
/*
    #] Multiply by the inverse content :
    AT.WorkPointer = tterm + *tterm;
*/
    AT.WorkPointer = oldworkpointer;

    return(in);
OverWork:
    MLOCK(ErrorMessageLock);
    MesPrint("Term too complex. Maybe increasing MaxTermSize can help");
    MUNLOCK(ErrorMessageLock);
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("TakeSymbolContent");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(0);
}

/*
        #] TakeSymbolContent :
        #[ GCDclean :

        Takes a numerator and a denominator that each consist of a
        single term with only a coefficient and symbols and makes them
        into a proper fraction. Output overwrites input.
*/

void GCDclean(PHEAD WORD *num, WORD *den)
{
    WORD *out1 = TermMalloc("GCDclean");
    WORD *out2 = TermMalloc("GCDclean");
    WORD *t1, *t2, *r1, *r2, *t1stop, *t2stop, csize1, csize2, csize3, pow, sign;
    int i;
    t1stop = num+*num; sign = ( t1stop[-1] < 0 ) ? -1 : 1;
    csize1 = ABS(t1stop[-1]); t1stop -= csize1;
    t2stop = den+*den; if ( t2stop[-1] < 0 ) sign = -sign;
    csize2 = ABS(t2stop[-1]); t2stop -= csize2;
    t1 = num+1; t2 = den+1;
    r1 = out1+3; r2 = out2+3;
    if ( t1 == t1stop ) {
        if ( t2 < t2stop ) {
            for ( i = 2; i < t2[1]; i += 2 ) {
                if ( t2[i+1] < 0 ) { *r1++ = t2[i]; *r1++ = -t2[i+1]; }
                else { *r2++ = t2[i]; *r2++ = t2[i+1]; }
            }
        }
    }
    else if ( t2 == t2stop ) {
        for ( i = 2; i < t1[1]; i += 2 ) {
            if ( t1[i+1] < 0 ) { *r2++ = t1[i]; *r2++ = -t1[i+1]; }
            else { *r1++ = t1[i]; *r1++ = t1[i+1]; }
        }
    }
    else {
        t1 += 2; t2 += 2;
        while ( t1 < t1stop && t2 < t2stop ) {
            if ( *t1 < *t2 ) {
                if ( t1[1] > 0 ) { *r1++ = *t1; *r1++ = t1[1]; t1 += 2; }
                else if ( t1[1] < 0 ) { *r2++ = *t1; *r2++ = -t1[1]; t1 += 2; }
            }
            else if ( *t1 > *t2 ) {
                if ( t2[1] > 0 ) { *r2++ = *t2; *r2++ = t2[1]; t2 += 2; }
                else if ( t2[1] < 0 ) { *r1++ = *t2; *r1++ = -t2[1]; t2 += 2; }
            }
            else {
```

```
                pow = t1[1]-t2[1];
                if ( pow > 0 ) { *r1++ = *t1; *r1++ = pow; }
                else if ( pow < 0 ) { *r2++ = *t1; *r2++ = -pow; }
                t1 += 2; t2 += 2;
            }
        }
        while ( t1 < t1stop ) {
            if ( t1[1] < 0 ) { *r2++ = *t1; *r2++ = -t1[1]; }
            else { *r1++ = *t1; *r1++ = t1[1]; }
            t1 += 2;
        }
        while ( t2 < t2stop ) {
            if ( t2[1] < 0 ) { *r1++ = *t2; *r1++ = -t2[1]; }
            else { *r2++ = *t2; *r2++ = t2[1]; }
            t2 += 2;
        }
    }
    if ( r1 > out1+3 ) { out1[1] = SYMBOL; out1[2] = r1 - out1 - 1; }
    else r1 = out1+1;
    if ( r2 > out2+3 ) { out2[1] = SYMBOL; out2[2] = r2 - out2 - 1; }
    else r2 = out2+1;
/*
    Now the coefficients.
*/
    csize1 = REDLENG(csize1);
    csize2 = REDLENG(csize2);
    if ( DivRat(BHEAD (UWORD *)t1stop,csize1,(UWORD *)t2stop,csize2,(UWORD *)r1,&csize3) ) {
        MLOCK(ErrorMessageLock);
        MesCall("GCDclean");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    UnPack((UWORD *)r1,csize3,&csize2,&csize1);
    t2 = r1+ABS(csize3);
    for ( i = 0; i < csize2; i++ ) r2[i] = t2[i];
    r2 += csize2; *r2++ = 1;
    for ( i = 1; i < csize2; i++ ) *r2++ = 0;
    csize2 = INCLENG(csize2); *r2++ = csize2; *out2 = r2-out2;
    r1 += ABS(csize1); *r1++ = 1;
    for ( i = 1; i < ABS(csize1); i++ ) *r1++ = 0;
    csize1 = INCLENG(csize1); *r1++ = csize1; *out1 = r1-out1;

    t1 = num; t2 = out1; i = *out1; NCOPY(t1,t2,i); *t1 = 0;
    if ( sign < 0 ) t1[-1] = -t1[-1];
    t1 = den; t2 = out2; i = *out2; NCOPY(t1,t2,i); *t1 = 0;

    TermFree(out2,"GCDclean");
    TermFree(out1,"GCDclean");
}

/*
        #] GCDclean :
        #[ PolyDiv :

        Special stub function for polynomials that should fit inside a term.
        We make sure that the space is allocated by TermMalloc.
        This makes things much easier on the calling routines.
*/

WORD *PolyDiv(PHEAD WORD *a,WORD *b,char *text)
{
/*
    Probably the following would work now
*/
    DUMMYUSE(text);
    return(poly_div(BHEAD a,b,1));
/*
    WORD *quo, *qq;
    WORD *x, *xx;
    LONG i;
    quo = poly_div(BHEAD a,b,1);
    x = TermMalloc(text);
    qq = quo; while ( *qq ) qq += *qq;
    i = (qq-quo+1);
    if ( i*sizeof(WORD) > (size_t)(AM.MaxTer) ) {
        DUMMYUSE(text);
        MLOCK(ErrorMessageLock);
        MesPrint("PolyDiv: Term too complex. Maybe increasing MaxTermSize can help");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    xx = x; qq = quo;
    NCOPY(xx,qq,i)
    TermFree(quo,"poly_div");
    return(x);
*/
}
```

```

/*
		#] PolyDiv :
	#] GCDfunction :
	#[ DIVfunction :

	Input: a div_ function that has two arguments inside a term.
	Action: Calculates [arg1/arg2] using polynomial techniques if needed.
	Output: The output result is combined with the remainder of the term
	and sent to Generator for further processing.
	Note that the output can be just a number or many terms.
	In case par == 0 the output is [arg1/arg2]
	In case par == 1 the output is [arg1%arg2]
	In case par == 2 the output is [inverse of arg1 modulus arg2]
	In case par == 3 the output is [arg1*arg2]
*/

WORD divrem[4] = { DIVFUNCTION, REMFUNCTION, INVERSEFUNCTION, MULFUNCTION };

int DIVfunction(PHEAD WORD *term,WORD level,int par)
{
	GETBIDENTITY
	WORD *t, *tt, *r, *arg1 = 0, *arg2 = 0, *arg3 = 0, *termout;
	WORD *tstop, *tend, *r3, *rr, *rstop, tlength, rlength, newlength;
	WORD *proper1, *proper2, *proper3 = 0, numdol = -1;
	int numargs = 0, type1, type2, actionflag1, actionflag2;
	WORD startebuf = cbuf[AT.ebufnum].numrhs;
	int division = ( par <= 2 );  /* false for mul_ */
	if ( par < 0 || par > 3 ) {
		MLOCK(ErrorMessageLock);
		MesPrint("Internal error. Illegal parameter %d in DIVfunction.",par);
		MUNLOCK(ErrorMessageLock);
		Terminate(-1);
	}
/*
	Find the function
*/
	tend = term + *term; tstop = tend - ABS(tend[-1]);
	t = term+1;
	while ( t < tstop ) {
		if ( *t != divrem[par] ) { t += t[1]; continue; }
		r = t + FUNHEAD;
		tt = t + t[1]; numargs = 0;
		while ( r < tt ) {
			if ( numargs == 0 ) { arg1 = r; }
			if ( numargs == 1 ) { arg2 = r; }
			if ( numargs == 2 && *r == -DOLLAREXPRESSION ) { numdol = r[1]; }
			numargs++;
			NEXTARG(r);
		}
		if ( numargs == 2 ) break;
		if ( division && numargs == 3 ) break;
		t = tt;
	}
	if ( t >= tstop ) {
		MLOCK(ErrorMessageLock);
		MesPrint("Internal error. Indicated div_ or rem_ function not encountered.");
		MUNLOCK(ErrorMessageLock);
		Terminate(-1);
	}
/*
	We have two arguments in arg1 and arg2.
*/
	if ( division && *arg1 == -SNUMBER && arg1[1] == 0 ) {
		if ( *arg2 == -SNUMBER && arg2[1] == 0 ) {
zerozero:;
			MLOCK(ErrorMessageLock);
			MesPrint("0/0 in either div_ or rem_ function.");
			MUNLOCK(ErrorMessageLock);
			Terminate(-1);
		}
		if ( numdol >= 0 ) PutTermInDollar(0,numdol);
		return(0);
	}
	if ( division && *arg2 == -SNUMBER && arg2[1] == 0 ) {
divzero:;
		MLOCK(ErrorMessageLock);
		MesPrint("Division by zero in either div_ or rem_ function.");
		MUNLOCK(ErrorMessageLock);
		Terminate(-1);
	}
	if ( !division ) {
		if ( (*arg1 == -SNUMBER && arg1[1] == 0) ||
		     (*arg2 == -SNUMBER && arg2[1] == 0) ) {
			return(0);
		}
	}
```

```
    if ( ( arg1 = ConvertArgument(BHEAD arg1, &type1) ) == 0 ) goto CalledFrom;
    if ( ( arg2 = ConvertArgument(BHEAD arg2, &type2) ) == 0 ) goto CalledFrom;
    if ( division && *arg1 == 0 ) {
        if ( *arg2 == 0 ) {
            M_free(arg2,"DIVfunction");
            M_free(arg1,"DIVfunction");
            goto zerozero;
        }
        M_free(arg2,"DIVfunction");
        M_free(arg1,"DIVfunction");
        if ( numdol >= 0 ) PutTermInDollar(0,numdol);
        return(0);
    }
    if ( division && *arg2 == 0 ) {
        M_free(arg2,"DIVfunction");
        M_free(arg1,"DIVfunction");
        goto divzero;
    }
    if ( !division && (*arg1 == 0 || *arg2 == 0) ) {
        M_free(arg2,"DIVfunction");
        M_free(arg1,"DIVfunction");
        return(0);
    }
    if ( ( proper1 = PutExtraSymbols(BHEAD arg1,startebuf,&actionflag1) ) == 0 ) goto CalledFrom;
    if ( ( proper2 = PutExtraSymbols(BHEAD arg2,startebuf,&actionflag2) ) == 0 ) goto CalledFrom;
/*
    if ( type2 == 0 ) M_free(arg2,"DIVfunction");
    else {
        DOLLARS d = ((DOLLARS)arg2)-1;
        if ( d->factors ) M_free(d->factors,"Dollar factors");
        M_free(d,"Copy of dollar variable");
    }
*/
    M_free(arg2,"DIVfunction");
/*
    if ( type1 == 0 ) M_free(arg1,"DIVfunction");
    else {
        DOLLARS d = ((DOLLARS)arg1)-1;
        if ( d->factors ) M_free(d->factors,"Dollar factors");
        M_free(d,"Copy of dollar variable");
    }
*/
    M_free(arg1,"DIVfunction");
    if ( par == 0 )      proper3 = poly_div(BHEAD proper1, proper2,0);
    else if ( par == 1 ) proper3 = poly_rem(BHEAD proper1, proper2,0);
    else if ( par == 2 ) proper3 = poly_inverse(BHEAD proper1, proper2);
    else if ( par == 3 ) proper3 = poly_mul(BHEAD proper1, proper2);
    if ( proper3 == 0 ) goto CalledFrom;
    if ( actionflag1 || actionflag2 ) {
        if ( ( arg3 = TakeExtraSymbols(BHEAD proper3,startebuf) ) == 0 ) goto CalledFrom;
        M_free(proper3,"DIVfunction");
    }
    else {
        arg3 = proper3;
    }
    M_free(proper2,"DIVfunction");
    M_free(proper1,"DIVfunction");
    cbuf[AT.ebufnum].numrhs = startebuf;
    if ( *arg3 ) {
        termout = AT.WorkPointer;
        tlength = tend[-1];
        tlength = REDLENG(tlength);
        r3 = arg3;
        while ( *r3 ) {
            tt = term + 1; rr = termout + 1;
            while ( tt < t ) *rr++ = *tt++;
            r = r3 + 1;
            r3 = r3 + *r3;
            rstop = r3 - ABS(r3[-1]);
            while ( r < rstop ) *rr++ = *r++;
            tt += t[1];
            while ( tt < tstop ) *rr++ = *tt++;
            rlength = r3[-1];
            rlength = REDLENG(rlength);
            if ( MulRat(BHEAD (UWORD *)tstop,tlength,(UWORD *)rstop,rlength,
                    (UWORD *)rr,&newlength) < 0 ) goto CalledFrom;
            rlength = INCLENG(newlength);
            rr += ABS(rlength);
            rr[-1] = rlength;
            *termout = rr - termout;
            AT.WorkPointer = rr;
            if ( Generator(BHEAD termout,level) ) goto CalledFrom;
        }
        AT.WorkPointer = termout;
    }
    M_free(arg3,"DIVfunction");
    return(0);
```

```
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("DIVfunction");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
    #] DIVfunction :
    #[ MULfunc :

    Multiplies two polynomials and puts the results in TermMalloc space.
*/

WORD *MULfunc(PHEAD WORD *p1, WORD *p2)
{
    WORD *prod,size1,size2,size3,*t,*tfill,*ps1,*ps2,sign1,sign2, error, *p3;
    UWORD *num1, *num2, *num3;
    int i;
    WORD oldsorttype = AR.SortType;
    COMPARE oldcompareroutine = (COMPARE)(AR.CompareRoutine);
    AR.SortType = SORTHIGHFIRST;
    AR.CompareRoutine = (COMPAREDUMMY)(&CompareSymbols);
    num3 = NumberMalloc("MULfunc");
    prod = TermMalloc("MULfunc");
    NewSort(BHEAD0);
    while ( *p1 ) {
        ps1 = p1+*p1; num1 = (UWORD *)(ps1 - ABS(ps1[-1])); size1 = ps1[-1];
        if ( size1 < 0 ) { sign1 = -1; size1 = -size1; }
        else               sign1 = 1;
        size1 = (size1-1)/2;
        p3 = p2;
        while ( *p3 ) {
            ps2 = p3+*p3; num2 = (UWORD *)(ps2 - ABS(ps2[-1])); size2 = ps2[-1];
            if ( size2 < 0 ) { sign2 = -1; size2 = -size2; }
            else               sign2 = 1;
            size2 = (size2-1)/2;
            if ( MulLong(num1,size1,num2,size2,num3,&size3) ) {
                error = 1;
CalledFrom:
                MLOCK(ErrorMessageLock);
                MesPrint(" Error %d",error);
                MesCall("MulFunc");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            tfill = prod+1;
            t = p1+1; while ( t < (WORD *)num1 ) *tfill++ = *t++;
            t = p3+1; while ( t < (WORD *)num2 ) *tfill++ = *t++;
            t = (WORD *)num3;
            for ( i = 0; i < size3; i++ ) *tfill++ = *t++;
            *tfill++ = 1;
            for ( i = 1; i < size3; i++ ) *tfill++ = 0;
            *tfill++ = (2*size3+1)*sign1*sign2;
            prod[0] = tfill - prod;
            if ( SymbolNormalize(prod) ) { error = 2; goto CalledFrom; }
            if ( StoreTerm(BHEAD prod) ) { error = 3; goto CalledFrom; }
            p3 += *p3;
        }
        p1 += *p1;
    }
    NumberFree(num3,"MULfunc");
    EndSort(BHEAD prod,1);
    AR.CompareRoutine = (COMPAREDUMMY)oldcompareroutine;
    AR.SortType = oldsorttype;
    return(prod);
}

/*
    #] MULfunc :
    #[ ConvertArgument :

    Converts an argument to a general notation in allocated space.
*/

WORD *ConvertArgument(PHEAD WORD *arg, int *type)
{
    WORD *output, *t, *r;
    int i, size;
    if ( *arg > 0 ) {
        output = (WORD *)Malloc1((*arg)*sizeof(WORD),"ConvertArgument");
        i = *arg - ARGHEAD; t = arg + ARGHEAD; r = output;
        NCOPY(r,t,i);
        *r = 0; *type = 0;
        return(output);
    }
    if ( *arg == -EXPRESSION ) {
```

```
        *type = 0;
        return(CreateExpression(BHEAD arg[1]));
    }
    if ( *arg == -DOLLAREXPRESSION ) {
        DOLLARS d;
        *type = 1;
        d = DolToTerms(BHEAD arg[1]);
/*
        The problem is that DolToTerms creates a copy of the dollar variable.
        If we just return d->where we create a memory leak. Hence we have to
        copy the contents of d->where to a new buffer
*/
        output = (WORD *)Malloc1((d->size+1)*sizeof(WORD),"Copy of dollar content");
        WCOPY(output,d->where,d->size+1);
        if ( d->factors ) { M_free(d->factors,"Dollar factors"); d->factors = 0; }
        M_free(d,"Copy of dollar variable");
        return(output);
    }
#if ( FUNHEAD > 4 )
    size = FUNHEAD+5;
#else
    size = 9;
#endif
    output = (WORD *)Malloc1(size*sizeof(WORD),"ConvertArgument");
    switch(*arg) {
        case -SYMBOL:
            output[0] = 8; output[1] = SYMBOL; output[2] = 4; output[3] = arg[1];
            output[4] = 1; output[5] = 1; output[6] = 1; output[7] = 3;
            output[8] = 0;
            break;
        case -INDEX:
        case -VECTOR:
        case -MINVECTOR:
            output[0] = 7; output[1] = INDEX; output[2] = 3; output[3] = arg[1];
            output[4] = 1; output[5] = 1;
            if ( *arg == -MINVECTOR ) output[6] = -3;
            else output[6] = 3;
            output[7] = 0;
            break;
        case -SNUMBER:
            output[0] = 4;
            if ( arg[1] < 0 ) {
                output[1] = -arg[1]; output[2] = 1; output[3] = -3;
            }
            else {
                output[1] =  arg[1]; output[2] = 1; output[3] =  3;
            }
            output[4] = 0;
            break;
        default:
            output[0] = FUNHEAD+4;
            output[1] = -*arg;
            output[2] = FUNHEAD;
            for ( i = 3; i <= FUNHEAD; i++ ) output[i] = 0;
            output[FUNHEAD+1] = 1;
            output[FUNHEAD+2] = 1;
            output[FUNHEAD+3] = 3;
            output[FUNHEAD+4] = 0;
            break;
    }
    *type = 0;
    return(output);
}

/*
    #] ConvertArgument :
    #[ ExpandRat :

    Expands the denominator of a PolyRatFun in the variable PolyFunVar.
    The output is a polyratfun with a single argument.
    In the case that there is a polyratfun with more than one argument
    or the dirtyflag is on, the argument(s) is/are normalized.
    The output overwrites the input.
*/

char *TheErrorMessage[] = {
     "PolyRatFun not of a type that FORM will expand: incorrect variable inside."
    ,"Division by zero in PolyRatFun encountered in ExpandRat."
    ,"Irregular code in PolyRatFun encountered in ExpandRat."
    ,"Called from ExpandRat."
    ,"WorkSpace overflow. Change parameter WorkSpace in setup file?"
    };

int ExpandRat(PHEAD WORD *fun)
{
    WORD *r, *rr, *rrr, *tt, *tnext, *arg1, *arg2, *rmin = 0, *rmininv;
    WORD *rcoef, rsize, rcopy, *ow = AT.WorkPointer;
```

```
    WORD *numerator, *denominator, *rnext;
    WORD *thecopy, *rc, ncoef, newcoef, *m, *mm, nco, *outarg = 0;
    UWORD co[2], co1[2], co2[2];
    WORD OldPolyFunPow = AR.PolyFunPow;
    int i, j, minpow = 0, eppow, first, error = 0, ipoly;
    if ( fun[1] == FUNHEAD ) { return(0); }
    tnext = fun + fun[1];
    if ( fun[1] == fun[FUNHEAD]+FUNHEAD ) { /* Single argument */
        if ( fun[2] == 0 ) { goto done; }
/*
        We have to normalize the argument. This could make it shorter.
*/
NormArg:;
        if ( outarg == 0 ) outarg = TermMalloc("ExpandRat")+ARGHEAD;
        AT.TrimPower = 1;
        NewSort(BHEAD0);
        r = fun+FUNHEAD+ARGHEAD;
        if ( AR.PolyFunExp ==  2 ) {     /* Find minimum power */
            WORD minpow2 = MAXPOWER, *rrm;
            rrm = r;
            while ( rrm < tnext ) {
                if ( *rrm == 4 ) {
                    if ( minpow2 > 0 ) minpow2 = 0;
                }
                else if ( ABS(rrm[*rrm-1]) == (*rrm-1) ) {
                    if ( minpow2 > 0 ) minpow2 = 0;
                }
                else {
                    if ( rrm[1] == SYMBOL && rrm[2] == 4 && rrm[3] == AR.PolyFunVar ) {
                        if ( rrm[4] < minpow2 ) minpow2 = rrm[4];
                    }
                    else {
                        MesPrint("Illegal term in expanded polyratfun.");
                        goto onerror;
                    }
                }
                rrm += *rrm;
            }
            AR.PolyFunPow += minpow2;
        }
        while ( r < tnext ) {
            rr = r + *r;
            i = *r; rrr = outarg; NCOPY(rrr,r,i);
            Normalize(BHEAD outarg);
            if ( *outarg > 0 ) StoreTerm(BHEAD outarg);
        }
        r = fun+FUNHEAD+ARGHEAD;
        EndSort(BHEAD r,1);
        AT.TrimPower = 0;
        if ( *r == 0 ) {
            fun[FUNHEAD] = -SNUMBER; fun[FUNHEAD+1] = 0;
            fun[1] = FUNHEAD+2;
        }
        else {
            rr = fun+FUNHEAD;
            if ( ToFast(rr,rr) ) {
                NEXTARG(rr); fun[1] = rr - fun;
            }
            else {
                while ( *r ) r += *r;
                *rr = r-rr; rr[1] = CLEANFLAG;
                fun[1] = r - fun;
            }
        }
        fun[2] = CLEANFLAG;
        goto done;
    }
/*
    First test whether we have only AR.PolyFunVar in the denominator
*/
    tt = fun + FUNHEAD;
    arg1 = arg2 = 0;
    if ( tt < tnext ) {
        arg1 = tt; NEXTARG(tt);
        if ( tt < tnext ) {
            arg2 = tt; NEXTARG(tt);
            if ( tt != tnext ) { arg1 = arg2 = 0; } /* more than two arguments */
        }
    }
    if ( arg2 == 0 ) {
        if ( *arg1 < 0 ) { fun[2] = CLEANFLAG; goto done; }
        if ( fun[2] == CLEANFLAG ) goto done;
        goto NormArg;    /* Note: should not come here */
    }
/*
    Produce the output argument in outarg
*/
```

```
	if ( outarg == 0 ) outarg = TermMalloc("ExpandRat")+ARGHEAD;

	if ( *arg2 <= 0 ) {
/*
		These cases are trivial.
		We try as much as possible to write the output directly into the
		function. We just have to be extremely careful not to overwrite
		relevant information before we are finished with it.
*/
		if ( *arg2 == -SYMBOL && arg2[1] == AR.PolyFunVar ) {
			rr = r = fun+FUNHEAD+ARGHEAD;
			if ( *arg1 < 0 ) {
				if ( *arg1 == -SYMBOL ) {
					if ( arg1[1] == AR.PolyFunVar ) {
						*r++ = 4; *r++ = 1; *r++ = 1; *r++ = 3; *r = 0;
					}
					else {
						*r++ = 10; *r++ = SYMBOL; *r++ = 6;
						*r++ = arg1[1]; *r++ = 1;
						*r++ = AR.PolyFunVar; *r++ = -1;
						*r++ = 1; *r++ = 1; *r++ = 3; *r = 0;
						Normalize(BHEAD rr);
					}
				}
				else if ( *arg1 == -SNUMBER ) {
					nco = arg1[1];
					if ( nco == 0 ) { *r++ = 0; }
					else {
						*r++ = 8; *r++ = SYMBOL; *r++ = 4;
						*r++ = AR.PolyFunVar; *r++ = -1;
						*r++ = ABS(nco); *r++ = 1;
						if ( nco < 0 ) *r++ = -3;
						else *r++ = 3;
						*r = 0;
					}
				}
				else { error = 2; goto onerror; }  /* should not happen! */
			}
			else {	/* Multi-term numerator. */
				m = arg1+ARGHEAD;
				NewSort(BHEAD0); /* Technically maybe not needed */
				if ( AR.PolyFunExp ==  2 ) {	/* Find minimum power */
					WORD minpow2 = MAXPOWER, *rrm;
					rrm = m;
					while ( rrm < arg2 ) {
						if ( *rrm == 4 ) {
							if ( minpow2 > 0 ) minpow2 = 0;
						}
						else if ( ABS(rrm[*rrm-1]) == (*rrm-1) ) {
							if ( minpow2 > 0 ) minpow2 = 0;
						}
						else {
							if ( rrm[1] == SYMBOL && rrm[2] == 4 && rrm[3] == AR.PolyFunVar ) {
								if ( rrm[4] < minpow2 ) minpow2 = rrm[4];
							}
							else {
								MesPrint("Illegal term in expanded polyratfun.");
								goto onerror;
							}
						}
						rrm += *rrm;
					}
					AR.PolyFunPow += minpow2-1;
				}
				while ( m < arg2 ) {
					r = outarg;
					rrr = r++; mm = m + *m;
					*r++ = SYMBOL; *r++ = 4; *r++ = AR.PolyFunVar; *r++ = -1;
					m++; while ( m < mm ) *r++ = *m++;
					*rrr = r-rrr;
					Normalize(BHEAD rrr);
					StoreTerm(BHEAD rrr);
				}
				EndSort(BHEAD rr,1);
				r = rr; while ( *r ) r += *r;
			}
			if ( *rr == 0 ) {
				fun[FUNHEAD] = -SNUMBER; fun[FUNHEAD+1] = CLEANFLAG;
				fun[1] = FUNHEAD+2;
			}
			else {
				rr = fun+FUNHEAD;
				*rr = r-rr;
				rr[1] = CLEANFLAG;
				if ( ToFast(rr,rr) ) {
					NEXTARG(rr);
					fun[1] = rr - fun;
```

```
                }
                else { fun[1] = r - fun; }
            }
            fun[2] = CLEANFLAG;
            goto done;
        }
        else if ( *arg2 == -SNUMBER ) {
            rr = r = outarg;
            if ( arg2[1] == 0 ) { error = 1; goto onerror; }
            if ( *arg1  == -SNUMBER ) { /* Things may not be normalized */
                if ( arg1[1] == 0 ) { *r++ = 0; }
                else {
                    co1[0] = ABS(arg1[1]); co1[1] = 1;
                    co2[0] = 1; co2[1] = ABS(arg2[1]);
                    MulRat(BHEAD co1,1,co2,1,co,&nco);
                    *r++ = 4; *r++ = (WORD)(co[0]); *r++ = (WORD)(co[1]);
                    if ( ( arg1[1] < 0 && arg2[1] > 0 ) ||
                         ( arg1[1] > 0 && arg2[1] < 0 ) ) *r++ = -3;
                    else *r++ = 3;
                    *r = 0;
                }
            }
            else if ( *arg1 == -SYMBOL ) {
                *r++ = 8; *r++ = SYMBOL; *r++ = 4;
                *r++ = arg1[1]; *r++ = 1;
                *r++ = 1; *r++ = ABS(arg2[1]);
                if ( arg2[1] < 0 ) *r++ = -3;
                else *r++ = 3;
                *r = 0;
            }
            else if ( *arg1 < 0 ) { error = 2; goto onerror; }
            else {  /* Multi-term numerator. */
                m = arg1+ARGHEAD;
                NewSort(BHEAD0); /* Technically maybe not needed */
                if ( AR.PolyFunExp ==  2 ) {    /* Find minimum power */
                    WORD minpow2 = MAXPOWER, *rrm;
                    rrm = m;
                    while ( rrm < arg2 ) {
                        if ( *rrm == 4 ) {
                            if ( minpow2 > 0 ) minpow2 = 0;
                        }
                        else if ( ABS(rrm[*rrm-1]) == (*rrm-1) ) {
                            if ( minpow2 > 0 ) minpow2 = 0;
                        }
                        else {
                            if ( rrm[1] == SYMBOL && rrm[2] == 4 && rrm[3] == AR.PolyFunVar ) {
                                if ( rrm[4] < minpow2 ) minpow2 = rrm[4];
                            }
                            else {
                                MesPrint("Illegal term in expanded polyratfun.");
                                goto onerror;
                            }
                        }
                        rrm += *rrm;
                    }
                    AR.PolyFunPow += minpow2;
                }
                while ( m < arg2 ) {
                    r = rr;
                    rrr = r++; mm = m + *m;
                    *r++ = DENOMINATOR; *r++ = FUNHEAD + 2; *r++ = DIRTYFLAG;
                    FILLFUN3(r);
                    *r++ = arg2[0]; *r++ = arg2[1];
                    m++; while ( m < mm ) *r++ = *m++;
                    *rrr = r-rrr;
                    if ( r < AT.WorkTop && r >= AT.WorkSpace )
                            AT.WorkPointer = r;
                    Normalize(BHEAD rrr);
                    if ( ABS(rrr[*rrr-1]) == *rrr-1 ) {
                        if ( AR.PolyFunPow >= 0 ) {
                            StoreTerm(BHEAD rrr);
                        }
                    }
                    else if ( rrr[1] == SYMBOL && rrr[2] == 4 &&
                    rrr[3] == AR.PolyFunVar && rrr[4] <= AR.PolyFunPow ) {
                        StoreTerm(BHEAD rrr);
                    }
                }
                EndSort(BHEAD rr,1);
            }
            r = rr; while ( *r ) r += *r;
            i = r-rr;
            r = fun + FUNHEAD + ARGHEAD;
            NCOPY(r,rr,i);
            rr = fun + FUNHEAD;
            *rr = r - rr; rr[1] = CLEANFLAG;
            if ( ToFast(rr,rr) ) {
```

```
                NEXTARG(rr);
                fun[1] = rr - fun;
            }
            else { fun[1] = r - fun; }
            fun[2] = CLEANFLAG;
            goto done;
        }
        else { error = 0; goto onerror; }
    }
    else {
        r = arg2+ARGHEAD; /* The argument ends at tnext */
        first = 1;
        while ( r < tnext ) {
            rr = r + *r; rr -= ABS(rr[-1]);
            if ( r+1 == rr ) {
                if ( first ) { minpow = 0; first = 0; rmin = r; }
                else if ( minpow > 0 ) { minpow = 0; rmin = r; }
            }
            else if ( r[1] != SYMBOL || r[2] != 4 || r[3] != AR.PolyFunVar
                || r[4] > MAXPOWER ) { error = 0; goto onerror; }
            else if ( first ) { minpow = r[4]; first = 0; rmin = r; }
            else if ( r[4] < minpow ) { minpow = r[4]; rmin = r; }
            r += *r;
        }
/*
        We have now:
        1: a numerator in arg1 which can contain several variables.
        2: a denominator in arg2 with at most only AR.PolyFunVar (ep).
        3: the minimum power in the denominator is minpow and the
           term with that minimum power is in rmin.
        Divide numerator and denominator by this minimum power.
        Determine the power range in the numerator.
        Call InvPoly.
        Multiply by the inverse in such a way that we never take more
        powers of ep than necessary.
*/
/*
        One: put 1/rmin in AT.WorkPointer -> rmininv
*/
        AT.WorkPointer += AM.MaxTer/sizeof(WORD);
        if ( AT.WorkPointer + (AM.MaxTer/sizeof(WORD)) >= AT.WorkTop ) {
            error = 4; goto onerror;
        }
        rmininv = r = AT.WorkPointer;
        rr = rmin; i = *rmin; NCOPY(r,rr,i)
        if ( minpow != 0 ) { rmininv[4] = -rmininv[4]; }
        rsize = ABS(r[-1]);
        rcoef = r - rsize;
        rsize = (rsize-1)/2; rr = rcoef + rsize;
        for ( i = 0; i < rsize; i++ ) {
            rcopy = rcoef[i]; rcoef[i] = rr[i]; rr[i] = rcopy;
        }
        AT.WorkPointer = r;
        if ( *arg1 < 0 ) {
            ToGeneral(arg1,r,0);
            arg1 = r; r += *r; *r++ = 0; rcopy = 0;
            AT.WorkPointer = r;
        }
        else {
            r = arg1 + *arg1;
            rcopy = *r; *r++ = 0;
        }
/*
        We can use MultiplyWithTerm.
*/
        AT.LeaveNegative = 1;
        numerator = MultiplyWithTerm(BHEAD arg1+ARGHEAD,rmininv,0);
        AT.LeaveNegative = 0;
        r[-1] = rcopy;
        r = numerator; while ( *r ) r += *r;
        AT.WorkPointer = r+1;
        rcopy = arg2[*arg2]; arg2[*arg2] = 0;
        denominator = MultiplyWithTerm(BHEAD arg2+ARGHEAD,rmininv,1);
        arg2[*arg2] = rcopy;
        r = denominator; while ( *r ) r += *r;
        AT.WorkPointer = r+1;
/*
        Now find the minimum power of ep in the numerator.
*/
        r = numerator;
        first = 1;
        while ( *r ) {
            rr = r + *r; rr -= ABS(rr[-1]);
            if ( r+1 == rr ) {
                if ( first ) { minpow = 0; first = 0; }
                else if ( minpow > 0 ) { minpow = 0; }
            }
```

```
            else if ( r[1] != SYMBOL ) { error = 0; goto onerror; }
            else {
                for ( i = 3; i < r[2]; i += 2 ) {
                    if ( r[i] == AR.PolyFunVar ) {
                        if ( first ) { minpow = r[i+1]; first = 0; }
                        else if ( r[i+1] < minpow ) minpow = r[i+1];
                        break;
                    }
                }
                if ( i >= r[2] ) {
                    if ( first ) { minpow = 0; first = 0; }
                    else if ( minpow > 0 ) minpow = 0;
                }
            }
            r += *r;
        }
/*
        We can invert the denominator.
        Note that the return value is an offset in AT.pWorkSpace.
        Hence there is no need to free memory afterwards.
*/
        if ( AR.PolyFunExp == 3 ) {
            ipoly = InvPoly(BHEAD denominator,AR.PolyFunPow-minpow,AR.PolyFunVar);
        }
        else {
            ipoly = InvPoly(BHEAD denominator,AR.PolyFunPow,AR.PolyFunVar);
        }
/*
        Now we start the multiplying
*/
        NewSort(BHEAD0);
        r = numerator;
        while ( *r ) {
/*
            1: Find power of ep.
*/
            rnext = r + *r;
            rrr = rnext - ABS(rnext[-1]);
            rr = r+1;
            eppow = 0;
            if ( rr < rrr ) {
                j = rr[1] - 2; rr += 2;
                while ( j > 0 ) {
                    if ( *rr == AR.PolyFunVar ) { eppow = rr[1]; break; }
                    j -= 2; rr += 2;
                }
            }
/*
            2: Multiply by the proper terms in ipoly
*/
            for ( i = 0; i <= AR.PolyFunPow-eppow+minpow; i++ ) {
                if ( AT.pWorkSpace[ipoly+i] == 0 ) continue;
/*
                Copy the term, add i to the power of ep and multiply coef.
*/
                rc = r;
                rr = thecopy = AT.WorkPointer;
                while ( rc < rrr ) *rr++ = *rc++;
                if ( i != 0 ) {
                    *rr++ = SYMBOL; *rr++ = 4; *rr++ = AR.PolyFunVar; *rr++ = i;
                }
                ncoef = REDLENG(rnext[-1]);
                MulRat(BHEAD (UWORD *)rrr,ncoef,
                    (UWORD *)(AT.pWorkSpace[ipoly+i])+1,AT.pWorkSpace[ipoly+i][0]
                    ,(UWORD *)rr,&newcoef);
                ncoef = ABS(newcoef); rr += 2*ncoef;
                newcoef = INCLENG(newcoef);
                *rr++ = newcoef;
                *thecopy = rr - thecopy;
                AT.WorkPointer = rr;
                Normalize(BHEAD thecopy);
                if ( *thecopy > 0 ) StoreTerm(BHEAD thecopy);
                AT.WorkPointer = thecopy;
            }
            r = rnext;
        }
/*
        Now we have all.
*/
        rr = fun + FUNHEAD; r = rr + ARGHEAD;
        EndSort(BHEAD r,1);
        if ( *r == 0 ) {
            fun[1] = FUNHEAD+2; fun[2] = CLEANFLAG;
            fun[FUNHEAD] = -SNUMBER; fun[FUNHEAD+1] = 0;
        }
        else {
            while ( *r ) r += *r;
```

```
            rr[0] = r-rr; rr[1] = CLEANFLAG;
            if ( ToFast(rr,rr) ) { NEXTARG(rr); fun[1] = rr-fun; }
            else { fun[1] = r-fun; }
            fun[2] = CLEANFLAG;
        }
    }
done:
    if ( outarg ) TermFree(outarg-ARGHEAD,"ExpandRat");
    AR.PolyFunPow = OldPolyFunPow;
    AT.WorkPointer = ow;
    AN.PolyNormFlag = 1;
    return(0);
onerror:
    if ( outarg ) TermFree(outarg-ARGHEAD,"ExpandRat");
    AR.PolyFunPow = OldPolyFunPow;
    AT.WorkPointer = ow;
    MLOCK(ErrorMessageLock);
    MesPrint(TheErrorMessage[error]);
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(-1);
}

/*
    #] ExpandRat :
    #[ InvPoly :

    The input polynomial is represented as a sequence of terms in ascending
    power. The first coefficient is 1. If we call this 1-a and
    a = sum_(j,1,n,x^j*a(j)), and b = 1/(1-a) we can find the coefficients
    of b with the recursion
        b(0) = 1, b(n) = sum_(j,1,n,a(j)*b(n-j))
    The variable is the symbol sym and we need maxpow powers in the answer.
    The answer is an array of pointers to the coefficients of the various
    powers as rational numbers in the notation signedsize,numerator,denominator
    We put these powers in the workspace and the answer is in AT.pWorkSpace.
    Hence the return value is an offset in the pWorkSpace.
    A zero pointer indicates that this coefficient is zero.
*/

int InvPoly(PHEAD WORD *inpoly, WORD maxpow, WORD sym)
{
    int needed, inpointers, outpointers, maxinput = 0, i, j;
    WORD *t, *tt, *ttt, *w, *c, *cc, *ccc, lenc, lenc1, lenc2, rc, *c1, *c2;
/*
    Step 0: allocate the space
*/
    needed = (maxpow+1)*2;
    WantAddPointers(needed);
    inpointers = AT.pWorkPointer;
    outpointers = AT.pWorkPointer+maxpow+1;
    for ( i = 0; i < needed; i++ ) AT.pWorkSpace[inpointers+i] = 0;
/*
    Step 1: determine the coefficients in inpoly
            often there is a maximum power that is much smaller than maxpow.
            keeping track of this can speed up things.
*/
    t = inpoly;
    w = AT.WorkPointer;
    while ( *t ) {
        if ( *t == 4 ) {
            if ( t[1] != 1 || t[2] != 1 || t[3] != 3 ) goto onerror;
            AT.pWorkSpace[inpointers] = 0;
        }
        else if ( t[1] != SYMBOL || t[2] != 4 || t[3] != sym || t[4] < 0 ) goto onerror;
        else if ( t[4] > maxpow ) {} /* power outside useful range */
        else {
            if ( t[4] > maxinput ) maxinput = t[4];
            AT.pWorkSpace[inpointers+t[4]] = w;
            tt = t + *t; rc = -*--tt; /* we need - the coefficient! */
            rc = REDLENG(rc); *w++ = rc;
            ttt = t+5;
            while ( ttt < tt ) *w++ = *ttt++;
        }
        t += *t;
    }
/*
    Step 2: compute the output. b(0) = 1.
    then the recursion starts.
*/
    AT.pWorkSpace[outpointers] = w;
    *w++ = 1; *w++ = 1; *w++ = 1;
    c  = TermMalloc("InvPoly");
    c1 = TermMalloc("InvPoly");
    c2 = TermMalloc("InvPoly");
    for ( j = 1; j <= maxpow; j++ ) {
/*
```

```
        Start at c = a(j)*b(0) = a(j)
*/
        if ( ( cc = AT.pWorkSpace[inpointers+j] ) != 0 ) {
            lenc = *cc++; /* reduced length */
            i = 2*ABS(lenc); ccc = c;
            NCOPY(ccc,cc,i);
        }
        else { lenc = 0; }
        for ( i = MiN(j-1,maxinput); i > 0; i-- ) {
/*
            c -> c + a(i)*b(j-i)
*/
            if ( AT.pWorkSpace[inpointers+i] == 0
                 || AT.pWorkSpace[outpointers+j-i] == 0 ) {
            }
            else {
                if ( MulRat(BHEAD (UWORD *)(AT.pWorkSpace[inpointers+i]+1),AT.pWorkSpace[inpointers+i][0],
                 (UWORD *)(AT.pWorkSpace[outpointers+j-i]+1),AT.pWorkSpace[outpointers+j-i][0],
                 (UWORD *)c1,&lenc1) ) goto calcerror;
                if ( lenc == 0 ) {
                    cc = c; c = c1; c1 = cc;
                    lenc = lenc1;
                }
                else {
                    if ( AddRat(BHEAD (UWORD *)c,lenc,(UWORD *)c1,lenc1,(UWORD *)c2,&lenc2) )
                        goto calcerror;
                    cc = c; c = c2; c2 = cc;
                    lenc = lenc2;
                }
            }
        }
/*
        Copy c to the proper location
*/
        if ( lenc == 0 ) AT.pWorkSpace[outpointers+j] = 0;
        else {
            AT.pWorkSpace[outpointers+j] = w;
            *w++ = lenc;
            i = 2*ABS(lenc); ccc = c;
            NCOPY(w,ccc,i);
        }
    }
    AT.WorkPointer = w;
    TermFree(c2,"InvPoly");
    TermFree(c1,"InvPoly");
    TermFree(c ,"InvPoly");

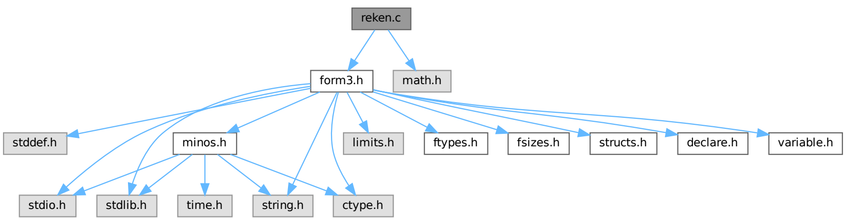
    return(outpointers);
onerror:
    MLOCK(ErrorMessageLock);
    MesPrint("Incorrect symbol field in InvPoly.");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(-1);
calcerror:
    MLOCK(ErrorMessageLock);
    MesPrint("Called from InvPoly.");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(-1);
}

/*
    #] InvPoly :
*/
```

4.118 reken.c File Reference

```
#include "form3.h"
#include <math.h>
```

Include dependency graph for reken.c:



Macros

- #define GCDMAX 3
- #define NEWTRICK 1
- #define COPYLONG(x1, nx1, x2, nx2) { int i; for(i=0;i<ABS(nx2);i++)x1[i]=x2[i];nx1=nx2; }
- #define WARMUP 6

Functions

- void Pack (UWORD ∗a, WORD ∗na, UWORD ∗b, WORD nb)
- void UnPack (UWORD ∗a, WORD na, WORD ∗denom, WORD ∗numer)
- WORD Mully (PHEAD UWORD ∗a, WORD ∗na, UWORD ∗b, WORD nb)
- WORD Divvy (PHEAD UWORD ∗a, WORD ∗na, UWORD ∗b, WORD nb)
- WORD AddRat (PHEAD UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- WORD MulRat (PHEAD UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- WORD DivRat (PHEAD UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- WORD Simplify (PHEAD UWORD ∗a, WORD ∗na, UWORD ∗b, WORD ∗nb)
- WORD AccumGCD (PHEAD UWORD ∗a, WORD ∗na, UWORD ∗b, WORD nb)
- int TakeRatRoot (UWORD ∗a, WORD ∗n, WORD power)
- WORD AddLong (UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- WORD AddPLon (UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- void SubPLon (UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- WORD MulLong (UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- WORD BigLong (UWORD ∗a, WORD na, UWORD ∗b, WORD nb)
- WORD DivLong (UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc, UWORD ∗d, WORD ∗nd)
- WORD RaisPow (PHEAD UWORD ∗a, WORD ∗na, UWORD b)
- void RaisPowCached (PHEAD WORD x, WORD n, UWORD ∗∗c, WORD ∗nc)
- WORD RaisPowMod (WORD x, WORD n, WORD m)
- int NormalModulus (UWORD ∗a, WORD ∗na)
- int MakeInverses (void)
- int GetModInverses (WORD m1, WORD m2, WORD ∗im1, WORD ∗im2)
- int GetLongModInverses (PHEAD UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗ia, WORD ∗nia, UWORD ∗ib, WORD ∗nib)
- WORD Product (UWORD ∗a, WORD ∗na, WORD b)
- UWORD Quotient (UWORD ∗a, WORD ∗na, WORD b)
- WORD Remain10 (UWORD ∗a, WORD ∗na)
- WORD Remain4 (UWORD ∗a, WORD ∗na)
- void PrtLong (UWORD ∗a, WORD na, UBYTE ∗s)
- WORD GetLong (UBYTE ∗s, UWORD ∗a, WORD ∗na)

- WORD GcdLong (PHEAD UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- WORD GetBinom (UWORD ∗a, WORD ∗na, WORD i1, WORD i2)
- WORD LcmLong (PHEAD UWORD ∗a, WORD na, UWORD ∗b, WORD nb, UWORD ∗c, WORD ∗nc)
- int TakeLongRoot (UWORD ∗a, WORD ∗n, WORD power)
- int MakeRational (WORD a, WORD m, WORD ∗b, WORD ∗c)
- int MakeLongRational (PHEAD UWORD ∗a, WORD na, UWORD ∗m, WORD nm, UWORD ∗b, WORD ∗nb)
- WORD CompCoef (WORD ∗term1, WORD ∗term2)
- WORD Modulus (WORD ∗term)
- WORD TakeModulus (UWORD ∗a, WORD ∗na, UWORD ∗cmodvec, WORD ncmod, WORD par)
- WORD TakeNormalModulus (UWORD ∗a, WORD ∗na, UWORD ∗c, WORD nc, WORD par)
- WORD MakeModTable (void)
- int Factorial (PHEAD WORD n, UWORD ∗a, WORD ∗na)
- int Bernoulli (WORD n, UWORD ∗a, WORD ∗na)
- WORD NextPrime (PHEAD WORD num)
- WORD Moebius (PHEAD WORD nn)
- void iniwranf (PHEAD0)
- UWORD wranf (PHEAD0)
- UWORD iranf (PHEAD UWORD imax)
- UBYTE ∗ PreRandom (UBYTE ∗s)

4.118.1 Detailed Description

This file contains the numerical routines. The arithmetic in FORM is normally over the rational numbers. Hence there are routines for dealing with integers and with rational of 'arbitrary precision' (within limits) There are also routines for that calculus modulus an integer. In addition there are the routines for factorials and Bernoulli numbers. The random number function is currently only for internal purposes.

Definition in file reken.c.

4.118.2 Macro Definition Documentation

4.118.2.1 GCDMAX

```
#define GCDMAX 3
```

Definition at line 49 of file reken.c.

4.118.2.2 NEWTRICK

```
#define NEWTRICK 1
```

Definition at line 51 of file reken.c.

4.118.2.3 COPYLONG

```
#define COPYLONG(
              x1,
              nx1,
              x2,
              nx2 ) { int i; for(i=0;i<ABS(nx2);i++)x1[i]=x2[i];nx1=nx2; }
```

Definition at line 2901 of file reken.c.

4.118.2.4 WARMUP

```
#define WARMUP 6
```

Definition at line 3801 of file reken.c.

4.118.3 Function Documentation

4.118.3.1 Pack()

```
void Pack (
            UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb )
```

Definition at line 62 of file reken.c.

4.118.3.2 UnPack()

```
void UnPack (
            UWORD * a,
            WORD na,
            WORD * denom,
            WORD * numer )
```

Definition at line 100 of file reken.c.

4.118.3.3 Mully()

```
WORD Mully (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb )
```

Definition at line 126 of file reken.c.

4.118.3.4 Divvy()

```
WORD Divvy (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb )
```

Definition at line 172 of file reken.c.

4.118.3.5 AddRat()

```
WORD AddRat (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 210 of file reken.c.

4.118.3.6 MulRat()

```
WORD MulRat (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 378 of file reken.c.

4.118.3.7 DivRat()

```
WORD DivRat (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 500 of file reken.c.

4.118.3.8 Simplify()

```
WORD Simplify (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD * nb )
```

Definition at line 530 of file reken.c.

4.118.3.9 AccumGCD()

```
WORD AccumGCD (
            PHEAD UWORD * a,
            WORD * na,
            UWORD * b,
            WORD nb )
```

Definition at line 647 of file reken.c.

4.118.3.10 TakeRatRoot()

```
int TakeRatRoot (
            UWORD * a,
            WORD * n,
            WORD power )
```

Definition at line 680 of file reken.c.

4.118.3.11 AddLong()

```
WORD AddLong (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 705 of file reken.c.

4.118.3.12 AddPLon()

```
WORD AddPLon (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 750 of file reken.c.

4.118.3.13 SubPLon()

```
void SubPLon (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 803 of file reken.c.

4.118.3.14 MulLong()

```
WORD MulLong (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 841 of file reken.c.

4.118.3.15 BigLong()

```
WORD BigLong (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb )
```

Definition at line 944 of file reken.c.

4.118.3.16 DivLong()

```
WORD DivLong (
            UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc,
            UWORD * d,
            WORD * nd )
```

Definition at line 972 of file reken.c.

4.118.3.17 RaisPow()

```
WORD RaisPow (
            PHEAD UWORD * a,
            WORD * na,
            UWORD b )
```

Definition at line 1228 of file reken.c.

4.118.3.18 RaisPowCached()

```
void RaisPowCached (
            PHEAD WORD x,
            WORD n,
            UWORD ** c,
            WORD * nc )
```

Computes power x^n and caches the value

4.118.4 Description

Calculates the power x^n and stores the results for caching purposes. The pointer c (i.e., the pointer, and not what it points to) is overwritten. What it points to should not be overwritten in the calling function.

4.118.5 Notes

- Caching is done in AT.small_power[]. This array is extended if necessary.

Definition at line 1296 of file reken.c.

Referenced by poly::divmod_heap(), poly::divmod_univar(), and poly::mul_heap().

4.118.5.1 RaisPowMod()

```
WORD RaisPowMod (
            WORD x,
            WORD n,
            WORD m )
```

Definition at line 1383 of file reken.c.

4.118.5.2 NormalModulus()

```
int NormalModulus (
            UWORD * a,
            WORD * na )
```

Brings a modular representation in the range -p/2 to +p/2 The return value tells whether anything was done. Routine made in the general modulus revamp of July 2008 (JV).

Definition at line 1403 of file reken.c.

Referenced by AddCoef(), and MergePatches().

4.118.5.3 MakeInverses()

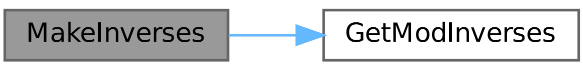
```
int MakeInverses (
            void )
```

Makes a table of inverses in modular calculus The modulus is in AC.cmod and AC.ncmod One should notice that the table of inverses can only be made if the modulus fits inside a single FORM word. Otherwise the table lookup becomes too difficult and the table too long.

Definition at line 1440 of file reken.c.

References GetModInverses().

Here is the call graph for this function:



4.118.5.4 GetModInverses()

```
int GetModInverses (
            WORD m1,
            WORD m2,
            WORD * im1,
            WORD * im2 )
```

Input m1 and m2, which are relative prime. determines a∗m1+b∗m2 = 1 (and 1 is the gcd of m1 and m2) then a∗m1 = 1 mod m2 and hence im1 = a. and b∗m2 = 1 mod m1 and hence im2 = b. Set m1 = 0∗m1+1∗m2 = a1∗m1+b1∗m2 m2 = 1∗m1+0∗m2 = a2∗m1+b2∗m2 If everything is OK, the return value is zero

Definition at line 1476 of file reken.c.

Referenced by poly::divmod_heap(), poly::divmod_univar(), MakeDollarMod(), MakeInverses(), MakeMod(), TakeContent(), and TakeSymbolContent().

4.118.5.5 GetLongModInverses()

```
int GetLongModInverses (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * ia,
            WORD * nia,
            UWORD * ib,
            WORD * nib )
```

Definition at line 1508 of file reken.c.

4.118.5.6 Product()

```
WORD Product (
            UWORD * a,
            WORD * na,
            WORD b )
```

Definition at line 1585 of file reken.c.

4.118.5.7 Quotient()

```
UWORD Quotient (
            UWORD * a,
            WORD * na,
            WORD b )
```

Definition at line 1620 of file reken.c.

4.118.5.8 Remain10()

```
WORD Remain10 (
        UWORD * a,
        WORD * na )
```

Definition at line 1659 of file reken.c.

4.118.5.9 Remain4()

```
WORD Remain4 (
        UWORD * a,
        WORD * na )
```

Definition at line 1686 of file reken.c.

4.118.5.10 PrtLong()

```
void PrtLong (
        UWORD * a,
        WORD na,
        UBYTE * s )
```

Definition at line 1712 of file reken.c.

4.118.5.11 GetLong()

```
WORD GetLong (
        UBYTE * s,
        UWORD * a,
        WORD * na )
```

Definition at line 1774 of file reken.c.

4.118.5.12 GcdLong()

```
WORD GcdLong (
        PHEAD UWORD * a,
        WORD na,
        UWORD * b,
        WORD nb,
        UWORD * c,
        WORD * nc )
```

Definition at line 2276 of file reken.c.

4.118.5.13 GetBinom()

```
WORD GetBinom (
            UWORD * a,
            WORD * na,
            WORD i1,
            WORD i2 )
```

Definition at line 2667 of file reken.c.

4.118.5.14 LcmLong()

```
WORD LcmLong (
            PHEAD UWORD * a,
            WORD na,
            UWORD * b,
            WORD nb,
            UWORD * c,
            WORD * nc )
```

Definition at line 2701 of file reken.c.

4.118.5.15 TakeLongRoot()

```
int TakeLongRoot (
            UWORD * a,
            WORD * n,
            WORD power )
```

Definition at line 2735 of file reken.c.

4.118.5.16 MakeRational()

```
int MakeRational (
            WORD a,
            WORD m,
            WORD * b,
            WORD * c )
```

Definition at line 2855 of file reken.c.

4.118.5.17 MakeLongRational()

```
int MakeLongRational (
            PHEAD UWORD * a,
            WORD na,
            UWORD * m,
            WORD nm,
            UWORD * b,
            WORD * nb )
```

Definition at line 2903 of file reken.c.

4.118.5.18 CompCoef()

```
WORD CompCoef (
            WORD * term1,
            WORD * term2 )
```

```
Routine takes a1 mod m1 and a2 mod m2 and returns a mod m1*m2 with
a mod m1 = a1 and a mod m2 = a2
```

Chinese remainder: a%(m1*m2) = q1*m1+a1 a%(m1*m2) = q2*m2+a2 Compute n1 such that (n1*m1)m2 is one Compute n2 such that (n2*m2)m1 is one Then (a1*n2*m2+a2*n1*m1)%(m1*m2) is a%(m1*m2)

Definition at line 3047 of file reken.c.

Referenced by Compare1().

4.118.5.19 Modulus()

```
WORD Modulus (
            WORD * term )
```

Definition at line 3102 of file reken.c.

4.118.5.20 TakeModulus()

```
WORD TakeModulus (
            UWORD * a,
            WORD * na,
            UWORD * cmodvec,
            WORD ncmod,
            WORD par )
```

Definition at line 3149 of file reken.c.

4.118.5.21 TakeNormalModulus()

```
WORD TakeNormalModulus (
            UWORD * a,
            WORD * na,
            UWORD * c,
            WORD nc,
            WORD par )
```

Definition at line 3321 of file reken.c.

4.118.5.22 MakeModTable()

```
WORD MakeModTable (
            void )
```

Definition at line 3363 of file reken.c.

4.118.5.23 Factorial()

```
int Factorial (
            PHEAD WORD n,
            UWORD * a,
            WORD * na )
```

Definition at line 3449 of file reken.c.

4.118.5.24 Bernoulli()

```
int Bernoulli (
            WORD n,
            UWORD * a,
            WORD * na )
```

Definition at line 3529 of file reken.c.

4.118.5.25 NextPrime()

```
WORD NextPrime (
            PHEAD WORD num )
```

Gives the next prime number in the list of prime numbers.

If the list isn't long enough we expand it. For ease in ParForm and because these lists shouldn't be very big we let each worker keep its own list.

The list is cut off at MAXPOWER, because we don't want to get into trouble that the power of a variable gets larger than the prime number.

Definition at line 3664 of file reken.c.

4.118.5.26 Moebius()

```
WORD Moebius (
            PHEAD WORD nn )
```

Definition at line 3728 of file reken.c.

4.118.5.27 iniwranf()

```
void iniwranf (
            PHEAD0 )
```

Definition at line 3819 of file reken.c.

4.118.5.28 wranf()

```
UWORD wranf (
            PHEAD0 )
```

Definition at line 3868 of file reken.c.

4.118.5.29 iranf()

```
UWORD iranf (
            PHEAD UWORD imax )
```

Definition at line 3887 of file reken.c.

4.118.5.30 PreRandom()

```
UBYTE * PreRandom (
            UBYTE * s )
```

Definition at line 3912 of file reken.c.

4.119 reken.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : reken.c
*/

#include "form3.h"
#include <math.h>

#ifdef WITHGMP
#include <gmp.h>
#define GMPSPREAD (GMP_LIMB_BITS/BITSINWORD)
#endif

#define GCDMAX 3

#define NEWTRICK 1
```

```
/*
    #] Includes :
    #[ RekenRational :
        #[ Pack :           void Pack(a,na,b,nb)

    Packs the contents of the numerator a and the denominator b into
    one normalized fraction a.

*/

void Pack(UWORD *a, WORD *na, UWORD *b, WORD nb)
{
    WORD c, sgn = 1, i;
    UWORD *to,*from;
    if ( (c = *na) == 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Caught a zero in Pack");
        MUNLOCK(ErrorMessageLock);
        return;
    }
    if ( nb == 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Division by zero in Pack");
        MUNLOCK(ErrorMessageLock);
        return;
    }
    if ( *na < 0 ) { sgn = -sgn; c = -c; }
    if ( nb < 0 ) { sgn = -sgn; nb = -nb; }
    *na = MaX(c,nb);
    to = a + c;
    i = *na - c;
    while ( --i >= 0 ) *to++ = 0;
    i = *na - nb;
    from = b;
    NCOPY(to,from,nb);
    while ( --i >= 0 ) *to++ = 0;
    if ( sgn < 0 ) *na = -*na;
}

/*
        #] Pack :
        #[ UnPack :         void UnPack(a,na,denom,numer)

    Determines the sizes of the numerator and the denominator in the
    normalized fraction a with length na.

*/

void UnPack(UWORD *a, WORD na, WORD *denom, WORD *numer)
{
    UWORD *pos;
    WORD i, sgn = na;
    if ( na < 0 ) { na = -na; }
    i = na;
    if ( i > 1 ) {      /* Find the respective leading words */
        a += i;
        a--;
        pos = a + i;
        while ( !(*a) ) { i--; a--; }
        while ( !(*pos) ) { na--; pos--; }
    }
    *denom = na;
    if ( sgn < 0 ) i = -i;
    *numer = i;
}

/*
        #] UnPack :
        #[ Mully :          WORD Mully(a,na,b,nb)

    Multiplies the rational a by the Long b.

*/

WORD Mully(PHEAD UWORD *a, WORD *na, UWORD *b, WORD nb)
{
    GETBIDENTITY
    UWORD *d, *e;
    WORD i, sgn = 1;
    WORD nd, ne, adenom, anumer;
    if ( !nb ) { *na = 0; return(0); }
    else if ( *b == 1 ) {
        if ( nb == 1 ) return(0);
        else if ( nb == -1 ) { *na = -*na; return(0); }
    }
    if ( *na < 0 ) { sgn = -sgn; *na = -*na; }
    if ( nb < 0 ) { sgn = -sgn; nb = -nb; }
```

```
    UnPack(a,*na,&adenom,&anumer);
    d = NumberMalloc("Mully"); e = NumberMalloc("Mully");
    for ( i = 0; i < nb; i++ ) { e[i] = *b++; }
    ne = nb;
    if ( Simplify(BHEAD a+*na,&adenom,e,&ne) ) goto MullyEr;
    if ( MulLong(a,anumer,e,ne,d,&nd) ) goto MullyEr;
    b = a+*na;
    for ( i = 0; i < *na; i++ ) { e[i] = *b++; }
    ne = adenom;
    *na = nd;
    b = d;
    *na = nd;
    for ( i = 0; i < *na; i++ ) { a[i] = *b++; }
    Pack(a,na,e,ne);
    if ( sgn < 0 ) *na = -*na;
    NumberFree(d,"Mully"); NumberFree(e,"Mully");
    return(0);
MullyEr:
    MLOCK(ErrorMessageLock);
    MesCall("Mully");
    MUNLOCK(ErrorMessageLock);
    NumberFree(d,"Mully"); NumberFree(e,"Mully");
    SETERROR(-1)
}

/*
        #] Mully :
        #[ Divvy :          WORD Divvy(a,na,b,nb)

    Divides the rational a by the Long b.

*/

WORD Divvy(PHEAD UWORD *a, WORD *na, UWORD *b, WORD nb)
{
    GETBIDENTITY
    UWORD *d,*e;
    WORD i, sgn = 1;
    WORD nd, ne, adenom, anumer;
    if ( !nb ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Division by zero in Divvy");
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    d = NumberMalloc("Divvy"); e = NumberMalloc("Divvy");
    if ( nb < 0 ) { sgn = -sgn; nb = -nb; }
    if ( *na < 0 ) { sgn = -sgn; *na = -*na; }
    UnPack(a,*na,&adenom,&anumer);
    for ( i = 0; i < nb; i++ ) { e[i] = *b++; }
    ne = nb;
    if ( Simplify(BHEAD a,&anumer,e,&ne) ) goto DivvyEr;
    if ( MulLong(a+*na,adenom,e,ne,d,&nd) ) goto DivvyEr;
    *na = anumer;
    Pack(a,na,d,nd);
    if ( sgn < 0 ) *na = -*na;
    NumberFree(d,"Divvy"); NumberFree(e,"Divvy");
    return(0);
DivvyEr:
    MLOCK(ErrorMessageLock);
    MesCall("Divvy");
    MUNLOCK(ErrorMessageLock);
    NumberFree(d,"Divvy"); NumberFree(e,"Divvy");
    SETERROR(-1)
}

/*
        #] Divvy :
        #[ AddRat :         WORD AddRat(a,na,b,nb,c,nc)
*/

WORD AddRat(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    GETBIDENTITY
    UWORD *d, *e, *f, *g;
    WORD nd, ne, nf, ng, adenom, anumer, bdenom, bnumer;
    if ( !na ) {
        WORD i;
        *nc = nb;
        if ( nb < 0 ) nb = -nb;
        nb *= 2;
        for ( i = 0; i < nb; i++ ) *c++ = *b++;
        return(0);
    }
    else if ( !nb ) {
        WORD i;
        *nc = na;
```

```
        if ( na < 0 ) na = -na;
        na *= 2;
        for ( i = 0; i < na; i++ ) *c++ = *a++;
        return(0);
    }
    else if ( b[1] == 1 && a[1] == 1 ) {
        if ( na == 1 ) {
            if ( nb == 1 ) {
                *c = *a + *b;
                c[1] = 1;
                if ( *c < *a ) { c[2] = 1; c[3] = 0; *nc = 2; }
                else { *nc = 1; }
                return(0);
            }
            else if ( nb == -1 ) {
                if ( *b > *a ) {
                    *c = *b - *a; *nc = -1;
                }
                else if ( *b < *a ) {
                    *c = *a - *b; *nc = 1;
                }
                else *nc = 0;
                c[1] = 1;
                return(0);
            }
        }
        else if ( na == -1 ){
            if ( nb == -1 ) {
                c[1] = 1;
                *c = *a + *b;
                if ( *c < *a ) { c[2] = 1; c[3] = 0; *nc = -2; }
                else { *nc = -1; }
                return(0);
            }
            else if ( nb == 1 ) {
                if ( *b > *a ) {
                    *c = *b - *a; *nc = 1;
                }
                else if ( *b < *a ) {
                    *c = *a - *b; *nc = -1;
                }
                else *nc = 0;
                c[1] = 1;
                return(0);
            }
        }
    }
    UnPack(a,na,&adenom,&anumer);
    UnPack(b,nb,&bdenom,&bnumer);
    if ( na < 0 ) na = -na;
    if ( nb < 0 ) nb = -nb;
    if ( na == 1 && nb == 1 ) {
        RLONG t1, t2, t3;
        t3 = ((RLONG)a[1])*((RLONG)b[1]);
        t1 = ((RLONG)a[0])*((RLONG)b[1]);
        t2 = ((RLONG)a[1])*((RLONG)b[0]);
        if ( ( anumer > 0 && bnumer > 0 ) || ( anumer < 0 && bnumer < 0 ) ) {
            if ( ( t1 = t1 + t2 ) < t2 ) {
                c[2] = 1;
                c[0] = (UWORD)t1;
                c[1] = (UWORD)(t1 » BITSINWORD);
                *nc = 3;
            }
            else {
                c[0] = (UWORD)t1;
                if ( ( c[1] = (UWORD)(t1 » BITSINWORD) ) != 0 ) *nc = 2;
                else *nc = 1;
            }
        }
        else {
            if ( t1 == t2 ) { *nc = 0; return(0); }
            if ( t1 > t2 ) {
                t1 -= t2;
            }
            else {
                t1 = t2 - t1;
                anumer = -anumer;
            }
            c[0] = (UWORD)t1;
            if ( ( c[1] = (UWORD)(t1 » BITSINWORD) ) != 0 ) *nc = 2;
            else *nc = 1;
        }
        if ( anumer < 0 ) *nc = -*nc;
        d = NumberMalloc("AddRat");
        d[0] = (UWORD)t3;
        if ( ( d[1] = (UWORD)(t3 » BITSINWORD) ) != 0 ) nd = 2;
        else nd = 1;
```

```
        if ( Simplify(BHEAD c,nc,d,&nd) ) goto AddRer1;
    }
/*
    else if ( a[na] == 1 && b[nb] == 1 && adenom == 1 && bdenom == 1 ) {
        if ( AddLong(a,na,b,nb,c,&nc) ) goto AddRer2;
        i = ABS(nc); d = c + i; *d++ = 1;
        while ( --i > 0 ) *d++ = 0 ;
        return(0);
    }
*/
    else {
        d = NumberMalloc("AddRat"); e = NumberMalloc("AddRat");
        f = NumberMalloc("AddRat"); g = NumberMalloc("AddRat");
        if ( GcdLong(BHEAD a+na,adenom,b+nb,bdenom,d,&nd) ) goto AddRer;
        if ( *d == 1 && nd == 1 ) nd = 0;
        if ( nd ) {
            if ( DivLong(a+na,adenom,d,nd,e,&ne,c,nc) ) goto AddRer;
            if ( DivLong(b+nb,bdenom,d,nd,f,&nf,c,nc) ) goto AddRer;
            if ( MulLong(a,anumer,f,nf,c,nc) ) goto AddRer;
            if ( MulLong(b,bnumer,e,ne,g,&ng) ) goto AddRer;
        }
        else {
            if ( MulLong(a+na,adenom,b,bnumer,c,nc) ) goto AddRer;
            if ( MulLong(b+nb,bdenom,a,anumer,g,&ng) ) goto AddRer;
        }
        if ( AddLong(c,*nc,g,ng,c,nc) ) goto AddRer;
        if ( !*nc ) {
            NumberFree(g,"AddRat"); NumberFree(f,"AddRat");
            NumberFree(e,"AddRat"); NumberFree(d,"AddRat");
            return(0);
        }
        if ( nd ) {
            if ( Simplify(BHEAD c,nc,d,&nd) ) goto AddRer;
            if ( MulLong(e,ne,d,nd,g,&ng) ) goto AddRer;
            if ( MulLong(g,ng,f,nf,d,&nd) ) goto AddRer;
        }
        else {
            if ( MulLong(a+na,adenom,b+nb,bdenom,d,&nd) ) goto AddRer;
        }
        NumberFree(g,"AddRat"); NumberFree(f,"AddRat"); NumberFree(e,"AddRat");
    }
    Pack(c,nc,d,nd);
    NumberFree(d,"AddRat");
    return(0);
AddRer:
    NumberFree(g,"AddRat"); NumberFree(f,"AddRat"); NumberFree(e,"AddRat");
AddRer1:
    NumberFree(d,"AddRat");
/* AddRer2: */
    MLOCK(ErrorMessageLock);
    MesCall("AddRat");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] AddRat :
        #[ MulRat :          WORD MulRat(a,na,b,nb,c,nc)

    Multiplies the rationals a and b. The Gcd of the individual
    pieces is divided out first to minimize the chances of spurious
    overflows.

*/

WORD MulRat(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    WORD i;
    WORD sgn = 1;
    if ( *b == 1 && b[1] == 1 ) {
        if ( nb == 1 ) {
            *nc = na;
            i = ABS(na); i *= 2;
            while ( --i >= 0 ) *c++ = *a++;
            return(0);
        }
        else if ( nb == -1 ) {
            *nc = - na;
            i = ABS(na); i *= 2;
            while ( --i >= 0 ) *c++ = *a++;
            return(0);
        }
    }
    if ( *a == 1 && a[1] == 1 ) {
        if ( na == 1 ) {
            *nc = nb;
            i = ABS(nb); i *= 2;
```

```
            while ( --i >= 0 ) *c++ = *b++;
            return(0);
        }
        else if ( na == -1 ) {
            *nc = - nb;
            i = ABS(nb); i *= 2;
            while ( --i >= 0 ) *c++ = *b++;
            return(0);
        }
    }
    if ( na < 0 ) { na = -na; sgn = -sgn; }
    if ( nb < 0 ) { nb = -nb; sgn = -sgn; }
    if ( !na || !nb ) { *nc = 0; return(0); }
    if ( na != 1 || nb != 1 ) {
        GETBIDENTITY
        UWORD *xd,*xe, *xf,*xg;
        WORD dden, dnumr, eden, enumr;
        UnPack(a,na,&dden,&dnumr);
        UnPack(b,nb,&eden,&enumr);
        xd = NumberMalloc("MulRat"); xf = NumberMalloc("MulRat");
        for ( i = 0; i < dnumr; i++ ) xd[i] = a[i];
        a += na;
        for ( i = 0; i < dden; i++ ) xf[i] = a[i];
        xe = NumberMalloc("MulRat"); xg = NumberMalloc("MulRat");
        for ( i = 0; i < enumr; i++ ) xe[i] = b[i];
        b += nb;
        for ( i = 0; i < eden; i++ ) xg[i] = b[i];
        if ( Simplify(BHEAD xd,&dnumr,xg,&eden) ||
             Simplify(BHEAD xe,&enumr,xf,&dden) ||
             MulLong(xd,dnumr,xe,enumr,c,nc) ||
             MulLong(xf,dden,xg,eden,xd,&dnumr) ) {
            MLOCK(ErrorMessageLock);
            MesCall("MulRat");
            MUNLOCK(ErrorMessageLock);
            NumberFree(xd,"MulRat"); NumberFree(xe,"MulRat"); NumberFree(xf,"MulRat");
NumberFree(xg,"MulRat");
            SETERROR(-1)
        }
        Pack(c,nc,xd,dnumr);
        NumberFree(xd,"MulRat"); NumberFree(xe,"MulRat"); NumberFree(xf,"MulRat");
NumberFree(xg,"MulRat");
    }
    else {
        UWORD y;
        UWORD a0,a1,b0,b1;
        RLONG xx;
        y = a[0]; b1=b[1];
        do { a0 = y % b1; y = b1; } while ( ( b1 = a0 ) != 0 );
        if ( y != 1 ) {
            a0 = a[0] / y;
            b1 = b[1] / y;
        }
        else {
            a0 = a[0];
            b1 = b[1];
        }
        y=b[0]; a1=a[1];
        do { b0 = y % a1; y = a1; } while ( ( a1 = b0 ) != 0 );
        if ( y != 1 ) {
            a1 = a[1] / y;
            b0 = b[0] / y;
        }
        else {
            a1 = a[1];
            b0 = b[0];
        }
        xx = ((RLONG)a0)*b0;
        if ( xx & AWORDMASK ) {
            *nc = 2;
            c[0] = (UWORD)xx;
            c[1] = (UWORD)(xx >> BITSINWORD);
            xx = ((RLONG)a1)*b1;
            c[2] = (UWORD)xx;
            c[3] = (UWORD)(xx >> BITSINWORD);
        }
        else {
            c[0] = (UWORD)xx;
            xx = ((RLONG)a1)*b1;
            if ( xx & AWORDMASK ) {
                c[1] = 0;
                c[2] = (UWORD)xx;
                c[3] = (UWORD)(xx >> BITSINWORD);
                *nc = 2;
            }
            else {
                c[1] = (UWORD)xx;
                *nc = 1;
```

```
            }
        }
    }
    if ( sgn < 0 ) *nc = -*nc;
    return(0);
}

/*
        #] MulRat :
        #[ DivRat :         WORD DivRat(a,na,b,nb,c,nc)

    Divides the rational a by the rational b.

*/

WORD DivRat(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    GETBIDENTITY
    WORD i, j;
    UWORD *xd,*xe,xx;
    if ( !nb ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Rational division by zero");
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    j = i = (nb >= 0)? nb: -nb;
    xd = b; xe = b + i;
    do { xx = *xd; *xd++ = *xe; *xe++ = xx; } while ( --j > 0 );
    j = MulRat(BHEAD a,na,b,nb,c,nc);
    xd = b; xe = b + i;
    do { xx = *xd; *xd++ = *xe; *xe++ = xx; } while ( --i > 0 );
    return(j);
}

/*
        #] DivRat :
        #[ Simplify :       WORD Simplify(a,na,b,nb)

    Determines the greatest common denominator of a and b and
    divides both by it. A possible sign is put in a. This is
    the simplification of the fraction a/b.

*/

WORD Simplify(PHEAD UWORD *a, WORD *na, UWORD *b, WORD *nb)
{
    GETBIDENTITY
    UWORD *x1,*x2,*x3;
    UWORD *x4;
    WORD n1,n2,n3,n4,sgn = 1;
    WORD i;
    UWORD *Siscrat5, *Siscrat6, *Siscrat7, *Siscrat8;
    if ( *na < 0 ) { *na = -*na; sgn = -sgn; }
    if ( *nb < 0 ) { *nb = -*nb; sgn = -sgn; }
    Siscrat5 = NumberMalloc("Simplify"); Siscrat6 = NumberMalloc("Simplify");
    Siscrat7 = NumberMalloc("Simplify"); Siscrat8 = NumberMalloc("Simplify");
    x1 = Siscrat8; x2 = Siscrat7;
    if ( *nb == 1 ) {
        x3 = Siscrat6;
        if ( DivLong(a,*na,b,*nb,x1,&n1,x2,&n2) ) goto SimpErr;
        if ( !n2 ) {
            for ( i = 0; i < n1; i++ ) *a++ = *x1++;
            *na = n1;
            *b = 1;
        }
        else {
            UWORD y1, y2, y3;
            y2 = *b;
            y3 = *x2;
            do { y1 = y2 % y3; y2 = y3; } while ( ( y3 = y1 ) != 0 );
            if ( ( *x2 = y2 ) != 1 ) {
                *b /= y2;
                if ( DivLong(a,*na,x2,(WORD)1,x1,&n1,x3,&n3) ) goto SimpErr;
                for ( i = 0; i < n1; i++ ) *a++ = *x1++;
                *na = n1;
            }
        }
    }
#ifdef NEWTRICK
    else if ( *na >= GCDMAX && *nb >= GCDMAX ) {
        n1 = i = *na; x3 = a;
        NCOPY(x1,x3,i);
        x3 = b; n2 = i = *nb;
        NCOPY(x2,x3,i);
        x4 = Siscrat5;
        x2 = Siscrat6;
```

```
        x3 = Siscrat7;
        if ( GcdLong(BHEAD Siscrat8,n1,Siscrat7,n2,x2,&n3) ) goto SimpErr;
        n2 = n3;
        if ( *x2 != 1 || n2 != 1 ) {
            DivLong(a,*na,x2,n2,x1,&n1,x4,&n4);
            *na = i = n1;
            NCOPY(a,x1,i);
            DivLong(b,*nb,x2,n2,x3,&n3,x4,&n4);
            *nb = i = n3;
            NCOPY(b,x3,i);
        }
    }
#endif
    else {
        x4 = Siscrat5;
        n1 = i = *na; x3 = a;
        NCOPY(x1,x3,i);
        x3 = b; n2 = i = *nb;
        NCOPY(x2,x3,i);
        x1 = Siscrat8; x2 = Siscrat7; x3 = Siscrat6;
        for(;;){
            if ( DivLong(x1,n1,x2,n2,x4,&n4,x3,&n3) ) goto SimpErr;
            if ( !n3 ) break;
            if ( n2 == 1 ) {
                while ( ( *x1 = (*x2) % (*x3) ) != 0 ) { *x2 = *x3; *x3 = *x1; }
                *x2 = *x3;
                break;
            }
            if ( DivLong(x2,n2,x3,n3,x4,&n4,x1,&n1) ) goto SimpErr;
            if ( !n1 ) { x2 = x3; n2 = n3; x3 = Siscrat7; break; }
            if ( n3 == 1 ) {
                while ( ( *x2 = (*x3) % (*x1) ) != 0 ) { *x3 = *x1; *x1 = *x2; }
                *x2 = *x1;
                n2 = 1;
                break;
            }
            if ( DivLong(x3,n3,x1,n1,x4,&n4,x2,&n2) ) goto SimpErr;
            if ( !n2 ) { x2 = x1; n2 = n1; x1 = Siscrat7; break; }
            if ( n1 == 1 ) {
                while ( ( *x3 = (*x1) % (*x2) ) != 0 ) { *x1 = *x2; *x2 = *x3; }
                break;
            }
        }
        if ( *x2 != 1 || n2 != 1 ) {
            DivLong(a,*na,x2,n2,x1,&n1,x4,&n4);
            *na = i = n1;
            NCOPY(a,x1,i);
            DivLong(b,*nb,x2,n2,x3,&n3,x4,&n4);
            *nb = i = n3;
            NCOPY(b,x3,i);
        }
    }
    if ( sgn < 0 ) *na = -*na;
    NumberFree(Siscrat5,"Simplify"); NumberFree(Siscrat6,"Simplify");
    NumberFree(Siscrat7,"Simplify"); NumberFree(Siscrat8,"Simplify");
    return(0);
SimpErr:
    MLOCK(ErrorMessageLock);
    MesCall("Simplify");
    MUNLOCK(ErrorMessageLock);
    NumberFree(Siscrat5,"Simplify"); NumberFree(Siscrat6,"Simplify");
    NumberFree(Siscrat7,"Simplify"); NumberFree(Siscrat8,"Simplify");
    SETERROR(-1)
}

/*
        #] Simplify :
        #[ AccumGCD :       WORD AccumGCD(PHEAD a,na,b,nb)

        Routine takes the rational GCD of the fractions in a and b and
        replaces a by the GCD of the two.
        The rational GCD is defined as the rational that consists of
        the GCD of the numerators divided by the GCD of the denominators
*/

WORD AccumGCD(PHEAD UWORD *a, WORD *na, UWORD *b, WORD nb)
{
    GETBIDENTITY
    WORD nna,nnb,numa,numb,dena,denb,numc,denc;
    UWORD *GCDbuffer = NumberMalloc("AccumGCD");
    int i;
    nna = *na; if ( nna < 0 ) nna = -nna; nna = (nna-1)/2;
    nnb = nb;  if ( nnb < 0 ) nnb = -nnb; nnb = (nnb-1)/2;
    UnPack(a,nna,&dena,&numa);
    UnPack(b,nnb,&denb,&numb);
    if ( GcdLong(BHEAD a,numa,b,numb,GCDbuffer,&numc) ) goto AccErr;
    numa = numc;
```

```
    for ( i = 0; i < numa; i++ ) a[i] = GCDbuffer[i];
    if ( GcdLong(BHEAD a+nna,dena,b+nnb,denb,GCDbuffer,&denc) ) goto AccErr;
    dena = denc;
    for ( i = 0; i < dena; i++ ) a[i+nna] = GCDbuffer[i];
    Pack(a,&numa,a+nna,dena);
    *na = INCLENG(numa);
    NumberFree(GCDbuffer,"AccumGCD");
    return(0);
AccErr:
    MLOCK(ErrorMessageLock);
    MesCall("AccumGCD");
    MUNLOCK(ErrorMessageLock);
    NumberFree(GCDbuffer,"AccumGCD");
    SETERROR(-1)
}

/*
        #] AccumGCD :
        #[ TakeRatRoot:
*/

int TakeRatRoot(UWORD *a, WORD *n, WORD power)
{
    WORD numer,denom, nn;
    if ( ( power & 1 ) == 0 && *n < 0 ) return(1);
    if ( ABS(*n) == 1 && a[0] == 1 && a[1] == 1 ) return(0);
    nn = ABS(*n);
    UnPack(a,nn,&denom,&numer);
    if ( TakeLongRoot(a+nn,&denom,power) ) return(1);
    if ( TakeLongRoot(a,&numer,power) ) return(1);
    Pack(a,&numer,a+nn,denom);
    if ( *n < 0 ) *n = -numer;
    else          *n = numer;
    return(0);
}

/*
        #] TakeRatRoot:
    #] RekenRational :
    #[ RekenLong :
        #[ AddLong :          WORD AddLong(a,na,b,nb,c,nc)

    Long addition. Uses addition and subtraction of positive numbers.

*/

WORD AddLong(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    WORD sgn, res;
    if ( na < 0 ) {
        if ( nb < 0 ) {
            if ( AddPLon(a,-na,b,-nb,c,nc) ) return(-1);
            *nc = -*nc;
            return(0);
        }
        else {
            na = -na;
            sgn = -1;
        }
    }
    else {
        if ( nb < 0 ) {
            nb = -nb;
            sgn = 1;
        }
        else { return( AddPLon(a,na,b,nb,c,nc) ); }
    }
    if ( ( res = BigLong(a,na,b,nb) ) > 0 ) {
        SubPLon(a,na,b,nb,c,nc);
        if ( sgn < 0 ) *nc = -*nc;
    }
    else if ( res < 0 ) {
        SubPLon(b,nb,a,na,c,nc);
        if ( sgn > 0 ) *nc = -*nc;
    }
    else {
        *nc = 0;
        *c = 0;
    }
    return(0);
}

/*
        #] AddLong :
        #[ AddPLon :          WORD AddPLon(a,na,b,nb,c,nc)

    Adds two long integers a and b and puts the result in c.
```

```
    The length of a and b are na and nb. The length of c is returned in nc.
    c can be a or b.
*/

WORD AddPLon(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    UWORD carry = 0, e, nd = 0;
    while ( na && nb ) {
        e = *a;
        *c = e + *b + carry;
        if ( carry ) {
            if ( e < *c ) carry = 0;
        }
        else {
            if ( e > *c ) carry = 1;
        }
        a++; b++; c++; nd++; na--; nb--;
    }
    while ( na ) {
        if ( carry ) {
            *c = *a++ + carry;
            if ( *c++ ) carry = 0;
        }
        else *c++ = *a++;
        nd++; na--;
    }
    while ( nb ) {
        if ( carry ) {
            *c = *b++ + carry;
            if ( *c++ ) carry = 0;
        }
        else *c++ = *b++;
        nd++; nb--;
    }
    if ( carry ) {
        nd++;
        if ( nd > (UWORD)AM.MaxTal ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Overflow in addition");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        *c++ = carry;
    }
    *nc = nd;
    return(0);
}

/*
        #] AddPLon :
        #[ SubPLon :        void SubPLon(a,na,b,nb,c,nc)

    Subtracts b from a. Assumes that a > b. Result in c.
    c can be a or b.

*/

void SubPLon(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    UWORD borrow = 0, e, nd = 0;
    while ( nb ) {
        e = *a;
        if ( borrow ) {
            *c = e - *b - borrow;
            if ( *c < e ) borrow = 0;
        }
        else {
            *c = e - *b;
            if ( *c > e ) borrow = 1;
        }
        a++; b++; c++; na--; nb--; nd++;
    }
    while ( na ) {
        if ( borrow ) {
            if ( *a ) { *c++ = *a++ - 1; borrow = 0; }
            else { *c++ = (UWORD)(-1); a++; }
        }
        else *c++ = *a++;
        na--; nd++;
    }
    while ( nd && !*--c ) { nd--; }
    *nc = (WORD)nd;
}

/*
        #] SubPLon :
        #[ MulLong :        WORD MulLong(a,na,b,nb,c,nc)
```

```

    Does a Long multiplication. Assumes that WORD is half the size
    of a LONG to work out the scheme! The number of operations is
    the canonical na*nm multiplications.
    c should not overlap with a or b.

*/

WORD MulLong(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    WORD sgn = 1;
    UWORD i, *ic, *ia;
    RLONG t, bb;
    if ( !na || !nb ) { *nc = 0; return(0); }
    if ( na < 0 ) { na = -na; sgn = -sgn; }
    if ( nb < 0 ) { nb = -nb; sgn = -sgn; }
    *nc = i = na + nb;
    if ( i > (UWORD)(AM.MaxTal+1) ) goto MulLov;
    ic = c;
/*
    #[ GMP stuff :
*/
#ifdef WITHGMP
    if (na > 3 && nb > 3) {
/*      mp_limb_t res;  */
        UWORD *to, *from;
        int j;
        GETIDENTITY
        UWORD *DLscrat9 = NumberMalloc("MulLong"), *DLscratA = NumberMalloc("MulLong"), *DLscratB =
NumberMalloc("MulLong");
#if ( GMPSPREAD != 1 )
        if ( na & 1 ) {
            from = a; a = to = DLscrat9; j = na; NCOPY(to, from, j);
            a[na++] = 0;
            ++*nc;
        } else
#endif
        if ( (LONG)a & (sizeof(mp_limb_t)-1) ) {
            from = a; a = to = DLscrat9; j = na; NCOPY(to, from, j);
        }

#if ( GMPSPREAD != 1 )
        if ( nb & 1 ) {
            from = b; b = to = DLscratA; j = nb; NCOPY(to, from, j);
            b[nb++] = 0;
            ++*nc;
        } else
#endif
        if ( (LONG)b & (sizeof(mp_limb_t)-1) ) {
            from = b; b = to = DLscratA; j = nb; NCOPY(to, from, j);
        }

        if ( ( *nc > (WORD)i ) || ( (LONG)c & (LONG)(sizeof(mp_limb_t)-1) ) ) {
            ic = DLscratB;
        }
        if ( na < nb ) {
            /* res = */
            mpn_mul((mp_ptr)ic, (mp_srcptr)b, nb/GMPSPREAD, (mp_srcptr)a, na/GMPSPREAD);
        } else {
            /* res = */
            mpn_mul((mp_ptr)ic, (mp_srcptr)a, na/GMPSPREAD, (mp_srcptr)b, nb/GMPSPREAD);
        }
        while ( ic[i-1] == 0 ) i--;
        *nc = i;
/*
        if ( res == 0 ) *nc -= GMPSPREAD;
        else if ( res <= WORDMASK ) --*nc;
*/
        if ( ic != c ) {
            j = *nc; NCOPY(c, ic, j);
        }
        if ( sgn < 0 ) *nc = -(*nc);
        NumberFree(DLscrat9,"MulLong"); NumberFree(DLscratA,"MulLong");
NumberFree(DLscratB,"MulLong");
        return(0);
    }
#endif
/*
    #] GMP stuff :
*/
    do { *ic++ = 0; } while ( --i > 0 );
    do {
        ia = a;
        ic = c++;
        t = 0;
        i = na;
        bb = (RLONG)(*b++);
```

```
        do {
            t = (*ia++) * bb + t + *ic;
            *ic++ = (WORD)t;
            t »= BITSINWORD;        /* should actually be a swap */
        } while ( --i > 0 );
        if ( t ) *ic = (UWORD)t;
    } while ( --nb > 0 );
    if ( !*ic ) (*nc)--;
    if ( *nc > AM.MaxTal ) goto MulLov;
    if ( sgn < 0 ) *nc = -(*nc);
    return(0);
MulLov:
    MLOCK(ErrorMessageLock);
    MesPrint("Overflow in Multiplication");
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] MulLong :
        #[ BigLong :        WORD BigLong(a,na,b,nb)

    Returns > 0 if a > b, < 0 if b > a and 0 if a == b

*/

WORD BigLong(UWORD *a, WORD na, UWORD *b, WORD nb)
{
    a += na;
    b += nb;
    while ( na && !*--a ) na--;
    while ( nb && !*--b ) nb--;
    if ( nb < na ) return(1);
    if ( nb > na ) return(-1);
    while ( --na >= 0 ) {
        if ( *a > *b ) return(1);
        else if ( *b > *a ) return(-1);
        a--; b--;
    }
    return(0);
}

/*
        #] BigLong :
        #[ DivLong :        WORD DivLong(a,na,b,nb,c,nc,d,nd)

    This is the long division which knows a couple of exceptions.
    It uses therefore a recursive call for the renormalization.
    The quotient comes in c and the remainder in d.
    d may be overlapping with b. It may also be identical to a.
    c should not overlap with a, but it can overlap with b.

*/

WORD DivLong(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c,
             WORD *nc, UWORD *d, WORD *nd)
{
    WORD sgna = 1, sgnb = 1, ne, nf, ng, nh;
    WORD i, ni;
    UWORD *w1, *w2;
    RLONG t, v;
    UWORD *e, *f, *ff, *g, norm, estim;
#ifdef WITHGMP
    UWORD *DLscrat9, *DLscratA, *DLscratB, *DLscratC;
#endif
    RLONG esthelp;
    if ( !nb ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Division by zero");
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    if ( !na ) { *nc = *nd = 0; return(0); }
    if ( na < 0 ) { sgna = -sgna; na = -na; }
    if ( nb < 0 ) { sgnb = -sgnb; nb = -nb; }
    if ( na < nb ) {
        for ( i = 0; i < na; i++ ) *d++ = *a++;
        *nd = na;
        *nc = 0;
    }
    else if ( nb == na && ( i = BigLong(b,nb,a,na) ) >= 0 ) {
        if ( i > 0 ) {
            for ( i = 0; i < na; i++ ) *d++ = *a++;
            *nd = na;
            *nc = 0;
        }
        else {
```

```
            *c = 1;
            *nc = 1;
            *nd = 0;
        }
    }
    else if ( nb == 1 ) {
        if ( *b == 1 ) {
            for ( i = 0; i < na; i++ ) *c++ = *a++;
            *nc = na;
            *nd = 0;
        }
        else {
            w1 = a+na;
            *nc = ni = na;
            *nd = 1;
            w2 = c+ni;
            v = (RLONG)(*b);
            t = (RLONG)(*--w1);
            while ( --ni >= 0 ) {
                *--w2 = t / v;
                t -= v * (*w2);
                if ( ni ) {
                    t «= BITSINWORD;
                    t += *--w1;
                }
            }
            if ( ( *d = (UWORD)t ) == 0 ) *nd = 0;
            if ( !*(c+na-1) ) (*nc)--;
        }
    }
    else {
        GETIDENTITY
/*
        #[ GMP stuff :

        We start with copying a and b.
        Then we make space for c and d.
        Next we call mpn_tdiv_qr
        We adjust sizes and copy to c and d if needed.
        Finally the signs are settled.
*/
#ifdef WITHGMP
        if ( na > 4 && nb > 3 ) {
          UWORD *ic, *id, *to, *from;
          int j = na - nb;
          DLscrat9 = NumberMalloc("DivLong"); DLscratA = NumberMalloc("DivLong");
          DLscratB = NumberMalloc("DivLong"); DLscratC = NumberMalloc("DivLong");

#if ( GMPSPREAD != 1 )
          if ( na & 1 ) {
            from = a; a = to = DLscrat9; i = na; NCOPY(to, from, i);
            a[na++] = 0;
          } else
#endif
          if ( (LONG)a & (sizeof(mp_limb_t)-1) ) {
            from = a; a = to = DLscrat9; i = na; NCOPY(to, from, i);
          }

#if ( GMPSPREAD != 1 )
          if ( nb & 1 ) {
            from = b; b = to = DLscratA; i = nb; NCOPY(to, from, i);
            b[nb++] = 0;
          } else
#endif
          if ( ( (LONG)b & (sizeof(mp_limb_t)-1) ) != 0 ) {
            from = b; b = to = DLscratA; i = nb; NCOPY(to, from, i);
          }
#if ( GMPSPREAD != 1 )
/*
          Not recognizing this case was a nasty and extremely rare bug.
          In the case that c == a and na is odd and nc == na and b
          still needs to be used, the least significant UWORD of b got
          overwritten by zero. (22-mar-2023)
*/
          ic = DLscratB; id = DLscratC;
#else
          if ( ( (LONG)c & (sizeof(mp_limb_t)-1) ) != 0 ) ic = DLscratB;
          else                                            ic = c;

          if ( ( (LONG)d & (sizeof(mp_limb_t)-1) ) != 0 ) id = DLscratC;
          else                                            id = d;
#endif
          mpn_tdiv_qr((mp_limb_t *)ic,(mp_limb_t *)id,(mp_size_t)0,
            (const mp_limb_t *)a,(mp_size_t)(na/GMPSPREAD),
            (const mp_limb_t *)b,(mp_size_t)(nb/GMPSPREAD));
          while ( j >= 0 && ic[j] == 0 ) j--;
          j++; *nc = j;
```

```
            if ( c != ic ) { NCOPY(c,ic,j); }
            j = nb-1;
            while ( j >= 0 && id[j] == 0 ) j--;
            j++; *nd = j;
            if ( d != id ) { NCOPY(d,id,j); }
            if ( sgna < 0 ) { *nc = -(*nc); *nd = -(*nd); }
            if ( sgnb < 0 ) { *nc = -(*nc); }
            NumberFree(DLscrat9,"DivLong"); NumberFree(DLscratA,"DivLong");
            NumberFree(DLscratB,"DivLong"); NumberFree(DLscratC,"DivLong");
            return(0);
        }
#endif
/*
        #] GMP stuff :
*/
        /* Start with normalization operation */

        e = NumberMalloc("DivLong"); f = NumberMalloc("DivLong"); g = NumberMalloc("DivLong");
        if ( b[nb-1] == (FULLMAX-1) ) norm = 1;
        else {
            norm = (UWORD)(((ULONG)FULLMAX) / (ULONG)((b[nb-1]+1L)));
        }
        f[na] = 0;
        if ( MulLong(b,nb,&norm,1,e,&ne) ||
             MulLong(a,na,&norm,1,f,&nf) ) {
            NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
            return(-1);
        }
        if ( BigLong(f+nf-ne,ne,e,ne) >= 0 ) {
            SubPLon(f+nf-ne,ne,e,ne,f+nf-ne,&nh);
            w1 = c + (nf-ne);
            *nc = nf-ne+1;
        }
        else {
            nh = ne;
            *nc = nf-ne;
            w1 = 0;
        }
        w2 = c; i = *nc; do { *w2++ = 0; } while ( --i > 0 );
        nf = na;
        ni = nf-ne;
        esthelp = (RLONG)(e[ne-1]) + 1L;
        while ( nf >= ne ) {
            if ( (WORD)esthelp == 0 ) {
                estim = (WORD)(((((RLONG)(f[nf]))«BITSINWORD)+f[nf-1])»BITSINWORD);
            }
            else {
                estim = (WORD)(((((RLONG)(f[nf]))«BITSINWORD)+f[nf-1])/esthelp);
            }
            /* This estimate may be up to two too small */
            if ( estim ) {
                MulLong(e,ne,&estim,1,g,&ng);
                nh = ne + 1; if ( !f[ni+ne] ) nh--;
                SubPLon(f+ni,nh,g,ng,f+ni,&nh);
            }
            else {
                w2 = f+ni+ne; nh = ne+1;
                while ( ( nh > 0 ) && !*w2 ) { nh--; w2--; }
            }
            if ( BigLong(f+ni,nh,e,ne) >= 0 ) {
                estim++;
                SubPLon(f+ni,nh,e,ne,f+ni,&nh);
                if ( BigLong(f+ni,nh,e,ne) >= 0 ) {
                    estim++;
                    SubPLon(f+ni,nh,e,ne,f+ni,&nh);
                    if ( BigLong(f+ni,nh,e,ne) >= 0 ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Problems in DivLong");
                        AO.OutSkip = 3;
                        FiniLine();
                        i = na;
                        while ( --i >= 0 ) { TalToLine((UWORD)(*a++)); TokenToLine((UBYTE *)"  "); }
                        FiniLine();
                        i = nb;
                        while ( --i >= 0 ) { TalToLine((UWORD)(*b++)); TokenToLine((UBYTE *)"  "); }
                        AO.OutSkip = 0;
                        FiniLine();
                        MUNLOCK(ErrorMessageLock);
                        NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
                        return(-1);
                    }
                }
            }
            c[ni] = estim;
            nf--;
            ni--;
        }
```

```
        if ( w1 ) *w1 = 1;

        /* Finish with the renormalization operation */

        if ( nh > 0 ) {
            if ( norm == 1 ) {
                *nd = i = nh; ff = f;
                NCOPY(d,ff,i);
            }
            else {
                w1 = f+nh;
                *nd = ni = nh;
                w2 = d+ni;
                v = norm;
                t = (RLONG)(*--w1);
                while ( --ni >= 0 ) {
                    *--w2 = t / v;
                    t -= v * (*w2);
                    if ( ni ) {
                        t «= BITSINWORD;
                        t += *--w1;
                    }
                }
                if ( t ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Error in DivLong");
                    MUNLOCK(ErrorMessageLock);
                    NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
                    return(-1);
                }
                if ( !*(d+nh-1) ) (*nd)--;
            }
        }
        else { *nd = 0; }
        NumberFree(e,"DivLong"); NumberFree(f,"DivLong"); NumberFree(g,"DivLong");
    }
    if ( sgna < 0 ) { *nc = -(*nc); *nd = -(*nd); }
    if ( sgnb < 0 ) { *nc = -(*nc); }
    return(0);
}

/*
        #] DivLong :
        #[ RaisPow :        WORD RaisPow(a,na,b)

    Raises a to the power b. a is a Long integer and b >= 0.
    The method that is used works with a bitdecomposition of b.
*/

WORD RaisPow(PHEAD UWORD *a, WORD *na, UWORD b)
{
    GETBIDENTITY
    WORD i, nu;
    UWORD *it, *iu, c;
    UWORD *is, *iss;
    WORD ns, nt, nmod;
    nmod = ABS(AN.ncmod);
    if ( !*na || ( ( *na == 1 ) && ( *a == 1 ) ) ) return(0);
    if ( !b ) { *na=1; *a=1; return(0); }
    is = NumberMalloc("RaisPow");
    it = NumberMalloc("RaisPow");
    for ( i = 0; i < ABS(*na); i++ ) is[i] = a[i];
    ns = *na;
    c = b;
    for ( i = 0; i < BITSINWORD; i++ ) {
        if ( !c ) break;
        c /= 2;
    }
    i--;
    c = 1 « i;
    while ( --i >= 0 ) {
        c /= 2;
        if(MulLong(is,ns,is,ns,it,&nt)) goto RaisOvl;
        if ( b & c ) {
            if ( MulLong(it,nt,a,*na,is,&ns) ) goto RaisOvl;
        }
        else {
            iu = is; is = it; it = iu;
            nu = ns; ns = nt; nt = nu;
        }
        if ( nmod != 0 ) {
            if ( DivLong(is,ns,(UWORD *)AC.cmod,nmod,it,&nt,is,&ns) ) goto RaisOvl;
        }
    }
    if ( ( nmod != 0 ) && ( ( AC.modmode & POSNEG ) != 0 ) ) {
        NormalModulus(is,&ns);
    }
```

```
    if ( ( *na = i = ns ) != 0 ) { iss = is; i=ABS(i); NCOPY(a,iss,i); }
    NumberFree(is,"RaisPow"); NumberFree(it,"RaisPow");
    return(0);
RaisOvl:
    MLOCK(ErrorMessageLock);
    MesCall("RaisPow");
    MUNLOCK(ErrorMessageLock);
    NumberFree(is,"RaisPow"); NumberFree(it,"RaisPow");
    SETERROR(-1)
}

/*
        #] RaisPow :
        #[ RaisPowCached :
*/

void RaisPowCached (PHEAD WORD x, WORD n, UWORD **c, WORD *nc) {

    int i,j;
    WORD new_small_power_maxx, new_small_power_maxn, ID;
    WORD *new_small_power_n;
    UWORD **new_small_power;

    /* check whether to extend the array */
    if (x>=AT.small_power_maxx || n>=AT.small_power_maxn) {

        new_small_power_maxx = AT.small_power_maxx;
        if (x>=AT.small_power_maxx)
            new_small_power_maxx = MaX(2*AT.small_power_maxx, x+1);

        new_small_power_maxn = AT.small_power_maxn;
        if (n>=AT.small_power_maxn)
            new_small_power_maxn = MaX(2*AT.small_power_maxn, n+1);

        new_small_power_n = (WORD*)
Malloc1(new_small_power_maxx*new_small_power_maxn*sizeof(WORD),"RaisPowCached");
        new_small_power = (UWORD **) Malloc1(new_small_power_maxx*new_small_power_maxn*sizeof(UWORD
*),"RaisPowCached");

        for (i=0; i<new_small_power_maxx * new_small_power_maxn; i++) {
            new_small_power_n[i] = 0;
            new_small_power  [i] = NULL;
        }

        for (i=0; i<AT.small_power_maxx; i++)
            for (j=0; j<AT.small_power_maxn; j++) {
                new_small_power_n[i*new_small_power_maxn+j] =
AT.small_power_n[i*AT.small_power_maxn+j];
                new_small_power  [i*new_small_power_maxn+j] = AT.small_power
[i*AT.small_power_maxn+j];
            }

        if (AT.small_power_n != NULL) {
            M_free(AT.small_power_n,"RaisPowCached");
            M_free(AT.small_power,"RaisPowCached");
        }

        AT.small_power_maxx = new_small_power_maxx;
        AT.small_power_maxn = new_small_power_maxn;
        AT.small_power_n    = new_small_power_n;
        AT.small_power      = new_small_power;
    }

    /* check whether the results is already calculated */
    ID = x * AT.small_power_maxn + n;

    if (AT.small_power[ID] == NULL) {
#ifdef OLDRAISPOWCACHED
        AT.small_power[ID] = NumberMalloc("RaisPowCached");
        AT.small_power_n[ID] = 1;
        AT.small_power[ID][0] = x;
        RaisPow(BHEAD AT.small_power[ID],&AT.small_power_n[ID],n);
#else
        UWORD *c = NumberMalloc("RaisPowCached");
        WORD i, k = 1;
        c[0] = x;
        RaisPow(BHEAD c,&k,n);
/*
        And now get the proper amount.
*/
        if ( AT.InNumMem < k ) {    /* We should start a new buffer */
            AT.InNumMem = 5*AM.MaxTal;
            AT.NumMem = (UWORD *)Malloc1(AT.InNumMem*sizeof(UWORD),"RaisPowCached");
/*
            MesPrint("  Got an extra %l UWORDS in RaisPowCached",AT.InNumMem);
*/
        }
```

```
        for ( i = 0; i < k; i++ ) AT.NumMem[i] = c[i];
        AT.small_power[ID] = AT.NumMem;
        AT.small_power_n[ID] = k;
        AT.NumMem += k;
        AT.InNumMem -= k;
        NumberFree(c,"RaisPowCached");
#endif
    }

    /* return the result */
    *c  = AT.small_power[ID];
    *nc = AT.small_power_n[ID];
}

/*
        #] RaisPowCached :
        #[ RaisPowMod :

   Computes the power x^n mod m
 */
WORD RaisPowMod (WORD x, WORD n, WORD m) {
    LONG y=1, z=x;
    while (n) {
        if (n&1) { y*=z; y%=m; }
        z*=z; z%=m;
        n /= 2;
    }
    return (WORD)y;
}

/*
    #] RaisPowMod :
        #[ NormalModulus :  int NormalModulus(UWORD *a,WORD *na)
*/
int NormalModulus(UWORD *a,WORD *na)
{
    WORD n;
    if ( AC.halfmod == 0 ) {
      LOCK(AC.halfmodlock);
      if ( AC.halfmod == 0 ) {
        UWORD two[1],remain[1];
        WORD dummy;
        two[0] = 2;
        AC.halfmod = (UWORD *)Malloc1((ABS(AC.ncmod))*sizeof(UWORD),"halfmod");
        DivLong((UWORD *)AC.cmod,(ABS(AC.ncmod)),two,1
                ,(UWORD *)AC.halfmod,&(AC.nhalfmod),remain,&dummy);
      }
      UNLOCK(AC.halfmodlock);
    }
    n = ABS(*na);
    if ( BigLong(a,n,AC.halfmod,AC.nhalfmod) > 0 ) {
        SubPLon((UWORD *)AC.cmod,(ABS(AC.ncmod)),a,n,a,&n);
        if ( *na > 0 ) { *na = -n; }
        else { *na = n; }
        return(1);
    }
    return(0);
}

/*
        #] NormalModulus :
        #[ MakeInverses :
*/
int MakeInverses(void)
{
    WORD n = AC.cmod[0], i, inv2;
    if ( AC.ncmod != 1 ) return(1);
    if ( AC.modinverses == 0 ) {
      LOCK(AC.halfmodlock);
      if ( AC.modinverses == 0 ) {
        AC.modinverses = (UWORD *)Malloc1(n*sizeof(UWORD),"modinverses");
        AC.modinverses[0] = 0;
        AC.modinverses[1] = 1;
        for ( i = 2; i < n; i++ ) {
            if ( GetModInverses(i,n,
                        (WORD *)(&(AC.modinverses[i])),&inv2) ) {
                SETERROR(-1)
            }
        }
      }
      UNLOCK(AC.halfmodlock);
    }
    return(0);
}

/*
        #] MakeInverses :
```

```
        #[ GetModInverses :
*/
int GetModInverses(WORD m1, WORD m2, WORD *im1, WORD *im2)
{
    WORD a1, a2, a3;
    WORD b1, b2, b3;
    WORD x = m1, y, c, d = m2;
    if ( x < 1 || d <= 1 ) goto somethingwrong;
    a1 = 0; a2 = 1;
    b1 = 1; b2 = 0;
    for(;;) {
        c = d/x; y = d%x; /* a good compiler makes this faster than y=d-c*x */
        if ( y == 0 ) break;
        a3 = a1-c*a2; a1 = a2; a2 = a3;
        b3 = b1-c*b2; b1 = b2; b2 = b3;
        d = x; x = y;
    }
    if ( x != 1 ) goto somethingwrong;
    if ( a2 < 0 ) a2 += m2;
    if ( b2 < 0 ) b2 += m1;
    if (im1!=NULL) *im1 = a2;
    if (im2!=NULL) *im2 = b2;
    return(0);
somethingwrong:
    MLOCK(ErrorMessageLock);
    MesPrint("Error trying to determine inverses in GetModInverses");
    MUNLOCK(ErrorMessageLock);
    return(-1);
}
/*
        #] GetModInverses :
        #[ GetLongModInverses :
*/

int GetLongModInverses(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *ia, WORD *nia, UWORD *ib,
WORD *nib) {

    UWORD *s, *t, *sa, *sb, *ta, *tb, *x, *y, *swap1;
    WORD ns, nt, nsa, nsb, nta, ntb, nx, ny, swap2;

    s = NumberMalloc("GetLongModInverses");
    ns = na;
    WCOPY(s, a, ABS(ns));

    t = NumberMalloc("GetLongModInverses");
    nt = nb;
    WCOPY(t, b, ABS(nt));

    sa = NumberMalloc("GetLongModInverses");
    nsa = 1;
    sa[0] = 1;

    sb = NumberMalloc("GetLongModInverses");
    nsb = 0;

    ta = NumberMalloc("GetLongModInverses");
    nta = 0;

    tb = NumberMalloc("GetLongModInverses");
    ntb = 1;
    tb[0] = 1;

    x = NumberMalloc("GetLongModInverses");
    y = NumberMalloc("GetLongModInverses");

    while (nt != 0) {
        DivLong(s,ns,t,nt,x,&nx,y,&ny);
        swap1=s; s=y; y=swap1;
        ns=ny;
        MulLong(x,nx,ta,nta,y,&ny);
        AddLong(sa,nsa,y,-ny,sa,&nsa);
        MulLong(x,nx,tb,ntb,y,&ny);
        AddLong(sb,nsb,y,-ny,sb,&nsb);

        swap1=s; s=t; t=swap1;
        swap2=ns; ns=nt; nt=swap2;
        swap1=sa; sa=ta; ta=swap1;
        swap2=nsa; nsa=nta; nta=swap2;
        swap1=sb; sb=tb; tb=swap1;
        swap2=nsb; nsb=ntb; ntb=swap2;
    }

    if (ia!=NULL) {
        *nia = nsa*ns;
        WCOPY(ia,sa,ABS(*nia));
    }

```

```
    if (ib!=NULL) {
        *nib = nsb*ns;
        WCOPY(ib,sb,ABS(*nib));
    }

    NumberFree(s,"GetLongModInverses");
    NumberFree(t,"GetLongModInverses");
    NumberFree(sa,"GetLongModInverses");
    NumberFree(sb,"GetLongModInverses");
    NumberFree(ta,"GetLongModInverses");
    NumberFree(tb,"GetLongModInverses");
    NumberFree(x,"GetLongModInverses");
    NumberFree(y,"GetLongModInverses");

    return 0;
}

/*
        #] GetLongModInverses :
        #[ Product :        WORD Product(a,na,b)

    Multiplies the Long number in a with the WORD b.

*/

WORD Product(UWORD *a, WORD *na, WORD b)
{
    WORD i, sgn = 1;
    RLONG t, u;
    if ( *na < 0 ) { *na = -(*na); sgn = -sgn; }
    if ( b < 0 ) { b = -b; sgn = -sgn; }
    t = 0;
    u = (RLONG)b;
    for ( i = 0; i < *na; i++ ) {
        t += *a * u;
        *a++ = (UWORD)t;
        t »= BITSINWORD;
    }
    if ( t > 0 ) {
        if ( ++(*na) > AM.MaxTal ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Overflow in Product");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        *a = (UWORD)t;
    }
    if ( sgn < 0 ) *na = -(*na);
    return(0);
}

/*
        #] Product :
        #[ Quotient :       UWORD Quotient(a,na,b)

        Routine divides the long number a by b with the assumption that
        there is no remainder (like while computing binomials).

*/

UWORD Quotient(UWORD *a, WORD *na, WORD b)
{
    RLONG v, t;
    WORD i, j, sgn = 1;
    if ( ( i = *na ) < 0 ) { sgn = -1; i = -i; }
    if ( b < 0 ) { b = -b; sgn = -sgn; }
    if ( i == 1 ) {
        if ( ( *a /= (UWORD)b ) == 0 ) *na = 0;
        if ( sgn < 0 ) *na = -*na;
        return(0);
    }
    a += i;
    j = i;
    v = (RLONG)b;
    t = (RLONG)(*--a);
    while ( --i >= 0 ) {
        *a = t / v;
        t -= v * (*a);
        if ( i ) {
            t «= BITSINWORD;
            t += *--a;
        }
    }
    a += j - 1;
    if ( !*a ) j--;
    if ( sgn < 0 ) j = -j;
    *na = j;
```

```
    return(0);
}

/*
        #] Quotient :
        #[ Remain10 :       WORD Remain10(a,na)

    Routine divides a by 10 and gives the remainder as return value.
    The value of a will be the quotient! a must be positive.

*/

WORD Remain10(UWORD *a, WORD *na)
{
    WORD i;
    RLONG t, u;
    UWORD *b;
    i = *na;
    t = 0;
    b = a + i - 1;
    while ( --i >= 0 ) {
        t += *b;
        *b-- = u = t / 10;
        t -= u * 10;
        if ( i > 0 ) t «= BITSINWORD;
    }
    if ( ( *na > 0 ) && !a[*na-1] ) (*na)--;
    return((WORD)t);
}

/*
        #] Remain10 :
        #[ Remain4 :        WORD Remain4(a,na)

    Routine divides a by 10000 and gives the remainder as return value.
    The value of a will be the quotient! a must be positive.

*/

WORD Remain4(UWORD *a, WORD *na)
{
    WORD i;
    RLONG t, u;
    UWORD *b;
    i = *na;
    t = 0;
    b = a + i - 1;
    while ( --i >= 0 ) {
        t += *b;
        *b-- = u = t / 10000;
        t -= u * 10000;
        if ( i > 0 ) t «= BITSINWORD;
    }
    if ( ( *na > 0 ) && !a[*na-1] ) (*na)--;
    return((WORD)t);
}

/*
        #] Remain4 :
        #[ PrtLong :        void PrtLong(a,na,s)

    Puts the long number a in string s.

*/

void PrtLong(UWORD *a, WORD na, UBYTE *s)
{
    GETIDENTITY
    WORD q, i;
    UBYTE *sa, *sb;
    UBYTE c;
    UWORD *bb, *b;

    if ( na < 0 ) {
        *s++ = '-';
        na = -na;
    }

    b = NumberMalloc("PrtLong");
    bb = b;
    i = na; while ( --i >= 0 ) *bb++ = *a++;
    a = b;
    if ( na > 2 ) {
        sa = s;
        do {
            q = Remain4(a,&na);
            *sa++ = (UBYTE)('0' + (q%10));
```

```
            q /= 10;
            *sa++ = (UBYTE)('0' + (q%10));
            q /= 10;
            *sa++ = (UBYTE)('0' + (q%10));
            q /= 10;
            *sa++ = (UBYTE)('0' + (q%10));
        } while ( na );
        while ( sa[-1] == '0' ) sa--;
        sb = s;
        s = sa;
        sa--;
        while ( sa > sb ) { c = *sa; *sa = *sb; *sb = c; sa--; sb++; }
    }
    else if ( na ) {
        sa = s;
        do {
            q = Remain10(a,&na);
            *sa++ = (UBYTE)('0' + q);
        } while ( na );
        sb = s;
        s = sa;
        sa--;
        while ( sa > sb ) { c = *sa; *sa = *sb; *sb = c; sa--; sb++; }
    }
    else *s++ = '0';
    *s = '\0';
    NumberFree(b,"PrtLong");
}

/*
        #] PrtLong :
        #[ GetLong :         WORD GetLong(s,a,na)

    Reads a long number from a string.
    The string is zero terminated and contains only digits!

    New algorithm: try to read 4 digits together before the result
    is accumulated.
*/

WORD GetLong(UBYTE *s, UWORD *a, WORD *na)
{
/*
    UWORD digit;
    *a = 0;
    *na = 0;
    while ( FG.cTable[*s] == 1 ) {
        digit = *s++ - '0';
        if ( *na && Product(a,na,(WORD)10) ) return(-1);
        if ( digit && AddLong(a,*na,&digit,(WORD)1,a,na) ) return(-1);
    }
    return(0);
*/
    UWORD digit, x = 0, y = 0;
    *a = 0;
    *na = 0;
    while ( FG.cTable[*s] == 1 ) {
        x = *s++ - '0';
        if ( FG.cTable[*s] != 1 ) { y = 10; break; }
        x = 10*x + *s++ - '0';
        if ( FG.cTable[*s] != 1 ) { y = 100; break; }
        x = 10*x + *s++ - '0';
        if ( FG.cTable[*s] != 1 ) { y = 1000; break; }
        x = 10*x + *s++ - '0';
        if ( *na && Product(a,na,(WORD)10000) ) return(-1);
        if ( ( digit = x ) != 0 && AddLong(a,*na,&digit,(WORD)1,a,na) )
            return(-1);
        y = 0;
    }
    if ( y ) {
        if ( *na && Product(a,na,(WORD)y) ) return(-1);
        if ( ( digit = x ) != 0 && AddLong(a,*na,&digit,(WORD)1,a,na) )
            return(-1);
    }
    return(0);
}

/*
        #] GetLong :
        #[ GCD :             WORD GCD(a,na,b,nb,c,nc)

    Algorithm to compute the GCD of two long numbers.
    See Knuth, sec 4.5.2 algorithm L.

    We assume that both numbers are positive

    NOTE!!!!!. NumberMalloc gets called and it may not be freed
```

```
*/

#ifdef EXTRAGCD

#define Convert(ia,aa,naa) \
    if ( (LONG)ia < 0 ) {         \
        ia = (ULONG)(-(LONG)ia); \
        aa[0] = ia;               \
        if ( ( aa[1] = ia » BITSINWORD ) != 0 ) naa = -2; \
        else naa = -1;            \
    }                             \
    else if ( ia == 0 ) { aa[0] = 0; naa = 0; } \
    else {                        \
        aa[0] = ia;               \
        if ( ( aa[1] = ia » BITSINWORD ) != 0 ) naa = 2; \
        else naa = 1;             \
    }

void GCD(UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    int ja = 0, jb = 0, j;
    UWORD *r,*t;
    UWORD *x1, *x2, *x3;
    WORD nd,naa,nbb;
    ULONG ia,ib,ic,id,u,v,w,q,T;
    UWORD aa[2], bb[2];
/*
    First eliminate easy powers of 2^...
*/
    while ( a[0] == 0 ) { na--; ja++; a++; }
    while ( b[0] == 0 ) { nb--; jb++; b++; }
    if ( ja > jb ) ja = jb;
    if ( ja > 0 ) {
        j = ja;
        do { *c++ = 0; } while ( --j > 0 );
    }
/*
    Now arrange things such that a >= b
*/
    if ( na < nb ) {
        jb = na; na = nb; nb = jb;
exch:
        r = a; a = b; b = r;
    }
    else if ( na == nb ) {
        r = a+na;
        t = b+nb;
        j = na;
        while ( --j >= 0 ) {
            if ( *--r > *--t ) break;
            if ( *r < *t ) goto exch;
        }
        if ( j < 0 ) {
out:
            j = nb;
            NCOPY(c,b,j);
            *nc = nb+ja;
            return;
        }
    }
/*
    {
        MLOCK(ErrorMessageLock);
        MesPrint("Ordered input,  ja = %d",(WORD)ja);
        AO.OutSkip = 3;
        FiniLine();
        j = na; r = a;
        while ( --j >= 0 ) { TalToLine((UWORD)(*r++)); TokenToLine((UBYTE *)"  "); }
        FiniLine();
        j = nb; r = b;
        while ( --j >= 0 ) { TalToLine((UWORD)(*r++)); TokenToLine((UBYTE *)"  "); }
        AO.OutSkip = 0;
        FiniLine();
        MUNLOCK(ErrorMessageLock);
    }
*/
/*
    We have now that A > B
    The loop recognizes the case that na-nb >= 1
    In that case we just have to divide!
*/
    r = x1 = NumberMalloc("GCD"); t = x2 = NumberMalloc("GCD"); x3 = NumberMalloc("GCD");
    j = na;
    NCOPY(r,a,j);
    j = nb;
    NCOPY(t,b,j);

```

```
    for(;;) {

        while ( na > nb ) {
toobad:
            DivLong(x1,na,x2,nb,c,nc,x3,&nd);
            if ( nd == 0 ) { b = x2; goto out; }
            t = x1; x1 = x2; x2 = x3; x3 = t; na = nb; nb = nd;
            if ( na == 2 ) break;
        }
/*
    Here we can use the shortcut.
*/
    if ( na == 2 ) {
        v = x1[0] + ( (ULONG)x1[1]) « BITSINWORD );
        w = x2[0];
        if ( nb == 2 ) w += ((ULONG)x2[1]) « BITSINWORD;
#ifdef EXTRAGCD2
        v = GCD2(v,w);
#else
        do { u = v%w; v = w; w = u; } while ( w );
#endif
        c[0] = (UWORD)v;
        if ( ( c[1] = (UWORD)(v » BITSINWORD) ) != 0 ) *nc = 2+ja;
        else *nc = 1+ja;
        NumberFree(x1,"GCD"); NumberFree(x2,"GCD"); NumberFree(x3,"GCD");
        return;
    }
    if ( na == 1 ) {
        UWORD ui, uj;
        ui = x1[0]; uj = x2[0];
#ifdef EXTRAGCD2
        ui = (UWORD)GCD2((ULONG)ui,(ULONG)uj);
#else
        do { nd = ui%uj; ui = uj; uj = nd; } while ( nd );
#endif
        c[0] = ui;
        *nc = 1 + ja;
        NumberFree(x1,"GCD"); NumberFree(x2,"GCD"); NumberFree(x3,"GCD");
        return;
    }
    ia = 1; ib = 0; ic = 0; id = 1;
    u = ( ((ULONG)x1[na-1]) « BITSINWORD ) + x1[na-2];
    v = ( ((ULONG)x2[nb-1]) « BITSINWORD ) + x2[nb-2];

    while ( v+ic != 0 && v+id != 0 &&
         ( q = (u+ia)/(v+ic) ) == (u+ib)/(v+id) ) {
        T = ia-q*ic; ia = ic; ic = T;
        T = ib-q*id; ib = id; id = T;
        T = u - q*v; u = v; v = T;
    }
    if ( ib == 0 ) goto toobad;
    Convert(ia,aa,naa);
    Convert(ib,bb,nbb);
    MulLong(x1,na,aa,naa,x3,&nd);
    MulLong(x2,nb,bb,nbb,c,nc);
    AddLong(x3,nd,c,*nc,c,nc);
    Convert(ic,aa,naa);
    Convert(id,bb,nbb);
    MulLong(x1,na,aa,naa,x3,&nd);
    t = c; na = j = *nc; r = x1;
    NCOPY(r,t,j);
    MulLong(x2,nb,bb,nbb,c,nc);
    AddLong(x3,nd,c,*nc,x2,&nb);
    }
}

#endif

/*
        #] GCD :
        #[ GcdLong :        WORD GcdLong(a,na,b,nb,c,nc)

    Returns the Greatest Common Divider of a and b in c.
    If a and or b are zero an error message will be returned.
    The answer is always positive.
    In principle a and c can be the same.
*/

#ifndef NEWTRICK
/*
    #[ Old Routine :
*/

WORD GcdLong(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    GETBIDENTITY
    if ( !na || !nb ) {
```

```
        if ( !na && !nb ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Cannot take gcd");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }

        if ( !na ) {
            *nc = abs(nb);
            NCOPY(c,b,*nc);
            *nc = abs(nb);
            return(0);
        }

        *nc = abs(na);
        NCOPY(c,a,*nc);
        *nc = abs(na);
        return(0);
    }
    if ( na < 0 ) na = -na;
    if ( nb < 0 ) nb = -nb;
    if ( na == 1 && nb == 1 ) {
#ifdef EXTRAGCD2
        *c = (UWORD)GCD2((ULONG)*a,(ULONG)*b);
#else
        UWORD x,y,z;
        x = *a;
        y = *b;
        do { z = x % y; x = y; } while ( ( y = z ) != 0 );
        *c = x;
#endif
        *nc = 1;
    }
    else if ( na <= 2 && nb <= 2 ) {
        RLONG lx,ly,lz;
        if ( na == 2 ) { lx = (((RLONG)(a[1]))«BITSINWORD) + *a; }
        else { lx = *a; }
        if ( nb == 2 ) { ly = (((RLONG)(b[1]))«BITSINWORD) + *b; }
        else { ly = *b; }
#ifdef LONGWORD
#ifdef EXTRAGCD2
        lx = GCD2(lx,ly);
#else
        do { lz = lx % ly; lx = ly; } while ( ( ly = lz ) != 0 );
#endif
#else
          if ( lx < ly ) { lz = lx; lx = ly; ly = lz; }
        do {
            lz = lx % ly; lx = ly;
        } while ( ( ly = lz ) != 0 && ( lx & AWORDMASK ) != 0 );
        if ( ly ) {
            do { *c = ((UWORD)lx)%((UWORD)ly); lx = ly; } while ( ( ly = *c ) != 0 );
            *c = (UWORD)lx;
            *nc = 1;
        }
        else
#endif
        {
            *c++ = (UWORD)lx;
            if ( ( *c = (UWORD)(lx » BITSINWORD) ) != 0 ) *nc = 2;
            else *nc = 1;
        }
    }
    else {
#ifdef EXTRAGCD
        GCD(a,na,b,nb,c,nc);
#else
#ifdef NEWGCD
        UWORD *x3,*x1,*x2, *GLscrat7, *GLscrat8;
        WORD n1,n2,n3,n4;
        WORD i, j;
        x1 = c; x3 = a; n1 = i = na;
        NCOPY(x1,x3,i);
        GLscrat7 = NumberMalloc("GcdLong"); GLscrat8 = NumberMalloc("GcdLong");
        x2 = GLscrat8; x3 = b; n2 = i = nb;
        NCOPY(x2,x3,i);
        x1 = c; i = 0;
        while ( x1[0] == 0 ) { i += BITSINWORD; x1++; n1--; }
        while ( ( x1[0] & 1 ) == 0 ) { i++; SCHUIF(x1,n1) }
        x2 = GLscrat8; j = 0;
        while ( x2[0] == 0 ) { j += BITSINWORD; x2++; n2--; }
        while ( ( x2[0] & 1 ) == 0 ) { j++; SCHUIF(x2,n2) }
        if ( j > i ) j = i;     /* powers of two in GCD */
        for(;;){
            if ( n1 > n2 ) {
firstbig:
                SubPLon(x1,n1,x2,n2,x1,&n3);
```

```
                n1 = n3;
                if ( n1 == 0 ) {
                    x1 = c;
                    n1 = i = n2; NCOPY(x1,x2,i);
                    break;
                }
                while ( ( x1[0] & 1 ) == 0 ) SCHUIF(x1,n1)
                if ( n1 == 1 ) {
                    if ( DivLong(x2,n2,x1,n1,GLscrat7,&n3,x2,&n4) ) goto GcdErr;
                    n2 = n4;
                    if ( n2 == 0 ) {
                        i = n1; x2 = c; NCOPY(x2,x1,i);
                        break;
                    }
#ifdef EXTRAGCD2
                    *c = (UWORD)GCD2((ULONG)x1[0],(ULONG)x2[0]);
#else
                    {
                        UWORD x,y,z;
                        x = x1[0];
                        y = x2[0];
                        do { z = x % y; x = y; } while ( ( y = z ) != 0 );
                        *c = x;
                    }
#endif
                    n1 = 1;
                    break;
                }
            }
            else if ( n1 < n2 ) {
lastbig:
                SubPLon(x2,n2,x1,n1,x2,&n3);
                n2 = n3;
                if ( n2 == 0 ) {
                    i = n1; x2 = c; NCOPY(x2,x1,i);
                    break;
                }
                while ( ( x2[0] & 1 ) == 0 ) SCHUIF(x2,n2)
                if ( n2 == 1 ) {
                    if ( DivLong(x1,n1,x2,n2,GLscrat7,&n3,x1,&n4) ) goto GcdErr;
                    n1 = n4;
                    if ( n1 == 0 ) {
                        x1 = c;
                        n1 = i = n2; NCOPY(x1,x2,i);
                        break;
                    }
#ifdef EXTRAGCD2
                    *c = (UWORD)GCD2((ULONG)x2[0],(ULONG)x1[0]);
#else
                    {
                        UWORD x,y,z;
                        x = x2[0];
                        y = x1[0];
                        do { z = x % y; x = y; } while ( ( y = z ) != 0 );
                        *c = x;
                    }
#endif
                    n1 = 1;
                    break;
                }
            }
            else {
                for ( i = n1-1; i >= 0; i-- ) {
                    if ( x1[i] > x2[i] ) goto firstbig;
                    else if ( x1[i] < x2[i] ) goto lastbig;
                }
                i = n1; x2 = c; NCOPY(x2,x1,i);
                break;
            }
        }
/*
        Now the GCD is in c but still needs j powers of 2.
*/
        x1 = c;
        while ( j >= BITSINWORD ) {
            for ( i = n1; i > 0; i-- ) x1[i] = x1[i-1];
            x1[0] = 0; n1++;
            j -= BITSINWORD;
        }
        if ( j > 0 ) {
            ULONG a1,a2 = 0;
            for ( i = 0; i < n1; i++ ) {
                a1 = x1[i]; a1 «= j;
                a2 += a1;
                x1[i] = a2;
                a2 »= BITSINWORD;
            }
```

```
            if ( a2 != 0 ) {
                x1[n1++] = a2;
            }
        }
        *nc = n1;
        NumberFree(GLscrat7,"GcdLong"); NumberFree(GLscrat8,"GcdLong");
#else
        UWORD *x1,*x2,*x3,*x4,*c1,*c2;
        WORD n1,n2,n3,n4,i;
        x1 = c; x3 = a; n1 = i = na;
        NCOPY(x1,x3,i);
        x1 = c; c1 = x2 = NumberMalloc("GcdLong"); x3 = NumberMalloc("GcdLong"); x4 = NumberMalloc("GcdLong");
        c2 = b; n2 = i = nb;
        NCOPY(c1,c2,i);
        for(;;){
            if ( DivLong(x1,n1,x2,n2,x4,&n4,x3,&n3) ) goto GcdErr;
            if ( !n3 ) { x1 = x2; n1 = n2; break; }
            if ( DivLong(x2,n2,x3,n3,x4,&n4,x1,&n1) ) goto GcdErr;
            if ( !n1 ) { x1 = x3; n1 = n3; break; }
            if ( DivLong(x3,n3,x1,n1,x4,&n4,x2,&n2) ) goto GcdErr;
            if ( !n2 ) {
                *nc = n1;
                NumberFree(x2,"GcdLong"); NumberFree(x3,"GcdLong"); NumberFree(x4,"GcdLong");
                return(0);
            }
        }
        *nc = i = n1;
        NCOPY(c,x1,i);
        NumberFree(x2,"GcdLong"); NumberFree(x3,"GcdLong"); NumberFree(x4,"GcdLong");
#endif
#endif
    }
    return(0);
GcdErr:
    MLOCK(ErrorMessageLock);
    MesCall("GcdLong");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}
/*
    #] Old Routine :
*/
#else

/*
    New routine for GcdLong that uses smart shortcuts.
    Algorithm by J. Vermaseren 15-nov-2006.
    It runs faster for very big numbers but only by a fixed factor.
    There is no improvement in the power behaviour.
    Improvement on the whole of hf9 (multiple zeta values at weight 9):
        Better than a factor 2 on a 32 bits architecture and 2.76 on a
        64 bits architecture.
    On hf10 (MZV's at weight 10), 64 bits architecture: factor 7.

    If we have two long numbers (na,nb > GCDMAX) we will work in a
    truncated way. At the moment of writing (15-nov-2006) it isn't
    clear whether this algorithm is an invention or a reinvention.
    A short search on the web didn't show anything.

    31-jul-2007:
    A better search shows that this is an adaptation of the Lehmer-Euclid
    algorithm, already described in Knuth. Here we can work without upper
    and lower limit because we are only interested in the GCD, not the
    extra numbers. Also it takes already some features of the double
    digit Lehmer-Euclid algorithm of Jebelean it seems.

    Maybe this can be programmed slightly better and we can get another
    few percent speed increase. Further improvements for the asymptotic
    case come from splitting the calculation as in Karatsuba and working
    with FFT divisions and multiplications etc. But this is when hundreds
    of words are involved at the least.

    Algorithm

    1: while ( na > nb || nb < GCDMAX ) {
        if ( nb == 0 ) { result in a }
        c = a % b;
        a = b;
        b = c;
       }
    2: Make the truncated values in which a and b are the combinations
       of the top two words of a and b. The whole numbers are aa and bb now.
    3: ma1 = 1; ma2 = 0; mb1 = 0; mb2 = 1;
    4: A = a; B = b; m = a/b; c = a - m*b;
       c = ma1*a+ma2*b-m*(mb1*a+mb2*b) = (ma1-m*mb1)*a+(ma2-m*mb2)*b
       mc1 = ma1-m*mb1; mc2 = ma2-m*mb2;
```

```
    5: a = b; ma1 = mb1; ma2 = mb2;
       b = c; mb1 = mc1; mb2 = mc2;
    6: if ( b != 0 && nb >= FULLMAX ) goto 4;
    7: Now construct the new quantities
        ma1*aa+ma2*bb and mb1*aa+mb2*bb
    8: goto 1;

    The essence of the above algorithm is that we do the divisions only
    on relatively short numbers. Also usually there are many steps 4&5
    for each step 7. This eliminates many operations.
    The termination at FULLMAX is that we make errors by not considering
    the tail of the number. If we run b down all the way, the errors combine
    in such a way that the new numbers may be of the same order as the old
    numbers. By stopping halfway we don't get the error beyond halfway
    either. Unfortunately this means that a >= FULLMAX and hence na > nb
    which means that next we will have a complete division. But just once.
    Running the steps 4-6 till a < FULLMAX runs already into problems.
    It may be necessary to experiment a bit to obtain the optimum value
    of GCDMAX.
*/

WORD GcdLong(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    GETBIDENTITY
    UWORD x,y,z;
    UWORD *x1,*x2,*x3,*x4,*x5,*d;
    UWORD *GLscrat6, *GLscrat7, *GLscrat8, *GLscrat9, *GLscrat10;
    WORD n1,n2,n3,n4,n5,i;
    RLONG lx,ly,lz;
    LONG ma1, ma2, mb1, mb2, mc1, mc2, m;
    if ( !na || !nb ) {
        if ( !na && !nb ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Cannot take gcd");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }

        if ( !na ) {
            *nc = abs(nb);
            NCOPY(c,b,*nc);
            *nc = abs(nb);
            return(0);
        }

        *nc = abs(na);
        NCOPY(c,a,*nc);
        *nc = abs(na);
        return(0);
    }
    if ( na < 0 ) na = -na;
    if ( nb < 0 ) nb = -nb;
/*
    #[ GMP stuff :
*/
#ifdef WITHGMP
    if ( na > 3 && nb > 3 ) {
        int ii;
        mp_limb_t *upa, *upb, *upc, xx;
        UWORD *uw, *u1, *u2;
        unsigned int tcounta, tcountb, tcounta1, tcountb1;
        mp_size_t ana, anb, anc;

        u1 = uw = NumberMalloc("GcdLong");
        upa = (mp_limb_t *)u1;
        ana = na; tcounta1 = 0;
        while ( a[0] == 0 ) { a++; ana--; tcounta1++; }
        for ( ii = 0; ii < ana; ii++ ) { *uw++ = *a++; }
        if ( ( ana & 1 ) != 0 ) { *uw = 0; ana++; }
        ana /= 2;

        u2 = uw = NumberMalloc("GcdLong");
        upb = (mp_limb_t *)u2;
        anb = nb; tcountb1 = 0;
        while ( b[0] == 0 ) { b++; anb--; tcountb1++; }
        for ( ii = 0; ii < anb; ii++ ) { *uw++ = *b++; }
        if ( ( anb & 1 ) != 0 ) { *uw = 0; anb++; }
        anb /= 2;

        xx = upa[0]; tcounta = 0;
        while ( ( xx & 15 ) == 0 ) { tcounta += 4; xx >>= 4; }
        while ( ( xx &  1 ) == 0 ) { tcounta += 1; xx >>= 1; }
        xx = upb[0]; tcountb = 0;
        while ( ( xx & 15 ) == 0 ) { tcountb += 4; xx >>= 4; }
        while ( ( xx &  1 ) == 0 ) { tcountb += 1; xx >>= 1; }

        if ( tcounta ) {
```

```
            mpn_rshift(upa,upa,ana,tcounta);
            if ( upa[ana-1] == 0 ) ana--;
        }
        if ( tcountb ) {
            mpn_rshift(upb,upb,anb,tcountb);
            if ( upb[anb-1] == 0 ) anb--;
        }

        upc = (mp_limb_t *)(NumberMalloc("GcdLong"));
        if ( ( ana > anb ) || ( ( ana == anb ) && ( upa[ana-1] >= upb[ana-1] ) ) ) {
            anc = mpn_gcd(upc,upa,ana,upb,anb);
        }
        else {
            anc = mpn_gcd(upc,upb,anb,upa,ana);
        }

        tcounta = tcounta1*BITSINWORD + tcounta;
        tcountb = tcountb1*BITSINWORD + tcountb;
        if ( tcountb > tcounta ) tcountb = tcounta;
        tcounta = tcountb/BITSINWORD;
        tcountb = tcountb%BITSINWORD;

        if ( tcountb ) {
            xx = mpn_lshift(upc,upc,anc,tcountb);
            if ( xx ) { upc[anc] = xx; anc++; }
        }

        uw = (UWORD *)upc; anc *= 2;
        while ( uw[anc-1] == 0 ) anc--;
        for ( ii = 0; ii < (int)tcounta; ii++ ) *c++ = 0;
        for ( ii = 0; ii < anc; ii++ ) *c++ = *uw++;
        *nc = anc + tcounta;
        NumberFree(u1,"GcdLong"); NumberFree(u2,"GcdLong"); NumberFree((UWORD *)(upc),"GcdLong");
        return(0);
    }
#endif
/*
    #] GMP stuff :
*/
/*
    #[ Easy cases :
*/
    if ( na == 1 && nb == 1 ) {
        x = *a;
        y = *b;
        do { z = x % y; x = y; } while ( ( y = z ) != 0 );
        *c = x;
        *nc = 1;
        return(0);
    }
    else if ( na <= 2 && nb <= 2 ) {
        if ( na == 2 ) { lx = (((RLONG)(a[1]))<<BITSINWORD) + *a; }
        else { lx = *a; }
        if ( nb == 2 ) { ly = (((RLONG)(b[1]))<<BITSINWORD) + *b; }
        else { ly = *b; }
        if ( lx < ly ) { lz = lx; lx = ly; ly = lz; }
#if ( BITSINWORD == 16 )
        do {
            lz = lx % ly; lx = ly;
        } while ( ( ly = lz ) != 0 );
#else
        do {
            lz = lx % ly; lx = ly;
        } while ( ( ly = lz ) != 0 && ( lx & AWORDMASK ) != 0 );
        if ( ly ) {
            x = (UWORD)lx; y = (UWORD)ly;
            do { *c = x % y; x = y; } while ( ( y = *c ) != 0 );
            *c = x;
            *nc = 1;
        }
        else
#endif
        {
            *c++ = (UWORD)lx;
            if ( ( *c = (UWORD)(lx >> BITSINWORD) ) != 0 ) *nc = 2;
            else *nc = 1;
        }
        return(0);
    }
/*
    #] Easy cases :
*/
    GLscrat6 = NumberMalloc("GcdLong"); GLscrat7 = NumberMalloc("GcdLong");
    GLscrat8 = NumberMalloc("GcdLong");
    GLscrat9 = NumberMalloc("GcdLong"); GLscrat10 = NumberMalloc("GcdLong");
restart:;
/*
```

```
    #[ Easy cases :
*/
    if ( na == 1 && nb == 1 ) {
        x = *a;
        y = *b;
        do { z = x % y; x = y; } while ( ( y = z ) != 0 );
        *c = x;
        *nc = 1;
    }
    else if ( na <= 2 && nb <= 2 ) {
        if ( na == 2 ) { lx = (((RLONG)(a[1]))«BITSINWORD) + *a; }
        else { lx = *a; }
        if ( nb == 2 ) { ly = (((RLONG)(b[1]))«BITSINWORD) + *b; }
        else { ly = *b; }
        if ( lx < ly ) { lz = lx; lx = ly; ly = lz; }
#if ( BITSINWORD == 16 )
        do {
            lz = lx % ly; lx = ly;
        } while ( ( ly = lz ) != 0 );
#else
        do {
            lz = lx % ly; lx = ly;
        } while ( ( ly = lz ) != 0 && ( lx & AWORDMASK ) != 0 );
        if ( ly ) {
            x = (UWORD)lx; y = (UWORD)ly;
            do { *c = x % y; x = y; } while ( ( y = *c ) != 0 );
            *c = x;
            *nc = 1;
        }
        else
#endif
        {
            *c++ = (UWORD)lx;
            if ( ( *c = (UWORD)(lx » BITSINWORD) ) != 0 ) *nc = 2;
            else *nc = 1;
        }
    }
/*
    #] Easy cases :
    #[ Original code :
*/
    else if ( na < GCDMAX || nb < GCDMAX || na != nb ) {
        if ( na < nb ) {
            x2 = GLscrat8; x3 = a; n2 = i = na;
            NCOPY(x2,x3,i);
            x1 = c; x3 = b; n1 = i = nb;
            NCOPY(x1,x3,i);
        }
        else {
            x1 = c; x3 = a; n1 = i = na;
            NCOPY(x1,x3,i);
            x2 = GLscrat8; x3 = b; n2 = i = nb;
            NCOPY(x2,x3,i);
        }
        x1 = c; x2 = GLscrat8; x3 = GLscrat7; x4 = GLscrat6;
        for(;;){
            if ( DivLong(x1,n1,x2,n2,x4,&n4,x3,&n3) ) goto GcdErr;
            if ( !n3 ) { x1 = x2; n1 = n2; break; }
            if ( n2 <= 2 ) { a = x2; b = x3; na = n2; nb = n3; goto restart; }
            if ( n3 >= GCDMAX && n2 == n3 ) {
                a = GLscrat9; b = GLscrat10; na = n2; nb = n3;
                for ( i = 0; i < na; i++ ) a[i] = x2[i];
                for ( i = 0; i < nb; i++ ) b[i] = x3[i];
                goto newtrick;
            }
            if ( DivLong(x2,n2,x3,n3,x4,&n4,x1,&n1) ) goto GcdErr;
            if ( !n1 ) { x1 = x3; n1 = n3; break; }
            if ( n3 <= 2 ) { a = x3; b = x1; na = n3; nb = n1; goto restart; }
            if ( n1 >= GCDMAX && n1 == n3 ) {
                a = GLscrat9; b = GLscrat10; na = n3; nb = n1;
                for ( i = 0; i < na; i++ ) a[i] = x3[i];
                for ( i = 0; i < nb; i++ ) b[i] = x1[i];
                goto newtrick;
            }
            if ( DivLong(x3,n3,x1,n1,x4,&n4,x2,&n2) ) goto GcdErr;
            if ( !n2 ) { *nc = n1; goto normalend; }
            if ( n1 <= 2 ) { a = x1; b = x2; na = n1; nb = n2; goto restart; }
            if ( n2 >= GCDMAX && n2 == n1 ) {
                a = GLscrat9; b = GLscrat10; na = n1; nb = n2;
                for ( i = 0; i < na; i++ ) a[i] = x1[i];
                for ( i = 0; i < nb; i++ ) b[i] = x2[i];
                goto newtrick;
            }
        }
        *nc = i = n1;
        NCOPY(c,x1,i);
    }
```

```
/*
    #] Original code :
    #[ New code :
*/
    else {
/*
        This is the new algorithm starting at step 3.

    3: ma1 = 1; ma2 = 0; mb1 = 0; mb2 = 1;
    4: A = a; B = b; m = a/b; c = a - m*b;
       c = ma1*a+ma2*b-m*(mb1*a+mb2*b) = (ma1-m*mb1)*a+(ma2-m*mb2)*b
       mc1 = ma1-m*mb1; mc2 = ma2-m*mb2;
    5: a = b; ma1 = mb1; ma2 = mb2;
       b = c; mb1 = mc1; mb2 = mc2;
    6: if ( b != 0 ) goto 4;
*/
newtrick:;
        ma1 = 1; ma2 = 0; mb1 = 0; mb2 = 1;
        lx = (((RLONG)(a[na-1]))«BITSINWORD) + a[na-2];
        ly = (((RLONG)(b[nb-1]))«BITSINWORD) + b[nb-2];
        if ( ly > lx ) { lz = lx; lx = ly; ly = lz; d = a; a = b; b = d; }
        do {
            m = lx/ly;
            mc1 = ma1-m*mb1; mc2 = ma2-m*mb2;
            ma1 = mb1; ma2 = mb2; mb1 = mc1; mb2 = mc2;
            lz = lx - m*ly; lx = ly; ly = lz;
        } while ( ly >= FULLMAX );
/*
        Next the construction of the two new numbers

    7: Now construct the new quantities
        a = ma1*aa+ma2*bb and b = mb1*aa+mb2*bb
*/
        x1 = GLscrat6;
        x2 = GLscrat7;
        x3 = GLscrat8;
        x5 = GLscrat10;
        if ( ma1 < 0 ) {
            ma1 = -ma1;
            x1[0] = (UWORD)ma1;
            x1[1] = (UWORD)(ma1 » BITSINWORD);
            if ( x1[1] ) n1 = -2;
            else         n1 = -1;
        }
        else {
            x1[0] = (UWORD)ma1;
            x1[1] = (UWORD)(ma1 » BITSINWORD);
            if ( x1[1] ) n1 = 2;
            else         n1 = 1;
        }
        if ( MulLong(a,na,x1,n1,x2,&n2) ) goto GcdErr;
        if ( ma2 < 0 ) {
            ma2 = -ma2;
            x1[0] = (UWORD)ma2;
            x1[1] = (UWORD)(ma2 » BITSINWORD);
            if ( x1[1] ) n1 = -2;
            else         n1 = -1;
        }
        else {
            x1[0] = (UWORD)ma2;
            x1[1] = (UWORD)(ma2 » BITSINWORD);
            if ( x1[1] ) n1 = 2;
            else         n1 = 1;
        }
        if ( MulLong(b,nb,x1,n1,x3,&n3) ) goto GcdErr;
        if ( AddLong(x2,n2,x3,n3,c,&n4) ) goto GcdErr;
        if ( mb1 < 0 ) {
            mb1 = -mb1;
            x1[0] = (UWORD)mb1;
            x1[1] = (UWORD)(mb1 » BITSINWORD);
            if ( x1[1] ) n1 = -2;
            else         n1 = -1;
        }
        else {
            x1[0] = (UWORD)mb1;
            x1[1] = (UWORD)(mb1 » BITSINWORD);
            if ( x1[1] ) n1 = 2;
            else         n1 = 1;
        }
        if ( MulLong(a,na,x1,n1,x2,&n2) ) goto GcdErr;
        if ( mb2 < 0 ) {
            mb2 = -mb2;
            x1[0] = (UWORD)mb2;
            x1[1] = (UWORD)(mb2 » BITSINWORD);
            if ( x1[1] ) n1 = -2;
            else         n1 = -1;
        }
```

```
        else {
            x1[0] = (UWORD)mb2;
            x1[1] = (UWORD)(mb2 >> BITSINWORD);
            if ( x1[1] ) n1 = 2;
            else         n1 = 1;
        }
        if ( MulLong(b,nb,x1,n1,x3,&n3) ) goto GcdErr;
        if ( AddLong(x2,n2,x3,n3,x5,&n5) ) goto GcdErr;
        a = c; na = n4; b = x5; nb = n5;
        if ( nb == 0 ) { *nc = n4; goto normalend; }
        x4 = GLscrat9;
        for ( i = 0; i < na; i++ ) x4[i] = a[i];
        a = x4;
        if ( na < 0 ) na = -na;
        if ( nb < 0 ) nb = -nb;
/*
        The typical case now is that in a we have the last step to go
        to loose the leading word, while in b we have lost the leading word.
        We could go to DivLong now but we can also add an extra step that
        is less wasteful.
        In the case that the new leading word of b is extrememly short (like 1)
        we make a rather large error of course. In the worst case the whole
        will be intercepted by DivLong after all, but that is so rare that
        it shouldn't influence any timing in a measurable way.
*/
        if ( nb >= GCDMAX && na == nb+1 && b[nb-1] >= HALFMAX && b[nb-1] > a[na-1] ) {
            lx = (((RLONG)(a[na-1]))<<BITSINWORD) + a[na-2];
            x1[0] = lx/b[nb-1]; n1 = 1;
            MulLong(b,nb,x1,n1,x2,&n2);
            n2 = -n2;
            AddLong(a,na,x2,n2,x4,&n4);
            if ( n4 == 0 ) {
                *nc = nb;
                for ( i = 0; i < nb; i++ ) c[i] = b[i];
                goto normalend;
            }
            if ( n4 < 0 ) n4 = -n4;
            a = b; na = nb; b = x4; nb = n4;
        }
        goto restart;
/*
    #] New code :
*/
    }
normalend:
    NumberFree(GLscrat6,"GcdLong"); NumberFree(GLscrat7,"GcdLong"); NumberFree(GLscrat8,"GcdLong");
    NumberFree(GLscrat9,"GcdLong"); NumberFree(GLscrat10,"GcdLong");
    return(0);
GcdErr:
    MLOCK(ErrorMessageLock);
    MesCall("GcdLong");
    MUNLOCK(ErrorMessageLock);
    NumberFree(GLscrat6,"GcdLong"); NumberFree(GLscrat7,"GcdLong"); NumberFree(GLscrat8,"GcdLong");
    NumberFree(GLscrat9,"GcdLong"); NumberFree(GLscrat10,"GcdLong");
    SETERROR(-1)
}

#endif

/*
        #] GcdLong :
        #[ GetBinom :       WORD GetBinom(a,na,i1,i2)
*/

WORD GetBinom(UWORD *a, WORD *na, WORD i1, WORD i2)
{
    GETIDENTITY
    WORD j, k, l;
    UWORD *GBscrat3, *GBscrat4;
    if ( i1-i2 < i2 ) i2 = i1-i2;
    if ( i2 == 0 ) { *a = 1; *na = 1; return(0); }
    if ( i2 > i1 ) { *a = 0; *na = 0; return(0); }
    *a = i1; *na = 1;
    GBscrat3 = NumberMalloc("GetBinom"); GBscrat4 = NumberMalloc("GetBinom");
    for ( j = 2; j <= i2; j++ ) {
        GBscrat3[0] = i1+1-j;
        if ( MulLong(a,*na,GBscrat3,(WORD)1,GBscrat4,&k) ) goto CalledFrom;
        GBscrat3[0] = j;
        if ( DivLong(GBscrat4,k,GBscrat3,(WORD)1,a,na,GBscrat3,&l) ) goto CalledFrom;
    }
    NumberFree(GBscrat3,"GetBinom"); NumberFree(GBscrat4,"GetBinom");
    return(0);
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("GetBinom");
    MUNLOCK(ErrorMessageLock);
    NumberFree(GBscrat3,"GetBinom"); NumberFree(GBscrat4,"GetBinom");
```

```
    SETERROR(-1)
}

/*
        #] GetBinom :
        #[ LcmLong :        WORD LcmLong(a,na,b,nb)

        Computes the LCM of the long numbers a and b and puts the result
        in c. c is allowed to be equal to a.
*/

WORD LcmLong(PHEAD UWORD *a, WORD na, UWORD *b, WORD nb, UWORD *c, WORD *nc)
{
    WORD error = 0;
    UWORD *d = NumberMalloc("LcmLong");
    UWORD *e = NumberMalloc("LcmLong");
    UWORD *f = NumberMalloc("LcmLong");
    WORD nd, ne, nf;
    GcdLong(BHEAD a, na, b, nb, d, &nd);
    DivLong(a,na,d,nd,e,&ne,f,&nf);
    if ( MulLong(b,nb,e,ne,c,nc) ) {
        MLOCK(ErrorMessageLock);
        MesCall("LcmLong");
        MUNLOCK(ErrorMessageLock);
        error = -1;
    }
    NumberFree(f,"LcmLong");
    NumberFree(e,"LcmLong");
    NumberFree(d,"LcmLong");
    return(error);
}

/*
        #] LcmLong :
        #[ TakeLongRoot:    int TakeLongRoot(a,n,power)

    Takes the 'power'-root of the long number in a.
    If the root could be taken the return value is zero.
    If the root could not be taken, the return value is 1.
    The root will be in a if it could be taken, otherwise there will be garbage
    Algorithm: (assume b is guess of root, b' better guess)
        b' = (a-(power-1)*b^power)/(n*b^(power-1))
    Note: power should be positive!
*/

int TakeLongRoot(UWORD *a, WORD *n, WORD power)
{
    GETIDENTITY
    int numbits, guessbits, i, retval = 0;
    UWORD x, *b, *c, *d, *e;
    WORD na, nb, nc, nd, ne;
    if ( *n < 0 && ( power & 1 ) == 0 ) return(1);
    if ( power == 1 ) return(0);
    if ( *n < 0 ) { na = -*n; }
    else          { na =  *n; }
    if ( na == 1 ) {
/*          Special cases that are the most frequent */
        if ( a[0] == 1 ) return(0);
        if ( power < BITSINWORD && na == 1 && a[0] == (UWORD)(1«power) ) {
            a[0] = 2; return(0);
        }
        if ( 2*power < BITSINWORD && na == 1 && a[0] == (UWORD)(1«(2*power)) ) {
            a[0] = 4; return(0);
        }
    }
/*
    Step 1: make a guess. We count the number of bits.
    numbits will be the 1+2log(a)
*/
    numbits = BITSINWORD*(na-1);
    x = a[na-1];
    while ( ( x » 8 ) != 0 ) { numbits += 8; x »= 8; }
    if ( ( x » 4 ) != 0 ) { numbits += 4; x »= 4; }
    if ( ( x » 2 ) != 0 ) { numbits += 2; x »= 2; }
    if ( ( x » 1 ) != 0 ) numbits++;
    guessbits = numbits / power;
    if ( guessbits <= 0 ) return(1); /* root < 2 and 1 we did already */
    nb = guessbits/BITSINWORD;
/*
    The recursion is:
    (b'-b) = (a/b^(power-1)-b)/n
           = (a/c-b)/n
           = (d-b)/n     (remainder of a/c is e)
           = c/n   (we reuse the scratch array c)
    Termination can be tricky. When a/c has no remainder and = b we have a root.
    When d = b but the remainder of a/c != 0, there is definitely no root.
*/
```

```
    b = NumberMalloc("TakeLongRoot"); c = NumberMalloc("TakeLongRoot");
    d = NumberMalloc("TakeLongRoot"); e = NumberMalloc("TakeLongRoot");
    for ( i = 0; i < nb; i++ ) { b[i] = 0; }
    b[nb] = 1 << (guessbits%BITSINWORD);
    nb++;
    for(;;) {
        nc = nb;
        for ( i = 0; i < nb; i++ ) c[i] = b[i];
        if ( RaisPow(BHEAD c,&nc,power-1) ) goto TLcall;
        if ( DivLong(a,na,c,nc,d,&nd,e,&ne) ) goto TLcall;
        nb = -nb;
        if ( AddLong(d,nd,b,nb,c,&nc) ) goto TLcall;
        nb = -nb;
        if ( nc == 0 ) {
            if ( ne == 0 ) break;
            retval = 1; break;
/*
            else {
                NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
                NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
                return(1);
            }
*/
        }
        DivLong(c,nc,(UWORD *)(&power),1,d,&nd,e,&ne);
        if ( nd == 0 ) {
            retval = 1;
            break;
/*
            NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
            NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
            return(1);
*/
/*
            This code tries b+1 as a final possibility.
            We believe this is not needed
            UWORD one = 1;
            if ( AddLong(b,nb,&one,1,c,&nc) ) goto TLcall;
            if ( RaisPow(BHEAD c,&nc,power-1) ) goto TLcall;
            if ( DivLong(a,na,c,nc,d,&nd,e,&ne) ) goto TLcall;
            if ( ne != 0 ) return(1);
            nb = -nb;
            if ( SubLong(d,nd,b,nb,c,&nc) ) goto TLcall;
            nb = -nb;
            if ( nc != 0 ) {
                NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
                NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
                return(1);
            }
            break;
*/
        }
        if ( AddLong(b,nb,d,nd,b,&nb) ) goto TLcall;
    }
    for ( i = 0; i < nb; i++ ) a[i] = b[i];
    if ( *n < 0 ) *n = -nb;
    else          *n =  nb;
    NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
    NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
    return(retval);
TLcall:
    MLOCK(ErrorMessageLock);
    MesCall("TakeLongRoot");
    MUNLOCK(ErrorMessageLock);
    NumberFree(b,"TakeLongRoot"); NumberFree(c,"TakeLongRoot");
    NumberFree(d,"TakeLongRoot"); NumberFree(e,"TakeLongRoot");
    Terminate(-1);
    return(-1);
}

/*
        #] TakeLongRoot:
        #[ MakeRational:

        Makes the integer a mod m into a traction b/c with |b|,|c| < sqrt(m)
        For the algorithm, see MakeLongRational.
*/

int MakeRational(WORD a,WORD m, WORD *b, WORD *c)
{
    LONG x1,x2,x3,x4,y1,y2;
    if ( a < 0 ) { a = a+m; }
    if ( a <= 1 ) {
        if ( a > m/2 ) a = a-m;
        *b = a; *c = 1; return(0);
    }
    x1 = m; x2 = a;
```

```
    if ( x2*x2 >= m ) {
        y1 = x1/x2; y2 = x1%x2; x3 = 1; x4 = -y1; x1 = x2; x2 = y2;
        while ( x2*x2 >= m ) {
            y1 = x1/x2; y2 = x1%x2; x1 = x2; x2 = y2; y2 = x3-y1*x4; x3 = x4; x4 = y2;
        }
    }
    else x4 = 1;
    if ( x2 == 0 ) { return(1); }
    if ( x2 > m/2 ) *b = x2-m;
    else            *b = x2;
    if ( x4 > m/2 ) { *c = x4-m; *c = -*c; *b = -*b; }
    else if ( x4 <= -m/2 ) { x4 += m; *c = x4; }
    else if ( x4 < 0 ) { x4 = -x4; *c = x4; *b = -*b; }
    else            *c = x4;
    return(0);
}

/*
        #] MakeRational:
        #[ MakeLongRational:

        Converts the long number a mod m into the fraction b
        One of the properties of b is that num,den < sqrt(m)
        The algorithm: Start with:   m 0
                                     a 1
        Make now c=m%a, c1=m/a       c c2=0-c1*1
        Make now d=a%c  d1=a/c       d d2=1-d1*c2
        Make now e=c%d  e1=c/d       e e2=1-e1*d2
            etc till in the first column we get a number < sqrt(m)
            We have then f,f2 and the fraction is f/f2.
        If at any moment we get a zero, m contained an unlucky prime.

        Note that this can be made a lot faster when we make the same
        improvements as in the GCD routine. That is something for later.
#ifdef WITHMAKERATIONAL
*/

#define COPYLONG(x1,nx1,x2,nx2) { int i; for(i=0;i<ABS(nx2);i++)x1[i]=x2[i];nx1=nx2; }

int MakeLongRational(PHEAD UWORD *a, WORD na, UWORD *m, WORD nm, UWORD *b, WORD *nb)
{
    UWORD *root = NumberMalloc("MakeRational");
    UWORD *x1 = NumberMalloc("MakeRational");
    UWORD *x2 = NumberMalloc("MakeRational");
    UWORD *x3 = NumberMalloc("MakeRational");
    UWORD *x4 = NumberMalloc("MakeRational");
    UWORD *y1 = NumberMalloc("MakeRational");
    UWORD *y2 = NumberMalloc("MakeRational");
    WORD nroot,nx1,nx2,nx3,nx4,ny1,ny2 = 0,retval = 0;
    WORD sign = 1;
/*
    Step 1: Take the square root of m
*/
    COPYLONG(root,nroot,m,nm)
    TakeLongRoot(root,&nroot,2);
/*
    Step 2: Set the start values
*/
    if ( na < 0 ) { na = -na; sign = -sign; }
    COPYLONG(x1,nx1,m,nm)
    COPYLONG(x2,nx2,a,na)
/*
    x3[0] = 0, nx3 = 0;
    x4[0] = 1, nx4 = 1;
*/
/*
    The start operation needs some special attention because of the zero.
*/
    if ( BigLong(x2,nx2,root,nroot) <= 0 ) {
        x4[0] = 1, nx4 = 1;
        goto gottheanswer;
    }
    DivLong(x1,nx1,x2,nx2,y1,&ny1,y2,&ny2);
    if ( ny2 == 0 ) { retval = 1; goto cleanup; }
    COPYLONG(x1,nx1,x2,nx2)
    COPYLONG(x2,nx2,y2,ny2)
    x3[0] = 1; nx3 = 1;
    COPYLONG(x4,nx4,y1,ny1)
    nx4 = -nx4;
/*
    Now the loop.
*/
    while ( BigLong(x2,nx2,root,nroot) > 0 ) {
        DivLong(x1,nx1,x2,nx2,y1,&ny1,y2,&ny2);
        if ( ny2 == 0 ) { retval = 1; goto cleanup; }
        COPYLONG(x1,nx1,x2,nx2)
        COPYLONG(x2,nx2,y2,ny2)
```

```
        MulLong(y1,ny1,x4,nx4,y2,&ny2);
        ny2 = -ny2;
        AddLong(x3,nx3,y2,ny2,y1,&ny1);
        COPYLONG(x3,nx3,x4,nx4)
        COPYLONG(x4,nx4,y1,ny1)
    }
/*
    Now we have the answer. It is x2/x4. It has to be packed into b.
*/
gottheanswer:
    if ( nx4 < 0 ) { sign = -sign; nx4 = -nx4; }
    COPYLONG(b,*nb,x2,nx2)
    Pack(b,nb,x4,nx4);
    if ( sign < 0 ) *nb = -*nb;
cleanup:
    NumberFree(y2,"MakeRational");
    NumberFree(y1,"MakeRational");
    NumberFree(x4,"MakeRational");
    NumberFree(x3,"MakeRational");
    NumberFree(x2,"MakeRational");
    NumberFree(x1,"MakeRational");
    NumberFree(root,"MakeRational");
    return(retval);
}

/*
#endif
        #] MakeLongRational:
        #[ ChineseRemainder:
*/
#ifdef WITHCHINESEREMAINDER

int ChineseRemainder(PHEAD MODNUM *a1, MODNUM *a2, MODNUM *a)
{
    UWORD *inv1 = NumberMalloc("ChineseRemainder");
    UWORD *inv2 = NumberMalloc("ChineseRemainder");
    UWORD *fac1 = NumberMalloc("ChineseRemainder");
    UWORD *fac2 = NumberMalloc("ChineseRemainder");
    UWORD two[1];
    WORD ninv1, ninv2, nfac1, nfac2;
    if ( a1->na < 0 ) {
        AddLong(a1->a,a1->na,a1->m,a1->nm,a1->a,&(a1->na));
    }
    if ( a2->na < 0 ) {
        AddLong(a2->a,a2->na,a2->m,a2->nm,a2->a,&(a2->na));
    }
    MulLong(a1->m,a1->nm,a2->m,a2->nm,a->m,&(a->nm));

    GetLongModInverses(BHEAD a1->m,a1->nm,a2->m,a2->nm,inv1,&ninv1,inv2,&ninv2);
    MulLong(inv1,ninv1,a1->m,a1->nm,fac1,&nfac1);
    MulLong(inv2,ninv2,a2->m,a2->nm,fac2,&nfac2);

    MulLong(fac1,nfac1,a2->a,a2->na,inv1,&ninv1);
    MulLong(fac2,nfac2,a1->a,a1->na,inv2,&ninv2);
    AddLong(inv1,ninv1,inv2,ninv2,a->a,&(a->na));

    two[0] = 2;
    MulLong(a->a,a->na,two,1,fac1,&nfac1);
    if ( BigLong(fac1,nfac1,a->m,a->nm) > 0 ) {
        a->nm = -a->nm;
        AddLong(a->a,a->na,a->m,a->nm,a->a,&(a->na));
        a->nm = -a->nm;
    }
    NumberFree(fac2,"ChineseRemainder");
    NumberFree(fac1,"ChineseRemainder");
    NumberFree(inv2,"ChineseRemainder");
    NumberFree(inv1,"ChineseRemainder");
    return(0);
}

#endif

/*
        #] ChineseRemainder:
    #] RekenLong :
    #[ RekenTerms :
        #[ CompCoef :       WORD CompCoef(term1,term2)

    Compares the coefficients of term1 and term2 by subtracting them.
    This does more work than needed but this routine is only called
    when sorting functions and function arguments.
    (and comparing values
*/
/* #define 64SAVE */

WORD CompCoef(WORD *term1, WORD *term2)
{
```

```
    GETIDENTITY
    UWORD *c;
    WORD n1,n2,n3,*a;
    GETCOEF(term1,n1);
    GETCOEF(term2,n2);
    if ( term1[1] == 0 && n1 == 1 ) {
        if ( term2[1] == 0 && n2 == 1 ) return(0);
        if ( n2 < 0 ) return(1);
        return(-1);
    }
    else if ( term2[1] == 0 && n2 == 1 ) {
        if ( n1 < 0 ) return(-1);
        return(1);
    }
    if ( n1 > 0 ) {
        if ( n2 < 0 ) return(1);
    }
    else {
        if ( n2 > 0 ) return(-1);
        a = term1; term1 = term2; term2 = a;
        n3 = -n1; n1 = -n2; n2 = n3;
    }
    if ( term1[1] == 1 && term2[1] == 1 && n1 == 1 && n2 == 1 ) {
        if ( (UWORD)*term1 > (UWORD)*term2 ) return(1);
        else if ( (UWORD)*term1 < (UWORD)*term2 ) return(-1);
        else return(0);
    }

/*
    The next call should get dedicated code, as AddRat does more than
    strictly needed. Also more attention should be given to overflow.
*/
    c = NumberMalloc("CompCoef");
    if ( AddRat(BHEAD (UWORD *)term1,n1,(UWORD *)term2,-n2,c,&n3) ) {
        MLOCK(ErrorMessageLock);
        MesCall("CompCoef");
        MUNLOCK(ErrorMessageLock);
        NumberFree(c,"CompCoef");
        SETERROR(-1)
    }
    NumberFree(c,"CompCoef");
    return(n3);
}

/*
        #] CompCoef :
        #[ Modulus :        WORD Modulus(term)

    Routine takes the coefficient of term modulus b. The answer
    is in term again and the length of term is adjusted.

*/

WORD Modulus(WORD *term)
{
    WORD *t;
    WORD n1;
    t = term;
    GETCOEF(t,n1);
    if ( TakeModulus((UWORD *)t,&n1,AC.cmod,AC.ncmod,UNPACK) ) {
        MLOCK(ErrorMessageLock);
        MesCall("Modulus");
        MUNLOCK(ErrorMessageLock);
        SETERROR(-1)
    }
    if ( !n1 ) {
        *term = 0;
        return(0);
    }
    else if ( n1 > 0 ) {
        n1 *= 2;
        t += n1;        /* Note that n1 >= 0 */
        n1++;
    }
    else if ( n1 < 0 ) {
        n1 *= 2;
        t += -n1;
        n1--;
    }
    *t++ = n1;
    *term = WORDDIF(t,term);
    return(0);
}

/*
        #] Modulus :
        #[ TakeModulus :    WORD TakeModulus(a,na,cmodvec,ncmod,par)
```

```

        Routine gets the rational number in a with reduced length na.
        It is called when AC.ncmod != 0 and the number in AC.cmod is the
        number wrt which we need the modulus.
        The result is returned in a and na again.

        If par == NOUNPACK we only do a single number, not a fraction.
        In addition we don't do fancy. We want a positive number and
        the input was supposed to be positive.
        We don't pack the result. The calling routine is responsible for that.
        This may not be a good idea. To be checked.
*/

WORD TakeModulus(UWORD *a, WORD *na, UWORD *cmodvec, WORD ncmod, WORD par)
{
    GETIDENTITY
    UWORD *c, *d, *e, *f, *g, *h;
    UWORD *x4,*x2;
    UWORD *x3,*x1,*x5,*x6,*x7,*x8;
    WORD y3,y1,y5,y6;
    WORD n1, i, y2, y4;
    WORD nh, tdenom, tnumer, nmod;
    LONG x;
    if ( ncmod == 0 ) return(0);        /* No modulus operation */
    nmod = ABS(ncmod);
    n1 = *na;
    if ( ( par & UNPACK ) != 0 ) UnPack(a,n1,&tdenom,&tnumer);
    else { tnumer = n1; }
/*
    We fish out the special case that the coefficient is short as well.
    There is no need to make lots of calls etc
*/
    if ( ( ( par & UNPACK ) == 0 ) && nmod == 1 && ( n1 == 1 || n1 == -1 ) ) {
        goto simplecase;
    }
    else if ( nmod == 1 && ( n1 == 1 || n1 == -1 ) ) {
        if ( a[1] != 1 ) {
            a[1] = a[1] % cmodvec[0];
            if ( a[1] == 0 ) {
                MesPrint("Division by zero in short modulus arithmetic");
                return(-1);
            }
            y1 = 0;
            if ( ( AC.modinverses != 0 ) && ( ( par & NOINVERSES ) == 0 ) ) {
                y1 = AC.modinverses[a[1]];
            }
            else {
                GetModInverses(a[1],cmodvec[0],&y1,&y2);
            }
            x = a[0];
            a[0] = (x*y1) % cmodvec[0];
            a[1] = 1;
        }
        else {
simplecase:
            a[0] = a[0] % cmodvec[0];
        }
        if ( a[0] == 0 ) { *na = 0; return(0); }
        if ( ( AC.modmode & POSNEG ) != 0 ) {
            if ( a[0] > (UWORD)(cmodvec[0]/2) ) {
                a[0] =  cmodvec[0] - a[0];
                *na = -*na;
            }
        }
        else if ( *na < 0 ) {
            *na = 1; a[0] = cmodvec[0] - a[0];
        }
        return(0);
    }
    c = NumberMalloc("TakeModulus"); d = NumberMalloc("TakeModulus"); e = NumberMalloc("TakeModulus");
    f = NumberMalloc("TakeModulus"); g = NumberMalloc("TakeModulus"); h = NumberMalloc("TakeModulus");
    n1 = ABS(n1);
    if ( DivLong(a,tnumer,(UWORD *)cmodvec,nmod,
        c,&nh,a,&tnumer) ) goto ModErr;
    if ( tnumer == 0 ) { *na = 0; goto normalreturn; }
    if ( ( par & UNPACK ) == 0 ) {
        if ( ( AC.modmode & POSNEG ) != 0 ) {
            NormalModulus(a,&tnumer);
        }
        else if ( tnumer < 0 ) {
            SubPLon((UWORD *)cmodvec,nmod,a,-tnumer,a,&tnumer);
        }
        *na = tnumer;
        goto normalreturn;
    }
    if ( tdenom == 1 && a[n1] == 1 ) {
        if ( ( AC.modmode & POSNEG ) != 0 ) {
```

```
            NormalModulus(a,&tnumer);
        }
        else if ( tnumer < 0 ) {
            SubPLon((UWORD *)cmodvec,nmod,a,-tnumer,a,&tnumer);
        }
        *na = tnumer;
        i = ABS(tnumer);
        a += i;
        *a++ = 1;
        while ( --i > 0 ) *a++ = 0;
        goto normalreturn;
    }
    if ( DivLong(a+n1,tdenom,(UWORD *)cmodvec,nmod,c,&nh,a+n1,&tdenom) ) goto ModErr;
    if ( !tdenom ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Division by zero in modulus arithmetic");
        if ( AP.DebugFlag ) {
            AO.OutSkip = 3;
            FiniLine();
            i = *na;
            if ( i < 0 ) i = -i;
            while ( --i >= 0 ) { TalToLine((UWORD)(*a++)); TokenToLine((UBYTE *)"  "); }
            i = *na;
            if ( i < 0 ) i = -i;
            while ( --i >= 0 ) { TalToLine((UWORD)(*a++)); TokenToLine((UBYTE *)"  "); }
            TalToLine((UWORD)(*na));
            AO.OutSkip = 0;
            FiniLine();
        }
        MUNLOCK(ErrorMessageLock);
        NumberFree(c,"TakeModulus"); NumberFree(d,"TakeModulus"); NumberFree(e,"TakeModulus");
        NumberFree(f,"TakeModulus"); NumberFree(g,"TakeModulus"); NumberFree(h,"TakeModulus");
        return(-1);
    }
    if ( ( AC.modinverses != 0 ) && ( ( par & NOINVERSES ) == 0 )
    && ( tdenom == 1 || tdenom == -1 ) ) {
        *d = AC.modinverses[a[n1]]; y1 = 1; y2 = tdenom;
        if ( MulLong(a,tnumer,d,y1,c,&y3) ) goto ModErr;
        if ( DivLong(c,y3,(UWORD *)cmodvec,nmod,d,&y5,a,&tdenom) ) goto ModErr;
        if ( y2 < 0 ) tdenom = -tdenom;
    }
    else {
      x2 = (UWORD *)cmodvec; x1 = c; i = nmod; while ( --i >= 0 ) *x1++ = *x2++;
      x1 = c; x2 = a+n1; x3 = d; x4 = e; x5 = f; x6 = g;
      y1 = nmod; y2 = tdenom; y4 = 0; y5 = 1; *x5 = 1;
      for(;;) {
        if ( DivLong(x1,y1,x2,y2,h,&nh,x3,&y3) ) goto ModErr;
        if ( MulLong(x5,y5,h,nh,x6,&y6) ) goto ModErr;
        if ( AddLong(x4,y4,x6,-y6,x6,&y6) ) goto ModErr;
        if ( !y3 ) {
            if ( y2 != 1 || *x2 != 1 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Inverse in modulus arithmetic doesn't exist");
                MesPrint("Denominator and modulus are not relative prime");
                MUNLOCK(ErrorMessageLock);
                goto ModErr;
            }
            break;
        }
        x7 = x1; x1 = x2; y1 = y2; x2 = x3; y2 = y3; x3 = x7;
        x8 = x4; x4 = x5; y4 = y5; x5 = x6; y5 = y6; x6 = x8;
      }
      if ( y5 < 0 && AddLong((UWORD *)cmodvec,nmod,x5,y5,x5,&y5) ) goto ModErr;
      if ( MulLong(a,tnumer,x5,y5,c,&y3) ) goto ModErr;
      if ( DivLong(c,y3,(UWORD *)cmodvec,nmod,d,&y5,a,&tdenom) ) goto ModErr;
    }
    if ( !tdenom ) { *na = 0; goto normalreturn; }
    if ( ( ( AC.modmode & POSNEG ) != 0 ) && ( ( par & FROMFUNCTION ) == 0 ) ) {
        NormalModulus(a,&tdenom);
    }
    else if ( tdenom < 0 ) {
        SubPLon((UWORD *)cmodvec,nmod,a,-tdenom,a,&tdenom);
    }
    *na = tdenom;
    i = ABS(tdenom);
    a += i;
    *a++ = 1;
    while ( --i > 0 ) *a++ = 0;
normalreturn:
    NumberFree(c,"TakeModulus"); NumberFree(d,"TakeModulus"); NumberFree(e,"TakeModulus");
    NumberFree(f,"TakeModulus"); NumberFree(g,"TakeModulus"); NumberFree(h,"TakeModulus");
    return(0);
ModErr:
    MLOCK(ErrorMessageLock);
    MesCall("TakeModulus");
    MUNLOCK(ErrorMessageLock);
    NumberFree(c,"TakeModulus"); NumberFree(d,"TakeModulus"); NumberFree(e,"TakeModulus");
```

```
    NumberFree(f,"TakeModulus"); NumberFree(g,"TakeModulus"); NumberFree(h,"TakeModulus");
    SETERROR(-1)
}

/*
        #] TakeModulus :
        #[ TakeNormalModulus :  WORD TakeNormalModulus(a,na,par)

        added by Jan [01-09-2010]
*/

WORD TakeNormalModulus (UWORD *a, WORD *na, UWORD *c, WORD nc, WORD par)
{
    WORD n;
    WORD nhalfc;
    UWORD *halfc;

    GETIDENTITY;

    /* determine c/2 by right shifting */
    halfc = NumberMalloc("TakeNormalModulus");
    nhalfc=nc;
    WCOPY(halfc,c,nc);

    for (n=0; n<nhalfc; n++) {
        halfc[n] /= 2;
        if (n+1<nc) halfc[n] |= c[n+1] « (BITSINWORD-1);
    }

    if (halfc[nhalfc-1]==0)
        nhalfc--;

    /* takes care of the number never expanding, e.g., -1(mod 100) -> 99 -> -1 */
    if (BigLong(a,ABS(*na),halfc,nhalfc) > 0) {

        TakeModulus(a,na,c,nc,par);

        n = ABS(*na);
        if (BigLong(a,n,halfc,nhalfc) > 0) {
            SubPLon(c,nc,a,n,a,&n);
            *na = (*na > 0 ? -n : n);
        }
    }

    NumberFree(halfc,"TakeNormalModulus");
    return(0);
}

/*
        #] TakeNormalModulus :
        #[ MakeModTable :   WORD MakeModTable()
*/

WORD MakeModTable(void)
{
    LONG size, i, j, n;
    n = ABS(AC.ncmod);
    if ( AC.modpowers ) {
        M_free(AC.modpowers,"AC.modpowers");
        AC.modpowers = NULL;
    }
    if ( n > 2 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("&No memory for modulus generator power table");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( n == 0 ) return(0);
    size = (LONG)(*AC.cmod);
    if ( n == 2 ) size += (((LONG)AC.cmod[1])«BITSINWORD);
    AC.modpowers = (UWORD *)Malloc1(size*n*sizeof(UWORD),"table for powers of modulus");
    if ( n == 1 ) {
        j = 1;
        for ( i = 0; i < size; i++ ) AC.modpowers[i] = 0;
        for ( i = 0; i < size; i++ ) {
            AC.modpowers[j] = (WORD)i;
            j *= *AC.powmod;
            j %= *AC.cmod;
        }
        for ( i = 2; i < size; i++ ) {
            if ( AC.modpowers[i] == 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("&improper generator for this modulus");
                MUNLOCK(ErrorMessageLock);
                M_free(AC.modpowers,"AC.modpowers");
                return(-1);
            }
```

```
        }
        AC.modpowers[1] = 0;
    }
    else {
        GETIDENTITY
        WORD nScrat, n2;
        UWORD *MMscrat7 = NumberMalloc("MakeModTable"), *MMscratC = NumberMalloc("MakeModTable");
        *MMscratC = 1;
        nScrat = 1;
        j = size * 2;
        for ( i = 0; i < j; i+=2 ) { AC.modpowers[i] = 0; AC.modpowers[i+1] = 0; }
        for ( i = 0; i < size; i++ ) {
            j = *MMscratC + (((LONG)MMscratC[1])«BITSINWORD);
            j *= 2;
            AC.modpowers[j] = (WORD)(i & WORDMASK);
            AC.modpowers[j+1] = (WORD)(i » BITSINWORD);
            MulLong((UWORD *)MMscratC,nScrat,(UWORD *)AC.powmod,
            AC.npowmod,(UWORD *)MMscrat7,&n2);
            TakeModulus(MMscrat7,&n2,AC.cmod,AC.ncmod,NOUNPACK);
            *MMscratC = *MMscrat7; MMscratC[1] = MMscrat7[1]; nScrat = n2;
        }
        NumberFree(MMscrat7,"MakeModTable"); NumberFree(MMscratC,"MakeModTable");
        j = size * 2;
        for ( i = 4; i < j; i+=2 ) {
            if ( AC.modpowers[i] == 0 && AC.modpowers[i+1] == 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("&improper generator for this modulus");
                MUNLOCK(ErrorMessageLock);
                M_free(AC.modpowers,"AC.modpowers");
                return(-1);
            }
        }
        AC.modpowers[2] = AC.modpowers[3] = 0;
    }
    return(0);
}

/*
        #] MakeModTable :
    #] RekenTerms :
    #[ Functions :
        #[ Factorial :      WORD Factorial(n,a,na)

    Starts with only the value of fac_(0).
    Builds up what is needed and remembers it for the next time.

    We have:
        AT.nfac: the number of the highest stored factorial
        AT.pfac: the array of locations in the array of stored factorials
        AT.factorials: the array with stored factorials
*/

int Factorial(PHEAD WORD n, UWORD *a, WORD *na)
{
    GETBIDENTITY
    UWORD *b, *c;
    WORD nc;
    int i, j;
    LONG ii;
    if ( n > AT.nfac ) {
        if ( AT.factorials == 0 ) {
            AT.nfac = 0; AT.mfac = 50; AT.sfact = 400;
            AT.pfac = (LONG *)Malloc1((AT.mfac+2)*sizeof(LONG),"factorials");
            AT.factorials = (UWORD *)Malloc1(AT.sfact*sizeof(UWORD),"factorials");
            AT.factorials[0] = 1; AT.pfac[0] = 0; AT.pfac[1] = 1;
        }
        b = a;
        c = AT.factorials+AT.pfac[AT.nfac];
        nc = i = AT.pfac[AT.nfac+1] - AT.pfac[AT.nfac];
        while ( --i >= 0 ) *b++ = *c++;
        for ( j = AT.nfac+1; j <= n; j++ ) {
            Product(a,&nc,j);
            if ( nc > AM.MaxTal ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Overflow in factorial. MaxTal = %d",AM.MaxTal);
                MesPrint("Increase MaxTerm in %s",setupfilename);
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            if ( j > AT.mfac ) {  /* double the pfac buffer */
                LONG *p;
                p = (LONG *)Malloc1((AT.mfac*2+2)*sizeof(LONG),"factorials");
                i = AT.mfac;
                for ( i = AT.mfac+1; i >= 0; i-- ) p[i] = AT.pfac[i];
                M_free(AT.pfac,"factorial offsets"); AT.pfac = p; AT.mfac *= 2;
            }
            if ( AT.pfac[j] + nc >= AT.sfact ) { /* double the factorials buffer */
```

```
                UWORD *f;
                f = (UWORD *)Malloc1(AT.sfact*2*sizeof(UWORD),"factorials");
                ii = AT.sfact;
                c = AT.factorials; b = f;
                while ( --ii >= 0 ) *b++ = *c++;
                M_free(AT.factorials,"factorials");
                AT.factorials = f;
                AT.sfact *= 2;
            }
            b = a; c = AT.factorials + AT.pfac[j]; i = nc;
            while ( --i >= 0 ) *c++ = *b++;
            AT.pfac[j+1] = AT.pfac[j] + nc;
        }
        *na = nc;
        AT.nfac = n;
    }
    else if ( n == 0 ) {
        *a = 1; *na = 1;
    }
    else {
        *na = i = AT.pfac[n+1] - AT.pfac[n];
        b = AT.factorials + AT.pfac[n];
        while ( --i >= 0 ) *a++ = *b++;
    }
    return(0);
}

/*
        #] Factorial :
        #[ Bernoulli :      WORD Bernoulli(n,a,na)

    Starts with only the value of bernoulli_(0).
    Builds up what is needed and remembers it for the next time.
    b_0 = 1
    (n+1)*b_n = -b_{n-1}-sum_(i,1,n-1,b_i*b_{n-i})
    The n-1 plays only a role for b_2.
    We have hard coded b_0,b_1,b_2 and b_odd. After that:
    (2n+1)*b_2n = -sum_(i,1,n-1,b_2i*b_{2n-2i})

    We have:
        AT.nBer: the number of the highest stored Bernoulli number
        AT.pBer: the array of locations in the array of stored Bernoulli numbers
        AT.bernoullis: the array with stored Bernoulli numbers
*/

int Bernoulli(WORD n, UWORD *a, WORD *na)
{
    GETIDENTITY
    UWORD *b, *c, *scrib, *ntop, *ntop1;
    WORD i, i1, i2, nhalf, nqua, nscrib, nntop, nntop1, *oldworkpointer;
    UWORD twee = 2, twonplus1;
    int j;
    LONG ii;
    if ( n <= 1 ) {
        if ( n == 0 ) { a[0] = a[1] = 1; *na = 3; }
        else if ( n == 1 ) { a[0] = 1; a[1] = 2; *na = 3; }
        return(0);
    }
    if ( ( n & 1 ) != 0 ) { a[0] = a[1] = 0; *na = 0; return(0); }
    nhalf = n/2;
    if ( nhalf > AT.nBer ) {
        oldworkpointer = AT.WorkPointer;
        if ( AT.bernoullis == 0 ) {
            AT.nBer = 1; AT.mBer = 50; AT.sBer = 400;
            AT.pBer = (LONG *)Malloc1((AT.mBer+2)*sizeof(LONG),"bernoullis");
            AT.bernoullis = (UWORD *)Malloc1(AT.sBer*sizeof(UWORD),"bernoullis");
            AT.pBer[1] = 0; AT.pBer[2] = 3;
            AT.bernoullis[0] = 3; AT.bernoullis[1] = 1; AT.bernoullis[2] = 12;
            if ( nhalf == 1 ) {
                a[0] = 1; a[1] = 12; *na = 3; return(0);
            }
        }
        while ( nhalf > AT.mBer ) {
            LONG *p;
            p = (LONG *)Malloc1((AT.mBer*2+1)*sizeof(LONG),"bernoullis");
            i = AT.mBer;
            for ( i = AT.mBer; i >= 0; i-- ) p[i] = AT.pBer[i];
            M_free(AT.pBer,"factorial pointers"); AT.pBer = p; AT.mBer *= 2;
        }
        for ( n = AT.nBer+1; n <= nhalf; n++ ) {
            scrib = (UWORD *)(AT.WorkPointer);
            nqua = n/2;
            if ( ( n & 1 ) == 1 ) {
                nscrib = 0; ntop = scrib;
            }
            else {
                b = AT.bernoullis + AT.pBer[nqua];
```

```
                    nscrib = *b++;
                    i = (WORD)(REDLENG(nscrib));
                    MulRat(BHEAD b,i,b,i,scrib,&nscrib);
                    ntop = scrib + 2*nscrib;
                    nqua--;
                }
                for ( j = 1; j <= nqua; j++ ) {
                    b = AT.bernoullis + AT.pBer[j];
                    c = AT.bernoullis + AT.pBer[n-j];
                    i1 = (WORD)(*b); i2 = (WORD)(*c);
                    i1 = REDLENG(i1);
                    i2 = REDLENG(i2);
                    MulRat(BHEAD b+1,i1,c+1,i2,ntop,&nntop);
                    Mully(BHEAD ntop,&nntop,&twee,1);
                    if ( nscrib ) {
                        i = (WORD)nntop; if ( i < 0 ) i = -i;
                        ntop1 = ntop + 2*i;
                        AddRat(BHEAD ntop,nntop,scrib,nscrib,ntop1,&nntop1);
                    }
                    else {
                        ntop1 = ntop; nntop1 = nntop;
                    }
                    nscrib = i1 = (WORD)nntop1;
                    if ( i1 < 0 ) i1 = - i1;
                    i1 = 2*i1;
                    for ( i = 0; i < i1; i++ ) scrib[i] = ntop1[i];
                    ntop = scrib + i1;
                }
                twonplus1 = 2*n+1;
                Divvy(BHEAD scrib,&nscrib,&twonplus1,-1);
                i1 = INCLENG(nscrib);
                i2 = i1; if ( i2 < 0 ) i2 = -i2;
                i = (WORD)(AT.bernoullis[AT.pBer[n-1]]);
                if ( i < 0 ) i = -i;
                AT.pBer[n] = AT.pBer[n-1]+i;
                if ( AT.pBer[n] + i2 >= AT.sBer ) {
                    UWORD *f;
                    f = (UWORD *)Malloc1(AT.sBer*2*sizeof(UWORD),"bernoullis");
                    ii = AT.sBer;
                    c = AT.bernoullis; b = f;
                    while ( --ii >= 0 ) *b++ = *c++;
                    M_free(AT.bernoullis,"bernoullis");
                    AT.bernoullis = f;
                    AT.sBer *= 2;
                }
                c = AT.bernoullis + AT.pBer[n]; b = scrib;
                *c++ = i1;
                for ( i = 1; i < i2; i++ ) *c++ = *b++;
            }
            AT.nBer = nhalf;
            AT.WorkPointer = oldworkpointer;
        }
        b = AT.bernoullis + AT.pBer[nhalf];
        *na = i = (WORD)(*b++);
        if ( i < 0 ) i = -i;
        i--;
        while ( --i >= 0 ) *a++ = *b++;
        return(0);
}

/*
        #] Bernoulli :
        #[ NextPrime :
*/
#if ( BITSINWORD == 32 )

void StartPrimeList(PHEAD0)
{
    int i, j;
    AR.PrimeList[AR.numinprimelist++] = 3;
    for ( i = 5; i < 46340; i += 2 ) {
        for ( j = 0; j < AR.numinprimelist && AR.PrimeList[j]*AR.PrimeList[j] <= i; j++ ) {
            if ( i % AR.PrimeList[j] == 0 ) goto nexti;
        }
        AR.PrimeList[AR.numinprimelist++] = i;
nexti:;
    }
    AR.notfirstprime = 1;
}

#endif

WORD NextPrime(PHEAD WORD num)
{
    int i, j;
    WORD *newpl;
    LONG newsize, x;
```

```
#if ( BITSINWORD == 32 )
    if ( AR.notfirstprime == 0 ) StartPrimeList(BHEAD0);
#endif
    if ( num > AT.inprimelist ) {
        while ( AT.inprimelist < num ) {
            if ( num >= AT.sizeprimelist ) {
                if ( AT.sizeprimelist == 0 ) newsize = 32;
                else newsize = 2*AT.sizeprimelist;
                while ( num >= newsize ) newsize = newsize*2;
                newpl = (WORD *)Malloc1(newsize*sizeof(WORD),"NextPrime");
                for ( i = 0; i < AT.sizeprimelist; i++ ) {
                    newpl[i] = AT.primelist[i];
                }
                if ( AT.sizeprimelist > 0 ) {
                    M_free(AT.primelist,"NextPrime");
                }
                AT.sizeprimelist = newsize;
                AT.primelist = newpl;
            }
            if ( AT.inprimelist < 0 ) { i = MAXPOSITIVE; }
            else { i = AT.primelist[AT.inprimelist]; }
            while ( i > MAXPOWER ) {
                i -= 2; x = i;
#if ( BITSINWORD == 32 )
                for ( j = 0; j < AR.numinprimelist && AR.PrimeList[j]*(LONG)(AR.PrimeList[j]) <= x; j++ ) {
                    if ( x % AR.PrimeList[j] == 0 ) goto nexti;
                }
#else
                for ( j = 3; j*((LONG)j) <= x; j += 2 ) {
                    if ( x % j == 0 ) goto nexti;
                }
#endif
                AT.inprimelist++;
                AT.primelist[AT.inprimelist] = i;
                break;
nexti:;
            }
            if ( i < MAXPOWER ) {
                MLOCK(ErrorMessageLock);
                MesPrint("There are not enough short prime numbers for this calculation");
                MesPrint("Try to use a computer with a %d-bits architecture",
                    (int)(BITSINWORD*4));
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
        }
    }
    return(AT.primelist[num]);
}

/*
        #] NextPrime :
        #[ Moebius :

        Returns the value of the Moebius function if n fits inside
        a WORD.
        The method we use is a bit like the sieve of Erathostenes.
*/

WORD Moebius(PHEAD WORD nn)
{
    WORD i,n = nn, x;
    LONG newsize;
    SBYTE *newtable, mu;
#if ( BITSINWORD == 32 )
    if ( AR.notfirstprime == 0 ) StartPrimeList(BHEAD0);
#endif
/*
    First we make sure that:
        a: the table is big enough.
        b: the number is not already in the table.
*/
    if ( nn >= AR.moebiustablesize ) {
        if ( AR.moebiustablesize <= 0 ) { newsize = (LONG)nn + 20; }
        else { newsize = (LONG)nn*2; }
        if ( newsize > MAXPOSITIVE ) newsize = MAXPOSITIVE;
        newtable = (SBYTE *)Malloc1(newsize*sizeof(SBYTE),"Moebius");
        for ( i = 0; i < AR.moebiustablesize; i++ ) newtable[i] = AR.moebiustable[i];
        for ( ; i < newsize; i++ ) newtable[i] = 2;
        if ( AR.moebiustablesize > 0 ) M_free(AR.moebiustable,"Moebius");
        AR.moebiustable = newtable;
        AR.moebiustablesize = newsize;
    }
    /* NOTE: nn == MAXPOSITIVE never fits in moebiustable. */
    if ( nn != MAXPOSITIVE && AR.moebiustable[nn] != 2 ) return((WORD)AR.moebiustable[nn]);
    mu = 1;
```

```
    if ( n == 1 ) goto putvalue;
    if ( n % 2 == 0 ) {
        n /= 2;
        if ( n % 2 == 0 ) { mu = 0; goto putvalue; }
        if ( AR.moebiustable[n] != 2 ) { mu = -AR.moebiustable[n]; goto putvalue; }
        mu = -mu;
        if ( n == 1 ) goto putvalue;
    }
#if ( BITSINWORD == 32 )
    for ( i = 0; i < AR.numinprimelist; i++ ) {
        x = AR.PrimeList[i];
#else
    for ( x = 3; x < MAXPOSITIVE; x += 2 ) {
#endif
        if ( n % x == 0 ) {
            n /= x;
            if ( n % x == 0 ) { mu = 0; goto putvalue; }
            if ( AR.moebiustable[n] != 2 ) { mu = -AR.moebiustable[n]; goto putvalue; }
            mu = -mu;
            if ( n == 1 ) goto putvalue;
        }
        if ( n < x*x ) break; /* notice that x*x always fits inside a WORD */
    }
    mu = -mu;
putvalue:
    if ( nn != MAXPOSITIVE ) AR.moebiustable[nn] = mu;
    return((WORD)mu);
}

/*
        #] Moebius :
        #[ wranf :

        A random number generator that generates random WORDs with a very
        long sequence. It is based on the Knuth generator.

        We take some care that each thread can run its own, but each
        uses its own startup. Hence the seed includes the identity of
        the thread.

        For NPAIR1, NPAIR2 we can use any pair from the table on page 28.
        Candidates are 24,55 (the example on the pages 171,172)
        or (33,97) or (38,89)
        These values are defined in fsizes.h and used in startup.c and threads.c
*/

#define WARMUP 6

static void wranfnew(PHEAD0)
{
    int i;
    LONG j;
    for ( i = 0; i < AR.wranfnpair1; i++ ) {
        j = AR.wranfia[i] - AR.wranfia[i+(AR.wranfnpair2-AR.wranfnpair1)];
        if ( j < 0 ) j += (LONG)1 « (2*BITSINWORD-2);
        AR.wranfia[i] = j;
    }
    for ( i = AR.wranfnpair1; i < AR.wranfnpair2; i++ ) {
        j = AR.wranfia[i] - AR.wranfia[i-AR.wranfnpair1];
        if ( j < 0 ) j += (LONG)1 « (2*BITSINWORD-2);
        AR.wranfia[i] = j;
    }
}

void iniwranf(PHEAD0)
{
    int imax = AR.wranfnpair2-1;
    ULONG i, ii, seed = AR.wranfseed;
    LONG j, k;
    ULONG offset = 12345;
#ifdef PARALLELCODE
    int id;
#if defined(WITHPTHREADS)
    id = AT.identity;
#elif defined(WITHMPI)
    id = PF.me;
#endif
    seed += id;
    i = id + 1;
    if ( i > 1 ) {
        ULONG pow, accu;
        pow = offset; accu = 1;
        while ( i ) {
            if ( ( i & 1 ) != 0 ) accu *= pow;
            i /= 2; pow = pow*pow;
        }
        offset = accu;
```

```
    }
#endif
    if ( seed < ((LONG)1«(BITSINWORD-1)) ) {
        j = ( (seed+31459L) « (BITSINWORD-2))+offset;
    }
    else if ( seed < ((LONG)1«(BITSINWORD+10-1)) ) {
        j = ( (seed+31459L) « (BITSINWORD-10-2))+offset;
    }
    else {
        j = ( (seed+31459L) « 1)+offset;
    }
    if ( ( seed & 1 ) == 1 ) seed++;
    j += seed;
    AR.wranfia[imax] = j;
    k = 1;
    for ( i = 0; i <= (ULONG)(imax); i++ ) {
        ii = (AR.wranfnpair1*i)%AR.wranfnpair2;
        AR.wranfia[ii] = k;
        k = ULongToLong((ULONG)j - (ULONG)k);
        if ( k < 0 ) k += (LONG)1 « (2*BITSINWORD-2);
        j = AR.wranfia[ii];
    }
    for ( i = 0; i < WARMUP; i++ ) wranfnew(BHEAD0);
    AR.wranfcall = 0;
}

UWORD wranf(PHEAD0)
{
    UWORD wval;
    if ( AR.wranfia == 0 ) {
        AR.wranfia = (ULONG *)Malloc1(AR.wranfnpair2*sizeof(ULONG),"wranf");
        iniwranf(BHEAD0);
    }
    if ( AR.wranfcall >= AR.wranfnpair2) {
        wranfnew(BHEAD0);
        AR.wranfcall = 0;
    }
    wval = (UWORD)(AR.wranfia[AR.wranfcall++]»(BITSINWORD-1));
    return(wval);
}

/*
    Returns a random UWORD in the range (0,...,imax-1)
*/

UWORD iranf(PHEAD UWORD imax)
{
    UWORD i;
    ULONG x, xmax;
    if (imax < 2) return 0;
    x = (LONG)1 « BITSINWORD;
    xmax = x - x%imax;
    while ( ( i = wranf(BHEAD0) ) >= xmax ) {}
    return(i%imax);
}

/*
        #] wranf :
        #[ PreRandom :

        The random number generator of the preprocessor.
        This one is completely different from the execution time generator
        random_(number). In the preprocessor we generate a floating point
        number in a string according to a distribution.
        Currently allowed are:
            RANDOM_(log,min,max)
            RANDOM_(lin,min,max)
        The return value is a string with the floating point number.
*/

UBYTE *PreRandom(UBYTE *s)
{
    GETIDENTITY
    UBYTE *mode,*mins = 0,*maxs = 0, *outval;
    float num;
    double minval, maxval, value = 0;
    int linlog = -1;
    mode = s;
    while ( FG.cTable[*s] <= 1 ) s++;
    if ( *s == ',' ) { *s = 0; s++; }
    mins = s;
    while ( *s && *s != ',' ) s++;
    if ( *s == ',' ) { *s = 0; s++; }
    maxs = s;
    while ( *s && *s != ',' ) s++;
    if ( *s || *maxs == 0 || *mins == 0 ) {
        MesPrint("@Illegal arguments in macro RANDOM_");
```

```
        Terminate(-1);
    }
    if ( StrICmp(mode,(UBYTE *)"lin") == 0 ) {
        linlog = 0;
    }
    else if ( StrICmp(mode,(UBYTE *)"log") == 0 ) {
        linlog = 1;
    }
    else {
        MesPrint("@Illegal mode argument in macro RANDOM_");
        Terminate(-1);
    }

    sscanf((char *)mins,"%f",&num); minval = num;
    sscanf((char *)maxs,"%f",&num); maxval = num;

    /*
     * Note on ParFORM: we should use the same random number on all the
     * processes in the complication phase. The random number is generated
     * on the master and broadcast to the other processes.
     */
    {
        UWORD x;
        double xx;
#ifdef WITHMPI
        x = 0;
        if ( PF.me == MASTER ) {
            x = wranf(BHEAD0);
        }
        x = (UWORD)PF_BroadcastNumber((LONG)x);
#else
        x = wranf(BHEAD0);
#endif
        xx = x/pow(2.0,(double)(BITSINWORD-1));
        if ( linlog == 0 ) {
            value = minval + (maxval-minval)*xx;
        }
        else if ( linlog == 1 ) {
            value = minval * pow(maxval/minval,xx);
        }
    }

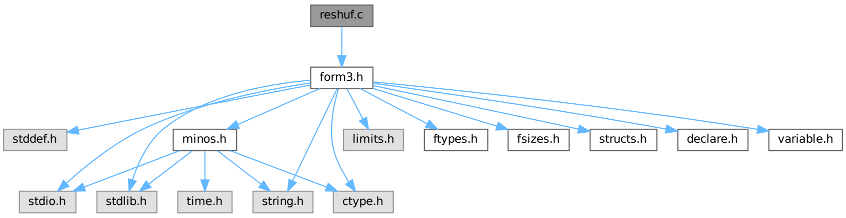
    outval = (UBYTE *)Malloc1(64,"PreRandom");
    if ( ABS(value) < 0.00001 || ABS(value) > 1000000. ) {
        snprintf((char *)outval,64,"%e",value);
    }
    else if ( ABS(value) < 0.0001 ) { snprintf((char *)outval,64,"%10f",value); }
    else if ( ABS(value) < 0.001 ) { snprintf((char *)outval,64,"%9f",value); }
    else if ( ABS(value) < 0.01 ) { snprintf((char *)outval,64,"%8f",value); }
    else if ( ABS(value) < 0.1 ) { snprintf((char *)outval,64,"%7f",value); }
    else if ( ABS(value) < 1. ) { snprintf((char *)outval,64,"%6f",value); }
    else if ( ABS(value) < 10. ) { snprintf((char *)outval,64,"%5f",value); }
    else if ( ABS(value) < 100. ) { snprintf((char *)outval,64,"%4f",value); }
    else if ( ABS(value) < 1000. ) { snprintf((char *)outval,64,"%3f",value); }
    else if ( ABS(value) < 10000. ) { snprintf((char *)outval,64,"%2f",value); }
    else { snprintf((char *)outval,64,"%1f",value); }
    return(outval);
}

/*
        #] PreRandom :
    #] Functions :
*/
```

4.120 reshuf.c File Reference

`#include "form3.h"`

Include dependency graph for reshuf.c:



Functions

- WORD ReNumber (PHEAD WORD ∗term)
- void FunLevel (PHEAD WORD ∗term)
- WORD DetCurDum (PHEAD WORD ∗t)
- int FullRenumber (PHEAD WORD ∗term, WORD par)
- void MoveDummies (PHEAD WORD ∗term, WORD shift)
- void AdjustRenumScratch (PHEAD0)
- WORD CountDo (WORD ∗term, WORD ∗instruct)
- WORD CountFun (WORD ∗term, WORD ∗countfun)
- WORD DimensionSubterm (WORD ∗subterm)
- WORD DimensionTerm (WORD ∗term)
- WORD DimensionExpression (PHEAD WORD ∗expr)
- WORD MultDo (PHEAD WORD ∗term, WORD ∗pattern)
- WORD TryDo (PHEAD WORD ∗term, WORD ∗pattern, WORD level)
- WORD DoDistrib (PHEAD WORD ∗term, WORD level)
- WORD EqualArg (WORD ∗parms, WORD num1, WORD num2)
- WORD DoDelta3 (PHEAD WORD ∗term, WORD level)
- WORD TestPartitions (WORD ∗tfun, PARTI ∗parti)
- WORD DoPartitions (PHEAD WORD ∗term, WORD level)
- WORD DoPermutations (PHEAD WORD ∗term, WORD level)
- WORD DoShuffle (WORD ∗term, WORD level, WORD fun, WORD option)
- int Shuffle (WORD ∗from1, WORD ∗from2, WORD ∗to)
- int FinishShuffle (WORD ∗fini)
- WORD DoStuffle (WORD ∗term, WORD level, WORD fun, WORD option)
- int Stuffle (WORD ∗from1, WORD ∗from2, WORD ∗to)
- int FinishStuffle (WORD ∗fini)
- WORD ∗ StuffRootAdd (WORD ∗t1, WORD ∗t2, WORD ∗to)

4.120.1 Detailed Description

Mixed routines: Routines for relabelling dummy indices. The multiply command The distrib_ function The tryreplace statement

Definition in file reshuf.c.

4.120.2 Macro Definition Documentation

4.120.2.1 NEWCODE

```
#define NEWCODE
```

Definition at line 35 of file reshuf.c.

4.120.3 Function Documentation

4.120.3.1 ReNumber()

```
WORD ReNumber (
             PHEAD WORD * term )
```

Definition at line 80 of file reshuf.c.

4.120.3.2 FunLevel()

```
void FunLevel (
             PHEAD WORD * term )
```

Definition at line 130 of file reshuf.c.

4.120.3.3 DetCurDum()

```
WORD DetCurDum (
             PHEAD WORD * t )
```

Definition at line 252 of file reshuf.c.

4.120.3.4 FullRenumber()

```
int FullRenumber (
             PHEAD WORD * term,
             WORD par )
```

Definition at line 324 of file reshuf.c.

4.120.3.5 MoveDummies()

```
void MoveDummies (
             PHEAD WORD * term,
             WORD shift )
```

Definition at line 436 of file reshuf.c.

4.120.3.6 AdjustRenumScratch()

```
void AdjustRenumScratch (
            PHEAD0  )
```

Definition at line 506 of file reshuf.c.

4.120.3.7 CountDo()

```
WORD CountDo (
            WORD * term,
            WORD * instruct )
```

Definition at line 552 of file reshuf.c.

4.120.3.8 CountFun()

```
WORD CountFun (
            WORD * term,
            WORD * countfun )
```

Definition at line 706 of file reshuf.c.

4.120.3.9 DimensionSubterm()

```
WORD DimensionSubterm (
            WORD * subterm )
```

Definition at line 867 of file reshuf.c.

4.120.3.10 DimensionTerm()

```
WORD DimensionTerm (
            WORD * term )
```

Definition at line 968 of file reshuf.c.

4.120.3.11 DimensionExpression()

```
WORD DimensionExpression (
            PHEAD WORD * expr )
```

Definition at line 1000 of file reshuf.c.

4.120.3.12 MultDo()

```
WORD MultDo (
            PHEAD WORD * term,
            WORD * pattern )
```

Definition at line 1044 of file reshuf.c.

4.120.3.13 TryDo()

```
WORD TryDo (
            PHEAD WORD * term,
            WORD * pattern,
            WORD level )
```

Definition at line 1072 of file reshuf.c.

4.120.3.14 DoDistrib()

```
WORD DoDistrib (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1119 of file reshuf.c.

4.120.3.15 EqualArg()

```
WORD EqualArg (
            WORD * parms,
            WORD num1,
            WORD num2 )
```

Definition at line 1379 of file reshuf.c.

4.120.3.16 DoDelta3()

```
WORD DoDelta3 (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1405 of file reshuf.c.

4.120.3.17 TestPartitions()

```
WORD TestPartitions (
            WORD * tfun,
            PARTI * parti )
```

Definition at line 1607 of file reshuf.c.

4.120.3.18 DoPartitions()

```
WORD DoPartitions (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 1722 of file reshuf.c.

4.120.3.19 DoPermutations()

```
WORD DoPermutations (
            PHEAD WORD * term,
            WORD level )
```

Definition at line 2007 of file reshuf.c.

4.120.3.20 DoShuffle()

```
WORD DoShuffle (
            WORD * term,
            WORD level,
            WORD fun,
            WORD option )
```

Definition at line 2095 of file reshuf.c.

4.120.3.21 Shuffle()

```
int Shuffle (
            WORD * from1,
            WORD * from2,
            WORD * to )
```

Definition at line 2229 of file reshuf.c.

4.120.3.22 FinishShuffle()

```
int FinishShuffle (
            WORD * fini )
```

Definition at line 2421 of file reshuf.c.

4.120.3.23 DoStuffle()

```
WORD DoStuffle (
            WORD * term,
            WORD level,
            WORD fun,
            WORD option )
```

Definition at line 2487 of file reshuf.c.

4.120.3.24 Stuffle()

```
int Stuffle (
            WORD * from1,
            WORD * from2,
            WORD * to )
```

Definition at line 2702 of file reshuf.c.

4.120.3.25 FinishStuffle()

```
int FinishStuffle (
            WORD * fini )
```

Definition at line 2829 of file reshuf.c.

4.120.3.26 StuffRootAdd()

```
WORD * StuffRootAdd (
            WORD * t1,
            WORD * t2,
            WORD * to )
```

Definition at line 2876 of file reshuf.c.

4.121 reshuf.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 */
/* #] License : */
#define NEWCODE
/*
    #[ Includes : reshuf.c
*/

#include "form3.h"

/*
    #] Includes :
```

```
    #[ Reshuf :

    Routines to rearrange dummy indices, so that
    a:  The notation becomes reasonably unique (the perfect job
        may consume very much time).
    b:  The value of AR.CurDum is reset.

    Also some routines are needed to aid in the reading of stored
    expressions. Also in those expressions there can be dummy
    indices, and there should be no conflict with the already
    existing dummies.

        #[ ReNumber :

        Reads the term, tests for dummies, and puts them in order.
        Note that this is kind of a first order approximation.
        There is quite some room to make this routine 'smart'
        First order:
            First index will be lowest, second will be next etc.
        Second order:
            Functions with more than one index and symmetry properties
            have some look ahead to see which index is the first to
            find its partner.
        Third order:
            Take the ordering of the functions into account.
        Fourth order:
            Try all permutations and see which one gives the 'minimal' term.
        Currently we use only the first order.

        We need a scratch array for the numbers we find, and one for
        the addresses at which these numbers are.
        We can use the space for the Scrat arrays. There are 13 of those
        and each is AM.MaxTal UWORDs long.

*/

WORD ReNumber(PHEAD WORD *term)
{
    GETBIDENTITY
    WORD *d, *e, **p, **f;
    WORD n, i, j, old;
    AN.DumFound = AN.RenumScratch;
    AN.DumPlace = AN.PoinScratch;
    AN.DumFunPlace = AN.FunScratch;
    AN.NumFound = 0;
    FunLevel(BHEAD term);
    d = AN.RenumScratch;
    p = AN.PoinScratch;
    f = AN.FunScratch;
/*
    Now the first level sorting.
*/
    i = AN.IndDum;
    n = AN.NumFound;
    while ( --n >= 0 ) {
        if ( *d > 0 ) {
            old = **p;
            **p = ++i;
            if ( *f ) **f |= DIRTYSYMFLAG;
            e = d;
            e++;
            for ( j = 1; j <= n; j++ ) {
                if ( *e && *(p[j]) == old ) {
                    *(p[j]) = i;
                    if ( f[j] ) *(f[j]) |= DIRTYSYMFLAG;
                    *e = 0;
                }
                e++;
            }
        }
        p++;
        d++;
        f++;
    }
    return(i);
}

/*
        #] ReNumber :
        #[ FunLevel :

        Does one term in determining where the dummies are.
        Made to work recursively for functions.

*/

void FunLevel(PHEAD WORD *term)
```

```
{
    GETBIDENTITY
    WORD *t, *tstop, *r, *fun;
    WORD *m;
    t = r = term;
    r += *r;
    tstop = r - ABS(r[-1]);
    t++;
    if ( t < tstop ) do {
        r = t + t[1];
        switch ( *t ) {
            case SYMBOL:
            case DOTPRODUCT:
                break;
            case VECTOR:
                t += 3;
                do {
                    if ( *t > AN.IndDum ) {
                        if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
                        AN.NumFound++;
                        *AN.DumFound++ = *t;
                        *AN.DumPlace++ = t;
                        *AN.DumFunPlace++ = 0;
                    }
                    t += 2;
                } while ( t < r );
                break;
            case SUBEXPRESSION:
/*
        Still must hunt down the wildcards(?)
*/
                break;
            case GAMMA:
                t += FUNHEAD-2;
                /* fall through */
            case DELTA:
            case INDEX:
                t += 2;
                while ( t < r ) {
                    if ( *t > AN.IndDum ) {
                        if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
                        AN.NumFound++;
                        *AN.DumFound++ = *t;
                        *AN.DumPlace++ = t;
                        *AN.DumFunPlace++ = 0;
                    }
                    t++;
                }
                break;
            case HAAKJE:
            case EXPRESSION:
            case SNUMBER:
            case LNUMBER:
                break;
            default:
                if ( *t < FUNCTION ) {
                  MLOCK(ErrorMessageLock);
                  MesPrint("Unexpected code in ReNumber");
                  MUNLOCK(ErrorMessageLock);
                  Terminate(-1);
                }
                fun = t+2;
                if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
                >= TENSORFUNCTION ) {
                    t += FUNHEAD;
                    while ( t < r ) {
                        if ( *t > AN.IndDum ) {
                            if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
                            AN.NumFound++;
                            *AN.DumFound++ = *t;
                            *AN.DumPlace++ = t;
                            *AN.DumFunPlace++ = fun;
                        }
                        t++;
                    }
                    break;
                }

                t += FUNHEAD;
                while ( t < r ) {
                    if ( *t > 0 ) {

                        /* General function. Enter 'recursion'. */

                        m = t + *t;
                        t += ARGHEAD;
                        while ( t < m ) {
```

```
                            FunLevel(BHEAD t);
                            t += *t;
                        }
                    }
                    else {
                        if ( *t == -INDEX ) {
                            t++;
                            if ( *t >= AN.IndDum ) {
                                if ( AN.NumFound >= AN.MaxRenumScratch ) AdjustRenumScratch(BHEAD0);
                                AN.NumFound++;
                                *AN.DumFound++ = *t;
                                *AN.DumPlace++ = t;
                                *AN.DumFunPlace++ = fun;
                            }
                            t++;
                        }
                        else if ( *t <= -FUNCTION ) t++;
                        else t += 2;
                    }
                }
                break;
        }
        t = r;
    } while ( t < tstop );
}
/*
        #] FunLevel :
        #[ DetCurDum :

        We look for indices in the range AM.IndDum to AM.IndDum+MAXDUMMIES.
        The maximum value is returned.
*/

WORD DetCurDum(PHEAD WORD *t)
{
    GETBIDENTITY
    WORD maxval = AN.IndDum;
    WORD maxtop = AM.IndDum + WILDOFFSET;
    WORD *tstop, *m, *r, i;
    tstop = t + *t - 1;
    tstop -= ABS(*tstop);
    t++;
    while ( t < tstop ) {
        if ( *t == VECTOR ) {
            m = t + 3;
            t += t[1];
            while ( m < t ) {
                if ( *m > maxval && *m < maxtop ) maxval = *m;
                m += 2;
            }
        }
        else if ( *t == DELTA || *t == INDEX ) {
            m = t + 2;
Singles:
            t += t[1];
            while ( m < t ) {
                if ( *m > maxval && *m < maxtop ) maxval = *m;
                m++;
            }
        }
        else if ( *t >= FUNCTION ) {
            if ( functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
                m = t + FUNHEAD;
                goto Singles;
            }
            r = t + FUNHEAD;
            t += t[1];
            while ( r < t ) {       /* The arguments */
                if ( *r < 0 ) {
                    if ( *r <= -FUNCTION ) r++;
                    else if ( *r == -INDEX ) {
                        if ( r[1] > maxval && r[1] < maxtop ) maxval = r[1];
                        r += 2;
                    }
                    else r += 2;
                }
                else {
                    m = r + ARGHEAD;
                    r += *r;
                    while ( m < r ) {    /* Terms in the argument */
                        i = DetCurDum(BHEAD m);
                        if ( i > maxval && i < maxtop ) maxval = i;
                        m += *m;
                    }
                }
            }
```

```
        }
        else {
            t += t[1];
        }
    }
    return(maxval);
}

/*
        #] DetCurDum :
        #[ FullRenumber :

        Does a full renumbering. May be slow if there are many indices.
        par = 1 Goes with a factorial!
        par = 0 All single exchanges only till there is no more improvement.
        Notice that there is a hole in the defense with respect to
        arguments inside functions inside functions.
*/

int FullRenumber(PHEAD WORD *term, WORD par)
{
    GETBIDENTITY
    WORD *d, **p, **f, *w, *t, *best, *stac, *perm, a, *termtry;
    WORD n, i, j, k, ii;
    WORD *oldworkpointer = AT.WorkPointer;
    n = ReNumber(BHEAD term) - AM.IndDum;
    if ( n <= 1 ) return(0);
    Normalize(BHEAD term);
    if ( *term == 0 ) return(0);
    n = ReNumber(BHEAD term) - AM.IndDum;
    d = AN.RenumScratch;
    p = AN.PoinScratch;
    f = AN.FunScratch;
    if ( AT.WorkPointer < term + *term ) AT.WorkPointer = term + *term;
    k = AN.NumFound;
    best = w = AT.WorkPointer; t = term;
    for ( i = *term; i > 0; i-- ) *w++ = *t++;
    AT.WorkPointer = w;
    Normalize(BHEAD best);
    AT.WorkPointer = w = best + *best;
    stac = w+100;
    perm = stac + n + 1;
    termtry = perm + n + 1;
    for ( i = 1; i <= n; i++ ) perm[i] = i + AM.IndDum;
    for ( i = 1; i <= n; i++ ) stac[i] = i;
    for ( i = 0; i < k; i++ ) d[i] = *(p[i]) - AM.IndDum;
    if ( par == 0 ) {    /* All single exchanges */
        for ( i = 1; i < n; i++ ) {
            for ( j = i+1; j <= n; j++ ) {
                a = perm[j]; perm[j] = perm[i]; perm[i] = a;
                for ( ii = 0; ii < k; ii++ ) {
                    *(p[ii]) = perm[d[ii]];
                    if ( f[ii] ) *(f[ii]) |= DIRTYSYMFLAG;
                }
                t = term; w = termtry;
                for ( ii = 0; ii < *term; ii++ ) *w++ = *t++;
                AT.WorkPointer = w;
                if ( Normalize(BHEAD termtry) == 0 ) {
                    if ( *termtry == 0 ) goto Return0;
                    if ( ( ii = CompareTerms(BHEAD termtry,best,0) ) > 0 ) {
                        t = termtry; w = best;
                        for ( ii = 0; ii < *termtry; ii++ ) *w++ = *t++;
                        i = 0; break; /* restart from beginning */
                    }
                    else if ( ii == 0 && CompCoef(termtry,best) != 0 )
                        goto Return0;
                }
                /* if no success, set back */
                a = perm[j]; perm[j] = perm[i]; perm[i] = a;
            }
        }
    }
    else if ( par == 1 ) {  /* all permutations */
        j = n-1;
        for(;;) {
            if ( stac[j] == n ) {
                a = perm[j]; perm[j] = perm[n]; perm[n] = a;
                stac[j] = j;
                j--;
                if ( j <= 0 ) break;
                continue;
            }
            if ( j != stac[j] ) {
                a = perm[j]; perm[j] = perm[stac[j]]; perm[stac[j]] = a;
            }
            (stac[j])++;
            a = perm[j]; perm[j] = perm[stac[j]]; perm[stac[j]] = a;
```

```
            {
                for ( i = 0; i < k; i++ ) {
                    *(p[i]) = perm[d[i]];
                    if ( f[i] ) *(f[i]) |= DIRTYSYMFLAG;
                }
                t = term; w = termtry;
                for ( i = 0; i < *term; i++ ) *w++ = *t++;
                AT.WorkPointer = w;
                if ( Normalize(BHEAD termtry) == 0 ) {
                    if ( *termtry == 0 ) goto Return0;
                    if ( ( ii = CompareTerms(BHEAD termtry,best,0) ) > 0 ) {
                        t = termtry; w = best;
                        for ( i = 0; i < *termtry; i++ ) *w++ = *t++;
                    }
                    else if ( ii == 0 && CompCoef(termtry,best) != 0 )
                        goto Return0;
                }
            }
            if ( j < n-1 ) { j = n-1; }
        }
    }
    t = term; w = best;
    n = *best;
    for ( i = 0; i < n; i++ ) *t++ = *w++;
    AT.WorkPointer = oldworkpointer;
    return(0);
Return0:
    *term = 0;
    AT.WorkPointer = oldworkpointer;
    return(0);
}

/*
        #] FullRenumber :
        #[ MoveDummies :

        Routine shifts the dummy indices by an amount 'shift'.
        It is an adaptation of DetCurDum.
        Needed for  = ...*expression^power*...
        in which expression contains dummy indices.
        Note that this code should have been in ver1 already and has
        always been missing. Routine made 29-jan-2007.
*/

void MoveDummies(PHEAD WORD *term, WORD shift)
{
    GETBIDENTITY
    WORD maxval = AN.IndDum;
    WORD maxtop = AM.IndDum + WILDOFFSET;
    WORD *tstop, *m, *r;
    tstop = term + *term - 1;
    tstop -= ABS(*tstop);
    term++;
    while ( term < tstop ) {
        if ( *term == VECTOR ) {
            m = term + 3;
            term += term[1];
            while ( m < term ) {
                if ( *m > maxval && *m < maxtop ) *m += shift;
                m += 2;
            }
        }
        else if ( *term == DELTA || *term == INDEX ) {
            m = term + 2;
Singles:
            term += term[1];
            while ( m < term ) {
                if ( *m > maxval && *m < maxtop ) *m += shift;
                m++;
            }
        }
        else if ( *term >= FUNCTION ) {
            if ( functions[*term-FUNCTION].spec >= TENSORFUNCTION ) {
                m = term + FUNHEAD;
                goto Singles;
            }
            r = term + FUNHEAD;
            term += term[1];
            while ( r < term ) {          /* The arguments */
                if ( *r < 0 ) {
                    if ( *r <= -FUNCTION ) r++;
                    else if ( *r == -INDEX ) {
                        if ( r[1] > maxval && r[1] < maxtop ) r[1] += shift;
                        r += 2;
                    }
                    else r += 2;
                }
```

```
                else {
                    m = r + ARGHEAD;
                    r += *r;
                    while ( m < r ) {    /* Terms in the argument */
                        MoveDummies(BHEAD m,shift);
                        m += *m;
                    }
                }
            }
        }
        else {
            term += term[1];
        }
    }
}

/*
        #] MoveDummies :
        #[ AdjustRenumScratch :

        Extends the buffer for number of dummies that can be found in
        a term. Originally we had a fixed buffer at size 300 in the AN
        struct. Thomas Hahn ran out of that. Hence we have now made it
        a dynamical buffer.
        Note that the pointers used in FunLevel need adjustment as well.
*/

void AdjustRenumScratch(PHEAD0)
{
    GETBIDENTITY
    WORD newsize;
    int i;
    WORD **newpoin, *newnum;
    if ( AN.MaxRenumScratch == 0 ) newsize = 100;
    else newsize = AN.MaxRenumScratch*2;
    if ( newsize > MAXPOSITIVE/2 ) newsize = MAXPOSITIVE/2+1;

    newpoin = (WORD **)Malloc1(newsize*sizeof(WORD *),"PoinScratch");
    for ( i = 0; i < AN.NumFound; i++ ) newpoin[i] = AN.PoinScratch[i];
    for ( ; i < newsize; i++ ) newpoin[i] = 0;
    if ( AN.PoinScratch ) M_free(AN.PoinScratch,"PoinScratch");
    AN.PoinScratch = newpoin;
    AN.DumPlace = newpoin + AN.NumFound;

    newpoin = (WORD **)Malloc1(newsize*sizeof(WORD *),"FunScratch");
    for ( i = 0; i < AN.NumFound; i++ ) newpoin[i] = AN.FunScratch[i];
    for ( ; i < newsize; i++ ) newpoin[i] = 0;
    if ( AN.FunScratch ) M_free(AN.FunScratch,"FunScratch");
    AN.FunScratch = newpoin;
    AN.DumFunPlace = newpoin + AN.NumFound;

    newnum = (WORD *)Malloc1(newsize*sizeof(WORD),"RenumScratch");
    for ( i = 0; i < AN.NumFound; i++ ) newnum[i] = AN.RenumScratch[i];
    for ( ; i < newsize; i++ ) newnum[i] = 0;
    if ( AN.RenumScratch ) M_free(AN.RenumScratch,"RenumScratch");
    AN.RenumScratch = newnum;
    AN.DumFound = newnum + AN.NumFound;

    AN.MaxRenumScratch = newsize;
}

/*
        #] AdjustRenumScratch :
    #] Reshuf :
    #[ Count :
        #[ CountDo :

        This function executes the counting action in a count
        operation. The return value is the count of the term.
        Input is the term and a pointer to the instruction.

*/

WORD CountDo(WORD *term, WORD *instruct)
{
    WORD *m, *r, i, j, count = 0;
    WORD *stopper, *tstop, *r1 = 0, *r2 = 0;
    m = instruct;
    stopper = m + m[1];
    instruct += 3;
    tstop = term + *term; tstop -= ABS(tstop[-1]); term++;
    while ( term < tstop ) {
        switch ( *term ) {
            case SYMBOL:
                i = term[1] - 2;
                term += 2;
                while ( i > 0 ) {
```

```
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == SYMBOL && m[2] == *term ) {
                            count += m[3] * term[1];
                        }
                        m += m[1];
                    }
                    term += 2;
                    i -= 2;
                }
                break;
            case DOTPRODUCT:
                i = term[1] - 2;
                term += 2;
                while ( i > 0 ) {
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == DOTPRODUCT && (( m[2] == *term &&
                        m[3] == term[1]) || ( m[2] == term[1] &&
                        m[3] == *term )) ) {
                            count += m[4] * term[2];
                            break;
                        }
                        m += m[1];
                    }
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == VECTOR && m[2] == *term &&
                        ( m[3] & DOTPBIT ) != 0 ) {
                            count += m[m[1]-1] * term[2];
                        }
                        m += m[1];
                    }
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == VECTOR && m[2] == term[1] &&
                        ( m[3] & DOTPBIT ) != 0 ) {
                            count += m[m[1]-1] * term[2];
                        }
                        m += m[1];
                    }
                    term += 3;
                    i -= 3;
                }
                break;
            case INDEX:
                j = 1;
                goto VectInd;
            case VECTOR:
                j = 2;
VectInd:        i = term[1] - 2;
                term += 2;
                while ( i > 0 ) {
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == VECTOR && m[2] == *term &&
                        ( m[3] & VECTBIT ) != 0 ) {
                            count += m[m[1]-1];
                        }
                        m += m[1];
                    }
                    term += j;
                    i -= j;
                }
                break;
            default:
                if ( *term >= FUNCTION ) {
                    i = *term;
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == FUNCTION && m[2] == i ) count += m[3];
                        m += m[1];
                    }
                    if ( functions[i-FUNCTION].spec >= TENSORFUNCTION ) {
                        i = term[1] - FUNHEAD;
                        term += FUNHEAD;
                        while ( i > 0 ) {
                            if ( *term < 0 ) {
                                m = instruct;
                                while ( m < stopper ) {
                                    if ( *m == VECTOR && m[2] == *term &&
                                    ( m[3] & FUNBIT ) != 0 ) {
                                        count += m[m[1]-1];
                                    }
                                    m += m[1];
                                }
                            }
```

```
                                    term++;
                                    i--;
                                }
                            }
                            else {
                                r = term + term[1];
                                term += FUNHEAD;
                                while ( term < r ) {
                                    if ( ( *term == -INDEX || *term == -VECTOR
                                    || *term == -MINVECTOR ) && term[1] < MINSPEC ) {
                                        m = instruct;
                                        while ( m < stopper ) {
                                            if ( *m == VECTOR && term[1] == m[2]
                                            && ( m[3] & SETBIT ) != 0 ) {
                                                r1 = SetElements + Sets[m[4]].first;
                                                r2 = SetElements + Sets[m[4]].last;
                                                while ( r1 < r2 ) {
                                                    if ( *r1 == i ) {
                                                        count += m[m[1]-1];
                                                        goto NextFF;
                                                    }
                                                    r1++;
                                                }
                                            }
                                            m += m[1];
                                        }
NextFF:
                                        term += 2;
                                    }
                                    else { NEXTARG(term) }
                                }
                            }
                            break;
                        }
                        else {
                            term += term[1];
                        }
                        break;
                }
        }
        return(count);
}

/*
                #] CountDo :
                #[ CountFun :

                This is the count function.
                The return value is the count of the term.
                Input is the term and a pointer to the count function.

*/

WORD CountFun(WORD *term, WORD *countfun)
{
        WORD *m, *r, i, j, count = 0, *instruct, *stopper, *tstop;
        m = countfun;
        stopper = m + m[1];
        instruct = countfun + FUNHEAD;
        tstop = term + *term; tstop -= ABS(tstop[-1]); term++;
        while ( term < tstop ) {
                switch ( *term ) {
                        case SYMBOL:
                                i = term[1] - 2;
                                term += 2;
                                while ( i > 0 ) {
                                        m = instruct;
                                        while ( m < stopper ) {
                                                if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                                                if ( *m == -SYMBOL && m[1] == *term
                                                && m[2] == -SNUMBER && ( m + 2 ) < stopper ) {
                                                        count += m[3] * term[1]; m += 4;
                                                }
                                                else { NEXTARG(m) }
                                        }
                                        term += 2;
                                        i -= 2;
                                }
                                break;
                        case DOTPRODUCT:
                                i = term[1] - 2;
                                term += 2;
                                while ( i > 0 ) {
                                        m = instruct;
                                        while ( m < stopper ) {
                                                if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                                                if ( *m == 9+ARGHEAD && m[ARGHEAD] == 9
```

```
                        && m[ARGHEAD+1] == DOTPRODUCT
                        && m[ARGHEAD+9] == -SNUMBER && ( m + ARGHEAD+9 ) < stopper
                        && (( m[ARGHEAD+3] == *term &&
                        m[ARGHEAD+4] == term[1]) ||
                         ( m[ARGHEAD+3] == term[1] &&
                        m[ARGHEAD+4] == *term )) ) {
                            count += m[ARGHEAD+10] * term[2];
                            m += ARGHEAD+11;
                        }
                        else { NEXTARG(m) }
                    }
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                        if ( ( *m == -VECTOR || *m == -MINVECTOR )
                        && m[1] == *term &&
                        m[2] == -SNUMBER && ( m+2 ) < stopper ) {
                            count += m[3] * term[2]; m += 4;
                        }
                        NEXTARG(m)
                    }
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                        if ( ( *m == -VECTOR || *m == -MINVECTOR )
                        && m[1] == term[1] &&
                        m[2] == -SNUMBER && ( m+2 ) < stopper ) {
                            count += m[3] * term[2];
                            m += 4;
                        }
                        NEXTARG(m)
                    }
                    term += 3;
                    i -= 3;
                }
                break;
            case INDEX:
                j = 1;
                goto VectInd;
            case VECTOR:
                j = 2;
VectInd:        i = term[1] - 2;
                term += 2;
                while ( i > 0 ) {
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                        if ( ( *m == -VECTOR || *m == -MINVECTOR )
                        && m[1] == *term &&
                        m[2] == -SNUMBER && (m+2) < stopper ) {
                            count += m[3]; m += 4;
                        }
                        NEXTARG(m)
                    }
                    term += j;
                    i -= j;
                }
                break;
            default:
                if ( *term >= FUNCTION ) {
                    i = *term;
                    m = instruct;
                    while ( m < stopper ) {
                        if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                        if ( *m == -i && m[1] == -SNUMBER && (m+1) < stopper ) {
                            count += m[2]; m += 3;
                        }
                        NEXTARG(m)
                    }
                    if ( functions[i-FUNCTION].spec >= TENSORFUNCTION ) {
                        i = term[1] - FUNHEAD;
                        term += FUNHEAD;
                        while ( i > 0 ) {
                            if ( *term < 0 ) {
                                m = instruct;
                                while ( m < stopper ) {
                                    if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                                    if ( ( *m == -VECTOR || *m == -INDEX
                                    || *m == -MINVECTOR ) && m[1] == *term &&
                                    m[2] == -SNUMBER && (m+2) < stopper ) {
                                        count += m[3]; m += 4;
                                    }
                                    else { NEXTARG(m) }
                                }
                            }
                            term++;
                            i--;
```

```
                        }
                    }
                    else {
                        r = term + term[1];
                        term += FUNHEAD;
                        while ( term < r ) {
                            if ( ( *term == -INDEX || *term == -VECTOR
                            || *term == -MINVECTOR ) && term[1] < MINSPEC ) {
                                m = instruct;
                                while ( m < stopper ) {
                                    if ( *m == -SNUMBER ) { NEXTARG(m) continue; }
                                    if ( *m == -VECTOR && m[1] == term[1]
                                    && m[2] == -SNUMBER && (m+2) < stopper ) {
                                        count += m[3];
                                        m += 4;
                                    }
                                    else { NEXTARG(m) }
                                }
                                term += 2;
                            }
                            else { NEXTARG(term) }
                        }
                    }
                    break;
                }
                else {
                    term += term[1];
                }
                break;
        }
    }
    return(count);
}

/*
        #] CountFun :
    #] Count :
    #[ DimensionSubterm :
*/

WORD DimensionSubterm(WORD *subterm)
{
    WORD *r, *rstop, dim, i;
    LONG x = 0;
    rstop = subterm + subterm[1];
    if ( *subterm == SYMBOL ) {
        r = subterm + 2;
        while ( r < rstop ) {
            if ( *r <= NumSymbols && *r > -MAXPOWER ) {
                dim = symbols[*r].dimension;
                if ( dim == MAXPOSITIVE ) goto undefined;
                x += dim * r[1];
                if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
                r += 2;
            }
            else if ( *r <= MAXVARIABLES ) {
/*
                Here we have an extra symbol. Store dimension in the compiler buffer
*/
                i = MAXVARIABLES - *r;
                dim = cbuf[AM.sbufnum].dimension[i];
                if ( dim ==  MAXPOSITIVE ) goto undefined;
                if ( dim == -MAXPOSITIVE ) goto outofrange;
                x += dim * r[1];
                if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
                r += 2;
            }
            else { r += 2; }
        }
    }
    else if ( *subterm == DOTPRODUCT ) {
        r = subterm + 2;
        while ( r < rstop ) {
            dim = vectors[*r-AM.OffsetVector].dimension;
            if ( dim == MAXPOSITIVE ) goto undefined;
            x += dim * r[2];
            if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
            dim = vectors[r[1]-AM.OffsetVector].dimension;
            if ( dim == MAXPOSITIVE ) goto undefined;
            x += dim * r[2];
            if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
            r += 3;
        }
    }
    else if ( *subterm == VECTOR ) {
        r = subterm + 2;
        while ( r < rstop ) {
```

```
            dim = vectors[*r-AM.OffsetVector].dimension;
            if ( dim == MAXPOSITIVE ) goto undefined;
            x += dim;
            if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
            r += 2;
        }
    }
    else if ( *subterm == INDEX ) {
        r = subterm + 2;
        while ( r < rstop ) {
            if ( *r < 0 ) {
                dim = vectors[*r-AM.OffsetVector].dimension;
                if ( dim == MAXPOSITIVE ) goto undefined;
                x += dim;
                if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
            }
            r++;
        }
    }
    else if ( *subterm >= FUNCTION ) {
        dim = functions[*subterm-FUNCTION].dimension;
        if ( dim == MAXPOSITIVE ) goto undefined;
        x += dim;
        if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
        if ( functions[*subterm-FUNCTION].spec > 0 ) {  /* tensor */
            r = subterm + FUNHEAD;
            while ( r < rstop ) {
                if ( *r < MINSPEC ) {
                    dim = vectors[*r-AM.OffsetVector].dimension;
                    if ( dim == MAXPOSITIVE ) goto undefined;
                    x += dim;
                    if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
                }
                r++;
            }
        }
    }
    return((WORD)x);
undefined:
    return((WORD)MAXPOSITIVE);
outofrange:
    return(-(WORD)MAXPOSITIVE);
}

/*
    #] DimensionSubterm :
    #[ DimensionTerm :

    Returns the dimension of the given term.
    If there is any variable of which the dimension is not defined
    we return the code for undefined which is MAXPOSITIVE
    When the value is out of range we return -MAXPOSITIVE
*/

WORD DimensionTerm(WORD *term)
{
    WORD *t, *tstop, dim;
    LONG x = 0;
    tstop = term + *term; tstop -= ABS(tstop[-1]);
    t = term+1;
    while ( t < tstop ) {
        dim = DimensionSubterm(t);
        if ( dim ==  MAXPOSITIVE ) goto undefined;
        if ( dim == -MAXPOSITIVE ) goto outofrange;
        x += dim;
        if ( x >= MAXPOSITIVE || x <= -MAXPOSITIVE ) goto outofrange;
        t += t[1];
    }
    return((WORD)x);
undefined:
    return((WORD)MAXPOSITIVE);
outofrange:
    return(-(WORD)MAXPOSITIVE);
}

/*
    #] DimensionTerm :
    #[ DimensionExpression :

    Returns the dimension of the given expression.
    If there is any variable of which the dimension is not defined
    we return the code for undefined which is MAXPOSITIVE
    When the value is out of range we return -MAXPOSITIVE
    When the value is not consistent we return -MAXPOSITIVE.
*/

WORD DimensionExpression(PHEAD WORD *expr)
```

```
{
    WORD dim, *term, *old, x = 0;
    int first = 1;
    term = expr;
    while ( *term ) {
        dim = DimensionTerm(term);
        if ( dim ==  MAXPOSITIVE ) goto undefined;
        if ( dim == -MAXPOSITIVE ) goto outofrange;
        if ( first ) { x = dim; }
        else if ( x != dim ) {
            old  = AN.currentTerm;
            MLOCK(ErrorMessageLock);
            MesPrint("Dimension is not the same in the terms of the expression");
            term = expr;
            while ( *term ) {
                AN.currentTerm = term;
                MesPrint("   %T");
            }
            MUNLOCK(ErrorMessageLock);
            AN.currentTerm = old;
            return(-(WORD)MAXPOSITIVE);
        }
        term += *term;
    }
    return((WORD)x);
undefined:
    return((WORD)MAXPOSITIVE);
outofrange:
    old  = AN.currentTerm;
    AN.currentTerm = term;
    MLOCK(ErrorMessageLock);
    MesPrint("Dimension out of range in %t in subexpression");
    MUNLOCK(ErrorMessageLock);
    AN.currentTerm = old;
    return(-(WORD)MAXPOSITIVE);
}

/*
    #] DimensionExpression :
    #[ Multiply Term :
        #[ MultDo :
*/

WORD MultDo(PHEAD WORD *term, WORD *pattern)
{
    GETBIDENTITY
    WORD *t, *r, i;
    t = term + *term;
    if ( pattern[2] > 0 ) {           /* Left multiply */
        i = *term - 1;
    }
    else {                            /* Right multiply */
        i = ABS(t[-1]);
    }
    *term += SUBEXPSIZE;
    r = t + SUBEXPSIZE;
    do { *--r = *--t; } while ( --i > 0 );
    r = pattern + 3;
    i = r[1];
    while ( --i >= 0 ) *t++ = *r++;
    AT.WorkPointer = term + *term;
    return(0);
}

/*
        #] MultDo :
    #] Multiply Term :
    #[ Try Term(s) :
        #[ TryDo :
*/

WORD TryDo(PHEAD WORD *term, WORD *pattern, WORD level)
{
    GETBIDENTITY
    WORD *t, *r, *m, i, j;
    ReNumber(BHEAD term);
    Normalize(BHEAD term);
    m = r = term + *term;
    m++;
    i = pattern[2];
    t = pattern + 3;
    NCOPY(m,t,i)
    j = *term - 1;
    t = term + 1;
    NCOPY(m,t,j)
    *r = WORDDIF(m,r);
    AT.WorkPointer = m;
```

```
    if ( ( j = Normalize(BHEAD r) ) == 0 || j == 1 ) {
        if ( *r == 0 ) return(0);
        ReNumber(BHEAD r); Normalize(BHEAD r);
        if ( *r == 0 ) return(0);
        if ( ( i = CompareTerms(BHEAD term,r,0) ) < 0 ) {
            *AN.RepPoint = 1;
            AR.expchanged = 1;
            return(Generator(BHEAD r,level));
        }
        if ( i == 0 && CompCoef(term,r) != 0 ) { return(0); }
    }
    AT.WorkPointer = r;
    return(Generator(BHEAD term,level));
}

/*
        #] TryDo :
    #] Try Term(s) :
    #[ Distribute :
        #[ DoDistrib :

        The routine that generates the terms ordered by a distrib_ function.
        The presence of a replaceable distrib_ function has been sensed
        in the routine TestSub and has been passed on to Generator.
        It is then Generator that calls this function in a way that is
        similar to calling the trace routines, except for that for the
        trace routines and the Levi-Civita tensors the arguments are put
        in temporary storage and here we leave them inside the term,
        because there is no knowing how long the field will be.
*/

WORD DoDistrib(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD *t, *m, *r = 0, *stop, *tstop, *termout, *endhead, *starttail, *parms;
    WORD i, j, k, n, nn, ntype, fun1 = 0, fun2 = 0, typ1 = 0, typ2 = 0;
    WORD *arg, *oldwork, *mf, ktype = 0, atype = 0;
    WORD sgn, dirtyflag;
    AN.TeInFun = AR.TePos = 0;
    t = term;
    tstop = t + *t;
    stop = tstop - ABS(tstop[-1]);
    t++;
    while ( t < stop ) {
        r = t + t[1];
        if ( *t == DISTRIBUTION && t[FUNHEAD] == -SNUMBER
        && t[FUNHEAD+1] >= -2 && t[FUNHEAD+1] <= 2
        && t[FUNHEAD+2] == -SNUMBER
        && t[FUNHEAD+4] <= -FUNCTION
        && t[FUNHEAD+5] <= -FUNCTION ) {
            WORD *ttt = t+FUNHEAD+6, *tttstop = t+t[1];
            while ( ttt < tttstop ) {
                if ( *ttt == -DOLLAREXPRESSION ) break;
                NEXTARG(ttt);
            }
            if ( ttt >= tttstop ) {
                fun1 = -t[FUNHEAD+4];
                fun2 = -t[FUNHEAD+5];
                typ1 = functions[fun1-FUNCTION].spec;
                typ2 = functions[fun2-FUNCTION].spec;
                if ( typ1 > 0 || typ2 > 0 ) {
                    m = t + FUNHEAD+6;
                    r = t + t[1];
                    while ( m < r ) {
                        if ( *m != -INDEX && *m != -VECTOR && *m != -MINVECTOR )
                            break;
                        m += 2;
                    }
                    if ( m < r ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Incompatible function types and arguments in distrib_");
                        MUNLOCK(ErrorMessageLock);
                        SETERROR(-1)
                    }
                }
                break;
            }
        }
        t = r;
    }
    dirtyflag = t[2];
    ntype = t[FUNHEAD+1];
    n = t[FUNHEAD+3];
/*
    t points at the distrib_ function to be expanded.
    fun1,fun2 and typ1,typ2 are the two functions and their types.
    ntype indicates the action:
```

```
        0:  Make all possible divisions:  2^nargs
        1:  fun1 should get n arguments:  nargs! / ( n! (nargs-n)! )
        2:  fun2 should get n arguments:  nargs! / ( n! (nargs-n)! )
    The distinction between 1 and two is for noncommuting objects.
        3:  fun1 should get n arguments. Super symmetric option.
        4:  fun2 idem
    The super symmetric option involves:
        a:  arguments get sorted
        b:  identical arguments are seen as such. Hence not all their
            distributions are taken into account. It is as if after the
            distrib there is a symmetrize fun1; symmetrize fun2;
        c:  Hence if the occurrence of each argument is a,b,c,...
            and their occurrence in fun1 is a1,b1,c1,... and in fun2
            is a2,b2,c2,... then each term is generated (a1+a2)!/a1!/a2!
            (b1+b2)!/b1!/b2! (c1+c2)!/c1!/c2! ... times.
        d:  We have to make an array of occurrences and positions.
        e:  Then we sort the arguments indirectly.
        f:  Next we generate the argument lists in the same way as we
            generate powers of expressions with binomials. Hence we need
            a third array to keep track of the `powers'
*/
    endhead = t;
    starttail = r;
    parms = m = t + FUNHEAD+6;
    i = 0;
    while ( m < r ) {   /* Count arguments */
        i++;
        NEXTARG(m);
    }
    oldwork = AT.WorkPointer;
    arg = AT.WorkPointer + 1;
    arg[-1] = 0;
    termout = arg + i;
    switch ( ntype ) {
        case  0: ktype = 1; atype = n < 0 ? 1: 0; n = 0; break;
        case  1: ktype = 1; atype = 0; break;
        case  2: ktype = 0; atype = 0; break;
        case -1: ktype = 1; atype = 1; break;
        case -2: ktype = 0; atype = 1; break;
    }
    do {
/*
        All distributions with n elements. We generate the array arg with
        all possible 1 and 0 patterns. 1 means in fun1 and 0 means in fun2.
*/
        if ( n > i ) return(0);     /* 0 elements */

        for ( j = 0; j < n; j++ ) arg[j] = 1;
        for ( j = n; j < i; j++ ) arg[j] = 0;
        for(;;) {
            sgn = 0;
            t = term;
            m = termout;
            while ( t < endhead ) *m++ = *t++;
            mf = m;
            *m++ = fun1;
            *m++ = FUNHEAD;
            *m++ = dirtyflag;
#if FUNHEAD > 3
            k = FUNHEAD -3;
            while ( k-- > 0 ) *m++ = 0;
#endif
            r = parms;
            for ( k = 0; k < i; k++ ) {
                if ( arg[k] == ktype ) {
                    if ( *r <= -FUNCTION ) *m++ = *r++;
                    else if ( *r < 0 ) {
                        if ( typ1 > 0 ) {
                            if ( *r == -MINVECTOR ) sgn ^= 1;
                            r++;
                            *m++ = *r++;
                        }
                        else { *m++ = *r++; *m++ = *r++; }
                    }
                    else {
                        nn = *r;
                        NCOPY(m,r,nn);
                    }
                }
                else { NEXTARG(r) }
            }
            mf[1] = WORDDIF(m,mf);
            mf = m;
            *m++ = fun2;
            *m++ = FUNHEAD;
            *m++ = dirtyflag;
#if FUNHEAD > 3
```

```
            k = FUNHEAD -3;
            while ( k-- > 0 ) *m++ = 0;
#endif
            r = parms;
            for ( k = 0; k < i; k++ ) {
                if ( arg[k] != ktype ) {
                    if ( *r <= -FUNCTION ) *m++ = *r++;
                    else if ( *r < 0 ) {
                        if ( typ2 > 0 ) {
                            if ( *r == -MINVECTOR ) sgn ^= 1;
                            r++;
                            *m++ = *r++;
                        }
                        else { *m++ = *r++; *m++ = *r++; }
                    }
                    else {
                        nn = *r;
                        NCOPY(m,r,nn);
                    }
                }
                else { NEXTARG(r) }
            }
            mf[1] = WORDDIF(m,mf);
#ifndef NUOVO
            if ( atype == 0 ) {
                WORD k1,k2;
                for ( k = 0; k < i-1; k++ ) {
                    if ( arg[k] == 0 ) continue;
                    k1 = 1; k2 = k;
                    while ( k < i-1 && EqualArg(parms,k,k+1) ) { k++; k1++; }
                    while ( k2 <= k && arg[k2] == 1 ) k2++;
                    k2 = k-k2+1;
/*
                    Now we need k1!/(k2! (k1-k2)!)
*/
                    if ( k2 != k1 && k2 != 0 ) {
                        if ( GetBinom((UWORD *)m+3,m+2,k1,k2) ) {
                            MLOCK(ErrorMessageLock);
                            MesCall("DoDistrib");
                            MUNLOCK(ErrorMessageLock);
                            SETERROR(-1)
                        }
                        m[1] = ( m[2] < 0 ? -m[2]: m[2] ) + 3;
                        *m = LNUMBER;
                        m += m[1];
                    }
                }
            }
#endif
            r = starttail;
            while ( r < tstop ) *m++ = *r++;

            if ( atype ) {      /* antisymmetric field */
                k = n;
                nn = 0;
                for ( j = 0; j < i && k > 0; j++ ) {
                    if ( arg[j] == 1 ) k--;
                    else nn += k;
                }
                sgn ^= nn & 1;
            }

            if ( sgn ) m[-1] = -m[-1];
            *termout = WORDDIF(m,termout);
            AT.WorkPointer = m;
            if ( AT.WorkPointer > AT.WorkTop ) {
                MLOCK(ErrorMessageLock);
                MesWork();
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            *AN.RepPoint = 1;
            AR.expchanged = 1;
            if ( Generator(BHEAD termout,level) ) Terminate(-1);
#ifndef NUOVO
            {
                WORD k1;
            j = i - 1;
            k = 0;
redok:      while ( arg[j] == 1 && j >= 0 ) { j--; k++; }
            while ( arg[j] == 0 && j >= 0 ) j--;
            if ( j < 0 ) break;
            k1 = j;
            arg[j] = 0;
            while ( !atype && EqualArg(parms,j,j+1) ) {
                j++;
                if ( j >= i - k - 1 ) { j = k1; k++; goto redok; }
```

```
                arg[j] = 0;
            }
            while ( k >= 0 ) { j++; arg[j] = 1; k--; }
            j++;
            while ( j < i ) { arg[j] = 0; j++; }
            }
#else
            j = i - 1;
            k = 0;
            while ( arg[j] == 1 && j >= 0 ) { j--; k++; }
            while ( arg[j] == 0 && j >= 0 ) j--;
            if ( j < 0 ) break;
            arg[j] = 0;
            while ( k >= 0 ) { j++; arg[j] = 1; k--; }
            j++;
            while ( j < i ) { arg[j] = 0; j++; }
#endif
        }
    } while ( ntype == 0 && ++n <= i );
    AT.WorkPointer = oldwork;
    return(0);
}

/*
        #] DoDistrib :
        #[ EqualArg :

        Returns 1 if the arguments in the field are identical.
*/

WORD EqualArg(WORD *parms, WORD num1, WORD num2)
{
    WORD *t1, *t2;
    WORD i;
    t1 = parms;
    while ( --num1 >= 0 ) { NEXTARG(t1); }
    t2 = parms;
    while ( --num2 >= 0 ) { NEXTARG(t2); }
    if ( *t1 != *t2 ) return(0);
    if ( *t1 < 0 ) {
        if ( *t1 <= -FUNCTION || t1[1] == t2[1] ) return(1);
        return(0);
    }
    i = *t1;
    while ( --i >= 0 ) {
        if ( *t1 != *t2 ) return(0);
        t1++; t2++;
    }
    return(1);
}

/*
        #] EqualArg :
        #[ DoDelta3 :
*/

WORD DoDelta3(PHEAD WORD *term, WORD level)
{
    GETBIDENTITY
    WORD *t, *m, *m1, *m2, *stopper, *tstop, *termout, *dels, *taken;
    WORD *ic, *jc, *factors;
    WORD num, num2, i, j, k, knum, a;
    AN.TeInFun = AR.TePos = 0;
    tstop = term + *term;
    stopper = tstop - ABS(tstop[-1]);
    t = term+1;
    while ( ( *t != DELTA3 || ((t[1]-FUNHEAD) & 1 ) != 0 ) && t < stopper )
        t += t[1];
    if ( t >= stopper ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Internal error with dd_ function");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    m1 = t; m2 = t + t[1];
    num = t[1] - FUNHEAD;
    if ( num == 0 ) {
        termout = t = AT.WorkPointer;
        m = term;
        while ( m < m1 ) *t++ = *m++;
        m = m2; while ( m < tstop ) *t++ = *m++;
        *termout = WORDDIF(t,termout);
        AT.WorkPointer = t;
        *AN.RepPoint = 1;
        AR.expchanged = 1;
        if ( Generator(BHEAD termout,level) ) {
            MLOCK(ErrorMessageLock);
```

```
            MesCall("Do dd_");
            MUNLOCK(ErrorMessageLock);
            SETERROR(-1)
        }
        AT.WorkPointer = termout;
        return(0);
    }
    t += FUNHEAD;
/*
    Step 1: sort the arguments
*/
    for ( i = 1; i < num; i++ ) {
        if ( t[i] < t[i-1] ) {
            a = t[i]; t[i] = t[i-1]; t[i-1] = a;
            j = i - 1;
            while ( j > 0 ) {
                if ( t[j] >= t[j-1] ) break;
                a = t[j]; t[j] = t[j-1]; t[j-1] = a;
                j--;
            }
        }
    }
/*
    Step 2: Order them by occurrence
    In 'taken' we have the array with the number of occurrences.
    in 'dels' is the type of object.
*/
    m = taken = AT.WorkPointer;
    for ( i = 0; i < num; i++ ) *m++ = 0;
    dels = m; knum = 0;
    for ( i = 0; i < num; knum++ ) {
        *m++ = t[i]; i++; taken[knum] = 1;
        while ( i < num ) {
            if ( t[i] != t[i-1] ) break;
            i++; (taken[knum])++;
        }
    }
    for ( i = 0; i < knum; i++ ) *m++ = taken[i];
    ic = m; num2 = num/2;
    jc = ic + num2;
    factors = jc + num2;
    termout = factors + num2;
/*
    The recursion has num/2 steps
*/
    k = 0;
    while ( k >= 0 ) {
        if ( k >= num2 ) {
            t = termout; m = term;
            while ( m < m1 ) *t++ = *m++;
            *t++ = DELTA; *t++ = num+2;
            for ( i = 0; i < num2; i++ ) {
                *t++ = dels[ic[i]]; *t++ = dels[jc[i]];
            }
            for ( i = 0; i < num2; i++ ) {
                if ( ic[i] == jc[i] ) {
                    j = 1;
                    while ( i < num2-1 && ic[i] == ic[i+1] && ic[i] == jc[i+1] )
                        { i++; j++; }
                    for ( a = 1; a < j; a++ ) {
                        *t++ = SNUMBER; *t++ = 4; *t++ = 2*a+1; *t++ = 1;
                    }
                    for ( a = 0; a+1+i < num2; a++ ) {
                        if ( ic[a+i] != ic[a+i+1] ) break;
                    }
                    if ( a > 0 ) {
                        if ( GetBinom((UWORD *)(t+3),t+2,2*j+a,a) ) {
                            MLOCK(ErrorMessageLock);
                            MesCall("Do dd_");
                            MUNLOCK(ErrorMessageLock);
                            SETERROR(-1)
                        }
                        t[1] = ( t[2] < 0 ? -t[2]: t[2] ) + 3;
                        *t = LNUMBER;
                        t += t[1];
                    }
                }
                else if ( factors[i] != 1 ) {
                    *t++ = SNUMBER; *t++ = 4; *t++ = factors[i]; *t++ = 1;
                }
            }
            for ( i = 0; i < num2-1; i++ ) {
                if ( ic[i] == jc[i] ) continue;
                j = 1;
                while ( i < num2-1 && jc[i] == jc[i+1] && ic[i] == ic[i+1] ) {
                    i++; j++;
                }
```

```
                for ( a = 0; a+i < num2-1; a++ ) {
                    if ( ic[i+a] != ic[i+a+1] ) break;
                }
                if ( a > 0 ) {
                    if ( GetBinom((UWORD *)(t+3),t+2,j+a,a) ) {
                        MLOCK(ErrorMessageLock);
                        MesCall("Do dd_");
                        MUNLOCK(ErrorMessageLock);
                        SETERROR(-1)
                    }
                    t[1] = ( t[2] < 0 ? -t[2]: t[2] ) + 3;
                    *t = LNUMBER;
                    t += t[1];
                }
            }
            m = m2;
            while ( m < tstop ) *t++ = *m++;
            *termout = WORDDIF(t,termout);
            AT.WorkPointer = t;
            *AN.RepPoint = 1;
            AR.expchanged = 1;
            if ( Generator(BHEAD termout,level) ) {
                MLOCK(ErrorMessageLock);
                MesCall("Do dd_");
                MUNLOCK(ErrorMessageLock);
                SETERROR(-1)
            }
            k--;
            if ( k >= 0 ) goto nextj;
            else break;
        }
        for ( ic[k] = 0; ic[k] < knum; ic[k]++ ) {
            if ( taken[ic[k]] > 0 ) break;
        }
        if ( k > 0 && ic[k-1] == ic[k] ) jc[k] = jc[k-1];
        else jc[k] = ic[k];
        for ( ; jc[k] < knum; jc[k]++ ) {
            if ( taken[jc[k]] <= 0 ) continue;
            if ( ic[k] == jc[k] ) {
                if ( taken[jc[k]] <= 1 ) continue;
/*
                factors[k] = taken[ic[k]];
                if ( ( factors[k] & 1 ) == 0 ) (factors[k])--;
*/
                taken[ic[k]] -= 2;
            }
            else {
                factors[k] = taken[jc[k]];
                (taken[ic[k]])--; (taken[jc[k]])--;
            }
            k++;
            goto nextk;  /* This is the simulated recursion */
nextj:;
            (taken[ic[k]])++; (taken[jc[k]])++;
        }
        k--;
        if ( k >= 0 ) goto nextj;
nextk:;
    }
    AT.WorkPointer = taken;
    return(0);
}

/*
        #] DoDelta3 :
        #[ TestPartitions :

        Checks whether the function in tfun is a partitions_ function
        that can be expanded. If it can a number of relevant objects is
        inside the struct parti.
        This test is not entirely trivial because there are many restrictions
        w.r.t. the arguments.
        Syntax (still to be implemented)
        partitions_(number_of_partition_entries,[function,number,]^nope,arguments)
        [function,number,]: can be
            f,3     for a partition of 3 arguments
            f,0     for the remaining arguments (should be last)
            num1,f,num2 with num1 effectively a number of partitions but this
                    counts as num1 entries.
            0,f,num2: all partitions have num2 arguments. No number of partition
                    entries needed. If num2 does not divide the number of
                    arguments there will be no action.
*/

WORD TestPartitions(WORD *tfun, PARTI *parti)
{
    WORD *tnext = tfun + tfun[1];
```

```
    WORD *t, *tt;
    WORD argcount = 0, sum = 0, i, ipart, argremain;
    WORD tensorflag = 0;
    parti->psize = parti->nfun = parti->args = parti->nargs = 0;
    parti->numargs = parti->numpart = parti->where = 0;
    tt = t = tfun + FUNHEAD;
    while ( t < tnext ) { argcount++; NEXTARG(t); }
    if ( argcount < 1 ) goto No;
    t = tt;
    if ( *t != -SNUMBER ) goto No;
    if ( t[1] == 0 ) {
        t += 2;
        if ( *t <= -FUNCTION && t[1] == -SNUMBER && t[2] > 0 ) {
            if ( functions[-*t-FUNCTION].spec > 0 ) tensorflag = 1;
            if ( argcount-3 < 0 ) goto No;
            if ( ( (argcount-3) % t[2] ) != 0 ) goto No;
        }
        else goto No;
        parti->numpart = (argcount-3)/t[2];
        parti->numargs = argcount - 3;
        parti->psize = (WORD *)Malloc1((parti->numpart*2+parti->numargs*2+2)
                    *sizeof(WORD),"partitions");
        parti->nfun = parti->psize + parti->numpart;
        parti->args = parti->nfun + parti->numpart;
        parti->nargs = parti->args + parti->numargs;
        for ( i = 0; i < parti->numpart; i++ ) {
            parti->psize[i] =  t[2];
            parti->nfun[i]  = -t[0];
        }
        t += 3;
    }
    else if ( t[1] > 0 ) { /* Number of partitions */
/*
        We can have sequences of function,number for one partition
        or number1,function,number2 for number1 partitions of size number2.
        The last partition can have number=0. It must be a single partition
        and it will take all remaining arguments.
        If any of the functions is a tensor, all arguments must be either
        vector or index.
*/
        parti->numpart = t[1]; t += 2;
        ipart = sum = 0; argremain = argcount - 1;
/*
        At this point is seems better to make an allocation already that
        may be too big. The alternative is having to pass this code twice.
*/
        parti->psize = (WORD *)Malloc1((argcount*4+2)*sizeof(WORD),"partitions");
        parti->nfun = parti->psize+argcount;
        parti->args = parti->nfun+argcount;
        parti->nargs = parti->args+argcount;
        while ( ipart < parti->numpart ) {
            if ( *t <= -FUNCTION && t[1] == -SNUMBER && t[2] >= 0 ) {
                if ( functions[-*t-FUNCTION].spec > 0 ) tensorflag = 1;
                if ( t[2] == 0 ) {
                    if ( ipart+1 != parti->numpart ) goto WhatAPity;
                    argremain -= 2;
                    parti->nfun[ipart] = -*t;
                    parti->psize[ipart++] = argremain-sum;
                    ipart++;
                    sum = argremain;
                }
                else {
                    parti->nfun[ipart] = -*t;
                    parti->psize[ipart++] = t[2];
                    argremain -= 2;
                    sum += t[2];
                }
                t += 3;
            }
            else if ( *t == -SNUMBER && t[1] > 0 && ipart+t[1] <= parti->numpart
            && t[2] <= -FUNCTION && t[3] == -SNUMBER && t[4] > 0 ) {
                if ( functions[-t[2]-FUNCTION].spec > 0 ) tensorflag = 1;
                argremain -= 3;
                for ( i = 0; i < t[1]; i++ ) {
                    parti->nfun[ipart] = -t[2];
                    parti->psize[ipart++] = t[4];
                    sum += t[4];
                }
                if ( sum > argremain ) goto WhatAPity;
                t += 5;
            }
            else goto WhatAPity;
        }
        if ( sum != argremain ) goto WhatAPity;
        parti->numargs = argremain;
    }
    else goto No;
```

```
/*
    Now load the offsets of the arguments and check if needed whether OK with tensor
*/
    for ( i = 0; i < parti->numargs; i++ ) {
        parti->args[i] = t - tfun;
        if ( tensorflag && ( *t != -VECTOR && *t != -INDEX ) ) goto WhatAPity;
        NEXTARG(t);
    }
    return(1);
WhatAPity:
    M_free(parti->psize,"partitions");
    parti->psize = parti->nfun = parti->args = parti->nargs = 0;
    parti->numargs = parti->numpart = parti->where = 0;
No:
    return(0);
}

/*
        #] TestPartitions :
        #[ DoPartitions :

    As we have only one AT.partitions we need to copy it locally
    if we keep needing it.
*/

WORD DoPartitions(PHEAD WORD *term, WORD level)
{
    WORD x, i, j, im, *fun, ndiff, siz, tensorflag = 0;
    PARTI part = AT.partitions;
    WORD *array, **j3, **j3fill, **j3where;
    WORD a, pfill, *j2, *j2fill, j3size, ncoeff, ncoeffnum, nfac, ncoeff2, ncoeff3, n;
    UWORD *coeff, *coeffnum, *cfac, *coeff2, *coeff3, *c;
    /* Make AT.partitions ready for future use (if there is another function) */
    AT.partitions.psize = AT.partitions.nfun = AT.partitions.args = AT.partitions.nargs = 0;
    AT.partitions.numargs = AT.partitions.numpart = AT.partitions.where = 0;
/*
    Start with bubble sorting the list of arguments. And the list of partitions.
*/
    fun = term + part.where;
    if ( functions[*fun-FUNCTION].spec > 0 ) tensorflag = 1;
    for ( i = 1; i < part.numargs; i++ ) {
        for ( j = i-1; j >= 0; j-- ) {
            if ( CompArg(fun+part.args[j+1],fun+part.args[j]) >= 0 ) break;
            x = part.args[j+1]; part.args[j+1] = part.args[j]; part.args[j] = x;
        }
    }
    for ( i = 1; i < part.numpart; i++ ) {
        for ( j = i-1; j >= 0; j-- ) {
            if ( part.psize[j+1] < part.psize[j] ) break;
            if ( part.psize[j+1] == part.psize[j] && part.nfun[j+1] <= part.nfun[j] ) break;
            x = part.psize[j+1]; part.psize[j+1] = part.psize[j]; part.psize[j] = x;
            x = part.nfun[j+1]; part.nfun[j+1] = part.nfun[j]; part.nfun[j] = x;
        }
    }
/*
    Now we have the partitions sorted from high to low and the arguments
    have been sorted the regular way arguments are sorted in a symmetrize.
    The important thing is that identical arguments are adjacent.
    Assign the numbers (identical arguments have identical numbers).
*/
    ndiff = 1; part.nargs[0] = ndiff;
    for ( i = 1; i < part.numargs; i++ ) {
        if ( CompArg(fun+part.args[i],fun+part.args[i-1]) != 0 ) ndiff++;
        part.nargs[i] = ndiff;
    }
    part.nargs[part.numargs] = 0;
    coeffnum = NumberMalloc("partitionsn");
    coeff = NumberMalloc("partitions");
    coeff2 = NumberMalloc("partitions2");
    coeff3 = NumberMalloc("partitions3");
    cfac = NumberMalloc("partitions!");
    ncoeffnum = 1; coeffnum[0] = 1;
/*
    The numerator of the coefficient will be n1!*n2!*...*n(ndiff)!
    We compute it only once.
*/
    j = 0;
    for ( i = 1; i <= ndiff; i++ ) {
        n = 0;
        while ( part.nargs[j] == i ) { n++; j++; }
        if ( n > 1 ) { /* 1! needs no attention */
            if ( Factorial(BHEAD n, cfac, &nfac) ) Terminate(-1);
            if ( MulLong(coeffnum,ncoeffnum,cfac,nfac,coeff2,&ncoeff2) ) Terminate(-1);
            c = coeffnum; coeffnum = coeff2; coeff2 = c;
            n = ncoeffnum; ncoeffnum = ncoeff2; ncoeff2 = n;
        }
    }
```

```
/*
    Now comes the part where we have to make sure that
    a: we generate all partitions.
    b: we generate only different partitions.
    c: we get the proper combinatorics factor.
    Method:
    Suppose the largest partition needs n objects and there are m partitions.
    We allocate m arrays of n 'digits'. Make in the smaller partitions the
    appropriate leading digits zero.
    Divide the largest numbers (of the arguments) over the partitions as
    leftmost digits (after possible zeroes). The arrays, seen as numbers,
    should be such that each is less or equal to its left neighbour. Take the
    next largest numbers, etc. This generates unique partitions and all of
    them. Because we have a formula for the multiplicity, this should do it.

    The general case. At a later stage we might put in a more economical
    version for special cases.
*/
    siz = part.psize[0];
    j3size = 2*(part.numpart+1)+2*(part.numargs+1);
    array = (WORD *)Malloc1((part.numpart+1)*siz*sizeof(WORD),"parts");
    j3 = (WORD **)Malloc1(j3size*sizeof(WORD *),"parts3");
    j2 = (WORD *)Malloc1((part.numpart+part.numargs+2)*sizeof(WORD),"parts2");
    j3fill = j3+(part.numpart+1);
    j3where = j3fill+(part.numpart+1);
    for ( i = 0; i < j3size; i++ ) j3[i] = 0;
    j2fill = j2+(part.numpart+1);
    for ( i = 0; i < part.numargs; i++ ) j2fill[i] = 0;
    for ( i = 0; i < part.numpart; i++ ) {
        j3[i] = array+i*siz;
        for ( j = 0; j < siz; j++ ) j3[i][j] = 0;
        j3fill[i] = j3[i]+(siz-part.psize[i]);
        j2[i] = part.psize[i];  /* Number of places still available */
    }
    j3[part.numpart] = array+part.numpart*siz;
    j2[part.numpart] = 0;
/*
    Now comes a complicated two-level recursion in a and pfill.
*/
    a = part.numargs-1;
    pfill = 0;
/*
    We start putting the last number in part.nargs in the first partition in array.
    For backtracking we need to know where we put this number. Hence j3where.
*/
    while ( a < part.numargs ) {
        while ( j2[pfill] <= 0 ) {
            pfill++;
            while ( pfill >= part.numpart ) { /* we have to pop */
                a++;
                if ( a >= part.numargs ) goto Done;
                pfill = j2fill[a];
                j2[pfill]++;
                j3where[a][0] = 0;
                j3fill[pfill]--;
                pfill++;
            }
        }
        j3where[a] = j3fill[pfill];
        *(j3fill[pfill])++ = part.nargs[a];
        j2[pfill]--; j2fill[a] = pfill;
/*
        Now test whether this is allowed.
*/
        if ( pfill > 0 && part.psize[pfill] == part.psize[pfill-1]
             && part.nfun[pfill] == part.nfun[pfill-1] ) { /* First check whether allowed */
            for ( im = 0; im < siz; im++ ) {
                if ( j3[pfill-1][im] < j3[pfill][im] ) break;
                if ( j3[pfill-1][im] > j3[pfill][im] ) im = siz;
            }
            if ( im < siz ) { /* not ordered. undo and raise pfill */
                pfill = j2fill[a];
                j2[pfill]++;
                j3where[a][0] = 0;
                j3fill[pfill]--;
                pfill++;
                continue; /* Note that j2[part.numpart] = 0 */
            }
        }
        a--;
        if ( a < 0 ) {  /* Solution */
/*
            #[ Solution :

            Now we compose the output term. The input term contains
            three parts: head, partitions_, tail.
            partitions_ starts at term+part.where.
```

```
            We first put the function parts and worry about the coefficient later.
*/
            WORD *t, *to, *twhere = term+part.where, *t2, *tend = term+*term, *termout;
            WORD num, jj, *targ, *tfun;
            t2 = twhere+twhere[1];
            to = termout = AT.WorkPointer;
            if ( termout + *term + part.numpart*FUNHEAD + AM.MaxTal >= AT.WorkTop ) {
                return(MesWork());
            }
            for ( i = 0; i < ncoeffnum; i++ ) coeff[i] = coeffnum[i];
            ncoeff = ncoeffnum;
            t = term; while ( t < twhere ) *to++ = *t++;
/*
            Now the partitions
*/
            for ( i = 0; i < part.numpart; i++ ) {
                tfun = to;
                *to++ = part.nfun[i]; to++; FILLFUN(to);
                for ( j = 1; j <= part.psize[i]; j++ ) {
                    num = j3[i][siz-j]; /* now we need an argument with this number */
                    for ( jj = num-1; jj < part.numargs; jj++ ) {
                        if ( part.nargs[jj] == num ) break;
                    }
                    targ = part.args[jj]+twhere;
                    if ( *targ < 0 ) {
                        if ( tensorflag ) targ++;
                        else if ( *targ > -FUNCTION ) *to++ = *targ++;
                        *to++ = *targ++;
                    }
                    else { jj = *targ; NCOPY(to,targ,jj); }
                }
                tfun[1] = to - tfun;
            }
/*
            Now the denominators of the coefficient
            First identical functions/partitions
*/
            j = 1; n = 1;
            while ( j < part.numpart ) {
                for ( im = 0; im < siz; im++ ) {
                    if ( part.nfun[j-1] != part.nfun[j] ) break;
                    if ( j3[j-1][im] < j3[j][im] ) break;
                    if ( j3[j-1][im] > j3[j][im] ) im = 2*siz+2;
                }
                if ( im == siz ) { n++; j++; continue; }
                if ( n > 1 ) {
div1:
                    if ( Factorial(BHEAD n, cfac, &nfac) ) Terminate(-1);
                    if ( DivLong(coeff,ncoeff,cfac,nfac,coeff2,&ncoeff2,coeff3,&ncoeff3) ) Terminate(-1);
                    c = coeff; coeff = coeff2; coeff2 = c;
                    n = ncoeff; ncoeff = ncoeff2; ncoeff2 = n;
                }
                n = 1; j++;
            }
            if ( n > 1 ) goto div1;
/*
            Now identical elements inside the partitions
*/
            for ( i = 0; i < part.numpart; i++ ) {
                j = 0; while ( j3[i][j] == 0 ) j++;
                n = 1; j++;
                while ( j < siz ) {
                    if ( j3[i][j-1] == j3[i][j] ) { n++; j++; }
                    else {
                        if ( n > 1 ) {
div2:
                            if ( Factorial(BHEAD n, cfac, &nfac) ) Terminate(-1);
                            if ( DivLong(coeff,ncoeff,cfac,nfac,coeff2,&ncoeff2,coeff3,&ncoeff3) ) Terminate(-1);
                            c = coeff; coeff = coeff2; coeff2 = c;
                            n = ncoeff; ncoeff = ncoeff2; ncoeff2 = n;
                        }
                        n = 1; j++;
                    }
                }
                if ( n > 1 ) goto div2;
            }
/*
            And put this inside the term. Normalize will take care of it.
*/
            if ( ncoeff != 1 || coeff[0] > 1 ) {
                if ( ncoeff == 1 && coeff[0] <= MAXPOSITIVE ) {
                    *to++ = SNUMBER; *to++ = 4; *to++ = (WORD)(coeff[0]); *to++ = 1;
                }
                else {
                    *to++ = LNUMBER; *to++ = ncoeff+3; *to++ = ncoeff;
                    for ( i = 0; i < ncoeff; i++ ) *to++ = ((WORD *)coeff)[i];
                }
```

```
            }
/*
            And the tail
*/
            while ( t2 < tend ) *to++ = *t2++;
            *termout = to-termout;
            AT.WorkPointer = to;
            if ( Generator(BHEAD termout,level) ) Terminate(-1);
            AT.WorkPointer = termout;
/*
            #] Solution :

            Now we can pop all a with the lowest value and one more.
*/
            a = 0;
            while ( part.nargs[a] == 1 ) {
                pfill = j2fill[a]; j2[pfill]++; j3where[a][0] = 0; j3fill[pfill]--; a++;
            }
            if ( a < part.numargs ) {
                pfill = j2fill[a]; j2[pfill]++; j3where[a][0] = 0; j3fill[pfill]--; a++;
            }
            a--;
            pfill++;
        }
        else if ( part.nargs[a] == part.nargs[a+1] ) {}
        else { pfill = 0; }
    }
Done:
    M_free(j2,"parts2");
    M_free(j3,"parts3");
    M_free(array,"parts");
    NumberFree(cfac,"partitions!");
    NumberFree(coeff3,"partitions3");
    NumberFree(coeff2,"partitions2");
    NumberFree(coeff,"partitions");
    NumberFree(coeffnum,"partitionsn");
    M_free(part.psize,"partitions");
    part.psize = part.nfun = part.args = part.nargs = 0;
    part.numargs = part.numpart = part.where = 0;
    return(0);
}

/*
        #] DoPartitions :
    #] Distribute :
    #[ DoPermutations :

    Routine replaces the function perm_(f,args) by occurrences of f with
    all permutations of the args. This should always fit!
*/

WORD DoPermutations(PHEAD WORD *term, WORD level)
{
    PERMP perm;
    WORD *oldworkpointer = AT.WorkPointer, *termout = AT.WorkPointer;
    WORD *t, *tstop, *tt, *ttstop, odd = 0;
    WORD *args[MAXMATCH], nargs, i, first, skip, *to, *from;
/*
    Find function and count arguments. Check for odd/even
*/
    tstop = term+*term; tstop -= ABS(tstop[-1]);
    t = term+1;
    while ( t < tstop ) {
        if ( *t == PERMUTATIONS ) {
            if ( t[1] >= FUNHEAD+1 && t[FUNHEAD] <= -FUNCTION ) {
                odd = 0; skip = 1;
            }
            else if ( t[1] >= FUNHEAD+3 && t[FUNHEAD] == -SNUMBER && t[FUNHEAD+2] <= -FUNCTION ) {
                if ( t[FUNHEAD+1] % 2 == 1 ) odd = -1;
                else odd = 0;
                skip = 3;
            }
            else { t += t[1]; continue; }
            tt = t+FUNHEAD+skip; ttstop = t + t[1];
            nargs = 0;
            while ( tt < ttstop ) { NEXTARG(tt); nargs++; }
            tt = t+FUNHEAD+skip;
            if ( nargs > MAXMATCH ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Too many arguments in function perm_. %d! is way too big",(WORD)MAXMATCH);
                MUNLOCK(ErrorMessageLock);
                SETERROR(-1)
            }
            i = 0;
            while ( tt < ttstop ) { args[i++] = tt; NEXTARG(tt); }
            perm.n = nargs;
            perm.sign = 0;
```

```
            perm.objects = args;
            first = 1;
            while ( (first = PermuteP(&perm,first) ) == 0 ) {
/*
                Compose the output term
*/
                to = termout; from = term;
                while ( from < t ) *to++ = *from++;
                *to++ = -t[FUNHEAD+skip-1];
                *to++ = t[1] - skip;
                for ( i = 2; i < FUNHEAD; i++ ) *to++ = t[i];
                for ( i = 0; i < nargs; i++ ) {
                    from = args[i];
                    COPY1ARG(to,from);
                }
                from = t+t[1];
                tstop = term + *term;
                while ( from < tstop ) *to++ = *from++;
                if ( odd && ( ( perm.sign & 1 ) != 0 ) ) to[-1] = -to[-1];
                *termout = to - termout;
                AT.WorkPointer = to;
                if ( Generator(BHEAD termout,level) ) Terminate(-1);
                AT.WorkPointer = oldworkpointer;
            }
            return(0);
        }
        t += t[1];
    }
    return(0);
}

/*
    #] DoPermutations :
    #[ DoShuffle :

    Merges the arguments of all occurrences of function fun into a
    single occurrence of fun. The opposite of Distrib_
    Syntax:
        Shuffle[,once|all],fun;
        Shuffle[,once|all],$fun;
    The expansion of the dollar should give a single function.
    The dollar is indicated as usual with a negative value.
    option = 1 (once): generate identical results only once
    option = 0 (all): generate identical results with combinatorics (default)
*/

/*
    We use the Shuffle routine which has a large amount of combinatorics.
    It doesn't have grouped combinatorics as in (0,1,2)*(0,1,3) where the
    groups (0,1) also cause double terms.
*/

WORD DoShuffle(WORD *term, WORD level, WORD fun, WORD option)
{
    GETIDENTITY
    SHvariables SHback, *SH = &(AN.SHvar);
    WORD *t1, *t2, *tstop, ncoef, n = fun, *to, *from;
    int i, error;
    LONG k;
    UWORD *newcombi;

    if ( n < 0 ) {
        if ( ( n = DolToFunction(BHEAD -n) ) == 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("$-variable in merge statement did not evaluate to a function.");
            MUNLOCK(ErrorMessageLock);
            return(1);
        }
    }
    if ( AT.WorkPointer + 3*(*term) + AM.MaxTal > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }

    tstop = term + *term;
    ncoef = tstop[-1];
    tstop -= ABS(ncoef);
    t1 = term + 1;
    while ( t1 < tstop ) {
        if ( ( *t1 == n ) && ( t1+t1[1] < tstop ) && ( t1[1] > FUNHEAD ) ) {
            t2 = t1 + t1[1];
            if ( t2 >= tstop ) {
                return(Generator(BHEAD term,level));
            }
            while ( t2 < tstop ) {
```

```
                if ( ( *t2 == n ) && ( t2[1] > FUNHEAD ) ) break;
                t2 += t2[1];
            }
            if ( t2 < tstop ) break;
        }
        t1 += t1[1];
    }
    if ( t1 >= tstop ) {
        return(Generator(BHEAD term,level));
    }
    *AN.RepPoint = 1;
/*
    Now we have two occurrences of the function.
    Back up all relevant variables and load all the stuff that needs to be
    passed on.
*/
    SHback = AN.SHvar;
    SH->finishuf = &FinishShuffle;
    SH->do_uffle = &DoShuffle;
    SH->outterm = AT.WorkPointer;
    AT.WorkPointer += *term;
    SH->stop1 = t1 + t1[1];
    SH->stop2 = t2 + t2[1];
    SH->thefunction = n;
    SH->option = option;
    SH->level = level;
    SH->incoef = tstop;
    SH->nincoef = ncoef;

    if ( AN.SHcombi == 0 || AN.SHcombisize == 0 ) {
        AN.SHcombisize = 200;
        AN.SHcombi = (UWORD *)Malloc1(AN.SHcombisize*sizeof(UWORD),"AN.SHcombi");
        SH->combilast = 0;
        SHback.combilast = 0;
    }
    else {
        SH->combilast += AN.SHcombi[SH->combilast]+1;
        if ( SH->combilast >= AN.SHcombisize - 100 ) {
            newcombi = (UWORD *)Malloc1(2*AN.SHcombisize*sizeof(UWORD),"AN.SHcombi");
            for ( k = 0; k < AN.SHcombisize; k++ ) newcombi[k] = AN.SHcombi[k];
            M_free(AN.SHcombi,"AN.SHcombi");
            AN.SHcombi = newcombi;
            AN.SHcombisize *= 2;
        }
    }
    AN.SHcombi[SH->combilast] = 1;
    AN.SHcombi[SH->combilast+1] = 1;

    i = t1-term; to = SH->outterm; from = term;
    NCOPY(to,from,i)
    SH->outfun = to;
    for ( i = 0; i < FUNHEAD; i++ ) { *to++ = t1[i]; }

    error = Shuffle(t1+FUNHEAD,t2+FUNHEAD,to);

    AT.WorkPointer = SH->outterm;
    AN.SHvar = SHback;
    if ( error ) {
        MesCall("DoShuffle");
        return(-1);
    }
    return(0);
}

/*
    #] DoShuffle :
    #[ Shuffle :

    How to make shuffles:

    We have two lists of arguments. We have to make a single
    shuffle of them. All combinations. Doubles should have as
    much as possible a combinatorics factor. Sometimes this is
    very difficult as in:
        (0,1,2)x(0,1,3) = -> (0,1) is a repeated pattern and the
        factor on that is difficult
    Simple way: (without combinatorics)
        repeat id  f0(?c)*f(x1?,?a)*f(x2?,?b) =
                        +f0(?c,x1)*f(?a)*f(x2,?b)
                        +f0(?c,x2)*f(x1,?a)*f(?b);
    Refinement:
            if ( x1 == x2 ) check how many more there are of the same.
            --> (n1,x) and (n2,x)
            id  f0(?c)*f1((n1,x),?b)*f2((n2,x),?c) =
                    +binom_(n1+n2,n1)*f0(?c,(n1+n2,x))*f1(?a)*f2(?b)
                    +sum_(j,0,n1-1,binom_(n2+j,j)*f0(?c,(j+n2,x))
                            *f1((n1-j),?a)*f2(?b))*force2
```

```
                    +sum_(j,0,n2-1,binom_(n1+j,j)*f0(?c,(j+n1,x))
                            *f1(?a)*f2((n2-j),?b))*force1
    The force operation can be executed directly

    The next question is how to program this: recursively or linearly
    which would require simulation of a recursion. Recursive is clearest
    but we need to pass a number of arguments from the calling routine
    to the final routine. This is done with AN.SHvar.

    We need space for the accumulation of the combinatoric factors.
*/

int Shuffle(WORD *from1, WORD *from2, WORD *to)
{
    GETIDENTITY
    WORD *t, *fr, *next1, *next2, na, *fn1, *fn2, *tt;
    int i, n, n1, n2, j;
    LONG combilast;
    SHvariables *SH = &(AN.SHvar);
    if ( from1 == SH->stop1 && from2 == SH->stop2 ) {
        return(FiniShuffle(to));
    }
    else if ( from1 == SH->stop1 ) {
        i = SH->stop2 - from2; t = to; tt = from2; NCOPY(t,tt,i)
        return(FiniShuffle(t));
    }
    else if ( from2 == SH->stop2 ) {
        i = SH->stop1 - from1; t = to; tt = from1; NCOPY(t,tt,i)
        return(FiniShuffle(t));
    }
/*
    Compare lead arguments
*/
    if ( AreArgsEqual(from1,from2) ) {
/*
        First find out how many of each
*/
        next1 = from1; n1 = 1; NEXTARG(next1)
        while ( ( next1 < SH->stop1 ) && AreArgsEqual(from1,next1) ) {
            n1++; NEXTARG(next1)
        }
        next2 = from2; n2 = 1; NEXTARG(next2)
        while ( ( next2 < SH->stop2 ) && AreArgsEqual(from2,next2) ) {
            n2++; NEXTARG(next2)
        }
        combilast = SH->combilast;
/*
            +binom_(n1+n2,n1)*f0(?c,(n1+n2,x))*f1(?a)*f2(?b)
*/
        t = to;
        n = n1 + n2;
        while ( --n >= 0 ) { fr = from1; CopyArg(t,fr) }
        if ( GetBinom((UWORD *)(t),&na,n1+n2,n1) ) goto shuffcall;
        if ( combilast + AN.SHcombi[combilast] + na + 2 >= AN.SHcombisize ) {
/*
            We need more memory in this stack. Fortunately this is the
            only place where we have to do this, because the other factors
            are definitely smaller.
            Layout:   size, LongInteger, size, LongInteger, .....
            We start pointing at the last one.
*/
            UWORD *combi = (UWORD *)Malloc1(2*AN.SHcombisize*2,"AN.SHcombi");
            LONG jj;
            for ( jj = 0; jj < AN.SHcombisize; jj++ ) combi[jj] = AN.SHcombi[jj];
            AN.SHcombisize *= 2;
            M_free(AN.SHcombi,"AN.SHcombi");
            AN.SHcombi = combi;
        }
        if ( MulLong((UWORD *)(AN.SHcombi+combilast+1),AN.SHcombi[combilast],
                     (UWORD *)(t),na,
                     (UWORD *)(AN.SHcombi+combilast+AN.SHcombi[combilast]+2),
                     (WORD *)(AN.SHcombi+combilast+AN.SHcombi[combilast]+1)) ) goto shuffcall;
        SH->combilast = combilast + AN.SHcombi[combilast] + 1;
        if ( next1 >= SH->stop1 ) {
            fr = next2; i = SH->stop2 - fr;
            NCOPY(t,fr,i)
            if ( FiniShuffle(t) ) goto shuffcall;
        }
        else if ( next2 >= SH->stop2 ) {
            fr = next1; i = SH->stop1 - fr;
            NCOPY(t,fr,i)
            if ( FiniShuffle(t) ) goto shuffcall;
        }
        else {
            if ( Shuffle(next1,next2,t) ) goto shuffcall;
        }
        SH->combilast = combilast;
```

```
/*
            +sum_(j,0,n1-1,binom_(n2+j,j)*f0(?c,(j+n2,x))
                    *f1((n1-j),?a)*f2(?b))*force2
*/
        if ( next2 < SH->stop2 ) {
         t = to;
         n = n2;
         while ( --n >= 0 ) { fr = from1; CopyArg(t,fr) }
         for ( j = 0; j < n1; j++ ) {
          if ( GetBinom((UWORD *)(t),&na,n2+j,j) ) goto shuffcall;
          if ( MulLong((UWORD *)(AN.SHcombi+combilast+1),AN.SHcombi[combilast],
                       (UWORD *)(t),na,
                       (UWORD *)(AN.SHcombi+combilast+AN.SHcombi[combilast]+2),
                       (WORD *)(AN.SHcombi+combilast+AN.SHcombi[combilast]+1)) ) goto shuffcall;
          SH->combilast = combilast + AN.SHcombi[combilast] + 1;
          if ( j > 0 ) { fr = from1; CopyArg(t,fr) }
          fn2 = next2; tt = t;
          CopyArg(tt,fn2)

          if ( fn2 >= SH->stop2 ) {
            n = n1-j;
            while ( --n >= 0 ) { fr = from1; CopyArg(tt,fr) }
            fr = next1; i = SH->stop1 - fr;
            NCOPY(tt,fr,i)
            if ( FiniShuffle(tt) ) goto shuffcall;
          }
          else {
            n = j; fn1 = from1; while ( --n >= 0 ) { NEXTARG(fn1) }
            if ( Shuffle(fn1,fn2,tt) ) goto shuffcall;
          }
          SH->combilast = combilast;
         }
        }
/*
            +sum_(j,0,n2-1,binom_(n1+j,j)*f0(?c,(j+n1,x))
                    *f1(?a)*f2((n2-j),?b))*force1
*/
        if ( next1 < SH->stop1 ) {
         t = to;
         n = n1;
         while ( --n >= 0 ) { fr = from1; CopyArg(t,fr) }
         for ( j = 0; j < n2; j++ ) {
          if ( GetBinom((UWORD *)(t),&na,n1+j,j) ) goto shuffcall;
          if ( MulLong((UWORD *)(AN.SHcombi+combilast+1),AN.SHcombi[combilast],
                       (UWORD *)(t),na,
                       (UWORD *)(AN.SHcombi+combilast+AN.SHcombi[combilast]+2),
                       (WORD *)(AN.SHcombi+combilast+AN.SHcombi[combilast]+1)) ) goto shuffcall;
          SH->combilast = combilast + AN.SHcombi[combilast] + 1;
          if ( j > 0 ) { fr = from1; CopyArg(t,fr) }
          fn1 = next1; tt = t;
          CopyArg(tt,fn1)

          if ( fn1 >= SH->stop1 ) {
            n = n2-j;
            while ( --n >= 0 ) { fr = from1; CopyArg(tt,fr) }
            fr = next2; i = SH->stop2 - fr;
            NCOPY(tt,fr,i)
            if ( FiniShuffle(tt) ) goto shuffcall;
          }
          else {
            n = j; fn2 = from2; while ( --n >= 0 ) { NEXTARG(fn2) }
            if ( Shuffle(fn1,fn2,tt) ) goto shuffcall;
          }
          SH->combilast = combilast;
         }
        }
    }
    else {
/*
        Argument from first list
*/
        t = to;
        fr = from1;
        CopyArg(t,fr)
        if ( fr >= SH->stop1 ) {
            fr = from2; i = SH->stop2 - fr;
            NCOPY(t,fr,i)
            if ( FiniShuffle(t) ) goto shuffcall;
        }
        else {
            if ( Shuffle(fr,from2,t) ) goto shuffcall;
        }
/*
        Argument from second list
*/
        t = to;
        fr = from2;
```

```
        CopyArg(t,fr)
        if ( fr >= SH->stop2 ) {
            fr = from1; i = SH->stop1 - fr;
            NCOPY(t,fr,i)
            if ( FiniShuffle(t) ) goto shuffcall;
        }
        else {
            if ( Shuffle(from1,fr,t) ) goto shuffcall;
        }
    }
    return(0);
shuffcall:
    MesCall("Shuffle");
    return(-1);
}

/*
    #] Shuffle :
    #[ FinishShuffle :

    The complications here are:
    1: We want to save space. We put the output term in 'out' straight
       on top of what we produced thusfar. We have to copy the early
       piece because once the term goes back to Generator, Normalize can
       change it in situ
    2: There can be other occurrence of the function between the two
       that we did. For shuffles that isn't likely, but we use this
       routine also for the stuffles and there it can happen.
*/

int FinishShuffle(WORD *fini)
{
    GETIDENTITY
    WORD *t, *t1, *oldworkpointer = AT.WorkPointer, *tcoef, ntcoef, *out;
    int i;
    SHvariables *SH = &(AN.SHvar);
    SH->outfun[1] = fini - SH->outfun;
    if ( functions[SH->outfun[0]-FUNCTION].symmetric != 0 )
                    SH->outfun[2] |= DIRTYSYMFLAG;
    out = fini; i = fini - SH->outterm; t = SH->outterm;
    NCOPY(fini,t,i)
    t = SH->stop1;
    t1 = t + t[1];
    while ( t1 < SH->stop2 ) { t = t1; t1 = t + t[1]; }
    t1 = SH->stop1;
    while ( t1 < t ) *fini++ = *t1++;
    t = SH->stop2;
    while ( t < SH->incoef ) *fini++ = *t++;
    tcoef = fini;
    ntcoef = SH->nincoef;
    i = ABS(ntcoef);
    NCOPY(fini,t,i);
    ntcoef = REDLENG(ntcoef);
    Mully(BHEAD (UWORD *)tcoef,&ntcoef,
        (UWORD *)(AN.SHcombi+SH->combilast+1),AN.SHcombi[SH->combilast]);
    ntcoef = INCLENG(ntcoef);
    fini = tcoef + ABS(ntcoef);
    if ( ( ( SH->option & 2 ) != 0 ) && ( ( SH->option & 256 ) != 0 ) ) ntcoef = -ntcoef;
    fini[-1] = ntcoef;
    i = *out = fini - out;
/*
    Now check whether we have to do more
*/
    AT.WorkPointer = out + *out;
    if ( ( SH->option & 1 ) == 1 ) {
        if ( Generator(BHEAD out,SH->level) ) goto Finicall;
    }
    else {
        if ( DoShtuffle(out,SH->level,SH->thefunction,SH->option) ) goto Finicall;
    }
    AT.WorkPointer = oldworkpointer;
    return(0);
Finicall:
    AT.WorkPointer = oldworkpointer;
    MesCall("FinishShuffle");
    return(-1);
}

/*
    #] FinishShuffle :
    #[ DoStuffle :

    Stuffling is a variation of shuffling.
    In the stuffling we insist that the arguments are (short) integers. nonzero.
    The stuffle sum is x st y = sig_(x)*sig_(y)*(abs(x)+abs(y))
    The way we do this is:
        1: count the arguments in each function: n1, n2
```

```
        2: take the minimum minval = min(n1,n2).
        3: for ( j = 0; j <= min; j++ ) take j elements in each of the lists.
        4: the j+1 groups of remaining arguments have to each be shuffled
        5: the j selected pairs have to be stuffle added.
    We can use many of the shuffle things.
    Considering the recursive nature of the generation we actually don't
    need to know n1, n2, minval.
*/

WORD DoStuffle(WORD *term, WORD level, WORD fun, WORD option)
{
    GETIDENTITY
    SHvariables SHback, *SH = &(AN.SHvar);
    WORD *t1, *t2, *tstop, *t1stop, *t2stop, ncoef, n = fun, *to, *from;
    WORD *r1, *r2;
    int i, error;
    LONG k;
    UWORD *newcombi;
#ifdef NEWCODE
    WORD *rr1, *rr2, i1, i2;
#endif
    if ( n < 0 ) {
        if ( ( n = DolToFunction(BHEAD -n) ) == 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("$-variable in merge statement did not evaluate to a function.");
            MUNLOCK(ErrorMessageLock);
            return(1);
        }
    }
    if ( AT.WorkPointer + 3*(*term) + AM.MaxTal > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }

    tstop = term + *term;
    ncoef = tstop[-1];
    tstop -= ABS(ncoef);
    t1 = term + 1;
retry1:;
    while ( t1 < tstop ) {
        if ( ( *t1 == n ) && ( t1+t1[1] < tstop ) && ( t1[1] > FUNHEAD ) ) {
            t2 = t1 + t1[1];
            if ( t2 >= tstop ) {
                return(Generator(BHEAD term,level));
            }
retry2:;
            while ( t2 < tstop ) {
                if ( ( *t2 == n ) && ( t2[1] > FUNHEAD ) ) break;
                t2 += t2[1];
            }
            if ( t2 < tstop ) break;
        }
        t1 += t1[1];
    }
    if ( t1 >= tstop ) {
        return(Generator(BHEAD term,level));
    }
/*
    Next we have to check that the arguments are of the correct type
    At the same time we can count them.
*/
#ifndef NEWCODE
    t1stop = t1 + t1[1];
    r1 = t1 + FUNHEAD;
    while ( r1 < t1stop ) {
        if ( *r1 != -SNUMBER ) break;
        if ( r1[1] == 0 ) break;
        r1 += 2;
    }
    if ( r1 < t1stop ) { t1 = t2; goto retry1; }
    t2stop = t2 + t2[1];
    r2 = t2 + FUNHEAD;
    while ( r2 < t2stop ) {
        if ( *r2 != -SNUMBER ) break;
        if ( r2[1] == 0 ) break;
        r2 += 2;
    }
    if ( r2 < t2stop ) { t2 = t2 + t2[1]; goto retry2; }
#else
    t1stop = t1 + t1[1];
    r1 = t1 + FUNHEAD;
    while ( r1 < t1stop ) {
        if ( *r1 == -SNUMBER ) {
            if ( r1[1] == 0 ) break;
            r1 += 2; continue;
```

```
        }
        else if ( *r1 == -SYMBOL ) {
            if ( ( symbols[r1[1]].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
                        break;
            r1 += 2; continue;
        }
        if ( *r1 > 0 && *r1 == r1[ARGHEAD]+ARGHEAD ) {
            if ( ABS(r1[r1[0]-1]) == r1[0]-ARGHEAD-1 ) {}
            else if ( r1[ARGHEAD+1] == SYMBOL ) {
                rr1 = r1 + ARGHEAD + 3;
                i1 = rr1[-1]-2;
                while ( i1 > 0 ) {
                    if ( ( symbols[*rr1].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
                        break;
                    i1 -= 2; rr1 += 2;
                }
                if ( i1 > 0 ) break;
            }
            else break;
            rr1 = r1+*r1-1;
            i1 = (ABS(*rr1)-1)/2;
            while ( i1 > 1 ) {
                if ( rr1[-1] ) break;
                i1--; rr1--;
            }
            if ( i1 > 1 || rr1[-1] != 1 ) break;
            r1 += *r1;
        }
        else break;
    }
    if ( r1 < t1stop ) { t1 = t2; goto retry1; }
    t2stop = t2 + t2[1];
    r2 = t2 + FUNHEAD;

    while ( r2 < t2stop ) {
        if ( *r2 == -SNUMBER ) {
            if ( r2[1] == 0 ) break;
            r2 += 2; continue;
        }
        else if ( *r2 == -SYMBOL ) {
            if ( ( symbols[r2[1]].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
                        break;
            r2 += 2; continue;
        }
        if ( *r2 > 0 && *r2 == r2[ARGHEAD]+ARGHEAD ) {
            if ( ABS(r2[r2[0]-1]) == r2[0]-ARGHEAD-1 ) {}
            else if ( r2[ARGHEAD+1] == SYMBOL ) {
                rr2 = r2 + ARGHEAD + 3;
                i2 = rr2[-1]-2;
                while ( i2 > 0 ) {
                    if ( ( symbols[*rr2].complex & VARTYPEROOTOFUNITY ) != VARTYPEROOTOFUNITY )
                        break;
                    i2 -= 2; rr2 += 2;
                }
                if ( i2 > 0 ) break;
            }
            else break;
            rr2 = r2+*r2-1;
            i2 = (ABS(*rr2)-1)/2;
            while ( i2 > 1 ) {
                if ( rr2[-1] ) break;
                i2--; rr2--;
            }
            if ( i2 > 1 || rr2[-1] != 1 ) break;
            r2 += *r2;
        }
        else break;
    }
    if ( r2 < t2stop ) { t2 = t2 + t2[1]; goto retry2; }
#endif
/*
    OK, now we got two objects that can be used.
*/
    *AN.RepPoint = 1;

    SHback = AN.SHvar;
    SH->finishuf = &FinishStuffle;
    SH->do_uffle = &DoStuffle;
    SH->outterm = AT.WorkPointer;
    AT.WorkPointer += *term;
    SH->ststop1 = t1 + t1[1];
    SH->ststop2 = t2 + t2[1];
    SH->thefunction = n;
    SH->option = option;
    SH->level = level;
    SH->incoef = tstop;
    SH->nincoef = ncoef;
```

```
    if ( AN.SHcombi == 0 || AN.SHcombisize == 0 ) {
        AN.SHcombisize = 200;
        AN.SHcombi = (UWORD *)Malloc1(AN.SHcombisize*sizeof(UWORD),"AN.SHcombi");
        SH->combilast = 0;
        SHback.combilast = 0;
    }
    else {
        SH->combilast += AN.SHcombi[SH->combilast]+1;
        if ( SH->combilast >= AN.SHcombisize - 100 ) {
            newcombi = (UWORD *)Malloc1(2*AN.SHcombisize*sizeof(UWORD),"AN.SHcombi");
            for ( k = 0; k < AN.SHcombisize; k++ ) newcombi[k] = AN.SHcombi[k];
            M_free(AN.SHcombi,"AN.SHcombi");
            AN.SHcombi = newcombi;
            AN.SHcombisize *= 2;
        }
    }
    AN.SHcombi[SH->combilast] = 1;
    AN.SHcombi[SH->combilast+1] = 1;

    i = t1-term; to = SH->outterm; from = term;
    NCOPY(to,from,i)
    SH->outfun = to;
    for ( i = 0; i < FUNHEAD; i++ ) { *to++ = t1[i]; }

    error = Stuffle(t1+FUNHEAD,t2+FUNHEAD,to);

    AT.WorkPointer = SH->outterm;
    AN.SHvar = SHback;
    if ( error ) {
        MesCall("DoStuffle");
        return(-1);
    }
    return(0);
}

/*
    #] DoStuffle :
    #[ Stuffle :

    The way to generate the stuffles
    1: select an argument in the first  list (for(j1=0;j1<last;j1++))
    2: select an argument in the second list (for(j2=0;j2<last;j2++))
    3: put values for SH->ststop1 and SH->ststop2 at these arguments.
    4: generate all shuffles of the arguments in front.
    5: Then put the stuffle sum of arg(j1) and arg(j2)
    6: Then continue calling Stuffle
    7: Once one gets exhausted, we can clean up the list and call FinishShuffle
    8: if ( ( SH->option & 2 ) != 0 ) the stuffle sum is negative.
*/

int Stuffle(WORD *from1, WORD *from2, WORD *to)
{
    GETIDENTITY
    WORD *t, *tf, *next1, *next2, *st1, *st2, *save1, *save2;
    SHvariables *SH = &(AN.SHvar);
    int i, retval;
/*
    First the special cases (exhausted list(s)):
*/
    save1 = SH->stop1; save2 = SH->stop2;
    if ( from1 >= SH->ststop1 && from2 == SH->ststop2 ) {
        SH->stop1 = SH->ststop1;
        SH->stop2 = SH->ststop2;
        retval = FinishShuffle(to);
        SH->stop1 = save1; SH->stop2 = save2;
        return(retval);
    }
    else if ( from1 >= SH->ststop1 ) {
        i = SH->ststop2 - from2; t = to; tf = from2; NCOPY(t,tf,i)
        SH->stop1 = SH->ststop1;
        SH->stop2 = SH->ststop2;
        retval = FinishShuffle(t);
        SH->stop1 = save1; SH->stop2 = save2;
        return(retval);
    }
    else if ( from2 >= SH->ststop2 ) {
        i = SH->ststop1 - from1; t = to; tf = from1; NCOPY(t,tf,i)
        SH->stop1 = SH->ststop1;
        SH->stop2 = SH->ststop2;
        retval = FinishShuffle(t);
        SH->stop1 = save1; SH->stop2 = save2;
        return(retval);
    }
/*
    Now the case that we have no stuffle sums.
*/
    SH->stop1 = SH->ststop1;
```

```
    SH->stop2 = SH->ststop2;
    SH->finishuf = &FinishShuffle;
    if ( Shuffle(from1,from2,to) ) goto stuffcall;
    SH->finishuf = &FinishStuffle;
/*
    Now we have to select a pair, one from 1 and one from 2.
*/
#ifndef NEWCODE
    st1 = from1; next1 = st1+2;       /* <----- */
#else
    st1 = next1 = from1;
    NEXTARG(next1)
#endif
    while ( next1 <= SH->ststop1 ) {
#ifndef NEWCODE
        st2 = from2; next2 = st2+2;       /* <----- */
#else
        next2 = st2 = from2;
        NEXTARG(next2)
#endif
        while ( next2 <= SH->ststop2 ) {
            SH->stop1 = st1;
            SH->stop2 = st2;
            if ( st1 == from1 && st2 == from2 ) {
                t = to;
#ifndef NEWCODE
                *t++ = -SNUMBER; *t++ = StuffAdd(st1[1],st2[1]);
#else
                t = StuffRootAdd(st1,st2,t);
#endif
                SH->option ^= 256;
                if ( Stuffle(next1,next2,t) ) goto stuffcall;
                SH->option ^= 256;
            }
            else if ( st1 == from1 ) {
                i = st2-from2;
                t = to; tf = from2; NCOPY(t,tf,i)
#ifndef NEWCODE
                *t++ = -SNUMBER; *t++ = StuffAdd(st1[1],st2[1]);
#else
                t = StuffRootAdd(st1,st2,t);
#endif
                SH->option ^= 256;
                if ( Stuffle(next1,next2,t) ) goto stuffcall;
                SH->option ^= 256;
            }
            else if ( st2 == from2 ) {
                i = st1-from1;
                t = to; tf = from1; NCOPY(t,tf,i)
#ifndef NEWCODE
                *t++ = -SNUMBER; *t++ = StuffAdd(st1[1],st2[1]);
#else
                t = StuffRootAdd(st1,st2,t);
#endif
                SH->option ^= 256;
                if ( Stuffle(next1,next2,t) ) goto stuffcall;
                SH->option ^= 256;
            }
            else {
                if ( Shuffle(from1,from2,to) ) goto stuffcall;
            }
#ifndef NEWCODE
            st2 = next2; next2 += 2;       /* <----- */
#else
            st2 = next2;
            NEXTARG(next2)
#endif
        }
#ifndef NEWCODE
        st1 = next1; next1 += 2;       /* <----- */
#else
        st1 = next1;
        NEXTARG(next1)
#endif
    }
    SH->stop1 = save1; SH->stop2 = save2;
    return(0);
stuffcall:;
    MesCall("Stuffle");
    return(-1);
}

/*
    #] Stuffle :
    #[ FinishStuffle :

    The program only comes here from the Shuffle routine.
```

```
    It should add the stuffle sum and then call Stuffle again.
*/

int FinishStuffle(WORD *fini)
{
    GETIDENTITY
    SHvariables *SH = &(AN.SHvar);
#ifdef NEWCODE
    WORD *next1 = SH->stop1, *next2 = SH->stop2;
    fini = StuffRootAdd(next1,next2,fini);
#else
    *fini++ = -SNUMBER; *fini++ = StuffAdd(SH->stop1[1],SH->stop2[1]);
#endif
    SH->option ^= 256;
#ifdef NEWCODE
    NEXTARG(next1)
    NEXTARG(next2)
    if ( Stuffle(next1,next2,fini) ) goto stuffcall;
#else
    if ( Stuffle(SH->stop1+2,SH->stop2+2,fini) ) goto stuffcall;
#endif
    SH->option ^= 256;
    return(0);
stuffcall:;
    MesCall("FinishStuffle");
    return(-1);
}

/*
    #] FinishStuffle :
    #[ StuffRootAdd :

    Makes the stuffle sum of two arguments.
    The arguments can be of one of three types:
    1: -SNUMBER,num
    2: -SYMBOL,symbol
    3: Numerical (long) argument.
    4: Generic argument with (only) symbols that are roots of unity and
       a coefficient.
    We have excluded the case that both t1 and t2 are of type 1:
    The output should be written to 'to' and the new fill position should
    be the return value.
    `to' is inside the workspace.

    The stuffle sum is sig_(t2)*t1+sig_(t1)*t2
    or sig_(t1)*sig_(t2)*(abs_(t1)+abs_(t2))
*/

#ifdef NEWCODE

WORD *StuffRootAdd(WORD *t1, WORD *t2, WORD *to)
{
    int type1, type2, type3, sgn, sgn1, sgn2, sgn3, pow, root, nosymbols, i;
    WORD *tt1, *tt2, it1, it2, *t3, *r, size1, size2, size3;
    WORD scratch[2];
    LONG x;
    if ( *t1 == -SNUMBER ) { type1 = 1; if ( t1[1] < 0 ) sgn1 = -1; else sgn1 = 1; }
    else if ( *t1 == -SYMBOL ) { type1 = 2; sgn1 = 1; }
    else if ( ABS(t1[*t1-1]) == *t1-ARGHEAD-1 ) {
        type1 = 3; if ( t1[*t1-1] < 0 ) sgn1 = -1; else sgn1 = 1; }
    else { type1 = 4; if ( t1[*t1-1] < 0 ) sgn1 = -1; else sgn1 = 1; }
    if ( *t2 == -SNUMBER ) { type2 = 1; if ( t2[1] < 0 ) sgn2 = -1; else sgn2 = 1; }
    else if ( *t2 == -SYMBOL ) { type2 = 2; sgn2 = 1; }
    else if ( ABS(t2[*t2-1]) == *t2-ARGHEAD-1 ) {
        type2 = 3; if ( t2[*t2-1] < 0 ) sgn2 = -1; else sgn2 = 1; }
    else { type2 = 4; if ( t2[*t2-1] < 0 ) sgn2 = -1; else sgn2 = 1; }
    if ( type1 > type2 ) {
        t3 = t1; t1 = t2; t2 = t3;
        type3 = type1; type1 = type2; type2 = type3;
        sgn3 = sgn1; sgn1 = sgn2; sgn2 = sgn3;
    }
    nosymbols = 1; sgn3 = 1;
    switch ( type1 ) {
        case 1:
            if ( type2 == 1 ) {
                x  = sgn2 * t1[1];
                x += sgn1 * t2[1];
                if ( x > MAXPOSITIVE || x < -(MAXPOSITIVE+1) ) {
                    if ( x < 0 ) { sgn1 = -3; x = -x; }
                    else sgn1 = 3;
                    *to++ = ARGHEAD+4;
                    *to++ = 0;
                    FILLARG(to)
                    *to++ = 4; *to++ = (UWORD)x; *to++ = 1; *to++ = sgn1;
                }
                else { *to++ = -SNUMBER; *to++ = (WORD)x; }
            }
```

```
            else if ( type2 == 2 ) {
                *to++ = ARGHEAD+8; *to++ = 0; FILLARG(to)
                *to++ = 8; *to++ = SYMBOL; *to++ = 4; *to++ = t2[1]; *to++ = 1;
                *to++ = ABS(t1[1])+1;
                *to++ = 1;
                *to++ = 3*sgn1;
            }
            else if ( type2 == 3 ) {
                tt1 = (WORD *)scratch; tt1[0] = ABS(t1[1]); size1 = 1;
                tt2 = t2+ARGHEAD+1; size2 = (ABS(t2[*t2-1])-1)/2;
                t3 = to;
                *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
                goto DoCoeffi;
            }
            else {
/*
                t1 is (short) numeric, t2 has the symbol(s).
*/
                tt1 = (WORD *)scratch; tt1[0] = ABS(t1[1]); size1 = 1;
                tt2 = t2+ARGHEAD+1; tt2 += tt2[1]; size2 = (ABS(t2[*t2-1])-1)/2;
                t3 = to; i = tt2 - t2; r = t2;
                NCOPY(to,r,i)
                nosymbols = 0;
                goto DoCoeffi;
            }
        break;
        case 2:
            if ( type2 == 2 ) {
                if ( t1[1] == t2[1] ) {
                    if ( ( symbols[t1[1]].maxpower == 4 )
                    && ( ( symbols[t1[1]].complex & VARTYPEMINUS ) == VARTYPEMINUS ) ) {
                        *to++ = -SNUMBER; *to++ = -2;
                    }
                    else if ( symbols[t1[1]].maxpower == 2 ) {
                        *to++ = -SNUMBER; *to++ = 2;
                    }
                    else {
                        *to++ = ARGHEAD+8; *to++ = 0; FILLARG(to)
                        *to++ = 8; *to++ = SYMBOL; *to++ = 4;
                        *to++ = t1[1]; *to++ = 2;
                        *to++ = 2; *to++ = 1; *to++ = 3;
                    }
                }
                else {
                    *to++ = ARGHEAD+10; *to++ = 0; FILLARG(to)
                    *to++ = 10; *to++ = SYMBOL; *to++ = 6;
                    if ( t1[1] < t2[1] ) {
                        *to++ = t1[1]; *to++ = 1; *to++ = t2[1]; *to++ = 1;
                    }
                    else {
                        *to++ = t2[1]; *to++ = 1; *to++ = t1[1]; *to++ = 1;
                    }
                    *to++ = 2; *to++ = 1; *to++ = 3;
                }
            }
            else if ( type2 == 3 ) {
                t3 = to;
                *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
                *to++ = SYMBOL; *to++ = 4; *to++ = t1[1]; *to++ = 1;
                tt1 = scratch; tt1[1] = 1; size1 = 1;
                tt2 = t2+ARGHEAD+1; size2 = (ABS(t2[*t2-1])-1)/2;
                nosymbols = 0;
                goto DoCoeffi;
            }
            else {
                tt1 = scratch; tt1[0] = 1; size1 = 1;
                t3 = to;
                *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
                *to++ = SYMBOL; *to++ = 0;
                tt2 = t2 + ARGHEAD+3; it2 = tt2[-1]-2;
                while ( it2 > 0 ) {
                    if ( *tt2 == t1[1] ) {
                        pow = tt2[1]+1;
                        root = symbols[*tt2].maxpower;
                        if ( pow >= root ) pow -= root;
                        if ( ( symbols[*tt2].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
                            if ( ( root & 1 ) == 0 && pow >= root/2 ) {
                                pow -= root/2; sgn3 = -sgn3;
                            }
                        }
                        if ( pow != 0 ) {
                            *to++ = *tt2; *to++ = pow;
                        }
                        tt2 += 2; it2 -= 2;
                        break;
                    }
                    else if ( t1[1] < *tt2 ) {
```

```
                        *to++ = t1[1]; *to++ = 1; break;
                    }
                    else {
                        *to++ = *tt2++; *to++ = *tt2++; it2 -= 2;
                        if ( it2 <= 0 ) { *to++ = t1[1]; *to++ = 1; }
                    }
                }
                while ( it2 > 0 ) { *to++ = *tt2++; *to++ = *tt2++; it2 -= 2; }
                if ( (to - t3) > ARGHEAD+3 ) {
                    t3[ARGHEAD+2] = (to-t3)-ARGHEAD-1; /* size of the SYMBOL field */
                    nosymbols = 0;
                }
                else {
                    to = t3+ARGHEAD+1; /* no SYMBOL field */
                }
                size2 = (ABS(t2[*t2-1])-1)/2;
                goto DoCoeffi;
            }
        break;
        case 3:
            if ( type2 == 3 ) {
/*
                Both are numeric
*/
                tt1 = t1+ARGHEAD+1; size1 = (ABS(t1[*t1-1])-1)/2;
                tt2 = t2+ARGHEAD+1; size2 = (ABS(t2[*t2-1])-1)/2;
                t3 = to;
                *to++ = 0; *to++ = 0; FILLARG(to) *to++ = 0;
                goto DoCoeffi;
            }
            else {
/*
                t1 is (long) numeric, t2 has the symbol(s).
*/
                tt1 = t1+ARGHEAD+1; size1 = (ABS(t1[*t1-1])-1)/2;
                tt2 = t2+ARGHEAD+1; tt2 += tt2[1]; size2 = (ABS(t2[*t2-1])-1)/2;
                t3 = to; i = tt2 - t2; r = t2;
                NCOPY(to,r,i)
                nosymbols = 0;
                goto DoCoeffi;
            }
        break;
        case 4:
/*
            Both have roots of unity
            1: Merge the lists and simplify if possible
*/
            tt1 = t1+ARGHEAD+3; it1 = tt1[-1]-2;
            tt2 = t2+ARGHEAD+3; it2 = tt2[-1]-2;
            t3 = to;
            *to++ = 0; *to++ = 0; FILLARG(to)
            *to++ = 0; *to++ = SYMBOL; *to++ = 0;
            while ( it1 > 0 && it2 > 0 ) {
                if ( *tt1 == *tt2 ) {
                    pow = tt1[1]+tt2[1];
                    root = symbols[*tt1].maxpower;
                    if ( pow >= root ) pow -= root;
                    if ( ( symbols[*tt1].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
                        if ( ( root & 1 ) == 0 && pow >= root/2 ) {
                            pow -= root/2; sgn3 = -sgn3;
                        }
                    }
                    if ( pow != 0 ) {
                        *to++ = *tt1; *to++ = pow;
                    }
                    tt1 += 2; tt2 += 2; it1 -= 2; it2 -= 2;
                }
                else if ( *tt1 < *tt2 ) {
                    *to++ = *tt1++; *to++ = *tt1++; it1 -= 2;
                }
                else {
                    *to++ = *tt2++; *to++ = *tt2++; it2 -= 2;
                }
            }
            while ( it1 > 0 ) { *to++ = *tt1++; *to++ = *tt1++; it1 -= 2; }
            while ( it2 > 0 ) { *to++ = *tt2++; *to++ = *tt2++; it2 -= 2; }
            if ( (to - t3) > ARGHEAD+3 ) {
                t3[ARGHEAD+2] = (to-t3)-ARGHEAD-1; /* size of the SYMBOL field */
                nosymbols = 0;
            }
            else {
                to = t3+ARGHEAD+1; /* no SYMBOL field */
            }
            size1 = (ABS(t1[*t1-1])-1)/2;
            size2 = (ABS(t2[*t2-1])-1)/2;
/*
            Now tt1 and tt2 are pointing at their coefficients.
```

```
                sgn1 is the sign of 1, sgn2 is the sign of 2 and sgn3 is an extra
                overall sign.
*/
DoCoeffi:
                if ( AddLong((UWORD *)tt1,size1,(UWORD *)tt2,size2,(UWORD *)to,&size3) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Called from StuffRootAdd");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                sgn = sgn1*sgn2*sgn3;
                if ( nosymbols && size3 == 1 ) {
                    if ( (UWORD)(to[0]) <= MAXPOSITIVE && sgn > 0 ) {
                        sgn1 = to[0];
                        to = t3; *to++ = -SNUMBER; *to++ = sgn1;
                    }
                    else if ( (UWORD)(to[0]) <= (MAXPOSITIVE+1) && sgn < 0 ) {
                        sgn1 = to[0];
                        to = t3; *to++ = -SNUMBER; *to++ = -sgn1;
                    }
                    else goto genericcoef;
                }
                else {
genericcoef:
                    to += size3;
                    sgn = sgn*(2*size3+1);
                    *to++ = 1;
                    while ( size3 > 1 ) { *to++ = 0; size3--; }
                    *to++ = sgn;
                    t3[0] = to - t3;
                    t3[ARGHEAD] = t3[0] - ARGHEAD;
                }
            break;
    }
    return(to);
}

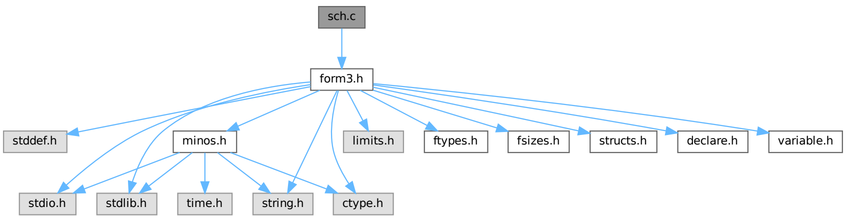
#endif

/*
    #] StuffRootAdd :
*/
```

4.122 sch.c File Reference

`#include "form3.h"`

Include dependency graph for sch.c:



Macros

- #define va_dcl int va_alist;
- #define va_start(list) list = (UBYTE ∗) &va_alist
- #define va_end(list)
- #define va_arg(list, mode) (((mode ∗)(list += sizeof(mode)))[-1])

Typedefs

- typedef UBYTE * va_list

Functions

- UBYTE * StrCopy (UBYTE *from, UBYTE *to)
- void AddToLine (UBYTE *s)
- void FiniLine (void)
- void IniLine (WORD extrablank)
- void LongToLine (UWORD *a, WORD na)
- void RatToLine (UWORD *a, WORD na)
- void TalToLine (UWORD x)
- void TokenToLine (UBYTE *s)
- UBYTE * CodeToLine (WORD number, UBYTE *Out)
- void MultiplyToLine (void)
- UBYTE * AddArrayIndex (WORD num, UBYTE *out)
- void PrtTerms (void)
- UBYTE * WrtPower (UBYTE *Out, WORD Power)
- void PrintTime (UBYTE *mess)
- void WriteLists (void)
- void WriteDictionary (DICTIONARY *dict)
- void WriteArgument (WORD *t)
- WORD WriteSubTerm (WORD *sterm, WORD first)
- WORD WriteInnerTerm (WORD *term, WORD first)
- WORD WriteTerm (WORD *term, WORD *lbrac, WORD first, WORD prtf, WORD br)
- WORD WriteExpression (WORD *terms, LONG ltot)
- WORD WriteAll (void)
- WORD WriteOne (UBYTE *name, int alreadyinline, int nosemi, WORD plus)

4.122.1 Detailed Description

Contains the functions that deal with the writing of expressions/terms in a textual representation. (Dutch schrijven = to write)

Definition in file sch.c.

4.122.2 Macro Definition Documentation

4.122.2.1 va_dcl

```
#define va_dcl int va_alist;
```

Definition at line 48 of file sch.c.

4.122.2.2 va_start

```
#define va_start(
              list ) list = (UBYTE *) &va_alist
```

Definition at line 49 of file sch.c.

4.122.2.3 va_end

```
#define va_end(
             list )
```

Definition at line 50 of file sch.c.

4.122.2.4 va_arg

```
#define va_arg(
             list,
             mode ) (((mode *)(list += sizeof(mode)))[-1])
```

Definition at line 51 of file sch.c.

4.122.3 Typedef Documentation

4.122.3.1 va_list

```
typedef UBYTE* va_list
```

Definition at line 47 of file sch.c.

4.122.4 Function Documentation

4.122.4.1 StrCopy()

```
UBYTE * StrCopy (
             UBYTE * from,
             UBYTE * to )
```

Definition at line 67 of file sch.c.

4.122.4.2 AddToLine()

```
void AddToLine (
             UBYTE * s )
```

Definition at line 82 of file sch.c.

4.122.4.3 FiniLine()

```
void FiniLine (
             void )
```

Definition at line 182 of file sch.c.

4.122.4.4 IniLine()

```
void IniLine (
            WORD extrablank )
```

Definition at line 267 of file sch.c.

4.122.4.5 LongToLine()

```
void LongToLine (
            UWORD * a,
            WORD na )
```

Definition at line 303 of file sch.c.

4.122.4.6 RatToLine()

```
void RatToLine (
            UWORD * a,
            WORD na )
```

Definition at line 337 of file sch.c.

4.122.4.7 TalToLine()

```
void TalToLine (
            UWORD x )
```

Definition at line 522 of file sch.c.

4.122.4.8 TokenToLine()

```
void TokenToLine (
            UBYTE * s )
```

Definition at line 551 of file sch.c.

4.122.4.9 CodeToLine()

```
UBYTE * CodeToLine (
            WORD number,
            UBYTE * Out )
```

Definition at line 658 of file sch.c.

4.122.4.10 MultiplyToLine()

```
void MultiplyToLine (
            void )
```

Definition at line 671 of file sch.c.

4.122.4.11 AddArrayIndex()

```
UBYTE * AddArrayIndex (
            WORD num,
            UBYTE * out )
```

Definition at line 696 of file sch.c.

4.122.4.12 PrtTerms()

```
void PrtTerms (
            void )
```

Definition at line 716 of file sch.c.

4.122.4.13 WrtPower()

```
UBYTE * WrtPower (
            UBYTE * Out,
            WORD Power )
```

Definition at line 738 of file sch.c.

4.122.4.14 PrintTime()

```
void PrintTime (
            UBYTE * mess )
```

Definition at line 784 of file sch.c.

4.122.4.15 WriteLists()

```
void WriteLists (
            void )
```

Definition at line 810 of file sch.c.

4.122.4.16 WriteDictionary()

```
void WriteDictionary (
            DICTIONARY * dict )
```

Definition at line 1307 of file sch.c.

4.122.4.17 WriteArgument()

```
void WriteArgument (
            WORD * t )
```

Definition at line 1424 of file sch.c.

4.122.4.18 WriteSubTerm()

```
WORD WriteSubTerm (
            WORD * sterm,
            WORD first )
```

Definition at line 1542 of file sch.c.

4.122.4.19 WriteInnerTerm()

```
WORD WriteInnerTerm (
            WORD * term,
            WORD first )
```

Definition at line 1951 of file sch.c.

4.122.4.20 WriteTerm()

```
WORD WriteTerm (
            WORD * term,
            WORD * lbrac,
            WORD first,
            WORD prtf,
            WORD br )
```

Definition at line 2148 of file sch.c.

4.122.4.21 WriteExpression()

```
WORD WriteExpression (
            WORD * terms,
            LONG ltot )
```

Definition at line 2470 of file sch.c.

4.122.4.22 WriteAll()

```
WORD WriteAll (
            void )
```

Definition at line 2501 of file sch.c.

4.122.4.23 WriteOne()

```
WORD WriteOne (
            UBYTE * name,
            int alreadyinline,
            int nosemi,
            WORD plus )
```

Definition at line 2727 of file sch.c.

4.123 sch.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : sch.c
*/

#include "form3.h"

#ifdef ANSI
#include <stdarg.h>
#else
#ifdef mBSD
#include <varargs.h>
#else
#ifdef VMS
#include <varargs.h>
#else
typedef UBYTE *va_list;
#define va_dcl int va_alist;
#define va_start(list) list = (UBYTE *) &va_alist
#define va_end(list)
#define va_arg(list,mode) (((mode *)(list += sizeof(mode)))[-1])
#endif
#endif
#endif

static int startinline = 0;
static char fcontchar = '&';
static int noextralinefeed = 0;
static int lowestlevel = 1;

/*
    #] Includes :
    #[ schryf-Utilities :
        #[ StrCopy :            UBYTE *StrCopy(from,to)
*/

UBYTE *StrCopy(UBYTE *from, UBYTE *to)
```

```
{
    while( ( *to++ = *from++ ) != 0 );
    return(to-1);
}

/*
        #] StrCopy :
        #[ AddToLine :          void AddToLine(s)

    Puts the characters of s in the outputline. If the line becomes
    filled it is written.

*/

void AddToLine(UBYTE *s)
{
    UBYTE *Out;
    LONG num;
    int i;
    if ( AO.OutInBuffer ) { AddToDollarBuffer(s); return; }
    Out = AO.OutFill;
    while ( *s ) {
        if ( Out >= AO.OutStop ) {
            if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                *Out++ = fcontchar;
            }
#ifdef WITHRETURN
            *Out++ = CARRIAGERETURN;
#endif
            *Out++ = LINEFEED;
            AO.FortFirst = 0;
            num = Out - AO.OutputLine;

            if ( AC.LogHandle >= 0 ) {
                if ( WriteFile(AC.LogHandle,AO.OutputLine+startinline
                                ,num-startinline) != (num-startinline) ) {
/*
                    We cannot write to an otherwise open log file.
                    The disk could be full of course.
*/
#ifdef DEBUGGER
                    if ( BUG.logfileflag == 0 ) {
                        fprintf(stderr,"Panic: Cannot write to log file! Disk full?\n");
                        BUG.logfileflag = 1;
                    }
                    BUG.eflag = 1; BUG.printflag = 1;
#else
                    Terminate(-1);
#endif
                }
            }

            if ( ( AO.PrintType & PRINTLFILE ) == 0 ) {
#ifdef WITHRETURN
                if ( num > 1 && AO.OutputLine[num-2] == CARRIAGERETURN ) {
                    AO.OutputLine[num-2] = LINEFEED;
                    num--;
                }
#endif
                if ( WriteFile(AM.StdOut,AO.OutputLine+startinline
                        ,num-startinline) != (num-startinline) ) {
#ifdef DEBUGGER
                    if ( BUG.stdoutflag == 0 ) {
                        fprintf(stderr,"Panic: Cannot write to standard output!\n");
                        BUG.stdoutflag = 1;
                    }
                    BUG.eflag = 1; BUG.printflag = 1;
#else
                    Terminate(-1);
#endif
                }
            }
            /* thomasr 23/04/09: A continuation line has been started.
             * In Fortran90 we do not want a space after the initial
             * '&' character otherwise we might end up with something
             * like:
             *     ...  2.&
             *   & 0 ...
             */
            startinline = 0;
            for ( i = 0; i < AO.OutSkip; i++ ) AO.OutputLine[i] = ' ';
            Out = AO.OutputLine + AO.OutSkip;
            if ( ( AC.OutputMode == FORTRANMODE
             || AC.OutputMode == PFORTRANMODE ) && AO.OutSkip == 7 ) {
                /* thomasr 23/04/09: fix leading blank in fortran90 mode */
                if(AC.IsFortran90 == ISFORTRAN90) {
                    Out[-1] = fcontchar;
```

```
                }
                else {
                    Out[-2] = fcontchar;
                    Out[-1] = ' ';
                }
            }
            if ( AO.IsBracket ) { *Out++ = ' ';
                if ( AC.OutputSpaces == NORMALFORMAT ) {
                     *Out++ = ' '; *Out++ = ' '; }
            }
            *Out = '\0';
            if ( AC.OutputMode == FORTRANMODE
             || ( AC.OutputMode == CMODE && AO.FactorMode == 0 )
             || AC.OutputMode == PFORTRANMODE )
                AO.InFbrack++;
        }
        *Out++ = *s++;
    }
    *Out = '\0';
    AO.OutFill = Out;
}

/*
        #] AddToLine :
        #[ FiniLine :           void FiniLine()
*/

void FiniLine(void)
{
    UBYTE *Out;
    WORD i;
    LONG num;
    if ( AO.OutInBuffer ) return;
    Out = AO.OutFill;
    while ( Out > AO.OutputLine ) {
        if ( Out[-1] == ' ' ) Out--;
        else break;
    }
    i = (WORD)(Out-AO.OutputLine);
    if ( noextralinefeed == 0 ) {
        if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90
            && Out > AO.OutputLine ) {
/*
            *Out++ = fcontchar;
*/
        }
#ifdef WITHRETURN
        *Out++ = CARRIAGERETURN;
#endif
        *Out++ = LINEFEED;
        AO.FortFirst = 0;
    }
    num = Out - AO.OutputLine;

    if ( AC.LogHandle >= 0 ) {
        if ( WriteFile(AC.LogHandle,AO.OutputLine+startinline
                ,num-startinline) != (num-startinline) ) {
#ifdef DEBUGGER
            if ( BUG.logfileflag == 0 ) {
                fprintf(stderr,"Panic: Cannot write to log file! Disk full?\n");
                BUG.logfileflag = 1;
            }
            BUG.eflag = 1; BUG.printflag = 1;
#else
            Terminate(-1);
#endif
        }
    }

    if ( ( AO.PrintType & PRINTLFILE ) == 0 ) {
#ifdef WITHRETURN
        if ( num > 1 && AO.OutputLine[num-2] == CARRIAGERETURN ) {
            AO.OutputLine[num-2] = LINEFEED;
            num--;
        }
#endif
        if ( WriteFile(AM.StdOut,AO.OutputLine+startinline,
                num-startinline) != (num-startinline) ) {
#ifdef DEBUGGER
            if ( BUG.stdoutflag == 0 ) {
                fprintf(stderr,"Panic: Cannot write to standard output!\n");
                BUG.stdoutflag = 1;
            }
            BUG.eflag = 1; BUG.printflag = 1;
#else
            Terminate(-1);
#endif
```

```
        }
    }
    startinline = 0;
    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
         || ( AC.OutputMode == CMODE && AO.FactorMode == 0 ) ) AO.InFbrack++;
    Out = AO.OutputLine;
    AO.OutStop = Out + AC.LineLength;
    i = AO.OutSkip;
    while ( --i >= 0 ) *Out++ = ' ';
    if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
     && AO.OutSkip == 7 ) {
        Out[-2] = fcontchar;
        Out[-1] = ' ';
    }
    AO.OutFill = Out;
}

/*
        #] FiniLine :
        #[ IniLine :            void IniLine(extrablank)

    Initializes the output line for the type of output

*/

void IniLine(WORD extrablank)
{
    UBYTE *Out;
    Out = AO.OutputLine;
    AO.OutStop = Out + AC.LineLength;
    *Out++ = ' ';
    *Out++ = ' ';
    *Out++ = ' ';
    *Out++ = ' ';
    *Out++ = ' ';
    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
        *Out++ = fcontchar;
        AO.OutSkip = 7;
    }
    else
        AO.OutSkip = 6;
    *Out++ = ' ';
    while ( extrablank > 0 ) {
        *Out++ = ' ';
        extrablank--;
    }
    AO.OutFill = Out;
}

/*
        #] IniLine :
        #[ LongToLine :         void LongToLine(a,na)

    Puts a Long integer in the output line. If it is only a single
    word long it is put in the line as a single token.
    The sign of a is ignored.

*/

static UBYTE *LLscratch = 0;

void LongToLine(UWORD *a, WORD na)
{
    UBYTE *OutScratch;
    if ( LLscratch == 0 ) {
        LLscratch = (UBYTE *)Malloc1(4*(AM.MaxTal*sizeof(WORD)+2)*sizeof(UBYTE),"LongToLine");
    }
    OutScratch = LLscratch;
    if ( na < 0 ) na = -na;
    if ( na > 1 ) {
        PrtLong(a,na,OutScratch);
        if ( AO.NoSpacesInNumbers || AC.OutputMode == REDUCEMODE ) {
            AO.BlockSpaces = 1;
            TokenToLine(OutScratch);
            AO.BlockSpaces = 0;
        }
        else {
            TokenToLine(OutScratch);
        }
    }
    else if ( !na ) TokenToLine((UBYTE *)"0");
    else TalToLine(*a);
}

/*
        #] LongToLine :
        #[ RatToLine :          void RatToLine(a,na)
```

```

    Puts a rational number in the output line. The sign is ignored.

*/

static UBYTE *RLscratch = 0;
static UWORD *RLscratE = 0;

void RatToLine(UWORD *a, WORD na)
{
    GETIDENTITY
    WORD adenom, anumer;
    UWORD maxInt;

    if ( AC.OutputMode == CMODE ) {
        // In C, integer literals over 2^32-1 are automatically promoted to longer types up to
        // unsigned long long, however FORM has always given integers over 2^32-1 a floating suffix
        // in C mode. Retain that behaviour here for now. In the future, maxInt could be a user-
        // configurable parameter.
        maxInt = 4294967295;
    }
    else {
        // In Fortran modes, literals over 2^31-1 should be printed with a float suffix
        maxInt = 2147483647;
    }

    if ( na < 0 ) na = -na;

    if ( AC.OutNumberType == RATIONALMODE ) {

        // Work out what is to be printed:
        WORD isOne = 0, isOneOver = 0, isIntegral = 0;
        WORD isLongNum = 0, isLongDen = 0;
        UnPack(a,na,&adenom,&anumer);
        if ( na == 1 && a[0] == 1 && a[1] == 1 ) { isOne = 1; isIntegral = 1; }
        else if ( adenom == 1 && a[na] == 1 ) { isIntegral = 1; }
        else if ( anumer == 1 && a[0] == 1 ) { isOneOver = 1; }
        if ( anumer > 1 || ( anumer == 1 && a[0] > maxInt ) ) { isLongNum = 1; };
        if ( adenom > 1 || ( adenom == 1 && a[na] > maxInt ) ) { isLongDen = 1; };

        // Now sort out the float suffix for the numerator:
        UBYTE* suffNum = (UBYTE*)"";
        UBYTE* suffDen = (UBYTE*)"";
        if ( isLongNum || !isIntegral || AC.Fortran90Kind || ( AO.DoubleFlag & 4 ) == 4 ) {
            if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                if ( AC.Fortran90Kind ) { suffNum = AC.Fortran90Kind; }
                else { suffNum = (UBYTE*)"."; }
            }
            else if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == CMODE ) {
                if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffNum = (UBYTE*)".Q0"; }
                else if ( ( AO.DoubleFlag & 1 ) == 1 ) { suffNum = (UBYTE*)".D0"; }
                else { suffNum = (UBYTE*)"."; }
            }
        }
        // The same again, for the denominator:
        if ( isLongDen || !isIntegral || AC.Fortran90Kind || ( AO.DoubleFlag & 4 ) == 4 ) {
            if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
                if ( AC.Fortran90Kind ) { suffDen = AC.Fortran90Kind; }
                else { suffDen = (UBYTE*)"."; }
            }
            else if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == CMODE ) {
                if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffDen = (UBYTE*)".Q0"; }
                else if ( ( AO.DoubleFlag & 1 ) == 1 ) { suffDen = (UBYTE*)".D0"; }
                else { suffDen = (UBYTE*)"."; }
            }
        }
        // In PFORTRAN mode, rationals don't get a suffix if the integers are not long
        // Also the suffix is at least ".D0" (and never ".").
        if ( AC.OutputMode == PFORTRANMODE ) {
            if ( isLongNum ) {
                if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffNum = (UBYTE*)".Q0"; }
                else { suffNum = (UBYTE*)".D0"; }
            }
            if ( isLongDen ) {
                if ( ( AO.DoubleFlag & 2 ) == 2 ) { suffDen = (UBYTE*)".Q0"; }
                else { suffDen = (UBYTE*)".D0"; }
            }
        }

        // Finally, print the number:
        // In PFORTRAN we use
        //     one      if denom = numerator = 1
        //     integer      if denom = 1
        //     (one/integer) if numerator = 1
        //     ((one*integer)/integer) in the general case
        if ( AC.OutputMode == PFORTRANMODE ) {
            if ( isOne ) {
```

```
                AddToLine((UBYTE *)"one");
            }
            else if ( isOneOver ) {
                AddToLine((UBYTE *)"(one/");
                LongToLine(a+na,adenom);
                AddToLine(suffDen);
                AddToLine((UBYTE*)")");
            }
            else if ( isIntegral ) {
                LongToLine(a,anumer);
                AddToLine(suffNum);
            }
            else {
                // rational
                AddToLine((UBYTE *)"((one*");
                LongToLine(a,anumer);
                AddToLine(suffNum);
                AddToLine((UBYTE*)")/");
                LongToLine(a+na,adenom);
                AddToLine(suffDen);
                AddToLine((UBYTE*)")");
            }
        }
        // All other modes use the same printing code:
        else {
            // Numerator:
            LongToLine(a,anumer);
            AddToLine(suffNum);
            // Denominator, if it is not 1:
            if (!isIntegral) {
                AddToLine((UBYTE*)"/");
                LongToLine(a+na,adenom);
                AddToLine(suffDen);
            }
        }
    }

    else {
/*
        This is the float mode
*/
        UBYTE *OutScratch;
        WORD exponent = 0, i, ndig, newl;
        UWORD *c, *den, b = 10, dig[10];
        UBYTE *o, *out, cc;
/*
        First we have to adjust the numerator and denominator
*/
        if ( RLscratch == 0 ) {
            RLscratch = (UBYTE *)Malloc1(4*(AM.MaxTal+2)*sizeof(UBYTE),"RatToLine");
            RLscratE = (UWORD *)Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD),"RatToLine");
        }
        out = OutScratch = RLscratch;
        c = RLscratE; for ( i = 0; i < 2*na; i++ ) c[i] = a[i];
        UnPack(c,na,&adenom,&anumer);
        while ( BigLong(c,anumer,c+na,adenom) >= 0 ) {
            Divvy(BHEAD c,&na,&b,1);
            UnPack(c,na,&adenom,&anumer);
            exponent++;
        }
        while ( BigLong(c,anumer,c+na,adenom) < 0 ) {
            Mully(BHEAD c,&na,&b,1);
            UnPack(c,na,&adenom,&anumer);
            exponent--;
        }
/*
        Now division will give a number between 1 and 9
*/
        den = c + na; i = 1;
        DivLong(c,anumer,den,adenom,dig,&ndig,c,&newl);
        *out++ = (UBYTE)(dig[0]+'0'); *out++ = '.';
        while ( newl && i < AC.OutNumberType ) {
            Pack(c,&newl,den,adenom);
            Mully(BHEAD c,&newl,&b,1);
            na = newl;
            UnPack(c,na,&adenom,&anumer);
            den = c + na;
            DivLong(c,anumer,den,adenom,dig,&ndig,c,&newl);
            if ( ndig == 0 ) *out++ = '0';
            else *out++ = (UBYTE)(dig[0]+'0');
            i++;
        }
        *out++ = 'E';
        if ( exponent < 0 ) { exponent = -exponent; *out++ = '-'; }
        else { *out++ = '+'; }
        o = out;
        do {
```

```
        *out++ = (UBYTE)((exponent % 10)+'0');
        exponent /= 10;
    } while ( exponent );
    *out = 0; out--;
    while ( o < out ) { cc = *o; *o = *out; *out = cc; o++; out--; }
    TokenToLine(OutScratch);
  }
}

/*
    #] RatToLine :
    #[ TalToLine :          void TalToLine(x)

  Writes the unsigned number x to the output as a single token.
  Par indicates the number of leading blanks in the line.
  This parameter is needed here for the WriteLists routine.

*/

void TalToLine(UWORD x)
{
  UBYTE t[BITSINWORD/3+1];
  UBYTE *s;
  WORD i = 0, j;
  s = t;
  do { *s++ = (UBYTE)((x % 10)+'0'); i++; } while ( ( x /= 10 ) != 0 );
  *s-- = '\0';
  j = ( i - 1 ) » 1;
  while ( j >= 0 ) {
    i = t[j]; t[j] = s[-j]; s[-j] = (UBYTE)i; j--;
  }
  TokenToLine(t);
}

/*
    #] TalToLine :
    #[ TokenToLine :        void TokenToLine(s)

  Puts s in the output buffer. If it doesn't fit the buffer is
  flushed first. This routine keeps tokens as one unit.
  Par indicates the number of leading blanks in the line.
  This parameter is needed here for the WriteLists routine.

  Remark (27-oct-2007): i and j must be longer than WORD!
  It can happen that a number is so long that it has more than 2^15 or 2^31
  digits!
*/

void TokenToLine(UBYTE *s)
{
  UBYTE *t, *Out;
  LONG num, i = 0, j;
  if ( AO.OutInBuffer ) { AddToDollarBuffer(s); return; }
  t = s; Out = AO.OutFill;
  while ( *t++ ) i++;
  while ( i > 0 ) {
    if ( ( Out + i ) >= AO.OutStop && ( ( i < ((AC.LineLength-AO.OutSkip)»1) )
    || ( (AO.OutStop-Out) < (i»2) ) ) ) {
      if ( AC.OutputMode == FORTRANMODE && AC.IsFortran90 == ISFORTRAN90 ) {
        *Out++ = fcontchar;
      }
#ifdef WITHRETURN
      *Out++ = CARRIAGERETURN;
#endif
      *Out++ = LINEFEED;
      AO.FortFirst = 0;
      num = Out - AO.OutputLine;
      if ( AC.LogHandle >= 0 ) {
        if ( WriteFile(AC.LogHandle,AO.OutputLine+startinline,
          num-startinline) != (num-startinline) ) {
#ifdef DEBUGGER
          if ( BUG.logfileflag == 0 ) {
            fprintf(stderr,"Panic: Cannot write to log file! Disk full?\n");
            BUG.logfileflag = 1;
          }
          BUG.eflag = 1; BUG.printflag = 1;
#else
          Terminate(-1);
#endif
        }
      }
      if ( ( AO.PrintType & PRINTLFILE ) == 0 ) {
#ifdef WITHRETURN
        if ( num > 1 && AO.OutputLine[num-2] == CARRIAGERETURN ) {
          AO.OutputLine[num-2] = LINEFEED;
          num--;
        }
```

```
#endif
                if ( WriteFile(AM.StdOut,AO.OutputLine+startinline,
                    num-startinline) != (num-startinline) ) {
#ifdef DEBUGGER
                    if ( BUG.stdoutflag == 0 ) {
                        fprintf(stderr,"Panic: Cannot write to standard output!\n");
                        BUG.stdoutflag = 1;
                    }
                    BUG.eflag = 1; BUG.printflag = 1;
#else
                    Terminate(-1);
#endif
                }
            }
            startinline = 0;
            Out = AO.OutputLine;
            if ( AO.BlockSpaces == 0 ) {
                for ( j = 0; j < AO.OutSkip; j++ ) { *Out++ = ' '; }
                if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) ) {
                  if ( AO.OutSkip == 7 ) {
                    Out[-2] = fcontchar;
                    Out[-1] = ' ';
                  }
                }
            }
/*
            Out = AO.OutputLine + AO.OutSkip;
            if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
             && AO.OutSkip == 7 ) {
                Out[-2] = fcontchar;
                Out[-1] = ' ';
            }
            else {
                for ( j = 0; j < AO.OutSkip; j++ ) { AO.OutputLine[j] = ' '; }
            }
*/
            if ( AO.IsBracket ) { *Out++ = ' '; *Out++ = ' '; *Out++ = ' '; }
            *Out = '\0';
            if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
             || ( AC.OutputMode == CMODE && AO.FactorMode == 0 ) ) AO.InFbrack++;
        }
        if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE ) {
                   /* Very long numbers */
            if ( i > (WORD)(AO.OutStop-Out) ) j = (WORD)(AO.OutStop - Out);
            else                           j = i;
            i -= j;
            NCOPYB(Out,s,j);
        }
        else {
            if ( i > (WORD)(AO.OutStop-Out) ) j = (WORD)(AO.OutStop - Out - 1);
            else                           j = i;
            i -= j;
            NCOPYB(Out,s,j);
            if ( i > 0 ) *Out++ = '\\';
        }
    }
    *Out = '\0';
    AO.OutFill = Out;
}

/*
        #] TokenToLine :
        #[ CodeToLine :         void CodeToLine(name,number,mode)

    Writes a name and possibly its number to output as a single token.

*/

UBYTE *CodeToLine(WORD number, UBYTE *Out)
{
    Out = StrCopy((UBYTE *)"(",Out);
    Out = NumCopy(number,Out);
    Out = StrCopy((UBYTE *)")",Out);
    return(Out);
}

/*
        #] CodeToLine :
        #[ MultiplyToLine :
*/

void MultiplyToLine(void)
{
    int i;
    if ( AO.CurrentDictionary > 0 && AO.CurDictSpecials > 0
     && AO.CurDictSpecials == DICT_DOSPECIALS ) {
        DICTIONARY *dict = AO.Dictionaries[AO.CurrentDictionary-1];
```

```
/*
        Find the star:
*/
        for ( i = 0; i < dict->numelements; i++ ) {
            if ( dict->elements[i]->type != DICT_SPECIALCHARACTER ) continue;
            if ( (UBYTE)dict->elements[i]->lhs[0] == (UBYTE)('*') ) {
                TokenToLine((UBYTE *)(dict->elements[i]->rhs));
                return;
            }
        }
    }
    TokenToLine((UBYTE *)"*");
}

/*
        #] MultiplyToLine :
        #[ AddArrayIndex :
*/

UBYTE *AddArrayIndex(WORD num,UBYTE *out)
{
    if ( AC.OutputMode == CMODE ) {
        out = StrCopy((UBYTE *)"[",out);
        out = NumCopy(num,out);
        out = StrCopy((UBYTE *)"]",out);
    }
    else {
        out = StrCopy((UBYTE *)"(",out);
        out = NumCopy(num,out);
        out = StrCopy((UBYTE *)")",out);
    }
    return(out);
}

/*
        #] AddArrayIndex :
        #[ PrtTerms :            void PrtTerms()
*/

void PrtTerms(void)
{
    UWORD a[2];
    WORD na;
    a[0] = (UWORD)AO.NumInBrack;
    a[1] = (UWORD)(AO.NumInBrack » BITSINWORD);
    if ( a[1] ) na = 2;
    else na = 1;
    TokenToLine((UBYTE *)" ");
    LongToLine(a,na);
    if ( a[0] == 1 && na == 1 ) {
        TokenToLine((UBYTE *)" term");
    }
    else TokenToLine((UBYTE *)" terms");
    AO.NumInBrack = 0;
}

/*
        #] PrtTerms :
        #[ WrtPower :
*/

UBYTE *WrtPower(UBYTE *Out, WORD Power)
{
    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
      || AC.OutputMode == REDUCEMODE ) {
        *Out++ = '*'; *Out++ = '*';
    }
    else if ( AC.OutputMode == CMODE ) *Out++ = ',';
    else {
        UBYTE *Out1 = IsExponentSign();
        if ( Out1 == 0 ) *Out++ = '^';
        else {
            while ( *Out1 ) *Out++ = *Out1++;
            *Out = 0;
        }
    }
    if ( Power >= 0 ) {
        if ( Power < 2*MAXPOWER )
            Out = NumCopy(Power,Out);
        else
            Out = StrCopy(FindSymbol((WORD)((LONG)Power-2*MAXPOWER)),Out);
/*          Out = StrCopy(VARNAME(symbols,(LONG)Power-2*MAXPOWER),Out); */
        if ( AC.OutputMode == CMODE ) *Out++ = ')';
        *Out = 0;
    }
    else {
        if ( ( AC.OutputMode >= FORTRANMODE || AC.OutputMode >= PFORTRANMODE
```

```
         || AC.OutputMode >= REDUCEMODE ) && AC.OutputMode != CMODE )
            *Out++ = '(';
        *Out++ = '-';
        if ( Power > -2*MAXPOWER )
            Out = NumCopy(-Power,Out);
        else
            Out = StrCopy(FindSymbol((WORD)((LONG)Power-2*MAXPOWER)),Out);
/*          Out = StrCopy(VARNAME(symbols,(LONG)(-Power)-2*MAXPOWER),Out); */
        if ( AC.OutputMode >= FORTRANMODE || AC.OutputMode >= PFORTRANMODE
         || AC.OutputMode >= REDUCEMODE) *Out++ = ')';
        *Out = 0;
    }
    return(Out);
}

/*
        #] WrtPower :
        #[ PrintTime :
*/

void PrintTime(UBYTE *mess)
{
    LONG millitime = TimeCPU(1);
    WORD timepart = (WORD)(millitime%1000);
    millitime /= 1000;
    timepart /= 10;
    MesPrint("At %s: Time = %7l.%2i sec",mess,millitime,timepart);
}

/*
        #] PrintTime :
    #] schryf-Utilities :
    #[ schryf-Writes :
        #[ WriteLists :         void WriteLists()

    Writes the namelists. If mode > 0 also the internal codes are given.

*/

static UBYTE *symname[] = {
     (UBYTE *)"(cyclic)",(UBYTE *)"(reversecyclic)"
    ,(UBYTE *)"(symmetric)",(UBYTE *)"(antisymmetric)" };
static UBYTE *rsymname[] = {
     (UBYTE *)"(-cyclic)",(UBYTE *)"(-reversecyclic)"
    ,(UBYTE *)"(-symmetric)",(UBYTE *)"(-antisymmetric)" };

void WriteLists(void)
{
    GETIDENTITY
    WORD i, j, k, *skip;
    int first, startvalue;
    UBYTE *OutScr, *Out;
    EXPRESSIONS e;
    CBUF *C = cbuf+AC.cbufnum;
    int olddict = AO.CurrentDictionary;
    skip = &AO.OutSkip;
    *skip = 0;
    AO.OutputLine = AO.OutFill = (UBYTE *)AT.WorkPointer;
    AO.CurrentDictionary = 0;
    FiniLine();
    OutScr = (UBYTE *)AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) /2;
    if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
    else startvalue = FIRSTUSERSYMBOL;
/*
        #[ Symbols :
*/
    if ( ( j = NumSymbols ) > startvalue ) {
        TokenToLine((UBYTE *)" Symbols");
        *skip = 3;
        FiniLine();
        for ( i = startvalue; i < j; i++ ) {
            if ( i >= BUILTINSYMBOLS && i < FIRSTUSERSYMBOL ) continue;
            Out = StrCopy(VARNAME(symbols,i),OutScr);
            if ( symbols[i].minpower > -MAXPOWER || symbols[i].maxpower < MAXPOWER ) {
                Out = StrCopy((UBYTE *)"(",Out);
                if ( symbols[i].minpower > -MAXPOWER )
                    Out = NumCopy(symbols[i].minpower,Out);
                Out = StrCopy((UBYTE *)":",Out);
                if ( symbols[i].maxpower < MAXPOWER )
                    Out = NumCopy(symbols[i].maxpower,Out);
                Out = StrCopy((UBYTE *)")",Out);
            }
            if ( ( symbols[i].complex & VARTYPEIMAGINARY ) == VARTYPEIMAGINARY ) {
                Out = StrCopy((UBYTE *)"#i",Out);
            }
            else if ( ( symbols[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) {
                Out = StrCopy((UBYTE *)"#c",Out);
```

```
            }
            else if ( ( symbols[i].complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
                Out = StrCopy((UBYTE *)"#",Out);
                if ( ( symbols[i].complex & VARTYPEMINUS ) == VARTYPEMINUS ) {
                    Out = StrCopy((UBYTE *)"-",Out);
                }
                else {
                    Out = StrCopy((UBYTE *)"+",Out);
                }
                Out = NumCopy(symbols[i].maxpower,Out);
            }
            if ( AC.CodesFlag ) Out = CodeToLine(i,Out);
            if ( ( symbols[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) i++;
            StrCopy((UBYTE *)" ",Out);
            TokenToLine(OutScr);
        }
        *skip = 0;
        FiniLine();
    }
/*
        #] Symbols :
        #[ Indices :
*/
    if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
    else startvalue = BUILTININDICES;
    if ( ( j = NumIndices ) > startvalue ) {
        TokenToLine((UBYTE *)" Indices");
        *skip = 3;
        FiniLine();
        for ( i = startvalue; i < j; i++ ) {
            Out = StrCopy(FindIndex(i+AM.OffsetIndex),OutScr);
            Out = StrCopy(VARNAME(indices,i),OutScr);
            if ( indices[i].dimension >= 0 ) {
                if ( indices[i].dimension != AC.lDefDim ) {
                    Out = StrCopy((UBYTE *)"=",Out);
                    Out = NumCopy(indices[i].dimension,Out);
                }
            }
            else if ( indices[i].dimension < 0 ) {
                Out = StrCopy((UBYTE *)"=",Out);
                Out = StrCopy(VARNAME(symbols,-indices[i].dimension),Out);
                if ( indices[i].nmin4 < -NMIN4SHIFT ) {
                    Out = StrCopy((UBYTE *)":",Out);
                    Out = StrCopy(VARNAME(symbols,-indices[i].nmin4-NMIN4SHIFT),Out);
                }
            }
            if ( AC.CodesFlag ) Out = CodeToLine(i+AM.OffsetIndex,Out);
            StrCopy((UBYTE *)" ",Out);
            TokenToLine(OutScr);
        }
        *skip = 0;
        FiniLine();
    }
/*
        #] Indices :
        #[ Vectors :
*/
    if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
    else startvalue = BUILTINVECTORS;
    if ( ( j = NumVectors ) > startvalue ) {
        TokenToLine((UBYTE *)" Vectors");
        *skip = 3;
        FiniLine();
        for ( i = startvalue; i < j; i++ ) {
            Out = StrCopy(VARNAME(vectors,i),OutScr);
            if ( AC.CodesFlag ) Out = CodeToLine(i+AM.OffsetVector,Out);
            StrCopy((UBYTE *)" ",Out);
            TokenToLine(OutScr);
        }
        *skip = 0;
        FiniLine();
    }
/*
        #] Vectors :
        #[ Functions :
*/
    if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
    else startvalue = AM.NumFixedFunctions;
    for ( k = 0; k < 2; k++ ) {
        first = 1;
        j = NumFunctions;
        for ( i = startvalue; i < j; i++ ) {
            if ( i > MAXBUILTINFUNCTION-FUNCTION
             && i < FIRSTUSERFUNCTION-FUNCTION ) continue;
            if ( ( k == 0 && functions[i].commute )
            || ( k != 0 && !functions[i].commute ) ) {
                if ( first ) {
```

```
                    TokenToLine((UBYTE *)(FG.FunNam[k]));
                    *skip = 3;
                    FiniLine();
                    first = 0;
                }
                Out = StrCopy(VARNAME(functions,i),OutScr);
                if ( ( functions[i].complex & VARTYPEIMAGINARY ) == VARTYPEIMAGINARY ) {
                    Out = StrCopy((UBYTE *)"#i",Out);
                }
                else if ( ( functions[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) {
                    Out = StrCopy((UBYTE *)"#c",Out);
                }
                if ( functions[i].spec == VERTEXFUNCTION ) {
                    Out = StrCopy((UBYTE *)"(Particle)",Out);
                }
                else if ( functions[i].spec >= TENSORFUNCTION ) {
                    Out = StrCopy((UBYTE *)"(Tensor)",Out);
                }
                if ( functions[i].symmetric > 0 ) {
                    if ( ( functions[i].symmetric & REVERSEORDER ) != 0 ) {
                        Out = StrCopy((UBYTE *)(rsymname[(functions[i].symmetric &
~REVERSEORDER)-1]),Out);
                    }
                    else {
                        Out = StrCopy((UBYTE *)(symname[functions[i].symmetric-1]),Out);
                    }
                }
                if ( AC.CodesFlag ) Out = CodeToLine(i+FUNCTION,Out);
                if ( ( functions[i].complex & VARTYPECOMPLEX ) == VARTYPECOMPLEX ) i++;
                StrCopy((UBYTE *)" ",Out);
                TokenToLine(OutScr);
            }
        }
        *skip = 0;
        if ( first == 0 ) FiniLine();
    }
/*
        #] Functions :
        #[ Sets :
*/
    if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
    else startvalue = AM.NumFixedSets;
    if ( ( j = AC.SetList.num ) > startvalue ) {
        WORD element, LastElement, type, number;
        TokenToLine((UBYTE *)" Sets");
        for ( i = startvalue; i < j; i++ ) {
            *skip = 3;
            FiniLine();
            if ( Sets[i].name < 0 ) {
                Out = StrCopy((UBYTE *)"{}",OutScr);
            }
            else {
                Out = StrCopy(VARNAME(Sets,i),OutScr);
            }
            if ( AC.CodesFlag ) Out = CodeToLine(i,Out);
            StrCopy((UBYTE *)":",Out);
            TokenToLine(OutScr);
            if ( i < AM.NumFixedSets ) {
                TokenToLine((UBYTE *)" ");
                TokenToLine((UBYTE *)fixedsets[i].description);
            }
            else if ( Sets[i].type == CRANGE ) {
                int iflag = 0;
                if ( Sets[i].first == 3*MAXPOWER ) {
                }
                else if ( Sets[i].first >= MAXPOWER ) {
                    TokenToLine((UBYTE *)"<=");
                    NumCopy(Sets[i].first-2*MAXPOWER,OutScr);
                    TokenToLine(OutScr);
                    iflag = 1;
                }
                else {
                    TokenToLine((UBYTE *)"<");
                    NumCopy(Sets[i].first,OutScr);
                    TokenToLine(OutScr);
                    iflag = 1;
                }
                if ( Sets[i].last == -3*MAXPOWER ) {
                }
                else if ( Sets[i].last <= -MAXPOWER ) {
                    if ( iflag ) TokenToLine((UBYTE *)",");
                    TokenToLine((UBYTE *)">=");
                    NumCopy(Sets[i].last+2*MAXPOWER,OutScr);
                    TokenToLine(OutScr);
                }
                else {
                    if ( iflag ) TokenToLine((UBYTE *)",");
```

```
                    TokenToLine((UBYTE *)">");
                    NumCopy(Sets[i].last,OutScr);
                    TokenToLine(OutScr);
                }
            }
            else {
                element = Sets[i].first;
                LastElement = Sets[i].last;
                type = Sets[i].type;
                while ( element < LastElement ) {
                    TokenToLine((UBYTE *)" ");
                    number = SetElements[element++];
                    switch ( type ) {
                        case CSYMBOL:
                            if ( number < 0 ) {
                                StrCopy(VARNAME(symbols,-number),OutScr);
                                StrCopy((UBYTE *)"?",Out);
                                TokenToLine(OutScr);
                            }
                            else if ( number < MAXPOWER )
                                TokenToLine(VARNAME(symbols,number));
                            else {
                                NumCopy(number-2*MAXPOWER,OutScr);
                                TokenToLine(OutScr);
                            }
                            break;
                        case CINDEX:
                            if ( number >= AM.IndDum ) {
                                Out = StrCopy((UBYTE *)"N",OutScr);
                                Out = NumCopy(number-(AM.IndDum),Out);
                                StrCopy((UBYTE *)"_?",Out);
                                TokenToLine(OutScr);
                            }
                            else if ( number >= AM.OffsetIndex + (WORD)WILDMASK ) {
                                Out = StrCopy(VARNAME(indices,number
                                -AM.OffsetIndex-WILDMASK),OutScr);
                                StrCopy((UBYTE *)"?",Out);
                                TokenToLine(OutScr);
                            }
                            else if ( number >= AM.OffsetIndex ) {
                                TokenToLine(VARNAME(indices,number-AM.OffsetIndex));
                            }
                            else {
                                NumCopy(number,OutScr);
                                TokenToLine(OutScr);
                            }
                            break;
                        case CVECTOR:
                            Out = OutScr;
                            if ( number < AM.OffsetVector ) {
                                number += WILDMASK;
                                Out = StrCopy((UBYTE *)"-",Out);
                            }
                            if ( number >= AM.OffsetVector + WILDOFFSET ) {
                                Out = StrCopy(VARNAME(vectors,number
                                -AM.OffsetVector-WILDOFFSET),Out);
                                StrCopy((UBYTE *)"?",Out);
                            }
                            else {
                                Out = StrCopy(VARNAME(vectors,number-AM.OffsetVector),Out);
                            }
                            TokenToLine(OutScr);
                            break;
                        case CFUNCTION:
                            if ( number >= FUNCTION + (WORD)WILDMASK ) {
                                Out = StrCopy(VARNAME(functions,number
                                -FUNCTION-WILDMASK),OutScr);
                                StrCopy((UBYTE *)"?",Out);
                                TokenToLine(OutScr);
                            }
                            TokenToLine(VARNAME(functions,number-FUNCTION));
                            break;
                        default:
                            NumCopy(number,OutScr);
                            TokenToLine(OutScr);
                            break;
                    }
                }
            }
        }
        *skip = 0;
        FiniLine();
    }
/*
        #] Sets :
        #[ Expressions :
*/
```

```
    if ( AS.ExecMode ) {
        e = Expressions;
        j = NumExpressions;
        first = 1;
        for ( i = 0; i < j; i++, e++ ) {
            if ( e->status >= 0 ) {
                if ( first ) {
                    TokenToLine((UBYTE *)" Expressions");
                    *skip = 3;
                    FiniLine();
                    first = 0;
                }
                Out = StrCopy(AC.exprnames->namebuffer+e->name,OutScr);
                Out = StrCopy((UBYTE *)(FG.ExprStat[e->status]),Out);
                if ( AC.CodesFlag ) Out = CodeToLine(i,Out);
                StrCopy((UBYTE *)" ",Out);
                TokenToLine(OutScr);
            }
        }
        if ( !first ) {
            *skip = 0;
            FiniLine();
        }
    }
    e = Expressions;
    j = NumExpressions;
    first = 1;
    for ( i = 0; i < j; i++ ) {
        if ( e->printflag && ( e->status == LOCALEXPRESSION ||
        e->status == GLOBALEXPRESSION || e->status == UNHIDELEXPRESSION
        || e->status == UNHIDEGEXPRESSION ) ) {
            if ( first ) {
                TokenToLine((UBYTE *)" Expressions to be printed");
                *skip = 3;
                FiniLine();
                first = 0;
            }
            Out = StrCopy(AC.exprnames->namebuffer+e->name,OutScr);
            StrCopy((UBYTE *)" ",Out);
            TokenToLine(OutScr);
        }
        e++;
    }
    if ( !first ) {
        *skip = 0;
        FiniLine();
    }
/*
        #] Expressions :
        #[ Dollars :
*/

    if ( AC.CodesFlag || AC.NamesFlag > 1 ) startvalue = 0;
    else startvalue = BUILTINDOLLARS;
    if ( ( j = NumDollars ) > startvalue ) {
        TokenToLine((UBYTE *)" Dollar variables");
        *skip = 3;
        FiniLine();
        for ( i = startvalue; i < j; i++ ) {
            Out = StrCopy((UBYTE *)"$", OutScr);
            Out = StrCopy(DOLLARNAME(Dollars, i), Out);
            if ( AC.CodesFlag ) Out = CodeToLine(i, Out);
            StrCopy((UBYTE *)" ", Out);
            TokenToLine(OutScr);
        }
        *skip = 0;
        FiniLine();
    }

    if ( ( j = NumPotModdollars ) > 0 ) {
        TokenToLine((UBYTE *)" Dollar variables to be modified");
        *skip = 3;
        FiniLine();
        for ( i = 0; i < j; i++ ) {
            Out = StrCopy((UBYTE *)"$", OutScr);
            Out = StrCopy(DOLLARNAME(Dollars, PotModdollars[i]), Out);
            for ( k = 0; k < NumModOptdollars; k++ )
                if ( ModOptdollars[k].number == PotModdollars[i] ) break;
            if ( k < NumModOptdollars ) {
                switch ( ModOptdollars[k].type ) {
                    case MODSUM:
                        Out = StrCopy((UBYTE *)"(sum)", Out);
                        break;
                    case MODMAX:
                        Out = StrCopy((UBYTE *)"(maximum)", Out);
                        break;
                    case MODMIN:
```

```
                        Out = StrCopy((UBYTE *)"(minimum)", Out);
                        break;
                    case MODLOCAL:
                        Out = StrCopy((UBYTE *)"(local)", Out);
                        break;
                    default:
                        Out = StrCopy((UBYTE *)"(?)", Out);
                        break;
                }
            }
            StrCopy((UBYTE *)" ", Out);
            TokenToLine(OutScr);
        }
        *skip = 0;
        FiniLine();
    }
/*
        #] Dollars :
*/

    if ( AC.ncmod != 0 ) {
        TokenToLine((UBYTE *)"All arithmetic is modulus ");
        LongToLine((UWORD *)AC.cmod,ABS(AC.ncmod));
        if ( AC.ncmod > 0 ) TokenToLine((UBYTE *)" with powerreduction");
        else                TokenToLine((UBYTE *)" without powerreduction");
        if ( ( AC.modmode & POSNEG ) != 0 ) TokenToLine((UBYTE *)" centered around 0");
        else                                TokenToLine((UBYTE *)" positive numbers only");
        FiniLine();
    }
    if ( AC.lDefDim != 4 ) {
        TokenToLine((UBYTE *)"The default dimension is ");
        if ( AC.lDefDim >= 0 ) {
            NumCopy(AC.lDefDim,OutScr);
            TokenToLine(OutScr);
        }
        else {
            TokenToLine(VARNAME(symbols,-AC.lDefDim));
            if ( AC.lDefDim4 != -NMIN4SHIFT ) {
                TokenToLine((UBYTE *)":");
                if ( AC.lDefDim4 >= -NMIN4SHIFT ) {
                    NumCopy(AC.lDefDim4,OutScr);
                    TokenToLine(OutScr);
                }
                else {
                    TokenToLine(VARNAME(symbols,-AC.lDefDim4-NMIN4SHIFT));
                }
            }
        }
        FiniLine();
    }
    if ( AC.lUnitTrace != 4 ) {
        TokenToLine((UBYTE *)"The trace of the unit matrix is ");
        if ( AC.lUnitTrace >= 0 ) {
            NumCopy(AC.lUnitTrace,OutScr);
            TokenToLine(OutScr);
        }
        else {
            TokenToLine(VARNAME(symbols,-AC.lUnitTrace));
        }
        FiniLine();
    }
    if ( AO.NumDictionaries > 0 ) {
        for ( i = 0; i < AO.NumDictionaries; i++ ) {
            WriteDictionary(AO.Dictionaries[i]);
        }
        if ( olddict > 0 )
            MesPrint("\nCurrently dictionary %s is active\n",
                AO.Dictionaries[olddict-1]->name);
        else
            MesPrint("\nCurrently there is no active dictionary\n");
    }
    if ( AC.CodesFlag ) {
        if ( C->numlhs > 0 ) {
            TokenToLine((UBYTE *)" Left Hand Sides:");
            AO.OutSkip = 3;
            for ( i = 1; i <= C->numlhs; i++ ) {
                FiniLine();
                skip = C->lhs[i];
                j = skip[1];
                while ( --j >= 0 ) { TalToLine((UWORD)(*skip++)); TokenToLine((UBYTE *)"  "); }
            }
            AO.OutSkip = 0;
            FiniLine();
        }
        if ( C->numrhs > 0 ) {
            TokenToLine((UBYTE *)" Right Hand Sides:");
            AO.OutSkip = 3;
```

```
            for ( i = 1; i <= C->numrhs; i++ ) {
                FiniLine();
                skip = C->rhs[i];
                while ( ( j = skip[0] ) != 0 ) {
                    while ( --j >= 0 ) { TalToLine((UWORD)(*skip++)); TokenToLine((UBYTE *)"  "); }
                }
                FiniLine();
            }
            AO.OutSkip = 0;
            FiniLine();
        }
    }
    AO.CurrentDictionary = olddict;
}

/*
        #] WriteLists :
        #[ WriteDictionary :

    This routine is part of WriteLists and should be called from there.
*/

void WriteDictionary(DICTIONARY *dict)
{
    GETIDENTITY
    int i, first;
    WORD *skip, na, *a, spec, *t, *tstop, j;
    UBYTE str[2], *OutScr, *Out;
    WORD oldoutputmode = AC.OutputMode, oldoutputspaces = AC.OutputSpaces;
    WORD oldoutskip = AO.OutSkip;
    AC.OutputMode = NORMALFORMAT;
    AC.OutputSpaces = NOSPACEFORMAT;
    MesPrint("===Contents of dictionary %s===",dict->name);
    skip = &AO.OutSkip;
    *skip = 3;
    AO.OutputLine = AO.OutFill = (UBYTE *)AT.WorkPointer;
    for ( j = 0; j < *skip; j++ ) *(AO.OutFill)++ = ' ';

    OutScr = (UBYTE *)AT.WorkPointer + ( TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer) ) /2;
    for ( i = 0; i < dict->numelements; i++ ) {
        switch ( dict->elements[i]->type ) {
            case DICT_INTEGERNUMBER:
                LongToLine((UWORD *)(dict->elements[i]->lhs),dict->elements[i]->size);
                Out = OutScr; *Out = 0;
                break;
            case DICT_RATIONALNUMBER:
                a = dict->elements[i]->lhs;
                na = a[a[0]-1]; na = (ABS(na)-1)/2;
                RatToLine((UWORD *)(a+1),na);
                Out = OutScr; *Out = 0;
                break;
            case DICT_SYMBOL:
                na = dict->elements[i]->lhs[0];
                Out = StrCopy(VARNAME(symbols,na),OutScr);
                break;
            case DICT_VECTOR:
                na = dict->elements[i]->lhs[0]-AM.OffsetVector;
                Out = StrCopy(VARNAME(vectors,na),OutScr);
                break;
            case DICT_INDEX:
                na = dict->elements[i]->lhs[0]-AM.OffsetIndex;
                Out = StrCopy(VARNAME(indices,na),OutScr);
                break;
            case DICT_FUNCTION:
                na = dict->elements[i]->lhs[0]-FUNCTION;
                Out = StrCopy(VARNAME(functions,na),OutScr);
                break;
            case DICT_FUNCTION_WITH_ARGUMENTS:
                t = dict->elements[i]->lhs;
                na = *t-FUNCTION;
                Out = StrCopy(VARNAME(functions,na),OutScr);
                spec = functions[*t - FUNCTION].spec;
                tstop = t + t[1];
                first = 1;
                if ( t[1] <= FUNHEAD ) {}
                else if ( spec >= TENSORFUNCTION ) {
                    t += FUNHEAD; *Out++ = (UBYTE)'(';
                    while ( t < tstop ) {
                        if ( first == 0 ) *Out++ = (UBYTE)(',');
                        else first = 0;
                        j = *t++;
                        if ( j >= 0 ) {
                            if ( j < AM.OffsetIndex ) { Out = NumCopy(j,Out); }
                            else if ( j < AM.IndDum ) {
                                Out = StrCopy(VARNAME(indices,j-AM.OffsetIndex),Out);
                            }
                            else {
```

```
                                MesPrint("Currently wildcards are not allowed in dictionary elements");
                                Terminate(-1);
                            }
                        }
                        else {
                            Out = StrCopy(VARNAME(vectors,j-AM.OffsetVector),Out);
                        }
                    }
                    *Out++ = (UBYTE)')'; *Out = 0;
                }
                else {
                    t += FUNHEAD; *Out++ = (UBYTE)'('; *Out = 0;
                    TokenToLine(OutScr);
                    while ( t < tstop ) {
                        if ( !first ) TokenToLine((UBYTE *)",");
                        WriteArgument(t);
                        NEXTARG(t)
                        first = 0;
                    }
                    Out = OutScr;
                    *Out++ = (UBYTE)')'; *Out = 0;
                }
                break;
            case DICT_SPECIALCHARACTER:
                str[0] = (UBYTE)(dict->elements[i]->lhs[0]);
                str[1] = 0;
                Out = StrCopy(str,OutScr);
                break;
            default:
                Out = OutScr; *Out = 0;
                break;
        }
        Out = StrCopy((UBYTE *)": \"",Out);
        Out = StrCopy((UBYTE *)(dict->elements[i]->rhs),Out);
        Out = StrCopy((UBYTE *)"\"",Out);
        TokenToLine(OutScr);
        FiniLine();
    }
    MesPrint("========End of dictionary %s===",dict->name);
    AC.OutputMode = oldoutputmode;
    AC.OutputSpaces = oldoutputspaces;
    AO.OutSkip = oldoutskip;
}

/*
        #] WriteDictionary :
        #[ WriteArgument :      void WriteArgument(WORD *t)

        Write a single argument field. The general field goes to
        WriteExpression and the fast field is dealt with here.
*/

void WriteArgument(WORD *t)
{
    UBYTE buffer[180];
    UBYTE *Out;
    WORD i;
    int oldoutsidefun, oldlowestlevel = lowestlevel;
    lowestlevel = 0;
    if ( *t > 0 ) {
        oldoutsidefun = AC.outsidefun; AC.outsidefun = 0;
        WriteExpression(t+ARGHEAD,(LONG)(*t-ARGHEAD));
        AC.outsidefun = oldoutsidefun;
        goto CleanUp;
    }
    Out = buffer;
    if ( *t == -SNUMBER) {
        NumCopy(t[1],Out);
    }
    else if ( *t == -SYMBOL ) {
        if ( t[1] >= MAXVARIABLES-cbuf[AM.sbufnum].numrhs ) {
            Out = StrCopy(FindExtraSymbol(MAXVARIABLES-t[1]),Out);
/*
            Out = StrCopy((UBYTE *)AC.extrasym,Out);
            if ( AC.extrasymbols == 0 ) {
                Out = NumCopy((MAXVARIABLES-t[1]),Out);
                Out = StrCopy((UBYTE *)"_",Out);
            }
            else if ( AC.extrasymbols == 1 ) {
                Out = AddArrayIndex((MAXVARIABLES-t[1]),Out);
            }
*/
/*
            else if ( AC.extrasymbols == 2 ) {
                Out = NumCopy((MAXVARIABLES-t[1]),Out);
            }
```

```
*/
        }
        else {
            StrCopy(FindSymbol(t[1]),Out);
/*          StrCopy(VARNAME(symbols,t[1]),Out); */
        }
    }
    else if ( *t == -VECTOR ) {
        if ( t[1] == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
        else
            StrCopy(FindVector(t[1]),Out);
/*          StrCopy(VARNAME(vectors,t[1] - AM.OffsetVector),Out); */
    }
    else if ( *t == -MINVECTOR ) {
        *Out++ = '-';
        StrCopy(FindVector(t[1]),Out);
/*      StrCopy(VARNAME(vectors,t[1] - AM.OffsetVector),Out); */
    }
    else if ( *t == -INDEX ) {
        if ( t[1] >= 0 ) {
            if ( t[1] < AM.OffsetIndex ) { NumCopy(t[1],Out); }
            else {
                i = t[1];
                if ( i >= AM.IndDum ) {
                    i -= AM.IndDum;
                    *Out++ = 'N';
                    Out = NumCopy(i,Out);
                    *Out++ = '_';
                    *Out++ = '?';
                    *Out = 0;
                }
                else {
                    i -= AM.OffsetIndex;
                    Out = StrCopy(FindIndex(i%WILDOFFSET+AM.OffsetIndex),Out);
/*                  Out = StrCopy(VARNAME(indices,i%WILDOFFSET),Out); */
                    if ( i >= WILDOFFSET ) { *Out++ = '?'; *Out = 0; }
                }
            }
        }
        else if ( t[1] == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
        else
            StrCopy(FindVector(t[1]),Out);
/*          StrCopy(VARNAME(vectors,t[1] - AM.OffsetVector),Out); */
    }
    else if ( *t == -DOLLAREXPRESSION ) {
        DOLLARS d = Dollars + t[1];
        *Out++ = '$';
        StrCopy(AC.dollarnames->namebuffer+d->name,Out);
    }
    else if ( *t == -EXPRESSION ) {
        StrCopy(EXPRNAME(t[1]),Out);
    }
    else if ( *t == -SETSET ) {
        StrCopy(VARNAME(Sets,t[1]),Out);
    }
    else if ( *t <= -FUNCTION ) {
        StrCopy(FindFunction(-*t),Out);
/*      StrCopy(VARNAME(functions,-*t-FUNCTION),Out); */
    }
    else {
        MesPrint("Illegal function argument while writing");
        goto CleanUp;
    }
    TokenToLine(buffer);
CleanUp:
    lowestlevel = oldlowestlevel;
    return;
}

/*
        #] WriteArgument :
        #[ WriteSubTerm :       WORD WriteSubTerm(sterm,first)

    Writes a single subterm field to the output line.
    There is a recursion for functions.


#define NUMSPECS 8
UBYTE *specfunnames[NUMSPECS] = {
      (UBYTE *)"fac" , (UBYTE *)"nargs", (UBYTE *)"binom"
    , (UBYTE *)"sign", (UBYTE *)"mod", (UBYTE *)"min", (UBYTE *)"max"
    , (UBYTE *)"invfac" };
*/

WORD WriteSubTerm(WORD *sterm, WORD first)
{
    UBYTE buffer[80];
```

```
UBYTE *Out, closepar[2] = { (UBYTE)')', 0};
WORD *stopper, *t, *tt, i, j, po = 0;
int oldoutsidefun;
stopper = sterm + sterm[1];
t = sterm + 2;
switch ( *sterm ) {
    case SYMBOL :
        while ( t < stopper ) {
            if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
                FiniLine();
                if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                    else IniLine(3);
                if ( first ) TokenToLine((UBYTE *)" ");
            }
            if ( !first ) MultiplyToLine();
            if ( AC.OutputMode == CMODE && t[1] != 1 ) {
                if ( AC.Cnumpows >= t[1] && t[1] > 0 ) {
                    po = t[1];
                    Out = StrCopy((UBYTE *)"POW",buffer);
                    Out = NumCopy(po,Out);
                    Out = StrCopy((UBYTE *)"(",Out);
                    TokenToLine(buffer);
                }
                else {
                    TokenToLine((UBYTE *)"pow(");
                }
            }
            if ( *t < NumSymbols ) {
                Out = StrCopy(FindSymbol(*t),buffer); t++;
/*              Out = StrCopy(VARNAME(symbols,*t),buffer); t++; */
            }
            else {
/*
                see also routine PrintSubtermList.
*/
                Out = StrCopy(FindExtraSymbol(MAXVARIABLES-*t),buffer);
/*
                Out = StrCopy((UBYTE *)AC.extrasym,buffer);
                if ( AC.extrasymbols == 0 ) {
                    Out = NumCopy((MAXVARIABLES-*t),Out);
                    Out = StrCopy((UBYTE *)"_",Out);
                }
                else if ( AC.extrasymbols == 1 ) {
                    Out = AddArrayIndex((MAXVARIABLES-*t),Out);
                }
*/
/*
                else if ( AC.extrasymbols == 2 ) {
                    Out = NumCopy((MAXVARIABLES-*t),Out);
                }
*/
                t++;
            }
            if ( AC.OutputMode == CMODE && po > 1
              && AC.Cnumpows >= po ) {
                Out = StrCopy((UBYTE *)")",Out);
                po = 0;
            }
            else if ( *t != 1 ) WrtPower(Out,*t);
            TokenToLine(buffer);
            t++;
            first = 0;
        }
        break;
    case VECTOR :
        while ( t < stopper ) {
            if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
                FiniLine();
                if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                    else IniLine(3);
                if ( first ) TokenToLine((UBYTE *)" ");
            }
            if ( !first ) MultiplyToLine();

            Out = StrCopy(FindVector(*t),buffer);
/*          Out = StrCopy(VARNAME(vectors,*t - AM.OffsetVector),buffer); */
            t++;
            if ( AC.OutputMode == MATHEMATICAMODE ) *Out++ = '[';
            else *Out++ = '(';
            if ( *t >= AM.OffsetIndex ) {
                i = *t++;
                if ( i >= AM.IndDum ) {
                    i -= AM.IndDum;
                    *Out++ = 'N';
                    Out = NumCopy(i,Out);
                    *Out++ = '_';
                    *Out++ = '?';
```

```
                        *Out = 0;
                    }
                    else
                        Out = StrCopy(FindIndex(i),Out);
/*                      Out = StrCopy(VARNAME(indices,i - AM.OffsetIndex),Out); */
                }
                else if ( *t == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
                else {
                    Out = NumCopy(*t++,Out);
                }
                if ( AC.OutputMode == MATHEMATICAMODE ) *Out++ = ']';
                else *Out++ = ')';
                *Out = 0;
                TokenToLine(buffer);
                first = 0;
            }
            break;
        case INDEX :
            while ( t < stopper ) {
                if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
                    FiniLine();
                    if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                        else IniLine(3);
                    if ( first ) TokenToLine((UBYTE *)" ");
                }
                if ( !first ) MultiplyToLine();
            if ( *t >= 0 ) {
                if ( *t < AM.OffsetIndex ) {
                    TalToLine((UWORD)(*t++));
                }
                else {
                    i = *t++;
                    if ( i >= AM.IndDum ) {
                        i -= AM.IndDum;
                        Out = buffer;
                        *Out++ = 'N';
                        Out = NumCopy(i,Out);
                        *Out++ = '_';
                        *Out++ = '?';
                        *Out = 0;
                    }
                    else {
                        i -= AM.OffsetIndex;
                        Out = StrCopy(FindIndex(i%WILDOFFSET+AM.OffsetIndex),buffer);
/*                      Out = StrCopy(VARNAME(indices,i%WILDOFFSET),buffer); */
                        if ( i >= WILDOFFSET ) { *Out++ = '?'; *Out = 0; }
                    }
                    TokenToLine(buffer);
                }
            }
            else {
                TokenToLine(FindVector(*t)); t++;
/*              TokenToLine(VARNAME(vectors,*t - AM.OffsetVector)); t++; */
            }
            first = 0;
            }
            break;
        case DOLLAREXPRESSION:
            {
                DOLLARS d = Dollars + sterm[2];
                Out = StrCopy((UBYTE *)"$",buffer);
                Out = StrCopy(AC.dollarnames->namebuffer+d->name,Out);
                if ( sterm[3] != 1 ) WrtPower(Out,sterm[3]);
                TokenToLine(buffer);
            }
            first = 0;
            break;
        case DELTA :
            while ( t < stopper ) {
                if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
                    FiniLine();
                    if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                        else IniLine(3);
                    if ( first ) TokenToLine((UBYTE *)" ");
                }
                if ( !first ) MultiplyToLine();
                Out = StrCopy((UBYTE *)"d_(",buffer);
                if ( *t >= AM.OffsetIndex ) {
                    if ( *t < AM.IndDum ) {
                        Out = StrCopy(FindIndex(*t),Out);
/*                      Out = StrCopy(VARNAME(indices,*t - AM.OffsetIndex),Out); */
                        t++;
                    }
                    else {
                        *Out++ = 'N';
                        Out = NumCopy( *t++ - AM.IndDum, Out);
                        *Out++ = '_';
```

```
                        *Out++ = '?';
                        *Out = 0;
                    }
                }
                else if ( *t == FUNNYVEC ) { *Out++ = '?'; *Out = 0; }
                else {
                    Out = NumCopy(*t++,Out);
                }
                *Out++ = ',';
                if ( *t >= AM.OffsetIndex ) {
                    if ( *t < AM.IndDum ) {
                        Out = StrCopy(FindIndex(*t),Out);
/*                      Out = StrCopy(VARNAME(indices,*t - AM.OffsetIndex),Out); */
                        t++;
                    }
                    else {
                        *Out++ = 'N';
                        Out = NumCopy(*t++ - AM.IndDum,Out);
                        *Out++ = '_';
                        *Out++ = '?';
                    }
                }
                else {
                    Out = NumCopy(*t++,Out);
                }
                *Out++ = ')';
                *Out = 0;
                TokenToLine(buffer);
                first = 0;
            }
            break;
        case DOTPRODUCT :
            while ( t < stopper ) {
                if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
                    FiniLine();
                    if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                        else IniLine(3);
                    if ( first ) TokenToLine((UBYTE *)" ");
                }
                if ( !first ) MultiplyToLine();
                if ( AC.OutputMode == CMODE && t[2] != 1 )
                    TokenToLine((UBYTE *)"pow(");
                if ( AC.OutputMode == MATHEMATICAMODE )
                    TokenToLine((UBYTE *)"(");
                Out = StrCopy(FindVector(*t),buffer);
/*              Out = StrCopy(VARNAME(vectors,*t - AM.OffsetVector),buffer); */
                t++;
                if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
                 || AC.OutputMode == CMODE )
                    *Out++ = AO.FortDotChar;
                else *Out++ = '.';
                Out = StrCopy(FindVector(*t),Out);
/*              Out = StrCopy(VARNAME(vectors,*t - AM.OffsetVector),Out); */
                if ( AC.OutputMode == MATHEMATICAMODE ) {
                    *Out++ = ')';
                    *Out = 0;
                }
                t++;
                if ( *t != 1 ) WrtPower(Out,*t);
                t++;
                TokenToLine(buffer);
                first = 0;
            }
            break;
        case EXPONENT :
#if FUNHEAD != 2
            t += FUNHEAD - 2;
#endif
            if ( !first ) MultiplyToLine();
            if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *)"pow(");
            else TokenToLine((UBYTE *)"(");
            WriteArgument(t);
            if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
             || AC.OutputMode == REDUCEMODE )
                TokenToLine((UBYTE *)")**(");
            else if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *)",");
            else {
                UBYTE *Out1 = IsExponentSign();
                if ( Out1 ) {
                    TokenToLine((UBYTE *)")");
                    TokenToLine(Out1);
                    TokenToLine((UBYTE *)"(");
                }
                else TokenToLine((UBYTE *)")^(");
            }
            NEXTARG(t)
            WriteArgument(t);
```

```
            TokenToLine((UBYTE *)")");
            break;
        case DENOMINATOR :
#if FUNHEAD != 2
            t += FUNHEAD - 2;
#endif
            if ( first ) TokenToLine((UBYTE *)"1/(");
            else TokenToLine((UBYTE *)"/(");
            WriteArgument(t);
            TokenToLine((UBYTE *)")");
            break;
        case SUBEXPRESSION:
            if ( !first ) MultiplyToLine();
            TokenToLine((UBYTE *)"(");
            t = cbuf[sterm[4]].rhs[sterm[2]];
            tt = t;
            while ( *tt ) tt += *tt;
            oldoutsidefun = AC.outsidefun; AC.outsidefun = 0;
            if ( *t ) {
                WriteExpression(t,(LONG)(tt-t));
            }
            else {
                TokenToLine((UBYTE *)"0");
            }
            AC.outsidefun = oldoutsidefun;
            TokenToLine((UBYTE *)")");
            if ( sterm[3] != 1 ) {
                UBYTE *Out1 = IsExponentSign();
                if ( Out1 ) TokenToLine(Out1);
                else TokenToLine((UBYTE *)"^");
                Out = buffer;
                NumCopy(sterm[3],Out);
                TokenToLine(buffer);
            }
            break;
        default :
            if ( lowestlevel && ( ( AO.PrintType & PRINTALL ) != 0 ) ) {
                FiniLine();
                if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                    else IniLine(3);
                if ( first ) TokenToLine((UBYTE *)" ");
            }
            if ( *sterm < FUNCTION ) {
            return(MesPrint("Illegal subterm while writing"));
            }
            if ( !first ) MultiplyToLine();
            first = 1;
            { UBYTE *tmp;
                if ( ( tmp = FindFunWithArgs(sterm) ) != 0 ) {
                    TokenToLine(tmp);
                    break;
                }
            }
            t += FUNHEAD-2;

            if ( *sterm == GAMMA && t[-FUNHEAD+1] == FUNHEAD+1 ) {
                TokenToLine((UBYTE *)"gi_(");
            }
            else {
                if ( *sterm != DUMFUN ) {
                    Out = StrCopy(FindFunction(*sterm),buffer);
/*                  Out = StrCopy(VARNAME(functions,*sterm - FUNCTION),buffer); */
                }
                else { Out = buffer; *Out = 0; }
                if ( t >= stopper ) {
                    TokenToLine(buffer);
                    break;
                }
                if ( AC.OutputMode == MATHEMATICAMODE ) { *Out++ = '['; closepar[0] = (UBYTE)']'; }
                else { *Out++ = '('; }
                *Out = 0;
                TokenToLine(buffer);
            }
            i = functions[*sterm - FUNCTION].spec;
            if ( i >= TENSORFUNCTION ) {
                int curdict = AO.CurrentDictionary;
                if ( AO.CurrentDictionary && AO.CurDictNotInFunctions > 0 )
                    AO.CurrentDictionary = 0;
                t = sterm + FUNHEAD;
                while ( t < stopper ) {
                    if ( !first ) TokenToLine((UBYTE *)",");
                    else first = 0;
                    j = *t++;
                    if ( j >= 0 ) {
                        if ( j < AM.OffsetIndex ) TalToLine((UWORD)(j));
                        else if ( j < AM.IndDum ) {
                            i = j - AM.OffsetIndex;
```

```
                            Out = StrCopy(FindIndex(i%WILDOFFSET+AM.OffsetIndex),buffer);
/*                          Out = StrCopy(VARNAME(indices,i%WILDOFFSET),buffer); */
                            if ( i >= WILDOFFSET ) { *Out++ = '?'; *Out = 0; }
                            TokenToLine(buffer);
                        }
                        else {
                            Out = buffer;
                            *Out++ = 'N';
                            Out = NumCopy(j - AM.IndDum,Out);
                            *Out++ = '_';
                            *Out++ = '?';
                            *Out = 0;
                            TokenToLine(buffer);
                        }
                    }
                    else if ( j == FUNNYVEC ) { TokenToLine((UBYTE *)"?"); }
                    else if ( j > -WILDOFFSET ) {
                        Out = buffer;
                        Out = NumCopy((UWORD)(-j + 4),Out);
                        *Out++ = '_';
                        *Out = 0;
                        TokenToLine(buffer);
                    }
                    else {
                        TokenToLine(FindVector(j));
/*                      TokenToLine(VARNAME(vectors,j - AM.OffsetVector)); */
                    }
                }
                AO.CurrentDictionary = curdict;
            }
            else {
                int curdict = AO.CurrentDictionary;
                if ( AO.CurrentDictionary && AO.CurDictNotInFunctions > 0 )
                    AO.CurrentDictionary = 0;
                while ( t < stopper ) {
                    if ( !first ) TokenToLine((UBYTE *)",");
                    WriteArgument(t);
                    NEXTARG(t)
                    first = 0;
                }
                AO.CurrentDictionary = curdict;
            }
            TokenToLine(closepar);
            closepar[0] = (UBYTE)')';
            break;
    }
    return(0);
}

/*
        #] WriteSubTerm :
        #[ WriteInnerTerm :     WORD WriteInnerTerm(term,first)

    Writes the contents of term to the output.
    Only the part that is inside parentheses is written.

*/

WORD WriteInnerTerm(WORD *term, WORD first)
{
    WORD *t, *s, *s1, *s2, n, i, pow;
#ifdef WITHFLOAT
    int FloatChars = 0;
    GETIDENTITY
#endif
    t = term;
    s = t+1;
    GETCOEF(t,n);
    while ( s < t ) {
        if ( *s == HAAKJE ) break;
        s += s[1];
    }
    if ( s < t ) { s += s[1]; }
    else { s = term+1; }

    if ( n < 0 || !first ) {
        if ( n > 0 ) { TOKENTOLINE(" + ","+") }
        else if ( n < 0 ) { n = -n; TOKENTOLINE(" - ","-") }
    }
    if ( AC.modpowers ) {
        if ( n == 1 && *t == 1 && t > s ) first = 1;
        else if ( ABS(AC.ncmod) == 1 ) {
            UBYTE *Out1 = IsExponentSign();
            LongToLine((UWORD *)AC.powmod,AC.npowmod);
            if ( Out1 ) TokenToLine(Out1);
            else TokenToLine((UBYTE *)"^");
            TalToLine(AC.modpowers[(LONG)((UWORD)*t)]);
```

```
                first = 0;
            }
            else {
                LONG jj;
                UBYTE *Out1 = IsExponentSign();
                LongToLine((UWORD *)AC.powmod,AC.npowmod);
                if ( Out1 ) TokenToLine(Out1);
                else TokenToLine((UBYTE *)"^");
                jj = (UWORD)*t;
                if ( n == 2 ) jj += ((LONG)t[1])«BITSINWORD;
                if ( AC.modpowers[jj+1] == 0 ) {
                    TalToLine(AC.modpowers[jj]);
                }
                else {
                    LongToLine(AC.modpowers+jj,2);
                }
                first = 0;
            }
        }
        else if ( n != 1 || *t != 1 || t[1] != 1 || t <= s ) {
            if ( lowestlevel && ( ( AO.PrintType & PRINTONEFUNCTION ) != 0 ) ) {
                    FiniLine();
                    if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                    else IniLine(3);
            }
            if ( AO.CurrentDictionary > 0 ) TransformRational((UWORD *)t,n);
            else                            RatToLine((UWORD *)t,n);
            first = 0;
        }
#ifdef WITHFLOAT
/*
        Check whether there is a 'proper' float_ function and no raw mode.
        If so, print as float.
        Raw mode is indicated as AO.FloatPrec < 0.
        AO.FloatPrec == 0 indicated as many digits as the precision allows.
*/
        else if ( AO.FloatPrec >= 0 && AT.aux_ != 0 ) {
            WORD *ss = s;
            while ( ss < t ) {
                if ( *ss == FLOATFUN ) {
                    if ( ( FloatChars = PrintFloat(ss,AO.FloatPrec) ) != 0 ) {
                        TokenToLine(AO.floatspace);
                        first = 0;
                    }
                    break;
                }
                ss += ss[1];
            }
            if ( ss >= t ) first = 1;
        }
#endif
        else first = 1;
        while ( s < t ) {
            if ( lowestlevel && ( (AO.PrintType & (PRINTONEFUNCTION | PRINTALL)) == PRINTONEFUNCTION ) ) {
                FiniLine();
                if ( AC.OutputSpaces == NOSPACEFORMAT ) IniLine(1);
                else IniLine(3);
            }

/*
            #[ NEWGAMMA :
*/
#ifdef NEWGAMMA
            if ( *s == GAMMA ) {     /* String them up */
                WORD *tt,*ss;
                ss = AT.WorkPointer;
                *ss++ = GAMMA;
                *ss++ = s[1];
                FILLFUN(ss)
                *ss++ = s[FUNHEAD];
                tt = s + FUNHEAD + 1;
                n = s[1] - FUNHEAD-1;
                do {
                    while ( --n >= 0 ) *ss++ = *tt++;
                    tt = s + s[1];
                    while ( *tt == GAMMA && tt[FUNHEAD] == s[FUNHEAD] && tt < t ) {
                        s = tt;
                        tt += FUNHEAD + 1;
                        n = s[1] - FUNHEAD-1;
                        if ( n > 0 ) break;
                    }
                } while ( n > 0 );
                tt = AT.WorkPointer;
                AT.WorkPointer = ss;
                tt[1] = WORDDIF(ss,tt);
                if ( WriteSubTerm(tt,first) ) {
                    MesCall("WriteInnerTerm");
```

```
                SETERROR(-1)
            }
            AT.WorkPointer = tt;
        }
        else
#endif
/*
        #] NEWGAMMA :
*/
#ifdef WITHFLOAT
        if ( *s == FLOATFUN && AO.FloatPrec >= 0 && AT.aux_ != 0 ) {
        }
        else
#endif
        {
            if ( *s >= FUNCTION && AC.funpowers > 0
            && functions[*s-FUNCTION].spec == 0 && ( AC.funpowers == ALLFUNPOWERS ||
            ( AC.funpowers == COMFUNPOWERS && functions[*s-FUNCTION].commute == 0 ) ) ) {
                pow = 1;
                for(;;) {
                    s1 = s; s2 = s + s[1]; i = s[1];
                    if ( s2 < t ) {
                        while ( --i >= 0 && *s1 == *s2 ) { s1++; s2++; }
                        if ( i < 0 ) {
                            pow++; s = s+s[1];
                        }
                        else break;
                    }
                    else break;
                }
                if ( pow > 1 ) {
                    if ( AC.OutputMode == CMODE ) {
                        if ( !first ) MultiplyToLine();
                        TokenToLine((UBYTE *)"pow(");
                        first = 1;
                    }
                    if ( WriteSubTerm(s,first) ) {
                        MesCall("WriteInnerTerm");
                        SETERROR(-1)
                    }
                    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE
                     || AC.OutputMode == REDUCEMODE ) { TokenToLine((UBYTE *)"**"); }
                    else if ( AC.OutputMode == CMODE ) { TokenToLine((UBYTE *)","); }
                    else {
                        UBYTE *Out1 = IsExponentSign();
                        if ( Out1 ) TokenToLine(Out1);
                        else TokenToLine((UBYTE *)"^");
                    }
                    TalToLine(pow);
                    if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *)")");
                }
                else if ( WriteSubTerm(s,first) ) {
                    MesCall("WriteInnerTerm");
                    SETERROR(-1)
                }
            }
            else if ( WriteSubTerm(s,first) ) {
                MesCall("WriteInnerTerm");
                SETERROR(-1)
            }
        }
        first = 0;
        s += s[1];
    }
    return(0);
}

/*
        #] WriteInnerTerm :
        #[ WriteTerm :          WORD WriteTerm(term,lbrac,first,prtf,br)

    Writes a term to output. It tests the bracket information first.
    If there are no brackets or the bracket is the same all is passed
    to WriteInnerTerm. If there are brackets and the bracket is not
    the same as for the predecessor the old bracket is closed and
    a new one is opened.
    br indicates whether we are in a subexpression, barring zeroing
    AO.IsBracket

*/

WORD WriteTerm(WORD *term, WORD *lbrac, WORD first, WORD prtf, WORD br)
{
    WORD *t, *stopper, *b, n;
    int oldIsFortran90 = AC.IsFortran90, i;
    if ( *lbrac >= 0 ) {
        t = term + 1;
```

```
        stopper = (term + *term - 1);
        stopper -= ABS(*stopper) - 1;
        while ( t < stopper ) {
            if ( *t == HAAKJE ) {
                stopper = t;
                t = term+1;
                if ( *lbrac == ( n = WORDDIF(stopper,t) ) ) {
                    b = AO.bracket + 1;
                    t = term + 1;
                    while ( n > 0 && ( *b++ == *t++ ) ) { n--; }
                    if ( n <= 0 && ( ( AM.FortranCont <= 0 || AO.InFbrack < AM.FortranCont )
                    || ( lowestlevel == 0 ) ) ) {
/*
                        We continue inside a bracket.
*/
                        AO.IsBracket = 1;
                        if ( ( prtf & PRINTCONTENTS ) != 0 ) {
                            AO.NumInBrack++;
                        }
                        else {
                            if ( WriteInnerTerm(term,0) ) goto WrtTmes;
                            if ( ( AO.PrintType & PRINTONETERM ) != 0 ) {
                                FiniLine();
                                TokenToLine((UBYTE *)"   ");
                            }
                        }
                        return(0);
                    }
                    t = term + 1;
                    n = WORDDIF(stopper,t);
                }
/*
                Close the bracket
*/
                if ( *lbrac ) {
                    if ( ( prtf & PRINTCONTENTS ) ) PrtTerms();
                    TOKENTOLINE(" )",")")
                    if ( AC.OutputMode == CMODE && AO.FactorMode == 0 )
                        TokenToLine((UBYTE *)";");
                    else if ( AO.FactorMode && ( n == 0 ) ) {
/*
                        This should not happen.
*/
                        return(0);
                    }
                    AC.IsFortran90 = ISNOTFORTRAN90;
                    FiniLine();
                    AC.IsFortran90 = oldIsFortran90;
                    if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE
                        && AC.OutputSpaces == NORMALFORMAT
                        && AO.FactorMode == 0 ) FiniLine();
                }
                else {
                    if ( AC.OutputMode == CMODE && AO.FactorMode == 0 )
                        TokenToLine((UBYTE *)";");
                    if ( AO.FortFirst == 0 ) {
                        if ( !first ) {
                            AC.IsFortran90 = ISNOTFORTRAN90;
                            FiniLine();
                            AC.IsFortran90 = oldIsFortran90;
                        }
                    }
                }
                if ( AO.FactorMode == 0 ) {
                  if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
                   && !first ) {
                    WORD oldmode = AC.OutputMode;
                    AC.OutputMode = 0;
                    IniLine(0);
                    AC.OutputMode = oldmode;
                    AO.OutSkip = 7;

                    if ( AO.FortFirst == 0 ) {
                        TokenToLine(AO.CurBufWrt);
                        TOKENTOLINE(" = ","=")
                        TokenToLine(AO.CurBufWrt);
                    }
                    else {
                        AO.FortFirst = 0;
                        TokenToLine(AO.CurBufWrt);
                        TOKENTOLINE(" = ","=")
                    }
                  }
                  else if ( AC.OutputMode == CMODE && !first ) {
                    IniLine(0);
                    if ( AO.FortFirst == 0 ) {
                        TokenToLine(AO.CurBufWrt);
```

```
                        TOKENTOLINE(" += ","+=")
                    }
                    else {
                        AO.FortFirst = 0;
                        TokenToLine(AO.CurBufWrt);
                        TOKENTOLINE(" = ","=")
                    }
                  }
                  else if ( startinline == 0 ) {
                    IniLine(0);
                  }
                  AO.InFbrack = 0;
                  if ( ( *lbrac = n ) > 0 ) {
                    b = AO.bracket;
                    *b++ = n + 4;
                    while ( --n >= 0 ) *b++ = *t++;
                    *b++ = 1; *b++ = 1; *b = 3;
                    AO.IsBracket = 0;
                    if ( WriteInnerTerm(AO.bracket,0) ) {
                        /* Error message */
                        WORD i;
WrtTmes:                t = term;
                        AO.OutSkip = 3;
                        FiniLine();
                        i = *t;
                        while ( --i >= 0 ) { TalToLine((UWORD)(*t++));
                            if ( AC.OutputSpaces == NORMALFORMAT )
                                        TokenToLine((UBYTE *)"  "); }
                        AO.OutSkip = 0;
                        FiniLine();
                        MesCall("WriteTerm");
                        SETERROR(-1)
                    }
                    TOKENTOLINE(" * ( ","*(")
                    AO.NumInBrack = 0;
                    AO.IsBracket = 1;
                    if ( ( prtf & PRINTONETERM ) != 0 ) {
                        first = 0;
                        FiniLine();
                        TokenToLine((UBYTE *)"   ");
                    }
                    else first = 1;
                  }
                  else {
                    AO.IsBracket = 0;
                    first = 0;
                  }
                }
                else {
/*
                    Here is the code that writes the glue between two factors.
                    We should not forget factors that are zero!
*/
                    if ( ( *lbrac = n ) > 0 ) {
                        b = AO.bracket;
                        *b++ = n + 4;
                        while ( --n >= 0 ) *b++ = *t++;
                        *b++ = 1; *b++ = 1; *b = 3;
                        for ( i = AO.FactorNum+1; i < AO.bracket[4]; i++ ) {
                            if ( first ) {
                                TOKENTOLINE("   ( 0 )"," (0)")
                                first = 0;
                            }
                            else {
                                TOKENTOLINE(" * ( 0 )","*(0)")
                            }
                            FiniLine();
                            IniLine(0);
                        }
                        AO.FactorNum = AO.bracket[4];
                    }
                    else {
                        AO.NumInBrack = 0;
                        return(0);
                    }
                    if ( first == 0 ) { TOKENTOLINE(" * ( ","*(") }
                    else              { TOKENTOLINE("   ( "," (") }
                    AO.NumInBrack = 0;
                    first = 1;
                }
                if ( ( prtf & PRINTCONTENTS ) != 0 ) AO.NumInBrack++;
                else if ( WriteInnerTerm(term,first) ) goto WrtTmes;
                if ( ( AO.PrintType & PRINTONETERM ) != 0 ) {
                    FiniLine();
                    TokenToLine((UBYTE *)"   ");
                }
                return(0);
```

```
            }
            else t += t[1];
        }
        if ( *lbrac > 0 ) {
            if ( ( prtf & PRINTCONTENTS ) != 0 ) PrtTerms();
            TokenToLine((UBYTE *)" )");
            if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *)";");
            if ( AO.FortFirst == 0 ) {
                AC.IsFortran90 = ISNOTFORTRAN90;
                FiniLine();
                AC.IsFortran90 = oldIsFortran90;
            }
            if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE
                 && AC.OutputSpaces == NORMALFORMAT ) FiniLine();
            if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
             && !first ) {
                WORD oldmode = AC.OutputMode;
                AC.OutputMode = 0;
                IniLine(0);
                AC.OutputMode = oldmode;
                AO.OutSkip = 7;
                if ( AO.FortFirst == 0 ) {
                    TokenToLine(AO.CurBufWrt);
                    TOKENTOLINE(" = ","=")
                    TokenToLine(AO.CurBufWrt);
                }
                else {
                    AO.FortFirst = 0;
                    TokenToLine(AO.CurBufWrt);
                    TOKENTOLINE(" = ","=")
                }
/*
                TokenToLine(AO.CurBufWrt);
                TOKENTOLINE(" = ","=")
                if ( AO.FortFirst == 0 )
                    TokenToLine(AO.CurBufWrt);
                else AO.FortFirst = 0;
*/
            }
            else if ( AC.OutputMode == CMODE && !first ) {
                IniLine(0);
                if ( AO.FortFirst == 0 ) {
                    TokenToLine(AO.CurBufWrt);
                    TOKENTOLINE(" += ","+=")
                }
                else {
                    AO.FortFirst = 0;
                    TokenToLine(AO.CurBufWrt);
                    TOKENTOLINE(" = ","=")
                }
/*
                TokenToLine(AO.CurBufWrt);
                if ( AO.FortFirst == 0 ) { TOKENTOLINE(" += ","+=") }
                else {
                    TOKENTOLINE(" = ","=")
                    AO.FortFirst = 0;
                }
*/
            }
            else IniLine(0);
            *lbrac = 0;
            first = 1;
        }
    }
    if ( !br ) AO.IsBracket = 0;
    if ( ( AM.FortranCont > 0 && AO.InFbrack >= AM.FortranCont ) && lowestlevel ) {
        if ( AC.OutputMode == CMODE ) TokenToLine((UBYTE *)";");
        if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
         && !first ) {
            WORD oldmode = AC.OutputMode;
            if ( AO.FortFirst == 0 ) {
                AC.IsFortran90 = ISNOTFORTRAN90;
                FiniLine();
                AC.IsFortran90 = oldIsFortran90;
                AC.OutputMode = 0;
                IniLine(0);
                AC.OutputMode = oldmode;
                AO.OutSkip = 7;
                TokenToLine(AO.CurBufWrt);
                TOKENTOLINE(" = ","=")
                TokenToLine(AO.CurBufWrt);
            }
            else {
                AO.FortFirst = 0;
/*
                TokenToLine(AO.CurBufWrt);
                TOKENTOLINE(" = ","=")
```

```
*/
            }
/*
            TokenToLine(AO.CurBufWrt);
            TOKENTOLINE(" = ","=")
            if ( AO.FortFirst == 0 )
                TokenToLine(AO.CurBufWrt);
            else AO.FortFirst = 0;
*/
        }
        else if ( AC.OutputMode == CMODE && !first ) {
            FiniLine();
            IniLine(0);
            if ( AO.FortFirst == 0 ) {
                TokenToLine(AO.CurBufWrt);
                TOKENTOLINE(" += ","+=")
            }
            else {
                AO.FortFirst = 0;
                TokenToLine(AO.CurBufWrt);
                TOKENTOLINE(" = ","=")
            }
/*
            TokenToLine(AO.CurBufWrt);
            if ( AO.FortFirst == 0 ) { TOKENTOLINE(" += ","+=") }
            else {
                TOKENTOLINE(" = ","=")
                AO.FortFirst = 0;
            }
*/
        }
        else {
            FiniLine();
            IniLine(0);
        }
        AO.InFbrack = 0;
    }
    if ( WriteInnerTerm(term,first) ) goto WrtTmes;
    if ( ( AO.PrintType & PRINTONETERM ) != 0 ) {
        FiniLine();
        IniLine(0);
    }
    return(0);
}

/*
        #] WriteTerm :
        #[ WriteExpression :    WORD WriteExpression(terms,ltot)

    Writes a subexpression to output.
    The subexpression is in terms and contains ltot words.
    This is only used for function arguments.

*/

WORD WriteExpression(WORD *terms, LONG ltot)
{
    WORD *stopper;
    WORD first, btot;
    WORD OldIsBracket = AO.IsBracket, OldPrintType = AO.PrintType;
    if ( !AC.outsidefun ) { AO.PrintType &= ~PRINTONETERM; first = 1; }
    else first = 0;
    stopper = terms + ltot;
    btot = -1;
    while ( terms < stopper ) {
        AO.IsBracket = OldIsBracket;
        if ( WriteTerm(terms,&btot,first,0,1) ) {
            MesCall("WriteExpression");
            SETERROR(-1)
        }
        first = 0;
        terms += *terms;
    }
/*  AO.IsBracket = 0; */
    AO.IsBracket = OldIsBracket;
    AO.PrintType = OldPrintType;
    return(0);
}

/*
        #] WriteExpression :
        #[ WriteAll :           WORD WriteAll()

        Writes all expressions that should be written
*/

WORD WriteAll(void)
```

```
{
    GETIDENTITY
    WORD lbrac, first;
    WORD *t, *stopper, n, prtf;
    int oldIsFortran90 = AC.IsFortran90, i;
    POSITION pos;
    FILEHANDLE *f;
    EXPRESSIONS e;
    if ( AM.exitflag ) return(0);
#ifdef WITHMPI
    if ( PF.me != MASTER ) {
        /*
         * For the slaves, we need to call Optimize() the same number of times
         * as the master. The first argument doesn't have any important role.
         */
        for ( n = 0; n < NumExpressions; n++ ) {
            e = &Expressions[n];
            if ( (!e->printflag) & PRINTON ) continue;
            switch ( e->status ) {
                case LOCALEXPRESSION:
                case GLOBALEXPRESSION:
                case UNHIDELEXPRESSION:
                case UNHIDEGEXPRESSION:
                    break;
                default:
                    continue;
            }
            e->printflag = 0;
            PutPreVar(AM.oldnumextrasymbols, GetPreVar((UBYTE *)"EXTRASYMBOLS_", 0), 0, 1);
            if ( AO.OptimizationLevel > 0 ) {
                if ( Optimize(0, 1) ) return(-1);
            }
        }
        return(0);
    }
#endif
    SeekScratch(AR.outfile,&pos);
    if ( ResetScratch() ) {
        MesCall("WriteAll");
        SETERROR(-1)
    }
    AO.termbuf = AT.WorkPointer;
    AO.bracket = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer*2);
    AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer;
    AT.WorkPointer += 2*AC.LineLength;
    *(AR.CompressBuffer) = 0;
    first = 0;
    for ( n = 0; n < NumExpressions; n++ ) {
        if ( ( Expressions[n].printflag & PRINTON ) != 0 ) { first = 1; break; }
    }
    if ( !first ) goto EndWrite;
    AO.IsBracket = 0;
    AO.OutSkip = 3;
    AR.DeferFlag = 0;
    while ( GetTerm(BHEAD AO.termbuf) ) {
        t = AO.termbuf + 1;
        e = Expressions + AO.termbuf[3];
        n = e->status;
        if ( ( n == LOCALEXPRESSION || n == GLOBALEXPRESSION
        || n == UNHIDELEXPRESSION || n == UNHIDEGEXPRESSION ) &&
        ( ( prtf = e->printflag ) & PRINTON ) != 0 ) {
            e->printflag = 0;
            AO.NumInBrack = 0;
            PutPreVar(AM.oldnumextrasymbols,
                    GetPreVar((UBYTE *)"EXTRASYMBOLS_",0),0,1);
            if ( ( prtf & PRINTLFILE ) != 0 ) {
                if ( AC.LogHandle < 0 ) prtf &= ~PRINTLFILE;
            }
            AO.PrintType = prtf;
/*
            if ( AC.OutputMode == VORTRANMODE ) {
                UBYTE *oldOutFill = AO.OutFill, *oldOutputLine = AO.OutputLine;
                AO.OutSkip = 6;
                if ( Optimize(AO.termbuf[3], 1) ) goto AboWrite;
                AO.OutSkip = 3;
                AO.OutFill = oldOutFill; AO.OutputLine = oldOutputLine;
                FiniLine();
                continue;
            }
            else
*/
            if ( AO.OptimizationLevel > 0 ) {
                UBYTE *oldOutFill = AO.OutFill, *oldOutputLine = AO.OutputLine;
                AO.OutSkip = 6;
                if ( Optimize(AO.termbuf[3], 1) ) goto AboWrite;
                AO.OutSkip = 3;
```

```
                    AO.OutFill = oldOutFill; AO.OutputLine = oldOutputLine;
                    FiniLine();
                    continue;
                }
                if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
                         AO.OutSkip = 6;
                FiniLine();
                AO.CurBufWrt = EXPRNAME(AO.termbuf[3]);
                TokenToLine(AO.CurBufWrt);
                stopper = t + t[1];
                t += SUBEXPSIZE;
                if ( t < stopper ) {
                    TokenToLine((UBYTE *)"(");
                    first = 1;
                    while ( t < stopper ) {
                        n = *t;
                        if ( !first ) TokenToLine((UBYTE *)",");
                        switch ( n ) {
                            case SYMTOSYM :
                                TokenToLine(FindSymbol(t[2]));
/*                              TokenToLine(VARNAME(symbols,t[2])); */
                                break;
                            case VECTOVEC :
                                TokenToLine(FindVector(t[2]));
/*                              TokenToLine(VARNAME(vectors,t[2] - AM.OffsetVector)); */
                                break;
                            case INDTOIND :
                                TokenToLine(FindIndex(t[2]));
/*                              TokenToLine(VARNAME(indices,t[2] - AM.OffsetIndex)); */
                                break;
                            default :
                                TokenToLine(FindFunction(t[2]));
/*                              TokenToLine(VARNAME(functions,t[2] - FUNCTION)); */
                                break;
                        }
                        t += t[1];
                        first = 0;
                    }
                    TokenToLine((UBYTE *)")");
                }
                TOKENTOLINE(" =","=");
                if ( AC.OutputMode == MATHEMATICAMODE ) {
                    TOKENTOLINE(" (","(");
                }
                lbrac = 0;
                AO.InFbrack = 0;
                if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
                    AO.FortFirst = 1;
                else
                    AO.FortFirst = 0;
                first = 1;
                if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
                    AO.FactorMode = 1+e->numfactors;
                    AO.FactorNum = 0; /* Which factor are we doing. For factors that are zero */
                }
                else {
                    AO.FactorMode = 0;
                }
                while ( GetTerm(BHEAD AO.termbuf) ) {
                    WORD *m;
                    GETSTOP(AO.termbuf,m);
                    if ( ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
                    && ( ( prtf & PRINTONETERM ) != 0 ) ) {}
                    else {
                        if ( first ) {
                            FiniLine();
                            IniLine(0);
                        }
                    }
                    if ( ( prtf & PRINTONETERM ) != 0 ) first = 0;
                    if ( WriteTerm(AO.termbuf,&lbrac,first,prtf,0) )
                        goto AboWrite;
                    first = 0;
                }
                if ( AO.FactorMode ) {
                    if ( first ) { AO.FactorNum = 1; TOKENTOLINE("    ( 0 )","  (0)") }
                    else TOKENTOLINE(" )",")");
                    for ( i = AO.FactorNum+1; i <= e->numfactors; i++ ) {
                        FiniLine();
                        IniLine(0);
                        TOKENTOLINE(" * ( 0 )","*(0)");
                    }
                    AO.FactorNum = e->numfactors;
                    if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE )
                        TokenToLine((UBYTE *)";");
                }
                else if ( AO.FactorMode == 0 || first ) {
```

```
                    if ( first ) { TOKENTOLINE(" 0","0") }
                    else if ( lbrac ) {
                        if ( ( prtf & PRINTCONTENTS ) != 0 ) PrtTerms();
                        TOKENTOLINE(" )",")")
                    }
                    else if ( ( prtf & PRINTCONTENTS ) != 0 ) {
                        TOKENTOLINE(" + 1 * ( ","+1*(")
                        PrtTerms();
                        TOKENTOLINE(" )",")")
                    }
                    if ( AC.OutputMode == MATHEMATICAMODE ) {
                        TokenToLine((UBYTE *)")");
                    }
                    if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE )
                        TokenToLine((UBYTE *)";");
                }
                AO.OutSkip = 3;
                AC.IsFortran90 = ISNOTFORTRAN90;
                FiniLine();
                AC.IsFortran90 = oldIsFortran90;
                AO.FactorMode = 0;
            }
            else {
                do { } while ( GetTerm(BHEAD AO.termbuf) );
            }
        }
        if ( AC.OutputSpaces == NORMALFORMAT ) FiniLine();
EndWrite:
        if ( AR.infile->handle >= 0 ) {
            SeekFile(AR.infile->handle,&(AR.infile->filesize),SEEK_SET);
        }
        AO.IsBracket = 0;
        AT.WorkPointer = AO.termbuf;
        SetScratch(AR.infile,&pos);
        f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
        return(0);
AboWrite:
        SetScratch(AR.infile,&pos);
        f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
        MesCall("WriteAll");
        Terminate(-1);
        return(-1);
}

/*
            #] WriteAll :
            #[ WriteOne :           WORD WriteOne(name,alreadyinline)

            Writes one expression from the preprocessor
*/

WORD WriteOne(UBYTE *name, int alreadyinline, int nosemi, WORD plus)
{
        GETIDENTITY
        WORD number;
        WORD lbrac, first;
        POSITION pos;
        FILEHANDLE *f;
        WORD prf;

        if ( GetName(AC.exprnames,name,&number,NOAUTO) != CEXPRESSION ) {
            MesPrint("@%s is not an expression",name);
            return(-1);
        }
        switch ( Expressions[number].status ) {
            case HIDDENLEXPRESSION:
            case HIDDENGEXPRESSION:
            case HIDELEXPRESSION:
            case HIDEGEXPRESSION:
            case UNHIDELEXPRESSION:
            case UNHIDEGEXPRESSION:
/*
            case DROPHLEXPRESSION:
            case DROPHGEXPRESSION:
*/
                AR.GetFile = 2;
                break;
            case LOCALEXPRESSION:
            case GLOBALEXPRESSION:
            case SKIPLEXPRESSION:
            case SKIPGEXPRESSION:
/*
            case DROPLEXPRESSION:
            case DROPGEXPRESSION:
*/
                AR.GetFile = 0;
                break;
```

```
        default:
            MesPrint("@expressions %s is not active. It cannot be written",name);
            return(-1);
    }
    SeekScratch(AR.outfile,&pos);

    f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
/*
    if ( ResetScratch() ) {
        MesCall("WriteOne");
        SETERROR(-1)
    }
*/
    if ( AR.GetFile == 2 ) f = AR.hidefile;
    else f = AR.infile;
    prf = Expressions[number].printflag;
    if ( plus ) prf |= PRINTONETERM;
/*
        Now position the file
*/
    if ( f->handle >= 0 ) {
        SetScratch(f,&(Expressions[number].onfile));
    }
    else {
        f->POfill = (WORD *)((UBYTE *)(f->PObuffer)
                 + BASEPOSITION(Expressions[number].onfile));
    }
    AO.termbuf = AT.WorkPointer;
    AO.bracket = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer);
    AT.WorkPointer = (WORD *)(((UBYTE *)(AT.WorkPointer)) + AM.MaxTer*2);

    AO.OutFill = AO.OutputLine = (UBYTE *)AT.WorkPointer;
    AT.WorkPointer += 2*AC.LineLength;
    *(AR.CompressBuffer) = 0;

    AO.IsBracket = 0;
    AO.OutSkip = 3;
    AR.DeferFlag = 0;

    if ( AC.OutputMode == FORTRANMODE || AC.OutputMode == PFORTRANMODE )
            AO.OutSkip = 6;
    if ( GetTerm(BHEAD AO.termbuf) <= 0 ) {
        MesPrint("@ReadError in expression %s",name);
        goto AboWrite;
    }
/*
    PutPreVar(AM.oldnumextrasymbols,
            GetPreVar((UBYTE *)"EXTRASYMBOLS_",0),0,1);
*/
    /*
     * Currently WriteOne() is called only from writeToChannel() with setting
     * AO.OptimizationLevel = 0, which means Optimize() is never called here.
     * So we don't need to think about how to ensure that the master and the
     * slaves call Optimize() at the same time. (TU 26 Jul 2013)
     */
    if ( AO.OptimizationLevel > 0 ) {
        AO.OutSkip = 6;
        if ( Optimize(AO.termbuf[3], 1) ) goto AboWrite;
        AO.OutSkip = 3;
        FiniLine();
    }
    else {
        lbrac = 0;
        AO.InFbrack = 0;
        AO.FortFirst = 0;
        first = 1;
        while ( GetTerm(BHEAD AO.termbuf) ) {
            WORD *m;
            GETSTOP(AO.termbuf,m);
            if ( first ) {
                IniLine(0);
                startinline = alreadyinline;
                AO.OutFill = AO.OutputLine + startinline;
                if ( WriteTerm(AO.termbuf,&lbrac,first,0,0) )
                    goto AboWrite;
                first = 0;
            }
            else {
                if ( ( prf & PRINTONETERM ) != 0 ) first = 1;
                if ( first ) {
                    FiniLine();
                    IniLine(0);
                }
                first = 0;
                if ( WriteTerm(AO.termbuf,&lbrac,first,0,0) )
                    goto AboWrite;
            }
```

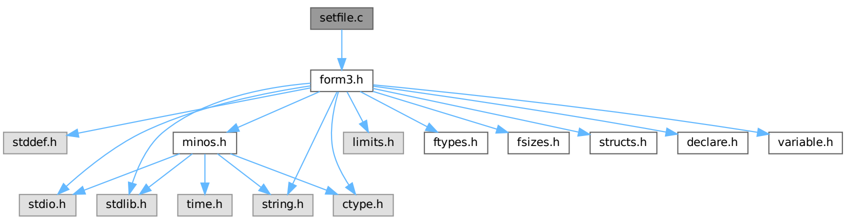
```
        }
        if ( first ) {
            IniLine(0);
            startinline = alreadyinline;
            AO.OutFill = AO.OutputLine + startinline;
            TOKENTOLINE(" 0","0");
        }
        else if ( lbrac ) {
            TOKENTOLINE(" )",")");
        }
        if ( AC.OutputMode != FORTRANMODE && AC.OutputMode != PFORTRANMODE
             && nosemi == 0 ) TokenToLine((UBYTE *)";");
        AO.OutSkip = 3;
        if ( AC.OutputSpaces == NORMALFORMAT && nosemi == 0 ) {
            FiniLine();
        }
        else {
            noextralinefeed = 1;
            FiniLine();
            noextralinefeed = 0;
        }
    }
    AO.IsBracket = 0;
    AT.WorkPointer = AO.termbuf;
    SetScratch(f,&pos);
    f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
    AO.InFbrack = 0;
    return(0);
AboWrite:
    SetScratch(AR.infile,&pos);
    f->POposition = pos;
    f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
    MesCall("WriteOne");
    Terminate(-1);
    return(-1);
}

/*
        #] WriteOne :
    #] schryf-Writes :
*/
```

4.124 setfile.c File Reference

`#include "form3.h"`

Include dependency graph for setfile.c:



Macros

- #define NUMERICALVALUE 0
- #define STRINGVALUE 1
- #define PATHVALUE 2
- #define ONOFFVALUE 3
- #define DEFINEVALUE 4
- #define SETBUFSIZE 257

Functions

- int DoSetups (void)
- int ProcessOption (UBYTE ∗s1, UBYTE ∗s2, int filetype)
- SETUPPARAMETERS ∗ GetSetupPar (UBYTE ∗s)
- int RecalcSetups (void)
- int AllocSetups (void)
- void WriteSetup (void)
- SORTING ∗ AllocSort (LONG inLargeSize, LONG inSmallSize, LONG inSmallEsize, LONG inTermsInSmall, int inMaxPatches, int inMaxFpatches, LONG inIOsize, int level)
- void AllocSortFileName (SORTING ∗sort)
- FILEHANDLE ∗ AllocFileHandle (WORD par, char ∗name)
- void DeAllocFileHandle (FILEHANDLE ∗fh)
- int MakeSetupAllocs (void)
- int TryFileSetups (void)
- int TryEnvironment (void)

Variables

- char curdirp [ ] = "."
- char cursortdirp [ ] = "."
- char commentchar [ ] = "∗"
- char dotchar [ ] = "_"
- char highfirst [ ] = "highfirst"
- char lowfirst [ ] = "lowfirst"
- char procedureextension [ ] = "prc"
- SETUPPARAMETERS setupparameters [ ]

4.124.1 Detailed Description

The routines that deal with the setup parameters.

Definition in file setfile.c.

4.124.2 Macro Definition Documentation

4.124.2.1 NUMERICALVALUE

```
#define NUMERICALVALUE 0
```

Definition at line 47 of file setfile.c.

4.124.2.2 STRINGVALUE

```
#define STRINGVALUE 1
```

Definition at line 48 of file setfile.c.

4.124.2.3 PATHVALUE

```
#define PATHVALUE 2
```

Definition at line 49 of file setfile.c.

4.124.2.4 ONOFFVALUE

```
#define ONOFFVALUE 3
```

Definition at line 50 of file setfile.c.

4.124.2.5 DEFINEVALUE

```
#define DEFINEVALUE 4
```

Definition at line 51 of file setfile.c.

4.124.2.6 SETBUFSIZE

```
#define SETBUFSIZE 257
```

Definition at line 1140 of file setfile.c.

4.124.3 Function Documentation

4.124.3.1 DoSetups()

```
int DoSetups (
             void )
```

Definition at line 132 of file setfile.c.

4.124.3.2 ProcessOption()

```
int ProcessOption (
             UBYTE * s1,
             UBYTE * s2,
             int filetype )
```

Definition at line 182 of file setfile.c.

4.124.3.3 GetSetupPar()

```
SETUPPARAMETERS * GetSetupPar (
             UBYTE * s )
```

Definition at line 337 of file setfile.c.

4.124.3.4 RecalcSetups()

```
int RecalcSetups (
            void )
```

Definition at line 357 of file setfile.c.

4.124.3.5 AllocSetups()

```
int AllocSetups (
            void )
```

Definition at line 404 of file setfile.c.

4.124.3.6 WriteSetup()

```
void WriteSetup (
            void )
```

Definition at line 811 of file setfile.c.

4.124.3.7 AllocSort()

```
SORTING * AllocSort (
            LONG inLargeSize,
            LONG inSmallSize,
            LONG inSmallEsize,
            LONG inTermsInSmall,
            int inMaxPatches,
            int inMaxFpatches,
            LONG inIOsize,
            int level )
```

Definition at line 869 of file setfile.c.

4.124.3.8 AllocSortFileName()

```
void AllocSortFileName (
            SORTING * sort )
```

Definition at line 1019 of file setfile.c.

4.124.3.9 AllocFileHandle()

```
FILEHANDLE * AllocFileHandle (
            WORD par,
            char * name )
```

Definition at line 1044 of file setfile.c.

4.124.3.10 DeAllocFileHandle()

```
void DeAllocFileHandle (
            FILEHANDLE * fh )
```

Definition at line 1105 of file setfile.c.

4.124.3.11 MakeSetupAllocs()

```
int MakeSetupAllocs (
            void )
```

Definition at line 1122 of file setfile.c.

4.124.3.12 TryFileSetups()

```
int TryFileSetups (
            void )
```

Definition at line 1142 of file setfile.c.

4.124.3.13 TryEnvironment()

```
int TryEnvironment (
            void )
```

Definition at line 1228 of file setfile.c.

4.124.4 Variable Documentation

4.124.4.1 curdirp

```
char curdirp[ ] = "."
```

Definition at line 39 of file setfile.c.

4.124.4.2 cursortdirp

```
char cursortdirp[ ] = "."
```

Definition at line 40 of file setfile.c.

4.124.4.3 commentchar

```
char commentchar[ ] = "*"
```

Definition at line 41 of file setfile.c.

4.124.4.4 dotchar

```
char dotchar[ ] = "_"
```

Definition at line 42 of file setfile.c.

4.124.4.5 highfirst

```
char highfirst[ ] = "highfirst"
```

Definition at line 43 of file setfile.c.

4.124.4.6 lowfirst

```
char lowfirst[ ] = "lowfirst"
```

Definition at line 44 of file setfile.c.

4.124.4.7 procedureextension

```
char procedureextension[ ] = "prc"
```

Definition at line 45 of file setfile.c.

4.124.4.8 setupparameters

```
SETUPPARAMETERS setupparameters[ ]
```

Definition at line 53 of file setfile.c.

4.125 setfile.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
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```
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :

    Routines that deal with settings and the setup file
*/

#include "form3.h"

char curdirp[] = ".";
char cursortdirp[] = ".";
char commentchar[] = "*";
char dotchar[] = "_";
char highfirst[] = "highfirst";
char lowfirst[] = "lowfirst";
char procedureextension[] = "prc";

#define NUMERICALVALUE 0
#define STRINGVALUE 1
#define PATHVALUE 2
#define ONOFFVALUE 3
#define DEFINEVALUE 4

SETUPPARAMETERS setupparameters[] =
{
     {(UBYTE *)"bracketindexsize",      NUMERICALVALUE, 0, (LONG)MAXBRACKETBUFFERSIZE}
    ,{(UBYTE *)"commentchar",              STRINGVALUE, 0, (LONG)commentchar}
    ,{(UBYTE *)"compresssize",          NUMERICALVALUE, 0, (LONG)COMPRESSBUFFER}
    ,{(UBYTE *)"constindex",            NUMERICALVALUE, 0, (LONG)NUMFIXED}
    ,{(UBYTE *)"continuationlines",     NUMERICALVALUE, 0, (LONG)FORTRANCONTINUATIONLINES}
#ifdef WITHFLOAT
    ,{(UBYTE *)"defaultprecision",      NUMERICALVALUE, 0, (LONG)DEFAULTPRECISION}
#endif
    ,{(UBYTE *)"define",                   DEFINEVALUE, 0, (LONG)0}
    ,{(UBYTE *)"dotchar",                  STRINGVALUE, 0, (LONG)dotchar}
    ,{(UBYTE *)"factorizationcache",    NUMERICALVALUE, 0, (LONG)FBUFFERSIZE}
    ,{(UBYTE *)"filepatches",           NUMERICALVALUE, 0, (LONG)MAXFPATCHES}
    ,{(UBYTE *)"functionlevels",        NUMERICALVALUE, 0, (LONG)MAXFLEVELS}
    ,{(UBYTE *)"hidesize",              NUMERICALVALUE, 0, (LONG)0}
    ,{(UBYTE *)"incdir",                     PATHVALUE, 0, (LONG)curdirp}
    ,{(UBYTE *)"indentspace",           NUMERICALVALUE, 0, (LONG)INDENTSPACE}
    ,{(UBYTE *)"insidefirst",               ONOFFVALUE, 0, (LONG)1}
    ,{(UBYTE *)"jumpratio",             NUMERICALVALUE, 0, (LONG)JUMPRATIO}
    ,{(UBYTE *)"largepatches",          NUMERICALVALUE, 0, (LONG)MAXPATCHES}
    ,{(UBYTE *)"largesize",             NUMERICALVALUE, 0, (LONG)LARGEBUFFER}
    ,{(UBYTE *)"maxnumbersize",         NUMERICALVALUE, 0, (LONG)0}
/*  ,{(UBYTE *)"maxnumbersize",         NUMERICALVALUE, 0, (LONG)MAXNUMBERSIZE} */
    ,{(UBYTE *)"maxtermsize",           NUMERICALVALUE, 0, (LONG)MAXTER}
#ifdef WITHFLOAT
    ,{(UBYTE *)"maxweight",             NUMERICALVALUE, 0, (LONG)MAXWEIGHT}
#endif
    ,{(UBYTE *)"maxwildcards",          NUMERICALVALUE, 0, (LONG)MAXWILDC}
    ,{(UBYTE *)"nospacesinnumbers",         ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"numstorecaches",        NUMERICALVALUE, 0, (LONG)NUMSTORECACHES}
    ,{(UBYTE *)"nwritefinalstatistics",     ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"nwriteprocessstatistics",   ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"nwritestatistics",          ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"nwritethreadstatistics",    ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"oldfactarg",                ONOFFVALUE, 0, (LONG)NEWFACTARG}
    ,{(UBYTE *)"oldgcd",                    ONOFFVALUE, 0, (LONG)1}
    ,{(UBYTE *)"oldorder",                  ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"oldparallelstatistics",     ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"parentheses",           NUMERICALVALUE, 0, (LONG)MAXPARLEVEL}
    ,{(UBYTE *)"path",                       PATHVALUE, 0, (LONG)curdirp}
    ,{(UBYTE *)"procedureextension",       STRINGVALUE, 0, (LONG)procedureextension}
    ,{(UBYTE *)"processbucketsize",     NUMERICALVALUE, 0, (LONG)DEFAULTPROCESSBUCKETSIZE}
    ,{(UBYTE *)"resettimeonclear",          ONOFFVALUE, 0, (LONG)1}
    ,{(UBYTE *)"scratchsize",           NUMERICALVALUE, 0, (LONG)SCRATCHSIZE}
    ,{(UBYTE *)"shmwinsize",            NUMERICALVALUE, 0, (LONG)SHMWINSIZE}
    ,{(UBYTE *)"sizestorecache",        NUMERICALVALUE, 0, (LONG)SIZESTORECACHE}
    ,{(UBYTE *)"smallextension",        NUMERICALVALUE, 0, (LONG)SMALLOVERFLOW}
    ,{(UBYTE *)"smallsize",             NUMERICALVALUE, 0, (LONG)SMALLBUFFER}
    ,{(UBYTE *)"sortiosize",            NUMERICALVALUE, 0, (LONG)SORTIOSIZE}
    ,{(UBYTE *)"sorttype",                 STRINGVALUE, 0, (LONG)lowfirst}
    ,{(UBYTE *)"spectatorsize",         NUMERICALVALUE, 0, (LONG)SPECTATORSIZE}
    ,{(UBYTE *)"subfilepatches",        NUMERICALVALUE, 0, (LONG)SMAXFPATCHES}
    ,{(UBYTE *)"sublargepatches",       NUMERICALVALUE, 0, (LONG)SMAXPATCHES}
    ,{(UBYTE *)"sublargesize",          NUMERICALVALUE, 0, (LONG)SLARGEBUFFER}
    ,{(UBYTE *)"subsmallextension",     NUMERICALVALUE, 0, (LONG)SSMALLOVERFLOW}
    ,{(UBYTE *)"subsmallsize",          NUMERICALVALUE, 0, (LONG)SSMALLBUFFER}
    ,{(UBYTE *)"subsortiosize",         NUMERICALVALUE, 0, (LONG)SSORTIOSIZE}
    ,{(UBYTE *)"subtermsinsmall",       NUMERICALVALUE, 0, (LONG)STERMSSMALL}
    ,{(UBYTE *)"tempdir",                  STRINGVALUE, 0, (LONG)curdirp}
    ,{(UBYTE *)"tempsortdir",              STRINGVALUE, 0, (LONG)cursortdirp}
```

```
    ,{(UBYTE *)"termsinsmall",          NUMERICALVALUE, 0, (LONG)TERMSSMALL}
    ,{(UBYTE *)"threadbucketsize",      NUMERICALVALUE, 0, (LONG)DEFAULTTHREADBUCKETSIZE}
    ,{(UBYTE *)"threadloadbalancing",       ONOFFVALUE, 0, (LONG)DEFAULTTHREADLOADBALANCING}
    ,{(UBYTE *)"threads",               NUMERICALVALUE, 0, (LONG)DEFAULTTHREADS}
    ,{(UBYTE *)"threadscratchoutsize",  NUMERICALVALUE, 0, (LONG)THREADSCRATCHOUTSIZE}
    ,{(UBYTE *)"threadscratchsize",     NUMERICALVALUE, 0, (LONG)THREADSCRATCHSIZE}
    ,{(UBYTE *)"threadsortfilesynch",       ONOFFVALUE, 0, (LONG)0}
    ,{(UBYTE *)"totalsize",                 ONOFFVALUE, 0, (LONG)2}
    ,{(UBYTE *)"workspace",             NUMERICALVALUE, 0, (LONG)WORKBUFFER}
    ,{(UBYTE *)"wtimestats",                ONOFFVALUE, 0, (LONG)2}
};

/*
    #] Includes :
    #[ Setups :
        #[ DoSetups :
*/

int DoSetups(void)
{
    UBYTE *setbuffer, *s, *t, *u /*, c */;
    int errors = 0;
    setbuffer = LoadInputFile((UBYTE *)setupfilename,SETUPFILE);
    if ( setbuffer ) {
/*
        The contents of the file are now in setbuffer.
        Each line is commentary or a single command.
        The buffer is terminated with a zero.
*/
        s = setbuffer;
        while ( *s ) {
            if ( *s == ' ' || *s == '\t' || *s == '*' || *s == '#' || *s == '\n' ) {
                while ( *s && *s != '\n' ) s++;
            }
            else if ( tolower(*s) < 'a' || tolower(*s) > 'z' ) {
                t = s;
                while ( *s && *s != '\n' ) s++;
/*
                c = *s; *s = 0;
                Error1("Setup file: Illegal statement: ",t);
                errors++; *s = c;
*/
            }
            else {
                t = s; /* name of the option */
                while ( tolower(*s) >= 'a' && tolower(*s) <= 'z' ) s++;
                *s++ = 0;
                while ( *s == ' ' || *s == '\t' ) s++;
                u = s; /* 'value' of the option */
                while ( *s && *s != '\n' && *s != '\r' ) s++;
                if ( *s ) *s++ = 0;
                errors += ProcessOption(t,u,0);
            }
            while ( *s == '\n' || *s == '\r' ) s++;
        }
        M_free(setbuffer,"setup file buffer");
    }
    if ( errors ) return(1);
    else return(0);
}

/*
        #] DoSetups :
        #[ ProcessOption :
*/

static char *proop1[3] = { "Setup file", "Setups in .frm file", "Setup in environment" };

int ProcessOption(UBYTE *s1, UBYTE *s2, int filetype)
{
    SETUPPARAMETERS *sp;
    int n, giveback = 0, error = 0;
    UBYTE *s, *t, *s2ret;
    LONG x;
    sp = GetSetupPar(s1);
    if ( sp ) {
/*
        We check now whether there are `' variables to be looked up in the
        environment. This is new (30-may-2008). This is only allowed in s2.
*/
restart:;
        {
            UBYTE *s3,*s4,*s5,*s6, c, *start;
            int n1,n2,n3;
            s = s2;
            while ( *s ) {
                if ( *s == '\\' ) s += 2;
```

```
                else if ( *s == '`' ) {
                    start = s; s++;
                    while ( *s && *s != '`' ) {
                        if ( *s == '\\' ) s++;
                        s++;
                    }
                    if ( *s == 0 ) {
                        MesPrint("%s: Illegal use of ` character for parameter %s"
                        ,proop1[filetype],s1);
                        return(1);
                    }
                    c = *s; *s = 0;
                    s3 = (UBYTE *)getenv((char *)(start+1));
                    if ( s3 == 0 ) {
                        MesPrint("%s: Cannot find environment variable %s for parameter %s"
                        ,proop1[filetype],start+1,s1);
                        return(1);

                    }
                    *s = c; s++;
                    n1 = start - s2; s4 = s3; n2 = 0;
                    while ( *s4 ) {
                        if ( *s4 == '\\' ) { s4++; n2++; }
                        s4++; n2++;
                    }
                    s4 = s; n3 = 0;
                    while ( *s4 ) {
                        if ( *s4 == '\\' ) { s4++; n3++; }
                        s4++; n3++;
                    }
                    s4 = (UBYTE *)Malloc1((n1+n2+n3+1)*sizeof(UBYTE),"environment in setup");
                    s5 = s2; s6 = s4;
                    while ( n1-- > 0 ) *s6++ = *s5++;
                    s5 = s3;
                    while ( n2-- > 0 ) *s6++ = *s5++;
                    s5 = s;
                    while ( n3-- > 0 ) *s6++ = *s5++;
                    *s6 = 0;
                    if ( giveback ) M_free(s2,"environment in setup");
                    s2 = s4;
                    giveback = 1;
                    goto restart;
                }
                else s++;
            }
        }
        n = sp->type;
        s2ret = s2;
        switch ( n ) {
            case NUMERICALVALUE:
                    ParseNumber(x,s2);
                if ( *s2 == 'K' ) { x = x * 1000; s2++; }
                else if ( *s2 == 'M' ) { x = x * 1000000; s2++; }
                else if ( *s2 == 'G' ) { x = x * 1000000000; s2++; }
                else if ( *s2 == 'T' ) { x = x * 1000000000000; s2++; }
                if ( *s2 && *s2 != ' ' && *s2 != '\t' ) {
                    MesPrint("%s: Numerical value expected for parameter %s"
                    ,proop1[filetype],s1);
                    error = 1; break;
                }
                sp->value = x;
                sp->flags = USEDFLAG;
                break;
            case STRINGVALUE:
                if ( StrICmp(s1,(UBYTE *)"tempsortdir") == 0 ) AM.havesortdir = 1;
                s = s2; t = s2;
                while ( *s ) {
                    if ( *s == ' ' || *s == '\t' ) break;
                    if ( *s == '\\' ) s++;
                    *t++ = *s++;
                }
                *t = 0;
                if ( sp->flags == USEDFLAG && sp->value != 0 )
                        M_free((void *)(sp->value),"Process option");
                sp->value = (LONG)strDup1(s2,"Process option");
                sp->flags = USEDFLAG;
                break;
            case PATHVALUE:
                if ( StrICmp(s1,(UBYTE *)"incdir") == 0 ) {
                    AM.IncDir = 0;
                }
                else if ( StrICmp(s1,(UBYTE *)"path") == 0 ) {
                    if ( AM.Path ) M_free(AM.Path,"path");
                    AM.Path = 0;
                }
                else {
                    MesPrint("Setups: %s not yet implemented",s1);
```

```
                    error = 1;
                    break;
                }
                if ( sp->flags == USEDFLAG && sp->value != 0 )
                    M_free((void *)(sp->value),"Process option");
                sp->value = (LONG)strDup1(s2,"Process option");
                sp->flags = USEDFLAG;
                break;
            case ONOFFVALUE:
                if ( tolower(*s2) == 'o' && tolower(s2[1]) == 'n'
                && ( s2[2] == 0 || s2[2] == ' ' || s2[2] == '\t' ) )
                    sp->value = 1;
                else if ( tolower(*s2) == 'o' && tolower(s2[1]) == 'f'
                && tolower(s2[2]) == 'f'
                && ( s2[3] == 0 || s2[3] == ' ' || s2[3] == '\t' ) )
                    sp->value = 0;
                else {
                    MesPrint("%s: Unrecognized option for parameter %s: %s"
                    ,proop1[filetype],s1,s2);
                    error = 1; break;
                }
                sp->flags = USEDFLAG;
                break;
            case DEFINEVALUE:
/*
                if ( sp->value ) M_free((UBYTE *)(sp->value),"Process option");
                sp->value = (LONG)strDup1(s2,"Process option");
*/
                if ( TheDefine(s2,2) ) error = 1;
                break;
            default:
                Error1("Error in setupparameter table for:",s1);
                error = 1;
                break;
        }
    }
    else {
        MesPrint("%s: Keyword not recognized: %s",proop1[filetype],s1);
        error = 1;
    }
    if ( giveback ) M_free(s2ret,"environment in setup");
    return(error);
}

/*
        #] ProcessOption :
        #[ GetSetupPar :
*/

SETUPPARAMETERS *GetSetupPar(UBYTE *s)
{
    int hi, med, lo, i;
    lo = 0;
    hi = sizeof(setupparameters)/sizeof(SETUPPARAMETERS);
    do {
        med = ( hi + lo ) / 2;
        i = StrICmp(s,(UBYTE *)setupparameters[med].parameter);
        if ( i == 0 ) return(setupparameters+med);
        if ( i < 0 ) hi = med-1;
        else         lo = med+1;
    } while ( hi >= lo );
    return(0);
}

/*
        #] GetSetupPar :
        #[ RecalcSetups :
*/

int RecalcSetups(void)
{
    SETUPPARAMETERS *sp, *sp1;

    sp1 = GetSetupPar((UBYTE *)"threads");
    if ( AM.totalnumberofthreads > 1 ) sp1->value = AM.totalnumberofthreads - 1;
    else sp1->value = 0;
/*
    if ( sp1->value > 0 ) AM.totalnumberofthreads = sp1->value+1;
    if ( AM.totalnumberofthreads == 0 ) AM.totalnumberofthreads = 1;
*/
    sp  = GetSetupPar((UBYTE *)"filepatches");
    if ( sp->value < AM.totalnumberofthreads-1 )
        sp->value = AM.totalnumberofthreads - 1;

    sp  = GetSetupPar((UBYTE *)"smallsize");
    sp1 = GetSetupPar((UBYTE *)"smallextension");
    if ( 6*sp1->value < 7*sp->value ) sp1->value = (7*sp->value)/6;
```

```
    sp = GetSetupPar((UBYTE *)"termsinsmall");
    sp->value = ( sp->value + 15 ) & (-16L);
#ifdef WITHPTHREADS
    {
    SETUPPARAMETERS *sp2;
    LONG totalsize, minimumsize;
    sp = GetSetupPar((UBYTE *)"largesize");
    totalsize = sp1->value+sp->value;
    sp2 = GetSetupPar((UBYTE *)"maxtermsize");
    AM.MaxTer = sp2->value*sizeof(WORD);
    if ( AM.MaxTer < 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = 200*(LONG)(sizeof(WORD));
    if ( AM.MaxTer > MAXPOSITIVE - 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = MAXPOSITIVE -
200*(LONG)(sizeof(WORD));
    AM.MaxTer /= sizeof(WORD);
    AM.MaxTer *= sizeof(WORD);
    minimumsize = (AM.totalnumberofthreads-1)*(AM.MaxTer+
        NUMBEROFBLOCKSINSORT*MINIMUMNUMBEROFTERMS*AM.MaxTer);
    if ( totalsize < minimumsize ) {
        sp->value = minimumsize - sp1->value;
    }
    }
#endif
    return(0);
}

/*
        #] RecalcSetups :
        #[ AllocSetups :
*/

int AllocSetups(void)
{
    SETUPPARAMETERS *sp;
    LONG LargeSize, SmallSize, SmallEsize, TermsInSmall, IOsize;
    int MaxPatches, MaxFpatches, error = 0, i, size;
    UBYTE *s;
#ifndef WITHPTHREADS
    int j;
#endif
    sp = GetSetupPar((UBYTE *)"threads");
    if ( sp->value > 0 ) AM.totalnumberofthreads = sp->value+1;

    AM.OutBuffer = (UBYTE *)Malloc1(AM.OutBufSize+1,"OutputBuffer");
    AP.PreAssignStack =(LONG *)Malloc1(AP.MaxPreAssignLevel*sizeof(LONG *),"PreAssignStack");
    for ( i = 0; i < AP.MaxPreAssignLevel; i++ ) AP.PreAssignStack[i] = 0;
    AC.iBuffer = (UBYTE *)Malloc1(AC.iBufferSize+1,"statement buffer");
    AC.iStop = AC.iBuffer + AC.iBufferSize-2;
    AP.preStart = (UBYTE *)Malloc1(AP.pSize,"instruction buffer");
    AP.preStop = AP.preStart + AP.pSize - 3;
    /* AP.PreIfStack is already allocated in StartPrepro(), but to be sure we
       "if" the freeing */
    if ( AP.PreIfStack ) M_free(AP.PreIfStack,"PreIfStack");
    AP.PreIfStack = (int *)Malloc1(AP.MaxPreIfLevel*sizeof(int),
                "Preprocessor if stack");
    AP.PreIfStack[0] = EXECUTINGIF;
    sp = GetSetupPar((UBYTE *)"insidefirst");
    AM.ginsidefirst = AC.minsidefirst = AC.insidefirst = sp->value;
/*
    We need to consider eliminating this variable
*/
    sp = GetSetupPar((UBYTE *)"maxtermsize");
    AM.MaxTer = sp->value*sizeof(WORD);
    if ( AM.MaxTer < 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = 200*(LONG)(sizeof(WORD));
    if ( AM.MaxTer > MAXPOSITIVE - 200*(LONG)(sizeof(WORD)) ) AM.MaxTer = MAXPOSITIVE -
200*(LONG)(sizeof(WORD));
    AM.MaxTer /= (LONG)sizeof(WORD);
    AM.MaxTer *= (LONG)sizeof(WORD);
/*
    Allocate workspace.
*/
    sp = GetSetupPar((UBYTE *)"workspace");
    AM.WorkSize = sp->value;
#ifdef WITHPTHREADS
#else
    AT.WorkSpace = (WORD *)Malloc1(AM.WorkSize*sizeof(WORD),(char *)(sp->parameter));
    AT.WorkTop = AT.WorkSpace + AM.WorkSize;
    AT.WorkPointer = AT.WorkSpace;
#endif
/*
    Fixed indices
*/
    sp = GetSetupPar((UBYTE *)"constindex");
    if ( ( sp->value+100+5*WILDOFFSET ) > MAXPOSITIVE ) {
        MesPrint("Setting of %s in setupfile too large","constindex");
        AM.OffsetIndex = MAXPOSITIVE - 5*WILDOFFSET - 100;
        MesPrint("value corrected to maximum allowed: %d",AM.OffsetIndex);
    }
```

```
    else AM.OffsetIndex = sp->value + 1;
    AC.FixIndices = (WORD *)Malloc1((AM.OffsetIndex)*sizeof(WORD),(char *)(sp->parameter));
    AM.WilInd = AM.OffsetIndex + WILDOFFSET;
    AM.DumInd = AM.OffsetIndex + 2*WILDOFFSET;
    AM.IndDum = AM.DumInd + WILDOFFSET;
#ifndef WITHPTHREADS
    AR.CurDum = AN.IndDum = AM.IndDum;
#endif
    AM.mTraceDum = AM.IndDum + 2*WILDOFFSET;

    sp = GetSetupPar((UBYTE *)"parentheses");
    AM.MaxParLevel = sp->value+1;
    AC.tokenarglevel = (WORD *)Malloc1((sp->value+1)*sizeof(WORD),(char *)(sp->parameter));
/*
    Space during calculations
*/
    sp = GetSetupPar((UBYTE *)"maxnumbersize");
/*
    size = ( sp->value + 11 ) & (-4);
    AM.MaxTal = size - 2;
    if ( AM.MaxTal > (AM.MaxTer/sizeof(WORD)-2)/2 )
                AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/2;
    if ( AM.MaxTal < (AM.MaxTer/sizeof(WORD)-2)/4 )
                AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/4;
*/
/*
    There is too much confusion about MaxTal cq maxnumbersize.
    It seems better to fix it at its maximum value. This way we only worry
    about maxtermsize. This can be understood better by the 'innocent' user.
*/
    if ( sp->value == 0 ) {
        AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/2;
    }
    else {
        size = ( sp->value + 11 ) & (-4);
        AM.MaxTal = size - 2;
        if ( (size_t)AM.MaxTal > (size_t)((AM.MaxTer/sizeof(WORD)-2)/2) )
                    AM.MaxTal = (AM.MaxTer/sizeof(WORD)-2)/2;
    }
    AM.MaxTal &= -sizeof(WORD)*2;

    sp->value = AM.MaxTal;
    AC.cmod = (UWORD *)Malloc1(AM.MaxTal*4*sizeof(UWORD),(char *)(sp->parameter));
    AM.gcmod = AC.cmod + AM.MaxTal;
    AC.powmod = AM.gcmod + AM.MaxTal;
    AM.gpowmod = AC.powmod + AM.MaxTal;
/*
    The IO buffers for the input and output expressions.
    Fscr[2] will be assigned in a later stage for hiding expressions from
    the regular action. That will make the program faster.
*/
    sp = GetSetupPar((UBYTE *)"scratchsize");
    AM.ScratSize = sp->value/sizeof(WORD);
    if ( AM.ScratSize < 4*AM.MaxTer ) AM.ScratSize = 4*AM.MaxTer;
    AM.HideSize = AM.ScratSize;
    sp = GetSetupPar((UBYTE *)"hidesize");
    if ( sp->value > 0 ) {
        AM.HideSize = sp->value/sizeof(WORD);
        if ( AM.HideSize < 4*AM.MaxTer ) AM.HideSize = 4*AM.MaxTer;
    }
    sp = GetSetupPar((UBYTE *)"factorizationcache");
    AM.fbuffersize = sp->value;
#ifdef WITHPTHREADS
    sp = GetSetupPar((UBYTE *)"threadscratchsize");
    AM.ThreadScratSize = sp->value/sizeof(WORD);
    sp = GetSetupPar((UBYTE *)"threadscratchoutsize");
    AM.ThreadScratOutSize = sp->value/sizeof(WORD);
#endif
#ifndef WITHPTHREADS
    for ( j = 0; j < 2; j++ ) {
        WORD *ScratchBuf;
        ScratchBuf = (WORD *)Malloc1(AM.ScratSize*sizeof(WORD),"scratchsize");
        AR.Fscr[j].POsize = AM.ScratSize * sizeof(WORD);
        AR.Fscr[j].POfull = AR.Fscr[j].POfill = AR.Fscr[j].PObuffer = ScratchBuf;
        AR.Fscr[j].POstop = AR.Fscr[j].PObuffer + AM.ScratSize;
        PUTZERO(AR.Fscr[j].POposition);
    }
    AR.Fscr[2].PObuffer = 0;
#endif
    sp = GetSetupPar((UBYTE *)"threadbucketsize");
    AC.ThreadBucketSize = AM.gThreadBucketSize = AM.ggThreadBucketSize = sp->value;
    sp = GetSetupPar((UBYTE *)"threadloadbalancing");
    AC.ThreadBalancing = AM.gThreadBalancing = AM.ggThreadBalancing = sp->value;
    sp = GetSetupPar((UBYTE *)"threadsortfilesynch");
    AC.ThreadSortFileSynch = AM.gThreadSortFileSynch = AM.ggThreadSortFileSynch = sp->value;
/*
     The size for shared memory window for oneside MPI2 communications
```

```
*/
    sp = GetSetupPar((UBYTE *)"shmwinsize");
    AM.shmWinSize = sp->value/sizeof(WORD);
    if ( AM.shmWinSize < 4*AM.MaxTer ) AM.shmWinSize = 4*AM.MaxTer;
/*
    The sort buffer
*/
    sp = GetSetupPar((UBYTE *)"smallsize");
    SmallSize = sp->value;
    sp = GetSetupPar((UBYTE *)"smallextension");
    SmallEsize = sp->value;
    sp = GetSetupPar((UBYTE *)"largesize");
    LargeSize = sp->value;
    sp = GetSetupPar((UBYTE *)"termsinsmall");
    TermsInSmall = sp->value;
    sp = GetSetupPar((UBYTE *)"largepatches");
    MaxPatches = sp->value;
    sp = GetSetupPar((UBYTE *)"filepatches");
    MaxFpatches = sp->value;
    sp = GetSetupPar((UBYTE *)"sortiosize");
    IOsize = sp->value;
    if ( IOsize < AM.MaxTer ) { IOsize = AM.MaxTer; sp->value = IOsize; }
#ifndef WITHPTHREADS
#ifdef WITHZLIB
    for ( j = 0; j < 2; j++ ) { AR.Fscr[j].ziosize = IOsize; }
#endif
#endif
    AM.S0 = 0;
    AM.S0 = AllocSort(LargeSize,SmallSize,SmallEsize,TermsInSmall
                    ,MaxPatches,MaxFpatches,IOsize,0);
    /* AM.S0->file.ziosize was already set to a (larger) value by AllocSort, here it is re-set. */
#ifdef WITHZLIB
    AM.S0->file.ziosize = IOsize;
#ifndef WITHPTHREADS
    AR.FoStage4[0].ziosize = IOsize;
    AR.FoStage4[1].ziosize = IOsize;
    AT.S0 = AM.S0;
#endif
#else
#ifndef WITHPTHREADS
    AT.S0 = AM.S0;
#endif
#endif
#ifndef WITHPTHREADS
    AR.FoStage4[0].POsize   = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
    AR.FoStage4[1].POsize   = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
#endif
    sp = GetSetupPar((UBYTE *)"subsmallsize");
    AM.SSmallSize = sp->value;
    sp = GetSetupPar((UBYTE *)"subsmallextension");
    AM.SSmallEsize = sp->value;
    sp = GetSetupPar((UBYTE *)"sublargesize");
    AM.SLargeSize = sp->value;
    sp = GetSetupPar((UBYTE *)"subtermsinsmall");
    AM.STermsInSmall = sp->value;
    sp = GetSetupPar((UBYTE *)"sublargepatches");
    AM.SMaxPatches = sp->value;
    sp = GetSetupPar((UBYTE *)"subfilepatches");
    AM.SMaxFpatches = sp->value;
    sp = GetSetupPar((UBYTE *)"subsortiosize");
    AM.SIOsize = sp->value;
    /* As for IOsize, make sure SIOsize is at least as large as AM.MaxTer. */
    if ( AM.SIOsize < AM.MaxTer ) { AM.SIOsize = AM.MaxTer; sp->value = AM.SIOsize; }
    sp = GetSetupPar((UBYTE *)"spectatorsize");
    AM.SpectatorSize = sp->value;
/*
    The next code is just for the moment (26-jan-1997) because we have
    the new parts combined with the old. Once the old parts are gone
    from the program, we can eliminate this code too.
*/
    sp = GetSetupPar((UBYTE *)"functionlevels");
    AM.maxFlevels = sp->value + 1;
#ifdef WITHPTHREADS
#else
    AT.Nest = (NESTING)Malloc1((LONG)sizeof(struct NeStInG)*AM.maxFlevels,"functionlevels");
    AT.NestStop = AT.Nest + AM.maxFlevels;
    AT.NestPoin = AT.Nest;
#endif

    sp = GetSetupPar((UBYTE *)"maxwildcards");
    AM.MaxWildcards = sp->value;
#ifdef WITHPTHREADS
#else
    AT.WildMask = (WORD *)Malloc1((LONG)AM.MaxWildcards*sizeof(WORD),"maxwildcards");
#endif

    sp = GetSetupPar((UBYTE *)"compresssize");
```

```
    if ( sp->value < 2*AM.MaxTer ) sp->value = 2*AM.MaxTer;
    AM.CompressSize = sp->value;
#ifndef WITHPTHREADS
    AR.CompressBuffer = (WORD *)Malloc1((AM.CompressSize+10)*sizeof(WORD),"compresssize");
    AR.CompressPointer = AR.CompressBuffer;
    AR.ComprTop = AR.CompressBuffer + AM.CompressSize;
#endif
    sp = GetSetupPar((UBYTE *)"bracketindexsize");
    if ( sp->value < 20*AM.MaxTer ) sp->value = 20*AM.MaxTer;
    AM.MaxBracketBufferSize = sp->value/sizeof(WORD);

    sp = GetSetupPar((UBYTE *)"dotchar");
    AO.FortDotChar = ((UBYTE *)(sp->value))[0];
    sp = GetSetupPar((UBYTE *)"commentchar");
    AP.cComChar = AP.ComChar = ((UBYTE *)(sp->value))[0];
    sp = GetSetupPar((UBYTE *)"procedureextension");
/*
    Check validity first.
*/
    s = (UBYTE *)(sp->value);
    if ( FG.cTable[*s] != 0 ) {
        MesPrint("  Illegal string for procedure extension %s",(UBYTE *)sp->value);
        error = -2;
    }
    else {
        s++;
        while ( *s ) {
            if ( *s == ' ' || *s == '\t' || *s == '\n' ) {
                MesPrint("  Illegal string for procedure extension %s",(UBYTE *)sp->value);
                error = -2;
                break;
            }
            s++;
        }
    }
    AP.cprocedureExtension = strDup1((UBYTE *)(sp->value),"procedureExtension");
    AP.procedureExtension = strDup1(AP.cprocedureExtension,"procedureExtension");

    sp = GetSetupPar((UBYTE *)"totalsize");
    if ( sp->value != 2 ) AM.PrintTotalSize = sp->value;

    sp = GetSetupPar((UBYTE *)"continuationlines");
    AM.FortranCont = sp->value;
    sp = GetSetupPar((UBYTE *)"oldorder");
    AM.OldOrderFlag = sp->value;
    sp = GetSetupPar((UBYTE *)"resettimeonclear");
    AM.resetTimeOnClear = sp->value;
    sp = GetSetupPar((UBYTE *)"nospacesinnumbers");
    AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers = AM.ggNoSpacesInNumbers = sp->value;
    sp = GetSetupPar((UBYTE *)"indentspace");
    AO.IndentSpace = AM.gIndentSpace = AM.ggIndentSpace = sp->value;
    sp = GetSetupPar((UBYTE *)"jumpratio");
    AM.jumpratio = sp->value;
    sp = GetSetupPar((UBYTE *)"nwritestatistics");
    AC.StatsFlag = AM.gStatsFlag = AM.ggStatsFlag = 1-sp->value;
    sp = GetSetupPar((UBYTE *)"nwritefinalstatistics");
    AC.FinalStats = AM.gFinalStats = AM.ggFinalStats = 1-sp->value;
    sp = GetSetupPar((UBYTE *)"nwritethreadstatistics");
    AC.ThreadStats = AM.gThreadStats = AM.ggThreadStats = 1-sp->value;
    sp = GetSetupPar((UBYTE *)"nwriteprocessstatistics");
    AC.ProcessStats = AM.gProcessStats = AM.ggProcessStats = 1-sp->value;
    sp = GetSetupPar((UBYTE *)"oldparallelstatistics");
    AC.OldParallelStats = AM.gOldParallelStats = AM.ggOldParallelStats = sp->value;
    sp = GetSetupPar((UBYTE *)"oldfactarg");
    AC.OldFactArgFlag = AM.gOldFactArgFlag = AM.ggOldFactArgFlag = sp->value;
    sp = GetSetupPar((UBYTE *)"oldgcd");
    AC.OldGCDflag = AM.gOldGCDflag = AM.ggOldGCDflag = sp->value;
    sp = GetSetupPar((UBYTE *)"wtimestats");
    if ( sp->value == 2 ) sp->value = AM.ggWTimeStatsFlag;
    AC.WTimeStatsFlag = AM.gWTimeStatsFlag = AM.ggWTimeStatsFlag = sp->value;
#ifdef WITHFLOAT
    sp = GetSetupPar((UBYTE *)"maxweight");
    AC.MaxWeight = AM.gMaxWeight = AM.ggMaxWeight = sp->value;
    sp = GetSetupPar((UBYTE *)"defaultprecision");
    AC.DefaultPrecision = AM.gDefaultPrecision = AM.ggDefaultPrecision = sp->value;
#endif
    sp = GetSetupPar((UBYTE *)"sorttype");
    if ( StrICmp((UBYTE *)"lowfirst",(UBYTE *)sp->value) == 0 ) {
        AC.lSortType = SORTLOWFIRST;
    }
    else if ( StrICmp((UBYTE *)"highfirst",(UBYTE *)sp->value) == 0 ) {
        AC.lSortType = SORTHIGHFIRST;
    }
    else {
        MesPrint("  Illegal SortType specification: %s",(UBYTE *)sp->value);
        error = -2;
    }
```

```

    sp = GetSetupPar((UBYTE *)"processbucketsize");
    AM.hProcessBucketSize = AM.gProcessBucketSize =
    AC.ProcessBucketSize = AC.mProcessBucketSize = sp->value;
/*
    The store caches (code installed 15-aug-2006 JV)
*/
    sp = GetSetupPar((UBYTE *)"numstorecaches");
    AM.NumStoreCaches = sp->value;
    sp = GetSetupPar((UBYTE *)"sizestorecache");
    AM.SizeStoreCache = sp->value;
    /* Make sure this is a multiple of sizeof(WORD). */
    AM.SizeStoreCache = ((AM.SizeStoreCache+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
#ifndef WITHPTHREADS
/*
    Install the store caches (15-aug-2006 JV)
    Note that in the case of PTHREADS this is done in InitializeOneThread
*/
    AT.StoreCache = AT.StoreCacheAlloc = 0;
    if ( AM.NumStoreCaches > 0 ) {
        STORECACHE sa, sb;
        size = sizeof(struct StOrEcAcHe)+AM.SizeStoreCache;
        size = ((size-1)/sizeof(size_t)+1)*sizeof(size_t);
        AT.StoreCacheAlloc = (STORECACHE)Malloc1(size*AM.NumStoreCaches,"StoreCaches");
        AT.StoreCache = AT.StoreCacheAlloc;
        sa = AT.StoreCache;
        for ( j = 0; j < AM.NumStoreCaches; j++ ) {
            sb = (STORECACHE)(void *)((UBYTE *)sa+size);
            if ( j == AM.NumStoreCaches-1 ) {
                sa->next = 0;
            }
            else {
                sa->next = sb;
            }
            SETBASEPOSITION(sa->position,-1);
            SETBASEPOSITION(sa->toppos,-1);
            sa = sb;
        }
    }
#endif

/*
    And now some order sensitive things
*/
    if ( AM.Path == 0 ) {
        sp = GetSetupPar((UBYTE *)"path");
        AM.Path = strDup1((UBYTE *)(sp->value),"path");
    }
    if ( AM.IncDir == 0 ) {
        sp = GetSetupPar((UBYTE *)"incdir");
        AM.IncDir = strDup1((UBYTE *)(sp->value),"incdir");
    }
/*
    if ( AM.TempDir == 0 ) {
        sp = GetSetupPar((UBYTE *)"tempdir");
        AM.TempDir = strDup1((UBYTE *)(sp->value),"tempdir");
    }
*/

/*
    As part of the setup, we possibly need to work around a flint multi-threading
    bug which was fixed in version 3.1.0. This function should be called by the
    master only, before the worker threads start processing terms.
*/
#ifdef WITHFLINT
#if __FLINT_RELEASE < 30100
    flint_startup_init();
#endif
#endif

    return(error);
}

/*
        #] AllocSetups :
        #[ WriteSetup :

    The routine writes the values of the setup parameters.
    We should do this better. (JV, 21-may-2008)
    The way it should be done is:
        a: write the raw values.
        b: give readjusted values.
        c: give derived values.
    Because this is a difficult subject, it would be nice to have a LaTeX
    document that explains this all exactly. There should then be a
    mechanism to poke the values of the setup into the LaTeX document.
    probably the easiest way is to make a file with lots of \def definitions
```

```
    and have that included into the LaTeX file.
*/

void WriteSetup(void)
{
    int n = sizeof(setupparameters)/sizeof(SETUPPARAMETERS);
    SETUPPARAMETERS *sp;
    MesPrint(" The setup parameters are:");
    for ( sp = setupparameters; n > 0; n--, sp++ ) {
        switch(sp->type){
            case NUMERICALVALUE:
                MesPrint("   %s: %l",sp->parameter,sp->value);
                break;
            case PATHVALUE:
                if ( StrICmp(sp->parameter,(UBYTE *)"path") == 0 && AM.Path ) {
                    MesPrint("   %s: '%s'",sp->parameter,(UBYTE *)(AM.Path));
                    break;
                }
                if ( StrICmp(sp->parameter,(UBYTE *)"incdir") == 0 && AM.IncDir ) {
                    MesPrint("   %s: '%s'",sp->parameter,(UBYTE *)(AM.IncDir));
                    break;
                }
                /* fall through */
            case STRINGVALUE:
                if ( StrICmp(sp->parameter,(UBYTE *)"tempdir") == 0 && AM.TempDir ) {
                    MesPrint("   %s: '%s'",sp->parameter,(UBYTE *)(AM.TempDir));
                }
                else if ( StrICmp(sp->parameter,(UBYTE *)"tempsortdir") == 0 && AM.TempSortDir ) {
                    MesPrint("   %s: '%s'",sp->parameter,(UBYTE *)(AM.TempSortDir));
                }
                else {
                    MesPrint("   %s: '%s'",sp->parameter,(UBYTE *)(sp->value));
                }
                break;
            case ONOFFVALUE:
                if ( sp->value == 0 )
                    MesPrint("   %s: OFF",sp->parameter);
                else if ( sp->value == 1 )
                    MesPrint("   %s: ON",sp->parameter);
                break;
            case DEFINEVALUE:
/*
                MesPrint("   %s: '%s'",sp->parameter,(UBYTE *)(sp->value));
*/
                break;
        }
    }
    AC.SetupFlag = 0;
}

/*
        #] WriteSetup :
        #[ AllocSort :

        Routine allocates a complete struct for sorting.
        To be used for the main allocation of the sort buffers, and
        in a later stage for the function and subroutine sort buffers.
        The level arg denotes a main buffer allocation (0) or a sub-buffer
        allocation (1), used only for printing warning messages for buffer
        size adjustments in DEBUGGING mode.
*/
SORTING *AllocSort(LONG inLargeSize, LONG inSmallSize, LONG inSmallEsize, LONG inTermsInSmall,
                   int inMaxPatches, int inMaxFpatches, LONG inIOsize, int level)
{
    DUMMYUSE(level); /* This is only used in DEBUGGING mode */
    LONG LargeSize = inLargeSize;
    LONG SmallSize = inSmallSize;
    LONG SmallEsize = inSmallEsize;
    LONG TermsInSmall = inTermsInSmall;
    int MaxPatches = inMaxPatches;
    int MaxFpatches = inMaxFpatches;
    LONG IOsize = inIOsize;

    LONG longer,terms2insmall,sortsize,longerp;
    LONG IObuffersize = IOsize;
    LONG IOtry;
    SORTING *sort;
    int fname2Size = 0, j = 0;
    char *s;
    if ( AM.S0 != 0 ) {
        s = FG.fname2; fname2Size = 0;
        while ( *s ) { s++; fname2Size++; }
        fname2Size += 16;
    }
    if ( MaxFpatches < 4 ) MaxFpatches = 4;
    longer = MaxPatches > MaxFpatches ? MaxPatches : MaxFpatches;
    longerp = longer;
```

```
    while ( (1 « j) < longerp ) j++;
    longerp = (1 « j) + 1;
    longerp += sizeof(WORD*) - (longerp%sizeof(WORD *));
    longer++;
    longer += sizeof(WORD*) - (longer%sizeof(WORD *));
    if ( SmallSize < 16*AM.MaxTer ) SmallSize = 16*AM.MaxTer+16;
    TermsInSmall = (TermsInSmall+15) & (-16L);
    terms2insmall = 2*TermsInSmall;  /* Used to be just + 100 rather than *2 */
    if ( SmallEsize < (SmallSize*3)/2 ) SmallEsize = (SmallSize*3)/2;
    if ( LargeSize > 0 && LargeSize < 2*SmallSize ) LargeSize = 2*SmallSize;
    SmallEsize = (SmallEsize+15) & (-16L);
    if ( LargeSize < 0 ) LargeSize = 0;
    sortsize = sizeof(SORTING);
    sortsize = (sortsize+15)&(-16L);
    IObuffersize = (IObuffersize+sizeof(WORD)-1)/sizeof(WORD);
/*
    The next statement fixes a bug. In the rare case that we have a
    problem here, we expand the size of the large buffer or the
    small extension
*/
    if ( (ULONG)( LargeSize+SmallEsize ) < MaxFpatches*((IObuffersize
        +COMPINC)*sizeof(WORD)+2*AM.MaxTer) ) {
        if ( LargeSize == 0 )
            SmallEsize = MaxFpatches*((IObuffersize+COMPINC)*sizeof(WORD)+2*AM.MaxTer);
        else
            LargeSize  = MaxFpatches*((IObuffersize+COMPINC)*sizeof(WORD)+2*AM.MaxTer)
                - SmallEsize;
    }

    IOtry = ((LargeSize+SmallEsize)/MaxFpatches-2*AM.MaxTer)/sizeof(WORD)-COMPINC;

    /* Here both IObuffersize and IOtry are in units of sizeof(WORD). */
    if ( IObuffersize < IOtry ) {
        IObuffersize = IOtry;
    }

#if DEBUGGING
    char *prefix;
    if ( level == 0 ) { prefix = ""; }
    else { prefix = "Sub"; }
    if ( LargeSize != inLargeSize ) { MesPrint("Warning: %sLargeSize adjusted: %l -> %l", prefix, inLargeSize, LargeSize); }
    if ( SmallSize != inSmallSize ) { MesPrint("Warning: %sSmallSize adjusted: %l -> %l", prefix, inSmallSize, SmallSize); }
    if ( SmallEsize != inSmallEsize ) {MesPrint("Warning: %sSmallEsize adjusted: %l -> %l", prefix, inSmallEsize, SmallEsize); }
    if ( TermsInSmall != inTermsInSmall ) { MesPrint("Warning: %sTermsInSmall adjusted: %l -> %l", prefix, inTermsInSmall, TermsInSmall); }
    if ( MaxPatches != inMaxPatches ) { MesPrint("Warning: MaxPatches adjusted: %d -> %d", inMaxPatches, MaxPatches); }
    if ( MaxFpatches != inMaxFpatches ) {MesPrint("Warning: MaxFPatches adjusted: %d -> %d", inMaxFpatches, MaxFpatches); }
    /* This one is always changed if the LargeSize has not been... */
    /* if ( IObuffersize != inIOsize/(LONG)sizeof(WORD) ) { MesPrint("Warning: IOsize adjusted: %l -> %l", inIOsize/sizeof(WORD), IObuffersize); } */
#endif

    /* Allocate separate buffers for most struct members, for better testing and debugging with valgrind. */
    sort = Malloc1(sizeof(*sort), "AllocSort: sorting struct");

    sort->LargeSize = LargeSize/sizeof(WORD);
    sort->SmallSize = SmallSize/sizeof(WORD);
    sort->SmallEsize = SmallEsize/sizeof(WORD);
    sort->MaxPatches = MaxPatches;
    sort->MaxFpatches = MaxFpatches;
    sort->TermsInSmall = TermsInSmall;
    sort->Terms2InSmall = terms2insmall;

    sort->sPointer     = Malloc1(sizeof(*(sort->sPointer    ))*terms2insmall, "AllocSort: sPointer");
    sort->Patches      = Malloc1(sizeof(*(sort->Patches     ))*longer, "AllocSort: Patches");
    sort->pStop        = Malloc1(sizeof(*(sort->pStop       ))*longer, "AllocSort: pStop");
    sort->poina        = Malloc1(sizeof(*(sort->poina       ))*longerp, "AllocSort: poina");
    sort->poin2a       = Malloc1(sizeof(*(sort->poin2a      ))*longerp, "AllocSort: poin2a");

    sort->fPatches     = Malloc1(sizeof(*(sort->fPatches    ))*longer, "AllocSort: fPatches");
    sort->fPatchesStop = Malloc1(sizeof(*(sort->fPatchesStop))*longer, "AllocSort: fPatchesStop");
    sort->inPatches    = Malloc1(sizeof(*(sort->inPatches   ))*longer, "AllocSort: inPatches");
    sort->tree         = Malloc1(sizeof(*(sort->tree        ))*longerp, "AllocSort: tree");
    sort->used         = Malloc1(sizeof(*(sort->used        ))*longerp, "AllocSort: used");

#ifdef WITHZLIB
    sort->fpcompressed   = Malloc1(sizeof(*(sort->fpcompressed  ))*(longerp+2), "AllocSort: fpcompressed");
    sort->fpincompressed = Malloc1(sizeof(*(sort->fpincompressed))*(longerp+2), "AllocSort: fpincompressed");
    sort->zsparray = 0;
```

```
#endif

    sort->ktoi          = Malloc1(sizeof(*(sort->ktoi))*(longerp+2), "AllocSort: ktoi");

    // The combined Large buffer and Small buffer (+ extension) are used.
    // They must be allocated together.
    sort->lBuffer       = Malloc1(sizeof(*(sort->lBuffer))*(sort->LargeSize+sort->SmallEsize),
"AllocSort: lBuffer+sBuffer");
    sort->lTop = sort->lBuffer+sort->LargeSize;

    sort->sBuffer = sort->lTop;
    if ( sort->LargeSize == 0 ) { sort->lBuffer = 0; sort->lTop = 0; }
    sort->sTop = sort->sBuffer + sort->SmallSize;
    sort->sTop2 = sort->sBuffer + sort->SmallEsize;
    sort->sHalf = sort->sBuffer + (LONG)((sort->SmallSize+sort->SmallEsize)»1);

    sort->file.PObuffer = Malloc1(IObuffersize*sizeof(*(sort->file.PObuffer))+fname2Size+16,
"AllocSort: PObuffer");
    sort->file.POstop = sort->file.PObuffer+IObuffersize;
    sort->file.POsize = IObuffersize * sizeof(WORD);
    sort->file.POfill = sort->file.POfull = sort->file.PObuffer;
    sort->file.active = 0;
    sort->file.handle = -1;
    PUTZERO(sort->file.POposition);
#ifdef WITHPTHREADS
    sort->file.pthreadslock = dummylock;
#endif
#ifdef WITHZLIB
    sort->file.ziosize = IObuffersize*sizeof(WORD);
    sort->file.ziobuffer = 0;
#endif
    if ( AM.S0 != 0 ) {
        sort->file.name = (char *)(sort->file.PObuffer + IObuffersize);
        AllocSortFileName(sort);
    }
    else sort->file.name = 0;
    sort->cBuffer = 0;
    sort->cBufferSize = 0;
    sort->f = 0;
    sort->PolyWise = 0;

    return(sort);
}

/*
        #] AllocSort :
        #[ AllocSortFileName :
*/

void AllocSortFileName(SORTING *sort)
{
    GETIDENTITY
    char *s, *t;
/*
        This is not the allocation before the tempfiles are determined.
        Hence we can use the name in FG.fname2 and modify the tail
*/
    s = FG.fname2; t = sort->file.name;
    while ( *s ) *t++ = *s++;
#ifdef WITHPTHREADS
    t[-2] = 'F';
    snprintf(t-1,FG.fname2size-(t-1-sort->file.name),"%d.%d",identity,AN.filenum);
#else
    t[-2] = 'f';
    snprintf(t-1,FG.fname2size-(t-1-sort->file.name),"%d",AN.filenum);
#endif
    AN.filenum++;
}

/*
        #] AllocSortFileName :
        #[ AllocFileHandle :
*/

FILEHANDLE *AllocFileHandle(WORD par,char *name)
{
    GETIDENTITY
    LONG allocation, Ssize;
    FILEHANDLE *fh;
    int i = 0;
    char *s, *t;

    s = FG.fname2; i = 0;
    while ( *s ) { s++; i++; }
    if ( par == 0 ) { i += 16; Ssize = AM.SIOsize; }
    else { s = name; while ( *s ) { i++; s++; } i+= 2; Ssize = AM.SpectatorSize; }

```

```
    allocation = sizeof(FILEHANDLE) + (Ssize+1)*sizeof(WORD) + i*sizeof(char)
            +FG.fname2size+20;
    fh = (FILEHANDLE *)Malloc1(allocation,"FileHandle");

    fh->PObuffer = (WORD *)(fh+1);
    fh->POstop = fh->PObuffer+Ssize;
    fh->POsize = Ssize * sizeof(WORD);
    fh->active = 0;
    fh->handle = -1;
    PUTZERO(fh->POposition);
#ifdef WITHPTHREADS
    fh->pthreadslock = dummylock;
#endif
    if ( par == 0 ) {   /* sort file */
      if ( AM.S0 != 0 ) {
        fh->name = (char *)(fh->POstop + 1);
        s = FG.fname2; t = fh->name;
        while ( *s ) *t++ = *s++;
#ifdef WITHPTHREADS
        t[-2] = 'F';
        snprintf(t-1,FG.fname2size-(t-1-fh->name),"%d-%d",identity,AN.filenum);
#else
        t[-2] = 'f';
        snprintf(t-1,FG.fname2size-(t-1-fh->name),"%d",AN.filenum);
#endif
        AN.filenum++;
      }
      else fh->name = 0;
    }
    else {  /* Spectator file */
        fh->name = (char *)(fh->POstop + 1);
        s = FG.fname; t = fh->name;
        for ( i  = 0; i < FG.namebase; i++ ) *t++ = *s++;
        s = name;
        while ( *s ) *t++ = *s++;
        *t = 0;
    }
    fh->POfill = fh->POfull = fh->PObuffer;
    return(fh);
}

/*
        #] AllocFileHandle :
        #[ DeAllocFileHandle :

        Made to repair deallocation of AN.filenum. 21-sep-2000
*/

void DeAllocFileHandle(FILEHANDLE *fh)
{
    GETIDENTITY
    if ( fh->handle >= 0 ) {
        CloseFile(fh->handle);
        fh->handle = -1;
        remove(fh->name);
    }
    AN.filenum--; /* free namespace. was forgotten in first reading */
    M_free(fh,"Temporary FileHandle");
}

/*
        #] DeAllocFileHandle :
        #[ MakeSetupAllocs :
*/

int MakeSetupAllocs(void)
{
    if ( RecalcSetups() || AllocSetups() ) return(1);
    else return(0);
}

/*
        #] MakeSetupAllocs :
        #[ TryFileSetups :

        Routine looks in the input file for a start of the type
        [#-]
        #: setupparameter value
        It keeps looking until the first line that does not start with
        #-, #+ or #:
        Then it rewinds the input.
*/

#define SETBUFSIZE 257

int TryFileSetups(void)
{
```

```
    LONG oldstreamposition;
    int oldstream;
    int error = 0, eqnum;
    int oldNoShowInput = AC.NoShowInput;
    UBYTE buff[SETBUFSIZE+1], *s, *t, *u, *settop, c;
    LONG linenum, prevline;

    if ( AC.CurrentStream == 0 ) return(error);
    oldstream = AC.CurrentStream - AC.Streams;
    oldstreamposition = GetStreamPosition(AC.CurrentStream);
    linenum = AC.CurrentStream->linenumber;
    prevline = AC.CurrentStream->prevline;
    eqnum = AC.CurrentStream->eqnum;
    AC.NoShowInput = 1;
    settop = buff + SETBUFSIZE;
    if ( AC.CurrentStream->type == INPUTSTREAM && oldstreamposition == 0 )
AC.CurrentStream->fileposition = 0;
    for(;;) {
        c = GetInput();
        if ( c == '*' || c == '\n' ) {
            while ( c != '\n' && c != ENDOFINPUT ) c = GetInput();
            if ( c == ENDOFINPUT ) goto eoi;
            continue;
        }
        if ( c == ENDOFINPUT ) goto eoi;
        if ( c != '#' ) break;
        c = GetInput();
        if ( c == ENDOFINPUT ) goto eoi;
        if ( c != '-' && c != '+' && c != ':' ) break;
        if ( c != ':' ) {
            while ( c != '\n' && c != ENDOFINPUT ) c = GetInput();
            continue;
        }
        s = buff;
        while ( ( c = GetInput() ) == ' ' || c == '\t' || c == '\r' ) {}
        if ( c == ENDOFINPUT ) break;
        if ( c == LINEFEED ) continue;
        if ( c == 0 || c == ENDOFINPUT ) break;
        while ( c != LINEFEED ) {
            *s++ = c;
            c = GetInput();
            if ( c != LINEFEED && c != '\r' ) continue;
            if ( s >= settop ) {
                while ( c != '\n' && c != ENDOFINPUT ) c = GetInput();
                MesPrint("Setups in .frm file: Line too long. setup ignored");
                error++; goto nextline;
            }
        }
        *s++ = '\n';
        t = s = buff; /* name of the option */
        while ( tolower(*s) >= 'a' && tolower(*s) <= 'z' ) s++;
        if ( *s != '\n' ) {
            *s++ = 0;
            while ( *s == ' ' || *s == '\t' ) s++;
            u = s; /* 'value' of the option */
            while ( *s && *s != '\n' && *s != '\r' ) s++;
            if ( *s ) *s++ = 0;
        }
        else {
            /* The value is empty. */
            u = s;
            *s++ = 0;
        }
        error += ProcessOption(t,u,1);
nextline:;
    }
    AC.NoShowInput = oldNoShowInput;
    AC.CurrentStream = AC.Streams + oldstream;
    PositionStream(AC.CurrentStream,oldstreamposition);
    AC.CurrentStream->linenumber = linenum;
    AC.CurrentStream->prevline = prevline;
    AC.CurrentStream->eqnum = eqnum;
    ClearPushback();
    if ( AC.CurrentStream->type == INPUTSTREAM ) AC.CurrentStream->fileposition = -1;
    return(error);
eoi:
    MesPrint("Input file without a program.");
    return(-1);
}

/*
        #] TryFileSetups :
        #[ TryEnvironment :
*/

int TryEnvironment(void)
{
```

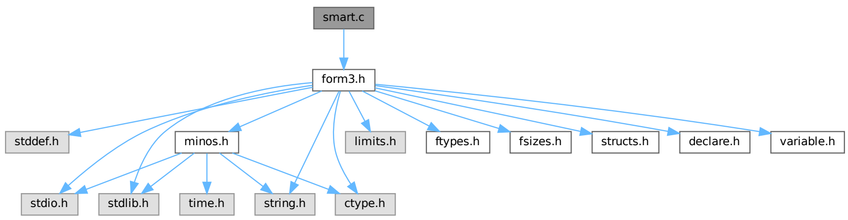
```
    char *s, *t, *u, varname[100];
    int i,imax = sizeof(setupparameters)/sizeof(SETUPPARAMETERS);
    int error = 0;
    varname[0] = 'F'; varname[1] = 'O'; varname[2] = 'R'; varname[3] = 'M';
    varname[4] = '_'; varname[5] = 0;
    for ( i = 0; i < imax; i++ ) {
        t = s = (char *)(setupparameters[i].parameter);
        u = varname+5;
        while ( *s ) { *u++ = (char)(toupper((unsigned char)*s)); s++; }
        *u = 0;
        s = (char *)(getenv(varname));
        if ( s ) {
            error += ProcessOption((UBYTE *)t,(UBYTE *)s,2);
        }
    }
    return(error);
}

/*
        #] TryEnvironment :
    #] Setups :
*/
```

4.126 smart.c File Reference

`#include "form3.h"`

Include dependency graph for smart.c:



Functions

- int StudyPattern (WORD ∗lhs)
- int MatchIsPossible (WORD ∗pattern, WORD ∗term)

4.126.1 Detailed Description

The functions for smart pattern searches in combinations of functions. When many wildcards are involved and the functions are (anti)symmetric an exhaustive search for all possibilities may take very much time (like factorial in the number of wildcards) while a human can often see immediately that there cannot be a match. The routines here try to make FORM a bit smarter in this respect.

This is just the beginning. It still needs lots of work!

Definition in file smart.c.

4.126.2 Function Documentation

4.126.2.1 StudyPattern()

```
int StudyPattern (
            WORD * lhs )
```

Definition at line 62 of file smart.c.

4.126.2.2 MatchIsPossible()

```
int MatchIsPossible (
            WORD * pattern,
            WORD * term )
```

Definition at line 299 of file smart.c.

4.127 smart.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
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 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : function.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ StudyPattern :

    Argument is a complete lhs of an id-statement
    Its last word is a zero (new convention(18-may-1997)) to indicate
    that no extra information is following. We can add there information
    about the pattern that will help during the actual pattern matching.
    Currently the WorkPointer points to just after this lhs.

    Task of this routine: To study the functions, their symmetry properties
    and their wildcards to determine in which order the functions can
    be matched best. If the order should be different we can change it here.

    Problem encountered 22-dec-2008 (JV): we don't take noncommuting
    functions into account.
```

```
*/

int StudyPattern(WORD *lhs)
{
    GETIDENTITY
    WORD *fullproto, *pat, *p, *p1, *p2, *pstop, *info, f, nn;
    int numfun = 0, numsym = 0, allwilds = 0, i, j, k, nc;
    FUN_INFO *finf, *fmin, *f1, *f2, funscratch;

    fullproto = lhs + IDHEAD;
/*  if ( *lhs == TYPEIF ) fullproto--; */
    pat = fullproto + fullproto[1];
    info = pat + *pat;

    p = pat + 1;
    while ( p < info ) {
        if ( *p >= FUNCTION ) {
            numfun++;
            nn = *p - FUNCTION;
            if ( nn >= WILDOFFSET ) nn -= WILDOFFSET;
/*
            We check here for cases that are not allowed like ?a inside
            symmetric functions or tensors.
*/
            if ( ( functions[nn].symmetric == SYMMETRIC ) ||
                 ( functions[nn].symmetric == ANTISYMMETRIC ) ) {
              p2 = p+p[1]; p1 = p+FUNHEAD;
              if ( functions[nn].spec > 0 ) {
                while ( p1 < p2 ) {
                    if ( *p1 == FUNNYWILD ) {
                        MesPrint("&Argument field wildcards are not allowed inside (anti)symmetric functions or tensors");
                        return(1);
                    }
                    p1++;
                }
              }
              else {
                while ( p1 < p2 ) {
                    if ( *p1 == -ARGWILD ) {
                        MesPrint("&Argument field wildcards are not allowed inside (anti)symmetric functions or tensors");
                        return(1);
                    }
                    NEXTARG(p1);
                }
              }
            }
        }
        p += p[1];
    }
    if ( numfun == 0 ) return(0);
    if ( ( lhs[2] & SUBMASK ) == SUBALL ) {
        p = pat + 1;
        while ( p < info ) {
            if ( *p == SYMBOL || *p == VECTOR || *p == DOTPRODUCT || *p == INDEX ) {
                MesPrint("&id,all can have only functions and/or tensors in the lhs.");
                return(1);
            }
            p += p[1];
        }
    }
/*
    We need now some room for the information about the functions
*/
    if ( numfun > AN.numfuninfo ) {
        if ( AN.FunInfo ) M_free(AN.FunInfo,"funinfo");
        AN.numfuninfo = numfun + 10;
        AN.FunInfo = (FUN_INFO *)Malloc1(AN.numfuninfo*sizeof(FUN_INFO),"funinfo");
    }
/*
    Now collect the information. First the locations.
*/
    p = pat + 1; i = 0;
    while ( p < info ) {
        if ( *p >= FUNCTION ) AN.FunInfo[i++].location = p;
        p += p[1];
    }
    for ( i = 0, finf = AN.FunInfo; i < numfun; i++, finf++ ) {
        p = finf->location;
        pstop = p + p[1];
        f = *p;
        if ( f > FUNCTION+WILDOFFSET ) f -= WILDOFFSET;
        finf->numargs = finf->numfunnies = finf->numwildcards = 0;
        finf->symmet = functions[f-FUNCTION].symmetric;
        finf->tensor = functions[f-FUNCTION].spec;
        finf->commute = functions[f-FUNCTION].commute;
```

```
        if ( finf->tensor >= TENSORFUNCTION ) {
            p += FUNHEAD;
            while ( p < pstop ) {
                if ( *p == FUNNYWILD ) {
                    finf->numfunnies++; p+= 2; continue;
                }
                else if ( *p < 0 ) {
                    if ( *p >= AM.OffsetVector + WILDOFFSET && *p < MINSPEC ) {
                        finf->numwildcards++;
                    }
                }
                else {
                    if ( *p >= AM.OffsetIndex + WILDOFFSET &&
                    *p <= AM.OffsetIndex + 2*WILDOFFSET ) finf->numwildcards++;
                }
                finf->numargs++;
                p++;
            }
        }
        else {
            p += FUNHEAD;
            while ( p < pstop ) {
                if ( *p > 0 ) { finf->numargs++; p += *p; continue; }
                if ( *p <= -FUNCTION ) {
                    if ( *p <= -FUNCTION - WILDOFFSET ) finf->numwildcards++;
                    p++;
                }
                else if ( *p == -SYMBOL ) {
                    if ( p[1] >= 2*MAXPOWER ) finf->numwildcards++;
                    p += 2;
                }
                else if ( *p == -INDEX ) {
                    if ( p[1] >= AM.OffsetIndex + WILDOFFSET &&
                    p[1] <= AM.OffsetIndex + 2*WILDOFFSET ) finf->numwildcards++;
                    p += 2;
                }
                else if ( *p == -VECTOR || *p == -MINVECTOR ) {
                    if ( p[1] >= AM.OffsetVector + WILDOFFSET && p[1] < MINSPEC ) {
                        finf->numwildcards++;
                    }
                    p += 2;
                }
                else if ( *p == -ARGWILD ) {
                    finf->numfunnies++;
                    p += 2;
                }
                else { p += 2; }
                finf->numargs++;
            }
        }
        if ( finf->symmet ) {
            numsym++;
            allwilds += finf->numwildcards + finf->numfunnies;
        }
    }
    if ( numsym == 0 ) return(0);
    if ( allwilds == 0 ) return(0);
/*
    We have the information in the array AN.FunInfo.
    We sort things and then write the sorted pattern.
    Of course we may not play with the order of the noncommuting functions.
    Of course we have to become even smarter in the future and look during
    the sorting which wildcards are assigned when.
    But for now this should do.
*/
    for ( nc = numfun-1; nc >= 0; nc-- ) { if ( AN.FunInfo[nc].commute ) break; }

    finf = AN.FunInfo;
    for ( i = nc+2; i < numfun; i++ ) {
        fmin = finf; finf++;
        if ( ( finf->symmet < fmin->symmet ) || (
        ( finf->symmet == fmin->symmet ) &&
        ( ( finf->numwildcards+finf->numfunnies < fmin->numwildcards+fmin->numfunnies )
        || ( ( finf->numwildcards+finf->numfunnies == fmin->numwildcards+fmin->numfunnies )
        && ( finf->numwildcards < fmin->numfunnies ) ) ) ) ) {
            funscratch = AN.FunInfo[i];
            AN.FunInfo[i] = AN.FunInfo[i-1];
            AN.FunInfo[i-1] = funscratch;
            for ( j = i-1; j > nc && j > 0; j-- ) {
                f1 = AN.FunInfo+j;
                f2 = f1-1;
                if ( ( f1->symmet < f2->symmet ) || (
                ( f1->symmet == f2->symmet ) &&
                ( ( f1->numwildcards+f1->numfunnies < f2->numwildcards+f2->numfunnies )
                || ( ( f1->numwildcards+f1->numfunnies == f2->numwildcards+f2->numfunnies )
                && ( f1->numwildcards < f2->numfunnies ) ) ) ) ) {
                    funscratch = AN.FunInfo[j];
```

```
                    AN.FunInfo[j] = AN.FunInfo[j-1];
                    AN.FunInfo[j-1] = funscratch;
                }
                else break;
            }
        }
    }
/*
    Now we rewrite the pattern. First into the space after it and then we
    copy it back. Be careful with the non-commutative functions. There the
    worst one should decide.
*/
    p = pat + 1;
    p2 = info;
    for ( i = 0; i < numfun; i++ ) {
        if ( i == nc ) {
            for ( k = 0; k <= nc; k++ ) {
                if ( AN.FunInfo[k].commute ) {
                    p1 = AN.FunInfo[k].location; j = p1[1];
                    NCOPY(p2,p1,j)
                }
            }
        }
        else if ( AN.FunInfo[i].commute == 0 ) {
            p1 = AN.FunInfo[i].location; j = p1[1];
            NCOPY(p2,p1,j)
        }
    }
    p = pat + 1; p1 = info;
    while ( p1 < p2 ) *p++ = *p1++;
/*
    And now we place the relevant information in the info part
*/
    p2 = info+1;
    for ( i = 0; i < numfun; i++ ) {
        if ( i == nc ) {
            for ( k = 0; k <= nc; k++ ) {
                if ( AN.FunInfo[k].commute ) {
                    finf = AN.FunInfo + k;
                    *p2++ = finf->numargs;
                    *p2++ = finf->numwildcards;
                    *p2++ = finf->numfunnies;
                    *p2++ = finf->symmet;
                }
            }
        }
        else if ( AN.FunInfo[i].commute == 0 ) {
            finf = AN.FunInfo + i;
            *p2++ = finf->numargs;
            *p2++ = finf->numwildcards;
            *p2++ = finf->numfunnies;
            *p2++ = finf->symmet;
        }
    }
    *info = p2-info;
    lhs[1] = p2-lhs;
    return(0);
}

/*
    #] StudyPattern :
    #[ MatchIsPossible :

    We come here when there are functions and there is nontrivial
    symmetry related wildcarding.
*/

int MatchIsPossible(WORD *pattern, WORD *term)
{
    GETIDENTITY
    WORD *info = pattern + *pattern;
    WORD *t, *tstop, *tt, *inf, *p;
    int numfun = 0, inpat, i, j, numargs;
    FUN_INFO *finf;
/*
    We need a list of functions and their number of arguments
*/
    GETSTOP(term,tstop);
    t = term + 1;
    while ( t < tstop ) {
        if ( *t >= FUNCTION ) numfun++;
        t += t[1];
    }
    if ( numfun == 0 ) goto NotPossible;
    if ( *info == SETSET ) info += info[1];
    inpat = (*info-1)/4;
    if ( inpat > numfun ) goto NotPossible;
```

```
/*
    We need now some room for the information about the functions
*/
    if ( numfun > AN.numfuninfo ) {
        if ( AN.FunInfo ) M_free(AN.FunInfo,"funinfo");
        AN.numfuninfo = numfun + 10;
        AN.FunInfo = (FUN_INFO *)Malloc1(AN.numfuninfo*sizeof(FUN_INFO),"funinfo");
    }
    t = term + 1; finf = AN.FunInfo;
    while ( t < tstop ) {
        if ( *t >= FUNCTION ) {
            finf->location = t;
            if ( functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
                numargs = t[1]-FUNHEAD;
                t += t[1];
            }
            else {
                numargs = 0;
                tt = t + t[1];
                t += FUNHEAD;
                while ( t < tt ) {
                    numargs++;
                    NEXTARG(t)
                }
            }
            finf->numargs = numargs;
            finf++;
        }
        else t += t[1];
    }
/*
    Now we first find a partner for each function in the pattern
    with a fixed number of arguments
*/
    for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
        if ( inf[2] ) continue;
        if ( *p >= (FUNCTION+WILDOFFSET) ) continue;
        for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
            if ( *p == *(finf->location) && *inf == finf->numargs ) {
                finf->numargs = -finf->numargs-1;
                break;
            }
        }
        if ( j >= numfun ) goto NotPossible;
    }
    for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
        if ( inf[2] ) continue;
        if ( *p < (FUNCTION+WILDOFFSET) ) continue;
        for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
            if ( *inf == finf->numargs ) {
                finf->numargs = -finf->numargs-1;
                break;
            }
        }
        if ( j >= numfun ) goto NotPossible;
    }
    for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
        if ( inf[2] == 0 ) continue;
        if ( *p >= (FUNCTION+WILDOFFSET) ) continue;
        for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
            if ( *p == *(finf->location) && (*inf-inf[2]) <= finf->numargs ) {
                finf->numargs = -finf->numargs-1;
                break;
            }
        }
        if ( j >= numfun ) goto NotPossible;
    }
    for ( i = 0, inf = info+1, p = pattern+1; i < inpat; i++, inf += 4, p+=p[1] ) {
        if ( inf[2] == 0 ) continue;
        if ( *p < (FUNCTION+WILDOFFSET) ) continue;
        for ( j = 0, finf = AN.FunInfo; j < numfun; j++, finf++ ) {
            if ( (*inf-inf[2]) <= finf->numargs ) {
                finf->numargs = -finf->numargs-1;
                break;
            }
        }
        if ( j >= numfun ) goto NotPossible;
    }
/*
    Thus far we have determined that for each function in the pattern
    there is a potential partner with enough arguments.
    For now the rest is up to the real pattern matcher.

    To undo the disabling of the number of arguments we need this code
    for ( i = 0, finf = AN.FunInfo; i < numfun; i++, finf++ ) {
        if ( finf->numargs < 0 ) finf->numargs = -finf->numargs-1;
    }
```

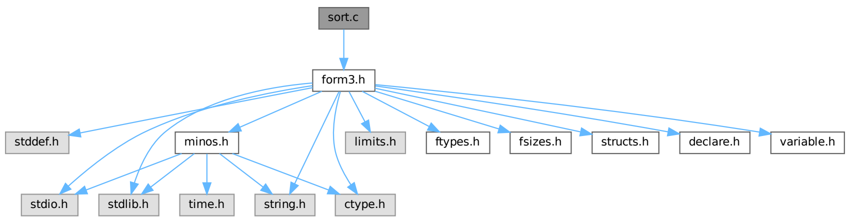
```
*/
    return(1);
NotPossible:
/*
    PrintTerm(pattern,"p");
    PrintTerm(term,"t");
*/
    return(0);
}

/*
    #] MatchIsPossible :
*/
```

4.128 sort.c File Reference

`#include "form3.h"`

Include dependency graph for sort.c:



Macros

- #define NEWSPLITMERGE
- #define NEWSPLITMERGETIMSORT
- #define HUMANSTRLEN 12
- #define HUMANSUFFLEN 4
- #define INSLENGTH(x) w[1] = FUNHEAD+ARGHEAD+x; w[FUNHEAD] = ARGHEAD+x;

Functions

- void HumanString (char ∗string, float input, const char suffix[HUMANSUFFLEN][4])
- void WriteStats (POSITION ∗plspace, WORD par, WORD checkLogType)
- WORD NewSort (PHEAD0)
- LONG EndSort (PHEAD WORD ∗buffer, int par)
- LONG PutIn (FILEHANDLE ∗file, POSITION ∗position, WORD ∗buffer, WORD ∗∗take, int npat)
- WORD Sflush (FILEHANDLE ∗fi)
- WORD PutOut (PHEAD WORD ∗term, POSITION ∗position, FILEHANDLE ∗fi, WORD ncomp)
- WORD FlushOut (POSITION ∗position, FILEHANDLE ∗fi, int compr)
- WORD AddCoef (PHEAD WORD ∗∗ps1, WORD ∗∗ps2)
- WORD AddPoly (PHEAD WORD ∗∗ps1, WORD ∗∗ps2)
- void AddArgs (PHEAD WORD ∗s1, WORD ∗s2, WORD ∗m)
- WORD Compare1 (PHEAD WORD ∗term1, WORD ∗term2, WORD level)
- WORD CompareSymbols (PHEAD WORD ∗term1, WORD ∗term2, WORD par)
- WORD CompareHSymbols (PHEAD WORD ∗term1, WORD ∗term2, WORD par)
- LONG ComPress (WORD ∗∗ss, LONG ∗n)

- LONG SplitMerge (PHEAD WORD **Pointer, LONG number)
- void GarbHand (void)
- WORD MergePatches (WORD par)
- WORD StoreTerm (PHEAD WORD *term)
- void StageSort (FILEHANDLE *fout)
- WORD SortWild (WORD *w, WORD nw)
- void CleanUpSort (int num)
- void LowerSortLevel (void)
- WORD * PolyRatFunSpecial (PHEAD WORD *t1, WORD *t2)
- void SimpleSplitMergeRec (WORD *array, WORD num, WORD *auxarray)
- void SimpleSplitMerge (WORD *array, WORD num)
- WORD BinarySearch (WORD *array, WORD num, WORD x)

Variables

- LONG numcompares
- char * toterms [ ] = { " ", " >>", "-->" }
- const char humanTermsSuffix [HUMANSUFFLEN][4] = {"K ","M ","B ","T "}
- const char humanBytesSuffix [HUMANSUFFLEN][4] = {"KiB","MiB","GiB","TiB"}

4.128.1 Detailed Description

Contains the sort routines. We distinguish levels of sorting. The ground level is the sorting of terms in an expression. When a term has functions, the arguments can contain terms that need sorting, which this then done by raising the level. This can happen recursively. NewSort and EndSort automatically raise and lower the level. Because the ground level does some special things, sometimes we have to raise immediately to the second level skipping the ground level.

Special routines for the parallel sorting are in the file threads.c Also the sorting of terms in polynomials is special but most of that is controlled by changing the address of the compare routine. Other routines relevant for adding rational polynomials are in the file polynito.c

Definition in file sort.c.

4.128.2 Macro Definition Documentation

4.128.2.1 NEWSPLITMERGE

```
#define NEWSPLITMERGE
```

Definition at line 53 of file sort.c.

4.128.2.2 NEWSPLITMERGETIMSORT

```
#define NEWSPLITMERGETIMSORT
```

Definition at line 55 of file sort.c.

4.128.2.3 HUMANSTRLEN

```
#define HUMANSTRLEN 12
```

Definition at line 85 of file sort.c.

4.128.2.4 HUMANSUFFLEN

```
#define HUMANSUFFLEN 4
```

Definition at line 86 of file sort.c.

4.128.2.5 INSLENGTH

```
#define INSLENGTH(
             x ) w[1] = FUNHEAD+ARGHEAD+x; w[FUNHEAD] = ARGHEAD+x;
```

Definition at line 2334 of file sort.c.

4.128.3 Function Documentation

4.128.3.1 HumanString()

```
void HumanString (
            char * string,
            float input,
            const char suffix[HUMANSUFFLEN][4] )
```

Definition at line 89 of file sort.c.

4.128.3.2 WriteStats()

```
void WriteStats (
            POSITION * plspace,
            WORD par,
            WORD checkLogType )
```

Writes the statistics.

Parameters

plspace	The size in bytes currently occupied
par	par = 0 = STATSSPLITMERGE after a splitmerge. par = 1 = STATSMERGETOFILE after merge to sortfile. par = 2 = STATSPOSTSORT after the sort
checkLogType	== CHECKLOGTYPE: The output should not go to the log file if AM.LogType is 1.

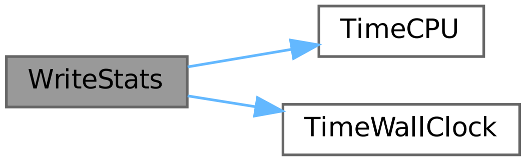
current expression is to be found in AR.CurExpr. terms are in S->TermsLeft. S->GenTerms.

Definition at line 121 of file sort.c.

References TimeCPU(), and TimeWallClock().

Referenced by EndSort(), PF_Processor(), and StoreTerm().

Here is the call graph for this function:



4.128.3.3 NewSort()

```
WORD NewSort (
            PHEAD0 )
```

Starts a new sort. At the lowest level this is a 'main sort' with the struct according to the parameters in S0. At higher levels this is a sort for functions, subroutines or dollars. We prepare the arrays and structs.

Returns

Regular convention (OK -> 0)

Definition at line 643 of file sort.c.

Referenced by generate_expression(), get_expression(), InFunction(), MakeDollarInteger(), MakeDollarMod(), PF_CollectModifiedDollars(), PF_Processor(), poly_factorize_expression(), poly_sort(), poly_unfactorize_expression(), PolyFunMul(), Processor(), TakeArgContent(), and TestMatch().

4.128.3.4 EndSort()

```
LONG EndSort (
            PHEAD WORD * buffer,
            int par )
```

Finishes a sort. At AR.sLevel == 0 the output is to the regular output stream. When AR.sLevel > 0, the parameter par determines the actual output. The AR.sLevel will be popped. All ongoing stages are finished and if the sortfile is open it is closed. The statistics are printed when AR.sLevel == 0 par == 0 Output to the buffer. par == 1 Sort for function arguments. The output will be copied into the buffer. It is assumed that this is in the WorkSpace. par == 2 Sort for $-variable. We return the address of the buffer that contains the output in buffer (treated like WORD **). We first catch the output in a file (unless we can intercept things after the small buffer has been sorted) Then we read from the file into a buffer. Only when par == 0 data compression can be attempted at AT.SS==AT.S0.

Parameters

buffer	buffer for output when needed
par	See above

Returns

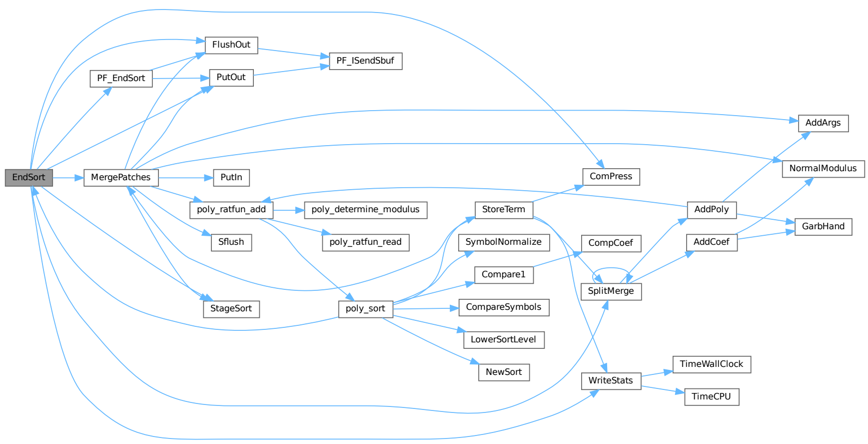
If negative: error. If positive: number of words in output.

Definition at line 733 of file sort.c.

References ComPress(), FlushOut(), MergePatches(), PF_EndSort(), PutOut(), SplitMerge(), StageSort(), and WriteStats().

Referenced by generate_expression(), get_expression(), InFunction(), MakeDollarInteger(), MakeDollarMod(), PF_CollectModifiedDollars(), PF_Processor(), poly_factorize_expression(), poly_sort(), poly_unfactorize_expression(), PolyFunMul(), Processor(), TakeArgContent(), and TestMatch().

Here is the call graph for this function:



4.128.3.5 PutIn()

```
LONG PutIn (
            FILEHANDLE * file,
            POSITION * position,
            WORD * buffer,
            WORD ** take,
            int npat )
```

Reads a new patch from position in file handle. It is put at buffer, anything after take is moved forward. This would be part of a term that hasn't been used yet. Because of this there should be some space before the start of the buffer

Parameters

file	The file system from which to read
position	The position from which to read
buffer	The buffer into which to read
take	The unused tail should be moved before the buffer
npat	The number of the patch. Is needed if the information was compressed with gzip, because each patch has its own independent gzip encoding.

Definition at line 1320 of file sort.c.

References FiLe::handle.

Referenced by MergePatches().

4.128.3.6 Sflush()

```
WORD Sflush (
            FILEHANDLE * fi )
```

Puts the contents of a buffer to output Only to be used when there is a single patch in the large buffer.

Parameters

fi	The filesystem (or its cache) to which the patch should be written

Definition at line 1380 of file sort.c.

References FiLe::handle.

Referenced by MergePatches().

4.128.3.7 PutOut()

```
WORD PutOut (
            PHEAD WORD * term,
            POSITION * position,
            FILEHANDLE * fi,
            WORD ncomp )
```

Routine writes one term to file handle at position. It returns the new value of the position.

NOTE: For 'final output' we may have to index the brackets. See the struct BRACKETINDEX. We should maintain: 1: a list with brackets array with the brackets 2: a list of objects of type BRACKETINDEX. It contains array with either pointers or offsets to the list of brackets. starting positions in the file. The index may be tied to a maximum size. In that case we may have to prune the list occasionally.

Parameters

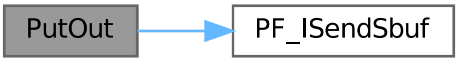
term	The term to be written
position	The position in the file. Afterwards it is updated
fi	The file (or its cache) to which should be written
ncomp	Information about what type of compression should be used

Definition at line 1466 of file sort.c.

References FiLe::handle, and PF_ISendSbuf().

Referenced by EndSort(), generate_expression(), MergePatches(), PF_EndSort(), PF_Processor(), and Processor().

Here is the call graph for this function:



4.128.3.8 FlushOut()

```
WORD FlushOut (
            POSITION * position,
            FILEHANDLE * fi,
            int compr )
```

Completes output to an output file and writes the trailing zero.

Parameters

position	The position in the file after writing
fi	The file (or its cache)
compr	Indicates whether there should be compression with gzip.

Returns

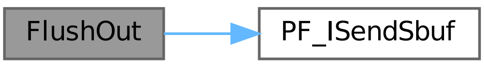
Regular conventions (OK -> 0).

Definition at line 1828 of file sort.c.

References FiLe::handle, BrAcKeTiNfO::indexbuffer, and PF_ISendSbuf().

Referenced by EndSort(), MergePatches(), PF_EndSort(), and Processor().

Here is the call graph for this function:



4.128.3.9 AddCoef()

```
WORD AddCoef (
            PHEAD WORD ** ps1,
            WORD ** ps2 )
```

Adds the coefficients of the terms ∗ps1 and ∗ps2. The problem comes when there is not enough space for a new longer coefficient. First a local solution is tried. If this is not successful we need to move terms around. The possibility of a garbage collection should not be ignored, as avoiding this costs very much extra space which is nearly wasted otherwise.

If the return value is zero the terms cancelled.

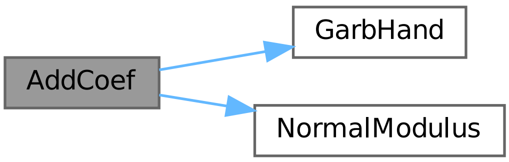
The resulting term is left in ∗ps1.

Definition at line 2042 of file sort.c.

References GarbHand(), and NormalModulus().

Referenced by SplitMerge().

Here is the call graph for this function:



4.128.3.10 AddPoly()

```
WORD AddPoly (
            PHEAD WORD ** ps1,
            WORD ** ps2 )
```

Routine should be called when S->PolyWise != 0. It points then to the position of AR.PolyFun in both terms.

We add the contents of the arguments of the two polynomials. Special attention has to be given to special arguments. We have to reserve a space equal to the size of one term + the size of the argument of the other. The addition has to be done in this routine because not all objects are reentrant.

Newer addition (12-nov-2007). The PolyFun can have two arguments. In that case S->PolyFlag is 2 and we have to call the routine for adding rational polynomials. We have to be rather careful what happens with: The location of the output The order of the terms in the arguments At first we allow only univariate polynomials in the PolyFun. This restriction will be lifted a.s.a.p.

Parameters

ps1	A pointer to the position of the first term
ps2	A pointer to the position of the second term

Returns

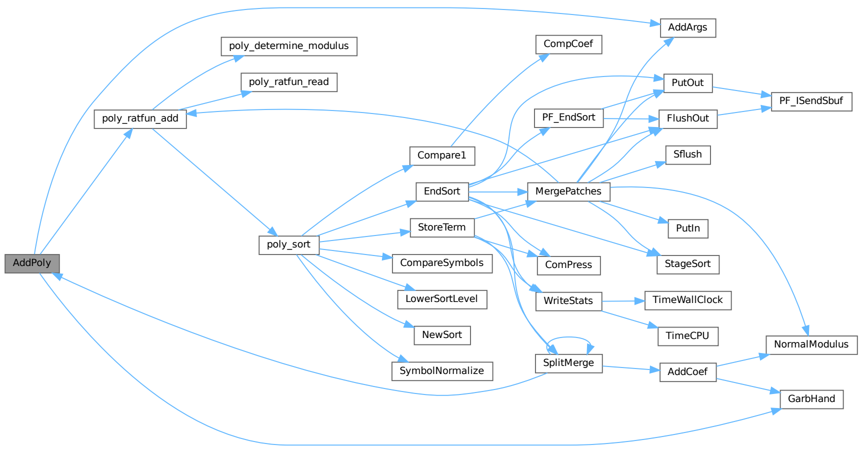
If zero the terms cancel. Otherwise the new term is in ∗ps1.

Definition at line 2171 of file sort.c.

References AddArgs(), GarbHand(), and poly_ratfun_add().

Referenced by SplitMerge().

Here is the call graph for this function:



4.128.3.11 AddArgs()

```
void AddArgs (
            PHEAD WORD * s1,
            WORD * s2,
            WORD * m )
```

Adds the arguments of two occurrences of the PolyFun.

Parameters

s1	Pointer to the first occurrence.
s2	Pointer to the second occurrence.
m	Pointer to where the answer should be.

Definition at line 2343 of file sort.c.

Referenced by AddPoly(), and MergePatches().

4.128.3.12 Compare1()

```
WORD Compare1 (
            PHEAD WORD * term1,
            WORD * term2,
            WORD level )
```

Compares two terms. The answer is: 0 equal (with exception of the coefficient if level == 0.) >0 term1 comes first. <0 term2 comes first. Some special precautions may be needed to keep the CompCoef routine from generating overflows, although this is very unlikely in subterms. This routine should not return an error condition.

Originally this routine was called Compare. With the treatment of special polynomials with terms that contain only symbols and the need for extreme speed for the polynomial routines we made a special compare routine and now we store the address of the current compare routine in AR.CompareRoutine and have a macro Compare which makes all existing code work properly and we can just replace the routine on a thread by thread basis (each thread has its own AR struct).

Parameters

term1	First input term
term2	Second input term
level	The sorting level (may influence on the result)

Returns

0 equal (with exception of the coefficient if level == 0.) >0 term1 comes first. <0 term2 comes first.

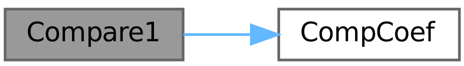
When there are floating point numbers active (float_ = FLOATFUN) the presence of one or more float_ functions is returned in AT.SortFloatMode: 0: no float_ 1: float_ in term1 only 2: float_ in term2 only 3: float_ in both terms

Definition at line 2636 of file sort.c.

References CompCoef().

Referenced by InFunction(), poly_sort(), and StartVariables().

Here is the call graph for this function:



4.128.3.13 CompareSymbols()

```
WORD CompareSymbols (
            PHEAD WORD * term1,
            WORD * term2,
            WORD par )
```

Compares the terms, based on the value of AN.polysortflag. If term1 $<$ term2 the return value is -1 If term1 $>$ term2 the return value is 1 If term1 = term2 the return value is 0 The coefficients may differ. The terms contain only a single subterm of type SYMBOL. If AN.polysortflag = 0 it is a 'regular' compare. If AN.polysortflag = 1 the sum of the powers is more important par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in sort.c

Definition at line 3100 of file sort.c.

Referenced by InFunction(), and poly_sort().

4.128.3.14 CompareHSymbols()

```
WORD CompareHSymbols (
            PHEAD WORD * term1,
            WORD * term2,
            WORD par )
```

Compares terms that can have only SYMBOL and HAAKJE subterms. If term1 $<$ term2 the return value is -1 If term1 $>$ term2 the return value is 1 If term1 = term2 the return value is 0 par is a dummy parameter to make the parameter field identical to that of Compare1 which is the regular compare routine in sort.c

Definition at line 3143 of file sort.c.

4.128.3.15 ComPress()

```
LONG ComPress (
            WORD ** ss,
            LONG * n )
```

Gets a list of pointers to terms and compresses the terms. In n it collects the number of terms and the return value of the function is the space that is occupied.

We have to pay some special attention to the compression of terms with a PolyFun. This PolyFun should occur only straight before the coefficient, so we can use the same trick as for the coefficient to sabotage compression of this object (Replace in the history the function pointer by zero. This is safe, because terms that would be identical otherwise would have been added).

Parameters

ss	Array of pointers to terms to be compressed.
n	Number of pointers in ss.

Returns

Total number of words needed for the compressed result.

Definition at line 3196 of file sort.c.

Referenced by EndSort(), and StoreTerm().

4.128.3.16 SplitMerge()

```
LONG SplitMerge (
            PHEAD WORD ** Pointer,
            LONG number )
```

Algorithm by J.A.M.Vermaseren (31-7-1988)

Note that AN.SplitScratch and AN.InScratch are used also in GarbHand

Merge sort in memory. The input is an array of pointers. Sorting is done recursively by dividing the array in two equal parts and calling SplitMerge for each. When the parts are small enough we can do the compare and take the appropriate action. An addition is that we look for 'runs'. Sequences that are already ordered. This happens a lot when there is very little action in a module. This made FORM faster by a few percent.

Parameters

Pointer	The array of pointers to the terms to be sorted.
number	The number of pointers in Pointer.

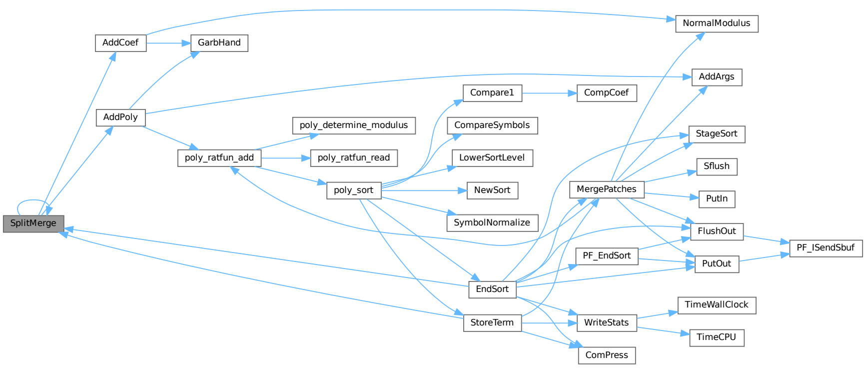
The terms are supposed to be sitting in the small buffer and there is supposed to be an extension to this buffer for when there are two terms that should be added and the result takes more space than each of the original terms. The notation guarantees that the result never needs more space than the sum of the spaces of the original terms.

Definition at line 3362 of file sort.c.

References AddCoef(), AddPoly(), and SplitMerge().

Referenced by EndSort(), SplitMerge(), and StoreTerm().

Here is the call graph for this function:



4.128.3.17 GarbHand()

```
void GarbHand (
            void )
```

Garbage collection that takes place when the small extension is full and we need to place more terms there. When this is the case there are many holes in the small buffer and the whole can be compactified. The major complication is the buffer for SplitMerge. There are to options for temporary memory: 1: find some buffer that has enough space (maybe in the large buffer). 2: allocate a buffer. Give it back afterwards of course. If the small extension is properly dimensioned this routine should be called very rarely. Most of the time it will be called when the polyfun or polyratfun is active.

Definition at line 3642 of file sort.c.

Referenced by AddCoef(), and AddPoly().

4.128.3.18 MergePatches()

```
WORD MergePatches (
            WORD par )
```

The general merge routine. Can be used for the large buffer and the file merging. The array S->Patches tells where the patches start S->pStop tells where they end (has to be computed first). The end of a 'line to be merged' is indicated by a zero. If the end is reached without running into a zero or a term runs over the boundary of a patch it is a file merging operation and a new piece from the file is read in.

Parameters

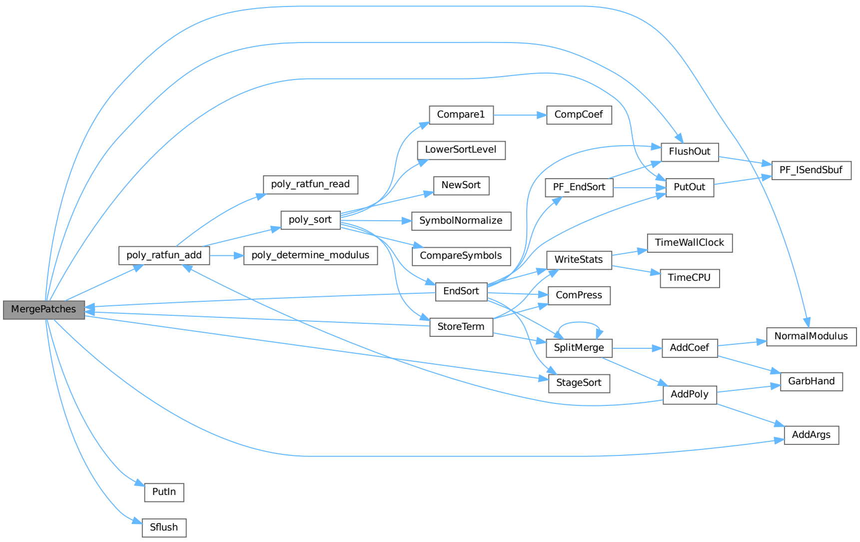
par	If par == 0 the sort is for file -> outputfile. If par == 1 the sort is for large buffer -> sortfile. If par == 2 the sort is for large buffer -> outputfile.

Definition at line 3757 of file sort.c.

References AddArgs(), FlushOut(), FiLe::handle, NormalModulus(), poly_ratfun_add(), PutIn(), PutOut(), Sflush(), and StageSort().

Referenced by EndSort(), and StoreTerm().

Here is the call graph for this function:



4.128.3.19 StoreTerm()

```
WORD StoreTerm (
            PHEAD WORD * term )
```

The central routine to accept terms, store them and keep things at least partially sorted. A call to EndSort will then complete storing and sorting.

Parameters

term	The term to be stored

Returns

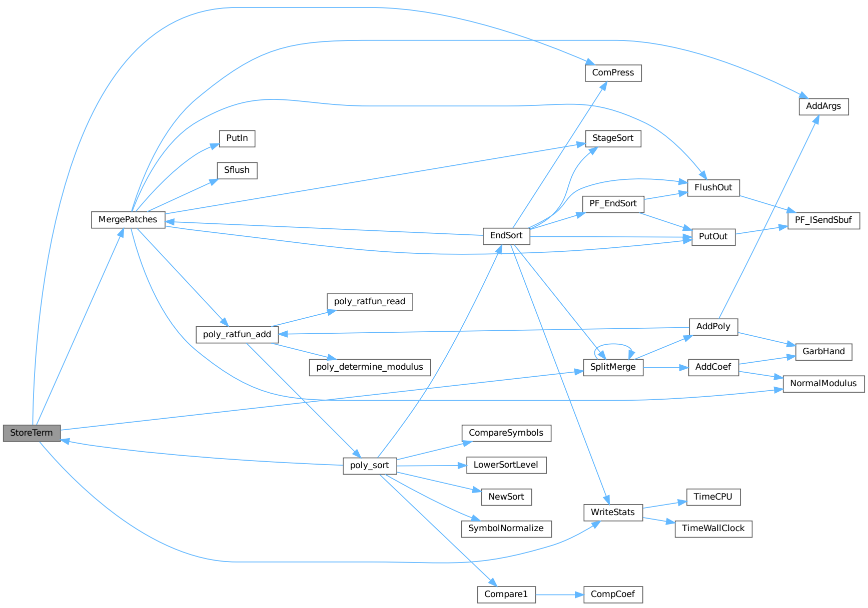
Regular return conventions (OK -> 0)

Definition at line 4540 of file sort.c.

References ComPress(), MergePatches(), SplitMerge(), and WriteStats().

Referenced by Generator(), get_expression(), InFunction(), PF_Processor(), poly_sort(), PolyFunMul(), Processor(), and TakeArgContent().

Here is the call graph for this function:



4.128.3.20 StageSort()

```
void StageSort (
            FILEHANDLE * fout )
```

Prepares a stage 4 or higher sort. Stage 4 sorts occur when the sort file contains more patches than can be merged in one pass.

Definition at line 4654 of file sort.c.

References FiLe::handle.

Referenced by EndSort(), and MergePatches().

4.128.3.21 SortWild()

```
WORD SortWild (
            WORD * w,
            WORD nw )
```

Sorts the wildcard entries in the parameter w. Double entries are removed. Full space taken is nw words. Routine serves for the reading of wildcards in the compiler. The entries come in the format: (type,4,number,0) in which the zero is reserved for the future replacement of 'number'.

Parameters

w	buffer with wildcard entries.
nw	number of wildcard entries.

Returns

Normal conventions (OK -> 0)

Definition at line 4753 of file sort.c.

4.128.3.22 CleanUpSort()

```
void CleanUpSort (
            int num )
```

Partially or completely frees function sort buffers.

Definition at line 4845 of file sort.c.

4.128.3.23 LowerSortLevel()

```
void LowerSortLevel (
            void )
```

Lowers the level in the sort system.

Definition at line 4945 of file sort.c.

Referenced by generate_expression(), get_expression(), InFunction(), PF_CollectModifiedDollars(), PF_Processor(), poly_factorize_expression(), poly_sort(), poly_unfactorize_expression(), PolyFunMul(), Processor(), and TestMatch().

4.128.3.24 PolyRatFunSpecial()

```
WORD * PolyRatFunSpecial (
            PHEAD WORD * t1,
            WORD * t2 )
```

Definition at line 4962 of file sort.c.

4.128.3.25 SimpleSplitMergeRec()

```
void SimpleSplitMergeRec (
            WORD * array,
            WORD num,
            WORD * auxarray )
```

Definition at line 5064 of file sort.c.

4.128.3.26 SimpleSplitMerge()

```
void SimpleSplitMerge (
            WORD * array,
            WORD num )
```

Definition at line 5092 of file sort.c.

4.128.3.27 BinarySearch()

```
WORD BinarySearch (
            WORD * array,
            WORD num,
            WORD x )
```

Definition at line 5110 of file sort.c.

4.128.4 Variable Documentation

4.128.4.1 numcompares

```
LONG numcompares
```

Definition at line 75 of file sort.c.

4.128.4.2 toterms

```
char* toterms[ ] = { " ", " >>", "-->" }
```

Definition at line 83 of file sort.c.

4.128.4.3 humanTermsSuffix

```
const char humanTermsSuffix[HUMANSUFFLEN][4] = {"K ","M ","B ","T "}
```

Definition at line 87 of file sort.c.

4.128.4.4 humanBytesSuffix

```
const char humanBytesSuffix[HUMANSUFFLEN][4] = {"KiB","MiB","GiB","TiB"}
```

Definition at line 88 of file sort.c.

4.129 sort.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : sort.c

    Sort routines according to new conventions (25-jun-1997).
    This is more object oriented.
    The active sort is indicated by AT.SS which should agree with
    AN.FunSorts[AR.sLevel];

#define GZIPDEBUG
*/
#define NEWSPLITMERGE
/* Comment to turn off Timsort in SplitMerge for debugging: */
#define NEWSPLITMERGETIMSORT
/* During SplitMerge, print pointer array state on entry. Very spammy, for debugging. */
/* #define SPLITMERGEDEBUG */
/* Debug printing for GarbHand */
/* #define TESTGARB */

#include "form3.h"

#ifdef WITHPTHREADS
UBYTE THRbuf[100];
#endif

#ifdef WITHSTATS
extern LONG numwrites;
extern LONG numreads;
extern LONG numseeks;
extern LONG nummallocs;
extern LONG numfrees;
#endif

LONG numcompares;

/*
    #] Includes :
    #[ SortUtilities :
        #[ WriteStats :             void WriteStats(lspace,par,checkLogType)
*/

char *toterms[] = { "   ", " >>", "-->" };

#define HUMANSTRLEN 12
#define HUMANSUFFLEN 4
const char humanTermsSuffix[HUMANSUFFLEN][4] = {"K  ","M  ","B  ","T  "};
const char humanBytesSuffix[HUMANSUFFLEN][4] = {"KiB","MiB","GiB","TiB"};
void HumanString(char* string, float input, const char suffix[HUMANSUFFLEN][4]) {
    int ind = -1;
    while (ind < 0 || (input >= 1000.0 && ind < HUMANSUFFLEN) ) {
        input /= 1000.0;
        ind++;
    }
    if ( input < 0.5 ) {
        snprintf(string, HUMANSTRLEN,
            "   ( <1 %s)", suffix[ind]);
```

```
    }
    else {
        snprintf(string, HUMANSTRLEN,
            "  (%3.f %s)", input, suffix[ind]);
    }
}

void WriteStats(POSITION *plspace, WORD par, WORD checkLogType)
{
    GETIDENTITY
    LONG millitime, y = 0x7FFFFFFFL » 1;
    WORD timepart;
    SORTING *S;
    POSITION pp;
    int use_wtime;
    if ( AT.SS == AT.S0 && AC.StatsFlag ) {
#ifdef WITHPTHREADS
        if ( AC.ThreadStats == 0 && identity > 0 ) return;
#elif defined(WITHMPI)
        if ( AC.OldParallelStats ) return;
        if ( ! AC.ProcessStats && PF.me != MASTER ) return;
#endif
        if ( Expressions == 0 ) return;

        if ( par == STATSSPLITMERGE ) {
            if ( AC.ShortStatsMax == 0 ) return;
            AR.ShortSortCount++;
            if ( AR.ShortSortCount < AC.ShortStatsMax ) return;
        }
        AR.ShortSortCount = 0;

        S = AT.SS;

        char humanGenTermsText[HUMANSTRLEN] = "";
        char humanTermsLeftText[HUMANSTRLEN] = "";
        char humanBytesText[HUMANSTRLEN] = "";
        if ( AC.HumanStatsFlag ) {
            HumanString(humanGenTermsText, (float)(S->GenTerms), humanTermsSuffix);
            HumanString(humanTermsLeftText, (float)(S->TermsLeft), humanTermsSuffix);
            HumanString(humanBytesText, (float)(BASEPOSITION(*plspace)), humanBytesSuffix);
        }

        MLOCK(ErrorMessageLock);

        /* If the statistics should not go to the log file, temporarily hide the
         * LogHandle. This must be done within the ErrorMessageLock region to
         * avoid a data race between threads. */
        const WORD oldLogHandle = AC.LogHandle;
        if ( checkLogType && AM.LogType ) {
            AC.LogHandle = -1;
        }

        if ( AC.ShortStats ) {}
        else {
#ifdef WITHPTHREADS
            if ( identity > 0 ) {
                MesPrint("              Thread %d reporting",identity);
            }
            else {
                MesPrint("");
            }
#elif defined(WITHMPI)
            if ( PF.me != MASTER ) {
                MesPrint("              Process %d reporting",PF.me);
            }
            else {
                MesPrint("");
            }
#else
            MesPrint("");
#endif
        }
        /*
         * We define WTimeStatsFlag as a flag to print the wall-clock time on
         * the *master*, not in workers. This can be confusing in thread
         * statistics when short statistics is used. Technically,
         * TimeWallClock() is not thread-safe in TFORM.
         */
        use_wtime = AC.WTimeStatsFlag;
#if defined(WITHPTHREADS)
        if ( use_wtime && identity > 0 ) use_wtime = 0;
#elif defined(WITHMPI)
        if ( use_wtime && PF.me != MASTER ) use_wtime = 0;
#endif
        millitime = use_wtime ? TimeWallClock(1) * 10 : TimeCPU(1);
        timepart = (WORD)(millitime%1000);
        millitime /= 1000;
```

```
        timepart /= 10;
        if ( AC.ShortStats ) {
#if defined(WITHPTHREADS) || defined(WITHMPI)
#ifdef WITHPTHREADS
          if ( identity > 0 ) {
#else
          if ( PF.me != MASTER ) {
            const int identity = PF.me;
#endif
            if ( par == STATSSPLITMERGE || par == STATSPOSTSORT ) {
                SETBASEPOSITION(pp,y);
                if ( ISLESSPOS(*plspace,pp) ) {
                    MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%10p %s %s",identity,
                    millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                    S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
/*
                    MesPrint("%d: %14s %17s %7l.%2is %8l>%10l%3s%10l:%10p",identity,
                    EXPRNAME(AR.CurExpr),AC.Commercial,millitime,timepart,
                    AN.ninterms,S->GenTerms,toterms[par],S->TermsLeft,plspace);
*/
                }
                else {
                    y = 1000000000L;
                    SETBASEPOSITION(pp,y);
                    MULPOS(pp,100);
                    if ( ISLESSPOS(*plspace,pp) ) {
                        MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%11p %s %s",identity,
                        millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                        S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                    }
                    else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%12p %s %s",identity,
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%13p %s %s",identity,
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%14p %s %s",identity,
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%15p %s %s",identity,
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%16p %s %s",identity,
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %8l>%10l%3s%10l:%17p %s %s",identity,
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        } } } } }
                    }
                }
            }
            else if ( par == STATSMERGETOFILE ) {
                SETBASEPOSITION(pp,y);
                if ( ISLESSPOS(*plspace,pp) ) {
                    MesPrint("%d: %7l.%2is %10l:%10p",identity,millitime,timepart,
                    S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                }
                else {
                    y = 1000000000L;
                    SETBASEPOSITION(pp,y);
                    MULPOS(pp,100);
                    if ( ISLESSPOS(*plspace,pp) ) {
```

```
                        MesPrint("%d: %7l.%2is %10l:%11p",identity,millitime,timepart,
                        S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                    }
                    else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %10l:%12p",identity,millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %10l:%13p",identity,millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %10l:%14p",identity,millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %10l:%15p",identity,millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %10l:%16p",identity,millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%d: %7l.%2is %10l:%17p",identity,millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        } } } } }
                    }
                }
            } } else
#endif
            {
            if ( par == STATSSPLITMERGE || par == STATSPOSTSORT ) {
                SETBASEPOSITION(pp,y);
                if ( ISLESSPOS(*plspace,pp) ) {
                    MesPrint("%7l.%2is %8l>%10l%3s%10l:%10p %s %s",
                    millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                    S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
/*
                    MesPrint("%14s %17s %7l.%2is %8l>%10l%3s%10l:%10p",
                    EXPRNAME(AR.CurExpr),AC.Commercial,millitime,timepart,
                    AN.ninterms,S->GenTerms,toterms[par],S->TermsLeft,plspace);
*/
                }
                else {
                    y = 1000000000L;
                    SETBASEPOSITION(pp,y);
                    MULPOS(pp,100);
                    if ( ISLESSPOS(*plspace,pp) ) {
                        MesPrint("%7l.%2is %8l>%10l%3s%10l:%11p %s %s",
                        millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                        S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                    }
                    else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %8l>%10l%3s%10l:%12p %s %s",
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %8l>%10l%3s%10l:%13p %s %s",
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %8l>%10l%3s%10l:%14p %s %s",
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
```

```
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %8l>%10l%3s%10l:%15p %s %s",
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %8l>%10l%3s%10l:%16p %s %s",
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %8l>%10l%3s%10l:%17p %s %s",
                            millitime,timepart,AN.ninterms,S->GenTerms,toterms[par],
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        } } } } }
                    }
                }
            }
            else if ( par == STATSMERGETOFILE ) {
                SETBASEPOSITION(pp,y);
                if ( ISLESSPOS(*plspace,pp) ) {
                    MesPrint("%7l.%2is %10l:%10p",millitime,timepart,
                    S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                }
                else {
                    y = 1000000000L;
                    SETBASEPOSITION(pp,y);
                    MULPOS(pp,100);
                    if ( ISLESSPOS(*plspace,pp) ) {
                        MesPrint("%7l.%2is %10l:%11p",millitime,timepart,
                        S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                    }
                    else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %10l:%12p",millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %10l:%13p",millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %10l:%14p",millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %10l:%15p",millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %10l:%16p",millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        else {
                        MULPOS(pp,10);
                        if ( ISLESSPOS(*plspace,pp) ) {
                            MesPrint("%7l.%2is %10l:%17p",millitime,timepart,
                            S->TermsLeft,plspace,EXPRNAME(AR.CurExpr),AC.Commercial);
                        }
                        } } } } }
                    }
                }
            }
        } }
        else {
        if ( par == STATSMERGETOFILE ) {
            if ( use_wtime ) {
                MesPrint("WTime = %7l.%2i sec",millitime,timepart);
            }
            else {
                MesPrint("Time = %7l.%2i sec",millitime,timepart);
```

```
            }
        }
        else {
#if ( BITSINLONG > 32 )
            if ( S->GenTerms >= 10000000000L ) {
                if ( use_wtime ) {
                    MesPrint("WTime = %7l.%2i sec   Generated terms = %16l%s",
                        millitime,timepart,S->GenTerms,humanGenTermsText);
                }
                else {
                    MesPrint("Time = %7l.%2i sec    Generated terms = %16l%s",
                        millitime,timepart,S->GenTerms,humanGenTermsText);
                }
            }
            else {
                if ( use_wtime ) {
                    MesPrint("WTime = %7l.%2i sec   Generated terms = %10l%s",
                        millitime,timepart,S->GenTerms,humanGenTermsText);
                }
                else {
                    MesPrint("Time = %7l.%2i sec    Generated terms = %10l%s",
                        millitime,timepart,S->GenTerms,humanGenTermsText);
                }
            }
#else
            if ( use_wtime ) {
                MesPrint("WTime = %7l.%2i sec   Generated terms = %10l%s",
                    millitime,timepart,S->GenTerms,humanGenTermsText);
            }
            else {
                MesPrint("Time = %7l.%2i sec    Generated terms = %10l%s",
                    millitime,timepart,S->GenTerms,humanGenTermsText);
            }
#endif
        }
#if ( BITSINLONG > 32 )
        if ( par == STATSSPLITMERGE )
            if ( S->TermsLeft >= 10000000000L ) {
                MesPrint("%16s%8l Terms %s = %16l%s",EXPRNAME(AR.CurExpr),
                AN.ninterms,FG.swmes[par],S->TermsLeft,humanTermsLeftText);
            }
            else {
                MesPrint("%16s%8l Terms %s = %10l%s",EXPRNAME(AR.CurExpr),
                AN.ninterms,FG.swmes[par],S->TermsLeft,humanTermsLeftText);
            }
        else {
            if ( S->TermsLeft >= 10000000000L ) {
#ifdef WITHPTHREADS
                if ( identity > 0 && par == STATSPOSTSORT ) {
                    MesPrint("%16s         Terms in thread = %16l%s",
                    EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
                }
                else
#elif defined(WITHMPI)
                if ( PF.me != MASTER && par == STATSPOSTSORT ) {
                    MesPrint("%16s         Terms in process= %16l%s",
                    EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
                }
                else
#endif
                {
                    MesPrint("%16s         Terms %s = %16l%s",
                    EXPRNAME(AR.CurExpr),FG.swmes[par],S->TermsLeft,humanTermsLeftText);
                }
            }
            else {
#ifdef WITHPTHREADS
                if ( identity > 0 && par == STATSPOSTSORT ) {
                    MesPrint("%16s         Terms in thread = %10l%s",
                    EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
                }
                else
#elif defined(WITHMPI)
                if ( PF.me != MASTER && par == STATSPOSTSORT ) {
                    MesPrint("%16s         Terms in process= %10l%s",
                    EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
                }
                else
#endif
                {
                    MesPrint("%16s         Terms %s = %10l%s",
                    EXPRNAME(AR.CurExpr),FG.swmes[par],S->TermsLeft,humanTermsLeftText);
                }
            }
        }
#else
        if ( par == STATSSPLITMERGE )
```

```
            MesPrint("%16s%8l Terms %s = %10l%s",EXPRNAME(AR.CurExpr),
            AN.ninterms,FG.swmes[par],S->TermsLeft,humanTermsLeftText);
        else {
#ifdef WITHPTHREADS
            if ( identity > 0 && par == STATSPOSTSORT ) {
                MesPrint("%16s         Terms in thread = %10l%s",
                EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
            }
            else
#elif defined(WITHMPI)
            if ( PF.me != MASTER && par == STATSPOSTSORT ) {
                MesPrint("%16s         Terms in process= %10l%s",
                EXPRNAME(AR.CurExpr),S->TermsLeft,humanTermsLeftText);
            }
            else
#endif
            {
                MesPrint("%16s         Terms %s = %10l%s",
                EXPRNAME(AR.CurExpr),FG.swmes[par],S->TermsLeft,humanTermsLeftText);
            }
        }
#endif
        SETBASEPOSITION(pp,y);
        if ( ISLESSPOS(*plspace,pp) ) {
            MesPrint("%24s Bytes used      = %10p%s",AC.Commercial,plspace,humanBytesText);
        }
        else {
            y = 1000000000L;
            SETBASEPOSITION(pp,y);
            MULPOS(pp,100);
            if ( ISLESSPOS(*plspace,pp) ) {
                MesPrint("%24s Bytes used      =%11p%s",AC.Commercial,plspace,humanBytesText);
            }
            else {
                MULPOS(pp,10);
                if ( ISLESSPOS(*plspace,pp) ) {
                MesPrint("%24s Bytes used     =%12p%s",AC.Commercial,plspace,humanBytesText);
                }
                else {
                MULPOS(pp,10);
                if ( ISLESSPOS(*plspace,pp) ) {
                MesPrint("%24s Bytes used    =%13p%s",AC.Commercial,plspace,humanBytesText);
                }
                else {
                MULPOS(pp,10);
                if ( ISLESSPOS(*plspace,pp) ) {
                MesPrint("%24s Bytes used   =%14p%s",AC.Commercial,plspace,humanBytesText);
                }
                else {
                MULPOS(pp,10);
                if ( ISLESSPOS(*plspace,pp) ) {
                MesPrint("%24s Bytes used  =%15p%s",AC.Commercial,plspace,humanBytesText);
                }
                else {
                MULPOS(pp,10);
                if ( ISLESSPOS(*plspace,pp) ) {
                MesPrint("%24s Bytes used =%16p%s",AC.Commercial,plspace,humanBytesText);
                }
                else {
                MULPOS(pp,10);
                if ( ISLESSPOS(*plspace,pp) ) {
                MesPrint("%24s Bytes used=%17p%s",AC.Commercial,plspace,humanBytesText);
                }
                } } } } }
            }
        } }
#ifdef WITHSTATS
        MesPrint("Total number of writes: %l, reads: %l, seeks, %l"
            ,numwrites,numreads,numseeks);
        MesPrint("Total number of mallocs: %l, frees: %l"
            ,nummallocs,numfrees);
#endif
        /* Put back the original LogHandle, it was changed if the statistics were
         * not printed in the log file. */
        AC.LogHandle = oldLogHandle;

        MUNLOCK(ErrorMessageLock);
    }
}

/*
        #] WriteStats :
        #[ NewSort :                WORD NewSort()
*/
WORD NewSort(PHEAD0)
{
    GETBIDENTITY
```

```
    SORTING *S, **newFS;
    int i, newsize;
    if ( AN.SoScratC == 0 )
        AN.SoScratC = (UWORD *)Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD),"NewSort");
    AR.sLevel++;
    if ( AR.sLevel >= AN.NumFunSorts ) {
        if ( AN.NumFunSorts == 0 ) newsize = 100;
        else newsize = 2*AN.NumFunSorts;
        newFS = (SORTING **)Malloc1((newsize+1)*sizeof(SORTING *),"FunSort pointers");
        for ( i = 0; i < AN.NumFunSorts; i++ ) newFS[i] = AN.FunSorts[i];
        for ( ; i <= newsize; i++ ) newFS[i] = 0;
        if ( AN.FunSorts ) M_free(AN.FunSorts,"FunSort pointers");
        AN.FunSorts = newFS; AN.NumFunSorts = newsize;
    }
    if ( AR.sLevel == 0 ) {

        numcompares = 0;

        AN.FunSorts[0] = AT.S0;
        if ( AR.PolyFun == 0 ) { AT.S0->PolyFlag = 0; }
        else if ( AR.PolyFunType == 1 ) { AT.S0->PolyFlag = 1; }
        else if ( AR.PolyFunType == 2 ) {
            if ( AR.PolyFunExp == 2
              || AR.PolyFunExp == 3 ) AT.S0->PolyFlag = 1;
            else                      AT.S0->PolyFlag = 2;
        }
        AR.ShortSortCount = 0;
    }
    else {
        if ( AN.FunSorts[AR.sLevel] == 0 ) {
            AN.FunSorts[AR.sLevel] = AllocSort(
                AM.SLargeSize,AM.SSmallSize,AM.SSmallEsize,AM.STermsInSmall
                    ,AM.SMaxPatches,AM.SMaxFpatches,AM.SIOsize,1);
        }
        AN.FunSorts[AR.sLevel]->PolyFlag = 0;
    }
    AT.SS = S = AN.FunSorts[AR.sLevel];
    S->sFill = S->sBuffer;
    S->lFill = S->lBuffer;
    S->lPatch = 0;
    S->fPatchN = 0;
    S->GenTerms = S->TermsLeft = S->GenSpace = S->SpaceLeft = 0;
    S->PoinFill = S->sPointer;
    *S->PoinFill = S->sFill;
    if ( AR.sLevel > 0 ) { S->PolyWise = 0; }
    PUTZERO(S->SizeInFile[0]); PUTZERO(S->SizeInFile[1]); PUTZERO(S->SizeInFile[2]);
    S->sTerms = 0;
    PUTZERO(S->file.POposition);
    S->stage4 = 0;
    if ( AR.sLevel > AN.MaxFunSorts ) AN.MaxFunSorts = AR.sLevel;
/*
    The next variable is for the staged sort only.
    It should be treated differently

    PUTZERO(AN.OldPosOut);
*/
    return(0);
}

/*
        #] NewSort :
        #[ EndSort :                WORD EndSort(PHEAD buffer,par)
*/
LONG EndSort(PHEAD WORD *buffer, int par)
{
  GETBIDENTITY
  SORTING *S = AT.SS;
  WORD j, **ss, *to, *t;
  LONG sSpace, over, tover, spare, retval = 0, jj;
  POSITION position, pp;
  off_t lSpace;
  FILEHANDLE *fout = 0, *oldoutfile = 0, *newout = 0;

  if ( AM.exitflag && AR.sLevel == 0 ) return(0);
#ifdef WITHMPI
  if( (retval = PF_EndSort()) > 0){
    oldoutfile = AR.outfile;
    retval = 0;
    goto RetRetval;
  }
  else if(retval < 0){
    retval = -1;
    goto RetRetval;
  }
    /* PF_EndSort returned 0: for S != AM.S0 and slaves still do the regular sort */
#endif /* WITHMPI */
    oldoutfile = AR.outfile;
```

```
/*      PolyFlag repair action
    if ( S == AT.S0 ) {
        if ( AR.PolyFun == 0 ) { S->PolyFlag = 0; }
        else if ( AR.PolyFunType == 1 ) { S->PolyFlag = 1; }
        else if ( AR.PolyFunType == 2 ) {
            if ( AR.PolyFunExp == 2
              || AR.PolyFunExp == 3 ) S->PolyFlag = 1;
            else                      S->PolyFlag = 2;
        }
        S->PolyWise = 0;
    }
    else {
        S->PolyFlag = S->PolyWise = 0;
    }
*/
    S->PolyWise = 0;
    *(S->PoinFill) = 0;
#ifdef SPLITTIME
        PrintTime((UBYTE *)"EndSort, before SplitMerge");
#endif
    S->sPointer[SplitMerge(BHEAD S->sPointer,S->sTerms)] = 0;
#ifdef SPLITTIME
        PrintTime((UBYTE *)"Endsort,  after SplitMerge");
#endif
    sSpace = 0;
    tover = over = S->sTerms;
    ss = S->sPointer;
    if ( over >= 0 ) {
        if ( S->lPatch > 0 || S->file.handle >= 0 ) {
            ss[over] = 0;
            sSpace = ComPress(ss,&spare);
            S->TermsLeft -= over - spare;
            if ( par == 1 ) { AR.outfile = newout = AllocFileHandle(0,(char *)0); }
        }
        else if ( S != AT.S0 ) {
            ss[over] = 0;
            if ( par == 2 ) {
                sSpace = 3;
                while ( ( t = *ss++ ) != 0 ) { sSpace += *t; }
                if ( AN.tryterm > 0 && ( (sSpace+1)*sizeof(WORD) < (size_t)(AM.MaxTer) ) ) {
                    to = TermMalloc("$-sort space");
                }
                else {
                    LONG allocsp = sSpace+1;
                    if ( allocsp < MINALLOC ) allocsp = MINALLOC;
                    allocsp = ((allocsp+7)/8)*8;
                    to = (WORD *)Malloc1(allocsp*sizeof(WORD),"$-sort space");
                    if ( AN.tryterm > 0 ) AN.tryterm = 0;
                }
                *((WORD **)buffer) = to;
                ss = S->sPointer;
                while ( ( t = *ss++ ) != 0 ) {
                    j = *t; while ( --j >= 0 ) *to++ = *t++;
                }
                *to = 0;
                retval = sSpace + 1;
            }
            else {
                to = buffer;
                sSpace = 0;
                while ( ( t = *ss++ ) != 0 ) {
                    j = *t;
                    if ( ( sSpace += j ) > AM.MaxTer/((LONG)sizeof(WORD)) ) {
                        /* Too big! Get the total size for useful error message */
                        while ( ( t = *ss++ ) != 0 ) {
                            sSpace += *t;
                        }
                        MLOCK(ErrorMessageLock);
                        MesPrint("Sorted function argument too long (%d words). Increase MaxTermSize
    (%l words).", sSpace, AM.MaxTer/((LONG)sizeof(WORD)));
                        MUNLOCK(ErrorMessageLock);
                        retval = -1; goto RetRetval;
                    }
                    while ( --j >= 0 ) *to++ = *t++;
                }
                *to = 0;
                retval = to - buffer;
            }
            goto RetRetval;
        }
        else {
            POSITION oldpos;
            if ( S == AT.S0 ) {
                fout = AR.outfile;
                *AR.CompressPointer = 0;
                SeekScratch(AR.outfile,&position);
            }
```

```
            else {
                fout = &(S->file);
                PUTZERO(position);
            }
            oldpos = position;
            S->TermsLeft = 0;
/*
            Here we can go directly to the output.
*/
#ifdef WITHZLIB
            { int oldgzipCompress = AR.gzipCompress;
                AR.gzipCompress = 0;
#endif
            if ( tover > 0 ) {
                ss = S->sPointer;
                while ( ( t = *ss++ ) != 0 ) {
                    if ( *t ) S->TermsLeft++;
#ifdef WITHPTHREADS
                    if ( AS.MasterSort && ( fout == AR.outfile ) ) { PutToMaster(BHEAD t); }
                    else
#endif
                    if ( PutOut(BHEAD t,&position,fout,1) < 0 ) {
                        retval = -1; goto RetRetval;
                    }
                }
            }
#ifdef WITHPTHREADS
            if ( AS.MasterSort && ( fout == AR.outfile ) ) { PutToMaster(BHEAD 0); }
            else
#endif
            if ( FlushOut(&position,fout,1) ) {
                retval = -1; goto RetRetval;
            }
#ifdef WITHZLIB
                AR.gzipCompress = oldgzipCompress;
            }
#endif
#ifdef WITHPTHREADS
            if ( AS.MasterSort && ( fout == AR.outfile ) ) goto RetRetval;
#endif
#ifdef WITHMPI
            if ( PF.me != MASTER && PF.exprtodo < 0 ) goto RetRetval;
#endif
            DIFPOS(oldpos,position,oldpos);
            S->SpaceLeft = BASEPOSITION(oldpos);
            WriteStats(&oldpos,STATSPOSTSORT,NOCHECKLOGTYPE);
            pp = oldpos;
            goto RetRetval;
        }
    }
    else if ( par == 1 && newout == 0 ) { AR.outfile = newout = AllocFileHandle(0,(char *)0); }
    sSpace++;
    lSpace = sSpace + (S->lFill - S->lBuffer) - (LONG)S->lPatch*(AM.MaxTer/sizeof(WORD));
/*          Note wrt MaxTer and lPatch: each patch starts with space for decompression */
/*          Not needed if only large buffer, but needed when using files (?) */
    SETBASEPOSITION(pp,lSpace);
    MULPOS(pp,sizeof(WORD));
    if ( S->file.handle >= 0 ) {
        ADD2POS(pp,S->fPatches[S->fPatchN]);
    }
    if ( S == AT.S0 ) {
        if ( S->lPatch > 0 || S->file.handle >= 0 ) {
            WriteStats(&pp,STATSSPLITMERGE,CHECKLOGTYPE);
        }
    }
    if ( par == 2 ) { AR.outfile = newout = AllocFileHandle(0,(char *)0); }
    if ( S->lPatch > 0 ) {
        if ( ( S->lPatch >= S->MaxPatches ) ||
            ( ( (WORD *)(((UBYTE *)(S->lFill + sSpace)) + 2*AM.MaxTer) ) >= S->lTop ) ) {
/*
            The large buffer is too full. Merge and write it
*/
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%w EndSort: lPatch = %d, MaxPatches = %d,lFill = %x, sSpace = %ld, MaxTer = %d,
lTop = %x"
                    ,S->lPatch,S->MaxPatches,S->lFill,sSpace,AM.MaxTer/sizeof(WORD),S->lTop);
            MUNLOCK(ErrorMessageLock);
#endif

            if ( MergePatches(1) ) {
                MLOCK(ErrorMessageLock);
                MesCall("EndSort");
                MUNLOCK(ErrorMessageLock);
                retval = -1; goto RetRetval;
            }
            S->lPatch = 0;
```

```
            pp = S->SizeInFile[1];
            MULPOS(pp,sizeof(WORD));
#ifndef WITHPTHREADS
            if ( S == AT.S0 )
#endif
            {
                POSITION pppp;
                SETBASEPOSITION(pppp,0);
                SeekFile(S->file.handle,&pppp,SEEK_CUR);
                SeekFile(S->file.handle,&pp,SEEK_END);
                SeekFile(S->file.handle,&pppp,SEEK_SET);
                WriteStats(&pp,STATSMERGETOFILE,CHECKLOGTYPE);
                UpdateMaxSize();
            }
        }
        else {
            S->Patches[S->lPatch++] = S->lFill;
            to = (WORD *)(((UBYTE *)(S->lFill)) + AM.MaxTer);
            if ( tover > 0 ) {
                ss = S->sPointer;
                while ( ( t = *ss++ ) != 0 ) {
                    j = *t;
                    if ( j < 0 ) j = t[1] + 2;
                    while ( --j >= 0 ) *to++ = *t++;
                }
            }
            *to++ = 0;
            S->lFill = to;
            if ( S->file.handle < 0 ) {
                if ( MergePatches(2) ) {
                    MLOCK(ErrorMessageLock);
                    MesCall("EndSort");
                    MUNLOCK(ErrorMessageLock);
                    retval = -1; goto RetRetval;
                }
                if ( S == AT.S0 ) {
                    pp = S->SizeInFile[2];
                    MULPOS(pp,sizeof(WORD));
#ifdef WITHPTHREADS
                    if ( AS.MasterSort && ( fout == AR.outfile ) ) goto RetRetval;
#endif
                    WriteStats(&pp,STATSPOSTSORT,NOCHECKLOGTYPE);
                    UpdateMaxSize();
                }
                else {
                    if ( par == 2 && newout->handle >= 0 ) {
                        POSITION zeropos;
                        PUTZERO(zeropos);
#ifdef ALLLOCK
                        LOCK(newout->pthreadslock);
#endif
                        SeekFile(newout->handle,&zeropos,SEEK_SET);
                        to = (WORD *)Malloc1(BASEPOSITION(newout->filesize)+sizeof(WORD)*2
                                ,"$-buffer reading");
                        if ( AN.tryterm > 0 ) AN.tryterm = 0;
                        if ( ( retval = ReadFile(newout->handle,(UBYTE
*)to,BASEPOSITION(newout->filesize)) ) !=
                                BASEPOSITION(newout->filesize) ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Error reading information for $ variable");
                            MUNLOCK(ErrorMessageLock);
                            M_free(to,"$-buffer reading");
                            retval = -1;
                        }
                        else {
                            *((WORD **)buffer) = to;
                            retval /= sizeof(WORD);
                        }
#ifdef ALLLOCK
                        UNLOCK(newout->pthreadslock);
#endif
                    }
                    else if ( newout->handle >= 0 ) {
/*
                        We land here if par == 1 (function arg sort) and PutOut has created a file.
                        This means that the term is larger than SIOsize, which we ensure is at least
                        as large as MaxTermSize in setfile.c. Thus we know already that the term won't
fit.
*/
TooLarge:
                        MLOCK(ErrorMessageLock);
                        MesPrint("(1)Output should fit inside a single term. Increase MaxTermSize?");
                        MesCall("EndSort");
                        MUNLOCK(ErrorMessageLock);
                        retval = -1; goto RetRetval;
                    }
                    else {
```

```
                        t = newout->PObuffer;
                        if ( par == 2 ) {
                            jj = newout->POfill - t;
                            if ( AN.tryterm > 0 && ( (jj+2)*sizeof(WORD) < (size_t)(AM.MaxTer) ) ) {
                                to = TermMalloc("$-sort space");
                            }
                            else {
                                LONG allocsp = jj+2;
                                if ( allocsp < MINALLOC ) allocsp = MINALLOC;
                                allocsp = ((allocsp+7)/8)*8;
                                to = (WORD *)Malloc1(allocsp*sizeof(WORD),"$-sort space");
                                if ( AN.tryterm > 0 ) AN.tryterm = 0;
                            }
                            *((WORD **)buffer) = to;
                            NCOPY(to,t,jj);
                            /* we should not set retval here in this par==2 case, it being 0 has meaning
                               after RetRetval, and it is set properly there. */
                        }
                        else {
                            j = newout->POfill - t;
                            to = buffer;
                            if ( to >= AT.WorkSpace && to < AT.WorkTop && to+j > AT.WorkTop )
                                goto WorkSpaceError;
                            if ( j > AM.MaxTer ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Encountered term of size: %d words.", j/(LONG)sizeof(WORD));
                                MUNLOCK(ErrorMessageLock);
                                goto TooLarge;
                            }
                            NCOPY(to,t,j);
                            retval = to - buffer - 1;
                        }
                    }
                }
                goto RetRetval;
            }
            if ( MergePatches(1) ) { /* --> SortFile */
                MLOCK(ErrorMessageLock);
                MesCall("EndSort");
                MUNLOCK(ErrorMessageLock);
                retval = -1; goto RetRetval;
            }
            UpdateMaxSize();
            pp = S->SizeInFile[1];
            MULPOS(pp,sizeof(WORD));
#ifndef WITHPTHREADS
            if ( S == AT.S0 )
#endif
            {
                POSITION pppp;
                SETBASEPOSITION(pppp,0);
                SeekFile(S->file.handle,&pppp,SEEK_CUR);
                SeekFile(S->file.handle,&pp,SEEK_END);
                SeekFile(S->file.handle,&pppp,SEEK_SET);
                WriteStats(&pp,STATSMERGETOFILE,CHECKLOGTYPE);
            }
#ifdef WITHERRORXXX
            if ( S != AT.S0 ) {
/*
                This is wrong! We have sorted to the sort file.
                Things are not sitting in the output yet.
*/
                if ( newout->handle >= 0 ) goto TooLarge;
                t = newout->PObuffer;
                j = newout->POfill - t;
                to = buffer;
                if ( to >= AT.WorkSpace && to < AT.WorkTop && to+j > AT.WorkTop )
                    goto WorkSpaceError;
                if ( j > AM.MaxTer ) goto TooLarge;
                NCOPY(to,t,j);
                goto RetRetval;
            }
#endif
        }
    }
    if ( S->file.handle >= 0 ) {
#ifdef GZIPDEBUG
        MLOCK(ErrorMessageLock);
        MesPrint("%w EndSort: fPatchN = %d, lPatch = %d, position = %12p"
            ,S->fPatchN,S->lPatch,&(S->fPatches[S->fPatchN]));
        MUNLOCK(ErrorMessageLock);
#endif
        if ( S->lPatch <= 0 ) {
            StageSort(&(S->file));
            position = S->fPatches[S->fPatchN];
```

```
            ss = S->sPointer;
            if ( *ss ) {
                *AR.CompressPointer = 0;
#ifdef WITHZLIB
                if ( S == AT.S0 && AR.NoCompress == 0 && AR.gzipCompress > 0 )
                    S->fpcompressed[S->fPatchN] = 1;
                else
                    S->fpcompressed[S->fPatchN] = 0;
                SetupOutputGZIP(&(S->file));
#endif
                while ( ( t = *ss++ ) != 0 ) {
                    if ( PutOut(BHEAD t,&position,&(S->file),1) < 0 ) {
                        retval = -1; goto RetRetval;
                    }
                }
                if ( FlushOut(&position,&(S->file),1) ) {
                    retval = -1; goto RetRetval;
                }
                ++(S->fPatchN);
                S->fPatches[S->fPatchN] = position;
                UpdateMaxSize();
#ifdef GZIPDEBUG
                MLOCK(ErrorMessageLock);
                MesPrint("%w EndSort+: fPatchN = %d, lPatch = %d, position = %12p"
                    ,S->fPatchN,S->lPatch,&(S->fPatches[S->fPatchN]));
                MUNLOCK(ErrorMessageLock);
#endif
            }
        }
        AR.Stage4Name = 0;
#ifdef WITHPTHREADS
        if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
            if ( S->file.handle >= 0 ) {
                SynchFile(S->file.handle);
            }
        }
#endif
        UpdateMaxSize();
        if ( MergePatches(0) ) {
            MLOCK(ErrorMessageLock);
            MesCall("EndSort");
            MUNLOCK(ErrorMessageLock);
            retval = -1; goto RetRetval;
        }
        S->stage4 = 0;
#ifdef WITHPTHREADS
        if ( AS.MasterSort && ( fout == AR.outfile ) ) goto RetRetval;
#endif
        pp = S->SizeInFile[0];
        MULPOS(pp,sizeof(WORD));
        WriteStats(&pp,STATSPOSTSORT,NOCHECKLOGTYPE);
        UpdateMaxSize();
    }
RetRetval:

#ifdef WITHMPI
    /* NOTE: PF_EndSort has been changed such that it sets S->TermsLeft. (TU 30 Jun 2011) */
    if ( AR.sLevel == 0 && (PF.me == MASTER || PF.exprtodo >= 0) ) {
        Expressions[AR.CurExpr].counter = S->TermsLeft;
        Expressions[AR.CurExpr].size = pp;
    }
#else
    if ( AR.sLevel == 0 ) {
        Expressions[AR.CurExpr].counter = S->TermsLeft;
        Expressions[AR.CurExpr].size = pp;
    }/*if ( AR.sLevel == 0 )*/
#endif
/*:[25nov2003 mt]*/
    if ( S->file.handle >= 0 && ( par != 1 ) && ( par != 2 ) ) {
                /* sortfile is still open */
        UpdateMaxSize();
#ifdef WITHZLIB
        ClearSortGZIP(&(S->file));
#endif
        CloseFile(S->file.handle);
        S->file.handle = -1;
        remove(S->file.name);
#ifdef GZIPDEBUG
        MLOCK(ErrorMessageLock);
        MesPrint("%wEndSort: sortfile %s removed",S->file.name);
        MUNLOCK(ErrorMessageLock);
#endif
    }
    AR.outfile = oldoutfile;
    AR.sLevel--;
    if ( AR.sLevel >= 0 ) AT.SS = AN.FunSorts[AR.sLevel];
    if ( par == 1 ) {
```

```
        if ( retval < 0 ) {
            UpdateMaxSize();
            if ( newout ) {
                DeAllocFileHandle(newout);
                newout = 0;
            }
        }
        else if ( newout ) {
          if ( newout->handle >= 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("(2)Output should fit inside a single term. Increase MaxTermSize?");
            MesCall("EndSort");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
          }
          else if ( newout->POfill > newout->PObuffer ) {
/*
            Here we have to copy the contents of the 'file' into
            the buffer. We assume that this buffer lies in the WorkSpace.
            Hence
*/
            j = newout->POfill-newout->PObuffer;
            if ( buffer >= AT.WorkSpace && buffer < AT.WorkTop && buffer+j > AT.WorkTop )
                goto WorkSpaceError;
            else {
                to = buffer; t = newout->PObuffer;
                while ( j-- > 0 ) *to++ = *t++;
            }
            UpdateMaxSize();
          }
          DeAllocFileHandle(newout);
          newout = 0;
        }
    }
    else if ( par == 2 ) {
        if ( newout ) {
            if ( retval == 0 ) {
              if ( newout->handle >= 0 ) {
/*
                output resides at the moment in a file
                Find the size, make a buffer, copy into the buffer and clean up.
*/
                POSITION zeropos;
                PUTZERO(position);
#ifdef ALLLOCK
                LOCK(newout->pthreadslock);
#endif
                SeekFile(newout->handle,&position,SEEK_END);
                PUTZERO(zeropos);
                SeekFile(newout->handle,&zeropos,SEEK_SET);
                to = (WORD *)Malloc1(BASEPOSITION(position)+sizeof(WORD)*3
                        ,"$-buffer reading");
                if ( AN.tryterm > 0 ) AN.tryterm = 0;
                if ( ( retval = ReadFile(newout->handle,(UBYTE *)to,BASEPOSITION(position)) ) !=
                BASEPOSITION(position) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Error reading information for $ variable");
                    MUNLOCK(ErrorMessageLock);
                    M_free(to,"$-buffer reading");
                    retval = -1;
                }
                else {
                    *((WORD **)buffer) = to;
                    retval /= sizeof(WORD);
                }
#ifdef ALLLOCK
                UNLOCK(newout->pthreadslock);
#endif
              }
              else {
/*
                output resides in the cache buffer and the file was never opened
*/
                LONG wsiz = newout->POfill - newout->PObuffer;
                if ( AN.tryterm > 0 && ( (wsiz+2)*sizeof(WORD) < (size_t)(AM.MaxTer) ) ) {
                    to = TermMalloc("$-sort space");
                }
                else {
                    LONG allocsp = wsiz+2;
                    if ( allocsp < MINALLOC ) allocsp = MINALLOC;
                    allocsp = ((allocsp+7)/8)*8;
                    to = (WORD *)Malloc1(allocsp*sizeof(WORD),"$-buffer reading");
                    if ( AN.tryterm > 0 ) AN.tryterm = 0;
                }
                *((WORD **)buffer) = to; t = newout->PObuffer;
                retval = wsiz;
                NCOPY(to,t,wsiz);
```

```
              }
            }
            UpdateMaxSize();
            DeAllocFileHandle(newout);
            newout = 0;
        }
    }
    else {
        if ( newout ) {
            DeAllocFileHandle(newout);
            newout = 0;
        }
    }
/*
    if ( AR.sLevel < 0 ) {
        MesPrint(" number of calls to compare was %l",numcompares);
    }
*/
    return(retval);
WorkSpaceError:
    MLOCK(ErrorMessageLock);
    MesWork();
    MesCall("EndSort");
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(-1);
}

/*
        #] EndSort :
        #[ PutIn :                  LONG PutIn(handle,position,buffer,take,npat)
*/
LONG PutIn(FILEHANDLE *file, POSITION *position, WORD *buffer, WORD **take, int npat)
{
    LONG i, RetCode;
    WORD *from, *to;
#ifndef WITHZLIB
    DUMMYUSE(npat);
#endif
    from = buffer + ( file->POsize * sizeof(UBYTE) )/sizeof(WORD);
    i = from - *take;
    if ( i*((LONG)(sizeof(WORD))) > AM.MaxTer ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Problems in PutIn");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    to = buffer;
    while ( --i >= 0 ) *--to = *--from;
    *take = to;
#ifdef WITHZLIB
    if ( ( RetCode = FillInputGZIP(file,position,(UBYTE *)buffer
                            ,file->POsize,npat) ) < 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("PutIn: We have RetCode = %x while reading %x bytes",
            RetCode,file->POsize);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
#else
#ifdef ALLLOCK
    LOCK(file->pthreadslock);
#endif
    SeekFile(file->handle,position,SEEK_SET);
    if ( ( RetCode = ReadFile(file->handle,(UBYTE *)buffer,file->POsize) ) < 0 ) {
#ifdef ALLLOCK
        UNLOCK(file->pthreadslock);
#endif
        MLOCK(ErrorMessageLock);
        MesPrint("PutIn: We have RetCode = %x while reading %x bytes",
            RetCode,file->POsize);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
#ifdef ALLLOCK
    UNLOCK(file->pthreadslock);
#endif
#endif
    return(RetCode);
}

/*
        #] PutIn :
        #[ Sflush :                 WORD Sflush(file)
*/
WORD Sflush(FILEHANDLE *fi)
{
```

```
    LONG size, RetCode;
#ifdef WITHZLIB
    GETIDENTITY
    int dobracketindex = 0;
    if ( AR.sLevel <= 0 && Expressions[AR.CurExpr].newbracketinfo
        && ( fi == AR.outfile || fi == AR.hidefile ) ) dobracketindex = 1;
#endif
    if ( fi->handle < 0 ) {
        if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%w Sflush created scratch file %s",fi->name);
            MUNLOCK(ErrorMessageLock);
#endif
            fi->handle = (WORD)RetCode;
            PUTZERO(fi->filesize);
            PUTZERO(fi->POposition);
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Cannot create scratch file %s",fi->name);
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
    }
#ifdef WITHZLIB
    if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
    && dobracketindex == 0 ) {
        if ( FlushOutputGZIP(fi) ) return(-1);
        fi->POfill = fi->PObuffer;
    }
    else
#endif
    {
#ifdef ALLLOCK
      LOCK(fi->pthreadslock);
#endif
      size = (fi->POfill-fi->PObuffer)*sizeof(WORD);
      SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
      if ( WriteFile(fi->handle,(UBYTE *)(fi->PObuffer),size) != size ) {
#ifdef ALLLOCK
        UNLOCK(fi->pthreadslock);
#endif
        MLOCK(ErrorMessageLock);
        MesPrint("Write error while finishing sort. Disk full?");
        MUNLOCK(ErrorMessageLock);
        return(-1);
      }
      ADDPOS(fi->filesize,size);
      ADDPOS(fi->POposition,size);
      fi->POfill = fi->PObuffer;
#ifdef ALLLOCK
      UNLOCK(fi->pthreadslock);
#endif
    }
    return(0);
}

/*
        #] Sflush :
        #[ PutOut :                   WORD PutOut(term,position,file,ncomp)
*/
WORD PutOut(PHEAD WORD *term, POSITION *position, FILEHANDLE *fi, WORD ncomp)
{
    GETBIDENTITY
    WORD i, *p, ret, *r, *rr, j, k, first;
    int dobracketindex = 0;
    LONG RetCode;

    if ( AT.SS != AT.S0 ) {
/*
        For this case no compression should be used
*/
        if ( ( i = *term ) <= 0 ) return(0);
        ret = i;
        ADDPOS(*position,i*sizeof(WORD));
        p = fi->POfill;
        do {
            if ( p >= fi->POstop ) {
                if ( fi->handle < 0 ) {
                    if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
#ifdef GZIPDEBUG
                        MLOCK(ErrorMessageLock);
                        MesPrint("%w PutOut created sortfile %s",fi->name);
                        MUNLOCK(ErrorMessageLock);
#endif
                        fi->handle = (WORD)RetCode;
```

```
                        PUTZERO(fi->filesize);
                        PUTZERO(fi->POposition);
/*
                        Should not be here anymore?
#ifdef WITHZLIB
                        fi->ziobuffer = 0;
#endif
*/
                    }
                    else {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Cannot create scratch file %s",fi->name);
                        MUNLOCK(ErrorMessageLock);
                        return(-1);
                    }
                }
#ifdef ALLLOCK
                LOCK(fi->pthreadslock);
#endif
                if ( fi == AR.hidefile ) {
                    LOCK(AS.inputslock);
                }
                SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
                if ( ( RetCode = WriteFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize) ) != fi->POsize ) {
                    if ( fi == AR.hidefile ) {
                        UNLOCK(AS.inputslock);
                    }
#ifdef ALLLOCK
                    UNLOCK(fi->pthreadslock);
#endif
                    MLOCK(ErrorMessageLock);
                    MesPrint("Write error during sort. Disk full?");
                    MesPrint("Attempt to write %l bytes on file %d at position %15p",
                                fi->POsize,fi->handle,&(fi->POposition));
                    MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG)(fi->PObuffer));
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                ADDPOS(fi->filesize,fi->POsize);
                p = fi->PObuffer;
                ADDPOS(fi->POposition,fi->POsize);
                if ( fi == AR.hidefile ) {
                    UNLOCK(AS.inputslock);
                }
#ifdef ALLLOCK
                UNLOCK(fi->pthreadslock);
#endif
#ifdef WITHPTHREADS
                if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
                    if ( fi->handle >= 0 ) SynchFile(fi->handle);
                }
#endif
            }
            *p++ = *term++;
        } while ( --i > 0 );
        fi->POfull = fi->POfill = p;
        return(ret);
    }
    if ( ( AP.PreDebug & DUMPOUTTERMS ) == DUMPOUTTERMS ) {
            MLOCK(ErrorMessageLock);
#ifdef WITHPTHREADS
            snprintf((char *)(THRbuf),100,"PutOut(%d)",AT.identity);
            PrintTerm(term,(char *)(THRbuf));
#else
            PrintTerm(term,"PutOut");
#endif
            MesPrint("ncomp = %d, AR.NoCompress = %d, AR.sLevel = %d",ncomp,AR.NoCompress,AR.sLevel);
            MesPrint("File %s, position %p",fi->name,position);
            MUNLOCK(ErrorMessageLock);
    }

    if ( AR.sLevel <= 0 && Expressions[AR.CurExpr].newbracketinfo
        && ( fi == AR.outfile || fi == AR.hidefile ) ) dobracketindex = 1;
    r = rr = AR.CompressPointer;
    first = j = k = ret = 0;
    if ( ( i = *term ) != 0 ) {
        if ( i < 0 ) {          /* Compressed term */
            i = term[1] + 2;
            if ( fi == AR.outfile || fi == AR.hidefile ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Ran into precompressed term");
                MUNLOCK(ErrorMessageLock);
                Crash();
                return(-1);
            }
        }
```

```
        else if ( !AR.NoCompress && ( ncomp > 0 ) && AR.sLevel <= 0 ) { /* Must compress */
            if ( dobracketindex ) {
                PutBracketInIndex(BHEAD term,position);
            }
            j = *r++ - 1;
            p = term + 1;
            i--;
            if ( AR.PolyFun ) {
                WORD *polystop, *sa;
                sa = p + i;
                sa -= ABS(sa[-1]);
                polystop = p;
                while ( polystop < sa && *polystop != AR.PolyFun ) {
                    polystop += polystop[1];
                }
                if ( polystop < sa ) {
                    if ( AR.PolyFunType == 2 ) polystop[2] &= ~MUSTCLEANPRF;
                    while ( i > 0 && j > 0 && *p == *r && p < polystop ) {
                        i--; j--; k--; p++; r++;
                    }
                }
                else {
                    while ( i > 0 && j > 0 && *p == *r && p < sa ) { i--; j--; k--; p++; r++; }
                }
            }
#ifdef WITHFLOAT
            else if ( AC.DefaultPrecision ) {
                WORD *floatstop, *sa;
                sa = p + i;
                sa -= ABS(sa[-1]);
                floatstop = p;
                while ( floatstop < sa && *floatstop != FLOATFUN ) {
                    floatstop += floatstop[1];
                }
                if ( floatstop < sa ) {
                    while ( i > 0 && j > 0 && *p == *r && p < floatstop ) {
                        i--; j--; k--; p++; r++;
                    }
                }
                else {
                    while ( i > 0 && j > 0 && *p == *r && p < sa ) { i--; j--; k--; p++; r++; }
                }
            }
#endif
            else {
                WORD *sa;
                sa = p + i;
                sa -= ABS(sa[-1]);
                while ( i > 0 && j > 0 && *p == *r && p < sa ) { i--; j--; k--; p++; r++; }
            }
            if ( k > -2 ) {
nocompress:
                j = i = *term;
                k = 0;
                p = term;
                r = rr;
                NCOPY(r,p,j);
            }
            else {
                *rr = *term;
                term = p;
                j = i;
                NCOPY(r,p,j);
                j = i;
                i += 2;
                first = 2;
            }
/*                  Sabotage getting into the coefficient next time */
            r[-(ABS(r[-1]))] = 0;
            if ( r >= AR.ComprTop ) {
                MLOCK(ErrorMessageLock);
                MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
                MUNLOCK(ErrorMessageLock);
                Crash();
                return(-1);
            }
        }
        else if ( !AR.NoCompress && ( ncomp < 0 ) && AR.sLevel <= 0 ) {
                /* No compress but put in compress buffer anyway */
            if ( dobracketindex ) {
                PutBracketInIndex(BHEAD term,position);
            }
            j = *r++ - 1;
            p = term + 1;
            i--;
            if ( AR.PolyFun ) {
                WORD *polystop, *sa;
```

```
                sa = p + i;
                sa -= ABS(sa[-1]);
                polystop = p;
                while ( polystop < sa && *polystop != AR.PolyFun ) {
                    polystop += polystop[1];
                }
                if ( polystop < sa ) {
                    if ( AR.PolyFunType == 2 ) polystop[2] &= ~MUSTCLEANPRF;
                    while ( i > 0 && j > 0 && *p == *r && p < polystop ) {
                        i--; j--; k--; p++; r++;
                    }
                }
                else {
                    while ( i > 0 && j > 0 && *p == *r ) { i--; j--; k--; p++; r++; }
                }
            }
            else {
                while ( i > 0 && j > 0 && *p == *r ) { i--; j--; k--; p++; r++; }
            }
            goto nocompress;
        }
        else {
            if ( AR.PolyFunType == 2 ) {
                WORD *t, *tstop;
                tstop = term + *term;
                tstop -= ABS(tstop[-1]);
                t = term+1;
                while ( t < tstop ) {
                    if ( *t == AR.PolyFun ) {
                        t[2] &= ~MUSTCLEANPRF;
                    }
                    t += t[1];
                }
            }
            if ( dobracketindex ) {
                PutBracketInIndex(BHEAD term,position);
            }
        }
        ret = i;
        ADDPOS(*position,i*sizeof(WORD));
        p = fi->POfill;
        do {
            if ( p >= fi->POstop ) {
#ifdef WITHMPI /* [16mar1998 ar] */
              if ( PF.me != MASTER && AR.sLevel <= 0 && (fi == AR.outfile || fi == AR.hidefile) &&
PF.parallel && PF.exprtodo < 0 ) {
                PF_BUFFER *sbuf = PF.sbuf;
                sbuf->fill[sbuf->active] = fi->POstop;
                PF_ISendSbuf(MASTER,PF_BUFFER_MSGTAG);
                p = fi->PObuffer = fi->POfill = fi->POfull =
                  sbuf->buff[sbuf->active];
                fi->POstop = sbuf->stop[sbuf->active];
              }
              else
#endif /* WITHMPI [16mar1998 ar] */
              {
                if ( fi->handle < 0 ) {
                    if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
#ifdef GZIPDEBUG
                        MLOCK(ErrorMessageLock);
                        MesPrint("%w PutOut created sortfile %s",fi->name);
                        MUNLOCK(ErrorMessageLock);
#endif
                        fi->handle = (WORD)RetCode;
                        PUTZERO(fi->filesize);
                        PUTZERO(fi->POposition);
/*
                        Should not be here?
#ifdef WITHZLIB
                        fi->ziobuffer = 0;
#endif
*/
                    }
                    else {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Cannot create scratch file %s",fi->name);
                        MUNLOCK(ErrorMessageLock);
                        return(-1);
                    }
                }
#ifdef WITHZLIB
                if ( !AR.NoCompress && ncomp > 0 && AR.gzipCompress > 0
                    && dobracketindex == 0 && fi->zsp != 0 ) {
                    fi->POfill = p;
                    if ( PutOutputGZIP(fi) ) return(-1);
                    p = fi->PObuffer;
                }
```

```
                else
#endif
                {
#ifdef ALLLOCK
                  LOCK(fi->pthreadslock);
#endif
                  if ( fi == AR.hidefile ) {
                    LOCK(AS.inputslock);
                  }
                  SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
                  if ( ( RetCode = WriteFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize) ) != fi->POsize ) {
                    if ( fi == AR.hidefile ) {
                        UNLOCK(AS.inputslock);
                    }
#ifdef ALLLOCK
                    UNLOCK(fi->pthreadslock);
#endif
                    MLOCK(ErrorMessageLock);
                    MesPrint("Write error during sort. Disk full?");
                    MesPrint("Attempt to write %l bytes on file %d at position %15p",
                                fi->POsize,fi->handle,&(fi->POposition));
                    MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG)(fi->PObuffer));
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                  }
                  ADDPOS(fi->filesize,fi->POsize);
                  p = fi->PObuffer;
                  ADDPOS(fi->POposition,fi->POsize);
                  if ( fi == AR.hidefile ) {
                    UNLOCK(AS.inputslock);
                  }
#ifdef ALLLOCK
                  UNLOCK(fi->pthreadslock);
#endif
#ifdef WITHPTHREADS
                  if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
                    if ( fi->handle >= 0 ) SynchFile(fi->handle);
                  }
#endif
                }
              }
            }
            if ( first ) {
                if ( first == 2 ) *p++ = k;
                else *p++ = j;
                first--;
            }
            else *p++ = *term++;
/*
            if ( AP.DebugFlag ) {
                TalToLine((UWORD)(p[-1])); TokenToLine((UBYTE *)"  ");
            }
*/
        } while ( --i > 0 );
        fi->POfull = fi->POfill = p;
    }
/*
    if ( AP.DebugFlag ) {
        AO.OutSkip = 0;
        FiniLine();
    }
*/
    return(ret);
}

/*
        #] PutOut :
        #[ FlushOut :                   WORD FlushOut(position,file,compr)
*/
WORD FlushOut(POSITION *position, FILEHANDLE *fi, int compr)
{
    GETIDENTITY
    LONG size, RetCode;
    int dobracketindex = 0;
#ifndef WITHZLIB
    DUMMYUSE(compr);
#endif
    if ( AR.sLevel <= 0 && Expressions[AR.CurExpr].newbracketinfo
        && ( fi == AR.outfile || fi == AR.hidefile ) ) dobracketindex = 1;
#ifdef WITHMPI /* [16mar1998 ar] */
    if ( PF.me != MASTER && AR.sLevel <= 0 && (fi == AR.outfile || fi == AR.hidefile) && PF.parallel && PF.exprtodo < 0 ) {
        PF_BUFFER *sbuf = PF.sbuf;
        if ( fi->POfill >= fi->POstop ){
          sbuf->fill[sbuf->active] = fi->POstop;
          PF_ISendSbuf(MASTER,PF_BUFFER_MSGTAG);
```

```
          fi->POfull = fi->POfill = fi->PObuffer = sbuf->buff[sbuf->active];
          fi->POstop = sbuf->stop[sbuf->active];
        }
        *(fi->POfill)++ = 0;
        sbuf->fill[sbuf->active] = fi->POfill;
        PF_ISendSbuf(MASTER,PF_ENDBUFFER_MSGTAG);
        fi->PObuffer = fi->POfill = fi->POfull = sbuf->buff[sbuf->active];
        fi->POstop = sbuf->stop[sbuf->active];
        return(0);
    }
#endif /* WITHMPI [16mar1998 ar] */
    if ( fi->POfill >= fi->POstop ) {
        if ( fi->handle < 0 ) {
            if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
#ifdef GZIPDEBUG
                MLOCK(ErrorMessageLock);
                MesPrint("%w FlushOut created scratch file %s",fi->name);
                MUNLOCK(ErrorMessageLock);
#endif
                PUTZERO(fi->filesize);
                PUTZERO(fi->POposition);
                fi->handle = (WORD)RetCode;
/*
                Should not be here?
#ifdef WITHZLIB
                fi->ziobuffer = 0;
#endif
*/
            }
            else {
                MLOCK(ErrorMessageLock);
                MesPrint("Cannot create scratch file %s",fi->name);
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
        }
#ifdef WITHZLIB
        if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
        && dobracketindex == 0 && ( compr > 0 ) && fi->zsp != 0 ) {
            if ( PutOutputGZIP(fi) ) return(-1);
            fi->POfill = fi->PObuffer;
        }
        else
#endif
        {
#ifdef ALLLOCK
          LOCK(fi->pthreadslock);
#endif
          if ( fi == AR.hidefile ) {
            LOCK(AS.inputslock);
          }
          SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
          if ( ( RetCode = WriteFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize) ) != fi->POsize )
{
#ifdef ALLLOCK
            UNLOCK(fi->pthreadslock);
#endif
            if ( fi == AR.hidefile ) {
                UNLOCK(AS.inputslock);
            }
            MLOCK(ErrorMessageLock);
            MesPrint("Write error while sorting. Disk full?");
            MesPrint("Attempt to write %l bytes on file %d at position %15p",
                        fi->POsize,fi->handle,&(fi->POposition));
            MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG)(fi->PObuffer));
            MUNLOCK(ErrorMessageLock);
            return(-1);
          }
          ADDPOS(fi->filesize,fi->POsize);
          fi->POfill = fi->PObuffer;
          ADDPOS(fi->POposition,fi->POsize);
          if ( fi == AR.hidefile ) {
            UNLOCK(AS.inputslock);
          }
#ifdef ALLLOCK
          UNLOCK(fi->pthreadslock);
#endif
#ifdef WITHPTHREADS
          if ( AS.MasterSort && AC.ThreadSortFileSynch && fi != AR.hidefile ) {
            if ( fi->handle >= 0 ) SynchFile(fi->handle);
          }
#endif
        }
    }
    *(fi->POfill)++ = 0;
    fi->POfull = fi->POfill;
/*
```

```
    {
        UBYTE OutBuf[140];
        if ( AP.DebugFlag ) {
            AO.OutFill = AO.OutputLine = OutBuf;
            AO.OutSkip = 3;
            FiniLine();
            TokenToLine((UBYTE *)"End of expression written");
            FiniLine();
        }
    }
*/
    size = (fi->POfill-fi->PObuffer)*sizeof(WORD);
    if ( fi->handle >= 0 ) {
#ifdef WITHZLIB
        if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
        && dobracketindex == 0 && ( compr > 0 ) && fi->zsp != 0 ) {
            if ( FlushOutputGZIP(fi) ) return(-1);
            fi->POfill = fi->PObuffer;
        }
        else
#endif
        {
#ifdef ALLLOCK
          LOCK(fi->pthreadslock);
#endif
          if ( fi == AR.hidefile ) {
            LOCK(AS.inputslock);
          }
          SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
/*
          MesPrint("FlushOut: writing %l bytes to position %12p",size,&(fi->POposition));
*/
          if ( ( RetCode = WriteFile(fi->handle,(UBYTE *)(fi->PObuffer),size) ) != size ) {
#ifdef ALLLOCK
            UNLOCK(fi->pthreadslock);
#endif
            if ( fi == AR.hidefile ) {
                UNLOCK(AS.inputslock);
            }
            MLOCK(ErrorMessageLock);
            MesPrint("Write error while finishing sorting. Disk full?");
            MesPrint("Attempt to write %l bytes on file %d at position %15p",
                        size,fi->handle,&(fi->POposition));
            MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG)(fi->PObuffer));
            MUNLOCK(ErrorMessageLock);
            return(-1);
          }
          ADDPOS(fi->filesize,size);
          ADDPOS(fi->POposition,size);
          fi->POfill = fi->PObuffer;
          if ( fi == AR.hidefile ) {
            UNLOCK(AS.inputslock);
          }
#ifdef ALLLOCK
          UNLOCK(fi->pthreadslock);
#endif
#ifdef WITHPTHREADS
          if ( AS.MasterSort && AC.ThreadSortFileSynch ) {
            if ( fi->handle >= 0 ) SynchFile(fi->handle);
          }
#endif
        }
    }
    if ( dobracketindex ) {
        BRACKETINFO *b = Expressions[AR.CurExpr].newbracketinfo;
        if ( b->indexfill > 0 ) {
            DIFPOS(b->indexbuffer[b->indexfill-1].next,*position,Expressions[AR.CurExpr].onfile);
        }
    }
#ifdef WITHZLIB
    if ( AT.SS == AT.S0 && !AR.NoCompress && AR.gzipCompress > 0
        && dobracketindex == 0 && ( compr > 0 ) && fi->zsp != 0 ) {
        PUTZERO(*position);
        if ( fi->handle >= 0 ) {
#ifdef ALLLOCK
            LOCK(fi->pthreadslock);
#endif
            SeekFile(fi->handle,position,SEEK_END);
#ifdef ALLLOCK
            UNLOCK(fi->pthreadslock);
#endif
        }
        else {
            ADDPOS(*position,((UBYTE *)fi->POfill-(UBYTE *)fi->PObuffer));
        }
    }
    else
```

```
#endif
    {
        ADDPOS(*position,sizeof(WORD));
    }
    return(0);
}

/*
        #] FlushOut :
        #[ AddCoef :                WORD AddCoef(pterm1,pterm2)
*/
WORD AddCoef(PHEAD WORD **ps1, WORD **ps2)
{
    GETBIDENTITY
    SORTING *S = AT.SS;
    WORD *s1, *s2;
    WORD l1, l2, i;
    WORD OutLen, *t, j;
    UWORD *OutCoef;
#ifdef WITHFLOAT
    if ( AT.SortFloatMode ) return(AddWithFloat(BHEAD ps1,ps2));
#endif
    OutCoef = AN.SoScratC;
    s1 = *ps1; s2 = *ps2;
    GETCOEF(s1,l1);
    GETCOEF(s2,l2);
    if ( AddRat(BHEAD (UWORD *)s1,l1,(UWORD *)s2,l2,OutCoef,&OutLen) ) {
        MLOCK(ErrorMessageLock);
        MesCall("AddCoef");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( AN.ncmod != 0 ) {
        if ( ( AC.modmode & POSNEG ) != 0 ) {
            NormalModulus(OutCoef,&OutLen);
/*
            We had forgotten that this can also become smaller but the
            denominator isn't there. Correct in the other case
            17-may-2009 [JV]
*/
            j = ABS(OutLen); OutCoef[j] = 1;
            for ( i = 1; i < j; i++ ) OutCoef[j+i] = 0;
        }
        else if ( BigLong(OutCoef,OutLen,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
            SubPLon(OutCoef,OutLen,(UWORD *)AC.cmod,ABS(AN.ncmod),OutCoef,&OutLen);
            OutCoef[OutLen] = 1;
            for ( i = 1; i < OutLen; i++ ) OutCoef[OutLen+i] = 0;
        }
    }
    if ( !OutLen ) { *ps1 = *ps2 = 0; return(0); }
    OutLen *= 2;
    if ( OutLen < 0 ) i = - ( --OutLen );
    else              i = ++OutLen;
    if ( l1 < 0 ) l1 = -l1;
    l1 *= 2; l1++;
    if ( i <= l1 ) {    /* Fits in 1 */
        l1 -= i;
        **ps1 -= l1;
        s2 = (WORD *)OutCoef;
        while ( --i > 0 ) *s1++ = *s2++;
        *s1++ = OutLen;
        while ( --l1 >= 0 ) *s1++ = 0;
        goto RegEnd;
    }
    if ( l2 < 0 ) l2 = -l2;
    l2 *= 2; l2++;
    if ( i <= l2 ) {    /* Fits in 2 */
        l2 -= i;
        **ps2 -= l2;
        s1 = (WORD *)OutCoef;
        while ( --i > 0 ) *s2++ = *s1++;
        *s2++ = OutLen;
        while ( --l2 >= 0 ) *s2++ = 0;
        *ps1 = *ps2;
        goto RegEnd;
    }

    /* Doesn't fit. Make a new term. */

    t = s1;
    s1 = *ps1;
    j = *s1++ + i - l1;     /* Space needed */
    if ( (S->sFill + j) >= S->sTop2 ) {
        GarbHand();
        s1 = *ps1;
        t = s1 + *s1 - 1;
        j = *s1++ + i - l1;     /* Space needed */
```

```
        l1 = *t;
        if ( l1 < 0 ) l1  = - l1;
        t -= l1-1;
    }
    s2 = S->sFill;
    *s2++ = j;
    while ( s1 < t ) *s2++ = *s1++;
    s1 = (WORD *)OutCoef;
    while ( --i > 0 ) *s2++ = *s1++;
    *s2++ = OutLen;
    *ps1 = S->sFill;
    S->sFill = s2;
RegEnd:
    *ps2 = 0;
    if ( **ps1 > AM.MaxTer/((LONG)(sizeof(WORD))) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Term too complex after addition in sort. MaxTermSize = %10l",
        AM.MaxTer/sizeof(WORD));
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    return(1);
}

/*
        #] AddCoef :
        #[ AddPoly :                WORD AddPoly(pterm1,pterm2)
*/
WORD AddPoly(PHEAD WORD **ps1, WORD **ps2)
{
    GETBIDENTITY
    SORTING *S = AT.SS;
    WORD i;
    WORD *s1, *s2, *m, *w, *t, oldpw = S->PolyWise;
    s1 = *ps1 + S->PolyWise;
    s2 = *ps2 + S->PolyWise;
    w = AT.WorkPointer;
/*
    Add here the two arguments. Is a straight merge.
*/
    if ( S->PolyFlag == 2 && AR.PolyFunExp != 2 && AR.PolyFunExp != 3 ) {
        WORD **oldSplitScratch = AN.SplitScratch;
        LONG oldSplitScratchSize = AN.SplitScratchSize;
        LONG oldInScratch = AN.InScratch;
        WORD oldtype = AR.SortType;
        if ( (WORD *)((UBYTE *)w + AM.MaxTer) >= AT.WorkTop ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Program was adding polyratfun arguments");
            MesWork();
            MUNLOCK(ErrorMessageLock);
        }
        AR.SortType = SORTHIGHFIRST;
        S->PolyWise = 0;
        AN.SplitScratch = AN.SplitScratch1;
        AN.SplitScratchSize = AN.SplitScratchSize1;
        AN.InScratch = AN.InScratch1;
        poly_ratfun_add(BHEAD s1,s2);
        S->PolyWise = oldpw;
        AN.SplitScratch1 = AN.SplitScratch;
        AN.SplitScratchSize1 = AN.SplitScratchSize;
        AN.InScratch1 = AN.InScratch;
        AN.SplitScratch = oldSplitScratch;
        AN.SplitScratchSize = oldSplitScratchSize;
        AN.InScratch = oldInScratch;
        AT.WorkPointer = w;
        AR.SortType = oldtype;
        if ( w[1] <= FUNHEAD ||
            ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) ) {
            *ps1 = *ps2 = 0; return(0);
        }
    }
    else {
        if ( w + s1[1] + s2[1] + 12 + ARGHEAD >= AT.WorkTop ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Program was adding polyfun arguments");
            MesWork();
            MUNLOCK(ErrorMessageLock);
        }
        AddArgs(BHEAD s1,s2,w);
    }
/*
    Now we need to store the result in a convenient place.
*/
    if ( w[1] <= FUNHEAD ) { *ps1 = *ps2 = 0; return(0); }
    if ( w[1] <= s1[1] || w[1] <= s2[1] ) {   /* Fits in place. */
        if ( w[1] > s1[1] ) {
            *ps1 = *ps2;
```

```
            s1 = s2;
        }
        t = s1 + s1[1];
        m = *ps1 + **ps1;
        i = w[1];
        NCOPY(s1,w,i);
        if ( s1 != t ) {
            while ( t < m ) *s1++ = *t++;
            **ps1 = WORDDIF(s1,(*ps1));
        }
        *ps2 = 0;
    }
    else {      /* Make new term */
#ifdef TESTGARB
        s2 = *ps2;
#endif
        *ps2 = 0;
        if ( (S->sFill + (**ps1 + w[1] - s1[1])) >= S->sTop2 ) {
#ifdef TESTGARB
            MesPrint("------Garbage collection-------");
#endif
            AT.WorkPointer += w[1];
            GarbHand();
            AT.WorkPointer = w;
            s1 = *ps1;
            if ( (S->sFill + (**ps1 + w[1] - s1[1])) >= S->sTop2 ) {
#ifdef TESTGARB
                UBYTE OutBuf[140];
                MLOCK(ErrorMessageLock);
                AO.OutFill = AO.OutputLine = OutBuf;
                AO.OutSkip = 3;
                FiniLine();
                i = *s2;
                while ( --i >= 0 ) {
                    TalToLine((UWORD)(*s2++)); TokenToLine((UBYTE *)"  ");
                }
                FiniLine();
                AO.OutFill = AO.OutputLine = OutBuf;
                AO.OutSkip = 3;
                FiniLine();
                s2 = *ps1;
                i = *s2;
                while ( --i >= 0 ) {
                    TalToLine((UWORD)(*s2++)); TokenToLine((UBYTE *)"  ");
                }
                FiniLine();
                AO.OutFill = AO.OutputLine = OutBuf;
                AO.OutSkip = 3;
                FiniLine();
                s2 = w;
                i = w[1];
                while ( --i >= 0 ) {
                    TalToLine((UWORD)(*s2++)); TokenToLine((UBYTE *)"  ");
                }
                FiniLine();
                if ( AR.sLevel > 0 ) {
                    MesPrint("Please increase SubSmallExtension setup parameter.");
                }
                else {
                    MesPrint("Please increase SmallExtension setup parameter.");
                }
                MUNLOCK(ErrorMessageLock);
#else
                MLOCK(ErrorMessageLock);
                if ( AR.sLevel > 0 ) {
                    MesPrint("Please increase SubSmallExtension setup parameter.");
                }
                else {
                    MesPrint("Please increase SmallExtension setup parameter.");
                }
                MUNLOCK(ErrorMessageLock);
#endif
                Terminate(-1);
            }
        }
        t = *ps1;
        s2 = S->sFill;
        m = s2;
        i = S->PolyWise;
        NCOPY(s2,t,i);
        i = w[1];
        NCOPY(s2,w,i);
        t = t + t[1];
        w = *ps1 + **ps1;
        while ( t < w ) *s2++ = *t++;
        *m = WORDDIF(s2,m);
        *ps1 = m;
```

```
        S->sFill = s2;
        if ( *m > AM.MaxTer/((LONG)sizeof(WORD)) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Term too complex after polynomial addition. MaxTermSize = %10l",
            AM.MaxTer/sizeof(WORD));
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    }
    return(1);
}

/*
        #] AddPoly :
        #[ AddArgs :                void AddArgs(arg1,arg2,to)
*/

#define INSLENGTH(x)  w[1] = FUNHEAD+ARGHEAD+x; w[FUNHEAD] = ARGHEAD+x;

void AddArgs(PHEAD WORD *s1, WORD *s2, WORD *m)
{
    GETBIDENTITY
    WORD i1, i2;
    WORD *w = m, *mm, *t, *t1, *t2, *tstop1, *tstop2;
    WORD tempterm[8+FUNHEAD];

    *m++ = AR.PolyFun; *m++ = 0; FILLFUN(m)
    *m++ = 0; *m++ = 0; FILLARG(m)
    if ( s1[FUNHEAD] < 0 || s2[FUNHEAD] < 0 ) {
        if ( s1[FUNHEAD] < 0 ) {
            if ( s2[FUNHEAD] < 0 ) {    /* Both are special */
                if ( s1[FUNHEAD] <= -FUNCTION ) {
                    if ( s2[FUNHEAD] == s1[FUNHEAD] ) {
                        *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
                        FILLFUN(m)
                        *m++ = 2; *m++ = 1; *m++ = 3;
                        INSLENGTH(4+FUNHEAD)
                    }
                    else if ( s2[FUNHEAD] <= -FUNCTION ) {
                        i1 = functions[-FUNCTION-s1[FUNHEAD]].commute != 0;
                        i2 = functions[-FUNCTION-s2[FUNHEAD]].commute != 0;
                        if ( ( !i1 && i2 ) || ( i1 == i2 && i1 > i2 ) ) {
                            i1 = s2[FUNHEAD];
                            s2[FUNHEAD] = s1[FUNHEAD];
                            s1[FUNHEAD] = i1;
                        }
                        *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
                        FILLFUN(m)
                        *m++ = 1; *m++ = 1; *m++ = 3;
                        *m++ = 4+FUNHEAD; *m++ = -s2[FUNHEAD]; *m++ = FUNHEAD;
                        FILLFUN(m)
                        *m++ = 1; *m++ = 1; *m++ = 3;
                        INSLENGTH(8+2*FUNHEAD)
                    }
                    else if ( s2[FUNHEAD] == -SYMBOL ) {
                        *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s2[FUNHEAD+1]; *m++ = 1;
                        *m++ = 1; *m++ = 1; *m++ = 3;
                        *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
                        FILLFUN(m)
                        *m++ = 1; *m++ = 1; *m++ = 3;
                        INSLENGTH(12+FUNHEAD)
                    }
                    else {      /* number */
                        *m++ = 4;
                        *m++ = ABS(s2[FUNHEAD+1]); *m++ = 1; *m++ = s2[FUNHEAD+1] < 0 ? -3: 3;
                        *m++ = 4+FUNHEAD; *m++ = -s1[FUNHEAD]; *m++ = FUNHEAD;
                        FILLFUN(m)
                        *m++ = 1; *m++ = 1; *m++ = 3;
                        INSLENGTH(8+FUNHEAD)
                    }
                }
                else if ( s1[FUNHEAD] == -SYMBOL ) {
                    if ( s2[FUNHEAD] == s1[FUNHEAD] ) {
                        if ( s1[FUNHEAD+1] == s2[FUNHEAD+1] ) {
                            *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s1[FUNHEAD+1];
                            *m++ = 1; *m++ = 2; *m++ = 1; *m++ = 3;
                            INSLENGTH(8)
                        }
                        else {
                            if ( s1[FUNHEAD+1] > s2[FUNHEAD+1] )
                                { i1 = s2[FUNHEAD+1]; i2 = s1[FUNHEAD+1]; }
                            else { i1 = s1[FUNHEAD+1]; i2 = s2[FUNHEAD+1]; }
                            *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = i1;
                            *m++ = 1; *m++ = 1; *m++ = 1; *m++ = 3;
                            *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = i2;
                            *m++ = 1; *m++ = 1; *m++ = 1; *m++ = 3;
                            INSLENGTH(16)
```

```
                    }
                }
                else if ( s2[FUNHEAD] <= -FUNCTION ) {
                    *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s1[FUNHEAD+1]; *m++ = 1;
                    *m++ = 1; *m++ = 1; *m++ = 3;
                    *m++ = 4+FUNHEAD; *m++ = -s2[FUNHEAD]; *m++ = FUNHEAD;
                    FILLFUN(m)
                    *m++ = 1; *m++ = 1; *m++ = 3;
                    INSLENGTH(12+FUNHEAD)
                }
                else {
                    *m++ = 4;
                    *m++ = ABS(s2[FUNHEAD+1]); *m++ = 1; *m++ = s2[FUNHEAD+1] < 0 ? -3: 3;
                    *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s1[FUNHEAD+1]; *m++ = 1;
                    *m++ = 1; *m++ = 1; *m++ = 3;
                    INSLENGTH(12)
                }
            }
            else {  /* Must be -SNUMBER! */
                if ( s2[FUNHEAD] <= -FUNCTION ) {
                    *m++ = 4;
                    *m++ = ABS(s1[FUNHEAD+1]); *m++ = 1; *m++ = s1[FUNHEAD+1] < 0 ? -3: 3;
                    *m++ = 4+FUNHEAD; *m++ = -s2[FUNHEAD]; *m++ = FUNHEAD;
                    FILLFUN(m)
                    *m++ = 1; *m++ = 1; *m++ = 3;
                    INSLENGTH(8+FUNHEAD)
                }
                else if ( s2[FUNHEAD] == -SYMBOL ) {
                    *m++ = 4;
                    *m++ = ABS(s1[FUNHEAD+1]); *m++ = 1; *m++ = s1[FUNHEAD+1] < 0 ? -3: 3;
                    *m++ = 8; *m++ = SYMBOL; *m++ = 4; *m++ = s2[FUNHEAD+1]; *m++ = 1;
                    *m++ = 1; *m++ = 1; *m++ = 3;
                    INSLENGTH(12)
                }
                else {      /* Both are numbers. add. */
                    LONG x1;
                    x1 = (LONG)s1[FUNHEAD+1] + (LONG)s2[FUNHEAD+1];
                    if ( x1 < 0 ) { i1 = (WORD)(-x1); i2 = -3; }
                    else { i1 = (WORD)x1; i2 = 3; }
                    if ( x1 && AN.ncmod != 0 ) {
                        m[0] = 4;
                        m[1] = i1;
                        m[2] = 1;
                        m[3] = i2;
                        if ( Modulus(m) ) Terminate(-1);
                        if ( *m == 0 ) w[1] = 0;
                        else {
                            if ( *m == 4 && ( m[1] & MAXPOSITIVE ) == m[1]
                            && m[3] == 3 ) {
                                i1 = m[1];
                                m -= ARGHEAD;
                                *m++ = -SNUMBER;
                                *m++ = i1;
                                INSLENGTH(4)
                            }
                            else {
                                INSLENGTH(*m)
                                m += *m;
                            }
                        }
                    }
                    else {
                        if ( x1 == 0 ) {
                            w[1] = FUNHEAD;
                        }
                        else if ( ( i1 & MAXPOSITIVE ) == i1 ) {
                            m -= ARGHEAD;
                            *m++ = -SNUMBER;
                            *m++ = (WORD)x1;
                            w[1] = FUNHEAD+2;
                        }
                        else {
                            *m++ = 4; *m++ = i1; *m++ = 1; *m++ = i2;
                            INSLENGTH(4)
                        }
                    }
                }
            }
        }
        else {  /* Only s1 is special */
s1only:
/*
        Compose a term in `tempterm'
*/
            t = tempterm;
            if ( s1[FUNHEAD] <= -FUNCTION ) {
                *t++ = 4+FUNHEAD; *t++ = -s1[FUNHEAD]; *t++ = FUNHEAD;
```

```
                    FILLFUN(t)
                    *t++ = 1; *t++ = 1; *t++ = 3;
                }
                else if ( s1[FUNHEAD] == -SYMBOL ) {
                    *t++ = 8; *t++ = SYMBOL; *t++ = 4;
                    *t++ = s1[FUNHEAD+1]; *t++ = 1;
                    *t++ = 1; *t++ = 1; *t++ = 3;
                }
                else {
                    *t++ = 4;   *t++ = ABS(s1[FUNHEAD+1]);
                    *t++ = 1;   *t++ = s1[FUNHEAD+1] < 0 ? -3: 3;
                }
                tstop1 = t;
                s1 = tempterm;
                goto twogen;
            }
        }
        else {      /* Only s2 is special */
            t = s1;
            s1 = s2;
            s2 = t;
            goto s1only;
        }
    }
    else {
        int oldPolyFlag;
        tstop1 = s1 + s1[1];
        s1 += FUNHEAD+ARGHEAD;
twogen:
        tstop2 = s2 + s2[1];
        s2 += FUNHEAD+ARGHEAD;
/*
        Now we should merge the expressions in s1 and s2 into m.
*/
        oldPolyFlag = AT.SS->PolyFlag;
        AT.SS->PolyFlag = 0;
        while ( s1 < tstop1 && s2 < tstop2 ) {
            i1 = CompareTerms(BHEAD s1,s2,(WORD)(-1));
            if ( i1 > 0 ) {
                i2 = *s1;
                NCOPY(m,s1,i2);
            }
            else if ( i1 < 0 ) {
                i2 = *s2;
                NCOPY(m,s2,i2);
            }
            else {  /* Coefficients should be added. */
                WORD i;
                t = s1+*s1;
                i1 = t[-1];
                i2 = *s1 - ABS(i1);
                t2 = s2 + i2;
                s2 += *s2;
                mm = m;
                NCOPY(m,s1,i2);
                t1 = s1;
                s1 = t;
                i2 = s2[-1];
/*
                t1,i1 is the first coefficient
                t2,i2 is the second coefficient
                It should be placed at m,i1
*/
                i1 = REDLENG(i1);
                i2 = REDLENG(i2);
                if ( AddRat(BHEAD (UWORD *)t1,i1,(UWORD *)t2,i2,(UWORD *)m,&i) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Addition of coefficients of PolyFun");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                if ( i == 0 ) {
                    m = mm;
                }
                else {
                    i1 = INCLENG(i);
                    m += ABS(i1);
                    m[-1] = i1;
                    *mm = WORDDIF(m,mm);
                    if ( AN.ncmod != 0 ) {
                        if ( Modulus(mm) ) Terminate(-1);
                        if ( !*mm ) m = mm;
                        else m = mm + *mm;
                    }
                }
            }
        }
```

```
        while ( s1 < tstop1 ) *m++ = *s1++;
        while ( s2 < tstop2 ) *m++ = *s2++;
        w[1] = WORDDIF(m,w);
        w[FUNHEAD] = w[1] - FUNHEAD;
        if ( ToFast(w+FUNHEAD,w+FUNHEAD) ) {
            if ( w[FUNHEAD] <= -FUNCTION ) w[1] = FUNHEAD+1;
            else w[1] = FUNHEAD+2;
            if ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) w[1] = FUNHEAD;
        }
/*      AT.SS->PolyFlag = AR.PolyFunType;*/
        AT.SS->PolyFlag = oldPolyFlag;
    }
}

/*
        #] AddArgs :
        #[ Compare1 :               WORD Compare1(term1,term2,level)
*/
WORD Compare1(PHEAD WORD *term1, WORD *term2, WORD level)
{
    SORTING *S = AT.SS;
    WORD *stopper1, *stopper2, *t2;
    WORD *s1, *s2, *t1;
    WORD *stopex1, *stopex2;
    WORD c1, c2;
    WORD prevorder;
    WORD count = -1, localPoly, polyhit = -1;

    if ( AR.sLevel == 0 ) {
        numcompares++;
    }

    if ( S->PolyFlag ) {
/*
        if ( S->PolyWise != 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("S->PolyWise is not zero!!!!!");
            MUNLOCK(ErrorMessageLock);
        }
*/
        count = 0; localPoly = 1; S->PolyWise = polyhit = 0;
        S->PolyFlag = AR.PolyFunType;
        if ( AR.PolyFunType == 2 &&
             ( AR.PolyFunExp == 2 || AR.PolyFunExp == 3 ) ) S->PolyFlag = 1;
    }
    else { localPoly = 0; }
#ifdef WITHFLOAT
    AT.SortFloatMode = 0;
#endif
    prevorder = 0;
    GETSTOP(term1,s1);
    stopper1 = s1;
    GETSTOP(term2,stopper2);
    t1 = term1 + 1;
    t2 = term2 + 1;
    while ( t1 < stopper1 && t2 < stopper2 ) {
        if ( *t1 != *t2 ) {
            if ( *t1 == HAAKJE ) return(PREV(-1));
            if ( *t2 == HAAKJE ) return(PREV(1));
            if ( *t1 >= (FUNCTION-1) ) {
                if ( *t2 < (FUNCTION-1) ) return(PREV(-1));
                if ( *t1 < FUNCTION && *t2 < FUNCTION ) return(PREV(*t2-*t1));
                if ( *t1 < FUNCTION ) return(PREV(1));
                if ( *t2 < FUNCTION ) return(PREV(-1));
                c1 = functions[*t1-FUNCTION].commute;
                c2 = functions[*t2-FUNCTION].commute;
                if ( !c1 ) {
                    if ( c2 ) return(PREV(1));
                    else return(PREV(*t2-*t1));
                }
                else {
                    if ( !c2 ) return(PREV(-1));
                    else return(PREV(*t2-*t1));
                }
            }
            else return(PREV(*t2-*t1));
        }
        s1 = t1 + 2;
        s2 = t2 + 2;
        c1 = *t1;
        t1 += t1[1];
        t2 += t2[1];
        if ( localPoly && c1 < FUNCTION ) {
            polyhit = 1;
        }
        if ( c1 <= (FUNCTION-1)
        || ( c1 >= FUNCTION && functions[c1-FUNCTION].spec > 0 ) ) {
```

```
            if ( c1 == SYMBOL ) {
                if ( *s1 == FACTORSYMBOL && *s2 == FACTORSYMBOL
                 && s1[-1] == 4 && s2[-1] == 4
                 && ( ( t1 < stopper1 && *t1 == HAAKJE )
                 || ( t1 == stopper1 && AT.fromindex ) ) ) {
/*
                    We have to be very careful with the criteria here, because
                    Compare1 is called both in the regular sorting and by the
                    routine that makes the bracket index. In the last case
                    there is no HAAKJE subterm.
*/
                    if ( s1[1] != s2[1] ) return(s2[1]-s1[1]);
                    s1 += 2; s2 += 2;
                }
                else if ( AR.SortType >= SORTPOWERFIRST ) {
                    WORD i1 = 0, *r1;
                    r1 = s1;
                    while ( s1 < t1 ) { i1 += s1[1]; s1 += 2; }
                    s1 = r1; r1 = s2;
                    while ( s2 < t2 ) { i1 -= s2[1]; s2 += 2; }
                    s2 = r1;
                    if ( i1 ) {
                        if ( AR.SortType >= SORTANTIPOWER ) i1 = -i1;
                        return(PREV(i1));
                    }
                }
                while ( s1 < t1 ) {
                    if ( s2 >= t2 ) {
/*                      return(PREV(1));   */
                        if ( AR.SortType==SORTLOWFIRST ) {
                            return(PREV((s1[1]>0?-1:1)));
                        }
                        else {
                            return(PREV((s1[1]<0?-1:1)));
                        }
                    }
                    if ( *s1 != *s2 ) {
/*                      return(PREV(*s2-*s1)); */
                        if ( AR.SortType==SORTLOWFIRST ) {
                            if ( *s1 < *s2 ) {
                                return(PREV((s1[1]<0?1:-1)));
                            }
                            else {
                                return(PREV((s2[1]<0?-1:1)));
                            }
                        }
                        else {
                            if ( *s1 < *s2 ) {
                                return(PREV((s1[1]<0?-1:1)));
                            }
                            else {
                                return(PREV((s2[1]<0?1:-1)));
                            }
                        }
                    }
                    s1++; s2++;
                    if ( *s1 != *s2 ) return(
                        PREV((AR.SortType==SORTLOWFIRST?*s2-*s1:*s1-*s2)));
                    s1++; s2++;
                }
                if ( s2 < t2 ) {
/*                  return(PREV(-1));   */
                    if ( AR.SortType==SORTLOWFIRST ) {
                        return(PREV((s2[1]<0?-1:1)));
                    }
                    else {
                        return(PREV((s2[1]<0?1:-1)));
                    }
                }
            }
            else if ( c1 == DOTPRODUCT ) {
                if ( AR.SortType >= SORTPOWERFIRST ) {
                    WORD i1 = 0, *r1;
                    r1 = s1;
                    while ( s1 < t1 ) { i1 += s1[2]; s1 += 3; }
                    s1 = r1; r1 = s2;
                    while ( s2 < t2 ) { i1 -= s2[2]; s2 += 3; }
                    s2 = r1;
                    if ( i1 ) {
                        if ( AR.SortType >= SORTANTIPOWER ) i1 = -i1;
                        return(PREV(i1));
                    }
                }
                while ( s1 < t1 ) {
                    if ( s2 >= t2 ) return(PREV(1));
                    if ( *s1 != *s2 ) return(PREV(*s2-*s1));
                    s1++; s2++;
```

```
                    if ( *s1 != *s2 ) return(PREV(*s2-*s1));
                    s1++; s2++;
                    if ( *s1 != *s2 ) return(
                        PREV((AR.SortType==SORTLOWFIRST?*s2-*s1:*s1-*s2)));
                    s1++; s2++;
                }
                if ( s2 < t2 ) return(PREV(-1));
            }
            else {
                while ( s1 < t1 ) {
                    if ( s2 >= t2 ) return(PREV(1));
                    if ( *s1 != *s2 ) return(PREV(*s2-*s1));
                    s1++; s2++;
                }
                if ( s2 < t2 ) return(PREV(-1));
            }
        }
        else {
#if FUNHEAD != 2
            s1 += FUNHEAD-2;
            s2 += FUNHEAD-2;
#endif
            if ( localPoly && c1 == AR.PolyFun ) {
                if ( count == 0 ) {
                  if ( S->PolyFlag == 1 ) {
                    WORD i1, i2;
                    if ( *s1 > 0 ) i1 = *s1;
                    else if ( *s1 <= -FUNCTION ) i1 = 1;
                    else i1 = 2;
                    if ( *s2 > 0 ) i2 = *s2;
                    else if ( *s2 <= -FUNCTION ) i2 = 1;
                    else i2 = 2;
                    if ( s1+i1 == t1 && s2+i2 == t2 ) { /* This is the stuff */
/*
                        Test for scalar nature
*/
                        if ( !polyhit ) {
                            WORD *u1, *u2, *ustop;
                            if ( *s1 < 0 ) {
                                if ( *s1 != -SNUMBER && *s1 != -SYMBOL && *s1 > -FUNCTION )
                                    goto NoPoly;
                            }
                            else {
                                u1 = s1 + ARGHEAD;
                                while ( u1 < t1 ) {
                                    u2 = u1 + *u1;
                                    ustop = u2 - ABS(u2[-1]);
                                    u1++;
                                    while ( u1 < ustop ) {
                                        if ( *u1 == INDEX ) goto NoPoly;
                                        u1 += u1[1];
                                    }
                                    u1 = u2;
                                }
                            }
                            if ( *s2 < 0 ) {
                                if ( *s2 != -SNUMBER && *s2 != -SYMBOL && *s2 > -FUNCTION )
                                    goto NoPoly;
                            }
                            else {
                                u1 = s2 + ARGHEAD;
                                while ( u1 < t2 ) {
                                    u2 = u1 + *u1;
                                    ustop = u2 - ABS(u2[-1]);
                                    u1++;
                                    while ( u1 < ustop ) {
                                        if ( *u1 == INDEX ) goto NoPoly;
                                        u1 += u1[1];
                                    }
                                    u1 = u2;
                                }
                            }
                        }
                        S->PolyWise = WORDDIF(s1,term1);
                        S->PolyWise -= FUNHEAD;
                        count = 1;
                        continue;
                    }
                    else {
NoPoly:
                        S->PolyWise = localPoly = 0;
                    }
                  }
                  else if ( AR.PolyFunType == 2 ) {
                    WORD i1, i2, i1a, i2a;
                    if ( *s1 > 0 ) i1 = *s1;
                    else if ( *s1 <= -FUNCTION ) i1 = 1;
```

```
                    else i1 = 2;
                    if ( *s2 > 0 ) i2 = *s2;
                    else if ( *s2 <= -FUNCTION ) i2 = 1;
                    else i2 = 2;
                    if ( s1[i1] > 0 ) i1a = s1[i1];
                    else if ( s1[i1] <= -FUNCTION ) i1a = 1;
                    else i1a = 2;
                    if ( s2[i2] > 0 ) i2a = s2[i2];
                    else if ( s2[i2] <= -FUNCTION ) i2a = 1;
                    else i2a = 2;
                    if ( s1+i1+i1a == t1 && s2+i2+i2a == t2 ) { /* This is the stuff */
/*
                        Test for scalar nature
*/
                        if ( !polyhit ) {
                            WORD *u1, *u2, *ustop;
                            if ( *s1 < 0 ) {
                                if ( *s1 != -SNUMBER && *s1 != -SYMBOL && *s1 > -FUNCTION )
                                    goto NoPoly;
                            }
                            else {
                                u1 = s1 + ARGHEAD;
                                while ( u1 < s1+i1 ) {
                                    u2 = u1 + *u1;
                                    ustop = u2 - ABS(u2[-1]);
                                    u1++;
                                    while ( u1 < ustop ) {
                                        if ( *u1 == INDEX ) goto NoPoly;
                                        u1 += u1[1];
                                    }
                                    u1 = u2;
                                }
                            }
                            if ( s1[i1] < 0 ) {
                                if ( s1[i1] != -SNUMBER && s1[i1] != -SYMBOL && s1[i1] > -FUNCTION )
                                    goto NoPoly;
                            }
                            else {
                                u1 = s1 +i1 + ARGHEAD;
                                while ( u1 < t1 ) {
                                    u2 = u1 + *u1;
                                    ustop = u2 - ABS(u2[-1]);
                                    u1++;
                                    while ( u1 < ustop ) {
                                        if ( *u1 == INDEX ) goto NoPoly;
                                        u1 += u1[1];
                                    }
                                    u1 = u2;
                                }
                            }
                            if ( *s2 < 0 ) {
                                if ( *s2 != -SNUMBER && *s2 != -SYMBOL && *s2 > -FUNCTION )
                                    goto NoPoly;
                            }
                            else {
                                u1 = s2 + ARGHEAD;
                                while ( u1 < s2+i2 ) {
                                    u2 = u1 + *u1;
                                    ustop = u2 - ABS(u2[-1]);
                                    u1++;
                                    while ( u1 < ustop ) {
                                        if ( *u1 == INDEX ) goto NoPoly;
                                        u1 += u1[1];
                                    }
                                    u1 = u2;
                                }
                            }
                            if ( s2[i2] < 0 ) {
                                if ( s2[i2] != -SNUMBER && s2[i2] != -SYMBOL && s2[i2] > -FUNCTION )
                                    goto NoPoly;
                            }
                            else {
                                u1 = s2 + i2 + ARGHEAD;
                                while ( u1 < t2 ) {
                                    u2 = u1 + *u1;
                                    ustop = u2 - ABS(u2[-1]);
                                    u1++;
                                    while ( u1 < ustop ) {
                                        if ( *u1 == INDEX ) goto NoPoly;
                                        u1 += u1[1];
                                    }
                                    u1 = u2;
                                }
                            }
                        }
                        S->PolyWise = WORDDIF(s1,term1);
                        S->PolyWise -= FUNHEAD;
```

```
                        count = 1;
                        continue;
                    }
                    else {
                        S->PolyWise = localPoly = 0;
                    }
                  }
                  else {
                    S->PolyWise = localPoly = 0;
                  }
                }
                else {
                    t1 = term1 + S->PolyWise;
                    t2 = term2 + S->PolyWise;
                    S->PolyWise = 0;
                    localPoly = 0;
                    continue;
                }
            }
#ifdef WITHFLOAT
            if ( c1 == FLOATFUN && t1 == stopper1 && t2 == stopper2 && AT.aux_ != 0 ) {
/*
                We have two FLOATFUN's. Test whether they are 'legal'
*/
                if ( TestFloat(s1-FUNHEAD) ) {
                    if ( TestFloat(s2-FUNHEAD) ) { AT.SortFloatMode = 3; return(0); }
                    else { return(1); }
                }
                else if ( TestFloat(s2-FUNHEAD) ) { return(-1); }
            }
#endif
            while ( s1 < t1 ) {
/*
                The next statement was added 9-nov-2001. It repaired a bad error
*/
                if ( s2 >= t2 ) return(PREV(-1));
/*
                There is a little problem here with fast arguments
                We don't want to sacrifice speed, but we like to
                keep a rational ordering. This last one suffers in
                the solution that has been chosen here.
*/
                if ( AC.properorderflag ) {
                    WORD oldpolyflag;
                    oldpolyflag = S->PolyFlag;
                    S->PolyFlag = 0;
                    if ( ( c2 = -CompArg(s1,s2) ) != 0 ) {
                        S->PolyFlag = oldpolyflag; return(PREV(c2));
                    }
                    S->PolyFlag = oldpolyflag;
                    NEXTARG(s1)
                    NEXTARG(s2)
                }
                else {
                    if ( *s1 > 0 ) {
                        if ( *s2 > 0 ) {
                            WORD oldpolyflag;
                            stopex1 = s1 + *s1;
                            if ( s2 >= t2 ) return(PREV(-1));
                            stopex2 = s2 + *s2;
                            s1 += ARGHEAD; s2 += ARGHEAD;
                            oldpolyflag = S->PolyFlag;
                            S->PolyFlag = 0;
                            while ( s1 < stopex1 ) {
                                if ( s2 >= stopex2 ) {
                                    S->PolyFlag = oldpolyflag; return(PREV(-1));
                                }
                                if ( ( c2 = CompareTerms(BHEAD s1,s2,(WORD)1) ) != 0 ) {
                                    S->PolyFlag = oldpolyflag; return(PREV(c2));
                                }
                                s1 += *s1;
                                s2 += *s2;
                            }
                            S->PolyFlag = oldpolyflag;
                            if ( s2 < stopex2 ) return(PREV(1));
                        }
                        else return(PREV(1));
                    }
                    else {
                        if ( *s2 > 0 ) return(PREV(-1));
                        if ( *s1 != *s2 ) { return(PREV(*s1-*s2)); }
                        if ( *s1 > -FUNCTION ) {
                            if ( *++s1 != *++s2 ) { return(PREV(*s2-*s1)); }
                        }
                        s1++; s2++;
                    }
                }
```

```
            }
            if ( s2 < t2 ) return(PREV(1));
        }
    }
#ifdef WITHFLOAT
    if ( t1 < stopper1 && *t1 == FLOATFUN && t1+t1[1] == stopper1
            && TestFloat(t1) ) {
        AT.SortFloatMode = 1; return(0);
    }
    else if ( t2 < stopper2 && *t2 == FLOATFUN && t2+t2[1] == stopper2
            && TestFloat(t2) ) {
        AT.SortFloatMode = 2; return(0);
    }
#endif
    {
        if ( AR.SortType != SORTLOWFIRST ) {
            if ( t1 < stopper1 ) return(PREV(1));
            if ( t2 < stopper2 ) return(PREV(-1));
        }
        else {
            if ( t1 < stopper1 ) return(PREV(-1));
            if ( t2 < stopper2 ) return(PREV(1));
        }
    }
    if ( level == 3 ) return(CompCoef(term1,term2));
    if ( level >= 1 )
        return(CompCoef(term2,term1));
    return(0);
}

/*
        #] Compare1 :
        #[ CompareSymbols :          WORD CompareSymbols(term1,term2,par)
*/
WORD CompareSymbols(PHEAD WORD *term1, WORD *term2, WORD par)
{
    int sum1, sum2;
    WORD *t1, *t2, *tt1, *tt2;
    int low, high;
    DUMMYUSE(par);
    if ( AR.SortType == SORTLOWFIRST ) { low = 1; high = -1; }
    else { low = -1; high = 1; }
    t1 = term1 + 1; tt1 = term1+*term1; tt1 -= ABS(tt1[-1]); t1 += 2;
    t2 = term2 + 1; tt2 = term2+*term2; tt2 -= ABS(tt2[-1]); t2 += 2;
    if ( AN.polysortflag > 0 ) {
        sum1 = 0; sum2 = 0;
        while ( t1 < tt1 ) { sum1 += t1[1]; t1 += 2; }
        while ( t2 < tt2 ) { sum2 += t2[1]; t2 += 2; }
        if ( sum1 < sum2 ) return(low);
        if ( sum1 > sum2 ) return(high);
        t1 = term1+3; t2 = term2 + 3;
    }
    while ( t1 < tt1 && t2 < tt2 ) {
        if ( *t1 > *t2 ) return(low);
        if ( *t1 < *t2 ) return(high);
        if ( t1[1] < t2[1] ) return(low);
        if ( t1[1] > t2[1] ) return(high);
        t1 += 2; t2 += 2;
    }
    if ( t1 < tt1 ) return(high);
    if ( t2 < tt2 ) return(low);
    return(0);
}

/*
        #] CompareSymbols :
        #[ CompareHSymbols :         WORD CompareHSymbols(term1,term2,par)
*/
WORD CompareHSymbols(PHEAD WORD *term1, WORD *term2, WORD par)
{
    WORD *t1, *t2, *tt1, *tt2, *ttt1, *ttt2;
    DUMMYUSE(par);
    DUMMYUSE(AT.WorkPointer);
    t1 = term1 + 1; tt1 = term1+*term1; tt1 -= ABS(tt1[-1]); t1 += 2;
    t2 = term2 + 1; tt2 = term2+*term2; tt2 -= ABS(tt2[-1]); t2 += 2;
    while ( t1 < tt1 && t2 < tt2 ) {
        if ( *t1 != *t2 ) {
            if ( t1[0] < t2[0] ) return(-1);
            return(1);
        }
        else if ( *t1 == HAAKJE ) {
            t1 += 3; t2 += 3; continue;
        }
        ttt1 = t1+t1[1]; ttt2 = t2+t2[1];
        while ( t1 < ttt1 && t2 < ttt2 ) {
            if ( *t1 > *t2 ) return(-1);
            if ( *t1 < *t2 ) return(1);
```

```
            if ( t1[1] < t2[1] ) return(-1);
            if ( t1[1] > t2[1] ) return(1);
            t1 += 2; t2 += 2;
        }
        if ( t1 < ttt1 ) return(1);
        if ( t2 < ttt2 ) return(-1);
    }
    if ( t1 < tt1 ) return(1);
    if ( t2 < tt2 ) return(-1);
    return(0);
}

/*
        #] CompareHSymbols :
        #[ ComPress :               LONG ComPress(ss,n)
*/
LONG ComPress(WORD **ss, LONG *n)
{
    GETIDENTITY
    WORD *t, *s, j, k;
    LONG size = 0;
    int newsize, i;
/*
            #[ debug :

    WORD **sss = ss;

    if ( AP.DebugFlag ) {
        UBYTE OutBuf[140];
        MLOCK(ErrorMessageLock);
        MesPrint("ComPress:");
        AO.OutFill = AO.OutputLine = OutBuf;
        AO.OutSkip = 3;
        FiniLine();
        ss = sss;
        while ( *ss ) {
            s = *ss++;
            j = *s;
            if ( j < 0 ) {
                j = s[1] + 2;
            }
            while ( --j >= 0 ) {
                TalToLine((UWORD)(*s++)); TokenToLine((UBYTE *)"  ");
            }
            FiniLine();
        }
        AO.OutSkip = 0;
        FiniLine();
        MUNLOCK(ErrorMessageLock);
        ss = sss;
    }

            #] debug :
*/
    *n = 0;
    if ( AT.SS == AT.S0 && !AR.NoCompress ) {
        if ( AN.compressSize == 0 ) {
            if ( *ss ) { AN.compressSize = **ss + 64; }
            else       { AN.compressSize = AM.MaxTer/sizeof(WORD) + 2; }
            AN.compressSpace = (WORD *)Malloc1(AN.compressSize*sizeof(WORD),"Compression");
        }
        AN.compressSpace[0] = 0;
        while ( *ss ) {
            k = 0;
            s = *ss;
            j = *s++;
            if ( j > AN.compressSize ) {
                newsize = j + 64;
                t = (WORD *)Malloc1(newsize*sizeof(WORD),"Compression");
                t[0] = 0;
                if ( AN.compressSpace ) {
                    for ( i = 0; i < *AN.compressSpace; i++ ) t[i] = AN.compressSpace[i];
                    M_free(AN.compressSpace,"Compression");
                }
                AN.compressSpace = t;
                AN.compressSize = newsize;
            }
            t = AN.compressSpace;
            i = *t - 1;
            *t++ = j; j--;
            if ( AR.PolyFun ) {
                WORD *polystop, *sa;
                sa = s + j;
                sa -= ABS(sa[-1]);
                polystop = s;
                while ( polystop < sa && *polystop != AR.PolyFun ) {
                    polystop += polystop[1];
```

```
                }
                while ( i > 0 && j > 0 && *s == *t && s < polystop ) {
                    i--; j--; s++; t++; k--;
                }
            }
            else {
                WORD *sa;
                sa = s + j;
                sa -= ABS(sa[-1]);
                while ( i > 0 && j > 0 && *s == *t && s < sa ) { i--; j--; s++; t++; k--; }
            }
            if ( k < -1 ) {
                s[-1] = j;
                s[-2] = k;
                *ss = s-2;
                size += j + 2;
            }
            else {
                size += *AN.compressSpace;
                if ( k == -1 ) { t--; s--; j++; }
            }
            while ( --j >= 0 ) *t++ = *s++;
/*              Sabotage getting into the coefficient next time */
            t = AN.compressSpace + *AN.compressSpace;
            t[-(ABS(t[-1]))] = 0;
            ss++;
            (*n)++;
        }
    }
    else {
        while ( *ss ) {
            size += *(*ss++);
            (*n)++;
        }
    }
/*
            #[ debug :

    if ( AP.DebugFlag ) {
        UBYTE OutBuf[140];
        AO.OutFill = AO.OutputLine = OutBuf;
        AO.OutSkip = 3;
        FiniLine();
        ss = sss;
        while ( *ss ) {
            s = *ss++;
            j = *s;
            if ( j < 0 ) {
                j = s[1] + 2;
            }
            while ( --j >= 0 ) {
                TalToLine((UWORD)(*s++)); TokenToLine((UBYTE *)"  ");
            }
            FiniLine();
        }
        AO.OutSkip = 0;
        FiniLine();
    }

            #] debug :
*/
    return(size);
}

/*
        #] ComPress :
        #[ SplitMerge :             void SplitMerge(Point,number)
*/
#ifdef NEWSPLITMERGE

LONG SplitMerge(PHEAD WORD **Pointer, LONG number)
{
    GETBIDENTITY
    SORTING *S = AT.SS;
    WORD **pp3, **pp1, **pp2, **pptop;
    LONG i, newleft, newright, split;

#ifdef SPLITMERGEDEBUG
    /* Print current array state on entry. */
    printf("%4ld: ", number);
    for (int ii = 0; ii < S->sTerms; ii++) {
        if ( (S->sPointer)[ii] ) {
            printf("%4d ", (unsigned)(S->sPointer[ii]-S->sBuffer));
        }
        else {
            printf(".... ");
        }
```

```
    }
    printf("\n");
    fflush(stdout);
#endif

    if ( number < 2 ) return(number);
    if ( number == 2 ) {
        pp1 = Pointer; pp2 = pp1 + 1;
        if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
            pp3 = (WORD **)(*pp1); *pp1 = *pp2; *pp2 = (WORD *)pp3;
        }
        else if ( i == 0 ) {
            number--;
            if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) == 0 ) number = 0; }
            else               { if ( AddCoef(BHEAD pp1,pp2) == 0 ) number = 0; }
        }
        return(number);
    }
    pptop = Pointer + number;
    split = number/2;
    newleft  = SplitMerge(BHEAD Pointer,split);
    newright = SplitMerge(BHEAD Pointer+split,number-split);
    if ( newright == 0 ) return(newleft);
/*
    We compare the last of the left with the first of the right
    If they are already in order, we will be done quickly.
    We may have to compactify the buffer because the recursion may
    have created holes. Also this compare may result in equal terms.
    Addition of 23-jul-1999. It makes things a bit faster.
*/
    if ( newleft > 0 && newright > 0 &&
    ( i = CompareTerms(BHEAD Pointer[newleft-1],Pointer[split],(WORD)0) ) >= 0 ) {
        pp2 = Pointer+split; pp1 = Pointer+newleft-1;
        if ( i == 0 ) {
            if ( S->PolyWise ) {
                if ( AddPoly(BHEAD pp1,pp2) > 0 ) pp1++;
                else newleft--;
            }
            else {
                if ( AddCoef(BHEAD pp1,pp2) > 0 ) pp1++;
                else newleft--;
            }
            *pp2++ = 0; newright--;
        }
        else pp1++;
        newleft += newright;
        if ( pp1 < pp2 ) {
            while ( --newright >= 0 ) *pp1++ = *pp2++;
            while ( pp1 < pptop ) *pp1++ = 0;
        }
        return(newleft);
    }

    if ( split >= AN.SplitScratchSize ) {
        AN.SplitScratchSize = (split*3)/2+100;
        if ( AN.SplitScratchSize > S->Terms2InSmall/2 )
            AN.SplitScratchSize = S->Terms2InSmall/2;
        if ( AN.SplitScratch ) M_free(AN.SplitScratch,"AN.SplitScratch");
        AN.SplitScratch = (WORD **)Malloc1(AN.SplitScratchSize*sizeof(WORD *),"AN.SplitScratch");
    }

    pp3 = AN.SplitScratch; pp1 = Pointer;
    /* Move rather than copy, so GarbHand can't double-count. */
    for ( i = 0; i < newleft; i++ ) { *pp3++ = *pp1; *pp1++ = 0; }
    AN.InScratch = newleft;
    pp1 = AN.SplitScratch; pp2 = Pointer + split; pp3 = Pointer;

#ifdef NEWSPLITMERGETIMSORT
/*
        An improvement in the style of Timsort
*/
    while ( newleft > 8 ) {
        /* Check the middle of the LHS terms */
        LONG nnleft = newleft/2;
        if ( ( i = CompareTerms(BHEAD pp1[nnleft],*pp2,(WORD)0) ) < 0 ) {
            /* The terms are not in order. Break out and continue as normal. */
            break;
        }
        /* The terms merge or are in order. Copy pointers up to this point. */
        /* In the copy, zero the skipped pointers so GarbHand can't double-count. */
        for (int iii = 0; iii < nnleft; iii++) {
            *pp3++ = *pp1;
            *pp1++ = 0;
        }
        newleft -= nnleft;
        if ( i == 0 ) {
            if ( S->PolyWise ) { i = AddPoly(BHEAD pp1,pp2); }
```

```
            else                { i = AddCoef(BHEAD pp1,pp2); }
            if ( i == 0 ) {
                /* The terms cancelled. The next term goes in *pp3. Don't move. */
            }
            else {
                /* The terms added. Advance pp3. */
                *pp3++ = *pp1;
            }
            /* We have taken a LHS (copy) and RHS term. */
            *pp2++ = 0;
            newright--;
            *pp1++ = 0;
            newleft--;
            break;
        }
    }
#endif

    while ( newleft > 0 && newright > 0 ) {
        if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
            *pp3++ = *pp2;
            *pp2++ = 0;
            newright--;
        }
        else if ( i > 0 ) {
            *pp3++ = *pp1;
            *pp1++ = 0;
            newleft--;
        }
        else {
            if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
            else {               if ( AddCoef(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
            *pp1++ = 0; *pp2++ = 0; newleft--; newright--;
        }
    }
    for ( i = 0; i < newleft; i++ ) { *pp3++ = *pp1; *pp1++ = 0; }
    if ( pp3 == pp2 ) {
        pp3 += newright;
    } else {
        for ( i = 0; i < newright; i++ ) { *pp3++ = *pp2++; }
    }
    newleft = pp3 - Pointer;
    while ( pp3 < pptop ) *pp3++ = 0;
    AN.InScratch = 0;
    return(newleft);
}

#else

LONG SplitMerge(PHEAD WORD **Pointer, LONG number)
{
    GETBIDENTITY
    SORTING *S = AT.SS;
    WORD **pp3, **pp1, **pp2;
    LONG nleft, nright, i, newleft, newright;
    WORD **pptop;

#ifdef SPLITMERGEDEBUG
    /* Print current array state on entry. */
    printf("%4ld: ", number);
    for (int ii = 0; ii < S->sTerms; ii++) {
        if ( (S->sPointer)[ii] ) {
            printf("%4d ", (unsigned)(S->sPointer[ii]-S->sBuffer));
        }
        else {
            printf(".... ");
        }
    }
    printf("\n");
    fflush(stdout);
#endif

    if ( number < 2 ) return(number);
    if ( number == 2 ) {
        pp1 = Pointer; pp2 = pp1 + 1;
        if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
            pp3 = (WORD **)(*pp1); *pp1 = *pp2; *pp2 = (WORD *)pp3;
        }
        else if ( i == 0 ) {
          number--;
          if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) == 0 ) { number = 0; } }
          else {               if ( AddCoef(BHEAD pp1,pp2) == 0 ) { number = 0; } }
        }
        return(number);
    }
    pptop = Pointer + number;
    nleft = number » 1; nright = number - nleft;
```

```
    newleft  = SplitMerge(BHEAD Pointer,nleft);
    newright = SplitMerge(BHEAD Pointer+nleft,nright);
/*
    We compare the last of the left with the first of the right
    If they are already in order, we will be done quickly.
    We may have to compactify the buffer because the recursion may
    have created holes. Also this compare may result in equal terms.
    Addition of 23-jul-1999. It makes things a bit faster.
*/
    if ( newleft > 0 && newright > 0 &&
    ( i = CompareTerms(BHEAD Pointer[newleft-1],Pointer[nleft],(WORD)0) ) >= 0 ) {
        pp2 = Pointer+nleft; pp1 = Pointer+newleft-1;
        if ( i == 0 ) {
          if ( S->PolyWise ) {
            if ( AddPoly(BHEAD pp1,pp2) > 0 ) pp1++;
            else newleft--;
          }
          else {
            if ( AddCoef(BHEAD pp1,pp2) > 0 ) pp1++;
            else newleft--;
          }
          *pp2++ = 0; newright--;
        }
        else pp1++;
        newleft += newright;
        if ( pp1 < pp2 ) {
            while ( --newright >= 0 ) *pp1++ = *pp2++;
            while ( pp1 < pptop ) *pp1++ = 0;
        }
        return(newleft);
    }
    if ( nleft > AN.SplitScratchSize ) {
        AN.SplitScratchSize = (nleft*3)/2+100;
        if ( AN.SplitScratchSize > S->Terms2InSmall/2 )
             AN.SplitScratchSize = S->Terms2InSmall/2;
        if ( AN.SplitScratch ) M_free(AN.SplitScratch,"AN.SplitScratch");
        AN.SplitScratch = (WORD **)Malloc1(AN.SplitScratchSize*sizeof(WORD *),"AN.SplitScratch");
    }
    pp3 = AN.SplitScratch; pp1 = Pointer; i = nleft;
    do { *pp3++ = *pp1; *pp1++ = 0; } while ( *pp1 && --i > 0 );
    if ( i > 0 ) { *pp3 = 0; i--; }
    AN.InScratch = nleft - i;
    pp1 = AN.SplitScratch; pp2 = Pointer + nleft; pp3 = Pointer;
    while ( nleft > 0 && nright > 0 && *pp1 && *pp2 ) {
        if ( ( i = CompareTerms(BHEAD *pp1,*pp2,(WORD)0) ) < 0 ) {
            *pp3++ = *pp2;
            *pp2++ = 0;
            nright--;
        }
        else if ( i > 0 ) {
            *pp3++ = *pp1;
            *pp1++ = 0;
            nleft--;
        }
        else {
          if ( S->PolyWise ) { if ( AddPoly(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
          else {               if ( AddCoef(BHEAD pp1,pp2) > 0 ) *pp3++ = *pp1; }
          *pp1++ = 0; *pp2++ = 0; nleft--; nright--;
        }
    }
    while ( --nleft  >= 0 && *pp1 ) { *pp3++ = *pp1; *pp1++ = 0; }
    while ( --nright >= 0 && *pp2 ) { *pp3++ = *pp2++; }
    nleft = pp3 - Pointer;
    while ( pp3 < pptop ) *pp3++ = 0;
    AN.InScratch = 0;
    return(nleft);
}

#endif

/*
        #] SplitMerge :
        #[ GarbHand :               void GarbHand()
*/
void GarbHand(void)
{
    GETIDENTITY
    SORTING *S = AT.SS;
    WORD **Point, *s2, *t, *garbuf, i;
    LONG k, total = 0;
    int tobereturned = 0;
/*
    Compute the size needed. Put it in total.
*/
#ifdef TESTGARB
    MLOCK(ErrorMessageLock);
    MesPrint("in:  S->sFill = %l, S->sTop2 = %l",S->sFill-S->sBuffer,S->sTop2-S->sBuffer);
```

```
#endif
    Point = S->sPointer;
    k = S->sTerms;
    while ( --k >= 0 ) {
        if ( ( s2 = *Point++ ) != 0 ) { total += *s2; }
    }
    Point = AN.SplitScratch;
    k = AN.InScratch;
    while ( --k >= 0 ) {
        if ( ( s2 = *Point++ ) != 0 ) { total += *s2; }
    }
#ifdef TESTGARB
    MesPrint("total = %l, nterms = %l",total,AN.InScratch);
    MUNLOCK(ErrorMessageLock);
#endif
/*
    Test now whether it fits. If so deal with the problem inside
    the memory at the tail of the large buffer.
*/
    if ( S->lBuffer != 0 && S->lFill + total <= S->lTop ) {
        garbuf = S->lFill;
    }
    else {
        garbuf = (WORD *)Malloc1(total*sizeof(WORD),"Garbage buffer");
        tobereturned = 1;
    }
    t = garbuf;
    Point = S->sPointer;
    k = S->sTerms;
    while ( --k >= 0 ) {
        if ( *Point ) {
            s2 = *Point++;
            i = *s2;
            NCOPY(t,s2,i);
        }
        else { Point++; }
    }
    Point = AN.SplitScratch;
    k = AN.InScratch;
    while ( --k >= 0 ) {
        if ( *Point ) {
            s2 = *Point++;
            i = *s2;
            NCOPY(t,s2,i);
        }
        else Point++;
    }
    s2 = S->sBuffer;
    t = garbuf;
    Point = S->sPointer;
    k = S->sTerms;
    while ( --k >= 0 ) {
        if ( *Point ) {
            *Point++ = s2;
            i = *t;
            NCOPY(s2,t,i);
        }
        else { Point++; }
    }
    Point = AN.SplitScratch;
    k = AN.InScratch;
    while ( --k >= 0 ) {
        if ( *Point ) {
            *Point++ = s2;
            i = *t;
            NCOPY(s2,t,i);
        }
        else Point++;
    }
    S->sFill = s2;
#ifdef TESTGARB
    MLOCK(ErrorMessageLock);
    MesPrint("out: S->sFill = %l, S->sTop2 = %l",S->sFill-S->sBuffer,S->sTop2-S->sBuffer);
    if ( S->sFill >= S->sTop2 ) {
        MesPrint("We are in deep trouble");
    }
    MUNLOCK(ErrorMessageLock);
#endif
    if ( tobereturned ) M_free(garbuf,"Garbage buffer");
    return;
}

/*
        #] GarbHand :
        #[ MergePatches :           WORD MergePatches(par)
*/
WORD MergePatches(WORD par)
```

```
{
    GETIDENTITY
    SORTING *S = AT.SS;
    WORD **poin, **poin2, ul, k, i, im, *m1;
    WORD *p, lpat, mpat, level, l1, l2, r1, r2, r3, c;
    WORD *m2, *m3, r31, r33, ki, *rr;
    UWORD *coef;
    POSITION position;
    FILEHANDLE *fin, *fout;
    int fhandle;
/*
    UBYTE *s;
*/
#ifdef WITHZLIB
    POSITION position2;
    int oldgzipCompress = AR.gzipCompress;
    if ( par == 2 ) {
        AR.gzipCompress = 0;
    }
#endif
    fin = &S->file;
    fout = &(AR.FoStage4[0]);
NewMerge:
    coef = AN.SoScratC;
    poin = S->poina; poin2 = S->poin2a;
    rr = AR.CompressPointer;
    *rr = 0;
/*
        #[ Setup :
*/
    if ( par == 1 ) {
        fout = &(S->file);
        if ( fout->handle < 0 ) {
FileMake:
            PUTZERO(AN.OldPosOut);
            if ( ( fhandle = CreateFile(fout->name) ) < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Cannot create file %s",fout->name);
                MUNLOCK(ErrorMessageLock);
                goto ReturnError;
            }
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%w MergePatches created output file %s",fout->name);
            MUNLOCK(ErrorMessageLock);
#endif
            fout->handle = fhandle;
            PUTZERO(fout->filesize);
            PUTZERO(fout->POposition);
/*
            Should not be here?
#ifdef WITHZLIB
            fout->ziobuffer = 0;
#endif
*/
#ifdef ALLLOCK
            LOCK(fout->pthreadslock);
#endif
            SeekFile(fout->handle,&(fout->filesize),SEEK_SET);
#ifdef ALLLOCK
            UNLOCK(fout->pthreadslock);
#endif
            S->fPatchN = 0;
            PUTZERO(S->fPatches[0]);
            fout->POfill = fout->PObuffer;
            PUTZERO(fout->POposition);
        }
ConMer:
        StageSort(fout);
#ifdef WITHZLIB
        if ( S == AT.S0 && AR.NoCompress == 0 && AR.gzipCompress > 0 )
            S->fpcompressed[S->fPatchN] = 1;
        else
            S->fpcompressed[S->fPatchN] = 0;
        SetupOutputGZIP(fout);
#endif
    }
    else if ( par == 0 && S->stage4 > 0 ) {
/*
        We will have to do our job more than once.
        Input is from S->file and output will go to AR.FoStage4.
        The file corresponding to this last one must be made now.
*/
        AR.Stage4Name ^= 1;
/*
        s = (UBYTE *)(fout->name); while ( *s ) s++;
        if ( AR.Stage4Name ) s[-1] += 1;
```

```
        else                s[-1] -= 1;
*/
        S->iPatches = S->fPatches;
        S->fPatches = S->inPatches;
        S->inPatches = S->iPatches;
        (S->inNum) = S->fPatchN;
        AN.OldPosIn = AN.OldPosOut;
#ifdef WITHZLIB
        m1 = S->fpincompressed;
        S->fpincompressed = S->fpcompressed;
        S->fpcompressed = m1;
        for ( i = 0; i < S->inNum; i++ ) {
            S->fPatchesStop[i] = S->iPatches[i+1];
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%w fPatchesStop[%d] = %10p",i,&(S->fPatchesStop[i]));
            MUNLOCK(ErrorMessageLock);
#endif
        }
#endif
        S->stage4 = 0;
        goto FileMake;
    }
    else {
#ifdef WITHZLIB
/*
        The next statement is just for now
*/
        AR.gzipCompress = 0;
#endif
        if ( par == 0 ) {
            S->iPatches = S->fPatches;
            S->inNum = S->fPatchN;
#ifdef WITHZLIB
            m1 = S->fpincompressed;
            S->fpincompressed = S->fpcompressed;
            S->fpcompressed = m1;
            for ( i = 0; i < S->inNum; i++ ) {
                S->fPatchesStop[i] = S->fPatches[i+1];
#ifdef GZIPDEBUG
                MLOCK(ErrorMessageLock);
                MesPrint("%w fPatchesStop[%d] = %10p",i,&(S->fPatchesStop[i]));
                MUNLOCK(ErrorMessageLock);
#endif
            }
#endif
        }
        fout = AR.outfile;
    }
    if ( par ) {                    /* Mark end of patches */
        S->Patches[S->lPatch] = S->lFill;
        for ( i = 0; i < S->lPatch; i++ ) {
            S->pStop[i] = S->Patches[i+1]-1;
            S->Patches[i] = (WORD *)(((UBYTE *)(S->Patches[i])) + AM.MaxTer);
        }
    }
    else {  /* Load the patches */
        S->lPatch = (S->inNum);
#ifdef WITHMPI
        if ( S->lPatch > 1 || ( (PF.exprtodo <0) && (fout == AR.outfile || fout == AR.hidefile ) ) ) {
#else
        if ( S->lPatch > 1 ) {
#endif
#ifdef WITHZLIB
            SetupAllInputGZIP(S);
#endif
            p = S->lBuffer;
            for ( i = 0; i < S->lPatch; i++ ) {
                p = (WORD *)(((UBYTE *)p)+2*AM.MaxTer+COMPINC*sizeof(WORD));
                S->Patches[i] = p;
                p = (WORD *)(((UBYTE *)p) + fin->POsize);
                S->pStop[i] = m2 = p;
#ifdef WITHZLIB
                PutIn(fin,&(S->iPatches[i]),S->Patches[i],&m2,i);
#else
                ADDPOS(S->iPatches[i],PutIn(fin,&(S->iPatches[i]),S->Patches[i],&m2,i));
#endif
            }
        }
    }
    if ( fout->handle >= 0 ) {
        PUTZERO(position);
#ifdef ALLLOCK
        LOCK(fout->pthreadslock);
#endif
        SeekFile(fout->handle,&position,SEEK_END);
        ADDPOS(position,((fout->POfill-fout->PObuffer)*sizeof(WORD)));
```

```
#ifdef ALLLOCK
        UNLOCK(fout->pthreadslock);
#endif
    }
    else {
        SETBASEPOSITION(position,(fout->POfill-fout->PObuffer)*sizeof(WORD));
    }
/*
        #] Setup :

    The old code had to be replaced because all output needs to go
    through PutOut. For this we have to go term by term and keep
    track of the compression.
*/
    if ( S->lPatch == 1 ) { /* Single patch --> direct copy. Very rare. */
        LONG length;

        if ( fout->handle < 0 ) if ( Sflush(fout) ) goto PatCall;
        if ( par ) {          /* Memory to file */
#ifdef WITHZLIB
/*
            We fix here the problem that the thing needs to go through PutOut
*/
            m2 = m1 = *S->Patches; /* The m2 is to keep the compiler from complaining */
            while ( *m1 ) {
                if ( *m1 < 0 ) { /* Need to uncompress */
                    i = -(*m1++); m2 += i; im = *m1+i+1;
                    while ( i > 0 ) { *m1-- = *m2--; i--; }
                    *m1 = im;
                }
#ifdef WITHPTHREADS
                /* Check here (and in the following) that we are at ground level
                   (so S == AT.S0) to use PutToMaster. Control can reach here,
                   with AS.MasterSort, but with S != AT.S0, when the SortBots are
                   adding PolyRatFun and the sorting of their arguments requires
                   large buffer patches to be merged. In this case, terms escape
                   the PolyRatFun argument and end up at ground level, if we use
                   PutToMaster. */
                if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
                    im = PutToMaster(BHEAD m1);
                }
                else
#endif
                if ( ( im = PutOut(BHEAD m1,&position,fout,1) ) < 0 ) goto ReturnError;
                ADDPOS(S->SizeInFile[par],im);
                m2 = m1;
                m1 += *m1;
            }
#ifdef WITHPTHREADS
            if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
                PutToMaster(BHEAD 0);
            }
            else
#endif
            if ( FlushOut(&position,fout,1) ) goto ReturnError;
            ADDPOS(S->SizeInFile[par],1);
#else
/* old code */
            length = (LONG)(*S->pStop)-(LONG)(*S->Patches)+sizeof(WORD);
            if ( WriteFile(fout->handle,(UBYTE *)(*S->Patches),length) != length )
                goto PatwCall;
            ADDPOS(position,length);
            ADDPOS(fout->POposition,length);
            ADDPOS(fout->filesize,length);
            ADDPOS(S->SizeInFile[par],length/sizeof(WORD));
#endif
        }
        else {                /* File to file */
#ifdef WITHZLIB
/*
            Note: if we change FRONTSIZE we need to make the minimum value
            of SmallEsize in AllocSort correspondingly larger or smaller.
            Theoretically we could get close to 2*AM.MaxTer!
*/
            #define FRONTSIZE (2*AM.MaxTer)
            WORD *copybuf = (WORD *)(((UBYTE *)(S->sBuffer)) + FRONTSIZE);
            WORD *copytop;
            SetupAllInputGZIP(S);
            m1 = m2 = copybuf;
            position2 = S->iPatches[0];
            while ( ( length = FillInputGZIP(fin,&position2,
                    (UBYTE *)copybuf,
                    (S->SmallEsize*sizeof(WORD)-FRONTSIZE),0) ) > 0 ) {
                copytop = (WORD *)(((UBYTE *)copybuf)+length);
                while ( *m1 && ( ( *m1 > 0 && m1+*m1 < copytop ) ||
                ( *m1 < 0 && ( m1+1 < copytop ) && ( m1+m1[1]+1 < copytop ) ) ) )
/*
```

```
    22-jun-2013 JV  Extremely nasty bug that has been around for a while.
                    What if the end is in the remaining part? We will loose terms!
                while ( *m1 && ( (WORD *)(((UBYTE *)(m1)) + AM.MaxTer ) < S->sTop2 ) )
*/
                {
                    if ( *m1 < 0 ) { /* Need to uncompress */
                        i = -(*m1++); m2 += i; im = *m1+i+1;
                        while ( i > 0 ) { *m1-- = *m2--; i--; }
                        *m1 = im;
                    }
#ifdef WITHPTHREADS
                    if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
                        im = PutToMaster(BHEAD m1);
                    }
                    else
#endif
                    if ( ( im = PutOut(BHEAD m1,&position,fout,1) ) < 0 ) goto ReturnError;
                    ADDPOS(S->SizeInFile[par],im);
                    m2 = m1;
                    m1 += *m1;
                }
                if ( m1 < copytop && *m1 == 0 ) break;
/*
                Now move the remaining part 'back'
*/
                m3 = copybuf;
                m1 = copytop;
                while ( m1 > m2 ) *--m3 = *--m1;
                m2 = m3;
                m1 = m2 + *m2;
            }
            if ( length < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Readerror");
                goto PatCall2;
            }
#ifdef WITHPTHREADS
            if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
                PutToMaster(BHEAD 0);
            }
            else
#endif
            if ( FlushOut(&position,fout,1) ) goto ReturnError;
            ADDPOS(S->SizeInFile[par],1);
#else
/* old code */
            SeekFile(fin->handle,&(S->iPatches[0]),SEEK_SET); /* needed for stage4 */
            while ( ( length = ReadFile(fin->handle,
                    (UBYTE *)(S->sBuffer),S->SmallEsize*sizeof(WORD)) ) > 0 ) {
                if ( WriteFile(fout->handle,(UBYTE *)(S->sBuffer),length) != length )
                    goto PatwCall;
                ADDPOS(position,length);
                ADDPOS(fout->POposition,length);
                ADDPOS(fout->filesize,length);
                ADDPOS(S->SizeInFile[par],length/sizeof(WORD));
            }
            if ( length < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Readerror");
                goto PatCall2;
            }
#endif
        }
        goto EndOfAll;
    }
    else if ( S->lPatch > 0 ) {

        /* More than one patch. Construct the tree. */

        lpat = 1;
        do { lpat *= 2; } while ( lpat < S->lPatch );
        mpat = ( lpat >> 1 ) - 1;
        k = lpat - S->lPatch;

        /* k is the number of empty places in the tree. they will
           be at the even positions from 2 to 2*k */

        for ( i = 1; i < lpat; i++ ) {
            S->tree[i] = -1;
        }
        for ( i = 1; i <= k; i++ ) {
            im = ( i * 2 ) - 1;
            poin[im] = S->Patches[i-1];
            poin2[im] = poin[im] + *(poin[im]);
            S->used[i] = im;
            S->ktoi[im] = i-1;
            S->tree[mpat+i] = 0;
```

```
            poin[im-1] = poin2[im-1] = 0;
        }
        for ( i = (k*2)+1; i <= lpat; i++ ) {
            S->used[i-k] = i;
            S->ktoi[i] = i-k-1;
            poin[i] = S->Patches[i-k-1];
            poin2[i] = poin[i] + *(poin[i]);
        }
/*
        the array poin tells the position of the i-th element of the S->tree
        'S->used' is a stack with the S->tree elements that need to be entered
        into the S->tree. at the beginning this is S->lPatch. during the
        sort there will be only very few elements.
        poin2 is the next value of poin. it has to be determined
        before the comparisons as the position or the size of the
        term indicated by poin may change.
        S->ktoi translates a S->tree element back to its stream number.

        start the sort
*/
        level = S->lPatch;

        /* introduce one term */
OneTerm:
        k = S->used[level];
        i = k + lpat - 1;
        if ( !*(poin[k]) ) {
            do { if ( !( i >>= 1 ) ) goto EndOfMerge; } while ( !S->tree[i] );
            if ( S->tree[i] == -1 ) {
                S->tree[i] = 0;
                level--;
                goto OneTerm;
            }
            k = S->tree[i];
            S->used[level] = k;
            S->tree[i] = 0;
        }
/*
        move terms down the tree
*/
        while ( i >>= 1 ) {
            if ( S->tree[i] > 0 ) {
                if ( ( c = CompareTerms(BHEAD poin[S->tree[i]],poin[k],(WORD)0) ) > 0 ) {
/*
                    S->tree[i] is the smaller. Exchange and go on.
*/
                    S->used[level] = S->tree[i];
                    S->tree[i] = k;
                    k = S->used[level];
                }
                else if ( !c ) {    /* Terms are equal */
                    S->TermsLeft--;
/*
                        Here the terms are equal and their coefficients
                        have to be added.
*/
                    l1 = *( m1 = poin[S->tree[i]] );
                    l2 = *( m2 = poin[k] );
                    if ( S->PolyWise ) {  /* Here we work with PolyFun */
                        WORD *tt1, *w;
                        tt1 = m1;
                        m1 += S->PolyWise;
                        m2 += S->PolyWise;
                        if ( S->PolyFlag == 2 ) {
                            w = poly_ratfun_add(BHEAD m1,m2);
                            if ( *tt1 + w[1] - m1[1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l is too small",AM.MaxTer);
                                MUNLOCK(ErrorMessageLock);
                                Terminate(-1);
                            }
                            AT.WorkPointer = w;
                        }
                        else {
                            w = AT.WorkPointer;
                            if ( w + m1[1] + m2[1] > AT.WorkTop ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("A WorkSpace of %10l is too small",AM.WorkSize);
                                MUNLOCK(ErrorMessageLock);
                                Terminate(-1);
                            }
                            AddArgs(BHEAD m1,m2,w);
                        }
                        r1 = w[1];
                        if ( r1 <= FUNHEAD
                            || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) )
```

```
                                { goto cancelled; }
                        if ( r1 == m1[1] ) {
                            NCOPY(m1,w,r1);
                        }
                        else if ( r1 < m1[1] ) {
                            r2 = m1[1] - r1;
                            m2 = w + r1;
                            m1 += m1[1];
                            while ( --r1 >= 0 ) *--m1 = *--m2;
                            m2 = m1 - r2;
                            r1 = S->PolyWise;
                            while ( --r1 >= 0 ) *--m1 = *--m2;
                            *m1 -= r2;
                            poin[S->tree[i]] = m1;
                        }
                        else {
                            r2 = r1 - m1[1];
                            m2 = tt1 - r2;
                            r1 = S->PolyWise;
                            m1 = tt1;
                            *m1 += r2;
                            poin[S->tree[i]] = m2;
                            NCOPY(m2,m1,r1);
                            r1 = w[1];
                            NCOPY(m2,w,r1);
                        }
                    }
#ifdef WITHFLOAT
                    else if ( AT.SortFloatMode ) {
                        WORD *term1, *term2;
                        term1 = poin[S->tree[i]];
                        term2 = poin[k];
                        if ( MergeWithFloat(BHEAD &term1,&term2) == 0 )
                            goto cancelled;
                        poin[S->tree[i]] = term1;
                    }
#endif
                    else {
                      r1 = *( m1 += l1 - 1 );
                      m1 -= ABS(r1) - 1;
                      r1 = ( ( r1 > 0 ) ? (r1-1) : (r1+1) ) >> 1;
                      r2 = *( m2 += l2 - 1 );
                      m2 -= ABS(r2) - 1;
                      r2 = ( ( r2 > 0 ) ? (r2-1) : (r2+1) ) >> 1;

                      if ( AddRat(BHEAD (UWORD *)m1,r1,(UWORD *)m2,r2,coef,&r3) ) {
                        MLOCK(ErrorMessageLock);
                        MesCall("MergePatches");
                        MUNLOCK(ErrorMessageLock);
                        SETERROR(-1)
                      }

                      if ( AN.ncmod != 0 ) {
                        if ( ( AC.modmode & POSNEG ) != 0 ) {
                            NormalModulus(coef,&r3);
                        }
                        else if ( BigLong(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
                            WORD ii;
                            SubPLon(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod),coef,&r3);
                            coef[r3] = 1;
                            for ( ii = 1; ii < r3; ii++ ) coef[r3+ii] = 0;
                        }
                      }
                      r3 *= 2;
                      r33 = ( r3 > 0 ) ? ( r3 + 1 ) : ( r3 - 1 );
                      if ( r3 < 0 ) r3 = -r3;
                      if ( r1 < 0 ) r1 = -r1;
                      r1 *= 2;
                      r31 = r3 - r1;
                      if ( !r3 ) {      /* Terms cancel */
cancelled:
                        ul = S->used[level] = S->tree[i];
                        S->tree[i] = -1;
/*
                        We skip to the next term in stream ul
*/
                        im = *poin2[ul];
                        if ( im < 0 ) {
                            r1 = poin2[ul][1] - im + 1;
                            m1 = poin2[ul] + 2;
                            m2 = poin[ul] - im + 1;
                            while ( ++im <= 0 ) *--m1 = *--m2;
                            *--m1 = r1;
                            poin2[ul] = m1;
                            im = r1;
                        }
                        poin[ul] = poin2[ul];
```

```
                        ki = S->ktoi[ul];
                        if ( !par && (poin[ul] + im + COMPINC) >= S->pStop[ki]
                        && im > 0 ) {
#ifdef WITHZLIB
                            PutIn(fin,&(S->iPatches[ki]),S->Patches[ki],&(poin[ul]),ki);
#else
                            ADDPOS(S->iPatches[ki],PutIn(fin,&(S->iPatches[ki]),
                            S->Patches[ki],&(poin[ul]),ki));
#endif
                            poin2[ul] = poin[ul] + im;
                        }
                        else {
                            poin2[ul] += im;
                        }
                        S->used[++level] = k;
                        S->TermsLeft--;
                      }
                      else if ( !r31 ) {          /* copy coef into term1 */
                        goto CopCof2;
                      }
                      else if ( r31 < 0 ) {       /* copy coef into term1
                                              and adjust the length of term1 */
                        goto CopCoef;
                      }
                      else {
/*
                            this is the dreaded calamity.
                            is there enough space?
*/
                        if( (poin[S->tree[i]]+l1+r31) >= poin2[S->tree[i]] ) {
/*
                                no space! now the special trick for which
                                we left 2*maxlng spaces open at the beginning
                                of each patch.
*/
                            if ( (l1 + r31) > AM.MaxTer/((LONG)sizeof(WORD)) ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("Coefficient overflow during sort");
                                MUNLOCK(ErrorMessageLock);
                                goto ReturnError;
                            }
                            m2 = poin[S->tree[i]];
                            m3 = ( poin[S->tree[i]] -= r31 );
                            do { *m3++ = *m2++; } while ( m2 < m1 );
                            m1 = m3;
                        }
CopCoef:
                        *(poin[S->tree[i]]) += r31;
CopCof2:
                        m2 = (WORD *)coef; im = r3;
                        NCOPY(m1,m2,im);
                        *m1 = r33;
                      }
                    }
/*
                    Now skip to the next term in stream k.
*/
NextTerm:
                    im = poin2[k][0];
                    if ( im < 0 ) {
                        r1 = poin2[k][1] - im + 1;
                        m1 = poin2[k] + 2;
                        m2 = poin[k] - im + 1;
                        while ( ++im <= 0 ) *--m1 = *--m2;
                        *--m1 = r1;
                        poin2[k] = m1;
                        im = r1;
                    }
                    poin[k] = poin2[k];
                    ki = S->ktoi[k];
                    if ( !par && ( (poin[k] + im + COMPINC) >= S->pStop[ki] )
                    && im > 0 ) {
#ifdef WITHZLIB
                        PutIn(fin,&(S->iPatches[ki]),S->Patches[ki],&(poin[k]),ki);
#else
                        ADDPOS(S->iPatches[ki],PutIn(fin,&(S->iPatches[ki]),
                        S->Patches[ki],&(poin[k]),ki));
#endif
                        poin2[k] = poin[k] + im;
                    }
                    else {
                        poin2[k] += im;
                    }
                    goto OneTerm;
                }
            }
            else if ( S->tree[i] < 0 ) {
```

```
                S->tree[i] = k;
                level--;
                goto OneTerm;
            }
        }
/*
            found the smallest in the set. indicated by k.
            write to its destination.
*/
#ifdef WITHPTHREADS
        if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
            im = PutToMaster(BHEAD poin[k]);
        }
        else
#endif
        if ( ( im = PutOut(BHEAD poin[k],&position,fout,1) ) < 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Called from MergePatches with k = %d (stream %d)",k,S->ktoi[k]);
            MUNLOCK(ErrorMessageLock);
            goto ReturnError;
        }
        ADDPOS(S->SizeInFile[par],im);
        goto NextTerm;
    }
    else {
        goto NormalReturn;
    }
EndOfMerge:
#ifdef WITHPTHREADS
        if ( AS.MasterSort && ( fout == AR.outfile ) && S == AT.S0 ) {
            PutToMaster(BHEAD 0);
        }
        else
#endif
    if ( FlushOut(&position,fout,1) ) goto ReturnError;
    ADDPOS(S->SizeInFile[par],1);
EndOfAll:
    if ( par == 1 ) {   /* Set the fpatch pointers */
#ifdef WITHZLIB
        SeekFile(fout->handle,&position,SEEK_CUR);
#endif
        (S->fPatchN)++;
        S->fPatches[S->fPatchN] = position;
    }
    if ( par == 0 && fout != AR.outfile ) {
/*
            Output went to sortfile. We have two possibilities:
            1:  We are not finished with the current in-out cycle
                In that case we should pop to the next set of patches
            2:  We finished a cycle and should clean up the in file
                Then we restart the sort.
*/
        (S->fPatchN)++;
        S->fPatches[S->fPatchN] = position;
        if ( ISNOTZEROPOS(AN.OldPosIn) ) {      /* We are not done */

            SeekFile(fin->handle,&(AN.OldPosIn),SEEK_SET);
/*
            We don't need extra provisions for the zlib compression here.
            If part of an expression has been sorted, the whole has been so.
            This means that S->fpincompressed[] will remain the same
*/
            if ( (ULONG)ReadFile(fin->handle,(UBYTE *)(&(S->inNum)),(LONG)sizeof(WORD)) !=
                sizeof(WORD)
              || (ULONG)ReadFile(fin->handle,(UBYTE *)(&AN.OldPosIn),(LONG)sizeof(POSITION)) !=
                sizeof(POSITION)
              || (ULONG)ReadFile(fin->handle,(UBYTE *)S->iPatches,(LONG)((S->inNum)+1)
                    *sizeof(POSITION)) != ((S->inNum)+1)*sizeof(POSITION) ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Read error fourth stage sorting");
                MUNLOCK(ErrorMessageLock);
                goto ReturnError;
            }
            *rr = 0;
#ifdef WITHZLIB
            for ( i = 0; i < S->inNum; i++ ) {
                S->fPatchesStop[i] = S->iPatches[i+1];
#ifdef GZIPDEBUG
                MLOCK(ErrorMessageLock);
                MesPrint("%w fPatchesStop[%d] = %10p",i,&(S->fPatchesStop[i]));
                MUNLOCK(ErrorMessageLock);
#endif
            }
#endif
            goto ConMer;
        }
        else {
```

```
/*
            if ( fin == &(AR.FoStage4[0]) ) {
                s = (UBYTE *)(fin->name); while ( *s ) s++;
                if ( AR.Stage4Name == 1 ) s[-1] -= 1;
                else                      s[-1] += 1;
            }
*/
/*          TruncateFile(fin->handle); */
            UpdateMaxSize();
#ifdef WITHZLIB
            ClearSortGZIP(fin);
#endif
            CloseFile(fin->handle);
            remove(fin->name);      /* Gives diskspace free again. */
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%w MergePatches removed in file %s",fin->name);
            MUNLOCK(ErrorMessageLock);
#endif
/*
            if ( fin == &(AR.FoStage4[0]) ) {
                s = (UBYTE *)(fin->name); while ( *s ) s++;
                if ( AR.Stage4Name == 1 ) s[-1] += 1;
                else                      s[-1] -= 1;
            }
*/
            fin->handle = -1;
            { FILEHANDLE *ff = fin; fin = fout; fout = ff; }
            PUTZERO(S->SizeInFile[0]);
            goto NewMerge;
        }
    }
    if ( par == 0 ) {
/*      TruncateFile(fin->handle); */
        UpdateMaxSize();
#ifdef WITHZLIB
        ClearSortGZIP(fin);
#endif
        CloseFile(fin->handle);
        remove(fin->name);
        fin->handle = -1;
#ifdef GZIPDEBUG
        MLOCK(ErrorMessageLock);
        MesPrint("%w MergePatches removed in file %s",fin->name);
        MUNLOCK(ErrorMessageLock);
#endif
    }
NormalReturn:
#ifdef WITHZLIB
    AR.gzipCompress = oldgzipCompress;
#endif
    return(0);
ReturnError:
#ifdef WITHZLIB
    AR.gzipCompress = oldgzipCompress;
#endif
    return(-1);
#ifndef WITHZLIB
PatwCall:
    MLOCK(ErrorMessageLock);
    MesPrint("Error while writing to file.");
    goto PatCall2;
#endif
PatCall:;
    MLOCK(ErrorMessageLock);
PatCall2:;
    MesCall("MergePatches");
    MUNLOCK(ErrorMessageLock);
#ifdef WITHZLIB
    AR.gzipCompress = oldgzipCompress;
#endif
    SETERROR(-1)
}

/*
        #] MergePatches :
        #[ StoreTerm :             WORD StoreTerm(term)
*/
WORD StoreTerm(PHEAD WORD *term)
{
    GETBIDENTITY
    SORTING *S = AT.SS;
    WORD **ss, *lfill, j, *t;
    POSITION pp;
    LONG lSpace, sSpace, RetCode, over, tover;

    if ( ( ( AP.PreDebug & DUMPTOSORT ) == DUMPTOSORT ) && AR.sLevel == 0 ) {
```

```
#ifdef WITHPTHREADS
        snprintf((char *)(THRbuf),100,"StoreTerm(%d)",AT.identity);
        PrintTerm(term,(char *)(THRbuf));
#else
        PrintTerm(term,"StoreTerm");
#endif
    }
    if ( AM.exitflag && AR.sLevel == 0 ) return(0);
    S->sFill = *(S->PoinFill);
    if ( S->sTerms >= S->TermsInSmall || ( S->sFill + *term ) >= S->sTop ) {
/*
    The small buffer is full. It has to be sorted and written.
*/
        tover = over = S->sTerms;
        ss = S->sPointer;
        ss[over] = 0;
#ifdef SPLITTIME
        PrintTime((UBYTE *)"Before SplitMerge");
#endif
        ss[SplitMerge(BHEAD ss,over)] = 0;
#ifdef SPLITTIME
        PrintTime((UBYTE *)"After SplitMerge");
#endif
        sSpace = 0;
        if ( over > 0 ) {
            sSpace = ComPress(ss,&RetCode);
            S->TermsLeft -= over - RetCode;
        }
        sSpace++;

        lSpace = sSpace + (S->lFill - S->lBuffer)
                 - (AM.MaxTer/sizeof(WORD))*((LONG)S->lPatch);
        SETBASEPOSITION(pp,lSpace);
        MULPOS(pp,sizeof(WORD));
        if ( S->file.handle >= 0 ) {
            ADD2POS(pp,S->fPatches[S->fPatchN]);
        }
        if ( S == AT.S0 ) { /* Only statistics at ground level */
            WriteStats(&pp,STATSSPLITMERGE,CHECKLOGTYPE);
        }
        if ( ( S->lPatch >= S->MaxPatches ) ||
            ( ( (WORD *)(((UBYTE *)(S->lFill + sSpace)) + 2*AM.MaxTer ) ) >= S->lTop ) ) {
/*
            The large buffer is too full. Merge and write it
*/
            if ( MergePatches(1) ) goto StoreCall;
/*
            pp = S->SizeInFile[1];
            ADDPOS(pp,sSpace);
            MULPOS(pp,sizeof(WORD));
*/
            SETBASEPOSITION(pp,sSpace);
            MULPOS(pp,sizeof(WORD));
            ADD2POS(pp,S->fPatches[S->fPatchN]);

            if ( S == AT.S0 ) { /* Only statistics at ground level */
                WriteStats(&pp,STATSMERGETOFILE,CHECKLOGTYPE);
            }
            S->lPatch = 0;
            S->lFill = S->lBuffer;
        }
        S->Patches[S->lPatch++] = S->lFill;
        lfill = (WORD *)(((UBYTE *)(S->lFill)) + AM.MaxTer);
        if ( tover > 0 ) {
            ss = S->sPointer;
            while ( ( t = *ss++ ) != 0 ) {
                j = *t;
                if ( j < 0 ) j = t[1] + 2;
                while ( --j >= 0 ){
                  *lfill++ = *t++;
                }
            }
        }
        *lfill++ = 0;
        S->lFill = lfill;
        S->sTerms = 0;
        S->PoinFill = S->sPointer;
        *(S->PoinFill) = S->sFill = S->sBuffer;
    }
    j = *term;
    while ( --j >= 0 ) *S->sFill++ = *term++;
    S->sTerms++;
    S->GenTerms++;
    S->TermsLeft++;
    *++S->PoinFill = S->sFill;

    return(0);
```

```

StoreCall:
    MLOCK(ErrorMessageLock);
    MesCall("StoreTerm");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] StoreTerm :
        #[ StageSort :              void StageSort(FILEHANDLE *fout)
*/
void StageSort(FILEHANDLE *fout)
{
    GETIDENTITY
    SORTING *S = AT.SS;
    if ( S->fPatchN >= S->MaxFpatches ) {
        POSITION position;
        if ( S != AT.S0 ) {
/*
            There are no proper provisions for stage 4 or higher sorts
            for function arguments and $ variables. The reason:
            The current code maps out the patches, based on the size of
            the buffers in the FoStage4 structs, while they are used
            inside the S->file struct that may have far smaller buffers.
            By itself that might still be repairable, but it goes completely
            wrong when during the sort polyRatFuns have to be added and they
            would go into stage4 (very rare but possible).
            The only really correct solution would be to put FoStage4 structs
            in all sort levels. Messy. (JV 8-oct-2018).
*/
            MLOCK(ErrorMessageLock);
            MesPrint("Currently Stage 4 sorts are not allowed for function arguments or $ variables.");
            MesPrint("Please increase correspondingsorting parameters (sub-) in the setup.");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        PUTZERO(position);
        MLOCK(ErrorMessageLock);
#ifdef WITHPTHREADS
        MesPrint("StageSort in thread %d",identity);
#elif defined(WITHMPI)
        MesPrint("StageSort in process %d",PF.me);
#else
        MesPrint("StageSort");
#endif
        MUNLOCK(ErrorMessageLock);
        SeekFile(fout->handle,&position,SEEK_END);
/*
        No extra compression data has to be written.
        S->fpincompressed should remain valid.
*/
        if ( (ULONG)WriteFile(fout->handle,(UBYTE *)(&(S->fPatchN)),(LONG)sizeof(WORD)) !=
            sizeof(WORD)
          || (ULONG)WriteFile(fout->handle,(UBYTE *)(&(AN.OldPosOut)),(LONG)sizeof(POSITION)) !=
            sizeof(POSITION)
          || (ULONG)WriteFile(fout->handle,(UBYTE *)(S->fPatches),(LONG)(S->fPatchN+1)
                    *sizeof(POSITION)) != (S->fPatchN+1)*sizeof(POSITION) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Write error while staging sort. Disk full?");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        AN.OldPosOut = position;
        fout->filesize = position;
        ADDPOS(fout->filesize,(S->fPatchN+2)*sizeof(POSITION) + sizeof(WORD));
        fout->POposition = fout->filesize;
        S->fPatches[0] = fout->filesize;
        S->fPatchN = 0;

        if ( AR.FoStage4[0].PObuffer == 0 ) {
            AR.FoStage4[0].PObuffer = (WORD *)Malloc1(AR.FoStage4[0].POsize*sizeof(WORD)
                                    ,"Stage 4 buffer");
            AR.FoStage4[0].POfill   = AR.FoStage4[0].PObuffer;
            AR.FoStage4[0].POstop   = AR.FoStage4[0].PObuffer
                        + AR.FoStage4[0].POsize/sizeof(WORD);
#ifdef WITHPTHREADS
            AR.FoStage4[0].pthreadslock = dummylock;
#endif
        }
        if ( AR.FoStage4[1].PObuffer == 0 ) {
            AR.FoStage4[1].PObuffer = (WORD *)Malloc1(AR.FoStage4[1].POsize*sizeof(WORD)
                                    ,"Stage 4 buffer");
            AR.FoStage4[1].POfill   = AR.FoStage4[1].PObuffer;
            AR.FoStage4[1].POstop   = AR.FoStage4[1].PObuffer
                        + AR.FoStage4[1].POsize/sizeof(WORD);
```

```
#ifdef WITHPTHREADS
            AR.FoStage4[1].pthreadslock = dummylock;
#endif
        }
        S->stage4 = 1;
    }
}

/*
        #] StageSort :
        #[ SortWild :               WORD SortWild(w,nw)
*/
WORD SortWild(WORD *w, WORD nw)
{
    GETIDENTITY
    WORD *v, *s, *m, k, i;
    WORD *pScrat, *stop, *sv, error = 0;
    pScrat = AT.WorkPointer;
    if ( ( AT.WorkPointer + 8 * AM.MaxWildcards ) >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    stop = w + nw;
    i = 0;
    while ( i < nw ) {
        m = w + i;
        v = m + m[1];
        while ( v < stop && (
             *v == FROMSET || *v == SETTONUM || *v == LOADDOLLAR ) ) v += v[1];
        while ( v < stop ) {
            if ( *v >= 0 ) {
                if ( AM.Ordering[*v] < AM.Ordering[*m] ) {
                    m = v;
                }
                else if ( *v == *m ) {
                    if ( v[2] < m[2] ) {
                        m = v;
                    }
                    else if ( v[2] == m[2] ) {
                        s = m + m[1];
                        sv = v + v[1];
                        if ( s < stop && ( *s == FROMSET
                         || *s == SETTONUM || *s == LOADDOLLAR ) ) {
                            if ( sv < stop && ( *sv == FROMSET
                            || *sv == SETTONUM || *sv == LOADDOLLAR ) ) {
                                if ( s[2] != sv[2] ) {
                                    error = -1;
                                    MLOCK(ErrorMessageLock);
                                    MesPrint("&Wildcard set conflict");
                                    MUNLOCK(ErrorMessageLock);
                                }
                            }
                            *v = -1;
                        }
                        else {
                            if ( sv < stop && ( *sv == FROMSET
                            || *sv == SETTONUM || *sv == LOADDOLLAR ) ) {
                                *m = -1;
                                m = v;
                            }
                            else {
                                *v = -1;
                            }
                        }
                    }
                }
            }
            v += v[1];
            while ( v < stop && ( *v == FROMSET
             || *v == SETTONUM || *v == LOADDOLLAR ) ) v += v[1];
        }
        s = pScrat;
        v = m;
        k = m[1];
        NCOPY(s,m,k);
        while ( m < stop && ( *m == FROMSET
         || *m == SETTONUM || *m == LOADDOLLAR ) ) {
            k = m[1];
            NCOPY(s,m,k);
        }
        *v = -1;
        pScrat = s;
        i = 0;
        while ( i < nw && ( w[i] < 0 || w[i] == FROMSET
        || w[i] == SETTONUM || w[i] == LOADDOLLAR ) ) i += w[i+1];
```

```
    }
    AC.NwildC = k = WORDDIF(pScrat,AT.WorkPointer);
    s = AT.WorkPointer;
    m = w;
    NCOPY(m,s,k);
    AC.WildC = m;
    return(error);
}

/*
        #] SortWild :
        #[ CleanUpSort :            void CleanUpSort(num)
*/
void CleanUpSort(int num)
{
    GETIDENTITY
    SORTING *S;
    int minnum = num, i;
    if ( AN.FunSorts ) {
        if ( num == -1 ) {
            if ( AN.MaxFunSorts > 3 ) {
                minnum = (AN.MaxFunSorts+4)/2;
            }
            else minnum = 4;
        }
        else if ( minnum == 0 ) minnum = 1;
        for ( i = minnum; i < AN.NumFunSorts; i++ ) {
            S = AN.FunSorts[i];
            if ( S ) {
                if ( S->file.handle >= 0 ) {
/*                  TruncateFile(S->file.handle); */
                    UpdateMaxSize();
#ifdef WITHZLIB
                    ClearSortGZIP(&(S->file));
#endif
                    CloseFile(S->file.handle);
                    S->file.handle = -1;
                    remove(S->file.name);
#ifdef GZIPDEBUG
                    MLOCK(ErrorMessageLock);
                    MesPrint("%w CleanUpSort removed file %s",S->file.name);
                    MUNLOCK(ErrorMessageLock);
#endif
                }
                M_free(S->sPointer, "CleanUpSort: sPointer");
                M_free(S->Patches, "CleanUpSort: Patches");
                M_free(S->pStop, "CleanUpSort: pStop");
                M_free(S->poina, "CleanUpSort: poina");
                M_free(S->poin2a, "CleanUpSort: poin2a");
                M_free(S->fPatches, "CleanUpSort: fPatches");
                M_free(S->fPatchesStop, "CleanUpSort: fPatchesStop");
                M_free(S->inPatches, "CleanUpSort: inPatches");
                M_free(S->tree, "CleanUpSort: tree");
                M_free(S->used, "CleanUpSort: used");
#ifdef WITHZLIB
                M_free(S->fpcompressed, "CleanUpSort: fpcompressed");
                M_free(S->fpincompressed, "CleanUpSort: fpincompressed");
#endif
                M_free(S->ktoi, "CleanUpSort: ktoi");
                M_free(S->lBuffer, "CleanUpSort: lBuffer+sBuffer");
                M_free(S->file.PObuffer, "CleanUpSort: PObuffer");
                M_free(S, "CleanUpSort: sorting struct");
            }
            AN.FunSorts[i] = 0;
        }
        AN.MaxFunSorts = minnum;
        if ( num == 0 ) {
            S = AN.FunSorts[0];
            if ( S ) {
                if ( S->file.handle >= 0 ) {
/*                  TruncateFile(S->file.handle); */
                    UpdateMaxSize();
#ifdef WITHZLIB
                    ClearSortGZIP(&(S->file));
#endif
                    CloseFile(S->file.handle);
                    S->file.handle = -1;
                    remove(S->file.name);
#ifdef GZIPDEBUG
                    MLOCK(ErrorMessageLock);
                    MesPrint("%w CleanUpSort removed file %s",S->file.name);
                    MUNLOCK(ErrorMessageLock);
#endif
                }
            }
        }
    }
```

```
    for ( i = 0; i < 2; i++ ) {
        if ( AR.FoStage4[i].handle >= 0 ) {
            UpdateMaxSize();
#ifdef WITHZLIB
            ClearSortGZIP(&(AR.FoStage4[i]));
#endif
            CloseFile(AR.FoStage4[i].handle);
            remove(AR.FoStage4[i].name);
            AR.FoStage4[i].handle = -1;
#ifdef GZIPDEBUG
            MLOCK(ErrorMessageLock);
            MesPrint("%w CleanUpSort removed stage4 file %s",AR.FoStage4[i].name);
            MUNLOCK(ErrorMessageLock);
#endif
        }
    }
}

/*
        #] CleanUpSort :
        #[ LowerSortLevel :         void LowerSortLevel()
*/
void LowerSortLevel(void)
{
    GETIDENTITY
    if ( AR.sLevel >= 0 ) {
        AR.sLevel--;
        if ( AR.sLevel >= 0 ) AT.SS = AN.FunSorts[AR.sLevel];
    }
}

/*
        #] LowerSortLevel :
        #[ PolyRatFunSpecial :

        Keeps only the most divergent term in AR.PolyFunVar
        We assume that the terms are already in that notation.
*/

WORD *PolyRatFunSpecial(PHEAD WORD *t1, WORD *t2)
{
    WORD *oldworkpointer = AT.WorkPointer, *t, *r;
    WORD exp1, exp2;
    int i;
    t = t1+FUNHEAD;
    if ( *t == -SYMBOL ) {
        if ( t[1] != AR.PolyFunVar ) goto Illegal;
        exp1 = 1;
        if ( t[2] != -SNUMBER ) goto Illegal;
        t[3] = 1;
    }
    else if ( *t == -SNUMBER ) {
        t[1] = 1;
        t += 2;
        if ( *t == -SYMBOL ) {
            if ( t[1] != AR.PolyFunVar ) goto Illegal;
            exp1 = -1;
        }
        else if ( *t == -SNUMBER ) {
            t[1] = 1;
            exp1 = 0;
        }
        else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
            && t[ARGHEAD+3] == AR.PolyFunVar ) {
            t[ARGHEAD+5] = 1;
            t[ARGHEAD+6] = 1;
            t[ARGHEAD+7] = 3;
            exp1 = -t[ARGHEAD+4];
        }
        else goto Illegal;
    }
    else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
        && t[ARGHEAD+3] == AR.PolyFunVar ) {
        t[ARGHEAD+5] = 1;
        t[ARGHEAD+6] = 1;
        t[ARGHEAD+7] = 3;
        exp1 = t[ARGHEAD+4];
        t += *t;
        if ( *t != -SNUMBER ) goto Illegal;
        t[1] = 1;
    }
    else goto Illegal;

    t = t2+FUNHEAD;
    if ( *t == -SYMBOL ) {
        if ( t[1] != AR.PolyFunVar ) goto Illegal;
        exp2 = 1;
```

```
        if ( t[2] != -SNUMBER ) goto Illegal;
        t[3] = 1;
    }
    else if ( *t == -SNUMBER ) {
        t[1] = 1;
        t += 2;
        if ( *t == -SYMBOL ) {
            if ( t[1] != AR.PolyFunVar ) goto Illegal;
            exp2 = -1;
        }
        else if ( *t == -SNUMBER ) {
            t[1] = 1;
            exp2 = 0;
        }
        else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
            && t[ARGHEAD+3] == AR.PolyFunVar ) {
            t[ARGHEAD+5] = 1;
            t[ARGHEAD+6] = 1;
            t[ARGHEAD+7] = 3;
            exp2 = -t[ARGHEAD+4];
        }
        else goto Illegal;
    }
    else if ( *t == ARGHEAD+8 && t[ARGHEAD] == 8 && t[ARGHEAD+1] == SYMBOL
        && t[ARGHEAD+3] == AR.PolyFunVar ) {
        t[ARGHEAD+5] = 1;
        t[ARGHEAD+6] = 1;
        t[ARGHEAD+7] = 3;
        exp2 = t[ARGHEAD+4];
        t += *t;
        if ( *t != -SNUMBER ) goto Illegal;
        t[1] = 1;
    }
    else goto Illegal;

    if ( exp1 <= exp2 ) { i = t1[1]; r = t1; }
    else                { i = t2[1]; r = t2; }
    t = oldworkpointer;
    NCOPY(t,r,i)

    return(oldworkpointer);
Illegal:
    MesPrint("Illegal occurrence of PolyRatFun with divergent option");
    Terminate(-1);
    return(0);
}

/*
        #] PolyRatFunSpecial :
        #[ SimpleSplitMerge :

        Sorts an array of WORDs. No adding of equal objects.
*/

void SimpleSplitMergeRec(WORD *array,WORD num,WORD *auxarray)
{
    WORD n1,n2,i,j,k,*t1,*t2;
    if ( num < 2 ) return;
    if ( num == 2 ) {
        if ( array[0] > array[1] ) {
            EXCH(array[0],array[1])
        }
        return;
    }
    n1 = num/2;
    n2 = num - n1;
    SimpleSplitMergeRec(array,n1,auxarray);
    SimpleSplitMergeRec(array+n1,n2,auxarray);
    if ( array[n1-1] <= array[n1] ) return;

    t1 = array; t2 = auxarray; i = n1; NCOPY(t2,t1,i);
    i = 0; j = n1; k = 0;
    while ( i < n1 && j < num ) {
        if ( auxarray[i] <= array[j] ) { array[k++] = auxarray[i++]; }
        else { array[k++] = array[j++]; }
    }
    while ( i < n1 ) array[k++] = auxarray[i++];
/*
    Remember: remnants of j are still in place!
*/
}

void SimpleSplitMerge(WORD *array,WORD num)
{
    WORD *auxarray = Malloc1(sizeof(WORD)*num/2,"SimpleSplitMerge");
    SimpleSplitMergeRec(array,num,auxarray);
    M_free(auxarray,"SimpleSplitMerge");
```

```
}

/*
        #] SimpleSplitMerge :
        #[ BinarySearch :

        Searches in the sorted array with length num for the object x.
        If x is in the list, it returns the number of the array element
        that matched. If it is not in the list, it returns -1.
        If there are identical objects in the list, which one will
        match is quasi random.
*/

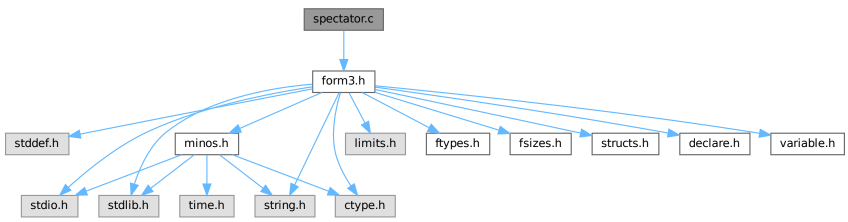
WORD BinarySearch(WORD *array,WORD num,WORD x)
{
    WORD i, bot, top, med;
    if ( num < 8 ) {
        for ( i = 0; i < num; i++ ) if ( array[i] == x ) return(i);
        return(-1);
    }
    if ( array[0] > x || array[num-1] < x ) return(-1);
    bot = 0; top = num-1; med = (top+bot)/2;
    do {
        if ( array[med] == x ) return(med);
        if ( array[med] < x ) { bot = med+1; }
        else { top = med-1; }
        med = (top+bot)/2;
    } while ( med >= bot && med <= top );
    return(-1);
}

/*
        #] BinarySearch :
    #] SortUtilities :
*/
```

4.130 spectator.c File Reference

`#include "form3.h"`

Include dependency graph for spectator.c:



Functions

- int CoCreateSpectator (UBYTE ∗inp)
- int CoToSpectator (UBYTE ∗inp)
- int CoRemoveSpectator (UBYTE ∗inp)
- int CoEmptySpectator (UBYTE ∗inp)
- int PutInSpectator (WORD ∗term, WORD specnum)
- void FlushSpectators (void)
- int CoCopySpectator (UBYTE ∗inp)
- WORD GetFromSpectator (WORD ∗term, WORD specnum)
- void ClearSpectators (WORD par)

4.130.1 Detailed Description

File contains the code for the spectator files and their control.

Definition in file spectator.c.

4.130.2 Function Documentation

4.130.2.1 CoCreateSpectator()

```
int CoCreateSpectator (
            UBYTE * inp )
```

Definition at line 89 of file spectator.c.

4.130.2.2 CoToSpectator()

```
int CoToSpectator (
            UBYTE * inp )
```

Definition at line 181 of file spectator.c.

4.130.2.3 CoRemoveSpectator()

```
int CoRemoveSpectator (
            UBYTE * inp )
```

Definition at line 233 of file spectator.c.

4.130.2.4 CoEmptySpectator()

```
int CoEmptySpectator (
            UBYTE * inp )
```

Definition at line 293 of file spectator.c.

4.130.2.5 PutInSpectator()

```
int PutInSpectator (
            WORD * term,
            WORD specnum )
```

Definition at line 345 of file spectator.c.

4.130.2.6 FlushSpectators()

```
void FlushSpectators (
             void )
```

Definition at line 402 of file spectator.c.

4.130.2.7 CoCopySpectator()

```
int CoCopySpectator (
             UBYTE * inp )
```

Definition at line 459 of file spectator.c.

4.130.2.8 GetFromSpectator()

```
WORD GetFromSpectator (
             WORD * term,
             WORD specnum )
```

Definition at line 601 of file spectator.c.

4.130.2.9 ClearSpectators()

```
void ClearSpectators (
             WORD par )
```

Definition at line 675 of file spectator.c.

4.131 spectator.c

Go to the documentation of this file.

```
00001
00006 /* #[ License : */
00007 /*
00008  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009  *   When using this file you are requested to refer to the publication
00010  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011  *   This is considered a matter of courtesy as the development was paid
00012  *   for by FOM the Dutch physics granting agency and we would like to
00013  *   be able to track its scientific use to convince FOM of its value
00014  *   for the community.
00015  *
00016  *   This file is part of FORM.
00017  *
00018  *   FORM is free software: you can redistribute it and/or modify it under the
00019  *   terms of the GNU General Public License as published by the Free Software
00020  *   Foundation, either version 3 of the License, or (at your option) any later
00021  *   version.
00022  *
00023  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
00024  *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
00025  *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
00026  *   details.
00027  *
00028  *   You should have received a copy of the GNU General Public License along
00029  *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
00030  */
00031 /* #] License : */
```

```
/*
    #[ includes :
*/

#include "form3.h"

/*
    #] includes :
    #[ Commentary :

    We use an array of SPECTATOR structs in AM.SpectatorFiles.
    When a spectator is removed this leaves a hole. This means that
    we cannot use AM.NumSpectatorFiles but always have to scan up to
    AM.SizeForSpectatorFiles which is the size of the array.
    An element is in use when it has a name. This is the name of the
    expression that is associated with it. There is also the number of
    the expression, but because the expressions are frequently renumbered
    at the end of a module, we always search for the spectators by name.
    The expression number is only valid in the current module.
    During execution we use the number of the spectator.

    The FILEHANDLE struct is borrowed from the structs for the scratch
    files, but we do not keep copies for all workers as with the scratch
    files. This brings some limitations (but saves much space). Basically
    the reading can only be done by one master or worker. And because
    we use the buffer both for writing and for reading we cannot read and
    write in the same module.

    Processor can see that an expression is to be filled with a spectator
    because we replace the compiler buffer number in the prototype by
    -specnum-1. Of course, after this filling has taken place we should
    make sure that in the next module there is a nonnegative number there.
    The input is then obtained from GetFromSpectator instead from GetTerm.
    This needed an extra argument in ThreadsProcessor. InParallelProcessor
    can figure it out by itself. ParFORM still needs to be treated for this.

    The writing is to a single buffer. Hence it needs a lock. It is possible
    to give all workers their own buffers (at great memory cost) and merge
    the results when needed. That would be friendlier on ParFORM. We ALWAYS
    assume that the order of the terms in the spectator file is random.

    In the first version there is no compression in the file. This could
    change in later versions because both the writing and the reading are
    purely sequential. Brackets are not possible.

    Currently, ParFORM allows use of spectators only in the sequential
    mode. The parallelization is switched off in modules containing
    ToSpectator or CopySpectator. Workers never create or access to
    spectator files. Their file handles are always -1. We leave the
    parallelization of modules with spectators for future work.

    #] Commentary :
    #[ CoCreateSpectator :

    Syntax: CreateSpectator name_of_expr "filename";
*/

int CoCreateSpectator(UBYTE *inp)
{
    UBYTE *p, *q, *filename, c, cc;
    WORD c1, c2, numexpr = 0, specnum, HadOne = 0;
    FILEHANDLE *fh;
    while ( *inp == ',' ) inp++;
    if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
        MesPrint("&Illegal name for expression");
        return(1);
    }
    c = *q; *q = 0;
    if ( GetVar(inp,&c1,&c2,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
        if ( c2 == CEXPRESSION &&
                Expressions[c1].status == DROPSPECTATOREXPRESSION ) {
            numexpr = c1;
            Expressions[numexpr].status = SPECTATOREXPRESSION;
            HadOne = 1;
        }
        else {
            MesPrint("&The name %s has been used already.",inp);
            *q = c;
            return(1);
        }
    }
    p = q+1;
    while ( *p == ',' ) p++;
    if ( *p != '"' ) goto Syntax;
    p++; filename = p;
    while ( *p && *p != '"' ) {
        if ( *p == '\\' ) p++;
```

```
        p++;
    }
    if ( *p != '"' ) goto Syntax;
    q = p+1;
    while ( *q && ( *q == ',' || *q == ' ' || *q == '\t' ) ) q++;
    if ( *q ) goto Syntax;
    cc = *p; *p = 0;
/*
    Now we need to: create a struct for the spectator file.
*/
    if ( HadOne == 0 )
        numexpr = EntVar(CEXPRESSION,inp,SPECTATOREXPRESSION,0,0,0);
    fh = AllocFileHandle(1,(char *)filename);
/*
    Make sure there is space in the AM.spectatorfiles array
*/
    if ( AM.NumSpectatorFiles >= AM.SizeForSpectatorFiles || AM.SpectatorFiles == 0 ) {
        int newsize, i;
        SPECTATOR *newspectators;
        if ( AM.SizeForSpectatorFiles == 0 ) {
            newsize = 10;
            AM.NumSpectatorFiles = AM.SizeForSpectatorFiles = 0;
        }
        else newsize = AM.SizeForSpectatorFiles*2;
        newspectators = (SPECTATOR *)Malloc1(newsize*sizeof(SPECTATOR),"Spectators");
        for ( i = 0; i < AM.NumSpectatorFiles; i++ )
            newspectators[i] = AM.SpectatorFiles[i];
        for ( ; i < newsize; i++ ) {
            newspectators[i].fh = 0;
            newspectators[i].name = 0;
            newspectators[i].exprnumber = -1;
            newspectators[i].flags = 0;
            PUTZERO(newspectators[i].position);
            PUTZERO(newspectators[i].readpos);
        }
        AM.SizeForSpectatorFiles = newsize;
        if ( AM.SpectatorFiles != 0 ) M_free(AM.SpectatorFiles,"Spectators");
        AM.SpectatorFiles = newspectators;
        specnum = AM.NumSpectatorFiles++;
    }
    else {
        for ( specnum = 0; specnum < AM.SizeForSpectatorFiles; specnum++ ) {
            if ( AM.SpectatorFiles[specnum].name == 0 ) break;
        }
        AM.NumSpectatorFiles++;
    }
    PUTZERO(AM.SpectatorFiles[specnum].position);
    AM.SpectatorFiles[specnum].name = (char *)(strDup1(inp,"Spectator expression name"));
    AM.SpectatorFiles[specnum].fh = fh;
    AM.SpectatorFiles[specnum].exprnumber = numexpr;
    *p = cc;
    return(0);
Syntax:
    MesPrint("&Proper syntax is: CreateSpectator,exprname,\"filename\";");
    return(-1);
}

/*
    #] CoCreateSpectator :
    #[ CoToSpectator :
*/

int CoToSpectator(UBYTE *inp)
{
    UBYTE *q;
    WORD c1, numexpr;
    int i;
    while ( *inp == ',' ) inp++;
    if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
        MesPrint("&Illegal name for expression");
        return(1);
    }
    if ( *q != 0 ) goto Syntax;
    if ( GetVar(inp,&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
            c1 != CEXPRESSION ) {
        MesPrint("&%s is not a valid expression.",inp);
        return(1);
    }
    if ( Expressions[numexpr].status != SPECTATOREXPRESSION ) {
        MesPrint("&%s is not an active spectator.",inp);
        return(1);
    }
    for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
      if ( AM.SpectatorFiles[i].name != 0 ) {
        if ( StrCmp((UBYTE *)(AM.SpectatorFiles[i].name),(UBYTE *)(inp)) == 0 ) break;
      }
    }
```

```
    if ( i >= AM.SizeForSpectatorFiles ) {
        MesPrint("&Spectator %s not found.",inp);
        return(1);
    }
    if ( ( AM.SpectatorFiles[i].flags & READSPECTATORFLAG ) != 0 ) {
        MesPrint("&Spectator %s: It is not permitted to read from and write to the same spectator in
one module.",inp);
        return(1);
    }
    AM.SpectatorFiles[i].exprnumber = numexpr;
    Add3Com(TYPETOSPECTATOR,i);
#ifdef WITHMPI
    /*
     * In ParFORM, ToSpectator has to be executed on the master.
     */
    AC.mparallelflag |= NOPARALLEL_SPECTATOR;
#endif
    return(0);
Syntax:
    MesPrint("&Proper syntax is: ToSpectator,exprname;");
    return(-1);
}

/*
    #] CoToSpectator :
    #[ CoRemoveSpectator :
*/

int CoRemoveSpectator(UBYTE *inp)
{
    UBYTE *q;
    WORD c1, numexpr;
    int i;
    SPECTATOR *sp;
    while ( *inp == ',' ) inp++;
    if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
        MesPrint("&Illegal name for expression");
        return(1);
    }
    if ( *q != 0 ) goto Syntax;
    if ( GetVar(inp,&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
            c1 != CEXPRESSION ) {
        MesPrint("&%s is not a valid expression.",inp);
        return(1);
    }
    if ( Expressions[numexpr].status != SPECTATOREXPRESSION ) {
        MesPrint("&%s is not a spectator.",inp);
        return(1);
    }
    for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
        if ( AM.SpectatorFiles[i].name != 0 ) {
            if ( StrCmp((UBYTE *)(AM.SpectatorFiles[i].name),(UBYTE *)(inp)) == 0 ) {
                break;
            }
        }
    }
    if ( i >= AM.SizeForSpectatorFiles ) {
        MesPrint("&Spectator %s not found.",inp);
        return(1);
    }
    sp = AM.SpectatorFiles+i;
    Expressions[numexpr].status = DROPSPECTATOREXPRESSION;
    if ( sp->fh->handle != -1 ) {
        CloseFile(sp->fh->handle);
        sp->fh->handle = -1;
        remove(sp->fh->name);
    }
    M_free(sp->fh,"Temporary FileHandle");
    M_free(sp->name,"Spectator expression name");

    PUTZERO(sp->position);
    PUTZERO(sp->readpos);
    sp->fh = 0;
    sp->name = 0;
    sp->exprnumber = -1;
    sp->flags = 0;
    AM.NumSpectatorFiles--;
    return(0);
Syntax:
    MesPrint("&Proper syntax is: RemoveSpectator,exprname;");
    return(-1);
}

/*
    #] CoRemoveSpectator :
    #[ CoEmptySpectator :
*/
```

```
int CoEmptySpectator(UBYTE *inp)
{
    UBYTE *q;
    WORD c1, numexpr;
    int i;
    SPECTATOR *sp;
    while ( *inp == ',' ) inp++;
    if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
        MesPrint("&Illegal name for expression");
        return(1);
    }
    if ( *q != 0 ) goto Syntax;
    if ( GetVar(inp,&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
            c1 != CEXPRESSION ) {
        MesPrint("&%s is not a valid expression.",inp);
        return(1);
    }
    if ( Expressions[numexpr].status != SPECTATOREXPRESSION ) {
        MesPrint("&%s is not a spectator.",inp);
        return(1);
    }
    for ( i = 0; i < AM.SizeForSpectatorFiles; i++ ) {
        if ( StrCmp((UBYTE *)(AM.SpectatorFiles[i].name),(UBYTE *)(inp)) == 0 ) break;
    }
    if ( i >= AM.SizeForSpectatorFiles ) {
        MesPrint("&Spectator %s not found.",inp);
        return(1);
    }
    sp = AM.SpectatorFiles+i;
    if ( sp->fh->handle != -1 ) {
        CloseFile(sp->fh->handle);
        sp->fh->handle = -1;
        remove(sp->fh->name);
    }
    sp->fh->POfill = sp->fh->POfull = sp->fh->PObuffer;
    PUTZERO(sp->position);
    PUTZERO(sp->readpos);
    return(0);
Syntax:
    MesPrint("&Proper syntax is: EmptySpectator,exprname;");
    return(-1);
}

/*
    #] CoEmptySpectator :
    #[ PutInSpectator :

    We need to use locks! There is only one file.
    The code was copied (and modified) from PutOut.
    Here we use no compression.
*/

int PutInSpectator(WORD *term,WORD specnum)
{
    GETBIDENTITY
    WORD i, *p, ret;
    LONG RetCode;
    SPECTATOR *sp = &(AM.SpectatorFiles[specnum]);
    FILEHANDLE *fi = sp->fh;

    if ( ( i = *term ) <= 0 ) return(0);
    LOCK(fi->pthreadslock);
    ret = i;
    p = fi->POfill;
    do {
        if ( p >= fi->POstop ) {
            if ( fi->handle < 0 ) {
                if ( ( RetCode = CreateFile(fi->name) ) >= 0 ) {
                    fi->handle = (WORD)RetCode;
                    PUTZERO(fi->filesize);
                    PUTZERO(fi->POposition);
                }
                else {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Cannot create spectator file %s",fi->name);
                    MUNLOCK(ErrorMessageLock);
                    UNLOCK(fi->pthreadslock);
                    return(-1);
                }
            }
            SeekFile(fi->handle,&(sp->position),SEEK_SET);
            if ( ( RetCode = WriteFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize) ) != fi->POsize
) {
                MLOCK(ErrorMessageLock);
                MesPrint("Error during spectator write. Disk full?");
                MesPrint("Attempt to write %l bytes on file %d at position %15p",
```

```
                            fi->POsize,fi->handle,&(fi->POposition));
                MesPrint("RetCode = %l, Buffer address = %l",RetCode,(LONG)(fi->PObuffer));
                MUNLOCK(ErrorMessageLock);
                UNLOCK(fi->pthreadslock);
                return(-1);
            }
            ADDPOS(fi->filesize,fi->POsize);
            p = fi->PObuffer;
            ADDPOS(sp->position,fi->POsize);
            fi->POposition = sp->position;
        }
        *p++ = *term++;
    } while ( --i > 0 );
    fi->POfull = fi->POfill = p;
    Expressions[AM.SpectatorFiles[specnum].exprnumber].counter++;
    UNLOCK(fi->pthreadslock);
    return(ret);
}

/*
    #] PutInSpectator :
    #[ FlushSpectators :
*/

void FlushSpectators(void)
{
    SPECTATOR *sp = AM.SpectatorFiles;
    FILEHANDLE *fh;
    LONG RetCode;
    int i;
    LONG size;
    if ( AM.NumSpectatorFiles <= 0 ) return;
    for ( i = 0; i < AM.SizeForSpectatorFiles; i++, sp++ ) {
        if ( sp->name == 0 ) continue;
        fh = sp->fh;
        if ( ( sp->flags & READSPECTATORFLAG ) != 0 ) { /* reset for writing */
            sp->flags &= ~READSPECTATORFLAG;
            fh->POfill = fh->PObuffer;
            if ( fh->handle >= 0 ) {
                SeekFile(fh->handle,&(sp->position),SEEK_SET);
                fh->POposition = sp->position;
            }
            continue;
        }
        if ( fh->POfill <= fh->PObuffer ) continue; /* is clean */
        if ( fh->handle < 0 ) {    /* File needs to be created */
            if ( ( RetCode = CreateFile(fh->name) ) >= 0 ) {
                PUTZERO(fh->filesize);
                PUTZERO(fh->POposition);
                fh->handle = (WORD)RetCode;
            }
            else {
                MLOCK(ErrorMessageLock);
                MesPrint("Cannot create spectator file %s",fh->name);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            PUTZERO(sp->position);
        }
        SeekFile(fh->handle,&(sp->position),SEEK_SET);
        size = (fh->POfill - fh->PObuffer)*sizeof(WORD);
        if ( ( RetCode = WriteFile(fh->handle,(UBYTE *)(fh->PObuffer),size) ) != size ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Write error synching spectator file. Disk full?");
            MesPrint("Attempt to write %l bytes on file %s at position %15p",
                        size,fh->name,&(sp->position));
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        fh->POfill = fh->PObuffer;
        SeekFile(fh->handle,&(sp->position),SEEK_END);
        fh->POposition = sp->position;
    }
    return;
}

/*
    #] FlushSpectators :
    #[ CoCopySpectator :
*/

int CoCopySpectator(UBYTE *inp)
{
    GETIDENTITY
    UBYTE *q, c, *exprname, *p;
    WORD c1, c2, numexpr;
    int specnum, error = 0;
```

```
    SPECTATOR *sp;
    while ( *inp == ',' ) inp++;
    if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
        MesPrint("&Illegal name for expression");
        return(1);
    }
    if ( *q == 0 ) goto Syntax;
    c = *q; *q = 0;
    if ( GetVar(inp,&c1,&c2,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
        MesPrint("&%s is the name of an existing variable.",inp);
        return(1);
    }
    numexpr = EntVar(CEXPRESSION,inp,LOCALEXPRESSION,0,0,0);
    p = q;
    exprname = inp;
    *q = c;
    while ( *q == ' ' || *q == ',' || *q == '\t' ) q++;
    if ( *q != '=' ) goto Syntax;
    q++;
    while ( *q == ' ' || *q == ',' || *q == '\t' ) q++;
    inp = q;
    if ( ( q = SkipAName(inp) ) == 0 || q[-1] == '_' ) {
        MesPrint("&Illegal name for spectator expression");
        return(1);
    }
    if ( *q != 0 ) goto Syntax;
    if ( AM.NumSpectatorFiles <= 0 ) {
        MesPrint("&CopySpectator: There are no spectator expressions!");
        return(1);
    }
    sp = AM.SpectatorFiles;
    for ( specnum = 0; specnum < AM.SizeForSpectatorFiles; specnum++, sp++ ) {
      if ( sp->name != 0 ) {
        if ( StrCmp((UBYTE *)(sp->name),(UBYTE *)(inp)) == 0 ) break;
      }
    }
    if ( specnum >= AM.SizeForSpectatorFiles ) {
        MesPrint("&Spectator %s not found.",inp);
        return(1);
    }
    sp->flags |= READSPECTATORFLAG;
    PUTZERO(sp->fh->POposition);
    PUTZERO(sp->readpos);
    sp->fh->POfill = sp->fh->PObuffer;
    if ( sp->fh->handle >= 0 ) {
        SeekFile(sp->fh->handle,&(sp->fh->POposition),SEEK_SET);
    }
/*
    Now we have:
    1: The name of the target expression: numexpr
    2: The spectator: sp (or specnum).
    Time for some action. We need:
    a: Write a prototype to create the expression
    b: Signal to Processor that this is a spectator.
       We do this by giving a negative compiler buffer number.
*/
    {
        WORD *OldWork, *w;
        POSITION pos;
        OldWork = w = AT.WorkPointer;
        *w++ = TYPEEXPRESSION;
        *w++ = 3+SUBEXPSIZE;
        *w++ = numexpr;
        AC.ProtoType = w;
        AR.CurExpr = numexpr;               /* Block expression numexpr */
        *w++ = SUBEXPRESSION;
        *w++ = SUBEXPSIZE;
        *w++ = numexpr;
        *w++ = 1;
        *w++ = -specnum-1;       /* Indicates "spectator" to Processor */
        FILLSUB(w)
        *w++ = 1;
        *w++ = 1;
        *w++ = 3;
        *w++ = 0;
        SeekScratch(AR.outfile,&pos);
        Expressions[numexpr].counter = 1;
        Expressions[numexpr].onfile = pos;
        Expressions[numexpr].whichbuffer = 0;
#ifdef PARALLELCODE
        Expressions[numexpr].partodo = AC.inparallelflag;
#endif
        OldWork[2] = w - OldWork - 3;
        AT.WorkPointer = w;

        if ( PutOut(BHEAD OldWork+2,&pos,AR.outfile,0) < 0 ) {
            c = *p; *p = 0;
```

```
            MesPrint("&Cannot create expression %s",exprname);
            *p = c;
            error = -1;
        }
        else {
            OldWork[2] = 4+SUBEXPSIZE;
            OldWork[4] = SUBEXPSIZE;
            OldWork[5] = numexpr;
            OldWork[SUBEXPSIZE+3] = 1;
            OldWork[SUBEXPSIZE+4] = 1;
            OldWork[SUBEXPSIZE+5] = 3;
            OldWork[SUBEXPSIZE+6] = 0;
            if ( PutOut(BHEAD OldWork+2,&pos,AR.outfile,0) < 0
            || FlushOut(&pos,AR.outfile,0) ) {
                c = *p; *p = 0;
                MesPrint("&Cannot create expression %s",exprname);
                *p = c;
                error = -1;
            }
            AR.outfile->POfull = AR.outfile->POfill;
        }
        OldWork[2] = numexpr;
/*
        Seems unnecessary (13-feb-2018)

        AddNtoL(OldWork[1],OldWork);
*/
        AT.WorkPointer = OldWork;
        if ( AC.dumnumflag ) Add2Com(TYPEDETCURDUM)
    }
#ifdef WITHMPI
    /*
     * In ParFORM, substitutions of spectators has to be done on the master.
     */
    AC.mparallelflag |= NOPARALLEL_SPECTATOR;
#endif
    return(error);
Syntax:
    MesPrint("&Proper syntax is: CopySpectator,exprname=spectatorname;");
    return(-1);
}

/*
    #] CoCopySpectator :
    #[ GetFromSpectator :

    Note that if we did things right, we do not need a lock for the reading.
*/

WORD GetFromSpectator(WORD *term,WORD specnum)
{
    SPECTATOR *sp = &(AM.SpectatorFiles[specnum]);
    FILEHANDLE *fh = sp->fh;
    WORD i, size, *t = term;
    LONG InIn;
    if ( fh-> handle < 0 ) {
        *term = 0;
        return(0);
    }
/*
    sp->position marks the 'end' of the file: the point where writing should
    take place. sp->readpos marks from where to read.
    fh->POposition marks where the file is currently positioned.
    Note that when we read, we need to
*/
    if ( ISZEROPOS(sp->readpos) ) { /* we start reading. Fill buffer. */
FillBuffer:
        SeekFile(fh->handle,&(sp->readpos),SEEK_SET);
        InIn = ReadFile(fh->handle,(UBYTE *)(fh->PObuffer),fh->POsize);
        if ( InIn < 0 || ( InIn & 1 ) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Error reading information for %s spectator",sp->name);
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        InIn /= sizeof(WORD);
        if ( InIn == 0 ) { *term = 0; return(0); }
        SeekFile(fh->handle,&(sp->readpos),SEEK_CUR);
        fh->POposition = sp->readpos;
        fh->POfull = fh->PObuffer+InIn;
        fh->POfill = fh->PObuffer;
    }
    if ( fh->POfill == fh->POfull ) { /* not even the size of the term! */
        if ( ISLESSPOS(sp->readpos,sp->position) ) goto FillBuffer;
        *term = 0;
        return(0);
    }
```

```
    size = *fh->POfill++; *t++ = size;
    for ( i = 1; i < size; i++ ) {
        if ( fh->POfill >= fh->POfull ) {
            SeekFile(fh->handle,&(sp->readpos),SEEK_SET);
            InIn = ReadFile(fh->handle,(UBYTE *)(fh->PObuffer),fh->POsize);
            if ( InIn < 0 || ( InIn & 1 ) ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Error reading information for %s spectator",sp->name);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            InIn /= sizeof(WORD);
            if ( InIn == 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Reading incomplete information for %s spectator",sp->name);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            SeekFile(fh->handle,&(sp->readpos),SEEK_CUR);
            fh->POposition = sp->readpos;
            fh->POfull = fh->PObuffer+InIn;
            fh->POfill = fh->PObuffer;
        }
        *t++ = *fh->POfill++;
    }
    return(size);
}

/*
    #] GetFromSpectator :
    #[ ClearSpectators :

    Removes all spectators.
    In case of .store, the ones that are protected by .global stay.
*/

void ClearSpectators(WORD par)
{
    SPECTATOR *sp = AM.SpectatorFiles;
    WORD numexpr, c1;
    int i;
    if ( AM.NumSpectatorFiles > 0 ) {
        for ( i = 0; i < AM.SizeForSpectatorFiles; i++, sp++ ) {
            if ( sp->name == 0 ) continue;
            if ( ( sp->flags & GLOBALSPECTATORFLAG ) == 1 && par == STOREMODULE ) continue;

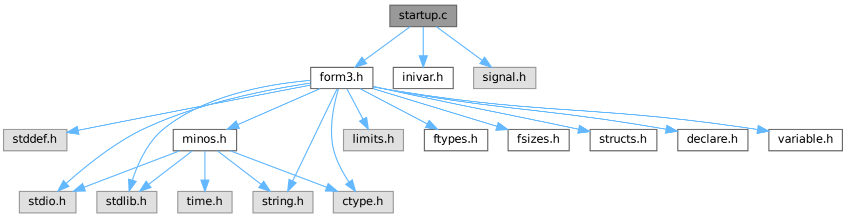
            if ( GetVar((UBYTE *)(sp->name),&c1,&numexpr,ALLVARIABLES,NOAUTO) == NAMENOTFOUND ||
                c1 != CEXPRESSION ) {
                MesPrint("&%s is not a valid expression.",sp->name);
                continue;
            }
            Expressions[numexpr].status = DROPPEDEXPRESSION;
            if ( sp->fh->handle != -1 ) {
                CloseFile(sp->fh->handle);
                sp->fh->handle = -1;
                remove(sp->fh->name);
            }
            M_free(sp->fh,"Temporary FileHandle");
            M_free(sp->name,"Spectator expression name");
            PUTZERO(sp->position);
            sp->fh = 0;
            sp->name = 0;
            sp->exprnumber = -1;
            sp->flags = 0;
            AM.NumSpectatorFiles--;
        }
    }
}

/*
    #] ClearSpectators :
*/
```

4.132 startup.c File Reference

```
#include "form3.h"
#include "inivar.h"
#include <signal.h>
```

Include dependency graph for startup.c:



Macros

- #define STRINGIFY(x) STRINGIFY__(x)
- #define STRINGIFY__(x) #x
- #define FORMNAME "FORM"
- #define VERSIONSTR__ STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION) "Beta"
- #define VERSIONSTR FORMNAME " " VERSIONSTR__ " (" PRODUCTIONDATE ")"
- #define TAKEPATH(x) if(s[1]== '='){x=s+2;} else{x=*argv++;argc--;}
- #define DEFAULTFNAMELENGTH 16

Functions

- int DoTail (int argc, UBYTE **argv)
- int OpenInput (void)
- void ReserveTempFiles (int par)
- void StartVariables (void)
- void StartMore (void)
- WORD IniVars (void)
- int main (int argc, char **argv)
- void CleanUp (WORD par)
- void TerminateImpl (int errorcode, const char *file, int line, const char *function)
- void PrintDeprecation (const char *feature, const char *issue)
- void PrintRunningTime (void)
- LONG GetRunningTime (void)

Variables

- UBYTE * emptystring = (UBYTE *)"."
- UBYTE * defaulttempfilename = (UBYTE *)"xformxxx.str"

4.132.1 Detailed Description

This file contains the main program. It also deals with the very early stages of the startup of FORM and the final stages when the program attempts some cleanup. Here is the routine that analyses the command tail.

Definition in file startup.c.

4.132.2 Macro Definition Documentation

4.132.2.1 STRINGIFY

```
#define STRINGIFY(
              x ) STRINGIFY__(x)
```

Definition at line 57 of file startup.c.

4.132.2.2 STRINGIFY__

```
#define STRINGIFY__(
              x ) #x
```

Definition at line 58 of file startup.c.

4.132.2.3 FORMNAME

```
#define FORMNAME "FORM"
```

Definition at line 68 of file startup.c.

4.132.2.4 VERSIONSTR__

```
#define VERSIONSTR__ STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION) "Beta"
```

Definition at line 97 of file startup.c.

4.132.2.5 VERSIONSTR

```
#define VERSIONSTR FORMNAME " " VERSIONSTR__ " (" PRODUCTIONDATE ")"
```

Definition at line 101 of file startup.c.

4.132.2.6 TAKEPATH

```
#define TAKEPATH(
              x ) if(s[1]== '=' ){x=s+2;} else{x=*argv++;argc--;}
```

Definition at line 235 of file startup.c.

4.132.2.7 DEFAULTFNAMELENGTH

```
#define DEFAULTFNAMELENGTH 16
```

Definition at line 664 of file startup.c.

4.132.3 Function Documentation

4.132.3.1 DoTail()

```
int DoTail (
            int argc,
            UBYTE ** argv )
```

Definition at line 237 of file startup.c.

4.132.3.2 OpenInput()

```
int OpenInput (
            void )
```

Definition at line 542 of file startup.c.

4.132.3.3 ReserveTempFiles()

```
void ReserveTempFiles (
            int par )
```

Definition at line 666 of file startup.c.

4.132.3.4 StartVariables()

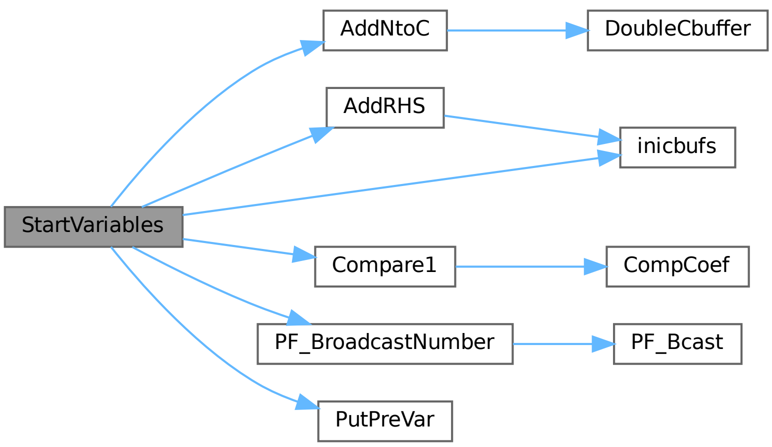
```
void StartVariables (
            void )
```

All functions (well, nearly all) are declared here.

Definition at line 905 of file startup.c.

References AddNtoC(), AddRHS(), Compare1(), inicbufs(), FuNcTiOn::name, PF_BroadcastNumber(), PutPreVar(), and FuNcTiOn::symmetric.

Here is the call graph for this function:



4.132.3.5 StartMore()

```
void StartMore (
            void )
```

Definition at line 1316 of file startup.c.

4.132.3.6 IniVars()

```
WORD IniVars (
            void )
```

Definition at line 1354 of file startup.c.

4.132.3.7 main()

```
int main (
            int argc,
            char ** argv )
```

Definition at line 1662 of file startup.c.

4.132.3.8 CleanUp()

```
void CleanUp (
            WORD par )
```

Definition at line 1761 of file startup.c.

4.132.3.9 TerminateImpl()

```
void TerminateImpl (
            int errorcode,
            const char * file,
            int line,
            const char * function )
```

Definition at line 1850 of file startup.c.

4.132.3.10 PrintDeprecation()

```
void PrintDeprecation (
            const char * feature,
            const char * issue )
```

Prints a deprecation warning for a given feature.

Parameters

feature	The name of the deprecated feature.
issue	The associated issue, e.g., "issues/700".

Definition at line 2063 of file startup.c.

4.132.3.11 PrintRunningTime()

```
void PrintRunningTime (
            void   )
```

Definition at line 2087 of file startup.c.

4.132.3.12 GetRunningTime()

```
LONG GetRunningTime (
            void   )
```

Definition at line 2128 of file startup.c.

4.132.4 Variable Documentation

4.132.4.1 emptystring

```
UBYTE* emptystring = (UBYTE *)"."
```

Definition at line 660 of file startup.c.

4.132.4.2 defaulttempfilename

```
UBYTE* defaulttempfilename = (UBYTE *)"xformxxx.str"
```

Definition at line 661 of file startup.c.

4.133 startup.c

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
        #[ includes :
*/

#include "form3.h"
#include "inivar.h"

#ifdef TRAPSIGNALS
#include "portsignals.h"
#else
#include <signal.h>
#endif
#ifdef ENABLE_BACKTRACE
    #include <execinfo.h>
#ifdef LINUX
    #include <stdint.h>
    #include <inttypes.h>
#endif
#endif

/*
 * A macro for translating the contents of `x' into a string after expanding.
 */
#define STRINGIFY(x)  STRINGIFY__(x)
#define STRINGIFY__(x) #x

/*
 * FORMNAME = "FORM" or "TFORM" or "ParFORM".
 */
#if defined(WITHPTHREADS)
    #define FORMNAME "TFORM"
#elif defined(WITHMPI)
    #define FORMNAME "ParFORM"
#else
    #define FORMNAME "FORM"
#endif

/*
 * VERSIONSTR is the version information printed in the header line.
 */
#ifdef HAVE_CONFIG_H
    /* We have also version.h. */
    #include "version.h"
    #ifndef REPO_VERSION
        #define REPO_VERSION STRINGIFY(REPO_MAJOR_VERSION) "." STRINGIFY(REPO_MINOR_VERSION)
    #endif
    #ifndef REPO_DATE
        /* The build date, instead of the repo date. */
        #define REPO_DATE __DATE__
    #endif
    #ifdef REPO_REVISION
        #define VERSIONSTR FORMNAME " " REPO_VERSION " (" REPO_DATE ", " REPO_REVISION ")"
    #else
        #define VERSIONSTR FORMNAME " " REPO_VERSION " (" REPO_DATE ")"
    #endif
```

```
    #define MAJORVERSION REPO_MAJOR_VERSION
    #define MINORVERSION REPO_MINOR_VERSION
#else
    /*
     * Otherwise, form3.h defines MAJORVERSION, MINORVERSION and PRODUCTIONDATE,
     * possibly BETAVERSION.
     */
    #ifdef BETAVERSION
        #define VERSIONSTR__ STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION) "Beta"
    #else
        #define VERSIONSTR__ STRINGIFY(MAJORVERSION) "." STRINGIFY(MINORVERSION)
    #endif
    #define VERSIONSTR FORMNAME " " VERSIONSTR__ " (" PRODUCTIONDATE ")"
#endif

/*
        #] includes :
        #[ PrintHeader :
*/

static void PrintHeader(int with_full_info)
{
#ifdef WITHMPI
    if ( PF.me == MASTER && !AM.silent ) {
#else
    if ( !AM.silent ) {
#endif
        char buffer1[250], buffer2[80], *s = buffer1, *t = buffer2;
        WORD length, n;
        for ( n = 0; n < 250; n++ ) buffer1[n] = ' ';
        /*
         * NOTE: we expect that the compiler optimizes strlen("string literal")
         * to just a number.
         */
        if ( strlen(VERSIONSTR) <= 100 ) {
            strcpy(s,VERSIONSTR);
            s += strlen(VERSIONSTR);
            *s = 0;
        }
        else {
            /*
             * Truncate when it is too long.
             */
            strncpy(s,VERSIONSTR,97);
            s[97] = '.';
            s[98] = '.';
            s[99] = ')';
            s[100] = '\0';
            s += 100;
        }
/*
        By now we omit the message about 32/64 bits. It should all be 64.

        s += snprintf(s,250-(s-buffer1)," %d-bits",(WORD)(sizeof(WORD)*16));
        *s = 0;
*/
        if ( with_full_info ) {
#if defined(WITHPTHREADS) || defined(WITHMPI)
#if defined(WITHPTHREADS)
            int nworkers = AM.totalnumberofthreads-1;
#elif defined(WITHMPI)
            int nworkers = PF.numtasks-1;
#endif
            s += snprintf(s,250-(s-buffer1)," %d worker",nworkers);
            *s = 0;
/*          while ( *s ) s++; */
            if ( nworkers != 1 ) {
                *s++ = 's';
                *s = '\0';
            }
#endif

            snprintf(t,80-(t-buffer2),"Run: %s",MakeDate());
            while ( *t ) t++;

            /*
             * Align the date to the right, if it fits in a line.
             */
            length = (s-buffer1) + (t-buffer2);
            if ( length+2 <= AC.LineLength ) {
                for ( n = AC.LineLength-length; n > 0; n-- ) *s++ = ' ';
                *s = 0;
                strcat(s,buffer2);
                while ( *s ) s++;
            }
            else {
                *s = 0;
```

```
                strcat(s,"  ");
                while ( *s ) s++;
                *s = 0;
                strcat(s,buffer2);
                while ( *s ) s++;
            }
        }

        /*
         * If the header information doesn't fit in a line, we need to extend
         * the line length temporarily.
         */
        length = s-buffer1;
        if ( length <= AC.LineLength ) {
            MesPrint("%s",buffer1);
        }
        else {
            WORD oldLineLength = AC.LineLength;
            AC.LineLength = length;
            MesPrint("%s",buffer1);
            AC.LineLength = oldLineLength;
        }
    }
#ifdef WINDOWS
    PrintDeprecation("the native Windows version", "issues/623");
#endif
#ifdef ILP32
    PrintDeprecation("the 32-bit version", "issues/624");
#endif
#ifdef WITHMPI
    PrintDeprecation("the MPI version (ParFORM)", "issues/625");
#endif
    if ( AC.CheckpointFlag ) {
        PrintDeprecation("the checkpoint mechanism", "issues/626");
    }
}

/*
        #] PrintHeader :
        #[ DoTail :

        Routine reads the command tail and handles the commandline options.
        It sets the flags for later actions and stored pathnames for
        the setup file, include/prc/sub directories etc.
        Finally the name of the program is passed on.
        Note that we do not support interactive use yet. This will come
        to pass in the distant future when we can couple STedi to FORM.
        Routine made 23-feb-1993 by J.Vermaseren
*/

#ifdef WITHINTERACTION
static UBYTE deflogname[] = "formsession.log";
#endif

#define TAKEPATH(x) if(s[1]== '=' ){x=s+2;} else{x=*argv++;argc--;}

int DoTail(int argc, UBYTE **argv)
{
    int errorflag = 0, onlyversion = 1;
    UBYTE *s, *t, *copy;
    int threadnum = 0;
    argc--; argv++;
    AM.ClearStore = 0;
    AM.TimeLimit = 0;
    AM.LogType = -1;
    AM.HoldFlag = AM.qError = AM.Interact = AM.FileOnlyFlag = 0;
    AM.InputFileName = AM.LogFileName = AM.IncDir = AM.TempDir = AM.TempSortDir =
    AM.SetupDir = AM.SetupFile = AM.Path = 0;
    AM.FromStdin = 0;
    /* Always use MultiRun, "-M" option is now ignored. */
    AM.MultiRun = 1;
    if ( argc < 1 ) {
        onlyversion = 0;
        goto printversion;
    }
    while ( argc >= 1 ) {
        s = *argv++; argc--;
        if ( *s == '-' || ( *s == '/' && ( argc > 0 || AM.Interact ) ) ) {
            s++;
            switch (*s) {
                case 'c': /* Error checking only */
                            AM.qError = 1;   break;
                case 'C': /* Next arg is filename of log file */
                            TAKEPATH(AM.LogFileName)  break;
                case 'D':
                case 'd': /* Next arg is define preprocessor var. */
                            t = copy = strDup1(*argv,"Dotail");
```

```
                            while ( *t && *t != '=' ) t++;
                            if ( *t == 0 ) {
                                if ( PutPreVar(copy,(UBYTE *)"1",0,0) < 0 ) return(-1);
                            }
                            else {
                                *t++ = 0;
                                if ( PutPreVar(copy,t,0,0) < 0 ) return(-1);
                                t[-1] = '=';
                            }
                            M_free(copy,"-d prevar");
                            argv++; argc--; break;
                case 'f': /* Output only to regular log file */
                            AM.FileOnlyFlag = 1; AM.LogType = 0; break;
                case 'F': /* Output only to log file. Further like L. */
                            AM.FileOnlyFlag = 1; AM.LogType = 1; break;
                case 'h': /* For old systems: wait for key before exit */
                            AM.HoldFlag = 1; break;
                case 'i':
                            if ( StrCmp(s, (UBYTE *)"ignore-deprecation") == 0 ) {
                                AM.IgnoreDeprecation = 1;
                                break;
                            }
#ifdef WITHINTERACTION
                            /* Interactive session (not used yet) */
                            AM.Interact = 1;
                            break;
#else
                            goto IllegalOption;
#endif
                case 'I': /* Next arg is dir for inc/prc/sub files */
                            TAKEPATH(AM.IncDir)  break;
                case 'l': /* Make regular log file */
                            if ( s[1] == 'l' ) AM.LogType = 1; /*compatibility! */
                            else               AM.LogType = 0;
                            break;
                case 'L': /* Make log file with only final statistics */
                            AM.LogType = 1;  break;
                case 'M': /* Multirun. Name of tempfiles will contain PID */
                            /* This option is now ignored. We always use MultiRun. */
                            /* AM.MultiRun = 1; */
                            break;
                case 'm': /* Read number of threads */
                case 'w': /* Read number of workers */
                            t = s++;
                            threadnum = 0;
                            while ( *s >= '0' && *s <= '9' )
                                threadnum = 10*threadnum + *s++ - '0';
                            if ( *s ) {
#ifdef WITHMPI
                                if ( PF.me == MASTER )
#endif
                                printf("Illegal value for option m or w: %s\n",t);
                                errorflag++;
                            }
/*                          if ( threadnum == 1 ) threadnum = 0; */
                            threadnum++;
                            break;
                case 'W': /* Print the wall-clock time on the master. */
                            AM.ggWTimeStatsFlag = 1;
                            break;
/*
                case 'n':
                            Reserved for number of slaves without MPI
*/
                case 'p':
#ifdef WITHEXTERNALCHANNEL
                    /*There are two possibilities: -p|-pipe*/
                    if(s[1]=='i'){
                        if( (s[2]=='p')&&(s[3]=='e')&&(s[4]=='\0') ){
                            argc--;
                            /*Initialize pre-set external channels, see
                                the file extcmd.c:*/
                            if(initPresetExternalChannels(*argv++,AX.timeout)<1){
#ifdef WITHMPI
                                if ( PF.me == MASTER )
#endif
                                printf("Error initializing preset external channels\n");
                                errorflag++;
                            }
                            AX.timeout=-1;/*This indicates that preset channels
                                                    are initialized from cmdline*/
                        }else{
#ifdef WITHMPI
                            if ( PF.me == MASTER )
#endif
                            printf("Illegal option in call of FORM: %s\n",s);
                            errorflag++;
```

```
                            }
                        }else
#else
                        if ( s[1] ) {
                            if ( ( s[1]=='i' ) && ( s[2] == 'p' ) && (s[3] == 'e' )
                            && ( s[4] == '\0' ) ){
#ifdef WITHMPI
                                if ( PF.me == MASTER )
#endif
                                printf("Illegal option: Pipes not supported on this system.\n");
                            }
                            else {
#ifdef WITHMPI
                                if ( PF.me == MASTER )
#endif
                                printf("Illegal option: %s\n",s);
                            }
                            errorflag++;
                        }
                        else
#endif
                        {
                                 /* Next arg is a path variable like in environment */
                            TAKEPATH(AM.Path)
                        }
                        break;
                    case 'q': /* Quiet option. Only output. Same as -si */
                                AM.silent = 1; break;
                    case 'R': /* recover from saved snapshot */
                                AC.CheckpointFlag = -1;
                                break;
                    case 's': /* Next arg is dir with form.set to be used */
                                if ( ( s[1] == 'o' ) && ( s[2] == 'r' ) && ( s[3] == 't' ) ) {
                                    if(s[4]== '=' ) {
                                        AM.TempSortDir = s+5;
                                    }
                                    else {
                                        AM.TempSortDir = *argv++;
                                        argc--;
                                    }
                                }
                                else if ( s[1] == 'i' ) { /* compatibility: silent/quiet */
                                    AM.silent = 1;
                                }
                                else {
                                    TAKEPATH(AM.SetupDir)
                                }
                                break;
                    case 'S': /* Next arg is setup file */
                                TAKEPATH(AM.SetupFile) break;
                    case 't': /* Next arg is directory for temp files */
                                if ( s[1] == 's' ) {
                                    s++;
                                    AM.havesortdir = 1;
                                    TAKEPATH(AM.TempSortDir)
                                }
                                else {
                                    TAKEPATH(AM.TempDir)
                                }
                                break;
                    case 'T': /* Print the total size used at end of job */
                                AM.PrintTotalSize = 1; break;
                    case 'v':
printversion:;
#ifdef WITHMPI
                                if ( PF.me == MASTER )
#endif
                                    PrintHeader(0);
                                if ( onlyversion ) return(1);
                                goto NoFile;
                    case 'y': /* Preprocessor dumps output. No compilation. */
                                AP.PreDebug = PREPROONLY;   break;
                    case 'z': /* The number following is a time limit in sec. */
                                t = s++;
                                AM.TimeLimit = 0;
                                while ( *s >= '0' && *s <= '9' )
                                    AM.TimeLimit = 10*AM.TimeLimit + *s++ - '0';
                                break;
                    case 'Z': /* Removes the .str file on crash, no matter its contents */
                                AM.ClearStore = 1;   break;
                    case '\0': /* "-" to use STDIN for the input. */
#ifdef WITHMPI
                                /* At the moment, ParFORM doesn't implement STDIN broadcasts. */
                                if ( PF.me == MASTER )
                                    printf("Sorry, reading STDIN as input is currently not supported by ParFORM\n");
                                errorflag++;
```

```
#endif
                            AM.FromStdin = 1;
                            AC.NoShowInput = 1; // disable input echoing by default
                            break;
                default:
                        if ( FG.cTable[*s] == 1 ) {
                            AM.SkipClears = 0; t = s;
                            while ( FG.cTable[*t] == 1 )
                                AM.SkipClears = 10*AM.SkipClears + *t++ - '0';
                            if ( *t != 0 ) {
#ifdef WITHMPI
                                if ( PF.me == MASTER )
#endif
                                printf("Illegal numerical option in call of FORM: %s\n",s);
                                errorflag++;
                            }
                        }
                        else {
IllegalOption:
#ifdef WITHMPI
                            if ( PF.me == MASTER )
#endif
                            printf("Illegal option in call of FORM: %s\n",s);
                            errorflag++;
                        }
                        break;
            }
        }
        else if ( argc == 0 && !AM.Interact ) AM.InputFileName = argv[-1];
        else {
#ifdef WITHMPI
            if ( PF.me == MASTER )
#endif
            printf("Illegal option in call of FORM: %s\n",s);
            errorflag++;
        }
    }
    AM.totalnumberofthreads = threadnum;
    if ( AM.InputFileName ) {
        if ( AM.FromStdin ) {
            printf("STDIN and the input filename cannot be specified simultaneously\n");
            errorflag++;
        }
        s = AM.InputFileName;
        while ( *s ) s++;
        if ( s < AM.InputFileName+4 ||
        s[-4] != '.' || s[-3] != 'f' || s[-2] != 'r' || s[-1] != 'm' ) {
            t = (UBYTE *)Malloc1((s-AM.InputFileName)+5,"adding .frm");
            s = AM.InputFileName;
            AM.InputFileName = t;
            while ( *s ) *t++ = *s++;
            *t++ = '.'; *t++ = 'f'; *t++ = 'r'; *t++ = 'm'; *t = 0;
        }
        if ( AM.LogType >= 0 && AM.LogFileName == 0 ) {
            AM.LogFileName = strDup1(AM.InputFileName,"name of logfile");
            s = AM.LogFileName;
            while ( *s ) s++;
            s[-3] = 'l'; s[-2] = 'o'; s[-1] = 'g';
        }
    }
#ifdef WITHINTERACTION
    else if ( AM.Interact ) {
        if ( AM.LogType >= 0 ) {
/*
            We may have to do better than just taking a name.
            It is not unique! This will be left for later.
*/
            AM.LogFileName = deflogname;
        }
    }
#endif
    else if ( AM.FromStdin ) {
        /* Do nothing. */
    }
    else {
NoFile:
#ifdef WITHMPI
        if ( PF.me == MASTER )
#endif
        printf("No filename specified in call of FORM\n");
        errorflag++;
    }
    if ( AM.Path == 0 ) AM.Path = (UBYTE *)getenv("FORMPATH");
    if ( AM.Path ) {
        /*
         * AM.Path is taken from argv or getenv. Reallocate it to avoid invalid
         * frees when AM.Path has to be changed.
```

```
         */
        AM.Path = strDup1(AM.Path,"DoTail Path");
    }
    return(errorflag);
}

/*
        #] DoTail :
        #[ OpenInput :

        Major task here after opening is to skip the proper number of
        .clear instructions if so desired without using interpretation
*/

int OpenInput(void)
{
    int oldNoShowInput = AC.NoShowInput;
    UBYTE c;
    if ( !AM.Interact ) {
        if ( AM.FromStdin ) {
            if ( OpenStream(0,INPUTSTREAM,0,PRENOACTION) == 0 ) {
                Error0("Cannot open STDIN");
                return(-1);
            }
        }
        else {
        if ( OpenStream(AM.InputFileName,FILESTREAM,0,PRENOACTION) == 0 ) {
            Error1("Cannot open file",AM.InputFileName);
            return(-1);
        }
        if ( AC.CurrentStream->inbuffer <= 0 ) {
            Error1("No input in file",AM.InputFileName);
            return(-1);
        }
        }
        AC.NoShowInput = 1;
        while ( AM.SkipClears > 0 ) {
            c = GetInput();
            if ( c == ENDOFINPUT ) {
                Error0("Not enough .clear instructions in input file");
            }
            if ( c == '\\' ) {
                c = GetInput();
                if ( c == ENDOFINPUT )
                    Error0("Not enough .clear instructions in input file");
                continue;
            }
            if ( c == ' ' || c == '\t' ) continue;
            if ( c == '.' ) {
                c = GetInput();
                if ( tolower(c) == 'c' ) {
                    c = GetInput();
                    if ( tolower(c) == 'l' ) {
                        c = GetInput();
                        if ( tolower(c) == 'e' ) {
                            c = GetInput();
                            if ( tolower(c) == 'a' ) {
                                c = GetInput();
                                if ( tolower(c) == 'r' ) {
                                    c = GetInput();
                                    if ( FG.cTable[c] > 2 ) {
                                        AM.SkipClears--;
                                    }
                                }
                            }
                        }
                    }
                }
                while ( c != '\n' && c != '\r' && c != ENDOFINPUT ) {
                    c = GetInput();
                    if ( c == '\\' ) c = GetInput();
                }
            }
            else if ( c == '\n' || c == '\r' ) continue;
            else {
                while ( ( c = GetInput() ) != '\n' && c != '\r' ) {
                    if ( c == ENDOFINPUT ) {
                        Error0("Not enough .clear instructions in input file");
                    }
                }
            }
        }
        AC.NoShowInput = oldNoShowInput;
    }
    if ( AM.LogFileName ) {
#ifdef WITHMPI
        if ( PF.me != MASTER ) {
```

```
            /*
             * Only the master writes to the log file. On slaves, we need
             * a dummy handle, without opening the file.
             */
            extern FILES **filelist;  /* in tools.c */
            int i = CreateHandle();
            RWLOCKW(AM.handlelock);
            filelist[i] = (FILES *)123;  /* Must be nonzero to prevent a reuse in CreateHandle. */
            UNRWLOCK(AM.handlelock);
            AC.LogHandle = i;
        }
        else
#endif
        if ( AC.CheckpointFlag != -1 ) {
            if ( ( AC.LogHandle = CreateLogFile((char *)(AM.LogFileName)) ) < 0 ) {
                Error1("Cannot create logfile",AM.LogFileName);
                return(-1);
            }
        }
        else {
            if ( ( AC.LogHandle = OpenAddFile((char *)(AM.LogFileName)) ) < 0 ) {
                Error1("Cannot re-open logfile",AM.LogFileName);
                return(-1);
            }
        }
    }
    return(0);
}

/*
        #] OpenInput :
        #[ ReserveTempFiles :

        Order of preference:
        a: if there is a path in the commandtail, take that.
        b: if none, try in the form.set file.
        c: if still none, try in the environment for the variable FORMTMP
        d: if still none, try the current directory.

        The parameter indicates action in the case of multithreaded running.
        par = 0 : We just run on a single processor. Keep everything normal.
        par = 1 : Multithreaded running startup phase 1.
        par = 2 : Multithreaded running startup phase 2.
*/

UBYTE *emptystring = (UBYTE *)".";
UBYTE *defaulttempfilename = (UBYTE *)"xformxxx.str";
/* This is the length of the above default, but with 7 spaces for PID digits
   instead of the "xxx". (Previously FORM used 5 digits and this value was 14) */
#define DEFAULTFNAMELENGTH 16

void ReserveTempFiles(int par)
{
    GETIDENTITY
    SETUPPARAMETERS *sp;
    UBYTE *s, *t, *tenddir, *tenddir2, c;
    int i = 0;
    WORD j;
    if ( par == 0 || par == 1 ) {
    if ( AM.TempDir == 0 ) {
        sp = GetSetupPar((UBYTE *)"tempdir");
        if ( ( sp->flags & USEDFLAG ) != USEDFLAG ) {
            AM.TempDir = (UBYTE *)getenv("FORMTMP");
            if ( AM.TempDir == 0 ) AM.TempDir = emptystring;
        }
        else AM.TempDir = (UBYTE *)(sp->value);
    }
    if ( AM.TempSortDir == 0 ) {
        if ( AM.havesortdir ) {
            sp = GetSetupPar((UBYTE *)"tempsortdir");
            AM.TempSortDir = (UBYTE *)(sp->value);
        }
        else {
            AM.TempSortDir = (UBYTE *)getenv("FORMTMPSORT");
            if ( AM.TempSortDir == 0 ) AM.TempSortDir = AM.TempDir;
        }
    }
/*
    We have now in principle a path but we will use its first element only.
    Later that should become more complicated. Then we will use a path and
    when one device is full we can continue on the next one.
*/
    s = AM.TempDir; i = 200;   /* Some extra for VMS */
    while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\' ) s++; s++; i++; }
    FG.fnamesize = sizeof(UBYTE)*(i+DEFAULTFNAMELENGTH);
    FG.fname = (char *)Malloc1(FG.fnamesize,"name for temporary files");
    s = AM.TempDir; t = (UBYTE *)FG.fname;
```

```
    while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\' ) s++; *t++ = *s++; }
    if ( (char *)t > FG.fname && t[-1] != SEPARATOR && t[-1] != ALTSEPARATOR )
        *t++ = SEPARATOR;
    *t = 0;
    tenddir = t;
    FG.fnamebase = t-(UBYTE *)(FG.fname);

    s = AM.TempSortDir; i = 200;   /* Some extra for VMS */
    while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\' ) s++; s++; i++; }

    FG.fname2size = sizeof(UBYTE)*(i+DEFAULTFNAMELENGTH);
    FG.fname2 = (char *)Malloc1(FG.fname2size,"name for sort files");
    s = AM.TempSortDir; t = (UBYTE *)FG.fname2;
    while ( *s && *s != PATHSEPARATOR ) { if ( *s == '\\' ) s++; *t++ = *s++; }
    if ( (char *)t > FG.fname2 && t[-1] != SEPARATOR && t[-1] != ALTSEPARATOR )
        *t++ = SEPARATOR;
    *t = 0;
    tenddir2 = t;
    FG.fname2base = t-(UBYTE *)(FG.fname2);

    t = tenddir;
    s = defaulttempfilename;
#ifdef WITHMPI
    {
        int iii;
        iii = snprintf((char*)t,FG.fnamesize-((char*)t-FG.fname),"%d",PF.me);
        t+= iii;
        s+= iii; /* in case defaulttmpfilename is too short */
    }
#endif
    while ( *s ) *t++ = *s++;
    *t = 0;
/*
        There are problems when running many FORM jobs at the same time
        from make or minos. If they start up simultaneously, occasionally
        they can make the same .str file. We prevent this with first trying
        a file that contains the digits of the pid. If this file
        has already been taken we fall back on the old scheme.
        The whole is controlled with the -M (MultiRun) parameter in the
        command tail.
*/
    if ( AM.MultiRun ) {
        int num = ((int)GetPID())%10000000;
        t += 4;
        *t = 0;
        t[-1] = 'r';
        t[-2] = 't';
        t[-3] = 's';
        t[-4] = '.';
        t[-5] = (UBYTE)('0' + num%10);
        t[-6] = (UBYTE)('0' + (num/10)%10);
        t[-7] = (UBYTE)('0' + (num/100)%10);
        t[-8] = (UBYTE)('0' + (num/1000)%10);
        t[-9] = (UBYTE)('0' + (num/10000)%10);
        t[-10] = (UBYTE)('0' + (num/100000)%10);
        t[-11] = (UBYTE)('0' + num/1000000);
        if ( ( AC.StoreHandle = CreateFile((char *)FG.fname) ) < 0 ) {
            t[-5] = 'x'; t[-6] = 'x'; t[-7] = 'x'; t[-8] = 'x'; t[-9] = 'x'; t[-10] = 'x'; t[-11] = 'x';
            goto classic;
        }
    }
    else
    {
classic:;
      for(;;) {
        if ( ( AC.StoreHandle = OpenFile((char *)FG.fname) ) < 0 ) {
            if ( ( AC.StoreHandle = CreateFile((char *)FG.fname) ) >= 0 ) break;
        }
        else CloseFile(AC.StoreHandle);
        c = t[-5];
        if ( c == 'x' ) t[-5] = '0';
        else if ( c == '9' ) {
            t[-5] = '0';
            c = t[-6];
            if ( c == 'x' ) t[-6] = '0';
            else if ( c == '9' ) {
                t[-6] = '0';
                c = t[-7];
                if ( c == 'x' ) t[-7] = '0';
                else if ( c == '9' ) {
/*
                    Note that we tried 1111 names!
*/
                    MesPrint("Name space for temp files exhausted");
                    t[-7] = 0;
                    MesPrint("Please remove files of the type %s or try a different directory"
```

```
                        ,FG.fname);
                    Terminate(-1);
                }
                else t[-7] = (UBYTE)(c+1);
            }
            else t[-6] = (UBYTE)(c+1);
        }
        else t[-5] = (UBYTE)(c+1);
      }
    }
/*
    Now we should make sure that the tempsortdir cq tempsortfilename makes it
    into a similar construction.
*/
    s = tenddir; t = tenddir2; while ( *s ) *t++ = *s++;
    *t = 0;

/*
    Now we should assign a name to the main sort file and the two stage 4 files.
*/
    AM.S0->file.name = (char *)Malloc1(sizeof(char)*(i+DEFAULTFNAMELENGTH),"name for temporary files");
    s = (UBYTE *)AM.S0->file.name;
    t = (UBYTE *)FG.fname2;
    i = 1;
    while ( *t ) { *s++ = *t++; i++; }
    s[-2] = 'o'; *s = 0;
    }
/*
    Try to create the sort file already, so we can Terminate earlier if this fails.
*/
#ifdef WITHPTHREADS
    if ( par <= 1 ) {
#endif
    if ( ( AM.S0->file.handle = CreateFile((char *)AM.S0->file.name) ) < 0 ) {
        MesPrint("Could not create sort file: %s", AM.S0->file.name);
        Terminate(-1);
    };
    /* Close and clean up the test file */
    CloseFile(AM.S0->file.handle);
    AM.S0->file.handle = -1;
    remove(AM.S0->file.name);
#ifdef WITHPTHREADS
    }
#endif
/*
    With the stage4 and scratch file names we have to be a bit more careful.
    They are to be allocated after the threads are initialized when there
    are threads of course.
*/
    if ( par == 0 ) {
        s = (UBYTE *)((void *)(FG.fname2)); i = 0;
        while ( *s ) { s++; i++; }
        /* +1 for null terminator */
        s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for stage4 file a");
        AR.FoStage4[1].name = (char *)s;
        t = (UBYTE *)FG.fname2;
        while ( *t ) *s++ = *t++;
        s[-2] = '4'; s[-1] = 'a'; *s = 0;
        s = (UBYTE *)((void *)(FG.fname)); i = 0;
        while ( *s ) { s++; i++; }
        /* +1 for null terminator */
        s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for stage4 file b");
        AR.FoStage4[0].name = (char *)s;
        t = (UBYTE *)FG.fname;
        while ( *t ) *s++ = *t++;
        s[-2] = '4'; s[-1] = 'b'; *s = 0;
        for ( j = 0; j < 3; j++ ) {
            /* +1 for null terminator */
            s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for scratch file");
            AR.Fscr[j].name = (char *)s;
            t = (UBYTE *)FG.fname;
            while ( *t ) *s++ = *t++;
            s[-2] = 'c'; s[-1] = (UBYTE)('0'+j); *s = 0;
        }
    }
#ifdef WITHPTHREADS
    else if ( par == 2 ) {
        size_t tname;
        s = (UBYTE *)((void *)(FG.fname2)); i = 0;
        while ( *s ) { s++; i++; }
        /* +1 for null terminator, +10 for 32bit int, +1 for "." */
        tname = sizeof(char)*(i+12);
        s = (UBYTE *)Malloc1(tname,"name for stage4 file a");
        snprintf((char *)s,tname,"%s.%d",FG.fname2,AT.identity);
        s[i-2] = '4'; s[i-1] = 'a';
        AR.FoStage4[1].name = (char *)s;
```

```
        s = (UBYTE *)((void *)(FG.fname)); i = 0;
        while ( *s ) { s++; i++; }
        /* +1 for null terminator, +10 for 32bit int, +1 for "." */
        tname = sizeof(char)*(i+12);
        s = (UBYTE *)Malloc1(tname,"name for stage4 file b");
        snprintf((char *)s,tname,"%s.%d",FG.fname,AT.identity);
        s[i-2] = '4'; s[i-1] = 'b';
        AR.FoStage4[0].name = (char *)s;
        if ( AT.identity == 0 ) {
            for ( j = 0; j < 3; j++ ) {
                /* +1 for null terminator */
                s = (UBYTE *)Malloc1(sizeof(char)*(i+1),"name for scratch file");
                AR.Fscr[j].name = (char *)s;
                t = (UBYTE *)FG.fname;
                while ( *t ) *s++ = *t++;
                s[-2] = 'c'; s[-1] = (UBYTE)('0'+j); *s = 0;
            }
        }
    }
#endif
}

/*
        #] ReserveTempFiles :
        #[ StartVariables :
*/

#ifdef WITHPTHREADS
ALLPRIVATES *DummyPointer = 0;
#endif

void StartVariables(void)
{
    int i, ii;
    PUTZERO(AM.zeropos);
    StartPrepro();
/*
    The module counter:
*/
    AC.CModule=0;
#ifdef WITHPTHREADS
/*
    We need a value in AB because in the startup some routines may call AB[0].
*/
    AB = (ALLPRIVATES **)&DummyPointer;
#endif
/*
    separators used to delimit arguments in #call and #do, by default ',' and '|':
    Be sure, it is en empty set:
*/
    set_sub(AC.separators,AC.separators,AC.separators);
    set_set(',',AC.separators);
    set_set('|',AC.separators);

    AM.BracketFactors[0] = 8;
    AM.BracketFactors[1] = SYMBOL;
    AM.BracketFactors[2] = 4;
    AM.BracketFactors[3] = FACTORSYMBOL;
    AM.BracketFactors[4] = 1;
    AM.BracketFactors[5] = 1;
    AM.BracketFactors[6] = 1;
    AM.BracketFactors[7] = 3;

    AM.SkipClears = 0;
    AC.Cnumpows = 0;
    AC.OutputMode = 72;
    AC.OutputSpaces = NORMALFORMAT;
    AC.LineLength = 79;
    AM.gIsFortran90 = AC.IsFortran90 = ISNOTFORTRAN90;
    AM.gFortran90Kind = AC.Fortran90Kind = 0;
    AM.gCnumpows = 0;
    AC.exprfillwarning = 0;
    AM.gLineLength = 79;
    AM.OutBufSize = 80;
    AM.MaxStreamSize = MAXFILESTREAMSIZE;
    AP.MaxPreAssignLevel = 4;
    AC.iBufferSize = 512;
    AP.pSize = 128;
    AP.MaxPreIfLevel = 10;
    AP.cComChar = AP.ComChar = '*';
    AP.firstnamespace = 0;
    AP.lastnamespace = 0;
    AP.fullnamesize = 127;
    AP.fullname = (UBYTE *)Malloc1((AP.fullnamesize+1)*sizeof(UBYTE),"AP.fullname");
    AM.OffsetVector = -2*WILDOFFSET+MINSPEC;
    AC.cbufList.num = 0;
    AM.hparallelflag = AM.gparallelflag =
```

```
    AC.parallelflag = AC.mparallelflag = PARALLELFLAG;
#ifdef WITHMPI
    if ( PF.numtasks < 2 ) AM.hparallelflag |= NOPARALLEL_NPROC;
#endif
    AC.tablefilling = 0;
    AC.models = 0;
    AC.nummodels = 0;
    AC.ModelLevel = 0;
    AC.modelspace = 0;
    AM.resetTimeOnClear = 1;
    AM.gnumextrasym = AM.ggnumextrasym = 0;
    AM.havesortdir = 0;
    AM.SpectatorFiles = 0;
    AM.NumSpectatorFiles = 0;
    AM.SizeForSpectatorFiles = 0;
/*
    Information for the lists of variables. Part of error message and size:
*/
    AP.ProcList.message = "procedure";
    AP.ProcList.size = sizeof(PROCEDURE);
    AP.LoopList.message = "doloop";
    AP.LoopList.size = sizeof(DOLOOP);
    AP.PreVarList.message = "PreVariable";
    AP.PreVarList.size = sizeof(PREVAR);
    AC.SymbolList.message = "symbol";
    AC.SymbolList.size = sizeof(struct SyMbOl);
    AC.IndexList.message = "index";
    AC.IndexList.size = sizeof(struct InDeX);
    AC.VectorList.message = "vector";
    AC.VectorList.size = sizeof(struct VeCtOr);
    AC.FunctionList.message = "function";
    AC.FunctionList.size = sizeof(struct FuNcTiOn);
    AC.SetList.message = "set";
    AC.SetList.size = sizeof(struct SeTs);
    AC.SetElementList.message = "set element";
    AC.SetElementList.size = sizeof(WORD);
    AC.ExpressionList.message = "expression";
    AC.ExpressionList.size = sizeof(struct ExPrEsSiOn);
    AC.cbufList.message = "compiler buffer";
    AC.cbufList.size = sizeof(CBUF);
    AC.ChannelList.message = "channel buffer";
    AC.ChannelList.size = sizeof(CHANNEL);
    AP.DollarList.message = "$-variable";
    AP.DollarList.size = sizeof(struct DoLlArS);
    AC.DubiousList.message = "ambiguous variable";
    AC.DubiousList.size = sizeof(struct DuBiOuS);
    AC.TableBaseList.message = "list of tablebases";
    AC.TableBaseList.size = sizeof(DBASE);
    AC.TestValue = 0;
    AC.InnerTest = 0;

    AC.AutoSymbolList.message = "autosymbol";
    AC.AutoSymbolList.size = sizeof(struct SyMbOl);
    AC.AutoIndexList.message = "autoindex";
    AC.AutoIndexList.size = sizeof(struct InDeX);
    AC.AutoVectorList.message = "autovector";
    AC.AutoVectorList.size = sizeof(struct VeCtOr);
    AC.AutoFunctionList.message = "autofunction";
    AC.AutoFunctionList.size = sizeof(struct FuNcTiOn);
    AC.PotModDolList.message = "potentially modified dollar";
    AC.PotModDolList.size = sizeof(WORD);
    AC.ModOptDolList.message = "moduleoptiondollar";
    AC.ModOptDolList.size = sizeof(MODOPTDOLLAR);

    AO.FortDotChar = '_';
    AO.ErrorBlock = 0;
    AC.firstconstindex = 1;
    AO.Optimize.mctsconstant.fval = 1.0;
    AO.Optimize.horner = O_MCTS;
    AO.Optimize.hornerdirection = O_FORWARDORBACKWARD;
    AO.Optimize.method = O_GREEDY;
    AO.Optimize.mctstimelimit = 0;
    AO.Optimize.mctsnumexpand = 1000;
    AO.Optimize.mctsnumkeep = 10;
    AO.Optimize.mctsnumrepeat = 1;
    AO.Optimize.greedytimelimit = 0;
    AO.Optimize.greedyminnum = 10;
    AO.Optimize.greedymaxperc = 5;
    AO.Optimize.printstats = 0;
    AO.Optimize.debugflags = 0;
    AO.OptimizeResult.code = NULL;
    AO.inscheme = 0;
    AO.schemenum = 0;
    AO.wpos = 0;
    AO.wpoin = 0;
    AO.wlen = 0;
    AM.dollarzero = 0;
```

```
    AC.doloopstack = 0;
    AC.doloopstacksize = 0;
    AC.dooplevel = 0;
/*
    Set up the main name trees:
*/
    AC.varnames  = MakeNameTree();
    AC.exprnames = MakeNameTree();
    AC.dollarnames = MakeNameTree();
    AC.autonames = MakeNameTree();
    AC.activenames = &(AC.varnames);
    AP.preError = 0;
/*
    Initialize the compiler:
*/
    inictable();
    AM.rbufnum = inicbufs();          /* Regular compiler buffer */
#ifndef WITHPTHREADS
    AT.ebufnum = inicbufs();          /* Buffer for extras during execution */
    AT.fbufnum = inicbufs();          /* Buffer for caching in factorization */
    AT.allbufnum = inicbufs();        /* Buffer for id,all */
    AT.aebufnum = inicbufs();         /* Buffer for id,all */
    AN.tryterm = 0;
#else
    AS.MasterSort = 0;
#endif
    AM.dbufnum = inicbufs();          /* Buffer for dollar variables */
    AM.sbufnum = inicbufs();          /* Subterm buffer for polynomials and optimization */
    AC.ffbufnum = inicbufs();         /* Buffer number for user defined factorizations */
    AM.zbufnum = inicbufs();          /* For very special values */
    {
        CBUF *C = cbuf+AM.zbufnum;
        WORD one[5] = {4,1,1,3,0};
        WORD zero = 0;
        AddRHS(AM.zbufnum,1);
        AM.zerorhs = C->numrhs;
        AddNtoC(AM.zbufnum,1,&zero,17);
        AddRHS(AM.zbufnum,1);
        AM.onerhs = C->numrhs;
        AddNtoC(AM.zbufnum,5,one,17);
    }
    AP.inside.inscbuf = inicbufs(); /* For the #inside instruction */
/*
    Enter the built in objects
*/
    AC.Symbols = &(AC.SymbolList);
    AC.Indices = &(AC.IndexList);
    AC.Vectors = &(AC.VectorList);
    AC.Functions = &(AC.FunctionList);
    AC.vetofilling = 0;

    AddDollar((UBYTE *)"$",DOLUNDEFINED,0,0);

    cbuf[AM.dbufnum].mnumlhs = cbuf[AM.dbufnum].numlhs;
    cbuf[AM.dbufnum].mnumrhs = cbuf[AM.dbufnum].numrhs;

    AddSymbol((UBYTE *)"i_",-MAXPOWER,MAXPOWER,VARTYPEIMAGINARY,0);
    AddSymbol((UBYTE *)"pi_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
/*
    coeff_ should have the number COEFFSYMBOL and den_ the number DENOMINATOR
    and the three should be in this order!
*/
    AddSymbol((UBYTE *)"coeff_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"num_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"den_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"xarg_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"dimension_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"factor_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"sep_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"ee_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    AddSymbol((UBYTE *)"em_",-MAXPOWER,MAXPOWER,VARTYPENONE,0);
    i = BUILTINSYMBOLS;  /* update this in ftypes.h when we add new symbols */
/*
    Next we add a number of dummy symbols for ensuring that the user defined
    symbols start at a fixed given number FIRSTUSERSYMBOL
    We do want to give them unique names though that the user cannot access.
*/
    {
        char dumstr[20];
        for ( ; i < FIRSTUSERSYMBOL; i++ ) {
            snprintf(dumstr,20,":%d:",i);
            AddSymbol((UBYTE *)dumstr,-MAXPOWER,MAXPOWER,VARTYPENONE,0);
        }
    }

    AddIndex((UBYTE *)"iarg_",4,0);
    AddVector((UBYTE *)"parg_",VARTYPENONE,0);
```

```

    AM.NumFixedFunctions = sizeof(fixedfunctions)/sizeof(struct fixedfun);
    for ( i = 0; i < AM.NumFixedFunctions; i++ ) {
        ii = AddFunction((UBYTE *)fixedfunctions[i].name
                                 ,fixedfunctions[i].commu
                                 ,fixedfunctions[i].tensor
                                 ,fixedfunctions[i].complx
                                 ,fixedfunctions[i].symmetric
                                 ,0,-1,-1);
        if ( fixedfunctions[i].tensor == GAMMAFUNCTION )
                            functions[ii].flags |= COULDCOMMUTE;
    }
/*
    Next we add a number of dummy functions for ensuring that the user defined
    functions start at a fixed given number FIRSTUSERFUNCTION.
    We do want to give them unique names though that the user cannot access.
*/
    {
        char dumstr[20];
        for ( ; i < FIRSTUSERFUNCTION-FUNCTION; i++ ) {
            snprintf(dumstr,20,"::%d::",i);
            AddFunction((UBYTE *)dumstr,0,0,0,0,0,-1,-1);
        }
    }
    AM.NumFixedSets = sizeof(fixedsets)/sizeof(struct fixedset);
    for ( i = 0; i < AM.NumFixedSets; i++ ) {
        ii = AddSet((UBYTE *)fixedsets[i].name,fixedsets[i].dimension);
        Sets[ii].type = fixedsets[i].type;
    }
    AM.RepMax = MAXREPEAT;
#ifndef WITHPTHREADS
    AT.RepCount = (int *)Malloc1((LONG)((AM.RepMax+3)*sizeof(int)),"repeat buffers");
    AN.RepPoint = AT.RepCount;
    AT.RepTop = AT.RepCount + AM.RepMax;
    AN.polysortflag = 0;
    AN.subsubveto = 0;
#endif
    AC.NumWildcardNames = 0;
    AC.WildcardBufferSize = 50;
    AC.WildcardNames = (UBYTE *)Malloc1((LONG)AC.WildcardBufferSize,"argument list names");
#ifndef WITHPTHREADS
    AT.WildArgTaken = (WORD *)Malloc1((LONG)AC.WildcardBufferSize*sizeof(WORD)/2
                ,"argument list names");
    AT.WildcardBufferSize = AC.WildcardBufferSize;
    AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
    AT.nfac = AT.nBer = 0;
    AT.factorials = 0;
    AT.bernoullis = 0;
    AR.wranfia = 0;
    AR.wranfcall = 0;
    AR.wranfnpair1 = NPAIR1;
    AR.wranfnpair2 = NPAIR2;
    AR.wranfseed = 0;
#endif
    AM.atstartup = 1;
    AM.oldnumextrasymbols = strDup1((UBYTE *)"OLDNUMEXTRASYMBOLS_","oldnumextrasymbols");
    PutPreVar((UBYTE *)"VERSION_",(UBYTE *)STRINGIFY(MAJORVERSION),0,0);
    PutPreVar((UBYTE *)"SUBVERSION_",(UBYTE *)STRINGIFY(MINORVERSION),0,0);
    PutPreVar((UBYTE *)"DATE_",(UBYTE *)MakeDate(),0,0);
    PutPreVar((UBYTE *)"random_",(UBYTE *)"________",(UBYTE *)"?a",0);
    PutPreVar((UBYTE *)"optimminvar_",(UBYTE *)("0"),0,0);
    PutPreVar((UBYTE *)"optimmaxvar_",(UBYTE *)("0"),0,0);
    PutPreVar(AM.oldnumextrasymbols,(UBYTE *)("0"),0,0);
    PutPreVar((UBYTE *)"optimvalue_",(UBYTE *)("0"),0,0);
    PutPreVar((UBYTE *)"optimscheme_",(UBYTE *)("0"),0,0);
    PutPreVar((UBYTE *)"tolower_",(UBYTE *)("0"),(UBYTE *)("?a"),0);
    PutPreVar((UBYTE *)"toupper_",(UBYTE *)("0"),(UBYTE *)("?a"),0);
    PutPreVar((UBYTE *)"takeleft_",(UBYTE *)("0"),(UBYTE *)("?a"),0);
    PutPreVar((UBYTE *)"takeright_",(UBYTE *)("0"),(UBYTE *)("?a"),0);
    PutPreVar((UBYTE *)"keepleft_",(UBYTE *)("0"),(UBYTE *)("?a"),0);
    PutPreVar((UBYTE *)"keepright_",(UBYTE *)("0"),(UBYTE *)("?a"),0);
    PutPreVar((UBYTE *)"SYSTEMERROR_",(UBYTE *)("0"),0,0);
/*
    Next are a few 'constants' for diagram generation
*/
    PutPreVar((UBYTE *)"ONEPI_",(UBYTE *)("1"),0,0);
    PutPreVar((UBYTE *)"WITHOUTINSERTIONS_",(UBYTE *)("2"),0,0);
    PutPreVar((UBYTE *)"NOTADPOLES_",(UBYTE *)("4"),0,0);
    PutPreVar((UBYTE *)"SYMMETRIZE_",(UBYTE *)("8"),0,0);
    PutPreVar((UBYTE *)"TOPOLOGIESONLY_",(UBYTE *)("16"),0,0);
    PutPreVar((UBYTE *)"NONODES_",(UBYTE *)("32"),0,0);
    PutPreVar((UBYTE *)"WITHEDGES_",(UBYTE *)("64"),0,0);
/*      Note that CHECKEXTERN is 128 */
    PutPreVar((UBYTE *)"WITHBLOCKS_",(UBYTE *)("256"),0,0);
        PutPreVar((UBYTE *)"WITHONEPISETS_",(UBYTE *)("512"),0,0);
    PutPreVar((UBYTE *)"NOSNAILS_",(UBYTE *)("1024"),0,0);
    PutPreVar((UBYTE *)"NOEXTSELF_",(UBYTE *)("2048"),0,0);
```

```
    {
        char buf[41];  /* up to 128-bit */
        LONG pid;
#ifndef WITHMPI
        pid = GetPID();
#else
        pid = ( PF.me == MASTER ) ? GetPID() : (LONG)0;
        pid = PF_BroadcastNumber(pid);
#endif
        LongCopy(pid,buf);
        PutPreVar((UBYTE *)"PID_",(UBYTE *)buf,0,0);
    }
    AM.atstartup = 0;
    AP.MaxPreTypes = 10;
    AP.NumPreTypes = 0;
    AP.PreTypes = (int *)Malloc1(sizeof(int)*(AP.MaxPreTypes+1),"preprocessor types");
    AP.inside.buffer = 0;
    AP.inside.size = 0;

    AC.SortType = AC.lSortType = AM.gSortType = SORTLOWFIRST;
#ifdef WITHPTHREADS
#else
    AR.SortType = AC.SortType;
#endif
    AC.LogHandle = -1;
    AC.SetList.numtemp        = AC.SetList.num;
    AC.SetElementList.numtemp = AC.SetElementList.num;

    GetName(AC.varnames,(UBYTE *)"exp_",&AM.expnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"denom_",&AM.denomnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"fac_",&AM.facnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"invfac_",&AM.invfacnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"sum_",&AM.sumnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"sump_",&AM.sumpnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"term_",&AM.termfunnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"match_",&AM.matchfunnum,NOAUTO);
    GetName(AC.varnames,(UBYTE *)"count_",&AM.countfunnum,NOAUTO);
    AM.termfunnum += FUNCTION;
    AM.matchfunnum += FUNCTION;
    AM.countfunnum += FUNCTION;

    AC.ThreadStats = AM.gThreadStats = AM.ggThreadStats = 1;
    AC.FinalStats = AM.gFinalStats = AM.ggFinalStats = 1;
    AC.StatsFlag = AM.gStatsFlag = AM.ggStatsFlag = 1;
    AC.ThreadsFlag = AM.gThreadsFlag = AM.ggThreadsFlag = 1;
    AC.ThreadBalancing = AM.gThreadBalancing = AM.ggThreadBalancing = 1;
    AC.ThreadSortFileSynch = AM.gThreadSortFileSynch = AM.ggThreadSortFileSynch = 0;
    AC.ProcessStats = AM.gProcessStats = AM.ggProcessStats = 1;
    AC.OldParallelStats = AM.gOldParallelStats = AM.ggOldParallelStats = 0;
    AC.OldFactArgFlag = AM.gOldFactArgFlag = AM.ggOldFactArgFlag = NEWFACTARG;
    AC.OldGCDflag = AM.gOldGCDflag = AM.ggOldGCDflag = 1;
    AC.WTimeStatsFlag = AM.gWTimeStatsFlag = AM.ggWTimeStatsFlag = 0;
    AM.gcNumDollars = AP.DollarList.num;
    AC.SizeCommuteInSet = AM.gSizeCommuteInSet = 0;
#ifdef WITHFLOAT
    AC.MaxWeight = AM.gMaxWeight = AM.ggMaxWeight = MAXWEIGHT;
    AC.DefaultPrecision = AM.gDefaultPrecision = AM.ggDefaultPrecision = DEFAULTPRECISION;
#endif
    AC.CommuteInSet = 0;

    AM.PrintTotalSize = 0;

    AO.NoSpacesInNumbers = AM.gNoSpacesInNumbers = AM.ggNoSpacesInNumbers = 0;
    AO.IndentSpace = AM.gIndentSpace = AM.ggIndentSpace = INDENTSPACE;
    AO.BlockSpaces = 0;
    AO.OptimizationLevel = 0;
    PUTZERO(AS.MaxExprSize);
    PUTZERO(AC.StoreFileSize);

#ifdef WITHPTHREADS
    AC.inputnumbers = 0;
    AC.pfirstnum = 0;
    AC.numpfirstnum = AC.sizepfirstnum = 0;
#endif
    AC.MemDebugFlag = 1;

#ifdef WITHEXTERNALCHANNEL
    AX.currentExternalChannel=0;
    AX.killSignal=SIGKILL;
    AX.killWholeGroup=1;
    AX.daemonize=1;
    AX.currentPrompt=0;
    AX.timeout=1000;/*One second to initialize preset channels*/
    AX.shellname=strDup1((UBYTE *)"/bin/sh -c","external channel shellname");
    AX.stderrname=strDup1((UBYTE *)"/dev/null","external channel stderrname");
#endif
```

```
}

/*
        #] StartVariables :
        #[ StartMore :
*/

void StartMore(void)
{
#ifdef WITHEXTERNALCHANNEL
    /*If env.variable "FORM_PIPES" is defined, we have to initialize
        corresponding pre-set external channels, see file extcmd.c.*/
    /*This line must be after all setup settings: in future, timeout
        could be changed at setup.*/
    if(AX.timeout>=0)/*if AX.timeout<0, this was done by cmdline option -pipe*/
        initPresetExternalChannels((UBYTE*)getenv("FORM_PIPES"),AX.timeout);
#endif

#ifdef WITHMPI
/*
    Define preprocessor variable PARALLELTASK_ as a process number, 0 is the master
    Define preprocessor variable NPARALLELTASKS_ as a total number of processes
*/
    {
        UBYTE buf[32];
        snprintf((char*)buf,32,"%d",PF.me);
        PutPreVar((UBYTE *)"PARALLELTASK_",buf,0,0);
        snprintf((char*)buf,32,"%d",PF.numtasks);
        PutPreVar((UBYTE *)"NPARALLELTASKS_",buf,0,0);
    }
#else
    PutPreVar((UBYTE *)"PARALLELTASK_",(UBYTE *)"0",0,0);
    PutPreVar((UBYTE *)"NPARALLELTASKS_",(UBYTE *)"1",0,0);
#endif

    PutPreVar((UBYTE *)"NAME_",AM.InputFileName ? AM.InputFileName : (UBYTE *)"STDIN",0,0);
}

/*
        #] StartMore :
        #[ IniVars :

        This routine initializes the parameters that may change during the run.
*/

WORD IniVars(void)
{
#ifdef WITHPTHREADS
    GETIDENTITY
#else
    WORD *t;
#endif
    WORD *fi, i, one = 1;
    CBUF *C = cbuf+AC.cbufnum;

#ifdef WITHPTHREADS
    UBYTE buf[32];
    snprintf((char*)buf,32,"%d",AM.totalnumberofthreads);
    PutPreVar((UBYTE *)"NTHREADS_",buf,0,1);
#else
    PutPreVar((UBYTE *)"NTHREADS_",(UBYTE *)"1",0,1);
#endif

    AC.ShortStats = 0;
    AC.WarnFlag = 1;
    AR.SortType = AC.SortType = AC.lSortType = AM.gSortType;
    AC.OutputMode = 72;
    AC.OutputSpaces = NORMALFORMAT;
    AR.Eside = 0;
    AC.DumNum = 0;
    AC.ncmod = AM.gncmod = 0;
    AC.modmode = AM.gmodmode = 0;
    AC.npowmod = AM.gnpowmod = 0;
    AC.halfmod = 0; AC.nhalfmod = 0;
    AC.modinverses = 0;
    AC.lPolyFun = AM.gPolyFun = 0;
    AC.lPolyFunInv = AM.gPolyFunInv = 0;
    AC.lPolyFunType = AM.gPolyFunType = 0;
    AC.lPolyFunExp = AM.gPolyFunExp = 0;
    AC.lPolyFunVar = AM.gPolyFunVar = 0;
    AC.lPolyFunPow = AM.gPolyFunPow = 0;
#ifdef WITHFLINT
    AC.FlintPolyFlag = 1;
#else
    AC.FlintPolyFlag = 0;
#endif
    AC.DirtPow = 0;
```

```
    AC.lDefDim = AM.gDefDim = 4;
    AC.lDefDim4 = AM.gDefDim4 = 0;
    AC.lUnitTrace = AM.gUnitTrace = 4;
    AC.NamesFlag = AM.gNamesFlag = 0;
    AC.CodesFlag = AM.gCodesFlag = 0;
    /* Printing a backtrace on crash is on by default for both normal and debug
        modes if FORM has been compiled with backtrace support. */
#ifdef ENABLE_BACKTRACE
    AC.PrintBacktraceFlag = 1;
#else
    AC.PrintBacktraceFlag = 0;
#endif
    /* Human-readable statistics are off by default */
    AC.HumanStatsFlag = 0;
    AC.extrasymbols = AM.gextrasymbols = AM.ggextrasymbols = 0;
    AC.extrasym = (UBYTE *)Malloc1(2*sizeof(UBYTE),"extrasym");
    AM.gextrasym = (UBYTE *)Malloc1(2*sizeof(UBYTE),"extrasym");
    AM.ggextrasym = (UBYTE *)Malloc1(2*sizeof(UBYTE),"extrasym");
    AC.extrasym[0] = AM.gextrasym[0] = AM.ggextrasym[0] = 'Z';
    AC.extrasym[1] = AM.gextrasym[1] = AM.ggextrasym[1] = 0;
    AC.TokensWriteFlag = AM.gTokensWriteFlag = 0;
    AC.SetupFlag = 0;
    AC.LineLength = AM.gLineLength = 79;
    AC.NwildC = 0;
    AC.OutputMode = 0;
    AM.gOutputMode = 0;
    AC.OutputSpaces = NORMALFORMAT;
    AM.gOutputSpaces = NORMALFORMAT;
    AC.OutNumberType = RATIONALMODE;
    AM.gOutNumberType = RATIONALMODE;
#ifdef WITHZLIB
    AR.gzipCompress = GZIPDEFAULT;
    AR.FoStage4[0].ziobuffer = 0;
    AR.FoStage4[1].ziobuffer = 0;
#ifdef WITHZSTD
    /* Zstd compression is on by default, if we have compiled with it */
    ZWRAP_useZSTDcompression(1);
#endif
#endif
    AR.BracketOn = 0;
    AC.bracketindexflag = 0;
    AT.bracketindexflag = 0;
    AT.bracketinfo = 0;
    AO.IsBracket = 0;
    AM.gfunpowers = AC.funpowers = COMFUNPOWERS;
    AC.parallelflag = AM.gparallelflag;
    AC.properorderflag = AM.gproperorderflag = PROPERORDERFLAG;
    AC.ProcessBucketSize = AC.mProcessBucketSize = AM.gProcessBucketSize;
    AC.ThreadBucketSize = AM.gThreadBucketSize;
    AC.ShortStatsMax = 0;
    AM.gShortStatsMax = 0;
    AM.ggShortStatsMax = 0;

    GlobalSymbols     = NumSymbols;
    GlobalIndices     = NumIndices;
    GlobalVectors     = NumVectors;
    GlobalFunctions   = NumFunctions;
    GlobalSets        = NumSets;
    GlobalSetElements = NumSetElements;
    AC.modpowers = (UWORD *)0;

    i = AM.OffsetIndex;
    fi = AC.FixIndices;
    if ( i > 0 ) do { *fi++ = one; } while ( --i >= 0 );
    AR.sLevel = -1;
    AM.Ordering[0] = 5;
    AM.Ordering[1] = 6;
    AM.Ordering[2] = 7;
    AM.Ordering[3] = 0;
    AM.Ordering[4] = 1;
    AM.Ordering[5] = 2;
    AM.Ordering[6] = 3;
    AM.Ordering[7] = 4;
    for ( i = 8; i < 15; i++ ) AM.Ordering[i] = i;
    AM.gUniTrace[0] =
    AC.lUniTrace[0] = SNUMBER;
    AM.gUniTrace[1] =
    AC.lUniTrace[1] =
    AM.gUniTrace[2] =
    AC.lUniTrace[2] = 4;
    AM.gUniTrace[3] =
    AC.lUniTrace[3] = 1;
#ifdef WITHPTHREADS
    AS.Balancing = 0;
#else
    AT.MinVecArg[0] = 7+ARGHEAD;
    AT.MinVecArg[ARGHEAD] = 7;
```

```
    AT.MinVecArg[1+ARGHEAD] = INDEX;
    AT.MinVecArg[2+ARGHEAD] = 3;
    AT.MinVecArg[3+ARGHEAD] = 0;
    AT.MinVecArg[4+ARGHEAD] = 1;
    AT.MinVecArg[5+ARGHEAD] = 1;
    AT.MinVecArg[6+ARGHEAD] = -3;
    t = AT.FunArg;
    *t++ = 4+ARGHEAD+FUNHEAD;
    for ( i = 1; i < ARGHEAD; i++ ) *t++ = 0;
    *t++ = 4+FUNHEAD;
    *t++ = 0;
    *t++ = FUNHEAD;
    for ( i = 2; i < FUNHEAD; i++ ) *t++ = 0;
    *t++ = 1; *t++ = 1; *t++ = 3;

#ifdef WITHMPI
    AS.printflag = 0;
#endif

    AT.comsym[0] = 8;
    AT.comsym[1] = SYMBOL;
    AT.comsym[2] = 4;
    AT.comsym[3] = 0;
    AT.comsym[4] = 1;
    AT.comsym[5] = 1;
    AT.comsym[6] = 1;
    AT.comsym[7] = 3;
    AT.comnum[0] = 4;
    AT.comnum[1] = 1;
    AT.comnum[2] = 1;
    AT.comnum[3] = 3;
    AT.comfun[0] = FUNHEAD+4;
    AT.comfun[1] = FUNCTION;
    AT.comfun[2] = FUNHEAD;
    AT.comfun[3] = 0;
#if FUNHEAD == 4
    AT.comfun[4] = 0;
#endif
    AT.comfun[FUNHEAD+1] = 1;
    AT.comfun[FUNHEAD+2] = 1;
    AT.comfun[FUNHEAD+3] = 3;
    AT.comind[0] = 7;
    AT.comind[1] = INDEX;
    AT.comind[2] = 3;
    AT.comind[3] = 0;
    AT.comind[4] = 1;
    AT.comind[5] = 1;
    AT.comind[6] = 3;
    AT.locwildvalue[0] = SUBEXPRESSION;
    AT.locwildvalue[1] = SUBEXPSIZE;
    for ( i = 2; i < SUBEXPSIZE; i++ ) AT.locwildvalue[i] = 0;
    AT.mulpat[0] = TYPEMULT;
    AT.mulpat[1] = SUBEXPSIZE+3;
    AT.mulpat[2] = 0;
    AT.mulpat[3] = SUBEXPRESSION;
    AT.mulpat[4] = SUBEXPSIZE;
    AT.mulpat[5] = 0;
    AT.mulpat[6] = 1;
    for ( i = 7; i < SUBEXPSIZE+5; i++ ) AT.mulpat[i] = 0;
    AT.proexp[0] = SUBEXPSIZE+4;
    AT.proexp[1] = EXPRESSION;
    AT.proexp[2] = SUBEXPSIZE;
    AT.proexp[3] = -1;
    AT.proexp[4] = 1;
    for ( i = 5; i < SUBEXPSIZE+1; i++ ) AT.proexp[i] = 0;
    AT.proexp[SUBEXPSIZE+1] = 1;
    AT.proexp[SUBEXPSIZE+2] = 1;
    AT.proexp[SUBEXPSIZE+3] = 3;
    AT.proexp[SUBEXPSIZE+4] = 0;
    AT.dummysubexp[0] = SUBEXPRESSION;
    AT.dummysubexp[1] = SUBEXPSIZE+4;
    for ( i = 2; i < SUBEXPSIZE; i++ ) AT.dummysubexp[i] = 0;
    AT.dummysubexp[SUBEXPSIZE] = WILDDUMMY;
    AT.dummysubexp[SUBEXPSIZE+1] = 4;
    AT.dummysubexp[SUBEXPSIZE+2] = 0;
    AT.dummysubexp[SUBEXPSIZE+3] = 0;

    AT.inprimelist = -1;
    AT.sizeprimelist = 0;
    AT.primelist = 0;
    AT.LeaveNegative = 0;
    AT.TrimPower = 0;
    AN.SplitScratch = 0;
    AN.SplitScratchSize = AN.InScratch = 0;
    AN.SplitScratch1 = 0;
    AN.SplitScratchSize1 = AN.InScratch1 = 0;
    AN.idfunctionflag = 0;
```

```
#endif
    AO.OutputLine = AO.OutFill = BufferForOutput;
    AO.FactorMode = 0;
    C->Pointer = C->Buffer;

    AP.PreOut = 0;
    AP.ComChar = AP.cComChar;
    AC.cbufnum = AM.rbufnum;        /* Select the default compiler buffer */
    AC.HideLevel = 0;
    AP.PreAssignFlag = 0;
    return(0);
}

/*
        #] IniVars :
        #[ Signal handlers :
*/
/*[28apr2004 mt]:*/
#ifdef TRAPSIGNALS

static int exitInProgress = 0;
static int trappedTerminate = 0;

/*INTSIGHANDLER : some systems require a signal handler to return an integer,
  so define the macro INTSIGHANDLER if compiler fails:*/
#ifdef INTSIGHANDLER
static int onErrSig(int i)
#else
static void onErrSig(int i)
#endif
{
    if (exitInProgress){
        signal(i,SIG_DFL);/* Use default behaviour*/
        raise (i);/*reproduce trapped signal*/
#ifdef INTSIGHANDLER
        return(i);
#else
        return;
#endif
    }
    trappedTerminate = 1;
    /*[13jul2005 mt]*//*TerminateImpl(-1) on signal is here:*/
    Terminate(-1);
}

#ifdef INTSIGHANDLER
static void setNewSig(int i, int (*handler)(int))
#else
static void setNewSig(int i, void (*handler)(int))
#endif
{
    if(! (i<NSIG) )/* Invalid signal -- see comments in the file */
        return;
    if ( signal(i,SIG_IGN) !=SIG_IGN)
    /* if compiler fails here, try to define INTSIGHANDLER):*/
        signal(i,handler);
}

void setSignalHandlers(void)
{
    /* Reset various unrecoverable error signals:*/
    setNewSig(SIGSEGV,onErrSig);
    setNewSig(SIGFPE,onErrSig);
    setNewSig(SIGILL,onErrSig);
    setNewSig(SIGEMT,onErrSig);
    setNewSig(SIGSYS,onErrSig);
    setNewSig(SIGPIPE,onErrSig);
    setNewSig(SIGLOST,onErrSig);
    setNewSig(SIGXCPU,onErrSig);
    setNewSig(SIGXFSZ,onErrSig);

    /* Reset interrupt signals:*/
    setNewSig(SIGTERM,onErrSig);
    setNewSig(SIGINT,onErrSig);
    setNewSig(SIGQUIT,onErrSig);
    setNewSig(SIGHUP,onErrSig);
    setNewSig(SIGALRM,onErrSig);
    setNewSig(SIGVTALRM,onErrSig);
/*  setNewSig(SIGPROF,onErrSig); */  /* Why did Tentukov forbid profilers?? */
}

#endif
/*:[28apr2004 mt]*/
/*
        #] Signal handlers :
        #[ main :
*/
```

```

#ifdef WITHPTHREADS
ALLPRIVATES *ABdummy[10];
#endif

int main(int argc, char **argv)
{
    int retval;
    bzero((void *)(&A),sizeof(A)); /* make sure A is initialized at zero */
    iniTools();
#ifdef TRAPSIGNALS
    setSignalHandlers();
#endif
#ifdef WINDOWS
    _setmode(_fileno(stdout),O_BINARY);
#endif

#ifdef WITHPTHREADS
    AB = ABdummy;
    StartHandleLock();
    BeginIdentities();
#else
    AM.SumTime = TimeCPU(0);
    TimeWallClock(0);
#endif

#ifdef WITHMPI
    if ( PF_Init(&argc,&argv) ) exit(-1);
#endif

    StartFiles();
    StartVariables();
#ifdef WITHMPI
    /*
     * Here MesPrint() is ready. We turn on AS.printflag to print possible
     * errors occurring on slaves in the initialization. With AS.printflag = -1
     * MesPrint() does not use the synchronized output. This may lead broken
     * texts in the output somewhat, but it is safer to implement in this way
     * for the situation in which some of MesPrint() calls use MLOCK()-MUNLOCK()
     * and some do not. In future if we set AS.printflag = 1 and modify the
     * source code such that all MesPrint() calls are sandwiched by MLOCK()-
     * MUNLOCK(), we need also to modify the code for the master to catch
     * messages corresponding to MUNLOCK() calls at some point.
     *
     * AS.printflag will be set to 0 in IniVars() to prevent slaves from
     * printing redundant errors in the preprocessor and compiler (e.g., syntax
     * errors).
     */
    AS.printflag = -1;
#endif

    if ( ( retval = DoTail(argc,(UBYTE **)argv) ) != 0 ) {
        if ( retval > 0 ) Terminate(0);
        else              Terminate(-1);
    }
    if ( DoSetups() ) Terminate(-2);
#ifdef WITHMPI
    /* It is messy if all errors in OpenInput() on slaves are printed. */
    AS.printflag = 0;
#endif
    if ( OpenInput() ) Terminate(-3);
#ifdef WITHMPI
    AS.printflag = -1;
#endif
    if ( TryEnvironment() ) Terminate(-2);
    if ( TryFileSetups() ) Terminate(-2);
    if ( MakeSetupAllocs() ) Terminate(-2);
    StartMore();
    InitRecovery();
    CheckRecoveryFile();
    if ( AM.totalnumberofthreads == 0 ) AM.totalnumberofthreads = 1;
    AS.MultiThreaded = 0;
#ifdef WITHPTHREADS
    if ( AM.totalnumberofthreads > 1 ) AS.MultiThreaded = 1;
    ReserveTempFiles(1);
    StartAllThreads(AM.totalnumberofthreads);
    IniFbufs();
#else
    ReserveTempFiles(0);
    IniFbuffer(AT.fbufnum);
#endif
    if ( !AM.FromStdin ) PrintHeader(1);
    IniVars();
    Globalize(1);
#ifdef WITH_ALARM
    if ( AM.TimeLimit > 0 ) alarm(AM.TimeLimit);
#endif
```

```
    TimeCPU(0);
    TimeChildren(0);
    TimeWallClock(0);
    PreProcessor();
    Terminate(0);
    return(0);
}
/*
        #] main :
        #[ CleanUp :

        if par < 0 we have to keep the storage file.
        when par > 0 we ran into a .clear statement.
        In that case we keep the zero level input and the log file.

*/

void CleanUp(WORD par)
{
    GETIDENTITY
    int i;

    if ( FG.fname ) {
#ifdef WITHPTHREADS
    if ( B ) {
#endif
    CleanUpSort(0);
    for ( i = 0; i < 3; i++ ) {
        if ( AR.Fscr[i].handle >= 0 ) {
            if ( AR.Fscr[i].name ) {
/*
                If there are more threads referring to the same file
                only the one with the name is the owner of the file.
*/
                CloseFile(AR.Fscr[i].handle);
                remove(AR.Fscr[i].name);
            }
            AR.Fscr[i].handle = - 1;
            AR.Fscr[i].POfill = 0;
        }
    }
#ifdef WITHPTHREADS
    }
#endif
    if ( par > 0 ) {
/*
    Close all input levels above the lowest?
*/
    }
    if ( AC.StoreHandle >= 0 && par <= 0 ) {
#ifdef TRAPSIGNALS
        if ( trappedTerminate ) { /* We don't throw .str if it has contents */
            POSITION pos;
            PUTZERO(pos);
            SeekFile(AC.StoreHandle,&pos,SEEK_END);
            if ( ISNOTZEROPOS(pos) ) {
                CloseFile(AC.StoreHandle);
                goto dontremove;
            }
        }
        CloseFile(AC.StoreHandle);
#ifdef WITHPTHREADS
    if ( B )
#endif
        if ( par >= 0 || AR.StoreData.Handle < 0 || AM.ClearStore ) {
            remove(FG.fname);
        }
dontremove:;
#else
        CloseFile(AC.StoreHandle);
#ifdef WITHPTHREADS
    if ( B )
#endif
        if ( par >= 0 || AR.StoreData.Handle < 0 || AM.ClearStore > 0 ) {
            remove(FG.fname);
        }
#endif
    }
    }
    ClearSpectators(CLEARMODULE);
/*
    Remove recovery file on exit if everything went well
*/
    if ( par == 0 ) {
        DeleteRecoveryFile();
    }
/*
```

```
    Now the final message concerning the total time
*/
    if ( AC.LogHandle >= 0 && par <= 0 ) {
        WORD lh = AC.LogHandle;
        AC.LogHandle = -1;
#ifdef WITHMPI
        if ( PF.me == MASTER )  /* Only the master opened the real file. */
#endif
        CloseFile(lh);
    }
}

/*
        #] CleanUp :
        #[ TerminateImpl :
*/

static int firstterminate = 1;

void TerminateImpl(int errorcode, const char* file, int line, const char* function)
{
    if ( errorcode && firstterminate ) {
        firstterminate = 0;

        MLOCK(ErrorMessageLock);
#ifdef WITHPTHREADS
        MesPrint("Program terminating in thread %w at &");
#elif defined(WITHMPI)
        MesPrint("Program terminating in process %w at &");
#else
        MesPrint("Program terminating at &");
#endif
        MesPrint("Terminate called from %s:%d (%s)", file, line, function);

        if ( AC.PrintBacktraceFlag ) {
#ifdef ENABLE_BACKTRACE
            void *stack[64];
            int stacksize, stop = 0;
            stacksize = backtrace(stack, sizeof(stack)/sizeof(stack[0]));

            /* First check whether eu-addr2line is available */
            if ( !system("command -v eu-addr2line > /dev/null 2>&1") ) {
                MesPrint("Backtrace:");
                for (int i = 0; i < stacksize && !stop; i++) {
                    FILE *fp;
                    char cmd[512];
                    // Leave an initial space
                    cmd[0] = ' ';
                    MesPrint("%#%2d:%", i);
                    snprintf(cmd+1, sizeof(cmd)-1, "eu-addr2line -s --pretty-print -f -i '%p' --pid=%d\n", stack[i], getpid());
                    fp = popen(cmd+1, "r");
                    while ( fgets(cmd+1, sizeof(cmd)-1, fp) != NULL ) {
                        MesPrint("%s", cmd);
                        /* Don't show functions lower than "main" (or thread equivalent) */
                        if ( strstr(cmd, " main ") || strstr(cmd, " RunThread ") || strstr(cmd, " RunSortBot ") ) {
                            stop = 1;
                        }
                    }
                    pclose(fp);
                }
            }
#ifdef LINUX
            else if ( !system("command -v addr2line > /dev/null 2>&1") ) {
                /* Get the executable path. */
                char exe_path[PATH_MAX];
                {
                    ssize_t len = readlink("/proc/self/exe", exe_path, sizeof(exe_path) - 1);
                    if ( len != -1 ) {
                        exe_path[len] = '\0';
                    }
                    else {
                        goto backtrace_fallback;
                    }
                }
                /* Assume PIE binary and get the base address. */
                uintptr_t base_address = 0;
                {
                    char line[256];
                    FILE *maps = fopen("/proc/self/maps", "r");
                    if ( !maps ) {
                        goto backtrace_fallback;
                    }
                    /* See the format used by nommu_region_show() in fs/proc/nommu.c of the Linux source. */
                    if ( fgets(line, sizeof(line), maps) ) {
```

```
                        sscanf(line, "%" SCNxPTR "-", &base_address);
                    }
                    else {
                        fclose(maps);
                        goto backtrace_fallback;
                    }
                    fclose(maps);
                }
                char **strings;
                strings = backtrace_symbols(stack, stacksize);
                MesPrint("Backtrace:");
                for ( int i = 0; i < stacksize && !stop; i++ ) {
                    FILE *fp;
                    char cmd[PATH_MAX + 512];
                    // Leave an initial space
                    cmd[0] = ' ';
                    uintptr_t addr = (uintptr_t)stack[i] - base_address;
                    MesPrint("%#%2d:%", i);
                    snprintf(cmd+1, sizeof(cmd)-1, "addr2line -e \"%s\" -i -p -s -f -C 0x%" PRIxPTR, exe_path, addr);
                    fp = popen(cmd+1, "r");
                    while ( fgets(cmd+1, sizeof(cmd)-1, fp) != NULL ) {
                        MesPrint("%s", cmd);
                        /* Don't show functions lower than "main" */
                        if ( strstr(cmd, " main ") || strstr(cmd, " RunThread ") || strstr(cmd, " RunSortBot ") ) {
                            stop = 1;
                        }
                    }
                    pclose(fp);
                }
                free(strings);
            }
#endif
            else {
                /* eu-addr2line not found */
#ifdef LINUX
backtrace_fallback: ;
#endif
                char **strings;
                strings = backtrace_symbols(stack, stacksize);
                MesPrint("Backtrace:");
                for ( int i = 0; i < stacksize && !stop; i++ ) {
                    char *p = strings[i];
                    while ( *p && *p != '(' ) p++;
                    MesPrint("%#%2d: %s\n", i, p);
                    /* Don't show functions lower than "main" (or thread equivalent) */
                    if ( strstr(p, "(main+") || strstr(p, "(RunThread+") || strstr(p, "(RunSortBot+") ) {
                        stop = 1;
                    }
                }
#ifdef LINUX
                MesPrint("Please install addr2line or eu-addr2line for readable stack information.");
#else
                MesPrint("Please install eu-addr2line for readable stack information.");
#endif
                free(strings);
            }
#else
            MesPrint("FORM compiled without backtrace support.");
#endif
        } /* if ( AC.PrintBacktraceFlag) { */

        MUNLOCK(ErrorMessageLock);

        Crash();
    }
#ifdef TRAPSIGNALS
    exitInProgress=1;
#endif
#ifdef WITHEXTERNALCHANNEL
/*
    This function can be called from the error handler, so it is better to
    clean up all started processes before any activity:
*/
    closeAllExternalChannels();
    AX.currentExternalChannel=0;
    /*[08may2006 mt]:*/
    AX.killSignal=SIGKILL;
    AX.killWholeGroup=1;
    AX.daemonize=1;
    /*:[08may2006 mt]*/
    if(AX.currentPrompt){
        M_free(AX.currentPrompt,"external channel prompt");
        AX.currentPrompt=0;
    }
```

```
    /*[08may2006 mt]:*/
    if(AX.shellname){
        M_free(AX.shellname,"external channel shellname");
        AX.shellname=0;
    }
    if(AX.stderrname){
        M_free(AX.stderrname,"external channel stderrname");
        AX.stderrname=0;
    }
    /*:[08may2006 mt]*/
#endif
#ifdef WITHPTHREADS
    if ( !WhoAmI() && !errorcode ) {
        TerminateAllThreads();
    }
#endif
    if ( AC.FinalStats ) {
        if ( AM.PrintTotalSize ) {
            MesPrint("Max. space for expressions: %19p bytes",&(AS.MaxExprSize));
        }
        PrintRunningTime();
    }
#ifdef WITHMPI
    if ( AM.HoldFlag && PF.me == MASTER ) {
        WriteFile(AM.StdOut,(UBYTE *)("Hit any key "),12);
        PF_FlushStdOutBuffer();
        getchar();
    }
#else
    if ( AM.HoldFlag ) {
        WriteFile(AM.StdOut,(UBYTE *)("Hit any key "),12);
        getchar();
    }
#endif
#ifdef WITHMPI
    PF_Terminate(errorcode);
#endif
/*
    We are about to terminate the program. If we are using flint, call the cleanup function.
    This keeps valgrind happy.
*/
#ifdef WITHFLINT
    flint_final_cleanup_master();
#endif
    CleanUp(errorcode);
    M_print();
#ifdef VMS
    P_term(errorcode? 0: 1);
#else
    P_term(errorcode);
#endif
}

/*
        #] TerminateImpl :
        #[ PrintDeprecation :
*/

void PrintDeprecation(const char *feature, const char *issue) {
#ifdef WITHMPI
    if ( PF.me != MASTER ) return;
#endif
    if ( AM.IgnoreDeprecation ) return;

    UBYTE *e = (UBYTE *)getenv("FORM_IGNORE_DEPRECATION");
    if ( e && e[0] && StrCmp(e, (UBYTE *)"0") && StrICmp(e, (UBYTE *)"false") && StrICmp(e, (UBYTE *)"no") ) return;

    MesPrint("DeprecationWarning: We are considering deprecating %s.", feature);
    MesPrint("If you want this support to continue, leave a comment at:");
    MesPrint("");
    MesPrint("    https://github.com/vermaseren/form/%s", issue);
    MesPrint("");
    MesPrint("Otherwise, it will be discontinued in the future.");
    MesPrint("To suppress this warning, use the -ignore-deprecation command line option or");
    MesPrint("set the environment variable FORM_IGNORE_DEPRECATION=1.");
}

/*
        #] PrintDeprecation :
        #[ PrintRunningTime :
*/

void PrintRunningTime(void)
{
#if (defined(WITHPTHREADS) && (defined(WITHPOSIXCLOCK) || defined(WINDOWS))) || defined(WITHMPI)
    LONG mastertime;
```

```
    LONG workertime;
    LONG wallclocktime;
    LONG totaltime;
#if defined(WITHPTHREADS)
    if ( AB[0] != 0 ) {
        workertime = GetWorkerTimes();
#else
    workertime = PF_GetSlaveTimes();  /* must be called on all processors */
    if ( PF.me == MASTER ) {
#endif
        mastertime = AM.SumTime + TimeCPU(1);
        wallclocktime = TimeWallClock(1);
        totaltime = mastertime+workertime;
        if ( !AM.silent ) {
        MesPrint("  %l.%2i sec + %l.%2i sec: %l.%2i sec out of %l.%2i sec",
            mastertime/1000,(WORD)((mastertime%1000)/10),
            workertime/1000,(WORD)((workertime%1000)/10),
            totaltime/1000,(WORD)((totaltime%1000)/10),
            wallclocktime/100,(WORD)(wallclocktime%100));
        }
    }
#else
    LONG mastertime = AM.SumTime + TimeCPU(1);
    LONG wallclocktime = TimeWallClock(1);
    if ( !AM.silent ) {
    MesPrint("  %l.%2i sec out of %l.%2i sec",
        mastertime/1000,(WORD)((mastertime%1000)/10),
        wallclocktime/100,(WORD)(wallclocktime%100));
    }
#endif
}

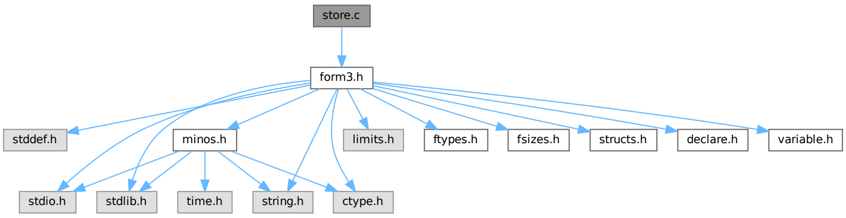
/*
        #] PrintRunningTime :
        #[ GetRunningTime :
*/

LONG GetRunningTime(void)
{
#if defined(WITHPTHREADS) && (defined(WITHPOSIXCLOCK) || defined(WINDOWS))
    LONG mastertime;
    if ( AB[0] != 0 ) {
/*
#if ( defined(APPLE64) || defined(APPLE32) )
        mastertime = AM.SumTime + TimeCPU(1);
        return(mastertime);
#else
*/
        LONG workertime = GetWorkerTimes();
        mastertime = AM.SumTime + TimeCPU(1);
        return(mastertime+workertime);
/*
#endif
*/
    }
    else {
        return(AM.SumTime + TimeCPU(1));
    }
#elif defined(WITHMPI)
    LONG mastertime, t = 0;
    LONG workertime = PF_GetSlaveTimes();  /* must be called on all processors */
    if ( PF.me == MASTER ) {
        mastertime = AM.SumTime + TimeCPU(1);
        t = mastertime + workertime;
    }
    return PF_BroadcastNumber(t);  /* must be called on all processors */
#else
    return(AM.SumTime + TimeCPU(1));
#endif
}

/*
        #] GetRunningTime :
*/
```

4.134 store.c File Reference

`#include "form3.h"`
Include dependency graph for store.c:



Macros

- #define SAVEREVISION 0x02

Functions

- WORD OpenTemp (void)
- void SeekScratch (FILEHANDLE ∗fi, POSITION ∗pos)
- void SetEndScratch (FILEHANDLE ∗f, POSITION ∗position)
- void SetEndHScratch (FILEHANDLE ∗f, POSITION ∗position)
- void SetScratch (FILEHANDLE ∗f, POSITION ∗position)
- WORD RevertScratch (void)
- WORD ResetScratch (void)
- int ReadFromScratch (FILEHANDLE ∗fi, POSITION ∗pos, UBYTE ∗buffer, POSITION ∗length)
- int AddToScratch (FILEHANDLE ∗fi, POSITION ∗pos, UBYTE ∗buffer, POSITION ∗length, int withflush)
- int CoSave (UBYTE ∗inp)
- int CoLoad (UBYTE ∗inp)
- WORD DeleteStore (WORD par)
- WORD PutInStore (INDEXENTRY ∗ind, WORD num)
- WORD GetTerm (PHEAD WORD ∗term)
- WORD GetOneTerm (PHEAD WORD ∗term, FILEHANDLE ∗fi, POSITION ∗pos, int par)
- WORD GetMoreTerms (WORD ∗term)
- WORD GetMoreFromMem (WORD ∗term, WORD ∗∗tpoin)
- WORD GetFromStore (WORD ∗to, POSITION ∗position, RENUMBER renumber, WORD ∗InCompState, WORD nexpr)
- void DetVars (WORD ∗term, WORD par)
- WORD ToStorage (EXPRESSIONS e, POSITION ∗length)
- INDEXENTRY ∗ NextFileIndex (POSITION ∗indexpos)
- WORD SetFileIndex (void)
- WORD VarStore (UBYTE ∗s, WORD n, WORD name, WORD namesize)
- WORD TermRenumber (WORD ∗term, RENUMBER renumber, WORD nexpr)
- WORD FindrNumber (WORD n, VARRENUM ∗v)
- INDEXENTRY ∗ FindInIndex (WORD expr, FILEDATA ∗f, WORD par, WORD mode)
- RENUMBER GetTable (WORD expr, POSITION ∗position, WORD mode)
- int CopyExpression (FILEHANDLE ∗from, FILEHANDLE ∗to)
- WORD WriteStoreHeader (WORD handle)
- WORD ReadSaveHeader (void)

- WORD ReadSaveIndex (FILEINDEX *fileind)
- WORD ReadSaveVariables (UBYTE *buffer, UBYTE *top, LONG *size, LONG *outsize, INDEXENTRY *ind, LONG *stage)
- UBYTE * ReadSaveTerm32 (UBYTE *bin, UBYTE *binend, UBYTE **bout, UBYTE *boutend, UBYTE *top, int terminbuf)
- WORD ReadSaveExpression (UBYTE *buffer, UBYTE *top, LONG *size, LONG *outsize)

4.134.1 Detailed Description

Contains all functions that deal with store-files and the system independent save-files.

Definition in file store.c.

4.134.2 Macro Definition Documentation

4.134.2.1 SAVEREVISION

```
#define SAVEREVISION 0x02
```

Definition at line 3961 of file store.c.

4.134.3 Function Documentation

4.134.3.1 OpenTemp()

```
WORD OpenTemp (
            void )
```

Definition at line 48 of file store.c.

4.134.3.2 SeekScratch()

```
void SeekScratch (
            FILEHANDLE * fi,
            POSITION * pos )
```

Definition at line 64 of file store.c.

4.134.3.3 SetEndScratch()

```
void SetEndScratch (
            FILEHANDLE * f,
            POSITION * position )
```

Definition at line 75 of file store.c.

4.134.3.4 SetEndHScratch()

```
void SetEndHScratch (
             FILEHANDLE * f,
             POSITION * position )
```

Definition at line 89 of file store.c.

4.134.3.5 SetScratch()

```
void SetScratch (
             FILEHANDLE * f,
             POSITION * position )
```

Definition at line 115 of file store.c.

4.134.3.6 RevertScratch()

```
WORD RevertScratch (
             void )
```

Definition at line 190 of file store.c.

4.134.3.7 ResetScratch()

```
WORD ResetScratch (
             void )
```

Definition at line 235 of file store.c.

4.134.3.8 ReadFromScratch()

```
int ReadFromScratch (
             FILEHANDLE * fi,
             POSITION * pos,
             UBYTE * buffer,
             POSITION * length )
```

Definition at line 273 of file store.c.

4.134.3.9 AddToScratch()

```
int AddToScratch (
             FILEHANDLE * fi,
             POSITION * pos,
             UBYTE * buffer,
             POSITION * length,
             int withflush )
```

Definition at line 300 of file store.c.

4.134.3.10 CoSave()

```
int CoSave (
            UBYTE * inp )
```

Definition at line 363 of file store.c.

4.134.3.11 CoLoad()

```
int CoLoad (
            UBYTE * inp )
```

Definition at line 573 of file store.c.

4.134.3.12 DeleteStore()

```
WORD DeleteStore (
            WORD par )
```

Definition at line 773 of file store.c.

4.134.3.13 PutInStore()

```
WORD PutInStore (
            INDEXENTRY * ind,
            WORD num )
```

Definition at line 874 of file store.c.

4.134.3.14 GetTerm()

```
WORD GetTerm (
            PHEAD WORD * term )
```

Definition at line 993 of file store.c.

4.134.3.15 GetOneTerm()

```
WORD GetOneTerm (
            PHEAD WORD * term,
            FILEHANDLE * fi,
            POSITION * pos,
            int par )
```

Definition at line 1268 of file store.c.

4.134.3.16 GetMoreTerms()

```
WORD GetMoreTerms (
            WORD * term )
```

Definition at line 1426 of file store.c.

4.134.3.17 GetMoreFromMem()

```
WORD GetMoreFromMem (
            WORD * term,
            WORD ** tpoin )
```

Definition at line 1522 of file store.c.

4.134.3.18 GetFromStore()

```
WORD GetFromStore (
            WORD * to,
            POSITION * position,
            RENUMBER renumber,
            WORD * InCompState,
            WORD nexpr )
```

Definition at line 1629 of file store.c.

4.134.3.19 DetVars()

```
void DetVars (
            WORD * term,
            WORD par )
```

Definition at line 1830 of file store.c.

4.134.3.20 ToStorage()

```
WORD ToStorage (
            EXPRESSIONS e,
            POSITION * length )
```

Definition at line 2017 of file store.c.

4.134.3.21 NextFileIndex()

```
INDEXENTRY * NextFileIndex (
            POSITION * indexpos )
```

Definition at line 2258 of file store.c.

4.134.3.22 SetFileIndex()

```
WORD SetFileIndex (
            void )
```

Reads the next file index and puts it into AR.StoreData.Index. TODO

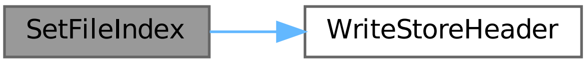
Returns

= 0 everything okay, != 0 an error occurred

Definition at line 2321 of file store.c.

References WriteStoreHeader().

Here is the call graph for this function:



4.134.3.23 VarStore()

```
WORD VarStore (
            UBYTE * s,
            WORD n,
            WORD name,
            WORD namesize )
```

Definition at line 2370 of file store.c.

4.134.3.24 TermRenumber()

```
WORD TermRenumber (
            WORD * term,
            RENUMBER renumber,
            WORD nexpr )
```

!! WORD *memterm=term; static LONG ctrap=0; !!!

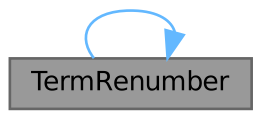
!! ctrap++; !!!

Definition at line 2428 of file store.c.

References ReNuMbEr::func, ReNuMbEr::funnum, ReNuMbEr::indi, ReNuMbEr::indnum, ReNuMbEr::symb, ReNuMbEr::symnum, TermRenumber(), ReNuMbEr::vecnum, and ReNuMbEr::vect.

Referenced by TermRenumber().

Here is the call graph for this function:



4.134.3.25 FindrNumber()

```
WORD FindrNumber (
            WORD n,
            VARRENUM * v )
```

Definition at line 2588 of file store.c.

4.134.3.26 FindInIndex()

```
INDEXENTRY * FindInIndex (
            WORD expr,
            FILEDATA * f,
            WORD par,
            WORD mode )
```

Definition at line 2665 of file store.c.

4.134.3.27 GetTable()

```
RENUMBER GetTable (
            WORD expr,
            POSITION * position,
            WORD mode )
```

Definition at line 2826 of file store.c.

4.134.3.28 CopyExpression()

```
int CopyExpression (
            FILEHANDLE * from,
            FILEHANDLE * to )
```

Definition at line 3308 of file store.c.

4.134.3.29 WriteStoreHeader()

```
WORD WriteStoreHeader (
            WORD handle )
```

Writes header with information about system architecture and FORM revision to an open store file.

Called by SetFileIndex().

Parameters

handle	specifies open file to which header will be written

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 3972 of file store.c.

References STOREHEADER::endianness, STOREHEADER::lenLONG, STOREHEADER::lenPOINTER, STOREHEADER::lenPOS, STOREHEADER::lenWORD, STOREHEADER::maxpower, STOREHEADER::sFun, STOREHEADER::sInd, STOREHEADER::sSym, STOREHEADER::sVec, and STOREHEADER::wildoffset.

Referenced by SetFileIndex().

4.134.3.30 ReadSaveHeader()

```
WORD ReadSaveHeader (
            void )
```

Reads the header in the save file and sets function pointers and flags according to the information found there. Must be called before any other ReadSave... function.

Currently works only for the exchange between 32bit and 64bit machines (WORD size must be 2 or 4 bytes)!

It is called by CoLoad().

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4065 of file store.c.

4.134.3.31 ReadSaveIndex()

```
WORD ReadSaveIndex (
            FILEINDEX * fileind )
```

Reads a FILEINDEX from the open save file specified by AO.SaveData.Handle. Translations for adjusting endianness and data sizes are done if necessary.

Depends on the assumption that sizeof(FILEINDEX) is the same everywhere. If FILEINDEX or INDEXENTRY change, then this functions has to be adjusted.

Called by CoLoad() and FindInIndex().

Parameters

fileind	contains the read FILEINDEX after successful return. must point to allocated, big enough memory.

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4183 of file store.c.

References INFILEINDEX, FiLeInDeX::next, FiLeInDeX::number, and FuNcTiOn::number.

4.134.3.32 ReadSaveVariables()

```
WORD ReadSaveVariables (
            UBYTE * buffer,
            UBYTE * top,
            LONG * size,
            LONG * outsize,
            INDEXENTRY * ind,
            LONG * stage )
```

Reads the variables from the open file specified by AO.SaveData.Handle. It reads the ∗size bytes and writes them to the ∗buffer. It is called by PutInStore().

If translation is necessary, the data might shrink or grow in size, then ∗*size* is adjusted so that the reading and writing fits into the memory from the buffer to the top. The actual number of read bytes is returned in ∗*size*, the number of written bytes is returned in ∗*outsize*.

If the ∗size is smaller than the actual size of the variables, this function will be called several times and needs to remember the current position in the variable structure. The parameter *stage* does this job. When ReadSaveVariables() is called for the first time, this parameter should have the value -1.

The parameter *ind* is used to get the number of variables.

Parameters

buffer	read variables are written into this allocated memory
top	upper end of allocated memory
size	number of bytes to read. might return a smaller number of read bytes if translation was necessary
outsize	if translation has be done, outsize contains the number of written bytes
ind	pointer of INDEXENTRY for the current expression. read-only
stage	should be -1 for the first call, will be increased by ReadSaveVariables to memorize the position in the variable structure

Returns

= 0 everything okay, != 0 an error occurred

Definition at line 4369 of file store.c.

References InDeXeNtRy::nfunctions, InDeXeNtRy::nindices, InDeXeNtRy::nsymbols, InDeXeNtRy::nvectors, and FuNcTiOn::spec.

4.134.3.33 ReadSaveTerm32()

```
UBYTE * ReadSaveTerm32 (
            UBYTE * bin,
            UBYTE * binend,
            UBYTE ** bout,
            UBYTE * boutend,
            UBYTE * top,
            int terminbuf )
```

Reads a single term from the given buffer at *bin* and write the translated term back to this buffer at *bout*.

ReadSaveTerm32() is currently the only instantiation of a ReadSaveTerm-function. It only deals with data that already has the correct endianness and that is resized to 32bit words but without being renumbered or translated in any other way. It uses the compress buffer AR.CompressBuffer.

The function is reentrant in order to cope with nested function arguments. It is called by ReadSaveExpression() and itself.

The *return value* indicates the position in the input buffer up to which the data has already been successfully processed. The parameter *bout* returns the corresponding position in the output buffer.

Parameters

bin	start of the input buffer
binend	end of the input buffer
bout	as input points to the beginning of the output buffer, as output points behind the already translated data in the output buffer
boutend	end of already decompressed data in output buffer
top	end of output buffer
terminbuf	flag whether decompressed data is already in the output buffer. used in recursive calls

Returns

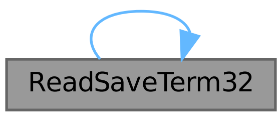
pointer to the next unprocessed data in the input buffer

Definition at line 4736 of file store.c.

References ReadSaveTerm32().

Referenced by ReadSaveExpression(), and ReadSaveTerm32().

Here is the call graph for this function:



4.134.3.34 ReadSaveExpression()

```
WORD ReadSaveExpression (
            UBYTE * buffer,
            UBYTE * top,
            LONG * size,
            LONG * outsize )
```

Reads an expression from the open file specified by AO.SaveData.Handle. The endianness flip and a resizing without renumbering is done in this function. Thereafter the buffer consists of chunks with a uniform maximal word size (32bit at the moment). The actual renumbering is then done by calling the function ReadSaveTerm32(). The result is returned in *buffer*.

If the translation at some point doesn't fit into the buffer anymore, the function returns and must be called again. In any case *size* returns the number of successfully read bytes, *outsize* returns the number of successfully written bytes, and the file will be positioned at the next byte after the successfully read data.

It is called by PutInStore().

Parameters

buffer	output buffer, holds the (translated) expression
top	end of buffer
size	number of read bytes
outsize	number of written bytes

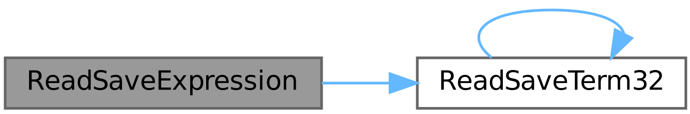
Returns

= 0 everything okay, != 0 an error occurred

Definition at line 5146 of file store.c.

References ReadSaveTerm32().

Here is the call graph for this function:



4.135 store.c

Go to the documentation of this file.

```
00001
00006 /* #[ License : */
```

```
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
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 *
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
#define HIDEDEBUG
    #[ Includes : store.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ StoreExpressions :
        #[ OpenTemp :

        Opens the scratch files for the input -> output operations.

*/

WORD OpenTemp(void)
{
    GETIDENTITY
    if ( AR.outfile->handle >= 0 ) {
        SeekFile(AR.outfile->handle,&(AR.outfile->filesize),SEEK_SET);
        AR.outfile->POposition = AR.outfile->filesize;
        AR.outfile->POfill = AR.outfile->PObuffer;
    }
    return(0);
}

/*
        #] OpenTemp :
        #[ SeekScratch :
*/

void SeekScratch(FILEHANDLE *fi, POSITION *pos)
{
    *pos = fi->POposition;
    ADDPOS(*pos,(TOLONG(fi->POfill)-TOLONG(fi->PObuffer)));
}

/*
        #] SeekScratch :
        #[ SetEndScratch :
*/

void SetEndScratch(FILEHANDLE *f, POSITION *position)
{
    if ( f->handle < 0 ) {
        SETBASEPOSITION(*position,(f->POfull-f->PObuffer)*sizeof(WORD));
    }
    else *position = f->filesize;
    SetScratch(f,position);
}

/*
        #] SetEndScratch :
        #[ SetEndHScratch :
*/

void SetEndHScratch(FILEHANDLE *f, POSITION *position)
{
    if ( f->handle < 0 ) {
        SETBASEPOSITION(*position,(f->POfull-f->PObuffer)*sizeof(WORD));
        f->POfill = f->POfull;
```

```
    }
    else {
#ifdef HIDEDEBUG
        POSITION possize;
        PUTZERO(possize);
        SeekFile(f->handle,&possize,SEEK_END);
        MesPrint("SetEndHScratch: filesize(th) = %12p, filesize(ex) = %12p",&(f->filesize),
                &(possize));
#endif
        *position = f->filesize;
        f->POposition = f->filesize;
        f->POfill = f->POfull = f->PObuffer;
    }
/*  SetScratch(f,position); */
}

/*
        #] SetEndHScratch :
        #[ SetScratch :
*/

void SetScratch(FILEHANDLE *f, POSITION *position)
{
    GETIDENTITY
    POSITION possize;
    LONG size, *whichInInBuf;
    if ( f == AR.hidefile ) whichInInBuf = &(AR.InHiBuf);
    else                    whichInInBuf = &(AR.InInBuf);
#ifdef HIDEDEBUG
    if ( f == AR.hidefile ) MesPrint("In the hide file");
    else MesPrint("In the input file");
    MesPrint("SetScratch to position %15p",position);
    MesPrint("POposition = %15p, full = %l, fill = %l"
        ,&(f->POposition),(f->POfull-f->PObuffer)*sizeof(WORD)
        ,(f->POfill-f->PObuffer)*sizeof(WORD));
#endif
    if ( ISLESSPOS(*position,f->POposition) ||
    ISGEPOSINC(*position,f->POposition,(f->POfull-f->PObuffer)*sizeof(WORD)) ) {
        if ( f->handle < 0 ) {
            if ( ISEQUALPOSINC(*position,f->POposition,
                (f->POfull-f->PObuffer)*sizeof(WORD)) ) goto endpos;
            MesPrint("Illegal position in SetScratch");
            Terminate(-1);
        }
        possize = *position;
        LOCK(AS.inputslock);
        SeekFile(f->handle,&possize,SEEK_SET);
        if ( ISNOTEQUALPOS(possize,*position) ) {
            UNLOCK(AS.inputslock);
            MesPrint("Cannot position file in SetScratch");
            Terminate(-1);
        }
#ifdef HIDEDEBUG
            MesPrint("SetScratch1(%w): position = %12p, size = %l, address =
%x",position,f->POsize,f->PObuffer);
#endif
        if ( ( size = ReadFile(f->handle,(UBYTE *)(f->PObuffer),f->POsize) ) < 0
        || ( size & 1 ) != 0 ) {
            UNLOCK(AS.inputslock);
            MesPrint("Read error in SetScratch");
            Terminate(-1);
        }
        UNLOCK(AS.inputslock);
        if ( size == 0 ) {
            f->PObuffer[0] = 0;
        }
        f->POfill = f->PObuffer;
        f->POposition = *position;
#ifdef WORD2
        *whichInInBuf = size » 1;
#else
        *whichInInBuf = size / TABLESIZE(WORD,UBYTE);
#endif
        f->POfull = f->PObuffer + *whichInInBuf;
#ifdef HIDEDEBUG
            MesPrint("SetScratch2: size = %l, InInBuf = %l, fill = %l, full = %l"
            ,size,*whichInInBuf,(f->POfill-f->PObuffer)*sizeof(WORD)
            ,(f->POfull-f->PObuffer)*sizeof(WORD));
#endif
    }
    else {
endpos:
        DIFPOS(possize,*position,f->POposition);
        f->POfill = (WORD *)(BASEPOSITION(possize)+(UBYTE *)(f->PObuffer));
        *whichInInBuf = f->POfull-f->POfill;
    }
}
```

```
/*
        #] SetScratch :
        #[ RevertScratch :

        Reverts the input/output directions. This way input comes
        always from AR.infile

*/

WORD RevertScratch(void)
{
    GETIDENTITY
    FILEHANDLE *f;
    if ( AR.infile->handle >= 0 && AR.infile->handle != AR.outfile->handle ) {
        CloseFile(AR.infile->handle);
        AR.infile->handle = -1;
        remove(AR.infile->name);
    }
    f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
    AR.infile->POfull = AR.infile->POfill;
    AR.infile->POfill = AR.infile->PObuffer;
    if ( AR.infile->handle >= 0 ) {
        POSITION scrpos;
        PUTZERO(scrpos);
        SeekFile(AR.infile->handle,&scrpos,SEEK_SET);
        if ( ISNOTZEROPOS(scrpos) ) {
            return(MesPrint("Error with scratch output."));
        }
        if ( ( AR.InInBuf = ReadFile(AR.infile->handle,(UBYTE *)(AR.infile->PObuffer)
            ,AR.infile->POsize) ) < 0 || AR.InInBuf & 1 ) {
            return(MesPrint("Error while reading from scratch file"));
        }
        else {
            AR.InInBuf /= TABLESIZE(WORD,UBYTE);
        }
        AR.infile->POfull = AR.infile->PObuffer + AR.InInBuf;
    }
    PUTZERO(AR.infile->POposition);
    AR.outfile->POfill = AR.outfile->POfull = AR.outfile->PObuffer;
    PUTZERO(AR.outfile->POposition);
    PUTZERO(AR.outfile->filesize);
    return(0);
}

/*
        #] RevertScratch :
        #[ ResetScratch :

        Resets the output scratch file to its beginning in such a way
        that the write routines can read it. The output buffers are
        left untouched as they may still be needed for extra declarations.

*/

WORD ResetScratch(void)
{
    GETIDENTITY
    FILEHANDLE *f;
    if ( AR.infile->handle >= 0 ) {
        CloseFile(AR.infile->handle); AR.infile->handle = -1;
        remove(AR.infile->name);
        PUTZERO(AR.infile->POposition);
        AR.infile->POfill = AR.infile->POfull = AR.infile->PObuffer;
    }
    if ( AR.outfile->handle >= 0 ) {
        POSITION scrpos;
        PUTZERO(scrpos);
        SeekFile(AR.outfile->handle,&scrpos,SEEK_SET);
        if ( ISNOTZEROPOS(scrpos) ) {
            return(MesPrint("Error with scratch output."));
        }
        if ( ( AR.InInBuf = ReadFile(AR.outfile->handle,(UBYTE *)(AR.outfile->PObuffer)
        ,AR.outfile->POsize) ) < 0 || AR.InInBuf & 1 ) {
            return(MesPrint("Error while reading from scratch file"));
        }
        else AR.InInBuf /= TABLESIZE(WORD,UBYTE);
        AR.outfile->POfull = AR.outfile->PObuffer + AR.InInBuf;
    }
    else AR.outfile->POfull = AR.outfile->POfill;
    AR.outfile->POfill = AR.outfile->PObuffer;
    PUTZERO(AR.outfile->POposition);
    f = AR.outfile; AR.outfile = AR.infile; AR.infile = f;
    return(0);
}

/*
```

```
        #] ResetScratch :
        #[ ReadFromScratch :

    Routine is used to copy files from scratch to hide.
*/

int ReadFromScratch(FILEHANDLE *fi, POSITION *pos, UBYTE *buffer, POSITION *length)
{
    GETIDENTITY
    LONG l = BASEPOSITION(*length);
    if ( fi->handle < 0 ) {
        memcpy(buffer,fi->POfill,l);
    }
    else {
        SeekFile(fi->handle,pos,SEEK_SET);
        if ( ReadFile(fi->handle,buffer,l) != l ) {
            if ( fi == AR.hidefile )
                MesPrint("Error reading from hide file.");
            else
                MesPrint("Error reading from scratch file.");
            return(-1);
        }
    }
    return(0);
}

/*
        #] ReadFromScratch :
        #[ AddToScratch :

    Routine is used to copy files from scratch to hide.
*/

int AddToScratch(FILEHANDLE *fi, POSITION *pos, UBYTE *buffer, POSITION *length,
              int withflush)
{
    GETIDENTITY
    LONG l = BASEPOSITION(*length), avail;
    DUMMYUSE(pos)
    fi->POfill = fi->POfull;
    while ( fi->POfill+l/sizeof(WORD) > fi->POstop ) {
        avail = (fi->POstop-fi->POfill)*sizeof(WORD);
        if ( avail > 0 ) {
            memcpy(fi->POfill,buffer,avail);
            l -= avail; buffer += avail;
        }
        if ( fi->handle < 0 ) {
            if ( ( fi->handle = (WORD)CreateFile(fi->name) ) < 0 ) {
                if ( fi == AR.hidefile )
                    MesPrint("Cannot create hide file %s",fi->name);
                else
                    MesPrint("Cannot create scratch file %s",fi->name);
                return(-1);
            }
            PUTZERO(fi->POposition);
        }
        SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
        if ( WriteFile(fi->handle,(UBYTE *)fi->PObuffer,fi->POsize) != fi->POsize )
            goto writeerror;
        ADDPOS(fi->POposition,fi->POsize);
        fi->POfill = fi->POfull = fi->PObuffer;
    }
    if ( l > 0 ) {
        memcpy(fi->POfill,buffer,l);
        fi->POfill += l/sizeof(WORD);
        fi->POfull = fi->POfill;
    }
    if ( withflush && fi->handle >= 0 && fi->POfill > fi->PObuffer ) {    /* flush */
        l = (LONG)fi->POfill - (LONG)fi->PObuffer;
        SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
        if ( WriteFile(fi->handle,(UBYTE *)fi->PObuffer,l) != l ) goto writeerror;
        ADDPOS(fi->POposition,fi->POsize);
        fi->POfill = fi->POfull = fi->PObuffer;
    }
    if ( withflush && fi->handle >= 0 )
        SETBASEPOSITION(fi->filesize,TellFile(fi->handle));
    return(0);
writeerror:
    if ( fi == AR.hidefile )
        MesPrint("Error writing to hide file. Disk full?");
    else
        MesPrint("Error writing to scratch file. Disk full?");
    return(-1);
}

/*
        #] AddToScratch :
```

```
        #[ CoSave :

        The syntax of the save statement is:

        save filename
        save filename expr1 expr2

*/

int CoSave(UBYTE *inp)
{
    GETIDENTITY
    UBYTE *p, c;
    WORD n = 0, i;
    WORD error = 0, type, number;
    WORD exprInStorageFlag = 0;
    LONG RetCode = 0, wSize;
    EXPRESSIONS e;
    INDEXENTRY *ind;
    INDEXENTRY *indold;
    WORD TMproto[SUBEXPSIZE];
    POSITION scrpos, scrpos1, filesize;
    int ii, j = sizeof(FILEINDEX)/(sizeof(LONG));
    LONG *lo;
    while ( *inp == ',' ) inp++;
    p = inp;

#ifdef WITHMPI
    if( PF.me != MASTER) return(0);
#endif

    if ( !*p ) return(MesPrint("No filename in save statement"));
    if ( FG.cTable[*p] > 1 && ( *p != '.' ) && ( *p != SEPARATOR ) && ( *p != ALTSEPARATOR ) )
                return(MesPrint("Illegal filename"));
    while ( *++p && *p != ',' ) {}
    c = *p;
    *p = 0;
    if ( !AP.preError ) {
        if ( ( RetCode = CreateFile((char *)inp) ) < 0 ) {
            return(MesPrint("Cannot open file %s",inp));
        }
    }
    AO.SaveData.Handle = (WORD)RetCode;
    PUTZERO(filesize);

    e = Expressions;
    n = NumExpressions;
    if ( c ) {          /* There follows a list of expressions */
        *p++ = c;
        inp = p;
        i = (WORD)(INFILEINDEX);
        if ( WriteStoreHeader(AO.SaveData.Handle) ) return(MesPrint("Error writing storage file header"));
/*      PUTZERO(AO.SaveData.Index.number); */
/*      PUTZERO(AO.SaveData.Index.next); */
        lo = (LONG *)(&AO.SaveData.Index);
        for ( ii = 0; ii < j; ii++ ) *lo++ = 0;
        SETBASEPOSITION(AO.SaveData.Position,(LONG)sizeof(STOREHEADER));
        ind = AO.SaveData.Index.expression;
        if ( !AP.preError && WriteFile(AO.SaveData.Handle,(UBYTE *)(&(AO.SaveData.Index))
        ,(LONG)sizeof(struct FiLeInDeX))!= (LONG)sizeof(struct FiLeInDeX) ) goto SavWrt;
        SeekFile(AO.SaveData.Handle,&(filesize),SEEK_END);
/*      ADDPOS(filesize,sizeof(struct FiLeInDeX)); */

        do {            /* Scan the list */
            if ( !FG.cTable[*p] || *p == '[' ) {
                p = SkipAName(p);
                if ( p == 0 ) return(-1);
            }
            c = *p; *p = 0;
            if ( GetVar(inp,&type,&number,CEXPRESSION,NOAUTO) != NAMENOTFOUND ) {
                if ( e[number].status == STOREDEXPRESSION ) {
                    if ( StrLen(AC.exprnames->namebuffer+e[number].name) > MAXENAME ) {
                        char msg[100];
                        snprintf(msg, sizeof(msg), "saved expr name over %d char: %s", MAXENAME, AC.exprnames->namebuffer+e[number].name);
                        Warning(msg);
                    }
/*
        Here we have to locate the stored expression, copy its index entry
        possibly after making a new fileindex and then copy the whole
        expression.
*/
                    if ( AP.preError ) goto NextExpr;
                    TMproto[0] = EXPRESSION;
                    TMproto[1] = SUBEXPSIZE;
                    TMproto[2] = number;
```

```
                TMproto[3] = 1;
                { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
                AT.TMaddr = TMproto;
                if ( ( indold = FindInIndex(number,&AR.StoreData,0,0) ) != 0 ) {
                    if ( i <= 0 ) {
/*
                        AO.SaveData.Index.next = filesize;
*/
                        SeekFile(AO.SaveData.Handle,&(AO.SaveData.Index.next),SEEK_END);
                        scrpos = AO.SaveData.Position;
                        SeekFile(AO.SaveData.Handle,&scrpos,SEEK_SET);
                        if ( ISNOTEQUALPOS(scrpos,AO.SaveData.Position) ) goto SavWrt;
                        if ( WriteFile(AO.SaveData.Handle,(UBYTE *)(&(AO.SaveData.Index))
                                ,(LONG)sizeof(struct FiLeInDeX)) != (LONG)sizeof(struct FiLeInDeX)
)
                            goto SavWrt;
                        i = (WORD)(INFILEINDEX);
                        AO.SaveData.Position = AO.SaveData.Index.next;
                        lo = (LONG *)(&AO.SaveData.Index);
                        for ( ii = 0; ii < j; ii++ ) *lo++ = 0;
                        ind = AO.SaveData.Index.expression;
                        scrpos = AO.SaveData.Position;
                        SeekFile(AO.SaveData.Handle,&scrpos,SEEK_SET);
                        if ( ISNOTEQUALPOS(scrpos,AO.SaveData.Position) ) goto SavWrt;
                        if ( WriteFile(AO.SaveData.Handle,(UBYTE *)(&(AO.SaveData.Index))
                        ,(LONG)sizeof(struct FiLeInDeX)) != (LONG)sizeof(struct FiLeInDeX) )
                            goto SavWrt;
                        ADDPOS(filesize,sizeof(struct FiLeInDeX));
                    }
                    *ind = *indold;
/*
                    ind->variables = SeekFile(AO.SaveData.Handle,&(AM.zeropos),SEEK_END);
*/
                    ind->variables = filesize;
                    ind->position = ind->variables;
                    ADDPOS(ind->position,DIFBASE(indold->position,indold->variables));
                    SeekFile(AR.StoreData.Handle,&(indold->variables),SEEK_SET);
                    wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
                    scrpos = ind->length;
                    ADDPOS(scrpos,DIFBASE(ind->position,ind->variables));
                    ADD2POS(filesize,scrpos);
                    SETBASEPOSITION(scrpos1,wSize);
                    do {
                        if ( ISLESSPOS(scrpos,scrpos1) ) wSize = BASEPOSITION(scrpos);
                        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,wSize)
                        != wSize ) {
                            MesPrint("ReadError");
                            error = -1;
                            goto EndSave;
                        }
                        if ( WriteFile(AO.SaveData.Handle,(UBYTE *)AT.WorkPointer,wSize)
                        != wSize ) goto SavWrt;
                        ADDPOS(scrpos,-wSize);
                    } while ( ISPOSPOS(scrpos) );
                    ADDPOS(AO.SaveData.Index.number,1);
                    ind++;
                }
                else error = -1;
                i--;
            }
            else {
                MesPrint("%s is not a stored expression",inp);
                error = -1;
            }
NextExpr:;
        }
        else {
            MesPrint("%s is not an expression",inp);
            error = -1;
        }
        *p = c;
        if ( c != ',' && c ) {
            MesComp("Illegal character",inp,p);
            error = -1;
            goto EndSave;
        }
        if ( c ) c = *++p;
        inp = p;
    } while ( c );
    if ( !AP.preError ) {
        scrpos = AO.SaveData.Position;
        SeekFile(AO.SaveData.Handle,&scrpos,SEEK_SET);
        if ( ISNOTEQUALPOS(scrpos,AO.SaveData.Position) ) goto SavWrt;
    }
    if ( !AP.preError &&
    WriteFile(AO.SaveData.Handle,(UBYTE *)(&(AO.SaveData.Index))
        ,(LONG)sizeof(struct FiLeInDeX)) != (LONG)sizeof(struct FiLeInDeX) ) goto SavWrt;
```

```
    }
    else if ( !AP.preError ) {              /* All stored expressions should be saved. Easy */
        /* Make sure there is at least one stored expression: */
        if ( n > 0 ) { do {
            if ( StrLen(AC.exprnames->namebuffer+e->name) > MAXENAME ) {
                char msg[100];
                snprintf(msg, sizeof(msg), "saved expr name over %d char: %s", MAXENAME,
AC.exprnames->namebuffer+e->name);
                Warning(msg);
            }
            if ( e->status == STOREDEXPRESSION ) exprInStorageFlag = 1;
            e++;
        } while ( --n > 0 ); }
        if ( exprInStorageFlag ) {
            wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
            PUTZERO(scrpos);
            SeekFile(AR.StoreData.Handle,&scrpos,SEEK_SET);     /* Start at the beginning */
            scrpos = AR.StoreData.Fill;         /* Number of bytes to be copied */
            SETBASEPOSITION(scrpos1,wSize);
            do {
                if ( ISLESSPOS(scrpos,scrpos1) ) wSize = BASEPOSITION(scrpos);
                if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,wSize) != wSize ) {
                    MesPrint("ReadError");
                    error = -1;
                    goto EndSave;
                }
                if ( WriteFile(AO.SaveData.Handle,(UBYTE *)AT.WorkPointer,wSize) != wSize )
                    goto SavWrt;
                ADDPOS(scrpos,-wSize);
            } while ( ISPOSPOS(scrpos) );
        }
    }
EndSave:
    if ( !AP.preError ) {
        CloseFile(AO.SaveData.Handle);
        AO.SaveData.Handle = -1;
    }
    return(error);
SavWrt:
    MesPrint("WriteError");
    error = -1;
    goto EndSave;
}

/*
        #] CoSave :
        #[ CoLoad :
*/

int CoLoad(UBYTE *inp)
{
    GETIDENTITY
    INDEXENTRY *ind;
    LONG RetCode;
    UBYTE *p, c;
    WORD num, i, error = 0;
    WORD type, number, silentload = 0;
    WORD TMproto[SUBEXPSIZE];
    POSITION scrpos,firstposition;
    while ( *inp == ',' ) inp++;
    p = inp;
    if ( ( *p == ',' && p[1] == '-' ) || *p == '-' ) {
        if ( *p == ',' ) p++;
        p++;
        if ( *p == 's' || *p == 'S' ) {
            silentload = 1;
            while ( *p && ( *p != ',' && *p != '-' && *p != '+'
            && *p != SEPARATOR && *p != ALTSEPARATOR && *p != '.' ) ) p++;
        }
        else if ( *p != ',' ) {
            return(MesPrint("Illegal option in Load statement"));
        }
        while ( *p == ',' ) p++;
    }
    inp = p;
    if ( !*p ) return(MesPrint("No filename in load statement"));
    if ( FG.cTable[*p] > 1 && ( *p != '.' ) && ( *p != SEPARATOR ) && ( *p != ALTSEPARATOR ) )
                return(MesPrint("Illegal filename"));
    while ( *++p && *p != ',' ) {}
    c = *p;
    *p = 0;
    if ( ( RetCode = OpenFile((char *)inp) ) < 0 ) {
        return(MesPrint("Cannot open file %s",inp));
    }

    if ( SetFileIndex() ) {
        MesCall("CoLoad");
```

```
        SETERROR(-1)
    }

    AO.SaveData.Handle = (WORD)(RetCode);

#ifdef SYSDEPENDENTSAVE
    if ( ReadFile(AO.SaveData.Handle,(UBYTE *)(&(AO.SaveData.Index)),
        (LONG)sizeof(struct FiLeInDeX)) != (LONG)sizeof(struct FiLeInDeX) ) goto LoadRead;
#else
    if ( ReadSaveHeader() ) goto LoadRead;
    TELLFILE(AO.SaveData.Handle,&firstposition);
    if ( ReadSaveIndex(&AO.SaveData.Index) ) goto LoadRead;
#endif
    if ( c ) {          /* There follows a list of expressions */
        *p++ = c;
        inp = p;

        do {            /* Scan the list */
            if ( !FG.cTable[*p] || *p == '[' ) {
                p = SkipAName(p);
                if ( p == 0 ) return(-1);
            }
            c = *p; *p = 0;
            if ( GetVar(inp,&type,&number,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
                MesPrint("Conflicting name: %s",inp);
                error = -1;
            }
            else {
                if ( ( num = EntVar(CEXPRESSION,inp,STOREDEXPRESSION,0,0,0) ) >= 0 ) {
                    TMproto[0] = EXPRESSION;
                    TMproto[1] = SUBEXPSIZE;
                    TMproto[2] = num;
                    TMproto[3] = 1;
                    { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
                    AT.TMaddr = TMproto;
                    SeekFile(AO.SaveData.Handle,&firstposition,SEEK_SET);
                    AO.SaveData.Position = firstposition;
                    if ( ReadSaveIndex(&AO.SaveData.Index) ) goto LoadRead;
                    if ( ( ind = FindInIndex(num,&AO.SaveData,1,0) ) != 0 ) {
                        if ( !error ) {
                            if ( PutInStore(ind,num) ) error = -1;
                            else if ( !AM.silent && silentload == 0 )
                                MesPrint(" %s loaded",ind->name);
                        }
/*
!!! Added 1-feb-1998
*/
                        Expressions[num].counter = -1;
                    }
                    else {
                        MesPrint(" %s not found",inp);
                        error = -1;
                    }
                }
                else error = -1;
            }
            *p = c;
            if ( c != ',' && c ) {
                MesComp("Illegal character",inp,p);
                error = -1;
                goto EndLoad;
            }
            if ( c ) c = *++p;
            inp = p;
        } while ( c );
        scrpos = AR.StoreData.Position;
        SeekFile(AR.StoreData.Handle,&scrpos,SEEK_SET);
        if ( ISNOTEQUALPOS(scrpos,AR.StoreData.Position) ) goto LoadWrt;
        if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&(AR.StoreData.Index))
            ,(LONG)sizeof(struct FiLeInDeX)) != (LONG)sizeof(struct FiLeInDeX) ) goto LoadWrt;
    }
    else {              /* All saved expressions should be stored. Easy */
        i = (WORD)BASEPOSITION(AO.SaveData.Index.number);
        ind = AO.SaveData.Index.expression;
#ifdef SYSDEPENDENTSAVE
        if ( i > 0 ) { do {
            if ( GetVar((UBYTE *)(ind->name),&type,&number,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
                MesPrint("Conflicting name: %s",ind->name);
                error = -1;
            }
            else {
                if ( ( num = EntVar(CEXPRESSION,(UBYTE *)(ind->name),STOREDEXPRESSION,0,0,0) ) >= 0 )
{
                    if ( !error ) {
                        if ( PutInStore(ind,num) ) error = -1;
                        else if ( !AM.silent && silentload == 0 )
                            MesPrint(" %s loaded",ind->name);
```

```
                    }
                }
                else error = -1;
            }
            i--;
            if ( i == 0 && ISNOTZEROPOS(AO.SaveData.Index.next) ) {
                SeekFile(AO.SaveData.Handle,&(AO.SaveData.Index.next),SEEK_SET);
                if ( ReadFile(AO.SaveData.Handle,(UBYTE *)(&(AO.SaveData.Index)),
                    (LONG)sizeof(struct FiLeInDeX)) != (LONG)sizeof(struct FiLeInDeX) ) goto LoadRead;
                i = (WORD)BASEPOSITION(AO.SaveData.Index.number);
                ind = AO.SaveData.Index.expression;
            }
            else ind++;
        } while ( i > 0 ); }
#else
        if ( i > 0 ) {
            do {
                if ( GetVar((UBYTE *)(ind->name),&type,&number,ALLVARIABLES,NOAUTO) != NAMENOTFOUND )
{
                    MesPrint("Conflicting name: %s",ind->name);
                    error = -1;
                }
                else {
                    if ( ( num = EntVar(CEXPRESSION,(UBYTE *)(ind->name),STOREDEXPRESSION,0,0,0) ) >=
0 ) {
                        if ( !error ) {
                            if ( PutInStore(ind,num) ) error = -1;
                            else if ( !AM.silent && silentload == 0 )
                                MesPrint(" %s loaded",ind->name);
                        }
                    }
                    else error = -1;
                }
                i--;
                if ( i == 0 && (ISNOTZEROPOS(AO.SaveData.Index.next) || AO.bufferedInd) ) {
                    SeekFile(AO.SaveData.Handle,&(AO.SaveData.Index.next),SEEK_SET);
                    if ( ReadSaveIndex(&AO.SaveData.Index) ) goto LoadRead;
                    i = (WORD)BASEPOSITION(AO.SaveData.Index.number);
                    ind = AO.SaveData.Index.expression;
                }
                else ind++;
            } while ( i > 0 );
        }
#endif
    }
EndLoad:
#ifndef SYSDEPENDENTSAVE
    if ( AO.powerFlag ) {
        MesPrint("WARNING: min-/maxpower had to be adjusted!");
    }
    if ( AO.resizeFlag ) {
        MesPrint("ERROR: could not downsize data!");
        return ( -2 );
    }
#endif
    CloseFile(AO.SaveData.Handle);
    AO.SaveData.Handle = -1;
    SeekFile(AR.StoreData.Handle,&(AC.StoreFileSize),SEEK_END);
    return(error);
LoadWrt:
    MesPrint("WriteError");
    error = -1;
    goto EndLoad;
LoadRead:
    MesPrint("ReadError");
    error = -1;
    goto EndLoad;
}

/*
        #] CoLoad :
        #[ DeleteStore :

        Routine deletes the contents of the entire storage file.
        We close the file and recreate it.
        If par > 0 we have to remove the expressions from the namelists.
*/

WORD DeleteStore(WORD par)
{
    GETIDENTITY
    char *s;
    WORD j, n = 0;
    EXPRESSIONS e_in, e_out;
    WORD DidClean = 0;
    if ( AR.StoreData.Handle >= 0 ) {
        if ( par > 0 ) {
```

```
            n = NumExpressions;
            j = 0;
            e_in = e_out = Expressions;
            if ( n > 0 ) { do {
                if ( e_in->status == STOREDEXPRESSION ) {
                    NAMENODE *node = GetNode(AC.exprnames,
                            AC.exprnames->namebuffer+e_in->name);
                    node->type = CDELETE;
                    DidClean = 1;
                }
                else {
                    if ( e_out != e_in ) {
                        NAMENODE *node;
                        node = GetNode(AC.exprnames,
                            AC.exprnames->namebuffer+e_in->name);
                        node->number = (WORD)(e_out - Expressions);
                        e_out->onfile = e_in->onfile;
                        e_out->prototype = e_in->prototype;
                        e_out->printflag = 0;
                        e_out->status = e_in->status;
                        e_out->name = e_in->name;
                        e_out->inmem = e_in->inmem;
                        e_out->counter = e_in->counter;
                        e_out->numfactors = e_in->numfactors;
                        e_out->numdummies = e_in->numdummies;
                        e_out->compression = e_in->compression;
                        e_out->namesize = e_in->namesize;
                        e_out->whichbuffer = e_in->whichbuffer;
                        e_out->hidelevel = e_in->hidelevel;
                        e_out->node = e_in->node;
                        e_out->replace = e_in->replace;
                        e_out->vflags = e_in->vflags;
                        e_out->uflags = e_in->uflags;
#ifdef PARALLELCODE
                        e_out->partodo = e_in->partodo;
#endif
                    }
                    e_out++;
                    j++;
                }
                e_in++;
            } while ( --n > 0 ); }
            NumExpressions = j;
            if ( DidClean ) CompactifyTree(AC.exprnames,EXPRNAMES);
        }
        AR.StoreData.Handle = -1;
        CloseFile(AC.StoreHandle);
        AC.StoreHandle = -1;
        {
/*
            Knock out the storage caches (25-apr-1990!)
*/
            STORECACHE st;
            st = (STORECACHE)(AT.StoreCache);
            while ( st ) {
                SETBASEPOSITION(st->position,-1);
                SETBASEPOSITION(st->toppos,-1);
                st = st->next;
            }
#ifdef WITHPTHREADS
            for ( j = 1; j < AM.totalnumberofthreads; j++ ) {
                st = (STORECACHE)(AB[j]->T.StoreCache);
                while ( st ) {
                    SETBASEPOSITION(st->position,-1);
                    SETBASEPOSITION(st->toppos,-1);
                    st = st->next;
                }
            }
#endif
        }
        PUTZERO(AC.StoreFileSize);
        s = FG.fname; while ( *s ) s++;
#ifdef VMS
        *s = ';'; s[1] = '*'; s[2] = 0;
        remove(FG.fname);
        *s = 0;
#endif
        return(AC.StoreHandle = CreateFile(FG.fname));
    }
    else return(0);
}

/*
        #] DeleteStore :
        #[ PutInStore :

        Copies the expression indicated by ind from a load file to the
```

```
        internal storage file. A return value of zero indicates that
        everything is OK.

*/

WORD PutInStore(INDEXENTRY *ind, WORD num)
{
    GETIDENTITY
    INDEXENTRY *newind;
    LONG wSize;
#ifndef SYSDEPENDENTSAVE
    LONG wSizeOut;
    LONG stage;
#endif
    POSITION scrpos,scrpos1;
    newind = NextFileIndex(&(Expressions[num].onfile));
    *newind = *ind;
#ifndef SYSDEPENDENTSAVE
    SETBASEPOSITION(newind->length, 0);
#endif
    newind->variables = AR.StoreData.Fill;
    SeekFile(AR.StoreData.Handle,&(newind->variables),SEEK_SET);
    if ( ISNOTEQUALPOS(newind->variables,AR.StoreData.Fill) ) goto PutErrS;
    newind->position = newind->variables;
#ifdef SYSDEPENDENTSAVE
    ADDPOS(newind->position,DIFBASE(ind->position,ind->variables));
#endif
    /* set read position to ind->variables */
    scrpos = ind->variables;
    SeekFile(AO.SaveData.Handle,&scrpos,SEEK_SET);
    if ( ISNOTEQUALPOS(scrpos,ind->variables) ) goto PutErrS;
    /* set max size for read-in */
    wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
#ifdef SYSDEPENDENTSAVE
    scrpos = ind->length;
    ADDPOS(scrpos,DIFBASE(ind->position,ind->variables));
    ADD2POS(AR.StoreData.Fill,scrpos);
#endif
    SETBASEPOSITION(scrpos1,wSize);
#ifndef SYSDEPENDENTSAVE
    /* prepare look-up table for tensor functions */
    if ( ind->nfunctions ) {
        AO.tensorList = (UBYTE *)Malloc1(MAXSAVEFUNCTION,"PutInStore");
    }
    SETBASEPOSITION(scrpos, DIFBASE(ind->position,ind->variables));
    /* copy variables first */
    stage = -1;
    do {
        wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
        if ( ISLESSPOS(scrpos,scrpos1) ) wSize = BASEPOSITION(scrpos);
        wSizeOut = wSize;
        if ( ReadSaveVariables(
            (UBYTE *)AT.WorkPointer, (UBYTE *)AT.WorkTop, &wSize, &wSizeOut, ind, &stage) ) {
            goto PutErrS;
        }
        if ( WriteFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, wSizeOut)
        != wSizeOut ) goto PutErrS;
        ADDPOS(scrpos,-wSize);
        ADDPOS(newind->position, wSizeOut);
        ADDPOS(AR.StoreData.Fill, wSizeOut);
    } while ( ISPOSPOS(scrpos) );
    /* then copy the expression itself */
    wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
    scrpos = ind->length;
#endif
    do {
        wSize = TOLONG(AT.WorkTop) - TOLONG(AT.WorkPointer);
        if ( ISLESSPOS(scrpos,scrpos1) ) wSize = BASEPOSITION(scrpos);
#ifdef SYSDEPENDENTSAVE
        if ( ReadFile(AO.SaveData.Handle,(UBYTE *)AT.WorkPointer,wSize)
        != wSize ) goto PutErrS;
        if ( WriteFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,wSize)
        != wSize ) goto PutErrS;
        ADDPOS(scrpos,-wSize);
#else
        wSizeOut = wSize;

        if ( ReadSaveExpression((UBYTE *)AT.WorkPointer, (UBYTE *)AT.WorkTop, &wSize, &wSizeOut) ) {
            goto PutErrS;
        }

        if ( WriteFile(AR.StoreData.Handle, (UBYTE *)AT.WorkPointer, wSizeOut)
        != wSizeOut ) goto PutErrS;
        ADDPOS(scrpos,-wSize);
        ADDPOS(AR.StoreData.Fill, wSizeOut);
        ADDPOS(newind->length, wSizeOut);
#endif
```

```
    } while ( ISPOSPOS(scrpos) );
    /* free look-up table for tensor functions */
    if ( ind->nfunctions ) {
        M_free(AO.tensorList,"PutInStore");
    }
    scrpos = AR.StoreData.Position;
    SeekFile(AR.StoreData.Handle,&scrpos,SEEK_SET);
    if ( ISNOTEQUALPOS(scrpos,AR.StoreData.Position) ) goto PutErrS;
    if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&AR.StoreData.Index),(LONG)sizeof(FILEINDEX))
    == (LONG)sizeof(FILEINDEX) ) return(0);
PutErrS:
    return(MesPrint("File error"));
}

/*
        #] PutInStore :
        #[ GetTerm :

        Gets one term from input scratch stream.
        Puts it in 'term'.
        Returns the length of the term.

        Used by Processor        (proces.c)
                WriteAll         (sch.c)
                WriteOne         (sch.c)
                GetMoreTerms     (store.c)
                ToStorage        (store.c)
                CoFillExpression (comexpr.c)
                FactorInExpr     (factor.c)
                LoadOpti         (optim.c)
                PF_Processor     (parallel.c)
                ThreadsProcessor (threads.c)
        In multi thread/processor mode all calls are done by the master.
        Note however that other routines, used by the threads, can use
        the same file. Hence we need to be careful about SeekFile and locks.
*/

WORD GetTerm(PHEAD WORD *term)
{
    GETBIDENTITY
    WORD *inp, i, j = 0, len;
    LONG InIn, *whichInInBuf;
    WORD *r, *m, *mstop = 0, minsiz = 0, *bra = 0, *from;
    WORD first, *start = 0, testing = 0;
    FILEHANDLE *fi;
    AN.deferskipped = 0;
    if ( AR.GetFile == 2 ) {
        fi = AR.hidefile;
        whichInInBuf = &(AR.InHiBuf);
    }
    else {
        fi = AR.infile;
        whichInInBuf = &(AR.InInBuf);
    }
    InIn = *whichInInBuf;
    from = term;
    if ( AR.KeptInHold ) {
        r = AR.CompressBuffer;
        i = *r;
        AR.KeptInHold = 0;
        if ( i <= 0 ) { *term = 0; goto RegRet; }
        m = term;
        NCOPY(m,r,i);
        goto RegRet;
    }
    if ( AR.DeferFlag ) {
        m = AR.CompressBuffer;
        if ( *m > 0 ) {
            mstop = m + *m;
            mstop -= ABS(mstop[-1]);
            m++;
            while ( m < mstop ) {
                if ( *m == HAAKJE ) {
                    testing = 1;
                    mstop = m + m[1];
                    bra = (WORD *)(((UBYTE *)(term)) + 2*AM.MaxTer);
                    m = AR.CompressBuffer+1;
                    r = bra;
                    while ( m < mstop ) *r++ = *m++;
                    mstop = r;
                    minsiz = WORDDIF(mstop,bra);
                    goto ReStart;
/*
                    We have the bracket to be tested in bra till mstop
*/
                }
                m += m[1];
```

```
            }
        }
        bra = (WORD *)(((UBYTE *)(term)) + 2*AM.MaxTer);
        mstop = bra+1;
        *bra = 0;
        minsiz = 1;
        testing = 1;
    }
ReStart:
    first = 0;
    r = AR.CompressBuffer;
    if ( fi->handle >= 0 ) {
        if ( InIn <= 0 ) {
            ADDPOS(fi->POposition,(fi->POfull-fi->PObuffer)*sizeof(WORD));
            LOCK(AS.inputslock);
            SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
            InIn = ReadFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize);
            UNLOCK(AS.inputslock);
            if ( ( InIn < 0 ) || ( InIn & 1 ) ) {
                goto GTerr;
            }
#ifdef WORD2
            InIn >>= 1;
#else
            InIn /= TABLESIZE(WORD,UBYTE);
#endif
            *whichInInBuf = InIn;
            if ( !InIn ) { *r = 0; *from = 0; goto RegRet; }
            fi->POfill = fi->PObuffer;
            fi->POfull = fi->PObuffer + InIn;
        }
        inp = fi->POfill;
        if ( ( len = i = *inp ) == 0 ) {
            (*whichInInBuf)--;
            (fi->POfill)++;
            *r = 0;
            *from = 0;
            goto RegRet;
        }
        if ( i < 0 ) {
            InIn--;
            inp++;
            r++;
            start = term;
            *term++ = -i + 1;
            while ( ++i <= 0 ) *term++ = *r++;
            if ( InIn > 0 ) {
                i = *inp++;
                InIn--;
                *start += i;
                *(AR.CompressBuffer) = len = *start;
            }
            else {
                first = 1;
                goto NewIn;
            }
        }
        InIn -= i;
        if ( InIn < 0 ) {
            j = (WORD)(- InIn);
            i -= j;
        }
        else j = 0;
        while ( --i >= 0 ) {
                *r++ = *term++ = *inp++;
        }
        if ( j ) {
NewIn:
            ADDPOS(fi->POposition,(fi->POfull-fi->PObuffer)*sizeof(WORD));
            LOCK(AS.inputslock);
            SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
            InIn = ReadFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize);
            UNLOCK(AS.inputslock);
            if ( ( InIn <= 0 ) || ( InIn & 1 ) ) {
                goto GTerr;
            }
#ifdef WORD2
            InIn >>= 1;
#else
            InIn /= TABLESIZE(WORD,UBYTE);
#endif
            inp = fi->PObuffer;
            fi->POfull = inp + InIn;

            if ( first ) {
                j = *inp++;
                InIn--;
```

```
                *start += j;
                *(AR.CompressBuffer) = len = *start;
            }
            InIn -= j;
            while ( --j >= 0 ) { *r++ = *term++ = *inp++; }
        }
        fi->POfill = inp;
        *whichInInBuf = InIn;
        AR.DefPosition = fi->POposition;
        ADDPOS(AR.DefPosition,((UBYTE *)(fi->POfill)-(UBYTE *)(fi->PObuffer)));
    }
    else {
        inp = fi->POfill;
        if ( inp >= fi->POfull ) { *from = 0; goto RegRet; }
        len = j = *inp;
        if ( j < 0 ) {
            inp++;
            *term++ = *r++ = len = - j + 1 + *inp;
            while ( ++j <= 0 ) *term++ = *r++;
            j = *inp++;
        }
        else if ( !j ) j = 1;
        while ( --j >= 0 ) { *r++ = *term++ = *inp++; }
        fi->POfill = inp;
/*%%%%%ADDED 7-apr-2006 for Keep Brackets in bucket */
        SETBASEPOSITION(AR.DefPosition,((UBYTE *)(fi->POfill)-(UBYTE *)(fi->PObuffer)));
        if ( inp > fi->POfull ) {
            goto GTerr;
        }
    }
    if ( r >= AR.ComprTop ) {
        MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
        Terminate(-1);
    }
    AR.CompressPointer = r; *r = 0;
/*
        The next *from is a bug fix that made the program read in forbidden
        territory.
*/
    if ( testing && *from != 0 ) {
        WORD jj;
        r = from;
        jj = *r - 1 - ABS(*(r+*r-1));
        if ( jj < minsiz ) goto strip;
        r++;
        m = bra;
        while ( m < mstop ) {
            if ( *m != *r ) {
strip:          r = from;
                m = r + *r;
                mstop = m - ABS(m[-1]);
                r++;
                while ( r < mstop ) {
                    if ( *r == HAAKJE ) {
                        *r++ = 1;
                        *r++ = 1;
                        *r++ = 3;
                        len = WORDDIF(r,from);
                        *from = len;
                        goto RegRet;
                    }
                    r += r[1];
                }
                goto RegRet;
            }
            m++;
            r++;
        }
        term = from;
        AN.deferskipped++;
        goto ReStart;
    }
RegRet:;
/*
            #[ debug :
*/
    {
        UBYTE OutBuf[140];
/*      if ( AP.DebugFlag ) { */
        if ( ( AP.PreDebug & DUMPINTERMS ) == DUMPINTERMS ) {
            MLOCK(ErrorMessageLock);
            AO.OutFill = AO.OutputLine = OutBuf;
            AO.OutSkip = 3;
            FiniLine();
            r = from;
            i = *r;
            TokenToLine((UBYTE *)("Input: "));
```

```
            if ( i == 0 ) {
                TokenToLine((UBYTE *)"zero");
            }
            else if ( i < 0 ) {
                TokenToLine((UBYTE *)"negative!!");
            }
            else {
                while ( --i >= 0 ) {
                    TalToLine((UWORD)(*r++)); TokenToLine((UBYTE *)"  ");
                }
            }
            FiniLine();
            MUNLOCK(ErrorMessageLock);
        }
    }
/*
            #] debug :
*/
    return(*from);
GTerr:
    MesPrint("Error while reading scratch file in GetTerm");
    Terminate(-1);
    return(-1);
}

/*
        #] GetTerm :
        #[ GetOneTerm :

        Gets one term from stream AR.infile->handle.
        Puts it in 'term'.
        Returns the length of the term.
        Input is unbuffered.
        Compression via AR.CompressPointer
        par is actually in all calls a file handle

        Routine is called from
            DoOnePow           Get one power of an expression
            Deferred           Get the contents of a bracket
            GetFirstBracket
            FindBracket
        We should do something about the lack of buffering.
        Maybe a buffer of a few times AM.MaxTer (MaxTermSize*sizeof(WORD)).
        Each thread will need its own buffer!

        If par == 0 we use ReadPosFile which can fill the whole buffer.
        If par == 1 we use ReadFile and do actual read operations.

        Note: we cannot use ReadPosFile when running in the master thread.
*/

WORD GetOneTerm(PHEAD WORD *term, FILEHANDLE *fi, POSITION *pos, int par)
{
    GETBIDENTITY
    WORD i, *p;
    LONG j, siz;
    WORD *r, *rr = AR.CompressPointer;
    int error = 0;
    r = rr;
    if ( fi->handle >= 0 ) {
#ifdef READONEBYONE
#ifdef WITHPTHREADS
/*
        This code needs some investigation.
        It may be that we should do this always.
        It may be that even for workers it is no good.
        We may have to make a variable like AM.ReadDirect with
        if ( AM.ReadDirect ) par = 1;
        and a user command like
        On ReadDirect;
*/
        if ( AT.identity > 0 ) par = 1;
#endif
#endif
/*
        To be changed:
        1: check first whether the term lies completely inside the buffer
        2: if not a: use old strategy for AT.identity == 0 (master)
                  b: for workers, position file and read buffer
*/
        if ( par == 0 ) {
            siz = ReadPosFile(BHEAD fi,(UBYTE *)term,1L,pos);
        }
        else {
            LOCK(AS.inputslock);
            SeekFile(fi->handle,pos,SEEK_SET);
            siz = ReadFile(fi->handle,(UBYTE *)term,sizeof(WORD));
```

```
            UNLOCK(AS.inputslock);
            ADDPOS(*pos,siz);
        }
        if ( siz == sizeof(WORD) ) {
            p = term;
            j = i = *term++;
            if ( ( i > AM.MaxTer/((WORD)sizeof(WORD)) ) || ( -i >= AM.MaxTer/((WORD)sizeof(WORD)) ) )
            {
                error = 1;
                goto ErrGet;
            }
            r++;
            if ( i < 0 ) { /* Loading a compressed term */
                *p = -i + 1;
                while ( ++i <= 0 ) *term++ = *r++;
                if ( par == 0 ) {
                    siz = ReadPosFile(BHEAD fi,(UBYTE *)term,1L,pos);
                }
                else {
                    LOCK(AS.inputslock);
                    SeekFile(fi->handle,pos,SEEK_SET);
                    siz = ReadFile(fi->handle,(UBYTE *)term,sizeof(WORD));
                    UNLOCK(AS.inputslock);
                    ADDPOS(*pos,sizeof(WORD));
                }
                if ( siz != sizeof(WORD) ) {
                    error = 2;
                    goto ErrGet;
                }
                *p += *term;
                j = *term;
                if ( ( j > AM.MaxTer/((WORD)sizeof(WORD)) ) || ( j <= 0 ) )
                {
                    error = 3;
                    goto ErrGet;
                }
                *rr = *p; /* Write the proper term size to *AR.CompressPointer */
            }
            else { /* Loading a regular term */
                if ( !j ) return(0);
                *rr = i; /* Write the proper term size to *AR.CompressPointer */
                j--;
            }
            i = (WORD)j;
            if ( par == 0 ) {
                siz = ReadPosFile(BHEAD fi,(UBYTE *)term,j,pos);
                j *= TABLESIZE(WORD,UBYTE);
            }
            else {
                j *= TABLESIZE(WORD,UBYTE);
                LOCK(AS.inputslock);
                SeekFile(fi->handle,pos,SEEK_SET);
                siz = ReadFile(fi->handle,(UBYTE *)term,j);
                UNLOCK(AS.inputslock);
                ADDPOS(*pos,j);
            }
            if ( siz != j ) {
                error = 4;
                goto ErrGet;
            }
            while ( --i >= 0 ) *r++ = *term++;
            if ( r >= AR.ComprTop ) {
                MLOCK(ErrorMessageLock);
                MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            AR.CompressPointer = r; *r = 0;
            return(*p);
        }
        error = 5;
    }
    else {
/*
        Here the whole expression is in the buffer.
*/
        fi->POfill = (WORD *)((UBYTE *)(fi->PObuffer) + BASEPOSITION(*pos));
        p = fi->POfill;
        if ( p >= fi->POfull ) { *term = 0; return(0); }
        j = i = *p;
        if ( i < 0 ) {
            p++;
            j = *r++ = *term++ = -i + 1 + *p;
            while ( ++i <= 0 ) *term++ = *r++;
            i = *p++;
        }
        if ( i == 0 ) { i = 1; *r++ = 0; *term++ = 0; }
```

```
        else { while ( --i >= 0 ) { *r++ = *term++ = *p++; } }
        fi->POfill = p;
        SETBASEPOSITION(*pos,(UBYTE *)(fi->POfill)-(UBYTE *)(fi->PObuffer));
        if ( p <= fi->POfull ) {
            if ( r >= AR.ComprTop ) {
                MLOCK(ErrorMessageLock);
                MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
            AR.CompressPointer = r; *r = 0;
            return((WORD)j);
        }
        error = 6;
    }
ErrGet:
    MLOCK(ErrorMessageLock);
    MesPrint("Error while reading scratch file in GetOneTerm (%d)",error);
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(-1);
}

/*
        #] GetOneTerm :
        #[ GetMoreTerms :
    Routine collects more contents of brackets inside a function,
    indicated by the number in AC.CollectFun.
    The first term is in term already.
    We can keep calling GetTerm either till a bracket is finished
    or till it would make the term too long (> AM.MaxTer/2)
    In all cases this function makes that the routine GetTerm
    has a term in 'hold', so the AR.KeptInHold flag must be turned on.
*/

WORD GetMoreTerms(WORD *term)
{
    GETIDENTITY
    WORD *t, *r, *m, *h, *tstop, i, inc, same;
    WORD extra;
    WORD retval = 0;
/*
    We use 23% as a quasi-random default value.
*/
    extra = ((AM.MaxTer/sizeof(WORD))*((LONG)100-AC.CollectPercentage))/100;
    if ( extra < 23 ) extra = 23;
/*
    First find the bracket pointer
*/
    t = term + *term;
    tstop = t - ABS(t[-1]);
    h = term+1;
    while ( *h != HAAKJE && h < tstop ) h += h[1];
    if ( h >= tstop ) return(retval);
    inc = FUNHEAD+ARGHEAD+1-h[1];
    same = WORDDIF(h,term) + h[1] - 1;
    r = m = t + inc;
    tstop = h + h[1];
    while ( t > tstop ) *--r = *--t;
    r--;
    *r = WORDDIF(m,r);
    while ( GetTerm(BHEAD m) > 0 ) {
        r = m + 1;
        t = m + *m - 1;
        if ( same > ( i = ( *m - ABS(*t) -1 ) ) ) { /* Must fail */
            if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
            break;
        }
        t = term+1;
        i = same;
        while ( --i >= 0 ) {
            if ( *r != *t ) {
                if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
                goto FullTerm;
            }
            r++; t++;
        }
        if ( ( WORDDIF(m,term) + i + extra ) > (WORD)(AM.MaxTer/sizeof(WORD)) ) {
/* 23 = 3 +20. The 20 is to have some extra for substitutions or whatever */
            if ( AS.CollectOverFlag == 0 && AC.AltCollectFun == 0 ) {
                Warning("Bracket contents too long in Collect statement");
                Warning("Contents spread over more than one term");
                Warning("If possible: increase MaxTermSize in setfile");
                AS.CollectOverFlag = 1;
            }
            else if ( AC.AltCollectFun ) {
                AS.CollectOverFlag = 2;
```

```
            }
            break;
        }
        tstop = m + *m;
        *m -= same;
        m++;
        while ( r < tstop ) *m++ = *r++;
        retval++;
        if ( extra == 23 ) extra = ((AM.MaxTer/sizeof(WORD))/6);
    }
FullTerm:
    h[1] = WORDDIF(m,h);
    if ( AS.CollectOverFlag > 1 ) {
        *h = AC.AltCollectFun;
        if ( AS.CollectOverFlag == 3 ) AS.CollectOverFlag = 1;
    }
    else *h = AC.CollectFun;
    h[2] |= DIRTYFLAG;
    h[FUNHEAD] = h[1] - FUNHEAD;
    h[FUNHEAD+1] = 0;
    if ( ToFast(h+FUNHEAD,h+FUNHEAD) ) {
        if ( h[FUNHEAD] <= -FUNCTION ) {
            h[1] = FUNHEAD+1;
            m = h + FUNHEAD+1;
        }
        else {
            h[1] = FUNHEAD+2;
            m = h + FUNHEAD+2;
        }
    }
    *m++ = 1;
    *m++ = 1;
    *m++ = 3;
    *term = WORDDIF(m,term);
    AR.KeptInHold = 1;
    return(retval);
}

/*
        #] GetMoreTerms :
        #[ GetMoreFromMem :

*/

WORD GetMoreFromMem(WORD *term, WORD **tpoin)
{
    GETIDENTITY
    WORD *t, *r, *m, *h, *tstop, i, j, inc, same;
    LONG extra = 23;
/*
    First find the bracket pointer
*/
    t = term + *term;
    tstop = t - ABS(t[-1]);
    h = term+1;
    while ( *h != HAAKJE && h < tstop ) h += h[1];
    if ( h >= tstop ) return(0);
    inc = FUNHEAD+ARGHEAD+1-h[1];
    same = WORDDIF(h,term) + h[1] - 1;
    r = m = t + inc;
    tstop = h + h[1];
    while ( t > tstop ) *--r = *--t;
    r--;
    *r = WORDDIF(m,r);
    while ( **tpoin ) {
        r = *tpoin; j = *r;
        for ( i = 0; i < j; i++ ) m[i] = *r++;
        *tpoin = r;
        r = m + 1;
        t = m + *m - 1;
        if ( same > ( i = ( *m - ABS(*t) -1 ) ) ) { /* Must fail */
            if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
            break;
        }
        t = term+1;
        i = same;
        while ( --i >= 0 ) {
            if ( *r != *t ) {
                if ( AC.AltCollectFun && AS.CollectOverFlag == 2 ) AS.CollectOverFlag = 3;
                goto FullTerm;
            }
            r++; t++;
        }
        if ( ( WORDDIF(m,term) + i + extra ) > (LONG)(AM.MaxTer/(2*sizeof(WORD))) ) {
/* 23 = 3 +20. The 20 is to have some extra for substitutions or whatever */
            if ( AS.CollectOverFlag == 0 && AC.AltCollectFun == 0 ) {
                Warning("Bracket contents too long in Collect statement");
```

```
                Warning("Contents spread over more than one term");
                Warning("If possible: increase MaxTermSize in setfile");
                AS.CollectOverFlag = 1;
            }
            else if ( AC.AltCollectFun ) {
                AS.CollectOverFlag = 2;
            }
            break;
        }
        tstop = m + *m;
        *m -= same;
        m++;
        while ( r < tstop ) *m++ = *r++;
        if ( extra == 23 ) extra = ((AM.MaxTer/sizeof(WORD))/6);
    }
FullTerm:
    h[1] = WORDDIF(m,h);
    if ( AS.CollectOverFlag > 1 ) {
        *h = AC.AltCollectFun;
        if ( AS.CollectOverFlag == 3 ) AS.CollectOverFlag = 1;
    }
    else *h = AC.CollectFun;
    h[2] |= DIRTYFLAG;
    h[FUNHEAD] = h[1] - FUNHEAD;
    h[FUNHEAD+1] = 0;
    if ( ToFast(h+FUNHEAD,h+FUNHEAD) ) {
        if ( h[FUNHEAD] <= -FUNCTION ) {
            h[1] = FUNHEAD+1;
            m = h + FUNHEAD+1;
        }
        else {
            h[1] = FUNHEAD+2;
            m = h + FUNHEAD+2;
        }
    }
    *m++ = 1;
    *m++ = 1;
    *m++ = 3;
    *term = WORDDIF(m,term);
    AR.KeptInHold = 1;
    return(0);
}

/*
        #] GetMoreFromMem :
        #[ GetFromStore :

        Gets a single term from the storage file at position and puts
        it at 'to'.
        The value to be returned is the number of words read.
        Renumbering is done also.
        This is controlled by the renumber table, given in 'renumber'

        This routine should work with a number of cache buffers. The
        exact number should be definable in form.set.
        The parameters are:
        AM.SizeStoreCache    (4096)
        The numbers are the proposed default values.

        The cache is a pure read cache.
*/

static int gfs = 0;

WORD GetFromStore(WORD *to, POSITION *position, RENUMBER renumber, WORD *InCompState, WORD nexpr)
{
    GETIDENTITY
    LONG RetCode, num, first = 0;
    WORD *from, *m;
    struct StOrEcAcHe sc;
    STORECACHE s;
    STORECACHE snext, sold;
    WORD *r, *rr = AR.CompressPointer;
    r = rr;
    gfs++;
    sc.next = AT.StoreCache;
    sold = s = &sc;
    snext = s->next;
    while ( snext ) {
        sold = s;
        s = snext;
        snext = s->next;
        if ( BASEPOSITION(s->position) == -1 ) break;
        if ( ISLESSPOS(*position,s->toppos) &&
        ISGEPOS(*position,s->position) ) {  /* Hit */
            if ( AT.StoreCache != s ) {
                sold->next = s->next;
```

```
                s->next = AT.StoreCache->next;
                AT.StoreCache = s;
            }
            from = (WORD *)(((UBYTE *)(s->buffer)) + DIFBASE(*position,s->position));
            num = *from;
            if ( !num ) { return(*to = 0); }
            *InCompState = (WORD)num;
            m = to;
            if ( num < 0 ) {
                from++;
                ADDPOS(*position,sizeof(WORD));
                *m++ = (WORD)(-num+1);
                r++;
                while ( ++num <= 0 ) *m++ = *r++;
                if ( ISLESSPOS(*position,s->toppos) ) {
                    num = *from++;
                    *to += (WORD)num;
                    ADDPOS(*position,sizeof(WORD));
                    *InCompState = (WORD)(num + 2);
                }
                else {
                    first = 1;
                    goto InNew;
                }
            }
PastCon:;
            while ( num > 0 && ISLESSPOS(*position,s->toppos) ) {
                *r++ = *m++ = *from++; ADDPOS(*position,sizeof(WORD)); num--;
            }
            if ( num > 0 ) {
InNew:
                SETBASEPOSITION(s->position,-1);
                SETBASEPOSITION(s->toppos,-1);
                LOCK(AM.storefilelock);
                SeekFile(AR.StoreData.Handle,position,SEEK_SET);
                RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)(s->buffer),AM.SizeStoreCache);
                UNLOCK(AM.storefilelock);
                if ( RetCode < 0 ) goto PastErr;
                if ( !RetCode ) return( *to = 0 );
                s->position = *position;
                s->toppos = *position;
                ADDPOS(s->toppos,RetCode);
                from = s->buffer;
                if ( first ) {
                    num = *from++;
                    ADDPOS(*position,sizeof(WORD));
                    *to += (WORD)num;
                    /* This next line has always been commented, but uncommenting it
                       fixes a rare bug when loading certain save files. */
                    first = 0;
                    *InCompState = (WORD)(num + 2);
                }
                goto PastCon;
            }
            goto PastEnd;
        }
    }
    if ( AT.StoreCache ) {      /* Fill the last buffer */
        s->position = *position;
        LOCK(AM.storefilelock);
        SeekFile(AR.StoreData.Handle,position,SEEK_SET);
        RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)(s->buffer),AM.SizeStoreCache);
        UNLOCK(AM.storefilelock);
        if ( RetCode < 0 ) goto PastErr;
        if ( !RetCode ) return( *to = 0 );
        s->toppos = *position;
        ADDPOS(s->toppos,RetCode);
        if ( AT.StoreCache != s ) {
            sold->next = s->next;
            s->next = AT.StoreCache->next;
            AT.StoreCache = s;
        }
        m = to;
        from = s->buffer;
        num = *from;
        if ( !num ) { return( *to = 0 ); }
        *InCompState = (WORD)num;
        if ( num < 0 ) {
            *m++ = (WORD)(-num+1);
            r++;
            from++;
            ADDPOS(*position,sizeof(WORD));
            while ( ++num <= 0 ) *m++ = *r++;
            num = *from++;
            *to += (WORD)num;
            ADDPOS(*position,sizeof(WORD));
            *InCompState = (WORD)(num+2);
```

```
        }
        goto PastCon;
    }
/*      No caching available */
    LOCK(AM.storefilelock);
    SeekFile(AR.StoreData.Handle,position,SEEK_SET);
    RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)to,(LONG)sizeof(WORD));
    SeekFile(AR.StoreData.Handle,position,SEEK_CUR);
    UNLOCK(AM.storefilelock);
    if ( RetCode != sizeof(WORD) ) {
        *to = 0;
        return((WORD)RetCode);
    }
    if ( !*to ) return(0);
    m = to;
    if ( *to < 0 ) {
        num = *m++;
        *to = *r++ = (WORD)(-num + 1);
        while ( ++num <= 0 ) *m++ = *r++;
        LOCK(AM.storefilelock);
        SeekFile(AR.StoreData.Handle,position,SEEK_SET);
        RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)m,(LONG)sizeof(WORD));
        SeekFile(AR.StoreData.Handle,position,SEEK_CUR);
        UNLOCK(AM.storefilelock);
        if ( RetCode != sizeof(WORD) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("@Error in compression of store file");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        num = *m;
        *to += (WORD)num;
        *InCompState = (WORD)(num + 2);
    }
    else {
        *InCompState = *to;
        num = *to - 1; m = to + 1; r = rr + 1;
    }
    first = num;
    num *= wsizeof(WORD);
    if ( num < 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("@Error in stored expressions file at position %9p",position);
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    LOCK(AM.storefilelock);
    SeekFile(AR.StoreData.Handle,position,SEEK_SET);
    RetCode = ReadFile(AR.StoreData.Handle,(UBYTE *)m,num);
    SeekFile(AR.StoreData.Handle,position,SEEK_CUR);
    UNLOCK(AM.storefilelock);
    if ( RetCode != num ) {
        MLOCK(ErrorMessageLock);
        MesPrint("@Error in stored expressions file at position %9p",position);
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    NCOPY(r,m,first);
PastEnd:
    *rr = *to;
    if ( r >= AR.ComprTop ) {
        MLOCK(ErrorMessageLock);
        MesPrint("CompressSize of %10l is insufficient",AM.CompressSize);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    AR.CompressPointer = r; *r = 0;
    if ( !TermRenumber(to,renumber,nexpr) ) {
        MarkDirty(to,DIRTYSYMFLAG);
        if ( AR.CurDum > AM.IndDum && Expressions[nexpr].numdummies > 0 )
            MoveDummies(BHEAD to,AR.CurDum - AM.IndDum);
        return((WORD)*to);
    }
PastErr:
    MLOCK(ErrorMessageLock);
    MesCall("GetFromStore");
    MUNLOCK(ErrorMessageLock);
    SETERROR(-1)
}

/*
        #] GetFromStore :
        #[ DetVars :            void DetVars(term)

    Determines which variables are used in term.

    When par = 1 we are scanning a prototype expression which involves
```

```
    completely different rules.

*/

void DetVars(WORD *term, WORD par)
{
    GETIDENTITY
    WORD *stopper;
    WORD *t, sym;
    WORD *sarg;
    stopper = term + *term - 1;
    stopper = stopper - ABS(*stopper) + 1;
    term++;
    if ( par ) {          /* Prototype expression */
        WORD n;
        if ( ( n = NumSymbols ) > 0 ) {
            SYMBOLS tt;
            tt = symbols;
            do {
                (tt++)->flags &= ~INUSE;
            } while ( --n > 0 );
        }
        if ( ( n = NumIndices ) > 0 ) {
            INDICES tt;
            tt = indices;
            do {
                (tt++)->flags &= ~INUSE;
            } while ( --n > 0 );
        }
        if ( ( n = NumVectors ) > 0 ) {
            VECTORS tt;
            tt = vectors;
            do {
                (tt++)->flags &= ~INUSE;
            } while ( --n > 0 );
        }
        if ( ( n = NumFunctions ) > 0 ) {
            FUNCTIONS tt;
            tt = functions;
            do {
                (tt++)->flags &= ~INUSE;
            } while ( --n > 0 );
        }
        term += SUBEXPSIZE;
        while ( term < stopper ) {
            if ( *term == SYMTOSYM || *term == SYMTONUM ) {
                term += 2;
                AN.UsedSymbol[*term] = 1;
                symbols[*term].flags |= INUSE;
            }
            else if ( *term == VECTOVEC ) {
                term += 2;
                AN.UsedVector[*term-AM.OffsetVector] = 1;
                vectors[*term-AM.OffsetVector].flags |= INUSE;
            }
            else if ( *term == INDTOIND ) {
                term += 2;
                sym = indices[*term - AM.OffsetIndex].dimension;
                if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
                AN.UsedIndex[(*term) - AM.OffsetIndex] = 1;
                sym = indices[*term-AM.OffsetIndex].nmin4;
                if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
                indices[*term-AM.OffsetIndex].flags |= INUSE;
            }
            else if ( *term == FUNTOFUN ) {
                term += 2;
                AN.UsedFunction[*term-FUNCTION] = 1;
                functions[*term-FUNCTION].flags |= INUSE;
            }
            term += 2;
        }
    }
    else {
        while ( term < stopper ) {
            t = term + term[1];
            if ( *term == SYMBOL ) {
                term += 2;
                do {
                    AN.UsedSymbol[*term] = 1;
                    term += 2;
                } while ( term < t );
            }
            else if ( *term == DOTPRODUCT ) {
                term += 2;
                do {
                    AN.UsedVector[(*term++) - AM.OffsetVector] = 1;
                    AN.UsedVector[(*term) - AM.OffsetVector] = 1;
```

```
                    term += 2;
                } while ( term < t );
            }
            else if ( *term == VECTOR ) {
                term += 2;
                do {
                    AN.UsedVector[(*term++) - AM.OffsetVector] = 1;
                    if ( *term >= AM.OffsetIndex && *term < AM.DumInd ) {
                        sym = indices[*term - AM.OffsetIndex].dimension;
                        if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
                        AN.UsedIndex[*term - AM.OffsetIndex] = 1;
                        sym = indices[(*term++)-AM.OffsetIndex].nmin4;
                        if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
                    }
                    else term++;
                } while ( term < t );
            }
            else if ( *term == INDEX || *term == LEVICIVITA || *term == GAMMA
            || *term == DELTA ) {
/*
Tensors:
                term += 2;
*/
                if ( *term == INDEX || *term == DELTA ) term += 2;
                else {
Tensors:
                    term += FUNHEAD;
                }
                while ( term < t ) {
                    if ( *term  >= AM.OffsetIndex && *term < AM.DumInd ) {
                        sym = indices[*term - AM.OffsetIndex].dimension;
                        if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
                        AN.UsedIndex[(*term) - AM.OffsetIndex] = 1;
                        sym = indices[*term-AM.OffsetIndex].nmin4;
                        if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
                    }
                    else if ( *term < (WILDOFFSET+AM.OffsetVector) )
                        AN.UsedVector[(*term) - AM.OffsetVector] = 1;
                    term++;
                }
            }
            else if ( *term == HAAKJE ) term = t;
            else {
                if ( *term > MAXBUILTINFUNCTION )
                    AN.UsedFunction[(*term)-FUNCTION] = 1;
                if ( *term >= FUNCTION && functions[*term-FUNCTION].spec
                >= TENSORFUNCTION && term[1] > FUNHEAD ) goto Tensors;
                term += FUNHEAD;            /* First argument */
                while ( term < t ) {
                    sarg = term;
                    NEXTARG(sarg)
                    if ( *term > 0 ) {
                        sarg = term + *term;    /* End of argument */
                        term += ARGHEAD;        /* First term in argument */
                        if ( term < sarg ) { do {
                            DetVars(term,par);
                            term += *term;
                        } while ( term < sarg ); }
                    }
                    else {
                        if ( *term < -MAXBUILTINFUNCTION ) {
                            AN.UsedFunction[-*term-FUNCTION] = 1;
                        }
                        else if ( *term == -SYMBOL ) {
                            AN.UsedSymbol[term[1]] = 1;
                        }
                        else if ( *term == -INDEX ) {
                            if ( term[1] < (WILDOFFSET+AM.OffsetVector) ) {
                                AN.UsedVector[term[1]-AM.OffsetVector] = 1;
                            }
                            else if ( term[1] >= AM.OffsetIndex && term[1] < AM.DumInd ) {
                                sym = indices[term[1] - AM.OffsetIndex].dimension;
                                if ( sym < 0 ) AN.UsedSymbol[-sym] = 1;
                                AN.UsedIndex[term[1] - AM.OffsetIndex] = 1;
                                sym = indices[term[1]-AM.OffsetIndex].nmin4;
                                if ( sym < -NMIN4SHIFT ) AN.UsedSymbol[-sym-NMIN4SHIFT] = 1;
                            }
                        }
                        else if ( *term == -VECTOR || *term == -MINVECTOR ) {
                            AN.UsedVector[term[1]-AM.OffsetVector] = 1;
                        }
                    }
                    term = sarg;            /* Next argument */
                }
                term = t;
            }
        }
```

```
    }
}

/*
        #] DetVars :
        #[ ToStorage :

    This routine takes an expression in the scratch buffer (indicated by e)
    and puts it in the storage file. The necessary actions are:

    1:  determine the list of the used variables.
    2:  make an index entry.
    3:  write the namelists.
    4:  copy the 'length' bytes of the expression.

*/

WORD ToStorage(EXPRESSIONS e, POSITION *length)
{
    GETIDENTITY
    WORD *w, i, j;
    WORD *term;
    INDEXENTRY *indexent;
    LONG size;
    POSITION indexpos, scrpos;
    FILEHANDLE *f;
    if ( ( indexent = NextFileIndex(&indexpos) ) == 0 ) {
        MesCall("ToStorage");
        SETERROR(-1)
    }
    indexent->CompressSize = 0;     /* thus far no compression */
    f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
    if ( e->status == HIDDENGEXPRESSION ) {
        AR.InHiBuf = 0; f = AR.hidefile; AR.GetFile = 2;
    }
    else {
        AR.InInBuf = 0; f = AR.infile;   AR.GetFile = 0;
    }
    if ( f->handle >= 0 ) {
        scrpos = e->onfile;
        SeekFile(f->handle,&scrpos,SEEK_SET);
        if ( ISNOTEQUALPOS(scrpos,e->onfile) ) {
            MesPrint(":::Error in Scratch file");
            goto ErrReturn;
        }
        f->POposition = e->onfile;
        f->POfull = f->PObuffer;
        if ( e->status == HIDDENGEXPRESSION ) AR.InHiBuf = 0;
        else                                  AR.InInBuf = 0;
    }
    else {
        f->POfill = (WORD *)((UBYTE *)(f->PObuffer)+BASEPOSITION(e->onfile));
    }
    w = AT.WorkPointer;
    AN.UsedSymbol   = w;    w += NumSymbols;
    AN.UsedVector   = w;    w += NumVectors;
    AN.UsedIndex    = w;    w += NumIndices;
    AN.UsedFunction = w;    w += NumFunctions;
    term = w;
    w = (WORD *)(((UBYTE *)(w)) + AM.MaxTer);
    if ( w > AT.WorkTop ) {
        MesWork();
        goto ErrReturn;
    }
    w = AN.UsedSymbol;
    i = NumSymbols + NumVectors + NumIndices + NumFunctions;
    do { *w++ = 0; } while ( --i > 0 );
    if ( GetTerm(BHEAD term) > 0 ) {
        DetVars(term,1);
        if ( GetTerm(BHEAD term) ) {
            do { DetVars(term,0); } while ( GetTerm(BHEAD term) > 0 );
        }
    }
    j = 0;
    w = AN.UsedSymbol;
    i = NumSymbols;
    while ( --i >= 0 ) { if ( *w++ ) j++; }
    indexent->nsymbols = j;
/*  size = j * sizeof(struct SyMbOl);  */
    j = 0;
    w = AN.UsedIndex;
    i = NumIndices;
    while ( --i >= 0 ) { if ( *w++ ) j++; }
    indexent->nindices = j;
/*  size += j * sizeof(struct InDeX);  */
    j = 0;
    w = AN.UsedVector;
```

```
    i = NumVectors;
    while ( --i >= 0 ) { if ( *w++ ) j++; }
    indexent->nvectors = j;
/*  size += j * sizeof(struct VeCtOr);  */
    j = 0;
    w = AN.UsedFunction;
    i = NumFunctions;
    while ( --i >= 0 ) { if ( *w++ ) j++; }
    indexent->nfunctions = j;
/*  size += j * sizeof(struct FuNcTiOn); */
    indexent->length = *length;
    indexent->variables = AR.StoreData.Fill;
/*  indexent->position = AR.StoreData.Fill + size; */
    StrCopy(AC.exprnames->namebuffer+e->name,(UBYTE *)(indexent->name));
    SeekFile(AR.StoreData.Handle,&(AR.StoreData.Fill),SEEK_SET);
    AO.wlen = 100000;
    AO.wpos = (UBYTE *)Malloc1(AO.wlen,"AO.wpos buffer");
    AO.wpoin = AO.wpos;
    {
         SYMBOLS a;
        w = AN.UsedSymbol;
        a = symbols;
        j = 0;
        i = indexent->nsymbols;
        while ( --i >= 0 ) {
            while ( !*w ) { w++; a++; j++; }
            a->number = j;
            if ( VarStore((UBYTE *)a,(WORD)(sizeof(struct SyMbOl)),a->name,
                a->namesize) ) goto ErrToSto;
            w++; j++; a++;
        }
    }
    {
         INDICES a;
        w = AN.UsedIndex;
        a = indices;
        j = 0;
        i = indexent->nindices;
        while ( --i >= 0 ) {
            while ( !*w ) { w++; a++; j++; }
            a->number = j;
            if ( VarStore((UBYTE *)a,(WORD)(sizeof(struct InDeX)),a->name,
                a->namesize) ) goto ErrToSto;
            w++; j++; a++;
        }
    }
    {
         VECTORS a;
        w = AN.UsedVector;
        a = vectors;
        j = 0;
        i = indexent->nvectors;
        while ( --i >= 0 ) {
            while ( !*w ) { w++; a++; j++; }
            a->number = j;
            if ( VarStore((UBYTE *)a,(WORD)(sizeof(struct VeCtOr)),a->name,
                a->namesize) ) goto ErrToSto;
            w++; j++; a++;
        }
    }
    {
         FUNCTIONS a;
        w = AN.UsedFunction;
        a = functions;
        j = 0;
        i = indexent->nfunctions;
        while ( --i >= 0 ) {
            while ( !*w ) { w++; a++; j++; }
            a->number = j;
            if ( VarStore((UBYTE *)a,(WORD)(sizeof(struct FuNcTiOn)),a->name,
                a->namesize) ) goto ErrToSto;
            w++; a++; j++;
        }
    }
    if ( VarStore((UBYTE *)0L,(WORD)0,(WORD)0,(WORD)0) ) goto ErrToSto; /* Flush buffer */
    TELLFILE(AR.StoreData.Handle,&(indexent->position));
    indexent->size = (WORD)DIFBASE(indexent->position,indexent->variables);
/*
        The following code was added when it became apparent (30-jan-2007)
        that we need provisions for extra space without upsetting existing
        .sav files. Here we can put as much as we want.
        Look in GetTable on how to recover numdummies.
        Forgetting numdummies has been in there from the beginning.
*/
    if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&(e->numdummies)),(LONG)sizeof(WORD)) !=
        sizeof(WORD) ) {
            MesPrint("Error while writing storage file");
```

```
            goto ErrReturn;
    }
    if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&(e->numfactors)),(LONG)sizeof(WORD)) !=
        sizeof(WORD) ) {
            MesPrint("Error while writing storage file");
            goto ErrReturn;
    }
    if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&(e->vflags)),(LONG)sizeof(WORD)) !=
        sizeof(WORD) ) {
            MesPrint("Error while writing storage file");
            goto ErrReturn;
    }
    if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&(e->uflags)),(LONG)sizeof(WORD)) !=
        sizeof(WORD) ) {
            MesPrint("Error while writing storage file");
            goto ErrReturn;
    }
    TELLFILE(AR.StoreData.Handle,&(indexent->position));
    if ( f->handle >= 0 ) {
        POSITION llength;
        llength = *length;
        SeekFile(f->handle,&(e->onfile),SEEK_SET);
        while ( ISPOSPOS(llength) ) {
            SETBASEPOSITION(scrpos,AO.wlen);
            if ( ISLESSPOS(llength,scrpos) ) size = BASEPOSITION(llength);
            else                             size = AO.wlen;
            if ( ReadFile(f->handle,AO.wpos,size) != size ) {
                MesPrint("Error while reading scratch file");
                goto ErrReturn;
            }
            if ( WriteFile(AR.StoreData.Handle,AO.wpos,size) != size ) {
                MesPrint("Error while writing storage file");
                goto ErrReturn;
            }
            ADDPOS(llength,-size);
        }
    }
    else {
        WORD *ppp;
        ppp = (WORD *)((UBYTE *)(f->PObuffer) + BASEPOSITION(e->onfile));
        if ( WriteFile(AR.StoreData.Handle,(UBYTE *)ppp,BASEPOSITION(*length)) !=
        BASEPOSITION(*length) ) {
            MesPrint("Error while writing storage file");
            goto ErrReturn;
        }
    }
    ADD2POS(*length,indexent->position);
    e->onfile = indexpos;
/*
    AR.StoreData.Fill = SeekFile(AR.StoreData.Handle,&(AM.zeropos),SEEK_END);
*/
    AR.StoreData.Fill = *length;
    SeekFile(AR.StoreData.Handle,&(AR.StoreData.Fill),SEEK_SET);
    scrpos = AR.StoreData.Position;
    ADDPOS(scrpos,sizeof(POSITION));
    SeekFile(AR.StoreData.Handle,&scrpos,SEEK_SET);
    if ( WriteFile(AR.StoreData.Handle,((UBYTE *)&(AR.StoreData.Index.number))
        ,(LONG)(sizeof(POSITION))) != sizeof(POSITION) ) goto ErrInSto;
    SeekFile(AR.StoreData.Handle,&indexpos,SEEK_SET);
    if ( WriteFile(AR.StoreData.Handle,(UBYTE *)indexent,(LONG)(sizeof(INDEXENTRY))) !=
        sizeof(INDEXENTRY) ) goto ErrInSto;
    FlushFile(AR.StoreData.Handle);
    SeekFile(AR.StoreData.Handle,&(AC.StoreFileSize),SEEK_END);
    f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
    if ( AO.wpos ) M_free(AO.wpos,"AO.wpos buffer");
    AO.wpos = AO.wpoin = 0;
    return(0);
ErrToSto:
    MesPrint("---Error while storing namelists");
    goto ErrReturn;
ErrInSto:
    MesPrint("Error in storage");
ErrReturn:
    if ( AO.wpos ) M_free(AO.wpos,"AO.wpos buffer");
    AO.wpos = AO.wpoin = 0;
    f = AR.infile; AR.infile = AR.outfile; AR.outfile = f;
    return(-1);
}

/*
        #] ToStorage :
        #[ NextFileIndex :
*/

INDEXENTRY *NextFileIndex(POSITION *indexpos)
{
    GETIDENTITY
```

```
    INDEXENTRY *ind;
    int i, j = sizeof(FILEINDEX)/(sizeof(LONG));
    LONG *lo;
    if ( AR.StoreData.Handle <= 0 ) {
        if ( SetFileIndex() ) {
            MesCall("NextFileIndex");
            return(0);
        }
        SETBASEPOSITION(AR.StoreData.Index.number,1);
#ifdef SYSDEPENDENTSAVE
        SETBASEPOSITION(*indexpos,(2*sizeof(POSITION)));
#else
        SETBASEPOSITION(*indexpos,(2*sizeof(POSITION)+sizeof(STOREHEADER)));
#endif
        return(AR.StoreData.Index.expression);
    }
    while ( BASEPOSITION(AR.StoreData.Index.number) >= (LONG)(INFILEINDEX) ) {
        if ( ISNOTZEROPOS(AR.StoreData.Index.next) ) {
            SeekFile(AR.StoreData.Handle,&(AR.StoreData.Index.next),SEEK_SET);
            AR.StoreData.Position = AR.StoreData.Index.next;
            if ( ReadFile(AR.StoreData.Handle,(UBYTE *)(&AR.StoreData.Index),(LONG)(sizeof(FILEINDEX))) !=
             (LONG)(sizeof(FILEINDEX)) ) goto ErrNextS;
        }
        else {
            PUTZERO(AR.StoreData.Index.number);
            SeekFile(AR.StoreData.Handle,&(AR.StoreData.Position),SEEK_SET);
            if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&(AR.StoreData.Fill)),(LONG)(sizeof(POSITION)))
            != (LONG)(sizeof(POSITION)) ) goto ErrNextS;
            PUTZERO(AR.StoreData.Index.next);
            SeekFile(AR.StoreData.Handle,&(AR.StoreData.Fill),SEEK_SET);
            AR.StoreData.Position = AR.StoreData.Fill;
            lo = (LONG *)(&AR.StoreData.Index);
            for ( i = 0; i < j; i++ ) *lo++ = 0;
            if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&AR.StoreData.Index),(LONG)(sizeof(FILEINDEX))) !=
             (LONG)(sizeof(FILEINDEX)) ) goto ErrNextS;
            ADDPOS(AR.StoreData.Fill,sizeof(FILEINDEX));
        }
    }
    *indexpos = AR.StoreData.Position;
    ADDPOS(*indexpos,(2*sizeof(POSITION)) +
        BASEPOSITION(AR.StoreData.Index.number) * sizeof(INDEXENTRY));
    ind = &AR.StoreData.Index.expression[BASEPOSITION(AR.StoreData.Index.number)];
    ADDPOS(AR.StoreData.Index.number,1);
    return(ind);
ErrNextS:
    MesPrint("Error in storage file");
    return(0);
}

/*
        #] NextFileIndex :
        #[ SetFileIndex :
*/

WORD SetFileIndex(void)
{
    GETIDENTITY
    int i, j = sizeof(FILEINDEX)/(sizeof(LONG));
    LONG *lo;
    if ( AR.StoreData.Handle < 0 ) {
        AR.StoreData.Handle = AC.StoreHandle;
        PUTZERO(AR.StoreData.Index.next);
        PUTZERO(AR.StoreData.Index.number);
#ifdef SYSDEPENDENTSAVE
        SETBASEPOSITION(AR.StoreData.Fill,sizeof(FILEINDEX));
#else
        if ( WriteStoreHeader(AR.StoreData.Handle) ) return(MesPrint("Error writing storage file header"));
        SETBASEPOSITION(AR.StoreData.Fill, (LONG)sizeof(FILEINDEX)+(LONG)sizeof(STOREHEADER));
#endif
        lo = (LONG *)(&AR.StoreData.Index);
        for ( i = 0; i < j; i++ ) *lo++ = 0;
        if ( WriteFile(AR.StoreData.Handle,(UBYTE *)(&AR.StoreData.Index),(LONG)(sizeof(FILEINDEX))) !=
         (LONG)(sizeof(FILEINDEX)) ) return(MesPrint("Error writing storage file"));
    }
    else {
        POSITION scrpos;
#ifdef SYSDEPENDENTSAVE
        PUTZERO(scrpos);
#else
        SETBASEPOSITION(scrpos, (LONG)(sizeof(STOREHEADER)));
#endif
        SeekFile(AR.StoreData.Handle,&scrpos,SEEK_SET);
```

```
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)(&AR.StoreData.Index),(LONG)(sizeof(FILEINDEX))) !=
        (LONG)(sizeof(FILEINDEX)) ) return(MesPrint("Error reading storage file"));
    }
#ifdef SYSDEPENDENTSAVE
    PUTZERO(AR.StoreData.Position);
#else
    SETBASEPOSITION(AR.StoreData.Position, (LONG)(sizeof(STOREHEADER)));
#endif
    return(0);
}

/*
        #] SetFileIndex :
        #[ VarStore :

        The n -= sizeof(WORD); makes that the real length comes in the
        padding space, provided there is padding space (it seems so).
        The reading of the information assumes this is the case and hence
        things work....
*/

WORD VarStore(UBYTE *s, WORD n, WORD name, WORD namesize)
{
    GETIDENTITY
    UBYTE *t, *u;
    if ( s ) {
        n -= sizeof(WORD);
        t = (UBYTE *)AO.wpoin;
/*
        u = (UBYTE *)AT.WorkTop;
*/
        u = AO.wpos+AO.wlen;
        while ( n > 0 && t < u ) { *t++ = *s++; n--; }
        while ( t >= u ) {
            if ( WriteFile(AR.StoreData.Handle,AO.wpos,AO.wlen) != AO.wlen ) return(-1);
            t = AO.wpos;
            while ( n > 0 && t < u ) { *t++ = *s++; n--; }
        }
        s = AC.varnames->namebuffer + name;
        n = namesize;
        n += sizeof(void *)-1; n &= -(sizeof(void *));
        *((WORD *)t) = n;
        t += sizeof(WORD);
        while ( n > 0 && t < u ) {
            if ( namesize > 0 ) { *t++ = *s++; namesize--; }
            else { *t++ = 0; }
            n--;
        }
        while ( t >= u ) {
            if ( WriteFile(AR.StoreData.Handle,AO.wpos,AO.wlen) != AO.wlen ) return(-1);
            t = AO.wpos;
            while ( n > 0 && t < u ) {
                if ( namesize > 0 ) { *t++ = *s++; namesize--; }
                else { *t++ = 0; }
                n--;
            }
        }
        AO.wpoin = t;
    }
    else {
        LONG size;
        size = AO.wpoin - AO.wpos;
        if ( WriteFile(AR.StoreData.Handle,AO.wpos,size) != size ) return(-1);
        AO.wpoin = AO.wpos;
    }
    return(0);
}

/*
        #] VarStore :
        #[ TermRenumber :

        renumbers the variables inside term according to the information
        in struct renumber.
        The search is binary. This avoided having to read/write the
        expression twice when it was stored.

*/

WORD TermRenumber(WORD *term, RENUMBER renumber, WORD nexpr)
{
    WORD *stopper;
    WORD *t, *sarg, n;
    stopper = term + *term - 1;
    stopper = stopper - ABS(*stopper) + 1;
    term++;
    while ( term < stopper ) {
```

```
        if ( *term == SYMBOL ) {
            t = term + term[1];
            term += 2;
            do {
                if ( ( n = FindrNumber(*term,&(renumber->symb)) ) < 0 ) goto ErrR;
                *term = renumber->symnum[n];
                term += 2;
            } while ( term < t );
        }
        else if ( *term == DOTPRODUCT ) {
            t = term + term[1];
            term += 2;
            do {
                if ( ( n = FindrNumber(*term,&(renumber->vect)) )
                     < 0 ) goto ErrR;
                *term++ = renumber->vecnum[n];
                if ( ( n = FindrNumber(*term,&(renumber->vect)) )
                     < 0 ) goto ErrR;
                *term = renumber->vecnum[n];
                term += 2;
            } while ( term < t );
        }
        else if ( *term == VECTOR ) {
            t = term + term[1];
            term += 2;
            do {
                if ( ( n = FindrNumber(*term,&(renumber->vect)) )
                     < 0 ) goto ErrR;
                *term++ = renumber->vecnum[n];
                if ( ( *term >= AM.OffsetIndex ) && ( *term < AM.IndDum ) ) {
                    if ( ( n = FindrNumber(*term,&(renumber->indi)) )
                         < 0 ) goto ErrR;
                    *term++ = renumber->indnum[n];
                }
                else term++;
            } while ( term < t );
        }
        else if ( *term == INDEX || *term == LEVICIVITA || *term == GAMMA
        || *term == DELTA ) {
Tensors:
            t = term + term[1];
            if ( *term == INDEX || * term == DELTA ) term += 2;
            else term += FUNHEAD;
/*
            term += 2;
*/
            while ( term < t ) {
                if ( *term >= AM.OffsetIndex + WILDOFFSET ) {
/*
        Still TOBEDONE
*/
                }
                else if ( ( *term  >= AM.OffsetIndex ) && ( *term < AM.IndDum ) ) {
                    if ( ( n = FindrNumber(*term,&(renumber->indi)) )
                         < 0 ) goto ErrR;
                    *term = renumber->indnum[n];
                }
                else if ( *term < (WILDOFFSET+AM.OffsetVector) ) {
                    if ( ( n = FindrNumber(*term,&(renumber->vect)) )
                         < 0 ) goto ErrR;
                    *term = renumber->vecnum[n];
                }
                term++;
            }
        }
        else if ( *term == HAAKJE ) term += term[1];
        else {
            if ( *term > MAXBUILTINFUNCTION ) {
                if ( ( n = FindrNumber(*term,&(renumber->func)) )
                     < 0 ) goto ErrR;
                *term = renumber->funnum[n];
            }
            if ( *term >= FUNCTION && functions[*term-FUNCTION].spec
                >= TENSORFUNCTION && term[1] > FUNHEAD ) goto Tensors;
            t = term + term[1];         /* General stopper */
            term += FUNHEAD;            /* First argument */
            while ( term < t ) {
                sarg = term;
                NEXTARG(sarg)
                if ( *term > 0 ) {
/*
                    Problem here:
                    Marking the argument as dirty attacks the heap
                    very heavily and costs much computer time.
*/
                    *++term = 1;
                    term += ARGHEAD-1;
```

```
                    while ( term < sarg ) {
                        if ( TermRenumber(term,renumber,nexpr) ) goto ErrR;
                        term += *term;
                    }
                }
                else {
                    if ( *term <= -MAXBUILTINFUNCTION ) {
                        if ( ( n = FindrNumber(-*term,&(renumber->func)) )
                         < 0 ) goto ErrR;
                        *term = -renumber->funnum[n];
                    }
                    else if ( *term == -SYMBOL ) {
                        term++;
                        if ( ( n = FindrNumber(*term,
                            &(renumber->symb)) ) < 0 ) goto ErrR;
                        *term = renumber->symnum[n];
                    }
                    else if ( *term == -INDEX ) {
                        term++;
                        if ( *term >= AM.OffsetIndex + WILDOFFSET ) {
/*
            Still TOBEDONE
*/
                        }
                        else if ( ( *term  >= AM.OffsetIndex ) && ( *term < AM.IndDum ) ) {
                            if ( ( n = FindrNumber(*term,&(renumber->indi)) )
                                 < 0 ) goto ErrR;
                            *term = renumber->indnum[n];
                        }
                        else if ( *term < (WILDOFFSET+AM.OffsetVector) ) {
                            if ( ( n = FindrNumber(*term,&(renumber->vect)) )
                                 < 0 ) goto ErrR;
                            *term = renumber->vecnum[n];
                        }
                    }
                    else if ( *term == -VECTOR || *term == -MINVECTOR ) {
                        term++;
                        if ( ( n = FindrNumber(*term,&(renumber->vect)) )
                         < 0 ) goto ErrR;
                        *term = renumber->vecnum[n];
                    }
                }
                term = sarg;            /* Next argument */
            }
            term = t;
        }
    }
    return(0);
ErrR:
    MesCall("TermRenumber");
    SETERROR(-1)
}

/*
        #] TermRenumber :
        #[ FindrNumber :
*/

WORD FindrNumber(WORD n, VARRENUM *v)
{
    WORD *hi,*med,*lo;
    hi = v->hi;
    lo = v->lo;
    med = v->start;
    if ( *hi == 0 ) {
        if ( n != *hi ) {
            MesPrint("Serious problems coming up in FindrNumber");
            return(-1);
        }
        return(*hi);
    }
    while ( *med != n ) {
        if ( *med < n ) {
            if ( med == hi ) goto ErrFindr;
            lo = med;
            med = hi - ((WORDDIF(hi,med))/2);
        }
        else {
            if ( med == lo ) goto ErrFindr;
            hi = med;
            med = lo + ((WORDDIF(med,lo))/2);
        }
    }
    return(WORDDIF(med,v->lo));
ErrFindr:
/*
    Reconstruction:
```

```
*/
    {
        int i;
        i = WORDDIF(v->hi,v->lo);
        MesPrint("FindrNumber: n = %d, list has %d members",n,i);
        while ( i >= 0 ) {
            MesPrint("v->lo[%d] = %d",i,v->lo[i]); i--;
        }
        hi = v->hi;
        lo = v->lo;
        med = v->start;
        MesPrint("Start with %d,%d,%d",0,WORDDIF(med,v->lo),WORDDIF(hi,v->lo));
        while ( *med != n ) {
            if ( *med < n ) {
                if ( med == hi ) goto ErrFindr2;
                lo = med;
                med = hi - ((WORDDIF(hi,med))/2);
            }
            else {
                if ( med == lo ) goto ErrFindr2;
                hi = med;
                med = ((WORDDIF(med,lo))/2) + lo;
            }
            MesPrint("New: %d,%d,%d, *med =
%d",WORDDIF(lo,v->lo),WORDDIF(med,v->lo),WORDDIF(hi,v->lo),*med);
        }
    }
    return(WORDDIF(med,v->lo));
ErrFindr2:
    return(MesPrint("Renumbering problems"));
}

/*
        #] FindrNumber :
        #[ FindInIndex :

        Finds an expression in the storage index if it exists.
        If found it returns a pointer to the index entry, otherwise zero.
        par = 0     Search by address (--> f == &AR.StoreData, called by GetTable, CoSave )
        par = 1     Search by name    (--> f == &AO.SaveData,  called by CoLoad )

        When comparing parameter fields the parameters of the expression
        to be searched are in AT.TMaddr. This includes the primary expression
        and a possible FROMBRAC information. The FROMBRAC is always last.

        The parameter mode tells whether we should worry about arguments of
        a stored expression.
*/

INDEXENTRY *FindInIndex(WORD expr, FILEDATA *f, WORD par, WORD mode)
{
    GETIDENTITY
    INDEXENTRY *ind;
    WORD i, hand, *m;
    WORD *start, *stop, *stop2, *m2, nomatch = 0;
    POSITION stindex, indexpos, scrpos;
    LONG number, num;
    stindex = f->Position;
    m = AT.TMaddr;
    stop = m + m[1];
    m += SUBEXPSIZE;
    start = m;
    while ( m < stop ) {
        if ( *m == FROMBRAC || *m == WILDCARDS ) break;
        m += m[1];
    }
    stop = m;
    if ( !par ) hand = AR.StoreData.Handle;
    else        hand = AO.SaveData.Handle;
    for(;;) {
        if ( ( i = (WORD)BASEPOSITION(f->Index.number) ) != 0 ) {
            indexpos = f->Position;
            ADDPOS(indexpos,(2*sizeof(POSITION)));
            ind = f->Index.expression;
            do {
                if ( ( !par && ISEQUALPOS(indexpos,Expressions[expr].onfile) )
                || ( par && !StrCmp(EXPRNAME(expr),(UBYTE *)(ind->name)) ) ) {
                    nomatch = 1;
/*
MesPrint("index: position: %8p",&(ind->position));
MesPrint("index: length: %8p",&(ind->length));
MesPrint("index: variables: %8p",&(ind->variables));
MesPrint("index: nsymbols: %d",ind->nsymbols);
MesPrint("index: nindices: %d",ind->nindices);
MesPrint("index: nvectors: %d",ind->nvectors);
MesPrint("index: nfunctions: %d",ind->nfunctions);
MesPrint("index: size: %d",ind->size);
```

```
*/
                    if ( par ) return(ind);
                    scrpos = ind->position;
                    SeekFile(hand,&scrpos,SEEK_SET);
                    if ( ISNOTEQUALPOS(scrpos,ind->position) ) goto ErrGt2;
                    if ( ReadFile(hand,(UBYTE *)AT.WorkPointer,(LONG)sizeof(WORD)) !=
                    sizeof(WORD) || !*AT.WorkPointer ) goto ErrGt2;
                    num = *AT.WorkPointer - 1;
                    num *= wsizeof(WORD);
                    if ( *AT.WorkPointer < 0 ||
                    ReadFile(hand,(UBYTE *)(AT.WorkPointer+1),num) != num ) goto ErrGt2;
                    m = start;  /* start of parameter field to be searched */
                    m2 = AT.WorkPointer + 1;
                    stop2 = m2 + m2[1];
                    m2 += SUBEXPSIZE;
                    while ( m < stop && m2 < stop2 ) {
                        if ( *m == SYMBOL ) {
                            if ( *m2 != SYMTOSYM ) break;
                            m2[3] = m[2];
                        }
                        else if ( *m == INDEX ) {
                            if ( m[2] >= 0 ) {
                                if ( *m2 != INDTOIND ) break;
                            }
                            else {
                                if ( *m2 != VECTOVEC ) break;
                            }
                            m2[3] = m[2];
                        }
                        else if ( *m >= FUNCTION ) {
                            if ( *m2 != FUNTOFUN ) break;
                            m2[3] = *m;
                        }
                        else {}
                        m += m[1];
                        m2 += m2[1];
                    }
                    if ( ( m >= stop && m2 >= stop2 ) || mode == 0 ) {
                        AT.WorkPointer = stop2;

                        return(ind);
                    }
                }
                ind++;
                ADDPOS(indexpos,sizeof(INDEXENTRY));
            } while ( --i > 0 );
        }
        f->Position = f->Index.next;
#ifndef SYSDEPENDENTSAVE
        if ( !ISNOTZEROPOS(f->Position) ) ADDPOS(f->Position,sizeof(STOREHEADER));
        number = sizeof(struct FiLeInDeX);
#endif
        if ( ISEQUALPOS(f->Position,stindex) && !AO.bufferedInd ) goto ErrGetTab;
        if ( !par ) {
            SeekFile(AR.StoreData.Handle,&(f->Position),SEEK_SET);
            if ( ISNOTEQUALPOS(f->Position,AR.StoreData.Position) ) goto ErrGt2;
#ifndef SYSDEPENDENTSAVE
            if ( ReadFile(f->Handle, (UBYTE *)(&(f->Index)), number) != number ) goto ErrGt2;
#endif
        }
        else {
            SeekFile(AO.SaveData.Handle,&(f->Position),SEEK_SET);
            if ( ISNOTEQUALPOS(f->Position,AO.SaveData.Position) ) goto ErrGt2;
#ifndef SYSDEPENDENTSAVE
            if ( ReadSaveIndex(&f->Index) ) goto ErrGt2;
#endif
        }
#ifdef SYSDEPENDENTSAVE
        number = sizeof(struct FiLeInDeX);
        if ( ReadFile(f->Handle,(UBYTE *)(&(f->Index)),number) !=
        number ) goto ErrGt2;
#endif
    }
ErrGetTab:
    if ( nomatch ) {
        MesPrint("Parameters of expression %s don't match."
        ,EXPRNAME(expr));
    }
    else {
        MesPrint("Cannot find expression %s",EXPRNAME(expr));
    }
    return(0);
ErrGt2:
    MesPrint("Readerror in IndexSearch");
    return(0);
}

```

```
/*
        #] FindInIndex :
        #[ GetTable :

        Locates stored files and constructs the renumbering tables.
        They are allocated in the WorkSpace.
        First the expression data are located. The Index is treated
        as a circularly linked buffer which is paged forwardly.
        If the indexentry is located (in ind) the two renumber tables
        have to be constructed.
        Finally the prototype has to be put in the proper buffer, so
        that wildcards can be passed. There should be a test with
        an already existing prototype that is constructed by the
        pattern matcher. This has not been put in yet.

        There is a problem with the parallel processing.
        Feeding in the variables that were erased by a .store could in
        principle happen in different orders (ParFORM) or simultaneously
        (TFORM). The proper resolution is to have the compiler call GetTable
        when a stored expression is encountered.

        This has been mended in development of TFORM by reading the
        symbol tables during compilation. See the call to GetTable
        in the CodeGenerator.

        Next is the problem of FindInIndex which writes in AR.StoreData
        Copying this is expensive!

        This Doesn't work well for TFORM yet.!!!!!!!!
        e[x1,x2] versus e[x2,x1] messes up.
        For the rest is the reloading during execution not thread safe.

        The parameter mode tells whether we should worry about arguments
        of a stored expression.
*/

RENUMBER GetTable(WORD expr, POSITION *position, WORD mode)
{
    GETIDENTITY
    WORD i, j;
    WORD *w;
    RENUMBER r;
    LONG num, nsize, xx;
    WORD jsym, jind, jvec, jfun;
    WORD k, type, error = 0, *oldw, *neww, *oldwork = AT.WorkPointer;
    struct SyMbOl SyM;
    struct InDeX InD;
    struct VeCtOr VeC;
    struct FuNcTiOn FuN;
    INDEXENTRY *ind;
/*
    Prepare for FindInIndex to put the prototype in the WorkSpace.
    oldw will point at the "wildcards"
*/
/*
        Bug fix. Look also in Generator.
#ifndef WITHPTHREADS

    if ( ( r = Expressions[expr].renum ) != 0 ) { }
    else {
        Expressions[expr].renum =
        r = (RENUMBER)Malloc1(sizeof(struct ReNuMbEr),"Renumber");
    }
#else
        r = (RENUMBER)Malloc1(sizeof(struct ReNuMbEr),"Renumber");
#endif
*/
        r = (RENUMBER)Malloc1(sizeof(struct ReNuMbEr),"Renumber");

    oldw = AT.WorkPointer + 1 + SUBEXPSIZE;
/*
    The prototype is loaded in the WorkSpace by the Index routine.
    After all it has to find an occurrence with the proper arguments.
    This sets the WorkPointer. Hence be careful now.
*/
    LOCK(AM.storefilelock);
    if ( ( ind = FindInIndex(expr,&AR.StoreData,0,mode) ) == 0 ) {
        UNLOCK(AM.storefilelock);
        return(0);
    }

    xx = ind->nsymbols+ind->nindices+ind->nvectors+ind->nfunctions;
    if ( xx == 0 ) {
        Expressions[expr].renumlists =
        w = AN.dummyrenumlist;
    }
    else {
```

```
/*
#ifndef WITHPTHREADS
        Expressions[expr].renumlists =
#endif
*/
        w = (WORD *)Malloc1(sizeof(WORD)*(xx*2),"VarSpace");
    }
    r->symb.lo = w;
    r->symb.start = w + ind->nsymbols/2;
    w += ind->nsymbols;
    r->symb.hi = w - 1;
    r->symnum = w;
    w += ind->nsymbols;

    r->indi.lo = w;
    r->indi.start = w + ind->nindices/2;
    w += ind->nindices;
    r->indi.hi = w - 1;
    r->indnum = w;
    w += ind->nindices;

    r->vect.lo = w;
    r->vect.start = w + ind->nvectors/2;
    w += ind->nvectors;
    r->vect.hi = w - 1;
    r->vecnum = w;
    w += ind->nvectors;

    r->func.lo = w;
    r->func.start = w + ind->nfunctions/2;
    w += ind->nfunctions;
    r->func.hi = w - 1;
    r->funnum = w;
/*  w += ind->nfunctions; */

    SeekFile(AR.StoreData.Handle,&(ind->variables),SEEK_SET);
    *position = ind->position;
    jsym = ind->nsymbols;
    jvec = ind->nvectors;
    jind = ind->nindices;
    jfun = ind->nfunctions;
/*
            #[ Symbols :
*/
    {
    SYMBOLS s = &SyM;
    w = r->symb.lo; j = jsym;
    for ( i = 0; i < j; i++ ) {
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)s,(LONG)(sizeof(struct SyMbOl)))
        != sizeof(struct SyMbOl) ) goto ErrGt2;
        nsize = s->namesize; nsize += sizeof(void *)-1;
        nsize &= -sizeof(void *);
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)(AT.WorkPointer),nsize)
        != nsize ) goto ErrGt2;
        *w = s->number;
        if ( ( s->flags & INUSE ) != 0 ) {
            /* Find the replacement. It must exist! */
            neww = oldw;
            while ( *neww != SYMTOSYM || neww[2] != *w ) neww += neww[1];
            k = neww[3];
        }
        else if ( GetVar((UBYTE *)AT.WorkPointer,&type,&k,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
            if ( type != CSYMBOL ) {
                MesPrint("Error: Conflicting types for %s",(AT.WorkPointer));
                error = -1;
            }
            else {
                if ( ( s->complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) !=
                ( symbols[k].complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) ) {
                    MesPrint("Warning: Conflicting complexity for %s",AT.WorkPointer);
                    error = -1;
                }
                if ( ( s->complex & (VARTYPEROOTOFUNITY) ) !=
                ( symbols[k].complex & (VARTYPEROOTOFUNITY) ) ) {
                    MesPrint("Warning: Conflicting root of unity properties for %s",AT.WorkPointer);
                    error = -1;
                }
                if ( ( s->complex & VARTYPEROOTOFUNITY ) == VARTYPEROOTOFUNITY ) {
                    if ( s->maxpower != symbols[k].maxpower ) {
                        MesPrint("Warning: Conflicting n in n-th root of unity properties for
%s",AT.WorkPointer);
                        error = -1;
                    }
                }
                else if ( ( s->minpower !=
                symbols[k].minpower || s->maxpower !=
                symbols[k].maxpower ) && AC.WarnFlag ) {
```

```
                    MesPrint("Warning: Conflicting power restrictions for %s",AT.WorkPointer);
                }
            }
        }
        else {
            if ( ( k = EntVar(CSYMBOL,(UBYTE *)(AT.WorkPointer),s->complex,s->minpower,
            s->maxpower,s->dimension) ) < 0 ) goto GetTcall;
        }
        *(w+j) = k;
        w++;
    }
    }
/*
            #] Symbols :
            #[ Indices :
*/
    {
    INDICES s = &InD;
    w = r->indi.lo; j = jind;
    for ( i = 0; i < j; i++ ) {
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)s,(LONG)(sizeof(struct InDeX)))
        != sizeof(struct InDeX) ) goto ErrGt2;
        nsize = s->namesize; nsize += sizeof(void *)-1;
        nsize &= -sizeof(void *);
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)(AT.WorkPointer),nsize)
        != nsize ) goto ErrGt2;
        *w = s->number + AM.OffsetIndex;
        if ( s->dimension < 0 ) {   /* Relabel the dimension */
            s->dimension = -r->symnum[FindrNumber(-s->dimension,&(r->symb))];
            if ( s->nmin4 < -NMIN4SHIFT ) { /* Relabel n-4 */
                s->nmin4 = -r->symnum[FindrNumber(-s->nmin4-NMIN4SHIFT
                ,&(r->symb))]-NMIN4SHIFT;
            }
        }
        if ( ( s->flags & INUSE ) != 0 ) {
            /* Find the replacement. It must exist! */
            neww = oldw;
            while ( *neww != INDTOIND || neww[2] != *w ) neww += neww[1];
            k = neww[3] - AM.OffsetIndex;
        }
        else if ( s->type == DUMMY ) {
/*
-------->           Here we may have to execute some renumbering
*/
        }
        else if ( GetVar((UBYTE *)(AT.WorkPointer),&type,&k,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
            if ( type != CINDEX ) {
                MesPrint("Error: Conflicting types for %s",(AT.WorkPointer));
                error = -1;
            }
            else {
                if ( s->type !=
                indices[k].type ) {
                    MesPrint("Warning: %s is also a dummy index",(AT.WorkPointer));
                    error = -1;
                    goto GetTb3;
                }
                if ( s->dimension != indices[k].dimension ) {
                    MesPrint("Warning: Conflicting dimensions for %s",(AT.WorkPointer));
                    error = -1;
                }
            }
        }
        else {
GetTb3:
            if ( ( k = EntVar(CINDEX,(UBYTE *)(AT.WorkPointer),
            s->dimension,0,s->nmin4,0) ) < 0 ) goto GetTcall;

        }
        *(w+j) = k + AM.OffsetIndex;
        w++;
    }
    }
/*
            #] Indices :
            #[ Vectors :
*/
    {
    VECTORS s = &VeC;
    w = r->vect.lo; j = jvec;
    for ( i = 0; i < j; i++ ) {
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)s,(LONG)(sizeof(struct VeCtOr)))
        != sizeof(struct VeCtOr) ) goto ErrGt2;
        nsize = s->namesize; nsize += sizeof(void *)-1;
        nsize &= -sizeof(void *);
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)(AT.WorkPointer),nsize)
        != nsize ) goto ErrGt2;
```

```
        *w = s->number + AM.OffsetVector;
        if ( ( s->flags & INUSE ) != 0 ) {
            /* Find the replacement. It must exist! */
            neww = oldw;
            while ( *neww != VECTOVEC || neww[2] != *w ) neww += neww[1];
            k = neww[3] - AM.OffsetVector;
        }
        else if ( GetVar((UBYTE *)(AT.WorkPointer),&type,&k,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
            if ( type != CVECTOR ) {
                MesPrint("Error: Conflicting types for %s",(AT.WorkPointer));
                error = -1;
            }
            else {
                if ( ( s->complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) !=
                ( vectors[k].complex & (VARTYPEIMAGINARY|VARTYPECOMPLEX) ) ) {
                    MesPrint("Warning: Conflicting complexity for %s",(AT.WorkPointer));
                    error = -1;
                }
            }
        }
        else {
            if ( ( k = EntVar(CVECTOR,(UBYTE *)(AT.WorkPointer),
            s->complex,0,0,s->dimension) ) < 0 ) goto GetTcall;
        }
        *(w+j) = k + AM.OffsetVector;
        w++;
    }
    }
/*
            #] Vectors :
            #[ Functions :
*/
    {
    FUNCTIONS s = &FuN;
    w = r->func.lo; j = jfun;
    for ( i = 0; i < j; i++ ) {
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)s,(LONG)(sizeof(struct FuNcTiOn)))
        != sizeof(struct FuNcTiOn) ) goto ErrGt2;
        nsize = s->namesize; nsize += sizeof(void *)-1;
        nsize &= -sizeof(void *);
        if ( ReadFile(AR.StoreData.Handle,(UBYTE *)(AT.WorkPointer),nsize)
        != nsize ) goto ErrGt2;
        *w = s->number + FUNCTION;
        if ( ( s->flags & INUSE ) != 0 ) {
            /* Find the replacement. It must exist! */
            neww = oldw;
            while ( *neww != FUNTOFUN || neww[2] != *w ) neww += neww[1];
            k = neww[3] - FUNCTION;
        }
        else if ( GetVar((UBYTE *)(AT.WorkPointer),&type,&k,ALLVARIABLES,NOAUTO) != NAMENOTFOUND ) {
            if ( type != CFUNCTION ) {
                MesPrint("Error: Conflicting types for %s",(AT.WorkPointer));
                error = -1;
            }
            else {
                if ( s->complex != functions[k].complex ) {
                    MesPrint("Warning: Conflicting complexity for %s",(AT.WorkPointer));
                    error = -1;
                }
                else if ( s->symmetric != functions[k].symmetric ) {
                    MesPrint("Warning: Conflicting symmetry properties for %s",(AT.WorkPointer));
                    error = -1;
                }
                else if ( ( s->maxnumargs != functions[k].maxnumargs )
                || ( s->minnumargs != functions[k].minnumargs ) ) {
                    MesPrint("Warning: Conflicting argument restriction properties for
%s",(AT.WorkPointer));
                    error = -1;
                }
            }
        }
        else {
            if ( ( k = EntVar(CFUNCTION,(UBYTE *)(AT.WorkPointer),
            s->complex,s->commute,s->spec,s->dimension) ) < 0 ) goto GetTcall;
            functions[k].symmetric = s->symmetric;
            functions[k].maxnumargs = s->maxnumargs;
            functions[k].minnumargs = s->minnumargs;
        }
        *(w+j) = k + FUNCTION;
        w++;
    }
    }
/*
            #] Functions :

    Now we skip the prototype. This sets the start position at the first term
*/
```

```
    if ( error ) {
        UNLOCK(AM.storefilelock);
        AT.WorkPointer = oldwork;
        return(0);
    }

    {
/*
        For clarity we look where we are.
        We want to know: is this position already known?
        Could we have inserted extra information here?

        nummystery indicates extra words. We have currently in order
        (if they exist)
            numdummies
            numfactors
            vflags
            uflags
*/
        POSITION pos;
        int nummystery;
        TELLFILE(AR.StoreData.Handle,&pos);
        nummystery = DIFBASE(ind->position,pos);
/*
        MesPrint("--> We are at position       %8p",&pos);
        MesPrint("--> The index says        at %8p",&(ind->position));
        MesPrint("--> There are %d mystery bytes",nummystery);
*/
        if ( nummystery > 0 ) {
            if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,(LONG)sizeof(WORD)) !=
            sizeof(WORD) ) {
                UNLOCK(AM.storefilelock);
                AT.WorkPointer = oldwork;
                return(0);
            }
            Expressions[expr].numdummies = *AT.WorkPointer;
/*
            MesPrint("--> numdummies = %d",Expressions[expr].numdummies);
*/
            nummystery -= sizeof(WORD);
        }
        else {
            Expressions[expr].numdummies = 0;
        }
        if ( nummystery > 0 ) {
            if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,(LONG)sizeof(WORD)) !=
            sizeof(WORD) ) {
                UNLOCK(AM.storefilelock);
                AT.WorkPointer = oldwork;
                return(0);
            }
            if ( ( AS.OldNumFactors == 0 ) || ( AS.NumOldNumFactors < NumExpressions ) ) {
                WORD *buffer;
                int capacity = 20;
                if (capacity < NumExpressions) capacity = NumExpressions * 2;

                buffer = (WORD *)Malloc1(capacity * sizeof(WORD), "numfactors pointers");
                if (AS.OldNumFactors) {
                    WCOPY(buffer, AS.OldNumFactors, AS.NumOldNumFactors);
                    M_free(AS.OldNumFactors, "numfactors pointers");
                }
                AS.OldNumFactors = buffer;

                buffer = (WORD *)Malloc1(capacity * sizeof(WORD), "vflags pointers");
                if (AS.Oldvflags) {
                    WCOPY(buffer, AS.Oldvflags, AS.NumOldNumFactors);
                    M_free(AS.Oldvflags, "vflags pointers");
                }
                AS.Oldvflags = buffer;

                buffer = (WORD *)Malloc1(capacity * sizeof(WORD), "uflags pointers");
                if (AS.Olduflags) {
                    WCOPY(buffer, AS.Olduflags, AS.NumOldNumFactors);
                    M_free(AS.Olduflags, "uflags pointers");
                }
                AS.Oldvflags = buffer;

                AS.NumOldNumFactors = capacity;
            }

            AS.OldNumFactors[expr] =
            Expressions[expr].numfactors = *AT.WorkPointer;
/*
            MesPrint("--> numfactors = %d",Expressions[expr].numfactors);
*/
            nummystery -= sizeof(WORD);
        }
```

```
        else {
            Expressions[expr].numfactors = 0;
        }
        if ( nummystery > 0 ) {
            if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,(LONG)sizeof(WORD)) !=
            sizeof(WORD) ) {
                UNLOCK(AM.storefilelock);
                AT.WorkPointer = oldwork;
                return(0);
            }
            AS.Oldvflags[expr] =
            Expressions[expr].vflags = *AT.WorkPointer;
/*
            MesPrint("--> vflags = %d",Expressions[expr].vflags);
*/
            nummystery -= sizeof(WORD);
        }
        else {
            Expressions[expr].vflags = 0;
        }
        if ( nummystery > 0 ) {
            if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,(LONG)sizeof(WORD)) !=
            sizeof(WORD) ) {
                UNLOCK(AM.storefilelock);
                AT.WorkPointer = oldwork;
                return(0);
            }
            AS.Olduflags[expr] =
            Expressions[expr].uflags = *AT.WorkPointer;
/*
            MesPrint("--> uflags = %d",Expressions[expr].uflags);
*/
            nummystery -= sizeof(WORD);
        }
        else {
            Expressions[expr].uflags = 0;
        }
    }

    SeekFile(AR.StoreData.Handle,&(ind->position),SEEK_SET);
    if ( ReadFile(AR.StoreData.Handle,(UBYTE *)AT.WorkPointer,(LONG)sizeof(WORD)) !=
    sizeof(WORD) || !*AT.WorkPointer ) {
        UNLOCK(AM.storefilelock);
        AT.WorkPointer = oldwork;
        return(0);
    }
    num = *AT.WorkPointer - 1;
    num *= sizeof(WORD);
    if ( *AT.WorkPointer < 0 ||
    ReadFile(AR.StoreData.Handle,(UBYTE *)(AT.WorkPointer+1),num) != num ) {
        MesPrint("@Error in stored expressions file at position %10p",*position);
        UNLOCK(AM.storefilelock);
        AT.WorkPointer = oldwork;
        return(0);
    }
    UNLOCK(AM.storefilelock);
    ADDPOS(*position,num+sizeof(WORD));
    r->startposition = *position;
    AT.WorkPointer = oldwork;
    return(r);
GetTcall:
    UNLOCK(AM.storefilelock);
    AT.WorkPointer = oldwork;
    MesCall("GetTable");
    return(0);
ErrGt2:
    UNLOCK(AM.storefilelock);
    AT.WorkPointer = oldwork;
    MesPrint("Readerror in GetTable");
    return(0);
}

/*
        #] GetTable :
        #[ CopyExpression :

        Copies from one scratch buffer to another.
        We assume here that the complete 'from' scratch buffer is taken.
        We also assume that the 'from' buffer is positioned at the end of
        the expression.

        The locks should be placed in the calling routine. We need basically
        AS.outputslock.
*/

int CopyExpression(FILEHANDLE *from, FILEHANDLE *to)
{
```

```
    POSITION posfrom, poscopy;
    LONG fullsize,i;
    WORD *t1, *t2;
    int RetCode;
    SeekScratch(from,&posfrom);
    if ( from->handle < 0 ) { /* input is in memory */
        fullsize = (BASEPOSITION(posfrom))/sizeof(WORD);
        if ( ( to->POstop - to->POfull ) >= fullsize ) {
/*
            Fits inside the buffer of the output. This will be fast.
*/
            t1 = from->PObuffer;
            t2 = to->POfull;
            NCOPY(t2,t1,fullsize)
            to->POfull = to->POfill = t2;
            goto WriteTrailer;
        }
        if ( to->handle < 0 ) {       /* First open the file */
            if ( ( RetCode = CreateFile(to->name) ) >= 0 ) {
                to->handle = (WORD)RetCode;
                PUTZERO(to->filesize);
                PUTZERO(to->POposition);
            }
            else {
                MLOCK(ErrorMessageLock);
                MesPrint("Cannot create scratch file %s",to->name);
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
        }
        t1 = from->PObuffer;
        while ( fullsize > 0 ) {
            i = to->POstop - to->POfull;
            if ( i > fullsize ) i = fullsize;
            fullsize -= i;
            t2 = to->POfull;
            NCOPY(t2,t1,i)
            if ( fullsize > 0 ) {
                SeekFile(to->handle,&(to->POposition),SEEK_SET);
                if ( WriteFile(to->handle,((UBYTE *)(to->PObuffer)),to->POsize) != to->POsize ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Error while writing to disk. Disk full?");
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                ADDPOS(to->POposition,to->POsize);
/*              SeekFile(to->handle,&(to->POposition),SEEK_CUR); */
                to->filesize = to->POposition;
                to->POfill = to->POfull = to->PObuffer;
            }
            else {
                to->POfill = to->POfull = t2;
            }
        }
        goto WriteTrailer;
    }
/*
    Now the input involves a file. This needs the use of the PObuffer of from.
    First make sure the tail of the buffer has been written
*/
    if ( ((UBYTE *)(from->POfill)-(UBYTE *)(from->PObuffer)) > 0 ) {
        if ( WriteFile(from->handle,((UBYTE *)(from->PObuffer)),((UBYTE *)(from->POfill)-(UBYTE
*)(from->PObuffer)))
        != ((UBYTE *)(from->POfill)-(UBYTE *)(from->PObuffer)) ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Error while writing to disk. Disk full?");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        SeekFile(from->handle,&(from->POposition),SEEK_CUR);
        posfrom = from->filesize = from->POposition;
        from->POfill = from->POfull = from->PObuffer;
    }
/*
    Now copy the complete contents
*/
    PUTZERO(poscopy);
    SeekFile(from->handle,&poscopy,SEEK_SET);
    while ( ISLESSPOS(poscopy,posfrom) ) {
        fullsize = ReadFile(from->handle,((UBYTE *)(from->PObuffer)),from->POsize);
        if ( fullsize < 0 || ( fullsize % sizeof(WORD) ) != 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Error while reading from disk while copying expression.");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        fullsize /= sizeof(WORD);
```

```
        from->POfull = from->PObuffer + fullsize;
        t1 = from->PObuffer;

        if ( ( to->POstop - to->POfull ) >= fullsize ) {
/*
            Fits inside the buffer of the output. This will be fast.
*/
            t2 = to->POfull;
            NCOPY(t2,t1,fullsize)
            to->POfill = to->POfull = t2;
        }
        else {
          if ( to->handle < 0 ) {      /* First open the file */
            if ( ( RetCode = CreateFile(to->name) ) >= 0 ) {
                to->handle = (WORD)RetCode;
                PUTZERO(to->POposition);
                PUTZERO(to->filesize);
            }
            else {
                MLOCK(ErrorMessageLock);
                MesPrint("Cannot create scratch file %s",to->name);
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
          }
          while ( fullsize > 0 ) {
            i = to->POstop - to->POfull;
            if ( i > fullsize ) i = fullsize;
            fullsize -= i;
            t2 = to->POfull;
            NCOPY(t2,t1,i)
            if ( fullsize > 0 ) {
                SeekFile(to->handle,&(to->POposition),SEEK_SET);
                if ( WriteFile(to->handle,((UBYTE *)(to->PObuffer)),to->POsize) != to->POsize ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Error while writing to disk. Disk full?");
                    MUNLOCK(ErrorMessageLock);
                    return(-1);
                }
                ADDPOS(to->POposition,to->POsize);
/*              SeekFile(to->handle,&(to->POposition),SEEK_CUR); */
                to->filesize = to->POposition;
                to->POfill = to->POfull = to->PObuffer;
            }
            else {
                to->POfill = to->POfull = t2;
            }
          }
        }
        SeekFile(from->handle,&poscopy,SEEK_CUR);
    }
WriteTrailer:
    if ( ( to->handle >= 0 ) && ( to->POfill > to->PObuffer ) ) {
        fullsize = (UBYTE *)(to->POfill) - (UBYTE *)(to->PObuffer);
/*
        PUTZERO(to->POposition);
        SeekFile(to->handle,&(to->POposition),SEEK_END);
*/
        SeekFile(to->handle,&(to->filesize),SEEK_SET);
        if ( WriteFile(to->handle,((UBYTE *)(to->PObuffer)),fullsize) != fullsize ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Error while writing to disk. Disk full?");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        ADDPOS(to->filesize,fullsize);
        to->POposition = to->filesize;
        to->POfill = to->POfull = to->PObuffer;
    }

    return(0);
}

/*
        #] CopyExpression :
        #[ ExprStatus :
*/

#ifdef HIDEDEBUG

static UBYTE *statusexpr[] = {
     (UBYTE *)"LOCALEXPRESSION"
    ,(UBYTE *)"SKIPLEXPRESSION"
    ,(UBYTE *)"DROPLEXPRESSION"
    ,(UBYTE *)"DROPPEDEXPRESSION"
    ,(UBYTE *)"GLOBALEXPRESSION"
    ,(UBYTE *)"SKIPGEXPRESSION"
```

```
    ,(UBYTE *)"DROPGEXPRESSION"
    ,(UBYTE *)"UNKNOWN"
    ,(UBYTE *)"STOREDEXPRESSION"
    ,(UBYTE *)"HIDDENLEXPRESSION"
    ,(UBYTE *)"HIDELEXPRESSION"
    ,(UBYTE *)"DROPHLEXPRESSION"
    ,(UBYTE *)"UNHIDELEXPRESSION"
    ,(UBYTE *)"HIDDENGEXPRESSION"
    ,(UBYTE *)"HIDEGEXPRESSION"
    ,(UBYTE *)"DROPHGEXPRESSION"
    ,(UBYTE *)"UNHIDEGEXPRESSION"
    ,(UBYTE *)"INTOHIDELEXPRESSION"
    ,(UBYTE *)"INTOHIDEGEXPRESSION"
};

void ExprStatus(EXPRESSIONS e)
{
    MesPrint("Expression %s(%d) has status %s(%d,%d). Buffer: %d, Position: %15p",
        AC.exprnames->namebuffer+e->name,(WORD)(e-Expressions),
        statusexpr[e->status],e->status,e->hidelevel,
        e->whichbuffer,&(e->onfile));
}

#endif

/*
        #] ExprStatus :
    #] StoreExpressions :
    #[ System Independent Saved Expressions :

    All functions concerned with the system independent reading of save-files
    are here. They are called by the functions CoLoad, PutInStore,
    SetFileIndex, FindInIndex. In case no translation (endianness flip,
    resizing of words, renumbering) has to be done, they just do simple file
    reading. The function SaveFileHeader() for writing a header with
    information about the system architecture, FORM version, etc. is also
    located here.

        #[ Flip :
*/

#ifndef INT16
#error "INT16 not defined!"
#endif
#ifndef INT32
#error "INT32 not defined!"
#endif

static void FlipN(UBYTE *p, int length)
{
    UBYTE *q, buf;
    q = p + length;
    do {
        --q;
        buf = *p; *p = *q; *q = buf;
    } while ( ++p != q );
}

static void Flip16(UBYTE *p)
{
    INT16 in = *((INT16 *)p);
    INT16 out = (INT16)( (((in) » 8) & 0x00FF) | (((in) « 8) & 0xFF00) );
    *((INT16 *)p) = out;
}

static void Flip32(UBYTE *p)
{
    INT32 in = *((INT32 *)p);
    INT32 out =
        ( (((in) » 24) & 0x000000FF) | (((in) »  8) & 0x0000FF00) | \
          (((in) «  8) & 0x00FF0000) | (((in) « 24) & 0xFF000000) );
    *((INT32 *)p) = out;
}

#ifdef INT64
static void Flip64(UBYTE *p)
{
    INT64 in = *((INT64 *)p);
    INT64 out =
        ( (((in) » 56) & (INT64)0x00000000000000FFLL) | (((in) » 40) & (INT64)0x000000000000FF00LL) | \
          (((in) » 24) & (INT64)0x0000000000FF0000LL) | (((in) »  8) & (INT64)0x00000000FF000000LL) | \
          (((in) «  8) & (INT64)0x000000FF00000000LL) | (((in) « 24) & (INT64)0x0000FF0000000000LL) | \
          (((in) « 40) & (INT64)0x00FF000000000000LL) | (((in) « 56) & (INT64)0xFF00000000000000LL) );
    *((INT64 *)p) = out;
```

```
}
#else
static void Flip64(UBYTE *p) { FlipN(p, 8); }
#endif /* INT64 */

static void Flip128(UBYTE *p) { FlipN(p, 16); }

/*
        #] Flip :
        #[ Resize :
*/

static void ResizeDataBE(UBYTE *src, int slen, UBYTE *dst, int dlen)
{
    if ( slen > dlen ) {
        src += slen - dlen;
        while ( dlen-- ) { *dst++ = *src++; }
    }
    else {
        int i = dlen - slen;
        while ( i-- ) { *dst++ = 0; }
        while ( slen-- ) { *dst++ = *src++; }
    }
}

static void ResizeDataLE(UBYTE *src, int slen, UBYTE *dst, int dlen)
{
    if ( slen > dlen ) {
        while ( dlen-- ) { *dst++ = *src++; }
    }
    else {
        int i = dlen - slen;
        while ( slen-- ) { *dst++ = *src++; }
        while ( i-- ) { *dst++ = 0; }
    }
}

static void Resize16t16(UBYTE *src, UBYTE *dst)
{
    *((INT16 *)dst) = *((INT16 *)src);
}

static void Resize16t32(UBYTE *src, UBYTE *dst)
{
    INT16 in = *((INT16 *)src);
    INT32 out = (INT32)in;
    *((INT32 *)dst) = out;
}

#ifdef INT64
static void Resize16t64(UBYTE *src, UBYTE *dst)
{
    INT16 in = *((INT16 *)src);
    INT64 out = (INT64)in;
    *((INT64 *)dst) = out;
}
#else
static void Resize16t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 2, dst, 8); }
#endif /* INT64 */

static void Resize16t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 2, dst, 16); }

static void Resize32t32(UBYTE *src, UBYTE *dst)
{
    *((INT32 *)dst) = *((INT32 *)src);
}

#ifdef INT64
static void Resize32t64(UBYTE *src, UBYTE *dst)
{
    INT32 in = *((INT32 *)src);
    INT64 out = (INT64)in;
    *((INT64 *)dst) = out;
}
#else
static void Resize32t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 4, dst, 8); }
#endif /* INT64 */

static void Resize32t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 4, dst, 16); }

#ifdef INT64
static void Resize64t64(UBYTE *src, UBYTE *dst)
{
    *((INT64 *)dst) = *((INT64 *)src);
}
#else
static void Resize64t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 8); }
```

```
#endif /* INT64 */

static void Resize64t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 16); }

static void Resize128t128(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 16); }

static void Resize32t16(UBYTE *src, UBYTE *dst)
{
    INT32 in = *((INT32 *)src);
    INT16 out = (INT16)in;
    if ( in > (1«15)-1 || in < -(1«15)+1 ) AO.resizeFlag |= 1;
    *((INT16 *)dst) = out;
}

static void Resize32t16NC(UBYTE *src, UBYTE *dst)
{
    INT32 in = *((INT32 *)src);
    INT16 out = (INT16)in;
    *((INT16 *)dst) = out;
}

#ifdef INT64
static void Resize64t16(UBYTE *src, UBYTE *dst)
{
    INT64 in = *((INT64 *)src);
    INT16 out = (INT16)in;
    if ( in > (1«15)-1 || in < -(1«15)+1 ) AO.resizeFlag |= 1;
    *((INT16 *)dst) = out;
}
static void Resize64t16NC(UBYTE *src, UBYTE *dst)
{
    INT64 in = *((INT64 *)src);
    INT16 out = (INT16)in;
    *((INT16 *)dst) = out;
}
#else
static void Resize64t16(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 2); }
static void Resize64t16NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 2); }
#endif /* INT64 */

#ifdef INT64
static void Resize64t32(UBYTE *src, UBYTE *dst)
{
    INT64 in = *((INT64 *)src);
    INT32 out = (INT32)in;
    if ( in > ((INT64)1«31)-1 || in < -((INT64)1«31)+1 ) AO.resizeFlag |= 1;
    *((INT32 *)dst) = out;
}
static void Resize64t32NC(UBYTE *src, UBYTE *dst)
{
    INT64 in = *((INT64 *)src);
    INT32 out = (INT32)in;
    *((INT32 *)dst) = out;
}
#else
static void Resize64t32(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 4); }
static void Resize64t32NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 8, dst, 4); }
#endif /* INT64 */

static void Resize128t16(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 2); }

static void Resize128t16NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 2); }

static void Resize128t32(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 4); }

static void Resize128t32NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 4); }

static void Resize128t64(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 8); }

static void Resize128t64NC(UBYTE *src, UBYTE *dst) { AO.ResizeData(src, 16, dst, 8); }

/*
        #] Resize :
        #[ CheckPower and RenumberVec :
*/

static void CheckPower32(UBYTE *p)
{
    if ( *((INT32 *)p) < -MAXPOWER ) {
        AO.powerFlag |= 0x01;
        *((INT32 *)p) = -MAXPOWER;
    }
    p += sizeof(INT32);
    if ( *((INT32 *)p) > MAXPOWER ) {
        AO.powerFlag |= 0x02;
        *((INT32 *)p) = MAXPOWER;
    }
```

```
}

static void RenumberVec32(UBYTE *p)
{
/*  INT32 wildoffset = *((INT32 *)AO.SaveHeader.wildoffset); */
    void *dummy = (void *)AO.SaveHeader.wildoffset;  /* to remove a warning about strict-aliasing rules in gcc */
    INT32 wildoffset = *(INT32 *)dummy;
    INT32 in = *((INT32 *)p);
    in = in + 2*wildoffset;
    in = in - 2*WILDOFFSET;
    *((INT32 *)p) = in;
}

/*
        #] CheckPower and RenumberVec :
        #[ ResizeCoeff :
*/

static void ResizeCoeff32(UBYTE **bout, UBYTE *bend, UBYTE *top)
{
    int i;
    INT32 sign;
    INT32 *in, *p;
    INT32 *out = (INT32 *)*bout;
    INT32 *end = (INT32 *)bend;

    if ( sizeof(WORD) == 2 ) {
        /* 4 -> 2 */
        INT32 len = (end - 1 - out) / 2;
        int zeros = 2;
        p = out + len - 1;

        if ( *p & 0xFFFF0000 ) --zeros;
        p += len;
        if ( *p & 0xFFFF0000 ) --zeros;

        in = end - 1;
        sign = ( *in-- > 0 ) ? 1 : -1;
        p = out + 4*len;
        if ( zeros == 2 ) p -= 2;
        out = p--;

        if ( zeros < 2 ) *p-- = *in >> 16;
        *p-- = *in-- & 0x0000FFFF;
        for ( i = 1; i < len; ++i ) {
            *p-- = *in >> 16;
            *p-- = *in-- & 0x0000FFFF;
        }
        if ( zeros < 2 ) *p-- = *in >> 16;
        *p-- = *in-- & 0x0000FFFF;
        for ( i = 1; i < len; ++i ) {
            *p-- = *in >> 16;
            *p-- = *in-- & 0x0000FFFF;
        }

        *out = (out - p) * sign;
        *bout = (UBYTE *)(out+1);

    }
    else {
        /* 2 -> 4 */
        INT32 len = (end - 1 - out) / 2;
        if ( len == 1 ) {
            *out = *(unsigned INT16 *)out;
            ++out;
            *out = *(unsigned INT16 *)out;
            ++out;
            ++out;
        }
        else {
            p = out;
            *out = *(unsigned INT16 *)out;
            in = out + 1;
            for ( i = 1; i < len; ++i ) {
                /* shift */
                *out = (unsigned INT32)(*(unsigned INT16 *)out)
                    + ((unsigned INT32)(*(unsigned INT16 *)in) << 16);
                ++in;
                if ( ++i == len ) break;
                /* copy */
                ++out;
                *out = *(unsigned INT16 *)in;
                ++in;
            }
            ++out;
            *out = *(unsigned INT16 *)in;
```

```
            ++in;
            for ( i = 1; i < len; ++i ) {
                /* shift */
                *out = (unsigned INT32)(*(unsigned INT16 *)out)
                    + ((unsigned INT32)(*(unsigned INT16 *)in) « 16);
                ++in;
                if ( ++i == len ) break;
                /* copy */
                ++out;
                *out = *(unsigned INT16 *)in;
                ++in;
            }
            ++out;
            if ( *in < 0 ) *out = -(out - p + 1);
            else *out = out - p + 1;
            ++out;
        }

        if ( out > (INT32 *)top ) {
            MesPrint("Error in resizing coefficient!");
        }

        *bout = (UBYTE *)out;
    }
}

/*
        #] ResizeCoeff :
        #[ WriteStoreHeader :
*/

#define SAVEREVISION 0x02

WORD WriteStoreHeader(WORD handle)
{
    /* template of the STOREHEADER */
    static STOREHEADER sh = {
        { 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF }, /* store header mark */
        0, 0, 0, 0,                                         /* sizeof of WORD,LONG,POSITION,void* */
        { 0 },                                              /* endianness check number */
        0, 0, 0, 0,                                         /* sizeof variable structs */
        { 0 },                                              /* maxpower */
        { 0 },                                              /* wildoffset */
        SAVEREVISION,                                       /* revision */
        { 0 } };                                            /* reserved */
    int endian, i;

    /* if called for the first time ... */
    if ( sh.lenWORD == 0 ) {
        sh.lenWORD = sizeof(WORD);
        sh.lenLONG = sizeof(LONG);
        sh.lenPOS = sizeof(POSITION);
        sh.lenPOINTER = sizeof(void *);

        endian = 1;
        for ( i = 1; i < (int)sizeof(int); ++i ) {
            endian «= 8;
            endian += i+1;
        }
        for ( i = 0; i < (int)sizeof(int); ++i ) sh.endianness[i] = ((char *)&endian)[i];

        sh.sSym = sizeof(struct SyMbOl);
        sh.sInd = sizeof(struct InDeX);
        sh.sVec = sizeof(struct VeCtOr);
        sh.sFun = sizeof(struct FuNcTiOn);

/*      *((WORD *)sh.maxpower) = MAXPOWER;
        *((WORD *)sh.wildoffset) = WILDOFFSET; */
        {
            WORD dumw[8];
            UBYTE *dummy;
            for ( i = 0; i < 8; i++ ) dumw[i] = 0;
            dummy = (UBYTE *)dumw;
            dumw[0] = (WORD)MAXPOWER;
            for ( i = 0; i < 16; i++ ) sh.maxpower[i] = dummy[i];
            dumw[0] = (WORD)WILDOFFSET;
            for ( i = 0; i < 16; i++ ) sh.wildoffset[i] = dummy[i];
        }
    }

    return ( WriteFile(handle,(UBYTE *)(&sh),(LONG)(sizeof(STOREHEADER)))
             != (LONG)(sizeof(STOREHEADER)) );
}

/*
        #] WriteStoreHeader :
        #[ CompactifySizeof :
```

```
*/

static unsigned int CompactifySizeof(unsigned int size)
{
    switch ( size ) {
        case  2: return 0;
        case  4: return 1;
        case  8: return 2;
        case 16: return 3;
        default: MesPrint("Error compactifying size.");
                 return 3;
    }
}

/*
        #] CompactifySizeof :
        #[ ReadSaveHeader :
*/

WORD ReadSaveHeader(void)
{
    /* Read-only tables of function pointers for conversions. */
    static void (*flipJumpTable[4])(UBYTE *) =
        { Flip16, Flip32, Flip64, Flip128 };
    static void (*resizeJumpTable[4][4])(UBYTE *, UBYTE *) = /* "own x saved"-sizes  */
        { { Resize16t16,  Resize32t16,  Resize64t16,  Resize128t16  },
          { Resize16t32,  Resize32t32,  Resize64t32,  Resize128t32  },
          { Resize16t64,  Resize32t64,  Resize64t64,  Resize128t64  },
          { Resize16t128, Resize32t128, Resize64t128, Resize128t128 } };
    static void (*resizeNCJumpTable[4][4])(UBYTE *, UBYTE *) = /* "own x saved"-sizes  */
        { { Resize16t16,  Resize32t16NC,  Resize64t16NC,  Resize128t16NC  },
          { Resize16t32,  Resize32t32,  Resize64t32NC,  Resize128t32NC  },
          { Resize16t64,  Resize32t64,  Resize64t64,  Resize128t64NC  },
          { Resize16t128, Resize32t128, Resize64t128, Resize128t128 } };

    int endian, i;
    WORD idxW = CompactifySizeof(sizeof(WORD));
    WORD idxL = CompactifySizeof(sizeof(LONG));
    WORD idxP = CompactifySizeof(sizeof(POSITION));
    WORD idxVP = CompactifySizeof(sizeof(void *));

    AO.transFlag = 0;
    AO.powerFlag = 0;
    AO.resizeFlag = 0;
    AO.bufferedInd = 0;

    if ( ReadFile(AO.SaveData.Handle,(UBYTE *)(&AO.SaveHeader),
         (LONG)sizeof(STOREHEADER)) != (LONG)sizeof(STOREHEADER) )
        return(MesPrint("Error reading save file header"));

    /* check whether save-file has no header. if yes then it is an old version
       of FORM -> go back to position 0 in file which then contains the first
       index and skip the rest. */
    for ( i = 0; i < 8; ++i ) {
        if ( AO.SaveHeader.headermark[i] != 0xFF ) {
            POSITION p;
            PUTZERO(p);
            SeekFile(AO.SaveData.Handle, &p, SEEK_SET);
            return ( 0 );
        }
    }

    if ( AO.SaveHeader.revision != SAVEREVISION ) {
        return(MesPrint("Save file header from an old version. Cannot read this file."));
    }

    endian = 1;
    for ( i = 1; i < (int)sizeof(int); ++i ) {
        endian «= 8;
        endian += i+1;
    }
    if ( ((char *)&endian)[0] < ((char *)&endian)[1] ) {
        /* this machine is big-endian */
        AO.ResizeData = ResizeDataBE;
    }
    else {
        /* this machine is little-endian */
        AO.ResizeData = ResizeDataLE;
    }

    /* set AO.transFlag if ANY conversion has to be done later */
    if ( AO.SaveHeader.endianness[0] > AO.SaveHeader.endianness[1] ) {
        AO.transFlag = ( ((char *)&endian)[0] < ((char *)&endian)[1] );
    }
    else {
        AO.transFlag = ( ((char *)&endian)[0] > ((char *)&endian)[1] );
    }
```

```
    if ( (WORD)AO.SaveHeader.lenWORD != sizeof(WORD) ) AO.transFlag |= 0x02;
    if ( (WORD)AO.SaveHeader.lenLONG != sizeof(LONG) ) AO.transFlag |= 0x04;
    if ( (WORD)AO.SaveHeader.lenPOS != sizeof(POSITION) ) AO.transFlag |= 0x08;
    if ( (WORD)AO.SaveHeader.lenPOINTER != sizeof(void *) ) AO.transFlag |= 0x10;

    AO.FlipWORD = flipJumpTable[idxW];
    AO.FlipLONG = flipJumpTable[idxL];
    AO.FlipPOS = flipJumpTable[idxP];
    AO.FlipPOINTER = flipJumpTable[idxVP];

    /* Works only for machines where WORD is not greater than 32bit ! */
    AO.CheckPower = CheckPower32;
    AO.RenumberVec = RenumberVec32;

    AO.ResizeWORD = resizeJumpTable[idxW][CompactifySizeof(AO.SaveHeader.lenWORD)];
    AO.ResizeNCWORD = resizeNCJumpTable[idxW][CompactifySizeof(AO.SaveHeader.lenWORD)];
    AO.ResizeLONG = resizeJumpTable[idxL][CompactifySizeof(AO.SaveHeader.lenLONG)];
    AO.ResizePOS = resizeJumpTable[idxP][CompactifySizeof(AO.SaveHeader.lenPOS)];
    AO.ResizePOINTER = resizeJumpTable[idxVP][CompactifySizeof(AO.SaveHeader.lenPOINTER)];

    {
        WORD dumw[8];
        UBYTE *dummy;
        for ( i = 0; i < 8; i++ ) dumw[i] = 0;
        dummy = (UBYTE *)dumw;
        for ( i = 0; i < 16; i++ ) dummy[i] = AO.SaveHeader.maxpower[i];
        AO.mpower = dumw[0];
    }

    return ( 0 );
}

/*
        #] ReadSaveHeader :
        #[ ReadSaveIndex :
*/

WORD ReadSaveIndex(FILEINDEX *fileind)
{
    /* do we need some translation for the FILEINDEX? */
    if ( AO.transFlag ) {
        /* if a translated FILEINDEX can hold less entries than the original
           FILEINDEX, then we need to buffer the extra entries in this static
           variable (can happen going from 32bit to 64bit */
        static FILEINDEX sbuffer;

        FILEINDEX buffer;
        UBYTE *p, *q;
        int i;

        /* shortcuts */
        int lenW = AO.SaveHeader.lenWORD;
        int lenL = AO.SaveHeader.lenLONG;
        int lenP = AO.SaveHeader.lenPOS;

        /* if we have a buffered FILEINDEX then just return it */
        if ( AO.bufferedInd ) {
            *fileind = sbuffer;
            AO.bufferedInd = 0;
            return ( 0 );
        }

        if ( ReadFile(AO.SaveData.Handle, (UBYTE *)fileind, sizeof(FILEINDEX))
                != sizeof(FILEINDEX) ) {
            return ( MesPrint("Error(1) reading stored expression.") );
        }

        /* do we need to flip the endianness? */
        if ( AO.transFlag & 1 ) {
            LONG number;
            /* padding bytes */
            int padp = lenL - ((lenW*5+(MAXENAME + 1)) & (lenL-1));
            p = (UBYTE *)fileind;
            AO.FlipPOS(p); p += lenP;           /* next */
            AO.FlipPOS(p);                      /* number */
            AO.ResizePOS(p, (UBYTE *)&number);
            p += lenP;
            for ( i = 0; i < number; ++i ) {
                AO.FlipPOS(p); p += lenP;       /* position */
                AO.FlipPOS(p); p += lenP;       /* length */
                AO.FlipPOS(p); p += lenP;       /* variables */
                AO.FlipLONG(p); p += lenL;      /* CompressSize */
                AO.FlipWORD(p); p += lenW;      /* nsymbols */
                AO.FlipWORD(p); p += lenW;      /* nindices */
                AO.FlipWORD(p); p += lenW;      /* nvectors */
                AO.FlipWORD(p); p += lenW;      /* nfunctions */
                AO.FlipWORD(p); p += lenW;      /* size */
```

```
                p += padp;
            }
        }

        /* do we need to resize data? */
        if ( AO.transFlag > 1 ) {
            LONG number, maxnumber;
            int n;
            /* padding bytes */
            int padp = lenL - ((lenW*5+(MAXENAME + 1)) & (lenL-1));
            int padq = sizeof(LONG) - ((sizeof(WORD)*5+(MAXENAME + 1)) & (sizeof(LONG)-1));

            p = (UBYTE *)fileind; q = (UBYTE *)&buffer;
            AO.ResizePOS(p, q);                     /* next */
            p += lenP; q += sizeof(POSITION);
            AO.ResizePOS(p, q);                 /* number */
            p += lenP;
            number = BASEPOSITION(*((POSITION *)q));
            /* if FILEINDEX in file contains more entries than the FILEINDEX in
               memory can contain, then adjust the numbers and prepare for
               buffering */
            if ( number > (LONG)INFILEINDEX ) {
                AO.bufferedInd = number-INFILEINDEX;
                if ( AO.bufferedInd > (WORD)INFILEINDEX ) {
                    /* can happen when reading 32bit and writing >=128bit.
                       Fix: more than one static buffer for FILEINDEX */
                    return ( MesPrint("Too many index entries.") );
                }
                maxnumber = INFILEINDEX;
                SETBASEPOSITION(*((POSITION *)q),INFILEINDEX);
            }
            else {
                maxnumber = number;
            }
            q += sizeof(POSITION);
            /* read all INDEXENTRY that fit into the output buffer */
            for ( i = 0; i < maxnumber; ++i ) {
                AO.ResizePOS(p, q);                 /* position */
                p += lenP; q += sizeof(POSITION);
                AO.ResizePOS(p, q);                 /* length */
                p += lenP; q += sizeof(POSITION);
                AO.ResizePOS(p, q);                 /* variables */
                p += lenP; q += sizeof(POSITION);
                AO.ResizeLONG(p, q);                /* CompressSize */
                p += lenL; q += sizeof(LONG);
                AO.ResizeWORD(p, q);                /* nsymbols */
                p += lenW; q += sizeof(WORD);
                AO.ResizeWORD(p, q);                /* nindices */
                p += lenW; q += sizeof(WORD);
                AO.ResizeWORD(p, q);                /* nvectors */
                p += lenW; q += sizeof(WORD);
                AO.ResizeWORD(p, q);                /* nfunctions */
                p += lenW; q += sizeof(WORD);
                AO.ResizeWORD(p, q);                /* size (unchanged!) */
                p += lenW; q += sizeof(WORD);
                n = MAXENAME + 1;
                NCOPYB(q, p, n)
                p += padp;
                q += padq;
            }
            /* read all the remaining INDEXENTRY and put them into the static buffer */
            if ( AO.bufferedInd ) {
                sbuffer.next = buffer.next;
                SETBASEPOSITION(sbuffer.number,AO.bufferedInd);
                q = (UBYTE *)&sbuffer + sizeof(POSITION) + sizeof(LONG);
                for ( i = maxnumber; i < number; ++i ) {
                    AO.ResizePOS(p, q);             /* position */
                    p += lenP; q += sizeof(POSITION);
                    AO.ResizePOS(p, q);             /* length */
                    p += lenP; q += sizeof(POSITION);
                    AO.ResizePOS(p, q);             /* variables */
                    p += lenP; q += sizeof(POSITION);
                    AO.ResizeLONG(p, q);            /* CompressSize */
                    p += lenL; q += sizeof(LONG);
                    AO.ResizeWORD(p, q);            /* nsymbols */
                    p += lenW; q += sizeof(WORD);
                    AO.ResizeWORD(p, q);            /* nindices */
                    p += lenW; q += sizeof(WORD);
                    AO.ResizeWORD(p, q);            /* nvectors */
                    p += lenW; q += sizeof(WORD);
                    AO.ResizeWORD(p, q);            /* nfunctions */
                    p += lenW; q += sizeof(WORD);
                    AO.ResizeWORD(p, q);            /* size (unchanged!) */
                    p += lenW; q += sizeof(WORD);
                    n = MAXENAME + 1;
                    NCOPYB(q, p, n)
                    p += padp;
```

```
                    q += padq;
                }
            }
            /* copy to output */
            p = (UBYTE *)fileind; q = (UBYTE *)&buffer; n = sizeof(FILEINDEX);
            NCOPYB(p, q, n)
        }
        return ( 0 );
    } else {
        return ( ReadFile(AO.SaveData.Handle, (UBYTE *)fileind, sizeof(FILEINDEX))
                 != sizeof(FILEINDEX) );
    }
}

/*
        #] ReadSaveIndex :
        #[ ReadSaveVariables :
*/

WORD ReadSaveVariables(UBYTE *buffer, UBYTE *top, LONG *size, LONG *outsize,\
                       INDEXENTRY *ind, LONG *stage)
{
    /* do we need some translation for the variables? */
    if ( AO.transFlag ) {
        /* counters for the number of already read symbols, indices, ... that
           need to remain valid between different calls to ReadSaveVariables().
           are initialized if stage == -1 */
        static WORD numReadSym;
        static WORD numReadInd;
        static WORD numReadVec;
        static WORD numReadFun;

        POSITION pos;
        UBYTE *in, *out, *pp = 0, *end, *outbuf;
        LONG numread;
        WORD namelen, realnamelen;
        /* shortcuts */
        WORD lenW = AO.SaveHeader.lenWORD;
        WORD lenL = AO.SaveHeader.lenLONG;
        WORD lenP = AO.SaveHeader.lenPOINTER;
        WORD flip = AO.transFlag & 1;

        /* remember file position in case we have to rewind */
        TELLFILE(AO.SaveData.Handle,&pos);

        /* decide on the position of the in and out buffers.
           if the input is "bigger" than the output, we resize in-place, i.e.
           we immediately overwrite the source data by the translated data. in
           and out buffers start at the same place.
           if not, we read from the end of the given buffer and write at the
           beginning. */
        if ( (lenW > (WORD)sizeof(WORD))
        || ( (lenW == (WORD)sizeof(WORD))
             && ( (lenL > (WORD)sizeof(LONG))
                  || ( (lenL == (WORD)sizeof(LONG)) && lenP > (WORD)sizeof(void *))
                )
           ) ) {
            in = out = buffer;
            end = buffer + *size;
        }
        else {
            /* data will grow roughly by sizeof(WORD)/lenW. the exact value is
               not important. if reading and writing areas start to overlap, the
               reading will already be near the end of the data and overwriting
               doesn't matter. */
            LONG newsize = (top - buffer) / (1 + sizeof(WORD)/lenW);
            end = top;
            out = buffer;
            in = end - newsize;
            if ( *size > newsize ) *size = newsize;
        }

        if ( ( numread = ReadFile(AO.SaveData.Handle, in, *size) ) != *size ) {
            return ( MesPrint("Error(2) reading stored expression.") );
        }

        *size = 0;
        *outsize = 0;

        /* first time in ReadSaveVariables(). initialize counters. */
        if ( *stage == -1 ) {
            numReadSym = 0;
            numReadInd = 0;
            numReadVec = 0;
            numReadFun = 0;
            ++*stage;
        }
```

```

        while ( in < end ) {
            /* Symbols */
            if ( *stage == 0 ) {
                if ( ind->nsymbols <= numReadSym ) {
                    ++*stage;
                    continue;
                }
                if ( end - in < AO.SaveHeader.sSym ) {
                    goto RSVEnd;
                }
                if ( flip ) {
                    pp = in;
                    AO.FlipLONG(pp); pp += lenL;
                    while ( pp < in + AO.SaveHeader.sSym ) {
                        AO.FlipWORD(pp); pp += lenW;
                    }
                }
                pp = in + AO.SaveHeader.sSym;
                AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name     */
                AO.CheckPower(in);
                AO.ResizeWORD(in, out); in += lenW;
                if ( *((WORD *)out) == -AO.mpower ) *((WORD *)out) = -MAXPOWER;
                out += sizeof(WORD);                                     /* minpower */
                AO.ResizeWORD(in, out); in += lenW;
                if ( *((WORD *)out) == AO.mpower ) *((WORD *)out) = MAXPOWER;
                out += sizeof(WORD);                                     /* maxpower */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* complex  */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number   */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags    */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node     */
                AO.ResizeWORD(in, out); in += lenW;                      /* namesize */
                realnamelen = *((WORD *)out);
                realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
                out += sizeof(WORD);
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
                while ( in < pp ) {
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
                }
                namelen = *((WORD *)out-1); /* cares for padding "bug" */
                if ( end - in < namelen ) {
                    goto RSVEnd;
                }
                *((WORD *)out-1) = realnamelen;
                *size += AO.SaveHeader.sSym + namelen;
                *outsize += sizeof(struct SyMbOl) + realnamelen;
                if ( realnamelen > namelen ) {
                    int j = namelen;
                    NCOPYB(out, in, j);
                    out += realnamelen - namelen;
                }
                else {
                    int j = realnamelen;
                    NCOPYB(out, in, j);
                    in += namelen - realnamelen;
                }
                ++numReadSym;
                continue;
            }
            /* Indices */
            if ( *stage == 1 ) {
                if ( ind->nindices <= numReadInd ) {
                    ++*stage;
                    continue;
                }
                if ( end - in < AO.SaveHeader.sInd ) {
                    goto RSVEnd;
                }
                if ( flip ) {
                    pp = in;
                    AO.FlipLONG(pp); pp += lenL;
                    while ( pp < in + AO.SaveHeader.sInd ) {
                        AO.FlipWORD(pp); pp += lenW;
                    }
                }
                pp = in + AO.SaveHeader.sInd;
                AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name      */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* type      */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number    */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags     */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* nmin4     */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node      */
                AO.ResizeWORD(in, out); in += lenW;                      /* namesize  */
                realnamelen = *((WORD *)out);
                realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
                out += sizeof(WORD);
```

```
                while ( in < pp ) {
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
                }
                namelen = *((WORD *)out-1); /* cares for padding "bug" */
                if ( end - in < namelen ) {
                    goto RSVEnd;
                }
                *((WORD *)out-1) = realnamelen;
                *size += AO.SaveHeader.sInd + namelen;
                *outsize += sizeof(struct InDeX) + realnamelen;
                if ( realnamelen > namelen ) {
                    int j = namelen;
                    NCOPYB(out, in, j);
                    out += realnamelen - namelen;
                }
                else {
                    int j = realnamelen;
                    NCOPYB(out, in, j);
                    in += namelen - realnamelen;
                }
                ++numReadInd;
                continue;
            }
            /* Vectors */
            if ( *stage == 2 ) {
                if ( ind->nvectors <= numReadVec ) {
                    ++*stage;
                    continue;
                }
                if ( end - in < AO.SaveHeader.sVec ) {
                    goto RSVEnd;
                }
                if ( flip ) {
                    pp = in;
                    AO.FlipLONG(pp); pp += lenL;
                    while ( pp < in + AO.SaveHeader.sVec ) {
                        AO.FlipWORD(pp); pp += lenW;
                    }
                }
                pp = in + AO.SaveHeader.sVec;
                AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name     */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* complex  */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number   */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags    */
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node     */
                AO.ResizeWORD(in, out); in += lenW;                      /* namesize */
                realnamelen = *((WORD *)out);
                realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
                out += sizeof(WORD);
                AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
                while ( in < pp ) {
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
                }
                namelen = *((WORD *)out-1); /* cares for padding "bug" */
                if ( end - in < namelen ) {
                    goto RSVEnd;
                }
                *((WORD *)out-1) = realnamelen;
                *size += AO.SaveHeader.sVec + namelen;
                *outsize += sizeof(struct VeCtOr) + realnamelen;
                if ( realnamelen > namelen ) {
                    int j = namelen;
                    NCOPYB(out, in, j)
                    out += realnamelen - namelen;
                }
                else {
                    int j = realnamelen;
                    NCOPYB(out, in, j)
                    in += namelen - realnamelen;
                }
                ++numReadVec;
                continue;
            }
            /* Functions */
            if ( *stage == 3 ) {
                if ( ind->nfunctions <= numReadFun ) {
                    ++*stage;
                    continue;
                }
                if ( end - in < AO.SaveHeader.sFun ) {
                    goto RSVEnd;
                }
                if ( flip ) {
                    pp = in;
                    AO.FlipPOINTER(pp); pp += lenP;
                    AO.FlipLONG(pp); pp += lenL;
                    AO.FlipLONG(pp); pp += lenL;
```

```
                        while ( pp < in + AO.SaveHeader.sFun ) {
                            AO.FlipWORD(pp); pp += lenW;
                        }
                    }
                    pp = in + AO.SaveHeader.sFun;
                    outbuf = out;
                    AO.ResizePOINTER(in, out); in += lenP; out += sizeof(void *); /* tabl */
                    AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* symminfo  */
                    AO.ResizeLONG(in, out); in += lenL; out += sizeof(LONG); /* name      */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* commute   */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* complex   */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* number    */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* flags     */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* spec      */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* symmetric */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* numargs */
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* node      */
                    AO.ResizeWORD(in, out); in += lenW;                      /* namesize  */
                    realnamelen = *((WORD *)out);
                    realnamelen += sizeof(void *)-1; realnamelen &= -(sizeof(void *));
                    out += sizeof(WORD);
                    AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD); /* dimension */
                    while ( in < pp ) {
                        AO.ResizeWORD(in, out); in += lenW; out += sizeof(WORD);
                    }
                    namelen = *((WORD *)out-1); /* cares for padding "bug" */
                    if ( end - in < namelen ) {
                        goto RSVEnd;
                    }
                    *((WORD *)out-1) = realnamelen;
                    *size += AO.SaveHeader.sFun + namelen;
                    *outsize += sizeof(struct FuNcTiOn) + realnamelen;
                    if ( realnamelen > namelen ) {
                        int j = namelen;
                        NCOPYB(out, in, j);
                        out += realnamelen - namelen;
                    }
                    else {
                        int j = realnamelen;
                        NCOPYB(out, in, j);
                        in += namelen - realnamelen;
                    }
                    ++numReadFun;
                    /* we use the information whether a function is tensorial later in ReadSaveTerm */
                    AO.tensorList[((FUNCTIONS)outbuf)->number+FUNCTION] =
                        (UBYTE)(((FUNCTIONS)outbuf)->spec == TENSORFUNCTION);
                    continue;
                }
                /* handle numdummies */
                if ( end - in >= lenW ) {
                    if ( flip ) AO.FlipWORD(in);
                    AO.ResizeWORD(in, out);
                    *size += lenW;
                    *outsize += sizeof(WORD);
                }
                /* handle numfactors */
                if ( end - in >= lenW ) {
                    if ( flip ) AO.FlipWORD(in);
                    AO.ResizeWORD(in, out);
                    *size += lenW;
                    *outsize += sizeof(WORD);
                }
                /* handle vflags */
                if ( end - in >= lenW ) {
                    if ( flip ) AO.FlipWORD(in);
                    AO.ResizeWORD(in, out);
                    *size += lenW;
                    *outsize += sizeof(WORD);
                }
                /* handle uflags */
                if ( end - in >= lenW ) {
                    if ( flip ) AO.FlipWORD(in);
                    AO.ResizeWORD(in, out);
                    *size += lenW;
                    *outsize += sizeof(WORD);
                }
                return ( 0 );
            }

RSVEnd:
            /* we are here because the remaining buffer cannot hold the next
               struct. we position the file behind the last successfully translated
               struct and return. */
            ADDPOS(pos, *size);
            SeekFile(AO.SaveData.Handle, &pos, SEEK_SET);
            return ( 0 );
        } else {
```

```
        return ( ReadFile(AO.SaveData.Handle, buffer, *size) != *size );
    }
}

/*
        #] ReadSaveVariables :
        #[ ReadSaveTerm :
*/

UBYTE *
ReadSaveTerm32(UBYTE *bin, UBYTE *binend, UBYTE **bout, UBYTE *boutend, UBYTE *top, int terminbuf)
{
    GETIDENTITY

    UBYTE *boutbuf;
    INT32 len, j, id;
    INT32 *r, *t, *coeff, *end, *newtermsize, *rend;
    INT32 *newsubtermp;
    INT32 *in = (INT32 *)bin;
    INT32 *out = (INT32 *)*bout;

    /* if called recursively the term is already decompressed in buffer.
       is this the case? */
    if ( terminbuf ) {
        /* don't do any decompression, just adjust the pointers */
        len = *out;
        end = out + len;
        r = in + 1;
        rend = (INT32 *)boutend;
        coeff = end - ABS(*(end-1));
        newtermsize = (INT32 *)*bout;
        out = newtermsize + 1;
    }
    else {
        /* do decompression of necessary. always return if the space in the
           buffer is not sufficient */
        INT32 rbuf;
        r = (INT32 *)AR.CompressBuffer;
        rbuf = *r;
        len = j = *in;
        /* first copy from AR.CompressBuffer if necessary */
        if ( j < 0 ) {
            ++in;
            if ( (UBYTE *)in >= binend ) {
                return ( bin );
            }
            *out = len = -j + 1 + *in;
            end = out + *out;
            if ( (UBYTE *)end >= top ) {
                return ( bin );
            }
            ++out;
            *r++ = len;
            while ( ++j <= 0 ) {
                INT32 bb = *r++;
                *out++ = bb;
            }
            j = *in++;
        }
        else if ( j == 0 ) {
            /* care for padding words */
            while ( (UBYTE *)in < binend ) {
                *out++ = 0;
                if ( (UBYTE *)out > top ) {
                    return ( (UBYTE *)bin );
                }

                *r++ = 0;
                ++in;
            }
            *bout = (UBYTE *)out;
            return ( (UBYTE *)in );
        }
        else {
            end = out + len;
            if ( (UBYTE *)end >= top ) {
                return ( bin );
            }
        }
        if ( (UBYTE *)(in + j) >= binend ) {
            *(AR.CompressBuffer) = rbuf;
            return ( bin );
        }
        if ( (UBYTE *)out + j >= top ) {
            return ( bin );
        }
        /* second copy from input buffer */
```

```
        while ( --j >= 0 ) {
            INT32 bb = *in++;
            *r++ = *out++ = bb;
        }

        rend = r;
        r = (INT32 *)AR.CompressBuffer + 1;
        coeff = end - ABS(*(end-1));
        newtermsize = (INT32 *)*bout;
        out = newtermsize + 1;
    }

    /* iterate over subterms */
    while ( out < coeff ) {

        id = *out++;
        ++r;
        t = out + *out - 1;
        newsubtermp = out;
        ++out; ++r;

        if ( id == SYMBOL ) {
            while ( out < t ) {
                ++out; ++r; /* symbol number */
                /* if exponent is too big, rewrite as exponent function */
                if ( ABS(*out) >= MAXPOWER ) {
                    INT32 *a, *b;
                    INT32 n;
                    INT32 num = *(out-1);
                    INT32 exp = *out;
                    coeff += 9;
                    end += 9;
                    t += 9;
                    if ( (UBYTE *)end > top ) return ( bin );
                    out -= 3;
                    *out++ = EXPONENT;      /* id */
                    *out++ = 13;            /* size */
                    *out++ = 1;             /* dirtyflag */
                    *out++ = -SYMBOL;       /* first short arg */
                    *out++ = num;
                    *out++ = 8;             /* second arg, size */
                    *out++ = 0;             /* dirtyflag */
                    *out++ = 6;             /* term size */
                    *out++ = ABS(exp) & 0x0000FFFF;
                    *out++ = ABS(exp) » 16;
                    *out++ = 1;
                    *out++ = 0;
                    *out++ = ( exp < 0 ) ? -5 : 5;
                    a = ++r;
                    b = out;
                    n = rend - r;
                    NCOPYI32(b, a, n)
                }
                else {
                    ++out; ++r;
                }
            }
        }
        else if ( id == DOTPRODUCT ) {
            while ( out < t ) {
                AO.RenumberVec((UBYTE *)out); /* vector 1 */
                ++out; ++r;
                AO.RenumberVec((UBYTE *)out); /* vector 2 */
                ++out; ++r;
                /* if exponent is too big, rewrite as exponent function */
                if ( ABS(*out) >= MAXPOWER ) {
                    INT32 *a, *b;
                    INT32 n;
                    INT32 num1 = *(out-2);
                    INT32 num2 = *(out-1);
                    INT32 exp = *out;
                    coeff += 17;
                    end += 17;
                    t += 17;
                    if ( (UBYTE *)end > top ) return ( bin );
                    out -= 4;
                    *out++ = EXPONENT;      /* id */
                    *out++ = 22;            /* size */
                    *out++ = 1;             /* dirtyflag */
                    *out++ = 11;            /* first arg, size */
                    *out++ = 0;             /* dirtyflag */
                    *out++ = 9;             /* term size */
                    *out++ = DOTPRODUCT;    /* p1.p2 */
                    *out++ = 5;             /* subterm size */
                    *out++ = num1;          /* p1 */
                    *out++ = num2;          /* p2 */
                    *out++ = 1;             /* exponent */
```

```
                    *out++ = 1;             /* coeff */
                    *out++ = 1;
                    *out++ = 3;
                    *out++ = 8;             /* second arg, size */
                    *out++ = 0;             /* dirtyflag */
                    *out++ = 6;             /* term size */
                    *out++ = ABS(exp) & 0x0000FFFF;
                    *out++ = ABS(exp) » 16;
                    *out++ = 1;
                    *out++ = 0;
                    *out++ = ( exp < 0 ) ? -5 : 5;
                    a = ++r;
                    b = out;
                    n = rend - r;
                    NCOPYI32(b, a, n)
                }
                else {
                    ++out; ++r;
                }
            }
        }
        else if ( id == VECTOR ) {
            while ( out < t ) {
                AO.RenumberVec((UBYTE *)out); /* vector number */
                ++out; ++r;
                ++out; ++r; /* index, do nothing */
            }
        }
        else if ( id == INDEX ) {
/*          INT32 vectoroffset = -2 * *((INT32 *)AO.SaveHeader.wildoffset); */
            void *dummy = (void *)AO.SaveHeader.wildoffset;  /* to remove a warning about strict-aliasing rules in gcc */
            INT32 vectoroffset = -2 * *((INT32 *)dummy);
            while ( out < t ) {
                /* if there is a vector, renumber it */
                if ( *out < vectoroffset ) {
                    AO.RenumberVec((UBYTE *)out);
                }
                ++out; ++r;
            }
        }
        else if ( id == SUBEXPRESSION ) {
            /* nothing to translate */
            while ( out < t ) {
                ++out; ++r;
            }
        }
        else if ( id == DELTA ) {
            /* nothing to translate */
            r += t - out;
            out = t;
        }
        else if ( id == HAAKJE ) {
            /* nothing to translate */
            r += t - out;
            out = t;
        }
        else if ( id == GAMMA || id == LEVICIVITA || (id >= FUNCTION && AO.tensorList[id]) ) {
/*          INT32 vectoroffset = -2 * *((INT32 *)AO.SaveHeader.wildoffset); */
            void *dummy = (void *)AO.SaveHeader.wildoffset;  /* to remove a warning about strict-aliasing rules in gcc */
            INT32 vectoroffset = -2 * *((INT32 *)dummy);
            while ( out < t ) {
                /* if there is a vector as an argument, renumber it */
                if ( *out < vectoroffset ) {
                    AO.RenumberVec((UBYTE *)out);
                }
                ++out; ++r;
            }
        }
        else if ( id >= FUNCTION ) {
            INT32 *argEnd;
            UBYTE *newbin;

            ++out; ++r; /* dirty flags */

            /* loop over arguments */
            while ( out < t ) {
                if ( *out < 0 ) {
                    /* short notation arguments */
                    switch ( -*out ) {
                        case SYMBOL:
                            ++out; ++r;
                            ++out; ++r;
                            break;
                        case SNUMBER:
                            argEnd = out+2;
```

```
                        ++out; ++r;
                        if ( sizeof(WORD) == 2 ) {
                            /* resize if needed */
                            if ( *out > (1«15)-1 || *out < -(1«15)+1 ) {
                                INT32 *a, *b;
                                INT32 n;
                                INT32 num = *out;
                                coeff += 6;
                                end += 6;
                                argEnd += 6;
                                t += 6;
                                if ( (UBYTE *)end > top ) return ( bin );
                                --out;
                                *out++ = 8;          /* argument size */
                                *out++ = 0;          /* dirtyflag */
                                *out++ = 6;          /* term size */
                                *out++ = ABS(num) & 0x0000FFFF;
                                *out++ = ABS(num) » 16;
                                *out++ = 1;
                                *out++ = 0;
                                *out++ = ( num < 0 ) ? -5 : 5;
                                a = ++r;
                                b = out;
                                n = rend - r;
                                NCOPYI32(b, a, n)
                            }
                            else {
                                ++out; ++r;
                            }
                        }
                        else {
                            ++out; ++r;
                        }
                        break;
                    case VECTOR:
                        ++out; ++r;
                        AO.RenumberVec((UBYTE *)out);
                        ++out; ++r;
                        break;
                    case INDEX:
                        ++out; ++r;
                        ++out; ++r;
                        break;
                    case MINVECTOR:
                        ++out; ++r;
                        AO.RenumberVec((UBYTE *)out);
                        ++out; ++r;
                        break;
                    default:
                        if ( -*out >= FUNCTION ) {
                            ++out; ++r;
                            break;
                        } else {
                            MesPrint("short function code %d not implemented.", *out);
                            return ( (UBYTE *)in );
                        }
                }
            }
            else {
                /* long arguments */
                INT32 *newargsize = out;
                argEnd = out + *out;
                ++out; ++r;
                ++out; ++r; /* dirty flags */
                while ( out < argEnd ) {
                    INT32 *keepsizep = out + *out;
                    INT32 lenbuf = *out;
                    INT32 **ppp = &out; /* to avoid a compiler warning */
                    /* recursion */
                    newbin = ReadSaveTerm32((UBYTE *)r, binend, (UBYTE **)ppp, (UBYTE *)rend, top,
1);
                    r += lenbuf;
                    if ( newbin == (UBYTE *)r ) {
                        return ( (UBYTE *)in );
                    }
                    /* if the term done by recursion has changed in size,
                       we need to move the rest of the data accordingly */
                    if ( out > keepsizep ) {
                        INT32 *a, *b;
                        INT32 n;
                        INT32 extention = out - keepsizep;
                        a = r;
                        b = out;
                        n = rend - r;
                        NCOPYI32(b, a, n)
                        coeff += extention;
                        end += extention;
```

```
                            argEnd += extention;
                            t += extention;
                        }
                        else if ( out < keepsizep ) {
                            INT32 *a, *b;
                            INT32 n;
                            INT32 extention = keepsizep - out;
                            a = keepsizep;
                            b = out;
                            n = rend - r;
                            NCOPYI32(b, a, n)
                            coeff -= extention;
                            end -= extention;
                            argEnd -= extention;
                            t -= extention;
                        }
                    }
                    *newargsize = out - newargsize;
                }
            }
        }
        else {
            MesPrint("ID %d not recognized.", id);
            return ( (UBYTE *)in );
        }

        *newsubtermp = out - newsubtermp + 1;
    }

    if ( (UBYTE *)end >= top ) {
        return ( bin );
    }

    /* do coefficient and adjust term size */
    boutbuf = *bout;
    *bout = (UBYTE *)out;

    ResizeCoeff32(bout, (UBYTE *)end, top);

    if ( *bout >= top ) {
        *bout = boutbuf;
        return ( bin );
    }

    *newtermsize = (INT32 *)*bout - newtermsize;

    return ( (UBYTE *)in );
}

/*
        #] ReadSaveTerm :
        #[ ReadSaveExpression :
*/

WORD ReadSaveExpression(UBYTE *buffer, UBYTE *top, LONG *size, LONG *outsize)
{
    if ( AO.transFlag ) {
        UBYTE *in, *end, *out, *outend, *p;
        POSITION pos;
        LONG half;
        WORD lenW = AO.SaveHeader.lenWORD;

        /* remember the last file position in case an expression cannot be
           fully processed */
        TELLFILE(AO.SaveData.Handle,&pos);

        /* adjust 'size' depending on whether the translated data is bigger or
           smaller */
        half = (top-buffer)/2;
        if ( *size > half ) *size = half;
        if ( lenW < (WORD)sizeof(WORD) ) {
            if ( *size * (LONG)sizeof(WORD)/lenW > half ) *size = half*lenW/(LONG)sizeof(WORD);
        }
        else {
            if ( *size > half ) *size = half;
        }

        /* depending on the necessary resizing we position the input pointer
           either at the start of the buffer or in the middle. if the data will
           roughly remain the same size, we need only one processing step, so
           we put the 'in' at the middle and 'out' and the beginning. in the
           other cases we need two processing steps, so first we put 'in' at
           the beginning and write at the middle. the second step can then read
           from the middle and put its results at the beginning. */
        in = out = buffer;
        if ( lenW == sizeof(WORD) ) in += half;
        else out += half;
```

```
        end = in + *size;
        outend = out + *size;

        if ( ReadFile(AO.SaveData.Handle, in, *size) != *size ) {
            return ( MesPrint("Error(3) reading stored expression.") );
        }

        if ( AO.transFlag & 1 ) {
            p = in;
            end -= lenW;
            while ( p <= end ) {
                AO.FlipWORD(p); p += lenW;
            }
            end += lenW;
        }

        if ( lenW > (WORD)sizeof(WORD) ) {
            /* renumber first */
            do {
                outend = out+*size;
                if ( outend > top ) outend = top;
                p = ReadSaveTerm32(in, end, &out, outend, top, 0);
                if ( p == in ) break;
                in = p;
            } while ( in <= end - lenW );
            /* then resize */
            *size = in - buffer;
            in = buffer + half;
            end = out;
            out = buffer;

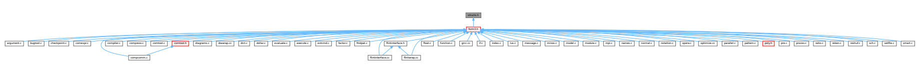
            while ( in < end ) {
                /* resize without checking */
                AO.ResizeNCWORD(in, out);
                in += lenW; out += sizeof(WORD);
            }
        }
        else {
            if ( lenW < (WORD)sizeof(WORD) ) {
                /* resize first */
                while ( in < end ) {
                    AO.ResizeWORD(in, out);
                    in += lenW; out += sizeof(WORD);
                }
                in = buffer + half;
                end = out;
                out = buffer;
            }
            /* then renumber */
            do {
                p = ReadSaveTerm32(in, end, &out,  buffer+half, buffer+half, 0);
                if ( p == in ) break;
                in = p;
            } while ( in <= end - sizeof(WORD) );
            *size = (in - buffer - half) * lenW / (ULONG)sizeof(WORD);
        }
        *outsize = out - buffer;
        ADDPOS(pos, *size);
        SeekFile(AO.SaveData.Handle, &pos, SEEK_SET);

        return ( 0 );
    }
    else {
        return ( ReadFile(AO.SaveData.Handle, buffer, *size) != *size );
    }
}

/*
        #] ReadSaveExpression :
    #] System Independent Saved Expressions :
*/
```

4.136 structs.h File Reference

This graph shows which files directly or indirectly include this file:



Data Structures

- struct PoSiTiOn
- struct STOREHEADER
- struct InDeXeNtRy
- struct FiLeInDeX
- struct FiLeDaTa
- struct VaRrEnUm
- struct ReNuMbEr
- struct LIST
- struct KEYWORD
- struct KEYWORDV
- struct NaMeNode
- struct NaMeTree
- struct tree
- struct MiNmAx
- struct BrAcKeTiNdEx
- struct BrAcKeTiNfO
- struct TaBlEs
- struct ExPrEsSiOn
- struct SyMbOl
- struct InDeX
- struct VeCtOr
- struct FuNcTiOn
- struct SeTs
- struct DuBiOuS
- struct FaCdOlLaR
- struct DoLlArS
- struct MoDoPtDoLlArS
- struct fixedset
- struct TaBlEbAsEsUbInDeX
- struct TaBlEbAsE
- struct FUN_INFO
- struct PaRtIcLe
- struct VeRtEx
- struct MoDeL
- struct NaMeSpAcE
- struct FiLe
- struct StreaM
- struct SpecTatoR
- struct TrAcEs
- struct TrAcEn
- struct pReVaR
- struct INSIDEINFO
- struct PRELOAD
- struct PROCEDURE
- struct DoLoOp
- struct bit_field
- struct HANDLERS
- struct CbUf
- struct ChAnNeL
- struct SETUPPARAMETERS
- struct NeStInG
- struct StOrEcAcHe
- struct PeRmUtE

- struct PeRmUtEp
- struct DiStRiBuTe
- struct PaRtI
- struct sOrT
- struct POLYMOD
- struct SHvariables
- struct COST
- struct MODNUM
- struct OPTIMIZE
- struct OPTIMIZERESULT
- struct DICTIONARY_ELEMENT
- struct DICTIONARY
- struct SWITCHTABLE
- struct SWITCH
- struct TERMINFO
- struct M_const
- struct P_const
- struct C_const
- struct S_const
- struct R_const
- struct T_const
- struct N_const
- struct O_const
- struct X_const
- struct AllGlobals
- struct FixedGlobals

Macros

- #define INFILEINDEX ((512-2∗sizeof(POSITION))/sizeof(INDEXENTRY))
- #define EMPTYININDEX (512-2∗sizeof(POSITION)-INFILEINDEX∗sizeof(INDEXENTRY))
- #define PHEAD
- #define PHEAD0 void
- #define BHEAD
- #define BHEAD0

Typedefs

- typedef struct PoSiTiOn **POSITION**
- typedef struct InDeXeNtRy INDEXENTRY
- typedef struct FiLeInDeX FILEINDEX
- typedef struct FiLeDaTa **FILEDATA**
- typedef struct VaRrEnUm VARRENUM
- typedef struct ReNuMbEr ∗ RENUMBER
- typedef struct NaMeNode NAMENODE
- typedef struct NaMeTree NAMETREE
- typedef struct tree COMPTREE
- typedef struct MiNmAx **MINMAX**
- typedef struct BrAcKeTiNdEx **BRACKETINDEX**
- typedef struct BrAcKeTiNfO **BRACKETINFO**
- typedef struct TaBlEs ∗ TABLES
- typedef struct ExPrEsSiOn ∗ **EXPRESSIONS**
- typedef struct SyMbOl ∗ **SYMBOLS**

- typedef struct InDeX * **INDICES**
- typedef struct VeCtOr * **VECTORS**
- typedef struct FuNcTiOn * FUNCTIONS
- typedef struct SeTs * **SETS**
- typedef struct DuBiOuS * **DUBIOUSV**
- typedef struct FaCdOlLaR **FACDOLLAR**
- typedef struct DoLlArS * **DOLLARS**
- typedef struct MoDoPtDoLlArS **MODOPTDOLLAR**
- typedef struct fixedset **FIXEDSET**
- typedef struct TaBlEbAsEsUbInDeX **TABLEBASESUBINDEX**
- typedef struct TaBlEbAsE **TABLEBASE**
- typedef struct PaRtIcLe **PARTICLE**
- typedef struct VeRtEx **VERTEX**
- typedef struct MoDeL **MODEL**
- typedef struct NaMeSpAcE **NAMESPACE**
- typedef struct FiLe FILEHANDLE
- typedef struct StreaM STREAM
- typedef struct SpecTatoR **SPECTATOR**
- typedef struct TrAcEs TRACES
- typedef struct TrAcEn * TRACEN
- typedef struct pReVaR PREVAR
- typedef struct DoLoOp DOLOOP
- typedef struct bit_field set_of_char[32]
- typedef struct bit_field * one_byte
- typedef WORD(* FINISHUFFLE) (WORD *)
- typedef WORD(* DO_UFFLE) (WORD *, WORD, WORD, WORD)
- typedef WORD(* COMPAREDUMMY) (WORD *, WORD *, WORD)
- typedef struct CbUf CBUF
- typedef struct ChAnNeL CHANNEL
- typedef struct NeStInG * NESTING
- typedef struct StOrEcAcHe * STORECACHE
- typedef struct PeRmUtE PERM
- typedef struct PeRmUtEp PERMP
- typedef struct DiStRiBuTe DISTRIBUTE
- typedef struct PaRtI PARTI
- typedef struct sOrT SORTING
- typedef struct AllGlobals ALLGLOBALS
- typedef WORD(* WCN) ()
- typedef WORD(* WCN2) ()
- typedef WORD(* COMPARE) (PHEAD WORD *, WORD *, WORD)
- typedef struct FixedGlobals FIXEDGLOBALS

Functions

- **STATIC_ASSERT** (sizeof(STOREHEADER)==512)
- **STATIC_ASSERT** (sizeof(FILEINDEX)==512)

4.136.1 Detailed Description

Contains definitions for global structs.

!!!CAUTION!!! Changes in this file will most likely have consequences for the recovery mechanism (see checkpoint.c). You need to care for the code in checkpoint.c as well and modify the code there accordingly!

The marker [D] is used in comments in this file to mark pointers to which dynamically allocated memory is assigned by a call to malloc() during runtime (in contrast to pointers that point into already allocated memory). This information is especially helpful if one needs to know which pointers need to be freed (cf. checkpoint.c).

Definition in file structs.h.

4.136.2 Macro Definition Documentation

4.136.2.1 INFILEINDEX

```
#define INFILEINDEX ((512-2*sizeof(POSITION))/sizeof(INDEXENTRY))
```

Maximum number of entries (struct InDeXeNtRy) in a file index (struct FiLeInDeX). Number is calculated such that the size of a file index is no more than 512 bytes.

Definition at line 119 of file structs.h.

4.136.2.2 EMPTYININDEX

```
#define EMPTYININDEX (512-2*sizeof(POSITION)-INFILEINDEX*sizeof(INDEXENTRY))
```

Number of empty filling bytes for a file index (struct FiLeInDeX). It is calculated such that the size of a file index is always 512 bytes.

Definition at line 124 of file structs.h.

4.136.2.3 PHEAD

```
#define PHEAD
```

Definition at line 2647 of file structs.h.

4.136.2.4 PHEAD0

```
#define PHEAD0 void
```

Definition at line 2648 of file structs.h.

4.136.2.5 BHEAD

```
#define BHEAD
```

Definition at line 2649 of file structs.h.

4.136.2.6 BHEAD0

```
#define BHEAD0
```

Definition at line 2650 of file structs.h.

4.136.3 Typedef Documentation

4.136.3.1 INDEXENTRY

```
typedef struct InDeXeNtRy INDEXENTRY
```

Defines the structure of an entry in a file index (see struct FiLeInDeX).

It represents one expression in the file.

4.136.3.2 FILEINDEX

```
typedef struct FiLeInDeX FILEINDEX
```

Defines the structure of a file index in store-files and save-files.

It contains several entries (see struct InDeXeNtRy) up to a maximum of INFILEINDEX.

The variable number has been made of type POSITION to avoid padding problems with some types of computers/↩ OS and keep system independence of the .sav files.

This struct is always 512 bytes long.

4.136.3.3 VARRENUM

```
typedef struct VaRrEnUm VARRENUM
```

Contains the pointers to an array in which a binary search will be performed.

4.136.3.4 RENUMBER

```
typedef struct ReNuMbEr * RENUMBER
```

Only symb.lo gets dynamically allocated. All other pointers points into this memory.

4.136.3.5 NAMENODE

```
typedef struct NaMeNode NAMENODE
```

The names of variables are kept in an array. Elements of type NAMENODE define a tree (that is kept balanced) that make it easy and fast to look for variables. See also NAMETREE.

4.136.3.6 NAMETREE

```
typedef struct NaMeTree NAMETREE
```

A struct of type NAMETREE controls a complete (balanced) tree of names for the compiler. The compiler maintains several of such trees and the system has been set up in such a way that one could define more of them if we ever want to work with local name spaces.

4.136.3.7 COMPTREE

```
typedef struct tree COMPTREE
```

The subexpressions in the compiler are kept track of in a (balanced) tree to reduce the need for subexpressions and hence save much space in large rhs expressions (like when we have xxxxxxx occurrences of objects like f(x+1,x+1) in which each x+1 becomes a subexpression. The struct that controls this tree is COMPTREE.

4.136.3.8 TABLES

```
typedef struct TaBlEs * TABLES
```

buffers, mm, flags, and prototype are always dynamically allocated, tablepointers only if needed (=0 if unallocated), boomlijst and argtail only for sparse tables.

Allocation is done for both the normal and the stub instance (spare), except for prototype and argtail which share memory.

4.136.3.9 FUNCTIONS

```
typedef struct FuNcTiOn * FUNCTIONS
```

Contains all information about a function. Also used for tables. It is used in the LIST elements of AC.

4.136.3.10 FILEHANDLE

```
typedef struct FiLe FILEHANDLE
```

The type FILEHANDLE is the struct that controls all relevant information of a file, whether it is open or not. The file may even not yet exist. There is a system of caches (PObuffer) and as long as the information to be written still fits inside the cache the file may never be created. There are variables that can store information about different types of files, like scratch files or sort files. Depending on what is available in the system we may also have information about gzip compression (currently sort file only) or locks (TFORM).

4.136.3.11 STREAM

```
typedef struct StreaM STREAM
```

Input is read from 'streams' which are represented by objects of type STREAM. A stream can be a file, a do-loop, a procedure, the string value of a preprocessor variable ..... When a new stream is opened we have to keep information about where to fall back in the parent stream to allow this to happen even in the middle of reading names etc as would be the case with a`i'b

4.136.3.12 TRACES

```
typedef struct TrAcEs TRACES
```

The struct TRACES keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue. The 4-dimensional traces are more complicated than the n-dimensional traces (see TRACEN) because of the extra tricks that can be used. They are responsible for the shorter final expressions.

4.136.3.13 TRACEN

```
typedef struct TrAcEn * TRACEN
```

The struct TRACEN keeps track of the progress during the expansion of a 4-dimensional trace. Each time a term gets generated the expansion tree continues in the next statement. When it returns it has to know where to continue.

4.136.3.14 PREVAR

```
typedef struct pReVaR PREVAR
```

An element of the type PREVAR is needed for each preprocessor variable.

4.136.3.15 DOLOOP

```
typedef struct DoLoOp DOLOOP
```

Each preprocessor do loop has a struct of type DOLOOP to keep track of all relevant parameters like where the beginning of the loop is, what the boundaries, increment and value of the loop parameter are, etc. Also we keep the whole loop inside a buffer of type PRELOAD

4.136.3.16 set_of_char

```
typedef struct bit_field set_of_char[32]
```

Used in set_in, set_set, set_del and set_sub.

Definition at line 952 of file structs.h.

4.136.3.17 one_byte

```
typedef struct bit_field* one_byte
```

Used in set_in, set_set, set_del and set_sub.

Definition at line 958 of file structs.h.

4.136.3.18 FINISHUFFLE

```
typedef WORD(* FINISHUFFLE) (WORD *)
```

Definition at line 978 of file structs.h.

4.136.3.19 DO_UFFLE

```
typedef WORD(* DO_UFFLE) (WORD *, WORD, WORD, WORD)
```

Definition at line 979 of file structs.h.

4.136.3.20 COMPAREDUMMY

```
typedef WORD(* COMPAREDUMMY) (WORD *, WORD *, WORD)
```

Definition at line 984 of file structs.h.

4.136.3.21 CBUF

```
typedef struct CbUf CBUF
```

The CBUF struct is used by the compiler. It is a compiler buffer of which since version 3.0 there can be many.

4.136.3.22 CHANNEL

```
typedef struct ChAnNeL CHANNEL
```

When we read input from text files we have to remember not only their handle but also their name. This is needed for error messages. Hence we call such a file a channel and reserve a struct of type CHANNEL to allow to lay this link.

4.136.3.23 NESTING

```
typedef struct NeStInG * NESTING
```

The NESTING struct is used when we enter the argument of functions and there is the possibility that we have to change something there. Because functions can be nested we have to keep track of all levels of functions in case we have to move the outer layers to make room for a larger function argument.

4.136.3.24 STORECACHE

```
typedef struct StOrEcAcHe * STORECACHE
```

The struct of type STORECACHE is used by a caching system for reading terms from stored expressions. Each thread should have its own system of caches.

4.136.3.25 PERM

```
typedef struct PeRmUtE PERM
```

The struct PERM is used to generate all permutations when the pattern matcher has to try to match (anti)symmetric functions.

4.136.3.26 PERMP

```
typedef struct PeRmUtEp PERMP
```

Like struct PERM but works with pointers.

4.136.3.27 DISTRIBUTE

```
typedef struct DiStRiBuTe DISTRIBUTE
```

The struct DISTRIBUTE is used to help the pattern matcher when matching antisymmetric tensors.

4.136.3.28 PARTI

```
typedef struct PaRtI PARTI
```

The struct PARTI is used to help determining whether a partition_ function can be replaced.

4.136.3.29 SORTING

```
typedef struct sOrT SORTING
```

The struct SORTING is used to control a sort operation. It includes a small and a large buffer and arrays for keeping track of various stages of the (merge) sorts. Each sort level has its own struct and different levels can have different sizes for its arrays. Also different threads have their own set of SORTING structs.

4.136.3.30 ALLGLOBALS

```
typedef struct AllGlobals ALLGLOBALS
```

Without pthreads (FORM) the ALLGLOBALS struct has all the global variables

4.136.3.31 WCN

```
typedef WORD(* WCN) ()
```

Definition at line 2657 of file structs.h.

4.136.3.32 WCN2

```
typedef WORD(* WCN2) ()
```

Definition at line 2658 of file structs.h.

4.136.3.33 COMPARE

```
typedef WORD(* COMPARE) (PHEAD WORD *, WORD *, WORD)
```

Definition at line 2661 of file structs.h.

4.136.3.34 FIXEDGLOBALS

```
typedef struct FixedGlobals FIXEDGLOBALS
```

The FIXEDGLOBALS struct is an anachronism. It started as the struct with global variables that needed initialization. It contains the elements Operation and OperaFind which define a very early way of automatically jumping to the proper operation. We find the results of it in parts of the file opera.c Later operations were treated differently in a more transparent way. We never changed the existing code. The most important part is currently the cTable which is used intensively in the compiler.

4.137 structs.h

Go to the documentation of this file.

```
/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
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 *   This file is part of FORM.
 *
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 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#ifndef __STRUCTS__

#define __STRUCTS__
#ifdef _MSC_VER
#include <wchar.h>  /* off_t */
#endif

/*
    #[ sav&store :
*/

typedef struct PoSiTiOn {
    off_t p1;
} POSITION;

/*  Next are the index structs for stored and saved expressions */

typedef struct {
    UBYTE headermark[8];
    UBYTE lenWORD;
    UBYTE lenLONG;
    UBYTE lenPOS;
    UBYTE lenPOINTER;
    UBYTE endianness[16];
    UBYTE sSym;
    UBYTE sInd;
    UBYTE sVec;
    UBYTE sFun;
    UBYTE maxpower[16];
    UBYTE wildoffset[16];
    UBYTE revision;
    UBYTE reserved[512-8-4-16-4-16-16-1];
} STOREHEADER;

STATIC_ASSERT(sizeof(STOREHEADER) == 512);
```

```
typedef struct InDeXeNtRy {
    POSITION    position;
    POSITION    length;
    POSITION    variables;
    LONG    CompressSize;
    WORD    nsymbols;
    WORD    nindices;
    WORD    nvectors;
    WORD    nfunctions;
    WORD    size;
    SBYTE   name[MAXENAME+1];
    PADPOSITION(0,1,0,5,MAXENAME+1);
} INDEXENTRY;

#define INFILEINDEX ((512-2*sizeof(POSITION))/sizeof(INDEXENTRY))
#define EMPTYININDEX (512-2*sizeof(POSITION)-INFILEINDEX*sizeof(INDEXENTRY))

typedef struct FiLeInDeX {
    POSITION    next;
    POSITION    number;
    INDEXENTRY expression[INFILEINDEX];
    SBYTE   empty[EMPTYININDEX];
} FILEINDEX;

STATIC_ASSERT(sizeof(FILEINDEX) == 512);

typedef struct FiLeDaTa {
    FILEINDEX Index;
    POSITION Fill;
    POSITION Position;
    WORD Handle;
    WORD dirtyflag;
    PADPOSITION(0,0,0,2,0);
} FILEDATA;

typedef struct VaRrEnUm {
    WORD  *start;
    WORD  *lo;
    WORD  *hi;
} VARRENUM;

typedef struct ReNuMbEr {
    POSITION   startposition;
    /* First stage renumbering */
    VARRENUM   symb;
    VARRENUM   indi;
    VARRENUM   vect;
    VARRENUM   func;
    /* Second stage renumbering */
    WORD       *symnum;
    WORD       *indnum;
    WORD       *vecnum;
    WORD       *funnum;
    PADPOSITION(4,0,0,0,sizeof(VARRENUM)*4);
} *RENUMBER;

/*
    #] sav&store :
    #[ Variables :
*/

typedef struct {
    void *lijst;
    char *message;
    int num;
    int maxnum;
    int size;
    int numglobal;
    int numtemp;
    int numclear;
    PADPOINTER(0,6,0,0);
} LIST;

typedef struct {
    char *name;
    TFUN func;
    int type;
    int flags;
} KEYWORD;

typedef struct {
    char *name;
    int *var;
    int type;
    int flags;
} KEYWORDV;
```

```c

typedef struct NaMeNode {
    LONG name;
    WORD parent;
    WORD left;
    WORD right;
    WORD balance;
    WORD type;
    WORD number;
    PADLONG(0,6,0);
} NAMENODE;

typedef struct NaMeTree {
    NAMENODE *namenode;
    UBYTE    *namebuffer;
    LONG      nodesize;
    LONG      nodefill;
    LONG      namesize;
    LONG      namefill;
    LONG      oldnamefill;
    LONG      oldnodefill;
    LONG      globalnamefill;
    LONG      globalnodefill;
    LONG      clearnamefill;
    LONG      clearnodefill;
    WORD      headnode;
    PADPOINTER(10,0,1,0);
} NAMETREE;

typedef struct tree {
    int parent;
    int left;
    int right;
    int value;
    int blnce;
    int usage;
} COMPTREE;

typedef struct MiNmAx {
    WORD mini;
    WORD maxi;
    WORD size;
} MINMAX;

typedef struct BrAcKeTiNdEx {   /* For indexing brackets in local expressions */
    POSITION start;             /* Place where bracket starts - start of expr */
    POSITION next;              /* Place of next indexed bracket in expr */
    LONG bracket;               /* Offset of position in bracketbuffer */
    LONG termsinbracket;
    PADPOSITION(0,2,0,0,0);
} BRACKETINDEX;

typedef struct BrAcKeTiNfO {
    BRACKETINDEX *indexbuffer;
    WORD  *bracketbuffer;
    LONG  bracketbuffersize;
    LONG  indexbuffersize;
    LONG  bracketfill;
    LONG  indexfill;
    WORD  SortType;
    PADPOINTER(4,0,1,0);
} BRACKETINFO;

typedef struct TaBlEs {
    WORD    *tablepointers;
#ifdef WITHPTHREADS
    WORD    **prototype;
    WORD    **pattern;
#else
    WORD    *prototype;
    WORD    *pattern;
#endif
    MINMAX  *mm;
    WORD    *flags;
    COMPTREE *boomlijst;
    UBYTE   *argtail;
    struct TaBlEs *spare;
    WORD    *buffers;
    LONG    totind;
    LONG    reserved;
    LONG    defined;
    LONG    mdefined;
    int     prototypeSize;
    int     numind;
    int     bounds;
    int     strict;
    int     sparse;
```

```
    int     numtree;
    int     rootnum;
    int     MaxTreeSize;
    WORD    bufnum;
    WORD    bufferssize;
    WORD    buffersfill;
    WORD    tablenum;
    WORD    mode;
    WORD    numdummies;
    PADPOINTER(4,8,6,0);
} *TABLES;

typedef struct ExPrEsSiOn {
    POSITION    onfile;
    POSITION    prototype;
    POSITION    size;
    RENUMBER renum;          /* For Renumbering of global stored expressions */
    BRACKETINFO *bracketinfo;
    BRACKETINFO *newbracketinfo;
    WORD    *renumlists;
    WORD    *inmem;          /* If in memory like e.g. a polynomial */
    LONG    counter;
    LONG    name;
    WORD    hidelevel;
    WORD    vflags;          /* Various flags */
    WORD    printflag;
    WORD    status;
    WORD    replace;
    WORD    node;
    WORD    whichbuffer;
    WORD    namesize;
    WORD    compression;
    WORD    numdummies;
    WORD    numfactors;
    WORD    sizeprototype;
    WORD    uflags;          /* User flags */
#ifdef PARALLELCODE
    WORD    partodo;         /* Whether to be done in parallel mode */
    PADPOSITION(5,2,0,14,0);
#else
    PADPOSITION(5,2,0,13,0);
#endif
} *EXPRESSIONS;

typedef struct SyMbOl {            /* Don't change unless altering .sav too */
    LONG    name;                /* Location in names buffer */
    WORD    minpower;            /* Minimum power admissible */
    WORD    maxpower;            /* Maximum power admissible */
    WORD    complex;             /* Properties wrt complex conjugation */
    WORD    number;              /* Number when stored in file */
    WORD    flags;               /* Used to indicate usage when storing */
    WORD    node;
    WORD    namesize;
    WORD    dimension;           /* For dimensionality checks */
    PADLONG(0,8,0);
} *SYMBOLS;

typedef struct InDeX {          /* Don't change unless altering .sav too */
    LONG    name;                /* Location in names buffer */
    WORD    type;                /* Regular or dummy */
    WORD    dimension;           /* Value of d_(n,n) or -number of symbol */
    WORD    number;              /* Number when stored in file */
    WORD    flags;               /* Used to indicate usage when storing */
    WORD    nmin4;               /* Used for n-4 if dimension < 0 */
    WORD    node;
    WORD    namesize;
    PADLONG(0,7,0);
} *INDICES;

typedef struct VeCtOr {            /* Don't change unless altering .sav too */
    LONG    name;                /* Location in names buffer */
    WORD    complex;             /* Properties under complex conjugation */
    WORD    number;              /* Number when stored in file */
    WORD    flags;               /* Used to indicate usage when storing */
    WORD    node;
    WORD    namesize;
    WORD    dimension;           /* For dimensionality checks */
    PADLONG(0,6,0);
} *VECTORS;

typedef struct FuNcTiOn {  /* Don't change unless altering .sav too */
    TABLES  tabl;
    LONG    symminfo;
    LONG    name;
    WORD    commute;
    WORD    complex;
    WORD    number;
```

```
    WORD    flags;
    WORD    spec;
    WORD    symmetric;
    WORD    node;
    WORD    namesize;
    WORD    dimension;     /* For dimensionality checks */
    WORD    maxnumargs;
    WORD    minnumargs;
    PADPOINTER(2,0,11,0);
} *FUNCTIONS;

typedef struct SeTs {
    LONG    name;               /* Location in names buffer */
    WORD    type;               /* Symbol, vector, index or function */
    WORD    first;              /* First element in setstore */
    WORD    last;               /* Last element in setstore (excluding) */
    WORD    node;
    WORD    namesize;
    WORD    dimension;          /* For dimensionality checks */
    WORD    flags;              /* Like: ordered */
    PADLONG(0,7,0);
} *SETS;

typedef struct DuBiOuS {      /* Undeclared objects. Just for compiler. */
    LONG    name;               /* Location in names buffer */
    WORD    node;
    WORD    dummy;
    PADLONG(0,2,0);
} *DUBIOUSV;

typedef struct FaCdOlLaR {
    WORD    *where;             /* A pointer(!) to the content */
    LONG    size;
    WORD    type;               /* Type can be DOLNUMBER or DOLTERMS */
    WORD    value;              /* in case it is a (short) number */
    PADPOINTER(1,0,2,0);
} FACDOLLAR;

typedef struct DoLlArS {
    WORD    *where;             /* A pointer(!) to the object */
    FACDOLLAR *factors;            /* an array of factors. nfactors elements */
#ifdef WITHPTHREADS
    pthread_mutex_t pthreadslockread;
    pthread_mutex_t pthreadslockwrite;
#endif
    LONG    size;               /* The number of words */
    LONG    name;
    WORD    type;
    WORD    node;
    WORD    index;
    WORD    zero;
    WORD    numdummies;
    WORD    nfactors;
#ifdef WITHPTHREADS
    PADPOINTER(2,0,6,sizeof(pthread_mutex_t)*2);
#else
    PADPOINTER(2,0,6,0);
#endif
} *DOLLARS;

typedef struct MoDoPtDoLlArS {
#ifdef WITHPTHREADS
    DOLLARS  dstruct;    /* If local dollar: list of DOLLARS for each thread */
#endif
    WORD    number;
    WORD    type;
#ifdef WITHPTHREADS
    PADPOINTER(0,0,2,0);
#endif
} MODOPTDOLLAR;

typedef struct fixedset {
    char *name;
    char *description;
    int type;
    int dimension;
} FIXEDSET;

typedef struct TaBlEbAsEsUbInDeX {
    POSITION where;
    LONG size;
    PADPOSITION(0,1,0,0,0);
} TABLEBASESUBINDEX;

typedef struct TaBlEbAsE {
    POSITION fillpoint;
    POSITION current;
```

```
    UBYTE *name;
    int *tablenumbers;      /* Number of each table */
    TABLEBASESUBINDEX *subindex;       /* For each table */
    int numtables;
    PADPOSITION(3,0,1,0,0);
} TABLEBASE;

typedef struct {
    WORD *location;
    int numargs;
    int numfunnies;
    int numwildcards;
    int symmet;
    int tensor;
    int commute;
    PADPOINTER(0,6,0,0);
} FUN_INFO;

typedef struct PaRtIcLe {
    WORD number; /* Number of the function */
    WORD spin;   /* +/- dimension of SU(2) representation */
    WORD mass;   /* Number of symbol or 0 */
    WORD type;   /* -1: anti particle, 1: particle, 0: own antiparticle */
} PARTICLE;

typedef struct VeRtEx {
    PARTICLE particles[MAXPARTICLES];
    WORD couplings[2*MAXCOUPLINGS];
    WORD nparticles;
    WORD ncouplings;
    WORD type;
    WORD error;
    WORD externonly;
    WORD spare;
} VERTEX;

typedef struct MoDeL {
    VERTEX **vertices;
    WORD *couplings;
    UBYTE *name;
    void *grccmodel;
    WORD legcouple[MAXLEGS+2];
    WORD nparticles;
    WORD nvertices;
    WORD invertices;
    WORD sizevertices;
    WORD sizecouplings;
    WORD ncouplings;
    WORD error;
    WORD dummy;
} MODEL;

/*
 *  The struct NAMESPACE is used for the namespace info. It is a
 *  doubly linked list of nested namespaces.
 */

typedef struct NaMeSpAcE {
    struct NaMeSpAcE *previous;
    struct NaMeSpAcE *next;
    NAMETREE  *usenames;
    UBYTE     *name;
} NAMESPACE;


/*
    #] Variables :
    #[ Files :
*/

typedef struct FiLe {
    POSITION POposition;         /* File position */
    POSITION filesize;           /* Because SEEK_END is unsafe on IBM */
    WORD *PObuffer;              /* Address of the intermediate buffer */
    WORD *POstop;                /* End of the buffer */
    WORD *POfill;                /* Fill position of the buffer */
    WORD *POfull;                /* Full buffer when only cached */
#ifdef WITHPTHREADS
    WORD *wPObuffer;              /* Address of the intermediate worker buffer */
    WORD *wPOstop;                /* End of the worker buffer */
    WORD *wPOfill;                /* Fill position of the worker buffer */
    WORD *wPOfull;                /* Full buffer when only worker cached */
#endif
    char *name;                  /* name of the file */
#ifdef WITHZLIB
    z_streamp zsp;               /* The pointer to the stream struct for gzip */
    Bytef *ziobuffer;            /* The output buffer for compression */
```

```
#endif
    ULONG numblocks;            /* Number of blocks in file */
    ULONG inbuffer;             /* Block in the buffer */
    LONG POsize;                /* size of the buffer */
#ifdef WITHZLIB
    LONG ziosize;               /* size of the zoutbuffer */
#endif
#ifdef WITHPTHREADS
    LONG wPOsize;                /* size of the worker buffer */
    pthread_mutex_t pthreadslock;
#endif
    int handle;
    int active;                 /* File is open or closed. Not used. */
#ifdef WITHPTHREADS
#ifdef WITHZLIB
    PADPOSITION(11,5,2,0,sizeof(pthread_mutex_t));
#else
    PADPOSITION(9,4,2,0,sizeof(pthread_mutex_t));
#endif
#else
#ifdef WITHZLIB
    PADPOSITION(7,4,2,0,0);
#else
    PADPOSITION(5,3,2,0,0);
#endif
#endif
} FILEHANDLE;

typedef struct StreaM {
    off_t fileposition;
    off_t linenumber;
    off_t prevline;
    UBYTE *buffer;
    UBYTE *pointer;
    UBYTE *top;
    UBYTE *FoldName;
    UBYTE *name;
    UBYTE *pname;
    LONG buffersize;
    LONG bufferposition;
    LONG inbuffer;
    int previous;
    int handle;
    int type;
    int prevars;
    int previousNoShowInput;
    int eqnum;
    int afterwards;
    int olddelay;
    int oldnoshowinput;
    UBYTE isnextchar;
    UBYTE nextchar[2];
    UBYTE reserved;
    PADPOSITION(6,3,9,0,4);
} STREAM;

typedef struct SpecTatoR {
    POSITION position;    /* The place where we will be writing */
    POSITION readpos;     /* The place from which we read */
    FILEHANDLE *fh;
    char *name;           /* We identify the spectator by the name of the expression */
    WORD exprnumber;      /* During running we use the number. */
    WORD flags;           /* local, global? */
    PADPOSITION(2,0,0,2,0);
} SPECTATOR;

/*
    #] Files :
    #[ Traces :
*/

typedef struct TrAcEs {            /* For computing 4 dimensional traces */
    WORD        *accu;      /* NUMBER * 2 */
    WORD        *accup;
    WORD        *termp;
    WORD        *perm;      /* number */
    WORD        *inlist;    /* number */
    WORD        *nt3;       /* number/2 */
    WORD        *nt4;       /* number/2 */
    WORD        *j3;        /* number*2 */
    WORD        *j4;        /* number*2 */
    WORD        *e3;        /* number*2 */
    WORD        *e4;        /* number */
    WORD        *eers;      /* number/2 */
    WORD        *mepf;      /* number/2 */
    WORD        *mdel;      /* number/2 */
    WORD        *pepf;      /* number*2 */
```

```
    WORD        *pdel;      /* number*3/2 */
    WORD        sgn;
    WORD        stap;
    WORD        step1,kstep,mdum;
    WORD        gamm,ad,a3,a4,lc3,lc4;
    WORD        sign1,sign2,gamma5,num,level,factor,allsign;
    WORD        finalstep;
    PADPOINTER(0,0,19,0);
} TRACES;

typedef struct TrAcEn {            /* For computing n dimensional traces */
    WORD        *accu;      /* NUMBER */
    WORD        *accup;
    WORD        *termp;
    WORD        *perm;      /* number */
    WORD        *inlist;    /* number */
    WORD        sgn,num,level,factor,allsign;
    PADPOINTER(0,0,5,0);
} *TRACEN;

/*
    #] Traces :
    #[ Preprocessor :
*/

typedef struct pReVaR {
    UBYTE *name;
    UBYTE *value;
    UBYTE *argnames;
    int nargs;
    int wildarg;
    PADPOINTER(0,2,0,0);
} PREVAR;

typedef struct {
    WORD *buffer;
    int oldcompiletype;
    int oldparallelflag;
    int oldnumpotmoddollars;
    WORD size;
    WORD numdollars;
    WORD oldcbuf;
    WORD oldrbuf;
    WORD inscbuf;
    WORD oldcnumlhs;
    PADPOINTER(0,3,6,0);
} INSIDEINFO;

typedef struct {
    UBYTE   *buffer;
    LONG    size;
    PADPOINTER(1,0,0,0);
} PRELOAD;

typedef struct {
    PRELOAD p;
    UBYTE   *name;
    int     loadmode;
    PADPOINTER(0,1,0,0);
} PROCEDURE;

typedef struct DoLoOp {
    PRELOAD p;
    UBYTE *name;
    UBYTE *vars;        /* for {} or name of expression */
    UBYTE *contents;
    UBYTE *dollarname;
    LONG startlinenumber;
    LONG firstnum;
    LONG lastnum;
    LONG incnum;
    int type;
    int NoShowInput;
    int errorsinloop;
    int firstloopcall;
    WORD firstdollar;   /* When >= 0 we have to get the value from a dollar */
    WORD lastdollar;    /* When >= 0 we have to get the value from a dollar */
    WORD incdollar;     /* When >= 0 we have to get the value from a dollar */
    WORD NumPreTypes;
    WORD PreIfLevel;
    WORD PreSwitchLevel;
    PADPOINTER(4,4,6,0);
} DOLOOP;

struct  bit_field {    /* Assume 8 bits per byte */
    UBYTE bit_0        : 1;
    UBYTE bit_1        : 1;
```

```
    UBYTE bit_2         : 1;
    UBYTE bit_3         : 1;
    UBYTE bit_4         : 1;
    UBYTE bit_5         : 1;
    UBYTE bit_6         : 1;
    UBYTE bit_7         : 1;
/*
    UINT bit_0         : 1;
    UINT bit_1         : 1;
    UINT bit_2         : 1;
    UINT bit_3         : 1;
    UINT bit_4         : 1;
    UINT bit_5         : 1;
    UINT bit_6         : 1;
    UINT bit_7         : 1;
*/
};

typedef struct bit_field set_of_char[32];

typedef struct bit_field *one_byte;

typedef struct {
    WORD newlogonly;
    WORD newhandle;
    WORD oldhandle;
    WORD oldlogonly;
    WORD oldprinttype;
    WORD oldsilent;
} HANDLERS;

/*
    #] Preprocessor :
    #[ Varia :
*/

typedef WORD (*FINISHUFFLE)(WORD *);
typedef WORD (*DO_UFFLE)(WORD *,WORD,WORD,WORD);

#ifdef WITHPTHREADS
typedef WORD (*COMPAREDUMMY)(void *,WORD *,WORD *,WORD);
#else
typedef WORD (*COMPAREDUMMY)(WORD *,WORD *,WORD);
#endif

typedef struct CbUf {
    WORD *Buffer;
    WORD *Top;
    WORD *Pointer;
    WORD **lhs;
    WORD **rhs;
    LONG *CanCommu;
    LONG *NumTerms;
    WORD *numdum;
    WORD *dimension;
    COMPTREE *boomlijst;
    LONG BufferSize;
    int numlhs;
    int numrhs;
    int maxlhs;
    int maxrhs;
    int mnumlhs;
    int mnumrhs;
    int numtree;
    int rootnum;
    int MaxTreeSize;
    PADPOINTER(1,9,0,0);
} CBUF;

typedef struct ChAnNeL {
    char *name;
    int handle;
    PADPOINTER(0,1,0,0);
} CHANNEL;

typedef struct {
    UBYTE *parameter;
    int type;
    int flags;
    LONG value;
} SETUPPARAMETERS;

typedef struct NeStInG {
    WORD *termsize;
    WORD *funsize;
    WORD *argsize;
} *NESTING;
```

```
typedef struct StOrEcAcHe {
    POSITION position;
    POSITION toppos;
    struct StOrEcAcHe *next;
    WORD buffer[2];
    PADPOSITION(1,0,0,2,0);
} *STORECACHE;

typedef struct PeRmUtE {
    WORD *objects;
    WORD sign;
    WORD n;
    WORD cycle[MAXMATCH];
    PADPOINTER(0,0,MAXMATCH+2,0);
} PERM;

typedef struct PeRmUtEp {
    WORD **objects;
    WORD sign;
    WORD n;
    WORD cycle[MAXMATCH];
    PADPOINTER(0,0,MAXMATCH+2,0);
} PERMP;

typedef struct DiStRiBuTe {
    WORD *obj1;
    WORD *obj2;
    WORD *out;
    WORD sign;
    WORD n1;
    WORD n2;
    WORD n;
    WORD cycle[MAXMATCH];
    PADPOINTER(0,0,(MAXMATCH+4),0);
} DISTRIBUTE;

typedef struct PaRtI {
    WORD *psize;  /* the sizes of the partitions */
    WORD *args;   /* the offsets of the arguments to be partitioned */
    WORD *nargs;  /* argument numbers (different number = different argument) */
    WORD *nfun;   /* the functions into which the partitions go */
    WORD numargs; /* the number of arguments to be partitioned */
    WORD numpart; /* the number of partitions */
    WORD where;   /* offset of the function in the term */
    PADPOINTER(0,0,3,0);
} PARTI;

typedef struct sOrT {
    FILEHANDLE file;            /* The own sort file */
    POSITION SizeInFile[3];     /* Sizes in the various files */
    WORD *lBuffer;              /* The large buffer */
    WORD *lTop;                 /* End of the large buffer */
    WORD *lFill;                /* The filling point of the large buffer */
    WORD *used;                 /*  auxiliary during actual sort */
    WORD *sBuffer;              /* The small buffer */
    WORD *sTop;                 /* End of the small buffer */
    WORD *sTop2;                /* End of the extension of the small buffer */
    WORD *sHalf;                /* Halfway point in the extension */
    WORD *sFill;                /* Filling point in the small buffer */
    WORD **sPointer;            /* Pointers to terms in the small buffer */
    WORD **PoinFill;            /* Filling point for pointers to the sm.buf */
    WORD *cBuffer;              /* Compress buffer (if it exists) */
    WORD **Patches;             /* Positions of patches in large buffer */
    WORD **pStop;               /* Ends of patches in the large buffer */
    WORD **poina;               /*  auxiliary during actual sort */
    WORD **poin2a;              /*  auxiliary during actual sort */
    WORD *ktoi;                 /*  auxiliary during actual sort */
    WORD *tree;                 /*  auxiliary during actual sort */
#ifdef WITHZLIB
    WORD *fpcompressed;         /*  is output filepatch compressed? */
    WORD *fpincompressed;       /*  is input filepatch compressed? */
    z_streamp zsparray;         /*  the array of zstreams for decompression */
#endif
    POSITION *fPatches;         /* Positions of output file patches */
    POSITION *inPatches;        /* Positions of input file patches */
    POSITION *fPatchesStop;     /* Positions of output file patches */
    POSITION *iPatches;         /* Input file patches, Points to fPatches or inPatches */
    FILEHANDLE *f;              /* The actual output file */
    FILEHANDLE **ff;            /* Handles for a staged sort */
    LONG sTerms;                /* Terms in small buffer */
    LONG LargeSize;             /* Size of large buffer (in words) */
    LONG SmallSize;             /* Size of small buffer (in words) */
    LONG SmallEsize;            /* Size of small + extension (in words) */
    LONG TermsInSmall;          /* Maximum number of terms in small buffer */
    LONG Terms2InSmall;         /* with extension for polyfuns etc. */
    LONG GenTerms;              /* Number of generated terms */
```

```
    LONG TermsLeft;             /* Number of terms still in existence */
    LONG GenSpace;              /* Amount of space of generated terms */
    LONG SpaceLeft;             /* Space needed for still existing terms */
    LONG putinsize;             /* Size of buffer in putin */
    LONG ninterms;              /* Which input term ? */
    int MaxPatches;             /* Maximum number of patches in large buffer */
    int MaxFpatches;            /* Maximum number of patches in one filesort */
    int type;                   /* Main, function or sub(routine) */
    int lPatch;                 /* Number of patches in the large buffer */
    int fPatchN1;               /* Number of patches in input file */
    int PolyWise;               /* Is there a polyfun and if so, where? */
    int PolyFlag;               /*  */
    int cBufferSize;            /* Size of the compress buffer */
    int maxtermsize;            /* Keeps track for buffer allocations */
    int newmaxtermsize;         /* Auxiliary for maxtermsize */
    int outputmode;             /* Tells where the output is going */
    int stagelevel;             /* In case we have a 'staged' sort */
    WORD fPatchN;               /* Number of patches on file (output) */
    WORD inNum;                 /* Number of patches on file (input) */
    WORD stage4;                /* Are we using stage4? */
#ifdef WITHZLIB
    PADPOSITION(27,12,12,3,0);
#else
    PADPOSITION(24,12,12,3,0);
#endif
} SORTING;

#ifdef WITHPTHREADS

typedef struct SoRtBlOcK {
    pthread_mutex_t *MasterBlockLock;
    WORD    **MasterStart;
    WORD    **MasterFill;
    WORD    **MasterStop;
    int     MasterNumBlocks;
    int     MasterBlock;
    int     FillBlock;
    PADPOINTER(0,3,0,0);
} SORTBLOCK;
#endif

#ifdef DEBUGGER
typedef struct DeBuGgInG {
    int eflag;
    int printflag;
    int logfileflag;
    int stdoutflag;
} DEBUGSTR;
#endif

#ifdef WITHPTHREADS

typedef struct ThReAdBuCkEt {
    POSITION *deferbuffer;      /* For Keep Brackets: remember position */
    WORD *threadbuffer;         /* Here are the (primary) terms */
    WORD *compressbuffer;       /* For keep brackets we need the compressbuffer */
    LONG threadbuffersize;      /* Number of words in threadbuffer */
    LONG ddterms;               /* Number of primary+secondary terms represented */
    LONG firstterm;             /* The number of the first term in the bucket */
    LONG firstbracket;          /* When doing complete brackets */
    LONG lastbracket;           /* When doing complete brackets */
    pthread_mutex_t lock;       /* For the load balancing phase */
    int  free;                  /* Status of the bucket */
    int  totnum;                /* Total number of primary terms */
    int  usenum;                /* Which is the term being used at the moment */
    int  busy;                  /*  */
    int  type;                  /* Doing brackets? */
    PADPOINTER(5,5,0,sizeof(pthread_mutex_t));
} THREADBUCKET;

#endif

typedef struct {
    WORD    *coefs;             /* The array of coefficients */
    WORD    numsym;             /* The number of the symbol in the polynomial */
    WORD    arraysize;          /* The size of the allocation of coefs */
    WORD    polysize;           /* The maximum power in the polynomial */
    WORD    modnum;             /* The prime number of the modulus */
} POLYMOD;

typedef struct {
    WORD    *outterm;            /* Used in DoShuffle/Merge/FinishShuffle system */
    WORD    *outfun;
    WORD    *incoef;
    WORD    *stop1;
    WORD    *stop2;
    WORD    *ststop1;
```

```
    WORD    *ststop2;
    FINISHUFFLE finishuf;
    DO_UFFLE    do_uffle;
    LONG    combilast;
    WORD    nincoef;
    WORD    level;
    WORD    thefunction;
    WORD    option;
    PADPOINTER(1,0,4,0);
} SHvariables;

typedef struct {                /* Used for computing calculational cost in optim.c */
    LONG    add;
    LONG    mul;
    LONG    div;
    LONG    pow;
} COST;

typedef struct {
    UWORD   *a;     /* The number array */
    UWORD   *m;     /* The modulus array */
    WORD    na;     /* Size of the number */
    WORD    nm;     /* size of the number in the modulus array */
} MODNUM;

/*
    Struct for optimizing outputs. If changed, do not forget to change
    the padding information in the AO struct.
*/
typedef struct {
    union { /* we do this to allow padding */
        float fval;
        int ival[2]; /* This should be enough */
    } mctsconstant;
    int   horner;
    int   hornerdirection;
    int   method;
    int   mctstimelimit;
    int   mctsnumexpand;
    int   mctsnumkeep;
    int   mctsnumrepeat;
    int   greedytimelimit;
    int   greedyminnum;
    int   greedymaxperc;
    int   printstats;
    int   debugflags;
    int   schemeflags;
    int   mctsdecaymode;
    int   saIter; /* Simulated annealing updates */
    union {
        float fval;
        int ival[2];
    } saMaxT; /* Maximum temperature of SA */
    union {
        float fval;
        int ival[2];
    } saMinT; /* Minimum temperature of SA */
    int    spare;
} OPTIMIZE;

typedef struct {
    WORD  *code;
    UBYTE *nameofexpr;  /* It is easier to remember an expression by name */
    LONG  codesize;     /* We need this for the checkpoints */
    WORD  exprnr;       /* Problem here is: we renumber them in execute.c */
    WORD  minvar;
    WORD  maxvar;

    PADPOSITION(2,1,0,3,0);
} OPTIMIZERESULT;

typedef struct {
    WORD *lhs;      /* Object to be replaced */
    WORD *rhs;      /* Depending on the type it will be UBYTE* or WORD* */
    int type;
    int size;       /* Size of the lhs */
} DICTIONARY_ELEMENT;

typedef struct {
    DICTIONARY_ELEMENT **elements;
    UBYTE *name;
    int sizeelements;
    int numelements;
    int numbers;        /* deal with numbers */
    int variables;      /* deal with single variables */
    int characters;     /* deal with special characters */
    int funwith;        /* deal with functions with arguments */
```

```
    int gnumelements;   /* if .global shrinks the dictionary */
    int ranges;
} DICTIONARY;

typedef struct {
    WORD ncase;
    WORD value;
    WORD compbuffer;
} SWITCHTABLE;

typedef struct {
    SWITCHTABLE *table;
    SWITCHTABLE defaultcase;
    SWITCHTABLE endswitch;
    WORD typetable;
    WORD maxcase;
    WORD mincase;
    WORD numcases;
    WORD tablesize;
    WORD caseoffset;
    WORD iflevel;
    WORD whilelevel;
    WORD nestingsum;
    WORD padding;
} SWITCH;

typedef struct {
    WORD *term;
    void *currentModel;
    void *currentMODEL;
    WORD *legcouple[MAXLEGS+1];
    LONG numtopo;
    LONG numdia;
    WORD diaoffset;
    WORD level;
    WORD externalset;
    WORD internalset;
    WORD numextern;
    WORD flags;
} TERMINFO;

/*
    #] Varia :
    #[ A :
        #[ M : The M struct is for global settings at startup or .clear
*/

struct M_const {
    POSITION zeropos;            /* (M) is zero */
    SORTING *S0;
    UWORD   *gcmod;
    UWORD   *gpowmod;
    UBYTE   *TempDir;            /* (M) Path with where the temporary files go */
    UBYTE   *TempSortDir;        /* (M) Path with where the sort files go */
    UBYTE   *IncDir;             /* (M) Directory path for include files */
    UBYTE   *InputFileName;      /* (M) */
    UBYTE   *LogFileName;        /* (M) */
    UBYTE   *OutBuffer;          /* (M) Output buffer in pre.c */
    UBYTE   *Path;               /* (M) */
    UBYTE   *SetupDir;           /* (M) Directory with setup file */
    UBYTE   *SetupFile;          /* (M) Name of setup file */
    UBYTE   *gFortran90Kind;
    UBYTE   *gextrasym;
    UBYTE   *ggextrasym;
    UBYTE   *oldnumextrasymbols;
    SPECTATOR *SpectatorFiles;
#ifdef WITHPTHREADS
    pthread_rwlock_t handlelock;  /* (M) */
    pthread_mutex_t storefilelock; /* (M) */
    pthread_mutex_t sbuflock;     /* (M) Lock for writing in the AM.sbuffer */
    LONG    ThreadScratSize;      /* (M) Size of Fscr[0/2] buffers of the workers */
    LONG    ThreadScratOutSize;   /* (M) Size of Fscr[1] buffers of the workers */
#endif
    LONG    MaxTer;              /* (M) Maximum term size. Fixed at setup. In Bytes!!!*/
    LONG    CompressSize;        /* (M) Size of Compress buffer */
    LONG    ScratSize;           /* (M) Size of Fscr[] buffers */
    LONG    HideSize;            /* (M) Size of Fscr[2] buffer */
    LONG    SizeStoreCache;      /* (M) Size of the caches for reading global expr. */
    LONG    MaxStreamSize;       /* (M) Maximum buffer size in reading streams */
    LONG    SIOsize;             /* (M) Sort InputOutput buffer size */
    LONG    SLargeSize;          /* (M) */
    LONG    SSmallEsize;         /* (M) */
    LONG    SSmallSize;          /* (M) */
    LONG    STermsInSmall;       /* (M) */
    LONG    MaxBracketBufferSize; /* (M) Max Size for B+ or AB+ per expression */
    LONG    hProcessBucketSize;  /* (M) */
    LONG    gProcessBucketSize;  /* (M) */
```

```
    LONG    shmWinSize;            /* (M) size for shared memory window used in communications */
    LONG    OldChildTime;          /* (M) Zero time. Needed in timer. */
    LONG    OldSecTime;            /* (M) Zero time for measuring wall clock time */
    LONG    OldMilliTime;          /* (M) Same, but milli seconds */
    LONG    WorkSize;              /* (M) Size of WorkSpace */
    LONG    gThreadBucketSize;     /* (C) */
    LONG    ggThreadBucketSize;    /* (C) */
    LONG    SumTime;               /*     Used in .clear */
    LONG    SpectatorSize;         /*     Size of the buffer in bytes */
    LONG    TimeLimit;             /*     Limit in sec to the total real time */
#ifdef WITHFLOAT
    LONG    gDefaultPrecision;     /* (M) Default precision in bits for float_ */
    LONG    gMaxWeight;            /* (M) Maximum weight for MZV or Euler */
    LONG    ggDefaultPrecision;    /* (M) Default precision in bits for float_ */
    LONG    ggMaxWeight;           /* (M) Maximum weight for MZV or Euler */
#endif
    int     FileOnlyFlag;          /* (M) Writing only to file */
    int     Interact;              /* (M) Interactive mode flag */
    int     MaxParLevel;           /* (M) Maximum nesting of parentheses */
    int     OutBufSize;            /* (M) size of OutBuffer */
    int     SMaxFpatches;          /* (M) */
    int     SMaxPatches;           /* (M) */
    int     StdOut;                /* (M) Regular output channel */
    int     ginsidefirst;          /* (M) Not used yet */
    int     gDefDim;               /* (M) */
    int     gDefDim4;              /* (M) */
    int     NumFixedSets;          /* (M) Number of the predefined sets */
    int     NumFixedFunctions;     /* (M) Number of built in functions */
    int     rbufnum;               /* (M) startup compiler buffer */
    int     dbufnum;               /* (M) dollar variables */
    int     sbufnum;               /* (M) subterm variables */
    int     zbufnum;               /* (M) special values */
    int     SkipClears;            /* (M) Number of .clear to skip at start */
    int     gTokensWriteFlag;      /* (M) */
    int     gfunpowers;            /* (M) */
    int     gStatsFlag;            /* (M) */
    int     gNamesFlag;            /* (M) */
    int     gCodesFlag;            /* (M) */
    int     gSortType;             /* (M) */
    int     gproperorderflag;      /* (M) */
    int     hparallelflag;         /* (M) */
    int     gparallelflag;         /* (M) */
    int     totalnumberofthreads;  /* (M) */
    int     gSizeCommuteInSet;
    int     gThreadStats;
    int     ggThreadStats;
    int     gFinalStats;
    int     ggFinalStats;
    int     gThreadsFlag;
    int     ggThreadsFlag;
    int     gThreadBalancing;
    int     ggThreadBalancing;
    int     gThreadSortFileSynch;
    int     ggThreadSortFileSynch;
    int     gProcessStats;
    int     ggProcessStats;
    int     gOldParallelStats;
    int     ggOldParallelStats;
    int     maxFlevels;            /* () maximum function levels */
    int     resetTimeOnClear;      /* (M) */
    int     gcNumDollars;          /* () number of dollars for .clear */
    int     MultiRun;
    int     gNoSpacesInNumbers;    /* For very long numbers */
    int     ggNoSpacesInNumbers;   /* For very long numbers */
    int     gIsFortran90;
    int     PrintTotalSize;
    int     fbuffersize;           /* Size for the AT.fbufnum factorization caches */
    int     gOldFactArgFlag;
    int     ggOldFactArgFlag;
    int     gnumextrasym;
    int     ggnumextrasym;
    int     NumSpectatorFiles;     /* Elements used in AM.spectatorfiles; */
    int     SizeForSpectatorFiles; /* Size in AM.spectatorfiles; */
    int     gOldGCDflag;
    int     ggOldGCDflag;
    int     gWTimeStatsFlag;
    int     ggWTimeStatsFlag;
    int     jumpratio;
    WORD    MaxTal;                /* (M) Maximum number of words in a number */
    WORD    IndDum;                /* (M) Basis value for dummy indices */
    WORD    DumInd;                /* (M) */
    WORD    WilInd;                /* (M) Offset for wildcard indices */
    WORD    gncmod;                /* (M) Global setting of modulus. size of gcmod */
    WORD    gnpowmod;              /* (M) Global printing as powers. size gpowmod */
    WORD    gmodmode;              /* (M) Global mode for modulus */
    WORD    gUnitTrace;            /* (M) Global value of Tr[1] */
    WORD    gOutputMode;           /* (M) */
```

```
    WORD    gOutputSpaces;          /* (M) */
    WORD    gOutNumberType;         /* (M) */
    WORD    gCnumpows;              /* (M) */
    WORD    gUniTrace[4];           /* (M) */
    WORD    MaxWildcards;           /* (M) Maximum number of wildcards */
    WORD    mTraceDum;              /* (M) Position/Offset for generated dummies */
    WORD    OffsetIndex;            /* (M) */
    WORD    OffsetVector;           /* (M) */
    WORD    RepMax;                 /* (M) Max repeat levels */
    WORD    LogType;                /* (M) Type of writing to log file */
    WORD    ggStatsFlag;            /* (M) */
    WORD    gLineLength;            /* (M) */
    WORD    qError;                 /* (M) Only error checking {-c option} */
    WORD    FortranCont;            /* (M) Fortran Continuation character */
    WORD    HoldFlag;               /* (M) Exit on termination? */
    WORD    Ordering[15];           /* (M) Auxiliary for ordering wildcards */
    WORD    silent;                 /* (M) Silent flag. Only results in output. */
    WORD    tracebackflag;          /* (M) For tracing errors */
    WORD    expnum;                 /* (M) internal number of ^ function */
    WORD    denomnum;               /* (M) internal number of / function */
    WORD    facnum;                 /* (M) internal number of fac_ function */
    WORD    invfacnum;              /* (M) internal number of invfac_ function */
    WORD    sumnum;                 /* (M) internal number of sum_ function */
    WORD    sumpnum;                /* (M) internal number of sump_ function */
    WORD    OldOrderFlag;           /* (M) Flag for allowing old statement order */
    WORD    termfunnum;             /* (M) internal number of term_ function */
    WORD    matchfunnum;            /* (M) internal number of match_ function */
    WORD    countfunnum;            /* (M) internal number of count_ function */
    WORD    gPolyFun;               /* (M) global value of PolyFun */
    WORD    gPolyFunInv;            /* (M) global value of Inverse of PolyFun */
    WORD    gPolyFunType;           /* (M) global value of PolyFun */
    WORD    gPolyFunExp;
    WORD    gPolyFunVar;
    WORD    gPolyFunPow;
    WORD    dollarzero;             /* (M) for dollars with zero value */
    WORD    atstartup;              /* To protect against DATE_ ending in \n */
    WORD    exitflag;               /* (R) For the exit statement */
    WORD    NumStoreCaches;         /* () Number of storage caches per processor */
    WORD    gIndentSpace;           /* For indentation in output */
    WORD    ggIndentSpace;
    WORD    gShortStatsMax;
    WORD    ggShortStatsMax;
    WORD    gextrasymbols;
    WORD    ggextrasymbols;
    WORD    zerorhs;
    WORD    onerhs;
    WORD    havesortdir;
    WORD    vectorzero;             /* p0_ */
    WORD    ClearStore;
    WORD    BracketFactors[8];
    BOOL    FromStdin;              /* read the input from STDIN */
    BOOL    IgnoreDeprecation;      /* ignore deprecation warning */
#ifdef WITHFLOAT
#ifdef WITHPTHREADS
    PADPOSITION(17,30,62,84,(sizeof(pthread_rwlock_t)+sizeof(pthread_mutex_t)*2)+2);
#else
    PADPOSITION(17,28,62,84,2);
#endif
#else
#ifdef WITHPTHREADS
    PADPOSITION(17,30,62,84,(sizeof(pthread_rwlock_t)+sizeof(pthread_mutex_t)*2)+2);
#else
    PADPOSITION(17,28,62,84,2);
#endif
#endif
};
/*
        #] M :
        #[ P : The P struct defines objects set by the preprocessor
*/
struct P_const {
    LIST DollarList;                /* (R) Dollar variables. Contains pointers
                                       to contents of the variables.*/
    LIST PreVarList;                /* (R) List of preprocessor variables
                                       Points to contents. Can be changed */
    LIST LoopList;                  /* (P) List of do loops */
    LIST ProcList;                  /* (P) List of procedures */
    INSIDEINFO inside;              /*     Information during #inside/#endinside */
    NAMESPACE *firstnamespace;      /* is zero when no namespace specified */
    NAMESPACE *lastnamespace;       /* is zero when no namespace specified */
    UBYTE *fullname;                /* buffer for namespace expanded names */
    UBYTE **PreSwitchStrings;       /* (P) The string in a switch */
    UBYTE *preStart;                /* (P) Preprocessor instruction buffer */
    UBYTE *preStop;                 /* (P) end of preStart */
    UBYTE *preFill;                 /* (P) Filling point in preStart */
    UBYTE *procedureExtension;      /* (P) Extension for procedure files (prc) */
    UBYTE *cprocedureExtension;     /* (P) Extension after .clear */
```

```
    LONG  *PreAssignStack;          /* For nesting #$name assignments */
    int *PreIfStack;                /* (P) Tracks nesting of #if */
    int *PreSwitchModes;            /* (P) Stack of switch status */
    int *PreTypes;                  /* (P) stack of #call, #do etc nesting */
#ifdef WITHPTHREADS
    pthread_mutex_t PreVarLock;     /* (P) */
#endif
    LONG    StopWatchZero;          /* For `timer_' and #reset timer */
    LONG    InOutBuf;               /* (P) Characters in the output buf in pre.c */
    LONG    pSize;                  /* (P) size of preStart */
    int     PreAssignFlag;          /* (C) Indicates #assign -> catch dollar */
    int     PreContinuation;        /* (C) Indicates whether the statement is new */
    int     PreproFlag;             /* (P) Internal use to mark work on prepro instr. */
    int     iBufError;              /* (P) Flag for errors with input buffer */
    int     PreOut;                 /* (P) Flag for #+ #- */
    int     PreSwitchLevel;         /* (P) Nesting of #switch */
    int     NumPreSwitchStrings;    /* (P) Size of PreSwitchStrings */
    int     MaxPreTypes;            /* (P) Size of PreTypes */
    int     NumPreTypes;            /* (P) Number of nesting objects in PreTypes */
    int     MaxPreIfLevel;          /* (C) Maximum number of nested #if. Dynamic */
    int     PreIfLevel;             /* (C) Current position if PreIfStack */
    int     PreInsideLevel;         /* (C) #inside active? */
    int     DelayPrevar;            /* (P) Delaying prevar substitution */
    int     AllowDelay;             /* (P) Allow delayed prevar substitution */
    int     lhdollarerror;          /* (R) */
    int     eat;                    /* () */
    int     gNumPre;                /* (P) Number of preprocessor variables for .clear */
    int     PreDebug;               /* (C) */
    int     OpenDictionary;
    int     PreAssignLevel;         /* For nesting #$name = ...; assignments */
    int     MaxPreAssignLevel;      /* For nesting #$name = ...; assignments */
    int     fullnamesize;           /* size of the fullname buffer */
    WORD    DebugFlag;              /* (P) For debugging purposes */
    WORD    preError;               /* (P) Blocks certain types of execution */
    UBYTE   ComChar;                /* (P) Commentary character */
    UBYTE   cComChar;               /* (P) Old commentary character for .clear */
    PADPOINTER(3,22,2,2);
};

/*
        #] P :
        #[ C : The C struct defines objects changed by the compiler
*/

struct C_const {
    set_of_char separators;
    POSITION StoreFileSize;         /* () Size of store file */
    NAMETREE *dollarnames;
    NAMETREE *exprnames;
    NAMETREE *varnames;
    LIST    ChannelList;
                                    /*     Later also for write? */
    LIST    DubiousList;
    LIST    FunctionList;
    LIST    ExpressionList;
    LIST    IndexList;
    LIST    SetElementList;
    LIST    SetList;
    LIST    SymbolList;
    LIST    VectorList;
    LIST    PotModDolList;
    LIST    ModOptDolList;
    LIST    TableBaseList;
/*
    Compile buffer variables
*/
    LIST    cbufList;
/*
    Objects for auto declarations
*/
    LIST    AutoSymbolList;         /* (C) */
    LIST    AutoIndexList;          /* (C) */
    LIST    AutoVectorList;         /* (C) */
    LIST    AutoFunctionList;       /* (C) */
    NAMETREE *autonames;
    LIST    *Symbols;               /* (C) Pointer for autodeclare. Which list is
                                       it searching. Later also for subroutines */
    LIST    *Indices;               /* (C) id. */
    LIST    *Vectors;               /* (C) id. */
    LIST    *Functions;             /* (C) id. */
    NAMETREE **activenames;
    STREAM  *Streams;
    STREAM  *CurrentStream;
    SWITCH  *SwitchArray;
    MODEL   **models;
    WORD    *SwitchHeap;
    LONG    *termstack;
```

```
    LONG    *termsortstack;
    UWORD   *cmod;
    UWORD   *powmod;
    UWORD   *modpowers;
    UWORD   *halfmod;               /* (C) half the modulus when not zero */
    WORD    *ProtoType;             /* (C) The subexpression prototype {wildcards} */
    WORD    *WildC;                 /* (C) Filling point for wildcards. */
    LONG    *IfHeap;
    LONG    *IfCount;
    LONG    *IfStack;
    UBYTE   *iBuffer;
    UBYTE   *iPointer;
    UBYTE   *iStop;
    UBYTE   **LabelNames;
    WORD    *FixIndices;
    WORD    *termsumcheck;
    UBYTE   *WildcardNames;
    int     *Labels;
    SBYTE   *tokens;
    SBYTE   *toptokens;
    SBYTE   *endoftokens;
    WORD    *tokenarglevel;
    UWORD   *modinverses;           /* Table for inverses of primes */
    UBYTE   *Fortran90Kind;         /* The kind of number in Fortran 90 as in _ki */
    WORD    **MultiBracketBuf;      /* Array of buffers for multi-level brackets */
    UBYTE   *extrasym;              /* Array with the name for extra symbols in ToPolynomial */
    WORD    *doloopstack;           /* To keep track of begin and end of doloops */
    WORD    *doloopnest;            /* To keep track of nesting of doloops etc */
    char    *CheckpointRunAfter;
    char    *CheckpointRunBefore;
    WORD    *IfSumCheck;
    WORD    *CommuteInSet;          /* groups of noncommuting functions that can commute */
    UBYTE   *TestValue;             /* For debugging */
#ifdef PARALLELCODE
    LONG    *inputnumbers;
    WORD    *pfirstnum;
#endif
#ifdef WITHPTHREADS
    pthread_mutex_t halfmodlock;    /* () Lock for adding buffer for halfmod */
#endif
    LONG    argstack[MAXNEST];      /* (C) {contents} Stack for nesting of Argument */
    LONG    insidestack[MAXNEST];   /* (C) {contents} Stack for Argument or Inside. */
    LONG    inexprstack[MAXNEST];   /* (C) {contents} Stack for Argument or Inside. */
    LONG    iBufferSize;            /* (C) Size of the input buffer */
    LONG    TransEname;             /* (C) Used when a new definition overwrites
                                       an old expression. */
    LONG    ProcessBucketSize;      /* (C) */
    LONG    mProcessBucketSize;     /* (C) */
    LONG    CModule;                /* (C) Counter of current module */
    LONG    ThreadBucketSize;       /* (C) Roughly the maximum number of input terms */
    LONG    CheckpointStamp;
    LONG    CheckpointInterval;
#ifdef WITHFLOAT
    LONG    DefaultPrecision;       /* (C) Default precision in bits for float_ */
    LONG    MaxWeight;              /* (C) Maximum weight for MZV or Euler */
    LONG    tDefaultPrecision;      /* (C) Default precision in bits for float_ */
    LONG    tMaxWeight;             /* (C) Maximum weight for MZV or Euler */
#endif
    int     cbufnum;
    int     AutoDeclareFlag;
    int     NoShowInput;            /* (C) No listing of input as in .prc, #do */
    int     ShortStats;             /* (C) */
    int     compiletype;            /* (C) type of statement {DECLARATION etc} */
    int     firstconstindex;        /* (C) flag for giving first error message */
    int     insidefirst;            /* (C) Not used yet */
    int     minsidefirst;           /* (?) Not used yet */
    int     wildflag;               /* (C) Flag for marking use of wildcards */
    int     NumLabels;              /* (C) Number of labels {in Labels} */
    int     MaxLabels;              /* (C) Size of Labels array */
    int     lDefDim;                /* (C) */
    int     lDefDim4;               /* (C) */
    int     NumWildcardNames;       /* (C) Number of ?a variables */
    int     WildcardBufferSize;     /* (C) size of WildcardNames buffer */
    int     MaxIf;                  /* (C) size of IfHeap, IfSumCheck, IfCount */
    int     NumStreams;             /* (C) */
    int     MaxNumStreams;          /* (C) */
    int     firstctypemessage;      /* (C) Flag for giving first st order error */
    int     tablecheck;             /* (C) For table checking */
    int     idoption;               /* (C) */
    int     BottomLevel;            /* (C) For propercount. Not used!!! */
    int     CompileLevel;           /* (C) Subexpression level */
    int     TokensWriteFlag;        /* (C) */
    int     UnsureDollarMode;       /* (C)?Controls error messages undefined $'s */
    int     outsidefun;             /* (C) Used for writing Tables to file */
    int     funpowers;              /* (C) */
    int     WarnFlag;               /* (C) */
    int     StatsFlag;              /* (C) */
```

```
    int     NamesFlag;              /* (C) */
    int     CodesFlag;              /* (C) */
    int     SetupFlag;              /* (C) */
    int     SortType;               /* (C) */
    int     lSortType;              /* (C) */
    int     ThreadStats;            /* (C) */
    int     FinalStats;             /* (C) */
    int     OldParallelStats;       /* (C) */
    int     ThreadsFlag;
    int     ThreadBalancing;
    int     ThreadSortFileSynch;
    int     ProcessStats;           /* (C) */
    int     BracketNormalize;       /* (C) Indicates whether the bracket st is normalized */
    int     maxtermlevel;           /* (C) Size of termstack */
    int     dumnumflag;             /* (C) Where there dummy indices in tokenizer? */
    int     bracketindexflag;       /* (C) Are brackets going to be indexed? */
    int     parallelflag;           /* (C) parallel allowed? */
    int     mparallelflag;          /* (C) parallel allowed in this module? */
    int     inparallelflag;         /* (C) inparallel allowed? */
    int     partodoflag;            /* (C) parallel allowed? */
    int     properorderflag;        /* (C) clean normalizing. */
    int     vetofilling;            /* (C) vetoes overwriting in tablebase stubs */
    int     tablefilling;           /* (C) to notify AddRHS we are filling a table */
    int     vetotablebasefill;      /* (C) For the load in tablebase */
    int     exprfillwarning;        /* (C) Warning has been printed for expressions in fill statements */
    int     lhdollarflag;           /* (R) left hand dollar present */
    int     NoCompress;             /* (R) Controls native compression */
    int     IsFortran90;            /* Tells whether the Fortran is Fortran90 */
    int     MultiBracketLevels;     /* Number of elements in MultiBracketBuf */
    int     topolynomialflag;       /* To avoid ToPolynomial and FactArg together */
    int     ffbufnum;               /* Buffer number for user defined factorizations */
    int     OldFactArgFlag;
    int     MemDebugFlag;           /* Only used when MALLOCDEBUG in tools.c */
    int     OldGCDflag;
    int     WTimeStatsFlag;
    int     SortReallocateFlag;     /* Controls reallocation of large+small buffer at module end.
                                         0 : Off
                                         1 : On, every module (set by On sortreallocate;)
                                         2 : On, single module (set by #sortreallocate) */
    int     PrintBacktraceFlag;     /* Print backtrace on terminate? */
    int    FlintPolyFlag;           /* Use Flint for polynomial arithmetic */
    int     HumanStatsFlag;         /* Print human-readable stats in the stats print? */
    int     doloopstacksize;
    int     dolooplevel;
    int     CheckpointFlag;
    int     SizeCommuteInSet;       /* Size of the CommuteInSet buffer */
#ifdef PARALLELCODE
    int     numpfirstnum;           /* For redefine */
    int     sizepfirstnum;          /* For redefine */
#endif
    int     origin;                 /* Determines whether .sort or ModuleOption */
    int     vectorlikeLHS;
    int     nummodels;
    int     modelspace;
    int     ModelLevel;
    WORD    argsumcheck[MAXNEST];   /* (C) Checking of nesting */
    WORD    insidesumcheck[MAXNEST];/* (C) Checking of nesting */
    WORD    inexprsumcheck[MAXNEST];/* (C) Checking of nesting */
    WORD    RepSumCheck[MAXREPEAT];/* (C) Checks nesting of repeat, if, argument */
    WORD    lUniTrace[4];           /* (C) */
    WORD    RepLevel;               /* (C) Tracks nesting of repeat. */
    WORD    arglevel;               /* (C) level of nested argument statements */
    WORD    insidelevel;            /* (C) level of nested inside statements */
    WORD    inexprlevel;            /* (C) level of nested inexpr statements */
    WORD    termlevel;              /* (C) level of nested term statements */
    WORD    MustTestTable;          /* (C) Indicates whether tables have been changed */
    WORD    DumNum;                 /* (C) */
    WORD    ncmod;                  /* (C) Local setting of modulus. size of cmod */
    WORD    npowmod;                /* (C) Local printing as powers. size powmod */
    WORD    modmode;                /* (C) Mode for modulus calculus */
    WORD    nhalfmod;               /* relevant word size of halfmod when defined */
    WORD    DirtPow;                /* (C) Flag for changes in printing mod powers */
    WORD    lUnitTrace;             /* (C) Local value of Tr[1] */
    WORD    NwildC;                 /* (C) Wildcard counter */
    WORD    ComDefer;               /* (C) defer brackets */
    WORD    CollectFun;             /* (C) Collect function number */
    WORD    AltCollectFun;          /* (C) Alternate Collect function number */
    WORD    OutputMode;             /* (C) */
    WORD    Cnumpows;
    WORD    OutputSpaces;           /* (C) */
    WORD    OutNumberType;          /* (C) Controls RATIONAL/FLOAT output */
    WORD    DidClean;               /* (C) Test whether nametree needs cleaning */
    WORD    IfLevel;                /* (C) */
    WORD    WhileLevel;             /* (C) */
    WORD    SwitchLevel;
    WORD    SwitchInArray;
```

```
    WORD    MaxSwitch;
    WORD    LogHandle;              /* (C) The Log File */
    WORD    LineLength;             /* (C) */
    WORD    StoreHandle;            /* (C) Handle of .str file */
    WORD    HideLevel;              /* (C) Hiding indicator */
    WORD    lPolyFun;               /* (C) local value of PolyFun */
    WORD    lPolyFunInv;            /* (C) local value of Inverse of PolyFun */
    WORD    lPolyFunType;           /* (C) local value of PolyFunType */
    WORD    lPolyFunExp;
    WORD    lPolyFunVar;
    WORD    lPolyFunPow;
    WORD    SymChangeFlag;          /* (C) */
    WORD    CollectPercentage;      /* (C) Collect function percentage */
    WORD    ShortStatsMax;          /* For  On FewerStatistics 10; */
    WORD    extrasymbols;           /* Flag for the extra symbols output mode */
    WORD    PolyRatFunChanged;      /* Keeps track whether we changed in the compiler */
    WORD    ToBeInFactors;
    WORD    InnerTest;             /* For debugging */
#ifdef WITHMPI
    WORD    RhsExprInModuleFlag;    /* (C) Set by the compiler if RHS expressions exists. */
#endif
    UBYTE   Commercial[COMMERCIALSIZE+2]; /* (C) Message to be printed in statistics */
    UBYTE   debugFlags[MAXFLAGS+2];    /* On/Off Flag number(s) */
#ifdef WITHFLOAT
#if defined(WITHPTHREADS)
PADPOSITION(50,12+3*MAXNEST,77,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17+sizeof(pthread_mutex_t)
#elif defined(WITHMPI)
    PADPOSITION(50,12+3*MAXNEST,77,49+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
#else
    PADPOSITION(48,12+3*MAXNEST,75,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
#endif
#else
#if defined(WITHPTHREADS)
PADPOSITION(50,8+3*MAXNEST,77,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17+sizeof(pthread_mutex_t))
#elif defined(WITHMPI)
    PADPOSITION(50,8+3*MAXNEST,77,49+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
#else
    PADPOSITION(48,8+3*MAXNEST,75,48+3*MAXNEST+MAXREPEAT,COMMERCIALSIZE+MAXFLAGS+4+sizeof(LIST)*17);
#endif
#endif
};
/*
        #] C :
        #[ S : The S struct defines objects changed at the start of the run (Processor)
               Basically only set by the master.
*/
struct S_const {
    POSITION MaxExprSize;           /* ( ) Maximum size of in/out/sort */
#ifdef WITHPTHREADS
    pthread_mutex_t inputslock;
    pthread_mutex_t outputslock;
    pthread_mutex_t MaxExprSizeLock;
#endif
    POSITION *OldOnFile;            /* (S) File positions of expressions */
    WORD    *OldNumFactors;         /* ( ) NumFactors in (old) expression */
    WORD    *Oldvflags;             /* ( ) vflags in (old) expression */
    WORD    *Olduflags;             /* ( ) uflags in (old) expression */
#ifdef WITHFLOAT
    void    *delta_1;
#endif
    int     NumOldOnFile;           /* (S) Number of expressions in OldOnFile */
    int     NumOldNumFactors;       /* (S) Number of expressions in OldNumFactors */
    int     MultiThreaded;          /* (S) Are we running multi-threaded? */
#ifdef WITHPTHREADS
    int     MasterSort;             /* Final stage of sorting to the master */
#endif
#ifdef WITHMPI
    int     printflag;              /* controls MesPrint() on each slave */
#endif
    int     Balancing;              /* For second stage loadbalancing */
    WORD    ExecMode;               /* (S) */

    WORD    CollectOverFlag;        /* (R) Indicates overflow at Collect */
#ifdef WITHPTHREADS
    WORD    sLevel;                 /* Copy of AR0.sLevel because it can get messy */
#endif
#if defined(WITHPTHREADS)
    PADPOSITION(3,0,5,3,sizeof(pthread_mutex_t)*3);
#elif defined(WITHMPI)
    PADPOSITION(4,0,5,2,0);
#else
    PADPOSITION(4,0,4,2,0);
#endif
};
/*
```

```
        #] S :
        #[ R : The R struct defines objects changed at run time.
               They determine the environment that has to be transferred
               together with a term during multithreaded execution.
*/
struct R_const {
    FILEDATA    StoreData;          /* (O) */
    FILEHANDLE  Fscr[3];            /* (R) Dollars etc play with it too */
    FILEHANDLE  FoStage4[2];        /* (R) In Sort. Stage 4. */
    POSITION DefPosition;           /* (R) Deferred position of keep brackets. */
    FILEHANDLE *infile;             /* (R) Points alternatingly to Fscr[0] or Fscr[1] */
    FILEHANDLE *outfile;            /* (R) Points alternatingly to Fscr[1] or Fscr[0] */
    FILEHANDLE *hidefile;           /* (R) Points to Fscr[2] */

    WORD    *CompressBuffer;        /* (M) */
    WORD    *ComprTop;              /* (M) */
    WORD    *CompressPointer;       /* (R) */
    COMPAREDUMMY CompareRoutine;
    ULONG   *wranfia;
    SBYTE   *moebiustable;

    LONG    OldTime;                /* (R) Zero time. Needed in timer. */
    LONG    InInBuf;                /* (R) Characters in input buffer. Scratch files. */
    LONG    InHiBuf;                /* (R) Characters in hide buffer. Scratch file. */
    LONG    pWorkSize;              /* (R) Size of pWorkSpace */
    LONG    lWorkSize;              /* (R) Size of lWorkSpace */
    LONG    posWorkSize;            /* (R) Size of posWorkSpace */
    ULONG   wranfseed;
    int     NoCompress;             /* (R) Controls native compression */
    int     gzipCompress;           /* (R) Controls gzip compression */
    int     Cnumlhs;                /* Local copy of cbuf[rbufnum].numlhs */
    int     outtohide;              /* Indicates that output is directly to hide */
#ifdef WITHPTHREADS
    int     exprtodo;               /* The expression to do in parallel mode */
#endif
    int     wranfcall;
    int     wranfnpair1;
    int     wranfnpair2;
#if ( BITSINWORD == 32 )
    WORD    PrimeList[5000];
    WORD    numinprimelist;
    WORD    notfirstprime;
#endif
    WORD    GetFile;                /* (R) Where to get the terms {like Hide} */
    WORD    KeptInHold;             /* (R) */
    WORD    BracketOn;              /* (R) Intensely used in poly_ */
    WORD    MaxBracket;             /* (R) Size of BrackBuf. Changed by poly_ */
    WORD    CurDum;                 /* (R) Current maximum dummy number */
    WORD    DeferFlag;              /* (R) For deferred brackets */
    WORD    TePos;                  /* (R) */
    WORD    sLevel;                 /* (R) Sorting level */
    WORD    Stage4Name;             /* (R) Sorting only */
    WORD    GetOneFile;             /* (R) Getting from hide or regular */
    WORD    PolyFun;                /* (C) Number of the PolyFun function */
    WORD    PolyFunInv;             /* (C) Number of the Inverse of the PolyFun function */
    WORD    PolyFunType;            /* () value of PolyFunType */
    WORD    PolyFunExp;
    WORD    PolyFunVar;
    WORD    PolyFunPow;
    WORD    Eside;                  /* () Tells which side of = sign */
    WORD    MaxDum;                 /* Maximum dummy value in an expression */
    WORD    level;                  /* Running level in Generator */
    WORD    expchanged;             /* (R) Info about expression */
    WORD    expflags;               /* (R) Info about expression */
    WORD    CurExpr;                /* (S) Number of current expression */
    WORD    SortType;               /* A copy of AC.SortType to play with */
    WORD    ShortSortCount;         /* For On FewerStatistics 10; */
    WORD    modeloptions;
    WORD    funoffset;
    WORD    moebiustablesize;
#if ( BITSINWORD == 32 )
#ifdef WITHPTHREADS
    PADPOSITION(8,7,8,5029,0);
#else
    PADPOSITION(8,7,7,5029,0);
#endif
#else
#ifdef WITHPTHREADS
    PADPOSITION(8,7,8,27,0);
#else
    PADPOSITION(8,7,7,27,0);
#endif
#endif
};

/*
        #] R :
```

```
       #[ T : These are variables that stay in each thread during multi threaded execution.
*/
struct T_const {
#ifdef WITHPTHREADS
    SORTBLOCK SB;
#endif
    SORTING *S0;                    /* (-) The thread specific sort buffer */
    SORTING *SS;                    /* (R) Current sort buffer */
    NESTING     Nest;               /* (R) Nesting of function levels etc. */
    NESTING     NestStop;           /* (R) */
    NESTING     NestPoin;           /* (R) */
    WORD    *BrackBuf;              /* (R) Bracket buffer. Used by poly_ at runtime. */
    STORECACHE  StoreCache;         /* (R) Cache for picking up stored expr. */
    STORECACHE  StoreCacheAlloc;    /* (R) Permanent address of StoreCache to keep valgrind happy */
    WORD    **pWorkSpace;           /* (R) Workspace for pointers. Dynamic. */
    LONG    *lWorkSpace;            /* (R) WorkSpace for LONG. Dynamic. */
    POSITION *posWorkSpace;         /* (R) WorkSpace for file positions */
    WORD    *WorkSpace;             /* (M) */
    WORD    *WorkTop;               /* (M) */
    WORD    *WorkPointer;           /* (R) Pointer in the WorkSpace heap. */
    int     *RepCount;              /* (M) Buffer for repeat nesting */
    int     *RepTop;                /* (M) Top of RepCount buffer */
    WORD    *WildArgTaken;          /* (N) Stack for wildcard pattern matching */
    UWORD   *factorials;            /* (T) buffer of factorials. Dynamic. */
    WORD    *small_power_n;         /*     length of the number */
    UWORD  **small_power;           /*     the number*/
    UWORD   *bernoullis;            /* (T) The buffer with Bernoulli numbers. Dynamic. */
    WORD    *primelist;
    LONG    *pfac;                  /* (T) array of positions of factorials. Dynamic. */
    LONG    *pBer;                  /* (T) array of positions of Bernoulli's. Dynamic. */
    WORD    *TMaddr;                /* (R) buffer for TestSub */
    WORD    *WildMask;              /* (N) Wildcard info during pattern matching */
    WORD    *previousEfactor;       /* () Cache for factors in expressions */
    WORD    **TermMemHeap;         /* For TermMalloc. Set zero in Checkpoint */
    UWORD    **NumberMemHeap;       /* For NumberMalloc. Set zero in Checkpoint */
    UWORD    **CacheNumberMemHeap; /* For CacheNumberMalloc. Set zero in Checkpoint */
    BRACKETINFO *bracketinfo;
    WORD    **ListPoly;
    WORD    *ListSymbols;
    UWORD   *NumMem;
    WORD    *TopologiesTerm;
    WORD    *TopologiesStart;
#ifdef WITHFLOAT
    int     *indi1;
    int     *indi2;
    void    *mpf_tab1;
    void    *mpf_tab2;
    void    *aux_;
    void    *auxr_;
#endif
    PARTI   partitions;
    LONG    sBer;                   /* (T) Size of the Bernoullis buffer */
    LONG    pWorkPointer;           /* (R) Offset-pointer in pWorkSpace */
    LONG    lWorkPointer;           /* (R) Offset-pointer in lWorkSpace */
    LONG    posWorkPointer;         /* (R) Offset-pointer in posWorkSpace */
    LONG    InNumMem;
    int     sfact;                  /* (T) size of the factorials buffer */
    int     mfac;                   /* (T) size of the pfac array. */
    int     ebufnum;                /* (R) extra compiler buffer */
    int     fbufnum;                /* extra compiler buffer for factorization cache */
    int     allbufnum;              /* extra compiler buffer for id,all */
    int     aebufnum;               /* extra compiler buffer for id,all */
    int     idallflag;              /* indicates use of id,all buffers */
    int     idallnum;
    int     idallmaxnum;
    int     WildcardBufferSize;     /* () local copy for updates */
#ifdef WITHPTHREADS
    int     identity;               /* () When we work with B->T */
    int     LoadBalancing;          /* Needed for synchronization */
#ifdef WITHSORTBOTS
    int     SortBotIn1;             /* Input stream 1 for a SortBot */
    int     SortBotIn2;             /* Input stream 2 for a SortBot */
#endif
#endif
    int     TermMemMax;             /* For TermMalloc. Set zero in Checkpoint */
    int     TermMemTop;             /* For TermMalloc. Set zero in Checkpoint */
    int     NumberMemMax;           /* For NumberMalloc. Set zero in Checkpoint */
    int     NumberMemTop;           /* For NumberMalloc. Set zero in Checkpoint */
    int     CacheNumberMemMax;      /* For CacheNumberMalloc. Set zero in Checkpoint */
    int     CacheNumberMemTop;      /* For CacheNumberMalloc. Set zero in Checkpoint */
    int     bracketindexflag;       /* Are brackets going to be indexed? */
    int     optimtimes;             /* Number of the evaluation of the MCTS tree */
    int     ListSymbolsSize;
    int     NumListSymbols;
    int     numpoly;
    int     LeaveNegative;
    int     TrimPower;              /* Indicates trimming in polyratfun expansion */
```

```
    WORD    small_power_maxx;       /*     size of the cache for small powers  */
    WORD    small_power_maxn;       /*     size of the cache for small powers */
    WORD    dummysubexp[SUBEXPSIZE+4]; /* () used in normal.c */
    WORD    comsym[8];              /* () Used in tools.c = {8,SYMBOL,4,0,1,1,1,3} */
    WORD    comnum[4];              /* () Used in tools.c = { 4,1,1,3 } */
    WORD    comfun[FUNHEAD+4];      /* () Used in tools.c = {7,FUNCTION,3,0,1,1,3} */
                                    /*             or { 8,FUNCTION,4,0,0,1,1,3 } */
    WORD    comind[7];              /* () Used in tools.c = {7,INDEX,3,0,1,1,3} */
    WORD    MinVecArg[7+ARGHEAD];   /* (N) but should be more local */
    WORD    FunArg[4+ARGHEAD+FUNHEAD]; /* (N) but can be more local */
    WORD    locwildvalue[SUBEXPSIZE]; /* () Used in argument.c = {SUBEXPRESSION,SUBEXPSIZE,0,0,0} */
    WORD    mulpat[SUBEXPSIZE+5];   /* () Used in argument.c = {TYPEMULT, SUBEXPSIZE+3, 0, */
                                    /* SUBEXPRESSION, SUBEXPSIZE, 0, 1, 0, 0, 0 } */
    WORD    proexp[SUBEXPSIZE+5];   /* () Used in poly.c */
    WORD    TMout[40];              /* (R) Passing info */
    WORD    TMbuff;                 /* (R) Communication between TestSub and Genera */
    WORD    TMdolfac;               /* factor number for dollar */
    WORD    nfac;                   /* (T) Number of highest stored factorial */
    WORD    nBer;                   /* (T) Number of highest Bernoulli number. */
    WORD    mBer;                   /* (T) Size of buffer pBer. */
    WORD    PolyAct;                /* (R) Used for putting the PolyFun at end. ini at 0 */
    WORD    RecFlag;                /* (R) Used in TestSub. ini at zero. */
    WORD    inprimelist;
    WORD    sizeprimelist;
    WORD    fromindex;              /* Tells the compare routine whether call from index */
    WORD    setinterntopo;          /* Set of internal momenta for topogen */
    WORD    setexterntopo;          /* Set of external momenta for topogen */
    WORD    TopologiesLevel;
    WORD    TopologiesOptions[2];
#ifdef WITHFLOAT
    WORD    FloatPos;
    WORD    SortFloatMode;
#endif
#ifdef WITHFLOAT
 #ifdef WITHPTHREADS
  #ifdef WITHSORTBOTS
    PADPOINTER(5,27,107+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
  #else
    PADPOINTER(5,25,107+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
  #endif
 #else
    PADPOINTER(5,23,107+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
 #endif
#else
 #ifdef WITHPTHREADS
  #ifdef WITHSORTBOTS
    PADPOINTER(5,27,105+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
  #else
    PADPOINTER(5,25,105+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
  #endif
 #else
    PADPOINTER(5,23,105+SUBEXPSIZE*4+FUNHEAD*2+ARGHEAD*2,0);
 #endif
#endif
};
/*
        #] T :
        #[ N : The N struct contains variables used in running information
               that is inside blocks that should not be split, like pattern
               matching, traces etc. They are local for each thread.
               They don't need initializations.
*/
struct N_const {
    POSITION OldPosIn;              /* (R) Used in sort. */
    POSITION OldPosOut;             /* (R) Used in sort */
    POSITION theposition;           /* () Used in index.c */
    WORD    *EndNest;               /* (R) Nesting of function levels etc. */
    WORD    *Frozen;                /* (R) Bracket info */
    WORD    *FullProto;             /* (R) Prototype of a subexpression or table */
    WORD    *cTerm;                 /* (R) Current term for coef_ and term_ */
    int     *RepPoint;              /* (R) Pointer in RepCount buffer. Tracks repeat */
    WORD    *WildValue;             /* (N) Wildcard info during pattern matching */
    WORD    *WildStop;              /* (N) Wildcard info during pattern matching */
    WORD    *argaddress;            /* (N) Used in pattern matching of arguments */
    WORD    *RepFunList;            /* (N) For pattern matching */
    WORD    *patstop;               /* (N) Used in pattern matching */
    WORD    *terstop;               /* (N) Used in pattern matching */
    WORD    *terstart;              /* (N) Used in pattern matching */
    WORD    *terfirstcomm;          /* (N) Used in pattern matching */
    WORD    *DumFound;              /* (N) For renumbering indices {make local?} */
    WORD    **DumPlace;             /* (N) For renumbering indices {make local?} */
    WORD    **DumFunPlace;          /* (N) For renumbering indices {make local?} */
    WORD    *UsedSymbol;            /* (N) When storing terms of a global expr. */
    WORD    *UsedVector;            /* (N) When storing terms of a global expr. */
    WORD    *UsedIndex;             /* (N) When storing terms of a global expr. */
    WORD    *UsedFunction;          /* (N) When storing terms of a global expr. */
    WORD    *MaskPointer;           /* (N) For wildcard pattern matching */
```

```
    WORD    *ForFindOnly;           /* (N) For wildcard pattern matching */
    WORD    *findTerm;              /* (N) For wildcard pattern matching */
    WORD    *findPattern;           /* (N) For wildcard pattern matching */
#ifdef WITHZLIB
    Bytef   **ziobufnum;            /* () Used in compress.c */
    Bytef   *ziobuffers;            /* () Used in compress.c */
#endif
    WORD    *dummyrenumlist;        /* () Used in execute.c and store.c */
    int     *funargs;               /* () Used in lus.c */
    WORD    **funlocs;              /* () Used in lus.c */
    int     *funinds;               /* () Used in lus.c */
    UWORD   *NoScrat2;              /* () Used in normal.c */
    WORD    *ReplaceScrat;          /* () Used in normal.c */
    TRACES    *tracestack;            /* () used in opera.c */
    WORD    *selecttermundo;        /* () Used in pattern.c */
    WORD    *patternbuffer;         /* () Used in pattern.c */
    WORD    *termbuffer;            /* () Used in pattern.c */
    WORD    **PoinScratch;          /* () used in reshuf.c */
    WORD    **FunScratch;           /* () used in reshuf.c */
    WORD    *RenumScratch;          /* () used in reshuf.c */
    FUN_INFO *FunInfo;              /* () Used in smart.c */
    WORD    **SplitScratch;         /* () Used in sort.c */
    WORD    **SplitScratch1;        /* () Used in sort.c */
    SORTING **FunSorts;             /* () Used in sort.c */
    UWORD   *SoScratC;              /* () Used in sort.c */
    WORD    *listinprint;           /* () Used in proces.c and message.c */
    WORD    *currentTerm;           /* () Used in proces.c and message.c */
    WORD    **arglist;              /* () Used in function.c */
    int     *tlistbuf;              /* () used in lus.c */
#ifdef WHICHSUBEXPRESSION
    UWORD   *BinoScrat;             /* () Used in proces.c */
#endif
    WORD    *compressSpace;         /* () Used in sort.c */
#ifdef WITHPTHREADS
    THREADBUCKET *threadbuck;
    EXPRESSIONS expr;
#endif
    UWORD   *SHcombi;
    WORD    *poly_vars;
    UWORD   *cmod;                  /* Local setting of modulus. Pointer to value. */
    SHvariables SHvar;
    LONG    deferskipped;           /* () Used in proces.c store.c and parallel.c */
    LONG    InScratch;              /* () Used in sort.c */
    LONG    SplitScratchSize;       /* () Used in sort.c */
    LONG    InScratch1;             /* () Used in sort.c */
    LONG    SplitScratchSize1;      /* () Used in sort.c */
    LONG    ninterms;               /* () Used in proces.c and sort.c */
#ifdef WITHPTHREADS
    LONG    inputnumber;            /* () For use in redefine */
    LONG    lastinindex;
#endif
#ifdef WHICHSUBEXPRESSION
    LONG    last2;                  /* () Used in proces.c */
    LONG    last3;                  /* () Used in proces.c */
#endif
    LONG    SHcombisize;
    int     NumTotWildArgs;         /* (N) Used in pattern matching */
    int     UseFindOnly;            /* (N) Controls pattern routines */
    int     UsedOtherFind;          /* (N) Controls pattern routines */
    int     ErrorInDollar;          /* (R) */
    int     numfargs;               /* () Used in lus.c */
    int     numflocs;               /* () Used in lus.c */
    int     nargs;                  /* () Used in lus.c */
    int     tohunt;                 /* () Used in lus.c */
    int     numoffuns;              /* () Used in lus.c */
    int     funisize;               /* () Used in lus.c */
    int     RSsize;                 /* () Used in normal.c */
    int     numtracesctack;         /* () used in opera.c */
    int     intracestack;           /* () used in opera.c */
    int     numfuninfo;             /* () Used in smart.c */
    int     NumFunSorts;            /* () Used in sort.c */
    int     MaxFunSorts;            /* () Used in sort.c */
    int     arglistsize;            /* () Used in function.c */
    int     tlistsize;              /* () used in lus.c */
    int     filenum;                /* () used in setfile.c */
    int     compressSize;           /* () Used in sort.c */
    int     polysortflag;
    int     nogroundlevel;          /* () Used to see whether pattern matching at groundlevel */
    int     subsubveto;             /* () Sabotage combining subexpressions in TestSub */
    WORD    MaxRenumScratch;        /* () used in reshuf.c */
    WORD    oldtype;                /* (N) WildCard info at pattern matching */
    WORD    oldvalue;               /* (N) WildCard info at pattern matching */
    WORD    NumWild;                /* (N) Used in Wildcard */
    WORD    RepFunNum;              /* (N) Used in pattern matching */
    WORD    DisOrderFlag;           /* (N) Disorder option? Used in pattern matching */
    WORD    WildDirt;               /* (N) dirty in wildcard substitution. */
    WORD    NumFound;               /* (N) in reshuf only. Local? */
```

```
    WORD    WildReserve;            /* (N) Used in the wildcards */
    WORD    TeInFun;                /* (R) Passing type of action */
    WORD    TeSuOut;                /* (R) Passing info. Local? */
    WORD    WildArgs;               /* (R) */
    WORD    WildEat;                /* (R) */
    WORD    PolyNormFlag;           /* (R) For polynomial arithmetic */
    WORD    PolyFunTodo;            /* deals with expansions and multiplications */
    WORD    sizeselecttermundo;     /* () Used in pattern.c */
    WORD    patternbuffersize;      /* () Used in pattern.c */
    WORD    numlistinprint;         /* () Used in process.c */
    WORD    ncmod;                  /* () used as some type of flag to disable */
    WORD    ExpectedSign;
    WORD    SignCheck;
    WORD    IndDum;                 /* Active dummy indices */
    WORD    poly_num_vars;
    WORD    idfunctionflag;
    WORD    poly_vars_type;         /* type of allocation. For free. */
    WORD    tryterm;                /* For EndSort(...,2) */
#ifdef WHICHSUBEXPRESSION
    WORD    nbino;                  /* () Used in proces.c */
    WORD    last1;                  /* () Used in proces.c */
#endif
#ifdef WITHPTHREADS
#ifdef WHICHSUBEXPRESSION
#ifdef WITHZLIB
    PADPOSITION(55,11,23,28,sizeof(SHvariables));
#else
    PADPOSITION(53,11,23,28,sizeof(SHvariables));
#endif
#else
#ifdef WITHZLIB
    PADPOSITION(54,9,23,26,sizeof(SHvariables));
#else
    PADPOSITION(52,9,23,26,sizeof(SHvariables));
#endif
#endif
#else
#ifdef WHICHSUBEXPRESSION
#ifdef WITHZLIB
    PADPOSITION(53,9,23,28,sizeof(SHvariables));
#else
    PADPOSITION(51,9,23,28,sizeof(SHvariables));
#endif
#else
#ifdef WITHZLIB
    PADPOSITION(52,7,23,26,sizeof(SHvariables));
#else
    PADPOSITION(50,7,23,26,sizeof(SHvariables));
#endif
#endif
#endif
};

/*
        #] N :
        #[ O : The O struct concerns output variables
*/
struct O_const {
    FILEDATA     SaveData;          /* (O) */
    STOREHEADER SaveHeader;         /* ()  System Independent save-Files */
    OPTIMIZERESULT OptimizeResult;
    UBYTE   *OutputLine;            /* (O) Sits also in debug statements */
    UBYTE   *OutStop;               /* (O) Top of OutputLine buffer */
    UBYTE   *OutFill;               /* (O) Filling point in OutputLine buffer */
    WORD    *bracket;               /* (O) For writing brackets */
    WORD    *termbuf;               /* (O) For writing terms */
    WORD    *tabstring;
    UBYTE   *wpos;                  /* (O) Only when storing file {local?} */
    UBYTE   *wpoin;                 /* (O) Only when storing file {local?} */
    UBYTE   *DollarOutBuffer;       /* (O) Outputbuffer for Dollars */
    UBYTE   *CurBufWrt;             /* (O) Name of currently written expr. */
    void    (*FlipWORD)(UBYTE *);   /* ()  Function pointers for translations. Initialized by
ReadSaveHeader() */
    void    (*FlipLONG)(UBYTE *);
    void    (*FlipPOS)(UBYTE *);
    void    (*FlipPOINTER)(UBYTE *);
    void    (*ResizeData)(UBYTE *,int,UBYTE *,int);
    void    (*ResizeWORD)(UBYTE *,UBYTE *);
    void    (*ResizeNCWORD)(UBYTE *,UBYTE *);
    void    (*ResizeLONG)(UBYTE *,UBYTE *);
    void    (*ResizePOS)(UBYTE *,UBYTE *);
    void    (*ResizePOINTER)(UBYTE *,UBYTE *);
    void    (*CheckPower)(UBYTE *);
    void    (*RenumberVec)(UBYTE *);
    DICTIONARY **Dictionaries;
    UBYTE   *tensorList;            /* Dynamically allocated list with functions that are tensorial. */
    WORD    *inscheme;              /* for feeding a Horner scheme to Optimize */
```

```
#ifdef WITHFLOAT
    UBYTE   *floatspace;
    LONG    floatsize;
#endif
/*----Leave NumInBrack as first non-pointer. This is used by the checkpoints--*/
    LONG    NumInBrack;             /* (O) For typing [] option in print */
    LONG    wlen;                   /* (O) Used to store files. */
    LONG    DollarOutSizeBuffer;    /* (O) Size of DollarOutBuffer */
    LONG    DollarInOutBuffer;      /* (O) Characters in DollarOutBuffer */
#if defined(mBSD) && defined(MICROTIME)
    LONG    wrap;                   /* (O) For statistics time. wrap around */
    LONG    wrapnum;                /* (O) For statistics time. wrap around */
#endif
    OPTIMIZE Optimize;
    int     OutInBuffer;            /* (O) Which routine does the writing */
    int     NoSpacesInNumbers;      /*     For very long numbers */
    int     BlockSpaces;            /*     For very long numbers */
    int     CurrentDictionary;
    int     SizeDictionaries;
    int     NumDictionaries;
    int     CurDictNumbers;
    int     CurDictVariables;
    int     CurDictSpecials;
    int     CurDictFunWithArgs;
    int     CurDictNumberWarning;
    int     CurDictNotInFunctions;
    int     CurDictInDollars;
    int     gNumDictionaries;
#ifdef WITHFLOAT
    int     FloatPrec;
#endif
    WORD    schemenum;              /* for feeding a Horner scheme to Optimize */
    WORD    transFlag;              /* ()  >0 indicates that translations have to be done */
    WORD    powerFlag;              /* ()  >0 indicates that some exponents/powers had to be adjusted */
    WORD    mpower;                 /*     For maxpower adjustment to larger value */
    WORD    resizeFlag;             /* ()  >0 indicates that something went wrong when resizing words */
    WORD    bufferedInd;            /* ()  Contains extra INDEXENTRIES, see ReadSaveIndex() for an explanation */
    WORD    OutSkip;                /* (O) How many chars to skip in output line */
    WORD    IsBracket;              /* (O) Controls brackets */
    WORD    InFbrack;               /* (O) For writing only */
    WORD    PrintType;              /* (O) */
    WORD    FortFirst;              /* (O) Only in sch.c */
    WORD    DoubleFlag;             /* (O) Output in double precision */
    WORD    IndentSpace;            /* For indentation in output */
    WORD    FactorMode;             /* When the output should be written as factors */
    WORD    FactorNum;              /* Number of factor currently treated */
    WORD    ErrorBlock;
    WORD    OptimizationLevel;      /* Level of optimization in the output */
    UBYTE   FortDotChar;            /* (O) */
/*
    For the padding, please count also the number of int's in the OPTIMIZE struct.
*/
#ifdef WITHFLOAT
#if defined(mBSD) && defined(MICROTIME)
    PADPOSITION(25,7,36,18,1);
#else
    PADPOSITION(25,5,36,18,1);
#endif
#else
#if defined(mBSD) && defined(MICROTIME)
    PADPOSITION(25,6,36,17,1);
#else
    PADPOSITION(25,4,36,17,1);
#endif
#endif
};
/*
        #] O :
        #[ X : The X struct contains variables that deal with the external channel
*/
struct X_const {
    UBYTE *currentPrompt;
    UBYTE *shellname;           /* if !=NULL (default is "/bin/sh -c"), start in
                                   the specified subshell*/
    UBYTE *stderrname;          /* If !=NULL (default if "/dev/null"), stderr is
                                   redirected to the specified file*/
    int timeout;                /* timeout to initialize preset channels.
                                   If timeout<0, the preset channels are
                                   already initialized*/
    int killSignal;             /* signal number, SIGKILL by default*/
    int killWholeGroup;         /* if 0, the signal is sent only to a process,
                                   if !=0 (default) is sent to a whole process group*/
    int daemonize;              /* if !=0 (default), start in a daemon mode */
    int currentExternalChannel;
```

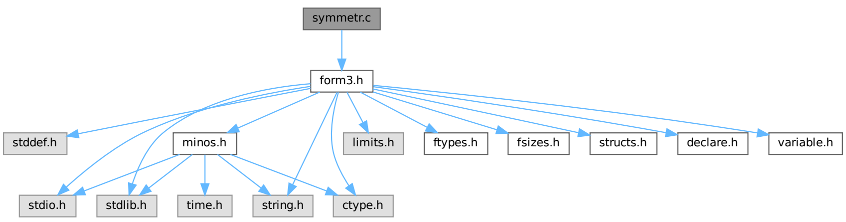
```
    PADPOINTER(0,5,0,0);
};
/*
        #] X :
        #[ Definitions :

    Note: we changed the definition from C to Cc in version 5.
    The reason is that everywhere the pointer to the compiler
    buffer was called C as well. Somehow that gave no problems but
    who knows what the future might bring.
*/

#ifdef WITHPTHREADS

typedef struct AllGlobals {
    struct M_const M;
    struct C_const Cc;
    struct S_const S;
    struct O_const O;
    struct P_const P;
    struct X_const X;
    PADPOSITION(0,0,0,0,sizeof(struct P_const)+sizeof(struct X_const));
} ALLGLOBALS;

typedef struct AllPrivates {
    struct R_const R;
    struct N_const N;
    struct T_const T;
    PADPOSITION(0,0,0,0,sizeof(struct T_const));
} ALLPRIVATES;

#else

typedef struct AllGlobals {
    struct M_const M;
    struct C_const Cc;
    struct S_const S;
    struct R_const R;
    struct N_const N;
    struct O_const O;
    struct P_const P;
    struct T_const T;
    struct X_const X;
    PADPOSITION(0,0,0,0,sizeof(struct P_const)+sizeof(struct T_const)+sizeof(struct X_const));
} ALLGLOBALS;

#endif

/*
        #] Definitions :
    #] A :
    #[ FG :
*/

#ifdef WITHPTHREADS
#define PHEAD  ALLPRIVATES *B,
#define PHEAD0 ALLPRIVATES *B
#define BHEAD  B,
#define BHEAD0 B
#else
#define PHEAD
#define PHEAD0 void
#define BHEAD
#define BHEAD0
#endif

#ifdef ANSI
typedef WORD (*WCN)(PHEAD WORD *,WORD *,WORD,WORD);
typedef WORD (*WCN2)(PHEAD WORD *,WORD *);
#else
typedef WORD (*WCN)();
typedef WORD (*WCN2)();
#endif

typedef WORD (*COMPARE)(PHEAD WORD *,WORD *,WORD);

typedef struct FixedGlobals {
    WCN     Operation[8];
    WCN2    OperaFind[6];
    char    *VarType[10];
    char    *ExprStat[21];
    char    *FunNam[2];
    char    *swmes[3];
    char    *fname;
    char    *fname2;
    UBYTE   *s_one;
    WORD    fnamebase;
```

```
    WORD    fname2base;
    WORD    fnamesize;
    WORD    fname2size;
    UINT    cTable[256];
} FIXEDGLOBALS;

/*
    #] FG :
*/

#endif
```

4.138 symmetr.c File Reference

`#include "form3.h"`

Include dependency graph for symmetr.c:



Functions

- WORD MatchE (PHEAD WORD ∗pattern, WORD ∗fun, WORD ∗inter, WORD par)
- WORD Permute (PERM ∗perm, WORD first)
- WORD PermuteP (PERMP ∗perm, WORD first)
- WORD Distribute (DISTRIBUTE ∗d, WORD first)
- int MatchCy (PHEAD WORD ∗pattern, WORD ∗fun, WORD ∗inter, WORD par)
- int FunMatchCy (PHEAD WORD ∗pattern, WORD ∗fun, WORD ∗inter, WORD par)
- int FunMatchSy (PHEAD WORD ∗pattern, WORD ∗fun, WORD ∗inter, WORD par)
- int MatchArgument (PHEAD WORD ∗arg, WORD ∗pat)

4.138.1 Detailed Description

The routines that deal with the pattern matching of functions with symmetric properties.

Definition in file symmetr.c.

4.138.2 Function Documentation

4.138.2.1 MatchE()

```
WORD MatchE (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )
```

Definition at line 56 of file symmetr.c.

4.138.2.2 Permute()

```
WORD Permute (
            PERM * perm,
            WORD first )
```

Definition at line 443 of file symmetr.c.

4.138.2.3 PermuteP()

```
WORD PermuteP (
            PERMP * perm,
            WORD first )
```

Definition at line 477 of file symmetr.c.

4.138.2.4 Distribute()

```
WORD Distribute (
            DISTRIBUTE * d,
            WORD first )
```

Definition at line 509 of file symmetr.c.

4.138.2.5 MatchCy()

```
int MatchCy (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )
```

Definition at line 570 of file symmetr.c.

4.138.2.6 FunMatchCy()

```
int FunMatchCy (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )
```

Definition at line 1066 of file symmetr.c.

4.138.2.7 FunMatchSy()

```
int FunMatchSy (
            PHEAD WORD * pattern,
            WORD * fun,
            WORD * inter,
            WORD par )
```

Definition at line 1524 of file symmetr.c.

4.138.2.8 MatchArgument()

```
int MatchArgument (
            PHEAD WORD * arg,
            WORD * pat )
```

Definition at line 2051 of file symmetr.c.

4.139 symmetr.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   for the community.
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 */
/* #] License : */
/*
    #[ Includes : function.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ MatchE :                 WORD MatchE(pattern,fun,inter,par)

        Matches symmetric and antisymmetric tensors.
        Pattern and fun point at a tensor.
        Problem is the wildcarding and all its possible permutations.
        This routine loops over all of them and calls for each
        possible wildcarding the recursion in ScanFunctions.
        Note that this can be very costly.

        Originally this routine did only Levi Civita tensors and hence
        it dealt only with commuting objects.
        Because of the backtracking we cannot fall back to the calling
        ScanFunctions routine and check the sequence of functions when
        non-commuting objects are involved.
*/

WORD MatchE(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
{
    GETBIDENTITY
    WORD *m, *t, *r, i, retval;
    WORD *mstop, *tstop, j, newvalue, newfun;
    WORD fixvec[MAXMATCH],wcvec[MAXMATCH],fixind[MAXMATCH],wcind[MAXMATCH];
    WORD tfixvec[MAXMATCH],tfixind[MAXMATCH];
    WORD vwc,vfix,ifix,iwc,tvfix,tifix,nv,ni;
    WORD sign = 0, *rstop, first1, first2, first3, funwild;
    WORD *OldWork, nwstore, oRepFunNum;
    PERM perm1,perm2;
    DISTRIBUTE distr;
    WORD *newpat, /* *newter, *instart, */ offset;
/*  instart = fun; */
    offset = WORDDIF(fun,AN.terstart);
```

```
    if ( pattern[1] != fun[1] ) return(0);
    if ( *pattern >= FUNCTION+WILDOFFSET ) {
        if ( CheckWild(BHEAD *pattern-WILDOFFSET,FUNTOFUN,*fun,&newfun) ) return(0);
        funwild = 1;
    }
    else funwild = 0;
    mstop = pattern + pattern[1];
    tstop = fun + fun[1];
    m = pattern + FUNHEAD;
    t = fun + FUNHEAD;
    while ( m < mstop ) {
        if ( *m != *t ) break;
        m++; t++;
    }
    if ( m >= mstop ) {
        AN.RepFunList[AN.RepFunNum++] = offset;
        AN.RepFunList[AN.RepFunNum++] = 0;
        newpat = pattern + pattern[1];
        if ( funwild ) {
            m = AN.WildValue;
            t = OldWork = AT.WorkPointer;
            nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
            r = AT.WildMask;
            if ( i > 0 ) {
                do {
                    *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
                } while ( --i > 0 );
            }
            if ( t >= AT.WorkTop ) {
                MLOCK(ErrorMessageLock);
                MesWork();
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            AT.WorkPointer = t;
            AddWild(BHEAD *pattern-WILDOFFSET,FUNTOFUN,newfun);
            if ( newpat >= AN.patstop ) {
                if ( AN.UseFindOnly == 0 ) {
                    if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                        AN.UsedOtherFind = 1;
                        return(1);
                    }
                    retval = 0;
                }
                else return(1);
            }
            else {
/*              newter = instart; */
                retval = ScanFunctions(BHEAD newpat,inter,par);
            }
            if ( retval == 0 ) {
                m = AN.WildValue;
                t = OldWork; r = AT.WildMask; i = nwstore;
                if ( i > 0 ) {
                    do {
                        *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                    } while ( --i > 0 );
                }
            }
            AT.WorkPointer = OldWork;
            return(retval);
        }
        else {
            if ( newpat >= AN.patstop ) {
                if ( AN.UseFindOnly == 0 ) {
                    if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                        AN.UsedOtherFind = 1;
                        return(1);
                    }
                    else return(0);
                }
                else return(1);
            }
/*          newter = instart; */
            i = ScanFunctions(BHEAD newpat,inter,par);
            return(i);
        }
/*

        Now the recursion
*/
    }
/*
    Strategy:
    1: match the fixed arguments
    2: match, permuting the wildcards if needed.
    3: keep track of sign.
*/
```

```
    vwc = 0;
    vfix = 0;
    ifix = 0;
    iwc = 0;
    r = pattern+FUNHEAD;
    while ( r < mstop ) {
        if ( *r < (AM.OffsetVector+WILDOFFSET) ) {
            fixvec[vfix++] = *r;            /* Fixed vectors */
            sign += vwc + ifix + iwc;
        }
        else if ( *r < MINSPEC ) {
            wcvec[vwc++] = *r;              /* Wildcard vectors */
            sign += ifix + iwc;
        }
        else if ( *r < (AM.OffsetIndex+WILDOFFSET) ) {
            fixind[ifix++] = *r;            /* Fixed indices */
            sign += iwc;
        }
        else if ( *r < (AM.OffsetIndex+(WILDOFFSET«1)) ) {
            wcind[iwc++] = *r;              /* Wildcard indices */
        }
        else {
            fixind[ifix++] = *r;            /* Generated indices ~ fixed */
            sign += iwc;
        }
        r++;
    }
    if ( iwc == 0 && vwc == 0 ) return(0);
    tvfix = tifix = 0;
    t = fun + FUNHEAD;
    m = fixvec;
    mstop = m + vfix;
    r = fixind;
    rstop = r + ifix;
    nv = 0; ni = 0;
    while ( t < tstop ) {
        if ( *t < 0 ) {
            nv++;
            if ( m < mstop && *t == *m ) {
                m++;
            }
            else {
                sign += WORDDIF(mstop,m);
                tfixvec[tvfix++] = *t;
            }
        }
        else {
            ni++;
            if ( r < rstop && *r == *t ) {
                r++;
            }
            else {
                sign += WORDDIF(rstop,r);
                tfixind[tifix++] = *t;
            }
        }
        t++;
    }
    if ( m < mstop || r < rstop ) return(0);
    if ( tvfix < vwc || (tvfix+tifix) < (vwc+iwc) ) return(0);
    sign += ( nv - vfix - vwc ) & ni;
/*
    Take now the wildcards that have an assignment already.
    See whether they match.
*/
    {
        WORD *wv, *wm, n;
        wm = AT.WildMask;
        wv = AN.WildValue;
        n = AN.NumWild;
        do {
            if ( *wm ) {
                if ( *wv == VECTOVEC ) {
                    for ( ni = 0; ni < vwc; ni++ ) {
                        if ( wcvec[ni]-WILDOFFSET == wv[2] ) {  /* Has been assigned */
                            sign += ni;
                            vwc--;
                            while ( ni < vwc ) {
                                wcvec[ni] = wcvec[ni+1];
                                ni++;
                            }
/* TryVect: */
                            for ( ni = 0; ni < tvfix; ni++ ) {
                                if ( tfixvec[ni] == wv[3] ) {
                                    sign += ni;
                                    tvfix--;
                                    while ( ni < tvfix ) {
```

```
                                tfixvec[ni] = tfixvec[ni+1];
                                ni++;
                            }
                            goto NextWV;
                        }
                    }
                    return(0);
                }
            }
        }
        else if ( *wv == INDTOIND ) {
            for ( ni = 0; ni < iwc; ni++ ) {
                if ( wcind[ni]-WILDOFFSET == wv[2] ) {  /* Has been assigned */
                    sign += ni;
                    iwc--;
                    while ( ni < iwc ) {
                        wcind[ni] = wcind[ni+1];
                        ni++;
                    }
                    for ( ni = 0; ni < tifix; ni++ ) {
                        if ( tfixind[ni] == wv[3] ) {
                            sign += ni;
                            tifix--;
                            while ( ni < tifix ) {
                                tfixind[ni] = tfixind[ni+1];
                                ni++;
                            }
                            goto NextWV;
                        }
                    }
/*                  goto TryVect; */
                    return(0);

                }
            }
        }
        else if ( *wv == VECTOSUB ) {
            for ( ni = 0; ni < vwc; ni++ ) {
                if ( wcvec[ni]-WILDOFFSET == wv[2] ) return(0);
            }
        }
        else if ( *wv == INDTOSUB ) {
            for ( ni = 0; ni < iwc; ni++ ) {
                if ( wcind[ni]-WILDOFFSET == wv[2] ) return(0);
            }
        }
    }
NextWV:
        wm++;
        wv += wv[1];
        n--;
        if ( n > 0 ) {
            while ( n > 0 && ( *wv == FROMSET || *wv == SETTONUM
            || *wv == LOADDOLLAR ) ) { wv += wv[1]; wm++; n--; }
/*
            Freak problem: doesn't test for n and ran into a remaining
            code equal to SETTONUM followed by a big number and then
            ran out of the memory.

            while ( *wv == FROMSET || *wv == SETTONUM
            || ( *wv == LOADDOLLAR && n > 0 ) ) { wv += wv[1]; wm++; n--; }
*/
        }
    } while ( n > 0 );
}
/*
    Now there are only free wildcards left.
    Possibly the assigned values ate too many vectors.
    The rest has to be done the 'hard way' via permutations.
    This is too bad when there are 10 indices.
    This could cause 10! tries.
    We try to avoid the worst case by using a very special
    (somewhat slow) permutation routine that has as its worst
    cases some rather unlikely configurations, rather than some
    common ones (as would have been the case with the conventional
    permutation routine).
        assume:
        vvvvvvvvvvvviiiiiii     (tvfix in tfixvec and tifix in tfixind)
        VVVVVVVVVIIIIIIIIII     (vwc in wcvec and iwc in wcind)
    Note: all further assignments are possible at this point!
    Strategy:
        permute v
        permute i
        loop over the ordered distribution of the leftover v's
        through the i's.
*/
    if ( tvfix < vwc ) { return(0); }
```

```
    perm1.n = tvfix;
    perm1.sign = 0;
    perm1.objects = tfixvec;
    perm2.n = tifix;
    perm2.sign = 0;
    perm2.objects = tfixind;
    distr.n1 = tvfix - vwc;
    distr.n2 = tifix;
    distr.obj1 = tfixvec + vwc;
    distr.obj2 = tfixind;
    distr.out = fixvec;             /* For scratch */
    first1 = 1;
/*
            Store the current Wildcard assignments
*/
    m = AN.WildValue;
    t = OldWork = AT.WorkPointer;
    nwstore = i = (m[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
    r = AT.WildMask;
    if ( i > 0 ) {
        do {
            *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
        } while ( --i > 0 );
    }
    if ( t >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AT.WorkPointer = t;
    while ( (first1 = Permute(&perm1,first1) ) == 0 ) {
        first2 = 1;
        while ( (first2 = Permute(&perm2,first2) ) == 0 ) {
            first3 = 1;
            while ( (first3 = Distribute(&distr,first3) ) == 0 ) {
/*
                Make now the wildcard assignments
*/
                for ( i = 0; i < vwc; i++ ) {
                    j = wcvec[i] - WILDOFFSET;
                    if ( CheckWild(BHEAD j,VECTOVEC,tfixvec[i],&newvalue) )
                        goto NoCaseB;
                    AddWild(BHEAD j,VECTOVEC,newvalue);
                }
                for ( i = 0; i < iwc; i++ ) {
                    j = wcind[i] - WILDOFFSET;
                    if ( CheckWild(BHEAD j,INDTOIND,fixvec[i],&newvalue) )
                        goto NoCaseB;
                    AddWild(BHEAD j,INDTOIND,newvalue);
                }
/*
                Go into the recursion
*/
                oRepFunNum = AN.RepFunNum;
                AN.RepFunList[AN.RepFunNum++] = offset;
                AN.RepFunList[AN.RepFunNum++] =
                    ( perm1.sign + perm2.sign + distr.sign + sign ) & 1;
                newpat = pattern + pattern[1];
                if ( funwild ) AddWild(BHEAD *pattern-WILDOFFSET,FUNTOFUN,newfun);
                if ( newpat >= AN.patstop ) {
                    if ( AN.UseFindOnly == 0 ) {
                        if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                            AN.UsedOtherFind = 1;
                            return(1);
                        }
                    }
                    else return(1);
                }
                else {
/*                  newter = instart; */
                    if ( ScanFunctions(BHEAD newpat,inter,par) ) { return(1); }
                }
/*
                Restore the old Wildcard assignments
*/
                AN.RepFunNum = oRepFunNum;
NoCaseB:        m = AN.WildValue;
                t = OldWork; r = AT.WildMask; i = nwstore;
                if ( i > 0 ) {
                    do {
                        *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                    } while ( --i > 0 );
                }
                AT.WorkPointer = t;
            }
        }
```

```
    }
    AT.WorkPointer = OldWork;
    return(0);
}

/*
    #] MatchE :
    #[ Permute :                WORD Permute(perm,first)

        Special permutation function.
        Works recursively.
        The aim is to cycle through in as fast a way as possible,
        to take care that each object hits the various positions
        already early in the game.

        Start at two: -> cycle of two
        then three -> cycle of three
        etc;
        The innermost cycle is the longest. This is the opposite
        of the usual way of generating permutations and it is
        certainly not the fastest one. It allows for the fastest
        hit in the assignment of wildcards though.
*/

WORD Permute(PERM *perm, WORD first)
{
    WORD *s, c, i, j;
    if ( first ) {
        perm->sign = ( perm->sign <= 1 ) ? 0: 1;
        for ( i = 0; i < perm->n; i++ ) perm->cycle[i] = 0;
        return(0);
    }
    i = perm->n;
    while ( --i > 0 ) {
        s = perm->objects;
        c = s[0];
        j = i;
        while ( --j >= 0 ) { *s = s[1]; s++; }
        *s = c;
        if ( ( i & 1 ) != 0 ) perm->sign ^= 1;
        if ( perm->cycle[i] < i ) {
            (perm->cycle[i])++;
            return(0);
        }
        else {
            perm->cycle[i] = 0;
        }
    }
    return(1);
}

/*
    #] Permute :
    #[ PermuteP :               WORD PermuteP(perm,first)

    Like Permute, but works on an array of pointers
*/

WORD PermuteP(PERMP *perm, WORD first)
{
    WORD **s, *c, i, j;
    if ( first ) {
        perm->sign = ( perm->sign <= 1 ) ? 0: 1;
        for ( i = 0; i < perm->n; i++ ) perm->cycle[i] = 0;
        return(0);
    }
    i = perm->n;
    while ( --i > 0 ) {
        s = perm->objects;
        c = s[0];
        j = i;
        while ( --j >= 0 ) { *s = s[1]; s++; }
        *s = c;
        if ( ( i & 1 ) != 0 ) perm->sign ^= 1;
        if ( perm->cycle[i] < i ) {
            (perm->cycle[i])++;
            return(0);
        }
        else {
            perm->cycle[i] = 0;
        }
    }
    return(1);
}

/*
    #] PermuteP :
```

```
    #[ Distribute :
*/

WORD Distribute(DISTRIBUTE *d, WORD first)
{
    WORD *to, *from, *inc, *from2, i, j;
    if ( first ) {
        d->n = d->n1 + d->n2;
        to = d->out;
        from = d->obj2;
        for ( i = 0; i < d->n2; i++ ) {
            d->cycle[i] = 0;
            *to++ = *from++;
        }
        from = d->obj1;
        while ( i < d->n ) {
            d->cycle[i++] = 1;
            *to++ = *from++;
        }
        d->sign = 0;
        return(0);
    }
    if ( d->n1 == 0 || d->n2 == 0 ) return(1);
    j = 0;
    i = 0;
    inc = d->cycle;
    from = inc + d->n;
    while ( *inc ) { j++; inc++; }
    while ( inc < from && !*inc ) { i++; inc++; }
    if ( inc >= from ) return(1);
    d->sign ^= ((i&j)-j+1) & 1;
    *inc = 0;
    *--inc = 1;
    while ( --j >= 0 ) *--inc = 1;
    while ( --i > 0 ) *--inc = 0;
    to = d->out;
    from = d->obj1;
    from2 = d->obj2;
    for ( i = 0; i < d->n; i++ ) {
        if ( *inc++ ) {
            *to++ = *from++;
        }
        else {
            *to++ = *from2++;
        }
    }
    return(0);
}

/*
    #] Distribute :
    #[ MatchCy :

        Matching of (r)cyclic tensors.
        Parameters like in MatchE.
        The structure of the routine is much simpler, because the number
        of possibilities is much more limited.
        The major complication is the ?a-type wildcards
        We need a strategy for T(i1?,?a,i1?,?b). Which is the shorter
        match: ?a or ?b ? (if possible of course)
        This is also relevant in the case of the shortest match if there
        is more than one choice for i1.
*/

int MatchCy(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
{
    GETBIDENTITY
    WORD *t, *tstop, *p, *pstop, *m, *r, *oldworkpointer = AT.WorkPointer;
    WORD *thewildcards, *multiplicity, *renum, wc, newvalue, oldwilval = 0;
    WORD *params, *lowlevel = 0;
    int argcount = 0, funnycount = 0, tcount = fun[1] - FUNHEAD;
    int type = 0, pnum, i, j, k, nwstore, iraise, itop, sumeat;
    CBUF *C = cbuf+AT.ebufnum;
    int ntwa = 3*AN.NumTotWildArgs+1;
    LONG oldcpointer = C->Pointer - C->Buffer;
    WORD offset = fun-AN.terstart, *newpat;

    if ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
    pnum = pattern[0];
    nwstore = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
    if ( pnum > FUNCTION + WILDOFFSET ) {
        pnum -= WILDOFFSET;
        if ( CheckWild(BHEAD pnum,FUNTOFUN,fun[0],&newvalue) ) return(0);
        oldwilval = 1;
        t = lowlevel = AT.WorkPointer;
        m = AN.WildValue;
        i = nwstore;
```

```
        r = AT.WildMask;
        if ( i > 0 ) {
            do {
                *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
            } while ( --i > 0 );
        }
        *t++ = C->numrhs;
        if ( t >= AT.WorkTop ) {
            MLOCK(ErrorMessageLock);
            MesWork();
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        AT.WorkPointer = t;
        AddWild(BHEAD pnum,FUNTOFUN,newvalue);
    }
    if ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;

    /* First we have to make an inventory. Are there FUNNYWILD pointers? */

    p = pattern + FUNHEAD;
    pstop = pattern + pattern[1];
    while ( p < pstop ) {
        if ( *p == FUNNYWILD ) { p += 2; funnycount++; }
        else { p++; argcount++; }
    }
    if ( argcount > tcount ) goto NoSuccess;
    if ( argcount < tcount && funnycount == 0 ) goto NoSuccess;
    if ( argcount == 0 && tcount == 0 && funnycount == 0 ) {
        AN.RepFunList[AN.RepFunNum++] = offset;
        AN.RepFunList[AN.RepFunNum++] = 0;
        newpat = pattern + pattern[1];
        if ( newpat >= AN.patstop ) {
            if ( AN.UseFindOnly == 0 ) {
                if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                    AT.WorkPointer = oldworkpointer;
                    AN.UsedOtherFind = 1;
                    return(1);
                }
                j = 0;
            }
            else {
                AT.WorkPointer = oldworkpointer;
                return(1);
            }
        }
        else j = ScanFunctions(BHEAD newpat,inter,par);
        if ( j ) return(j);
        goto NoSuccess;
    }
    tstop = fun + fun[1];

    /* Store the wildcard assignments */

    params = AT.WorkPointer;
    thewildcards = t = params + tcount;
    t += ntwa;
    if ( oldwilval ) lowlevel = oldworkpointer;
    else lowlevel = t;
    m = AN.WildValue;
    i = nwstore;
    if ( i > 0 ) {
        r = AT.WildMask;
        do {
            *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
        } while ( --i > 0 );
        *t++ = C->numrhs;
    }
    if ( t >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AT.WorkPointer = t;
/*
    #[ Case 1: no funnies or all funnies must be empty. We just cycle through.
*/
    if ( argcount == tcount ) {
        if ( funnycount > 0 ) { /* Test all funnies first */
            p = pattern + FUNHEAD;
            t = fun + FUNHEAD;
            while ( p < pstop ) {
                if ( *p != FUNNYWILD ) { p++; continue; }
                AN.argaddress = t;
                if ( CheckWild(BHEAD p[1],ARGTOARG,0,t) ) goto nomatch;
                AddWild(BHEAD p[1],ARGTOARG,0);
```

```
                p += 2;
            }
            oldwilval = 1;
        }
        for ( k = 0; k <= type; k++ ) {
            if ( k == 0 ) {
                p = params; t = fun + FUNHEAD;
                while ( t < tstop ) *p++ = *t++;
            }
            else {
                p = params+tcount; t = fun + FUNHEAD;
                while ( t < tstop ) *--p = *t++;
            }
            for ( i = 0; i < tcount; i++ ) { /* The various cycles */
                p = pattern + FUNHEAD;
                wc = 0;
                for ( j = 0; j < tcount; j++, p++ ) { /* The arguments */
                    while ( *p == FUNNYWILD ) p += 2;
                    t = params + (i+j)%tcount;
                    if ( *t == *p ) continue;
                    if ( *p >= AM.OffsetIndex + WILDOFFSET
                    && *p < AM.OffsetIndex + 2*WILDOFFSET ) {

                        /* Test wildcard index */

                        wc = *p - WILDOFFSET;
                        if ( CheckWild(BHEAD wc,INDTOIND,*t,&newvalue) ) break;
                        AddWild(BHEAD wc,INDTOIND,newvalue);
                    }
                    else if ( *t < MINSPEC && p[j] < MINSPEC
                        && *p >= AM.OffsetVector + WILDOFFSET ) {

                        /* Test wildcard vector */

                        wc = *p - WILDOFFSET;
                        if ( CheckWild(BHEAD wc,VECTOVEC,*t,&newvalue) ) break;
                        AddWild(BHEAD wc,VECTOVEC,newvalue);
                    }
                    else break;
                }
                if ( j >= tcount ) { /* Match! */

                    /* Continue with other functions. Make sure of the funnies */

                    AN.RepFunList[AN.RepFunNum++] = offset;
                    AN.RepFunList[AN.RepFunNum++] = 0;

                    if ( funnycount > 0 ) {
                        p = pattern + FUNHEAD;
                        t = fun + FUNHEAD;
                        while ( p < pstop ) {
                            if ( *p != FUNNYWILD ) { p++; continue; }
                            AN.argaddress = t;
                            AddWild(BHEAD p[1],ARGTOARG,0);
                            p += 2;
                        }
                    }
                    newpat = pattern + pattern[1];
                    if ( newpat >= AN.patstop ) {
                        if ( AN.UseFindOnly == 0 ) {
                            if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                                AT.WorkPointer = oldworkpointer;
                                AN.UsedOtherFind = 1;
                                return(1);
                            }
                            j = 0;
                        }
                        else {
                            AT.WorkPointer = oldworkpointer;
                            return(1);
                        }
                    }
                    else j = ScanFunctions(BHEAD newpat,inter,par);
                    if ( j ) {
                        AT.WorkPointer = oldworkpointer;
                        return(j); /* Full match. Return our success */
                    }
                    AN.RepFunNum -= 2;
                }

                /* No (deeper) match. -> reset wildcards and continue */

                if ( wc && nwstore > 0 ) {
                    j = nwstore;
                    m = AN.WildValue;
                    t = thewildcards + ntwa; r = AT.WildMask;
                    if ( j > 0 ) {
```

```
                            do {
                                *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                            } while ( --j > 0 );
                        }
                        C->numrhs = *t++;
                        C->Pointer = C->Buffer + oldcpointer;
                    }
                }
            }
            goto NoSuccess;
        }
/*
    #] Case 1:
    #[ Case 2: One FUNNYWILD. Fix its length.
*/
    if ( funnycount == 1 ) {
        funnycount = tcount - argcount; /* Number or arguments to be eaten */
        for ( k = 0; k <= type; k++ ) {
            if ( k == 0 ) {
                p = params; t = fun + FUNHEAD;
                while ( t < tstop ) *p++ = *t++;
            }
            else {
                p = params+tcount; t = fun + FUNHEAD;
                while ( t < tstop ) *--p = *t++;
            }
            for ( i = 0; i < tcount; i++ ) { /* The various cycles */
                p = pattern + FUNHEAD;
                t = params;
                wc = 0;
                for ( j = 0; j < tcount; j++, p++, t++ ) { /* The arguments */
                    if ( *t == *p ) continue;
                    if ( *p == FUNNYWILD ) {
                        p++; wc = 1;
                        AN.argaddress = t;
                        if ( CheckWild(BHEAD *p,ARGTOARG|EATTENSOR,funnycount,t) ) break;
                        AddWild(BHEAD *p,ARGTOARG|EATTENSOR,funnycount);
                        j += funnycount-1; t += funnycount-1;
                    }
                    else if ( *p >= AM.OffsetIndex + WILDOFFSET
                    && *p < AM.OffsetIndex + 2*WILDOFFSET ) {

                        /* Test wildcard index */

                        wc = *p - WILDOFFSET;
                        if ( CheckWild(BHEAD wc,INDTOIND,*t,&newvalue) ) break;
                        AddWild(BHEAD wc,INDTOIND,newvalue);
                    }
                    else if ( *t < MINSPEC && *p < MINSPEC
                        && *p >= AM.OffsetVector + WILDOFFSET ) {

                        /* Test wildcard vector */

                        wc = *p - WILDOFFSET;
                        if ( CheckWild(BHEAD wc,VECTOVEC,*t,&newvalue) ) break;
                        AddWild(BHEAD wc,VECTOVEC,newvalue);
                    }
                    else break;
                }
                if ( j >= tcount ) { /* Match! */

                    /* Continue with other functions. Make sure of the funnies */

                    AN.RepFunList[AN.RepFunNum++] = offset;
                    AN.RepFunList[AN.RepFunNum++] = 0;
                    newpat = pattern + pattern[1];
                    if ( newpat >= AN.patstop ) {
                        if ( AN.UseFindOnly == 0 ) {
                            if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                                AT.WorkPointer = oldworkpointer;
                                AN.UsedOtherFind = 1;
                                return(1);
                            }
                            j = 0;
                        }
                        else {
                            AT.WorkPointer = oldworkpointer;
                            return(1);
                        }
                    }
                    else j = ScanFunctions(BHEAD newpat,inter,par);
                    if ( j ) {
                        AT.WorkPointer = oldworkpointer;
                        return(j); /* Full match. Return our success */
                    }
                    AN.RepFunNum -= 2;
                }
```

```

                /* No (deeper) match. -> reset wildcards and continue */

                if ( wc ) {
                    j = nwstore;
                    m = AN.WildValue;
                    t = thewildcards + ntwa; r = AT.WildMask;
                    if ( j > 0 ) {
                        do {
                            *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                        } while ( --j > 0 );
                    }
                    C->numrhs = *t++;
                    C->Pointer = C->Buffer + oldcpointer;
                }
                t = params;
                wc = *t;
                for ( j = 1; j < tcount; j++ ) { *t = t[1]; t++; }
                *t = wc;
            }
        }
        goto NoSuccess;
    }
/*
    #] Case 2:
    #[ Case 3: More than one FUNNYWILD. Complicated.
*/

    sumeat = tcount - argcount; /* Total number to be eaten by Funnies */
/*
    In the first funnycount elements of 'thewildcards' we arrange
    for the summing over the various possibilities.
    The renumbering table is in thewildcards[2*funnycount]
    The multiplicity table is in thewildcards[funnycount]
    The number of arguments for each is in thewildcards[]
*/
    p = pattern+FUNHEAD;
    for ( i = funnycount; i < ntwa; i++ ) thewildcards[i] = -1;
    multiplicity = thewildcards + funnycount;
    renum = multiplicity + funnycount;
    j = 0;
    while ( p < pstop ) {
        if ( *p != FUNNYWILD ) { p++; continue; }
        p++;
        if ( renum[*p] < 0 ) {
            renum[*p] = j;
            multiplicity[j] = 1;
            j++;
        }
        else multiplicity[renum[*p]]++;
        p++;
    }
/*
    Strategy: First 'declared' has a tendency to be smaller
*/
    for ( i = 1; i < AN.NumTotWildArgs; i++ ) {
        if ( renum[i] < 0 ) continue;
        for ( j = i+1; j <= AN.NumTotWildArgs; j++ ) {
            if ( renum[j] < 0 ) continue;
            if ( renum[i] < renum[j] ) continue;
            k = multiplicity[renum[i]];
            multiplicity[renum[i]] = multiplicity[renum[j]];
            multiplicity[renum[j]] = k;
            k = renum[i]; renum[i] = renum[j]; renum[j] = k;
        }
    }
    for ( i = 0; i < funnycount; i++ ) thewildcards[i] = 0;
    iraise = funnycount-1;
    for ( ;; ) {
        for ( i = 0, j = sumeat; i < iraise; i++ )
                j -= thewildcards[i]*multiplicity[i];
        if ( j < 0 || j % multiplicity[iraise] != 0 ) {
            if ( j > 0 ) {
                thewildcards[iraise-1]++;
                continue;
            }
            itop = iraise-1;
            while ( itop > 0 && j < 0 ) {
                j += thewildcards[itop]*multiplicity[itop];
                thewildcards[itop] = 0;
                itop--;
            }
            if ( itop <= 0 && j <= 0 ) break;
            thewildcards[itop]++;
            continue;
        }
        thewildcards[iraise] = j / multiplicity[iraise];
```

```

        for ( k = 0; k <= type; k++ ) {
            if ( k == 0 ) {
                p = params; t = fun + FUNHEAD;
                while ( t < tstop ) *p++ = *t++;
            }
            else {
                p = params+tcount; t = fun + FUNHEAD;
                while ( t < tstop ) *--p = *t++;
            }
            for ( i = 0; i < tcount; i++ ) { /* The various cycles */
                p = pattern + FUNHEAD;
                t = params;
                wc = 0;
                for ( j = 0; j < tcount; j++, p++, t++ ) { /* The arguments */
                    if ( *t == *p ) continue;
                    if ( *p == FUNNYWILD ) {
                        p++; wc = thewildcards[renum[*p]];
                        AN.argaddress = t;
                        if ( CheckWild(BHEAD *p,ARGTOARG|EATTENSOR,wc,t) ) break;
                        AddWild(BHEAD *p,ARGTOARG|EATTENSOR,wc);
                        j += wc-1; t += wc-1; wc = 1;
                    }
                    else if ( *p >= AM.OffsetIndex + WILDOFFSET
                    && *p < AM.OffsetIndex + 2*WILDOFFSET ) {

                        /* Test wildcard index */

                        wc = *p - WILDOFFSET;
                        if ( CheckWild(BHEAD wc,INDTOIND,*t,&newvalue) ) break;
                        AddWild(BHEAD wc,INDTOIND,newvalue);
                    }
                    else if ( *t < MINSPEC && *p < MINSPEC
                        && *p >= AM.OffsetVector + WILDOFFSET ) {

                        /* Test wildcard vector */

                        wc = *p - WILDOFFSET;
                        if ( CheckWild(BHEAD wc,VECTOVEC,*t,&newvalue) ) break;
                        AddWild(BHEAD wc,VECTOVEC,newvalue);
                    }
                    else break;
                }
                if ( j >= tcount ) { /* Match! */

                    /* Continue with other functions. Make sure of the funnies */

                    AN.RepFunList[AN.RepFunNum++] = offset;
                    AN.RepFunList[AN.RepFunNum++] = 0;
                    newpat = pattern + pattern[1];
                    if ( newpat >= AN.patstop ) {
                        if ( AN.UseFindOnly == 0 ) {
                            if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                                AT.WorkPointer = oldworkpointer;
                                AN.UsedOtherFind = 1;
                                return(1);
                            }
                            j = 0;
                        }
                        else {
                            AT.WorkPointer = oldworkpointer;
                            return(1);
                        }
                    }
                    else j = ScanFunctions(BHEAD newpat,inter,par);
                    if ( j ) {
                        AT.WorkPointer = oldworkpointer;
                        return(j); /* Full match. Return our success */
                    }
                    AN.RepFunNum -= 2;
                }

                /* No (deeper) match. -> reset wildcards and continue */

                if ( wc ) {
                    j = nwstore;
                    m = AN.WildValue;
                    t = thewildcards + ntwa; r = AT.WildMask;
                    if ( j > 0 ) {
                        do {
                            *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                        } while ( --j > 0 );
                    }
                    C->numrhs = *t++;
                    C->Pointer = C->Buffer + oldcpointer;
                }
                t = params;
```

```
                wc = *t;
                for ( j = 1; j < tcount; j++ ) { *t = t[1]; t++; }
                *t = wc;
            }
        }
        (thewildcards[iraise-1])++;
    }
/*
    #] Case 3:
*/
NoSuccess:
    if ( oldwilval > 0 ) {
nomatch:;
        j = nwstore;
        if ( j > 0 ) {
            m = AN.WildValue;
            t = lowlevel; r = AT.WildMask;
            if ( j > 0 ) {
                do {
                    *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                } while ( --j > 0 );
            }
            C->numrhs = *t++;
            C->Pointer = C->Buffer + oldcpointer;
        }
    }
    AT.WorkPointer = oldworkpointer;
    return(0);
}

/*
    #] MatchCy :
    #[ FunMatchCy :

        Matching of (r)cyclic functions.
        Like MatchCy, but now for general functions.
*/

int FunMatchCy(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
{
    GETBIDENTITY
    WORD *t, *tstop, *p, *pstop, *m, *r, *oldworkpointer = AT.WorkPointer;
    WORD **a, *thewildcards, *multiplicity, *renum, wc, wcc, oldwilval = 0;
    LONG oww = AT.pWorkPointer;
    WORD newvalue, *lowlevel = 0;
    int argcount = 0, funnycount = 0, tcount = 0;
    int type = 0, pnum, i, j, k, nwstore, iraise, itop, sumeat;
    CBUF *C = cbuf+AT.ebufnum;
    int ntwa = 3*AN.NumTotWildArgs+1;
    LONG oldcpointer = C->Pointer - C->Buffer;
    WORD offset = fun-AN.terstart, *newpat;

    if ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
    pnum = pattern[0];
    nwstore = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
    if ( pnum > FUNCTION + WILDOFFSET ) {
        pnum -= WILDOFFSET;
        if ( CheckWild(BHEAD pnum,FUNTOFUN,fun[0],&newvalue) ) return(0);
        oldwilval = 1;
        t = lowlevel = oldworkpointer;
        m = AN.WildValue;
        i = nwstore;
        r = AT.WildMask;
        if ( i > 0 ) {
            do {
                *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
            } while ( --i > 0 );
        }
        *t++ = C->numrhs;
        if ( t >= AT.WorkTop ) {
            MLOCK(ErrorMessageLock);
            MesWork();
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        AT.WorkPointer = t;
        AddWild(BHEAD pnum,FUNTOFUN,newvalue);
    }
    if ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;

    /* First we have to make an inventory. Are there -ARGWILD pointers? */

    p = pattern + FUNHEAD;
    pstop = pattern + pattern[1];
    while ( p < pstop ) {
        if ( *p == -ARGWILD ) { p += 2; funnycount++; }
        else { NEXTARG(p); argcount++; }
```

```
    }
    t = fun + FUNHEAD;
    tstop = fun + fun[1];
    while ( t < tstop ) { NEXTARG(t); tcount++; }

    if ( argcount > tcount ) return(0);
    if ( argcount < tcount && funnycount == 0 ) return(0);
    if ( argcount == 0 && tcount == 0 && funnycount == 0 ) {
        AN.RepFunList[AN.RepFunNum++] = offset;
        AN.RepFunList[AN.RepFunNum++] = 0;
        newpat = pattern + pattern[1];
        if ( newpat >= AN.patstop ) {
            if ( AN.UseFindOnly == 0 ) {
                if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                    AT.WorkPointer = oldworkpointer;
                    AN.UsedOtherFind = 1;
                    return(1);
                }
                j = 0;
            }
            else {
                AT.WorkPointer = oldworkpointer;
                return(1);
            }
        }
        else j = ScanFunctions(BHEAD newpat,inter,par);
        if ( j ) return(j);
        goto NoSuccess;
    }

    /* Store the wildcard assignments */

    WantAddPointers(tcount);
    AT.pWorkPointer += tcount;
    thewildcards = t = AT.WorkPointer;
    t += ntwa;
    if ( oldwilval ) lowlevel = oldworkpointer;
    else lowlevel = t;
    m = AN.WildValue;
    i = nwstore;
    if ( i > 0 ) {
        r = AT.WildMask;
        do {
            *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
        } while ( --i > 0 );
        *t++ = C->numrhs;
    }
    if ( t >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AT.WorkPointer = t;
/*
    #[ Case 1: no funnies or all funnies must be empty. We just cycle through.
*/
    if ( argcount == tcount ) {
        if ( funnycount > 0 ) { /* Test all funnies first */
            p = pattern + FUNHEAD;
            t = fun + FUNHEAD;
            while ( p < pstop ) {
                if ( *p != -ARGWILD ) { p++; continue; }
                AN.argaddress = t;
                if ( CheckWild(BHEAD p[1],ARGTOARG,0,t) ) goto nomatch;
                AddWild(BHEAD p[1],ARGTOARG,0);
                p += 2;
            }
            oldwilval = 1;
        }
        for ( k = 0; k <= type; k++ ) {
            if ( k == 0 ) {
                a = AT.pWorkSpace+oww; t = fun + FUNHEAD;
                while ( t < tstop ) { *a++ = t; NEXTARG(t); }
            }
            else {
                a = AT.pWorkSpace+oww+tcount; t = fun + FUNHEAD;
                while ( t < tstop ) { *--a = t; NEXTARG(t); }
            }
            for ( i = 0; i < tcount; i++ ) { /* The various cycles */
                p = pattern + FUNHEAD;
                wc = 0;
                for ( j = 0; j < tcount; j++ ) { /* The arguments */
                    while ( *p == -ARGWILD ) p += 2;
                    t = AT.pWorkSpace[oww+((i+j)%tcount)];
                    if ( ( wcc =  MatchArgument(BHEAD t,p) ) == 0 ) break;
                    if ( wcc > 1 ) wc = 1;
```

```
                    NEXTARG(p);
                }
                if ( j >= tcount ) { /* Match! */

                    /* Continue with other functions. Make sure of the funnies */

                    AN.RepFunList[AN.RepFunNum++] = offset;
                    AN.RepFunList[AN.RepFunNum++] = 0;

                    if ( funnycount > 0 ) {
                        p = pattern + FUNHEAD;
                        t = fun + FUNHEAD;
                        while ( p < pstop ) {
                            if ( *p != -ARGWILD ) { p++; continue; }
                            AN.argaddress = t;
                            AddWild(BHEAD p[1],ARGTOARG,0);
                            p += 2;
                        }
                    }
                    newpat = pattern + pattern[1];
                    if ( newpat >= AN.patstop ) {
                        if ( AN.UseFindOnly == 0 ) {
                            if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                                AT.WorkPointer = oldworkpointer;
                                AT.pWorkPointer = oww;
                                AN.UsedOtherFind = 1;
                                return(1);
                            }
                            j = 0;
                        }
                        else {
                            AT.WorkPointer = oldworkpointer;
                            AT.pWorkPointer = oww;
                            return(1);
                        }
                    }
                    else j = ScanFunctions(BHEAD newpat,inter,par);
                    if ( j ) {
                        AT.WorkPointer = oldworkpointer;
                        AT.pWorkPointer = oww;
                        return(j); /* Full match. Return our success */
                    }
                    AN.RepFunNum -= 2;
                }

                /* No (deeper) match. -> reset wildcards and continue */

                if ( wc && nwstore > 0 ) {
                    j = nwstore;
                    m = AN.WildValue;
                    t = thewildcards + ntwa; r = AT.WildMask;
                    if ( j > 0 ) {
                        do {
                            *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                        } while ( --j > 0 );
                    }
                    C->numrhs = *t++;
                    C->Pointer = C->Buffer + oldcpointer;
                }
            }
        }
        goto NoSuccess;
    }
/*
    #] Case 1:
    #[ Case 2: One -ARGWILD. Fix its length.
*/
    if ( funnycount == 1 ) {
        funnycount = tcount - argcount; /* Number or arguments to be eaten */
        for ( k = 0; k <= type; k++ ) {
            if ( k == 0 ) {
                a =  AT.pWorkSpace+oww; t = fun + FUNHEAD;
                while ( t < tstop ) { *a++ = t; NEXTARG(t); }
            }
            else {
                a =  AT.pWorkSpace+oww+tcount; t = fun + FUNHEAD;
                while ( t < tstop ) { *--a = t; NEXTARG(t); }
            }
            for ( i = 0; i < tcount; i++ ) { /* The various cycles */
                p = pattern + FUNHEAD;
                a = AT.pWorkSpace+oww;
                wc = 0;
                for ( j = 0; j < tcount; j++, a++ ) { /* The arguments */
                    t = *a;
                    if ( *p == -ARGWILD ) {
                        wc = 1;
                        AN.argaddress = (WORD *)a;
```

```
                        if ( CheckWild(BHEAD p[1],ARLTOARL,funnycount,(WORD *)a) ) break;
                        AddWild(BHEAD p[1],ARLTOARL,funnycount);
                        j += funnycount-1; a += funnycount-1;
                    }
                    else if ( MatchArgument(BHEAD t,p) == 0 ) break;
                    NEXTARG(p);
                }
                if ( j >= tcount ) { /* Match! */

                    /* Continue with other functions. Make sure of the funnies */

                    AN.RepFunList[AN.RepFunNum++] = offset;
                    AN.RepFunList[AN.RepFunNum++] = 0;
                    newpat = pattern + pattern[1];
                    if ( newpat >= AN.patstop ) {
                        if ( AN.UseFindOnly == 0 ) {
                            if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                                AT.WorkPointer = oldworkpointer;
                                AT.pWorkPointer = oww;
                                AN.UsedOtherFind = 1;
                                return(1);
                            }
                            j = 0;
                        }
                        else {
                            AT.WorkPointer = oldworkpointer;
                            AT.pWorkPointer = oww;
                            return(1);
                        }
                    }
                    else j = ScanFunctions(BHEAD newpat,inter,par);
                    if ( j ) {
                        AT.WorkPointer = oldworkpointer;
                        AT.pWorkPointer = oww;
                        return(j); /* Full match. Return our success */
                    }
                    AN.RepFunNum -= 2;
                }

                /* No (deeper) match. -> reset wildcards and continue */

                if ( wc ) {
                    j = nwstore;
                    m = AN.WildValue;
                    t = thewildcards + ntwa; r = AT.WildMask;
                    if ( j > 0 ) {
                        do {
                            *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                        } while ( --j > 0 );
                    }
                    C->numrhs = *t++;
                    C->Pointer = C->Buffer + oldcpointer;
                }
                a = AT.pWorkSpace+oww;
                t = *a;
                for ( j = 1; j < tcount; j++ ) { *a = a[1]; a++; }
                *a = t;
            }
        }
        goto NoSuccess;
    }
/*
    #] Case 2:
    #[ Case 3: More than one -ARGWILD. Complicated.
*/

    sumeat = tcount - argcount; /* Total number to be eaten by Funnies */
/*
    In the first funnycount elements of 'thewildcards' we arrange
    for the summing over the various possibilities.
    The renumbering table is in thewildcards[2*funnycount]
    The multiplicity table is in thewildcards[funnycount]
    The number of arguments for each is in thewildcards[]
*/
    p = pattern+FUNHEAD;
    for ( i = funnycount; i < ntwa; i++ ) thewildcards[i] = -1;
    multiplicity = thewildcards + funnycount;
    renum = multiplicity + funnycount;
    j = 0;
    while ( p < pstop ) {
        if ( *p != -ARGWILD ) { p++; continue; }
        p++;
        if ( renum[*p] < 0 ) {
            renum[*p] = j;
            multiplicity[j] = 1;
            j++;
        }
```

```
        else multiplicity[renum[*p]]++;
        p++;
    }
/*
    Strategy: First 'declared' has a tendency to be smaller
*/
    for ( i = 1; i < AN.NumTotWildArgs; i++ ) {
        if ( renum[i] < 0 ) continue;
        for ( j = i+1; j <= AN.NumTotWildArgs; j++ ) {
            if ( renum[j] < 0 ) continue;
            if ( renum[i] < renum[j] ) continue;
            k = multiplicity[renum[i]];
            multiplicity[renum[i]] = multiplicity[renum[j]];
            multiplicity[renum[j]] = k;
            k = renum[i]; renum[i] = renum[j]; renum[j] = k;
        }
    }
    for ( i = 0; i < funnycount; i++ ) thewildcards[i] = 0;
    iraise = funnycount-1;
    for ( ;; ) {
        for ( i = 0, j = sumeat; i < iraise; i++ )
                j -= thewildcards[i]*multiplicity[i];
        if ( j < 0 || j % multiplicity[iraise] != 0 ) {
            if ( j > 0 ) {
                thewildcards[iraise-1]++;
                continue;
            }
            itop = iraise-1;
            while ( itop > 0 && j < 0 ) {
                j += thewildcards[itop]*multiplicity[itop];
                thewildcards[itop] = 0;
                itop--;
            }
            if ( itop <= 0 && j <= 0 ) break;
            thewildcards[itop]++;
            continue;
        }
        thewildcards[iraise] = j / multiplicity[iraise];

        for ( k = 0; k <= type; k++ ) {
            if ( k == 0 ) {
                a = AT.pWorkSpace+oww; t = fun + FUNHEAD;
                while ( t < tstop ) { *a++ = t; NEXTARG(t); }
            }
            else {
                a = AT.pWorkSpace+oww+tcount; t = fun + FUNHEAD;
                while ( t < tstop ) { *--a = t; NEXTARG(t); }
            }
            for ( i = 0; i < tcount; i++ ) { /* The various cycles */
                p = pattern + FUNHEAD;
                a = AT.pWorkSpace+oww;
                wc = 0;
                for ( j = 0; j < tcount; j++, a++ ) { /* The arguments */
                    t = *a;
                    if ( *p == -ARGWILD ) {
                        wc = thewildcards[renum[p[1]]];
                        AN.argaddress = (WORD *)a;
                        if ( CheckWild(BHEAD p[1],ARLTOARL,wc,(WORD *)a) ) break;
                        AddWild(BHEAD p[1],ARLTOARL,wc);
                        j += wc-1; a += wc-1; wc = 1;
                    }
                    else if ( MatchArgument(BHEAD t,p) == 0 ) break;
                    NEXTARG(p);
                }
                if ( j >= tcount ) { /* Match! */

                    /* Continue with other functions. Make sure of the funnies */

                    AN.RepFunList[AN.RepFunNum++] = offset;
                    AN.RepFunList[AN.RepFunNum++] = 0;
                    newpat = pattern + pattern[1];
                    if ( newpat >= AN.patstop ) {
                        if ( AN.UseFindOnly == 0 ) {
                            if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                                AT.WorkPointer = oldworkpointer;
                                AT.pWorkPointer = oww;
                                AN.UsedOtherFind = 1;
                                return(1);
                            }
                            j = 0;
                        }
                        else {
                            AT.WorkPointer = oldworkpointer;
                            AT.pWorkPointer = oww;
                            return(1);
                        }
                    }
```

```
                    else j = ScanFunctions(BHEAD newpat,inter,par);
                    if ( j ) {
                        AT.WorkPointer = oldworkpointer;
                        AT.pWorkPointer = oww;
                        return(j); /* Full match. Return our success */
                    }
                    AN.RepFunNum -= 2;
                }

                /* No (deeper) match. -> reset wildcards and continue */

                if ( wc ) {
                    j = nwstore;
                    m = AN.WildValue;
                    t = thewildcards + ntwa; r = AT.WildMask;
                    if ( j > 0 ) {
                        do {
                            *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                        } while ( --j > 0 );
                    }
                    C->numrhs = *t++;
                    C->Pointer = C->Buffer + oldcpointer;
                }
                a = AT.pWorkSpace+oww;
                t = *a;
                for ( j = 1; j < tcount; j++ ) { *a = a[1]; a++; }
                *a = t;
            }
        }
        (thewildcards[iraise-1])++;
    }
/*
    #] Case 3:
*/
NoSuccess:
    if ( oldwilval > 0 ) {
nomatch:;
        j = nwstore;
        m = AN.WildValue;
        t = lowlevel; r = AT.WildMask;
        if ( j > 0 ) {
            do {
                *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
            } while ( --j > 0 );
        }
        C->numrhs = *t++;
        C->Pointer = C->Buffer + oldcpointer;
    }
    AT.WorkPointer = oldworkpointer;
    AT.pWorkPointer = oww;
    return(0);
}

/*
    #] FunMatchCy :
    #[ FunMatchSy :

        Matching of (anti)symmetric functions.
        Like MatchE, but now for general functions.
*/

int FunMatchSy(PHEAD WORD *pattern, WORD *fun, WORD *inter, WORD par)
{
    GETBIDENTITY
    WORD *t, *tstop, *p, *pstop, *m, *r, *oldworkpointer = AT.WorkPointer;
    WORD **a, *thewildcards, oldwilval = 0;
    WORD newvalue, *lowlevel = 0, num, assig;
    WORD *cycles;
    LONG oww = AT.pWorkPointer, lhpars, lhfunnies;
    int argcount = 0, funnycount = 0, tcount = 0, signs = 0, signfun = 0, signo;
    int type = 0, pnum, i, j, k, nwstore, iraise, cou2;
    CBUF *C = cbuf+AT.ebufnum;
    int ntwa = 3*AN.NumTotWildArgs+1;
    LONG oldcpointer = C->Pointer - C->Buffer;
    WORD offset = fun-AN.terstart, *newpat;

    if ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;
    pnum = pattern[0];
    nwstore = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;
    if ( pnum > FUNCTION + WILDOFFSET ) {
        pnum -= WILDOFFSET;
        if ( CheckWild(BHEAD pnum,FUNTOFUN,fun[0],&newvalue) ) return(0);
        oldwilval = 1;
        t = lowlevel = oldworkpointer;
        m = AN.WildValue;
        i = nwstore;
        r = AT.WildMask;
```

```
        if ( i > 0 ) {
            do {
                *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
            } while ( --i > 0 );
        }
        *t++ = C->numrhs;
        if ( t >= AT.WorkTop ) {
            MLOCK(ErrorMessageLock);
            MesWork();
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        AT.WorkPointer = t;
        AddWild(BHEAD pnum,FUNTOFUN,newvalue);
    }
    if ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == RCYCLESYMMETRIC ) type = 1;

    /* Try for a straight match. After all, both have been normalized */

    if ( fun[1] == pattern[1] ) {
        i = fun[1]-FUNHEAD; p = pattern+FUNHEAD; t = fun + FUNHEAD;
        while ( --i >= 0 ) { if ( *p++ != *t++ ) break; }
        if ( i < 0 ) goto quicky;
    }

    /* First we have to make an inventory. Are there -ARGWILD pointers? */

    p = pattern + FUNHEAD;
    pstop = pattern + pattern[1];
    while ( p < pstop ) {
        if ( *p == -ARGWILD ) { p += 2; funnycount++; }
        else { NEXTARG(p); argcount++; }
    }
    t = fun + FUNHEAD;
    tstop = fun + fun[1];
    while ( t < tstop ) { NEXTARG(t); tcount++; }

    if ( argcount > tcount ) return(0);
    if ( argcount < tcount && funnycount == 0 ) return(0);
    if ( argcount == 0 && tcount == 0 && funnycount == 0 ) {
quicky:
        if ( AN.SignCheck && signs != AN.ExpectedSign ) goto NoSuccess;
        AN.RepFunList[AN.RepFunNum++] = offset;
        AN.RepFunList[AN.RepFunNum++] = signs;
        newpat = pattern + pattern[1];
        if ( newpat >= AN.patstop ) {
            if ( AN.UseFindOnly == 0 ) {
                if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                    AT.WorkPointer = oldworkpointer;
                    AN.UsedOtherFind = 1;
                    return(1);
                }
                j = 0;
            }
            else {
                AT.WorkPointer = oldworkpointer;
                return(1);
            }
        }
        else j = ScanFunctions(BHEAD newpat,inter,par);
        if ( j ) {
            AT.WorkPointer = oldworkpointer;
            return(j);
        }
        goto NoSuccess;
    }

    /* Store the wildcard assignments */

    WantAddPointers(tcount+argcount+funnycount);
    AT.pWorkPointer += tcount+argcount+funnycount;
    thewildcards = t = AT.WorkPointer;
    t += ntwa;
    if ( oldwilval ) lowlevel = oldworkpointer;
    else lowlevel = t;
    m = AN.WildValue;
    i = nwstore; assig = 0;
    if ( i > 0 ) {
        r = AT.WildMask;
        do {
            assig += *r;
            *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *m++; *t++ = *r++;
        } while ( --i > 0 );
        *t++ = C->numrhs;
    }
    if ( t >= AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
```

```
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AT.WorkPointer = t;

    /* Store pointers to the arguments */

    t = fun + FUNHEAD; a = AT.pWorkSpace+oww;
    while ( t < tstop ) { *a++ = t; NEXTARG(t) }
    lhpars = a-AT.pWorkSpace;
    t = pattern + FUNHEAD;
    while ( t < pstop ) {
        if ( *t != -ARGWILD ) *a++ = t;
        NEXTARG(t)
    }
    lhfunnies = a-AT.pWorkSpace;
    t = pattern + FUNHEAD; cou2 = 0;
    while ( t < pstop ) {
        cou2++;
        if ( *t == -ARGWILD ) {
            *a++ = t;
/*
    signfun: last ?a: tcount-argcount: number of arguments in ?a (assume one ?a)
        argcount+funnycount-cou2: arguments after ?a.
    Together tells whether moving ?a to end of list is even or odd
*/
            signfun = ((argcount+funnycount-cou2)*(tcount-argcount)) & 1;
        }
        NEXTARG(t)
    }
    signs += signfun;
    if ( funnycount > 0 ) {
        if ( ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == SYMMETRIC )
            || ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
            || ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == SYMMETRIC )
            || ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC ) ) {
                AT.WorkPointer = oldworkpointer;
                AT.pWorkPointer = oww;
                MLOCK(ErrorMessageLock);
                MesPrint("Sorry: no argument field wildcards yet in (anti)symmetric functions");
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
        }
    }
/*
    Sort the regular arguments by
    1: no wildcards, fast.
    2: wildcards that have been assigned.
    3: general arguments.
    4: wildcards without an assignment.
*/
    iraise = argcount;
    for ( i = 0; i < iraise; i++ ) {
        t = AT.pWorkSpace[i+lhpars];
        if ( *t > 0 ) { /* Category 3: general argument */
            continue;
        }
        else if ( *t <= -FUNCTION ) {
            if ( *t > -FUNCTION - WILDOFFSET ) goto cat1;
            type = FUNTOFUN; num = -*t - WILDOFFSET;
        }
        else if ( *t == -SYMBOL ) {
            if ( t[1] < 2*MAXPOWER ) goto cat1;
            type = SYMTOSYM; num = t[1] - 2*MAXPOWER;
        }
        else if ( *t == -INDEX ) {
            if ( t[1] < AM.OffsetIndex + WILDOFFSET ) goto cat1;
            type = INDTOIND; num = t[1] - WILDOFFSET;
        }
        else if ( *t == -VECTOR || *t == -MINVECTOR ) {
            if ( t[1] < AM.OffsetVector + WILDOFFSET ) goto cat1;
            type = VECTOVEC; num = t[1] - WILDOFFSET;
        }
        else goto cat1; /* Things like -SNUMBER etc. */
/*
        Now we have a wildcard and have to see whether it was assigned
*/
        m = AN.WildValue;
        j = nwstore;
        r = AT.WildMask;
        while ( --j >= 0 ) {
            if ( m[2] == num && *r ) {
                if ( type == *m ) break;
                if ( type == SYMTOSYM ) {
                    if ( *m == SYMTONUM || *m == SYMTOSUB ) break;
                }
```

```
                else if ( type == INDTOIND ) {
                    if ( *m == INDTOSUB ) break;
                }
                else if ( type == VECTOVEC ) {
                    if ( *m == VECTOMIN || *m == VECTOSUB ) break;
                }
            }
            m += 4; r++;
        }
        if ( j < 0 ) {  /* Category 4: Wildcard that was not assigned */
            a = AT.pWorkSpace+lhpars;
            iraise--;
            if ( iraise != i ) signs++;
            m = a[iraise];
            a[iraise] = a[i];
            a[i] = m; i--;
        }
        else {  /* Category 2: Wildcard that was assigned */
            for ( j = 0; j < tcount; j++ ) {
                if ( MatchArgument(BHEAD AT.pWorkSpace[oww+j],t) ) {
                    k = nwstore;
                    r = AT.WildMask;
                    num = 0;
                    while ( --k >= 0 ) num += *r++;
                    if ( num == assig ) { /* no wildcards were changed */
                        goto oneless;
                    }
                    break;
                }
            }
            if ( j >= tcount ) goto NoSuccess;
            j = nwstore;
            m = AN.WildValue;
            t = thewildcards + ntwa; r = AT.WildMask;
            if ( j > 0 ) {
                do {    /* undo assignment */
                    *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
                } while ( --j > 0 );
            }
            C->numrhs = *t++;
        }
        continue;
cat1:
        for ( j = 0; j < tcount; j++ ) {
            m = AT.pWorkSpace[j+oww];
            if ( *t != *m ) continue;
            if ( *t < 0 ) {
                if ( *t <= -FUNCTION ) break;
                if ( t[1] == m[1] ) break;
            }
            else {
                k = *t; r = t;
                while ( --k >= 0 && *m++ == *r++ ) {}
                if ( k < 0 ) break;
            }
        }
        if ( j >= tcount ) goto NoSuccess;  /* Even the fixed ones don't match */
oneless:
        signs += j - i;
/*
        The next statements replace the one that is commented out
*/
        tcount--;
        while ( j < tcount ) {
            AT.pWorkSpace[oww+j] = AT.pWorkSpace[oww+j+1]; j++;
        }
/*
        AT.pWorkSpace[oww+j] = AT.pWorkSpace[oww+(--tcount)];
*/
        argcount--; j = i;
        while ( j < argcount ) {
            AT.pWorkSpace[lhpars+j] = AT.pWorkSpace[lhpars+j+1]; j++;
        }
        iraise--; i--;
    }
/*
    Now we see whether there are any ARGWILD objects that have been
    assigned already. In that case the work simplifies considerably.
    Currently (12-nov-2001) only in (R)CYCLIC functions; hence we do not
    test the sign!
*/
    for ( i = 0; i < funnycount; i++ ) {
        k = AT.pWorkSpace[lhfunnies+i][1];
        m = AN.WildValue;
        j = nwstore;
        r = AT.WildMask;
        while ( --j >= 0 ) {
```

```
                if ( *m == ARGTOARG && m[2] == k ) break;
                m += 4; r++;
            }
            if ( *r == 0 ) continue; /* not assigned yet */
            m = cbuf[AT.ebufnum].rhs[m[3]];
            if ( *m > 0 ) {      /* Tensor arguments */
                j = *m;
                if ( j > tcount - argcount ) goto NoSuccess;
                while ( --j >= 0 ) {
                    m++;
                    if ( *m < 0 ) type = -VECTOR;
                    else if ( *m < AM.OffsetIndex ) type = -SNUMBER;
                    else type = -INDEX;
                    a = AT.pWorkSpace+oww;
                    for ( k = 0; k < tcount; k++ ) {
                        if ( a[k][0] != type || a[k][1] != *m ) continue;
                        a[k] = a[--tcount];
                        goto nextjarg;
                    }
                    goto NoSuccess;
nextjarg:;
                }
            }
            else {
                m++;
                while ( *m ) {
                    for ( k = 0; k < tcount; k++ ) {
                        t = AT.pWorkSpace[oww+k];
                        if ( *t != *m ) continue;
                        r = m;
                        if ( *r < 0 ) {
                            if ( *r < -FUNCTION ) goto nextargw;
                            else if ( r[1] == t[1] ) goto nextargw;
                        }
                        else {
                            j = *r;
                            while ( --j >= 0 && *r++ == *t++ ) {}
                            if ( j < 0 ) goto nextargw;
                        }
                    }
                    goto NoSuccess;
nextargw:;
                    AT.pWorkSpace[oww+k] = AT.pWorkSpace[oww+(--tcount)];
                    NEXTARG(m)
                }
            }
            AT.pWorkSpace[lhfunnies+i] = AT.pWorkSpace[lhfunnies+(--funnycount)];
        }
        if ( tcount == 0 ) {
            if ( argcount > 0 ) goto NoSuccess;
            for ( i = 0; i < funnycount; i++ ) {
                AddWild(BHEAD AT.pWorkSpace[lhfunnies+i][1],ARGTOARG,0);
            }
            goto quicky;
        }
/*
        We have now in lhpars first iraise elements with a dubious nature.
        Then argcount-iraise wildcards that have not been assigned.
        In lhfunnies we have funnycount ARGTOARG objects. ( (R)CyCLIC only )

        First work our way through the 'dubious' objects
        We check whether assig changes.
*/
        for ( i = 0; i < iraise; i++ ) {
            for ( j = 0; j < tcount; j++ ) {
                if ( MatchArgument(BHEAD AT.pWorkSpace[oww+j],AT.pWorkSpace[lhpars+i]) ) {
                    k = nwstore;
                    r = AT.WildMask;
                    num = 0;
                    while ( --k >= 0 ) num += *r++;
                    if ( num == assig ) { /* no wildcards were changed */
                        signs += j-i;
                        AT.pWorkSpace[oww+j] = AT.pWorkSpace[oww+(--tcount)];
                        if ( tcount > j ) signs += tcount-j-1;
                        argcount--;
                        a = AT.pWorkSpace + lhpars;
                        for ( j = i; j < argcount; j++ ) a[j] = a[j+1];
                        iraise--;
                        goto nextiraise;
                    }
                    else {  /* We cannot use this yet */
                        j = nwstore;
                        m = AN.WildValue;
                        t = thewildcards + ntwa; r = AT.WildMask;
                        if ( j > 0 ) {
                            do {     /* undo assignment */
                                *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
```

```
                        } while ( --j > 0 );
                    }
                    C->numrhs = *t++;
                    C->Pointer = C->Buffer + oldcpointer;
                    goto nextiraise;
                }
            }
        }
        goto NoSuccess;
nextiraise:;
    }
/*
    Now all leftover patterns have unassigned wildcards in them.
    From now on we are in potential factorial territory.

    Strategy:
        1: cycle through the regular objects.
        2: save wildcard settings
        3: divide the ARGWILDs
        4: make permutations of leftover arguments
        5: try them all
*/
    cycles = AT.WorkPointer;
    for ( i = 0; i < tcount; i++ ) cycles[i] = tcount-i;
    AT.WorkPointer += tcount;
    signo = 0;
/*MesPrint("<1> signs = %d",signs);*/
    for (;;) {
        WORD oRepFunNum = AN.RepFunNum;
        for ( j = 0; j < argcount; j++ ) {
            if ( MatchArgument(BHEAD AT.pWorkSpace[oww+j],AT.pWorkSpace[lhpars+j]) == 0 ) {
                break;
            }
        }
        if ( j >= argcount ) {
/*
            Thus far we have a match. Now the funnies
*/
            if ( funnycount ) {
                AT.WorkPointer = oldworkpointer;
                AT.pWorkPointer = oww;
                MLOCK(ErrorMessageLock);
                MesPrint("Sorry: no argument field wildcards yet in (anti)symmetric functions");
                MUNLOCK(ErrorMessageLock);
/*
                Bugfix 31-oct-2001, reported by Kasper Peeters
                We returned here with value -1 but that is not caught.
                Extra note (12-nov-2001): the sign becomes a bit problematic
                if we have funnies. No more than one allowed in antisymmetric
                functions, or we have serious problems.
*/
                Terminate(-1);
            }

            AN.RepFunList[AN.RepFunNum++] = offset;
            if ( ( (functions[fun[0]-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC )
                || ( (functions[pnum-FUNCTION].symmetric & ~REVERSEORDER) == ANTISYMMETRIC ) ) {
                AN.RepFunList[AN.RepFunNum++] = ( signs + signo ) & 1;
            }
            else {
                AN.RepFunList[AN.RepFunNum++] = 0;
            }
            newpat = pattern + pattern[1];
            if ( newpat >= AN.patstop ) {
                WORD countsgn, sgn = 0;
                for ( countsgn = oRepFunNum+1; countsgn < AN.RepFunNum; countsgn += 2 ) {
                    if ( AN.RepFunList[countsgn] ) sgn ^= 1;
                }
                if ( AN.SignCheck == 0 || sgn == AN.ExpectedSign ) {
                    AT.WorkPointer = oldworkpointer;
                    AT.pWorkPointer = oww;
                    return(1);
                }
                if ( AN.UseFindOnly == 0 ) {
                    if ( FindOnce(BHEAD AN.findTerm,AN.findPattern) ) {
                        AT.WorkPointer = oldworkpointer;
                        AT.pWorkPointer = oww;
                        AN.UsedOtherFind = 1;
                        return(1);
                    }
                }
                j = 0;
            }
            else j = ScanFunctions(BHEAD newpat,inter,par);
            if ( j ) {
                WORD countsgn, sgn = 0;
                for ( countsgn = oRepFunNum+1; countsgn < AN.RepFunNum; countsgn += 2 ) {
```

```
                    if ( AN.RepFunList[countsgn] ) sgn ^= 1;
                }
                if ( AN.SignCheck == 0 || sgn == AN.ExpectedSign ) {
                    AT.WorkPointer = oldworkpointer;
                    AT.pWorkPointer = oww;
                    return(j);
                }
            }
            AN.RepFunNum = oRepFunNum;
            i = argcount - 1;
        }
        else i = j;
        j = nwstore;
        m = AN.WildValue;
        t = thewildcards + ntwa; r = AT.WildMask;
        if ( j > 0 ) {
            do {
                *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
            } while ( --j > 0 );
        }
        C->numrhs = *t++;
        C->Pointer = C->Buffer + oldcpointer;
/*
        On to the next cycle
*/
        a = AT.pWorkSpace + oww;
        for ( j = i+1, t = a[i]; j < tcount; j++ ) a[j-1] = a[j];
        a[tcount-1] = t; cycles[i]--;
        signo += tcount - i - 1;
        while ( cycles[i] <= 0 ) {
            cycles[i] = tcount - i;
            i--;
            if ( i < 0 ) goto NoSuccess;
/*
            MLOCK(ErrorMessageLock);
            MesPrint("Cycle i = %d",i);
            MUNLOCK(ErrorMessageLock);
*/
            for ( j = i+1, t = a[i]; j < tcount; j++ ) a[j-1] = a[j];
            a[tcount-1] = t; cycles[i]--;
            signo += tcount - i - 1;
        }
    }
NoSuccess:
    if ( oldwilval > 0 ) {
        j = nwstore;
        m = AN.WildValue;
        t = lowlevel; r = AT.WildMask;
        if ( j > 0 ) {
            do {
                *m++ = *t++; *m++ = *t++; *m++ = *t++; *m++ = *t++; *r++ = *t++;
            } while ( --j > 0 );
        }
        C->numrhs = *t++;
        C->Pointer = C->Buffer + oldcpointer;
    }
    AT.WorkPointer = oldworkpointer;
    AT.pWorkPointer = oww;
    return(0);
}

/*
    #] FunMatchSy :
    #[ MatchArgument :
*/

int MatchArgument(PHEAD WORD *arg, WORD *pat)
{
    GETBIDENTITY
    WORD *m = pat, *t = arg, i, j, newvalue;
    WORD *argmstop = pat, *argtstop = arg;
    WORD *cto, *cfrom, *csav, ci;
    WORD oRepFunNum, *oRepFunList;
    WORD *oterstart,*oterstop,*opatstop;
    WORD wildargs, wildeat;
    WORD *mtrmstop, *ttrmstop, *msubstop, msizcoef;
    WORD *wildargtaken;
    int wc = 1;

    NEXTARG(argmstop);
    NEXTARG(argtstop);
/*
    #[ Both fast :
*/
    if ( *m < 0 && *t < 0 ) {
        if ( *t <= -FUNCTION ) {
            if ( *t == *m ) {}
```

```
            else if ( *m <= -FUNCTION-WILDOFFSET
            && functions[-*t-FUNCTION].spec
            == functions[-*m-FUNCTION-WILDOFFSET].spec ) {
                i = -*m - WILDOFFSET; wc = 2;
                if ( CheckWild(BHEAD i,FUNTOFUN,-*t,&newvalue) ) {
                    return(0);
                }
                AddWild(BHEAD i,FUNTOFUN,newvalue);
            }
            else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER ) {
                i = m[1] - 2*MAXPOWER;
                AN.argaddress = AT.FunArg;
                AT.FunArg[ARGHEAD+1] = -*t;
                if ( CheckWild(BHEAD i,SYMTOSUB,1,AN.argaddress) ) return(0);
                AddWild(BHEAD i,SYMTOSUB,0);
            }
            else return(0);
        }
        else if ( *t == *m ) {
            if ( t[1] == m[1] ) {}
            else if ( *t == -SYMBOL ) {
                j = SYMTOSYM;
SymAll:         if ( ( i = m[1] - 2*MAXPOWER ) < 0 ) return(0);
                wc = 2;
                if ( CheckWild(BHEAD i,j,t[1],&newvalue) ) return(0);
                AddWild(BHEAD i,j,newvalue);
            }
            else if ( *t == -INDEX ) {
IndAll:         i = m[1] - WILDOFFSET;
                if ( i < AM.OffsetIndex || i >= WILDOFFSET+AM.OffsetIndex )
                                                return(0);
                                /* We kill the summed over indices here */
                wc = 2;
                if ( CheckWild(BHEAD i,INDTOIND,t[1],&newvalue) ) return(0);
                AddWild(BHEAD i,INDTOIND,newvalue);
            }
            else if ( *t == -VECTOR || *t == -MINVECTOR ) {
                i = m[1] - WILDOFFSET;
                if ( i < AM.OffsetVector ) return(0);
                wc = 2;
                if ( CheckWild(BHEAD i,VECTOVEC,t[1],&newvalue) ) return(0);
                AddWild(BHEAD i,VECTOVEC,newvalue);
            }
            else return(0);
        }
        else if ( *m == -INDEX && m[1] >= AM.OffsetIndex+WILDOFFSET
        && m[1] < AM.OffsetIndex+(WILDOFFSET«1) ) {
            if ( *t == -VECTOR ) goto IndAll;
            if ( *t == -SNUMBER && t[1] >= 0 && t[1] < AM.OffsetIndex ) goto IndAll;
            if ( *t == -MINVECTOR ) {
                i = m[1] - WILDOFFSET;
                AN.argaddress = AT.MinVecArg;
                AT.MinVecArg[ARGHEAD+3] = t[1];
                wc = 2;
                if ( CheckWild(BHEAD i,INDTOSUB,1,AN.argaddress) ) return(0);
                AddWild(BHEAD i,INDTOSUB,(WORD)0);
            }
            else return(0);
        }
        else if ( *m == -SYMBOL && m[1] >= 2*MAXPOWER && *t == -SNUMBER ) {
            j = SYMTONUM;
            goto SymAll;
        }
        else if ( *m == -VECTOR && *t == -MINVECTOR &&
        ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
            wc = 2;
/*
            AN.argaddress = AT.MinVecArg;
            AT.MinVecArg[ARGHEAD+3] = t[1];
            if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) return(0);
            AddWild(BHEAD i,VECTOSUB,(WORD)0);
*/
            if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) return(0);
            AddWild(BHEAD i,VECTOMIN,newvalue);

        }
        else if ( *m == -MINVECTOR && *t == -VECTOR &&
        ( i = m[1] - WILDOFFSET ) >= AM.OffsetVector ) {
            wc = 2;
/*
            AN.argaddress = AT.MinVecArg;
            AT.MinVecArg[ARGHEAD+3] = t[1];
            if ( CheckWild(BHEAD i,VECTOSUB,1,AN.argaddress) ) return(0);
            AddWild(BHEAD i,VECTOSUB,(WORD)0);
*/
            if ( CheckWild(BHEAD i,VECTOMIN,t[1],&newvalue) ) return(0);
            AddWild(BHEAD i,VECTOMIN,newvalue);
```

```
        }
        else return(0);
    }
/*
    #] Both fast :
    #[ Fast arg :
*/
    else if ( *m > 0 && *t <= -FUNCTION ) {
        if ( ( m[ARGHEAD]+ARGHEAD == *m ) && m[*m-1] == 3
        && m[*m-2] == 1 && m[*m-3] == 1 && m[ARGHEAD+1] >= FUNCTION
        && m[ARGHEAD+2] == *m-ARGHEAD-4 ) { /* Check for f(?a) etc */
            WORD *mmmst, *mmm, mmmi;
            if ( m[ARGHEAD+1] >= FUNCTION+WILDOFFSET ) {
                mmmi = *m - WILDOFFSET;
                wc = 2;
                if ( CheckWild(BHEAD mmmi,FUNTOFUN,-*t,&newvalue) ) return(0);
                AddWild(BHEAD mmmi,FUNTOFUN,newvalue);
            }
            else if ( m[ARGHEAD+1] != -*t ) return(0);
/*
                Only arguments allowed are ?a etc.
*/
            mmmst = m+*m-3;
            mmm = m + ARGHEAD + FUNHEAD + 1;
            while ( mmm < mmmst ) {
                if ( *mmm != -ARGWILD ) return(0);
                mmmi = 0;
                AN.argaddress = t; wc = 2;
                if ( CheckWild(BHEAD mmm[1],ARGTOARG,mmmi,t) ) return(0);
                AddWild(BHEAD mmm[1],ARGTOARG,mmmi);
                mmm += 2;
            }
        }
        else return(0);
    }
/*
    #] Fast arg :
    #[ Fast pat :
*/
    else if ( *m < 0 && *t > 0 ) {
        if ( *m == -SYMBOL ) {          /* SYMTOSUB */
            if ( m[1] < 2*MAXPOWER ) return(0);
            i = m[1] - 2*MAXPOWER;
            AN.argaddress = t; wc = 2;
            if ( CheckWild(BHEAD i,SYMTOSUB,1,AN.argaddress) ) return(0);
            AddWild(BHEAD i,SYMTOSUB,0);
        }
        else if ( *m == -VECTOR ) {
            if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetVector ) return(0);
            AN.argaddress = t; wc = 2;
            if ( CheckWild(BHEAD i,VECTOSUB,1,t) ) return(0);
            AddWild(BHEAD i,VECTOSUB,(WORD)0);
        }
        else if ( *m == -INDEX ) {
            if ( ( i = m[1] - WILDOFFSET ) < AM.OffsetIndex ) return(0);
            if ( i >= AM.OffsetIndex + WILDOFFSET ) return(0);
            AN.argaddress = t; wc = 2;
            if ( CheckWild(BHEAD i,INDTOSUB,1,AN.argaddress) ) return(0);
            AddWild(BHEAD i,INDTOSUB,(WORD)0);
        }
        else return(0);
    }
/*
    #] Fast pat :
    #[ Both general :
*/
    else if ( *m > 0 && *t > 0 ) {
        i = *m;
        do { if ( *m++ != *t++ ) break; } while ( --i > 0 );
        if ( i > 0 ) {
/*
            Not an exact match here.
            We have to hope that the pattern contains a composite wildcard.
*/
            m = pat; t = arg;
            m += ARGHEAD; t += ARGHEAD;         /* Point at (first?) term */
            mtrmstop = m + *m;
            ttrmstop = t + *t;
            if ( mtrmstop < argmstop ) return(0);/* More than one term */
            msizcoef = mtrmstop[-1];
            if ( msizcoef < 0 ) msizcoef = -msizcoef;
            msubstop = mtrmstop - msizcoef;
            m++;
            if ( m >= msubstop ) return(0); /* Only coefficient */
/*
            Here we have a composite term. It can match provided it
            matches the entire argument. This argument must be a
```

```
            single term also and the coefficients should match
            (more or less).
            The matching takes:
            1:  Match the functions etc. Nothing can be left.
            2:  Match dotproducts and symbols. ONLY must match
                and nothing may be left.
            For safety it is best to take the term out and put it
            in workspace.
*/
            if ( argtstop > ttrmstop ) return(0);
            m--;

            oterstart = AN.terstart;
            oterstop = AN.terstop;
            opatstop = AN.patstop;
            oRepFunList = AN.RepFunList;
            oRepFunNum = AN.RepFunNum;
            AN.RepFunNum = 0;
            wildargtaken = AT.WorkPointer;
            AN.RepFunList = wildargtaken + AN.NumTotWildArgs;
            AT.WorkPointer = (WORD *)(((UBYTE *)(AN.RepFunList)) + AM.MaxTer/2);
            csav = cto = AT.WorkPointer;
            cfrom = t;
            ci = *t;
            while ( --ci >= 0 ) *cto++ = *cfrom++;
            AT.WorkPointer = cto;
            ci = msizcoef;
            cfrom = mtrmstop;
            while ( --ci >= 0 ) {
                if ( *--cfrom != *--cto ) {
                    AT.WorkPointer = wildargtaken;
                    AN.RepFunList = oRepFunList;
                    AN.RepFunNum = oRepFunNum;
                    AN.terstart = oterstart;
                    AN.terstop = oterstop;
                    AN.patstop = opatstop;
                    return(0);
                }
            }
            *m -= msizcoef;
            wildargs = AN.WildArgs;
            wildeat = AN.WildEat;
            for ( i = 0; i < wildargs; i++ ) wildargtaken[i] = AT.WildArgTaken[i];
            AN.ForFindOnly = 0; AN.UseFindOnly = 1;
            AN.nogroundlevel++;
            if ( FindRest(BHEAD csav,m) && ( AN.UsedOtherFind || FindOnly(BHEAD csav,m) ) ) { }
            else {
                *m += msizcoef;
                AT.WorkPointer = wildargtaken;
                AN.RepFunList = oRepFunList;
                AN.RepFunNum = oRepFunNum;
                AN.terstart = oterstart;
                AN.terstop = oterstop;
                AN.patstop = opatstop;
                AN.WildArgs = wildargs;
                AN.WildEat = wildeat;
                AN.nogroundlevel--;
                for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
                return(0);
            }
            AN.nogroundlevel--;
            AN.WildArgs = wildargs;
            AN.WildEat = wildeat;
            for ( i = 0; i < wildargs; i++ ) AT.WildArgTaken[i] = wildargtaken[i];
            Substitute(BHEAD csav,m,1);
            cto = csav;
            cfrom = cto + *cto - msizcoef;
            cto++;
            *m += msizcoef;
            AT.WorkPointer = wildargtaken;
            AN.RepFunList = oRepFunList;
            AN.RepFunNum = oRepFunNum;
            AN.terstart = oterstart;
            AN.terstop = oterstop;
            AN.patstop = opatstop;
            if ( *cto != SUBEXPRESSION ) return(0);
            cto += cto[1];
            if ( cto < cfrom ) return(0);
        }
    }
/*
    #] Both general :
*/
    else return(0);
/*
    And now the success: (wc = 2 means that there was a wildcard involved)
*/
```

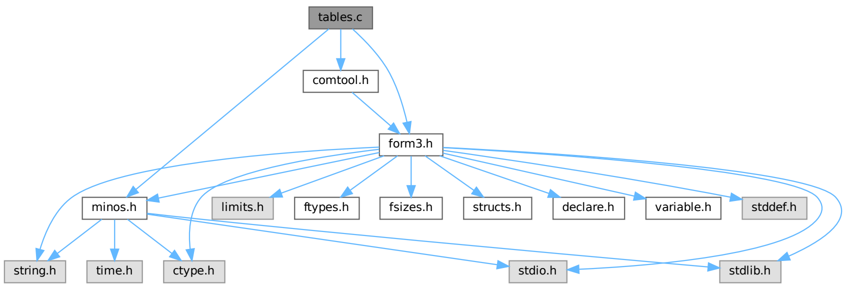
```
    return(wc);
}

/*
    #] MatchArgument :
*/
```

4.140 tables.c File Reference

```
#include "form3.h"
#include "minos.h"
#include "comtool.h"
```

Include dependency graph for tables.c:



Functions

- void ClearTableTree (TABLES T)
- int InsTableTree (TABLES T, WORD ∗tp)
- void RedoTableTree (TABLES T, int newsize)
- int FindTableTree (TABLES T, WORD ∗tp, int inc)
- WORD DoTableExpansion (WORD ∗term, WORD level)
- int CoTableBase (UBYTE ∗s)
- int FlipTable (FUNCTIONS f, int type)
- int SpareTable (TABLES TT)
- DBASE ∗ FindTB (UBYTE ∗name)
- int CoTBcreate (UBYTE ∗s)
- int CoTBopen (UBYTE ∗s)
- int CoTBaddto (UBYTE ∗s)
- int CoTBenter (UBYTE ∗s)
- int CoTestUse (UBYTE ∗s)
- int CheckTableDeclarations (DBASE ∗d)
- int CoTBload (UBYTE ∗ss)
- WORD TestUse (WORD ∗term, WORD level)
- int CoTBaudit (UBYTE ∗s)
- int CoTBon (UBYTE ∗s)
- int CoTBoff (UBYTE ∗s)
- int CoTBcleanup (UBYTE ∗s)
- int CoTBreplace (UBYTE ∗s)
- int CoTBuse (UBYTE ∗s)
- int CoApply (UBYTE ∗s)
- int CoTBhelp (UBYTE ∗s)

- void ReWorkT (WORD ∗term, WORD ∗funs, WORD numfuns)
- WORD Apply (WORD ∗term, WORD level)
- int ApplyExec (WORD ∗term, int maxtogo, WORD level)
- WORD ApplyReset (WORD level)
- WORD TableReset (void)
- int ReleaseTB (void)

Variables

- char ∗ helptb [ ]

4.140.1 Detailed Description

Contains all functions that deal with the table bases on the 'FORM level' The low level database routines are in minos.c

Definition in file tables.c.

4.140.2 Function Documentation

4.140.2.1 ClearTableTree()

```
void ClearTableTree (
            TABLES T )
```

Definition at line 72 of file tables.c.

4.140.2.2 InsTableTree()

```
int InsTableTree (
            TABLES T,
            WORD * tp )
```

Definition at line 103 of file tables.c.

4.140.2.3 RedoTableTree()

```
void RedoTableTree (
            TABLES T,
            int newsize )
```

Definition at line 266 of file tables.c.

4.140.2.4 FindTableTree()

```
int FindTableTree (
            TABLES T,
            WORD * tp,
            int inc )
```

Definition at line 291 of file tables.c.

4.140.2.5 DoTableExpansion()

```
WORD DoTableExpansion (
            WORD * term,
            WORD level )
```

Definition at line 334 of file tables.c.

4.140.2.6 CoTableBase()

```
int CoTableBase (
            UBYTE * s )
```

Definition at line 627 of file tables.c.

4.140.2.7 FlipTable()

```
int FlipTable (
            FUNCTIONS f,
            int type )
```

Definition at line 671 of file tables.c.

4.140.2.8 SpareTable()

```
int SpareTable (
            TABLES TT )
```

Definition at line 694 of file tables.c.

4.140.2.9 FindTB()

```
DBASE * FindTB (
            UBYTE * name )
```

Definition at line 734 of file tables.c.

4.140.2.10 CoTBcreate()

```
int CoTBcreate (
            UBYTE * s )
```

Definition at line 756 of file tables.c.

4.140.2.11 CoTBopen()

```
int CoTBopen (
            UBYTE * s )
```

Definition at line 772 of file tables.c.

4.140.2.12 CoTBaddto()

```
int CoTBaddto (
            UBYTE * s )
```

Definition at line 801 of file tables.c.

4.140.2.13 CoTBenter()

```
int CoTBenter (
            UBYTE * s )
```

Definition at line 974 of file tables.c.

4.140.2.14 CoTestUse()

```
int CoTestUse (
            UBYTE * s )
```

Definition at line 1133 of file tables.c.

4.140.2.15 CheckTableDeclarations()

```
int CheckTableDeclarations (
            DBASE * d )
```

Definition at line 1178 of file tables.c.

4.140.2.16 CoTBload()

```
int CoTBload (
            UBYTE * ss )
```

Definition at line 1252 of file tables.c.

4.140.2.17 TestUse()

```
WORD TestUse (
            WORD * term,
            WORD level )
```

Definition at line 1414 of file tables.c.

4.140.2.18 CoTBaudit()

```
int CoTBaudit (
            UBYTE * s )
```

Definition at line 1473 of file tables.c.

4.140.2.19 CoTBon()

```
int CoTBon (
            UBYTE * s )
```

Definition at line 1513 of file tables.c.

4.140.2.20 CoTBoff()

```
int CoTBoff (
            UBYTE * s )
```

Definition at line 1544 of file tables.c.

4.140.2.21 CoTBcleanup()

```
int CoTBcleanup (
            UBYTE * s )
```

Definition at line 1575 of file tables.c.

4.140.2.22 CoTBreplace()

```
int CoTBreplace (
            UBYTE * s )
```

Definition at line 1587 of file tables.c.

4.140.2.23 CoTBuse()

```
int CoTBuse (
            UBYTE * s )
```

Definition at line 1602 of file tables.c.

4.140.2.24 CoApply()

```
int CoApply (
            UBYTE * s )
```

Definition at line 1796 of file tables.c.

4.140.2.25 CoTBhelp()

```
int CoTBhelp (
            UBYTE * s )
```

Definition at line 1883 of file tables.c.

4.140.2.26 ReWorkT()

```
void ReWorkT (
            WORD * term,
            WORD * funs,
            WORD numfuns )
```

Definition at line 1899 of file tables.c.

4.140.2.27 Apply()

```
WORD Apply (
            WORD * term,
            WORD level )
```

Definition at line 1980 of file tables.c.

4.140.2.28 ApplyExec()

```
int ApplyExec (
            WORD * term,
            int maxtogo,
            WORD level )
```

Definition at line 2039 of file tables.c.

4.140.2.29 ApplyReset()

```
WORD ApplyReset (
            WORD level )
```

Definition at line 2256 of file tables.c.

4.140.2.30 TableReset()

```
WORD TableReset (
            void )
```

Definition at line 2292 of file tables.c.

4.140.2.31 ReleaseTB()

```
int ReleaseTB (
            void )
```

Definition at line 2319 of file tables.c.

4.140.3 Variable Documentation

4.140.3.1 helptb

```
char* helptb[]
```

Definition at line 1851 of file tables.c.

4.141 tables.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 */
/* #] License : */
/*
    #[ Includes :

    File contains the routines for the tree structure of sparse tables
    We insert elements by
    InsTableTree(T,tp) with T the TABLES element and tp the pointer
        to the indices.
    We look for elements with
    FindTableTree(T,tp,inc) with T the TABLES element, tp the pointer to the
        indices or the function arguments and inc tells which of these options.
    The tree is cleared with ClearTableTree(T) and we rebuild the tree
    after a .store in which we lost a part of the table with
    RedoTableTree(T,newsize)

    In T->tablepointers we have the lists of indices for each element.
    Additionally for each element there is an extension. There are
    TABLEEXTENSION WORDs reserved for that. The old system had two words
    One for the element in the rhs of the compile buffer and one for
    an additional rhs in case the original would be overwritten by a new
    definition, but the old was fixed by .global and hence it should be possible
    to restore it.
    New use (new = 24-sep-2001)
        rhs1,numCompBuffer1,rhs2,numCompBuffer2,usage
    Hence TABLEEXTENSION will be 5. Note that for 64 bits the use of the
    compiler buffer is overdoing it a bit, but it would be too complicated
    to try to give it special code.
*/

#include "form3.h"
#include "minos.h"
#include "comtool.h"

/* static UBYTE *sparse = (UBYTE *)"sparse"; */
static UBYTE *tablebase = (UBYTE *)"tablebase";

/*
    #] Includes :
    #[ ClearTableTree :
*/
```

```

void ClearTableTree(TABLES T)
{
    COMPTREE *root;
    if ( T->boomlijst == 0 ) {
        T->MaxTreeSize = 125;
        T->boomlijst = (COMPTREE *)Malloc1(T->MaxTreeSize*sizeof(COMPTREE),
                "ClearTableTree");
    }
    root = T->boomlijst;
    T->numtree = 0;
    T->rootnum = 0;
    root->left = -1;
    root->right = -1;
    root->parent = -1;
    root->blnce = 0;
    root->value = -1;
    root->usage = 0;
}

/*
    #] ClearTableTree :
    #[ InsTableTree :

    int InsTableTree(TABLES T,WORD *,arglist)
        Searches for the element specified by the list of arguments.
        If found, it returns -(the offset in T->tablepointers)
        If not found, it will allocate a new element, balance the tree if
        necessary and return the number of the element in the boomlijst
        This number is always > 0, because we start from 1.
*/

int InsTableTree(TABLES T, WORD *tp)
{
    COMPTREE *boomlijst, *q, *p, *s;
    WORD *v1, *v2, *v3, xstop;
    int ip, iq, is;
    if ( T->numtree + 1 >= T->MaxTreeSize ) {
        if ( T->MaxTreeSize == 0 ) ClearTableTree(T);
        else {
            is = T->MaxTreeSize * 2;
            s  = (COMPTREE *)Malloc1(is*sizeof(COMPTREE),"InsTableTree");
            for ( ip = 0; ip < T->MaxTreeSize; ip++ ) { s[ip] = T->boomlijst[ip]; }
            if ( T->boomlijst ) M_free(T->boomlijst,"InsTableTree");
            T->boomlijst = s;
            T->MaxTreeSize = is;
        }
    }
    boomlijst = T->boomlijst;
    q = boomlijst + T->rootnum;
    if ( q->right == -1 ) { /* First element */
        T->numtree++;
        s = boomlijst+T->numtree;
        q->right = T->numtree;
        s->parent = T->rootnum;
        s->left = s->right = -1;
        s->blnce = 0;
        s->value = tp - T->tablepointers;
        s->usage = 0;
        return(T->numtree);
    }
    ip = q->right;
    if ( T->numind >= 0 ) xstop = T->numind;
    else xstop = *tp + 1;
    while ( ip >= 0 ) {
        p = boomlijst + ip;
        v1 = T->tablepointers + p->value;
        v2 = tp; v3 = tp + xstop;
        while ( *v1 == *v2 && v2 < v3 ) { v1++; v2++; }
        if ( v2 >= v3 ) return(-p->value);
        if ( *v1 > *v2 ) {
            iq = p->right;
            if ( iq >= 0 ) { ip = iq; }
            else {
                T->numtree++;
                is = T->numtree;
                p->right = is;
                s = boomlijst + is;
                s->parent = ip; s->left = s->right = -1;
                s->blnce = 0;   s->value = tp - T->tablepointers;
                s->usage = 0;
                p->blnce++;
                if ( p->blnce == 0 ) return(T->numtree);
                goto balance;
            }
        }
        else if ( *v1 < *v2 ) {
```

```
            iq = p->left;
            if ( iq >= 0 ) { ip = iq; }
            else {
                T->numtree++;
                is = T->numtree;
                s = boomlijst+is;
                p->left = is;
                s->parent = ip; s->left = s->right = -1;
                s->blnce = 0;   s->value = tp - T->tablepointers;
                s->usage = 0;
                p->blnce--;
                if ( p->blnce == 0 ) return(T->numtree);
                goto balance;
            }
        }
    }
    MesPrint("Serious problems in InsTableTree!\n");
    Terminate(-1);
    return(0);
balance:;
    for (;;) {
        p = boomlijst + ip;
        iq = p->parent;
        if ( iq == T->rootnum ) break;
        q = boomlijst + iq;
        if ( ip == q->left ) q->blnce--;
        else                 q->blnce++;
        if ( q->blnce == 0 ) break;
        if ( q->blnce == -2 ) {
            if ( p->blnce == -1 ) { /* single rotation */
                q->left = p->right;
                p->right = iq;
                p->parent = q->parent;
                q->parent = ip;
                if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
                else                                   boomlijst[p->parent].right = ip;
                if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
                q->blnce = p->blnce = 0;
            }
            else {  /* double rotation */
                s = boomlijst + is;
                q->left = s->right;
                p->right = s->left;
                s->right = iq;
                s->left = ip;
                if ( p->right >= 0 ) boomlijst[p->right].parent = ip;
                if ( q->left >= 0 ) boomlijst[q->left].parent = iq;
                s->parent = q->parent;
                q->parent = is;
                p->parent = is;
                if ( boomlijst[s->parent].left == iq )
                       boomlijst[s->parent].left = is;
                else   boomlijst[s->parent].right = is;
                if ( s->blnce > 0 ) { q->blnce = s->blnce = 0; p->blnce = -1; }
                else if ( s->blnce < 0 ) { p->blnce = s->blnce = 0; q->blnce = 1; }
                else { p->blnce = s->blnce = q->blnce = 0; }
            }
            break;
        }
        else if ( q->blnce == 2 ) {
            if ( p->blnce == 1 ) {  /* single rotation */
                q->right = p->left;
                p->left = iq;
                p->parent = q->parent;
                q->parent = ip;
                if ( boomlijst[p->parent].left == iq ) boomlijst[p->parent].left = ip;
                else                                   boomlijst[p->parent].right = ip;
                if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
                q->blnce = p->blnce = 0;
            }
            else {  /* double rotation */
                s = boomlijst + is;
                q->right = s->left;
                p->left = s->right;
                s->left = iq;
                s->right = ip;
                if ( p->left >= 0 ) boomlijst[p->left].parent = ip;
                if ( q->right >= 0 ) boomlijst[q->right].parent = iq;
                s->parent = q->parent;
                q->parent = is;
                p->parent = is;
                if ( boomlijst[s->parent].left == iq ) boomlijst[s->parent].left = is;
                else                                   boomlijst[s->parent].right = is;
                if ( s->blnce < 0 ) { q->blnce = s->blnce = 0; p->blnce = 1; }
                else if ( s->blnce > 0 ) { p->blnce = s->blnce = 0; q->blnce = -1; }
                else { p->blnce = s->blnce = q->blnce = 0; }
            }
```

```
            break;
        }
        is = ip; ip = iq;
    }
    return(T->numtree);
}

/*
    #] InsTableTree :
    #[ RedoTableTree :

    To be used when a sparse table is trimmed due to a .store
    We rebuild the tree. In the future one could try to become faster
    at the cost of quite some complexity.
    We need to keep the first 'size' elements in the boomlijst.
    Kill all others and reconstruct the tree with the original ordering.
    This is very complicated! Because .store will either keep the whole
    table or remove the whole table we should not come here often.
    Hence we choose the slow solution for now.
*/

void RedoTableTree(TABLES T, int newsize)
{
    WORD *tp;
    int i;
    ClearTableTree(T);
    for ( i = 0, tp = T->tablepointers; i < newsize; i++ ) {
        InsTableTree(T,tp);
        tp += ABS(T->numind)+TABLEEXTENSION;
    }
}

/*
    #] RedoTableTree :
    #[ FindTableTree :

    int FindTableTree(TABLES T,WORD *,arglist,int,inc)
        Searches for the element specified by the list of arguments.
        If found, it returns the offset in T->tablepointers
        If not found, it will return -1
        The list here is from the list of function arguments. Hence it
        has pairs of numbers -SNUMBER,index
        Actually inc says how many numbers there are and the above case is
        for inc = 2. For inc = 1 we have just a list of indices.
*/

int FindTableTree(TABLES T, WORD *tp, int inc)
{
    COMPTREE *boomlijst = T->boomlijst, *q = boomlijst + T->rootnum, *p;
    WORD *v1, *v2, *v3, xstop;
    int ip, iq;
    if ( q->right == -1 ) return(-1);
    ip = q->right;
    if ( inc > 1 ) tp += inc-1;
    if ( T->numind >= 0 ) xstop = T->numind;
    else {  /* We have to read the number of arguments first */
        if ( *tp <= 0 ) return(-1); /* Cannot be! */
        xstop = *tp+1;
    }
    while ( ip >= 0 ) {
        p = boomlijst + ip;
        v1 = T->tablepointers + p->value;
        v2 = tp; v3 = v1 + xstop;
        while ( *v1 == *v2 && v1 < v3 ) { v1++; v2 += inc; }
        if ( v1 == v3 ) {
            p->usage++;
            return(p->value);
        }
        if ( *v1 > *v2 ) {
            iq = p->right;
            if ( iq >= 0 ) { ip = iq; }
            else return(-1);
        }
        else if ( *v1 < *v2 ) {
            iq = p->left;
            if ( iq >= 0 ) { ip = iq; }
            else return(-1);
        }
    }
    MesPrint("Serious problems in FindTableTree\n");
    Terminate(-1);
    return(-1);
}

/*
    #] FindTableTree :
    #[ DoTableExpansion :
```

```
*/

WORD DoTableExpansion(WORD *term, WORD level)
{
    GETIDENTITY
    WORD *t, *tstop, *stopper, *termout, *m, *mm, *tp, *r, xx;
    WORD numsubexp, numbuf;
    TABLES T = 0;
    int i, j, num;
    AN.TeInFun = AR.TePos = 0;
    tstop = term + *term;
    stopper = tstop - ABS(tstop[-1]);
    t = term+1;
    while ( t < stopper ) {
        if ( *t != TABLEFUNCTION ) { t += t[1]; continue; }
        if ( t[FUNHEAD] > -FUNCTION ) { t += t[1]; continue; }
        T = functions[-t[FUNHEAD]-FUNCTION].tabl;
        if ( T == 0 ) { t += t[1]; continue; }
        if ( T->spare ) T = T->spare;
        if ( t[1] == FUNHEAD+2 && t[FUNHEAD+1] <= -FUNCTION ) break;
        if ( t[1] < FUNHEAD+1+2*ABS(T->numind) ) { t += t[1]; continue; }
        for ( i = 0; i < ABS(T->numind); i++ ) {
            if ( t[FUNHEAD+1+2*i] != -SYMBOL ) break;
        }
        if ( i >= ABS(T->numind) ) break;
        t += t[1];
    }
    if ( t >= stopper ) {
        MesPrint("Internal error: Missing table_ function");
        Terminate(-1);
    }
/*
    Table in T. Now collect the numbers of the symbols;
*/
    termout = AT.WorkPointer;
    if ( T->sparse ) {
        for ( i = 0; i < T->totind; i++ ) {
/*
            Loop over all table elements
*/
            m = termout + 1; mm = term + 1;
            while ( mm < t ) *m++ = *mm++;
            r = m;
            if ( t[1] == FUNHEAD+2 && t[FUNHEAD+1] <= -FUNCTION ) {
                *m++ = -t[FUNHEAD+1];
                tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i;
                if ( T->numind < 0 ) {
                    xx = tp[0]+1;
                    *m++ = FUNHEAD+xx*2;
                    for ( j = 2; j < FUNHEAD; j++ ) *m++ = 0;
                    for ( j = 0; j < xx; j++ ) {
                        *m++ = -SNUMBER; *m++ = *tp++;
                    }
                }
                else {
                    *m++ = FUNHEAD+T->numind*2;
                    for ( j = 2; j < FUNHEAD; j++ ) *m++ = 0;
                    for ( j = 0; j < T->numind; j++ ) {
                        *m++ = -SNUMBER; *m++ = *tp++;
                    }
                }
            }
            else if ( T->numind < 0 ) {
                tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i;
                xx = tp[0]+1;
                *m++ = SYMBOL; *m++ = 2+xx*2; mm = t + FUNHEAD+1;
                for ( j = 0; j < xx; j++, mm += 2, tp++ ) {
                    if ( *tp != 0 ) { *m++ = mm[1]; *m++ = *tp; }
                }
                r[1] = m-r;
                if ( r[1] == 2 ) m = r;
            }
            else {
                *m++ = SYMBOL; *m++ = 2+T->numind*2; mm = t + FUNHEAD+1;
                tp = T->tablepointers + (T->numind+TABLEEXTENSION)*i;
                for ( j = 0; j < T->numind; j++, mm += 2, tp++ ) {
                    if ( *tp != 0 ) { *m++ = mm[1]; *m++ = *tp; }
                }
                r[1] = m-r;
                if ( r[1] == 2 ) m = r;
            }
/*
            The next code replaces this old code

            *m++ = SUBEXPRESSION;
            *m++ = SUBEXPSIZE;
            *m++ = *tp;
```

```
            *m++ = 1;
            *m++ = T->bufnum;
            FILLSUB(m);
            mm = t + t[1];

            We had forgotten to take the parameters into account.
            Hence the subexpression prototype for wildcards was missed
            Now we slow things down a little bit, but we do not run
            any risks. There is still one problem. We have not checked
            that the prototype matches.
*/
            tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i
                    +ABS(T->numind);
            numsubexp = tp[0]; numbuf = tp[1];
            r = m;
#ifdef WITHPTHREADS
            tp = T->prototype[identity];
#else
            tp = T->prototype;
#endif
            for ( j = 0; j < tp[1]; j++ ) *m++ = tp[j];
            r[2] = numsubexp; r[4] = numbuf;
/*
            r = m;
            tp = T->tablepointers + (ABS(T->numind)+TABLEEXTENSION)*i;
            *m++ = -t[FUNHEAD];
            if ( T->numind < 0 ) {
                xx = tp[0]+1;
                *m++ = t[1] - xx - T->numind - 1;
            }
            else {
                xx = T->numind;
                *m++ = t[1] - 1;
            }
            for ( j = 2; j < FUNHEAD; j++ ) *m++ = t[j];
            for ( j = 0; j < xx; j++ ) {
                *m++ = -SNUMBER; *m++ = *tp++;
            }
            tp = t + FUNHEAD + 1 + 2*T->numind;
            mm = t + t[1];
            while ( tp < mm ) *m++ = *tp++;
            r[1] = m-r;
*/
/*
            From now on is old code
*/
            mm = t + t[1];
            while ( mm < tstop ) *m++ = *mm++;
            *termout = m - termout;
            AT.WorkPointer = m;
            if ( Generator(BHEAD termout,level) ) {
                MesCall("DoTableExpand");
                return(-1);
            }
            AT.WorkPointer = termout;
        }
    }
    else {
        for ( i = 0; i < T->totind; i++ ) {
#if TABLEEXTENSION == 2
            if ( T->tablepointers[i] < 0 ) continue;
#else
            if ( T->tablepointers[TABLEEXTENSION*i] < 0 ) continue;
#endif
            m = termout + 1; mm = term + 1;
            while ( mm < t ) *m++ = *mm++;
            r = m;
            if ( t[1] == FUNHEAD+2 && t[FUNHEAD+1] <= -FUNCTION ) {
                *m++ = -t[FUNHEAD+1];
                *m++ = FUNHEAD+T->numind*2;
                for ( j = 2; j < FUNHEAD; j++ ) *m++ = 0;
                tp = T->tablepointers + (T->numind+TABLEEXTENSION)*i;
                for ( j = 0; j < T->numind; j++ ) {
                    if ( j > 0 ) {
                        num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;
                    }
                    else {
                        num = T->mm[j].mini + i / T->mm[j].size;
                    }
                    *m++ = -SNUMBER; *m++ = num;
                }
            }
            else {
                *m++ = SYMBOL; *m++ = 2+T->numind*2; mm = t + FUNHEAD+1;
                for ( j = 0; j < T->numind; j++, mm += 2 ) {
                    if ( j > 0 ) {
                        num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;
```

```
                    }
                    else {
                        num = T->mm[j].mini + i / T->mm[j].size;
                    }
                    if ( num != 0 ) { *m++ = mm[1]; *m++ = num; }
                }
                r[1] = m-r;
                if ( r[1] == 2 ) m = r;
            }
/*
            The next code replaces this old code

            *m++ = SUBEXPRESSION;
            *m++ = SUBEXPSIZE;
            *m++ = *tp;
            *m++ = 1;
            *m++ = T->bufnum;
            FILLSUB(m);
            mm = t + t[1];

            We had forgotten to take the parameters into account.
            Hence the subexpression prototype for wildcards was missed
            Now we slow things down a little bit, but we do not run
            any risks. There is still one problem. We have not checked
            that the prototype matches.
*/
            r = m;
            *m++ = -t[FUNHEAD];
            *m++ = t[1] - 1;
            for ( j = 2; j < FUNHEAD; j++ ) *m++ = t[j];
            for ( j = 0; j < T->numind; j++ ) {
                if ( j > 0 ) {
                    num = T->mm[j].mini + ( i % T->mm[j-1].size ) / T->mm[j].size;
                }
                else {
                    num = T->mm[j].mini + i / T->mm[j].size;
                }
                *m++ = -SNUMBER; *m++ = num;
            }
            tp = t + FUNHEAD + 1 + 2*T->numind;
            mm = t + t[1];
            while ( tp < mm ) *m++ = *tp++;
            r[1] = m - r;
/*
            From now on is old code
*/
            while ( mm < tstop ) *m++ = *mm++;
            *termout = m - termout;
            AT.WorkPointer = m;
            if ( Generator(BHEAD termout,level) ) {
                MesCall("DoTableExpand");
                return(-1);
            }
        }
    }
    return(0);
}

/*
    #] DoTableExpansion :
    #[ TableBase :

    File with all the database related things.
    We have the routines for the generic database command
    TableBase,options;
    TB,options;
    Options are:
        Open "File.tbl";                   Open for R/W
        Open "File.tbl", readonly;         Open for R
        Create "File.tbl";                 Create for write
        Load "File.tbl", tablename;        Loads stubs of table
        Load "File.tbl";                   Loads stubs of all tables
        Enter "File.tbl", tablename;       Loads whole table
        Enter "File.tbl";                  Loads all tables
        Audit "File.tbl", options;         Print list of contents
        Replace "File.tbl", tablename;     Saves a table (with overwrite)
        Replace "File.tbl", table element; Saves a table element    „
        Cleanup "File.tbl";                Makes tables contingent
        AddTo "File.tbl" tablename;        Add if not yet there.
        AddTo "File.tbl" table element;    Add if not yet there.
        Delete "File.tbl" tablename;
        Delete "File.tbl" table element;

        On/Off substitute;
        On/Off compress "File.tbl";
    id tbl_(f?,?a) = f(?a);
    When a tbl_ is used, automatically the corresponding element is compiled
```

```
    at the start of the next module.
    if TB,On,substitute [tablename], use of table RHS (if loaded)
    if TB,Off,substitute [tablename], use of tbl_(table,...);


    Still needed: Something like OverLoad to allow loading parts of a table
    from more than one file. Date stamps needed? In that case we need a touch
    command as well.

    If we put all our diagrams inside, we have to go outside the concept
    of tables.

    #] TableBase :
    #[ CoTableBase :

    To be followed by ,subkey
*/
static KEYWORD tboptions[] = {
     {"addto",           (TFUN)CoTBaddto,      0,    PARTEST}
    ,{"audit",           (TFUN)CoTBaudit,      0,    PARTEST}
    ,{"cleanup",         (TFUN)CoTBcleanup,    0,    PARTEST}
    ,{"create",          (TFUN)CoTBcreate,     0,    PARTEST}
    ,{"enter",           (TFUN)CoTBenter,      0,    PARTEST}
    ,{"help",            (TFUN)CoTBhelp,       0,    PARTEST}
    ,{"load",            (TFUN)CoTBload,       0,    PARTEST}
    ,{"off",             (TFUN)CoTBoff,        0,    PARTEST}
    ,{"on",              (TFUN)CoTBon,         0,    PARTEST}
    ,{"open",            (TFUN)CoTBopen,       0,    PARTEST}
    ,{"replace",         (TFUN)CoTBreplace,    0,    PARTEST}
    ,{"use",             (TFUN)CoTBuse,        0,    PARTEST}
};

static UBYTE *tablebasename = 0;

int CoTableBase(UBYTE *s)
{
    UBYTE *option, c, *t;
    int i,optlistsize = sizeof(tboptions)/sizeof(KEYWORD), error = 0;
    while ( *s == ' ' ) s++;
    if ( *s != '"' ) {
        if ( ( tolower(*s) == 'h' ) && ( tolower(s[1]) == 'e' )
         && ( tolower(s[2]) == 'l' ) && ( tolower(s[3]) == 'p' )
         && ( FG.cTable[s[4]] > 1 ) ) {
            CoTBhelp(s);
            return(0);
        }
proper:;
        MesPrint("&Proper syntax: TableBase \"filename\" options");
        return(1);
    }
    s++; tablebasename = s;
    while ( *s && *s != '"' ) s++;
    if ( *s != '"' ) goto proper;
    t = s; s++; *t = 0;
    while ( *s == ' ' || *s == '\t' || *s == ',' ) s++;
    option = s;
    while ( FG.cTable[*s] == 0 ) s++;
    c = *s; *s = 0;
    for ( i = 0; i < optlistsize; i++ ) {
        if ( StrICmp(option,(UBYTE *)(tboptions[i].name)) == 0 ) {
            *s = c;
            while ( *s == ',' ) s++;
            error = (tboptions[i].func)(s);
            *t = '"';
            return(error);
        }
    }
    MesPrint("&Unrecognized option %s in TableBase statement",option);
    return(1);
}

/*
    #] CoTableBase :
    #[ FlipTable :

    Flips the table between use as 'stub' and regular use
*/

int FlipTable(FUNCTIONS f, int type)
{
    TABLES T, TT;
    T = f->tabl;
    if ( ( TT = T->spare ) == 0 ) {
        MesPrint("Error: trying to change mode on a table that has no tablebase");
        return(-1);
    }
    if ( TT->mode == type ) f->tabl = TT;
```

```
    return(0);
}

/*
    #] FlipTable :
    #[ SpareTable :

    Creates a spare element for a table. This is used in the table bases.
    It is a (thus far) empty copy of the TT table.
    By using FlipTable we can switch between them and alter which version of
    a table we will be using. Note that this also causes some extra work in the
    ResetVariables and the Globalize routines.
*/

int SpareTable(TABLES TT)
{
    TABLES T;
    T = (TABLES)Malloc1(sizeof(struct TaBlEs),"table");
    T->defined = T->mdefined = 0; T->sparse = TT->sparse; T->mm = 0; T->flags = 0;
    T->numtree = 0; T->rootnum = 0; T->MaxTreeSize = 0;
    T->boomlijst = 0;
    T->strict = TT->strict;
    T->bounds = TT->bounds;
    T->bufnum = inicbufs();
    T->argtail = TT->argtail;
    T->spare = TT;
    T->bufferssize = 8;
    T->buffers = (WORD *)Malloc1(sizeof(WORD)*T->bufferssize,"SpareTable buffers");
    T->buffersfill = 0;
    T->buffers[T->buffersfill++] = T->bufnum;
    T->mode = 0;
    T->numind = TT->numind;
    T->totind = 0;
    T->prototype = TT->prototype;
    T->pattern = TT->pattern;
    T->tablepointers = 0;
    T->reserved = 0;
    T->tablenum = 0;
    T->numdummies = 0;
    T->mm = (MINMAX *)Malloc1(ABS(T->numind)*sizeof(MINMAX),"table dimensions");
    T->flags = (WORD *)Malloc1(ABS(T->numind)*sizeof(WORD),"table flags");
    ClearTableTree(T);
    TT->spare = T;
    TT->mode = 1;
    return(0);
}

/*
    #] SpareTable :
    #[ FindTB :

    Looks for a tablebase with the given name in the active tablebases.
*/

DBASE *FindTB(UBYTE *name)
{
    DBASE *d;
    int i;
    for ( i = 0; i < NumTableBases; i++ ) {
        d = tablebases+i;
        if ( d->name && ( StrCmp(name,(UBYTE *)(d->name)) == 0 ) ) { return(d); }
    }
    return(0);
}

/*
    #] FindTB :
    #[ CoTBcreate :

    Creates a new tablebase.
    Error is when there is already an active tablebase by this name.
    If a file with the given name exists already, but it does not correspond
    to an active table base, its contents will be lost.
    Note that tablebasename is a static variable, defined in CoTableBase
*/

int CoTBcreate(UBYTE *s)
{
    DUMMYUSE(s);
    if ( FindTB(tablebasename) != 0 ) {
        MesPrint("&There is already an open TableBase with the name %s",tablebasename);
        return(-1);
    }
    NewDbase((char *)tablebasename,0);
    return(0);
}
```

```
/*
    #] CoTBcreate :
    #[ CoTBopen :
*/

int CoTBopen(UBYTE *s)
{
    DBASE *d;
    MLONG rw = 1;

    SkipSpaces(&s);

    if ( *s ) {
        if ( ConsumeOption(&s,"readonly") != 0 ) {
            rw = 0;
        } else {
            MesPrint("&Invalid option for TableBase open: %s, ignoring", s);
        }
    }

    if ( ( d = FindTB(tablebasename) ) != 0 ) {
        MesPrint("&There is already an open TableBase with the name %s",tablebasename);
        return(-1);
    }
    d = GetDbase((char *)tablebasename, rw);
    if ( CheckTableDeclarations(d) ) return(-1);
    return(0);
}

/*
    #] CoTBopen :
    #[ CoTBaddto :
*/

int CoTBaddto(UBYTE *s)
{
    GETIDENTITY
    DBASE *d;
    UBYTE *tablename, c, *t, elementstring[ELEMENTSIZE+20], *ss, *es;
    WORD type, funnum, lbrac, first, num, *expr, *w;
    TABLES T = 0;
    MLONG basenumber;
    LONG x;
    int i, j, error = 0, sum;
    if ( ( d = FindTB(tablebasename) ) == 0 ) {
        MesPrint("&No open tablebase with the name %s",tablebasename);
        return(-1);
    }

    if ( ( d->rwmode ) == 0 ) {
        MesPrint("&Tablebase with the name %s opened in read only mode",tablebasename);
        return(-1);
    }
    AO.DollarOutSizeBuffer = 32;
    AO.DollarOutBuffer = (UBYTE *)Malloc1(AO.DollarOutSizeBuffer,
                            "TableOutBuffer");
/*
    Now loop through the names and start adding
*/
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    while ( *s ) {
        tablename = s;
        if ( ( s = SkipAName(s) ) == 0 ) goto tableabort;
        c = *s; *s = 0;
        if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
            || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",tablename);
            *s = c; goto tableabort;
        }
        if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",tablename);
            *s = c; goto tableabort;
        }
        basenumber = AddTableName(d,(char *)tablename,T);
        if ( T->spare && ( T->mode ==  1 ) ) T = T->spare;
        if ( basenumber < 0 ) basenumber = -basenumber;
        else if ( basenumber == 0 ) { *s = c; goto tableabort; }
        *s = c;
        if ( *s == '(' ) { /* Addition of single element */
            s++; es = s;
            for ( i = 0, w = AT.WorkPointer; i < ABS(T->numind); i++ ) {
              ParseSignedNumber(x,s);
              if ( FG.cTable[s[-1]] != 1 || ( *s != ',' && *s != ')' ) ) {
                MesPrint("&Table arguments in TableBase addto statement should be numbers");
                return(1);
              }
              *w++ = x;
```

```
              if ( *s == ')' ) break;
              s++;
            }
            if ( *s != ')' || i < ( ABS(T->numind) - 1 ) ) {
              MesPrint("&Incorrect number of table arguments in TableBase addto statement. Should be %d"
                ,ABS(T->numind));
              error = 1;
            }
            c = *s; *s = 0;
            i = FindTableTree(T,AT.WorkPointer,1);
            if ( i < 0 ) {
                MesPrint("&Element %s has not been defined",es);
                error = 1;
                *s++ = c;
            }
            else if ( ExistsObject(d,basenumber,(char *)es) ) {}
            else {
              int dict = AO.CurrentDictionary;
              AO.CurrentDictionary = 0;
              sum = i + ABS(T->numind);
/*
              See also commentary below
*/
              AO.DollarInOutBuffer = 1;
              AO.PrintType = 1;
              ss = AO.DollarOutBuffer;
              *ss = 0;
              AO.OutInBuffer = 1;
#if ( TABLEEXTENSION == 2 )
              expr = cbuf[T->bufnum].rhs[T->tablepointers[sum]];
#else
              expr = cbuf[T->tablepointers[sum+1]].rhs[T->tablepointers[sum]];
#endif
              lbrac = 0; first = 0;
              while ( *expr ) {
                if ( WriteTerm(expr,&lbrac,first,PRINTON,0) ) {
                        error = 1; break;
                }
                expr += *expr;
              }
              AO.OutInBuffer = 0;
              AddObject(d,basenumber,(char *)es,(char *)(AO.DollarOutBuffer));
              *s++ = c;
              AO.CurrentDictionary = dict;
            }
        }
        else {
/*
        Now we have to start looping through all defined elements of this table.
        We have to construct the arguments in text format.
*/
          for ( i = 0; i < T->totind; i++ ) {
#if ( TABLEEXTENSION == 2 )
            if ( !T->sparse && T->tablepointers[i] < 0 ) continue;
#else
            if ( !T->sparse && T->tablepointers[TABLEEXTENSION*i] < 0 ) continue;
#endif
            sum = i * ( ABS(T->numind) + TABLEEXTENSION );
            t = elementstring;
            for ( j = 0; j < ABS(T->numind); j++, sum++ ) {
                if ( j > 0 ) *t++ = ',';
                num = T->tablepointers[sum];
                t = NumCopy(num,t);
                if ( ( t - elementstring ) >= ELEMENTSIZE ) {
                    MesPrint("&Table element specification takes more than %ld characters and cannot be handled",
                        (MLONG)ELEMENTSIZE);
                    goto tableabort;
                }
            }
            if ( ExistsObject(d,basenumber,(char *)elementstring) ) { continue; }
/*
            We have the number in basenumber and the element in elementstring.
            Now we need the rhs. We can use the code from WriteDollarToBuffer.
            Main complication: in the table compiler buffer there can be
            brackets. The dollars do not have those......
*/
            AO.DollarInOutBuffer = 1;
            AO.PrintType = 1;
            ss = AO.DollarOutBuffer;
            *ss = 0;
            AO.OutInBuffer = 1;
#if ( TABLEEXTENSION == 2 )
            expr = cbuf[T->bufnum].rhs[T->tablepointers[sum]];
#else
            expr = cbuf[T->tablepointers[sum+1]].rhs[T->tablepointers[sum]];
```

```
#endif
            lbrac = 0; first = 0;
            while ( *expr ) {
                if ( WriteTerm(expr,&lbrac,first,PRINTON,0) ) {
                    error = 1; break;
                }
                expr += *expr;
            }
            AO.OutInBuffer = 0;
            AddObject(d,basenumber,(char *)elementstring,(char *)(AO.DollarOutBuffer));
          }
        }
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    }
    if ( WriteIniInfo(d) ) goto tableabort;
    M_free(AO.DollarOutBuffer,"DollarOutBuffer");
    AO.DollarOutBuffer = 0;
    AO.DollarOutSizeBuffer = 0;
    return(error);
tableabort:;
    M_free(AO.DollarOutBuffer,"DollarOutBuffer");
    AO.DollarOutBuffer = 0;
    AO.DollarOutSizeBuffer = 0;
    AO.OutInBuffer = 0;
    return(1);
}

/*
    #] CoTBaddto :
    #[ CoTBenter :

    Loads the elements of the tables specified into memory and sends them
    one by one to the compiler as Fill statements.
*/

int CoTBenter(UBYTE *s)
{
    DBASE *d;
    MLONG basenumber;
    UBYTE *arguments, *rhs, *buffer, *t, *u, c, *tablename;
    LONG size;
    int i, j, error = 0, error1 = 0, printall = 0;
    TABLES T = 0;
    WORD type, funnum;
    int dict = AO.CurrentDictionary;
    AO.CurrentDictionary = 0;
    if ( ( d = FindTB(tablebasename) ) == 0 ) {
        MesPrint("&No open tablebase with the name %s, check for existence of file or try readonly
mode when opening.",tablebasename);
        error = -1;
        goto Endofall;
    }
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    if ( *s == '!' ) { printall = 1; s++; }
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    if ( *s ) {
      while ( *s ) {
        tablename = s;
        if ( ( s = SkipAName(s) ) == 0 ) { error = 1; goto Endofall; }
        c = *s; *s = 0;
        if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
            || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",tablename);
            basenumber = 0;
        }
        else if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",tablename);
            basenumber = 0;
        }
        else { basenumber = GetTableName(d,(char *)tablename); }
        if ( T->spare == 0 ) { SpareTable(T); }
        if ( basenumber > 0 ) {
            for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
                for ( j = 0; j < NUMOBJECTS; j++ ) {
                    if ( basenumber != d->iblocks[i]->objects[j].tablenumber )
                        continue;
                    arguments = (UBYTE *)(d->iblocks[i]->objects[j].element);
                    rhs = (UBYTE *)ReadObject(d,basenumber,(char *)arguments);
                    if ( printall ) {
                        if ( rhs ) {
                            MesPrint("%s(%s) = %s",tablename,arguments,rhs);
                        }
                        else {
                            MesPrint("%s(%s) = 0",tablename,arguments);
                        }
                    }
                    if ( rhs ) {
```

```
                        u = rhs; while ( *u ) u++;
                        size = u-rhs;
                        u = arguments; while ( *u ) u++;
                        size += u-arguments;
                        u = tablename; while ( *u ) u++;
                        size += u-tablename;
                        size += 6;
                        buffer  = (UBYTE *)Malloc1(size,"TableBase copy");
                        t = tablename; u = buffer;
                        while ( *t ) *u++ = *t++;
                        *u++ = '(';
                        t = arguments;
                        while ( *t ) *u++ = *t++;
                        *u++ = ')'; *u++ = '=';
                        t = rhs;
                        while ( *t ) *u++ = *t++;
                        if ( t == rhs ) *u++ = '0';
                        *u++ = 0; *u = 0;
                        M_free(rhs,"rhs in TBenter");

                        error1 = CoFill(buffer);

                        if ( error1 < 0 ) goto Endofall;
                        if ( error1 != 0 ) error = error1;
                        M_free(buffer,"TableBase copy");
                    }
                }
            }
        }
        *s = c;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
      }
    }
    else {
      s = (UBYTE *)(d->tablenames); basenumber = 0;
      while ( *s ) {
        basenumber++;
        tablename = s; while ( *s ) s++; s++;
        while ( *s ) s++;
        s++;
        if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
            || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",tablename);
        }
        else if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",tablename);
        }
        if ( T->spare == 0 ) { SpareTable(T); }
        for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
            for ( j = 0; j < NUMOBJECTS; j++ ) {
                if ( d->iblocks[i]->objects[j].tablenumber == basenumber ) {
                    arguments = (UBYTE *)(d->iblocks[i]->objects[j].element);
                    rhs = (UBYTE *)ReadObject(d,basenumber,(char *)arguments);
                    if ( printall ) {
                        if ( rhs ) {
                            MesPrint("%s%s = %s",tablename,arguments,rhs);
                        }
                        else {
                            MesPrint("%s%s = 0",tablename,arguments);
                        }
                    }
                    if ( rhs ) {
                        u = rhs; while ( *u ) u++;
                        size = u-rhs;
                        u = arguments; while ( *u ) u++;
                        size += u-arguments;
                        u = tablename; while ( *u ) u++;
                        size += u-tablename;
                        size += 6;
                        buffer  = (UBYTE *)Malloc1(size,"TableBase copy");
                        t = tablename; u = buffer;
                        while ( *t ) *u++ = *t++;
                        *u++ = '(';
                        t = arguments;
                        while ( *t ) *u++ = *t++;
                        *u++ = ')'; *u++ = '=';
                        t = rhs;
                        while ( *t ) *u++ = *t++;
                        if ( t == rhs ) *u++ = '0';
                        *u++ = 0; *u = 0;
                        M_free(rhs,"rhs in TBenter");

                        error1 = CoFill(buffer);

                        if ( error1 < 0 ) goto Endofall;
                        if ( error1 != 0 ) error = error1;
                        M_free(buffer,"TableBase copy");
```

```
                        }
                    }
                }
            }
        }
    }
Endofall:;
    AO.CurrentDictionary = dict;
    return(error);
}

/*
    #] CoTBenter :
    #[ CoTestUse :

    Possibly to be followed by names of tables.
    We make an array of TABLES structs to be tested in AC.usedtables.
    Note: only sparse tables are allowed.
    No arguments means all tables.
*/

int CoTestUse(UBYTE *s)
{
    GETIDENTITY
    UBYTE *tablename, c;
    WORD type, funnum, *w;
    TABLES T;
    int error = 0;
    w = AT.WorkPointer;
    *w++ = TYPETESTUSE; *w++ = 2;
    while ( *s ) {
        tablename = s;
        if ( ( s = SkipAName(s) ) == 0 ) return(1);
        c = *s; *s = 0;
        if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
            || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",tablename);
            error = 1;
        }
        else if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",tablename);
            error = 1;
        }
        *w++ = funnum + FUNCTION;
        *s = c;
        while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    }
    AT.WorkPointer[1] = w - AT.WorkPointer;
/*
    if ( AT.WorkPointer[1] > 2 ) {
        AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    }
*/
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    return(error);
}

/*
    #] CoTestUse :
    #[ CheckTableDeclarations :

        Checks that all tables in a tablebase  have identical properties to
        possible previous declarations. If they have not been declared
        before, they are declared here.
*/

int CheckTableDeclarations(DBASE *d)
{
    WORD type, funnum;
    UBYTE *s, *ss, *t, *command = 0;
    int k, error = 0, error1, i;
    TABLES T;
    LONG commandsize = 0;

    s = (UBYTE *)(d->tablenames);
    for ( k = 0; k < d->topnumber; k++ ) {
        if ( GetVar(s,&type,&funnum,ANYTYPE,NOAUTO) == NAMENOTFOUND ) {
/*
            We have to declare the table
*/
            ss = s; i = 0; while ( *ss ) { ss++; i++; } /* name */
            ss++; while ( *ss ) { ss++; i++; } /* tail */
            if ( commandsize == 0 ) {
                commandsize = i + 15;
                if ( commandsize < 100 ) commandsize = 100;
            }
            if ( (i+11) > commandsize ) {
```

```
                if ( command ) { M_free(command,"table command"); command = 0; }
                commandsize = i+10;
            }
            if ( command == 0 ) {
                command = (UBYTE *)Malloc1(commandsize,"table command");
            }
            t = command; ss = tablebase; while ( *ss ) *t++ = *ss++;
            *t++ = ','; while ( *s ) *t++ = *s++;
            s++; while ( *s ) *t++ = *s++;
            *t++ = ')'; *t = 0; s++;
            error1 = DoTable(command,1);
            if ( error1 ) error = error1;
        }
        else if ( ( type != CFUNCTION )
             || ( ( T = functions[funnum].tabl ) == 0 )
             || ( T->sparse == 0 ) ) {
            MesPrint("&%s has been declared previously, but not as a sparse table.",s);
            error = 1;
            while ( *s ) s++;
            s++;
            while ( *s ) s++;
            s++;
        }
        else {
/*
            Test dimension and argtail. There should be an exact match.
            We are not going to rename arguments when reading the elements.
*/
            ss = s;
            while ( *s ) s++;
            s++;
            if ( StrCmp(s,T->argtail) ) {
                MesPrint("&Declaration of table %s in %s different from previous declaration",ss,d->name);
                error = 1;
            }
            while ( *s ) s++;
            s++;
        }
    }
    if ( command ) { M_free(command,"table command"); }
    return(error);
}

/*
    #] CheckTableDeclarations :
    #[ CoTBload :

        Loads the table stubbs of the specified tables in the indicated
        tablebase. Syntax:
        TableBase "tablebasename.tbl" load [tablename(s)];
        If no tables are specified all tables are taken.
*/

int CoTBload(UBYTE *ss)
{
    DBASE *d;
    UBYTE *s, *name, *t, *r, *command, *arguments, *tail;
    LONG commandsize;
    int num, cs, es, ns, ts, i, j, error = 0, error1;
    if ( ( d = FindTB(tablebasename) ) == 0 ) {
        MesPrint("&No open tablebase with the name %s",tablebasename);
        return(-1);
    }
    commandsize = 120;
    command = (UBYTE *)Malloc1(commandsize,"Fill command");
    AC.vetofilling = 1;
    if ( *ss ) {
      while ( *ss == ',' || *ss == ' ' || *ss == '\t' ) ss++;
      while ( *ss ) {
        name = ss; ss = SkipAName(ss); *ss = 0;
        s = (UBYTE *)(d->tablenames);
        num = 0; ns = 0;
        while ( *s ) {
            num++;
            if ( StrCmp(s,name) ) {
                while ( *s ) s++;
                s++;
                while ( *s ) s++;
                s++;
                num++;
                continue;
            }
            name = s; while ( *s ) s++; ns = s-name; s++;
            tail = s; while ( *s ) s++; ts = s-tail; s++;
            tail++; while ( FG.cTable[*tail] == 1 ) tail++;
/*
```

```
            Go through all elements
*/
            for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
                for ( j = 0; j < NUMOBJECTS; j++ ) {
                    if ( d->iblocks[i]->objects[j].tablenumber == num ) {
                        t = arguments = (UBYTE *)(d->iblocks[i]->objects[j].element);
                        while ( *t ) t++;
                        es = t - arguments;
                        cs = 2*es + 2*ns + ts + 10;
                        if ( cs > commandsize ) {
                            commandsize = 2*cs;
                            if ( command ) M_free(command,"Fill command");
                            command = (UBYTE *)Malloc1(commandsize,"Fill command");
                        }
                        r = command; t = name; while ( *t ) *r++ = *t++;
                        *r++ = '('; t = arguments; while ( *t ) *r++ = *t++;
                        *r++ = ')'; *r++ = '='; *r++ = 't'; *r++ = 'b'; *r++ = 'l';
                        *r++ = '_'; *r++ = '('; t = name; while ( *t ) *r++ = *t++;
                        *r++ = ','; t = arguments; while ( *t ) *r++ = *t++;
                        t = tail; while ( *t ) {
                            if ( *t == '?' && r[-1] != ',' ) {
                                t++;
                                if ( FG.cTable[*t] == 0 || *t == '$' || *t == '[' ) {
                                    t = SkipAName(t);
                                    if ( *t == '[' ) {
                                        SKIPBRA1(t);
                                    }
                                }
                                else if ( *t == '{' ) {
                                    SKIPBRA2(t);
                                }
                                else if ( *t ) { *r++ = *t++; continue; }
                            }
                            else *r++ = *t++;
                        }
                        *r++ = ')'; *r = 0;
/*
                        Still to do: replacemode or no replacemode?
*/
                        AC.vetotablebasefill = 1;
                        error1 = CoFill(command);
                        AC.vetotablebasefill = 0;
                        if ( error1 < 0 ) goto finishup;
                        if ( error1 != 0 ) error = error1;
                    }
                }
            }
            break;
        }
        while ( *ss == ',' || *ss == ' ' || *ss == '\t' ) ss++;
      }
    }
    else { /* do all of them */
        s = (UBYTE *)(d->tablenames);
        num = 0; ns = 0;
        while ( *s ) {
            num++;
            name = s; while ( *s ) s++; ns = s-name; s++;
            tail = s; while ( *s ) s++; ts = s-tail; s++;
            tail++; while ( FG.cTable[*tail] == 1 ) tail++;
/*
            Go through all elements
*/
            for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
                for ( j = 0; j < NUMOBJECTS; j++ ) {
                    if ( d->iblocks[i]->objects[j].tablenumber == num ) {
                        t = arguments = (UBYTE *)(d->iblocks[i]->objects[j].element);
                        while ( *t ) t++;
                        es = t - arguments;
                        cs = 2*es + 2*ns + ts + 10;
                        if ( cs > commandsize ) {
                            commandsize = 2*cs;
                            if ( command ) M_free(command,"Fill command");
                            command = (UBYTE *)Malloc1(commandsize,"Fill command");
                        }
                        r = command; t = name; while ( *t ) *r++ = *t++;
                        *r++ = '('; t = arguments; while ( *t ) *r++ = *t++;
                        *r++ = ')'; *r++ = '='; *r++ = 't'; *r++ = 'b'; *r++ = 'l';
                        *r++ = '_'; *r++ = '('; t = name; while ( *t ) *r++ = *t++;
                        *r++ = ','; t = arguments; while ( *t ) *r++ = *t++;
                        t = tail; while ( *t ) {
                            if ( *t == '?' && r[-1] != ',' ) {
                                t++;
                                if ( FG.cTable[*t] == 0 || *t == '$' || *t == '[' ) {
                                    t = SkipAName(t);
                                    if ( *t == '[' ) {
                                        SKIPBRA1(t);
```

```
                                    }
                                }
                                else if ( *t == '{' ) {
                                    SKIPBRA2(t);
                                }
                                else if ( *t ) { *r++ = *t++; continue; }
                            }
                            else *r++ = *t++;
                        }
                        *r++ = ')'; *r = 0;
/*
                        Still to do: replacemode or no replacemode?
*/
                        AC.vetotablebasefill = 1;
                        error1 = CoFill(command);
                        AC.vetotablebasefill = 0;
                        if ( error1 < 0 ) goto finishup;
                        if ( error1 != 0 ) error = error1;
                    }
                }
            }
        }
    }
finishup:;
    AC.vetofilling = 0;
    if ( command ) M_free(command,"Fill command");
    return(error);
}

/*
    #] CoTBload :
    #[ TestUse :

    Look for tbl_(tablename,arguments)
    if tablename is encountered, check first whether the element is in
    use already. If not, check in the tables in AC.usedtables.
    If the element is not there, add it to AC.usedtables.


    We need the arguments of TestUse to see for which tables it is to be done
*/

WORD TestUse(WORD *term, WORD level)
{
    WORD *tstop, *t, *m, *tstart, tabnum;
    WORD *funs, numfuns, error = 0;
    TABLES T;
    LONG i;
    CBUF *C = cbuf+AM.rbufnum;
    int isp;

    numfuns = C->lhs[level][1] - 2;
    funs = C->lhs[level] + 2;
    GETSTOP(term,tstop);
    t = term+1;
    while ( t < tstop ) {
        if ( *t != TABLESTUB ) { t += t[1]; continue; }
        tstart = t;
        m = t + FUNHEAD;
        t += t[1];
        if ( *m >= -FUNCTION ) continue;
        tabnum = -*m;
        if ( ( T = functions[tabnum-FUNCTION].tabl ) == 0 ) continue;
        if ( T->sparse == 0 ) continue;
/*
        Check whether we have to test this one
*/
        if ( numfuns > 0 ) {
            for ( i = 0; i < numfuns; i++ ) {
                if ( tabnum == funs[i] ) break;
            }
            if ( i >= numfuns && numfuns > 0 ) continue;
        }
/*
        Test whether the element has been defined already.
            If not, mark it as used.
        Note: we only allow sparse tables (for now)
*/
        m++;
        for ( i = 0; i < ABS(T->numind); i++, m += 2 ) {
            if ( m >= t || *m != -SNUMBER ) break;
        }
        if ( ( i == ABS(T->numind) ) &&
         ( ( isp = FindTableTree(T,tstart+FUNHEAD+1,2) ) >= 0 ) ) {
            if ( ( T->tablepointers[isp+ABS(T->numind)+4] & ELEMENTLOADED ) == 0 ) {
                    T->tablepointers[isp+ABS(T->numind)+4] |= ELEMENTUSED;
            }
```

```
        }
        else {
            MesPrint("TestUse: Encountered a table element inside tbl_ that does not correspond to a tablebase element");
            error = -1;
        }
    }
    return(error);
}

/*
    #] TestUse :
    #[ CoTBaudit :
*/

int CoTBaudit(UBYTE *s)
{
    DBASE *d;
    UBYTE *name, *tail;
    int i, j, error = 0, num;

    if ( ( d = FindTB(tablebasename) ) == 0 ) {
        MesPrint("&No open tablebase with the name %s",tablebasename);
        return(-1);
    }
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    while ( *s ) {
/*
        Get the options here
        They will mainly involve the sorting of the output.
*/
        s++;
    }
    s = (UBYTE *)(d->tablenames); num = 0;
    while ( *s ) {
        num++;
        name = s; while ( *s ) s++; s++;
        tail = s; while ( *s ) s++; s++;
        MesPrint("Table,sparse,%s%s)",name,tail);
        for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
            for ( j = 0; j < NUMOBJECTS; j++ ) {
                if ( d->iblocks[i]->objects[j].tablenumber == num ) {
                    MesPrint("    %s(%s)",name,d->iblocks[i]->objects[j].element);
                }
            }
        }
    }
    return(error);
}

/*
    #] CoTBaudit :
    #[ CoTBon :
*/

int CoTBon(UBYTE *s)
{
    DBASE *d;
    UBYTE *ss, c;
    int error = 0;
    if ( ( d = FindTB(tablebasename) ) == 0 ) {
        MesPrint("&No open tablebase with the name %s",tablebasename);
        return(-1);
    }
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    while ( *s ) {
        ss = SkipAName(s);
        c = *ss; *ss = 0;
        if ( StrICmp(s,(UBYTE *)("compress")) == 0 ) {
            d->mode &= ~NOCOMPRESS;
        }
        else {
            MesPrint("&subkey %s not defined in TableBase On statement");
            error = 1;
        }
        *ss = c; s = ss;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    }
    return(error);
}

/*
    #] CoTBon :
    #[ CoTBoff :
*/

int CoTBoff(UBYTE *s)
```

```
{
    DBASE *d;
    UBYTE *ss, c;
    int error = 0;
    if ( ( d = FindTB(tablebasename) ) == 0 ) {
        MesPrint("&No open tablebase with the name %s",tablebasename);
        return(-1);
    }
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    while ( *s ) {
        ss = SkipAName(s);
        c = *ss; *ss = 0;
        if ( StrICmp(s,(UBYTE *)("compress")) == 0 ) {
            d->mode |= NOCOMPRESS;
        }
        else {
            MesPrint("&subkey %s not defined in TableBase Off statement");
            error = 1;
        }
        *ss = c; s = ss;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    }
    return(error);
}

/*
    #] CoTBoff :
    #[ CoTBcleanup :
*/

int CoTBcleanup(UBYTE *s)
{
    DUMMYUSE(s);
    MesPrint("&TableBase Cleanup statement not yet implemented");
    return(1);
}

/*
    #] CoTBcleanup :
    #[ CoTBreplace :
*/

int CoTBreplace(UBYTE *s)
{
    DUMMYUSE(s);
    MesPrint("&TableBase Replace statement not yet implemented");
    return(1);
}

/*
    #] CoTBreplace :
    #[ CoTBuse :

    Here the actual table use as determined in TestUse causes the needed
    table elements to be loaded
*/

int CoTBuse(UBYTE *s)
{
    GETIDENTITY
    DBASE *d;
    MLONG basenumber;
    UBYTE *arguments, *rhs, *buffer, *t, *u, c, *tablename, *p;
    LONG size, sum, x;
    int i, j, error = 0, error1 = 0, k;
    TABLES T = 0;
    WORD type, funnum, mode, *w;
    if ( ( d = FindTB(tablebasename) ) == 0 ) {
        MesPrint("&No open tablebase with the name %s",tablebasename);
        return(-1);
    }
    while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
    if ( *s ) {
      while ( *s ) {
        tablename = s;
        if ( ( s = SkipAName(s) ) == 0 ) return(1);
        c = *s; *s = 0;
        if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
            || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",tablename);
            basenumber = 0;
        }
        else if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",tablename);
            basenumber = 0;
        }
        else { basenumber = GetTableName(d,(char *)tablename); }
```

```
/*      if ( T->spare == 0 ) { SpareTable(T); } */
        if ( basenumber > 0 ) {
            for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
                for ( j = 0; j < NUMOBJECTS; j++ ) {
                    if ( d->iblocks[i]->objects[j].tablenumber != basenumber ) continue;
                    arguments = p = (UBYTE *)(d->iblocks[i]->objects[j].element);
/*
                    Now translate the arguments and see whether we need
                    this one....
*/
                    for ( k = 0, w = AT.WorkPointer; k < ABS(T->numind); k++ ) {
                        ParseSignedNumber(x,p);
                        *w++ = x; p++;
                    }
                    sum = FindTableTree(T,AT.WorkPointer,1);
                    if ( sum < 0 ) {
                        MesPrint("Table %s in tablebase %s has not been loaded properly"
                                ,tablename,tablebasename);
                        error = 1;
                        continue;
                    }
                    sum += ABS(T->numind) + 4;
                    mode = T->tablepointers[sum];
                    if ( ( mode & ELEMENTLOADED ) == ELEMENTLOADED ) {
                        T->tablepointers[sum] &= ~ELEMENTUSED;
                        continue;
                    }
                    if ( ( mode & ELEMENTUSED ) == 0 ) continue;
/*
                    We need this one!
*/
                    rhs = (UBYTE *)ReadijObject(d,i,j,(char *)arguments);
                    if ( rhs ) {
                        u = rhs; while ( *u ) u++;
                        size = u-rhs;
                        u = arguments; while ( *u ) u++;
                        size += u-arguments;
                        u = tablename; while ( *u ) u++;
                        size += u-tablename;
                        size += 6;
                        buffer  = (UBYTE *)Malloc1(size,"TableBase copy");
                        t = tablename; u = buffer;
                        while ( *t ) *u++ = *t++;
                        *u++ = '(';
                        t = arguments;
                        while ( *t ) *u++ = *t++;
                        *u++ = ')'; *u++ = '=';
                        t = rhs;
                        while ( *t ) *u++ = *t++;
                        if ( t == rhs ) { *u++ = '0'; }
                        *u++ = 0; *u = 0;
                        M_free(rhs,"rhs in TBuse xxx");

                        error1 = CoFill(buffer);

                        if ( error1 < 0 ) { return(error); }
                        if ( error1 != 0 ) error = error1;
                        M_free(buffer,"TableBase copy");
                    }
                    T->tablepointers[sum] &= ~ELEMENTUSED;
                    T->tablepointers[sum] |= ELEMENTLOADED;
                }
            }
        }
        *s = c;
        while ( *s == ',' || *s == ' ' || *s == '\t' ) s++;
      }
    }
    else {
      s = (UBYTE *)(d->tablenames); basenumber = 0;
      while ( *s ) {
        basenumber++;
        tablename = s;
        while ( *s ) s++;
        s++;
        while ( *s ) s++;
        s++;
        if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
            || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",tablename);
        }
        else if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",tablename);
        }
    if ( T->spare && T->mode == 0 ) {
        MesPrint("In table %s we have a problem with stubb orders in CoTBuse",tablename);
        error = -1;
```

```
    }
/*      if ( T->spare == 0 ) { SpareTable(T); } */
        for ( i = 0; i < d->info.numberofindexblocks; i++ ) {
            for ( j = 0; j < NUMOBJECTS; j++ ) {
                if ( d->iblocks[i]->objects[j].tablenumber == basenumber ) {
                    arguments = p = (UBYTE *)(d->iblocks[i]->objects[j].element);
/*

                    Now translate the arguments and see whether we need
                    this one....
*/
                    for ( k = 0, w = AT.WorkPointer; k < ABS(T->numind); k++ ) {
                        ParseSignedNumber(x,p);
                        *w++ = x; p++;
                    }
                    sum = FindTableTree(T,AT.WorkPointer,1);
                    if ( sum < 0 ) {
                        MesPrint("Table %s in tablebase %s has not been loaded properly"
                                ,tablename,tablebasename);
                        error = 1;
                        continue;
                    }
                    sum += ABS(T->numind) + 4;
                    mode = T->tablepointers[sum];
                    if ( ( mode & ELEMENTLOADED ) == ELEMENTLOADED ) {
                        T->tablepointers[sum] &= ~ELEMENTUSED;
                        continue;
                    }
                    if ( ( mode & ELEMENTUSED ) == 0 ) continue;
/*

                    We need this one!
*/

                    rhs = (UBYTE *)ReadijObject(d,i,j,(char *)arguments);
                    if ( rhs ) {
                        u = rhs; while ( *u ) u++;
                        size = u-rhs;
                        u = arguments; while ( *u ) u++;
                        size += u-arguments;
                        u = tablename; while ( *u ) u++;
                        size += u-tablename;
                        size += 6;
                        buffer  = (UBYTE *)Malloc1(size,"TableBase copy");
                        t = tablename; u = buffer;
                        while ( *t ) *u++ = *t++;
                        *u++ = '(';
                        t = arguments;
                        while ( *t ) *u++ = *t++;
                        *u++ = ')'; *u++ = '=';

                        t = rhs;
                        while ( *t ) *u++ = *t++;
                        if ( t == rhs ) { *u++ = '0'; }
                        *u++ = 0; *u = 0;
                        M_free(rhs,"rhs in TBuse");

                        error1 = CoFill(buffer);

                        if ( error1 < 0 ) { return(error); }
                        if ( error1 != 0 ) error = error1;
                        M_free(buffer,"TableBase copy");
                    }
                    T->tablepointers[sum] &= ~ELEMENTUSED;
                    T->tablepointers[sum] |= ELEMENTLOADED;
                }
            }
        }
      }
    }
    return(error);
}

/*
    #] CoTBuse :
    #[ CoApply :

    Possibly to be followed by names of tables.
*/

int CoApply(UBYTE *s)
{
    GETIDENTITY
    UBYTE *tablename, c;
    WORD type, funnum, *w;
    TABLES T;
    LONG maxtogo = MAXPOSITIVE;
    int error = 0;
    w = AT.WorkPointer;
    if ( FG.cTable[*s] == 1 ) {
```

```
        maxtogo = 0;
        while ( FG.cTable[*s] == 1 ) {
            maxtogo = maxtogo*10 + (*s-'0');
            s++;
        }
        while ( *s == ',' ) s++;
        if ( maxtogo > MAXPOSITIVE || maxtogo < 0 ) maxtogo = MAXPOSITIVE;
    }
    *w++ = TYPEAPPLY; *w++ = 3; *w++ = maxtogo;
    while ( *s ) {
        tablename = s;
        if ( ( s = SkipAName(s) ) == 0 ) return(1);
        c = *s; *s = 0;
        if ( ( GetVar(tablename,&type,&funnum,CFUNCTION,NOAUTO) == NAMENOTFOUND )
            || ( T = functions[funnum].tabl ) == 0 ) {
            MesPrint("&%s should be a previously declared table",tablename);
            error = 1;
        }
        else if ( T->sparse == 0 ) {
            MesPrint("&%s should be a sparse table",tablename);
            error = 1;
        }
        *w++ = funnum + FUNCTION;
        *s = c;
        while ( *s == ' ' || *s == ',' || *s == '\t' ) s++;
    }
    AT.WorkPointer[1] = w - AT.WorkPointer;
/*
    if ( AT.WorkPointer[1] > 2 ) {
        AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
    }
*/
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
/*
    AT.WorkPointer[0] = TYPEAPPLYRESET;
    AddNtoL(AT.WorkPointer[1],AT.WorkPointer);
*/
    return(error);
}

/*
    #] CoApply :
    #[ CoTBhelp :
*/

char *helptb[] = {
     "The TableBase statement is used as follows:"
    ,"TableBase \"file.tbl\" keyword subkey(s)"
    ,"    in which we have"
    ,"Keyword   Subkey(s)   Action"
    ,"open                  Opens file.tbl for R/W"
    ,"create                Creates file.tbl for R/W. Old contents are lost"
    ,"load                  Loads all stubs of all tables"
    ,"load    tablename(s)  Loads all stubs the tables mentioned"
    ,"enter                 Loads all stubs and rhs of all tables"
    ,"enter   tablename(s)  Loads all stubs and rhs of the tables mentioned"
    ,"audit                 Prints list of contents"
/*  ,"replace tablename     saves a table (with overwrite)" */
/*  ,"replace tableelement  saves a table element (with overwrite)" */
/*  ,"cleanup               makes tables contingent" */
    ,"addto   tablename     adds all elements if not yet there"
    ,"addto   tableelement  adds element if not yet there"
/*  ,"delete  tablename     removes table from tablebase" */
/*  ,"delete  tableelement  removes element from tablebase" */
    ,"on      compress      elements are stored in gzip format (default)"
    ,"off     compress      elements are stored in uncompressed format"
    ,"use                   compiles all needed elements"
    ,"use     tablename(s)  compiles all needed elements of these tables"
    ,""
    ,"Related commands are:"
    ,"testuse               marks which tbl_ elements occur for all tables"
    ,"testuse tablename(s)  marks which tbl_ elements occur for given tables"
    ,"apply                 replaces tbl_ if rhs available"
    ,"apply   tablename(s)  replaces tbl_ for given tables if rhs available"
    ,""
        };

int CoTBhelp(UBYTE *s)
{
    int i, ii = sizeof(helptb)/sizeof(char *);
    DUMMYUSE(s);
    for ( i = 0; i < ii; i++ ) MesPrint("%s",helptb[i]);
    return(0);
}

/*
    #] CoTBhelp :
```

```
    #[ ReWorkT :

    Replaces the STUBBS of the functions in the list.
    This gains one space. Hence we have to be very careful
*/

void ReWorkT(WORD *term, WORD *funs, WORD numfuns)
{
    WORD *tstop, *tend, *m, *t, *tt, *mm, *mmm, *r, *rr;
    int i, j;
    tend = term + *term; tstop = tend - ABS(tend[-1]);
    m = t = term+1;
    while ( t < tstop ) {
        if ( *t == TABLESTUB ) {
            for ( i = 0; i < numfuns; i++ ) {
                if ( -t[FUNHEAD] == funs[i] ) break;
            }
            if ( numfuns == 0 || i < numfuns ) { /* Hit */
                i = t[1] - 1;
                *m++ = -t[FUNHEAD]; *m++ = i; t += 2; i -= FUNHEAD;
                if ( m < t ) { for ( j = 0; j < FUNHEAD-2; j++ ) *m++ = *t++; }
                else { m += FUNHEAD-2; t += FUNHEAD-2; }
                t++;
                while ( i-- > 0 ) { *m++ = *t++; }
                tt = t; mm = m;
                if ( mm < tt ) {
                    while ( tt < tend ) *mm++ = *tt++;
                    *term = mm - term;
                    tend = term + *term; tstop = tend - ABS(tend[-1]);
                    t = m;
                }
            }
            else { goto inc; }
        }
        else if ( *t >= FUNCTION ) {
            tt = t + t[1];
            mm = m;
            for ( j = 0; j < FUNHEAD; j++ ) {
                if ( m == t ) { m++; t++; }
                else *m++ = *t++;
            }
            while ( t < tt ) {
                if ( *t <= -FUNCTION ) {
                    if ( m == t ) { m++; t++; }
                    else *m++ = *t++;
                }
                else if ( *t < 0 ) {
                    if ( m == t ) { m += 2; t += 2; }
                    else { *m++ = *t++; *m++ = *t++; }
                }
                else {
                    rr = t + *t; mmm = m;
                    for ( j = 0; j < ARGHEAD; j++ ) {
                        if ( m == t ) { m++; t++; }
                        else *m++ = *t++;
                    }
                    while ( t < rr ) {
                        r = t + *t;
                        ReWorkT(t,funs,numfuns);
                        j = *t;
                        if ( m == t ) { m += j; t += j; }
                        else { while ( j-- >= 0 ) *m++ = *t++; }
                        t = r;
                    }
                    *mmm = m-mmm;
                }
            }
            mm[1] = m - mm;
            t = tt;
        }
        else {
inc:        j = t[1];
            if ( m < t ) { while ( j-- >= 0 ) *m++ = *t++; }
            else { m += j; t += j; }
        }
    }
    if ( m < t ) {
        while ( t < tend ) *m++ = *t++;
        *term = m - term;
    }
}

/*
    #] ReWorkT :
    #[ Apply :
*/
```

```
WORD Apply(WORD *term, WORD level)
{
    WORD *funs, numfuns;
    TABLES T;
    int i, j;
    CBUF *C = cbuf+AM.rbufnum;
/*
    Point the tables in the proper direction
*/
    numfuns = C->lhs[level][1] - 2;
    funs = C->lhs[level] + 2;
    if ( numfuns > 0 ) {
        for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
            if ( ( T = functions[i].tabl ) != 0 ) {
                for ( j = 0; j < numfuns; j++ ) {
                    if ( i == (funs[j]-FUNCTION) && T->spare ) {
                        FlipTable(&(functions[i]),0);
                        break;
                    }
                }
            }
        }
    }
    else {
        for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
            if ( ( T = functions[i].tabl ) != 0 ) {
                if ( T->spare ) FlipTable(&(functions[i]),0);
            }
        }
    }
/*
    Now the replacements everywhere of
        id tbl_(table,?a) = table(?a);
    Actually, this has to be done recursively.
    Note that we actually gain one space.
*/
    ReWorkT(term,funs,numfuns);
    return(0);
}

/*
    #] Apply :
    #[ ApplyExec :

    Replaces occurrences of tbl_(table,indices,pattern) by the proper
    rhs of table(indices,pattern). It does this up to maxtogo times
    in the given term. It starts with the occurrences inside the
    arguments of functions. If necessary it finishes at groundlevel.
    An infinite number of tries is indicated by maxtogo = 2^15-1 or 2^31-1.
    The occurrences are replaced by subexpressions. This allows TestSub
    to finish the job properly.

    The main trick here is T = T->spare which turns to the proper rhs.

    The return value is the number of substitutions that can still be made
    based on maxtogo. Hence, if the returnvalue is different from maxtogo
    there was a substitution.
*/

int ApplyExec(WORD *term, int maxtogo, WORD level)
{
    GETIDENTITY
    WORD rhsnumber, *Tpattern, *funs, numfuns, funnum;
    WORD ii, *t, *t1, *w, *p, *m, *m1, *u, *r, tbufnum, csize, wilds;
    NESTING NN;
    int i, j, isp, stilltogo;
    CBUF *C;
    TABLES T;
/*
    Startup. We need NestPoin for when we have to replace something deep down.
*/
    t = term;
    m = t + *t;
    csize = ABS(m[-1]);
    m -= csize;
    AT.NestPoin->termsize = t;
    if ( AT.NestPoin == AT.Nest ) AN.EndNest = t + *t;
    t++;
/*
    First we look inside function arguments. Also when clean!
*/
    while ( t < m ) {
        if ( *t < FUNCTION ) { t += t[1]; continue; }
        if ( functions[*t-FUNCTION].spec > 0 ) { t += t[1]; continue; }
        AT.NestPoin->funsize = t;
        r = t + t[1];
        t += FUNHEAD;
```

```
        while ( t < r ) {
            if ( *t < 0 ) { NEXTARG(t); continue; }
            AT.NestPoin->argsize = t1 = t;
            u = t + *t;
            t += ARGHEAD;
            AT.NestPoin++;
            while ( t < u ) {
/*
                Now we loop over the terms inside a function argument
                This defines a recursion and we have to call ApplyExec again.
                The real problem is when we catch something and we have
                to insert a subexpression pointer. This may use more or
                less space and the whole term has to be readjusted.
                This is why we have the NestPoin variables. They tell us
                where the sizes of the term, the function and the arguments
                are sitting, and also where the dirty flags are.
                This readjusting is of course done in the groundlevel code.
                Here we worry abound the maxtogo count.
*/
                stilltogo = ApplyExec(t,maxtogo,level);
                if ( stilltogo != maxtogo ) {
                    if ( stilltogo <= 0 ) {
                        AT.NestPoin--;
                        return(stilltogo);
                    }
                    maxtogo = stilltogo;
                    u = t1 + *t1;
                    m = term + *term - csize;
                }
                t += *t;
            }
            AT.NestPoin--;
        }
    }
/*
    Now we look at the ground level
*/
    C = cbuf+AM.rbufnum;
    t = term + 1;
    while ( t < m ) {
        if ( *t != TABLESTUB ) { t += t[1]; continue; }
        funnum = -t[FUNHEAD];
        if ( ( funnum < FUNCTION )
          || ( funnum >= FUNCTION+WILDOFFSET )
          || ( ( T = functions[funnum-FUNCTION].tabl ) == 0 )
          || ( T->sparse == 0 )
          || ( T->spare == 0 ) ) { t += t[1]; continue; }
        numfuns = C->lhs[level][1] - 3;
        funs = C->lhs[level] + 3;
        if ( numfuns > 0 ) {
            for ( i = 0; i < numfuns; i++ ) {
                if ( funs[i] == funnum ) break;
            }
            if ( i >= numfuns ) { t += t[1]; continue; }
        }
        r = t + t[1];
        AT.NestPoin->funsize = t + 1;
        t1 = t;
        t += FUNHEAD + 1;
/*
        Test whether the table catches
        Test 1: index arguments and range. isp will be the number
            of the element in the table.
*/
        T = T->spare;
#ifdef WITHPTHREADS
        Tpattern = T->pattern[identity];
#else
        Tpattern = T->pattern;
#endif
        p = Tpattern+FUNHEAD+1;
        for ( i = 0; i < ABS(T->numind); i++, t += 2 ) {
            if ( *t != -SNUMBER ) break;
        }
        if ( i < ABS(T->numind) ) { t = r; continue; }
        isp = FindTableTree(T,t1+FUNHEAD+1,2);
        if ( isp < 0 ) { t = r; continue; }
        rhsnumber = T->tablepointers[isp+ABS(T->numind)];
#if ( TABLEEXTENSION == 2 )
        tbufnum = T->bufnum;
#else
        tbufnum = T->tablepointers[isp+ABS(T->numind)+1];
#endif
        t = t1+FUNHEAD+2;
        ii = ABS(T->numind);
        while ( --ii >= 0 ) {
            *p = *t; t += 2; p += 2;
```

```
        }
/*
        If there are more arguments we have to do some
        pattern matching. This should be easy. We adapted the
        pattern, so that the array indices match already.
*/
#ifdef WITHPTHREADS
        AN.FullProto = T->prototype[identity];
#else
        AN.FullProto = T->prototype;
#endif
        AN.WildValue = AN.FullProto + SUBEXPSIZE;
        AN.WildStop = AN.FullProto+AN.FullProto[1];
        ClearWild(BHEAD0);
        AN.RepFunNum = 0;
        AN.RepFunList = AN.EndNest;
        AT.WorkPointer = (WORD *)(((UBYTE *)(AN.EndNest)) + AM.MaxTer/2);
/*
        The RepFunList is after the term but not very relevant.
        We need because MatchFunction uses it
*/
        if ( AT.WorkPointer + t1[1] >= AT.WorkTop ) { MesWork(); }
        wilds = 0;
        w = AT.WorkPointer;
        *w++ = -t1[FUNHEAD];
        *w++ = t1[1] - 1;
        for ( i = 2; i < FUNHEAD; i++ ) *w++ = t1[i];
        t = t1 + FUNHEAD+1;
        while ( t < r ) *w++ = *t++;
        t = AT.WorkPointer;
        AT.WorkPointer = w;
        if ( MatchFunction(BHEAD Tpattern,t,&wilds) > 0 ) {
/*
            Here we caught one. Now we should worry about:
            1: inserting the subexpression pointer with its wildcards
            2: NestPoin because we may not be at the lowest level
            The function starts at t1.
*/
#ifdef WITHPTHREADS
            m1 = T->prototype[identity];
#else
            m1 = T->prototype;
#endif
            m1[2] = rhsnumber;
            m1[4] = tbufnum;
            t = t1;
            j = t[1];
            i = m1[1];
            if ( j > i ) {
                j = i - j;
                NCOPY(t,m1,i);
                m1 = AN.EndNest;
                while ( r < m1 ) *t++ = *r++;
                AN.EndNest = t;
                *term += j;
                NN = AT.NestPoin;
                while ( NN > AT.Nest ) {
                    NN--;
                    NN->termsize[0] += j;
                    NN->funsize[1] += j;
                    NN->argsize[0] += j;
                    NN->funsize[2] |= DIRTYFLAG;
                    NN->argsize[1] |= DIRTYFLAG;
                }
                m += j;
            }
            else if ( j < i ) {
                j = i-j;
                t = AN.EndNest;
                while ( t >= r ) { t[j] = *t; t--; }
                t = t1;
                NCOPY(t,m1,i);
                AN.EndNest += j;
                *term += j;
                NN = AT.NestPoin;
                while ( NN > AT.Nest ) {
                    NN--;
                    NN->termsize[0] += j;
                    NN->funsize[1] += j;
                    NN->argsize[0] += j;
                    NN->funsize[2] |= DIRTYFLAG;
                    NN->argsize[1] |= DIRTYFLAG;
                }
                m += j;
            }
            else {
                NCOPY(t,m1,j);
```

```
            }
            r = t1 + t1[1];
            maxtogo--;
            if ( maxtogo <= 0 ) return(maxtogo);
        }
        t = r;
    }
    return(maxtogo);
}

/*
    #] ApplyExec :
    #[ ApplyReset :
*/

WORD ApplyReset(WORD level)
{
    WORD *funs, numfuns;
    TABLES T;
    int i, j;
    CBUF *C = cbuf+AM.rbufnum;

    numfuns = C->lhs[level][1] - 2;
    funs = C->lhs[level] + 2;
    if ( numfuns > 0 ) {
        for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
            if ( ( T = functions[i].tabl ) != 0 ) {
                for ( j = 0; j < numfuns; j++ ) {
                    if ( i == (funs[j]-FUNCTION) && T->spare ) {
                        FlipTable(&(functions[i]),1);
                        break;
                    }
                }
            }
        }
    }
    else {
        for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
            if ( ( T = functions[i].tabl ) != 0 ) {
                if ( T->spare ) FlipTable(&(functions[i]),1);
            }
        }
    }
    return(0);
}

/*
    #] ApplyReset :
    #[ TableReset :
*/

WORD TableReset(void)
{
    TABLES T;
    int i;

    for ( i = MAXBUILTINFUNCTION-FUNCTION; i < AC.FunctionList.num; i++ ) {
        if ( ( T = functions[i].tabl ) != 0 && T->spare && T->mode == 0 ) {
            functions[i].tabl = T->spare;
        }
    }
    return(0);
}

/*
    #] TableReset :
    #[ LoadTableElement :
?????
int LoadTableElement(DBASE *d, TABLE *T, WORD num)
{
}

    #] LoadTableElement :
    #[ ReleaseTB :

    Releases all TableBases
*/

int ReleaseTB(void)
{
    DBASE *d;
    int i;
    for ( i = NumTableBases - 1; i >= 0; i-- ) {
        d = tablebases+i;
        fclose(d->handle);
        FreeTableBase(d);
    }
```

```
    return(0);
}

/*
    #] ReleaseTB :
*/
```

4.142 threads.c File Reference

4.142.1 Detailed Description

Routines for the interface of FORM with the pthreads library

This is the main part of the parallelization of TFORM. It is important to also look in the files structs.h and variable.h because the treatment of the A and B structs is essential (these structs are used by means of the macros AM, AP, AC, AS, AR, AT, AN, AO and AX). Also the definitions and use of the macros BHEAD and PHEAD should be looked up.

The sources are set up in such a way that if WITHPTHREADS isn't defined there is no trace of pthread parallelization. The reason is that TFORM is far more memory hungry than sequential FORM.

Special attention should also go to the locks. The proper use of the locks is essential and determines whether TFORM can work at all. We use the LOCK/UNLOCK macros which are empty in the case of sequential FORM These locks are at many places in the source files when workers can interfere with each others data or with the data of the master.

Definition in file threads.c.

4.143 threads.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#ifdef WITHPTHREADS

#define WHOLEBRACKETS
/*
    #[ Variables :

    The sortbot additions are from 17-may-2007 and after. They constitute
```

```
    an attempt to make the final merge sorting faster for the master.
    This way the master has only one compare per term.
    It does add some complexity, but the final merge routine (MasterMerge)
    is much simpler for the sortbots. On the other hand the original merging is
    for a large part a copy of the MergePatches routine in sort.c and hence
    even though complex the bad part has been thoroughly debugged.
*/

#include "form3.h"

#ifdef WITH_ALARM
// This is only required if we are blocking SIG_ALRM in the worker threads.
#include <signal.h>
#endif

#ifdef WITHFLOAT
#include <gmp.h>

int PackFloat(WORD *,mpf_t);
int UnpackFloat(mpf_t, WORD *);
void RatToFloat(mpf_t result, UWORD *formrat, int ratsize);
#endif

static int numberofthreads;
static int numberofworkers;
static int identityofthreads = 0;
static int *listofavailables;
static int topofavailables = 0;
static pthread_key_t identitykey;
static INILOCK(numberofthreadslock)
static INILOCK(availabilitylock)
static pthread_t *threadpointers = 0;
static pthread_mutex_t *wakeuplocks;
static pthread_mutex_t *wakeupmasterthreadlocks;
static pthread_cond_t *wakeupconditions;
static pthread_condattr_t *wakeupconditionattributes;
static int *wakeup;
static int *wakeupmasterthread;
static INILOCK(wakeupmasterlock)
static pthread_cond_t wakeupmasterconditions = PTHREAD_COND_INITIALIZER;
static pthread_cond_t *wakeupmasterthreadconditions;
static int wakeupmaster = 0;
static int identityretval;
/* static INILOCK(clearclocklock) */
static LONG *timerinfo;
static LONG *sumtimerinfo;
static int numberclaimed;

static THREADBUCKET **threadbuckets, **freebuckets;
static int numthreadbuckets;
static int numberoffullbuckets;

/* static int numberbusy = 0; */

INILOCK(dummylock)
INIRWLOCK(dummyrwlock)
static pthread_cond_t dummywakeupcondition = PTHREAD_COND_INITIALIZER;

#ifdef WITHSORTBOTS
static POSITION SortBotPosition;
static int numberofsortbots;
static INILOCK(wakeupsortbotlock)
static pthread_cond_t wakeupsortbotconditions = PTHREAD_COND_INITIALIZER;
static int topsortbotavailables = 0;
static LONG numberofterms;
#endif

/*
    #] Variables :
    #[ Identity :
        #[ StartIdentity :
*/
void StartIdentity(void)
{
    pthread_key_create(&identitykey,FinishIdentity);
}

/*
        #] StartIdentity :
        #[ FinishIdentity :
*/
void FinishIdentity(void *keyp)
{
    DUMMYUSE(keyp);
/*  free(keyp); */
}

```

```
/*
        #] FinishIdentity :
        #[ SetIdentity :
*/
int SetIdentity(int *identityretval)
{
/*
#ifdef _MSC_VER
    printf("addr %d\n",&numberofthreadslock);
    printf("size %d\n",sizeof(numberofthreadslock));
#endif
*/
    LOCK(numberofthreadslock);
    *identityretval = identityofthreads++;
    UNLOCK(numberofthreadslock);
    pthread_setspecific(identitykey,(void *)identityretval);
    return(*identityretval);
}

/*
        #] SetIdentity :
        #[ WhoAmI :
*/

int WhoAmI(void)
{
    int *identity;
/*
    First a fast exit for when there is at most one thread
*/
    if ( identityofthreads <= 1 ) return(0);
/*
    Now the reading of the key.
    According to the book the statement should read:

    pthread_getspecific(identitykey,(void **)(&identity));

    but according to the information in pthread.h it is:
*/
    identity = (int *)pthread_getspecific(identitykey);
    return(*identity);
}

/*
        #] WhoAmI :
        #[ BeginIdentities :
*/
void BeginIdentities(void)
{
    StartIdentity();
    SetIdentity(&identityretval);
}

/*
        #] BeginIdentities :
    #] Identity :
    #[ StartHandleLock :
*/
void StartHandleLock(void)
{
    AM.handlelock = dummyrwlock;
}

/*
    #] StartHandleLock :
    #[ StartAllThreads :
*/
int StartAllThreads(int number)
{
    int identity, j, dummy, mul;
    ALLPRIVATES *B;
    pthread_t thethread;
    identity = WhoAmI();

#ifdef WITHSORTBOTS
    timerinfo = (LONG *)Malloc1(sizeof(LONG)*number*2,"timerinfo");
    sumtimerinfo = (LONG *)Malloc1(sizeof(LONG)*number*2,"sumtimerinfo");
    for ( j = 0; j < number*2; j++ ) { timerinfo[j] = 0; sumtimerinfo[j] = 0; }
    mul = 2;
#else
    timerinfo = (LONG *)Malloc1(sizeof(LONG)*number,"timerinfo");
    sumtimerinfo = (LONG *)Malloc1(sizeof(LONG)*number,"sumtimerinfo");
    for ( j = 0; j < number; j++ ) { timerinfo[j] = 0; sumtimerinfo[j] = 0; }
    mul = 1;
#endif

    listofavailables = (int *)Malloc1(sizeof(int)*(number+1),"listofavailables");
```

```
    threadpointers = (pthread_t *)Malloc1(sizeof(pthread_t)*number*mul,"threadpointers");
    AB = (ALLPRIVATES **)Malloc1(sizeof(ALLPRIVATES *)*number*mul,"Private structs");

    wakeup = (int *)Malloc1(sizeof(int)*number*mul,"wakeup");
    wakeuplocks = (pthread_mutex_t *)Malloc1(sizeof(pthread_mutex_t)*number*mul,"wakeuplocks");
    wakeupconditions = (pthread_cond_t *)Malloc1(sizeof(pthread_cond_t)*number*mul,"wakeupconditions");
    wakeupconditionattributes = (pthread_condattr_t *)
            Malloc1(sizeof(pthread_condattr_t)*number*mul,"wakeupconditionattributes");

    wakeupmasterthread = (int *)Malloc1(sizeof(int)*number*mul,"wakeupmasterthread");
    wakeupmasterthreadlocks = (pthread_mutex_t *)Malloc1(sizeof(pthread_mutex_t)*number*mul,"wakeupmasterthreadlocks");
    wakeupmasterthreadconditions = (pthread_cond_t *)Malloc1(sizeof(pthread_cond_t)*number*mul,"wakeupmasterthread");

    numberofthreads = number;
    numberofworkers = number - 1;
    threadpointers[identity] = pthread_self();
    topofavailables = 0;

#ifdef WITH_ALARM
    /* During thread creation, we block SIGALRM on the main thread. The created
       threads will inherit this. This is required for #timeout to work properly
       in TFORM: only the main thread should recieve SIGALRM. */
    sigset_t sig_set;
    sigemptyset(&sig_set);
    sigaddset(&sig_set, SIGALRM);
    pthread_sigmask(SIG_BLOCK, &sig_set, NULL);
#endif

    for ( j = 1; j < number; j++ ) {
        if ( pthread_create(&thethread,NULL,RunThread,(void *)(&dummy)) )
            goto failure;
    }
/*
    Now we initialize the master at the same time that the workers are doing so.
*/
    B = InitializeOneThread(identity);
    AR.infile = &(AR.Fscr[0]);
    AR.outfile = &(AR.Fscr[1]);
    AR.hidefile = &(AR.Fscr[2]);
    AM.sbuflock = dummylock;
    AS.inputslock = dummylock;
    AS.outputslock = dummylock;
    AS.MaxExprSizeLock = dummylock;
    AP.PreVarLock = dummylock;
    AC.halfmodlock = dummylock;
    MakeThreadBuckets(number,0);
/*
    Now we wait for the workers to finish their startup.
    We don't want to initialize the sortbots yet and run the risk that
    some of them may end up with a lower number than one of the workers.
*/
    MasterWaitAll();
#ifdef WITHSORTBOTS
    if ( numberofworkers > 2 ) {
        numberofsortbots = numberofworkers-2;
        for ( j = numberofworkers+1; j < 2*numberofworkers-1; j++ ) {
            if ( pthread_create(&thethread,NULL,RunSortBot,(void *)(&dummy)) )
                goto failure;
        }
    }
    else {
        numberofsortbots = 0;
    }
    MasterWaitAllSortBots();
    DefineSortBotTree();
#endif
    IniSortBlocks(number-1);
    AS.MasterSort = 0;
    AM.storefilelock = dummylock;

#ifdef WITH_ALARM
    /* Now we allow the main thread to recieve SIGALRM again. */
    pthread_sigmask(SIG_UNBLOCK, &sig_set, NULL);
#endif

/*
MesPrint("AB = %x %x %x  %d",AB[0],AB[1],AB[2], identityofthreads);
*/
    return(0);
failure:
    MesPrint("Cannot start %d threads",number);
    Terminate(-1);
    return(-1);
}
```

```
/*
    #] StartAllThreads :
    #[ InitializeOneThread :
*/
UBYTE *scratchname[] = { (UBYTE *)"scratchsize",
                         (UBYTE *)"scratchsize",
                         (UBYTE *)"hidesize" };
ALLPRIVATES *InitializeOneThread(int identity)
{
    WORD *t, *ScratchBuf;
    int i, j, bsize, *bp;
    LONG ScratchSize[3], IOsize;
    ALLPRIVATES *B;
    UBYTE *s;

    wakeup[identity] = 0;
    wakeuplocks[identity] = dummylock;
    pthread_condattr_init(&(wakeupconditionattributes[identity]));
    pthread_condattr_setpshared(&(wakeupconditionattributes[identity]),PTHREAD_PROCESS_PRIVATE);
    wakeupconditions[identity] = dummywakeupcondition;
    pthread_cond_init(&(wakeupconditions[identity]),&(wakeupconditionattributes[identity]));
    wakeupmasterthread[identity] = 0;
    wakeupmasterthreadlocks[identity] = dummylock;
    wakeupmasterthreadconditions[identity] = dummywakeupcondition;

    bsize = sizeof(ALLPRIVATES);
    bsize = (bsize+sizeof(int)-1)/sizeof(int);
    B = (ALLPRIVATES *)Malloc1(sizeof(int)*bsize,"B struct");
    for ( bp = (int *)B, j = 0; j < bsize; j++ ) *bp++ = 0;

    AB[identity] = B;
/*
            12-jun-2007 JV:
    For the timing one has to know a few things:
    The POSIX standard is that there is only a single process ID and that
    getrusage returns the time of all the threads together.
    Under Linux there are two methods though: The older LinuxThreads and NPTL.
    LinuxThreads gives each thread its own process id. This makes that we
    can time the threads with getrusage, and hence this was done. Under NPTL
    this has been 'corrected' and suddenly getruage doesn't work anymore the
    way it used to. Now we need
        clock_gettime(CLOCK_THREAD_CPUTIME_ID,&timing)
    which is declared in <time.h> and we need -lrt extra in the link statement.
    (this is at least the case on blade02 at DESY-Zeuthen).
    See also the code in tools.c at the routine Timer.
    We may still have to include more stuff there.
*/
    if ( identity > 0 ) TimeCPU(0);

#ifdef WITHSORTBOTS

    if ( identity > numberofworkers ) {
/*
        Some workspace is needed when we have a PolyFun and we have to add
        two terms and the new result is going to be longer than the old result.
*/
        LONG length = AM.WorkSize*sizeof(WORD)/8+AM.MaxTer*2;
        AT.WorkSpace = (WORD *)Malloc1(length,"WorkSpace");
        AT.WorkTop = AT.WorkSpace + length/sizeof(WORD);
        AT.WorkPointer = AT.WorkSpace;
        AT.identity = identity;
/*
        The SB struct gets treated in IniSortBlocks.
        The SortBotIn variables will be defined DefineSortBotTree.
*/
        if ( AN.SoScratC == 0 ) {
            AN.SoScratC = (UWORD *)Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD),"Scratch in SortBot");
        }
        AT.SS = (SORTING *)Malloc1(sizeof(SORTING),"dummy sort buffer");
        AT.SS->PolyFlag = 0;

        AT.comsym[0] = 8;
        AT.comsym[1] = SYMBOL;
        AT.comsym[2] = 4;
        AT.comsym[3] = 0;
        AT.comsym[4] = 1;
        AT.comsym[5] = 1;
        AT.comsym[6] = 1;
        AT.comsym[7] = 3;
        AT.comnum[0] = 4;
        AT.comnum[1] = 1;
        AT.comnum[2] = 1;
        AT.comnum[3] = 3;
        AT.comfun[0] = FUNHEAD+4;
        AT.comfun[1] = FUNCTION;
        AT.comfun[2] = FUNHEAD;
```

```
        AT.comfun[3] = 0;
#if FUNHEAD > 3
        for ( i = 4; i <= FUNHEAD; i++ )
            AT.comfun[i] = 0;
#endif
        AT.comfun[FUNHEAD+1] = 1;
        AT.comfun[FUNHEAD+2] = 1;
        AT.comfun[FUNHEAD+3] = 3;
        AT.comind[0] = 7;
        AT.comind[1] = INDEX;
        AT.comind[2] = 3;
        AT.comind[3] = 0;
        AT.comind[4] = 1;
        AT.comind[5] = 1;
        AT.comind[6] = 3;

        AT.inprimelist = -1;
        AT.sizeprimelist = 0;
        AT.primelist = 0;
        AT.bracketinfo = 0;

        AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);

        AR.sLevel = 0;
        AR.wranfia = 0;
        AR.wranfcall = 0;
        AR.wranfnpair1 = NPAIR1;
        AR.wranfnpair2 = NPAIR2;
        AN.NumFunSorts = 5;
        AN.MaxFunSorts = 5;
        AN.SplitScratch = 0;
        AN.SplitScratchSize = AN.InScratch = 0;
        AN.SplitScratch1 = 0;
        AN.SplitScratchSize1 = AN.InScratch1 = 0;

        AN.FunSorts = (SORTING **)Malloc1((AN.NumFunSorts+1)*sizeof(SORTING *),"FunSort pointers");
        for ( i = 0; i <= AN.NumFunSorts; i++ ) AN.FunSorts[i] = 0;
        AN.FunSorts[0] = AT.S0 = AT.SS;
        AN.idfunctionflag = 0;
        AN.tryterm = 0;

        return(B);
    }
    if ( identity == 0 && AN.SoScratC == 0 ) {
        AN.SoScratC = (UWORD *)Malloc1(2*(AM.MaxTal+2)*sizeof(UWORD),"Scratch in SortBot");
    }
#endif
    AR.CurDum = AM.IndDum;
    for ( j = 0; j < 3; j++ ) {
        if ( identity == 0 ) {
            if ( j == 2 ) {
                ScratchSize[j] = AM.HideSize;
            }
            else {
                ScratchSize[j] = AM.ScratSize;
            }
            if ( ScratchSize[j] < 10*AM.MaxTer ) ScratchSize[j] = 10 * AM.MaxTer;
        }
        else {
/*
            ScratchSize[j] = AM.ScratSize / (numberofthreads-1);
            ScratchSize[j] = ScratchSize[j] / 20;
            if ( ScratchSize[j] < 10*AM.MaxTer ) ScratchSize[j] = 10 * AM.MaxTer;
*/
            if ( j == 1 ) ScratchSize[j] = AM.ThreadScratOutSize;
            else          ScratchSize[j] = AM.ThreadScratSize;
            if ( ScratchSize[j] < 4*AM.MaxTer ) ScratchSize[j] = 4 * AM.MaxTer;
            AR.Fscr[j].name = 0;
        }
        ScratchSize[j] = ( ScratchSize[j] + 255 ) / 256;
        ScratchSize[j] = ScratchSize[j] * 256;
        ScratchBuf = (WORD *)Malloc1(ScratchSize[j]*sizeof(WORD),(char *)(scratchname[j]));
        AR.Fscr[j].POsize = ScratchSize[j] * sizeof(WORD);
        AR.Fscr[j].POfull = AR.Fscr[j].POfill = AR.Fscr[j].PObuffer = ScratchBuf;
        AR.Fscr[j].POstop = AR.Fscr[j].PObuffer + ScratchSize[j];
        PUTZERO(AR.Fscr[j].POposition);
        AR.Fscr[j].pthreadslock = dummylock;
        AR.Fscr[j].wPOsize = AR.Fscr[j].POsize;
        AR.Fscr[j].wPObuffer = AR.Fscr[j].PObuffer;
        AR.Fscr[j].wPOfill = AR.Fscr[j].POfill;
        AR.Fscr[j].wPOfull = AR.Fscr[j].POfull;
        AR.Fscr[j].wPOstop = AR.Fscr[j].POstop;
    }
    AR.InInBuf = 0;
    AR.InHiBuf = 0;
    AR.Fscr[0].handle = -1;
    AR.Fscr[1].handle = -1;
```

```
    AR.Fscr[2].handle = -1;
    AR.FoStage4[0].handle = -1;
    AR.FoStage4[1].handle = -1;
    IOsize = AM.S0->file.POsize;
#ifdef WITHZLIB
    AR.FoStage4[0].ziosize = IOsize;
    AR.FoStage4[1].ziosize = IOsize;
    AR.FoStage4[0].ziobuffer = 0;
    AR.FoStage4[1].ziobuffer = 0;
#endif
    AR.FoStage4[0].POsize  = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);
    AR.FoStage4[1].POsize  = ((IOsize+sizeof(WORD)-1)/sizeof(WORD))*sizeof(WORD);

    AR.hidefile = &(AR.Fscr[2]);
    AR.StoreData.Handle = -1;
    AR.SortType = AC.SortType;

    AN.IndDum = AM.IndDum;

    if ( identity > 0 ) {
        s = (UBYTE *)(FG.fname); i = 0;
        while ( *s ) { s++; i++; }
        s = (UBYTE *)Malloc1(sizeof(char)*(i+12),"name for Fscr[0] file");
        snprintf((char *)s,i+12,"%s.%d",FG.fname,identity);
        s[i-3] = 's'; s[i-2] = 'c'; s[i-1] = '0';
        AR.Fscr[0].name = (char *)s;
        s = (UBYTE *)(FG.fname); i = 0;
        while ( *s ) { s++; i++; }
        s = (UBYTE *)Malloc1(sizeof(char)*(i+12),"name for Fscr[1] file");
        snprintf((char *)s,i+12,"%s.%d",FG.fname,identity);
        s[i-3] = 's'; s[i-2] = 'c'; s[i-1] = '1';
        AR.Fscr[1].name = (char *)s;
    }

    AR.CompressBuffer = (WORD *)Malloc1((AM.CompressSize+10)*sizeof(WORD),"compresssize");
    AR.ComprTop = AR.CompressBuffer + AM.CompressSize;
    AR.CompareRoutine = (COMPAREDUMMY)(&Compare1);
/*
    Here we make all allocations for the struct AT
    (which is AB[identity].T or B->T with B = AB+identity).
*/
    AT.WorkSpace = (WORD *)Malloc1(AM.WorkSize*sizeof(WORD),"WorkSpace");
    AT.WorkTop = AT.WorkSpace + AM.WorkSize;
    AT.WorkPointer = AT.WorkSpace;

    AT.Nest = (NESTING)Malloc1((LONG)sizeof(struct NeStInG)*AM.maxFlevels,"functionlevels");
    AT.NestStop = AT.Nest + AM.maxFlevels;
    AT.NestPoin = AT.Nest;

    AT.WildMask = (WORD *)Malloc1((LONG)AM.MaxWildcards*sizeof(WORD),"maxwildcards");

    LOCK(availabilitylock);
    AT.ebufnum = inicbufs();        /* Buffer for extras during execution */
    AT.fbufnum = inicbufs();        /* Buffer for caching in factorization */
    AT.allbufnum = inicbufs();      /* Buffer for id,all */
    AT.aebufnum = inicbufs();       /* Buffer for id,all */
    UNLOCK(availabilitylock);

    AT.RepCount = (int *)Malloc1((LONG)((AM.RepMax+3)*sizeof(int)),"repeat buffers");
    AN.RepPoint = AT.RepCount;
    AN.polysortflag = 0;
    AN.subsubveto = 0;
    AN.tryterm = 0;
    AT.RepTop = AT.RepCount + AM.RepMax;

    AT.WildArgTaken = (WORD *)Malloc1((LONG)AC.WildcardBufferSize*sizeof(WORD)/2
                ,"argument list names");
    AT.WildcardBufferSize = AC.WildcardBufferSize;
    AT.previousEfactor = 0;

    AT.identity = identity;
    AT.LoadBalancing = 0;
/*
    Still to do: the SS stuff.
                 the Fscr[3]
                 the FoStage4[2]
*/
    if ( AT.WorkSpace == 0 ||
         AT.Nest == 0 ||
         AT.WildMask == 0 ||
         AT.RepCount == 0 ||
         AT.WildArgTaken == 0 ) goto OnError;
/*
    And initializations
*/
    AT.comsym[0] = 8;
    AT.comsym[1] = SYMBOL;
```

```
    AT.comsym[2] = 4;
    AT.comsym[3] = 0;
    AT.comsym[4] = 1;
    AT.comsym[5] = 1;
    AT.comsym[6] = 1;
    AT.comsym[7] = 3;
    AT.comnum[0] = 4;
    AT.comnum[1] = 1;
    AT.comnum[2] = 1;
    AT.comnum[3] = 3;
    AT.comfun[0] = FUNHEAD+4;
    AT.comfun[1] = FUNCTION;
    AT.comfun[2] = FUNHEAD;
    AT.comfun[3] = 0;
#if FUNHEAD > 3
    for ( i = 4; i <= FUNHEAD; i++ )
        AT.comfun[i] = 0;
#endif
    AT.comfun[FUNHEAD+1] = 1;
    AT.comfun[FUNHEAD+2] = 1;
    AT.comfun[FUNHEAD+3] = 3;
    AT.comind[0] = 7;
    AT.comind[1] = INDEX;
    AT.comind[2] = 3;
    AT.comind[3] = 0;
    AT.comind[4] = 1;
    AT.comind[5] = 1;
    AT.comind[6] = 3;
    AT.locwildvalue[0] = SUBEXPRESSION;
    AT.locwildvalue[1] = SUBEXPSIZE;
    for ( i = 2; i < SUBEXPSIZE; i++ ) AT.locwildvalue[i] = 0;
    AT.mulpat[0] = TYPEMULT;
    AT.mulpat[1] = SUBEXPSIZE+3;
    AT.mulpat[2] = 0;
    AT.mulpat[3] = SUBEXPRESSION;
    AT.mulpat[4] = SUBEXPSIZE;
    AT.mulpat[5] = 0;
    AT.mulpat[6] = 1;
    for ( i = 7; i < SUBEXPSIZE+5; i++ ) AT.mulpat[i] = 0;
    AT.proexp[0] = SUBEXPSIZE+4;
    AT.proexp[1] = EXPRESSION;
    AT.proexp[2] = SUBEXPSIZE;
    AT.proexp[3] = -1;
    AT.proexp[4] = 1;
    for ( i = 5; i < SUBEXPSIZE+1; i++ ) AT.proexp[i] = 0;
    AT.proexp[SUBEXPSIZE+1] = 1;
    AT.proexp[SUBEXPSIZE+2] = 1;
    AT.proexp[SUBEXPSIZE+3] = 3;
    AT.proexp[SUBEXPSIZE+4] = 0;
    AT.dummysubexp[0] = SUBEXPRESSION;
    AT.dummysubexp[1] = SUBEXPSIZE+4;
    for ( i = 2; i < SUBEXPSIZE; i++ ) AT.dummysubexp[i] = 0;
    AT.dummysubexp[SUBEXPSIZE] = WILDDUMMY;
    AT.dummysubexp[SUBEXPSIZE+1] = 4;
    AT.dummysubexp[SUBEXPSIZE+2] = 0;
    AT.dummysubexp[SUBEXPSIZE+3] = 0;

    AT.MinVecArg[0] = 7+ARGHEAD;
    AT.MinVecArg[ARGHEAD] = 7;
    AT.MinVecArg[1+ARGHEAD] = INDEX;
    AT.MinVecArg[2+ARGHEAD] = 3;
    AT.MinVecArg[3+ARGHEAD] = 0;
    AT.MinVecArg[4+ARGHEAD] = 1;
    AT.MinVecArg[5+ARGHEAD] = 1;
    AT.MinVecArg[6+ARGHEAD] = -3;
    t = AT.FunArg;
    *t++ = 4+ARGHEAD+FUNHEAD;
    for ( i = 1; i < ARGHEAD; i++ ) *t++ = 0;
    *t++ = 4+FUNHEAD;
    *t++ = 0;
    *t++ = FUNHEAD;
    for ( i = 2; i < FUNHEAD; i++ ) *t++ = 0;
    *t++ = 1; *t++ = 1; *t++ = 3;

    AT.inprimelist = -1;
    AT.sizeprimelist = 0;
    AT.primelist = 0;
    AT.nfac = AT.nBer = 0;
    AT.factorials = 0;
    AT.bernoullis = 0;
    AR.wranfia = 0;
    AR.wranfcall = 0;
    AR.wranfnpair1 = NPAIR1;
    AR.wranfnpair2 = NPAIR2;
    AR.wranfseed = 0;
    AN.SplitScratch = 0;
    AN.SplitScratchSize = AN.InScratch = 0;
```

```
    AN.SplitScratch1 = 0;
    AN.SplitScratchSize1 = AN.InScratch1 = 0;
/*
    Now the sort buffers. They depend on which thread. The master
    inherits the sortbuffer from AM.S0
*/
    if ( identity == 0 ) {
        AT.S0 = AM.S0;
    }
    else {
/*
        For the moment we don't have special settings.
        They may become costly in virtual memory.
*/
        AT.S0 = AllocSort(AM.S0->LargeSize*sizeof(WORD)/numberofworkers
                         ,AM.S0->SmallSize*sizeof(WORD)/numberofworkers
                         ,AM.S0->SmallEsize*sizeof(WORD)/numberofworkers
                         ,AM.S0->TermsInSmall
                         ,AM.S0->MaxPatches
/*                       ,AM.S0->MaxPatches/numberofworkers  */
                         ,AM.S0->MaxFpatches/numberofworkers
                         ,AM.S0->file.POsize
                         ,0);
    }
    AR.CompressPointer = AR.CompressBuffer;
/*
    Install the store caches (15-aug-2006 JV)
*/
    AT.StoreCache = AT.StoreCacheAlloc = 0;
    if ( AM.NumStoreCaches > 0 ) {
        STORECACHE sa, sb;
        LONG size;
        size = sizeof(struct StOrEcAcHe)+AM.SizeStoreCache;
        size = ((size-1)/sizeof(size_t)+1)*sizeof(size_t);
        AT.StoreCacheAlloc = (STORECACHE)Malloc1(size*AM.NumStoreCaches,"StoreCaches");
        sa = AT.StoreCache = AT.StoreCacheAlloc;
        for ( i = 0; i < AM.NumStoreCaches; i++ ) {
            sb = (STORECACHE)(void *)((UBYTE *)sa+size);
            if ( i == AM.NumStoreCaches-1 ) {
                sa->next = 0;
            }
            else {
                sa->next = sb;
            }
            SETBASEPOSITION(sa->position,-1);
            SETBASEPOSITION(sa->toppos,-1);
            sa = sb;
        }
    }

    ReserveTempFiles(2);
    return(B);
OnError:;
    MLOCK(ErrorMessageLock);
    MesPrint("Error initializing thread %d",identity);
    MUNLOCK(ErrorMessageLock);
    Terminate(-1);
    return(B);
}

/*
    #] InitializeOneThread :
    #[ FinalizeOneThread :
*/
void FinalizeOneThread(int identity)
{
    timerinfo[identity] = TimeCPU(1);
}

/*
    #] FinalizeOneThread :
    #[ ClearAllThreads :
*/
void ClearAllThreads(void)
{
    int i;
    MasterWaitAll();
    for ( i = 1; i <= numberofworkers; i++ ) {
        WakeupThread(i,CLEARCLOCK);
    }
#ifdef WITHSORTBOTS
    for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
        WakeupThread(i,CLEARCLOCK);
    }
#endif
}
```

```
/*
    #] ClearAllThreads :
    #[ TerminateAllThreads :
*/
void TerminateAllThreads(void)
{
    int i;
    for ( i = 1; i <= numberofworkers; i++ ) {
        GetThread(i);
        WakeupThread(i,TERMINATETHREAD);
    }
#ifdef WITHSORTBOTS
    for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
        WakeupThread(i,TERMINATETHREAD);
    }
#endif
    for ( i = 1; i <= numberofworkers; i++ ) {
        pthread_join(threadpointers[i],NULL);
    }
#ifdef WITHSORTBOTS
    for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
        pthread_join(threadpointers[i],NULL);
    }
#endif
}

/*
    #] TerminateAllThreads :
    #[ MakeThreadBuckets :
*/
int MakeThreadBuckets(int number, int par)
{
    int i;
    LONG sizethreadbuckets;
    THREADBUCKET *thr;
/*
    First we need a decent estimate. Not all terms should be maximal.
    Note that AM.MaxTer is in bytes!!!
    Maybe we should try to limit the size here a bit more effectively.
    This is a great consumer of memory.
*/
    sizethreadbuckets = ( AC.ThreadBucketSize + 1 ) * AM.MaxTer + 2*sizeof(WORD);
    if ( AC.ThreadBucketSize >= 250 )      sizethreadbuckets /= 4;
    else if ( AC.ThreadBucketSize >= 90 )  sizethreadbuckets /= 3;
    else if ( AC.ThreadBucketSize >= 40 )  sizethreadbuckets /= 2;
    sizethreadbuckets /= sizeof(WORD);

    if ( par == 0 ) {
        numthreadbuckets = 2*(number-1);
        threadbuckets = (THREADBUCKET **)Malloc1(numthreadbuckets*sizeof(THREADBUCKET *),"threadbuckets");
        freebuckets = (THREADBUCKET **)Malloc1(numthreadbuckets*sizeof(THREADBUCKET *),"threadbuckets");
    }
    if ( par > 0 ) {
        if ( sizethreadbuckets <= threadbuckets[0]->threadbuffersize ) return(0);
        for ( i = 0; i < numthreadbuckets; i++ ) {
            thr = threadbuckets[i];
            M_free(thr->deferbuffer,"deferbuffer");
        }
    }
    else {
        for ( i = 0; i < numthreadbuckets; i++ ) {
            threadbuckets[i] = (THREADBUCKET *)Malloc1(sizeof(THREADBUCKET),"threadbuckets");
            threadbuckets[i]->lock = dummylock;
        }
    }
    for ( i = 0; i < numthreadbuckets; i++ ) {
        thr = threadbuckets[i];
        thr->threadbuffersize = sizethreadbuckets;
        thr->free = BUCKETFREE;
        thr->deferbuffer = (POSITION *)Malloc1(2*sizethreadbuckets*sizeof(WORD)
                    +(AC.ThreadBucketSize+1)*sizeof(POSITION),"deferbuffer");
        thr->threadbuffer = (WORD *)(thr->deferbuffer+AC.ThreadBucketSize+1);
        thr->compressbuffer = (WORD *)(thr->threadbuffer+sizethreadbuckets);
        thr->busy = BUCKETPREPARINGTERM;
        thr->usenum = thr->totnum = 0;
        thr->type = BUCKETDOINGTERMS;
    }
    return(0);
}

/*
    #] MakeThreadBuckets :
    #[ GetTimerInfo :
*/

```

```
int GetTimerInfo(LONG** ti,LONG** sti)
{
    *ti = timerinfo;
    *sti = sumtimerinfo;
#ifdef WITHSORTBOTS
    return AM.totalnumberofthreads*2;
#else
    return AM.totalnumberofthreads;
#endif
}

/*
    #] GetTimerInfo :
    #[ WriteTimerInfo :
*/

void WriteTimerInfo(LONG* ti,LONG* sti)
{
    int i;
#ifdef WITHSORTBOTS
    int max = AM.totalnumberofthreads*2;
#else
    int max = AM.totalnumberofthreads;
#endif
    for ( i=0; i<max; ++i ) {
        timerinfo[i] = ti[i];
        sumtimerinfo[i] = sti[i];
    }
}

/*
    #] WriteTimerInfo :
    #[ GetWorkerTimes :
*/
LONG GetWorkerTimes(void)
{
    LONG retval = 0;
    int i;
    for ( i = 1; i <= numberofworkers; i++ ) retval += timerinfo[i] + sumtimerinfo[i];
#ifdef WITHSORTBOTS
    for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ )
        retval += timerinfo[i] + sumtimerinfo[i];
#endif
    return(retval);
}

/*
    #] GetWorkerTimes :
    #[ UpdateOneThread :
*/
int UpdateOneThread(int identity)
{
    ALLPRIVATES *B = AB[identity], *B0 = AB[0];
    AR.GetFile = AR0.GetFile;
    AR.KeptInHold = AR0.KeptInHold;
    AR.CurExpr = AR0.CurExpr;
    AR.SortType = AC.SortType;
    if ( AT.WildcardBufferSize < AC.WildcardBufferSize ) {
        M_free(AT.WildArgTaken,"argument list names");
        AT.WildcardBufferSize = AC.WildcardBufferSize;
        AT.WildArgTaken = (WORD *)Malloc1((LONG)AC.WildcardBufferSize*sizeof(WORD)/2
                ,"argument list names");
        if ( AT.WildArgTaken == 0 ) return(-1);
    }
    return(0);
}

/*
    #] UpdateOneThread :
    #[ LoadOneThread :
*/
int LoadOneThread(int from, int identity, THREADBUCKET *thr, int par)
{
    WORD *t1, *t2;
    ALLPRIVATES *B = AB[identity], *B0 = AB[from];

    AR.DefPosition = AR0.DefPosition;
    AR.NoCompress = AR0.NoCompress;
    AR.gzipCompress = AR0.gzipCompress;
    AR.BracketOn = AR0.BracketOn;
    AR.CurDum = AR0.CurDum;
    AR.DeferFlag = AR0.DeferFlag;
    AR.TePos = 0;
    AR.sLevel = AR0.sLevel;
    AR.Stage4Name = AR0.Stage4Name;
    AR.GetOneFile = AR0.GetOneFile;
    AR.PolyFun = AR0.PolyFun;
```

```
    AR.PolyFunInv = AR0.PolyFunInv;
    AR.PolyFunType = AR0.PolyFunType;
    AR.PolyFunExp = AR0.PolyFunExp;
    AR.PolyFunVar = AR0.PolyFunVar;
    AR.PolyFunPow = AR0.PolyFunPow;
    AR.Eside = AR0.Eside;
    AR.Cnumlhs = AR0.Cnumlhs;
/*
    AR.MaxBracket = AR0.MaxBracket;

    The compressbuffer contents are mainly relevant for keep brackets
    We should do this only if there is a keep brackets statement
    We may however still need the compressbuffer for expressions in the rhs.
*/
    if ( par >= 1 ) {
/*
        We may not need this %%%%% 7-apr-2006
*/
        t1 = AR.CompressBuffer; t2 = AR0.CompressBuffer;
        while ( t2 < AR0.CompressPointer ) *t1++ = *t2++;
        AR.CompressPointer = t1;

    }
    else {
        AR.CompressPointer = AR.CompressBuffer;
    }
    if ( AR.DeferFlag ) {
        if ( AR.infile->handle < 0 ) {
            AR.infile->POfill = AR0.infile->POfill;
        }
        else {
/*
            We have to set the value of POposition to something that will
            force a read in the first try.
*/
            AR.infile->POfull = AR.infile->POfill = AR.infile->PObuffer;
        }
    }
    if ( par == 0 ) {
        AN.threadbuck = thr;
        AN.ninterms = thr->firstterm;
    }
    else if ( par == 1 ) {
        WORD *tstop;
        t1 = thr->threadbuffer; tstop = t1 + *t1;
        t2 = AT.WorkPointer;
        while ( t1 < tstop ) *t2++ = *t1++;
        AN.ninterms = thr->firstterm;
    }
    AN.TeInFun = 0;
    AN.ncmod = AC.ncmod;
    AT.BrackBuf = AT0.BrackBuf;
    AT.bracketindexflag = AT0.bracketindexflag;
    AN.PolyFunTodo = 0;
/*
    The relevant variables and the term are in their place.
    There is nothing more to do.
*/
    return(0);
}

/*
    #] LoadOneThread :
    #[ BalanceRunThread :
*/
int BalanceRunThread(PHEAD int identity, WORD *term, WORD level)
{
    GETBIDENTITY
    ALLPRIVATES *BB;
    WORD *t, *tt;
    int i, *ti, *tti;

    LoadOneThread(AT.identity,identity,0,2);
/*
    Extra loading if needed. Quantities changed in Generator.
    Like the level that has to be passed.
*/
    BB = AB[identity];
    BB->R.level = level;
    BB->T.TMbuff = AT.TMbuff;
    ti = AT.RepCount; tti = BB->T.RepCount;
    i = AN.RepPoint - AT.RepCount;
    BB->N.RepPoint = BB->T.RepCount + i;
    for ( ; i >= 0; i-- ) tti[i] = ti[i];

    t = term; i = *term;
    tt = BB->T.WorkSpace;
```

```
    NCOPY(tt,t,i);
    BB->T.WorkPointer = tt;

    WakeupThread(identity,HIGHERLEVELGENERATION);

    return(0);
}

/*
    #] BalanceRunThread :
    #[ SetWorkerFiles :
*/
void SetWorkerFiles(void)
{
    int id;
    ALLPRIVATES *B, *B0 = AB[0];
    for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
        B = AB[id];
        AR.infile = &(AR.Fscr[0]);
        AR.outfile = &(AR.Fscr[1]);
        AR.hidefile = &(AR.Fscr[2]);
        AR.infile->handle = AR0.infile->handle;
        AR.hidefile->handle = AR0.hidefile->handle;
        if ( AR.infile->handle < 0 ) {
            AR.infile->PObuffer = AR0.infile->PObuffer;
            AR.infile->POstop = AR0.infile->POstop;
            AR.infile->POfill = AR0.infile->POfill;
            AR.infile->POfull = AR0.infile->POfull;
            AR.infile->POsize = AR0.infile->POsize;
            AR.InInBuf = AR0.InInBuf;
            AR.infile->POposition = AR0.infile->POposition;
            AR.infile->filesize = AR0.infile->filesize;
        }
        else {
            AR.infile->PObuffer = AR.infile->wPObuffer;
            AR.infile->POstop = AR.infile->wPOstop;
            AR.infile->POfill = AR.infile->wPOfill;
            AR.infile->POfull = AR.infile->wPOfull;
            AR.infile->POsize = AR.infile->wPOsize;
            AR.InInBuf = 0;
            PUTZERO(AR.infile->POposition);
        }
/*
        If there is some writing, it betters happens to ones own outfile.
        Currently this is to be done only for InParallel.
        Merging of the outputs is then done by the CopyExpression routine.
*/
        {
            AR.outfile->PObuffer = AR.outfile->wPObuffer;
            AR.outfile->POstop = AR.outfile->wPOstop;
            AR.outfile->POfill = AR.outfile->wPOfill;
            AR.outfile->POfull = AR.outfile->wPOfull;
            AR.outfile->POsize = AR.outfile->wPOsize;
            PUTZERO(AR.outfile->POposition);
        }
        if ( AR.hidefile->handle < 0 ) {
            AR.hidefile->PObuffer = AR0.hidefile->PObuffer;
            AR.hidefile->POstop = AR0.hidefile->POstop;
            AR.hidefile->POfill = AR0.hidefile->POfill;
            AR.hidefile->POfull = AR0.hidefile->POfull;
            AR.hidefile->POsize = AR0.hidefile->POsize;
            AR.InHiBuf = AR0.InHiBuf;
            AR.hidefile->POposition = AR0.hidefile->POposition;
            AR.hidefile->filesize = AR0.hidefile->filesize;
        }
        else {
            AR.hidefile->PObuffer = AR.hidefile->wPObuffer;
            AR.hidefile->POstop = AR.hidefile->wPOstop;
            AR.hidefile->POfill = AR.hidefile->wPOfill;
            AR.hidefile->POfull = AR.hidefile->wPOfull;
            AR.hidefile->POsize = AR.hidefile->wPOsize;
            AR.InHiBuf = 0;
            PUTZERO(AR.hidefile->POposition);
        }
    }
    if ( AR0.StoreData.dirtyflag ) {
        for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
            B = AB[id];
            AR.StoreData = AR0.StoreData;
        }
    }
}

/*
    #] SetWorkerFiles :
    #[ RunThread :
*/
```

```
void *RunThread(void *dummy)
{
    WORD *term, *ttin, *tt, *ttco, *oldwork;
    int identity, wakeupsignal, identityretv, i, tobereleased, errorcode;
    ALLPRIVATES *B;
    THREADBUCKET *thr;
    POSITION *ppdef;
    EXPRESSIONS e;
    DUMMYUSE(dummy);
    identity = SetIdentity(&identityretv);
    threadpointers[identity] = pthread_self();
    B = InitializeOneThread(identity);
    while ( ( wakeupsignal = ThreadWait(identity) ) > 0 ) {
        switch ( wakeupsignal ) {
/*
            #[ STARTNEWEXPRESSION :
*/
            case STARTNEWEXPRESSION:
/*
                Set up the sort routines etc.
                Start with getting some buffers synchronized with the compiler
*/
                if ( UpdateOneThread(identity) ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Update error in starting expression in thread %d in module %d",identity,AC.CModule);
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                AR.DeferFlag = AC.ComDefer;
                AR.sLevel = AS.sLevel;
                AR.MaxDum = AM.IndDum;
                AR.expchanged = AB[0]->R.expchanged;
                AR.expflags = AB[0]->R.expflags;
                AR.PolyFun = AB[0]->R.PolyFun;
                AR.PolyFunInv = AB[0]->R.PolyFunInv;
                AR.PolyFunType = AB[0]->R.PolyFunType;
                AR.PolyFunExp = AB[0]->R.PolyFunExp;
                AR.PolyFunVar = AB[0]->R.PolyFunVar;
                AR.PolyFunPow = AB[0]->R.PolyFunPow;
/*
                Now fire up the sort buffer.
*/
                NewSort(BHEAD0);
                break;
/*
            #] STARTNEWEXPRESSION :
            #[ LOWESTLEVELGENERATION :
*/
            case LOWESTLEVELGENERATION:
#ifdef INNERTEST
                if ( AC.InnerTest ) {
                    if ( StrCmp(AC.TestValue,(UBYTE *)INNERTEST) == 0 ) {
                        MesPrint("Testing(Worker%d): value = %s",AT.identity,AC.TestValue);
                    }
                }
#endif
                e = Expressions + AR.CurExpr;
                thr = AN.threadbuck;
                ppdef = thr->deferbuffer;
                ttin = thr->threadbuffer;
                ttco = thr->compressbuffer;
                term = AT.WorkPointer;
                thr->usenum = 0;
                tobereleased = 0;
                AN.inputnumber = thr->firstterm;
                AN.ninterms = thr->firstterm;
                do {
                  thr->usenum++;    /* For if the master wants to steal the bucket */
                  tt = term; i = *ttin;
                  NCOPY(tt,ttin,i);
                  AT.WorkPointer = tt;
                  if ( AR.DeferFlag ) {
                    tt = AR.CompressBuffer; i = *ttco;
                    NCOPY(tt,ttco,i);
                    AR.CompressPointer = tt;
                    AR.DefPosition = ppdef[0]; ppdef++;
                  }
                  if ( thr->free == BUCKETTERMINATED ) {
/*
                    The next statement allows the master to steal the bucket
                    for load balancing purposes. We do still execute the current
                    term, but afterwards we drop out.
                    Once we have written the release code, we cannot use this
                    bucket anymore. Hence the exit to the label bucketstolen.
*/
                    if ( thr->usenum == thr->totnum ) {
```

```
                            thr->free = BUCKETCOMINGFREE;
                        }
                        else {
                            thr->free = BUCKETRELEASED;
                            tobereleased = 1;
                        }
                      }
/*
                        What if we want to steal and we set thr->free while
                        the thread is inside the next code for a long time?
                      if ( AT.LoadBalancing ) {
*/
                        LOCK(thr->lock);
                        thr->busy = BUCKETDOINGTERM;
                        UNLOCK(thr->lock);
/*
                      }
                      else {
                        thr->busy = BUCKETDOINGTERM;
                      }
*/
                      AN.RepPoint = AT.RepCount + 1;

                      if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
                        StoreTerm(BHEAD term);
                      }
                      else {
                      if ( AR.DeferFlag ) {
                        AR.CurDum = AN.IndDum = Expressions[AR.CurExpr].numdummies + AM.IndDum;
                      }
                      else {
                        AN.IndDum = AM.IndDum;
                        AR.CurDum = ReNumber(BHEAD term);
                      }
                      if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
                      if ( AN.ncmod ) {
                        if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
                        else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
                      }
                      else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
                      if ( ( AP.PreDebug & THREADSDEBUG ) != 0 ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Thread %w executing term:");
                        PrintTerm(term,"LLG");
                        MUNLOCK(ErrorMessageLock);
                      }
                      if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
                            && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
                            PolyFunClean(BHEAD term);
                      }
                      if ( Generator(BHEAD term,0) ) {
                        LowerSortLevel();
                        MLOCK(ErrorMessageLock);
                        MesPrint("Error in processing one term in thread %d in module
%d",identity,AC.CModule);
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                      }
                      AN.ninterms++;
                      }
/*                    if ( AT.LoadBalancing ) { */
                        LOCK(thr->lock);
                        thr->busy = BUCKETPREPARINGTERM;
                        UNLOCK(thr->lock);
/*
                      }
                      else {
                        thr->busy = BUCKETPREPARINGTERM;
                      }
*/
                      if ( thr->free == BUCKETTERMINATED ) {
                        if ( thr->usenum == thr->totnum ) {
                            thr->free = BUCKETCOMINGFREE;
                        }
                        else {
                            thr->free = BUCKETRELEASED;
                            tobereleased = 1;
                        }
                      }
                      if ( tobereleased ) goto bucketstolen;
                    } while ( *ttin );
                    thr->free = BUCKETCOMINGFREE;
bucketstolen:;
/*                  if ( AT.LoadBalancing ) { */
                        LOCK(thr->lock);
                        thr->busy = BUCKETTOBERELEASED;
                        UNLOCK(thr->lock);
```

```
/*              }
                else {
                    thr->busy = BUCKETTOBERELEASED;
                }
*/
                AT.WorkPointer = term;
                break;
/*
            #] LOWESTLEVELGENERATION :
            #[ FINISHEXPRESSION :
*/
#ifdef WITHSORTBOTS
            case CLAIMOUTPUT:
                LOCK(AT.SB.MasterBlockLock[1]);
                break;
#endif
            case FINISHEXPRESSION:
/*
                Finish the sort

                Start with claiming the first block
                Once we have claimed it we can let the master know that
                everything is all right.
*/
                LOCK(AT.SB.MasterBlockLock[1]);
                ThreadClaimedBlock(identity);
/*
                Entry for when we work with sortbots
*/
#ifdef WITHSORTBOTS
                /* fall through */
            case FINISHEXPRESSION2:
#endif
/*
                Now we may need here an fsync on the sort file
*/
                if ( AC.ThreadSortFileSynch ) {
                  if ( AT.S0->file.handle >= 0 ) {
                    SynchFile(AT.S0->file.handle);
                  }
                }
                AT.SB.FillBlock = 1;
                AT.SB.MasterFill[1] = AT.SB.MasterStart[1];
                errorcode = EndSort(BHEAD AT.S0->sBuffer,0);
                UNLOCK(AT.SB.MasterBlockLock[AT.SB.FillBlock]);
                UpdateMaxSize();
                if ( errorcode ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Error terminating sort in thread %d in module %d",identity,AC.CModule);
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                break;
/*
            #] FINISHEXPRESSION :
            #[ CLEANUPEXPRESSION :
*/
            case CLEANUPEXPRESSION:
/*
                Cleanup everything and wait for the next expression
*/
                if ( AR.outfile->handle >= 0 ) {
                    CloseFile(AR.outfile->handle);
                    AR.outfile->handle = -1;
                    remove(AR.outfile->name);
                    AR.outfile->POfill = AR.outfile->POfull = AR.outfile->PObuffer;
                    PUTZERO(AR.outfile->POposition);
                    PUTZERO(AR.outfile->filesize);
                }
                else {
                    AR.outfile->POfill = AR.outfile->POfull = AR.outfile->PObuffer;
                    PUTZERO(AR.outfile->POposition);
                    PUTZERO(AR.outfile->filesize);
                }
                {
                    CBUF *C = cbuf+AT.ebufnum;
                    WORD **w, ii;
                    if ( C->numrhs > 0 || C->numlhs > 0 ) {
                        if ( C->rhs ) {
                            w = C->rhs; ii = C->numrhs;
                            do { *w++ = 0; } while ( --ii > 0 );
                        }
                        if ( C->lhs ) {
                            w = C->lhs; ii = C->numlhs;
                            do { *w++ = 0; } while ( --ii > 0 );
                        }
                        C->numlhs = C->numrhs = 0;
```

```
                        ClearTree(AT.ebufnum);
                        C->Pointer = C->Buffer;
                    }
                }
                break;
/*
            #] CLEANUPEXPRESSION :
            #[ HIGHERLEVELGENERATION :
*/
            case HIGHERLEVELGENERATION:
/*
                When foliating halfway the tree.
                This should only be needed in a second level load balancing
*/
                term = AT.WorkSpace; AT.WorkPointer = term + *term;
                if ( Generator(BHEAD term,AR.level) ) {
                    LowerSortLevel();
                    MLOCK(ErrorMessageLock);
                    MesPrint("Error in load balancing one term at level %d in thread %d in module
%d",AR.level,AT.identity,AC.CModule);
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                AT.WorkPointer = term;
                break;
/*
            #] HIGHERLEVELGENERATION :
            #[ STARTNEWMODULE :
*/
            case STARTNEWMODULE:
/*
                For resetting variables.
*/
                SpecialCleanup(B);
                break;
/*
            #] STARTNEWMODULE :
            #[ TERMINATETHREAD :
*/
            case TERMINATETHREAD:
                goto EndOfThread;
/*
            #] TERMINATETHREAD :
            #[ DOONEEXPRESSION :

                When a thread has to do a complete (not too big) expression.
                The number of the expression to be done is in AR.exprtodo.
                The code is mostly taken from Processor. The only difference
                is with what to do with the output.
                The output should go to the scratch buffer of the worker
                (which is free at the right moment). If this buffer is too
                small we have a problem. We could write to file or give the
                master what we have and from now on the master has to collect
                pieces until things are complete.
                Note: this assumes that the expressions don't keep their order.
                If they have to keep their order, don't use this feature.
*/
            case DOONEEXPRESSION: {

                POSITION position, outposition;
                FILEHANDLE *fi, *fout, *oldoutfile;
                LONG dd = 0;
                WORD oldBracketOn = AR.BracketOn;
                WORD *oldBrackBuf = AT.BrackBuf;
                WORD oldbracketindexflag = AT.bracketindexflag;
                WORD fromspectator = 0;
                e = Expressions + AR.exprtodo;
                i = AR.exprtodo;
                AR.CurExpr = i;
                AR.SortType = AC.SortType;
                AR.expchanged = 0;
                if ( ( e->vflags & ISFACTORIZED ) != 0 ) {
                    AR.BracketOn = 1;
                    AT.BrackBuf = AM.BracketFactors;
                    AT.bracketindexflag = 1;
                }

                position = AS.OldOnFile[i];
                if ( e->status == HIDDENLEXPRESSION || e->status == HIDDENGEXPRESSION ) {
                    AR.GetFile = 2; fi = AR.hidefile;
                }
                else {
                    AR.GetFile = 0; fi = AR.infile;
                }
/*
                PUTZERO(fi->POposition);
                if ( fi->handle >= 0 ) {
```

```
                    fi->POfill = fi->POfull = fi->PObuffer;
                }
*/
                SetScratch(fi,&position);
                term = oldwork = AT.WorkPointer;
                AR.CompressPointer = AR.CompressBuffer;
                AR.CompressPointer[0] = 0;
                AR.KeptInHold = 0;
                if ( GetTerm(BHEAD term) <= 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Expression %d has problems in scratchfile (t)",i);
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                if ( AT.bracketindexflag > 0 ) OpenBracketIndex(i);
                term[3] = i;
                if ( term[5] < 0 ) {
                    fromspectator = -term[5];
                    PUTZERO(AM.SpectatorFiles[fromspectator-1].readpos);
                    term[5] = AC.cbufnum;
                }
                PUTZERO(outposition);
                fout = AR.outfile;
                fout->POfill = fout->POfull = fout->PObuffer;
                fout->POposition = outposition;
                if ( fout->handle >= 0 ) {
                    fout->POposition = outposition;
                }
/*
                The next statement is needed because we need the system
                to believe that the expression is at position zero for
                the moment. In this worker, with no memory of other expressions,
                it is. This is needed for when a bracket index is made
                because there e->onfile is an offset. Afterwards, when the
                expression is written to its final location in the masters
                output e->onfile will get its real value.
*/
                PUTZERO(e->onfile);
                if ( PutOut(BHEAD term,&outposition,fout,0) < 0 ) goto ProcErr;

                AR.DeferFlag = AC.ComDefer;

                AR.sLevel = AB[0]->R.sLevel;
                term = AT.WorkPointer;
                NewSort(BHEAD0);
                AR.MaxDum = AM.IndDum;
                AN.ninterms = 0;
                if ( fromspectator ) {
                 while ( GetFromSpectator(term,fromspectator-1) ) {
                  AT.WorkPointer = term + *term;
                  AN.RepPoint = AT.RepCount + 1;
                  AN.IndDum = AM.IndDum;
                  AR.CurDum = ReNumber(BHEAD term);
                  if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
                  if ( AN.ncmod ) {
                    if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
                    else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
                  }
                  else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
                  if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
                        && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
                        PolyFunClean(BHEAD term);
                  }
                  if ( Generator(BHEAD term,0) ) {
                    LowerSortLevel(); goto ProcErr;
                  }
                 }
                }
                else {
                 while ( GetTerm(BHEAD term) ) {
                  SeekScratch(fi,&position);
                  AN.ninterms++; dd = AN.deferskipped;
                  if ( ( e->vflags & ISFACTORIZED ) != 0 && term[1] == HAAKJE ) {
                      StoreTerm(BHEAD term);
                  }
                  else {
                  if ( AC.CollectFun && *term <= (AM.MaxTer/(2*(LONG)sizeof(WORD))) ) {
                    if ( GetMoreTerms(term) < 0 ) {
                      LowerSortLevel(); goto ProcErr;
                    }
                    SeekScratch(fi,&position);
                  }
                  AT.WorkPointer = term + *term;
                  AN.RepPoint = AT.RepCount + 1;
                  if ( AR.DeferFlag ) {
                    AR.CurDum = AN.IndDum = Expressions[AR.exprtodo].numdummies;
                  }
```

```
                  else {
                    AN.IndDum = AM.IndDum;
                    AR.CurDum = ReNumber(BHEAD term);
                  }
                  if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
                  if ( AN.ncmod ) {
                    if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
                    else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
                  }
                  else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
                  if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
                        && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
                        PolyFunClean(BHEAD term);
                  }
                  if ( Generator(BHEAD term,0) ) {
                    LowerSortLevel(); goto ProcErr;
                  }
                  AN.ninterms += dd;
                  }
                  SetScratch(fi,&position);
                  if ( fi == AR.hidefile ) {
                    AR.InHiBuf = (fi->POfull-fi->PObuffer)
                        -DIFBASE(position,fi->POposition)/sizeof(WORD);
                  }
                  else {
                    AR.InInBuf = (fi->POfull-fi->PObuffer)
                        -DIFBASE(position,fi->POposition)/sizeof(WORD);
                  }
                 }
                }
                AN.ninterms += dd;
                if ( EndSort(BHEAD AT.S0->sBuffer,0) < 0 ) goto ProcErr;
                e->numdummies = AR.MaxDum - AM.IndDum;
                AR.BracketOn = oldBracketOn;
                AT.BrackBuf = oldBrackBuf;
                if ( ( e->vflags & TOBEFACTORED ) != 0 )
                        poly_factorize_expression(e);
                else if ( ( ( e->vflags & TOBEUNFACTORED ) != 0 )
                 && ( ( e->vflags & ISFACTORIZED ) != 0 ) )
                        poly_unfactorize_expression(e);
                if ( AT.S0->TermsLeft )   e->vflags &= ~ISZERO;
                else                      e->vflags |= ISZERO;
                if ( AR.expchanged == 0 ) e->vflags |= ISUNMODIFIED;
                if ( AT.S0->TermsLeft ) AR.expflags |= ISZERO;
                if ( AR.expchanged )    AR.expflags |= ISUNMODIFIED;
                AR.GetFile = 0;
                AT.bracketindexflag = oldbracketindexflag;
/*
                Now copy the whole thing from fout to AR0.outfile
                Do this in one go to keep the lock occupied as short as possible
*/
                SeekScratch(fout,&outposition);
                LOCK(AS.outputslock);
                oldoutfile = AB[0]->R.outfile;
                if ( e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION ) {
                    AB[0]->R.outfile = AB[0]->R.hidefile;
                }
                SeekScratch(AB[0]->R.outfile,&position);
                e->onfile = position;
                if ( CopyExpression(fout,AB[0]->R.outfile) < 0 ) {
                    AB[0]->R.outfile = oldoutfile;
                    UNLOCK(AS.outputslock);
                    MLOCK(ErrorMessageLock);
                    MesPrint("Error copying output of 'InParallel' expression to master. Thread: %d",identity);
                    MUNLOCK(ErrorMessageLock);
                    goto ProcErr;
                }
                AB[0]->R.outfile = oldoutfile;
                AB[0]->R.hidefile->POfull = AB[0]->R.hidefile->POfill;
                AB[0]->R.expflags = AR.expflags;
                UNLOCK(AS.outputslock);

                if ( fout->handle >= 0 ) {     /* Now get rid of the file */
                    CloseFile(fout->handle);
                    fout->handle = -1;
                    remove(fout->name);
                    PUTZERO(fout->POposition);
                    PUTZERO(fout->filesize);
                    fout->POfill = fout->POfull = fout->PObuffer;
                }
                UpdateMaxSize();

                AT.WorkPointer = oldwork;

                } break;
/*
```

```
            #] DOONEEXPRESSION :
            #[ DOBRACKETS :

                In case we have a bracket index we can have the worker treat
                one or more of the entries in the bracket index.
                The advantage is that identical terms will meet each other
                sooner in the sorting and hence fewer compares will be needed.
                Also this way the master doesn't need to fill the buckets.
                The main problem is the load balancing which can become very
                bad when there is a long tail without things outside the bracket.

                We get sent:
                1: The number of the first bracket to be done
                2: The number of the last bracket to be done
*/
            case DOBRACKETS: {
                BRACKETINFO *binfo;
                BRACKETINDEX *bi;
                FILEHANDLE *fi;
                POSITION stoppos,where;
                e = Expressions + AR.CurExpr;
                binfo = e->bracketinfo;
                thr = AN.threadbuck;
                bi = &(binfo->indexbuffer[thr->firstbracket]);
                if ( AR.GetFile == 2 ) fi = AR.hidefile;
                else                   fi = AR.infile;
                where = bi->start;
                ADD2POS(where,AS.OldOnFile[AR.CurExpr]);
                SetScratch(fi,&(where));
                stoppos = binfo->indexbuffer[thr->lastbracket].next;
                ADD2POS(stoppos,AS.OldOnFile[AR.CurExpr]);
                AN.ninterms = thr->firstterm;
/*
                Now we have to put the 'value' of the bracket in the
                Compress buffer.
*/
                ttco = AR.CompressBuffer;
                tt = binfo->bracketbuffer + bi->bracket;
                i = *tt;
                NCOPY(ttco,tt,i)
                AR.CompressPointer = ttco;
                term = AT.WorkPointer;
                while ( GetTerm(BHEAD term) ) {
                    SeekScratch(fi,&where);
                    AT.WorkPointer = term + *term;
                    AN.IndDum = AM.IndDum;
                    AR.CurDum = ReNumber(BHEAD term);
                    if ( AC.SymChangeFlag ) MarkDirty(term,DIRTYSYMFLAG);
                    if ( AN.ncmod ) {
                        if ( ( AC.modmode & ALSOFUNARGS ) != 0 ) MarkDirty(term,DIRTYFLAG);
                        else if ( AR.PolyFun ) PolyFunDirty(BHEAD term);
                    }
                    else if ( AC.PolyRatFunChanged ) PolyFunDirty(BHEAD term);
                    if ( ( AR.PolyFunType == 2 ) && ( AC.PolyRatFunChanged == 0 )
                        && ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION ) ) {
                        PolyFunClean(BHEAD term);
                    }
                    if ( ( AP.PreDebug & THREADSDEBUG ) != 0 ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Thread %w executing term:");
                        PrintTerm(term,"DoBrackets");
                        MUNLOCK(ErrorMessageLock);
                    }
                    AT.WorkPointer = term + *term;
                    if ( Generator(BHEAD term,0) ) {
                        LowerSortLevel();
                        MLOCK(ErrorMessageLock);
                        MesPrint("Error in processing one term in thread %d in module %d",identity,AC.CModule);
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                    }
                    AN.ninterms++;
                    SetScratch(fi,&(where));
                    if ( ISGEPOS(where,stoppos) ) break;
                }
                AT.WorkPointer = term;
                thr->free = BUCKETCOMINGFREE;
                break;
            }
/*
            #] DOBRACKETS :
            #[ CLEARCLOCK :

            The program only comes here after a .clear
*/
            case CLEARCLOCK:
```

```
/*              LOCK(clearclocklock); */
                sumtimerinfo[identity] += TimeCPU(1);
                timerinfo[identity] = TimeCPU(0);
/*              UNLOCK(clearclocklock); */
                break;
/*
            #] CLEARCLOCK :
            #[ MCTSEXPANDTREE :
*/
            case MCTSEXPANDTREE:
                AT.optimtimes = AB[0]->T.optimtimes;
                find_Horner_MCTS_expand_tree();
                break;
/*
            #] MCTSEXPANDTREE :
            #[ OPTIMIZEEXPRESSION :
*/
            case OPTIMIZEEXPRESSION:
                optimize_expression_given_Horner();
                break;
/*
            #] OPTIMIZEEXPRESSION :
*/
            default:
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal wakeup signal %d for thread %d",wakeupsignal,identity);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
                break;
        }
        /* we need the following update in case we are using checkpoints. then we
           need to readjust the clocks when recovering using this information */
        timerinfo[identity] = TimeCPU(1);
    }
EndOfThread:;
/*
    This is the end of the thread. We cleanup and exit.
    If we are using flint, call the per-thread cleanup function. This keep valgrind happy.
*/
#ifdef WITHFLINT
    flint_final_cleanup_thread();
#endif
    FinalizeOneThread(identity);
    return(0);
ProcErr:
    Terminate(-1);
    return(0);
}

/*
    #] RunThread :
    #[ RunSortBot :
*/
#ifdef WITHSORTBOTS

void *RunSortBot(void *dummy)
{
    int identity, wakeupsignal, identityretv;
    ALLPRIVATES *B, *BB;
    DUMMYUSE(dummy);
    identity = SetIdentity(&identityretv);
    threadpointers[identity] = pthread_self();
    B = InitializeOneThread(identity);
    while ( ( wakeupsignal = SortBotWait(identity) ) > 0 ) {
        switch ( wakeupsignal ) {
/*
            #[ INISORTBOT :
*/
            case INISORTBOT:
                AR.CompressBuffer = AB[0]->R.CompressBuffer;
                AR.ComprTop = AB[0]->R.ComprTop;
                AR.CompressPointer = AB[0]->R.CompressPointer;
                AR.CurExpr = AB[0]->R.CurExpr;
                AR.PolyFun = AB[0]->R.PolyFun;
                AR.PolyFunInv = AB[0]->R.PolyFunInv;
                AR.PolyFunType = AB[0]->R.PolyFunType;
                AR.PolyFunExp = AB[0]->R.PolyFunExp;
                AR.PolyFunVar = AB[0]->R.PolyFunVar;
                AR.PolyFunPow = AB[0]->R.PolyFunPow;
                AR.SortType = AC.SortType;
                if ( AR.PolyFun == 0 ) { AT.SS->PolyFlag = 0; }
                else if ( AR.PolyFunType == 1 ) { AT.SS->PolyFlag = 1; }
                else if ( AR.PolyFunType == 2 ) {
                    if ( AR.PolyFunExp == 2
                      || AR.PolyFunExp == 3 ) AT.SS->PolyFlag = 1;
                    else                      AT.SS->PolyFlag = 2;
                }
```

```
                AT.SS->PolyWise = 0;
                AN.ncmod = AC.ncmod;
                LOCK(AT.SB.MasterBlockLock[1]);
                BB = AB[AT.SortBotIn1];
                LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);
                BB = AB[AT.SortBotIn2];
                LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);
                AT.SB.FillBlock = 1;
                AT.SB.MasterFill[1] = AT.SB.MasterStart[1];
                SETBASEPOSITION(AN.theposition,0);
                break;
/*
            #] INISORTBOT :
            #[ RUNSORTBOT :
*/
            case RUNSORTBOT:
                SortBotMerge(B);
                break;
/*
            #] RUNSORTBOT :
            #[ TERMINATETHREAD :
*/
            case TERMINATETHREAD:
                goto EndOfThread;
/*
            #] TERMINATETHREAD :
            #[ CLEARCLOCK :

            The program only comes here after a .clear
*/
            case CLEARCLOCK:
/*              LOCK(clearclocklock); */
                sumtimerinfo[identity] += TimeCPU(1);
                timerinfo[identity] = TimeCPU(0);
/*              UNLOCK(clearclocklock); */
                break;
/*
            #] CLEARCLOCK :
*/
            default:
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal wakeup signal %d for thread %d",wakeupsignal,identity);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
                break;
        }
    }
EndOfThread:;
/*
    This is the end of the thread. We cleanup and exit.
    If we are using flint, call the per-thread cleanup function. This keep valgrind happy.
*/
#ifdef WITHFLINT
    flint_final_cleanup_thread();
#endif
    FinalizeOneThread(identity);
    return(0);
}

#endif

/*
    #] RunSortBot :
    #[ IAmAvailable :
*/
void IAmAvailable(int identity)
{
    int top;
    LOCK(availabilitylock);
    top = topofavailables;
    listofavailables[topofavailables++] = identity;
    if ( top == 0 ) {
        UNLOCK(availabilitylock);
        LOCK(wakeupmasterlock);
        wakeupmaster = identity;
        pthread_cond_signal(&wakeupmasterconditions);
        UNLOCK(wakeupmasterlock);
    }
    else {
        UNLOCK(availabilitylock);
    }
}

/*
    #] IAmAvailable :
    #[ GetAvailableThread :
*/
```

```
int GetAvailableThread(void)
{
    int retval = -1;
    LOCK(availabilitylock);
    if ( topofavailables > 0 ) retval = listofavailables[--topofavailables];
    UNLOCK(availabilitylock);
    if ( retval >= 0 ) {
/*
        Make sure the thread is indeed waiting and not between
        saying that it is available and starting to wait.
*/
        LOCK(wakeuplocks[retval]);
        UNLOCK(wakeuplocks[retval]);
    }
    return(retval);
}

/*
    #] GetAvailableThread :
    #[ ConditionalGetAvailableThread :
*/
int ConditionalGetAvailableThread(void)
{
    int retval = -1;
    if ( topofavailables > 0 ) {
        LOCK(availabilitylock);
        if ( topofavailables > 0 ) {
            retval = listofavailables[--topofavailables];
        }
        UNLOCK(availabilitylock);
        if ( retval >= 0 ) {
/*
            Make sure the thread is indeed waiting and not between
            saying that it is available and starting to wait.
*/
            LOCK(wakeuplocks[retval]);
            UNLOCK(wakeuplocks[retval]);
        }
    }
    return(retval);
}

/*
    #] ConditionalGetAvailableThread :
    #[ GetThread :
*/
int GetThread(int identity)
{
    int retval = -1, j;
    LOCK(availabilitylock);
    for ( j = 0; j < topofavailables; j++ ) {
        if ( identity == listofavailables[j] ) break;
    }
    if ( j < topofavailables ) {
        --topofavailables;
        for ( ; j < topofavailables; j++ ) {
            listofavailables[j] = listofavailables[j+1];
        }
        retval = identity;
    }
    UNLOCK(availabilitylock);
    return(retval);
}

/*
    #] GetThread :
    #[ ThreadWait :
*/
int ThreadWait(int identity)
{
    int retval, top, j;
    LOCK(wakeuplocks[identity]);
    LOCK(availabilitylock);
    top = topofavailables;
    for ( j = topofavailables; j > 0; j-- )
        listofavailables[j] = listofavailables[j-1];
    listofavailables[0] = identity;
    topofavailables++;
    if ( top == 0 || topofavailables == numberofworkers ) {
        UNLOCK(availabilitylock);
        LOCK(wakeupmasterlock);
        wakeupmaster = identity;
        pthread_cond_signal(&wakeupmasterconditions);
        UNLOCK(wakeupmasterlock);
    }
    else {
        UNLOCK(availabilitylock);
```

```
    }
    while ( wakeup[identity] == 0 ) {
        pthread_cond_wait(&(wakeupconditions[identity]),&(wakeuplocks[identity]));
    }
    retval = wakeup[identity];
    wakeup[identity] = 0;
    UNLOCK(wakeuplocks[identity]);
    return(retval);
}

/*
    #] ThreadWait :
    #[ SortBotWait :
*/

#ifdef WITHSORTBOTS
int SortBotWait(int identity)
{
    int retval;
    LOCK(wakeuplocks[identity]);
    LOCK(availabilitylock);
    topsortbotavailables++;
    if ( topsortbotavailables >= numberofsortbots ) {
        UNLOCK(availabilitylock);
        LOCK(wakeupsortbotlock);
        wakeupmaster = identity;
        pthread_cond_signal(&wakeupsortbotconditions);
        UNLOCK(wakeupsortbotlock);
    }
    else {
        UNLOCK(availabilitylock);
    }
    while ( wakeup[identity] == 0 ) {
        pthread_cond_wait(&(wakeupconditions[identity]),&(wakeuplocks[identity]));
    }
    retval = wakeup[identity];
    wakeup[identity] = 0;
    UNLOCK(wakeuplocks[identity]);
    return(retval);
}

#endif

/*
    #] SortBotWait :
    #[ ThreadClaimedBlock :
*/
int ThreadClaimedBlock(int identity)
{
    LOCK(availabilitylock);
    numberclaimed++;
    if ( numberclaimed >= numberofworkers ) {
        UNLOCK(availabilitylock);
        LOCK(wakeupmasterlock);
        wakeupmaster = identity;
        pthread_cond_signal(&wakeupmasterconditions);
        UNLOCK(wakeupmasterlock);
    }
    else {
        UNLOCK(availabilitylock);
    }
    return(0);
}

/*
    #] ThreadClaimedBlock :
    #[ MasterWait :
*/
int MasterWait(void)
{
    int retval;
    LOCK(wakeupmasterlock);
    while ( wakeupmaster == 0 ) {
        pthread_cond_wait(&wakeupmasterconditions,&wakeupmasterlock);
    }
    retval = wakeupmaster;
    wakeupmaster = 0;
    UNLOCK(wakeupmasterlock);
    return(retval);
}

/*
    #] MasterWait :
    #[ MasterWaitThread :
*/
int MasterWaitThread(int identity)
{
```

```
    int retval;
    LOCK(wakeupmasterthreadlocks[identity]);
    while ( wakeupmasterthread[identity] == 0 ) {
        pthread_cond_wait(&(wakeupmasterthreadconditions[identity])
                ,&(wakeupmasterthreadlocks[identity]));
    }
    retval = wakeupmasterthread[identity];
    wakeupmasterthread[identity] = 0;
    UNLOCK(wakeupmasterthreadlocks[identity]);
    return(retval);
}

/*
    #] MasterWaitThread :
    #[ MasterWaitAll :
*/
void MasterWaitAll(void)
{
    LOCK(wakeupmasterlock);
    while ( topofavailables < numberofworkers ) {
        pthread_cond_wait(&wakeupmasterconditions,&wakeupmasterlock);
    }
    UNLOCK(wakeupmasterlock);
    return;
}

/*
    #] MasterWaitAll :
    #[ MasterWaitAllSortBots :
*/

#ifdef WITHSORTBOTS

void MasterWaitAllSortBots(void)
{
    LOCK(wakeupsortbotlock);
    while ( topsortbotavailables < numberofsortbots ) {
        pthread_cond_wait(&wakeupsortbotconditions,&wakeupsortbotlock);
    }
    UNLOCK(wakeupsortbotlock);
    return;
}

#endif

/*
    #] MasterWaitAllSortBots :
    #[ MasterWaitAllBlocks :
*/
void MasterWaitAllBlocks(void)
{
    LOCK(wakeupmasterlock);
    while ( numberclaimed < numberofworkers ) {
        pthread_cond_wait(&wakeupmasterconditions,&wakeupmasterlock);
    }
    UNLOCK(wakeupmasterlock);
    return;
}

/*
    #] MasterWaitAllBlocks :
    #[ WakeupThread :
*/
void WakeupThread(int identity, int signalnumber)
{
    if ( signalnumber == 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal wakeup signal for thread %d",identity);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    LOCK(wakeuplocks[identity]);
    wakeup[identity] = signalnumber;
    pthread_cond_signal(&(wakeupconditions[identity]));
    UNLOCK(wakeuplocks[identity]);
}

/*
    #] WakeupThread :
    #[ WakeupMasterFromThread :
*/
void WakeupMasterFromThread(int identity, int signalnumber)
{
    if ( signalnumber == 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal wakeup signal for master %d",identity);
        MUNLOCK(ErrorMessageLock);
```

```
        Terminate(-1);
    }
    LOCK(wakeupmasterthreadlocks[identity]);
    wakeupmasterthread[identity] = signalnumber;
    pthread_cond_signal(&(wakeupmasterthreadconditions[identity]));
    UNLOCK(wakeupmasterthreadlocks[identity]);
}

/*
    #] WakeupMasterFromThread :
    #[ SendOneBucket :
*/
int SendOneBucket(int type)
{
    ALLPRIVATES *B0 = AB[0];
    THREADBUCKET *thr = 0;
    int j, k, id;
    for ( j = 0; j < numthreadbuckets; j++ ) {
        if ( threadbuckets[j]->free == BUCKETFILLED ) {
            thr = threadbuckets[j];
            for ( k = j+1; k < numthreadbuckets; k++ )
                threadbuckets[k-1] = threadbuckets[k];
            threadbuckets[numthreadbuckets-1] = thr;
            break;
        }
    }
    AN0.ninterms++;
    while ( ( id = GetAvailableThread() ) < 0 ) { MasterWait(); }
/*
    Prepare the thread. Give it the term and variables.
*/
    LoadOneThread(0,id,thr,0);
    thr->busy = BUCKETASSIGNED;
    thr->free = BUCKETINUSE;
    numberoffullbuckets--;
/*
    And signal the thread to run.
    Form now on we may only interfere with this bucket
    1: after it has been marked BUCKETCOMINGFREE
    2: when thr->busy == BUCKETDOINGTERM and then only when protected by
       thr->lock. This would be for load balancing.
*/
    WakeupThread(id,type);
/*  AN0.ninterms += thr->ddterms; */
    return(0);
}

/*
    #] SendOneBucket :
    #[ InParallelProcessor :
*/
int InParallelProcessor(void)
{
    GETIDENTITY
    int i, id, retval = 0, num = 0;
    EXPRESSIONS e;
    if ( numberofworkers >= 2 ) {
        SetWorkerFiles();
        for ( i = 0; i < NumExpressions; i++ ) {
            e = Expressions+i;
            if ( e->partodo <= 0 ) continue;
            if ( e->status == LOCALEXPRESSION || e->status == GLOBALEXPRESSION
            || e->status == UNHIDELEXPRESSION || e->status == UNHIDEGEXPRESSION
            || e->status == INTOHIDELEXPRESSION || e->status == INTOHIDEGEXPRESSION ) {
            }
            else {
                e->partodo = 0;
                continue;
            }
            if ( e->counter == 0 ) { /* Expression with zero terms */
                e->partodo = 0;
                continue;
            }
/*
            This expression should go to an idle worker
*/
            while ( ( id = GetAvailableThread() ) < 0 ) { MasterWait(); }
            LoadOneThread(0,id,0,-1);
            AB[id]->R.exprtodo = i;
            WakeupThread(id,DOONEEXPRESSION);
            num++;
        }
/*
        Now we have to wait for all workers to finish
*/
        if ( num > 0 ) MasterWaitAll();

```

```
        if ( AC.CollectFun ) AR.DeferFlag = 0;
    }
    else {
        for ( i = 0; i < NumExpressions; i++ ) {
            Expressions[i].partodo = 0;
        }
    }
    return(retval);
}

/*
    #] InParallelProcessor :
    #[ ThreadsProcessor :
*/
int ThreadsProcessor(EXPRESSIONS e, WORD LastExpression, WORD fromspectator)
{
    ALLPRIVATES *B0 = AB[0], *B = B0;
    int id, oldgzipCompress, endofinput = 0, j, still, k, defcount = 0, bra = 0, first = 1;
    LONG dd = 0, ddd, thrbufsiz, thrbufsiz0, thrbufsiz2, numbucket = 0, numpasses;
    LONG num, i;
    WORD *oldworkpointer = AT0.WorkPointer, *tt, *ttco = 0, *t1 = 0, ter, *tstop = 0, *t2;
    THREADBUCKET *thr = 0;
    FILEHANDLE *oldoutfile = AR0.outfile;
    GETTERM GetTermP = &GetTerm;
    POSITION eonfile = AS.OldOnFile[e-Expressions];
    numberoffullbuckets = 0;
/*
    Start up all threads. The lock needs to be around the whole loop
    to keep processes from terminating quickly and putting themselves
    in the list of available threads again.
*/
    AM.tracebackflag = 1;

    AS.sLevel = AR0.sLevel;
    LOCK(availabilitylock);
    topofavailables = 0;
    for ( id = 1; id <= numberofworkers; id++ ) {
        WakeupThread(id,STARTNEWEXPRESSION);
    }
    UNLOCK(availabilitylock);
    NewSort(BHEAD0);
    AN0.ninterms = 1;
/*
    Now for redefine
*/
    if ( AC.numpfirstnum > 0 ) {
        for ( j = 0; j < AC.numpfirstnum; j++ ) {
            AC.inputnumbers[j] = -1;
        }
    }
    MasterWaitAll();
/*
    Determine a reasonable bucketsize.
    This is based on the value of AC.ThreadBucketSize and the number
    of terms. We want at least 5 buckets per worker at the moment.
    Some research should show whether this is reasonable.

    The number of terms in the expression is in e->counter
*/
    thrbufsiz2 = thrbufsiz = AC.ThreadBucketSize-1;
    if ( ( e->counter / ( numberofworkers * 5 ) ) < thrbufsiz ) {
        thrbufsiz = e->counter / ( numberofworkers * 5 ) - 1;
        if ( thrbufsiz < 0 ) thrbufsiz = 0;
    }
    thrbufsiz0 = thrbufsiz;
    numpasses = 5; /* this is just for trying */
    thrbufsiz = thrbufsiz0 / (2 « numpasses);
/*
    Mark all buckets as free and take the first.
*/
    for ( j = 0; j < numthreadbuckets; j++ )
        threadbuckets[j]->free = BUCKETFREE;
    thr = threadbuckets[0];
/*
    #[ Whole brackets :

    First we look whether we have to work with entire brackets
    This is the case when there is a non-NULL address in e->bracketinfo.
    Of course we shouldn't have interference from a collect or keep statement.
*/
#ifdef WHOLEBRACKETS
    if ( e->bracketinfo && AC.CollectFun == 0 && AR0.DeferFlag == 0 ) {
        FILEHANDLE *curfile;
        int didone = 0;
        LONG num, n;
        AN0.expr = e;
        for ( n = 0; n < e->bracketinfo->indexfill; n++ ) {
```

```
            num = TreatIndexEntry(B0,n);
            if ( num > 0 ) {
                didone = 1;
/*
                This bracket can be sent off.
                1: Look for an empty bucket
*/
ReTry:;
                for ( j = 0; j < numthreadbuckets; j++ ) {
                    switch ( threadbuckets[j]->free ) {
                        case BUCKETFREE:
                            thr = threadbuckets[j];
                            goto Found1;
                        case BUCKETCOMINGFREE:
                            thr = threadbuckets[j];
                            thr->free = BUCKETFREE;
                            for ( k = j+1; k < numthreadbuckets; k++ )
                                threadbuckets[k-1] = threadbuckets[k];
                            threadbuckets[numthreadbuckets-1] = thr;
                            j--;
                            break;
                        default:
                            break;
                    }
                }
Found1:;
                if ( j < numthreadbuckets ) {
/*
                    Found an empty bucket. Fill it.
*/
                    thr->firstbracket = n;
                    thr->lastbracket = n + num - 1;
                    thr->type = BUCKETDOINGBRACKET;
                    thr->free = BUCKETFILLED;
                    thr->firstterm = AN0.ninterms;
                    for ( j = n; j < n+num; j++ ) {
                        AN0.ninterms += e->bracketinfo->indexbuffer[j].termsinbracket;
                    }
                    n += num-1;
                    numberoffullbuckets++;
                    if ( topofavailables > 0 ) {
                        SendOneBucket(DOBRACKETS);
                    }
                }
/*
                    All buckets are in use.
                    Look/wait for an idle worker. Give it a bucket.
                    After that, retry for a bucket
*/
                else {
                    while ( topofavailables <= 0 ) {
                        MasterWait();
                    }
                    SendOneBucket(DOBRACKETS);
                    goto ReTry;
                }
            }
        }
        if ( didone ) {
/*
            And now put the input back in the original position.
*/
            switch ( e->status ) {
                case UNHIDELEXPRESSION:
                case UNHIDEGEXPRESSION:
                case DROPHLEXPRESSION:
                case DROPHGEXPRESSION:
                case HIDDENLEXPRESSION:
                case HIDDENGEXPRESSION:
                    curfile = AR0.hidefile;
                    break;
                default:
                    curfile = AR0.infile;
                    break;
            }
            SetScratch(curfile,&eonfile);
            GetTerm(B0,AT0.WorkPointer);
/*
            Now we point the GetTerm that is used to the one that is selective
*/
            GetTermP = &GetTerm2;
/*
            Next wait till there is a bucket available and initialize thr to it.
*/
            for(;;) {
                for ( j = 0; j < numthreadbuckets; j++ ) {
                    switch ( threadbuckets[j]->free ) {
```

```
                        case BUCKETFREE:
                            thr = threadbuckets[j];
                            goto Found2;
                        case BUCKETCOMINGFREE:
                            thr = threadbuckets[j];
                            thr->free = BUCKETFREE;
                            for ( k = j+1; k < numthreadbuckets; k++ )
                                threadbuckets[k-1] = threadbuckets[k];
                            threadbuckets[numthreadbuckets-1] = thr;
                            j--;
                            break;
                        default:
                            break;
                    }
                }
                while ( topofavailables <= 0 ) {
                    MasterWait();
                }
                while ( topofavailables > 0 && numberoffullbuckets > 0 ) {
                    SendOneBucket(DOBRACKETS);
                }
            }
Found2:;
            while ( numberoffullbuckets > 0 ) {
                while ( topofavailables <= 0 ) {
                    MasterWait();
                }
                while ( topofavailables > 0 && numberoffullbuckets > 0 ) {
                    SendOneBucket(DOBRACKETS);
                }
            }
/*
            Disable the 'warming up' with smaller buckets.

            numpasses = 0;
            thrbufsiz = thrbufsiz0;
*/
            AN0.lastinindex = -1;
        }
        MasterWaitAll();
    }
#endif
/*
    #] Whole brackets :

    Now the loop to start a bucket
*/
    for(;;) {
        if ( fromspectator ) {
            ter = GetFromSpectator(thr->threadbuffer,fromspectator-1);
            if ( ter == 0 ) fromspectator = 0;
        }
        else {
            ter = GetTermP(B0,thr->threadbuffer);
/*
            At this point we could check whether the input term is
            just an expression that resides in a scratch file.
            If this is the case we should store the current input info
            (file and buffer content) and redirect the input.
            At the end we can go back to where we were.
            There are two possibilities: we are in the same scratchfile
            as the main input and the other expression is still in the
            input buffer, or we have to do some reading. The reading is
            of course done in GetTerm.

            How to set this up can also be studied in TestSub (in file proces.c)
            where it checks for EXPRESSION.
*/
        }
        if ( ter < 0 ) break;
        if ( ter == 0 ) { endofinput = 1; goto Finalize; }
        dd = AN0.deferskipped;
        if ( AR0.DeferFlag ) {
            defcount = 0;
            thr->deferbuffer[defcount++] = AR0.DefPosition;
            ttco = thr->compressbuffer; t1 = AR0.CompressBuffer; j = *t1;
            NCOPY(ttco,t1,j);
        }
        else if ( first && ( AC.CollectFun == 0 ) ) { /* Brackets ? */
            first = 0;
            t1 = tstop = thr->threadbuffer;
            tstop += *tstop; tstop -= ABS(tstop[-1]);
            t1++;
            while ( t1 < tstop ) {
                if ( t1[0] == HAAKJE ) { bra = 1; break; }
                t1 += t1[1];
            }
```

```
            t1 = thr->threadbuffer;
        }
/*
        Check whether we have a collect,function going. If so execute it.
*/
        if ( AC.CollectFun && *(thr->threadbuffer) < (AM.MaxTer/((LONG)sizeof(WORD))-10) ) {
            if ( ( dd = GetMoreTerms(thr->threadbuffer) ) < 0 ) {
                LowerSortLevel(); goto ProcErr;
            }
        }
/*
        Check whether we have a priority task:
*/
        if ( topofavailables > 0 && numberoffullbuckets > 0 ) SendOneBucket(LOWESTLEVELGENERATION);
/*
        Now put more terms in the bucket. Position tt after the first term
*/
        tt = thr->threadbuffer; tt += *tt;
        thr->totnum = 1;
        thr->usenum = 0;
/*
        Next we worry about the 'slow startup' in which we make the initial
        buckets smaller, so that we get all threads busy as soon as possible.
*/
        if ( numpasses > 0 ) {
            numbucket++;
            if ( numbucket >= numberofworkers ) {
                numbucket = 0;
                numpasses--;
                if ( numpasses == 0 ) thrbufsiz = thrbufsiz0;
                else                  thrbufsiz = thrbufsiz0 / (2 « numpasses);
            }
            thrbufsiz2 = thrbufsiz + thrbufsiz/5; /* for completing brackets */
        }
/*
        we have already 1+dd terms
*/
        while ( ( dd < thrbufsiz ) &&
             ( tt - thr->threadbuffer ) < ( thr->threadbuffersize - AM.MaxTer/((LONG)sizeof(WORD)) - 2
) ) {
/*
            First check:
*/
            if ( topofavailables > 0 && numberoffullbuckets > 0 )
SendOneBucket(LOWESTLEVELGENERATION);
/*
            There is room in the bucket. Fill yet another term.
*/
            if ( GetTermP(B0,tt) == 0 ) { endofinput = 1; break; }
            dd++;
            thr->totnum++;
            dd += AN0.deferskipped;
            if ( AR0.DeferFlag ) {
                thr->deferbuffer[defcount++] = AR0.DefPosition;
                t1 = AR0.CompressBuffer; j = *t1;
                NCOPY(ttco,t1,j);
            }
            if ( AC.CollectFun && *tt < (AM.MaxTer/((LONG)sizeof(WORD))-10) ) {
                if ( ( ddd = GetMoreTerms(tt) ) < 0 ) {
                    LowerSortLevel(); goto ProcErr;
                }
                dd += ddd;
            }
            t1 = tt;
            tt += *tt;
        }
/*
        Check whether there are regular brackets and if we have no DeferFlag
        and no collect, we try to add more terms till we finish the current
        bracket. We should however not overdo it. Let us say: up to 20%
        more terms are allowed.
*/
        if ( bra ) {
            tstop = t1 + *t1; tstop -= ABS(tstop[-1]);
            t2 = t1+1;
            while ( t2 < tstop ) {
                if ( t2[0] == HAAKJE ) { break; }
                t2 += t2[1];
            }
            if ( t2[0] == HAAKJE ) {
              t2 += t2[1]; num = t2 - t1;
              while ( ( dd < thrbufsiz2 ) &&
                ( tt - thr->threadbuffer ) < ( thr->threadbuffersize - AM.MaxTer - 2 ) ) {
/*
                First check:
*/
                if ( topofavailables > 0 && numberoffullbuckets > 0 )
```

```
SendOneBucket(LOWESTLEVELGENERATION);
/*
                There is room in the bucket. Fill yet another term.
*/
                if ( GetTermP(B0,tt) == 0 ) { endofinput = 1; break; }
/*
                Same bracket?
*/
                tstop = tt + *tt; tstop -= ABS(tstop[-1]);
                if ( tstop-tt < num ) { /* Different: abort */
                    AR0.KeptInHold = 1;
                    break;
                }
                for ( i = 1; i < num; i++ ) {
                    if ( t1[i] != tt[i] ) break;
                }
                if ( i < num ) { /* Different: abort */
                    AR0.KeptInHold = 1;
                    break;
                }
/*
                Same bracket. We need this term.
*/
                dd++;
                thr->totnum++;
                tt += *tt;
              }
            }
        }
        thr->ddterms = dd; /* total number of terms including keep brackets */
        thr->firstterm = AN0.ninterms;
        AN0.ninterms += dd;
        *tt = 0;           /* mark end of bucket */
        thr->free = BUCKETFILLED;
        thr->type = BUCKETDOINGTERMS;
        numberoffullbuckets++;
        if ( topofavailables <= 0 && endofinput == 0 ) {
/*
            Problem: topofavailables may already be > 0, but the
            thread has not yet gone into waiting. Can the signal get lost?
            How can we tell that a thread is waiting for a signal?

            All threads are busy. Try to load up another bucket.
            In the future we could be more sophisticated.
            At the moment we load a complete bucket which could be
            1000 terms or even more.
            In principle it is better to keep a full bucket ready
            and check after each term we put in the next bucket. That
            way we don't waste time of the workers.
*/
            for ( j = 0; j < numthreadbuckets; j++ ) {
                switch ( threadbuckets[j]->free ) {
                    case BUCKETFREE:
                        thr = threadbuckets[j];
                        if ( !endofinput ) goto NextBucket;
/*
                        If we are at the end of the input we mark
                        the free buckets in a special way. That way
                        we don't keep running into them.
*/
                        thr->free = BUCKETATEND;
                        break;
                    case BUCKETCOMINGFREE:
                        thr = threadbuckets[j];
                        thr->free = BUCKETFREE;
/*
                        Bucket has just been finished.
                        Put at the end of the list. We don't want
                        an early bucket to wait to be treated last.
*/
                        for ( k = j+1; k < numthreadbuckets; k++ )
                            threadbuckets[k-1] = threadbuckets[k];
                        threadbuckets[numthreadbuckets-1] = thr;
                        j--;   /* we have to redo the same number j. */
                        break;
                    default:
                        break;
                }
            }
/*
            We have no free bucket or we are at the end.
            The only thing we can do now is wait for a worker to come free,
            provided there are still buckets to send.
*/
        }
/*
        Look for the next bucket to send. There is at least one full bucket!
```

```
*/
        for ( j = 0; j < numthreadbuckets; j++ ) {
            if ( threadbuckets[j]->free == BUCKETFILLED ) {
                thr = threadbuckets[j];
                for ( k = j+1; k < numthreadbuckets; k++ )
                    threadbuckets[k-1] = threadbuckets[k];
                threadbuckets[numthreadbuckets-1] = thr;
                break;
            }
        }
/*
        Wait for a thread to become available
        The bucket we are going to use is in thr.
*/
DoBucket:;
        AN0.ninterms++;
        while ( ( id = GetAvailableThread() ) < 0 ) { MasterWait(); }
/*
        Prepare the thread. Give it the term and variables.
*/
        LoadOneThread(0,id,thr,0);
        LOCK(thr->lock);
        thr->busy = BUCKETASSIGNED;
        UNLOCK(thr->lock);
        thr->free = BUCKETINUSE;
        numberoffullbuckets--;
/*
        And signal the thread to run.
        Form now on we may only interfere with this bucket
        1: after it has been marked BUCKETCOMINGFREE
        2: when thr->busy == BUCKETDOINGTERM and then only when protected by
           thr->lock. This would be for load balancing.
*/
        WakeupThread(id,LOWESTLEVELGENERATION);
/*      AN0.ninterms += thr->ddterms; */
/*
        Now look whether there is another bucket filled and a worker available
*/
        if ( topofavailables > 0 ) {  /* there is a worker */
            for ( j = 0; j < numthreadbuckets; j++ ) {
                if ( threadbuckets[j]->free == BUCKETFILLED ) {
                    thr = threadbuckets[j];
                    for ( k = j+1; k < numthreadbuckets; k++ )
                        threadbuckets[k-1] = threadbuckets[k];
                    threadbuckets[numthreadbuckets-1] = thr;
                    goto DoBucket; /* and we found a bucket */
                }
            }
/*
            no bucket is loaded but there is a thread available
            find a bucket to load. If there is none (all are USED or ATEND)
            we jump out of the loop.
*/
            for ( j = 0; j < numthreadbuckets; j++ ) {
                switch ( threadbuckets[j]->free ) {
                    case BUCKETFREE:
                        thr = threadbuckets[j];
                        if ( !endofinput ) goto NextBucket;
                        thr->free = BUCKETATEND;
                        break;
                    case BUCKETCOMINGFREE:
                        thr = threadbuckets[j];
                        if ( endofinput ) {
                            thr->free = BUCKETATEND;
                        }
                        else {
                            thr->free = BUCKETFREE;
                            for ( k = j+1; k < numthreadbuckets; k++ )
                                threadbuckets[k-1] = threadbuckets[k];
                            threadbuckets[numthreadbuckets-1] = thr;
                            j--;
                        }
                        break;
                    default:
                        break;
                }
            }
            if ( j >= numthreadbuckets ) break;
        }
        else {
/*
            No worker available.
            Look for a bucket to load.
            Its number will be in "still"
*/
Finalize:;
            still = -1;
```

```
            for ( j = 0; j < numthreadbuckets; j++ ) {
                switch ( threadbuckets[j]->free ) {
                    case BUCKETFREE:
                        thr = threadbuckets[j];
                        if ( !endofinput ) goto NextBucket;
                        thr->free = BUCKETATEND;
                        break;
                    case BUCKETCOMINGFREE:
                        thr = threadbuckets[j];
                        if ( endofinput ) thr->free = BUCKETATEND;
                        else {
                            thr->free = BUCKETFREE;
                            for ( k = j+1; k < numthreadbuckets; k++ )
                                threadbuckets[k-1] = threadbuckets[k];
                            threadbuckets[numthreadbuckets-1] = thr;
                            j--;
                        }
                        break;
                    case BUCKETFILLED:
                        if ( still < 0 ) still = j;
                        break;
                    default:
                        break;
                }
            }
            if ( still < 0 ) {
/*
                No buckets to be executed and no buckets FREE.
                We must be at the end. Break out of the loop.
*/
                break;
            }
            thr = threadbuckets[still];
            for ( k = still+1; k < numthreadbuckets; k++ )
                threadbuckets[k-1] = threadbuckets[k];
            threadbuckets[numthreadbuckets-1] = thr;
            goto DoBucket;
        }
NextBucket:;
    }
/*
    Now the stage one load balancing.
    If the load has been readjusted we have again filled buckets.
    In that case we jump back in the loop.

    Tricky point: when do the workers see the new value of AT.LoadBalancing?
    It should activate the locks on thr->busy
*/
    if ( AC.ThreadBalancing ) {
        for ( id = 1; id <= numberofworkers; id++ ) {
            AB[id]->T.LoadBalancing = 1;
        }
        if ( LoadReadjusted() ) goto Finalize;
        for ( id = 1; id <= numberofworkers; id++ ) {
            AB[id]->T.LoadBalancing = 0;
        }
    }
    if ( AC.ThreadBalancing ) {
/*
        The AS.Balancing flag should have Generator look for
        free workers and apply the "buro" method.

        There is still a serious problem.
        When for instance a sum_, there may be space created in a local
        compiler buffer for a wildcard substitution or whatever.
        Compiler buffer execution scribble space.....
        This isn't copied along?
        Look up ebufnum. There are 12 places with AddRHS!
        Problem: one process allocates in ebuf. Then term is given to
        other process. It would like to use from this ebuf, but the sender
        finishes first and removes the ebuf (and/or overwrites it).

        Other problem: local $ variables aren't copied along.
*/
        AS.Balancing = 0;
    }
    MasterWaitAll();
    AS.Balancing = 0;
/*
    When we deal with the last expression we can now remove the input
    scratch file. This saves potentially much disk space (up to 1/3)
*/
    if ( LastExpression ) {
        UpdateMaxSize();
        if ( AR0.infile->handle >= 0 ) {
            CloseFile(AR0.infile->handle);
            AR0.infile->handle = -1;
```

```
            remove(AR0.infile->name);
            PUTZERO(AR0.infile->POposition);
            AR0.infile->POfill = AR0.infile->POfull = AR0.infile->PObuffer;
        }
    }
/*
    We order the threads to finish in the MasterMerge routine
    It will start with waiting for all threads to finish.
    One could make an administration in which threads that have
    finished can start already with the final sort but
    1: The load balancing should not make this super urgent
    2: It would definitely not be very compatible with the second
       stage load balancing.
*/
    oldgzipCompress = AR0.gzipCompress;
    AR0.gzipCompress = 0;
    if ( AR0.outtohide ) AR0.outfile = AR0.hidefile;
    if ( MasterMerge() < 0 ) {
        if ( AR0.outtohide ) AR0.outfile = oldoutfile;
        AR0.gzipCompress = oldgzipCompress;
        goto ProcErr;
    }
    if ( AR0.outtohide ) AR0.outfile = oldoutfile;
    AR0.gzipCompress = oldgzipCompress;
/*
    Now wait for all threads to be ready to give them the cleaning up signal.
    With the new MasterMerge routine we can do the cleanup already automatically
    avoiding having to send these signals.
*/
    MasterWaitAll();
    AR0.sLevel--;
    for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
        if ( GetThread(id) > 0 ) WakeupThread(id,CLEANUPEXPRESSION);
    }
    e->numdummies = 0;
    for ( id = 1; id < AM.totalnumberofthreads; id++ ) {
        if ( AB[id]->R.MaxDum - AM.IndDum > e->numdummies )
            e->numdummies = AB[id]->R.MaxDum - AM.IndDum;
        AR0.expchanged |= AB[id]->R.expchanged;
    }
/*
    And wait for all to be clean.
*/
    MasterWaitAll();
    AT0.WorkPointer = oldworkpointer;
    return(0);
ProcErr:;
    return(-1);
}

/*
    #] ThreadsProcessor :
    #[ LoadReadjusted :
*/
int LoadReadjusted(void)
{
    ALLPRIVATES *B0 = AB[0];
    THREADBUCKET *thr = 0, *thrtogo = 0;
    int numtogo, numfree, numbusy, n, nperbucket, extra, i, j, u, bus;
    LONG numinput;
    WORD *t1, *c1, *t2, *c2, *t3;
/*
    Start with waiting for at least one free processor.
    We don't want the master competing for time when all are busy.
*/
    while ( topofavailables <= 0 ) MasterWait();
/*
    Now look for the fullest bucket and make a list of free buckets
    The bad part is that most numbers can change at any moment.
*/
restart:;
    numtogo = 0;
    numfree = 0;
    numbusy = 0;
    for ( j = 0; j < numthreadbuckets; j++ ) {
        thr = threadbuckets[j];
        if ( thr->free == BUCKETFREE || thr->free == BUCKETATEND
        || thr->free == BUCKETCOMINGFREE ) {
            freebuckets[numfree++] = thr;
        }
        else if ( thr->type != BUCKETDOINGTERMS ) {}
        else if ( thr->totnum > 1 ) { /* never steal from a bucket with one term */
            LOCK(thr->lock);
            bus = thr->busy;
            UNLOCK(thr->lock);
            if ( thr->free == BUCKETINUSE ) {
                n = thr->totnum-thr->usenum;
```

```
                if ( bus == BUCKETASSIGNED ) numbusy++;
                else if ( ( bus != BUCKETASSIGNED )
                       && ( n > numtogo ) ) {
                    numtogo = n;
                    thrtogo = thr;
                }
            }
            else if ( bus == BUCKETTOBERELEASED
             && thr->free == BUCKETRELEASED ) {
                freebuckets[numfree++] = thr;
                thr->free = BUCKETATEND;
                LOCK(thr->lock);
                thr->busy = BUCKETPREPARINGTERM;
                UNLOCK(thr->lock);
            }
        }
    }
    if ( numfree == 0 ) return(0); /* serious problem */
    if ( numtogo > 0 ) {   /* provisionally there is something to be stolen */
        thr = thrtogo;
/*
        If the number has changed there is good progress.
        Maybe there is another thread that needs assistance.
        We start all over.
*/
        if ( thr->totnum-thr->usenum < numtogo ) goto restart;
/*
        If the thread is in the term loading phase
        (thr->busy == BUCKETPREPARINGTERM) we better stay away from it.
        We wait now for the thread to be busy, and don't allow it
        now to drop out of this state till we are done here.
        This all depends on whether AT.LoadBalancing == 1 is seen by
        the thread.
*/
        LOCK(thr->lock);
        if ( thr->busy != BUCKETDOINGTERM ) {
            UNLOCK(thr->lock);
            goto restart;
        }
        if ( thr->totnum-thr->usenum < numtogo ) {
            UNLOCK(thr->lock);
            goto restart;
        }
        thr->free = BUCKETTERMINATED;
/*
        The above will signal the thread we want to terminate.
        Next all effort goes into making sure the landing is soft.
        Unfortunately we don't want to wait for a signal, because the thread
        may be working for a long time on a single term.
*/
        if ( thr->usenum == thr->totnum ) {
/*
            Terminated in the mean time or by now working on the
            last term. Try again.
*/
            thr->free = BUCKETATEND;
            UNLOCK(thr->lock);
            goto restart;
        }
        goto intercepted;
    }
/* This has always been commented. Indeed no lock is held here. */
/* UNLOCK(thr->lock); */
    if ( numbusy > 0 ) {
        /* JD: this avoids large runtimes for tform tests under valgrind.
           What seems to happen is we return from here, goto Finalize, and
           end up in LoadReadjusted again without the threads having a
           chance to update their busy status. Then we end up here again.
           Sleep the thread for, say, 1us to allow threads to aquire the lock. */
        struct timespec sleeptime;
        sleeptime.tv_sec = 0;
        sleeptime.tv_nsec = 1000L;
        nanosleep(&sleeptime, NULL);
        return(1); /* Wait a bit.... */
    }
    return(0);
intercepted:;
/*
    We intercepted one successfully. Now it becomes interesting. Action:
    1: determine how many terms per free bucket.
    2: find the first untreated term.
    3: put the terms in the free buckets.

    Remember: we still have the lock to avoid interference from the thread
    that is being robbed. We were holding it and then jumped here with
    goto intercepted.
*/
```

```
    numinput = thr->firstterm + thr->usenum;
    nperbucket = numtogo / numfree;
    extra = numtogo - nperbucket*numfree;
    if ( AR0.DeferFlag ) {
        t1 = thr->threadbuffer; c1 = thr->compressbuffer; u = thr->usenum;
        for ( n = 0; n < thr->usenum; n++ ) { t1 += *t1; c1 += *c1; }
        t3 = t1;
        if ( extra > 0 ) {
          for ( i = 0; i < extra; i++ ) {
            thrtogo = freebuckets[i];
            t2 = thrtogo->threadbuffer;
            c2 = thrtogo->compressbuffer;
            thrtogo->free = BUCKETFILLED;
            thrtogo->type = BUCKETDOINGTERMS;
            thrtogo->totnum = nperbucket+1;
            thrtogo->ddterms = 0;
            thrtogo->usenum = 0;
            thrtogo->busy = BUCKETASSIGNED;
            thrtogo->firstterm = numinput;
            numinput += nperbucket+1;
            for ( n = 0; n <= nperbucket; n++ ) {
                j = *t1; NCOPY(t2,t1,j);
                j = *c1; NCOPY(c2,c1,j);
                thrtogo->deferbuffer[n] = thr->deferbuffer[u++];
            }
            *t2 = *c2 = 0;
          }
        }
        if ( nperbucket > 0 ) {
          for ( i = extra; i < numfree; i++ ) {
            thrtogo = freebuckets[i];
            t2 = thrtogo->threadbuffer;
            c2 = thrtogo->compressbuffer;
            thrtogo->free = BUCKETFILLED;
            thrtogo->type = BUCKETDOINGTERMS;
            thrtogo->totnum = nperbucket;
            thrtogo->ddterms = 0;
            thrtogo->usenum = 0;
            thrtogo->busy = BUCKETASSIGNED;
            thrtogo->firstterm = numinput;
            numinput += nperbucket;
            for ( n = 0; n < nperbucket; n++ ) {
                j = *t1; NCOPY(t2,t1,j);
                j = *c1; NCOPY(c2,c1,j);
                thrtogo->deferbuffer[n] = thr->deferbuffer[u++];
            }
            *t2 = *c2 = 0;
          }
        }
    }
    else {
        t1 = thr->threadbuffer;
        for ( n = 0; n < thr->usenum; n++ ) { t1 += *t1; }
        t3 = t1;
        if ( extra > 0 ) {
          for ( i = 0; i < extra; i++ ) {
            thrtogo = freebuckets[i];
            t2 = thrtogo->threadbuffer;
            thrtogo->free = BUCKETFILLED;
            thrtogo->type = BUCKETDOINGTERMS;
            thrtogo->totnum = nperbucket+1;
            thrtogo->ddterms = 0;
            thrtogo->usenum = 0;
            thrtogo->busy = BUCKETASSIGNED;
            thrtogo->firstterm = numinput;
            numinput += nperbucket+1;
            for ( n = 0; n <= nperbucket; n++ ) {
                j = *t1; NCOPY(t2,t1,j);
            }
            *t2 = 0;
          }
        }
        if ( nperbucket > 0 ) {
          for ( i = extra; i < numfree; i++ ) {
            thrtogo = freebuckets[i];
            t2 = thrtogo->threadbuffer;
            thrtogo->free = BUCKETFILLED;
            thrtogo->type = BUCKETDOINGTERMS;
            thrtogo->totnum = nperbucket;
            thrtogo->ddterms = 0;
            thrtogo->usenum = 0;
            thrtogo->busy = BUCKETASSIGNED;
            thrtogo->firstterm = numinput;
            numinput += nperbucket;
            for ( n = 0; n < nperbucket; n++ ) {
                j = *t1; NCOPY(t2,t1,j);
            }
```

```
            *t2 = 0;
          }
        }
    }
    *t3 = 0;   /* This is some form of extra insurance */
    if ( thr->free == BUCKETRELEASED && thr->busy == BUCKETTOBERELEASED ) {
        thr->free = BUCKETATEND; thr->busy = BUCKETPREPARINGTERM;
    }
    UNLOCK(thr->lock);
    return(1);
}

/*
    #] LoadReadjusted :
    #[ SortStrategy :
*/
/*
    #] SortStrategy :
    #[ PutToMaster :
*/
int PutToMaster(PHEAD WORD *term)
{
    int i,j,nexti,ret = 0;
    WORD *t, *fill, *top, zero = 0;
    if ( term == 0 ) { /* Mark the end of the expression */
        t = &zero; j = 1;
    }
    else {
        t = term; ret = j = *term;
        if ( j == 0 ) { j = 1; } /* Just in case there is a spurious end */
    }
    i = AT.SB.FillBlock;           /* The block we are working at */
    fill = AT.SB.MasterFill[i];      /* Where we are filling */
    top = AT.SB.MasterStop[i];       /* End of the block */
    while ( j > 0 ) {
        while ( j > 0 && fill < top ) {
            *fill++ = *t++; j--;
        }
        if ( j > 0 ) {
/*
            We reached the end of the block.
            Get the next block and release this block.
            The order is important. This is why there should be at least
            4 blocks or deadlocks can occur.
*/
            nexti = i+1;
            if ( nexti > AT.SB.MasterNumBlocks ) {
                nexti = 1;
            }
            LOCK(AT.SB.MasterBlockLock[nexti]);
            UNLOCK(AT.SB.MasterBlockLock[i]);
            AT.SB.MasterFill[i] = AT.SB.MasterStart[i];
            AT.SB.FillBlock = i = nexti;
            fill = AT.SB.MasterStart[i];
            top = AT.SB.MasterStop[i];
        }
    }
    AT.SB.MasterFill[i] = fill;
    return(ret);
}

/*
    #] PutToMaster :
    #[ SortBotOut :
*/

#ifdef WITHSORTBOTS

int
SortBotOut(PHEAD WORD *term)
{
    WORD im;

    if ( AT.identity != 0 ) return(PutToMaster(BHEAD term));

    if ( term == 0 ) {
        if ( FlushOut(&SortBotPosition,AR.outfile,1) ) return(-1);
        ADDPOS(AT.SS->SizeInFile[0],1);
        return(0);
    }
    else {
        numberofterms++;
        if ( ( im = PutOut(BHEAD term,&SortBotPosition,AR.outfile,1) ) < 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Called from MasterMerge/SortBotOut");
            MUNLOCK(ErrorMessageLock);
            return(-1);
```

```
        }
        ADDPOS(AT.SS->SizeInFile[0],im);
        return(im);
    }
}

#endif

/*
    #] SortBotOut :
    #[ MasterMerge :
*/
int MasterMerge(void)
{
    ALLPRIVATES *B0 = AB[0], *B = 0;
    SORTING *S = AT0.SS;
    WORD **poin, **poin2, ul, k, i, im, *m1, j;
    WORD lpat, mpat, level, l1, l2, r1, r2, r3, c;
    WORD *m2, *m3, r31, r33, ki, *rr;
    UWORD *coef;
    POSITION position;
    FILEHANDLE *fin, *fout;
#ifdef WITHSORTBOTS
    if ( numberofworkers > 2 ) return(SortBotMasterMerge());
#endif
    fin = &S->file;
    if ( AR0.PolyFun == 0 ) { S->PolyFlag = 0; }
    else if ( AR0.PolyFunType == 1 ) { S->PolyFlag = 1; }
    else if ( AR0.PolyFunType == 2 ) {
        if ( AR0.PolyFunExp == 2
          || AR0.PolyFunExp == 3 ) S->PolyFlag = 1;
        else                       S->PolyFlag = 2;
    }
    S->TermsLeft = 0;
    coef = AN0.SoScratC;
    poin = S->poina; poin2 = S->poin2a;
    rr = AR0.CompressPointer;
    *rr = 0;
/*
        #[ Setup :
*/
    S->inNum = numberofthreads;
    fout = AR0.outfile;
/*
    Load the patches. The threads have to finish their sort first.
*/
    S->lPatch = S->inNum - 1;
/*
    Claim all zero blocks. We need them anyway.
    In principle the workers should never get into these.
    We also claim all last blocks. This is a safety procedure that
    should prevent the workers from working their way around the clock
    before the master gets started again.
*/
    AS.MasterSort = 1;
    numberclaimed = 0;
    for ( i = 1; i <= S->lPatch; i++ ) {
        B = AB[i];
        LOCK(AT.SB.MasterBlockLock[0]);
        LOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
    }
/*
    Now wake up the threads and have them start their final sorting.
    They should start with claiming their block and the master is
    not allowed to continue until that has been done.
    This waiting of the master will be done below in MasterWaitAllBlocks
*/
    for ( i = 0; i < S->lPatch; i++ ) {
        GetThread(i+1);
        WakeupThread(i+1,FINISHEXPRESSION);
    }
/*
    Prepare the output file.
*/
    if ( fout->handle >= 0 ) {
        PUTZERO(position);
        SeekFile(fout->handle,&position,SEEK_END);
        ADDPOS(position,((fout->POfill-fout->PObuffer)*sizeof(WORD)));
    }
    else {
        SETBASEPOSITION(position,(fout->POfill-fout->PObuffer)*sizeof(WORD));
    }
/*
    Wait for all threads to finish loading their first block.
*/
    MasterWaitAllBlocks();
/*
```

```
    Claim all first blocks.
    We don't release the last blocks.
    The strategy is that we always keep the previous block.
    In principle it looks like it isn't needed for the last block but
    actually it is to keep the front from overrunning the tail when writing.
*/
    for ( i = 1; i <= S->lPatch; i++ ) {
        B = AB[i];
        LOCK(AT.SB.MasterBlockLock[1]);
        AT.SB.MasterBlock = 1;
    }
/*
        #] Setup :

    Now construct the tree:
*/
    lpat = 1;
    do { lpat «= 1; } while ( lpat < S->lPatch );
    mpat = ( lpat » 1 ) - 1;
    k = lpat - S->lPatch;
/*
    k is the number of empty places in the tree. they will
    be at the even positions from 2 to 2*k
*/
    for ( i = 1; i < lpat; i++ ) { S->tree[i] = -1; }
    for ( i = 1; i <= k; i++ ) {
        im = ( i * 2 ) - 1;
        poin[im] = AB[i]->T.SB.MasterStart[AB[i]->T.SB.MasterBlock];
        poin2[im] = poin[im] + *(poin[im]);
        S->used[i] = im;
        S->ktoi[im] = i-1;
        S->tree[mpat+i] = 0;
        poin[im-1] = poin2[im-1] = 0;
    }
    for ( i = (k*2)+1; i <= lpat; i++ ) {
        S->used[i-k] = i;
        S->ktoi[i] = i-k-1;
        poin[i] = AB[i-k]->T.SB.MasterStart[AB[i-k]->T.SB.MasterBlock];
        poin2[i] = poin[i] + *(poin[i]);
    }
/*
    the array poin tells the position of the i-th element of the S->tree
    'S->used' is a stack with the S->tree elements that need to be entered
    into the S->tree. at the beginning this is S->lPatch. during the
    sort there will be only very few elements.
    poin2 is the next value of poin. it has to be determined
    before the comparisons as the position or the size of the
    term indicated by poin may change.
    S->ktoi translates a S->tree element back to its stream number.

    start the sort
*/
    level = S->lPatch;
/*
    introduce one term
*/
OneTerm:
    k = S->used[level];
    i = k + lpat - 1;
    if ( !*(poin[k]) ) {
        do { if ( !( i »= 1 ) ) goto EndOfMerge; } while ( !S->tree[i] );
        if ( S->tree[i] == -1 ) {
            S->tree[i] = 0;
            level--;
            goto OneTerm;
        }
        k = S->tree[i];
        S->used[level] = k;
        S->tree[i] = 0;
    }
/*
    move terms down the tree
*/
    while ( i »= 1 ) {
        if ( S->tree[i] > 0 ) {
            if ( ( c = CompareTerms(B0, poin[S->tree[i]],poin[k],(WORD)0) ) > 0 ) {
/*
                S->tree[i] is the smaller. Exchange and go on.
*/
                S->used[level] = S->tree[i];
                S->tree[i] = k;
                k = S->used[level];
            }
            else if ( !c ) {    /* Terms are equal */
/*
                S->TermsLeft--;
                    Here the terms are equal and their coefficients
```

```
                    have to be added.
*/
                l1 = *( m1 = poin[S->tree[i]] );
                l2 = *( m2 = poin[k] );
                if ( S->PolyWise ) {  /* Here we work with PolyFun */
                    WORD *tt1, *w;
                    tt1 = m1;
                    m1 += S->PolyWise;
                    m2 += S->PolyWise;
                    if ( S->PolyFlag == 2 ) {
                        w = poly_ratfun_add(B0,m1,m2);
                        if ( *tt1 + w[1] - m1[1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l is too small",AM.MaxTer);
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        }
                        AT0.WorkPointer = w;
                        if ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 && w[1] > FUNHEAD ) {
                            goto cancelled;
                        }
                    }
                    else {
                        w = AT0.WorkPointer;
                        if ( w + m1[1] + m2[1] > AT0.WorkTop ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("MasterMerge: A WorkSpace of %10l is too small",AM.WorkSize);
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        }
                        AddArgs(B0,m1,m2,w);
                    }
                    r1 = w[1];
                    if ( r1 <= FUNHEAD
                        || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) )
                             { goto cancelled; }
                    if ( r1 == m1[1] ) {
                        NCOPY(m1,w,r1);
                    }
                    else if ( r1 < m1[1] ) {
                        r2 = m1[1] - r1;
                        m2 = w + r1;
                        m1 += m1[1];
                        while ( --r1 >= 0 ) *--m1 = *--m2;
                        m2 = m1 - r2;
                        r1 = S->PolyWise;
                        while ( --r1 >= 0 ) *--m1 = *--m2;
                        *m1 -= r2;
                        poin[S->tree[i]] = m1;
                    }
                    else {
                        r2 = r1 - m1[1];
                        m2 = tt1 - r2;
                        r1 = S->PolyWise;
                        m1 = tt1;
                        *m1 += r2;
                        poin[S->tree[i]] = m2;
                        NCOPY(m2,m1,r1);
                        r1 = w[1];
                        NCOPY(m2,w,r1);
                    }
                }
#ifdef WITHFLOAT
                else if ( AT.SortFloatMode ) {
                    WORD *term1, *term2;
                    term1 = poin[S->tree[i]];
                    term2 = poin[k];
                    if ( MergeWithFloat(B0, &term1,&term2) == 0 )
                        goto cancelled;
                    poin[S->tree[i]] = term1;
                }
#endif
                else {
                    r1 = *( m1 += l1 - 1 );
                    m1 -= ABS(r1) - 1;
                    r1 = ( ( r1 > 0 ) ? (r1-1) : (r1+1) ) » 1;
                    r2 = *( m2 += l2 - 1 );
                    m2 -= ABS(r2) - 1;
                    r2 = ( ( r2 > 0 ) ? (r2-1) : (r2+1) ) » 1;

                    if ( AddRat(B0,(UWORD *)m1,r1,(UWORD *)m2,r2,coef,&r3) ) {
                        MLOCK(ErrorMessageLock);
                        MesCall("MasterMerge");
                        MUNLOCK(ErrorMessageLock);
                        SETERROR(-1)
                    }
```

```
                    if ( AN.ncmod != 0 ) {
                        if ( ( AC.modmode & POSNEG ) != 0 ) {
                            NormalModulus(coef,&r3);
                        }
                        else if ( BigLong(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
                            WORD ii;
                            SubPLon(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod),coef,&r3);
                            coef[r3] = 1;
                            for ( ii = 1; ii < r3; ii++ ) coef[r3+ii] = 0;
                        }
                    }
                    r3 *= 2;
                    r33 = ( r3 > 0 ) ? ( r3 + 1 ) : ( r3 - 1 );
                    if ( r3 < 0 ) r3 = -r3;
                    if ( r1 < 0 ) r1 = -r1;
                    r1 *= 2;
                    r31 = r3 - r1;
                    if ( !r3 ) {        /* Terms cancel */
cancelled:
                        ul = S->used[level] = S->tree[i];
                        S->tree[i] = -1;
/*
                        We skip to the next term in stream ul
*/
                        im = *poin2[ul];
                        poin[ul] = poin2[ul];
                        ki = S->ktoi[ul];
                        if ( (poin[ul] + im + COMPINC) >=
                        AB[ki+1]->T.SB.MasterStop[AB[ki+1]->T.SB.MasterBlock]
                        && im > 0 ) {
/*
                            We made it to the end of the block. We have to
                            release the previous block and claim the next.
*/
                            B = AB[ki+1];
                            i = AT.SB.MasterBlock;
                            if ( i == 1 ) {
                                UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
                            }
                            else {
                                UNLOCK(AT.SB.MasterBlockLock[i-1]);
                            }
                            if ( i == AT.SB.MasterNumBlocks ) {
/*
                                Move the remainder down into block 0
*/
                                WORD *from, *to;
                                to = AT.SB.MasterStart[1];
                                from = AT.SB.MasterStop[i];
                                while ( from > poin[ul] ) *--to = *--from;
                                poin[ul] = to;
                                i = 1;
                            }
                            else { i++; }
                            LOCK(AT.SB.MasterBlockLock[i]);
                            AT.SB.MasterBlock = i;
                            poin2[ul] = poin[ul] + im;
                        }
                        else {
                            poin2[ul] += im;
                        }
                        S->used[++level] = k;
                    }
                    else if ( !r31 ) {      /* copy coef into term1 */
                        goto CopCof2;
                    }
                    else if ( r31 < 0 ) {       /* copy coef into term1
                                            and adjust the length of term1 */
                        goto CopCoef;
                    }
                    else {
/*
                            this is the dreaded calamity.
                            is there enough space?
*/
                        if( (poin[S->tree[i]]+l1+r31) >= poin2[S->tree[i]] ) {
/*
                                no space! now the special trick for which
                                we left 2*maxlng spaces open at the beginning
                                of each patch.
*/
                            if ( (l1 + r31)*((LONG)sizeof(WORD)) >= AM.MaxTer ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("MasterMerge: Coefficient overflow during sort");
                                MUNLOCK(ErrorMessageLock);
                                goto ReturnError;
```

```
                                }
                                m2 = poin[S->tree[i]];
                                m3 = ( poin[S->tree[i]] -= r31 );
                                do { *m3++ = *m2++; } while ( m2 < m1 );
                                m1 = m3;
                            }
CopCoef:
                            *(poin[S->tree[i]]) += r31;
CopCof2:
                            m2 = (WORD *)coef; im = r3;
                            NCOPY(m1,m2,im);
                            *m1 = r33;
                        }
                    }
/*
                    Now skip to the next term in stream k.
*/
NextTerm:
                    im = poin2[k][0];
                    poin[k] = poin2[k];
                    ki = S->ktoi[k];
                    if ( (poin[k] + im + COMPINC) >=
                    AB[ki+1]->T.SB.MasterStop[AB[ki+1]->T.SB.MasterBlock]
                    && im > 0 ) {
/*
                    We made it to the end of the block. We have to
                    release the previous block and claim the next.
*/
                        B = AB[ki+1];
                        i = AT.SB.MasterBlock;
                        if ( i == 1 ) {
                            UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
                        }
                        else {
                            UNLOCK(AT.SB.MasterBlockLock[i-1]);
                        }
                        if ( i == AT.SB.MasterNumBlocks ) {
/*
                            Move the remainder down into block 0
*/
                            WORD *from, *to;
                            to = AT.SB.MasterStart[1];
                            from = AT.SB.MasterStop[i];
                            while ( from > poin[k] ) *--to = *--from;
                            poin[k] = to;
                            i = 1;
                        }
                        else { i++; }
                        LOCK(AT.SB.MasterBlockLock[i]);
                        AT.SB.MasterBlock = i;
                        poin2[k] = poin[k] + im;
                    }
                    else {
                        poin2[k] += im;
                    }
                    goto OneTerm;
                }
            }
            else if ( S->tree[i] < 0 ) {
                S->tree[i] = k;
                level--;
                goto OneTerm;
            }
        }
/*
        found the smallest in the set. indicated by k.
        write to its destination.
*/
        S->TermsLeft++;
        if ( ( im = PutOut(B0,poin[k],&position,fout,1) ) < 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Called from MasterMerge with k = %d (stream %d)",k,S->ktoi[k]);
            MUNLOCK(ErrorMessageLock);
            goto ReturnError;
        }
        ADDPOS(S->SizeInFile[0],im);
        goto NextTerm;
EndOfMerge:
        if ( FlushOut(&position,fout,1) ) goto ReturnError;
        ADDPOS(S->SizeInFile[0],1);
        CloseFile(fin->handle);
        remove(fin->name);
        fin->handle = -1;
        position = S->SizeInFile[0];
        MULPOS(position,sizeof(WORD));
        S->GenTerms = 0;
        for ( j = 1; j <= numberofworkers; j++ ) {
```

```
        S->GenTerms += AB[j]->T.SS->GenTerms;
    }
    WriteStats(&position,STATSPOSTSORT,NOCHECKLOGTYPE);
    Expressions[AR0.CurExpr].counter = S->TermsLeft;
    Expressions[AR0.CurExpr].size = position;
/*
    Release all locks
*/
    for ( i = 1; i <= S->lPatch; i++ ) {
        B = AB[i];
        UNLOCK(AT.SB.MasterBlockLock[0]);
        if ( AT.SB.MasterBlock == 1 ) {
            UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
        }
        else {
            UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock-1]);
        }
        UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock]);
    }
    AS.MasterSort = 0;
    return(0);
ReturnError:
    for ( i = 1; i <= S->lPatch; i++ ) {
        B = AB[i];
        UNLOCK(AT.SB.MasterBlockLock[0]);
        if ( AT.SB.MasterBlock == 1 ) {
            UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterNumBlocks]);
        }
        else {
            UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock-1]);
        }
        UNLOCK(AT.SB.MasterBlockLock[AT.SB.MasterBlock]);
    }
    AS.MasterSort = 0;
    return(-1);
}

/*
    #] MasterMerge :
    #[ SortBotMasterMerge :
*/

#ifdef WITHSORTBOTS

int SortBotMasterMerge(void)
{
    FILEHANDLE *fin, *fout;
    ALLPRIVATES *B = AB[0], *BB;
    POSITION position;
    SORTING *S = AT.SS;
    int i, j;
/*
    Get the sortbots get to claim their writing blocks.
    We have to wait till all have been claimed because they also have to
    claim the last writing blocks of the workers to prevent the head of
    the circular buffer to overrun the tail.

    Before waiting we can do some needed initializations.
    Also the master has to claim the last writing blocks of its input.
*/
    topsortbotavailables = 0;
    for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
        WakeupThread(i,INISORTBOT);
    }

    AS.MasterSort = 1;
    fout = AR.outfile;
    numberofterms = 0;
    AR.CompressPointer[0] = 0;
    numberclaimed = 0;
    BB = AB[AT.SortBotIn1];
    LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);
    BB = AB[AT.SortBotIn2];
    LOCK(BB->T.SB.MasterBlockLock[BB->T.SB.MasterNumBlocks]);

    MasterWaitAllSortBots();
/*
    Now we can start up the workers. They will claim their writing blocks.
    Here the master will wait till all writing blocks have been claimed.
*/
    for ( i = 1; i <= numberofworkers; i++ ) {
        j = GetThread(i);
        WakeupThread(i,FINISHEXPRESSION);
    }
/*
    Prepare the output file in the mean time.
*/
```

```
    if ( fout->handle >= 0 ) {
        PUTZERO(SortBotPosition);
        SeekFile(fout->handle,&SortBotPosition,SEEK_END);
        ADDPOS(SortBotPosition,((fout->POfill-fout->PObuffer)*sizeof(WORD)));
    }
    else {
        SETBASEPOSITION(SortBotPosition,(fout->POfill-fout->PObuffer)*sizeof(WORD));
    }
    MasterWaitAllBlocks();
/*
    Now we can start the sortbots after which the master goes in
    sortbot mode to do its part of the job (the very final merge and
    the writing to output file).
*/
    topsortbotavailables = 0;
    for ( i = numberofworkers+1; i <= numberofworkers+numberofsortbots; i++ ) {
        WakeupThread(i,RUNSORTBOT);
    }
    if ( SortBotMerge(BHEAD0) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Called from SortBotMasterMerge");
        MUNLOCK(ErrorMessageLock);
        AS.MasterSort = 0;
        return(-1);
    }
/*
    And next the cleanup
*/
    if ( S->file.handle >= 0 )
    {
        fin = &S->file;
        CloseFile(fin->handle);
        remove(fin->name);
        fin->handle = -1;
    }
    position = S->SizeInFile[0];
    MULPOS(position,sizeof(WORD));
    S->GenTerms = 0;
    for ( j = 1; j <= numberofworkers; j++ ) {
        S->GenTerms += AB[j]->T.SS->GenTerms;
    }
    S->TermsLeft = numberofterms;
    WriteStats(&position,STATSPOSTSORT,NOCHECKLOGTYPE);
    Expressions[AR.CurExpr].counter = S->TermsLeft;
    Expressions[AR.CurExpr].size = position;
    AS.MasterSort = 0;
/*
    The next statement is to prevent one of the sortbots not having
    completely cleaned up before the next module starts.
    If this statement is omitted every once in a while one of the sortbots
    is still running when the next expression starts and misses its
    initialization. The result is usually disastrous.
*/
    MasterWaitAllSortBots();

    return(0);
}

#endif

/*
    #] SortBotMasterMerge :
    #[ SortBotMerge :
*/

#ifdef WITHSORTBOTS

int SortBotMerge(PHEAD0)
{
    GETBIDENTITY
    ALLPRIVATES *Bin1 = AB[AT.SortBotIn1],*Bin2 = AB[AT.SortBotIn2];
    WORD *term1, *term2, *next, *wp;
    int blin1, blin2;    /* Current block numbers */
    int error = 0;
    WORD l1, l2, *m1, *m2, *w, r1, r2, r3, r33, r31, *tt1, ii;
    WORD *to, *from, im, c;
    UWORD *coef;
    SORTING *S = AT.SS;
#ifdef WITHFLOAT
    WORD *fun1, *fun2, *fun3, *tstop1, *tstop2, size1, size2, size3, l3, jj, *m3;
#endif
/*
    Set the pointers to the input terms and the output space
*/
    coef = AN.SoScratC;
    blin1 = 1;
    blin2 = 1;
```

```
    if ( AT.identity == 0 ) {
        wp = AT.WorkPointer;
    }
    else {
        wp = AT.WorkPointer = AT.WorkSpace;
    }
/*
    Get the locks for reading the input
    This means that we can start once these locks have been cleared
    which means that there will be input.
*/
    LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
    LOCK(Bin2->T.SB.MasterBlockLock[blin2]);

    term1 = Bin1->T.SB.MasterStart[blin1];
    term2 = Bin2->T.SB.MasterStart[blin2];
    AT.SB.FillBlock = 1;
/*
    Now the main loop. Keep going until one of the two hits the end.
*/
    while ( *term1 && *term2 ) {
        if ( ( c = CompareTerms(BHEAD term1,term2,(WORD)0) ) > 0 ) {
/*
            #[ One is smallest :
*/
            if ( SortBotOut(BHEAD term1) < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
                MUNLOCK(ErrorMessageLock);
                error = -1;
                goto ReturnError;
            }
            im = *term1;
            next = term1 + im;
            if ( next >= Bin1->T.SB.MasterStop[blin1] || ( *next &&
            next+*next+COMPINC > Bin1->T.SB.MasterStop[blin1] ) ) {
                if ( blin1 == 1 ) {
                    UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
                }
                else {
                    UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
                }
                if ( blin1 == Bin1->T.SB.MasterNumBlocks ) {
/*
                    Move the remainder down into block 0
*/
                    to = Bin1->T.SB.MasterStart[1];
                    from = Bin1->T.SB.MasterStop[Bin1->T.SB.MasterNumBlocks];
                    while ( from > next ) *--to = *--from;
                    next = to;
                    blin1 = 1;
                }
                else {
                    blin1++;
                }
                LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
                Bin1->T.SB.MasterBlock = blin1;
            }
            term1 = next;
/*
            #] One is smallest :
*/
        }
        else if ( c < 0 ) {
/*
            #[ Two is smallest :
*/
            if ( SortBotOut(BHEAD term2) < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
                MUNLOCK(ErrorMessageLock);
                error = -1;
                goto ReturnError;
            }
next2:      im = *term2;
            next = term2 + im;
            if ( next >= Bin2->T.SB.MasterStop[blin2] || ( *next
            && next+*next+COMPINC > Bin2->T.SB.MasterStop[blin2] ) ) {
                if ( blin2 == 1 ) {
                    UNLOCK(Bin2->T.SB.MasterBlockLock[Bin2->T.SB.MasterNumBlocks]);
                }
                else {
                    UNLOCK(Bin2->T.SB.MasterBlockLock[blin2-1]);
                }
                if ( blin2 == Bin2->T.SB.MasterNumBlocks ) {
/*
                    Move the remainder down into block 0
```

```
*/
                    to = Bin2->T.SB.MasterStart[1];
                    from = Bin2->T.SB.MasterStop[Bin2->T.SB.MasterNumBlocks];
                    while ( from > next ) *--to = *--from;
                    next = to;
                    blin2 = 1;
                }
                else {
                    blin2++;
                }
                LOCK(Bin2->T.SB.MasterBlockLock[blin2]);
                Bin2->T.SB.MasterBlock = blin2;
            }
            term2 = next;
/*
            #] Two is smallest :
*/
        }
        else {
/*
            #[ Equal :
*/
            l1 = *( m1 = term1 );
            l2 = *( m2 = term2 );
            if ( S->PolyWise ) {  /* Here we work with PolyFun */
                tt1 = m1;
                m1 += S->PolyWise;
                m2 += S->PolyWise;
                if ( S->PolyFlag == 2 ) {
                    AT.WorkPointer = wp;
                    w = poly_ratfun_add(BHEAD m1,m2);
                    if ( *tt1 + w[1] - m1[1] > AM.MaxTer/((LONG)sizeof(WORD)) ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Term too complex in PolyRatFun addition. MaxTermSize of %10l is too small",AM.MaxTer);
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                    }
                    AT.WorkPointer = wp;
                    if ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 && w[1] > FUNHEAD ) {
                        goto cancelled;
                    }
                }
                else {
                    w = wp;
                    if ( w + m1[1] + m2[1] > AT.WorkTop ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("SortBotMerge(%d): A Maxtermsize of %10l is too small",
                                AT.identity,AM.MaxTer/sizeof(WORD));
                        MesPrint("m1[1] = %d, m2[1] = %d, Space = %l",m1[1],m2[1],(LONG)(AT.WorkTop-wp));
                        PrintTerm(term1,"term1");
                        PrintTerm(term2,"term2");
                        MesPrint("PolyWise = %d",S->PolyWise);
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                    }
                    AddArgs(BHEAD m1,m2,w);
                }
                r1 = w[1];
                if ( r1 <= FUNHEAD
                    || ( w[FUNHEAD] == -SNUMBER && w[FUNHEAD+1] == 0 ) )
                         { goto cancelled; }
                if ( r1 == m1[1] ) {
                    NCOPY(m1,w,r1);
                }
                else if ( r1 < m1[1] ) {
                    r2 = m1[1] - r1;
                    m2 = w + r1;
                    m1 += m1[1];
                    while ( --r1 >= 0 ) *--m1 = *--m2;
                    m2 = m1 - r2;
                    r1 = S->PolyWise;
                    while ( --r1 >= 0 ) *--m1 = *--m2;
                    *m1 -= r2;
                    term1 = m1;
                }
                else {
                    r2 = r1 - m1[1];
                    m2 = tt1 - r2;
                    r1 = S->PolyWise;
                    m1 = tt1;
                    *m1 += r2;
                    term1 = m2;
                    NCOPY(m2,m1,r1);
                    r1 = w[1];
                    NCOPY(m2,w,r1);
```

```
                }
            }
#ifdef WITHFLOAT
            else if ( AT.SortFloatMode ) {
/*
                The terms are in m1/term1 and m2/term2 and their length in l1 and l2.
                We have to locate the floats which have already been
                verified in the compare routine. Once we have the new
                term we can jump to the code that writes away the
                result after adding the rationals.
*/
                tstop1 = m1+l1; size1 = tstop1[-1]; tstop1 -= ABS(size1);
                tstop2 = m2+l2; size2 = tstop2[-1]; tstop2 -= ABS(size2);
                if ( AT.SortFloatMode == 3 ) {
                    fun1 = m1+1;
                    while ( fun1[0] != FLOATFUN && fun1+fun1[1] < tstop1 ) {
                        fun1 += fun1[1];
                    }
                    if ( size1 < 0 ) fun1[FUNHEAD+3] = -fun1[FUNHEAD+3];
                    UnpackFloat(aux1,fun1);
                    fun2 = m2+1;
                    while ( fun2[0] != FLOATFUN && fun2+fun2[1] < tstop2 ) {
                        fun2 += fun2[1];
                    }
                    if ( size2 < 0 ) fun2[FUNHEAD+3] = -fun2[FUNHEAD+3];
                    UnpackFloat(aux2,fun2);
                }
                else if ( AT.SortFloatMode == 1 ) {
                    fun1 = m1+1;
                    while ( fun1[0] != FLOATFUN && fun1+fun1[1] < tstop1 ) {
                        fun1 += fun1[1];
                    }
                    if ( size1 < 0 ) fun1[FUNHEAD+3] = -fun1[FUNHEAD+3];
                    UnpackFloat(aux1,fun1);
                    fun2 = tstop2;
                    RatToFloat(aux2,(UWORD *)fun2,size2);
                }
                else if ( AT.SortFloatMode == 2 ) {
                    fun1 = tstop1;
                    RatToFloat(aux1,(UWORD *)fun1,size1);
                    fun2 = m2+1;
                    while ( fun2[0] != FLOATFUN && fun2+fun2[1] < tstop2 ) {
                        fun2 += fun2[1];
                    }
                    if ( size2 < 0 ) fun2[FUNHEAD+3] = -fun2[FUNHEAD+3];
                    UnpackFloat(aux2,fun2);
                }
                else {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Illegal value %d for AT.SortFloatMode in
SortBotMerge.",AT.SortFloatMode);
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                    return(0);
                }
                mpf_add(aux3,aux1,aux2);
                size3 = mpf_sgn(aux3);
                if ( size3 == 0 ) { /* Cancelling! Rare! */
                    goto cancelled;
                }
                else if ( size3 < 0 ) mpf_neg(aux3,aux3);
                fun3 = TermMalloc("MasterMerge");
                PackFloat(fun3,aux3);
                l3 = fun3[1]+(fun1-m1)+3; /* new size */

                if ( l3 != l1 ) {
/*
                    Copy it all to wp
*/
                    if ( (l3)*((LONG)sizeof(WORD)) >= AM.MaxTer ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("MasterMerge: Coefficient overflow during sort");
                        MUNLOCK(ErrorMessageLock);
                        goto ReturnError;
                    }
                    m3 = wp; m2 = term1;
                    while ( m2 < fun1 ) *m3++ = *m2++;
                    for ( jj = 0; jj < fun3[1]; jj++ ) *m3++ = fun3[jj];
                    *m3++ = 1; *m3++ = 1;
                    *m3++ = size3 < 0 ? -3: 3;
                    *wp = m3-wp;
                    TermFree(fun3,"MasterMerge");
                    goto PutOutwp;
                }
                for ( jj = 0; jj < fun3[1]; jj++ ) fun1[jj] = fun3[jj];
                fun1 += fun3[1];
                *fun1++ = 1; *fun1++ = 1;
```

```
                *fun1++ = size3 < 0 ? -3: 3;
                *term1 = fun1-term1;
                TermFree(fun3,"MasterMerge");
            }
#endif
            else {
                r1 = *( m1 += l1 - 1 );
                m1 -= ABS(r1) - 1;
                r1 = ( ( r1 > 0 ) ? (r1-1) : (r1+1) ) >> 1;
                r2 = *( m2 += l2 - 1 );
                m2 -= ABS(r2) - 1;
                r2 = ( ( r2 > 0 ) ? (r2-1) : (r2+1) ) >> 1;

                if ( AddRat(BHEAD (UWORD *)m1,r1,(UWORD *)m2,r2,coef,&r3) ) {
                    MLOCK(ErrorMessageLock);
                    MesCall("SortBotMerge");
                    MUNLOCK(ErrorMessageLock);
                    SETERROR(-1)
                }

                if ( AN.ncmod != 0 ) {
                    if ( ( AC.modmode & POSNEG ) != 0 ) {
                        NormalModulus(coef,&r3);
                    }
                    else if ( BigLong(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod)) >= 0 ) {
                        SubPLon(coef,r3,(UWORD *)AC.cmod,ABS(AN.ncmod),coef,&r3);
                        coef[r3] = 1;
                        for ( ii = 1; ii < r3; ii++ ) coef[r3+ii] = 0;
                    }
                }
                if ( !r3 ) { goto cancelled; }
                r3 *= 2;
                r33 = ( r3 > 0 ) ? ( r3 + 1 ) : ( r3 - 1 );
                if ( r3 < 0 ) r3 = -r3;
                if ( r1 < 0 ) r1 = -r1;
                r1 *= 2;
                r31 = r3 - r1;
                if ( !r31 ) {       /* copy coef into term1 */
                    m2 = (WORD *)coef; im = r3;
                    NCOPY(m1,m2,im);
                    *m1 = r33;
                }
                else {
                    to = wp; from = term1;
                    while ( from < m1 ) *to++ = *from++;
                    from = (WORD *)coef; im = r3;
                    NCOPY(to,from,im);
                    *to++ = r33;
                    wp[0] = to - wp;
PutOutwp:
                    if ( SortBotOut(BHEAD wp) < 0 ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
                        MUNLOCK(ErrorMessageLock);
                        error = -1;
                        goto ReturnError;
                    }
                    goto cancelled;
                }
            }
            if ( SortBotOut(BHEAD term1) < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
                MUNLOCK(ErrorMessageLock);
                error = -1;
                goto ReturnError;
            }
cancelled:;     /* Now we need two new terms */
            im = *term1;
            next = term1 + im;
            if ( next >= Bin1->T.SB.MasterStop[blin1] || ( *next &&
            next+*next+COMPINC > Bin1->T.SB.MasterStop[blin1] ) ) {
                if ( blin1 == 1 ) {
                    UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
                }
                else {
                    UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
                }
                if ( blin1 == Bin1->T.SB.MasterNumBlocks ) {
/*
                    Move the remainder down into block 0
*/
                    to = Bin1->T.SB.MasterStart[1];
                    from = Bin1->T.SB.MasterStop[Bin1->T.SB.MasterNumBlocks];
                    while ( from > next ) *--to = *--from;
                    next = to;
                    blin1 = 1;
```

```
                }
                else {
                    blin1++;
                }
                LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
                Bin1->T.SB.MasterBlock = blin1;
            }
            term1 = next;
            goto next2;
/*
            #] Equal :
*/
        }
    }
/*
    Copy the tail
*/
    if ( *term1 ) {
/*
            #[ Tail in one :
*/
        while ( *term1 ) {
            if ( SortBotOut(BHEAD term1) < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
                MUNLOCK(ErrorMessageLock);
                error = -1;
                goto ReturnError;
            }
            im = *term1;
            next = term1 + im;
            if ( next >= Bin1->T.SB.MasterStop[blin1] || ( *next &&
            next+*next+COMPINC > Bin1->T.SB.MasterStop[blin1] ) ) {
                if ( blin1 == 1 ) {
                    UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
                }
                else {
                    UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
                }
                if ( blin1 == Bin1->T.SB.MasterNumBlocks ) {
/*
                    Move the remainder down into block 0
*/
                    to = Bin1->T.SB.MasterStart[1];
                    from = Bin1->T.SB.MasterStop[Bin1->T.SB.MasterNumBlocks];
                    while ( from > next ) *--to = *--from;
                    next = to;
                    blin1 = 1;
                }
                else {
                    blin1++;
                }
                LOCK(Bin1->T.SB.MasterBlockLock[blin1]);
                Bin1->T.SB.MasterBlock = blin1;
            }
            term1 = next;
        }
/*
            #] Tail in one :
*/
    }
    else if ( *term2 ) {
/*
            #[ Tail in two :
*/
        while ( *term2 ) {
            if ( SortBotOut(BHEAD term2) < 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Called from SortBotMerge with thread = %d",AT.identity);
                MUNLOCK(ErrorMessageLock);
                error = -1;
                goto ReturnError;
            }
            im = *term2;
            next = term2 + im;
            if ( next >= Bin2->T.SB.MasterStop[blin2] || ( *next
            && next+*next+COMPINC > Bin2->T.SB.MasterStop[blin2] ) ) {
                if ( blin2 == 1 ) {
                    UNLOCK(Bin2->T.SB.MasterBlockLock[Bin2->T.SB.MasterNumBlocks]);
                }
                else {
                    UNLOCK(Bin2->T.SB.MasterBlockLock[blin2-1]);
                }
                if ( blin2 == Bin2->T.SB.MasterNumBlocks ) {
/*
                    Move the remainder down into block 0
*/
```

```
                    to = Bin2->T.SB.MasterStart[1];
                    from = Bin2->T.SB.MasterStop[Bin2->T.SB.MasterNumBlocks];
                    while ( from > next ) *--to = *--from;
                    next = to;
                    blin2 = 1;
                }
                else {
                    blin2++;
                }
                LOCK(Bin2->T.SB.MasterBlockLock[blin2]);
                Bin2->T.SB.MasterBlock = blin2;
            }
            term2 = next;
        }
/*
            #] Tail in two :
*/
    }
    SortBotOut(BHEAD 0);
ReturnError:;
/*
    Release all locks
*/
    UNLOCK(Bin1->T.SB.MasterBlockLock[blin1]);
    if ( blin1 > 1 ) {
        UNLOCK(Bin1->T.SB.MasterBlockLock[blin1-1]);
    }
    else {
        UNLOCK(Bin1->T.SB.MasterBlockLock[Bin1->T.SB.MasterNumBlocks]);
    }
    UNLOCK(Bin2->T.SB.MasterBlockLock[blin2]);
    if ( blin2 > 1 ) {
        UNLOCK(Bin2->T.SB.MasterBlockLock[blin2-1]);
    }
    else {
        UNLOCK(Bin2->T.SB.MasterBlockLock[Bin2->T.SB.MasterNumBlocks]);
    }
    if ( AT.identity > 0 ) {
        UNLOCK(AT.SB.MasterBlockLock[AT.SB.FillBlock]);
    }
/*
    And that was all folks
*/
    return(error);
}

#endif

/*
    #] SortBotMerge :
    #[ IniSortBlocks :
*/

static int SortBlocksInitialized = 0;

int IniSortBlocks(int numworkers)
{
    ALLPRIVATES *B;
    SORTING *S;
    LONG totalsize, workersize, blocksize, numberofterms;
    int maxter, id, j;
    int numberofblocks = NUMBEROFBLOCKSINSORT, numparts;
    WORD *w;

    if ( SortBlocksInitialized ) return(0);
    SortBlocksInitialized = 1;
    if ( numworkers == 0 ) return(0);

#ifdef WITHSORTBOTS
    if ( numworkers > 2 ) {
        numparts = 2*numworkers - 2;
        numberofblocks = numberofblocks/2;
    }
    else {
        numparts = numworkers;
    }
#else
    numparts = numworkers;
#endif
    S = AM.S0;
    totalsize = S->LargeSize + S->SmallEsize;
    workersize = totalsize / numparts;
    maxter = AM.MaxTer/sizeof(WORD);
    blocksize = ( workersize - maxter )/numberofblocks;
    numberofterms = blocksize / maxter;
    if ( numberofterms < MINIMUMNUMBEROFTERMS ) {
/*
```

```
        This should have been taken care of in RecalcSetups.
*/
        MesPrint("We have a problem with the size of the blocks in IniSortBlocks");
        Terminate(-1);
    }
/*
    Layout:  For each worker
                block 0: size is maxter WORDS
                numberofblocks blocks of size blocksize WORDS
*/
    w = S->lBuffer;
    if ( w == 0 ) w = S->sBuffer;
    for ( id = 1; id <= numparts; id++ ) {
        B = AB[id];
        AT.SB.MasterBlockLock = (pthread_mutex_t *)Malloc1(
            sizeof(pthread_mutex_t)*(numberofblocks+1),"MasterBlockLock");
        AT.SB.MasterStart = (WORD **)Malloc1(sizeof(WORD *)*(numberofblocks+1)*3,"MasterBlock");
        AT.SB.MasterFill = AT.SB.MasterStart + (numberofblocks+1);
        AT.SB.MasterStop = AT.SB.MasterFill  + (numberofblocks+1);
        AT.SB.MasterNumBlocks = numberofblocks;
        AT.SB.MasterBlock = 0;
        AT.SB.FillBlock = 0;
        AT.SB.MasterFill[0] = AT.SB.MasterStart[0] = w;
        w += maxter;
        AT.SB.MasterStop[0] = w;
        AT.SB.MasterBlockLock[0] = dummylock;
        for ( j = 1; j <= numberofblocks; j++ ) {
            AT.SB.MasterFill[j] = AT.SB.MasterStart[j] = w;
            w += blocksize;
            AT.SB.MasterStop[j] = w;
            AT.SB.MasterBlockLock[j] = dummylock;
        }
    }
    if ( w > S->sTop2 ) {
        MesPrint("Counting problem in IniSortBlocks");
        Terminate(-1);
    }
    return(0);
}

/*
    #] IniSortBlocks :
    #[ UpdateSortBlocks :
*/

int UpdateSortBlocks(int numworkers)
{
    ALLPRIVATES *B;
    SORTING *S;
    LONG totalsize, workersize, blocksize, numberofterms;
    int maxter, id, j;
    int numberofblocks = NUMBEROFBLOCKSINSORT, numparts;
    WORD *w;

    if ( numworkers == 0 ) return(0);

#ifdef WITHSORTBOTS
    if ( numworkers > 2 ) {
        numparts = 2*numworkers - 2;
        numberofblocks = numberofblocks/2;
    }
    else {
        numparts = numworkers;
    }
#else
    numparts = numworkers;
#endif
    S = AM.S0;
    totalsize = S->LargeSize + S->SmallEsize;
    workersize = totalsize / numparts;
    maxter = AM.MaxTer/sizeof(WORD);
    blocksize = ( workersize - maxter )/numberofblocks;
    numberofterms = blocksize / maxter;
    if ( numberofterms < MINIMUMNUMBEROFTERMS ) {
/*
        This should have been taken care of in RecalcSetups.
*/
        MesPrint("We have a problem with the size of the blocks in UpdateSortBlocks");
        Terminate(-1);
    }
/*
    Layout:  For each worker
                block 0: size is maxter WORDS
                numberofblocks blocks of size blocksize WORDS
*/
    w = S->lBuffer;
    if ( w == 0 ) w = S->sBuffer;
```

```
    for ( id = 1; id <= numparts; id++ ) {
        B = AB[id];
        AT.SB.MasterFill[0] = AT.SB.MasterStart[0] = w;
        w += maxter;
        AT.SB.MasterStop[0] = w;
        for ( j = 1; j <= numberofblocks; j++ ) {
            AT.SB.MasterFill[j] = AT.SB.MasterStart[j] = w;
            w += blocksize;
            AT.SB.MasterStop[j] = w;
        }
    }
    if ( w > S->sTop2 ) {
        MesPrint("Counting problem in UpdateSortBlocks");
        Terminate(-1);
    }
    return(0);
}

/*
    #] UpdateSortBlocks :
    #[ DefineSortBotTree :
*/

#ifdef WITHSORTBOTS

void DefineSortBotTree(void)
{
    ALLPRIVATES *B;
    int n, i, from;
    if ( numberofworkers <= 2 ) return;
    n = numberofworkers*2-2;
    for ( i = numberofworkers+1, from = 1; i <= n; i++ ) {
        B = AB[i];
        AT.SortBotIn1 = from++;
        AT.SortBotIn2 = from++;
    }
    B = AB[0];
    AT.SortBotIn1 = from++;
    AT.SortBotIn2 = from++;
}

#endif

/*
    #] DefineSortBotTree :
    #[ GetTerm2 :

    Routine does a GetTerm but only when a bracket index is involved and
    only from brackets that have been judged not suitable for treatment
    as complete brackets by a single worker. Whether or not a bracket should
    be treated by a single worker is decided by TreatIndexEntry
*/

WORD GetTerm2(PHEAD WORD *term)
{
    WORD *ttco, *tt, retval;
    LONG n,i;
    FILEHANDLE *fi;
    EXPRESSIONS e = AN.expr;
    BRACKETINFO *b  = e->bracketinfo;
    BRACKETINDEX *bi = b->indexbuffer;
    POSITION where, eonfile = AS.OldOnFile[e-Expressions], bstart, bnext;
/*
    1: Get the current position.
*/
    switch ( e->status ) {
        case UNHIDELEXPRESSION:
        case UNHIDEGEXPRESSION:
        case DROPHLEXPRESSION:
        case DROPHGEXPRESSION:
        case HIDDENLEXPRESSION:
        case HIDDENGEXPRESSION:
            fi = AR.hidefile;
            break;
        default:
            fi = AR.infile;
            break;
    }
    if ( AR.KeptInHold ) {
        retval = GetTerm(BHEAD term);
        return(retval);
    }
    SeekScratch(fi,&where);
    if ( AN.lastinindex < 0 ) {
/*
        We have to test whether we have to do the first bracket
*/
```

```
        if ( ( n = TreatIndexEntry(BHEAD 0) ) <= 0 ) {
            AN.lastinindex = n;
            where = bi[n].start;
            ADD2POS(where,eonfile);
            SetScratch(fi,&where);
/*
            Put the bracket in the Compress buffer.
*/
            ttco = AR.CompressBuffer;
            tt = b->bracketbuffer + bi[0].bracket;
            i = *tt;
            NCOPY(ttco,tt,i)
            AR.CompressPointer = ttco;
            retval = GetTerm(BHEAD term);
            return(retval);
        }
        else AN.lastinindex = n-1;
    }
/*
    2: Find the corresponding index number
       a: test whether it is in the current bracket
*/
    n = AN.lastinindex;
    bstart = bi[n].start;
    ADD2POS(bstart,eonfile);
    bnext = bi[n].next;
    ADD2POS(bnext,eonfile);
    if ( ISLESSPOS(bstart,where) && ISLESSPOS(where,bnext) ) {
        retval = GetTerm(BHEAD term);
        return(retval);
    }
    for ( n++ ; n < b->indexfill; n++ ) {
        i = TreatIndexEntry(BHEAD n);
        if ( i <= 0 ) {
/*
            Put the bracket in the Compress buffer.
*/
            ttco = AR.CompressBuffer;
            tt = b->bracketbuffer + bi[n].bracket;
            i = *tt;
            NCOPY(ttco,tt,i)
            AR.CompressPointer = ttco;
            AN.lastinindex = n;
            where = bi[n].start;
            ADD2POS(where,eonfile);
            SetScratch(fi,&(where));
            retval = GetTerm(BHEAD term);
            return(retval);
        }
        else n += i - 1;
    }
    return(0);
}

/*
    #] GetTerm2 :
    #[ TreatIndexEntry :
*/
int TreatIndexEntry(PHEAD LONG n)
{
    BRACKETINFO *b  = AN.expr->bracketinfo;
    LONG numbra = b->indexfill - 1, i;
    LONG totterms;
    BRACKETINDEX *bi;
    POSITION pos1, average;
/*
    1: number of the bracket should be such that there is one bucket
       for each worker remaining.
*/
    if ( ( numbra - n ) <= numberofworkers ) return(0);
/*
    2: size of the bracket contents should be less than what remains in
       the expression divided by the number of workers.
*/
    bi = b->indexbuffer;
    DIFPOS(pos1,bi[numbra].next,bi[n].next);  /* Size of what remains */
    BASEPOSITION(average) = DIVPOS(pos1,(3*numberofworkers));
    DIFPOS(pos1,bi[n].next,bi[n].start);      /* Size of the current bracket */
    if ( ISLESSPOS(average,pos1) ) return(0);
/*
    It passes.
    Now check whether we can do more brackets
*/
    totterms = bi->termsinbracket;
    if ( totterms > 2*AC.ThreadBucketSize ) return(1);
    for ( i = 1; i < numbra-n; i++ ) {
        DIFPOS(pos1,bi[n+i].next,bi[n].start); /* Size of the combined brackets */
```

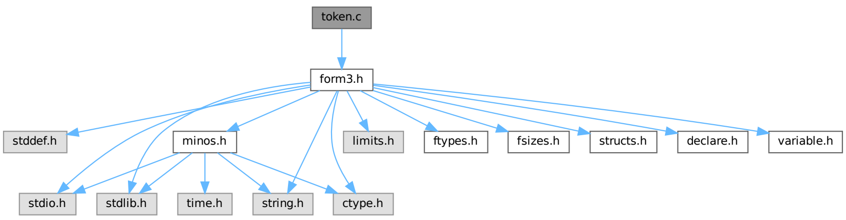
```
        if ( ISLESSPOS(average,pos1) ) return(i);
        totterms += bi->termsinbracket;
        if ( totterms > 2*AC.ThreadBucketSize ) return(i+1);
    }
/*
    We have a problem at the end of the system. Just do one.
*/
    return(1);
}

/*
    #] TreatIndexEntry :
    #[ SetHideFiles :
*/

void SetHideFiles(void) {
    int i;
    ALLPRIVATES *B, *B0 = AB[0];
    for ( i = 1; i <= numberofworkers; i++ ) {
        B = AB[i];
        AR.hidefile->handle = AR0.hidefile->handle;
        if ( AR.hidefile->handle < 0 ) {
            AR.hidefile->PObuffer = AR0.hidefile->PObuffer;
            AR.hidefile->POstop = AR0.hidefile->POstop;
            AR.hidefile->POfill = AR0.hidefile->POfill;
            AR.hidefile->POfull = AR0.hidefile->POfull;
            AR.hidefile->POsize = AR0.hidefile->POsize;
            AR.hidefile->POposition = AR0.hidefile->POposition;
            AR.hidefile->filesize = AR0.hidefile->filesize;
        }
        else {
            AR.hidefile->PObuffer = AR.hidefile->wPObuffer;
            AR.hidefile->POstop = AR.hidefile->wPOstop;
            AR.hidefile->POfill = AR.hidefile->wPOfill;
            AR.hidefile->POfull = AR.hidefile->wPOfull;
            AR.hidefile->POsize = AR.hidefile->wPOsize;
            PUTZERO(AR.hidefile->POposition);
        }
    }
}

/*
    #] SetHideFiles :
    #[ IniFbufs :
*/

void IniFbufs(void)
{
    int i;
    for ( i = 0; i < AM.totalnumberofthreads; i++ ) {
        IniFbuffer(AB[i]->T.fbufnum);
    }
}

/*
    #] IniFbufs :
    #[ SetMods :
*/

void SetMods(void)
{
    ALLPRIVATES *B;
    int i, n, j;
    for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
        B = AB[j];
        AN.ncmod = AC.ncmod;
        if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
        n = ABS(AN.ncmod);
        AN.cmod = (UWORD *)Malloc1(sizeof(WORD)*n,"AN.cmod");
        for ( i = 0; i < n; i++ ) AN.cmod[i] = AC.cmod[i];
    }
}

/*
    #] SetMods :
    #[ UnSetMods :
*/

void UnSetMods(void)
{
    ALLPRIVATES *B;
    int j;
    for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
        B = AB[j];
        if ( AN.cmod != 0 ) M_free(AN.cmod,"AN.cmod");
        AN.cmod = 0;
    }
```

```
}

/*
    #] UnSetMods :
    #[ find_Horner_MCTS_expand_tree_threaded :
*/

void find_Horner_MCTS_expand_tree_threaded(void) {
    int id;
    while (( id = GetAvailableThread() ) < 0)
        MasterWait();
    WakeupThread(id,MCTSEXPANDTREE);
}

/*
    #] find_Horner_MCTS_expand_tree_threaded :
    #[ optimize_expression_given_Horner_threaded :
*/

extern void optimize_expression_given_Horner_threaded(void) {
    int id;
    while (( id = GetAvailableThread() ) < 0)
        MasterWait();
    WakeupThread(id,OPTIMIZEEXPRESSION);
}

/*
    #] optimize_expression_given_Horner_threaded :
*/

#endif
```

4.144 token.c File Reference

`#include "form3.h"`

Include dependency graph for token.c:



Macros

- #define CHECKPOLY {if(polyflag)MesPrint("&Illegal use of polynomial function"); polyflag = 0; }

Functions

- int tokenize (UBYTE ∗in, WORD leftright)
- void WriteTokens (SBYTE ∗in)
- int simp1token (SBYTE ∗s)
- int simpwtoken (SBYTE ∗s)
- int simp2token (SBYTE ∗s)
- int simp3atoken (SBYTE ∗s, int mode)
- int simp3btoken (SBYTE ∗s, int mode)
- int simp4token (SBYTE ∗s)
- int simp5token (SBYTE ∗s, int mode)
- int simp6token (SBYTE ∗tokens, int mode)

Variables

- char ∗ ttypes [ ]

4.144.1 Detailed Description

The tokenizer. This is a part of the compiler that does an intermediate type of translation. It does look up the names etc and can do a number of optimizations. The resulting output is a stream of bytes which can be processed by the code generator (in the file compiler.c)

Definition in file token.c.

4.144.2 Macro Definition Documentation

4.144.2.1 CHECKPOLY

```
#define CHECKPOLY {if(polyflag)MesPrint("&Illegal use of polynomial function"); polyflag = 0;
}
```

Definition at line 56 of file token.c.

4.144.3 Function Documentation

4.144.3.1 tokenize()

```
int tokenize (
            UBYTE * in,
            WORD leftright )
```

Definition at line 58 of file token.c.

4.144.3.2 WriteTokens()

```
void WriteTokens (
            SBYTE * in )
```

Definition at line 657 of file token.c.

4.144.3.3 simp1token()

```
int simp1token (
            SBYTE * s )
```

Definition at line 705 of file token.c.

4.144.3.4 simpwtoken()

```
int simpwtoken (
             SBYTE * s )
```

Definition at line 794 of file token.c.

4.144.3.5 simp2token()

```
int simp2token (
             SBYTE * s )
```

Definition at line 948 of file token.c.

4.144.3.6 simp3atoken()

```
int simp3atoken (
             SBYTE * s,
             int mode )
```

Definition at line 1196 of file token.c.

4.144.3.7 simp3btoken()

```
int simp3btoken (
             SBYTE * s,
             int mode )
```

Definition at line 1437 of file token.c.

4.144.3.8 simp4token()

```
int simp4token (
             SBYTE * s )
```

Definition at line 1822 of file token.c.

4.144.3.9 simp5token()

```
int simp5token (
             SBYTE * s,
             int mode )
```

Definition at line 1982 of file token.c.

4.144.3.10 simp6token()

```
int simp6token (
            SBYTE * tokens,
            int mode )
```

Definition at line 2028 of file token.c.

4.144.4 Variable Documentation

4.144.4.1 ttypes

```
char* ttypes[]
```

Initial value:

```
= { "\n", "S", "I", "V", "F", "set", "E", "dotp", "#",
   "sub", "d_", "$", "dub", "(", ")", "?", "??", ".", "[", "]",
   ",", "((", "))", "*", "/", "^", "+", "-", "!", "end", "{{", "}}",
   "N_?", "conj", "()", "#d", "^d", "_", "snum"  }
```

Definition at line 652 of file token.c.

4.145 token.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
*/

#include "form3.h"

/*
    #] Includes :
    #[ Compiler :
        #[ tokenize :

        Takes the input in 'in' and translates it into tokens.
        The tokens are put in the token buffer which starts at 'AC.tokens'
        and runs till 'AC.toptokens'
        We may assume that the various types of brackets match properly.
        object = -1: after , or (
        object = 0: name/variable/number etc is allowed
```

```
        object = 1: variable.
        object = 2: number
        object = 3: ) after subexpression
*/

#define CHECKPOLY {if(polyflag)MesPrint("&Illegal use of polynomial function"); polyflag = 0; }

int tokenize(UBYTE *in, WORD leftright)
{
    int error = 0, object, funlevel = 0, bracelevel = 0, explevel = 0, numexp;
    int polyflag = 0;
    WORD number, type;
    UBYTE *s = in, c;
    SBYTE *out, *outtop, num[MAXNUMSIZE], *t;
    LONG i;
    if ( AC.tokens == 0 ) {
        SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
        SBYTE **pppp = &(AC.toptokens);
        DoubleBuffer((void **)ppp,(void **)pppp,sizeof(SBYTE),"start tokens");
    }
    out = AC.tokens;
    outtop = AC.toptokens - MAXNUMSIZE;
    AC.dumnumflag = 0;
    object = 0;
    while ( *in ) {
        if ( out > outtop ) {
            LONG oldsize = (LONG)(out - AC.tokens);
            SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
            SBYTE **pppp = &(AC.toptokens);
            DoubleBuffer((void **)ppp,(void **)pppp,sizeof(SBYTE),"expand tokens");
            out = AC.tokens + oldsize;
            outtop = AC.toptokens - MAXNUMSIZE;
        }
        switch ( FG.cTable[*in] ) {
            case 0:     /* a-zA-Z */
                CHECKPOLY
                s = in++;
                while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1
                || *in == '_' ) in++;
dovariable:     c = *in; *in = 0;
                if ( object > 0 ) {
                    MesPrint("&Illegal position for %s",s);
                    if ( !error ) error = 1;
                }
                if ( out > AC.tokens && ( out[-1] == TWILDCARD || out[-1] == TNOT ) ) {
                    type = GetName(AC.varnames,s,&number,NOAUTO);
                }
                else {
                    type = GetName(AC.varnames,s,&number,WITHAUTO);
                }
                if ( type < 0 )
                    type = GetName(AC.exprnames,s,&number,NOAUTO);
                switch ( type ) {
                    case CSYMBOL:       *out++ = TSYMBOL;     break;
                    case CINDEX:
                        if ( number >= (AM.IndDum-AM.OffsetIndex) ) {
                            if ( c != '?' ) {
                                MesPrint("&Generated indices should be of the type Nnumber_?");
                                error = 1;
                            }
                            else {
                                *in++ = c; c = *in; *in = 0;
                                AC.dumnumflag = 1;
                            }
                        }
                        *out++ = TINDEX;
                        break;
                    case CVECTOR:       *out++ = TVECTOR;     break;
                    case CFUNCTION:
#ifdef WITHMPI
                        /*
                         * In the preprocessor, random functions in #$var=... and #inside
                         * may cause troubles, because the program flow on a slave may be
                         * different from those on others. We set AC.RhsExprInModuleFlag in order
                         * to make the change of $-variable be done on the master and thus keep the
                         * consistency among the master and all slave processes. The previous value
                         * of AC.RhsExprInModuleFlag will be restored after #$var=... and #inside.
                         */
                        if ( AP.PreAssignFlag || AP.PreInsideLevel ) {
                            switch ( number + FUNCTION ) {
                                case RANDOMFUNCTION:
                                case RANPERM:
                                    AC.RhsExprInModuleFlag = 1;
                            }
                        }
#endif
                        *out++ = TFUNCTION;
```

```
                        break;
                    case CSET:          *out++ = TSET;          break;
                    case CEXPRESSION:   *out++ = TEXPRESSION;
                                        if ( leftright == LHSIDE ) {
                                            if ( !error ) error = 1;
                                            MesPrint("&Expression not allowed in LH-side of substitution: %s",s);
                                        }
/*[06nov2003 mt]:*/
#ifdef WITHMPI
                                        else { /*RHSide*/
                                            /* NOTE: We always set AC.RhsExprInModuleFlag regardless of
                                             *       AP.PreAssignFlag or AP.PreInsideLevel because we have to detect
                                             *       RHS expressions even in those cases. */
                                            AC.RhsExprInModuleFlag = 1;
                                        }
                                        if ( !AP.PreAssignFlag && !AP.PreInsideLevel )
                                            Expressions[number].vflags |= ISINRHS;
#endif
/*:[06nov2003 mt]*/
                                        if ( AC.exprfillwarning == 0 ) {
                                                AC.exprfillwarning = 1;
                                        }
                                        break;
                    case CDELTA:        *out++ = TDELTA;        *in = c;
                                        object = 1; continue;
                    case CDUBIOUS:      *out++ = TDUBIOUS;      break;
                    default:            *out++ = TDUBIOUS;
                                        if ( !error ) error = 1;
                                        MesPrint("&Undeclared variable %s",s);
                                        number = AddDubious(s);
                                        break;
                }
                object = 1;
donumber:       i = 0;
                do { num[i++] = (SBYTE)(number & 0x7F); number >>= 7; } while ( number );
                while ( --i >= 0 ) *out++ = num[i];
                *in = c;
                break;
            case 1:     /* 0-9 */
              {
#ifdef WITHFLOAT
                int spec;
                UBYTE *in2;
#endif
                CHECKPOLY
                s = in;
                while ( *s == '0' && FG.cTable[s[1]] == 1 ) s++;
                in = s+1; i = 1;
                while ( FG.cTable[*in] == 1 ) { in++; i++; }
                if ( object > 0 ) {
                    c = *in; *in = 0;
                    MesPrint("&Illegal position for %s",s);
                    *in = c;
                    if ( !error ) error = 1;
                }
                if ( i == 1 && *in == '_' && ( *s == '5' || *s == '6'
                || *s == '7' ) ) {
                    in++; *out++ = TSGAMMA; *out++ = (SBYTE)(*s - '4');
                    object = 1;
                    break;
                }
#ifdef WITHFLOAT
                in2 = CheckFloat(in,&spec);
                if ( in2 > in ) {
                    if ( spec == 1 ) {
                        *out++ = TNUMBER; *out++ = 0; s = in2;
                    }
                    else if ( spec == -1 ) {
                        MesPrint("&The floating point system has not been started: %s",in);
                        if ( !error ) error = 1;
                    }
                    else {
                      in = in2;
dofloat:
                      while ( out + (in-s) >= AC.toptokens ) {
                        LONG oldsize = (LONG)(out - AC.tokens);
                        SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
                        SBYTE **pppp = &(AC.toptokens);
                        DoubleBuffer((void **)ppp,(void **)pppp,sizeof(SBYTE),"more tokens");
                        out = AC.tokens + oldsize;
                        outtop = AC.toptokens - MAXNUMSIZE;
                      }
                      *out++ = TFLOAT;
                      while ( s < in ) *out++ = *s++;
```

```
                    }
                }
                else
#endif
                {
                *out++ = TNUMBER;
                if ( ( i & 1 ) != 0 ) *out++ = (SBYTE)(*s++ - '0');
                while ( out + (in-s)/2 >= AC.toptokens ) {
                    LONG oldsize = (LONG)(out - AC.tokens);
                    SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
                    SBYTE **pppp = &(AC.toptokens);
                    DoubleBuffer((void **)ppp,(void **)pppp,sizeof(SBYTE),"more tokens");
                    out = AC.tokens + oldsize;
                    outtop = AC.toptokens - MAXNUMSIZE;
                }
                while ( s < in ) {    /* We store in base 100 */
                    *out++ = (SBYTE)(( *s - '0' ) * 10 + ( s[1] - '0' ));
                    s += 2;
                }
                }
                object = 2;
              }
              break;
            case 2:     /* . $ _ ? # ' */
                CHECKPOLY
                if ( *in == '?' ) {
                    if ( leftright == LHSIDE ) {
                        if ( object == 1 ) {     /* follows a name */
                            in++; *out++ = TWILDCARD;
                            if ( FG.cTable[in[0]] == 0 || in[0] == '[' || in[0] == '{' ) object = 0;
                        }
                        else if ( object == -1 ) {  /* follows comma or ( */
                            in++; s = in;
                            while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
                            c = *in; *in = 0;
                            if ( FG.cTable[*s] != 0 ) {
                                MesPrint("&Illegal name for argument list variable %s",s);
                                error = 1;
                            }
                            else {
                                i = AddWildcardName((UBYTE *)s);
                                *in = c;
                                *out++ = TWILDARG;
                                *out++ = (SBYTE)i;
                            }
                            object = 1;
                        }
                        else {
                            MesPrint("&Illegal position for ?");
                            error = 1;
                            in++;
                        }
                    }
                    else {
                        if ( object != -1 ) goto IllPos;
                        in++;
                        if ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) {
                            s = in;
                            while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
                            c = *in; *in = 0;
                            i = GetWildcardName((UBYTE *)s);
                            if ( i <= 0 ) {
                                MesPrint("&Undefined argument list variable %s",s);
                                error = 1;
                            }
                            *in = c;
                            *out++ = TWILDARG;
                            *out++ = (SBYTE)i;
                        }
                        else {
                            if ( AC.vectorlikeLHS == 0 ) {
                                MesPrint("&Generated index ? only allowed in vector substitution",s);
                                error = 1;
                            }
                            *out++ = TGENINDEX;
                        }
                        object = 1;
                    }
                }
                else if ( *in == '.' ) {
                    if ( object == 1 ) {     /* follows a name */
                        *out++ = TDOT;
                        object = 0;
                        in++;
                    }
#ifdef WITHFLOAT
                    else if ( object == 0 || object == -1 ) {
```

```
/*
                        Test for floating point number
*/
                        int spec;
                        s = CheckFloat(in,&spec);
                        if ( s > in ) {
                            if ( spec == 1 ) {
                                *out++ = TNUMBER; *out++ = 0;
                                object = 2; in = s;
                            }
                            else if ( spec == -1 ) {
                                MesPrint("&The floating point system has not been started: %s",in);
                                if ( !error ) error = 1;
                            }
                            else {
                                UBYTE *a = s; s = in; in = a;
                                goto dofloat;
                            }
                        }
                        else goto IllPos;
                    }
#endif
                    else goto IllPos;
                }
                else if ( *in == '$' ) {     /* $ variable */
                    in++;
                    s = in;
                    if ( FG.cTable[*in] == 0 ) {
                        while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
                        if ( *in == '_' && AP.PreAssignFlag == 2 ) in++;
                        c = *in; *in = 0;
                        if ( object > 0 ) {
                            if ( object != 1 || leftright == RHSIDE ) {
                                MesPrint("&Illegal position for $%s",s);
                                if ( !error ) error = 1;
                            }   /* else can be assignment in wildcard */
                            else {
                                if ( ( number = GetDollar(s) ) < 0 ) {
                                    number = AddDollar(s,0,0,0);
                                }
                            }
                        }
                        else if ( ( number = GetDollar(s) ) < 0 ) {
                            MesPrint("&Undefined variable $%s",s);
                            if ( !error ) error = 1;
                            number = AddDollar(s,0,0,0);
                        }
                        *out++ = TDOLLAR;
                        object = 1;
                        if ( ( AC.exprfillwarning == 0 ) &&
                             ( ( out > AC.tokens+1 ) && ( out[-2] != TWILDCARD ) ) ) {
                            AC.exprfillwarning = 1;
                        }
                        goto donumber;
                    }
                    else {
                        MesPrint("Illegal name for $ variable after %s",in);
                        if ( !error ) error = 1;
                    }
                }
                else if ( *in == '#' ) {
                    in++;
                    if ( object == 1 ) {     /* follows a name */
                        *out++ = TCONJUGATE;
                    }
                    else {
                        MesPrint("&Illegal position for %#");
                        error = 1;
                    }
                }
                else goto IllPos;
                break;
            case 3:      /* [ ] */
                CHECKPOLY
                if ( *in == '[' ) {
                    if ( object == 1 ) {     /* after name */
                        t = out-1;
                        if ( *t == RPARENTHESIS ) {
                            *out++ = LBRACE; *out++ = LPARENTHESIS;
                            bracelevel++; explevel = bracelevel;
                        }
                        else {
                            while ( *t >= 0 && t > AC.tokens ) t--;
                            if ( *t == TEXPRESSION ) {
                                *out++ = LBRACE; *out++ = LPARENTHESIS;
                                bracelevel++; explevel = bracelevel;
                            }
```

```
                            else {*out++ = LBRACE; bracelevel++; }
                        }
                        object = 0;
                    }
                    else {                  /* name. find matching ] */
                        s = in;
                        in = SkipAName(in);
                        goto dovariable;
                    }
                }
                else {
                    if ( explevel > 0 && explevel == bracelevel ) {
                        *out++ = RPARENTHESIS; explevel = 0;
                    }
                    *out++ = RBRACE; object = 1; bracelevel--;
                }
                in++;
                break;
            case 4:     /* ( ) = ; , */
                if ( *in == '(' ) {
                    if ( funlevel >= AM.MaxParLevel ) {
                        MesPrint("&More than %d levels of parentheses",AM.MaxParLevel);
                        return(-1);
                    }
                    if ( object == 1 ) {    /* After name -> function,vector */
                        AC.tokenarglevel[funlevel++] = TYPEISFUN;
                        *out++ = TFUNOPEN;
                        if ( polyflag ) {
                            if ( in[1] != ')' && in[1] != ',' ) {
                                *out++ = TNUMBER; *out++ = (SBYTE)(polyflag);
                                *out++ = TCOMMA;
                                *out++ = LPARENTHESIS;
                            }
                            else {
                                *out++ = LPARENTHESIS;
                                *out++ = TNUMBER; *out++ = (SBYTE)(polyflag);
                            }
                            polyflag = 0;
                        }
                        else if ( in[1] != ')' && in[1] != ',' ) {
                            *out++ = LPARENTHESIS;
                        }
                    }
                    else if ( object <= 0 ) {
                        CHECKPOLY
                        AC.tokenarglevel[funlevel++] = TYPEISSUB;
                        *out++ = LPARENTHESIS;
                    }
                    else {
                        polyflag = 0;
                        AC.tokenarglevel[funlevel++] = TYPEISMYSTERY;
                        MesPrint("&Illegal position for (: %s",in);
                        if ( error >= 0 ) error = -1;
                    }
                    object = -1;
                }
                else if ( *in == ')' ) {
                    funlevel--;
                    if ( funlevel < 0 ) {
/*                      if ( funflag == 0 ) { */
                            MesPrint("&There is an unmatched parenthesis");
                            if ( error >= 0 ) error = -1;
/*                      } */
                    }
                    else if ( object <= 0
                    && ( AC.tokenarglevel[funlevel] != TYPEISFUN
                    || out[-1] != TFUNOPEN ) ) {
                        MesPrint("&Illegal position for closing parenthesis.");
                        if ( error >= 0 ) error = -1;
                        if ( AC.tokenarglevel[funlevel] == TYPEISFUN ) object = 1;
                        else object = 3;
                    }
                    else {
                        if ( AC.tokenarglevel[funlevel] == TYPEISFUN ) {
                            if ( out[-1] == TFUNOPEN ) out--;
                            else {
                                if ( out[-1] != TCOMMA ) *out++ = RPARENTHESIS;
                                *out++ = TFUNCLOSE;
                            }
                            object = 1;
                        }
                        else if ( AC.tokenarglevel[funlevel] == TYPEISSUB ) {
                            *out++ = RPARENTHESIS;
                            object = 3;
                        }
                    }
                }
```

```
                else if ( *in == ',' ) {
                    if ( /* object > 0 && */ funlevel > 0 &&
                    AC.tokenarglevel[funlevel-1] == TYPEISFUN ) {
                        if ( out[-1] != TFUNOPEN && out[-1] != TCOMMA )
                            *out++ = RPARENTHESIS;
                        else { *out++ = TNUMBER; *out++ = 0; }
                        *out++ = TCOMMA;
                        if ( in[1] != ',' && in[1] != ')' )
                            *out++ = LPARENTHESIS;
                        else if ( in[1] == ')' ) {
                            *out++ = TNUMBER; *out++ = 0;
                        }
                    }
/*
                    else if ( object > 0 ) {
                    }
*/
                    else {
                        MesPrint("&Illegal position for comma: %s",in);
                        MesPrint("&Forgotten ; ?");
                        if ( error >= 0 ) error = -1;
                    }
                    object = -1;
                }
                else goto IllPos;
                in++;
                break;
            case 5:     /* + - * % / ^ : */
                CHECKPOLY
                if ( *in == ':' || *in == '%' ) goto IllPos;
                if ( *in == '*' || *in == '/' || *in == '^' ) {
                    if ( object <= 0 ) {
                        MesPrint("&Illegal position for operator: %s",in);
                        if ( error >= 0 ) error = -1;
                    }
                    else if ( *in == '*' ) *out++ = TMULTIPLY;
                    else if ( *in == '/' ) *out++ = TDIVIDE;
                    else                   *out++ = TPOWER;
                    in++;
                }
                else {
                    i = 1;
                    while ( *in == '+' || *in == '-' ) {
                        if ( *in == '-' ) i = -i;
                        in++;
                    }
                    if ( i == 1 ) {
                        if ( out > AC.tokens && out[-1] != TFUNOPEN &&
                        out[-1] != LPARENTHESIS && out[-1] != TCOMMA
                        && out[-1] != LBRACE )
                            *out++ = TPLUS;
                    }
                    else *out++ = TMINUS;
                }
                object = 0;
                break;
            case 6:     /* Whitespace */
                in++; break;
            case 7:     /* { | } */
                CHECKPOLY
                if ( *in == '{' ) {
                    if ( object > 0 ) {
                        MesPrint("&Illegal position for %s",in);
                        if ( !error ) error = 1;
                    }
                    s = in+1;
                    SKIPBRA2(in)
                    number = DoTempSet(s,in);
                    in++;
                    if ( number >= 0 ) {
                        *out++ = TSET;
                        i = 0;
                        do { num[i++] = (SBYTE)(number & 0x7F); number »= 7; } while ( number );
                        while ( --i >= 0 ) *out++ = num[i];
                    }
                    else if ( error == 0 ) error = 1;
                    object = 1;
                }
                else goto IllPos;
                break;
            case 8:     /* ! & < > */
                CHECKPOLY
                if ( *in != '!' || leftright == RHSIDE
                || object != 1 || out[-1] != TWILDCARD ) goto IllPos;
                *out++ = TNOT;
                if ( FG.cTable[in[1]] == 0 || in[1] == '[' || in[1] == '{' ) object = 0;
                in++;
```

```
                break;
            default:
IllPos:         MesPrint("&Illegal character at this position: %s",in);
                if ( error >= 0 ) error = -1;
                in++;
                polyflag = 0;
                break;
        }
    }
    *out++ = TENDOFIT;
    AC.endoftokens = out;
    if ( funlevel > 0 || bracelevel != 0 ) {
        if ( funlevel > 0 ) MesPrint("&Unmatched parentheses");
        if ( bracelevel != 0 ) MesPrint("&Unmatched braces");
        return(-1);
    }
    if ( AC.TokensWriteFlag ) WriteTokens(AC.tokens);
/*
    Simplify fixed set elements
*/
    if ( error == 0 && simp1token(AC.tokens) ) error = 1;
/*
    Collect wildcards for the prototype. Simplify the leftover wildcards
*/
    if ( error == 0 && leftright == LHSIDE && simpwtoken(AC.tokens) )
                error = 1;
/*
    Now prepare the set[n] objects in the RHS.
*/
    if ( error == 0 && leftright == RHSIDE && simp4token(AC.tokens) )
                error = 1;
/*
    Simplify simple function arguments (and 1/fac_ and 1/invfac_)
*/
    if ( error == 0 && simp2token(AC.tokens) ) error = 1;
/*
    Next we try to remove composite denominators or exponents and
    replace them by their internal functions. This may involve expanding
    the buffer. The return code of 3a is negative if there is an error
    and positive if indeed we need to do some work.
    simp3btoken does the work
*/
    numexp = 0;
    if ( error == 0 && ( numexp = simp3atoken(AC.tokens,leftright) ) < 0 )
        error = 1;
    if ( numexp > 0 ) {
        SBYTE *tt;
        out = AC.tokens;
        while ( *out != TENDOFIT ) out++;
        while ( out+numexp*9+20 > outtop ) {
            LONG oldsize = (LONG)(out - AC.tokens);
            SBYTE **ppp = &(AC.tokens); /* to avoid a compiler warning */
            SBYTE **pppp = &(AC.toptokens);
            DoubleBuffer((void **)ppp,(void **)pppp,sizeof(SBYTE),"out tokens");
            out = AC.tokens + oldsize;
            outtop = AC.toptokens - MAXNUMSIZE;
        }
        tt = out + numexp*9+20;
        while ( out >= AC.tokens ) { *tt-- = *out--; }
        while ( tt >= AC.tokens ) { *tt-- = TEMPTY; }
        if ( error == 0 && simp3btoken(AC.tokens,leftright) ) error = 1;
        if ( error == 0 && simp2token(AC.tokens) ) error = 1;
    }
/*
    In simp5token we test for special cases like sumvariables that are
    already wildcards, etc.
*/
    if ( error == 0 && simp5token(AC.tokens,leftright) ) error = 1;
/*
    In simp6token we test for special cases like factorized expressions
    that occur in the RHS in an improper way.
*/
    if ( error == 0 && simp6token(AC.tokens,leftright) ) error = 1;

    return(error);
}

/*
        #] tokenize :
        #[ WriteTokens :
*/

char *ttypes[] = { "\n", "S", "I", "V", "F", "set", "E", "dotp", "#",
   "sub", "d_", "$", "dub", "(", ")", "?", "??", ".", "[", "]",
   ",", "((", "))", "*", "/", "^", "+", "-", "!", "end", "{{", "}}",
   "N_?", "conj", "()", "#d", "^d", "_", "snum"  };
```

```
void WriteTokens(SBYTE *in)
{
    int numinline = 0, x, n = sizeof(ttypes)/sizeof(char *);
    char outbuf[81], *s, *out, c;
    out = outbuf;
    while ( *in != TENDOFIT ) {
        if ( *in < 0 ) {
            if ( *in >= -n ) {
                s = ttypes[-*in];
                while ( *s ) { *out++ = *s++; numinline++; }
            }
            else {
                *out++ = '-'; x = -*in; numinline++;
                goto writenumber;
            }
        }
        else {
            x = *in;
writenumber:
            s = out;
            do {
                *out++ = (char)(( x % 10 ) + '0');
                numinline++;
                x = x / 10;
            } while ( x );
            c = out[-1]; out[-1] = *s; *s = c;
        }
        if ( numinline > 70 ) {
            *out = 0;
            MesPrint("%s",outbuf);
            out = outbuf; numinline = 0;
        }
        else {
            *out++ = ' '; numinline++;
        }
        in++;
    }
    if ( numinline > 0 ) { *out = 0; MesPrint("%s",outbuf); }
}

/*
        #] WriteTokens :
        #[ simp1token :

        Routine substitutes set elements if possible.
        This means sets with a fixed argument like setname[3].
*/

int simp1token(SBYTE *s)
{
    int error = 0, n, i, base;
    WORD numsub;
    SBYTE *fill = s, *start, *t, numtab[10];
    SETS set;
    while ( *s != TENDOFIT ) {
        if ( *s == RBRACE ) {
            start = fill-1;
            while ( *start != LBRACE ) start--;
            t = start - 1;
            while ( *t >= 0 ) t--;
            if ( *t == TSET && ( start[1] == TNUMBER || start[1] == TNUMBER1 ) ) {
                base = start[1] == TNUMBER ? 100: 128;
                start += 2;
                numsub = *start++;
                while ( *start >= 0 && start < fill )
                    { numsub = base*numsub + *start++; }
                if ( start == fill ) {
                    start = t;
                    t++; n = *t++; while ( *t >= 0 ) { n = 128*n + *t++; }
                    set = Sets+n;
                    if ( ( set->type != CRANGE )
                    && ( numsub > 0 && numsub <= set->last-set->first ) ) {
                        fill = start;
                        n = SetElements[set->first+numsub-1];
                        switch (set->type) {
                            case CSYMBOL:
                                if ( n > MAXPOWER ) {
                                    n -= 2*MAXPOWER;
                                    if ( n < 0 ) { n = -n; *fill++ = TMINUS; }
                                    *fill++ = TNUMBER1;
                                }
                                else *fill++ = TSYMBOL;
                                break;
                            case CINDEX:
                                if ( n < AM.OffsetIndex ) *fill++ = TNUMBER1;
                                else {
                                    *fill++ = TINDEX;
```

```
                                    n -= AM.OffsetIndex;
                                }
                                break;
                            case CVECTOR:   *fill++ = TVECTOR;
                                n -= AM.OffsetVector;   break;
                            case CFUNCTION: *fill++ = TFUNCTION;
                                n -= FUNCTION; break;
                            case CNUMBER:   *fill++ = TNUMBER1;  break;
                            case CDUBIOUS:  *fill++ = TDUBIOUS; n = 1; break;
                        }
                        i = 0;
if ( n < 0 ) {
    MesPrint("Value of n = %d",n);
}
                        do { numtab[i++] = (SBYTE)(n & 0x7F); n >>= 7; } while ( n );
                        while ( --i >= 0 ) *fill++ = numtab[i];
                    }
                    else {
                        MesPrint("&Illegal element %d in set",numsub);
                        error++;
                    }
                    s++; continue;
                }
            }
            *fill++ = *s++;
        }
        else *fill++ = *s++;
    }
    *fill++ = TENDOFIT;
    return(error);
}

/*
        #] simp1token :
        #[ simpwtoken :

        Only to be called in the LHS.
        Hunts down the wildcards and writes them to the wildcardbuffer.
        Next it causes the ProtoType to be constructed.
        All wildcards are simplified into the trailing TWILDCARD,
        because the specifics are stored in the prototype.
        These specifics also include the transfer of wildcard values
        to $variables.

        Types of wildcards:
        a?, a?set, a?!set, a?set[i], A?set1?set2, ?a
        After this we can strip the set information.
        We still need the ? because of the wildcarding offset in code generation
*/

int simpwtoken(SBYTE *s)
{
    int error = 0, first = 1, notflag;
    WORD num, numto, numdollar, *w = AC.WildC, *wstart, *wtop;
    SBYTE *fill = s, *t, *v, *s0 = s;
    while ( *s != TENDOFIT ) {
        if ( *s == TWILDCARD ) {
            notflag = 0; t = fill;
            while ( t > s0 && t[-1] >= 0 ) t--;
            v = t; num = 0; *fill++ = *s++;
            while ( *v >= 0 ) num = 128*num + *v++;
            if ( t > s0 ) t--;
            AC.NwildC += 4;
            if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
            switch ( *t ) {
                case TSYMBOL:
                case TDUBIOUS:
                    *w++ = SYMTOSYM; *w++ = 4; *w++ = num; *w++ = num; break;
                case TINDEX:
                    num += AM.OffsetIndex;
                    *w++ = INDTOIND; *w++ = 4; *w++ = num; *w++ = num;  break;
                case TVECTOR:
                    num += AM.OffsetVector;
                    *w++ = VECTOVEC; *w++ = 4; *w++ = num; *w++ = num;  break;
                case TFUNCTION:
                    num += FUNCTION;
                    *w++ = FUNTOFUN; *w++ = 4; *w++ = num; *w++ = num;  break;
                default:
                    MesPrint("&Illegal type of wildcard in LHS");
                    error = -1;
                    *w++ = SYMTOSYM; *w++ = 4; *w++ = num; *w++ = num;  break;
                    break;
            }
/*
            Now the sets. The s pointer sits after the ?
*/
            wstart = w;
```

```
            if ( *s == TNOT && s[1] == TSET ) { notflag = 1; s++; }
            if ( *s == TSET ) {
                s++; num = 0; while ( *s >= 0 ) num = 128*num + *s++;
                if ( notflag == 0 && *s == TWILDCARD && s[1] == TSET ) {
                    s += 2; numto = 0; while ( *s >= 0 ) numto = 128*numto + *s++;
                    if ( num < AM.NumFixedSets || numto < AM.NumFixedSets
                    || Sets[num].type == CRANGE || Sets[numto].type == CRANGE ) {
                        MesPrint("&This type of set not allowed in this wildcard construction");
                        error = 1;
                    }
                    else {
                    AC.NwildC += 4;
                    if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
                    *w++ = FROMSET; *w++ = 4; *w++ = num; *w++ = numto;
                    wstart = w;
                    }
                }
                else if ( notflag == 0 && *s == LBRACE && s[1] == TSYMBOL ) {
                    if ( num < AM.NumFixedSets || Sets[num].type == CRANGE ) {
                        MesPrint("&This type of set not allowed in this wildcard construction");
                        error = 1;
                    }
                    v = s; s += 2;
                    numto = 0; while ( *s >= 0 ) numto = 128*numto + *s++;
                    if ( *s == TWILDCARD ) s++; /* most common mistake */
                    if ( *s == RBRACE ) {
                        s++;
                        AC.NwildC += 8;
                        if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
                        *w++ = SETTONUM; *w++ = 4; *w++ = num; *w++ = numto;
                        wstart = w;
                        *w++ = SYMTOSYM; *w++ = 4; *w++ = numto; *w++ = 0;
                    }
                    else if ( *s == TDOLLAR ) {
                        s++; numdollar = 0;
                        while ( *s >= 0 ) numdollar = 128*numdollar + *s++;
                        if ( *s == RBRACE ) {
                            s++;
                            AC.NwildC += 12;
                            if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
                            *w++ = SETTONUM; *w++ = 4; *w++ = num; *w++ = numto;
                            wstart = w;
                            *w++ = SYMTOSYM; *w++ = 4; *w++ = numto; *w++ = 0;
                            *w++ = LOADDOLLAR; *w++ = 4; *w++ = numdollar;
                            *w++ = numdollar;
                        }
                        else { s = v; goto singlewild; }
                    }
                    else { s = v; goto singlewild; }
                }
                else {
singlewild:         num += notflag * 2*WILDOFFSET;
                    AC.NwildC += 4;
                    if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
                    *w++ = FROMSET; *w++ = 4; *w++ = num; *w++ = -WILDOFFSET;
                    wstart = w;
                }
            }
            else if ( *s != TDOLLAR && *s != TENDOFIT && *s != RPARENTHESIS
            && *s != RBRACE && *s != TCOMMA && *s != TFUNCLOSE && *s != TMULTIPLY
            && *s != TPOWER && *s != TDIVIDE && *s != TPLUS && *s != TMINUS
            && *s != TPOWER1 && *s != TEMPTY && *s != TFUNOPEN && *s != TDOT ) {
                MesPrint("&Illegal type of wildcard in LHS");
                error = -1;
            }
            if ( *s == TDOLLAR ) {
                s++; numdollar = 0;
                while ( *s >= 0 ) numdollar = 128*numdollar + *s++;
                AC.NwildC += 4;
                if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
                wtop = w + 4;
                if ( wstart < w ) {
                    while ( w > wstart ) { w[4] = w[0]; w--; }
                }
                *w++ = LOADDOLLAR; *w++ = 4; *w++ = numdollar; *w++ = numdollar;
                w = wtop;
            }
        }
        else if ( *s == TWILDARG ) {
            *fill++ = *s++;
            num = 0;
            while ( *s >= 0 ) { num = 128*num + *s;  *fill++ = *s++; }
            AC.NwildC += 4;
            if ( AC.NwildC > 4*AM.MaxWildcards ) {
firsterr:       if ( first ) {
                    MesPrint("&More than %d wildcards",AM.MaxWildcards);
                    error = -1;
```

```
                    first = 0;
                }
            }
            else { *w++ = ARGTOARG; *w++ = 4; *w++ = num; *w++ = -1; }
            if ( *s == TDOLLAR ) {
                s++; num = 0; while ( *s >= 0 ) num = 128*num + *s++;
                AC.NwildC += 4;
                if ( AC.NwildC > 4*AM.MaxWildcards ) goto firsterr;
                *w++ = LOADDOLLAR; *w++ = 4; *w++ = num; *w++ = num;
            }
        }
        else *fill++ = *s++;
    }
    *fill++ = TENDOFIT;
    AC.WildC = w;
    return(error);
}

/*
        #] simpwtoken :
        #[ simp2token :

        Deals with function arguments.
        The tokenizer has given function arguments extra parentheses.
        We remove the double parentheses.
        Next we remove the parentheses around the simple arguments.

        It also replaces /fac_() by *invfac_() and /invfac_() by *fac_()
*/

int simp2token(SBYTE *s)
{
    SBYTE *to, *fill, *t, *v, *w, *s0 = s, *vv;
    int error = 0, n;
/*
    Set substitutions
*/
    fill = to = s;
    while ( *s != TENDOFIT ) {
        if ( *s == LPARENTHESIS && s[1] == LPARENTHESIS ) {
            t = s+1; n = 0;
            while ( n >= 0 ) {
                t++;
                if ( *t == LPARENTHESIS ) n++;
                else if ( *t == RPARENTHESIS ) n--;
            }
            if ( t[1] == RPARENTHESIS ) {
                *t = TEMPTY; s++;
            }
            *fill++ = *s++;
        }
        else if ( *s == TEMPTY ) s++;
        else if ( *s == AM.facnum && ( fill > (s0+1) ) && fill[-2] == TDIVIDE
         && fill[-1] == TFUNCTION ) {
            fill[-2] = TMULTIPLY; *fill++ = (SBYTE)(AM.invfacnum); s++;
        }
        else if ( *s == AM.invfacnum && ( fill > (s0+1) ) && fill[-2] == TDIVIDE
         && fill[-1] == TFUNCTION ) {
            fill[-2] = TMULTIPLY; *fill++ = (SBYTE)(AM.facnum); s++;
        }
        else *fill++ = *s++;
    }
    *fill++ = TENDOFIT;
/*
    Second round: try to locate 'simple' arguments and strip their brackets

    We add (9-feb-2010) to the simple arguments integers of any size
*/
    fill = s = to;
    while ( *s != TENDOFIT ) {
        if ( *s == LPARENTHESIS ) {
            t = s; n = 0;
            while ( n >= 0 ) {
                t++;
                if ( *t == LPARENTHESIS ) n++;
                else if ( *t == RPARENTHESIS ) n--;
            }
            if ( t[1] == TFUNCLOSE && s[1] != TWILDARG ) { /* Check for last argument in sum */
                v = fill - 1; n = 0;
                while ( n >= 0 && v >= to ) {
                    if ( *v == TFUNOPEN ) n--;
                    else if ( *v == TFUNCLOSE ) n++;
                    v--;
                }
                if ( v > to ) {
                    while ( *v >= 0 ) v--;
                    if ( *v == TFUNCTION ) { v++;
```

```
                        n = 0; while ( *v >= 0 && v < fill ) n = 128*n + *v++;
                        if ( n == AM.sumnum || n == AM.sumpnum ) {
                            *fill++ = *s++; continue;
                        }
                        else if ( ( n == (FIRSTBRACKET-FUNCTION)
                        || n == (TERMSINEXPR-FUNCTION)
                        || n == (SIZEOFFUNCTION-FUNCTION)
                        || n == (NUMFACTORS-FUNCTION)
                        || n == (GCDFUNCTION-FUNCTION)
                        || n == (DIVFUNCTION-FUNCTION)
                        || n == (REMFUNCTION-FUNCTION)
                        || n == (INVERSEFUNCTION-FUNCTION)
                        || n == (MULFUNCTION-FUNCTION)
                        || n == (FACTORIN-FUNCTION)
                        || n == (FIRSTTERM-FUNCTION)
                        || n == (CONTENTTERM-FUNCTION) )
                        && fill[-1] == TFUNOPEN ) {
                            v = s+1;
                            if ( *v == TEXPRESSION ) {
                                v++;
                                n = 0; while ( *v >= 0 ) n = 128*n + *v++;
                                if ( v == t ) {
                                    *t = TEMPTY; s++;
                                }
                            }
                        }
                    }
                }
            }
            if ( ( fill > to )
             && ( ( fill[-1] == TFUNOPEN || fill[-1] == TCOMMA )
               && ( t[1] == TFUNCLOSE || t[1] == TCOMMA ) ) ) {
                v = s + 1;
                switch ( *v ) {
                    case TMINUS:
                        v++;
                        if ( *v == TVECTOR ) {
                            w = v+1; while ( *w >= 0 ) w++;
                            if ( w == t ) {
                                *t = TEMPTY; s++;
                            }
                        }
                        else {
                            if ( *v == TNUMBER || *v == TNUMBER1 ) {
                              if ( BITSINWORD == 16 ) { ULONG x; WORD base;
                                base = ( *v == TNUMBER ) ? 100: 128;
                                vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
                                if ( ( vv != t ) || ( ( vv - v ) > 4 ) || ( x > (MAXPOSITIVE+1) ) )
                                    *fill++ = *s++;
                                else { *t = TEMPTY; s++; break; }
                              }
                              else if ( BITSINWORD == 32 ) { ULONG x; WORD base;
                                base = ( *v == TNUMBER ) ? 100: 128;
                                vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
                                if ( ( vv != t ) || ( ( vv - v ) > 6 ) || ( x > (MAXPOSITIVE+1) ) )
                                    *fill++ = *s++;
                                else { *t = TEMPTY; s++; break; }
                              }
                              else {
                                if ( ( v+2 == t ) || ( v+3 == t && v[2] >= 0 ) )
                                    { *t = TEMPTY; s++; break; }
                                else *fill++ = *s++;
                              }
                            }
                            else if ( *v == LPARENTHESIS && t[-1] == RPARENTHESIS ) {
                                w = v; n = 0;
                                while ( n >= 0 ) {
                                    w++;
                                    if ( *w == LPARENTHESIS ) n++;
                                    else if ( *w == RPARENTHESIS ) n--;
                                }
                                if ( w == ( t-1 ) ) { *t = TEMPTY; s++; }
                                else *fill++ = *s++;
                            }
                            else *fill++ = *s++;
                            break;
                        }
                        /* fall through */
                    case TSETNUM:
                        v++; while ( *v >= 0 ) v++;
                        goto tcommon;
                    case TSYMBOL:
                        if ( ( v[1] == COEFFSYMBOL || v[1] == NUMERATORSYMBOL
                        || v[1] == DENOMINATORSYMBOL ) && v[2] < 0 ) {
                            *fill++ = *s++; break;
                        }
                        /* fall through */
```

```
                    case TSET:
                    case TVECTOR:
                    case TINDEX:
                    case TFUNCTION:
                    case TDOLLAR:
                    case TDUBIOUS:
                    case TSGAMMA:
tcommon:                v++; while ( *v >= 0 ) v++;
                        if ( v == t || ( v[0] == TWILDCARD && v+1 == t ) )
                        { *t = TEMPTY; s++; }
                        else *fill++ = *s++;
                        break;
                    case TGENINDEX:
                        v++;
                        if ( v == t ) { *t = TEMPTY; s++; }
                        else *fill++ = *s++;
                        break;
                    case TNUMBER:
                    case TNUMBER1:
                        if ( BITSINWORD == 16 ) { ULONG x; WORD base;
                            base = ( *v == TNUMBER ) ? 100: 128;
                            vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
                            if ( ( vv != t ) || ( ( vv - v ) > 4 ) || ( x > MAXPOSITIVE ) )
                                *fill++ = *s++;
                            else { *t = TEMPTY; s++; break; }
                        }
                        else if ( BITSINWORD == 32 ) { ULONG x; WORD base;
                            base = ( *v == TNUMBER ) ? 100: 128;
                            vv = v+1; x = 0; while ( *vv >= 0 ) { x = x*base + *vv++; }
                            if ( ( vv != t ) || ( ( vv - v ) > 6 ) || ( x > MAXPOSITIVE ) )
                                *fill++ = *s++;
                            else { *t = TEMPTY; s++; break; }
                        }
                        else {
                            if ( ( v+2 == t ) || ( v+3 == t && v[2] >= 0 ) )
                                { *t = TEMPTY; s++; break; }
                            else *fill++ = *s++;
                        }
                        break;
                    case TWILDARG:
                        v++; while ( *v >= 0 ) v++;
                        if ( v == t ) { *t = TEMPTY; s++; }
                        else *fill++ = *s++;
                        break;
                    case TEXPRESSION:
/*
                        First establish that there is only the expression
                        in this argument.
*/
                        vv = s+1;
                        while ( vv < t ) {
                            if ( *vv != TEXPRESSION ) break;
                            vv++; while ( *vv >= 0 ) vv++;
                        }
                        if ( vv < t ) { *fill++ = *s++; break; }
/*
                        Find the function
*/
                        w = fill-1; n = 0;
                        while ( n >= 0 && w >= to ) {
                            if ( *w == TFUNOPEN ) n--;
                            else if ( *w == TFUNCLOSE ) n++;
                            w--;
                        }
                        w--; while ( w > to && *w >= 0 ) w--;
                        if ( *w != TFUNCTION ) { *fill++ = *s++; break; }
                        w++; n = 0;
                        while ( *w >= 0 ) { n = 128*n + *w++; }
                        if ( n == GCDFUNCTION-FUNCTION
                        || n == DIVFUNCTION-FUNCTION
                        || n == REMFUNCTION-FUNCTION
                        || n == INVERSEFUNCTION-FUNCTION
                        || n == MULFUNCTION-FUNCTION ) {
                            *t = TEMPTY; s++;
                        }
                        else *fill++ = *s++;
                        break;
                    default: *fill++ = *s++; break;
                }
            }
            else *fill++ = *s++;
        }
        else if ( *s == TEMPTY ) s++;
        else *fill++ = *s++;
    }
    *fill++ = TENDOFIT;
    return(error);
```

```
}

/*
        #] simp2token :
        #[ simp3atoken :

        We hunt for denominators and exponents that seem hidden.
        For the denominators we have to recognize:
            /fun  /fun()  /fun^power  /fun()^power
            /set[n]  /set[n]()  /set[n]^power  /set[n]()^power
            /symbol^power (power no number or symbol wildcard)
            /dotpr^power (id)
            /#^power (id)
            /()  /()^power
            /vect /index /vect(anything) /vect(anything)^power
*/

int simp3atoken(SBYTE *s, int mode)
{
    int error = 0, n, numexp = 0, denom, base, numprot, i;
    SBYTE *t, c;
    LONG num;
    WORD *prot;
    if ( mode == RHSIDE ) {
        prot = AC.ProtoType;
        numprot = prot[1] - SUBEXPSIZE;
        prot += SUBEXPSIZE;
    }
    else { prot = 0; numprot = 0; }
    while ( *s != TENDOFIT ) {
        denom = 1;
        if ( *s == TDIVIDE ) { denom = -1; s++; }
        c = *s;
        switch(c) {
            case TSYMBOL:
            case TNUMBER:
            case TNUMBER1:
                s++; while ( *s >= 0 ) s++; /* skip the object */
                if ( *s == TWILDCARD ) s++; /* and the possible wildcard */
dosymbol:
                if ( *s != TPOWER ) continue;   /* No power -> done */
                s++;                            /* Skip the power */
                if ( *s == TMINUS ) s++;        /* negative: no difference here */
                if ( *s == TNUMBER || *s == TNUMBER1 ) {
                    base = *s == TNUMBER ? 100: 128; /* NUMBER = base 100 */
                    s++;                /* Now we compose the power */
                    num = *s++;         /* If the number is way too large */
                    while ( *s >= 0 ) { /* it may look like not too big */
                        if ( num > MAXPOWER ) break;    /* Hence... */
                        num = base*num + *s++;
                    }
                    while ( *s >= 0 ) s++;  /* Finish the number if needed */
                    if ( *s == TPOWER ) goto doublepower;
                    if ( num <= MAXPOWER ) continue;    /* Simple case */
                }
                else if ( *s == TSYMBOL && c != TNUMBER && c != TNUMBER1 ) {
                    s++; n = 0; while ( *s >= 0 ) { n = 128*n + *s++; }
                    if ( *s == TWILDCARD ) { s++;
                        if ( *s == TPOWER ) goto doublepower;
                        continue; }
/*
                    Now we have to test whether n happens to be a wildcard
*/
                    if ( mode == RHSIDE ) {
                        n += 2*MAXPOWER;
                        for ( i = 0; i < numprot; i += 4 ) {
                            if ( prot[i+2] == n && prot[i] == SYMTOSYM ) break;
                        }
                        if ( i < numprot ) break;
                    }
                    if ( *s == TPOWER ) goto doublepower;
                }
                numexp++;
                break;
            case TINDEX:
                s++; while ( *s >= 0 ) s++;
                if ( *s == TWILDCARD ) s++;
doindex:
                if ( denom < 0 || *s == TPOWER ) {
                    MesPrint("&Index to a power or in denominator is illegal");
                    error = 1;
                }
                break;
            case TVECTOR:
                s++; while ( *s >= 0 ) s++;
                if ( *s == TWILDCARD ) s++;
dovector:
```

```
                if ( *s == TFUNOPEN ) {
                    s++; n = 1;
                    for(;;) {
                        if ( *s == TFUNOPEN ) {
                            n++;
                            MesPrint("&Illegal vector index");
                            error = 1;
                        }
                        else if ( *s == TFUNCLOSE ) {
                            n--;
                            if ( n <= 0 ) break;
                        }
                        s++;
                    }
                    s++;
                }
                else if ( *s == TDOT ) goto dodot;
                if ( denom < 0 || *s == TPOWER || *s == TPOWER1 ) numexp++;
                break;
            case TFUNCTION:
                s++; while ( *s >= 0 ) s++;
                if ( *s == TWILDCARD ) s++;
dofunction:
                t = s;
                if ( *t == TFUNOPEN ) {
                    t++; n = 1;
                    for(;;) {
                        if ( *t == TFUNOPEN ) n++;
                        else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
                        t++;
                    }
                    t++; s++;
                }
                if ( denom < 0 || *t == TPOWER || *t == TPOWER1 ) numexp++;
                break;
#ifdef WITHFLOAT
            case TFLOAT:
                s++; while ( *s >= 0 ) s++;
                if ( denom < 0 || *s == TPOWER || *s == TPOWER1 ) numexp++;
                break;
#endif
            case TEXPRESSION:
                s++; while ( *s >= 0 ) s++;
                t = s;
                if ( *t == TFUNOPEN ) {
                    t++; n = 1;
                    for(;;) {
                        if ( *t == TFUNOPEN ) n++;
                        else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
                        t++;
                    }
                    t++;
                }
                if ( *t == LBRACE ) {
                    t++; n = 1;
                    for(;;) {
                        if ( *t == LBRACE ) n++;
                        else if ( *t == RBRACE ) { if ( --n <= 0 ) break; }
                        t++;
                    }
                    t++;
                }
                if ( denom < 0 || ( ( *t == TPOWER || *t == TPOWER1 )
                && t[1] == TMINUS ) ) numexp++;
                break;
            case TDOLLAR:
                s++; while ( *s >= 0 ) s++;
                if ( denom < 0 || ( ( *s == TPOWER || *s == TPOWER1 )
                && s[1] == TMINUS ) ) numexp++;
                break;
            case LPARENTHESIS:
                s++; n = 1; t = s;
                for(;;) {
                    if ( *t == LPARENTHESIS ) n++;
                    else if ( *t == RPARENTHESIS ) { if ( --n <= 0 ) break; }
                    t++;
                }
                t++;
                if ( denom > 0 && ( *t == TPOWER || *t == TPOWER1 ) ) {
                    if ( ( t[1] == TNUMBER || t[1] == TNUMBER1 ) && t[2] >= 0
                    && t[3] < 0 ) break;
                    numexp++;
                }
                else if ( denom < 0 && ( *t == TPOWER || *t == TPOWER1 ) ) {
                    if ( t[1] == TMINUS && ( t[2] == TNUMBER
                     || t[2] == TNUMBER1 ) && t[3] >= 0
                    && t[4] < 0 ) break;
```

```
                numexp++;
            }
            else if ( denom < 0 || ( ( *t == TPOWER || *t == TPOWER1 )
            && ( t[1] == TMINUS || t[1] == LPARENTHESIS ) ) ) numexp++;
            break;
        case TSET:
            s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
            n = Sets[n].type;
            switch ( n ) {
                case CSYMBOL: goto dosymbol;
                case CINDEX: goto doindex;
                case CVECTOR: goto dovector;
                case CFUNCTION: goto dofunction;
                case CNUMBER: goto dosymbol;
                case CMODEL:
                    if ( denom < 0 || *s == TPOWER ) {
                        MesPrint("&A model to a power or in denominator is illegal");
                        error = 1;
                    }
                    break;
                case ANYTYPE:
                    if ( denom < 0 || *s == TPOWER ) {
                        MesPrint("&A set without type to a power or in denominator is illegal");
                        error = 1;
                    }
                    break;
                default: error = 1; break;
            }
            break;
        case TDOT:
dodot:      s++;
            if ( *s == TVECTOR ) { s++; while ( *s >= 0 ) s++; }
            else if ( *s == TSET ) {
                s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
                if ( Sets[n].type != CVECTOR ) {
                    MesPrint("&Set in dotproduct is not a set of vectors");
                    error = 1;
                }
                if ( *s == LBRACE ) {
                    s++; n = 1;
                    for(;;) {
                        if ( *s == LBRACE ) n++;
                        else if ( *s == RBRACE ) { if ( --n <= 0 ) break; }
                        s++;
                    }
                    s++;
                }
                else {
                    MesPrint("&Set without argument in dotproduct");
                    error = 1;
                }
            }
            else if ( *s == TSETNUM ) {
                s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
                if ( *s != TVECTOR ) goto nodot;
                s++; n = *s++; while ( *s >= 0 ) { n = 128*n + *s++; }
                if ( Sets[n].type != CVECTOR ) {
                    MesPrint("&Set in dotproduct is not a set of vectors");
                    error = 1;
                }
            }
            else {
nodot:          MesPrint("&Illegal second element in dotproduct");
                error = 1;
                s++; while ( *s >= 0 ) s++;
            }
            goto dosymbol;
        default:
            s++; while ( *s >= 0 ) s++;
            break;
    }
  }
  if ( error ) return(-1);
  return(numexp);
doublepower:
  MesPrint("&Dubious notation with object^power1^power2");
  return(-1);
}

/*
        #] simp3atoken :
        #[ simp3btoken :
*/

int simp3btoken(SBYTE *s, int mode)
{
  int error = 0, i, numprot, n, denom, base, inset = 0, dotp, sube = 0;
```

```
    SBYTE *t, c, *fill, *ff, *ss;
    LONG num;
    WORD *prot;
    if ( mode == RHSIDE ) {
        prot = AC.ProtoType;
        numprot = prot[1] - SUBEXPSIZE;
        prot += SUBEXPSIZE;
    }
    else { prot = 0; numprot = 0; }
    fill = s;
    while ( *s == TEMPTY ) s++;
    while ( *s != TENDOFIT ) {
        if ( *s == TEMPTY ) { s++; continue; }
        denom = 1;
        if ( *s == TDIVIDE ) { denom = -1; *fill++ = *s++; }
        ff = fill; ss = s; c = *s;
        if ( c == TSETNUM ) {
            *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
            c = *s;
        }
        dotp = 0;
        switch(c) {
            case TSYMBOL:
            case TNUMBER:
            case TNUMBER1:
                *fill++ = *s++;
                while ( *s >= 0 ) *fill++ = *s++;
                if ( *s == TWILDCARD ) *fill++ = *s++;
dosymbol:
                t = s;
                if ( *s != TPOWER ) continue;
                *fill++ = *s++;
                if ( *s == TMINUS ) *fill++ = *s++;
                if ( *s == TPLUS ) s++;
                if ( *s == TSETNUM ) {
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                    inset = 1;
                }
                else inset = 0;
                if ( *s == TNUMBER || *s == TNUMBER1 ) {
                    base = *s == TNUMBER ? 100: 128;
                    *fill++ = *s++;
                    num = *s++; *fill++ = num;
                    while ( *s >= 0 ) {
                        if ( num > MAXPOWER ) break;
                        *fill++ = *s;
                        num = base*num + *s++;
                    }
                    while ( *s >= 0 ) *fill++ = *s++;
                    if ( num <= MAXPOWER ) continue;
                    goto putexp1;
                }
                else if ( *s == TSYMBOL && c != TNUMBER && c != TNUMBER1 ) {
                    *fill++ = *s++;
                    n = 0; while ( *s >= 0 ) { n = 128*n + *s; *fill++ = *s++; }
                    if ( *s == TWILDCARD ) { *fill++ = *s++;
                        if ( *s == TPOWER ) goto doublepower;
                    break; }
/*
                    Now we have to test whether n happens to be a wildcard
*/
                    if ( mode == RHSIDE && inset == 0 ) {
/*                      n += WILDOFFSET;*/
                        for ( i = 0; i < numprot; i += 4 ) {
                            if ( prot[i+2] == n && prot[i] == SYMTOSYM ) break;
                        }
                        if ( i < numprot ) break;
                    }

putexp1:            fill = ff;
                    if ( denom < 0 ) fill[-1] = TMULTIPLY;
                    *fill++ = TFUNCTION; *fill++ = (SBYTE)(AM.expnum); *fill++ = TFUNOPEN;
                    if ( dotp ) *fill++ = LPARENTHESIS;
                    while ( ss < t ) *fill++ = *ss++;
                    if ( dotp ) *fill++ = RPARENTHESIS;
                    *fill++ = TCOMMA;
                    ss++;                       /* Skip TPOWER */
                    if ( *ss == TMINUS ) { denom = -denom; ss++; }
                    if ( denom < 0 ) {
                        *fill++ = LPARENTHESIS;
                        *fill++ = TMINUS;
                        while ( ss < s ) *fill++ = *ss++;
                        *fill++ = RPARENTHESIS;
                    }
                    else {
                        while ( ss < s ) *fill++ = *ss++;
                    }
```

```
                        *fill++ = TFUNCLOSE;
                        if ( *ss == TPOWER ) goto doublepower;
                    }
                    else {  /* other objects can be composite */
                        goto dofunpower;
                    }
                    break;
                case TINDEX:
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                    if ( *s == TWILDCARD ) *fill++ = *s++;
                    break;
                case TVECTOR:
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                    if ( *s == TWILDCARD ) *fill++ = *s++;
dovector:
                    if ( *s == TFUNOPEN ) {
                        while ( *s != TFUNCLOSE ) *fill++ = *s++;
                        *fill++ = *s++;
                    }
                    else if ( *s == TDOT ) goto dodot;
                    t = s;
                    goto dofunpower;
#ifdef WITHFLOAT
                case TFLOAT:
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                    t = s;
                    sube = 0;
                    goto dofunpower;
#endif
                case TFUNCTION:
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                    if ( *s == TWILDCARD ) *fill++ = *s++;
dofunction:
                    t = s;
                    if ( *t == TFUNOPEN ) {
                        t++; n = 1;
                        for(;;) {
                            if ( *t == TFUNOPEN ) n++;
                            else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
                            t++;
                        }
                        t++; *fill++ = *s++;
                    }
                    sube = 0;
dofunpower:
                    if ( *t == TPOWER || *t == TPOWER1 ) {
                        if ( sube ) {
                            if ( ( t[1] == TNUMBER || t[1] == TNUMBER1 )
                             && denom > 0 ) {
                                if ( t[2] >= 0 && t[3] < 0 ) { sube = 0; break; }
                            }
                            else if ( t[1] == TMINUS && denom < 0 &&
                            ( t[2] == TNUMBER || t[2] == TNUMBER1 ) ) {
                                if ( t[2] >= 0 && t[3] < 0 ) { sube = 0; break; }
                            }
                            sube = 0;
                        }
                        fill = ff;
                        *fill++ = TFUNCTION; *fill++ = (SBYTE)(AM.expnum); *fill++ = TFUNOPEN;
                        *fill++ = LPARENTHESIS;
                        while ( ss < t ) *fill++ = *ss++;
                        t++;
                        *fill++ = RPARENTHESIS; *fill++ = TCOMMA;
                        if ( *t == TMINUS ) { t++; denom = -denom; }
                        *fill++ = LPARENTHESIS;
                        if ( denom < 0 ) *fill++ = TMINUS;
                        if ( *t == LPARENTHESIS ) {
                            *fill++ = *t++; n = 0;
                            while ( n >= 0 ) {
                                if ( *t == LPARENTHESIS ) n++;
                                else if ( *t == RPARENTHESIS ) n--;
                                *fill++ = *t++;
                            }
                        }
                        else if ( *t == TFUNCTION || *t == TDUBIOUS ) {
                            *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
                            if ( *t == TWILDCARD ) *fill++ = *t++;
                            if ( *t == TFUNOPEN ) {
                                *fill++ = *t++; n = 0;
                                while ( n >= 0 ) {
                                    if ( *t == TFUNOPEN ) n++;
                                    else if ( *t == TFUNCLOSE ) n--;
                                    *fill++ = *t++;
                                }
                            }
                        }
                        else if ( *t == TSET ) {
```

```
                        *fill++ = *t++; n = 0;
                        while ( *t >= 0 ) { n = 128*n + *t; *fill++ = *t++; }
                        if ( *t == LBRACE ) {
                            if ( n < AM.NumFixedSets || Sets[n].type == CRANGE ) {
                                MesPrint("&This type of usage of sets is not allowed");
                                error = 1;
                            }
                            *fill++ = *t++; n = 0;
                            while ( n >= 0 ) {
                                if ( *t == LBRACE ) n++;
                                else if ( *t == RBRACE ) n--;
                                *fill++ = *t++;
                            }
                        }
                    }
                    else if ( *t == TEXPRESSION ) {
                        *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
                        if ( *t == TFUNOPEN ) {
                            *fill++ = *t++; n = 0;
                            while ( n >= 0 ) {
                                if ( *t == TFUNOPEN ) n++;
                                else if ( *t == TFUNCLOSE ) n--;
                                *fill++ = *t++;
                            }
                        }
                        if ( *t == LBRACE ) {
                            *fill++ = *t++; n = 0;
                            while ( n >= 0 ) {
                                if ( *t == LBRACE ) n++;
                                else if ( *t == RBRACE ) n--;
                                *fill++ = *t++;
                            }
                        }
                    }
                    else if ( *t == TVECTOR ) {
                        *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
                        if ( *t == TFUNOPEN ) {
                            *fill++ = *t++; n = 0;
                            while ( n >= 0 ) {
                                if ( *t == TFUNOPEN ) n++;
                                else if ( *t == TFUNCLOSE ) n--;
                                *fill++ = *t++;
                            }
                        }
                        else if ( *t == TDOT ) {
                            *fill++ = *t++;
                            if ( *t == TVECTOR || *t == TDUBIOUS ) {
                                *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
                            }
                            else if ( *t == TSET ) {
                                *fill++ = *t++; num = 0;
                                while ( *t >= 0 ) { num = 128*num + *t; *fill++ = *t++; }
                                if ( Sets[num].type != CVECTOR ) {
                                    MesPrint("&Illegal set type in dotproduct");
                                    error = 1;
                                }
                                if ( *t == LBRACE ) {
                                    *fill++ = *t++; n = 0;
                                    while ( n >= 0 ) {
                                        if ( *t == LBRACE ) n++;
                                        else if ( *t == RBRACE ) n--;
                                        *fill++ = *t++;
                                    }
                                }
                            }
                            else if ( *t == TSETNUM ) {
                                *fill++ = *t++;
                                while ( *t >= 0 ) { *fill++ = *t++; }
                                *fill++ = *t++;
                                while ( *t >= 0 ) { *fill++ = *t++; }
                            }
                        }
                        else {
                            MesPrint("&Illegal second element in dotproduct");
                            error = 1;
                        }
                    }
                    else {
                        *fill++ = *t++; while ( *t >= 0 ) *fill++ = *t++;
                        if ( *t == TWILDCARD ) *fill++ = *t++;
                    }
                    *fill++ = RPARENTHESIS; *fill++ = TFUNCLOSE;
                    if ( *t == TPOWER ) goto doublepower;
                    while ( fill > ff ) *--t = *--fill;
                    s = t;
                }
                else if ( denom < 0 ) {
```

```
                    fill = ff; ff[-1] = TMULTIPLY;
                    *fill++ = TFUNCTION; *fill++ = (SBYTE)(AM.denomnum);
                    *fill++ = TFUNOPEN; *fill++ = LPARENTHESIS;
                    while ( ss < t ) *fill++ = *ss++;
                    *fill++ = RPARENTHESIS; *fill++ = TFUNCLOSE;
                    while ( fill > ff ) *--t = *--fill;
                    s = t; denom = 1; sube = 0;
                    break;
                }
                sube = 0;
                break;
            case TEXPRESSION:
                *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                t = s;
                if ( *t == TFUNOPEN ) {
                    t++; n = 1;
                    for(;;) {
                        if ( *t == TFUNOPEN ) n++;
                        else if ( *t == TFUNCLOSE ) { if ( --n <= 0 ) break; }
                        t++;
                    }
                    t++;
                }
                if ( *t == LBRACE ) {
                    t++; n = 1;
                    for(;;) {
                        if ( *t == LBRACE ) n++;
                        else if ( *t == RBRACE ) { if ( --n <= 0 ) break; }
                        t++;
                    }
                    t++;
                }
                if ( t > s || denom < 0 || ( ( *t == TPOWER || *t == TPOWER1 )
                && t[1] == TMINUS ) ) goto dofunpower;
                else goto dosymbol;
            case TDOLLAR:
                *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                goto dosymbol;
            case LPARENTHESIS:
                *fill++ = *s++; n = 1; t = s;
                for(;;) {
                    if ( *t == LPARENTHESIS ) n++;
                    else if ( *t == RPARENTHESIS ) { if ( --n <= 0 ) break; }
                    t++;
                }
                t++; sube = 1;
                goto dofunpower;
            case TSET:
                *fill++ = *s++; n = *s++; *fill++ = (SBYTE)n;
                while ( *s >= 0 ) { *fill++ = *s; n = 128*n + *s++; }
                n = Sets[n].type;
                switch ( n ) {
                    case CSYMBOL: goto dosymbol;
                    case CINDEX: break;
                    case CVECTOR: goto dovector;
                    case CFUNCTION: goto dofunction;
                    case CNUMBER: goto dosymbol;
                    case CMODEL: break;
                    case ANYTYPE: break;
                    default: error = 1; break;
                }
                break;
            case TDOT:
dodot:          *fill++ = *s++;
                if ( *s == TVECTOR ) {
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                }
                else if ( *s == TSET ) {
                    *fill++ = *s++; n = *s++; *fill++ = (SBYTE)n;
                    while ( *s >= 0 ) { *fill++ = *s; n = 128*n + *s++; }
                    if ( *s == LBRACE ) {
                        if ( n < AM.NumFixedSets || Sets[n].type == CRANGE ) {
                            MesPrint("&This type of usage of sets is not allowed");
                            error = 1;
                        }
                        *fill++ = *s++; n = 1;
                        for(;;) {
                            if ( *s == LBRACE ) n++;
                            else if ( *s == RBRACE ) { if ( --n <= 0 ) break; }
                            *fill++ = *s++;
                        }
                        *fill++ = *s++;
                    }
                    else {
                        MesPrint("&Set without argument in dotproduct");
                        error = 1;
                    }
```

```
                }
                else if ( *s == TSETNUM ) {
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                    if ( *s != TVECTOR ) goto nodot;
                    *fill++ = *s++; while ( *s >= 0 ) *fill++ = *s++;
                }
                else {
nodot:              MesPrint("&Illegal second element in dotproduct");
                    error = 1;
                    *fill++ = *s++;
                    while ( *s >= 0 ) *fill++ = *s++;
                }
                dotp = 1;
                goto dosymbol;
            default:
                *fill++ = *s++;
                while ( *s >= 0 ) *fill++ = *s++;
                break;
        }
    }
    *fill = TENDOFIT;
    return(error);
doublepower:;
    MesPrint("&Dubious notation with power of power");
    return(-1);
}

/*
        #] simp3btoken :
        #[ simp4token :

        Deal with the set[n] objects in the RHS.
*/

int simp4token(SBYTE *s)
{
    int error = 0, n, nsym, settype;
    WORD i, *w, *wstop, level;
    SBYTE *const s0 = s;
    SBYTE *fill = s, *s1, *s2, *s3, type, s1buf[10];
    SBYTE *tbuf = s, *t, *t1;

    while ( *s != TENDOFIT ) {
        if ( *s != TSET ) {
            if ( *s == TEMPTY ) s++;
            else *fill++ = *s++;
            continue;
        }
        if ( fill >= (s0+1) && fill[-1] == TWILDCARD ) { *fill++ = *s++; continue; }
        if ( fill >= (s0+2) && fill[-1] == TNOT && fill[-2] == TWILDCARD ) { *fill++ = *s++; continue;
}
        s1 = s++; n = 0; while ( *s >= 0 ) { n = 128*n + *s++; }
        i = Sets[n].type;
        if ( *s != LBRACE ) { while ( s1 < s ) *fill++ = *s1++; continue; }
        if ( n < AM.NumFixedSets || i == CRANGE ) {
            MesPrint("&It is not allowed to refer to individual elements of built in or ranged sets");
            error = 1;
        }
        s++;
        if ( *s != TSYMBOL && *s != TDOLLAR ) {
            MesPrint("&Set index in RHS is not a wildcard symbol or $-variable");
            error = 1;
            while ( s1 < s ) *fill++ = *s1++;
            continue;
        }
        settype = ( *s == TDOLLAR );
        s++; nsym = 0; s2 = s;
        while ( *s >= 0 ) nsym = 128*nsym + *s++;
        if ( *s != RBRACE ) {
            MesPrint("&Improper set argument in RHS");
            error = 1;
            while ( s1 < s ) *fill++ = *s1++;
            continue;
        }
        s++;
/*
        Verify that nsym is a wildcard
*/
        if ( !settype ) {
            w = AC.ProtoType; wstop = w + w[1]; w += SUBEXPSIZE;
            while ( w < wstop ) {
                if ( *w == SYMTOSYM && w[2] == nsym ) break;
                w += w[1];
            }
            if ( w >= wstop ) {
/*
                It could still be a summation parameter!
```

```
*/
                t = fill - 1;
                while ( t >= tbuf ) {
                    if ( *t == TFUNCLOSE ) {
                        level = 1; t--;
                        while ( t >= tbuf ) {
                            if ( *t == TFUNCLOSE ) level++;
                            else if ( *t == TFUNOPEN ) {
                                level--;
                                if ( level == 0 ) break;
                            }
                            t--;
                        }
                    }
                    else if ( *t == RBRACE ) {
                        level = 1; t--;
                        while ( t >= tbuf ) {
                            if ( *t == RBRACE ) level++;
                            else if ( *t == LBRACE ) {
                                level--;
                                if ( level == 0 ) break;
                            }
                            t--;
                        }
                    }
                    else if ( *t == RPARENTHESIS ) {
                        level = 1; t--;
                        while ( t >= tbuf ) {
                            if ( *t == RPARENTHESIS ) level++;
                            else if ( *t == LPARENTHESIS ) {
                                level--;
                                if ( level == 0 ) break;
                            }
                            t--;
                        }
                    }
                    else if ( *t == TFUNOPEN ) {
                        t1 = t-1;
                        while ( *t1 > 0 && t1 > tbuf ) t1--;
                        if ( *t1 == TFUNCTION ) {
                            t1++; level = 0;
                            while ( *t1 > 0 ) level = level*128+*t1++;
                            if ( level == (SUMF1-FUNCTION)
                             || level == (SUMF2-FUNCTION) ) {
                                t1 = t + 1;
                                if ( *t1 == LPARENTHESIS ) t1++;
                                if ( *t1 == TSYMBOL ) {
                                  if ( ( t1[1] == COEFFSYMBOL
                                  || t1[1] == NUMERATORSYMBOL
                                  || t1[1] == DENOMINATORSYMBOL )
                                  && t1[2] < 0 ) {}
                                  else {
                                    t1++; level = 0;
                                    while ( *t1 >= 0 && t1 < fill ) level = 128*level + *t1++;
                                    if ( level == nsym && t1 < fill ) {
                                        if ( t[1] == LPARENTHESIS
                                         && *t1 == RPARENTHESIS && t1[1] == TCOMMA ) break;
                                        if ( t[1] != LPARENTHESIS && *t1 == TCOMMA ) break;
                                  }
                                    }
                                }
                            }
                        }
                    }
                    t--;
                }
                if ( t < tbuf ) {
                    fill--;
                    MesPrint("&Set index in RHS is not a wildcard symbol");
                    error = 1;
                    while ( s1 < s ) *fill++ = *s1++;
                    continue;
                }
            }
        }
/*
        Now replace by a set marker: TSETNUM,nsym,TYPE,setnumber
*/
        switch ( i ) {
            case CSYMBOL: type = TSYMBOL; break;
            case CINDEX: type = TINDEX; break;
            case CVECTOR: type = TVECTOR; break;
            case CFUNCTION: type = TFUNCTION; break;
            case CNUMBER: type = TNUMBER1; break;
            case CDUBIOUS: type = TDUBIOUS; break;
            default:
                MesPrint("&Unknown set type in simp4token");
```

```
                error = 1; type = CDUBIOUS; break;
        }
        s3 = s1buf; s1++;
        while ( *s1 >= 0 ) *s3++ = *s1++;
        *s3 = -1; s1 = s1buf;
        if ( settype ) *fill++ = TSETDOL;
        else           *fill++ = TSETNUM;
        while ( *s2 >= 0 ) *fill++ = *s2++;
        *fill++ = type; while ( *s1 >= 0 ) *fill++ = *s1++;
    }
    *fill++ = TENDOFIT;
    return(error);
}

/*
        #] simp4token :
        #[ simp5token :

    Making sure that first argument of sumfunction is not a wildcard already
*/

int simp5token(SBYTE *s, int mode)
{
    int error = 0, n, type;
    WORD *w, *wstop;
    if ( mode == RHSIDE ) {
        while ( *s != TENDOFIT ) {
            if ( *s == TFUNCTION ) {
                s++; n = 0; while ( *s >= 0 ) n = 128*n + *s++;
                if ( n == AM.sumnum || n == AM.sumpnum ) {
                    if ( *s != TFUNOPEN ) continue;
                    s++;
                    if ( *s != TSYMBOL && *s != TINDEX ) continue;
                    type = *s++;
                    n = 0; while ( *s >= 0 ) n = 128*n + *s++;
                    if ( type == TINDEX ) n += AM.OffsetIndex;
                    if ( *s != TCOMMA ) continue;
                    w = AC.ProtoType;
                    wstop = w + w[1];
                    w += SUBEXPSIZE;
                    while ( w < wstop ) {
                        if ( w[2] == n ) {
                            if ( ( type == TSYMBOL && ( w[0] == SYMTOSYM
                            || w[0] == SYMTONUM || w[0] == SYMTOSUB ) ) || (
                            type == TINDEX && ( w[0] == INDTOIND
                            || w[0] == INDTOSUB ) ) ) {
                                error = 1;
                                MesPrint("&Parameter of sum function is already a wildcard");
                            }
                        }
                        w += w[1];
                    }
                }
            }
            else s++;
        }
    }
    return(error);
}

/*
        #] simp5token :
        #[ simp6token :

    Making sure that factorized expressions are used properly
*/

int simp6token(SBYTE *tokens, int mode)
{
/*  EXPRESSIONS e = Expressions; */
    int error = 0, n;
    int level = 0, haveone = 0;
    SBYTE *s = tokens, *ss;
    LONG numterms;
    WORD funnum = 0;
    GETIDENTITY
    if ( mode == RHSIDE ) {
        while ( *s == TPLUS || *s == TMINUS ) s++;
        numterms = 1;
        while ( *s != TENDOFIT ) {
            if ( *s == LPARENTHESIS ) level++;
            else if ( *s == RPARENTHESIS ) level--;
            else if ( *s == TFUNOPEN ) level++;
            else if ( *s == TFUNCLOSE ) level--;
            else if ( ( *s == TPLUS || *s == TMINUS ) && level == 0 ) {
/*
                Special exception: x^-1 etc.
```

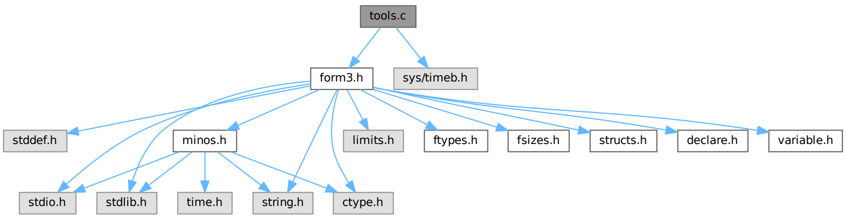
```
*/
                if ( s[-1] != TPOWER && s[-1] != TPLUS && s[-1] != TMINUS ) {
                    numterms++;
                }
            }
            else if ( *s == TEXPRESSION ) {
                ss = s;
                s++; n = 0; while ( *s >= 0 ) n = 128*n + *s++;

                if ( Expressions[n].status == STOREDEXPRESSION ) {
                    POSITION position;
/*
#ifdef WITHPTHREADS
                    RENUMBER renumber;
#endif
*/
                    RENUMBER renumber;

                    WORD TMproto[SUBEXPSIZE];
                    TMproto[0] = EXPRESSION;
                    TMproto[1] = SUBEXPSIZE;
                    TMproto[2] = n;
                    TMproto[3] = 1;
                    { int ie; for ( ie = 4; ie < SUBEXPSIZE; ie++ ) TMproto[ie] = 0; }
                    AT.TMaddr = TMproto;
                    PUTZERO(position);
/*
                    if ( (
#ifdef WITHPTHREADS
                        renumber =
#endif
                            GetTable(n,&position,0) ) == 0 )
*/
                    if ( ( renumber = GetTable(n,&position,0) ) == 0 )
                    {
                        error = 1;
                        MesPrint("&Problems getting information about stored expression %s(4)"
                        ,EXPRNAME(n));
                    }
/*
#ifdef WITHPTHREADS
*/
                    if ( renumber->symb.lo != AN.dummyrenumlist )
                        M_free(renumber->symb.lo,"VarSpace");
                    M_free(renumber,"Renumber");
/*
#endif
*/
                }

                if ( ( ( AS.Oldvflags[n] & ISFACTORIZED ) != 0 ) && *s != LBRACE ) {
                    if ( level == 0 ) {
                        haveone = 1;
                    }
                    else if ( error == 0 ) {
                        if ( ss[-1] != TFUNOPEN || funnum != NUMFACTORS-FUNCTION ) {
                            MesPrint("&Illegal use of factorized expression(s) in RHS");
                            error = 1;
                        }
                    }
                }
                continue;
            }
            else if ( *s == TFUNCTION ) {
                s++; funnum = 0; while ( *s >= 0 ) funnum = 128*funnum + *s++;
                continue;
            }
            s++;
        }
        if ( haveone ) {
            if ( numterms > 1 ) {
                MesPrint("&Factorized expression in RHS in an expression of more than one term.");
                error = 1;
            }
            else if ( AC.ToBeInFactors == 0 ) {
                MesPrint("&Attempt to put a factorized expression inside an unfactorized expression.");
                error = 1;
            }
        }
    }
    return(error);
}

/*
        #] simp6token :
    #] Compiler :
```

```
02134 */
```

4.146 tools.c File Reference

```
#include "form3.h"
#include <sys/timeb.h>
```

Include dependency graph for tools.c:



Macros

- #define FILLVALUE 0
- #define **COPYFILEBUFSIZE** 40960L
- #define TERMMEMSTARTNUM 16
- #define TERMEXTRAWORDS 10
- #define NUMBERMEMSTARTNUM 16
- #define NUMBEREXTRAWORDS 10L
- #define DODOUBLE(x)
- #define DOEXPAND(x)

Functions

- UBYTE ∗ LoadInputFile (UBYTE ∗filename, int type)
- UBYTE ReadFromStream (STREAM ∗stream)
- UBYTE GetFromStream (STREAM ∗stream)
- UBYTE LookInStream (STREAM ∗stream)
- STREAM ∗ OpenStream (UBYTE ∗name, int type, int prevarmode, int raiselow)
- int LocateFile (UBYTE ∗∗name, int type)
- STREAM ∗ CloseStream (STREAM ∗stream)
- STREAM ∗ CreateStream (UBYTE ∗where)
- LONG GetStreamPosition (STREAM ∗stream)
- void PositionStream (STREAM ∗stream, LONG position)
- int ReverseStatements (STREAM ∗stream)
- void StartFiles (void)
- int OpenFile (char ∗name)
- int OpenAddFile (char ∗name)
- int ReOpenFile (char ∗name)
- int CreateFile (char ∗name)
- int CreateLogFile (char ∗name)
- void CloseFile (int handle)
- int CopyFile (char ∗source, char ∗dest)

- int CreateHandle (void)
- LONG ReadFile (int handle, UBYTE ∗buffer, LONG size)
- LONG ReadPosFile (PHEAD FILEHANDLE ∗fi, UBYTE ∗buffer, LONG size, POSITION ∗pos)
- LONG WriteFileToFile (int handle, UBYTE ∗buffer, LONG size)
- void SeekFile (int handle, POSITION ∗offset, int origin)
- LONG TellFile (int handle)
- void TELLFILE (int handle, POSITION ∗position)
- void FlushFile (int handle)
- int GetPosFile (int handle, fpos_t ∗pospointer)
- int SetPosFile (int handle, fpos_t ∗pospointer)
- void SynchFile (int handle)
- void TruncateFile (int handle)
- int GetChannel (char ∗name, int mode)
- int GetAppendChannel (char ∗name)
- int CloseChannel (char ∗name)
- void UpdateMaxSize (void)
- int StrCmp (UBYTE ∗s1, UBYTE ∗s2)
- int StrICmp (UBYTE ∗s1, UBYTE ∗s2)
- int StrHICmp (UBYTE ∗s1, UBYTE ∗s2)
- int StrICont (UBYTE ∗s1, UBYTE ∗s2)
- int CmpArray (WORD ∗t1, WORD ∗t2, WORD n)
- int ConWord (UBYTE ∗s1, UBYTE ∗s2)
- int StrLen (UBYTE ∗s)
- void NumToStr (UBYTE ∗s, LONG x)
- void WriteString (int type, UBYTE ∗str, int num)
- void WriteUnfinString (int type, UBYTE ∗str, int num)
- UBYTE ∗ AddToString (UBYTE ∗outstring, UBYTE ∗extrastring, int par)
- UBYTE ∗ strDup1 (UBYTE ∗instring, char ∗ifwrong)
- UBYTE ∗ EndOfToken (UBYTE ∗s)
- UBYTE ∗ ToToken (UBYTE ∗s)
- UBYTE ∗ SkipField (UBYTE ∗s, int level)
- WORD ReadSnum (UBYTE ∗∗p)
- UBYTE ∗ NumCopy (WORD y, UBYTE ∗to)
- char ∗ LongCopy (LONG y, char ∗to)
- char ∗ LongLongCopy (off_t ∗y, char ∗to)
- UBYTE ∗ MakeDate (void)
- int set_in (UBYTE ch, set_of_char set)
- one_byte set_set (UBYTE ch, set_of_char set)
- one_byte set_del (UBYTE ch, set_of_char set)
- one_byte set_sub (set_of_char set, set_of_char set1, set_of_char set2)
- void iniTools (void)
- void ∗ Malloc (LONG size)
- void ∗ Malloc1 (LONG size, const char ∗messageifwrong)
- void M_free (void ∗x, const char ∗where)
- void M_check1 (void)
- void M_print (void)
- void TermMallocAddMemory (PHEAD0)
- void NumberMallocAddMemory (PHEAD0)
- void CacheNumberMallocAddMemory (PHEAD0)
- void ∗ FromList (LIST ∗L)
- void ∗ From0List (LIST ∗L)
- void ∗ FromVarList (LIST ∗L)
- int DoubleList (void ∗∗∗lijst, int ∗oldsize, int objectsize, char ∗nameoftype)
- int DoubleLList (void ∗∗∗lijst, LONG ∗oldsize, int objectsize, char ∗nameoftype)
- void DoubleBuffer (void ∗∗start, void ∗∗stop, int size, char ∗text)

- void ExpandBuffer (void ∗∗buffer, LONG ∗oldsize, int type)
- LONG iexp (LONG x, int p)
- void ToGeneral (WORD ∗r, WORD ∗m, WORD par)
- int ToFast (WORD ∗r, WORD ∗m)
- WORD ToPolyFunGeneral (PHEAD WORD ∗term)
- int IsLikeVector (WORD ∗arg)
- int AreArgsEqual (WORD ∗arg1, WORD ∗arg2)
- int CompareArgs (WORD ∗arg1, WORD ∗arg2)
- int CompArg (WORD ∗s1, WORD ∗s2)
- LONG TimeWallClock (WORD par)
- LONG TimeChildren (WORD par)
- LONG TimeCPU (WORD par)
- int Crash (void)
- int TestTerm (WORD ∗term)
- int DistrN (int n, int ∗cpl, int ncpl, int ∗scratch)

Variables

- FILES ∗∗ filelist
- int numinfilelist = 0
- int filelistsize = 0
- WRITEFILE WriteFile = &WriteFileToFile

4.146.1 Detailed Description

Low level routines for many types of task. There are routines for manipulating the input system (streams and files) routines for string manipulation, the memory allocation interface, and the clock. The last is the most sensitive to ports. In the past nearly every port to another OS or computer gave trouble. Nowadays it is slightly better but the poor POSIX compliance of LINUX again gave problems for the multithreaded version.

Definition in file tools.c.

4.146.2 Macro Definition Documentation

4.146.2.1 FILLVALUE

```
#define FILLVALUE 0
```

Definition at line 49 of file tools.c.

4.146.2.2 TERMMEMSTARTNUM

```
#define TERMMEMSTARTNUM 16
```

Provides memory for one term (or one small polynomial) This means that the memory is limited to a buffer of size AM.MaxTer plus a few extra words. In parallel versions, each worker has its own memory pool.

The way we use the memory is by: term = TermMalloc(BHEAD0); and later we free it by TermFree(BHEAD term);

Layout: We have a list of available pointers to buffers: AT.TermMemHeap Its size is AT.TermMemMax We take from the top (indicated by AT.TermMemTop). When we run out of buffers we assign new ones (doubling the amount) and we have to extend the AT.TermMemHeap array. Important: There is no checking that the returned memory is legal, ie is memory that was handed out earlier.

Definition at line 2547 of file tools.c.

4.146.2.3 TERMEXTRAWORDS

```
#define TERMEXTRAWORDS 10
```

Definition at line 2548 of file tools.c.

4.146.2.4 NUMBERMEMSTARTNUM

```
#define NUMBERMEMSTARTNUM 16
```

Provides memory for one Long number This means that the memory is limited to a buffer of size AM.MaxTal In parallel versions, each worker has its own memory pool.

The way we use the memory is by: num = NumberMalloc(BHEAD0); Number = AT.NumberMemHeap[num]; and later we free it by NumberFree(BHEAD num);

Layout: We have a list of available pointers to buffers: AT.NumberMemHeap Its size is AT.NumberMemMax We take from the top (indicated by AT.NumberMemTop). When we run out of buffers we assign new ones (doubling the amount) and we have to extend the AT.NumberMemHeap array. Important: There is no checking on the returned memory!!!!

Definition at line 2647 of file tools.c.

4.146.2.5 NUMBEREXTRAWORDS

```
#define NUMBEREXTRAWORDS 10L
```

Definition at line 2648 of file tools.c.

4.146.2.6 DODOUBLE

```
#define DODOUBLE(
                x )
```

Value:

```
{ x *s, *t, *u; if ( *start ) { \
oldsize = *(x **)stop - *(x **)start; newsize = 2*oldsize; \
t = u = (x *)Malloc1(newsize*sizeof(x),text); s = *(x **)start; \
for ( i = 0; i < oldsize; i++ ) {*t++ = *s++;} M_free(*start,"double"); } \
else { newsize = 100; u = (x *)Malloc1(newsize*sizeof(x),text); } \
*start = (void *)u; *stop = (void *)(u+newsize); }
```

Definition at line 2970 of file tools.c.

4.146.2.7 DOEXPAND

```
#define DOEXPAND(
                x )
```

Value:

```
{ x *newbuffer, *t, *m;                              \
t = newbuffer = (x *)Malloc1((newsize+2)*type,"ExpandBuffer");       \
if ( *buffer ) { m = (x *)*buffer; i = *oldsize;                     \
    while ( --i >= 0 ) {*t++ = *m++;} M_free(*buffer,"ExpandBuffer"); \
} *buffer = newbuffer; *oldsize = newsize; }
```

Definition at line 2996 of file tools.c.

4.146.3 Function Documentation

4.146.3.1 LoadInputFile()

```
UBYTE * LoadInputFile (
             UBYTE * filename,
             int type )
```

Definition at line 112 of file tools.c.

4.146.3.2 ReadFromStream()

```
UBYTE ReadFromStream (
             STREAM * stream )
```

Definition at line 151 of file tools.c.

4.146.3.3 GetFromStream()

```
UBYTE GetFromStream (
             STREAM * stream )
```

Definition at line 286 of file tools.c.

4.146.3.4 LookInStream()

```
UBYTE LookInStream (
             STREAM * stream )
```

Definition at line 309 of file tools.c.

4.146.3.5 OpenStream()

```
STREAM * OpenStream (
             UBYTE * name,
             int type,
             int prevarmode,
             int raiselow )
```

Definition at line 321 of file tools.c.

4.146.3.6 LocateFile()

```
int LocateFile (
             UBYTE ** name,
             int type )
```

Definition at line 576 of file tools.c.

4.146.3.7 CloseStream()

```
STREAM * CloseStream (
             STREAM * stream )
```

Definition at line 663 of file tools.c.

4.146.3.8 CreateStream()

```
STREAM * CreateStream (
             UBYTE * where )
```

Definition at line 782 of file tools.c.

4.146.3.9 GetStreamPosition()

```
LONG GetStreamPosition (
             STREAM * stream )
```

Definition at line 813 of file tools.c.

4.146.3.10 PositionStream()

```
void PositionStream (
             STREAM * stream,
             LONG position )
```

Definition at line 823 of file tools.c.

4.146.3.11 ReverseStatements()

```
int ReverseStatements (
             STREAM * stream )
```

Definition at line 855 of file tools.c.

4.146.3.12 StartFiles()

```
void StartFiles (
             void  )
```

Definition at line 956 of file tools.c.

4.146.3.13 OpenFile()

```
int OpenFile (
             char * name )
```

Definition at line 983 of file tools.c.

4.146.3.14 OpenAddFile()

```
int OpenAddFile (
            char * name )
```

Definition at line 1002 of file tools.c.

4.146.3.15 ReOpenFile()

```
int ReOpenFile (
            char * name )
```

Definition at line 1023 of file tools.c.

4.146.3.16 CreateFile()

```
int CreateFile (
            char * name )
```

Definition at line 1043 of file tools.c.

4.146.3.17 CreateLogFile()

```
int CreateLogFile (
            char * name )
```

Definition at line 1060 of file tools.c.

4.146.3.18 CloseFile()

```
void CloseFile (
            int handle )
```

Definition at line 1078 of file tools.c.

4.146.3.19 CopyFile()

```
int CopyFile (
            char * source,
            char * dest )
```

Copies a file with name ∗source to a file named ∗dest. The involved files must not be open. Returns non-zero if an error occurred. Uses if possible the combined large and small sorting buffers as cache.

Definition at line 1101 of file tools.c.

Referenced by DoRecovery().

4.146.3.20 CreateHandle()

```
int CreateHandle (
            void )
```

Definition at line 1162 of file tools.c.

4.146.3.21 ReadFile()

```
LONG ReadFile (
            int handle,
            UBYTE * buffer,
            LONG size )
```

Definition at line 1217 of file tools.c.

4.146.3.22 ReadPosFile()

```
LONG ReadPosFile (
            PHEAD FILEHANDLE * fi,
            UBYTE * buffer,
            LONG size,
            POSITION * pos )
```

Definition at line 1267 of file tools.c.

4.146.3.23 WriteFileToFile()

```
LONG WriteFileToFile (
            int handle,
            UBYTE * buffer,
            LONG size )
```

Definition at line 1333 of file tools.c.

4.146.3.24 SeekFile()

```
void SeekFile (
            int handle,
            POSITION * offset,
            int origin )
```

Definition at line 1371 of file tools.c.

4.146.3.25 TellFile()

```
LONG TellFile (
            int handle )
```

Definition at line 1396 of file tools.c.

4.146.3.26 TELLFILE()

```
void TELLFILE (
            int handle,
            POSITION * position )
```

Definition at line 1406 of file tools.c.

4.146.3.27 FlushFile()

```
void FlushFile (
            int handle )
```

Definition at line 1420 of file tools.c.

4.146.3.28 GetPosFile()

```
int GetPosFile (
            int handle,
            fpos_t * pospointer )
```

Definition at line 1434 of file tools.c.

4.146.3.29 SetPosFile()

```
int SetPosFile (
            int handle,
            fpos_t * pospointer )
```

Definition at line 1448 of file tools.c.

4.146.3.30 SynchFile()

```
void SynchFile (
            int handle )
```

Definition at line 1468 of file tools.c.

4.146.3.31 TruncateFile()

```
void TruncateFile (
            int handle )
```

Definition at line 1490 of file tools.c.

4.146.3.32 GetChannel()

```
int GetChannel (
            char * name,
            int mode )
```

Definition at line 1510 of file tools.c.

4.146.3.33 GetAppendChannel()

```
int GetAppendChannel (
            char * name )
```

Definition at line 1546 of file tools.c.

4.146.3.34 CloseChannel()

```
int CloseChannel (
            char * name )
```

Definition at line 1576 of file tools.c.

4.146.3.35 UpdateMaxSize()

```
void UpdateMaxSize (
            void )
```

Definition at line 1610 of file tools.c.

4.146.3.36 StrCmp()

```
int StrCmp (
            UBYTE * s1,
            UBYTE * s2 )
```

Definition at line 1700 of file tools.c.

4.146.3.37 StrICmp()

```
int StrICmp (
            UBYTE * s1,
            UBYTE * s2 )
```

Definition at line 1711 of file tools.c.

4.146.3.38 StrHICmp()

```
int StrHICmp (
            UBYTE * s1,
            UBYTE * s2 )
```

Definition at line 1722 of file tools.c.

4.146.3.39 StrICont()

```
int StrICont (
            UBYTE * s1,
            UBYTE * s2 )
```

Definition at line 1733 of file tools.c.

4.146.3.40 CmpArray()

```
int CmpArray (
            WORD * t1,
            WORD * t2,
            WORD n )
```

Definition at line 1745 of file tools.c.

4.146.3.41 ConWord()

```
int ConWord (
            UBYTE * s1,
            UBYTE * s2 )
```

Definition at line 1759 of file tools.c.

4.146.3.42 StrLen()

```
int StrLen (
            UBYTE * s )
```

Definition at line 1771 of file tools.c.

4.146.3.43 NumToStr()

```
void NumToStr (
            UBYTE * s,
            LONG x )
```

Definition at line 1783 of file tools.c.

4.146.3.44 WriteString()

```
void WriteString (
            int type,
            UBYTE * str,
            int num )
```

Definition at line 1807 of file tools.c.

4.146.3.45 WriteUnfinString()

```
void WriteUnfinString (
            int type,
            UBYTE * str,
            int num )
```

Definition at line 1841 of file tools.c.

4.146.3.46 AddToString()

```
UBYTE * AddToString (
            UBYTE * outstring,
            UBYTE * extrastring,
            int par )
```

Definition at line 1866 of file tools.c.

4.146.3.47 strDup1()

```
UBYTE * strDup1 (
            UBYTE * instring,
            char * ifwrong )
```

Definition at line 1904 of file tools.c.

4.146.3.48 EndOfToken()

```
UBYTE * EndOfToken (
            UBYTE * s )
```

Definition at line 1919 of file tools.c.

4.146.3.49 ToToken()

```
UBYTE * ToToken (
            UBYTE * s )
```

Definition at line 1931 of file tools.c.

4.146.3.50 SkipField()

```
UBYTE * SkipField (
            UBYTE * s,
            int level )
```

Definition at line 1946 of file tools.c.

4.146.3.51 ReadSnum()

```
WORD ReadSnum (
            UBYTE ** p )
```

Definition at line 1973 of file tools.c.

4.146.3.52 NumCopy()

```
UBYTE * NumCopy (
            WORD y,
            UBYTE * to )
```

Definition at line 1997 of file tools.c.

4.146.3.53 LongCopy()

```
char * LongCopy (
            LONG y,
            char * to )
```

Definition at line 2024 of file tools.c.

4.146.3.54 LongLongCopy()

```
char * LongLongCopy (
            off_t * y,
            char * to )
```

Definition at line 2051 of file tools.c.

4.146.3.55 MakeDate()

```
UBYTE * MakeDate (
            void )
```

Definition at line 2090 of file tools.c.

4.146.3.56 set_in()

```
int set_in (
              UBYTE ch,
              set_of_char set )
```

Definition at line 2112 of file tools.c.

4.146.3.57 set_set()

```
one_byte set_set (
              UBYTE ch,
              set_of_char set )
```

Definition at line 2132 of file tools.c.

4.146.3.58 set_del()

```
one_byte set_del (
              UBYTE ch,
              set_of_char set )
```

Definition at line 2153 of file tools.c.

4.146.3.59 set_sub()

```
one_byte set_sub (
              set_of_char set,
              set_of_char set1,
              set_of_char set2 )
```

Definition at line 2175 of file tools.c.

4.146.3.60 iniTools()

```
void iniTools (
              void  )
```

Definition at line 2201 of file tools.c.

4.146.3.61 Malloc()

```
void * Malloc (
              LONG size )
```

Definition at line 2221 of file tools.c.

4.146.3.62 Malloc1()

```
void * Malloc1 (
            LONG size,
            const char * messageifwrong )
```

Definition at line 2303 of file tools.c.

4.146.3.63 M_free()

```
void M_free (
            void * x,
            const char * where )
```

Definition at line 2383 of file tools.c.

4.146.3.64 M_check1()

```
void M_check1 (
            void  )
```

Definition at line 2516 of file tools.c.

4.146.3.65 M_print()

```
void M_print (
            void  )
```

Definition at line 2517 of file tools.c.

4.146.3.66 TermMallocAddMemory()

```
void TermMallocAddMemory (
            PHEAD0  )
```

Definition at line 2550 of file tools.c.

4.146.3.67 NumberMallocAddMemory()

```
void NumberMallocAddMemory (
            PHEAD0  )
```

Definition at line 2654 of file tools.c.

4.146.3.68 CacheNumberMallocAddMemory()

```
void CacheNumberMallocAddMemory (
            PHEAD0  )
```

Definition at line 2728 of file tools.c.

4.146.3.69 FromList()

```
void * FromList (
            LIST * L )
```

Definition at line 2780 of file tools.c.

4.146.3.70 From0List()

```
void * From0List (
            LIST * L )
```

Definition at line 2806 of file tools.c.

4.146.3.71 FromVarList()

```
void * FromVarList (
            LIST * L )
```

Definition at line 2835 of file tools.c.

4.146.3.72 DoubleList()

```
int DoubleList (
            void *** lijst,
            int * oldsize,
            int objectsize,
            char * nameoftype )
```

Definition at line 2877 of file tools.c.

4.146.3.73 DoubleLList()

```
int DoubleLList (
            void *** lijst,
            LONG * oldsize,
            int objectsize,
            char * nameoftype )
```

Definition at line 2929 of file tools.c.

4.146.3.74 DoubleBuffer()

```
void DoubleBuffer (
            void ** start,
            void ** stop,
            int size,
            char * text )
```

Definition at line 2977 of file tools.c.

4.146.3.75 ExpandBuffer()

```
void ExpandBuffer (
            void ** buffer,
            LONG * oldsize,
            int type )
```

Definition at line 3002 of file tools.c.

4.146.3.76 iexp()

```
LONG iexp (
            LONG x,
            int p )
```

Definition at line 3026 of file tools.c.

4.146.3.77 ToGeneral()

```
void ToGeneral (
            WORD * r,
            WORD * m,
            WORD par )
```

Definition at line 3057 of file tools.c.

4.146.3.78 ToFast()

```
int ToFast (
            WORD * r,
            WORD * m )
```

Definition at line 3101 of file tools.c.

4.146.3.79 ToPolyFunGeneral()

```
WORD ToPolyFunGeneral (
            PHEAD WORD * term )
```

Definition at line 3154 of file tools.c.

4.146.3.80 IsLikeVector()

```
int IsLikeVector (
            WORD * arg )
```

Definition at line 3229 of file tools.c.

4.146.3.81 AreArgsEqual()

```
int AreArgsEqual (
            WORD * arg1,
            WORD * arg2 )
```

Definition at line 3257 of file tools.c.

4.146.3.82 CompareArgs()

```
int CompareArgs (
            WORD * arg1,
            WORD * arg2 )
```

Definition at line 3276 of file tools.c.

4.146.3.83 CompArg()

```
int CompArg (
            WORD * s1,
            WORD * s2 )
```

Definition at line 3307 of file tools.c.

4.146.3.84 TimeWallClock()

```
LONG TimeWallClock (
            WORD par )
```

Returns the wall-clock time.

Parameters

par	If zero, the wall-clock time will be reset to 0.

Returns

The wall-clock time in centiseconds.

Definition at line 3477 of file tools.c.

Referenced by DoCheckpoint(), DoRecovery(), find_Horner_MCTS(), optimize_expression_given_Horner(), optimize_greedy(), and WriteStats().

4.146.3.85 TimeChildren()

```
LONG TimeChildren (
            WORD par )
```

Definition at line 3533 of file tools.c.

4.146.3.86 TimeCPU()

```
LONG TimeCPU (
            WORD par )
```

Returns the CPU time.

Parameters

par	If zero, the CPU time will be reset to 0.

Returns

The CPU time in milliseconds.

Definition at line 3551 of file tools.c.

Referenced by PF_GetSlaveTimes(), PF_Processor(), and WriteStats().

4.146.3.87 Crash()

```
int Crash (
            void )
```

Definition at line 3834 of file tools.c.

4.146.3.88 TestTerm()

```
int TestTerm (
            WORD * term )
```

Tests the consistency of the term. Returns 0 when the term is OK. Any nonzero value is trouble. In the current version the testing isn't 100% complete. For instance, we don't check the validity of the symbols nor do we check the range of their powers. Etc. This should be extended when the need is there.

Parameters

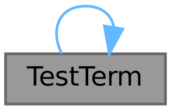
term	the term to be tested

Definition at line 3862 of file tools.c.

References TestTerm().

Referenced by TestTerm().

Here is the call graph for this function:



4.146.3.89 DistrN()

```
int DistrN (
            int n,
            int * cpl,
            int ncpl,
            int * scratch )
```

Definition at line 4048 of file tools.c.

4.146.4 Variable Documentation

4.146.4.1 filelist

```
FILES** filelist
```

Definition at line 65 of file tools.c.

4.146.4.2 numinfilelist

```
int numinfilelist = 0
```

Definition at line 66 of file tools.c.

4.146.4.3 filelistsize

```
int filelistsize = 0
```

Definition at line 67 of file tools.c.

4.146.4.4 WriteFile

```
WRITEFILE WriteFile = &WriteFileToFile
```

Definition at line 1357 of file tools.c.

4.147 tools.c

Go to the documentation of this file.

```c

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
    Note: TERMMALLOCDEBUG tests part of the TermMalloc and NumberMalloc
          system. To work properly it needs MEMORYMACROS in declare.h
          not to be defined to make sure that all calls will be diverted
          to the routines here.
#define TERMMALLOCDEBUG
#define FILLVALUE 126
#define MALLOCDEBUGOUTPUT
#define MALLOCDEBUG 1
*/
#ifndef FILLVALUE
    #define FILLVALUE 0
#endif

/*
    The enhanced malloc debugger, see comments in the beginning of the
    file mallocprotect.h
    MALLOCPROTECT == -1  -- protect left side, used block is left-aligned.
    MALLOCPROTECT == 0  -- protect both sides, used block is left-aligned;
    MALLOCPROTECT == 1  -- protect both sides, used block is right-aligned;
    ATTENTION! The macro MALLOCPROTECT must be defined
    BEFORE #include mallocprotect.h
#define MALLOCPROTECT 1
*/

#include "form3.h"

FILES **filelist;
int numinfilelist = 0;
int filelistsize = 0;
#ifdef MALLOCDEBUG
#define BANNER (4*sizeof(LONG))
void *malloclist[60000];
LONG mallocsizes[60000];
char *mallocstrings[60000];
int nummalloclist = 0;
#endif

#ifdef GPP
extern "C" getdtablesize();
#endif

#ifdef WITHSTATS
LONG numwrites = 0;
LONG numreads = 0;
LONG numseeks = 0;
LONG nummallocs = 0;
LONG numfrees = 0;
#endif

#ifdef MALLOCPROTECT
#ifdef TRAPSIGNALS
#error "MALLOCPROTECT":  undefine "TRAPSIGNALS" in unix.h first!
#endif
```

```
#include "mallocprotect.h"

#ifdef M_alloc
#undef M_alloc
#endif

#define M_alloc mprotectMalloc

#endif

#ifdef TERMMALLOCDEBUG
WORD **DebugHeap1, **DebugHeap2;
#endif

/*
    #] Includes :
    #[ Streams :
        #[ LoadInputFile :
*/

UBYTE *LoadInputFile(UBYTE *filename, int type)
{
    int handle;
    LONG filesize;
    UBYTE *buffer, *name = filename;
    POSITION scrpos;
    handle = LocateFile(&name,type);
    if ( handle < 0 ) return(0);
    PUTZERO(scrpos);
    SeekFile(handle,&scrpos,SEEK_END);
    TELLFILE(handle,&scrpos);
    filesize = BASEPOSITION(scrpos);
    PUTZERO(scrpos);
    SeekFile(handle,&scrpos,SEEK_SET);
    buffer = (UBYTE *)Malloc1(filesize+2,"LoadInputFile");
    if ( ReadFile(handle,buffer,filesize) != filesize ) {
        Error1("Read error for file ",name);
        M_free(buffer,"LoadInputFile");
        if ( name != filename ) M_free(name,"FromLoadInputFile");
        CloseFile(handle);
        return(0);
    }
    CloseFile(handle);
    if ( type == PROCEDUREFILE || type == SETUPFILE ) {
        buffer[filesize] = '\n';
        buffer[filesize+1] = 0;
    }
    else {
        buffer[filesize] = 0;
    }
    if ( name != filename ) M_free(name,"FromLoadInputFile");
    return(buffer);
}

/*
        #] LoadInputFile :
        #[ ReadFromStream :
*/

UBYTE ReadFromStream(STREAM *stream)
{
    UBYTE c;
    POSITION scrpos;
#ifdef WITHPIPE
    if ( stream->type == PIPESTREAM ) {
#ifndef WITHMPI
        FILE *f;
        int cc;
        RWLOCKR(AM.handlelock);
        f = (FILE *)(filelist[stream->handle]);
        UNRWLOCK(AM.handlelock);
        cc = getc(f);
        if ( cc == EOF ) return(ENDOFSTREAM);
        c = (UBYTE)cc;
#else
        if ( stream->pointer >= stream->top ) {
            /* The master reads the pipe and broadcasts it to the slaves. */
            LONG len;
            if ( PF.me == MASTER ) {
                FILE *f;
                UBYTE *p, *end;
                RWLOCKR(AM.handlelock);
                f = (FILE *)filelist[stream->handle];
                UNRWLOCK(AM.handlelock);
                p = stream->buffer;
                end = stream->buffer + stream->buffersize;
                while ( p < end ) {
```

```
                    int cc = getc(f);
                    if ( cc == EOF ) {
                        break;
                    }
                    *p++ = (UBYTE)cc;
                }
                len = p - stream->buffer;
                PF_BroadcastNumber(len);
            }
            else {
                len = PF_BroadcastNumber(0);
            }
            if ( len > 0 ) {
                PF_Bcast(stream->buffer, len);
            }
            stream->pointer = stream->buffer;
            stream->inbuffer = len;
            stream->top = stream->buffer + stream->inbuffer;
            if ( stream->pointer == stream->top ) return ENDOFSTREAM;
        }
        c = (UBYTE)*stream->pointer++;
#endif
        if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
        if ( c == LINEFEED ) stream->eqnum = 1;
        return(c);
    }
#endif
/*[14apr2004 mt]:*/
#ifdef WITHEXTERNALCHANNEL
    if ( stream->type == EXTERNALCHANNELSTREAM ) {
        int cc;
        cc = getcFromExtChannel();
        /*[18may20006 mt]:*/
        /*if ( cc == EOF ) return(ENDOFSTREAM);*/
        if ( cc < 0 ){
            if( cc == EOF )
             return(ENDOFSTREAM);
            else{
                Error0("No current external channel");
                Terminate(-1);
            }
        }/*if ( cc < 0 )*/
        /*:[18may20006 mt]*/
        c = (UBYTE)cc;
        if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
        if ( c == LINEFEED ) stream->eqnum = 1;
        return(c);
    }
#endif /*ifdef WITHEXTERNALCHANNEL*/
/*:[14apr2004 mt]*/
    if ( stream->type == INPUTSTREAM ) {
        if ( stream->pointer < stream->top ) {
            c = *stream->pointer++;
        }
        else {
            if ( ReadFile(stream->handle,&c,1) != 1 ) {
                return(ENDOFSTREAM);
            }
            if ( stream->fileposition == 0 ) {
                if ( !stream->buffer ) {
                    stream->buffersize = 32;
                    stream->buffer = (UBYTE *)Malloc1(stream->buffersize,"input stream buffer");
                    stream->pointer = stream->top = stream->buffer;
                }
                else {
                    if ( stream->top - stream->buffer >= stream->buffersize ) {
                        LONG oldsize = stream->buffersize;
DoubleBuffer((void**)&stream->buffer,(void**)&stream->top,sizeof(UBYTE),"double input stream buffer");
                        stream->buffersize = stream->top - stream->buffer;
                        stream->pointer = stream->top = stream->buffer + oldsize;
                    }
                }
                *stream->pointer = c;
                stream->pointer = ++stream->top;
                stream->inbuffer++;
            }
        }
        if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
        if ( c == LINEFEED ) stream->eqnum = 1;
        return(c);
    }
    if ( stream->pointer >= stream->top ) {
        if ( stream->type != FILESTREAM ) return(ENDOFSTREAM);
        if ( stream->fileposition != stream->bufferposition+stream->inbuffer ) {
            stream->fileposition = stream->bufferposition+stream->inbuffer;
            SETBASEPOSITION(scrpos,stream->fileposition);
```

```
            SeekFile(stream->handle,&scrpos,SEEK_SET);
        }
        stream->bufferposition = stream->fileposition;
        stream->inbuffer = ReadFile(stream->handle,
                stream->buffer,stream->buffersize);
        if ( stream->inbuffer <= 0 ) return(ENDOFSTREAM);
        stream->top = stream->buffer + stream->inbuffer;
        stream->pointer = stream->buffer;
        stream->fileposition = stream->bufferposition + stream->inbuffer;
    }
    if ( stream->eqnum == 1 ) { stream->eqnum = 0; stream->linenumber++; }
    c = *(stream->pointer)++;
    if ( c == LINEFEED ) stream->eqnum = 1;
    return(c);
}

/*
        #] ReadFromStream :
        #[ GetFromStream :
*/

UBYTE GetFromStream(STREAM *stream)
{
    UBYTE c1, c2;
    if ( stream->isnextchar > 0 ) {
        return(stream->nextchar[--stream->isnextchar]);
    }
    c1 = ReadFromStream(stream);
    if ( c1 == LINEFEED || c1 == CARRIAGERETURN ) {
        c2 = ReadFromStream(stream);
        if ( c2 == c1 || ( c2 != LINEFEED && c2 != CARRIAGERETURN ) ) {
            stream->isnextchar = 1;
            stream->nextchar[0] = c2;
        }
        return(LINEFEED);
    }
    else return(c1);
}

/*
        #] GetFromStream :
        #[ LookInStream :
*/

UBYTE LookInStream(STREAM *stream)
{
    UBYTE c = GetFromStream(stream);
    UngetFromStream(stream,c);
    return(c);
}

/*
        #] LookInStream :
        #[ OpenStream :
*/

STREAM *OpenStream(UBYTE *name, int type, int prevarmode, int raiselow)
{
    STREAM *stream;
    UBYTE *rhsofvariable, *s, *newname, c;
    POSITION scrpos;
    int handle, num;
    LONG filesize;
    switch ( type ) {
        case REVERSEFILESTREAM:
        case FILESTREAM:
/*
            Notice that FILESTREAM is only used for text files:
            The #include files and the main input file (.frm)
            Hence we do not worry about files longer than 2 Gbytes.
*/
            newname = name;
            handle = LocateFile(&newname,-1);
            if ( handle < 0 ) return(0);
            PUTZERO(scrpos);
            SeekFile(handle,&scrpos,SEEK_END);
            TELLFILE(handle,&scrpos);
            filesize = BASEPOSITION(scrpos);
            PUTZERO(scrpos);
            SeekFile(handle,&scrpos,SEEK_SET);
            if ( filesize > AM.MaxStreamSize && type == FILESTREAM )
                    filesize = AM.MaxStreamSize;
            stream = CreateStream((UBYTE *)"filestream");
/*
            The extra +1 in the Malloc1 is potentially needed in ReverseStatements!
*/
            stream->buffer = (UBYTE *)Malloc1(filesize+1,"name of input stream");
```

```
            stream->inbuffer = ReadFile(handle,stream->buffer,filesize);
            if ( type == REVERSEFILESTREAM ) {
                if ( ReverseStatements(stream) ) {
                    M_free(stream->buffer,"name of input stream");
                    return(0);
                }
            }
            stream->top = stream->buffer + stream->inbuffer;
            stream->pointer = stream->buffer;
            stream->handle = handle;
            stream->buffersize = filesize;
            stream->fileposition = stream->inbuffer;
            if ( newname != name ) stream->name = newname;
            else if ( name ) stream->name = strDup1(name,"name of input stream");
            else
                stream->name = 0;
            stream->prevline = stream->linenumber = 1;
            stream->eqnum = 0;
            break;
        case PREVARSTREAM:
            if ( ( rhsofvariable = GetPreVar(name,WITHERROR) ) == 0 ) return(0);
            stream = CreateStream((UBYTE *)"var-stream");
            stream->buffer = stream->pointer = s = rhsofvariable;
            while ( *s ) s++;
            stream->top = s;
            stream->inbuffer = s - stream->buffer;
            stream->name = AC.CurrentStream->name;
            stream->linenumber = AC.CurrentStream->linenumber;
            stream->prevline = AC.CurrentStream->prevline;
            stream->eqnum = AC.CurrentStream->eqnum;
            stream->pname = strDup1(name,"stream->pname");
            stream->olddelay = AP.AllowDelay;
            s = stream->pname; while ( *s ) s++;
            while ( s[-1] == '+' || s[-1] == '-' ) s--;
            *s = 0;
            UnsetAllowDelay();
            break;
        case DOLLARSTREAM:
            if ( ( num = GetDollar(name) ) < 0 ) {
                WORD numfac = 0;
/*
                Here we have to test first whether we have $x[1], $x[0]
                or just an undefined $x.
*/
                s = name; while ( *s && *s != '[' ) s++;
                if ( *s == 0 ) return(0);
                c = *s; *s = 0;
                if ( ( num = GetDollar(name) ) < 0 ) return(0);
                *s = c;
                s++;
                if ( *s == 0 || FG.cTable[*s] != 1 || *s == ']' ) {
                    MesPrint("@Illegal factor number for dollar variable");
                    return(0);
                }
                while ( *s && FG.cTable[*s] == 1 ) {
                    numfac = 10*numfac+*s++-'0';
                }
                if ( *s != ']' || s[1] != 0 ) {
                    MesPrint("@Illegal factor number for $ variable");
                    return(0);
                }
                stream = CreateStream((UBYTE *)"dollar-stream");
                stream->buffer = stream->pointer = s = WriteDollarFactorToBuffer(num,numfac,1);
            }
            else {
                stream = CreateStream((UBYTE *)"dollar-stream");
                stream->buffer = stream->pointer = s = WriteDollarToBuffer(num,1);
            }
            while ( *s ) s++;
            stream->top = s;
            stream->inbuffer = s - stream->buffer;
            stream->name = AC.CurrentStream->name;
            stream->linenumber = AC.CurrentStream->linenumber;
            stream->prevline= AC.CurrentStream->prevline;
            stream->eqnum = AC.CurrentStream->eqnum;
            stream->pname = strDup1(name,"stream->pname");
            s = stream->pname; while ( *s ) s++;
            while ( s[-1] == '+' || s[-1] == '-' ) s--;
            *s = 0;
            /* We 'stole' the buffer. Later we can free it. */
            AO.DollarOutSizeBuffer = 0;
            AO.DollarOutBuffer = 0;
            AO.DollarInOutBuffer = 0;
            break;
        case PREREADSTREAM:
        case PREREADSTREAM2:
        case PREREADSTREAM3:
```

```
        case PRECALCSTREAM:
            stream = CreateStream((UBYTE *)"calculator");
            stream->buffer = stream->pointer = s = name;
            while ( *s ) s++;
            stream->top = s;
            stream->inbuffer = s - stream->buffer;
            stream->name = AC.CurrentStream->name;
            stream->linenumber = AC.CurrentStream->linenumber;
            stream->prevline = AC.CurrentStream->prevline;
            stream->eqnum = 0;
            break;
#ifdef WITHPIPE
        case PIPESTREAM:
            stream = CreateStream((UBYTE *)"pipe");
#ifndef WITHMPI
            {
                FILE *f;
                if ( ( f = popen((char *)name,"r") ) == 0 ) {
                    Error0("@Cannot create pipe");
                }
                stream->handle = CreateHandle();
                RWLOCKW(AM.handlelock);
                filelist[stream->handle] = (FILES *)f;
                UNRWLOCK(AM.handlelock);
            }
            stream->buffer = stream->top = 0;
            stream->inbuffer = 0;
#else
            {
                /* Only the master opens the pipe. */
                FILE *f;
                if ( PF.me == MASTER ) {
                    f = popen((char *)name, "r");
                    PF_BroadcastNumber(f == 0);
                    if ( f == 0 ) Error0("@Cannot create pipe");
                }
                else {
                    if ( PF_BroadcastNumber(0) ) Error0("@Cannot create pipe");
                    f = (FILE *)123;  /* dummy */
                }
                stream->handle = CreateHandle();
                RWLOCKW(AM.handlelock);
                filelist[stream->handle] = (FILES *)f;
                UNRWLOCK(AM.handlelock);
            }
            /* stream->buffer as a send/receive buffer. */
            stream->buffersize = AM.MaxStreamSize;
            stream->buffer = (UBYTE *)Malloc1(stream->buffersize, "pipe buffer");
            stream->inbuffer = 0;
            stream->top = stream->buffer;
            stream->pointer = stream->buffer;
#endif
            stream->name = strDup1((UBYTE *)"pipe","pipe");
            stream->prevline = stream->linenumber = 1;
            stream->eqnum = 0;
            break;
#endif
/*[14apr2004 mt]:*/
#ifdef WITHEXTERNALCHANNEL
        case EXTERNALCHANNELSTREAM:
            {/*Block*/
                int n, *tmpn;
                if( (n=getCurrentExternalChannel()) == 0 )
                    Error0("@No current external channel");
                stream = CreateStream((UBYTE *)"externalchannel");
                stream->handle = CreateHandle();
                tmpn = (int *)Malloc1(sizeof(int),"external channel handle");
                *tmpn = n;
                RWLOCKW(AM.handlelock);
                filelist[stream->handle] = (FILES *)tmpn;
                UNRWLOCK(AM.handlelock);
            }/*Block*/
            stream->buffer = stream->top = 0;
            stream->inbuffer = 0;
            stream->name = strDup1((UBYTE *)"externalchannel","externalchannel");
            stream->prevline = stream->linenumber = 1;
            stream->eqnum = 0;
            break;
#endif /*ifdef WITHEXTERNALCHANNEL*/
/*:[14apr2004 mt]*/
        case INPUTSTREAM:
            /*
             * Assume that "name" stores a file descriptor (UNIX) or a FILE
             * pointer (Windows). We don't close it automatically on closing
             * the INPUTSTREAM stream (e.g., for stdin).
             */
            stream = CreateStream((UBYTE *)"input stream");
```

```
            stream->handle = CreateHandle();
            {
                FILES *f = (FILES *)Malloc1(sizeof(int),"input stream handle");
                /* NOTE: in both cases name=0 indicates stdin. */
#ifdef UNIX
                f->descriptor = (int)(ssize_t)name;
#else
                f = name ? (FILES *)name : stdin;
#endif
                RWLOCKW(AM.handlelock);
                filelist[stream->handle] = f;
                UNRWLOCK(AM.handlelock);
            }
            stream->buffer = stream->pointer = stream->top = 0;
            stream->inbuffer = 0;
            stream->name = strDup1((UBYTE *)(name ? "INPUT" : "STDIN"),"input stream name");
            stream->prevline = stream->linenumber = 1;
            stream->eqnum = 0;
            /*
             * fileposition == -1: default
             * fileposition ==  0: cache the input
             * See also: ReadFromStream, TryFileSetups
             */
            stream->fileposition = -1;
            break;
        default:
            return(0);
    }
    stream->bufferposition = 0;
    stream->isnextchar = 0;
    stream->type = type;
    stream->previousNoShowInput = AC.NoShowInput;
    stream->afterwards = raiselow;
    if ( AC.CurrentStream ) stream->previous = AC.CurrentStream - AC.Streams;
    else stream->previous = -1;
    stream->FoldName = 0;
    if ( prevarmode == 0 ) stream->prevars = -1;
    else if ( prevarmode > 0 ) stream->prevars = NumPre;
    else if ( prevarmode < 0 ) stream->prevars = -prevarmode-1;
    AC.CurrentStream = stream;
    if ( type == PREREADSTREAM || type == PREREADSTREAM3 || type == PRECALCSTREAM
        || type == DOLLARSTREAM ) AC.NoShowInput = 1;
    return(stream);
}

/*
        #] OpenStream :
        #[ LocateFile :
*/

int LocateFile(UBYTE **name, int type)
{
    int handle, namesize, i;
    UBYTE *s, *to, *u1, *u2, *newname, *indir;
    handle = OpenFile((char *)(*name));
    if ( handle >= 0 ) return(handle);
    if ( type == SETUPFILE && AM.SetupFile ) {
        handle = OpenFile((char *)(AM.SetupFile));
        if ( handle >= 0 ) return(handle);
        MesPrint("Could not open setup file %s",(char *)(AM.SetupFile));
    }
    namesize = 4; s = *name;
    while ( *s ) { s++; namesize++; }
    if ( type == SETUPFILE ) indir = AM.SetupDir;
    else indir = AM.IncDir;
    if ( indir ) {

        s = indir; i = 0;
        while ( *s ) { s++; i++; }
        newname = (UBYTE *)Malloc1(namesize+i,"LocateFile");
        s = indir; to = newname;
        while ( *s ) *to++ = *s++;
        if ( to > newname && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
        s = *name;
        while ( *s ) *to++ = *s++;
        *to = 0;
        handle = OpenFile((char *)newname);
        if ( handle >= 0 ) {
            *name = newname;
            return(handle);
        }
        M_free(newname,"LocateFile, incdir/file");
    }
    if ( type == SETUPFILE ) {
        handle = OpenFile(setupfilename);
        if ( handle >= 0 ) return(handle);
        s = (UBYTE *)getenv("FORMSETUP");
```

```
        if ( s ) {
            handle = OpenFile((char *)s);
            if ( handle >= 0 ) return(handle);
            MesPrint("Could not open setup file %s",s);
        }
    }
    if ( type != SETUPFILE && AM.Path ) {
        u1 = AM.Path;
        while ( *u1 ) {
            u2 = u1; i = 0;
#ifdef WINDOWS
            while ( *u1 && *u1 != ';' ) {
                u1++; i++;
            }
#else
            while ( *u1 && *u1 != ':' ) {
                if ( *u1 == '\\' ) u1++;
                u1++; i++;
            }
#endif
            newname = (UBYTE *)Malloc1(namesize+i,"LocateFile");
            s = u2; to = newname;
            while ( s < u1 ) {
#ifndef WINDOWS
                if ( *s == '\\' ) s++;
#endif
                *to++ = *s++;
            }
            if ( to > newname && to[-1] != SEPARATOR ) *to++ = SEPARATOR;
            s = *name;
            while ( *s ) *to++ = *s++;
            *to = 0;
            handle = OpenFile((char *)newname);
            if ( handle >= 0 ) {
                *name = newname;
                return(handle);
            }
            M_free(newname,"LocateFile Path/file");
            if ( *u1 ) u1++;
        }
    }
    if ( type != SETUPFILE && type >= -1 ) Error1("LocateFile: Cannot find file",*name);
    return(-1);
}

/*
        #] LocateFile :
        #[ CloseStream :
*/

STREAM *CloseStream(STREAM *stream)
{
    int newstr = stream->previous, sgn;
    UBYTE *t, numbuf[24];
    LONG x;
    if ( stream->FoldName ) {
        M_free(stream->FoldName,"stream->FoldName");
        stream->FoldName = 0;
    }
    if ( stream->type == FILESTREAM || stream->type == REVERSEFILESTREAM ) {
        CloseFile(stream->handle);
        if ( stream->buffer != 0 ) M_free(stream->buffer,"name of input stream");
        stream->buffer = 0;
    }
#ifdef WITHPIPE
    else if ( stream->type == PIPESTREAM ) {
        RWLOCKW(AM.handlelock);
#ifdef WITHMPI
        if ( PF.me == MASTER )
#endif
        pclose((FILE *)(filelist[stream->handle]));
        filelist[stream->handle] = 0;
        numinfilelist--;
        UNRWLOCK(AM.handlelock);
#ifdef WITHMPI
        if ( stream->buffer != 0 ) {
            M_free(stream->buffer, "pipe buffer");
            stream->buffer = 0;
        }
#endif
    }
#endif
/*[14apr2004 mt]:*/
#ifdef WITHEXTERNALCHANNEL
    else if ( stream->type == EXTERNALCHANNELSTREAM ) {
        int *tmpn;
        RWLOCKW(AM.handlelock);
```

```
        tmpn = (int *)(filelist[stream->handle]);
        filelist[stream->handle] = 0;
        numinfilelist--;
        UNRWLOCK(AM.handlelock);
        M_free(tmpn,"external channel handle");
    }
#endif /*ifdef WITHEXTERNALCHANNEL*/
/*:[14apr2004 mt]*/
    else if ( stream->type == INPUTSTREAM ) {
        FILES *f;
        RWLOCKW(AM.handlelock);
        f = filelist[stream->handle];
        filelist[stream->handle] = 0;
        numinfilelist--;
        UNRWLOCK(AM.handlelock);
        M_free(f,"input stream handle");
    }
    else if ( stream->type == PREVARSTREAM && (
    stream->afterwards == PRERAISEAFTER || stream->afterwards == PRELOWERAFTER ) ) {
        t = stream->buffer; x = 0; sgn = 1;
        while ( *t == '-' || *t == '+' ) {
            if ( *t == '-' ) sgn = -sgn;
            t++;
        }
        if ( FG.cTable[*t] == 1 ) {
            while ( *t && FG.cTable[*t] == 1 ) x = 10*x + *t++ - '0';
            if ( *t == 0 ) {
                if ( stream->afterwards == PRERAISEAFTER ) x = sgn*x + 1;
                else x = sgn*x - 1;
                NumToStr(numbuf,x);
                PutPreVar(stream->pname,numbuf,0,1);
            }
        }
    }
    else if ( stream->type == DOLLARSTREAM && (
    stream->afterwards == PRERAISEAFTER || stream->afterwards == PRELOWERAFTER ) ) {
        if ( stream->afterwards == PRERAISEAFTER ) x = 1;
        else x = -1;
        DollarRaiseLow(stream->pname,x);
    }
    else if ( stream->type == PRECALCSTREAM || stream->type == DOLLARSTREAM ) {
        if ( stream->buffer ) M_free(stream->buffer,"stream->buffer");
        stream->buffer = 0;
    }
    if ( stream->name && stream->type != PREVARSTREAM
    && stream->type != PREREADSTREAM && stream->type != PREREADSTREAM2 && stream->type !=
PREREADSTREAM3
    && stream->type != PRECALCSTREAM && stream->type != DOLLARSTREAM ) {
        M_free(stream->name,"stream->name");
    }
    stream->name = 0;
/*  if ( stream->type != FILESTREAM )  */
    AC.NoShowInput = stream->previousNoShowInput;
    stream->buffer = 0;        /* To make sure we will not reuse it */
    stream->pointer = 0;
/*
    Look whether we have to pop preprocessor variables.
*/
    if ( stream->prevars >= 0 ) {
        while ( NumPre > stream->prevars ) {
            NumPre--;
            M_free(PreVar[NumPre].name,"PreVar[NumPre].name");
            PreVar[NumPre].name = PreVar[NumPre].value = 0;
        }
    }
    if ( stream->type == PREVARSTREAM ) {
        AP.AllowDelay = stream->olddelay;
        ClearMacro(stream->pname);
        M_free(stream->pname,"stream->pname");
    }
    else if ( stream->type == DOLLARSTREAM ) {
        M_free(stream->pname,"stream->pname");
    }
    AC.NumStreams--;
    if ( newstr >= 0 ) return(AC.Streams + newstr);
    else return(0);
}

/*
        #] CloseStream :
        #[ CreateStream :
*/

STREAM *CreateStream(UBYTE *where)
{
    STREAM *newstreams;
    int numnewstreams,i;
```

```
    int offset;
    if ( AC.NumStreams >= AC.MaxNumStreams ) {
        if ( AC.MaxNumStreams == 0 ) numnewstreams = 10;
        else                        numnewstreams = 2*AC.MaxNumStreams;
        newstreams = (STREAM *)Malloc1(sizeof(STREAM)*(numnewstreams+1),"CreateStream");
        if ( AC.MaxNumStreams > 0 ) {
            offset = AC.CurrentStream - AC.Streams;
            for ( i = 0; i < AC.MaxNumStreams; i++ ) {
                newstreams[i] = AC.Streams[i];
            }
            AC.CurrentStream = newstreams + offset;
        }
        else newstreams[0].previous = -1;
        AC.MaxNumStreams = numnewstreams;
        if ( AC.Streams ) M_free(AC.Streams,(char *)where);
        AC.Streams = newstreams;
    }
    newstreams = AC.Streams+AC.NumStreams++;
    newstreams->name = 0;
    return(newstreams);
}

/*
        #] CreateStream :
        #[ GetStreamPosition :
*/

LONG GetStreamPosition(STREAM *stream)
{
    return(stream->bufferposition + ((LONG)stream->pointer-(LONG)stream->buffer));
}

/*
        #] GetStreamPosition :
        #[ PositionStream :
*/

void PositionStream(STREAM *stream, LONG position)
{
    POSITION scrpos;
    if ( position >= stream->bufferposition
    && position < stream->bufferposition + stream->inbuffer ) {
        stream->pointer = stream->buffer + (position-stream->bufferposition);
    }
    else if ( stream->type == FILESTREAM ) {
        SETBASEPOSITION(scrpos,position);
        SeekFile(stream->handle,&scrpos,SEEK_SET);
        stream->inbuffer = ReadFile(stream->handle,stream->buffer,stream->buffersize);
        stream->pointer = stream->buffer;
        stream->top = stream->buffer + stream->inbuffer;
        stream->bufferposition = position;
        stream->fileposition = position + stream->inbuffer;
        stream->isnextchar = 0;
    }
    else {
        Error0("Illegal position for stream");
        Terminate(-1);
    }
}

/*
        #] PositionStream :
        #[ ReverseStatements :

        Reverses the order of the statements in the buffer.
        We allocate an extra buffer and copy a bit to and from.
        Note that there are some nasties that cannot be resolved.
*/

int ReverseStatements(STREAM *stream)
{
    UBYTE *spare = (UBYTE *)Malloc1((stream->inbuffer+1)*sizeof(UBYTE),"Reverse copy");
    UBYTE *top = stream->buffer + stream->inbuffer, *in, *s, *ss, *out;
    out = spare+stream->inbuffer+1;
    in = stream->buffer;
    while ( in < top ) {
        s = in;
        if ( *s == AP.ComChar ) {
toeol:;
            for(;;) {
                if ( s == top ) { *--out = '\n'; break; }
                if ( *s == '\\' ) {
                    s++;
                    if ( s >= top ) { /* This is an error! */
irrend:                 MesPrint("@Irregular end of reverse include file.");
                        return(1);
                    }
```

```
                }
                else if ( *s == '\n' ) {
                    s++; ss = s;
                    while ( ss > in ) *--out = *--ss;
                    in = s;
                    if ( out[0] == AP.ComChar && ss+6 < s && out[3] == '#' ) {
/*
                        For folds we have to exchange begin and end
*/
                        if ( out[4] == '[' ) out[4] = ']';
                        else if ( out[4] == ']' ) out[4] = '[';
                    }
                    break;
                }
                s++;
            }
            continue;
        }
        while ( s < top && ( *s == ' ' || *s == '\t' ) ) s++;
        if ( *s == '#' ) {  /* preprocessor instruction */
            goto toeol;     /* read to end of line */
        }
        if ( *s == '.' ) {  /* end-of-module instruction */
            goto toeol;     /* read to end of line */
        }
/*
        Here we have a regular statement. In principle we scan to ; and its \n
        but there are special cases.
        1: ; inside a string (in print "......;";)
        2: multiple statements on one line.
        3: ; + commentary after some blanks.
        4: `var' can cause problems.....
*/
        while ( s < top ) {
            if ( *s == ';' ) {
                s++;
                while ( s < top && ( *s == ' ' || *s == '\t' ) ) s++;
                while ( s < top && *s == '\n' ) s++;
                if ( s >= top && s[-1] != '\n' ) *s++ = '\n';
                ss = s;
                while ( ss > in ) *--out = *--ss;
                in = s;
                break;
            }
            else if ( *s == '"' ) {
                s++;
                while ( s < top ) {
                    if ( *s == '"' ) break;
                    if ( *s == '\\' ) { s++; }
                    s++;
                }
                if ( s >= top ) goto irrend;
            }
            else if ( *s == '\\' ) {
                s++;
                if ( s >= top ) goto irrend;
            }
            s++;
        }
        if ( in < top ) { /* Like blank lines at the end */
            if ( s >= top && s[-1] != '\n' ) *s++ = '\n';
            ss = s;
            while ( ss > in ) *--out = *--ss;
            in = s;
        }
    }
    if ( out == spare ) stream->inbuffer++;
    if ( out > spare+1 ) {
        MesPrint("@Internal error in #reverseinclude instruction.");
        return(1);
    }
    memcpy((void *)(stream->buffer),(void *)out,(size_t)(stream->inbuffer*sizeof(UBYTE)));
    M_free(spare,"Reverse copy");
    return(0);
}

/*
        #] ReverseStatements :
    #] Streams :
    #[ Files :
        #[ StartFiles :
*/

void StartFiles(void)
{
    int i = CreateHandle();
    filelist[i] = Ustdout;
```

```
    AM.StdOut = i;
    AC.StoreHandle = -1;
    AC.LogHandle = -1;
#ifndef WITHPTHREADS
    AR.Fscr[0].handle = -1;
    AR.Fscr[1].handle = -1;
    AR.Fscr[2].handle = -1;
    AR.FoStage4[0].handle = -1;
    AR.FoStage4[1].handle = -1;
    AR.infile = &(AR.Fscr[0]);
    AR.outfile = &(AR.Fscr[1]);
    AR.hidefile = &(AR.Fscr[2]);
    AR.StoreData.Handle = -1;
#endif
    AC.Streams = 0;
    AC.MaxNumStreams = 0;
}

/*
        #] StartFiles :
        #[ OpenFile :
*/

int OpenFile(char *name)
{
    FILES *f;
    int i;

    if ( ( f = Uopen(name,"rb") ) == 0 ) return(-1);
/*  Usetbuf(f,0); */
    i = CreateHandle();
    RWLOCKW(AM.handlelock);
    filelist[i] = f;
    UNRWLOCK(AM.handlelock);
    return(i);
}

/*
        #] OpenFile :
        #[ OpenAddFile :
*/

int OpenAddFile(char *name)
{
    FILES *f;
    int i;
    POSITION scrpos;
    if ( ( f = Uopen(name,"a+b") ) == 0 ) return(-1);
/*  Usetbuf(f,0); */
    i = CreateHandle();
    RWLOCKW(AM.handlelock);
    filelist[i] = f;
    UNRWLOCK(AM.handlelock);
    TELLFILE(i,&scrpos);
    SeekFile(i,&scrpos,SEEK_SET);
    return(i);
}

/*
        #] OpenAddFile :
        #[ ReOpenFile :
*/

int ReOpenFile(char *name)
{
    FILES *f;
    int i;
    POSITION scrpos;
    if ( ( f = Uopen(name,"r+b") ) == 0 ) return(-1);
    i = CreateHandle();
    RWLOCKW(AM.handlelock);
    filelist[i] = f;
    UNRWLOCK(AM.handlelock);
    TELLFILE(i,&scrpos);
    SeekFile(i,&scrpos,SEEK_SET);
    return(i);
}

/*
        #] ReOpenFile :
        #[ CreateFile :
*/

int CreateFile(char *name)
{
    FILES *f;
    int i;
```

```
    if ( ( f = Uopen(name,"w+b") ) == 0 ) return(-1);
    i = CreateHandle();
    RWLOCKW(AM.handlelock);
    filelist[i] = f;
    UNRWLOCK(AM.handlelock);
    return(i);
}

/*
        #] CreateFile : 
        #[ CreateLogFile :
*/

int CreateLogFile(char *name)
{
    FILES *f;
    int i;
    if ( ( f = Uopen(name,"w+b") ) == 0 ) return(-1);
    Usetbuf(f,0);
    i = CreateHandle();
    RWLOCKW(AM.handlelock);
    filelist[i] = f;
    UNRWLOCK(AM.handlelock);
    return(i);
}

/*
        #] CreateLogFile : 
        #[ CloseFile :
*/

void CloseFile(int handle)
{
    if ( handle >= 0 ) {
        FILES *f;   /* we need this variable to be thread-safe */
        RWLOCKW(AM.handlelock);
        f = filelist[handle];
        filelist[handle] = 0;
        numinfilelist--;
        UNRWLOCK(AM.handlelock);
        Uclose(f);
    }
}

/*
        #] CloseFile : 
        #[ CopyFile :
*/

int CopyFile(char *source, char *dest)
{
    #define COPYFILEBUFSIZE 40960L
    FILE *in, *out;
    size_t countin, countout, sumcount;
    char *buffer = NULL;

    sumcount = (AM.S0->LargeSize+AM.S0->SmallEsize)*sizeof(WORD);
    if ( sumcount <= COPYFILEBUFSIZE ) {
        sumcount = COPYFILEBUFSIZE;
        buffer = (char*)Malloc1(sumcount, "file copy buffer");
    }
    else {
        buffer = (char *)(AM.S0->lBuffer);
    }

    in = fopen(source, "rb");
    if ( in == NULL ) {
        perror("CopyFile: ");
        return(1);
    }
    out = fopen(dest, "wb");
    if ( out == NULL ) {
        perror("CopyFile: ");
        return(2);
    }

    while ( !feof(in) ) {
        countin = fread(buffer, 1, sumcount, in);
        if ( countin != sumcount ) {
            if ( ferror(in) ) {
                perror("CopyFile: ");
                return(3);
            }
        }
        countout = fwrite(buffer, 1, countin, out);
        if ( countin != countout ) {
            perror("CopyFile: ");
```

```
            return(4);
        }
    }

    fclose(in);
    fclose(out);
    if ( sumcount <= COPYFILEBUFSIZE ) {
        M_free(buffer, "file copy buffer");
    }
    return(0);
}

/*
        #] CopyFile :
        #[ CreateHandle :

        We need a lock here.
        Problem: the same lock is needed inside Malloc1 and M_free which
        is used in DoubleList when we use MALLOCDEBUG

        Conclusion: MALLOCDEBUG will have to be a bit unsafe
*/

int CreateHandle(void)
{
    int i, j;
#ifndef MALLOCDEBUG
    RWLOCKW(AM.handlelock);
#endif
    if ( filelistsize == 0 ) {
        filelistsize = 10;
        filelist = (FILES **)Malloc1(sizeof(FILES *)*filelistsize,"file handle");
        for ( j = 0; j < filelistsize; j++ ) filelist[j] = 0;
        numinfilelist = 1;
        i = 0;
    }
    else if ( numinfilelist >= filelistsize ) {
        void **fl = (void **)filelist;
        i = filelistsize;
        if ( DoubleList((void ***)(&fl),&filelistsize,(int)sizeof(FILES *),
            "list of open files") != 0 ) Terminate(-1);
        filelist = (FILES **)fl;
        for ( j = i; j < filelistsize; j++ ) filelist[j] = 0;
        numinfilelist = i + 1;
    }
    else {
        i = filelistsize;
        for ( j = 0; j < filelistsize; j++ ) {
            if ( filelist[j] == 0 ) { i = j; break; }
        }
        numinfilelist++;
    }
    filelist[i] = (FILES *)(filelist); /* Just for now to not get into problems */
/*
    The next code is not needed when we use open.
    It may be needed when we use fopen.
    fopen is used in minos.c without this central administration.
*/
    if ( numinfilelist > MAX_OPEN_FILES ) {
#ifndef MALLOCDEBUG
        UNRWLOCK(AM.handlelock);
#endif
        MesPrint("More than %d open files",MAX_OPEN_FILES);
        Error0("System limit. This limit is not due to FORM!");
    }
    else {
#ifndef MALLOCDEBUG
        UNRWLOCK(AM.handlelock);
#endif
    }
    return(i);
}

/*
        #] CreateHandle :
        #[ ReadFile :
*/

LONG ReadFile(int handle, UBYTE *buffer, LONG size)
{
    LONG inbuf = 0, r;
    FILES *f;
    char *b;
    b = (char *)buffer;
    for(;;) {   /* Gotta do difficult because of VMS! */
        RWLOCKR(AM.handlelock);
        f = filelist[handle];
```

```
        UNRWLOCK(AM.handlelock);
#ifdef WITHSTATS
        numreads++;
#endif
        r = Uread(b,1,size,f);
        if ( r < 0 ) return(r);
        if ( r == 0 ) return(inbuf);
        inbuf += r;
        if ( r == size ) return(inbuf);
        if ( r > size ) return(-1);
        size -= r;
        b += r;
    }
}

/*
        #] ReadFile :
        #[ ReadPosFile :

        Gets words from a file(handle).
        First tries to get the information from the buffers.
        Reads a file at a position. Updates the position.
        Places a lock in the case of multithreading.
        Exists for multiple reading from the same file.
        size is the number of WORDs to read!!!!

        We may need some strategy in the caching. This routine is used from
        GetOneTerm only. The problem is when it reads brackets and the
        brackets are read backwards. This is very uneconomical because
        each time it may read a large buffer.
        On the other hand, reading piece by piece in GetOneTerm takes
        much overhead as well.
        Two strategies come to mind:
        1: keep things as they are but limit the size of the buffers.
        2: have the position of 'pos' at about 1/3 of the buffer.
           this is of course guess work.
        Currently we have implemented the first method by creating the
        setup parameter threadscratchsize with the default value 100K.
        In the test program much bigger values gave a slower program.
*/

LONG ReadPosFile(PHEAD FILEHANDLE *fi, UBYTE *buffer, LONG size, POSITION *pos)
{
    GETBIDENTITY
    LONG i, retval = 0;
    WORD *b = (WORD *)buffer, *t;

    if ( fi->handle < 0 ) {
        fi->POfill = (WORD *)((UBYTE *)(fi->PObuffer) + BASEPOSITION(*pos));
        t = fi->POfill;
        while ( size > 0 && fi->POfill < fi->POfull ) { *b++ = *t++; size--; }
    }
    else {
        if ( ISLESSPOS(*pos,fi->POposition) || ISGEPOSINC(*pos,fi->POposition,
            ((UBYTE *)(fi->POfull)-(UBYTE *)(fi->PObuffer))) ) {
/*
            The start is not inside the buffer. Fill the buffer.
*/

            fi->POposition = *pos;
            LOCK(AS.inputslock);
            SeekFile(fi->handle,pos,SEEK_SET);
            retval = ReadFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize);
            UNLOCK(AS.inputslock);
            fi->POfull = fi->PObuffer+retval/sizeof(WORD);
            fi->POfill = fi->PObuffer;
            if ( fi != AR.hidefile ) AR.InInBuf = retval/sizeof(WORD);
            else                     AR.InHiBuf = retval/sizeof(WORD);
        }
        else {
            fi->POfill = (WORD *)((UBYTE *)(fi->PObuffer) + DIFBASE(*pos,fi->POposition));
        }
        if ( fi->POfill + size <= fi->POfull ) {
            t = fi->POfill;
            while ( size > 0 ) { *b++ = *t++; size--; }
        }
        else {
          for (;;) {
            i = fi->POfull - fi->POfill; t = fi->POfill;
            if ( i > size ) i = size;
            size -= i;
            while ( --i >= 0 ) *b++ = *t++;
            if ( size == 0 ) break;
            ADDPOS(fi->POposition,(UBYTE *)(fi->POfull)-(UBYTE *)(fi->PObuffer));
            LOCK(AS.inputslock);
            SeekFile(fi->handle,&(fi->POposition),SEEK_SET);
            retval = ReadFile(fi->handle,(UBYTE *)(fi->PObuffer),fi->POsize);
```

```
            UNLOCK(AS.inputslock);
            fi->POfull = fi->PObuffer+retval/sizeof(WORD);
            fi->POfill = fi->PObuffer;
            if ( fi != AR.hidefile ) AR.InInBuf = retval/sizeof(WORD);
            else                     AR.InHiBuf = retval/sizeof(WORD);
            if ( retval == 0 ) { t = fi->POfill; break; }
          }
        }
    }
    retval = (UBYTE *)b - buffer;
    fi->POfill = t;
    ADDPOS(*pos,retval);
    return(retval);
}

/*
        #] ReadPosFile :
        #[ WriteFile :
*/

LONG WriteFileToFile(int handle, UBYTE *buffer, LONG size)
{
    FILES *f;
    LONG retval, totalwritten = 0, stilltowrite;
    RWLOCKR(AM.handlelock);
    f = filelist[handle];
    UNRWLOCK(AM.handlelock);
    while ( totalwritten < size ) {
        stilltowrite = size - totalwritten;
#ifdef WITHSTATS
        numwrites++;
#endif
        retval = Uwrite((char *)buffer+totalwritten,1,stilltowrite,f);
        if ( retval < 0 ) return(retval);
        if ( retval == 0 ) return(totalwritten);
        totalwritten += retval;
    }
/*
if ( handle == AC.LogHandle || handle == ERROROUT ) FlushFile(handle);
*/
    return(totalwritten);
}
#ifndef WITHMPI
/*[17nov2005]:*/
WRITEFILE WriteFile = &WriteFileToFile;
/*
LONG (*WriteFile)(int handle, UBYTE *buffer, LONG size) = &WriteFileToFile;
*/
/*:[17nov2005]*/
#else
WRITEFILE WriteFile = &PF_WriteFileToFile;
#endif

/*
        #] WriteFile :
        #[ SeekFile :
*/

void SeekFile(int handle, POSITION *offset, int origin)
{
    FILES *f;
    RWLOCKR(AM.handlelock);
    f = filelist[handle];
    UNRWLOCK(AM.handlelock);
#ifdef WITHSTATS
    numseeks++;
#endif
    if ( origin == SEEK_SET ) {
        Useek(f,BASEPOSITION(*offset),origin);
        SETBASEPOSITION(*offset,(Utell(f)));
        return;
    }
    else if ( origin == SEEK_END ) {
        Useek(f,0,origin);
    }
    SETBASEPOSITION(*offset,(Utell(f)));
}

/*
        #] SeekFile :
        #[ TellFile :
*/

LONG TellFile(int handle)
{
    POSITION pos;
    TELLFILE(handle,&pos);
```

```
#ifdef WITHSTATS
    numseeks++;
#endif
    return(BASEPOSITION(pos));
}

void TELLFILE(int handle, POSITION *position)
{
    FILES *f;
    RWLOCKR(AM.handlelock);
    f = filelist[handle];
    UNRWLOCK(AM.handlelock);
    SETBASEPOSITION(*position,(Utell(f)));
}

/*
        #] TellFile :
        #[ FlushFile :
*/

void FlushFile(int handle)
{
    FILES *f;
    RWLOCKR(AM.handlelock);
    f = filelist[handle];
    UNRWLOCK(AM.handlelock);
    Uflush(f);
}

/*
        #] FlushFile :
        #[ GetPosFile :
*/

int GetPosFile(int handle, fpos_t *pospointer)
{
    FILES *f;
    RWLOCKR(AM.handlelock);
    f = filelist[handle];
    UNRWLOCK(AM.handlelock);
    return(Ugetpos(f,pospointer));
}

/*
        #] GetPosFile :
        #[ SetPosFile :
*/

int SetPosFile(int handle, fpos_t *pospointer)
{
    FILES *f;
    RWLOCKR(AM.handlelock);
    f = filelist[handle];
    UNRWLOCK(AM.handlelock);
    return(Usetpos(f,(fpos_t *)pospointer));
}

/*
        #] SetPosFile :
        #[ SynchFile :

        It may be that when we use many sort files at the same time there
        is a big traffic jam in the cache. This routine is experimental,
        just to see whether this improves the situation.
        It could also be that the internal disk of the Quad opteron norma
        is very slow.
*/

void SynchFile(int handle)
{
    FILES *f;
    if ( handle >= 0 ) {
        RWLOCKR(AM.handlelock);
        f = filelist[handle];
        UNRWLOCK(AM.handlelock);
        Usync(f);
    }
}

/*
        #] SynchFile :
        #[ TruncateFile :

        It may be that when we use many sort files at the same time there
        is a big traffic jam in the cache. This routine is experimental,
        just to see whether this improves the situation.
        It could also be that the internal disk of the Quad opteron norma
```

```
        is very slow.
*/

void TruncateFile(int handle)
{
    FILES *f;
    if ( handle >= 0 ) {
        RWLOCKR(AM.handlelock);
        f = filelist[handle];
        UNRWLOCK(AM.handlelock);
        Utruncate(f);
    }
}

/*
        #] TruncateFile :
        #[ GetChannel :

        Checks whether we have this file already. If so, we return its
        handle. If not and mode == 0, we open the file first and add it
        to the buffers.
*/

int GetChannel(char *name,int mode)
{
    CHANNEL *ch;
    int i;
    FILES *f;
    for ( i = 0; i < NumOutputChannels; i++ ) {
        if ( channels[i].name == 0 ) continue;
        if ( StrCmp((UBYTE *)name,(UBYTE *)(channels[i].name)) == 0 ) return(channels[i].handle);
    }
    if ( mode == 1 ) {
        MesPrint("&File %s in print statement is not open",name);
        MesPrint("   You should open it first with a #write or #append instruction");
        return(-1);
    }
    for ( i = 0; i < NumOutputChannels; i++ ) {
        if ( channels[i].name == 0 ) break;
    }
    if ( i < NumOutputChannels ) { ch = &(channels[i]); }
    else { ch = (CHANNEL *)FromList(&AC.ChannelList); }
    ch->name = (char *)strDup1((UBYTE *)name,"name of channel");
    ch->handle = CreateFile(name);
    RWLOCKR(AM.handlelock);
    f = filelist[ch->handle];
    UNRWLOCK(AM.handlelock);
    Usetbuf(f,0);     /* We turn the buffer off!!!!!!*/
    return(ch->handle);
}

/*
        #] GetChannel :
        #[ GetAppendChannel :

        Checks whether we have this file already. If so, we return its
        handle. If not, we open the file first and add it to the buffers.
*/

int GetAppendChannel(char *name)
{
    CHANNEL *ch;
    int i;
    FILES *f;
    for ( i = 0; i < NumOutputChannels; i++ ) {
        if ( channels[i].name == 0 ) continue;
        if ( StrCmp((UBYTE *)name,(UBYTE *)(channels[i].name)) == 0 ) return(channels[i].handle);
    }
    for ( i = 0; i < NumOutputChannels; i++ ) {
        if ( channels[i].name == 0 ) break;
    }
    if ( i < NumOutputChannels ) { ch = &(channels[i]); }
    else { ch = (CHANNEL *)FromList(&AC.ChannelList); }
    ch->name = (char *)strDup1((UBYTE *)name,"name of channel");
    ch->handle = OpenAddFile(name);
    RWLOCKR(AM.handlelock);
    f = filelist[ch->handle];
    UNRWLOCK(AM.handlelock);
    Usetbuf(f,0);     /* We turn the buffer off!!!!!!*/
    return(ch->handle);
}

/*
        #] GetAppendChannel :
        #[ CloseChannel :

        Checks whether we have this file already. If so, we close it.
```

```
*/

int CloseChannel(char *name)
{
    int i;
    for ( i = 0; i < NumOutputChannels; i++ ) {
        if ( channels[i].name == 0 ) continue;
        if ( channels[i].name[0] == 0 ) continue;
        if ( StrCmp((UBYTE *)name,(UBYTE *)(channels[i].name)) == 0 ) {
            CloseFile(channels[i].handle);
            M_free(channels[i].name,"CloseChannel");
            channels[i].name = 0;
            return(0);
        }
    }
    return(0);
}

/*
        #] CloseChannel :
        #[ UpdateMaxSize :

        Updates the maximum size of the combined input/output/hide scratch
        files, the sort files and the .str file.
        The result becomes only visible with either
            ON totalsize;
            #: totalsize ON;
        or the -T in the command tail.

        To be called, whenever a file is closed/removed or truncated to zero.

        We have no provisions yet for expressions that remain inside the
        small or large buffer during the sort. The space they use there is
        currently ignored.
*/

void UpdateMaxSize(void)
{
    POSITION position, sumsize;
    int i;
    FILEHANDLE *scr;
#ifdef WITHMPI
    /* Currently, it works only on the master. The sort files on the slaves
     * are ignored. (TU 11 Oct 2011) */
    if ( PF.me != MASTER ) return;
#endif
    PUTZERO(sumsize);
    if ( AM.PrintTotalSize ) {
/*
        First the three scratch files
*/
#ifdef WITHPTHREADS
    scr = AB[0]->R.Fscr;
#else
    scr = AR.Fscr;
#endif
    for ( i = 0; i <=2; i++ ) {
        if ( scr[i].handle < 0 ) {
            SETBASEPOSITION(position,(scr[i].POfull-scr[i].PObuffer)*sizeof(WORD));
        }
        else {
            position = scr[i].filesize;
        }
        ADD2POS(sumsize,position);
    }
/*
        Now the sort file(s)
*/
#ifdef WITHPTHREADS
    {
        int j;
        ALLPRIVATES *B;
        for ( j = 0; j < AM.totalnumberofthreads; j++ ) {
            B = AB[j];
            if ( AT.SS && AT.SS->file.handle >= 0 ) {
                position = AT.SS->file.filesize;
/*
MLOCK(ErrorMessageLock);
MesPrint("%d: %10p",j,&(AT.SS->file.filesize));
MUNLOCK(ErrorMessageLock);
*/
                ADD2POS(sumsize,position);
            }
            if ( AR.FoStage4[0].handle >= 0 ) {
                position = AR.FoStage4[0].filesize;
                ADD2POS(sumsize,position);
            }
```

```
        }
    }
#else
    if ( AT.SS && AT.SS->file.handle >= 0 ) {
        position = AT.SS->file.filesize;
        ADD2POS(sumsize,position);
    }
    if ( AR.FoStage4[0].handle >= 0 ) {
        position = AR.FoStage4[0].filesize;
        ADD2POS(sumsize,position);
    }
#endif
/*
        And of course the str file.
*/
    ADD2POS(sumsize,AC.StoreFileSize);
/*
        Finally the test whether it is bigger
*/
    if ( ISLESSPOS(AS.MaxExprSize,sumsize) ) {
#ifdef WITHPTHREADS
        LOCK(AS.MaxExprSizeLock);
        if ( ISLESSPOS(AS.MaxExprSize,sumsize) ) AS.MaxExprSize = sumsize;
        UNLOCK(AS.MaxExprSizeLock);
#else
        AS.MaxExprSize = sumsize;
#endif
    }
    }
    return;
}

/*
        #] UpdateMaxSize :
    #] Files :
    #[ Strings :
        #[ StrCmp :
*/

int StrCmp(UBYTE *s1, UBYTE *s2)
{
    while ( *s1 && *s1 == *s2 ) { s1++; s2++; }
    return((int)*s1-(int)*s2);
}

/*
        #] StrCmp :
        #[ StrICmp :
*/

int StrICmp(UBYTE *s1, UBYTE *s2)
{
    while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
    return((int)tolower(*s1)-(int)tolower(*s2));
}

/*
        #] StrICmp :
        #[ StrHICmp :
*/

int StrHICmp(UBYTE *s1, UBYTE *s2)
{
    while ( *s1 && tolower(*s1) == *s2 ) { s1++; s2++; }
    return((int)tolower(*s1)-(int)(*s2));
}

/*
        #] StrHICmp :
        #[ StrICont :
*/

int StrICont(UBYTE *s1, UBYTE *s2)
{
    while ( *s1 && tolower(*s1) == tolower(*s2) ) { s1++; s2++; }
    if ( *s1 == 0 ) return(0);
    return((int)tolower(*s1)-(int)tolower(*s2));
}

/*
        #] StrICont :
        #[ CmpArray :
*/

int CmpArray(WORD *t1, WORD *t2, WORD n)
{
    int i,x;
```

```
    for ( i = 0; i < n; i++ ) {
        if ( ( x = (int)(t1[i]-t2[i]) ) != 0 ) return(x);
    }
    return(0);
}

/*
        #] CmpArray :
        #[ ConWord :
*/

int ConWord(UBYTE *s1, UBYTE *s2)
{
    while ( *s1 && ( tolower(*s1) == tolower(*s2) ) ) { s1++; s2++; }
    if ( *s1 == 0 ) return(1);
    return(0);
}

/*
        #] ConWord :
        #[ StrLen :
*/

int StrLen(UBYTE *s)
{
    int i = 0;
    while ( *s ) { s++; i++; }
    return(i);
}

/*
        #] StrLen :
        #[ NumToStr :
*/

void NumToStr(UBYTE *s, LONG x)
{
    UBYTE *t, str[24];
    ULONG xx;
    t = str;
    if ( x < 0 ) { *s++ = '-'; xx = -x; }
    else xx = x;
    do {
        *t++ = xx % 10 + '0';
        xx /= 10;
    } while ( xx );
    while ( t > str ) *s++ = *--t;
    *s = 0;
}

/*
        #] NumToStr :
        #[ WriteString :

        Writes a characterstring to the various outputs.
        The action may depend on the flags involved.
        The type of output is given by type, the string by str and the
        number of characters in it by num
*/
void WriteString(int type, UBYTE *str, int num)
{
    int error = 0;

    if ( num > 0 && str[num-1] == 0 ) { num--; }
    else if ( num <= 0 || str[num-1] != LINEFEED ) {
        AddLineFeed(str,num);
    }
    /*[15apr2004 mt]:*/
    if(type == EXTERNALCHANNELOUT){
        if(WriteFile(0,str,num) != num) error = 1;
    }else
    /*:[15apr2004 mt]*/
    if ( AM.silent == 0 || type == ERROROUT ) {
        if ( type == INPUTOUT ) {
            if ( !AM.FileOnlyFlag && WriteFile(AM.StdOut,(UBYTE *)"    ",4) != 4 ) error = 1;
            if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle,(UBYTE *)"    ",4) != 4 ) error = 1;
        }
        if ( !AM.FileOnlyFlag && WriteFile(AM.StdOut,str,num) != num ) error = 1;
        if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle,str,num) != num ) error = 1;
    }
    if ( error ) Terminate(-1);
}

/*
        #] WriteString :
        #[ WriteUnfinString :

```

```
        Writes a characterstring to the various outputs.
        The action may depend on the flags involved.
        The type of output is given by type, the string by str and the
        number of characters in it by num
*/

void WriteUnfinString(int type, UBYTE *str, int num)
{
    int error = 0;

    /*[15apr2004 mt]:*/
    if(type == EXTERNALCHANNELOUT){
        if(WriteFile(0,str,num) != num) error = 1;
    }else
    /*:[15apr2004 mt]*/
    if ( AM.silent == 0 || type == ERROROUT ) {
        if ( type == INPUTOUT ) {
            if ( !AM.FileOnlyFlag && WriteFile(AM.StdOut,(UBYTE *)"    ",4) != 4 ) error = 1;
            if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle,(UBYTE *)"    ",4) != 4 ) error = 1;
        }
        if ( !AM.FileOnlyFlag && WriteFile(AM.StdOut,str,num) != num ) error = 1;
        if ( AC.LogHandle >= 0 && WriteFile(AC.LogHandle,str,num) != num ) error = 1;
    }
    if ( error ) Terminate(-1);
}

/*
        #] WriteUnfinString :
        #[ AddToString :
*/

UBYTE *AddToString(UBYTE *outstring, UBYTE *extrastring, int par)
{
    UBYTE *s = extrastring, *t, *newstring;
    int n, nn;
    while ( *s ) { s++; }
    n = s-extrastring;
    if ( outstring == 0 ) {
        s = extrastring;
        t = outstring = (UBYTE *)Malloc1(n+1,"AddToString");
        NCOPY(t,s,n)
        *t++ = 0;
        return(outstring);
    }
    else {
        t = outstring;
        while ( *t ) t++;
        nn = t - outstring;
        t = newstring = (UBYTE *)Malloc1(n+nn+2,"AddToString");
        s = outstring;
        NCOPY(t,s,nn)
        if ( par == 1 ) *t++ = ',';
        s = extrastring;
        NCOPY(t,s,n)
        *t = 0;
        M_free(outstring,"AddToString");
        return(newstring);
    }
}

/*
        #] AddToString :
        #[ strDup1 :

        string duplication with message passing for Malloc1, allowing
        this routine to give a more detailed error message if there
        is not enough memory.
*/

UBYTE *strDup1(UBYTE *instring, char *ifwrong)
{
    UBYTE *s = instring, *to;
    while ( *s ) s++;
    to = s = (UBYTE *)Malloc1((s-instring)+1,ifwrong);
    while ( *instring ) *to++ = *instring++;
    *to = 0;
    return(s);
}

/*
        #] strDup1 :
        #[ EndOfToken :
*/

UBYTE *EndOfToken(UBYTE *s)
{
    UBYTE c;
```

```
    while ( ( c = (UBYTE)(FG.cTable[*s]) ) == 0 || c == 1 ) s++;
    return(s);
}

/*
        #] EndOfToken :
        #[ ToToken :
*/

UBYTE *ToToken(UBYTE *s)
{
    UBYTE c;
    while ( *s && ( c = (UBYTE)(FG.cTable[*s]) ) != 0 && c != 1 ) s++;
    return(s);
}

/*
        #] ToToken :
        #[ SkipField :

    Skips from s to the end of a declaration field.
    par is the number of parentheses that still has to be closed.
*/

UBYTE *SkipField(UBYTE *s, int level)
{
    while ( *s ) {
        if ( *s == ',' && level == 0 ) return(s);
        if ( *s == '(' ) level++;
        else if ( *s == ')' ) { level--; if ( level < 0 ) level = 0; }
        else if ( *s == '[' ) {
            SKIPBRA1(s)
        }
        else if ( *s == '{' ) {
            SKIPBRA2(s)
        }
        s++;
    }
    return(s);
}

/*
        #] SkipField :
        #[ ReadSnum :           WORD ReadSnum(p)

        Reads a number that should fit in a word.
        The number should be unsigned and a negative return value
        indicates an irregularity.

*/

WORD ReadSnum(UBYTE **p)
{
    LONG x = 0;
    UBYTE *s;
    s = *p;
    if ( FG.cTable[*s] == 1 ) {
        do {
            x = ( x << 3 ) + ( x << 1 ) + ( *s++ - '0' );
            if ( x > MAXPOSITIVE ) return(-1);
        } while ( FG.cTable[*s] == 1 );
        *p = s;
        return((WORD)x);
    }
    else return(-1);
}

/*
        #] ReadSnum :
        #[ NumCopy :

    Adds the decimal representation of a number to a string.

*/

UBYTE *NumCopy(WORD y, UBYTE *to)
{
    UBYTE *s;
    WORD i = 0, j;
    UWORD x;
    if ( y < 0 ) {
        *to++ = '-';
    }
    x = WordAbs(y);
    s = to;
    do { *s++ = (UBYTE)((x % 10)+'0'); i++; } while ( ( x /= 10 ) != 0 );
    *s-- = '\0';
```

```
    j = ( i - 1 ) » 1;
    while ( j >= 0 ) {
        i = to[j]; to[j] = s[-j]; s[-j] = (UBYTE)i; j--;
    }
    return(s+1);
}

/*
        #] NumCopy :
        #[ LongCopy :

    Adds the decimal representation of a number to a string.

*/

char *LongCopy(LONG y, char *to)
{
    char *s;
    WORD i = 0, j;
    ULONG x;
    if ( y < 0 ) {
        *to++ = ’-’;
    }
    x = LongAbs(y);
    s = to;
    do { *s++ = (x % 10)+’0’; i++; } while ( ( x /= 10 ) != 0 );
    *s-- = ’\0’;
    j = ( i - 1 ) » 1;
    while ( j >= 0 ) {
        i = to[j]; to[j] = s[-j]; s[-j] = (char)i; j--;
    }
    return(s+1);
}

/*
        #] LongCopy :
        #[ LongLongCopy :

    Adds the decimal representation of a number to a string.
    Bugfix feb 2003. y was not pointer!
*/

char *LongLongCopy(off_t *y, char *to)
{
    /*
     * This code fails to print the maximum negative value on systems with two’s
     * complement. To fix this, we need the unsigned version of off_t with the
     * same size, but unfortunately it is undefined. On the other hand, if a
     * system is configured with a 64-bit off_t, in practice one never reaches
     * 2^63 ~ 10^18 as of 2016. If one really reach such a big number, then it
     * would be the time to move on a 128-bit off_t.
     */
    off_t x = *y;
    char *s;
    WORD i = 0, j;
    if ( x < 0 ) { x = -x; *to++ = ’-’; }
    s = to;
    do { *s++ = (x % 10)+’0’; i++; } while ( ( x /= 10 ) != 0 );
    *s-- = ’\0’;
    j = ( i - 1 ) » 1;
    while ( j >= 0 ) {
        i = to[j]; to[j] = s[-j]; s[-j] = (char)i; j--;
    }
    return(s+1);
}

/*
        #] LongLongCopy :
        #[ MakeDate :

        Routine produces a string with the date and time of the run
*/

#ifdef ANSI
#else
#ifdef mBSD
#else
static char notime[] = "";
#endif
#endif

UBYTE *MakeDate(void)
{
#ifdef ANSI
    time_t tp;
    time(&tp);
    return((UBYTE *)ctime(&tp));
```

```
#else
#ifdef mBSD
    time_t tp;
    time(&tp);
    return((UBYTE *)ctime(&tp));
#else
    return((UBYTE *)notime);
#endif
#endif
}

/*
        #] MakeDate :
        #[ set_in :
         Returns 1 if ch is in set ; 0 if ch is not in set:
*/
int set_in(UBYTE ch, set_of_char set)
{
    set += ch/8;
    switch (ch % 8){
        case 0: return(set->bit_0);
        case 1: return(set->bit_1);
        case 2: return(set->bit_2);
        case 3: return(set->bit_3);
        case 4: return(set->bit_4);
        case 5: return(set->bit_5);
        case 6: return(set->bit_6);
        case 7: return(set->bit_7);
    }/*switch (ch % 8)*/
    return(-1);
}/*set_in*/
/*
        #] set_in :
        #[ set_set :
            sets ch into set; returns *set:
*/
one_byte set_set(UBYTE ch, set_of_char set)
{
    one_byte tmp=(one_byte)set;
    set += ch/8;
    switch (ch % 8){
        case 0: set->bit_0=1;break;
        case 1: set->bit_1=1;break;
        case 2: set->bit_2=1;break;
        case 3: set->bit_3=1;break;
        case 4: set->bit_4=1;break;
        case 5: set->bit_5=1;break;
        case 6: set->bit_6=1;break;
        case 7: set->bit_7=1;break;
    }
    return(tmp);
}/*set_set*/
/*
        #] set_set :
        #[ set_del :
            deletes ch from set; returns *set:
*/
one_byte set_del(UBYTE ch, set_of_char set)
{
    one_byte tmp=(one_byte)set;
    set += ch/8;
    switch (ch % 8){
        case 0: set->bit_0=0;break;
        case 1: set->bit_1=0;break;
        case 2: set->bit_2=0;break;
        case 3: set->bit_3=0;break;
        case 4: set->bit_4=0;break;
        case 5: set->bit_5=0;break;
        case 6: set->bit_6=0;break;
        case 7: set->bit_7=0;break;
    }
    return(tmp);
}/*set_del*/
/*
        #] set_del :
        #[ set_sub :
            returns *set = set1\set2. This function may be usd for initialising,
                set_sub(a,a,a) => now a is empty set :
*/
one_byte set_sub(set_of_char set, set_of_char set1, set_of_char set2)
{
    one_byte tmp=(one_byte)set;
    int i=0,j=0;
    while(j=0,i++<32)
    while(j<9)
        switch (j++){
            case 0: set->bit_0=(set1->bit_0&&(!set2->bit_0));break;
```

```
            case 1: set->bit_1=(set1->bit_1&&(!set2->bit_1));break;
            case 2: set->bit_2=(set1->bit_2&&(!set2->bit_2));break;
            case 3: set->bit_3=(set1->bit_3&&(!set2->bit_3));break;
            case 4: set->bit_4=(set1->bit_4&&(!set2->bit_4));break;
            case 5: set->bit_5=(set1->bit_5&&(!set2->bit_5));break;
            case 6: set->bit_6=(set1->bit_6&&(!set2->bit_6));break;
            case 7: set->bit_7=(set1->bit_7&&(!set2->bit_7));break;
            case 8: set++;set1++;set2++;
     };
    return(tmp);
}/*set_sub*/
/*
        #] set_sub :
    #] Strings :
    #[ Mixed :
        #[ iniTools :
*/

void iniTools(void)
{
#ifdef MALLOCPROTECT
    if ( mprotectInit() ) exit(0);
#endif
    return;
}

/*
        #] iniTools :
        #[ Malloc :

        Malloc routine with built in error checking.
        This saves lots of messages.
*/
#ifdef MALLOCDEBUG
char *dummymessage = "Malloc";
INILOCK(MallocLock)
#endif

void *Malloc(LONG size)
{
    void *mem;
#ifdef MALLOCDEBUG
    char *t, *u;
    int i;
    LOCK(MallocLock);
/*  MLOCK(ErrorMessageLock); */
    if ( size == 0 ) {
        MesPrint("Asking for 0 bytes in Malloc");
    }
#endif
    if ( ( size & 7 ) != 0 ) { size = size - ( size&7 ) + 8; }
#ifdef MALLOCDEBUG
    size += 2*BANNER;
#endif
    mem = (void *)M_alloc(size);
    if ( mem == 0 ) {
#ifndef MALLOCDEBUG
        MLOCK(ErrorMessageLock);
#endif
        Error0("No memory!");
#ifndef MALLOCDEBUG
        MUNLOCK(ErrorMessageLock);
#else
/*      MUNLOCK(ErrorMessageLock); */
#endif
#ifdef MALLOCDEBUG
        UNLOCK(MallocLock);
#endif
        Terminate(-1);
    }
#ifdef MALLOCDEBUG
    mallocsizes[nummalloclist] = size;
    mallocstrings[nummalloclist] = dummymessage;
    malloclist[nummalloclist++] = mem;
    if ( filelist ) MesPrint("Mem0 at 0x%x, %l bytes",mem,size);
    {
        int i = nummalloclist-1;
        while ( --i >= 0 ) {
            if ( (char *)mem < (((char *)malloclist[i]) + mallocsizes[i])
            && (char *)(malloclist[i]) < ((char *)mem + size) ) {
                if ( filelist ) MesPrint("This memory overlaps with the block at 0x%x"
                    ,malloclist[i]);
            }
        }
    }
    t = (char *)mem;
    u = t + size;
```

```
    for ( i = 0; i < (int)BANNER; i++ ) { *t++ = FILLVALUE; *--u = FILLVALUE; }
    mem = (void *)t;
    {
        int j = nummalloclist-1, i;
        while ( --j >= 0 ) {
            t = (char *)(malloclist[j]);
            u = t + mallocsizes[j];
            for ( i = 0; i < (int)BANNER; i++ ) {
                u--;
                if ( *t != FILLVALUE || *u != FILLVALUE ) {
                    MesPrint("Writing outside memory for %s",malloclist[i]);
/*                  MUNLOCK(ErrorMessageLock); */
                    UNLOCK(MallocLock);
                    Terminate(-1);
                }
                t--;
            }
        }
    }
/*  MUNLOCK(ErrorMessageLock); */
    UNLOCK(MallocLock);
#endif
    return(mem);
}

/*
        #] Malloc :
        #[ Malloc1 :

        Malloc with more detailed error message.
        Gives the user some idea of what is happening.
*/

void *Malloc1(LONG size, const char *messageifwrong)
{
    void *mem;
#ifdef MALLOCDEBUG
    char *t, *u;
    int i;
    LOCK(MallocLock);
/*  MLOCK(ErrorMessageLock); */
    if ( size == 0 ) {
        MesPrint("%wAsking for 0 bytes in Malloc1");
    }
#endif
#ifdef WITHSTATS
    nummallocs++;
#endif
    if ( ( size & 7 ) != 0 ) { size = size - ( size&7 ) + 8; }
#ifdef MALLOCDEBUG
    size += 2*BANNER;
#endif
    mem = (void *)M_alloc(size);
    if ( mem == 0 ) {
#ifndef MALLOCDEBUG
        MLOCK(ErrorMessageLock);
#endif
        MesPrint("Attempted to allocate %l bytes.", size);
        Error1("No memory while allocating ",(UBYTE *)messageifwrong);
#ifndef MALLOCDEBUG
        MUNLOCK(ErrorMessageLock);
#else
/*      MUNLOCK(ErrorMessageLock); */
#endif
#ifdef MALLOCDEBUG
        UNLOCK(MallocLock);
#endif
        Terminate(-1);
    }
#ifdef MALLOCDEBUG
    mallocsizes[nummalloclist] = size;
    mallocstrings[nummalloclist] = (char *)messageifwrong;
    malloclist[nummalloclist++] = mem;
    if ( AC.MemDebugFlag && filelist ) MesPrint("%wMem1 at 0x%x: %l bytes.
%s",mem,size,messageifwrong);
    {
        int i = nummalloclist-1;
        while ( --i >= 0 ) {
            if ( (char *)mem < (((char *)malloclist[i]) + mallocsizes[i])
            && (char *)(malloclist[i]) < ((char *)mem + size) ) {
                if ( filelist ) MesPrint("This memory overlaps with the block at 0x%x"
                    ,malloclist[i]);
            }
        }
    }

#ifdef MALLOCDEBUGOUTPUT
```

```
    printf ("Malloc1: %s, allocated %li bytes at %.8lx\n",messageifwrong,size,(unsigned long)mem);
    fflush (stdout);
#endif

    t = (char *)mem;
    u = t + size;
    for ( i = 0; i < (int)BANNER; i++ ) { *t++ = FILLVALUE; *--u = FILLVALUE; }
    mem = (void *)t;
    M_check();
/*  MUNLOCK(ErrorMessageLock); */
    UNLOCK(MallocLock);
#endif
/*
    if ( size > 500000000L ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Malloc1: %s, allocated %l bytes\n",messageifwrong,size);
        MUNLOCK(ErrorMessageLock);
    }
*/
    return(mem);
}

/*
        #] Malloc1 :
        #[ M_free :
*/

void M_free(void *x, const char *where)
{
#ifdef MALLOCDEBUG
    char *t = (char *)x;
    int i, j, k;
    LONG size = 0;
    x = (void *)(((char *)x)-BANNER);
/*  MLOCK(ErrorMessageLock); */
    if ( AC.MemDebugFlag ) MesPrint("%wFreeing 0x%x: %s",x,where);
    LOCK(MallocLock);
    for ( i = nummalloclist-1; i >= 0; i-- ) {
        if ( x == malloclist[i] ) {
            size = mallocsizes[i];
            for ( j = i+1; j < nummalloclist; j++ ) {
                malloclist[j-1] = malloclist[j];
                mallocsizes[j-1] = mallocsizes[j];
                mallocstrings[j-1] = mallocstrings[j];
            }
            nummalloclist--;
            break;
        }
    }
    if ( i < 0 ) {
        unsigned int xx = ((ULONG)x);
        printf("Error returning non-allocated address: 0x%x from %s\n"
            ,xx,where);
/*      MUNLOCK(ErrorMessageLock); */
        UNLOCK(MallocLock);
        exit(-1);
    }
    else {
        for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
            if ( *--t != FILLVALUE ) j++;
        }
        if ( j ) {
            LONG *tt = (LONG *)x;
            MesPrint("%w!!!!! Banner has been written in !!!!!: %x %x %x %x",
            tt[0],tt[1],tt[2],tt[3]);
        }
        t += size;
        for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
            if ( *--t != FILLVALUE ) j++;
        }
        if ( j ) {
            LONG *tt = (LONG *)x;
            MesPrint("%w!!!!! Tail has been written in !!!!!: %x %x %x %x",
            tt[0],tt[1],tt[2],tt[3]);
        }
        M_check();
/*      MUNLOCK(ErrorMessageLock); */
        UNLOCK(MallocLock);
    }
#else
    DUMMYUSE(where);
#endif
#ifdef WITHSTATS
    numfrees++;
#endif
    if ( x ) {
#ifdef MALLOCDEBUGOUTPUT
```

```
        printf ("M_free: %s, memory freed at %.8lx\n",where,(unsigned long)x);
        fflush(stdout);
#endif

#ifdef MALLOCPROTECT
        mprotectFree((void *)x);
#else
        free(x);
#endif
    }
}

/*
        #] M_free :
        #[ M_check :
*/

#ifdef MALLOCDEBUG

void M_check1() { MesPrint("Checking Malloc"); M_check(); }

void M_check()
{
    int i,j,k,error = 0;
    char *t;
    LONG *tt;
    for ( i = 0; i < nummalloclist; i++ ) {
        t = (char *)(malloclist[i]);
        for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
            if ( *t++ != FILLVALUE ) j++;
        }
        if ( j ) {
            tt = (LONG *)(malloclist[i]);
            MesPrint("%w!!!!! Banner %d (%s) has been written in !!!!!: %x %x %x %x",
            i,mallocstrings[i],tt[0],tt[1],tt[2],tt[3]);
            tt[0] = tt[1] = tt[2] = tt[3] = 0;
            error = 1;
        }
        t = (char *)(malloclist[i]) + mallocsizes[i];
        for ( k = 0, j = 0; k < (int)BANNER; k++ ) {
            if ( *--t != FILLVALUE ) j++;
        }
        if ( j ) {
            tt = (LONG *)t;
            MesPrint("%w!!!!! Tail %d (%s) has been written in !!!!!: %x %x %x %x",
            i,mallocstrings[i],tt[0],tt[1],tt[2],tt[3]);
            tt[0] = tt[1] = tt[2] = tt[3] = 0;
            error = 1;
        }
        if ( ( mallocstrings[i][0] == ' ' ) || ( mallocstrings[i][0] == '#' ) ) {
            MesPrint("%w!!!!! Funny mallocstring");
            error = 1;
        }
    }
    if ( error ) {
        M_print();
/*      MUNLOCK(ErrorMessageLock); */
        UNLOCK(MallocLock);
        Terminate(-1);
    }
}

void M_print()
{
    int i;
    MesPrint("We have the following memory allocations left:");
    for ( i = 0; i < nummalloclist; i++ ) {
        MesPrint("0x%x: %l bytes. number %d: '%s'",malloclist[i],mallocsizes[i],i,mallocstrings[i]);
    }
}

#else

void M_check1(void) {}
void M_print(void) {}

#endif

/*
        #] M_check :
        #[ TermMalloc :
*/
#define TERMMEMSTARTNUM 16
#define TERMEXTRAWORDS 10

void TermMallocAddMemory(PHEAD0)
{
```

```
    WORD *newbufs;
    int i, extra;
    if ( AT.TermMemMax == 0 ) extra = TERMMEMSTARTNUM;
    else                      extra = AT.TermMemMax;
    if ( AT.TermMemHeap ) M_free(AT.TermMemHeap,"TermMalloc");
    newbufs = (WORD *)Malloc1(extra*(AM.MaxTer+TERMEXTRAWORDS*sizeof(WORD)),"TermMalloc");
    AT.TermMemHeap = (WORD **)Malloc1((extra+AT.TermMemMax)*sizeof(WORD *),"TermMalloc");
    for ( i = 0; i < extra; i++ ) {
        AT.TermMemHeap[i] = newbufs + i*(AM.MaxTer/sizeof(WORD)+TERMEXTRAWORDS);
    }
#ifdef TERMMALLOCDEBUG
    DebugHeap2 = (WORD **)Malloc1((extra+AT.TermMemMax)*sizeof(WORD *),"TermMalloc");
    for ( i = 0; i < AT.TermMemMax; i++ ) { DebugHeap2[i] = DebugHeap1[i]; }
    for ( i = 0; i < extra; i++ ) {
        DebugHeap2[i+AT.TermMemMax] = newbufs + i*(AM.MaxTer/sizeof(WORD)+TERMEXTRAWORDS);
    }
    if ( DebugHeap1 ) M_free(DebugHeap1,"TermMalloc");
    DebugHeap1 = DebugHeap2;
#endif
    AT.TermMemTop = extra;
    AT.TermMemMax += extra;
#ifdef TERMMALLOCDEBUG
    MesPrint("AT.TermMemMax is now %l",AT.TermMemMax);
#endif
}

#ifndef MEMORYMACROS

WORD *TermMalloc2(PHEAD char *text)
{
    if ( AT.TermMemTop <= 0 ) TermMallocAddMemory(BHEAD0);

#ifdef TERMMALLOCDEBUG
    MesPrint("TermMalloc: %s, %d",text,(AT.TermMemMax-AT.TermMemTop));
#endif

#ifdef MALLOCDEBUGOUTPUT
    MesPrint("TermMalloc: %s, %l/%l
(%x)",text,AT.TermMemTop,AT.TermMemMax,AT.TermMemHeap[AT.TermMemTop-1]);
#endif

    DUMMYUSE(text);
    return(AT.TermMemHeap[--AT.TermMemTop]);
}

void TermFree2(PHEAD WORD *TermMem, char *text)
{
#ifdef TERMMALLOCDEBUG

    int i;

    for ( i = 0; i < AT.TermMemMax; i++ ) {
        if ( TermMem == DebugHeap1[i] ) break;
    }
    if ( i >= AT.TermMemMax ) {
        MesPrint(" ERROR: TermFree called with an address not given by TermMalloc.");
        Terminate(-1);
    }
#endif
    DUMMYUSE(text);
    AT.TermMemHeap[AT.TermMemTop++] = TermMem;

#ifdef TERMMALLOCDEBUG
    MesPrint("TermFree: %s, %d",text,(AT.TermMemMax-AT.TermMemTop));
#endif
#ifdef MALLOCDEBUGOUTPUT
    MesPrint("TermFree: %s, %l/%l (%x)",text,AT.TermMemTop,AT.TermMemMax,TermMem);
#endif
}

#endif

/*
        #] TermMalloc :
        #[ NumberMalloc :
*/
#define NUMBERMEMSTARTNUM 16
#define NUMBEREXTRAWORDS 10L

#ifdef TERMMALLOCDEBUG
UWORD **DebugHeap3, **DebugHeap4;
#endif

void NumberMallocAddMemory(PHEAD0)
{
    UWORD *newbufs;
    WORD extra;
```

```
    int i;
    if ( AT.NumberMemMax == 0 ) extra = NUMBERMEMSTARTNUM;
    else                        extra = AT.NumberMemMax;
    if ( AT.NumberMemHeap ) M_free(AT.NumberMemHeap,"NumberMalloc");
    newbufs = (UWORD *)Malloc1(extra*(AM.MaxTal+NUMBEREXTRAWORDS)*sizeof(UWORD),"NumberMalloc");
    AT.NumberMemHeap = (UWORD **)Malloc1((extra+AT.NumberMemMax)*sizeof(UWORD *),"NumberMalloc");
    for ( i = 0; i < extra; i++ ) {
        AT.NumberMemHeap[i] = newbufs + i*(LONG)(AM.MaxTal+NUMBEREXTRAWORDS);
    }
#ifdef TERMMALLOCDEBUG
    DebugHeap4 = (UWORD **)Malloc1((extra+AT.NumberMemMax)*sizeof(WORD *),"NumberMalloc");
    for ( i = 0; i < AT.NumberMemMax; i++ ) { DebugHeap4[i] = DebugHeap3[i]; }
    for ( i = 0; i < extra; i++ ) {
        DebugHeap4[i+AT.NumberMemMax] = newbufs + i*(LONG)(AM.MaxTal+NUMBEREXTRAWORDS);
    }
    if ( DebugHeap3 ) M_free(DebugHeap3,"NumberMalloc");
    DebugHeap3 = DebugHeap4;
#endif
    AT.NumberMemTop = extra;
    AT.NumberMemMax += extra;
/*
MesPrint("AT.NumberMemMax is now %l",AT.NumberMemMax);
*/
}

#ifndef MEMORYMACROS

UWORD *NumberMalloc2(PHEAD char *text)
{
    if ( AT.NumberMemTop <= 0 ) NumberMallocAddMemory(BHEAD text);

#ifdef MALLOCDEBUGOUTPUT
    if ( (AT.NumberMemMax-AT.NumberMemTop) > 10 )
    MesPrint("NumberMalloc: %s, %l/%l
(%x)",text,AT.NumberMemTop,AT.NumberMemMax,AT.NumberMemHeap[AT.NumberMemTop-1]);
#endif

    DUMMYUSE(text);
    return(AT.NumberMemHeap[--AT.NumberMemTop]);
}

void NumberFree2(PHEAD UWORD *NumberMem, char *text)
{
#ifdef TERMMALLOCDEBUG
    int i;
    for ( i = 0; i < AT.NumberMemMax; i++ ) {
        if ( NumberMem == DebugHeap3[i] ) break;
    }
    if ( i >= AT.NumberMemMax ) {
        MesPrint(" ERROR: NumberFree called with an address not given by NumberMalloc.");
        Terminate(-1);
    }
#endif
    DUMMYUSE(text);
    AT.NumberMemHeap[AT.NumberMemTop++] = NumberMem;

#ifdef MALLOCDEBUGOUTPUT
    if ( (AT.NumberMemMax-AT.NumberMemTop) > 10 )
    MesPrint("NumberFree: %s, %l/%l (%x)",text,AT.NumberMemTop,AT.NumberMemMax,NumberMem);
#endif
}

#endif

/*
        #] NumberMalloc :
        #[ CacheNumberMalloc :

    Similar to NumberMalloc
 */

void CacheNumberMallocAddMemory(PHEAD0)
{
    UWORD *newbufs;
    WORD extra;
    int i;
    if ( AT.CacheNumberMemMax == 0 ) extra = NUMBERMEMSTARTNUM;
    else                             extra = AT.CacheNumberMemMax;
    if ( AT.CacheNumberMemHeap ) M_free(AT.CacheNumberMemHeap,"NumberMalloc");
    newbufs = (UWORD *)Malloc1(extra*(AM.MaxTal+NUMBEREXTRAWORDS)*sizeof(UWORD),"CacheNumberMalloc");
    AT.CacheNumberMemHeap = (UWORD **)Malloc1((extra+AT.NumberMemMax)*sizeof(UWORD
*),"CacheNumberMalloc");
    for ( i = 0; i < extra; i++ ) {
        AT.CacheNumberMemHeap[i] = newbufs + i*(LONG)(AM.MaxTal+NUMBEREXTRAWORDS);
    }
    AT.CacheNumberMemTop = extra;
    AT.CacheNumberMemMax += extra;
```

```
}

#ifndef MEMORYMACROS

UWORD *CacheNumberMalloc2(PHEAD char *text)
{
    if ( AT.CacheNumberMemTop <= 0 ) CacheNumberMallocAddMemory(BHEAD0);

#ifdef MALLOCDEBUGOUTPUT
    MesPrint("NumberMalloc: %s, %l/%l
(%x)",text,AT.NumberMemTop,AT.NumberMemMax,AT.NumberMemHeap[AT.NumberMemTop-1]);
#endif

    DUMMYUSE(text);
    return(AT.CacheNumberMemHeap[--AT.CacheNumberMemTop]);
}

void CacheNumberFree2(PHEAD UWORD *NumberMem, char *text)
{
    DUMMYUSE(text);
    AT.CacheNumberMemHeap[AT.CacheNumberMemTop++] = NumberMem;

#ifdef MALLOCDEBUGOUTPUT
    MesPrint("NumberFree: %s, %l/%l (%x)",text,AT.NumberMemTop,AT.NumberMemMax,NumberMem);
#endif
}

#endif

/*
        #] CacheNumberMalloc :
        #[ FromList :

    Returns the next object in a list.
    If the list has exhausted we double it (like a realloc)
    If the list has not been initialized yet we start with 12 elements.
*/

void *FromList(LIST *L)
{
    void *newlist;
    int i, *old, *newL;
    if ( L->num >= L->maxnum || L->lijst == 0 ) {
        if ( L->maxnum == 0 ) L->maxnum = 12;
        else if ( L->lijst ) L->maxnum *= 2;
        newlist = Malloc1(L->maxnum * L->size,L->message);
        if ( L->lijst ) {
            i = ( L->num * L->size ) / sizeof(int);
            old = (int *)L->lijst; newL = (int *)newlist;
            while ( --i >= 0 ) *newL++ = *old++;
            if ( L->lijst ) M_free(L->lijst,"L->lijst FromList");
        }
        L->lijst = newlist;
    }
    return( ((char *)(L->lijst)) + L->size * (L->num)++ );
}

/*
        #] FromList :
        #[ From0List :

        Same as FromList, but we zero excess variables.
*/

void *From0List(LIST *L)
{
    void *newlist;
    int i, *old, *newL;
    if ( L->num >= L->maxnum || L->lijst == 0 ) {
        if ( L->maxnum == 0 ) L->maxnum = 12;
        else if ( L->lijst ) L->maxnum *= 2;
        newlist = Malloc1(L->maxnum * L->size,L->message);
        i = ( L->num * L->size ) / sizeof(int);
        old = (int *)(L->lijst); newL = (int *)newlist;
        while ( --i >= 0 ) *newL++ = *old++;
        i = ( L->maxnum - L->num ) / sizeof(int);
        while ( --i >= 0 ) *newL++ = 0;
        if ( L->lijst ) M_free(L->lijst,"L->lijst From0List");
        L->lijst = newlist;
    }
    return( ((char *)(L->lijst)) + L->size * (L->num)++ );
}

/*
        #] From0List :
        #[ FromVarList :

```

```
    Returns the next object in a list of variables.
    If the list has been exhausted we double it (like a realloc)
    If the list has not been initialized yet we start with 12 elements.
    We allow at most MAXVARIABLES elements!
*/

void *FromVarList(LIST *L)
{
    void *newlist;
    int i, *old, *newL;
    if ( L->num >= L->maxnum || L->lijst == 0 ) {
        if ( L->maxnum == 0 ) L->maxnum = 12;
        else if ( L->lijst ) {
            L->maxnum *= 2;
            if ( L == &(AP.DollarList) ) {
                if ( L->maxnum > MAXDOLLARVARIABLES ) L->maxnum = MAXDOLLARVARIABLES;
                if ( L->num >= MAXDOLLARVARIABLES ) {
                    MesPrint("!!!More than %l objects in list of $-variables",
                        MAXDOLLARVARIABLES);
                    Terminate(-1);
                }
            }
            else {
                if ( L->maxnum > MAXVARIABLES ) L->maxnum = MAXVARIABLES;
                if ( L->num >= MAXVARIABLES ) {
                    MesPrint("!!!More than %l objects in list of variables",
                        MAXVARIABLES);
                    Terminate(-1);
                }
            }
        }
        newlist = Malloc1(L->maxnum * L->size,L->message);
        if ( L->lijst ) {
            i = ( L->num * L->size ) / sizeof(int);
            old = (int *)(L->lijst); newL = (int *)newlist;
            while ( --i >= 0 ) *newL++ = *old++;
            if ( L->lijst ) M_free(L->lijst,"L->lijst from VarList");
        }
        L->lijst = newlist;
    }
    return( ((char *)(L->lijst)) + L->size * ((L->num)++) );
}

/*
        #] FromVarList :
        #[ DoubleList :
*/

int DoubleList(void ***lijst, int *oldsize, int objectsize, char *nameoftype)
{
    void **newlist;
    LONG i, newsize, fullsize;
    void **to, **from;
    static LONG maxlistsize = (LONG)(MAXPOSITIVE);
    if ( *lijst == 0 ) {
        if ( *oldsize > 0 ) newsize = *oldsize;
        else newsize = 100;
    }
    else newsize = *oldsize * 2;
    if ( newsize > maxlistsize ) {
        if ( *oldsize == maxlistsize ) {
            MesPrint("No memory for extra space in %s",nameoftype);
            return(-1);
        }
        newsize = maxlistsize;
    }
    fullsize = ( newsize * objectsize + sizeof(void *)-1 ) & (-sizeof(void *));
    newlist = (void **)Malloc1(fullsize,nameoftype);
    if ( *lijst ) { /* Now some punning. DANGEROUS CODE in principle */
        to = newlist; from = *lijst; i = (*oldsize * objectsize)/sizeof(void *);
/*
#ifdef MALLOCDEBUG
if ( filelist ) MesPrint("    oldsize: %l, objectsize: %d, fullsize: %l"
        ,*oldsize,objectsize,fullsize);
#endif
*/
        while ( --i >= 0 ) *to++ = *from++;
    }
    if ( *lijst ) M_free(*lijst,"DoubleLList");
    *lijst = newlist;
    *oldsize = newsize;
    return(0);
/*
    int error;
    LONG lsize = *oldsize;

    maxlistsize = (LONG)(MAXPOSITIVE);
```

```
    error = DoubleLList(lijst,&lsize,objectsize,nameoftype);
    *oldsize = lsize;
    maxlistsize = (LONG)(MAXLONG);

    return(error);
*/
}

/*
        #] DoubleList :
        #[ DoubleLList :
*/

int DoubleLList(void ***lijst, LONG *oldsize, int objectsize, char *nameoftype)
{
    void **newlist;
    LONG i, newsize, fullsize;
    void **to, **from;
    static LONG maxlistsize = (LONG)(MAXLONG);
    if ( *lijst == 0 ) {
        if ( *oldsize > 0 ) newsize = *oldsize;
        else newsize = 100;
    }
    else newsize = *oldsize * 2;
    if ( newsize > maxlistsize ) {
        if ( *oldsize == maxlistsize ) {
            MesPrint("No memory for extra space in %s",nameoftype);
            return(-1);
        }
        newsize = maxlistsize;
    }
    fullsize = ( newsize * objectsize + sizeof(void *)-1 ) & (-sizeof(void *));
    newlist = (void **)Malloc1(fullsize,nameoftype);
    if ( *lijst ) { /* Now some punning. DANGEROUS CODE in principle */
        to = newlist; from = *lijst; i = (*oldsize * objectsize)/sizeof(void *);
/*
#ifdef MALLOCDEBUG
if ( filelist ) MesPrint("    oldsize: %l, objectsize: %d, fullsize: %l"
        ,*oldsize,objectsize,fullsize);
#endif
*/
        while ( --i >= 0 ) *to++ = *from++;
    }
    if ( *lijst ) M_free(*lijst,"DoubleLList");
    *lijst = newlist;
    *oldsize = newsize;
    return(0);
}

/*
        #] DoubleLList :
        #[ DoubleBuffer :
*/

#define DODOUBLE(x) { x *s, *t, *u; if ( *start ) { \
    oldsize = *(x **)stop - *(x **)start; newsize = 2*oldsize; \
    t = u = (x *)Malloc1(newsize*sizeof(x),text); s = *(x **)start; \
    for ( i = 0; i < oldsize; i++ ) {*t++ = *s++;} M_free(*start,"double"); } \
    else { newsize = 100; u = (x *)Malloc1(newsize*sizeof(x),text); } \
    *start = (void *)u; *stop = (void *)(u+newsize); }

void DoubleBuffer(void **start, void **stop, int size, char *text)
{
    LONG oldsize, newsize, i;
    if ( size == sizeof(char) ) DODOUBLE(char)
    else if ( size == sizeof(short) ) DODOUBLE(short)
    else if ( size == sizeof(int) ) DODOUBLE(int)
    else if ( size == sizeof(LONG) ) DODOUBLE(LONG)
    else if ( size % sizeof(int) == 0 ) DODOUBLE(int)
    else {
        MesPrint("---Cannot handle doubling buffers of size %d",size);
        Terminate(-1);
    }
}

/*
        #] DoubleBuffer :
        #[ ExpandBuffer :
*/

#define DOEXPAND(x) { x *newbuffer, *t, *m;                              \
    t = newbuffer = (x *)Malloc1((newsize+2)*type,"ExpandBuffer");      \
    if ( *buffer ) { m = (x *)*buffer; i = *oldsize;                    \
        while ( --i >= 0 ) {*t++ = *m++;} M_free(*buffer,"ExpandBuffer"); \
    } *buffer = newbuffer; *oldsize = newsize; }

void ExpandBuffer(void **buffer, LONG *oldsize, int type)
```

```
{
    LONG newsize, i;
    if ( *oldsize <= 0 ) { newsize = 100; }
    else newsize = 2*(*oldsize);
    if ( type == sizeof(char) ) DOEXPAND(char)
    else if ( type == sizeof(short) ) DOEXPAND(short)
    else if ( type == sizeof(int) ) DOEXPAND(int)
    else if ( type == sizeof(LONG) ) DOEXPAND(LONG)
    else if ( type == sizeof(POSITION) ) DOEXPAND(POSITION)
    else {
        MesPrint("---Cannot handle expanding buffers with objects of size %d",type);
        Terminate(-1);
    }
}

/*
        #] ExpandBuffer :
        #[ iexp :

        Raises the long integer y to the power p.
        Returnvalue is long, regardless of overflow.
*/

LONG iexp(LONG x, int p)
{
    int sign;
    ULONG y;
    ULONG ux;
    if ( x == 0 ) return(0);
    if ( p == 0 ) return(1);
    sign = x < 0 ? -1 : 1;
    if ( sign < 0 && ( p & 1 ) == 0 ) sign = 1;
    ux = LongAbs(x);
    if ( ux == 1 ) return(sign);
    if ( p < 0 ) return(0);
    y = 1;
    while ( p ) {
        if ( ( p & 1 ) != 0 ) y *= ux;
        p >>= 1;
        ux = ux*ux;
    }
    if ( sign < 0 ) y = -y;
    return ULongToLong(y);
}

/*
        #] iexp :
        #[ ToGeneral :

        Convert a fast argument to a general argument
        Input in r, output in m.
        If par == 0 we need the argument header also.
*/

void ToGeneral(WORD *r, WORD *m, WORD par)
{
    WORD *mm = m, j, k;
    if ( par ) m++;
    else { m[1] = 0; m += ARGHEAD + 1; }
    j = -*r++;
    k = 3;
/*      JV: Bugfix 1-feb-2016. Old code assumed FUNHEAD to be 2 */
    if ( j >= FUNCTION ) { *m++ = j; *m++ = FUNHEAD; FILLFUN(m) }
    else {
        switch ( j ) {
            case SYMBOL: *m++ = j; *m++ = 4; *m++ = *r++; *m++ = 1; break;
            case SNUMBER:
                if ( *r > 0 ) { *m++ =  *r; *m++ = 1; *m++ =  3; }
                else if ( *r == 0 ) { m--; }
                else              { *m++ = -*r; *m++ = 1; *m++ = -3; }
                goto MakeSize;
            case MINVECTOR:
                k = -k;
                /* fall through */
            case INDEX:
            case VECTOR:
                *m++ = INDEX; *m++ = 3; *m++ = *r++;
                break;
        }
    }
    *m++ = 1; *m++ = 1; *m++ = k;
MakeSize:
    *mm = m-mm;
    if ( !par ) mm[ARGHEAD] = *mm-ARGHEAD;
}

/*
```

```
        #] ToGeneral :
        #[ ToFast :

        Checks whether an argument can be converted to fast notation
        If this can be done it does it.
        Important: m should be allowed to be equal to r!
        Return value is 1 if conversion took place.
        If there was conversion the answer is in m.
        If there was no conversion m hasn't been touched.
*/

int ToFast(WORD *r, WORD *m)
{
    WORD i;
    if ( *r == ARGHEAD ) { *m++ = -SNUMBER; *m++ = 0; return(1); }
    if ( *r != r[ARGHEAD]+ARGHEAD ) return(0);  /* > 1 term */
    r += ARGHEAD;
    if ( *r == 4 ) {
        if ( r[2] != 1 || r[1] <= 0 ) return(0);
        *m++ = -SNUMBER; *m = ( r[3] < 0 ) ? -r[1] : r[1]; return(1);
    }
    i = *r - 1;
    if ( r[i-1] != 1 || r[i-2] != 1 ) return(0);
    if ( r[i] != 3 ) {
        if ( r[i] == -3 && r[2] == *r-4 && r[2] == 3 && r[1] == INDEX
        && r[3] < MINSPEC ) {}
        else return(0);
    }
    else if ( r[2] != *r - 4 ) return(0);
    r++;
    if ( *r >= FUNCTION ) {
        if ( r[1] <= FUNHEAD ) { *m++ = -*r; return(1); }
    }
    else if ( *r == SYMBOL ) {
        if ( r[1] == 4 && r[3] == 1 )
            { *m++ = -SYMBOL; *m++ = r[2]; return(1); }
    }
    else if ( *r == INDEX ) {
        if ( r[1] == 3 ) {
            if ( r[2] >= MINSPEC ) {
                if ( r[2] >= 0 && r[2] < AM.OffsetIndex ) *m++ = -SNUMBER;
                else *m++ = -INDEX;
            }
            else {
                if ( r[5] == -3 ) *m++ = -MINVECTOR;
                else *m++ = -VECTOR;
            }
            *m++ = r[2];
            return(1);
        }
    }
    return(0);
}

/*
        #] ToFast :
        #[ ToPolyFunGeneral :

    Routine forces a polyratfun into general notation if needed.
    If no action was needed, the return value is zero.
    A positive return value indicates how many arguments were converted.
    The new term overwrite the old.
*/

WORD ToPolyFunGeneral(PHEAD WORD *term)
{
    WORD *t = term+1, *tt, *to, *to1, *termout, *tstop, *tnext;
    WORD numarg, i, change = 0;
    tstop = term + *term; tstop -= ABS(tstop[-1]);
    termout = to = AT.WorkPointer;
    to++;
    while ( t < tstop ) { /* go through the subterms */
        if ( *t == AR.PolyFun ) {
            tt = t+FUNHEAD; tnext = t + t[1];
            numarg = 0;
            while ( tt < tnext ) { numarg++; NEXTARG(tt); }
            if ( numarg == 2 ) { /* this needs attention */
                tt = t + FUNHEAD;
                to1 = to;
                i = FUNHEAD; NCOPY(to,t,i);
                while ( tt < tnext ) { /* Do the arguments */
                    if ( *tt > 0 ) {
                        i = *tt; NCOPY(to,tt,i);
                    }
                    else if ( *tt == -SYMBOL ) {
                        to1[1] += 6+ARGHEAD; to1[2] |= MUSTCLEANPRF; change++;
                        *to++ = 8+ARGHEAD; *to++ = 0; FILLARG(to);
```

```
                        *to++ = 8; *to++ = SYMBOL; *to++ = 4; *to++ = tt[1];
                        *to++ = 1; *to++ = 1; *to++ = 1; *to++ = 3;
                        tt += 2;
                    }
                    else if ( *tt == -SNUMBER ) {
                        if ( tt[1] > 0 ) {
                            to1[1] += 2+ARGHEAD; to1[2] |= MUSTCLEANPRF; change++;
                            *to++ = 4+ARGHEAD; *to++ = 0; FILLARG(to);
                            *to++ = 4; *to++ = tt[1]; *to++ = 1; *to++ = 3;
                            tt += 2;
                        }
                        else if ( tt[1] < 0 ) {
                            to1[1] += 2+ARGHEAD; to1[2] |= MUSTCLEANPRF; change++;
                            *to++ = 4+ARGHEAD; *to++ = 0; FILLARG(to);
                            *to++ = 4; *to++ = -tt[1]; *to++ = 1; *to++ = -3;
                            tt += 2;
                        }
                        else {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Internal error: Zero in PolyRatFun");
                            MUNLOCK(ErrorMessageLock);
                            Terminate(-1);
                        }
                    }
                }
                t = tnext;
                continue;
            }
        }
        i = t[1]; NCOPY(to,t,i)
    }
    if ( change ) {
        tt = term + *term;
        while ( t < tt ) *to++ = *t++;
        *termout = to - termout;
        t = term; i = *termout; tt = termout;
        NCOPY(t,tt,i)
        AT.WorkPointer = term + *term;
    }
    return(change);
}

/*
        #] ToPolyFunGeneral :
        #[ IsLikeVector :

        Routine determines whether a function argument is like a vector.
        Returnvalue: 1: is vector or index
                     0: is not vector or index
                    -1: may be an index
*/

int IsLikeVector(WORD *arg)
{
    WORD *sstop, *t, *tstop;
    if ( *arg < 0 ) {
        if ( *arg == -VECTOR || *arg == -INDEX ) return(1);
        if ( *arg == -SNUMBER && arg[1] >= 0 && arg[1] < AM.OffsetIndex )
            return(-1);
        return(0);
    }
    sstop = arg + *arg; arg += ARGHEAD;
    while ( arg < sstop ) {
        t = arg + *arg;
        tstop = t - ABS(t[-1]);
        arg++;
        while ( arg < tstop ) {
            if ( *arg == INDEX ) return(1);
            arg += arg[1];
        }
        arg = t;
    }
    return(0);
}

/*
        #] IsLikeVector :
        #[ AreArgsEqual :
*/

int AreArgsEqual(WORD *arg1, WORD *arg2)
{
    int i;
    if ( *arg2 != *arg1 ) return(0);
    if ( *arg1 > 0 ) {
        i = *arg1;
        while ( --i > 0 ) { if ( arg1[i] != arg2[i] ) return(0); }
```

```
            return(1);
        }
        else if ( *arg1 <= -FUNCTION ) return(1);
        else if ( arg1[1] == arg2[1] ) return(1);
        return(0);
}

/*
        #] AreArgsEqual :
        #[ CompareArgs :
*/

int CompareArgs(WORD *arg1, WORD *arg2)
{
    int i1,i2;
    if ( *arg1 > 0 ) {
        if ( *arg2 < 0 ) return(-1);
        i1 = *arg1-ARGHEAD; arg1 += ARGHEAD;
        i2 = *arg2-ARGHEAD; arg2 += ARGHEAD;
        while ( i1 > 0 && i2 > 0 ) {
            if ( *arg1 != *arg2 ) return((int)(*arg1)-(int)(*arg2));
            i1--; i2--; arg1++; arg2++;
        }
        return(i1-i2);
    }
    else if ( *arg2 > 0 ) return(1);
    else {
        if ( *arg1 != *arg2 ) {
            if ( *arg1 < *arg2 ) return(-1);
            else return(1);
        }
        if ( *arg1 <= -FUNCTION ) return(0);
        return((int)(arg1[1])-(int)(arg2[1]));
    }
}

/*
        #] CompareArgs :
        #[ CompArg :

    returns 1 if arg1 comes first, -1 if arg2 comes first, 0 if equal
*/

int CompArg(WORD *s1, WORD *s2)
{
    GETIDENTITY
    WORD *st1, *st2, x[7];
    int k;
    if ( *s1 < 0 ) {
        if ( *s2 < 0 ) {
            if ( *s1 <= -FUNCTION && *s2 <= -FUNCTION ) {
                if ( *s1 > *s2 ) return(-1);
                if ( *s1 < *s2 ) return(1);
                return(0);
            }
            if ( *s1 > *s2 ) return(1);
            if ( *s1 < *s2 ) return(-1);
            if ( *s1 <= -FUNCTION ) return(0);
            s1++; s2++;
            if ( *s1 > *s2 ) return(1);
            if ( *s1 < *s2 ) return(-1);
            return(0);
        }
        x[1] = AT.comsym[3];
        x[2] = AT.comnum[1];
        x[3] = AT.comnum[3];
        x[4] = AT.comind[3];
        x[5] = AT.comind[6];
        x[6] = AT.comfun[1];
        if ( *s1 == -SYMBOL ) {
            AT.comsym[3] = s1[1];
            st1 = AT.comsym+8; s1 = AT.comsym;
        }
        else if ( *s1 == -SNUMBER ) {
            if ( s1[1] < 0 ) {
                AT.comnum[1] = -s1[1]; AT.comnum[3] = -3;
            }
            else {
                AT.comnum[1] = s1[1]; AT.comnum[3] = 3;
            }
            st1 = AT.comnum+4;
            s1 = AT.comnum;
        }
        else if ( *s1 == -INDEX || *s1 == -VECTOR ) {
            AT.comind[3] = s1[1]; AT.comind[6] = 3;
            st1 = AT.comind+7; s1 = AT.comind;
        }
```

```
        else if ( *s1 == -MINVECTOR ) {
            AT.comind[3] = s1[1]; AT.comind[6] = -3;
            st1 = AT.comind+7; s1 = AT.comind;
        }
        else if ( *s1 <= -FUNCTION ) {
            AT.comfun[1] = -*s1;
            st1 = AT.comfun+FUNHEAD+4; s1 = AT.comfun;
        }
/*
            Symmetrize during compilation of id statement when properorder
            needs this one. Code added 10-nov-2001
*/
        else if ( *s1 == -ARGWILD ) {
             return(-1);
        }
        else { goto argerror; }
        st2 = s2 + *s2; s2 += ARGHEAD;
        goto docompare;
    }
    else if ( *s2 < 0 ) {
        x[1] = AT.comsym[3];
        x[2] = AT.comnum[1];
        x[3] = AT.comnum[3];
        x[4] = AT.comind[3];
        x[5] = AT.comind[6];
        x[6] = AT.comfun[1];
        if ( *s2 == -SYMBOL ) {
            AT.comsym[3] = s2[1];
            st2 = AT.comsym+8; s2 = AT.comsym;
        }
        else if ( *s2 == -SNUMBER ) {
            if ( s2[1] < 0 ) {
                AT.comnum[1] = -s2[1]; AT.comnum[3] = -3;
                st2 = AT.comnum+4;
            }
            else if ( s2[1] == 0 ) {
                st2 = AT.comnum+4; s2 = st2;
            }
            else {
                AT.comnum[1] = s2[1]; AT.comnum[3] = 3;
                st2 = AT.comnum+4;
            }
            s2 = AT.comnum;
        }
        else if ( *s2 == -INDEX || *s2 == -VECTOR ) {
            AT.comind[3] = s2[1]; AT.comind[6] = 3;
            st2 = AT.comind+7; s2 = AT.comind;
        }
        else if ( *s2 == -MINVECTOR ) {
            AT.comind[3] = s2[1]; AT.comind[6] = -3;
            st2 = AT.comind+7; s2 = AT.comind;
        }
        else if ( *s2 <= -FUNCTION ) {
            AT.comfun[1] = -*s2;
            st2 = AT.comfun+FUNHEAD+4; s2 = AT.comfun;
        }
/*
            Symmetrize during compilation of id statement when properorder
            needs this one. Code added 10-nov-2001
*/
        else if ( *s2 == -ARGWILD ) {
             return(1);
        }
        else { goto argerror; }
        st1 = s1 + *s1; s1 += ARGHEAD;
        goto docompare;
    }
    else {
        x[1] = AT.comsym[3];
        x[2] = AT.comnum[1];
        x[3] = AT.comnum[3];
        x[4] = AT.comind[3];
        x[5] = AT.comind[6];
        x[6] = AT.comfun[1];
        st1 = s1 + *s1; st2 = s2 + *s2;
        s1 += ARGHEAD; s2 += ARGHEAD;
docompare:
        while ( s1 < st1 && s2 < st2 ) {
            if ( ( k = CompareTerms(BHEAD s1,s2,(WORD)2) ) != 0 ) {
                AT.comsym[3] = x[1];
                AT.comnum[1] = x[2];
                AT.comnum[3] = x[3];
                AT.comind[3] = x[4];
                AT.comind[6] = x[5];
                AT.comfun[1] = x[6];
                return(-k);
            }
```

```
            s1 += *s1; s2 += *s2;
        }
        AT.comsym[3] = x[1];
        AT.comnum[1] = x[2];
        AT.comnum[3] = x[3];
        AT.comind[3] = x[4];
        AT.comind[6] = x[5];
        AT.comfun[1] = x[6];
        if ( s1 < st1 ) return(1);
        if ( s2 < st2 ) return(-1);
    }
    return(0);

argerror:
    MesPrint("Illegal type of short function argument in Normalize");
    Terminate(-1); return(0);
}

/*
        #] CompArg :
        #[ TimeWallClock :
*/

#ifdef HAVE_CLOCK_GETTIME
#include <time.h> /* for clock_gettime() */
#else
#ifdef HAVE_GETTIMEOFDAY
#include <sys/time.h> /* for gettimeofday() */
#else
#include <sys/timeb.h> /* for ftime() */
#endif
#endif

LONG TimeWallClock(WORD par)
{
    /*
     * NOTE: this function is not thread-safe. Operations on tp are not atomic.
     */

#ifdef HAVE_CLOCK_GETTIME
    struct timespec ts;
    clock_gettime(CLOCK_MONOTONIC, &ts);

    if ( par ) {
        return(((LONG)(ts.tv_sec)-AM.OldSecTime)*100 +
            ((LONG)(ts.tv_nsec / 1000000)-AM.OldMilliTime)/10);
    }
    else {
        AM.OldSecTime   = (LONG)(ts.tv_sec);
        AM.OldMilliTime = (LONG)(ts.tv_nsec / 1000000);
        return(0L);
    }
#else
#ifdef HAVE_GETTIMEOFDAY
    struct timeval t;
    LONG sec, msec;
    gettimeofday(&t, NULL);
    sec = (LONG)t.tv_sec;
    msec = (LONG)(t.tv_usec/1000);
    if ( par ) {
        return (sec-AM.OldSecTime)*100 + (msec-AM.OldMilliTime)/10;
    }
    else {
        AM.OldSecTime   = sec;
        AM.OldMilliTime = msec;
        return(0L);
    }
#else
    struct timeb tp;
    ftime(&tp);

    if ( par ) {
        return(((LONG)(tp.time)-AM.OldSecTime)*100 +
            ((LONG)(tp.millitm)-AM.OldMilliTime)/10);
    }
    else {
        AM.OldSecTime   = (LONG)(tp.time);
        AM.OldMilliTime = (LONG)(tp.millitm);
        return(0L);
    }
#endif
#endif
}

/*
        #] TimeWallClock :
        #[ TimeChildren :
```

```
*/

LONG TimeChildren(WORD par)
{
    if ( par ) return(Timer(1)-AM.OldChildTime);
    AM.OldChildTime = Timer(1);
    return(0L);
}

/*
        #] TimeChildren :
        #[ TimeCPU :
*/

LONG TimeCPU(WORD par)
{
    GETIDENTITY
    if ( par ) return(Timer(0)-AR.OldTime);
    AR.OldTime = Timer(0);
    return(0L);
}

/*
        #] TimeCPU :
        #[ Timer :
*/
#if defined(WINDOWS)

LONG Timer(int par)
{
#ifndef WITHPTHREADS
    static int initialized = 0;
    static HANDLE hProcess;
    FILETIME ftCreate, ftExit, ftKernel, ftUser;
    DUMMYUSE(par);

    if ( !initialized ) {
        hProcess = OpenProcess(PROCESS_QUERY_INFORMATION, FALSE, GetCurrentProcessId());
    }
    if ( GetProcessTimes(hProcess, &ftCreate, &ftExit, &ftKernel, &ftUser) ) {
        PFILETIME pftKernel = &ftKernel;  /* to avoid strict-aliasing rule warnings */
        PFILETIME pftUser   = &ftUser;
        __int64 t = *(__int64 *)pftKernel + *(__int64 *)pftUser;  /* in 100 nsec. */
        return (LONG)(t / 10000);  /* in msec. */
    }
    return 0;
#else
    LONG lResult = 0;
    HANDLE hThread;
    FILETIME ftCreate, ftExit, ftKernel, ftUser;
    DUMMYUSE(par);

    hThread = OpenThread(THREAD_QUERY_INFORMATION, FALSE, GetCurrentThreadId());
    if ( hThread ) {
        if ( GetThreadTimes(hThread, &ftCreate, &ftExit, &ftKernel, &ftUser) ) {
            PFILETIME pftKernel = &ftKernel;  /* to avoid strict-aliasing rule warnings */
            PFILETIME pftUser   = &ftUser;
            __int64 t = *(__int64 *)pftKernel + *(__int64 *)pftUser;  /* in 100 nsec. */
            lResult = (LONG)(t / 10000);  /* in msec. */
        }
        CloseHandle(hThread);
    }
    return lResult;
#endif
}

#elif defined(UNIX)
#include <sys/time.h>
#include <sys/resource.h>
#ifdef WITHPOSIXCLOCK
#include <time.h>
/*
    And include -lrt in the link statement (on blade02)
*/
#endif

LONG Timer(int par)
{
#ifdef WITHPOSIXCLOCK
/*
    Only to be used in combination with WITHPTHREADS
    This clock seems to be supported by the standard.
    The getrusage clock returns according to the standard only the combined
    time of the whole process. But in older versions of Linux LinuxThreads
    is used which gives a separate id to each thread and individual timings.
    In NPTL we get, according to the standard, one combined timing.
    To get individual timings we need to use
```

```
        clock_gettime(CLOCK_THREAD_CPUTIME_ID, &timing)
    with timing of the time
    struct timespec {
        time_t tv_sec;            Seconds.
        long   tv_nsec;           Nanoseconds.
    };

*/
    struct timespec t;
    if ( par == 0 ) {
        if ( clock_gettime(CLOCK_THREAD_CPUTIME_ID, &t) ) {
            MesPrint("Error in getting timing information");
        }
        return (LONG)t.tv_sec * 1000 + (LONG)t.tv_nsec / 1000000;
    }
    return(0);
#else
    struct rusage rusage;
    if ( par == 1 ) {
        getrusage(RUSAGE_CHILDREN,&rusage);
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
    else {
        getrusage(RUSAGE_SELF,&rusage);
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
#endif
}

#elif defined(SUN)
#define _TIME_T_
#include <sys/time.h>
#include <sys/resource.h>

LONG Timer(int par)
{
    struct rusage rusage;
    if ( par == 1 ) {
        getrusage(RUSAGE_CHILDREN,&rusage);
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
    else {
        getrusage(RUSAGE_SELF,&rusage);
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
}

#elif defined(RS6K)
#include <sys/time.h>
#include <sys/resource.h>

LONG Timer(int par)
{
    struct rusage rusage;
    if ( par == 1 ) {
        getrusage(RUSAGE_CHILDREN,&rusage);
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
    else {
        getrusage(RUSAGE_SELF,&rusage);
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
}

#elif defined(ANSI)
LONG Timer(int par)
{
#ifdef ALPHA
/*  clock_t t,tikken = clock();                                    */
/*  MesPrint("ALPHA-clock = %l",(LONG)tikken);                     */
/*  t = tikken % CLOCKS_PER_SEC;                                   */
/*  tikken /= CLOCKS_PER_SEC;                                      */
/*  tikken *= 1000;                                                */
/*  tikken += (t*1000)/CLOCKS_PER_SEC;                             */
/*  return((LONG)tikken);                                          */
/* #define _TIME_T_                                                */
#include <sys/time.h>
#include <sys/resource.h>
    struct rusage rusage;
    if ( par == 1 ) {
        getrusage(RUSAGE_CHILDREN,&rusage);
```

```
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
               +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
    else {
        getrusage(RUSAGE_SELF,&rusage);
        return(((LONG)(rusage.ru_utime.tv_sec)+(LONG)(rusage.ru_stime.tv_sec))*1000
               +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
#else
#ifdef DEC_STATION
    clock_t tikken = clock();
    return((LONG)tikken/1000);
#else
    clock_t t, tikken = clock();
    t = tikken % CLK_TCK;
    tikken /= CLK_TCK;
    tikken *= 1000;
    tikken += (t*1000)/CLK_TCK;
    return(tikken);
#endif
#endif
}
#elif defined(VMS)

#include <time.h>
void times(tbuffer_t *buffer);

LONG
Timer(int par)
{
    tbuffer_t buffer;
    if ( par == 1 ) { return(0); }
    else {
        times(&buffer);
        return(buffer.proc_user_time * 10);
    }
}

#elif defined(mBSD)

#ifdef MICROTIME
/*
    There is only a CP time clock in microseconds here
    This can cause problems with AO.wrap around
*/
#else
#ifdef mBSD2
#include <sys/types.h>
#include <sys/times.h>
#include <time.h>
LONG pretime = 0;
#else
#define _TIME_T_
#include <sys/time.h>
#include <sys/resource.h>
#endif
#endif

LONG Timer(int par)
{
#ifdef MICROTIME
    LONG t;
    if ( par == 1 ) { return(0); }
    t = clock();
    if ( ( AO.wrapnum & 1 ) != 0 ) t ^= 0x80000000;
    if ( t < 0 ) {
        t ^= 0x80000000;
        warpnum++;
        AO.wrap += 2147584;
    }
    return(AO.wrap+(t/1000));
#else
#ifdef mBSD2
    struct tms buffer;
    LONG ret;
    ULONG a1, a2, a3, a4;
    if ( par == 1 ) { return(0); }
    times(&buffer);
    a1 = (ULONG)buffer.tms_utime;
    a2 = a1 » 16;
    a3 = a1 & 0xFFFFL;
    a3 *= 1000;
    a2 = 1000*a2 + (a3 » 16);
    a3 &= 0xFFFFL;
    a4 = a2/CLK_TCK;
    a2 %= CLK_TCK;
    a3 += a2 « 16;
```

```
    ret = (LONG)((a4 « 16) + a3 / CLK_TCK);
/*  ret = ((LONG)buffer.tms_utime * 1000)/CLK_TCK; */
    return(ret);
#else
#ifdef REALTIME
    struct timeval tp;
    struct timezone tzp;
    if ( par == 1 ) { return(0); }
    gettimeofday(&tp,&tzp); */
    return(tp.tv_sec*1000+tp.tv_usec/1000);
#else
    struct rusage rusage;
    if ( par == 1 ) {
        getrusage(RUSAGE_CHILDREN,&rusage);
        return((rusage.ru_utime.tv_sec+rusage.ru_stime.tv_sec)*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
    else {
        getrusage(RUSAGE_SELF,&rusage);
        return((rusage.ru_utime.tv_sec+rusage.ru_stime.tv_sec)*1000
              +(rusage.ru_utime.tv_usec/1000+rusage.ru_stime.tv_usec/1000));
    }
#endif
#endif
#endif
}

#endif

/*
        #] Timer :
        #[ Crash :

        Routine for debugging purposes
*/

int Crash(void)
{
    int retval;
#ifdef DEBUGGING
    int *zero = 0;
    retval = *zero;
#else
    retval = 0;
#endif
    return(retval);
}

/*
        #] Crash :
        #[ TestTerm :
*/

int TestTerm(WORD *term)
{
    int errorcode = 0, coeffsize;
    WORD *t, *tt, *tstop, *endterm, *targ, *targstop, *funstop, *argterm;
    endterm = term + *term;
    coeffsize = ABS(endterm[-1]);
    if ( coeffsize >= *term ) {
        MLOCK(ErrorMessageLock);
        MesPrint("TestTerm: Internal inconsistency in term. Coefficient too big.");
        MUNLOCK(ErrorMessageLock);
        errorcode = 1;
        goto finish;
    }
    if ( ( coeffsize < 3 ) || ( ( coeffsize & 1 ) != 1 ) ) {
        MLOCK(ErrorMessageLock);
        MesPrint("TestTerm: Internal inconsistency in term. Wrong size coefficient.");
        MUNLOCK(ErrorMessageLock);
        errorcode = 2;
        goto finish;
    }
    t = term+1;
    tstop = endterm - coeffsize;
    while ( t < tstop ) {
        switch ( *t ) {
            case SYMBOL:
            case DOTPRODUCT:
            case INDEX:
            case VECTOR:
            case DELTA:
            case HAAKJE:
                break;
            case SNUMBER:
            case LNUMBER:
                MLOCK(ErrorMessageLock);
```

```
                MesPrint("TestTerm: Internal inconsistency in term. L or S number");
                MUNLOCK(ErrorMessageLock);
                errorcode = 3;
                goto finish;
                break;
            case EXPRESSION:
            case SUBEXPRESSION:
            case DOLLAREXPRESSION:
/*
                MLOCK(ErrorMessageLock);
                MesPrint("TestTerm: Internal inconsistency in term. Expression survives.");
                MUNLOCK(ErrorMessageLock);
                errorcode = 4;
                goto finish;
*/
                break;
            case SETSET:
            case MINVECTOR:
            case SETEXP:
            case ARGFIELD:
                MLOCK(ErrorMessageLock);
                MesPrint("TestTerm: Internal inconsistency in term. Illegal subterm.");
                MUNLOCK(ErrorMessageLock);
                errorcode = 5;
                goto finish;
                break;
            case ARGWILD:
                break;
            default:
                if ( *t <= 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("TestTerm: Internal inconsistency in term. Illegal subterm number.");
                    MUNLOCK(ErrorMessageLock);
                    errorcode = 6;
                    goto finish;
                }
/*
                This is a regular function.
*/
                if ( *t-FUNCTION >= NumFunctions ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("TestTerm: Internal inconsistency in term. Illegal function number");
                    MUNLOCK(ErrorMessageLock);
                    errorcode = 7;
                    goto finish;
                }
                funstop = t + t[1];
                if ( funstop > tstop ) goto subtermsize;
                if ( t[2] != 0 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("TestTerm: Internal inconsistency in term. Dirty flag nonzero.");
                    MUNLOCK(ErrorMessageLock);
                    errorcode = 8;
                    goto finish;
                }
                targ = t + FUNHEAD;
                if ( targ > funstop ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("TestTerm: Internal inconsistency in term. Illegal function size.");
                    MUNLOCK(ErrorMessageLock);
                    errorcode = 9;
                    goto finish;
                }
                if ( functions[*t-FUNCTION].spec >= TENSORFUNCTION ) {
                }
                else {
                  while ( targ < funstop ) {
                    if ( *targ < 0 ) {
                        if ( *targ <= -(FUNCTION+NumFunctions) ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("TestTerm: Internal inconsistency in term. Illegal function number in argument.");
                            MUNLOCK(ErrorMessageLock);
                            errorcode = 10;
                            goto finish;
                        }
                        if ( *targ <= -FUNCTION ) { targ++; }
                        else {
                            if ( ( *targ != -SYMBOL ) && ( *targ != -VECTOR )
                            && ( *targ != -MINVECTOR )
                            && ( *targ != -SNUMBER )
                            && ( *targ != -ARGWILD )
                            && ( *targ != -INDEX ) ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("TestTerm: Internal inconsistency in term. Illegal object in argument.");
                                MUNLOCK(ErrorMessageLock);
```

```
                                errorcode = 11;
                                goto finish;
                            }
                            targ += 2;
                        }
                    }
                    else if ( ( *targ < ARGHEAD ) || ( targ+*targ > funstop ) ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("TestTerm: Internal inconsistency in term. Illegal size of argument.");
                        MUNLOCK(ErrorMessageLock);
                        errorcode = 12;
                        goto finish;
                    }
                    else if ( targ[1] != 0 ) {
                        MLOCK(ErrorMessageLock);
                        MesPrint("TestTerm: Internal inconsistency in term. Dirty flag in argument.");
                        MUNLOCK(ErrorMessageLock);
                        errorcode = 13;
                        goto finish;
                    }
                    else {
                        targstop = targ + *targ;
                        argterm = targ + ARGHEAD;
                        while ( argterm < targstop ) {
                            if ( ( *argterm < 4 ) || ( argterm + *argterm > targstop ) ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("TestTerm: Internal inconsistency in term. Illegal termsize in argument.");
                                MUNLOCK(ErrorMessageLock);
                                errorcode = 14;
                                goto finish;
                            }
                            if ( TestTerm(argterm) != 0 ) {
                                MLOCK(ErrorMessageLock);
                                MesPrint("TestTerm: Internal inconsistency in term. Called from TestTerm.");
                                MUNLOCK(ErrorMessageLock);
                                errorcode = 15;
                                goto finish;
                            }
                            argterm += *argterm;
                        }
                        targ = targstop;
                    }
                  }
                }
                break;
        }
        tt = t + t[1];
        if ( tt > tstop ) {
subtermsize:
            MLOCK(ErrorMessageLock);
            MesPrint("TestTerm: Internal inconsistency in term. Illegal subterm size.");
            MUNLOCK(ErrorMessageLock);
            errorcode = 100;
            goto finish;
        }
        t = tt;
    }
    return(errorcode);
finish:
    return(errorcode);
}

/*
        #] TestTerm :
        #[ DistrN :
*/

int DistrN(int n, int *cpl, int ncpl, int *scratch)
{
/*
    Divides n objects over ncpl bins (cpl), each time returning one
    of those distributions until there are no more after which the
    routine returns the value zero (otherwise one).
    The array scratch (size n) is kept for the intermediate information.
    The whole starts with scratch[0] == -2;
*/
    int i, j;
    if ( ncpl == 0 ) {
        if ( scratch[0] == -2 ) { scratch[0] = 0; return(1); }
        else return(0);
    }
    if ( scratch[0] == ncpl-1 ) {
        return(0);
    }
```

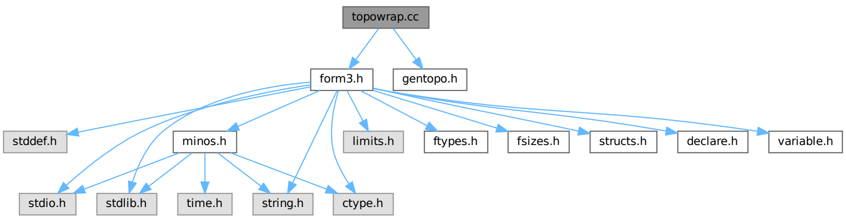
```
    else if ( scratch[0] == -2 ) {
        for ( i = 0; i < n; i++ ) scratch[i] = 0;
    }
    else {
        j = n-1;
        while ( j >= 0 ) {
            scratch[j]++;
            if ( scratch[j] < ncpl ) break;
            j--;
        }
        j++;
        while ( j < n ) { scratch[j] = scratch[j-1]; j++; }
    }
    for ( i = 0; i < ncpl; i++ ) cpl[i] = 0;
    for ( i = 0; i < n; i++ ) { cpl[scratch[i]]++; }
    return(1);
}

/*
        #] DistrN :
    #] Mixed :
*/
```

4.148 topowrap.cc File Reference

```
#include "form3.h"
#include "gentopo.h"
```

Include dependency graph for topowrap.cc:



Data Structures

- struct ToPoTyPe

Macros

- #define MAXPOINTS 120

Typedefs

- typedef struct ToPoTyPe **TOPOTYPE**

Functions

- int GenerateVertices (TOPOTYPE ∗TopoInf, int pointsremaining, int level)
- WORD GenerateTopologies (PHEAD WORD nloops, WORD nlegs, WORD setvert, WORD setmax)
- void toForm (T_EGraph ∗egraph)

4.148.1 Detailed Description

routines for conversion of topology and diagram output to FORM notation

Definition in file topowrap.cc.

4.148.2 Macro Definition Documentation

4.148.2.1 MAXPOINTS

```
#define MAXPOINTS 120
```

Definition at line 44 of file topowrap.cc.

4.148.3 Function Documentation

4.148.3.1 GenerateVertices()

```
int GenerateVertices (
            TOPOTYPE * TopoInf,
            int pointsremaining,
            int level )
```

Definition at line 72 of file topowrap.cc.

4.148.3.2 GenerateTopologies()

```
WORD GenerateTopologies (
            PHEAD WORD nloops,
            WORD nlegs,
            WORD setvert,
            WORD setmax )
```

Definition at line 123 of file topowrap.cc.

4.148.3.3 toForm()

```
void toForm (
            T_EGraph * egraph )
```

Definition at line 178 of file topowrap.cc.

4.149 topowrap.cc

Go to the documentation of this file.

```
/* #[ License : */
/*
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 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
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 */
/*
    #] License :
    #[ includes :
*/

extern "C" {
#include "form3.h"
}

#include "gentopo.h"

//using namespace std;

#define MAXPOINTS 120

typedef struct ToPoTyPe {
    int cldeg[MAXPOINTS], clnum[MAXPOINTS], clext[MAXPOINTS];
    int ncl, nloops, nlegs, npadding;
    WORD *vert;
    WORD *vertmax;
    WORD nvert;
    WORD sopi;
} TOPOTYPE;

/*
    #] includes :
    #[ GenerateVertices :

    Routine is to be used recursively to work its way through a list
    of possible vertices. The array of vertices is in TopoInf->vert
    with TopoInf->nvert the number of possible vertices.
    Currently we allow in TopoInf->vert only vertices with 3 or more edges.

    We work with a point system. Each n-point vertex contributes n-2 points.
    When all points are assigned, we can call mgraph->generate().

    The number of vertices of a given number of edges is stored in
    TopoInf->clnum[..] but the loop that determines how many there are
    may be limited by the corresponding element in TopoInf->vertmax[level]
*/

int GenerateVertices(TOPOTYPE *TopoInf, int pointsremaining, int level)
{
    int i, j;

    for ( i = pointsremaining, j = 0; i >= 0; i -= TopoInf->vert[level]-2, j++ ) {
        if ( TopoInf->vertmax && TopoInf->vertmax[level] >= 0
                && j > TopoInf->vertmax[level] ) break;
        if ( i == 0 ) { // We got one!
            T_MGraph *mgraph;
            TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
            TopoInf->clnum[TopoInf->ncl] = j;
            TopoInf->clext[TopoInf->ncl] = 0;
            TopoInf->ncl++;

            mgraph = new T_MGraph(0, TopoInf->ncl, TopoInf->cldeg,
```

```
                         TopoInf->clnum, TopoInf->clext, TopoInf->sopi);

            mgraph->generate();

            delete mgraph;

            TopoInf->ncl--;

            break;
        }
        if ( level < TopoInf->nvert-1 ) {
            if ( j > 0 ) {
                TopoInf->cldeg[TopoInf->ncl] = TopoInf->vert[level];
                TopoInf->clnum[TopoInf->ncl] = j;
                TopoInf->clext[TopoInf->ncl] = 0;
                TopoInf->ncl++;
            }
            if ( GenerateVertices(TopoInf,i,level+1) < 0 ) return(-1);
            if ( j > 0 ) { TopoInf->ncl--; }
        }
    }
    return(0);
}

/*
    #] GenerateVertices :
    #[ GenerateTopologies :

    Note that setmax, option1 and option2 are optional.
    Default values are -1,0,0

    vert is a pointer to a set of numbers indicating the types of vertices
         that are allowed.
    nvert is the number of elements in vert.
*/

WORD GenerateTopologies(PHEAD WORD nloops, WORD nlegs, WORD setvert, WORD setmax)
{
    TOPOTYPE TopoInf;
    int i, points, identical = 0;
    DUMMYUSE(AT.nfac);

    if ( nlegs == -2 ) {
        nlegs = 2;
        identical = 1;
    }
    TopoInf.vert = &(SetElements[Sets[setvert].first]);
    TopoInf.nvert = Sets[setvert].last-Sets[setvert].first;

    if ( setmax >= 0 ) TopoInf.vertmax = &(SetElements[Sets[setmax].first]);
    else               TopoInf.vertmax = 0;

//  point counting: an n-point vertex counts for n-2 points.

    points = 2*nloops-2+nlegs;
    if ( points >= MAXPOINTS ) {
        MLOCK(ErrorMessageLock);
        MesPrint("GenerateTopologies: %d loops and %d legs considered excessive",nloops,nlegs);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

//  First the external nodes.

    for ( i = 0; i < nlegs; i++ ) {
        TopoInf.cldeg[i] = 1; TopoInf.clnum[i] = 1; TopoInf.clext[i] = 1;
    }
    if ( identical == 1 ) { /* Only propagator topologies..... */
        nlegs = 1;
        TopoInf.clnum[0] = 2;
    }
    TopoInf.ncl = nlegs;
    TopoInf.sopi = 1;  // For now
    if ( GenerateVertices(&TopoInf,points,0) != 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Called from GenerateTopologies with %d loops and %d legs considered excessive",nloops,nlegs);
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    return(0);
}

/*
    #] GenerateTopologies :
    #[ toForm :
*/
```

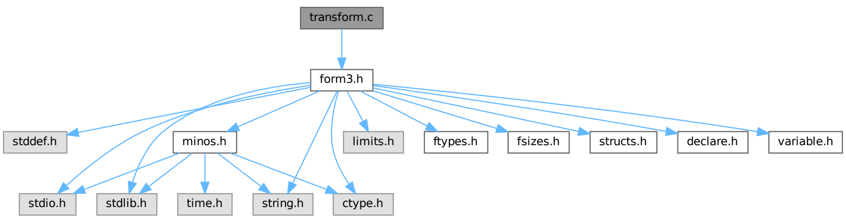
```

//=============================================================
// Output for FROM called by genTopo each time a new graph is
// generated
//
void toForm(T_EGraph *egraph)
{
    GETIDENTITY;
//
//  skipext : boolean; whether to skip external node/line in the output.
//
//  const int skipext = 1;

    int n, lg, ed, i, fromset;

    WORD *termout = AT.WorkPointer;
    WORD *t, *tt, *ttstop, *ttend, *ttail;

    t = termout+1;

//  First pick up part of the original term

    tt = AT.TopologiesTerm + 1;
    i = AT.TopologiesStart - tt;
    NCOPY(t,tt,i)
    ttail = tt+tt[1];

//  Now the vertices/nodes
//  Options are in AT.TopologiesOptions[]

    for ( n = 0; n < egraph->nNodes; n++ ) {
        if ( ( AT.TopologiesOptions[1] &1 ) == 1 && egraph->nodes[n].ext ) continue;
        tt = t;
        *t++ = NODEFUNCTION; t++; FILLFUN(t);
        *t++ = -SNUMBER; *t++ = n;
        for ( lg = 0; lg < egraph->nodes[n].deg; lg++ ) {
            ed = egraph->nodes[n].edges[lg];
            if ( ed >= 0 ) { *t++ = -VECTOR; }
            else { ed = -ed; *t++ = -MINVECTOR; }

//          Now we need to pick up the proper set element.

            fromset =  egraph->edges[ed].ext ? AT.setexterntopo : AT.setinterntopo;
            *t++ = SetElements[Sets[fromset].first+egraph->edges[ed].momn-1];
        }
        tt[1] = t-tt;
    }

    if ( ( AT.TopologiesOptions[0] & 1 ) == 1 ) {
//      Note that the edges count from 1.
        for ( n = 1; n <= egraph->nEdges; n++ ) {
            if ( ( AT.TopologiesOptions[1] & 1 ) == 1 && egraph->edges[n].ext ) continue;
            tt = t;
            *t++ = EDGE; t++; FILLFUN(t);
            *t++ = -SNUMBER; *t++ = egraph->edges[n].nodes[0];
            *t++ = -SNUMBER; *t++ = egraph->edges[n].nodes[1];
            *t++ = -VECTOR;
            fromset =  egraph->edges[n].ext ? AT.setexterntopo : AT.setinterntopo;
            *t++ = SetElements[Sets[fromset].first+egraph->edges[n].momn-1];
            tt[1] = t-tt;
        }
    }

//  Now the tail end

    ttend = AT.TopologiesTerm; ttend = ttend+ttend[0];
    ttstop = ttend - ABS(ttend[-1]);
    i = ttstop - ttail;
    NCOPY(t,ttail,i)

//  Finally the coefficient
//  The topological coefficient should be in egraph->wsum, egraph->nwsum
//  as a FORM Long number

#ifdef WITHFACTOR
    if ( egraph->nwsum == 1 && egraph->wsum[0] == 1 ) {
        while (ttstop < ttend ) *t++ = *ttstop++;
    }
    else {
        WORD newsize;
        newsize = ttend[-1];
        newsize = REDLENG(newsize);
        if ( Divvy(BHEAD (UWORD *)ttstop,&newsize,egraph->wsum,egraph->nwsum) )
            goto OnError;
        newsize = INCLENG(newsize);
        i = ABS(newsize)-1;
        NCOPY(t,ttstop,i)
```

```
        *t++ = newsize;
    }
#else
    while (ttstop < ttend ) *t++ = *ttstop++;
#endif
    *termout = t - termout;

    AT.WorkPointer = t;

    if ( Generator(BHEAD termout,AT.TopologiesLevel) < 0 ) {
// OnError:
        MLOCK(ErrorMessageLock);
        MesPrint("Called from the topologies routine toForm");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }

    AT.WorkPointer = termout;
}

/*
    #] toForm :
*/

```

4.150 transform.c File Reference

`#include "form3.h"`

Include dependency graph for transform.c:



Functions

- int CoTransform (UBYTE *in)
- WORD RunTransform (PHEAD WORD *term, WORD *params)
- WORD RunEncode (PHEAD WORD *fun, WORD *args, WORD *info)
- WORD RunDecode (PHEAD WORD *fun, WORD *args, WORD *info)
- WORD RunReplace (PHEAD WORD *fun, WORD *args, WORD *info)
- WORD RunImplode (WORD *fun, WORD *args)
- WORD RunExplode (PHEAD WORD *fun, WORD *args)
- WORD RunPermute (PHEAD WORD *fun, WORD *args, WORD *info)
- WORD RunReverse (PHEAD WORD *fun, WORD *args)
- WORD RunDedup (PHEAD WORD *fun, WORD *args)
- WORD RunCycle (PHEAD WORD *fun, WORD *args, WORD *info)
- WORD RunAddArg (PHEAD WORD *fun, WORD *args)
- WORD RunMulArg (PHEAD WORD *fun, WORD *args)
- WORD RunIsLyndon (PHEAD WORD *fun, WORD *args, int par)
- WORD RunToLyndon (PHEAD WORD *fun, WORD *args, int par)
- WORD RunDropArg (PHEAD WORD *fun, WORD *args)
- WORD RunSelectArg (PHEAD WORD *fun, WORD *args)

- WORD RunZtoHArg (PHEAD WORD ∗fun, WORD ∗args)
- WORD RunHtoZArg (PHEAD WORD ∗fun, WORD ∗args)
- int TestArgNum (int n, int totarg, WORD ∗args)
- WORD PutArgInScratch (WORD ∗arg, UWORD ∗scrat)
- UBYTE ∗ ReadRange (UBYTE ∗s, WORD ∗out, int par)
- int FindRange (PHEAD WORD ∗args, WORD ∗arg1, WORD ∗arg2, WORD totarg)

4.150.1 Detailed Description

Routines that deal with the transform statement and its varieties.

Definition in file transform.c.

4.150.2 Function Documentation

4.150.2.1 CoTransform()

```
int CoTransform (
            UBYTE * in )
```

Definition at line 87 of file transform.c.

4.150.2.2 RunTransform()

```
WORD RunTransform (
            PHEAD WORD * term,
            WORD * params )
```

Definition at line 732 of file transform.c.

4.150.2.3 RunEncode()

```
WORD RunEncode (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )
```

Definition at line 982 of file transform.c.

4.150.2.4 RunDecode()

```
WORD RunDecode (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )
```

Definition at line 1169 of file transform.c.

4.150.2.5 RunReplace()

```
WORD RunReplace (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )
```

Definition at line 1339 of file transform.c.

4.150.2.6 RunImplode()

```
WORD RunImplode (
            WORD * fun,
            WORD * args )
```

Definition at line 1817 of file transform.c.

4.150.2.7 RunExplode()

```
WORD RunExplode (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 2018 of file transform.c.

4.150.2.8 RunPermute()

```
WORD RunPermute (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )
```

Definition at line 2132 of file transform.c.

4.150.2.9 RunReverse()

```
WORD RunReverse (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 2359 of file transform.c.

4.150.2.10 RunDedup()

```
WORD RunDedup (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 2446 of file transform.c.

4.150.2.11 RunCycle()

```
WORD RunCycle (
            PHEAD WORD * fun,
            WORD * args,
            WORD * info )
```

Definition at line 2535 of file transform.c.

4.150.2.12 RunAddArg()

```
WORD RunAddArg (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 2687 of file transform.c.

4.150.2.13 RunMulArg()

```
WORD RunMulArg (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 2776 of file transform.c.

4.150.2.14 RunIsLyndon()

```
WORD RunIsLyndon (
            PHEAD WORD * fun,
            WORD * args,
            int par )
```

Definition at line 2917 of file transform.c.

4.150.2.15 RunToLyndon()

```
WORD RunToLyndon (
            PHEAD WORD * fun,
            WORD * args,
            int par )
```

Definition at line 2995 of file transform.c.

4.150.2.16 RunDropArg()

```
WORD RunDropArg (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 3107 of file transform.c.

4.150.2.17 RunSelectArg()

```
WORD RunSelectArg (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 3133 of file transform.c.

4.150.2.18 RunZtoHArg()

```
WORD RunZtoHArg (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 3161 of file transform.c.

4.150.2.19 RunHtoZArg()

```
WORD RunHtoZArg (
            PHEAD WORD * fun,
            WORD * args )
```

Definition at line 3216 of file transform.c.

4.150.2.20 TestArgNum()

```
int TestArgNum (
            int n,
            int totarg,
            WORD * args )
```

Definition at line 3281 of file transform.c.

4.150.2.21 PutArgInScratch()

```
WORD PutArgInScratch (
            WORD * arg,
            UWORD * scrat )
```

Definition at line 3350 of file transform.c.

4.150.2.22 ReadRange()

```
UBYTE * ReadRange (
            UBYTE * s,
            WORD * out,
            int par )
```

Definition at line 3386 of file transform.c.

4.150.2.23 FindRange()

```
int FindRange (
            PHEAD WORD * args,
            WORD * arg1,
            WORD * arg2,
            WORD totarg )
```

Definition at line 3546 of file transform.c.

4.151 transform.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 */
/* #] License : */
/*
    #[ Includes : transform.c
*/

#include "form3.h"

/*
    #] Includes :
    #[ Transform :
        #[ Intro :

        Here are the routines for the transform statement. This is a
        group of transformations on function arguments or groups of
        function arguments. The purpose of this command is that it
        avoids repetitive pattern matching.
        Syntax:
            Transform,SetOfFunctions,OneOrMoreTransformations;
        Each transformation is given by
            Replace(argfirst,arglast)=(„,)
            Encode(argfirst,arglast):base=#
            Decode(argfirst,arglast):base=#
            Implode(argfirst,arglast)
            Explode(argfirst,arglast)
            Permute(cycle)(cycle)(cycle)...(cycle)
            Reverse(argfirst,arglast)
            Dedup(argfirst,arglast)
            Cycle(argfirst,arglast)=+/-num
            IsLyndon(argfirst,arglast)=(yes,no)
            ToLyndon(argfirst,arglast)=(yes,no)
        In replace the extra information is
            a replace_() without the name of the replace_ function.
            This can be as in (0,1,1,0) or (xarg_,1-xarg_) to indicate
            a symbolic argument or (x,y,y,x) to exchange x and y, etc.
        In Encode and Decode argfirst is the most significant 'word' and
        arglast is the least significant 'word'.
        Note that we need to introduce the generic symbolic arguments xarg_,
```

```
        parg_, iarg_ and farg_.
        Examples:
            Transform,{H,E}
                    ,Replace(1:`WEIGHT')=(0,1,1,0)
                    ,Encode(1:`WEIGHT')=base(2);
            Transform,{H,E}
                    ,Decode(1:`WEIGHT')=base(3)
                    ,Replace(1:`WEIGHT')=(2,-1,1,0,0,1);
        Others that can be added:
            symmetrize?

        6-may-2016: Changed MAXPOSITIVE2 into MAXPOSITIVE4. This makes room
                    for the use of dollar variables as arguments.

        #] Intro :
        #[ CoTransform :
*/

static WORD tranarray[10] = { SUBEXPRESSION, SUBEXPSIZE, 0, 1, 0, 0, 0, 0, 0, 0 };

int CoTransform(UBYTE *in)
{
    GETIDENTITY
    UBYTE *s = in, c, *ss, *Tempbuf;
    WORD number, type, i, *work = AT.WorkPointer+2, *wp, range[2], one = 1;
    WORD numdol, *wstart;
    int error = 0, irhs;
    LONG x;
    while ( *in == ',' ) in++;
    wp = work + 1;
/*
    #[ Sets :

    First the set specification(s). No sets means all functions (dangerous!)
*/
    for(;;) {
        if ( *in == '{' ) {
            s = in+1;
            SKIPBRA2(in)
            number = DoTempSet(s,in);
            in++;
            if ( *in != ',' ) {
                c = in[1]; in[1] = 0;
                MesPrint("& %s: A set in a transform statement should be followed by a comma",s);
                in[1] = c; in++;
                if ( error == 0 ) error = 1;
            }
        }
        else if ( *in == '[' || FG.cTable[*in] == 0 ) {
            s = in;
            in = SkipAName(in);
            if ( *in != ',' ) break;
            c = *in; *in = 0;
            type = GetName(AC.varnames,s,&number,NOAUTO);
            if ( type == CFUNCTION ) { number += MAXVARIABLES + FUNCTION; }
            else if ( type != CSET ) {
                MesPrint("& %s: A transform statement starts with sets of functions",s);
                if ( error == 0 ) error = 1;
            }
            *in++ = c;
        }
        else {
            MesPrint("&Illegal syntax in Transform statement",s);
            if ( error == 0 ) error = 1;
            return(error);
        }
        if ( number >= 0 ) {
          if ( number < MAXVARIABLES ) {
/*
            Check that this is a set of functions
*/
            if ( Sets[number].type != CFUNCTION ) {
                MesPrint("&A set in a transform statement should be a set of functions");
                if ( error == 0 ) error = 1;
            }
          }
        }
        else if ( error == 0 ) error = 1;
/*
        Now write the number to the right place
*/
        *wp++ = number;
        while ( *in == ',' ) in++;
    }
    *work = wp - work;
    work = wp; wp++;
/*
```

```
    #] Sets :

    Now we should loop over the various transformations
*/
    while ( *s ) {
        in = s;
        if ( FG.cTable[*in] != 0 ) {
            MesPrint("&Illegal character in Transform statement");
            if ( error == 0 ) error = 1;
            return(error);
        }
        in = SkipAName(in);
        if ( *in == '>' || *in == '<' || *in == '+' || *in == '-' ) in++;
        ss = in;
        c = *ss; *ss = 0;
        if ( c != '(' ) {
            MesPrint("&Illegal syntax in specifying a transformation inside a Transform statement");
            if ( error == 0 ) error = 1;
            return(error);
        }
/*
        #[ replace :
*/
        if ( StrICmp(s,(UBYTE *)"replace") == 0 ) {
/*
                Subkeys: (,,,) as in replace_(,,,)
            The idea here is to read the subkeys as the argument
            of a replace_ function.
            We put the whole together as in the multiply statement (which
            could just be a replace_(....)) and compile it.
            Then we expand the tree with Generator and check the complete
            expression for legality.
*/
            type = REPLACEARG;
doreplace:
            *ss = c;
            if ( ( in = ReadRange(in,range,0) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            in++;
/*
            We have replace(#,#)=(...), and we want dum_(...)  (DUMFUN)
            to send to the compiler. The pointer is after the '=';
*/
            s = in;
            if ( *s != '(' ) {
                MesPrint("&");
                if ( error == 0 ) error = 1;
                return(error);
            }
            SKIPBRA3(in);
            if ( *in != ')' ) {
                MesPrint("&");
                if ( error == 0 ) error = 1;
                return(error);
            }
            in++;
            if ( *in != ',' && *in != '\0' ) {
                MesPrint("&");
                if ( error == 0 ) error = 1;
                return(error);
            }
            i = in - s;
            ss = Tempbuf = (UBYTE *)Malloc1(i+5,"CoTransform/replace");
            *ss++ = 'd'; *ss++ = 'u'; *ss++ = 'm'; *ss++ = '_';
            NCOPY(ss,s,i)
            *ss++ = 0;
            AC.ProtoType = tranarray;
            tranarray[4] = AC.cbufnum;
            irhs = CompileAlgebra(Tempbuf,RHSIDE,AC.ProtoType);
            M_free(Tempbuf,"CoTransform/replace");
            if ( irhs < 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            tranarray[2] = irhs;
/*
            The result of the compilation goes through Generator during
            execution, because that takes care of $-variables.
            This is why we could not use replace_ and had to use dum_.
*/
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *wp++ = SUBEXPSIZE+4;
```

```
            for ( i = 0; i < SUBEXPSIZE; i++ ) *wp++ = tranarray[i];
            *wp++ = 1;
            *wp++ = 1;
            *wp++ = 3;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] replace :
        #[ encode/decode :
*/
        else if ( StrICmp(s,(UBYTE *)"decode" ) == 0 ) {
            type = DECODEARG;
            goto doencode;
        }
        else if ( StrICmp(s,(UBYTE *)"encode" ) == 0 ) {
            type = ENCODEARG;
doencode:   *ss = c;
            if ( ( in = ReadRange(in,range,2) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            in++;
            s = in; while ( FG.cTable[*in] == 0 ) in++;
            c = *in; *in = 0;
/*
                Subkeys: base=# or base=$var
*/
            if ( StrICmp(s,(UBYTE *)"base") == 0 ) {
                *in = c;
                if ( *in != '=' ) {
                    MesPrint("&Illegal base specification in encode/decode transformation");
                    if ( error == 0 ) error = 1;
                    return(error);
                }
                in++;
                if ( *in == '$' ) {
                    in++; ss = in;
                    in = SkipAName(in);
                    c = *in; *in = 0;
                    if ( GetName(AC.dollarnames,ss,&numdol,NOAUTO) != CDOLLAR ) {
                        MesPrint("&%s is undefined",ss-1);
                        numdol = AddDollar(ss,DOLINDEX,&one,1);
                        return(1);
                    }
                    *in = c;
                    x = -numdol;
                }
                else {
                    x = 0;
                    while ( FG.cTable[*in] == 1 ) {
                        x = 10*x + *in++ - '0';
                        if ( x > MAXPOSITIVE4 ) {
illsize:                    MesPrint("&Illegal value for base in encode/decode transformation");
                            if ( error == 0 ) error = 1;
                            return(error);
                        }
                    }
                    if ( x <= 1 ) goto illsize;
                }
                if ( *in != ',' && *in != '\0' ) {
                    MesPrint("&Illegal termination of transformation");
                    if ( error == 0 ) error = 1;
                    return(error);
                }
            }
            else {
                MesPrint("&Illegal option in encode/decode transformation");
                if ( error == 0 ) error = 1;
                return(error);
            }
/*
            Now we can put the whole statement together
            We have the set(s) in work up to wp and the range in range.
            The base is in x and the type tells whether it is encode or decode.
*/
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *wp++ = 4;
            *wp++ = BASECODE;
            *wp++ = (WORD)x;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
```

```
        }
/*
        #] encode/decode :
        #[ implode :
*/
        else if ( StrICmp(s,(UBYTE *)"implode") == 0
               || StrICmp(s,(UBYTE *)"tosumnotation") == 0 ) {
/*
                Subkeys: ?
*/
            type = IMPLODEARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] implode :
        #[ explode :
*/
        else if ( StrICmp(s,(UBYTE *)"explode") == 0
               || StrICmp(s,(UBYTE *)"tointegralnotation") == 0 ) {
/*
                Subkeys: ?
*/
            type = EXPLODEARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] explode :
        #[ permute :
*/
        else if ( StrICmp(s,(UBYTE *)"permute") == 0 ) {
            type = PERMUTEARG;
            *ss = c;
            *wp++ = ARGRANGE;
            *wp++ = 1;
            *wp++ = MAXPOSITIVE4;
            *wp++ = type;
/*
            Now a sequence of cycles
*/
            do {
              wstart = wp; wp++;
              do {
                in++;
                if ( *in == '$' ) {
                    WORD number; UBYTE *t;
                    in++; t = in;
                    while ( FG.cTable[*in] < 2 ) in++;
                    c = *in; *in = 0;
                    if ( ( number = GetDollar(t) ) < 0 ) {
                        MesPrint("&Undefined variable $%s",t);
                        if ( !error ) error = 1;
                        number = AddDollar(t,0,0,0);
                    }
                    *in = c;
                    *wp++ = -number-1;
                }
                else {
                  x = 0;
                  while ( FG.cTable[*in] == 1 ) {
                    x = 10*x + *in++ - '0';
                    if ( x > MAXPOSITIVE4 ) {
                        MesPrint("&value in permute transformation too large");
                        if ( error == 0 ) error = 1;
                        return(error);
                    }
```

```
                  }
                  if ( x == 0 ) {
                    MesPrint("&value 0 in permute transformation not allowed");
                    if ( error == 0 ) error = 1;
                    return(error);
                  }
                  *wp++ = (WORD)x-1;
                }
              } while ( *in == ',' );
              if ( *in != ')' ) {
                MesPrint("&Illegal syntax in permute transformation");
                if ( error == 0 ) error = 1;
                return(error);
              }
              in++;
              if ( *in != ',' && *in != '(' && *in != '\0' ) {
                MesPrint("&Illegal ending in permute transformation");
                if ( error == 0 ) error = 1;
                return(error);
              }
              *wstart = wp-wstart;
              if ( *wstart == 1 ) wstart--;
            } while ( *in == '(' );
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] permute :
        #[ reverse :
*/
        else if ( StrICmp(s,(UBYTE *)"reverse") == 0 ) {
            type = REVERSEARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] reverse :
        #[ dedup :
*/
        else if ( StrICmp(s,(UBYTE *)"dedup") == 0 ) {
            type = DEDUPARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] dedup :
        #[ cycle :
*/
        else if ( StrICmp(s,(UBYTE *)"cycle") == 0 ) {
            type = CYCLEARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,0) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
/*
            Now a sequence of cycles
*/
            in++;
            if ( *in == '+' ) {
            }
            else if ( *in == '-' ) {
```

```
                one = -1;
            }
            else {
                MesPrint("&Cycle in a Transform statement should be followed by =+/-number/$");
                if ( error == 0 ) error = 1;
                return(error);
            }
            in++; x = 0;
            if ( *in == '$' ) {
                UBYTE *si = in;
                in++; si = in;
                while ( FG.cTable[*in] == 0 || FG.cTable[*in] == 1 ) in++;
                c = *in; *in = 0;
                if ( ( x = GetDollar(si) ) < 0 ) {
                    MesPrint("&Undefined $-variable in transform,cycle statement.");
                    error = 1;
                }
                *in = c;
                if ( one < 0 ) x += MAXPOSITIVE4;
                x += MAXPOSITIVE2;
                *wp++ = x;
            }
            else {
              while ( FG.cTable[*in] == 1 ) {
                x = 10*x + *in++ - '0';
                if ( x > MAXPOSITIVE4 ) {
                    MesPrint("&Number in cycle in a Transform statement too big");
                    if ( error == 0 ) error = 1;
                    return(error);
                }
              }
              *wp++ = x*one;
            }
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] cycle :
        #[ islyndon/tolyndon :
*/
        else if ( StrICmp(s,(UBYTE *)"islyndon" ) == 0 ) {
            type = ISLYNDON;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"islyndon<" ) == 0 ) {
            type = ISLYNDON;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"islyndon-" ) == 0 ) {
            type = ISLYNDON;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"islyndon>" ) == 0 ) {
            type = ISLYNDONR;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"islyndon+" ) == 0 ) {
            type = ISLYNDONR;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"tolyndon" ) == 0 ) {
            type = TOLYNDON;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"tolyndon<" ) == 0 ) {
            type = TOLYNDON;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"tolyndon-" ) == 0 ) {
            type = TOLYNDON;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"tolyndon>" ) == 0 ) {
            type = TOLYNDONR;
            goto doreplace;
        }
        else if ( StrICmp(s,(UBYTE *)"tolyndon+" ) == 0 ) {
            type = TOLYNDONR;
            goto doreplace;
        }
/*
        #] islyndon/tolyndon :
        #[ addarg :
*/
        else if ( StrICmp(s,(UBYTE *)"addargs" ) == 0 ) {
            type = ADDARG;
```

```
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] addarg :
        #[ mularg :
*/
        else if ( ( StrICmp(s,(UBYTE *)"mulargs" ) == 0 )
            || ( StrICmp(s,(UBYTE *)"multiplyargs" ) == 0 ) ) {
            type = MULTIPLYARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] mularg :
        #[ droparg :
*/
        else if ( StrICmp(s,(UBYTE *)"dropargs" ) == 0 ) {
            type = DROPARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] droparg :
        #[ selectarg :
*/
        else if ( StrICmp(s,(UBYTE *)"selectargs" ) == 0 ) {
            type = SELECTARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] selectarg :
        #[ ZtoH :
*/
        else if ( StrICmp(s,(UBYTE *)"ztoh") == 0 ) {
/*
                Subkeys: ?
*/
            type = ZTOHARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
```

```
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] ZtoH :
        #[ HtoZ :
*/
        else if ( StrICmp(s,(UBYTE *)"htoz") == 0 ) {
/*
                Subkeys: ?
*/
            type = HTOZARG;
            *ss = c;
            if ( ( in = ReadRange(in,range,1) ) == 0 ) {
                if ( error == 0 ) error = 1;
                return(error);
            }
            *wp++ = ARGRANGE;
            *wp++ = range[0];
            *wp++ = range[1];
            *wp++ = type;
            *work = wp-work;
            work = wp; *wp++ = 0;
            s = in;
        }
/*
        #] HtoZ :
*/
        else {
            MesPrint("&Unknown transformation inside a Transform statement: %s",s);
            *ss = c;
            if ( error == 0 ) error = 1;
            return(error);
        }
        while ( *s == ',') s++;
    }
    AT.WorkPointer[0] = TYPETRANSFORM;
    AT.WorkPointer[1] = i = wp - AT.WorkPointer;
    AddNtoL(i,AT.WorkPointer);
    return(error);
}

/*
        #] CoTransform :
        #[ RunTransform :

        Executes the transform statement.
        This routine hunts down the functions and sends them to the various
        action routines.
        params: size,#set1,...,#setn, transformations

*/

WORD RunTransform(PHEAD WORD *term, WORD *params)
{
    WORD *t, *tstop, *w, *m, *out, *in, *tt, retval;
    WORD *fun, *args, *info, *infoend, *onetransform, *funs, *endfun;
    WORD *thearg = 0, *iterm, *newterm, *nt, *oldwork = AT.WorkPointer, sign = 1;
    int i;
    out = tstop = term + *term;
    tstop -= ABS(tstop[-1]);
    in = term;
    t = term + 1;
    while ( t < tstop ) {
        endfun = onetransform = params + *params;
        funs = params + 1;
        if ( *t < FUNCTION ) {}
        else if ( funs == endfun ) {  /* we do all functions */
hit:;
            while ( in < t ) *out++ = *in++;
            tt = t + t[1]; fun = out;
            while ( in < tt ) *out++ = *in++;
            do {
                args = onetransform + 1;
                info = args; while ( *info <= MAXRANGEINDICATOR ) {
                    if ( *info == ALLARGS ) info++;
                    else if ( *info == NUMARG ) info += 2;
                    else if ( *info == ARGRANGE ) info += 3;
                    else if ( *info == MAKEARGS ) info += 3;
                }
                switch ( *info ) {
                    case REPLACEARG:
                        if ( RunReplace(BHEAD fun,args,info) ) goto abo;
                        out = fun + fun[1];
```

```
                        break;
                    case ENCODEARG:
                        if ( RunEncode(BHEAD fun,args,info) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case DECODEARG:
                        if ( RunDecode(BHEAD fun,args,info) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case IMPLODEARG:
                        if ( RunImplode(fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case EXPLODEARG:
                        if ( RunExplode(BHEAD fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case PERMUTEARG:
                        if ( RunPermute(BHEAD fun,args,info) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case REVERSEARG:
                        if ( RunReverse(BHEAD fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case DEDUPARG:
                        if ( RunDedup(BHEAD fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case CYCLEARG:
                        if ( RunCycle(BHEAD fun,args,info) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case ADDARG:
                        if ( RunAddArg(BHEAD fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case MULTIPLYARG:
                        if ( RunMulArg(BHEAD fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case ISLYNDON:
                        if ( ( retval = RunIsLyndon(BHEAD fun,args,1) ) < -1 ) goto abo;
                        goto returnvalues;
                        break;
                    case ISLYNDONR:
                        if ( ( retval = RunIsLyndon(BHEAD fun,args,-1) ) < -1 ) goto abo;
                        goto returnvalues;
                        break;
                    case TOLYNDON:
                        if ( ( retval = RunToLyndon(BHEAD fun,args,1) ) < -1 ) goto abo;
                        goto returnvalues;
                        break;
                    case TOLYNDONR:
                        if ( ( retval = RunToLyndon(BHEAD fun,args,-1) ) < -1 ) goto abo;
returnvalues:;
                        out = fun + fun[1];
                        if ( retval == -1 ) break;
/*
                        Work out the yes/no stuff
*/
                        AT.WorkPointer += 2*AM.MaxTer;
                        if ( AT.WorkPointer > AT.WorkTop ) {
                            MLOCK(ErrorMessageLock);
                            MesWork();
                            MUNLOCK(ErrorMessageLock);
                            return(-1);
                        }
                        iterm = AT.WorkPointer;
                        info++;
                        for ( i = 0; i < *info; i++ ) iterm[i] = info[i];
                        AT.WorkPointer = iterm + *iterm;
                        AR.Eside = LHSIDEX;
                        NewSort(BHEAD0);
                        if ( Generator(BHEAD iterm,AR.Cnumlhs) ) {
                            LowerSortLevel();
                            AT.WorkPointer = oldwork;
                            return(-1);
                        }
                        newterm = AT.WorkPointer;
                        if ( EndSort(BHEAD newterm,1) < 0 ) {}
                        if ( ( *newterm && *(newterm+*newterm) != 0 ) || *newterm == 0 ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("&yes/no information in islyndon/tolyndon does not evaluate into a single term");
                            MUNLOCK(ErrorMessageLock);
                            return(-1);
```

```
                        }
                        AR.Eside = RHSIDE;
                        i = *newterm; tt = iterm; nt = newterm;
                        NCOPY(tt,nt,i);
                        AT.WorkPointer = iterm + *iterm;
                        info = iterm + 1;
                        infoend = info+info[1];
                        info += FUNHEAD;

                        if ( retval == 0 ) {
/*
                            Need second argument (=no)
*/
                            if ( info >= infoend ) {
abortlyndon:;
                                MLOCK(ErrorMessageLock);
                                MesPrint("There should be a yes and a no argument in
      islyndon/tolyndon");
                                MUNLOCK(ErrorMessageLock);
                                Terminate(-1);
                            }
                            NEXTARG(info)
                            if ( info >= infoend ) goto abortlyndon;
                            thearg = info;
                        }
                        else if ( retval == 1 ) {
/*
                            Need first argument (=yes)
*/
                            if ( info >= infoend ) goto abortlyndon;
                            thearg = info;
                            NEXTARG(info)
                            if ( info >= infoend ) goto abortlyndon;
                        }
                        NEXTARG(info)
                        if ( info < infoend ) goto abortlyndon;
/*
                        The argument in thearg needs to be copied
                        We did not pull it through generator to guarantee
                        that it is a single argument.
                        The easiest way is to let the routine Normalize
                        do the job and put everything in an exponent function
                        with the power one.
*/
                        if ( *thearg == -SNUMBER && thearg[1] == 0 ) {
                            *term = 0; return(0);
                        }
                        if ( *thearg == -SNUMBER && thearg[1] == 1 ) { }
                        else {
                            fun = out;
                            *out++ = EXPONENT; out++; *out++ = 1; FILLFUN3(out);
                            COPY1ARG(out,thearg);
                            *out++ = -SNUMBER; *out++ = 1;
                            fun[1] = out-fun;
                        }
                        break;
                    case DROPARG:
                        if ( RunDropArg(BHEAD fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case SELECTARG:
                        if ( RunSelectArg(BHEAD fun,args) ) goto abo;
                        out = fun + fun[1];
                        break;
                    case ZTOHARG:
                        {
                          WORD s = RunZtoHArg(BHEAD fun,args);
                          if ( s < 0 ) goto abo;
                          if ( s == 1 ) sign = -sign;
                          out = fun + fun[1];
                        }
                        break;
                    case HTOZARG:
                        {
                          WORD s = RunHtoZArg(BHEAD fun,args);
                          if ( s < 0 ) goto abo;
                          if ( s == 1 ) sign = -sign;
                          out = fun + fun[1];
                        }
                        break;
                    default:
                        MLOCK(ErrorMessageLock);
                        MesPrint("Irregular code in execution of transform statement");
                        MUNLOCK(ErrorMessageLock);
                        Terminate(-1);
                }
                onetransform += *onetransform;
```

```
            } while ( *onetransform );
        }
        else {
          while ( funs < endfun ) {  /* sum over sets */
            if ( *funs > MAXVARIABLES ) {
              if ( *t == *funs-MAXVARIABLES ) goto hit;
            }
            else {
              w = SetElements + Sets[*funs].first;
              m = SetElements + Sets[*funs].last;
              while ( w < m ) {  /* sum over set elements */
                if ( *w == *t ) goto hit;
                w++;
              }
            }
            funs++;
          }
        }
        t += t[1];
    }
    tt = term + *term; while ( in < tt ) *out++ = *in++;
    if ( sign == -1 ) out[-1] = -out[-1];
    *tt = i = out - tt;
/*
    Now copy the whole thing back
*/
    NCOPY(term,tt,i)
    return(0);
abo:
    MLOCK(ErrorMessageLock);
    MesCall("RunTransform");
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] RunTransform :
        #[ RunEncode :

        The info is given by
            ENCODEARG,size,BASECODE,num
        and possibly more codes to follow.
        Only one range is allowed and for now, it should be fully numerical
        If the range is in reverse order, we need to either revert it
        first or work with an array of pointers.
*/

WORD RunEncode(PHEAD WORD *fun, WORD *args, WORD *info)
{
    WORD base, *f, *funstop, *fun1, *t, size1, size2, size3, *arg;
    int num, num1, num2, n, i, i1, i2;
    UWORD *scrat1, *scrat2, *scrat3;
    WORD *tt, *tstop, totarg, arg1, arg2;
    if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunEncode");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
    if ( arg1 > totarg || arg2 > totarg ) return(0);

    if ( info[2] == BASECODE ) {
        base = info[3];
        if ( base <= 0 ) { /* is a dollar variable */
            i1 = -base;
            base = DolToNumber(BHEAD i1);
            if ( AN.ErrorInDollar || base < 2 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("$%s does not have a number value > 1 in base/encode/transform statement in module %l",
                    DOLLARNAME(Dollars,i1),AC.CModule);
                MUNLOCK(ErrorMessageLock);
                Terminate(-1);
            }
        }
/*
        Compute number of pointers needed and make sure there is space
*/
        if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; }
        else { num1 = arg1; num2 = arg2; }
        num = num2-num1+1;
        WantAddPointers(num);
/*
```

```
        Collect the pointers in pWorkSpace
*/
        n = 1; funstop = fun+fun[1]; f = fun+FUNHEAD;
        while ( n < num1 ) {
            if ( f >= funstop ) return(0);
            NEXTARG(f);
            n++;
        }
        fun1 = f; i = 0;
        while ( n <= num2 ) {
            if ( f >= funstop ) return(0);
            if ( *f != -SNUMBER ) {
                if ( *f < 0 ) return(0);
                t = f + *f - 1;
                i1 = ABS(*t);
                if ( (*f-i1) != (ARGHEAD+1) ) return(0); /* Not numerical */
                i1 = (i1-1)/2 - 1;
                t--;
                while ( i1 > 0 ) {
                    if ( *t != 0 ) return(0); /* Not an integer */
                    t--; i1--;
                }
            }
            AT.pWorkSpace[AT.pWorkPointer+i] = f;
            i++;
            NEXTARG(f);
            n++;
        }
/*
        f points now to after the arguments; fun1 at the first.
        Now check whether we need to revert the order
*/
        if ( arg1 > arg2 ) {
            i1 = 0; i2 = i-1;
            while ( i1 < i2 ) {
                t = AT.pWorkSpace[AT.pWorkPointer+i1];
                AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
                AT.pWorkSpace[AT.pWorkPointer+i2] = t;
                i1++; i2--;
            }
        }
/*
        Now we can put the thing together.
        x = arg1;
        x = base*x+arg2
        x = base*x+arg3  etc.
        We need three scratch arrays for long integers
                (see NumberMalloc in tools.c).
*/
        scrat1 = NumberMalloc("RunEncode");
        scrat2 = NumberMalloc("RunEncode");
        scrat3 = NumberMalloc("RunEncode");
        arg = AT.pWorkSpace[AT.pWorkPointer];
        size1 = PutArgInScratch(arg,scrat1);
        i--;
        while ( i > 0 ) {
            if ( MulLong(scrat1,size1,(UWORD *)(&base),1,scrat2,&size2) ) {
                NumberFree(scrat3,"RunEncode");
                NumberFree(scrat2,"RunEncode");
                NumberFree(scrat1,"RunEncode");
                goto CalledFrom;
            }
            NEXTARG(arg);
            size3 = PutArgInScratch(arg,scrat3);
            if ( AddLong(scrat2,size2,scrat3,size3,scrat1,&size1) ) {
                NumberFree(scrat3,"RunEncode");
                NumberFree(scrat2,"RunEncode");
                NumberFree(scrat1,"RunEncode");
                goto CalledFrom;
            }
            i--;
        }
/*
        Now put the output in place. There are two cases, one being much
        faster than the other. Hence we program both.
        Fast: it fits inside the old location.
        Slow: it does not.
        The total space is f-fun1
*/
        if ( size1 == 0 ) { /* Fits! */
            *fun1++ = -SNUMBER; *fun1++ = 0;
            while ( f < funstop ) *fun1++ = *f++;
            fun[1] = funstop-fun;
        }
        else if ( size1 == 1 && scrat1[0] <= MAXPOSITIVE ) { /* Fits! */
            *fun1++ = -SNUMBER; *fun1++ = scrat1[0];
            while ( f < funstop ) *fun1++ = *f++;
```

```
                fun[1] = fun1-fun;
            }
            else if ( size1 == -1 && scrat1[0] <= MAXPOSITIVE+1 ) { /* Fits! */
                *fun1++ = -SNUMBER;
                if ( scrat1[0] < MAXPOSITIVE ) *fun1++ = scrat1[0];
                else *fun1++ = (WORD)(MAXPOSITIVE+1);
                while ( f < funstop ) *fun1++ = *f++;
                fun[1] = fun1-fun;
            }
            else if ( ABS(size1)*2+2+ARGHEAD <= f-fun1 ) { /* Fits! */
                if ( size1 < 0 ) { size2 = size1*2-1; size1 = -size1; size3 = -size2; }
                else { size2 = 2*size1+1; size3 = size2; }
                *fun1++ = size3+ARGHEAD+1;
                *fun1++ = 0; FILLARG(fun1);
                *fun1++ = size3+1;
                for ( i = 0; i < size1; i++ ) *fun1++ = scrat1[i];
                *fun1++ = 1;
                for ( i = 1; i < size1; i++ ) *fun1++ = 0;
                *fun1++ = size2;
                while ( f < funstop ) *fun1++ = *f++;
                fun[1] = fun1-fun;
            }
            else {  /* Does not fit */
                t = funstop;
                if ( size1 < 0 ) { size2 = size1*2-1; size1 = -size1; size3 = -size2; }
                else { size2 = 2*size1+1; size3 = size2; }
                *t++ = size3+ARGHEAD+1;
                *t++ = 0; FILLARG(t);
                *t++ = size3+1;
                for ( i = 0; i < size1; i++ ) *t++ = scrat1[i];
                *t++ = 1;
                for ( i = 1; i < size1; i++ ) *t++ = 0;
                *t++ = size2;
                while ( f < funstop ) *t++ = *f++;
                f = funstop;
                while ( f < t ) *fun1++ = *f++;
                fun[1] = fun1-fun;
            }
            NumberFree(scrat3,"RunEncode");
            NumberFree(scrat2,"RunEncode");
            NumberFree(scrat1,"RunEncode");
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("Unimplemented type of encoding encountered in RunEncode");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        return(0);
CalledFrom:
        MLOCK(ErrorMessageLock);
        MesCall("RunEncode");
        MUNLOCK(ErrorMessageLock);
        return(-1);
}

/*
            #] RunEncode :
            #[ RunDecode :
*/

WORD RunDecode(PHEAD WORD *fun, WORD *args, WORD *info)
{
        WORD base, num, num1, num2, n, *f, *funstop, *fun1, size1, size2, size3, *t;
        WORD i1, i2, i, sig;
        UWORD *scrat1, *scrat2, *scrat3;
        WORD *tt, *tstop, totarg, arg1, arg2;
        if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
        if ( *args != ARGRANGE ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Illegal range encountered in RunDecode");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
        tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
        while ( tt < tstop ) { totarg++; NEXTARG(tt); }
        if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
        if ( arg1 > totarg && arg2 > totarg ) return(0);
        if ( info[2] == BASECODE ) {
            base = info[3];
            if ( base <= 0 ) { /* is a dollar variable */
                i1 = -base;
                base = DolToNumber(BHEAD i1);
                if ( AN.ErrorInDollar || base < 2 ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("$%s does not have a number value > 1 in base/decode/transform statement in
module %l",
```

```
                DOLLARNAME(Dollars,i1),AC.CModule);
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
        }
    }
/*
        Compute number of output arguments needed
*/
    if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; }
    else { num1 = arg1; num2 = arg2; }
    num = num2-num1+1;
    if ( num <= 1 ) return(0);
/*
        Find argument num1
*/
    funstop = fun + fun[1];
    f = fun + FUNHEAD; n = 1;
    while ( f < funstop ) {
        if ( n == num1 ) break;
        NEXTARG(f); n++;
    }
    if ( f >= funstop ) return(0);  /* not enough arguments */
/*
        Check that f is integer
*/
    if ( *f == -SNUMBER ) {}
    else if ( *f < 0 ) return(0);
    else {
        t = f + *f - 1;
        i1 = ABS(*t);
        if ( (*f-i1) != (ARGHEAD+1) ) return(0); /* Not numerical */
        i1 = (i1-1)/2 - 1;
        t--;
        while ( i1 > 0 ) {
            if ( *t != 0 ) return(0); /* Not an integer */
            t--; i1--;
        }
    }
    fun1 = f;
/*
        The argument that should be decoded is in fun1
        We have to copy it to scratch
*/
    scrat1 = NumberMalloc("RunEncode");
    scrat2 = NumberMalloc("RunEncode");
    scrat3 = NumberMalloc("RunEncode");
    size1 = PutArgInScratch(fun1,scrat1);
    if ( size1 < 0 ) { sig = -1; size1 = -size1; }
    else sig = 1;
/*
        We can check first whether this number can be decoded
*/
    scrat2[0] = base; size2 = 1;
    if ( RaisPow(BHEAD scrat2,&size2,num) ) {
        NumberFree(scrat3,"RunEncode");
        NumberFree(scrat2,"RunEncode");
        NumberFree(scrat1,"RunEncode");
        goto CalledFrom;
    }
    if ( BigLong(scrat1,size1,scrat2,size2) >= 0 ) { /* Number too big */
        NumberFree(scrat3,"RunEncode");
        NumberFree(scrat2,"RunEncode");
        NumberFree(scrat1,"RunEncode");
        return(0);
    }
/*
        We need num*2 spaces
*/
    if ( *fun1 > num*2 ) {  /* shrink space */
        t = fun1 + 2*num; f = fun1 + *fun1;
        while ( f < funstop ) *t++ = *f++;
        fun[1] = t - fun;
    }
    else if ( *fun1 < num*2 ) { /* case includes -SNUMBER */
        if ( *fun1 < 0 ) { /* expand space from -SNUMBER */
            fun[1] += (num-1)*2;
            t = funstop + (num-1)*2;
        }
        else { /* expand space from general argument */
            fun[1] += 2*num - *fun1;
            t = funstop +2*num - *fun1;
        }
        f = funstop;
        while ( f > fun1 ) *--t = *--f;
    }
/*
        Now there is space for num -SNUMBER arguments filled from the top.
```

```
*/
        for ( i = num-1; i >= 0; i-- ) {
            DivLong(scrat1,size1,(UWORD *)(&base),1,scrat2,&size2,scrat3,&size3);
            fun1[2*i]   = -SNUMBER;
            if ( size3 == 0 ) fun1[2*i+1] = 0;
            else fun1[2*i+1] = (WORD)(scrat3[0])*sig;
            for ( i1 = 0; i1 < size2; i1++ ) scrat1[i1] = scrat2[i1];
            size1 = size2;
        }
        if ( size2 != 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("RunDecode: number to be decoded is too big");
            MUNLOCK(ErrorMessageLock);
            NumberFree(scrat3,"RunEncode");
            NumberFree(scrat2,"RunEncode");
            NumberFree(scrat1,"RunEncode");
            goto CalledFrom;
        }
/*
        Now check whether we should change the order of the arguments
*/
        if ( arg1 > arg2 ) {
            i1 = 1; i2 = 2*num-1;
            while ( i2 > i1 ) {
                i = fun1[i1]; fun1[i1] = fun1[i2]; fun1[i2] = i;
                i1 += 2; i2 -= 2;
            }
        }
        NumberFree(scrat3,"RunEncode");
        NumberFree(scrat2,"RunEncode");
        NumberFree(scrat1,"RunEncode");
    }
    else {
        MLOCK(ErrorMessageLock);
        MesPrint("Unimplemented type of encoding encountered in RunDecode");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    return(0);
CalledFrom:
    MLOCK(ErrorMessageLock);
    MesCall("RunDecode");
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] RunDecode :
        #[ RunReplace :

        Gets the function, passes the arguments and looks whether they
        need to be treated. If so, the exact treatment is found in info.
        The info is given as if it is a function of type REPLACEMENT but
        its name is REPLACEARG (which is NOT a function).
        It is performed on the arguments.
        The output is at first written after fun and in the end overwrites fun.
*/

WORD RunReplace(PHEAD WORD *fun, WORD *args, WORD *info)
{
    int n = 0, i, dirty = 0, totarg, nfix, nwild, ngeneral;
    WORD *t, *tt, *u, *tstop, *info1, *infoend, *oldwork = AT.WorkPointer;
    WORD *term, *newterm, *nt, *term1, *term2;
    WORD wild[4], mask, *term3, *term4, *oldmask = AT.WildMask;
    WORD n1, n2, doanyway;
    info++;
    t = fun; tstop = fun + fun[1]; u = tstop;
    for ( i = 0; i < FUNHEAD; i++ ) *u++ = *t++;
    tt = t;
    if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
        totarg = 0;
        while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    }
    else {
        totarg = tstop - tt;
    }
/*
    Now get the info through Generator to bring it to standard form.
    info points at a single term that should be sent to Generator.

    We want to put the information in the WorkSpace but fun etc lies there
    already. This means that we have to move the WorkPointer quite high up.
*/
    AT.WorkPointer += 2*AM.MaxTer;
    if ( AT.WorkPointer > AT.WorkTop ) {
        MLOCK(ErrorMessageLock);
        MesWork();
```

```
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    term = AT.WorkPointer;
    for ( i = 0; i < *info; i++ ) term[i] = info[i];
    AT.WorkPointer = term + *term;
    AR.Eside = LHSIDEX;
    NewSort(BHEAD0);
    if ( Generator(BHEAD term,AR.Cnumlhs) ) {
        LowerSortLevel();
        AT.WorkPointer = oldwork;
        return(-1);
    }
    newterm = AT.WorkPointer;
    if ( EndSort(BHEAD newterm,1) < 0 ) {}
    if ( ( *newterm && *(newterm+*newterm) != 0 ) || *newterm == 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("&information in replace transformation does not evaluate into a single term");
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    AR.Eside = RHSIDE;
    i = *newterm; tt = term; nt = newterm;
    NCOPY(tt,nt,i);
    AT.WorkPointer = term + *term;
    info = term + 1;

    term1 = term + *term;
    term2 = term1+1;
    *term2++ = REPLACEMENT;
    term2++; FILLFUN(term2)
/*
    First we count the different types of objects
*/
    infoend = info + info[1];
    info1 = info + FUNHEAD;
    nfix = nwild = ngeneral = 0;
    while ( info1 < infoend ) {
        if ( *info1 == -SNUMBER ) {
            nfix++;
            info1 += 2; NEXTARG(info1)
        }
        else if ( *info1 <= -FUNCTION ) {
            if ( *info1 == -WILDARGFUN ) {
                nwild++;
                info1++; NEXTARG(info1)
            }
            else {
                *term2++ = *info1++; COPY1ARG(term2,info1)
                ngeneral++;
            }
        }
        else if ( *info1 == -INDEX ) {
            if ( info1[1] == WILDARGINDEX + AM.OffsetIndex ) {
                nwild++;
                info1 += 2; NEXTARG(info1)
            }
            else {
                *term2++ = *info1++; *term2++ = *info1++; COPY1ARG(term2,info1)
                ngeneral++;
            }
        }
        else if ( *info1 == -SYMBOL ) {
            if ( info1[1] == WILDARGSYMBOL ) {
                nwild++;
                info1 += 2; NEXTARG(info1)
            }
            else {
                *term2++ = *info1++; *term2++ = *info1++; COPY1ARG(term2,info1)
                ngeneral++;
            }
        }
        else if ( *info1 == -MINVECTOR || *info1 == -VECTOR ) {
            if ( info1[1] == WILDARGVECTOR + AM.OffsetVector ) {
                nwild++;
                info1 += 2; NEXTARG(info1)
            }
            else {
                *term2++ = *info1++; *term2++ = *info1++; COPY1ARG(term2,info1)
                ngeneral++;
            }
        }
        else {
            MLOCK(ErrorMessageLock);
            MesPrint("&irregular code found in replace transformation (RunReplace)");
            MUNLOCK(ErrorMessageLock);
            Terminate(-1);
```

```
        }
    }
    AT.WorkPointer = term2;
    *term1 = term2 - term1;
    term1[2] = *term1 - 1;
/*
    And now stepping through the arguments
*/
    while ( t < tstop ) {
        n++;     /* The number of the argument. Now check whether we need it */
        if ( TestArgNum(n,totarg,args) == 0 ) {
            if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
                if ( *t <= -FUNCTION ) { *u++ = *t++; }
                else if ( *t < 0 ) { *u++ = *t++; *u++ = *t++; }
                else { i = *t; NCOPY(u,t,i) }
            }
            else *u++ = *t++;
            continue;
        }
/*
        Here we have in info effectively a replace_ function, but with
        additionally the possibility of integer arguments. We treat those first
        and for the rest we have to do some pattern matching.
        Note that the compilation routine should check that there is an
        even number of arguments in the replace function.

        First we go for number -> something
*/
        doanyway = 0;
        if ( nfix > 0 ) {
          if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
            if ( *t == -SNUMBER ) {
              info1 = info + FUNHEAD;
              while ( info1 < infoend ) {
                if ( *info1 == -SNUMBER ) {
                    if ( info1[1] == t[1] ) {
                      if ( info1[2] == -SNUMBER ) {
                        *u++ = -SNUMBER; *u++ = info1[3];
                        info1 += 4;
                      }
                      else {
                        info1 += 2;
                        if ( info1[0] <= -FUNCTION ) i = 1;
                        else if ( info1[0] < 0 ) i = 2;
                        else i = *info1;
                        NCOPY(u,info1,i)
                      }
                      t += 2; goto nextt;
                    }
                    info1 += 2;
                    NEXTARG(info1);
                }
                else {
                    NEXTARG(info1);
                    NEXTARG(info1);
                }
              }
/*
              Here we had no match in the style of 1->2. It could however
              be that xarg_ does something
*/
              doanyway = 1; n2 = t[1];
            }
          }
          else {  /* Tensor */
            if ( *t < AM.OffsetIndex && *t >= 0 ) {
              info1 = info + FUNHEAD;
              while ( info1 < infoend ) {
                if ( ( *info1 == -SNUMBER ) && ( info1[1] == *t )
                 && ( ( ( info1[2] == -SNUMBER ) && ( info1[3] >= 0 )
                 && ( info1[3] < AM.OffsetIndex ) )
                 || ( info1[2] == -INDEX || info1[2] == -VECTOR
                 || info1[2] == -MINVECTOR ) ) ) {
                    *u++ = info1[3];
                    info1 += 4;
                    t++; goto nextt;
                }
                else {
                    NEXTARG(info1);
                    NEXTARG(info1);
                }
              }
            }
          }
        }
        else if ( *t == -SNUMBER ) {
          doanyway = 1; n2 = t[1];
```

```
        }
/*
        First we try to catch those elements that have an exact match
        in the traditional replace_ part.
        This means that *t should be less than zero and match an entry
        in the replace_ function that we prepared.
*/
        if ( ngeneral > 0 ) {
          if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
            if ( *t < 0 ) {
                term3 = term1 + *term1;
                term4 = term1 + FUNHEAD;
                while ( term4 < term3 ) {
                    if ( *term4 == *t && ( *t <= -FUNCTION ||
                    ( t[1] == term4[1] ) ) ) break;
                    NEXTARG(term4)
                }
                if ( term4 < term3 ) goto dothisnow;
            }
          }
          else {
            term3 = term1 + *term1;
            term4 = term1 + FUNHEAD;
            while ( term4 < term3 ) {
                if ( ( term4[1] == *t ) &&
                    ( ( *term4 == -INDEX || *term4 == -VECTOR ||
                     ( *term4 == -SYMBOL && term4[1] < AM.OffsetIndex
                        && term4[1] >= 0 ) ) ) ) break;
                NEXTARG(term4)
            }
            if ( term4 < term3 ) goto dothisnow;
          }
        }
/*
        First we eliminate the fixed arguments and make a 'new info'
        If there is anything left we can continue.
        Now we look for whole argument wildcards (arg_, parg_, iarg_ or farg_)
*/
        if ( nwild > 0 ) {
/*
            If we have f(a)*replace_(xarg_,b(xarg_)) this gives f(b(a))
            In testing the wildcard we have CheckWild do the work.
            This means that we have to set op the special variables
            (AT.WildMask,AN.WildValue,AN.NumWild)

*/
            wild[1] = 4;
            info1 = info + FUNHEAD;
            while ( info1 < infoend ) {
                if ( *info1 == -SYMBOL && info1[1] == WILDARGSYMBOL
                && ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) ) {
                    wild[0] = SYMTOSUB;
                    wild[2] = WILDARGSYMBOL;
                    wild[3] = 0;
                    AN.WildValue = wild;
                    AT.WildMask = &mask;
                    mask = 0;
                    AN.NumWild = 1;
                    if ( *t == -SYMBOL || ( *t > 0 && CheckWild(BHEAD WILDARGSYMBOL,SYMTOSUB,1,t) == 0 )
                    || doanyway ) {
/*
                        We put the part in replace in a function and make
                        a replace_(xarg_,(t argument)).
*/
                        n1 = SYMBOL; n2 = WILDARGSYMBOL;
                        info1 += 2;
getthisone:;
                        term3 = term2+1;
                        if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
                            *term3++ = DUMFUN; term3++; FILLFUN(term3)
                            COPY1ARG(term3,info1)
                        }
                        else {
                            *term3++ = fun[0]; term3++; FILLFUN(term3)
                            *term3++ = *info1;
                        }
                        term2[2] = term3 - term2 - 1;
                        tt = term3;
                        *term3++ = REPLACEMENT;
                        term3++; FILLFUN(term3)
                        *term3++ = -n1;
                        if ( n1 < FUNCTION ) *term3++ = n2;
                        if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
                            term4 = t;
                            COPY1ARG(term3,term4)
                        }
```

```
                        else {
                            *term3++ = *t;
                        }
                        tt[1] = term3 - tt;
                        *term3++ = 1; *term3++ = 1; *term3++ = 3;
                        *term2 = term3 - term2;

                        AT.WorkPointer = term3;
                        NewSort(BHEAD0);
                        if ( Generator(BHEAD term2,AR.Cnumlhs) ) {
                            LowerSortLevel();
                            AT.WorkPointer = oldwork;
                            AT.WildMask = oldmask;
                            return(-1);
                        }
                        term4 = AT.WorkPointer;
                        if ( EndSort(BHEAD term4,1) < 0 ) {}
                        if ( ( *term4 && *(term4+*term4) != 0 ) || *term4 == 0 ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("&information in replace transformation does not evaluate into a single term");
                            MUNLOCK(ErrorMessageLock);
                            return(-1);
                        }
/*
                        Now we can copy the new function argument to the output u
*/
                        i = term4[2]-FUNHEAD;
                        term3 = term4+FUNHEAD+1;
                        NCOPY(u,term3,i)
                        if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
                            NEXTARG(t)
                        }
                        else t++;
                        AT.WorkPointer = term2;

                        goto nextt;
                    }
                    info1 += 2; NEXTARG(info1)
                }
                else if ( ( *info1 == -INDEX )
                    && ( info[1] == WILDARGINDEX + AM.OffsetIndex ) ) {
                    wild[0] = INDTOSUB;
                    wild[2] = WILDARGINDEX+AM.OffsetIndex;
                    wild[3] = 0;
                    AN.WildValue = wild;
                    AT.WildMask = &mask;
                    mask = 0;
                    AN.NumWild = 1;
                    if ( ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION )
                    || ( *t == -INDEX || ( *t > 0 && CheckWild(BHEAD WILDARGINDEX,INDTOSUB,1,t) == 0 ) ) ) {
/*
                        We put the part in replace in a function and make
                        a replace_(xarg_,(t argument)).
*/
                        n1 = INDEX; n2 = WILDARGINDEX+AM.OffsetIndex;
                        info1 += 2;
                        goto getthisone;
                    }
                    info1 += 2; NEXTARG(info1)
                }
                else if ( ( *info1 == -VECTOR )
                    && ( info1[1] == WILDARGVECTOR + AM.OffsetVector ) ) {
                    wild[0] = VECTOSUB;
                    wild[2] = WILDARGVECTOR+AM.OffsetVector;
                    wild[3] = 0;
                    AN.WildValue = wild;
                    AT.WildMask = &mask;
                    mask = 0;
                    AN.NumWild = 1;
                    if ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION ) {
                        if ( *t < MINSPEC ) {
                            n1 = VECTOR; n2 = WILDARGVECTOR+AM.OffsetVector;
                            info1 += 2;
                            goto getthisone;
                        }
                    }
                    else if ( *t == -VECTOR || *t == -MINVECTOR ||
                    ( *t > 0 && CheckWild(BHEAD WILDARGVECTOR,VECTOSUB,1,t) == 0 ) ) {
/*
                        We put the part in replace in a function and make
                        a replace_(xarg_,(t argument)).
*/
                        n1 = VECTOR; n2 = WILDARGVECTOR+AM.OffsetVector;
                        info1 += 2;
                        goto getthisone;
```

```
                    }
                    info1 += 2; NEXTARG(info1)
                }
                else if ( *info1 == -WILDARGFUN ) {
                    wild[0] = FUNTOFUN;
                    wild[2] = WILDARGFUN;
                    wild[3] = 0;
                    AN.WildValue = wild;
                    AT.WildMask = &mask;
                    mask = 0;
                    AN.NumWild = 1;
                    if ( *t <= -FUNCTION || ( *t > 0 && CheckWild(BHEAD WILDARGFUN,FUNTOFUN,1,t) == 0 ) ) {
/*
                        We put the part in replace in a function and make
                        a replace_(xarg_,(t argument)).
*/
                        n2 = n1 = -WILDARGFUN; /* n2 is to keep the compiler quiet */
                        info1++;
                        goto getthisone;
                    }
                    info1++; NEXTARG(info1)
                }
                else {
                    NEXTARG(info1) NEXTARG(info1)
                }
            }
        }
        if ( ngeneral > 0 ) {
/*
            They are all in a replace_ function.
            Compose the whole thing into a term with replace_()*dum_(arg)
            which will be given to Generator.
            If we have f(a(x))*replace_(x,b) this gives f(a(b))
*/
dothisnow:;
            term3 = term2; term4 = term1; i = *term1;
            NCOPY(term3,term4,i)
            term4 = term3;
            if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
                *term3++ = DUMFUN; term3++; FILLFUN(term3);
                tt = t;
                COPY1ARG(term3,tt)
            }
            else {
                *term3++ = fun[0]; term3++; FILLFUN(term3); *term3++ = *t;
            }
            term4[1] = term3-term4;
            *term3++ = 1; *term3++ = 1; *term3++ = 3;
            *term2 = term3-term2;
            AT.WorkPointer = term3;
            NewSort(BHEAD0);
            if ( Generator(BHEAD term2,AR.Cnumlhs) ) {
                LowerSortLevel();
                AT.WorkPointer = oldwork;
                AT.WildMask = oldmask;
                return(-1);
            }
            term4 = AT.WorkPointer;
            if ( EndSort(BHEAD term4,1) < 0 ) {}
            if ( ( *term4 && *(term4+*term4) != 0 ) || *term4 == 0 ) {
                MLOCK(ErrorMessageLock);
                MesPrint("&information in replace transformation does not evaluate into a single term");
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
/*
            Now we can copy the new function argument to the output u
*/
            i = term4[2]-FUNHEAD;
            term3 = term4+FUNHEAD+1;
            NCOPY(u,term3,i)
            NEXTARG(t)
            AT.WorkPointer = term2;

            goto nextt;
        }

/*
        No catch. Copy the argument and continue.
*/
        if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
            if ( *t <= -FUNCTION ) { *u++ = *t++; }
            else if ( *t < 0 ) { *u++ = *t++; *u++ = *t++; }
            else { i = *t; NCOPY(u,t,i) }
        }
```

```
        else {
            *u++ = *t++;
        }
nextt:;
    }
    i = u - tstop; tstop[1] = i; tstop[2] = dirty;
    t = fun; u = tstop; NCOPY(t,u,i)
    AT.WorkPointer = oldwork;
    AT.WildMask = oldmask;
    return(0);
}

/*
        #] RunReplace :
        #[ RunImplode :

        Note that we restrict ourselves to short integers and/or single symbols
*/

WORD RunImplode(WORD *fun, WORD *args)
{
    GETIDENTITY
    WORD *tt, *tstop, totarg, arg1, arg2, num1, num2, i1, n;
    WORD *f, *t, *ttt, *t4, *ff, *fff;
    WORD moveup, numzero, outspace;
    if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunImplode");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
/*
    Get the proper range in forward direction and the number of arguments
*/
    if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; }
    else { num1 = arg1; num2 = arg2; }
    if ( num1 > totarg || num2 > totarg ) return(0);
/*
    We need, for the most general case 4 spots for each:
            x,pow,coef,sign
    Hence we put these in the workspace above the term after tstop
*/
    n = 1; f = fun+FUNHEAD;
    while ( n < num1 ) {
        if ( f >= tstop ) return(0);
        NEXTARG(f);
        n++;
    }
    ff = f;
/*
    We are now at the first argument to be done
    Go through the terms and test their validity.
    If one of them doesn't conform to the rules we don't do anything.
    The terms to be done are put in special notation after the function.
    Notation: numsymbol, power, |coef|, sign
    If numsymbol is negative there is no symbol.
    We do it this way because otherwise stepping backwards (as in range=(4,1))
    would be very difficult.
*/
    tt = tstop;
    while ( n <= num2 ) {
        if ( f >= tstop ) return(0);
        if ( *f == -SNUMBER ) { *tt++ = -1; *tt++ = 0;
            if ( f[1] < 0 ) { *tt++ = -f[1]; *tt++ = -1; }
            else { *tt++ = f[1]; *tt++ = 1; }
            f += 2;
        }
        else if ( *f == -SYMBOL ) { *tt++ = f[1]; *tt++ = 1; *tt++ = 1; *tt++ = 1; f += 2; }
        else if ( *f < 0 ) return(0);
        else {
            if ( *f != ( f[ARGHEAD]+ARGHEAD ) ) return(0); /* Not a single term */
            t = f + *f - 1;
            i1 = ABS(*t);
            if ( ( i1 > 3 ) || ( t[-1] != 1 ) ) return(0); /* Not an integer or too big */
            if ( (UWORD)(t[-2]) > MAXPOSITIVE4 ) return(0); /* number too big */
            if ( f[ARGHEAD] == i1+1 ) { /* numerical which is fine */
                *tt++ = -1; *tt++ = 0; *tt++ = t[-2];
                if ( *t < 0 ) { *tt++ = -1; }
                else { *tt++ = 1; }
            }
            else if ( ( f[ARGHEAD+1] != SYMBOL )
                || ( f[ARGHEAD+2] != 4 )
                || ( ( f+ARGHEAD+1+f[ARGHEAD+2] ) < ( t-i1 ) ) ) return(0);
```

```
                    /* not a single symbol with a coefficient */
            else {
                *tt++ = f[ARGHEAD+3];
                *tt++ = f[ARGHEAD+4];
                *tt++ = t[-2];
                if ( *t < 0 ) { *tt++ = -1; }
                else { *tt++ = 1; }
            }
            f += *f;
        }
        n++;
    }
    fff = f;
/*
    At this point we can do the implosion.
    Requirement: no coefficient shall take more than one word.
    (a stricter requirement may be needed to keep the explosion contained)
*/
    if ( arg1 > arg2 ) {
/*
        Work backward.
*/
        t = tt - 4; numzero = 0;
        while ( t >= tstop ) {
            if ( t[2] == 0 ) numzero++;
            else {
                if ( numzero > 0 ) {
                    t[2] += numzero;
                    t4 = t+4;
                    ttt = t4 + 4*numzero;
                    while ( ttt < tt ) *t4++ = *ttt++;
                    tt -= 4*numzero;
                    numzero = 0;
                }
            }
            t -= 4;
        }
    }
    else {
        t = tstop;
        numzero = 0; ttt = t;
        while ( t < tt ) {
            if ( t[2] == 0 ) numzero++;
            else {
                if ( numzero > 0 ) {
                    t[2] += numzero;
                    t4 = t;
                    while ( t4 < tt ) *ttt++ = *t4++;
                    tt -= 4*numzero;
                    t -= 4*numzero;
                    ttt = t + 4;
                    numzero = 0;
                }
                else {
                    ttt = t + 4;
                }
            }
            t += 4;
        }
/*
        We may have numzero > 0 at the end. We leave them.
        Output space is currently from tstop to tt
*/
    }
/*
    Now we compute the real output space needed
*/
    t = tstop; outspace = 0;
    while ( t < tt ) {
        if ( t[0] == -1 ) {
            if ( t[2] > MAXPOSITIVE4 ) { return(0); /* Number too big */ }
            outspace += 2;
        }
        else if ( t[1] == 1 && t[2] == 1 && t[3] == 1 ) { outspace += 2; }
        else { outspace += 8 + ARGHEAD; }
        t += 4;
    }
    if ( outspace < (fff-ff) ) {
        t = tstop;
        while ( t < tt ) {
            if ( t[0] == -1 ) { *ff++ = -SNUMBER; *ff++ = t[2]*t[3]; }
            else if ( t[1] == 1 && t[2] == 1 && t[3] == 1 ) {
                *ff++ = -SYMBOL; *ff++ = t[0];
            }
            else {
                *ff++ = 8+ARGHEAD; *ff++ = 0; FILLARG(ff);
                *ff++ = 8; *ff++ = SYMBOL; *ff++ = 4; *ff++ = t[0]; *ff++ = t[1];
```

```
                *ff++ = t[2]; *ff++ = 1; *ff++ = t[3] > 0 ? 3: -3;
            }
            t += 4;
        }
        while ( fff < tstop ) *ff++ = *fff++;
        fun[1] = ff - fun;
    }
    else if ( outspace > (fff-ff) ) {
/*
        Move the answer up by the required amount.
        Move the tail to its new location
        Move in things as for outspace == (fff-ff)
*/
        moveup = outspace-(fff-ff);
        ttt = tt + moveup;
        t = tt;
        while ( t > fff ) *--ttt = *--t;
        tt += moveup; tstop += moveup;
        fff += moveup;
        fun[1] += moveup;
        goto moveinto;
    }
    else {
moveinto:
        t = tstop;
        while ( t < tt ) {
            if ( t[0] == -1 ) { *ff++ = -SNUMBER; *ff++ = t[2]*t[3]; }
            else if ( t[1] == 1 && t[2] == 1 && t[3] == 1 ) {
                *ff++ = -SYMBOL; *ff++ = t[0];
            }
            else {
                *ff++ = 8+ARGHEAD; *ff++ = 0; FILLARG(ff);
                *ff++ = 8; *ff++ = SYMBOL; *ff++ = 4; *ff++ = t[0]; *ff++ = t[1];
                *ff++ = t[2]; *ff++ = 1; *ff++ = t[3] > 0 ? 3: -3;
            }
            t += 4;
        }
    }
    return(0);
}

/*
        #] RunImplode :
        #[ RunExplode :
*/

WORD RunExplode(PHEAD WORD *fun, WORD *args)
{
    WORD arg1, arg2, num1, num2, *tt, *tstop, totarg, *tonew, *newfun;
    WORD *ff, *f;
    int reverse = 0, iarg, i, numzero;
    if ( functions[fun[0]-FUNCTION].spec != 0 ) return(0);
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunExplode");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
/*
    Get the proper range in forward direction and the number of arguments
*/
    if ( arg1 > arg2 ) { num1 = arg2; num2 = arg1; reverse = 1; }
    else { num1 = arg1; num2 = arg2; }
    if ( num1 > totarg || num2 > totarg ) return(0);
    if ( tstop + AM.MaxTer > AT.WorkTop ) goto OverWork;
/*
    We will make the new function after the old one in the workspace
    Find the first argument
*/
    tonew = newfun = tstop;
    ff = fun + FUNHEAD; iarg = 0;
    while ( ff < tstop ) {
        iarg++;
        if ( iarg == num1 ) {
            i = ff - fun; f = fun;
            NCOPY(tonew,f,i)
            break;
        }
        NEXTARG(ff)
    }
/*
    We have reached the first argument to be done
*/
    while ( iarg <= num2 ) {
```

```
        if ( *ff == -SYMBOL || ( *ff == -SNUMBER && ff[1] == 0 ) )
            { *tonew++ = *ff++; *tonew++ = *ff++; }
        else if ( *ff == -SNUMBER ) {
            numzero = ABS(ff[1])-1;
            if ( reverse ) {
                *tonew++ = -SNUMBER; *tonew++ = ff[1] < 0 ? -1: 1;
                while ( numzero > 0 ) {
                    *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
                }
            }
            else {
                while ( numzero > 0 ) {
                    *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
                }
                *tonew++ = -SNUMBER; *tonew++ = ff[1] < 0 ? -1: 1;
            }
            ff += 2;
        }
        else if ( *ff < 0 ) { return(0); }
        else {
            if ( *ff != ARGHEAD+8 || ff[ARGHEAD] != 8
                 || ff[ARGHEAD+1] != SYMBOL || ABS(ff[ARGHEAD+7]) != 3
                 || ff[ARGHEAD+6] != 1 ) return(0);
            numzero = ff[ARGHEAD+5];
            if ( numzero >= MAXPOSITIVE4 ) return(0);
            numzero--;
            if ( reverse ) {
                if ( ff[ARGHEAD+7] > 0 ) { *tonew++ = -SNUMBER; *tonew++ = 1; }
                else {
                    *tonew++ = ARGHEAD+8; *tonew++ = 0; FILLARG(tonew)
                    *tonew++ = 8; *tonew++ = SYMBOL; *tonew++ = ff[ARGHEAD+3];
                    *tonew++ = ff[ARGHEAD+4]; *tonew++ = 1; *tonew++ = 1;
                    *tonew++ = -3;
                }
                while ( numzero > 0 ) {
                    *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
                }
            }
            else {
                while ( numzero > 0 ) {
                    *tonew++ = -SNUMBER; *tonew++ = 0; numzero--;
                }
                *tonew++ = ARGHEAD+8; *tonew++ = 0; FILLARG(tonew)
                *tonew++ = 8; *tonew++ = SYMBOL; *tonew++ = 4;
                *tonew++ = ff[ARGHEAD+3]; *tonew++ = ff[ARGHEAD+4];
                *tonew++ = 1; *tonew++ = 1;
                if ( ff[ARGHEAD+7] > 0 ) *tonew++ = 3;
                else                     *tonew++ = -3;
            }
            ff += *ff;
        }
        if ( tonew > AT.WorkTop ) goto OverWork;
        iarg++;
    }
/*
    Copy the tail, settle the size and copy the whole thing back.
*/
    while ( ff < tstop ) *tonew++ = *ff++;
    i = newfun[1] = tonew-newfun;
    NCOPY(fun,newfun,i)
    return(0);
OverWork:;
    MLOCK(ErrorMessageLock);
    MesWork();
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] RunExplode :
        #[ RunPermute :
*/

WORD RunPermute(PHEAD WORD *fun, WORD *args, WORD *info)
{
    WORD *tt, totarg, *tstop, arg1, arg2, n, num, i, *f, *f1, *f2, *infostop;
    WORD *in, *iw, withdollar;
    DOLLARS d;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunPermute");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
      while ( tt < tstop ) { totarg++; NEXTARG(tt); }
```

```
        arg1 = 1; arg2 = totarg;
/*
        We need to:
          1: get pointers to the arguments
          2: permute the pointers
          3: copy the arguments to safe territory in the new order
          4: copy this new order back in situ.
*/
        num = arg2-arg1+1;
        WantAddPointers(num); /* Guarantees the presence of enough pointers */
        f = fun+FUNHEAD; n = 1; i = 0;
        while ( n < arg1 ) { n++; NEXTARG(f) }
        f1 = f;
        while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
/*
        Now the permutations
*/
        info++;
        while ( *info ) {
          infostop = info + *info;
          info++;
          if ( *info > totarg ) return(0);
/*
          Now we have a look whether there are dollar variables to be expanded
          We also shift out out all values that are out of range.
*/
          withdollar = 0;  in = info;
          while ( in < infostop ) {
              if ( *in < 0 ) { /* Dollar variable -(number+1) */
                  d = Dollars - *in - 1;
#ifdef WITHPTHREADS
                  {
                      int nummodopt, dtype = -1, numdollar = -*in-1;
                      if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
                          for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                              if ( numdollar == ModOptdollars[nummodopt].number ) break;
                          }
                          if ( nummodopt < NumModOptdollars ) {
                              dtype = ModOptdollars[nummodopt].type;
                              if ( dtype == MODLOCAL ) {
                                  d = ModOptdollars[nummodopt].dstruct+AT.identity;
                              }
                              else {
                                  LOCK(d->pthreadslockread);
                              }
                          }
                      }
                  }
#endif
                  if ( ( d->type == DOLNUMBER || d->type == DOLTERMS )
                   && d->where[0] == 4 && d->where[4] == 0 ) {
                      if ( d->where[3] < 0 || d->where[2] != 1 || d->where[1] > totarg ) return(0);
                  }
                  else if ( d->type == DOLWILDARGS ) {
                      iw = d->where+1;
                      while ( *iw ) {
                          if ( *iw == -SNUMBER ) {
                              if ( iw[1] <= 0 || iw[1] > totarg ) return(0);
                          }
                          else goto IllType;
                          iw += 2;
                      }
                  }
                  else {
IllType:
                      MLOCK(ErrorMessageLock);
                      MesPrint("Illegal type of $-variable in RunPermute");
                      MUNLOCK(ErrorMessageLock);
                      Terminate(-1);
                  }
                  withdollar++;
              }
              else if ( *in > totarg ) return(0);
              in++;
          }
          if ( withdollar ) { /* We need some space for a copy */
              WORD *incopy, *tocopy;
              incopy = TermMalloc("RunPermute");
              tocopy = incopy+1; in = info;
              while ( in < infostop ) {
                  if ( *in < 0 ) {
                      d = Dollars - *in - 1;
#ifdef WITHPTHREADS
                  {
                      int nummodopt, dtype = -1, numdollar = -*in-1;
                      if ( AS.MultiThreaded && ( AC.mparallelflag == PARALLELFLAG ) ) {
                          for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
```

```
                            if ( numdollar == ModOptdollars[nummodopt].number ) break;
                        }
                        if ( nummodopt < NumModOptdollars ) {
                            dtype = ModOptdollars[nummodopt].type;
                            if ( dtype == MODLOCAL ) {
                                d = ModOptdollars[nummodopt].dstruct+AT.identity;
                            }
                            else {
                                LOCK(d->pthreadslockread);
                            }
                        }
                    }
                }
#endif
                    if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
                        *tocopy++ = d->where[1] - 1;
                    }
                    else if ( d->type == DOLWILDARGS ) {
                        iw = d->where+1;
                        while ( *iw ) {
                            *tocopy++ = iw[1] - 1;
                            iw += 2;
                        }
                    }
                    in++;
                }
                else *tocopy++ = *in++;
            }
            *tocopy = 0;
            *incopy = tocopy - incopy;
            in = incopy+1;
            tt = AT.pWorkSpace[AT.pWorkPointer+*in];
            in++;
            while ( in < tocopy ) {
                if ( *in > totarg ) return(0);
                AT.pWorkSpace[AT.pWorkPointer+in[-1]] = AT.pWorkSpace[AT.pWorkPointer+*in];
                in++;
            }
            AT.pWorkSpace[AT.pWorkPointer+in[-1]] = tt;
            TermFree(incopy,"RunPermute");
            info = infostop;
        }
        else {
            tt = AT.pWorkSpace[AT.pWorkPointer+*info];
            info++;
            while ( info < infostop ) {
                if ( *info > totarg ) return(0);
                AT.pWorkSpace[AT.pWorkPointer+info[-1]] = AT.pWorkSpace[AT.pWorkPointer+*info];
                info++;
            }
            AT.pWorkSpace[AT.pWorkPointer+info[-1]] = tt;
        }
      }
/*
      info++;
      while ( *info ) {
        infostop = info + *info;
        info++;
        if ( *info > totarg ) return(0);
        tt = AT.pWorkSpace[AT.pWorkPointer+*info];
        info++;
        while ( info < infostop ) {
            if ( *info > totarg ) return(0);
            AT.pWorkSpace[AT.pWorkPointer+info[-1]] = AT.pWorkSpace[AT.pWorkPointer+*info];
            info++;
        }
        AT.pWorkSpace[AT.pWorkPointer+info[-1]] = tt;
      }
*/
/*
      And the final cleanup
*/
      if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
      f2 = tstop;
      for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
      i = f2 - tstop;
      NCOPY(f1,tstop,i)
    }
    else {  /* tensors */
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop-tt;
      arg1 = 1; arg2 = totarg;
      num = arg2-arg1+1;
      WantAddPointers(num); /* Guarantees the presence of enough pointers */
      f = fun+FUNHEAD; n = 1; i = 0;
      while ( n < arg1 ) { n++; f++; }
      f1 = f;
      while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; f++; }
```

```
/*
      Now the permutations
*/
      info++;
      while ( *info ) {
        infostop = info + *info;
        info++;
        if ( *info > totarg ) return(0);
        tt = AT.pWorkSpace[AT.pWorkPointer+*info];
        info++;
        while ( info < infostop ) {
            if ( *info > totarg ) return(0);
            AT.pWorkSpace[AT.pWorkPointer+info[-1]] = AT.pWorkSpace[AT.pWorkPointer+*info];
            info++;
        }
        AT.pWorkSpace[AT.pWorkPointer+info[-1]] = tt;
      }
/*
      And the final cleanup
*/
      if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
      f2 = tstop;
      for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; *f2++= *f++; }
      i = f2 - tstop;
      NCOPY(f1,tstop,i)
    }
    return(0);
OverWork:;
    MLOCK(ErrorMessageLock);
    MesWork();
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] RunPermute :
        #[ RunReverse :
*/

WORD RunReverse(PHEAD WORD *fun, WORD *args)
{
    WORD *tt, totarg, *tstop, arg1, arg2, n, num, i, *f, *f1, *f2, i1, i2;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunReverse");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
      while ( tt < tstop ) { totarg++; NEXTARG(tt); }
      if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
/*
      We need to:
        1: get pointers to the arguments
        2: reverse the order of the pointers
        3: copy the arguments to safe territory in the new order
        4: copy this new order back in situ.
*/
      if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
      if ( arg2 > totarg ) return(0);

      num = arg2-arg1+1;
      WantAddPointers(num); /* Guarantees the presence of enough pointers */
      f = fun+FUNHEAD; n = 1; i = 0;
      while ( n < arg1 ) { n++; NEXTARG(f) }
      f1 = f;
      while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
      i1 = i-1; i2 = 0;
      while ( i1 > i2 ) {
        tt = AT.pWorkSpace[AT.pWorkPointer+i1];
        AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
        AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
        i1--; i2++;
      }
      if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
      f2 = tstop;
      for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
      i = f2 - tstop;
      NCOPY(f1,tstop,i)
    }
    else {  /* Tensors */
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop - tt;
      if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
/*
      We need to:
        1: get pointers to the arguments
```

```
        2: reverse the order of the pointers
        3: copy the arguments to safe territory in the new order
        4: copy this new order back in situ.
*/
      if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
      if ( arg2 > totarg ) return(0);

      num = arg2-arg1+1;
      WantAddPointers(num); /* Guarantees the presence of enough pointers */
      f = fun+FUNHEAD; n = 1; i = 0;
      while ( n < arg1 ) { n++; f++; }
      f1 = f;
      while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; f++; }
      i1 = i-1; i2 = 0;
      while ( i1 > i2 ) {
        tt = AT.pWorkSpace[AT.pWorkPointer+i1];
        AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
        AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
        i1--; i2++;
      }
      if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
      f2 = tstop;
      for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; *f2++ = *f++; }
      i = f2 - tstop;
      NCOPY(f1,tstop,i)
    }
    return(0);
OverWork:;
    MLOCK(ErrorMessageLock);
    MesWork();
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] RunReverse :
        #[ RunDedup :
*/

WORD RunDedup(PHEAD WORD *fun, WORD *args)
{
    WORD *tt, totarg, *tstop, arg1, arg2, n, i, j,k, *f, *f1, *f2, *fd, *fstart;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunDedup");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
      while ( tt < tstop ) { totarg++; NEXTARG(tt); }
      if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);

      if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
      if ( arg2 > totarg ) return(0);

      f = fun+FUNHEAD; n = 1;
      while ( n < arg1 ) { n++; NEXTARG(f) }
      f1 = f; // fast forward to first element in range
      i = 0; // new argument count
      fstart = f1;

      for (; n <= arg2; n++ ) {
        f2 = fstart;
        for ( j = 0; j < i; j++ ) { // check all previous terms
            fd = f2;
            NEXTARG(fd)
            for ( k = 0; k < fd-f2; k++ ) // byte comparison of args
                if ( f2[k] != f[k] ) break;

            if ( k == fd-f2 ) break; // duplicate arg
            f2 = fd;
        }

        if ( j == i ) {
            // unique factor, copy in situ
            COPY1ARG(f1,f)
            i++;
        } else {
            NEXTARG(f)
        }
      }

      // move the terms from after the range
      for (j = n; j <= totarg; j++) {
        COPY1ARG(f1,f)
      }
```

```

      fun[1] = f1 - fun; // resize function
    }
    else {  /* Tensors */
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop - tt;
      if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);

      if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
      if ( arg2 > totarg ) return(0);

      f = fun+FUNHEAD;
      i = arg1; // new argument count
      n = i;

      for (; n <= arg2; n++ ) {
        for ( j = arg1; j < i; j++ ) { // check all previous terms
            if ( f[n-1] == f[j-1] ) break; // duplicate arg
        }

        if ( j == i ) {
            // unique factor, copy in situ
            f[i-1] = f[n-1];
            i++;
        }
      }

      // move the terms from after the range
      for (j = n; j <= totarg; j++, i++) {
        f[i-1] = f[j-1];
      }

      fun[1] = f + i - 1 - fun; // resize function
    }
    return(0);
}

/*
        #] RunDedup :
        #[ RunCycle :
*/

WORD RunCycle(PHEAD WORD *fun, WORD *args, WORD *info)
{
    WORD *tt, totarg, *tstop, arg1, arg2, n, num, i, j, *f, *f1, *f2, x, ncyc, cc;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunCycle");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    ncyc = info[1];
    if ( ncyc >= MAXPOSITIVE2 ) { /* $ variable */
        ncyc -= MAXPOSITIVE2;
        if ( ncyc >= MAXPOSITIVE4 ) {
            ncyc -= MAXPOSITIVE4; /* -$ */
            cc = -1;
        }
        else cc = 1;
        ncyc = DolToNumber(BHEAD ncyc);
        if ( AN.ErrorInDollar ) {
            MesPrint(" Error in Dollar variable in transform,cycle()=$");
            return(-1);
        }
        if ( ncyc >= MAXPOSITIVE4 || ncyc <= -MAXPOSITIVE4 ) {
            MesPrint(" Illegal value from Dollar variable in transform,cycle()=$");
            return(-1);
        }
        ncyc *= cc;
    }
    if ( functions[fun[0]-FUNCTION].spec != TENSORFUNCTION ) {
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
      while ( tt < tstop ) { totarg++; NEXTARG(tt); }
      if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
      if ( arg1 > arg2 ) { n = arg1; arg1 = arg2; arg2 = n; }
      if ( arg2 > totarg ) return(0);
/*
      We need to:
        1: get pointers to the arguments
        2: cycle the pointers
        3: copy the arguments to safe territory in the new order
        4: copy this new order back in situ.
*/
      num = arg2-arg1+1;
      WantAddPointers(num); /* Guarantees the presence of enough pointers */
      f = fun+FUNHEAD; n = 1; i = 0;
      while ( n < arg1 ) { n++; NEXTARG(f) }
      f1 = f;
```

```
      while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
/*
      Now the cycle(s). First minimize the number of cycles.
*/
      x = ncyc;
      if ( x >= i ) {
        x %= i;
        if ( x > i/2 ) x -= i;
      }
      else if ( x <= -i ) {
        x = -((-x) % i);
        if ( x <= -i/2 ) x += i;
      }
      while ( x ) {
        if ( x > 0 ) {
            tt = AT.pWorkSpace[AT.pWorkPointer+i-1];
            for ( j = i-1; j > 0; j-- )
                AT.pWorkSpace[AT.pWorkPointer+j] = AT.pWorkSpace[AT.pWorkPointer+j-1];
            AT.pWorkSpace[AT.pWorkPointer] = tt;
            x--;
        }
        else {
            tt = AT.pWorkSpace[AT.pWorkPointer];
            for ( j = 1; j < i; j++ )
                AT.pWorkSpace[AT.pWorkPointer+j-1] = AT.pWorkSpace[AT.pWorkPointer+j];
            AT.pWorkSpace[AT.pWorkPointer+j-1] = tt;
            x++;
        }
      }
/*
      And the final cleanup
*/
      if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
      f2 = tstop;
      for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
      i = f2 - tstop;
      NCOPY(f1,tstop,i)
    }
    else {  /* Tensors */
      tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = tstop - tt;
      if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
      if ( arg1 > arg2 ) { n = arg1; arg1 = arg2; arg2 = n; }
      if ( arg2 > totarg ) return(0);
/*
      We need to:
        1: get pointers to the arguments
        2: cycle the pointers
        3: copy the arguments to safe territory in the new order
        4: copy this new order back in situ.
*/
      num = arg2-arg1+1;
      WantAddPointers(num); /* Guarantees the presence of enough pointers */
      f = fun+FUNHEAD; n = 1; i = 0;
      while ( n < arg1 ) { n++; f++; }
      f1 = f;
      while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; f++; }
/*
      Now the cycle(s). First minimize the number of cycles.
*/
      x = ncyc;
      if ( x >= i ) {
        x %= i;
        if ( x > i/2 ) x -= i;
      }
      else if ( x <= -i ) {
        x = -((-x) % i);
        if ( x <= -i/2 ) x += i;
      }
      while ( x ) {
        if ( x > 0 ) {
            tt = AT.pWorkSpace[AT.pWorkPointer+i-1];
            for ( j = i-1; j > 0; j-- )
                AT.pWorkSpace[AT.pWorkPointer+j] = AT.pWorkSpace[AT.pWorkPointer+j-1];
            AT.pWorkSpace[AT.pWorkPointer] = tt;
            x--;
        }
        else {
            tt = AT.pWorkSpace[AT.pWorkPointer];
            for ( j = 1; j < i; j++ )
                AT.pWorkSpace[AT.pWorkPointer+j-1] = AT.pWorkSpace[AT.pWorkPointer+j];
            AT.pWorkSpace[AT.pWorkPointer+j-1] = tt;
            x++;
        }
      }
/*
      And the final cleanup
*/
```

```
      if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
      f2 = tstop;
      for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; *f2++ = *f++; }
      i = f2 - tstop;
      NCOPY(f1,tstop,i)
    }
    return(0);
OverWork:;
    MLOCK(ErrorMessageLock);
    MesWork();
    MUNLOCK(ErrorMessageLock);
    return(-1);
}

/*
        #] RunCycle :
        #[ RunAddArg :
*/

WORD RunAddArg(PHEAD WORD *fun, WORD *args)
{
    WORD *tt, totarg, *tstop, arg1, arg2, n, num, *f, *f1, *f2;
    WORD scribble[10+ARGHEAD];
    LONG space;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunAddArg");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal attempt to add arguments of a tensor in AddArg");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    /* ignore functions with no arguments */
    if ( totarg == 0 ) return(0);
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
/*
    We need to:
        1: establish that we actually need to add something
        2: start a sort
        3: if needed, convert arguments to long arguments
        4: send (terms in) argument to StoreTerm
        5: EndSort and copy the result back into the function
    Note that the function is in the workspace, above the term and no
    relevant information is trailing it.
*/
    if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
    if ( arg2 > totarg ) return(0);
    num = arg2-arg1+1;
    if ( num == 1 ) return(0);
    f = fun+FUNHEAD; n = 1;
    while ( n < arg1 ) { n++; NEXTARG(f) }
    f1 = f;
    NewSort(BHEAD0);
    while ( n <= arg2 ) {
        if ( *f > 0 ) {
            f2 = f + *f; f += ARGHEAD;
            while ( f < f2 ) { StoreTerm(BHEAD f); f += *f; }
        }
        else if ( *f == -SNUMBER && f[1] == 0 ) {
            f+= 2;
        }
        else {
            ToGeneral(f,scribble,1);
            StoreTerm(BHEAD scribble);
            NEXTARG(f);
        }
        n++;
    }
    if ( EndSort(BHEAD tstop+ARGHEAD,1) < 0 ) return(-1);
    num = 0;
    f2 = tstop+ARGHEAD;
    while ( *f2 ) { f2 += *f2; num++; }
    *tstop = f2-tstop;
    for ( n = 1; n < ARGHEAD; n++ ) tstop[n] = 0;
    if ( num == 1 && ToFast(tstop,tstop) == 1 ) {
        f2 = tstop; NEXTARG(f2);
    }
    if ( *tstop == ARGHEAD ) {
        *tstop = -SNUMBER; tstop[1] = 0;
        f2 = tstop+2;
    }
```

```
/*
    Copy the trailing arguments after the new argument, then copy the whole back.
*/
    while ( f < tstop ) *f2++ = *f++;
    while ( f < f2 ) *f1++ = *f++;
    space = f1 - fun;
    if ( (space+8)*sizeof(WORD) > (UWORD)AM.MaxTer ) {
        MLOCK(ErrorMessageLock);
        MesWork();
        MUNLOCK(ErrorMessageLock);
        return(-1);
    }
    fun[1] = (WORD)space;
    return(0);
}

/*
        #] RunAddArg :
        #[ RunMulArg :
*/

WORD RunMulArg(PHEAD WORD *fun, WORD *args)
{
    WORD *t, totarg, *tstop, arg1, arg2, n, *f, nb, *m, i, *w;
    WORD *scratch, argbuf[20], argsize, *where, *newterm;
    LONG oldcpointer_pos;
    CBUF *C = cbuf + AT.ebufnum;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunMulArg");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( functions[fun[0]-FUNCTION].spec == TENSORFUNCTION ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal attempt to multiply arguments of a tensor in MulArg");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    t = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( t < tstop ) { totarg++; NEXTARG(t); }
    /* ignore functions with no arguments */
    if ( totarg == 0 ) return(0);
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
    if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
    if ( arg1 > totarg ) return(0);
    if ( arg2 < 1 ) return(0);
    if ( arg1 < 1 ) arg1 = 1;
    if ( arg2 > totarg ) arg2 = totarg;
    if ( arg1 == arg2 ) return(0);
/*
    Now we move the arguments to a compiler buffer
    Then we create a term in the workspace that is the product of
    subexpression pointers to the objects in the compiler buffer.
    Next we let Generator work out that term.
    Finally we pick up the results from EndSort and put it in the function.
*/
    f = fun+FUNHEAD; n = 1;
    while ( n < arg1 ) { n++; NEXTARG(f) }
    t = f;
    if ( fun >= AT.WorkSpace && fun < AT.WorkTop ) {
        if ( AT.WorkPointer < fun+fun[1] ) AT.WorkPointer = fun+fun[1];
    }
    scratch = AT.WorkPointer;
    w = scratch+1;
    oldcpointer_pos = C->Pointer-C->Buffer;
    nb = C->numrhs;
    while ( n <= arg2 ) {
        if ( *t > 0 ) {
            argsize = *t - ARGHEAD; where = t + ARGHEAD; t += *t;
        }
        else if ( *t <= -FUNCTION ) {
            argbuf[0] = FUNHEAD+4; argbuf[1] = -*t++; argbuf[2] = FUNHEAD;
            for ( i = 2; i < FUNHEAD; i++ ) argbuf[i+1] = 0;
            argbuf[FUNHEAD+1] = 1;
            argbuf[FUNHEAD+2] = 1;
            argbuf[FUNHEAD+3] = 3;
            argsize = argbuf[0];
            where = argbuf;
        }
        else if ( *t == -SYMBOL ) {
            argbuf[0] = 8; argbuf[1] = SYMBOL; argbuf[2] = 4;
            argbuf[3] = t[1]; argbuf[4] = 1;
            argbuf[5] = 1; argbuf[6] = 1; argbuf[7] = 3;
            argsize = 8; t += 2;
            where = argbuf;
        }
```

```
        else if ( *t == -VECTOR || *t == -MINVECTOR ) {
            argbuf[0] = 7; argbuf[1] = INDEX; argbuf[2] = 3;
            argbuf[3] = t[1];
            argbuf[4] = 1; argbuf[5] = 1;
            if ( *t == -MINVECTOR ) argbuf[6] = -3;
            else argbuf[6] = 3;
            argsize = 7; t += 2;
            where = argbuf;
        }
        else if ( *t == -INDEX ) {
            argbuf[0] = 7; argbuf[1] = INDEX; argbuf[2] = 3;
            argbuf[3] = t[1];
            argbuf[4] = 1; argbuf[5] = 1; argbuf[6] = 3;
            argsize = 7; t += 2;
            where = argbuf;
        }
        else if ( *t == -SNUMBER ) {
            if ( t[1] < 0 ) {
                argbuf[0] = 4; argbuf[1] = -t[1]; argbuf[2] = 1; argbuf[3] = -3;
            }
            else {
                argbuf[0] = 4; argbuf[1] = t[1]; argbuf[2] = 1; argbuf[3] = 3;
            }
            argsize = 4; t += 2;
            where = argbuf;
        }
        else {
            /* unreachable */
            return(1);
        }
/*
        Now add the argbuf to AT.ebufnum
*/
        m = AddRHS(AT.ebufnum,1);
        while ( (m + argsize + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,17);
        for ( i = 0; i < argsize; i++ ) m[i] = where[i];
        m[i] = 0;
        C->Pointer = m + i + 1;
        n++;
        *w++ = SUBEXPRESSION; *w++ = SUBEXPSIZE; *w++ = C->numrhs; *w++ = 1;
        *w++ = AT.ebufnum; FILLSUB(w);
    }
    *w++ = 1; *w++ = 1; *w++ = 3;
    *scratch = w-scratch;
    AT.WorkPointer = w;
    NewSort(BHEAD0);
    Generator(BHEAD scratch,AR.Cnumlhs);
    newterm = AT.WorkPointer;
    EndSort(BHEAD newterm+ARGHEAD,1);
    C->Pointer = C->Buffer+oldcpointer_pos;
    C->numrhs = nb;
    w = newterm+ARGHEAD; while ( *w ) w += *w;
    *newterm = w-newterm; newterm[1] = 0;
    if ( ToFast(newterm,newterm) ) {
        if ( *newterm <= -FUNCTION ) w = newterm+1;
        else w = newterm+2;
    }
    while ( t < tstop ) *w++ = *t++;
    i = w - newterm;
    t = newterm; NCOPY(f,t,i);
    fun[1] = f-fun;
    AT.WorkPointer = scratch;
    if ( AT.WorkPointer > AT.WorkSpace && AT.WorkPointer < f ) AT.WorkPointer = f;
    return(0);
}

/*
        #] RunMulArg :
        #[ RunIsLyndon :

        Determines whether the range constitutes a Lyndon word.
        The two cases of ordering are distinguished by the order of
        the numbers of the arguments in the range.
*/

WORD RunIsLyndon(PHEAD WORD *fun, WORD *args, int par)
{
    WORD *tt, totarg, *tstop, arg1, arg2, arg, num, *f, n, i;
/*  WORD *f1; */
    WORD sign, i1, i2, retval;
    if ( fun[0] <= GAMMASEVEN && fun[0] >= GAMMA ) return(0);
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunIsLyndon");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
```

```
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
    if ( arg1 > totarg || arg2 > totarg ) return(-1);
/*
    Now make a list of the relevant arguments.
*/
    if ( arg1 == arg2 ) return(1);
    if ( arg2 < arg1 ) {    /* greater, rather than smaller */
        arg = arg1; arg1 = arg2; arg2 = arg; sign = 1;
    }
    else sign = 0;

    num = arg2-arg1+1;
    WantAddPointers(num);   /* Guarantees the presence of enough pointers */
    f = fun+FUNHEAD; n = 1; i = 0;
    while ( n < arg1 ) { n++; NEXTARG(f) }
/*  f1 = f; */
    while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
/*
    If sign == 1 we should alter the order of the pointers first
*/
    if ( sign ) {
        i1 = i-1; i2 = 0;
        while ( i1 > i2 ) {
            tt = AT.pWorkSpace[AT.pWorkPointer+i1];
            AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
            AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
            i1--; i2++;
        }
    }
/*
    The argument range is from f1 to f and the num pointers to the arguments
    are in AT.pWorkSpace[AT.pWorkPointer] to AT.pWorkSpace[AT.pWorkPointer+num-1]
*/
    for ( i1 = 1; i1 < num; i1++ ) {
        retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+i1],
                    AT.pWorkSpace[AT.pWorkPointer]);
        if ( retval > 0 ) continue;
        if ( retval < 0 ) return(0);
        for ( i2 = 1; i2 < num; i2++ ) {
            retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+(i1+i2)%num],
                    AT.pWorkSpace[AT.pWorkPointer+i2]);
            if ( retval < 0 ) return(0);
            if ( retval > 0 ) goto nexti1;
        }
/*
        If we come here the sequence is not unique.
*/
        return(0);
nexti1:;
    }
    return(1);
}

/*
        #] RunIsLyndon :
        #[ RunToLyndon :

        Determines whether the range constitutes a Lyndon word.
        If not, we rotate it to a Lyndon word. If this is not possible
        we return the noLyndon condition.
        The two cases of ordering are distinguished by the order of
        the numbers of the arguments in the range.
*/

WORD RunToLyndon(PHEAD WORD *fun, WORD *args, int par)
{
    WORD *tt, totarg, *tstop, arg1, arg2, arg, num, *f, *f1, *f2, n, i;
    WORD sign, i1, i2, retval, unique;
    if ( fun[0] <= GAMMASEVEN && fun[0] >= GAMMA ) return(0);
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunToLyndon");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
    if ( arg1 > totarg || arg2 > totarg ) return(-1);
/*
    Now make a list of the relevant arguments.
*/
    if ( arg1 == arg2 ) return(1);
    if ( arg2 < arg1 ) {    /* greater, rather than smaller */
        arg = arg1; arg1 = arg2; arg2 = arg; sign = 1;
```

```
    }
    else sign = 0;

    num = arg2-arg1+1;
    WantAddPointers((2*num));   /* Guarantees the presence of enough pointers */
    f = fun+FUNHEAD; n = 1; i = 0;
    while ( n < arg1 ) { n++; NEXTARG(f) }
    f1 = f;
    while ( n <= arg2 ) { AT.pWorkSpace[AT.pWorkPointer+i++] = f; n++; NEXTARG(f) }
/*
    If sign == 1 we should alter the order of the pointers first
*/
    if ( sign ) {
        i1 = i-1; i2 = 0;
        while ( i1 > i2 ) {
            tt = AT.pWorkSpace[AT.pWorkPointer+i1];
            AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
            AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
            i1--; i2++;
        }
    }
/*
    The argument range is from f1 to f and the num pointers to the arguments
    are in AT.pWorkSpace[AT.pWorkPointer] to AT.pWorkSpace[AT.pWorkPointer+num-1]
*/
    unique = 1;
    for ( i1 = 1; i1 < num; i1++ ) {
        retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+i1],
                            AT.pWorkSpace[AT.pWorkPointer]);
        if ( retval > 0 ) continue;
        if ( retval < 0 ) {
Rotate:;
/*
            Rotate so that i1 becomes the zero element. Then start again.
*/
            for ( i2 = 0; i2 < num; i2++ ) {
                AT.pWorkSpace[AT.pWorkPointer+num+i2] =
                    AT.pWorkSpace[AT.pWorkPointer+(i1+i2)%num];
            }
            for ( i2 = 0; i2 < num; i2++ ) {
                AT.pWorkSpace[AT.pWorkPointer+i2] =
                    AT.pWorkSpace[AT.pWorkPointer+i2+num];
            }
            i1 = 0;
            goto nexti1;
        }
        for ( i2 = 1; i2 < num; i2++ ) {
            retval = par * CompArg(AT.pWorkSpace[AT.pWorkPointer+(i1+i2)%num],
                            AT.pWorkSpace[AT.pWorkPointer+i2]);
            if ( retval < 0 ) goto Rotate;
            if ( retval > 0 ) goto nexti1;
        }
/*
        If we come here the sequence is not unique.
*/
        unique = 0;
nexti1:;
    }
    if ( sign ) {
        i1 = i-1; i2 = 0;
        while ( i1 > i2 ) {
            tt = AT.pWorkSpace[AT.pWorkPointer+i1];
            AT.pWorkSpace[AT.pWorkPointer+i1] = AT.pWorkSpace[AT.pWorkPointer+i2];
            AT.pWorkSpace[AT.pWorkPointer+i2] = tt;
            i1--; i2++;
        }
    }
/*
    Now rewrite the arguments into the proper order
*/
    if ( tstop+(f-f1) > AT.WorkTop ) goto OverWork;
    f2 = tstop;
    for ( i = 0; i < num; i++ ) { f = AT.pWorkSpace[AT.pWorkPointer+i]; COPY1ARG(f2,f) }
    i = f2 - tstop;
    NCOPY(f1,tstop,i)
/*
    The return value indicates whether we have a Lyndon word
*/
    return(unique);
OverWork:;
    MLOCK(ErrorMessageLock);
    MesWork();
    MUNLOCK(ErrorMessageLock);
    return(-2);
}

/*
```

```
        #] RunToLyndon :
        #[ RunDropArg :
*/

WORD RunDropArg(PHEAD WORD *fun, WORD *args)
{
    WORD *t, *tstop, *f, totarg, arg1, arg2, n;

    t = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( t < tstop ) { totarg++; NEXTARG(t); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
    if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
    if ( arg1 > totarg ) return(0);
    if ( arg2 < 1 ) return(0);
    if ( arg1 < 1 ) arg1 = 1;
    if ( arg2 > totarg ) arg2 = totarg;
    f = fun+FUNHEAD; n = 1;
    while ( n < arg1 ) { n++; NEXTARG(f) }
    t = f;
    while ( n <= arg2 ) { n++; NEXTARG(t) }
    while ( t < tstop ) *f++ = *t++;
    fun[1] = f-fun;
    return(0);
}

/*
        #] RunDropArg :
        #[ RunSelectArg :
*/

WORD RunSelectArg(PHEAD WORD *fun, WORD *args)
{
    WORD *t, *tstop, *f, *tt, totarg, arg1, arg2, n;

    t = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( t < tstop ) { totarg++; NEXTARG(t); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
    if ( arg2 < arg1 ) { n = arg1; arg1 = arg2; arg2 = n; }
    if ( arg1 > totarg ) return(0);
    if ( arg2 < 1 ) return(0);
    if ( arg1 < 1 ) arg1 = 1;
    if ( arg2 > totarg ) arg2 = totarg;
    f = fun+FUNHEAD; n = 1; t = f;
    while ( n < arg1 ) { n++; NEXTARG(t) }
    while ( n <= arg2 ) {
        tt = t; NEXTARG(tt)
        while ( t < tt ) *f++ = *t++;
        n++;
    }
    fun[1] = f-fun;
    return(0);
}

/*
        #] RunSelectArg :
        #[ RunZtoHArg :
*/

WORD RunZtoHArg(PHEAD WORD *fun, WORD *args)
{
    WORD *tt, totarg, *tstop, arg1, arg2, n, i, *f, *f1, sign = 0;
    WORD *t, *t1, *t2, *t3;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunZtoHArg.");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( functions[fun[0]-FUNCTION].spec != 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("The ZtoH transformation can only be executed on regular functions with nonzero integer arguments.");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
/*
    Check the arguments. Should be -SNUMBER x!=0
*/
    f = fun+FUNHEAD; n = 1;
    while ( n < arg1 ) { n++; NEXTARG(f) }
    f1 = f;
    for ( i = arg1; i <= arg2; i++, f += 2 ) {
        if ( *f != -SNUMBER || f[1] == 0 ) return(-1);
    }
```

```
/*
    Now we need a copy.
*/
    t = f1; t1 = t2 = tt = TermMalloc("RunZtoHArg");
    while ( t < f ) { *t1++ = *t++; *t1++ = *t++; }
    t = f1;
    while ( t2 < t1 ) {
        t += 2;
        if ( t2[1] < 0 ) {
            t3 = t;
            while ( t3 < f ) { t3[1] = -t3[1]; t3 += 2; }
        }
        t2 += 2;
    }
    TermFree(tt,"RunZtoHArg");
/*
    Now the overall sign.
*/
    while ( f1 < f ) { if ( f1[1] < 0 ) sign = 1-sign; f1 += 2; }
    return(sign);
}

/*
        #] RunZtoHArg :
        #[ RunHtoZArg :
*/

WORD RunHtoZArg(PHEAD WORD *fun, WORD *args)
{
    WORD *tt, totarg, *tstop, arg1, arg2, n, i, *f, *f1, *f2, sign = 0;
    WORD *t, *t1, *t2;
    if ( *args != ARGRANGE ) {
        MLOCK(ErrorMessageLock);
        MesPrint("Illegal range encountered in RunZtoHArg.");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    if ( functions[fun[0]-FUNCTION].spec != 0 ) {
        MLOCK(ErrorMessageLock);
        MesPrint("The HtoZ transformation can only be executed on regular functions with nonzero
integer arguments.");
        MUNLOCK(ErrorMessageLock);
        Terminate(-1);
    }
    tt = fun+FUNHEAD; tstop = fun+fun[1]; totarg = 0;
    while ( tt < tstop ) { totarg++; NEXTARG(tt); }
    if ( FindRange(BHEAD args,&arg1,&arg2,totarg) ) return(-1);
/*
    Check the arguments. Should be -SNUMBER x!=0
*/
    f = fun+FUNHEAD; n = 1;
    while ( n < arg1 ) { n++; NEXTARG(f) }
    f2 = f1 = f;
    for ( i = arg1; i <= arg2; i++, f += 2 ) {
        if ( *f != -SNUMBER || f[1] == 0 ) return(-1);
    }
/*
    First the overall sign.
*/
    while ( f2 < f ) { if ( f2[1] < 0 ) sign = 1-sign; f2 += 2; }
/*
    Now we need a copy.
*/
    t = f1; t1 = tt = TermMalloc("RunHtoZArg");
    while ( t < f ) { *t1++ = *t++; *t1++ = *t++; }
/*
    Now the transformation.
*/
    t = f1; t2 = tt + 2;
    while ( t2 < t1 ) {
        t += 2;
        if ( t2[-1] < 0 ) t[1] = -t[1];
        t2 += 2;
    }
    TermFree(tt,"RunHtoZArg");
    return(sign);
}

/*
        #] RunHtoZArg :
        #[ TestArgNum :

        Looks whether argument n is contained in any of the ranges
        specified in args. Args contains objects of the types
            ALLARGS
            NUMARG,num
            ARGRANGE,num1,num2
```

```
        The object MAKEARGS,num1,num2 is skipped
        Any other object terminates the range specifications.

        Currently only ARGRANGE is used (10-may-2016)
*/

int TestArgNum(int n, int totarg, WORD *args)
{
    GETIDENTITY
    WORD x1, x2;
    for(;;) {
        switch ( *args ) {
            case ALLARGS:
                return(1);
            case NUMARG:
                if ( n == args[1] ) return(1);
                if ( args[1] >= MAXPOSITIVE4 ) {
                    x1 = args[1]-MAXPOSITIVE4;
                    if ( totarg-x1 == n ) return(1);
                }
                args += 2;
                break;
            case ARGRANGE:
                if ( args[1] >= MAXPOSITIVE2 ) {
                    x1 = args[1] - MAXPOSITIVE2;
                    if ( x1 > MAXPOSITIVE4 ) {
                        x1 = x1 - MAXPOSITIVE4;
                        x1 = DolToNumber(BHEAD x1);
                        x1 = totarg - x1;
                    }
                    else {
                        x1 = DolToNumber(BHEAD x1);
                    }
                }
                else if ( args[1] >= MAXPOSITIVE4 ) {
                    x1 = totarg-(args[1]-MAXPOSITIVE4);
                }
                else x1 = args[1];
                if ( args[2] >= MAXPOSITIVE2 ) {
                    x2 = args[2] - MAXPOSITIVE2;
                    if ( x2 > MAXPOSITIVE4 ) {
                        x2 = x2 - MAXPOSITIVE4;
                        x2 = DolToNumber(BHEAD x2);
                        x2 = totarg - x2;
                    }
                    else {
                        x2 = DolToNumber(BHEAD x2);
                    }
                }
                else if ( args[2] >= MAXPOSITIVE4 ) {
                    x2 = totarg-(args[2]-MAXPOSITIVE4);
                }
                else x2 = args[2];
                if ( x1 >= x2 ) {
                    if ( n >= x2 && n <= x1 ) return(1);
                }
                else {
                    if ( n >= x1 && n <= x2 ) return(1);
                }
                args += 3;
                break;
            case MAKEARGS:
                args += 3;
                break;
            default:
                return(0);
        }
    }
}

/*
        #] TestArgNum :
        #[ PutArgInScratch :
*/

WORD PutArgInScratch(WORD *arg,UWORD *scrat)
{
    WORD size, *t, i;
    if ( *arg == -SNUMBER ) {
        scrat[0] = ABS(arg[1]);
        if ( arg[1] < 0 ) size = -1;
        else              size =  1;
    }
    else {
        t = arg+*arg-1;
        if ( *t < 0 ) { i = ((-*t)-1)/2; size = -i; }
        else          { i = (  *t -1)/2; size =  i; }
```

```
        t = arg+ARGHEAD+1;
        NCOPY(scrat,t,i);
    }
    return(size);
}

/*
        #] PutArgInScratch :
        #[ ReadRange :

        Comes in at the bracket and leaves at the = sign
        Ranges can be:
            #1,#2  with # numbers. If the second is smaller than the
                    first we work it backwards.
            first,#2 or #2,first
            #1,last  or last,#1
            first,last or last,first
        First is represented by 1. Last is represented by MAXPOSITIVE4.

        par = 0: we need the = after.
        par = 1: we need a , or '\0' after.
        par = 2: we need a :
*/

UBYTE *ReadRange(UBYTE *s, WORD *out, int par)
{
    UBYTE *in = s, *ss, c;
    LONG x1, x2;

    SKIPBRA3(in)
    if ( par == 0 && in[1] != '=' ) {
        MesPrint("&A range in this type of transform statement should be followed by an = sign");
        return(0);
    }
    else if ( par == 1 && in[1] != ',' && in[1] != '\0' ) {
        MesPrint("&A range in this type of transform statement should be followed by a comma or
end-of-statement");
        return(0);
    }
    else if ( par == 2 && in[1] != ':' ) {
        MesPrint("&A range in this type of transform statement should be followed by a :");
        return(0);
    }
    s++;
    if ( FG.cTable[*s] == 0 ) {
        ss = s; while ( FG.cTable[*s] == 0 ) s++;
        c = *s; *s = 0;
        if ( StrICmp(ss,(UBYTE *)"first") == 0 ) {
            *s = c;
            x1 = 1;
        }
        else if ( StrICmp(ss,(UBYTE *)"last") == 0 ) {
            *s = c;
            if ( c == '-' ) {
                s++;
                if ( *s == '$' ) {
                    s++; ss = s;
                    while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
                    c = *s; *s = 0;
                    if ( ( x1 = GetDollar(ss) ) < 0 ) goto Error;
                    *s = c;
                    x1 += MAXPOSITIVE2;
                }
                else {
                    x1 = 0;
                    while ( *s >= '0' && *s <= '9' ) {
                        x1 = 10*x1 + *s++ - '0';
                        if ( x1 >= MAXPOSITIVE4 ) {
                            MesPrint("&Fixed range indicator bigger than %l",(LONG)MAXPOSITIVE4);
                            return(0);
                        }
                    }
                }
                x1 += MAXPOSITIVE4;
            }
            else x1 = MAXPOSITIVE4;
        }
        else {
            MesPrint("&Illegal keyword inside range specification");
            return(0);
        }
    }
    else if ( FG.cTable[*s] == 1 ) {
        x1 = 0;
        while ( *s >= '0' && *s <= '9' ) {
            x1 = x1*10 + *s++ - '0';
            if ( x1 >= MAXPOSITIVE4 ) {
```

```
                MesPrint("&Fixed range indicator bigger than %l",(LONG)MAXPOSITIVE4);
                return(0);
            }
        }
    }
    else if ( *s == '$' ) {
        s++; ss = s;
        while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
        c = *s; *s = 0;
        if ( ( x1 = GetDollar(ss) ) < 0 ) goto Error;
        *s = c;
        x1 += MAXPOSITIVE2;
    }
    else {
        MesPrint("&Illegal character in range specification");
        return(0);
    }
    if ( *s != ',' ) {
        MesPrint("&A range is two indicators, separated by a comma or blank");
        return(0);
    }
    s++;
    if ( FG.cTable[*s] == 0 ) {
        ss = s; while ( FG.cTable[*s] == 0 ) s++;
        c = *s; *s = 0;
        if ( StrICmp(ss,(UBYTE *)"first") == 0 ) {
            *s = c;
            x2 = 1;
        }
        else if ( StrICmp(ss,(UBYTE *)"last") == 0 ) {
            *s = c;
            if ( c == '-' ) {
                s++;
                if ( *s == '$' ) {
                    s++; ss = s;
                    while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
                    c = *s; *s = 0;
                    if ( ( x2 = GetDollar(ss) ) < 0 ) goto Error;
                    *s = c;
                    x2 += MAXPOSITIVE2;
                }
                else {
                    x2 = 0;
                    while ( *s >= '0' && *s <= '9' ) {
                        x2 = 10*x2 + *s++ - '0';
                        if ( x2 >= MAXPOSITIVE4 ) {
                            MesPrint("&Fixed range indicator bigger than %l",(LONG)MAXPOSITIVE4);
                            return(0);
                        }
                    }
                }
                x2 += MAXPOSITIVE4;
            }
            else x2 = MAXPOSITIVE4;
        }
        else {
            MesPrint("&Illegal keyword inside range specification");
            return(0);
        }
    }
    else if ( FG.cTable[*s] == 1 ) {
        x2 = 0;
        while ( *s >= '0' && *s <= '9' ) {
            x2 = x2*10 + *s++ - '0';
            if ( x2 >= MAXPOSITIVE4 ) {
                MesPrint("&Fixed range indicator bigger than %l",(LONG)MAXPOSITIVE4);
                return(0);
            }
        }
    }
    else if ( *s == '$' ) {
        s++; ss = s;
        while ( FG.cTable[*s] == 0 || FG.cTable[*s] == 1 ) s++;
        c = *s; *s = 0;
        if ( ( x2 = GetDollar(ss) ) < 0 ) goto Error;
        *s = c;
        x2 += MAXPOSITIVE2;
    }
    else {
        MesPrint("&Illegal character in range specification");
        return(0);
    }
    if ( s < in ) {
        MesPrint("&A range is two indicators, separated by a comma or blank between parentheses");
        return(0);
    }
    out[0] = x1; out[1] = x2;
```

```
    return(in+1);
Error:
    MesPrint("&Undefined variable $%s in range",ss);
    return(0);
}

/*
        #] ReadRange :
        #[ FindRange :
*/

int FindRange(PHEAD WORD *args, WORD *arg1, WORD *arg2, WORD totarg)
{
    WORD n[2], fromlast, i;
    for ( i = 0; i < 2; i++ ) {
        n[i] = args[i+1];
        fromlast = 0;
        if ( n[i] >= MAXPOSITIVE2 ) { /* This is a dollar variable */
            n[i] -= MAXPOSITIVE2;
            if ( n[i] >= MAXPOSITIVE4 ) {
                fromlast = 1;
                n[i] -= MAXPOSITIVE4; /* Now we have the number of the dollar variable   */
            }
            n[i] = DolToNumber(BHEAD n[i]);
            if ( AN.ErrorInDollar ) {
                MLOCK(ErrorMessageLock);
                MesPrint("Illegal $ value in range while executing transform statement.");
                MUNLOCK(ErrorMessageLock);
                return(-1);
            }
            if ( fromlast ) n[i] = totarg-n[i];
        }
        else if ( n[i] >= MAXPOSITIVE4 ) { n[i] = totarg-(n[i]-MAXPOSITIVE4); }
        if ( n[i] <= 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Illegal non-positive value in range (%d) while executing transform statement.",
i+1);
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
    }
    *arg1 = n[0];
    *arg2 = n[1];
    return(0);
}

/*
        #] FindRange :
    #] Transform :
*/
```

4.152 unix.h File Reference

Macros

- #define LINEFEED '\n'
- #define CARRIAGERETURN 0x0D
- #define WITHPIPE
- #define WITHSYSTEM
- #define WITHEXTERNALCHANNEL
- #define TRAPSIGNALS
- #define P_term(code) exit((int)(code<0?-code:code))
- #define SEPARATOR '/'
- #define ALTSEPARATOR '/'
- #define PATHSEPARATOR ':'
- #define WITH_ENV
- #define WITH_ALARM

4.152.1 Detailed Description

Settings for Unix-like systems.

Definition in file unix.h.

4.152.2 Macro Definition Documentation

4.152.2.1 LINEFEED

```
#define LINEFEED '\n'
```

Definition at line 33 of file unix.h.

4.152.2.2 CARRIAGERETURN

```
#define CARRIAGERETURN 0x0D
```

Definition at line 34 of file unix.h.

4.152.2.3 WITHPIPE

```
#define WITHPIPE
```

Definition at line 36 of file unix.h.

4.152.2.4 WITHSYSTEM

```
#define WITHSYSTEM
```

Definition at line 37 of file unix.h.

4.152.2.5 WITHEXTERNALCHANNEL

```
#define WITHEXTERNALCHANNEL
```

Definition at line 46 of file unix.h.

4.152.2.6 TRAPSIGNALS

```
#define TRAPSIGNALS
```

Definition at line 49 of file unix.h.

4.152.2.7 P_term

```
#define P_term(
            code ) exit((int)(code<0?-code:code))
```

Definition at line 52 of file unix.h.

4.152.2.8 SEPARATOR

```
#define SEPARATOR '/'
```

Definition at line 54 of file unix.h.

4.152.2.9 ALTSEPARATOR

```
#define ALTSEPARATOR '/'
```

Definition at line 55 of file unix.h.

4.152.2.10 PATHSEPARATOR

```
#define PATHSEPARATOR ':'
```

Definition at line 56 of file unix.h.

4.152.2.11 WITH_ENV

```
#define WITH_ENV
```

Definition at line 57 of file unix.h.

4.152.2.12 WITH_ALARM

```
#define WITH_ALARM
```

Definition at line 58 of file unix.h.

4.153 unix.h

Go to the documentation of this file.

```
00001
00006 /* #[ License : */
00007 /*
00008  *   Copyright (C) 1984-2023 J.A.M. Vermaseren
00009  *   When using this file you are requested to refer to the publication
00010  *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
00011  *   This is considered a matter of courtesy as the development was paid
00012  *   for by FOM the Dutch physics granting agency and we would like to
00013  *   be able to track its scientific use to convince FOM of its value
00014  *   for the community.
00015  *
00016  *   This file is part of FORM.
00017  *
00018  *   FORM is free software: you can redistribute it and/or modify it under the
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00020  *   Foundation, either version 3 of the License, or (at your option) any later
00021  *   version.
00022  *
00023  *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
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00025  *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
00026  *   details.
00027  *
```
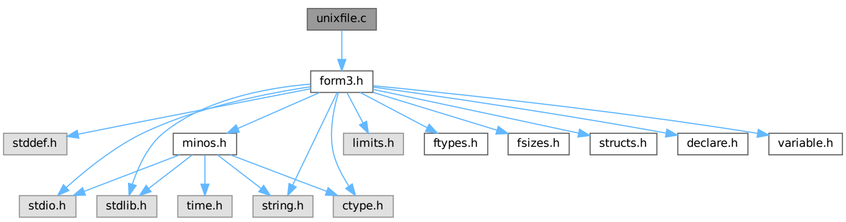

```
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

#define LINEFEED '\n'
#define CARRIAGERETURN 0x0D

#define WITHPIPE
#define WITHSYSTEM

/*[13jul2005 mt]:*/
/*With SAFESIGNAL defined, write() and read() syscalls are wrapped by
the errno checkup*/
/*#define SAFESIGNAL*/
/*:[13jul2005 mt]*/

/*[29apr2004 mt]:*/
#define WITHEXTERNALCHANNEL
/*
*/
#define TRAPSIGNALS
/*:[29apr2004 mt]*/

#define P_term(code)    exit((int)(code<0?-code:code))

#define SEPARATOR '/'
#define ALTSEPARATOR '/'
#define PATHSEPARATOR ':'
#define WITH_ENV
#define WITH_ALARM
```

4.154 unixfile.c File Reference

`#include "form3.h"`

Include dependency graph for unixfile.c:



4.154.1 Detailed Description

The interface to a fast variety of file routines in the unix system.

Definition in file unixfile.c.

4.155 unixfile.c

Go to the documentation of this file.

```

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
```

```
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
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 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :

    File with direct interface to the UNIX functions.
    This makes a big difference in speed!
*/
#include "form3.h"


/*[13jul2005 mt]:*/
#ifdef SAFESIGNAL
/*[15oct2004 mt]:*/
/*To access errno variable and EINTR constant:*/
#include <errno.h>
/*:[15oct2004 mt]*/
#endif
/*:[13jul2005 mt]*/

#ifdef UFILES

FILES TheStdout = { 1 };
FILES *Ustdout = &TheStdout;

#ifdef GPP
extern "C" open();
extern "C" close();
extern "C" read();
extern "C" write();
extern "C" lseek();
#endif

/*
    #] Includes :
    #[ Uopen :
*/

FILES *Uopen(char *filename, char *mode)
{
    FILES *f = (FILES *)Malloc1(sizeof(FILES),"Uopen");
    int flags = 0, rights = 0644;
    while ( *mode ) {
        if      ( *mode == 'r' ) { flags |= O_RDONLY; }
        else if ( *mode == 'w' ) { flags |= O_CREAT | O_TRUNC; }
        else if ( *mode == 'a' ) { flags |= O_APPEND; }
        else if ( *mode == 'b' ) { }
        else if ( *mode == '+' ) { flags |= O_RDWR; }
        mode++;
    }
    f->descriptor = open(filename,flags,rights);
    if ( f->descriptor >= 0 ) return(f);
    if ( ( flags & O_APPEND ) != 0 ) {
        flags |= O_CREAT;
        f->descriptor = open(filename,flags,rights);
        if ( f->descriptor >= 0 ) return(f);
    }
    M_free(f,"Uopen");
    return(0);
}

/*
    #] Uopen :
    #[ Uclose :
*/

int Uclose(FILES *f)
{
```
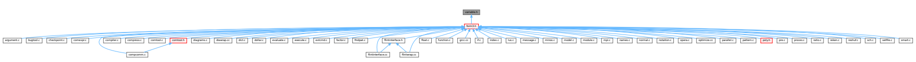

```
    int retval;
    if ( f ) {
        retval = close(f->descriptor);
        M_free(f,"Uclose");
        return(retval);
    }
    return(EOF);
}

/*
    #] Uclose :
    #[ Uread :
*/


size_t Uread(char *ptr, size_t size, size_t nobj, FILES *f)
{
/*[13jul2005 mt]:*/
#ifdef SAFESIGNAL

/*[15oct2004 mt]:*/
/*Operation read() can be interrupted by a signal. Note, this is rather unlikely,
so we do not save size*nobj for future attempts*/
size_t ret;
    /*If the syscall is interrupted by a signal before it
        succeeded in getting any progress, it must be repeated:*/
    while( ( (ret=read(f->descriptor,ptr,size*nobj))<1)&&(errno == EINTR) );
    return(ret);
#else
#ifdef DEEPDEBUG
    {
        POSITION pos;
        SETBASEPOSITION(pos,lseek(f->descriptor,0L,SEEK_CUR));
        MesPrint("handle %d: reading %ld bytes from position %p\n",f->descriptor,size*nobj,pos);
    }
#endif

    return(read(f->descriptor,ptr,size*nobj));

#endif
/*:[15oct2004 mt]*/
/*:[13jul2005 mt]*/
}

/*
    #] Uread :
    #[ Uwrite :
*/

size_t Uwrite(char *ptr, size_t size, size_t nobj, FILES *f)
{
/*[13jul2005 mt]:*/
#ifdef SAFESIGNAL
/*[15oct2004 mt]:*/
/*Operation write() can be interrupted by a signal. */
size_t ret;
size_t thesize=size*nobj;
    /*If the syscall is interrupted by a signal before it
        succeeded in getting any progress, it must be repeated:*/
    while( ( (ret=write(f->descriptor,ptr,thesize))<1 )&&(errno == EINTR) );

    return(ret);
#else
#ifdef DEEPDEBUG
    {
        POSITION pos;
        SETBASEPOSITION(pos,lseek(f->descriptor,0L,SEEK_CUR));
        MesPrint("handle %d: writing %ld bytes to position %p\n",f->descriptor,size*nobj,pos);
    }
#endif
    return(write(f->descriptor,ptr,size*nobj));

/*:[15oct2004 mt]*/
#endif
/*:[13jul2005 mt]*/
}

/*
    #] Uwrite :
    #[ Useek :
*/

int Useek(FILES *f, off_t offset, int origin)
{
    if ( f && ( lseek(f->descriptor,offset,origin) >= 0 ) ) return(0);
    return(-1);
}
```

```
/*
    #] Useek :
    #[ Utell :
*/

off_t Utell(FILES *f)
{
    if ( f ) return((off_t)lseek(f->descriptor,0L,SEEK_CUR));
    else return(-1);
}

/*
    #] Utell :
    #[ Uflush :
*/

void Uflush(FILES *f) { DUMMYUSE(f); }

/*
    #] Uflush :
    #[ Ugetpos :
*/

int Ugetpos(FILES *f, fpos_t *ptr)
{
    DUMMYUSE(f); DUMMYUSE(ptr);
    return(-1);
}

/*
    #] Ugetpos :
    #[ Usetpos :
*/

int Usetpos(FILES *f,fpos_t *ptr)
{
    DUMMYUSE(f); DUMMYUSE(ptr);
    return(-1);
}

/*
    #] Usetpos :
    #[ Usetbuf :
*/

void Usetbuf(FILES *f, char *ptr) { DUMMYUSE(f); DUMMYUSE(ptr); }

/*
    #] Usetbuf :
*/
#endif
```

4.156 variable.h File Reference

This graph shows which files directly or indirectly include this file:



Macros

- #define chartype FG.cTable
- #define Procedures ((PROCEDURE ∗)(AP.ProcList.lijst))
- #define NumProcedures AP.ProcList.num
- #define MaxProcedures AP.ProcList.maxnum
- #define DoLoops ((DOLOOP ∗)(AP.LoopList.lijst))
- #define NumDoLoops AP.LoopList.num
- #define MaxDoLoops AP.LoopList.maxnum
- #define PreVar ((PREVAR ∗)(AP.PreVarList.lijst))

- #define NumPre AP.PreVarList.num
- #define MaxNumPre AP.PreVarList.maxnum
- #define SetElements ((WORD ∗)(AC.SetElementList.lijst))
- #define Sets ((SETS)(AC.SetList.lijst))
- #define functions ((FUNCTIONS)(AC.FunctionList.lijst))
- #define indices ((INDICES)(AC.IndexList.lijst))
- #define symbols ((SYMBOLS)(AC.SymbolList.lijst))
- #define vectors ((VECTORS)(AC.VectorList.lijst))
- #define tablebases ((DBASE ∗)(AC.TableBaseList.lijst))
- #define NumFunctions AC.FunctionList.num
- #define NumIndices AC.IndexList.num
- #define NumSymbols AC.SymbolList.num
- #define NumVectors AC.VectorList.num
- #define NumSets AC.SetList.num
- #define NumSetElements AC.SetElementList.num
- #define NumTableBases AC.TableBaseList.num
- #define GlobalFunctions AC.FunctionList.numglobal
- #define GlobalIndices AC.IndexList.numglobal
- #define GlobalSymbols AC.SymbolList.numglobal
- #define GlobalVectors AC.VectorList.numglobal
- #define GlobalSets AC.SetList.numglobal
- #define GlobalSetElements AC.SetElementList.numglobal
- #define cbuf ((CBUF ∗)(AC.cbufList.lijst))
- #define channels ((CHANNEL ∗)(AC.ChannelList.lijst))
- #define NumOutputChannels AC.ChannelList.num
- #define Dollars ((DOLLARS)(AP.DollarList.lijst))
- #define NumDollars AP.DollarList.num
- #define Dubious ((DUBIOUSV)(AC.DubiousList.lijst))
- #define NumDubious AC.DubiousList.num
- #define Expressions ((EXPRESSIONS)(AC.ExpressionList.lijst))
- #define NumExpressions AC.ExpressionList.num
- #define autofunctions ((FUNCTIONS)(AC.AutoFunctionList.lijst))
- #define autoindices ((INDICES)(AC.AutoIndexList.lijst))
- #define autosymbols ((SYMBOLS)(AC.AutoSymbolList.lijst))
- #define autovectors ((VECTORS)(AC.AutoVectorList.lijst))
- #define xsymbol (cbuf[AM.sbufnum].rhs)
- #define numxsymbol (cbuf[AM.sbufnum].numrhs)
- #define PotModdollars ((WORD ∗)(AC.PotModDolList.lijst))
- #define NumPotModdollars AC.PotModDolList.num
- #define ModOptdollars ((MODOPTDOLLAR ∗)(AC.ModOptDolList.lijst))
- #define NumModOptdollars AC.ModOptDolList.num
- #define BUG A.bug;
- #define AC A.Cc
- #define AM A.M
- #define AN A.N
- #define AO A.O
- #define AP A.P
- #define AR A.R
- #define AS A.S
- #define AT A.T
- #define AX A.X

Variables

- WRITEBUFTOEXTCHANNEL **writeBufToExtChannel**
- GETCFROMEXTCHANNEL **getcFromExtChannel**
- SETTERMINATORFOREXTERNALCHANNEL **setTerminatorForExternalChannel**
- SETKILLMODEFOREXTERNALCHANNEL **setKillModeForExternalChannel**
- WRITEFILE WriteFile
- ALLGLOBALS A
- FIXEDGLOBALS FG
- FIXEDSET fixedsets [ ]
- char ∗ setupfilename

4.156.1 Detailed Description

Contains a number of defines to make the coding easier. Especially the defines for the use of the lists are very nice. And of course the AC for A.Cc and AT for either A.T of B->T are indispensable to keep FORM and TFORM in one set of sources.

Definition in file variable.h.

4.156.2 Macro Definition Documentation

4.156.2.1 chartype

```
#define chartype FG.cTable
```

Definition at line 77 of file variable.h.

4.156.2.2 Procedures

```
#define Procedures ((PROCEDURE *)(AP.ProcList.lijst))
```

Definition at line 79 of file variable.h.

4.156.2.3 NumProcedures

```
#define NumProcedures AP.ProcList.num
```

Definition at line 80 of file variable.h.

4.156.2.4 MaxProcedures

```
#define MaxProcedures AP.ProcList.maxnum
```

Definition at line 81 of file variable.h.

4.156.2.5 DoLoops

```
#define DoLoops ((DOLOOP *)(AP.LoopList.lijst))
```

Definition at line 82 of file variable.h.

4.156.2.6 NumDoLoops

```
#define NumDoLoops AP.LoopList.num
```

Definition at line 83 of file variable.h.

4.156.2.7 MaxDoLoops

```
#define MaxDoLoops AP.LoopList.maxnum
```

Definition at line 84 of file variable.h.

4.156.2.8 PreVar

```
#define PreVar ((PREVAR *)(AP.PreVarList.lijst))
```

Definition at line 85 of file variable.h.

4.156.2.9 NumPre

```
#define NumPre AP.PreVarList.num
```

Definition at line 86 of file variable.h.

4.156.2.10 MaxNumPre

```
#define MaxNumPre AP.PreVarList.maxnum
```

Definition at line 87 of file variable.h.

4.156.2.11 SetElements

```
#define SetElements ((WORD *)(AC.SetElementList.lijst))
```

Definition at line 88 of file variable.h.

4.156.2.12 Sets

```
#define Sets ((SETS)(AC.SetList.lijst))
```

Definition at line 89 of file variable.h.

4.156.2.13 functions

```
#define functions ((FUNCTIONS)(AC.FunctionList.lijst))
```

Definition at line 90 of file variable.h.

4.156.2.14 indices

```
#define indices ((INDICES)(AC.IndexList.lijst))
```

Definition at line 91 of file variable.h.

4.156.2.15 symbols

```
#define symbols ((SYMBOLS)(AC.SymbolList.lijst))
```

Definition at line 92 of file variable.h.

4.156.2.16 vectors

```
#define vectors ((VECTORS)(AC.VectorList.lijst))
```

Definition at line 93 of file variable.h.

4.156.2.17 tablebases

```
#define tablebases ((DBASE *)(AC.TableBaseList.lijst))
```

Definition at line 94 of file variable.h.

4.156.2.18 NumFunctions

```
#define NumFunctions AC.FunctionList.num
```

Definition at line 95 of file variable.h.

4.156.2.19 NumIndices

```
#define NumIndices AC.IndexList.num
```

Definition at line 96 of file variable.h.

4.156.2.20 NumSymbols

```
#define NumSymbols AC.SymbolList.num
```

Definition at line 97 of file variable.h.

4.156.2.21 NumVectors

```
#define NumVectors AC.VectorList.num
```

Definition at line 98 of file variable.h.

4.156.2.22 NumSets

```
#define NumSets AC.SetList.num
```

Definition at line 99 of file variable.h.

4.156.2.23 NumSetElements

```
#define NumSetElements AC.SetElementList.num
```

Definition at line 100 of file variable.h.

4.156.2.24 NumTableBases

```
#define NumTableBases AC.TableBaseList.num
```

Definition at line 101 of file variable.h.

4.156.2.25 GlobalFunctions

```
#define GlobalFunctions AC.FunctionList.numglobal
```

Definition at line 102 of file variable.h.

4.156.2.26 GlobalIndices

```
#define GlobalIndices AC.IndexList.numglobal
```

Definition at line 103 of file variable.h.

4.156.2.27 GlobalSymbols

```
#define GlobalSymbols AC.SymbolList.numglobal
```

Definition at line 104 of file variable.h.

4.156.2.28 GlobalVectors

```
#define GlobalVectors AC.VectorList.numglobal
```

Definition at line 105 of file variable.h.

4.156.2.29 GlobalSets

```
#define GlobalSets AC.SetList.numglobal
```

Definition at line 106 of file variable.h.

4.156.2.30 GlobalSetElements

```
#define GlobalSetElements AC.SetElementList.numglobal
```

Definition at line 107 of file variable.h.

4.156.2.31 cbuf

```
#define cbuf ((CBUF *)(AC.cbufList.lijst))
```

Definition at line 108 of file variable.h.

4.156.2.32 channels

```
#define channels ((CHANNEL *)(AC.ChannelList.lijst))
```

Definition at line 109 of file variable.h.

4.156.2.33 NumOutputChannels

```
#define NumOutputChannels AC.ChannelList.num
```

Definition at line 110 of file variable.h.

4.156.2.34 Dollars

```
#define Dollars ((DOLLARS)(AP.DollarList.lijst))
```

Definition at line 111 of file variable.h.

4.156.2.35 NumDollars

```
#define NumDollars AP.DollarList.num
```

Definition at line 112 of file variable.h.

4.156.2.36 Dubious

```
#define Dubious ((DUBIOUSV)(AC.DubiousList.lijst))
```

Definition at line 113 of file variable.h.

4.156.2.37 NumDubious

```
#define NumDubious AC.DubiousList.num
```

Definition at line 114 of file variable.h.

4.156.2.38 Expressions

```
#define Expressions ((EXPRESSIONS)(AC.ExpressionList.lijst))
```

Definition at line 115 of file variable.h.

4.156.2.39 NumExpressions

```
#define NumExpressions AC.ExpressionList.num
```

Definition at line 116 of file variable.h.

4.156.2.40 autofunctions

```
#define autofunctions ((FUNCTIONS)(AC.AutoFunctionList.lijst))
```

Definition at line 117 of file variable.h.

4.156.2.41 autoindices

```
#define autoindices ((INDICES)(AC.AutoIndexList.lijst))
```

Definition at line 118 of file variable.h.

4.156.2.42 autosymbols

```
#define autosymbols ((SYMBOLS)(AC.AutoSymbolList.lijst))
```

Definition at line 119 of file variable.h.

4.156.2.43 autovectors

```
#define autovectors ((VECTORS)(AC.AutoVectorList.lijst))
```

Definition at line 120 of file variable.h.

4.156.2.44 xsymbol

```
#define xsymbol (cbuf[AM.sbufnum].rhs)
```

Definition at line 121 of file variable.h.

4.156.2.45 numxsymbol

```
#define numxsymbol (cbuf[AM.sbufnum].numrhs)
```

Definition at line 122 of file variable.h.

4.156.2.46 PotModdollars

```
#define PotModdollars ((WORD *)(AC.PotModDolList.lijst))
```

Definition at line 124 of file variable.h.

4.156.2.47 NumPotModdollars

```
#define NumPotModdollars AC.PotModDolList.num
```

Definition at line 125 of file variable.h.

4.156.2.48 ModOptdollars

```
#define ModOptdollars ((MODOPTDOLLAR *)(AC.ModOptDolList.lijst))
```

Definition at line 126 of file variable.h.

4.156.2.49 NumModOptdollars

```
#define NumModOptdollars AC.ModOptDolList.num
```

Definition at line 127 of file variable.h.

4.156.2.50 BUG

```
#define BUG A.bug;
```

Definition at line 129 of file variable.h.

4.156.2.51 AC

```
#define AC A.Cc
```

Definition at line 145 of file variable.h.

4.156.2.52 AM

```
#define AM A.M
```

Definition at line 146 of file variable.h.

4.156.2.53 AN

```
#define AN A.N
```

Definition at line 147 of file variable.h.

4.156.2.54 AO

```
#define AO A.O
```

Definition at line 148 of file variable.h.

4.156.2.55 AP

```
#define AP A.P
```

Definition at line 149 of file variable.h.

4.156.2.56 AR

```
#define AR A.R
```

Definition at line 150 of file variable.h.

4.156.2.57 AS

```
#define AS A.S
```

Definition at line 151 of file variable.h.

4.156.2.58 AT

```
#define AT A.T
```

Definition at line 152 of file variable.h.

4.156.2.59 AX

```
#define AX A.X
```

Definition at line 153 of file variable.h.

4.156.3 Variable Documentation

4.156.3.1 WriteFile

```
WRITEFILE WriteFile [extern]
```

Definition at line 1357 of file tools.c.

4.156.3.2 A

```
ALLGLOBALS A [extern]
```

Definition at line 133 of file inivar.h.

4.156.3.3 FG

```
FIXEDGLOBALS FG [extern]
```

Definition at line 34 of file inivar.h.

4.156.3.4 fixedsets

```
FIXEDSET fixedsets[ ] [extern]
```

Definition at line 258 of file inivar.h.

4.156.3.5 setupfilename

```
char* setupfilename [extern]
```

Definition at line 276 of file inivar.h.

4.157 variable.h

Go to the documentation of this file.

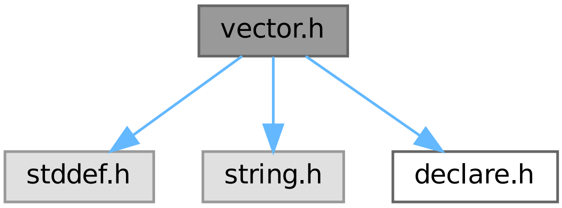
```
#ifndef __VARIABLE__

#define __VARIABLE__

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
 *   This file is part of FORM.
 *
 *   FORM is free software: you can redistribute it and/or modify it under the
 *   terms of the GNU General Public License as published by the Free Software
 *   Foundation, either version 3 of the License, or (at your option) any later
 *   version.
 *
 *   FORM is distributed in the hope that it will be useful, but WITHOUT ANY
 *   WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS
 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */

/*See the file extcmd.c*/

#ifdef REMOVEDBY_MT
extern int (*writeBufToExtChannel)(char *buf, size_t n);
extern int (*getcFromExtChannel)();
extern int (*setTerminatorForExternalChannel)(char *newterminator);
#endif
extern WRITEBUFTOEXTCHANNEL writeBufToExtChannel;
extern GETCFROMEXTCHANNEL getcFromExtChannel;
extern SETTERMINATORFOREXTERNALCHANNEL setTerminatorForExternalChannel;
extern SETKILLMODEFOREXTERNALCHANNEL setKillModeForExternalChannel;

/*
extern LONG (*WriteFile)(int handle, UBYTE *buffer, LONG number);
*/
extern WRITEFILE WriteFile;
/*:[17nov2005 mt]*/

extern ALLGLOBALS A;
#ifdef WITHPTHREADS
extern ALLPRIVATES **AB;
#endif

extern FIXEDGLOBALS FG;
extern FIXEDSET fixedsets[];

extern char *setupfilename;

EXTERNLOCK(ErrorMessageLock)
EXTERNLOCK(FileReadLock)
EXTERNLOCK(dummylock)

#ifdef VMS
#include <stdio.h>
extern FILE **FileStructs;
#endif

#define chartype FG.cTable

#define Procedures ((PROCEDURE *)(AP.ProcList.lijst))
#define NumProcedures AP.ProcList.num
#define MaxProcedures AP.ProcList.maxnum
#define DoLoops ((DOLOOP *)(AP.LoopList.lijst))
#define NumDoLoops AP.LoopList.num
#define MaxDoLoops AP.LoopList.maxnum
#define PreVar ((PREVAR *)(AP.PreVarList.lijst))
#define NumPre AP.PreVarList.num
#define MaxNumPre AP.PreVarList.maxnum
#define SetElements ((WORD *)(AC.SetElementList.lijst))
#define Sets ((SETS)(AC.SetList.lijst))
#define functions ((FUNCTIONS)(AC.FunctionList.lijst))
```

```
#define indices ((INDICES)(AC.IndexList.lijst))
#define symbols ((SYMBOLS)(AC.SymbolList.lijst))
#define vectors ((VECTORS)(AC.VectorList.lijst))
#define tablebases ((DBASE *)(AC.TableBaseList.lijst))
#define NumFunctions AC.FunctionList.num
#define NumIndices AC.IndexList.num
#define NumSymbols AC.SymbolList.num
#define NumVectors AC.VectorList.num
#define NumSets AC.SetList.num
#define NumSetElements AC.SetElementList.num
#define NumTableBases AC.TableBaseList.num
#define GlobalFunctions AC.FunctionList.numglobal
#define GlobalIndices AC.IndexList.numglobal
#define GlobalSymbols AC.SymbolList.numglobal
#define GlobalVectors AC.VectorList.numglobal
#define GlobalSets AC.SetList.numglobal
#define GlobalSetElements AC.SetElementList.numglobal
#define cbuf ((CBUF *)(AC.cbufList.lijst))
#define channels ((CHANNEL *)(AC.ChannelList.lijst))
#define NumOutputChannels AC.ChannelList.num
#define Dollars ((DOLLARS)(AP.DollarList.lijst))
#define NumDollars AP.DollarList.num
#define Dubious ((DUBIOUSV)(AC.DubiousList.lijst))
#define NumDubious AC.DubiousList.num
#define Expressions ((EXPRESSIONS)(AC.ExpressionList.lijst))
#define NumExpressions AC.ExpressionList.num
#define autofunctions ((FUNCTIONS)(AC.AutoFunctionList.lijst))
#define autoindices ((INDICES)(AC.AutoIndexList.lijst))
#define autosymbols ((SYMBOLS)(AC.AutoSymbolList.lijst))
#define autovectors ((VECTORS)(AC.AutoVectorList.lijst))
#define xsymbol (cbuf[AM.sbufnum].rhs)
#define numxsymbol (cbuf[AM.sbufnum].numrhs)

#define PotModdollars ((WORD *)(AC.PotModDolList.lijst))
#define NumPotModdollars AC.PotModDolList.num
#define ModOptdollars ((MODOPTDOLLAR *)(AC.ModOptDolList.lijst))
#define NumModOptdollars AC.ModOptDolList.num

#define BUG A.bug;

#ifdef WITHPTHREADS
#define AC A.Cc
#define AM A.M
#define AO A.O
#define AP A.P
#define AS A.S
#define AX A.X
#define AN B->N
#define AR B->R
#define AT B->T
#define AN0 B0->N
#define AR0 B0->R
#define AT0 B0->T
#else
#define AC A.Cc
#define AM A.M
#define AN A.N
#define AO A.O
#define AP A.P
#define AR A.R
#define AS A.S
#define AT A.T
#define AX A.X
#endif

#endif
```

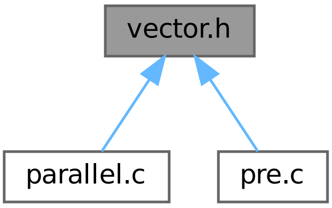
4.158 vector.h File Reference

```
#include <stddef.h>
#include <string.h>
#include "declare.h"
```

Include dependency graph for vector.h:



This graph shows which files directly or indirectly include this file:



Macros

- #define VectorStruct(T)
- #define Vector(T, X) VectorStruct(T) X = { NULL, 0, 0 }
- #define DeclareVector(T, X) VectorStruct(T) X
- #define VectorInit(X)
- #define VectorFree(X)
- #define VectorPtr(X) ((X).ptr)
- #define VectorFront(X) ((X).ptr[0])
- #define VectorBack(X) ((X).ptr[(X).size - 1])
- #define VectorSize(X) ((X).size)
- #define VectorCapacity(X) ((X).capacity)
- #define VectorEmpty(X) ((X).size == 0)
- #define VectorClear(X) do { (X).size = 0; } while (0)
- #define VectorReserve(X, newcapacity)
- #define VectorPushBack(X, x)
- #define VectorPushBacks(X, src, n)
- #define VectorPopBack(X) do { (X).size --; } while (0)
- #define VectorInsert(X, index, x)
- #define VectorInserts(X, index, src, n)
- #define VectorErase(X, index)
- #define VectorErases(X, index, n)

4.158.1 Detailed Description

An implementation of dynamic array.

Example:
```
size_t i;
Vector(int, vec);
VectorPushBack(vec, 1);
VectorPushBack(vec, 2);
VectorPushBack(vec, 3);
for ( i = 0; i < VectorSize(vec); i++ )
  printf("%d\n", VectorPtr(vec)[i]);
VectorFree(vec);
```

Definition in file vector.h.

4.158.2 Macro Definition Documentation

4.158.2.1 VectorStruct

```
#define VectorStruct(
                T )
```

Value:
```
    struct { \
        T *ptr; \
        size_t size; \
        size_t capacity; \
    }
```

A struct for vector objects.

Parameters

T	the type of elements.

Definition at line 65 of file vector.h.

4.158.2.2 Vector

```
#define Vector(
                T,
                X )   VectorStruct(T) X = { NULL, 0, 0 }
```

Defines and initialises a vector X of the type T. The user must call VectorFree(X) after the use of X.

Parameters

T	the type of elements.
X	the name of vector object

Definition at line 84 of file vector.h.

4.158.2.3 DeclareVector

```
#define DeclareVector(
              T,
              X )  VectorStruct(T) X
```

Declares a vector X of the type T. The user must call VectorInit(X) before the use of X.

Parameters

T	the type of elements.
X	the name of vector object

Definition at line 99 of file vector.h.

4.158.2.4 VectorInit

```
#define VectorInit(
              X )
```

Value:

```
do { \
    (X).ptr = NULL; \
    (X).size = 0; \
    (X).capacity = 0; \
} while (0)
```

Initialises a vector X of the type T. The user must call VectorFree(X) after the use of X.

Parameters

X	the vector object.

Definition at line 113 of file vector.h.

4.158.2.5 VectorFree

```
#define VectorFree(
              X )
```

Value:

```
do { \
    M_free((X).ptr, "VectorFree:" #X); \
    (X).ptr = NULL; \
    (X).size = 0; \
    (X).capacity = 0; \
} while (0)
```

Frees the buffer allocated by the vector X.

Parameters

X	the vector object.

Definition at line 130 of file vector.h.

4.158.2.6 VectorPtr

```
#define VectorPtr(
            X )   ((X).ptr)
```

Returns the pointer to the buffer for the vector X. NULL when VectorCapacity(X) == 0.

Parameters

X	the vector object.

Returns

the pointer to the allocated buffer for the vector.

Definition at line 150 of file vector.h.

4.158.2.7 VectorFront

```
#define VectorFront(
            X )   ((X).ptr[0])
```

Returns the first element of the vector X. Undefined when VectorSize(X) == 0.

Parameters

X	the vector object.

Returns

the first element of the vector.

Definition at line 165 of file vector.h.

4.158.2.8 VectorBack

```
#define VectorBack(
            X )   ((X).ptr[(X).size - 1])
```

Returns the last element of the vector X. Undefined when VectorSize(X) == 0.

Parameters

X	the vector object.

Returns

the last element of the vector.

Definition at line 180 of file vector.h.

4.158.2.9 VectorSize

```
#define VectorSize(
              X )   ((X).size)
```

Returns the size of the vector X.

Parameters

X	the vector object.

Returns

the size of the vector.

Definition at line 194 of file vector.h.

4.158.2.10 VectorCapacity

```
#define VectorCapacity(
              X )   ((X).capacity)
```

Returns the capacity (the number of the already allocated elements) of the vector X.

Parameters

X	the vector object.

Returns

the capacity of the vector.

Definition at line 208 of file vector.h.

4.158.2.11 VectorEmpty

```
#define VectorEmpty(
              X )   ((X).size == 0)
```

Returns true the size of the vector X is zero.

Parameters

X	the vector object.

Returns

true if the vector has no elements, false otherwise.

Definition at line 222 of file vector.h.

4.158.2.12 VectorClear

```
#define VectorClear(
                X )   do { (X).size = 0; } while (0)
```

Sets the size of the vector X to zero.

Parameters

X	the vector object.

Definition at line 235 of file vector.h.

4.158.2.13 VectorReserve

```
#define VectorReserve(
                X,
                newcapacity )
```

Value:

```
do { \
    size_t v_tmp_newcapacity_ = (newcapacity); \
    if ( (X).capacity < v_tmp_newcapacity_ ) { \
        void *v_tmp_newptr_; \
        v_tmp_newcapacity_ = (v_tmp_newcapacity_ * 3) / 2; \
        if ( v_tmp_newcapacity_ < 4 ) v_tmp_newcapacity_ = 4; \
        v_tmp_newptr_ = Malloc1(sizeof((X).ptr[0]) * v_tmp_newcapacity_, "VectorReserve:" #X); \
        if ( (X).ptr != NULL ) { \
            memcpy(v_tmp_newptr_, (X).ptr, (X).size * sizeof((X).ptr[0])); \
            M_free((X).ptr, "VectorReserve:" #X); \
        } \
        (X).ptr = v_tmp_newptr_; \
        (X).capacity = v_tmp_newcapacity_; \
    } \
} while (0)
```

Requires that the capacity of the vector X is equal to or lager than newcapacity.

Parameters

X	the vector object.
newcapacity	the capacity to be reserved.

Definition at line 249 of file vector.h.

4.158.2.14 VectorPushBack

```
#define VectorPushBack(
                X,
                x )
```

Value:
```
do { \
    VectorReserve((X), (X).size + 1); \
    (X).ptr[(X).size++] = (x); \
} while (0)
```

Adds an element x at the end of the vector X.

Parameters

X	the vector object.
x	the element to be added.

Definition at line 277 of file vector.h.

4.158.2.15 VectorPushBacks

```
#define VectorPushBacks(
                X,
                src,
                n )
```

Value:
```
do { \
    size_t v_tmp_n_ = (n); \
    VectorReserve((X), (X).size + v_tmp_n_); \
    memcpy((X).ptr + (X).size, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
    (X).size += v_tmp_n_; \
} while (0)
```

Adds an n elements of src at the end of the vector X.

Parameters

X	the vector object.
src	the starting address of the buffer storing elements to be added.
n	the number of elements to be added.

Definition at line 295 of file vector.h.

4.158.2.16 VectorPopBack

```
#define VectorPopBack(
                X )  do { (X).size --; } while (0)
```

Removes the last element of the vector X. VectorSize(X) must be $>$ 0.

Parameters

X	the vector object.

Definition at line 314 of file vector.h.

4.158.2.17 VectorInsert

```
#define VectorInsert(
                X,
                index,
                x )
```

Value:

```
do { \
    size_t v_tmp_index_ = (index); \
    VectorReserve((X), (X).size + 1); \
    memmove((X).ptr + v_tmp_index_ + 1, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) *
  sizeof((X).ptr[0])); \
    (X).ptr[v_tmp_index_] = (x); \
    (X).size++; \
} while (0)
```

Inserts an element x at the specified index of the vector X. The index must be 0 <= index < VectorSize(X).

Parameters

X	the vector object.
index	the position at which the element will be inserted.
x	the element to be inserted.

Definition at line 330 of file vector.h.

4.158.2.18 VectorInserts

```
#define VectorInserts(
                X,
                index,
                src,
                n )
```

Value:

```
do { \
    size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
    VectorReserve((X), (X).size + v_tmp_n_); \
    memmove((X).ptr + v_tmp_index_ + v_tmp_n_, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) *
  sizeof((X).ptr[0])); \
    memcpy((X).ptr + v_tmp_index_, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
    (X).size += v_tmp_n_; \
} while (0)
```

Inserts an n elements of src at the specified index of the vector X. The index must be 0 <= index < VectorSize(X).

Parameters

X	the vector object.
index	the position at which the elements will be inserted.
src	the starting address of the buffer storing elements to be inserted.
n	the number of elements to be inserted.

Definition at line 353 of file vector.h.

4.158.2.19 VectorErase

```
#define VectorErase(
                X,
                index )
```

Value:
```
do { \
    size_t v_tmp_index_ = (index); \
    memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + 1, ((X).size - v_tmp_index_ - 1) *
  sizeof((X).ptr[0])); \
    (X).size--; \
} while (0)
```

Removes an element at the specified index of the vector X. The index must be 0 <= index < VectorSize(X).

Parameters

X	the vector object.
index	the position of the element to be removed.

Definition at line 374 of file vector.h.

4.158.2.20 VectorErases

```
#define VectorErases(
                X,
                index,
                n )
```

Value:
```
do { \
    size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
    memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + v_tmp_n_, ((X).size - v_tmp_index_ - 1) *
  sizeof((X).ptr[0])); \
    (X).size -= v_tmp_n_; \
} while (0)
```

Removes an n elements at the specified index of the vector X. The index must be 0 <= index < VectorSize(X) - n + 1.

Parameters

X	the vector object.
index	the starting position of the elements to be removed.
n	the number of elements to be removed.

Definition at line 394 of file vector.h.

4.159 vector.h

Go to the documentation of this file.

```c
#ifndef VECTOR_H_
#define VECTOR_H_

/* #[ License : */
/*
 *   Copyright (C) 1984-2023 J.A.M. Vermaseren
 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
 *
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 *   FOR A PARTICULAR PURPOSE.  See the GNU General Public License for more
 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes :
*/

#include <stddef.h>
#include <string.h>
#include "declare.h"

/*
    #] Includes :
    #[ Vector :
        #[ VectorStruct :
*/

#define VectorStruct(T) \
    struct { \
        T *ptr; \
        size_t size; \
        size_t capacity; \
    }

/*
        #] VectorStruct :
        #[ Vector :
*/

#define Vector(T, X) \
    VectorStruct(T) X = { NULL, 0, 0 }

/*
        #] Vector :
        #[ DeclareVector :
*/

#define DeclareVector(T, X) \
    VectorStruct(T) X

/*
        #] DeclareVector :
        #[ VectorInit :
*/

#define VectorInit(X) \
    do { \
        (X).ptr = NULL; \
        (X).size = 0; \
        (X).capacity = 0; \
    } while (0)

/*
        #] VectorInit :
        #[ VectorFree :
*/

#define VectorFree(X) \
    do { \
        M_free((X).ptr, "VectorFree:" #X); \
        (X).ptr = NULL; \
```

```
        (X).size = 0; \
        (X).capacity = 0; \
    } while (0)

/*
        #] VectorFree :
        #[ VectorPtr :
*/

#define VectorPtr(X) \
    ((X).ptr)

/*
        #] VectorPtr :
        #[ VectorFront :
*/

#define VectorFront(X) \
    ((X).ptr[0])

/*
        #] VectorFront :
        #[ VectorBack :
*/

#define VectorBack(X) \
    ((X).ptr[(X).size - 1])

/*
        #] VectorBack :
        #[ VectorSize :
*/

#define VectorSize(X) \
    ((X).size)

/*
        #] VectorSize :
        #[ VectorCapacity :
*/

#define VectorCapacity(X) \
    ((X).capacity)

/*
        #] VectorCapacity :
        #[ VectorEmpty :
*/

#define VectorEmpty(X) \
    ((X).size == 0)

/*
        #] VectorEmpty :
        #[ VectorClear :
*/

#define VectorClear(X) \
    do { (X).size = 0; } while (0)

/*
        #] VectorClear :
        #[ VectorReserve :
*/

#define VectorReserve(X, newcapacity) \
    do { \
        size_t v_tmp_newcapacity_ = (newcapacity); \
        if ( (X).capacity < v_tmp_newcapacity_ ) { \
            void *v_tmp_newptr_; \
            v_tmp_newcapacity_ = (v_tmp_newcapacity_ * 3) / 2; \
            if ( v_tmp_newcapacity_ < 4 ) v_tmp_newcapacity_ = 4; \
            v_tmp_newptr_ = Malloc1(sizeof((X).ptr[0]) * v_tmp_newcapacity_, "VectorReserve:" #X); \
            if ( (X).ptr != NULL ) { \
                memcpy(v_tmp_newptr_, (X).ptr, (X).size * sizeof((X).ptr[0])); \
                M_free((X).ptr, "VectorReserve:" #X); \
            } \
            (X).ptr = v_tmp_newptr_; \
            (X).capacity = v_tmp_newcapacity_; \
        } \
    } while (0)

/*
        #] VectorReserve :
        #[ VectorPushBack :
*/

```

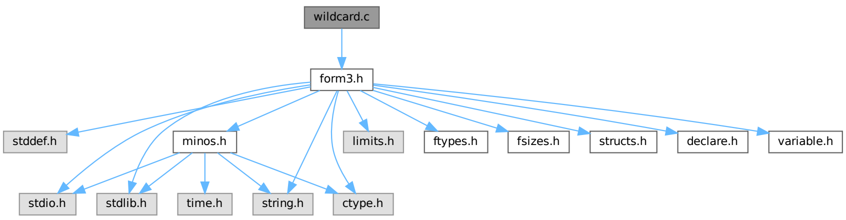
```
#define VectorPushBack(X, x) \
    do { \
        VectorReserve((X), (X).size + 1); \
        (X).ptr[(X).size++] = (x); \
    } while (0)

/*
        #] VectorPushBack :
        #[ VectorPushBacks :
*/

#define VectorPushBacks(X, src, n) \
    do { \
        size_t v_tmp_n_ = (n); \
        VectorReserve((X), (X).size + v_tmp_n_); \
        memcpy((X).ptr + (X).size, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
        (X).size += v_tmp_n_; \
    } while (0)

/*
        #] VectorPushBacks :
        #[ VectorPopBack :
*/

#define VectorPopBack(X) \
    do { (X).size --; } while (0)

/*
        #] VectorPopBack :
        #[ VectorInsert :
*/

#define VectorInsert(X, index, x) \
    do { \
        size_t v_tmp_index_ = (index); \
        VectorReserve((X), (X).size + 1); \
        memmove((X).ptr + v_tmp_index_ + 1, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) * sizeof((X).ptr[0])); \
        (X).ptr[v_tmp_index_] = (x); \
        (X).size++; \
    } while (0)

/*
        #] VectorInsert :
        #[ VectorInserts :
*/

#define VectorInserts(X, index, src, n) \
    do { \
        size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
        VectorReserve((X), (X).size + v_tmp_n_); \
        memmove((X).ptr + v_tmp_index_ + v_tmp_n_, (X).ptr + v_tmp_index_, ((X).size - v_tmp_index_) * sizeof((X).ptr[0])); \
        memcpy((X).ptr + v_tmp_index_, (src), v_tmp_n_ * sizeof((X).ptr[0])); \
        (X).size += v_tmp_n_; \
    } while (0)

/*
        #] VectorInserts :
        #[ VectorErase :
*/

#define VectorErase(X, index) \
    do { \
        size_t v_tmp_index_ = (index); \
        memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + 1, ((X).size - v_tmp_index_ - 1) * sizeof((X).ptr[0])); \
        (X).size--; \
    } while (0)

/*
        #] VectorErase :
        #[ VectorErases :
*/

#define VectorErases(X, index, n) \
    do { \
        size_t v_tmp_index_ = (index), v_tmp_n_ = (n); \
        memmove((X).ptr + v_tmp_index_, (X).ptr + v_tmp_index_ + v_tmp_n_, ((X).size - v_tmp_index_ - 1) * sizeof((X).ptr[0])); \
        (X).size -= v_tmp_n_; \
    } while (0)

/*
        #] VectorErases :
    #] Vector :
*/
```

```
00405 #endif  /* VECTOR_H_ */
```

4.160 wildcard.c File Reference

`#include "form3.h"`

Include dependency graph for wildcard.c:



Macros

- #define DEBUG(x)

Functions

- WORD WildFill (PHEAD WORD ∗to, WORD ∗from, WORD ∗sub)
- WORD ResolveSet (PHEAD WORD ∗from, WORD ∗to, WORD ∗subs)
- void ClearWild (PHEAD0)
- WORD AddWild (PHEAD WORD oldnumber, WORD type, WORD newnumber)
- WORD CheckWild (PHEAD WORD oldnumber, WORD type, WORD newnumber, WORD ∗newval)
- int DenToFunction (WORD ∗term, WORD numfun)

4.160.1 Detailed Description

Contains the functions that deal with the wildcards. During the pattern matching there are two steps: 1: check that a wildcard substitution is correct (if there was already an assignment for this variable, it is the same; it is part of the proper set; it is the proper type of variables, etc.) 2: make the assignment In addition we have to be able to clear assignments. During execution we have to make the actual replacements (WildFill)

Definition in file wildcard.c.

4.160.2 Macro Definition Documentation

4.160.2.1 DEBUG

```
#define DEBUG(
                x )
```

Definition at line 44 of file wildcard.c.

4.160.3 Function Documentation

4.160.3.1 WildFill()

```
WORD WildFill (
             PHEAD WORD * to,
             WORD * from,
             WORD * sub )
```

Definition at line 65 of file wildcard.c.

4.160.3.2 ResolveSet()

```
WORD ResolveSet (
             PHEAD WORD * from,
             WORD * to,
             WORD * subs )
```

Definition at line 1361 of file wildcard.c.

4.160.3.3 ClearWild()

```
void ClearWild (
             PHEAD0 )
```

Definition at line 1518 of file wildcard.c.

4.160.3.4 AddWild()

```
WORD AddWild (
             PHEAD WORD oldnumber,
             WORD type,
             WORD newnumber )
```

Definition at line 1544 of file wildcard.c.

4.160.3.5 CheckWild()

```
WORD CheckWild (
             PHEAD WORD oldnumber,
             WORD type,
             WORD newnumber,
             WORD * newval )
```

Definition at line 1791 of file wildcard.c.

4.160.3.6 DenToFunction()

```
int DenToFunction (
            WORD * term,
            WORD numfun )
```

Definition at line 2556 of file wildcard.c.

4.161 wildcard.c

Go to the documentation of this file.

```

/* #[ License : */
/*
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 *   When using this file you are requested to refer to the publication
 *   J.A.M.Vermaseren "New features of FORM" math-ph/0010025
 *   This is considered a matter of courtesy as the development was paid
 *   for by FOM the Dutch physics granting agency and we would like to
 *   be able to track its scientific use to convince FOM of its value
 *   for the community.
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 *   details.
 *
 *   You should have received a copy of the GNU General Public License along
 *   with FORM.  If not, see <http://www.gnu.org/licenses/>.
 */
/* #] License : */
/*
    #[ Includes : wildcard.c
*/

#include "form3.h"

#define DEBUG(x)

/*
#define DEBUG(x) x

    #] Includes :
    #[ Wildcards :
        #[ WildFill :           WORD WildFill(to,from,sub)

        Takes the term in from and puts it into to while
        making wildcard substitutions.
        The return value is the number of words put in to.
        The length as the first word of from is not copied.

        There are two possible algorithms:
        1:  For each element in `from': scan sub.
        2:  For each wildcard in sub replace elements in term.
        The original algorithm used 1:

*/

WORD WildFill(PHEAD WORD *to, WORD *from, WORD *sub)
{
    GETBIDENTITY
    WORD i, j, *s, *t, *m, len, dflag, odirt, adirt;
    WORD *r, *u, *v, *w, *z, *zst, *zz, *subs, *accu, na, dirty = 0, *tstop;
    WORD *temp = 0, *uu, *oldcpointer, sgn;
    WORD subcount, setflag, *setlist = 0, si;
    accu = oldcpointer = AR.CompressPointer;
    t = sub;
    t += sub[1];
    s = sub + SUBEXPSIZE;
    i = 0;
```

```
    while ( s < t && *s != FROMBRAC ) {
        i++; s += s[1];
    }
    if ( !i ) {          /* No wildcards -> done quickly */
        j = i = *from;
        NCOPY(to,from,i);
        if ( dirty ) AN.WildDirt = dirty;
        return(j);
    }
    sgn = 0;
    subs = sub + SUBEXPSIZE;
    t = from;
    GETSTOP(t,r);
    t++;
    m = to + 1;
    if ( t < r ) do {
        uu = u = t + t[1];
        setflag = 0;
ReSwitch:
        switch ( *t ) {
            case SYMBOL:
/*
            #[ SYMBOLS :
*/
                z = accu;
                *m++ = *t++;
                *m++ = *t++;
                v = m;
                while ( t < u ) {
                    *m = *t;
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] ) goto sspow;
                    }
                    s = subs;
                    for ( j = 0; j < i; j++ ) {
                        if ( *t == s[2] ) {
                            if ( *s == SYMTOSYM ) {
                                *m = s[3]; dirty = 1;
                                break;
                            }
                            else if ( *s == SYMTONUM ) {
                                dirty = 1;
                                zst = z;
                                *z++ = SNUMBER;
                                *z++ = 4;
                                *z++ = s[3];
                                w = z;
                                *z++ = *++t;
                                if ( ABS(*t) >= 2*MAXPOWER) {
DoPow:                              s = subs;
                                for ( j = 0; j < i; j++ ) {
                                    if ( ( *s == SYMTONUM ) &&
                                    ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
                                        dirty = 1;
                                        *w = s[3];
                                        if ( *t < 0 ) *w = -*w;
                                        break;
                                    }
                                    if ( ( *s == SYMTOSYM ) &&
                                    ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
                                        dirty = 1;
                                        zz = z;
                                        while ( --zz >= zst ) {
                                            zz[1+FUNHEAD+ARGHEAD] = *zz;
                                        }
                                        w += 1+FUNHEAD+ARGHEAD;
                                        *zst = EXPONENT;
                                        zst[2] = DIRTYFLAG;
                                        zst[FUNHEAD+ARGHEAD] = WORDDIF(z,zst)+4;
                                        zst[1+FUNHEAD] = 1;
                                        zst[FUNHEAD] = WORDDIF(z,zst)+4+ARGHEAD;
                                        z += FUNHEAD+ARGHEAD+1;
                                        *w = 1; /* exponent -> 1 */
                                        *z++ = 1;
                                        *z++ = 1;
                                        *z++ = 3;
                                        if ( *t > 0 ) {
                                            *z++ = -SYMBOL;
                                            *z++ = s[3];
                                        }
                                        else {
                                            *z++ = ARGHEAD+8;
                                            *z++ = 1;
                                            *z++ = 8;
                                            *z++ = SYMBOL;
                                            *z++ = 4;
                                            *z++ = s[3];
```

```
                                            *z++ = 1;
                                            *z++ = 1;
                                            *z++ = 1;
                                            *z++ = -3;
                                        }
                                        zst[1] = WORDDIF(z,zst);
                                        break;
                                    }
                                    if ( *s == SYMTOSUB &&
                                    ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
MakeExp:                                dirty = 1;
                                        zz = z;
                                        while ( --zz >= zst ) {
                                            zz[1+FUNHEAD+ARGHEAD] = *zz;
                                        }
                                        w += 1+FUNHEAD+ARGHEAD;
                                        *zst = EXPONENT;
                                        zst[2] = DIRTYFLAG;
                                        zst[FUNHEAD+ARGHEAD] = WORDDIF(z,zst)+4;
                                        zst[1+FUNHEAD] = 1;
                                        zst[FUNHEAD] = WORDDIF(z,zst)+4+ARGHEAD;
                                        z += FUNHEAD+ARGHEAD+1;
                                        *w = 1; /* exponent -> 1 */
                                        *z++ = 1;
                                        *z++ = 1;
                                        *z++ = 3;
                                        *z++ = 4+SUBEXPSIZE+ARGHEAD;
                                        *z++ = 1;
                                        *z++ = 4+SUBEXPSIZE;
                                        *z++ = SUBEXPRESSION;
                                        *z++ = SUBEXPSIZE;
                                        *z++ = s[3];
                                        *z++ = 1;
                                        *z++ = AT.ebufnum;
                                        FILLSUB(z)
                                        *z++ = 1;
                                        *z++ = 1;
                                        *z++ = *t > 0 ? 3: -3;
                                        zst[1] = WORDDIF(z,zst);
                                        break;
                                    }
                                    s += s[1];
                                }
                                }
                                if ( !*w ) z = w - 3;
                                t++;
                                goto Seven;
                            }
                            else if ( *s == SYMTOSUB ) {
                                dirty = 1;
                                zst = z;
                                *z++ = SUBEXPRESSION;
                                *z++ = SUBEXPSIZE;
                                *z++ = s[3];
                                w = z;
                                *z++ = *++t;
                                *z++ = AT.ebufnum;
                                FILLSUB(z)
                                goto DoPow;
                            }
                        }
                        s += s[1];
                    }
sspow:
                    s = subs;
                    *++m = *++t;
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] ) {
                            t++; m++;
                            goto Seven;
                        }
                    }
                    for ( j = 0; j < i; j++ ) {
                        if ( ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
                            if ( *s == SYMTONUM ) {
                                dirty = 1;
                                *m = s[3];
                                if ( *t < 0 ) *m = -*m;
                                break;
                            }
                            else if ( *s == SYMTOSYM ) {
                                dirty = 1;
                                *z++ = EXPONENT;
                                if ( *t < 0 ) *z++ = FUNHEAD+ARGHEAD+10;
                                else *z++ = 4+FUNHEAD;
                                *z++ = 0;
                                FILLFUN3(z)
```

```
                                    *z++ = -SYMBOL;
                                    *z++ = m[-1];
                                    if ( *t < 0 ) {
                                        *z++ = ARGHEAD+8;
                                        *z++ = 0;
                                        *z++ = 8;
                                        *z++ = SYMBOL;
                                        *z++ = 4;
                                        *z++ = s[3];
                                        *z++ = 1;
                                        *z++ = 1;
                                        *z++ = 1;
                                        *z = -3;
                                    }
                                    else {
                                        *z++ = -SYMBOL;
                                        *z++ = s[3];
                                    }
                                    m -= 2;
                                    break;
                                }
                                else if ( *s == SYMTOSUB ) {
                                    zst = z;
                                    *z++ = SYMBOL;
                                    *z++ = 4;
                                    *z++ = *--m;
                                    w = z;
                                    *z++ = *t;
                                    goto MakeExp;
                                }
                            }
                            s += s[1];
                        }
                        t++;
                        if ( *m ) m++;
                        else m--;
Seven:;
                    }
                    j = WORDDIF(m,v);
                    if ( !j ) m -= 2;
                    else v[-1] = j + 2;
                    s = accu;
                    while ( s < z ) *m++ = *s++;
                    break;
/*
                #] SYMBOLS :
*/
                case DOTPRODUCT:
/*
                #[ DOTPRODUCTS :
*/
                    *m++ = *t++;
                    *m++ = *t++;
                    v = m;
                    z = accu;
                    while ( t < u ) {
                        *m = *t;
                        subcount = 0;
                        /* Process the first vector of the DOTPRODUCT */
                        for ( si = 0; si < setflag; si += 2 ) {
                            if ( t == temp + setlist[si] ) goto ss2;
                        }
                        s = subs;
                        for ( j = 0; j < i; j++ ) {
                            if ( *t == s[2] ) {
                                if ( *s == VECTOVEC ) {
                                    *m = s[3]; dirty = 1; break;
                                }
                                if ( *s == VECTOMIN ) {
                                    *m = s[3];
                                    dirty = 1;
                                    if ( ( ABS(t[2]) - 2*MAXPOWER ) < 0 ) {
                                        /* The power is a number */
                                        sgn += t[2];
                                    }
                                    else {
                                        /* The power is a wildcard. Put a -, resolve later. */
                                        sgn++;
                                    }
                                    break;
                                }
                                if ( *s == VECTOSUB ) {
                                    *m = s[3]; dirty = 1; subcount = 1; break;
                                }
                            }
                            s += s[1];
                        }
```

```
ss2:
                    *++m = *++t;
                    s = subs;
                    /* Process the second vector of the DOTPRODUCT */
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] ) goto ss3;
                    }
                    for ( j = 0; j < i; j++ ) {
                        if ( *t == s[2] ) {
                            if ( *s == VECTOVEC ) {
                                *m = s[3]; dirty = 1; break;
                            }
                            if ( *s == VECTOMIN ) {
                                *m = s[3];
                                dirty = 1;
                                if ( ( ABS(t[1]) - 2*MAXPOWER ) < 0 ) {
                                    /* The power is a number */
                                    sgn += t[1];
                                }
                                else {
                                    /* The power is a wildcard. Put a -, resolve later. */
                                    sgn++;
                                }
                                break;
                            }
                            if ( *s == VECTOSUB ) {
                                *m = s[3]; dirty = 1; subcount += 2; break;
                            }
                        }
                        s += s[1];
                    }
ss3:                *++m = *++t;
                    /* Process the power */
                    if ( ( ABS(*t) - 2*MAXPOWER ) < 0 ) goto RegPow;
                    s = subs;
                    for ( j = 0; j < i; j++ ) {
                        if ( ( ABS(*t) - 2*MAXPOWER ) == s[2] ) {
                            if ( *s == SYMTONUM ) {
                                *m = s[3];
                                /* Since the power is a wildcard, sgn is 0,1,2 depending whether
                                    there were 0,1,2 VECTOMIN in the DOTPRODUCT. Multiply by the
                                    power, which is positive currently. */
                                sgn *= *m;
                                /* Now flip the sign of the power, if the wildcard came with a - */
                                if ( *t < 0 ) *m = -*m;
                                dirty = 1;
                                break;
                            }
                            if ( *s <= SYMTOSUB ) {
/*
                Here we put together a power function with the proper
                arguments. Note that a p?.q? resolves to a single power.
*/
                                m -= 2;
                                *z++ = EXPONENT;
                                w = z;
                                if ( subcount == 0 ) {
                                    *z++ = 17+FUNHEAD+2*ARGHEAD;
                                    *z++ = DIRTYFLAG;
                                    FILLFUN3(z)
                                    *z++ = 9+ARGHEAD;
                                    *z++ = 0;
                                    FILLARG(z)
                                    *z++ = 9;
                                    *z++ = DOTPRODUCT;
                                    *z++ = 5;
                                    *z++ = *m;
                                    *z++ = m[1];
                                    *z++ = 1;
                                    *z++ = 1;
                                    *z++ = 1;
                                    *z++ = 3;
                                    if ( *s == SYMTOSYM ) {
                                        *z++ = 8+ARGHEAD;
                                        *z++ = 0;
                                        FILLARG(z)
                                        *z++ = 8;
                                        *z++ = SYMBOL;
                                        *z++ = 4;
                                        *z++ = s[3];
                                        *z++ = 1;
                                    }
                                    else {
                                        *z++ = 4+SUBEXPSIZE+ARGHEAD;
                                        *z++ = 1;
                                        FILLARG(z)
                                        *z++ = 4+SUBEXPSIZE;
```

```
                *z++ = SUBEXPRESSION;
                *z++ = SUBEXPSIZE;
                *z++ = s[3];
                *z++ = 1;
                *z++ = AT.ebufnum;
                FILLSUB(z)
            }
            *z++ = 1; *z++ = 1;
            *z++ = ( s[2] > 0 ) ? 3: -3;
        }
        else if ( subcount == 3 ) {
            *z++ = 20+2*SUBEXPSIZE+FUNHEAD+2*ARGHEAD;
            *z++ = DIRTYFLAG;
            FILLFUN3(z)
            *z++ = 12+2*SUBEXPSIZE+ARGHEAD;
            *z++ = 1;
            *z++ = 12+2*SUBEXPSIZE;
            *z++ = SUBEXPRESSION;
            *z++ = 4+SUBEXPSIZE;
            *z++ = *m + 1;
            *z++ = 1;
            *z++ = AT.ebufnum;
            FILLSUB(z)
            *z++ = INDTOIND;
            *z++ = 4;
            *z++ = FUNNYVEC;
            *z++ = ++AR.CurDum;

            *z++ = SUBEXPRESSION;
            *z++ = 4+SUBEXPSIZE;
            *z++ = m[1] + 1;
            *z++ = 1;
            *z++ = AT.ebufnum;
            FILLSUB(z)
            *z++ = INDTOIND;
            *z++ = 4;
            *z++ = FUNNYVEC;
            *z++ = AR.CurDum;
            *z++ = 1; *z++ = 1; *z++ = 3;
        }
        else {
            if ( subcount == 2 ) {
                j = *m; *m = m[1]; m[1] = j;
            }
            *z++ = 16+SUBEXPSIZE+FUNHEAD+2*ARGHEAD;
            *z++ = DIRTYFLAG;
            FILLFUN3(z)
            *z++ = 8+SUBEXPSIZE+ARGHEAD;
            *z++ = 1;
            *z++ = 8+SUBEXPSIZE;
            *z++ = SUBEXPRESSION;
            *z++ = 4+SUBEXPSIZE;
            *z++ = *m + 1;
            *z++ = 1;
            *z++ = AT.ebufnum;
            FILLSUB(z)
            *z++ = INDTOIND;
            *z++ = 4;
            *z++ = FUNNYVEC;
            *z++ = m[1];
            *z++ = 1; *z++ = 1; *z++ = 3;
        }
        if ( *s == SYMTOSYM ) {
            if ( s[2] > 0 ) {
                *z++ = -SYMBOL;
                *z++ = s[3];
                t++;
                *w = z-w+1;
                goto NextDot;
            }
            *z++ = 8+ARGHEAD;
            *z++ = 0;
            *z++ = 8;
            *z++ = SYMBOL;
            *z++ = 4;
            *z++ = s[3];
            *z++ = 1;
        }
        else {
            *z++ = 4+SUBEXPSIZE+ARGHEAD;
            *z++ = 1;
            *z++ = 4+SUBEXPSIZE;
            *z++ = SUBEXPRESSION;
            *z++ = SUBEXPSIZE;
            *z++ = s[3];
            *z++ = 1;
            *z++ = AT.ebufnum;
```

```
                                    FILLSUB(z)
                                }
                                *z++ = 1; *z++ = 1;
                                *z++ = ( s[2] > 0 ) ? 3: -3;
                                t++;
                                *w = z-w+1;
                                goto NextDot;
                            }
                        }
                        s += s[1];
                    }
RegPow:             if ( *m ) m++;
                    else { m -= 2; subcount = 0; }
                    t++;
                    if ( subcount ) {
                        m -= 3;
                        if ( subcount == 3 ) {
                            if ( m[2] < 0 ) {
                                j = (-m[2]) * (2*SUBEXPSIZE+8);
                                *z++ = DENOMINATOR;
                                *z++ = j + 8 + FUNHEAD + ARGHEAD;
                                *z++ = DIRTYFLAG;
                                FILLFUN3(z)
                                *z++ = j + 8 + ARGHEAD;
                                *z++ = 1;
                                *z++ = j + 8;
                                while ( m[2] < 0 ) {
                                    (m[2])++;
                                    *z++ = SUBEXPRESSION;
                                    *z++ = 4+SUBEXPSIZE;
                                    *z++ = *m + 1;
                                    *z++ = 1;
                                    *z++ = AT.ebufnum;
                                    FILLSUB(z)
                                    *z++ = INDTOIND;
                                    *z++ = 4;
                                    *z++ = FUNNYVEC;
                                    *z++ = ++AR.CurDum;
                                    *z++ = SUBEXPRESSION;
                                    *z++ = 8+SUBEXPSIZE;
                                    *z++ = m[1] + 1;
                                    *z++ = 1;
                                    *z++ = AT.ebufnum;
                                    FILLSUB(z)
                                    *z++ = INDTOIND;
                                    *z++ = 4;
                                    *z++ = FUNNYVEC;
                                    *z++ = AR.CurDum;
                                    *z++ = SYMTOSYM;    /* Needed to avoid */
                                    *z++ = 4;           /* problems with   */
                                    *z++ = 1000;        /* conversion to   */
                                    *z++ = 1000;        /* square of subexp*/
                                }
                                *z++ = 1; *z++ = 1; *z++ = 3;
                            }
                            else {
                                while ( m[2] > 0 ) {
                                    (m[2])--;
                                    *z++ = SUBEXPRESSION;
                                    *z++ = 4+SUBEXPSIZE;
                                    *z++ = *m + 1;
                                    *z++ = 1;
                                    *z++ = AT.ebufnum;
                                    FILLSUB(z)
                                    *z++ = INDTOIND;
                                    *z++ = 4;
                                    *z++ = FUNNYVEC;
                                    *z++ = ++AR.CurDum;
                                    *z++ = SUBEXPRESSION;
                                    *z++ = 4+SUBEXPSIZE;
                                    *z++ = m[1] + 1;
                                    *z++ = 1;
                                    *z++ = AT.ebufnum;
                                    FILLSUB(z)
                                    *z++ = INDTOIND;
                                    *z++ = 4;
                                    *z++ = FUNNYVEC;
                                    *z++ = AR.CurDum;
                                }
                            }
                        }
                        else {
                            if ( subcount == 2 ) {
                                j = *m; *m = m[1]; m[1] = j;
                            }
                            if ( m[2] < 0 ) {
                                *z++ = DENOMINATOR;
```

```
                                *z++ = 8+SUBEXPSIZE+FUNHEAD+ARGHEAD;
                                *z++ = DIRTYFLAG;
                                FILLFUN3(z)
                                *z++ = 8+SUBEXPSIZE+ARGHEAD;
                                *z++ = 1;
                                *z++ = 8+SUBEXPSIZE;
                            }
                            *z++ = SUBEXPRESSION;
                            *z++ = 4+SUBEXPSIZE;
                            *z++ = *m + 1;
                            *z++ = ABS(m[2]);
                            *z++ = AT.ebufnum;
                            FILLSUB(z)
                            *z++ = INDTOIND;
                            *z++ = 4;
                            *z++ = FUNNYVEC;
                            *z++ = m[1];
                            if ( m[2] < 0 ) {
                                *z++ = 1; *z++ = 1; *z++ = 3;
                            }
                        }
                    }
NextDot:;
                }
                if ( m <= v ) m = v - 2;
                else v[-1] = WORDDIF(m,v) + 2;
                if ( z > accu ) {
                    j = WORDDIF(z,accu);
                    z = accu;
                    NCOPY(m,z,j);
                }
                break;
/*
            #] DOTPRODUCTS :
*/
            case SETSET:
/*
            #[ SETS :
*/
                temp = accu + (((AR.ComprTop - accu)»1)&(-2));
                if ( ResolveSet(BHEAD t,temp,sub) ) {
                    Terminate(-1);
                }
                setlist = t + 2 + t[3];
                setflag = t[1] - 2 - t[3];  /* Number of elements * 2 */
                t = temp; u = t + t[1];
                goto ReSwitch;
/*
            #] SETS :
*/
            case VECTOR:
/*
            #[ VECTORS :
*/
                *m++ = *t++;
                *m++ = *t++;
                v = m;
                z = accu;
                while ( t < u ) {
                    *m = *t;
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] ) goto ss4;
                    }
                    s = subs;
                    for ( j = 0; j < i; j++ ) {
                        if ( *t == s[2] ) {
                            if ( *s == INDTOIND || *s == VECTOVEC ) {
                                *m = s[3]; dirty = 1; break;
                            }
                            if ( *s == VECTOMIN ) {
                                *m = s[3]; dirty = 1; sgn++; break;
                            }
                            else if ( *s == VECTOSUB ) {
                                *z++ = SUBEXPRESSION;
                                *z++ = 4+SUBEXPSIZE;
                                *z++ = s[3]+1;
                                *z++ = 1;
                                *z++ = AT.ebufnum;
                                FILLSUB(z)
                                *z++ = VECTOVEC;
                                *z++ = 4;
                                *z++ = FUNNYVEC;
                                *z++ = *++t;
                                m--;
                                s = subs;
                                for ( j = 0; j < i; j++ ) {
                                    if ( z[-1] == s[2] ) {
```

```
                                        if ( *s == INDTOIND || *s == VECTOVEC ) {
                                            z[-1] = s[3];
                                            break;
                                        }
                                        if ( *s == INDTOSUB || *s == VECTOSUB ) {
                                            z[-1] = ++AR.CurDum;
                                            *z++ = SUBEXPRESSION;
                                            *z++ = 4+SUBEXPSIZE;
                                            *z++ = s[3]+1;
                                            *z++ = 1;
                                            *z++ = AT.ebufnum;
                                            FILLSUB(z)
                                            if ( *s == INDTOSUB ) *z++ = INDTOIND;
                                            else *z++ = VECTOSUB;
                                            *z++ = 4;
                                            *z++ = FUNNYVEC;
                                            *z++ = AR.CurDum;
                                            break;
                                        }
                                    }
                                    s += s[1];
                                }
                                dirty = 1;
                                break;
                            }
                            else if ( *s == INDTOSUB ) {
                                *z++ = SUBEXPRESSION;
                                *z++ = 4+SUBEXPSIZE;
                                *z++ = s[3]+1;
                                *z++ = 1;
                                *z++ = AT.ebufnum;
                                FILLSUB(z)
                                *z++ = INDTOIND;
                                *z++ = 4;
                                *z++ = FUNNYVEC;
                                m -= 2;
                                *z++ = m[1];
                                dirty = 1;
                                t++;
                                break;
                            }
                        }
                        s += s[1];
                    }
ss4:                m++; t++;
                }
                if ( m <= v ) m = v-2;
                else v[-1] = WORDDIF(m,v)+2;
                if ( z > accu ) {
                    j = WORDDIF(z,accu); z = accu;
                    NCOPY(m,z,j);
                }
                break;
/*
            #] VECTORS :
*/
            case INDEX:
/*
            #[ INDEX :
*/
                *m++ = *t++;
                *m++ = *t++;
                v = m;
                z = accu;
                while ( t < u ) {
                    *m = *t;
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] ) goto ss5;
                    }
                    s = subs;
                    for ( j = 0; j < i; j++ ) {
                        if ( *t == s[2] ) {
                            if ( *s == INDTOIND || *s == VECTOVEC )
                                { *m = s[3]; dirty = 1; break; }
                            if ( *s == VECTOMIN )
                                { *m = s[3]; dirty = 1; sgn++; break; }
                            else if ( *s == VECTOSUB || *s == INDTOSUB ) {
                                *z++ = SUBEXPRESSION;
                                *z++ = SUBEXPSIZE;
                                *z++ = s[3];
                                *z++ = 1;
                                *z++ = AT.ebufnum;
                                FILLSUB(z)
                                m--;
                                dirty = 1;
                                break;
                            }
```

```
                        }
                        s += s[1];
                    }
ss5:                m++; t++;
                }
                if ( m <= v ) m = v-2;
                else v[-1] = WORDDIF(m,v)+2;
                if ( z > accu ) {
                    j = WORDDIF(z,accu); z = accu;
                    NCOPY(m,z,j);
                }
                break;
/*
            #] INDEX :
*/
            case DELTA:
            case LEVICIVITA:
            case GAMMA:
/*
            #[ SPECIALS :
*/
                v = m;
                *m++ = *t++;
                *m++ = *t++;
#if FUNHEAD > 2
                if ( t[-2] != DELTA ) *m++ = *t++;
#endif
Tensors:
                COPYFUN3(m,t)
                z = accu;
                while ( t < u ) {
                    *m = *t;
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] ) goto ss6;
                    }
                    s = subs;
                    if ( *m == FUNNYWILD ) {
                        CBUF *C = cbuf+AT.ebufnum;
                        t++;
                        for ( j = 0; j < i; j++ ) {
                            if ( *s == ARGTOARG && *t == s[2] ) {
                                v[2] |= DIRTYFLAG;
                                if ( s[3] < 0 ) { /* empty */
                                    t++; break;
                                }
                                w = C->rhs[s[3]];
DEBUG(MesPrint("Thread %w(a): s[3] = %d, w=(%d,%d,%d,%d)",s[3],w[0],w[1],w[2],w[3]);)
                                j = *w++;
                                if ( j > 0 ) {
                                    NCOPY(m,w,j);
                                }
                                else {
                                    while ( *w ) {
                                        if ( *w == -INDEX || *w == -VECTOR
                                        || *w == -MINVECTOR
                                        || ( *w == -SNUMBER && w[1] >= 0
                                        && w[1] < AM.OffsetIndex ) ) {
                                            if ( *w == -MINVECTOR ) sgn++;
                                            w++;
                                            *m++ = *w++;
                                        }
                                        else {
                                            MLOCK(ErrorMessageLock);
DEBUG(MesPrint("Thread %w(aa): *w = %d",*w);)
                                            MesPrint("Illegal substitution of argument field in tensor");
                                            MUNLOCK(ErrorMessageLock);
                                            SETERROR(-1)
                                        }
                                    }
                                }
                                t++;
                                break;
                            }
                            s += s[1];
                        }
                    }
                    else {
                        for ( j = 0; j < i; j++ ) {
                            if ( *t == s[2] ) {
                                if ( *s == INDTOIND || *s == VECTOVEC )
                                    { *m = s[3]; dirty = 1; break; }
                                if ( *s == VECTOMIN )
                                    { *m = s[3]; dirty = 1; sgn++; break; }
                                else if ( *s == VECTOSUB || *s == INDTOSUB ) {
                                    *m = ++AR.CurDum;
                                    *z++ = SUBEXPRESSION;
```

```
                                    *z++ = 4+SUBEXPSIZE;
                                    *z++ = s[3]+1;
                                    *z++ = 1;
                                    *z++ = AT.ebufnum;
                                    FILLSUB(z)
                                    *z++ = INDTOIND;
                                    *z++ = 4;
                                    *z++ = FUNNYVEC;
                                    *z++ = AR.CurDum;
                                    dirty = 1;
                                    break;
                                }
                            }
                            s += s[1];
                        }
                        if ( j < i && *v != DELTA ) v[2] |= DIRTYFLAG;
ss6:                    m++; t++;
                    }
                }
                v[1] = WORDDIF(m,v);
                if ( z > accu ) {
                    j = WORDDIF(z,accu); z = accu;
                    NCOPY(m,z,j);
                }
                break;
/*
            #] SPECIALS :
*/
            case SUBEXPRESSION:
/*
            #[ SUBEXPRESSION :
*/
                dirty = 1;
                tstop = t + t[1];
                *m++ = *t++;
                *m++ = *t++;
                *m++ = *t++;
                *m++ = *t++;
                if ( t[-1] >= 2*MAXPOWER || t[-1] <= -2*MAXPOWER ) {
                    s = subs;
                    for ( j = 0; j < i; j++ ) {
                        if ( *s == SYMTONUM &&
                                ( ABS(t[-1]) - 2*MAXPOWER ) == s[2] ) {
                            m[-1] = s[3];
                            if ( t[-1] < 0 ) m[-1] = -m[-1];
                            break;
                        }
                        s += s[1];
                    }
                }
                *m++ = *t++;
                COPYSUB(m,t)
                while ( t < tstop ) {
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] - 2 ) goto ss7;
                    }
                    s = subs;
                    for ( j = 0; j < i; j++ ) {
                        if ( s[2] == t[2] ) {
                            if ( ( *s <= SYMTOSUB && *t <= SYMTOSUB )
                            || ( *s == *t && *s < FROMBRAC )
                            || ( *s == VECTOVEC && ( *t == VECTOSUB || *t == VECTOMIN ) )
                            || ( *s == VECTOSUB && ( *t == VECTOVEC || *t == VECTOMIN ) )
                            || ( *s == VECTOMIN && ( *t == VECTOSUB || *t == VECTOVEC ) )
                            || ( *s == INDTOIND && *t == INDTOSUB )
                            || ( *s == INDTOSUB && *t == INDTOIND ) ) {
                                WORD *vv = m;
/*                              *t = *s;  Wrong!!! Overwrites compiler buffer  */
                                j = t[1];
                                NCOPY(m,t,j);
                                vv[0] = s[0];
                                vv[3] = s[3];
                                goto sr7;
                            }
                        }
                        s += s[1];
                    }
ss7:                j = t[1];
                    NCOPY(m,t,j);
sr7:;
                }
                break;
/*
            #] SUBEXPRESSION :
*/
            case EXPRESSION:
/*
```

```
            #[ EXPRESSION :
*/
                dirty = 1;
                tstop = t + t[1];
                v = m;
                *m++ = *t++;
                *m++ = *t++;
                *m++ = *t++;
                *m++ = *t++;
                s = subs;
                for ( j = 0; j < i; j++ ) {
                    if ( ( ABS(t[-1]) - 2*MAXPOWER ) == s[2] ) {
                        if ( *s == SYMTONUM ) {
                            m[-1] = s[3];
                            if ( t[-1] < 0 ) m[-1] = -m[-1];
                            break;
                        }
                        else if ( *s <= SYMTOSUB ) {
                            MLOCK(ErrorMessageLock);
                            MesPrint("Wildcard power of expression should be a number");
                            MUNLOCK(ErrorMessageLock);
                            SETERROR(-1)
                        }
                    }
                    s += s[1];
                }
                *m++ = *t++;
                COPYSUB(m,t)
                while ( t < tstop && *t != WILDCARDS ) {
                    j = t[1];
                    NCOPY(m,t,j);
                }
                if ( t < tstop && *t == WILDCARDS ) {
                    *m++ = *t;
                    s = sub;
                    j = s[1];
                    *m++ = j+2;
                    NCOPY(m,s,j);
                    t += t[1];
                }
                if ( t < tstop && *t == FROMBRAC ) {
                    w = m;
                    *m++ = *t;
                    *m++ = t[1];
                    if ( WildFill(BHEAD m,t+2,sub) < 0 ) {
                        MLOCK(ErrorMessageLock);
                        MesCall("WildFill");
                        MUNLOCK(ErrorMessageLock);
                        SETERROR(-1)
                    }
                    m += *m;
                    w[1] = m - w;
                    t += t[1];
                }
                while ( t < tstop ) {
                    j = t[1];
                    NCOPY(m,t,j);
                }
                v[1] = m-v;
                break;
/*
            #] EXPRESSION :
*/
            default:
/*
            #[ FUNCTIONS :
*/
                if ( *t >= FUNCTION ) {
                    dflag = 0;
                    na = 0;
                    *m = *t;
                    for ( si = 0; si < setflag; si += 2 ) {
                        if ( t == temp + setlist[si] ) {
                            dflag = DIRTYFLAG;  goto ss8;
                        }
                    }
                    s = subs;
                    for ( j = 0; j < i; j++ ) {
                        if ( *s == FUNTOFUN && *t == s[2] )
                            { *m = s[3]; dirty = 1; dflag = DIRTYFLAG; break; }
                        s += s[1];
                    }
ss8:                v = m;
                    if ( *t >= FUNCTION && functions[*t-FUNCTION].spec
                    >= TENSORFUNCTION ) {
                        if ( *m < FUNCTION || functions[*m-FUNCTION].spec
                        < TENSORFUNCTION ) {
```

```
                                MLOCK(ErrorMessageLock);
                                MesPrint("Illegal wildcarding of regular function to tensorfunction");
                                MUNLOCK(ErrorMessageLock);
                                SETERROR(-1)
                            }
                            m++; t++;
                            *m++ = *t++;
                            *m++ = *t++ | dflag;
                            goto Tensors;
                        }
                        m++; t++;
                        *m++ = *t++;
                        *m++ = *t++ | dflag;
                        COPYFUN3(m,t)
                        z = accu;
                        while ( t < u ) {       /* do an argument */
                            if ( *t < 0 ) {
/*
            #[ Simple arguments :
*/
                                CBUF *C = cbuf+AT.ebufnum;
                                for ( si = 0; si < setflag; si += 2 ) {
                                    if ( *t <= -FUNCTION ) {
                                        if ( t == temp + setlist[si] ) {
                                            v[2] |= DIRTYFLAG; goto ss10; }
                                    }
                                    else {
                                        if ( t == temp + setlist[si]-1 ) {
                                            v[2] |= DIRTYFLAG; goto ss9; }
                                    }
                                }
                                if ( *t == -ARGWILD ) {
                                    s = subs;
                                    for ( j = 0; j < i; j++ ) {
                                        if ( *s == ARGTOARG && s[2] == t[1] ) break;
                                        s += s[1];
                                    }
                                    v[2] |= DIRTYFLAG;
                                    w = C->rhs[s[3]];
DEBUG(MesPrint("Thread %w(b): s[3] = %d, w=(%d,%d,%d,%d)",s[3],w[0],w[1],w[2],w[3]);)
                                    if ( *w == 0 ) {
                                        w++;
                                        while ( *w ) {
                                            if ( *w > 0 ) j = *w;
                                            else if ( *w <= -FUNCTION ) j = 1;
                                            else j = 2;
                                            NCOPY(m,w,j);
                                        }
                                    }
                                    else {
                                        j = *w++;
                                        while ( --j >= 0 ) {
                                            if ( *w < MINSPEC ) *m++ = -VECTOR;
                                            else if ( *w >= 0 && *w < AM.OffsetIndex )
                                                *m++ = -SNUMBER;
                                            else *m++ = -INDEX;
                                            *m++ = *w++;
                                        }
                                    }
                                    t += 2;
                                    dirty = 1;
                                    if ( ( *v == NUMARGSFUN || *v == NUMTERMSFUN )
                                    && t >= u && m == v + FUNHEAD ) {
                                        m = v;
                                        *m++ = SNUMBER; *m++ = 3; *m++ = 0;
                                        break;
                                    }
                                }
                                else if ( *t <= -FUNCTION ) {
                                    *m = *t;
                                    s = subs;
                                    for ( j = 0; j < i; j++ ) {
                                        if ( -*t == s[2] ) {
                                            if ( *s == FUNTOFUN )
                                                { *m = -s[3]; dirty = 1; v[2] |= DIRTYFLAG; break; }
                                        }
                                        s += s[1];
                                    }
                                    m++; t++;
                                }
                                else if ( *t == -SYMBOL ) {
                                    *m++ = *t++;
                                    *m = *t;
                                    s = subs;
                                    for ( j = 0; j < i; j++ ) {
                                        if ( *t == s[2] && *s <= SYMTOSUB ) {
                                            dirty = 1; v[2] |= DIRTYFLAG;
```

```
                                        if ( AR.PolyFunType == 2 && v[0] == AR.PolyFun )
                                            v[2] |= MUSTCLEANPRF;
                                        if ( *s == SYMTOSYM ) *m = s[3];
                                        else if ( *s == SYMTONUM ) {
                                            m[-1] = -SNUMBER;
                                            *m = s[3];
                                        }
                                        else if ( *s == SYMTOSUB ) {
ToSub:                                      m--;
                                            w = C->rhs[s[3]];
DEBUG(MesPrint("Thread %w(c): s[3] = %d, w=(%d,%d,%d,%d)",s[3],w[0],w[1],w[2],w[3]);)
                                            s = m;
                                            m += 2;
                                            while ( *w ) {
                                                j = *w;
                                                NCOPY(m,w,j);
                                            }
                                            *s = WORDDIF(m,s);
                                            s[1] = 0;
                                            *m = 0;
                                            if ( t[-1] == -MINVECTOR ) {
                                                w = s+2;
                                                while ( *w ) {
                                                    w += *w;
                                                    w[-1] = -w[-1];
                                                }
                                            }
                                            if ( ToFast(s,s) ) {
                                                if ( *s <= -FUNCTION ) m = s;
                                                else m = s + 1;
                                            }
                                            else m--;
                                        }
                                        break;
                                    }
                                    s += s[1];
                                }
                                m++; t++;
                            }
                            else if ( *t == -INDEX ) {
                                *m++ = *t++;
                                *m = *t;
                                s = subs;
                                for ( j = 0; j < i; j++ ) {
                                    if ( *t == s[2] ) {
                                        if ( *s == INDTOIND || *s == VECTOVEC ) {
                                            *m = s[3];
                                            if ( *m < MINSPEC ) m[-1] = -VECTOR;
                                            else if ( *m >= 0 && *m < AM.OffsetIndex )
                                                m[-1] = -SNUMBER;
                                            else m[-1] = -INDEX;
                                        }
                                        else if ( *s == VECTOSUB || *s == INDTOSUB ) {
                                            m[-1] = -INDEX;
                                            *m = ++AR.CurDum;
                                            *z++ = SUBEXPRESSION;
                                            *z++ = 4+SUBEXPSIZE;
                                            *z++ = s[3]+1;
                                            *z++ = 1;
                                            *z++ = AT.ebufnum;
                                            FILLSUB(z)
                                            *z++ = INDTOIND;
                                            *z++ = 4;
                                            *z++ = FUNNYVEC;
                                            *z++ = AR.CurDum;
                                        }
                                        v[2] |= DIRTYFLAG; dirty = 1;
                                        break;
                                    }
                                    s += s[1];
                                }
                                m++; t++;
                            }
                            else if ( *t == -VECTOR || *t == -MINVECTOR ) {
                                *m++ = *t++;
                                *m = *t;
                                s = subs;
                                for ( j = 0; j < i; j++ ) {
                                    if ( *t == s[2] ) {
                                        if ( *s == VECTOVEC ) *m = s[3];
                                        else if ( *s == VECTOMIN ) {
                                            *m = s[3];
                                            if ( t[-1] == -VECTOR )
                                                m[-1] = -MINVECTOR;
                                            else
                                                m[-1] = -VECTOR;
                                        }
```

```
                                        else if ( *s == VECTOSUB ) goto ToSub;
                                        dirty = 1; v[2] |= DIRTYFLAG;
                                        break;
                                    }
                                    s += s[1];
                                }
                                m++; t++;
                            }
                            else if ( *t == -SNUMBER ) {
                                *m++ = *t++;
                                *m = *t;
                                s = subs;
                                for ( j = 0; j < i; j++ ) {
                                    if ( *t == s[2] && *s >= NUMTONUM && *s <= NUMTOSUB ) {
                                        dirty = 1; v[2] |= DIRTYFLAG;
                                        if ( *s == NUMTONUM ) *m = s[3];
                                        else if ( *s == NUMTOSYM ) {
                                            m[-1] = -SYMBOL;
                                            *m = s[3];
                                        }
                                        else if ( *s == NUMTOIND ) {
                                            m[-1] = -INDEX;
                                            *m = s[3];
                                        }
                                        else if ( *s == NUMTOSUB ) goto ToSub;
                                        break;
                                    }
                                    s += s[1];
                                }
                                m++; t++;
                            }
                            else {
ss9:
                                *m++ = *t++;
ss10:
                                *m++ = *t++;
                            }
                            na = WORDDIF(z,accu);
/*
            #] Simple arguments :
*/
                        }
                        else {
                            w = m;
                            zz = t;
                            NEXTARG(zz)
                            odirt = AN.WildDirt; AN.WildDirt = 0;
                            AR.CompressPointer = accu + na;
                            for ( j = 0; j < ARGHEAD; j++ ) *m++ = *t++;
                            j = 0;
                            adirt = 0;
                            while ( t < zz ) {  /* do a term */
                                if ( ( len = WildFill(BHEAD m,t,sub) ) < 0 ) {
                                    MLOCK(ErrorMessageLock);
                                    MesCall("WildFill");
                                    MUNLOCK(ErrorMessageLock);
                                    SETERROR(-1)
                                }
                                if ( AN.WildDirt ) {
                                    adirt = AN.WildDirt;
                                    AN.WildDirt = 0;
                                }
                                m += len;
                                t += *t;
                            }
                            *w = WORDDIF(m,w);  /* Fill parameter length */
                            if ( adirt ) {
                                dirty = w[1] = 1; v[2] |= DIRTYFLAG;
                                if ( AR.PolyFunType == 2 && v[0] == AR.PolyFun )
                                    v[2] |= MUSTCLEANPRF;
                                AN.WildDirt = adirt;
                            }
                            else {
                                AN.WildDirt = odirt;
                            }
                            if ( ToFast(w,w) ) {
                                if ( *w <= -FUNCTION ) {
                                    if ( *w == NUMARGSFUN || *w == NUMTERMSFUN ) {
                                        *w = -SNUMBER; w[1] = 0; m = w + 2;
                                    }
                                    else m = w+1;
                                }
                                else m = w+2;
                            }
                            AR.CompressPointer = oldcpointer;
                        }
                    }
                    v[1] = WORDDIF(m,v);        /* Fill function length */
                    s = accu;
```

```
                    NCOPY(m,s,na);
/*
                    Now some code to speed up a few special cases
*/
                    if ( v[0] == EXPONENT ) {
                        if ( v[1] == FUNHEAD+4 && v[FUNHEAD] == -SYMBOL &&
                        v[FUNHEAD+2] == -SNUMBER && v[FUNHEAD+3] < MAXPOWER
                        && v[FUNHEAD+3] > -MAXPOWER ) {
                            v[0] = SYMBOL;
                            v[1] = 4;
                            v[2] = v[FUNHEAD+1];
                            v[3] = v[FUNHEAD+3];
                            m = v+4;
                        }
                        else if ( v[1] == FUNHEAD+ARGHEAD+11
                         && v[FUNHEAD] == ARGHEAD+9
                         && v[FUNHEAD+ARGHEAD] == 9
                         && v[FUNHEAD+ARGHEAD+1] == DOTPRODUCT
                         && v[FUNHEAD+ARGHEAD+8] == 3
                         && v[FUNHEAD+ARGHEAD+7] == 1
                         && v[FUNHEAD+ARGHEAD+6] == 1
                         && v[FUNHEAD+ARGHEAD+5] == 1
                         && v[FUNHEAD+ARGHEAD+9] == -SNUMBER
                         && v[FUNHEAD+ARGHEAD+10] < MAXPOWER
                         && v[FUNHEAD+ARGHEAD+10] > -MAXPOWER ) {
                            v[0] = DOTPRODUCT;
                            v[1] = 5;
                            v[2] = v[FUNHEAD+ARGHEAD+3];
                            v[3] = v[FUNHEAD+ARGHEAD+4];
                            v[4] = v[FUNHEAD+ARGHEAD+10];
                            m = v+5;
                        }
                    }
                }
                else { while ( t < u ) *m++ = *t++; }
/*
            #] FUNCTIONS :
*/
        }
        t = uu;
    } while ( t < r );
    t = from;           /* Copy coefficient */
    t += *t;
    if ( r < t ) do { *m++ = *r++; } while ( r < t );
    if ( ( sgn & 1 ) != 0 ) m[-1] = -m[-1];
    *to = WORDDIF(m,to);
    if ( dirty ) AN.WildDirt = dirty;
    return(*to);
}

/*
        #] WildFill :
        #[ ResolveSet :         WORD ResolveSet(from,to,subs)

        The set syntax is:
        SET,length,subterm,where,whichmember[,where,whichmember]

        setlength is 2*n+1 with n the number of set substitutions.
        length = setlength + subtermlength + 2

        At `where' is the number of the set and `whichmember' is the
        number of the element. This is still a symbol/dollar and we
        have to find the substitution in the wildcards.
        The output is the subterm in which the setelements have been
        substituted. This is ready for further wildcard substitutions.
*/

WORD ResolveSet(PHEAD WORD *from, WORD *to, WORD *subs)
{
    GETBIDENTITY
    WORD *m, *s, *w, j, i, ii, i3, flag, num;
    DOLLARS d = 0;
#ifdef WITHPTHREADS
    int nummodopt, dtype = -1;
#endif
    m = to;                     /* pointer in output */
    s = from + 2;
    w = s + s[1];
    while ( s < w ) *m++ = *s++;
    j = (from[1] - WORDDIF(w,from) ) >> 1;
    m = subs + subs[1];
    subs += SUBEXPSIZE;
    s = subs;
    i = 0;
    while ( s < m ) { i++; s += s[1]; }
    m = to;
    if ( *m >= FUNCTION && functions[*m-FUNCTION].spec
```

```
    >= TENSORFUNCTION ) flag = 0;
    else flag = 1;
    while ( --j >= 0 ) {
        if ( w[1] >= 0 ) {
            s = subs;
            for ( ii = 0; ii < i; ii++ ) {
                if ( *s == SYMTONUM && s[2] == w[1] ) { num = s[3]; goto GotOne; }
                s += s[1];
            }
            MLOCK(ErrorMessageLock);
            MesPrint(" Unresolved setelement during substitution");
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        else {  /* Dollar ! */
            d = Dollars - w[1];
#ifdef WITHPTHREADS
            if ( AS.MultiThreaded ) {
                for ( nummodopt = 0; nummodopt < NumModOptdollars; nummodopt++ ) {
                    if ( -w[1] == ModOptdollars[nummodopt].number ) break;
                }
                if ( nummodopt < NumModOptdollars ) {
                    dtype = ModOptdollars[nummodopt].type;
                    if ( dtype == MODLOCAL ) {
                        d = ModOptdollars[nummodopt].dstruct+AT.identity;
                    }
                    else {
                        LOCK(d->pthreadslockread);
                    }
                }
            }
#endif
            if ( d->type == DOLNUMBER || d->type == DOLTERMS ) {
                if ( d->where[0] == 4 && d->where[3] == 3 && d->where[2] == 1
                && d->where[1] > 0 && d->where[4] == 0 ) {
                    num = d->where[1]; goto GotOne;
                }
            }
            else if ( d->type == DOLINDEX ) {
                if ( d->index > 0 && d->index < AM.OffsetIndex ) {
                    num = d->index; goto GotOne;
                }
            }
            else if ( d->type == DOLARGUMENT ) {
                if ( d->where[0] == -SNUMBER && d->where[1] > 0 ) {
                    num = d->where[1]; goto GotOne;
                }
            }
            else if ( d->type == DOLWILDARGS ) {
                if ( d->where[0] == 1 &&
                d->where[1] > 0 && d->where[1] < AM.OffsetIndex ) {
                    num = d->where[1]; goto GotOne;
                }
                if ( d->where[0] == 0 && d->where[1] < 0 && d->where[3] == 0 ) {
                    if ( ( d->where[1] == -SNUMBER && d->where[2] > 0 )
                    || ( d->where[1] == -INDEX && d->where[2] > 0
                    && d->where[2] < AM.OffsetIndex ) ) {
                        num = d->where[2]; goto GotOne;
                    }
                }
            }
#ifdef WITHPTHREADS
            if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
            MLOCK(ErrorMessageLock);
            MesPrint("Unusable type of variable $%s in set substitution",
                AC.dollarnames->namebuffer+d->name);
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
GotOne:;
#ifdef WITHPTHREADS
        if ( dtype > 0 && dtype != MODLOCAL ) { UNLOCK(d->pthreadslockread); }
#endif
        ii = m[*w];
        if ( ii >= 2*MAXPOWER ) i3 = ii - 2*MAXPOWER;
        else if ( ii <= -2*MAXPOWER ) i3 = -ii - 2*MAXPOWER;
        else i3 = ( ii >= 0 ) ? ii: -ii - 1;

        if ( num > ( Sets[i3].last - Sets[i3].first ) || num <= 0 ) {
            MLOCK(ErrorMessageLock);
            MesPrint("Array bound check during set substitution");
            MesPrint("    value is %d",num);
            MUNLOCK(ErrorMessageLock);
            return(-1);
        }
        m[*w] = (SetElements+Sets[i3].first)[num-1];
```

```
        if ( Sets[i3].type == CSYMBOL && m[*w] > MAXPOWER ) {
            if ( ii >= 2*MAXPOWER ) m[*w] -= 2*MAXPOWER;
            else if ( ii <= -2*MAXPOWER ) m[*w] = -(m[*w] - 2*MAXPOWER);
            else {
                m[*w] -= MAXPOWER;
                if ( m[*w] < MAXPOWER ) m[*w] -= 2*MAXPOWER;
                if ( flag ) MakeDirty(m,m+*w,1);
            }
        }
        else if ( Sets[i3].type == CSYMBOL ) {
            if ( ii >= 2*MAXPOWER ) m[*w] += 2*MAXPOWER;
            else if ( ii <= -2*MAXPOWER ) m[*w] = -m[*w] - 2*MAXPOWER;
            else if ( ii < 0 ) m[*w] = - m[*w];
        }
        else if ( ii < 0 ) m[*w] = - m[*w];
        w += 2;
    }
    m = to;
    if ( *m >= FUNCTION && functions[*m-FUNCTION].spec
    >= TENSORFUNCTION ) {
        w = from + 2 + from[3];
        if ( *w == 0 ) {     /* We had function -> tensor */
            m = from + 2 + FUNHEAD; s = to + FUNHEAD;
            while ( m < w ) {
                if ( *m == -INDEX || *m == -VECTOR ) {}
                else if ( *m == -ARGWILD ) { *s++ = FUNNYWILD; }
                else {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Illegal argument in tensor after set substitution");
                    MUNLOCK(ErrorMessageLock);
                    SETERROR(-1)
                }
                *s++ = m[1];
                m += 2;
            }
            to[1] = WORDDIF(s,to);
        }
    }
    return(0);
}

/*
        #] ResolveSet :
        #[ ClearWild :           void ClearWild()

    Clears the current wildcard settings and makes them ready for
    CheckWild and AddWild.

*/

void ClearWild(PHEAD0)
{
    GETBIDENTITY
    WORD n, nn, *w;
    n = (AN.WildValue[-SUBEXPSIZE+1]-SUBEXPSIZE)/4;      /* Number of wildcards */
    AN.NumWild = nn = n;
    if ( n > 0 ) {
        w = AT.WildMask;
        do { *w++ = 0; } while ( --n > 0 );
        w = AN.WildValue;
        do {
            if ( *w == SYMTONUM ) *w = SYMTOSYM;
            w += w[1];
        } while ( --nn > 0 );
    }
}

/*
        #] ClearWild :
        #[ AddWild :            WORD AddWild(oldnumber,type,newnumber)

        Adds a wildcard assignment.
        Extra parameter in AN.argaddress;

*/

WORD AddWild(PHEAD WORD oldnumber, WORD type, WORD newnumber)
{
    GETBIDENTITY
    WORD *w, *m, n, k, i = -1;
    CBUF *C = cbuf+AT.ebufnum;
    WORD eattensor = type & EATTENSOR;
    type = type & ~EATTENSOR;
DEBUG(WORD *mm;)
    AN.WildReserve = 0;
    m = AT.WildMask;
    w = AN.WildValue;
```

```
    n = AN.NumWild;
    if ( n <= 0 ) { return(-1); }
    if ( type <= SYMTOSUB ) {
        do {
            if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
                if ( n > 1 && w[4] == SETTONUM ) i = w[7];
                *w = type;
                if ( *m != 2 ) *m = 1;
                if ( type != SYMTOSUB ) {
                    if ( type == SYMTONUM ) AN.MaskPointer = m;
                    w[3] = newnumber;
                    goto FlipOn;
                }
                m = AddRHS(AT.ebufnum,1);
                w[3] = C->numrhs;
                w = AN.argaddress;
DEBUG(mm = m;)
                n = *w - ARGHEAD;
                w += ARGHEAD;
                while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,4);
                while ( --n >= 0 ) *m++ = *w++;
                *m++ = 0;
                C->rhs[C->numrhs+1] = m;
DEBUG(MesPrint("Thread %w(d): m=(%d,%d,%d,%d)(%d)",mm[0],mm[1],mm[2],mm[3],C->numrhs);)
                C->Pointer = m;
                goto FlipOn;
            }
            m++; w += w[1];
        } while ( --n > 0 );
    }
    else if ( type == ARGTOARG ) {
        do {
            if ( w[2] == oldnumber && *w == ARGTOARG ) {
                *m = 1;
                m = AddRHS(AT.ebufnum,1);
                w[3] = C->numrhs;
                w = AN.argaddress;
DEBUG(mm=m;)
                if ( eattensor ) {
                    n = newnumber;
                    *m++ = n;
                    w = AN.argaddress;
                }
                else {
                    while ( --newnumber >= 0 ) { NEXTARG(w) }
                    n = WORDDIF(w,AN.argaddress);
                    w = AN.argaddress;
                    *m++ = 0;
                }
                while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,5);
DEBUG(if ( mm != m-1 ) MesPrint("Thread %w(e): Alarm!"); mm = m-1;)
                while ( --n >= 0 ) *m++ = *w++;
                *m++ = 0;
                C->rhs[C->numrhs+1] = m;
                C->Pointer = m;
DEBUG(MesPrint("Thread %w(e): w=(%d,%d,%d,%d)(%d)",mm[0],mm[1],mm[2],mm[3],C->numrhs);)
                return(0);
            }
            m++; w += w[1];
        } while ( --n > 0 );
    }
    else if ( type == ARLTOARL ) {
        do {
            if ( w[2] == oldnumber && *w == ARGTOARG ) {
                WORD **a;
                *m = 1;
                m = AddRHS(AT.ebufnum,1);
                w[3] = C->numrhs;
DEBUG(mm=m;)
                a = (WORD **)(AN.argaddress); n = 0; k = newnumber;
                while ( --newnumber >= 0 ) {
                    w = *a++;
                    if ( *w > 0 ) n += *w;
                    else if ( *w <= -FUNCTION ) n++;
                    else n += 2;
                }
                *m++ = 0;
                while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,6);
DEBUG(if ( mm != m-1 ) MesPrint("Thread %w(f): Alarm!"); mm = m-1;)
                a = (WORD **)(AN.argaddress);
                while ( --k >= 0 ) {
                    w = *a++;
                    if ( *w > 0 ) { n = *w; NCOPY(m,w,n); }
                    else if ( *w <= -FUNCTION ) *m++ = *w++;
                    else { *m++ = *w++; *m++ = *w++; }
                }
                *m++ = 0;
```

```
                C->rhs[C->numrhs+1] = m;
DEBUG(MesPrint("Thread %w(f): w=(%d,%d,%d,%d)(%d)",mm[0],mm[1],mm[2],mm[3],C->numrhs);)
                C->Pointer = m;
                return(0);
            }
            m++; w += w[1];
        } while ( --n > 0 );
    }
    else if ( type == VECTOSUB || type == INDTOSUB ) {
        WORD *ss, *sstop, *tt, *ttstop, j, *v1, *v2 = 0;
        do {
            if ( w[2] == oldnumber && ( *w == type ||
            ( type == VECTOSUB && ( *w == VECTOVEC || *w == VECTOMIN ) )
            || ( type == INDTOSUB && *w == INDTOIND ) ) ) {
                if ( n > 1 && w[4] == SETTONUM ) i = w[7];
                *w = type;
                *m = 1;
                m = AddRHS(AT.ebufnum,1);
                w[3] = C->numrhs;
                w = AN.argaddress;
                n = *w - ARGHEAD;
                w += ARGHEAD;
                while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,7);
                while ( --n >= 0 ) *m++ = *w++;
                *m++ = 0;
                C->rhs[C->numrhs+1] = m;
                C->Pointer = m;
                m = AddRHS(AT.ebufnum,1);
                w = AN.argaddress;
                n = *w - ARGHEAD;
                w += ARGHEAD;
                while ( (m + n + 10) > C->Top ) m = DoubleCbuffer(AT.ebufnum,m,8);
                sstop = w + n;
                while ( w < sstop ) {            /* Run over terms */
                    tt = w + *w; ttstop = tt - ABS(tt[-1]);
                    ss = m; m++; w++;
                    while ( w < ttstop ) {       /* Subterms */
                        if ( *w != INDEX ) {
                            j = w[1];
                            NCOPY(m,w,j);
                        }
                        else {
                            v1 = m;
                            *m++ = *w++;
                            *m++ = j = *w++;
                            j -= 2;
                            while ( --j >= 0 ) {
                                if ( *w >= MINSPEC ) *m++ = *w++;
                                else v2 = w++;
                            }
                            j = WORDDIF(m,v1);
                            if ( j != v1[1] ) {
                                if ( j <= 2 ) m -= 2;
                                else v1[1] = j;
                                *m++ = VECTOR;
                                *m++ = 4;
                                *m++ = *v2;
                                *m++ = FUNNYVEC;
                            }
                        }
                    }
                    while ( w < tt ) *m++ = *w++;
                    *ss = WORDDIF(m,ss);
                }
                *m++ = 0;
                C->rhs[C->numrhs+1] = m;
                C->Pointer = m;
                if ( m > C->Top ) {
                    MLOCK(ErrorMessageLock);
                    MesPrint("Internal problems with extra compiler buffer");
                    MUNLOCK(ErrorMessageLock);
                    Terminate(-1);
                }
                goto FlipOn;
            }
            m++; w += w[1];
        } while ( --n > 0 );
    }
    else {
        do {
            if ( w[2] == oldnumber && ( *w == type || ( type == VECTOVEC
            && ( *w == VECTOMIN || *w == VECTOSUB ) ) || ( type == VECTOMIN
            && ( *w == VECTOVEC || *w == VECTOSUB ) )
            || ( type == INDTOIND && *w == INDTOSUB ) ) ) {
                if ( n > 1 && w[4] == SETTONUM ) i = w[7];
                *w = type;
                w[3] = newnumber;
```

```
                *m = 1;
                goto FlipOn;
            }
            m++; w += w[1];
        } while ( --n > 0 );
    }
    MLOCK(ErrorMessageLock);
    MesPrint("Bug in AddWild.");
    MUNLOCK(ErrorMessageLock);
    return(-1);
FlipOn:
    if ( i >= 0 ) {
        m = AT.WildMask;
        w = AN.WildValue;
        n = AN.NumWild;
        while ( --n >= 0 ) {
            if ( w[2] == i && *w == SYMTONUM ) {
                *m = 2;
                return(0);
            }
            m++; w += w[1];
        }
        MLOCK(ErrorMessageLock);
        MesPrint(" Bug in AddWild with passing set[i]");
        MUNLOCK(ErrorMessageLock);
/*
        For the moment we want to crash here. That is easier with debugging.
*/
#ifdef WITHPTHREADS
        { WORD *s = 0;
            *s++ = 1;
        }
#endif
        Terminate(-1);
    }
    return(0);
}

/*
        #] AddWild :
        #[ CheckWild :          WORD CheckWild(oldnumber,type,newnumber,newval)

        Tests whether a wildcard assignment is allowed.
        A return value of zero means that it is allowed (nihil obstat).
        If the variable has been assigned already its existing
        assignment is returned in AN.oldvalue and AN.oldtype, which are
        global variables.

        Note the special problem with name?set[i]. Here we have to pass
        an extra assignment. This cannot be done via globals as we
        call CheckWild sometimes twice before calling AddWild.
        Trick: Check the assignment of the number and if OK put it
        in place, but don't alter the used flag (if needed).
        Then AddWild can alter the used flag but the value is there.
        As long as this trick is `hanging' we turn on the flag:
        `AN.WildReserve' which is either turned off by AddWild or by
        a failing call to CheckWild.

        With ARGTOARG the tensors give the number of arguments
        or-ed with EATTENSOR which is at least 8192.
*/

WORD CheckWild(PHEAD WORD oldnumber, WORD type, WORD newnumber, WORD *newval)
{
    GETBIDENTITY
    WORD *w, *m, *s, n, old2, inset;
    WORD n2, oldval, dirty, i, j, notflag = 0, retblock = 0;
    CBUF *C = cbuf+AT.ebufnum;
    WORD eattensor = type & EATTENSOR;
    type = type & ~EATTENSOR;
    m = AT.WildMask;
    w = AN.WildValue;
    n = AN.NumWild;
    if ( n <= 0 ) { AN.oldtype = -1; AN.WildReserve = 0; return(-1); }
    switch ( type ) {
        case SYMTONUM :
            *newval = newnumber;
            do {
                if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
                    old2 = *w;
                    if ( !*m ) goto TestSet;
                    AN.MaskPointer = m;
                    if ( *w == SYMTONUM  && w[3] == newnumber ) {
                        return(0);
                    }
                    AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
                }
```

```
                m++; w += w[1];
            } while ( --n > 0 );
            break;
        case SYMTOSYM :
            *newval = newnumber;
            do {
                if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
                    old2 = *w;
                    if ( *w == SYMTOSYM ) {
                        if ( !*m ) goto TestSet;
                        if ( newnumber >= 0 && (w+4) < AN.WildStop
                        && ( w[4] == FROMSET || w[4] == SETTONUM )
                        && w[7] >= 0 ) goto TestSet;
                        if ( w[3] == newnumber ) return(0);
                    }
                    else {
                        if ( !*m ) goto TestSet;
                    }
                    goto NoM;
                }
                m++; w += w[1];
            } while ( --n > 0 );
            break;
        case SYMTOSUB :
/*
            Now newval contains the pointer to the argument.
*/
            {
/*
                Search for vector or index nature. If so: reject.
*/
                WORD *ss, *sstop, *tt, *ttstop;
                ss = newval;
                sstop = ss + *ss;
                ss += ARGHEAD;
                while ( ss < sstop ) {
                    tt = ss + *ss;
                    ttstop = tt - ABS(tt[-1]);
                    ss++;
                    while ( ss < ttstop ) {
                        if ( *ss == INDEX ) goto NoMatch;
                        ss += ss[1];
                    }
                    ss = tt;
                }
            }
            do {
                if ( w[2] == oldnumber && *w <= SYMTOSUB ) {
                    old2 = *w;
                    if ( *w == SYMTONUM || *w == SYMTOSYM ) {
                        if ( !*m ) {
                            s = w + w[1];
                            if ( s >= AN.WildStop || *s != SETTONUM )
                                    goto TestSet;
                        }
                    }
                    else if ( *w == SYMTOSUB ) {
                        if ( !*m ) {
                            s = w + w[1];
                            if ( s >= AN.WildStop || *s != SETTONUM )
                                    goto TestSet;
                        }
                        n = *newval - 2;
                        newval += 2;
                        m = C->rhs[w[3]];
                        if ( (C->rhs[w[3]+1] - m - 1) == n ) {
                            while ( n > 0 ) {
                                if ( *m != *newval ) {
                                    m++; newval++; break;
                                }
                                m++; newval++;
                                n--;
                            }
                            if ( n <= 0 ) return(0);
                        }
                    }
                    AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
                }
                m++; w += w[1];
            } while ( --n > 0 );
            break;
        case ARGTOARG :
            do {
                if ( w[2] == oldnumber && *w == ARGTOARG ) {
                    if ( !*m ) return(0);       /* nihil obstat */
                    m = C->rhs[w[3]];
                    if ( eattensor ) {
```

```
                        n = newnumber;
                        if ( *m != 0 ) {
                            if ( n == *m ) {
                                m++;
                                while ( --n >= 0 ) {
                                    if ( *m != *newval ) {
                                        m++; newval++; break;
                                    }
                                    m++; newval++;
                                }
                                if ( n < 0 ) return(0);
                            }
                        }
                        else {
                            m++;
                            while ( --n >= 0 ) {
                                if ( *newval != m[1] || ( *m != -INDEX
                                && *m != -VECTOR && *m != -SNUMBER ) ) break;
                                m += 2;
                                newval++;
                            }
                            if ( n < 0 && *m == 0 ) return(0);
                        }
                    }
                    else {
                        i = newnumber;
                        if ( *m != 0 ) {     /* Tensor field */
                            if ( *m == i ) {
                                m++;
                                while ( --i >= 0 ) {
                                    if ( *m != newval[1]
                                    || ( *newval != -VECTOR
                                      && *newval != -INDEX
                                      && *newval != -SNUMBER ) ) break;
                                    newval += 2;
                                    m++;
                                }
                                if ( i < 0 ) return(0);
                            }
                        }
                        else {
                            m++;
                            s = newval;
                            while ( --i >= 0 ) { NEXTARG(s) }
                            n = WORDDIF(s,newval);
                            while ( --n >= 0 ) {
                                if ( *m != *newval ) {
                                    m++; newval++; break;
                                }
                                m++; newval++;
                            }
                            if ( n < 0 && *m == 0 ) return(0);
                        }
                    }
                    AN.oldtype = *w; AN.oldvalue = w[3]; goto NoMatch;
                }
                m++; w += w[1];
            } while ( --n > 0 );
            break;
        case ARLTOARL :
            do {
                if ( w[2] == oldnumber && *w == ARGTOARG ) {
                    WORD **a;
                    if ( !*m ) return(0);       /* nihil obstat */
                    m = C->rhs[w[3]];
                    i = newnumber;
                    a = (WORD **)newval;
                    if ( *m != 0 ) {     /* Tensor field */
                        if ( *m == i ) {
                            m++;
                            while ( --i >= 0 ) {
                                s = *a++;
                                if ( *m != s[1]
                                || ( *s != -VECTOR
                                  && *s != -INDEX
                                  && *s != -SNUMBER ) ) break;
                                m++;
                            }
                            if ( i < 0 ) return(0);
                        }
                    }
                    else {
                        m++;
                        while ( --i >= 0 ) {
                            s = *a++;
                            if ( *s > 0 ) {
                                n = *s;
```

```
                                while ( --n >= 0 ) {
                                    if ( *s != *m ) {
                                        s++; m++; break;
                                    }
                                    s++; m++;
                                }
                                if ( n >= 0 ) break;
                            }
                            else if ( *s <= -FUNCTION ) {
                                if ( *s != *m ) {
                                    s++; m++; break;
                                }
                                s++; m++;
                            }
                            else {
                                if ( *s != *m ) {
                                    s++; m++; break;
                                }
                                s++; m++;
                                if ( *s != *m ) {
                                    s++; m++; break;
                                }
                                s++; m++;
                            }
                        }
                        if ( i < 0 && *m == 0 ) return(0);
                    }
                    AN.oldtype = *w; AN.oldvalue = w[3]; goto NoMatch;
                }
                m++; w += w[1];
            } while ( --n > 0 );
            break;
        case VECTOSUB :
        case INDTOSUB :
/*
            Now newval contains the pointer to the argument(s).
*/
            {
/*
                Search for vector or index nature. If not so: reject.
*/
                WORD *ss, *sstop, *tt, *ttstop, count, jt;
                ss = newval;
                sstop = ss + *ss;
                ss += ARGHEAD;
                while ( ss < sstop ) {
                    tt = ss + *ss;
                    ttstop = tt - ABS(tt[-1]);
                    ss++;
                    count = 0;
                    while ( ss < ttstop ) {
                        if ( *ss == INDEX ) {
                            jt = ss[1] - 2; ss += 2;
                            while ( --jt >= 0 ) {
                                if ( *ss < MINSPEC ) count++;
                                ss++;
                            }
                        }
                        else ss += ss[1];
                    }
                    if ( count != 1 ) goto NoMatch;
                    ss = tt;
                }
            }
            do {
                if ( w[2] == oldnumber ) {
                    old2 = *w;
                    if ( ( type == VECTOSUB && ( *w == VECTOVEC || *w == VECTOMIN ) )
                    || ( type == INDTOSUB && *w == INDTOIND ) ) {
                        if ( !*m ) goto TestSet;
                        AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
                    }
                    else if ( *w == type ) {
                        if ( !*m ) goto TestSet;
                        if ( type != INDTOIND && type != INDTOSUB ) {   /* Prevent double index */
                            n = *newval - 2;
                            newval += 2;
                            m = C->rhs[w[3]];
                            if ( (C->rhs[w[3]+1] - m - 1) == n ) {
                                while ( n > 0 ) {
                                    if ( *m != *newval ) {
                                        m++; newval++; break;
                                    }
                                    m++; newval++;
                                    n--;
                                }
                                if ( n <= 0 ) return(0);
```

```
                            }
                        }
                        AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
                    }
                }
                m++; w += w[1];
            } while ( --n > 0 );
            break;
        default :
            *newval = newnumber;
            do {
                if ( w[2] == oldnumber ) {
                    if ( *w == type ) {
                        old2 = *w;
                        if ( !*m ) goto TestSet;
                        if ( newnumber >= 0 && (w+4) < AN.WildStop &&
                            ( w[4] == FROMSET || w[4] == SETTONUM )
                            && w[7] >= 0 ) goto TestSet;
                        if ( newnumber < 0 && *w == VECTOVEC
                            && (w+4) < AN.WildStop && ( w[4] == FROMSET
                            || w[4] == SETTONUM ) && w[7] >= 0 ) goto TestSet;
/*
                        The next statement kills multiple indices -> vector
*/
                        if ( *w == INDTOIND && w[3] < 0 ) goto NoMatch;
                        if ( w[3] == newnumber ) {
                            if ( *w != FUNTOFUN || newnumber < FUNCTION
                            || functions[newnumber-FUNCTION].spec ==
                            functions[oldnumber-FUNCTION].spec )
                                return(0);
                        }
                        AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
                    }
                    else if ( ( type == VECTOVEC &&
                            ( *w == VECTOSUB || *w == VECTOMIN ) )
                           || ( type == INDTOIND && *w == INDTOSUB ) ) {
                        if ( *m ) goto NoMatch;
                        old2 = *w;
                        goto TestSet;
                    }
                    else if ( type == VECTOMIN &&
                            ( *w == VECTOSUB || *w == VECTOVEC ) ) {
                        if ( *m ) goto NoMatch;
                        old2 = *w;
                        goto TestSet;
                    }
                }
                m++; w += w[1];
                if ( n > 1 && ( *w == FROMSET
                || *w == SETTONUM ) ) { n--; m++; w += w[1]; }
            } while ( --n > 0 );
            break;
    }
    AN.oldtype = -1;
    AN.oldvalue = -1;
    AN.WildReserve = 0;
    MLOCK(ErrorMessageLock);
    MesPrint("Inconsistency in Wildcard prototype.");
    MUNLOCK(ErrorMessageLock);
    return(-1);
NoMatch:
    AN.WildReserve = 0;
    return(1+retblock);
/*
    Here we test the compatibility with a set specification.
*/
TestSet:
    dirty = *m;
    oldval = w[3];
    w += w[1];
    if ( w < AN.WildStop && ( *w == FROMSET || *w == SETTONUM ) ) {
        WORD k;
        s = w;
        j = w[2]; n2 = w[3];
/*
            if SETTONUM:  x?j[n2]
            if FROMSET:   x?j?n2    or x?j and n2 = -WOLDOFFSET.
*/
        if ( j > WILDOFFSET ) {
            j -= 2*WILDOFFSET;
            notflag = 1;
/*
        ???????
*/
            AN.oldtype = -1;
            AN.oldvalue = -1;
        }
```

```
        if ( j < AM.NumFixedSets ) {    /* special set */
            retblock = 1;
            switch ( j ) {
                case POS_:
                    if ( type != SYMTONUM ||
                    newnumber <= 0 ) goto NoMnot;
                    break;
                case POS0_:
                    if ( type != SYMTONUM ||
                    newnumber < 0 ) goto NoMnot;
                    break;
                case NEG_:
                    if ( type != SYMTONUM ||
                    newnumber >= 0 ) goto NoMnot;
                    break;
                case NEG0_:
                    if ( type != SYMTONUM ||
                    newnumber > 0 ) goto NoMnot;
                    break;
                case EVEN_:
                    if ( type != SYMTONUM ||
                    ( newnumber & 1 ) != 0 ) goto NoMnot;
                    break;
                case ODD_:
                    if ( type != SYMTONUM ||
                    ( newnumber & 1 ) == 0 ) goto NoMnot;
                    break;
                case Z_:
                    if ( type != SYMTONUM ) goto NoMnot;
                    break;
                case SYMBOL_:
                    if ( type != SYMTOSYM ) goto NoMnot;
                    break;
                case FIXED_:
                    if ( type != INDTOIND ||
                    newnumber >= AM.OffsetIndex ||
                    newnumber < 0 ) goto NoMnot;
                    break;
                case INDEX_:
                    if ( type != INDTOIND ||
                    newnumber < 0 ) goto NoMnot;
                    break;
                case Q_:
                    if ( type == SYMTONUM ) break;
                    if ( type == SYMTOSUB ) {
                        WORD *ss, *sstop;
                        ss = newval;
                        sstop = ss + *ss;
                        ss += ARGHEAD;
                        if ( ss >= sstop ) break;
                        if ( ss + *ss < sstop ) goto NoMnot;
                        if ( ABS(sstop[-1]) == ss[0]-1 ) break;
                    }
                    goto NoMnot;
                case DUMMYINDEX_:
                    if ( type != INDTOIND ||
                    newnumber < AM.IndDum || newnumber >= AM.IndDum+MAXDUMMIES ) goto NoMnot;
                    break;
                case VECTOR_:
                    if ( type != VECTOVEC ) goto NoMnot;
                    break;
                default:
                    goto NoMnot;
            }
Mnot:
            if ( notflag ) goto NoM;
            return(0);
NoMnot:
            if ( !notflag ) goto NoM;
            return(0);
        }
        else if ( Sets[j].type == CRANGE ) {
            if ( ( type == SYMTONUM )
            || ( type == INDTOIND && ( newnumber > 0
             && newnumber <= AM.OffsetIndex ) ) ) {
                if ( Sets[j].first < MAXPOWER ) {
                    if ( newnumber >= Sets[j].first ) goto NoMnot;
                }
                else if ( Sets[j].first < 3*MAXPOWER ) {
                    if ( newnumber+2*MAXPOWER > Sets[j].first ) goto NoMnot;
                }
                if ( Sets[j].last > -MAXPOWER ) {
                    if ( newnumber <= Sets[j].last ) goto NoMnot;
                }
                else if ( Sets[j].last > -3*MAXPOWER ) {
                    if ( newnumber-2*MAXPOWER < Sets[j].last ) goto NoMnot;
                }
```

```
                goto Mnot;
            }
            goto NoMnot;
        }
/*
        Now we have to determine which set element
*/
        w = SetElements + Sets[j].first;
        m = SetElements + Sets[j].last;
        if ( w == m ) {
/*
            The set is empty! This is not a match unless we have !{}, in
            which case it is.
*/
            if ( notflag ) return 0;
            AN.oldtype = old2;
            AN.oldvalue = oldval;
            goto NoMatch;
        }

        if ( ( Sets[j].flags & ORDEREDSET ) == ORDEREDSET ) {
/*
            We search first and ask questions later
*/
            i = BinarySearch(w,Sets[j].last-Sets[j].first,newnumber);
            if ( i < 0 ) {  /* no matter what, it is not in the set. */
                goto NoMnot;
            }
            else {
/*
                We can set the proper parameters now to make only the
                checks for the given set element.
                After that we jump into the appropriate loop.
*/
                w = m = SetElements + i;
                i++;
                if ( Sets[j].type == -1 || Sets[j].type == CNUMBER ) {
                    goto insideloop1;
                }
                else {
                    goto insideloop2;
                }
            }
        }
        i = 1;
        if ( Sets[j].type == -1 || Sets[j].type == CNUMBER ) {
          do {
            insideloop1:
            if ( notflag ) {
                switch ( type ) {
                    case SYMTOSYM:
                        if ( Sets[j].type == CNUMBER ) {}
                        else {
                            if ( *w == newnumber ) goto NoMatch;
                        }
                        break;
                    case SYMTONUM:
                    case INDTOIND:
                        if ( *w == newnumber ) goto NoMatch;
                        break;
                    default:
                        break;
                }
            }
            else if ( type != SYMTONUM && type != INDTOIND
            && type != SYMTOSYM ) goto NoMatch;
            else if ( type == SYMTOSYM && Sets[j].type == CNUMBER ) goto NoMatch;
            else if ( *w == newnumber ) {
                if ( *s == SETTONUM ) {
                    if ( n2 == oldnumber && type
                    <= SYMTOSUB ) goto NoMatch;
                    m = AT.WildMask;
                    w = AN.WildValue;
                    n = AN.NumWild;
                    while ( --n >= 0 ) {
                        if ( w[2] == n2 && *w <= SYMTOSUB ) {
                            if ( !*m ) {
                                *w = SYMTONUM;
                                w[3] = i;
                                AN.WildReserve = 1;
                                return(0);
                            }
                            if ( *w != SYMTONUM )
                                goto NoMatch;
                            if ( w[3] == i ) return(0);
                            i = w[3];
                            j = (SetElements + Sets[j].first)[i];
```

```
                            if ( j == n2 ) return(0);
                            goto NoMatch;
                        }
                        m++; w += w[1];
                    }
                }
                else if ( n2 >= 0 ) {
                    *newval = *(w - Sets[j].first + Sets[n2].first);
                    if ( *newval > MAXPOWER ) *newval -= 2*MAXPOWER;
                    if ( dirty && *newval != oldval ) {
                        *newval = oldval; goto NoMatch;
                    }
                }
                return(0);
            }
            i++;
          } while ( ++w < m );
        }
        else {
          do {
            insideloop2:
            inset = *w;
            if ( notflag ) {
            switch ( type ) {
                case SYMTONUM:
                case SYMTOSYM:
                    if ( ( type == SYMTOSYM && *w == newnumber )
                    || ( type == SYMTONUM && *w-2*MAXPOWER == newnumber ) ) {
                        goto NoMatch;
                    }
                    /* fall through */
                case SYMTOSUB:
                    if ( *w < 0 ) {
                        WORD *mm = AT.WildMask, *mmm, *part;
                        WORD *ww = AN.WildValue;
                        WORD nn = AN.NumWild;
                        k = -*w;
                        while ( --nn >= 0 ) {
                            if ( *mm && ww[2] == k && ww[0] == type ) {
                                if ( type != SYMTOSUB ) {
                                    if ( ww[3] == newnumber ) goto NoMatch;
                                }
                                else {
                                    mmm = C->rhs[ww[3]];
                                    nn = *newval-2;
                                    part = newval+2;
                                    if ( (C->rhs[ww[3]+1]-mmm-1) == nn ) {
                                        while ( --nn >= 0 ) {
                                            if ( *mmm != *part ) {
                                                mmm++; part++; break;
                                            }
                                            mmm++; part++;
                                        }
                                        if ( nn < 0 ) goto NoMatch;
                                    }
                                }
                                break;
                            }
                            mm++; ww += ww[1];
                        }
                    }
                    break;
                case VECTOMIN:
                    if ( type == VECTOMIN ) {
                        if ( inset >= AM.OffsetVector ) { i++; continue; }
                        inset += WILDMASK;
                    }
                    /* fall through */
                case VECTOVEC:
                    if ( inset == newnumber ) goto NoMatch;
                    /* fall through */
                case VECTOSUB:
                    if ( inset - WILDOFFSET >= AM.OffsetVector ) {
                        WORD *mm = AT.WildMask, *mmm, *part;
                        WORD *ww = AN.WildValue;
                        WORD nn = AN.NumWild;
                        k = inset - WILDOFFSET;
                        while ( --nn >= 0 ) {
                            if ( *mm && ww[2] == k && ww[0] == type ) {
                                if ( type == VECTOVEC ) {
                                    if ( ww[3] == newnumber ) goto NoMatch;
                                }
                                else {
                                    mmm = C->rhs[ww[3]];
                                    nn = *newval-2;
                                    part = newval+2;
                                    if ( (C->rhs[ww[3]+1]-mmm-1) == nn ) {
```

```
                            while ( --nn >= 0 ) {
                                if ( *mmm != *part ) {
                                    mmm++; part++; break;
                                }
                                mmm++; part++;
                            }
                            if ( nn < 0 ) goto NoMatch;
                        }
                    }
                    break;
                }
                mm++; ww += ww[1];
            }
        }
        break;
    case INDTOIND:
        if ( *w == newnumber ) goto NoMatch;
        /* fall through */
    case INDTOSUB:
        if ( *w - (WORD)WILDMASK >= AM.OffsetIndex ) {
            WORD *mm = AT.WildMask, *mmm, *part;
            WORD *ww = AN.WildValue;
            WORD nn = AN.NumWild;
            k = *w - WILDMASK;
            while ( --nn >= 0 ) {
                if ( *mm && ww[2] == k && ww[0] == type ) {
                    if ( type == INDTOIND ) {
                        if ( ww[3] == newnumber ) goto NoMatch;
                    }
                    else {
                        mmm = C->rhs[ww[3]];
                        nn = *newval-2;
                        part = newval+2;
                        if ( (C->rhs[ww[3]+1]-mmm-1) == nn ) {
                            while ( --nn >= 0 ) {
                                if ( *mmm != *part ) {
                                    mmm++; part++; break;
                                }
                                mmm++; part++;
                            }
                            if ( nn < 0 ) goto NoMatch;
                        }
                    }
                    break;
                }
                mm++; ww += ww[1];
            }
        }
        break;
    case FUNTOFUN:
        if ( *w == newnumber ) goto NoMatch;
        if ( ( type == FUNTOFUN &&
        ( k = *w - WILDMASK ) > FUNCTION ) ) {
            WORD *mm = AT.WildMask;
            WORD *ww = AN.WildValue;
            WORD nn = AN.NumWild;
            while ( --nn >= 0 ) {
                if ( *mm && ww[2] == k && ww[0] == type ) {
                    if ( ww[3] == newnumber ) goto NoMatch;
                    break;
                }
                mm++; ww += ww[1];
            }
        }
    default:
        break;
}
}
else {
  if ( type == VECTOMIN ) {
    if ( inset >= AM.OffsetVector ) { i++; continue; }
    inset += WILDMASK;
  }
  if ( ( inset == newnumber && type != SYMTONUM ) ||
  ( type == SYMTONUM && inset-2*MAXPOWER == newnumber ) ) {
    if ( *s == SETTONUM ) {
        if ( n2 == oldnumber && type
        <= SYMTOSUB ) goto NoMatch;
        m = AT.WildMask;
        w = AN.WildValue;
        n = AN.NumWild;
        while ( --n >= 0 ) {
            if ( w[2] == n2 && *w <= SYMTOSUB ) {
                if ( !*m ) {
                    *w = SYMTONUM;
                    w[3] = i;
                    AN.WildReserve = 1;
```

```
                            return(0);
                        }
                        if ( *w != SYMTONUM )
                            goto NoMatch;
                        if ( w[3] == i ) return(0);
                        i = w[3];
                        j = (SetElements + Sets[j].first)[i];
                        if ( j == n2 ) return(0);
                        goto NoMatch;
                    }
                    m++; w += w[1];
                }
            }
            else if ( n2 >= 0 ) {
                *newval = *(w - Sets[j].first + Sets[n2].first);
                if ( *newval > MAXPOWER ) *newval -= 2*MAXPOWER;
                if ( dirty && *newval != oldval ) {
                    *newval = oldval; goto NoMatch;
                }
            }
            return(0);
          }
        }
        i++;
      } while ( ++w < m );
    }
    if ( notflag ) return(0);
    AN.oldtype = old2; AN.oldvalue = oldval; goto NoMatch;
  }
  else { return(0); }

NoM:
  AN.oldtype = old2; AN.oldvalue = w[3]; goto NoMatch;
}

/*
      #] CheckWild :
  #] Wildcards :
  #[ DenToFunction :

  Renames the denominator function into a function with the given number.
  For the syntax see   Denominators,function;
*/

int DenToFunction(WORD *term, WORD numfun)
{
  int action = 0;
  WORD *t, *tstop, *tnext, *arg, *argstop, *targ;
  t = term+1;
  tstop = term + *term; tstop -= ABS(tstop[-1]);
  while ( t < tstop ) {
    if ( *t == DENOMINATOR ) {
      *t = numfun; t[2] |= DIRTYFLAG; action = 1;
    }
    tnext = t + t[1];
    if ( *t >= FUNCTION && functions[*t-FUNCTION].spec <= 0 ) {
      arg = t + FUNHEAD;
      while ( arg < tnext ) {
        if ( *arg > 0 ) {
          targ = arg + ARGHEAD; argstop = arg + *arg;
          while ( targ < argstop ) {
            if ( DenToFunction(targ,numfun) ) {
              arg[1] |= DIRTYFLAG; t[2] |= DIRTYFLAG; action = 1;
            }
            targ += *targ;
          }
          arg = argstop;
        }
        else if ( *arg <= -FUNCTION ) arg++;
        else arg += 2;
      }
    }
    t = tnext;
  }
  return(action);
}

/*
  #] DenToFunction :
*/
```

Index